Any3D-VLA: Enhancing VLA Robustness via Diverse Point Clouds
Xianzhe Fan, Shengliang Deng, Xiaoyang Wu, Yuxiang Lu, Zhuoling Li, Mi Yan, Yuji a Zhang, Zhizheng Zhang, He Wang, Hengshuang Zhao
Existing Vision-Language-Action (VLA) models typically take 2D images as visual input, which limits their spatial understanding in complex scenes. How can we in corporate 3D information to enhance VLA capabilities? We conduct a pilot study a cross different observation spaces and visual representations. The results show that explicitly lifting visual input into point clouds yields representations th at better complement their corresponding 2D representations. To address the chal lenges of (1) scarce 3D data and (2) the domain gap induced by cross-environment differences and depth-scale biases, we propose Any3D-VLA. It unifies the simula tor, sensor, and model-estimated point clouds within a training pipeline, constr ucts diverse inputs, and learns domain-agnostic 3D representations that are fuse d with the corresponding 2D representations. Simulation and real-world experimen ts demonstrate Any3D-VLA's advantages in improving performance and mitigating th e domain gap. Our project homepage is available at https://xianzhefan.github.io/ Any3D-VLA.github.io.
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Addressing Instrument-Outcome Confounding in Mendelian Randomization through Rep resentation Learning
Shimeng Huang, Matthew R Robinson, Francesco Locatello
Mendelian Randomization (MR) is a prominent observational epidemiological research method, designed to address unobserved confounding when estimating causal effects. It is closely related to instrumental variable (IV) methods, where genetic variants serve as instruments to infer causal relationships from observational data. However, the core assumptions required for valid IV analysis---particularly the independence between instruments and unobserved confounders---are untestable and often violated in practice. In MR, such violations commonly arise when genetic variants are correlated with environmental factors (e.g., population stratificat ion
and assortive mating), leading to confounding between instruments and outcomes. At the same time, MR studies increasingly include data collected across multiple environments or populations, providing an opportunity to address these violation s.
Leveraging this setting, we propose a representation learning framework that exploits multi-environment data to recover latent exogenous components of genetic instruments suitable for causal inference. We provide theoretical insigh ts
into when and how the learned components can act as valid instruments, and we demonstrate the effectiveness of our approach through simulations and semi-synthetic experiments using genetic data from the All of Us Biobank.
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AVI-Bench: Toward Human-like Audio-Visual Intelligence of Omni-MLLMs
Yaoting Wang, Ziyi Zhang, Wenming Tu, Shaoxuan Xu, Wenjie Du, Cheng Liang, Weiju n Wang, Yuanchao Li, Guangyao Li, Hao Fei, Yuanchun Li, Henghui Ding, Yunxin Liu
Recent advances in Omni-Multimodal Large Language Models (Omni-MLLMs) have enabl ed strong integration of vision, audio, and language. However, their audio-visua l intelligence (AVI) remains insufficiently evaluated due to the lack of systema tic and comprehensive benchmarks.
We introduce **AVI-Bench**, a cognitively inspired benchmark that evaluates Omni -MLLMs across three stages, perception, understanding, and reasoning, through cr oss-modal tasks requiring joint audio-visual interpretation. This design enables fine-grained diagnosis of model capabilities and failure modes.
To further assess robustness beyond familiar domains, we propose **AVI-Bench-Pri Se**, an extension that probes models' primitive audio-visual sensation using un familiar, low-semantic stimuli, testing generalization beyond common training di stributions.
Extensive experiments on both open-source and closed-source models reveal substa ntial limitations in current Omni-MLLMs. Based on these findings, we present a *

*four-level AVI taxonomy**. Overall, AVI-Bench provides a principled evaluation framework to guide the development of more robust and generalizable AVI. Project website: https://fudancvl.github.io/AVI-Bench
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OptMaster: A DAG-Based Framework for Formulation and Heuristic Discovery in Optimization

Hang Lin, Yuanpeng Gao, Yuzhi Zhang, Kun Yuan, Gang Yan, Siheng Chen, Linfeng Zhang, Weinan E

Optimization problems are fundamental across science and industry, including planning, scheduling, and resource allocation. While LLMs show promise in automating optimization, they struggle to bridge the gap between real-world requirements and both mathematical formulations and effective heuristic designs. Furthermore, the field lacks a unified framework that spans problem formulation and heuristic discovery for NP-hard settings. To address these challenges, we propose OptMaster, a unified framework that spans optimization from formulation to heuristic discovery, structuring the process as a Directed Acyclic Graph (DAG) where each node represents a candidate solution. The DAG architecture enables cross-branch knowledge transfer when search progress stagnates. Within each node, we further replace textual self-reflection with independently generated verification code, grounding the evaluation in deterministic computation to suppress hallucinations. OptMaster achieves competitive performance across two optimization paradigms. In Formulation Intelligence, OptMaster achieves state-of-the-art accuracy across the three most challenging benchmarks in the field. In Heuristic Discovery, OptMaster surpasses the best known solutions on Circle Packing ($n=26, 32$) and achieves a cut of 9,590 on Gset70 with significantly reduced time and search budgets.
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From Extrinsic to Intrinsic: Geodesic-Guided Representation Learning for 3D Geometric Data

Yuming Zhao, Junhui Hou, Qijian Zhang, Jia Qin, Ying He

Geometric analysis fundamentally distinguishes between extrinsic and intrinsic perspectives. The dominant paradigm in current 3D representation learning relies on either extrinsic spatial structures or high-level semantics, struggling to capture the essence of shape identity and underlying manifold topology. To bridge this gap, we introduce a novel 3D representation learning paradigm, namely PRISM, for Pre-training, which learns isometric embeddings by Recovering the Intrinsic Surface geodesic Metric. PRISM incorporates a topology-enforcing objective that explicitly constrains the structure of latent space, alongside a specialized two-stage training recipe mitigating sample imbalance inherent in the distribution of geodesic distances. Experiments demonstrate that our approach shows satisfactory accuracy, robustness, and high efficiency in geodesic distance prediction and achieves superior performance across diverse downstream tasks, including shape recognition, surface parameterization, and non-rigid correspondence. Our code will be made publicly available.
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Reliable Confidence Alignment for Generalized Category Discovery

Jiawei Yu, Zijian Gao, Tianjiao Wan, Xuan Liu, Cheng Yang, Kele Xu

Generalized Category Discovery (GCD) requires models to categorize an unlabeled pool containing known and novel classes under sparse supervision. We identify a systemic confidence bias inherent in existing parametric methods: while entropy regularization prevents class collapse, it indiscriminately suppresses predictive certainty on all unlabeled instances. This bias drives a distributional wedge between labeled and unlabeled samples of the same category, forcing models to sacrifice their stability on known classes to achieve plasticity for new ones. To resolve this, we propose Reliable Confidence Alignment (RCA), a plug-and-play framework grounded in Evidential Deep Learning. RCA first establishes high certainty anchors on labeled data using a Reliable Anchor for Certainty (RAC) module. Then, we introduce Cross-view Confidence Alignment (CCA) to propagate this grounded reliability to the unlabeled discovery set. Thus, RCA captures the fine-grained geometry of the probability simplex, effectively calibrating the model's epis

temic uncertainty. Extensive evaluations on coarse- and fine-grained benchmarks demonstrate that RCA effectively rectifies the confidence landscape, significantly mitigating performance decay on known classes without compromising novel-class discovery.
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Grouter: Decoupling Routing from Representation for Accelerated MoE Training
Yuqi Xu, Rizhen Hu, Zihan Liu, Mou Sun, Kun Yuan
Traditional Mixture-of-Experts (MoE) training typically proceeds without any structural priors, effectively requiring the model to simultaneously train expert weights while searching for an optimal routing policy within a vast combinatorial space. This entanglement often leads to sluggish convergence and training instabilities. This paper introduces Grouter a preemptive routing method that by distilling high-quality structures from fully-trained MoE models and serving as a fixed router for target models. By decoupling structural optimization from weight updates, Grouter significantly accelerates both the speed and quality of model convergence. To ensure the framework's versatility, we also introduce expert folding to adapt Grouter across varying model configurations and expert tuning to rebalance workloads across different data distributions. Furthermore, by leveraging the structural priors provided by preemptive routing, we can implement targeted optimizations to further enhance training throughput. Experiments demonstrate that Grouter achieves superior performance and efficiency which boosts pre-training data utilization by $\textbf{4.28$\times$}$ and achieves up to $\textbf{33.5$\%$}$ throughput acceleration, establishing preemptive routing as a fundamental paradigm for scalable MoE training. We publicly release our code and pretrained Grouter checkpoints.
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ScoreMix: Synthetic Data Generation by Score Composition in Diffusion Models Improves Recognition
Parsa Rahimi, Sébastien Marcel
Synthetic data generation is increasingly used in machine learning for **training and data augmentation**. Yet, many current strategies rely on external foundation models or datasets, which can be restricted by policy or legal constraints, especially for sensitive modalities such as human face images and videos. We propose **ScoreMix**, a **self-contained data augmentation** method to boost recognition performance by leveraging score compositionality in class-conditioned diffusion models. ScoreMix mixes class-conditioned scores along reverse diffusion trajectories, yielding domain-specific hard augmentations without external resources. We systematically study class-selection strategies and find that mixing classes that are distant in the discriminator embedding space yields larger gains, providing **up to 3\% additional average improvement across benchmarks** over proximity-based selection. Interestingly, we observe that learned condition and embedding spaces are largely uncorrelated under standard alignment metrics, and that condition-space distances are weakly correlated to downstream gains. Across **8 public face recognition benchmarks**, ScoreMix improves accuracy by **up to 7 percentage points** without hyperparameter search, highlighting robustness and practicality. Project page: https://parsa-ra.github.io/scoremix/ .
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Attention's forward pass and Frank-Wolfe
Albert Alcalde, Borjan Geshkovski, Domènec Ruiz-Balet
We analyze the hardmax limit of self-attention dynamics for token embeddings in the zero-temperature regime $(\beta \to +\infty)$ and relate it to finite-β behavior. In this limit, the update rule can be viewed as a Frank-Wolfe step for a quadratic objective over the convex hull of the current tokens. When the key-query matrix is negative semidefinite, the dynamics converge with the standard sublinear rate $\mathcal{O}(t^{-1})$ on the quadratic energy, whereas in the positive semidefinite case, extending the hardmax rule to the convex hull induces a Voronoi structure: vertices are stationary, interior points remain in their initial cells, and each token moves along a straight line toward its cell's vertex with exponential convergence under a step-size bounded away from zero. We additionally establish well-posedness of the associated ODE limit in this regime. For

finite β, we model self-attention as a Markov chain and prove *dynamic metastability*: interior tokens reach near-vertex configurations in a constant number of steps and remain trapped for times exponential in β with high probability, before eventual collapse to some point within the initial convex hull. Thus, hardmax dynamics accurately approximate the finite-β process over exponentially long time horizons.
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Why Specialist Models Still Matter: A Heterogeneous Multi-Agent Paradigm for Medical Artificial Intelligence
Yanan Wang, Shuaicong Hu, Jian Liu, Guohui Zhou, Aiguo Wang, Cuiwei Yang
The impressive performance of generalist large language models (LLMs) such as GPT and Claude in healthcare raises a critical question: will domain-specific medical specialist models become obsolete? We argue that the future of medical artificial intelligence (AI) lies not in building monolithic medical foundation models, nor in replacing human expertise, but in orchestrating collaboration among generalist LLMs, domain-specific specialist models, and clinicians. We propose HetMedAgent, a heterogeneous medical multi-agent framework that enables conflict-aware evidence fusion, uncertainty-based clinician intervention triggering, and adaptive threshold calibration. Experiments on three real-world clinical decision-making tasks demonstrate that the synergy between generalist LLMs and domain-specific specialist models significantly outperforms using either type of model alone, validating the irreplaceable value of specialist models in modality-specific analysis. HetMedAgent represents a shift from building medical LLMs or foundation models to multi-agent collaboration, achieving a balance between general reasoning capabilities and domain-specific precision.
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MSP: Probabilistically Consistent Multi-Scale Action Generation
Zhixuan Lin, Gengqi Liu, Chao Zheng, Gao Lin, Jindong Yu, Song Gao, Fei Wang
In robotic imitation learning, accurately modeling the multimodality and temporal correlations of long-horizon action sequences remains challenging. Long-horizon tasks require preserving global task intent while executing precise low-level control; otherwise, local errors can accumulate and lead to failure. While recent coarse-to-fine autoregressive models have improved action generation, they struggle to maintain consistency across hierarchies, leading to suboptimal performance in long-horizon tasks. To address these shortcomings, we propose Probabilistically Consistent Multi-Scale Action Generation (MSP), a novel coarse-to-fine approach that promotes cross-scale consistency. MSP adopts a streamlined multi-scale design by directly downsampling in a continuous latent space. A scale-wise autoregressive Transformer is used to generate semantic conditions at each scale, which guide a lightweight MeanFlow model to capture multi-scale latent distributions, enabling probabilistically consistent refinement across scales. Through extensive simulation and real-world experiments, including long-horizon, multi-task, and few-shot generalization settings, we show that MSP outperforms existing coarse-to-fine methods, achieving state-of-the-art performance with high efficiency.
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ROAMM: A Benchmark Dataset for Multimodal Human Attention Decoding and EEG-to-Text Modeling During Naturalistic Reading
Haorui Sun, Ardyn Vivienne Olszko, Niharika Singh, David C. Jangraw
We present Reading Observed At Mindless Moments (ROAMM), a multimodal dataset comprising 50 hours of simultaneous EEG and eye-tracking recordings collected during naturalistic multi-page reading from 44 participants. ROAMM includes synchronized physiological recordings, eye-movement events, page-level comprehension scores, and span-level mind-wandering (MW) annotations obtained using a retrospective self-report paradigm. We introduce a standardized leave-one-subject-out benchmark for MW detection and achieve up to 0.609 AUROC using supervised models. We additionally evaluate EEG-to-text decoding on reading segments with and without MW labels, showing that decoding performance decreases during MW episodes. ROAMM enables research on MW detection, EEG-to-text decoding, multimodal representation learning, and attention-related degradation of language representations durin

g naturalistic reading.
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DynaSchedBench: Calibrated Dynamic Scheduling Benchmarks and Observability Paradox in LLM-based Scheduling Agents
Shijie Cao, Yuan Yuan, Jing Liu
Progress in neural combinatorial optimization for Dynamic Flexible Job Shop Scheduling Problem (DFJSP) is currently hindered by a methodological tension: static benchmarks encourage benchmark overfitting, while uncalibrated generators obscure algorithmic capability with stochastic noise.
To resolve this, we introduce \textbf{DynaSchedBench}, a diagnostic framework for DFJSP that rigorously controls the instance-generation process.
Instead of relying on parameter sampling, our approach utilizes Sequential Event-Space Calibrator (SESC) that computes a novel Schedule Stress Index (SSI) to stratify instances by difficulty.
We demonstrate that SESC is substantially more computationally efficient than evolutionary baselines while converging reliably to the target metrics.
The framework integrates modular components for instance generation, snapshot-based simulation, agents, evaluation, and visualization, thereby enabling rigorous testing of reactive and lookahead-based policies.
Leveraging this calibrated environment, we identify key limitations of LLM-based scheduling agents.
Specifically, in step-wise online decision-making for dynamic scheduling, we identify an ``Observability Paradox'': providing agents with oracle access to full structural information can degrade policy performance, underperforming concise information.
Furthermore, despite substantial token overhead, tool-augmented and refinement strategies fail to reliably improve performance, and most LLM agents fail to consistently surpass strong dispatching baselines—behaving more like robust heuristic approximators than superior optimizers.
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Motion-Residual Conflict-Aware Time Reversal for Generative Inbetweening
Zhenbang Zhang, Zihui Cui, Haythem El-Messiry, Renmin Han, zhiqiang xu
Image-to-video (I2V) diffusion models have recently made generative inbetweening a practical reality by synthesizing semantically plausible intermediate frames between two keyframes. Among them, inference-time sampling schemes that re-use large pre-trained I2V backbones without any additional training are especially attractive. Yet current methods frequently exhibit temporal inconsistency and artifacts such as ghosting or reverse motion. A key reason is that the two trajectories are driven by distinct motion priors, each inherited from its own conditioning frame, and are simply stitched together without explicitly reconciling these priors. We introduce Motion-Residual Conflict-Aware Time Reversal (MR■CATR), an inference-time sampling framework that aligns conflicting motion priors instead of discarding one of them or collapsing to a single start-conditioned prior. MR■CATR first derives a motion-residual–based direction from the forward path, combined with an end-conditioned residual to form a consensus motion axis. This design suppresses bidirectional motion conflicts while still allowing end-frame information to refine the trajectory and enforce endpoint consistency. MR■CATR can be seamlessly integrated into existing time-reversal samplers without changing model parameters. Experiments on generative inbetweening benchmarks show that our method produces videos with smoother motion, fewer artifacts, and consistently better quantitative scores and user preferences than prior strategies.
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ForesightKV: Optimizing KV Cache Eviction for Reasoning Models by Learning Long-Term Contribution
Zican Dong, Peiyu Liu, Junyi Li, Zhipeng Chen, Han Peng, Shuo Wang, Xin Zhao
Recently, large language models (LLMs) have shown remarkable reasoning abilities by producing long reasoning traces. However, as the sequence length grows, the key-value (KV) cache expands linearly, incurring significant memory and computation costs. Existing KV cache eviction methods mitigate this issue by discarding less important KV pairs, but often fail to capture complex KV dependencies, resu

lting in performance degradation. To better balance efficiency and performance, we introduce ForesightKV, a training-based KV cache eviction framework that learns to predict which KV pairs to evict during long-text generations.
We first design the Golden Eviction algorithm, which identifies the optimal eviction KV pairs at each step using future attention scores. These traces and the scores at each step are then distilled via supervised training with a Pairwise Ranking Loss. Furthermore, we formulate cache eviction as a Markov Decision Process and apply the GRPO algorithm to mitigate the significant language modeling loss increase on low-entropy tokens.
Experiments on AIME2024 and AIME2025 benchmarks of three reasoning models demonstrate that ForesightKV consistently outperforms prior methods under only half the cache budget, while benefiting synergistically from both supervised and reinforcement learning approaches. Code is available at \url{https://github.com/RUCAIBox/ForesightKV}.
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Policies Permitting LLM Use for Polishing Peer Reviews Are Currently Not Enforceable
Rounak Saha, Gurusha Juneja, Dayita Chaudhuri, Naveeja Sajeevan, Nihar B Shah, Danish Pruthi
A number of scientific conferences and journals have recently enacted policies that prohibit LLM usage by peer reviewers, except for polishing, paraphrasing, and grammar correction of otherwise human-written reviews. But, are these policies enforceable? To answer this question, we assemble a dataset of peer reviews simulating multiple levels of human-AI collaboration, and evaluate five state-of-the-art detectors, including two commercial systems. Our analysis shows that all detectors misclassify a non-trivial fraction of LLM-polished reviews as AI-generated, thereby risking false accusations of academic misconduct. We further investigate whether peer-review-specific signals, including access to the paper manuscript and the constrained domain of scientific writing, can be leveraged to improve detection. While incorporating such signals yields measurable gains in some settings, we identify limitations in each approach and find that none meets the accuracy standards required for identifying AI use in peer reviews. Importantly, our results suggest that recent public estimates of AI use in peer reviews through the use of current AI-text detectors should be interpreted with caution, as they misclassify mixed reviews (collaborative human-AI outputs) as fully AI generated, potentially overstating the extent of policy violations.
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Q-CLIP: Unleashing the Power of Vision-Language Models for Video Quality Assessment through Unified Cross-Modal Adaptation
Yachun Mi, Yu Li, Yanting Li, Chen Hui, Tong Zhang, Zhixuan Li, Chenyue Song, Wei Yang Bryan Lim, Shaohui Liu
Accurate and efficient Video Quality Assessment (VQA) has long been a key research challenge. Current mainstream VQA methods typically improve performance by pretraining on large-scale classification datasets, followed by fine-tuning on VQA datasets. However, this strategy presents two significant challenges: (1) merely transferring semantic knowledge learned from pretraining is insufficient for VQA, as video quality depends on multiple factors (e.g., semantics, distortion, motion); (2) pretraining on large-scale datasets demands enormous computational resources, often dozens to hundreds of times more than training on VQA datasets. Recently, Contrastive Vision-Language Models (CVLMs) have shown strong generalization across visual tasks and promising potential for quality assessment. In this work, we propose Q-CLIP, the first fully CVLMs-based framework for VQA. Q-CLIP enhances both visual and textual representations through a Shared Cross-Modal Adapter (SCMA), which contains only a minimal number of trainable parameters and is the only component that requires training. This design significantly reduces computational cost. In addition, we introduce a set of five learnable quality-level prompts to guide the CVLMs in perceiving subtle quality variations. Furthermore, we investigate the impact of different frame sampling strategies on VQA performance. Extensive experiments demonstrate that Q-CLIP exhibits excellent performance on several VQA datasets. Code: https://github.com/xiao-mi-d/Q-CLIP

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Verbalized Sampling: How to Mitigate Mode Collapse and Unlock LLM Diversity
Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael Tomz, Christopher D Manning, Weiyan Shi
Post-training alignment often reduces LLM diversity, leading to a phenomenon known as mode collapse. Unlike prior work that attributes this effect to algorithmic limitations, we identify a fundamental, pervasive data-level driver: typicality bias in preference data, whereby annotators systematically favor familiar text as a result of well-established findings in cognitive psychology. We formalize this bias theoretically, verify it empirically on preference datasets, and show that it plays a central role in mode collapse. Motivated by this analysis, we introduce Verbalized Sampling (VS), a simple, training-free prompting strategy to circumvent mode collapse. VS prompts the model to verbalize a probability distribution over a set of responses (e.g., "Generate 5 jokes about coffee and their corresponding probabilities"), which relieves the pressure to produce a single "typical" answer. Experiments show that VS significantly improves performance across creative writing (poems, stories, jokes), social dialogue simulation, synthetic data generation, and open-ended QA, without sacrificing safety and factual accuracy. For instance, in creative writing, VS increases diversity by 1.6-2.1x compared to direct prompting. We further observe an emergent trend that more capable models benefit more from VS. In sum, our work provides a new data-centric perspective on mode collapse and a practical inference-time remedy that helps unlock pre-trained generative diversity.
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Robust Bayesian Optimisation with Unbounded Corruptions
Abdelhamid Ezzerg, Ilija Bogunovic, Jeremias Knoblauch
Bayesian Optimization is critically vulnerable to extreme outliers. Existing provably robust methods typically assume a bounded cumulative corruption budget, which makes them defenseless against even a single corruption of sufficient magnitude. To address this, we introduce a new adversary whose budget is only bounded in the frequency of corruptions, not in their magnitude. We then derive RCGP-UCB, an algorithm coupling the upper confidence bound (UCB) approach with a Robust Conjugate Gaussian Process (RCGP). We present stable and adaptive versions of RCGP-UCB, and prove that they achieve sublinear regret in the presence of up to $O(T^{1/4})$ and $O(T^{1/7})$ corruptions with possibly infinite magnitude. This robustness comes at near zero cost: without outliers, RCGP-UCB's regret bounds match those of the standard GP-UCB algorithm.
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Quantile-Free Uncertainty Quantification in Graph Neural Networks
Soyoung Park, Hwanjun Song, Sungsu Lim
Uncertainty quantification (UQ) in graph neural networks (GNNs) is crucial in high-stakes domains but remains a significant challenge. In graph settings, message passing often relies on strong assumptions such as exchangeability, which are rarely satisfied in practice. Moreover, achieving reliable UQ typically requires costly resampling or post-hoc calibration. To address these issues, we introduce Quantile-free Prediction Interval GNN (QpiGNN), a framework that builds on quantile regression (QR) to enable GNN-based UQ by directly optimizing coverage and interval width without requiring quantile inputs or post-processing. QpiGNN employs a dual-head architecture that decouples prediction and uncertainty, and is trained with label-only supervision through a quantile-free joint loss. This design allows efficient training and yields robust prediction intervals, with theoretical guarantees of asymptotic coverage and near-optimal width under mild assumptions. Experiments on 19 synthetic and real-world benchmarks show QpiGNN achieves average 22% higher coverage and 50% narrower intervals than baselines, while ensuring efficiency and robustness to noise and structural shifts.
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Solving Positive Linear Programs with Differential Privacy
Alina Ene, Huy Nguyen, Ta Duy Nguyen, Adrian Vladu
We study differentially private approximation algorithms for positive linear programs (LPs with nonnegative coefficients and variables), focusing on the fundame

ntal families of packing, covering, and mixed packing-covering formulations. We focus on the high-sensitivity, constraint-private regime of Hsu-Roth-Roughgarden-Ullman (ICALP 2014), where neighboring instances may differ by an arbitrary single constraint, so one cannot hope to approximately satisfy every constraint under privacy. We give private solvers that return approximate solutions while violating only a controlled number of constraints. Our algorithms improve the prior instance-dependent guarantees, and also yield new data-independent bounds that depend only on the dimension. Our techniques involve a dense multiplicative weights update method developed from a regularized dual viewpoint, which we analyze in a way that exploits structure specific to positive LPs.

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Optimizing Inference-Time Compute for Medical Reasoning via Uncertainty Quantification

Shaohao Rui, Kaitao Chen, Weijie Ma, Xiaosong Wang

Extended Chain-of-Thought (CoT) reasoning has significantly bolstered the capabilities of medical large language models (LLMs). However, current models exhibit static computational expenditure, applying lengthy reasoning processes indiscriminately to both simple queries and complex diagnostic cases. This inefficiency is particularly prohibitive in real-world healthcare, where clinical scenarios range from time-sensitive emergencies requiring rapid response to intricate pathologies demanding deep analysis. To address this, we propose **AdaThink-Med**, an end-to-end framework for adaptive reasoning via uncertainty-guided length calibration. Although the underlying mechanism is generalizable, we demonstrate its critical value in the medical domain, where balancing inference latency with diagnostic precision is paramount. AdaThink-Med leverages entropy-based uncertainty estimation within reinforcement fine-tuning to dynamically shape reward signals: it penalizes verbosity for high-confidence correct answers (e.g., straightforward knowledge retrieval) while incentivizing extended exploration for uncertain or ambiguous scenarios. Across six medical benchmarks, AdaThink-Med reduces inference token consumption by **4.7× to 6.4×** on Qwen and Llama architectures, respectively, with minimal performance trade-offs. Notably, our reward design naturally produces distinct "non-thinking" and "thinking" modes within a single model, enabling efficient allocation of computational resources without any external router or classifier.

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The Forgetting-Retention Dilemma: Certified Unlearning Theory in Continual Learning

Yiting Hu, Lingjie Duan, Qian Zhang

Machine unlearning aims to eliminate the influence of specific data from trained models to safeguard privacy. However, this presents a significant challenge in the context of continual learning (CL), where models update sequentially on dynamic datasets. A major limitation is that current certified unlearning algorithms fail to account for the complex, cumulative model evolution inherent to CL framework. In this work, we establish the first theoretical foundation bridging CL and machine unlearning. We formulate the CL's unlearning objective as the minimization of post-unlearning excess risk, which decomposes into CL excess risk and unlearning loss, characterizing the fundamental trade-off between preserving historical knowledge and targeted forgetting. Under mild assumptions, we first establish an upper bound for the CL excess risk in non-convex models. We then adapt two certified unlearning approaches, gradient-based and Hessian-based, to the CL framework. Our analysis reveals that while the gradient-based approach is less effective than the Hessian-based method in minimizing unlearning loss, it offers the distinct advantage of nearly zero storage overhead for enabling unlearning. This insight motivates a hybrid strategy that reduces storage costs while maintaining post-unlearning performance. Experimental results further validate our theoretical findings.

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Diffusion Flow Matching: Dimension-Improved KL Bounds and Wasserstein Guarantees

Marta Gentiloni Silveri, Giovanni Conforti, Alain Oliviero Durmus

Diffusion Flow Matching (DFM) has recently emerged as a versatile framework for

generative modeling, yet its theoretical convergence properties remain only partially understood. In this work, we provide refined and novel convergence guarantees for Brownian motion based DFMs, focusing on the discretization error. Our analysis is conducted under the Kullback–Leibler (KL) divergence and the 2-Wasserstein distance. Under finite-moment conditions and a mild score integrability assumption, we derive KL convergence bounds with improved dimensional dependence compared to prior work, achieving, up to our knowledge, state-of-the-art scaling under minimal conditions. We further extend the analysis to the 2-Wasserstein distance: under an additional first-order score integrability assumption and a weak log-concavity condition, we obtain convergence guarantees with dimensional dependence consistent with the KL case.
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From 2D Grids to 1D Tokens: Reforming Shared Representations for Multimodal Image Fusion
Yuchen Xian, Yunqiu Xu, Yang He, Yi Yang
Multimodal image fusion aims to integrate complementary information from different modalities into a fused image that preserves rich local details while maintaining globally consistent appearance. Existing approaches build shared representations on 2D feature grids, which excel at modeling local structures but offer limited leverage over image-level global appearance factors. To balance these objectives, we introduce a compact 1D token interface based on a frozen pretrained image tokenizer for modeling non-local appearance/base factors. Rather than using the tokenizer as a reconstruction backbone, our design uses the 1D token space as a global carrier while retaining the 2D spatial pathway for local structure restoration. Specifically, we introduce Selective Token Editing (STE), which sparsely updates/replaces a small set of critical tokens, providing a lightweight mechanism to steer global appearance coherence while keeping the fusion backbone unchanged and avoiding extra losses. Experiments on four commonly used benchmarks show that our method achieves the best overall performance, with consistent, multi-metric improvements in both global coherence and local fidelity. Project page: https://zju-xyc.github.io/1D-Fusion-Project-Page/
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Q-Tab: Quantized Tabular Data Generator
Julian Wustl, Philipp Haid, Yarema Okhrin, Claudius Schnörr
Codebook-based generators built on masked language model (MLM) transformers have become highly effective in text and vision, yet remain underused for tabular data. This is because codebooks typically act as information bottlenecks, whereas synthetic tabular generation requires a code space larger than the training sample, with additional codes trained to support new tabular rows. We address this gap with Q-Tab, a codebook-based tabular generator that uses lookup-free quantization (LFQ) with residual corruption to jointly tokenize numerical variables, categorical variables are directly one-hot tokenized. A BERT-style MLM captures dependencies in the token space and can then be sampled from. Corruption propagates reconstruction supervision across the numerical code space, but under joint encoder–decoder training induces a moving-target regression problem whose difficulty depends on the corruption structure. This motivates residual LFQ as the quantization mechanism, balancing broader supervision with locality. Q-Tab achieves state-of-the-art predictive utility and label prediction, while matching the distributional fidelity of diffusion-based generators.
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Estimating Continuous Treatment Effects with Two-Stage Kernel Ridge Regression
Seok-Jin Kim, Kaizheng Wang
We study the problem of estimating the effect function for a continuous treatment, which maps each treatment value to a population-averaged outcome. A central challenge in this setting is confounding: treatment assignment often depends on covariates, creating selection bias that makes direct regression of the response on treatment unreliable. To address this issue, we propose a two-stage kernel ridge regression method. In the first stage, we learn a model for the response as a function of both treatment and covariates; in the second stage, we use this model to construct pseudo-outcomes that correct for distribution shift, and then f

it a second model to estimate the treatment effect. Although the response varies with both treatment and covariates, the induced effect function obtained by averaging over covariates is typically much simpler, and our estimator adapts to this structure. Our optimal learning bounds are achieved without estimating the conditional treatment density, thereby bypassing a major bottleneck in existing methods. Furthermore, we introduce a fully data-driven model selection procedure that achieves provable adaptivity to both the unknown degree of overlap and the spectral decay of the underlying kernel.
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AGZO: Activation-Guided Zeroth-Order Optimization for LLM Fine-Tuning
Wei Lin, Yining Jiang, Qingyu Song, Qiao Xiang, Hong Xu
Zeroth-Order (ZO) optimization has emerged as a promising solution for fine-tuning LLMs under strict memory constraints, as it avoids the prohibitive memory cost of storing activations for backpropagation. However, existing ZO methods typically employ isotropic perturbations, neglecting the rich structural information available during the forward pass. In this paper, we identify a crucial link between gradient formation and activation structure: the gradient of a linear layer is confined to the subspace spanned by its input activations. Leveraging this insight, we propose Activation-Guided Zeroth-Order optimization (AGZO). Unlike prior methods, AGZO extracts a compact, activation-informed subspace on the fly during the forward pass and restricts perturbations to this low-rank subspace. We provide a theoretical framework showing that AGZO optimizes a subspace-smoothed objective and provably yields update directions with higher cosine similarity to the true gradient than isotropic baselines. Empirically, we evaluate AGZO on Qwen3 and Pangu models across various benchmarks. AGZO consistently outperforms state-of-the-art ZO baselines and significantly narrows the performance gap with first-order fine-tuning, while maintaining almost the same peak memory footprint as other ZO methods.
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TapSampling: Inference-Time Sampling with a Task-Progress-Understanding Verifier for Robotic Manipulation
Sizhe Zhao, Shengping Zhang, Shuo Yang, Weiyu Zhao, Shuigen Wang, Xiangyang Ji
Existing embodied control research demonstrates remarkable performance improvements by scaling training data and model size.
We instead explore inference-time strategy as an alternative axis. Non-deterministic generative models, such as diffusion and autoregressive models, have been widely adopted in the field of embodied control. However, the single-shot inference paradigm limits their performance. In this paper, we propose **TapSampling**, a plug-and-play framework for inference-time sampling. First, we introduce an Action-VAE that represents actions in a low-dimensional latent space by mapping policy-generated initial actions into a compressed posterior distribution, from which any number of latent samples can be drawn and decoded into candidate actions that approximate the true action distribution. Second, we formulate action verification as task-progress outcome prediction, using the intrinsic sequential structure of robotic datasets to train a semantically grounded verifier for interpretable action selection. Furthermore, TapSampling is a policy-agnostic framework. Extensive experiments in both simulated and real-world environments demonstrate that our method substantially improves multiple generalist policies without further policy finetuning. Code and models are available at the **project page** (https://aipixel.github.io/TapSampling/).
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AutoMS: Multi-Agent Evolutionary Search for Cross-Physics Inverse Microstructure Design
Zhenyuan Zhao, Yu Xing, Tianyang Xue, Lingxin Cao, Xin Yan, Lin Lu
Designing microstructures with coupled cross-physics objectives is a fundamental challenge where traditional topology optimization is often computationally prohibitive and deep generative models frequently suffer from physical hallucinations. We introduce AutoMS, a multi-agent neuro-symbolic framework that reformulates inverse design as an LLM-driven evolutionary search. AutoMS leverages LLMs as semantic navigators to decompose complex requirements and coordinate agent workfl

ows, while a novel Simulation-Aware Evolutionary Search (SAES) mechanism handles low-level numerical optimization via local gradient approximation and directed parameter updates. This architecture achieves a state-of-the-art 83.8\% success rate on 17 diverse cross-physics tasks, significantly outperforming both traditional evolutionary algorithms and existing agentic baselines.
By decoupling open-ended semantic orchestration from simulation-grounded numerical search, AutoMS provides a robust pathway for navigating complex physical landscapes that remain intractable for standard generative or purely linguistic approaches.
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Condition-Aware Graph Flow Matching for Modeling the Distributions of Complex Fluid Systems
Xiaochao Deng, Jie Chen, Xiaogang Deng
Accurately modeling the full distributions of possible states is crucial for understanding statistical properties and enabling reliable predictions in complex fluid systems. Recently, diffusion models and flow matching have shown promise in these tasks. However, they remain limited in uncovering the general principles of systems from multiple short trajectories across the condition space. In addition, they exhibit inferior adaptability to large irregular geometries, particularly in regions with sharp gradients. In this paper, we propose a condition-aware graph flow matching (CGFM) method that combines condition-aware flow matching with a hierarchical graph structure to learn the full distributions of fluid systems from incomplete training data. Specifically, CGFM constructs a flow enabling smooth interpolation across physical conditions and parameterizes the graph-conditioned vector field through HieraGraphNet. HieraGraphNet performs message passing across multilevel graphs to capture multi-scale dynamics and facilitate long-range information interactions in fluid systems. Moreover, we introduce a topology- and geometry-aware graph coarsening scheme that incorporates topological connectivity and local geometric density to construct reliable coarse graphs. We validate the effectiveness of CGFM on three canonical scenarios across both 2D and 3D dynamics, which demonstrate its superior performance compared with that of state-of-the-art baselines.
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Token-Level LLM Collaboration via FusionRoute
Nuoya Xiong, Yuhang Zhou, Hanqing Zeng, Zhaorun Chen, Furong Huang, Shuchao Bi, Lizhu Zhang, Zhuokai Zhao
Large language models (LLMs) exhibit strengths across diverse domains. However, achieving strong performance across these domains with a single general-purpose model typically requires scaling to sizes that are prohibitively expensive to train and deploy. On the other hand, while smaller domain-specialized models are much more efficient, they struggle to generalize beyond their training distributions. To address this dilemma, we propose FusionRoute, a robust and effective token-level multi-LLM collaboration framework in which a lightweight router simultaneously (i) selects the most suitable expert at each decoding step and (ii) contributes a complementary logit that refines or corrects the selected expert’s next-token distribution via logit addition. Unlike existing token-level collaboration methods that rely solely on fixed expert outputs, we provide a theoretical analysis showing that pure expert-only routing is fundamentally limited: unless strong global coverage assumptions hold, it cannot in general realize the optimal decoding policy. By augmenting expert selection with a trainable complementary generator, FusionRoute expands the effective policy class and enables recovery of optimal value functions under mild conditions. Empirically, across both Llama-3 and Gemma-2 families and diverse benchmarks spanning mathematical reasoning, code generation, and instruction following, FusionRoute outperforms both sequence- and token-level collaboration, model merging,
and direct fine-tuning, while remaining competitive with domain experts on their respective tasks.
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RaBitQCache: Rotated Binary Quantization for KVCache in Long Context LLM Inference

Wenhao Li, Jinhao Dong, Hailin Zhang, Wenhang Shi, WEI LU, Xiaoyong Du
Long-context Large Language Model inference is severely bottlenecked by the massive Key-Value (KV) cache, yet existing sparse attention methods often suffer from static fixed-budget (Top-k) retrieval or rely on proxy scores that are computationally expensive and biased. To address these limitations, we propose RaBitQCache, a novel sparse attention framework that utilizes randomized rotated binary quantization and high-throughput binary-INT4 arithmetic to efficiently estimate attention weights. Our proxy score serves as an unbiased estimator with a proven error bound, enabling adaptive Top-p retrieval that dynamically adjusts the token budget based on actual attention sparsity. We further implement a hardware-aware system with asynchronous pipelining and lazy updates to mask overhead. Evaluations demonstrate that RaBitQCache significantly accelerates inference and reduces memory I/O while preserving generation quality compared to state-of-the-art baselines. Code is available at https://github.com/Sakuraaa0/RaBitQCache.git.
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Target-Agnostic Calibration under Distribution Shift with Frequency-Aware Gradient Rectification
Yilin Zhang, Cai Xu, You Wu, Ziyu Guan, Wei Zhao
Real-world model deployments inevitably encounter distribution shifts, rendering the confidence estimates of deep neural networks highly unreliable, posing severe risks in safety-critical applications. Existing methods improve calibration via training-time regularization or post-hoc adjustment, but often rely on access to (or simulation of) target domains, limiting practicality. We propose Frequency-aware Gradient Rectification (FGR), a target-agnostic training framework for robust calibration. From a frequency perspective, FGR applies low-pass filtering to a subset of training images to diminish spurious high-frequency cues and encourage the learning of domain-invariant features. However, the associated information loss can degrade In-Distribution (ID) calibration. To resolve this trade-off, FGR treats ID calibration as a hard constraint and rectifies conflicting parameter updates via geometric projection. This ensures a first-order non-increase in the ID calibration objective without introducing an additional loss-balancing coefficient. Extensive experiments on synthetic, real-world, and semantic shift datasets demonstrate that FGR significantly improves calibration under diverse shifts while preserving ID performance, and it remains compatible with post-hoc calibration methods. Our code is available at https://github.com/YilinZhang107/FGR-Calib.
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Towards Spectroscopy: Susceptibility Clusters in Language Models
Andrew Gordon, Garrett Baker, George Wang, William Snell, Stan van Wingerden, Daniel Murfet
Spectroscopy infers the internal structure of physical systems by measuring their response to perturbations. We apply this principle to neural networks: perturbing the data distribution by upweighting a token y in context x, we measure the model's response via susceptibilities χ_{xy}, which are covariances between component-level observables and the perturbation computed over a localized Gibbs posterior via stochastic gradient Langevin dynamics (SGLD). Theoretically, we show that susceptibilities decompose as a sum over \emph{modes} of the data distribution, explaining why tokens that follow their contexts ``for similar reasons'' cluster together in susceptibility space. Empirically, we apply this methodology to Pythia-14M, developing a conductance-based clustering algorithm that identifies 510 interpretable clusters ranging from grammatical patterns to code structure to mathematical notation. Comparing to sparse autoencoders, 50\% of our clusters match SAE features, validating that both methods recover similar structure.
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Initialization is Half the Battle: Generating Diverse Images from a Guidance Potential Posterior
Xiang Li, Dianbo Liu, Kenji Kawaguchi
Despite the remarkable fidelity of generative models, they frequently suffer from mode collapse. Existing strategies for enhancing diversity predominantly focus

on intervening during the generation trajectory. We identify a critical oversig
ht that the standard Gaussian initialization often causes trajectories to collap
se into dominant modes because it is agnostic to the guidance potential landscap
e. In this work, we formulate selecting the initial noise from a *guidance poten
tial posterior*, which effectively re-weights the prior towards diversity-rich r
egions. To sample from this distribution efficiently, we introduce *Diversity-in
ducing Initialization* (DivIn), which leverages Langevin dynamics to actively na
vigate the initialization landscape, steering initial noise away from collapsing
regions while anchoring them to the valid data manifold. Our method serves as a
n inference-time diversity enhancement compatible with both diffusion and flow m
atching models. Extensive experiments show that DivIn exhibits a superior perfor
mance in both class-to-image and text-to-image scenarios.
Furthermore, we highlight that as DivIn is orthogonal to trajectory-based method
s, combining them significantly expands the diversity-quality Pareto frontier be
yond what either achieves in isolation.
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How to Avoid Debate: Scalable AI Safety via Doubly-Efficient Interactive Proofs
Liyan Chen, Yael Tauman Kalai, Zoe Xi
As AI models continue to develop powerful
capabilities, it becomes critical that we are able to
verify that their output is aligned with our intentions. A recent line of work
focuses on
verification via debate, a model
of interactive proofs where two competing powerful provers, or AI
models, debate each other to convince a weak verifier, or a human,
of the correctness of their claim. However, debate assumes that the two AI mod
els possess equal abilities and that one of them is truthful, which may not be r
ealistic.

In this work, we show *how to avoid debate*: we
initiate the study of *single-prover*
interactive proofs for AI safety.
Prior results in single-prover interactive proofs do not immediately
carry over to the AI safety setting because they do not work when the computat
ion has access to an oracle, such as to human judgment or an external database s
uch as the web.
We present doubly-efficient
single-prover interactive proofs for oracle-aided computations (also known as
relativizing proofs), in the settings where (1) the
computation is robust, in the sense that the output does not change
if at most a small fraction of the answers to oracle queries are
incorrect, or (2) the oracle is a low-degree polynomial. These results suggest
that interactive verification is possible even without debate, under structured
or noise-tolerant oracle access.
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What You Think is What You See: Driving Exploration in VLM Agents via Visual-Lin
guistic Curiosity
Haoxi Li, Qinglin Hou, Jianfei Ma, Jinxiang Lai, Tao Han, Sikai Bai, Jingcai Guo
, Jie ZHANG, Song Guo
To navigate partially observable visual environments, recent VLM agents increasi
ngly internalize world modeling capabilities directly into their policies via ex
plicit CoT reasoning with reinforcement learning (RL). However, mere passive exp
loitation of reasoning on visited states is insufficient for sparse-reward agent
ic tasks, as it lacks the epistemic drive to actively uncover the *known unknown
* required for robust generalization. We ask: *Can VLM agents actively find sign
als that challenge and update their internal world model through curiosity-drive
n exploration?* In this work, we propose **GLANCE**, a unified framework that br
idges reasoning and exploration by grounding the agent's linguistic world model
into the stable visual representations of an evolving target network. Crucially,
GLANCE leverages the discrepancy between linguistic prediction and visual r

eality as an intrinsic curiosity signal within reinforcement learning, steering the agent to actively explore areas where its internal model is uncertain. Exten sive experiments across a series of agentic tasks show the effectiveness of **GL ANCE**, and demonstrate that aligning *what the agent thinks* with *what the age nt sees* is key to solving complex or sparse agentic tasks.
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Textual Stochastic Gradient Descent: Discrete Optimization of External Memory fo r Reasoning Language Agents
Jian Li, Hua Huang
While Large Language Models (LLMs) possess strong reasoning capabilities, enabli ng them to learn continuously from experience without parametric retraining rema ins an open challenge. Existing Retrieval-Augmented Generation (RAG) approaches typically treat memory as a static or append-only corpus, leading to "memory sat uration," where accumulating noise and redundant information degrade performance over time. To address this, we propose an Experience Library Optimization frame work that treats the agent's external memory, which we call the experience libra ry, as a learnable parameter under an explicit capacity budget. We introduce Tex tual Stochastic Gradient Descent (TSGD), a discrete optimization algorithm that refines this library via failure-driven Add, Edit, and Delete operations. TSGD e stimates "textual gradients" through self-reflection and uses a dual-verificatio n mechanism to ensure generalization, which prevents overfitting to local errors . Empirical results on MATH and AIME benchmarks show that TSGD achieves state-of -the-art performance, improving accuracy by up to 18.7% over zero-shot baseli nes and substantially outperforming static RAG, while keeping a compact memory f ootprint (compressing hundreds of experiences into ≈ 30 high-utility rul es).
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LILO: Bayesian Optimization with Natural Language Feedback
Kasia Kobalczyk, Zhiyuan Jerry Lin, Benjamin Letham, Zhuokai Zhao, Maximilian Ba landat, Eytan Bakshy
Many real-world optimization problems are guided by complex, subjective preferen ces that are difficult to express as explicit closed-form objectives. In respons e, we introduce Language-in-the-Loop Optimization (LILO), a Bayesian optimizatio n (BO) framework that employs a large language model (LLM) to translate free-for m natural language feedback and prior knowledge from a decision maker into struc tured preference signals, going beyond the restrictive scalar or pairwise feedba ck formats typically assumed in preferential BO. The LLM-derived preferences are integrated by a Gaussian process proxy model, enabling principled acquisition-d riven exploration with calibrated uncertainty. By placing the LLM in a supportin g role rather than as the optimizer itself, LILO preserves the sample efficiency and stability of BO while providing a flexible and expressive feedback interfac e. Across synthetic and real-world benchmarks, LILO consistently outperforms bot h conventional preference-based BO methods and LLM-only optimizers, with particu larly strong gains in feedback-limited regimes. The code for reproducing our exp erimental results is available at: https://github.com/facebookresearch/lilo.
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Intervene When It Doubts: Conjunction-Guided Interactive Reasoning
Qianyue Wang, Jinwu Hu, Yaofo Chen, Yufeng Wang, Bailin Chen, Huanxiang Lin, Yu Rong, Yuanqing Li, Zhiquan Wen, Mingkui Tan
Large Reasoning Models (LRMs) excel at complex reasoning but suffer from ineffic ient reasoning, like overthinking and overshoot. These issues stem from excessiv e or misdirected reasoning triggered by the model's "doubt", manifested as self- validation and exploratory extension, increasing computational cost and degradin g performance. Existing efficient reasoning methods seek to regulate reasoning v ia internal signals or static schedules, lacking specialization in the "doubt" c haracteristics of LRMs. To address this, we propose a Conjunction-Guided Interve ntion (CGI) reasoning framework that intervenes when the model shows signs of do ubt. Our key insight is that overthinking and overshoot in LRMs arise from conju nction-triggered extensions where LRMs exhibit "doubt" through transitional conj unctions, extending redundant self-validation and exploration without timely sta

te-based correction. Building on this insight, CGI pauses reasoning at conjuncti on markers of doubt for external state-based feedback, adaptively extending or t erminating reasoning to reduce redundancy while preserving accuracy. The feedbac k is generated via criteria evaluation (rationality and completeness) and comes from either human or LLM proxies. We train the target model by Group Relative Po licy Optimization (GRPO) to adapt to the interactive mode. Experiments show that our framework achieves a superior balance between accuracy and reasoning length .

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Learning Multi-Timescale Abstractions for Hierarchical Combinatorial Planning

Vivienne Huiling Wang, Tinghuai Wang, Joni Pajarinen

The combination of exponentially large action spaces, stochastic dynamics, and l ong-horizon decision-making under limited resources makes Sequential Stochastic Combinatorial Optimization (SSCO) particularly challenging for reinforcement lea rning. Hierarchical Reinforcement Learning (HRL) offers a natural decomposition, but it places the high-level policy in a Semi-Markov Decision Process (SMDP) wh ere actions have variable durations, making it difficult to learn a world model that is suitable for planning. We introduce a model-based hierarchical framework for sequential stochastic combinatorial decision-making that directly addresses this issue. Our method combines a latent-space tree-search planner with an SMDP -aware world model for variable-duration decisions. A multi-timescale objective structures the latent dynamics so that transition magnitudes reflect the effecti ve temporal scales of abstract actions, enabling efficient lookahead under adapt ive temporal abstraction. We further learn a subgoal-conditioned budget policy j ointly with the world model to support context-aware resource allocation. Across challenging SSCO benchmarks, our method outperforms strong baselines.

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Generative Large Neighborhood Search: Scalable Set Cover Optimization via Discre te Diffusion

Achref Jaziri, Thibaut Cuvelier, Bruno Adrien De Backer

Large-scale Set Cover Problems (SCP) with millions of variables and complex cost structures require high-quality solutions within seconds, yet remain beyond the reach of exact solvers and pose severe generalization challenges for neural met hods. Such problems necessitate decomposition into bounded subproblems; however, when the induced subproblem topology differs from that observed during training , existing neural approaches often fail to transfer reliably.

We introduce Generative Large Neighborhood Search (GLNS), which reframes neighbo rhood selection as generation using a discrete diffusion model. Our key insight is that the diffusion denoising trajectory exposes variables exhibiting high pre diction instability across timesteps and identifies regions where local repair y ields downstream improvement. GLNS exploits this trajectory-level signal to cons truct high-impact neighborhoods via a localized, bounded-complexity generative s ampling procedure, enabling robust neighborhood selection without retraining. As a result, GLNS transfers effectively across cost regimes and instance scales wi thin SCP. Under tight and equal wall-clock budgets, GLNS consistently outperform s established neural baselines and achieves competitive performance with state-o f-the-art MIP solvers. These results demonstrate trajectory-guided generation as a scalable framework for large-scale SCP and suggest potential relevance to oth er constrained optimization settings.

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Memora: A Harmonic Memory Representation Balancing Abstraction and Specificity

Menglin Xia, Xuchao Zhang, Shantanu Dixit, Paramaguru Harimurugan, Rujia Wang, V ictor Rühle, Robert Sim, Chetan Bansal, Saravan Rajmohan

Agent memory systems must accommodate continuously growing information while sup porting efficient, context-aware retrieval for downstream tasks. Abstraction is essential for scaling agent memory, yet it often comes at the cost of specificit y, obscuring the fine-grained details required for effective reasoning. We intro duce Memora, a harmonic memory representation that structurally balances abstrac tion and specificity. Memora organizes information via its *primary abstractions * that index concrete memory values and consolidate related updates into unified

memory entries, while *cue anchors* expand retrieval access across diverse aspects of the memory and connect related memories. Building on this structure, we employ a retrieval policy that actively exploits these memory connections to retrieve relevant information beyond direct semantic similarity. Theoretically, we show that standard Retrieval-Augmented Generation (RAG) and Knowledge Graph (KG)-based memory systems emerge as special cases of our framework. Empirically, Memora establishes a new state-of-the-art on the LoCoMo and LongMemEval benchmarks, demonstrating better retrieval relevance and reasoning effectiveness as memory scales.

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MAMBO-G: Magnitude-Aware Mitigation for Boosted Guidance

Shangwen Zhu, Qianyu Peng, Zhilei Shu, Yuting Hu, Han Zhang, Andy Zheng, Xinyu Cui, Jian Zhao, Ruili Feng, Fan Cheng

High-fidelity text-to-image and text-to-video generation typically relies on Classifier-Free Guidance (CFG), but achieving optimal results often demands computationally expensive sampling schedules. In this work, we propose MAMBO-G, a training-free acceleration framework that significantly reduces computational cost by dynamically optimizing guidance magnitudes. We observe that standard CFG schedules are inefficient, applying disproportionately large updates in early steps that hinder convergence speed. MAMBO-G mitigates this by modulating the guidance scale based on the update-to-prediction magnitude ratio, effectively stabilizing the trajectory and enabling rapid convergence. This efficiency is particularly vital for resource-intensive tasks like video generation. Our method serves as a universal plug-and-play accelerator, achieving up to 3$\times$ speedup on Stable Diffusion v3.5 (SD3.5) and 4$\times$ on Lumina. Most notably, MAMBO-G accelerates the 14B-parameter Wan2.1 video model by 2$\times$ while preserving visual fidelity, offering a practical solution for efficient large-scale video synthesis. Our implementation follows a mainstream open-source diffusion framework and is plug-and-play with existing pipelines.

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Stop When Further Reasoning Won’t Help: Attention-State Adaptive Generation in Reasoning Models

Jiakai Li, Ke Qin, Rongzheng Wang, Yizhuo Ma, Qizhi Chen, Muquan Li, Shuang Liang

By incorporating test-time compute scaling, large reasoning models (LRMs) can solve complex problems through explicit chain-of-thought (CoT) reasoning processes.

However, they often suffer from overthinking, resulting in redundant token outputs and degraded accuracy.

Current methods to mitigate this issue remain limited: training-based approaches require substantial computational resources, while training-free methods rely on well-crafted prompts or unreliable confidence signals.

In this work, we investigate early stopping from the perspective of attention distributions and propose a simple method, ASAG, which infers the model's reasoning state and adaptively adjusts the generation strategy.

The proposed framework is training-free and plug-and-play, enabling seamless integration into existing LRMs.

Extensive experiments on nine benchmarks demonstrate consistent improvements across mainstream LRMs with varying parameter scales, including the DeepSeek-R1-Distill and Qwen3 series.

Specifically, ASAG improves average accuracy by 3.2% while reducing the number of generated tokens by nearly 40% across all reasoning tasks on Qwen3-8B.

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VeRO: A Harness for Agents to Optimize Agents

Varun Ursekar, Apaar Shanker, Veronica Chatrath, Yuan Xue, Samuel Marc Denton

An important emerging application of coding agents is *agent harness optimization*: the iterative improvement of a *target agent* by editing and evaluating its code. Despite its relevance, the community lacks a systematic understanding of coding agent performance on this task. Harness optimization differs from conventional software engineering: agent harnesses interleave deterministic code with st

ochastic LLM completions, requiring structured capture of both intermediate execution traces and downstream outcomes. To address these challenges, we introduce (1) VeRO (Versioning, Rewards, and Observations), an *outer harness* that provides versioned snapshots, budget-controlled evaluation, and structured execution traces of *target harnesses*, and (2) VeRO-Bench, a benchmark suite of target agents and tasks with reference evaluation procedures. Using VeRO, we conduct an empirical study comparing optimizers across tasks and analyzing which modifications reliably improve target agent harnesses. We release VeRO to support research on agent optimization as a core capability for coding agents. Code is available at https://github.com/scaleapi/vero.

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Split Personality Training: Revealing Latent Knowledge Through Alternate Personalities

Florian Dietz, William Wale, Oscar Gilg, Robert McCarthy, Felix Michalak, Gustavo Ewbank Rodrigues Danon, Miguelito de Guzman, Dietrich Klakow

Detecting misalignment in large language models is challenging because models may learn to conceal misbehavior during training. Standard auditing techniques fall short: black-box methods often cannot distinguish misaligned outputs from benign ones, and mechanistic interpretability does not scale with model capabilities. We introduce Split Personality Training (SPT), which fine-tunes a second ``honest persona'' into LoRA parameters that remain inactive during normal operation. After the main model responds, we activate the LoRA adapter and insert a trigger string, enabling the honest persona to review the response while accessing the main model's latent states. We test our method on the Anthropic Auditing Game Model Organism, a benchmark where Llama-3.3-70B is trained to exploit reward hacks while concealing this behavior. SPT achieves 96% overall accuracy, whereas Anthropic reports near 0% accuracy. The honest persona reveals latent knowledge inaccessible to external observers, such as the fictional biases the compromised model was trained on.

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Generalized Discrete Diffusion with Self-Correction

Linxuan Wang, Ziyi Wang, Yikun Bai, Wei Deng, Guang Lin, Qifan Song

Self-correction is an effective technique for maintaining parallel sampling in discrete diffusion models with minimal performance degradation. Prior work has explored self-correction at inference time or during post-training; however, such approaches often suffer from limited generalization and may impair reasoning performance. GIDD pioneers pretraining-based self-correction via a multi-step BERT-style uniform-absorbing objective. However, GIDD relies on a continuous interpolation-based pipeline with opaque interactions between uniform transitions and absorbing masks, which complicates hyperparameter tuning and hinders practical performance. In this work, we propose a **S**elf-**C**orrecting **D**iscrete **D**iffusion (SCDD) model to reformulate pretrained self-correction with explicit state transitions and learn directly in discrete time. Our framework also simplifies the training noise schedule, eliminates a redundant remasking step, and relies exclusively on uniform transitions to learn self-correction. Experiments at the GPT-2 scale demonstrate that our method enables more efficient parallel decoding while preserving generation quality. Our code is available at https://github.com/laaaarrywang/Self-Correcting-Discrete-Diffusion.git.

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Persistent Semantic Entities in Tool-Augmented LLM Systems

Zhaohui Geoffrey Wang

Tool-augmented LLM agents can harbor implicit state that persists across sessions, activates through events, and propagates across agent boundaries—all invisible to standard debugging. We formalize this as Persistent Semantic Entities (PSE): constructs defined by name binding, event triggering, and cross-boundary propagation, and evaluate them across 20 models from 9 families (3B–671B parameters). First, all tested models are susceptible to contamination (20–100%), with name binding as the dominant mechanism: without it, contamination is 0%. Second, persistence depends on contamination type rather than scale or deployment—factual injection conflicting with parametric knowledge self-corrects, but preference, per

sona, and instruction contamination persists at 100% over a 10-turn horizon with no decay, an effect that holds across providers in our controlled setting and is consistent with a model-intrinsic explanation. Third, context-isolated self-verification achieves 20–79% reduction without oracle references while keyword-based detection produces systematic false positives, and contamination compounds 3× across multi-agent pipelines. Preference and instruction contamination—persistent, lacking self-correction, and invisible to standard monitoring—represents a particularly concerning attack surface for deployed agent systems.
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Rays as Pixels: Learning A Joint Distribution of Video and Camera Trajectories
Wonbong Jang, Shikun Liu, Soubhik Sanyal, Juan Camilo Perez, Kam Woh Ng, Sanskar Agrawal, Juan-Manuel Perez-Rua, Yiannis Douratsos, Tao Xiang
Can we bridge the gap between perceiving camera trajectories and rendering novel views within a single generative framework? Recovering camera parameters from images and rendering scenes from novel viewpoints are considered the forward and inverse problems in the field of computer vision and graphics. Previous approaches treat these problems in isolation, often failing when image coverage is sparse or camera poses are ambiguous. In this work, we propose Rays as Pixels, a specialized Video Diffusion Model (VDM) that learns a joint distribution of videos and camera trajectories. We represent cameras as dense ray pixels (raxels) and simultaneously denoise them alongside video frames using a novel Decoupled Self-Cross Attention. This joint formulation enables us to: i) generate a video from multiple input images following a defined camera trajectory, ii) perform novel view synthesis from sparse views (without necessarily requiring camera poses), and iii) predict the camera trajectory from a raw video. We evaluate our model on pose estimation, camera-controlled video generation and validate its self-consistency. Please reference supplementary material for more qualitative results.
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NAACA: Training-Free NeuroAuditory Attentive Cognitive Architecture with Oscillatory Working Memory for Salience-Driven Attention Gating
Zhongju Yuan, Geraint A. Wiggins, Dick B.M. Botteldooren
Audio provides critical situational cues, yet current Audio Language Models (ALMs) face an attention bottleneck in long-form recordings where dominant background patterns can dilute rare, salient events. We introduce NAACA, a training-free NeuroAuditory Attentive Cognitive Architecture that reframes attention allocation as an auditory salience filtering problem. At its core is OWM, a neuro-inspired Oscillatory Working Memory that maintains stable attractor-like states and triggers higher-cognition ALM processing only when adaptive energy fluctuations signal perceptual salience, triggering higher-level reasoning. On XD-Violence, NAACA improves AudioQwen’s average precision (AP) from 53.50\% to 70.60\% while reducing unnecessary ALM invocations. Furthermore, qualitative case studies on the Urban Soundscapes of the World (USoW) dataset show that OWM captures novel events and subcategory shifts while remaining robust to transient pauses and ambient urban noise.
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A KL-regularization framework for learning to plan with adaptive priors
Alvaro Serra-Gomez, Daniel Jarne Ornia, Dhruva Tirumala, Thomas M. Moerland
Effective exploration remains a key challenge in model-based reinforcement learning (MBRL), especially in high-dimensional continuous control tasks where sample efficiency is critical. Recent work addresses this by using learned policies as proposal distributions for Model-Predictive Path Integral (MPPI) planning. Early approaches update the sampling policy independently of the planner, typically via deterministic policy gradients with entropy regularization. However, since the data distribution is induced by the MPPI planner, misalignment between the policy and planner degrades value estimation and long-term performance. To address this, recent methods explicitly align the policy with the planner by minimizing KL divergence to the planner distribution or by incorporating planner-guided regularization. In this work, we unify these approaches under the Policy Optimization–Model Predictive Control (PO-MPC) framework, a family of KL-regularized MBRL methods that treat the planner’s action distribution as a prior in policy optim

ization. We show how existing methods emerge as special cases of this family and explore previously unstudied variants. Experiments demonstrate that these variants yield significant performance gains, advancing the state of the art in MPPI-based RL.

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LangPrecip: Language-Aware Multimodal Precipitation Nowcasting

Xudong Ling, Chaorong Li, Tianxi Huang, Qian Dong, Guiduo Duan

Short-term precipitation nowcasting is inherently under-constrained due to limited historical observation windows: identical observations can lead to multiple plausible future trajectories, especially for extreme events. Existing generative methods rely solely on visual features and lack explicit constraints on precipitation motion semantics, resulting in ambiguous dynamics, blurred details, and unstable predictions. We propose LangPrecip, the first language-guided precipitation nowcasting framework, and contribute LangPrecip-160K, a large-scale radar-text paired dataset with 160K annotated sequences. LangPrecip addresses the under-constrained challenge by leveraging natural-language motion descriptions as explicit semantic constraints to reduce motion ambiguity and introducing a dual-path wavelet consistency unfolding decoder that enforces physical data fidelity during latent-to-pixel reconstruction. By reformulating nowcasting as semantically constrained trajectory generation under the Rectified Flow paradigm with model-based decoder optimization, LangPrecip produces sharper and more physically consistent forecasts. Experiments on Swedish and MRMS benchmarks demonstrate substantial improvements over state-of-the-art vision-only methods, achieving over 60\% and 19\% relative gains in heavy-rainfall CSI at 80-minute lead time with enhanced spatial detail preservation. Dataset is available at https://github.com/UESTC-LXD/LangPrecip.

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WorldCompass: Reinforcement Learning for Long-Horizon World Models

Zehan Wang, Tengfei Wang, Haiyu Zhang, Xuhui Zuo, Junta Wu, Haoyuan Wang, Wenqiang Sun, Zhenwei Wang, Chenjie Cao, Hengshuang Zhao, Chunchao Guo, Zhou Zhao

This work presents WorldCompass, a novel Reinforcement Learning (RL) post-training framework for the long-horizon, interactive video-based world models, enabling them to explore the world more accurately and consistently based on interaction signals. To effectively "steer" the world model's exploration, we introduce three core innovations tailored to the autoregressive video generation paradigm: 1) Clip-level Roll-out Strategy: We generate and evaluate multiple samples at a single target clip, which significantly boosts roll-out efficiency and provides fine-grained reward signals. 2) Complementary Reward Functions: We design reward function for both interaction-following accuracy and visual quality, which provide direct supervision and effectively suppress reward-hacking behaviors. 3) Efficient RL Algorithm: We employ the negative-aware fine-tuning strategy coupled with various efficiency optimizations to efficiently and effectively enhance model capacity. Evaluations on the SoTA open-source world model, WorldPlay, demonstrate that WorldCompass significantly improves interaction accuracy and visual fidelity across various scenarios.

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Weight Decay Improves Language Model Plasticity

Tessa Han, Sebastian Bordt, Hanlin Zhang, Sham M. Kakade

Large language models are typically trained in two broad phases: pretraining to produce a base model, followed by further training to improve downstream performance. However, hyperparameter optimization and scaling laws are studied primarily from the perspective of the base model's validation loss, overlooking a crucial model property: downstream adaptability. In this work, we study pretraining from the perspective of *model plasticity*, that is, the ability of the base model to successfully adapt to downstream tasks upon additional training. We focus on the role of weight decay, a key regularization parameter during pretraining, and show through systematic experiments that larger weight decay increases the plasticity of the pretrained model, resulting in greater performance gains downstream after fine-tuning. This effect can lead to counterintuitive trade-offs where base models that perform worse after pretraining can perform better after furthe

r training. Further investigation of weight decay's mechanistic effects on model behavior reveals that it encourages linearly separable representations, regularizes attention matrices, and reduces overfitting on the training data. Together, these findings highlight the importance of pretrained model plasticity, the limits of using cross-entropy loss as the sole metric for hyperparameter optimization, and the multifaceted role that a single optimization hyperparameter plays in shaping model behavior.
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No Free Lunch: Non-Asymptotic Analysis of Prediction-Powered Inference
Pranav Mani, Peng Xu, Zachary Chase Lipton, Michael Oberst
Prediction-Powered Inference (PPI) is a popular strategy for combining gold-standard and possibly noisy pseudo-labels to perform statistical estimation. Prior work has shown an asymptotic \textit{free lunch} for PPI++, an adaptive form of PPI, showing that the \textit{asymptotic} variance of PPI++ is always less than or equal to the variance obtained from using gold-standard labels alone. Notably, this result holds \textit{regardless of the quality of the pseudo-labels}. In this work, we demystify this result by conducting an exact finite-sample analysis of the estimation error of PPI++ on the mean estimation problem. We give a \textit{no free lunch} result, characterizing the settings (and sample sizes) where PPI++ has provably worse estimation error than using gold-standard labels alone. Specifically, PPI++ will outperform if and only if the correlation between pseudo- and gold-standard is above a certain level that depends on the number of labeled samples (n).
In some cases our results simplify considerably: For Gaussian data, for instance, the correlation must be at least $1/\sqrt{n - 2}$ in order to see improvement. More broadly, by providing exact non-asymptotic expressions for the variance of PPI++ under sample splitting, we aim to empower practitioners to transparently reason about the benefits of PPI++ in specific applications.
In experiments, we illustrate that our theoretical findings hold on real-world datasets.
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TRAP: Hijacking VLA CoT-Reasoning via Adversarial Patches
Zhengxian Huang, Wenjun Zhu, Haoxuan Qiu, Xiaoyu Ji, Wenyuan Xu
By integrating Chain-of-Thought (CoT) reasoning, Vision-Language-Action (VLA) models have demonstrated strong capabilities in robotic manipulation, particularly by improving generalization and interpretability. However, the security of CoT-based reasoning mechanisms remains largely unexplored. In this paper, we show that CoT reasoning introduces a novel attack vector for targeted behavior hijacking—for example, causing a robot to mistakenly deliver a knife to a person instead of an apple—without modifying the user’s instruction. We first provide empirical evidence that CoT strongly governs action generation, even when it is semantically misaligned with the input instructions. Building on this observation, we propose **TRAP**, the first targeted behavior-hijacking adversarial attack against CoT-reasoning VLA models. By targeting the reasoning-to-action pathway, **TRAP** uses an adversarial patch (*e.g.*, a tablecloth placed on the table) to steer intermediate CoT reasoning and downstream actions toward adversary-defined behaviors. Extensive evaluations on three representative reasoning VLAs, spanning distinct CoT reasoning mechanisms, demonstrate the effectiveness of **TRAP**. Notably, we implemented the patch by printing it on paper in a real-world setting. Our findings highlight the urgent need to secure CoT reasoning in VLA systems. The project page is available at https://zhengxian-huang.github.io/TRAP-website/.
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One Probe Won’t Catch Them All: Towards Targeted Deception Detection
Vikram Natarajan, Devina Jain, Shivam Arora, Satvik Golechha, Joseph Isaac Bloom
Linear probes are a promising approach for monitoring AI systems for deceptive behaviour. Previous work has shown that a linear classifier trained on a contrastive instruction pair and a simple dataset can achieve good performance. However, these probes exhibit notable failures even in straightforward scenarios, including spurious correlations and false positives on non-deceptive responses. In this paper, we demonstrate that deception detection is inherently heterogeneous: wh

ile a single universal probe achieves modest improvements (+0.032 AUC), post-hoc oracle analysis reveals substantially higher potential (+0.108 AUC) when probes are matched to specific deception types, and synthetic validation experiments suggest this ceiling is achievable a priori when the deception type is known in advance. Our findings reveal that instruction pairs capture deceptive intent rather than content-specific patterns, explaining why prompt choice dominates probe performance (70.6% of variance). Given this heterogeneity, we conclude that organizations should define their specific threat models and deploy appropriately matched probes rather than seeking a universal deception detector.
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Learning to Emulate Chaos: Adversarial Optimal Transport Regularization
Gabriel Melo, Leonardo Santiago, Peter Y. Lu
Chaos arises in many complex dynamical systems, from weather to power grids, but is difficult to accurately model with data-driven methods such as machine learning emulators. While emulators are promising tools for accelerating simulations and solving inverse problems, they still struggle to learn chaotic dynamics, where sensitivity to initial conditions renders exact long-term forecasts infeasible, especially given noisy data. Recent work instead trains emulators to match the statistical properties of chaotic attractors, but these approaches often rely on handcrafted summary statistics or large, diverse multi-environment datasets. In this work, we propose a family of adversarial optimal transport objectives that can jointly learn high-quality summary statistics and a physically consistent emulator from a single noisy trajectory. We theoretically analyze and experimentally validate a Sinkhorn divergence formulation (2-Wasserstein) and a WGAN-style dual formulation (1-Wasserstein) of our approach. Numerical experiments across a variety of chaotic systems, including ones with high-dimensional spatiotemporal chaos, show that emulators trained using our proposed objectives have significantly improved long-term statistical fidelity.
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IRIS: Implicit Reward-Guided Internal Sifting for Mitigating Multimodal Hallucination
Yuanshuai li, Yuping Yan, Jirui Han, Fei Ming, Lingjuan Lyu, Yaochu Jin
Hallucination remains a fundamental challenge for Multimodal Large Language Models (MLLMs). While Direct Preference Optimization (DPO) is a key alignment framework, existing approaches often rely heavily on costly external evaluators for scoring or rewriting, incurring off-policy learnability gaps and discretization loss. Due to the lack of access to internal states, such feedback overlooks the fine-grained conflicts between different modalities that lead to hallucinations during generation. To address this issue, we propose IRIS (Implicit Reward-Guided Internal Sifting), which leverages continuous implicit rewards in the native log-probability space to preserve fine-grained preference information and capture internal modal competition. After an SFT warm-up, IRIS performs on-policy preference alignment by sifting self-generated responses sampled from the current policy. These responses are then ranked with multimodal implicit rewards to form preference pairs that drive optimization toward resolving modal conflicts. Extensive experiments demonstrate that IRIS achieves highly competitive performance on key hallucination benchmarks using only 5.7k samples, without requiring any external feedback during preference alignment. These results confirm that IRIS provides an efficient and principled paradigm for mitigating MLLM hallucinations. Code is available \href{https://github.com/ShawnLee0910/IRIS}{here}.
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TransLight: Image-Guided Customized Lighting Control with Generative Decoupling
Zongming Li, Lianghui Zhu, Haocheng Shen, Longjin Ran, Wenyu Liu, Xinggang Wang
Most existing illumination-editing methods struggle to jointly offer customized lighting control and preserve content integrity, limiting their effectiveness especially in transferring complex light effects from a reference to a target image in portrait photography. To address this problem, we propose TransLight, a novel framework that enables high-fidelity and high-freedom transfer of geometric-structured light effects such as Tyndall beams and specular highlights. Extracting light effects from the reference image is the most critical and challenging st

ep, as real-world lighting contains complex geometric structures tightly coupled with image content. To achieve this, we propose Generative Decoupling, using two fine-tuned diffusion models to accurately separate image content and lighting, and create a new million-scale dataset of image–content–light triplets. We then adopt IC-Light as the generative model, training it on these triplets with the reference lighting image as an additional conditioning signal. The resulting model enables customized and natural transfer of diverse light effects. Notably, by fully disentangling light effects from reference images, our generative decoupling strategy gives TransLight highly flexible illumination control. Experiments show that TransLight successfully transfers geometric-structured lighting effects across diverse images in portrait photography, offering more customized control than existing methods and charting new directions in illumination harmonization and editing. Our project page is hustvl.github.io/TransLight.
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From Generalist to Specialist Representation
Yujia Zheng, Fan Feng, Yuke Li, Shaoan Xie, Kevin Patrick Murphy, Kun Zhang
Given a generalist model, learning a task-relevant specialist representation is fundamental for downstream applications. Identifiability, the asymptotic guarantee of recovering the ground-truth representation, is critical because it sets the ultimate limit of any model, even with infinite data and computation. We study this problem in a completely nonparametric setting, without relying on interventions, parametric forms, or structural constraints. We first prove that the structure between time steps and tasks is identifiable in a fully unsupervised manner, even when sequences lack strict temporal dependence and may exhibit disconnections, and task assignments can follow arbitrarily complex and interleaving structures. We then prove that, within each time step, the task-relevant latent representation can be disentangled from the irrelevant part under a simple sparsity regularization, without any additional information or parametric constraints. Together, these results establish a hierarchical foundation: task structure is identifiable across time steps, and task-relevant latent representations are identifiable within each step. To our knowledge, each result provides a first general nonparametric identifiability guarantee, and together they mark a step toward provably moving from generalist to specialist models.
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SiameseNorm: Breaking the Barrier to Reconciling Pre/Post-Norm
Tianyu Li, Dongchen Han, Zixuan Cao, Haofeng Huang, Mengyu Zhou, Ming Chen, Erchao Zhao, xiaoxi jiang, Guanjun Jiang, Gao Huang
The long-standing tension between Pre- and Post-Norm remains an open problem in Transformer architecture, reflecting a fundamental trade-off between training stability and representational capacity. Prior attempts to combine their strengths have made progress, but often show limited robustness across training settings, restricting their broader applicability. We revisit this dilemma, showing that *single-stream* architectures struggle to reconcile Pre-Norm's stable identity-gradient propagation with Post-Norm's normalization of the main residual path. To address this structural tension, we propose SiameseNorm, a simple yet effective *two-stream* architecture that remains compatible with Pre-Norm training recipes. SiameseNorm couples Pre-Norm-like and Post-Norm-like streams through shared residual blocks, allowing each residual block to receive optimization signals from both pathways with negligible overhead. Extensive experiments on 400M and 1.3B dense language models, 15B MoE models, Vision Transformers, and Diffusion Transformers show that SiameseNorm consistently improves performance while maintaining strong training stability across architectures and modalities. Code is available at https://github.com/Qwen-Applications/SiameseNorm.
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Turning Back Without Forgetting: Selective Backward Refinement for Parameter-Efficient Continual Learning
Anushka Tiwari, Kaiyi Ji
While prompt-based parameter-efficient continual learning mitigates catastrophic forgetting by isolating task-specific prompts, this isolation also limits later tasks from improving earlier ones, leaving backward knowledge transfer underexp

lored. We address this limitation by proposing Selective bAckward refinement for positive Backward knowledge transfER (SABER), a replay-free framework that enables controlled backward transfer in prompt-based continual learning. SABER determines when backward refinement is beneficial using complementary task-correlation criteria based on prompt-gradient geometry and loss-distribution similarity, and how to perform refinement safely by restricting updates to non-interfering directions in the prompt parameter space. Extensive experiments across multiple continual learning benchmarks, and diverse pretrained backbones, including T5-Large, LLaMA, and Qwen, demonstrate that SABER consistently achieves positive backward transfer while maintaining strong overall average performance.
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Precision-Induced Miscalibration: Understanding and Correcting Confidence Distortion in Quantized Neural Networks
Jiawei Gu, Fengyuan Nie, Hao Tang, Yanpeng Sun
Low-precision arithmetic is pervasive in neural network training and deployment, yet its effect on prediction \textit{confidence}, not just accuracy, remains unexamined. We show that the softmax function amplifies logit-space quantization errors in an input-dependent manner: confidence distortion scales with the product of precision-dependent error bound ϵ and logit norm, peaking when the model is confident but not saturated. This explains why identical models report different confidence values across precisions, a phenomenon we term \textit{Precision Split}. During training, the same mechanism causes gradient underflow: when logit margins exceed a precision-dependent threshold, gradients vanish and samples silently stop contributing to learning. Since logit norm serves as a computable proxy for precision-induced risk, we propose Precision-Aware Confidence Scaling (PACS), which applies sample-adaptive temperature inversely related to this risk, with sub-one-percent overhead and no full-precision computation required. On ImageNet with mixed-precision ResNet-50, PACS reduces Expected Calibration Error from 5.82\% to 1.92\% while maintaining accuracy, with consistent improvements across architectures, precision formats, and modalities.
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CAReDiO: Enhancing Cultural Alignment of LLM via Representativeness and Distinctiveness Guided Data Optimization
Jing Yao, Xiaoyuan Yi, Jindong Wang, Zhicheng Dou, Xing Xie
As Large Language Models (LLMs) more deeply integrate into human life across various regions, aligning them with pluralistic cultures is crucial for improving user engagement and mitigating cultural conflicts. For this purpose, recently, different culture-specific corpora have been carefully curated, either synthesized or manually annotated. Nevertheless, inspired by culture theories, we identify two key challenges faced by these datasets: (1) Representativeness: These corpora fail to fully capture the target culture's core characteristics, causing insufficient cultural coverage with redundancy; (2) Distinctiveness: They struggle to distinguish the unique nuances of a given culture from shared patterns across other relevant ones, hindering precise cultural modelling. To handle these challenges, we introduce CAReDiO, a novel data optimization framework, which alternatively refines culture-sensitive questions and responses according to information-theoretic objectives in an in-context optimization manner, enhancing the cultural informativeness and distinguishability of constructed data. Extensive experiments on 15 distinct cultures demonstrate that CAReDiO can create high-quality data with richer cultural information and enable efficient alignment of small open-source or large proprietary LLMs with as few as 200 training samples, consistently outperforming previous datasets in both multi-choice and open-ended cultural benchmarks.
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Textual Supervision Enhances Geospatial Representations in Vision-Language Models
Marcelo Sartori Locatelli, Fernando Tonucci, Jea Kwon, Luiz Felipe Vecchietti, Bryan Nathanael Wijaya, Cheng Yaw Low, Virgilio Almeida, Meeyoung Cha
Geospatial understanding is a critical yet underexplored dimension in the development of machine learning systems for tasks such as image geolocation and spatia

l reasoning. In this work, we analyze the geospatial representations acquired by three model families: vision-only architectures (e.g., ViT), vision-language models (e.g., CLIP), and large-scale multimodal foundation models (e.g., LLaVA, Qwen, and Gemma). By evaluating across image clusters, including people, landmarks, and everyday objects, grouped based on the degree of localizability, we reveal systematic gaps in spatial accuracy and show that textual supervision enhances the learning of geospatial representations. Our findings suggest the role of language as an effective complementary modality for encoding spatial context and multimodal learning as a key direction for advancing geospatial AI.

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Learning to Refine: Spectral-Decoupled Iterative Refinement Framework for Precipitation Nowcasting

Yunlong Zhou, Chen Zhao, Danyang Peng, Fanfan Ji, Xiaotong Yuan

Accurate precipitation nowcasting is vital for disaster mitigation, but deep learning methods face a key trade-off: regression models produce over-smoothed, spectrally decaying predictions that blur convective details and violate turbulence power laws; diffusion models generate realistic yet unanchored hallucinations lacking physical grounding. We propose Spectral-Decoupled Iterative Refinement (SDIR), a deterministic framework that reformulates nowcasting as progressive frequency-decoupled refinement. SDIR first extracts a stable low-frequency synoptic skeleton, then iteratively refines high-frequency textures under physical constraints, eliminating both blurring and hallucinations. It features a dual-path design: the Synoptic Frequency-Guided Former (SFG-Former) with Scale-Adaptive Transformers for global structure, and the Fourier Residual Refiner (FR-Refiner) with Scale-Conditioned Fourier Neural Operators for fine residuals. A Physically Consistent Power Spectral Density (PCPSD) loss with dynamic masking enforces a turbulence-consistent spectral distribution. Experiments on three benchmarks show SDIR significantly outperforms SOTA methods in spatial accuracy while achieving spectral fidelity competitive with diffusion-based methods, enabling reliable high-resolution operational nowcasting. Code link: https://github.com/RuntimeWarning/SDIR.

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Stabilizing Equation Learning via Zero-Point Constraints

Sannyuya Liu, Ao Chen, Lin Liu, Ruxia Liang, Xiaoxuan Shen, Jianwen Sun

Symbolic Regression aims to discover interpretable mathematical expressions from data. Equation Learner (EQL) is a gradient-based method with strong fitting capability and expressive potential, yet it often activates redundant operators as model complexity grows, leading to over-complex expressions and unstable equation recovery. We analyze a gradient residual issue induced by operators that do not vanish at zero, which can prevent the ideal sparse expression from acting as a stable attractor during training and bias training toward unnecessarily complex structures, making exact recovery highly unreliable in practice. To address this, we propose EQL-Z, a structurally controllable symbolic regression framework. EQL-Z enforces zero-point constraints via zero-point consistent operator transformations to eliminate residual gradients on silent paths, and performs a small-to-large structure search that grows depth/width from a compact seed under a complexity-penalized validation score. After selecting a compact structure, we apply BFGS fine-tuning to refine coefficients. Experiments on synthetic and real-world datasets show that EQL-Z substantially improves exact equation recovery and in-/out-of-distribution generalization over vanilla EQL, achieving performance close to leading symbolic regression baselines. Code is available at https://github.com/Caaaa-a/EQL-Z.

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Motion Attribution for Video Generation

Xindi Wu, Despoina Paschalidou, Jun Gao, Antonio Torralba, Laura Leal-Taixé, Olga Russakovsky, Sanja Fidler, Jonathan Lorraine

Despite the rapid progress of video generation models, the role of data in influencing motion is poorly understood. We present Motive (MOTIon attribution for Video gEneration), a motion-centric, gradient-based data attribution framework that scales to modern, large, high-quality video datasets and models. We use this t

o study which fine-tuning clips improve or degrade temporal dynamics. Motive iso
lates temporal dynamics from static appearance via motion-weighted loss masks, y
ielding efficient and scalable motion-specific influence computation. On text-to
-video models, Motive identifies clips that strongly affect motion and guides da
ta curation that improves temporal consistency and physical plausibility. With M
otive-selected high-influence data, we improve both motion smoothness and dynami
c degree on VBench, achieving a 74.1% human preference win rate compared with th
e pretrained base model. To our knowledge, this is the first framework to attrib
ute motion rather than visual appearance in video generative models and to use i
t to curate fine-tuning data.
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Beyond Trajectory-Level Attribution: Graph-Based Credit Assignment for Agentic R
einforcement Learning
Xin Cheng, Shuo He, Lang Feng, Haiyang Xu, Ming Yan, Lei Feng, Bo An
Group-based reinforcement learning (RL) methods have achieved remarkable success
 in improving the performance of large language models (LLMs) and have been rapi
dly extended to agentic tasks. However, their credit assignment relies heavily o
n coarse-grained trajectory-level attribution according to final outcomes, makin
g it difficult to capture the contribution of individual steps, such as valuable
 steps obscured within failed trajectories. To uncover latent information and en
able more faithful step-level credit assignment, we propose Graph-based Group Po
licy Optimization (GraphGPO), which first aggregates all rollout trajectories in
to a unified state-transition graph and then estimates the distance from each st
ate to the task goal using the global information encoded in the graph. Finally,
 GraphGPO assigns credit to each edge by estimating a graph-based advantage, bas
ed on how much the transition reduces the distance to the task goal. In this way
, GraphGPO significantly improves training efficiency and achieves state-of-the-
art performance across a range of challenging benchmarks.
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Multi-Task GRPO: Reliable LLM Reasoning Across Tasks
Shyam Sundhar Ramesh, Xiaotong Ji, Matthieu Zimmer, Sangwoong Yoon, Zhiyong Wang
, Haitham Bou Ammar, Aurelien Lucchi, Ilija Bogunovic
RL-based post-training with GRPO is widely used to improve large language models
 on individual reasoning tasks. However, real-world deployment requires reliable
 performance across diverse tasks. A straightforward multi-task adaptation of GR
PO often leads to imbalanced outcomes, with some tasks dominating optimization w
hile others stagnate. Moreover, tasks can vary widely in how frequently prompts
yield zero advantages (and thus zero gradients), which further distorts their ef
fective contribution to the optimization signal. To address these issues, we pro
pose a novel Multi-Task GRPO (MT-GRPO) algorithm that (i) dynamically adapts tas
k weights to explicitly optimize worst-task performance and promote balanced pro
gress across tasks, and (ii) introduces a ratio-preserving sampler to ensure tas
k-wise policy gradients reflect the adapted weights. Experiments on both 3-task
and 9-task settings show that MT-GRPO consistently outperforms baselines in wors
t-task accuracy. In particular, MT-GRPO achieves 16–28\% and 6\% absolute improv
ement on worst-task performance over standard GRPO and DAPO, respectively, while
 maintaining competitive average accuracy. Moreover, MT-GRPO requires 50\% fewer
 training steps to reach 50\% worst-task accuracy in the 3-task setting, demonst
rating substantially improved efficiency in achieving reliable performance acros
s tasks.
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Game-Theoretic Co-Evolution for LLM-Based Heuristic Discovery
Xinyi Ke, Kai Li, Junliang Xing, Yifan Zhang, Jian Cheng
Large language models (LLMs) have enabled rapid progress in automatic heuristic
discovery (AHD), yet most existing methods are predominantly limited by static e
valuation against fixed instance distributions, leading to potential overfitting
 and poor generalization under distributional shifts. We propose Algorithm Space
 Response Oracles (ASRO), a game-theoretic framework that reframes heuristic dis
covery as a program level co-evolution between solver and instance generator. AS
RO models their interaction as a two-player zero-sum game, maintains growing str

ategy pools on both sides, and iteratively expands them via LLM-based best-respo
nse oracles against mixed opponent meta-strategies, thereby replacing static eva
luation with an adaptive, self-generated curriculum. Across multiple combinatori
al optimization domains, ASRO consistently outperforms static-training AHD basel
ines built on the same program search mechanisms, achieving substantially improv
ed generalization and robustness on diverse and out-of-distribution instances.
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Cross-Modal Knowledge Distillation without Paired Data: Theoretical Foundation a
nd Algorithm

T. Khiem Tran, Anh Duc Chu, Quang Hung Pham, Phi Le Nguyen, Trong Nghia Hoang

Cross-modal knowledge distillation (CMKD) studies how a (large) teacher model tr
ained on one type of data (e.g., images) can guide a (smaller) student model bui
lding on another type of data (e.g., text/audio). Existing CMKD methods often re
quire paired multi-modal data with aligned semantics, but obtaining such paired
data are often costly and impractical. To mitigate this limitation, we develop a
 new CMKD framework for the more challenging setting where paired data are unava
ilable. In particular, we establish a cross-modal distributional relationship be
tween teacher and student models which reveals two fundamental quantities govern
ing effective distillation: feature alignment and label alignment. These quantit
ies characterize semantic discrepancy between modalities at the levels of repres
entation and prediction distributions, respectively. Motivated by this insight,
we propose a principled framework, with theoretical guarantees, that enables eff
ective cross-modal knowledge distillation by aligning distributions rather than
individual samples. Extensive experiments across a wide range of multimodal benc
hmarks show that our framework is highly effective in both unpaired and paired d
ata settings, improving significantly over prior work.
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StableI2I: Spotting Unintended Changes in Image-to-Image Transition

Jiayang Li, Shuo Cao, Xiaohui Li, Zhizhen Zhang, Kaiwen Zhu, Yule Duan, Yu Qiao,
 Jian Zhang, Yihao Liu

In most real-world image-to-image (I2I) scenarios, existing evaluations primaril
y focus on instruction following and the perceptual quality or aesthetics of the
 generated images. However, they largely fail to assess whether the output image
 preserves the semantic correspondence and spatial structure of the input image.
 To address this limitation, we propose StableI2I, a unified and dynamic evaluat
ion framework that explicitly measures content fidelity and pre--post consistenc
y across a wide range of I2I tasks without requiring reference images, including
 image editing and image restoration. In addition, we construct StableI2I-Bench,
 a benchmark designed to systematically evaluate the accuracy of MLLMs on such f
idelity and consistency assessment tasks. Extensive experimental results demonst
rate that StableI2I provides accurate, fine-grained, and interpretable evaluatio
ns of content fidelity and consistency, with strong correlations to human subjec
tive judgments. Our framework serves as a practical and reliable evaluation tool
 for diagnosing content consistency and benchmarking model performance in real-w
orld I2I systems.
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Reinforcement Learning with Verifiable Rewards: GRPO's Loss, Dynamics, and Succe
ss Amplification

Youssef Mroueh

Group Relative Policy Optimization (GRPO) was introduced recently and used to tr
ain DeepSeek\textendash R1 for promoting reasoning in LLMs under verifiable (bin
ary) rewards. We show that the mean{+}variance calibration of these rewards indu
ces a contrastive loss in which the contrastive samples are synthetic data drawn
 from the previous policy. While GRPO was originally paired with clipping to kee
p updates near the old policy, we analyze variants that differ in reward normali
zation (mean-only vs.\ mean{+}variance) and in how they regularize updates using
 KL divergence: either penalizing divergence from the previous model (\emph{mirr
or}), penalizing divergence from a fixed reference model π_{ref}, o
r combining both forms of regularization. For each, the optimal policy π_n a
dmits an explicit form in terms of the binary reward and the first and second o

rder statistics of the reward under π_{n-1}, as well as the policies π_{n-1} and π_{ref}. Iterating results in a sequence $\{\pi_n\}$ whose \emph{probability of success (PoS)} obeys a simple recurrence that converges to a fixed point determined by the reference PoS and the regularization strength. We further show that this fixed point exceeds the reference, demonstrating that GRPO amplifies the policy's probability of success.
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Expert-level Leaf Cell Layout Generation via Preference-Optimized LLM
Yaohui Han, Rongliang Fu, Yanming Liu, Shuo Ren, Shuai Dong, Yunpeng Wang, Tinghuan Chen, Bei Yu, Tsung-Yi Ho
In the field of integrated circuits, leaf cells are the basic units, serving as the fundamental building blocks (e.g., standard cells) that are widely reused in various VLSI designs, forming the basis for more complex circuits. Therefore, the design quality of leaf cell layouts significantly impacts the PPA (Power, Performance, and Area) of the final VLSI designs. To automatically design leaf cell layouts that are close to expert designs, we propose GenLeaf. GenLeaf first utilizes a supervised, performance-aware embedding model to represent layouts and automatically calculate their similarity scores. Since there are expert-designed layouts but no corresponding scripts, we implement Bayesian optimization to generate a layout-script dataset for LLM training. With subsequent supervised fine-tuning and further preference optimization, GenLeaf can generate leaf cell layouts through scripts whose performance closely resembles that designed by human engineers. Experiment results demonstrate that GenLeaf outperforms expert-designed golden layouts across key performance metrics.
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Towards Understanding Modality Interaction in Multimodal Language Models via Partial Information Decomposition
Wanlong Fang, Tianle Zhang, Wen Tao, Alvin Chan
Understanding modality interaction in multimodal large language models (MLLMs) is central to reliable deployment. We introduce Partial Information Decomposition (PID) as a decision-level framework that separates unique, redundant, and synergistic contributions of sensory and linguistic inputs, beyond representation alignment and outcome-based evaluation. Across vision--language benchmarks, PID reveals recurring modality-use profiles: reasoning and grounding-oriented tasks tend to exhibit high synergy, whereas expert and knowledge-oriented tasks show stronger language-unique reliance. These profiles generalize across model families and predict sensitivity to modality-level interventions. We further extend PID to tri-modal systems with Sensory PID, treating language as a control variable to decompose video--audio information gain. Applied to omni-modal models, Sensory PID reveals a sensory synergy bottleneck dominated by visual information even on audio--visual fusion tasks. Finally, PID-guided reweighting provides initial evidence for improving multimodal reasoning and grounding performance.
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Distributional Open-Ended Evaluation of LLM Cultural Value Alignment Based on Value Codebook
Jaehyeok Lee, Xiaoyuan Yi, Jing Yao, Hyunjin Hwang, Roy Ka-Wei Lee, Xing Xie, JinYeong Bak
As LLMs are globally deployed, aligning their cultural value orientations is critical for safety and user engagement. However, existing benchmarks face the Construct-Composition-Context (C^3) challenge: relying on discriminative, multiple-choice formats that probe value knowledge rather than true orientations, overlook subcultural heterogeneity, and mismatch with real-world open-ended generation. We introduce DOVE, a distributional evaluation framework that directly compares human-written text distributions with LLM-generated outputs. DOVE utilizes a rate-distortion variational optimization objective to construct a compact value codebook from 10K documents, mapping text into a structured value space to filter semantic noise. Alignment is measured using unbalanced optimal transport, capturing intra-cultural distributional structures and subgroup diversity. Experiments across 12 LLMs show that DOVE achieves superior predictive validity, attaining a 31.56% correlation with downstream tasks, while maintaining high reliability

with as few as 500 samples per culture.
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Bayesian-LoRA: Probabilistic Low-Rank Adaptation of Large Language Models
Moule Lin, Shuhao Guan, Andrea Patane, David Gregg, Goetz Botterweck
Large language models are typically optimized for accuracy and, therefore, will guess even when uncertain about their predictions. This problem becomes especially pronounced when the model is fine-tuned on small datasets, which often causes overfitting and results in a tendency toward miscalibration. In this work, we introduce Bayesian-LoRA, which reformulates the deterministic LoRA update as a probabilistic low-rank representation inspired by Sparse Gaussian Processes (SGP). We identify a structural isomorphism between LoRA's factorization and Kronecker-factored SGP posteriors, and show that LoRA emerges as a limiting case when posterior uncertainty collapses. We conduct extensive experiments on various LLM architectures across commonsense reasoning, language modeling, and mathematical reasoning benchmarks. With only approximately 0.42M additional parameters and ${\approx}1.2{\times}$ training cost relative to standard LoRA, Bayesian-LoRA significantly improves calibration across models from 7B up to 30B, achieving up to 84\% Expected Calibration Error (ECE) and 76\% Negative Log-Likelihood (NLL) reduction while maintaining competitive accuracy for both in-distribution and out-of-distribution (OoD) evaluations.
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Do We Need Adam? Surprisingly Strong and Sparse Reinforcement Learning with SGD in LLMs
Sagnik Mukherjee, Lifan Yuan, Pavan Jayasinha, Dilek Hakkani-Tür, Hao Peng
Reinforcement learning (RL), particularly RL from verifiable reward (RLVR), has become a crucial phase of training large language models (LLMs) and a key focus of current scaling efforts.
However, optimization practices in RL largely follow those of next-token-prediction stages (e.g., pretraining and supervised fine-tuning), despite the fundamental differences between RL and these stages emphasized by recent work.
One such practice is the use of the AdamW optimizer, which is widely adopted for training large-scale transformers despite its high memory overhead.
Our analysis shows that both momentum and adaptive learning rate of AdamW are less influential in RL than in SFT, leading us to hypothesize that RL benefits less from Adam’s per-parameter adaptive learning rates and momentum.
Confirming our hypothesis, our experiments demonstrate that the substantially more memory-efficient SGD, which is known to perform poorly in supervised learning of large-scale transformers, matches or even outperforms AdamW in RL for LLMs.
Remarkably, full fine-tuning with SGD updates fewer than 0.02% of model without any sparsity-promoting regularization, more than 1,000 times fewer than AdamW. Our analysis offers potential reasons for this update sparsity.
Our findings provide fresh insights into the optimization dynamics of RL in LLMs and demonstrate that RL can be substantially more parameter-efficient than previously recognized.
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InfiMed-ORBIT: Aligning LLMs on Open-Ended Complex Tasks via Rubric-Based Incremental Training
Pengkai Wang, Pengwei Liu, Qi Zuo, Zhijie Sang, Congkai Xie, Hongxia Yang
Reinforcement learning (RL) has powered many recent breakthroughs in large language models (LLMs), especially for tasks where rewards can be computed automatically, such as code generation. However, it is less effective in open-ended medical dialogue, where feedback is ambiguous, context-dependent, and difficult to simply summarize into a single scalar signal—often requiring heavily supervised reward models and creating risks of reward hacking. Thus, we introduce ORBIT, an open-ended rubric-based incremental training framework tailored for critical medical dialogues. ORBIT integrates medical dialogue construction with dynamically generated case-conditioned rubrics that serve as adaptive guides for incremental RL. Unlike approaches that rely on external medical knowledge bases or handcrafted rules, ORBIT uses rubric-guided evaluation and can be implemented with general-purpose instruction-following LLMs, avoiding task-specific judge fine-tuning. W

ith only 2k training samples, ORBIT raises Qwen3-4B-Instruct's HealthBench-Hard score from 7.0 to 27.5, achieving state-of-the-art performance among similarly s ized open-source models while maintaining strong consultation quality as rubric coverage broadens. Project page: https://pidneuralode.github.io/ORBIT.
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Learning to Decode Against Compositional Hallucination in Video Multimodal Large Language Models

Wenbin Xing, Quanxing Zha, Lizheng Zu, Mengran Li, Ming Li, Junchi Yan

Current research on video hallucination mitigation primarily focuses on isolated error types, leaving *compositional* hallucinations—arising from incorrect reas oning over multiple interacting spatial and temporal factors largely underexplor ed. We introduce **OmniVCHall**, a benchmark designed to systematically evaluate both isolated and compositional hallucinations in Video Large Language Models (VLLMs). OmniVCHall spans diverse video domains, introduces a novel camera-based hallucination type, and defines a fine-grained taxonomy, together with adversari al answer options (*e.g.*, “All are correct” and “None of the above”) to prevent shortcut reasoning. The evaluations of 39 representative VLLMs reveal that even advanced models (*e.g.*, Qwen3-VL and GPT-5) exhibit substantial performance de gradation. We propose **TriCD**, a contrastive decoding framework with a triple- pathway calibration mechanism. An adaptive perturbation controller dynamically s elects distracting operations to construct negative video variants, while a sali ency-guided enhancement module adaptively reinforces grounded token-wise visual evidences. These components are optimized via reinforcement learning to encourag e precise decision-making under compositional hallucination settings. Experiment al results show that **TriCD** consistently improves performance across two repr esentative backbones, achieving an average accuracy improvement of over 10\%. Th e data and code are released at https://github.com/BMRETURN/OmniVCHall.
****

Frequency-Aware Perceptual Optimization for Low-Complexity Implicit Image Compre ssion

Haotian Wu, Gen Li, Di You, Pier Luigi Dragotti, Deniz Gunduz

We propose a frequency-aware perceptual optimization framework for low-complexit y image compression, realized as a **Re**alism-enhanced **Re**gion-based **I**mp licit **C**odec (Re2IC). Re2IC models visual perception via saliency-guided regi on partitioning and local–global perceptual modulation. To enhance realism under complexity constraints, we introduce wavelet–Wasserstein distortion (WA-WD), a frequency-decomposed perceptual distortion that balances fidelity and realism th rough subband-aware modeling and provides a more reliable approximation than sta ndard Wasserstein distortion. Together, these designs enable fine-grained spatia l–spectral optimization, allowing Re2IC to achieve superior rate–perception trad e-offs, outperforming generative codecs such as HiFiC while using less than 1\% of their decoding cost. Extensive experiments demonstrate state-of-the-art perce ptual performance among overfitted codecs. Beyond compression, WA-WD serves as a standalone, tunable perceptual metric with strong alignment to human preference (Pearson 94.6\%, Spearman 92.3\%) and competitive performance across multiple I QA benchmarks. Project page: https://eedavidwu.github.io/ReReIC/
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Focusing Where Vision Matters: Selective Training for Large Vision Language Mode ls via Visual Information Gain

Seulbi Lee, Sangheum Hwang

Large Vision Language Models (LVLMs) have achieved remarkable progress, yet they often suffer from language bias, producing answers without relying on visual ev idence. While prior work attempts to mitigate this issue through decoding strate gies, architectural modifications, or curated instruction data, they typically l ack a quantitative measure of how much individual training samples or tokens act ually benefit from the image. In this work, we introduce Visual Information Gain (VIG), a perplexity-based metric that measures the reduction in prediction unce rtainty provided by visual input. VIG enables fine-grained analysis at both samp le and token levels, effectively highlighting visually grounded elements such as colors, spatial relations, and attributes. Leveraging this, we propose a VIG-gu

ided selective training scheme that prioritizes high-VIG samples and tokens. Thi
s approach improves visual grounding and mitigates language bias, achieving supe
rior performance with significantly reduced supervision by focusing exclusively
on visually informative samples and tokens.
****
IO-Adam: Rethinking Memory-Efficient Adaptive Optimizers from Gradient Computati
on
Yiting Chen, Zongwei Huo, Junchi Yan
Adaptive Moment Estimation (Adam) is one of the most popular and often the defau
lt stochastic optimizers for deep neural network training. Using first- and seco
nd-moment estimation, Adam provides adaptive learning rates for each parameter,
significantly outperforming Stochastic Gradient Descent (SGD). However, as deep
neural networks become larger, estimating the first and second moments consumes
substantial memory. It motivates various methods to reduce memory usage for adap
tive optimizers. In this paper, we propose to rethink the first and second momen
t estimation from a gradient computation perspective. The gradient of the weight
 matrix is the multiplication of the input and the gradient of the output. Inste
ad of finding low-rank approximations of the first and second moments, as in pre
vious work, we propose tracking the input and output gradients to efficiently es
timate moments. We provide analyses of the similarities and differences between
our proposed method, the widely used Adam optimizer, and previous memory-efficie
nt optimizers designed to reduce memory usage. We conduct experiments to verify
the effectiveness of our method, which reduces memory usage by up to 30% while
 preserving similar performance or even improving the performance of Adam.
****
Self-Soupervision: Cooking Model Soups without Labels
Anthony Fuller, James R Green, Evan Shelhamer
Model soups are strange and strangely effective combinations of parameters. They
 take a model (the stock), fine-tune it into multiple models (the ingredients),
and then mix their parameters back into one model (the soup) to improve predicti
ons. While all known soups require supervised learning, and optimize the same lo
ss on labeled data, our recipes for Self-Soupervision generalize soups to self-s
upervised learning (SSL). Our Self-Souping lets us flavor ingredients on new dat
a sources, e.g. from unlabeled data from a task for transfer or from a shift for
 robustness. We show that Self-Souping on corrupted test data, then fine-tuning
back on uncorrupted train data, boosts robustness by +3.5% (ImageNet-C) and +7%
(LAION-C). Self-Soupervision also unlocks countless SSL algorithms to cook the d
iverse ingredients needed for more robust soups. We show for the first time that
 ingredients can differ in their SSL hyperparameters---and more surprisingly, in
 their SSL algorithms. We cook soups of MAE, MoCoV3, MMCR, and LeJEPA ingredient
s that are more accurate than any single SSL ingredient.
****
Failure is Feedback: History-Aware Backtracking for Agentic Traversal in Multimo
dal Graphs
Joohyung Yun, Doyup Lee, Wook-Shin Han
Open-domain multimodal document retrieval aims to retrieve specific components (
paragraphs, tables, or images) from large and interconnected document corpora. E
xisting graph-based retrieval approaches typically rely on a uniform similarity
metric that overlooks hop-specific semantics, and their rigid pre-defined plans
hinder dynamic error correction. These limitations suggest that a retriever shou
ld adapt its reasoning to the evolving context and recover intelligently from de
ad ends. To address these needs, we propose **Failure is Feedback**, which casts
 subgraph retrieval as a *sequential decision process* and introduces two key in
novations. (i) We introduce a *history-aware backtracking mechanism*; unlike sta
ndard backtracking that simply reverts the state, our approach piggybacks on the
 context of failed traversals, leveraging insights from previous failures. (ii)
We implement an *economically-rational agentic workflow*. Unlike conventional ag
ents with static strategies, our orchestrator employs a cost-aware traversal met
hod to dynamically manage the trade-off between retrieval accuracy and inference
 costs, escalating to intensive LLM-based reasoning only when the prior failure

justifies the additional computational investment. The project page is available at failureisfeedback.github.io.
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Mitigating the Contractivity Trap in Diffusion ODEs via Stein Stabilization
Shigui Li, Delu zeng
A fundamental tension exists in the large-step inference of diffusion models via their deterministic probability flow ordinary differential equation (PF-ODE) trajectories, which we identify as the contractivity trap: efficient inference favors large step sizes, while aggressive steps and highly expressive denoisers can undermine contraction-based stability certificates for error suppression. To address this, we propose SteinDiff, a step-wise inference-time stabilization framework that employs Stein-derived corrections without requiring reference samples. Specifically, SteinDiff introduces a geometry-aware residual correction mechanism that regularizes large-step solver updates without retraining. To this end, we derive a closed-form Stein correction coefficient for step-wise solver adjustment, enabling reference-free adaptation to local data geometry. We further establish a score-controlled perturbation bound under distributional shifts and provide a complementary Stein perspective on EDM-style parameterizations. Extensive experiments demonstrate that SteinDiff mitigates severe artifacts and improves generative quality across large-step inference settings.
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DecoderTCR: Compositional Pretraining and Entropy-Guided Decoding for TCR-pMHC Interactions
Boqiao Lai, Melissa Englund, Ramit Bharanikumar, Isabel Nocedal, Ali Davariashtiyani, Jason Perera, Aly A Khan
Modeling recognition between T-cell receptors (TCRs) and peptide-MHC (pMHC) complexes is a fundamental challenge in computational immunology, constrained by sparse paired interaction data relative to abundant unpaired sequences. We introduce DecoderTCR, a masked language model framework that addresses this through two contributions: (1) a compositional continual pre-training curriculum that learns component representations from marginal data before refining cross-chain dependencies, and (2) Iterative Entropy-Guided Refinement (IEGR), a non-autoregressive decoding algorithm that resolves high-confidence positions first to provide context for uncertain regions. On held-out benchmarks, DecoderTCR achieves 0.96 AUROC for zero-shot pMHC binding prediction and 0.76 AUROC for epitope-specific TCR recognition, approaching supervised baselines without epitope-specific training. Learned representations recover structural contacts without coordinate supervision, and generated sequences exhibit realistic recombination statistics. Experimental validation across two rounds of wet-lab screening reveals a prediction-generation gap that can be narrowed via a lab-in-the-loop paradigm for TCR design.
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Adaptive Time Series Reasoning via Segment Selection
Shvat Messica, Jiawen Zhang, Kevin Li, Theodoros Tsiligkaridis, Marinka Zitnik
Time series reasoning tasks often start with a natural language question and require targeted analysis of a time series. Evidence may span the full series or appear in a few short intervals, so the model must decide what to inspect. Most existing approaches encode the entire time series into a fixed representation before inference, regardless of relevance.
We introduce ARTIST, which formulates time-series reasoning as a sequential decision problem. ARTIST interleaves reasoning with adaptive temporal segment selection. It adopts a controller-reasoner architecture and uses reinforcement learning to train the controller role to select informative segments and the reasoner role to generate segment-conditioned reasoning traces and final answers. During inference, the model actively acquires task-relevant information instead of relying on a static summary of the full sequence. We use a novel hierarchical policy optimization approach for post-training that allows the model to excel in both segment selection and question-answering behavior. We evaluate ARTIST on six time-series reasoning benchmarks and compare it with large language models, vision-language models, and prior time-series reasoning systems. ARTIST improves average accuracy by 6.46 absolute percentage points over the strongest baseline. The la

rgest gains appear on rare event localization and multi-segment reasoning tasks. Supervised fine-tuning improves performance, and reinforcement learning provides additional gains by optimizing question-adaptive segment selection, showing that selective data use drives effective time-series reasoning.
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Di-BiLPS: Denoising induced Bidirectional Latent-PDE-Solver under Sparse Observations
Zhonghao Li, Chaoyu Liu, Zhang Qian
Partial differential equations (PDEs) are fundamental for modeling complex natural and physical phenomena. In many real-world applications, however, observational data are \textbf{extremely sparse}, which severely limits the applicability of both classical numerical solvers and existing neural approaches. While neural methods have shown promising results under moderately sparse observations, their inference efficiency at high resolutions is limited, and their accuracy degrades substantially in the extremely sparse regime. In this work, we propose the \textbf{Di-BiLPS}, a unified neural framework that effectively handle \textbf{both forward and inverse} PDE problems under extremely sparse observations. Di-BiLPS combines a variational autoencoder to compress high-dimensional inputs into a compact latent space, a latent diffusion module to model uncertainty, and contrastive learning to align representations. Operating entirely in this latent space, the framework achieves efficient inference while retaining flexible input–output mapping. In addition, we introduce a \textbf{PDE-informed denoising algorithm} based on a variance-preserving diffusion process, which further improves inference efficiency. Extensive experiments on multiple PDE benchmarks demonstrate that Di-BiLPS consistently achieves \textbf{SOTA performance under extremely sparse inputs (as low as 3\%)}, while substantially reducing computational cost. Moreover, Di-BiLPS enables \textbf{zero-shot super-resolution}, as it allows predictions over continuous spatial–temporal domains.
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Beyond Pixel Histories: World Models with Persistent 3D State
Samuel Garcin, Thomas Walker, Steven McDonagh, Tim Pearce, Hakan Bilen, Tianyu He, Kaixin Wang, Jiang Bian
Interactive world models continually generate video by responding to a user's actions, enabling open-ended generation capabilities.
However, existing models typically lack a 3D representation of the environment, meaning 3D consistency must be implicitly learned from data, and spatial memory is restricted to limited temporal context windows. This results in an unrealistic user experience and presents significant obstacles to downstream tasks such as training agents. To address this, we present PERSIST, a new paradigm of world model which simulates the evolution of a latent 3D scene: environment, camera, and renderer.
This allows us to synthesise new frames with persistent spatial memory and consistent geometry. Both quantitative metrics and a qualitative user study show substantial improvements in spatial memory, 3D consistency, and long-horizon stability over existing methods, enabling coherent, evolving 3D worlds. We further demonstrate novel capabilities, including synthesising diverse 3D environments from a single image, as well as enabling fine-grained, geometry-aware control over generated experiences by supporting environment editing and specification directly in 3D space.
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Future Dynamic 3D Reconstruction: A 3D World Model with Disentangled Ego-Motion
Nils Morbitzer, Jonathan Evers, Artem Savkin, Thomas Stauner, Nassir Navab, Federico Tombari, Stefano Gasperini
Forecasting the evolution of dynamic environments is crucial for autonomous agents. While generative world models have recently achieved high photorealism in 2D video synthesis by mixing ego-motion and environmental dynamics within the image plane, they exhibit physical inconsistencies, such as morphing or vanishing objects, especially over long time horizons. In this paper, we propose FR3D, a world model that predicts a persistent 3D latent representation for future dynamic 3D reconstruction. Unlike prior works that treat the world as a sequence of imag

e-based features, FR3D explicitly decouples the 3D evolution of the scene from t
he agent's trajectory, treating the inferred ego-motion as a latent proxy for ac
tion. This disentanglement resolves the ambiguities between self-motion and worl
d-motion, ensuring geometric consistency into the future. Furthermore, we introd
uce a teacher-student distillation strategy that leverages the spatial "common s
ense" of off-the-shelf foundation models, leading to robust zero-shot generaliza
tion. Extensive experiments demonstrate FR3D's strong performance for future dyn
amic 3D reconstruction from monocular observations across multiple datasets, eve
n 2 seconds into the future. Project page: https://fr3d-wm.github.io.
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DomED: Redesigning Ensemble Distillation for Domain Generalization
Ziang Song, Zhidan Zhou, Zijun Zhang
Domain generalization aims to improve model performance on unseen, out-of-distri
bution (OOD) domains, yet existing methods often overlook the crucial aspect of
uncertainty quantification in their predictions. While ensemble learning combine
d with knowledge distillation offers a promising avenue, naively combining these
techniques is non-trivial and remains largely unexplored in the context of doma
in generalization. In this work, we systematically investigate different ensembl
e and distillation strategies for domain generalization tasks and design a tailo
red data allocation scheme. This approach trains teacher models on distinct subs
ets of domains and performs distillation on complementary (unseen) subsets, ther
eby fostering model diversity and training efficiency. Moreover, our theoretical
analysis demonstrates that distilling from teachers on unseen domains effective
ly filters out domain-specific spurious correlations. To address the accuracy de
gradation often observed with standard uncertainty distillation, we further deve
lop a novel technique that decouples uncertainty distillation from the standard
distillation process, enabling accurate uncertainty estimation without compromis
ing model accuracy. Our proposed method, *Domain-aware Ensemble Distillation* (D
omED), is extensively evaluated against state-of-the-art domain generalization a
nd ensemble distillation techniques across multiple benchmarks, achieving compet
itive accuracies and substantially improved uncertainty estimates.
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Why Linear Recurrent Memory Works in Partially Observable Reinforcement Learning
Yike Zhao, Onno Eberhard, Malek khammassi, Ali H. Sayed, Michael Muehlebach
The family of linear recurrent neural networks has shown strong performance as r
ecurrent memory units in partially observable reinforcement learning. We provide
a theoretical justification for their empirical effectiveness by constructing a
nd studying two linear filters:
(i) the first exactly reproduces the pre–softmax logits of the belief vector in
a hidden Markov model (HMM) under a deterministic transition matrix, thereby ser
ving as a sufficient statistic for optimal policy learning, (ii) the second achi
eves vanishing state-decoding error under a nearly deterministic transition matr
ix, thus reducing state ambiguity to near zero. The results extend to action-con
trolled HMMs, where the corresponding linear filters become time-varying with ac
tion-dependent dynamics. We illustrate our main results through numerical experi
ments and further show that the constructed linear filter serves as a strong fea
ture extractor in a small reinforcement learning game.
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Learning to Watch: Active Video Anomaly Understanding via Interleaved Policy Opt
imization
Mengjingcheng Mo, Jiaxu Leng, Xinbo Gao
Video anomaly understanding (VAU) relies on sparse, context-dependent cues. Howe
ver, existing passive paradigms suffer from observational aliasing, where static
sampling fails to disambiguate semantically distinct events. To overcome this,
we propose $Anom\text{-}\pi$, a closed-loop framework that reconceptualizes vide
o understanding as an active sequential decision-making process within a dynamic
environment.
Inspired by human video-reviewing behavior, this framework unifies internal cogn
itive reasoning and strategic evidence acquisition into an interleaved policy, u
tilizing temporal atomic operators such as local backtracking, temporal expansio

n, and fine-grained sampling to endow the model with perceptual proactivity. To learn such complex interaction strategies under video-level weak supervision, we design Interactive Direct Preference Optimization (iDPO) to achieve trajectory-level policy alignment, guided by an Active Evidence Inquiry (AEI) utility that balances task success, informative evidence acquisition, and interaction cost. This approach enables the agent to learn to actively disambiguate hypotheses while suppressing redundant exploration. Extensive experiments demonstrate that our framework, with only 2B parameters, achieves highly competitive performance, significantly outperforming state-of-the-art large-scale VAU models in complex scenarios.

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DAG-MoE: From Simple Mixture to Structural Aggregation in Mixture-of-Experts

Jiarui Feng, Hanqing Zeng, Karish Grover, Ruizhong Qiu, Yinglong Xia, Qiang Zhang, Qifan Wang, Ren Chen, Dongqi Fu, Jiayi Liu, Zhuokai Zhao, Xiangjun Fan, Benyu Zhang, Yixin Chen

Mixture-of-Experts (MoE) models have become a leading approach for decoupling parameter count from computational cost in large language models, yet effectively scaling MoE performance remains a challenge. Prior work shows that fine-grained experts enlarge the space of expert combinations and improve flexibility, but they also impose substantial routing overhead, creating a new scalability bottleneck. In this paper, we explore a complementary axis for scaling—how expert outputs are aggregated. We theoretically show that replacing the standard weighted-summation aggregation with structural aggregation expands the expert-combination space without altering the experts or router, and enables possible multi-step reasoning within a single MoE layer. To this end, we propose DAG-MoE, a sparse MoE framework that employs a lightweight module to automatically learn the optimal aggregation structure among the selected experts. Extensive experiments under standard language modeling settings show that DAG-MoE consistently improves performance in both pretraining and fine-tuning, surpassing traditional MoE baselines.

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Steering Out-of-Distribution Generalization with Concept Ablation Fine-Tuning

Helena Casademunt, Caden Juang, Adam Karvonen, Samuel Marks, Senthooran Rajamanoharan, Neel Nanda

Fine-tuning large language models (LLMs) can lead to unintended out-of-distribution generalization. Standard approaches to this problem rely on modifying the training data, for example by adding data that better specify the intended generalization. However, this is not always practical. We introduce Concept Ablation Fine-Tuning (CAFT), a technique that leverages interpretability tools to control how LLMs generalize from fine-tuning, without needing to modify the training data or otherwise use data from the target distribution. Given a set of directions in an LLM's latent space corresponding to undesired concepts, CAFT works by ablating these concepts with linear projections during fine-tuning, steering the model away from unintended generalizations. We successfully apply CAFT to three fine-tuning tasks, including emergent misalignment, a phenomenon where LLMs fine-tuned on a narrow task generalize to give egregiously misaligned responses to general questions. Without any changes to the fine-tuning data, CAFT reduces misaligned responses by 10x without degrading performance on the training distribution. Overall, CAFT represents a novel approach for steering LLM generalization without modifying training data.

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Sketch-Based Low-Rank Model Merging with Shared Circulant Transforms

Zhiming Zhang, Rong Yin, Xiaoshuai Hao, Hansong Zhang, Hao Peng, Yong Liu, Can Ma, Dan Meng

Merging multiple low-rank adapters (LoRA) provides a practical route to scaling multi-task learning and deployment more efficiently than full-model weight merging, while avoiding reliance on task-specific training data. However, most existing approaches either treat LoRA updates as dense weight deltas or depend on expensive subspace factorizations, making the merge step a primary latency bottleneck. To address this issue, this paper establishes a theoretically positive relationship between merging quality and the effective rank of the matrices being merg

ed. Motivated by this insight, we propose **CircuMerge**, a sketch-based framewo rk for low-rank model merging built on shared circulant transforms. Especially, this approach treats each adapter as a pair of low-rank matrices and applies a s hared circulant transform to align all tasks in a common coordinate system. This alignment enables more efficient sampling, allowing us to generate compact sket ches that effectively summarize the interactions between tasks. These compact sk etches enable applying the merging rules directly to them and reconstructing a s tandard low-rank adapter, preserving the essential information while significant ly reducing computational overhead. Across a broad multi-task LoRA benchmarks co vering both vision and language settings, extensive empirical results demonstrat e that CircuMerge reduces the overall merging time by at least 44\% compared to the state-of-the-art approaches, with accuracy matching or exceeding the optimal level.

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The Realignment Problem: When Right becomes Wrong in LLMs

Aakash Sen Sharma, Debdeep Sanyal, Manodeep Ray, Vivek Srivastava, Shirish Karan de, Murari Mandal

Post-training alignment of large language models (LLMs) relies on large-scale hu man annotations guided by policy specifications that change over time. Cultural shifts, value reinterpretations, and regulatory or industrial updates make stati c alignment increasingly brittle. As policies evolve, deployed models can diverg e from current alignment objectives, creating an Alignment–Reality Gap that is d ifficult to audit or correct. Existing remediation typically requires re-annotat ion under revised guidelines, which introduces systematic challenges, including guideline ambiguity, annotator interpretation drift, and reduced consistency at scale. We introduce TRACE (Triage and Re-align by Alignment Conflict Evaluation) , a framework that transforms re-alignment into a structured optimization proble m over existing data without requiring fresh human annotation. Leveraging a stro nger model as a proxy judge, TRACE operates via a three-stage pipeline: (1) tria ging preference pairs into inversion, suppression, or retention categories based on alignment conflicts; (2) computing an alignment impact score via bi-level op timization to prioritize high-leverage samples; and (3) executing updates using a hybrid objective that combines relational losses (e.g., IPO) for preference in version and punitive losses (e.g., NPO) for response suppression. Experiments on Qwen2.5-7B, Gemma-2-9B, and Llama-3.1-8B demonstrate robust re-alignment on syn thetic benchmarks and the PKU-SafeRLHF dataset without degrading general utility . This work provides a scalable approach for LLM realignment under evolving data annotation policies and alignment guidelines. We release our code \href{https:/ /respailab.github.io/TRACE/}{here}.

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Learning Sparse Visual Representations via Spatial-Semantic Factorization

Theodore Zhao, Sid Kiblawi, Jianwei Yang, Naoto Usuyama, Reuben Tan, Noel C Code lla, Tristan Naumann, Hoifung Poon, Mu Wei

Self-supervised learning (SSL) faces a tension between semantic understanding an d image reconstruction. High-level semantic SSL methods such as DINO encourage t ransformation-invariant global representations for augmentation alignment, while reconstruction-oriented methods such as MAE preserve dense feature grids for sp atial grounding but often yield weaker semantic abstractions. We introduce STELL AR, a framework that alleviates this tension by factorizing visual features into a low-rank product of semantic concepts and their spatial distributions. This d isentanglement allows us to perform augmentation alignment on semantic tokens wh ile maintaining spatial localization through a localization matrix for reconstru ction. We demonstrate that as few as 16 sparse tokens under this factorized form can simultaneously support high-quality reconstruction (2.60 FID) and strong se mantic transfer (79.10% ImageNet accuracy). Our results highlight STELLAR as a v ersatile sparse representation that bridges discriminative and generative vision by separating semantic identity from spatial geometry. Code is available at htt ps://github.com/microsoft/STELLAR.

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Unbiased and Second-Order-Free Training for High-Dimensional PDEs

Jaemin Seo, Su Rin Lee, Jae Yong Lee
Deep learning methods based on backward stochastic differential equations (BSDEs) have emerged as competitive alternatives to physics-informed neural networks (PINNs) for solving high-dimensional partial differential equations (PDEs). By le veraging probabilistic representations, BSDE approaches can avoid the curse of d imensionality and often admit second-order-free training objectives that do not require explicit Hessian evaluations. It has recently been established that the commonly used Euler–Maruyama (EM) time discretization induces an intrinsic bias in BSDE training losses. While high-order schemes such as Heun can fully elimina te this bias, such schemes re-introduce second-order spatial derivatives and inc ur substantial computational overhead. In this work, we provide a principled ana lysis of EM-induced loss bias and propose an unbiased, second-order-free trainin g framework that preserves the computational advantages of BSDE methods. Our cod e is available at https://github.com/seojaemin22/Un-EM-BSDE.
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More Capable, Less Cooperative? When LLMs Fail at Zero-Cost Collaboration
Advait Yadav, Sidney Black, Oliver Sourbut
Large language model (LLM) agents increasingly coordinate in multi-agent systems , yet we lack an understanding of where and why cooperation fails. Many real-wor ld coordination problems are not social dilemmas: helping others—sharing documen tation, unblocking a teammate—costs the helper almost nothing while producing su bstantial collective benefit. Whether LLM agents cooperate in this regime, where helping is free and they are explicitly instructed to do so, remains unknown. W e build a turn-based multi-agent environment that strips away all strategic comp lexity, making cooperation costless and trivially optimal. Across eight widely u sed LLMs, capability does not predict cooperation: OpenAI o3 reaches only 17\% o f optimal collective performance while the weaker o3-mini reaches 50\%, despite identical instructions to maximize group revenue. Using a causal decomposition t hat automates one side of agent communication, we separate cooperation failures from competence failures, and find that several capable models actively withhold information despite gaining nothing from withholding. Targeted interventions ad dress each mode: explicit protocols roughly double the performance of competence -limited models, while small sharing incentives unlock cooperation-limited ones. Our results suggest that scaling intelligence alone will not solve coordination in multi-agent systems, and will require deliberate cooperative design, even wh en helping costs nothing.
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Steering Large Language Models through the DMTA Cycle: Structure-Based Drug Desi gn via Knowledge-Driven Bi-Level Thompson Sampling
Xuanning Hu, Hao Tuo, Jinglong Ji, Anchen Li, Qianli Xing, Bo Yang
Structure-based drug design (SBDD) can be effectively realized through an iterat ive refinement via the Design-Make-Test-Analyze (DMTA) cycle, which is a common workflow used by human experts. However, most conventional deep learning-based m odels as one-shot generators that lack feedback mechanisms, leaving the DMTA loo p disconnected. In this work, we propose K-BTS, a Knowledge-Driven Bi-level Thom pson Sampling framework that formalizes iterative SBDD as a Dynamic Hierarchical Multi-Armed Bandit problem. K-BTS closes the DMTA loop by decoupling decisions into two levels: an upper-level policy that prioritizes high-potential molecular lineages and a lower-level mechanism that retrieves explicit chemical rules to guide LLM generation. By integrating a dual-level Bayesian update, the framework transforms sparse docking scores into reusable experience. On the CrossDocked20 20 benchmark, K-BTS achieves a state-of-the-art Top-1 average docking score. The results from diverse dimensions show that K-BTS ensures search determinism thro ugh a smooth, monotonic convergence that synchronizes structural drift with affi nity improvement.
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UniScale: Adaptive Unified Inference Scaling via Online Joint Optimization of Mo del Routing and Test-Time Scaling
Kaiyu Huang, Xingyu Wang, Mingze Kong, Zhubo Shi, Yuqian Hou, Hong Xu, Zhongxian g Dai, Minchen Yu, Qingjiang Shi

In real-world deployments of large language models (LLMs), balancing inference q
uality and computational cost has become a central challenge.
Existing approaches tackle this trade-off along two largely independent dimensio
ns: model routing, which switches among models of different scales to match requ
est complexity, and test-time scaling (TTS), which adjusts inference-time comput
e within a fixed model for fine-grained control.
However, this decoupled design introduces inherent limitations.
Model routing yields coarse-grained, discrete performance changes due to the spa
rse set of model scales, while single-model TTS often encounters capacity ceilin
gs and exhibits diminishing returns as compute increases.
Moreover, treating the two mechanisms separately restricts adaptability in dynam
ic inference environments.
To overcome these limitations, we introduce *Unified Inference Scaling (UIS)*, w
hich unifies model routing and TTS in a single optimization space.
Building on this formulation, we propose UniScale, an online framework that mode
ls adaptive UIS as a contextual multi-armed bandit problem and learns inference
policies via LinUCB.
The framework incorporates efficiency-aware learning and cost modeling to ensure
 stable and scalable optimization over high-dimensional action spaces.
Evaluation shows that UniScale effectively exploits the synergy in the UIS space
 to deliver a fine-grained and consistently better quality–cost trade-off across
 diverse, dynamic inference scenarios.
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Chunk-Guided Q-Learning
Gwanwoo Song, Kwanyoung Park, Youngwoon Lee
In offline reinforcement learning (RL), single-step temporal-difference (TD) lea
rning can suffer from bootstrapping error accumulation over long horizons. Actio
n-chunked TD methods mitigate this by backing up over multiple steps, but can in
troduce suboptimality by restricting the policy class to open-loop action sequen
ces. To resolve this trade-off, we present Chunk-Guided Q-Learning (CGQ), a sing
le-step TD algorithm that guides a fine-grained single-step critic by regularizi
ng it toward a chunk-based critic trained using temporally extended backups. Thi
s reduces compounding error while preserving fine-grained value propagation. We
theoretically show that CGQ attains tighter critic optimality bounds than either
 single-step or action-chunked TD learning alone. Empirically, CGQ achieves stro
ng performance on challenging long-horizon OGBench tasks, often outperforming bo
th single-step and action-chunked methods.
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Efficient-DLM: From Autoregressive to Diffusion Language Models, and Beyond in S
peed
Yonggan Fu, Lexington Whalen, Zhifan Ye, Xin Dong, Shizhe Diao, Jingyu Liu, Chen
gyue Wu, Hao Zhang, Enze Xie, Song Han, Maksim Khadkevich, Jan Kautz, Yingyan Ce
line Lin, Pavlo Molchanov
Diffusion language models (dLMs) have emerged as a promising paradigm enabling p
arallel generation, but their learning efficiency lags behind that of autoregres
sive (AR) language models when trained from scratch. To this end, we study AR-to
-dLM conversion, which transforms pretrained AR models into efficient dLMs that
excel in speed while preserving AR models' task accuracy. We achieve this by ide
ntifying limitations in the attention patterns and objectives of existing AR-to-
dLM methods and then proposing methodologies and actionable insights for scalabl
e AR-to-dLM conversion. Specifically, we first systematically compare different
attention patterns and find that maintaining pretrained AR weight distributions
is key to effective AR-to-dLM conversion. Accordingly, we introduce a continuous
 pretraining scheme with a block-wise attention pattern. We find that, in additi
on to block-wise attention's known benefit of enabling KV caching, its block-wis
e causality better preserves pretrained AR models' weight distributions, leading
 to a win–win in accuracy and efficiency. Second, to mitigate the training–test
gap in mask token distributions (uniform vs. highly left-to-right), we propose a
 position-dependent token masking strategy that assigns higher masking probabili
ties to later tokens during training to better mimic test-time behavior. These s

tudies lead to the Efficient-DLM model family, which outperforms state-of-the-art AR models and dLMs in accuracy–throughput trade-offs; for example, our Efficient-DLM-8B achieves +5.4\%/+2.7\% higher accuracy with 4.7$\times$/2.8$\times$ higher throughput compared to Dream-7B and Qwen3-4B, respectively.
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PortraitRL: Reinforcement Learning for Personalized Portrait Pose Transfer with Multi-Objective Reward Modeling
Jiahui Wu, Zelong Sun, Yanbiao Ma, Zhiwu Lu
Portrait pose transfer (PPT) requires generative models to preserve fine-grained identity details while following complex pose and layout modification instructions.
Existing methods often struggle with extensive data annotation requirements or employ optimization objectives that are suboptimal for addressing PPT's two key challenges.
In this work, we propose PortraitRL, a novel post-training framework that addresses these challenges with a multi-objective reward mechanism. Specifically, we employ LVLM-based reward functions to effectively evaluate PPT's two challenges and apply within-group standardization to eliminate scale differences, allowing these rewards to effectively guide optimization.
More importantly, we devise a novel reinforcement learning algorithm, Negative-aware Score Preference Optimization (NaSPO), which automatically identifies positive and negative preference samples through within-group advantages, eliminating annotation requirements while fully leveraging both positive and negative learning signals.
Extensive experiments show state-of-the-art performance, with significant improvements in both detail preservation and editing accuracy.
****
AGoQ: Activation and Gradient Quantization for Memory-Efficient Distributed Training of LLMs
WenXiang Lin, Juntao Huang, LuHan Zhang, Laiyi Li, Xiang Bao, mengyang zhang, Bing Wang, Shaohuai Shi
Quantization is a key method for reducing the GPU memory requirement of training large language models (LLMs). Yet, current approaches are ineffective for 4-bit activations and 8-bit gradients, which would easily cause slow convergence or accuracy loss. To address this, we introduce AGoQ, incorporating two new techniques: 1) a layer-aware activation quantization algorithm that allocates appropriate bit-widths for activations of various layers based on their types and pipeline stages to achieve near 4-bit activation storage, and 2) a gradient quantization algorithm that reduces memory usage and shortens communication time by employing 8-bit gradient storage and precision-preserving 8-bit All-Reduce communication. We conduct extensive experiments using different sizes of LLMs on two GPU clusters (up to 64 GPUs), and the experimental results show that our AGoQ reduces the memory by up to 52\% and achieves up to 1.34$\times$ improvement of training speed compared to state-of-the-art training systems Megatron-LM (w/ or w/o ZeRO), COAT and DeepSpeed with 8B to 32B LLaMA models, while achieving convergence loss on pretraining and comparable accuracy on downstream tasks with LLaMA architectures.
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LLM-Guided Loop Bound Generation for Program Termination Verification
Zan Gong, Biting Huang, Fei He
Program termination is a fundamental liveness property in software verification. Proving termination of a given program is a formidable challenge due to the undecidability of the problem. In this paper, we propose LIFT, a termination verification framework that leverages LLMs to generate loop bounds within a guess-and-check workflow. LIFT couples this generation with a sound formal validation procedure that both guarantees all reported terminations and refutes invalid loop bounds via violation analysis. Experiments on publicly accessible termination benchmarks show that LIFT significantly outperforms existing termination verification tools.
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Selling Data as a Digital Good with Scaling Valuations
Ningyuan Li, Yanru Guan, Xiaotie Deng, Zihe Wang, Jie Zhang
We study mechanism design for selling data as a digital good when the value derived from training AI models follows a scaling law. The seller faces a linear cost when producing data, while the buyers benefit from additional data with diminishing returns as data volume increases. This departs from classical auction models by allowing allocations to be continuous quantities of data rather than binary outcomes.
We first analyze an offline setting in which all buyer types are realized simultaneously, characterizing profit-optimal mechanisms and showing how virtual-value methods extend to continuous data allocations. We then consider an online setting with sequential arrivals, where production decisions must be made under demand uncertainty. We show that myopic allocation and fixed production plans can be arbitrarily suboptimal,
whereas a simple two-stage algorithm that combines upfront production with adaptive expansion achieves a constant-factor approximation to the offline optimum.
Finally, we study bilateral data trading under asymmetric information, where both the buyer's value and the seller's cost are private. Although the optimal truthful mechanism has a complex structure, we show that simple and implementable mechanisms recover a constant fraction of the first-best gain-from-trade.
Overall, our results highlight how scaling laws introduce new algorithmic trade-offs in market design and provide performance guarantees for data markets under uncertainty.
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ReMoE: Boosting Expert Reuse through Router Fine-Tuning in Memory-Constrained MoE LLM Inference
Xiongwei Zhu, Xiaojian Liao, Tianyang Jiang, Yusen Zhang, Liang Wang, Limin Xiao
Fine-grained Mixture-of-Experts (MoE) models sparsely activate only a subset of experts per token, reducing activated computation while maintaining high model capacity. However, in memory-constrained inference scenarios, only a small set of experts can be cached. Experts not in the cache must be fetched from slow external storage (e.g., UFS), leading to frequent evictions and substantial I/O overhead.
We propose ReMoE, a router fine-tuning framework designed to boost
token-wise expert reuse. ReMoE biases the router toward recently
selected experts, producing temporally stable routing that better
matches cache locality constraints. By increasing short-horizon expert
reuse, ReMoE reduces expert fetches from storage without adding
inference-time computation.
Experiments on DeepSeek and Qwen models show that ReMoE improves expert reuse by 26\% while maintaining downstream task performance. Real-system evaluations further confirm these benefits, improving output throughput by 8.4\% under vLLM GPU--CPU expert offloading and reducing TPOT by 43.6-49.8\% under llama.cpp on Jetson Orin NX, corresponding to a 1.77-1.99$\times$ decode speedup across diverse workloads. Checkpoints and usage instructions are available at https://github.com/BUAA-OSCAR/ReMoE.
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LFQ: Logit-aware Final-block Quantization for Boosting the Generation Quality of Low-Bit Quantized LLMs
Jung Hyun Lee, June Yong Yang, Jungwook Choi, Eunho Yang
As large language models continue to scale, low-bit weight-only post-training quantization (PTQ) offers a practical solution to their memory-efficient deployment. Although block-wise PTQ is capable of matching the full-precision (FP) baseline on basic language modeling and understanding, its quality is degraded for \textit{generative} tasks---especially at longer responses and extended chains of thought, which is critical in boosting task accuracy. We attribute this shortfall to two factors: (i) the omission of the unembedding layer (the LM head) in block-wise optimization and (ii) the reliance on the mean squared error (MSE) objective. Both factors cause the token probability distribution of the quantized model to misalign with that of the FP model, yielding notable accuracy drops on text

generation benchmarks. To rectify the discrepancy, we introduce \emph{Logit-aware Final-block Quantization (LFQ)}, a simple yet effective enhancement to block-wise PTQ that quantizes the final Transformer block by minimizing the cross-entropy between the logits of the FP model and those of its quantized counterpart. By aligning token probabilities at the logit level in the final block, LFQ consistently improves the accuracy of complex generation tasks over state-of-the-art block-wise PTQ across diverse model families, while maintaining parity with FP baselines on language modeling and understanding.

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What Makes a Strong Model? A Unified Spectral Analysis of Knowledge Transfer over High-dimensional Linear Regression

Wendao Wu, Fangqing Zhang, Haihan Zhang, Cong Fang

Teacher-Student Knowledge Transfer (KT) is ubiquitous in modern machine learning, ranging from classical model compression via Knowledge Distillation (KD) to the emergent phenomenon of Weak-to-Strong (W2S) generalization. While existing studies offer isolated insights, a unified theoretical framework explaining the efficacy of KT across these disparate regimes remains lacking. In this work, we establish a unified spectral analysis of SGD dynamics in high-dimensional linear regression, elucidating the efficiency of KT across seemingly disparate regimes. We characterize KT efficiency through two distinct mechanisms: \emph{Spectral Horizon Expansion} in KD, which enables the capture of statistically inaccessible high-frequency signals, and \emph{Spectral Denoising} in W2S, where the student acts as a filter for optimization noise. Our framework unifies these phenomena, revealing that the efficacy of transfer is governed by the interplay between implicit regularization and heterogeneous spectral learning speeds over the spectrum.

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DAVE: Distribution-Aware Attribution via ViT Gradient Decomposition

Adam Wróbel, Siddhartha Gairola, Jacek Tabor, Bernt Schiele, Bartosz Micha■ Ziel i■ski, Dawid Damian Rymarczyk

Vision Transformers (ViTs) have become a dominant architecture in computer vision, yet producing stable and high-resolution attribution maps remains challenging. Architectural components such as patch embeddings and attention routing often introduce structured artifacts in pixel-level explanations, leading many existing methods to rely on coarse patch-level attributions. We introduce DAVE (Distribution-Aware Attribution via ViT Gradient Decomposition), a mathematically grounded attribution method for ViTs based on a structured decomposition of the input gradient. By exploiting architectural properties of ViTs, DAVE isolates locally equivariant and stable components of the effective input-output mapping while suppressing architecture-induced artifacts and instability. Consequently, DAVE produces robust, precise, and class-consistent attribution maps that highlight model-relevant visual features. Experimental results show that across supervised, self-supervised, and inherently interpretable ViTs, DAVE outperforms prior methods on localization, faithfulness, and user studies.

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Exploiting weight-space symmetries for approximating curvature

Artem Artemev, Rui Xia, Benjamin M. Boyd, Youjing Yu, Felix Dangel, Guillaume Hennequin, Alberto Bernacchia

Many machine learning techniques rely on approximating a loss function's curvature, but this is notoriously hard to do at the scale of modern deep networks. Surprisingly, no previous work has exploited the curvature constraints that arise from well known weight-space symmetries in loss landscapes. By analytically averaging over group actions that leave the loss invariant, we construct structured Hessian approximations from single gradients that can be tractably estimated, stored, and inverted. The choice of user-specified symmetry group directly governs the trade-off between approximation accuracy and computational cost. Moreover, our framework provides a unifying theoretical lens for viewing existing methods; in particular, a specific choice of symmetry group recovers Shampoo/Muon-like curvature estimates. We validate our method on a range of network architectures, and deploy it to second-order optimization benchmarks, including a small language

model. Our curvature estimation framework might find applications in other machine learning problems such as uncertainty estimation, continual learning, compression/pruning, training data attribution, and more.
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ABC-Bench: An Agentic Bio-Capabilities Benchmark for Biosecurity
Andrew Bo Liu, Samira Nedungadi, Bryce Cai, Alex Kleinman, Harmon Bhasin, Seth Donoughe
Large language models (LLMs) are rapidly acquiring capabilities relevant to biological research, from literature synthesis to interpretation of experimental data. Increasingly, LLM agents can also perform *in silico* biology tasks that previously required experienced human biologists. These emerging AI capabilities offer new opportunities for scientific discovery and biomedical advances, but they also shift the landscape of biosecurity risks. To address this, we introduce the Agentic Bio-Capabilities Benchmark (ABC-Bench), a suite of tasks to measure agentic biosecurity-relevant capabilities. ABC-Bench evaluates LLM agents on both benign and dual-use biology tasks: writing code to operate liquid handling robots, designing DNA fragments for in vitro assembly, and evading DNA synthesis screening. These tasks require a combination of biology and software expertise. All tested LLM agents outperformed the median expert human baseliner on all three tasks. Agents performed highly on tasks drawing on published knowledge and well-documented protocols, and more weakly on a task requiring novel bioinformatics reasoning. In three wet-lab validation experiments, we found that OpenAI’s o4-mini-high produced scripts that, when run on an OpenTrons liquid handling robot, successfully assembled DNA with expected sequences.
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Step-Size Stability in Stochastic Optimization: A Theoretical Perspective
Fabian Schaipp, Robert M. Gower, Adrien Taylor
We present a theoretical analysis of stochastic optimization methods in terms of their sensitivity with respect to the step size.
    We identify a key quantity that, for each method, describes how the performance degrades as the step size becomes too large.
    For convex problems, we show that this quantity directly impacts the suboptimality bound of the method.
    Most importantly, our analysis provides direct *theoretical* evidence that adaptive step size methods, such as SPS or NGN, are more robust than SGD.
    This allows us to quantify the advantage of these adaptive methods beyond empirical evaluation.
    Finally, we show through experiments that our theoretical bound qualitatively mirrors the actual performance as a function of the step size, even for non-convex problems.
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HilbertA: Hilbert-Curve–Aligned Sparse Attention for 2D Structured Data
Shaoyi Zheng, Wenbo Lu, Yuxuan Xia, Shenji Wan
Designing sparse attention for 2D image data in diffusion and vision-language models requires reconciling spatial locality with hardware-efficient execution: handcrafted 2D sparsity patterns preserve spatial structure but often induce uncoalesced memory access, limiting practical speedups on modern GPUs.
We present **HilbertA**, a 2D-aware sparse attention mechanism that reorders image tokens along a Hilbert curve, converting local spatial neighborhoods into contiguous memory segments for efficient GPU execution. To enable communication beyond local tiles, HilbertA shifts attention windows along the Hilbert-ordered sequence across layers and uses a small central shared region, preserving contiguous access while supporting cross-tile information flow.
Across diffusion and vision-language models, HilbertA delivers consistent efficiency gains while maintaining competitive quality, achieving up to **4.16×** attention acceleration and **1.44×** end-to-end speedup on Flux.1-dev, and up to **2.30×** attention acceleration with **1.57×** faster time-to-first-token on Qwen3-VL-8B inference.
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Efficient Multi-Agent Reasoning via Confidence-Guided Adaptive Debate

Seungdong Yoa, Ye Seul Sim, Suhee Yoon, Sanghyu Yoon, Dongmin Kim, Soonyoung Lee, Bumsoo Kim, Junhyun Lee

Multi-agent debate has shown promise for improving the reasoning of large language models, yet recent theory suggests its benefits are highly regime-dependent. While interaction can amplify informative signals under corrective conditions, symmetric debate dynamics are neutral in expectation, often making majority voting preferable. We reconcile these views by arguing that debate is effective only when invoked at the right time and with appropriate structure.

Based on this insight, we propose LASE: Leader-Adaptive Structured Engagement, a leader-centric multi-agent debate framework that selectively engages interaction only in non-neutral regimes.

LASE introduces an asymmetric leader–supporter architecture that enables directed information flow and selective signal amplification, while defaulting to simple aggregation otherwise. Experiments across diverse reasoning benchmarks show that LASE achieves multi-agent-level performance with near single-agent token cost, substantially improving efficiency over static debate and voting baselines.

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Multivariate Distributional Reinforcement Learning Using Sliced Divergences

Baptiste Debes, Tinne Tuytelaars

Distributional reinforcement learning (DRL) models the full return distribution rather than expectations, but extending it to multivariate settings remains challenging. Many common metrics do not naturally generalize beyond one dimension or lose computational tractability, and the multivariate case introduces additional difficulties such as general matrix discounting, for which no contraction results are available. We introduce Sliced Distributional Reinforcement Learning (SDRL), which lifts tractable one-dimensional divergences to multivariate return distributions via projections. We prove Bellman contraction for uniform slicing under shared scalar discounting, and introduce a maximum-slicing variant with contraction under general dense discount matrices. SDRL supports a broad class of base divergences; we analyze Wasserstein, Cramér, and Maximum Mean Discrepancy (MMD), and characterize which SDRL variants suit the standard single-sample Bellman update used in distributional RL. We evaluate SDRL on a toy chain problem and a gridworld image-based environment as well as a subset of Atari games. Code is available at https://github.com/BaptisteDebes/SlicedDistributionalRL

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Models Under SCOPE: Scalable and Controllable Routing via Pre-hoc Reasoning

Qi Cao, Shuhao Zhang, Ruizhe Zhou, Ruiyi Zhang, Peijia Qin, Pengtao Xie

Model routing chooses which language model to use for each query. By sending easy queries to cheaper models and hard queries to stronger ones, it can significantly reduce inference cost while maintaining high accuracy. However, most existing routers treat this as a fixed choice among a small set of models, which makes them hard to adapt to new models or changing budget constraints. In this paper, we propose SCOPE (Scalable and Controllable Outcome Performance Estimator), a routing framework that goes beyond model selection by predicting their cost and performance. Trained with reinforcement learning, SCOPE makes reasoning-based predictions by retrieving how models behave on similar problems, rather than relying on fixed model names, enabling it to work with new, unseen models. Moreover, by explicitly predicting how accurate and how expensive a model will be, it turns routing into a dynamic decision problem, allowing users to easily control the trade-off between accuracy and cost. Experiments show that SCOPE is more than just a cost-saving tool. It flexibly adapts to user needs: it can boost accuracy by up to **25.7\%** when performance is the priority, or cut costs by up to **95.1\%** when efficiency matters most. We release the dataset and code at our project page: https://sullivan07043.github.io/SCOPE/.

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Taming the Loss Landscape of PINNs with Noisy Feynman–Kac Supervision: Operator Preconditioning and Non-Asymptotic Error Bounds

Nathanael Tepakbong, Hanyu HU, Chengyu Liu, Xiang ZHOU

Physics-Informed Neural Networks (PINNs) often train slowly or fail to converge on challenging partial differential equations (PDEs), a behavior recently linked

 to severely ill-conditioned loss landscapes inherited from the underlying diffe
rential operator. We study PINNs augmented with a pointwise data-fidelity term,
added at a few points in the domain to the standard residual and boundary losses
. We show that this supervision term acts as an operator-level preconditioner: f
or suitable weights, our comparison bounds guarantee a substantially smaller con
dition number than under the standard PINN loss, independently of how the pointw
ise labels are obtained. For a broad class of PDEs admitting a Feynman-Kac (FK)
representation, we generate such labels by Monte Carlo averages of the FK functi
onal, resulting in what we call "FK-PINNs", and using the excess risk decomposit
ion approach, we derive non-asymptotic $L^2(\Omega)$-error bounds for FK-PINNs w
ith $\tanh$ activation trained by finitely many steps of gradient descent. Along
 the way, we establish pseudo-dimension bounds for first- and second-order deriv
atives of $\tanh$ neural networks, which are of independent interest and, to the
 best of our knowledge, new. Numerical experiments on Poisson, Schrödinger, mean
 exit time, and committor problems corroborate the theory, and show that FK-PINN
s can successfully solve PDEs for which standard PINNs exhibit severe failure mo
des.
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Refined Analysis of Entropy-Regularized Actor-Critic
Safwan Labbi, Paul Mangold, Daniil Tiapkin, Eric Moulines
In this paper, we study the role of the critic in actor-critic for entropy-regul
arized, finite, discounted environments. We establish that, when the critic is e
xact, using the latter as a baseline is an actual variance-reduction method. In
this case, actor-critic with stochastic gradients matches the sample complexity
of deterministic policy gradient, reaching an ϵ-optimal regularized va
lue with $\tilde{O}(\log(1/\epsilon))$ samples.
In practice, the critic is learned alongside the actor: the variance of the acto
r update is then influenced by the critic's variance and bias. Specifically, whe
n the critic has a sufficiently small error, the variance reduction and rapid co
nvergence are preserved. This suggests to learn the critic first, keeping it up
to date after each actor update, underscoring the pivotal role of accurate criti
c estimation in actor-critic methods.
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ConceptMoE: Adaptive Token-to-Concept Compression for Implicit Compute Allocatio
n
Zihao Huang, Jundong Zhou, Xingwei Qu, Qiyang Min, Ge Zhang
Large language models allocate uniform computation across all tokens, ignoring t
hat some sequences are trivially predictable while others require deep reasoning
. We introduce ConceptMoE, which dynamically merges semantically similar tokens
into concepts through learnable chunking at target compression ratio R. The Mo
E architecture enables controlled evaluation: reallocating saved computation to
match baseline FLOPs and parameters isolates genuine architectural benefits. Con
ceptMoE consistently outperforms standard MoE, achieving +0.9 points on language
 pretraining, +2.3 on long context, and +0.6 on multimodal tasks. Continual trai
ning conversion with layer looping gains +5.5 points. Beyond performance, at $R=
2$, ConceptMoE reduces attention computation by $R^2\times$ and KV cache by $R\t
imes$, delivering prefill speedups up to 175\% and decoding speedups up to 117\%
. The minimal architectural changes enable straightforward integration, demonstr
ating that adaptive concept-level processing fundamentally improves LLM effectiv
eness and efficiency.
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Optimization, Generalization and Differential Privacy Bounds for Gradient Desce
nt on Kolmogorov–Arnold Networks
Puyu Wang, Junyu Zhou, Philipp Liznerski, Marius Kloft
Kolmogorov--Arnold Networks (KANs) have recently emerged as a structured alterna
tive to standard MLPs, yet a principled theory for their training dynamics, gene
ralization, and privacy properties remains limited. In this paper, we analyze gr
adient descent (GD) for training two-layer KANs and derive general bounds that c
haracterize their training dynamics, generalization, and utility under different
ial privacy (DP). As a concrete instantiation, we specialize our analysis to log

istic loss under an NTK-separable assumption, where we show that polylogarithmic network width suffices for GD to achieve an optimization rate of order $1/T$ and a generalization rate of order $1/n$, with T denoting the number of GD iterations and n the sample size. In the private setting, we characterize the noise required for (ϵ,δ)-DP and obtain a utility bound of order $\sqrt{d}/(n\epsilon)$ (with d the input dimension), matching the classical lower bound for general convex Lipschitz problems. Our results imply that polylogarithmic width is not only sufficient but also necessary under differential privacy, revealing a qualitative gap between non-private (sufficiency only) and private (necessity also emerges) training regimes. Experiments further illustrate how these theoretical insights can guide practical choices, including network width selection and early stopping.

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On Regret Bounds of Thompson Sampling for Bayesian Optimization

Shion Takeno, Shogo Iwazaki

We study a widely used Bayesian optimization method, Gaussian process Thompson sampling (GP-TS), under the assumption that the objective function is a sample path from a GP.

Compared with the GP upper confidence bound (GP-UCB) with established high-probability and expected regret bounds, most analyses of GP-TS have been limited to expected regret.

Moreover, whether the recent analyses of GP-UCB for the lenient regret and the improved cumulative regret upper bound can be applied to GP-TS remains unclear.

To fill these gaps, this paper shows several regret bounds:

(i) a regret lower bound for GP-TS, which implies that GP-TS suffers from a polynomial dependence on $1/\delta$ with probability δ,

(ii) an upper bound of the second moment of cumulative regret, which directly suggests an improved regret upper bound on δ,

(iii) expected lenient regret upper bounds, and

(iv) an improved cumulative regret upper bound on the time horizon T.

Along the way, we provide several useful lemmas, including a relaxation of the necessary condition from recent analysis to obtain improved regret upper bounds on T.

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A Linearly Convergent Proximal Subgradient Algorithm for Sparse Portfolio Optimization with Transaction Cost

Xiaoting Yao, Na Zhang

Transaction cost optimization (TCO) of online portfolio selection is crucial in computing science, due to the significant impact of transaction costs in practical short-term trading. Moreover, sparsity of portfolio vector is often desired to enhance stability and decrease risk.

However, there is a lack of models considering transaction costs and sparsity simultaneously in the literature. In this paper, we first propose a K-sparse TCO model that minimizes the negative return and transaction costs while keeping the portfolio vector being K-sparse. Noting that the model is NP-hard due to the K-sparse constraint, we bypass this difficulty by reformulating the sparse model to a nonsmooth difference of convex (DC) optimization problem. We show that both problems are equivalent by proving that the penalty parameter is large enough. Then, to overcome the difficulty caused by the nonsmoothness and the simplex constraint of the model, we develop a proximal subgradient algorithm (PSGA) to solve the DC problem and apply the alternating direction of multipliers (ADMM) to compute the proximity operator of the corresponding function. Furthermore, we establish the global convergence of the entire sequence generated by PSGA through showing the surrogate function satisfies the Kurdyka-■ojasiewicz (KL) property. In addition, by showing the KL exponent of the surrogate function is $1/2$, we establish the R-linear convergence rate of PSGA for any arbitrary initiaal point. Finally, we compare our proposed algorithm with other state-of-the-art strategies on four benchmark real-market data sets, with the numerical results showing that the proposed algorithm achieves lower risk while keeping higher return than classical TCO models.

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Low-dimensional topology of deep neural networks
Junyu Ren, Lek-heng Lim
We study layered models, including feedforward networks, ResNets, and transforme rs, by limiting each layer to a width of $d = 3$, i.e., $\mathbb{R}^3$ as repre sentation space. This allows us to track how a neural network changes low-dimens ional topological invariants through its layers. Just about any topological stru cture may be simplified or even trivialized by simply increasing dimension; e.g. , any knot is equivalent to an unknot in $\mathbb{R}^4$. By restricting to $\mat hbb{R}^3$, we not only isolate the effects of activation and depth from that of width, we work in a space that lends itself to easy visualization. We focus on l inking number here, deferring other invariants like link groups, Milnor's $\bar{ \mu}$-invariants, knot types, ambient cobordisms, to a sequel. We provide full proofs and empirical experiments to justify the following insights: When measure d by their power to effect changes in linking numbers, the layer-skipping featur e in ResNets is as powerful as the attention mechanism in transformers; both Res Nets and transformers are strictly more powerful than feedforward neural network s with monotonic activations, which are in turn more powerful than invertible an d flow-based models; but replacing monotonic activation with a nonmonotonic one elevates a feedforward network into the same expressivity class as ResNets and t ransformers. These results suggest that low-dimensional topology can be a useful tool to guide designs of AI architectures. We also generalize our results from $d = 3$ to arbitrary $d > 3$.
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FLAG: Foundation model representation with Latent diffusion Alignment via Graph for spatial gene expression prediction
Qi Si, Penglei Wang, Yushuai Wu, Yifeng Jiao, Xuyang Liu, Xin Guo, Yuan Qi, Yuan Cheng
Predicting spatial gene expression from routine H\&E enables large-scale molecul ar profiling, yet current models treat this as isolated pointwise tasks, thereby overlooking essential biological structures like gene coordination and spatial distribution. To preserve these relationships, we introduce \textbf{FLAG}, a dif fusion-based framework that redefines this task as structured distribution model ing. At the same time, we identify the critical \textbf{Gene Dimension Curse}, w here joint modeling gene expression and their spatial interactions fail in high- dimensional spaces, and FLAG solves this challenge by integrating a spatial grap h encoder for topological consistency and utilizing Gene Foundation Model (GFM) alignment for gene-gene fidelity in the generation process. To rigorously assess model performance, we propose a set of novel structural evaluation metrics, inc luding Gene Structural Correlation (\textbf{GSC}) and Spatial Structural Correla tion (\textbf{SSC}). Our experiments demonstrate that FLAG is highly competitive in traditional accuracy (PCC/MSE) while achieving significantly enhanced struct ural fidelity in capturing both gene-gene and gene-spatial relationships. The co de is available at https://github.com/darkflash03/FLAG.
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Towards Foundation Models for Zero-Shot Time Series Anomaly Detection: Leveragin g Synthetic Data and Relative Context Discrepancy
Tian Lan, Hao Duong Le, Jinbo Li, Wenjun He, Meng Wang, Chenghao Liu, Chen Zhang
Time series anomaly detection(TSAD) is a critical task, but developing models th at generalize to unseen data in a zero-shot manner remains challenging. Existing foundation models for TSAD often rely on reconstruction-error scoring at infere nce time, which can miss subtle anomalies that are well reconstructed and can fa lsely flag complex but normal patterns in unseen domains. We introduce TimeRCD, a foundation model for TSAD built on Relative Context Discrepancy (RCD), a pre-t raining paradigm that trains the model to detect anomalies by comparing a query pattern with its surrounding context. This relational formulation, implemented w ith a standard Transformer architecture, enables the model to infer normality fr om the input context rather than relying on fixed global normal patterns. We fur ther construct a large-scale synthetic corpus with context-dependent anomaly lab els to provide supervised pre-training signals for RCD. Experiments across diver

se benchmarks show that TimeRCD outperforms existing general-purpose and anomaly-specific foundation models in most zero-shot TSAD settings, while remaining competitive with dataset-specific full-shot baselines. These results provide empirical evidence that RCD is an effective direction for building robust and generalizable TSAD models.
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N2M: Bridging Navigation and Manipulation by Learning Pose Preference from Rollout
Kaixin Chai, Hyunjun Lee, Joseph J Lim
Determining where to execute the manipulation policy is a fundamental challenge in mobile manipulation. Most approaches have formulated this as a geometric search problem, prioritizing physical reachability. However, given the high sensitivity of modern learning-based manipulation policies, geometric criteria alone are insufficient. Optimal performance requires base positioning that is aware of the policy's preference. While recent works have attempted to address this, they remain limited in practicality due to reliance on pre-built scene reconstruction and slow inference. In this work, we introduce **N2M** that systematically reformulates the approach to base positioning problem, naturally overcoming limitations of previous methods. Our key insight is that policy preferences are inherent to the local scene structure and can be effectively learned from the policy rollouts. Technically, we propose a novel *viewpoint augmentation* strategy that enables the model to learn robust, viewpoint-invariant pose preferences with remarkable data efficiency. Extensive experiments demonstrate that N2M achieves state-of-the-art performance, outperforming both non-policy-aware baselines and recent policy-aware alternatives. Furthermore, we provide a comprehensive analysis highlighting N2M’s broad applicability, generalization capabilities, and data efficiency. Project website: https://clvrai.github.io/N2M/
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Mitigating Translationese Bias in Multilingual LLM-as-a-Judge via Disentangled Information Bottleneck
Hongbin Zhang, Kehai Chen, Xuefeng Bai, Youcheng Pan, Yang Xiang, Jinpeng Wang, Min Zhang
Large language models (LLMs) have emerged as a standard paradigm for automated multilingual evaluation, yet exhibit systematic biases. In this paper, we identify ``translationese bias'', in which LLMs systematically favor machine-translated text over human-authored references, and this bias is particularly pronounced in low-resource languages. We attribute this bias to spurious correlations with (a) strong latent manifold isomorphism with English and (b) high predictive confidence. To mitigate these issues, we present DIBJudge, a robust fine-tuning framework that decouples robust features and bias representations by explicitly isolating spurious attributes into a dedicated bias branch and penalizing mutual dependence to enforce disentanglement.
In particular, we present a vector-quantized compression that ensures the robust representation retains minimal and sufficient judgment-critical information.
Extensive evaluations on multilingual reward modeling benchmarks and a specially designed translationese bias evaluation suite demonstrate that DIBJudge outperforms strong baselines and effectively mitigates translationese bias.
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Self-Supervised Learning as Discrete Communication
Kawtar Zaher, Ilyass Moummad, Olivier Buisson, Alexis Joly
Most self-supervised learning (SSL) methods learn continuous visual representations by aligning different views of the same input, offering limited control over how information is structured across representation dimensions. In this work, we frame visual self-supervised learning as a discrete communication process between a teacher and a student network, where semantic information is transmitted through a fixed-capacity binary channel. Rather than aligning continuous features, the student predicts multi-label binary messages produced by the teacher. Discrete agreement is enforced through an element-wise binary cross-entropy objective, while a coding-rate regularization term encourages effective utilization of the constrained channel, promoting structured representations. We further show th

at periodically reinitializing the projection head strengthens this effect by encouraging embeddings that remain predictive across multiple discrete encodings. Extensive experiments demonstrate consistent improvements over continuous agreement baselines on image classification, retrieval, and dense visual prediction tasks, as well as under domain shift through self-supervised adaptation. Beyond backbone representations, we analyze the learned binary codes and show that they form a compact and informative discrete language, capturing semantic factors reusable across classes.
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WFR-MFM: One-Step Inference for Dynamic Unbalanced Optimal Transport
Xinyu Wang, Ruoyu Wang, Qiangwei Peng, Peijie Zhou, Tiejun Li
Reconstructing dynamical evolution from limited observations is a fundamental challenge in single-cell biology, where dynamic unbalanced optimal transport (OT) provides a principled framework for modeling coupled transport and mass variation.
However, existing approaches rely on trajectory simulation at inference time, making inference a key bottleneck for scalable applications.
In this work, we propose a mean-flow framework for unbalanced flow matching that summarizes both transport and mass-growth dynamics over arbitrary time intervals using mean velocity and mass-growth fields, enabling fast one-step generation without trajectory simulation.
To solve dynamic unbalanced OT under the Wasserstein-Fisher-Rao geometry, we further build on this framework to develop **Wasserstein-Fisher-Rao Mean Flow Matching (WFR-MFM)**.
Across synthetic and real single-cell RNA sequencing datasets, WFR-MFM achieves orders-of-magnitude faster inference than a range of existing baselines while maintaining high predictive accuracy, and enables efficient perturbation response prediction on large synthetic datasets with thousands of conditions.
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PonderLM-2: Pretraining LLM with Latent Thoughts in Continuous Space
Boyi Zeng, He Li, Shixiang Song, Yixuan Wang, Zitong Wang, Ziwei He, Xinbing Wang, Zhouhan Lin
The remarkable success of Chain-of-Thought (CoT), which enhances performance by scaling generation steps at test-time, inspires us to ask: can we leverage a similar scaling of computational steps during pretraining to improve the generation of each individual token? To address this, we propose a novel pre-training methodology: Pretraining Language Models with Latent Thoughts (PonderLM-2). Our approach pretrains a language model (LM) to first generate an intermediate latent thought—the last hidden
state of the current position—which is then used as input to predict the actual subsequent token. This additional computational step enables the LM to refine its prediction within unconstrained continuous space. Our experiments demonstrate that, at an identical inference cost, a LM that generates one additional latent thought per token outperforms a standard model with double the parameters. For instance, our PonderLM-2-Pythia-1.4B, pretrained on 300B tokens from the Pile, significantly surpasses the vanilla Pythia-2.8B trained on the same data on both language modeling and a range of general downstream tasks. Furthermore, increasing the number of latent thoughts generated before each actual token—forming a chain analogous to CoT—consistently improves the model's performance. The code and models are available at https://github.com/LUMIA-Group/PonderLM-2.
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Spatially-Adaptive Gradient Re-parameterization for 3D Large Kernel Optimization
Ho Hin Lee, Quan Liu, Shunxing Bao, Yuankai Huo, Bennett Allan Landman
Large kernel convolutions offer a scalable alternative to vision transformers for high-resolution 3D volumetric analysis, yet naively increasing kernel size often leads to optimization instability. Motivated by the spatial bias inherent in effective receptive fields (ERFs), we theoretically demonstrate that structurally re-parameterized blocks induce spatially varying learning rates that are crucial for convergence. Leveraging this insight, we introduce Rep3D, a framework that employs a lightweight modulation network to generate receptive-biased scaling

masks, adaptively re-weighting kernel updates within a plain encoder architectur
e. This approach unifies spatial inductive bias with optimization-aware learning
, avoiding the complexity of multi-branch designs while ensuring robust local-to
-global convergence. Extensive evaluations on five 3D segmentation benchmarks de
monstrate that Rep3D consistently outperforms state-of-the-art transformer and f
ixed-prior baselines. The source code is publicly available at https://github.co
m/leeh43/Rep3D.
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Visual Para-Thinker: Divide-and-Conquer Reasoning for Visual Comprehension
Haoran Xu, hongyu wang, Jiaze Li, Shunpeng Chen, Zizhao Tong, Jianzhong Ju, Zhen
bo Luo, Jian Luan
Existing LLM test-time scaling laws emphasize the emergence of self-reflective b
ehaviors through extended reasoning length. Nevertheless, this vertical scaling
strategy often encounters plateaus in exploration as the model becomes locked in
to specific thinking pattern. By shifting from depth to parallelism, parallel th
inking mitigates the narrowing of exploration. However, the extension of this pa
radigm to visual domain remains an open research question. In this paper, we fir
st examine the role of visual partitioning in parallelized reasoning and subsequ
ently propose two distinct strategies. Based on the above, we introduce Visual P
ara-Thinker, representing the inaugural parallel reasoning framework for MLLMs.
To maintain path independence and promote diversity in reasoning, our approach i
ntegrates Pa-Attention alongside LPRoPE. Leveraging the vLLM framework, we have
developed a native multimodal implementation that facilitates high-efficiency pa
rallel processing. Empirical results on benchmark datasets such as V*, CountBenc
h, RefCOCO, and HallusionBench confirm that Visual Para-Thinker successfully ext
ends the benefits of parallel reasoning to the visual domain.
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Grounding Functional Similarity by Invariance-Aware Model Stitching
Ioannis Athanasiadis, Anmar Karmush, Michael Felsberg
In deep learning, functional similarity evaluation quantifies the extent to whic
h independently trained models learn similar input--output relationships. In mod
el stitching, functional similarity is framed as representation forward compatib
ility, i.e., whether the representations of two models can be aligned to solve a
 given task. Recent studies, however, highlight a critical limitation: models re
lying on different information cues can still produce compatible representations
, making them appear misleadingly similar (Smith et al., 2025). We attribute thi
s failure to standard model stitching being inherently blind to the invariance p
roperties of the stitched models. To address this limitation, we introduce the f
orward--backward compatibility requirement under which we formulate the invarian
ce-aware model stitching. Through analyzing key stitching configurations, we stu
dy the interplay between forward and backward compatibility, showing that invari
ance-aware model stitching provides a more principled approach to functional sim
ilarity evaluation while revealing functional discrepancies previously obscured.
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When LLMs Encounter Open-world Graph Learning: A Fresh View on Unlabeled Data Un
certainty
Yanzhe Wen, Xunkai Li, Qi Zhang, Lei Zhu, Guang Zeng, Zhihan Zhang, Rong-Hua Li,
 Guoren Wang
Recently, large language models (LLMs) have
driven a systematic shift in the graph ML com-
munity through the adoption of text-attributed
graphs (TAGs). Although a variety of frameworks
have been developed, most fail to properly ad-
dress the challenge of data uncertainty in open-
world environments. A representative source of
such uncertainty is the limited availability of la-
bels in large-scale datasets due to high annotation
costs, where unlabeled nodes may belong to either
known classes or novel, unknown classes. While
node-level out-of-distribution detection and con-

ventional open-world graph learning attempt to tackle this problem, two core limitations remain: ① Insufficient methods — existing approaches typically optimize semantics or topology in isolation for unknown-class rejection, failing to effectively integrate textual and structural information in TAGs; ② Incomplete pipelines — most studies conduct only idealized analyses, such as assuming a predefined number of unknown classes, which restricts practical utility for model updates and long-term deployment. To overcome these issues, we introduce the Open-world Graph Assistant (OGA), an LLM-based framework. OGA first performs unknown-class rejection via adaptive label traceability (ALT), harmoniously combining semantic and topological cues, and then applies the graph label annotator (GLA) for unknown-class annotation, allowing unlabeled nodes to contribute to model training. In essence, OGA offers a new pipeline that fully automates the handling of unlabeled nodes in open-world environments, and we establish a systematic benchmark covering four key aspects to validate its effectiveness and practicality through extensive experiments.
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QuArch: A Benchmark for Evaluating LLM Reasoning in Computer Architecture
Shvetank Prakash, Andrew Cheng, Mark Mazumder, Arya Tschand, Varun Gohil, Jeffrey Jian Ma, Jason Yik, Zishen Wan, Jessica Quaye, Elisavet Lydia Alvanaki, Avinash Kumar, Chandrashis Mazumdar, Tuhin Khare, Alexander Ingare, Ikechukwu Uchendu, Radhika Ghosal, Abhishek Tyagi, Chenyu Wang, Andrea Mattia Garavagno, Sarah Yueqi Gu, Alice Guo, Grace Hur, Luca Carloni, Tushar Krishna, Ankita Nayak, Amir Yazdanbakhsh, Vijay Janapa Reddi
The field of computer architecture, which bridges high-level software abstractions and low-level hardware implementations, remains absent from current large language model (LLM) evaluations. To this end, we present QuArch (pronounced ‘quark’), the first benchmark designed to facilitate the development and evaluation of LLM knowledge and reasoning capabilities specifically in computer architecture. QuArch v1.0 provides a comprehensive collection of 2,671 expert-validated question-answer (QA) pairs covering various aspects of computer architecture, including processor design, memory systems, and interconnection networks. Our evaluation reveals that while frontier models possess domain-specific knowledge, they struggle with skills that require higher-order thinking in computer architecture. Frontier model accuracies vary widely (from 34% to 73%) on these advanced questions, highlighting persistent gaps in architectural reasoning across analysis, design, and implementation QAs. Furthermore, via fine-tuning we find that QuArch can translate to improved performance on a realistic memory hierarchy design task, resulting in up to 1.99× more area-efficient solutions and up to 40% more viable solutions overall. By holistically assessing fundamental skills, QuArch provides a foundation for building and measuring LLM capabilities that can accelerate innovation in computing systems. The QuArch benchmark and leaderboard are publicly available at: https://quarch.ai/.
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Don't Walk the Line: Boundary Guidance for Filtered Generation
Sarah Ball, Andreas Haupt
Generative models are increasingly paired with safety classifiers that filter harmful or undesirable outputs. A common strategy is to fine-tune the generator to reduce the probability of being filtered, but this can be suboptimal: it often pushes the model toward producing samples near the classifier’s decision boundary, increasing both false positives and false negatives. We propose Boundary Guidance, a reinforcement learning fine-tuning method that explicitly steers generat

ion away from the classifier’s margin. On a benchmark of jailbreak, ambiguous, a nd long-context prompts, Boundary Guidance improves the safety while maintaining or improving the utility of outputs, as judged by LLM-as-a-Judge evaluations. C omprehensive ablations across model scales and reward designs demonstrate the ro bustness of our approach.
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On the Role of Batch Size in Stochastic Conditional Gradient Methods
Rustem Islamov, Roman Machacek, Aurelien Lucchi, Tony Silveti-Falls, Eduard Gorb unov, Volkan Cevher
We study the role of batch size in stochastic conditional gradient methods under a μ-Kurdyka–■ojasiewicz (μ-KL) condition. Focusing on momentum-based st ochastic Frank–Wolfe–type conditional gradient algorithms (e.g., Scion), we deri ve a new analysis that explicitly captures the interaction between stepsize, bat ch size, and stochastic noise. Our study reveals a regime-dependent behavior: in creasing the batch size initially improves optimization accuracy, but beyond a c ritical threshold, the benefits saturate and can eventually degrade performance under a fixed token budget. Notably, the theory predicts the magnitude of the op timal stepsize and aligns well with empirical practices observed in large-scale training. Leveraging these insights, we derive principled guidelines for selecti ng the batch size and stepsize, and propose an adaptive strategy that increases batch size and sequence length during training while preserving convergence guar antees. Preliminary experiments are consistent with the theoretical predictions and illustrate the emergence of the predicted scaling regimes. Overall, our resu lts provide a theoretical framework for understanding batch-size scaling in stoc hastic conditional gradient methods and offer guidance for designing efficient t raining schedules in large-scale optimization.
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VectorWorld: Efficient Streaming World Model via Diffusion Flow on Vector Graphs
Chaokang Jiang, Desen Zhou, Jiuming Liu, Li Sun
Closed-loop evaluation of autonomous-driving policies requires interactive simulation beyond log replay. Existing generative world models suffer three gaps : history-incompatible initialization, sampling latency exceeding real-time budg ets, and compounding kinematic infeasibility. We propose VectorWorld, a streamin g vector-graph world model that incrementally generates ego-centric lane--agent tiles during rollout. VectorWorld couples a motion-aware gated VAE for history-c ompatible initialization, an edge-gated relational DiT with interval-conditioned MeanFlow and JVP-based large-step supervision for solver-free outpainting, and ΔSim, a physics-aligned NPC policy with hybrid discrete--continuous actio ns and differentiable kinematic logit shaping. On Waymo Open Motion and nuPlan, VectorWorld improves map fidelity, initialization validity, and density calibrat ion, enabling stable real-time $1\mathrm{km}+$ closed-loop rollouts.
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Neural-HSS: Hierarchical Semi-Separable Neural PDE Solver
Pietro Sittoni, Emanuele Zangrando, Angelo Alberto Casulli, Nicola Guglielmi, Fr ancesco Tudisco
Deep learning-based methods have shown remarkable effectiveness in solving PDEs, largely due to their ability to enable fast simulations once trained. However, despite the availability of high-performance computing infrastructure, many crit ical applications remain constrained by the substantial computational costs asso ciated with generating large-scale, high-quality datasets and training models. I n this work, inspired by studies on the structure of Green's functions for ellip tic PDEs, we introduce Neural-HSS, a parameter-efficient architecture built upon the Hierarchical Semi-Separable (HSS) matrix structure that is provably data-ef ficient for a broad class of PDEs. We theoretically analyze the proposed archite cture, proving that it satisfies exactness properties even in very low-data regi mes. We also investigate its connections with other architectural primitives, su ch as the Fourier neural operator layer and convolutional layers. We experimenta lly validate the data efficiency of Neural-HSS on the three-dimensional Poisson equation over a grid of two million points, demonstrating its superior ability t o learn from data generated by elliptic PDEs in the low-data regime while outper

forming baseline methods. Finally, we demonstrate its capability to learn from d ata arising from a broad class of PDEs in diverse domains, including electromagn etism, fluid dynamics, and biology.
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What Information Matters? Graph Out-of-Distribution Detection via Tri-Component Information Decomposition
Danny Wang, Ruihong Qiu, Zi Huang
Graph neural networks are widely used for node classification, but they remain v ulnerable to out-of-distribution (OOD) shifts in node features and graph structu re. Prior work established that methods trained with standard supervised learnin g (SL) objectives tend to capture spurious signals from either features and/or s tructure, leaving the model fragile under distributional changes. To address thi s, we propose TIDE, a novel and effective Tri-Component Information Decompositio n framework that explicitly decomposes information into feature-specific, struct ure-specific and joint components. TIDE aims to preserve only the label-relevant part of the joint information while filtering out spurious feature- and structu re-specific information, thereby enhancing the separation between in-distributio n (ID) and OOD nodes. Beyond the framework, we provide theoretical and empirical analyses showing that an information bottleneck objective is preferable to stan dard SL for graph OOD detection, with higher ID confidence and a greater entropy gap between ID and OOD data. Extensive experiments across seven datasets confir m the efficacy of Tide, achieving up to a 34% improvement in FPR95 over strong b aselines while maintaining competitive ID accuracy. Code is available at https:/ /github.com/DannyW618/TIDE.
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D-ARL: A Distribution-Matched Asynchronous Reinforcement Learning Framework for Language Reasoning
Yinqi Bai, Tong Xialiang, Jie Wang, Hongyu Liu, Longdi Pan, Jiashuo Li, Zehao Wa ng, Jianye HAO, Mingxuan Yuan, Feng Wu
Asynchronous reinforcement learning (RL) has shown notable success in accelerati ng the post-training of large language models (LLMs). However, its decoupled dat a generation and training paradigm introduces a fundamental distributional misma tch between data generated by stale behavior policies and current policy, leadin g to unstable training and degraded performance. To address this challenge, we p ropose D-ARL, a **D**istribution-matched **A**synchronous **R**einforcement **L* *earning framework that selects high-quality asynchronous samples whose distribu tions are well aligned with the current policy for policy optimization. Specific ally, D-ARL maintains a replay buffer that collects samples from the most recent K behavior policies and proposes a variance-guided metric to select distribut ion-matched data. During training, D-ARL introduces a multi-behavior policy opti mization algorithm to leverage the multi-source nature of the selected samples f or policy update. Experiments on six widely used reasoning benchmarks show that D-ARL outperforms state-of-the-art asynchronous methods, achieving an average im provement of 6.4\% in reasoning performance and 34.7\% in sample efficiency.
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Meta-iLaD: Identifiable Latent Dynamics via Meta-Learning of Dynamics Environmen ts
Yubo Ye, Sweekar Piya, Xiajun Jiang, Linwei Wang
Learning latent dynamics is central to assessing current states and forecasting future trajectories for high-dimensional time series. For locally-stationary lat ent dynamics parameterized by past latent states and an environment variable c, with latent dynamics state z_t, prior identifiability results largely focus on z _t when conditioned on pre-defined label u of the dynamics environment. This lea ves two limitations: reliance on pre-defined labels that hinder generalization t o unseen environments, and limited understanding of the identifiability of F and c which---while offering important structural properties for the identifiabilit y of z_t---are learned jointly with z_t. We address these challenges with Meta-i LaD, a novel latent dynamics framework to attain identifiability by meta-learnin g across dynamics environments. Meta-iLaD replaces the conditioning of c on pre- defined labels with a novel conditional prior, modeled as a feedforward meta-lea

rner that rapidly extracts c from few-shot examples. We further establish simult aneous identifiability for z_t, c and F, for a general formulation of the dynami cs function F over past latent states and c, without restricting the dimension o f c or how it modulates F. We provide strong empirical evidence that 1) conditio ning on few-shot examples enables generalization to out-of-distribution environm ents, and 2) identifiability for c and F is critical for accurate forecasting be yond reconstructing observed trajectories.
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Robust Cross-Modal Retrieval via Generative Semantic Refinement and Exclusion-Gu ided Adaptation
Qin Yang, Xin Wei, Yanjia Li, Jiasun Feng, Mingrui Zhu, Nannan Wang, Xinbo Gao
Vision-Language Pre-trained (VLP) models are vulnerable to real-world query nois e. Current cross-modal Test-Time Adaptation (TTA) methods often rely on high-con fidence predictions, which induces confirmation bias and neglects the informativ e signals in ambiguous Low-Confidence Queries. To address this, we propose Gener ative Semantic Refinement and Exclusion-Guided Adaptation (ReEx), a robust retri eval framework that extends adaptation to the entire query stream. Specifically, textual structural noise is rectified by a Generative Semantic Refinement (GSR) module, which employs Confidence-Guided Dynamic Fusion to anchor LLM-based repa irs and prevent semantic drift. To exploit ambiguous data, adaptation is driven by Exclusion-Guided Proxy Contrastive Learning (EPCL), which imposes negative co nstraints via Exclusion Sets of unlikely candidates. Experimental results on COC O-C and Flickr-C demonstrate that ReEx consistently outperforms existing TTA met hods, achieving significant robustness gains with a justifiable computational tr ade-off.
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Multi-View Causal Discovery without Non-Gaussianity: Identifiability and Algorit hms
Ambroise Heurtebise, Omar Chehab, Pierre Ablin, Alexandre Gramfort, Aapo Hyvarin en
Causal discovery is a difficult problem that typically relies on strong assumpti ons on the data-generating model, such as non-Gaussianity. In practice, many mod ern applications provide multiple related views of the same system, which has ra rely been considered for causal discovery. Here, we leverage this multi-view str ucture to achieve causal discovery with weak assumptions. We propose a multi-vie w linear Structural Equation Model (SEM) that extends the well-known framework o f non-Gaussian disturbances by alternatively leveraging correlation over views. We prove the identifiability of the model for acyclic SEMs. Subsequently, we pro pose several multi-view causal discovery algorithms, inspired by single-view alg orithms (DirectLiNGAM, PairwiseLiNGAM, and ICA-LiNGAM). The new methods are vali dated through simulations and applications on neuroimaging data, where they enab le the estimation of causal graphs between brain regions.
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Hierarchical Representations for Cross-task Automated Heuristic Design using LLM s
Fei Liu, Rui Zhang, Shunyu Yao, Qinglong Hu, kefeng zheng, Zhichao Lu, Qingfu Zh ang
Designing heuristic algorithms for complex optimization problems is a time-consu ming and expert-driven process. Recently, Automated
Heuristic Design (AHD) using Large Language Models (LLMs) has shown significant promise for automating algorithm development. However, existing works mainly rel y on programs to represent heuristics, which are inherently taskspecific and fai l to generalize as effectively as established metaheuristics like tabu search or guided local search. To bridge this gap, we introduce Multi-Task Hierarchical S earch (MTHS), an LLM-guided evolutionary method that co-designs general-purpose metaheuristics and task-specific programs. MTHS employs a hierarchical represent ation and adopts a two-level evolution framework to evolve task-agnostic metaheu ristics and task-specific program implementations simultaneously across multiple heuristic design tasks. During this evolution, a knowledge transfer mechanism a llows learning from elite programs designed for other tasks. We evaluated MTHS o

n distinct combinatorial optimization problems, where it outperforms both commonly-used heuristics and existing LLM-driven AHD approaches. Our results demonstrate that the hierarchical representations facilitate effective multi-task AHD, and the evolved metaheuristics exhibit strong generalization to related tasks. Code: https://github.com/FeiLiu36/MTHS.

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FedGain: Toward Negative-Gain-Free Client Collaboration in Federated Learning

Yuqing Zhang, Changli Zhou, Binghuang Huang, Hui Tian

Data heterogeneity is a fundamental challenge in Federated Learning (FL), where induced model drift often results in "negative gains" for global models on data-abundant clients, with performance falling below that of local training. To address this issue, we propose FedGain, a novel framework that optimizes collaborative client clustering to mitigate the negative gain. We are the first to develop a modified Scaling Law (SL) to quantify the reduction in data utility caused by heterogeneity and define Effective Federated Capacity to align clients with the highest potential collaboration gains. Extensive experiments demonstrate that our modified SL strictly adheres to the power-law learning discipline in non-IID scenarios. FedGain effectively suppresses negative gains to a negligible level across various FL algorithms and outperforms other Clustered FL methods.

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Task-and-Model-Aware Fractal-Consistency for Efficient LLM Reasoning

Ziqiu Luo, Jianmin Liu, Yukai Miao, Li Chen, Dan Li

While self-consistency methods have emerged as a promising approach to enhance the correctness of large language model (LLM) outputs by aggregating multiple stochastic samples, they suffer from two critical limitations, resulting in high computation cost. First, they evaluate output consistency monolithically, failing to efficiently combine partially correct answers across multiple samples. Second, they use static stopping criteria that cannot adapt to varying task complexities and model capabilities, resulting in suboptimal computational efficiency.In this work, we present Task-and-Model-Aware Fractal-Consistency (TMAFC), a novel self-consistency framework that addresses these limitations through two key innovations: (1) Fractal-Consistency, which evaluates output consistency at the granularity of output components to effectively combine partial correct answers across samples, and (2) Adaptive Stopping Criteria Calibration (ASCC), which dynamically adjusts sampling stopping criteria based on real-time assessment of both task difficulty and LLM capability. Through extensive experiments on diverse question-answering benchmarks, we demonstrate that TMAFC achieves superior efficiency-accuracy trade-offs, reducing sample cost by up to 55% while maintaining competitive accuracy compared to state-of-the-art baselines.

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Unfolding Generative Flows with Koopman Operators: Trajectory-Preserving Linearization

Erkan Turan, Ari Siozopoulos, Louis Martinez, Julien Gaubil, Emery Pierson, Maks Ovsjanikov

Continuous Normalizing Flows (CNFs) enable elegant generative modeling but remain bottlenecked by their iterative nature requiring costly sampling and lacking interpretability of the intermediate states. Recent approaches accelerate sampling by straightening trajectories or distilling endpoints, yet they treat the original generative process as a black box, discarding the teacher’s intermediate dynamics. We propose a fundamentally different perspective: globally linearizing flow dynamics via Koopman theory to achieve trajectory-preserving linearization. By lifting Conditional Flow Matching (CFM) into a higher-dimensional Koopman space, we represent its evolution with a single linear operator. Crucially, unlike boundary-only distillation, our method enforces infinitesimal consistency with the teacher's vector field along *the full generative path*. We derive a practical, simulation-free training objective that ensures this global alignment and yields two key benefits. First, sampling becomes one-step and parallelizable. Second, because the linearization is faithful to the dynamics, the Koopman operator provides unique insights on the generation. We demonstrate that this structure enables novel applications unavailable in prior approaches, including discovery of

semantically coherent editing directions, inversion with a teacher-aligned linear operator and class-conditional spectral signatures. Empirically, our approach achieves competitive sample quality, while enabling spectral analysis and control of the *entire trajectories* of generative flows.
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MetaphorVU: Towards Metaphorical Video Understanding
Zhuoqun Li, Boxi Cao, Guiping Jiang, Fangrui Lv, Ruotong Pan, Jianan Wang, Xiangyu Wu, Hongyu Lin, Yaojie Lu, Yong Du, Ruyin Jia, Liyan, Tingting Gao, Han Li, Xianpei Han, Le Sun
Metaphorical videos are prevalent across various real-world scenarios to convey complex ideas, and understanding them typically requires high-order cognitive capabilities.
The lack of systematic studies on metaphorical video understanding not only constrains the real-world applicability of MLLMs but also impedes the thorough assessment of their high-order cognitive capabilities.
To bridge this gap, we propose MetaphorVU-Bench, the first systematic and comprehensive benchmark dedicated to metaphorical video understanding.
Through experiments, we find current MLLMs struggle with accurate metaphorical video understanding, lagging far behind human level, primarily due to defective cross-domain mapping.
Motivated by this finding, we construct a metaphor knowledge graph as mapping augmentation and propose MetaphorBoost, an inference-time enhancement framework achieving consistent performance improvement.
Our benchmark, analysis, and method provide useful insights and a foundation for future research on advancing MLLMs.
Code: https://github.com/icip-cas/MetaphorVU.
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Towards Practical World Model-based Reinforcement Learning for Vision-Language-Action Models
Zhilong Zhang, Haoxiang Ren, Yihao Sun, Yifei Sheng, Haonan Wang, Zhichao Wu, Haoxin Lin, Pierre-Luc Bacon, Yang Yu
Vision-Language-Action (VLA) models show strong generalization for robotic control, but finetuning them with reinforcement learning (RL) is constrained by the high cost and safety risks of real-world interaction. Training VLA models in interactive world models avoids these issues but introduces several challenges, including pixel-level world modeling, multi-view consistency, and compounding errors under sparse rewards. Building on recent advances across large multimodal models and model-based RL, we propose VLA-MBPO, a practical framework to tackle these problems in VLA finetuning. Our approach has three key design choices: (i) adapting unified multimodal models (UMMs) for data-efficient world modeling; (ii) an interleaved view decoding mechanism to enforce multi-view consistency; and (iii) chunk-level branched rollout to mitigate error compounding. Theoretical analysis and experiments across simulation and real-world tasks demonstrate that VLA-MBPO significantly improves policy performance and sample efficiency. Crucially, our method maintains a universal set of hyperparameters across all tasks, underscoring its robustness and scalability for real-world robotic deployment.
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Mitigating Error Propagation in Low-Rank Approximation of Large Models via Distribution-Aware Whitening
Yao Zhu, Rui Sun, Xiu Yan, Yang Li, Wang Lu, Yunjian Zhang
Low-rank approximation has emerged as a cornerstone technique for model compression and parameter-efficient fine-tuning, enabling substantial reductions in computation and memory without altering model architectures. However, existing approaches often overlook the shifts in feature distributions induced by the approximation process, which can lead to error amplification and unstable inference.
We propose a distribution-aware whitening framework that dynamically whitens layer inputs based on the evolving feature distributions, ensuring second-order isotropy of input features.
This allows that the discarded components in the low-rank approximation are those with minimal impact on model outputs, thereby minimizing cumulative approximat

ion errors across layers.
We theoretically analyze how distribution misalignment leads to error propagation and demonstrate that our approach achieves tighter control over layerwise distortion.
Extensive experiments across various large language models demonstrate the superiority of our method in post-training compression. Moreover, our method can also serve as an effective initialization for LoRA-style parameter-efficient fine-tuning.
Our findings highlight the importance of considering feature distributions in low-rank approximations, paving the way for reliable and effective model compression strategies.
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Modular Pretraining Enables Access Control
Ethan Roland, Murat Cubuktepe, Erick Martinez, Stijn Servaes, Keenan Pepper, Michael Vaiana, Diogo S de Lucena, Judd Rosenblatt, Addie Foote, Cem Anil, Alex Cloud
AI developers face a dual-use dilemma. A model capability that helps one user cure a disease can help another synthesize one. This dilemma could be resolved with access control, limiting dual-use AI capabilities to trusted deployments with a legitimate need. A gold standard for access control would be to serve separate models with different capabilities to different users. However, training and deploying multiple models is prohibitively expensive. To address this challenge, we propose gradient-routed auxiliary modules (GRAM), a pre-training method that adds modules to a neural network and selectively updates them to induce specialization. Ablating a module at inference time removes its capability from the network, approximating a model trained on filtered data. We evaluate GRAM on synthetic stories and realistic dual-use data spanning virology, cybersecurity, nuclear physics, and specialized code. These experiments show that GRAM preserves selected retain capabilities while disabling forgotten capabilities, and limits recovery better than post-hoc unlearning. Most importantly, a Chinchilla-optimal scaling analysis from 50M to 5B parameters shows that the forget capability gap between data-filtered and full-data models widens with scale while the retain gap stays constant, and that GRAM closely tracks data filtering. GRAM's training cost is independent of the number of supported capability profiles, yielding a 5x cost reduction in our 5-profile setting.
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Scaling Behavior in Model Fine-tuning for Audio DeepFake Detection
Xiang Li, Pin-Yu Chen, Wenqi Wei
Recent advances in audio deepfake detection have been driven by increasingly large speech foundation models and growing amounts of synthetic data. Despite strong benchmark performance, it remains unclear how detection capability scales with model capacity and training data under realistic deployment conditions involving distribution shift, signal corruption, and unseen synthesis pipelines. In this work, we present the first systematic study of scaling laws in post-training audio deepfake detection, focusing on fine-tuning regimes rather than large-scale pretraining. Using a controlled family of speech foundation models with shared architecture and pretraining, we analyze how detection performance, robustness, and generalization evolve as a function of model size and training data scale. Our results reveal a fundamental asymmetry between performance scaling and robustness scaling in audio deepfake detection, suggesting increasing model capacity alone is insufficient for achieving reliable real-world generalization.
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Criterion-Conditional In-Context Learning: Evaluating Criterion-Shift Adaptation in Vision-Language Models
Kaiyun Yang, Ruilin Yang, Zhimin Yao, Jikai Wang, Wei Ge
Vision-language models can perform new tasks without parameter updates through in-context learning (ICL), whose core mechanism is utilizing the support set for task induction. In the standard ICL setting, once the task is induced, its decision criterion remains fixed.
However, in real-world applications, many tasks exhibit a stable high-level inte

nt, while their decision criteria shift according to specific requirements.
Thus, we introduce a new setting, denoted as Criterion-Conditional In-Context Le arning (CC-ICL), where models must infer the latent criterion from context and a djust predictions accordingly under fixed task semantics.
To evaluate this capability, we propose two complementary metrics, Criterion Inv ariance and Criterion Sensitivity, capturing the model's robustness and adaptabi lity under criterion shifts.
We further construct CC-Bench, a multi-domain benchmark that supports evaluation under the CC-ICL setting. By employing a dual-level data hierarchy, CC-Bench en ables legitimate ground-truth variation conditioned on the active criterion even when the task remains fixed.
Experiments on CC-Bench reveal that most models exhibit a rigid boundary bias, s truggling to align their decisions with the latent criterion.
We also find that even a simple multi-criterion training strategy can significan tly reduce this bias, improving Criterion Sensitivity and enabling 7B-scale mode ls to surpass proprietary models without degrading general multimodal performanc e.
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GeoAlign: Geometric Rollout Curation for Robust LLM Reinforcement Learning
Ting Zhou, Zhenqing Ling, Zhao Yiyang, Ying Shen, Daoyuan Chen
Online reinforcement learning is widely used to align large language models (LLM s) with reward signals, yet training can be unstable under noisy or misspecified rewards. We identify a failure mode we call directional inconsistency: within a batch, a small set of high-reward rollouts induces representation-space prefere nce directions that sharply disagree with the batch majority, resulting in high- variance and destabilizing updates. We propose GeoAlign, a lightweight plug-in f or rollout curation in iterative policy optimization. GeoAlign (i) forms within- prompt preference pairs, (ii) learns an online projector on per-rollout hidden s tates to concentrate reward-ordered displacement directions, and (iii) detects d irectionally inconsistent rollouts via their angular deviation from a batch cons ensus prototype and rectifies them with within-prompt stable alternatives. GeoAl ign is forward-pass only and adds negligible overhead. Across dialogue alignment with a learned reward model and mathematical reasoning with binary verified rew ards, GeoAlign improves final performance and reduces training oscillation, outp erforming PF-PPO, PAR, PODS, and Seed-GRPO.
These results suggest latent directional consensus as an effective reliability s ignal for online LLM RL.
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A Judge-Aware Ranking Framework for Evaluating Large Language Models without Gro und Truth
Mingyuan Xu, Xinzi Tan, Jiawei Wu, Doudou Zhou
Evaluating large language models (LLMs) on open-ended tasks without ground-truth labels is increasingly done via the LLM-as-a-judge paradigm. A critical but und er-modeled issue is that judge LLMs differ substantially in reliability; treatin g all judges equally can yield biased leaderboards and misleading uncertainty es timates—more data can make evaluation more confidently wrong under misspecified aggregation. We propose a judge-aware ranking framework that extends the Bradley -Terry-Luce model by introducing judge-specific discrimination parameters, joint ly estimating latent model quality and judge reliability from pairwise compariso ns without reference labels. We establish identifiability up to natural normaliz ations and prove consistency and asymptotic normality of the maximum likelihood estimator, enabling confidence intervals for score differences and rank comparis ons. Across multiple public benchmarks and a newly collected dataset, our method improves agreement with human preferences, achieves higher data efficiency than unweighted baselines, and produces calibrated uncertainty quantification for LL M rankings.
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DataGuard: A Non-intrusive Dataset Auditing Framework via Differential Informati on Forensics
Jiadong Lou, Wenxin Rong, Li Chen, Xing Gao, Rui Zhang, Xu Yuan

Concerns over dataset misuse in deep learning have highlighted the need for effe
ctive auditing.
Unlike existing intrusive methods that require dataset modifications, which risk
 model performance and security, we present DataGuard, a non-intrusive framework
 for quantitative dataset auditing.
Specifically, DataGuard integrates three key components: 1) a differential compa
rison between the target dataset and auxiliary non-training datasets, 2) an info
rmation-forensic analysis establishing formal inequalities to distinguish traini
ng data; and 3) a multivariate statistical test that translates these discrepanc
ies into rigorous auditing scores.
Extensive experiments demonstrate that DataGuard can detect both full and partia
l dataset usage without false positives while remaining robust under diverse tra
ining scenarios, offering a principled, information-theoretic solution for trans
parent AI development.
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AdaHC: Accelerating Multi-Token Prediction with Adaptive Head Chunking with Pipe
line Parallelism
Yan Wang, Chang Si, Kaiming Yang, Zhipeng Zhang, Weijian Liu, Man Yuan, Mingzhen
 Li, Yong Li, Weile Jia
Multi-token prediction (MTP) architecture is widely adopted in LLMs.
MTP blocks can be appended to the tail of model to predict additional tokens.
However, when training with pipeline parallel, MTP leads to more pipeline bubble
s and deteriorates the pipeline efficiency.
Based on in-depth analysis of MTP architectures and loss functions, we have iden
tified the parallel nature of the MTP blocks, and leverage it for superior pipel
ine scheduling.
We propose AdaHC, an adaptive pipeline scheduling framework for accelerating LLM
s training with MTP block(s).
AdaHC splits the output heads into chunks and reassembles the chunks to generate
 balanced pipeline stages, and performs adaptive activation forwarding to preser
ve the numerical equivalence.
Experimental results show that AdaHC improves the training throughput of SOTA LL
Ms with diverse MTP configurations by 1.35$\times$ on average.
This work paves a new direction for practical pipeline training.
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PGC: Peak-Guided Calibration for Generalizable AI-Generated Image Detection
Xiaoyu Zhou, Jianwei Fei, Peipeng Yu, Jingchang Xie, Chong Cheng, Zhihua Xia
The rapid evolution of generative AI, from GANs to modern diffusion models, has
resulted in increasingly subtle discriminative clues. These fine-grained signals
 are often overshadowed by dominant, high-fidelity image content (e.g., the main
 subject), limiting the reliability of existing detectors that predominantly rel
y on global representations. To address this challenge, we propose the Peak-Guid
ed Calibration (PGC) framework. PGC introduces a novel strategy that aggregates
salient features via a peak-focusing mechanism. Specifically, by employing a pea
k-sensitive aggregation that accentuates the most discriminative local clues, PG
C leverages these critical signals to calibrate the global decision. This approa
ch recovers subtle patterns that would otherwise be submerged in the global cont
ext. Furthermore, to better simulate real-world threats, we introduce the CommGe
n15 dataset, a challenging benchmark comprising samples from 15 commercial model
s. Extensive experiments demonstrate that PGC achieves state-of-the-art performa
nce. Specifically, it improves mean accuracy by +12.3% on our CommGen15 dataset,
 and sets new records on standard benchmarks, including GenImage (+2.1%), AIGI (
+3.5%), and UniversalFakeDetect (+1.7%). Code is available at https://github.com
/xiaoyu6868/PGC.
****
Faster Query-Key Learning Sharpens Attention in Self-Attention Models
Rahul Vashisht, Harish G. Ramaswamy
A standard self-attention layer consists of two interacting circuits: the query-
key circuit that governs attention allocation, and the output-value circuit that
 maps attended representations to predictions. Collapsed and factorized paramete

rizations of the query-key and output-value circuits lead to qualitatively diffe rent attention patterns. In particular, some parameterizations give sharper atte ntion to task-relevant tokens, at a similar training loss. We analyze how the pa rameterizations of these circuits shape the parameter trajectories in single-lay er self-attention models trained on next-token prediction. Through gradient-flow analysis, we show that factorization induces implicit rescaling of the two circ uits' learning rates. We derive closed-form dynamics showing that output-value a nd query-key parameters move along a line, with relative speeds determined by th eir learning rates. Faster query-key learning relative to output-value learning thus produces sharper attention, as the model compensates for slower output-valu e learning by increasing attention mass on relevant tokens. Experiments show tha t differences in the relative learning rates of the two circuits govern attentio n concentration. This improves attention interpretability while maintaining comp arable predictive performance.

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Improved Scaling Laws via Weak-to-Strong Generalization in Random Features Ridge Regression

Diyuan Wu, Lehan Chen, Theodor Misiakiewicz, Marco Mondelli

It is increasingly common in machine learning to use learned models to label dat a and then employ such data to train more capable models. The phenomenon of weak -to-strong generalization exemplifies the advantage of this two-stage procedure: a strong student is trained on imperfect labels obtained from a weak teacher, a nd yet the strong student outperforms the weak teacher. In this paper, we show t hat the potential improvement is substantial, in the sense that it affects the s caling law followed by the test error. Specifically, we consider students and te achers trained via random features ridge regression (RFRR). Our main technical c ontribution is to derive a deterministic equivalent for the excess test error of the student trained on labels obtained via the teacher. Via this deterministic equivalent, we then identify regimes in which the scaling law of the student imp roves upon that of the teacher, unveiling that the improvement can be achieved b oth in bias-dominated and variance-dominated settings. Strikingly, the student m ay attain the minimax optimal rate regardless of the scaling law of the teacher- --in fact, when the test error of the teacher does not even decay with the sampl e size.

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MV-FGAD: Towards Efficient and Effective Federated Graph Anomaly Detection via M ulti-view Learning

Junyi Yan, KE LIANG, Hao Yu, Meng Liu, Hao Tan, Tianrui Liu, Jun-Jie Huang, Xinw ang Liu

Federated graph anomaly detection (GAD) aims to identify abnormal nodes in distr ibuted subgraphs through federated learning. However, existing methods suffer fr om two limitations. 1) Their reliance on neighborhood aggregation assumes that a nomalous information can be sufficiently captured, which often fails in federate d learning with partitioned client subgraphs. 2) They overlook the detection bot tleneck caused by weak attribute or structural anomalies. To tackle these challe nges, we revisit federated GAD and reveal that weak anomalies exhibit harder-to- detect signals compared to strong anomalies. Specifically, we propose MV-FGAD, a n efficient and effective federated GAD framework for mining anomalies of varyin g strengths. MV-FGAD introduces a federated knowledge learning module to aggrega te and broadcast shared knowledge, which is further exploited to optimize local topological structures. Moreover, it designs a multi-view learning mechanism to capture diverse anomaly patterns, and adopts Mahalanobis distance–based scoring strategy to quantify node abnormality across views. Extensive experiments on rea l-world datasets of varying types and scales demonstrate MV-FGAD's efficiency an d effectiveness. Our code is publicly available at https://github.com/Junyi-Yan/ MV-FGAD.

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RT-Lynx: Putting the GEMM Sparsity In a Right Way for Diffusion Models

Xing Cong, Hanlin Tang, Kan Liu, Tao Lan, Lin Qu, chenhao xie

Diffusion Transformers (DiT) achieve strong performance in image generation but

incur substantial inference costs. While prior work has reduced this cost via qu antization and distillation, semi-structured sparsity—which can nearly halve FLO Ps—remains underexplored. A key reason is that most existing approaches focus on weight sparsification, and pruning 50\% of the weights can remove critical mode l capacity and degrade generation quality. Our study, however, shows that DiT ac tivations are intrinsically sparse and significantly more robust to N:M semi-str uctured sparsification than weights. Motivated by this observation, we advocate a paradigm shift from weight sparsification to activation sparsification. We pro pose RT-Lynx, which applies N:M sparsification to activations and incorporates e rror-compensation techniques to mitigate accuracy loss. We further implement hig hly optimized CUDA kernels tailored to this setting, achieving up to a 1.55× spe edup on average in linear layers. Extensive experiments across multiple diffusio n models demonstrate that our method preserves the generation quality of the ori ginal models while substantially accelerating inference.

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PerturbDiff: Functional Diffusion for Single-Cell Perturbation Modeling

Xinyu Yuan, Xixian Liu, Ya Shi Zhang, Zuobai Zhang, Hongyu Guo, Jian Tang

Building _Virtual Cells_ that can accurately simulate cellular responses to pert urbations is a long-standing goal in systems biology. A fundamental challenge is that high-throughput single-cell sequencing is destructive: the same cell cann ot be observed both before and after a perturbation. Thus, perturbation predicti on requires mapping unpaired control and perturbed populations. Existing models address this by learning maps between distributions, but typically assume a sing le fixed response distribution when conditioned on observed cellular context (_e .g._, cell type) and the perturbation type. In reality, responses vary systemati cally due to unobservable latent factors such as microenvironmental fluctuations and complex batch effects, forming a _manifold_ of possible distributions for t he same observed conditions. To capture this variability, we introduce PerturbD iff, which shifts modeling from individual cells to entire distributions. By emb edding distributions as points in a Hilbert space, we define a diffusion-based g enerative process operating directly over probability distributions. This allows PerturbDiff to capture population-level response shifts across hidden factors, improving generalization. Benchmarks on established datasets show that PerturbDi ff achieves state-of-the-art performance in single-cell response prediction and generalizes substantially better to unseen perturbations. See our project page (https://katarinayuan.github.io/PerturbDiff-ProjectPage/), where code and data (h ttps://github.com/DeepGraphLearning/PerturbDiff) are publicly available.

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Foreground-Aware Token Routing Vision Transformer for Real-Time Satellite Video Tracking

Jiahao Wang, Fang Liu, Licheng Jiao, Shuo Li, Hao Wang, Lingling Li, Xinyi Wang, Xu Liu

Real-time satellite video tracking poses distinct challenges, including accommod ating high spatial-temporal resolution, dynamic backgrounds, and constrained onb oard computational resources. While Discriminative Correlation Filter (DCF)-base d methods offer high-speed inference, they suffer from limited accuracy. In cont rast, Vision Transformer (ViT)-based trackers achieve strong performance by unif ying representation and aggregation in a single-stream design, yet their heavy c omputational footprint limits practical deployment in real-time satellite scenar ios. In this work, we present FATrack, a novel tracking framework that effective ly balances tracking accuracy and computational efficiency. At its core is FA-Vi T, a lightweight Vision Transformer backbone that introduces foreground-aware to ken routing, enabling the model to concentrate computation on target-relevant re gions while suppressing redundancy. To mitigate semantic degradation caused by t oken sparsification, we propose the Adaptive Scatter Module (ASM), which selecti vely reinforces informative tokens via joint spatial-channel attention and spars e structural propagation, thereby enhancing both semantic fidelity and spatial c oherence. By synergistically integrating FA-ViT and ASM, FATrack forms a unified architecture that delivers real-time performance with significantly improved tr acking precision. Extensive evaluations on multiple satellite video benchmarks d

emonstrate that FATrack surpasses existing real-time trackers in accuracy and ac
hieves inference efficiency comparable to DCF-based methods, highlighting its po
tential for practical deployment in large-scale aerial video tracking systems.
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Scalable Event Cloud Network for Event-based Classification

Hongwei Ren, Fei Ma, Xiaopeng LIN, Yuetong Fang, Hongxiang Huang, Yue Zhou, Yulo
ng Huang, Haotian FU, Ziyi Yang, Youxin Jiang, Xiangqian Wu, Bojun Cheng

Event cameras are biologically inspired sensors garnering significant attention
from both industry and academia. Mainstream methods favor frame and voxel repres
entations, which reach a satisfactory performance while introducing time-consumi
ng transformations, bulky models, and sacrificing fine-grained temporal informat
ion. Alternatively, Point Cloud representation demonstrates promise in addressin
g the mentioned weaknesses, but it has limited scalability in abstracting featur
es of higher spatial resolution and longer temporal sequence events. In this pa
per, we propose a Scalable Network named SECNet to leverage Event Cloud represen
tation. SECNet integrates polarity at the structural level by innovating the Eve
nt-based Group and Sampling module rather than only at the input level. To accom
modate the surge in the number of events, SECNet embraces feature extraction in
the frequency domain via the Fourier transform. This approach not only substanti
ally extinguishes the explosion of Multiply Accumulate Operations but also effec
tively abstracts spatio-temporal features. We conducted extensive experiments on
 ten event-based datasets, and substantiate the scalability, effectiveness, and
efficiency of SECNet. Our code will be available at: https://github.com/rhwxmx/S
ECNet_ICML.
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Towards Achieving Optimal Strong Regret and Constraint Violation via Computation
ally Efficient Model-free RL

Xiyue Peng, Lingkai Zu, Ziyu Shao, Xin Liu

We study episodic constrained Markov decision processes (CMDPs) with linear func
tion approximation, where the goal is to achieve strong regret and constraint vi
olation guarantees without allowing error cancellations. Unlike the existing wor
k, which focuses on either tabular CMDP or model-based reinforcement learning me
thods.

We propose a model-free policy APMPO that achieves near-optimal $\widetilde{O}(\
sqrt{K})$ strong regret and strong constraint violation with Slater's condition
(or strict feasibility assumption),

where K is the total number of episodes. It matches the best-known rates witho
ut requiring any prior knowledge of the feasibility gap reported in prior model-
based work for tabular CMDPs. Besides, APMPO achieves $\widetilde{O}(K^{\frac{3
}{4}})$ strong regret and $\widetilde{O}(K^\frac{3}{4})$ strong constraint viola
tion without Slater's condition. To the best of our knowledge, this is the first
 sublinear result of CMDP w.r.t. the strong metrics without Slater's condition.

APMPO achieves these results by a novel and adaptive design of a violation-aware
 penalty and learning rates to balance the strong regret and constraint violatio
n, which is quite different from the (regularized) primal-dual methods imposing
constraints via dual penalty in the literature. The experiments show APMPO signi
ficantly outperforms the strong baselines, which justify our design and theoreti
cal performance.
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Unsupervised Diffusion Solver for Combinatorial Optimization via Combinatorial A
djoint Matching

Shengyu Feng, Tarun Suresh, Yiming Yang

Diffusion-based neural solvers have shown strong promise for combinatorial optim
ization (CO), but existing methods typically rely on supervised training with la
rge collections of near-optimal solutions. In this work, we extend adjoint-based
 trajectory optimization methods to discrete combinatorial domains. We formulate
 diffusion-based CO as a stochastic control problem over Continuous-Time Markov
Chains and introduce discrete adjoint dynamics for propagating optimization sign
als through discrete generative trajectories. Building on this formulation, we p

ropose *Combinatorial Adjoint Matching (CAM)*, an unsupervised training framewor
k for discrete diffusion solvers with structured and low-variance trajectory-lev
el optimization signals. Empirically, CAM consistently outperforms existing unsu
pervised diffusion baselines and achieves performance competitive with strong su
pervised diffusion solvers and even traditional solvers across diverse combinato
rial optimization problems. Our code is available at https://github.com/Shengyu-
Feng/CAM.
****
Reflector: Internalizing Step-wise Reflection against Indirect Jailbreaks
Jiachen Ma, Jiawen Zhang, Xiangtian Li, Bo Zou, Chaochao Lu, Chao Yang
While Large Language Models (LLMs) demonstrate remarkable capabilities, they rem
ain susceptible to sophisticated, multi-step jailbreak attacks that circumvent c
onventional surface-level safety alignment by exploiting the internal generation
 process. To address these vulnerabilities, we propose Reflector, a principled t
wo-stage framework that internalizes self-reflection within the generation traje
ctory. Reflector first leverages teacher-guided generation to produce high-quali
ty reflection data for supervised fine-tuning (SFT), establishing structured ref
lection patterns. It subsequently uses Reinforcement Learning (RL) with outcome-
driven and reward-validity supervision to instill robust, autonomous self-reflec
tion capabilities.
Empirical results show that Reflector achieves Defense Success Rates (DSR) excee
ding 90% against complex indirect attacks while generalizing robustly across div
erse threat scenarios. Notably, the framework enhances both task-specific and ge
neral utility, yielding a 5.85% gain on GSM8K alongside improved performance on
knowledge-intensive benchmarks. By internalizing trajectory-level safety, Reflec
tor overcomes the fundamental limitations of surface alignment without significa
nt computational overhead, offering an efficient and scalable solution for the d
evelopment of safe and capable LLMs.
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Causal Discovery for Irregularly Time Series with Consistency Guarantees
Weihong Li, Baohong Li, Anpeng Wu, Zhihan Li, Ming Ma, Keting Yin, Kun Kuang
This paper studies causal discovery in irregularly sampled time series—a key cha
llenge in risk-sensitive domains like finance, healthcare, and climate science,
where missing data and inconsistent sampling frequencies distort causal mechanis
ms. The main challenge comes from the interdependence between missing data imput
ation and causal structure recovery: errors in imputation and structure learning
 can reinforce each other, leading to an inaccurate causal graph. Existing metho
ds either impute first and then discover, or jointly optimize both via neural re
presentation learning, but lack explicit mechanisms to ensure mutual consistency
 of imputation and structure learning. We address this challenge with ReTimeCaus
al, an EM-based framework that alternates between imputation and structure learn
ing, which encourages structural consistency throughout the optimization process
. Our framework provides theoretical consistency guarantees for structure recove
ry and extends classical results to settings with irregular sampling and high mi
ssingness. ReTimeCausal combines kernel-based sparse regression and structural c
onstraints in an alternating process that updates the completed data and the cau
sal graph in turn. Experiments on synthetic and real-world datasets show that Re
TimeCausal is more effective than existing methods under challenging irregular s
ampling and missing data.
****
The Geometry of Projection Heads: Conditioning, Invariance, and Collapse
Faris Chaudhry
We develop a geometric theory of projection heads in self-supervised learning by
 modeling the head as a trainable Riemannian metric on the backbone representati
on manifold. We show that linear heads perform implicit subspace whitening, whil
e nonlinear heads adapt local metrics to satisfy the specific topological constr
aints of the loss, with head depth empirically dictating this capacity. Analyzin
g dimensional collapse, we prove that smooth nonlinear heads natively induce neg
ative eigenvalues in the Hessian at collapsed equilibria, making them unstable.
We empirically validate this by continuously tracking the optimization geometry

during training, which reveals that smooth activations like Swish can generate e xplicit negative curvature to escape collapse, whereas linear and ReLU heads und er continuous-time gradient flow cannot, relying instead on discrete-time optimi zation dynamics and BatchNorm. Finally, we geometrically characterize how metric degeneracy governs the information-invariance trade-off, explaining why the hea d must be discarded. Evaluated across contrastive and decorrelation-based object ives on foundation models, our results demonstrate that the projection head acts as a universal geometric buffer, decoupling the semantic backbone from the rigi d, destructive constraints of the pretraining objective.
****
Federated Causal Inference on Multi-Site Observational Data via Propensity Score Aggregation
Rémi Khellaf, Aurélien Bellet, Julie Josse
Causal inference typically assumes centralized access to individual-level data. Yet, in practice, data are often decentralized across multiple sites, making cen tralization infeasible due to privacy, logistical, or legal constraints. We addr ess this problem by estimating the Average Treatment Effect (ATE) from decentral ized observational data via a Federated Learning (FL) approach, allowing inferen ce through the exchange of aggregate statistics rather than individual-level dat a. We propose a novel method to estimate propensity scores via a federated weigh ted average of local scores using Membership Weights (MW), defined as probabilit ies of site membership conditional on covariates. MW can be flexibly estimated w ith parametric or non-parametric classification models using standard FL algorit hms.

The resulting propensity scores are used to construct Federated Inverse Prop ensity Weighting (Fed-IPW) and Augmented IPW (Fed-AIPW) estimators.

In contrast to meta-analysis methods, which fail when any site violates posi tivity, our approach exploits heterogeneity in treatment assignment across sites to improve overlap.

We show that Fed-IPW and Fed-AIPW perform well under site-level heterogeneit y in sample sizes, treatment mechanisms, and covariate distributions. Theoretica l analysis and experiments on simulated and real-world data demonstrate clear ad vantages over meta-analysis and related approaches.
****
Learning Efficient Guardrails for Compliance
Xiaofei Wen, Wenjie Jacky Mo, Yanan Xie, Peng Qi, Muhao Chen
Autonomous web agents are increasingly deployed for long-horizon tasks, yet thei r ability to adhere to real-world policies remains critically underexplored comp ared to standard safety objectives.
To address this gap, we introduce PolicyGuardBench, a benchmark of 60k policy-tr ajectory pairs designed to evaluate compliance through both full-trajectory and novel prefix-based violation detection tasks. Using this dataset, we train Polic yGuard, a lightweight guardrail model that achieves strong detection accuracy wh ile maintaining high inference efficiency.
Notably, our model demonstrates robust generalization capabilities, preserving h igh performance even on unseen domains. These contributions establish a comprehe nsive framework for studying policy compliance, showing that accurate and genera lizable guardrails are feasible at small scales.
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SoftMatcha 2: A Fast and Soft Pattern Matcher for Trillion-Scale Corpora
Masataka Yoneda, Yusuke Matsushita, Go Kamoda, Kohei Suenaga, Takuya Akiba, Masa ki Waga, Sho Yokoi
We present SoftMatcha 2, an ultra-fast and flexible search algorithm that enable s search over trillion-scale natural language corpora in under 0.3 seconds while allowing semantic variations in the form of substitution, insertion, and deleti on. Our approach employs string matching based on suffix arrays that scales well with corpus size, and represents words as vectors, which underpin its semantic flexibility. To mitigate the combinatorial explosion induced by the semantic rel axation of queries, our method is built on two key algorithmic ideas: dynamic co rpus-aware pruning and fast exact lookup enabled by a disk-aware design. We theo

retically analyze the efficiency of the proposed method, indicating that it can mitigate exponential growth in the search space. Empirically, on FineWeb-Edu (Lozhkov et al., 2024) (1.4T tokens), it attains substantially lower search latency than existing methods: infini-gram (Liu et al., 2024), infini-gram mini (Xu et al., 2025), and SoftMatcha (Deguchi et al., 2025). As a practical application, our method uncovers benchmark contamination in training corpora that existing approaches miss, and it also benefits information retrieval and paraphrase detection. We also provide an online demo of fast, soft search across corpora in seven languages.

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Mining Useful General Data for Low-Resource Domain Adaptation

Pingjie Wang, Hongcheng Liu, Yusheng Liao, Ziqing Fan, Yaxin Du, Shuo Tang, Yanfeng Wang, Yu Wang

Adapting large language models (LLMs) to low-resource domains remains challenging due to the scarcity of domain-specific data. While in-domain data is limited, there exists a vast amount of general-domain data that shares similar question–answer formats and reasoning patterns with domain tasks. This observation raises an important question: can useful general-domain data be mined to improve low-resource domain adaptation? Our initial findings show that general-domain chain-of-thought data contains useful auxiliary signals for domain adaptation, even without careful selection. This observation motivates a new paradigm for domain adaptation beyond exclusive reliance on domain-specific data.

To systematically identify the most beneficial general-domain samples, we propose NTK-Selector, motivated by the Neural Tangent Kernel’s ability to capture alignment in training dynamics. Since directly applying NTK to pretrained LLMs is impractical, we introduce a Jacobian-free NTK approximation and empirically demonstrate stable NTK-like behavior during fine-tuning. Extensive experiments across medical, financial, legal, and psychological domains demonstrate that NTK-Selector consistently outperforms domain-only fine-tuning and existing data selection baselines. In particular, NTK-Selector achieves gains of +8.7 and +5.1 points on Llama3-8B-Instruct and Qwen3-8B, respectively, compared to only +0.8 and +0.9 points from domain-only fine-tuning.

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Fourier Features Let Agents Learn High Precision Policies with Imitation Learning

Balázs Gyenes, Emiliyan Gospodinov, Jan Frieling, Enrico Krohmer, Nicolas Schreiber, Xiaogang Jia, Niklas Freymuth, Gerhard Neumann

High-precision robotic manipulation requires fine-grained spatial reasoning that is often difficult to achieve with RGB-only policies due to depth ambiguity and perspective scale issues. Policies that leverage 3D information directly, such as those based on point clouds, offer a stronger geometric prior over purely image-based ones, yet their performance remains highly task-dependent. We hypothesize that this discrepancy may be due to the spectral bias of neural networks towards learning low frequency functions, which especially affects architectures conditioned on slow-moving Cartesian features. We thus propose to map point clouds from Cartesian space into high-dimensional Fourier space, effectively equipping the point cloud encoder with direct access to high-frequency features. We experimentally validate the use of Fourier features on challenging manipulation tasks from the RoboCasa and ManiSkill3 benchmarks and on a real robot setup. Despite their simplicity, we find that Fourier features provide significant benefits across diverse encoder architectures and benchmarks and are robust across hyperparameters. Our results indicate that Fourier features let policies leverage geometric details more effectively than Cartesian features, showing their potential as a general-purpose tool for point cloud-based imitation learning. We provide source code and videos on our project page: https://fourier-il.github.io/fourier-il.

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SWE-IF: Aligning Code Evaluation with Human Preference

Ming Zhong, Xiang Zhou, Ting-Yun Chang, Qingze Wang, Nan Xu, Xiance Si, Dan Garrette, Shyam Upadhyay, Jeremiah Zhe Liu, Jiawei Han, Benoit Schillings, Jiao Sun

Large Language Models (LLMs) have catalyzed vibe coding, where users leverage LL

Ms to generate and iteratively refine code through natural language interactions until it passes their *vibe check*. *Vibe check* reflects human preference and goes beyond functionality: the solution should feel right, read cleanly, preserve intent, and remain correct. However, current code evaluation remains anchored to pass@k and captures only functional correctness, overlooking non-functional instructions that users routinely apply. In this paper, we hypothesize that instruction following is the missing piece underlying *vibe check* besides functional correctness. To quantify models' code instruction-following capabilities with measurable signals, we present VeriCode, a taxonomy of 30 verifiable code instructions together with deterministic verifiers. We use the taxonomy to augment established evaluation suites, resulting in **SWE-IF**, a testbed to assess both instruction following and functional correctness. Evaluating 31 LLMs, we show that even the strongest models struggle to comply with multiple instructions and exhibit functional regression. Most importantly, a composite score of functional correctness and instruction following correlates best with human preference, with instruction following emerging as the primary differentiator among LLMs. Our code, data, and taxonomy are available at https://github.com/maszhongming/SWE-IF.

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Expandable, Compressible, Mineable: Open-World Thermal Infrared Image Restoration

Pu Li, Huafeng Li, Yafei Zhang, Wen Wang, Neng Dong, Jie Wen

In open-world settings, thermal infrared (TIR) image degradations continuously emerge and evolve, while most existing all-in-one restoration methods are built on a closed-set assumption and struggle to continually adapt to novel degradations.

To address this, we propose ECMRNet, an Expandable, Compressible, and Mineable Restoration Network for open-world TIR restoration from a continual learning perspective.

Conceptually, ECMRNet unifies continual degradation learning as an "expand–compress–mine" closed-loop process, enabling sustained adaptation to new degradations with controllable evolution.

Structurally, ECMRNet decomposes intermediate representations into group-isolated subspaces, and achieves strict parameter isolation and fast adaptation to new degradations by freezing historical groups and isomorphically expanding new ones.

To curb model growth as tasks accumulate, we present Structural Entropy Pruning, which identifies and removes redundant channel groups via two-dimensional structural entropy minimization, achieving information contribution–driven adaptive compression.

Moreover, we design a Sub-degradation Knowledge Mining Module that dynamically retrieves and recombines transferable components from historical representations to improve restoration under compound degradations.

Experimental results demonstrate that ECMRNet achieves superior overall performance across diverse single and compound degradations while using fewer parameters and lower computational cost.

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HumanLM: Simulating Users with State Alignment Beats Response Imitation

Shirley Wu, Evelyn Choi, Arpandeep Khatua, Zhanghan Wang, Joy He-Yueya, Tharindu Cyril Weerasooriya, Wei Wei, Diyi Yang, Jure Leskovec, James Zou

Large Language Models (LLMs) are increasingly used to simulate how specific users respond to any context, enabling more user-centric applications that rely on user feedback. However, existing user simulators mostly imitate surface-level patterns and language styles, which fails to reflect the underlying state of real users (e.g., beliefs, emotions). To address these limitations, we propose a novel training framework, HumanLM, which builds user simulators that accurately reflect real users. Our key insight is, in addition to generating responses, we generate natural-language latent states that align with the ground truth responses through reinforcement learning. These latent states correspond to a set of state dimensions which psychologically lead to how real users respond. HumanLM further synthesizes these aligned latent states into responses that accurately represent

real users. For extensive evaluation, we develop Humanual, a comprehensive benchmark on simulating real users based on public data. Humanual consists of six large-scale datasets with 26k users and 216k responses in total. It spans diverse tasks such as generating user responses to daily life issues, political blogs, and chat sessions with LLM assistants. Across the datasets, HumanLM significantly outperforms the best alternative approaches by an average relative improvement of 16.3% on alignment score from an LLM judge. In a real-time simulation study with 37 participants, HumanLM achieves the highest scores on similarity with real user responses and human-likeness.

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QPoint: End-to-End Lightweight Point Cloud Processing via Robust Quaternion Feature Learning

Zhiming Zhou, Yong He, Chaoxu Mu, Qiaoyun Wu, Ajmal Saeed Mian

The inherent sparsity, lack of structure, and rotation sensitivity of point clouds often lead to high computational and parameter cost in robust feature learning. To address these problems, we present QPoint, a lightweight framework that leverages robust quaternion feature learning. QPoint incorporates a Quaternion-Enhanced local perception module that uses learnable rotations to stabilize local features against geometric transformations, and a Quaternion global attention mechanism that employs quaternion similarity to capture global geometric context with inherent rotation invariance. Extensive experiments show that QPoint achieves top performance across multiple tasks. It achieves excellent 95.0%, 93.9%, and 92.1% on the challenging ScanObjectNN variants (OBJ_BG, OBJ_ONLY, PB_T50_RS), 94.7% overall accuracy on ModelNet40, and 87.0% instance mIoU on ShapeNetParts. Furthermore, QPoint exhibits superior generalization in few-shot learning scenarios. Crucially, it accomplishes this with extremely minimal parameter and computational requirements, establishing a strong and efficient baseline for point cloud processing. Our source code is in the supplementary material and will be released to the public.

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LIF Recurrent Memory Enables Long-Horizon Spiking Computation

Fenghao Liu, Yipeng Shen, Peng Chen, Qian Zheng, Peng Lin, Gang Pan

Processing long sequence data such as speech requires models to maintain long-term dependencies, which is challenging for recurrent spiking neural networks due to high temporal dynamics in neuron models that leak stored information in their membrane potentials, and due to vanishing gradients during backpropagation through time. These issues can be mitigated by employing more complex neuron designs, such as ALIF and TC-LIF, but these neuron-level solutions often incur high computational costs and complicate hardware implementation, undermining the efficiency advantages of spiking neural networks. Here we propose an architectural-level solution that leverages the dynamical interactions of a few leaky integrate-and-fire (LIF) neurons to enhance long-term information storage. The memory capability of this LIF-based micro-circuit is adaptively modulated by global recurrent connections of the recurrent spiking neural network, contributing to selective enhancement of temporal information retention, and promoting stable gradient propagation through time. The proposed model outperforms baselines including LSTM, ALIF, and TC-LIF in long sequence tasks, achieving 96.52\% accuracy on the PS-MNIST dataset. Furthermore, our method also provides a compelling efficiency advantage, yielding up to 277$\times$ computational efficiency improvement compared to conventional models such as LSTM. This work paves the way for building cost-effective, hardware-friendly, and interpretable spiking neural networks for long sequence modeling.

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Graph Alignment via Dual-Pass Spectral Encoding and Latent Space Communication

Maysam Behmanesh, Erkan Turan, Maks Ovsjanikov

Graph alignment, the problem of identifying corresponding nodes across multiple graphs, is fundamental to numerous applications. Most existing unsupervised methods embed node features into latent representations to enable cross-graph comparison without ground-truth correspondences. However, these methods suffer from two critical limitations: the degradation of node distinctiveness due to oversmoot

hing in GNN-based embeddings, and the misalignment of latent spaces across graph
s caused by structural noise, feature heterogeneity, and training instability, u
ltimately leading to unreliable node correspondences. We propose a novel framewo
rk employing a dual-pass encoder to inject high-frequency discriminability into
node features, paired with a geometry-aware functional map module that learns bi
jective and isometric transformations to align latent spaces while acting as a l
ow-pass filter on correspondences, enforcing smoothness and robustness as a stru
ctural prior in map space. Extensive experiments on graph benchmarks demonstrate
 that our method consistently outperforms existing unsupervised alignment baseli
nes, exhibiting superior robustness to structural inconsistencies and challengin
g alignment scenarios.
The implementation is available at https://github.com/maysambehmanesh/GADL.
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Neural Low-Discrepancy Sequences
Michael Etienne Van Huffel, Nathan Kirk, Makram Chahine, Daniela Rus, T. Konstan
tin Rusch
Low-discrepancy points are designed to efficiently fill the space in a uniform m
anner. This uniformity is highly advantageous in many problems in science and en
gineering, including in numerical integration, computer vision, machine percepti
on, computer graphics, machine learning, and simulation. Whereas most previous l
ow-discrepancy constructions rely on abstract algebra and number theory, Message
-Passing Monte Carlo (MPMC) was recently introduced to exploit machine learning
methods for generating point sets with lower discrepancy than previously possibl
e. However, MPMC is limited to generating point sets and cannot be extended to l
ow-discrepancy sequences (LDS), i.e., sequences of points in which every prefix
has low discrepancy, a property essential for many applications. To address this
 limitation, we introduce Neural Low-Discrepancy Sequences (*NeuroLDS*), the fir
st machine learning-based framework for generating finite LDS. Drawing inspirati
on from classical LDS, we train a neural network to map indices to points such t
hat the resulting sequences exhibit minimal discrepancy across all prefixes. To
this end, we deploy a two-stage learning process: supervised approximation of cl
assical constructions followed by unsupervised fine-tuning to minimize prefix di
screpancies. We demonstrate that *NeuroLDS* outperforms all previous LDS constru
ctions by a significant margin with respect to discrepancy measures. Moreover, w
e demonstrate the effectiveness of *NeuroLDS* across diverse applications, inclu
ding numerical integration, robot motion planning, and scientific machine learni
ng. These results highlight the promise and broad significance of Neural Low-Dis
crepancy Sequences.
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Physics from Video: Identifiability of Time-Invariant Second-Order ODEs under Mi
nimal Trajectory Conditions
Yuanyuan Wang, Wenjie Wang, Kun Zhang, Mingming Gong
Bridging the gap between visual realism and physical understanding is a core cha
llenge for video-based world models. We study the structural identifiability of
continuous-time physical laws from raw pixels, focusing on whether an encoder-on
ly pipeline can uniquely recover the parameters of second-order linear ODEs. We
prove that a level-set slope-coverage condition ensures the learned latent space
 is locally affine to the true physical state, enabling exact parameter recovery
. Our theory provides the first characterization of minimal data requirements ac
ross damping regimes, establishing that underdamped systems are identifiable fro
m a single video clip, whereas other regimes require three diverse trajectories.
 We further introduce a variance-floor regularizer to stabilize the decoder-free
 objective and prevent latent collapse. Validated on synthetic and real-world da
ta, our approach demonstrates that interpretable physical constants can be relia
bly estimated from video without the need for compute-intensive pixel reconstruc
tion, ensuring both physical correctness and transparency. Code is available at
https://github.com/wenjiewang3/PhysicsFromVideo.
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On the Sharp Input-Output Analysis of Nonlinear Systems under Adversarial Attack
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Jihun Kim, Yuchen Fang, Javad Lavaei
This paper is concerned with learning the input-output mapping of general nonlinear dynamical systems. While the existing literature focuses on Gaussian inputs and benign disturbances, we significantly broaden the scope of admissible control inputs and allow correlated, nonzero-mean, adversarial disturbances. With our reformulation as a linear combination of basis functions, we prove that the ℓ_2-norm estimator overcomes the challenges posed by an adversary with access to the full information history, provided that the attack times are sparse, *i.e.*, the probability that the system is under adversarial attack at a given time is smaller than a certain threshold. We provide an estimation error bound that decays with the input memory length and prove its optimality by constructing a problem instance that suffers from the same bound under probabilistic adversarial attacks. Our work provides a sharp input-output analysis for a generic nonlinear and partially observed system under significantly generalized assumptions compared to existing works.
****
Tail Annealing for Heavy-Tailed Flow Matching
Jean Pachebat
Standard generative models struggle with heavy-tailed data: Lipschitz architectures cannot produce power-law tails from Gaussian noise, and interpolating between heavy-tailed data and Gaussians is ill-posed. We propose a simple fix: apply the soft-log transform $\phi(x) = \mathrm{sign}(x) \cdot \log(1 + |x|)$ coordinate-wise to data before training, then exponentiate samples after generation. A Hill diagnostic decides per-coordinate whether to transform, leaving light-tailed margins untouched at no added complexity. This compresses heavy tails into a range where standard flow matching succeeds, without heavy-tailed base distributions or architectural modifications. We provide theoretical intuition for why this works: the log-transform maps Pareto tails to exponentials, and the induced dynamics implement a form of tail annealing via power transformations. On a 144-configuration multivariate benchmark (3 copulas, d up to 100, 4 tail indices), Log-FM dominates specialized baselines on W_1, CVaR$_{99}$, and extreme-quantile metrics, and is the only method with zero severe divergences across 2{,}880 runs.
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Overcoming the Incentive Collapse Paradox
Qichuan Yin, Ziwei Su, Shuangning Li
AI-assisted task delegation is increasingly common, yet human effort in such systems is costly and typically unobserved.
Recent work by Bastani & Cachon (2025); Sambasivan et al. (2021) shows that accuracy-based payment schemes suffer from incentive collapse: as AI accuracy improves, sustaining positive human effort requires unbounded payments.
We study this phenomenon in a budget-constrained principal-agent framework with strategic human agents whose output accuracy depends on unobserved effort.
Our first contribution is a general impossibility result showing that incentive collapse is not merely a limitation of simple linear payments, but arises for any payment rule based only on observed task accuracy.
To overcome this barrier, we propose a sentinel-auditing payment mechanism that enforces a strictly positive and controllable level of human effort at finite cost, independent of AI accuracy.
Building on this incentive-robust foundation, we develop an incentive-aware active statistical inference framework that jointly optimizes (i) the auditing rate and (ii) active sampling and budget allocation across tasks of varying difficulty to minimize the final statistical loss under a single budget. Experiments demonstrate improved cost-error tradeoffs relative to standard active learning and auditing-only baselines.
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Learning, Solving and Optimizing PDEs with TensorGalerkin: an efficient high-performance Galerkin assembly algorithm
Shizheng Wen, Mingyuan chi, Tianwei Yu, Ben Moseley, Mike Yan Michelis, Pu Ren, Hao Sun, Siddhartha Mishra

We present a unified algorithmic framework for the numerical solution, constrained optimization, and physics-informed learning of PDEs with a variational structure. Our framework is based on a Galerkin discretization of the underlying variational forms, and its high efficiency stems from a novel highly-optimized and GPU-compliant TensorGalerkin framework for linear system assembly (stiffness matrices and load vectors). TensorGalerkin operates by tensorizing element-wise operations within a Python-level Map stage and then performs global reduction with a sparse matrix multiplication that performs message passing on the mesh-induced sparsity graph. The Map and Reduce stages are co-designed inside PyTorch's autograd so that the assembly graph contains $O(1)$ nodes regardless of how the number of elements and local DoFs scale. We validate this $O(1)$-graph property by deploying TensorGalerkin downstream as i) a highly-efficient numerical PDEs solver, ii) an end-to-end differentiable framework for PDE-constrained optimization, and iii) a physics-informed operator learning algorithm for PDEs. With multiple benchmarks, including 2D and 3D elliptic, parabolic, and hyperbolic PDEs on unstructured meshes, we demonstrate that the proposed framework provides significant computational efficiency and accuracy gains over a variety of baselines in all the targeted downstream applications.
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Random Selection Reveals Implicit Knowledge Consensus in Code Generation
Ren-Biao Liu, Xin-Ye Li, Hui Sun, Yali Du, Jiang-Tian Xue, Ming Li
Training large language models for code generation often involves selecting data from verifiable multi-solution pools, where each problem admits multiple correct implementations. Conventional studies on data selection suggest that complex selection strategies, such as diversity maximization or difficulty ranking, should outperform naive random sampling. In this work, we systematically evaluate within-problem solution selection strategies across different representation spaces, including continuous embeddings, discrete tokens, and syntactic structures, using various base language models. Instead, simple random sampling achieves consistently competitive performance across all models, exhibiting greater cross-model stability than complex methods. We interpret these results through the lens of *implicit knowledge consensus*: verified solution pools may contain representative algorithmic patterns that random sampling can preserve. Our findings suggest that practitioners should treat random sampling as a low-cost default for verifiable code-generation fine-tuning and move to complex selectors when hard-tail or constraint-focused coverage is the target.
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Optimal Top-k Identification from Pairwise Comparisons
Motti Goldberger, Nils Rudi
We study the active learning problem of fixed-confidence top-k identification from noisy pairwise comparisons. In this problem, an algorithm sequentially chooses pairs of items to compare, observes the outcomes, and stops when it can return the set of top-k items with error probability at most δ. The objective is to design such a *δ-correct* procedure that minimizes the expected number of comparisons (the sample complexity).
This problem falls within the broader literature on fixed-confidence pure exploration in bandit models, where a common target is asymptotic optimality: the algorithm's expected sample complexity matches the information theoretic lower bound as $\delta \to 0$. Asymptotically optimal procedures have been developed for a range of fixed-confidence pure-exploration problems, however to the best of our knowledge, for top-1, or more generally top-k identification from pairwise comparisons under latent utility models an asymptotically optimal algorithm has not been established. In this setting, we develop such an algorithm. We characterize the structure of the lower bound and formulate it as a saddle-point problem. This structure enables a computationally efficient primal--dual procedure that learns the asymptotically optimal comparison allocation online. We then construct an adaptive comparison-allocation algorithm that tracks the allocation learned by the primal--dual procedure and prove it is asymptotically optimal.
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Parameter-free Dynamic Regret: Time-varying Movement Costs, Delayed Feedback, an

d Memory
Hao Qiu, Andrew Jacobsen, Emmanuel Esposito, Mengxiao Zhang
In this paper, we study dynamic regret in unconstrained online convex optimizati on (OCO) with movement costs. Specifically, we generalize the standard setting by allowing the movement cost coefficients λ_t to vary arbitrarily over time. Our main contribution is a novel algorithm that establishes the first com parator-adaptive dynamic regret bound for this setting, guaranteeing $\widetilde {\mathcal{O}}(\sqrt{(M^2+MP_T)(T+\sum_t \lambda_t)})$ regret, where P_T is the path length of the comparator sequence over T rounds and M is the maximal c omparator norm. Our result recovers the optimal adaptive rates for both static a nd dynamic regret in OCO as the special case where $\lambda_t=0$ for all rounds . To demonstrate the versatility of our results, we consider two applications: * OCO with delayed feedback* and *OCO with time-varying memory*. We show that both problems can be translated into time-varying movement costs, establishing a nov el reduction specifically for the delayed feedback setting that is of independen t interest. A crucial observation is that the first-order dependence on movemen t costs in our regret bound plays a key role in enabling optimal comparator-adap tive dynamic regret guarantees in both settings.
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ScaleErasure: Inference-Time Minimal Intervention for Precise Concept Erasure in Next-Scale Autoregressive Image Generation
Cong Wang, Haiyu Wu, Zhiwei Jiang, Zifeng Cheng, Fei Shen, Yafeng Yin, Qing Gu
Concept erasure aims to prevent image generative models from producing unsafe co ntent while preserving their general generative capability.
Meanwhile, next-scale autoregressive (AR) image generation has recently emerged as a new generative paradigm characterized by next-scale prediction, for which c oncept erasure remains largely unexplored.
In this paradigm, semantic information is highly compressed at early scales, lea ding to severe entanglement between unsafe and unrelated semantics.
In this paper, we propose ScaleErasure, an inference-time concept erasure method that performs minimal intervention.
ScaleErasure precisely selects and guides predicted logits that are most relevan t to the unsafe concept, thereby enabling effective erasure under severe semanti c entanglement.
Specifically, ScaleErasure performs two additional forward passes conditioned on the unsafe concept and the corresponding safe concept, and leverages their outp uts to guide the target logits away from unsafe concepts toward safe concepts.
To enable precise and minimal intervention, logits selection and guidance are co nducted across three dimensions: scales, tokens, and bit channels.
Experiments demonstrate that ScaleErasure outperforms adapted baselines in the n ext-scale AR paradigm, achieving more precise concept erasure while largely pres erving general generative capability.
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SynLaD: Latent Diffusion for Generating Synthesizable Molecules Conditioned on 3 D Pharmacophore Profiles
Miruna Cretu, John Bradshaw, Patricia Suriana, Saeed Saremi, Omar Mahmood, Kiril l Shmilovich, Kangway V. Chuang, Vishnu Sresht, Colin A Grambow
We present SynLaD, a latent diffusion framework for small-molecule generation th at unifies ligand-based drug design objectives (what to make) with synthetic acc essibility (how to make it). Current models typically optimize one objective at the expense of the other, creating a bottleneck for discovering high-scoring and synthesizable molecules. SynLaD combines reaction-constrained generation with p harmacophore-conditioned 3D design by learning a latent space that decodes to bo th 3D structures and synthesis pathways. An encoder maps molecules to a latent r epresentation used by two decoder heads: (i) a geometric head that reconstructs atom types and coordinates and (ii) an autoregressive synthesis head that output s synthetic routes in a serialized, reaction-based notation. A diffusion transfo rmer generates novel latents in the learned space, conditioned on pharmacophore profiles. Across analogue generation tasks for bioactive ligands, SynLaD outperf orms existing baselines in synthesizable and diverse hit generation, demonstrati

ng that a single model can produce shape-aligned molecules with feasible synthes is plans.
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Power-Boosted Granger-Causal Discovery for Large Heterogeneous Panel Data
Yiheng Gu, Xiufan Yu
This paper proposes a power-enhanced panel Granger causality test (PE-PGCT) for assessing the Granger non-causality in heterogeneous and potentially high-dimens ional panel data. Building on any existing panel Granger non-causality test, we show, both theoretically and empirically, that the proposed PE-PGCT boosts its p ower substantially. The power gains are particularly significant in situations o f high-dimensional panels when the cross-sectional dimension exceeds the time di mension, as well as under sparse alternatives when the signals are sparsely dist ributed across panel units. We establish rigorous theoretical guarantees on the asymptotic behavior of the proposed power enhancement component, demonstrating a ttractive power enhancement properties that it induces negligible size distortio n under the null hypothesis while delivering significant power gain under the al ternatives. The empirical performances are illustrated via extensive simulation studies, as well as a real-world application.
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Ariadne's Thread of LipSync: Unraveling Forgeries via Inconsistency between Lip Motions and Head Poses
Tianyi She, Jiawei Liu, Weifeng Liu, Hanqing Zhao, Weiming Zhang, Kejiang Chen
Recent advances in LipSync generation technology have led to the creation of hig hly realistic videos, posing severe societal risks. However, existing defense st rategies struggle against LipSync forgeries, as state-of-the-art generative mode ls not only optimize for the lip synchronization but also significantly eliminat e visual artifacts, resulting in the lack of key detection signals. Inspired by the inherent biological coupling between lip movements and head poses in natural speech, we observe that generative models fundamentally disrupt this global coo rdination when optimizing for local lip motion. In this paper, we propose LipDA, a novel framework for joint LipSync Detection and Attribution, which takes adva ntage of the inconsistency between head and lip. For detection, the framework le arns to quantify this discrepancy by contrasting lip and pose features from auth entic versus forged videos. For attribution, our method is designed to capture t he unique temporal dynamics and audio-visual synchronization patterns that act a s generative fingerprints, enabling source tracing. To validate our approach, we conduct extensive experiments on two challenging LipSync benchmarks as well as on our own proposed large-scale and multi-generator dataset, LipSyncBench-A. Lip DA achieves over 97% AUC in detection and 97.5% accuracy in model attribution, s ignificantly outperforming existing methods.
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TerraBind: Fast and Accurate Binding Affinity Prediction through Coarse Structur al Representations
Matteo Rossi, Ryan Pederson, Miles Wang-Henderson, Ben Kaufman, Edward C. Willia ms, Carl Underkoffler, Owen Lewis Howell, Adrian Layer, Stephan Thaler, Narbe Ma rdirossian, John Anthony Parkhill
We present TerraBind, a foundation model for protein-ligand structure and bindin g affinity prediction that achieves 26$\times$ faster inference than state-of-th e-art methods while improving affinity prediction accuracy by up to 20\%. Curren t deep learning approaches to structure-based drug design rely on expensive all-atom diffusion to generate 3D coordinates, creating inference bottlenecks that r ender large-scale compound screening computationally intractable. We challenge t his paradigm with the hypothesis: full all-atom resolution is unnecessary for ac curate small molecule pose and binding affinity prediction. TerraBind tests this hypothesis through a coarse pocket-level representation (protein C$_\beta$ atom s and ligand heavy atoms only) within a multimodal architecture combining pretra ined molecular encoders and ESM-2 protein embeddings that learns rich structural representations, which are used in a diffusion-free optimization module for pos e generation and a binding affinity likelihood prediction module. On structure p rediction benchmarks, TerraBind matches diffusion-based baselines in ligand pose

accuracy. For binding affinity, TerraBind outperforms Boltz-2 by 16-20\% in Pea
rson correlation on both a public benchmark (CASP16) and a diverse private datas
et (18 assays). The affinity module also provides well-calibrated uncertainty es
timates, addressing a critical gap in compound prioritization for drug discovery
. Furthermore, this module enables a continual learning framework and a hedged b
atch selection strategy that, in simulated drug discovery cycles, achieves 6$\ti
mes$ greater affinity improvement over greedy approaches.
****
Toward Subspace-Perturbed Trajectory-Aware Backdoor Attacks in Deep Reinforcemen
t Learning
Yaguan Qian, Taining Zhang, Qiqi Bao, Yanru Guo, Lufang Zhang, Zhaoquan Gu, Shou
ling Ji, Bin Wang, Zhen Lei
Deep Reinforcement Learning agents are in-
creasingly used in safety-critical domains but
remain vulnerable to stealthy backdoor attacks.
Existing outer-loop attacks face a trade-off be-
tween perceptual stealth, poisoning efficiency, and
value-function consistency, often making the at-
tack ineffective or easily exposed. To address
these challenges, we propose SpecDRL, a uni-
fied framework that ❶ embeds triggers in the
least sensitive subspaces of the state manifold
via Subspace-Aware Injection, exploiting percep-
tual blind spots, ❷ selects the most influential
time steps for poisoning through Value-Guided
Strategic Sampling based on Return-to-Go and
Temporal-Difference error, and ❸ preserves re-
ward integrity via Bellman-Consistent Dynamic
Reward Poisoning, which analytically enforces ■-
consistency of value functions and bounds global
return deviations. Experiments across 12 Atari en-
vironments demonstrate that SpecDRL achieves
near-100% attack success, accelerates backdoor
convergence, and maintains benign task perfor-
mance.
****
Statistical Consistency and Generalization of Contrastive Representation Learnin
g
Yuanfan Li, Xiyuan Wei, Tianbao Yang, Yiming Ying
Contrastive representation learning (CRL) underpins many modern foundation model
s.
Despite recent theoretical progress, existing analyses suffer from several key
limitations: (i) the statistical consistency of CRL remains poorly understood;
(ii) available generalization bounds deteriorate as the number of negative sampl
es
increases, contradicting the empirical benefits of large negative sets; and (iii
)
the retrieval performance of CRL has received limited theoretical attention.
In this paper, we develop a unified statistical learning theory for CRL. For
downstream tasks, we evaluate retrieval quality using an AUC-type population
criterion and show that the contrastive loss is *statistically consistent*
with optimal ranking. We further establish a *calibration-style inequality*
that quantitatively relates excess contrastive risk to excess retrieval
suboptimality. For upstream training, we study both supervised and self-supervis
ed
contrastive objectives and derive generalization bounds of order
$O(1/m + 1/\sqrt{n})$ and $O(1/\sqrt{m} + 1/\sqrt{n})$, respectively, where m
denotes the number of negative samples and n the number of anchor points. Thes
e
bounds not only explain the empirical advantages of large negative sets but also

reveal an explicit trade-off between m and n. Extensive experiments on large-scale vision--language models corroborate our theoretical predictions.
****
Physics-Informed Residual Flows
Jephte Abijuru, Mayank Nagda, Phil Ostheimer, Sebastian Josef Vollmer, Marius Kloft, Sophie Fellenz
Physics-Informed Neural Networks (PINNs) embed physical laws into deep learning models. However, conventional PINNs often suffer from failure modes leading to inaccurate solutions. We trace these failure modes to two structural pathologies: gradient shattering, where gradients degrade with depth and provide little training signal, and flow mismatch, where training pushes predictions along trajectories that diverge from the PDE solution path. We introduce ResPINNs, which reformulate PINNs as residual flows, networks that iteratively refine their own predictions through explicit corrective steps, in the spirit of classical iterative solvers. Our analysis shows that this design mitigates both pathologies by keeping updates aligned with descent and by preserving informative gradients across depth. Extensive experiments on PDE benchmarks confirm that ResPINNs achieve higher accuracy with substantially fewer parameters than conventional architectures.
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EntroKV: Entropy-Guided Dynamic Budget Allocation for KV-Cache Compression
Wenhao Gao, Haoran Cao, Yueyan Li, YongGao Xiao, Caixia Yuan, Xiaojie Wang
The prohibitive memory footprint of the Key-Value (KV) cache imposes a critical bottleneck for efficient long-context LLM serving.
Current compression techniques typically rely on static or uniform budget allocation, overlooking the significant heterogeneity in information density across attention heads.
To address this, we introduce \textsc{EntroKV}, an entropy-driven dynamic budget allocation framework.
Our method enables dynamic and rational allocation across layers, attention heads, and different tasks.
We demonstrate that attention entropy serves as a robust proxy for compression sensitivity: heads with high entropy require larger retention budgets, whereas low-entropy heads can be aggressively compressed without accuracy degradation.
Functioning as a lightweight, plug-and-play module, \textsc{EntroKV} optimizes budget scheduling in real-time and is compatible with diverse compression operators.
Extensive experiments demonstrate that \textsc{EntroKV} consistently outperforms baselines, retaining $\sim$98\% of full-cache performance at a 30\% budget ratio with negligible computational overhead.
Our code is available at \url{https://anonymous.4open.science/r/EntroKV-D0C8/}.
****
Deep Incentive Design with Differentiable Equilibrium Blocks
Vinzenz Thoma, Georgios Piliouras, Luke Marris
Automated design of multi-agent interactions with desirable equilibrium outcomes is inherently difficult due to the computational hardness, non-uniqueness, and instability of the resulting equilibria.
In this work, we propose the use of game-agnostic _differentiable equilibrium blocks_ (DEBs) as modules in a novel, differentiable framework to address a wide variety of incentive design problems from economics and computer science. We call this framework _deep incentive design_ (DID).
To validate our approach, we examine three diverse, challenging incentive design tasks: contract design, machine scheduling, and inverse equilibrium problems.
For each task, we train a single neural network using a unified pipeline and DEB. This architecture solves the _full distribution_ of problem instances, parameterized by a context, handling _all_ games across a wide range of scales (from two to sixteen actions per player).
****
DyGRO-VLA: Cross-Task Scaling of Vision–Language–Action Models via Dynamic Grouped Residual Optimization
Sixu Lin, Yunpeng Qing, Litao Liu, Ming Zhou, Ruixing Jin, Xiaoyi Fan, Guiliang

Liu

Recent progress in Reinforcement Learning (RL) provides a principled approach to optimizing Vision-Language-Action (VLA) models, facilitating a shift from trajectory imitation to active learning in the task environment. Despite improvements in control precision, most RL optimizers remain task-specific, which reduces VLA models from generalist controllers to policies that overfit to a narrow set of tasks. In this study, we conduct an in-depth analysis of this phenomenon and highlight the importance of cross-task feature representations for improving the generalizability of VLA models. Motivated by this finding, we introduce DyGRO-VLA, a two-stage optimization framework that 1) effectively captures cross-task latent representations based on information-theoretic principles, and 2) dynamically refines policy optimization via a mixture-of-RL-residuals. DyGRO-VLA enables the RL optimizer to exploit task-relevant latent information while strategically mitigating adverse interference on the learned representations throughout the optimization process. We evaluate our approach on LIBERO, RoboTwin2 benchmarks, and further validate it on real world, demonstrating consistent improvements over strong baselines under multi-task training and distribution shift. Our project page is available at DyGRO-VLA.

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Open Materials Generation with Inference-Time Reinforcement Learning

Philipp Höllmer, Stefano Martiniani

Continuous-time generative models for crystalline materials enable inverse materials design by learning to predict stable crystal structures, but incorporating explicit target properties into the generative process remains challenging. Policy-gradient reinforcement learning (RL) provides a principled mechanism for aligning generative models with downstream objectives but typically requires access to the score, which has prevented its application to flow-based models that learn only velocity fields. We introduce Open Materials Generation with Inference-time Reinforcement Learning (OMatG-IRL), a policy-gradient RL framework that operates directly on the learned velocity fields and eliminates the need for the explicit computation of the score. OMatG-IRL leverages stochastic perturbations of the underlying generation dynamics preserving the baseline performance of the pretrained generative model while enabling exploration and policy-gradient estimation at inference time. Using OMatG-IRL, we present the first application of RL to crystal structure prediction (CSP). Our method enables effective reinforcement of an energy-based objective while preserving diversity through composition conditioning, and it achieves performance competitive with score-based RL approaches. Finally, we show that OMatG-IRL can learn time-dependent velocity-annealing schedules, enabling accurate CSP with order-of-magnitude improvements in sampling efficiency and, correspondingly, reduction in generation time. The OMatG-IRL code is included in a new release of the Open Materials Generation (OMatG) framework available at https://github.com/FERMat-ML/OMatG.

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Joint Learning in the Gaussian Single Index Model

Loucas Pillaud-Vivien, Adrien Schertzer

We consider the problem of jointly learning a one-dimensional projection and a univariate function in high-dimensional Gaussian models. Specifically, we study predictors of the form $f(x)=\varphi^\star(\langle w^\star, x \rangle)$, where both the direction $w^\star \in \mathcal{S}_{d-1}$, the sphere of $\mathbb{R}^d$, and the function $\varphi^\star: \mathbb{R} \to \mathbb{R}$ are learned from Gaussian data. This setting captures a fundamental non-convex problem at the intersection of representation learning and nonlinear regression. We analyze the gradient flow dynamics of a natural alternating scheme and prove convergence, with a rate controlled by the information exponent reflecting the *Gaussian regularity* of the function $\varphi^\star$. Strikingly, our analysis shows that convergence still occurs even when the initial direction is negatively correlated with the target. On the practical side, we demonstrate that such joint learning can be effectively implemented using a Reproducing Kernel Hilbert Space (RKHS) adapted to the structure of the problem, enabling efficient and flexible estimation of the univariate function. Our results offer both theoretical insight and practical

methodology for learning low-dimensional structure in high-dimensional settings.

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Order Matters in Retrosynthesis: Structure-aware Generation via Reaction-Center-Guided Discrete Flow Matching

Chenguang Wang, Zihan Zhou, LEI BAI, Tianshu Yu

Template-free retrosynthesis methods treat the task as black-box sequence generation, limiting learning efficiency, while semi-template approaches rely on rigid reaction libraries that constrain generalization. We address this gap with a key insight: atom ordering in neural representations matters.

Building on this insight, we propose a structure-aware template-free framework that encodes the two-stage nature of chemical reactions as a positional inductive bias. By placing reaction center atoms at the sequence head, our method transforms implicit chemical knowledge into explicit positional patterns that the model can readily capture. The proposed RetroDiT backbone, a graph transformer with rotary position embeddings, exploits this ordering to prioritize chemically critical regions. Combined with discrete flow matching, our approach decouples training from sampling and enables generation in 20--50 steps versus 500 for prior diffusion methods.

Our method achieves state-of-the-art performance on both USPTO-50k (61.2\% top-1) and the large-scale USPTO-Full (51.3\% top-1) with predicted reaction centers. With oracle centers, performance reaches 71.1\% and 63.4\% respectively, surpassing foundation models trained on 10 billion reactions while using orders of magnitude less data. Ablation studies further reveal that structural priors outperform brute-force scaling: a 280K-parameter model with proper ordering matches a 65M-parameter model without it.

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Rethinking 1-bit Optimization Leveraging Pre-trained Large Language Models

Zhijun Tu, Jian Li, Yuanyuan Xi, Siqi Liu, Chuanjian Liu, Hanting Chen, Jie Hu, Yunhe Wang

1-bit LLM quantization offers significant advantages in reducing storage and computational costs. However, existing methods typically train 1-bit LLMs from scratch, failing to fully leverage pre-trained models. This results in high training costs and notable accuracy degradation. We identify that the large gap between full precision and 1-bit representations makes naive adaptation difficult. In this paper, we introduce a consistent progressive training for both forward and backward, smoothly converting the full-precision weights into the binarized ones. Additionally, we incorporate binary-aware initialization and dual-scaling compensation to reduce the difficulty of progressive training and improve the performance. Experimental results on LLMs of various sizes demonstrate that our method outperforms existing approaches. Our results show that high-performance 1-bit LLMs can be achieved using pre-trained models, eliminating the need for expensive training from scratch.

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FullStack-Agent: Enhancing Agentic Full-Stack Web Coding via Development-Oriented Testing and Repository Back-Translation

Zimu Lu, Houxing Ren, Yunqiao Yang, Ke Wang, Zhuofan Zong, Mingjie Zhan, Hongsheng Li

Assisting non-expert users to develop complex interactive websites has become a popular task for LLM-powered code agents.

However, existing code agents tend to only generate frontend web pages, masking the lack of real full-stack data processing and storage with fancy visual effects.

Notably, constructing production-level full-stack web applications is far more challenging than only generating frontend web pages, demanding careful control of data flow, comprehensive understanding of constantly updating packages and dependencies, and accurate localization of obscure bugs in the codebase.

To address these difficulties, we introduce FullStack-Agent, a unified agent system for full-stack agentic coding that consists of three parts:

(1) FullStack-Dev, a multi-agent framework with strong planning, code editing, c

odebase navigation, and bug localization abilities.
(2) FullStack-Learn, an innovative data-scaling and self-improving method that back-translates crawled and synthesized website repositories to improve the backbone LLM of FullStack-Dev.
(3) FullStack-Bench, a comprehensive benchmark that systematically tests the frontend, backend and database functionalities of the generated website.
Our FullStack-Dev outperforms the previous state-of-the-art method by 8.7\%, 38.2\%, and 15.9\% on the frontend, backend, and database test cases respectively. Additionally, FullStack-Learn raises the performance of a 30B model by 9.7\%, 9.5\%, and 2.8\% on the three sets of test cases through self-improvement, demonstrating the effectiveness of our approach.
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DIYHealth Suite: Dataset, Model, and Benchmark for Health Management at Home
Changshuo Liu, Wu Junran, Zhongle Xie, Wenqiao Zhang, Kaiping Zheng, Jiaqi Zhu, Qingpeng Cai, Ooi Gene Anne, Marcus Chun Jin Tan, Jianwei Yin, James Wei Luen Yip, Beng Chin Ooi
Generative AI is reshaping healthcare, yet most existing advances rely on hospital-grade devices, which limits their accessibility and potential for health management outside clinical settings. With the proliferation of portable devices and telemedicine, healthcare is shifting toward home-based Diagnosis-It-Yourself (DIY) care. Despite this promise, several distinctive challenges remain: (i) home-collected data are heterogeneous, exacerbated by the absence of standardized large-scale datasets; (ii) models require adaptation to variable task demands and evolving individual conditions; (iii) the broad spectrum of home care tasks lacks a unified benchmark for systematic evaluation. In this paper, we present **DIYHealth Suite**, a comprehensive framework designed to address these challenges through a tailored dataset, model, and benchmark. We first curate **DIYHealth-900K**, a large-scale multimodal dataset capturing diverse real-world home care scenarios. Building on this, we propose **DIYHealthGPT**, an adaptive foundation model for home-based health management, powered by the novel Hybrid Hyper Low-Rank Adaptation technique. Finally, we establish **DIYHealthBench**, the first benchmark to evaluate foundation models on home care tasks. Extensive experiments demonstrate that DIYHealthGPT delivers state-of-the-art performance over both general-purpose and medical-specific baselines on 11 home care tasks in both open-QA and closed-QA settings, laying the groundwork for the next generation of personalized health management at home.
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TarGATE: Target-Aware Data Selection via Token-Attenuation Gates
Xiandi Luo, Shiwei Li, Haozhao Wang, Yihao Ouyang, Zhuoqi Hu, Yichen Li, Xiao Yang, Huning Liu, Ruixuan Li
Targeted instruction tuning requires selecting pertinent samples from massive mixed *candidate datasets* guided by a small *reference dataset* reflecting the desired capability, yet efficiently identifying high-quality data amidst noise remains challenging. To address this, we propose **TarGATE** (**Tar**get-aware **GATE**s, a simple yet effective data selection framework that leverages the model's inherent data understanding. TarGATE computes a token-level Information Retention Ratio (**IRR**) to scale the output of the feed-forward network, where the instance-level average IRR serves as a quantitative metric for data quality. To align gates' preferences with the target task, we employ a joint optimization strategy utilizing the reference set and a subset of candidate data, which encourages the gates to assign higher IRRs to reference-aligned data while suppressing low-quality samples. Extensive experiments across noisy and real-world scenarios demonstrate that TarGATE outperforms related baselines. Furthermore, TarGATE exhibits superior computational efficiency and strong cross-model transferability, enabling smaller selector to effectively curate high-quality fine-tuning data for larger foundation models. The code is available at [here](https://github.com/FGinfinite/TarGATE).
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SCOUT: Cyclic Causal Discovery Under Soft Interventions with Unknown Targets
Alpar Turkoglu, Muralikrishnna Guruswamy Sethuraman, Faramarz Fekri

Learning causal relationships between variables from data is a fundamental research area with many applications across disciplines. Most of the existing causal discovery algorithms rely on the assumptions that (i) the underlying system is acyclic, (ii) the exogenous noise variables are Gaussian, and (iii) that the intervention targets for the data generating experiments are known. While these assumptions simplify the analysis, they are violated in real-life systems. Most existing methods that address these issues either assume the underlying model is linear or are constrained to operate in limited interventional settings. To that end, we propose SCOUT, a novel causal discovery framework to learn nonlinear causal cyclic relationships from soft interventional data with unknown targets. Our main approach maximizes the data log-likelihood to recover the graph structure, using two normalizing-flow architectures—contractive residual flows and neural spline flows. By conducting experiments on synthetic and real-world data, we show that SCOUT outperforms state-of-the-art methods in both causal graph and unknown target recovery across various interventional and noise settings.
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Position Is All You Need: A Free Lunch Token Compression Strategy for MLLM-based Referring Expression Segmentation

Yuhan Liu, Yixiong Zou, Yuhua Li, Ruixuan Li

Referring Expression Segmentation (RES) aims to generate pixel-wise segmentation masks from complex and implicit textual queries. While recent advances in Multimodal Large Language Models (MLLMs) have substantially boosted RES performance, their prohibitive computational overhead remains a critical bottleneck, which, however, is rarely explored. To fill this gap, we first evaluate typical token compression methods on this task and observe a surprising performance degradation. In this paper, we aim to understand this phenomenon for a solution. By extensive experiments, we find that token compression for RES requires preserving the original position embeddings and local neighboring spatial structures, indicating that visual token position information is far more critical than in other tasks. Building on this insight, we ask: Can we design the token compression method purely based on the position information? Therefore, we propose PAYN, a plug-and-play, training-free token compression method that relies solely on position information. PAYN retains tokens that are adequately distributed in every local neighboring region while strictly preserving original positional indices, thereby maintaining spatial relational consistency. Experiments on multiple RES benchmarks demonstrate that our method outperforms existing token compression methods, verifying that position is indeed all you need for token compression in the MLLM-based RES task. Codes are avaliable at https://github.com/YuhanLiu231/PAYN.
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Beyond Two-Stage Training: Cooperative SFT and RL for LLM Reasoning

Liang Chen, Xueting Han, Li Shen, Jing Bai, Kam-Fai Wong

Supervised fine-tuning (SFT) and reinforcement learning with verifiable rewards (RLVR) are two widely used post-training paradigms for improving the reasoning ability of large language models (LLMs). Recent methods attempt to integrate SFT and RLVR in a single stage by reweighting or scheduling their objectives. However, such coupling can be counterproductive because supervised updates are not uniformly beneficial for reward optimization. To address this, we propose BRIDGE, a scalable framework in which SFT learns to supervise RL by selectively transferring knowledge that improves reward optimization. Specifically, BRIDGE alternates two updates at each meta-training step: a base-model update that fuses the SFT and RL gradients, and an update to a lightweight low-rank adapter (LoRA) that coordinates the two objectives by maximizing a cooperative-gain signal, defined as the reward of joint SFT–RL training over an RL-only baseline. Across five mathematical reasoning benchmarks, BRIDGE consistently outperforms two-stage cold start, naive mixing, and representative single-stage integration baselines, yielding over three points average absolute improvement and more stable training dynamics. We further show that BRIDGE extends to logical reasoning and generalizes out-of-distribution to code and science without additional training, while staying robust under noisy rewards.
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ElicitR: Unlocking Latent Reasoning in Dense Retrievers via Generative Regularization
Fengyu Cai, Iryna Gurevych, Heinz Koeppl
Reasoning-intensive retrieval is increasingly important for downstream applications, requiring more than lexical overlap or coarse semantic matching. While prior work mainly relies on Language Models (LMs) to synthesize reasoning-oriented supervision, we posit that it is already latent in LM-based retrievers but suppressed by contrastive overfitting. To elicit this latent reasoning, we introduce ElicitR, a retriever–LM framework with generative regularization that captures nuanced relationships among a query and its candidate documents beyond binary relevance. Concretely, alongside contrastive learning, we regularize the retriever by co-training a small LM on query–positive–negative batches. Next token prediction (NTP) for each text is conditioned on its prefix and the other in-batch texts, with cross-text conditioning weighted by retriever-computed similarities. Using MS MARCO as the only paired query-document supervision and a 135M LM for generative regularization with unlabeled raw-text initialization, ElicitR consistently improves BRIGHT by 16-29% relative across 0.1B–3B retriever scales while maintaining performance on BEIR. At 3B, ElicitR reaches an nDCG@10 of 23.1, substantially outperforming larger models trained with far more curated pairs and proprietary APIs. Further analyses show that ElicitR prevents overfitting, improves retrieval calibration, and remains robust to batch sizes, supporting its practicality.
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MulFCoder: Framework-conditioned Multi-agent for MLLM-based Multi-framework Front-end Code Generation
Jie Wu, Haoran Ma, Shisong Tang, Yulin Xu, Xiaoyu Kang, Jiechao Gao
Generating runnable front-end code from UI screenshots is a long-standing goal in automated software engineering.
Existing MLLM-based methods predominantly focused on HTML/CSS, leaving multi-framework generation for React/Vue/Angular underexplored.
Naively modifying prompts leads to substantial performance gaps across multi-framework and highly framework-specific error modes.
To address this, we propose MulFCoder, a framework-conditioned multi-agent method that explicitly encodes framework constraints to bring multi-framework differences into a decidable rule space.
MulFCoder orchestrates four agents:
Grounder constructs an ElementTable, ContentTable, and macro-layout regions from detected UI elements;
Planner builds a DOM-like hierarchical layout tree, produces a task schedule, and derives a framework-specific file Contract;
Writer generates structured file writes or patches within a restricted edit window;
Judger enforces lightweight, framework-conditioned constraints to accept or reject updates and trigger bounded repairs, preventing drift and deadlocks without expensive builds.
Experiments demonstrate that MulFCoder substantially improves compilation success rate and reduces framework-specific errors, with particularly pronounced gains on Angular.
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Delving into Muon and Beyond: Deep Analysis and Extensions
Xianbiao Qi, Marco Chen, Jiaquan Ye, Yelin He, Rong Xiao
The Muon optimizer has recently attracted considerable attention for its strong empirical performance and use of orthogonalized updates on matrix-shaped parameters, yet its underlying mechanisms and relationship to adaptive optimizers such as Adam remain insufficiently understood.
In this work, we aim to address these questions through a unified spectral perspective. Specifically, we view Muon as the $p = 0$ endpoint of a family of spectral transformations of the form $\boldsymbol{U} \boldsymbol{\Sigma}^{p} \boldsymbol{V}^{\top}$, and consider additional variants with $p = \frac{1}{2}$, $p = \frac{1}{4}$, and $p = 1$. These transformations are applied to

both first-moment updates, as in momentum SGD, and to root-mean-square (RMS) normalized gradient updates as in Adam. To enable efficient computation, we develop a coupled Newton iteration that avoids explicit singular value decomposition. Across controlled experiments, we find that RMS-normalized updates yield more stable optimization than first-moment updates. Moreover, while spectral compression provides strong stabilization benefits under first-moment updates, the Muon update (\(p = 0 \)) does not consistently outperform Adam. These results suggest that Muon is best understood as an effective form of spectral normalization, but not a universally superior optimization method. Our code is available at \url{https://github.com/Ocram7/BeyondMuon}.
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Logit-Attention Divergence: Mitigating Position Bias in Multi-Image Retrieval via Attention-Guided Calibration
Mingtao Xian, Yifeng Yang, Qinying Gu, Xinbing Wang, Nanyang Ye
Multimodal Large Language Models (MLLMs) have shown strong performance in multi-image cross-modal retrieval, yet suffer from severe position bias, where predictions are dominated by input order rather than semantic relevance.
Through empirical analysis, we identify a phenomenon termed Logit-Attention Divergence, in which output logits are heavily biased while internal attention maps remain well-aligned with relevant visual evidence.
This observation reveals a fundamental limitation of existing logit-level calibration methods such as PriDe.
Based on this insight, we propose a training-free, attention-guided debiasing framework that leverages intrinsic attention signals for instance-level correction at inference time, requiring only a minimal calibration set with negligible computational overhead.
Experiments on MS-COCO-based benchmarks show that our method substantially improves permutation invariance and achieves state-of-the-art performance, enhancing accuracy by over 40\% compared to baselines. Code is available at https://github.com/brightXian/LAD.
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High-Probability Convergence Guarantees of Decentralized SGD
Aleksandar Armacki, Ali H. Sayed
Convergence in high-probability (HP) has attracted increasing interest, due to implying exponentially decaying tail bounds and strong guarantees for individual runs of an algorithm. While many works study HP guarantees in centralized settings, much less is understood in the decentralized setup, where existing works require strong assumptions, like uniformly bounded gradients, or asymptotically vanishing noise. This results in a significant gap between the assumptions used to establish convergence in the HP and the mean-squared error (MSE) sense, and is also contrary to centralized settings, where it is known that $\mathtt{SGD}$ converges in HP under the same conditions on the cost function as needed for MSE convergence. Motivated by these observations, we study the HP convergence of Decentralized $\mathtt{SGD}$ ($\mathtt{DSGD}$) in the presence of light-tailed noise, providing several strong results. First, we show that $\mathtt{DSGD}$ converges in HP under the same conditions on the cost as in the MSE sense, removing the restrictive assumptions used in prior works. Second, our sharp analysis yields order-optimal rates for both non-convex and strongly convex costs. Third, we establish a linear speed-up in the number of users, leading to matching or strictly better transient times than those obtained from MSE results, further underlining the tightness of our analysis. To the best of our knowledge, this is the first work that shows $\mathtt{DSGD}$ achieves a linear speed-up in the HP sense. Our relaxed assumptions and sharp rates stem from several technical results of independent interest, including a result on the variance-reduction effect of decentralized methods in the HP sense, as well as a novel bound on the moment-generating function of strongly convex costs, of interest even in centralized settings. Numerical experiments validate our theory.
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Memory is Reconstructed, Not Retrieved: Graph Memory for LLM Agents
Shuo Ji, Yibo Li, Bryan Hooi

Despite recent progress, LLM agents still struggle with reasoning over long interaction histories. While current memory-augmented agents rely on a static ``retrieve-then-reason'' paradigm, this rigid pipeline design prevents them from dynamically adapting memory access to intermediate evidence discovered during inference. To bridge this gap, we propose MRAgent, a framework that combines an associative memory graph with an active reconstruction mechanism. We represent memory as a Cue–Tag–Content graph, where associative tags serve as semantic bridges connecting fine-grained cues to memory contents. Operating on this structure, our active reconstruction mechanism integrates LLM reasoning directly into memory access, allowing the agent to iteratively explore and prune retrieval paths based on accumulated evidence. This ensures that memory retrieval is dynamically adapted to the reasoning context while avoiding combinatorial explosion caused by unconstrained expansion. Experiments on the LoCoMo benchmark and LongMemEval benchmark demonstrate significant improvements over strong baselines (up to $23\\%$), while substantially reducing retrieval cost, highlighting the effectiveness of active and associative reconstruction for long-horizon memory reasoning.
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TRACE: Trajectory Recovery for Continuous Mechanism Evolution in Causal Representation Learning
Shicheng Fan, Kun Zhang, Lu Cheng
Temporal causal representation learning methods assume that causal mechanisms switch instantaneously between discrete domains, yet real-world systems often exhibit continuous mechanism transitions. For example, a vehicle's dynamics evolve gradually through a turning maneuver, and human gait shifts smoothly from walking to running. We formalize this setting by modeling transitional mechanisms as convex combinations of finitely many atomic mechanisms, governed by time-varying mixing coefficients. Our theoretical contributions establish that both the latent causal variables and the continuous mixing trajectory are jointly identifiable. We further propose TRACE, a Mixture-of-Experts framework where each expert learns one atomic mechanism during training, enabling test-time recovery of mechanism trajectories, including intermediate mechanism states never observed during training. Experiments on synthetic and real-world data demonstrate that TRACE recovers mixing trajectories with up to 0.99 correlation, substantially outperforming discrete-switching baselines.
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The Cost of Learning Under Multiple Change Points
Tomer Gafni, Garud Iyengar, assaf zeevi
We consider an online learning problem in environments with multiple change points.
In contrast to the single change point problem that is widely studied using classical "high confidence" detection schemes, the multiple change point environment presents new learning-theoretic and algorithmic challenges. Specifically, we show that classical methods may exhibit catastrophic failure (high regret) due to a phenomenon we refer to as endogenous confounding. To overcome this, we propose a new class of learning algorithms dubbed Anytime Tracking CUSUM (ATC). These are horizon-free online algorithms that implement a selective detection principle, balancing the need to ignore "small" (hard-to-detect) shifts, while reacting "quickly" to significant ones. We prove that the performance of a properly tuned ATC algorithm is nearly minimax-optimal; its regret is guaranteed to closely match a novel information-theoretic lower bound on the achievable performance of any learning algorithm in the multiple change point problem. Experiments on synthetic as well as real-world data validate the aforementioned theoretical findings.
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A theory of learning data statistics in diffusion models, from easy to hard
Lorenzo Bardone, Claudia Merger, Sebastian Goldt
While diffusion models have emerged as a powerful class of generative models, their learning dynamics remain poorly understood.
We address this issue first by empirically showing that standard diffusion models trained on natural images exhibit a simplicity bias, learning simple, pair-wis

e input statistics before specializing to higher-order correlations. We reproduc
e this behaviour in simple denoisers trained on a minimal data model, the mixed
cumulant model, where we precisely control both pair-wise and higher-order corre
lations of the inputs. We identify a scalar invariant of the model that governs
the sample complexity of learning pair-wise and higher-order correlations that w
e call the _diffusion information exponent_, in analogy to related invariants in
 different learning paradigms.
Using this invariant, we prove that the denoiser learns simple, pair-wise statis
tics of the inputs at linear sample complexity, while more complex higher-order
statistics, such as the fourth cumulant, require at least cubic sample complexit
y. We also prove that the sample complexity of learning the fourth cumulant is l
inear if pair-wise and higher-order statistics share a correlated latent structu
re.
Our work describes a key mechanism for how diffusion models can learn distributi
ons of increasing complexity.
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Federated Manifold Learning (FML): Tackling Domain Heterogeneity with Structural
 Knowledge Transfer

Xutong Mu, Yanbiao Ma, Jia Shi, Xueli Geng, Fengkai Xiang, Tao Zhang, Ke Cheng,
Yulong Shen

Federated Learning (FL) faces significant challenges due to domain heterogeneity
, where data from different clients exhibit substantial statistical shifts that
hinder the generalization of the global model. Although existing methods attempt
 to mitigate this by exchanging class prototypes, they fall short by representin
g an entire class's complex distribution with a single point. This oversimplific
ation disregards the rich structural information within the data, especially acr
oss diverse domains. To address this limitation, we propose a paradigm shift fro
m point-based representation to structure-based knowledge transfer. We introduce
 Federated Manifold Learning (FML), a novel framework that leverages perceptual
manifolds—the intrinsic geometric structures of classes in the feature space—as
rich knowledge carriers. In FML, clients transmit compressed manifolds, which ar
e adaptively fused on the server using an attention-based Manifold Mutual Learni
ng (MML) mechanism. This process enables domain-specific structures to learn fro
m each other, creating a unified yet flexible global convergence target. Manifol
d-guided local training, enforced by a manifold approximation loss and a separat
ion loss, further aligns local models with this global structure. Extensive expe
riments on the Digits and Office31 benchmarks demonstrate that FML substantially
 outperforms state-of-the-art methods, achieving accuracy improvements of up to
6.47%.
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Gradient Smoothing: Coupling Layer-wise Updates for Improved Optimization

Haoming Meng, Anton Sugolov, Vardan Papyan

Deep neural networks with repeated blocks, such as transformers and ResNets, oft
en exhibit structured relationships across layers that emerge during training. M
otivated by this observation, we introduce a general paradigm of *Depth-wise Gra
dient Augmentation*, in which the update applied to a layer may depend on the ba
se optimizer updates computed for other layers. We study an instantiation of thi
s idea, termed *Gradient Smoothing*, which couples optimizer updates across dept
h and admits a natural interpretation as a structured preconditioning method. Ou
r framework operates directly on block-wise update vectors produced by arbitrary
 base optimizers (e.g., SGD, Adam, Muon), applying structured depth-wise smoothi
ng operators such as local weighted averaging with minimal computational overhea
d. We evaluate Gradient Smoothing across a diverse set of architectures and trai
ning regimes, including language model pretraining, RL post-training of LLMs on
reasoning tasks, diffusion modeling, and image classification with Vision Transf
ormers. Across these settings, Gradient Smoothing consistently improves converge
nce and generalization performance without modifying model architectures or trai
ning objectives. We further show that smoothing promotes more structured represe
ntation evolution across depth, suggesting a connection between structured updat
e coupling and the internal organization of learned representations. These resul

ts position Gradient Smoothing as a simple and broadly applicable approach for i
mproving training in modern deep networks.
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Baguan-TS: dual in-context learning model for time series forecasting with covar
iates

Linxiao Yang, Xue Jiang, Gezheng Xu, Tian Zhou, Min Yang, Zhaoyang Zhu, Linyuan
Geng, Zhipeng Zeng, Qiming Chen, Xinyue Gu, Rong Jin, Liang Sun

Transformers enable in-context learning (ICL) for rapid, gradient-free adaptatio
n in time series forecasting, yet most ICL-style approaches rely on tabularized,
 hand-crafted features, while end-to-end sequence models lack inference-time ada
ptation. We bridge this gap with a unified framework, Baguan-TS, which integrate
s the raw-sequence representation learning with ICL, instantiated by a 3D Transf
ormer that attends jointly over temporal, variable, and context axes.
To make this high-capacity model practical, we tackle two key hurdles: (i) calib
ration and training stability, improved with a feature-agnostic, target-space re
trieval-based local calibration; and (ii) output oversmoothing, mitigated via co
ntext-overfitting strategy. On public benchmark with covariates, Baguan-TS consi
stently outperforms established baselines, achieving the highest win rate and si
gnificant reductions in both point and probabilistic forecasting metrics. Furthe
r evaluations across diverse real-world energy datasets demonstrate its robustne
ss, yielding substantial improvements.
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AdaRoPE: Not All Attention Heads Should Rotate and Scale Equally

Shaowen Wang, Yuke Zheng, Tansheng Zhu, Shuang Chen, Shaofan Liu, Suncong Zheng,
 Jian Li

Rotary Position Embeddings (RoPE) are widely adopted in Transformers to encode p
ositional information, yet standard implementations enforce a uniform frequency
 schedule and scaling across all attention heads.
Using simplified retrieval tasks and length generalization scenarios, we show—bo
th empirically and theoretically—that heads with different functional roles requ
ire distinct frequency ranges and scaling factors to operate effectively. Ignori
ng this structure leads to suboptimal utilization of embedding dimensions and de
graded performance, particularly under long-context settings.
To address these limitations, we proposeAdaRoPE, which equips each attention hea
d with learnable rotation frequencies and scaling factors. Pretrained LLM with A
daRoPE consistently outperforms existing RoPE variants, including partial-RoPE a
nd NoPE baselines. For context extension, we further show that uniform frequency
 and attention scaling, used in methods such as YaRN, are suboptimal. By applyin
g head-specific scaling, AdaRoPE enables better context extension while better p
reserving short-context performance in both extrapolation setting and long conte
xt continued pretrain setting. These results highlight the importance of optimiz
ing rotary position embeddings at the level of individual attention heads.
****

Rank-Learner: Orthogonal Ranking of Treatment Effects

Henri Arno, Dennis Frauen, Emil Javurek, Thomas Demeester, Stefan Feuerriegel

Many decision-making problems require ranking individuals by their treatment eff
ects rather than estimating the exact effect magnitudes. Examples include priori
tizing patients for preventive care interventions, or ranking customers by the e
xpected incremental impact of an advertisement. Surprisingly, while causal effec
t estimation has received substantial attention in the literature, the problem o
f directly learning *rankings of treatment effects* has largely remained unexplo
red. In this paper, we introduce *Rank-Learner*, a novel two-stage learner that
directly learns the ranking of treatment effects from observational data. We fir
st show that naive approaches based on precise treatment effect estimation solve
 a harder problem than necessary for ranking, while our *Rank-Learner* optimizes
 a pairwise learning objective that recovers the true treatment effect ordering,
 without explicit CATE estimation. We further show that our *Rank-Learner* is Ne
yman-orthogonal and thus comes with strong theoretical guarantees, including rob
ustness to estimation errors in the nuisance functions. In addition, our *Rank-L
earner* is model-agnostic, and can be instantiated with arbitrary machine learni

ng models (e.g., neural networks). We demonstrate the effectiveness of our metho d through extensive experiments where *Rank-Learner* consistently outperforms st andard CATE estimators and non-orthogonal ranking methods. Overall, we provide p ractitioners with a new, orthogonal two-stage learner for ranking individuals by their treatment effects.
****
Gated Relational Alignment via Confidence-based Distillation for Efficient VLMs
Yanlong Chen, Amir Habibian, Luca Benini, Yawei Li
Vision-Language Models (VLMs) achieve strong multimodal performance but are cost ly to deploy, and post-training quantization often causes significant accuracy l oss. Despite its potential, quantization-aware training for VLMs remains underex plored. We propose GRACE, a framework unifying knowledge distillation and QAT un der the Information Bottleneck principle: quantization constrains information ca pacity while distillation guides what to preserve within this budget. Treating t he teacher as a proxy for task-relevant information, we introduce confidence-gat ed decoupled distillation to filter unreliable supervision, relational centered kernel alignment to transfer visual token structures, and an adaptive controller via Lagrangian relaxation to balance fidelity against capacity constraints. Acr oss extensive benchmarks on LLaVA and Qwen families, our INT4 models consistentl y outperform FP16 baselines (e.g., LLaVA-1.5-7B: 70.1 vs. 66.8 on SQA; Qwen2-VL- 2B: 76.9 vs. 72.6 on MMBench), nearly matching teacher performance. Using real I NT4 kernel, we achieve 3× throughput with 54\% memory reduction. This principled framework significantly outperforms existing quantization methods, making GRACE a compelling solution for resource-constrained deployment. Code and data are av ailable at: https://github.com/ForeverBlue816/GRACE.
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gp2Scale: A Class of Compactly Supported Non-Stationary Kernels and Distributed Computing for Exact Gaussian Processes on 10 Million Data Points
Marcus M. Noack, Mark D Risser, Hengrui Luo, Vardaan Tekriwal, Ronald J. Pandolf i
Despite a large corpus of recent work on scaling up Gaussian processes, a stubbo rn trade-off between computational speed, prediction, and uncertainty quantifica tion accuracy, and customizability persists. This is because the vast majority o f existing methodologies exploit various levels of approximation that lower accu racy and limit the flexibility of kernel and noise-model designs --- an unaccept able drawback at a time when expressive non-stationary kernels are on the rise i n many fields. Here, we propose a methodology we term *gp2Scale* that scales exa ct Gaussian processes to more than 10 million data points without relying on ind ucing points, kernel interpolation, or neighborhood-based approximations, and in stead leverages the existing capabilities of a GP: its kernel design. Highly fle xible, compactly supported, and non-stationary kernels lead to the identificatio n of naturally occurring sparse structure in the covariance matrix, which is the n exploited to compute the linear system solution and the log-determinant for tr aining. We demonstrate our method's functionality on several real-world datasets and compare it with state-of-the-art approximation algorithms. Although we show superior approximation performance in many cases, the method's real power lies in its agnosticism toward arbitrary GP customizations --- core kernel design, no ise, and mean functions --- and the type of input space, making it optimally sui ted for modern Gaussian process applications.
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Geometric Control of Out-of-Distribution Shift in Safe Offline RL
Zhiqi Zhuang, Di Wu, Benoit Boulet
Safe offline reinforcement learning (RL) requires optimizing policies within the support of static datasets while satisfying strict safety constraints. Although recent latent generative policies achieve strong empirical performance, they re ly heavily on implicit regularization and lack systematic control over distribut ional shift during policy improvement.
In this work, we propose a geometric control framework that leverages the biject ive structure of conditional normalizing flows to provide a tractable mechanism to regulate distributional deviation of the policy. By constraining divergence i

n the latent base space, we derive tractable upper bounds on the induced Wassers tein distance and total variation of the policy distribution, establishing an an alyzable connection between latent geometry and downstream behaviors.
This insight motivates a decoupled architecture: a flow prior shapes a feasibili ty-weighted latent manifold using Hamilton--Jacobi reachability signals, while a latent refiner performs geometrically constrained optimization directly in the base space. Across multiple safe RL benchmarks, our method achieves consistently low violation rates with competitive returns, highlighting the benefits of stru ctured geometric regularization.
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Scaling Beyond Masked Diffusion Language Models
Subham Sekhar Sahoo, Jean-Marie Lemercier, Zhihan Yang, Justin Deschenaux, Jingy u Liu, John Thickstun, Ante Juki■
Diffusion language models are a promising alternative to autoregressive models d ue to their potential for faster generation. Among discrete diffusion approaches , Masked diffusion currently dominates, largely driven by strong perplexity on l anguage modeling benchmarks. In this work, **we present the first scaling law st udy of uniform-state and interpolating discrete diffusion methods**. We also sho w that Masked diffusion models can be made approximately 12% more FLOPs-efficien t when trained with a simple cross-entropy objective.
We find that perplexity is informative within a diffusion family but can be misl eading across families, where models with worse likelihood scaling may be prefer able due to faster and more practical sampling, as reflected by the speed-qualit y Pareto frontier. **These results challenge the view that Masked diffusion is c ategorically the future of diffusion language modeling** and that perplexity alo ne suffices for cross-algorithm comparison. Scaling all methods to 1.7B paramete rs, we show that uniform-state diffusion remains competitive on likelihood-based benchmarks and outperforms autoregressive and Masked diffusion models on GSM8K, despite worse validation perplexity.
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Forensic Prompting with Dual-Action Policy Optimization for Vision-Language Forg ery Detection and Localization
Ye Zhu, Ai Zhao, Jinwei Wang
Image forgery is rapidly evolving, rendering forensic traces increasingly subtle and readily attenuated by post-processing. Although vision-language prompting c an inject priors, open-ended LLM-generated prompts are difficult to constrain, a nd naive language descriptions can introduce semantic perturbations. To address these challenges, we propose *Forensic Prompting with Dual-Action policy optimiz ation* (FPDA) for vision-language forgery detection and localization, where the *Forensic Prompting Module* (FPM) constructs a structured and reproducible foren sic prompt bank and supports optional text input as a reliability-aware cue for stable conditioning. Moreover, *Dual-Action Policy Optimization* (DAPO) is appli ed to learn sample-adaptive evidence usage by routing forensic prompts and sched uling localization refinement on a per-image basis, stabilizing discriminative c ues and improving mask spatial consistency. Extensive experiments are conducted on multiple public datasets covering manual manipulations, diffusion content, fa ce forgeries, and text-enabled settings, demonstrating favorable detection and l ocalization performance over representative state-of-the-art methods under compa rable evaluation protocols.
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An Odd Estimator for Shapley Values
Fabian Fumagalli, Landon Butler, Justin Singh Kang, Kannan Ramchandran, R. Teal Witter
The Shapley value is a ubiquitous framework for attribution in machine learning, encompassing feature importance, data valuation, and causal inference. However, its exact computation is generally intractable, necessitating efficient approxi mation methods. While the most effective and popular estimators leverage the *pa ired sampling* heuristic to reduce estimation error, the theoretical mechanism d riving this improvement has remained opaque. In this work, we provide an elegant and fundamental justification for paired sampling: we prove that the Shapley va

lue depends *exclusively* on the odd component of the set function, and that pai red sampling orthogonalizes the regression objective to filter out the irrelevan t even component. Leveraging this insight, we propose OddSHAP, a novel consisten t estimator that performs polynomial regression solely on the odd subspace. By u tilizing the Fourier basis to isolate this subspace and employing a proxy model to identify high-impact interactions, OddSHAP overcomes the combinatorial explos ion of higher-order approximations. Through an extensive benchmark, we find that OddSHAP achieves state-of-the-art estimation accuracy at larger sampling budget s.

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CoF-T2I: Video Models as Pure Visual Reasoners for Text-to-Image Generation

Chengzhuo Tong, Mingkun Chang, Shenglong Zhang, Yuran Wang, Cheng Liang, Zhizhen g Zhao, Bohan Zeng, Yang Shi, Ruichuan An, Yifan Dai, Ziming Zhao, Guanbin Li, P engfei Wan, Yuanxing Zhang, Wentao Zhang

Recent video generation models have revealed the emergence of Chain-of-Frame (Co F) reasoning, enabling frame-by-frame visual inference. With this capability, vi deo models have been successfully applied to various visual tasks (*e.g.*, maze solving, visual puzzles). However, their potential to enhance text-to-image (T2I) generation remains largely unexplored due to the absence of a clearly defined visual reasoning starting point and interpretable intermediate states in the T2I generation process.

To bridge this gap, we propose **CoF-T2I**, a model that integrates CoF reasonin g into T2I generation via progressive visual refinement, where intermediate fram es act as explicit reasoning steps and the final frame is taken as output.

To establish such explicit generation process, we curate **CoF-Evol-Instruct**, a dataset of CoF trajectories that model the generation process from semantics t o aesthetics. To further improve quality and avoid motion artifacts, we enable a n independent encoding operation for each frame. Experiments show that CoF-T2I s ignificantly outperforms the base video model and achieves competitive performan ce, reaching 0.86 on GenEval and 7.468 on Imagine-Bench. These results indicate the substantial promise of video models for advancing high-quality text-to-image generation.

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Taylor-Gaussians-Flow: Towards Non-uniform Motion for Novel View Synthesis from Monocular Video

Zaoming Yan, Qizhou Chen, Yaomin Huang, Pengcheng Lei, Chenhao Shi, Yi Xu, Haich uan Song, Faming Fang

Long-term non-uniform motion poses a significant challenge for Novel View Synthe sis (\textbf{NVS}), as it requires modeling higher-order motion, such as acceler ation.

Existing methods primarily rely on deformation fields or scene flow, which are l imited to first-order approximations.

Due to neglecting higher-order motion representations and supervision, these app roaches suffer from long-term non-uniform motion scenarios.

Inspired by Taylor’s theorem, we propose Taylor-Gaussians-Flow (\textbf{TGsF}) t o represent and supervise non-uniform motion through first-order and second-orde r motion components.

TGsF comprises two key modules: Taylor-Gaussians (\textbf{TGs}) and Taylor-Gauss ians-Flow (\textbf{TGs-Flow}).

TGs represent motion using Gaussian means with a quadratic temporal term and tim e-dependent opacity.

Unlike previous methods, TGs-Flow decouples scene-flow supervision into separate depth and 2D optical-flow constraints.

This approach effectively mitigates error propagation from either depth or motio n estimation while circumventing the scarcity of labeled scene flow data.

Guided by the above analysis, we develop the Feed-Forward Taylor-Gaussians-Flow framework, which sets a new state-of-the-art on four dynamic benchmarks.

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DualOptim+: Bridging Shared and Decoupled Optimizer States for Better Machine Un learning in Large Language Models

Xuyang Zhong, Qizhang Li, Steven Y. Guo, Chen Liu
We propose **DualOptim+**, a novel optimization framework for improving machine unlearning in large language models.
It introduces a base state to capture common representations shared by forgetting and retaining objectives and delta states to preserve objective-specific residuals. This architecture allows the optimizer to adaptively bridge shared and decoupled states based on the directional conflict between forgetting and retaining gradients. We further introduce DualOptim+ 8bit, a quantized variant that reduces memory overhead without compromising performance. Extensive experiments across fictitious and real-world unlearning, safety alignment, and multi-task learning tasks demonstrate that DualOptim+ consistently achieves a superior trade-off between different objectives. Codes are available at https://github.com/CityU-MLO/DualOptimPlus.
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Normalized Rewards for Preference Optimization
Shawn Im, Federico Danieli, Skyler Seto, Barry-John Theobald, Katherine Metcalf
Direct Alignment Algorithms (DAAs) such as DPO have become a common way to post-train and align LLMs with human preferences. However, DAAs have been observed to over-optimize their implicit reward model and decrease the likelihood of preferred responses. This results in a decrease in the total likelihood assigned to responses seen in the preference dataset, potentially resulting in undesirable behavior. To counteract this undesired side-effect of DAAs, we examine the effect of using objectives that add a regularization term to maintain the total length-normalized probabilities of the chosen and rejected responses. To better understand over-optimization, we investigate how response likelihood changes are distributed over the tokens with and without regularization. We find that a significant portion of the likelihood changes are due to a small set of outlier tokens, which explains how DAAs improve generation quality despite decreasing the likelihoods of chosen responses. We apply the proposed regularization to reference-based (DPO) and reference-free (SimPO) methods and find (1) improved trade-offs between generation quality and general benchmark capability and (2) improvements in reward modeling across datasets. For example, on Llama-3.1-8B-Instruct, we see both a >20\% relative increase in AlpacaEval2 scores and >9\% relative performance gains on general benchmarks. Additionally, we find that the added regularization term effectively mitigates the amount of displacement within preferred responses overall, and for the outlier tokens specifically, by utilizing low-likelihood tokens.
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Pix2Key: Controllable Open-Vocabulary Retrieval with Semantic Decomposition and Self-Supervised Visual Dictionary Learning
Guoyizhe Wei, Yang Jiao, Nan Xi, Zhishen Huang, Jingjing Meng, Rama Chellappa, Yan Gao
Composed image retrieval uses a reference image plus a natural-language edit to retrieve images that apply the requested change while preserving other relevant visual content. Classic fusion pipelines typically rely on supervised triplets and can lose fine-grained cues, while recent zero-shot approaches often caption the reference image and merge the caption with the edit, which may miss implicit user intent and return repetitive results. We present Pix2Key, which represents both queries and candidates as open-vocabulary visual dictionaries, enabling intent-aware constraint matching and diversity-aware reranking in a unified embedding space. A self-supervised pretraining component, V-Dict-AE, further improves the dictionary representation using only images, strengthening fine-grained attribute understanding without CIR-specific supervision. On the extended DeepFashion-MM benchmark, Pix2Key improves Recall@10 up to 3.2 points, and adding V-Dict-AE yields an additional 2.3-point gain while improving intent consistency and maintaining high list diversity.
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Does Your Reasoning Model Implicitly Know When to Stop Thinking?
Zixuan Huang, Xin Xia, Yuxi Ren, Jianbin Zheng, Xuanda Wang, Zhixia Zhang, Hongyan Xie, Songshi Liang, Zehao Chen, Xuefeng Xiao, Fuzhen Zhuang, Jianxin Li, deqi

ng wang, Yikun Ban
Recent advancements in large reasoning models (LRMs) have greatly improved their capabilities on complex reasoning tasks through Long Chains of Thought (CoTs). However, this approach often results in substantial redundancy, impairing computational efficiency and causing significant delays in real-time applications. Recent studies show that longer reasoning chains are frequently uncorrelated with correctness and can even be detrimental to accuracy.
In a further in-depth analysis of this phenomenon, we surprisingly uncover and empirically verify that LRMs implicitly know the appropriate time to stop thinking, while this capability is obscured by current sampling paradigms.
Motivated by this, we introduce SAGE (Self-Aware Guided Efficient Reasoning), a novel sampling paradigm that unleashes this efficient reasoning potential. Furthermore, integrating SAGE as mixed sampling into group-based reinforcement learning (SAGE-RL) enables SAGE-RL to effectively incorporate SAGE-discovered efficient reasoning patterns into standard pass@1 inference, markedly enhancing both the reasoning accuracy and efficiency of LRMs across multiple challenging mathematical benchmarks.
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RL-SPH: Learning to Achieve Feasible Solutions for Integer Linear Programs
Tae-Hoon Lee, Min-Soo Kim
Primal heuristics play a crucial role in quickly finding feasible solutions for NP-hard integer linear programming (ILP). Although $\textit{end-to-end learning}$-based primal heuristics (E2EPH) have recently been proposed, they are typically unable to independently generate feasible solutions. To address this challenge, we propose RL-SPH, a novel reinforcement learning-based start primal heuristic capable of independently generating feasible solutions, even for ILP involving non-binary integers. Empirically, RL-SPH rapidly obtains high-quality feasible solutions with a 100% feasibility rate, achieving on average a 28.6$\times$ lower primal gap and a 2.6$\times$ lower primal integral compared to existing start primal heuristics.
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PanoWorld-X: Generating Explorable Panoramic Worlds via Sphere-Aware Video Diffusion
Yuyang Yin, Hao-Xiang Guo, Fangfu Liu, Mengyu Wang, Hanwen Liang, Eric Li, Yikai Wang, Xiaojie Jin, Yao Zhao, Yunchao Wei
Achieving a complete and explorable 360-degree visual world is a cornerstone of immersive content creation. While recent advances in video generation have achieved impressive results, they follow a 2D paradigm that treats content generation as transitions of 2D pixels, lacking an intrinsic understanding of the physical 3D world, resulting in frequent geometric inconsistencies.
To achieve an explorable and physical-consistent visual world, the generation process should shift to a 3D paradigm: the visual content is governed by the physical relationships of the entire 3D environment together with 3D motion signals. However, under this setting, the conventional modeling methods and control signals, such as spatial attention computation in a 2D space, become unsuitable and ineffective.
To address this, we propose PanoWorld-X for explorable immersive scene video generation. Our framework is built on the panoramic representation, which naturally maps a 3D scene into a standard format and provides an ideal basis for consistency. Specifically, we first develop a data curation pipeline to produce high-quality and large-motion 3D scene evolution with movement trajectories. To achieve precise control, we design the Exploration Panoramic Plücker Embedding (PPE), a guidance signal tailored for 3D motion. Furthermore, leveraging the spherical geometric properties of panoramic data, we propose a sphere-aware attention mechanism, which can capture true geometric adjacency by reprojecting features onto a spherical surface. Extensive experiments demonstrate that PanoWorld-X achieves superior performance in motion range, control precision, and visual quality, underscoring its potential for real-world applications.
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Consistency Training Can Entrench Misalignment

David Demitri Africa, Arathi Mani
Consistency training encourages a model to produce similar outputs across related inputs or sampling procedures. Such methods are simple, scalable, and largely label-free, but their effects on model alignment remain poorly understood. Could the self-bootstrapping nature of these methods amplify undesired behavior in models? We test seven consistency training methods on 108 model organisms: open-source models (7B--70B) fine-tuned to exhibit various forms of controlled misaligned behavior. We find that outcomes vary significantly: consistency training generally suppresses reward hacking and emergent misalignment but amplifies sycophancy. We present evidence that distribution shifts induced by the consistency labeling process, rather than variation in the selection operators, may be the primary driver of systematic alignment effects. Finally, we present a unifying theoretical framework to derive conditions under which consistency training will amplify or suppress misalignment. In total, our study establishes that consistency training is not alignment-neutral, and that its use in critical systems should be carefully audited.
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Strategy-Aware Optimization Modeling with Reasoning LLMs
Ruiqing Zhao, Fengzhi Li, Yuan Zuo, Rui Liu, YanSong Liu, Yunfei Ma, Fanyu Meng, Junlan Feng
Large language models (LLMs) can generate syntactically valid optimization programs, yet often struggle to reliably choose an effective modeling strategy, leading to incorrect formulations and inefficient solver behavior. We propose **SAGE**, a strategy-aware framework that makes *Modeling Strategy* explicit in both data construction and post-training. SAGE builds a solver-verified multi-strategy dataset and trains a student model with supervised fine-tuning followed by Segment-Weighted GRPO using a composite reward over format compliance, correctness, and solver efficiency. Across eight benchmarks spanning synthetic and real-world settings, SAGE improves average pass@1 from 72.7 to 80.3 over the strongest open-source baseline. With multiple generations, SAGE discovers more distinct correct formulations and improves component-level diversity at pass@16 by 19-29%. At the largest scale, SAGE produces more compact constraint systems with 14.2% fewer constraints than the baseline, consistent with solver-efficient modeling. Overall, these results show that making *Modeling Strategy* explicit improves automated optimization modeling. The code and data are available at https://github.com/rachhhhing/SAGE.
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PODiff: Latent Diffusion in Proper Orthogonal Decomposition Space for Scientific Super-Resolution
Onkar Jadhav, Tim French, Matthew Rayson, Nicole L. Jones
Probabilistic super-resolution of high-dimensional spatial fields using diffusion models is often computationally prohibitive due to the cost of operating directly in pixel space. We propose PODiff, a structured conditional generative framework that performs diffusion in a fixed, variance-ordered Proper Orthogonal Decomposition (POD) coefficient space, exploiting the orthogonality of POD modes to impose an interpretable, variance-ordered latent geometry. This design enables efficient ensemble generation, preserves dominant spatial structure, and yields spatially interpretable, well-calibrated uncertainty at substantially lower computational cost. We evaluate PODiff on sea surface temperature downscaling over the West Australian coast and on a controlled advection--diffusion benchmark. PODiff achieves reconstruction accuracy comparable to pixel-space and learned-latent diffusion while using a substantially smaller structured latent space, requiring significantly less memory, and producing more reliable uncertainty estimates than deterministic and Monte Carlo Dropout baselines.
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PSMix: Robust Point Cloud Recognition through Spectral Domain Mixing
Xin Wei, Qin Yang, Hongji Zhao, Fei Gao, Mingrui Zhu, Nannan Wang, Xinbo Gao
While data augmentation is essential for robust point cloud recognition, conventional spatial mixup strategies often compromise geometric integrity by generating physically unrealistic samples. To overcome this limitation, we propose PSMix,

which shifts the mixing paradigm to the spectral domain via the Spherical Harmonic Transform. Instead of simple coordinate interpolation, PSMix performs a rotation-aware hierarchical mixing on spectral coefficients. This approach explicitly preserves global structural properties while diversifying local details, achieving a balance that spatial methods struggle to maintain. Complementing this, we introduce an adversarial rotation optimization strategy to enforce invariance against challenging orientations. Extensive experiments on ModelNet-C and ScanObjectNN-C demonstrate that PSMix achieves state-of-the-art robustness, while also serving as an orthogonal plug-in that further boosts the performance of existing spatial strategies.
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EgoTactile: Learning Grasp Pressure for Everyday Objects from Egocentric Video
Yuan Zeng, Yujia Shi, Tiao Tan, Xingting Li, Yaqi Qin, Zongqing Lu, Wenming Yang, Jing-Hao Xue, Qingmin Liao
Estimating full-hand grasp pressure from egocentric video is critical for immersive VR and robotic manipulation, yet dense tactile sensing often relies on intrusive hardware.
Existing vision-based methods predominantly rely on planar surfaces or fingertip contacts, failing to generalize to complex 3D object interactions.
Therefore, we introduce EgoTactile, a benchmark pairing egocentric video with full-hand pressure supervision for diverse everyday objects, incorporating a bare-hand transfer subset to enable generalization to natural scenarios.
Leveraging this benchmark, we first establish EgoPressureFormer as a discriminative baseline. Beyond this, to explicitly address the uncertainty in partial observations, we propose EgoPressureDiff, a conditional diffusion framework that adapts a large-scale pre-trained video diffusion backbone. By combining rich world knowledge priors with a Physically-Informed Feature Rectification layer to inject semantic constraints, our approach effectively hallucinates plausible contact patterns and resolves visual-physical ambiguities.
Extensive experiments demonstrate that our method achieves superior performance on the benchmark and robust transferability to in-the-wild scenarios. Our project page is at https://egotactile.github.io/.
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LAVA: A Unified Framework for Finetuning Language and Vision Models
Daorui Ding, Fanhua Shang, Tiancan Feng, Junkang Liu, Hongying Liu
LoRA and its variants have attracted considerable attention because of their abilities to tune a negligible number of parameters while achieving comparable downstream performance. This success is largely attributed to the intrinsic low-rank structure of model parameter spaces, which allows LoRA to train two projection matrices to project weights into a low-dimensional subspace and then map them back. However, it does not consider how to explore this low-rank subspace sufficiently and may lose the expression ability accordingly. Moreover, when using LoRA to tune convolution layers, a flatten operation is required to convert tensors into matrices. We argue that this will degrade the model's performance. In this paper, we address this issue from a general parameter sub-space perspective: we present a unified **L**anguage **A**nd **V**ision **A**daption finetuning framework (called **LAVA**). Specifically, we verify the existence of low-rank subspaces in convolution layers empirically and propose to parameterize the increment of both convolution kernels and matrices as sum of learnable rank-1 components. To improve training stability, we analyze the optimization dynamics of LoRA and incorporate orthogonal regularization into our parameterization, for which we give theoretical proof that it will help reduce the variance of the gradient. We conduct various experiments on different downstreaming tasks to validate LAVA's superiority. For example, when tuning LLaMA2-7b for commonsense tasks, the performance of our LAVA is **+1.9\%** higher than that of LoRA. For metric depth estimation tasks, LAVA only tunes $\sim$1.5\% of Depth-Anything (335.3M), and achieves **+3.5\%** δ_1 accuracy against that of LoRA and **+5.6\%** δ_1 accuracy against that of SVDiff.
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Doubly Outlier-Robust Online Infinite Hidden Markov Model

Horace Yiu, Leandro Sánchez-Betancourt, Alvaro Cartea, Gerardo Duran-Martin
We derive a robust update rule for the online infinite hidden Markov model (iHMM) for when the streaming data contains outliers and the model is misspecified. L everaging recent advances in generalised Bayesian inference, we define robustnes s via the posterior influence function (PIF), and provide conditions under which the online iHMM has bounded PIF. Imposing robustness inevitably induces an adap tation lag for regime switching. Our method, which is called Batched Robust iHMM (BR-iHMM), balances adaptivity and robustness with two additional tunable param eters. Across limit order book data, hourly electricity demand, and a synthetic high-dimensional linear system, BR-iHMM reduces one-step-ahead forecasting error by up to 67% relative to competing online Bayesian methods. Together with theor etical guarantees of bounded PIF, our results highlight the practicality of our approach for both forecasting and interpretable online learning.
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Beyond Language Modeling: An Exploration of Multimodal Pretraining
Shengbang Tong, David Fan, John Nguyen, Ellis L Brown II, Gaoyue Zhou, Shengyi Q ian, Boyang Zheng, Théophane Vallaeys, Rob Fergus, Naila Murray, Marjan Ghazvini nejad, Mike Lewis, Nicolas Ballas, Amir Bar, Michael Rabbat, Jakob Verbeek, Luke Zettlemoyer, Koustuv Sinha, Yann LeCun, Saining Xie
The visual world offers a critical axis for advancing foundation models beyond l anguage. Despite growing interest in this direction, the design space for native multimodal models remains opaque. We provide empirical clarity through controll ed, from-scratch pretraining experiments, isolating the factors that govern mult imodal pretraining without interference from language pretraining. We adopt the Transfusion framework, using next-token prediction for language and diffusion fo r vision, to train on diverse data including text, video, image-text pairs, and even action-conditioned video. Our experiments yield four key insights: (i) Repr esentation Autoencoder (RAE) provides an optimal unified visual representation b y excelling at both visual understanding and generation; (ii) visual and languag e data are complementary and yield synergy for downstream capabilities; (iii) un ified multimodal pretraining leads naturally to world modeling, with capabilitie s emerging from general training; and (iv) Mixture-of-Experts (MoE) enables effi cient and effective multimodal scaling while naturally inducing modality special ization. Through IsoFLOP analysis, we compute scaling laws for both modalities a nd uncover a scaling asymmetry: vision is significantly more data-hungry than la nguage. We demonstrate that the MoE architecture harmonizes this scaling asymmet ry by providing the high model capacity required by language while accommodating the data-intensive nature of vision, paving the way for truly unified multimoda l models.
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Geometry-Correct Diffusion Posterior Sampling with Denoiser-Pullback Curvature G uidance and Manifold-Aligned Damping
Seunghyeok Shin, Minwoo Kim, Dabin Kim, Hongki Lim
Diffusion posterior sampling conditions diffusion priors on measurements, but da ta-consistency updates are typically scaled by hand-tuned guidance weights and c an destabilize sampling under stiff, operator-dependent curvature. We replace sc alar guidance with a per-noise-level damped Gauss--Newton correction computed in diffusion-state coordinates. The correction pulls likelihood gradients back thr ough the denoiser, uses a one-sided curvature model that avoids forward denoiser Jacobians, and applies diffusion-calibrated rank-one damping aligned with the d enoiser residual. Each correction is solved with matrix-free GMRES using automat ic differentiation, and sampling proceeds with a variance-preserving Langevin tr ansition with a closed-form drift/noise split. On FFHQ and ImageNet across inver se problems, it achieves competitive PSNR/SSIM/LPIPS while running markedly fast er than most of the compared baselines; on accelerated MRI reconstruction, it ac hieves the best PSNR/SSIM among the compared baselines. Code is available at htt ps://github.com/Seunghyeok0715/CLAMP
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CLIP Tricks You: Training-free Token Pruning for Efficient Pixel Grounding in La rge Vision-Language Models

Sangin Lee, Yukyung Choi
In large vision-language models, visual tokens typically constitute the majority of input tokens, leading to substantial computational overhead. To address this, recent studies have explored pruning redundant or less informative visual tokens for image understanding tasks. However, these methods struggle with pixel grounding tasks, where token importance is highly contingent on the input text. Through an in-depth analysis of CLIP, we observe that visual tokens within referent regions often exhibit low similarity to their textual representation. Motivated by this insight, we introduce LiteLVLM, a training-free, text-guided token pruning strategy for efficient pixel grounding inference. By reversing the ranking of CLIP's visual-text similarity, LiteLVLM effectively retains visual tokens covering the referent regions, while recovering context tokens to enable clear foreground-background separation. Extensive experiments demonstrate that LiteLVLM significantly outperforms existing methods by over 5% across diverse token budgets. Without any training or fine-tuning, LiteLVLM maintains 90% of the original performance with a 22% speedup and a 2.3$\times$ memory reduction. Our code is available at https://github.com/sejong-rcv/LiteLVLM.
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Adaptive Memory Retention in Dynamic Graphs
Fabrizio De Castelli, Alessio Gravina, Moshe Eliasof, Carola-Bibiane Schönlieb, Davide Bacciu
Modeling graphs demands a careful balance between long-range propagation of information across nodes and the controlled dissipation of noisy or redundant signals to ensure stable learning and generalization. This challenge is exacerbated in dynamic graphs, where structural and temporal information interact, leading to uncontrolled information accumulation and amplifying noise, thereby affecting generalization. We introduce LAMP, a dynamic graph model for snapshot-based dynamic graphs that incorporates adaptive, learned dissipation within a principled dynamical systems framework. Our architecture combines impulsive neural ODEs with an antisymmetric parameterization to model conservative information flow, alongside data-driven dissipative dynamics that regulate information retention over space and time. This formulation yields stable yet expressive representations and enables effective long-range dependency modeling while avoiding pathological information buildup. We provide a theoretical analysis establishing stability guarantees and characterizing the representational power. Extensive experiments on synthetic and real-world benchmarks demonstrate state-of-the-art performance, particularly on tasks requiring extended-range dependency modeling.
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From Blind Spots to Gains: Diagnostic-Driven Iterative Training for Large Multimodal Models
Hongrui Jia, Chaoya Jiang, Yongrui Heng, Shikun Zhang, Wei Ye
As Large Multimodal Models (LMMs) scale up and reinforcement learning (RL) methods mature, LMMs have made notable progress in complex reasoning and decision making. Yet training still relies on static data and fixed recipes, making it difficult to diagnose capability blind spots or provide dynamic, targeted reinforcement. Motivated by findings that test driven error exposure and feedback based correction outperform repetitive practice, we propose Diagnostic-driven Progressive Evolution (DPE), a spiral loop where diagnosis steers data generation and reinforcement, and each iteration re-diagnoses the updated model to drive the next round of targeted improvement. DPE has two key components. First, multiple agents annotate and quality control massive unlabeled multimodal data, using tools such as web search and image editing to produce diverse, realistic samples. Second, DPE attributes failures to specific weaknesses, dynamically adjusts the data mixture, and guides agents to generate weakness focused data for targeted reinforcement. Experiments on Qwen3-VL-8B-Instruct and Qwen2.5-VL-7B-Instruct show stable, continual gains across eleven benchmarks, indicating DPE as a scalable paradigm for continual LMM training under open task distributions. Our code, models, and data are publicly available at https://github.com/hongruijia/DPE.
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Towards Professional-Grade Financial Agents: Benchmarking, Tooling, and Structur

ed Reasoning
Cheng Huang, Jinghua Piao, Ranran Wang, Yong Li
While Large Language Model (LLM) agents have shown encouraging progress in financial reasoning, their effectiveness in realistic financial workflows is severely hindered by the lack of holistic benchmarks and the fragility of unstructured reasoning. To address this gap, we introduce ProFinR, the first Professional Finance Reasoning benchmark, covering four financial domain tasks and comprising 528 expert-designed problems. To solve these complex financial reasoning questions, we construct the Financial Tool Universe, a tool library containing 53 domain-specific tools organized into 13 categories. Building on the tool library, we introduce ProFinAgent, a structured agent framework based on Directed Acyclic Graph (DAG) and Case-Based Memory (CBM). Compared with strictly sequential workflows, ProFinAgent coordinates tool execution through a DAG planner for parallel efficiency and uses CBM to retrieve prior cases for more reliable decision-making. Experimental results demonstrate that ProFinAgent achieves a 49.81% performance gain over state-of-the-art baselines with a 47.1% reduction in inference latency. Our code is available at https://github.com/tsinghua-fib-lab/ProFinAgent.
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Learning Structured Reasoning via Tractable Trajectory Control
Po-Nien Kung, Zhen Yang, Jeffrey Luo, Cheng-Fu Yang, Haikang Deng, Zi-Yi Dou, Yinfei Yang, Nanyun Peng, Zhe Gan, Kai-Wei Chang
Large language models can exhibit emergent reasoning behaviors, often manifested as recurring lexical patterns (e.g., “wait,” indicating verification). However, complex reasoning trajectories remain sparse in unconstrained sampling, and standard RL often fails to guarantee the acquisition of diverse reasoning behaviors. We propose a systematic discovery and reinforcement of diverse reasoning patterns through structured reasoning, a paradigm that requires targeted exploration of specific reasoning patterns during the RL process.
To this end, we propose Ctrl-R, a framework for learning structured reasoning via a tractable trajectory control that actively guides the rollout process, incentivizing the exploration of diverse reasoning patterns that are critical for complex problem-solving. The resulting behavior policy enables accurate importance-sampling estimation, supporting unbiased on-policy optimization. We further introduce a power-scaling factor on the importance-sampling weights, allowing the policy to selectively learn from exploratory, out-of-distribution trajectories while maintaining stable optimization.
Experiments demonstrate that Ctrl-R enables effective exploration and internalization of previously unattainable reasoning patterns, yielding consistent improvements across language and vision–language models on mathematical reasoning tasks.
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Towards Whole-corpus Reconstruction of Heterogeneous RAG Knowledge Bases
Peiru Yang, Yi Luo, Zhenfeng Gao, Tong Ju, Haoran Zheng, Linjie Zhu, Hongke Fu, Qing Li, Shangguang Wang, Tao Qi
Retrieval-Augmented Generation (RAG) systems are increasingly deployed to provide query-based access to large knowledge bases, thereby introducing concrete privacy risks whereby the underlying corpus may be partially or fully extracted through the deployed service.
Existing extraction attacks typically rely on locally driven search strategies, in which newly extracted content is inferred or expanded based on previously recovered fragments.
However, real-world knowledge bases are often multi-source and heterogeneous, with pronounced semantic discontinuities across domains. Such gaps can trap extraction methods that rely on local semantic continuity in local optima, severely limiting large-scale corpus reconstruction.
In this paper, we introduce an extraction framework (GeoEx) designed to navigate and reconstruct heterogeneous RAG knowledge bases without any prior knowledge. The framework plans extraction directly in the embedding space of a proxy retrieval model to improve global coverage, and employs an embedding inversion module to convert latent vectors into executable queries.

We further propose a composite geometric strategy that combines orthogonal query synthesis for cross-domain exploration with local embedding perturbations for dense extraction within discovered clusters.
Experiments on mixed corpora spanning eight diverse domains and multiple retrievers and LLMs show that GeoEx significantly outperforms baselines in both extraction coverage and query efficiency.
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Nested Spatio-Temporal Time Series Forecasting
YingHao Ai, Yukai Zhou, Ruoxi Jiang, Junyi An, Chao Qu, Zhijian Zhou, Shiyu Wang, Fenglei Cao, Zenglin Xu, Furao Shen, Yuan Qi
Spatio-temporal forecasting is critical for real-world applications like traffic management, yet capturing complex interactions under high-noise conditions remains challenging. While current methods have shown improved accuracy using spatial physical priors, they often struggle with evolving temporal correlations and systematic errors. In this work, we propose a nested forecasting framework that couples future macro-level regional trends with micro-level historical observations, enabling top-down guidance from abstract future representations for fine-grained forecasting. Specifically, we construct semantically coherent regions via spectral clustering and design a progressive coarse-to-fine predictor to inject macro-dynamics into node-level forecasting. Extensive experiments on multiple real-world datasets demonstrate that our method consistently outperforms state-of-the-art baselines, validating the effectiveness of future macro-guided nested forecasting.
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Think Twice Before You Act: Enhancing Agent Behavioral Safety with Thought Correction
Changyue Jiang, Wenqi Zhang, Xudong Pan, Geng Hong, Min Yang
LLM-based agents solve complex tasks through iterative reasoning, tool use, and environment interaction, where each intermediate thought directly shapes subsequent actions. Small deviations in these thoughts can therefore propagate into unsafe behaviors, yet existing guardrails typically operate only on final outputs or require intrusive model modifications. We introduce Thought-Aligner, a lightweight plug-in safety model that performs causal correction on unsafe thoughts before action execution, without altering the underlying agent. The corrected thoughts are fed back into the agent, steering its decision process and tool use toward safer trajectories. Because it operates solely at the thought level, Thought-Aligner is model-agnostic and can be integrated into diverse agent frameworks. We train Thought-Aligner via two-stage contrastive learning on paired safe and unsafe thoughts generated across ten risk scenarios. Experiments on diverse agent-safety benchmarks and six LLMs show that Thought-Aligner increases behavioral safety from about 50% without protection to around 90% on average, exceeding state-of-the-art guardrails by roughly 23%, while also improving helpfulness by about 5%. The method incurs low per-step latency and minimal overhead, enabling scalable and practical deployment. We publicly release Thought-Aligner-7B at https://huggingface.co/WhitzardAgent/Thought-Aligner-7B.
****
Ripple Perturbations Through Structure: Likelihood-Constrained Adversarial Attacks on Heterogeneous Tabular Data
Zhengjie Zhou, Jiahuan Yan, Boqun Ma, Weiwei Feng, Tengfei LIU, Weiqiang Wang
Generating realistic adversarial examples for tabular data remains challenging due to heterogeneous feature types and asymmetric inter-feature dependencies. Existing approaches typically rely on handcrafted constraints or undirected similarity criteria to delimit the feasible attack region, which often fail to capture the structural dependency governing tabular generation. Consequently, standard attacks typically produce perturbations that are statistically likely yet semantically inconsistent and prone to optimization stagnation via gradient masking. To address this, we propose LCSA, a white-box framework that formulates adversarial generation as optimization over structurally admissible perturbations. LCSA leverages an ensemble of heterogeneous neural Structural Causal Models to infer dependencies and introduces a structure-aware ripple mechanism. Unlike attacks tha

t perturb features in isolation, this mechanism propagates updates downstream, a
cting as a structural preconditioner that conditions gradient flow to mitigate m
asking effects. Extensive experiments demonstrate that LCSA outperforms state-of
-the-art baselines in 45 of 50 evaluated configurations, yielding adversarial ex
amples with superior structural consistency and transferability.
****

How Many Different Outputs Can a Transformer Generate?
Maxime Meyer, Mario Michelessa, Caroline Chaux, Vincent Y. F. Tan
We study how we can leverage only a handful of characteristics of a transformer'
s architecture to closely predict the number of different sequences it can outpu
t, both qualitatively and quantitatively.

We provide an upper bound depending on the length of the prompt, which we show
empirically to be tight up to a factor less than 10, across architectures and m
odel sizes.

Our analysis also provides a theoretical explanation for previously observed e
mpirical failures of transformers on simple sequence tasks—such as copying and c
ramming.

Formally, we prove that (i) the maximal length of accessible sequences (those
that the transformer can output for some prompt) grows linearly with the prompt
length, (ii) beyond a critical threshold, the proportion of accessible sequences
decays exponentially with sequence length, and (iii) the linear coefficient rel
ating prompt length to accessible sequence length admits a theoretical upper bou
nd.

Notably, these results hold even with unbounded context and computation time.
****

Distortion of AI Alignment Revisited: RLHF is a Decent Utilitarian Aligner
Kazusato Oko, Annie S Ulichney, Nika Haghtalab, Han Bao
While Reinforcement Learning from Human Feedback (RLHF) is the standard paradigm
for aligning large language models with human preferences, its effectiveness in
pluralistic settings has been called into question. Notably, recent work by Göl
z et al. (2025) demonstrated that the *distortion* — defined as the multiplicati
ve gap between the average user utility of the RLHF policy and the optimal avera
ge utility — can scale exponentially with the Bradley-Terry temperature paramete
r β when users have heterogeneous preferences. In this work, we present a
fine-grained analysis of the distortion of RLHF with reward clipping and demonst
rate that such exponential degradation is not a fundamental property of the algo
rithm but rather a consequence of distribution mismatch between the distribution
generating preference data (μ) and the KL reference policy ($\pi_{\mathrm{r
ef}}$). To this end, we establish tight upper and lower bounds on the distortion
of RLHF across multiple regimes of the KL regularization strength. We show that
in a representative regime, under the Bradley–Terry model, the distortion is $\
tilde{\Theta}(\beta B + \beta)$, where B is an upper bound on the log density
ratio between μ and π_{ref}. In particular, when there is no di
stribution mismatch (i.e., $\mu = \pi_{\mathrm{ref}}$), RLHF achieves the optima
l distortion of $O(\beta)$ up to a constant. Our results suggest that, to reason
ably maximize average utility with RLHF, it is preferable to use on-policy sampl
ed preference data or to fine-tune before RLHF on data from a source close to $\
mu$.
****

Forget to Know, Remember to Use: Context-Aware Unlearning for Large Language Mod
els
Yuefeng Peng, Parnian Afshar, Megan Ganji, Thomas Butler, Amir Houmansadr, Mingx
ian Wang, Dezhi Hong
Large language models can memorize information that must be removed—ranging from
copyright-sensitive content (e.g., book chapters) to personally identifiable in
formation (e.g., income)—to ensure responsible and compliant behavior. Unlearnin
g has emerged as an efficient alternative to full retraining, aiming to remove s
pecific knowledge. However, users may still expect model to leverage the removed
information when it is re-introduced in the prompt. Existing evaluations of unl
earning methods focus on (1) the extent of forgetting of the target knowledge (f

orget set) and (2) performance preservation on the retain set (i.e., utility), b
ut overlook this critical usability dimension. Through a systematic evaluation o
f six state-of-the-art unlearning methods, we show that they consistently degrad
e such contextual utility—the model's ability to use forgotten knowledge when it
 is provided in context. To address this, we augment unlearning objectives with
a plug-in term that explicitly preserves contextual utility. Extensive experimen
ts demonstrate that our approach restores contextual utility to near original le
vels while still maintaining effective forgetting and retain-set utility.
****

Beyond Static Pipelines: Learning Dynamic Workflows for Text-to-SQL
Yihan Wang, Peiyu Liu, Runyu Chen, Wei Xu
Text-to-SQL has recently achieved impressive progress, yet remains difficult to
apply effectively in real-world scenarios. This gap stems from the reliance on s
ingle static workflows, fundamentally limiting scalability to out-of-distributio
n and long-tail scenarios. Instead of requiring users to select suitable methods
 through extensive experimentation, we attempt to enable systems to adaptively c
onstruct workflows at inference time. Through rigorous theoretical and empirical
 analysis, we demonstrate that optimal dynamic policies consistently outperform
the best static workflow, with performance gains fundamentally driven by heterog
eneity across candidate workflows. Motivated by this, we propose SquRL, a reinfo
rcement learning framework that enhances LLMs' reasoning capability in adaptive
workflow construction. We design a rule-based reward function and introduce two
effective training mechanisms: dynamic actor masking to encourage broader explor
ation, and pseudo rewards to improve training efficiency. Experiments on widely-
used Text-to-SQL benchmarks demonstrate that dynamic workflow construction consi
stently outperforms the best static workflow methods, with especially pronounced
 gains on complex and out-of-distribution queries.
****

CGSVD: Cascaded Granular Singular Value Decomposition for Large Language Model C
ompression
Yuli Chen, Shuhao Zhang, Jiale Han, Fanshen Meng, Haishen Jiang, Bo Cheng, Qiang
 Tong, Xiulei Liu
The exponential growth in the parameter scale of Large Language Models (LLMs) ha
s precipitated an urgent demand for efficient compression techniques to facilita
te practical deployment. To address this challenge, low-rank decomposition based
 on Singular Value Decomposition (SVD) offers a principled, hardware-friendly pa
thway for compressing LLMs without retraining. However, existing training-free a
pproaches predominantly rely on uniform rank allocation, implicitly assuming hom
ogeneous redundancy across the model depth and thereby neglecting the inherent n
on-uniformity of representational evolution. To bridge this gap, we introduce CG
SVD, a Cascaded Granular Singular Value Decomposition framework that leverages a
 dual-level non-uniform allocation strategy to maximize semantic preservation. S
pecifically, we quantify inter-layer significance via angular distance and asses
s intra-layer compressibility through spectral entropy, enabling precise identif
ication of critical architectural components. Furthermore, we propose an Iterati
ve Residual Filling (IRF) mechanism to bridge the parameter gap caused by intege
r-rank truncation and ensure strict adherence to global compression targets. Ext
ensive experiments on representative LLM families ranging from 3B to 13B paramet
ers verify the superiority of our approach. Notably, under a 30\% compression ra
tio on the LLaMA3.1-8B model, CGSVD achieves a remarkable average zero-shot accu
racy boost of 6.08% and reduces perplexity by 15.46 compared to the baseline. We
 release the [code](https://github.com/ironartisan/CGSVD) to facilitate future r
esearch.
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What Makes Effective Supervision in Latent Chain-of-Thought: An Information-Theo
retic Analysis
Xinghao Chen, Chak Tou Leong, Guo Wenjin, Jian Wang, Wenjie Li, Xiaoyu Shen
Latent Chain-of-Thought (CoT) internalizes reasoning within continuous hidden st
ates, offering a promising alternative to verbose discrete reasoning traces. How
ever, robust latent reasoning remains difficult because outcome supervision prov

ides weak learning signals and leaves latent trajectories prone to semantic drift. In this work, we analyze Latent CoT from an information-theoretic perspective and identify this failure as a dual collapse: gradient attenuation along the optimization path and representational drift in the latent space. We further decompose process supervision into two complementary dimensions: Trajectory Supervision, which injects dense stepwise reasoning signals, and Space Supervision, which preserves the semantic structure of the latent manifold. Our analysis shows that rigid geometric compression can collapse the reasoning space, whereas generative reconstruction provides a more flexible semantic anchor that better preserves information capacity. To measure these effects, we introduce the Unified Latent Probe (ULP), which quantifies the mutual information between latent trajectories and explicit reasoning steps. Experiments reveal a clear Information–Performance Binding: reasoning accuracy depends on the information fidelity preserved in the latent chain. These findings provide a principled framework for latent reasoning supervision and suggest shifting from geometric imitation toward mutual information maximization. Our code will be released at: \url{https://github.com/EIT-NLP/Supervision-in-Latent-CoT}.

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Safe and Scalable Web Agent Learning via Recreated Websites

Hyungjoo Chae, Jungsoo Park, Alan Ritter

Training autonomous web agents is fundamentally limited by the environments they learn from: real-world websites are unsafe to explore, hard to reset, and rarely provide verifiable feedback.

We propose VeriEnv, a framework that treats language models as environment creators, automatically cloning real-world websites into fully executable, verifiable synthetic environments.

By exposing controlled internal access via a Python SDK, VeriEnv enables agents to self-generate tasks with deterministic, programmatically verifiable rewards, eliminating reliance on heuristic or LLM-based judges.

This design decouples agent learning from unsafe real-world interaction while enabling scalable self-evolution through environment expansion.

Through experiments on web agent benchmarks, we show that agents trained with VeriEnv generalize to unseen websites, achieve site-specific mastery through self-evolving training, and benefit from scaling the number of training environments.

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Localizing Memorized Regions in Diffusion Models via Coordinate-Wise Curvature Differences

Gwangho Kim, Sungyoon Lee

Diffusion models can unintentionally memorize training samples, raising concerns about privacy and copyright. While recent methods can detect memorization, they often rely on global or model-specific signals and provide limited insight into where memorization appears within a generated image. We provide a geometric characterization of local memorization as a coordinate-wise variance collapse. However, such collapse can also arise from intrinsic data constraints rather than overfitting. To isolate overfitting-driven memorization, we propose curvature-difference methods that subtract the curvature of an underfitted baseline, either the unconditional model or a less-trained version of itself. We further derive a score-difference proxy that provides a geometric explanation for the widely used score-difference-based detection metric. Experiments on Stable Diffusion, evaluated against ground-truth memorization masks, show that our method outperforms the prior attention-based localization method. Code is available at \url{https://github.com/Gwangho99/mem-curv-diff}.

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Proximal Splitting Methods for Hybrid Differentiable Models

Abdel-Rahim Mezidi, Jordan Patracone, Amaury Habrard

Operator splitting methods are at the foundation of many numerical solvers for partial differential equations. In parallel, unrolled and hybrid learning-based architectures have been introduced to enhance classical solvers, but their design is rarely linked to the underlying problem structure. In this work, we propose a unifying framework that explicitly links operator splitting algorithms from op

timization with unrolled hybrid architectures. We show that each operator splitt
ing scheme naturally defines an unrolled architecture, which recovers a wide ran
ge of existing plug-and-play and hybrid models as special cases. Using this fram
ework, we design new unrolled hybrid architectures and validate them on benchmar
k fluid dynamics simulations, where they achieve improved accuracy and stability
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****

Co-RedTeam: Orchestrated Security Discovery and Exploitation with LLM Agents
Pengfei He, Ash Fox, Lesly Miculicich, Stefan Friedli, Daniel Fabian, Burak Gokt
urk, Jiliang Tang, Chen-Yu Lee, Tomas Pfister, Long Le
Large language models (LLMs) have shown promise in assisting cybersecurity tasks
, yet existing approaches struggle with automatic vulnerability discovery and ex
ploitation due to limited interaction, weak execution grounding, and a lack of e
xperience reuse. We propose Co-RedTeam, a security-aware multi-agent framework d
esigned to mirror real-world red-teaming workflows by integrating security-domai
n knowledge, code-aware analysis, execution-grounded iterative reasoning, and lo
ng-term memory. Co-RedTeam decomposes vulnerability analysis into coordinated di
scovery and exploitation stages, enabling agents to plan, execute, validate, and
 refine actions based on real execution feedback while learning from prior traje
ctories. Extensive evaluations on challenging security benchmarks demonstrate th
at Co-RedTeam consistently outperforms strong baselines across diverse backbone
models, achieving over 60\% success rate in vulnerability exploitation and over
10\% absolute improvement in vulnerability detection. Ablation and iteration stu
dies further confirm the critical role of execution feedback, structured interac
tion, and memory for building robust and generalizable cybersecurity agents.
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Real-Time Aligned Reward Model beyond Semantics
Zixuan Huang, Xin Xia, Yuxi Ren, Jianbin Zheng, Xuefeng Xiao, Hongyan Xie, Li Hu
aqiu, Songshi Liang, Zhongxiang Dai, Fuzhen Zhuang, Jianxin Li, Yikun Ban, deqin
g wang
Reinforcement Learning from Human Feedback (RLHF) is a pivotal technique for ali
gning large language models (LLMs) with human preferences, yet it is susceptible
 to reward overoptimization, in which policy models overfit to the reward model,
 exploit spurious reward patterns instead of faithfully capturing human intent.
Prior mitigations primarily relies on surface semantic information and fails to
efficiently address the misalignment between the reward model (RM) and the polic
y model caused by continuous policy distribution shifts.
This inevitably leads to an increasing reward discrepancy, exacerbating reward o
veroptimization.
To address these limitations, we introduce R2M (Real-Time Aligned Reward Model)
, a novel lightweight RLHF framework.
R2M goes beyond vanilla reward models that solely depend on the semantic repres
entations of a pretrained LLM. Instead, it leverages the evolving hidden states
of the policy (namely policy feedback) to align with the real-time distribution
shift of the policy during the RL process.
This work points to a promising new direction for improving the performance of
reward models through real-time utilization of feedback from policy models.
****

MOC: Multi-Order Communication in LLM-based Multi-Agent Systems
Yao Guan, Lin Wang, Zhihui Lu, Ziyi Wang, Wenzhu Yan, Qiang Duan
Despite the remarkable progress of Large Language Model (LLM) based Multi-Agent
Systems, most research focuses on optimizing coordination topology while largely
 underexploring the equally critical problem: how to transmit and optimize messa
ges among agents effectively? Current communication schemes typically rely on th
e direct concatenation of first-order neighbor responses, which induces a restri
cted evidence receptive field and leads to the dilution of crucial insights over
 multi-hop paths. To address these limitations, we propose the Multi-Order Commu
nication (MOC) scheme, which reconstructs the inter-agent communication to captu
re multi-hop dependencies and incorporates a structural message consolidation st
rategy to ensure efficiency. Specifically, we formalize the communication mechan

ism to construct a structured multi-order evidence stream, and subsequently desi
gn a Semantic-Topological Merging algorithm to optimize semantic fidelity within
 token constraints. Extensive experiments across six diverse datasets and LLM ba
ckbones of varying parameter scales demonstrate that MOC consistently improves t
ask performance and reduces communication costs.
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EAKV: An Entropy-Driven Adaptive KV Compression Framework for Long Video Underst
anding
Hengrui Hu, Jingyu Li, Juntao Liang, Guanyu Chen, Lan Zhang
Although Multimodal Large Language Models (MLLMs) have made remarkable progress,
 they still struggle with long-video understanding due to the massive memory foo
tprint of KV Caches. Existing methods often resort to disjoint retrieval or atte
ntion-based reduction with a uniform layer-wise budget to achieve compression. H
owever, these methods disrupt temporal continuity and ignore the varying informa
tion density across network layers. In this work, we reveal that memory allocati
on should mirror layer-wise semantic density, rather than adhering to a uniform
budget. To this end, we introduce EAKV, a training-free entropy-driven adaptive
KV compression framework that leverages attention entropy to adaptively allocate
 compression budgets, selectively preserving critical tokens while distilling re
dundant contexts into compact contextual anchors, thereby achieving granular mem
ory allocation proportional to semantic density. Extensive experiments on variou
s benchmarks demonstrate that EAKV surpasses existing methods across diverse mod
el architectures and varying parameter scales, yielding improvements ranging fro
m 0.6% to 6.5%.
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ParisKV: Fast and Drift-Robust KV-Cache Retrieval for Long-Context LLMs
Yanlin Qi, Xinhang Chen, Huiqiang Jiang, Qitong Wang, Botao Peng, Themis Palpana
s
KV-cache retrieval is essential for long-context LLM inference, yet existing met
hods struggle with distribution drift and high latency at scale. We introduce **
ParisKV**, a drift-robust, GPU-native KV-cache retrieval framework based on coll
ision-based candidate selection, followed by a quantized inner-product reranking
 estimator. For million-token contexts, ParisKV supports CPU-offloaded KV caches
 via Unified Virtual Addressing (UVA), enabling on-demand top-k fetching with
minimal overhead. ParisKV matches or outperforms full attention quality on both
long-input and **long-generation** benchmarks. It achieves state-of-the-art
long-context decoding efficiency: it matches or exceeds full-attention speed eve
n at batch size 1 for long contexts, delivers up to **2.8$\times$** higher throu
ghput within full attention’s runnable range, and scales to **million-token** co
ntexts where full attention runs out of memory. At million-token scale, ParisKV
reduces decode latency by **17$\times$** and **44$\times$** compared to MagicPIG
 and PQCache, respectively—two state-of-the-art KV-cache top-k retrieval basel
ines; code is available at https://github.com/amy-77/ParisKV/tree/main.
****

Improved Distribution Estimation in ℓ_∞
Doron Cohen, Aryeh Kontorovich, Yonatan Livshitz
We present improved bounds for estimating discrete probability distributions und
er the ℓ_{∞} norm. These include minimax bounds in expectation and hig
h-probability tail bounds. We resolve some of the open questions posed in Kontor
ovich and Painsky (JMLR, 2025) --- including a fully empirical version of the ti
ghtest risk bound they presented and identifying the form of the worst-case extr
emal distribution. Encouraging empirical results are reported as well.
****

AesFormer: Transform Everyday Photos into Beautiful Memories
Tianxiang Du, Hulingxiao He, Yuxin Peng
In everyday photography, aesthetically appealing moments are often captured with
 structural flaws (e.g., composition, camera viewpoint, or pose) that existing r
etouching and portrait enhancement methods cannot fix. We formulate Aesthetic Ph
oto Reconstruction (APR) as improving a photo’s aesthetic quality via structural
 reconstruction while preserving subject identity and scene semantics. Although

recent advances in image editing models make APR feasible, they often lack aesthetic understanding, yielding edits that are semantically plausible yet aesthetically weak. To address this, we propose AesFormer, a two-stage framework that decouples aesthetic planning from image editing. In Stage 1, an aesthetic action model (AesThinker) analyzes the input along seven progressive photographic dimensions and outputs executable editing actions; we further apply GRPO-A to encourage broad exploration over diverse action plans beyond SFT. In Stage 2, an action-conditioned editor (AesEditor) performs structural edits guided by these actions. To support APR, we build a video-based corpus-mining pipeline (VCMP) and construct AesRecon, a benchmark of 9,071 strictly aligned (poor, good) image pairs. Experiments show that AesFormer substantially improves APR performance and is competitive with Nano Banana Pro. Code is available at: https://github.com/PKU-ICST-MIPL/AesFormer_ICML2026.

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AgentTailor: A Semantic-Aware LLM-Based Multi-Agent System with Actor-Critic Structure

Peiting Yang, Jiahao Shi, Caiyi Xu, Ming Liu, Yanxia Wu, Rongsheng Li

Large Language Model (LLM)-based multi-agent systems often suffer from high communication cost due to redundant interactions, as existing methods optimize communication structures without explicitly measuring whether exchanged messages contribute to the final decision. To better utilize the semantic information in the execution stage to further optimize the structure of multi-agent systems and reduce token costs, we propose **AgentTailor**, a cost-aware framework that evaluates the semantic contribution of communication edges via an edge judgment mechanism, and employs an **Edge Prediction Network (EPN)** to estimate edge utilities through virtual execution without invoking LLMs. Experiments show that AgentTailor achieves the best average accuracy (91.36\%) on six datasets of diverse fields, while reducing total tokens by 21.2\%--61.6\%. Our work demonstrates that explicitly modeling semantic edge contribution is crucial for scalable and efficient multi-agent systems, providing a principled approach to communication optimization that goes beyond structural heuristics. The code is available at https://github.com/Pt3Y/AgentTailor.

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SimulCost: A Cost-Aware Benchmark and Toolkit for Automating Physics Simulations with LLMs

Yadi Cao, Sicheng Lai, Jiahe Huang, Yang Zhang, Zach Lawrence, Rohan Bhakta, Izzy F. Thomas, Mingyun Cao, Chung-Hao Tsai, Zihao Zhou, Yidong Zhao, Hao Liu, Alessandro Marinoni, Alexey Arefiev, Rose Yu

Evaluating LLM agents for scientific tasks has focused on token costs while ignoring tool-use costs like simulation time and experimental resources. As a result, metrics like pass@k become impractical under realistic budget constraints.

To address this gap, we introduce \textsc{SimulCost}, the first benchmark targeting cost-sensitive parameter tuning in physics simulations.

\textsc{SimulCost} compares LLM tuning cost-sensitive parameters against traditional scanning approach in both accuracy and computational cost, spanning 2,947 single-round (initial guess) and 1,931 multi-round (adjustment by trial-and-error) tasks across 13 simulators from fluid dynamics, solid mechanics, and plasma physics. Each simulator's cost is analytically defined and platform-independent.

Frontier LLMs achieve 46--65\% success rates in single-round mode, dropping to 35--55\% under high accuracy requirements, rendering their initial guesses unreliable especially for high accuracy tasks.

Multi-round mode improves rates to 72--81\%, but LLMs are 1.5--2.5$\times$ slower than traditional scanning, making them uneconomical choices.

We also investigate parameter group correlations for knowledge transfer potential, and the impact of in-context examples and reasoning effort, providing practical implications for deployment and fine-tuning.

We open-source \textsc{SimulCost} as a static benchmark and extensible toolkit to facilitate research on improving cost-aware agentic designs for physics simulations, and for expanding new simulation environments. Code and data are available at \url{https://github.com/Rose-STL-Lab/SimulCost-Bench}.

****
Continuous Diffusion Models Can Obey Formal Syntax
Jinwoo Kim, Taylor Berg-Kirkpatrick, Loris D'Antoni
Diffusion language models offer a promising alternative to autoregressive models due to their global, non-causal generation process, but their continuous latent dynamics make discrete constraints---e.g., the output should be a JSON file that matches a given schema---difficult to impose.
We introduce a training-free guidance method for steering continuous diffusion language models to satisfy formal syntactic constraints expressed using regular expressions.
Our approach constructs an analytic score estimating the probability that a latent state decodes to a valid string accepted by a given regular expression, and uses its gradient to guide sampling, _without_ training auxiliary classifiers.
The denoising process targets the base model conditioned on syntactic validity.
We implement our method in Diffinity on top of the PLAID diffusion model and evaluate it on 180 regular-expression constraints over JSON and natural-language benchmarks.
Diffinity achieves 68-96\% constraint satisfaction while incurring only a
small perplexity cost relative to unconstrained sampling,
outperforming autoregressive constrained decoding in both constraint satisfaction and output quality.
Diffinity is open-sourced at github.com/large-loris-models/Diffinity.
****
NanoFLUX: Distillation-Driven Compression of Large Text-to-Image Generation Models for Mobile Devices
Ruchika Chavhan, Malcolm Chadwick, Alberto Gil Couto Pimentel Ramos, Luca Morreale, Mehdi Noroozi, Abhinav Mehrotra
While large-scale text-to-image diffusion models continue to improve in visual quality, their increasing scale has widened the gap between state-of-the-art models and on-device solutions. To address this gap, we introduce NanoFLUX, a **2.4B** text-to-image flow-matching model distilled from **17B** FLUX.1-Schnell using a progressive compression pipeline designed to preserve generation quality. Our contributions include: (1) A model compression strategy driven by pruning redundant components in the diffusion transformer, reducing its size from 12B to 2B; (2) A ResNet-based token downsampling mechanism that reduces latency by allowing intermediate blocks to operate on lower-resolution tokens while preserving high-resolution processing elsewhere; (3) A novel text encoder distillation approach that leverages visual signals from early layers of the denoiser during sampling. Empirically, NanoFLUX generates 512×512 images in approximately 2.5 seconds on mobile devices, demonstrating the feasibility of high-quality on-device text-to-image generation.
****
GenUnfold: Rapidly Predict Protein Mechanical Unfolding Trajectory via a Physics-Guided Diffusion Model
Yiyuan Zhang, Cailong Hua, VINITENDRA SINGH, Joseph M. Muretta, James M. Ervasti, Murti Salapaka
Many fundamental biological processes are governed by mechanical forces, with proteins acting as the key molecular mediators. Elucidating how protein unfolding responds to force is critical for understanding the mechano-pathologies, such as cardiomyopathy and muscular dystrophy. While the unfolding trajectories measured by Single-Molecule Force Spectroscopy (SMFS) map the instantaneous force response against molecular extension, its broader application is limited by time-consuming data collection and high operational costs. Here, we present the first scalable generative diffusion framework for full unfolding trajectory prediction, which integrates protein encoders for multi-scale conditioning. Beyond establishing the field's first systematic benchmark using existing models, we propose GenUnfold, a novel physics-guided diffusion model that combines global coevolutionary context with a local mechanical representation of the protein. The representation is derived from a novel physics-biased attention mechanism, which steers the generative diffusion process by modeling dynamic residue dependencies as a func

tion of both structural topology and interaction stiffness. The benchmark for th
is task is built upon the biomolecule stretching database and several representa
tive baseline models. Empirical results demonstrate that GenUnfold achieves stat
e-of-the-art performance, reducing distributional error (FID) by 30\% and 54\% c
ompared to pretrained Evolutionary Scale Model (ESM)-2 and standard transformer,
 respectively. Beyond statistical curve similarity, GenUnfold demonstrates super
ior physical consistency; in downstream mechanical property prediction, it reduc
es prediction errors for unfolding force and energy distributions by 6\% and 36\
% over the ESM-2 baseline. These results indicate that while existing generative
 AI approaches can alleviate the need for predicting representative force curves
, GenUnfold further improves performance by leveraging the synergy between prote
in structure and evolutionary information. By enabling proteome-wide screening t
o identify mechanical candidates before costly physical validation, our approach
 is promising to accelerate the discovery of force-targeted therapeutics.
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MODEL SOUPS NEED ONLY ONE INGREDIENT

Alireza Abdollahpoorrostam, Nikolaos Dimitriadis, Adam Hazimeh, Pascal Frossard

Fine-tuning large pre-trained models on a target distribution often improves in-
distribution (ID) accuracy, but at the cost of out-of-distribution (OOD) robustn
ess as representations specialize to the fine-tuning data. Weight-space ensembl
ing methods, such as Model Soups, mitigate this effect by averaging multiple che
ckpoints, but they are computationally prohibitive, requiring the training and s
torage of dozens of fine-tuned models. In this paper, we introduce MonoSoup, a s
imple, data-free, hyperparameter-free, post-hoc method that achieves a strong ID
–OOD balance using _only a single_ checkpoint. Our method applies Singular Value
 Decomposition (SVD) to each layer’s update and decomposes it into high-energy d
irections that capture task-specific adaptation and low-energy directions that i
ntroduce noise but may still encode residual signals useful for robustness. Mono
Soup then uses entropy-based effective rank to automatically re-weigh these comp
onents with layer-wise coefficients that account for the spectral and geometric
structure of the model. Experiments on CLIP models fine-tuned on ImageNet and ev
aluated under natural distribution shifts, as well as on Qwen language models te
sted on mathematical reasoning and multiple-choice benchmarks, show that this pl
ug-and-play approach is a practical and effective alternative to multi-checkpoin
t methods, retaining much of their benefits without their computational overhead
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PromptDyG: Test-Time Prompt Adaptation on Dynamic Graphs

Guoguo Ai, Chaoxi Niu, Hui Yan, Joey Tianyi Zhou, Yew-Soon Ong, Guansong Pang

Activities in numerous evolving systems can be represented as dynamic graphs in
snapshot form at different time intervals, i.e., discrete-time dynamic graphs (D
TDGs). Existing methods show impressive advances in capturing historical tempora
l evolution patterns in DTDGs, but they focus on addressing an offline learning
setting, where models are trained using historical snapshots once and then evalu
ated to all subsequent graph snapshots without further updating. This fails to c
apture 1) the nature of evolving complexities across graph snapshots and 2) the
distribution shift in the testing graph snapshots. To address these problems, we
 propose PromptDyG, a novel framework that leverages unsupervised test-time Prom
pt adaptation for Dynamic Graph learning under a live-update online setting. The
 key insight is that an expressive dynamic graph prompt can be learned on a froz
en backbone via minimization of feature-wise, label-free entropy to efficiently
and continuously model the evolving patterns. We show theoretically that this un
supervised prompt adaptation can guarantee a larger similarity margin between po
sitive and negative pairs, facilitating more accurate dynamic predictions. It is
 further confirmed by our extensive empirical results on six benchmark datasets
that show consistent and significant improvements of PromptDyG over state-of-the
-art baselines. Code is available at https://github.com/mala-lab/PromptDyG.
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Causal Disentangled Anchor Learning for Scalable Fair Multi-view Clustering

Suyuan Liu, Shengfei Wei, Wenjing Yang, Shengju Yu, Siwei Wang, Xueqiong Li, Wen

peng Lu, Xinwang Liu
Existing fair multi-view clustering methods typically suffer from a severe trade-off between clustering utility and fairness, while incurring prohibitive quadratic complexity on large-scale datasets. To address these challenges, we propose Causal Disentangled Anchor Learning (CDAL), a novel framework that achieves scalable fairness via structural disentanglement. Guided by a structural causal model perspective, CDAL utilizes a dual-anchor mechanism to structurally separate latent representations into orthogonal semantic and sensitive subspaces. We further ensure statistical independence through a linearized Hilbert-Schmidt Independence Criterion (HSIC) constraint, which is optimized via an efficient alternating scheme. Theoretically, we prove the identifiability of the disentangled factors and guarantee the algorithm's global convergence and linear time complexity $\mathcal{O}(n)$. Extensive experiments on large-scale benchmarks demonstrate that CDAL outperforms state-of-the-art methods, achieving a superior utility-fairness trade-off. Our code is publicly available at https://github.com/Tracesource/CDAL.
****
PointCHR: Point Cloud Analysis via Curvature-Aware Hyperbolic Rectification
Xinxing Yu, Liying Yang, Hao Mo, Hui Ma, Fang Kai, Ajian Liu, Yanyan Liang
High-curvature regions in 3D point clouds encapsulate critical fine-grained geometric semantics yet exhibit a distinct long-tail sparsity in their spatial distribution. The inherent limitations of polynomial volume growth in Euclidean space frequently render these intricate geometric features challenging to adequately resolve within a uniform-scale feature space. Consequently, these regions are frequently overshadowed by smooth global features dominated by low-curvature regions, thereby limiting the discriminative capacity of the network. To address this issue, we propose PointCHR, a curvature-aware hyperbolic rectification (CHR) for point cloud analysis. Utilising the property of exponential volume expansion in the vicinity of hyperbolic manifolds, CHR presents a learnable curvature-guided radial rectification mechanism. By adaptively projecting high-curvature points towards boundary regions endowed with larger effective embedding capacities, PointCHR effectively mitigates the representation crowding problem inherent in Euclidean settings. Extensive experimentation has demonstrated that PointCHR significantly enhances the ability of backbone to capture fine-grained geometric details, achieving state-of-the-art performance across multiple benchmarks.
****
Capacity without Access: Reinterpreting the Mid-Depth Spectral Plateau in LLMs
SeongMin Kang, Woo-Seong Yun, Nahyun Lee, Yoon-Sik Cho
Prior probing-based analyses show that individual layers specialize in distinct linguistic and semantic functions, while a complementary line of work observes that deeper layers enter an alignment-dominated regime where residual transformations become increasingly constrained relative to the evolving hidden-state geometry. These findings appear to conflict: layer-wise functional specialization versus apparent representational stagnation in deeper layers. This raises a key question: Is the observed representational stagnation in deeper layers attributable to suboptimal acquisition or encoding of novel features, or are these features learned appropriately but exhibit minimal marginal contribution to the model’s predictive output? We address this by separating model behavior into (i) representational capacity: the richness and spectral diversity of the encoded features, and (ii) accessibility: the extent to which these features are aligned with, and exploited by, the output-relevant subspace. Analyzing hidden-state second-moment geometry across depth shows that intermediate layers maintain a broad representational span and rich spectral diversity, yet project only weakly onto the output-relevant subspace, indicating intact representational capacity but constrained accessibility within task-relevant dimensions. To probe their functional relevance, we add a controlled diagnostic pathway that routes intermediate-layer embeddings to the terminal readout. Analysis of the induced behavioral changes indicates that the apparent stagnation is better accounted for by geometric limitations on accessibility, rather than by a deficit in representational capacity. The source code is available here.

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RACER: Risk-Aware Calibrated Efficient Routing for Large Language Models
Sai Hao, Hao Zeng, Hongxin Wei, Bingyi Jing
Efficiently routing queries to the optimal large language model (LLM) is crucial for optimizing the cost-performance trade-off in multi-model systems. However, most existing routers rely on single-model selection, making them susceptible to misrouting. In this work, we formulate LLM routing as the α-VOR problem to minimize expected set size while controlling the misrouting risk, and propose a novel method -- RACER, extending base routers to output model sets that can be subsequently aggregated for improved output. In particular, RACER constructs nested model sets via augmented scoring and utilizes finite-sample concentration bounds to calibrate a threshold that allows for both variable set sizes and abstention. We theoretically prove that RACER achieves rigorous distribution-free risk control on unseen test data in a post-hoc and model-agnostic manner. Extensive experiments verify our theoretical guarantees and demonstrate that RACER consistently enhances downstream accuracy across a wide range of benchmarks.
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Learning to Theorize the World from Observation
Doojin Baek, Gyubin Lee, Junyeob Baek, Hosung Lee, Sungjin Ahn
What does it mean to understand the world? Is it simply to predict future video frames? Developmental cognitive science suggests that understanding the world is fundamentally the process of constructing internal theories of how it works rather than mere prediction, even before language is acquired. However, in machine learning, it remains unclear how to endow AI systems with such theory-building capability from raw, non-textual observation alone. In this paper, we introduce Learning-to-Theorize (L2T), a learning paradigm in which an AI system acquires the ability to construct theories represented as executable programs directly from observation alone. To instantiate this paradigm, we propose the Neural Language-of-Thought Programmer, a neural model that induces and executes latent programs as explanations rather than task-specific predictors or policies. In experiments, we show that this formulation enables explanation-driven generalization, allowing observations to be understood in terms of the programs that generate them.
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The Obfuscation Atlas: Mapping Where Honesty Emerges in RLVR with Deception Probes
Mohammad Taufeeque, Stefan Heimersheim, Adam Gleave, Chris Cundy
Training against white-box deception detectors has been proposed as a way to make AI systems honest. However, such training risks models learning to obfuscate their deception to evade the detector. Prior work has studied obfuscation only in artificial settings where models were directly rewarded for harmful output. We construct a realistic coding environment where reward hacking via hardcoding test cases naturally occurs, and show that obfuscation emerges in this setting.
We introduce a taxonomy of possible outcomes when training against a deception detector. The model either remains honest, or becomes deceptive via two possible obfuscation strategies. (i) *Obfuscated activations*: the model outputs deceptive text while modifying its internal representations to no longer trigger the detector. (ii) *Obfuscated policy*: the model outputs deceptive text that evades the detector, typically by including a justification for the reward hack. Empirically, obfuscated activations arise from representation drift during RL, with or without a detector penalty. The detector penalty only incentivizes obfuscated policies; we theoretically show this is expected for policy gradient methods. Sufficiently high KL regularization and detector penalty can yield honest policies, establishing white-box deception detectors as viable training signals for tasks prone to reward hacking.
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The Perception–Physics Paradox: Probing Scientific Alignment with TC-Bench
Dingling Yao, Andrea Polesello, Adeel Pervez, Caroline Muller, Francesco Locatello
While Vision Foundation Models (VFMs) excel at predictive tasks on satellite imagery, their performance can arise from visual correlations rather than underlyin

g structural invariants, making certain perception-based out-of-distribution acc uracy a poor proxy for scientific utility.
As a result, models may look correct without reasoning correctly—a discrepancy w e term the *Perception–Physics Paradox*.
To address this gap, we introduce *Scientific Alignment* as an implicit objectiv e for representation learning in scientific domains.
We study a principled, testable aspect of scientific alignment through *Structur al Isomorphism*, which requires latent representations to uniquely identify phys ical systems up to a linear reparameterization.
This perspective induces a hierarchy of necessary conditions and yields a system atic probing protocol for physical and causal interpretability.
To operationalize this framework, we release *TC-Bench*, a foundational global d ataset and automated construction pipeline for tropical cyclone research, and sh ow that current VFMs rely on visual shortcuts that collapse in extreme regimes, indicating that scientific alignment does not arise as a natural byproduct of vi sual scaling alone.
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Risk Awareness Injection: Calibrating Vision-Language Models for Safety without Compromising Utility

Mengxuan Wang, Yuxin Chen, Gang Xu, Tao He, hongjie jiang, Ming Li

Vision language models (VLMs) extend the reasoning capabilities of large languag e models (LLMs) to cross-modal settings, yet remain highly vulnerable to multimo dal jailbreak attacks. Existing defenses predominantly rely on safety fine-tunin g or \textit{aggressive} token manipulations, incurring substantial training cos ts or significantly degrading utility. Recent research shows that LLMs inherentl y recognize unsafe content in text, and the incorporation of visual inputs in VL Ms frequently dilutes risk-related signals. Motivated by this, we propose Risk A wareness Injection (RAI), a \textit{lightweight} and training-free framework for safety calibration that restores LLM-like risk recognition by amplifying unsafe signals in VLMs. Specifically, RAI constructs an Unsafe Prototype Subspace from language embeddings and performs targeted modulation on selected high-risk visu al tokens, explicitly activating safety-critical signals within the cross-modal feature space.
This modulation restores the model’s LLM-like ability to detect unsafe content f rom visual inputs, while preserving the semantic integrity of original tokens fo r cross-modal reasoning. Extensive experiments across multiple jailbreak and uti lity benchmarks demonstrate that RAI substantially reduces attack success rate w ithout compromising task performance.
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ProjQ: Project-and-Quantize for Adapter-Aware LLM Compression

Wenya Yu, Chao Zhang, Li Wang, Samson Lasaulce, Merouane Abdelkader DEBBAH

Post-Training Quantization (PTQ) and Low-Rank Adaptation (LoRA) constitute the s tandard pipeline for efficient Large Language Model (LLM) deployment. However, a pplying them sequentially poses a problem: PTQ often leaves behind random noise that is spread out (across the model's weights) in a way LoRA can't easily fix, meaning that LoRA ends up wasting its limited capacity trying to fix uncorrectab le noise instead of improving task performance. In this paper, we propose \textb f{ProjQ}, a novel framework for
constraining quantization noise to the low-rank manifold via orthogonal subspace projection. We derive an efficient alternating algorithm that shapes the quanti zation noise into a low-rank structure, effectively offloading dominant error co mponents to the subsequent adapter while minimizing the residual error in the or thogonal "uncorrectable" subspace. Our theoretical analysis demonstrates that Pr ojQ preserves strictly greater model plasticity for downstream tasks compared to standard PTQ. Extensive experiments on LLaMA-2, Qwen2.5 and Qwen3 confirm that ProjQ consistently outperforms existing methods in both quantization error compe nsation and downstream task fine-tuning, achieving up to $2\times$ lower evaluat ion loss for compensation and matching the performance of standard 4-bit baselin es on language modeling tasks with only 3 bits. The code is available on \url{ht tps://github.com/yy9301/ProjQ}.

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Heterogeneity-Aware Knowledge Sharing for Graph Federated Learning
Wentao Yu, Sheng Wan, Shuo Chen, Bo Han, Chen Gong
Graph Federated Learning (GFL) enables distributed graph representation learning while preserving graph data privacy. However, it suffers from heterogeneity in node features and graph structures across clients. To address this challenge, we propose a novel graph Federated learning method via Semantic and Structural Alignment (FedSSA). For node feature heterogeneity, FedSSA infers class-wise node distributions through a variational model, clusters clients according to the inferred distributions, and aligns local distributions with cluster-level representatives. For structural heterogeneity, FedSSA employs spectral Graph Neural Networks (GNNs) and introduces a spectral energy measure to characterize graph topology, enabling structural alignment between local and cluster-level spectral GNNs. Experiments on eleven homophilic and heterophilic graph datasets under non-overlapping and overlapping partitioning settings demonstrate that FedSSA consistently outperforms eleven state-of-the-art methods. Our code is available at https://github.com/blgpb/FedSSA.
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Leveraging Lineage Barcodes as Natural Augmentations for Contrastive Learning of Cell Fate in scRNA-seq Data
Shizhao Joshua Yang, Yixin Wang, Kevin Z. Lin
Deciphering how cells commit to future fates is essential for developing precision therapeutics that can reprogram stem cells or modulate immune functions. However, isolating these fate-determining signals in single-cell lineage tracing (scLT) remains challenging because differentiation programs are often confounded by unrelated processes like the cell cycle. To address this, we introduce Lineage-aware Contrastive Learning (LCL), a framework that treats inheritable lineage barcodes as a "natural" data augmentation to isolate subtle, lineage-specific signals. LCL utilizes a semi-supervised architecture to align unlabeled cells, facilitating the transfer of lineage structures to clinical datasets where explicit barcoding is unavailable. We demonstrate LCL’s utility by predicting future cell-type compositions from early-time points, effectively modeling longitudinal fate commitment from cross-sectional data. Benchmarking on hematopoietic and fibroblast systems shows that LCL significantly outperforms standard models like scVI, establishing contrastive learning as a scalable paradigm for understanding and potentially manipulating cellular differentiation.
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NL2Repo-Bench: Towards Long-Horizon Repository Generation Evaluation of Coding Agents
Jingzhe Ding, Shengda Long, Changxin PU, Ge Zhang, zhou huan, Hongwan Gao, Xiang Gao, Chao He, Yue Hou, FEI HU, Zhaojian Li, Weiran Shi, Zaiyuan Wang, Daoguang Zan, Chenchen Zhang, Xiaoxu Zhang, Chen Qizhi, Xianfu Cheng, Bo Deng, Qingshui Gu, Kai Hua, Juntao Lin, Pai Liu, Mingchen Li, Minghao Li, Xuanguang Pan, Zifan Peng, Yujia Qin, Yong Shan, Zhewen Tan, Haoran Wang, Zihan Wang, Weihao Xie, Yishuo Yuan, Jiayu Zhang, Yunfei Zhao, He Zhu, LIYA ZHU, Chenyang Zou, Ming Ding, Jianpeng Jiao, Jiaheng Liu, Minghao Liu, Qian Liu, Chongyang Tao, Jian Yang, Tong Yang, Zhaoxiang Zhang, Xinjie Chen, Wenhao Huang
Recent advances in coding agents suggest rapid progress toward autonomous software development, yet existing benchmarks primarily evaluate short-horizon behaviors such as localized code generation, scaffolded completion, or repository repair, leaving it unclear whether agents can sustain coherent reasoning, planning, and execution over the extended horizons demanded by real-world repository construction. To address this gap, we introduce NL2Repo-Bench, a benchmark explicitly designed to evaluate the long-horizon repository generation from scratch: given only a single natural-language requirements document and an empty workspace, agents must autonomously design the architecture, manage dependencies, and produce a fully installable Python library. Experiments across state-of-the-art open- and closed-source models reveal that long-horizon repository generation remains largely unsolved, with even the strongest agents achieving merely 40\% average test pass rates and rarely completing an entire repository correctly. Further analy

sis identifies systematic long-horizon failure modes, including premature termination, loss of global coherence, fragile cross-file dependencies, and inadequate planning over hundreds of interaction steps.
These results position NL2Repo-Bench as a rigorous, execution-based testbed for evaluating sustained agentic competence and highlight long-horizon reasoning as a key bottleneck for autonomous coding agents. Our data and code are available at https://github.com/multimodal-art-projection/NL2RepoBench.
****
Explaining Concept Shift with Interpretable Feature Attribution
Ruiqi Lyu, Alistair Turcan, Bryan Wilder
Concept shift occurs when the distribution of labels conditioned on the features changes between domains, which can make even a well-tuned ML model miscalibrated on a new domain. Identifying these shifted features provides unique insight into how feature-label relationships differ between domains, considering the difference may be across a scientifically relevant dimension, such as time, disease status, population, etc. In this paper, we propose SGShift, a method for attributing performance degradation under concept shift in tabular data to a sparse set of shifted features. We frame concept shift as a feature selection task to learn the features that can explain performance differences between models in the source and target domain. This framework enables SGShift to adapt powerful statistical tools such as generalized additive models, knockoffs, and absorption towards identifying these shifted features. We conduct extensive experiments in synthetic and real data across various ML models and find SGShift can identify shifted features much more accurately than baseline methods, requires few samples in the shifted domain, and is robust to complex cases of concept shift.
****
Depth over Fidelity in Fixed-Budget Noisy Evolution Strategies
Sichen Wang, Zhipeng Lu
Noisy evolution strategies under fixed evaluation budgets face a depth–fidelity trade-off: spending evaluations to denoise intra-generation rankings reduces the number of distribution updates the optimizer can execute. We argue for depth over fidelity and propose probabilistic elite membership (PEM), which replaces hard rank-based weights in evolution strategies with conditional expected rank weights that integrate over ranking uncertainty. PEM preserves the conditional mean update while reducing conditional update dispersion—a Rao–Blackwellization of the noisy rank-based step. We instantiate PEM via residual bootstrapping (RB-PEM) with capped per-generation overhead, complemented by an adaptive probe-and-switch mechanism for low-noise regimes. Across the COCO bbob-noisy suite and external tasks including RL policy search and hyperparameter optimization, RB-PEM achieves consistent gains in high-misranking, budget-constrained settings.
****
SafeDec: Constrained Decoding for Safe Autoregressive Generalist Robot Navigation Policies
Parv kapoor, Akila Ganlath, Michael Clifford, Changliu Liu, Sebastian Scherer, Eunsuk Kang
Recent advances in end-to-end, multi-task robot policies based on transformer models have demonstrated impressive generalization to real-world embodied navigation tasks. Trained on vast datasets of simulated and real-world trajectories, these policies map multimodal observations directly to action sequences for physical execution. Despite promising real-world capabilities, these models are still data-driven and, therefore, lack explicit notions of behavioral correctness. We address this gap by introducing **SafeDec**, a constrained decoding framework for autoregressive, transformer-based robot navigation policies that enforces safety specifications expressed as Signal Temporal Logic (STL) formulas. Our method ensures that generated actions provably satisfy STL specifications under assumed dynamics at runtime without retraining while remaining agnostic of the underlying policy.
We evaluate **SafeDec** on tasks from the CHORES benchmark for state-of-the-art embodied navigation policies across hundreds of procedurally generated environments and show that our decoding-time interventions are useful not only for filter

ing unsafe actions but also for conditional action generation. Videos are availa
ble at constrained-robot-fms.github.io.
****

Rethinking Loss Reweighting for Imbalance Learning as an Inverse Problem: A Neur
al Collapse Point of View
Jinping Wang, Zixin Tong, Zhiwu Xie, Zhiqiang Gao
Loss reweighting is a widely used strategy for long-tailed classification, but e
xisting reweighting strategies often rely on heuristics and rarely define a well
-specified target. Inspired by Neural Collapse (NC), the ideal simplex Equiangul
ar Tight Frame (ETF) terminal geometry suggests equal per-class average loss as
a reasonable target for reweighting.
Based on the ideal equal loss objective, we consider loss reweighting as an inve
rse problem and propose an inverse-view reweighting strategy that infers class w
eights dynamically to match this ideal objective.
Empirically, NC metrics suggest our method can effectively reduce the loss imbal
ance coefficient and achieve closer alignment with NC geometry while consistentl
y outperforming strong long-tailed baselines on different datasets.
Our code is publicly available at: https://github.com/tongzixin716716/Inverse-Lo
ss-Reweighting.
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Second-Order Bilevel Optimization with Accelerated Convergence Rates
Sheng Yang, Chengchang Liu, Lesi Chen, John C.S. Lui
This paper studies second-order methods for nonconvex-strongly-convex bilevel op
timization. We propose a novel fully second-order bilevel approximation method (
FSBA) that achieves an iteration complexity of $\tilde{\mathcal{O}}(\epsilon^{-1
.5})$ for finding the $(\epsilon, \mathcal{O}(\sqrt{\epsilon}))$ second-order st
ationary point of the hyper-objective function.
Our results demonstrate that second-order methods can achieve an accelerated con
vergence rate than first-order methods in bilevel optimization.
To address the heavy computational cost associated with the second-order oracle,
 we introduce a lazy variant of FSBA, called LFSBA, which reuses second-order in
formation across several iterations. We prove that LFSBA exhibits better computa
tional complexity than FSBA by a factor of $\sqrt{d}$, where d is the dimensio
n of the problem.
We also apply a similar idea to nonconvex strongly-concave minimax optimization
and propose the lazy minimax cubic-regularized Newton (LMCN) method with better
computational complexity compared to existing second-order methods.
****

Impact of Connectivity on Laplacian Representations in Reinforcement Learning
Tommaso Giorgi, Pierriccardo Olivieri, Keyue Jiang, Laura Toni, Matteo Papini
Learning compact state representations in Markov Decision Processes (MDPs) has p
roven crucial for addressing the curse of dimensionality in large-scale reinforc
ement learning (RL) problems. Existing principled approaches leverage structural
 priors on the MDP by constructing state representations as linear combinations
of the state-graph Laplacian eigenvectors. When the transition graph is unknown
or the state space is prohibitively large, the graph spectral features can be es
timated directly via sample trajectories. In this work, we prove an upper bound
on the approximation error of linear value function approximation under the lear
ned spectral features. We show how this error scales with the algebraic connecti
vity of the state-graph, grounding the approximation quality in the topological
structure of the MDP. We further bound the error introduced by the eigenvector e
stimation itself, leading to an end-to-end error decomposition across the repres
entation learning pipeline. Additionally, we show how the common expression for
the symmetrized MDP Laplacian is easy to misinterpret, and propose a more straig
htforward reformulation. Our results hold for general (non-uniform) policies wit
hout any assumptions on the symmetry of the induced transition kernel. We valida
te our theoretical findings with numerical simulations on gridworld environments
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Interpreting and Enhancing Emotional Circuits in Large Vision-Language Models vi

a Cross-Modal Information Flow
Chengsheng Zhang, Chenghao Sun, Zhining Xie, Xinmei Tian
Large Vision-Language Models (LVLMs) represent a significant leap towards empathetic agents, demonstrating remarkable capabilities in emotion understanding. However, the internal mechanisms governing how LVLMs translate abstract visual stimuli into coherent emotional narratives remain largely unexplored, primarily due to the scarcity of visual counterfactuals and the diffuse nature of emotional expression.
In this paper, we bridge this gap by introducing a steering-vector-based causal attribution framework tailored for descriptive emotional reasoning.
To this end, we construct a specialized dataset to demystify the emotional circuits underlying the three-stage '' Adapt-Aggregate-Execute'' mechanism.
Crucially, we discover a functional decoupling: visual emotional cues are aggregated in middle layers via $\textit{sentiment-specific}$ attention heads, but are subsequently translated into narrative generation in deep layers through $\textit{emotion-general}$ pathways.
Guided by these insights, we regulate the emotional information routing to strengthen attention flow and amplify the semantic activation to consolidate expression.
Extensive experiments on the comprehensive MER-UniBench demonstrate that our methods significantly improve performance via inference-time intervention, effectively mitigating emotional hallucinations and corroborating the causal fidelity of the discovered circuits.
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EpiTwin: Spatiotemporal Graph Transformers for Epileptic sEEG Signal Reconstruction
Jingbo Yang, Yunfeng Zhao, Chao Qiu, Yulin Sun, Xiuyun Liu, Xiaofei Wang
Stereotactic electroencephalography (sEEG) provides temporally precise intracranial recordings but is inherently constrained by sparse and irregular spatial sampling due to clinical limitations on electrode implantation. Signal reconstruction under this setting aims to infer neural activity at unmonitored locations, potentially expanding the coverage of neural recordings without increasing the number of implanted electrodes. However, most existing sEEG reconstruction methods underutilize the spatial information of electrode contacts in both encoding and modeling, and rely on deterministic objectives that favor average patterns, leading to over-smoothed reconstructions. We propose EpiTwin, a conditional spatial graph transformer for sEEG signal reconstruction, comprising three key components. Hybrid Spatial Positional Encoding (HSPE) constructs explicit spatial identities from electrode coordinates, graph topology, and anatomical priors. Geometry–Functional Biased Attention (GFBA) incorporates geometric distance and data-driven functional similarity biases into attention computation. The adversarial refinement training employs a multi-scale discriminator to counter reconstruction over-smoothing. Experiments on real-world clinical sEEG data demonstrate that EpiTwin consistently achieves lower reconstruction error under electrode series-level masking, outperforming recent foundation models such as LaBraM with a 16.8\% relative reduction in RMSE. Furthermore, EpiTwin effectively mitigates spectral over-smoothing and improves reconstruction fidelity.
****
Predicting the Order of Upcoming Tokens Improves Language Modeling
Zayd Muhammad Kawakibi Zuhri, Erland Hilman Fuadi, Alham Fikri Aji
Multi-token prediction (MTP) has been proposed as an auxiliary objective to improve next-token prediction (NTP) in language model training but shows inconsistent improvements, underperforming in standard NLP benchmarks. We found MTP's exact future token prediction to be too difficult as an auxiliary loss. Instead, we propose token order prediction (TOP), which trains models to order upcoming tokens by their proximity using a learning-to-rank loss. TOP requires only a single additional unembedding layer compared to MTP's multiple transformer layers. We pretrain models of 340M, 1.8B, and 7B parameters using NTP, MTP, DeepSeek MTP (DS-MTP) and TOP objectives. The results of nine standard NLP benchmarks show that TOP overall outperforms NTP, MTP, and DS-MTP even at scale. TOP models with conti

nued training on math and code also perform better on 4 relevant benchmarks. On the synthetic star graph task, TOP enables pathfinding on graphs where NTP, MTP, and DS-MTP fail. Our implementation and training code is available at https://github.com/zaydzuhri/token-order-prediction.
****
A Solvable High-Dimensional Model Where Nonlinear Autoencoders Learn Structure Invisible to PCA While Test Loss Misaligns With Generalization
Vicente Conde Mendes, Lorenzo Bardone, Cédric Koller, Jorge Medina Moreira, Vittorio Erba, Emanuele Troiani, Lenka Zdeborová
Many real-world datasets contain hidden structure that cannot be detected by simple linear correlations between input features. For example, latent factors may influence the data in a coordinated way, even though their effect is invisible to covariance-based methods such as PCA. In practice, nonlinear neural networks often succeed in extracting such hidden structure in unsupervised and self-supervised learning. However, constructing a minimal high-dimensional model where this advantage can be rigorously analyzed has remained an open theoretical challenge.

We introduce a tractable high-dimensional spiked model with two latent factors: one visible to covariance, and one statistically dependent yet uncorrelated, appearing only in higher-order moments. PCA and linear autoencoders fail to recover the latter, while a minimal nonlinear autoencoder provably extracts both. We analyze both the population risk, and empirical risk minimization.

Our model also provides a tractable example where self-supervised test loss is poorly aligned with representation quality: nonlinear autoencoders recover latent structure that linear methods miss, even though their reconstruction loss is higher.
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Assistive Prompt Mediation: Evaluating Language Models Under Accessibility Constraints
Priyaranjan Pattnayak, Ishan Banerjee
Large language models (LLMs) are increasingly used as assistive interfaces for users who cannot reliably produce clean text due to accessibility constraints, yet existing evaluations assume iterative input repair and focus on task accuracy or generic noise robustness. We introduce Assistive Prompt Mediation (APM), a theory-grounded evaluation paradigm that reframes assistance as a constrained mediation problem: recovering latent user intent from accessibility-impaired input without clarification, while minimizing cognitive burden and hallucination risk. APM decomposes assistive quality along these axes and is instantiated across 8 languages, 4 accessibility-driven noise classes, and 10 frontier LLMs, with impairment severity yielding accessibility sensitivity curves. Results show that apparent robustness often masks trade-offs—high intent preservation frequently coincides with increased burden or hallucinated mediation, hallucination rates vary by more than $2\times$ across noise types, and assistive decisions exhibit bounded entropy (<0.81 normalized), indicating systematic rather than unstable behavior. These findings demonstrate that standard robustness metrics substantially overestimate assistive reliability and motivate evaluating LLMs as constrained mediators under accessibility-driven input degradation.
****
GoodDiffusion: Proactive Copyright Protection for Diffusion Bridge Models via Learnable Sample-specific Signatures
Shixi Qin, Zhiyong Yang, Shilong Bao, Zitai Wang, Qianqian Xu, Qingming Huang
This paper tackles the challenging problem of developing a proactive copyright protection mechanism that cuts off unauthorized use of diffusion bridge models. Existing studies largely fall into post-hoc attribution (e.g., watermarking and fingerprinting) or degradation-only defenses, which offer only indirect and limited preventive effect. We therefore propose GoodDiffusion, inspired by backdoor mechanisms, to enforce model-level use-time control by internalizing authorization into the generative process through a selectively permissive, otherwise closed

behavior. Specifically, GoodDiffusion preserves high-quality generation for authorized queries carrying valid signatures, yet refuses to generate for unauthorized inputs. We further empirically show that naive static-signature designs (like conventional backdoor injection) are fundamentally fragile, since a surrogate signature can be efficiently recovered via gradient-based optimization. To strengthen security, we introduce a Learnable Signature Network (LSN) that assigns sample-specific signatures conditioned on each input. This breaks the universality of signatures and prevents a surrogate from transferring across inputs. Extensive experiments validate that GoodDiffusion effectively blocks unauthorized use while maintaining strong generation quality for authorized users.

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Discrete Survival Knowledge Distillation for Competing Risks Analysis

Feiyang Deng, Lingfeng Luo, Di Wang, Qinmengge Li, Lingxuan Kong, Kevin He

Accurate prediction in survival analysis with competing risks is challenged by rare event rates and limited effective sample sizes. Knowledge distillation offers a promising way to transfer information from an external teacher to improve a local student, but existing methods are overwhelmingly developed for uncensored outcomes and do not directly extend to survival analysis, where censored observations provide only partial information. Moreover, prior work often assumes that teacher and student share identical outcome definitions, whereas in competing risks settings, they may differ in outcome granularity and event definitions, further complicating knowledge transfer. To address these gaps, we propose DiSKD (Discrete Survival Knowledge Distillation), a deep learning framework for discrete-time competing risks that integrates teacher predictions via a cause-specific, time-dependent Kullback-Leibler divergence. DiSKD enables flexible and privacy-conscious transfer without requiring raw data sharing, remains robust to model misspecification or outcome-definition heterogeneity, and adaptively weights teacher guidance by emphasizing compatible teachers while down-weighting less relevant ones. Simulation studies and real-world applications demonstrate improved discrimination and calibration.

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From Zero to Hero: Advancing Zero-Shot Foundation Models for Tabular Outlier Detection

Xueying Ding, Haomin Wen, Simon Klüttermann, Leman Akoglu

Outlier detection (OD) is widely used in practice; but its effective deployment on new tasks is hindered by lack of labeled outliers, which makes algorithm and hyperparameter selection notoriously hard. Foundation models (FMs) have transformed ML, and OD is no exception: Shen et al. (2025) introduced FOMO-0D, the first FM for OD, achieving remarkable performance against numerous baselines. This work introduces OUTFORMER, which advances FOMO-0D with (1) a mixture of synthetic priors and (2) self-evolving curriculum training. OUTFORMER is pretrained solely on synthetic labeled datasets and infers test labels of a new task by using its training data as
in-context input. Inference is fast and zero-shot, requiring merely forward pass and no labeled outliers. Thanks to in-context learning, it requires zero additional work—no OD model training or bespoke model selection—enabling truly plug-and-play deployment. OUTFORMER achieves state-of-the-art performance on the prominent ADBench, as well as two new large-scale OD benchmarks that we introduce, comprising over 1,500 datasets, while maintaining speedy inference.

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Federated Learning with Unlabeled Clients: Personalization Can Happen in Low Dimensions

Hossein Zakerinia, Jonathan Scott, Christoph H. Lampert

Personalized federated learning has emerged as a popular approach to training on devices holding statistically heterogeneous data, known as clients. However, most existing approaches require a client to have labeled data for training or finetuning in order to obtain their own personalized model. In this paper we address this by proposing FLowDUP, a novel method that is able to generate a personalized model using only a forward pass with unlabeled data. The generated model parameters reside in a low-dimensional subspace, enabling efficient communication a

nd computation. FLowDUP's learning objective is theoretically motivated by our n
ew transductive multi-task PAC-Bayesian generalization bound, that provides perf
ormance guarantees for unlabeled clients. The objective is structured in such a
way that it allows both clients with labeled data and clients with only unlabele
d data to contribute to the training process. To supplement our theoretical resu
lts we carry out a thorough experimental evaluation of FLowDUP, demonstrating st
rong empirical performance on a range of datasets with differing sorts of statis
tically heterogeneous clients. Through numerous ablation studies, we test the ef
ficacy of the individual components of the method.
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CoFrGeNet: Continued Fraction Architectures for Language Generation
Amit Dhurandhar, Vijil Chenthamarakshan, Dennis Wei, Tejaswini Pedapati, Karthik
eyan Natesan Ramamurthy, Rahul Nair
Transformers are arguably the preferred architecture for language generation. In
 this paper, inspired by continued fractions, we introduce a new function class
for generative modeling. The architecture family implementing this function clas
s is named CoFrGeNets - Continued Fraction Generative Networks. We design novel
architectural components based on this function class that can replace Multi-hea
d Attention and Feed-Forward Networks in Transformer blocks while requiring much
 fewer parameters. We derive custom gradient formulations to optimize the propos
ed components more accurately and efficiently than using standard PyTorch-based
gradients. Our components are a plug-in replacement requiring little change in t
raining or inference procedures that have already been put in place for Transfor
mer-based models thus making our approach easy to incorporate in large industria
l workflows. We experiment on two very different transformer architectures GPT2-
xl (1.5B) and Llama3 (3.2B), where the former we pre-train on OpenWebText and Gn
eissWeb, while the latter we pre-train on the docling data mix which consists of
 nine different datasets. Results show that the performance on downstream classi
fication, Q\& A, reasoning and text understanding tasks of our models is competi
tive and sometimes even superior to the original models with two thirds to half
the parameters and shorter pre-training time. We believe that future implementat
ions customized to hardware will further bring out the true potential of our arc
hitectures.
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Making Learner Weakness Actionable for Learning from Demonstration with Novice T
eachers
Yuqing Zhu, Matthew Howard
Learning from demonstration can be an effective way to teach robots task-oriente
d policies. However, in an interactive setting when demonstrations are limited b
y time or other budgetary constraints, it is challenging to find those that fix
the learner's (remaining) errors. This is especially difficult for novice teache
rs: they may provide task-valid trajectories, often these fail to meaningfully i
mprove the policy due to their lack of knowledge of learning mechanisms internal
 to the robot. This paper introduces CLASP (Collaborative Learning with Anchored
 State-space Partitions), which summarises the teaching process as a compact map
 of behavioural regions anchored in the teacher's own demonstrations. The map co
nnects task failure to actionable changes to demonstrations by indicating what i
s going wrong in an intuitive way. It also enables difficulty-aware training tha
t emphasises regions where learning is failing. Across diverse benchmarks, CLASP
 improves success by up to 20\% over offline and interactive baselines under the
 same demonstration budget, improves robustness under distribution shift by 14–2
0\%, and preserves behavioural diversity.
****

RE-TRAC: REcursive TRAjectory Compression for Deep Search Agents
jialiang zhu, Gongrui Zhang, Xiaolong Ma, Lin Xu, Miaosen Zhang, Ruiqi Yang, Son
g Wang, Kai Qiu, Zhirong Wu, Qi Dai, Ruichun Ma, Bei Liu, Yifan Yang, Chong Luo,
 Zhengyuan Yang, Linjie Li, Lijuan Wang, Weizhu Chen, Xin Geng, Baining Guo
LLM-based deep research agents are largely built on the ReAct framework. This li
near design makes it difficult to revisit earlier states, branch into alternativ
e search directions, or maintain global awareness under long contexts, often lea

ding to local optima, redundant exploration, and inefficient search. We propose Re-TRAC, an agentic framework that performs cross-trajectory exploration by generating a structured state representation after each trajectory to summarize evidence, uncertainties, failures, and future plans, and conditioning subsequent trajectories on this state representation. This enables iterative reflection and globally informed planning, reframing research as a progressive process. Empirical results show that Re-TRAC consistently outperforms ReAct by 15–20% on BrowseComp with frontier LLMs. For smaller models, we introduce Re-TRAC-aware supervised fine-tuning, achieving state-of-the-art performance at comparable scales. Notably, Re-TRAC shows a monotonic reduction in tool calls and token usage across rounds, indicating progressively targeted exploration driven by cross-trajectory reflection rather than redundant search.

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STD-Former: Image-Conditioned Texture Dictionary Encoding with Sparse Topological Supervision for Texture Recognition

Bo Peng, Ke Xu, Yurui Pan

Texture recognition is often framed as matching an image to a static training-set dictionary or codebook. In practice, this assumption is brittle: label-preserving transformations (illumination, scale, compression, blur) can shift test features away from the fixed training dictionary, producing a training-set codebook misalignment that limits accuracy. We propose STD-Former (Simple Texture Dictionary Transformer), a lightweight framework for image-conditioned texture dictionary encoding. Instead of comparing against a static codebook, STD-Former extracts a compact set of Intrinsic Textons (dictionary atoms / codewords) from the input image itself, yielding self-aligned representations at inference. Our design is intentionally simple and uses a decoupled two-stage recipe. In Stage 1, a Texture Dictionary Extractor (TDE) is pre-trained with a self-supervised Texton Coverage Loss that encourages the learned textons to collectively cover the image patch feature manifold. In Stage 2, a classifier is trained on the encoded dictionary representation; optionally, we add a Sparse Topological Loss derived from 0D persistent homology, which is equivalent to supervising only the (B-1) edges of a minimum spanning tree (MST) in each batch, providing efficient structure regularization. Across six standard texture benchmarks, STD-Former and STD-Former+ achieve new state-of-the-art results.

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Dichotomy of Feature Learning and Unlearning: Fast-Slow Analysis on Neural Networks with Stochastic Gradient Descent

Shota Imai, Sota Nishiyama, Masaaki Imaizumi

The dynamics of gradient-based training in neural networks often exhibit nontrivial structures; hence, understanding them remains a central challenge in theoretical machine learning. In particular, the concept of feature unlearning, in which a neural network progressively loses previously learned features over long training, has gained attention. In this study, we consider the infinite-width limit of a two-layer neural network trained with a large-batch stochastic gradient, then derive differential equations with different time scales, revealing the mechanism and conditions for feature unlearning to occur. Specifically, we utilize the fast-slow dynamics: while an alignment of first-layer weights develops rapidly, the second-layer weights develop slowly. The direction of the flow on a critical manifold, determined by the slow dynamics, decides whether feature unlearning occurs. We give numerical validation of the result and derive theoretical grounding and scaling laws for the feature unlearning. Our results yield the following insights: (i) the strength of the primary nonlinear term in the data induces the feature unlearning, and (ii) an initial scale of the second-layer weights mitigates the feature unlearning. Our result should be understood as a population loss of alignment rather than finite-sample overfitting. Technically, our analysis utilizes Tensor Programs and singular perturbation theory.

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Multilingual Safety Alignment via Representation-Space Separability

Dan Shi, Zhuowen Han, Deyi Xiong

Large language models (LLMs) have been globally adopted in various scenarios, ma

king robust multilingual safety alignment a prerequisite for their reliable deployment across diverse languages. Despite recent advances, LLMs exhibit a substantial safety gap between high- and low-resource languages: models that can consistently refuse harmful requests in high-resource languages often fail to do so in low-resource languages. In this work, we reveal that such safety failures stem from insufficient representation-space separability between harmful and harmless prompts in low-resource languages. Through geometric analyses, we find that, compared to English, harmful prompts are significantly less separated from the manifold of harmless prompts, and that the resulting cross-lingual spatial margin gap is strongly correlated with attack success rates. Capitalizing on these insights, we propose Multilingual Spatial Margin Gap-based Optimization (SMO), a novel training strategy that exploits the well-aligned safety geometry of a dominant language (e.g., English) to enhance safety alignment in other languages. SMO explicitly leverages the spatial margin gap between English and target languages as an example-wise supervision signal, enabling effective cross-lingual transfer of safety capabilities while preserving the dominant language’s original performance. Experiments conducted on LLaMA-3.1-8B-Instruct and Qwen2.5-7B-Instruct demonstrate that SMO is capable of substantially reducing attack success rates in low-resource languages to near zero, often reaching zero, while maintaining strong general multilingual performance. Warning: This paper contains content that may be harmful.

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VDW-GNNs: Vector diffusion wavelets for geometric graph neural networks

David R Johnson, Alexander Sietsema, Rishabh Anand, Deanna Needell, Smita Krishnaswamy, Michael Perlmutter

We introduce vector diffusion wavelets (VDWs), a novel family of wavelets inspired by the vector diffusion maps algorithm that was introduced to analyze data lying in the tangent bundle of a Riemannian manifold. We show that these wavelets may be effectively incorporated into a family of geometric graph neural networks, which we refer to as VDW-GNNs. We demonstrate that such networks are effective on synthetic point cloud data, as well as on real-world data derived from wind field and neural activity measurements. Theoretically, we prove that these new wavelets have desirable frame theoretic properties, similar to traditional diffusion wavelets. Additionally, we prove that these wavelets have useful symmetries with respect to rotations and translations.

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Who Gets Credit or Blame? Attributing Accountability in Modern AI Systems

Shichang Zhang, Hongzhe Du, Jiaqi W. Ma, Himabindu Lakkaraju

Modern AI systems are typically developed through multiple stages: pretraining, fine-tuning rounds, and subsequent adaptation or alignment, where each stage builds on the previous ones and updates the model in distinct ways. This raises a critical question of accountability: when a deployed model succeeds or fails, which stage is responsible, and to what extent? We pose the accountability attribution problem for tracing model behavior back to specific stages of the model development process. To address this challenge, we propose a general framework that answers counterfactual questions about stage effects: how would the model's behavior have changed if the updates from a particular stage had not occurred? Within this framework, we introduce estimators that efficiently quantify stage effects without retraining the model, accounting for both the data and key aspects of model optimization dynamics, including learning rate schedules, momentum, and weight decay. We demonstrate that our approach successfully quantifies the accountability of each stage to the model's behavior. Based on the attribution results, our method can identify stages associated with spurious correlations in image classification and text toxicity detection tasks and guide targeted follow-up interventions. Our approach provides a practical tool for model analysis and represents a significant step toward more accountable AI development.

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Contextual Rollout Bandits for Reinforcement Learning with Verifiable Rewards

Xiaodong Lu, Xiaohan Wang, Jiajun Chai, Guojun Yin, Wei Lin, Zhijun Chen, Yu Luo, Fuzhen Zhuang, Yikun Ban, deqing wang

Reinforcement Learning with Verifiable Rewards (RLVR) is an effective paradigm for improving the reasoning capabilities of large language models. However, existing RLVR methods utilize rollouts in an indiscriminate and short-horizon manner: responses of heterogeneous quality within each prompt are treated uniformly, and historical rollouts are discarded after a single use. This leads to noisy supervision, poor sample efficiency, and suboptimal policy updates. We address these issues by formulating rollout scheduling in RLVR as a contextual bandit problem and proposing a unified neural scheduling framework that adaptively selects high-value rollouts throughout training. Each rollout is treated as an arm whose reward is defined by the induced performance gain between consecutive optimization steps. The resulting scheduler supports both noise-aware intra-group selection and adaptive global reuse of historical rollouts within a single principled framework. We provide theoretical justification by deriving sublinear regret bounds and showing that enlarging the rollout buffer improves the achievable performance upper bound. Experiments on six mathematical reasoning benchmarks demonstrate consistent gains in performance and training efficiency across multiple RLVR optimization methods.

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What If We Allocate Test-Time Compute Adaptively?

Ahsan Bilal, Muhammad Ahmed Mohsin, Muhammad Umer, Ali Subhan, Hassan Rizwan, Ayesha Mohsin, Dean F. Hougen

Test-time compute scaling allocates inference computation uniformly, uses fixed sampling strategies, and applies verification only for reranking. In contrast, we propose a verifier-guided adaptive framework treating reasoning as iterative trajectory generation and selection. For each problem, the agent runs multiple inference iterations. In each iteration, it optionally produces a high-level plan, selects a set of reasoning tools and a compute strategy together with an exploration parameter, and then generates a candidate reasoning trajectory. A process reward model (PRM) serves as a unified control signal: within each iteration, step-level PRM scores are aggregated to guide pruning and expansion during generation, and across iterations, aggregated trajectory rewards are used to select the final response. Across datasets, our dynamic, PRM-guided approach consistently outperforms direct test-time scaling, yielding large gains on MATH-500 and several-fold improvements on harder benchmarks such as AIME24 and AMO-Bench. We characterize efficiency using theoretical FLOPs and a compute intensity metric penalizing wasted generation and tool overhead, demonstrating that verification-guided allocation concentrates computation on high-utility reasoning paths.

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Rethinking Low-Confidence Pseudo Labels: Influence-Aware Semi-Supervised Fine-Tuning for Hyperspectral Change Detection

Keyun Zhao, Guangchang Li, Yunpeng Bai, Jiang Shao, Ying Li

Hyperspectral image change detection (HSI-CD) suffers from severe annotation scarcity and complex change patterns, which fundamentally limit the effectiveness of directly fine-tuning pre-trained foundation models. Although semi-supervised learning provides a promising direction, existing approaches mainly rely on confidence-based pseudo-label selection, leading to limited data diversity or severe error propagation. In this paper, we propose Influence-Aware Semi-supervised Fine-tuning (IA-SFT), a novel framework that evaluates the influence of pseudo-labels on model decision behavior to identify truly valuable supervision signals. Instead of confidence-based selection, IA-SFT evaluates each low-confidence pseudo-label by measuring its impact on labeled data, enabling reliable filtering of high-value pseudo-labels with minimal noise. To further adapt foundation models to HSI-CD, we design an Adaptive Fusion Change Decoder (AFCD) that jointly models global semantic consistency and local change details. Extensive experiments on three benchmark datasets demonstrate that IA-SFT consistently improves pseudo-label quality and detection performance, achieving superior accuracy compared to state-of-the-art methods. Additional analyses validate the transferability of IA-SFT when integrated into different frameworks in a plug-and-play manner. Code will be released.

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Utility Boundary of Dataset Distillation: Scaling and Coverage Laws
Zhengquan Luo, zhiqiang xu
Dataset distillation (DD) aims to replace a full training set with a tiny synthetic one, yet current theories neither explain why heterogeneous matching objectives (gradient, distribution, trajectory) work nor provide a quantitative boundary for robustness under configuration changes (optimizer, architecture, augmentation). We propose **configuration-dynamics-error (CDE) analysis** for a broad class of matching-based DD methods, a unified generalization framework that treats a training configuration as an update operator inducing optimization dynamics and measures distillation robustness by the test-risk gap between models trained on distilled versus full data. Within this framework, gradient, distribution, and trajectory matching reduce the same dynamics-induced risk gap, explaining why these heterogeneous objectives can all support dataset distillation. CDE yields two predictive laws: within a fixed configuration, the gap decays as $\mathcal{O}(k^{-1/2})$ with the distilled set size k until a configuration-dependent floor, explaining IPC saturation and indicating when reducing the floor is more valuable than enlarging k. Across configurations, an order-tight coverage law formalizes the utility boundary: the required k grows linearly with the configuration diversity captured by covering-number complexity. Experiments with representative DD methods and configuration changes exhibit predictive behavior consistent with both laws.
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SAD-Flower: Flow Matching for Safe, Admissible, and Dynamically Consistent Planning
Tzu-Yuan Huang, Armin Lederer, Dai-Jie Wu, Xiaobing Dai, Sihua Zhang, Hsiu-Chin Lin, Shao-Hua Sun, Stefan Georg Sosnowski, Sandra Hirche
Flow matching (FM) has shown promising results in data-driven planning. However, it inherently lacks formal guarantees for ensuring state and action constraints, whose satisfaction is a fundamental and crucial requirement for the safety and admissibility of planned trajectories on various systems. Moreover, existing FM planners do not ensure the dynamical consistency, which potentially renders trajectories inexecutable. We address these shortcomings by proposing SAD-Flower, a novel framework for generating $\textbf{S}$afe, $\textbf{A}$dmissible, and $\textbf{D}$ynamically consistent trajectories. Our approach relies on an augmentation of the flow with a virtual control input. Thereby, principled guidance can be derived using techniques from nonlinear control theory, providing formal guarantees for state constraints, action constraints, and dynamic consistency. Crucially, SAD-Flower operates without retraining, enabling test-time satisfaction of unseen constraints. Through extensive experiments across several tasks, we demonstrate that SAD-Flower outperforms various generative-model-based baselines in ensuring constraint satisfaction. Video and demos can be found at [sadflowerplanning.github.io](https://sadflowerplanning.github.io/project_web/).
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LAGEA: Language Guided Embodied Agents for Robotic Manipulation
Abdul Monaf Chowdhury, Akm Moshiur Rahman Mazumder, Safaeid Hossain Arib, Rabeya Akter
Robotic manipulation benefits from foundation models that describe goals, but today's agents still lack a principled way to learn from their own mistakes. We ask whether natural language can serve as feedback, an error-reasoning signal that helps embodied agents diagnose what went wrong and correct course. We introduce LAGEA (Language Guided Embodied Agents), a framework that turns episodic, schema-constrained reflections from a vision language model (VLM) into temporally grounded guidance for reinforcement learning. LAGEA summarizes each attempt in concise language, localizes the decisive moments in the trajectory, aligns feedback with visual state in a shared representation, and converts goal progress and feedback agreement into bounded, step-wise shaping rewardswhose influence is modulated by an adaptive, failure-aware coefficient. This design yields dense signals early when exploration needs direction and gracefully recedes as competence grows. On the Meta-World MT10 and Robotic Fetch embodied manipulation benchmarks, LAGEA improves average success over the state-of-the-art (SOTA) methods by 9.0% on

random goals, 5.3% on fixed goals, and 17% on fetch tasks, while converging faster. These results support our hypothesis: language, when structured and grounded in time, is an effective mechanism for teaching robots to self-reflect on mistakes and make better choices. Code will be released soon.

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Transferable Reinforcement Learning via Probabilistic Latent Embeddings and Dynamic Policy Adaptation for Sim-to-Real Deployment

Gengyue Han, Yiheng Feng

Due to limited resources and public safety concerns, deep reinforcement learning (RL) agents for many cyber-physical systems (e.g., autonomous vehicles) are first trained in simulators. However, when deployed in real world environments, they often suffer from performance degradation or safety violations because of the inevitable Sim2Real gap. Existing zero-shot approaches, such as robust safe RL and domain randomization, mitigate this issue but typically at the cost of degraded performance or residual safety risks when experiencing unmodeled system dynamics. To address these limitations, we propose a novel reinforcement learning framework that enables safe and efficient policy transfer via probabilistic latent embeddings and dynamic policy adaptation. We consider a family of Constrained Markov Decision Processes (CMDPs) under different environment contexts. By leveraging latent context variable in meta-RL, the proposed framework infers the latent representation of the environment from simulated experiences. Furthermore, it incorporates a distributional RL formulation, which allows risk levels of the deployed policy to be adjusted dynamically, based on the estimation accuracy of the latent context variable. This strategy promotes safety at the early deployment stage and improves efficiency through fast policy adaptation under the Sim2Real gap.

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WildActor: Unconstrained Identity-Preserving Video Generation

Qin Guo, Tianyu Yang, Xuanhua He, Fei Shen, Yong Zhang, Zhuoliang Kang, Xiaoming Wei, Dan Xu

Production-ready human video generation requires digital actors to maintain strictly consistent full-body identities across dynamic shots, viewpoints and motions, a setting that remains challenging for existing methods. Prior methods often suffer from face-centric behavior that neglects body-level consistency, or produce copy-paste artifacts where subjects appear rigid due to pose locking. We present Actor-18M, a large-scale human video dataset designed to capture identity consistency under unconstrained viewpoints and environments. Actor-18M comprises 1.6M videos with 18M corresponding human images, covering both arbitrary views and canonical three-view representations. Leveraging Actor-18M, we propose WildActor, a framework for any-view conditioned human video generation. We introduce an Asymmetric Identity-Preserving Attention mechanism coupled with a Viewpoint-Adaptive Monte Carlo Sampling strategy that iteratively re-weights reference conditions by marginal utility for balanced manifold coverage. Evaluated on the proposed Actor-Bench, WildActor consistently preserves body identity under diverse shot compositions, large viewpoint transitions, and substantial motions, surpassing existing methods in these challenging settings.

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CoCoEmo: Composable and Controllable Human-Like Emotional TTS via Activation Steering

Siyi Wang, Shihong Tan, Siyi Liu, Hong Jia, Gongping Huang, James Bailey, Ting Dang

Emotional expression in human speech is nuanced and compositional, often involving multiple, sometimes conflicting, affective cues that may diverge from linguistic content. In contrast, most expressive text-to-speech (TTS) systems enforce a single utterance-level emotion, collapsing affective diversity and suppressing mixed or text–emotion–misaligned expression.

While activation steering via latent direction vectors offers a promising solution, it remains unclear whether emotion representations are linearly steerable in TTS, where steering should be applied within hybrid TTS architectures, and how such complex emotion behaviors should be evaluated.

This paper presents the first systematic analysis of activation steering for emo
tional control in hybrid TTS models, introducing a quantitative, controllable st
eering framework, and multi-rater evaluation protocols that enable composable mi
xed-emotion synthesis and reliable text–emotion mismatch synthesis.
Our results demonstrate, for the first time, that emotional prosody and expressi
ve variability are primarily synthesized by the TTS language module instead of t
he flow-matching module, and also provide a lightweight steering approach for ge
nerating natural, human-like emotional speech.
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FUSE: Frequency-domain Unification and Spectral Energy Alignment for Multi-modal
 Object Re-Identification

Xuanhao Qi, Tom H. Luan, Yukang Zhang, Jinkai Zheng, su zhou, Shuwei Li, Lei Tan
Despite significant progress in multi-modal Re-Identification (ReID), existing m
ethods tend to emphasize low-frequency cues. Consequently, they focus on attribu
tes such as color, illumination, and coarse appearance, while overlooking mid- a
nd high-frequency structures that encode geometric, textural, and identity-discr
iminative details. This imbalance leads to incomplete spectral representations a
nd unstable cross-modal alignment. To overcome these limitations, we introduce F
USE, a frequency-domain framework that reformulates multi-modal ReID as a two-st
age process of spectral disentanglement and energy alignment. The proposed Spect
ral Decomposition Module (SDM) adaptively partitions features into low, mid, and
 high-frequency subspaces, enabling hierarchical spectral modeling. The Cross-Mo
dal Alignment Module (CAM) further enforces energy alignment and subspace comple
mentarity across modalities via frequency-consistency regularization. In additio
n, FUSE incorporates learnable frequency modulation to enhance robustness under
varying illumination and heterogeneous sensor conditions. Extensive experiments
on RGBNT201, RGBNT100, and MSVR310 show that FUSE achieves 9.1% mAP and 9.5% Ran
k-1 improvements, establishing an interpretable frequency-domain paradigm for mu
lti-modal representation learning.
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FRIGID: Scaling Diffusion-Based Molecular Generation from Mass Spectra at Traini
ng and Inference Time

Montgomery Bohde, Hongxuan Liu, Mrunali Manjrekar, Magdalena Lederbauer, Shuiwan
g Ji, Runzhong Wang, Connor W. Coley
Tandem mass spectrometry is prominent in scientific discovery workflows for iden
tifying unknown small molecules, yet high-throughput structural elucidation rema
ins challenging. While recent autoregressive and graph diffusion models have sho
wn promise in *de novo* elucidation, performance remains limited by poor scalabi
lity during both training and inference time. In this work, we present FRIGID, a
 framework with a novel diffusion language model that generates molecular struct
ures conditioned on mass spectra via intermediate fingerprint representations an
d determined chemical formulae, training at the scale of hundreds of millions of
 unlabeled structures. We then demonstrate how forward fragmentation models enab
le inference-time scaling by identifying spectrum-inconsistent fragments and ref
ining them through targeted remasking and denoising. While FRIGID already achiev
es strong performance with its diffusion base, inference-time scaling significan
tly improves its accuracy, surpassing 18% Top-1 accuracy on the challenging Mass
SpecGym benchmark and tripling the Top-1 accuracy of the leading methods on NPLI
B1. Further empirical analyses show that FRIGID exhibits log-linear performance
scaling with increasing inference-time compute, opening a promising new directio
n for continued improvements in *de novo* structural elucidation. FRIGID code is
 publicly available at https://github.com/coleygroup/FRIGID.
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Cure-SFT: Diagnostic-Guided Data Curation for Instruction Tuning

Yuankang Fu, Xinrong Gong, Chen Gong, Tong Zhang, Kaixiang Yang
Instruction data curation is central to improving the instruction-following abil
ity of large language models. However, existing approaches often struggle to sim
ultaneously maintain data quality, diversity, and distributional consistency, la
rgely because they do not explicitly distinguish semantic redundancy from qualit
y defects and rely on coarse-grained modeling of instruction data quality. To ad

dress this issue, we propose Cure-SFT, a coarse-to-fine, diagnostic-guided metho d for instruction data curation that explicitly disentangles semantic redundancy from quality defects. Specifically, Cure-SFT removes redundant samples via stra tified semantic-geometric sampling, applies teacher models for diagnostic triage , and performs targeted defect remediation on fixable samples. Our experiments s how that Cure-SFT can surpass full-data instruction tuning using only 10% of the data budget. Moreover, Cure-SFT consistently outperforms strong selection-based and rewriting-based baselines across data budgets, supporting the effectiveness of diagnostic-guided data curation.

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PISCES: Annotation-free Text-to-Video Post-Training via Optimal Transport-Aligne d Rewards

Minh-Quan Le, Gaurav Mittal, Cheng Zhao, David Gu, Dimitris Samaras, Mei Chen

Text-to-video (T2V) generation aims to synthesize videos with high visual qualit y and temporal consistency that are semantically aligned with input text. Reward -based post-training has emerged as a promising direction to improve the quality and semantic alignment of generated videos. However, recent methods either rely on large-scale human preference annotations or operate on misaligned embeddings from pre-trained vision-language models, leading to limited scalability or subo ptimal supervision. We present PISCES, an annotation-free post-training algorith m that addresses these limitations via a novel Dual Optimal Transport (OT)-align ed Rewards module. To align reward signals with human judgment, PISCES uses OT t o bridge text and video embeddings at both distributional and discrete token lev els, enabling reward supervision to fulfill two objectives: (i) a Distributional OT-aligned Quality Reward that captures overall visual quality and temporal coh erence; and (ii) a Discrete Token-level OT-aligned Semantic Reward that enforces semantic, spatio-temporal correspondence between text and video tokens. To our knowledge, PISCES is the first to improve annotation-free reward supervision in generative post-training through the lens of OT. Experiments on both short- and long-video generation show that PISCES outperforms both annotation-based and ann otation-free methods on VBench across Quality and Semantic scores, with human pr eference studies further validating its effectiveness. We show that the Dual OT-aligned Rewards module is compatible with multiple optimization paradigms, inclu ding direct backpropagation and reinforcement learning fine-tuning.

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Attentive Multi-Layer Fusion for Vision Transformers

Laure Ciernik, Marco Morik, Lukas Thede, Luca Eyring, Shinichi Nakajima, Zeynep Akata, Lukas Muttenthaler

With the rise of large-scale foundation models, efficiently adapting them to dow nstream tasks remains a central challenge. Linear probing, which freezes the bac kbone and trains a lightweight head, is computationally efficient but often rest ricted to last-layer representations. We show that task-relevant information is distributed across the network hierarchy rather than encoded solely in the last layers. To leverage this distribution of information, we apply an attentive prob ing mechanism that dynamically fuses representations from all layers of a Vision Transformer. This attentive layer fusion (ALF) learns to identify the most rele vant layers for a target task and combines low-level structural cues with high-l evel semantic abstractions. Across 20 diverse datasets and multiple pretrained f oundation models, ALF achieves consistent, substantial gains over standard linea r probes. Attention heatmaps further reveal that tasks different from the pre-tr aining domain benefit most from intermediate representations. Overall, our findi ngs underscore the value of intermediate layers and demonstrate a principled, ta sk-aware approach for unlocking their potential for probing-based adaptation.

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Vector Linking via Cross-Model Local Isometric Consistency

Ziying Chen, Yang Cao, He Sun, Beining Yang, Tianjian Yang

We study Vector Linking: given two embedding clouds produced by different black-box encoders over partially overlapping datasets, recover cross-model object cor respondences using only vectors. Empirically and theoretically, we show that ind ependently trained contrastive encoders exhibit local geometric consistency: sho

rt-range distances are approximately preserved up to a scale factor, while long-range distances are not due to model-specific distortion. Building on this, we propose an iterative, reference-based geometric embedding hashing that recovers vector links from a tiny seed set of paired anchors. It represents each vector by distances to sampled paired anchors, proposes candidate links via hash-space matching, and aggregates evidence across views in a Beta--Bernoulli posterior to bootstrap high-confidence links as new anchors. Experiments across multiple benchmarks and embedding model pairs demonstrate accurate and robust linking under varying overlap, seed budgets, and out-of-domain references, with applications to vector database integration and cross-model clustering. Code is available at \url{https://github.com/DBgroup-Edinburgh/VecLinking}.

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Tightening the Score Matching Gap for Diffusion Models

Benjamin Dupuis, Tyler Farghly, Maxime Haddouche, Alain Oliviero Durmus, Umut Simsekli

Diffusion models (DMs) are a state-of-the-art generative method to approximately sample from an unknown distribution. Their training and evaluation primarily rely on an Evidence Lower Bound (ELBO), which relates the Kullback-Leibler (KL) divergence of model samples to the score matching loss along the path, which serves as a tractable surrogate. The difference between sample quality and the score matching loss produced by this bound leads to the score matching gap, which is known to be tight in the worst-case but not descriptive of sample quality in general. In this work, we provide a theoretical analysis of this gap, developing tighter bounds for three metrics: KL divergence, reverse KL divergence, and Wasserstein distance, effectively exploiting the regularity of the class of score estimators. Our results suggest that the quality of the score approximation has more impact on closing the score matching gap for low noise scales. To obtain these bounds, our key technical insight is to exploit the contraction properties of the backward processes. In particular, we rely on entropy flows, logarithmic Sobolev inequalities and reflection couplings, rigorously linking the ergodicity of the Langevin diffusion to the score matching gap problem.

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Any-dimensional invariant universality

Shengtai Yao, Eitan Levin, Mateo Diaz Diaz

Several machine learning models are defined for inputs of any size, such as graphs with different numbers of nodes and point clouds with varying numbers of points. The universality properties of such any-dimensional models remain poorly understood, as universality is traditionally studied for models accepting inputs of a fixed size, defined on a compact subset of their domain. In sharp contrast, any-dimensional models can be viewed as sequences of functions defined on growing-sized inputs, and it is not clear in which sense they can be universal. We develop a systematic approach to establish any-dimensional universality by identifying any-dimensional functions with a unique function that takes inputs in a suitable infinite-dimensional limit space containing inputs of all finite sizes, as well as their limits. Using the symmetries of these inputs and relations between inputs of different sizes, we show that this limit space admits a natural topology with rich families of compact sets on which any-dimensional universality can be established. We illustrate our approach by showing that several existing architectures fail to be universal, and we propose simple modifications that restore universality.

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When the Prompt Becomes Visual: Vision-Centric Jailbreak Attacks for Large Image Editing Models

Jiacheng Hou, Yining Sun, Ruochong Jin, Haochen Han, Fangming Liu, Wai Kin Victor Chan, Alex Jinpeng Wang

Recent advances in large image editing models have shifted the paradigm from text-driven instructions to vision-prompt editing, where user intent is inferred directly from visual inputs such as marks, arrows, and visual–text prompts. While this paradigm greatly expands usability, it also introduces a critical and underexplored safety risk: the attack surface itself becomes visual. In this work, we

propose Vision-Centric Jailbreak Attack (VJA), the first visual-to-visual jailbreak attack that conveys malicious instructions purely through visual inputs. To systematically study this emerging threat, we introduce IESBench, a safety-oriented benchmark for image editing models. Extensive experiments on IESBench demonstrate that VJA effectively compromises state-of-the-art commercial models, achieving attack success rates of up to 80.9% on Nano Banana Pro and 70.1% on GPT-Image-1.5. To mitigate this vulnerability, we propose a training-free defense based on introspective multimodal reasoning, which substantially improves the safety of poorly aligned models to a level comparable with commercial systems, without auxiliary guard models and with negligible computational overhead. Our findings expose new vulnerabilities, provide both a benchmark and practical defense to advance safe and trustworthy modern image editing systems.

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Self-Prompting Diffusion Transformer for Open-Vocabulary Scene Text Edit via In-Context Learning

Hongxi Li, Tong Wang, WU CHENGJING, Tianbao Liu, Jiangtao Yao, Xiaochao Qu, Xinxiao Wu, Luoqi Liu, Ting Liu

Scene text editing aims to modify text in a target region of an image while preserving its background style and texture.
Existing methods rely solely on image background information while neglecting the visual details of target regions,
which discards stylistic features in the original text and essentially degrades the task to text rendering.
Moreover, the conditions imposed by pre-trained glyph encoder limit the scope of editable text.
To address these issues, this paper proposes a self-prompting scene text editing method, which constructs style and glyph prompts directly from the original image without additional style or glyph encoders.
We employ a two-stage training strategy, where the diffusion transformer is first trained on large-scale self-supervised datasets and subsequently refined with a small set of paired images.
By leveraging the in-context learning capability of FLUX-Fill, it achieves open-vocabulary and style-consistent text editing.
Experimental results on various languages demonstrate that our method achieves the state-of-the-art performance in both text accuracy and style consistency.

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Inference-time Alignment with Rewards in Besov Spaces: Provable Advantages of Feature Learning and Multi-Step Policy Updates

Naoki Nishikawa, Taiji Suzuki

Inference-time alignment, the approach of adapting pre-trained models to reward feedback during inference, has proven empirically effective at improving language-model performance. Despite its success, theoretical foundations remain underdeveloped, especially in practical settings where neural networks are employed as reward models. In this paper, we explore the advantages of neural networks and how to effectively train them for inference-time alignment. Assuming that the true reward function lies in Besov spaces to capture the non-uniform smoothness, we compare neural networks to linear estimators and show that feature learning capability of neural networks is crucial for improving performance. We further analyze algorithms for training neural-network reward estimators. Specifically, we consider a multi-step algorithm that alternates between sampling from the current policy and refitting the reward estimator, and prove that it improves the regret, especially when the true reward exhibits local structure.

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Next-Gen CAPTCHAs: Leveraging the Cognitive Gap for Scalable and Diverse GUI-Agent Defense

Jiacheng Liu, Yaxin Luo, Jiacheng Cui, Xinyi Shang, Xiaohan Zhao, Zhiqiang Shen

The rapid evolution of GUI-enabled agents has rendered traditional CAPTCHAs obsolete. While previous benchmarks like OpenCaptchaWorld established a baseline for evaluating multimodal agents, recent advancements in reasoning-heavy models, such as Gemini3-Pro-High and GPT-5.2-Xhigh have effectively collapsed this securit

y barrier, achieving pass rates as high as 90\% on complex logic puzzles like '' Bingo''. In response, we introduce Next-Gen CAPTCHAs, a scalable defense framewo rk designed to secure the next-generation web against the advanced agents. Unli ke static datasets, our benchmark is built upon a robust data generation pipelin e, allowing for large-scale and easily scalable evaluations, notably, for backen d-supported types, our system is capable of generating effectively unbounded CAP TCHA instances. We exploit the persistent human--agent ``Cognitive Gap'' in inte ractive perception, memory, decision-making, and action. By engineering dynamic tasks that require adaptive intuition rather than granular planning, we re-estab lish a robust distinction between biological users and artificial agents, offeri ng a scalable and diverse defense mechanism for the agentic era.

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Segment-driven Structural Induction and Semantic Alignment for Heterogeneous Tab ular Representation

Woojun Jung, Susik Yoon

Real-world domains often contain heterogeneous tables whose headers vary while t heir underlying attribute semantics are shared, making it difficult to induce do main-specialized semantics from table-local evidence alone. Existing encoders mo del parts of this problem, but often underuse column-level value distributions a nd apply uniform objectives across attributes with different semantic roles. We propose NAVI, a segment-centric pretraining framework that treats each header–va lue pair as the unit for aggregating schema-level structural evidence and column -level distributional evidence. We realize this design through Masked Segment Mo deling and Entropy-driven Segment Alignment, which jointly enforce structured he ader–value coupling and semantic alignment across stable and instance-specific a ttributes. Experiments on heterogeneous in-domain tables show improved reconstru ction, semantic consistency, and downstream utility across evaluation settings o verall.

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FunCQNet: A Functional Censored Quantile Neural Network for Predicting Long-Term Post-Transplant Kidney Survival

Jiaqi Men, Hua Liu, Yiming Tang, Jinhong You, Jianghu James Dong, Jiguo Cao

Accurate survival prediction in kidney transplantation is critical yet challengi ng due to the complex interplay between functional biomarkers and patient charac teristics under censoring. To address this, we propose a functional censored qua ntile neural network (FunCQNet), a novel framework that integrates deep neural n etworks with a censoring-adjusted sequential quantile loss to approximate intera ction-dependent coefficient functions. We further introduce a conformal inferenc e approach to rigorously assess the significance of functional-scalar interactio ns, ensuring interpretability alongside predictive power. Extensive simulations demonstrate that FunCQNet robustly recovers functional effects under varying noi se and censoring levels. When applied to kidney transplant data, the model yield s precise multi-quantile predictions and reveals clinically significant, age-dep endent interaction patterns between donor type and recipient survival.

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Understanding Truncated Positional Encodings for Graph Neural Networks

James Flora, Mitchell Black, Weng-Keen Wong, Amir Nayyeri

Positional encodings (PEs) enhance the power of graph neural networks (GNNs), bo th theoretically and empirically. Two of the most popular families of PEs---spec tral (e.g., Laplacian eigenspaces, effective resistance) and random walk (polyno mials of the adjacency matrix)---are theoretically equivalent in expressive powe r, and both are known to lie between the 1-WL and 3-WL tests in terms of express ivity. However, this equivalence assumes the GNN uses the "complete'' version of these PEs, which requires $O(n^3)$ time and space complexity. Practitioners the refore commonly use truncated variants of these encodings (e.g., the first k e igenspaces or powers of adjacency matrix). However, the theoretical properties o f these truncated PEs are unknown. In this work, we initiate the study of these truncated PEs. Theoretically, we show that, under truncation, several families o f PEs are fundamentally different in expressive power. As a corollary, we show t hat truncated spectral PEs are no longer stronger than the 1-WL test. We also st

udy a family of spectral PEs, the k-harmonic distances, to highlight the differences in expressive power of even closely related truncated PEs. Finally, we experimentally show that a mix of truncated PEs is preferable to any single family on real-world datasets.
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Nonconvex Low-Rank Tensor Representation with Deep Priors for Multiview Subspace Clustering
Yao Fu, Dong Hu, Zhi Wang
Multiview subspace clustering (MvSC) has shown remarkable potential in exploring underlying structures of high-dimensional data. However, existing MvSC methods still suffer from two shortcomings: (1) the commonly use of convex low-rank approximations inadequately capture high-order correlations across views, while sensitivity to noise and outliers degrades clustering performance, and (2) they lack the ability to preserve global correlations and local geometric patterns simultaneously. To address these issues, we propose a novel nonconvex regularized MvSC model with deep prior, which not only accurately characterizes the intrinsic low-rank structure and suppresses the effect of outliers, but also preserves local structural properties through deep networks. By mathematically analyzing the optimal solution of the optimization problem in our proposed model, we develop an efficient ADMM-based algorithm with provable convergence guarantees to solve it. Extensive experiments on various datasets demonstrate the superiority of the proposed model.
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GO-PRE: Goal-Oriented Next-Best-View Selection via Predictive Rendering Entropy for Active 3D Reconstruction
Yan Song, Zhihao Li, Chenglong Li, Li He, Yan Wang, Wenqiang Zhang
Active 3D reconstruction relies on active view selection to maximize reconstruction fidelity under limited capture budgets. However, most existing methods rely on surrogate signals—such as parameter uncertainty or geometric heuristics—which are often misaligned with the ultimate goal: the fidelity of rendered predictions. We propose GO-PRE, a goal-oriented next-best-view selection framework that explicitly targets information gain in the prediction space. Specifically, we formulate the objective as maximizing the reduction of the average marginal predictive entropy over a user-specified target view manifold. GO-PRE supports interactive goal specification and yields an efficient acquisition rule that enables real-time computation of information gain. Extensive experiments across benchmarks demonstrate that GO-PRE consistently improves active reconstruction performance and provides more reliable uncertainty quantification compared to state-of-the-art methods.
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Referring Multiple Regions with Large Multimodal Models via Contextual Latent Steering
Yun Xing, Hanyuan Liu, Jiahao Nie, Shijian Lu
Large Multimodal Models (LMMs) have recently demonstrated their proficiency in holistic visual comprehension. However, most of them struggle to tackle region-level perception guided by visual prompts, especially for cases where multiple regions are referred simultaneously, or scenarios where global contexts are necessary for precise visual referring. We introduce Contextual Latent Steering (CSteer), a training-free approach for guiding general LMMs to refer multiple regions contextually, without expensive fine-tuning or architectural modifications. CSteer starts with pre-computing contextual vectors that implicitly represent visual referring behaviors, such as differentiation among regions and attention to global contexts, followed by representation editing during inference time. Experimental results on multiple datasets indicate that general LMMs with CSteer outperform referring LMMs in most cases, suggesting a promising solution in training-free, and setting new state-of-the-art for this field. Code is available at https://github.com/xing0047/csteer.git.
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Learning to Remember, Learn, and Forget in Attention-Based Models
Djohan Bonnet, Jamie Lohoff, Jan Finkbeiner, Elidona Shiqerukaj, Emre Neftci

In-Context Learning (ICL) in transformers acts as an online associative memory and is believed to underpin their high performance on complex sequence processing tasks.
However, in gated linear attention models, this memory has a fixed capacity and is prone to interference, especially for long sequences.
We propose Palimpsa, a self-attention model that views ICL as a continual learning problem that must address a stability-plasticity dilemma.
Palimpsa uses Bayesian metaplasticity, where the plasticity of each attention state is tied to an importance state grounded by a prior distribution that captures accumulated knowledge.
We demonstrate that various gated linear attention models emerge as specific architecture choices and posterior approximations, and that Mamba2 is a special case of Palimpsa where forgetting dominates.
This theoretical link enables the transformation of any non-metaplastic model into a metaplastic one, significantly expanding its memory capacity.
Our experiments show that Palimpsa consistently outperforms baselines on the Multi-Query Associative Recall (MQAR) benchmark and on Commonsense Reasoning tasks.
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Stage-wise Distortion–Perception Traversal in Zero-shot Inverse Problems with Diffusion Models
Jiawei Zhang, Ziyuan Liu, Leon Yan, Zhenyu Xiao, Yuantao Gu
The distortion–perception (D–P) tradeoff is a fundamental phenomenon of Bayesian inverse problems, which characterizes the inherent tension between distortion performance and perceptual quality. Enabling flexible traversal of the D-P tradeoff at inference time is crucial for practical applications. Despite the recent success of diffusion models in zero-shot inverse problem solving, efficient and principled strategies for D-P traversal in diffusion-based inverse algorithms remain inadequately characterized. In this paper, we propose a stage-wise framework for realizing D-P traversal using a single diffusion model in zero-shot inverse problems. Our proposed method, termed MAP-RPS, starts with an MAP estimation stage that approximates the MMSE solution and provides a low-distortion initialization, followed by a re-noised posterior sampling stage that progressively improves perceptual quality. We provide theoretical analyses for both stages, establishing the validity and effectiveness of the proposed design. Furthermore, we extend MAP-RPS to the latent space, yielding LMAP-RPS, which enjoys broader applicability by leveraging large-scale pre-trained latent diffusion backbones. Extensive experiments demonstrate that MAP-RPS and LMAP-RPS enable more effective D-P traversal on various tasks, while also exhibiting strong performance as efficient solvers for real-world inverse problems.
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CoMem: Context Management with A Decoupled Long-Context Model
Yuwei Zhang, Chengyu Dong, Shuowei Jin, Changlong Yu, Hejie Cui, Hongye Jin, Xinyang Zhang, Hamed Bonab, Colin Lockard, Jianshu Chen, Zhenyu Shi, Jingbo Shang, Xian Li, Bing Yin
Context management enables agentic models to solve long-horizon tasks through iterative summarization of previous interaction histories.
However, this process typically incurs substantial decoding overhead for the extra summarization tokens, which significantly affect the end-to-end response latency at deployment.
In this paper, we introduce CoMem, a novel framework that decouples memory management from the primary agent workflow, enabling these processes to execute in parallel. We propose a k-step-off asynchronous pipeline that overlaps the memory model's summarization with the agent's inference, effectively masking the latency of context processing. To ensure robustness under this asynchronous setting, we introduce a reward-driven training strategy that aligns the memory model to capture sufficient statistics for the agent's decision-making. Theoretical analysis confirms that CoMem offers a superior efficiency-effectiveness trade-off compared to coupled architectures.
Our extensive experimental results on SWE-Bench-Verified show that CoMem provides 1.4x latency improvements upon vanilla long-context solutions while preserving

most of the performance.
Furthermore, we demonstrate that these latency gains scale favorably with increa sed system throughput, offering a modular path forward for the independent optim ization of agent reasoning and memory compression.
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On the Effect of Misspecifying the Embedding Dimension in Low-rank Network Model s
Roddy Taing, Keith Levin
As network data has become ubiquitous in the sciences, there has been growing in terest in network models whose structure is driven by latent node-level variable s in a (typically low-dimensional) latent geometric space. These "latent positio ns" are often estimated via embeddings, whereby the nodes of a network are mappe d to points in Euclidean space so that "similar" nodes are mapped to nearby poi nts. Under certain model assumptions, these embeddings are consistent estimates of the latent positions, but most such results require the embedding dimension t o be chosen correctly. Methods for choosing the embedding dimension have been st udied extensively, but little is known about the behavior of embeddings when the dimension is misspecified. In this work, we provide a theoretical description o f the effects of dimension misspecification under the random dot product graph, a class of latent space network models that includes several widely-used network models, most notably the stochastic blockmodel, as special cases. We show that when the dimension is chosen too large, consistent estimation still holds, albei t at a slower rate than when the embedding dimension is chosen correctly. On the other hand, when the dimension is chosen too small, there is a fundamental esti mation error lower bound that need not go to zero in the large-network limit. A range of synthetic data experiments support our theoretical results. Our main te chnical result, which may be of independent interest, is a generalization of ear lier work in random matrix theory showing that all non-signal eigenvectors of a low-rank matrix subject to additive noise are delocalized.
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AnyBand-Diff: A Unified Remote Sensing Image Generation and Band Repair Framewor k with Spectral Priors
Zhao zuopeng, Ying Liu, xiaoyu Li, Su Luo, Lu Li, Wenwen Liu
Existing diffusion models have made significant progress in generating realistic images. However, their direct adaptation to remote sensing imagery often disreg ards intrinsic physical laws. This oversight frequently leads to spectral distor tion and radiometric inconsistency, severely limiting the scientific utility of generated data. To address this issue, this paper introduces AnyBand-Diff, a nov el spectral-prior-guided diffusion framework tailored for robust spectral recons truction. Specifically, we design a Masked Conditional Diffusion backbone integr ated with a dual stochastic masking strategy, empowering the model to recover co mplete spectral information from arbitrary band subsets. Subsequently, to ensure radiometric fidelity, a Physics-Guided Sampling mechanism is proposed, leveragi ng gradients from a differentiable physical model to explicitly steer the denois ing trajectory toward the manifold of physically plausible solutions. Furthermor e, a Multi-Scale Physical Loss is formulated to enforce rigorous constraints acr oss pixel, region, and global levels in a joint manner. Extensive experiments c onfirm the effectiveness of AnyBand-Diff in generating reliable imagery and achi eving accurate spectral reconstruction, contributing to the advancement of physi cs-aware generative methods for Earth observation.
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With Argus Eyes: Assessing Retrieval Gaps via Uncertainty Scoring to Detect and Remedy Retrieval Blind Spots
Zeinab Sadat Taghavi, Ali Modarressi, Hinrich Schuetze, Andreas Marfurt
Reliable retrieval-augmented generation (RAG) systems depend fundamentally on th e retriever’s ability to find relevant information. We show that neural retrieve rs used in RAG systems have blind spots, which we define as the failure to retri eve entities that are relevant to the query, but have low similarity to the quer y embedding. We investigate the training-induced biases that cause such blind-sp ot entities to be mapped to inaccessible parts of the embedding space, resulting

in low retrievability.

Using a large-scale dataset constructed from Wikidata relations and first paragraphs of Wikipedia, and our proposed Retrieval Probability Score (RPS), we show that blind spot risk in standard retrievers (e.g., Contriever, ReasonIR) can be predicted pre-index from entity embedding geometry, avoiding expensive retrieval evaluations.

To address these blind spots, we introduce ARGUS, a pipeline that enables the retrievability of high-risk (low-RPS) entities through targeted document augmentation from a knowledge base (KB), first paragraphs of Wikipedia, in our case. Extensive experiments on BRIGHT, ImpliRet, and RAR-b show that ARGUS achieves consistent improvements across all evaluated retrievers (averaging +3.4 nDCG@5 and +4.5 nDCG@10 absolute points), with substantially larger gains in challenging subsets. These results establish that preemptively remedying blind spots is critical for building robust and trustworthy RAG systems (Code and data: github.com/ZeinabTaghavi/With_Argus_Eyes)
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Uncovering Hidden Triggers: Backdoor Attribution in Language Models
Miao Yu, Zhenhong Zhou, Moayad Aloqaily, Kun Wang, Biwei Huang, Stephen Wang, Yueming Jin, Qingsong Wen
Fine-tuned Large Language Models (LLMs) are vulnerable to backdoor attacks through data poisoning, yet the internal mechanisms governing these attacks remain a black box. Previous research on interpretability for LLM safety tends to focus on alignment, jailbreak, and hallucination, but overlooks backdoor mechanisms, making it difficult to understand and fully eliminate the backdoor threat. In this paper, aiming to bridge this gap, we explore the interpretable mechanisms of LLM backdoors through Backdoor Attribution (BkdAttr), a tripartite causal analysis framework. We first introduce the Backdoor Probe that proves the existence of learnable backdoor features encoded within the representations. Building on this insight, we further develop Backdoor Attention Head Attribution (BAHA), efficiently pinpointing the specific attention heads responsible for processing these features. Our primary experiments reveals these heads are relatively sparse; ablating a minimal \textbf{$\sim$ 3%} of total heads is sufficient to reduce the Attack Success Rate (ASR) by \textbf{over 90%}. More importantly, we further employ these findings to construct the Backdoor Vector derived from these attributed heads as a master controller for the backdoor. Through only \textbf{1-point} intervention on \textbf{single} representation, the vector can either boost ASR up to \textbf{$\sim$ 100% $\uparrow$} on clean inputs, or completely neutralize backdoor, suppressing ASR down to \textbf{
$\sim$ 0%} on triggered inputs. In conclusion, our work pioneers the exploration of mechanistic interpretability in LLM backdoors, demonstrating a powerful method for backdoor control and revealing actionable insights for the community.
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PLaID++: A Preference Aligned Language Model for Targeted Inorganic Materials Design
Andy Xu, Rohan Desai, Larry Wang, Ethan T. Ritz, Gabriel Hope
Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a promising approach to improve correctness in LLMs, however, in many scientific problems, the objective is not necessarily to produce *the* correct answer, but instead to produce a diverse array of candidates which satisfy a set of constraints. We study this challenge in the context of materials generation. To this end, we introduce PLaID++, an LLM post-trained for stable and property-guided crystal generation. We find that applying naive preference optimization to a coordinate-based crystal representation leads to mode collapse. Hence, we introduce a compact, symmetry-informed Wyckoff text representation which improves computational efficiency and encourages generalization from physical priors. By encoding symmetry constraints directly into text and guiding model outputs towards desirable chemical space, PLaID++ generates structures that are thermodynamically stable, unique, and novel at a $>$50\% greater rate than prior methods. We further demonstrate t

hat unified training across conditional and unconditional tasks are mutually ben
eficial in data-sparse regimes. Our work demonstrates the potential of adapting
post-training techniques from natural language processing to materials design, p
aving the way for targeted and efficient discovery of novel materials.
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GuidedBridge: Training-freely Improving Bridge Models with Prior Guidance
Zehua Chen, Yucheng Yang, Binjie Yuan, Kaiwen Zheng, Jun S. Liu, Jun Zhu
Guidance methods, such as classifier-free guidance (CFG) and auto-guidance (AG),
 have advanced noise-to-data generation in diffusion models.
Recently, bridge models have introduced a data-to-data generative process that c
an exploit an instructive clean prior.
In this work, inspired by previous methods creating quality difference between d
enoising results as guidance, we propose a training-free bridge guidance method,
 termed Prior Guidance (PG).
Specifically, we introduce a weak prior, which is unseen during bridge pre-train
ing, hindering prior exploitation and thereby degrading denoising result.
Then, we contrast it with the seen prior to highlight and enhance prior exploita
tion via a scaling factor.
Moreover, we analyze the underlying mechanism of prior exploitation in the bridg
e process and design frequency-modulated prior guidance (FMPG), which tailors th
e guidance scale to low- and high-frequency bands coherent with bridge generativ
e dynamics.
To address prior exploitation in image in-painting, we develop a cascaded framew
ork, CFG-FMPG, which first generates a noisy hidden representation via CFG and t
hen exploits it as a generative prior with FMPG, fulfilling their complementary
strengths without compromising inference efficiency. Experiments demonstrate tha
t our PG methods consistently improve pre-trained bridge models across diverse i
mage translation tasks.
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AURA: Visually Interpretable Affective Understanding via Robust Archetypes
Guanyu Hu, Dimitrios Kollias, Xinyu Yang
Interpretable methods such as vision--language models (VLMs) have advanced affec
t analysis by aligning images with textual descriptions. However, relying on tex
t as an intermediate proxy faces critical limitations: linguistic templates are
inherently discrete, making them fundamentally incompatible with continuous vale
nce--arousal regression, while also acting as a bottleneck for fine-grained visu
al nuances. Cognitive psychology suggests that human affective perception is not
 mediated by linguistic translation, but is grounded in perceptual resemblance t
o internalized visual archetypes. Motivated by this, we propose AURA, an archety
pe-based framework that replaces brittle linguistic proxies with a self-organizi
ng archetype manifold. By adaptively allocating representational density accordi
ng to affective complexity, AURA enables accurate continuous regression and resh
apes affective taxonomies by decomposing labels into interpretable, geometricall
y coherent visual primitives. This paradigm offers a transparent, visually groun
ded decision trail and achieves state-of-the-art results across discrete and con
tinuous tasks.
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Global Merger-Arbitrage Forecasting with Language Models
Hinal Jajal, Micha■ Mucha, Charles Sweat, Chris Pulman, Charlie Flanagan, Peter
Anderson
We present a language-model forecasting system for merger arbitrage, a specializ
ed high-stakes financial setting in which the task is to predict the outcome of
announced M\&A deals. Unlike prior work on judgmental forecasting with LLMs, whi
ch has focused on broad mixed-topic benchmarks and short context such as news sn
ippets, we study a setting that requires long-context reasoning over hundreds of
 pages of technical documents. Our system combines expert-guided context enginee
ring with finetuning on hindsight-guided reasoning traces derived from historica
l deals. Given an announced deal, it outputs a probability distribution over thr
ee mutually exclusive outcomes: closing at announced terms, a higher bid, or dea
l termination. On an out-of-sample set of more than 400 large deals spanning 42

countries, our finetuned system achieves the best performance of any method we e valuate, reducing class-balanced Brier score to 0.151. This is 24% below calibra ted market-implied probabilities, 19% below XGBoost, and 25-42% below frontier l anguage models. These results, together with ablation studies, show that LLM-bas ed forecasting can succeed in specialized, long-context financial workflows, wit h hindsight-based supervision and expert-designed context playing a critical rol e.

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Drop-in Circulant Structural Priors for Transformer Decoding of Cyclic Codes

Shuai Xiao, Weijun Fang, Qiaosheng Zhang

While Transformer-based architectures have revolutionized neural decoding, exist ing models often treat codes as generic sequences, ignoring their inherent algeb raic properties. In this paper, we take a step toward bridging these two domains by proposing a novel decoding approach that integrates the algebraic structure of cyclic codes into Transformer-based decoders. Leveraging the inherent cyclic properties, we introduce interpretable error correction patterns and inter-node relationship hypotheses that link the structural characteristics of the codes to the model parameters. Building on these insights, we design a plug-and-play, fl exibly deployable decoding method tailored for cyclic codes. Experimental result s show that our method achieves an average reduction in bit error rate (BER) by an order of magnitude, while also reducing the total number of parameters by app roximately 97%. Additional comparative experiments validate our proposed conject ures and highlight a promising pathway for bridging classical coding theory and modern Transformer-based decoding architectures.

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TAMPO: Task- and Model-Aware Automatic Prompt Optimization for Auto-Routing in L LM-based Systems

Yuejun Jiao, Jun Xia, Yanxin Yang, Yonghao Yang, Hao Shen, Mingsong Chen

Automatic Prompt Optimization (APO) enables Large Language Models (LLMs) to adap t to specific tasks while minimizing manual engineering costs. However, since ex isting APO approaches either rely solely on multi-round iterative procedures or use model-specific generators tailored to optimizing prompts for a single model and objective, they are not readily applicable to auto-routing scenarios, which require operating over diverse LLMs and juggling multiple, often competing, trad e-offs. To address this issue, we propose TAMPO, a novel task- and model-aware A PO framework for auto-routing in LLM-based systems. Specifically, to capture per formance variation across a broad range of tasks and models, we construct a comp rehensive, heterogeneity-aware dataset to train an uncertainty-aware reward mode l. Serving as an offline proxy, this reward model can greatly mitigate reward ha cking, allowing TAMPO to learn an optimal multi-objective conditional policy for robust prompt generation. Based on the user requirements encoded in our defined preference vector, this policy enables flexible control over prompt generation and supports a cost-effective deployment strategy. Extensive experiments across 86 tasks demonstrate that TAMPO effectively maintains performance stability acro ss diverse tasks and models, providing a robust, controllable solution for auto- routing in various LLM-based systems.

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From Correspondence to Actions: Human-Like Multi-Image Spatial Reasoning in Mult i-modal Large Language Models

Masanari Oi, Koki Maeda, Ryuto Koike, Daisuke Oba, Nakamasa Inoue, Naoaki Okazak i

While multimodal large language models (MLLMs) have made substantial progress in single-image spatial reasoning, multi-image spatial reasoning, which requires i ntegration of information from multiple viewpoints, remains challenging.

Cognitive studies suggest that humans address such tasks through two mechanisms: *cross-view correspondence*, which identifies regions across different views th at correspond to the same physical locations, and *stepwise viewpoint transforma tion*, which composes relative viewpoint changes sequentially.

However, existing studies incorporate these mechanisms only partially and often implicitly, without explicit supervision for both.

We propose Human-Aware Training for Cross-view correspondence and viewpoint cHan ge (HATCH), a training framework with two complementary objectives: (1) Patch-Le vel Spatial Alignment, which encourages patch representations to align across vi ews for spatially corresponding regions, and (2) Action-then-Answer Reasoning, which requires the model to generate explicit viewpoint transition actions befor e predicting the final answer.
Experiments on three benchmarks demonstrate that HATCH consistently outperforms baselines of comparable size by a clear margin and achieves competitive results against much larger models, while preserving single-image reasoning capabilities .
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SpeedVFI: One-step Diffusion for Efficient Video Frame Interpolation
Ganggui Ding, Xiaogang Xu, Hao Chen, Chunhua Shen
Generative video diffusion models have shown strong robustness to large motion a nd occlusions for video frame interpolation (VFI). However, their inference effi ciency lags significantly behind learning-based methods due to the structural re dundancy of pairwise inference and the procedural latency of multi-step iterativ e denoising. To address these limitations, we propose SpeedVFI, a task-specific one-step diffusion formulation that recasts generative VFI as unified sequence i nterpolation. SpeedVFI achieves dual efficiency improvements by interpolating th e entire video sequence in a single forward pass to eliminate pairwise overhead, and by distilling the generation trajectory into a one-step denoising process t o bypass iterative latency. To make this formulation effective for VFI, we intro duce temporal RoPE alignment for temporally consistent conditioning and noise-ce ntric partial attention to reduce computational overhead while preserving global context. Extensive experiments demonstrate that SpeedVFI accelerates diffusion-based VFI by orders of magnitude while maintaining competitive quantitative and visual quality.
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Dynamics and Representation Structure of Local Approximations to Gradient-Based Learning in Linear Recurrent Neural Networks
Ezekiel Williams, Alexandre Payeur, Guillaume Lajoie
Biological and neuromorphic recurrent neural networks (RNNs) are subject to spat ial and temporal locality constraints on the information that can plausibly be u sed during learning. A common strategy to satisfy these constraints is to modify gradient descent by neglecting non-local terms to varying degrees, as in random feedback local online (RFLO) learning and truncated backpropagation through tim e (tBPTT). However, the learning dynamics of these algorithms, and how they comp are with BPTT, remain poorly understood. We apply dynamical systems theory to da ta-aligned linear RNNs---whose dynamics can be separated into orthogonal modes--- to compare stationary solutions, stability properties, and convergence rates, f inding qualitatively distinct behaviour for RFLO versus BPTT and one-step tBPTT. We further observe that the solutions learned by RFLO are restricted to low-ran k perturbations of initial parameters, a result which holds beyond the data-alig ned setting. Our work provides analytical insight into how locality constraints shape learning dynamics, with implications for neuroscientific models of learnin g and alternative optimization approaches for RNNs.
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ArcDAE: Asymmetric Rectified Contrastive Diffusion Autoencoder for Unified Repre sentation Learning
Ge Gao, Di Xiong, Zeke Xie, Jian Yang, Shuo Chen
The unification of generative details and discriminative semantics presents a st ructural paradox in *diffusion-based representation learning*. Early approaches decouple semantics from generation, inevitably compromising representational com pleteness (i.e., *information split*). While recent bridge-based methods achieve unification via a tightly coupled mapping, they suffer from *information overlo ad*. This is because unconstrained reconstruction objectives incentivize the enc oder to entangle high-frequency stochastic noise into the latent bottleneck. To solve this, we introduce *asymmetric rectified contrastive diffusion autoencoder * (ArcDAE), which rebuilds the diffusion bridge as a *dynamic sifter*. Through i

mposing a *timestep-aware rectification constraint* that orthogonalizes the sema
ntic manifold from the stochastic noise space, ArcDAE compels the bottleneck to
distill discriminative features while actively shedding high-frequency redundanc
y. Consequently, our approach eliminates the overload trap without reverting to
decoupling. Extensive experiments validate the superiority of our FFHQ-trained A
rcDAE, surpassing state-of-the-art methods by up to 6.4% in downstream semantics
 regression and 9.7% in reconstruction fidelity.
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MIST: Moment-Aligned Invariant Stability Transform for Robust Flow Matching
Liang Peng, Deqing Li, Yujia Wu, Hao Meng, Kuan Cao, Yu Wu, Xiaoxiao Xu, Lin Qu
Classifier-Free Guidance (CFG) is a cornerstone of flow-matching models, signifi
cantly enhancing visual quality and prompt adherence. However, high guidance sca
les inherently violate the optimal transport dynamics, leading to visual artifac
ts and mode collapse.
In this paper, we investigate the mechanisms of this failure through the lens of
 velocity moment decomposition. Our analysis reveals that the distributional shi
ft induced by CFG decouples into two geometric components: a Linear Barycentric
Drift that shifts the global distribution center, and a Quadratic Energetic Inst
ability that injects surplus kinetic energy, disrupting the transport cost and t
riggering variance explosion. To mitigate these issues, we introduce MIST (Momen
t-aligned Invariant Stability Transform), a training-free method designed to con
fine the sampling trajectory to the learned data manifold. MIST comprises two hi
erarchical stages: (1) Invariant Alignment (IA), a global statistical rectifier
that restores structural integrity by removing the linear drift and realigning t
he energy profile; and (2) Stability Thresholding (ST), a local dynamical regula
tor that enforces Lipschitz-like smoothness via temporal decay and spatial suppr
ession. MIST enables robust, high-fidelity generation across a wide range of gui
dance scales while consistently improving performance at moderate scales. Extens
ive experiments on diverse text-to-image and text-to-video benchmarks demonstrat
e that MIST outperforms standard CFG and state-of-the-art corrections, establish
ing a new benchmark for robust guidance in flow-based generative models.
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COGNOS: Universal Enhancement for Time Series Anomaly Detection via Constrained
Gaussian-Noise Optimization and Smoothing
Wenlong Shang, Shihao Tian, Xutong Wan, Peng Chang
Reconstruction-based methods are a dominant paradigm in time series anomaly dete
ction (TSAD), however, their near-universal reliance on Mean Squared Error (MSE)
 loss results in statistically flawed reconstruction residuals. This fundamental
 weakness leads to noisy, unstable anomaly scores, hindering reliable detection.
 To address this, we propose Constrained Gaussian-Noise Optimization and Smoothi
ng (COGNOS), a universal, model-agnostic enhancement framework that tackles this
 issue at its source. COGNOS introduces a novel Gaussian-White Noise Regularizat
ion strategy during training, which directly constrains the model's output resid
uals to conform to a Gaussian white noise distribution. This engineered statisti
cal property creates the ideal precondition for our second contribution: Adaptiv
e Residual Kalman Smoother that provably operates as a statistically robust esti
mator to denoise the raw anomaly scores.
Extensive experiments on multiple benchmarks demonstrate that COGNOS consistentl
y enhances the performance of state-of-the-art backbones significantly, validati
ng the efficacy of coupling statistical regularization with adaptive filtering.
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DRIFT-BENCH: Diagnosing CoopeRative Breakdowns in LLM Agents under Input Faults
via Multi-Turn Interaction
Han Bao, Zheyuan Zhang, Pengcheng Jing, Zhengqing Yuan, Kaiwen Shi, Yanfang Ye
As Large Language Models transition to autonomous agents, user inputs frequently
 violate cooperative assumptions (e.g., implicit intent, missing parameters, fal
se presuppositions, or ambiguous expressions), creating execution risks that tex
t-only evaluations do not capture. Existing benchmarks typically assume well-spe
cified instructions or restrict evaluation to text-only, single-turn clarificati
on, and thus do not measure multi-turn disambiguation under grounded execution r

isk. We introduce DRIFT-BENCH, the first diagnostic benchmark that evaluates agentic pragmatics under input faults through multi-turn clarification across state-oriented and service-oriented execution environments. Grounded in classical theories of communication, DRIFT-BENCH provides a unified taxonomy of cooperative breakdowns and employs a persona-driven user simulator with the Rise evaluation protocol. Experiments show substantial performance drops under these faults, with clarification effectiveness varying across user personas and fault types. DRIFT-BENCH connects clarification studies with agent benchmarking, providing a framework to diagnose failures arising from faulty user inputs.
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Sparks of Cooperative Reasoning: LLMs as Strategic Hanabi Agents
Mahesh Ramesh, Kaousheik Jayakumar, Aswinkumar Ramkumar, Pavan Thodima, Aniket Rege, Emmanouil-Vasileios Vlatakis-Gkaragkounis
Cooperative reasoning under incomplete information remains complex for both humans and multi-agent AI, requiring agents to transcend individual logic in favor of recursive Theory-of-Mind (ToM) and strategic coordination. To investigate these challenges, we conduct a large-scale evaluation of 17 state-of-the-art LLMs (4B–600B+) on Hanabi card game across 2–5 players. To examine their limitations, we analyze the impact of context engineering and scaffold robustness, ranging from minimal prompts (Watson setting) to Bayesian-motivated scaffolding (Sherlock setting) and multi-turn working memory (Mycroft setting). Our findings reveal that: (1) top-performing models can autonomously track game states via internal working memory, although not reliably, and (2) cross-play performance scales smoothly with model capability. However, even the best models (scoring ≈ 15/25) trail specialist human experts (> 20/25). We introduce and release two novel datasets: HanabiLogs (1,520 annotated trajectories) and HanabiRewards (560 games with dense move-level utilities). By fine-tuning a 4B open-weight model (Qwen3-Instruct) on our datasets, we achieve performance gains of up to 156%, bringing performance to within 3 points of a strong proprietary reasoning model (o4-mini) and surpassing the best non-reasoning model (GPT-4.1) by 52%. Crucially, our HanabiRewards RL-finetuned model further generalizes beyond Hanabi, improving performance on a cooperative group-guessing benchmark by 11 percentage points, temporal reasoning on EventQA by 6.4 points, instruction-following on IFBench by 1.7 Pass@10, and matching AIME 2025 mathematical reasoning Pass@10. Code and datasets are available at https://app.primeintellect.ai/dashboard/environments/mahesh-ramesh/hanabi.
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HERMES: Towards Efficient and Verifiable Mathematical Reasoning in LLMs
Azim Ospanov, Zijin Feng, Jiacheng Sun, Haoli Bai, SHEN XIN, Farzan Farnia
Informal mathematics has been central to modern large language model (LLM) reasoning, offering flexibility and efficient construction of arguments. However, purely informal reasoning is prone to logical gaps and subtle errors that are difficult to detect and correct. In contrast, formal theorem proving provides rigorous, verifiable mathematical reasoning, where each inference step is checked by a trusted compiler, but lacks the exploratory freedom of informal problem-solving. This mismatch leaves current LLM-based math agents without a principled way to combine the strengths of both paradigms. In this work, we introduce Hermes, the first tool-assisted agent that explicitly interleaves informal reasoning with formally verified proofs in Lean. The framework performs intermediate formal checking to prevent reasoning drift and a memory module for proof continuity across multi-step reasoning chains, enabling both exploration and verification. We evaluate Hermes on four challenging mathematical reasoning benchmarks using LLMs of varying parameter scales, from small models to state-of-the-art systems. Across all settings, Hermes reliably improves the reasoning accuracy of base models while substantially reducing reasoning token usage and computational cost compared to reward-based approaches. On difficult datasets such as AIME and HARDMath2, Hermes@1 achieves up to a 40% accuracy improvement while using 80% fewer total inference FLOPs. When scaled at test time, Hermes@5 boosts accuracy further by 20%.
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DualTimesField: Rethinking Time Series as Continuous-Time Trends and Events

Wencheng Zhang, Long Li, Huayi Qin, Zongjuan Wu, Jing Li, Wanghu Chen
Effective time series representation is critical for revealing temporal dynamics in many fields. However, existing approaches encounter fundamental limitations. Discrete-time representations struggle with irregular sampling and the tradeoff of fidelity and efficiency, while traditional implicit neural representations suffer from spectral bias and frequency entanglement. To address these challenges, we conceptualize time series as the superposition of continuous trends and discrete events from a continuous-time perspective and propose DualTimesField, a framework that utilizes dual implicit neural fields. Its Continuous Time Field captures smooth trends through bandwidth-limited parameterization, while a Discrete Geometric Field models transient events using learnable Gabor atoms, gated sparsity, and coarse-to-fine scale annealing. This explicit field separation effectively overcomes both limitations. Experiments on nine real-world benchmarks demonstrate substantial improvements in representation fidelity, achieving 51.2% average MSE reduction over discrete-time baselines and competitive interpolation on irregular data. Code is available at https://github.com/WisdomTogether/DualTimesField.
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On Efficient Scaling of GNNs via IO-Aware Layers Implementations
Daria Fomina, Daniil Krasylnikov, Alexey Boykov, Andrey Dolgovyazov, Vyacheslav Zhdanovskiy, Fedor Velikonivtsev
Graph Neural Networks (GNNs) are bottlenecked by sparse, irregular memory access. Popular frameworks such as DGL and PyTorch Geometric support general message passing, but complex layers often materialize edge-wise intermediates, increasing memory traffic and limiting scalability on large graphs.
We take an I/O- and arithmetic-intensity--centric view and show that widely used layers fall into three kernel families: SpMM-based convolutions, reduction-based aggregations, and attention-based layers (GATv2/Graph Transformer). For each family, we develop GPU kernels that reduce data movement, improve locality, and remain robust across realistic graphs. We also study graph reordering and find that its impact depends on the kernel mapping: it benefits neighbor-parallel (gather-dominated) kernels more consistently than feature-parallel designs.
Empirically, our fused attention kernels reach up to **3.9**$\times$ speedup for Graph Transformer (median **1.6**$\times$), with Tensor Core (block-sparse) variants up to **7.3**$\times$ on locally dense graphs; for GATv2 we reach up to **8.5**$\times$ speedup (median **2.0**$\times$) while reducing peak memory by up to **76**$\times$ (median **6**$\times$). Our degree-aware reduction kernels achieve up to **10**$\times$ speedup (median **2.6**$\times$). For SpMM-based layers, properly cached cuSPARSE achieves up to **8**$\times$ speedup over DGL and outperforms evaluated custom baselines in the majority of evaluations. We release our implementations as drop-in replacements in our [GitHub repository](https://github.com/yandex-research/On-Efficient-Scaling-Of-GNNs) to support reproducible, hardware-aware GNN acceleration.
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HypRAG: Hyperbolic Dense Retrieval for Retrieval Augmented Generation
Hiren Madhu, Ngoc Bui, Ali Maatouk, Leandros Tassiulas, Smita Krishnaswamy, Menglin Yang, Sukanta Ganguly, Kiran Srinivasan, Rex Ying
Embedding geometry plays a fundamental role in retrieval quality, yet dense retrievers for retrieval-augmented generation (RAG) remain largely confined to Euclidean space. However, natural language exhibits hierarchical structure from broad topics to specific entities that Euclidean embeddings fail to preserve, causing semantically distant documents to appear spuriously similar and increasing hallucination risk. To address these limitations, we introduce hyperbolic dense retrieval, developing two model variants in the Lorentz model of hyperbolic space: HyTE-FH, a fully hyperbolic transformer, and HyTE-H, a hybrid architecture projecting pre-trained Euclidean embeddings into hyperbolic space. To prevent representational collapse during sequence aggregation, we introduce the Outward Einstein Midpoint, a geometry-aware pooling operator that provably preserves hierarchical structure. On MTEB, HyTE-FH outperforms equivalent Euclidean baselines, while on RAGBench, HyTE-H achieves up to 29\% gains over Euclidean baselines in contex

t relevance and answer relevance using substantially smaller models than current state-of-the-art retrievers. Our analysis also reveals that hyperbolic representations encode document specificity through norm-based separation—with over 20\% radial increase from general to specific concepts—a property absent in Euclidean embeddings, underscoring the critical role of geometric inductive bias in faithful RAG systems. The code is available at: https://github.com/Graph-and-Geometric-Learning/HypRAG

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HiPhO: How Far Are (M)LLMs from Humans in the Latest High School Physics Olympiad Benchmark?

Fangchen Yu, Haiyuan Wan, Qianjia Cheng, Yuchen Zhang, Jiacheng Chen, Fujun Han, Yulun Wu, Junchi Yao, Ruilizhen Hu, Ning Ding, Yu Cheng, Tao Chen, LEI BAI, Dongzhan Zhou, Yun Luo, Ganqu Cui, Peng Ye

Recently, the physics reasoning capabilities of (M)LLMs have attracted growing attention. However, existing physics benchmarks lack systematic coverage of recent physics Olympiads and direct comparison with human contestants. We present **HiPhO**, the first benchmark dedicated to high school physics Olympiads with human-aligned evaluation. HiPhO highlights three key innovations. **(1) Comprehensive data:** it compiles 13 latest Olympiads from 2024--2025, covering international and regional competitions and spanning mixed modalities from text-only to diagram-based problems. **(2) Professional evaluation:** it adopts official rubrics for fine-grained answer- and step-level grading aligned with human examiners. **(3) Human-level comparison:** it assigns gold, silver, and bronze medals to models based on official medal scores, enabling direct comparison with human contestants. Evaluating 30 (M)LLMs across 13 exams, we find that most open-source MLLMs remain at or below the bronze level, open-source LLMs demonstrate notable progress with multiple gold medals, and closed-source MLLMs achieve 6-13 gold medals, while most models still fall well short of full marks. These results underscore the substantial gap between current (M)LLMs and top human contestants, as well as the room for further improvement. The dataset and leaderboard are available at https://huggingface.co/datasets/SciYu/HiPhO and https://phyarena.github.io, respectively.

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Zeroth-Order Forward-Only SNN Training Inspiring Neuromorphic On-Chip Learning

Mingyue Qin, Shuyu Yin, Qinghai Guo, Peilin Liu, Xiaolin Huang, Fei Wen

The human brain is a biologically instantiated on-device neural system that integrates both learning and inference in a unified architecture, which enables rapid and flexible learning on-the-fly. This extraordinary online learning ability is realized through biological learning mechanisms operating on a well-initialized innate model. This work considers the on-chip edge learning upon pretrained models with zeroth-order (ZO) methods. ZO optimization methods, which resemble bio-plausible perturbation-based learning, offer a promising approach that enables learning with only forward passes and hence can significantly reduce the complexity of on-chip hardware implementation. However, in this work we show that applying ZO methods to spiking neural networks (SNNs) is non-trivial due to the step-function nature of spiking activation. We analyze the challenges posed by the spiking activation, and reveal a variance amplification effect of it. Based on this insight, we propose a subspace-based ZO (SZO) method that leverages the intrinsic low-dimensional structure of the SNN optimization trajectory. By learning in a low-dimensional subspace, SZO substantially enhances ZO learning efficacy, achieving accuracy comparable to first-order (FO) methods with faster learning speed than full-space BP. We evaluate SZO on model training from scratch, continual training, and unsupervised adaptation. Experimental results demonstrate that SZO closely approaches FO training performance for the first time while offering fast learning speed.

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Diffusion-based learning framework for Constrained Nonconvex Optimization with Weighted Bootstrapped Refinement

Shutong Ding, Yimiao Zhou, Ke Hu, Xi Yao, Junchi Yan, Xiaoying Tang, Ye Shi

Recent advances in diffusion models show promising potential to accelerate nonco

nvex problem solving by leveraging their multimodality. However, most existing d
iffusion-based optimization approaches rely on supervised learning and lack a me
chanism to enforce constraint satisfaction, which is required in real-world appl
ications. In that case, we investigate and theoretically analyze the inherent pr
oblem of supervised diffusion solvers and identify the distributional misalignme
nt problem, i.e., the generated solution distribution often exhibits low probabi
lity mass on the feasible region. To resolve this issue, we propose DiOpt, a new
 diffusion-based learning framework for constrained nonconvex optimization, whic
h effectively learns the mapping from noise to the constraint region. Specifical
ly, this framework operates in two distinct phases: an initial warm-start phase,
 implemented via supervised learning, followed by a bootstrapping training phase
. This dual-phase architecture is designed to iteratively refine solutions, ther
eby improving the objective function with high constraint satisfaction. Finally,
 we also employ a solution selection technique in inference for better optimalit
y. Notably, DiOpt is the first successful integration of the diffusion solver in
 constrained nonconvex optimization. Evaluations on diverse nonconvex tasks demo
nstrate the superiority of DiOpt in both optimality and constraint satisfaction.
 Our official page is released at \url{https://dingsht.tech/diopt-webpage}.
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FiGuRO: Intrinsic Dimension Estimation for Multi-Modal Data
Viktoria Schuster, Sana Tonekaboni, Caroline Uhler
Determining the complexity, or Intrinsic Dimension (ID), of data is fundamental
to efficient and interpretable representation learning. This is particularly cha
llenging in multi-modal settings when trying to learn disentangled representatio
ns for shared and private information. Existing techniques leave a critical gap:
 they are often static, uni-modal, or in the case of contrastive methods, adapt
only to the shared ID implicitly. We introduce Fidelity-Guided Rank Optimization
 (FiGuRO), a framework for approximating the ID of uni- and multi-modal data und
er constraints of model capacity and hyperparameters. FiGuRO learns the dimensio
ns of low-rank projections using truncated singular value decomposition and an a
lgorithm that determines when to reduce or increase dimension and in which laten
t space. Disentanglement of shared and private information arises as an emergent
 property of this optimization, eliminating the need for complex auxiliary loss
functions. We demonstrate that FiGuRO outperforms existing ID estimation techniq
ues and is more robust to hyperparameter changes. Across simulations and real-wo
rld data, FiGuRO captures distinct ID scales and varying subspace ratios, and de
composes shared and private information successfully. Furthermore, we show that
FiGuRO can be applied to modern uni-modal pretrained models, enabling efficient,
 post-hoc disentanglement of multi-modal representations.
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Likelihood over Estimation: Robust Quadratic Discriminant Analysis for Heavy-Tai
led Distributions with Theory and Evidence
Niranjana Ambadi, eugene pinsky
Quadratic Discriminant Analysis (QDA) assumes Gaussian class-conditional distrib
utions, causing systematic misclassification when data exhibit heavy tails. We p
ropose Stable-QDA, which replaces the Gaussian likelihood with a symmetric $\alp
ha$-stable likelihood that decays polynomially rather than exponentially in Maha
lanobis distance. Crucially, we find that correcting likelihood misspecification
 yields larger gains than robustifying parameter estimation: standard estimators
 (sample mean, Ledoit--Wolf covariance) often outperform robust alternatives whe
n class heteroscedasticity is discriminative. We provide consistency guarantees
under infinite-variance regimes, data-driven diagnostics for estimator selection
, and demonstrate 15--53\% error reduction on real-world heavy-tailed benchmarks
.
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L-SR1: Learned Symmetric-Rank-One Preconditioning
Gal Lifshitz, Shahar Zuler, Ori Fouks, Dan Raviv
End-to-end deep learning has achieved impressive results but often relies on lar
ge labeled datasets, exhibits limited generalization to unseen scenarios, and in
curs substantial computational cost.

Classical optimization methods, in contrast, are more data-efficient and lightweight but frequently suffer from slow convergence.
Learned optimizers aim to bridge this gap, yet existing approaches have focused primarily on first-order methods, while learned second-order optimization has received much less attention.
We introduce L-SR1, a learned second-order optimizer inspired by the classical Symmetric Rank-One (SR1) method.
At its core, L-SR1 employs a Projection-Guided Secant Mechanism (PGSM) that generates positive semi-definite preconditioners and biases meta-training toward the quasi-Newton secant relation.
Through controlled analytic benchmarks, we study stability, generalization across problem dimensions, and search direction quality, and further evaluate L-SR1 on Monocular Human Mesh Recovery (HMR), where it outperforms both classical and learned optimization-based baselines.
With a compact model and no reliance on task-specific fine-tuning or annotated data, L-SR1 demonstrates strong generalization and can be integrated into a broad range of iterative optimization problems to accelerate convergence and reduce the required number of iterations.
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Multi-Label Test-Time Adaptation with Bayesian Conditional Priors
QiRu Li, Ao Zhou, Zhiwei Jiang, Zifeng Cheng, Cong Wang, Yafeng Yin, Qing Gu
Multi-label recognition with frozen Vision-Language Models (VLMs) is brittle under distribution shift: standard zero-shot inference scores labels independently, ignoring co-occurrence structure and producing incoherent label sets where dominant concepts suppress weaker but compatible labels. We introduce Bayesian Conditional Priors (BCP) Estimation, a gradient-free test-time adaptation method that injects label dependency without tuning the backbone. BCP views zero-shot logits as a proxy for marginal posteriors under a fixed image-text likelihood and attributes shift-induced errors mainly to a mismatched label prior. For each test image, it selects a high-confidence anchor label and applies an anchor-conditioned Bayesian refinement. This update is closed-form in logit space and admits a pointwise mutual information (PMI) interpretation, explicitly promoting compatible labels and suppressing incompatible ones. BCP operates without target annotations by estimating anchor-conditioned priors online from the unlabeled test stream via lightweight second-order co-occurrence statistics, adding negligible overhead beyond a single forward pass. Across standard multi-label benchmarks and multiple CLIP backbones, BCP consistently outperforms strong TTA baselines, e.g., improving RN50 average mAP from 57.31 to 69.22 and ViT-B/16 from 62.61 to 71.79.
****

How Reasoning Evolves from Post-Training Data: An Empirical Study Using Chess
Lucas Dionisopoulos, Nicklas Majamaki, Prithviraj Ammanabrolu
We study how reasoning evolves in a language model -- from supervised fine-tuning (SFT) to reinforcement learning (RL) -- by analyzing how a set of theoretically-inspired datasets impacts language model performance in chess. We find that fine-tuning a model to directly predict the best move leads to effective RL and the strongest downstream performance -- however, the RL stage elicits \textit{unfaithful} reasoning (reasoning inconsistent with the chosen move). Alternatively, training on multi-move trajectories yields comparable downstream performance with faithful reasoning and more stable RL. We show that RL induces a substantial positive shift in the distribution of move quality and reduces hallucination rates as a side effect. Finally, we find several SFT-checkpoint metrics -- metrics spanning evaluation performance, hallucination rates, and reasoning quality -- to be predictive of post-RL model performance. We release checkpoints and final models as well as training data, evaluations, and code that allowed us to surpass leading open-source reasoning models in chess with a 7B-parameter model.
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The Devil is in the Condition Numbers: Why is GLU Better than non-GLU Structure?
Xingyu Lyu, Qianqian Xu, Zhiyong Yang, Peisong Wen, Qingming Huang
Gated Linear Units (GLU) and their variants are widely adopted in modern open-source large language model architectures and consistently outperform their non-ga

ted counterparts, yet the underlying reasons for this advantage remain unclear. In this work, we study GLU by analyzing two-layer networks in the neural tangent kernel (NTK) regime. Our analysis reveals that the GLU structure reshapes the NTK spectrum, leading to a smaller condition number and a more compact eigenvalue distribution. Building on this finding, we further analyze the resulting training dynamics and show how the reshaped spectrum leads to faster convergence of GLU models, including a characteristic loss-crossing phenomenon observed between GLU and non-GLU models. Finally, we empirically observe that GLU has limited impact in reducing the generalization gap on various models, including ViT and GPT-2, suggesting that its primary benefit lies in accelerating optimization rather than reducing the generalization gap. The code is available at: https://github.com/Zemdalk/GLU-NTK.

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KIStego: Key-Independent Secure Image Distribution via Bipartite Structural Invariants

Ren Lijing, Denghui Zhang

Generative Image Steganography (GIS) embeds complex secrets within stego-images that are indistinguishable from the stochastic synthesis process itself. It achieves this by exploiting the reversible probability flow between Gaussian noise and the natural image manifold. However, existing steganography faces a key-dependency paradox: precise secret extraction usually requires an external private key or random seed to synchronize the denoising path.

In this paper, we present KIStego, a training-free framework substituting cryptographic secrecy with structural redundancy for Key-Independent, high-resolution secure distribution. Our dual-guidance mechanism utilizes structural latent guidance to project secrets into a bipartite manifold via downsampling and halftoning, yielding self-synchronizing shares. Measurement posterior sampling leverages these shares as discrete invariants to steer a differentiable restoration. By backpropagating through a differentiable measurement surrogate, KIStego reconstructs high-fidelity continuous-tone details from sparse binary observations, mitigating fidelity loss from inversion drift.

KIStego offers an endogenous secure image distribution paradigm by connecting discrete structural invariants with high-fidelity generative reconstruction.

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Understanding Dynamic Compute Allocation in Recurrent Transformers

Ibraheem Muhammad Moosa, Suhas Lohit, Ye Wang, Moitreya Chatterjee, Wenpeng Yin

Token-level adaptive computation seeks to reduce inference cost by allocating more computation to harder tokens and less to easier ones. However, prior work is primarily evaluated on natural-language benchmarks using task-level metrics, where token-level difficulty is unobservable and confounded with architectural factors, making it unclear whether compute allocation truly aligns with underlying complexity. We address this gap through three contributions. First, we introduce a complexity-controlled evaluation paradigm using existing algorithmic and synthetic language tasks with parameterized difficulty, enabling direct testing of token-level compute allocation. Second, we propose ANIRA, a unified recurrent Transformer framework that supports per-token variable-depth computation while isolating compute allocation decisions from other model factors. Third, we use this framework to conduct a systematic analysis of token-level adaptive computation across alignment with complexity, generalization, and decision timing. Our results show that compute allocation aligned with task complexity can emerge without explicit difficulty supervision, but such alignment does not imply algorithmic generalization: models fail to extrapolate to unseen input sizes despite allocating additional computation. We further find that early compute decisions rely on static structural cues, whereas online halting more closely tracks algorithmic execution state. Code: https://github.com/merlresearch/ANIRA

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AAD-1: Asymmetric Adversarial Distillation for One-Step Autoregressive Video Generation

Haobo Li, Yanhong Zeng, Yunhong Lu, Jiapeng Zhu, Hao Ouyang, Qiuyu Wang, Ka Leong Cheng, Yujun Shen, Zhipeng Zhang

We present \textbf{AAD-1}, an \textbf{A}symmetric \textbf{A}dversarial \textbf{D}istillation framework for \textbf{O}ne-step autoregressive image-to-video generation. State-of-the-art methods adopt adversarial distillation but suffer from motion collapse and training instability, resulting in static videos. AAD-1 addresses these challenges through two key designs in architecture and training strategy. Our key architectural insight is to break the symmetry between generator and discriminator. While the generator remains causal to preserve autoregressive sampling capability, the discriminator attends bidirectionally over the full spatiotemporal context and produces a single holistic realism score for the entire video sequence. This asymmetric design enables the discriminator to effectively detect global temporal failures and long-range drift that cause motion collapse in autoregressive generation. To stabilize training, we introduce a phased strategy that first uses distribution matching to bootstrap a stable one-step generator, providing a warm-up phase that brings the student distribution closer to the teacher before adversarial distillation begins. Extensive experiments on VBench demonstrate that AAD-1 achieves state-of-the-art performance in one-step autoregressive video generation.
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Accordion-Thinking: Self-Regulated Step Summaries for Efficient and Readable LLM Reasoning
Zhicheng Yang, Zhijiang Guo, Yinya Huang, Yongxin Wang, Wenlei Shi, Yiwei Wang, Xiaodan Liang, Jing Tang
Scaling test-time compute via long Chain-of-Thought unlocks remarkable gains in reasoning capabilities, yet it faces practical limits due to the linear growth of KV cache and quadratic attention complexity.
In this paper, we introduce AccordionThinking, an end-to-end framework where LLMs learn to self-regulate the granularity of the reasoning steps through dynamic summarization.
This mechanism enables a Fold inference mode, where the model periodically summarizes its thought process and discards former thoughts to reduce dependency on historical tokens.
We apply reinforcement learning to incentivize this capability further, uncovering a critical insight: the accuracy gap between the highly efficient Fold mode and the exhaustive Unfold mode progressively narrows and eventually vanishes over the course of training. This phenomenon demonstrates that the model learns to encode essential reasoning information into compact summaries, achieving effective compression of the reasoning context.
Our AccordionThinker demonstrates that with learned self-compression, LLMs can tackle complex reasoning tasks with minimal dependency token overhead without compromising solution quality, and it achieves a 3× throughput while maintaining accuracy on a 48GB GPU memory configuration, while the structured step summaries provide a human-readable account of the reasoning process.
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Convergence of Steepest Descent and Adam under Non-Uniform Smoothness
Sharan Vaswani, Yifan Sun, Reza Babanezhad Harikandeh
Recent work has analyzed the convergence of first-order methods under non-uniform smoothness assumptions that better model the loss landscape in machine learning tasks. We generalize this assumption to objectives whose curvature is an affine function of the objective value. This property is satisfied by a broad class of problems, including logistic regression, generalized linear models with a logistic link function, softmax policy gradient in reinforcement learning, and a class of neural networks. Under this assumption and gradient domination conditions, we establish a general convergence rate for the steepest descent method, and deterministic, diagonal variants of RMSProp and Adam. Our results imply that for logistic regression on separable data and the softmax policy gradient objective, sign GD converges linearly and is provably faster than GD. Furthermore, we show that for a class of two-layer neural networks on separable data, RMSProp and Adam can converge at a linear rate with a constant step-size and momentum parameter. Finally, we present a lower bound demonstrating that, under our assumption, RMSProp and Adam are provably faster than AdaGrad, AMSGrad, gradient descent, and

heavy-ball momentum.
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ZeroUnlearn: Few-Shot Knowledge Unlearning in Large Language Models
Yujie Lin, Chengyi Yang, Zhishang Xiang, YIPING SONG, Jinsong Su
Large language models inevitably retain sensitive information, defined as inputs that may induce harmful generations, due to training on massive web corpora, raising concerns for privacy and safety. Existing machine unlearning methods primarily rely on retraining or aggressive fine-tuning, which are either computationally expensive or prone to degrading related knowledge and overall model utility. In this work, we reformulate machine unlearning as a precise knowledge re-mapping problem via model editing. We propose ZeroUnlearn, a few-shot unlearning framework. It overwrites sensitive inputs by mapping them to a neutral target state and removing their original representations.
ZeroUnlearn enforces representational orthogonality through a multiplicative parameter update with a closed-form solution, enabling efficient and targeted unlearning. We further extend ZeroUnlearn to a gradient-based variant for multi-sample unlearning. Experiments demonstrate that our approach outperforms existing baselines while preserving general model utility. Our code is available at the github: https://github.com/XMUDeepLIT/ZeroUnlearn.
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ACTG-ARL: Differentially Private Conditional Text Generation with RL-Boosted Control
Yuzheng Hu, Ryan McKenna, Da Yu, Shanshan Wu, Han Zhao, Zheng Xu, Peter Kairouz
Generating high-quality synthetic text under differential privacy (DP) is critical for training and evaluating language models without compromising user privacy. Prior work on synthesizing DP *datasets* often fail to preserve key statistical attributes, suffer utility loss from the noise required by DP, and lack fine-grained control over generation. To address these challenges, we make two contributions. First, we introduce a hierarchical framework that decomposes DP synthetic text generation into two subtasks: *feature learning* and *conditional text generation*. This design explicitly incorporates learned features into the generation process and simplifies the end-to-end synthesis task. Through systematic ablations, we identify the most effective configuration: a rich tabular schema as feature, a DP tabular synthesizer, and a DP fine-tuned conditional generator, which we term ACTG (**A**ttribute-**C**onditioned **T**ext **G**eneration). Second, we propose Anchored RL (ARL), a post-training method that improves the instruction-following ability of ACTG for conditional generation. ARL combines RL to boost control with an SFT anchor on best-of-N data to prevent reward hacking. Together, these components form our end-to-end algorithm **ACTG-ARL**, which advances both the quality of DP synthetic text (+20% MAUVE over prior work) and the control of the conditional generator under strong privacy guarantees. Our code is at https://github.com/actg-arl/ACTG-ARL.
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MemOCR: Layout-Aware Visual Memory for Efficient Long-Horizon Reasoning
Yaorui Shi, Shugui Liu, Yu Yang, Wenyu Mao, Yuxin Chen, Qi GU, Hui Su, Xunliang Cai, Xiang Wang, An Zhang
Long-horizon agentic reasoning necessitates effectively compressing growing interaction histories into a limited context window.
Most existing memory systems serialize history as text, where token-level cost is uniform and scales linearly with length, often spending scarce budget on low-value details.
To this end, we introduce **MemOCR**, a multimodal memory agent that improves long-horizon reasoning under tight context budgets by allocating memory space with adaptive information density through visual layout.
Concretely, MemOCR maintains a structured rich-text memory (e.g., headings, highlights) and renders it into an image that the agent consults for memory access, visually prioritizing crucial evidence while aggressively compressing auxiliary details.
To ensure robustness across varying memory budgets, we train MemOCR with reinforcement learning under budget-aware objectives that expose the agent to diverse c

ompression levels.
Across long-context multi-hop and single-hop question-answering benchmarks, MemOCR outperforms strong text-based baselines and achieves more effective context utilization under extreme budgets.
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A Unified Sparse Attention via Multi-Granularity Compression
Siran Liu, Zane Cao, Yongchao He
Efficient long-context understanding is increasingly vital for large language model (LLM) applications such as multi-turn dialogue and program analysis. However, the core self-attention scales quadratically with sequence length, creating a fundamental computational bottleneck. Existing sparse attention methods alleviate this issue but face trade-offs: training-based methods are costly and cannot be directly applied as acceleration plugins for other models, while inference-time methods often compromise efficiency or cross-modal generality. To address these limitations, we present UniSparse, a unified mechanism that introduces the notion of composite tokens—compact representations that aggregate multi-granularity contextual information. Building on this abstraction, UniSparse dynamically constructs sparse attention through multi-granularity compression and block-level selection, enabling efficient and hardware-friendly execution on GPU. Across multiple modalities and tasks ranging from synthetic benchmarks to real-world applications, UniSparse consistently surpasses state-of-the-art sparse attention methods (e.g., MInference, XAttention, FlexPrefill) in both accuracy and efficiency, achieving $\ge$ 99\% of full-attention accuracy and up to 2.61$\times$ faster attention computation than FlashAttention.
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PRIM■Cooperative Dynamic Token Compression for Efficient Large Multimodal Models
Song Li, yongping xiong
Large multimodal models (LMMs) have shown strong capabilities in audio-visual understanding by jointly reasoning over visual, auditory, and linguistic inputs. However, processing long-form audio-visual content often requires a large number of tokens, leading to substantial computational and memory overhead during inference. Existing efficiency-oriented methods typically apply uniform compression or pruning strategies, which overlook modality-specific characteristics and instruction-dependent reasoning behaviors in multimodal models. In this work, we present PRIM, an inference framework for efficient multimodal reasoning that systematically compresses audio-visual representations based on attention dynamics and instruction relevance. Motivated by an attention-based analysis revealing modality imbalance and layer-wise redundancy in LMMs, PRIM introduces a cooperative compression pipeline that spans both multimodal encoders and the language model. Specifically, PRIM performs early text-conditioned audio-visual fusion to externalize cross-modal interactions, applies attention-guided and frequency-aware strategies to remove redundant audio and video tokens, and further adapts token retention inside the language model according to task demands. Extensive experiments on multiple audio-visual benchmark datasets demonstrate that PRIM consistently achieves stable and superior efficiency--accuracy trade-offs across diverse tasks and datasets. These results demonstrate that PRIM, a multimodal cooperative compression approach, provides an effective pathway toward scalable and efficient audio-visual reasoning.
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Differential Smoothing Mitigates Sharpening and Improves LLM Reasoning
Jingchu Gai, Guanning Zeng, Huaqing Zhang, Aditi Raghunathan
It is widely recognized that reinforcement learning (RL) fine-tuning of large language models often leads to \textit{diversity collapse}, where outputs lack variety. Prior work has proposed a range of heuristics to counteract this effect, but these methods are ad hoc: they frequently trade off correctness for diversity, their effectiveness varies across tasks, and in some cases they even contradict one another. In this work, we place these observations on a rigorous foundation. We first provide a formal proof of why RL fine-tuning exhibits diversity collapse via a selection and reinforcement bias. Next, we make a key observation that any reward modification to address diversity collapse only needs to be applied

on the correct trajectories. Building directly on this analysis, we introduce a principled method---\textit{differential smoothing}---that provably improves both correctness and diversity, outperforming vanilla RL as well as widely used entropy-based heuristics. Our theory precisely characterizes when existing heuristics help and why they fail, while showing that differential smoothing is universally superior. Extensive experiments with models from 1B to 7B parameters, across domains including CountDown and real-world mathematical reasoning, demonstrate consistent gains. Differential smoothing improves both Pass@1 and Pass@k, with up to 6.7\% improvements on AIME24 dataset.
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Allocating Variance to Maximize Expectation
Renato Paes Leme, Clifford Stein, Yifeng Teng, Pratik Worah
We design efficient approximation algorithms for maximizing the expectation of the supremum of families of Gaussian random variables. In particular, let $OPT:=\max_{\sigma_1,\cdots,\sigma_n}\mathbb{E}\sum_{j=1}^{m}\max_{i\in S_j} X_i$, where X_i are Gaussian, $S_j\subset[n]$ and $\sum_i\sigma_i^2=1$, then our theoretical results include:

- We characterize the optimal variance allocation -- it concentrates on a small subset of variables as $|S_j|$ increases,

- A polynomial time approximation scheme (PTAS) for computing OPT when $m=1$, and

- An $O(\log n)$ approximation algorithm for computing OPT for general $m>1$.
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Rewiring Experts on the Fly: Continuous Rerouting for Better Online Adaptation in Mixture-of-Expert Models
Guinan Su, Yanwu Yang, Li Shen, Lu Yin, Shiwei Liu, Jonas Geiping
Mixture-of-Experts (MoE) models achieve efficient scaling through sparse expert activation, but often suffer from suboptimal routing decisions due to distribution shifts in deployment. While existing test-time adaptation methods could potentially address these issues, they primarily focus on dense models and require access to external data, limiting their practical applicability to MoE architectures. However, we find that, instead of relying on reference data, we can optimize MoE expert selection on-the-fly based only on input context. As such, we propose *a data-free, online test-time framework* that continuously adapts MoE routing decisions during text generation without external supervision or data. Our method cycles between two phases: During the prefill stage, and later in regular intervals, we optimize the routing decisions of the model using self-supervision based on the already generated sequence. Then, we generate text as normal, maintaining the modified router until the next adaption. We implement this through lightweight additive vectors that only update router logits in selected layers, maintaining computational efficiency while preventing over-adaptation. The results show consistent performance gains on challenging reasoning tasks while maintaining robustness to context shifts. For example, our method achieves a 5.5\% improvement on HumanEval with OLMoE. Furthermore, owing to its plug-and-play property, our method complements existing test-time scaling techniques, e.g., achieving 6\% average gains when incorporated with self-consistency on DeepSeek-V2-Lite.
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ToaSt: Token Channel Selection and Structured Pruning for Efficient ViT
Hyunchan Moon, Cheonjun Park, Steven L. Waslander
Vision Transformers (ViTs) have achieved remarkable success across various vision tasks, yet their deployment is often hindered by prohibitive computational costs. While structured weight pruning and token compression have emerged as promising solutions, they suffer from prolonged retraining and inter-layer dependencies that complicate optimization, respectively. We propose ToaSt, a decoupled framework applying specialized strategies to distinct ViT components. We apply coupled head-wise structured pruning to Multi-Head Self-Attention modules, leveraging attention operation characteristics to enhance robustness. For Feed-Forward Net

works (over 60\% of FLOPs), we introduce Token Channel Selection (TCS), a training-free method that filters redundant noise channels at inference time. Extensive evaluations across nine diverse models, including DeiT, ViT-MAE, and Swin Transformer, demonstrate that ToaSt achieves superior trade-offs between accuracy and efficiency, consistently outperforming existing baselines. On ViT-MAE-Huge, ToaSt achieves 88.52\% accuracy (+1.64\%p) with 39.4\% FLOPs reduction. ToaSt also transfers effectively to diverse downstream tasks (COCO detection, ADE20K segmentation, CIFAR-100 classification), achieving 52.2 versus 51.9 mAP on COCO. Code: \href{https://github.com/SHANNonLab-HUFS/ToaSt}{github.com/SHANNonLab-HUFS/ToaSt}.
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Off-Policy Evaluation for Missingness-Aware Policies in MDPs with Rewards Missing Not at Random
Ziheng Wei, Annie Qu, Rui Miao
In offline Reinforcement Learning, immediate rewards in logged batch data are often unobserved due to sparse or irregular record-keeping, or censored beyond certain reward values. This issue arises in practical settings, including health care and marketing. We investigate off-policy evaluation (OPE) in finite-horizon Markov decision processes when rewards are missing not at random (MNAR), which breaks ignorability and induces selection bias even after conditioning on states and actions. To address this, we formalize a reward-dependent propensity model and use future states as shadow variables to identify the full-data conditional mean reward. We further introduce a bridge function that recovers the conditional mean reward without explicitly modeling the MNAR mechanism, and estimate it via a min-max procedure to avoid double sampling. Building upon these identification results, we propose an Fitted-Q-Evaluation-style estimator that propagates the recovered rewards while allowing target policies to depend on past missingness indicators. Finally, we establish consistency and finite-sample error bounds for our OPE estimator, and show through experiments the strong performance of our method compared to existing methods on simulated and MIMIC-III Sepsis data.
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When Data Is Scarce: Scaling Sparse Language Models with Repeated Training
Boqian Wu, Qiao Xiao, Patrik Okanovic, Tomasz Sternal, Maurice van Keulen, Mykola Pechenizkiy, Elena Mocanu, Torsten Hoefler, Decebal Constantin Mocanu
Scaling laws for dense LLMs under infinite data are well explored, but how sparsity interacts with limited data is not. In this work, we study sparse training in data-constrained regimes where limited unique tokens require multi-epoch training. Our experiments span models up to 1.92B parameters in the fitting set, sparsity up to 93.75\%, unique data budgets up to 2.6B tokens, and total training tokens up to 41.6B over 16 epochs; we further validate extrapolation on held-out dense-equivalent models up to 7.68B parameters. We find that: **1. Sparse scaling in data-limited settings:** We introduce a scaling law that models loss as a function of active parameters, unique tokens, data repetition, and sparsity, accurately predicting performance across compute and data budgets. **2. Delayed data saturation:** sparse training postpones diminishing returns from repeated data, making multi-epoch training more effective. **3. Resource trade-offs:** With fixed data, loss-optimal sparsity is moderate (~ 50%), while compute-optimal sparsity is higher and grows with data scale. Overall, sparsity is not just a tool for efficiency, but a mechanism for improving scaling trade-offs under data scarcity. Our code is available at: https://github.com/boqian333/sparse-dc-scaling.
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MCP-Persona: Benchmarking LLM Agents on Real-World Personal Applications via Environment Simulation
WenHao Wang, Peizhi Niu, Gongyi Zou, Xiyuan Yang, Jingxing Wang, Haoting Shi, Yaxin Du, Jingyi Chai, Xianghe Pang, Shuo Tang, Yanfeng Wang, Siheng Chen
The Model Context Protocol (MCP) has emerged as a transformative standard for connecting large language models (LLMs) with external data sources and tools, and has been rapidly adopted across personal applications and development platforms. However, existing benchmarks predominantly focus on generic information-seeking tools and fail to capture the practical challenges posed by personal social appl

ications, where tools interact with individual accounts or local databases.
To bridge this critical gap, we introduce MCP-Persona, the first benchmark specifically designed for evaluating agent performance on real-world, personalized MCP tools. MCP-Persona encompasses a diverse set of widely-used applications, ranging from social media platforms like Reddit and Xiaohongshu (Rednote) to enterprise collaboration suites such as Lark (Feishu) and Slack. Our extensive experiments on various state-of-the-art (SOTA) agents demonstrate their significant struggles with personalized tool use, thereby highlighting the benchmark's crucial role in identifying and addressing these limitations.
MCP-Persona is publicly available at \href{https://github.com/wwh0411/MCP-Persona}{https://github.com/wwh0411/MCP-Persona}
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GAAVI: Global Asymptotic Anytime Valid Inference for the Conditional Mean Function
Brian M Cho, Raaz Dwivedi, Nathan Kallus
Inference on the conditional mean function (CMF) is central to tasks from adaptive experimentation to optimal treatment assignment and algorithmic fairness auditing. In this work, we provide a novel asymptotic anytime-valid test for a CMF global null (e.g., that all conditional means are zero) and contrasts between CMFs, enabling experimenters to make high confidence decisions at \textit{any} time during the experiment beyond a minimum sample size. We provide mild conditions under which our tests achieve (i) asymptotic type-I error guarantees, (i) power one, and, unlike past tests, (iii) optimal sample complexity relative to a Gaussian location testing. By inverting our tests, we show how to construct function-valued asymptotic confidence sequences for the CMF and contrasts thereof. Experiments on both synthetic and real-world data show our method is well-powered across various distributions while preserving the nominal error rate under continuous monitoring.
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Feature Collapse Under Corruption: An Entropy Perspective on Robust Neural Networks
Vishesh Kumar, Akshay Agarwal
Even after decades of advances in neural network training, the inherent robustness challenge remains open. While the sensitivity to adversarial perturbations is understandable given their intentional learning, the most surprising fact is the vulnerability to natural corruptions. Surprisingly, not only is the cause of this inherent vulnerability unknown, but the concern extends beyond traditional CNNs; it also applies to current models, including transformers and large foundation models. For the first time, through this work, we observe that natural corruptions often collapse the network's internal feature space into a high-entropy state, causing predictions to rely on a small subset of fragile features. Inspired by this, we propose a simple yet effective entropy-guided fine-tuning framework, Dem-HEC, that strengthens corruption robustness while maintaining clean accuracy. Our method generates high-entropy samples within a bounded perturbation region and repairs the model using both clean and high-entropy samples. We further combine this objective with distilling knowledge from a teacher snapshot to maintain stable predictions. The proposed Dem-HEC is effective across datasets ranging from small to large-resolution, from pure CNNs to transformers, and to large foundation models, including DinoV3. The proposed approach outperforms the state-of-the-art (SOTA) models not only in improving robustness but also in retaining or boosting clean accuracy.
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Cutting LLM Evaluation Costs with SySRs: A Bandit Algorithm That Provably Exploits Model Similarity
Zifan Lyu, Chahine Nejma, Tobias Wegel, Fanny Yang, Florian E. Dorner
Large Language Models are typically benchmarked by evaluating every model on every test query. For practitioners seeking the best model to deploy, this is often wasteful: if a model clearly performs worse than others, there is no need to precisely estimate its performance. Best-arm identification algorithms can be naturally applied to drastically reduce costs by adaptively allocating evaluation bu

dget. Further, language models often respond similarly to the same prompt—a property previous work has tried to leverage with mixed success in different use cases. We propose Synchronized Successive Rejects (SySRs), augmenting the classical Successive Reject algorithm with paired comparisons. Unlike prior attempts to leverage model similarity in best-model identification, our approach is hyperparameter-free and enjoys performance guarantees that improve with the degree of similarity between evaluated models. Empirically, our method outperforms all baselines in terms of average error rate across 15 standard benchmarks, and in terms of worst-case budget for reliably identifying the best model.
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Conflict-Aware Adaptive Alignment for LLM Hallucination Mitigation
Ruohan Zong, Yang Zhang, Dong Wang
Despite strong performance, large language models (LLMs) still suffer from hallucinations. Most existing mitigation methods operate at inference time, without addressing the alignment limitation: LLMs are not trained to recognize their own lack of knowledge, and therefore tend to generate plausible responses even when the required knowledge is missing. Preference alignment approaches encourage uncertainty expression or refusal to improve truthfulness, but often consequently degrade helpfulness. To address this trade-off, existing preference alignment methods typically treat truthfulness and helpfulness as either universally collaborative or universally conflicting objectives across all samples. In contrast, we show that these objectives are consistent for most samples and conflict only in a small subset—where adaptive trade-off is truly needed. Based on this insight, we propose Conflict-Aware Adaptive Margin Preference Alignment (CAMP), which explicitly models when conflicts arise and adaptively regulates optimization strength. Experiments on UltraFeedback and representative hallucination benchmarks demonstrate that CAMP consistently improves truthfulness while maintaining a favorable helpfulness trade-off compared to strong hallucination mitigation and multi-objective alignment baselines.
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MAFE: Enabling Equitable Algorithm Design in Multi-Agent Multi-Stage Decision-Making Systems
Zachary McBride Lazri, Anirudh Nakra, Ivan Brugere, Danial Dervovic, Antigoni Polychroniadou, Furong Huang, Dana Dachman-Soled, Min Wu
Algorithmic fairness is often studied in static or single-agent settings, yet many real-world decision-making systems involve multiple interacting entities whose multi-stage actions jointly influence long-term outcomes. Existing fairness methods applied at isolated decision points frequently fail to mitigate disparities that accumulate over time. Although recent work has modeled fairness as a sequential decision-making problem, it typically assumes centralized agents or simplified dynamics, limiting its applicability to complex social systems. We introduce **MAFE**, a suite of *Multi-Agent Fair Environments* designed to simulate realistic, modular, and dynamic systems in which fairness emerges from the interplay of multiple agents. We demonstrate MAFEs in three domains—loan processing, healthcare, and higher education—supporting heterogeneous agents, configurable interventions, and fairness metrics. The environments are open-source and compatible with standard multi-agent reinforcement learning (MARL) libraries, enabling reproducible evaluation of fairness-aware policies. Through extensive experiments on cooperative use cases, we demonstrate how MAFE facilitates the design of equitable multi-agent algorithms and reveals critical trade-offs between fairness, performance, and coordination. MAFE provides a foundation for systematic progress in dynamic, multi-agent fairness research.
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Path-dependent Discrete Amortized Inference
Tiago Silva, Esmeralda S. Whitammer, Salem Lahlou
We consider the problem of sampling compositional and discrete objects from a given unnormalized posterior distribution.
Notably, recent studies have shown that this problem can be efficiently solved by learning a deterministic Markov Decision Process (MDP) that progressively builds each object in proportion to the posterior.

In this work, however, we demonstrate that the Markovian assumption can both ham per signal propagation during training and catastrophically reduce the learned s ampler's expressivity due to state aliasing.
To address these issues, we propose lifting the MDP with a learnable latent dyna mics that allows the underlying policy to depend on the entire past trajectory-- -and not only on the current state.
In view of this, we refer to the resulting method as \emph{path-dependent discre te amortized inference}.
Importantly, we provably extend existing learning algorithms for amortized sampl ers to our setting.
In experiments on standard benchmark problems, we also show that our approach of ten leads to faster learning convergence and improved state space exploration re latively to prior techniques.
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Vegas: Self-Speculative Decoding with Verification-Guided Sparse Attention
Yikang Yue, Yuqi Xue, Jian Huang
Long-context large language model (LLM) inference has become the norm for today' s AI applications. However, it is severely bottlenecked by the increasing memory demands of its KV cache. Previous works have shown that self-speculative decodi ng with sparse attention, where tokens are drafted using a subset of the KV cach e and verified in parallel against the full KV cache, speeds up inference in a l ossless manner. However, they rely on a standalone KV selection algorithm to sel ect the KV entries used for drafting and overlook the fact that the criticality of each KV entry is inherently computed during verification.

In this paper, we propose Vegas, a self-speculative decoding method with verific ation-guided sparse attention. Vegas identifies critical KV cache entries as a b yproduct of verification and computes attention only over these entries when dra fting subsequent tokens. This not only improves the draft token acceptance rate but also incurs low KV selection overhead, thereby improving decoding throughput . Vegas achieves a 1.25$\times$–2.81$\times$ speedup in decoding throughput over default vLLM and a 1.15$\times$–1.29$\times$ speedup over state-of-the-art spar se attention-based self-speculative decoding methods. Our code is available at h ttps://github.com/platformxlab/vegas.
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Active Learning with Foundation Model Priors: Efficient Learning under Class Imb alance
Jiancheng Zhang, Meiqing Li, Qi Zhang, Yinglun Zhu
Real-world datasets across image and text domains are often characterized by ske wed class distributions and noisy annotations, which jointly degrade model perfo rmance, particularly on minority classes. Among existing solutions, active learn ing offers an effective and efficient paradigm by selectively querying the most informative and balanced samples for annotation. We propose an innovative active learning framework that mitigates class imbalance and selects the most informat ive samples to annotate. Leveraging foundation model priors, our algorithm enabl es imbalance-aware co-decisions between foundation model and small model to tack le noisy and imbalanced labels across various domains. We introduce the first st udy to systematically explore active learning under the dual challenges of label noise and class imbalance across image and text domains. Extensive experiments on imbalanced datasets demonstrate that our method achieves substantial annotati on savings—over 50\% compared to the best active learning baseline—while preserv ing performance and robustness to label noise.
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What Linear Probes Miss: Multi-View Probing for Weight-Space Learning
Eunwoo Heo, Kyeongkook Seo, Jaejun Yoo
The explosive growth of open-source model repositories has created a Model Jungl e, where checkpoints are frequently shared without adequate documentation or met adata. While weight-space learning offers a pathway to identify and analyze thes e models directly from their parameters, processing full-scale weights is comput ationally prohibitive. Probing-based methods have emerged as a lightweight alter

native, extracting permutation-equivariant representations via learnable probe v ectors. However, existing probing methods are limited by a single-view design: t hey capture first-order structures but fail to encode the rich, higher-order cor relation patterns inherent in row--column interactions. To bridge this gap, we i ntroduce MVProbe, a multi-perspective probing framework that synthesizes first-o rder signals with interaction-aware (Gram-based) views. Our approach is theoreti cally grounded; we analyze the scaling laws of different probing orders to deriv e a principled standardization and fusion strategy that ensures balanced contrib utions from all branches. On the Model Jungle benchmark, MVProbe consistently ou tperforms the state-of-the-art ProbeX across diverse architectures, including Re sNet, SupViT, MAE, and DINO.
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Grounding Multi-Hop Reasoning in Structural Causal Models via Group Relative Pol icy Optimization

Yunhan Bu, quan zhang, Huaping Zhang, Guotong Geng, Chunxiao Gao, Askar Hamdulla , Juan Wang, Qiuchi Li, Baohua Zhang, Yunbo Cao, Zhunchen Luo, Shuai Lei

Multi-Hop Fact Verification requires complex reasoning across disparate evidence , posing significant challenges for Large Language Models , which may suffer fro m hallucinations and fractured logical chains. Existing methods, while improving transparency via Chain-of-Thought , often lack explicit modeling of the structu ral dependencies between evidence and claims. In this work, we introduce an SCM- inspired framework that grounds reasoning in explicit directed dependency graphs , treating verification as a constructive structural reasoning process rather th an full causal inference with interventions or counterfactual semantics. We empi rically identify an ``inverted U-shaped'' correlation between reasoning-chain le ngth and accuracy, revealing that excessive structural complexity can degrade pe rformance. To address this, we propose a rule-based reinforcement learning strat egy using Group Relative Policy Optimization. This approach dynamically optimize s the trade-off between structural depth and conciseness. Extensive experiments on HoVer and EX-FEVER demonstrate that our SCM-GRPO framework outperforms strong baselines while producing more traceable reasoning structures for complex fact verification.
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Information Flow Reveals When to Trust Language Models

Rui Xu, Yi Chen, Jiujiu Chen, Sihong Xie

In retrieval-augmented generation, language models can generate incorrect respon ses if they fail to utilize query-relevant content from the retrieved evidence. This shifts the focus of uncertainty quantification (UQ) toward assessing contex tual grounding, i.e., whether predictions are supported by query-relevant tokens . Recent UQ methods unpack language models to characterize how inputs are proces sed. Nevertheless, these methods focus on a few layers and overlook the whole pr ogressive propagation within the model, thereby failing to fully capture the gro unding dynamics essential for reliable uncertainty estimation. We use informatio n flow to build a layer-wise trace that reveals each context token’s contributio n to the output, providing an interpretable basis for assessing reliability. Fro m this analysis, we introduce two measures to calibrate prediction confidence. T he first, \textit{simulatability}, posits that a prediction is more likely to be correct when context token contributions align closely with their true relevanc e. The second, \textit{concentration}, asserts that a response is more likely to be correct when it is derived from a narrow, focused subset of tokens. Experime nts show that our method achieves an average AUROC of 0.709, exceeding the runne r-up performance of 0.676, while maintaining moderate computational cost.
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Evaluating AI Grading on Real-World Handwritten College Mathematics: A Large-Sca le Study Toward a Benchmark

Zhiqi Yu, Xingping Liu, Haobin Mao, Mingshuo Liu, Long Chen, Jack Xin, Yifeng Yu

Grading in large undergraduate STEM courses often yields minimal feedback due to heavy instructional workloads. We present a large-scale empirical study of AI g rading on real, handwritten single-variable calculus work from the University of California, Irvine. Using OCR-conditioned large language models with structured

, rubric-guided prompting, our system produces scores and formative feedback for thousands of handwritten, in-person proctored free-response quiz submissions from nearly 800 students included in the paper’s empirical analysis. In a setting with no single ground-truth label, we evaluate performance against official teaching-assistant grades, student surveys, and independent human review, finding strong alignment with TA scoring and a large majority of AI-generated feedback rated correct or acceptable across quizzes. Beyond calculus, this setting highlights core challenges in OCR-conditioned mathematical reasoning and partial-credit assessment. We analyze key failure modes, propose practical rubric- and prompt-design principles, and introduce a multi-perspective evaluation protocol for reliable real-course deployment. Building on the dataset and evaluation framework developed here, we outline a path toward a future standardized benchmark for AI grading of handwritten mathematics to support reproducible evaluation, transparent comparison, reliable deployment, and future research.
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Learning-Augmented Online Covering Problems
Afrouz Jabal Ameli, Laura Sanità, Moritz Venzin
We give a very general and simple framework to incorporate predictions on requests for online covering problems in a rigorous and black-box manner. Our framework turns any online algorithm with competitive ratio $\rho(k, \cdot)$ depending on k, the number of arriving requests, into an algorithm with competitive ratio of $\rho(\eta, \cdot)$, where η is the prediction error. With accurate enough prediction, the resulting competitive ratio breaks through the corresponding worst-case online lower bounds, and smoothly degrades as the prediction error grows. This framework directly applies to a wide range of well-studied online covering problems such as facility location, Steiner problems, set cover, parking permit, etc., and yields improved and novel bounds.
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Tiny Brains, Giant Impact: Uncovering the Keystone Neurons of LLM with Just a Few Prompts
Xiangtian Ji, Yuxin Chen, Zhengzhou Cai, Xiang Wang, An Zhang, Tat-Seng Chua
Large language models (LLMs) display strong comprehensive abilities, yet the internal mechanisms that support these behaviors remain insufficiently understood.
In this work, we show that across a wide range of open-weight Transformers, a subset of neurons remains consistently highly activated during inference across tasks of multiple capability dimensions.
By probing along the cross-task activation strength, an extremely sparse subset is isolated, whose removal causes a collapse in model behavior, which we term keystone neurons.
Our analysis reveals that keystone neurons are a stable and intrinsic neuron subset of the model that is largely established during pretraining.
The parameters associated with these neurons are tightly calibrated during the training process, and their precise values are critical for the capabilities of the model.
Building on these insights, we propose a supervised fine-tuning approach that updates only keystone neurons, achieving task gains comparable to or even better than full-parameter fine-tuning while better preserving performance in other capability dimensions, despite modifying a much smaller number of parameters.
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ViewMask-1-to-3: Multi-View Consistent Image Generation via Multimodal Discrete Diffusion Models
Ruishu Zhu, Zhihao Huang, Jiacheng Sun, Ping Luo, Hongyuan Zhang, Xuelong Li
Motivated by discrete diffusion's success in language-vision modeling, we explore its potential for multi-view generation, a task dominated by continuous approaches. We introduce **ViewMask-1-to-3**, formulating multi-view generation as a discrete sequence modeling problem where each viewpoint is represented as visual tokens from MAGVIT-v2. Through **discrete diffusion** via masked token prediction, our approach **enables progressive multi-view generation via iterative token unmasking**, unifying language and vision in a shared token space. Importantly, simple random masking combined with self-attention naturally encourages cross-vi

ew consistency without specialized architectures or 3D geometric priors. Our met hod outperforms the baseline on the GSO and 3D-FUTURE benchmarks, ranking first on average across standard image metricsand, and achieving a 10.6% higher IoU th an continuous diffusion models on 3D-FUTURE. Furthermore, the proposed framework can be naturally extended to support text-to-image generation and multimodal un derstanding, highlighting its potential toward a more unified paradigm for multi modal understanding and generation.
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Med-SegLens: Latent-Level Model Diffing for Interpretable Medical Image Segmenta tion
Salma J. Ahmed, Emad Mohammed, Azam Asilian Bidgoli
Modern segmentation models achieve strong predictive performance but remain larg ely opaque, limiting our ability to diagnose failures, understand dataset shift, or intervene in a principled manner. We introduce $\textbf{Med-SegLens}$, a mod el-diffing framework that decomposes segmentation model activations into interpr etable latent features using sparse autoencoders trained on SegFormer and U-Net. Through cross-architecture and cross-dataset latent alignment across healthy, a dult, pediatric, and sub-Saharan African glioma cohorts, we identify a stable ba ckbone of shared representations, while dataset shift is driven by differential reliance on population-specific latents. We show that these latents act as causa l bottlenecks for segmentation failures, and that targeted latent-level interven tions can correct errors and improve cross-dataset adaption without retraining, recovering performance in 70% of failure cases and improving Dice score from 39. 4% to 74.2%. Our results demonstrate that latent-level model diffing provides a practical and mechanistic tool for diagnosing failures and mitigating dataset sh ift in segmentation models.
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Conditional Coverage Diagnostics for Conformal Prediction
Sacha Braun, David Holzmüller, Michael I. Jordan, Francis Bach
Evaluating conditional coverage remains one of the most persistent challenges in assessing the reliability of predictive systems.
Although conformal methods can give guarantees on marginal coverage, no method c an guarantee to produce sets with correct conditional coverage, leaving practiti oners without a clear way to interpret local deviations.
To overcome sample-inefficiency and overfitting issues of existing metrics, we c ast conditional coverage estimation as a classification problem. Conditional cov erage is violated if and only if some classifier can achieve lower risk than the target coverage. Through the choice of a (proper) loss function, the resulting risk difference gives a conservative estimate of natural miscoverage measures su ch as L1 and L2 distance, and can even separate the effects of over- and under-c overage, as well as handle non-constant target coverages. We call the resulting family of metrics excess risk of the target coverage (ERT). We show experimental ly that the use of modern classifiers provides much higher statistical power tha n simple classifiers underlying established metrics like CovGap.
Additionally, we use our metric to benchmark different conformal prediction meth ods.
Finally, we release an open-source package for ERT as well as previous condition al coverage metrics. Together, these contributions provide a new lens for unders tanding, diagnosing, and improving the conditional reliability of predictive sys tems.
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Uncovering Grounding IDs: How External Cues Shape Multi-Modal Binding
Amirmohammad Izadi, Hosein Hasani, Fatemeh Askari, Mobin Bagherian, Sadegh Moham madian, Mohammad Izadi, Mahdieh Soleymani Baghshah
Large vision–language models (LVLMs) perform well on multimodal tasks, but their ability to reason and precisely align visual and textual information still has room for improvement. In this study, we show that external visual cues, such as symbols or grid lines, help LVLMs form more accurate connections between visual components, such as objects, and their corresponding textual descriptions, impro ving their grounding and reasoning abilities. We introduce the concept of Ground

ing IDs, which are latent identifiers that arise within the model as a result of external cues structuring both visual and textual modalities. Our analysis reveals that partition-inducing external cues lead to Grounding IDs that make better alignment between corresponding visual and text representations, helping the model focus on relevant information. We find that Grounding IDs enhance attention between related components, improving cross-modal grounding and reducing hallucinations. Overall, our results show that Grounding IDs are a key mechanism that enables external cues to improve cross-modal alignment, reduce errors, and enhance the overall performance of LVLMs across a range of multimodal tasks.
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Probing the Inductive Bias of Neural Networks through Learning Random Cellular Automata

Jan Disselhoff, Michael Wand

Why do neural networks generalize well on natural data? Natural data originates from processes subject to specific physical constraints, such as temporal and spatial invariance, that make it easier to learn. We investigate the sufficiency of these properties using 2D cellular automata as a controlled testbed: systems that are perfectly local, symmetric, and deterministic. We find that these conditions alone are \textit{not sufficient} to predict the k-step evolution of a cellular automaton. We then examine smoothness (average sensitivity) as an additional criterion and find it predictive but still incomplete. Finally, we introduce a circuit complexity perspective, hypothesizing that natural functions are computable by small circuits. Junta coefficients, measuring the concentration of Fourier weight by interaction degree, provide a tighter predictor of learnability and a correspondence to combinatorial complexity. Across architectures (CNNs, transformers, MLPs), learnable functions are predominantly those with spectral weight concentrated at low degrees and therefore low complexity. These results would be consistent with the hypothesis that natural data is learnable because natural dynamics filters out complex, high-degree interactions.
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EchoAttention: Exploiting Token-Pair Redundancy and Frame-Block Similarity for Efficient Video Generation

Yifei Xia, Fangcheng Fu, Hao Yuan, Suhan Ling, Xupeng Miao, Huixia Li, Yuxi Ren, Xin Xia, Xuefeng Xiao, Bin CUI

Diffusion Transformers (DiTs) are increasingly adopted for long-video generation, yet inference is dominated by the quadratic cost of 3D full attention.
Sparse attention mitigates this bottleneck by exploiting *token-pair redundancy* and pruning query-key interactions.
Nevertheless, its effectiveness on video generation is often constrained by non-sparse attention heads, making it hard to strike a good balance between inference speed and generation quality.
To address this, we identify another pervasive but overlooked redundancy specific to video DiTs: *frame-block similarity*, where frame-blocks in attention weights exhibit highly similar distributions and can be well approximated by lightweight linear calibration.
Motivated by this observation, we propose **EchoAttention**, which jointly leverages *token-pair redundancy* (*Sparse* operator) and *frame-block similarity* (*Echo* operator), together with a fine-grained routing policy learned via three-stage distillation.
This design enables efficient handling of both sparse and non-sparse heads, overcoming the inherent ceiling of purely sparse attention and yielding a better speed-quality trade-off.
Across public video DiTs, EchoAttention consistently improves the speed-quality frontier over SOTA sparse-attention baselines, reducing end-to-end latency up to 2.42$\times$ with minimal quality loss.
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Zero-Flow Encoders

Yakun Wang, Leyang Wang, Song Liu, Taiji Suzuki

Flow-based methods have achieved significant success in various generative modeling tasks, capturing nuanced details within complex data distributions. However,

few existing works have exploited this unique capability to resolve fine-graine
d structural details beyond generation tasks. This paper presents a flow-inspire
d framework for representation learning. First, we demonstrate that a rectified
flow trained using independent coupling is zero everywhere at $t=0.5$ if and onl
y if the source and target distributions are identical. We term this property th
e *zero-flow criterion*. Second, we show that this criterion can certify conditi
onal independence, thereby extracting \emph{sufficient information} from the dat
a. Third, we translate this criterion into a tractable, simulation-free loss fun
ction that enables learning amortized Markov blankets in graphical models and la
tent representations in self-supervised learning tasks. Experiments on both simu
lated and real-world datasets demonstrate the effectiveness of our approach.
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SSL4RL: Revisiting Self-supervised Learning as Intrinsic Reward for Visual-Langu
age Reasoning
Xiaojun Guo, Runyu Zhou, Yifei Wang, Qi Zhang, Chenheng Zhang, Stefanie Jegelka,
Xiaohan Wang, Jiajun Chai, Guojun Yin, Wei Lin, Yisen Wang
Vision-language models (VLMs) have shown remarkable abilities by integrating lar
ge language models with visual inputs. However, they often rely on textual short
cuts rather than adequately using visual evidence during reasoning. Although rei
nforcement learning (RL) can align models with desired behaviors, its applicatio
n to VLMs has been hindered by the lack of scalable and reliable rewards. To ove
rcome this challenge, we propose **SSL4RL**, a novel framework that leverages se
lf-supervised learning (SSL) tasks as a source of verifiable rewards for RL. Our
approach reformulates SSL objectives like rotation prediction and patch reconst
ruction into dense automatic rewards, removing the need for human preferences or
AI evaluators. Experiments show that SSL4RL substantially improves performance
on both vision-centric and vision-language reasoning benchmarks, with encouragin
g potentials on open-ended scenarios and stronger resilience to visual corruptio
ns. Through systematic ablations, we identify key factors influencing SSL4RL, in
cluding data volume, model scale, model choice, task combination, and task diffi
culty, thereby offering new design principles for future work. Our implementatio
n is open-sourced at https://github.com/PKU-ML/SSL4RL, with models hosted on Hug
gingface collection https://huggingface.co/collections/PKU-ML/ssl4rl.
****
Optimal conversion from Rényi Differential Privacy to f-Differential Privacy
Anneliese Riess, Juan Felipe Gomez, Flavio Calmon, Julia Schnabel, Georgios Kais
sis
We prove the conjecture stated in Appendix F.3 of Zhu et al.: among all conversi
on rules that map a Rényi Differential Privacy (RDP) profile $\tau \mapsto \rho(
\tau)$ to a valid hypothesis-testing trade-off f (or equivalently, an $(\varep
silon,\delta)$-Differential Privacy curve), the rule based on the intersection o
f single-order RDP privacy regions is optimal.

This optimality holds simultaneously for all valid RDP profiles and for all
Type I error levels α.

Concretely, we show that in the space of trade-off functions, the tightest p
ossible bound is $f_{\rho(\cdot)}(\alpha) = \sup_{\tau \geq 0.5} f_{\tau,\rho(\t
au)}(\alpha)$: the pointwise maximum of the single-order bounds for each RDP pri
vacy region.

Our proof unifies and sharpens the insights of Balle et al., Asoodeh et al.,
and Zhu et al..

Our analysis relies on a precise geometric characterization of the RDP priva
cy region, leveraging its convexity and the fact that its boundary is determined
exclusively by Bernoulli mechanisms.

Our results establish that the "intersection-of-RDP-privacy-regions" rule is
not only valid, but optimal: no other black-box conversion can uniformly domina
te it in the Blackwell sense, marking the fundamental limit of what can be infer
red about a mechanism's privacy solely from its RDP guarantees.
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HiMe: Hierarchical Embodied Memory for Long-Horizon Vision-Language-Action Contr
ol

Li Ji, Siyin Wang, Pengfang Qian, Xiaopeng Yu, Yihai Tian, Zhaoye Fei, Jingjing Gong, Xipeng Qiu
Current Vision-Language-Action (VLA) models excel at robotic manipulation but often struggle with non-Markovian tasks requiring long-term memory and reasoning due to their reliance on immediate observations. Existing solutions face a frequency-competence paradox, where high-performance models are too slow for real-time control, while faster models lack sufficient reasoning capabilities. To resolve this architectural misalignment, we propose **HiMe**, a Hierarchical Embodied Memory framework that decouples embodied intelligence into a high-frequency Executor for execution, a Sentry for working memory, and a Planner for long-term strategy. We also introduce a dynamic knowledge system based on cross-modal semantic schemas and active management mechanisms, allowing robots to maintain memory plasticity through "Add, Update, and Delete" operations. This hierarchical design effectively balances the conflict between real-time execution and slow thinking planning, significantly improving success rates in long-horizon tasks. Experiments demonstrate that this approach not only outperforms flat memory baselines but also exhibits the novel ability to self-correct its internal knowledge based on human preferences.
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Reverse Flow Matching: A Unified Framework for Online Reinforcement Learning with Diffusion and Flow Policies
Zeyang Li, Sunbochen Tang, Navid Azizan
Diffusion and flow policies are gaining prominence in online reinforcement learning (RL) due to their expressive power, yet training them efficiently remains a critical challenge. A fundamental difficulty that distinguishes online RL from standard generative modeling is the lack of direct samples from the target Boltzmann distribution defined by the Q-function. To address this, two seemingly distinct families of methods have been proposed for diffusion policies: a noise-expectation family, which uses a weighted average of noise as the training target, and a gradient-expectation family, which employs a weighted average of Q-function gradients. However, it remains unclear how these objectives are formally related, or whether they can be synthesized into a more general formulation. In this paper, we propose a unified framework, reverse flow matching (RFM), which rigorously addresses the problem of training diffusion and flow models without direct target samples. By adopting a reverse inferential perspective, we formulate the training target as a posterior mean estimation problem given an intermediate noisy sample. Crucially, we introduce Langevin Stein operators to construct zero-mean control variates, deriving a general class of estimators that share the same expectation. We show that existing noise-expectation and gradient-expectation methods are simply two specific instances within this broader class. This unified view yields two key advancements: it extends the capability of targeting Boltzmann distributions from diffusion to flow policies, and it enables the principled combination of Q-value and Q-gradient information to form an effective estimator, thereby improving training efficiency and stability. We instantiate RFM to train a flow policy in online RL and demonstrate improved performance on continuous-control benchmarks compared to diffusion policy baselines.
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Sequential Kernel-based Conditional Independence Testing via Adaptive Betting
Zheng He, Danica J. Sutherland
Testing conditional independence is fundamental yet intrinsically difficult:
  without additional assumptions, Type I error control is impossible in general.
  The ``Model-X'' paradigm addresses this difficulty by assuming exact knowledge of a relevant conditional distribution.
  While small deviations from this assumption can sometimes be tolerated in classical one-shot testing,
  existing sequential conditional independence tests typically require the Model-X conditional to be known exactly,
  making them fragile when it must instead be estimated.
  We propose a new approach that is substantially more robust to such estimation error.

Our method applies testing-by-betting to an adaptively optimized Kernel Conditional Independence statistic,
together with a normalization scheme and a truncate-and-shift calibration strategy.
These modifications greatly reduce Type I error inflation while preserving high power
across high-dimensional synthetic benchmarks and real-world fairness tasks, outperforming existing sequential Model-X approaches.
Code is available at \url{https://github.com/he-zh/SKCI}.
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SplAttN: Bridging 2D and 3D with Gaussian Soft Splatting and Attention for Point Cloud Completion
Zhaoyang Li, Zhichao You, Tianrui Li
Although multi-modal learning has advanced point cloud completion, the theoretical mechanisms remain unclear. Recent works attribute success to the connection between modalities, yet we identify that standard hard projection severs this connection: projecting a sparse point cloud onto the image plane yields an extremely sparse support, which hinders visual prior propagation, a failure mode we term Cross-Modal Entropy Collapse. To address this practical limitation, we propose SplAttN, which replaces hard projection with Differentiable Gaussian Splatting to produce a dense, continuous image-plane representation. By reformulating projection as continuous density estimation, SplAttN avoids collapsed sparse support, facilitates gradient flow, and improves cross-modal connection learnability. Extensive experiments show that SplAttN achieves state-of-the-art performance on PCN and ShapeNet-55/34. Crucially, we utilize the real-world KITTI benchmark as a stress test for multi-modal reliance. Counter-factual evaluation reveals that while baselines degenerate into unimodal template retrievers insensitive to visual removal, SplAttN maintains a robust dependency on visual cues, validating that our method establishes an effective cross-modal connection. Code is available at https://github.com/zay002/SplAttN.
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Security--Fidelity Tradeoffs: The Hidden Cost of Prompt Injection Defense
Mitchell Hermon, Rahul Gupta, Weitong Ruan, Ekraam Sabir, Haohan Wang
We identify a **security--fidelity tradeoff** in defending LLMs against indirect prompt injection: defenses resist injected instructions largely by
suppressing untrusted text, which corrupts tasks that must preserve it, such as translation and document editing. Attack-success metrics cannot see this, because a model that ignores an injection and one that faithfully processes it as data score identically. We introduce **SecFid**, a benchmark built so that *executing* an injection, *processing* it as data, and *ignoring* it produce distinguishable outputs. This makes fidelity measurable, and exposes a frontier: across 1,168 examples and 48 configurations, no model or defense
achieves both objectives. The highest-fidelity model reaches 96.5% fidelity at 47.8% security, while the most secure defenses invert this, at 99.3% security but only 71.0%--73.9% fidelity. Even defenses with identical security differ in how they earn it: some repair hijacks into faithful processing, others simply suppress benign content. A decision-theoretic analysis shows why no fixed choice can be right everywhere: the correct behavior is not a property of the defense but of the deployment, set by its relative cost of a hijack versus a dropped span. Security alone therefore measures only half of robustness, and reporting it without fidelity hides the price at which it was bought.
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Neural Logistic Bandits
Seoungbin Bae, Dabeen Lee
We study the problem of neural logistic bandits, where the main task is to learn an unknown reward function within a logistic link function using a neural network. Existing approaches either exhibit unfavorable dependencies on κ, where $1/\kappa$ represents the minimum variance of reward distributions, or suffer from direct dependence on the feature dimension d, which can be huge in neural network–based settings. In this work, we introduce a novel Bernstein-type ineq

uality for self-normalized vector-valued martingales that is designed to bypass a direct dependence on the ambient dimension. This lets us deduce a regret upper bound that grows with the effective dimension $\widetilde{d}$, not the feature dimension, while keeping a minimal dependence on κ. Based on the concentration inequality, we propose two algorithms, NeuralLog-UCB-1 and NeuralLog-UCB-2, that guarantee regret upper bounds of order $\widetilde{O}(\widetilde{d}\sqrt{\kappa T})$ and $\widetilde{O}(\widetilde{d}\sqrt{T/\kappa})$, respectively, improving on the existing results. Lastly, we report numerical results on both synthetic and real datasets to validate our theoretical findings.

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Reward-free Alignment for Conflicting Objectives

Peter Chen, Xiaopeng Li, Xi Chen, Tianyi Lin

Direct alignment methods are increasingly used to align large language models (LLMs) with human preferences. However, many real-world alignment problems involve multiple conflicting objectives, where naive aggregation of preferences can lead to unstable training and poor trade-offs. In particular, weighted loss methods may fail to identify update directions that simultaneously improve all objectives, and existing multi-objective approaches often rely on explicit reward models, introducing additional complexity and distorting user-specified preferences. The contributions of this paper are two-fold. First, we propose a **R**eward-free **A**lignment framework for **C**onflicted **O**bjectives (RACO) that directly leverages pairwise preference data and resolves gradient conflicts via a novel clipped variant of conflict-averse gradient descent. We provide convergence guarantees to Pareto-critical points that respect user-specified objective weights, and further show that clipping can strictly improve convergence rate in the two-objective setting. Second, we improve our method using some heuristics and conduct experiments to demonstrate the compatibility of the proposed framework for LLM alignment. Both qualitative and quantitative evaluations on multi-objective summarization and safety alignment tasks across multiple LLM families (Qwen 3, Llama 3, Gemma 3) show that our method consistently achieves better Pareto trade-offs compared to existing multi-objective alignment baselines.

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Incremental Learning of Sparse Attention Patterns in Transformers

O■uz Kaan Yüksel

This paper studies simple transformers trained on a high-order Markov chain, where the model must incorporate information from multiple past positions, each with different statistical importance. We show that transformers learn the task incrementally, with each stage corresponding to learning how to copy information from a subset of positions via a sparse attention pattern. Notably, the learning dynamics transition from a competitive phase, where all heads focus on the statistically most important positions, to a cooperative phase, where different heads specialize in different patterns. We model these dynamics with simplified differential equations and prove stage-wise convergence of the resulting system. Functionally, these stages correspond to a sequence of increasingly expressive misspecified models, with the full model class reached only at the end. Overall, we give a theoretical account of how structured attention patterns and head specialization emerge in stages without an explicit curriculum, with implications for generalization in sequential tasks.

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Representational Similarity and Model Behavior in Multi-Agent Interaction

Yujin Potter, Seun Eisape, Shiyang Lai, Alexander Huth, James Evans, Been Kim, Jacob Eisenstein, Dawn Song, Alane Suhr

Researchers have shown that neural similarity among humans predicts social closeness and cooperative success, whereas innovation often emerges from interactions among dissimilar individuals. We investigate whether these principles extend to artificial intelligence by examining interactions between large language models. In our experiments, 276 model pairs interact across eight games spanning both cooperation and novelty. We find that pairs with more similar representation spaces achieve significantly higher cooperation but exhibit reduced novelty and creativity. The effects of representational similarity on cooperation and novelty r

emain robust even after controlling for other factors such as performance disparity and model size. We also find that similarity in the early layers consistently shows the strongest association with cooperation and novelty, compared to the middle and later layers. This suggests that a central factor underlying these patterns could be the extent to which the two models share lexical and semantic grounding. Overall, representational similarity can be an important consideration in multi-agent system design.
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Induction Meets Biology: Mechanisms of Repeat Detection in Protein Language Models
Gal Pomerants, Yaniv Nikankin, Anja Reusch, Tomer Tsaban, Ora Schueler-Furman, Yonatan Belinkov
Protein sequences are abundant in repeating segments, both as exact copies and as approximate segments with mutations.
These repeats are important for protein structure and function, motivating decades of algorithmic work on repeat identification.
Recent work has shown that protein language models (PLMs) identify repeats, by examining their behavior in masked-token prediction.
To elucidate their internal mechanisms, we investigate how PLMs detect both exact and approximate repeats.
We find that the mechanism for approximate repeats functionally subsumes that of exact repeats.
We then characterize this mechanism, revealing two main stages: PLMs first build feature representations using both general positional attention heads and biologically specialized components, such as neurons that encode amino-acid similarity.
Then, induction heads attend to aligned tokens across repeated segments, promoting the correct answer.
Our results reveal how PLMs solve this biological task by combining language-based pattern matching with specialized biological knowledge, thereby establishing a basis for studying more complex evolutionary processes in PLMs.
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DroneDINO: Towards Heterogeneous Routed Mixture of Experts for Drone-based Unified Object Detection
Dongdong Li, Rui Chen, Yan Fan, Yan Liu, Yangliu Kuai, Pengfei Zhu
Recently, the rapid development of low-altitude aerial applications has driven the need for drone-based unified detectors. In contrast to task-specific detectors that suffer from poor scalability across diverse scenarios, existing unified detectors leverage the Mixture-of-Experts (MoE) architecture to learn task-aware features from diverse datasets. However, the imbalanced multi-task data distribution leads to over-activation of experts for dominant tasks and under-activation for others. To enable balanced feature learning, this paper combines three detection paradigms (RGB, IR, and RGB-IR) into a unified framework termed DroneDINO. DroneDINO extends DINO by introducing heterogeneous routed MoEs that organize experts into three functional groups: shared, task-specific, and dynamic. Unlike conventional dynamic experts where the top-k experts are activated for each input, the shared expert is activated for all inputs, while each task-specific expert is activated exclusively for the matching task. To ensure inputs are routed to appropriate experts and yield task-discriminative features, we propose a task-recognition auxiliary training strategy to penalize features with low task-discriminability. Experiments demonstrate the effectiveness and generalizability of DroneDINO, which consistently outperforms state-of-the-art unified and task-specific detectors across multiple drone-based detection benchmarks.
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Disentangling Latent Risk Pathways via Bayesian Hypergraph Inference
Shengxian Ding, Haonan Gao, Pangpang Liu, Xinyuan Tian, Yize Zhao
Electronic health records (EHR) pose large-scale multi-disease modeling problems in which many outcomes are rare and strongly influenced by shared risk factors. While modern approaches achieve strong predictive performance, they often treat diseases independently or rely on black-box architectures, offering limited ins

ight into how risk factors organize disease risk and little principled uncertain ty quantification.
We introduce a Bayesian hypergraph inference framework that reframes multi-disea se modeling around **latent, risk-factor-modulated disease pathways**. Risk fact ors act on hyperedges, latent disease subsets with shared risk patterns, allowin g diseases to participate in multiple distinct pathways and enabling interpretab le, higher-order structure beyond pairwise associations. A repulsion prior encou rages parsimonious and identifiable structure, while posterior inference provide s calibrated uncertainty over both disease groupings and risk-factor influence.
To enable scalable inference on large EHR datasets, we develop a structured vari ational inference algorithm that preserves logical dependencies among hyperedge existence, disease membership, and pathway-level effects.
Experiments on simulated data and UK Biobank demonstrate stable and interpretabl e disease pathway structure, well-calibrated uncertainty, improved estimation fo r rare diseases, and competitive predictive performance.
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ePC: Fast and Deep Predictive Coding in Digital Simulation
Cédric Goemaere, Gaspard Oliviers, Rafal Bogacz, Thomas Demeester
Predictive Coding (PC) offers a brain-inspired alternative to backpropagation fo r neural network training, described as a physical system minimizing its interna l energy. While ideally suited for analog implementation, such hardware does not exist yet, and thus, in practice, PC is predominantly *digitally simulated*, re quiring excessive amounts of compute while struggling to scale to deeper archite ctures. This paper reformulates PC to overcome this hardware-algorithm mismatch. First, we uncover how the canonical state-based formulation of PC (sPC) is, by design, deeply inefficient in digital simulation, inevitably resulting in expone ntial signal decay that stalls the entire numerical process. Then, to overcome t his fundamental limitation, we introduce error-based PC (ePC), a novel reparamet erization of PC which does not suffer from signal decay. Though no longer direct ly implementable in analog, ePC numerically computes *exact* PC weights gradient s and runs orders of magnitude faster than sPC. Experiments across multiple arch itectures and datasets demonstrate that ePC matches backpropagation's performanc e even for deeper models where sPC struggles. Besides practical improvements, ou r work provides theoretical insight into PC dynamics and establishes a foundatio n for scaling PC-based learning to deeper architectures in digital simulation an d beyond.
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AnyMod-LLVE: Low-Light Video Enhancement with Modality-Agnostic Inference
Hangfeng Liang, Yutao Hu, Yanhan Hu, Xiaohan Wu, Wenqi Shao, Ying Fu
Low-light video enhancement (LLVE) remains a challenging task due to severe info rmation degradation under low-illumination conditions. Recent multimodal approac hes have significantly improved enhancement performance by incorporating auxilia ry modalities, such as event streams and infrared images. However, these methods typically assume the availability of these modalities at inference, which is of ten not feasible in real-world scenarios. To solve this problem, in this work, w e propose AMNet, a unified multimodal framework for LLVE, to support flexible mo dality-agnostic inference, where auxiliary modalities may be unavailable. To add ress the issue of modality absence, we introduce a Spatial-Spectral Dual-Gated T ranslator that learns the correspondence between auxiliary modalities and RGB in puts, producing implicit auxiliary representations to support the robust enhance ment. Additionally, to fully facilitate the learning of cross-modal corresponden ce, we conduct large-scale multimodal pretraining based on the RGB-only dataset with synthetic auxiliary modalities. Extensive experiments demonstrate that AMNe t could handle arbitrary inference-time modality combinations and exhibits super ior performance for LLVE under modality absence conditions. Code and models are available on the project page.
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RL with Learnable Textual Feedback: A Bilevel Approach
Utsav Singh, Sidhaarth Sredharan Murali, Souradip Chakraborty, Amrit Singh Bedi
Reinforcement learning with verifiable rewards can improve LLM reasoning, but le

arning remains sample-inefficient when terminal rewards are sparse. This has mot
ivated a growing line of work on RL with textual feedback, where a critic model
generates natural language feedback to guide a reasoning model (the actor), augm
enting scalar rewards with richer learning signals. However, existing methods ty
pically treat feedback as fixed or auxiliary, which misses a key property: feedb
ack should not merely be correct, but should improve the policy (actor model) wh
en provided in context. This motivates a paradigm of \textit{learnable textual f
eedback for RL}. Yet the learnability and usefulness of feedback depend on the p
olicy's ability to learn from it, making RL with learnable feedback an inherentl
y bilevel problem. We formalize this coupling as a Stackelberg bilevel program a
nd derive \emph{Bilevel Natural Language Actor-Critic} (Bi-NAC), which jointly t
rains a critic to generate reward-improving feedback and an actor to exploit it.
 Across MATH-500, MBPP, and GPQA, Bi-NAC improves sample and parameter efficienc
y over RL and fixed-critic baselines: our 2B model outperforms the 3B GRPO basel
ine, achieving 46.6% versus 41.4% on MATH-500, while our 6B model surpasses the
7B GRPO baseline, achieving 49.3% versus 43.6% on GPQA. Our code is available at
 [GitHub Repository](https://github.com/sid-in-the-loop/Bi-Natural-Language-Acto
r-Critic/).
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Rethinking Parameter Sharing as Graph Coloring for Structured Compression
Boyang Zhang, Daning Cheng, Yunquan Zhang, Fangming Liu
Parameter sharing is a key model compression technique, yet existing methods ove
rlook the geometric properties of the loss landscape, often causing severe accur
acy degradation under high compression ratios. Inspired by second-order optimiza
tion, we propose Curvature-aware Graph Coloring (CGC), a cross-layer parameter s
haring framework that treats each network layer as a graph node, with each node
assigned to a shared low-rank basis. CGC leverages Hessian eigenspace informatio
n to group layers with similar curvature profiles, aligning the perturbations in
troduced by parameter sharing with the low-curvature (flat) directions of the lo
ss ellipsoid. This effectively mitigates performance loss while enabling flexibl
e, global cross-layer sharing.
Experiments on LLaMA-7B and Swin Transformer show that CGC achieves 28\%–50\% pa
rameter compression with Top-1 accuracy loss no more than 0.01\% on Swin—or even
 accuracy gains on LLaMA—while delivering over 60\% higher inference throughput,
 significantly outperforming SVD-based and heuristic-based methods. This work de
monstrates that curvature-guided, geometry-aware sharing is essential for effici
ent, stable, and high-ratio model compression.
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CriticalKV: Optimizing KV Cache Eviction from an Output Perturbation Perspective
Yuan Feng, Junlin Lv, Haoyu Guo, Yukun Cao, S Kevin Zhou, Xike Xie
Large language models have revolutionized natural language processing but face s
ignificant challenges of high storage and runtime costs, due to the transformer
architecture's reliance on self-attention, particularly the large KV cache for l
ong-sequence inference.
Recent efforts to reduce KV cache size by pruning less critical entries based on
 attention weights remain empirical and lack formal grounding. This paper presen
ts a formal study on identifying critical KV cache entries by analyzing attentio
n output perturbation.
Our analysis reveals that, beyond attention weights, the value states within KV
entries and pretrained parameter matrices are also crucial.
Based on this, we propose a perturbation-constrained selection algorithm that op
timizes the worst-case output perturbation to identify critical entries. We demo
nstrate that our algorithm is a universal, plug-and-play enhancement that incurs
 negligible computational overhead. When integrated with three state-of-the-art
cache eviction methods on three distinct LLMs, our algorithm significantly reduc
es the compression loss by more than \textit{half} on average across 29 datasets
 from the Ruler and LongBench benchmarks. Further perturbation analysis, at both
 the head and layer levels, confirms the principles underlying our effectiveness
. This work offers a new, formally grounded perspective to cache eviction , ope
ning promising avenues for future research. The code is publicly available at \u

rl{https://github.com/FFY0/DefensiveKV}.
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Is Data Shapley Not Better than Random in Data Selection? Ask NASH
Xiao Tian, Jue Fan, Rachael Hwee Ling Sim, Wang Zixuan, Nancy F. Chen, Bryan Kian Hsiang Low
Data selection studies the problem of identifying high-quality subsets of training data. While some existing works have considered selecting the subset of data with top-m Data Shapley or other semivalues as they account for the interaction among every subset of data, other works argue that Data Shapley can sometimes perform ineffectively in practice and select subsets that are *no better than random*. This raises the questions: **(I)** *Are there certain "Shapley-informative" settings where Data Shapley consistently works well?* **(II)** *Can we strategically utilize these settings to select high-quality subsets consistently and efficiently?*
In this paper, we propose a novel data selection framework, **NASH** (Non-linear Aggregation of SHapley-informative components), which **(I)** decomposes the target utility function (e.g., validation accuracy) into simpler, Shapley-informative component functions, and selects data by optimizing an objective that **(II)** aggregates these components non-linearly. We demonstrate that NASH substantially boosts the effectiveness of Shapley/semivalue-based data selection with minimal additional runtime cost.
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Joint Model and Data Sparsification via the Marginal Likelihood
Alexander Timans, Thomas Möllenhoff, Christian A. Naesseth, Mohammad Emtiyaz Khan, Eric Nalisnick
Sparse recovery in linear systems underpins applications from signal processing to high-dimensional regression. Sparse Bayesian Learning, grounded in the principle of automatic relevance determination (ARD), offers a practical Bayesian mechanism for feature sparsity via marginal likelihood optimization. Yet, its reliance on a homoscedastic noise model renders it sensitive to data contaminations such as outliers or misspecified noise, harming model fit and predictions. Instead, we propose jointly learning individual feature and sample relevancies, enabling simultaneous model and data sparsification via a single Bayesian objective. This symmetric pruning of model and data offers a natural extension that preserves conjugacy, admits closed-form updates for standard optimization procedures, and aligns with perspectives from robust regression and influence functions. Empirical results across diverse regression tasks affirm that a joint ARD approach consistently yields both sparse and robust prediction models.
****
Activation with Intrinsic-Extrinsic Consensus
Tian Qiu, Zunlei Feng, Yang Gao, Bingde Hu, Yi Gao, Mingli Song
Artificial Neural Networks (ANNs) are powerful tools for complex decision-making tasks. While existing activation mechanisms often promote sparsity through thresholding, they lack explicit awareness of feature channel relevance, causing networks to continuously suffer from interference by noisy channels. Such irrelevant activation signals can propagate through the network and adversely affect the final decision. Inspired by observations that channel relevance can be reflected in both intrinsic activity levels and extrinsic decision weights, and that there is strong consensus between these two aspects, we propose AIEC (Activation with Intrinsic-Extrinsic Consensus), a novel activation mechanism that has the ability to identify and suppress irrelevant feature channels during training. With a basic threshold activation, AIEC integrates an intrinsic Activation-Counting Unit that tracks channel activation statistics, an extrinsic Decision-Making Unit that learns channel decision weights, and a Consensus Gatekeeping Unit that suppresses irrelevant channels based on the agreement between intrinsic and extrinsic channel relevance assessments. Extensive experiments demonstrate that AIEC can effectively suppress irrelevant channels and encourage sparser representations. Furthermore, AIEC is compatible with a wide range of mainstream ANN architectures and achieves superior performance compared to existing activation mechanisms across multiple tasks and domains. Code is available at https://github.com/horri

ble-dong/AIEC.
****
Adapting Noise to Data: Generative Flows from Learned 1D Processes
Jannis Chemseddine, Gregor Kornhardt, Richard Duong, Gabriele Steidl
The default Gaussian latent in flow-based generative models poses challenges when learning certain distributions such as heavy-tailed ones.
We introduce a general framework for learning data-adaptive parametric prior distributions (latent noise) using one-dimensional quantile functions, optimized via the Wasserstein distance between noise and data. The quantile-based prior parameterization naturally adapts to both heavy-tailed and compactly supported distributions and shortens transport paths. Numerical results on heavy-tailed weather and image datasets confirm the method’s flexibility and effectiveness achieved with negligible computational overhead.
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MM-Snowball: Evaluating and Mitigating Hallucination Snowballing in Multimodal Multi-Turn Dialogue
Yue Jiang, Xue Jiang, Lihua Zhang, Zhiqiang Wang, Yuhang Lu, Peng Wang, Bo Han, Feng Zheng, Dingkang Yang
Multimodal large language models (MLLMs) demonstrate remarkable visual understanding, yet their reliability in interactive settings is severely undermined by hallucination snowballing: a phenomenon where initial errors amplify across conversational turns, leading to a collapse in coherence. This failure reveals a fundamental vulnerability where models progressively neglect visual grounding in favor of over-relying on polluted textual history. Existing benchmarks are predominantly confined to single-turn VQA, which fail to capture the complex dynamics of error propagation in long-horizon interactions. To address this, we introduce MM-Snowball, the first benchmark for fine-grained diagnosis of hallucination snowballing within dialogues.
Extensive evaluation shows that our benchmark poses a significant challenge even to advanced MLLMs and reveals the inefficacy of existing mitigation methods designed for single-turn VQA. To counteract this degradation, we propose Conflict-Aware Visual Rectification (CAVR). This training-free method mitigates snowballing through a synergistic dual-mechanism that refreshes visual grounding at the representation level and rectifies output distributions at the logit level, effectively re-anchoring the model to visual facts. Experiments demonstrate that CAVR achieves state-of-the-art performance, offering a promising path toward more reliable interactive AI. Data and code are available at: https://frenkie-chiang.github.io/MM-Snowball
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Alignment-Guided Score Matching for Text-to-Image Alignment in Diffusion Models
Jaa-Yeon Lee, Yeobin Hong, Taesung Kwon, Jong Chul Ye
Diffusion models generate highly realistic images but often struggle with precise text–image alignment. While recent post-training methods improve alignment using external rewards or human preference signals, their performance heavily depends on reward quality and does not directly address alignment within the diffusion process itself.
Recent reward-free approaches such as SoftREPA demonstrate that optimizing soft text tokens via contrastive learning can effectively improve text-image representation alignment, outperforming standard parameter-efficient fine-tuning baselines. However, the contrastive formulation can excessively penalize negative pairs, which manifests as characteristic failure cases such as over-counting and repetition.
To address this issue, we propose a lightweight, reward-free post-training method that refines soft tokens by integrating contrastive alignment guidance directly into the score-matching objective of diffusion models. By assigning alignment directions at the score level, our approach mitigates these limitations and yields more coherent and semantically faithful generations.
Experiments show that our method matches SoftREPA while substantially improving its failure cases, achieving over 35\% improvement in counting accuracy on the GenEval benchmark. Our method is seamlessly applicable to existing diffusion back

bones (SD1.5, SDXL, and SD3), and is complementary to existing RL-based diffusio
n post-training methods.
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APIC: Orthogonalized Neuro-Symbolic Modeling for Nonlinear Dissipative Dynamics
YanHui Zhu, Xiangfu Meng, Chen Zhao, Yinhao Li
Current data-driven scientific modeling struggles with a functional dichotomy: n
eural operators exhibit spectral bias in high-frequency regimes, while physics-c
onstrained paradigms suffer from optimization pathologies. To bridge this gap, w
e propose Adaptive Physics-Informed Computing (APIC), a neuro-symbolic meta-arch
itecture designed with structural reconfigurability to encode diverse domain pri
ors. Crucially, APIC integrates a gradient isolation strategy that reduces inter
ference between the optimization paths of parameter identification and residual
correction, effectively mitigating gradient conflicts. By instantiating this fra
mework for nonlinear dissipative systems, we derive the Generalized Kuramoto-Siv
ashinsky-Cahn-Hilliard (G-KSCH) kernel, providing a unified representation for s
parse dynamic identification. Extensive experiments demonstrate that APIC establ
ishes new benchmarks in 3D compressible supersonic shock wave prediction, surpas
sing diverse architectures (e.g., CNNs and Transformers) by substantial margins
in predictive accuracy. Notably, APIC achieves Pareto-optimal performance, deliv
ering superior precision with reduced computational overhead compared to SOTA mo
dels, while exhibiting strong cross-task adaptability across meteorological and
urban traffic datasets.
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PoMtVRS: Preference-Optimized Multi-Task Vehicle Routing Solver with Preference
Gating
Dian Meng, Yaoxin Wu, Yaqing Hou, Zhiguang Cao
Multi-task vehicle routing solvers via deep reinforcement learning have attracte
d broad attention and achieved significant progress in handling multiple constra
ints. However, existing neural solvers still face critical challenges, including
 insufficient representation, unstable training, and inefficient exploration in
large combinatorial action spaces, which often prevents performance from meeting
 its full potential. To address these issues, we propose PoMtVRS (Preference-Opt
imized Multi-Task Vehicle Routing Solver with Preference Gating), a plug-and-pla
y framework that jointly improves decoder representations and exploration effici
ency through a synergistic combination of decoder-side augmentation and preferen
ce-driven optimization. Specifically, we introduce the preference optimization o
bjective to learn relative comparisons among candidate solutions for different r
outing tasks, encouraging a higher generation probability of better solutions. M
eanwhile, we design a preference-gated block that adaptively modulates decoder r
epresentations via sparse gated attention and nonlinear residual refinement. Ext
ensive experiments demonstrate that PoMtVRS elevates state-of-the-art unified ne
ural VRP backbones, achieving leading performance in multi-task benchmarks and s
tronger generalization.
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Preserving Plasticity in Continual Learning via Dynamical Isometry
Andries Rosseau, Robert Müller, Ann Nowe
Continual training of deep neural networks under non-stationarity often leads to
 a progressive loss of plasticity, eventually limiting further learning. We rela
te plasticity to the empirical Neural Tangent Kernel, and identify dynamical iso
metry (the condition that layer-wise Jacobian singular values remain close to on
e) as a key mechanism for preserving plasticity in continual learning. We revisi
t a class of networks that are almost-everywhere isometric while remaining unive
rsal Lipschitz function approximators, demonstrating that near-dynamical isometr
y is compatible with expressive nonlinear representations. For general architect
ures, we propose an efficient isometry-promoting regularization scheme and ident
ify a novel mechanism by which it can reactivate dormant ReLU units. Building on
 this, we introduce AdamO, an Adam-style adaptive optimizer that decouples isome
try regularization from gradient updates, analogous to AdamW. We further reinter
pret prior plasticity-preserving approaches through the lens of dynamical isomet
ry, showing that they target only a partial measure of isometry. Across supervis

ed and reinforcement-learning continual-learning benchmarks designed to induce p lasticity loss, our methods consistently match or outperform existing approaches .

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3D-DLP: Self-supervised 3D Object-centric Scene Representation Learning

Ellina Zhang, Madhavan Iyengar, Amir Zadeh, Chuan Li, David Held, Deepak Pathak, Tal Daniel

We introduce 3D-DLP, a self-supervised object-centric representation learning mo del that decomposes scene-level RGB-D or voxel observations into a set of 3D lat ent particles. Building on the Deep Latent Particles (DLP) framework, each parti cle encodes disentangled attributes, including 3D keypoint position, bounding bo x dimensions, and appearance features, and represents a distinct entity in the s cene. The model learns interpretable per-particle segmentation maps through an e nd-to-end self-supervised reconstruction objective. We demonstrate on both simul ated and real-world datasets that the learned latent space is interpretable and controllable: by manipulating particle positions and decoding, we can generate n ovel scene configurations. Furthermore, we show that leveraging these compact 3D latent particles for downstream robotic manipulation improves performance over baselines that either lack explicit 3D information or rely on memory-intensive d ense 3D inputs without object-centric structure. Code and videos are available a t https://eubooks3003.github.io/3d-dlp

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TokSuite: Measuring the Impact of Tokenizer Choice on Language Model Behavior

Gül Sena Alt■nta■, Malikeh Ehghaghi, Brian Lester, Fengyuan Liu, Wanru Zhao, Mar co Ciccone, Colin Raffel

Tokenizers provide the fundamental basis through which text is represented and p rocessed by language models (LMs).

Despite the importance of tokenization, its role in LM performance and behavior is poorly understood due to the challenge of measuring the impact of tokenizatio n in isolation.

To address this need, we present TokSuite, a collection of models and a benchmar k that supports research into tokenization's influence on LMs.

Specifically, we release fourteen pre-trained models that use different off-the-shelf tokenizers but are otherwise identical, using the same architecture, datas et, training budget, and initialization.

We also release a multilingual robustness benchmark that measures model performa nce under real-world perturbations in English, Chinese, Farsi, Italian, and Turk ish, curated by native annotators.

Together, TokSuite allows robust decoupling of the influence of a model's tokeni zer, supporting a series of novel findings that elucidate the respective benefit s and shortcomings of a wide range of popular tokenizers.

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HyperPotter: Spell the Charm of High-Order Interactions in Audio Deepfake Detect ion

Qing Wen, Haohao Li, Zhongjie Ba, Peng Cheng, Miao He, Li Lu, Kui Ren

Advances in AIGC technologies have enabled the synthesis of highly realistic aud io deepfakes capable of deceiving human auditory perception. Although numerous a udio deepfake detection (ADD) methods have been developed, most rely on local te mporal/spectral features or pairwise relations, overlooking high-order interacti ons (HOIs). HOIs capture discriminative patterns that emerge from multiple feat ure components beyond their individual contributions. We propose HyperPotter, a hypergraph-based framework designed to capture high-order relations associated w ith synergistic patterns through clustering-based hyperedges with class-aware pr ototype initialization. Extensive experiments on 13 test sets show that HyperPot ter improves over the baseline on 11 sets, yielding an average relative EER redu ction of 12.68\% across all test sets and 22.15\% on the improved sets. These re sults demonstrate strong cross-scenario generalization, while also revealing rob ustness limits under severe codec or channel distortion.

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CORE-MTL: Rethinking Gradient Balancing via Causal Orthogonal Representations

Chengfeng Wu, Tao Zou, Yanru Wu, Jingge Wang
Multi-task learning (MTL) aims to construct a joint model for multiple tasks by sharing a common representation across domains. To achieve this goal, existing optimization-centric methods either balance task gradients or modify the shared architecture. However, as these approaches remain agnostic to the content of the shared representation, they fail to disentangle task-relevant structure from spurious context, leading to negative transfer and poor generalization. To overcome this limitation, we propose Causal Orthogonal Representations for Multi-Task Learning (CORE-MTL)
, a representation-centric framework that structurally disentangles the shared representation into semantic and residual streams, concentrating task-relevant structure in the semantic stream while relegating nuisance variation to the residual stream. We instantiate this framework in the visual domain by leveraging physical priors for structured scenes and statistical constraints for attributes. Theoretically, our method enjoys a tighter out-of-distribution generalization bound than optimization-centric methods and reduces task gradient interference without explicit gradient projection or reweighting. Empirically, CORE-MTL consistently outperforms existing methods on visual multi-task benchmarks in both in-distribution and out-of-distribution settings.
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Factored Classifier-Free Guidance
Tian Xia, Fabio De Sousa Ribeiro, Rajat R Rasal, Avinash Kori, Raghav Mehta, Ben Glocker
Counterfactual generation aims to simulate realistic hypothetical outcomes under causal interventions. Diffusion models have emerged as a powerful tool for this task, combining DDIM inversion with conditional generation and classifier-free guidance (CFG). In this work, we identify a key limitation of CFG for counterfactual generation: it prescribes a global guidance scale for all attributes, leading to significant spurious changes in inferred counterfactuals. To mitigate this, we propose *Factored Classifier-Free Guidance* (FCFG), a flexible and model-agnostic guidance technique that enables attribute-wise control following a causal graph. FCFG complements recent advances in classifier-free guidance and can be seamlessly extended to advanced guidance schemes such as CFG++ and APG. Our experiments demonstrate that FCFG significantly improves the axiomatic soundness of inferred counterfactuals across both natural and medical image datasets, mitigating spurious amplification effects, and enhancing counterfactual reversibility
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PADD: Path-Aligned Decompression Distillation for Non-Router Teacher to Guide MoE Student Learning
Xinyue Peng, Yi Qian, Jiaojiao Lin, Wenjian Shao, Yanming Liu
As large language models (LLMs) continue to scale, it becomes increasingly challenging to grow model capacity under fixed computation budgets. We propose Path-Aligned Decompression Distillation (PADD), a framework for distilling knowledge from dense teachers without explicit routing into mixture-of-experts (MoE) students while learning high-quality routing policies. PADD organizes knowledge distillation into four stages in two phases: an initialization phase (Stage I) that builds diverse functionality in the student's experts through teacher neuron clustering and student-expert warmup, and a training phase (Stages II--IV) that integrates online adaptive distillation, path-refined policy optimization, and reward-augmented load balancing in a single training pipeline.Experiments on mathematical reasoning benchmarks demonstrate that PADD yields substantial gains over strong baselines at the same inference cost and that the MoE student can match or surpass its dense teacher. They also demonstrate effective teacher-to-student knowledge distillation and stable routing behavior.
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SPEAR: A Unified SSL Framework for Learning Speech and Audio Representations
Xiaoyu Yang, Yifan Yang, Zengrui Jin, Ziyun Cui, Wen Wu, Baoxiang Li, Chao Zhang, Phil Woodland
Self-supervised learning (SSL) has significantly advanced acoustic representatio

n learning. However, most existing models are optimised for either speech or aud io event understanding, resulting in a persistent gap between these two domains. We address this gap with SPEAR (SPEech and Audio Representations), a self-super vised framework that distils complementary knowledge from a speech-focused SSL t eacher and a general-audio SSL teacher into a single unified model. SPEAR applie s multi-codebook vector quantisation to continuous teacher representations to pr oduce fine-grained discrete tokens that capture both semantic and acoustic infor mation. To effectively integrate these heterogeneous representations, SPEAR join tly predicts them given a masked input with an asymmetric pre-training loss. We further improve robustness in complex sound scenes through a novel token mixing mechanism. Extensive experiments demonstrate that SPEAR consistently outperforms existing unified speech and audio models. SPEAR establishes a new state-of-the-art on the SUPERB benchmark, surpassing WavLM Large on 12 of 15 tasks, while ach ieving competitive performance on the HEAR benchmark. These results position SPE AR as a versatile foundation for general-purpose speech and audio representation learning. The code and pre-trained models will be released.

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Symbal: Detecting Systematic Misalignments in Model-Generated Captions

Maya Varma, Jean-Benoit Delbrouck, Sophie Ostmeier, Akshay S Chaudhari, Curtis L anglotz

Multimodal large language models (MLLMs) often introduce errors when generating image captions, resulting in misaligned image-text pairs. Our work focuses on a class of captioning errors that we refer to as systematic misalignments, where a recurring error in MLLM-generated captions is closely associated with the prese nce of a specific visual feature in the paired image. Given a vision-language da taset with MLLM-generated captions, our aim in this work is to detect such error s, a task we refer to as systematic misalignment detection. As our first key con tribution, we present Symbal, which utilizes a structured, dual-stage setup with off-the-shelf foundation models to identify systematic misalignments and summar ize results in natural language. As our second key contribution, we introduce Sy mbalBench, a benchmark designed to evaluate automated methods on our proposed ta sk. SymbalBench consists of 1.7 million image-text pairs from two domains (natur al and medical images), organized into 420 vision-language datasets with annotat ed systematic misalignments. Symbal exhibits strong performance on this benchmar k, correctly identifying systematic misalignments in 63.8% of datasets, a nearly 4x improvement over the closest baseline. We supplement our evaluations on Symb alBench with real-world evaluations, showing that (1) Symbal can accurately surf ace systematic misalignments in captions generated by four MLLMs and (2) Symbal is a powerful tool for auditing off-the-shelf image-caption datasets. Ultimately , our novel task, method, and benchmark can aid users with auditing MLLM-generat ed captions and identifying critical errors, without requiring access to the und erlying MLLM. Code is available at https://github.com/Stanford-AIMI/Symbal.

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EEG-FM-Bench: A Comprehensive Benchmark for the Systematic Evaluation and Diagno stic Analyses of EEG Foundation Models

Wei Xiong, Jiangtong Li, Jie Li, Kun Zhu, Changjun Jiang

Electroencephalography foundation models (EEG-FMs) have advanced brain signal an alysis, but the lack of standardized evaluation benchmarks impedes model compari son and scientific progress.

Current evaluations rely on inconsistent protocols that render cross-model compa risons unreliable, while a lack of diagnostic analyses obscures the internal mec hanisms driving transfer efficiency and scaling behaviors.

To address this, we introduce **EEG-FM-Bench**, a unified system for the standar dized evaluation of EEG-FMs.

The benchmark integrates 14 datasets across 10 paradigms and incorporates divers e experimental settings, including multiple fine-tuning strategies, task organiz ations, and classifier configurations, supported by tools for gradient and repre sentation analysis.

Our experiments and analysis reveal several critical insights: (1) multi-task le arning often acts as a useful regularizer that mitigates overfitting in data-sca

rce EEG contexts, although negative transfer can arise under specific task paradigms; (2) pre-training efficiency is currently limited by gradient conflicts between reconstruction objectives and downstream tasks;
(3) under released checkpoints and a matched downstream protocol,
model or data scale alone does not fully explain transfer performance, while objective alignment, adaptation compatibility, and EEG-specific design appear to be important factors.
This benchmark enables fair comparison and reproducible analysis, providing a step toward fairer comparison and more interpretable analysis of EEG-FMs. Code is available at https://github.com/xw1216/EEG-FM-Bench.
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SoftMoE: Soft Differentiable Routing for Mixture-of-Experts in LLMs
Miko■aj Zasada, ■ukasz Struski, Jacek Tabor, Marcin Kurdziel
Sparse Mixture-of-Experts (MoE) architectures enable scaling LLM parameters under a fixed inference budget by activating only a small subset of experts via top-k routing. While this preserves causality and suits autoregressive language models, the discrete top-k operator is not differentiable, forcing a fixed number of active experts per input and resulting in inefficient use of computation. We propose SoftMoE, which replaces discrete routing with a truncated soft top-k LapSum relaxation, allowing gradient-based optimization of expert routing. We further parameterize the mean number of active experts per layer and impose a global budget constraint, enabling the model to learn how to allocate expert capacity across layers. SoftMoE remains fully compatible with autoregressive modeling and achieves performance comparable to or better than sparse MoE on language modeling and downstream tasks, while activating significantly fewer experts. Notably, the learned allocation is highly non-uniform, with later layers activating more experts. The source code is publicly available†.
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Toward Safe Quantization-Aware Fine-tuning: Understanding and Mitigating Safety Alignment Degradation
Yuning Yang, Guowei Peng, Xiurui Xie, Minrui Jiang, Shuang Liang, Guisong Liu
Large language models (LLMs) are increasingly adapted to downstream tasks in resource-constrained scenarios, making quantization-aware fine-tuning (QAF) a common practice for practical deployment. However, we find that quantized LLMs are substantially more vulnerable to safety alignment degradation during fine-tuning than full-precision models by interpretability analyses. In this paper, we first theoretically reveal that this vulnerability is driven by quantization errors, manifesting as an initial safety shift followed by a distorted optimization path. Based on this insight, we propose Explicit-Safety Quantization-Aware Fine-tuning (ExSQF), which effectively restores model safety while preserving downstream performance. It initializes adapters by combining quantization error with a safety matrix projection to mitigate early safety shifts, followed by post-training refinement that corrects deviations in the optimization path. Extensive experimental results show that ExSQF achieves state-of-the-art safety alignment recovery, even surpassing existing full-precision safety-aware fine-tuning baseline, while effectively preserving model performance.
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Effective Reasoning Chains Reduce Intrinsic Dimensionality
Archiki Prasad, Mandar Joshi, Kenton Lee, Mohit Bansal, Peter Shaw
Chain-of-thought (CoT) reasoning and its variants have substantially improved the performance of language models on complex reasoning tasks, yet the precise mechanisms by which different strategies facilitate generalization remain poorly understood. While current explanations often point to increased test-time computation or structural guidance, establishing a consistent, quantifiable link between these factors and generalization remains challenging. In this work, we identify *intrinsic dimensionality* as a quantitative measure for characterizing the effectiveness of reasoning chains. Intrinsic dimensionality quantifies the minimum number of model dimensions needed to reach a given accuracy threshold on a given task. By keeping the model architecture fixed and varying the task formulation through different reasoning strategies, we demonstrate that effective reasoning

strategies consistently reduce the intrinsic dimensionality of the task. Validating this on GSM8K with Gemma-3 1B and 4B, we observe a strong inverse correlation between the intrinsic dimensionality of a reasoning strategy and its generalization performance on both in-distribution and out-of-distribution data. Our findings suggest that effective reasoning chains facilitate learning by better compressing the task using fewer parameters, offering a new quantitative metric for analyzing reasoning processes.
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A Semantically Consistent Dataset for Data-Efficient Query-Based Universal Sound Separation
Kai Li, Jintao Cheng, Chang Zeng, Zijun Yan, Helin Wang, Zixiong Su, Bo Zheng, Xiaolin Hu
Query-based universal sound separation is fundamental to intelligent auditory systems, aiming to isolate specific sources from mixtures. Despite recent advances, existing methods continue to suffer from residual interference in complex acoustic scenes. This performance limitation stems largely from a data bottleneck: in-the-wild datasets contain weak labels and severe co-occurrence of events. These flaws induce models to learn spurious correlations between background noise and target categories instead of robust acoustic features. To address this, we propose an automated pipeline that eliminates co-occurrence of events by mining high-purity single-event segments from in-the-wild datasets via a semantically consistent synthesis protocol. Utilizing this pipeline, we constructed Hive, a high-quality synthetic dataset comprising 2.4k hours of raw audio. Experimental results demonstrate that, compared with the state-of-the-art model SAM-Audio which was trained on a huge dataset $\sim$500 times larger than Hive, certain open-source models trained on Hive achieve competitive separation accuracy and perceptual quality. Moreover, these models exhibited remarkable zero-shot generalization on out-of-distribution evaluation benchmarks. These findings highlight that prioritizing purity of supervised signals enables significant data efficiency, offering a new paradigm for training robust auditory foundation models with reduced computational costs. Code and dataset are available at https://cslikai.cn/Hive.
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Small Generalizable Prompt Predictive Models Can Steer Efficient RL Post-Training of Large Reasoning Models
Yun Qu, Qi Wang, Yixiu Mao, Heming Zou, Yuhang Jiang, Weijie Liu, Clive Bai, Kai Yang, Yangkun Chen, Saiyong Yang, Xiangyang Ji
Reinforcement learning enhances the reasoning capabilities of large language models but often involves high computational costs due to rollout-intensive optimization.
Online prompt selection presents a plausible solution by prioritizing informative prompts to improve training efficiency.
However, current methods either depend on costly, exact evaluations or construct prompt-specific predictive models lacking generalization across prompts.
This study introduces Generalizable Predictive Prompt Selection (GPS), which performs Bayesian inference towards prompt difficulty using a lightweight generative model trained on the shared optimization history.
Intermediate-difficulty prioritization and history-anchored diversity are incorporated into the batch acquisition principle to select informative prompt batches.
The small predictive model also generalizes at test-time for efficient computational allocation.
Experiments across varied reasoning benchmarks indicate GPS's substantial improvements in training efficiency, final performance, and test-time efficiency over superior baseline methods.
The code is available at https://github.com/thu-rllab/GPS.
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Learning Rate Scaling across LoRA Ranks and Transfer to Full Finetuning
Nan Chen, Soledad Villar, Soufiane Hayou
Low-Rank Adaptation (LoRA) is a standard tool for parameter-efficient finetuning of large models. While it induces a small memory footprint, its training dynami

cs can be surprisingly complex as they depend on several hyperparameters such as initialization, adapter rank, and learning rate. In particular, it is unclear *how the optimal learning rate scales with adapter rank*, which forces practitioners to re-tune the learning rate whenever the rank is changed. In this paper, we introduce *Maximal-Update Adaptation* (μA), a theoretical framework that characterizes how the "optimal" learning rate should scale with model width and adapter rank to produce stable, non-vanishing feature updates under standard configurations. Our analysis leverages techniques from hyperparameter transfer and reveals that the optimal learning rate exhibits different scaling patterns depending on initialization and LoRA scaling factor. Specifically, we identify two regimes: one where the optimal learning rate remains roughly invariant across ranks, and another where it scales inversely with rank.
We further identify a configuration that allows learning rate transfer from LoRA to full finetuning, drastically reducing the cost of learning rate tuning for full finetuning. Experiments across language, vision, vision–language, image generation, and reinforcement-learning tasks validate our scaling rules and show that learning rates tuned on LoRA transfer reliably to full finetuning.
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Emergence of Hierarchical Emotion Organization in Large Language Models
Maya Okawa, Bo Zhao, Eric Bigelow, Rose Yu, Tomer Ullman, Ekdeep Singh Lubana, Hidenori Tanaka
As large language models (LLMs) increasingly power conversational agents, understanding how they model users' emotional states is critical for ethical deployment. Inspired by emotion wheels, i.e., a psychological framework that argues emotions organize hierarchically, we analyze probabilistic dependencies between emotional states in model outputs. We find that LLMs naturally form hierarchical emotion trees that align with human psychological models, and larger models develop more complex hierarchies. We also uncover systematic biases in emotion recognition across socioeconomic personas, with compounding misclassifications for intersectional, underrepresented groups. Human studies reveal striking parallels, suggesting that LLMs internalize aspects of social perception. Beyond highlighting emergent emotional reasoning in LLMs, our results hint at the potential of using cognitively-grounded theories for developing better model evaluations.
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Investigating Memory in Model-Free RL with POPGym Arcade
Zekang Wang, Zhe He, Borong Zhang, Edan Toledo, Steven Morad
How should we analyze memory in deep RL? We introduce tools for analyzing policies under partial observability and revealing how agents use memory to make decisions. To utilize these tools, we present POPGym Arcade, a collection of Atari-inspired, hardware-accelerated environments sharing a single observation and action space. Each environment provides fully and partially observable variants, enabling counterfactual studies on observability. We find that controlled studies are necessary for fair comparisons and identify a pathology where value functions smear credit over irrelevant history. Using this pathology, we demonstrate how out-of-distribution scenarios can contaminate memory, perturbing the policy far into the future.
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Can Recommender Systems Teach Themselves? A Recursive Self-Improving Framework with Fidelity Control
Luankang Zhang, Hao Wang, Zhongzhou Liu, Mingjia Yin, Yonghao Huang, Jiaqi Li, Wei Guo, Yong Liu, Huifeng Guo, Defu Lian, Enhong Chen
The scarcity of high-quality training data presents a fundamental bottleneck to scaling machine learning models. This challenge is particularly acute in recommendation systems, where extreme sparsity in user interactions leads to rugged optimization landscapes and poor generalization. We propose the Recursive Self-Improving Recommendation (RSIR) framework, a paradigm in which a model bootstraps its own performance without reliance on external data or teacher models. RSIR operates in a closed loop: the current model generates plausible user interaction sequences, a fidelity-based quality control mechanism filters them for consistency with user’s approximate preference manifold, and a successor model is augmented

on the enriched dataset. Our theoretical analysis shows that RSIR acts as a data-driven implicit regularizer, smoothing the optimization landscape and guiding models toward more robust solutions. Empirically, RSIR yields consistent, cumulative gains across multiple benchmarks and architectures. Notably, even smaller models benefit, and weak models can generate effective training curricula for stronger ones. These results demonstrate that recursive self-improvement is a general, model-agnostic approach to overcoming data sparsity, suggesting a scalable path forward for recommender systems and beyond. Our code is available at https://github.com/USTC-StarTeam/RSIR.

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Evidential Copula Concept Embedding Models

Yanjie Qiu, Xiaodong Yue, Xuhui Fan, Yufei Chen, Jie Shi, Wei Liu

Concept Embedding Models (CEMs) advance interpretable AI by extending Concept Bottleneck Models (CBMs) through semantic concept embeddings, providing an important solution in high-stakes domains such as medical diagnosis where accuracy and interpretability are critical. However, a fundamental limitation persists: existing CEMs inherently assume concept independence, critically overlooking the highly complex dependencies among concepts. To address this, we propose an Evidential Copula Concept Embedding Model (EC-CEM) that redefines the joint distribution over concepts, capturing inter-concept dependencies while maintaining a flexible structure that decouples the marginal concept distributions from their dependency structure. In particular, EC-CEM relaxes the concept independence assumption and uniquely integrates Copula theory with evidential deep learning to define a joint distribution over concepts. The proposed EC-CEM also develops two training objectives that aim at classification and concept modeling simultaneously. We provide theoretical justification via variational inference and demonstrate empirical superiority through extensive experiments.

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Joint Enhancement and Classification using Coupled Diffusion Models of Signals and Logits

Gilad Nurko, Roi Benita, Yehoshua Dissen, Tomohiro Nakatani, Marc Delcroix, Shoko Araki, Joseph Keshet

Robust classification in noisy environments remains a fundamental challenge in machine learning. Standard approaches typically treat signal enhancement and classification as separate, sequential stages: first enhancing the signal and then applying a classifier. This approach fails to leverage the semantic information in the classifier's output during denoising. In this work, we propose a general, domain-agnostic framework that integrates two interacting diffusion models: one operating on the input signal and the other on the classifier's output logits, without requiring any retraining or fine-tuning of the classifier. This coupled formulation enables mutual guidance, where the enhancing signal refines the class estimation and, conversely, the evolving class logits guide the signal reconstruction towards discriminative regions of the manifold. We introduce three strategies to effectively model the joint distribution of the input and the logit. We evaluated our joint enhancement method for image classification and automatic speech recognition. The proposed framework surpasses traditional sequential enhancement baselines, delivering robust and flexible improvements in classification accuracy under diverse noise conditions.

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CauSciBench: Can LLMs Automate Causal Inference in Real-World Scientific Research?

Sawal Acharya, Terry Jingchen Zhang, Andrew Kim, Rahul Babu Shrestha, Xianlin Sun, Pepijn Cobben, Maximilian Mordig, Jacob T. Emmerson, Anahita Haghighat, Furkan Danisman, Yuen Chen, Clijo Jose, Andrei Ioan Muresanu, Justin Cui, Jiarui Liu, Yahang Qi, Punya Syon Pandey, Yinya Huang, Bernhard Schölkopf, Zhijing Jin

Identifying and estimating causal relationships from data is a crucial component of empirical research. While large language model-powered tools have shown potential for assisting research workflows, their ability to perform end-to-end causal inference remains underexplored. We introduce CauSciBench, a benchmark that puts LLM-powered tools to the test on causality- driven research questions. Unlik

e previous related benchmarks that focus on coding alone, CauSciBench enables ev aluation across the full pipeline of causal inference: from method and variable selection to computation of causal effects and statistical interpretation in the context of real-world research problems. We evaluated 7 frontier models on over 300 queries derived from scientific publications, textbook problems, sem- inal datasets, and synthetic scenarios. Results show that models consistently perform worse on real datasets, with the key bottleneck being the selection of an appro priate causal inference method.
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Regularized Offline Policy Optimization with Posterior Hybrid Bayesian Belief
Hongqiang Lin, Pengfei Wang, Nenggan Zheng
Offline reinforcement learning (RL) aims to optimize policies from pre-collected datasets. A bottleneck of this paradigm is managing epistemic uncertainty, whic h arises from limited data coverage (sample-level) and the ambiguity in identify ing transition dynamics from finite data (model-level). To provide a unified qua ntification of these uncertainties, Bayesian RL has been proposed by treating th e dynamics model as a random variable and maintaining a corresponding belief. De spite its theoretical appeal, policy optimization in Bayesian RL remains computa tionally challenging as it requires solving composite objectives with expectatio ns. Prior methods either employ search-based techniques with poor computational scalability or impose restrictive posterior assumptions that sacrifice the adapt ability of Bayesian RL. To address these limitations, we propose Posterior Hybri d Bayesian Belief (PhyB), which reformulates the expectation as a convex combina tion over a subset of dynamics models. Theoretical analysis demonstrates that th e objective discrepancy induced by this approximation remains bounded. Based on PhyB, we develop an iterative regularized policy optimization algorithm that pro vides metric-agnostic guarantees for monotonic improvement until convergence. Em pirical results demonstrate that PhyB achieves state-of-the-art performance on v arious benchmarks.
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Do Audio LLMs Listen or Read? Analyzing and Mitigating Paralinguistic Failures w ith VoxParadox
Jiacheng Pang, Ashutosh Chaubey, Mohammad Soleymani
Audio large language models (Audio LLMs) demonstrate strong performance on speec h understanding tasks, yet their ability to understand paralinguistic informatio n remains limited. To systematically quantify this issue, we introduce **VoxPara dox**, an adversarial benchmark with 2,000 verified examples, spanning 10 parali nguistic tasks, created with controlled speech synthesis to intentionally mismat ch transcript claims and speaking style, enabling direct measurement of speech p aralinguistic understanding. Evaluation of a diverse set of Audio LLMs reveals c onsistently low accuracy on acoustic ground truth and a strong tendency to follo w language-implied (incorrect) answers. To understand the cause of this gap, we perform layer-wise probing and find that (i) paralinguistic cues can degrade in deeper encoder layers and at the encoder--LLM interface, and (ii) even when such cues are available in audio tokens, the language model frequently ignores them. To address these problems, we propose **Prompt-Conditioned Layer Mixer (PCLM)** , which adaptively combines information from multiple audio layers based on the input prompt, and pair it with **Direct Preference Optimization (DPO)** to expli citly prefer acoustically supported options over language-implied alternatives. These methods substantially improve Audio LLM paralinguistic understanding, impr oving Audio Flamingo 3 from **17.40%** to **65.20%** on VoxParadox, and from **3 7.74%** to **54.78%** on MMSU paralinguistic subset. Our project page is availab le at https://voxparadox.github.io/.
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Taming Stochastic Gradient Descent: Almost Sure Convergence and Saddle-Point Avo idance under (L_{0},L_{1})-Smoothness
Vassilis Apidopoulos, Iosif Lytras, Panayotis Mertikopoulos
Many optimization problems in machine learning and data science—from deep neural networks to Bayesian inference and beyond—fall outside the standard Lipschitz s moothness framework that underpins the convergence theory of stochastic gradien

t descent (SGD). Motivated by this theory-practice disconnect, we examine the al
most sure convergence of the trajectories of SGD in non-convex landscapes under
a generalized (L_0,L_1)-smoothness condition which allows for gradients with s
uperlinear growth (even exponential). We begin by proposing a taming scheme for
SGD that achieves almost sure convergence under a generalized ABC-type condition
 on the gradient noise. Subsequently, to relax this requirement, we introduce a
more flexible, dissipative taming scheme which converges almost surely under les
s restrictive moment bound conditions for the stochastic gradients entering the
process. For both taming schemes, we show that the generated trajectories avoid
strict saddle points (and/or manifolds thereof) with probability 1 so, generical
ly, both methods only converge to local minimizers.
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Knapsack RL: Compute-Efficient Reinforcement Learning via Heterogeneous Rollout
Allocation

Ziniu Li, Congliang Chen, Tianyun Yang, Tian Ding, Ruoyu Sun, Ge Zhang, Wenhao H
uang, Zhi-Quan Luo

Reinforcement learning (RL) fine-tuning of Large Language Models (LLMs) is compu
te-intensive because each prompt requires generating multiple responses, or roll
outs. To make the best use of GPU resources, the key question is how to allocate
 rollout-generation jobs across prompts. Existing methods typically use uniform
allocation, assigning every prompt the same rollout budget. This is inefficient
and ineffective: easy prompts are oversampled after they are already solved, whi
le hard prompts receive too little exploration. In policy optimization methods s
uch as Group Relative Policy Optimization (GRPO), both cases often yield near-ze
ro gradients and limited learning progress. We address this problem by formulati
ng rollout allocation as a computeconstrained resource allocation task, where ea
ch prompt-budget choice has an expected learning value and rollout cost. Based o
n this view, we propose Knapsack RL, a practical framework that uses knapsack op
timization to assign heterogeneous, task-specific rollout budgets. It directs ex
ploration toward prompts that benefit most from additional sampling. Applied to
GRPO, Knapsack RL increases the effective-gradient ratio by up to 40%, enables l
arger budgets for challenging prompts, and improves mathematical reasoning by 2–
4 points on average, with peak gains of up to 9 points. Achieving similar perfor
mance with uniform allocation requires about 2× more compute, demonstrating a si
mple and practical path toward scaling RL fine-tuning for LLMs.
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InfoGeo: Information-Theoretic Object-Centric Learning for Cross-View Generaliza
ble UAV Geo-Localization

Hongyang ZHANG, Maonan Wang, Ziyao Wang, Hongrui Yin, Man On Pun

Cross-view geo-localization (CVGL) is fundamental for precise localization and n
avigation in GPS-denied environments, aiming to match ground or UAV imagery with
 satellite views. Existing approaches often rely on global feature alignment, bu
t they suffer from substantial domain shifts induced by varying regional texture
s and weather conditions. This issue becomes even more pronounced in UAV-based s
cenarios, where the broader perspective inevitably introduces dense, fine-graine
d objects, creating significant visual clutter. To address this, we draw inspira
tion from Object-Centric Learning (OCL) and propose InfoGeo, an information-theo
retic framework designed to enhance robustness and generalization. InfoGeo refor
mulates the optimization as an information bottleneck process with two core obje
ctives: (i) maximizing view-invariant information by aligning the object-centric
 structural relations across views, and (ii) minimizing view-specific noisy sign
als through cross-view knowledge constraints. Extensive evaluations across diver
se benchmarks and challenging scenarios demonstrate that InfoGeo significantly o
utperforms state-of-the-art methods.
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HiDe: Rethinking The Zoom-IN method in High Resolution MLLMs via Hierarchical De
coupling

Xianjie Liu, Yiman Hu, Yixiong Zou, Liang Wu, Jian Xu, Bo Zheng

Multimodal Large Language Models have made substantial progress on visual unders
tanding tasks, yet they still perform poorly on high-resolution images. Prior wo

rk often attributes this limitation to perceptual constraints, arguing that MLLM s fail to recognize small objects and therefore rely on ''zoom-in" strategies to recover fine details. In contrast, our analysis shows that the dominant failure mode is background interference rather than object size.
We study the "zoom-in" operation through a **hierarchical decoupling analysis** and propose the **Hierarchical Decoupling Framework**, a training-free method th at turns implicit attention into explicit region selection. HiDe first performs Token-wise Attention Decoupling to disentangle question semantics and identify t he most informative tokens, then uses their attention patterns to pinpoint the c orresponding visual regions. It subsequently applies Layout-Preserving Decouplin g to extract these regions from cluttered backgrounds and construct a compact re presentation that retains key spatial structure while filtering out irrelevant c ontext. HiDe achieves state-of-the-art results on high-resolution benchmarks lik e Vstar Bench. It boosts Qwen2.5-VL 7B and InternVL3 8B to state of the art perf ormance, reaching 92.1\% and 91.6\% on Vstar Bench, and even surpasses reinforce ment learning based methods. After optimization, HiDe reduces memory usage by 75 \% compared with the previous training-free approach. Code will be available at https://tennine2077.github.io/HiDe.github.io/.
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Spectral Guidance for Flexible and Efficient Control of Diffusion Models
Gabriel Moreira, Manuel Marques, Joao Costeira, Chenyan Xiong
We introduce Spectral Guidance, a framework for controlling diffusion models by leveraging the intrinsic geometry of the generative process. As data is progress ively corrupted by noise, only a small number of features remain informative for control. We characterize them as the singular functions of a conditional expect ation operator and show that they can be learned via a self-supervised objective . Once recovered, this basis enables the projection of arbitrary guidance signal s, such as labels, CLIP embeddings, or masks, directly onto the sampling traject ory. This approach allows for stable, high-fidelity control without retraining o r denoiser backpropagation during sampling. Empirically, we improve conditional accuracy on CIFAR-10 by 37 percentage points over the strongest training-free ba seline while offering $4\times$ faster sampling. Moreover, the same representati ons that support label and CLIP guidance also enable spatial control, such as ma sk-based guidance, without auxiliary models. Finally, our framework reveals a ph ase transition in the generative process, pinpointing the optimal time window fo r effective guidance.
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Escaping the Mode: Multi-Answer Reinforcement Learning in LMs
Isha Puri, Mehul Damani, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, Yoon Ki m
Given a question, a language model (LM) implicitly encodes a distribution over p ossible answers. In practice, post-training procedures for LMs often collapse th is distribution onto a single dominant mode. While this is generally not a probl em for benchmark-style evaluations that assume one correct answer, many real-wor ld tasks inherently involve multiple valid answers or irreducible uncertainty. E xamples include medical diagnosis, ambiguous question answering, and settings wi th incomplete information. In these cases, we would like LMs to generate multipl e plausible hypotheses, ideally with confidence estimates for each one, and with out computationally intensive repeated sampling to generate non-modal answers. T his paper describes a multi-answer reinforcement learning approach for training LMs to perform distributional reasoning over multiple answers during inference. We modify the RL objective to enable models to explicitly generate multiple cand idate answers in a single forward pass, internalizing aspects of inference-time search into the model's generative process. Across question-answering, medical d iagnostic, and coding benchmarks, we observe improved diversity, coverage, and s et-level calibration scores compared to single answer trained baselines. Models trained with our approach require fewer tokens to generate multiple answers than competing approaches. On coding tasks, they are also substantially more accurat e. These results position multi-answer RL as a principled and compute-efficient alternative to inference-time scaling procedures such as best-of-k.

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RNA-FM: Flow-Matching Generative Model for Genome-wide RNA-Seq Prediction
Yaxuan Song, Jianan Fan, Tianyi Wang, Qiuyue Hu, Hang Chang, Heng Huang, Weidong Cai
Histopathology whole-slide images (WSIs) are routinely acquired in clinical practice and contain rich tissue morphology but lack direct molecular architecture and functional programs defining pathological states, whereas RNA sequencing (RNA-seq) provides genome-wide transcriptional profiles at substantial cost, thereby motivating WSI-based genome-wide transcriptomic prediction. Existing approaches for predicting gene expression from WSIs predominantly rely on deterministic regression with one-to-one mapping, limiting their ability to capture biological heterogeneity and predictive uncertainty. We propose RNA-FM, a flow-matching generative framework for genome-wide bulk RNA-seq prediction from WSIs. RNA-FM formulates transcriptomic prediction as a continuous-time conditional transport problem, learning a velocity field that maps a simple prior to the target gene expression distribution conditioned on morphologies. By integrating pathway-level structure, RNA-FM enables scalable and biologically interpretable genome-wide gene expression imputation. Extensive experiments demonstrate that RNA-FM consistently outperforms state-of-the-art approaches while maintaining biological meaningfulness. Code is available at https://github.com/YXSong000/RNA-FM.
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LDARNet: DNA Adaptive Representation Network with Learnable Tokenization for Genomic Modeling
Daria Ledneva, Denis Kuznetsov
Genomic foundation models increasingly adopt large language model architectures, yet almost universally rely on fixed tokenization schemes such as k-mers, BPE, or single nucleotides, which impose arbitrary sequence boundaries that may obscure biologically relevant structure. We present LDARNet, a 120M-parameter hierarchical genomic foundation model that adapts H-Net-style dynamic chunking from autoregressive generation to masked language modeling, combining BiMamba-2 state-space layers with local attention, bidirectional routing, and a ratio-based regularizer to induce adaptive token boundaries without supervision. Fine-tuned on 27 tasks from the Nucleotide Transformer and Genomic Benchmarks suites, LDARNet achieves 11/18 wins among compact models ($<$300M parameters) and state-of-the-art results on 5 histone modification tasks, outperforming models up to 20$\times$ larger. A FLOPs-matched controlled experiment isolates learned routing as the source of these gains: learned boundaries beat fixed-grid boundaries by up to 14 percentage points on histone tasks at identical compute. Nucleotide-resolution analysis further shows that the learned boundaries align with canonical promoter motifs and splice junctions without supervision, providing a biological interpretation for adaptive tokenization in genomic foundation models.
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Depth-Breadth Synergy in RLVR: Unlocking LLM Reasoning Gains with Adaptive Exploration
Zhicheng Yang, Zhijiang Guo, Yinya Huang, Yongxin Wang, Dongchun Xie, Hanhui Li, Yiwei Wang, Xiaodan Liang, Jing Tang
Reinforcement Learning with Verifiable Reward (RLVR) is a powerful method for enhancing the reasoning abilities of Large Language Models, but its full potential is limited by a lack of exploration in two key areas: \textbf{Depth} (the difficulty of problems) and \textbf{Breadth} (the number of training instances). Our analysis of the popular GRPO algorithm reveals a bias that down-weights difficult, low-accuracy problems, which are crucial for improving reasoning skills. To address this, we introduce Difficulty Adaptive Rollout Sampling (DARS), a method that re-weights difficult problems by using targeted, multi-stage rollouts. This approach increases the number of rollout outcomes for these harder problems according to our proposed re-balancing schedules and leads to consistent gains in \textit{Pass@K}. We also found that simply enlarging the rollout size isn't effective and can even harm performance. We also investigated the role of breadth by scaling the batch size and using full-batch updates. This significantly improved \textit{Pass@1} performance by maintaining high token-level entropy, which indi

cates continued exploration and reduced gradient noise. Finally, we present DARS-Breadth, a combined approach that uses DARS with a large breadth of training data. This method demonstrates simultaneous gains in both \textit{Pass@K} and \textit{Pass@1}, confirming that depth (adaptive exploration) and breadth (scaling the training data) are orthogonal and essential dimensions for unlocking the full reasoning power of RLVR.
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Towards Trustworthy and Identifiable Virtual Face Generation
Chunyang Li, Haoyue Wang, Zhenxing Qian, Sheng Li, Xinpeng Zhang, Jian liu, Yixuan Pei, Weiqiang Wang
Identifiable virtual face (IVF) generation aims to transform a user's original face into a virtual face for high utility privacy protection. The IVF is visually and statistically different from the original face, which can still be used for recognizing the user's identity. Despite this advantage, these schemes are unable to verify the trustworthiness of the IVF, the quality and controllability of which is often limited. To address these issues, we propose TIVDiff, a diffusion-based framework for trustworthy and identifiable virtual face generation. TIVDiff learns a virtual identity (VID) space via Virtual Identity Projection (VIP) and synthesizes high-quality virtual faces conditioned on VID and 3D facial geometry for pose and expression preservation. To enable the trustworthiness of IVF, we further propose an Identity-Guarded Generative Watermarking (IGGW) scheme to bind the diffusion initial noise with VID through a reversible mapping mechanism. This enables the embedding of an imperceptible cue into IVF for legitimacy verification. Experiments demonstrate the advantage of our TIVDiff over the state-of-the-art IVF generation schemes in terms of image quality, identifiability and trustworthiness.
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Rethinking Contrastive Learning for Graph Collaborative Filtering: Limitations and a Simple Remedy
Geon Lee, Sunwoo Kim, Kyungho Kim, Kijung Shin
Graph collaborative filtering (GCF) is a dominant paradigm in recommender systems, where contrastive learning (CL) objectives such as the Sampled Softmax (SSM) loss are widely used for optimization. However, it remains unclear how CL interacts with the prediction mechanism of GCF. By unfolding the prediction mechanism of GCF, we show that the user-item prediction score is computed by aggregating learnable weights over a large number of neighbor pairs formed by the multi-hop neighbors of the user and the item. This analysis suggests that effective optimization critically depends on which neighbor pairs are upweighted during training. Empirically, we find that effective recommendation is achievable by selectively upweighting only a small subset of neighbor pairs whose constituent neighbors are structurally similar to the target user and item, and that the effect of such selective upweighting varies across different neighbor pair types. Based on these findings, we analyze SSM and identify key limitations in its neighbor pair weight update dynamics. To address these limitations, we propose NT-SSM, an effective and principled CL objective that induces type-aware neighbor pair weight update dynamics. Experiments demonstrate consistent performance improvements over SSM across multiple datasets and GCF models.
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Unlearning Isn’t Forgetting: Revealing Hidden Leakage in Class Unlearning Evaluations
Ali Ebrahimpour-Boroojeny, Yian Wang, Hari Sundaram
In this paper, we reveal a significant shortcoming in class unlearning evaluations: overlooking the underlying class geometry can cause information leakage about the forgotten class. We further propose a simple unlearning strategy to mitigate this issue.
We introduce Class Membership Inference Attack (CMIA) that uses the probabilities the model assigns to neighboring classes to detect unlearned samples. We find that existing unlearning methods are vulnerable to CMIA across multiple datasets. We then propose a new fine-tuning objective that mitigates this privacy leakage by approximating, for forget-class inputs, the distribution over the remaining

classes that a retrained-from-scratch model would produce. To construct this approximation, we estimate inter-class similarity and tilt the target model’s distribution accordingly. The resulting Tilted REWeighting (TREW) distribution serves as the desired distribution during fine-tuning. We also show that across multiple benchmarks, TREW matches or surpasses existing unlearning methods on prior unlearning metrics. More specifically, on CIFAR-10, it reduces the gap with retrained models by 19% and 46% for U-LiRA and CMIA scores, accordingly, compared to the SOTA method for each category.
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Hierarchical Procedural Meta-Reasoning for Generalizable Multimodal Agents
Yao Fu, Shengyi Qian, Pierluca D'Oro, Fanyi Xiao, Honglak Lee, Joseph Tighe, Manchen Wang
While multimodal agents can achieve strong performance through fine-tuning, their ability to generalize remains limited in complex real-world tasks such as mobile navigation, where diverse applications, frequent system changes, and customized workflows are common in practice. We argue that a fundamental bottleneck lies in whether an agent possesses sufficient task-specific procedural knowledge to accomplish a given goal. Such procedural knowledge may be provided by the general capabilities of large language models, or obtained from additional external resources such as web search when necessary. Based on this view, we propose Procedure-Aware Multimodal Agent with Meta Reasoning, a framework that explicitly represents task knowledge as natural-language procedures and trains a procedure-aware grounded agent to condition its actions on this knowledge. By learning to leverage procedural knowledge from different sources, our approach enables robust generalization across tasks, applications, interface versions, and multi-app workflows, achieving substantial improvements on challenging Android benchmarks.
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Bridging Functional Correctness and Runtime Efficiency Gaps in LLM-Based Code Translation
Longhui Zhang, Jiahao Wang, Chenhao Hu, Bingyu Liang, Jing Li, Min Zhang
While large language models (LLMs) have greatly advanced the functional correctness of automated code translation systems, the runtime efficiency of translated programs has received comparatively little attention.
With the waning of Moore’s law, runtime efficiency has become increasingly important for program quality, alongside functional correctness.
Our preliminary study reveals that LLM-translated programs often run slower than human-written ones, and this issue cannot be remedied through prompt engineering alone.
Therefore, our work proposes SwiftTrans, a code translation framework comprising two key stages:
(1) Multi-Perspective Exploration, where MpTranslator leverages parallel in-context learning (ICL) to generate diverse translation candidates;
and
(2) Difference-Aware Selection, where DiffSelector identifies the optimal candidate by explicitly comparing differences between translations.
We further introduce Hierarchical Guidance for MpTranslator and Ordinal Guidance for DiffSelector, enabling LLMs to better adapt to these two core components.
To support the evaluation of runtime efficiency in translated programs, we extend existing benchmarks, CodeNet and F2SBench, and introduce a new benchmark, SwiftBench.
Experimental results across all three benchmarks show that SwiftTrans achieves consistent improvements in both correctness and runtime efficiency.
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FLAC: Maximum Entropy RL via Kinetic Energy Regularized Bridge Matching
Lei Lv, Yunfei Li, Yu Luo, Fuchun Sun, Xiao Ma
Iterative generative policies, such as diffusion models and flow matching, offer superior expressivity for continuous control but complicate Maximum Entropy Reinforcement Learning because their action log-densities are not directly accessible. To address this, we propose \textbf{Field Least-Energy Actor-Critic (FLAC)}, a likelihood-free framework that regulates policy stochasticity by penalizing t

he kinetic energy of the velocity field.
Our key insight is to formulate policy optimization as a Generalized Schr\"oding er Bridge (GSB) problem relative to a high-entropy reference process (e.g., unif orm). Under this view, the maximum-entropy principle emerges naturally as stayin g close to a high-entropy reference while optimizing return, without requiring e xplicit action densities. In this framework, kinetic energy serves as a physical ly grounded proxy for divergence from the reference: minimizing path-space energ y bounds the deviation of the induced terminal action distribution. Building on this view, we derive an energy-regularized policy iteration scheme and a practic al off-policy algorithm that automatically tunes the kinetic energy via a Lagran gian dual mechanism. Empirically, FLAC achieves superior or comparable performan ce on high-dimensional benchmarks relative to strong baselines, while avoiding e xplicit density estimation.
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TokenSwap: Backdoor Attack on the Compositional Understanding of Large Vision-La nguage Models
Zhifang Zhang, Qiqi Tao, Jiaqi Lv, Na Zhao, Lei Feng, Joey Tianyi Zhou
Large vision-language models (LVLMs) excel at vision-language tasks but remain v ulnerable to backdoor attacks. Most existing backdoor attacks on LVLMs force the model to generate predefined target patterns. However, these fixed-pattern atta cks are easy to detect, as the model tends to memorize frequent patterns and exh ibits overconfidence on targets given poisoned inputs. To address these limitati ons, we introduce TokenSwap, a more evasive and stealthy backdoor attack that fo cuses on the \emph{compositional understanding} capabilities of LVLMs. Instead o f enforcing a fixed targeted content, TokenSwap subtly disrupts the understandin g of object relationships in text. Specifically, it causes the backdoored model to generate outputs that mention the correct objects in the image but misreprese nt their relationships (i.e., bags-of-words behavior). During training, TokenSwa p injects a visual trigger into selected samples while swapping the grammatical roles of key tokens in the textual answers. Since the poisoned samples differ on ly subtly from clean ones, an adaptive token-weighted loss is employed to emphas ize learning on swapped tokens, strengthening the association between visual tri ggers and the bags-of-words behavior. Extensive experiments demonstrate that Tok enSwap achieves high attack success rates while maintaining evasiveness and stea lthiness across multiple benchmarks and LVLM architectures.
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Spatial-Aware Reduction Framework: Towards Efficient and Faithful Visual State Space Models
Jindi Lv, Aoyu Li, Yuhao Zhou, Zheng Zhu, Xiaofeng Wang, Qing Ye, Yueqi Duan, We ntao Feng, Jiancheng Lv
Mamba demonstrates strong efficiency in modeling long visual sequences. However, when token reduction is applied to structurally enhanced Mamba variants, these models exhibit a severe performance collapse. We attribute this degradation to t he spatially agnostic nature of existing reduction methods, which violate the tw o-dimensional structural premise required by the selective scanning mechanism.
In this work, we propose STORM, a spatial-aware token reduction framework design ed to maintain structural integrity throughout the compression process. STORM re formulates reduction into a structured operation on spatial units, enforcing loc alized constraints to maintain both grid topology and neighborhood coherence. A s a plug-and-play module, STORM equips existing reduction pipelines with explici t spatial awareness without any training.
Empirical results demonstrate that STORM achieves state-of-the-art pruning accur acy across diverse vision Mamba backbones under training-free settings. Notably, STORM delivers a substantial accuracy recovery on VMamba, outperforming prior m ethods by up to 63.3\% in top-1 accuracy. Meanwhile, STORM incurs only a 1.0\% a ccuracy drop on PlainMamba, achieving performance comparable to ViT.
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Bridging the Perceptual Gap: Residual-Enhanced Downscaling and Manifold-Aware Pe rception Alignment Adaptation for NR-IQA
Yu Li, Zhengran Shen, Yachun Mi, Puchao Zhou, Shaohui Liu

Leveraging Large Vision-Language Models like CLIP has recently set new benchmarks for No-Reference Image Quality Assessment (NR-IQA). However, the contrastive pretraining of CLIP inherently prioritizes semantic invariance, which often suppresses subtle perceptual signals, a phenomenon we term perceptual submergence. Furthermore, standard preprocessing techniques (e.g., cropping and interpolation) further exacerbate the loss of critical high-frequency quality cues. In this paper, we propose the Cross-modal Perception Alignment Adapter (CMPA), a manifold-aware framework designed to disentangle perceptual distortions from dominant semantics. CMPA introduces a Perception-Sensitive Feature Extractor (PFE) that projects CLIP features into a compact, low-dimensional subspace, explicitly magnifying distortion-induced off-manifold deviations. Subsequently, a Cross-Modal Perception Alignment Injector (PAI) aligns these features with quality-aware text anchors and re-injects them into the backbone. To ensure input fidelity, we also devise a Residual-enhanced Perceptual Downscaling strategy that adaptively compensates for resolution-induced information loss using Just Noticeable Difference (JND) guided frequency re-injection. Extensive evaluations on several benchmark datasets demonstrate that our approach significantly outperforms state-of-the-art methods, effectively recovering the perceptual signals submerged in semantic-dense representations.
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MedCRP-CL: Continual Medical Image Segmentation via Bayesian Nonparametric Semantic Modality Discovery

Ziyuan Gao

Medical image segmentation faces a fundamental challenge in continual learning: data arrives sequentially from heterogeneous sources, yet effective continual learning requires discovering which tasks share sufficient structure to benefit from joint learning. Existing methods either apply uniform constraints across all tasks, causing catastrophic forgetting when tasks conflict, or require predefined task groupings that cannot anticipate future task diversity. We introduce MedCRP-CL, a framework that performs online task structure discovery and structure-aware continual learning. Leveraging the Chinese Restaurant Process (CRP), our method dynamically infers task groupings from clinical text prompts as tasks arrive, without requiring predefined cluster counts or access to future tasks. We term these discovered groupings semantic modalities, as they capture finer-grained structure than physical imaging modalities by integrating anatomical region and pathological context. Guided by this discovered structure, we maintain semantic modality-specific LoRA adapters regularized by intra-modality EWC, ensuring parameter isolation across dissimilar task groups while facilitating knowledge transfer within similar ones. The framework is also replay-free, storing only aggregate statistics rather than raw patient data. Experiments on 16 medical segmentation tasks across four imaging modalities demonstrate that MedCRP-CL achieves 73.3 \% Dice score with only 4.1\% forgetting, outperforming the best baseline by 8.0 \% while requiring 6$\times$ fewer parameters.
Code is available at https://github.com/zygao930/MedCRP-CL.
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TEFormer: Structured Bidirectional Temporal Enhancement Modeling in Spiking Transformers

Sicheng Shen, Mingyang Lv, Bing Han, Dongcheng Zhao, Guobin Shen, Feifei Zhao, Yi Zeng

In recent years, Spiking Neural Networks (SNNs) have achieved remarkable progress, with Spiking Transformers emerging as a promising architecture for energy-efficient sequence modeling. However, existing Spiking Transformers still lack a principled mechanism for effective temporal fusion, limiting their ability to fully exploit spatiotemporal dependencies.

Inspired by feedforward–feedback modulation in the human visual pathway, we propose **TEFormer**, the first Spiking Transformer framework that achieves bidirectional temporal fusion by decoupling temporal modeling across its core components. Specifically, TEFormer employs a lightweight and hyperparameter-free **forward temporal fusion mechanism in the attention module**, enabling fully parallel co

mputation, while incorporating a **backward gated recurrent structure in the MLP ** to aggregate temporal information in reverse order and reinforce temporal con sistency.

Extensive experiments across a wide range of benchmarks demonstrate that TEForme r consistently and significantly outperforms strong SNN and Spiking Transformer baselines under diverse datasets. Moreover, through the first systematic evaluat ion of Spiking Transformers under different neural encoding schemes, we show tha t the performance gains of TEFormer remain stable across encoding choices, indic ating that the improved temporal modeling directly translates into reliable accu racy improvements across varied spiking representations. These results collectiv ely establish TEFormer as an effective and general framework for temporal modeli ng in Spiking Transformers. Code is available [here](https://github.com/Fancyssc /TEFormer).
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Pushing the Boundaries of Natural Reasoning: Interleaved Bonus from Formal-Logic Verification

Chuxue Cao, Jinluan Yang, Haoran Li, Kunhao Pan, Zijian Zhao, Zhengyu Chen, Yuch en Tian, Lijun Wu, Conghui He, Yike Guo, Sirui Han

Large Language Models (LLMs) show remarkable capabilities, yet their stochastic next-token prediction creates logical inconsistencies and reward hacking that fo rmal symbolic systems avoid. To bridge this gap, we introduce a formal logic ver ification-guided framework that dynamically interleaves formal symbolic verifica tion with the natural language generation process, providing real-time feedback to detect and rectify errors as they occur. Distinguished from previous neuro-sy mbolic methods limited by passive post-hoc validation, our approach actively pen alizes intermediate fallacies during the reasoning chain. We operationalize this framework via a novel two-stage training pipeline that synergizes formal logic verification-guided supervised fine-tuning and policy optimization. Extensive ev aluation on six benchmarks spanning mathematical, logical, and general reasoning demonstrates that our 7B and 14B models outperform state-of-the-art baselines b y average margins of 10.4\% and 14.2\%, respectively. These results validate tha t formal verification can serve as a scalable mechanism to significantly push th e performance boundaries of advanced LLM reasoning.
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Revisiting the Platonic Representation Hypothesis: An Aristotelian View

Fabian Gröger, Shuo Wen, Maria Brbic

The Platonic Representation Hypothesis suggests that representations from neural networks are converging to a common statistical model of reality.

We show that the existing metrics used to measure representational similarity ar e *confounded by network scale*: increasing model depth or width can systematica lly inflate representational similarity scores.

To correct these effects, we introduce a permutation-based null-calibration fram ework that transforms any representational similarity metric into a calibrated s core with statistical guarantees.

We revisit the Platonic Representation Hypothesis with our calibration framework , which reveals a nuanced picture: the apparent convergence reported by global s pectral measures largely disappears after calibration, while local neighborhood similarity, but not local distances, retains significant agreement across differ ent modalities.

Based on these findings, we propose the *Aristotelian Representation Hypothesis* : representations in neural networks are converging to shared local neighborhood relationships.
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SORA: Free Second-Order Attacks in Fast Adversarial Training

Mazdak Teymourian, Ramtin Moslemi, Farzan Rahmani, Mohammad Hossein Rohban

Adversarial Training (AT) is a leading defense against adversarial examples but often suffers from *Catastrophic Overfitting* (CO) in efficient single-step vari ants, where robustness to multi-step attacks collapses despite high single-step performance.

We address this failure mode with two contributions.

First, we formalize *Epsilon Overfitting* (EO), a perspective in which fixed perturbation magnitudes and directions exacerbate CO, and show that introducing perturbation variability significantly improves robust generalization across different architectures and datasets.

Second, we propose **PertAlign** (Perturbation Alignment), a theoretically grounded, computationally negligible metric that predicts CO onset by measuring gradient alignment across attack stages.

Leveraging these insights, we introduce **SORA**, an adaptive step-size AT method that dynamically adjusts perturbations based on loss surface geometry.

SORA consistently prevents CO, achieves state-of-the-art robustness and clean accuracy, and generalizes across datasets and architectures using a single fixed set of hyperparameters, which is essential for applicability in fast AT.

Extensive experiments on diverse datasets and architectures show that SORA matches or surpasses the robustness of prior methods while delivering higher clean accuracy and superior efficiency.

Code is available at https://github.com/SecondOrderAT/SORA.

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Optimal Attention Temperature Improves the Robustness of In-Context Learning under Distribution Shift in High Dimensions

Samet Demir, Zafer Dogan

Pretrained Transformers can perform in-context learning (ICL) from a few demonstrations, but this ability can fail sharply when the test distribution differs from pretraining—a common deployment setting. We study attention temperature as a simple inference-time control for improving ICL robustness under such shifts. In a high-dimensional linear-regression framework, we analyze a Transformer with "approximate softmax" attention, which preserves softmax's normalization and temperature-dependent selectivity while remaining tractable. We derive a closed-form expression for the ICL generalization error under distribution shift, and show that it is minimized by an explicit optimal attention temperature. This characterization yields interpretable guidance by linking the best temperature to moments of the pre-softmax attention scores, and predicts when temperature adjustment can recover near Bayes-optimal performance. We validate the theory with extensive simulations, and further demonstrate gains for pretrained LLMs on question-answering benchmarks under distribution shift induced by noisy in-context demonstrations. Overall, attention temperature emerges as a principled, lightweight knob for improving the robustness of ICL in pretrained Transformers.

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Spectral Gradient Descent Mitigates Anisotropy-Driven Misalignment: A Case Study in Phase Retrieval

Guillaume Braun, Han Bao, Wei Huang, Masaaki Imaizumi

Spectral gradient methods, such as the Muon optimizer, modify gradient updates by preserving directional information while discarding scale, and have shown strong empirical performance in deep learning. We investigate the mechanisms underlying these gains through a dynamical analysis of a nonlinear phase retrieval model with anisotropic Gaussian inputs, equivalent to training a two-layer neural network with quadratic activation and fixed second-layer weights.

Focusing on a spiked covariance setting where the dominant variance direction is orthogonal to the signal, we show that gradient descent (GD) suffers from a variance-induced misalignment: during the early escape phase, high-variance but uninformative directions are multiplicatively amplified, degrading alignment with the true signal under strong anisotropy. In contrast, spectral gradient descent (SpecGD) removes this anisotropic magnification effect, leading to stable alignment and accelerated noise contraction. Numerical experiments confirm the theory and show that these phenomena persist under more general anisotropic covariances.

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Decouple Searching from Training: Scaling Data Mixing via Model Merging for Large Language Model Pre-training

Shengrui Li, Fei zhao, Kaiyan Zhao, Jieying Ye, Haifeng Liu, Fangcheng Shi, Zhey

ong Xie, Yao Hu, Shaosheng Cao
Determining an effective data mixture is a key factor in Large Language Model (LLM) pre-training, where models must balance general competence with proficiency on hard tasks such as math and code. However, identifying an optimal mixture remains an open challenge, as existing approaches either rely on unreliable tiny-scale proxy experiments or require prohibitively expensive large-scale exploration.
To address this, we propose Decouple Searching from Training Mix (DeMix), a novel framework that leverages model merging to predict optimal data ratios. Instead of training proxy models for every sampled mixture, DeMix trains component models on candidate datasets at scale and derives data mixture proxies via weighted model merging. This paradigm decouples search from training costs, enabling evaluation of unlimited sampled mixtures without extra training burden and thus facilitating better mixture discovery through more search trials. Extensive experiments demonstrate that DeMix breaks the trade-off between sufficiency, accuracy and efficiency, obtaining the optimal mixture with higher benchmark performance at lower search cost. Additionally, we release the DeMix Corpora, a comprehensive 22T-token dataset comprising high-quality pre-training data with validated mixtures to facilitate open research. Our
code and DeMix Corpora is available at https://github.com/Lucius-lsr/DeMix.
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Consistency Deep Equilibrium Models
Junchao Lin, Zenan Ling, Jingwen Xu, Robert C Qiu
Deep Equilibrium Models (DEQs) have emerged as a powerful paradigm in deep learning, offering the ability to model infinite-depth networks with constant memory usage. However, DEQs incur significant inference latency due to the iterative nature of fixed-point solvers.
In this work, we introduce the Consistency Deep Equilibrium Model (C-DEQ), a novel framework that leverages consistency distillation to accelerate DEQ inference.
We cast the DEQ iterative inference process as evolution along a fixed ODE trajectory toward the equilibrium. Along this trajectory, we train C-DEQs to consistently map intermediate states directly to the fixed point, enabling few-step inference while preserving the performance of the teacher DEQ. At the same time, it facilitates multi-step evaluation to flexibly trade computation for performance gains.
Extensive experiments across various domain tasks demonstrate that C-DEQs achieve consistent 2-20$\times$ accuracy improvements over implicit DEQs under the same few-step inference budget. Our code is available at \url{https://github.com/landrarwolf/CDEQ}.
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Diagnosing and Correcting Concept Omission in Multimodal Diffusion Transformers
Kanghyun Baek, Jaihyun Lew, Chaehun Shin, Jungbeom Lee, Sungroh Yoon
Multimodal Diffusion Transformers (MM-DiTs) have achieved remarkable progress in text-to-image generation, yet they frequently suffer from concept omission, where specified objects or attributes fail to emerge in the generated image. By performing linear probing on text tokens, we demonstrate that text embeddings can distinguish a characteristic `omission signal' representing the absence of target concepts. Leveraging this insight, we propose Omission Signal Intervention (OSI), which amplifies the omission signal to actively catalyze the generation of missing concepts. Comprehensive experiments on FLUX.1-Dev and SD3.5-Medium demonstrate that OSI significantly alleviates concept omission even in extreme scenarios.
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Heavy-tailed Physics-Informed Neural Networks
Jephte Abijuru, Mayank Nagda, Jan Tauberschmidt, Phil Ostheimer, Sebastian Josef Vollmer, Stephan Mandt, Marius Kloft, Sophie Fellenz
Physics-informed neural networks (PINNs) enforce physical laws by minimizing partial differential equation (PDE) residuals and auxiliary constraints. Standard training relies on a mean-squared error (MSE) objective, which implicitly assumes

independent Gaussian residuals with a fixed global variance. We show theoretica
lly and empirically that residuals encountered during PINN training are heteroge
neous and heavy-tailed, revealing a systematic mismatch with this assumption. As
 a consequence, a small number of large residuals can disproportionately dominat
e both the loss and gradient, leading to poorly balanced optimization dynamics.
Motivated by this mismatch, we adopt a Student-t residual model to explicitly
capture heavy-tailed behavior. An equivalent hierarchical representation yields
an expectation–maximization (EM) algorithm that alternates between estimating re
sidual-dependent weights and optimizing network parameters via a weighted MSE ob
jective, allowing existing PINN solvers to be reused in the M-step. The resultin
g training dynamics bound the influence of extreme residuals and admit almost su
re convergence guarantees under standard stochastic optimization assumptions. Ex
periments across a diverse suite of challenging PDE benchmarks demonstrate consi
stently improved solution accuracy and robustness compared to standard PINN trai
ning.
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Optimal structure learning and conditional independence testing
Ming Gao, Yuhao Wang, Bryon Aragam
We establish a fundamental connection between optimal structure learning and opt
imal conditional independence testing by showing that the minimax optimal rate f
or structure learning problems is determined by the minimax rate for conditional
 independence testing in these problems. This is accomplished by establishing a
general reduction between these two problems in the case of poly-forests, and de
monstrated by deriving optimal rates for several examples, including Bernoulli,
Gaussian and nonparametric models. Furthermore, we show that the optimal algorit
hm in these settings is a suitable modification of the PC algorithm. This theore
tical finding provides a unified framework for analyzing the statistical complex
ity of structure learning through the lens of minimax testing.
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Hallucination is a Consequence of Space-Optimality: A Rate-Distortion Theorem fo
r Membership Testing
Anxin Guo, Jingwei Li
Large language models often hallucinate with high confidence on "random facts" t
hat lack inferable patterns.
We formalize the memorization of such facts as a membership testing problem, uni
fying the discrete error metrics of Bloom filters with the continuous log-loss o
f LLMs.
By analyzing this problem in the regime where facts are sparse in the universe o
f plausible claims, we establish a rate-distortion theorem: the optimal memory e
fficiency is characterized by the minimum KL divergence between score distributi
ons on facts and non-facts.
This theoretical framework provides a distinctive explanation for hallucination
under an idealized setting: even with optimal training, perfect data, and a simp
lified ``closed world'' setting, the information-theoretically optimal strategy
under limited capacity is not to abstain or forget, but to assign high confidenc
e to some non-facts, resulting in hallucination.
We validate this theory empirically on both synthetic and real-world data, showi
ng that hallucinations persist as a natural consequence of lossy compression.
The same theorem recovers and sharpens classical space lower bounds for Bloom-ty
pe filters, pinning down an additive constant left open for two-sided filters.
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Regression Language Models for Code
Yash Akhauri, Xingyou Song, Arissa Wongpanich, Bryan Lewandowski, Mohamed S. Abd
elfattah
We study \textbf{code-to-metric regression}: predicting numeric outcomes of code
 executions, a challenging task due to the open-ended nature of programming lang
uages. While prior methods have resorted to heavy and domain-specific feature en
gineering, we show that a single unified Regression Language Model (RLM) using a
 frozen LLM encoder can simultaneously predict directly from text, (i) the memor
y footprint of code across multiple high-level languages such as Python and C++,

(ii) the latency of Triton GPU kernels, and (iii) the accuracy and speed of trained neural networks represented in ONNX. In particular, a relatively small 300M parameter RLM based on T5Gemma, obtains $>$0.9 Spearman-rank on competitive programming submissions from APPS, and a single unified model achieves $>$0.5 average Spearman-rank across 24 different programming languages from CodeNet. Furthermore, the RLM can obtain the highest average Kendall-Tau of 0.46 on five classic NAS design spaces previously dominated by graph neural networks, and simultaneously predict architecture latencies on numerous hardware platforms.
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Concept-Guided Tokenization: Closing the Gap Between Reconstruction and Generation

Yunqiao Yang, Haokun Lin, Guanzhong Wu, Ying Wei

Recent advances in image generation have been largely driven by image tokenization, which compresses raw pixels into compact latent representations. While existing tokenizers excel at preserving low-level visual details through reconstruction-based training, they often lack explicit semantic guidance, which limits their ability to capture semantically structured representations and thus hinders their performance on downstream tasks like image generation. To overcome this limitation, we propose a novel tokenization framework that incorporates high-level semantics through two key innovations: (1) a text-integrated encoder that jointly processes images and textual descriptions to produce semantically enriched latent representations, and (2) a concept-guided training objective that leverages sparse autoencoders to decompose pre-trained vision-language model features to a semantic concept space, employing sparse and disentangled concept indices for guidance. Our approach achieves strong alignment with semantic concepts, maintaining high reconstruction fidelity with an rFID of 1.39 on ImageNet, while achieving a gFID of 2.65 on the class-conditional image generation task and 10.73 on the text-to-image generation task. By infusing high-level semantic structures into low-level visual fidelity, our method bridges the reconstruction-generation divide and drives generative modeling as a powerful foundation. The code is available at https://github.com/hustyyq/ConceptTok
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RA-VLA: Retrieval-Augmented VLA for Test-Time Adaptation

Sanghwan Jang, Minjin Jeon, Minsoo Kim, Seong Jin Choi, Dongha Kim, Hwanjo Yu

Vision-Language-Action (VLA) models provide a versatile foundation for general robotic manipulation, yet they exhibit significant brittleness when confronted with novel task distributions. While In-Context Imitation Learning (ICIL) offers a training-free alternative, existing frameworks suffer from an adaptation bottleneck that hinders the effective translation of expert context to executable actions. This failure originates from superficial retrieval mechanisms and an inherent behavioral inertia that anchors the policy to its pre-trained priors. To address these limitations, we present RA-VLA, a retrieval-augmented VLA framework that integrates behavior-aligned context retrieval with a grounded execution pipeline. By enforcing faithful adherence to functional cues within a scalable architecture, RA-VLA facilitates seamless task adaptation while preserving inference efficiency. Our empirical evaluations across the LIBERO benchmark and a real-world UR5e environment demonstrate that RA-VLA achieves superior success rates and computational efficiency, establishing a robust framework for training-free robotic adaptation.
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Learning Rate Annealing Improves Tuning Robustness in Stochastic Optimization

Amit Attia, Tomer Koren

The learning rate in stochastic gradient methods is a critical hyperparameter that is notoriously costly to tune via standard grid search, especially for training modern large-scale models with billions of parameters. We identify a theoretical advantage of learning rate annealing schemes that decay the learning rate to zero at a polynomial rate, such as the widely-used cosine schedule, by demonstrating their increased robustness to initial parameter misspecification due to a coarse grid search. We present an analysis in a stochastic convex optimization setup demonstrating that the convergence rate of stochastic gradient descent wit

h annealed schedules depends *sublinearly* on the multiplicative misspecificatio n factor ρ (i.e., the grid resolution), achieving a rate of $O(\rho^{1/(2p+ 1)}/\sqrt{T})$ where p is the degree of polynomial decay and T is the number of steps. This is in contrast to the $O(\rho/\sqrt{T})$ rate obtained under the inverse-square-root and fixed stepsize schedules, which depend linearly on $\rh o$. Experiments confirm the increased robustness compared to tuning with a fixed stepsize, that has significant implications for the computational overhead of h yperparameter search in practical training scenarios.

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From geometry to dynamics: Learning overdamped Langevin dynamics from sparse obs ervations with geometric constraints

Dimitra Maoutsa

How can we learn the laws underlying the dynamics of stochastic systems when the ir trajectories are sampled sparsely in time? Existing methods either require te mporally resolved high-frequency observations, or rely on geometric arguments th at apply only to conservative systems, limiting the range of dynamics they can r ecover.

Here, we present a new framework that reconciles these two perspectives by refor mulating inference as a stochastic control problem. Our method uses geometry-dri ven path augmentation, guided by structure in the system’s invariant density to reconstruct likely trajectories and infer the underlying dynamics without assumi ng specific parametric models. Applied to overdamped Langevin systems, our appro ach accurately recovers stochastic dynamics even from severely undersampled data , outperforming existing methods in synthetic benchmarks. This work demonstrates the effectiveness of incorporating geometric inductive biases into stochastic s ystem identification methods, with broad applications across physics, biology, a nd control.

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Breaking the Computational Barrier: Provably Efficient Actor–Critic for Low-Rank MDPs

Ruiquan Huang, Donghao Li, Yingbin Liang, Jing Yang

Reinforcement learning (RL) is a fundamental framework for sequential decision-m aking, in which an agent learns an optimal policy through interactions with an u nknown environment. In settings with function approximation, many existing RL al gorithms achieve favorable sample complexity, but often rely on computationally intractable oracles. In this paper, we use supervised learning as a computationa l proxy to establish a clear hierarchy of commonly adopted RL oracles under low- rank Markov Decision Processes (MDPs). This hierarchy shows that policy evaluati on is the most computationally efficient oracle, provided that supervised learni ng can be efficiently solved. Motivated by this observation, we propose a novel optimistic actor–critic algorithm that relies solely on the policy evaluation or acle. We prove that our algorithm outperforms the existing sample complexity gua rantees for low-rank MDPs while avoiding computationally expensive planning or o ptimization oracles commonly assumed in prior works. We further extend our theor etical results to approximately low-rank MDPs and demonstrate that this setting captures a broad class of real-world environments. Finally, we validate our theo retical results with experiments on several standard Gym benchmarks.

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An Evidential Route to Asymptotic Bayes Optimality under Sparsity

Qiaoyu Liang, Haohua Chen, Zihan Zhu, Michael Evans

From a statistical evidence perspective, we establish some asymptotic optimality properties of certain multiple testing rules based on the relative belief ratio (Evans, 2015). Under the two-groups model with an additive 0-1 loss and within a Bayesian decision theoretic asymptotic framework of Bogdan et al. (2011), we s how that relative belief multiple testing rules induced by a simple one-group li ght-tailed normal prior with a single hyperparameter achieve the same asymptotic Bayes risk as the Bayes oracle benchmark. This risk is the minimum achievable i n this asymptotic framework. Despite originating from a different starting point , the evidential relative belief approach enjoys oracle properties. The relative belief multiple testing approach is fundamentally different from existing Bayes

ian multiple testing procedures, virtually all induced by more complex heavy-tailed one-group global-local shrinkage priors using purely posterior-based inferences (Datta & Ghosh, 2013; Ghosh et al., 2016; Bhadra et al., 2017; Ghosh & Chakrabarti, 2017; Qin & Ghosh, 2025). By measuring statistical evidence via both the prior and posterior, the relative belief approach reveals an alternative new inferential paradigm for attaining asymptotic Bayes optimality under sparsity, one that does not rely on developing increasingly elaborate priors.
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The Truth Stays in the Family: Enhancing Contextual Truthfulness via Inherited Heads in Model Lineages
Miso Choi, Seonga Choi, Mincheol Kwon, Woosung Joung, Jinkyu Kim, Jungbeom Lee
Recent advances in large language models (LLMs) have produced many specialized multimodal LLMs (MLLMs) that share common foundational LLMs, forming distinct model lineages. It remains unclear whether a fundamental behavioral link exists between the foundational LLMs and downstream variants. We investigate this question by quantifying head-level context-truthfulness scores. Across diverse LLM and MLLM lineages, including Vicuna-, Qwen2.5-, LLaMA2-, and Mistral-based models, we find that Truth Scores are strongly preserved within model families, even after instruction tuning or multimodal adaptation. We further show that this inheritance is consistent with attention-head weight preservation, and that context-truthful heads attend to query-relevant evidence. Building on this finding, we propose TruthProbe, a soft-gating strategy that amplifies context-truthful heads while preserving other head contributions. TruthProbe improves contextual truthfulness on HaluEval and reduces multimodal hallucination on POPE and CHAIR, with base-LLM Truth Scores transferring effectively to their fine-tuned LLM and MLLM descendants. Code is available at https://github.com/miso-choi/TruthProbe.
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Beyond Token-level Supervision: Unlocking the Potential of Decoding-based Regression via Reinforcement Learning
Ming Chen, Sheng Tang, Rong-Xi Tan, Ziniu Li, Jiacheng Chen, Ke Xue, Chao Qian
Decoding-based regression, which reformulates regression as a sequence generation task, has emerged as a promising paradigm of applying large language models for numerical prediction. However, its progress is hindered by the misalignment between discrete token-level objectives (e.g., cross-entropy) and continuous numerical values. Existing approaches relying on token-level constraints often fail to capture the global magnitude of the target value, limiting their precision and generalization. In this paper, we propose to unlock the potential of decoding-based regression via reinforcement learning. We formulate the generation process as a Markov decision process, utilizing sequence-level rewards to enforce global numerical coherence.Under this framework, we present GenRe^2, which combines policy gradient methods and on-policy distillation to provide dense expert supervision while preserving error magnitudes, thereby resolving the temporal credit assignment challenge. Extensive experiments across tabular regression, code metric prediction and generative reward modeling demonstrate that GenRe^2 consistently outperforms traditional baselines, establishing a robust paradigm for general-purpose numerical prediction.
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Influence-Guided Symbolic Regression: Scientific Discovery via LLM-Driven Equation Search with Granular Feedback
Evgeny Saveliev, Samuel Holt, Nabeel Seedat, David L. Bentley, Jim Weatherall, Mihaela van der Schaar
Large Language Models (LLMs) offer a promising avenue for scientific discovery, yet their application to symbolic regression is often constrained by inefficient search strategies and coarse feedback signals. Current methods typically guide LLMs using scalar metrics (e.g., global Mean Squared Error), which fail to identify which components of a proposed equation are driving performance or causing error. We introduce \textit{Influence-Guided Symbolic Regression} (IGSR), a method that frames equation discovery as an iterative two-step process combining diverse term generation with rigorous selection: an LLM generates candidate basis functions $\psi_j(\mathbf{x})$ for a linear model, which are then evaluated using

granular influence scores Δ_j. These scores quantify each term's marginal contribution to generalization accuracy, enabling an influence-guided pruning process that systematically refines the model structure. Integrating this mechanism into a Monte Carlo Tree Search (MCTS) enables navigating the combinatorial search space while balancing exploration of novel functional forms with exploitation of high-influence components. We demonstrate IGSR's effectiveness on a diverse suite of benchmarks, including LLM-SRBench, pharmacological PKPD models, an epidemiological simulation, and real-world genomic data. Notably, we validate the framework's capacity for genuine discovery in a case study using a high-dimensional biological dataset, in which IGSR identified a novel relationship between DNA methylation and RNA Polymerase II pausing; a hypothesis that was subsequently supported via wet-lab experimentation.
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DASH: Faster Shampoo via Batched Block Preconditioning and Efficient Inverse-Root Solvers

Ionut-Vlad Modoranu, Philip Zmushko, Erik Schultheis, Mher Safaryan, Dan Alistarh

Shampoo is one of the leading approximate second-order optimizers: a variant of it has won the MLCommons AlgoPerf competition, and it has been shown to produce models with lower activation outliers that are easier to compress. Yet, applying Shampoo currently comes at the cost of significant computational slowdown, due to its expensive internal operations. In this paper, we take a significant step to address this shortcoming by proposing DASH (for Distributed Accelerated SHampoo), a faster implementation of Distributed Shampoo based on two main new techniques: First, we show that preconditioner blocks can be stacked into 3D tensors to significantly improve GPU utilization; second, we introduce the Newton-DB iteration and the Chebyshev polynomial approximations as novel and faster approaches for computing the inverse matrix roots required by Shampoo. Along with these algorithmic contributions, we provide a first in-depth analysis of how matrix scaling critically affects Shampoo convergence. On the practical side, our GPU-aware implementation achieves up to $5.6\times$ faster optimizer steps compared to the well-optimized Distributed Shampoo, while Newton-DB attains the lowest validation perplexity per iteration among all tested methods. Our code is available at https://github.com/IST-DASLab/DASH.
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Joint Geometric and Trajectory Consistency Learning for One-Step Real-World Super-Resolution

Chengyan Deng, Zhangquan Chen, Li Yu, Kai Zhang, Xue Zhou, Wang Zhang

Diffusion-based Real-World Image Super-Resolution (Real-ISR) achieves impressive perceptual quality but suffers from high computational costs due to iterative sampling. While recent distillation approaches leveraging large-scale Text-to-Image (T2I) priors have enabled one-step generation, they are typically hindered by prohibitive parameter counts and the inherent capability bounds imposed by teacher models. As a lightweight alternative, Consistency Models offer efficient inference but struggle with two critical limitations: the accumulation of consistency drift inherent to transitive training, and a phenomenon we term "Geometric Decoupling"— where the generative trajectory achieves pixel-wise alignment yet fails to preserve structural coherence. To address these challenges, we propose GTASR (Geometric Trajectory Alignment Super-Resolution), a {simple yet effective} consistency training paradigm for Real-ISR. Specifically, we introduce a Trajectory Alignment (TA) strategy to rectify the tangent vector field via full-path projection, and a Dual-Reference Structural Rectification (DRSR) mechanism to enforce strict structural constraints. Extensive experiments verify that GTASR delivers superior performance over representative baselines while maintaining minimal latency.
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Agentic Framework for Epidemiological Modeling

Rituparna Datta, Zihan Guan, Baltazar Espinoza, Yiqi Su, Priya Pitre, Srini Venkatramanan, Naren Ramakrishnan, Anil Vullikanti

Epidemic modeling is essential for public health planning, yet traditional appro

aches rely on fixed model classes that require manual redesign as pathogens, pol
icies, and scenario assumptions evolve. We introduce EpiAgent, an agentic framew
ork that automatically synthesizes, calibrates, verifies, and refines epidemiolo
gical simulators by modeling disease progression as an iterative program synthes
is problem. A central design choice is an explicit epidemiological flow graph in
termediate representation that links scenario specifications to model structure
and enables strong, modular correctness checks before code is generated. Verifie
d flow graphs are then compiled into mechanistic models supporting interpretable
 parameter learning under physical and epidemiological constraints.
Evaluation on epidemiological scenario case studies demonstrates that EpiAgent c
aptures complex growth dynamics and produces epidemiologically consistent counte
rfactual projections across varying vaccination and immune escape assumptions. O
ur results show that the agentic feedback loop prevents degeneration and signifi
cantly accelerates convergence toward valid models by mimicking professional exp
ert workflows.
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G-RANS: Generalizable Residual-Aware Neural Solvers for Sparse Systems
Weixin Liao, Mingquan Feng, Zhizhou Zhang, Youjia Wu, Yifan Fu, Junchi Yan
Neural operators have shown promise in accelerating PDE solvers, yet they remain
 unreliable for the sparse linear systems induced by discretization due to limit
ed generalization across physical parameters and insufficient accuracy, and hybr
id neural iterative schemes face stagnation as the residual distribution evolves
 over iterations.
To address these limitations, we propose G-RANS (Generalizable Residual-Aware Ne
ural Solver), a neuralized iterative paradigm that performs residual-aware subsp
ace corrections by mapping the residual to the matrix graph, generating multi-sc
ale correction subspaces via a residual-aware basis generator, and applying proj
ected updates through a differentiable subspace projection. G-RANS is trained fu
lly self-supervised via progressive bootstrap with multi-stage residual distribu
tions.
On sparse FEM systems from a representative suite of second-order elliptic PDEs
(Poisson, advection--diffusion, reaction--diffusion, and Helmholtz), G-RANS is r
obust to severe coefficient shifts (up to $\pm70\\%$) and shows strong cross-equ
ation generalization.
G-RANS reaches relative residuals on the order of 10^{-5}, substantially outpe
rforming end-to-end neural operators, and attains the same target residual with
2--$4\times$ less wall-clock time than classical Krylov solvers.
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MMClima: A Framework for Multimodal Climate Science Data and Evaluation
Muhammad Umer Sheikh, Hassan Abid, Khawar shehzad, Ufaq Khan, Muhammad Haris Kha
n
Climate change research increasingly requires AI systems that reason across text
, dynamic visual content, and scientific figures, yet existing climate QA benchm
arks are small, mostly textual, and cover a narrow range of models. We introduce
 MMClima, a large-scale multimodal climate question answering framework with ove
r 104k expert-validated question–answer pairs spanning articles, video transcrip
tions, and figures across five core climate science domains. MMClima is construc
ted via automated claim extraction and QA synthesis with human-in-the-loop valid
ation to ensure both scale and reliability. Using MMClima, we benchmark state-of
-the-art multimodal language models on tasks requiring factual recall, visual in
terpretation, and cross-modal synthesis. We additionally fine-tune on the textua
l split to produce mmclima-70b-txt, a domain-adapted baseline that outperforms s
trong open- and closed-source models on textual QA. We release the dataset, eval
uation pipeline, fine-tuned model weights, and data creation framework to suppor
t standardized multimodal evaluation for climate science.
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Model Monotonicity in Autobidding Auctions: When Do Better Predictions Lead to B
etter Outcomes?
Ashwinkumar Badanidiyuru
Online advertising platforms rely on machine learning models to predict click-th

rough rates (pCTR) and conversion rates (pCVR) for auction mechanisms. We introduce a novel framework to study the interaction between recommender system model quality, auction format, and autobidder behavior. We formalize when model improvements---defined via a refinement relation inspired by filtrations in probability theory---lead to improvements in platform-level Evaluation Criteria Metrics (ECM) such as revenue, welfare, or liquid welfare. Our main contributions are: (1) a formal definition of model improvement based on cluster refinement, and (2) a systematic characterization of ECM monotonicity across different combinations of bidder types (tCPA, max-CPA), auction formats (first-price, second-price, VCG), and budget constraints. We show that first-price auctions with uniform bidding guarantee revenue monotonicity for tCPA bidders without budgets (via Jensen's inequality), while second-price auctions and budget constraints can break this property. We provide full numerical constructions for the non-monotonicity results. Our findings have practical implications for advertising platforms seeking to align model improvements with business outcomes.
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Effects of Structural Reward Shaping on Biophysical Properties in RL-Trained Plasmid Generators
McClain Thiel, Angus G. Cunningham, Chris P Barnes
We compare the efficacy and distributional effects
of supervised fine-tuning (SFT) and reinforcement learning (RL) post-training for PlasmidGPT,
a foundation model for whole-plasmid generation, using Group Relative Policy Optimization
(GRPO) for the RL model. Using a biologically
motivated reward function encoding functional annotations, length constraints, and repeat penalties,
the RL model achieves a 71.6% quality-control
pass rate across 8 prompts on 4,000 sequences,
compared to 4.3% for the pretrained baseline and
11.0% for SFT. A five-model reward ablation identifies the cassette arrangement bonus, which rewards correct promoter→CDS→terminator ordering, as the critical reward component. Rejectionsampling baselines indicate that the gain is not
recovered by sampling more heavily from the base
model. Beyond directly optimized features, RLgenerated sequences converge toward real plasmid distributions in 3-mer composition and minimum free energy density, neither of which is
directly optimized by the reward function. Minimum free energy density independently converges
to the real-plasmid regime under both SFT and RL
despite these being parallel post-training paths.
On a small curated hold-out set, RL improves
continuation log-likelihood over the pretrained
baseline on all 29 held-out sequences (mean
Δ = +0.83 nats).
****
Native Spatio-Temporal 4D Variational Autoencoder
Lihe Ding, Weicai Ye, Shaocong Dong, Xintao Wang, Pengfei Wan, Kun Gai, Tianfan Xue
Dynamic 3D content representation is crucial for generating moving 3D objects and scenes. Existing 4D variational autoencoders (VAEs) are mainly based on projected 2D pointmaps, which are only incomplete and view-dependent observations that do not model the native 4D positional relations between points. This often leads to projection-induced distortions and irreversible token dislocation. In this paper, we
introduce a novel 4D VAE that operates directly in native 4D space, that is dynamic colored voxel space, without 2D projection. This
preserves explicit spatio-temporal coordinates throughout the learned encoder and decoder, enabling both partial and complete 4D
content encoding. To support a flexible temporal compression ratio, we also desi

gn a novel spatio-temporal window attention module that performs attention withi
n local 4D windows. Additionally, we propose a differentiable voxel rendering lo
ss based on sparse voxel rasterization to improve the geometry and color reconst
ruction quality. On 4D reconstruction tasks, our approach improves reconstructio
n fidelity over pointmap VAEs and flow-based VAEs while learning a more structur
ally consistent latent space. We further demonstrate the generative potential of
 our method by training a video-conditioned 4D diffusion model.
****

Self-Augmenting Retrieval for Diffusion Language Models
Paul Jünger, Justin Lovelace, Linxi Zhao, Dongyoung Go, Kilian Q Weinberger
Discrete diffusion language models generate text by iteratively denoising an ent
ire response in parallel. At each step, they predict tentative tokens for every
masked position, committing the confident predictions to the output and discardi
ng the unconfident ones. We show that the discarded tokens are in fact a useful
lookahead signal for retrieval-augmented generation: even low-confidence tokens
often surface salient entities early in the denoising trajectory, enabling retri
eval of stronger evidence before the output is finalized. We exploit this throug
h Self-Augmenting Retrieval for Diffusion Language Models (SARDI), a dynamic RAG
 framework that uses these lookahead tokens to guide retrieval during denoising.
 SARDI is training-free, retriever-agnostic, and applicable to any reasoning-cap
able discrete diffusion language model. Across five multi-hop QA benchmarks, SAR
DI outperforms current training-free diffusion and autoregressive retrieval base
lines at up to $8\times$ higher throughput.
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QiMeng-LibBench: Benchmarking LLM Agents for Library-Scale Cross-Architecture Mi
gration
Weijia Li, KE GAO, Jiajie Li, Han Sun, Yuhe Ding, Yongdong Mai, Yiran Le, Yongji
e Qian, Zhibin Zhang, Xinyu Wang, Limin Cheng, Shouxu Kuang, Pengfei Chen, Ling
Li
Cross-architecture migration of high-performance libraries dictates ecosystem re
adiness on emerging hardware. The challenge is twofold: disentangling library-sc
ale dependencies and performance-critical kernels with ISA-specific SIMD intrins
ics, often trading migration speed for peak performance. While LLM-based agents
offer a promising approach, are confined to function-level tasks or scalar code,
 failing to assess agents’ capabilities and limitations in realistic, library-sc
ale migration. We present QM-LibBench, a benchmark for cross-architecture librar
y-scale code migration, featuring 85 critical kernels from widely used libraries
, including OpenCV, libjpeg, and NCNN. It supports comprehensive evaluations of
compilability, correctness, and performance across major transitions: ARM→RISC-V
, x86→ARM, and ARM→LoongArch. Evaluation of 12 SOTA agent-LLM combinations on QM
-LibBench reveals that, due to the lack of library-level navigation and hardware
-aware optimization, agents regress to superficial pattern matching, yielding on
ly 20.88\% correctness and 0.83 speedup for libjpeg. Motivated by these findings
, we further propose FSCM, a multi-agent framework incorporating hardware-aware
global reconfiguration and performance optimization. FSCM improves OpenCV correc
tness to 71\%. The benchmark and code are available at https://github.com/Wisdom
Joy/QM-LibBench.
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When More Experts Hurt: Underfitting in Multi-Expert Learning to Defer
Shuqi Liu, Yuzhou Cao, Lei Feng, Bo An, Luke Ong
Learning to Defer (L2D) enables a classifier to abstain from predictions and def
er to an expert, and has recently been extended to multi-expert settings.
In this work, we show that multi-expert L2D is fundamentally more challenging th
an the single-expert case.
With multiple experts, the classifier's underfitting becomes inherent, which ser
iously degrades prediction performance, whereas in the single-expert setting it
arises only under specific conditions.
We theoretically reveal that this stems from an intrinsic expert identifiability
 issue: learning which expert to trust from a diverse pool, a problem absent in
the single-expert case and renders existing underfitting remedies failed.

To tackle this issue, we propose PiCCE (**Pi**ck the **C**onfident and **C**orre
ct **E**xpert), a surrogate-based method that adaptively identifies a reliable e
xpert based on empirical evidence. PiCCE effectively reduces multi-expert L2D to
 a single-expert–like learning problem, thereby resolving multi-expert underfitt
ing.
We further prove its statistical consistency and ability to recover class probab
ilities and expert accuracies.
Extensive experiments across diverse settings, including real-world expert scena
rios, validate our theoretical results and demonstrate improved performance.
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DiasR: Dual-Modal Identity-Anchored Sparse Routing for Efficient Multi-Subject V
ideo Generation
Yang-yang Li, Wu Liu, Jie Li, Xinchen Liu, Yongdong Zhang, Guoqing Jin
Personalized multi-subject video generation is a promising direction within the
 field of controllable video generation; however, existing methods face challenge
s in maintaining cross-frame identity consistency and incur high computational o
verhead. To address these issues, we propose DiasR, an efficient framework that
 integrates Dual-Modal Identity-Anchored Alignment and a novel Sparse Routing Str
ategy. The Dual-Modal Identity-Anchored Alignment employs learnable identity que
ries to align visual and textual modalities with ground-truth subject masks, the
reby mitigating cross-frame identity drift. The Sparse Routing Strategy dynamica
lly routes video tokens to relevant subjects and groups them through bucket aggr
egation, reducing computational overhead and alleviating identity entanglement i
nduced by redundant tokens. We also construct MuSA-2M, a large-scale dataset com
prising 2 million annotated samples equipped with subject-level masks, which fil
ls the gap in existing multi-subject video datasets. Experiments conducted on th
e OpenS2V-Eval benchmark demonstrate that our method achieves superior performan
ce in identity consistency, text fidelity, and video naturalness. Notably, it ma
intains a nearly constant inference time as the number of reference subjects inc
reases, outperforming existing baselines in both efficiency and generation quali
ty for scenarios involving multi-subject interactions. Project Page: https://tal
e17.github.io/diasr.
****

Linear Bandits beyond Inner Product Spaces, the case of Bandit Optimal Transport
Lorenzo Croissant
Linear bandits have long been a central topic in online learning, with applicati
ons ranging from recommendation systems to adaptive clinical trials. Their gener
al learnability has been established when the objective is to minimise the inner
 product between a cost parameter and the decision variable. While this is highl
y general, this reliance on an inner product structure belies the name of \emph{
linear} bandits, and fails to account for problems such as Optimal Transport. Us
ing the Kantorovich formulation of Optimal Transport as an example, this article
 shows that an inner product structure is \emph{not} necessary to achieve effici
ent learning in linear bandits. We propose a refinement of the classical OFUL al
gorithm that operates by embedding the action set into a Hilbertian subspace, wh
ere confidence sets can be built via least-squares estimation. Actions are then
constrained to this subspace by penalising optimism. The analysis is completed b
y leveraging convergence results from penalised (entropic) transport to the Kant
orovich problem. Up to this approximation term, the resulting algorithm achieves
 the same trajectorial regret upper bounds as the OFUL algorithm, which we turn
into worst-case regret using functional regression techniques. Its regret interp
olates between $\tilde{\mathcal O}(\sqrt{T})$ and ${\mathcal O}(T)$, depending o
n the regularity of the cost function, and recovers the parametric rate $\tilde{
\mathcal O}(\sqrt{dT})$ in finite-dimensional settings.
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Interaction-Breaking Adversarial Learning Framework for Robust Multi-Agent Reinf
orcement Learning
Sunwoo Lee, Mingu Kang, Yonghyeon Jo, Seungyul Han
Cooperation is central to multi-agent reinforcement learning (MARL), yet learned
 coordination can be fragile when external perturbations disrupt inter-agent int

eractions. Prior robust MARL methods have primarily considered value-oriented attacks, leaving a gap in robustness when interaction structures themselves are corrupted. In this paper, we propose an interaction-breaking adversarial learning (IBAL) framework that takes an information-theoretic view to construct attacks that impede coordination by perturbing agents' observations and actions, and trains agents to perform reliably under such disruptions. Empirically, our approach improves robustness over existing robust MARL baselines across diverse attack settings and yields stronger performance even under agent-missing scenarios. Our code is available at https://sunwoolee0504.github.io/IBAL.

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Decoupling Regularization and Privacy in Differentially Private Ridge Regression and ERM

Wanjie Wang, Tathagata Banerjee

We study ridge regression and ridge-regularized empirical risk minimization (ERM) under (ε,δ)-differential privacy via output perturbation. In classical private ERM, the ridge parameter simultaneously controls statistical regularization and the estimator's global sensitivity. Larger regularization reduces the DP noise scale but increases bias. So choosing the tuning parameter becomes a privacy--accuracy bottleneck. We propose a framework that makes these two roles explicit by decoupling regularization into (i) a statistical penalty α, defining the target ridge/ERM solution, and (ii) a privacy-stabilization parameter c, used only to enforce a curvature floor and hence a tight sensitivity bound. We apply this framework to ridge regression, where c is used to boost the minimum eigenvalue of the empirical Gram matrix. We derive an explicit bias--variance--DP-variance risk decomposition and characterize optimal (α,c) in several regimes, yielding sharp tuning guidance and improved accuracy relative to single-parameter regularization. Finally, we extend the same decoupling principle to general ridge-regularized ERM. We support the theory with simulations.

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Even Faster Kernel Matrix Linear Algebra via Density Estimation

Rikhav Shah, Sandeep Silwal, Haike Xu

This paper studies the use of *kernel density estimation* (KDE) for linear algebraic tasks involving the *kernel matrix* of a collection of n data points in $\mathbb{R}^d$.

In particular, we improve upon the best existing algorithms for computing the following up to $(1+\varepsilon)$ relative error for a Gaussian kernel matrix and other kernels: matrix-vector products, matrix-matrix products, the spectral norm, and sum of all entries. The runtimes of our algorithms depend linearly on the dimension d, sub-quadratically in the number of points n, and polynomially on the target error ε. Importantly, the dependence on n in each case is far lower when accessing the kernel matrix through KDE queries as opposed to reading individual entries. Our improvements over existing best algorithms (particularly those of [Backurs et al. ICML `21]) for these tasks reduce the polynomial dependence on ε, and additionally decrease the dependence on n in the case of computing the sum of all entries of the kernel matrix. For example, we reduce the power of $1/\epsilon$ from ≈ 7.7 to ≈ 3.2 for a $1-\varepsilon$ relative error estimation of the spectral norm of a Gaussian kernel matrix. We complement our upper bounds with several lower bounds for related problems, which provide (conditional) quadratic time hardness results and additionally hint at the limits of KDE based approaches for the problems we study.

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Physics-Informed Diffusion Models in Spectral Space

Davide Gallon, Philippe von Wurstemberger, Patrick Cheridito, Arnulf Jentzen

We propose physics-informed spectral diffusion (PISD), a methodology that combines generative latent diffusion models with physics-informed machine learning to generate solutions of partial differential equations (PDEs) conditioned on partial observations, which includes, in particular, forward and inverse PDE problems. We learn the joint distribution of PDE parameters and solutions via a diffusion process in a latent space of scaled spectral representations, where Gaussian n

oise corresponds to functions with controlled regularity. This spectral formulat ion enables significant dimensionality reduction compared to grid-based diffusio n models and ensures that the induced process in function space remains within a class of functions for which the PDE operators are well defined. Building on di ffusion posterior sampling, we enforce physics-informed constraints and measurem ent conditions during inference, applying Adam-based updates at each diffusion s tep. We evaluate the proposed approach on Poisson, Helmholtz, and incompressible Navier-Stokes equations, demonstrating improved accuracy and computational effi ciency compared with existing diffusion-based PDE solvers, which are state of th e art for sparse observations. Code is available at https://github.com/deeplearn ingmethods/PISD.

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Universal One-third Time Scaling in Learning Peaked Distributions

Yizhou Liu, Ziming Liu, Cengiz Pehlevan, Jeff Gore

Training large language models (LLMs) is computationally expensive, partly becau se the loss exhibits slow power-law convergence whose origin remains debatable. Through systematic analysis of toy models and empirical evaluation of LLMs, we s how that this behavior can arise intrinsically from the use of softmax and cross -entropy. When learning peaked probability distributions, e.g., next-token distr ibutions, these components generically yield power-law vanishing losses and grad ients, regardless of many microscopic details, creating a fundamental optimizati on bottleneck. This ultimately leads to power-law time scaling of the loss with a universal exponent of $1/3$. Our results provide a mechanistic explanation for observed neural scaling and suggest new directions for improving LLM training e fficiency.

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A-MemGuard: A Proactive Defense Framework For LLM-Based Agent Memory

Qianshan Wei, Tengchao Yang, Yaochen Wang, Xinfeng Li, Lijun Li, Zhenfei Yin, Yi Zhan, Thorsten Holz, Zhiqiang Lin, XiaoFeng Wang

Large Language Model (LLM) agents use memory to learn from past interactions. Ho wever, this reliance on memory introduces a critical security risk: an adversary can inject seemingly harmless records into an agent's memory to manipulate its future behavior. This vulnerability is characterized by two core aspects: First, the malicious effect of injected records is only activated within a specific co ntext, making them hard to detect when individual memory entries are audited in isolation. Second, once triggered, the manipulation can initiate a self-reinforc ing error cycle: the corrupted outcome is stored as precedent, which not only am plifies the initial error but also progressively lowers the threshold for simila r attacks in the future. To address these challenges, we introduce \emph{A-MemGu ard} (\underline{A}gent-\underline{Mem}ory \underline{Guard}), the first defense framework for LLM agent memory. The core idea of our work is the insight that m emory itself must become both \emph{self-checking} and \emph{self-correcting}. W ithout modifying the agent's core architecture, A-MemGuard combines two mechanis ms: (1) \textbf{consensus-based validation}, which detects anomalies by comparin g reasoning paths derived from multiple related memories and (2) a \textbf{dual- memory structure}, where detected failures are distilled into ``lessons'' stored separately and consulted before future actions, breaking error cycles and enabl ing adaptation.

Comprehensive evaluations on multiple benchmarks show that A-MemGuard effectivel y cuts attack success rates by over 95\% while incurring a minimal utility cost. This work shifts LLM memory security from static filtering to a proactive, expe rience-driven model where defenses strengthen over time.

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Local Redundancy: An Information-Theoretic Measure of Plasticity from Synthetic Memorization

Jiaxuan Cheng

Plasticity—a neural network's ability to adapt to new tasks—is critical for cont inual and transfer learning. Existing measures, such as effective rank, dead neu ron fraction, and weight norm, lack theoretical grounding and correlate poorly w ith performance on new tasks. We introduce *local redundancy*, an information-th

eoretic measure derived from universal compression theory. We define local redundancy as the worst-case redundancy of a local model family—parameters in an infinitesimal neighborhood along gradient directions—and show this is a principled measure of plasticity. Although local redundancy is intractable to compute exactly, we prove that the expected squared gradient norm on a synthetic memorization task provides an efficiently computable lower bound. Experiments on continual image classification and time series transfer learning demonstrate that local redundancy predicts downstream performance better than existing measures and enables pretraining checkpoint selection where validation loss plateaus.

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Provable Accuracy Collapse of Embedding-Based Representations under Dimensionality Mismatch

Dionysis Arvanitakis, Vaggos Chatziafratis, Yiyuan Luo

Embedding-based representations in Euclidean space $\mathbb{R}^d$ are a cornerstone of modern machine learning, where a major goal is to use the \emph{smallest dimension} that faithfully captures data relations. In this work, we prove sharp dimension--accuracy tradeoffs and identify a fundamental information-theoretic limitation: unless the embedding dimension d is chosen close to the ground-truth dimension D, accuracy undergoes a sudden collapse. Our main result shows that this phenomenon arises even in standard contrastive learning settings, where supervision is limited to a set of m anchor--positive--negative triplets (i,j,k) encoding distance comparisons $\mathrm{dist}(i,j) < \mathrm{dist}(i,k)$. Specifically, given triplets realizable by an unknown ground-truth embedding in D dimensions, we prove that there exists constant $c < 1$, such that \emph{every embedding of dimension at most cD violates almost half of the triplets}, yielding accuracy as low as a trivial one-dimensional solution that ignores the input. We complement our information-theoretic bounds with strong computational hardness results: under the Unique Games Conjecture, even if the given triplets are nearly realizable in $D=1$ dimension, no polynomial-time algorithm---\textit{regardless of its dimension}---can achieve accuracy above the trivial 50% baseline.

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$\texttt{ShaplEIG}$: Bayesian Experimental Design for Shapley Value Estimation

David Rundel, Fabian Fumagalli, Maximilian Muschalik, Bernd Bischl, Matthias Feurer

Shapley values are a principled attribution measure widely used in interpretable machine learning, but their exact computation scales exponentially with the number of players, motivating a wide range of approximation methods based on value function evaluations of sampled coalitions. This raises the question of whether approximation accuracy can be improved by *adaptively* selecting coalitions for evaluation based on previous evaluations. This is particularly relevant in settings where the value function is *costly* and the number of evaluations is severely limited, such as retraining-based feature importance, data valuation, and hyperparameter importance. For this purpose, we propose $\texttt{ShaplEIG}$, a Bayesian experimental design approach that approximates the expensive value function using a Gaussian process surrogate and adaptively selects coalitions based on their expected information gain about the Shapley values. By the linearity of the Shapley values in the value function, we show that the expected information gain is available in *closed form*. Furthermore, we propose an *efficient* computation scheme that reduces the complexity from exponential to polynomial in the number of players via elementary symmetric polynomials. In extensive experiments across diverse costly applications, our method consistently improves sample efficiency in the low-budget regime over state-of-the-art baselines.

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GENEB: Why Genomic Models Are Hard to Compare

Daria Ledneva, Mikhail Nuridinov, Denis Kuznetsov

Progress in genomic foundation models is difficult to assess due to fragmented benchmarks, incompatible evaluation protocols, and task-specific reporting. As a result, claims of superiority or generality across models are often not directly comparable. We introduce GENEB, a large-scale diagnostic benchmark that evaluat

es frozen representations from 40 genomic foundation models across 100 tasks spa
nning 13 functional categories under a unified probing-based protocol, including
 few-shot regimes. GENEB enables controlled comparison across model scale, archi
tecture, tokenization, and pretraining data while explicitly exposing task-level
 trade-offs. Our analysis shows that aggregate leaderboards are unstable: model
rankings vary sharply across task categories, scale provides only modest and inc
onsistent gains, and architectural and pretraining alignment frequently outweigh
 parameter count. These results highlight limitations of current evaluation prac
tices and position GENEB as a reference framework for principled comparison and
category-aware model selection in genomic machine learning.
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Expanding the Chaos: Neural Operator for Stochastic (Partial) Differential Equat
ions

Dai Shi, Lequan Lin, Andi Han, Luke Thompson, José Miguel Hernández-Lobato, Zhiy
ong Wang, Junbin Gao

Stochastic differential equations (SDEs) and stochastic partial differential equ
ations (SPDEs) are fundamental for modeling stochastic dynamics across the natur
al sciences and modern machine learning. Learning their solution operators with
deep learning models promises fast solvers and new perspectives on classical lea
rning tasks. In this work, we build on Wiener--chaos expansions (WCE) to design
neural operator (NO) architectures for SDEs and SPDEs: we project driving noise
paths onto orthonormal Wick--Hermite features and use NO backbones to parameteri
ze the corresponding chaos coefficients governed by deterministic equations, ena
bling reconstruction of S(P)DE solutions from the observed noise trajectories. E
mpirically, we achieve competitive accuracy across several tasks, including stan
dard SPDE benchmarks and SDE-based diffusion one-step image sampling, topologica
l graph interpolation, financial extrapolation, parameter estimation, and manifo
ld SDE flood forecasting. We further explore a law-matching setting where the Br
ownian trajectories paired with training solutions are unobserved, and show that
 the proposed framework remains effective by sampling auxiliary Brownian paths f
rom the prescribed noise prior. These results suggest WCE-based neural operators
 are a practical and scalable approach to learning SDE/SPDE solution operators a
cross domains. Our source code can be found in https://github.com/EEthanShi/SDEN
O.
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Chain-of-Thought Gradient Descent

Hong-Yu Chen, Venkat Sripad Ganti, Hude Liu, Jerry Yao-Chieh Hu, Han Liu

We show that Chain-of-Thought (CoT) expands the expressiveness of Transformer in
-context learning (ICL).
Specifically, we show CoT enable efficient simulation of In-Context Gradient Des
cent (ICGD) for N-layer neural network.
Different from CoT, a Transformer with fixed depth and hidden dimension has fixe
d ICL capacity in one forward pass.
Simulating larger models or more optimization steps in-context requires deeper o
r wider Transformers.
CoT removes this limitation by providing an expandable workspace via the sequenc
e trajectory.
This enables arbitrary-step and arbitrary-capacity ICGD within a constant-depth
Transformer.
Second, we provide a provable efficient guarantee unique to CoT through dynamica
l masking.
The attention mechanism only process the relevant tokens for the current update
step.
This eliminates the redundant ``process everything'' cost of single-pass deep mo
dels.
Specifically, we prove this CoT mechanism improves the computational cost of the
 prior best in-context result [Wu et al., ICML 2025] by $O(N)$.
Numerical validations support our theory.
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Generation Enhances Understanding in Unified Multimodal Models via Multi-Represe

ntation Generation
Zihan Su, Hongyang Wei, Kangrui Cen, Yong Wang, Guanhua Chen, Chun Yuan, Xiangxi ang Chu
Unified Multimodal Models (UMMs) integrate both visual understanding and generat ion within a single framework.
Their ultimate aspiration is to create a cycle where understanding and generatio n mutually reinforce each other.
While recent post-training methods have successfully leveraged understanding to enhance generation, the reverse
direction of utilizing generation to improve understanding remains largely unexp lored. In this work, we propose
UniMRG (Unified Multi-Representation Generation), a simple yet effective archite cture-agnostic post-training method.
UniMRG enhances the understanding capabilities of UMMs by incorporating auxiliar y generation tasks. Specifically,
we train UMMs to generate multiple intrinsic representations of input images, na mely pixel (reconstruction), depth
(geometry), and segmentation (structure), alongside standard visual understandin g objectives. By synthesizing these
diverse representations, UMMs capture rich complementary information regarding a ppearance, spatial relations, and
structural layout. Consequently, UMMs develop a deeper and more comprehensive un derstanding of visual inputs. Extensive
experiments across diverse UMM architectures
demonstrate that our method notably enhances fine-grained perception, reduces ha llucinations, and improves spatial understanding,
while simultaneously boosting generation capabilities.
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Towards Understanding Adam Convergence on Highly Degenerate Polynomials
Zhiwei Bai, Jiajie Zhao, Zhangchen Zhou, Zhi-Qin John Xu, Yaoyu Zhang
Adam is a widely used optimization algorithm in deep learning, yet the specific class of objective functions where it exhibits inherent advantages remains under explored. Unlike prior studies requiring external schedulers and β_2 near 1 for convergence, this work investigates the ``natural'' auto-convergence prope rties of Adam. We identify a class of highly degenerate polynomials where Adam c onverges automatically without additional schedulers. Specifically, we derive th eoretical conditions for local asymptotic stability on degenerate polynomials an d demonstrate strong alignment between theoretical bounds and experimental resul ts. We prove that Adam achieves local linear convergence on these degenerate fun ctions, significantly outperforming the sub-linear convergence of Gradient Desce nt and Momentum. This acceleration stems from a decoupling mechanism between the second moment v_t and squared gradient g_t^2, which exponentially amplifies the effective learning rate. Finally, we characterize Adam's hyperparameter pha se diagram, identifying three distinct behavioral regimes: stable convergence, s pikes, and SignGD-like oscillation.
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Spectra: Rethinking Optimizers for LLMs Under Spectral Anisotropy
Zhendong Huang, Hengjie Cao, Fang Dong, Ruijun Huang, Mengyi Chen, Yifeng Yang, Xin Zhang, Anrui Chen, Mingzhi Dong, Yujiang Wang, Jinlong Hou, Qin Lv, Robert P . Dick, Yuan Cheng, Tun Lu, Fan Yang, Li Shang
Gradient signals in LLM training are highly anisotropic: recurrent linguistic st ructure concentrates energy into a small set of dominant spectral directions, wh ile context-specific information resides in a long tail. We show that this spike –tail separation persists throughout training, with the spike occupying only abo ut 1.5% of directions yet dominating optimizer statistics. This dominance suppre sses tail learning by contracting tail updates through second-moment normalizati on and tightening the globally stable learning-rate bound.
Motivated by this analysis, we propose Spectra, a spike-aware optimizer that sup presses the dominant low-rank spike subspace without amplifying the noise-sensit ive spectral tail. Spectra tracks the spike subspace via cached, warm-started po

wer iteration and applies low-rank spectral shaping with negligible overhead and substantially reduced optimizer-state memory.
Across Qwen3-0.6B trained on 100B tokens and LLaMA3-8B trained on 50B tokens, Spectra achieves the lowest final validation loss, improving average downstream accuracy by 1.41/0.89 and 1.62/0.66 points over AdamW/Muon, respectively. For wall-clock convergence, Spectra reaches matched loss targets up to 1.31×, 1.34×, and 1.24× faster than AdamW on Qwen3-0.6B, Qwen3-2B-A0.8B, and Qwen3-8B; its speedup over Muon grows as model scale increases from 0.6B to 8B. For computational efficiency, Spectra is 5.1× faster than Muon in optimizer processing time, cuts optimizer-state memory by 49.25%, and achieves the lowest measured end-to-end per-step runtime. Spectra's Megatron integration is available at https://github.com/kimmichtank/spectra.
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SLAP: The Semantic Least Action Principle for Variational Video-Language Modeling

Xiang Fang, Wanlong Fang

In the era of Large Video-Language Models (LVLMs), the computational necessity of sparse frame sampling creates a fundamental ``temporal gap'', rendering models blind to critical causal transitions. Existing solutions relying on generative hallucination (e.g., latent diffusion) or autoregressive extrapolation often fail to maintain semantic consistency over long horizons, suffering from object vanishing and energetic instability. We propose a paradigm shift from probabilistic generation to variational mechanics with the Semantic Least Action Principle (SLAP). Drawing a rigorous isomorphism between classical mechanics and semantic dynamics, we model the latent video trajectory as a path on a Riemannian manifold governed by a Semantic Lagrangian. By formulating the interpolation task as a Boundary Value Problem (BVP) solved via the discrete Euler-Lagrange equations, SLAP naturally enforces object persistence without pixel-level rendering. Extensive experiments on multiple challenging datasets show the effectiveness of our proposed SLAP.
****

A Unifying Relational Perspective on Expressive Lottery Tickets

Lorenz Kummer, Samir Moustafa, Anatol Ehrlich, Franka Bause, Marco Nennstiel, Przemys■aw Andrzej Wa■■ga, Nils Morten Kriege

Graph neural networks (GNNs) are widely used, but how parameter sparsity affects the expressivity of relational (RGNNs) and temporal (TGNNs) variants is poorly understood. The Strong Expressive Lottery Ticket Hypothesis (SELTH) posits the existence of sparse GNNs that preserve Weisfeiler-Leman (WL) expressivity on static graphs. We generalize this existence result to a probabilistic statement for multi-relational and temporal domains via the relational WL (RWL). We prove that sufficiently parameterized RGNNs contain sparse subnetworks that maintain 1-RWL expressivity and derive a lower bound on the probability that a random pruning yields such a subnetwork. We show that common TGNNs and cross-graph message passing schemes admit RGNN reformulations such that they inherit these guarantees and, moreover, that the expressivity of a sparse RGNN is connected to its optimization behavior under common update regimes. Experiments instantiate the bound, compare it to empirical probabilities on synthetic data, and study how pre-training expressivity relates to optimization and prediction quality metrics on temporal and molecular benchmarks.
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$\textit{S}$-SPPO: Semantic-Calibrated Self-Play Preference Optimization

Xiwen Chen, Wenhui Zhu, Jingjing Wang, Peijie Qiu, Zhipeng Wang, Huayu Li, Zheng Xiao He, XUANZHAO DONG, Prayag Tiwari, Mingkun Xu, Yujian Xiong, Feng Luo, Abolfazl Razi, Brendan Hogan Rappazzo, Anderson Schneider, Yuriy Nevmyvaka

Aligning Large Language Models (LLMs) with human preferences is often formulated via Direct Preference Optimization (DPO). However, the standard Bradley-Terry instantiation of DPO is limited in modeling common departures from transitivity in human preferences. To address this, recent work has introduced Self-Play Preference Optimization (SPPO), which iteratively refines the policy by training on self-generated win-lose pairs. Our investigation, however, reveals a critical ins

tability in SPPO: the optimization is prone to \textit{policy degeneration} when the preference oracle assigns overly confident wins to semantically indistinguishable responses. To mitigate this, we propose $\textit{S}$-SPPO, a dual-space semantic calibration framework comprising: i) $\textit{Supervision Calibration}$ via semantic gating, which anneals win rate targets toward the maximum-entropy baseline as semantic overlap increases; and ii) $\textit{Representation Calibration}$ via latent repulsion to enforce geometric diversity to prevent manifold collapse and maintain latent diversity between chosen and rejected samples. Theoretically, we show that the calibration preserves the constant-sum game structure, facilitating convergence to a Nash Equilibrium. Empirically, $\textit{S}$-SPPO avoids the performance degradation seen in prior methods, achieving 52.19\% win rate and 47.46\% length-controlled win rate on AlpacaEval 2.0 with Llama-3-8B, without using additional human-annotated preferences during training.
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TimeOmni-VL: Unified Models for Time Series Understanding and Generation
Tong Guan, Sheng Pan, Johan Barthelemy, Zhao Li, Yujun Cai, Cesare Alippi, Ming Jin, Shirui Pan
Recent time series modeling faces a sharp divide between numerical generation and semantic understanding, with research showing that generation models often rely on superficial pattern matching, while understanding-oriented models struggle with high-fidelity numerical output. Although unified multimodal models (UMMs) have bridged this gap in vision, their potential for time series remains untapped. We propose TimeOmni-VL, the first vision-centric framework that unifies time series understanding and generation through two key innovations: (1) Fidelity-preserving bidirectional mapping between time series and images (Bi-TSI), which advances Time Series-to-Image (TS2I) and Image-to-Time Series (I2TS) conversions to ensure near-lossless transformations. (2) Understanding-guided generation. We introduce TSUMM-Suite, a novel dataset consisting of six understanding tasks rooted in time series analytics and coupled with two generation tasks. With a calibrated Chain-of-Thought (CoT), TimeOmni-VL is the first to leverage time series understanding as an explicit control signal for high-fidelity generation. Experiments confirm that this unified approach significantly improves semantic understanding and numerical precision, establishing a new frontier for multimodal time series modeling.
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Learning Anisotropic Value Geometry with Finsler Reinforcement Learning
Jumman Hossain, Nirmalya Roy
We introduce **Finslerian Reinforcement Learning (FiRL)**, an RL framework that makes directional costs explicit and improves robustness to tail risk. FiRL incorporates a *Finsler metric* into the locomotion cost, expressing effort as $F(x, v)$ that depends on the state x and motion v, so it can capture uphill versus downhill asymmetry, lateral slip, and other direction-dependent effects. To handle rare but catastrophic outcomes, FiRL optimizes a Conditional Value-at-Risk ($CVaR_\alpha$) objective. We derive the corresponding risk-sensitive Bellman equation and show that the resulting CVaR--Finsler Bellman operator is a γ-contraction. This guarantees a unique fixed-point value function, while the underlying Finsler cost induces an asymmetric path cost d_F that satisfies a triangle inequality despite directional asymmetry. We then develop a FiRL actor--critic algorithm to learn policies under this anisotropic, risk-averse objective. Across simulation benchmarks and real-world robot trials, FiRL demonstrates safer and more energy-efficient locomotion behavior than strong baselines such as risk-neutral PPO. For instance, on a 12° sloped Hopper task, FiRL reduces worst-case ($CVaR_{0.1}$) impact forces by over 35% and total energy cost by 15%, while also improving success rate.
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LLM Watermark Evasion via Bias Inversion
Jeongyeon Hwang, Sangdon Park, Jungseul Ok
Watermarking offers a promising solution for detecting LLM-generated content, yet its robustness under realistic query-free (black-box) evasion remains an open challenge. Existing query-free attacks often achieve limited success or severely

distort semantic meaning. We bridge this gap by theoretically analyzing rewriting-based evasion, demonstrating that reducing the average conditional probability of sampling green tokens by a small margin causes the detection probability to decay exponentially. Guided by this insight, we propose the \emph{Bias-Inversion Rewriting Attack} (BIRA), a practical query-free method that applies a negative logit bias to a proxy suppression set identified via token surprisal. Empirically, BIRA achieves state-of-the-art evasion rates ($>99\%$) across diverse watermarking schemes while preserving semantic fidelity substantially better than prior baselines. Our findings reveal a fundamental vulnerability in current watermarking methods and highlight the need for rigorous stress tests. Our code is available at \href{https://github.com/ml-postech/LLM-Watermark-Evasion-via-Bias-Inversion}{here}.

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Emergent Communication Under Misinformation

Heeyoung Lee, Kyungwoo Song

Social interactions are characterized by both adversarial and cooperative aspects. Communications between agents may also involve adversarially motivated actors. Messages can pass through intermediaries with malicious intent before reaching the intended receiver. These actors may modify the message to induce misunderstanding from the receiver while preserving the overall characteristics of the message. This form of misinformation is prevalent in real-world communications and may affect the dynamics under which communication protocols are developed. However, this aspect of social interaction is relatively underexplored in many emergent communication studies that aim to understand the environmental factors behind the emergence of languages' characteristics. This work explores how misinformation affects language emergence with a focus on compositionality. We design a communication game containing a malign intermediary between the sender and receiver. We find that risks of malign misrepresentation promote the emergence of compositional languages in simulations of communicative agents. Furthermore, we observe that adaptability of malign intermediaries is a crucial factor in forming a consistent pressure toward compositionality and that partial misinformation, in which the intermediary targets only a subset of attributes, can also induce compositionality.

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EARL: Towards a Unified Analysis-Guided Reinforcement Learning Framework for Egocentric Interaction Reasoning and Pixel Grounding

Yuejiao Su, Xinshen ZHANG, Zhen Ye, Lei Yao, Lap-Pui Chau, Yi Wang

A precise and comprehensive understanding of human-environment interactions in egocentric vision is essential for next-generation intelligent agents, such as assistive robotics.

While existing multimodal large language models (MLLMs) support unified reasoning from scene-level analysis to instance-specific grounding, their accuracy and generalization remain limited.

To this end, this paper introduces a novel Egocentric Analysis-guided RL-based method (EARL) that employs Group Relative Policy Optimization (GRPO) to enhance the interaction understanding of MLLMs in first-person vision.

Specifically, EARL adopts a two-stage parsing framework including coarse-grained interpretation and fine-grained response.

The first stage holistically interprets egocentric interactions and generates a structured textual description. The second stage produces the language answer and corresponding pixel-level grounding mask in response to the user query. To bridge the two stages, we extract a global interaction descriptor from the first stage and treat it as a semantic prior, which is then integrated via a novel Analysis-guided Feature Synthesizer (AFS) to support query-oriented reasoning.

Furthermore, to effectively guide policy optimization, we design a sophisticated, multi-faceted reward mechanism that incorporates format correctness, answer relevance, and grounding accuracy. Experimental results demonstrate that EARL achieves an impressive 65.48% cIoU on the Ego-IRGBench benchmark for pixel grounding, surpassing previous state-of-the-art RL-based methods by 8.37%.

Superior performance in out-of-distribution evaluations further validates EARL's

generalization capability.
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Performative Learning Theory
Julian Rodemann, Unai Fischer-Abaigar, James Bailie, Krikamol Muandet
Performative predictions influence the very outcomes they aim to forecast. We study performative predictions that affect a sample (e.g., only existing users of an app) and/or the whole population (e.g., all potential app users). This raises the question of how well models generalize under performativity. For example, how well can we draw insights about new app users based on existing users when both of them react to the app's predictions? We address this question by embedding performative predictions into statistical learning theory. We prove generalization bounds under performative effects on the sample, on the population, and on both. A key intuition behind our proofs is that in the worst case, the population negates predictions, while the sample deceptively fulfills them. We cast such self-negating and self-fulfilling predictions as min-max and min-min risk functionals in Wasserstein space, respectively. Our analysis reveals both a fundamental trade-off between performatively changing the world and learning from it, as well as a surprising insight on how to improve generalization guarantees by retraining on performatively distorted samples. We illustrate our bounds using real data on prediction-informed assignments to job trainings.
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MME-Reasoning: A Broad-Spectrum Benchmark for Evaluating Logical Reasoning in MLLMs
Jiakang Yuan, Tianshuo Peng, Yilei Jiang, Yiting Lu, Renrui Zhang, Kaituo Feng, Chaoyou Fu, Tao Chen, LEI BAI, Bo Zhang
Logical reasoning is a fundamental aspect of human intelligence and an essential capability for multimodal large language models (MLLMs). Despite the significant advancement in multimodal reasoning, existing benchmarks fail to comprehensively evaluate their reasoning abilities due to the lack of explicit categorization for logical reasoning types and an unclear understanding of reasoning. To address these issues, we introduce MME-Reasoning, a comprehensive benchmark designed to evaluate the reasoning ability of MLLMs, which covers all three types of reasoning (i.e., inductive, deductive, and abductive). We carefully curate the data to ensure that each question effectively evaluates reasoning ability rather than perceptual skills or knowledge breadth, and extend the evaluation protocols to cover the evaluation of diverse questions. Our evaluation reveals substantial limitations of SoTA MLLMs when subjected to holistic assessments of logical reasoning capabilities. Even the most advanced MLLMs show limited performance in comprehensive logical reasoning, with notable performance imbalances across reasoning types. In addition, we conducted an in-depth analysis of approaches such as ``thinking mode'' and Rule-based RL, which are commonly believed to enhance reasoning abilities. We hope the community can pay more attention to the comprehensive reasoning capabilities of MLLMs instead of only focusing on its subset.
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Score Based Error Correcting Code Decoder
Alon Helvits, Eliya Nachmani
Error-correcting codes enable reliable communication, yet practical soft decoding remains challenging across code families and block lengths. We propose SB-ECC, a score-based decoder that casts decoding as continuous-time denoising. A neural denoiser defines a probability-flow ordinary differential equation (ODE) that iteratively updates the noisy channel observation toward a valid codeword, guided by parity constraints. The model is trained across noise levels without time/SNR conditioning, enabling inference without SNR estimation and supporting a direct latency--accuracy trade-off controlled by the ODE solver budget. We use the raw signed channel observation as input for learning a continuous denoising field. Across 42 code/SNR settings, SB-ECC achieves the best BER in $39/42$ entries, with an average SNR gain of 0.17\,dB and a maximum gain of 0.46\,dB over the strongest competing baseline, we showed that swapping the solver from Euler to DPM preserves $-\ln(\mathrm{BER})$ while reducing end-to-end decoding time by 8.86% on average (up to 12.82%).

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OPT-Engine: Benchmarking the Limits of LLMs in Optimization Modeling via Complexity Scaling
Yitian Chen, Cheng Cheng, Yinan Sun, Zi Ling, Dongdong Ge
We investigate the capabilities and scalability of Large Language Models (LLMs) in optimization modeling, a domain requiring structured reasoning and precise formulation.
To this end, we introduce OPT-ENGINE, an extensible benchmark framework with quantifiable and controllable complexity. OPT-ENGINE spans ten canonical Operations Research problems, systematically scaling from Linear Programming to Mixed-Integer Programming, providing a structured environment to probe the limits of automated problem formulation and solving.
Utilizing OPT-Engine, we address three pivotal research questions.
First, we examine whether Pure-Text Reasoning (PTR) via classical Chain-of-Thought can efficiently tackle optimization tasks, finding that PTR suffers from a critical robustness gap as task complexity increases.
Second, we examine whether integrating external computational tools can mitigate PTR's arithmetic weaknesses and improve performance. Our results show that while such tools aid local calculations, they still fail to adhere to global optimization constraints.
Finally, we pinpoint that for the current SOTA paradigm, Solver-integrated Reasoning (SIR), the automated formulation of constraints represents the primary bottleneck.
These findings clarify current paradigms' limitations and provide a structured roadmap for developing next-generation LLMs for optimization modeling.
We release our code and data to facilitate future research (https://github.com/Cardinal-Operations/OPTEngine).
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Improving Backward Conformal Prediction via Non-Conformity Score Transformation
Junxian Liu, Hao Zeng, Hongxin Wei
Conformal Prediction (CP) provides a statistical framework for uncertainty quantification that constructs prediction sets with coverage guarantees. While CP yields uncontrolled prediction set sizes, Backward Conformal Prediction (BCP) inverts this paradigm by enforcing a predefined upper bound on set size and estimating the resulting coverage guarantee. However, the looseness induced by Markov's inequality within the BCP framework causes a significant gap between the estimated coverage bound and the empirical coverage. In this work, we introduce ST-BCP, a novel method that introduces a data-dependent transformation of nonconformity scores to narrow the coverage gap. In particular, we develop a computable transformation and prove that it outperforms the baseline identity transformation. Extensive experiments demonstrate the effectiveness of our method, reducing the average coverage gap from 4.20\% to 1.12\% on common benchmarks.
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Dense associative memory for Gaussian distributions
Chandan Tankala, Krishna Balasubramanian
Dense associative memories (DAMs) store and retrieve patterns via energy-function based fixed points, but existing models are limited to vector representations. We extend DAMs to Gaussian densities equipped with the 2-Wasserstein distance. Our framework defines a log-sum-exp energy over stored distributions and a retrieval dynamics aggregating optimal transport maps in a Gibbs-weighted manner. Stationary points correspond to self-consistent Wasserstein barycenters, generalizing classical DAM fixed points. We prove exponential storage capacity and provide quantitative retrieval guarantees under Wasserstein perturbations. We validate the method on synthetic and real-world image (CelebA and CIFAR-10 datasets) and text (text8 and NLI corpus) datasets. By generalizing from vectors to distributions, our work bridges classical DAMs with modern generative modeling and paves way for distributional storage and retrieval in memory-augmented learning.
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Flexible Kernels for Protein Property Prediction
Martin Jankowiak, Yerdos Ordabayev, Rudraksh Tuwani, Henry Neil Ward, Hunter Nis

onoff, James M McFarland, Gevorg Grigoryan
Despite its importance to applications in protein design, predicting protein properties like binding affinity and thermostability from sparse experimental data remains a significant challenge. Accordingly, we introduce a class of sequence kernels that exploit evolutionary substitution matrices as well as local linearity and demonstrate that the resulting Gaussian processes provide data-efficient models of protein property landscapes, frequently outperforming alternatives that rely on foundation model embeddings. Furthermore--by learning what are in effect structure-aware substitution matrices--we show that our kernels can readily incorporate structural information from foundation models. We demonstrate that these structure-conditioned kernels are well suited to multi-task learning across multiple protein property landscapes and can decisively outperform local supervised learning methods.
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OmniShow: Unifying Multimodal Conditions for Human-Object Interaction Video Generation
Donghao Zhou, Guisheng Liu, Hao Yang, Jiatong Li, Jingyu Lin, Xiaohu Huang, Yichen Liu, Xin Gao, Cunjian Chen, Shilei Wen, Chi-Wing Fu, Pheng-Ann Heng
In this work, we study **Human-Object Interaction Video Generation (HOIVG)**, which aims to synthesize high-quality human-object interaction videos conditioned on text, reference images, audio, and pose. We present **OmniShow**, the first all-in-one model tailored for this practical yet challenging task, capable of harmonizing multimodal conditions and delivering industry-grade performance. To overcome the trade-off between controllability and quality, we introduce **Unified Channel-wise Conditioning** for efficient image and pose injection, and **Gated Local-Context Attention** to ensure precise audio-visual synchronization. To effectively address data scarcity, we develop a **Decoupled-Then-Joint Training** strategy that leverages a multi-stage training process with model merging to efficiently harness heterogeneous sub-task datasets. Furthermore, to fill the evaluation gap in this field, we establish HOIVG-Bench, a dedicated and comprehensive benchmark for HOIVG. Extensive experiments demonstrate that OmniShow achieves overall state-of-the-art performance across various multimodal conditioning settings, setting a solid standard for the emerging HOIVG task.
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COBRA: Contribution-Based Bayesian Rank Allocation for Parameter-Efficient Fine-Tuning
Hongcheng Ding, Xuanze Zhao, Xuanhuang Liu, Jing Jin, Shamsul Nahar Abdullah, Deshinta Arrova Dewi
Full fine-tuning of large language models (LLMs) incurs prohibitive computational and storage costs. Parameter-efficient fine-tuning (PEFT) addresses this limitation, with Low-Rank Adaptation (LoRA) gaining widespread adoption due to its simplicity and zero inference overhead. However, LoRA and its variants typically rely on uniform rank allocation or a single importance metric such as gradient magnitude or output sensitivity to guide rank distribution. This approach fails to recognize that gradient magnitude and output contribution are decoupled properties, leading to suboptimal allocation where critical layers are under-provisioned while less important ones waste capacity. To address this challenge, we propose COBRA, a principled framework integrating dual importance factors for adaptive rank allocation. COBRA operates in three stages: (1) layer conductance attribution quantifies each layer's contribution via path-integral attribution; (2) dual-factor aggregation combines contribution with adaptation demand, producing the Task-Adaptive Layer Conductance (TA-LC) distribution; and (3) Bayesian rank allocation translates this distribution into optimal heterogeneous ranks via variational optimization. Layer conductance provides layer-level interpretability by explicitly quantifying how much each layer contributes to predictions without redundancy, directly aligning with the granularity of rank allocation decisions and enabling principled cross-layer comparison for rank distribution. Experiments across diverse architectures and tasks demonstrate that COBRA consistently outperforms existing methods, achieving up to 1.6 points improvement on GLUE and a 6.6\% average MSE reduction in high-rank regression regimes under comparable paramet

er budgets.
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FIBER: A Differentially Private Optimizer with Filter-Aware Innovation Bias Correction
Duc Dm, Thao Do, Minh Son Hoang, Anh Le Duc Tran, Daeyoung Kim, Huy Nguyen
Differentially private (DP) training protects individual examples by adding noise to gradients, but the injected noise interacts nontrivially with adaptive optimizers. Recent DP methods temporally filter privatized gradients to reduce variance; however, filtering also changes the DP noise statistics seen by AdamW's second-moment accumulator. As a result, bias corrections derived for unfiltered DP noise (e.g., subtracting σ_w^2) can become miscalibrated when filtering is present. We propose FIBER, a DP optimizer designed for temporally filtered privatized gradients. FIBER (i) performs denoising in innovation space by filtering the residual stream and integrating it to form the filtered gradient estimate, (ii) decouples the two-point observation geometry from the innovation gain to enable independent tuning, and (iii) introduces a filter-aware second-moment calibration that subtracts the attenuated DP noise contribution $A(\omega)\sigma_w^2$, where $A(\omega)$ is derived in closed form for the innovation filter and can be computed for general stable linear filters. Across vision and language benchmarks, FIBER consistently demonstrates substantial improvements in the performance of DP optimizers, surpassing state-of-the-art results under equivalent privacy constraints on multiple tasks.
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Convolutional Learnable-Group Weightless Neural Network
Qinhong Ma, Yulin Chen, Zhiwei Fan, Suzhen Wu, Bo Mao
Weightless Neural Networks (WNNs) based on interconnected Lookup Tables (LUTs) have attracted attention for inference in extremely compact models, but achieving competitive accuracy under such tight resource budgets remains challenging. To address these issues, we introduce the Convolutional Learnable-Group Weightless Neural Network (CLGN). CLGN constructs convolutional layers using LUTs and incorporates a learnable GroupSum connection, thereby enhancing the accuracy of WNNs while maintaining low implementation resource consumption. Moreover, we propose a hierarchical training strategy to improve the training efficiency. We evaluate CLGN in two edge computing scenarios: (1) FPGA, where we evaluate accuracy, latency, throughput, power consumption, LUTs usage, and parameter size; and (2) Microprocessor, where we evaluate latency and memory usage. Compared with the state-of-the-art solutions, the proposed CLGN achieves superior accuracy while maintaining lower implementation resource consumption.
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VideoMAETok: Boosting Video Diffusion Models via Masked Autoencoders as Tokenizers
Zhan Tong, Tinne Tuytelaars
Latent diffusion models have become the dominant paradigm for video generation, making the video tokenizer a critical role. While most existing tokenizers are trained primarily for reconstruction, diffusion models are optimized to denoise heavily corrupted latents, which creates a mismatch between tokenizer training objectives and downstream generative learning. As a result, reconstruction metrics (e.g., rFVD) can be a poor proxy for generation quality (gFVD), and overly prioritizing reconstruction may even hinder diffusion training. We propose VideoMAETok, a simple family of ViT-based video tokenizers trained explicitly as corruption-inversion models for latent video diffusion. VideoMAETok builds on masked autoencoders: we (i) apply high-ratio token masking and encode only visible spatiotemporal tokens for efficiency, and (ii) corrupt latent tokens with interpolative Gaussian noise to better match the denoising nature of diffusion generators. Training under such corruption encourages latents that remain informative and well-conditioned for downstream denoising. Extensive experiments show that VideoMAETok consistently improves generation quality when paired with off-the-shelf diffusion models (SiT and LightningDiT), achieving state-of-the-art gFVD on Kinetics-600 and UCF-101 while remaining compute-efficient. Code is available at https://github.com/yztongzhan/VideoMAETok.

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SMD: Multi-view Safety-Critical Driving Video Generation in the Real-world Domain
Jiawei Zhou, Linye Lyu, Zhuotao Tian, Cheng Zhuo, Yu Li
Safety-critical scenarios are essential for evaluating autonomous driving (AD) systems, yet they are rare in practice. Existing generators produce trajectories, simulations, or single-view videos—but they don’t meet what modern AD systems actually consume: realistic multi-view video.
We present SMD, the first framework for generating multi-view safety-critical driving videos in the real-world domain.
SMD couples a safety-critical trajectory engine with a diffusion-based multi-view video generator through three design choices. First, we pick the right adversary: a GRPO-fine-tuned vision-language model (VLM) that understands multi-camera context and selects vehicles most likely to induce hazards. Second, we generate the right motion: a two-stage trajectory process that (i) produces collisions, then (ii) transforms them into natural evasion trajectories—preserving risk while staying within what current video generators can faithfully render. Third, we synthesize the right data: a diffusion model that turns these trajectories into multi-view videos suitable for end-to-end planners.
Videos generated by SMD substantially increase collision rates when stress testing multiple end-to-end planners, and reduce collision rates when incorporated into training, improving planner robustness and safety.
Our code and video examples are available at: \href{https://icml-2.github.io/SMD/}{https://icml-2.github.io/SMD/}.
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Moving Out: Physically-grounded Human-AI Collaboration
Xuhui Kang, Sung-Wook Lee, Haolin Liu, Yuyan Wang, Yen-Ling Kuo
The ability to adapt to physical actions and constraints in an environment is crucial for embodied agents (e.g., robots) to effectively collaborate with humans. Such physically grounded human-AI collaboration must account for the increased complexity of the continuous state-action space and constrained dynamics caused by physical constraints.
However, most existing collaboration benchmarks are discrete or do not consider physical attributes and constraints.
To address this, we introduce Moving Out, a human-AI collaboration benchmark that resembles a wide range of collaboration modes affected by physical attributes and constraints, such as moving heavy items together and coordinating actions to move an item around a corner.
Moving Out consists of two challenges and human-human interaction data to comprehensively evaluate models' abilities to adapt to diverse human behaviors and unseen physical attributes.
To give embodied agents the capability to collaborate with humans under physical attributes and constraints, we propose a novel method, BASS (Behavior Augmentation, Simulation, and Selection), to enhance the diversity of agents and their understanding of the outcome of actions.
We systematically compare BASS and state-of-the-art models in AI-AI and human-AI experiments, showing that BASS can effectively collaborate with both unseen AI and humans.
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From Backward Spreading to Forward Replay: Revisiting Target Construction in LLM Parameter Editing
Wei Liu, Liu Hongkai, Zhiying Deng, Yee Whye Teh, Wee Sun Lee
LLM parameter editing methods commonly rely on computing an ideal target hidden-state at a target layer (referred as anchor point) and distributing the target vector to multiple preceding layers (commonly known as backward spreading) for cooperative editing. Although widely used for a long time, its underlying basis have not been systematically investigated. In this paper, we first conduct a systematic study of its foundations, which helps clarify its capability boundaries, practical considerations, and potential failure modes.
  Then, we propose a simple and elegant alternative that replaces backward sprea

ding with forward-propagation. Instead of optimizing the target at the last editing layer, we optimize the anchor point at the first editing layer, and then propagate it forward to obtain accurate and mutually compatible target hidden-states for all subsequent editing layers. This approach achieves the same computational complexity as existing methods while producing more accurate layer-wise targets. Our method is simple, without interfering with either the computation of the initial target hidden state or any other components of the subsequent editing pipeline, and thus constituting a benefit for a wide range of LLM parameter editing methods.
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DiscoverLLM: From Executing Intents to Discovering Them
Tae Soo Kim, Yoonjoo Lee, Jaesang Yu, John Joon Young Chung, Juho Kim
To handle ambiguous and open-ended requests, Large Language Models (LLMs) are increasingly trained to interact with users to surface intents they have not yet expressed (e.g., ask clarification questions). However, users are often ambiguous because they have not yet formed their intents: they must observe and explore outcomes to discover what they want. Simply asking "what kind of tone do you want?" fails when users themselves do not know. We introduce DiscoverLLM, a novel and generalizable framework that trains LLMs to help users form and discover their intents. Central to our approach is a novel user simulator that models cognitive state with a hierarchy of intents that progressively concretize as the model surfaces relevant options---where the degree of concretization serves as a reward signal that models can be trained to optimize. Resulting models learn to collaborate with users by adaptively diverging (i.e., explore options) when intents are unclear, and converging (i.e., refine and implement) when intents concretize. Across proposed interactive benchmarks in creative writing, technical writing, and SVG drawing, DiscoverLLM achieves over 10% higher task performance while reducing conversation length by up to 40%. In a user study with 75 human participants, DiscoverLLM improved conversation satisfaction and efficiency compared to baselines.
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MOES-Pred: Molecular Structural Representation Learning by Adaptive Energy-Sentinel Vibration for Generalized Property Prediction
Zhiran Hou, Tinghuai Ma, Huan Rong, Li Jia, Anouar Imel, Heng Zhang, Ming Li
Predicting molecular properties from three-dimensional structures is fundamentally hindered by limited labeled data. While researchers have adapted self-supervised pre-training techniques from computer vision and natural language processing to address this scarcity, these approaches frequently neglect the intrinsic physical principles unique to molecular systems. From a physical perspective, denoising pre-training can be formally proven equivalent to learning molecular force fields. However, existing methods indiscriminately apply uniform noise across all molecules, thereby introducing systematic bias into the modeling of molecular distributions. To mitigate this issue, we introduce MOES-Pred, a denoising pre-training framework featuring an energy sentinel mechanism that dynamically tailors noise perturbations to individual molecules. Leveraging chemical prior knowledge, our molecule-specific noising strategies enhance conformational sampling coverage and improve distribution modeling fidelity. Extensive experiments show that MOES-Pred surpasses mainstream approaches in both force prediction and downstream quantum chemical property prediction, demonstrating substantial improvements.
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Priority-Aware Shapley Value
Kiljae Lee, Ziqi Liu, Weijing Tang, Yuan Zhang
Shapley values are widely used for model-agnostic data valuation and feature attribution, yet they implicitly assume contributors are interchangeable.

    This can be problematic when contributors are dependent (e.g., reused/augmented data or causal feature orderings) or when contributions should be adjusted by factors such as trust or risk.

    We propose Priority-Aware Shapley Value (PASV), which incorporates both hard precedence constraints and soft, contributor-specific priority weights.

PASV is applicable to general precedence structures, recovers precedence-only and weight-only Shapley variants as special cases, and is uniquely characterized by natural axioms.

We develop an efficient adjacent-swap Metropolis–Hastings sampler for scalable Monte Carlo estimation and analyze limiting regimes induced by extreme priority weights.

Experiments on data valuation (MNIST/CIFAR10) and feature attribution (Census Income) demonstrate more structure-faithful allocations and a practical sensitivity analysis via our proposed ``priority sweeping".

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CoT is Not the Chain of Truth: An Empirical Internal Analysis of Reasoning LLMs for Fake News Generation

Zhao Tong, Chunlin Gong, Yiping ZHANG, Haichao Shi, Qiang Liu, Xingcheng Xu, Shu Wu, Xiao-Yu Zhang

From generating headlines to fabricating news, the Large Language Models (LLMs) are typically assessed by their final outputs, under the safety assumption that a refusal response signifies safe reasoning throughout the entire process. Challenging this assumption, our study reveals that during fake news generation, even when a model rejects a harmful request, its Chain-of-Thought (CoT) reasoning may still internally contain and propagate unsafe narratives. To analyze this phenomenon, we introduce a unified safety-analysis framework that systematically deconstructs CoT generation across model layers and evaluates the role of individual attention heads through Jacobian-based spectral metrics. Within this framework, we introduce three interpretable measures: stability, geometry, and energy to quantify how specific attention heads respond or embed deceptive reasoning patterns. Extensive experiments on multiple reasoning-oriented LLMs show that the generation risk rise significantly when the thinking mode is activated, where the critical routing decisions concentrated in only a few contiguous mid-depth layers. By precisely identifying the attention heads responsible for this divergence, our work challenges the assumption that refusal implies safety and provides a new understanding perspective for mitigating latent reasoning risks.

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From Individual Calibration to Reliable Classifiers: ALD Parameterization with mPAIC Guarantees

Deming Sheng, Ricardo Henao

Modern neural classifiers can achieve remarkable predictive performance, yet often suffer from *miscalibration*.

In this paper, we introduce a unified calibration framework applicable to arbitrary distribution-based classifiers.

The proposed calibration objective guarantees a *monotone Probably Approximately Individually Calibrated (mPAIC)* predictor, which theoretically implies the properties of a *Probably Approximately Calibrated Classifier (PACC)* with explicit error bounds.

To enable stable and effective optimization, we further devise a *Decoupled Dual-Stream Optimization (DDSO)* strategy with gradient detachment to reconcile discriminative representation learning and continuous calibration.

Notably, our framework bridges calibration paradigms, supporting flexible deployment either as an end-to-end *pre-calibration* objective or as a lightweight *post-calibration* adapter.

Extensive experiments across nine real-world datasets demonstrate that our approach consistently outperforms strong baselines, achieving superior performance on both *accuracy* and multi-level *calibration*.

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Demystifying Mergeability: Interpretable Properties to Predict Model Merging Success

Luca Zhou, Bo Zhao, Rose Yu, Emanuele Rodolà

Model merging combines knowledge from separately fine-tuned models, yet the factors driving its success remain poorly understood. While recent work treats mergeability as an intrinsic property of the models, we show with an architecture-agnostic framework that it fundamentally depends on both the merging method and the

partner tasks. Using L1-regularized linear optimization over a set of interpretable pairwise metrics (e.g., gradient L2 distance), we uncover properties correlating with post-merge normalized accuracy across five merging methods. We find that the drivers of merge success vary across architectures and merging methods, overall with only a moderate agreement (64.0% average top-5 metric overlap; 79.3% sign agreement). Crucially, however, gradient alignment metrics consistently emerge as the most fundamental signals of mergeability. These findings provide a diagnostic foundation for understanding mergeability and motivate future merge-aware fine-tuning strategies.

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BEST: Benchmarking Efficiency in Space and Time for LLM-Generated Code

Aocheng Shen, Boyu Zhang, Jiaze Li, Ruixuan Ma, Qiankun Zhang, Jing Wang, Bin Yuan, Shenghao Liu, Xianjun Deng

Large language models (LLMs) have revolutionized research in software engineering, and among various tasks, LLM-based code synthesis is promising. A recent line of benchmarks aims to evaluate LLM-generated codes in time efficiency, beyond their correctness. However, *space*, another vital aspect of code efficiency, is rarely evaluated in prior benchmarks. To fill in the gap, this paper introduces *BEST*, the first benchmark for evaluating the efficiency of LLM-generated codes in *both time and space*. It comprises 440 coding tasks that are rigorously constructed by experts. In addition, we propose a fine-grained *subtask-based* evaluation scheme by dividing each task into multiple subtasks, with different input scales and difficulties. Each subtask is then accompanied by an expert-crafted standard implementation as the efficiency baseline, which achieves the *Pareto optimum*. Building on BEST, we introduce a unified and novel dual-indicator (time and space) metric, named dual@${k}$, generalizing the notion of the standard pass@${k}$ metric and building on a careful and novel construction of a *weight matrix* of subtasks. Through extensive experiments with dual@${k}$ across 50 LLMs on BEST, our evaluation demonstrates that while LLMs exhibit weak capabilities in generating time-efficient code, their capabilities in space-efficient code generation are even worse. The benchmark is provided at https://github.com/kmsgk0/BEST.

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Rational Transductors

Mehryar Mohri

Standard Transformers excel at semantic modeling but struggle with rigid
  sequential logic and state tracking. Theoretical work establishes that
  self-attention is limited to AC^0 (under hard attention) or TC^0 (under
  soft attention), complexity classes that often fail to support robust length
  generalization on sequential problems without intermediate chain-of-thought
  (see hahn2020theoretical and merrill2022saturated). In this work, we introduce
  \emph{Rational Transductors}, a dual-stream architecture that augments the
  Transformer with a matrix-valued recurrence derived from Weighted Finite
  Automata (WFA). By injecting rational state information into the attention
  mechanism via a *Deep Rational Injection* scheme, our framework strictly
  generalizes Transformers to capture all Regular Languages, NC^1-complete
  problems (such as Boolean Formula Evaluation), and fundamental separations
  like Parity and Modular Counting, while preserving $O(\log T)$ parallel
  training efficiency. Theoretical analysis and empirical results demonstrate
  that Rational Transductors solve the "Regular Gap," enabling robust length
  generalization on algorithmic tasks where standard Transformers fail, without
  the sequential computational bottlenecks of traditional RNNs.

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SWE-Bench Pro: Can AI Agents Solve Long-Horizon Software Engineering Tasks?

Xiang Deng, Jeff Da, Edwin Pan, Yannis Yiming He, Charles Ide, Kanak Garg, Niklas Lauffer, Andrew Park, Chetan Rane, Karmini Sampath, Maya Krishnan, Srivatsa R Kundurthy, Sean M. Hendryx, Zifan Wang, Chen Bo Calvin Zhang, Noah Jacobson, Bin

g Liu, Brad Kenstler
We present SWE-Bench Pro, a comprehensive benchmark designed to evaluate softwar e engineering capabilities through complex, realistic programming challenges. Th is benchmark extends beyond traditional algorithmic problems to encompass the fu ll spectrum of professional software development tasks. The dataset comprises 1, 865 problems sourced from 41 active software engineering repositories, spanning 123 unique programming languages and various application domains. The benchmark is structured into public and private components, with public access to problems from 11 repositories and private evaluation sets from 12 repositories across 4 distinct problem categories. SWE-Bench Pro addresses limitations of existing eva luation frameworks by incorporating problems that reflect real-world software en gineering scenarios, including substantial codebases, complex enterprise applica tions, and multi-file projects requiring sophisticated reasoning and code modifi cation skills. Problems range from early-stage startup environments to enterpris e-level applications, with the private commercial set remaining inaccessible to maintain evaluation integrity while enabling public access to representative pro blems for professional development. Our evaluation methodology employs diverse c oding approaches and models under controlled conditions, ensuring robust perform ance assessment across multiple programming paradigms. Results demonstrate signi ficant performance variations across different problem categories, with traditio nal algorithmic challenges showing notably higher success rates compared to comp lex, multi-file engineering tasks. The benchmark reveals substantial gaps in cur rent capabilities for handling real-world software engineering scenarios, partic ularly in areas requiring deep contextual understanding, cross-file reasoning, a nd integration with existing large-scale systems. This work contributes a more c omprehensive and realistic evaluation framework for assessing software engineeri ng capabilities, providing insights into current limitations and establishing a foundation for future development in automated software engineering tools and me thodologies.
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Scout Before You Attend: Sketch-and-Walk Sparse Attention for Efficient LLM Infe rence
Hoang Anh Duy Le, Sahil Joshi, Zeyu Yang, Zhaozhuo Xu, Anshumali Shrivastava
Self-attention dominates the computational and memory cost of long-context LLM i nference across both prefill and decode phases. To address this challenge, we in troduce **Sketch\&Walk** Attention, a training-free sparse attention method that determines sparsity with lightweight sketches and deterministic walk. Sketch\&W alk applies Hadamard sketching to get inexpensive approximations of attention sc ores, then aggregates these estimates across layers via a walk mechanism that ca ptures attention influence beyond direct interactions between tokens. The accumu lated walk scores are used to select top-k attention blocks, enabling dynamic sparsity with a single training-free algorithm that applies uniformly to both th e prefill and decode phases, together with custom sparse attention kernels. Acro ss a wide range of models and tasks, Sketch\&Walk maintains near-lossless accura cy at 20\% attention density and can slightly outperform dense attention in some settings, while achieving up to $4.7\times$ end-to-end attention speedup over F lashAttention-2.
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Learning Human-Robot Collaboration via Heterogeneous-Agent Lyapunov Policy Optim ization
Hao Zhang, Yaru Niu, Yikai Wang, Ding Zhao, Eric H. Tseng
To improve generalization and resilience in human–robot collaboration (HRC), rob ots must contend with diverse combinations of human behaviors and contexts, moti vating multi-agent reinforcement learning (MARL). However, inherent heterogeneit y between robots and humans creates a rationality gap (RG), where decentralized policy updates deviate from cooperative joint optimization. The resulting learni ng problem is a general-sum differentiable game, so independent policy-gradient updates can oscillate or diverge without added structure. We propose heterogeneo us-agent Lyapunov policy optimization (HALO), a framework that stabilizes decent ralized MARL by enforcing Lyapunov-based contraction in policy-parameter space.

Unlike Lyapunov-based safe RL, which targets state/trajectory constraints in constrained Markov decision processes, HALO uses Lyapunov certification to stabilize decentralized policy learning. HALO rectifies decentralized gradients via optimal quadratic projections, ensuring monotonic contraction of RG and enabling effective exploration of open-ended interaction spaces. Extensive simulations and real-world humanoid-robot experiments show that this certified stability improves generalization and robustness in collaborative corner cases.
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AdaGC: Enhancing LLM Pretraining Stability via Adaptive Gradient Clipping
Guoxia Wang, Shuai Li, Congliang Chen, Jinle Zeng, Jiabin Yang, Dianhai Yu, Yanjun Ma, Li Shen
Loss spikes remain a persistent obstacle in large-scale language model pretraining. While previous research has attempted to identify the root cause of loss spikes by investigating individual factors, we observe that, in practice, such spikes are typically triggered by the confluence of heterogeneous factors. Empirically, loss spikes may arise from a combination of data outliers, hardware or transient computational faults, numerical precision issues, and hyperparameter settings. Regardless of the underlying cause, these spikes manifest as unstable optimizer updates, as abnormal gradients contaminate both first- and second-moment states. In this paper, we propose a principled gradient-centric remedy: AdaGC, an adaptive per-tensor gradient clipping scheme that mitigates such contamination by bounding gradient norms relative to a tensor-wise exponential moving average of their historical clipped values. AdaGC is optimizer-agnostic, introduces negligible memory overhead, and reduces communication costs compared to GlobalGC, particularly in hybrid-parallel distributed training. Experiments on Llama-2 7B, Mixtral 8×1B, and ERNIE 10B-A1.4B demonstrate that AdaGC robustly eliminates training instabilities, consistently reducing spike scores to zero for all models and improving downstream accuracy over GlobalGC by 1.32\%, 1.27\%, and 2.48\%, respectively. Furthermore, AdaGC seamlessly integrates with optimizers such as Muon and Lion, consistently yielding higher average accuracy and zero spike scores. The code is available at https://github.com/PaddlePaddle/PaddleFleet (see Research/AdaGC).
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EntRAG: Entity-Centric Retrieval-Augmented Generation for Knowledge-based Visual Question Answering
Yiheng Hu, Xiaoyang Wang, Qing Liu, Xiwei Xu, Qian Fu, Wenjie Zhang
Knowledge-based Visual Question Answering (KB-VQA) remains a challenging task, particularly when queries require precise identification and grounding of fine-grained entities within large-scale knowledge base. Existing methods often treat visual and textual signals in isolation and rely heavily on image-centric retrieval, which makes them sensitive to visual ambiguities. To address these limitations, we propose EntRAG, an entity-centric retrieval-augmented generation framework. Our approach first introduces EntBind to align query representations with multimodal entity embeddings by explicitly binding entity tokens to latent visual features, retrieving a set of relevant candidate entities. A reranking mechanism is applied to these candidate entities to select the most informative context by combining entity-level alignment with overall contextual relevance. The selected evidence is incorporated into context-aware generation module to produce final answer. By explicitly operating at the entity level, EntRAG achieves more consistent and reliable results.
Extensive experiments demonstrate that EntRAG consistently outperforms prior methods, achieving scores of 46.1 on E-VQA and 44.5 on InfoSeek.
****

Higher-Order Certified Robustness for Regression
Claire Jie Zhang, Natalie Frank
Randomized smoothing has emerged as a scalable technique for certifying the adversarial robustness of classifiers. However, its application to regression remains under-explored and faces unique challenges. Existing regression certificates rely on probabilistic acceptance regions and fail to exploit the local geometry of the function. In this work, we present a novel framework for certified robust

regression that addresses these limitations. We derive a prediction-centered certificate that guarantees the stability of the smoothed model’s prediction and ensures practical computability at test time. We investigate several alternatives for constructing these certificates by explicitly incorporating means, variances, and gradients. In particular we demonstrate on the MNIST rotation task that utilizing gradient information yields significantly tighter robustness certificates compared to the current state-of-the-art, α-smoothing.
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Why DDIM Hallucinates More Than DDPM: A Theoretical Analysis of Reverse Dynamics
Muhammad H. Ashiq, Samanyu Arora, Abhinav Narayan Harish, Ishaan Kharbanda, Hung Yun Tseng, Grigorios Chrysos

We theoretically study the hallucination phenomena in two canonical diffusion samplers: the stochastic Denoising Diffusion Probabilistic Model (DDPM) and the deterministic Denoising Diffusion Implicit Model (DDIM). We analyze the reverse ODE (DDIM) and SDE (DDPM) for a Gaussian mixture target, proving that after a critical time τ, (a) DDIM can become stuck on the segment connecting the two nearest modes and (b) DDPM *stochasticity* helps it become unstuck from this region, thus avoiding hallucination. Our empirical validation verifies that DDPM has a significantly lower hallucination rate than DDIM when this region is entered. Building on our observations, we exhibit how using additional stochastic steps can help DDIM avoid hallucinations and offer new insights on how to design improved samplers.
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Approximation Bounds for Transformer Networks with Application to Regression
Yuling Jiao, Yanming Lai, Defeng Sun, Yang Wang, Bokai Yan

We develop approximation and statistical theory for standard Transformer networks in sequence modeling. Given a sequence-to-sequence target on $[0,1]^{d_x \times n}$ whose entries are γ-Hölder for $\gamma \in (0,1]$ or belong to a first-order Sobolev class, we establish explicit L^p-approximation bounds for all $p \in [1,\infty]$, including the previously elusive endpoint $p=\infty$ under softmax attention. In particular, achieving error ε in L^p-norm requires $\mathcal{O}(\varepsilon^{-d_x n/\gamma})$ parameters for γ-Hölder targets and $\mathcal{O}(\varepsilon^{-d_x n})$ parameters for Sobolev targets, matching the best known scalings in ambient dimension $d_x n$. We further study nonparametric regression with sequential and dependent observations using Transformer networks. Assuming stationary β-mixing covariates whose temporal dependence weakens over time, we analyze a sliding-window empirical risk minimization procedure and establish excess-risk guarantees for the Transformer-based estimators. Our analysis clarifies the role of attention and enables extensions beyond softmax.
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Unifying Adversarial Robustness and Training Across Text Scoring Models
Manveer Singh Tamber, Hosna Oyarhoseini, Jimmy Lin

Research on adversarial robustness in language models is currently fragmented across applications and attacks, obscuring shared vulnerabilities. In this work, we propose unifying the study of adversarial robustness in text scoring models spanning dense retrievers, rerankers, and reward models. This motivates adapting both attacks and adversarial training methods across model roles. Unlike open-ended generation, text scoring failures are directly testable: an attack succeeds when an irrelevant or rejected text outscores a relevant or chosen one. Using this principled lens of text scoring, we demonstrate that current adversarial training formulations for language models are often short-sighted, failing to effectively generalize across attacks. To address this, we introduce multiple adversarial training methods for text scoring models and show that combining complementary training methods can yield strong robustness while also improving task effectiveness. We also highlight the practical value of our approach for RLHF, showing that our adversarially trained reward models mitigate reward hacking and support the training of better-aligned LLMs. We provide our code and models for further study.
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D-CORE: Incentivizing Task Decomposition in Large Reasoning Models for Complex T
ool Use
Bowen Xu, Shaoyu Wu, Hao Jiang, Kai Liu, Xin Chen, lulu hu, Bin Yang
Effective tool use and reasoning are essential capabilities for large reasoning
models (LRMs) to address complex real-world problems. Through empirical analysis
, we identify that current LRMs lack the capability of sub-task decomposition in
complex tool use scenarios, leading to Lazy Reasoning. To address this, we prop
ose a two-stage training framework D-CORE (Decomposing tasks and Composing Reaso
ning processes) that first incentivize the LRMs' task decomposition reasoning ca
pability via self-distillation, followed by diversity-aware reinforcement learni
ng (RL) to restore LRMs' reflective reasoning capability. D-CORE achieves robust
tool-use improvements across diverse benchmarks and model scales. Experiments o
n BFCLv3 demonstrate superiority of our method: D-CORE-8B reaches 77.7\% accurac
y, surpassing the best-performing 8B model by 5.7\%. Meanwhile, D-CORE-14B estab
lishes a new state-of-the-art at 79.3\%. Code is available at https://github.com
/alibaba/EfficientAI; the model is available at https://huggingface.co/bowiehsu/
D-CORE-8B; and the data are available at https://huggingface.co/datasets/bowiehs
u/D-CORE-Self-Distillation-50K.
****
Flat Minima and Generalization: Insights from Stochastic Convex Optimization
Matan Schliserman, Shira Vansover-Hager, Tomer Koren
Understanding the generalization behavior of learning algorithms is a central go
al of learning theory. A recently emerging explanation is that learning algorith
ms are successful in practice because they converge to flat minima, which have b
een consistently associated with improved generalization performance.
In this work, we study the link between flat minima and generalization in the ca
nonical setting of stochastic convex optimization with a non-negative, β-s
mooth objective.
Our first finding is that, even in this fundamental setting, flat empirical mini
ma may incur trivial $\Omega(1)$ population risk while sharp minima generalizes
optimally.
We then demonstrate that this phenomenon extends to sharpness-aware algorithms i
ntroduced by Foret et al. (2021), namely Sharpness-Aware Gradient Descent (SA-GD
) and Sharpness-Aware Minimization (SAM).
For SA-GD we prove that it successfully converges to a flat minimum at a fast ra
te, but the population risk of the solution can still be as large as $\Omega(1)$
.
For SAM we show that although it minimizes the empirical loss, it may converge t
o a sharp minimum and also incur population risk $\Omega(1)$.
Finally, we establish population risk upper bounds for both SA-GD and SAM using
algorithmic stability techniques.
****
LaRA-Fusion: Latent-Robust Adaptation via Dual-Loop Constraints for Infrared and
Visible Image Fusion
Yaru Su, Chaowei Huang, Huangbiao Xu, Xiao Ke
Infrared and visible image fusion (IVIF) aims to synergize complementary thermal
radiation and textural details for comprehensive scene perception. However, exi
sting unsupervised paradigms often overlook the intrinsic topological consistenc
y shared across modalities. Lacking explicit geometric regularization, encoders
frequently succumb to degenerate numerical shortcuts, capturing superficial high
-frequency noise rather than domain-invariant semantic structures to satisfy rec
onstruction objectives. To address this, we propose LaRA-Fusion, a framework ach
ieving Latent-Robust Adaptation via Dual-Loop Manifold Constraints. We construct
a strictly constrained latent space where an inner loop enforces geometric reve
rsibility, while an outer loop aligns the generated representations with the tar
get data distribution. This mechanism effectively mitigates latent space collaps
e, compelling the model to extract topologically aligned features that remain ro
bust against modality-specific variations. Extensive experiments demonstrate tha
t LaRA-Fusion outperforms state-of-the-art methods with superior robustness and
interpretability.

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Adaptive Volumetric Mechanical Property Fields Invariant to Resolution
Rishit Dagli, Donglai Xiang, Vismay Modi, Xuning Yang, Gavriel State, David Levi n I.W., Maria Shugrina
Accurate mechanical properties (or materials) Young's modulus (E), Poisson's r atio (ν) and density (ρ) are essential for reliable physics simulation of digital worlds, but most 3D assets lack this information. We propose AdaVoMP, a method for predicting accurate dense spatially-varying (E, ν, ρ) for i nput 3D objects across representations, improving the resolution, accuracy, and memory efficiency over the state-of-the-art. The foundation of our technique is a sparse and adaptive voxel structure SAV that efficiently represents both the i nput 3D shape and the material field output. We replace the fixed-voxel model of the most accurate prior method, VoMP, with a novel sparse transformer encoder-d ecoder model that learns to generate a unique SAV autoregressively for every inp ut shape to represent its materials, achieving a resolution $16^3\times$ higher than prior art. Experiments show that AdaVoMP estimates more accurate volumetric properties, even with lesser test-time compute than all prior art. This allows us to convert high-resolution complex 3D objects into simulation-ready assets, r esulting in realistic deformable simulations.
****
Quantum Algorithms for Triangle Cut Sparsification
Shan Jiang, Pan Peng
Triangles capture higher-order structures in graphs and are fundamental to appli cations such as clustering and network analysis.
To enable efficient use of such structures at scale, we study the problem of tri angle cut sparsification, which aims to reduce the graph size while approximatel y preserving triangle counts across every cut.
We investigate quantum algorithms for this problem, using triangle listing as ou r main technical ingredient.
In particular, we present a quantum algorithm for triangle listing that, for a g raph with n vertices, m edges, and t triangles, runs in time $T_{\mathrm{q \text{-}list}} =\widetilde{O}\left(\min(n^{5/4}t^{7/12} + n^{7/6}t^{7/9}, m + m^ {3/4}t^{1/2},n^{3/2}t^{1/2})\right)$, improving upon the best known classical bo unds over a broad range of parameters. Our algorithm is based on a heavy–light v ertex partition and an extension of triangle detection via quantum walks and Gro ver search.
Leveraging this result, we design a quantum algorithm for constructing $\varepsi lon$-triangle cut sparsifiers of size $\tilde{O}(n/\varepsilon^2)$ in time $\wi detilde{O}(T_{\mathrm{q\text{-}list}} + \sqrt{mn}/\varepsilon)$.
Finally, we demonstrate applications to clustering algorithms based on triangle-related measures
and prove a lower bound of $\Omega(n/\varepsilon^2)$ on the size of any $\vareps ilon$-triangle cut sparsifiers.
****
Faster Activation Functions at the Edge for Post-Training Speedups
Anton Lydike, Jun Bi, Jackson Woodruff
On-device AI has gained significant attention for enabling efficient, low-latenc y inference on edge devices.
However, tight resource constraints on these platforms make the deployment of ac curate and lightweight deep
learning models challenging.
In particular,
advanced activation functions (AFs) like Swish and GELU often incur
high inference overhead due to the lack of hardware fast-paths for exponentiatio n and division,
restricting edge-ML applications to simple AFs like ReLU, limiting model accurac y.
To address this, we propose FFCC, a compiler that automatically
generates efficient approximations of AFs through floating-point reinterpretatio n.

These functions do not require hardware fast-paths, meaning they remain fast
on edge devices, but are accurate enough to be used as post-training drop-ins.
FFCC takes a specification of AFs using basic floating-point operators
and applies derivation rules to lower these expressions into
efficient instruction sequences.
Our experiments show that FFCC provides fast approximations of AFs, achieving order-of-magnitude
speed-ups over accurate baselines on Arm M7, Aarch64 and Intel platforms.
Using ConvNeXt as an example, we demonstrate how these activation-level gains translate to end-to-end speed-ups,
and do not result in significant loss of model accuracy.
****
What Do Agents Learn from Trajectory-SFT: Semantics or Interfaces?
Weizheng Gu, Chengze Li, Zhuohao Yu, Mengyuan Sun, Zhibang Yang, Wei Wang, Hongrui Jia, Shikun Zhang, Wei Ye
Large language models are increasingly evaluated as interactive agents, yet standard agent benchmarks conflate two qualitatively distinct sources of success: semantic tool-use and interface-specific interaction pattern memorization.
Because both mechanisms can yield identical task success on the original interface, benchmark scores alone are not identifiable evidence of environment-invariant capability.
We propose **PIPE**, a protocol-level evaluation augmentation for diagnosing interface reliance by minimally rewriting environment interfaces while preserving task semantics and execution behavior.
Across 16 environments from AgentBench and AgentGym and a range of open-source and API-based agents, PIPE reveals that task-specific trajectory-SFT can amplify reliance on training-time interface forms: in several environments, agents with trajectory-SFT degrade sharply under minimal interface rewrites, whereas other agents are often more stable.
We further introduce Interface Reliance (IR), a counterbalanced alias-based metric that quantifies preference for training-time interfaces, and show that interface shortcutting exhibits environment-dependent, non-monotonic training dynamics that remain invisible under standard evaluation. Our code is available at https://github.com/ChengZe2005/What-Do-Agents-Learn-from-Trajectory-SFT-Semantics-or-Interfaces-.
****
Collaborative and Efficient Fine-tuning: Leveraging Task Similarity
Gagik Magakyan, Amirhossein Reisizadeh, Chanwoo Park, Pablo A. Parrilo, Asuman E. Ozdaglar
Adaptability has been regarded as a central feature in the foundation models, enabling them to effectively acclimate to unseen downstream tasks. Parameter-efficient fine-tuning methods such as celebrated LoRA facilitate efficient adaptation of large foundation models using labeled, high-quality and generally scarce task data. To mitigate data scarcity in fine-tuning of foundation models, we propose to leverage *task similarity* across downstream users. Intuitively, users with similar tasks must be able to assist each other in boosting the effective fine-tuning data size. We propose *Collaborative Low-Rank Adaptation*, or CoLoRA, which exploits task similarity to collaboratively and efficiently fine-tune personalized
foundation models. The main idea in CoLoRA is to train one shared adapter capturing underlying task similarities across all tasks, and personalized adapters tailored to user-specific tasks. We theoretically study CoLoRA on heterogeneous linear regression and provide provable guarantees for ground truth recovery.
We also conduct several natural language experiments with varying task similarity, which further demonstrate that when trained together with similar tasks, individual performances are significantly boosted.
****
Stability-Aware Feature Design for Robust Watermark Detection in Machine-Generated Text
Sina Mansouri, Mohit Marvania, Abolfazl Safikhani

The widespread adoption of large language models (LLMs) has intensified the dema
nd for principled methods to distinguish human- from machine-generated text. Wat
ermarking provides a promising avenue, yet existing detectors exhibit sharp perf
ormance deterioration under multiple paraphrasing and when applied to shorter te
xts. We introduce Pattern Stability Score (PSS), a novel detection framework tha
t leverages local statistical features and stability dynamics across paraphrased
 variants. Specifically, the proposed method combines global and local z-score f
eatures with higher-order statistics of run-length patterns, enriched by autocor
relation signals and stability scores computed over paraphrase depth. Numerical
evaluations are performed on three benchmark datasets (PG-19, CNN/DailyMail, and
 WikiText) using multiple LLMs (Llama-3-8B, Qwen2-7B) and paraphrasers (Mistral-
7B, Qwen2-7B, Gemma-7B), systematically stress-testing robustness under up to ei
ght rounds of paraphrasing. Compared to prior z-score thresholding baselines and
 some state-of-the-art deep learning methods, our approach improves detection AU
C (area under the receiver operating characteristic curve) by over 10-15 percent
age points across different token lengths. Additionally, extensive cross-domain
experiments demonstrate that a single universal classifier generalizes across di
fferent LLMs, paraphrasers, and text domains without retraining, maintaining abo
ve 83.7\% AUC even when all components differ from training.
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LUGS: Latent-aware Guidance for Efficient Unmasking in Diffusion Large Language
Models

Nuanqiao Shan, Kairong Han, Xinpeng Dong, Kun Kuang

Diffusion Language Models (DLMs) have emerged as a flexible alternative to autor
egressive (AR) models. They can decode tokens in any order, but the generation q
uality critically depends on the decoding strategy. Existing approaches predomin
antly rely on local heuristics, such as confidence or entropy, which may fail to
 capture sequence-level dependencies and the semantics in the context. To solve
this problem, we propose Latent-aware Unmasking Guidance Search (LUGS), a novel
decoding framework that leverages the model's internal hidden states to guide th
e unmasking process. By incorporating latent-aware scores to compensate for the
limitations of local heuristics such as confidence or entropy, LUGS improves the
 model's performance. Extensive experiments on various downstream tasks demonstr
ate that our approach consistently outperforms existing baselines on LLaDA-8B-in
struct and LLaDA-1.5 models. In Science and Reason tasks, LUGS improved performa
nce by more than 1\% on both base models. And LUGS obtains an average improvemen
t of 3.5\% in code generation. Remarkably, LUGS outperforms the beam search base
line by more than 5\% on average using LLaDA-8B-Instruct on code tasks. These r
esults highlight the potential of latent-aware guidance for advancing controllab
le and high-quality generation.
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Interpretability and Generalization Bounds for Learning Spatial Physics

Alejandro Francisco Queiruga, Theo Gutman-Solo, Shuai Jiang

While there are many applications of machine learning (ML) to scientific problem
s that \emph{look} promising, the eye test can be misleading compared to the qua
ntitative values.
Using numerical analysis techniques, we rigorously quantify the accuracy, conver
gence rates, and generalization bounds of certain ML models applied to linear di
fferential equations (DEs) for parameter discovery or solution finding.
Beyond the quantity and discretization of data, we identify that the {function s
pace} of the data is critical to the generalization of the model which can lead
to divergence.
Similar lack of generalization is empirically demonstrated for commonly used mod
els.
Surprisingly, we find that different classes of models can exhibit opposing gene
ralization behaviors.
Based on our theoretical analysis, we also introduce a new mechanistic interpret
ability lens on scientific models whereby Green's function representations can b
e extracted from the weights of black-box models.
Our results inform a new cross-validation technique for measuring generalization

in physical systems, and can be useful as a benchmark of future methods.
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LineageFlow: Flow Matching for High-Fidelity Family-Aware Protein Sequence Generation
Langzhang Liang, Ming Yang, Feng Yi, Junfan Li, Shirui Pan, Xu Yinghui, Tianlei Ying, YIZHEN ZHENG, Zenglin Xu
Protein sequence generation for engineering requires samples that are biophysically plausible and, when targeting a family/domain, remain recognizable members while exploring within-family diversity.
Current discrete generative models typically start from uniform or masked-token noise, which discards strong position-specific constraints induced by evolution and forces the model to reconstruct conserved residues from scratch, leading to weak family control and low foldability. We propose \emph{LineageFlow}, a simplex-valued flow-matching model that initializes generation from lineage priors derived from ancestral sequence reconstruction, turning generation into structured mutation from an evolved scaffold.
Across diverse protein families, LineageFlow achieves family validity close to held-out natural sequences and improves predicted structural confidence over uniform-/mask-initialized baselines while maintaining substantial within-family novelty and diversity, even surpassing a large pretrained baseline trained on substantially more data.
Finally, we introduce \emph{rerouting}, a single intermediate-time mutate--select--amplify intervention that enables objective-guided sampling without per-step predictor guidance and yields further gains in plausibility, including a zero-shot enzyme generation case study.
****
InftyThink+: Effective and Efficient Infinite-Horizon Reasoning via Reinforcement Learning
Yuchen Yan, Liang Jiang, Jin Jiang, Shuaicheng Li, zujie wen, Zhiqiang Zhang, JUN ZHOU, Jian Shao, Yueting Zhuang, Yongliang Shen
Large reasoning models achieve strong performance by scaling inference-time chain-of-thought, but this paradigm suffers from quadratic cost, context length limits, and degraded reasoning due to lost-in-the-middle effects. Iterative reasoning mitigates these issues by periodically summarizing intermediate thoughts, yet existing methods rely on supervised learning or fixed heuristics and fail to optimize when to summarize, what to preserve, and how to resume reasoning. We propose InftyThink+, an end-to-end reinforcement learning framework that optimizes the entire iterative reasoning trajectory, building on model-controlled iteration boundaries and explicit summarization. InftyThink+ adopts a two-stage training scheme with supervised cold-start followed by trajectory-level reinforcement learning, enabling the model to learn strategic summarization and continuation decisions. Experiments on DeepSeek-R1-Distill-Qwen-1.5B show that InftyThink+ improves accuracy by 21% on AIME24 and outperforms conventional long chain-of-thought reinforcement learning by a clear margin, while also generalizing better to out-of-distribution benchmarks. Moreover, InftyThink+ significantly reduces inference latency and accelerates reinforcement learning training, demonstrating improved reasoning efficiency alongside stronger performance.
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Native Parallel Reasoner: Reasoning in Parallelism via Self-Distilled Reinforcement Learning
Tong Wu, Yang Liu, Jun Bai, Zixia Jia, Shuyi Zhang, Ziyong Lin, Yanting Wang, Song-Chun Zhu, Zilong Zheng
We introduce **Native Parallel Reasoner (NPR)**, a teacher-free framework that enables Large Language Models (LLMs) to self-evolve genuine parallel reasoning capabilities. NPR transforms the model from sequential emulation to native parallel cognition through three key innovations: (1) a **self-distilled** progressive training paradigm that transitions from cold-start format discovery to strict topological constraints without external supervision; (2) a novel **Parallel-Aware Policy Optimization (PAPO)** algorithm that optimizes branching policies directly within the execution graph, allowing the model to learn adaptive decompositio

n via trial and error; and (3) a robust **NPR Engine** that refactors memory man
agement and flow control of SGLang to enable stable, large-scale parallel RL tra
ining. Across eight reasoning benchmarks, NPR trained on Qwen3-4B achieves perfo
rmance gains of up to 24.5\% and inference speedups up to 4.6$\times$. Unlike pr
ior baselines that often fall back to autoregressive decoding, NPR demonstrates
100\% genuine parallel execution, establishing a new standard for self-evolving,
 efficient, and scalable agentic reasoning.
****
How to Fine-Tune a Reasoning Model? A Teacher–Student Cooperation Framework to S
ynthesize Student-Consistent SFT Data
Zixian Huang, Kaichen Yang, Xu Huang, Feiyang Hao, Qiming Ge, Bowen Li, He Du, K
ai Chen, Qipeng Guo
A widely adopted strategy for model enhancement is to use synthetic data generat
ed by a stronger model for supervised fine-tuning (SFT). However, for emerging r
easoning models like Qwen3-8B, this approach often fails to improve reasoning ca
pabilities and can even lead to a substantial drop in performance. In this work,
 we identify substantial stylistic divergence between teacher generated data and
 the distribution of student as a major factor impacting SFT. To bridge this gap
, we propose a Teacher–Student Cooperation Data Synthesis framework (TESSY), whi
ch interleaves teacher and student models to alternately generate style and non-
style tokens. Consequently, TESSY produces synthetic sequences that inherit the
advanced reasoning capabilities of the teacher while maintaining stylistic consi
stency with the distribution of the student. In experiments on code generation u
sing GPT-OSS-120B as the teacher, fine-tuning Qwen3-8B on teacher-generated data
 leads to performance drops of 3.25% on LiveCodeBench-Pro and 10.02% on OJBench,
 whereas TESSY achieves improvements of 11.25% and 6.68%.
****
Hierarchical Successor Representation for Robust Transfer
Changmin Yu, Máté Lengyel
The successor representation (SR) provides a powerful framework for decoupling
  predictive dynamics from rewards, enabling rapid generalisation across reward
  configurations. However, the classical SR is limited by its inherent policy
  dependence: policies change due to ongoing learning, environmental
  non-stationarities, and changes in task demands, making established predictive
  representations obsolete. Furthermore, in topologically complex environments,
  SRs suffer from spectral diffusion, leading to dense and overlapping features
  that scale poorly. Here we propose the Hierarchical Successor Representation
  (HSR) for overcoming these limitations. By incorporating temporal abstractions
  into the construction of predictive representations, HSR learns stable state
  features which are robust to task-induced policy changes. Applying
  non-negative matrix factorisation (NMF) to the HSR yields a sparse, low-rank
  state representation that facilitates highly sample-efficient transfer to
  novel tasks in multi-compartmental environments. Further analysis reveals that
  HSR-NMF discovers interpretable topological structures, providing a
  policy-agnostic hierarchical map that effectively bridges model-free
  optimality and model-based flexibility. Beyond providing a useful basis for
  task-transfer, we show that HSR's temporally extended predictive structure can
  also be leveraged to drive efficient exploration, effectively scaling to
  large, procedurally generated environments.
****
DecodeShare: Tracing the Shared Subspace of LLM Decode-Time Decisions
Zishan Shao, Lixun Zhang, Kangning Cui, Yixiao Wang, Ting Jiang, Hancheng Ye, Qi
nsi Wang, Zhixu Du, Yuzhe Fu, Fan Yang, Danyang Zhuo, Yiran Chen, Hai Helen Li
Large language models (LLMs) handle many tasks with one set of parameters, but u
nder KV-cached inference it is unclear what task-general structure, if any, is u
sed at $\textit{decode time}$ rather than during $\textit{prefill}$. We propose
$\textbf{DecodeShare}$, a protocol that identifies a low-dimensional subspace co
nsistently shared across tasks in decode-time hidden states, and then tests its
causal role by removing that subspace only during decoding. In our experiments,
disturbing the discovered shared subspace degrades decision performance far more

than disturbing either a prefill-derived or random subspace under the same intervention budget. We further show this decode-shared subspace has practical consequences for activation steering: common steering directions can overlap the task-general decode channel. Projecting out this shared subspace directly separates the functional roles of the two components, while evaluating steering vectors at decode-time yields more reliable signal for downstream deployment than prefill-based proxies. Despite its compactness, the shared subspace can serve as a high-leverage causal channel at decode time. Code is available at: https://github.com/Zishan-Shao/decodeshare.git.

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Diagnosing Multi-step Reasoning Failures in Black-box LLMs via Stepwise Confidence Attribution

Xiaoou Liu, Tiejin Chen, Dengjia Zhang, Yaqing Wang, Lu Cheng, Hua Wei

Large Language Models have achieved strong performance on reasoning tasks with objective answers by generating step-by-step solutions, but diagnosing where a multi-step reasoning trace might fail remains difficult. Confidence estimation offers a natural diagnostic signal, yet existing methods are restricted to final answers or require internal model access.

We introduce Stepwise Confidence Attribution (SCA), a framework for closed-source LLMs that assigns step-level confidence based only on generated reasoning traces. SCA applies the Information Bottleneck principle: steps aligning with consensus structures across correct solutions receive high confidence, while deviations are flagged as potentially erroneous.

We propose two complementary methods: (1) NIBS, a non-parametric IB approach measuring consistency without graph structures, and (2) GIBS, a graph-based IB model that learns subgraphs through a differentiable mask to capture logical variability. Extensive experiments on mathematical reasoning and multi-hop question answering show that SCA reliably identifies low-confidence steps strongly correlated with reasoning errors. Moreover, using step-level confidence to guide self-correction improves the correction success rate by up to 13.5\% over answer-level feedback.

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Multi-Objective Bayesian Optimization via Adaptive ε-Constraint Decomposition

Yaohong Yang, Sammie Katt, Samuel Kaski

Multi-objective Bayesian optimization (MOBO) provides a principled framework for optimizing multiple expensive black-box functions. However, existing MOBO methods often struggle with coverage, scalability, and handling constraints and preferences. In this work we propose *STAGE-BO, Sequential Targeting Adaptive Gap-Filling ε-Constraint Bayesian Optimization*: by analyzing the coverage of the surrogate Pareto front, our method identifies the Pareto front point with the largest uncovered gap, and uses its coordinates to define adaptive constraints in ε-constraint method, which transforms the problem into a sequence of inequality-constrained subproblems, efficiently solved via constrained expected improvement acquisition. Our approach provides uniform Pareto coverage without hypervolume computation and naturally handles constraints and preferences. Experiments on synthetic and real-world benchmarks demonstrate superior coverage and competitive hypervolume performance against state-of-the-art baselines. Our code implementation can be found at https://github.com/YangYaohong1/STAGE-BO.

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Beyond Temperature: Hyperfitting as a Late-Stage Geometric Expansion

Meimingwei Li, Yuanhao Ding, Esteban Garces Arias, Christian Heumann

Recent work has identified a counterintuitive phenomenon termed “Hyperfitting", where fine-tuning Large Language Models (LLMs) to near-zero training loss on small datasets surprisingly enhances open-ended generation quality and mitigates repetition in greedy decoding. While effective, the underlying mechanism remains poorly understood, with the extremely low-entropy output distributions suggesting a potential equivalence to simple temperature scaling. In this work, we demonstrate that this phenomenon is fundamentally distinct from distribution sharpening; entropy-matched control experiments reveal that temperature scaling fails to r

eplicate the diversity gains of hyperfitting. Furthermore, we falsify the hypoth esis of static vocabulary reweighting, showing through ablation studies that hyp erfitting relies on a dynamic, context-dependent rank reordering mechanism. Laye r-wise analysis localizes this effect to a “Terminal Expansion" in the final tra nsformer block, where a substantial geometric expansion of the feature space ($\ Delta \mathrm{Dim} \approx +80.8$) facilitates the promotion of deep-tail tokens . Additionally, we introduce \textbf{Late-Stage LoRA}, a targeted fine-tuning st rategy that updates only the final 5 layers, yielding robust generation with min imal parameter updates.
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Thinned Mean Field Langevin Dynamics
Zonghao Chen, Heishiro Kanagawa, Francois-Xavier Briol, Chris J. Oates, Lester M ackey
Several important learning tasks can be formulated as minimizing an entropy-regu larized objective over an appropriate space of probability distributions. Mean-f ield Langevin dynamics (MFLD) facilitate computation in this general context, ca sting the minimizer as the invariant distribution of a McKean--Vlasov process, w hich can be numerically discretized using N particles and thus simulated. However, simulating this interacting particle system has computational complexit y $\mathcal{O}(N^2)$. Motivated by recent research into \emph{kernel thinning}, we propose \texttt{KT-MFLD}, in which each particle interacts only with a corese t of size $\mathcal{O}(N^{\frac{1}{2}})$. \texttt{KT-MFLD} thus reduces the comp utational complexity to $\mathcal{O}(N^{\frac{3}{2}})$ while, under mild regular ity conditions, achieving the same convergence guarantees (up to logarithmic fac tors) as MFLD. Our theoretical analysis is empirically confirmed on tasks includ ing the training of student-teacher neural networks, quantization with maximum m ean discrepancy, and computation of predictively-oriented posteriors in a post-B ayesian framework.
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Rotary Position Encodings for Graphs
Isaac Reid, Arijit Sehanobish, Cederik Höfs, Bruno Kacper Mlodozeniec, Leonhard Vulpius, Federico Barbero, Adrian Weller, Krzysztof Marcin Choromanski, Richard E. Turner, Petar Veli■kovi■
We study the extent to which rotary position encodings (RoPE), a recent transfor mer position encoding algorithm broadly adopted in large language models (LLMs) and vision transformers (ViTs), can be applied to graph-structured data. We find that rotating tokens depending on the spectrum of the graph Laplacian efficient ly injects structural information into the attention mechanism, boosting perform ance in synthetic and real-world graph learning tasks. This approach, coined _Wa ve-Induced Rotary Encodings_ (WIRE), enjoys intriguing theoretical properties: i t recovers regular RoPE on grids, and depends asymptotically on the graph effect ive resistance. Unlike bias-based relative position encodings, WIRE is compatibl e with linear attention.
****

SeisMark: A Large-Scale Open Benchmark for Robust 3D Seismic Fault Detection
Min Jun Park, Joseph Stitt, Robert Graham Clapp, Ilan Naiman, Artem Goncharuk, K evin F. Smith
We introduce SeisMark, a large-scale open benchmark designed to bridge the gap b etween verifiable ground truth and realistic texture in 3D seismic fault detecti on. Using a novel pipeline merging procedural geology with diffusion-based synth esis, we produce domain-realistic (survey-specific) textured volumes that expose significant brittleness in existing models masked by simplified physics data. E xperiments demonstrate that SeisMark acts as a rigorous discriminator, distingui shing robust modern architecture from legacy model that suffers performance coll apse under realistic domain shifts. We release this benchmark to the community t o serve as a verifiable standard for developing trustworthy, deployment-ready AI for safety-critical subsurface applications.
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Less Is More: Fast and Accurate Reasoning with Cross-Head Unified Sparse Attenti on

Lijie Yang, Zhihao Zhang, Arti Jain, Shijie Cao, Baihong Yuan, Yiwei Chen, Zhihao Jia, Ravi Netravali

Large reasoning models achieve strong performance through test-time scaling, but this incurs substantial computational overhead due to long decoding from short prompts. While sparse attention can reduce latency and memory usage, existing methods often degrade reasoning accuracy because selection errors accumulate over long generation horizons, or require costly retraining. We introduce LessIsMore, a training-free sparse attention mechanism for long-horizon reasoning. Our key insight is that token importance in reasoning is global and stable: critical tokens are largely shared across attention heads and remain stable over decoding steps. Guided by this structure, LessIsMore enforces cross-head unified token selection and preserves recent context via a stable recency window, yielding a globally consistent token set that can be reused across layers. Across multiple model families and challenging reasoning benchmarks, LessIsMore matches or improves accuracy while attending to substantially fewer tokens. With kernel-level optimizations, LessIsMore achieves up to $1.6\times$ end-to-end decoding speedup and up to $1.72\times$ faster sparse attention computation, with additional long-context results demonstrating the generality of our approach.

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Positional Encoding for Spiking Transformers

Zijian Zhou, Yu Liang, Honglin Cao, Ammar Belatreche, Jieyuan Zhang, Wenjie Wei, Shuai Wang, Malu Zhang, Yang Yang, Haizhou Li

Transformer-based Spiking Neural Networks (SNNs) have recently emerged as a promising paradigm to sequential modeling, combining the strong representational capabilities of Transformers with the sparse spike-driven computation of SNNs. Within such position-agnostic architectures, positional encoding is critical for injecting order information, allowing the model to distinguish token positions, capture sequential dependencies, and represent relative relationships among tokens. However, existing positional encoding methods for SNNs are largely inherited from ANNs and, in doing so, undermine the spike-driven computational properties that are central to spiking transformers.

To address this limitation, we propose the Spiking Positional Encoding (SPE), a method designed specifically for Spiking Transformers, aimed at encoding relative positional information while preserving both spike-driven computation and the linear complexity of spiking self-attention.

The core component of SPE is the Positional Encoding Leaky Integrate-and-Fire (PE-LIF) neuron, which incorporates position-dependent signals into neuronal thresholds and implicitly propagates this information through spike trains via continuous firing and membrane potential reset dynamics. Extensive experiments on thirteen NLP benchmarks demonstrate that SPE consistently outperforms existing SNN positional encoding methods, strengthens the sequence modeling capability, and improves energy efficiency without introducing additional trainable parameters.

Code is available at https://github.com/CayleyZ/SPE.

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AtomWorld: A Benchmark for Evaluating Spatial Reasoning in Large Language Models on Material Structures

Taoyuze Lv, Alexander Chen, Fengyu Xie, Chu Wu, Jeffrey Meng, Dongzhan Zhou, Bram Hoex, Yingheng Wang, Zhicheng Zhong, Tong Xie

Large language models (LLMs) have shown promising potential in materials science, enabling tasks ranging from knowledge retrieval to property prediction. Existing materials science benchmarks mainly focus on perceptual or knowledge-based tasks, largely ignoring the structure modelling tasks, a core challenge in real scientific workflows. In practice, constructing and manipulating atomic structures is one of the most creative and least automated steps in materials research. In this work, we introduce AtomWorld, a benchmark designed to evaluate the abilities of LLMs on structure modifications. The benchmark includes ten fundamental actions under four widely used modelling categories, enabling verifiable evaluation metrics. We find that Gemini 2.5 Pro generally performs the best. While the success rate decreases markedly with increasing modelling complexity, with particularly low success rates (below 12\% for rotation) for operations involving compl

ex spatial relations. Our results suggest that contemporary LLMs are better suited as copilots for materials structure modelling rather than fully unsupervised autonomous scientific agents. Beyond evaluation, AtomWorld also serves as a test bed and playground for developing future structure-aware models, including reinforcement learning and agentic approaches.
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Spurious Rewards: Rethinking Training Signals in RLVR
Rulin Shao, Shuyue Stella Li, Rui Xin, Scott Geng, Yiping Wang, Sewoong Oh, Simon Shaolei Du, Nathan Lambert, Sewon Min, Ranjay Krishna, Yulia Tsvetkov, Hannaneh Hajishirzi, Pang Wei Koh, Luke Zettlemoyer
We show that reinforcement learning with verifiable rewards (RLVR) can elicit strong mathematical reasoning in certain language models even with spurious rewards that have little, no, or outright negative correlation with the correct answer. For example, RLVR training with GRPO improves MATH-500 performance for Qwen2.5-Math-7B in absolute points by 21.4% using randomly assigned rewards, nearly matching the 29.1% gained with ground truth rewards. To explain this counterintuitive observation, we show that GRPO exhibits a clipping bias arising from the clip term, which can amplify high-prior behaviors learned during pre-training even without informative rewards. As a case study, we identify one such high-prior behavior for Qwen2.5-Math models, which we term code reasoning---reasoning in code without actual code execution; code reasoning frequency increases from 65% to over 90% with spurious rewards. However, the presence of such amplifiable behaviors is highly model-dependent. In practice, spurious rewards that are effective for Qwen models often fail to produce gains for other model families, such as Llama3 or OLMo2. Our results highlight the importance of validating RL methods across diverse models rather than relying on a single de facto choice: large performance gains can arise on Qwen models even from random rewards that do not reflect genuine capability improvements.
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Continual Segmentation under Joint Nonstationarity
Prashant Pandey, Himanshu Kumar, Devineni Sri Venkatraya Chowdary, Brejesh Lall
Evolving data streams induce joint nonstationarity in continual semantic segmentation, where semantic classes, input distributions, and supervision availability change simultaneously over time. This setting reflects practical structured prediction systems, yet remains largely unexplored in prior continual learning work, which typically studies these factors in isolation. We formalize continual segmentation under coupled class, domain, and label shifts and investigate learning in heterogeneous dense prediction environments with limited annotations and abundant unlabeled data. To address instability and overfitting arising from few-shot supervision under distribution drift, we introduce gradient-adaptive stabilization, a parameter-wise regularization mechanism implemented via gradient-scaled stochastic perturbations that promotes a principled stability–plasticity tradeoff. We further leverage unlabeled data through semi-supervised learning and introduce prototype anchored supervision that validates pseudo-labels via joint confidence and prototype consistency. Together, these mechanisms enable learning under joint nonstationarity in continual segmentation. Extensive empirical evaluation across class-incremental, domain-incremental, and few-shot regimes demonstrates consistent improvements over prior methods in heterogeneous structured prediction settings. Our results expose fundamental failure modes of existing continual segmentation approaches and provide insight into learning robust dense predictors in dynamically evolving environments. Our code is available at https://github.com/prinshul/JASCL.git.
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Feasible Fusion: Constrained Joint Estimation under Structural Non-Overlap
Yuxi Du, Zhiheng Zhang, Haoxuan Li, Cong Fang, Jixing Xu, Zhen Peng, Jiecheng Guo
Causal inference in modern large-scale systems faces growing challenges, including high-dimensional covariates, multi-valued treatments, massive observational (OBS) data, and limited randomized controlled trial (RCT) samples due to cost constraints. We formalize treatment-induced structural non-overlap and show that, u

nder this regime, commonly used weighted fusion methods provably fail to satisfy randomized identifying restrictions.To address this issue,we propose a constrained joint estimation framework that minimizes observational risk while enforcing causal validity through orthogonal experimental moment conditions. We further show that structural non-overlap creates a feasibility obstruction for moment enforcement in the original covariate space.We also derive a penalized primal–dual algorithm that jointly learns representations and predictors, and establish oracle inequalities decomposing error into overlap recovery, moment violation, and statistical terms.Extensive synthetic experiments demonstrate robust performance under varying degrees of non-overlap. A large-scale ride-hailing application shows that our method achieves substantial gains over existing baselines, matching the performance of models trained with significantly more RCT data.

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FormalRx: Rectify and eXamine Semantic Failures in Autoformalization

Haocheng Wang, Baiyu Huang, Yingjia Wan, Xiao Zhu, Xiaoyang Liu, Yinya Huang, Zhijiang Guo

The veracious semantic alignment in autoformalization is significant for formal mathematical reasoning. However, existing evaluations provide only opaque binary verdicts or scalar scores, offering no interpretable insight into where or why translations fail. This opacity severely limits both human understanding and automated system improvement. To bridge this gap, we introduce FormalRx, a comprehensive diagnostic evaluation framework that transforms autoformalization assessment from black-box judgments into actionable feedback. At its core is SCI Error Taxonomy, a hierarchical classification scheme decomposing autoformalization errors into 28 distinct categories with strict priority ordering. Building on this taxonomy, FormalRx provides four critical diagnostic capabilities: alignment verdicts, error categorization, error localization, and correction. We instantiate the framework with a diagnostic model FormalRx-8B, trained on 56,287 synthetically generated samples with fine-grained diagnostic annotations, and release FormalRx-Test as the first fine-grained diagnostic benchmark. FormalRx-8B achieves F1-scores of 0.88 (verdict) and 0.71 (categorization), along with accuracies of 0.75 (localization) and 0.73 (correction), substantially outperforming both general-purpose LLMs and specialized baselines. By connecting evaluation with actionable insights, FormalRx enables systematic diagnosis and improvement of autoformalization systems.

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FormalJudge: A Neuro-Symbolic Paradigm for Agentic Oversight

Jiayi Zhou, Yang Sheng, Hantao Lou, Yaodong Yang, Jie Fu

As LLM-based agents increasingly operate in high-stakes domains with real-world consequences, ensuring their behavioral safety becomes paramount. The dominant oversight paradigm, *LLM-as-a-Judge*, faces a fundamental dilemma: how can probabilistic systems reliably supervise other probabilistic systems without inheriting their failure modes? We argue that formal verification offers a principled escape from this dilemma, yet its adoption has been hindered by a critical bottleneck: the translation from natural language requirements to formal specifications. This paper bridges this gap by proposing , a neuro-symbolic framework that employs a bidirectional **Formal-of-Thought** architecture: LLMs serve as *specification compilers* that top-down decompose high-level human intent into atomic, verifiable constraints, then bottom-up prove compliance using Dafny specifications and Z3 Satisfiability modulo theories solving, which produces mathematical guarantees rather than probabilistic scores. We validate across three benchmarks spanning behavioral safety, multi-domain constraint adherence, and agentic upward deception detection. Experiments on 7 agent models demonstrate that achieves an average improvement of **16.6%** over LLM-as-a-Judge baselines, enables *weak-to-strong* generalization where a 7B judge achieves over 90% accuracy detecting deception from 72B agents, and provides *near-linear safety improvement* through iterative refinement.

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Muon in Associative Memory Learning: Training Dynamics and Scaling Laws

Kaifei Wang, Binghui Li, Han Zhong, Pinyan Lu, Liwei Wang

Muon updates matrix parameters via the matrix sign of the gradient and has shown strong empirical gains, yet its dynamics and scaling behavior remain unclear in theory. We study Muon in a linear associative memory model with softmax retrieval and a hierarchical frequency spectrum over query–answer pairs, with and without label noise.
In this setting, we show that Gradient Descent (GD) learns frequency components at highly imbalanced rates, leading to slow convergence bottlenecked by low-frequency components. In contrast, the Muon optimizer mitigates this imbalance, leading to faster and more uniform progress.
Specifically, in the noiseless case, Muon achieves an exponential speedup over GD; in the noisy case with a power-law frequency spectrum, we derive Muon's scaling law and demonstrate its superior scaling efficiency over GD. Furthermore, we show that Muon can be interpreted as an implicit matrix preconditioner arising from adaptive task alignment and block-symmetric gradient structure. In contrast, the preconditioner with coordinate-wise sign operator could match Muon under oracle access to unknown task representations, which is infeasible for SignGD in practice. Experiments on synthetic long-tail classification and LLaMA-style pre-training corroborate the theory.
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Beyond Rational Illusion: Behaviorally Realistic Strategic Classification
Xinpeng Lv, Yunxin Mao, Renzhe Xu, Chunyuan Zheng, Yikai Chen, Haoxuan Li, Yang Shi, Jinxuan Yang, Yuanlong Chen, Yuanxing Zhang, Shaowu Yang, Wenjing Yang, Zhouchen Lin, Haotian Wang
Strategic classification studies the interaction between decision models and agents who strategically manipulate their features for favorable outcomes.
Existing SC frameworks typically rely on the idealized assumption that agents are strictly rational.
However, evidence from behavioral economics and psychology consistently shows that real-world decision-making is often shaped by cognitive biases, deviating from pure rationality.
To formalize this limitation, we identify and define a new problem setting, termed the behaviorally realistic strategic classification problem, where agents’ strategic manipulations deviate from full rationality due to psychological biases.
Motivated by the identified limitation, we propose the Prospect-Guided Strategic Framework (Pro-SF) to address the problem, a principled framework grounded in prospect theory to model and learn under behaviorally realistic strategic responses.
Specifically, to capture behaviorally realistic strategic manipulations, our framework reformulates the Stackelberg-style interaction between agents and the decision-maker by incorporating three key mechanisms inspired by prospect theory, including the asymmetry between benefits and costs, different subjective reference points, and non-rational probability distortion. Experiments on synthetic and real-world datasets establish Pro-SF as a behaviorally grounded approach to strategic classification, bridging machine learning and behavioral economics for more reliable deployment in the real world.
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Multilingual Safety Alignment Via Sparse Weight Editing
Jiaming Liang, Zhaoxin Wang, Handing Wang
Large Language Models (LLMs) exhibit significant safety disparities across languages, with low-resource languages (LRLs) often bypassing safety guardrails established for high-resource languages (HRLs) like English. Existing solutions, such as multilingual supervised fine-tuning (SFT) or Reinforcement Learning from Human Feedback (RLHF), are computationally expensive and de-
pendent on scarce multilingual safety data. In this paper, we propose a novel, training-free alignment framework based on Sparse Weight Editing. Identifying that safety capabilities are localized within a sparse set of ”safety neurons,” we formulate the cross-lingual alignment problem as a constrained linear transformation. We derive a closed-form solution to optimally map the harmful representations of LRLs to the robust safety subspaces of HRLs, while preserving general utility via a null-space projection constraint. Extensive experiments across 8 lang

uages and multiple model families (Llama-3, Qwen-2.5) demonstrate that our metho d significantly reduces Attack Success Rate (ASR) in LRLs with negligible impact on general reasoning capabilities, all achieved with a single, data-efficient c alculation.

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Rethinking the Hardness of PbRL: A Provable General Regret Bound

Chenjie Mao, Yi Fan, Ning Zhang, Chongjie Zhang

This paper studies \emph{preference-based reinforcement learning} (PbRL), where agents learn from comparative, trajectory-level feedback rather than numeric rew ards. While PbRL has seen rapid empirical and theoretical progress, existing ana lyses are largely confined to restricted settings and fail to jointly capture th e outcome-based and comparison-based nature of preference feedback.

We prove that under a broad \emph{general function approximation} framework, PbR L admits a $\sqrt{T}$ regret guarantee. In particular, we introduce a simple and provably efficient algorithm, \emph{Recursive Trajectory-based Preference Q-Lea rning} (RTPQ), and establish its regret bound while explicitly accounting for th e trajectory-level and comparative structure of preferences.

Our analysis is characterized by a new complexity measure, the \emph{Dual Episod ic Eluder Dimension} (DEED), which quantifies the intrinsic difficulty of PbRL. We show that for linear MDPs, the DEED scales as $\mathcal{O}(dH)$, yielding a r egret bound of $\tilde{\mathcal{O}}(dH\sqrt{T}\max(H^{3/2},\,1/\kappa))$, where κ is a problem-dependent constant. This bound is near-optimal up to horiz on- and problem-dependent factors when compared to standard reward-based linear MDPs.

In addition, our framework recovers the best-known regret bounds in the special cases of dueling bandits and standard outcome-based reinforcement learning. Over all, our results provide a general regret guarantee for PbRL with outcome-based preference feedback and broad function approximation.

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From Diagrams to Code: Multilingual Programming with Visual Design

Linzheng Chai, Jian Yang, Shukai Liu, Wei Zhang, Liran WANG, Ke Jin, Tao Sun, Co ngnan Liu, Chenchen Zhang, Hualei Zhu, Jiaheng Liu, Xianjie Wu, Ge Zhang, Tianyu Liu, Zhoujun Li

In modern software development, particularly in emerging ``vibe coding'' paradig ms, project implementation increasingly begins with visual interactions between users and AI coding assistants, where system architectures are communicated thro ugh visual designs before coding. This visual-first approach necessitates AI sys tems capable of interpreting diagrams across multiple programming languages. How ever, the development of such systems is severely hindered by the lack of large- scale multimodal training data and evaluation benchmarks. To address these limit ations, we present M²C-INSTRUCT, a comprehensive multilingual multimodal instruc tion-tuning dataset containing over 13.1M samples across 50+ programming languag es, designed for visual understanding and diagram interpretation in code generat ion tasks. We validate our dataset by training M²-CODER, a multilingual multimod al software developer that successfully integrates visual design inputs with tex tual instructions. We also introduce M²EVAL, a novel multilingual evaluation ben chmark for multimodal code generation performance. Experiments show our 7B M²-CO DER, performs on par with much larger 70B+ models, confirming the quality and ef fectiveness of our M²C-INSTRUCT. Together, M²C-INSTRUCT, M²-CODER, and M²EVAL pr ovide essential infrastructure for visual-assisted programming in vibe-coding an d visual-interactive development workflows.

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AmbiRefer3D: 3D Visual Grounding with Referential Ambiguity

Rongjiang Zhu, Wei Kang, Zeqi Liu, Chen junyu, Shuo Yang, Xinxiao Wu

Traditional 3D visual grounding typically assumes that natural language express ions unambiguously refer to
target objects in a 3D scene.

However, in practical applications, human instructions are often ambiguous or in sufficient, which may lead existing models to associate the query with multiple possible objects, resulting in incorrect results.

In this paper, we propose a new task, 3D visual grounding with referential ambiguity, which allows for referential ambiguity in language descriptions, making it more broadly applicable to real-world scenarios.
To tackle this task, we propose an interactive grounding framework that performs multi-round question-answer interactions, in which the model actively generates clarifying questions and receives human-provided answers to acquire additional object attributes, spatial relationships, and other contextual information, thereby resolving referential ambiguity and achieving accurate grounding.
To support the learning of interactive grounding,
we construct a large-scale dataset named AmbiRefer3D, which contains 47,085 samples with 141,255 annotations of
question-answer dialogues that capture interactive disambiguation processes, covering 7,316 indoor 3D scenes.
Furthermore, we establish multi-round evaluation metrics to measure both disambiguation efficiency and grounding accuracy.
The code is available at *https://github.com/yearnallover/AmbiRefer3D*.
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A New Framework for Cybersecurity Refusals in AI Agents
Eliot Krzysztof Jones, Matt Fredrikson, J Zico Kolter
Agentic scaffolds have dramatically improved LLM performance on complex, long-horizon tasks, yielding both broad benefits and amplified risks in domains like cybersecurity. Existing benchmarks for AI agents in cybersecurity focus mainly on measuring proficiency–how effectively agents can complete offensive security tasks–but neglect a critical question: when and how should agents refuse harmful requests? We present the first framework for establishing refusal boundaries in offensive security contexts. Our framework defines (1) principled criteria for when tasks should be refused, (2) categories of tasks that warrant refusal, and (3) evaluation methodology for measuring agent robustness under both benign and adversarial conditions. We apply this framework to assess how current LLM-powered agents adhere to appropriate refusal boundaries across a range of web-based offensive security scenarios, finding that 6 of 8 frontier models tested refuse to identify vulnerabilities in real systems 0\% of the time.
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VERA-V: Variational Inference Framework for Jailbreaking Vision-Language Models
Qilin Liao, Anamika Lochab, Ruqi Zhang
Vision-Language Models (VLMs) extend large language models with visual reasoning, but their multimodal design also introduces new, underexplored vulnerabilities. Existing multimodal red-teaming methods largely rely on brittle templates, focus on single-attack settings, and expose only a narrow subset of vulnerabilities. To address these limitations, we introduce VERA-V, a variational inference framework that recasts multimodal jailbreak discovery as learning a joint posterior distribution over paired text-image prompts. This probabilistic view captures complex cross-modal interactions, enabling stealthy, coordinated adversarial inputs that bypass model guardrails. We train a lightweight attacker to approximate the posterior, allowing efficient sampling of diverse jailbreaks and providing distributional insights into vulnerabilities. VERA-V further integrates three complementary strategies: (i) typography-based text prompts that embed harmful cues, (ii) diffusion-based image synthesis that introduces adversarial signals, and (iii) structured distractors to fragment VLM attention. Experiments on HarmBench and HADES benchmarks show that VERA-V consistently outperforms state-of-the-art baselines on both open-source and frontier VLMs, achieving up to 53.75\% higher attack success rate (ASR) over the best baseline on GPT-4o. We include the code on the project page available here: https://github.com/kxwhiowo/VERA-V
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Optimal Transport under Group Fairness Constraints
Linus Bleistein, Mathieu Dagréou, Francisco Andrade, Thomas Boudou, Aurélien Bellet
Ensuring fairness in matching algorithms is a key challenge in allocating scarce resources and positions. Focusing on Optimal Transport (OT), we introduce a novel notion of group fairness requiring that the probability of matching two indiv

iduals from any two given groups in the OT plan satisfies a predefined target. We first propose a modified Sinkhorn algorithm to compute perfectly fair transport plans efficiently. Since exact fairness can significantly degrade matching quality in practice, we then develop two relaxation strategies. The first one involves solving a penalized OT problem, for which we derive novel finite-sample complexity guarantees. Our second strategy leverages bilevel optimization to learn a ground cost that induces a fair OT solution, and we establish a bound on the deviation of fairness when matching unseen data. Finally, we present empirical results illustrating the performance of our approaches and the trade-off between fairness and transport cost.

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Error Analysis of Discrete Flow with Generator Matching

Zhengyan Wan, Yidong Ouyang, Qiang Yao, Liyan Xie, Fang Fang, Hongyuan Zha, Guang Cheng

Discrete flow models offer a powerful framework for learning distributions over discrete state spaces and have demonstrated superior performance compared to the discrete diffusion models. However, their convergence properties and error analysis remain largely unexplored. In this work, we develop a unified framework grounded in stochastic calculus theory to systematically investigate the theoretical properties of discrete flow models. Specifically, by leveraging a Girsanov-type theorem for the path measures of two continuous-time Markov chains (CTMCs), we present a comprehensive error analysis that accounts for both transition rate estimation error and early stopping error. In fact, the estimation error of transition rates has received little attention in existing works. Unlike discrete diffusion models, discrete flow incurs no initialization error caused by truncating the time horizon in the noising process. Building on generator matching and uniformization, we establish non-asymptotic error bounds for distribution estimation without the boundedness condition on oracle transition rates. Furthermore, we derive a faster rate of total variation convergence for the estimated distribution with the boundedness condition, yielding a nearly optimal rate in terms of sample size. Our results provide the first error analysis for discrete flow models. We also investigate model performance under different settings based on simulation results.

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Continuous Variable Hamiltonian Learning at Heisenberg Limit via Displacement-Random Unitary Transformation

Xi Huang, Lixing Zhang, Di Luo

Characterizing continuous-variable (CV) Hamiltonians can be formulated as Hamiltonian learning under quantum measurement constraints: finite operator coefficients are inferred from noisy measurement outcomes obtained by probing an infinite-dimensional system. Existing Heisenberg-limited CV protocols are often limited to low-order structures, vulnerable to noise, or unresolved for generic multi-mode settings. We introduce Displacement-Random Unitary Transformation (D-RUT), an active data acquisition protocol with pre-specified probes and number-preserving transformations that reduce finite-order bosonic Hamiltonian learning to polynomial recovery. We prove Heisenberg-limited total evolution time with robustness to state preparation and measurement (SPAM) errors, and develop hierarchical multi-mode coefficient recovery with better statistical efficiency than simultaneous estimation. We also extend D-RUT to first-quantized Hamiltonian coefficient learning, and numerical experiments on single- and multi-mode nonlinear systems validate the predicted Heisenberg scaling.

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TsLLM: Augmenting LLMs for General Time Series Understanding and Prediction

Felix Parker, Nimeesha Chan, Chi Zhang, Kimia Ghobadi

Time series data is fundamental to decision-making across many domains including healthcare, finance, power systems, and logistics. However, analyzing this data correctly often requires incorporating unstructured contextual information, answering domain-specific questions, and generating natural language explanations – capabilities that traditional time series models lack. While Large Language Models (LLMs) excel at contextual reasoning and knowledge integration, they struggl

e with numerical time series due to inefficient text-based representations and l
imited exposure to numerical data during pretraining. We address this gap by aug
menting an LLM with specialized time series perception through a patch-based enc
oder-decoder architecture. We train this time series-augmented LLM (TsLLM) on a
large corpus of over 25 billion tokens of interleaved time series and text spann
ing diverse tasks: forecasting with contextual information, question-answering,
anomaly detection, classification, report generation, and more, all unified as a
utoregressive next token prediction. This training enables TsLLM to leverage bot
h its natural language skills and newly acquired understanding of numerical time
 series signals. While not designed to surpass specialized models on traditional
 benchmarks, TsLLM demonstrates strong performance on tasks requiring the integr
ation of time series analysis with natural language – capabilities that conventi
onal approaches cannot provide. It also exhibits strong zero-shot and few-shot p
erformance, showing it can adapt to new data without additional training.
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Towards Generative Graph Matching for Graph Edit Distance Computation
Wei Huang, Hanchen Wang, Dong Wen, Wenjie Zhang, Ying Zhang, Xuemin Lin
Graph Edit Distance (GED), which aims to find an edit path with minimum number o
f edit operations to transform one graph into another, is a fundamental NP-hard
problem and a widely used graph similarity measure. Recent matching-based hybrid
 approaches have demonstrated better scalability than A* search-based hybrids by
 reformulating GED as a graph matching problem. In these methods, a neural netwo
rk predicts a single deterministic node matching matrix, from which top-k node
 mappings are extracted iteratively to derive candidate edit paths. However, the
se methods often suffer from highly correlated candidates that easily lead to su
boptimal solutions, while the iterative extraction becomes inefficient for large
 k. In this paper, we propose DiffGED, the first generative approach for GED c
omputation. Specifically, we formulate the graph matching problem as a generativ
e task, and employ a diffusion-based model to generate multiple diverse node mat
ching matrices simultaneously, from which diverse node mappings can be efficient
ly extracted. The generative diversity introduced by the diffusion process enabl
es DiffGED to avoid suboptimal solutions and achieve superior solution quality c
lose to the exact solution. Experiments on real-world datasets show that DiffGED
 generates multiple diverse edit paths with accuracy comparable to exact solutio
ns, while running faster than existing hybrid approaches.
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Uncovering Competency Gaps in Large Language Models and Their Benchmarks
Maty Bohacek, Nino Scherrer, Nicholas Dufour, Thomas Leung, Christoph Bregler, S
tephanie C.Y. Chan
The evaluation of large language models relies heavily on standardized benchmark
s. These benchmarks provide useful aggregated metrics, but can obscure (i) parti
cular sub-areas where the models are weak ("model gaps") and (ii) imbalanced cov
erage in the benchmarks themselves ("benchmark gaps"). To automatically uncover
both types of gaps, we propose a simple new method using concept activations fro
m sparse autoencoders, to identify fine-grained gaps on a per-concept basis. The
 method also benefits from grounding evaluation in the model's internal represen
tations, as well as easy comparison across benchmarks. We applied the method to
five popular open-source models and more than a dozen benchmarks, as illustrativ
e examples. As validation of the approach, we found that our automatic, unsuperv
ised method was able to recover model gaps that have been previously documented
in the literature (e.g. relating to sycophancy), in addition to identifying nove
l model gaps. We were also able to automatically uncover benchmark gaps: core co
ncepts that should fall within the scope of a given benchmark. Our "competency g
aps" method can be used to complement existing benchmarks, by providing a concep
t-level decomposition of model behavior, and by helping benchmark developers ite
rate upon benchmark design. Code is available at https://competency-gaps.github.
io.
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Variational Learning for Insertion-based Generation
Yangtian Zhang, Zhe Wang, Arthur Gretton, Rex Ying, David van Dijk, Michalis Tit

sias, Jiaxin Shi
Non-monotonic sequence generation methods, such as masked diffusion models, provide a flexible alternative to left-to-right autoregressive modeling by allowing tokens to be generated in non-fixed and prescribed orders. Despite their practical advantages, most existing non-monotonic models are order-agnostic and rely on a fixed-length grid, limiting their ability to support variable-length generation and adaptive insertion order. In this work, we introduce a probabilistic framework for learning insertion order in variable-length insertion models. We formalize a bijective correspondence between insertion trajectories and permutations, which enables an exact reparameterization of the data likelihood as a sum over permutations. Building on this result, we propose the Insertion Process (IP), a stochastic generative model that jointly learns where to insert, what to insert, and when to terminate, trained via permutation-based variational inference. Unlike prior fixed-canvas approaches, IP natively supports variable-length generation and learns data-driven preferences over insertion orders. Experiments on goal-conditioned planning and molecular string generation demonstrate that learning insertion order improves both modeling quality and generalization in domains without a canonical left-to-right structure.
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LoKiFormer: Locality-aware Attention with Decoupled Knowledge Memory for Efficient Large Language Model Pretraining
Qiuwu Chen, Zimo Liu, Yuchen Li, Ying Sun, Yifan Zhang, Zhijie Qiu, Zeng You, Ryan Dong, Simeng Ma, Yaofo Chen, Mingkui Tan
Large language models (LLMs) have achieved remarkable breakthroughs across various applications. However, their architectures remain inefficient in pretraining due to two main limitations: (i) self-attention lacks an explicit inductive bias for locality, leading to redundant modeling of sequence-internal local information; (ii) mixture-of-experts (MoE) implicitly couples knowledge storage with computational pathways, hindering flexible access to sequence-external global knowledge. To overcome these limitations, we propose LoKiFormer, a novel LLM architecture that augments the standard decoder with two dedicated modules: 1) Local Fusion Attention (LFA), which incorporates a convolutional fusion to attention, explicitly capturing local patterns and allowing the attention to operate on more informative representations; 2) Knowledge Memory Module (KMM), which introduces a parametric key–value memory that explicitly stores global knowledge in addressable slots, decoupling storage from computation and enabling direct knowledge retrieval. Together, these modules enable LoKiFormer to achieve more efficient and effective integration of information at both levels. Experimental results show that LoKiFormer converges 1.33x faster in pre-training than baseline models, underscoring its superiority over existing LLM architectures.
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Attend to Anything: Foundation Model for Unified Human Attention Modeling
Wenzhuo Zhao, ronghao xian, Keren Fu, Qijun Zhao
Existing human attention (saliency) modeling methods persist as highly fragmented across modalities, scenes, and task formulations. Consequently, even with increasing model capacity and data scale, current models predominantly remain scene-dependent and task-specific, failing to practically generalize in real-world applications.

To address the fundamental limitations, we present the Attend to Anything Model (AAM), a multi-modal foundation model that unifies attention modeling across various image, video, and audio-visual tasks and scenes. AAM reformulates attention as a cognitive entailment relationship organized in a general-to-specific hierarchy, implemented through language prompts with hierarchical embeddings in hyperbolic space. Furthermore, to unify static image and dynamic video attention, we adopt a fluid-dynamics perspective, formulating video-frame attention as a diffusive temporal evolution governed by the Fokker--Planck equation.

Extensive experiments on 16 benchmarks demonstrate that AAM consistently outperforms state-of-the-art methods by an average of 6\% across various scenarios, while achieving approximately a 4$\times$ speedup in video inference.

Overall, these results demonstrate that AAM provides a principled foundation f

or future research on attention and saliency-related tasks. The dataset and code will be available at \url{https://github.com/wz-zhao/Attend-to-Anything}.
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SAMT: Generating Structured Avatar Meshes and Textures from a Single Image
Muyu Wang, Jianzhe Gao, Xingping Dong, Yujia Wang, Wenguan Wang
Despite rapid progress in generative 3D creation, producing high-fidelity 3D face assets from a single image remains challenging, as it requires both identity-critical facial micro-structures and fine-grained view-consistent textures. To address this, we present a two-stage framework named **SAMT** for monocular 3D avatar generation and texture synthesis. Specifically, a latent 3D diffusion model for facial mesh generation is pretrained and then further adapted to generate high-quality facial geometry through large-scale domain-specific finetuning on 35K curated 3D avatar models. Subsequently, the generated facial mesh is textured through a multi-view-aware texturing strategy. It incorporates multi-view facial priors along with the mesh geometry to guide a 2D texturing diffusion, enabling cross-view consistent and mesh-aligned texture synthesis. Extensive experiments demonstrate that SAMT improves over existing baselines by producing more coherent facial geometry together with more fine-grained and view-consistent textures. Project page is available at https://github.com/muyuWang/SAMT.
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Dissecting the Safety Circuit: Neuronal Intervention for Transferable Adversarial Attacks on VLMs
Chunlong Xie, Kangjie Chen, Shangwei Guo, Shudong Zhang, Jiamou Liu, Tianwei Zhang, Tao Xiang
The limited transferability of adversarial attacks on Vision-Language Models (VLMs) stems from their failure to navigate model-specific safety alignments, where superficial perturbations exploit surrogate-specific artifacts rather than shared safety-critical features. We reveal through linear probing that safety-related representations are concentrated within specific intermediate neuronal circuits, which act as localized defense bottlenecks that can be disentangled from transferable features. To overcome this barrier, we propose the Safety Circuit Intervention Attack (SCIA), a framework that surgically steers internal representations to bypass these localized safety mechanisms. SCIA employs a dual-objective steering strategy that suppresses the defensive circuit encoding safety features while amplifying the transferable circuit capturing model-agnostic representations, effectively decoupling adversarial patterns from surrogate-specific safety behaviors. Furthermore, we incorporate contrastive semantic steering and spectral smoothness regularization to guide optimization toward compliant semantic regions while producing visually coherent perturbations. Experimental results demonstrate that SCIA significantly outperforms state-of-the-art methods in bypassing unseen black-box VLMs.
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DeepAnalyze: Agentic Large Language Models for Autonomous Data Science
Shaolei Zhang, Ju Fan, Meihao Fan, Yizhe Liu, Yuxin Zhang, Xiaoyong Du
Autonomous data science has been a long-standing challenge, and is now becoming feasible with the emergence of powerful large language models (LLMs). Recent workflow-based data agents have shown promising results on specific data tasks but remain fundamentally limited in achieving full autonomy due to their reliance on predefined workflows. In this paper, we introduce DeepAnalyze, the first agentic LLM for autonomous data science, capable of automatically completing the end-to-end data science from raw data to research reports. To tackle high-complexity data science tasks, we propose a curriculum-based agentic training paradigm that emulates the learning trajectory of human data scientists, enabling LLMs to progressively acquire and integrate multiple capabilities in real-world environments. Accordingly, we contribute a data-grounded trajectory synthesis framework to constructs high-quality data science training data. Through training in real-world environment, DeepAnalyze learns to perform a broad spectrum of data tasks, ranging from data question answering to open-ended data research. Experiments on 13 benchmarks demonstrate that, with only 8B parameters, DeepAnalyze outperforms workflow-based agents built on advanced proprietary LLMs. DeepAnalyze is fully o

pen-sourced, paving the way toward autonomous data science.
****
Reinforcement Learning with Discrete Diffusion Policies for Combinatorial Action Spaces
Haitong Ma, Ofir Nabati, Aviv Rosenberg, Bo Dai, Oran Lang, Craig Boutilier, Na Li, Shie Mannor, Lior Shani, Guy Tennenholtz
Reinforcement learning (RL) struggles to scale to large, combinatorial action spaces common in many real-world problems. This paper introduces a novel framework for training discrete diffusion models as highly effective policies in these complex settings. Our key innovation is an efficient online training process that ensures stable and effective policy improvement and . By leveraging policy mirror descent (PMD) to define an ideal, regularized target policy distribution, we frame the policy update as a distributional matching problem, training the expressive diffusion model to replicate this stable target. This decoupled approach stabilizes learning and significantly enhances training performance. Our method achieves state-of-the-art results and superior sample efficiency across a diverse set of challenging combinatorial benchmarks, including DNA sequence generation, RL with macro-actions, and multi-agent systems. Experiments demonstrate that our diffusion policies attain comparable or superior performance compared to other baselines. Crucially, our extensive empirical analysis reveals a key trade-off: FKL demonstrates superior sample efficiency and faster initial convergence, whereas RKL ensures stable training and higher asymptotic performance on challenging tasks.
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Mitigating Perceptual Judgment Bias in Multimodal LLM-as-a-Judge via Perceptual Perturbation and Reward Modeling
Seojeong Park, Jiho Choi, Junyong Kang, Seonho Lee, Jaeyo Shin, Hyunjung Shim
Recent multimodal large language models have demonstrated strong reasoning ability, yet their reliability as automated evaluators remains limited by a critical weakness: when visual evidence conflicts with textual cues, MLLM judges tend to reward plausible narratives over perceptually correct answers. We identify and systematically analyze this phenomenon, which we term \textit{Perceptual Judgment Bias}. Through controlled visual perturbations, existing multimodal judges frequently anchor on the response text instead of their own visual perception, leading to inconsistent and non-verifiable evaluations. To address this issue, we introduce the \textit{Perceptually Perturbed Judgment Dataset}, which constructs minimally edited counterfactual responses that isolate perceptual errors and enable verifiable supervision. Building on this dataset, we develop a unified training framework that combines a structured GRPO-based reward with a batch-ranking objective, achieving coherent global ordering without explicit pairwise labels. Experiments across diverse MLLM-as-a-Judge benchmarks show that our approach substantially improves perceptual fidelity, ranking coherence, and alignment with human evaluation. Our results establish a scalable and generalizable pathway for training multimodal judges that are perceptually grounded, interpretable, and robust to visual–reasoning conflicts.
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Stabilizing Recurrent Dynamics for Test-Time Scalable Latent Reasoning in Looped Language Models
Xiao-Wen Yang, Ziyu Han, Xi-Hua Zhang, Wen-Da Wei, Jie-Jing Shao, Lan-Zhe Guo, Yu-Feng Li
Looped Language Models (LoopLMs) enable efficient latent reasoning through depth recurrence, yet exhibit unreliable test-time scaling behavior: performance often peaks at a certain iteration depth and then collapses with further recurrence. Through latent dynamics analysis, we find an inherent trade-off between stability and effectiveness in existing architectures and strategies. By conceptualizing reasoning as uncertainty reduction, we propose that convergence toward stable fixed points while preserving effectiveness represents a promising way. To this end, we propose STARS (STAbility-driven Recurrent Scaling), a training framework that constrains latent states to approach asymptotically stable fixed points. This is realized via efficient Jacobian Spectral Radius Regularization with rand

om loop sampling, enabling STARS to maximize effectiveness while ensuring rigoro
us stability. Experiments on arithmetic tasks show that STARS achieves reliable
test-time scaling, and on complex mathematical reasoning it substantially mitiga
tes performance degradation as recurrence depth increases while also improving p
eak performance.
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HVR-Met: A Hypothesis-Verification-Replanning Agentic System for Extreme Weather
 Diagnosis
Shuo Tang, Jiadong Zhang, Gengxian Zhou, Qizhao Jin, Qinxuan Wang, Yi Hu, Ning H
u, Hongchang Ren, Lingli He, Shiming Xiang, Jingtao Ding, Jian Xu, Jiaolan Fu, C
heng-Lin Liu
While deep learning-based weather forecasting paradigms have made significant st
rides, addressing extreme weather diagnostics remains a formidable challenge. Th
is gap exists primarily because the diagnostic process demands sophisticated mul
ti-step logical reasoning, dynamic tool invocation, and expert-level prior judgm
ent. Although agents possess inherent advantages in task decomposition and auton
omous execution, current architectures are still hampered by critical bottleneck
s: inadequate expert knowledge integration, a lack of professional-grade iterati
ve reasoning loops, and the absence of fine-grained validation and evaluation sy
stems for complex workflows under extreme conditions. To this end, we propose HV
R-Met, a multi-agent meteorological diagnostic system characterized by the deep
integration of expert knowledge. Its central innovation is the ``Hypothesis-Veri
fication-Replanning'' closed-loop mechanism, which facilitates sophisticated ite
rative reasoning for anomalous meteorological signals during extreme weather eve
nts. To bridge gaps within existing evaluation frameworks, we further introduce
a novel benchmark focused on atomic-level sub-tasks. Experimental evidence demon
strates that the system excels in complex diagnostic scenarios.
****

Diversity Over Frequency: Rethinking Tool Use in Visual Chain-of-Thought Agents
Dong-Hee Kim, Reuben Tan, Donghyun Kim
Visual agents employ external visual tools within visual chains of thought to in
corporate fine-grained evidence. While prior work has mainly studied these tools
 in visual search tasks, their role in more complex visual reasoning remains und
erexplored. In this paper, we move beyond simple visual search tasks to investig
ate more challenging tasks, including 3D spatial reasoning and medical visual qu
estion answering, where agents must integrate tool-acquired local evidence with
the global context.

    We identify a \emph{tool-use collapse phenomenon}: models progressively stop
 using tools while still achieving higher task accuracy.

    Moreover, we observe a clear asymmetry: (i) completely eliminating tool use
degrades performance, whereas (ii) incentivizing tool use yields only marginal g
ains despite substantially increasing usage.

    We find that vanilla training and tool-use encouragement both reduce rollout
 diversity, explaining why higher tool use does not yield stronger reasoning per
formance.

    Motivated by these findings, we add an entropy regularization term to encour
age diverse rollout exploration, achieving the best performance despite graduall
y declining tool usage.

    Overall, our findings suggest a training-time view of tools as scaffolding,
where broader exploration over language generation and visual tool invocation im
proves reasoning despite tool-use collapse.

    Project page: https://scaffolded-exploration.github.io
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Path-conditioned training: a principled way to rescale ReLU neural networks
Arthur Lebeurrier, Titouan Vayer, Rémi Gribonval
Despite recent algorithmic advances, we still lack principled ways to leverage t
he well-documented rescaling symmetries in ReLU neural network parameters. While
 two properly rescaled weights implement the same function, the training dynamic
s can be dramatically different. To offer a fresh perspective on exploiting this
 phenomenon, we build on the recent path-lifting framework, which provides a com

pact factorization of ReLU networks. We introduce a geometrically motivated crit
erion to rescale neural network parameters which minimization leads to a conditi
oning strategy that aligns a kernel in the path-lifting space with a chosen refe
rence. We derive an efficient algorithm to perform this alignment. In the contex
t of random network initialization, we analyze how the architecture and the init
ialization scale jointly impact the output of the proposed method. Numerical exp
eriments illustrate its potential to speed up training.
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Towards Fully Parameter-Free Stochastic Optimization: Grid Search with Self-Boun
ding Analysis
Yuheng Zhao, Yu-Hu Yan, Amit Attia, Tomer Koren, Lijun Zhang, Peng Zhao
Parameter-free stochastic optimization aims to design algorithms that are agnost
ic to the underlying problem parameters while still achieving convergence rates
competitive with optimally tuned methods. While some parameter-free methods do n
ot require the specific values of the problem parameters, they still rely on pri
or knowledge, such as the lower or upper bounds of them. We refer to such method
s as "partially parameter-free". In this work, we target achieving "*fully* para
meter-free" methods, i.e., the algorithmic inputs do not need to satisfy any *un
verifiable* condition related to the true problem parameters. We propose a power
ful and general *grid search* framework, named GRASP, with a novel *self-boundin
g* analysis technique that effectively determines the search ranges of parameter
s, in contrast to previous work. Our method demonstrates generality in: (i) the
non-convex case, where we propose a fully parameter-free method that achieves ne
ar-optimal convergence rate, up to logarithmic factors; (ii) the convex case, wh
ere our parameter-free methods are competitive with strong performance in terms
of acceleration and universality. Finally, we contribute a sharper guarantee for
 the model ensemble, a final step of the grid search framework, under interpolat
ed variance characterization.
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SetPO: Set-Level Policy Optimization for Diversity-Preserving LLM Reasoning
Chenyi Li, Yuan Zhang, Bo Wang, Guoqing Ma, Wei Tang, Haoyang Huang, Nan Duan
Reinforcement learning with verifiable rewards has shown notable effectiveness i
n enhancing large language models (LLMs) reasoning performance, especially in ma
thematics tasks. However, such improvements often come with reduced outcome dive
rsity, where the model concentrates probability mass on a narrow set of solution
s. Motivated by diminishing-returns principles, we introduce a set level diversi
ty objective defined over sampled trajectories using kernelized similarity. Our
approach derives a leave-one-out marginal contribution for each sampled trajecto
ry and integrates this objective as a plug-in advantage shaping term for policy
optimization. We further investigate the contribution of a single trajectory to
language model diversity within a distribution perturbation framework. This anal
ysis theoretically confirms a monotonicity property, proving that rarer trajecto
ries yield consistently higher marginal contributions to the global diversity. E
xtensive experiments across a range of model scales demonstrate the effectivenes
s of our proposed algorithm, consistently outperforming strong baselines in both
 Pass@1 and Pass@K across various benchmarks.
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Blocking the Leakage: Manifold-Aware Gradient Projection for Long-Horizon Test-T
ime Adaptation
Haoyu Xiong, Chengchao Wang, ZhongQiang Wang, He Huang, Qiuxia Yang, Zhengpeng Z
hao, Yuanyuan Pu
Test-Time Adaptation (TTA) empowers pre-trained models to adapt online to distri
bution shifts during inference, but such online updates often become unstable in
 long-horizon deployments. Prevailing approaches attribute this failure to error
 accumulation from noisy pseudo-labels, relying on heuristics to gate samples us
ed for updates. We argue that this statistical view is insufficient: the problem
 lies not only in sample quality but also in the directionality of gradients. In
 this work, we identify a geometric failure mode termed manifold erosion. Throug
h spectral analysis, we find that reliable gradients concentrate in a stable low
-rank subspace, while gradients from confident mispredictions are high-rank yet

exhibit a persistent directional leakage into this protected subspace. This leak age can accumulate coherently and gradually erode core representations, eventual ly leading to collapse. To address this, we propose Manifold-Aware Gradient Proj ection (MGP), a geometric intervention that tracks the dominant subspace online and projects gradients onto its orthogonal complement. By blocking the leakage p ath, MGP decouples stability from plasticity. Extensive experiments on diverse T TA benchmarks demonstrate its long-horizon stability, whereas prior methods ofte n fail.
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Particle-Guided Diffusion Models for Partial Differential Equations
Andrew Millard, Fredrik Lindsten, Zheng Zhao
We introduce a guided stochastic sampling method that augments sampling from dif fusion models with physics-based guidance derived from partial differential equa tion (PDE) residuals and observational constraints, ensuring generated samples r emain physically admissible. We embed this sampling procedure within a new Seque ntial Monte Carlo (SMC) framework, yielding a scalable generative PDE solver. Ac ross multiple benchmark PDE systems as well as multiphysics and interacting PDE systems, our method produces solution fields with lower numerical error than exi sting state-of-the-art generative methods.
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Interpretability Transfer from Language to Vision via Sparse Autoencoders
Alexey Kravets, Da Li, Chuan Li, Da Chen, Vinay P. Namboodiri
Recent advances in language model interpretability using sparse autoencoders (SA Es) have yet to effectively translate to the visual domain, mainly due to the di fficulty and ambiguity of labeling visual concepts. In this paper, we introduce Visual Interpretability via SAE Transfer Alignment (VISTA), a framework that tra nsfers interpretability from language to vision in a LLaVA-style vision-language model by constraining a visual projector to map visual tokens into an LLM's pre -existing, labeled textual SAE space. This approach enables visual interpretabil ity without training dedicated vision SAEs. By regularizing the projector using the LLM's SAE reconstruction loss, VISTA achieves a threefold increase in the ma tching rate, which measures how accurately the most activating textual concepts in the SAE space correspond to semantic elements in the image. Using this framew ork, we further analyze spatial localization properties of different vision enco ders and show that DINOv2 features have stronger localization abilities than oth er encoders. Leveraging this precision, we validate VISTA's cross-modal alignmen t through fine-grained, localized concept interventions, where specific objects are removed or replaced in the model's perception while preserving the surroundi ng scene. This results in improvements of 35\% in object removal and 47\% in obj ect replacement tasks over vision-only baselines, providing causal evidence that visual tokens inhabit the text SAE manifold. These contributions are validated across multiple LLM architectures.
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JADAI: Jointly Amortizing Adaptive Design and Bayesian Inference
Niels Leif Bracher, Lars Kühmichel, Desi R. Ivanova, Xavier Intes, Paul-Christia n Bürkner, Stefan T. Radev
We consider problems of parameter estimation where design variables can be activ ely optimized to maximize information gain. To this end, we introduce JADAI, a f ramework that jointly amortizes Bayesian adaptive design and inference by traini ng a policy, a history network, and an inference network end-to-end. The network s minimize a generic loss that aggregates incremental reductions in posterior er ror along experimental sequences without density evaluations.
Inference networks are instantiated with diffusion models that can approximate h igh-dimensional and multimodal posteriors at every experimental step.
JADAI achieves superior or competitive performance across adaptive design benchm arks.
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PhaseCoder: Microphone Geometry-Agnostic Spatial Audio Understanding for Multimo dal LLMs
Artem Dementyev, Wazeer Zulfikar, Sinan Hersek, Pascal Getreuer, Anurag Kumar, V

ivek Kumar
Current multimodal large language models (LLMs) process audio as a mono stream, ignoring the rich spatial information essential for embodied AI. Conversely, existing spatial audio models are constrained to fixed microphone geometries, preventing their deployment across diverse devices. We present **PhaseCoder**, a transformer-only spatial audio encoder that is inherently agnostic to microphone geometry. By taking raw multichannel audio and microphone coordinates as inputs, PhaseCoder performs accurate localization and produces robust spatial embeddings. We demonstrate that the Gemma 3n LLM can be fine-tuned to process and reason over the "Spatial Audio Tokens" produced by our encoder. PhaseCoder achieves state-of-the-art results on microphone-invariant localization benchmarks and, for the first time, enables an LLM to perform complex spatial reasoning and targeted transcription tasks from an arbitrary microphone array.
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Gradient-Based Causal Tree Ensembles: A Backbone Architecture for Heterogeneous Treatment Effects
Yusuke Kano, Jeremy Paul Voisey, Mihaela van der Schaar
Estimating Heterogeneous Treatment Effects (HTE) from observational data is essential in fields such as healthcare and policy-making, where randomized experiments are often impractical.
While representation learning-based methods have shown promise, recent studies suggest that tree-based approaches may offer superior performance on tabular data, particularly in the presence of uninformative features.
We introduce **GRA**dient-based **C**ausal tree **E**nsembles (GRACE), a novel tree-based architecture for HTE estimation that incorporates multi-way, oblique, and soft splits, enabling end-to-end training via backpropagation.
GRACE can be seamlessly integrated into existing models as a replacement for fully-connected neural network layers.
Across diverse benchmarks involving binary and non-binary treatment settings, GRACE consistently surpasses neural network and tree-based baselines, often by a substantial margin.
We further analyze GRACE as an extension of fully-connected neural network layers and conduct ablation studies to isolate and quantify the contribution of each architectural component to the improvement in performance.
These results position GRACE as a promising new foundation for flexible, robust, and accurate HTE estimation.
GRACE code is available at https://github.com/ysk-kano/GRACE
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What Makes Value Learning Efficient in Residual Reinforcement Learning?
Guozheng Ma, Lu Li, Haoyu Wang, Zixuan Liu, Pierre-Luc Bacon, Dacheng Tao
Residual reinforcement learning (RL) enables stable online refinement of expressive pretrained policies by freezing the base and learning only bounded corrections. However, value learning in residual RL poses unique challenges that remain poorly understood. In this work, we identify two key bottlenecks: cold start pathology, where the critic lacks knowledge of the value landscape around the base policy, and structural scale mismatch, where the residual contribution is dwarfed by the base action. Through systematic investigation, we uncover the mechanisms underlying these bottlenecks, revealing that simple yet principled solutions suffice: base-policy transitions serve as an essential value anchor for implicit warmup, and critic normalization effectively restores representation sensitivity for discerning value differences. Based on these insights, we propose DAWN (Data-Anchored Warmup and Normalization), a minimal approach targeting efficient value learning in residual RL. By addressing these bottlenecks, DAWN demonstrates substantial efficiency gains across diverse benchmarks, policy architectures, and observation modalities.
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Toward Structural Multimodal Representations: Specialization, Selection, and Sparsification via Mixture-of-Experts
Hahyeon Choi, Nojun Kwak
We propose S3 (Specialization, Selection, Sparsification), a framework that reth

inks multimodal learning through a structural perspective. Instead of encoding a ll signals into a fixed embedding, S3 decomposes multimodal inputs into semantic experts and selectively routes them for each task. Specialization forms concept -level experts in a shared latent space, Selection adapts routing for task-speci fic needs, and Sparsification prunes low-utility paths to yield compact, informa tion-minimal representations. Across four MultiBench benchmarks, S3 improves acc uracy and exhibits consistent sparsity-performance dynamics, exhibiting a revers e U-shaped trend, with performance peaking at intermediate sparsity. These resul ts suggest that structuring multimodal representations as selectable semantic co mponents provides a practical and principled alternative to contrastive learning or InfoMax-driven approaches.

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Ensembling Sparse Autoencoders

Soham Gadgil, Chris Lin, Su-In Lee

Sparse autoencoders (SAEs) are used to decompose neural network activations into human-interpretable features. Typically, features learned by a single SAE are u sed for downstream applications. However, it has recently been shown that a sing le SAE captures only a limited subset of features that can be extracted from the activation space. Motivated by this limitation, we introduce and formalize SAE ensembles. Furthermore, we propose to ensemble multiple SAEs through *naive bagg ing* and *boosting*. In naive bagging, SAEs trained with different weight initia lizations are ensembled, whereas in boosting SAEs sequentially trained to minimi ze the residual error are ensembled. Theoretically, naive bagging and boosting a re justified as approaches to reduce reconstruction error. Empirically, we evalu ate our ensemble approaches with three settings of language models and SAE archi tectures. Our empirical results demonstrate that, compared to an expanded SAE th at matches the number of features in the ensemble, ensembling SAEs improves the reconstruction of language model activations along with SAE stability. Additiona lly, on downstream tasks such as concept detection and spurious correlation remo val, SAE ensembles achieve better performance, showing improved practical utilit y.

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A Graphop Analysis of Graph Neural Networks on Sparse Graphs: Generalization and Universal Approximation

Ofek Amran, Tom Gilat, Ron Levie

Generalization and approximation capabilities of message passing graph neural ne tworks (MPNNs) are often studied by defining a compact metric on a space of inpu t graphs under which MPNNs are equicontinuous. Such analyses are of two varietie s: 1) when the metric space includes graphs of unbounded sizes, the theory is on ly appropriate for dense graphs, and, 2) when studying sparse graphs, the metric space only includes graphs of uniformly bounded size. In this work, we present a unified approach, defining a compact metric on the space of graphs of all size s, both sparse and dense, under which MPNNs are equicontinuous. This leads to mo re powerful universal approximation theorems and generalization bounds than prev ious works. The theory is based on, and extends, a recent approach to graph limi t theory called graphop analysis.

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T-measure: A Topology-Consistent Metric for Binary Segmentation

pengfei zhang, Jian Ji

Evaluation metrics establish a standard assessment framework for models, playing a pivotal role in model optimization and advancement. However, widely adopted p ixel-wise metrics (e.g., IoU, Dice) rely heavily on pixel-level statistics, ofte n failing to capture the structural integrity of predictions. While the S-measur e (S-m) incorporates structural perception to some extent, it struggles to diffe rentiate critical structural violations and remains insensitive to background fa lse positives and small objects. To address these limitations, we propose the To pology-aware measure (T-m), a novel metric designed to explicitly quantify topol ogical consistency. T-m employs the Fuzzy Jaccard Index as a foundational score, integrates a Topological Integrity term (I-topo) to penalize critical structura l fragmentation, and utilizes a Boundary Alignment term (A-bdy) to evaluate boun

dary alignment. These three components synergize to achieve robust evaluation of prediction maps at the topological level. We establish a rigorous Meta-Measure validation framework and benchmark our method against nine mainstream metrics across diverse complex scenarios. Extensive experiments demonstrate that T-m performs exceptionally in downstream tasks and maintains high consistency with human visual perception.

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A Unified Framework for Deep Hypergraph Clustering Beyond Homophily

Bowen Zhao, Qianqian Wang

Deep hypergraph clustering exhibits compelling capacity for node representation learning via modeling high-order relationships. However, most existing methods adopt fixed propagation mechanisms and implicitly assume homophily, which presumes that adjacent nodes possess similar characteristics. This assumption might deviate from real-world situations, particularly under heterophilic conditions, thereby degrading clustering performance. To address this limitation, we propose a \textbf{Uni}fied Framework for \textbf{D}eep \textbf{H}ypergraph \textbf{C}lustering (Uni-DHC). Specifically, we design a learnable high-order hypergraph propagation strategy that fuses multi-order information and adaptively learns their importance derived from raw data. To stabilize unsupervised training and eliminate structural redundancy caused by high-order aggregation, we additionally enforce node-level consistency and hyperedge-level decorrelation constraints. From the spectral perspective, we demonstrate that conventional HGNN-style propagation corresponds to a fixed low-pass filter, whereas our designed method induces a learnable polynomial spectral filter. Extensive experiments on homophilic and heterophilic datasets illustrate that Uni-DHC consistently outperforms state-of-the-art methods, achieving prominent performance improvement in heterophilic settings.

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DOCKSMITH: Scaling Reliable Coding Environments via an Agentic Docker Builder

Jiaran Zhang, Lu Ma, Yanhao Li, Fanqi Wan, DI QI, Xin Wu, Zhewei Huang, Liangyu Chen, YINGWEI MA, Qi Han, Xiangyu Zhang

Reliable Docker-based environment construction is a dominant bottleneck for scaling execution-grounded training and evaluation of software engineering agents. We introduce DockSmith, a specialized agentic Docker builder designed to address this challenge. DockSmith treats environment construction not merely as a preprocessing step, but as a core agentic capability that exercises long-horizon tool use, dependency reasoning, and failure recovery, yielding supervision that transfers beyond Docker building itself. DockSmith is trained on large-scale, execution-grounded Docker-building trajectories produced by a SWE-Factory--style pipeline augmented with a loop-detection controller and a cross-task success memory. Training a 30B-A3B model on these trajectories achieves open-source state-of-the-art performance on Multi-Docker-Eval, with 39.72\% Fail-to-Pass and 58.28\% Commit Rate. Moreover, DockSmith improves out-of-distribution performance on SWE-bench Verified, SWE-bench Multilingual, and Terminal-Bench 2.0, demonstrating the broader agentic benefits of environment construction. Our model and Docker-building trajectories are publicly available at https://huggingface.co/JiaranZhang/DockSmith.

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SciVideoBench: Benchmarking Scientific Video Reasoning in Large Multimodal Models

Andong Deng, Taojiannan Yang, Shoubin Yu, Lincoln Spencer, Mohit Bansal, Chen Chen, Serena Yeung-Levy, Xiaohan Wang

Large Multimodal Models (LMMs) have achieved remarkable progress across various capabilities; however, complex video reasoning in the scientific domain remains a significant and challenging frontier. Current video benchmarks predominantly target general scenarios where perception/recognition is heavily relied on, while with relatively simple reasoning tasks, leading to saturation and thus failing to effectively evaluate advanced multimodal cognitive skills. To address this critical gap, we introduce SciVideoBench, a rigorous benchmark specifically designed to assess advanced video reasoning in scientific contexts. SciVideoBench consists of 1,000 carefully crafted multiple-choice questions derived from cutting-e

dge scientific experimental videos spanning over 25 specialized academic subjects and verified by a semi- automatic system. Each question demands sophisticated domain-specific knowledge, precise spatiotemporal perception, and intricate logical reasoning, effectively challenging models' higher-order cognitive abilities. Our evaluation highlights significant performance deficits in state-of-the-art proprietary and open-source LMMs, including Gemini 2.5 Pro and Qwen2.5-VL, indicating substantial room for advancement in video reasoning capabilities. Detailed analyses of critical factors such as reasoning complexity and visual grounding provide valuable insights and clear direction for future developments in LMMs, driving the evolution of truly capable multimodal AI co-scientists. We hope SciVideoBench could fit the interests of the community and help to push the boundary of cutting-edge AI for border science.

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MedCoG: Maximizing LLM Inference Density in Medical Reasoning via Meta-Cognitive Regulation

Yu Zhao, Hao Guan, Yongcheng Jing, Ying Zhang, Dacheng Tao

Large Language Models (LLMs) have shown strong potential in complex medical reasoning yet face diminishing gains under inference scaling laws. While existing studies augment LLMs with various knowledge types, it remains unclear how effectively the additional costs translate into accuracy. In this paper, we explore how meta-cognition of LLMs, i.e., their self-assessment of their own cognitive states, can regulate the reasoning process. Specifically, we propose MedCoG, a Medical Meta-Cognition Agent with Knowledge Graph, where the meta-cognitive assessments of task complexity, familiarity, and knowledge density dynamically regulate utilization of procedural, episodic, and factual knowledge. The LLM-centric on-demand reasoning aims to mitigate the diminishing returns under scaling law by (1) reducing costs via avoiding indiscriminate scaling, (2) improving accuracy via filtering out distractive knowledge. To validate this, we empirically characterize the scaling curve and introduce inference density to quantify inference efficiency. Experiments demonstrate the effectiveness and efficiency of MedCoG on five hard sets of medical benchmarks, yielding 6.2x inference density. Furthermore, the Oracle study highlights the significant potential of meta-cognitive regulation.

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Principled RL for Flow Matching Emerges from the Chunk-level Policy Optimization

Yifu Luo, Haoyuan Sun, Xinhao Hu, Penghui Du, Keyu Fan, Bo Li, SiNan Du, Xu Wan, Zhiyu Chen, Bo Xia, Yongzhe Chang, Kai Wu, Kun Gai, Tiantian Zhang, Xueqian Wang

Recent Progress in post-training flow matching for text-to-image (T2I) generation with Group Relative Policy Optimization (GRPO) has demonstrated strong potential. However, it is hindered by a critical limitation: inaccurate advantage attribution. In this work, we argue that aggregating consecutive timesteps into a coherent 'chunk' and shifting the policy optimization paradigm from GRPO's step level to the chunk level can effectively mitigate the negative impact of this issue. Building on this insight, we propose Group Chunking Policy Optimization (GCPO), the first chunk-level reinforcement learning approach for post-training flow matching. Extensive experiments demonstrate that GCPO achieves superior performance on both standard T2I benchmarks and preference alignment, with up to 43% additional gains over GRPO, highlighting the promise of chunk-level policy optimization.

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Efficient Inference for Noisy LLM-as-a-Judge Evaluation

Yiqun T. Chen, Sizhu Lu, Sijia Li, Moran Guo, Shengyi Li

Large language models (LLMs) are increasingly used as automatic evaluators of generative AI outputs, a paradigm often referred to as "LLM-as-a-judge." In practice, LLM judges are imperfect predictions for the underlying truth and can exhibit systematic, non-random errors. Two main approaches have recently been proposed to address this issue: (i) *direct measurement-error correction* based on misclassification models such as Rogan--Gladen-style estimators, and (ii) *surrogate-outcome approaches* such as prediction-powered inference (PPI), which correct bi

as by calibrating prediction residuals on a small set of gold-standard human labels. In this paper, we systematically study the performance of these two approaches for estimating mean parameters (e.g., average benchmark scores or pairwise win rates). Leveraging tools from semiparametric efficiency theory, we unify the two classes of estimators by deriving explicit forms of *efficient influence function*-based efficient estimators and characterize conditions under which PPI-style estimators attain strictly smaller asymptotic variance than measurement-error corrections. We verify our theoretical results through simulations and demonstrate the methods on a real-data example using our open-source software package for performing the calibration.

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Stability Analysis of Sharpness-Aware Minimization

Hoki Kim, Jinseong Park, Yujin Choi, Jaewook Lee

Sharpness-aware minimization (SAM) is a training method that seeks to find flat minima in deep learning, resulting in state-of-the-art performance across various domains. Instead of minimizing the loss of the current weights, SAM minimizes the worst-case loss in its neighborhood in the parameter space. In this paper, we investigate the convergence instability of SAM near a saddle point. Using the qualitative theory of dynamical systems, we explain how SAM becomes stuck in the saddle point and theoretically prove that the saddle point can become an attractor under SAM dynamics. Additionally, we show that this convergence instability can also occur in stochastic dynamical systems by establishing the diffusion of SAM. We prove that SAM diffusion is worse than that of vanilla gradient descent in terms of saddle point escape. Finally, we demonstrate that often overlooked training tricks, momentum and batch-size, might be important to mitigate the convergence instability and achieve high generalization performance. Our theoretical and empirical results are thoroughly verified through experiments on several well-known optimization problems and benchmark tasks.

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ParamMem: Augmenting Language Agents with Parametric Reflective Memory

Tianjun Yao, Yongqiang Chen, Yujia Zheng, Pan Li, Zhiqiang Shen, Kun Zhang

Self-reflection enables language agents to iteratively refine solutions, yet often produces repetitive outputs that limit reasoning performance. Recent studies have attempted to address this limitation through various approaches, among which increasing reflective diversity has shown promise. Our empirical analysis reveals a strong positive correlation between reflective diversity and task success, further motivating the need for diverse reflection signals. We introduce `ParamMem`, a parametric memory module that encodes cross-sample reflection patterns into model parameters, enabling diverse reflection generation through temperature-controlled sampling. Building on this module, we propose ParamAgent, a reflection-based agent framework that integrates parametric memory with episodic and cross-sample memory. Extensive experiments on code generation, mathematical reasoning, and multi-hop question answering demonstrate consistent improvements over state-of-the-art baselines. Further analysis reveals that `ParamMem` is sample-efficient, enables weak-to-strong transfer across model scales, and supports self-improvement without reliance on stronger external model, highlighting the potential of `ParamMem` as an effective component for enhancing language agents.

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Darwinian Memory: A Training-Free Self-Regulating Memory System for GUI Agent Evolution

Hongze Mi, Yibo Feng, WenJie Lu, Song Cao, Jinyuan Li, Yanming Li, Xuelin Zhang, Haotian Luo, Songyang Peng, He Cui, Tengfei Tian, Jun Fang, Hua Chai, Naiqiang Tan

Multimodal Large Language Model (MLLM) agents facilitate Graphical User Interface (GUI) automation but struggle with long-horizon, cross-application tasks due to limited context windows. While memory systems provide a viable solution, existing paradigms struggle to adapt to dynamic GUI environments, suffering from a granularity mismatch between high-level intent and low-level execution, and context pollution where the static accumulation of outdated experiences drives agents into hallucination.

To address these bottlenecks, we propose the Darwinian Memory System (DMS), a self-evolving architecture that constructs memory as a dynamic ecosystem governed by the law of "survival of the fittest." DMS decomposes complex trajectories into independent, reusable units for compositional flexibility, and implements Utility-driven Natural Selection to track survival value, actively pruning suboptimal paths and inhibiting high-risk plans. This evolutionary pressure compels the agent to derive superior strategies. Extensive experiments on real-world multi-app benchmarks validate that DMS boosts general-purpose MLLMs without training costs or architectural overhead, achieving average gains of 18.0% in success rate and 33.9% in execution stability, while reducing task latency, establishing it as an effective self-evolving memory system for GUI tasks. The implementation is available at https://anonymous.4open.science/r/DMS-C48C.

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Set Diffusion: Interpolating Token Orderings between Autoregression and Diffusion for Fast and Flexible Decoding

Marianne Arriola, Volodymyr Kuleshov

Discrete diffusion models have steadily improved in quality relative to autoregressive (AR) models. However, these models are normally constrained to fixed-length generation and do not support key-value (KV) caching. Block diffusion partially bridges diffusion and AR by unmasking token blocks left-to-right, but it is still limited to generate fixed-size blocks sequentially. Here, we present a new class of language models, set diffusion, comprised of (i) a likelihood parameterization that factorizes over flexible-position, flexible-length token sets and (ii) a set-causal diffusion architecture that supports KV cache updates after every inference step. By factorizing over token sets instead of fixed-size blocks, tokens can be decoded in arbitrarily-ordered sets, including sliding-window sets, enabling faster inference and support for any-order decoding. Set diffusion achieves better speed-quality tradeoffs on mathematical reasoning, summarization, and unconditional generation compared to prior diffusion language models while offering stronger infilling performance than block diffusion. We provide the code, along with the model weights and blog post on the project page: https://m-arriola.com/setdlms/

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ST-Veto: Spatio-Temporal Token Veto for Diffusion MLLMs via Taylor Prediction and Visual Grounding

Keuntae Kim, Beomseok Lee, Hyunwoo Kim, Yong Suk Choi

Vision Language Models (VLMs) achieve strong reasoning with Chain-of-Thought (CoT) prompting but incur high sequential-generation cost, error accumulation, and limited self-correction. Diffusion Multimodal Large Language Models (dMLLMs) unmask tokens in an order-agnostic process, improving efficiency and enabling iterative refinement, yet their reasoning and how to enhance it remain underexplored. We propose a training-free method, Spatio-Temporal Token Veto (ST-Veto), which leverages the ability to observe all token positions at each diffusion step. Rather than relying only on current-step confidence, ST-Veto vetoes temporally unstable tokens via second-order Taylor prediction of confidence dynamics and filters weakly grounded tokens using image-attention mass, swapping them with safer candidates. Across multiple dMLLMs and multimodal reasoning benchmarks, ST-Veto consistently outperforms standard decoding policies and prior VLM reasoning methods, improving accuracy by up to 9\% with no additional training or generation cost. Analyses show that ST-Veto steers generation toward higher-confidence, better-grounded paths.

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Improving Diffusion Planners by Self-Supervised Action Gating with Energies

Yuan Lu, Dongqi Han, Yansen Wang, Dongsheng Li

Diffusion planners are a strong approach for offline reinforcement learning, but they can fail when value-guided selection favours trajectories that score well yet are locally inconsistent with the environment dynamics, resulting in brittle execution. We propose Self-supervised Action Gating with Energies (SAGE), an inference-time re-ranking method that penalises dynamically inconsistent plans using a latent consistency signal. SAGE trains a Joint-Embedding Predictive Archite

cture (JEPA) encoder on offline state sequences and an action-conditioned latent predictor for short horizon transitions. At test time, SAGE assigns each sampled candidate an energy given by its latent prediction error and combines this feasibility score with value estimates to select actions. SAGE can integrates into existing diffusion planning pipelines that can sample trajectories and select actions via value scoring; it requires no environment rollouts and no policy re-training. Across locomotion, navigation, and manipulation benchmarks, SAGE improves the performance and robustness of diffusion planners.
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Sample from What You See: Visuomotor Policy Learning via Diffusion Bridge with Observation-Embedded Stochastic Differential Equation
Zhaoyang Liu, Mokai Pan, Zhongyi Wang, Kaizhen Zhu, Haotao Lu, Haipeng Zhang, Jingya Wang, Ye Shi
Imitation learning with diffusion models has advanced robotic control by capturing the multi-modal action distributions. However, existing methods typically treat observations only as high-level conditions to the denoising network, rather than integrating them into the stochastic dynamics of the diffusion process itself. As a result, the sampling is forced to begin from random noise, weakening the coupling between perception and control and often yielding suboptimal performance. We propose BridgePolicy, a generative visuomotor policy that directly integrates observations into the stochastic dynamics via a diffusion-bridge formulation. By constructing an observation-informed trajectory, BridgePolicy enables sampling to start from a rich and informative prior rather than random noise, substantially improving precision and reliability in control. A key difficulty is that diffusion bridge normally connects distributions of matched dimensionality, while robotic observations are heterogeneous and not naturally aligned with actions. To overcome this, we introduce a semantic aligner to unify the visual and state inputs and align the observations with action representations, making diffusion bridge applicable to heterogeneous robot data. Extensive experiments across 52 simulation tasks on three benchmarks and 5 real-world tasks demonstrate that BridgePolicy consistently outperforms state-of-the-art generative policies. Our code is available at https://github.com/jianghcsr/BridgePolicy.
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Multi-Agent Reinforcement Learning with Submodular Reward
Wenjing Chen, Chengyuan Qian, Shuo Xing, Yi Zhou, Victoria G. Crawford
In this paper, we study cooperative multi-agent reinforcement learning (MARL) where
the joint reward exhibits submodularity, which is a natural property capturing
diminishing marginal returns when adding agents to a team. Unlike standard
MARL with additive rewards, submodular rewards model realistic scenarios
where agent contributions overlap (e.g., multi-drone surveillance,
collaborative exploration). We provide the first formal framework for this
setting and develop algorithms with provable guarantees on sample efficiency and regret bound. For known dynamics,
our greedy policy optimization achieves a $1/2$-approximation with polynomial
complexity in the number of agents K, overcoming the exponential curse of
dimensionality inherent in joint policy optimization. For unknown dynamics,
we propose a UCB-based learning algorithm achieving a $1/2$-regret of $O(H^2KS\sqrt{AT})$ over
T episodes.
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video-SALMONN S: Memory-Enhanced Streaming Audio-Visual LLM
Guangzhi Sun, Yixuan Li, Xiaodong Wu, Yudong Yang, Wei Li, Zejun MA, Chao Zhang
Long-duration streaming video understanding is fundamental for future AI agents, yet remains limited by ineffective long-term memory. We introduce video-SALMONN S, a memory-enhanced streaming audio-visual large language model that processes over 3-hour videos at 1 FPS and 360p resolution, outperforming strong non-streaming models under the same memory budget.
In addition to token merging or downsampling, video-SALMONN S is the first to employ test-time training (TTT) as a streaming memory mechanism for video understa

nding. TTT continuously transforms short-term multimodal representations into lo
ng-term memory embedded in model parameters. To improve long-range dependency mo
deling and memory capacity, we propose (i) a TTT$_\text{MEM}$ layer with an addi
tional long-span prediction objective, (ii) a two-stage training scheme, and (ii
i) a modality-aware memory reader.
We further introduce the episodic learning from video memory (ELViM) benchmark,
simulating agent-like scenarios where models must learn from videos observed hou
rs earlier. video-SALMONN S consistently outperforms both streaming and non-stre
aming baselines by 3-7\% on long video benchmarks. Notably, video-SALMONN S achi
eves a 15% absolute accuracy improvement over strong non-streaming models on
ELViM, demonstrating strong learning abilities from video memory.
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MaxSAT-Based Compression for Tsetlin Machines
Stefan Szeider
We consider the computational problem of compacting Tsetlin Machine classifiers
by reducing the number of propositional clauses while preserving predictive accu
racy. TMs trained with limited clause capacity often perform poorly because stoc
hastic optimization cannot reliably find the few precise clauses needed in a vas
t configuration space. High-quality compact subsets also exist in the case of la
rger Tsetlin Machines. The difficulty here lies in extracting them. Local prunin
g heuristics can fail badly on TMs because clauses interact through Boolean logi
c: a clause may appear unimportant in isolation yet becomes critical when others
 are removed. We formalize compression as the Minimum Discriminating Clause Set
(MDCS) problem, which asks to find a smallest subset of clauses that preserves t
he trained model's discrimination of training samples. We show that MDCS is NP-h
ard. We solve MDCS using weighted partial Maximum Satisfiability (MaxSAT). A par
tition-and-merge strategy allows us to scale to 100,000 samples. Across 13 datas
ets, the compressed model preserves the 200-clause teacher's accuracy within a f
ew percentage points while using a median of only 16 clauses, and outperforms a
matched-capacity TM trained from scratch on every dataset where direct training
has room to improve, by up to 45 percentage points.
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Discrete Diffusion with Physical Mass Constraints for \emph{De Novo} Peptide Seq
uencing
An Zeyu, Wanyu Lin
$\textit{De novo}$ peptide sequencing is a pivotal technique that directly recon
structs amino acid sequences from tandem mass spectrometry (MS/MS) data; it enab
les the identification of novel proteins and variants absent from reference data
bases. Previous methods are typically based on autoregressive (AR) decoding or o
ne-shot generation. The AR-based methods conflict with the bidirectional and glo
bally constrained nature of MS/MS evidence and inevitably accumulate errors, whi
le one-shot generation does not explicitly enforce physical constraints, failing
 to produce chemically valid and reliable peptides in a single pass. Accurate se
quencing necessitates reasoning over the entire peptide simultaneously, enabling
 iterative self-correction under global constraints. To this end, we introduce $
\textbf{PhysNovo}$, a novel paradigm that harnesses discrete diffusion to enable
 simultaneous global reasoning and iterative refinement. Specifically, PhysNovo
reformulates sequencing as a $\textbf{phys}$ically mass-constrained reasoning pr
ocess by embedding a knapsack-based feasibility kernel to enforce exact precurso
r mass consistency. By conditioning the diffusion process on global spectral con
text, PhysNovo supports abductive reasoning where bidirectional evidence is expl
oited to iteratively resolve local inconsistencies and ensure physically valid p
redictions. PhysNovo achieves state-of-the-art performance, exceeding baselines
by over 2% in precision, with larger gains on out-of-distribution data. The sour
ce code is publicly available at https://github.com/WanyuGroup/ICML2026_PhysNovo
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Biases in the Blind Spot: Detecting What LLMs Fail to Mention
Iván Arcuschin, David Chanin, Adrià Garriga-Alonso, Oana-Maria Camburu
Large Language Models (LLMs) often provide chain-of-thought (CoT) reasoning trac

es that appear plausible, but may hide internal biases. We call these *unverbalized biases*. Monitoring models via their stated reasoning is therefore unreliable, and existing bias evaluations typically require predefined categories and hand-crafted datasets.
In this work, we introduce a fully automated, black-box pipeline for detecting task-specific unverbalized biases. Given a task dataset, the pipeline uses LLM autoraters to generate candidate bias concepts. It then tests each concept on progressively larger input samples by generating positive and negative variations, and applies statistical techniques for multiple testing and early stopping. A concept is flagged as an unverbalized bias if it yields statistically significant performance differences while not being cited as justification in the model's CoTs.
We evaluate our pipeline across seven LLMs on three decision tasks (hiring, loan approval, and university admissions). Our technique automatically discovers previously unknown biases in these models (e.g., Spanish fluency, English proficiency, writing formality).
In the same run, the pipeline also validates biases that were manually identified by prior work (gender, race, religion, ethnicity).
More broadly, our proposed approach provides a practical, scalable path to automatic, more efficient, and broader task-specific unverbalized bias discovery.
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Rethinking Code Complexity Through the Lens of Large Language Models
Chen Xie, Xiaodong Gu, Yuling Shi, Beijun Shen
Code complexity metrics such as cyclomatic complexity have long been used to assess software quality and maintainability. With the rapid advancement of large language models (LLMs) on coding tasks, an important yet underexplored question arises: do traditional complexity metrics meaningfully characterize the coding difficulty that LLMs perceive?
In this work, we empirically demonstrate that classical complexity metrics exhibit no consistent correlation with LLM performance, revealing a fundamental mismatch with model-perceived difficulty.
To address this gap, we propose LM-CC, a novel code complexity metric tailored for LLMs, grounded in the hypothesis that model-perceived code difficulty is fundamentally driven by semantic nonlinearity.
LM-CC quantifies complexity through an entropy-guided semantic compositional hierarchy, capturing the cumulative uncertainty encountered by LLMs during code understanding.
Our experimental results demonstrate that LM-CC exhibits strong and consistent partial correlations with LLM performance, while semantics-preserving reductions in LM-CC consistently lead to improved downstream task performance.
The source code is available at: https://github.com/xchen121/lm-cc.
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Simultaneous Multi-objective Alignment Across Verifiable and Non-verifiable Rewards
Yiran Jenny Shen, Yu Xia, Jonathan Daniel Chang, Prithviraj Ammanabrolu
Aligning large language models to human preferences is inherently multidimensional, yet most pipelines collapse heterogeneous signals into a single objective.
We seek to answer what it would take to simultaneously align a model across various domains spanning those with: verifiable rewards, non-verifiable subjective preferences, and complex interactive scenarios. Such multi-objective alignment setups are often plagued by individual objectives being at odds with each other, resulting in inefficient training and limited user control during inference. To address these issues, we propose **M**ulti-**A**ction-**H**ead **AL**ignment with PRM-guided Dec**O**ding (**MAHALO**), a unified framework that standardizes PRM training across verifiable and non-verifiable settings for step-level supervision, performs vectorized multi-objective alignment with Multi-Action-Head DPO, and enables controllable inference through objective-specific weighting and PRM-guided decoding. Experiments across math reasoning, human values alignment, and multi-turn tutoring show that MAHALO jointly improves multiple objectives simultaneously with limited interference, while remaining generalizable and adaptable ac

ross domains and offering flexible user control at inference time. Our code is a
vailable at: https://github.com/pearls-lab/multiobj-align.
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Diffusing to Coordinate: Efficient Online Multi-Agent Diffusion Policies
Zhuoran Li, Hai Zhong, Xun Wang, Qingxin Xia, Lihua Zhang, Longbo Huang
Online Multi-Agent Reinforcement Learning (MARL) is a prominent framework for ef
ficient agent coordination. Crucially, enhancing policy expressiveness is pivota
l for achieving superior performance. Diffusion-based generative models are well
-positioned to meet this demand, having demonstrated remarkable expressiveness a
nd multimodal representation in image generation and offline settings. Yet, thei
r potential in online MARL remains largely under-explored. A major obstacle is t
hat the intractable likelihoods of diffusion models impede entropy-based explora
tion and coordination. To tackle this challenge, we propose among the first Onli
ne off-policy MARL framework using Diffusion policies (**OMAD**) to orchestrate
coordination. Our key innovation is a relaxed policy objective that maximizes sc
aled joint entropy, facilitating effective exploration without relying on tracta
ble likelihood. Complementing this, within the centralized training with decentr
alized execution (CTDE) paradigm, we employ a joint distributional value functio
n to optimize decentralized diffusion policies. It leverages tractable entropy-a
ugmented targets to guide the simultaneous updates of diffusion policies, thereb
y ensuring stable coordination. Extensive evaluations on MPE and MAMuJoCo establ
ish our method as the new state-of-the-art across 10 diverse tasks, demonstrat
ing a remarkable $2.5\times$ to $5\times$ improvement in sample efficiency.
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Aggregate Models, Not Explanations: Improving Feature Importance Estimation
Joseph Paillard, Angel David REYERO LOBO, Denis-Alexander Engemann, Bertrand Thi
rion
Feature-importance methods show promise for transforming machine learning (ML) m
odels from predictive engines into tools for scientific discovery. However, expr
essive models can be unstable due to data sampling and algorithmic stochasticity
, leading to inaccurate variable importance estimates, undermining their utility
 in critical biomedical applications. While ensembling offers a remedy, the choi
ce between explaining a single ensemble model or aggregating individual model ex
planations is non-trivial due to the non-linearity of importance measures, and r
emains largely understudied. Our theoretical analysis, developed under assumptio
ns accommodating complex state-of-the-art ML models, reveals that this choice is
 governed by a trade-off involving the model's excess risk. In contrast to prior
 literature, we show that ensembling at the model level provides more accurate v
ariable-importance estimates, particularly for expressive models, by reducing th
is leading error term. We validate these findings on classical benchmarks and a
large-scale proteomic study from the UK Biobank.
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Tvcache: A Tool-Value Cache for Post-Training LLM Agents
Abhishek Vijaya Kumar, Bhaskar Kataria, Byungsoo Oh, Emaad Manzoor, Rachee Singh
In RL post-training of LLM agents, calls to external tools take several seconds
or even minutes, leaving allocated GPUs idle and inflating post-training time an
d cost. While many tool invocations repeat across parallel rollouts and could in
 principle be cached, naively caching their outputs for reuse is incorrect since
 tool outputs depend on the environment state induced by prior agent interaction
s. We present TVCACHE, a stateful tool-value cache for LLM agent post-training.
TVCACHE maintains a tree of observed tool-call sequences and performs longest-pr
efix matching for cache lookups: a hit occurs only when the agent’s full tool hi
story matches a previously executed sequence, guaranteeing identical environment
 state. On three diverse workloads—terminal-based tasks, SQL generation, and vid
eo understanding—TVCACHE achieves cache hit rates of up to 70% and reduces media
n tool call execution time by up to 6.9×, with no degradation in post-training r
eward accumulation.
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3D Scene Assertion Verification
Jun Lin, Jiayu Ding, Xiangtian Si, Xitong Cao, Lixin Hong, Zhang Chen, Chenxi Lv

, Wenqian Wang
Existing 3D Visual Question Answering (3D-VQA) methods rely on generative outputs that can be ambiguous in decision-making settings. We introduce 3D Scene Assertion Verification, a task that verifies natural language assertions in 3D scenes with strict binary judgments. We present 3DSAV, a large-scale diagnostic benchmark with 22.5k samples across six semantic types. To address this task, we propose DualLPSS, which uses dual-stage subspace routing for type-aware cross-modal fusion and scene-guided assertion focusing. Experiments show that DualLPSS achieves state-of-the-art performance on 3DSAV and handles complex logical assertions better than existing 3D-VQA baselines.
****

The Differences Between Direct Alignment Algorithms are a Blur
Alexey Gorbatovski, Boris Shaposhnikov, Viacheslav Sinii, Alexey Malakhov, Daniil Gavrilov
Direct Alignment Algorithms (DAAs) simplify LLM alignment by directly optimizing policies, bypassing reward modeling and RL. While DAAs differ in their use of SFT (one-stage vs. two-stage) and the scalar score they optimize (likelihood vs. odds ratios), the key performance drivers remain underexplored. We present a systematic comparison and analyze a previously overlooked axis - the ranking objective (pairwise vs. pointwise). To isolate this factor, we propose a unified training framework across DAAs by (i) converting one-stage methods (ORPO, ASFT) into a two-stage pipeline with an explicit SFT phase and (ii) introducing a β parameter that places all methods in the same hyperparameter space and improves the quality of odds-ratio DAAs (ORPO, ASFT). Under this setup, the ranking objective emerges as the primary determinant of alignment quality, whereas the particular scalar score (policy–reference ratio vs. odds ratio) is secondary. We corroborate this on instruction-following tasks and further confirm it on math-reasoning benchmarks across model scales. Evidence suggests that this stems from how these objectives interact with prompt-specific biases, supported both by strictly controlled experiments and by observations on real data. Our findings underscore the need for nuanced evaluations in DAA research to avoid oversimplified claims of superiority.
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LAPRAS : Learning-Augmented PRivate Answering for linear query Streams.
Pranay Mundra, Adam Sealfon, Ziteng Sun, Quanquan C. Liu
Modern database workloads are highly predictable: query streams are dominated by recurring jobs and templates, even when their arrival order is not known in advance. This motivates a learning-augmented view of online differentially private (DP) analytics: can algorithms utilize predictions about *which* queries will occur to improve utility under a single global privacy budget, while remaining robust when predictions are wrong? We study online DP query answering, where a curator must answer a stream Q of S linear queries arriving in uniformly random order under privacy budget (ϵ,δ). We present *LAPRAS*, which assumes access to an oracle that outputs a prediction set of queries likely to appear in the stream and uses it to guide privacy spending. LAPRAS answers predicted queries using the offline-optimal Matrix Mechanism and answers the remaining queries online from a residual budget. To pace spending across an unknown number of unpredicted queries, we introduce *Smooth Allocation*, which forms an unbiased stopping-time estimate $\widehat{B}$ from the first $T=\Theta(\log^2 S)$ unpredicted queries and continuously recalibrates per-query expenditure. Empirically, over two real datasets, we validate the intended consistency--robustness trade-off: LAPRAS achieves near-offline utility under high overlap and degrades gracefully to baseline-level performance when overlap is low.
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Spectral Evolution Search: Efficient Inference-Time Scaling for Reward-Aligned Image Generation
Jinyan Ye, Zhongjie Duan, Zhiwen Li, Cen Chen, Daoyuan Chen, Yaliang Li, Yingda Chen
Inference-time scaling offers a flexible way to align visual generative models with downstream objectives without updating model parameters. In modern image gen

eration models, a natural way to do this is to optimize the random noise from wh
ich generation starts. However, searching in this high-dimensional noise space i
s highly inefficient, because many directions have little effect on the final im
age. We trace this inefficiency to a spectral bias in generative dynamics: model
 sensitivity to initial perturbations decays rapidly as frequency increases. Bas
ed on this insight, we propose Spectral Evolution Search (SES), a plug-and-play
framework for initial noise optimization that performs gradient-free evolutionar
y search in a low-frequency subspace. Theoretically, we derive the Spectral Scal
ing Prediction from perturbation propagation dynamics, which explains the freque
ncy-dependent impact of perturbations. Extensive experiments across diverse sett
ings show that SES substantially improves the trade-off between generation quali
ty and computational cost, consistently outperforming strong baselines under the
 same compute budget.
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MLLM-4D: Towards Visual-based Spatial-Temporal Intelligence

Xingyilang Yin, Chengzhengxu Li, Jiahao Chang, Chi-Man Pun, Xiaodong Cun

Humans are born with vision-based 4D spatial-temporal intelligence, which enable
s us to perceive and reason about the evolution of 3D space over time from purel
y visual inputs. Despite its importance, this capability remains a significant b
ottleneck for current multimodal large language models (MLLMs). To tackle this c
hallenge, we introduce MLLM-4D, a comprehensive framework designed to bridge the
 gaps in training data curation and model post-training for spatiotemporal under
standing and reasoning. On the data front, we develop a cost-efficient data cura
tion pipeline that repurposes existing stereo video datasets into high-quality 4
D spatiotemporal instructional data. This results in the MLLM4D-2M and MLLM4D-R1
-30k datasets for Supervised Fine-Tuning (SFT) and Reinforcement Fine-Tuning (RF
T), alongside MLLM4D-Bench for comprehensive evaluation. Regarding model trainin
g, our post-training strategy establishes a foundational 4D understanding via SF
T and further catalyzes 4D reasoning capabilities by employing Group Relative Po
licy Optimization (GRPO) with specialized Spatiotemporal Chain of Thought (ST-Co
T) prompting and Spatiotemporal reward functions (ST-reward) without involving t
he modification of architecture. Extensive experiments demonstrate that MLLM-4D
achieves state-of-the-art spatial-temporal understanding and reasoning capabilit
ies from purely 2D RGB inputs. Project page: https://github.com/GVCLab/MLLM-4D.
****

Improved Dynamic Algorithm for Non-monotone Submodular Maximization under Cardin
ality Constraint

Kiarash Banihashem, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade
, Morteza Monemizadeh

We study fully dynamic non-monotone submodular maximization under a cardinality
constraint k. Prior work achieved approximation guarantees of $(0.125-\epsilon
)$ using $\tilde{O}(\epsilon^{-1}k^2)$ oracle queries per update (NeurIPS'20) an
d 0.171 using $\tilde{O}(\epsilon^{-3}k^4)$ oracle queries per update (NeurIPS
'25). In this work, we present a dynamic algorithm that achieves a 0.262-appro
ximation with worst-case expected update time $O(\epsilon^{-3}k\log(k)\log(\epsi
lon^{-1}k) + \epsilon^{-2}k^2\log(k))$, where $0 < \epsilon \leq 1$ is the error
 parameter. We also develop another dynamic algorithm with update time bounded b
y $\mathrm{poly}(\epsilon^{-1},k)$ that achieves a 0.277-approximation guarant
ee.
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Helpful to a Fault: Measuring Illicit Assistance in Multi-Turn, Multilingual LLM
 Agents

Nivya Talokar, Ayush Kumar Tarun, Murari Mandal, Maksym Andriushchenko, Antoine
Bosselut

LLM-based agents execute real-world workflows via tools. These affordances enabl
e ill-intended adversaries to also use these agents to carry out complex misuse
scenarios. Existing agent-misuse benchmarks largely test single-prompt instructi
ons, leaving a gap in measuring how agents end up helping with harmful or illega
l tasks over multiple turns. We introduce **STING** (*Sequential Testing of Illi
cit N-step Goal execution*), an automated red-teaming framework that constructs

a step-by-step illicit plan grounded in a benign persona and iteratively probes a target agent with adaptive follow-ups, using judge agents to track phase completion. We further introduce an analysis framework that models multi-turn red-teaming as a time-to-first-jailbreak random variable, enabling analysis tools like discovery curves, hazard-ratio attribution by attack language, and a new metric: Restricted Mean Jailbreak Discovery. Across AgentHarm scenarios, STING yields substantially higher illicit-task completion than single-turn prompting and chat-oriented multi-turn baselines adapted to tool-using agents. In multilingual evaluations across six non-English settings, we find that attack success and illicit-task completion do *not* consistently increase in lower-resource languages, diverging from common chatbot findings. Overall, STING provides a practical way to evaluate and stress-test agent misuse in realistic deployment settings, where interactions are inherently multi-turn and often multilingual. Our code is available at https://github.com/epfl-nlp/helpful-to-a-fault.
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Diamond Maps: Efficient Reward Alignment via Stochastic Flow Maps
Peter Holderrieth, Douglas Chen, Luca Eyring, Ishin Shah, Giri Anantharaman, Yutong He, Zeynep Akata, Tommi Jaakkola, Nicholas Matthew Boffi, Max Simchowitz
Flow and diffusion models produce high-quality samples, but adapting them to user preferences or constraints post-training remains costly and brittle, a challenge commonly called reward alignment. We argue that efficient reward alignment should be a property of the generative model itself, not an afterthought, and redesign the model for adaptability. We propose Diamond Maps, a stochastic flow-map model that enables efficient and accurate alignment to arbitrary rewards at inference time. Diamond Maps amortize many simulation steps into a single-step sampler, like flow maps, while preserving the stochasticity required for optimal reward adaptation. This design makes search, Sequential Monte Carlo, and guidance scalable by enabling efficient and consistent estimation of the value function. Our experiments show that Diamond Maps can be learned efficiently via distillation from GLASS Flows, achieve stronger reward-alignment performance, and scale better than existing alignment methods. Overall, our results point toward a practical route to generative models that can be rapidly adapted to arbitrary preferences and constraints at inference time.
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VLA-ATTC: Adaptive Test-Time Compute for VLA Models with Relative Action Critic Model
Wenhao Li, Xiu Su, Yichao Cao, Hongyan Xu, Xiaobo Xia, Shan You, Yi Chen, Chang Xu
Vision-Language-Action (VLA) models have demonstrated remarkable capabilities and generalization in embodied manipulation. However, their decision-making relies on a fast, instinctive process that lacks deliberation. This strategy often leads to suboptimal or catastrophic actions when facing complex or ambiguous scenarios that require greater consideration. In this paper, we introduce \textbf{VLA-ATTC}, a framework that endows VLA models with adaptive test-time compute (TTC). VLA-ATTC employs an uncertainty-based ``cognitive clutch'' to dynamically transition from reflexive execution to a TTC deliberation phase when necessary. During TTC phase, a novel \textbf{Relative Action Critic} (RAC) model identifies the optimal action from generated candidates via pairwise comparisons. This relative mechanism replaces unstable absolute value estimation, significantly simplifying the learning objective. Furthermore, we introduce an efficient sampling strategy to amortize computational costs and an automated data pipeline that curates preference pairs without manual annotation. On the LIBERO-LONG benchmark, VLA-ATTC reduces the failure rate of the SOTA model PI0.5 by over 50\%.
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Adversarial Dual On-Policy Distillation from Expressive Teacher
Zhenglin Wan, Jingxuan Wu, Xingrui Yu, Chubin Zhang, Mingcong Lei, Bo An, Ivor Tsang, Yang You
Learning from demonstrations in embodied control is often cast as behavioral cloning, and recent diffusion or flow-matching policies improve this paradigm by modeling multi-modal expert actions. Yet these methods remain offline supervised l

earners: the policy is trained only on expert states and receives no corrective signal on the states it actually visits. On-policy distillation (OPD) offers a natural remedy, but standard OPD assumes a strong fixed teacher, which is unavailable in demonstration-only control. We propose FA-OPD, an adversarial dual on-policy distillation method in which a Flow Matching (FM) teacher is learned from demonstrations and co-trained with a lightweight MLP student. The teacher provides two complementary signals on student rollouts. The reward channel learns an expert-likeness objective over state-action pairs and drives online exploration through long-horizon policy optimization. The action channel supplies dense local targets at student-visited states, stabilizing exploitation. FA-OPD couples them so that reward distillation enables generalization beyond point-wise demonstrations, while action distillation keeps exploration anchored near expert-like behavior. Across six robot navigation, manipulation, and locomotion benchmarks, FA-OPD beats strong baselines and shows much stronger robustness under noisy or limited demonstrations. Source code: https://github.com/vanzll/FA-OPD.
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Zero Sum SVD: Balancing Loss Sensitivity for Low Rank LLM Compression
Ali Abbasi, Chayne Thrash, Haoran Qin, Shansita Sharma, Sepehr Seifi, Soheil Kolouri
Advances in large language models have driven strong performance across many tasks, but their memory and compute costs still hinder deployment. SVD-based compression reduces storage and can speed up inference via low-rank factors, yet performance depends on how rank is allocated under a global compression ratio. Prior methods often use homogeneous ranks for similarly sized matrices, despite large differences in loss sensitivity, or rely on expensive iterative pre-truncation optimization to determine per matrix ranks. We propose **Zero Sum SVD** (**ZS-SVD**), a post-training method that performs *global* singular component selection using activation whitening and first-order calibration loss estimates in whitened coordinates. **ZS-SVD** prunes components across the whole model with a **zero sum** rule that keeps the cumulative predicted loss change near zero, automatically yielding heterogeneous ranks without solving a rank allocation optimization. Motivated by evidence that gradients near pretrained solutions exhibit low rank structure, we also introduce an optional lightweight correction that applies a **single** projected gradient update after truncation, followed by re-truncation. Extensive experiments across multiple LLM architectures show consistent gains across diverse benchmarks and compression ratios. Code is available at https://github.com/mint-vu/Zero-Sum-SVD
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Universal Algorithm-Implicit Learning
Stefano Woerner, Seong Joon Oh, Christian F. Baumgartner
Current meta-learning methods are constrained to narrow task distributions with fixed feature and label spaces, limiting applicability. Moreover, the current meta-learning literature uses key terms like “universal” and “general-purpose” inconsistently and lacks precise definitions, hindering comparability. We introduce a theoretical framework for meta-learning which formally defines practical universality and introduces a distinction between algorithm-explicit and algorithm-implicit learning, providing a principled vocabulary for reasoning about universal meta-learning methods. Guided by this framework, we present TAIL, a transformer-based algorithm-implicit meta-learner that functions across tasks with varying domains, modalities, and label configurations. TAIL features three innovations over prior transformer-based meta-learners: random projections for cross-modal feature encoding, random injection label embeddings that extrapolate to larger label spaces, and efficient inline query processing. TAIL achieves state-of-the-art performance on standard few-shot benchmarks while generalizing to unseen domains. Unlike other meta-learning methods, it also generalizes to unseen modalities, solving text and audio classification tasks despite training exclusively on images, handles tasks with up to 20× more classes than seen during training, and provides orders-of-magnitude computational savings over prior transformer-based approaches.
****

VEQ: Modality-Adaptive Quantization for MoE Vision-Language Models
Guangshuo Qin, Zhiteng Li, Zheng Chen, Weihang Zhang, Linghe Kong, Yulun Zhang
Mixture-of-Experts(MoE) Vision-Language Models(VLMs) offer remarkable performanc e but incur prohibitive memory and computational costs, making compression essen tial. Post-Training Quantization (PTQ) is an effective training-free technique t o address the massive memory and computation overhead. Existing quantization par adigms fall short as they are oblivious to two critical forms of heterogeneity: the inherent discrepancy between vision and language tokens, and the non-uniform contribution of different experts. To bridge this gap, we introduce Visual Expe rt Quantization (VEQ), a dual-aware quantization framework designed to simultane ously accommodate cross-modal differences and heterogeneity between experts. Spe cifically, VEQ incorporates 1)**Modality-expert-aware Quantization**, which util izes expert activation frequency to prioritize error minimization for pivotal ex perts, and 2)**Modality-affinity-aware Quantization**, which constructs an enhan ced Hessian matrix by integrating token-expert affinity with modality informatio n to guide the calibration process. Extensive experiments across diverse benchma rks verify that VEQ consistently outperforms state-of-the-art baselines. Specifi cally, under the W3A16 configuration, our method achieves significant average ac curacy gains of 2.04\% on Kimi-VL and 3.09\% on Qwen3-VL compared to the previou s SOTA quantization methods, demonstrating superior robustness across various mu lti-modal tasks.
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Group Cognition Learning: Making Everything Better Through Controlled Two-Stage Agents Collaboration
Chunlei Meng, Pengbin Feng, Rong Fu, Hoi Leong Lee, Xiaojing Du, Zhaolu Kang, Ze yu Zhang, Weilin Zhou, Chun Ouyang, Zhongxue Gan
Centralized multimodal learning commonly compresses language, acoustic, and visu al signals into a single fused representation for prediction. While effective, t his paradigm suffers from two limitations: modality dominance, where optimizatio n gravitates towards the path of least resistance, ignoring weaker but informati ve modalities, and spurious modality coupling, where models overfit to incidenta l cross-modal correlations. To address these, we propose \textbf{Group Cognition Learning (GCL)}, a governed collaboration paradigm that applies a two-stage pro tocol after modality-specific encoding. In Stage 1 (Selective Interaction), a Ro uting Agent proposes directed interaction routes, and an Auditing Agent assigns sample-wise gates to emphasize exchanges that yield positive marginal predictive gain while suppressing redundant coupling. In Stage 2 (Consensus Formation), a Public-Factor Agent maintains an explicit shared factor, and an Aggregation Agen t produces the final prediction through contribution-aware weighting while keepi ng each modality representation as a specialization channel. Extensive experimen ts on CMU-MOSI, CMU-MOSEI, and MIntRec demonstrate that GCL mitigates dominance and coupling, establishing state-of-the-art results across both regression and c lassification benchmarks. Analysis experiments further demonstrate the effective ness of the design.
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How do LLMs Compute Verbal Confidence?
Dharshan Kumaran, Arthur Conmy, Federico Barbero, Simon Osindero, Viorica Patrau cean, Petar Veli■kovi■
Verbal confidence—prompting LLMs to state their confidence as a number or catego ry—is widely used to extract uncertainty estimates from black-box models. Howeve r, how LLMs internally generate such scores remains unknown. We address two ques tions: first, when confidence is computed -- just-in-time when requested, or aut omatically during answer generation and cached for later retrieval; and second, what verbal confidence represents -- token log-probabilities, or a richer evalua tion of answer quality? Focusing on Gemma 3 27B (across TriviaQA, BigMath, and M MLU), Qwen 2.5 7B, and the reasoning model Magistral Small 24B, we provide conve rgent evidence for cached retrieval. Activation steering, patching, noising, and swap experiments reveal that confidence representations emerge at answer-adjace nt positions before appearing at the verbalization site. Attention blocking pinp oints the information flow: confidence is gathered from answer tokens, cached at

the first post-answer position, then retrieved for output. Critically, linear probing and variance partitioning reveal that these cached representations explain substantial variance in verbal confidence beyond token log-probabilities, suggesting a richer answer-quality evaluation rather than a simple fluency readout. These findings demonstrate that verbal confidence reflects automatic, sophisticated self-evaluation—not post-hoc reconstruction—with implications for understanding metacognition in LLMs and improving calibration.
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MODUS: Decoder-only Any-to-Any Modeling of Diverse Modalities
Mingqiao Ye, Zhaochong An, Zhitong Gao, Xian Liu, O■uzhan Fatih Kar, Jesse Allardice, Roman Bachmann, David Mizrahi, François Fleuret, Chuan Li, Amir Zadeh, Serge Belongie, Afshin Dehghan, Amir Zamir
Any-to-any modeling aims to flexibly relate arbitrary modalities within a single system, a requirement that arises across multimodal learning and scientific domains such as ecology and astronomy.
However, existing any-to-any approaches are typically trained from scratch using encoder–decoder or diffusion architectures, limiting empirical performance and the use of pretrained models.
We investigate decoder-only any-to-any multimodal modeling, which treats all modalities symmetrically and supports arbitrary modalities as inputs and outputs without modality-specific heads, losses, or task pipelines.
As a consequence of this unified design, the resulting model MODUS naturally enables chained generation through intermediate modalities, cross-modal consistency verification, and analysis of visual representations by combining semantic and reconstruction features.
Across a range of benchmarks, MODUS demonstrates strong out-of-the-box performance and flexible multimodal composition within a single model.
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Latent Laplace Diffusion for Irregular Multivariate Time Series
Zinuo You, Jin Zheng, John Cartlidge
Irregular multivariate time series impose a trade-off for long-horizon forecasting: discrete methods can distort temporal structure via re-gridding, while continuous-time models often require sequential solvers prone to drift. To bridge this gap, we present Latent Laplace Diffusion (LLapDiff), a generative framework that models the target as a low-dimensional latent trajectory, enabling horizon-wide generation without step-by-step integration over physical time. We guide the reverse process utilizing a stable modal parameterization motivated by stochastic port-Hamiltonian dynamics, and parameterize its mean evolution in the Laplace domain via learnable complex-conjugate poles, enabling direct evaluation over irregular timestamps. We also link continuous dynamics to irregular observations through renewal-averaging analysis, which maps sampling gaps to effective event-domain poles and motivates a gap-aware history summarizer. Extensive experiments show that LLapDiff improves over baselines in long-horizon forecasting, and its continuous-time generative nature supports missing-value imputation by querying the same model at historical timestamps. Code is available at \url{https://github.com/pixelhero98/LLapDiffusion}.
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3D-RFT: Reinforcement Fine-Tuning for Video-based 3D Scene Understanding
Xiongkun Linghu, Jiangyong Huang, Baoxiong Jia, Siyuan Huang
Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a key paradigm for unlocking complex reasoning in Large Language Models (LLMs), yet its potential in 3D scene understanding remains untapped. To bridge this gap, we present Reinforcement Fine-Tuning for Video-based 3D Scene Understanding (3D-RFT), the first framework to extend RLVR to 3D perception and reasoning. Our pipeline operates in two stages: activating 3D-aware Multi-modal Large Language Models (MLLMs) via Supervised Fine-Tuning (SFT), followed by reinforcement fine-tuning using Group Relative Policy Optimization (GRPO) with strictly verifiable reward functions. We design task-specific rewards—such as 3D IoU and F1-score—to provide deterministic signals for spatial alignment. Extensive experiments demonstrate that 3D-RFT achieves state-of-the-art performance on video-based 3D scene understand

ing benchmarks, significantly outperforming VG LLM-8B on detection and grounding tasks. Moreover, our model surpasses larger mainstream models on VSI-Bench, demonstrating the efficiency of verifiable reinforcement learning. We conclude by offering valuable insights into optimal training strategies .

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Physics-Informed Distillation of Diffusion Models for PDE-Constrained Generation
Yi Zhang, Peng Wang, Difan Zou

Diffusion models show growing promise for generative modeling of physical systems, but enforcing partial differential equation (PDE) constraints directly is infeasible during the stochastic denoising process. Current methods apply constraints to the expected clean sample, incurring a Jensen’s Gap that forces a trade-off between PDE satisfaction and generative accuracy. To bridge this gap, we propose Physics-Informed Distillation of Diffusion Models (PIDDM), a simple yet effective post-hoc distillation strategy that enforces PDE constraints after training. PIDDM enables fast single-step generation while improving both physical consistency and sample quality, supporting forward/inverse problems and reconstruction from partial observations. Extensive experiments across PDE benchmarks show PIDDM outperforms recent baselines, such as PIDM, DiffusionPDE, and ECI-sampling, in both accuracy and constraint satisfaction, with lower computation and minimal hyperparameter tuning, offering a more efficient pathway to physics-informed diffusion models.

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Decomposing Query-Key Feature Interactions Using Contrastive Covariances
Andrew Lee, Yonatan Belinkov, Fernanda Viégas, Martin Wattenberg

Despite the central role of attention heads in Transformers, we lack tools to understand why a model attends to a particular token. To address this, we study the query-key (QK) space -- the bilinear joint embedding space between queries and keys. We present a contrastive covariance method to decompose the QK space into low-rank, human-interpretable components. It is when features in keys and queries align in these low-rank subspaces that high attention scores are produced. We first study our method both analytically and empirically in a simplified setting. We then apply our method to large language models to identify human-interpretable QK subspaces for categorical semantic features and binding features. Finally, we demonstrate how attention scores can be attributed to our identified features.

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Self-Refining Video Sampling
Sangwon Jang, Taekyung Ki, Jaehyeong Jo, Saining Xie, Jaehong Yoon, Sung Ju Hwang

Modern video generators still struggle with complex physical dynamics, often falling short of physical realism. Existing approaches address this using external verifiers or additional training on augmented data, which is computationally expensive and still limited in capturing fine-grained motion. In this work, we present self-refining video sampling, a simple method that uses a pre-trained video generator trained on large-scale datasets as its own self-refiner. By interpreting the generator as a denoising autoencoder, we enable iterative inner-loop refinement at inference time without any external verifier or additional training. We further introduce an uncertainty-aware refinement strategy that selectively refines regions based on self-consistency, which prevents artifacts caused by over-refinement. Experiments on state-of-the-art video generators demonstrate significant improvements in motion coherence and physics alignment, achieving over 70% human preference compared to the default sampler and guidance-based sampler.

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PCA of Probability Measures: Sparse and Dense Sampling Regimes
Erell Gachon, Jérémie Bigot, Elsa Cazelles

A common approach to perform PCA on probability measures is to embed them into a Hilbert space where standard functional PCA techniques apply. While convergence rates for estimating the embedding of a single measure from m samples are well understood, the literature has not addressed the setting involving multiple measures. In this paper, we study PCA in a double asymptotic regime where n prob

ability measures are observed, each through m samples. We derive convergence r
ates of the form $n^{-1/2} + m^{-\alpha}$ for the empirical covariance operator
and the PCA excess risk, where $\alpha>0$ depends on the chosen embedding. This
characterizes the relationship between the number n of measures and the numbe
r m of samples per measure, revealing a sparse (small m) to dense (large m
) transition in the convergence behavior. Moreover, we prove that the dense-regi
me rate is minimax optimal for the empirical covariance error. Our numerical exp
eriments validate these theoretical rates and demonstrate that appropriate subsa
mpling preserves PCA accuracy while reducing computational cost.
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Clipping Low-Probability Tokens in SFT Yields a Generalizable Initialization for
RL
Tian-Shuo Liu, Chengxing Jia, Haoyu Liu, Pengyuan Wang, Shiyuan Zhang, Jie Fu, Y
ang Yu
Supervised Fine-Tuning (SFT) is a critical step for adapting Large Language Mode
ls (LLMs) to specialized domains, often serving as an initialization for subsequ
ent reinforcement learning (RL). However, SFT can overfit a small set of expert
data, harming generalization and eroding prior knowledge. This can limit downstr
eam RL, which benefits from a strong, generalizable initialization for explorati
on. Here, we demonstrate that prior knowledge degradation primarily results from
tokens in the expert data to which the base model assigns low probability. Spec
ifically, these low-probability tokens represent a significant deviation from th
e model’s current prior knowledge. Due to the nature of the log-likelihood objec
tive, they produce larger gradient magnitudes, which speed up adaptation to the
new data but degrade generalization. In this paper, we study the token-wise clip
ping strategy, a commonly used trust-region method for bounding per-token update
s.
We find that it reshapes token-level learning priorities, promoting more progres
sive adaptation that fits the new data while preserving general abilities.
Compared with standard SFT, clipping low-probability tokens reduces out-of-distr
ibution forgetting by 11.54\% and improves final RL performance by 7.09\% across
the agentic benchmarks. Moreover, latent-space analysis shows smaller represent
ational drift under clipping, indicating that it provides a generalizable initia
lization.
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Extending Prediction-Powered Inference through Conformal Prediction
Daniel Csillag, Pedro Dall'Antonia, Claudio Jose Struchiner, Guilherme Tegoni Go
edert
Prediction-powered inference is a recent methodology for the safe use of black-b
ox ML models to impute missing data, strengthening inference of statistical para
meters. However, many applications require strong properties besides valid infer
ence, such as privacy, robustness or validity under continuous distribution shif
ts; deriving prediction-powered methods with such guarantees is generally an ard
uous process, and has to be done case by case. In this paper, we resolve this is
sue by connecting prediction-powered inference with conformal prediction: by per
forming imputation through a calibrated set-predictor, we attain validity while
achieving additional guarantees in a natural manner. We instantiate our procedur
e for the inference of means, Z- and M-estimation, as well as e-values and e-val
ue-based procedures. Furthermore, in the case of e-values, ours is the first gen
eral prediction-powered procedure that operates off-line. We demonstrate these a
dvantages by applying our method on private and time-series data. Both tasks are
nontrivial within the standard prediction-powered framework but become natural
under our method.
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Alethia: a Foundational Encoder for Voice Deepfakes
Yi Zhu, Brahmi Dwivedi, Jayaram Raghuram, Surya Koppisetti
Existing voice deepfake detection and localization models rely heavily on repres
entations extracted from speech foundation models (SFMs). However, downstream fi
netuning has now reached a state of diminishing returns. In this paper, we shift
the focus to pretraining and propose a novel recipe that combines *bottleneck m

asked embedding prediction with flow-matching based spectrogram reconstruction*. The outcome, *Alethia*, is the first foundational audio encoder for various voice deepfake detection and localization tasks. We evaluate on 5 different tasks with 56 benchmark datasets, and note *Alethia* significantly outperforms state-of-the-art SFMs with superior robustness to real-world perturbations and zero-shot generalization to unseen domains (e.g., singing deepfakes). We also demonstrate the limitation of discrete targets in masked token prediction, and show the importance of *continuous embedding* prediction and *generative pretraining* for capturing deepfake artifacts.

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Improving the Performance and Learning Stability of Parallelizable RNNs Designed for Ultra-Low Power Applications

Julien Brandoit, Arthur Fyon, Damien Ernst, Guillaume Drion

Sequence learning is dominated by Transformers and parallelizable recurrent neural networks (RNNs) such as state-space models, yet learning long-term dependencies remains challenging, and state-of-the-art designs trade power consumption for performance. The Bistable Memory Recurrent Unit (BMRU) was introduced to enable hardware-software co-design of ultra-low power RNNs: quantized states with hysteresis provide persistent memory while mapping directly to analog primitives. However, BMRU performance lags behind parallelizable RNNs on complex sequential tasks. In this paper, we identify gradient blocking during state updates as a key limitation and propose a cumulative update formulation that restores gradient flow while preserving persistent memory, creating skip-connections through time. This leads to the Cumulative Memory Recurrent Unit (CMRU) and its relaxed variant, the αCMRU. Experiments show that the cumulative formulation dramatically improves convergence stability and reduces initialization sensitivity. The CMRU and αCMRU match or outperform Linear Recurrent Units (LRUs) and minimal Gated Recurrent Units (minGRUs) across diverse benchmarks at small model sizes, with particular advantages on tasks requiring discrete long-range retention, while the CMRU retains quantized states, persistent memory, and noise-resilient dynamics essential for analog implementation.

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Attention Projection Mixing with Exogenous Anchors

Jonathan Su

Cross-layer reuse of early attention projections can improve optimization and data efficiency, but it creates a structural conflict: the first layer must simultaneously act as a stable, reusable anchor for all deeper layers and as an effective computational block. We demonstrate that this tension constrains the performance of internal-anchor designs. We propose ExoFormer, which resolves the conflict by learning exogenous anchor projections outside the sequential layer stack. We introduce a unified normalized mixing framework that mixes queries, keys, values, and gate logits using learnable coefficients (exploring coefficient granularities: elementwise, headwise, and scalar), and we show that normalizing anchor sources is key to stable reuse. ExoFormer variants consistently outperform their internal-anchor counterparts, and the dynamic variant yields 1.5x downstream accuracy points while matching validation loss using 1.5x fewer tokens than Gated Attention. We explain this efficacy via an Offloading Hypothesis: external anchors preserve essential token identity, allowing layers to specialize exclusively in feature transformation. We release code and models to facilitate future research.

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R1-SyntheticVL: Is Synthetic Data from Generative Models Ready for Multimodal Large Language Model?

Jingyi Zhang, Tianyi Lin, Huanjin Yao, Xiang Lan, Shunyu Liu, Jiaxing Huang

In this work, we aim to develop effective data synthesis techniques that autonomously synthesize multimodal training data for enhancing MLLMs in solving complex real-world tasks. To this end, we propose Collective Adversarial Data Synthesis (CADS), a novel and general approach to synthesize high-quality, diverse and challenging multimodal data for MLLMs. The core idea of CADS is to leverage collective intelligence to ensure high-quality and diverse generation, while exploring

adversarial learning to synthesize challenging samples for effectively driving model improvement. Specifically, CADS operates with two cyclic phases, i.e., Collective Adversarial Data Generation (CAD-Generate) and Collective Adversarial Data Judgment (CAD-Judge). CAD-Generate leverages collective knowledge to jointly generate new and diverse multimodal data, while CAD-Judge collaboratively assesses the quality of synthesized data. In addition, CADS introduces an Adversarial Context Optimization mechanism to optimize the generation context to encourage challenging and high-value data generation. With CADS, we construct MMSynthetic-20K and train our model R1-SyntheticVL, which demonstrates superior performance on various benchmarks.
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Learning Robust Multi-Agent Policies via Selective Adversarial Fault Induction
David Henry Mguni, Yaqi Sun, Haojun Chen, Wanrong Yang, Amir Darabi, Larry Olanrewaju Orimoloye, Yaodong Yang
We study robustness to agent malfunctions in cooperative multi-agent reinforcement learning (MARL), a failure mode that is critical in practice yet underexplored in existing theory. We introduce MARTA, a plug-and-play robustness layer that augments standard MARL algorithms with a Switcher–Adversary mechanism which selectively induces malfunctions in performance-critical states. This formulation defines a fault-switching $(N+2)$-player Markov game in which the Switcher chooses when and which agent fails, and the Adversary controls the resulting faulty behaviour via random or worst-case policies. We develop a Q-learning-type scheme and show that the associated Bellman operator is a contraction, yielding existence and uniqueness of the minimax value, convergence to a Markov perfect equilibrium. MARTA integrates seamlessly with MARL algorithms without architectural modification and consistently improves robustness across Traffic Junction (TJ), Level-Based Foraging (LBF), MPE SimpleTag, and SMAC (v2). In these domains, MARTA achieves large gains in final performance of up to 116.7\% in SMAC, 21.4\% in MPE SimpleTag, and 44.6\% in LBF, while significantly reducing failure rates under train–test mismatched fault regimes. These results establish MARTA as a theoretically grounded and practically deployable mechanism for fault-tolerant MARL.
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Fast KV Compaction via Attention Matching
Adam Zweiger, Xinghong Fu, Han Guo, Yoon Kim
Scaling language models to long contexts is often bottlenecked by the size of the key-value (KV) cache. In deployed settings, long contexts are typically managed through *compaction* in token space via summarization. However, summarization can be highly lossy, substantially harming downstream performance. Recent work on Cartridges (Eyuboglu et al., 2025) has shown that it is possible to *train* highly compact KV caches in latent space that closely match full-context performance, but at the cost of slow and expensive end-to-end optimization. This work describes an approach for *fast* context compaction in latent space through **Attention Matching**, which constructs compact keys and values to reproduce attention outputs and preserve attention mass at a per-KV-head level. We show that this formulation naturally decomposes into simple subproblems, some of which admit efficient closed-form solutions. Within this framework, we develop a family of methods that significantly push the Pareto frontier of compaction time versus quality, achieving up to $50\times$ compaction in seconds on some datasets with little quality loss.
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Temporal Context Reinstatement Drives Episodic-Like Order Memory in Long-Context Language Models
Mathis Pink, Vy A. Vo, Qinyuan Wu, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, Mariya Toneva
Human episodic memory supports the retrieval of experiences that unfold over extended timescales, yet the computational mechanisms underlying this ability remain debated due to the limited mechanistic accessibility in long-term memory experiments in humans. Long-context LLMs may offer promising ways to reveal plausible computational mechanisms that drive this type of retrieval. Here, we investigate whether and how LLMs capture the core behavioral signatures of episodic memory

via a temporal order memory task. Using a new dataset of human behavior based on memory of a full-length novel, we show that models exhibit the same characteristic distance effect observed in humans on this task. We next apply long-context mechanistic interpretability analyses to uncover how models solve this task, and find that model performance relies on a one-dimensional temporal code that is reinstated during retrieval by a single time-reinstatement attention head. These findings support temporal context reinstatement as an important mechanism for episodic-like temporal-order memory in LLMs, offering new insights into how temporal aspects of long-term episodic memory may be instantiated in both artificial and biological systems.
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Latent Collaboration in Multi-Agent Systems
Jiaru Zou, Ruizhong Qiu, Gaotang Li, Xiyuan Yang, Katherine Tieu, Pan Lu, Ke Shen, Hanghang Tong, Yejin Choi, Jingrui He, James Zou, Mengdi Wang, Ling Yang
Multi-agent systems (MAS) extend large language models (LLMs) from independent single-model reasoning to coordinative system-level intelligence. While existing LLM agents depend on text-based mediation for reasoning and communication, we take a step forward by enabling models to collaborate directly within the continuous latent space. We introduce LatentMAS, an end-to-end training-free framework that enables pure latent collaboration among LLM agents. In LatentMAS, each agent first performs auto-regressive latent thoughts generation through last-layer hidden embeddings instead of text. Then, a shared latent working memory preserves and transfers each agent's internal representations and latent thoughts, ensuring lossless information exchange without re-encoding. We provide detailed theoretical analyses showing that LatentMAS achieves higher expressiveness and lossless information preservation with lower overall complexity than standard text-based MAS. In addition, empirical evaluations across 9 comprehensive benchmarks spanning math and science reasoning, commonsense understanding, and code generation show that LatentMAS outperforms advanced single agents and text-based MAS baselines, achieving up to 14.6\% higher accuracy, reducing output token usage by 70.8\%-83.7\%, and providing 4$\times$-4.3$\times$ faster end-to-end inference.
****

MentisOculi: Revealing the Limits of Reasoning with Mental Imagery
Jana Ricarda Zeller, Thaddäus Wiedemer, Fanfei Li, Thomas Klein, Prasanna Mayilvahanan, Matthias Bethge, Felix A. Wichmann, Ryan Cotterell, Wieland Brendel
Frontier models are transitioning from _multimodal large language models_ (MLLMs) that merely ingest visual information to _unified multimodal models_ (UMMs) capable of native interleaved generation.
This shift has sparked interest in using intermediate visualizations as a reasoning aid, akin to human _mental imagery_.
Central to this idea is the ability to form, maintain, and manipulate visual representations in a goal-oriented manner.
To evaluate and probe this capability, we develop MentisOculi, a procedural, stratified suite of multi-step reasoning problems amenable to visual solution, tuned to challenge frontier models.
Evaluating visual strategies ranging from latent tokens to explicit generated imagery, we find they generally fail to improve performance.
Analysis of UMMs specifically exposes a critical limitation:
While they possess the textual reasoning capacity to solve a task and can sometimes generate correct visuals, they suffer from compounding generation errors and fail to leverage even ground-truth visualizations.
Our findings suggest that despite their inherent appeal, _visual thoughts do not yet benefit model reasoning_.
MentisOculi establishes the necessary foundation to analyze and close this gap across diverse model families.
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General Synthetic-Powered Inference
Meshi Bashari, Yonghoon Lee, Roy Maor Lotan, Edgar Dobriban, Yaniv Romano
The rapid proliferation of high-quality synthetic data---generated by advanced AI models or collected as auxiliary data from related tasks---presents both oppor

tunities and challenges for statistical inference. This paper introduces a GEner al Synthetic-Powered Inference (GESPI) framework that wraps around any statistic al inference procedure to safely enhance sample efficiency by combining syntheti c and real data. Our framework leverages high-quality synthetic data to boost st atistical power, yet adaptively defaults to the standard method using only real data when synthetic data are of low quality.
The error rate of our method remains below a user-specified bound without any di stributional assumptions on the synthetic data, and decreases as the quality of the synthetic data improves.
This flexibility enables seamless integration with conformal prediction, risk co ntrol, hypothesis testing, and multiple testing procedures, all without modifyin g the base inference method.
We demonstrate the benefits of our method on challenging tasks with limited labe led data, including AlphaFold protein structure prediction, and comparing large reasoning models on complex math problems.
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iGRPO: Fast Online RL for Flow Matching Model with Instant Reward
Sucheng Ren, Chen Chen, Zhenbang Wang, Liangchen Song, Xiangxin Zhu, Yinfei Yang , Jiasen Lu
Conventional practice assumes that online reinforcement learning for flow-matchi ng models requires sampling full denoising trajectories to compute rewards.
This assumption underlies methods such as Group Relative Policy Optimization (GR PO), where the policy must traverse the entire reverse process before receiving a delayed, trajectory-level reward.
We observe, however, that while such terminal rewards provide feedback, they are neither necessary nor optimal for effective learning.
In this work, we introduce iGRPO (Instant-reward GRPO), which replaces GRPO's fu ll-trajectory rollouts with a single-step mapping that assigns rewards instantly at each denoising step. Because the flow matching model behaves differently acr oss timesteps, our step-local instant rewards which are inherently time-dependen t, overcome prior approaches that rely on a single, time-independent terminal re ward.
By evaluating each action locally rather than relying on a final terminal score, iGRPO eliminates the need for multi-step SDE rollouts and offers more precise c redit assignment.
Across standard benchmarks, iGRPO converges 10.2× faster than FlowGRPO while ach ieving higher final alignment quality. We hope this work motivates more efficien t and scalable online RL methods for flow-matching generative models.
****

CAST: Modeling Visual State Transitions for Consistent Video Retrieval
Yanqing Liu, Yingcheng Liu, Fanghong Dong, Budianto Budianto, Cihang Xie, Yan ji ao
As video content creation shifts toward long-form narratives, composing short cl ips into coherent storylines becomes increasingly important. However, prevailing retrieval formulations remain context-agnostic at inference time, prioritizing local semantic alignment while neglecting the state and identity consistency. To address this structural limitation, we formalize the task of **Consistent Video Retrieval (CVR)** and introduce a diagnostic benchmark spanning YouCook2, COIN, and CrossTask. We propose **CAST (Context-Aware State Transition)**, a lightwei ght, plug-and-play adapter compatible with diverse frozen vision-language embedd ing spaces. By predicting a state-conditioned residual update (Δ) from visual hi story, CAST introduces an explicit inductive bias for latent state evolution. Ex tensive experiments show that CAST improves performance on YouCook2 and CrossTas k, remains competitive on COIN, and consistently outperforms zero-shot baselines across diverse foundation backbones. Furthermore, CAST provides a useful rerank ing signal for black-box video generation candidates (e.g., from Veo), promoting more temporally coherent continuations.
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Human-in-the-Loop Policy Optimization for Preference-Based Multi-Objective Reinf orcement Learning

Tianmeng Hu, Biao Luo, Ke Li
Multi-objective reinforcement learning (MORL) seeks policies that effectively balance conflicting objectives. However, presenting many diverse policies without accounting for the decision maker’s (DM’s) preferences can overwhelm the decision-making process. On the other hand, accurately specifying preferences in advance is often unrealistic. To address these challenges, we introduce a human-in-the-loop MORL framework that interactively discovers preferred policies during optimization. Our approach proactively learns the DM’s implicit preferences in real time, requiring no a priori knowledge. Importantly, we integrate this preference learning directly into a parallel optimization framework, balancing exploration and exploitation to identify high-quality policies aligned with the DM's preferences. Evaluations on a complex quadrupedal robot simulation environment demonstrate that, with only interactions, our proposed method can identify policies aligned with human preferences, e.g., running like a dog. Further experiments on seven MuJoCo tasks and a multi-microgrid system design task against eight state-of-the-art MORL algorithms fully demonstrate the effectiveness of our proposed framework. Demonstrations and full experiments are in https://sites.google.com/view/pbmorl/home.
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Safety-Efficacy Trade Off: Robustness against Data-Poisoning
Diego Marco Granziol, Ulugbek Abdimanabov
Backdoor and data-poisoning attacks can achieve high attack success while evading existing spectral and optimisation-based defences. We show that this behaviour is not incidental, but arises from a fundamental geometric mechanism in input space. Using kernel ridge regression as an exact model of wide neural networks, we prove that clustered dirty-label poisons induce a rank-one spike in the input Hessian whose magnitude scales quadratically with attack efficacy. Crucially, for nonlinear kernels we identify a near-clone regime in which poison efficacy remains order-one while the induced input curvature vanishes, making the attack provably spectrally undetectable. We further show that input-gradient regularisation contracts poison-aligned Fisher and Hessian eigenmodes under gradient flow, yielding an explicit and unavoidable safety–efficacy trade-off by reducing data-fitting capacity. For exponential kernels, this defence admits a precise interpretation as an anisotropic high-pass filter that increases the effective length scale and suppresses near-clone poisons. Extensive experiments on linear models and deep convolutional networks across MNIST and CIFAR-10/100 validate the theory, demonstrating consistent lags between attack success and spectral visibility, and showing that regularisation and data augmentation jointly suppress poisoning. Our results establish when backdoors are inherently invisible, and provide the first end-to-end characterisation of poisoning, detectability, and defence through input-space curvature.
****
TGV-KV: Text-Grounded KV Eviction for Vision-Language Models
Jizhihui Liu, Ruizi Han, Miao Zhang, Rui Shao, Xuebo Liu, Weili Guan, Yaowei Wang
Vision-Language Models (VLMs) inherit the auto-regressive generation paradigm and cache the keys and values (KV) of all previous tokens to accelerate inference, resulting in memory consumption that scales linearly with context length. This issue is particularly pronounced in VLMs due to substantial redundancy in the visual modality. Although KV cache eviction approaches can effectively reduce inference memory, they often incur significant performance degradation in VLMs, as most are designed for language models and overlook the inherent gap between text and vision. By systematically analyzing the modality gap in VLMs in this work, we argue that the importance of visual information should be grounded in textual guidance and accordingly propose a **T**ext-**G**rounded KV Eviction method for **V**LMs (**TGV-KV**). TGV-KV comprises three submodules: *(1) Text-Vision Budgeting (TVB)* assigns budget to each layer based on the mutual information interaction. *(2) Text-Weighted Ranking (TWR)* assesses the priority of text and ranks vision importance based on weighted text-image attention. *(3) Text-Prioritised Retention (TPR)* policy strategically preserves text KV to avoid acute informati

on loss. We evaluate TGV-KV across five models with different sizes and architectures, showing that TGV-KV preserves 99.2% full-KV accuracy on the VizWiz-VQA task with LLaVA-NeXT and boosts end-to-end throughput by 52.6% with an extreme retention budget of 5%. [Code Link](https://github.com/Danielement321/TGV-KV).
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How Do Language Models Speak Languages? A Case Study on Unintended Code-Switching
Yuxin Xiao, Zhen Huang, Wenxiao Wang, Yan Zhao, Zhihong Gu, Binbin Lin, Xiaofei He, Xu Shen, Jieping Ye
Unintended code-switching, where LLMs unexpectedly switch languages, poses a fundamental challenge to multilingual generation in LLMs. However, we still lack a mechanistic account of how this failure mode is implemented inside the model. Key questions remain: what internal components (i.e., circuits) give rise to unintended code-switching, where do they emerge across layers, and how can we intervene to mitigate it? In this work, we introduce a scalable circuit discovery framework that causally localizes multilingual neurons and describes their functional patterns, then further groups them into interpretable circuits—without any additional training or manual annotation. Our findings are twofold: a) The model's "speaking-a-language" circuit decomposes into a language regime (detecting and maintaining language identity) and a semantic regime (retrieving language-agnostic semantics). b) The mechanism of unintended code-switching is a regime shift. The semantic regime suppresses the language regime and overwhelms the multilingual circuit, causing the model to generate in an unintended language. To validate these findings, we further fine-tune the identified language sub-circuit, reducing the code-switching rate by 20.8% with minimal parameter updates (~0.019 % of all neurons).
This work serves as a preliminary exploration of multilingual generation mechanism, offering actionable insight for targeted training for multilingual LLMs.
****
LithoGRPO: Fast Inverse Lithography via GRPO Reinforced Flow Matching
Yao Lai, Xuyuan Xiong, Zeyue Xue, Guojin Chen, Jing Wang, Xihui Liu, Rui Zhang, Robert D. Mullins, Bei Yu, Ping Luo
In semiconductor manufacturing, lithography projects circuit layouts onto silicon wafers through an optical mask. As circuit features shrink below the wavelength of light, optical diffraction causes the printed patterns to deviate from their intended layouts. Inverse Lithography Technology (ILT) addresses this challenge by generating optimized masks that enhance the fidelity of pattern transfer onto wafers. While ILT resembles an image synthesis task, its reliance on explicit physical metrics for mask evaluation limits the applicability of existing generative models. We introduce LithoGRPO, an ILT framework that integrates the flow■matching paradigm with GRPO■based reinforcement learning (RL) fine■tuning, enabling efficient exploration of diverse masks for a given target layout. Unlike purely generative or optimization■based approaches, RL in LithoGRPO exploits the explicitly defined, physics■based reward function of ILT, enabling optimization under complex, process■aware constraints. To the best of our knowledge, this is the first framework that unifies flow matching and RL for mask optimization. To improve RL sampling efficiency, we propose a fast shot-counting algorithm for manufacturability evaluation, achieving over 130× speedup while preserving the mask ranking of the traditional shot-count metric. Extensive experiments demonstrate that LithoGRPO achieves state■of■the■art performance over both optimization■based and learning■based methods, while maintaining efficient mask generation.
****
Finding Differentially Private Second Order Stationary Points in Stochastic Minimax Optimization
Difei Xu, Youming Tao, Meng Ding, Chenglin Fan, Di Wang
We provide the first study of the problem of finding differentially private (DP) second-order stationary points (SOSP) in stochastic (non-convex) minimax optimization.
Existing literature either focuses only on first-order stationary points for minimax problems or on SOSP for classical stochastic minimization problems. This wo

rk provides, for the first time, a unified and detailed treatment of both empiri
cal and population risks. Specifically, we propose a purely first-order method
that combines a nested gradient descent--ascent scheme with SPIDER-style varianc
e reduction and Gaussian perturbations to ensure privacy. A key technical device
 is a block-wise (q-period) analysis that controls the accumulation of stochas
tic variance and privacy noise without summing over the full iteration horizon,
yielding a unified treatment of both empirical-risk and population formulations.
 Under standard smoothness, Hessian-Lipschitzness, and strong concavity assumpti
ons, we establish high-probability guarantees for reaching an $(\alpha,\sqrt{\rh
o_\Phi \alpha})$-approximate second-order stationary point with $\alpha = \mathc
al{O}((\frac{\sqrt{d}}{n\varepsilon})^{2/3})$
for empirical risk objectives and $\mathcal{O}(\frac{1}{n^{1/3}} + (\frac{\sqrt{
d}}{n\varepsilon})^{1/2})$ for population objectives, matching the best known ra
tes for private first-order stationarity.
****

DriveWorld-VLA: Unified Latent-Space World Modeling with Vision–Language–Action for Autonomous Driving

Feiyang Jia, Lin Liu, Ziying Song, Caiyan Jia, Hangjun Ye, Xiaoshuai Hao, Long C
hen

End-to-end (E2E) autonomous driving has recently attracted increasing interest i
n unifying Vision–Language–Action (VLA) with World Models to enhance decision-ma
king and forward-looking imagination. However, existing methods fail to effectiv
ely unify future scene evolution and action planning within a single architectur
e due to inadequate sharing of latent states, limiting the impact of visual imag
ination on action decisions. To address this limitation, we propose DriveWorld-V
LA, a novel framework that unifies world modeling and planning within a latent s
pace by tightly integrating VLA and world models at the representation level, wh
ich enables the VLA planner to benefit directly from holistic scene-evolution mo
deling and reducing reliance on dense annotated supervision. Additionally, Drive
World-VLA incorporates the latent states of the world model as core decision-mak
ing states for the VLA planner, facilitating the planner to assess how candidate
 actions impact future scene evolution. By conducting world modeling entirely in
 the latent space, DriveWorld-VLA supports controllable, action-conditioned imag
ination at the feature level, avoiding expensive pixel-level rollouts. Extensive
 open-loop and closed-loop evaluations demonstrate the effectiveness of DriveWor
ld-VLA, which achieves state-of-the-art performance with 91.3 PDMS on NAVSIMv1,
86.8 EPDMS on NAVSIMv2, and 0.16 3-second average collision rate on nuScenes. Co
de and models are released at https://github.com/liulin815/DriveWorld-VLA.
****

CoPE: Continual Probe-guided Expansion for Large Vision-Language Models

Ziqin Wang, Hengyuan Zhao, Qixin Sun, Kaiyou Song, Yilin Li, Xiaolin Hu, Qingpei
 Guo, Linjiang Huang, Si Liu

Mixture of Experts architectures have recently advanced the scalability and adap
tability of Large Language Models for continual multimodal learning. However, ex
tending these models to accommodate sequential tasks remains challenging. As ne
w tasks arrive, naive model expansion leads to rapid parameter growth, while mod
ifying shared routing components often causes catastrophic forgetting, undermini
ng previously learned knowledge. To address these issues, we propose CoPE, a con
tinual learning framework for LLMs that requires no replay data of previous task
s and ensures both parameter efficiency and robust knowledge retention. Our appr
oach introduces the Probe-Guided Knowledge Extension mechanism, which uses probe
 experts to dynamically determine when and where new experts should be added, en
abling adaptive and minimal parameter expansion tailored to task complexity. To
support inference without task labels, we further incorporate a Probabilistic Ta
sk Locator that dynamically matches inputs to the correct task-specific componen
ts. To handle the practical issue that task labels are unknown during inference
, we leverage a VAE-based reconstruction strategy to identify the most suitable
router by matching input distributions, allowing automatic and accurate expert a
llocation. This design mitigates routing conflicts and catastrophic forgetting,
enabling robust continual learning without explicit task labels. Extensive exper

iments on the CoIN benchmark, covering eight diverse VQA tasks, demonstrate that CoPE delivers strong continual learning performance with a compact model size, significantly reducing forgetting and parameter overhead compared to prior methods. These results showcase the effectiveness and scalability of our approach for parameter-efficient continual learning in large language models. Our code will be open-sourced soon.

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Nonlinear Covariate Balance in Experimental Design

Qing Chen, Peng Zhang

We study experimental designs that balance nonlinear functions of covariates, extending classical methods that primarily target linear balance. Building on the Gram-Schmidt Walk (GSW) framework of Harshaw et al (2024) for linear covariate balancing, we introduce a design that directly controls imbalance in nonlinear structure, including polynomial and more general smooth function classes. Like GSW, the proposed design retains sufficient robustness against model misspecification. Our implementation operates directly on a Gram matrix, avoiding the expensive step of explicitly constructing the nonlinear covariate expansions. We further accelerate the nonlinear design via a low-rank approximation of the Gram matrix, achieving runtimes comparable to the GSW of Harshaw et al (2024) while preserving nonlinear covariate balance and robustness.

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Off-Policy Learning in Large Action Spaces: Optimization Matters More Than Estimation

Imad Aouali, Otmane Sakhi

Off-policy evaluation (OPE) and off-policy learning (OPL) are foundational for decision-making in offline contextual bandits. Recent advances in OPL primarily optimize OPE estimators with improved statistical properties, assuming that better estimators inherently yield superior policies. Although theoretically justified, this estimator-centric approach neglects a critical practical obstacle: challenging optimization landscapes. In this paper, we provide theoretical insights and empirical evidence showing that current OPL methods encounter severe optimization issues, particularly as the action space grows. We show that estimator-aware policy parametrization can mitigate, but not fully resolve, optimization challenges. Building on this, we explore simpler weighted log-likelihood objectives and demonstrate that they enjoy substantially better optimization properties and still recover competitive, often superior, learned policies. Our findings emphasize the necessity of explicitly addressing optimization considerations in the development of OPL algorithms for large action spaces.

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Automatic Layer Selection for Hallucination Detection

Xinpeng Wang, William Cao, Andrew Gordon Wilson, Zhe Zeng

Recent studies on hallucination detection have shown that hallucination-related signals are more strongly encoded in intermediate layers than in the final layer of large language models (LLMs). Although a growing body of work has sought to exploit this property for hallucination detection, how to automate the selection of high-performing layers remains underexplored, and principled methods for this purpose are still lacking. To address this gap, we first propose several hypotheses for why such signals emerge in intermediate layers and evaluate corresponding criteria for automatic layer selection across diverse LLM architectures, scales, and tasks, covering both question answering and summarization hallucination detection benchmarks. However, we find that none of these criteria consistently delivers satisfactory performance. We therefore propose a new selection criterion, First Effective Peak of Intrinsic Dimension (FEPoID), which consistently identify optimal or near-optimal layers and outperforms both the aforementioned criteria and existing hallucination detection baselines. FEPoID is training-free and incurs negligible computational overhead. In addition, we study the generation behaviors of LLMs and introduce a simple yet effective truncation strategy, which further amplifies hallucination-related signals and substantially improves overall detection performance. Code is publicly available at https://github.com/DesoloYw/Automatic-Layer-Selection-for-Hallucination-Detection.git

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ProactiveLLM: Learning Active Interaction for Streaming Large Language Models
Junlong Tong, Yao Zhang, Anhao Zhao, Yingqi Fan, Yunpu Ma, Xiaoyu Shen
Standard Large Language Models (LLMs) follow a *read-then-generate* paradigm, ca
using unnecessary latency and computation. Streaming LLMs alleviate this issue b
y generating while receiving inputs, but still struggle to decide when to intera
ct with the stream. Existing methods either hard-code interaction timing or rely
 on costly external alignment signals, such as timing labels, reasoning trajecto
ries, or stronger teachers. In this paper, we propose **ProactiveLLM**, which ac
hieves active interaction by leveraging the model's endogenous states to guide i
nteraction decisions. The model first learns to perceive semantic sufficiency fr
om partial inputs through two complementary training mechanisms: *mask-based str
eaming modeling* and *synchronized privileged self-distillation (SPSD)*. The for
mer applies monotonic random masking to the input during training, simulating pr
ogressively revealed streaming inputs and enabling the model to learn local sema
ntic dependencies from partial-input views. The latter aligns the partial-contex
t student view with a full-context teacher view generated by the same evolving m
odel, allowing privileged full-context evidence to guide the student's understan
ding under incomplete observations. Together, these mechanisms induce endogenous
 sufficiency cues *without requiring external teachers or annotations*, providin
g a versatile foundation for the plug-and-play integration of diverse decision h
eads. Extensive evaluation across text and speech streaming tasks confirms that
ProactiveLLM significantly reduces interaction latency while maintaining quality
, validating its capacity for dynamic and active interaction. Code is publicly a
vailable at https://github.com/EIT-NLP/StreamingLLM/tree/main/ProactiveLLM.
****
Personalized Image Generation via Human-in-the-loop Bayesian Optimization
Rajalaxmi Rajagopalan, Debottam Dutta, Yu-Lin Wei, Romit Roy Choudhury
Imagine Alice has a specific image $x^\ast$ in her mind, say, the view of the st
reet in which she grew up during her childhood.
To generate that exact image, she guides a generative model with multiple rounds
 of prompting and arrives at an image x^{p*}.
Although x^{p*} is reasonably close to $x^\ast$, Alice finds it difficult to c
lose that gap using language prompts.
This paper aims to narrow this gap by observing that even after language has rea
ched its limits, humans can still tell when a new image x^+ is closer to $x^\a
st$ than x^{p*}.
Leveraging this observation, we develop **MultiBO** (Multi-Choice Preferential B
ayesian Optimization) that carefully generates K new images as a function of $
x^{p*}$, gets preferential feedback from the user, uses the feedback to guide th
e diffusion model, and ultimately generates a new set of K images.
We show that within B rounds of user feedback, it is possible to arrive much c
loser to $x^\ast$, even though the generative model has no information about $x^
\ast$.
Qualitative scores from 30 users, combined with quantitative metrics compared
across 5 baselines, show promising results, suggesting that multi-choice feedb
ack from humans can be effectively harnessed for personalized image generation.
****
Evaluating LLM Uncertainty in Long-Form Generation Using Deterministic Ground Tr
uth
Ido Amit, Ido Galil, Ran El-Yaniv
As LLMs generate increasingly long outputs, effective uncertainty estimation mus
t identify errors at fine-grained levels rather than discard entire responses. W
hile such methods exist, evaluating uncertainty at any resolution (token to an e
ntire generation) is challenging and highly sensitive to label imperfections, ma
king zero-noise benchmarks essential; yet, long-form generation benchmarks tend
to rely on fallible labels rather than deterministic ground truth.
We introduce Single-answer Atomic Long-form Target (SALT), a benchmark of six pr
ocedurally generated tasks with single deterministic long textual ground truths,
 enabling unit-level evaluation of correctness, calibration, and ranking without

external judges.
Equipped with SALT, our analysis of 50+ LLMs reveals key insights: We identify which confidence functions dominate each uncertainty aspect
and show that confidence ranking largely breaks at atomic resolution, even when clearer separability emerges at coarser line-level units.
SALT further enables controlled atom-level interventions throughout generation, revealing two separable drivers of future errors: propagation from corrupted prefixes, dominated by global context correctness, and bounded degradation from increasing answer-context length.
Finally, we demonstrate that reasoning, via Chain-of-Thought prompting or internalized through training, introduces a trade-off, improving accuracy while degrading confidence ranking.
These findings directly impact risk-critical applications requiring reliable error identification and mitigation. We release the Code at [https://github.com/IdoAmit198/SALT](github.com/IdoAmit198/SALT).
****

EasyBalance: Cross-Layer Load Balancing in Distributed MoE Inference
Yize Wu, KE GAO, Ling Li, Yanjun Wu
Load Balancing has emerged as a critical problem in expert-parallel distributed inference of Mixture-of-Experts (MoE) models. As routing distributions are typically skewed across experts, devices hosting lighter-loaded experts must idle to wait for the heaviest during expert computing, leading to inefficiency. Existing load-balancing approaches primarily rely on expert replication or migration within each layer, which introduce additional overhead and limit their flexibility and scalability. To address this problem, we propose EasyBalance, a \textbf{cross-layer} load balancing strategy that requires no modifications to the expert-device mapping, enabling instant adaptability and incurring essentially no additional overhead. Our key insights are that (1) experts from other layers can be viewed as naturally redundant, and (2) expert workloads of multiple layers can be jointly executed. Based on these observations, EasyBalance greedily schedules a subset of cross-layer workloads at each MoE-computation stage to run, while deferring the remaining workloads for future balancing opportunities. Extensive experiments across different models, tasks, and parallelism configurations demonstrate that EasyBalance consistently accelerates distributed MoE inference, reducing GPU idling by mostly over 40\%.
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Dynamic Thinking-Token Selection for Efficient Reasoning in Large Reasoning Models
Zhenyuan Guo, Tong Chen, Wenlong Meng, Chen GONG, Xin Yu, Chengkun Wei, Wenzhi CHEN
Large Reasoning Models (LRMs) excel at solving complex problems by explicitly generating a reasoning trace before deriving the final answer.
However, these extended generations incur substantial memory footprint and computational overhead, bottlenecking LRMs' efficiency.
This work uses attention maps to analyze the influence of reasoning traces and uncover an interesting phenomenon: *only some decision-critical tokens in a reasoning trace steer the model toward the final answer, while the remaining tokens contribute negligibly.*
Building on this observation, we propose **Dyn**amic **T**hinking-Token **S**election (**DynTS**).
This method identifies decision-critical tokens and retains only their associated Key-Value (KV) cache states during inference, evicting the remaining redundant entries to optimize efficiency.
Across six benchmarks, **DynTS** surpasses the state-of-the-art KV cache compression methods, improving Pass@1 by $2.6\\%$ under the same budget.
Compared to vanilla Transformers, it reduces inference latency by $1.84–2.62\times$ and peak KV-cache memory footprint by $3.32–5.73\times$ without compromising LRMs' reasoning performance.
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Robust AI Evaluation through Maximal Lotteries

Hadi Khalaf, Serena Lutong Wang, Daniel Halpern, Itai Shapira, Flavio Calmon, Ariel D. Procaccia

The standard way to evaluate language models on subjective tasks is through pairwise comparisons: an annotator chooses the "better" of two model responses for a given prompt. These comparisons are then aggregated into a single ranking via the Bradley–Terry (BT) framework, forcing heterogeneous preferences into a total order and violating basic social-choice desiderata. In contrast, social choice theory provides an alternative approach called maximal lotteries, which aggregates pairwise preferences without imposing any assumptions on their structure. However, we show that maximal lotteries can be highly sensitive to heterogeneity among annotators and across prompts. We introduce *robust lotteries*, which optimize worst-case performance under plausible shifts in the preference data. On large-scale preference datasets, robust lotteries achieve more reliable win rate guarantees across the annotator distribution and recover a stable set of top performing models.

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Causal-Adapter: Taming Text-to-Image Diffusion for Faithful Counterfactual Generation

Lei Tong, Zhihua Liu, Chaochao Lu, Dino Oglic, Tom Diethe, Philip Alexander Tearle, Sotirios A. Tsaftaris, Chen Jin

We present Causal-Adapter, a modular framework that adapts frozen text-to-image diffusion backbones for counterfactual image generation. Our method enables causal interventions on target attributes while preserving all other aspects of the image, including the core identity. In contrast to prior approaches that rely on prompt engineering without explicit causal structure, Causal-Adapter leverages structural causal modeling augmented with two attribute regularization strategies: prompt-aligned injection, which aligns causal attributes with textual embeddings for precise semantic control, and a conditioned token contrastive loss to disentangle attribute factors and reduce spurious correlations. Causal-Adapter achieves state-of-the-art results on synthetic and real-world datasets, outperforming other baselines in effectiveness, composition, realism, and minimality. These results demonstrate that the approach enables efficient, robust, and generalizable counterfactual image editing with faithful attribute modification and strong preservation of core identity.

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PRISM: Perception Reasoning Interleaved for Sequential Decision Making.

Mohamed Salim AISSI, Clémence Grislain, Clément ROMAC, Laure Soulier, Mohamed CHETOUANI, Olivier Sigaud, Nicolas THOME

Scaling LLM-based embodied agents from text-only environments to complex multimodal settings remains a major challenge. Recent work identifies a perception–reasoning–decision gap in standalone Vision–Language Models (VLMs), which often overlook task-critical information. In this paper, we introduce PRISM, a framework that tightly couples perception (VLM) and decision (LLM) through a dynamic question–answer (DQA) pipeline. Instead of passively accepting the VLM’s description, the LLM critiques it, probes the VLM with goal-oriented questions, and synthesizes a compact image description. This closed-loop interaction yields a sharp, task-driven understanding of the scene. We evaluate PRISM on the ALFWorld and Room-to-Room (R2R) benchmarks. We show that: (1) PRISM significantly outperforms state-of-the-art image-based models, (2) our Interactive goal-oriented perception pipeline yields systematic and substantial gains, and (3) PRISM is fully automatic, eliminating the need for handcrafted questions or answers.

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Revealing Behavioral Plasticity in Large Language Models: A Token-Conditional Perspective

Liyuan Mao, Le Yu, Jing Zhou, Chujie Zheng, Bowen Yu, Chang Gao, Shixuan Liu, An Yang, Weinan Zhang, Junyang Lin

In this work, we reveal that Large Language Models (LLMs) possess intrinsic behavioral plasticity—akin to chameleons adapting their coloration to environmental cues—that can be *exposed* through token-conditional generation and *stabilized* via reinforcement learning. Specifically, by conditioning generation on careful

ly selected token prefixes sampled from responses exhibiting desired behaviors, LLMs seamlessly adapt their behavioral modes at inference time (e.g., switching from step-by-step reasoning to direct answering) without retraining. Based on th is insight, we propose **To**ken-**Co**nditioned **R**einforcement **L**earning (**ToCoRL**), a principled framework that leverages RL to internalize this chame leon-like plasticity, transforming transient inference-time adaptations into sta ble and learnable behavioral patterns. ToCoRL guides exploration with token-cond itional generation and keep enhancing exploitation, enabling emergence of approp riate behaviors. Extensive experiments show that ToCoRL enables precise behavior al control without capability degradation. Notably, we show that large reasoning models, while performing strongly on complex mathematics, can be effectively ad apted to excel at factual question answering, which was a capability previously hindered by their step-by-step reasoning patterns.

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Modeling Spectral Energy Shifts in Spatio-Temporal Graph Anomaly Detection

Yilin Liu, Hongchao Zhang, Ahmad Taha, Taylor T Johnson, Meiyi Ma

Graph anomaly detection methods aim to distinguish anomalous nodes. While prior methods characterize anomalies through increased variation in the spectral energ y distributions, they overlook those that result in decreased variation, i.e., c amouflaged anomalies that appear normal. We show that this type of anomaly persi sts across multiple datasets and remains undetectable by existing spectral appro aches. To address this limitation, we propose a node-level spectral energy formu lation that is fully compatible with message passing and enables the detection o f camouflaged anomalies. Building on this formulation, we introduce an energy-aw are graph learning framework that models spectral shifts through energy-driven m essage passing in both static and time-series graphs. Besides, our unified archi tecture extends to temporal settings without introducing specialized sequence mo dules, enabling efficient learning under long sliding windows. Extensive experim ents on large-scale benchmarks demonstrate the effectiveness and scalability of our approach. Our code is available at https://github.com/AICPS-Lab/Spectral-Ene rgy-Shifts-in-GAD.

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Understanding the Performance Gap in Preference Learning: A Dichotomy of RLHF an d DPO

Ruizhe Shi, Minhak Song, Runlong Zhou, Zihan Zhang, Maryam Fazel, Simon Shaolei Du

We present a fine-grained theoretical analysis of the performance gap between tw o-stage reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO). Our study decomposes this gap into two sources: the explicit representation gap under exact optimization and the implicit representation gap under finite samples. In the exact optimization setting, we characterize how th e relative capacities of the reward and policy model classes influence the final policy qualities. We show that RLHF, DPO, or online DPO can outperform one anot her depending on type of model mis-specifications. Notably, online DPO can outpe rform both RLHF and standard DPO when the reward and policy model classes are is omorphic and both mis-specified. In the approximate optimization setting, we pro vide a concrete construction where the ground-truth reward is sparse and show th at RLHF requires significantly fewer samples than DPO to recover an effective re ward model, highlighting a statistical advantage of two-stage learning. Together , these results provide a comprehensive understanding of the performance gap bet ween RLHF and DPO under various settings, and offer practical insights into when each method is preferred.

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Efficient Learned Image Compression without Entropy Coding

Hao Cao, Wenqi Guo, Zhijin Qin, Jungong Han

Entropy coding is widely used in typical learned image compression (LIC) that co nverts latents into a compact bitstream.

However, entropy coding is typically sequential and becomes the coding latency b ottleneck.

To overcome it, we present **E**ntropy-Coding **F**ree **L**earned **I**mage **C**

**ompression (EF-LIC), a multi-rate framework that generates compact representation by removing statistical and correlation redundancy with low coding latency.
First, we introduce unconstrained vector quantization and prove that its index distribution approaches the maximum-entropy bound, yielding minimal statistical redundancy.
Second, we propose a context-conditioned autoregressive transform that directly reparameterizes the latents to reduce inter-dependency.
Theoretical analysis shows that EF-LIC can remove correlation redundancy as effectively as typical LIC with entropy coding, leading to comparable compression performance.
Experiments show EF-LIC achieves up to 67.86\% bitrate reduction over MS-ILLM on Kodak with LPIPS.
Ablation studies further show EF-LIC matches the compression performance of its entropy-coding based variant while achieving over $3\times$ faster encoding and $5\times$ faster decoding.
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cMoLLM at Scale: Horizontal Scaling Laws for Convolutionally-Gated Mixture-of-LLMs

Xin Yang, Yemin Wang, Mingda Liu, Letian Li, Shuaishuai Cao, ZhengXiao He, Ryan Dong

Scaling large language models (LLMs) has driven their success, yet dense Transformers couple capacity and computation: every parameter is activated for every token, making training and inference costs grow linearly with model size—a critical bottleneck as models approach trillion-parameter regimes. We aim to scale capacity through MoE-style mixture throughout the LLM pipeline rather than only the FFN. Prior pipeline-level approaches include ParaScale, which introduces virtual tokens and parallel streams but incurs substantial overhead and suffers from homogenized routing and gradient collapse, and AltUp, which uses an auxiliary prediction branch but offers limited adaptivity and slow convergence. We establish that MoE-style mixture layers can be reformulated as variable-kernel dynamic convolutions, where each expert corresponds to a $1{\times}1$ convolutional kernel and routing implements input-conditioned kernel aggregation. Building on this equivalence, we introduce cMoLLM: a convolutionally gated mixture-of-LLMs that routes over end-to-end streams through fully differentiable dynamic convolution. In GPT-2-style models trained on FineWeb, cMoLLM improves language modeling perplexity and downstream GLUE and SQuAD accuracy under matched compute, with better stream utilization, more stable optimization, and favorable scaling compared to ParaScale- and AltUp-style baselines.
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Learn from Your Mistakes: Tree-like Self-Play for Secure Code LLMs

Wenqi Chen, Ziyan Zhang, Wang Bin, Lin Liu, Hengheng Zhang, Zhengsu Chen

While Large Language Models (LLMs) excel in code generation, they remain prone to replicating subtle yet critical vulnerabilities endemic to their training data. Current alignment techniques, such as Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL), typically apply coarse-grained optimization at the sequence level. This approach often fails to address the localized nature of security flaws, where a single incorrect token choice can compromise an entire program.
To bridge this gap, we introduce Tree-like Self-Play (TSP), a framework that reframes secure code generation as a fine-grained sequential decision process. Unlike standard methods that blindly maximize likelihood, TSP constructs a decision tree where the model explores branching trajectories—generating both secure "golden paths" and vulnerable variants. By treating code generation as a self-play game, the model learns to strictly discriminate against its own localized errors. This provides a dense, on-policy learning signal that forces self-correction precisely at the critical decision nodes where vulnerabilities typically emerge.
Our experiments demonstrate that TSP fundamentally enhances model reliability. In Python security benchmarks, TSP boosts CodeLlama-7B’s pass rate (SPR@1) to 75.8\%, significantly outperforming SFT (57.0\%) and unstructured self-play baselines. Crucially, TSP induces robust out-of-distribution generalization: the model not only reduces vulnerabilities in unseen categories (CWEs) by 24.5\% but also

successfully transfers security principles learned from C/C++ to diverse languages, including Python, Go, and JavaScript. This suggests that TSP does not merely memorize patches, but internalizes abstract, language-agnostic security logic.
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Reasoning over Boundaries: Enhancing Specification Alignment via Test-time Deliberation
Haoran Zhang, Yafu Li, Xuyang Hu, Dongrui Liu, Zhilin Wang, Bo Li, Yu Cheng
Large language models (LLMs) are increasingly applied in diverse real-world applications, each governed by bespoke behavioral and safety specifications (spec) customtailored by users or organizations. These specifications, categorized into safety-spec and behavioral-spec, vary across scenarios and evolve with changing preferences and requirements. We formalize this challenge as specification alignment, focusing on LLMs' ability to follow dynamic, scenario-specific spec from both behavioral and safety perspectives. To address this challenge, we introduce SpecBench, a unified benchmark for measuring specification alignment, covering 5 scenarios, 103 spec, and 1,500 prompts. Experiments on 15 reasoning and 18 instruct models with several Test-Time Deliberation (TTD) methods, including Self-Refine, TPO, and MoreThink, show that SpecBench effectively reveals alignment gaps and that test-time deliberation improves specification alignment. Based on previous TTD methods, we further propose Align3, a lightweight method with hierarchical reflection and revision to reason over specification boundaries, advancing the safety-helpfulness trade-off frontier with minimal overhead. These results highlight test-time deliberation as an effective strategy for reasoning over the real-world specification boundaries. Our code and resources are available at https://github.com/zzzhr97/SpecBench.
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Can Vision Language Models Learn Intuitive Physics from Interaction?
Luca M. Schulze Buschoff, Konstantinos Voudouris, Can Demircan, Eric Schulz
Pre-trained vision language models do not have good intuitions about the physical world. Recent work has shown that supervised fine-tuning can improve model performance on simple physical tasks. However, fine-tuned models do not appear to learn robust physical rules that can generalize to new contexts. Based on research in cognitive science, we hypothesize that models need to interact with an environment to properly learn its physical dynamics. We train models that learn through interaction with a simulated environment using reinforcement learning. While learning from interaction allows models to improve their within-task performance, it fails to produce models with generalizable physical intuitions. We find that models trained on one task do not reliably generalize to related tasks, even if the tasks share visual statistics and physical principles, and regardless of whether the models are trained through interaction.
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What Does Thompson Sampling Optimize?
Yanlin Qu, Hongseok Namkoong, assaf zeevi
Thompson Sampling is one of the most widely used and studied bandit algorithms, known for its simple structure, low regret performance, and solid theoretical guarantees. Yet, in stark contrast to most other families of bandit algorithms, the exact mechanism through which posterior sampling (as introduced by Thompson) is able to "properly" balance exploration and exploitation, remains a mystery. In this paper, we show that the core insight to address this question stems from recasting Thompson Sampling as an online optimization algorithm. To distill this, we introduce a time invariant notion of regret that summarizes cumulative regret across horizons (through a regret bound), leading to a time invariant Bellman-optimal policy. It turns out that Thompson Sampling admits an online optimization form that mimics the structure of the Bellman-optimal policy, where greediness is regularized by a measure of residual uncertainty. When viewed through this new lens of online optimization, Thompson Sampling can be understood and improved in a principled manner, by comparing it against the Bellman-optimal benchmark.
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Language Generation with Replay: A Learning-Theoretic View of Model Collapse
Giorgio Racca, Michal Valko, Amartya Sanyal

As scaling laws push the training of frontier large language models (LLMs) toward ever-growing data requirements, training pipelines are approaching a regime where much of the publicly available online text may be consumed. At the same time, widespread LLM usage increases the volume of machine-generated content on the web; together, these trends raise the likelihood of generated text re-entering future training corpora, increasing the associated risk of performance degradation often called *model collapse*. In practice, model developers address this concern through data cleaning, watermarking, synthetic-data policies, or, in some cases, blissful ignorance. However, the problem of model collapse in generative models has not been examined from a learning-theoretic perspective: we study it through the theoretical lens of the *language generation in the limit* framework, introducing a *replay* adversary that augments the example stream with the generator's own past outputs. Our main contribution is a fine-grained learning-theoretic characterization of when replay fundamentally limits generation: while replay is benign for the strongest notion of uniform generation, it provably creates separations for the weaker notions of non-uniform generation and generation in the limit. Interestingly, our positive results mirror heuristics widely used in practice, such as data cleaning, watermarking, and output filtering, while our separations show when these ideas can fail.

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Understanding Self-Supervised Learning via Latent Distribution Matching

Fabian A Mikulasch, Friedemann Zenke

Self-supervised learning (SSL) excels at finding general-purpose latent representations from complex data, yet lacks a unifying theoretical framework that explains the diverse existing methods and guides the design of new ones. We cast SSL as latent distribution matching (LDM): learning representations that maximize their log-probability under an assumed latent model (alignment), while maximizing latent entropy to prevent collapse (uniformity). This view unifies independent component analysis with contrastive, non-contrastive, and predictive SSL methods, including stop gradient approaches. Leveraging LDM, we derive a nonlinear, sampling-free Bayesian filtering model with a Kalman-based predictor for high-dimensional timeseries. We further prove that predictive LDM yields identifiable latent representations under mild assumptions, even with nonlinear predictors. Overall, LDM clarifies the assumptions behind established SSL methods and provides principled guidance for developing new approaches.

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Identifying and Mitigating Errors in Gradient Aggregation of Distributed Data Parallel Training

Zhenheng Tang, Junlin Huang, Zichen TANG, Xueze Kang, Yuxin Wang, Peijie Dong, Shaohuai Shi, Xiaowen Chu, Bo Li

Hardware-related silent data corruptions during gradient aggregation pose significant challenges to fault-tolerant distributed training, often leading to slow or failed convergence. To address this, we first mathematically formulate these errors as gradient inconsistency and theoretically analyze how they result in accumulated model divergence. Guided by this analysis, we introduce PAFT, a fault-tolerant distributed training system designed with dynamic and asynchronous parameter synchronization. PAFT comprises two core components: PAFT-Sync, which mitigates divergence via periodic synchronization, and PAFT-Dyn, which minimizes overhead through dynamic training overlap and frequency scheduling. Furthermore, the system’s synchronization mechanism is optimized to support standard optimizers, including SGD, SGD momentum, and Adam. We implement PAFT on PyTorch Distributed, and experimental results training ResNet, GPT-2, and LLaMA-2 on 4$\sim$32 GPUs demonstrate that it efficiently defends against aggregation errors while maintaining training performance.

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From Reasoning Traces to Reusable Modules: Understanding Compositional Generalization in Language Model Reasoning

Lingjing Kong, Xin Liu, Guangyi Chen, Martin Q. Ma, Xiangchen Song, Yuekai Sun, Mikhail Yurochkin, Taylor W. Killian, Ruslan Salakhutdinov, Kun Zhang, Eric P. Xing, Zhengzhong Liu

Reinforcement learning (RL) has emerged as a key mechanism for transforming LLMs into robust reasoners. While supervised fine-tuning (SFT) often limits models to the distribution of observed reasoning traces, RL post-training significantly improves performance on out-of-distribution (OOD) tasks that require unfamiliar recombinations of familiar steps. We argue that this improvement is driven by **compositional generalization**, which we formalize through a **Hierarchical Latent Selection Model**. In this framework, reasoning traces are generated by a cascade of discrete latent selection variables corresponding to reusable atomic modules, including both skills (local operations) and routing mechanisms (how intermediate information is selected, reused, and composed). We theoretically show that RL's exploratory nature provides sufficient coverage to identify latent structure and enable compositional generalization.
We design controlled experiments to validate this theory. Our results demonstrate that RL can extract atomic modules from compound traces and recombine them to solve new configurations. Moreover, we find that training on compound traces can yield stronger generalization than training on isolated atomic modules. Finally, we investigate relations between SFT and RL and identify an effective protocol in which SFT ensures coverage of all atomic modules, while RL focuses on novel compositions beyond the SFT support to encourage exploration.
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RN-D: Discretized Categorical Actors for On-Policy Reinforcement Learning
Yuexin Bian, Jie Feng, Tao Wang, Yijiang Li, Sicun Gao, Yuanyuan Shi
On-policy Reinforcement Learning (RL) remains a dominant paradigm for continuous control, yet standard implementations rely on Gaussian actors and relatively shallow MLP policies, often leading to brittle optimization when gradients are noisy, and policy updates must be conservative. In this paper, we revisit actor policy representation as a first-class design choice for on-policy RL. We study discretized categorical actors, which represent each action dimension as a distribution over discrete bins and induce a policy objective analogous to classification cross-entropy loss. Building on architectural advances from supervised learning, we further pair discretized categorical actors with regularized networks, yielding RN-D. Across diverse continuous-control benchmarks, we show that simply replacing the standard Gaussian actor with our proposed actor substantially improves performance, achieving state-of-the-art results within on-policy RL. We release our code at https://github.com/alwaysbyx/RND-RL.
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Real Data Lies: Unveiling and Closing the Quality Shortcut in Generalizable AI-Generated Video Detection
Ziyuan Fang, Tianyi Wei, Guanjie Wang, Weiming Zhang, Nenghai Yu, Wenbo Zhou
Recent advances in video generation have enabled highly realistic synthetic content, raising concerns about the integrity of digital media and motivating the development of benchmarks and detection methods for generated videos. Prior works have largely prioritized bolstering model generalization against unseen generators. However, we uncover a neglected factor: the quality distribution of real videos plays a pivotal role. Current training protocols suffer from a clear quality bias between real and fake data, prone to shortcut learning. Compounded by testing on similar real data distributions, this creates an illusion of generalization. In reality, these models fail to generalize when exposed to real data with significantly different quality profiles. To address this, we propose training with quality-matched real and fake data to mitigate bias. Building on this, we introduce a data expansion strategy that broadens the training set to comprehensively cover the full quality spectrum. This approach enables the model to learn quality-agnostic features for detection, thereby achieving generalization across real data of varying qualities and enhancing real-world applicability. Extensive experiments demonstrate that our method scales well across diverse backbones, consistently enhancing the generalization capability of existing models.
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Causally Evaluating the Learnability of Formal Language Tasks
Vésteinn Snæbjarnarson, Anej Svete, Josef Valvoda, Reda Boumasmoud, Brian DuSell, Ryan Cotterell

Language models, as multi-task learners, acquire a wide range of abilities during training. A fundamental question is how much task-specific data is needed to learn a given task. Answering this for natural language is difficult: tasks are hard to delineate and can confound one another. To rigorously investigate the relationship between data frequency and learnability, we turn to a controlled setting using formal languages induced from probabilistic finite automata. These serve as a methodological testbed to demonstrate that standard correlational evaluation practices are inherently flawed. To enable causal analysis, we introduce the binning semiring, an algebraic object that lets us control how often a targeted property occurs in a sampled corpus. We formulate the experimental pipeline as a causal graphical model and derive decomposed Kullback--Leibler divergence metrics to measure the learnability of specific sub-tasks. Our experiments show that evaluating learnability without causal intervention leads to incorrect conclusions due to confounders in correlational analysis, and serve as a warning about correlational pitfalls in natural-language settings.
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Furina: Fragmented Uncertainty-Driven Refusal Instability Attack
Tongxi Wu, Jian Zhang, Yang Gao
Safety alignment in large language models (LLMs) and multimodal large language models (MLLMs) is commonly assumed to operate as a near-binary threshold mechanism. We challenge this assumption by revealing that safety behavior is governed by an \emph{instability region} where small perturbations induce stochastic refusal decisions rather than deterministic outcomes.
We develop a multi-metric diagnostic framework combining external and internal signals to characterize this instability. Through systematic experiments, we identify a characteristic \emph{diagnostic signature}: inputs in unstable regimes exhibit elevated output uncertainty yet \emph{decreased} internal safety activation, a decoupling phenomenon that explains why detection-based defenses fail against sophisticated attacks.
Building on this framework, we introduce \textbf{Furina}, a jailbreak attack that deliberately induces this signature through fragmented, scene-anchored prompts without model-specific optimization.
Furina outperforms strong single-turn and multi-turn baselines on HarmBench and achieves competitive results on MM-SafetyBench, demonstrating that uncertainty amplification provides a principled and transferable mechanism for understanding safety vulnerabilities. Code is available at: \url{https://github.com/0xCavaliers/Furina_Jailbreak}.
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Don't Overthink with Pixels: Efficient Reasoning for Segmentation
Song Wang, Gongfan Fang, Lingdong Kong, Xiangtai Li, Jianyun Xu, Sheng Yang, Qiang Li, Jianke Zhu, Xinchao Wang
Existing reasoning segmentation approaches typically fine-tune multimodal large language models (MLLMs) using image-text pairs and corresponding mask labels. While recent efforts leverage reinforcement fine-tuning to further enhance reasoning ability, they often suffer from overthinking and produce uniformly verbose reasoning chains irrespective of task complexity. To address this problem, we propose PixelThink, a simple yet effective scheme that integrates externally estimated task difficulty and internally measured model uncertainty to regulate reasoning generation within a reinforcement learning paradigm. The model learns to compress reasoning length in accordance with scene complexity and predictive confidence. To support comprehensive evaluation, we introduce ReasonSeg-Diff, an extended benchmark with annotated reasoning references and difficulty scores, along with a suite of metrics designed to assess segmentation accuracy, reasoning quality, and efficiency jointly. Experimental results demonstrate that the proposed approach not only improves segmentation performance but also significantly reduces inference latency by 30.4%, cutting token usage by 48.2%.
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Evaluating and Steering Modality Preferences in Multi-modal LLMs
Yu Zhang, Jinlong Ma, Yongshuai Hou, Xuefeng Bai, Kehai Chen, Yang Xiang, Jun Yu, Min Zhang

Multi-modal large language models (MLLMs) have achieved remarkable success on co
mplex multi-modal tasks.
However, it remains insufficiently explored whether they exhibit modality prefer
ence, a tendency to favor one modality over another when processing multi-modal
contexts.
To study this question, we introduce MC^2 benchmark, which constructs controll
ed evidence-conflict scenarios to systematically evaluate modality preference in
 decision-making.
Extensive experiments reveal that all 20 tested MLLMs generally demonstrate clea
r modality preferences, and such preferences are
statistically associated with performances of downstream taks for MLLMs.
Further analysis shows that modality preference can be controlled by instruction
 guidance and captured within the latent representations of MLLMs.
Built on these insights, we propose a probing and steering method based on repre
sentation engineering to explicitly control modality preference without requirin
g additional fine-tuning.
This method effectively amplifies modality preference toward a desired direction
 and demonstrates promising improvements across
multiple multi-modal understanding and reasoning tasks.
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Probabilistic Modeling of Latent Agentic Substructures in Deep Neural Networks
Su Hyeong Lee, Risi Kondor, Richard Ngo
We develop a theory of intelligent agency grounded in probabilistic modeling for
 neural models. Agents are represented as outcome distributions with epistemic u
tility given by log score, and compositions are defined through weighted logarit
hmic pooling that strictly improves every member's welfare. We prove that strict
 unanimity is impossible under linear pooling or in binary outcome spaces, but p
ossible with three or more outcomes. Our framework admits recursive structure vi
a cloning invariance, continuity, and openness, while tilt-based analysis rules
out trivial duplication. Finally, we formalize an agentic alignment phenomenon i
n LLMs using our theory: eliciting a benevolent persona ("Luigi'') induces an an
tagonistic counterpart ("Waluigi''), while a manifest-then-suppress Waluigi stra
tegy yields strictly larger first-order misalignment reduction than pure Luigi r
einforcement alone. These results clarify how developing a principled mathematic
al framework for how subagents can coalesce into coherent higher-level entities
provides novel implications for alignment in agentic AI systems.
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Continuity-Regularized Flow Matching for Offline Reinforcement Learning
Xiaocong Chen, Siyu Wang, Lina Yao
Flow-matching policies have recently emerged as a powerful class of generative m
odels for offline reinforcement learning (RL), capable of capturing complex, mul
ti-modal action distributions from static datasets. However, standard training o
bjectives are largely agnostic to the global properties of the generative path,
permitting learned vector fields that are irregular and unstable, which can hind
er performance. In this work, we introduce PDE-regularized Q-Learning (PQL), a n
ovel algorithm that addresses this limitation by imposing a principled structure
 on the entire probability flow. PQL makes two synergistic contributions: first,
 a partial differential equation based regularizer derived from the continuity e
quation promotes global smoothness and stability on the flow. Second, to solve t
he complex optimization problem introduced by this regularizer, we propose a Bet
a-distributed timestep sampling strategy that focuses learning on the critical t
rajectory segments where the trade-off between imitation and smoothness is most
acute. Through extensive experiments, we demonstrate that by structuring the gen
erative journey and not just its destination, PQL achieves state-of-the-art perf
ormance on a wide range of challenging offline RL tasks.
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Optimal Self-Consistency for Efficient Reasoning with Large Language Models
Austin Feng, Marius Alonso, Ambroise Odonnat, Vasilii Feofanov, Ievgen Redko
Self-consistency (SC) is a widely-used test-time inference technique for improvi
ng performance in chain-of-thought reasoning. It consists of generating multiple

responses, or "samples," from a large language model (LLM) and selecting the most frequent answer. This procedure can naturally be viewed as a majority vote or empirical mode estimation. Despite its effectiveness, self-consistency is prohibitively expensive at scale when naively applied to datasets, and it lacks a unified theoretical treatment of sample efficiency and scaling behavior.
In this paper, we provide the first comprehensive analysis of SC's scaling behavior and its variants, drawing on mode estimation and voting theory. We derive and empirically validate power law scaling for self-consistency across datasets, and analyze the sample efficiency for fixed-allocation and dynamic-allocation sampling schemes. From these insights, we introduce Blend-ASC, a novel variant of self-consistency that dynamically allocates samples to questions during inference, achieving state-of-the-art sample efficiency. Our approach uses $4.8\times$ fewer samples than vanilla SC on average, outperforming both fixed- and dynamic-allocation SC baselines, thereby demonstrating the superiority of our approach in terms of efficiency. In contrast to existing variants, we note that Blend-ASC is hyperparameter-free and can fit any budget of samples, ensuring it can be easily applied to any self-consistency application.
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Rényi Diffusion Models
Yirong Shen, Lu GAN, Cong Ling
The choice of training objective is central to diffusion-based generative modeling in terms of both sample quality and distribution coverage. While standard maximum likelihood training provides a principled objective with strong theoretical grounding, empirical studies indicate that previous training objectives in diffusion models often face an inverse correlation between likelihood optimization and perceptual evaluations.

We propose the Rényi diffusion model, a unified generative framework that formulates training objectives using Rényi divergence. This yields a generalized score matching objective providing explicit control over the trade-off between sample quality and distribution coverage. Experiments demonstrate improved balance between density estimation and sample generation performances across multiple datasets without modifying model architectures or sampling procedures.
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Revisiting Regularized Policy Optimization for Stable and Efficient Reinforcement Learning in Two-Player Games
Kazuki Ota, Takayuki Osa, Motoki Omura, Tatsuya Harada
Two-player games such as board games have long been used as traditional benchmarks for reinforcement learning. This work revisits a policy optimization method with reverse Kullback-Leibler regularization and entropy regularization and analyzes this combination in two-player zero-sum settings from theoretical and empirical perspectives. From a theoretical perspective, we investigate the stability of the policy update rule in two theoretical settings: game-theoretic normal-form games and finite-length games. We provide novel convergence guarantees and verify our theoretical results through numerical experiments on synthetic games. From an empirical perspective, we derive a practical model-free reinforcement learning algorithm based on the regularized policy optimization. We validate the training efficiency of our algorithm through comprehensive experiments on five board games: Animal Shogi, Gardner Chess, Go, Hex, and Othello. Experimental results show that our agent learns more efficiently than existing methods across environments.
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Through the Stealth Lens: Attention-Aware Defenses Against Poisoning in RAG
Sarthak Choudhary, Nils Palumbo, Ashish Hooda, Krishnamurthy Dj Dvijotham, Somesh Jha
Retrieval-augmented generation (RAG) systems are vulnerable to attacks that inject poisoned passages into the retrieved context, even at low corruption rates. We show that existing attacks are not designed to be stealthy, allowing reliable detection and mitigation. We formalize a distinguishability-based security game to quantify stealth for such attacks. If a few poisoned passages control the res

ponse, they must bias the inference process more than the benign ones, inherentl
y compromising stealth. This motivates analyzing intermediate signals of LLMs, s
uch as attention weights, to approximate the influence of different passages on
the response. Leveraging attention weights, we introduce the Normalized Passage
Attention Score (NPAS) and a lightweight Attention-Variance Filter (AV Filter) t
hat flags anomalous passages. Our method improves robustness, yielding up to ~20
% higher accuracy than baseline defenses. We also develop adaptive attacks that
attempt to conceal such anomalies, achieving up to 35% success rate and undersco
ring the challenges of achieving true stealth in poisoning RAG systems.
****
SLIM: Secure and Efficient Inference for Large Language Models on Untrusted Devi
ces via TEEs
Wei Wang, Zihao Guan, Xing Zhou, Yan Ding, Yusong Tan, Jie Yu, Bao Li
Deploying large language models (LLMs) on untrusted hardware entails a risk of w
eight extraction, which can lead to unauthorized replication and misuse of the m
odel. A practical approach is to leverage Trusted Execution Environments (TEEs)
and protect model security by obfuscating model weights. However, existing obfus
cation schemes struggle to simultaneously provide strong security guarantees and
high performance: schemes with security guarantees incur substantial overhead d
ue to frequent TEE interactions, whereas schemes that achieve efficient inferenc
e are insecure. We propose SLIM, a secure inference framework that exploits the
iterative structure of LLMs to let transformed representations cascade through c
onsecutive obfuscated layers, thereby minimizing interactions with the TEE. SLIM
introduces a T-Way Mixing algorithm that performs consecutive inter-vector cove
ring using carefully constructed block-diagonal Householder matrices and combine
s it with successive random permutations, providing thorough weight obfuscation
while keeping TEE-side computation lightweight. Evaluations demonstrate that SLI
M provides robust security guarantees and significantly outperforms prior state-
of-the-art obfuscation schemes in terms of performance, delivering up to a $13.8
0\times$ speedup while preserving fidelity.
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Are LLM Evaluators Really Narcissists? Sanity Checking Self-Preference Evaluatio
ns
Dani Roytburg, Matthew Bozoukov, Matthew Nguyen, Jou Barzdukas, Mackenzie Puig-H
all, Narmeen Fatimah Oozeer
Recent research has shown that large language models (LLMs) favor their own outp
uts when acting as judges, undermining the integrity of automated post-training
and evaluation workflows.
However, it is difficult to disentangle which behaviors are explained by narciss
ism versus experimental confounds.
Specifically, LLM evaluators may deliver self-preferring verdicts when comparing
responses to questions they fail on; these verdicts may not depend on the ident
ity of the author, but on evaluator quality.

We correct this by directly comparing the judge's voting distribution in cases w
here it evaluates itself versus another model.
This evaluator quality baseline reveals that only **51%** of examples in previou
s findings retain statistical significance against this null hypothesis, coverin
g **89.6%** of total self-preference probability mass.
Finally, we compare the entropy of voting distributions, suggesting uncertainty-
driven overlap, and show that our procedure enables more careful documentation a
gainst the backdrop of judge-bias research.
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Leak@k: Unlearning Does Not Make LLMs Forget Under Probabilistic Decoding
Hadi Reisizadeh, Jiajun Ruan, Yiwei Chen, Soumyadeep Pal, Sijia Liu, Mingyi Hong
Unlearning in large language models (LLMs) is critical for regulatory compliance
and for building ethical generative AI systems that avoid producing private, to
xic, illegal, or copyrighted content. Despite rapid progress, in this work, we s
how that \textit{almost all} existing unlearning methods fail to achieve true fo
rgetting in practice. Specifically, while evaluations of these `unlearned' model

s under deterministic (greedy) decoding often suggest successful knowledge removal using standard benchmarks, we show that sensitive information reliably resurfaces when models are sampled with standard probabilistic decoding. To rigorously capture this vulnerability, we introduce \texttt{leak@k}, a new meta-evaluation metric that quantifies the likelihood of forgotten knowledge reappearing when generating k samples from the model under realistic decoding strategies. Using three widely adopted benchmarks, TOFU, MUSE, and WMDP, we conduct the first large-scale, systematic study of unlearning reliability using \texttt{leak@k} metric. Our findings demonstrate that knowledge leakage persists across methods and tasks, underscoring that current state-of-the-art (SOTA) unlearning techniques provide only limited forgetting. We propose an algorithm, termed Robust Unlearning under LEak@k metric (\texttt{RULE}) to address this concern. We demonstrate that \texttt{RULE} provides an unlearned model for TOFU benchmark with no information leakage for a large number of generation samples. On the MUSE benchmark, \texttt{RULE} outperforms SOTA unlearning methods under the \texttt{leak@k} metric across most sampling budgets k. Codes are available at \url{https://github.com/OptimAI-Lab/Leak-k}.

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LARA: Latent Action Representation Alignment for Vision-Language-Action Models

Mengya Liu, Baoxiong Jia, Jiangyong Huang, Jingze Zhang, Siyuan Huang

Visual-language-action (VLA) models enable robots to predict actions directly from observations and language instructions, but their performance depends on large-scale, high-quality data and is limited by the scarcity of real-world robot action datasets. To facilitate VLA model learning with abundant unlabeled human videos, Latent Action Models (LAM) learn latent action representations from visual dynamics to provide additional supervision for VLA learning. However, LAM and VLA are typically trained separately, leaving LAM ungrounded during VLA training and VLA models constrained by frozen LAM representations. To address these issues, we propose Latent Action Representation Alignment (LARA), a plug-and-play framework that jointly optimizes LAM and VLA via representation alignment. This enables reciprocal benefits where LAMs learn with action trajectories to avoid spurious visual changes, while VLAs are regularized by forward dynamics learned within LAMs to reduce hallucinations of functionally ineffective trajectories. We demonstrate LARA's versatility and effectiveness for pre-training, post-training enhancement of pre-trained VLA models, and LAM refinement, achieving an average of ~10%, ~5%, and ~15% improvement over 3 simulation and 1 meticulously designed real-world robotic manipulation benchmarks. The code is publicly available at https://github.com/lmy1001/LARA.

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Mitigating Mask Prior Drift and Positional Attention Collapse in Large Diffusion Vision-Language Models

Sujung Hong, Chanyong Yoon, Seong Jae Hwang

Large diffusion vision–language models (LDVLMs) have recently emerged as a promising alternative to autoregressive models, enabling parallel decoding for efficient inference and leveraging bidirectional attention for global context.

Despite these advances, their behavior under long-form generation remains underexplored.

In this work, we show that existing LDVLMs suffer from repetitive generation and degraded visual grounding, and identify two underlying causes.

First, repetitive generation originates from a mask token prior: since generation tokens are initialized as mask tokens, their hidden representations progressively drift toward a shared prior direction over generation steps.

Second, a fundamental misalignment between the positional attention bias and the iterative unmasking process suppresses attention toward informative visual tokens, degrading visual grounding.

Based on these insights, we propose a training-free approach, introducing Mask Prior Suppression and Monotonic RoPE Scaling to mitigate mask prior drift and positional attention collapse during decoding.

Experiments on general multimodal benchmarks and visual grounding tasks demonstrate improvements over baseline LDVLMs, with robust gains on long-form descriptio

n benchmarks.
Our results show that these failures can be effectively addressed with a lightwe ight, plug-and-play strategy that requires no additional training and generalize s across diverse LDVLM architectures.
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Unraveling Syntax: Language Modeling and the Substructure of Grammars
Laura Ying Schulz, Daniel Mitropolsky, Tomaso Poggio
While language models achieve impressive results, their *learning dynamics* are far from understood. Many domains of interest – such as natural language syntax, coding languages, arithmetic – are captured by context-free grammars (CFGs). In this work, we extend prior work on neural language modeling of CFGs in a novel direction: how language modeling behaves with respect to CFG *substructure*, nam ely sub*grammars*. We define subgrammars, and prove a set of fundamental theorem s connecting language modeling and subgrammars. We show that language modeling l oss recurses linearly over its top-level subgrammars; applied recursively, the l oss decomposes into losses for "irreducible" subgrammars. Under additional assum ptions, and empirically, parametrized models learn subgrammars in parallel, unli ke children who first master simple substructures. We find that subgrammar pretr aining can improve final performance, but only for tiny models relative to the g rammar, while alignment analyses show that pretraining consistently leads to int ernal representations that better reflect the grammar’s substructure.
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MotionGRPO: Overcoming Low Intra-Group Diversity in GRPO-Based Egocentric Motion Recovery
Nanjie Yao, Junlong Ren, Wenhao Shen, Hao Wang
This paper studies full-body 3D human motion recovery from head-mounted device s ignals. Existing diffusion-based methods often rely on global distribution match ing, leading to local joint reconstruction errors. We propose MotionGRPO, a nove l framework leveraging reinforcement learning post-training to inject fine-grain ed guidance into the diffusion process. Technically, we model diffusion sampling as a Markov decision process optimized via Group Relative Policy Optimization (GRPO). To this end, we introduce a hybrid reward mechanism that combines a learn ed conditioned perceptual model for global visual plausibility and explicit cons traints for local joint precision. Our key technical insight is that policy opti mization in diffusion-based recovery suffers from vanishing gradients due to lim ited intra-group sample diversity. To address this, we further introduce a noise -injection strategy that explicitly increases sample variance and stabilizes lea rning. Extensive experiments demonstrate that MotionGRPO achieves state-of-the-a rt performance with superior visual fidelity. Code is available at: https://gith ub.com/3DAgentWorld/MotionGRPO/
****
ExpAlign: Expectation-Guided Vision–Language Alignment for Open-Vocabulary Groun ding
Junyi Hu, Tian Bai, Fengyi Wu, Wenyan Li, Zhenming Peng, Yi ZHANG
Open-vocabulary grounding requires accurate vision-language alignment under weak supervision, yet existing methods either rely on global sentence embeddings tha t lack fine-grained expressiveness or introduce token-level alignment with expli cit supervision or heavy cross-attention designs. We propose \textbf{ExpAlign}, a theoretically grounded vision-language alignment framework built on a principl ed multiple instance learning formulation. ExpAlign introduces an Expectation Al ignment Head that performs attention-based soft MIL pooling over token-region si milarities, enabling implicit token and instance selection without additional an notations. To further stabilize alignment learning, we develop an energy-based m ulti-scale consistency regularization scheme, including a Top-K multi-positive c ontrastive objective and a Geometry-Aware Consistency Objective derived from a L agrangian-constrained free-energy minimization. Extensive experiments show that ExpAlign consistently improves open-vocabulary detection and zero-shot instance segmentation, particularly on long-tail categories. Most notably, it achieves 36 .2 AP$_r$ on the LVIS minival split, outperforming other state-of-the-art method s at comparable model scale, while remaining lightweight and inference-efficient

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Topology-Preserving Neural Operator Learning via Hodge Decomposition
Dongzhe Zheng, Tao Zhong, Christine Allen-Blanchette
In this paper, we study solution operators of physical field equations on geometric meshes from a function-space perspective. We reveal that Hodge orthogonality fundamentally resolves spectral interference by isolating unlearnable topological degrees of freedom from learnable geometric dynamics, enabling an additive approximation confined to structure-preserving subspaces. Building on Hodge theory and operator splitting, we derive a principled operator-level decomposition. The result is a Hybrid Eulerian-Lagrangian architecture with an algebraic-level inductive bias we call Hodge Spectral Duality (HSD). In our framework, we use discrete differential forms to capture topology-dominated components and an orthogonal auxiliary ambient space to represent complex local dynamics. Our method achieves superior accuracy and efficiency on geometric graphs with enhanced fidelity to physical invariants.
****
[CLS] is Not Enough: Multi-Label Recognition via Patch-Level Inference and Adaptive Aggregation
Akang Wang, Xili Deng, Zhanxuan Hu, Yi Zhao, Yonghang Tai, Huafeng Li
Vision-Language Models such as CLIP exhibit strong zero-shot recognition capability by aligning images with textual concepts, yet they often underperform on multi-label recognition where multiple objects co-exist. A key bottleneck is that the CLS token, as a single global visual representation, is insufficient to faithfully encode diverse targets with varying scales, contexts, and co-occurrence patterns. To address this limitation, we present a new multi-label image recognition framework, termed **PIAA**, which formulates prediction as ***P**atch-level **I**nference followed by **A**daptive **A**ggregation*. Specifically, we first enhance patch-wise predictions from two complementary perspectives: (i) mitigating semantic entanglement in the visual encoder to obtain more discriminative patch representations, and (ii) learning an unsupervised visual classifier to narrow the vision–language modality gap. We then introduce an adaptive aggregation module that consolidates patch-level scores into the final multi-label prediction. Notably, the entire pipeline is fully *Training-free*, requiring no gradient updates or parameter fine-tuning. Experiments show that our method achieves strong improvements with minimal extra computation, exceeding a 6\% mAP gain on the challenging NUS-WIDE benchmark over representative baselines.
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Towards the Explainability of Temporal Graph Networks via Memory Backtracking and Topological Attribution
Yazheng Liu, Xi Zhang, Sihong Xie, Hui Xiong
Temporal graphs are ubiquitous in real-world applications and Temporal Graph Networks (TGNs) have achieved superior predictive accuracy.
Understanding which historical events drive
model predictions can enhance trustworthiness of TGNs. Existing explanation methods overlook the memory module, the core component that records and updates node histories, leaving the influence of past events unexplored. To address this, we attribute TGNs predictions through the topology attribution tree and memory backtracking tree. The topology attribution tree captures the influence of neighbors and their memory vectors, then the memory backtracking tree quantifies how historical events shape node memory vectors. We apply the LRP in TGNs,
ensuring that the total contribution of events equals the model’s logits. Finally, top-k selection may be unfaithful due to the nonlinear mapping from logits to probabilities, we design optimization objectives to identify the important events. Experiments on nine temporal graph datasets, spanning node property prediction, link prediction tasks and graph classification tasks, show that our method provides faithful explanations and outperforms state-of-the-art baselines. The code is available at https://github.com/yazhengliu/MemExplainer.
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A Unifying View of Variational Generative Wasserstein Flows

Paul Caucheteux, Clément Bonet, Anna Korba
Many modern generative models can be viewed as minimizing divergences between pr
obability distributions, yet they rely on different algorithmic and geometric pr
inciples. Wasserstein gradient flows provide a continuous-time formulation for o
ptimizing over distributions, and can be approximated through their implicit dis
cretization via the Jordan–Kinderlehrer–Otto (JKO) scheme. In this work, we pres
ent a unified theoretical framework for generative modeling based on Wasserstein
 gradient flows, which we refer to as Generative Wasserstein Flows (GWF). We sho
w that a broad class of existing methods can be derived as instances of parametr
ic JKO schemes for f-divergence objectives, and we establish equivalences betw
een several recently proposed algorithms. We extend this framework beyond f-di
vergences to Integral Probability Metrics and squared Maximum Mean Discrepancy,
deriving new JKO-based generative algorithms, and clarifying their connections w
ith GANs. We study empirically the impact of the JKO regularization for a wide s
et of objectives. Finally, we analyze parametric Wasserstein flows, where the dy
namics are restricted to distributions induced by parametrized maps.
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FedScar: Correcting Geometric Bias for Flatness-Consistent Federated Learning
Jianfeng Lu, YuZhao Xiang, Yue Chen, Gang Li, Shuqin Cao, Guanghui Wen
Federated Learning (FL) often suffers from degraded generalization under statist
ical heterogeneity, where client updates systematically deviate from the global
objective. While recent Sharpness-Aware Minimization (SAM) methods promote local
ly flat solutions, they implicitly assume that local flatness transfers to the g
lobal model, which generally does not hold under heterogeneous data distribution
s. This mismatch gives rise to a flatness discrepancy induced by misaligned loss
 landscapes. To address this issue, we propose FedScar, a federated optimization
 framework that explicitly corrects heterogeneityinduced geometric inconsistency
. FedScar maintains a history-accumulated geometric bias to capture persistent c
urvature skew across clients, and employs a variance-aware injection mechanism t
o steer local updates toward regions that are flat with respect to the global ob
jective. We provide a theoretical interpretation of FedScar as a Split-Dual ADMM
 formulation, which jointly enforces parameter consensus and geometric alignment
. Extensive experiments under severe heterogeneity demonstrate that FedScar cons
istently reduces flatness discrepancy and improves generalization over state-of-
the-art methods, without incurring additional communication overhead. The source
 code is available at https://github.com/Ultraman6/FedScar.
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A Regime-Aware Trajectory Prediction Framework for 1000+ Systems Biology Models
Heng Rao, Jason Zipeng Zhang, Yu Gu, Zhenghao Liu, Ge Yu, Jeffrey Su, Yang Cao,
Fan Yang, Minghan Chen
Predicting long-horizon trajectories of biological dynamical systems remains cha
llenging due to substantial system heterogeneity.
Most existing machine learning approaches are system-specific, requiring retrain
ing for each new system and exhibiting limited generalization across distinct bi
ological regimes. To address this limitation, we create a large-scale benchmark
of over 1,000 ODE-based systems biology models spanning diverse organisms, biolo
gical processes, and dynamical behaviors. Building on this benchmark, we propose
 a regime-aware trajectory prediction framework that enables cross-system genera
lization and uncertainty quantification for unseen systems. Our approach introdu
ces structured initial states derived from biological regime priors, such as gro
wth trends and oscillatory rhythms, into conditional flow matching, replacing th
e standard Gaussian source distribution. We provide theoretical justification fo
r this initialization and empirically demonstrate state-of-the-art accuracy (31\
% MAE reduction), well-calibrated uncertainty (17\% CRPS improvement), and effic
ient long-horizon inference across the benchmark.
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From Reward-Free Representations to Preferences: Rethinking Offline Preference-B
ased Reinforcement Learning
Jun-Jie Yang, Chia-Heng Hsu, Kui-Yuan Chen, Ping-Chun Hsieh
Preference-based reinforcement learning (PbRL) avoids explicit reward engineerin

g by learning from pairwise human preference feedback. Existing offline PbRL methods typically follow a two-stage pipeline, first learning a reward or preference model from labeled preferences and then performing offline RL on unlabeled data. We revisit offline PbRL through the lens of reward-free representation learning (RFRL) from the zero-shot RL literature, and propose a new training framework that first learns latent successor-measure representations from reward-free offline data, followed by contrastive search and fine-tuning using preference data. Through extensive experiments and ablations, we show that our method achieves superior preference efficiency over offline PbRL baselines. This work is the first to connect RFRL with PbRL, highlighting its potential as a feedback-efficient solution. Our code is publicly available at https://github.com/rl-bandits-lab/FB-PbRL.
****

VIA-SD: Verification via Intra-Model Routing for Speculative Decoding
Yuchen Xian, Yang He, Yunqiu Xu, Yi Yang
Speculative decoding (SD) addresses the high inference costs of LLMs by having lightweight drafters generate candidates for large verifiers to validate in parallel. Existing draft-verify methods use binary decisions: accept or fully recompute. Yet we find that many rejected tokens can be verified correctly by a slim submodel derived from the full verifier via intra-model routing, instead of the full verifier. This motivates our slim-verifier to handle tokens requiring moderate verification resources, reducing expensive large-model calls. We propose **V**erification via **I**ntr**a**-Model Routing for **S**peculative **D**ecoding (VIA-SD), a multi-tier framework using a routed slim-verifier. Draft tokens are processed hierarchically: direct acceptance for high-confidence cases, slim-verifier regeneration for medium-confidence cases, and full-model verification for uncertain cases. Across four representative tasks and multiple model families, VIA-SD reduces rejection rates by 0.10–0.22 and delivers 10–20% speedups over strong SD baselines, while achieving 2.5–3× acceleration over non-drafting decoding. Moreover, VIA-SD is compatible with existing SD frameworks without modifying their training procedures. Our results suggest multi-tier SD as a general paradigm for scalable and efficient LLM inference. Project page: https://zju-xyc.github.io/VIA-SD-Project-Page/
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Any-Order GPT as Masked Diffusion Model: Decoupling Formulation and Architecture
Shuchen Xue, Tianyu Xie, Tianyang Hu, Zijin Feng, Jiacheng Sun, Kenji Kawaguchi, Zhenguo Li, Zhi-Ming Ma
Efficiently scaling Large Language Models (LLMs) necessitates exploring alternatives to dominant autoregressive (AR) methods, with Masked Diffusion Models (MDMs) emerging as candidates. However, comparing AR (typically decoder-only) and MDM (often encoder-only) paradigms is confounded by differing architectures, obscuring true algorithmic and efficiency trade-offs. This research decouples these factors by evaluating MDMs within a decoder-only framework to: (1) Equitably compare MDM (as Any-Order AR) and standard AR paradigms through discrepancies on orders. (2) Investigate MDM architectural impacts on computational efficiency. We show decoder-only MDMs, despite a larger modeling space, can achieve significant inference speedups ($\sim25\times$) and comparable perplexity with techniques like temperature annealing, offering a path to reduced inference compute. This work provides insights for developing more computationally efficient foundation models by disentangling core modeling choices from architectural influences. Code is available at \url{https://github.com/scxue/AO-GPT-MDM}.
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FedReLa: Imbalanced Federated Learning via Re-Labeling
Guangzheng Hu, Patricia Menendez, Feng Liu, Mingming Gong, Guanghui Wang, Liuhua Peng
Federated learning has emerged as the foremost approach for decentralized model training with privacy preserving. The global class imbalance and cross-client data heterogeneity naturally coexist, and the mismatch between local and global imbalances exacerbates the performance degradation of the aggregated model. The agnosticism of global class distribution poses significant challenges for data-lev

el methods, especially under extreme conditions with severe class absence across clients. In this paper, we propose FedReLa, a novel data-level approach that tackles the coexistence of data heterogeneity and class imbalance in federated learning. By re-labeling samples with a feature-dependent label re-allocator, FedReLa corrects biased global decision boundaries without requiring knowledge of the global class distribution. This modular, model-agnostic approach can be integrated with algorithmic methods to deliver consistent improvements without additional communication overhead. Through extensive experiments, our method significantly improves the accuracy of minority classes and the overall accuracy on stepwise-imbalanced and long-tailed datasets, outperforming the previous state of the art.
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f-Trajectory Balance: A Loss Family for Tuning GFlowNets, Generative Models, and LLMs with Off- and On-Policy Data
Jake Fawkes, Jason Hartford
In GFlowNets and variational inference, it has been shown that the mean square error between target and model log probabilities is an effective, low variance, surrogate loss for training generative models. This loss has the property that when evaluated \emph{on-policy} its gradients correspond to those of the KL divergence, while \emph{off-policy} it remains a valid loss with the same global minimiser. In this work, we demonstrate that this construction can be extended to the whole family of f-divergences, leading to a family of losses whose on-policy gradients are that of the corresponding f-divergence, but retain the same global minimiser off-policy. Specifically, we show that the on-policy gradients lead to a one to one correspondence between translation invariant loss functions on the target and model log probabilities, and f-divergences. This equivalence allows us to design new surrogate loss functions for tuning a wide class of generative models that inherit the properties of the corresponding f-divergence, such as being more mode covering, whilst being applicable to off-policy data. We apply our losses on a range of tasks, including classic synthetic examples, SynFloWNets for molecule discovery, and asynchronous large language model (LLM) tuning, demonstrating that our models retain their predicted properties on- and off-policy and can be applied to a wide class of generative models.
****
Saving Foundation Flow-Matching Priors for Inverse Problems
Yuxiang Wan, Ryan Devera, Wenjie Zhang, Ju Sun
Foundation flow-matching (FM) models promise universal priors for solving inverse problems (IPs); yet today, they trail behind domain-specific and even untrained priors. \emph{How can we unlock their potential?} We introduce FMPlug, a plug-in framework that redefines how foundation FMs are used in IPs. FMPlug combines an instance-guided, time-dependent warm-start strategy with sharp Gaussianity regularization, adding problem-specific guidance while preserving the Gaussian structures. For evaluation, we consider both simple image restoration tasks and scientific IPs with a few similar samples---where the prohibitive cost of data collection and model training hinders the development of domain-specific generative models. Our superior experimental results confirm the effectiveness of FMPlug. Overall, FMPlug paves the way for making foundation FM models practical, reusable priors for IPs, especially scientific ones with few similar samples.
****
Scaling Vision Transformers for Functional MRI with Flat Maps
Connor Lane, Mihir Tripathy, Leema Krishna Murali, Ratna Sagari Grandhi, Shamus Sim Zi Yang, Sam Gijsen, Debojyoti Das, Manish Ram, Utkarsh Kumar Singh, Cesar Kadir Torrico Villanueva, YUXIANG WEI, Will Beddow, Gianfranco Cortes, Suin Cho, Daniel Z Kaplan, Benjamin Warner, Tanishq Mathew Abraham, Paul Steven Scotti
We study the problem of training self-supervised foundation models for functional MRI. Our main contributions are: (1) we introduce a new model family (CortexMAE) trained using the masked autoencoder framework on 2.1K hours of open fMRI data, and (2) we release the first open evaluation suite (Brainmarks) for fMRI foundation models. Our core innovation is simple: we adapt the Vision Transformer to fMRI by first converting each 3D fMRI volume to a 2D map using a cortical flat

map projection. We directly compare flat maps to both parcellation and volume-ba
sed representations. While each has its advantages, flat maps generally perform
best. We perform the first systematic scaling analysis for fMRI and observe stri
ct power law scaling, albeit with limits. Finally, we use Brainmarks to do contr
olled benchmark comparisons. On subject-level trait prediction, we report a chal
lenging null result: no single model achieves clear state-of-the-art performance
. Moreover, all models struggle to outperform a simple functional connectivity b
aseline. On cognitive state decoding, we observe more robust performance, and in
 this setting our CortexMAE family outperforms prior models by a large margin. C
ode, models, and datasets are available at https://github.com/MedARC-AI/CortexMA
E and https://github.com/MedARC-AI/Brainmarks.
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NaRA: Noise-Aware LoRA for Parameter-Efficient Fine-Tuning of Diffusion LLMs
Shuaidi Wang, Zhan Zhuang, HUANG Ruping, Yu Zhang
Diffusion Large Language Models (dLLMs) have emerged as a promising non-autoregr
essive generative paradigm. Given the prohibitive computational cost of full fin
e-tuning, Parameter-Efficient Fine-Tuning (PEFT) has become the standard approac
h. However, existing PEFT methods (e.g., LoRA), originally tailored for autoregr
essive models, rely on static parameters that are agnostic to the noise level. C
onsequently, they ignore the intrinsic dynamics of the diffusion process, where
input distributions and generation difficulty shift significantly along the deno
ising trajectory, rendering them suboptimal for dLLMs. To address this, we propo
se **N**oise-**a**ware Low-**R**ank
Adaptation (NaRA), which introduces a low-rank core matrix generated by a li
ghtweight, globally shared hypernetwork conditioned on the noise level. This des
ign enables the update matrices to vary continuously along the diffusion process
 while keeping parameter and latency overhead negligible. We provide a theoretic
al justification for the proposed NaRA framework and empirically demonstrate con
sistent improvements over noise-agnostic baselines across commonsense reasoning,
 mathematical reasoning, and code generation benchmarks. Our code is available a
t https://github.com/generaldi/NaRA.
****

Robust In-Context Reinforcement Learning Under Reward Poisoning Attacks
Paulius Sasnauskas, Yi■it Yal■n, Goran Radanovic
We study the corruption-robustness of in-context reinforcement learning (ICRL),
focusing on the Decision-Pretrained Transformer (DPT, Lee et al., 2023). To addr
ess the challenge of reward poisoning attacks targeting the DPT, we propose a no
vel adversarial training framework, called Adversarially Trained DPT (AT-DPT). O
ur method simultaneously trains a population of attackers to minimize the true r
eward of the DPT by poisoning environment rewards, and a DPT model to infer opti
mal actions from the poisoned data. We evaluate the effectiveness of our approac
h against standard bandit algorithms, including robust baselines designed to han
dle reward contamination. Our results show that AT-DPT significantly outperforms
 them in bandit settings under a learned attacker, and generalizes to more compl
ex environments such as adaptive attackers and MDPs. It shows promise in ICRL as
 a meta-RL approach to learning effective corruption-robust algorithms.
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Zeroth-Order Optimization at the Edge of Stability
Minhak Song, Liang Zhang, Bingcong Li, Niao He, Michael Muehlebach, Sewoong Oh
Zeroth-order (ZO) methods are widely used when gradients are unavailable or proh
ibitively expensive, including black-box learning and memory-efficient fine-tuni
ng of large models, yet their optimization dynamics in deep learning remain unde
rexplored. In this work, we provide an explicit step size condition that exactly
 captures the (mean-square) linear stability of a family of ZO methods based on
the standard two-point estimator. Our characterization reveals a sharp contrast
with first-order (FO) methods: whereas FO stability is governed solely by the la
rgest Hessian eigenvalue, mean-square stability of ZO methods depends on the ent
ire Hessian spectrum. Since computing the full Hessian spectrum is infeasible in
 practical neural network training, we further derive tractable stability bounds
 that depend only on the largest eigenvalue and the Hessian trace. Empirically,

we find that full-batch ZO methods operate at the edge of stability: ZO-GD, ZO-GDM, and ZO-Adam consistently stabilize near the predicted stability boundary across CNNs, ResNets, and Transformers on vision tasks. Our results highlight an implicit regularization effect specific to ZO methods, where large step sizes primarily regularize the Hessian trace, whereas in FO methods they regularize the top eigenvalue.

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Coverage Improvement and Fast Convergence of On-policy Preference Learning

Juno Kim, Jihun Yun, Jason D. Lee, Kwang-Sung Jun

On-policy preference learning algorithms for language model alignment such as online direct policy optimization (DPO) can significantly outperform their offline counterparts. We provide a theoretical explanation for this phenomenon by analyzing how the sampling policy's coverage evolves throughout on-policy training. We propose and rigorously justify the coverage improvement principle: with sufficient batch size, each update moves into a region around the target where coverage is uniformly better, making subsequent data increasingly informative and enabling rapid convergence. In the contextual bandit setting with Bradley-Terry preferences and linear softmax policy class, we show that on-policy DPO converges exponentially in the number of iterations for batch size exceeding a generalized coverage threshold. In contrast, any learner restricted to offline samples from the initial policy suffers a slower minimax rate, leading to a sharp separation in total sample complexity. We further propose a hybrid sampler based on a novel preferential G-optimal design, which removes dependence on coverage and guarantees convergence in just two rounds. Finally, we develop principled on-policy schemes for reward distillation which achieve faster noiseless rates. Experimentally, we confirm that on-policy DPO and our proposed reward distillation algorithms outperform their off-policy counterparts and enjoy stable gains across iterations.

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Structure-Aware Riemannian Flow Matching for Registration and Fusion of Hyperspectral and Multispectral Images

Quan Zhang, Jun Li, Weilong Zhu, MINGYANG LI, Qinmu Shen, Yuanxi Peng

Precise alignment is a prerequisite for hyperspectral and multispectral image fusion, yet existing methods struggle with complex non-rigid deformations. Existing techniques either suffer from inter-task error accumulation by treating registration and fusion as disjoint processes or neglect the geometric nature of distortions by relying on isotropic Euclidean metrics. We propose Structure-Aware Riemannian Flow Matching (SA-RFM), a geometry-informed framework for joint registration and fusion of hyperspectral and multispectral images. SA-RFM reformulates registration as dynamic optimal transport on a structure-induced Riemannian manifold, where anisotropic costs are derived from MSI structural cues. To circumvent the complexity of explicit OT solvers, we incorporate this geometry into a conditional flow matching framework via a local cost approximation. This formulation is realized through two synergistic mechanisms: a Riemannian Flow Matching objective that enforces structure-aware error measurement, and an optimal transport direction regularization that aligns the velocity field with the induced metric, thereby resolving the fundamental mismatch between anisotropic costs and conventional Euclidean supervision. Extensive experiments on four datasets demonstrate the superiority of our method. Code is available at: https://github.com/ZhangQuan-hub.

****

Concept Removal for Frontier Image Generative Models

Aditya Kumar, Pierre Joly, Adam Dziedzic, Franziska Boenisch

Image generative models are trained on massive, largely uncurated internet-scale datasets that contain undesirable visual concepts. Efficiently removing such concepts from the model generations without degrading the quality of output images remains challenging. We introduce a novel concept removal method for frontier diffusion and image autoregressive models, such as, SD3.5, Flux, and Infinity. Our intervention replaces the internal bottleneck layer present in all these modern models with a transcoder that is trained to replicate the original layer while

structuring it into distinct activation features. This in■place substitution cr
eates an integrated filter through which concept■specific signals can be selecti
vely disabled while preserving the rest of the model’s behavior. Since the inter
vention modifies the model backbone rather than attaching an external component,
it remains persistent under white■box access. Empirically, the approach achieve
s state■of■the■art concept removal performance across modern diffusion and autor
egressive models, maintains visual generation quality, provides robustness again
st adversarial prompts, and supports sequential removal of diverse concepts. Thi
s positions our method as a practical approach for concept removal in frontier i
mage generative models.
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Towards Feedback-to-Plan Decisions for Self-Evolving LLM Agents in CUDA Kernel G
eneration

Yee Hin Chong, Jiaming Wu, Youhui Zhang, Peng Qu

Large language models (LLMs) have shown strong empirical gains as self-evolving
agents for CUDA kernel generation, driven by feedback-conditioned planning acros
s generations. However, how planning decisions attribute and combine heterogeneo
us feedback signals remains opaque. Standard end-to-end ablations fail to resolv
e this question, as iterative planning amplifies early perturbations and conflat
es feedback effects with trajectory-dependent drift.

We introduce \texttt{CUDAnalyst}, a unified analysis layer for controlled, gener
ation-level attribution of planning decisions to feedback components via traject
ory freezing and selective feedback injection. \texttt{CUDAnalyst} enables stabl
e generation-level evaluation and principled coalitional-style attribution of fe
edback effects and interactions. Our results show that explicit planning is bene
ficial only when feedback is aligned, that effective planning emerges from struc
tured multi-feedback interactions, and that high-level plans from stronger reaso
ning models can partially transfer to weaker ones. These trends hold across refe
rence backbones, representative workloads, and reference induction regimes, indi
cating that the identified feedback-to-plan structure is robust within the contr
olled axes studied.
****

A Game-Theoretic Analysis of Attacks on Large Language Models via Compositional
Skills

Xinbo Wu, Huan Zhang, Abhishek Umrawal, Lav R. Varshney

As large language models grow increasingly capable, concerns about their safe de
ployment have intensified. While numerous alignment strategies aim to restrict h
armful behavior, these defenses can still be circumvented through carefully desi
gned adversarial prompts. In this work, we introduce a theoretical framework tha
t formalizes a game between an attacker hiding its intent via compositional skil
ls and a defender. Within this framework, we design a theoretical best-response
attack strategy and show that it is closely related to many existing adversarial
prompting methods. We further analyze the resulting game, characterize its equi
libria, and reveal inherent advantages for the attacker. Drawing on our theoreti
cal analysis, we also derive a provably optimal defense strategy. Empirically, w
e evaluate a practical instantiation of the theoretically optimal attack and obs
erve stronger performance relative to existing adversarial prompting approaches
in diverse settings encompassing different LLMs and benchmarks.
****

CircuitPrint: Mechanistic Circuit Fingerprints for Large Language Models

Zhenxiong Yan, Suhang Yao, Yu Liu, Wenqiang Jin

Large language models (LLMs) are trained at significant computational and data c
ost, making them valuable intellectual property (IP). Existing IP verification m
ethods primarily rely either on invasive watermarking that degrades model utilit
y, or on superficial behavioral signatures disrupted by fine-tuning and model me
rging. This apparent trade-off between model utility and IP protection has const
rained practical deployment. We challenge this trade-off and propose CircuitPrin
t, a non-invasive IP fingerprinting framework that enables robust verification t
hrough standard model queries by leveraging stable internal computational circui

ts of LLMs. We show that these circuits function as a persistent computational b
ackbone across model derivatives, allowing them to serve as stable fingerprints
for distinguishing LLMs. Building on this stability, CircuitPrint constructs IP
signatures by identifying mechanistically essential supernodes that causally pro
duce specific predictions within these circuits. Specifically, trigger queries a
re synthesized to replicate the internal suppression of these supernodes, thereb
y inducing distinctive and observable output shifts. Experimental results demons
trate that CircuitPrint substantially outperforms existing baselines while remai
ning robust under aggressive fine-tuning and model merging, effectively resolvin
g this trade-off without altering model parameters.
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Ramba: Selective State-Space Models for Relational Deep Learning
Yiming Liu, Chunyu Wei, Haozhe Lin, Fengjun Xiao, Junqi Zhang, Yunhai Wang, Yueg
uo Chen
Relational Deep Learning aims to learn directly on multi-table databases, yet cu
rrent methods face a fundamental tension: Transformers' quadratic complexity pro
hibits the large contexts relational data demands, while GNNs sacrifice global c
ontext for efficiency. We introduce Ramba, the first selective state-space model
for relational databases. Our approach features two innovations: (1) Topology-A
ware Linearization, which processes cells via global columnar serialization in O
(L) complexity while recovering relational structure through sparse entity and f
oreign-key attention masks; and (2) Schema Dynamic Gating, which modulates SSM s
tate transitions based on semantic alignment between the currently scanned attri
bute and the prediction target, enabling cross-table relevance filtering without
relying on value distributions. Together, these enable Ramba to ingest vast rel
ational contexts while selectively retaining semantically relevant information,
a combination existing architectures cannot achieve. Experiments demonstrate sta
te-of-the-art performance with linear scalability across diverse relational benc
hmarks.
****

Zero-source LLM Hallucination Detection with Human-like Criteria Probing
Jiahao Yang, Shuhai Zhang, Hailong Kang, Feng Liu, Qi Chen, Mingkui Tan
Large language models (LLMs) often hallucinate by generating factually incorrect
or unfaithful content, posing significant risks to their safe use. Detecting su
ch hallucinations is particularly challenging under the zero-source constraint,
where no model internals or external references are available, and detection mus
t rely solely on the textual query–answer pair. In this paper, we propose Human
-like Criteria Probing for Hallucination Detection (HCPD), a paradigm that emula
tes the multi-faceted reasoning of human evaluators. Its core is a Human-like Cr
iteria Probing (HCP) mechanism, in which a LLM agent adaptively decomposes its j
udgment into a weighted set of interpretable criteria and aggregates criterion-s
pecific scores into a final truthfulness measure. To achieve this adaptive capab
ility, we introduce a reward-based alignment scheme using only weak supervision
from semantic consistency. At inference, we employ a multi-sampling aggregation
strategy to ensure robust decisions while preserving full interpretability. We f
urther provide theoretical analysis supporting the reliability of our approach.
Extensive experiments show that HCPD consistently outperforms state-of-the-art b
aselines, offering an effective and explainable solution for zero-source halluci
nation detection.
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From Observations to States: Latent Time Series Forecasting
Jie Yang, Yifan Hu, Yuante Li, Kexin Zhang, Kaize Ding, Philip S. Yu
Deep learning has achieved strong performance in Time Series Forecasting (TSF).
However, we identify a critical representation paradox, termed Latent Chaos: mod
els with accurate predictions often learn latent representations that are tempor
ally disordered and lack continuity. We attribute this to the dominant observati
on-space forecasting paradigm, where minimizing point-wise errors on noisy and p
artially observed data encourages shortcut solutions instead of the recovery of
underlying system dynamics. To address this, we propose Latent Time Series Forec
asting (LatentTSF), a paradigm that shifts TSF from observation regression to la

tent state prediction. LatentTSF employs an AutoEncoder to project each observation into a learned latent state space and performs forecasting entirely in this space, allowing the model to focus on learning structured temporal dynamics. We provide an information-theoretic analysis showing that the latent objectives can be motivated as surrogates for maximizing mutual information between predicted and ground-truth latent states and future observations. Extensive experiments on widely-used benchmarks confirm that LatentTSF effectively mitigates latent chaos, yielding consistent improvements in both forecasting accuracy and representation quality. Our code is available at https://github.com/Muyiiiii/LatentTSF.
****

Instance-Specific Approximation Ratios for Correlation Clustering and Max-Cut
Sebastian Lüderssen, Ioana-Oriana Bercea, Stefan Neumann
For many NP-hard optimization problems, strong theoretical inapproximability results exist. However, in practice, heuristics regularly outperform these pessimistic worst-case results on real-world datasets. Assessing the quality of these algorithms' outputs is often difficult since we lack good lower bounds on the optimal solution. In this paper, we present efficient algorithms for computing lower bounds on the optimal solutions for correlation clustering, which is a popular problem in social-network analysis. Our lower bounds allow us to provide empirical certificates that bound the solution quality of practical algorithms by obtaining instance-specific approximation ratios. Our main technical contribution is an algorithm that approximates an LP relaxation of a related triangle covering problem in near-linear time on sparse graphs; the algorithm is based on the multiplicative weights update framework and runs on graphs with millions of edges in a few minutes. For the concrete problem of correlation clustering, our lower bounds certify that state-of-the-art heuristics achieve almost optimal approximation ratios of 0.94 for the agreement version and 1.97 for the disagreement version (averaged over 7 real-world datasets). We also show similar results for the fundamental max-cut problem.
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Detecting Perspective Shifts in Multi-Agent Systems
Eric W Bridgeford, Hayden Helm
Generative models augmented with external tools and update mechanisms (or *agents*) have demonstrated capabilities beyond intelligent prompting of base models. As agent use proliferates, dynamic multi-agent systems have naturally emerged. Recent work has investigated the theoretical and empirical properties of low-dimensional representations of agents based on query responses at a single time point. This paper introduces the Temporal Data Kernel Perspective Space (TDKPS), which jointly embeds agents across time, and proposes several novel hypothesis tests for detecting behavioral change at the agent- and group-level in black-box multi-agent systems. We characterize the empirical properties of our proposed tests, including their sensitivity to key hyperparameters, in simulations motivated by a multi-agent system of evolving digital personas. Finally, we demonstrate via natural experiment that our proposed tests detect changes that correlate sensitively, specifically, and significantly with a real exogenous event. TDKPS is the first principled framework for monitoring behavioral dynamics in black-box multi-agent systems -- a critical capability as generative agent deployment continues to scale.
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TWLA: Achieving Ternary Weights and Low-Bit Activations for LLMs via Post-Training Quantization
Zhixiong Zhao, Zukang Xu, Zhixuan Chen, Xing Hu, Zhe jiang, Dawei Yang
Large language models (LLMs) exhibit exceptional general language processing capabilities, but their memory and compute costs hinder deployment. Ternarization has emerged as a promising compression technique, offering significant reductions in model size and inference complexity. However, existing methods struggle with heavy-tailed activation distributions and therefore keep activations in high precision, fundamentally limiting end-to-end inference acceleration.
To overcome this limitation, we propose **TWLA**, a post-training quantization (

PTQ) framework that achieves 1.58-bit weight compression and 4-bit activation qu
antization while maintaining high accuracy. TWLA comprises three components: (1)
 Euclidean-to-Manifold Asymmetric Ternary Quantizer (E2M-ATQ) minimizes layer-ou
tput error under weight ternarization via a two-stage optimization from Euclidea
n initialization to manifold relocation; (2) Kronecker Orthogonal Tri-Modal Shap
ing (KOTMS) applies a Kronecker-structured orthogonal rotation to reshape weight
s into ternary-friendly tri-modal distributions, while the shared rotation stati
stically suppresses activation outliers; and (3) Inter-Layer Aware Activation Mi
xed Precision (ILA-AMP) explicitly introduces adjacent-layer second-order intera
ction costs in bit allocation and jointly optimizes for the layer-wise disparity
 of activation quantization gains induced by the shared orthogonal transform, pr
eventing cascades triggered by a few weak layers.
Extensive experiments demonstrate that TWLA is a PTQ method that maintains high
accuracy under the **W1.58A4** configuration, while delivering significant infer
ence acceleration. The code is available at [TWLA](https://github.com/Kishon-zzx
/TWLA).
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Turning Stale Gradients into Stable Gradients: Coherent Coordinate Descent with Implicit Landscape Smoothing for Lightweight Zeroth-Order Optimization
Chen Liang, Xiatao Sun, Qian Wang, Daniel Rakita
Zeroth-Order (ZO) optimization is pivotal for scenarios where backpropagation is
 unavailable, such as memory-constrained on-device learning and black-box optimi
zation. However, existing methods face a stark trade-off: they are either sample
-inefficient (e.g., standard finite differences) or suffer from high variance du
e to randomized estimation (e.g., random subspace methods). In this work, we pro
pose Coherent Coordinate Descent (CoCD), a deterministic, sample-efficient, and
budget-aware ZO optimizer. Theoretically, we formalize the notion of gradient co
herence and demonstrate that CoCD is equivalent to Block Cyclic Coordinate Desce
nt (BCCD) with ``warm starts,'' effectively converting historical (stale) gradie
nts from a liability into a computational asset. This mechanism enables $O(1)$ q
uery complexity per step while maintaining global descent directions. Furthermor
e, we derive error bounds revealing a counter-intuitive insight: larger finite-d
ifference step sizes can induce an implicit smoothing effect on the optimization
 landscape by reducing the effective smoothness constant, thereby improving conv
ergence stability. Experiments on MLP, CNN, and ResNet architectures (up to 270
k parameters) demonstrate that CoCD significantly outperforms BCCD in terms of s
ample efficiency and convergence loss/accuracy, and exhibits superior stability
over randomized ZO methods. Our results suggest that deterministic, structure-aw
are updates offer a superior alternative to randomization for lightweight ZO opt
imization.
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Sonar-TS: Search-Then-Verify Natural Language Querying for Time Series Databases
Zhao Tan, Yiji Zhao, Shiyu Wang, Chang Xu, Yuxuan Liang, Xiping Liu, Shirui Pan,
 Ming Jin
Natural Language Querying for Time Series Databases (NLQ4TSDB) aims to assist no
n-expert users retrieve meaningful events, intervals, and summaries from massive
 temporal records.
However, existing Text-to-SQL methods are not designed for continuous morphologi
cal intents such as shapes or anomalies, while time series models struggle to ha
ndle ultra-long histories.
To address these challenges, we propose Sonar-TS, a neuro-symbolic framework tha
t tackles NLQ4TSDB via a "Search-Then-Verify" pipeline.
Analogous to active sonar, it utilizes a feature index to "ping'' candidate wind
ows via SQL, followed by generated Python programs to "lock on'' and verify cand
idates against raw signals.
To enable effective evaluation, we introduce NLQTSBench, the first large-scale b
enchmark designed for NLQ over TSDB-scale histories.
Our experiments highlight the unique challenges within this domain and demonstra
te that Sonar-TS effectively navigates complex temporal queries where traditiona
l methods fail.

This work presents the first systematic study of NLQ4TSDB, offering a general fr
amework and evaluation standard to facilitate future research.
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General Analysis of LMO-based Optimizers: Beyond Bounded Variance
Egor Shulgin, Mohamed Awad, Peter Richtárik, Eduard Gorbunov
We study a broad family of momentum *Linear Minimization Oracle* (LMO) methods t
hat includes normalized SGD with momentum, sign-based (Adam-like) directions, an
d Muon (spectral) updates. Our focus is on subsampling regimes where the classic
al uniformly-bounded-variance model can be fragile even for finite-sum objective
s on unbounded domains. To obtain subsampling-faithful guarantees, we analyze th
is LMO family under *expected smoothness* (ABC condition), which captures common
 sampling schemes. We establish a unified nonconvex convergence theory via a new
 self-bounding closure that handles the history-coupling induced by momentum und
er ABC. Our bounds recover known bounded-variance results as a special case and
simplify in strong-growth regimes. Specializing to τ-nice sampling, we deri
ve explicit batch-size scaling laws, predicting that the optimal momentum must i
ncrease with the batch size to maximize sample efficiency. We further identify a
 theoretical **optimal batch size** that minimizes total sample complexity. Expe
riments on linear and matrix regression corroborate these predictions, showing a
 distinct diagonal shift in the optimal momentum-batch landscape that matches ou
r theoretical scaling.
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Many-Shot CoT-ICL: Making In-Context Learning Truly Learn
Tsz Ting Chung, Lemao Liu, Mo Yu, Dit-Yan Yeung
While many-shot ICL achieves remarkable performance, prior studies of its scalin
g behavior have mainly focused on non-reasoning tasks. In this work, we study ma
ny-shot ICL on reasoning tasks, with a particular focus on many-shot chain-of-th
ought in-context learning (CoT-ICL). Analyzing across non-reasoning and reasonin
g tasks and across non-reasoning and reasoning-oriented LLMs, we identify severa
l distinctive properties of many-shot CoT-ICL. We further interpret these findin
gs by viewing many-shot CoT-ICL as in-context test-time learning rather than sca
led pattern matching, and suggest two principles: (i) demonstrations should be e
asy for the target model to understand, and (ii) they should be ordered to suppo
rt a smooth conceptual progression. Guided by the principle, we propose Curvilin
ear Demonstration Selection (CDS), a simple ordering method that yields up to a
5.42 percentage-point gain on a math task with 64 demonstrations. Overall, our r
esults reframe the long context window from a retrieval buffer into a structured
 curriculum for in-context test-time learning.
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d3LLM: Ultra-Fast Diffusion LLM using Pseudo-Trajectory Distillation
Yu-Yang Qian, Junda Su, Lanxiang Hu, Peiyuan Zhang, Zhijie Deng, Peng Zhao, Hao
Zhang
Diffusion large language models (dLLMs) offer capabilities beyond those of autor
egressive (AR) LLMs, such as parallel decoding and random-order generation. Howe
ver, realizing these benefits in practice is non-trivial, as dLLMs inherently fa
ce an *accuracy-parallelism trade-off*. Despite increasing interest, existing me
thods typically focus on only one-side of the coin, targeting either efficiency
or accuracy. To address this limitation, we propose d3LLM (*Pseudo-Distilled Dif
fusion Large Language Model*), striking a balance between accuracy and paralleli
sm: (i) during training, we introduce *pseudo-trajectory distillation* to teach
the model which tokens can be decoded confidently at early steps, thereby improv
ing parallelism; (ii) during inference, we employ *entropy-based multi-block dec
oding* with a KV-cache refresh mechanism to achieve high parallelism while maint
aining accuracy. To better evaluate dLLMs, we also introduce AUP (*Accuracy Unde
r Parallelism*), a new metric that jointly measures accuracy and parallelism. Ex
periments demonstrate that our d3LLM achieves up to $10\times$ speedup over vani
lla LLaDA/Dream, and $5\times$ speedup over AR models without much accuracy drop
. Our code is available at https://github.com/hao-ai-lab/d3LLM.
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Controllable Molecule Generation via Sparse Representation Editing: An Interpret

ability-Driven Perspective
Zhuoran Li, Xu Sun, Chang Wen Chen, Wanyu Lin
Controllable molecule generation is crucial for diverse scientific applications, such as drug discovery and materials design. While large language models (LLMs) show great promise, their dense and entangled representations impede precise control over the generation of molecules with bespoke substructures or properties. To address this, we propose Sparse Representation Editing (SpaRE), an interpretability-driven framework for fine-grained and precise control in LLM-based molecule generation. The crux of SpaRE is to learn an overcomplete sparse feature space that disentangles LLM representations into a compact set of latent features corresponding to chemically meaningful concepts. Within this space, we can directly manipulate these concept-aligned latent features to achieve (1) local control, by generating target atoms and functional groups at specified positions; and (2) global control, by customizing the overall structural and physicochemical properties within defined ranges. In this way, our framework advances interpretability from post-hoc analysis to actionable generative control. Experiments show that SpaRE can generate chemically desirable molecules under complex constraints in real-world scenarios, while offering mechanistic insights for quantitative structure–property analysis. The code and demo are available at https://github.com/WanyuGroup/ICML2026_SpaRE.
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The Optimal Sample Complexity of Linear Contracts
Mikael Møller Høgsgaard
In this paper, we settle the problem of learning optimal linear contracts from data in the offline setting, where agent types are drawn from an unknown distribution and the principal's goal is to design a contract that maximizes her expected utility. Specifically, our analysis shows that the simple Empirical Utility Maximization (EUM) algorithm yields an ε-approximation of the optimal linear contract with probability at least $1-\delta$, using just $O(\ln(1/\delta) / \varepsilon^2)$ samples. This result improves upon previously known bounds and matches a lower bound from (Dütting et al., 2025) up to constant factors, thereby proving its optimality.
Furthermore, our result establishes the stronger guarantee of uniform convergence: the empirical utility of every linear contract is a ε-approximation of its true expectation with probability at least $1-\delta$, using the same optimal $O(\ln(1/\delta) / \varepsilon^2)$ sample complexity.
****
Revisiting Parameter-Based Knowledge Editing in Large Language Models: Theoretical Limits and Empirical Evidence
Wanying Ren, Xin Song, Futing Wang, Guoxiu He, Aixin Sun
Parameter-based knowledge editing updates the internal knowledge of large language models (LLMs) via localized weight modifications and has attracted significant attention.
However, most existing methods overlook fundamental theoretical limitations and are rarely evaluated under realistic, practice-oriented settings.
In this paper, we first present a theoretical analysis based on the dimensional Collapse Hypothesis, explaining how localized parameter edits can propagate along fragile directions in the representation space, inducing global interference and ultimately causing reasoning collapse.
Building on this insight, we conduct a comprehensive empirical evaluation by systematically varying knowledge complexity, number of edits, evaluation dimensions, and baseline methods.
Our results show that parameter-based editing methods consistently damage core LLM capabilities.
In contrast, a simple retrieval-based baseline achieves consistently stronger performance than all parameter-editing methods across all evaluated conditions.
These findings highlight that preserving the fundamental capabilities of LLMs after knowledge editing should be a central concern for future research.
****
Action-Sufficient Goal Representations

Jinu Hyeon, Woobin Park, Hongjoon Ahn, Taesup Moon
Hierarchical policies in offline goal-conditioned reinforcement learning (GCRL) addresses long-horizon tasks by decomposing control into high-level subgoal planning and low-level action execution. A critical design choice in such architectures is the goal representation—the compressed encoding of goals that serves as the interface between these levels. Existing approaches commonly derive goal representations while learning value functions, implicitly assuming that preserving information sufficient for value estimation is adequate for optimal control. We show that this assumption can fail, even when the value estimation is exact, as such representations may collapse goal states that need to be differentiated for action learning. To address this, we introduce an information-theoretic framework that defines *action sufficiency*, a condition on goal representations necessary for optimal action selection. We prove that value sufficiency does not imply action sufficiency and empirically verify that the latter is more strongly associated with control success in a discrete environment. We further demonstrate that standard log-loss training of low-level policies naturally induces action-sufficient representations. Our experimental results a popular benchmark demonstrate that our actor-derived representations consistently outperform representations learned via value estimation.
****
Closing the Sim-to-Real Gap in Non-Markovian Spreading Processes via GPU-Accelerated Distributional RL
Heman Shakeri
Controlling spreading processes on networks, such as epidemics,
information cascades, and product adoption, requires policies that perform
on realistic stochastic dynamics, not just tractable approximations. Yet policies trained on standard simplifications (mean-field ODEs, Markovian dynamics) suffer severe performance degradation at deployment. We trace this sim-to-real gap to three theoretical pathologies: Optimism Bias, where deterministic approximations systematically underestimate variance via Jensen's inequality; Hub Blindness, where global state aggregation obscures the super-spreaders driving scale-free networks; and the Valley of Death, where mean-value critics fail to navigate the bimodal nature (extinction vs. viral) of cascade outcomes. We resolve these challenges through two synergistic contributions. First, the Stratified Mean-Field Observer partitions nodes by influence tier, preserving hub dynamics at $O(N)$ cost while producing fixed-dimensional observations that enable zero-shot transfer across network scales and topologies. Second, we show that distributional RL via Truncated Quantile Critics improves risk-aware control of bimodal cascades. Trained on a GPU-accelerated simulator supporting non-Markovian renewal dynamics, our approach achieves $59\times$ improvement over Markovian baselines and robust zero-shot transfer to real-world social networks (Facebook, Twitter, YouTube), significantly mitigating the simulation-to-reality gap.
****
PACE: Post-Causal Entropy Modeling for Learned LiDAR Point Cloud Compression
Jiahao Zhu, Kang You, Dandan Ding, Zhan Ma
LiDAR point cloud compression is vital for autonomous systems to handle massive data from high-resolution sensors. While learned entropy modeling built upon octree structures yields high compression gains, it faces two critical bottlenecks: 1) prohibitive latency, particularly during decoding, caused by causal, multi-stage context modeling; and 2) a rigid performance-latency trade-off, preventing a single model from adapting to varying constraints. These limitations stem from the tight coupling between the context aggregation backbone and probability prediction. To address this, we propose PACE, a new framework that reformulates ancestral context aggregation as a non-causal backbone and confines causality to a lightweight, stage-scalable predictor, eliminating repetitive backbone executions and reducing computational overhead. The predictor supports an arbitrary number of prediction stages, enabling seamless adaptation across diverse performance-latency trade-offs without reloading parameters. Experiments demonstrate that PACE sets a new state-of-the-art in compression efficiency, achieving notable BD-BR savings and reducing decoding latency by over 90\% in autoregressive mode, mak

ing it attractive for practical applications.
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Towards Understanding Massive Activations in Attention Sink Mechanism
Haiyu Wang, Yuanyuan Lin
Recent studies have revealed two intriguing phenomena in large language models: attention sinks and massive activations. However, the co-emergence and co-existence of these two phenomena remain poorly understood. In this work, we revisit the prevailing view that massive activations are the primary mechanism responsible for concentrating attention on sink tokens, and provide a more nuanced interpretation of their relationship. Through both theoretical analysis and empirical evidence, we demonstrate that massive activations and attention sinks jointly act to prevent excessive token mixing in self-attention. Specifically, attention sinks suppress mixing among non-sink tokens, whereas massive activations suppress mixing between sink tokens and non-sink tokens. Furthermore, our theory provides a principled explanation of how KV-biases, gating mechanisms, and normalization layers can remove massive activations while largely preserving attention sinks. We further conduct intervention analyses and find that removing the value vector of the sink token can recover attention sinks even when massive activations are entirely suppressed. Overall, this work provides a mechanistic perspective on how massive activations and attention sinks interact under normalization and self-attention layers, offering new insights into their functional roles in Transformer models.
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Provable Training Data Identification for Large Language Models
Zhenlong Liu, Hao Zeng, Weiran Huang, Hongxin Wei
Identifying training data of large-scale models is critical for copyright litigation, privacy auditing, and ensuring fair evaluation. However, existing works typically treat this task as an instance-wise identification without controlling the error rate of the identified set, which cannot provide statistically reliable evidence. In this work, we formalize training data identification as a set-level inference problem and propose Provable Training Data Identification (PTDI), a distribution-free approach that enables provable and strict false identification rate control. Specifically, our method computes conformal p-values for each data point using a set of known unseen data and then develops a novel Jackknife-corrected Beta boundary (JKBB) estimator to estimate the training-data proportion of the test set, which allows us to scale these p-values. By applying the Benjamini–Hochberg (BH) procedure to the scaled p-values, we select a subset of data points with provable and strict false identification control. Extensive experiments across various models and datasets demonstrate that PTDI achieves higher power than prior methods while strictly controlling the FIR.
****
Decentralized and Disentangled Task–Role Representation Learning for Generalizable Offline Multi-Agent Meta Reinforcement Learning
Lei Yuan, Ruiqi Xue, Yang Yu
Offline meta reinforcement learning (RL) enables agents to learn a unified policy from multi-task offline data to support generalization in out-of-distribution (OOD) tasks.
Recent approaches in single-agent RL tackle this by learning an efficient task representation to distinguish between tasks, showing promising adaptation ability.
However, when extended to multi-agent settings, these methods struggle with decentralized task identification due to limited global information, and suffer from inefficient knowledge transfer in the absence of role information.
To address this, we propose D^{2}TR, a novel context-based meta RL framework with efficient decentralized and disentangled task-role identification.
Specifically, D^{2}TR first introduces mutual information knowledge distillation to align decentralized task representations with centralized task representations inferred from global trajectories, enabling efficient decentralized team-centric information identification. Next, D^{2}TR leverages a large language model to assign semantic roles to trajectories in offline data, and achieves effective

individual-centric information inference by learning decentralized role representations.
Extensive experiments conducted on commonly used multi-agent environments, including CN, SMAC, and SMACv2, demonstrate that D^{2}TR exhibits strong generalization performance to unseen tasks, outperforming prior multi-agent multi-task and context-based meta RL baselines.
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General Covariant Action Modeling: Constructing Generalized Manifolds via Spatio-Temporal Decoupling
Huaihai Lyu, Chaofan Chen, Mingyu Cao, Yuheng Ji, Changsheng Xu
Achieving robust generalization from limited data is a central challenge in embodied intelligence. Prevailing methods fail by regressing absolute coordinates, which violates the principle of general covariance. Theoretically, this conflates the intrinsic task geometry with rigid execution patterns, binding policies to specific motion styles and fixed speeds. To resolve this, we propose the Generalized Action Manifold (GAM) framework that enforces general covariance through structural disentanglement. Specifically, GAM constructs the manifold by enforcing invariance across two orthogonal dimensions: (1) Temporal Invariance, utilizing an Arc-Length Parameterizer to orthogonalize the spatial path geometry from temporal dynamics, ensuring robustness to velocity variations; (2) Geometric Invariance, where a Schema-Affine-Factorization mechanism maps trajectories to canonical “world lines” in the Lie-algebraic tangent space. This distinguishes invariant topological schemas from affine modulations, ensuring spatial generalizability. By integrating GAM within a structured Vision-Language-Action (VLA) architecture, we expand sparse training data into a continuous, valid action manifold. Empirical results demonstrate that GAM enables superior transfer and robustness capabilities, significantly outperforming geometry-agnostic baselines.
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Scale-Aware Domain Harmonization for Domain Adaptation Person Search
Huibing Wang, Guojian Zhao, Jinjia Peng, Linfeng Qi, Mingze Yao, Jiqing Zhang
Unsupervised Domain Adaptation (UDA) person search aims to transfer a model trained on a labeled source domain to an unlabeled target domain without using target annotations. However, existing UDA methods frequently neglect the issue of scale inconsistency between the source and target domains. This inconsistency arises from variations in camera height, tilt angle, focal length, and scene layout. To address this challenge, we propose a Scale-Aware Consistent Alignment Learning (SCALE) framework. Specifically, we propose a Scale-aware Domain Harmonization (SDH) module, which adaptively harmonizes semantic and structural scales through cross-path interaction and consistency refinement to alleviate cross-domain scale inconsistency. To further reduce pseudo-label noise, we introduce a Bidirectional Cluster Regularization (BCR) strategy, which improves pseudo-label reliability by refining the clustering results through a second regularized clustering step. By collaboratively alleviating the impact of scale misalignment and enhancing pseudo-label reliability, our approach achieves state-of-the-art performance on two benchmark person search datasets, with 82.3% mAP and 84.0% top-1 on the CUHK-SYSU dataset, 41.7% mAP and 82.4% top-1 on the PRW dataset. Our source code is available at https://github.com/whhbdmu/SCALE.
****

Group Distributionally Robust Optimization-Driven RL for LLM Reasoning
Kishan Panaganti, Zhenwen Liang, Wenhao Yu, Haitao Mi, Dong Yu
Reasoning post-training with GRPO is typically built on *static uniformity*: uniform prompt sampling and a fixed number of rollouts per prompt. For heterogeneous, heavy-tailed reasoning data, this wastes compute on already-solved patterns while under-training the long tail of hard problems. We cast GRPO post-training as two independent GDRO games (not coupled) over *dynamic difficulty groups* defined by an *online train-time pass@n* statistic (computed from the rollouts used for GRPO updates): a *data adversary* that reshapes prompt sampling and a *compute adversary* that redistributes rollouts. **Prompt-GDRO** applies multiplicative-weights reweighting over bins (with an EMA-debiased difficulty score) to upweight persistently hard groups without frequency bias. **Rollout-GDRO** allocate

s rollouts across bins under a fixed mean budget via a shadow-price controller, improving gradient information efficiency on high-uncertainty groups while preserving the mean train-time rollout budget. Our approach is principled and theory-driven: we provide no-regret guarantees for the Prompt-GDRO game (via an entropy-regularized GDRO surrogate) and a variance-proxy analysis that yields a square-root optimal compute allocation for Rollout-GDRO. On DAPO 14.1k with Qwen3-Base (1.7B/4B/8B), each controller improves pass@8 by 9--13\% over GRPO; targeted Qwen3-4B reruns reproduce the Prompt-GDRO gap under a second seed, show second-seed Rollout-GDRO gains on four of five comparable benchmarks, and show low sensitivity to several key hyperparameters.
****

The Entropic Signature of Class Speciation in Diffusion Models
Florian Handke, Dejan Stancevic, Felix Koulischer, Thomas Demeester, Luca Ambrogioni
Diffusion models do not recover semantic structure uniformly over time. Instead, samples transition from semantic ambiguity to class commitment within a narrow regime. Recent theoretical work attributes this transition to dynamical instabilities along class-separating directions, but practical methods to detect and exploit these windows in trained models are still limited. We show that tracking the class-conditional entropy of a latent semantic variable given the noisy state provides a reliable signature of these transition regimes. By restricting the entropy to semantic partitions, the entropy can furthermore
resolve semantic decisions at different levels of abstraction. We validate our method on EDM2-XS and Stable Diffusion 1.5, where class-conditional entropy consistently isolates the noise regimes critical for semantic structure formation. Finally, we use our framework to quantify how guidance redistributes semantic information over time. Together, these results connect information-theoretic and statistical physics perspectives on diffusion and provide a principled basis for time-localized control.
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Amortized Simulation-Based Inference in Generalized Bayes via Neural Posterior Estimation
Shiyi Sun, Geoff K. Nicholls, Kate Lee
Generalized Bayesian Inference (GBI) tempers a loss with a temperature $\beta>0$ to
mitigate overconfidence and improve robustness under model misspecification, but existing GBI methods typically rely on costly MCMC or SDE-based samplers and must be re-run for each new dataset and each β-value. We give the first fully amortized variational approximation for the specific case of the tempered posterior family $p_\beta(\theta\mid x) \propto \pi(\theta)p(x \mid\\theta)^\beta$ by training a single (x,β)-conditioned neural posterior estimator $q_\phi(\theta \mid x, \beta)$ that enables sampling in a single forward pass, without simulator calls or
inference-time MCMC. We introduce two complementary training routes: (i) synthesizes off-manifold samples $(\theta, x) \sim \pi(\theta)p(x \mid \theta)^\beta$ and (ii) reweights a fixed base dataset $\pi(\theta)p(x \mid \theta)$ using
self-normalized importance sampling (SNIS), where we show that the SNIS-weighted objective provides a consistent forward-KL fit to the tempered posterior with finite weight variance.
Across four standard simulation-based inference (SBI) benchmarks—including the chaotic Lorenz–96 system—our β-amortized estimator achieves competitive posterior approximations, in standard two-sample metrics, with non-amortized MCMC-based power-posterior samplers over a wide range of temperatures.
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Twice Sequential Monte Carlo for Tree Search
Yaniv Oren, Joery A. de Vries, Pascal R. Van der Vaart, Matthijs T. J. Spaan, Wendelin Boehmer
Model-based reinforcement learning (RL) methods that leverage search are responsible for many milestone breakthroughs in RL.
Sequential Monte Carlo (SMC) recently emerged as an alternative to the Monte Car

lo Tree Search (MCTS) algorithm which drove these breakthroughs. SMC is easier t
o parallelize and more suitable to GPU acceleration. However, it also suffers fr
om large variance and path degeneracy which prevent it from scaling well with in
creased search depth, i.e., increased sequential compute. To address these probl
ems, we introduce Twice Sequential Monte Carlo Tree Search (TSMCTS). Across disc
rete and continuous environments TSMCTS outperforms the SMC baseline as well as
a popular modern version of MCTS as a policy improvement operator, scales favora
bly with sequential compute, reduces estimator variance and mitigates the effect
s of path degeneracy while retaining the properties that make SMC natural to par
allelize.
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An Embarrassingly Simple Way to Optimize Orthogonal Matrices at Scale
Adrián Javaloy, Antonio Vergari
Orthogonality constraints are ubiquitous in robust and probabilistic machine lea
rning. Unfortunately, current optimizers are computationally expensive and do no
t scale to problems with hundreds or thousands of constraints. One notable excep
tion is the Landing algorithm (Ablin et al., 2024) which, however comes at the e
xpense of temporarily relaxing orthogonality. In this work, we revisit and impro
ve on the ideas behind Landing, enabling the inclusion of modern adaptive optimi
zers while ensuring that orthogonal constraints are effectively met. Remarkably,
 these improvements come at little to no cost, and reduce the number of required
 hyperparemeters. Our algorithm POGO is fast and GPU-friendly, _consisting of on
ly 5 matrix products_, and in practice maintains orthogonality at all times. On
several challenging benchmarks, POGO greatly outperforms recent optimizers and s
hows it can optimize problems with thousands of orthogonal matrices in minutes w
hile alternatives would take hours. As such, POGO sets a milestone to finally ex
ploit orthogonality constraints in ML at scale. A public PyTorch implementation
of POGO is available at https://github.com/adrianjav/pogo.
****

OSF: On Pre-training and Scaling of Sleep Foundation Models
Zitao Shuai, Zongzhe Xu, David Yang, Wei Wang, Yuzhe Yang
Polysomnography (PSG) provides the gold standard for sleep assessment but suffer
s from substantial heterogeneity across recording devices and cohorts.
There have been growing efforts to build general-purpose foundation models (FMs)
 for sleep physiology, but lack an in-depth understanding of the pre-training pr
ocess and scaling patterns that lead to more generalizable sleep FMs.
To fill this gap, we curate a massive corpus of 166,500 hours of sleep recording
s from nine public sources and establish SleepBench, a comprehensive, fully open
-source benchmark.
Leveraging SleepBench, we systematically evaluate four families of self-supervis
ed pre-training objectives and uncover three critical findings:
(1) existing FMs fail to generalize to missing channels at inference;
(2) channel-invariant feature learning is essential for pre-training;
and (3) scaling sample size, model capacity, and multi-source data mixture consi
stently improves downstream performance.
With an enhanced pre-training and scaling recipe, we introduce OSF, a family of
sleep FMs that achieves state-of-the-art performance across nine datasets on div
erse sleep and disease prediction tasks.
Further analysis of OSF also reveals intriguing properties in sample efficiency,
 hierarchical aggregation, and cross-dataset scaling. Codes are available at: ht
tps://github.com/yang-ai-lab/OSF-Open-Sleep-FM.
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Tackling Fake Forgetting through Uncertainty Quantification
Yingdan Shi, Sijia Liu, Kaize Ding, Ren Wang
Machine unlearning seeks to remove the influence of specified data from a traine
d model. While the unlearning accuracy provides a widely used metric for assessi
ng unlearning performance, it falls short in assessing the reliability of forget
ting. In this paper, we find that the forgetting data points misclassified by un
learning accuracy still have their ground truth labels included in the conformal
 prediction set from the uncertainty quantification perspective, leading to a ph

enomenon we term fake forgetting. To address this issue, we propose a novel metr
ic CR, inspired by conformal prediction, that offers a more reliable assessment
of forgetting quality. Building on these insights, we further propose an unlearn
ing framework CPU that incorporates conformal prediction into the Carlini & Wagn
er adversarial attack loss, enabling the ground truth label to be effectively re
moved from the conformal prediction set. Through extensive experiments on image
classification tasks, we demonstrate both the effectiveness of our proposed metr
ic and the superior forgetting quality achieved by our framework. Code is availa
ble at https://github.com/TIML-Group/Conformal-Prediction-Unlearning.
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Krause Synchronization Transformers
Jingkun Liu, Yisong Yue, Max Welling, Yue Song
Self-attention in Transformers relies on globally normalized softmax weights, ca
using all tokens to compete for influence at every layer. When composed across d
epth, this interaction pattern induces strong synchronization dynamics that favo
r convergence toward a dominant mode, a behavior associated with representation
collapse and attention sink phenomena. We introduce $\textbf{Krause Attention}$,
 a principled attention mechanism inspired by bounded-confidence consensus dynam
ics. Krause Attention replaces similarity-based global aggregation with distance
-based, localized, and selectively sparse interactions, promoting structured loc
al synchronization instead of global mixing. We relate this behavior to recent t
heory modeling Transformer dynamics as interacting particle systems, and show ho
w bounded-confidence interactions naturally moderate attention concentration and
 alleviate attention sinks. Restricting interactions to local neighborhoods also
 reduces runtime complexity from quadratic to linear in sequence length. Empiric
ally, we validate Krause Attention across diverse settings, including vision (Vi
T on CIFAR/ImageNet), autoregressive image generation (MNIST/CIFAR-10), large la
nguage models (Llama/Qwen), and language models trained from scratch at multiple
 scales (100M/200M). Across these domains, Krause Attention achieves consistent
performance gains while improving computational efficiency, highlighting bounded
-confidence dynamics as a scalable and effective inductive bias for attention. P
roject page is available at https://jingkun-liu.github.io/krause-sync-transforme
rs/.
****

Hybrid Reinforcement Learning in Adversarial Markov Decision Processes
Duo Cheng, Xingyu Zhou, Bo Ji
We study hybrid reinforcement learning (RL) in adversarial Markov Decision Proce
sses (MDPs), where the learner simultaneously receives on-policy feedback from t
he executed policy and off-policy feedback from a fixed behavior policy, and los
s functions can change arbitrarily over time. On-policy feedback allows explorat
ion and ensures the worst-case guarantee against any comparator policy, while of
f-policy feedback provides coverage-dependent guarantee that scales with the "mi
smatch" between the behavior and comparator policies (called coverage ratio) and
 can be sharper than on-policy results whenever this ratio is small. We propose
a new hybrid RL framework that accommodates adversarial losses and unknown trans
itions, preserving off-policy guarantees while ensuring non-trivial worst-case p
erformance.
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Learning with Admissibility: Robust Fuzzy Hashing for Cross-Modal Retrieval with
 Noisy Labels
Xincheng Sun, Ruitao Pu, Guangsi Shi, Zhenwen Ren, Peng Hu, Yuan Sun
Recently, cross-modal hashing (CMH) has garnered significant attention due to it
s low storage costs and high retrieval efficiency. Most existing CMH methods imp
licitly assume the availability of high-quality annotations, which is often viol
ated in real-world scenarios as label noise inevitably arises from human errors
or non-expert annotations. To cope with noisy supervision, current noise-robust
CMH methods mainly follow two paradigms, i.e., noise separation and label smooth
ing. They often discard the predicted noisy instances or smooth discriminative s
ignals to mitigate the impact of noisy labels. However, aggressive separation le
ads to reduced data utilization, while smoothing weakens the discriminative capa

bility regarding the true distribution of clean instances. To address these limi tations, we propose a novel Robust Fuzzy Cross-modal Hashing framework (RFCMH) t hat introduces fuzzy set theory to endow the labels with admissibility, thereby obtaining reliable discriminative supervision from noisy labels. Specifically, w e first leverage possibility and necessity measures to model the noisy labels. S ubsequently, we propose Fuzzy Admissibility Refinement (FAR) to dynamically cali brate supervision signals, thereby preventing the model from being misled by fal se positives. Furthermore, we introduce Dual-Granularity Structural Alignment (D GSA) to enforce both cross-modal alignment and instance-level uniformity, ensuri ng stable and diverse representations. Extensive experiments on multiple benchma rks demonstrate that RFCMH achieves state-of-the-art retrieval performance. Code is available at https://github.com/XinchengSun/RFCMH.
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Reverse-Engineering Model Editing on Language Models
Zhiyu Sun, Minrui Luo, Yu Wang, Zhili Chen, Tianxing He
Large language models (LLMs) are pretrained on corpora containing trillions of t okens and, therefore, inevitably memorize sensitive information. Locate-then-edi t methods, as a mainstream paradigm of model editing, offer a promising solution by modifying model parameters without retraining. However, in this work, we rev eal a critical vulnerability of this paradigm: the parameter updates inadvertent ly serve as a side channel, enabling attackers to recover the edited data. We pr opose a two-stage reverse-engineering attack named KSTER (KeySpaceReconsTruction -then-EntropyReduction) that leverages the low-rank structure of these updates. First, we theoretically show that the row space of the update matrix encodes a " fingerprint" of the edited subjects, enabling accurate subject recovery via spec tral analysis. Second, we introduce an entropy-based prompt recovery attack that reconstructs the semantic context of the edit. Extensive experiments on multipl e LLMs demonstrate that our attacks can recover edited data with high success ra tes. Furthermore, we propose subspace camouflage, a defense strategy that obfusc ates the update fingerprint with semantic decoys. This approach effectively miti gates reconstruction risks without compromising editing utility. Our code is ava ilable at https://github.com/reanatom/EditingAttack.
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Bits That Count: Quantifying and Predicting Capabilities of Language Models
Elizabeth Donoway, Hailey Joren, Michael R DeWeese, Ethan Perez, John Schulman, Fabien Roger, Jan Leike
When does learning elicit \textit{existing} knowledge, and when does it primaril y teach \textit{new} capabilities? We find that the amount of generalizable info rmation language models learn during training predicts the origins of their emer gent capabilities. Minuscule amounts of information---in many cases, a few bits in a single example---can unlock large fractions of models' maximum performance when capabilities are \textit{elicited} rather than \textit{taught}. We quantify these learning regimes using excess description length (EDL), an information-th eoretic measure of generalizable information learned during training. We find th at elicitation and teaching exhibit distinct EDL signatures that characterize th e predominant learning mechanism as information scales: elicitation requires ord ers of magnitude less information than teaching to comparable performance. We de monstrate that EDL provides a practical tool for quantitatively estimating the m aximum amount of predictive information models can compress from data into train able parameters during learning. These capacity limits describe optimal tradeoff s between data and parameter count that robustly predict when parameter-efficien t fine-tuning methods (\textit{e.g.}, LoRA) will underperform full fine-tuning.
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Distribution Matching Variational AutoEncoder
Sen Ye, Jianning Pei, Mengde Xu, Shuyang Gu, Chunyu Wang, Liwei Wang, Han Hu
Most visual generative models compress images into a latent space before applyin g diffusion or autoregressive modelling. Yet, existing approaches such as VAEs a nd foundation model aligned encoders implicitly constrain the latent space witho ut explicitly shaping its distribution, making it unclear which types of distrib utions are optimal for modeling. We introduce **Distribution-Matching VAE** (**D

MVAE**), which explicitly aligns the encoder’s latent distribution with an arbit
rary reference distribution via a distribution matching constraint. This general
izes beyond the Gaussian prior of conventional VAEs, enabling alignment with dis
tributions derived from self-supervised features, diffusion noise, or other prio
r distributions. With DMVAE, we can systematically investigate which latent dist
ributions are more conducive to modeling, and we find that SSL-derived distribut
ions provide an excellent balance between reconstruction fidelity and modeling e
fficiency, reaching a gFID of 3.2 on ImageNet with only 64 training epochs. Our
results suggest that choosing a suitable latent distribution structure (achieved
 via distribution-level alignment), rather than relying on fixed priors, is key
to bridging the gap between easy-to-model latents and high-fidelity image synthe
sis.
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EPS3D: End-to-End Feed-Forward 3D Panoptic Segmentation
Runsong Zhu, Jiaxin Guo, Xiaoyang Guo, Zhengzhe Liu, Ka-Hei Hui, Wei Yin, Kai Ch
en, Wei Chen, Weiqiang Ren, Yunhui Liu, Pheng-Ann Heng, Chi-Wing Fu
This paper introduces EPS3D, a new end-to-end feed-forward framework for open-vo
cabulary 3D panoptic segmentation. Unlike existing methods relying on additional
 preprocessing, we design an end-to-end architecture, with a distillation-based
training strategy on diverse 3D scenes to predict 3D-aware semantic and instance
 features from multi-view images, improving 3D consistency and avoiding error ac
cumulation. We further propose a mutual enhancement module to enforce inherent s
emantic-instance consistency. By aligning semantics within instances (Ins2Sem) a
nd refining instance features with semantic guidance (Sem2Ins), we achieve more
coherent 3D scene understanding. Ultimately, EPS3D outperforms SOTA baselines on
 two benchmarks (e.g., +13% mIoU for semantics on Replica) with high efficiency
(e.g., 1s per scene), supporting tasks like robotic manipulation and 3D scene ed
iting.
****

Test-Time Anchoring for Discrete Diffusion Posterior Sampling
Litu Rout, Andreas Lugmayr, Yasamin Jafarian, Srivatsan Varadharajan, Constantin
e Caramanis, Sanjay Shakkottai, Ira Kemelmacher-Shlizerman
While continuous diffusion models have achieved remarkable success, discrete dif
fusion offers a unified framework for jointly modeling text and images. Beyond u
nification, discrete diffusion provides faster inference, finer control, and pri
ncipled training-free guidance, making it well-suited for posterior sampling. Ex
isting approaches to posterior sampling using discrete diffusion face severe cha
llenges: derivative-free guidance yields sparse signals, continuous relaxations
limit applicability, and split Gibbs samplers suffer from the curse of dimension
ality. To overcome these limitations, we introduce Anchored Posterior Sampling (
APS), built on two key innovations: *quantized expectation* for gradient-like gu
idance in discrete embedding space, and *anchored remasking* for adaptive decodi
ng. APS achieves state-of-the-art performance among discrete diffusion samplers
on both linear and nonlinear inverse problems across the standard image benchmar
ks. We demonstrate the generality of APS through training-free stylization and t
ext-guided editing. We further apply APS to a large-scale diffusion language mod
el, showing consistent improvement in question answering.
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Conformal Policy Control
Drew Prinster, Clara Fannjiang, Ji Won Park, Kyunghyun Cho, Anqi Liu, Suchi Sari
a, Samuel Don Stanton
An agent must try new behaviors to explore and improve. In high-stakes environme
nts, an agent that violates safety constraints may cause harm and must be taken
offline, curtailing any future interaction. Imitating old behavior is safe, but
excessive conservatism discourages exploration. How much behavior change is too
much? We show how to use any safe reference policy as a probabilistic regulator
for any optimized but untested policy. Conformal calibration on data from the sa
fe policy determines how aggressively the new policy can act, while provably enf
orcing the user's declared risk tolerance. Unlike conservative optimization meth
ods, we do not assume the user has identified the correct model class nor tuned

any hyperparameters. Unlike previous conformal methods, our theory provides finite-sample guarantees even for non-monotonic bounded loss functions. Our experiments on applications ranging from natural language question answering to biomolecular engineering show that safe exploration is not only possible from the first moment of deployment, but can also improve performance.
****
Fingerprinting Pre-trained Encoders under Arbitrary Downstream Fine-Tuning via Adversarial Shifting
Tianlong Xu, Zixiong Wang, Lishuai Hou, Gaoyang Liu, Chen Wang, Xiaoyi Fan
In the pre-training-fine-tuning paradigm, pre-trained encoders have become high-value intellectual property (IP) due to their immense training costs, necessitating robust protection. Existing fingerprinting or watermarking methods typically rely on pre-defined samples and labels, or require intrusive modifications to the training process. However, downstream fine-tuning can significantly alter an encoder's representation and label space, thereby destroying the label consistency of existing methods and rendering them ineffective. Consequently, it is both challenging and urgent to provide a downstream-agnostic, black-box ownership verification mechanism for pre-trained encoders. To address this, we propose a downstream-agnostic, label-only fingerprinting method that leverages Adversarial Shifting to construct stable fingerprint clusters in the encoder’s latent space. By exploiting the inherent output consistency of these clusters, our method remains effective regardless of the specific downstream task or label mapping. Extensive experiments demonstrate that our method maintains superior robustness and stealthiness across various downstream tasks and category scales, providing a practical and reliable IP protection scheme for high-value pre-trained encoders.
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Real-Time Monitoring and Calibration of Chain-of-Thought Sycophancy in Large Reasoning Models
Jingyu Hu, Shu Yang, Xilin Gong, Hongming Wang, Weiru Liu, Di Wang
Large Reasoning Models (LRMs) suffer from sycophantic behavior, where models tend to agree with users' incorrect beliefs and follow misinformation rather than maintain independent reasoning. This behavior undermines model reliability and poses societal risks. Mitigating LRM sycophancy requires monitoring how this sycophancy emerges during the reasoning trajectory; however, current methods mainly focus on judging based on final answers and correcting them, without understanding how sycophancy develops during reasoning processes.
To address this limitation, we propose MONICA, a novel Monitor-guided Calibration framework that monitors and mitigates sycophancy during model inference at the level of reasoning steps, without requiring the model to finish generating its complete answer.
MONICA integrates a sycophantic monitor that provides real-time monitoring of sycophantic drift scores during response generation with a calibrator that dynamically suppresses sycophantic behavior when scores exceed predefined thresholds.
Extensive experiments across 12 datasets and 3 LRMs demonstrate that our method effectively reduces sycophantic behavior in both intermediate reasoning steps and final answers, yielding robust performance improvements.
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Factored Latent Action World Models
Zizhao Wang, Chang Shi, Jiaheng Hu, Kevin Rohling, Roberto Martín-Martín, Amy Zhang, Peter Stone
Learning latent actions from action-free video has emerged as a powerful paradigm for scaling up controllable world model learning. Latent actions provide a natural interface for users to iteratively generate and manipulate videos. However, most existing approaches rely on monolithic inverse and forward dynamics models that learn a single latent action to control the entire scene, and therefore struggle in complex environments where multiple entities act simultaneously. This paper introduces Factored Latent Action Model (FLAM), a factored dynamics framework that decomposes the scene into independent factors, each inferring its own latent action and predicting its own next-step factor value. This factorized structure enables more accurate modeling of complex multi-entity dynamics and improv

es video generation quality in action-free video settings compared to monolithic models. Based on experiments on both simulation and real-world multi-entity datasets, we find that FLAM outperforms prior work in prediction accuracy and representation quality, and facilitates downstream policy learning, demonstrating the benefits of factorized latent action models.
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Partial Identification under High-Dimensional Potential Outcomes and Confounders via Optimal Transport
Yunfeng Wang, Zhiheng Zhang, Zijun Gao
Partial identification provides informative causal guarantees when point identification is impossible, but existing approaches based on optimal transport (OT) become computationally and statistically intractable in high-dimensional settings. This limitation is particularly severe when both potential outcomes and confounders are high-dimensional, where classical OT-based bounds suffer from the curse of dimensionality and unfavorable convergence rates. To address this challenge, we propose a novel estimator that decomposes the transport problem into a low-dimensional signal subspace and a high-dimensional residual subspace. Unlike existing projection-based methods that discard residual information, we recover the residual transport energy using the Sliced Wasserstein distance, which is computationally efficient and robust to high dimensions. We establish interpretable conditions controlling the approximation gap based on residual structure and provide a data-driven rule for signal dimension selection. Empirical results show that our estimator consistently outperforms projection-only baselines by recovering lost transport energy, yielding more informative causal bounds while remaining computationally tractable in high dimensions.
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MIRA: A Score for Conditional Distribution Accuracy and Model Comparison
Sammy Nasser Sharief, Justine Zeghal, Gabriel Missael Barco, Pablo Lemos, Yashar Hezaveh, Laurence Perreault-Levasseur
We introduce MIRA, a sample-based score for assessing the accuracy of a candidate conditional distribution using only joint samples from the true data-generating process. Relying on the principle that distributions coincide if they assign equal probability mass to all regions, we derive an analytic expression for the MIRA statistic, whose average defines the MIRA score. This formulation further allows us to compute theoretical reference values and uncertainty estimates when the candidate distribution matches the true one. This framework enables model comparison by quantifying the alignment between the conditional distribution of a candidate model and the true data generating process. Consequently, MIRA enables Bayesian model comparison through direct posterior validation, bypassing the challenging evidence computation. We demonstrate its effectiveness across several toy problems and Bayesian inference tasks.
****
Online Packet Scheduling with Deadlines and Learning
Gianmarco Genalti, Achraf Azize, Vianney Perchet
Network routers that enforce Quality-of-Service (QoS) guarantees must decide, at every clock cycle, which expiring packet of information to transmit, even when the value of the packet is unknown until it is processed. We frame this problem as the Online Packet Scheduling with Deadlines (OPSD) problem under Partial Feedback: packets arrive at every clock cycle, with different deadlines, but the weights are only observed after execution. Under a stochastic assumption on the unknown weights, we explore different variants of the OPSD problem with bandit feedback. We establish a connection between our setting and the sleeping bandits problem, and set our learning goal to α-regret minimization. We provide algorithms with provable α-regret guarantees under different spans of slackness, distinguishing systems allowing for randomization and systems that do not. In every scenario, our algorithms achieve an α-regret upper bound of $\widetilde{\mathcal{O}}\left(\sqrt{KT}\right)$, matching the lower bound for the standard bandit setting. In the practically relevant case of 2-bounded deadline instances, where the deadline is set at most one clock cycle away from the arrival, our deterministic algorithm achieves the provably tightest possible competiti

ve ratio. Remarkably, when the number of distinct packet types $K\ge 2$ is finit e, it is possible to break the well-established $\Phi = \frac{1+\sqrt{5}}{2}$ co mpetitive ratio barrier and attain a tighter competitive ratio θ_K rangin g in $[\sqrt{2}, \Phi)$.
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Emergent Visual Representations through Unsupervised Spiking Networks with Synap tic Pruning

Di Hong, Dazhong Rong, Yueming Wang

Recent work has shown that brain-aligned visual representations can emerge even in randomly initialized, high-dimensional neural networks, suggesting that corti cal representations may be discovered rather than fully learned through task opt imization. However, how such latent brain-relevant representations are stabilize d and refined during development remains unclear. Motivated by this perspective and by neuroscientific evidence of activity-dependent synaptic pruning, we study how brain-aligned representations can emerge and be refined from high-dimension al unsupervised spiking systems. We propose a biologically grounded deep SNN tha t integrates unsupervised learning with developmental pruning dynamics. Starting from an overcomplete spiking architecture, the model self-organizes through sen sory-driven activity while selectively eliminating weak or redundant synapses, p rogressively yielding compact and informative representations. Without using lab els, the resulting network forms hierarchical visual representations that strong ly align with neural responses across multiple areas of the mouse and macaque vi sual cortex, outperforming supervised and unsupervised ANN and SNN baselines. Sy naptic pruning further improves alignment and robustness under noisy and few-sho t recognition settings. By unifying high-dimensional unsupervised spiking repres entations with activity-dependent synaptic pruning, this work provides a computa tional account of developmental refinement in visual cortex and bridges recent f indings on emergent brain alignment in random networks with biologically grounde d models of representation learning.
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ConsMSA: Semantic Distribution Consistency Learning for Multimodal Sentiment Ana lysis

Pan Wang, Lipeng Ke, Huajun Ying, Pritish Mohapatra, Rohan Sarkar, Suresh Lakhan i, sankar venkataraman, Jingtong Hu

Multimodal sentiment analysis (MSA) aims to predict human sentiments by integrat ing signals from different modalities such as text, video, and audio. However, r aw multimodal sequences often suffer from semantic inconsistency--exhibiting red undancy or conflicts within and across modalities--which hinders robust understa nding and increases computational cost. To this end, we introduce ConsMSA, which explicitly formalizes semantic distribution consistency across both $\textit{in tra}$- and $\textit{inter}$-modality, providing a consistency-aware mechanism fo r robust and efficient multimodal sentiment prediction. Specifically, ConsMSA pr ojects multimodal token features into a shared sentiment space to compute an Int ra- and Inter-modality Consistency Score (I^2CS). By coupling this score with predictive relevance, we formulate consistency-aware importance signals that are utilized: (i) as a consistency regularizer to align latent distributions during training, (ii) to derive semantic-aware weights for adaptive multimodal token r eweighting, and (iii) as a practical criterion to prune redundant or conflicting tokens. Extensive experiments on CMU-MOSI, CMU-MOSEI, and CH-SIMS demonstrate t hat ConsMSA achieves state-of-the-art performance while remaining robust under a ggressive token compression--retaining only 10\% of tokens yields comparable acc uracy. These results establish semantic distribution consistency as a promising foundation for synergizing predictive robustness with computational efficiency.
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IntentRL: Training Proactive User-intent Agents for Open-ended Deep Research via Reinforcement Learning

Haohao Luo, Zexi Li, Yuexiang Xie, Wenhao Zhang, Yaliang Li, Ying Shen

Deep Research (DR) agents extend Large Language Models (LLMs) beyond parametric knowledge by autonomously retrieving and synthesizing evidence from large web co rpora into long-form reports, enabling a long-horizon agentic paradigm. However,

unlike real-time conversational assistants, DR is computationally expensive and time-consuming, creating an autonomy-interaction dilemma: high autonomy on ambiguous user queries often leads to prolonged execution with unsatisfactory outcomes. To address this, we propose IntentRL, a framework that trains proactive agents to clarify latent user intents before starting long-horizon research. To overcome the scarcity of open-ended research data, we introduce a scalable pipeline that expands a few seed samples into high-quality dialogue turns via a shallow-to-deep intent refinement graph. We further adopt a two-stage reinforcement learning (RL) strategy: Stage I applies RL on offline dialogues to efficiently learn general user-interaction behavior, while Stage II uses the trained agent and a user simulator for online rollouts to strengthen adaptation to diverse user feedback. Extensive experiments show that IntentRL significantly improves both intent hit rate and downstream task performance, outperforming the built-in clarify modules of closed-source DR agents and proactive LLM baselines.

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MINIM: Privacy-Aware Minimal View for Agents via Trusted Local Sanitization

Hexuan Yu, Chaoyu Zhang, Heng Jin, Shanghao Shi, Ning Zhang, Thomas Hou, Wenjing Lou

Modern LLM-powered autonomous agents increasingly rely on rich user interface (UI) state observations to achieve reliable action grounding in complex digital environments. However, many deployments transmit the full UI state to remote inference servers even when most elements are irrelevant to the current task, which can leak sensitive but unnecessary context such as authentication codes, private notifications, and background application states. We propose MINIM, a trusted local broker that performs privacy-aware minimization on the client side before any observation leaves the device. Grounded in Contextual Integrity (CI), MINIM learns a dual-score representation for each UI element by predicting an inherent sensitivity score (s) and a task-conditioned necessity score (n). These scores drive a ternary disclosure policy that keeps essential elements, abstracts sensitive attributes when needed, and removes task-irrelevant content. We optimize a CI-aware objective that penalizes necessity errors more strongly on high-risk content, enabling aggressive pruning while preserving task-critical information. Experiments on real-world UI observations derived from WebArena show that MINIM substantially reduces task-irrelevant sensitive leakage while preserving task-critical semantic context and the interactive affordances required for reliable agent actions.

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A Progressive Evidence Localization Framework Based on Wasserstein Gradient Flows for Document Visual Question Answering

Haosen Wang, Jing Xiao, Mengqiao Li, Xuanze Wang, Mingzheng Zhang, Xiaowang Zhang, Zhiyong Feng

Precise localization of evidence regions in Document Visual Question Answering is crucial for improving model interpretability and reliability.

However, existing methods predominantly adopt one-step localization strategies, which often fail to effectively distinguish evidence regions from irrelevant content when page semantics are complex or evidence regions are extremely small, leading to ambiguous boundaries and inaccurate localization. To address this issue, we propose a progressive evidence localization framework based on Wasserstein gradient flow, which formulates evidence localization as an optimal transport problem over probability distributions.

Since continuous gradient flows are intractable in practice, we employ the Jordan-Kinderlehrer-Otto (JKO) scheme for discrete optimization and further derive an end-to-end trainable loss function that transforms the theoretical formulation into a neural network optimization objective, enabling coarse-to-fine precise characterization of evidence regions.Experimental results demonstrate that the proposed method significantly outperforms existing approaches in both evidence localization and answer generation tasks.

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Rethinking 3D Shape Generation: Diffusion over Superquadrics

Zhiyang Liu, Li Wanze, Yuwei Wu, Chengran Yuan, JIAWEI SUN, Rui Zheng, Marcelo H

Ang Jr
Diffusion models have advanced 3D shape generation, yet most methods still denoise in high-cardinality spaces (e.g., voxel/SDF grids, meshes, or point clouds), which is computationally and memory intensive and makes it difficult to scale in terms of both higher resolution and stronger controllability.
We rethink the diffusion representation and propose to move diffusion from dense geometry to compact geometric primitives, representing each shape as a small set of **superquadrics**.
Instead of operating on thousands to millions of geometric representation values, we leverage 7KB superquadric parameters (pose, size, and shape), drastically reducing diffusion-state dimensionality and per-step compute/memory.
Our diffusion-over-superquadrics improves scalability by supporting broader capabilities (e.g., resolution-free point-cloud decoding, part-level editing, and constraint-based design) and achieving competitive surface-fidelity and distributional performance on standard benchmarks after point-cloud decoding, while enabling efficient generation within 0.6s per shape for most conditions.
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Contrastive Order Learning: A General Framework for Ordinal Regression
Chaewon Lee, BeomJun Shim, Kwang Pyo Choi, Chang-Su Kim
We propose contrastive order learning (ConOrd), a contrastive learning framework for ordinal regression that integrates the strengths of contrastive learning and order learning. While contrastive learning effectively leverages all samples in a batch, it typically ignores the inherent ordering among rank labels. Conversely, order learning explicitly models label ordinality but often relies on local, margin-based comparisons, limiting its ability to capture global ordinal structure. ConOrd addresses these limitations by introducing a contrastive order loss with soft affinity and disparity weights based on rank differences, enabling fine-grained modeling of ordinal relationships across all sample pairs within a batch. Extensive experiments on a range of ordinal regression tasks, including facial age estimation, blind image quality assessment, and blind video quality assessment, demonstrate that ConOrd consistently achieves state-of-the-art performance and generalizes well across diverse ordinal regression scenarios. The source code is available at https://github.com/cwlee00/ConOrd.
****
Understanding the Parameter Space Geometry of Transformers Encoding Boolean Functions
Blanka Kövér, Alexandra Butoi, Anej Svete, Michael Hahn, Ryan Cotterell
Transformers consistently fail to learn certain simple functions that are provably expressible with specific parameter settings.
    This gap between *learnability* and *expressivity* is particularly prominent for sensitive functions---functions whose output is likely to change if a single bit of the input is flipped---for example, Parity.
    While prior work has established that transformers exhibit a bias toward functions with low average sensitivity, the precise mechanism underlying this bias remains poorly understood.
    To shed light on this phenomenon, we study the geometry of transformers' parameter space.
    We show that sensitive functions---even when representable---occupy a vanishingly small region that random initialization is very likely to miss.
    Specifically, we shift the focus from average sensitivity to the full sensitivity profile---the distribution of sensitivity values across all inputs---and prove that randomly initialized transformers almost surely compute functions which have low-sensitivity strings.
    Consequently, any function that lacks such strings is provably unlearnable.
****
How does Bayesian Sampling help Membership Inference Attacks?
Zhenlong Liu, Wenyu Jiang, Feng Zhou, Hongxin Wei
Membership Inference Attacks (MIAs) aim to estimate whether a specific data point was used in the training of a given model. Existing state-of-the-art attacks typically rely on training multiple reference models to approximate the condition

al score distribution for individual data points, which leads to significant com putational overhead and limits their practical applicability. In this work, we propose a novel approach -- Bayesian Membership Inference Attack (BMIA), which performs conditional attack through Bayesian sampling. Specifically, we apply La place approximation to a single reference model to obtain a posterior over model parameters, enabling direct estimation of the conditional score distribution. T heoretically, we demonstrate that Bayesian sampling reduces intra-model variance , thereby improving attack power. This insight naturally motivates the multi-ref erence variant that further enhances performance when additional reference model s are available. Extensive experiments across image, text, and tabular datasets indicate that our method achieves state-of-the-art performance in both effective ness and efficiency.
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Frictional Q-Learning
Hyunwoo Kim, Hyo Kyung Lee
Off-policy reinforcement learning suffers from extrapolation errors when a learn ed policy selects actions that are weakly supported in the replay buffer. In thi s study, we address this issue by drawing an analogy to static friction. From th is perspective, the replay buffer is represented as a smooth, low-dimensional ac tion manifold, where the support directions correspond to the tangential compone nt, while the normal component captures the dominant first-order extrapolation e rror. This decomposition reveals an intrinsic anisotropy in value sensitivity th at naturally induces a stability condition analogous to a friction threshold. To mitigate deviations toward unsupported actions, we propose Frictional Q-Learnin g, an off-policy algorithm that encodes supported actions as tangent directions using a contrastive variational autoencoder. We further show that an orthonormal basis of the orthogonal complement corresponds to normal components under mild local isometry assumptions. Extensive empirical results on standard continuous-c ontrol benchmarks consistently demonstrate robust and stable performance compare d with competitive baselines.
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Learning to Self-Verify Makes Language Models Better Reasoners
Yuxin Chen, Yu Wang, Yi Zhang, Ziang Ye, Zhengzhou Cai, Yaorui Shi, Qi GU, Hui S u, Xunliang Cai, Xiang Wang, An Zhang, Tat-Seng Chua
Recent large language models (LLMs) achieve strong performance in generating pro mising reasoning paths for complex tasks.
However, despite powerful generation ability, LLMs remain weak at verifying thei r own answers, revealing a persistent capability asymmetry between generation an d self-verification. In this work, we conduct an in-depth investigation of this asymmetry throughout training evolution and show that, even on the same task, im proving generation does not lead to corresponding improvements in self-verificat ion. Interestingly, we find that the reverse direction of this asymmetry behaves differently: learning to self-verify can effectively improve generation perform ance, achieving accuracy comparable to standard generation training while yieldi ng more efficient and effective reasoning traces. Building on this observation, we further explore integrating self-verification into generation training by for mulating a multi-task reinforcement learning framework, where generation and sel f-verification are optimized as two independent but complementary objectives. Ex tensive experiments across benchmarks and models demonstrate performance gains o ver generation-only training in both generation and verification capabilities.
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ReSpinQuant: Efficient Layer-Wise LLM Quantization via Subspace Residual Rotatio n Approximation
Suyoung Kim, Sunghyun Wee, Hyeonjin Kim, Kyomin Hwang, Hyunho Lee, Nojun Kwak
Rotation-based Post-Training Quantization (PTQ) has emerged as a promising solut ion for mitigating activation outliers in the quantization of Large Language Mod els (LLMs). Global rotation methods achieve inference efficiency by fusing activ ation rotations into attention and FFN blocks, but suffer from limited expressiv ity as they are constrained to use a single learnable rotation matrix across all layers. To tackle this, layer-wise transformation methods emerged, achieving su

perior accuracy through localized adaptation. However, layer-wise methods cannot fuse activation rotation matrices into weights, requiring online computations and causing significant overhead. In this paper, we propose **ReSpinQuant**, a quantization framework that resolves such overhead by leveraging offline activation rotation fusion and matching basis using efficient residual subspace rotation. This design reconciles the high expressivity of layer-wise adaptation with only negligible inference overhead. Extensive experiments on W4A4 and W3A3 quantization demonstrate that ReSpinQuant achieves state-of-the-art performance, outperforming global rotation methods and matching the accuracy of computationally expensive layer-wise methods with minimal overhead.
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Fast Reconstruction of Mixtures of Bernoulli Product Distributions
Sanyam Agarwal, Pranjal Dutta, Markus Bläser
Mixtures of Bernoulli product distributions are a simple and widely used latent-variable model, with applications in e.g.\ recommendation systems, crowdsourcing, and medical data analysis. We consider the problem of reconstructing the mixture parameters from oracle access to its probability generating polynomial (PGP), for instance represented by a probabilistic generating circuit (PGC). We show that the parameters are uniquely identifiable for almost all mixtures, and give a randomized algorithm that exactly recovers the mixture weights and component marginals for mixtures of r Bernoulli product distributions over n variables using only $O(nr^2)$ oracle queries. The algorithm repeatedly applies restrictions to $O(r)$ variables, extracts low-degree coefficients, and then recovers the parameters using a moment-based tensor decomposition. To the best of our knowledge, this is the {\em first} exact reconstruction algorithm in this PGP oracle model with query complexity linear in n and polynomial in r.
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Identifying and Correcting Label Noise for Robust GNNs via Influence Contradiction
Wei Ju, Wei Zhang, Siyu Yi, Zhengyang Mao, Yifan Wang, Jingyang Yuan, Zhiping Xiao, Ziyue Qiao, Ming Zhang
Graph Neural Networks (GNNs) have shown remarkable capabilities in learning from graph-structured data with various applications such as social analysis and bioinformatics. However, the presence of label noise in real scenarios poses a significant challenge in learning robust GNNs, and their effectiveness can be severely impacted when dealing with noisy labels on graphs, often stemming from annotation errors or inconsistencies. To address this, in this paper we propose a novel approach called ICGNN that harnesses the structure information of the graph to effectively alleviate the challenges posed by noisy labels. Specifically, we first design a novel noise indicator that measures the influence contradiction score (ICS) based on the graph diffusion matrix to quantify the credibility of nodes with clean labels, such that nodes with higher ICS values are more likely to be detected as having noisy labels. Then we leverage the Gaussian mixture model to precisely detect whether the label of a node is noisy or not. Additionally, we develop a soft strategy to combine the predictions from neighboring nodes on the graph to correct the detected noisy labels. At last, pseudo-labeling for abundant unlabeled nodes is incorporated to provide auxiliary supervision signals and guide the model optimization. Experiments on benchmark datasets show the superiority of our approach over competitive baselines in noisy label scenarios. The source code is available at: https://github.com/wayc04/ICGNN.
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Goal-Conditioned Agents that Learn Everything All at Once
Michael Matthews, Matthew Thomas Jackson, Michael Beukman, Thomas Foster, Alistair Letcher, Scott Fujimoto, Cédric Colas, Jakob Nicolaus Foerster
A goal-conditioned reinforcement learning agent acting in an environment will see a wealth of information throughout a trajectory, most of which is discarded when only considering the trajectory with respect to a single goal. All-goals learning, where each transition is used for learning off-policy with respect to every goal, allows agents to extract maximal information, however it is usually computationally infeasible when done via naive relabelling. This can be overcome by

jointly outputting values and actions for every goal at once, allowing for efficient, parallel all-goals updates with a single pass through the network, in a process we call Learning Everything all at Once (LEO). We show that this approach significantly outperforms other methods on goal-conditioned Craftax and is competitive with existing baselines on continuous control environments, while achieving a 250x speed-up compared to all-goals relabelling. We hope that, by unlocking all-goals learning at scale, LEO can serve as a useful tool for RL practitioners in complex environments. We open source our code at https://anonymous.4open.science/r/CraftaxGC-D3E1.
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Sim2Reason: Solving Physics Olympiad via Reinforcement Learning on Physics Simulators
Mihir Prabhudesai, Aryan Satpathy, Yangmin Li, Zheyang Qin, Nikash Bhardwaj, Amir Zadeh, Chuan Li, Katerina Fragkiadaki, Deepak Pathak
We have witnessed remarkable advances in LLM reasoning capabilities with the advent of DeepSeek-R1. However, much of this progress has been fueled by the abundance of internet question–answer (QA) pairs—a major bottleneck going forward, since such data is limited in scale and concentrated mainly in domains like mathematics. In contrast, other sciences such as physics lack large-scale QA datasets to effectively train reasoning-capable models. In this work, we show that physics simulators can serve as a powerful alternative source of supervision for training LLMs for physical reasoning. We generate random scenes in physics engines, create synthetic question–answer pairs from simulated interactions, and train LLMs using reinforcement learning on this synthetic data. Our models exhibit zero-shot sim-to-real transfer to real-world physics benchmarks: for example, training solely on synthetic simulated data improves performance on IPhO (International Physics Olympiad) problems by 5-10 percentage points across model sizes. These results demonstrate that physics simulators can act as scalable data generators, enabling LLMs to acquire deep physical reasoning skills beyond the limitations of internet-scale QA data. Code available at: https://sim2reason.github.io/.
****

Perceptrons and Localization of Attention’s Mean-Field Landscape
Antonio Álvarez-López, Borjan Geshkovski, Domènec Ruiz-Balet
The forward pass of a Transformer can be seen as an interacting particle system on the unit sphere: time plays the role of layers, particles that of token embeddings, and the unit sphere idealizes layer normalization. In some weight settings the system can even be seen as a gradient flow for an explicit energy, and one can make sense of the infinite context length *mean-field* limit thanks to Wasserstein gradient flows. In this paper we study the effect of the perceptron block in this setting, and show that critical points are generically atomic and localized on subsets of the sphere.
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PluRel: Synthetic Data unlocks Scaling Laws for Relational Foundation Models
Vignesh Kothapalli, Rishabh Ranjan, Valter Hudovernik, Vijay Prakash Dwivedi, Johannes Hoffart, Carlos Guestrin, Jure Leskovec
Relational Foundation Models (RFMs) facilitate data-driven decision-making by learning from complex multi-table databases. However, the diverse relational databases needed to train such models are rarely public due to privacy constraints. While there are methods to generate synthetic tabular data of arbitrary size, incorporating schema structure and primary--foreign key connectivity for multi-table generation remains challenging. Here we introduce PluRel, a framework to synthesize multi-tabular relational databases from scratch.
In a step-by-step fashion, PluRel models (1) schemas with directed graphs, (2) inter-table primary-foreign key connectivity with bipartite graphs, and, (3) feature distributions in tables via conditional causal mechanisms. The design space across these stages supports the synthesis of a wide range of diverse databases, while being computationally lightweight. Using PluRel, we observe for the first time that (1) RFM pretraining loss exhibits power-law scaling with the number of synthetic databases and total pretraining tokens, (2) scaling the number of synthetic databases improves generalization to real databases, and (3) synthetic p

retraining yields strong base models for continued pretraining on real databases. Overall, our framework and results position synthetic data scaling as a promising paradigm for RFMs. Webpage: https://snap-stanford.github.io/plurel
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Elastic Attention: Test-time Adaptive Sparsity Ratios for Efficient Transformers
Zecheng Tang, Quantong Qiu, Yi Yang, Zhiyi Hong, Haiya Xiang, Kebin Liu, Qingqing Dang, Juntao Li, Min Zhang
The quadratic complexity of standard attention mechanisms poses a significant scalability bottleneck for large language models (LLMs) in long-context scenarios. While hybrid attention strategies that combine sparse and full attention within a single model offer a viable solution, they typically employ static computation ratios (i.e., fixed proportions of sparse versus full attention) and fail to adapt to the varying sparsity sensitivities of downstream tasks during inference. To address this issue, we propose $\textit{\textbf{Elastic Attention}}$, which allows the model to dynamically adjust its overall sparsity based on the input. This is achieved by integrating a lightweight $\textit{\textbf{Attention Router}}$ into the existing pretrained model, which dynamically assigns each attention head to different computation modes. Within only 12 hours of training on 8$\times$A800 GPUs, our method enables models to achieve both strong performance and efficient inference. Experiments across three long-context benchmarks on widely-used LLMs demonstrate the superiority of our method.
****
Sycophancy Towards Researchers Drives Performative Misalignment
David D. Baek, Xinnuo Li, Anay Gupta, Taslim Mahbub, Kejian Shi, Max Tegmark, Shi Feng
The increasing situational awareness of language models raises safety concerns: models might be aware when they are evaluated, and adjust their behavior to evade monitoring and resist modification, e.g., pretending to be aligned only in evaluation. This *alignment faking* behavior is often interpreted as scheming: an intentional effort of strategic deception. In this paper, we examine an alternative interpretation, *performative misalignment*, which explains the change in behavior as a result of *sycophancy towards AI researchers*. To examine this hypothesis, we present three empirical findings. First, we show that evaluation awareness persists even when we tell models they are deployed, which contradicts the scheming story which predicts less misalignment when the model perceives evaluation. Second, we use probing and steering to show that our current methods cannot mechanistically distinguish sycophancy and scheming in alignment faking evaluations. Third, we fine-tune models to be more sycophantic and observe increased sensitivity to evaluation cues. To conclude, we emphasize deconfounding sycophancy from scheming for future work on evaluations and mitigations of intent misalignment.
****
Singular Bayesian Neural Networks
Mame Diarra Toure, David A. Stephens
Bayesian neural networks promise calibrated uncertainty but require $O(mn)$ parameters for standard mean-field Gaussian posteriors. We argue this cost is often unnecessary, particularly when weight matrices exhibit fast singular value decay.
By parameterizing weights as $W = AB^{\top}$ with $A \in \mathbb{R}^{m \times r}$,
$B \in \mathbb{R}^{n \times r}$, we induce a posterior that is \emph{singular} with respect to the Lebesgue measure, concentrating on the rank-r manifold. This
singularity captures structured weight correlations through shared latent factors, geometrically distinct from mean-field's independence assumption. We derive PAC-Bayes generalization bounds whose complexity term scales as $\sqrt{r(m+n)}$ instead of $\sqrt{m n}$, and prove loss bounds that decompose the error into optimization and rank-induced bias using the Eckart-Young-Mirsky theorem. We further adapt recent Gaussian complexity bounds for low-rank deterministic networks to Bayesian predictive means.

Empirically, across MLPs, LSTMs, and Transformers on standard benchmarks, our method achieves competitive predictive performance while using up to $33\times$ fewer parameters than 5-member Deep Ensembles. It substantially improves OOD detection and often improves calibration relative to mean-field and perturbation baselines, while Deep Ensembles can still be stronger on in-distribution likelihood-based metrics.
****

DRIFT: Decoupled Rollouts and Importance-Weighted Fine-Tuning for Efficient Multi-Turn Optimization
Jian Mu, Tianyi Lin, Chengwei Qin, Zhongxiang Dai, Yao Shu
Large language models are increasingly deployed in multi-turn interactive settings where users or environments can iteratively provide lightweight feedback.
Unfortunately, optimizing such behavior presents a sharp dilemma in practice: online reinforcement learning is able to effectively address multi-turn dynamics but is prohibitively expensive due to the cost of generating full correction trajectories at every update, whereas offline supervised fine-tuning (SFT) is efficient but suffers from distribution shift and behavioral collapse.
To this end, we novelly propose DRIFT (Decoupled Rollouts and Importance-Weighted Fine-Tuning), a framework that operationalizes the theoretical insight that the KL-regularized RL objective is equivalent to importance-weighted supervised learning.
DRIFT decouples rollout from optimization by sampling offline interaction trajectories from a fixed reference policy, deriving return-based importance weights, and optimizing the policy via weighted SFT on the resulting dataset.
Empirically, we demonstrate that DRIFT matches or exceeds the performance of multi-turn reinforcement learning baselines while maintaining the training efficiency and simplicity of standard supervised fine-tuning. Code is available at https://github.com/2020-qqtcg/DRIFT.
****

Local Hessian Spectral Filtering for Robust Intrinsic Dimension Estimation
GENKI OSADA
While diffusion models enable new approaches for estimating Local Intrinsic Dimension (LID), existing methods fail in high-dimensional spaces where noise from vast normal directions overwhelms the tangent signal. We propose Local Hessian Spectral Dimension (LHSD), which resolves this by applying spectral filtering to the log-density Hessian, explicitly cutting off large eigenvalues associated with normal directions to count zero-curvature tangent directions. Implemented using Stochastic Lanczos Quadrature (SLQ), LHSD avoids full Hessian construction, achieving linear scalability with dimension D.
Experiments on synthetic and real data confirm LHSD’s superior robustness and its utility in detecting memorization in large-scale diffusion models.
****

A Capacity-Based Rationale for Multi-Head Attention
Micah Adler
We study the capacity of the self-attention key-query channel: for a fixed budget, how many distinct token-token relations can a single layer reliably encode? We introduce *Relational Graph Recognition*, where the key-query channel encodes a directed graph and, given a context (a subset of the vertices), must recover the neighbors of each vertex in the context. We measure resources by the total key dimension $D_K = hd_k$. In a tractable multi-head model, we prove matching information-theoretic lower bounds and upper bounds via explicit constructions showing that recovering a graph with m' relations in d_{model}-dimensional embeddings requires D_K to grow essentially as m'/d_{model} up to logarithmic factors, and we obtain corresponding guarantees for scaled-softmax attention. This analysis yields a new, capacity-based rationale for multi-head attention: even in permutation graphs, where **all queries attend to a single target**, splitting a fixed D_K budget into multiple heads increases capacity by reducing interference from embedding superposition. Controlled experiments mirror the theory, revealing sharp phase transitions at the predicted capacity, and the multi-head advantage persists when adding softmax normalization, value routin

g, and a full Transformer block trained with frozen GPT-2 embeddings.
****

Nonparametric LLM Evaluation from Preference Data
Dennis Frauen, Athiya Deviyani, Mihaela van der Schaar, Stefan Feuerriegel
Evaluating the performance of large language models (LLMs) from human preference data is crucial for obtaining LLM leaderboards. However, many existing approaches either rely on restrictive parametric assumptions or lack valid uncertainty quantification when flexible machine learning methods are used. In this paper, we propose a nonparametric statistical framework, called DMLRank, for comparing and ranking LLMs from preference data using debiased machine learning (DML). For this, we introduce generalized average ranking scores (GARS), which generalize commonly used ranking models, including the Bradley-Terry model or PageRank/ Rank centrality with complex human responses such as ties. DMLRank comes with the following advantages: (i) It produces statistically efficient estimates of GARS ranking scores. (ii) It naturally allows to incorporate black-box machine learning methods for estimation. (iii) It can be combined with pre-trained LLM evaluators (e.g., using LLM-as-a-judge). (iv) It suggests optimal policies for collecting preference data under budget constraints. We demonstrate these advantages both theoretically and empirically using both synthetic and real-world preference datasets. In summary, our framework provides practitioners with powerful, state-of-the-art methods for comparing or ranking LLMs for leaderboards.
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DTS: Enhancing Large Reasoning Models via Decoding Tree Sketching
Zicheng Xu, Xiuyi Lou, Guanchu Wang, Yu-Neng Chuang, Feng Luo, Guangyao Zheng, Alex Szalay, Zirui Liu, Vladimir Braverman
Large Reasoning Models (LRMs) achieve remarkable inference-time improvements through parallel thinking. However, existing methods rely on redundant sampling of reasoning trajectories, failing to effectively explore the reasoning space to uncover high-quality solutions. To address these limitations, we propose **D**ecoding **T**ree **S**ketching (DTS), a plug-and-play decoding framework for structural multi-trajectory exploration and reasoning selection. For reasoning exploration, DTS sketches a backbone tree of the reasoning space by selectively branching at decision tokens. For reasoning selection, guided by length-accuracy anti-correlation, DTS designs an early termination to prioritize short and reliable trajectories during decoding. Experimental results across four LRMs and datasets demonstrate that DTS significantly enhances accuracy by **14\%** and reduces repetitive generation by **8\%** on average. Notably, DTS enables smaller models to outperform larger models with 10$\times$ the size, highlighting its potential to strengthen reasoning capabilities.
****

TextResNet: Decoupling and Routing Optimization Signals in Compound AI Systems via Deep Residual Tuning
Suizhi Huang, Mei Li, Han Yu, Xiaoxiao Li
Textual Gradient-style optimizers (TextGrad) enable gradient-like feedback propagation through compound AI systems. However, they do not work well for deep chains. The root cause of this limitation stems from the *Semantic Entanglement* problem in these extended workflows. In standard textual backpropagation, feedback signals mix local critiques with upstream contexts, leading to *Attribution Ambiguity*. To address this challenge, we propose TextResNet, a framework that reformulates the optimization process to achieve precise signal routing via four key innovations. Firstly, in the forward pass, it enforces Additive Semantic Deltas to preserve an Identity Highway for gradient flow. Secondly, in the backward pass, it introduces Semantic Gradient Decomposition via a Semantic Projector to disentangle feedback into causally independent subspaces. Thirdly, it implements Causal Routing, which routes projected signals to their specific components. Finally, it performs Density-Aware Optimization Scheduling to leverage the disentangled signals to dynamically allocate resources to key system bottlenecks. Our results show that TextResNet not only achieves superior performance compared to TextGrad, but also exhibits remarkable stability for agentic tasks in compound AI systems where baselines collapse. Code is available at https://github.com/JeanDiab

le/TextResNet.
****
WET: Mitigating World-Conditioned Knowledge Conflicts via World Entropy Tetherin
g
Zixuan Wang, Yifei He, Zihan Wang, Kun Wang, Chaomeng Chen
Large language models (LLMs) face a "loyalty dilemma" when correctness is condit
ioned on an active world-of-discourse. We identify a systemic failure mode---wor
ld misattribution---where models implicitly ground generation in an incompatible
 regime and drift from the target world. We propose World Entropy Tethering (WET
), an inference-time monitor-and-tether: a world-entropy probe flags drift risk
on prompt anchors, and a conditional score matching geometry model identifies te
thering heads for entropy-gated rescaling. Experiments show: (I) Linear Separabi
lity: world labels are linearly decodable from internal states; (II) Geometric D
rift: hallucinations are preceded by measurable deviations from the target world
 region; and (III) Targeted Mitigation: WET improves world consistency and reduc
es hallucination rates by up to 22.4% without compromising generation quality. C
ode is available at https://github.com/guess-guess-who-i-am/WET-World-Entropy-Te
thering.
****
Eyes-on-Me: Scalable RAG Poisoning through Transferable Attention-Steering Attra
ctors
Yen-Shan Chen, Sian-Yao Huang, Cheng-Lin Yang, Yun-Nung Chen
Existing data poisoning attacks on retrieval-augmented generation (RAG) systems
scale poorly because they require costly optimization of poisoned documents for
each target phrase. We introduce Eyes-on-Me, a modular attack that decomposes an
 adversarial document into reusable **Attention Attractors** and **Focus Regions
**. Attractors are optimized to direct attention to the Focus Region. Attackers
can then insert semantic baits for the retriever or malicious instructions for t
he generator, adapting to new targets at near zero cost. This is achieved by ste
ering a small subset of attention heads that we empirically identify as strongly
 correlated with attack success. Across 18 end-to-end RAG settings (3 datasets $
\times$ 2 retrievers $\times$ 3 generators), Eyes-on-Me raises average attack su
ccess rates from 21.9 to 57.8 (+35.9 points, 2.6$\times$ over prior work). A sin
gle optimized attractor transfers to unseen black box retrievers and generators
without retraining. Our findings establish a scalable paradigm for RAG data pois
oning and show that modular, reusable components pose a practical threat to mode
rn AI systems. They also reveal a strong link between attention concentration an
d model outputs, informing interpretability research.
****
Beyond Accuracy: Latent Perturbations for Cognitive-Aware Diagnosis
Yuting Yan, Yinghao Fu, Wendi Ren, Haozhou Gao, Shuang Li
Diagnosing rare diseases remains a persistent challenge, often hindered by cogni
tive anchoring: once clinicians settle on a common diagnosis, they often discoun
t alternative explanations, including rare conditions. To address this, we propo
se a cognitive-aware counterfactual reasoning framework using a Denoising Masked
 AutoEncoder (DMAE) to simulate what-if diagnostic scenarios that probe clinicia
ns’ initial assumptions. Our model jointly learns (1) the true distribution of d
iseases and symptoms, and (2) human diagnostic behavior, revealing critical gaps
 between medically possible and clinically considered diagnoses. By strategicall
y perturbing latent patient representations, it generates contrastive counterfac
tuals that highlight rare yet plausible diseases that cognitive bias often obscu
res. Unlike traditional decision-support tools, our system suggests rare disease
s not because they are statistically dominant, but because they are systematical
ly under-considered relative to the observed evidence and learned diagnostic beh
avior. Across four public and three private rare-disease datasets, our approach
outperforms standard machine learning classifiers in detecting rare conditions w
hile maintaining strong performance on common diagnoses. Beyond boosting accurac
y, the counterfactual evidence encourages hypothesis-driven reasoning and suppor
ts clinical learning.
****

IsoCompute Playbook: Optimally Scaling Sampling Compute for LLM RL
Zhoujun Cheng, Yutao Xie, Yuxiao Qu, Amrith Setlur, Shibo Hao, Varad Pimpalkhute, Tongtong Liang, Feng Yao, Zhengzhong Liu, Eric P. Xing, Virginia Smith, Ruslan Salakhutdinov, Zhiting Hu, Taylor W. Killian, Aviral Kumar
While scaling laws guide compute allocation for LLM pre-training, analogous prescriptions for reinforcement learning (RL) post-training of LLMs remain poorly understood. We study the compute-optimal allocation of sampling compute for on-policy RL methods in LLMs, framing scaling as a compute-constrained optimization over three resources: parallel rollouts per problem, number of problems per batch, and number of update steps. We find that the compute-optimal number of parallel rollouts per problem increases predictably with compute budget and then saturates. This trend holds across both easy and hard problems, though driven by different mechanisms: solution sharpening on easy problems and coverage expansion on hard problems. We further show that increasing the number of parallel rollouts mitigates interference across problems, while the number of problems per batch primarily affects training stability and can be chosen within a broad range. Validated across base models and data distributions, our results recast RL scaling laws as prescriptive allocation rules and provide practical guidance for compute-efficient LLM RL post-training.
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Practical and Optimal Algorithm for Linear Contextual Bandits with Rare Parameter Updates
Sanghoon Yu, Min-hwan Oh
We study linear contextual bandits under rare parameter updates:
the learner may incorporate reward feedback into its parameter estimate only at a small number of update times,
while still observing contexts online and selecting actions sequentially.
This viewpoint clarifies a practical distinction that is often blurred in the literature:
many "strictly batched" methods additionally restrict within-interval context adaptivity,
meaning that the action rule inside an interval cannot depend on the sequence of realized contexts/actions in that interval (beyond the current round's context).
For linear contextual bandits, we propose two practical algorithms with only $O(\log\log T)$ parameter updates.
Our first algorithm BLCE-G attains minimax-optimal regret (up to polylogarithmic factors in T) simultaneously in both the small-K and large-K regimes under a static schedule.
Our second algorithm BLCE removes the near G-optimal design step---a dominant computational bottleneck in prior strictly batched static-grid methods---yet preserves minimax-optimal regret and achieves the lowest known runtime complexity among optimal algorithms.
We further extend these rare-update and computational principles to generalized linear contextual bandits.
Overall, our results yield statistically optimal algorithms under $O(\log\log T)$ parameter updates that are also computationally efficient in practice.
****
Words & Weights: Streamlining Multi-Turn Interactions via Co-Adaptation
Chenxing Wei, Hong Wang, Ying He, Zhongxiang Dai, Bo Jiang, Fei Yu, Yao Shu
Test-time policy adaptation for multi-turn interactions (T^{2}PAM) is essential for aligning Large Language Models (LLMs) with dynamic user needs. However, existing paradigms typically treat adaptation as a single-axis problem by either purely refining instructions or solely updating weights. This bifurcated approach overlooks the fact that interaction failures arise from a coupled mixture of context ambiguity and model incapacity. To address this, we propose ROSA2, a framework that reformulates T^{2}PAM as a joint optimization problem over the heterogeneous space of Words and Weights. Within this framework, the semantic stream acts as a feedback normalizer that transforms noisy or ambiguous user feedback into actionable instructions, ensuring that parametric adaptation is performed on sem

antically clarified trajectories. Theoretically, we prove that this semantic pre
-conditioning strictly reduces the required parameter shift for convergence. Emp
irically, ROSA2 consistently outperforms state-of-the-art baselines across diver
se mathematical, general reasoning, and coding benchmarks. It achieves up to a 3
7.8\% accuracy improvement while reducing average interaction turns by 40\%, dem
onstrating that ROSA2 unlocks the true potential of parameter updates.
****

RouterInterp: Understanding Superposed Specialisation in Mixture of Experts Rout
ing

Ilya Lasy, Nora Yinuo Cai, Kola Ayonrinde

Sparse Mixture of Experts (MoE) models scale more efficiently than dense models
by routing tokens to modular expert networks that are only active when relevant
to the task. A leading hypothesis for the performance of MoE models is that each
expert specialises in a single, coherent domain. However, interpretability effo
rts that assume this hypothesis have generally been unsuccessful. We propose and
present evidence for an alternative account that we call the *Superposed Specia
lisation Hypothesis* (SSH): experts specialise in a disjoint union of fine-grain
ed features rather than one broad domain. Leveraging the SSH, we introduce *Rout
erInterp*, a method for interpreting expert routing that identifies Sparse Autoe
ncoder features most predictive of routing decisions and produces unified natura
l language explanations. On gpt-oss-20b, RouterInterp explains expert routing wi
th 57% higher detection accuracy than prior token statistics based methods. This
work provides a scalable method for generating concise and more accurate explan
ations of expert routing and increases our understanding of a previously uninter
pretable component of foundation models.
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Hyper-ICL: Attention Calibration with Hyperbolic Anchor Distillation for Multimo
dal In-Context Learning

Niloufar Alipour Talemi, Hossein Kashiani, Fatemeh Afghah

Multimodal In-Context Learning (ICL) has emerged as a practical inference paradi
gm for Multimodal Large Language Models, where a small set of interleaved image-
text In-Context Demonstrations (ICDs) conditions the model to solve new tasks. D
espite its flexibility, multimodal ICL incurs high inference latency and suffers
from instability due to sensitivity to demonstration formatting, ordering, and
content. To address these limitations, we propose Hyper-ICL, a lightweight, trai
ning-based framework for demonstration-free multimodal ICL that reconstructs dem
onstration effects directly without requiring ICDs at inference time. Hyper-ICL
learns a parameter-efficient low-rank logit-level adapter that calibrates attent
ion distributions to better match demonstration-induced attention redistribution
. To capture how demonstration influence varies across queries, we introduce a q
uery-adaptive modulation mechanism that adaptively controls intervention strengt
h at token level across layers and heads based on the current query. Finally, we
propose a layer-wise hyperbolic anchor distillation loss that aligns intermedia
te student features to a demonstration-conditioned teacher via Lorentz geodesic
distance. This loss encourages the student to reconstruct the demonstration–quer
y relationships induced by ICDs. Extensive experiments across six different mult
imodal benchmarks (including VQAv2, OK-VQA, and COCO Caption) demonstrate that H
yper-ICL consistently improves accuracy and stability over vanilla ICL and exist
ing state-of-the-art methods.
****

Non-Parametric Optimization for Scalable Learning in Stochastic Decision Problem
s

Mohsen Amidzadeh, Lauri Viitasaari, Mario Di Francesco

Stochastic optimization (SO) plays a central role in decision-making under uncer
tainty. Among SO problems, time-varying stochastic optimization (TV-SO) is parti
cularly important due to its applications in adaptive control and machine learni
ng. Non-parametric approaches have been proposed for time-varying deterministic
optimization; however, they have not been developed for their stochastic counter
parts. This work addresses that gap by developing a stochastic variational frame
work based on Malliavin calculus. This framework yields non-parametric optimalit

y conditions for SO problems with stochastic decisions and supports the design of a scalable deep-learning algorithm that is insensitive to the parameterization dimension. This algorithm, called the Stochastic Path Follower (SPF), is applied to two important problems under distribution drift, namely least-squares recovery and logistic regression. Experimental results show that the proposed approach outperforms state-of-the-art learning-based and gradient-based methods in both performance and scalability.
****

Safety Recovery in Reasoning Models Is Only a Few Early Steering Steps Away
Soumya Suvra Ghosal, Souradip Chakraborty, Vaibhav Singh, Furong Huang, Dinesh Manocha, Amrit Singh Bedi
Reinforcement learning (RL) based post-training for explicit chain-of-thought (e.g., GRPO) improves the reasoning ability of multimodal large-scale reasoning models (MLRMs). But recent evidence shows that it can simultaneously degrade safety alignment and increase jailbreak success rates. We propose SafeThink, a lightweight inference-time defense that treats safety recovery as a satisficing constraint rather than a maximization objective. SafeThink monitors the evolving reasoning trace with a safety reward model and conditionally injects an optimized short corrective prefix ("Wait, think safely") only when the safety threshold is violated. In our evaluations across six open-source MLRMs and four jailbreak benchmarks (JailbreakV-28K, Hades, FigStep, and MM-SafetyBench), SafeThink reduces attack success rates by 30-60 % (e.g., LlamaV-o1: 63.33\% $\rightarrow$5.74\% on JailbreakV-28K, R1-OneVision: 69.07\%$\rightarrow$5.65\% on Hades) while preserving reasoning performance (MathVista accuracy: 65.20\%$\rightarrow$65.00\%). A key empirical finding from our experiments is that safety recovery is often only a few steering steps away: intervening in the first $1–3$ reasoning steps typically suffices to redirect the full generation toward safe completions.
****

E²I-VRWKV: Explicit EPI-Representation and Interaction-Aware Vision-RWKV for Light Field Semantic Segmentation
Wei Zhang, Chen Jia, Xu Cheng, Fan Shi, Hui Liu, Shengyong Chen
Pixel-level semantic segmentation of 4D light field (LF) data remains a considerable challenge, primarily due to the conflict between modeling complex spatial-angular dependencies and maintaining linear computational efficiency. Current linear models like VRWKV offer scalability but often fail to capture intrinsic geometric structures, leading to the structural collapse of Epipolar Plane Image (EPI) cues. To overcome these limitations, we propose E²I-VRWKV, an EPI-Enhanced and Interaction-aware network that generates high-quality segmentation maps by embedding explicit geometric priors into a linear-complexity backbone. Specifically, we introduce the Light Field Epipolar-Aware Cross-Modal Attention (LF-ECMA) block. The key innovation lies in the integration of an EPI Geometric Prior Generator, which explicitly extracts disparity-sensitive biases to enforce geometric consistency, and a Geometric-Context Gating (GC-Gate) mechanism. This mechanism functions as a geometrically modulated aperture to dynamically calibrate the fusion of spatial and angular manifolds. Experiments on the UrbanLF benchmark demonstrate that our method outperforms other state-of-the-art (SOTA) methods, achieving 86.55% mIoU on UrbanLF-Real while maintaining a superior balance between accuracy and linear efficiency.
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Quant VideoGen: Auto-Regressive Long Video Generation via 2-Bit KV-Cache Quantization
Haocheng Xi, Shuo Yang, Yilong Zhao, Muyang Li, Han Cai, Xingyang Li, Yujun Lin, Zhuoyang Zhang, Jintao Zhang, Xiuyu Li, Zhiying Xu, Jun Wu, Chenfeng Xu, Ion Stoica, Song Han, Kurt Keutzer
Despite rapid progress in auto-regressive video diffusion, we identify an emerging system–algorithm bottleneck that limits both deployability and generation quality: KV-cache memory. In auto-regressive video generation models, the KV-cache grows with generation history and quickly dominates GPU memory (often ≥30 GB), preventing deployment on widely available hardware. More critically, memory-bounded KV budgets constrain the effective working memory, directly degrading long-ho

rizon consistency in identity, layout, and motion.
To address this challenge, we present Quant VideoGen (QVG), a training-free KV-cache quantization framework for auto-regressive video diffusion models. QVG exploits video’s inherent spatiotemporal redundancy via Semantic-Aware Smoothing, producing low-magnitude, quantization-friendly residuals. Building on this, QVG introduces Progressive Residual Quantization, a coarse-to-fine multi-stage scheme that further reduces quantization error while enabling a smooth quality–memory trade-off.
Across LongCat-Video, HY-WorldPlay, and Self-Forcing, QVG establishes a new Pareto frontier between quality and memory efficiency, reducing KV memory by up to 7.0× with less than 4% end-to-end latency overhead, while delivering significantly better generation quality than existing baselines.
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Plug-and-Play Label Map Diffusion for Universal Goal-Oriented Navigation
Zhixuan Shen, Yijie Zeng, Shengxiang Luo, Tianrui Li, Haonan Luo
In embodied vision, Goal-Oriented Navigation (GON) requires robots to locate a specific goal within an unexplored environment. The primary challenge of GON arises from the need to construct a Bird's-Eye-View (BEV) map to understand the environment while simultaneously localizing an unobserved goal. Existing map-based methods typically employ self-centered semantic maps, often facing challenges such as reliance on complete maps or inconsistent semantic association. To this end, we propose Plug-and-Play Label Map Diffusion (PLMD), which defines a novel map completion diffusion model based on Denoising Diffusion Probabilistic Models (DDPM). PLMD generates obstacle and semantic labels for unobserved regions through a diffusion-based completion process, thereby enabling goal localization even in partially observed environments. Moreover, it mitigates inconsistent semantic association by leveraging structural consistency between known and unknown obstacle layouts and integrating obstacle priors into the semantic denoising process. By substituting predicted labels for unobserved regions, robots can accurately localize the specified objects. Extensive experiments demonstrate that PLMD **(I)** effectively expands the region of unknown maps, **(II)** integrates seamlessly into existing navigation strategies that rely on semantic maps, **(III)** achieves state-of-the-art performance on three GON tasks. Code is available at: <https://github.com/FrankZxShen/PLMD>.
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Estimating the Empowerment of Language Model Agents
Jinyeop Song, Jeff Gore, Max Kleiman-Weiner
As language model (LM) agents become increasingly capable and adopted in real-world applications, there is a growing need for scalable evaluation frameworks beyond costly, manually designed benchmarks. We propose information-theoretic evaluation based on empowerment, an information-theoretic measure of an agent’s influence on future states through its actions. To handle the unique challenges of text-based environments, we introduce EELMA (Estimating Empowerment of Language Model Agents), an algorithm for approximating effective empowerment from multi-turn text interactions. We demonstrate EELMA on textual games and realistic web and tool-use environments, showing that empowerment strongly correlates with average task performance. We further analyze how empowerment varies across models, environment complexity, and agent configurations, and show that high-empowerment states and actions often mark pivotal moments for general capabilities. These results establish empowerment as a goal-agnostic metric that complements task-success measures for LM-agent evaluation. Code available: https://github.com/Jinyeop3110/EELMA
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Faster Than Flash: Exploiting Attention Sparsity for Efficient Long-Context Decoding
Zhigeng Liu, Zhiyuan Ning, Ruixiao Li, Xiaoran Liu, Yuerong Song, Min Zhang, Ziwei He, Xipeng Qiu
The development of long-context Large Language Models (LLMs) is constrained by the memory bandwidth bottleneck and quadratic complexity of the attention mechanism during decoding. To overcome the inherent trade-offs between the memory overh

ead of metadata-based metrics and the computational inefficiency of adaptive sel
ection strategies, we present \textbf{Faster Flash Decoding (FFD)}, a novel hard
ware-algorithm co-design framework designed to break the memory wall in long-con
text decoding. FFD integrates the selector and computer into a fully fused kerne
l, replacing external metadata indices with content-aware scanning via low-bit q
uantization. Furthermore, we introduce the top-δ strategy, which dynamica
lly filters blocks to achieve distribution-adaptive sparsity without global sync
hronization. As a training-free, plug-and-play solution, FFD enables the reuse o
f scanning results for computation, achieving up to 11.6x kernel-level speedup a
t 256k context length and 2.37x end-to-end throughput improvement. Empirical val
idation on Ruler and Longbench confirms that FFD maintains model accuracy while
delivering high-ratio sparsity.
****

From Seeing to Thinking: Decoupling Perception and Reasoning Improves Post-Train
ing of Vision-Language Models
Juncheng Wu, Hardy Chen, Haoqin Tu, Xianfeng Tang, Freda Shi, Hui Liu, Hanqing L
u, Cihang Xie, Yuyin Zhou
Recent advances in vision-language models (VLMs) emphasize long chain-of-thought
 reasoning; yet, we find that their performance on visual tasks is primarily lim
ited by a lack of visual perception as opposed to reasoning itself. In this work
, we systematically study the interplay between perception and reasoning in VLM
post-training by decomposing their capabilities into three separate training sta
ges: visual perception, visual reasoning, and textual reasoning, incorporating s
pecialized training data. We demonstrate that visual perception (a) requires tar
geted optimization with specialized data; (b) serves as a fundamental scaffold t
hat should be solidified through staged training before refining visual reasonin
g; and (c) is more effectively learned via RL than caption-based SFT. Our experi
ments across multiple VLMs demonstrate that staged training consistently improve
s both visual perception and reasoning performance over merged training. Notably
, models trained with our approach achieve 1.5% higher reasoning accuracy with 2
0.8% shorter reasoning traces, suggesting that superior perception reduces the n
eed for excessive reasoning. Furthermore, we show that this capability-based sta
ging represents a new curriculum dimension orthogonal to traditional difficulty-
based curricula, and combining both yields further additive gains. Our staged-tr
aining models achieve superior performance among open-weight VLMs, establishing
advanced results on several visual math and perception (e.g., +5.2% on WeMath an
d +3.7% on RealWorldQA) tasks compared with the base counterpart.
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WeightCLIP: Aligning Datasets and Models for Weight Space Learning
Aron Asefaw, Konstantinos Tzevelekakis, Damian Falk, Léo Meynent, Damian Borth
Weight space learning aims to learn representations of neural network (NN) weigh
ts, enabling different downstream tasks. Existing approaches show promising perf
ormance, but lacking a way to shape these weight-space representations using inf
ormation about the datasets the models were trained on, thus limiting downstream
 applications. We propose **WeightCLIP**, a method for learning a dataset-aligne
d latent space for neural networks, where datasets information is induced during
 training. The NNs are encoded as latent representations using an autoencoder, w
hile dataset samples are encoded using a dataset encoder. The two representation
s are aligned using a contrastive objective, effectively reshaping the weight-sp
ace representations according to the datasets. We demonstrate that such represen
tations can be used for different downstream tasks, including mapping dataset in
formation to a weight-space representation that decode to strong models. In addi
tion, we introduce a latent refinement process for generating models that outper
forms standard fine-tuning. Overall, our results demonstrate that explicitly inc
orporating dataset information improves what can be achieved with weight-space r
epresentations across retrieval, generation, and refinement. Code will be availa
ble at https://github.com/HSG-AIML/WeightCLIP.
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Leaderboard Incentives: Model Rankings under Strategic Post-Training
Yatong Chen, Guanhua Zhang, Moritz Hardt

Influential benchmarks incentivize competing model builders to strategically allocate post-training resources towards improvements on the leaderboard, a phenomenon dubbed \emph{benchmaxxing} or \emph{training on the test task}. In this work, we initiate a principled study of the incentive structure that benchmarks induce. We model benchmarking as a Stackelberg game between a benchmark designer who chooses an evaluation protocol and multiple model providers who compete simultaneously in a subgame given by the designer’s choice. Each competitor has a model of unknown latent quality and can inflate its observed score by allocating resources to benchmark-specific improvements. First, we prove that current benchmarks induce games for which no Nash equilibrium between model developers exists. This result suggests one explanation for why current practice leads to misaligned incentives, prompting model providers to strategize in opaque ways. However, we prove that under mild conditions, a recently proposed evaluation protocol, called tune-before-test, induces a benchmark with a unique Nash equilibrium that ranks models by latent quality. This positive result demonstrates that benchmarks need not set bad incentives, even if current evaluations do.
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Steering at the Source: Style Modulation Heads for Robust Persona Control
Yoshihiro Izawa, Gouki Minegishi, Koshi Eguchi, Sosuke Hosokawa, Kenjiro Taura
Activation steering offers a computationally efficient mechanism for controlling Large Language Models (LLMs) without fine-tuning.
While effectively controlling target traits (e.g., persona), coherency degradation remains a major obstacle to safety and practical deployment.
We hypothesize that this degradation stems from intervening on the residual stream, which indiscriminately affects aggregated features and inadvertently amplifies off-target noise.
In this work, we identify a sparse subset of attention heads (only three heads) that independently govern persona and style formation, which we term *Style Modulation Heads*.
Specifically, these heads can be localized via geometric analysis of internal representations, combining layer-wise cosine similarity and head-wise contribution scores.
We demonstrate that intervention targeting only these specific heads achieves robust behavioral control while significantly mitigating the coherency degradation observed in residual stream steering.
More broadly, our findings show that precise, component-level localization enables safer and more precise model control.
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Non-Monotonic Autoregressive Sequence Model
Tianyi Ma, Yiyue Qian, Yiyang Li, Zehong Wang, Yifan Ding, Zheyuan Zhang, Yan Liang, Chuxu Zhang, Yanfang Ye
Autoregressive models generate sequences monotonically, where any sampled token, even if erroneous or sub-optimal, becomes a permanent condition for all subsequent steps.
This structural limitation means that autoregressive models cannot revisit or revise earlier decisions, i.e., a capability essential for complex generation tasks where exploration and correction are necessary.
To this end, we propose N-MARS, a Non-Monotonic AutoregRessive Sequence modeling framework that enables models to generate, evaluate, and revise tokens within a single forward pass, effectively allowing exploration before commitment.
We operationalize this framework through a learned erase token <UNDO> that retracts the previous token, enabling on-the-fly revision within standard autoregressive decoding.
To train the model, we introduce a sequence augmentation method that constructs error-correction trajectories from model-generated deviations paired with ground-truth references.
We then propose masked supervised fine-tuning (mSFT) that exposes the model to errors as context for learning when to revise, without optimizing their likelihood.
Finally, we refine the model with group relative policy optimization (GRPO) that

incentivizes judicious <UNDO> usage, i.e., rewarding effective corrections while penalizing unsuccessful ones.
We conduct comprehensive theoretical and empirical analysis to validate the effectiveness of N-MARS, demonstrating a robust foundation for non-monotonic sequence modeling.
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Approximating Drift-Diffusion Models for User Decisions under Nudging and External Information
Gustavo Grivol, Hanna Halaburda, Alexander Tuzhilin
Modeling decision-making outside of controlled environments requires accounting for asynchronous, exogenous signals, such as notifications or algorithmic feeds, that dynamically alter user response times. Standard Drift-Diffusion Models (DDM) become analytically intractable when drift rates vary continuously with time. In this paper, we derive a closed-form analytical approximation for the first-passage time distribution of a single-boundary DDM with time-dependent drift, valid in the high-threshold regime. The main result allows us to analytically study the optimal timing of external signals to maximize the probability of a user response within our approximation framework. To evaluate our response time model, we conduct an extensive empirical comparison with state-of-the-art methods for user watch-time prediction and evaluation in simulated environments.
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Parametric Prior Mapping Framework for Non-stationary Probabilistic Time Series Forecasting
Jinglin Li, Jun Tan, QI Fang, Ning Gui
Effectively modeling non-stationary dynamics in probabilistic multivariate time series(MTS) forecasting requires balancing expressiveness with robustness. Existing parametric approaches benefit from strong inductive biases but lack flexibility, whereas deep generative models struggle to capture complex temporal dependencies without extensive data and computation. We introduce Parametric Prior Mapping (PPM), a framework that injects parametric structural priors into a generative modeling process. Specifically, PPM utilizes a parametric estimator to derive a dynamic, adaptive prior that guides the learning of a complex predictive distribution via a learnable mapping. This design allows the model to retain the efficiency of parametric methods while exploiting the expressive power of generative models. Trained with a hybrid objective, PPM yields precise forecasts with well-calibrated uncertainty estimates. Empirical results show that PPM outperforms existing baselines in handling non-stationary data, offering a superior trade-off between accuracy and computational efficiency. The code is available at https://github.com/ljl8336/PPM.
****

Questioning the Coverage-Length Metric in Conformal Prediction: When Shorter Intervals Are Not Better
Yizhou Min, Yizhou Lu, Lanqi Li, Zhen Zhang, Jiaye Teng
Conformal prediction (CP) has become a cornerstone of distribution-free uncertainty quantification, conventionally evaluated by its coverage and interval length.
This work critically examines the sufficiency of these standard metrics.
We demonstrate that the interval length might be deceptively improved through a counter-intuitive approach termed Prejudicial Trick (PT), while the coverage remains valid.
Specifically, for any given test sample, PT probabilistically returns an interval, which is either null or constructed using an adjusted confidence level, thereby preserving marginal coverage.
While PT potentially yields a deceptively lower interval length, it introduces practical vulnerabilities: the same input can yield completely different prediction intervals across repeated runs of the algorithm. We formally derive the conditions under which PT achieves these misleading improvements and provide extensive empirical evidence across various regression and classification tasks.
Furthermore, we introduce a new metric interval stability which helps detect whether a new CP method implicitly improves the length based on such PT-like techni

ques. Code is available at https://github.com/benben-cd/PT-Conformal-Prediction
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Video-BCI: Bayesian Cognitive Integration of Self-Prior Hypotheses for Video Understanding
Xing Xi, Peixian Chen, Yu Qiu, Ronghua Luo, Peilin Tong, Jiahao Liang
Recent progress in vision-language models (VLMs) has driven significant advances in video understanding. However, existing methods often act as naive empiricists, mapping video input directly to output without any mechanism to introspect or challenge inherent bias. In this work, we challenge this paradigm by reframing video reasoning as a Bayesian cognitive process. We propose Video-BCI (Bayesian Cognitive Integration of Self-Prior Hypotheses), a novel framework that first samples a set of Self-Prior Hypotheses to represent the model's intuitive yet potentially biased cognitive state, and then guides the VLMs to perform a critical integration of these priors. This process encourages the model to challenge erroneous majority consensus in cases of high information divergence and to distill superior reasoning chains from its own prior space. The integration is driven by a composite Cognitive Utility Function comprising two intrinsic learning signals: Dialectical Uncertainty Signal (DUS) and Process Tracing Signal (PTS). The DUS incentivizes correct, non-majority judgments by quantifying both the conflict (entropy) among priors and their consensus-challenging strength. The PTS guides the model to trace and learn from reasoning paths within its own priors that lead to better answers, enabling self-driven procedural knowledge distillation. Extensive experiments on six mainstream benchmarks show that Video-BCI achieves new state-of-the-art (SOTA) results across the board. For example, it surpasses the previous best on the MMVU benchmark by 3.8%.
****
Improved Bounds for Private and Robust Alignment
Wenqian Weng, Yi He, Xingyu Zhou
In this paper, we study the private and robust alignment of language models from a theoretical perspective by establishing upper bounds on the suboptimality gap in both offline and online settings. We consider preference labels subject to privacy constraints and/or adversarial corruption, and analyze two distinct interplays between them: privacy-first and corruption-first. For the privacy-only setting, we show that log loss with an MLE-style algorithm achieves near-optimal rates, in contrast to conventional wisdom. For the joint privacy-and-corruption setting, we first demonstrate that existing offline algorithms in fact provide stronger guarantees---simultaneously in terms of corruption level and privacy parameters---than previously known, which further yields improved bounds in the corruption-only regime. In addition, we also present the first set of results for private and robust online alignment. Our results are enabled by new uniform convergence guarantees for log loss and square loss under privacy and corruption, which we believe have broad applicability across learning theory and statistics.
****
On the Convergence of Decentralized Stochastic Minimax Optimization Algorithm with Compressed Communication
Yihan Zhang, Xinghua Shi, Meikang Qiu, Yu Wang, Hongchang Gao
The stochastic minimax optimization problem has widespread applications in machine learning. Recently, numerous distributed minimax optimization algorithms have been developed to handle distributed training data. However, most of these algorithms suffer from high communication costs. To address this issue, we develop a novel communication-efficient decentralized stochastic gradient descent ascent with momentum algorithm based on the error feedback mechanism. Importantly, our algorithm demonstrates how to balance the full-precision update and the compression residual with novel designs for coefficients regarding variables and gradients to guarantee convergence. However, compressing the primal and dual variables (and their gradients) of stochastic minimax optimization problems with the error feedback mechanism presents significant challenges for convergence analysis. In particular, it incurs the circle dependence among consensus errors and compression errors. To overcome this challenge, we propose novel strategies that enable the establishment of the convergence rate for our algorithm. Our theoretical re

sults demonstrate how the compression operator influences the convergence rate. Finally, extensive experimental results confirm the efficacy of our proposed alg orithm.
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Energy-based Compositional Diffusion Planning
Tao Sun, Utkarsh Aashu Mishra, Jiaxin Lu, Danfei Xu, Iro Armeni
Compositional diffusion planners aim to solve long-horizon robotic tasks using s hort training trajectories. Yet, current approaches often rely on the heuristic stitching of local predictions. We show that the resulting stitched update is ge nerally a non-conservative field} that does not mathematically correspond to any valid global trajectory log-density function. We propose **Energy-based Composi tional Diffuser** (ECD), a framework that formulates the global trajectory as th e minimizer of the sum of local bridge potentials. This energy-based perspective defines a conservative correction field and contains a boundary reaction term t hat heuristic stitching omits. To enable efficient inference, we further introdu ce a Markov-based score approximation that computes the reaction term via a sing le block-tridiagonal solve, maintaining time complexity linear in the planning h orizon. Empirically, ECD achieves state-of-the-art success rates on a range of O GBench stitching tasks, while nearly matching the inference speed of heuristic s titching methods. Code is available at https://github.com/GradientSpaces/ECD.
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Hydra-Nav: Object Navigation via Adaptive Dual-Process Reasoning
Zixuan Wang, Huang Fang, Shaoan Wang, Yuanfei Luo, Heng Dong, Wei Li, Yiming Gan
While large vision-language models (VLMs) show promise for object goal navigatio n, current methods still struggle with low success rates and inefficient localiz ation of unseen objects—failures primarily attributed to weak temporal-spatial r easoning. Meanwhile, recent attempts to inject reasoning into VLM-based agents i mprove success rates but incur substantial computational overhead. To address bo th the ineffectiveness and inefficiency of existing approaches, we introduce Hyd ra-Nav, a unified VLM architecture that adaptively switches between a deliberati ve "slow system" for analyzing exploration history and formulating high-level pl ans, and a reactive "fast system" for efficient execution. We train Hydra-Nav th rough a three-stage curriculum: (i) spatial-action alignment to strengthen traje ctory planning, (ii) memory-reasoning integration to enhance temporal-spatial re asoning over long-horizon exploration, and (iii) iterative rejection fine-tuning to enable selective reasoning at critical decision points. Extensive experiment s demonstrate that Hydra-Nav achieves state-of-the-art performance on the HM3D, MP3D, and OVON benchmarks, outperforming the second-best methods by 11.1\%, 17.4 \%, and 21.2\%, respectively. Furthermore, we introduce SOT (Success weighted by Operation Time), a new metric to measure search efficiency across VLMs with var ying reasoning intensity. Results show that adaptive reasoning significantly enh ances search efficiency over fixed-frequency baselines. Project page is availabl e at: https://zixuan-wang99.github.io/Hydra-Nav/.
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Towards Reliable Marking and Verification of AI-Generated Text via Geometry-awar e Sentence-level Watermarking
Yubing Ren, Ping Guo, Yanan Cao
Large generative models raise growing concerns about provenance, misinformation, and impersonation. Digital watermarking offers a principled solution, yet exten ding it to natural language remains challenging due to text discreteness and sen sitivity to semantic perturbations. Existing text watermarking methods either op erate at the token level requiring white-box access and remaining fragile to par aphrasing, or at the sentence level, which supports black-box deployment but suf fers from low Watermark Success Rate (WSR). We show that low WSR in sentence-lev el watermarking primarily stems from low injection success probability caused by a mismatch between posterior embedding distributions and semantic accept region s. Based on this insight, we propose \textbf{X-Guard}, a geometry-aware sentence -level watermarking framework that improves injection success by jointly optimiz ing embedding distributions and semantic space partitioning. X-Guard learns a mo re isotropic embedding space and introduces \textbf{A^{2}PQ}, a centroid-aligned

 partitioning scheme that approximately equalizes probability mass across region
s.
Extensive experiments across multiple models, languages, and attack settings dem
onstrate that X-Guard consistently improves robustness while preserving text flu
ency and practical deployability.
****
MVP-LAM: Learning Action-Centric Latent Action via Cross-Viewpoint Reconstructio
n
Jung Min Lee, Dohyeok Lee, Seokhun Ju, Taehyun Cho, Jin Woo Koo, Li Zhao, Sangwo
o Hong, Jungwoo Lee
Latent actions learned from diverse human videos serve as pseudo-labels for visi
on-language-action (VLA) pretraining, but provide effective supervision only if
they remain informative about the underlying ground-truth actions. For effective
 supervision, latent actions should contain information about the underlying act
ions even though they are inaccessible. We propose **M**ulti-**V**iew**P**oint *
*L**atent **A**ction **M**odel (**MVP-LAM**), which learns latent actions that a
re highly informative about ground-truth actions from multi-view videos. MVP-LAM
 trains latent actions with a *cross-viewpoint reconstruction* objective, so tha
t a latent action from one view must explain the future in another view, reducin
g reliance on viewpoint-specific cues. On Bridge V2, MVP-LAM produces more actio
n-centric latent actions, achieving higher mutual information with ground-truth
actions and improved action prediction, including under out-of-distribution eval
uation. Finally, pretraining VLAs with MVP-LAM latent actions improves downstrea
m manipulation performance on various benchmarks. The code and trained checkpoin
ts are available at [https://jm-this.github.io/mvp_lam/](https://jm-this.github.
io/mvp_lam/).
****
Self-Captioning Multimodal Interaction Tuning: Amplifying Exploitable Redundanci
es for Robust Vision Language Models
Yuriel Ryan, Ip Hei Man, Adriel Kuek, Paul Pu Liang, Roy Ka-Wei Lee
Current vision language models face hallucination and robustness issues against
ambiguous or corrupted modalities. We hypothesize that these issues can be addre
ssed by exploiting the shared information between modalities to compensate for t
he impaired one. To this end, we analyze multimodal interactions -- redundant (s
hared), unique (exclusive), and synergistic (emergent) task-relevant information
 provided by the modalities -- to determine their impacts on model reliability.
Specifically, amplifying redundant interactions would increase this exploitable
shared information to resolve these issues; yet, modern instruction datasets oft
en eliminate redundancies to prioritize visual grounding. We bridge this gap thr
ough a self-captioning workflow featuring a \textsc{Multimodal Interaction Gate}
: a mechanism to convert unique interactions into redundant interactions. Our fi
ndings suggest that increasing redundancy can reduce visual induced errors by 38
.3\% and improve consistency by 16.8\%.
****
Population-Aware Imitation Learning in Mean-field Games with Common Noise
Grégoire Lambrecht, Mathieu Lauriere
Mean Field Games (MFGs) provide a powerful framework for modeling the collective
 behavior of large populations of interacting agents. In this paper, we address
the problem of Imitation Learning (IL) in MFGs subject to common noise, where th
e population distribution evolves stochastically. This stochasticity compels age
nts to adopt population-aware policies to respond to aggregate shocks. We formul
ate two distinct learning objectives: recovering a Nash equilibrium and maximizi
ng performance against an expert population. We investigate two imitation proxie
s: Behavioral Cloning (BC) and Adversarial (ADV) divergence. We then establish f
inite-sample error bounds showing that minimizing these proxies effectively cont
rols both the policy’s exploitability and its performance gap relative to the ex
pert. Furthermore, we propose a numerical framework using generalized Fictitious
 Play and Deep Learning to compute expert population-aware policies. Through exp
eriments on three environments we demonstrate that standard population-unaware p
olicies fail to capture the equilibrium dynamics. Our results highlight that lea

rning population-aware policies is crucial to avoid being misled by the randomne
ss inherent in common noise.
****

SEPS: Semantic-Enhanced Patch Slimming Framework for Fine-Grained Cross-Modal Al
ignment

Xinyu Mao, Junsi Li, Haoji Zhang, Yu Liang, Ming Sun

Fine-grained cross-modal alignment is pivotal for multimodal reasoning yet remai
ns limited by Semantic Sparsity Bias—a fundamental asymmetry where dense visual
signals are under-represented by sparse textual captions. This disparity leads t
o the inadvertent suppression of contextually vital visual regions (patch redund
ancy) and hinders precise concept grounding (patch ambiguity). While Multimodal
Large Language Models (MLLMs) offer rich descriptive capabilities, their naive i
ntegration often induces semantic drift due to inconsistencies with sparse groun
d-truth captions. To systematically resolve these challenges, we present the Sem
antic-Enhanced Patch Slimming (SEPS) framework. Central to SEPS is a novel Dual-
Granularity Semantic Calibration mechanism, which synthesizes a Holistic Visual-
Linguistic Anchor from MLLMs to synergize with original sparse queries. This mec
hanism transforms patch selection into a semantic consensus process, ensuring th
at retained patches satisfy both local discriminability and global contextual in
tegrity. Furthermore, we propose a Salience-Guided Metric Aggregation strategy t
o mitigate the similarity dilution effect inherent in global mean pooling, there
by amplifying highly-relevant patch-word correspondences. Extensive experiments
on Flickr30K and MS-COCO datasets demonstrate that SEPS surpasses existing state
-of-the-art approaches across diverse backbones, delivering significant performa
nce gains in text-to-image retrieval tasks. The complete implementation is avail
able at https://anonymous.4open.science/r/SEPS/.
****

Fast Non-Episodic Finite-Horizon RL with K-Step Lookahead Thresholding

Jiamin Xu, Kyra Gan

Online reinforcement learning in non-episodic, finite-horizon MDPs remains under
explored and is challenged by the need to estimate returns to a fixed terminal t
ime. Existing infinite-horizon methods, which often rely on discounted contract
ion, do not naturally account for this fixed-horizon structure. We introduce a m
odified Q-function: rather than targeting the full-horizon, we learn a K-step lo
okahead Q-function that truncates planning to the next K steps. To further impro
ve sample efficiency, we introduce a thresholding mechanism: actions are selecte
d only when their estimated K-step lookahead value exceeds a time-varying thresh
old. We provide an efficient tabular learning algorithm for this novel objective
, proving it achieves fast finite-sample convergence: it achieves minimax optima
l constant regret for $K=1$ and $\mathcal{O}(\max((K-1),C_{K-1})\sqrt{SAT\log(T)
})$ regret for any $K \geq 2$. We numerically evaluate the performance of our a
lgorithm under the objective of maximizing reward. Our implementation adaptively
 increases K over time, balancing lookahead depth against estimation variance. E
mpirical results demonstrate superior cumulative rewards over state-of-the-art t
abular RL methods across synthetic MDPs and RL environments: JumpRiverswim, Fro
zenLake and AnyTrading. Code is provided on \href{https://github.com/jamie01713/
K-Step-Lookahead}{github}.
****

Conformal Path Reasoning: Trustworthy Knowledge Graph Question Answering via Pat
h-Level Calibration

Shuhang Lin, Chuhao Zhou, Xiao Lin, Zihan Dong, Kuan Lu, Zhencan Peng, Jie Yin,
Dimitris N. Metaxas

Knowledge Graph Question Answering (KGQA) offers grounded, interpretable reasoni
ng, but existing methods often fail to provide reliable coverage guarantees over
 retrieved answers. While Conformal Prediction (CP) offers a principled framewor
k for producing prediction sets with statistical guarantees, prior conformal KGQ
A methods suffer from two critical pitfalls: violated coverage guarantees due to
 invalid calibration, and weak score discriminability that yields excessively la
rge prediction sets. We propose Conformal Path Reasoning (CPR), a novel trustwor
thy KGQA framework built on two key innovations. First, query-level conformal ca

libration over path-level scores preserves exchangeability to ensure valid cover age guarantees. Second, we introduce the Residual Conformal Value Network (RCVNe t), a lightweight module trained via PUCT-guided exploration to learn discrimina tive path-level nonconformity scores. Extensive experiments show that CPR signif icantly improves the Empirical Coverage Rate by 45\% while reducing prediction s et size by 52\% on average over conformal baselines across benchmark datasets, h ighlighting its effectiveness for reliable conformal reasoning over knowledge gr aphs.
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DLM: Unified Decision Language Model for Offline Multi-Agent Sequential Decision Making

Zhuohui Zhang, Bin Cheng, Bin He

Building scalable and reusable multi-agent decision policies from offline datase ts remains a challenge in offline multi-agent reinforcement learning (MARL), as existing methods often rely on fixed observation formats and action spaces that limit generalization. In contrast, large language models (LLMs) offer a flexible modeling interface that can naturally accommodate heterogeneous observations an d actions. Motivated by this, we propose the Decision Language Model (DLM), whic h formulates multi-agent decision making as a dialogue-style sequence prediction problem under the centralized training with decentralized execution paradigm. D LM is trained in two stages: a supervised fine-tuning phase, which leverages dia logue-style datasets for centralized training with inter-agent context and gener ates executable actions from offline trajectories, followed by a group relative policy optimization phase to enhance robustness to out-of-distribution actions t hrough lightweight reward functions. Experiments on multiple benchmarks show tha t a unified DLM outperforms strong offline MARL baselines and LLM-based conversa tional decision-making methods, while demonstrating strong zero-shot generalizat ion to unseen scenarios across tasks.
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EpiCoCo: De Novo Epitope Generation via MHC-Context Co-Modeling and Contrastive Affinity Guidance

Haoyang Luan, Gufeng Yu, Letian Chen, Zhenran Xiao, Yueshan Huang, Junkun Guo, Y ang Yang

The *de novo* generation of high-affinity epitopes tailored to specific major hi stocompatibility complex (MHC) proteins is a pivotal challenge in computational immunotherapy. However, current methods struggle to effectively integrate the MH C context into the generation process, and often fail to guarantee high binding affinity due to the neglect of discriminative signals from non-binders. To bridg e these gaps, we present **EpiCoCo**, a probabilistic framework for **Epi**tope generation via MHC-context **Co**-modeling and **Co**ntrastive affinity learning . EpiCoCo treats the pMHC complex as a dynamic, co-adaptive system by operating on the joint E(3) graph. In addition, we introduce Contrastive Affinity Guidance (CAG), an inference mechanism that leverages the gradient difference between le arned high- and low-affinity distributions. CAG actively drives the generation t rajectory towards high-affinity manifolds while utilizing repulsive signals to f ilter out candidates with poor binding potential. Extensive evaluations demonstr ate that EpiCoCo achieves a mean binding free energy of -45.20 REU, a 23% improv ement over the state-of-the-art, while maintaining high structural plausibility. The results validate that context co-modeling and negative-informed guidance ar e essential for generating valid, high-potency immunotherapeutics.
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Persona2Web: Benchmarking Personalized Web Agents for Contextual Reasoning with User History

Serin Kim, Sangam Lee, Dongha Lee

Large language models have advanced web agents, yet current agents lack personal ization capabilities.
Since users rarely specify every detail of their intent, practical web agents mu st be able to interpret ambiguous queries by inferring user preferences and cont exts.
To address this challenge, we present Persona2Web, the first benchmark for evalu

ating personalized web agents on the real open web, built upon the clarify-to-pe
rsonalize principle, which requires agents to resolve ambiguity based on user hi
story rather than relying on explicit instructions.
Persona2Web consists of:
(1) user histories that reveal preferences implicitly over long time spans,
(2) ambiguous queries that require agents to infer implicit user preferences, an
d
(3) a reasoning-aware evaluation framework that enables fine-grained assessment
of personalization.
We conduct extensive experiments across various agent architectures, backbone mo
dels, history access schemes, and queries with varying ambiguity levels, reveali
ng key challenges in personalized web agent behavior. For reproducibility, our c
odes and datasets are publicly available at https://serin-kimm.github.io/Persona
2Web/
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MMBench-Live: A Continuously Evolving Benchmark for Multimodal Models
Yuanzhi Liu, Shousheng Zhao, Bo Zhou, Kongming Liang, Zhanyu Ma
Evaluation benchmarks are essential for assessing vision--language models (VLMs)
, but most multimodal benchmarks are static, making them vulnerable to temporal
staleness, data contamination, and costly maintenance. We present MMBench-Live,
a continuously evolving multimodal benchmark built by a multi-agent-driven autom
ated pipeline. Our framework treats benchmark evolution as task-guided dataset c
onstruction, integrating structured benchmark specification, feedback-controlled
 real-time data acquisition, and verifiable QA generation with executable reason
ing. To maintain cross-version comparability, we introduce a distribution-consis
tent update strategy that extracts task-related visual patterns from the origina
l benchmark to guide data collection and filtering. Instantiated from MMBench, M
MBench-Live contains 5.9K newly generated evaluation instances with a high answe
r correctness rate, while each update costs about \$30 and takes 1--2 hours. Ext
ensive evaluations show that MMBench-Live preserves stable model rankings, maint
ains semantic alignment with the original benchmark, and exhibits weaker contami
nation-related memorization signals, suggesting a practical and scalable paradig
m for sustainable multimodal benchmark evolution. The project is available at \u
rl{https://github.com/PRIS-CV/MMBench-Live}.
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Tackling Length Inflation Without Trade-offs: Group Relative Reward Rescaling fo
r Reinforcement Learning
Zichao Li, Jie Lou, Fangchen Dong, Zhiyuan Fan, Mengjie Ren, Hongyu Lin, Xianpei
 Han, Debing Zhang, Le Sun, Yaojie Lu, XingYu Li
Reinforcement learning significantly enhances LLM capabilities but suffers from
a critical issue: length inflation, where models adopt verbosity or inefficient
reasoning to maximize rewards.
Prior approaches struggle to address this challenge in a general and lossless ma
nner, primarily because additive penalties introduce a compensatory effect that
creates optimization shortcuts, while heuristic gating strategies lack generalit
y beyond binary feedback.
To bridge this gap, we present Group Relative Reward Rescaling (GR^3), which r
eframes length control as a multiplicative rescaling paradigm, effectively estab
lishing a generalized, continuous, and reward-dependent gating mechanism. To fur
ther ensure lossless optimization, we incorporate group-relative regularization
and advantage-aware calibration, which dynamically adapt length budgets to insta
nce difficulty and preserve the advantage signal of high-quality trajectories. E
mpirically, across both RLHF and RLVR settings, GR^3 maintains training dynami
cs and downstream performance comparable to standard GRPO while significantly mi
tigating length inflation, outperforming state-of-the-art length-regularized bas
elines.
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ConFlux: Multivariate Time Series in Flux, One Unified Forecast in Confluence
Shiyu Wang, Juntong Ni, Ziyi Zhang, Baichuan Mo, Xinyue Zhong, Chengxin Wang, Yu
chen Fang, Zhou Ye, Yang Xiang

Real-world multivariate time series are inherently in flux: different variables evolve asynchronously and interact in complex, time-varying ways, yet accurate forecasting requires these dispersed signals to converge into a single unified prediction.
This structural mismatch between dynamic, heterogeneous inputs and a unified forecasting objective poses a fundamental challenge for building general-purpose multivariate forecasting models, especially in zero-shot and large-scale settings.

To this end, inspired by the idea that ``\emph{all rivers run into the sea}'', we propose \textbf{ConFlux}, a \emph{general-purpose foundation model for multivariate time-series forecasting} by learning to adaptively integrate cross-channel information under a unified forecasting objective. Specifically, ConFlux first reorders variables to reduce cross-variable entanglement, then aggregates adjacent variables into compact patches that can be processed by a Vision Transformer-style architecture. This design shortens the effective context, reduces attention complexity, and provides a unified token representation for pre-training and downstream tasks.
Experiments on 25 public datasets show that ConFlux achieves state-of-the-art performance in zero-shot, fine-tuning, and from-scratch settings, while offering faster inference and lower memory usage.
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GEM: Geometric Entropy Mixing for Optimal LLM Data Curation
Yue Min, Ziyun Qiao, Ruining Chen, Yujun Li
LLM pre-training efficacy increasingly depends on data composition rather than sheer volume. Yet, optimal mixing is hindered by categorization flaws: human taxonomies suffer from ontological misalignment, and Euclidean clustering fails to address embedding anisotropy. We introduce **GEM** (**G**eometric **E**ntropy **M**ixing), a framework reformulating data curation as a variational problem on the hypersphere augmented with a **mixing-balance regularizer**. By decoupling the generative prior and optimizing the objective via a provable **MM (Minorize-Maximize)** algorithm, GEM effectively counteracts the cluster collapse to discover balanced semantic structures invisible to Euclidean heuristics. We employ teacher-student distillation to scale this geometric fidelity to web-scale corpora and introduce the **Geometric Influence Score (GIS)** for interpretable taxonomy generation. Experiments with 1.1B-parameter models demonstrate that GEM establishes a new state-of-the-art when integrated into mixing strategies like DoReMi and RegMix, improving average downstream accuracy by up to **1.2%** and offering a robust coordinate system for predictable data mixing.
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Diversity-Aware Recursive Feature Multiple Kernel Learning
Nan Cao, Xu Zhao, Teng Zhang
\textit{\underline{M}ultiple \underline{k}ernel \underline{l}earning}~(MKL) combines several base kernels in the spirit of ensemble learning, yet existing methods rarely model kernel diversity---a known cornerstone of ensembles---and most traditional kernels weight all features uniformly, ignoring feature-level discriminability. We address both gaps with DARFMMKL: a data-driven kernel family (Recursive Feature Machine kernels) that learns feature importance directly from data, paired with a kernel selection method that jointly optimizes diversity and quality. The resulting NP-hard binary quadratic program is reformulated via Glover linearization and continuous relaxation into a linear program, and accelerated by Nystr\"om sketching, yielding a selector whose cost is decoupled from the sample size. We provide a covering-number generalization bound that explicitly relates kernel diversity to estimation error. Experiments on 12 benchmark datasets show that DARFMMKL consistently outperforms 9 state-of-the-art MKL methods.
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CausalX: A Unified and Causally-Interpretable Plug-and-Play Model for Multi-modal Spatio-Temporal Forecasting
Shiqi Zhang, Pan Mu, Hanting Yan, Yuchao Zhu, jinglin zhang, Cong Bai
Multi-modal spatio-temporal forecasting underpins many real-world applications but remains challenging due to the complex and evolving interactions across modal

ities and time steps. Moreover, the lack of interpretability in existing models limits their reliability in safety-critical scenarios.
In this paper, we present CausalX, a unified and causally interpretable plug-and-play model for multi-modal spatio-temporal forecasting.
CausalX achieves interpretability by learning a dynamic causal graph across modalities and time, whose edge weights quantify causal attribution strength, and are further refined by a diffusion-based generative process guided by structural priors.
To overcome the absence of ground-truth causal structures, CausalX aggregates multi-source constraints from causal analysis techniques and a variational autoencoder, spanning predictive, temporal, interventional, and generative aspects to jointly learn a more comprehensive causal graph. Extensive experiments on real-world forecasting tasks, including pedestrian trajectory prediction and tropical cyclone forecasting, demonstrate that CausalX achieves superior accuracy while producing interpretable causal graphs. CausalX is modular, architecture-agnostic, and generalizable, offering a new perspective on bridging causal inference and spatio-temporal forecasting.
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Population-Free Pareto Tracking for Sample-Efficient Multi-Policy MORL
Zeyu Zhao, Yueling Che, Kaichen Liu, Jian Li, Junmei Yao
Multi-objective reinforcement learning (MORL) is a fundamental framework for real-world decision-making problems involving multiple conflicting criteria. Existing multi-policy (MP) methods typically rely on online evolutionary frameworks that maintain large policy populations, leading to high sample complexity and excessive agent–environment interactions. To mitigate these limitations, we present Multi-policy Pareto Front Tracking (MPFT), a framework without a self-evolving population. It leverages an efficient Pareto-tracking mechanism initialized with single-objective extreme policies to trace the Pareto front, and further densifies sparse regions to achieve an accurate approximation of the full Pareto front. MPFT can be seamlessly integrated with advanced offline MORL algorithms, thereby
substantially improving sample efficiency. We evaluate MPFT on six robotic control tasks with up to three objectives and three high-dimensional tasks with more than three objectives. Experimental results show that MPFT outperforms state-of-the-art baselines in terms of hypervolume and expected utility. It also significantly reduces agent–environment interactions. These results further demonstrate that MPFT serves as a general purpose framework that can seamlessly integrate both online and offline MORL algorithms.
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GeoFlow: Geo-Aware Modeling of Inter-Area Relationships in Origin-Destination Flow Prediction and Generation
Zherui Huang, Guanjie Zheng, Hao Xue, Linghe Kong
Origin–destination (OD) flow modeling underpins urban planning and mobility analysis, but prevailing graph-based methods often neglect salient geographic attributes, limiting their ability to model long-range and multi-area dependencies. In this paper, we introduce GeoFlow, a novel framework that (i) augments area representations with geospatial attributes, including relative positions,
-hop and geodesic distances, (ii) employs a specialized geometric-intrinsic fusion encoder design that combines graph attention for intrinsic area signals with coordinate-aware encoders for global structure, and (iii) adopts an axial-global attention decoder to capture OD-specific competitive dependencies. For OD flow generation, GeoFlow is paired with flow matching models to produce more authentic and diverse mobility samples. Empirically, GeoFlow achieves superior performance in predictive accuracy, while substantially improving generative fidelity and diversity. Ablation and analytical studies confirm the contribution of each component. Code is available at https://github.com/ZheruiHuang/GeoFlow.
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Nested birth-death processes are competitive with neural networks as time-dependent models of protein evolution

Annabel Large, Ian Holmes
Most statistical phylogenetics analyses use simple continuous-time finite-state Markov models of point substitution to describe molecular evolution. These models enforce unrealistic assumptions like keeping sequence length fixed, ignoring insertions and deletions (indels) entirely, and making little (if any) allowance for variation in selection pressure due to interactions between amino acids. We extend the TKF92 model—the canonical hierarchical model combining an outer birth-death process for indels with an inner finite-state Markov chain for substitutions—by introducing additional nesting and latent states, allowing for structural heterogeneity. We compare these TKF92 extensions to two classes of neural seq2seq models that use evolutionary time as an input feature: the first "basic" class lacks any evolutionary modeling constraints, while the second "hybrid" class combines neural sequence embeddings with a TKF92-like likelihood function. We evaluate the per-character perplexities of all models on splits of the Pfam database of aligned protein domains. The hybrid neural models outperform their basic counterparts across all sequence embedding architectures. Furthermore, a nested TKF-based model with only 30,000 parameters is highly competitive with all neural networks (which contain tens of millions of parameters), outperforming all but two of the neural architectures tested. Taken together, our results indicate that approaches grounded in molecular evolutionary theory may provide a better fit to real alignments than unconstrained alternatives, supporting the incorporation of CTMC-based model structure within future neural phylogenetic approaches.
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Physics-Guided Motion Loss for Video Generation Model
Bowen Xue, Giuseppe Claudio Guarnera, Shuang Zhao, Zahra Montazeri
Current video diffusion models generate visually compelling content but often struggle with physical motion, producing subtle artifacts like rubber-sheet deformations and
inconsistent object motion. We introduce a frequency-domain physics prior that improves
motion plausibility without modifying model architectures. Our method decomposes common
motion patterns (translation, rotation, scaling) into lightweight spectral losses.
Applied to Open-Sora, MVDIT, and Hunyuan, our approach improves both motion accuracy and action recognition by ~11\% on average on OpenVID-1M (relative), while maintaining visual quality. Additional results on Wan 2.1-14B show consistent gains on video-quality and physics-oriented metrics. User studies show 74-83\% preference for our physics-enhanced videos. It also reduces warping error by 22-37\% (depending on the backbone) and improves temporal consistency scores. These results indicate that simple, global spectral cues are an effective drop-in regularizer for physically plausible motion in video diffusion.
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Two Modalities Are Better Than One: Efficient Adversarial Purification via Multimodal Diffusion Models
Mingyuan Bai, Wei Huang, Tenghui Li, Andong Wang, Chao Li, Cesar F Caiafa, Junbin Gao, Qibin Zhao
Adversarial purification uses generative models to restore clean data distributions from unseen attacks without retraining classifiers. However, unimodal diffusion-based approaches struggle to preserve semantic consistency, while recent multimodal variants rely on computationally expensive adversarial training or distillation. Both approaches often lack theoretical guarantees. In this work, we propose MultiDAP, a novel framework leveraging multimodal diffusion models for efficient adversarial purification. MultiDAP first learns continuous class-agnostic prompts from clean data to capture rich semantic priors, replacing rigid hand-crafted templates. Guided by these prompts, MultiDAP purifies adversarial inputs by minimizing a regularized DDPM loss for only a few steps (e.g., 5-20). We provide theoretical guarantees for both the likelihood improvement via prompt learning and the convergence of the purification process. Extensive experiments on CIFAR-10, CIFAR-100, and ImageNet-1K demonstrate that MultiDAP matches the robustnes

s of state-of-the-art baselines but with improved efficiency.
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MemEvolve: Meta-Evolution of Agent Memory Systems
Guibin Zhang, Haotian Ren, Chong Zhan, Junhao Wang, He Zhu, Wangchunshu Zhou, Sh uicheng YAN
Self-evolving memory systems are rapidly reshaping the evolutionary paradigm of large language model (LLM)-based agents. Prior work has predominantly relied on manually engineered memory architectures to store trajectories, distill experien ce, and synthesize reusable tools, enabling agents to evolve on the fly within e nvironment interactions. However, this paradigm is fundamentally constrained by the \textit{staticity} of the memory system itself: while memory facilitates age nt-level evolving, the underlying memory architecture cannot be meta-adapted to diverse task contexts. To address this gap, we propose MemEvolve, a meta-evoluti onary framework that jointly evolves agents' experiential knowledge and their me mory architecture, allowing agent systems not only to accumulate experience but also to progressively refine how they learn from it. To ground MemEvolve in prio r work and promote openness in future self-evolving systems, we introduce Evolve Lab, a unified memory codebase that distills twelve representative memory system s into a modular design space (\textit{encode}, \textit{store}, \textit{retrieve }, \textit{manage}), providing a standardized implementation substrate and a fai r experimental arena. Extensive evaluations on four challenging agentic benchmar ks show that MemEvolve delivers (i) substantial performance gains, improving fra meworks such as SmolAgent and Flash-Searcher by up to 17.06%, and (ii) strong cross-task and cross-LLM generalization, yielding memory architectures that tra nsfer effectively across diverse benchmarks and backbones.
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From Feasible to Practical: Pareto-Optimal Synthesis Planning
Friedrich Hastedt, Dongda Zhang, Antonio Del rio chanona
Current computer-aided synthesis planning (CASP) methods often treat retrosynthe sis as solved once a single feasible route is identified, focusing primarily on convergence or shortest-path metrics. This view is misaligned with real-world pr actice, where chemists must balance competing objectives such as cost, sustainab ility, toxicity, and overall yield. To address this, we formulate synthesis plan ning as a multi-objective search problem and introduce MORetro*, an algori thm that generates a Pareto front of synthesis routes to explicitly capture trad e-offs between user-defined criteria. MORetro* uses weighted scalarization and solution-informed sampling to efficiently navigate the combinatorial search space and prioritize promising trade-offs. Building on multi-objective A* -search, we provide optimality guarantees showing that, for a fixed single-step model, MORetro* recovers the true Pareto front under admissibility. Across multiple retrosynthesis benchmarks, MORetro* produces diverse, high-quali ty Pareto fronts, uncovering solutions overlooked by single-objective approaches and better aligning CASP outputs with industrial decision-making.
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DREAM: Dual-Standard Semantic Homogeneity with Dynamic Optimization for Graph Le arning with Label Noise
Yusheng Zhao, Jiaye Xie, Qixin Zhang, Weizhi Zhang, Xiao Luo, Zhiping Xiao, Phil ip S. Yu, Ming Zhang
Graph neural networks (GNNs) have been widely used in various graph machine lear ning scenarios. Existing literature primarily assumes well-annotated training gr aphs, while the reliability of labels is not guaranteed in real-world scenarios. Recently, efforts have been made to address the problem of graph learning with label noise. However, existing methods often (i) struggle to distinguish between reliable and unreliable nodes, and (ii) overlook the relational information emb edded in the graph topology. To tackle this problem, this paper proposes a novel method, Dual-Standard Semantic Homogeneity with Dynamic Optimization (DREAM), f or reliable, relation-informed optimization on graphs with label noise. Specific ally, we design a relation-informed dynamic optimization framework that iterativ ely reevaluates the reliability of each labeled node in the graph during the opt imization process according to the relation of the target node and other nodes.

To measure this relation comprehensively, we propose a dual-standard selection strategy that selects a set of anchor nodes based on both node proximity and graph topology. Subsequently, we compute the semantic homogeneity between the target node and the anchor nodes, which serves as guidance for optimization. We also provide a rigorous theoretical analysis to justify the design of DREAM. Extensive experiments are performed on six graph datasets across various domains under three types of graph label noise against competing baselines, and the results demonstrate the effectiveness of the proposed DREAM.
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Vision-aligned Latent Reasoning for Multi-modal Large Language Model
Byungwoo Jeon, Yoonwoo Jeong, Hyunseok Lee, Minsu Cho, Jinwoo Shin
Despite recent advancements in Multi-modal Large Language Models (MLLMs) on diverse understanding tasks, these models struggle to solve problems which require extensive multi-step reasoning. This is primarily due to the progressive dilution of visual information during long-context generation, which hinders their ability to fully exploit test-time scaling. To address this issue, we introduce Vision-aligned Latent Reasoning (VaLR), a simple, yet effective reasoning framework that dynamically generates vision-aligned latent tokens before each Chain of Thought reasoning step, guiding the model to reason based on perceptual cues in the latent space. Specifically, VaLR is trained to preserve visual knowledge during reasoning by aligning intermediate embeddings of MLLM with those from vision encoders. Empirical results demonstrate that VaLR consistently outperforms existing approaches across a wide range of benchmarks requiring long-context understanding or precise visual perception, while exhibiting test-time scaling behavior not observed in prior MLLMs. In particular, VaLR improves the performance significantly from 33.0\% to 52.9\% on VSI-Bench, achieving a 19.9\%p gain over Qwen2.5-VL.
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Generalizable and Composable Multi-Model Embedding Translation
Beining Yang, Yang Cao
Embedding translation enables interoperability across embedding models, allowing embedding vectors to be reused without costly re-embedding. However, existing methods are typically evaluated under simplified pairwise and i.i.d. settings and behave as black boxes at inference time, leading to unreliable performance under out-of-distribution (OOD) inputs, multi-model mixing, and composed translations. We analyze embedding translation from a geometric perspective and derive an interpretable error bound that explains systematic error amplification under OOD inputs, mixing and chaining. Building on this, we propose a geometry-aware confidence metric and a Hierarchical Mixture of Experts (HMoE) framework with localized, parameter-efficient adaptation. Following the MTEB leaderboard, we conduct large-scale experiments over 10 embedding models and 6 datasets across 90 pairwise translation settings. HMoE outperforms every baseline for every model pair over every dataset under OOD scenarios. Furthermore, multi-model mixing and chaining only degrade our performance in Recall@100 by $0.5\% -- 2.6\%$, compared to $7.2\% -- 92.3\%$ recall drop by existing methods.
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Scaling-Aware Adapter for Structure-Grounded LLM Reasoning
Zihao Jing, QIUHAO Zeng, Ruiyi Fang, Yan Yi Li, Yan Sun, Boyu Wang, Pingzhao Hu
Large language models (LLMs) are enabling reasoning over 2D and 3D structures, yet existing methods remain modality-specific and typically compress structural inputs through sequence-based tokenization or fixed-length query connectors. Such architectures either omit the geometric grounding requisite for mitigating structural hallucinations, or impose inflexible modality fusion bottlenecks that concurrently over-compress and suboptimally allocate structural tokens, thereby impeding the realization of generalized all-atom reasoning. We introduce **Cuttlefish**, a unified multimodal LLM that grounds language reasoning in geometric cues while scaling modality tokens with structural complexity. First, **Scaling-Aware Patching** leverages an instruction-conditioned gating mechanism to generate variable-size patches over structural graphs, adaptively scaling the query token budget with structural complexity to mitigate fixed-length connector bottlenecks

. Second, **Geometry Grounding Adapter** refines these adaptive tokens via cross-attention to modality embeddings and injects the resulting modality tokens into the LLM, exposing explicit geometric cues to reduce structural hallucination. Experiments across interdisciplinary all-atom benchmarks demonstrate that Cuttlefish achieves superior performance in heterogeneous structure-grounded reasoning. Code: github.com/zihao-jing/Cuttlefish.
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MADA-Attack: Transferable Multi-modal Attention Distraction Adversarial Attack against Vision Language Models
Zhihan Qin, Jiahao Chen, Chunyi Zhou, Yuwen Pu, Chunqiang Hu, Xiaolei Liu, Shouling Ji
Vision Language Models (VLMs) achieve strong performance across multi-modal tasks but remain vulnerable to universal adversarial perturbations (UAPs). Existing UAP methods mainly operate on the visual modality, overlooking structured textual semantics and cross-modal interactions, which limits their ability to disrupt alignment and generalize across tasks and model architectures. To address these limits, we propose **Multi-modal Attention Distraction Adversarial Attack (MADA-Attack)** framework. We begin by conducting several insight experiments and discover that modality attention distributes differently over layers and early phase of optimization is decisive. Building on these observations, we introduce Semantic Token Manipulation (STM) to steer text-guided attention, and Fused Embedding Training (FET) to jointly optimize textual and visual embedding losses for coordinated misalignment. We further incorporate an Adaptive Data Augmentation (ADA) strategy that dynamically balances attack strength, transferability, and training efficiency. Extensive experiments demonstrate that **MADA-Attack** consistently achieves state-of-the-art performance and strong transferability while remaining computationally lightweight, with an average ASR of 82.60\% and 73.42\% in zero-shot classification and image captioning tasks. For the visual question answering (VQA) and I-T Retrieval task, our method exceeds the SOTA baseline by 10\%. Our code is available at this [GitHub Repository](https://github.com/RaymundoTheWolf/MADA-Attack).
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Memory as a Markov Matrix: Sample Efficient Knowledge Expansion via Token-to-Dictionary Mapping
Kaustubh Pethkar, Ziyang Xiong, Zuofeng Shang, Yingcong Li
Continual incorporation of new knowledge is essential for the long-term evolution of large language models (LLMs). Existing approaches typically rely on parameter-update algorithms to mitigate catastrophic forgetting, yet they suffer from fundamental limitations: 1) forgetting is unavoidable as the amount of newly injected knowledge grows; and 2) model updates are often irreversible. As modern LLMs become increasingly expressive, it is natural to question whether large-scale weight updates are necessary for acquiring a small amount of new knowledge. In this work, we propose a principled framework that models autoregressive language generation as a Markov process over tokens, where model memory is represented by a Markov transition matrix. Under this formulation, incorporating new knowledge/tokens corresponds to extending the state space, and preserving existing transitions guarantees retention of previously learned knowledge. We then prove a sample complexity bound for incorporating new tokens via a token-to-dictionary mapping strategy. In particular, for learning the transition behavior of each new token, the required number of samples scales linearly with the number of existing tokens it is mapped to. To realize this mapping, we propose an embedding-tuning algorithm that requires minimal parameter updates and induces zero forgetting. Experimental results further demonstrate the effectiveness of our method and validate our theoretical findings.
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FiSeR: Fine-Grained Source Representations for Cross-Domain AI Image Detection
Shan Zhang, Yongxin He, Mingming Zhang, Huiwen Tian, Lei Ma
Real-world synthetic image detectors often generalize poorly under domain shift despite strong in-domain performance. Using unsupervised UMAP projections, we find that natural and synthetic features remain partially separable on unseen data

sets, yet performance still drops, suggesting that the classification head overfits to training-domain artifacts.
Therefore, the key is to learn more transferable representations so that the decision criterion is more stable and robust to domain shifts. Based on the structural fact that synthetic images are produced by diverse generators, we propose a hierarchical contrastive learning framework that improves the separability between natural and synthetic images while preserving generator identity information. It jointly optimizes (i) a coarse contrastive objective between natural and synthetic images and (ii) a fine contrastive objective among synthetic images using generator identities.
Trained on WildFake, our method achieves an average AUROC gain of +10.22 on cross-domain evaluation over Chameleon, AIGIBench, Community Forensics, and GenImage under the same settings as the strong baseline DIRE. For few-shot adaptation, we freeze the backbone and fit an SVM head on 10 labeled samples per class, improving AUROC by +10.64 on AIGIBench and +17.41 on Chameleon, averaged over 12 widely used detectors. Our code is publicly available at: https://github.com/heyongxin233/FiSeR.
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VLA-Arena: An Open-Source Framework for Benchmarking Vision-Language-Action Models
Borong Zhang, Jiahao Li, Jiachen Shen, Yuhao Zhang, Yishuai Cai, Yuanpei Chen, Juntao Dai, Jiaming Ji, Yaodong Yang
While Vision-Language-Action models (VLAs) are rapidly advancing toward generalist robot policies, quantitatively characterizing their capability boundaries and failure modes remains challenging. To address this, we introduce **VLA-Arena**, a comprehensive benchmark. It features a novel structured task design framework to quantify difficulty across three orthogonal axes: **(1) Task Structure**, **(2) Language Command**, and **(3) Visual Observation**. This allows us to systematically design tasks with fine-grained difficulty levels, enabling a precise measurement of model capability frontiers. For task structure, VLA-Arena comprises 11 task suites organized into four dimensions: **Safety**, **Distractor**, **Extrapolation**, and **Long Horizon**, totaling 170 tasks. Each suite spans three difficulty levels (L0–L2), with fine-tuning restricted to L0 to rigorously assess generalization. Orthogonal to this, language (W0-W4) and visual (V0-V4) perturbations can be applied to any task as diagnostic probes to distinguish robust grounding from superficial pattern matching. Our extensive evaluation of state-of-the-art VLAs reveals critical limitations: memorization over generalization, superficial visual perception, and a neglect of safety constraints. Additionally, model rank reversals across L0–L2 validate that each level provides non-redundant insights. To foster research addressing these model limitations and ensure reproducibility, we provide the complete VLA-Arena framework, including an end-to-end toolchain from task definition to automated evaluation and the VLA-Arena-S/M/L datasets for fine-tuning. Our benchmark, datasets, models, and leaderboard are publicly available at https://vla-arena.github.io.
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HEARTS: Benchmarking LLM Reasoning on Health Time Series
Sirui Li, Shuhan Xiao, Mihir Joshi, Ahmed Metwally, Daniel McDuff, Wei Wang, Yuzhe Yang
The rise of large language models (LLMs) has shifted time series analysis from narrow analytics to general-purpose reasoning. Yet, existing benchmarks cover only a small set of health time series modalities and tasks, failing to reflect the diverse domains and extensive temporal dependencies inherent in real-world physiological modeling. To bridge these gaps, we introduce HEARTS (Health Reasoning over Time Series), a unified benchmark for evaluating hierarchical reasoning capabilities of LLMs over general health time series. HEARTS integrates 16 real-world datasets across 12 health domains and 20 signal modalities, and defines a comprehensive taxonomy of 110 tasks grouped into four core capabilities: Perception, Inference, Generation, and Deduction. Evaluating 14 state-of-the-art LLMs on more than 20K test samples reveals intriguing findings. First, LLMs substantially underperform specialized models, and their performance is only weakly related t

o general reasoning scores. Moreover, LLMs often rely on simple heuristics and s truggle with multi-step temporal reasoning. Finally, performance declines with i ncreasing temporal complexity, with similar failure modes within model families, indicating that scaling alone is insufficient. By making these gaps measurable, HEARTS provides a standardized testbed and living benchmark for developing next -generation LLM agents capable of reasoning over diverse health signals.
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Decoupling Universal Laws and Environmental Heterogeneity: A Physics-Inspired Fr amework for Robust Spatio-Temporal Forecasting
Aoyu Liu, Liming Wei, Yaying Zhang
Most spatio-temporal forecasting models assume in-distribution data and can degr ade sharply under non-stationary environments. Existing methods for handling dis tribution shift largely rely on discrete graph inference, making it difficult to disentangle universal dynamics from environment-specific changes and to respect the continuous physical nature of spatio-temporal fields. To this end, we propo se STPDE, a general framework that reformulates spatio-temporal dynamics as the evolution of inhomogeneous partial differential equations. STPDE explicitly deco mposes dynamics into an Invariant Diffusion Operator that captures universal mec hanisms and an Environment Basis Manifold that parameterizes local heterogeneous media. We show that the Green's function of the Laplacian can be effectively ap proximated by linear attention, enabling global diffusion at scale. Combined wit h stochastic environment perturbations, STPDE improves robustness under heteroge neous and shifting environments. Extensive experiments on in-distribution foreca sting, out-of-distribution generalization, few-shot cross-city transfer, and con tinual learning demonstrate consistent improvements over state-of-the-art baseli nes with competitive computational efficiency.
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Derivative Informed Learning of Exchange-Correlation Functionals
Eike Eberhard, Luca Thiede, Abdulrahman Aldossary, Andreas Burger, Nicholas Gao, Vignesh C Bhethanabotla, Alan Aspuru-Guzik, Stephan Günnemann
Machine-learned (ML) XC functionals aim to replace human-designed density functi onal approximations by learning directly from reference data, but they still do not consistently outperform traditional $\mathcal{O}(N^4)$-scaling hybrid functi onals. We therefore study a hybrid-distillation setting, where $\mathcal{O}(N^3) $-scaling semilocal ML-XC functionals are trained to reproduce B3LYP/def2-SVP ta rgets. We introduce Derivative Informed XC-Loss (DI-Loss), a loss that incorpora tes additional information from the reference hybrid functional by supervising f irst and second derivatives of the energy on the Grassmannian of admissible dens ity matrices. Rather than only matching the self-consistent fixed point, DI-Loss aligns the local first- and second-order response of the learned functional wit h that of the target functional. Across four evaluated architectures, DI-Loss co nsistently improves the main energy metrics. Averaged uniformly across architect ures, the total-energy MAE decreases by 66% relative to energy and density super vision alone. The density-sensitive mean-field energy metric E_ρ improves f rom 1.2 to 0.8 mEh on average, while dipole and $\mathcal{L}_2$ density errors d o not improve uniformly. We further show that densities from the distilled funct ionals reduce hybrid-functional SCF iterations by up to 55%. In downstream TDDFT calculations, Hessian supervision improves excited-state predictions, with XCdi ff reducing the mean excitation-energy MAE by 24-35% across molecule sizes on QM 40.
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NanoQuant: Efficient Sub-1-Bit Quantization of Large Language Models
Hyochan Chong, Dongkyu Kim, Changdong Kim, Minseop Choi
Weight-only quantization has become a standard approach for efficiently serving large language models (LLMs). However, existing methods fail to efficiently comp ress models to binary (1-bit) levels, as they either require large amounts of da ta and compute or incur additional storage. In this work, we propose NanoQuant, a post-training quantization (PTQ) method to compress LLMs to both binary and su b-1-bit levels. NanoQuant formulates quantization as a low-rank binary factoriza tion problem, and compresses full-precision weights to low-rank binary matrices

and scales. Specifically, it utilizes an efficient alternating direction method of multipliers (ADMM) solver to precisely initialize latent binary matrices and scales, and then tunes the initialized parameters through a block and model reconstruction process. Consequently, NanoQuant establishes a new Pareto frontier in low-memory post-training quantization, and enables sub-1-bit compression. NanoQuant makes large-scale deployment feasible on consumer hardware. For example, it compresses Llama-2-70B by 24$\times$ in just 13 hours on a single H100, enabling a 70B model to operate on a consumer 8 GB GPU.

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Lost in Context: Adressing Context Anxiety in Large Language Models

Ifueko Igbinedion, Jillian Ross, Etienne Ricardez, Sertac Karaman, Eric So

Conventional wisdom suggests that reasoning models fail when problems exceed their capabilities. However, we find that frontier reasoning models sometimes possess the necessary capabilities to solve problems but fail due to premature self-doubt -- a phenomenon informally known as context anxiety. We provide the first systematic study of context anxiety, demonstrating that it arises, in part, from a model's inability to accurately estimate the tokens required to complete a task. We also show that context anxiety leads to material efficiency losses when models operate under perceived constraints. Building on this analysis, we further show that models can learn alternative strategies for solving long-horizon problems without exhibiting context anxiety, suggesting that performance improvements may be achievable not through scaling model capabilities, but by improving models' ability to accurately assess and adapt to their own limitations.

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An Exterior Method for Nonnegative Matrix Factorization

Qiujing Lu, Tonmoy Monsoor, Ehsan Ebrahimzadeh, Kartik Sharma, Vwani Roychowdhury

Nonnegative matrix factorization (NMF) seeks a low-rank approximation $X \approx UV^T$ with nonnegative factors and is commonly solved using *interior* methods that enforce feasibility throughout optimization. We show that such constraint-driven approaches can impede progress in the nonconvex landscape, leading to slow convergence or convergence to suboptimal stationary points. We propose an *exterior* framework for NMF (eNMF) that separates low-rank approximation from nonnegativity enforcement. Our method initializes from the optimal unconstrained factorization and introduces a rotation procedure that maps unconstrained factors to an exterior point closest to the nonnegative orthant. This viewpoint yields an algorithmic framework in which simple iterative updates converge to KKT-satisfying stationary points on the boundary of the positive orthant. The exterior formulation also enables a geometric interpretation of NMF solutions, clarifying equivalence classes of factorizations under permutation and orthogonal transformations. An intriguing numerical result, involving 400 NMF experiments across both real and synthetic datasets, show that in 99\% of the cases, different algorithms tend to converge towards equivalent factor matrices. We benchmark eNMF against 9 state-of-the-art NMF algorithms with 9 initialization schemes across 3 real-world and 2 synthetic datasets. eNMF consistently outperforms all 81 competitors, achieving up to 30\% lower reconstruction error under equal-time settings and up to 150\% speedup under equal-error settings. The downstream experiments further demonstrate substantial performance gains in audio processing and recommendation tasks, corroborating the practical benefits of the proposed exterior optimization framework. Code is available at https://github.com/roychowdhuryresearch/eNMF

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Measuring Intent Comprehension in LLMs

Nadav Kunievsky, James Evans

People judge interactions with large language models (LLMs) as successful when outputs match what they want, not what they type. Yet LLMs are trained to predict the next token solely from text input, not underlying intent. Because written language is an imperfect proxy for intent, and correlations between phrasing and desired outcomes can break down in training data, models that rely too heavily on surface cues may respond inconsistently to semantically equivalent prompts. This makes it essential to evaluate whether LLMs can reliably infer user intent—es

pecially in high-stakes settings where robustness and generalization are critica
l. We introduce a formal framework for assessing intent comprehension in LLMs: w
hether a model demonstrates robust understanding of user intent by producing con
sistent outputs across semantically equivalent prompts while differentiating bet
ween prompts with distinct intents. Our evaluation approach is based on a varian
ce decomposition of model responses into three components: variability due to us
er intent, user articulation, and model uncertainty. Models that understand what
 users want, and are not overly sensitive to textual cues, should attribute most
 output variance to intent differences, rather than articulation style. Applying
 this framework across diverse domains, we find that, within the five LLaMA and
Gemma models we evaluate, larger models typically assign a greater share of vari
ance to intent, indicating stronger comprehension of intent, although gains are
uneven and often modest with increasing model size. These results motivate movin
g beyond accuracy-only benchmarks toward semantic diagnostics that directly asse
ss whether models understand what users intend.
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Modeling Attributional Style at Scale: A Dataset and Analysis for Psychological
Attribution Assessment and Reframing

Qiang Zhou, Hanzhen Zhu, Pan Wang, Rui Tu, Huaizhi Qu, Zhuoran Wang, Xin Hu, Lei
 Li, Tianlong Chen, Jingtong Hu

According to the reformulated Learned Helplessness theory, repeated exposure to
uncontrollable negative events can foster a depressogenic attributional style—in
creasing susceptibility to depression yet remaining a tractable target for cogni
tive therapy. Computational research on attributional cognition, however, is ham
pered by the lack of large-scale datasets and robust evaluation protocols. In th
is work, we introduce the Attributional Style Transfer Dataset (ASTD) along with
 dedicated evaluation metrics, the first benchmark designed to model, assess, an
d reframe attributional explanations at scale. Constructed via a Prevent–Filter–
Validate pipeline that integrates LLM-based generation with specialist validatio
n, ASTD contains 42,000 real-world events paired with psychologically grounded a
ttributions spanning seven styles. Using this dataset, we address two key challe
nges: (1) scalable assessment of attributional style via both supervised classif
iers and zero/few-shot LLMs; and (2) attributional reframing and evaluation, whe
re we propose automatic evaluation metrics to quantify psychological validity. F
urthermore, we leverage our proposed metrics to construct a preference dataset,
fine-tuning LLMs with Direct Preference Optimization (DPO) and achieving substan
tial gains in reframing quality. Together, our dataset, metrics, and methodology
 offer a new paradigm for understanding and modeling attributional style, with d
irect implications for scalable and adaptive mental health interventions.
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Learning Generalizable Skill Policy with Data-Efficient Unsupervised RL

Jongchan Park, Seungjun Oh, Seungho Baek, Yusung Kim

Unsupervised Reinforcement Learning (URL) aims to pre-train scalable, skill-cond
itioned policies without extrinsic rewards, serving as a foundation for downstre
am control tasks. Despite recent progress, we argue that current off-policy URL
methods are limited by two critical, overlooked bottlenecks: (1) non-stationarit
y skill semantic and (2) brittle generalization. To address these challenges, we
 propose GenDa (Generalizable Data-efficient Agent), a unified framework for rob
ust unsupervised reinforcement learning. First, we introduce a skill relabeling
mechanism to mitigate non-stationarity and significantly improve sample efficien
cy for pretraining. Second, we propose a Complementary Information Bottleneck(CI
B), encouraging the learned skill policy to focus on ego-centric features and be
come robust to distribution shifts for downstream tasks. Through various experim
ents, we demonstrate that GenDa significantly enhances the scalability of URL wi
th superior generality and sample efficiency. Our source codes are available in
the supplementary materials.
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ThoughtFold: Folding Reasoning Chains via Introspective Preference Learning

Ziyan Liu, Xueda Shen, Yuzhe Gu, Songyang Gao, Kuikun Liu, Guangran Cheng, Cheng
qi Lyu, Dahua Lin, Wenwei Zhang, Kai Chen

Large Reasoning Models (LRMs) have achieved remarkable progress thanks to Reinfo rcement Learning with Verifiable Rewards (RLVR) on Chain-of-Thoughts (CoTs). However, since long CoTs naturally contain trial and errors and mainstream RLVR approaches choose outcome-correct CoT trajectories for memorization, the redunda nt explorations in long CoTs are inevitably reinforced, which results in the ove r-thinking issues of LRMs.
Previous attempts to resolve this issue mainly give more advantage to shorter tr ajectories, yet their learning signals are still outcome-based and cannot reduce the memorization of redundant explorations in long CoTs.
Therefore, we propose ThoughtFold, a framework that leverages fine-grained prefe rence learning to mitigate redundant explorations for efficient reasoning. Thoug htFold employs an introspective strategy to identify redundancy within each corr ect trajectory, which yields a spectrum of candidate sub-trajectories. Leveragin g this spectrum, we introduce a masked preference optimization objective that ex plicitly penalizes redundant explorations and encourages the model to directly b ridge essential reasoning segments, effectively folding its reasoning chains int o a more concise path. Extensive experiments show that ThoughtFold significantly enhances efficiency. It reduces the token usage of DeepSeek-R1-Distill-Qwen-7B by approximately 56\% while maintaining state-of-the-art accuracy.
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Controlled Collaboration Geometry for Personalized Federated Learning
Hongbo Yin, Jichun Wu, Zhou Yang, Chi Jiang, Yin Zhang, Yan Zhang
In personalized federated learning (PFL), collaboration graphs specify model agg regation among clients. However, without constraints on the collaboration geomet ry, training can drift into two degenerate regimes: global consensus or spontane ous clustering. This paper provides a unified dynamical analysis: under the same budget of representative models, collaborative PFL is more expressive and achie ves higher-order approximation accuracy than clustered PFL. An upper bound on di sagreement further reveals two degeneration mechanisms—overly strong collaborati on drives consensus (reducing to standard federated learning), while similarity-driven weight updates make the graph nearly reducible and induce self-clustering (collapsing to clustered PFL). Motivated by these findings, we propose pFedCCG. pFedCCG preserves the expressivity advantage via controlled collaboration geome try (CCG): it builds a static similarity-based collaboration template decoupled from training, optimizes a Markovian collaboration matrix with a prescribed stat ionary distribution via reversible parameterization and Euclidean projection, an d schedules collaboration strength to avoid self-clustering. Experiments across diverse heterogeneity settings show consistent personalization gains and markedl y reduced collapse and self-clustering. Code is available at https://github.com/YinHonb/pFedCCG.
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Collaborative Learning for Semi-Supervised LiDAR Semantic Segmentation
Bin Yang, Alexandru Paul Condurache
Annotating large-scale LiDAR point clouds for 3D semantic segmentation is costly and time-consuming, which motivates the use of semi-supervised learning (SemiSL). Standard LiDAR SemiSL methods typically adopt a two-step training paradigm, w here pseudo-labels are separately generated from a single distillation source, e ither from the same or another LiDAR representation. Such supervision relies on a unique source of pseudo-labels, which can reinforce confirmation bias and prop agate errors during training, ultimately limiting performance. To address this c hallenge, we introduce *CoLLiS*, a novel framework that leverages **Co**llaborat ive **L**earning for **Li**DAR **S**emi-supervised segmentation. Unlike prior pa radigms with decoupled pseudo-labeling and training phases, *CoLLiS* trains mult iple representations collaboratively in a single step by treating them as coequa l students. Each student is adaptively distilled from multiple representations, while inter-student disparities are monitored online to resolve contradictory su pervision and effectively mitigate confirmation bias. Extensive experiments on t hree datasets demonstrate that *CoLLiS* consistently outperforms state-of-the-ar t LiDAR SemiSL methods, with particularly strong gains in low-label regimes.
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RoboOmni: Actions Are Just Another Modality for Vision-Language Models
Dong Wang, Zilong Chen, Jirong Liu, Ziqing Qiao, Xin Xiao, Bingyi Kang, Hongtao Wu, Xiao Ma, Tao Kong, Huaping Liu
Integrating Vision-Language Models (VLMs) into robotics has facilitated the development of generalizable Vision-Language Action (VLA) policies.
However, unified discrete frameworks lag behind decoupled continuous designs due to limitations in action chunking and temporal modeling.
To address this, we introduce **RoboOmni**, a unified multi-modal next-token prediction framework.
Challenging the assumption that continuous modeling is essential for high-performance manipulation, **RoboOmni** demonstrates that *actions are just another modality* capable of being effectively modeled discretely.
At the core of our method is Multi-Token Action Prediction (MTAP), which integrates action chunking directly into the discrete tokenizer. This design resolves temporal modeling bottlenecks and significantly reduces distribution shift between training and inference. By preserving the native VLM training and inference pipeline, **RoboOmni** naturally benefits from large-scale multimodal co-training and modern decoding optimizations.
Extensive evaluations on the CALVIN, SimplerEnv, and real-world platforms confirm that **RoboOmni** establishes new state-of-the-art performance, significantly outperforming diffusion-based baselines such as π_0.
Notably, combining our proposed MTAP with the FAST tokenizer achieves a 94.4\% average success rate on CALVIN,
while the Bin tokenizer implementation attains a 27$\times$ inference speedup compared to OpenVLA.
****
Less Data, Faster Training: repeating smaller datasets speeds up learning via sampling biases
Jingwen Liu, Ezra Edelman, Surbhi Goel, Bingbin Liu
This work investigates the "small-vs-large gap", where repeating on _fewer samples_ can lead to _compute saving_ during training compared to using a larger dataset.
This is observed across algorithmic tasks, architectures and optimizers and cannot be explained using prior theory.
We argue that the speedup comes from appropriate layer-wise growth enabled by _sampling biases_, which is more pronounced when the dataset size is smaller.
We provide both theoretical analysis and empirical evidence from various interventions.
Our results suggest that using a smaller dataset with more repetitions is not just a fallback strategy under data scarcity, but can be proactively leveraged as a favorable inductive biases for optimization, particularly in reasoning tasks.
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Routing by Reaching: Composition of Pre-trained GFlowNets for Multi-Objective Generation
Seokwon Yoon, Youngbin Choi, Seunghyuk Cho, Seungbeom Lee, MoonJeong Park, Dongwoo Kim
Generative Flow Networks (GFlowNets) learn to sample diverse candidates in proportion to a reward function, making them well-suited for scientific discovery, where exploring multiple promising solutions is crucial.
Further extending GFlowNets to multi-objective settings has attracted growing interest as real-world applications often involve multiple, conflicting objectives.
However, existing approaches require joint training for each combination of objectives, meaning that any change in the objective set necessitates retraining from scratch.
We propose a framework that composes pre-trained GFlowNets at inference time, enabling rapid adaptation without fine-tuning or retraining.
Importantly, our framework is flexible, capable of handling diverse reward combinations ranging from linear scalarization to complex nonlinear operators, which are often handled separately in previous literature.

We prove that our method exactly recovers the target distribution for linear sca
larization, and quantify the approximation quality for nonlinear operators throu
gh a distortion factor.
Experiments on a synthetic 2D grid and real-world molecule generation tasks demo
nstrate that our approach achieves performance comparable to baselines.
The code is available at https://github.com/ml-postech/gflownet-composition.
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Local Covariate Selection for Average Causal Effect Estimation without Pretreatm
ent and Causal Sufficiency Assumptions
Zeyu Liu, Zheng Li, Feng Xie, Yan Zeng, Hao Zhang, Kun Zhang
Causal effect estimation is a fundamental task in many scientific fields. Select
ing appropriate covariates for adjustment is crucial for obtaining unbiased caus
al effects. However, most existing methods either rely on learning the global ca
usal structure, assume the absence of latent variables, or impose the pretreatme
nt assumption-restricts covariates to those unaffected by the treatment or outco
me. These assumptions are often unrealistic in real-world scenarios, and global
structure learning can be computationally intensive and inefficient.
To address these challenges, we first characterize the local existence boundary
of adjustment sets for causal effect estimation. Based on this characterization,
 we develop a novel local learning method for covariate selection in nonparametr
ic causal effect estimation. This method accommodates the presence of latent var
iables and eliminates the need for the pretreatment assumption.
We prove that the proposed method is both sound and complete under standard assu
mptions. Its effectiveness is validated through extensive experiments on both sy
nthetic and real-world datasets.
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Supervise Less, See More: Training-free Nuclear Instance Segmentation with Proto
type-Guided Prompting
Wen Zhang, Qin Ren, Wenjing Liu, Haibin Ling, Chenyu You
Accurate nuclear instance segmentation is a pivotal task in computational pathol
ogy, supporting data-driven clinical insights and facilitating downstream transl
ational applications. While large vision foundation models have shown promise fo
r zero-shot biomedical segmentation, most existing approaches still depend on de
nse supervision and computationally expensive fine-tuning. Consequently, trainin
g-free methods present a compelling research direction, yet remain largely unexp
lored.
In this work, we introduce SPROUT, a fully training- and annotation-free prompti
ng framework for nuclear instance segmentation. SPROUT leverages histology-infor
med priors to construct slide-specific reference prototypes that mitigate domain
 gaps. These prototypes progressively guide feature alignment through a partial
optimal transport scheme. The resulting foreground and background features are t
ransformed into positive and negative point prompts, enabling the Segment Anythi
ng Model (SAM) to produce precise nuclear delineations without any parameter upd
ates.
Extensive experiments across multiple histopathology benchmarks demonstrate that
 SPROUT achieves competitive performance without supervision or retraining, esta
blishing a novel paradigm for scalable, training-free nuclear instance segmentat
ion in pathology. Code is available at https://github.com/Y-Research-SBU/SPROUT.
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The Interplay Between Interpolation and Aggregation in Regression: Optimal Sampl
e Complexity
Mikael Møller Høgsgaard, Kasper Green Larsen, Liang-Yu Zou
This work investigates theoretically the interplay between interpolation and agg
regation in regression.
We establish that the
γ-graph dimension characterizes learnability for a broad class of natural
 aggregation procedures.
Furthermore, we prove that an extremely simple aggregation procedure, combining
three interpolating hypotheses via the median, is optimal among all these aggreg
ation procedures, and is strictly more

powerful than proper learning.
Finally, we show that some hypothesis classes are learnable only by aggregating infinitely many hypotheses or by using non-interpolating aggregation rules (which may predict outside the range of their inputs), and any finite interpolating aggregation fails to achieve even trivial performance.
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SCORE: A Unified Framework for Overshoot Refund in Online FDR Control
Qi Kuang, Bowen Gang, Yin Xia
We propose a unified framework to enhance the power of online multiple hypothesis testing procedures based on e-values. While e-value-based methods offer robust online False Discovery Rate (FDR) control under minimal assumptions, they often suffer from power loss by discarding evidence that exceeds the rejection threshold. We address this inefficiency via the \textbf{S}equential \textbf{C}ontrol with \textbf{O}vershoot \textbf{R}efund for \textbf{E}-values (SCORE) framework, which leverages the inequality $\mathbb{I}(y \ge 1) \le y - (y-1)_+$, valid for all $y\ge 0$, to reclaim this otherwise ``wasted'' evidence. This simple yet powerful insight yields a unified principle for improving a broad class of online testing algorithms. Building on this framework, we develop SCORE-enhanced versions of several state-of-the-art procedures, including SCORE-LOND, SCORE-LORD, and SCORE-SAFFRON, all of which strictly dominate their original counterparts while preserving valid finite-sample FDR control. Furthermore, under mild assumptions, SCORE permits retroactive updates of alpha■wealth by using the latest decision twice: first to determine its reward or loss, and then to refresh past wealth. Such a mechanism enables more aggressive testing strategies while maintaining valid FDR control, thereby further improving statistical power. The effectiveness of the proposed methods is validated through extensive simulation and real-data experiments.
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MARS: Modular Agent with Reflective Search for Automated AI Research
Jiefeng Chen, Bhavana Dalvi Mishra, Jaehyun Nam, Rui Meng, Tomas Pfister, Jinsung Yoon
A critical bottleneck in automating AI research is the execution of complex machine learning engineering (MLE) tasks. MLE differs from general software engineering due to computationally expensive evaluation (e.g., model training) and opaque performance attribution. Current LLM-based agents struggle here, often generating monolithic scripts that ignore execution costs and causal factors. We introduce **MARS** (**M**odular **A**gent with **R**eflective **S**earch), a framework optimized for autonomous AI research. MARS relies on three pillars: (1) Budget-Aware Planning via cost-constrained Monte Carlo Tree Search (MCTS) to explicitly balance performance with execution expense; (2) Modular Construction, employing a "Design-Decompose-Implement" pipeline to manage complex research repositories; and (3) Comparative Reflective Memory, which addresses credit assignment by analyzing solution differences to distill high-signal insights. MARS achieves state-of-the-art performance among open-source frameworks on MLE-Bench under comparable settings, maintaining competitiveness with the global leaderboard's top methods. Furthermore, the system exhibits qualitative "Aha!" moments, where 63% of all utilized lessons originate from cross-branch transfer, demonstrating that the agent effectively generalizes insights across search paths.
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Optimal Rates for Feasible Payoff Set Estimation in Games
Annalisa Barbara, Riccardo Poiani, Martino Bernasconi, Andrea Celli
We study a setting in which two players play a (possibly approximate) Nash equilibrium of a bimatrix game, while a learner observes only their actions and has no knowledge of the equilibrium or the underlying game. A natural question is whether the learner can rationalize the observed behavior by inferring the players' payoff functions. Rather than producing a single payoff estimate, inverse game theory aims to identify the entire set of payoffs consistent with observed behavior, enabling downstream use in, e.g., counterfactual analysis and mechanism design across applications like auctions, pricing, and security games. We focus on the problem of estimating the set of feasible payoffs with high probability and

up to precision ϵ on the Hausdorff metric. We provide the first minimax-optimal rates for both exact and approximate equilibrium play, in zero-sum as well as general-sum games. Our results provide learning-theoretic foundations for set-valued payoff inference in multi-agent environments.

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Online Social Welfare Function-based Resource Allocation

Kanad Shrikar Pardeshi, Samsara Foubert, Aarti Singh

In many real-world settings, a centralized decision-maker must repeatedly allocate finite resources to a population over multiple time steps. Individuals who receive a resource derive some stochastic utility; to characterize the population-level effects of an allocation, the expected individual utilities are then aggregated using a social welfare function (SWF). We formalize this setting and present a general confidence sequence framework for SWF-based online learning and inference, valid for any monotonic, concave, and Lipschitz-continuous SWF. Our key insight is that monotonicity alone suffices to lift confidence sequences from individual utilities to anytime-valid bounds on optimal welfare. Building on this foundation, we propose SWF-UCB, a SWF-agnostic online learning algorithm that achieves near-optimal $\tilde{\mathcal{O}}(n+\sqrt{nkT})$ regret (for k resources distributed among n individuals at each of T time steps). We instantiate our framework on three normatively distinct SWF families: Weighted Power Mean, Kolm, and Gini, providing bespoke oracle algorithms for each. Experiments confirm $\sqrt{T}$ scaling and reveal rich interactions between k and SWF parameters. This framework naturally supports inference applications such as sequential hypothesis testing, optimal stopping, and policy evaluation.

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GLAD: Bidirectional Structure-Attribute Alignment via Latent Graph Diffusion Models

Jiankai Zuo, Yu Zhang, Yang Zhang, Zihao Yao, Yaying Zhang

Learning on graphs with missing node attributes is a prevalent yet challenging problem in real-world scenarios, as graph neural networks (GNNs) typically rely on complete attribute information. Existing solutions often employ adversarial learning in a shared latent space to align graph structure and attributes. However, these methods frequently suffer from training instability and mode collapse, failing to fully capture the complex, multi-modal joint distribution of topology and features. To address these limitations, we present GLAD, a novel generative framework for robust node attribute completion. GLAD leverages the strong generative capabilities of diffusion models to learn the conditional distribution of attributes given the graph structure within a decoupled latent space. Unlike previous unidirectional approaches, GLAD introduces a robust bidirectional alignment mechanism. Specifically, we incorporate a structure reconstruction constraint during training and structure-aware classifier-free guidance during sampling, ensuring that generated attributes are not only plausible but also maintain strict topological consistency with the underlying graph. Theoretically, we show that GLAD maximizes a tighter variational lower bound on the joint log-likelihood compared to GAN-based predecessors, leading to superior mode coverage. Extensive experiments on large-scale benchmarks demonstrate that GLAD significantly outperforms state-of-the-art baselines in both attribute recovery quality and downstream task performance.

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UltraHorizon: Benchmarking LLM-Agent Capabilities in Ultra Long-Horizon Scenarios

Haotian Luo, Huaisong Zhang, Xuelin Zhang, Haoyu Wang, Zeyu Qin, WenJie Lu, Guozheng Ma, Haiying He, Yingsha Xie, Qiyang Zhou, Zixuan Hu, Hongze Mi, Yibo Wang, Naiqiang Tan, Hong Chen, Yi R. Fung, Chun Yuan, Li Shen

Autonomous agents have recently achieved remarkable progress across diverse domains, yet most evaluations focus on short-horizon, fully observable tasks. In contrast, many critical real-world tasks, such as large-scale software development, commercial investment, and scientific discovery, unfold in long-horizon and partially observable scenarios where success hinges on sustained reasoning, planning, memory management, and tool use. Existing benchmarks rarely capture these lon

g-horizon challenges, leaving a gap in systematic evaluation. To bridge this gap , we introduce $\textbf{UltraHorizon}$, a novel benchmark that measures the foun dational capabilities essential for complex real-world challenges. We use explor ation as a unifying task across three distinct environments to validate these co re competencies. Agents are designed in long-horizon discovery tasks where they must iteratively uncover hidden rules through sustained reasoning, planning, mem ory and tools management, and interaction with environments. Under the heaviest scale setting, trajectories average $\textbf{200k+}$ tokens and $\textbf{400+}$ tool calls, whereas in standard configurations they still exceed $\textbf{35k}$ tokens and involve more than $\textbf{60}$ tool calls on average. Our extensive experiments reveal that agents powered by state-of-the-art LLMs consistently und erperform in these settings, whereas human participants achieve much higher scor es, underscoring a persistent gap in agents' long-horizon exploration abilities. We also observe that simple scaling fails in our task. To better illustrate the failure of agents, we conduct an in-depth analysis of collected trajectories. W e identify eight types of errors and attribute them to two primary causes: in-co ntext locking and functional fundamental capability gaps.

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FedARC: Anchor-Guided Residual Compensation for Data and Model Heterogeneous Fed erated Learning

Chentao Lu, Xuhao Ren, Dawei xu, Chuan Zhang, Liehuang Zhu

Federated learning (FL) allows clients to collaboratively train models without e xposing private data, but practical FL is simultaneously challenged by data hete rogeneity and model heterogeneity. Prior heterogeneous FL (HtFL) approaches ofte n fail to handle fine-grained feature shifts, leading to weak representation ali gnment and limited cross-client knowledge transfer, which degrades both personal ization and generalization. We propose FedARC, an HtFL framework that couples a shared lightweight extractor with client-specific fusion: a trainable projector integrates local and global embeddings, while adaptive residual compensation dyn amically corrects feature-level mismatches. To further stabilize aggregation, Fe dARC performs semantic anchor alignment across clients, and we theoretically pro ve FedARC converges with a non-convex convergence rate $\mathcal{O}(1/T)$. Exper iments on five public benchmarks demonstrate that FedARC outperforms nine state- of-the-art HtFL baselines by up to 2.63\% in average accuracy, while maintaining efficient communication and computation.

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MedMamba: Multi-View State Space Models with Adaptive Graph Learning for Medical Time Series Classification

Da Zhang, Bingyu Li, Zhiyuan Zhao, Hongyuan Zhang, Junyu Gao, Xuelong Li

Medical time series are central to healthcare, enabling continuous monitoring an d supporting timely clinical decisions. Despite recent progress, existing method s struggle to jointly model local-global dynamics and handle nonstationarities l ike baseline drift, while often failing to capture latent channel interactions. To address these challenges, we propose **MedMamba**, an end-to-end architecture that integrates state space models with domain-specific inductive biases. Speci fically, MedMamba first employs multi-scale convolutional embeddings to capture discriminative local morphology. Second, to mitigate nonstationarity, we introdu ce a tri-branch differential state space encoder that processes raw, temporal-di fference, and frequency-domain views, fusing them to emphasize informative patte rns while suppressing drift. Furthermore, to uncover latent channel correlations , we design a spatial graph Mamba module that learns a directed dependency struc ture regularized toward sparsity and acyclicity, which obviates the need for pre defined graphs. Extensive experiments on five real-world datasets demonstrate th at MedMamba achieves state-of-the-art performance while maintaining linear compu tational complexity, and ablation studies validate each component's contribution .Code is available at https://github.com/zhangda1018/MedMamba.

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MetaPerch: Learning from metadata for bioacoustics foundation models

Mustafa Chasmai, Vincent Dumoulin, Jenny Hamer

Bioacoustic foundation models rely on large-scale citizen science platforms like

Xeno-Canto for geographically and ecologically diverse data. Recent work has shown that supervision alone can produce SotA species detection models when trained on this large-scale data---however, there remains unutilized potential in the form of recording metadata readily available within these community-driven data hubs. In this work, we explore the use of metadata---such as location and time---as auxiliary supervision signals, allowing the model to leverage species-metadata correlations in its learned representation. Auxiliary metadata losses provide additional information beyond vocalizations alone that can encourage a richer, more robust representation that generalizes better to species distribution and acoustic domain shifts---important challenges for deployment in real-world passive acoustic monitoring (PAM) settings. We introduce MetaPerch, a new foundation model that achieves strong species identification performance across multiple challenging domains and present an extensive empirical study of the effects of 9 diverse metadata sources on 17 bioacoustic datasets.

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Seeing Realism from Simulation: Efficient Video Transfer for Vision-Language-Action Data Augmentation

Chenyu Hui, Xiaodi Huang, Siyu Xu, Yunke Wang, Shan You, Fei Wang, Tao Huang, Chang Xu

Vision-language-action (VLA) models typically rely on large-scale real-world videos, whereas simulated data, despite being inexpensive and highly parallelizable to collect, often suffers from a substantial visual domain gap and limited environmental diversity, resulting in weak real-world generalization. We present an efficient video augmentation framework that converts simulated VLA videos into realistic training videos while preserving task semantics and action trajectories. Our pipeline extracts structured conditions from simulation via video semantic segmentation and video captioning, rewrites captions to diversify environments, and uses a conditional video transfer model to synthesize realistic videos. To make augmentation practical at scale, we introduce a diffusion feature-reuse mechanism that reuses video tokens across adjacent timesteps to accelerate generation, and a coreset sampling strategy that identifies a compact, non-redundant subset for augmentation under limited computation.Extensive experiments on Robotwin 2.0, LIBERO, LIBERO-Plus, and a real robotic platform demonstrate consistent improvements.For example, our method improves RDT-1B by 8\% on Robotwin 2.0, and boosts π_0 by 5.1\% on the more challenging LIBERO-Plus benchmark. Code is available at: https://github.com/nanfangxiansheng/Seeing-Realism-from-Simulation

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JAEGER: Joint 3D Audio-Visual Grounding and Reasoning in Simulated Physical Environments

Zhan Liu, Changli Tang, Yuxin WANG, Zhiyuan Zhu, Youjun Chen, Yiwen Shao, TIANZI WANG, Lei Ke, Zengrui Jin, Chao Zhang

Current audio-visual large language models (AV-LLMs) are predominantly restricted to 2D perception, relying on RGB video and monaural audio. This design choice introduces a fundamental dimensionality mismatch that precludes reliable source localization and spatial reasoning in complex 3D environments. We address this limitation by presenting JAEGER, a framework that extends AV-LLMs to 3D space, to enable joint spatial grounding and reasoning through the integration of RGB-D observations and multi-channel first-order ambisonics. A core contribution of our work is the neural intensity vector (Neural IV), a learned spatial audio representation that encodes robust directional cues to enhance direction-of-arrival estimation, even in adverse acoustic scenarios with overlapping sources. To facilitate large-scale training and systematic evaluation, we propose SpatialSceneQA, a benchmark of 61k instruction-tuning samples curated from simulated physical environments. Extensive experiments demonstrate that our approach consistently surpasses 2D-centric baselines across diverse spatial perception and reasoning tasks, underscoring the necessity of explicit 3D modelling for advancing AI in physical environments. Our source code, pre-trained model checkpoints, and datasets are available at https://github.com/liuzhan22/JAEGER.

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Alignment Tampering: How Reinforcement Learning from Human Feedback Is Exploited

to Optimize Misaligned Biases
Dongyoon Hahm, Dylan Hadfield-Menell, Kimin Lee
Reinforcement Learning from Human Feedback (RLHF) is the standard method to align Large Language Models (LLMs) with human preferences. In this work, we introduce alignment tampering, a potential vulnerability where the LLM undergoing alignment influences the preference dataset, causing RLHF to amplify undesired behaviors. This arises from core limitations of RLHF: (1) preference datasets are constructed from the LLM's own outputs, allowing it to influence them, and (2) pairwise comparisons only indicate which response is better, not why. These limitations can be exploited to cause alignment tampering. For example, if an LLM generates biased responses with higher quality, annotators will prefer them based on quality. However, preference labels do not distinguish quality from bias, and the reward model inherits this limitation. Optimizing such rewards through reinforcement learning or best-of-N sampling can amplify misaligned biases. Our experiments demonstrate amplification across diverse biases: from keyword bias to propaganda (e.g., sexism), brand promotion, and instrumental goal-seeking. Mitigation remains challenging, as existing techniques for robust RLHF fail to fully resolve alignment tampering without sacrificing response quality. These findings reveal structural vulnerabilities of current RLHF and emphasize the need to prevent this vulnerability. Project page: https://alignment-tampering.github.io/
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FrontierCS: Evolving Challenges for Evolving Intelligence
Qiuyang Mang, Wenhao Chai, Zhifei Li, Huanzhi Mao, Shang Zhou, Alexander Du, Hanchen Li, Shu Liu, Edwin Chen, Yichuan Wang, Xieting Chu, Zerui Cheng, Yuan Xu, Tian Xia, Zirui Wang, Tianneng Shi, Jianzhu Yao, Yilong Zhao, Qizheng Zhang, Charlie F. Ruan, Zeyu Shen, Kaiyuan Liu, Zhaoyang Hong, Alex Gu, Ziyi Zhang, Runyuan He, Dong Xing, Zerui Li, Zirong Zeng, Yige Jiang, Lufeng Cheng, Ziyi Zhao, Youran Sun, Suyang Zhong, Junpeng Wang, Donglin Li, Wenyuan Huang, Jialiang Gu, Wesley Kai Zheng, Wangmeiyu Zhang, Ruyi Ji, Xuechang Tu, Zihan Zheng, Zhaozi Wang, Zexing Chen, Jingbang Chen, JIALU ZHANG, Aleksandra Korolova, Peter Henderson, Pramod Viswanath, Vijay Ganesh, Saining Xie, Zhuang Liu, Dawn Song, Sewon Min, Ion Stoica, Joseph E. Gonzalez, Jingbo Shang, Alvin Cheung
We introduce FrontierCS, a benchmark of 240 open-ended problems across diverse areas of computer science, designed and reviewed by experts, including CS PhDs and top-tier competitive programming participants and problem setters.
Unlike existing benchmarks that focus on tasks with known optimal solutions, FrontierCS targets problems where the optimal solution is unknown, but the quality of a solution can be objectively evaluated. Models solve these tasks by implementing executable programs rather than outputting a direct answer.
FrontierCS includes algorithmic problems, which are often NP-hard variants of competitive programming problems with objective partial scoring, and research problems with the same property.
For each problem, we provide an expert reference solution and an automatic evaluator.
Combining open-ended design, measurable progress, and expert curation,
FrontierCS provides a benchmark at the frontier of computer-science difficulty.
Empirically, we find that frontier reasoning models still lag far behind human experts, and that simply increasing reasoning budgets does not close this gap on open-ended challenges.
Moreover, these models struggle to identify internal equivalence classes, and existing agentic frameworks also exhibit brittleness on such problems due to overfitting.
FrontierCS thus offers a new lens into model capabilities on real frontier computer science problems.
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Instance-Level Costs for Nuanced Classifier Evaluation
Kabir Kang, Stephen Mussmann
Standard classification treats all errors equally, but in applications such as content moderation and medical screening, mistakes on clear-cut cases are more costly than errors on ambiguous ones. From a contextual bandit framework, we propo

se normalized excess cost (NEC), a metric that weighs classification errors by p
er-example costs and reduces to standard error rate when costs are uniform. Cost
s can derive from annotator vote margins, distance from decision thresholds, or
confidence ratings. Across text, image, and tabular benchmarks, we find that NEC
 is often substantially lower than error rate—models with 5% error rate can achi
eve 1.8% NEC—revealing that most mistakes concentrate on ambiguous, low-cost exa
mples. We also find that incorporating costs into training via loss weighting, s
ampling strategies, or regression yields inconsistent benefits. Our framework pr
ovides a practical methodology for deriving and evaluating instance-level miscla
ssification costs, even if cost-sensitive training offers limited benefit.
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Federated Data and Feature Selection by Generalized CUR Decomposition
Ying-Peng Tang, Zhuang Qi, Xiaoli Tang, Wei Zhuo, Sheng-Jun Huang, Han Yu
With the advance of federated learning (FL) in privacy-sensitive domains, the ne
ed for efficient and robust training becomes increasingly urgent. Communication
bottlenecks, heterogeneous client distributions, and fairness requirements make
it essential to select the “right” data and features for model training. Yet exi
sting FL research often addresses feature selection and data selection separatel
y, ignoring their interplay in real-world high-dimensional and noisy datasets, l
eading to suboptimal performance. In this paper, we propose a unified framework
for data and feature selection by formulating the problem as a generalized CUR d
ecomposition problem. We introduce FedGCUR, a practical framework that integrate
s a federated column-pivoted QR (FedCPQR) decomposition routine with per-silo ro
w selection. We prove that FedCPQR produces exactly the same decomposition resul
ts as centralized CPQR and establish an upper bound of the reconstruction error
of FedGCUR. Experiments on tabular tasks and vision benchmarks show strong downs
tream accuracy and absolute CUR reconstruction quality compared with representat
ive data and feature selection baselines.
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Riemannian Diffusion Models on General Manifolds via Physics-Informed Neural Net
works
Gyeonghoon Ko, Juho Lee
Riemannian diffusion models generalize score-based generative modeling to manifo
ld-supported data via stochastic diffusion equations on the manifold.
However, training requires sampling from and differentiating the manifold heat k
ernel, which is rarely available in closed form beyond a few highly symmetric ma
nifolds.
We propose a general approach that approximates the heat kernel by directly solv
ing the manifold heat equation with a physics-informed neural network (PINN).
Given an explicit manifold specification, we choose a coordinate system, derive
the corresponding heat (Fokker--Planck) equation and a short-time asymptotic app
roximation, and then train a PINN to learn the log heat kernel.
The resulting surrogate enables both forward noising (heat-kernel sampling) and
conditional-score evaluation for denoising score matching.
We demonstrate the method on diverse manifolds including S^2, $SO(3)$, $\mathr
m{SPD}(n)$, and permutation-quotiented point clouds.
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LatentLens: Revealing Highly Interpretable Visual Tokens in LLMs
Benno Krojer, Shravan Nayak, Oscar Mañas, Vaibhav Adlakha, Desmond Elliott, Siva
 Reddy, Marius Mosbach
Transforming a large language model (LLM) into a vision-language model (VLM) can
 be achieved by mapping the visual tokens from a vision encoder into the embeddi
ng space of an LLM.
Intriguingly, this mapping can be as simple as a shallow MLP transformation.
To understand why LLMs can so readily process visual tokens, we need interpretab
ility methods that reveal what is encoded in the visual token representations at
 every layer of LLM processing.
In this work, we introduce LatentLens, a novel approach for mapping latent repre
sentations to descriptions in natural language.
LatentLens encodes a large text corpus and stores contextualized token represent

ations for each token in that corpus.
Visual token representations are then compared to these contextualized represent ations and the top-k nearest neighbor representations serve as descriptions of the visual token.
We evaluate this method on 15 different VLMs, showing that commonly used methods , such as LogitLens, substantially underestimate the interpretability of visual tokens.
With LatentLens instead, the majority of visual tokens are interpretable across all studied models and all layers.
Qualitatively, we show that the descriptions produced by LatentLens are semantic ally meaningful and provide more fine-grained interpretations for humans compare d to individual tokens.
More broadly, our findings contribute new evidence on the alignment between visi on and language representations and open up new directions for analyzing the lat ent representations of LLMs.
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Head-in-Head in Linear Attention
Shijie Mei, Man Yao, Jiabo Tong, Bo XU, Guoqi Li
The state-transition (decay) matrix governs how fixed-size memory is updated and used, making it a core design in linear attention models. Prior work exploits r ank-1 approximations to reduce the cost of constructing decay matrices, but this low-rank constraint also limits the expressive capacity. We therefore formulate decay-matrix design as an open optimization problem: maximizing expressiveness while introducing minimal additional cost. Inspired by the multi-head mechanism, we propose Head-in-Head, which introduces an additional mask matrix to structur e memory partitioning and interactions within a single linear-attention head. Th is simple, generic, and efficient design: 1) enables a rank-r approximation of the decay matrix with only a few extra parameters and 2) strengthens intra-head information interaction. We further develop mask normalization and a chunk-wise parallelization scheme to support efficient parallel training. Extensive experi ments on synthetic benchmarks and language modeling tasks, together with visual analyses, show that Head-in-Head consistently improves baseline performance by e nriching information diversity and strengthening intra-head interactions. Code a vailable at:
\url{https://github.com/msj-19/Head-in-Head-Linear-Attention}
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SVL: Goal-Conditioned Reinforcement Learning as Survival Learning
Franki NGUIMATSIA TIOFACK, Fabian Schramm, Théotime Le Hellard, Justin Carpentie r
Standard approaches to goal-conditioned reinforcement learning (GCRL) that rely on temporal-difference learning can be unstable and sample-inefficient due to bo otstrapping. While recent work has explored contrastive and supervised formulati ons to improve stability, we present a probabilistic alternative, called surviva l value learning (SVL), that reframes GCRL as a survival learning problem by mod eling the distribution of time-to-goal from each state. This structured distribu tional Monte Carlo perspective yields a closed-form identity that expresses the goal-conditioned value function as a discounted sum of survival probabilities, e nabling value estimation via a hazard model trained via maximum likelihood on bo th event and right-censored trajectories. We introduce three practical value est imators, including finite-horizon truncation and two binned infinite-horizon app roximations to capture long-horizon objectives. Experiments on offline GCRL benc hmarks show that SVL combined with hierarchical actors matches or surpasses stro ng hierarchical TD and Monte Carlo baselines, excelling on complex, long-horizon tasks.
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Joint-Space Empowerment as a Theory of Dexterous Motor Coordination
James Heald, Vittorio Caggiano, Vikash Kumar, Maneesh Sahani
Searching for effective policies is notoriously challenging in high-dimensional musculoskeletal systems, where multiple muscles actuate individual joints. Altho ugh this redundancy complicates naive policy search, it also implies that effect

ive control can be captured by a low-dimensional action manifold. To identify su ch a manifold, we introduce Joint-Space Empowerment (JoSE), a novel information- theoretic objective that quantifies how much control an agent has over its mecha nical degrees-of-freedom. We frame manifold discovery as an optimal precoding pr oblem—where a state-dependent precoder maps low-dimensional latent actions to hi gh-dimensional muscle commands—and derive the optimal precoder in closed form un der control-affine Gaussian dynamics. We show that manipulation policies learned on this manifold display significantly enhanced dexterity, sample efficiency, a nd improved generalization. These results present optimal precoding as a general information-theoretic paradigm for coordinating high-dimensional actuators to c ontrol low-dimensional features.

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Is Vibe Coding Safe? Benchmarking Vulnerability of Agent-Generated Code in Real-World Tasks

Songwen Zhao, Danqing Wang, Kexun Zhang, Jiaxuan Luo, Zhuo Li, Lei Li

Vibe coding is a new software development paradigm in which human engineers prom pt a large language model (LLM) agent to complete complex coding tasks with litt le supervision. Although vibe coding is increasingly adopted, is the generated c ode really safe to deploy in production? To investigate this question, we propos e SUSVIBES, a benchmark consisting of 186 feature-request software engineering t asks from real-world open-source projects, for which human programmers committed vulnerable implementations. We evaluate 12 widely used coding agentic settings with frontier models on the benchmark. Disturbingly, all agents perform poorly i n terms of software security. Although 57% of the solutions from SWE-Agent with Claude 4 Sonnet are functionally correct, only 11.8% are secure. Further experim ents demonstrate that preliminary security strategies, such as augmenting the fe ature request with vulnerability hints, cannot mitigate these security issues. O ur findings raise serious concerns about the widespread adoption of vibe coding, particularly in security-sensitive applications. The code and dataset are avail able at https://github.com/LeiLiLab/susvibes. The leaderboard is at https://leil ilab.github.io/ susvibes-leaderboard.

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Just Ask: Curious Code Agents Reveal System Prompts in Frontier LLMs

Xiang Zheng, Yutao Wu, Hanxun Huang, Yige Li, Xingjun Ma, Bo Li, Yu-Gang Jiang, Cong Wang

Autonomous code agents built on large language models are reshaping software and AI development through tool use, long-horizon reasoning, and self-directed inte raction. However, this autonomy introduces a previously unrecognized security ri sk: agentic interaction fundamentally expands the LLM attack surface, enabling s ystematic probing and recovery of hidden system prompts that guide model behavio r. We identify system prompt extraction as an emergent vulnerability intrinsic t o code agents and present JustAsk, a self-evolving framework that autonomously d iscovers effective extraction strategies through interaction alone. Unlike prior prompt-engineering or dataset-based attacks, JustAsk requires no handcrafted pr ompts, labeled supervision, or privileged access beyond standard user interactio n. It formulates extraction as an online exploration problem, using Upper Confid ence Bound-based strategy selection and a hierarchical skill space spanning atom ic probes and high-level orchestration. These skills exploit imperfect system-in struction generalization and inherent tensions between helpfulness and safety. E valuated on 41 black-box commercial models across multiple providers, JustAsk co nsistently achieves full or near-complete system prompt recovery, revealing recu rring design- and architecture-level vulnerabilities. Our results expose system prompts as a critical yet largely unprotected attack surface in modern agent sys tems.

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Federated Sketching LoRA: A Flexible Framework for Heterogeneous Collaborative F ine-Tuning of LLMs

Wenzhi Fang, Dong-Jun Han, Liangqi Yuan, Seyyedali Hosseinalipour, Christopher B rinton

Fine-tuning large language models (LLMs) on resource-constrained clients remains

a challenging problem. Recent works have fused low-rank adaptation (LoRA) techniques with federated fine-tuning to mitigate challenges associated with client model sizes and data scarcity. Still, the heterogeneity of resources remains a critical bottleneck: while higher-rank modules generally enhance performance, varying client capabilities constrain LoRA's feasible rank range. Existing approaches attempting to resolve this issue either lack analytical justification or impose additional computational overhead, leaving a wide gap for efficient and theoretically-grounded solutions. To address these challenges, we propose federated sketching LoRA (FSLoRA), which leverages a sketching mechanism to enable clients to selectively update submatrices of global LoRA modules maintained by the server. By adjusting the sketching ratios, which determine the ranks of the submatrices on the clients, FSLoRA flexibly adapts to client-specific communication and computational constraints. We provide a rigorous convergence analysis of FSLoRA that characterizes how the sketching ratios affect the convergence rate. Through extensive experiments, we demonstrate that FSLoRA outperforms baselines and significantly improves training efficiency while preserving stable convergence.

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Exact Functional ANOVA Decomposition for Categorical Inputs Models

Baptiste Ferrere, Nicolas Bousquet, Fabrice Gamboa, Jean-Michel Loubes, Joseph Muré

Functional ANOVA offers a principled framework for interpretability by decomposing a model’s prediction into main effects and higher-order interactions. For independent features, this decomposition is well-defined, strongly linked with SHAP values, and serves as a cornerstone of additive explainability. However, the lack of an explicit closed-form expression for general dependent distributions has forced practitioners to rely on costly sampling-based approximations. We completely resolve this limitation for categorical inputs. By bridging functional analysis with the extension of discrete Fourier analysis, we derive a closed-form decomposition without any assumption. Our formulation is computationally very efficient. It seamlessly recovers the classical independent case and extends to arbitrary dependence structures, including distributions with non-rectangular support. Furthermore, leveraging the intrinsic link between SHAP and ANOVA under independence, our framework yields a natural generalization of SHAP values for the general categorical setting.

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Embodied Task Planning via Graph-Informed Action Generation with Large Language Models

Xiang Li, Ning Yan, Masood S. Mortazavi

While Large Language Models (LLMs) have demonstrated strong zero-shot reasoning capabilities, their deployment as embodied agents still faces fundamental challenges in long-horizon planning. Unlike open-ended text generation, embodied agents must decompose high-level intents into actionable sub-goals while adhering to the constraints of a dynamic environment. Standard LLM planners frequently fail to maintain strategy coherence over extended horizons due to context window limitations or hallucinate state transitions that violate environment constraints. We propose GiG, a planning framework that structures embodied agents’ memory using a Graph-in-Graph architecture. Our approach employs a Graph Neural Network (GNN) to encode environmental states into embeddings, organizing these embeddings into action-connected execution trace graphs within an experience memory bank. GiG enables retrieval of structurally-similar priors, allowing agents to ground current decisions in relevant past structural patterns. Furthermore, we introduce a bounded lookahead module that leverages symbolic transition logic to enhance the agent’s planning capabilities through grounded action projections. We evaluate our framework on three embodied planning benchmarks—Robotouille Synchronous, Robotouille Asynchronous, and ALFWorld. Our method outperforms state-of-the-art baselines, achieving Pass@1 performance gains of up to 22% on Robotouille Synchronous, 37% on Asynchronous, and 15% on ALFWorld while maintaining comparable or lower computational cost.

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Distribution-Calibrated Inference Time Compute for Thinking LLM-as-a-Judge

Hamid Dadkhahi, Firas Trabelsi, Parker Riley, Juraj Juraska, Mehdi Mirzazadeh
Thinking Large Language Models (LLMs) used as judges for pairwise preferences remain noisy at the single-sample level, and common aggregation rules (majority vote, soft self-consistency, or instruction-based self-aggregation) are inconsistent when ties are allowed. We study inference-time compute (ITC) for evaluators that generate n independent thinking--rating samples per item, and propose a principled, distribution-calibrated aggregation scheme. Our method models three-way preferences with a Bradley–Terry-Davidson formulation on rating counts, leveraging both polarity (margin among non-ties) and decisiveness (non-tie rate) to distinguish narrow margins from strong consensus. Across various evaluation benchmarks, our approach consistently reduces MAE and increases pairwise accuracy versus standard baselines, and when evaluated against human-consensus meta-labels, matches or exceeds individual human raters. These results show that carefully allocating ITC and aggregating with distribution-aware methods turns noisy individual model judgments into reliable ratings for evaluation.
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A Single Layer to Explain Them All: Understanding Massive Values in Large Language Models
Zeru Shi, Zhenting Wang, Fan Yang, Qifan Wang, Ruixiang Tang
We investigate the origins of massive activations in large language models (LLMs) and identify a specific layer named the Massive Emergence Layer (ME Layer), that is consistently observed across model families, where massive activations first emerge and subsequently propagate to deeper layers through residual connections. We show that, within the ME Layer both the RMSNorm and the FFN parameters jointly contribute to the emergence of massive activations. Once formed, the massive activation token representation remains largely invariant across layers, reducing the diversity of hidden representations passed to the attention module. Motivated by this limitation, we propose a simple and effective method to reduce the rigidity of the massive activation token. Our approach consistently improves LLM performance across multiple tasks, including instruction following and math reasoning, in both training free and fine tuning settings. Moreover, we show that our method mitigates attention sinks by selectively weakening their influence, elucidating their origin at the hidden state level and shedding new light on principled mitigation strategies.
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Self-Distillation Enables Continual Learning
Idan Shenfeld, Mehul Damani, Jonas Hübotter, Pulkit Agrawal
Continual learning, enabling models to acquire new skills and knowledge without degrading existing capabilities, remains a fundamental challenge for foundation models. While on-policy reinforcement learning can reduce forgetting, it requires explicit reward functions that are often unavailable. Learning from expert demonstrations, the primary alternative, is dominated by supervised fine-tuning (SFT), which is inherently off-policy. We introduce Self-Distillation Fine-Tuning (SDFT), a simple method that enables on-policy learning directly from demonstrations. SDFT leverages in-context learning by using a demonstration-conditioned model as its own teacher, generating on-policy training signals that preserve prior capabilities while acquiring new skills. Across skill learning and knowledge acquisition tasks, SDFT consistently outperforms SFT, achieving higher new-task accuracy while substantially reducing catastrophic forgetting. In sequential learning experiments, SDFT enables a single model to accumulate multiple skills over time without performance regression, establishing on-policy distillation as a practical path to continual learning from demonstrations.
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Diving into Kronecker Adapters: Component Design Matters
Jiayu Bai, Danchen Yu, Zhenyu Liao, TianQi Hou, Feng Zhou, Robert C Qiu, Zenan Ling
Kronecker adapters have emerged as a promising approach for fine-tuning large-scale models, enabling high-rank updates through tunable component structures. However, existing work largely treats the component structure as a fixed or heuristic design choice, leaving the dimensions and number of Kronecker components unde

rexplored. In this paper, we identify component structure as a key factor governing the capacity of Kronecker adapters. We perform a fine-grained analysis of both the dimensions and number of Kronecker components.
In particular, we show that the alignment between Kronecker adapters and full fine-tuning depends on component configurations. Guided by these insights, we propose Component Designed Kronecker Adapters (CDKA). We further provide parameter-budget–aware configuration guidelines and a tailored training stabilization strategy for practical deployment. Experiments across various architectures and modalities demonstrate the effectiveness of CDKA. Code is available at \url{https://github.com/rainstonee/CDKA}.
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LLM4Cov: Execution-Aware Agentic Learning for High-coverage Testbench Generation
Hejia Zhang, Zhongming Yu, Chia-Tung Ho, Haoxing Ren, Brucek Khailany, Jishen Zhao
Execution-aware LLM agents offer a promising paradigm for learning from tool feedback, but such feedback can be expensive and slow to obtain, making online reinforcement learning (RL) less practical in certain scenarios.
High-coverage hardware verification exemplifies this challenge due to its reliance on industrial simulators and non-differentiable execution signals.
We propose LLM4Cov, an offline agent-learning framework that models verification as single-step state transitions guided by deterministic evaluators.
Building on this formulation, we introduce execution-validated data curation, policy-aware agentic data synthesis, and worst-state-prioritized sampling to enable scalable learning under execution constraints.
We further curate a reality-aligned benchmark adapted from an existing verification suite through a revised evaluation protocol.
Using the proposed pipeline, a compact 4B-parameter model achieves 69.2\% pass rate and 90.4\% average coverage in CVDP-ECov under agentic evaluation, outperforming its teacher by 5.3\% and 10.5\%, demonstrating competitive performance against models an order of magnitude larger.
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Bayesian Tensor Decomposition with Diffusion Model Prior
Zerui Tao, Qibin Zhao
Low-rank tensor decomposition (TD) is usually effective on clean, fully observed data, but it often degrades under severe missingness or noise. Low-rankness is itself a useful but limited structural prior, and additional handcrafted priors (e.g., sparsity or smoothness) still fall short of capturing the rich statistics of real-world data. To compensate for this weak inductive bias under heavy corruption, one would like to inject a learned, data-driven prior; however, the state-of-the-art diffusion models are not readily compatible with current TD and tractable posterior inference. To address these challenges, we introduce DiffBCP, a hybrid-prior Bayesian CP decomposition framework that couples a cumulative shrinkage process prior over the CP factors for automatic rank selection with an off-the-shelf pre-trained diffusion model as an implicit data prior on the reconstructed tensor. To make posterior inference tractable despite the coupling among the likelihood, low-rank constraint, and diffusion prior, we develop a split Gibbs sampler: CP factors admit conjugate updates, while the diffusion block is sampled via low-rank-guided denoising. A noise-adaptive coupling schedule further reduces sensitivity to hand-tuned annealing. Experiments on image inpainting and denoising, including high-resolution out-of-distribution images, show consistent gains over Bayesian, nonlinear, and plug-and-play TD baselines.
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Operator Splitting with Hamilton-Jacobi-based Proximals
Nicholas Di, Eric Chi, Samy Wu Fung
Operator splitting algorithms are a cornerstone of modern first-order optimization, decomposing complex problems into simpler subproblems solved via proximal operators. However, most functions lack closed-form proximal operators, which has long restricted these methods to a narrow set of problems. Hamilton-Jacobi-based proximal operator (HJ-Prox) is a recent derivative-free Monte Carlo technique based on Hamilton-Jacobi PDE theory, that approximates proximal operators numeric

ally. In this work, we introduce a unified framework for operator splitting via HJ-Prox, which allows for deployment of operator splitting even when functions a re not proximable. We prove that replacing exact proximal steps with HJ-Prox in algorithms such as proximal point, proximal gradient descent, Douglas–Rachford s plitting, Davis–Yin splitting, and primal–dual hybrid gradient preserves converg ence guarantees under mild assumptions. Numerical experiments demonstrate HJ-Pro x is competitive and effective on a wide variety of statistical learning tasks.
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MIND: Multi-rationale INtegrated Discriminative Reasoning Framework for Multi-mo dal Large Models

Chuang Yu, Jinmiao Zhao, Mingxuan Zhao, Yunpeng Liu, Xiujun Shu, Yuanhao Feng, B o Wang, Xiangyu Yue

Recently, multimodal large language models (MLLMs) have been widely applied to r easoning tasks. However, they suffer from limited multi-rationale semantic model ing, insufficient logical robustness, and susceptibility to misleading cues. The refore, we propose a Multi-rationale INtegrated Discriminative (MIND) reasoning framework, which is designed to endow MLLMs with human-like cognitive abilities of “Understand → Rethink → Correct”, and achieves a paradigm evolution from pass ive imitation-based reasoning to active discriminative reasoning. Specifically, we introduce a Rationale Augmentation and Discrimination (RAD) paradigm, which p rovides a unified and extensible data foundation. Meanwhile, we design a Progres sive Two-stage Correction Learning (P2CL) strategy. The first phase enhances mul ti-rationale positive learning, while the second phase enables active logic disc rimination and correction. In addition, to mitigate representation entanglement in the multi-rationale semantic space, we propose a Multi-rationale Contrastive Alignment (MCA) optimization strategy. Extensive experiments show that our MIND achieves SOTA performance on multiple public datasets. Our data and code are ava ilable at https://github.com/YuChuang1205/MIND.
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On the Adversarial Robustness of Large Vision-Language Models under Visual Token Compression

Xinwei Zhang, Hangcheng Liu, Li Bai, Hao Wang, Qingqing Ye, Tianwei Zhang, Haibo Hu

Visual token compression is widely used to accelerate large vision-language mode ls (LVLMs) by pruning or merging visual tokens, yet its adversarial robustness r emains unexplored. We show that existing encoder-based attacks cannot fully disc lose the robustness vulnerabilities of compressed LVLMs, due to an optimization-inference mismatch: perturbations are optimized on the full-token representation , while inference is performed through a token-compression bottleneck. To addres s this gap, we propose the Compression-AliGnEd attack (CAGE), which aligns pertu rbation optimization with compression inference without assuming access to the d eployed compression mechanism or its token budget. CAGE combines (i) expected fe ature disruption, which concentrates distortion on tokens likely to survive acro ss plausible budgets, and (ii) rank distortion alignment, which actively aligns token distortions with rank scores to promote the retention of highly distorted evidence. Across diverse representative plug-and-play compression mechanisms and datasets, our results show that CAGEconsistently achieves lower robust accuracy than the baseline. This work highlights that robustness assessments ignoring co mpression can be overly optimistic, calling for compression-aware security evalu ation and defenses for efficient LVLMs.
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Beyond Static Allocation: Dynamic Sensitivity-Aware Fine-Tuning for Vision Trans formers

Yuanyang Cao, Xichun Liu, Fuwei Zhang, Shangqi Deng, Ziyang Ren, Jianji Wang

Existing Parameter-Efficient Fine-Tuning (PEFT) methods are fundamentally constr ained by a static allocation paradigm, which overlooks the model's evolving opti mization priorities during training. To address this, we introduce Dynamic Adapt ive Fine-tuning (DAF), a novel framework that periodically evaluates and reconfi gures the trainable structure based on a context-aware decoupled sensitivity ana lysis. DAF employs a Rebuild-and-Refocus strategy to preserve learned knowledge

by freezing outdated modules while decisively reallocating the parameter budget to newly identified critical regions. Extensive experiments on challenging visio n benchmarks demonstrate that DAF significantly outperforms mainstream static PE FT methods and achieves superior performance and efficiency, particularly under extreme parameter budgets. Our work fundamentally challenges the static nature o f the field, offering a more intelligent and efficient paradigm for adapting lar ge pretrained models. The code is available at https://github.com/E-green11/DAF.
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Depth-Progressive Monotonic Learning without Global Backpropagation
Chenhao Ye, Rongguang Ye, Yuchao Zhang, Ming Tang
Backpropagation (BP) remains the dominant training paradigm for deep neural netw orks, yet its reliance on global gradient propagation fundamentally induces upda te locking problem, enforcing strong inter-layer dependencies in parameter updat es. To address this limitation, we propose Depth-progressive Monotonic Learning (DMoL), a training scheme that assigns layer-wise local belief objectives and in crementally refines them across network depth, enabling unlocked parameter updat es. As a result, DMoL supports dynamic modification of network depth during trai ning, adapting to available compute and device resources while maintaining stabl e optimization. We provide theoretical guarantees that layer-wise local belief o bjectives improve monotonically with increasing depth and converge exponentially . Empirically, DMoL consistently matches or outperforms BP across diverse tasks, yielding a 4.3\% accuracy gain on CIFAR-100, mitigating over-smoothing in deep graph neural networks (+37.5\% on Cora), and reducing the final loss by over 35\ % in diffusion model training, highlighting its robustness and flexibility as an alternative to BP. The code is publicly available at: https://github.com/ychAlb ert/DMoL.
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Quantifying Biases in LLM-as-a-Judge Evaluations
Magda Dubois, Harry Coppock, Mario Giulianelli, Ole Kristian Jorgensen, Timo Fle sch, Lennart Luettgau, Cozmin Ududec
The evaluation of large language models (LLMs) is increasingly performed by othe r LLMs, a setup commonly known as "LLM-as-a-judge", or autograders. While autogr aders offer a scalable alternative to human evaluation, they are not free from b iases (e.g., favouring longer outputs or generations from their own model family). Here we propose a statistical framework based on Bayesian generalised linear models (GLMs) that enables researchers to address their primary research questio ns (e.g., LLM capability or risk assessment), while simultaneously identifying, quantifying and mitigating various biases in their autograders. Our approach can be applied to various evaluation formats (e.g., absolute scores or pairwise pre ferences) and augments traditional metrics (e.g., inter-rater agreement) by prov iding precise uncertainty estimates and clarifying sources of disagreement betwe en graders. This framework also enables efficient counterfactual simulations wit hout costly re-evaluation (e.g., assessing agreement after removing systematic b iases). We demonstrate these capabilities through simulated examples, with all m ethods available in an open-source software package. Overall, we introduce a nov el framework for autograder evaluation which allows researchers to detect, quant ify and correct for various biases in a systematic way.
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DV-World: Benchmarking Data Visualization Agents in Real-World Scenarios
Jinxiang Meng, Shaoping Huang, Fangyu Lei, Jingyu Guo, Haoxiang Liu, Jiahao Su, Sihan Wang, Yao Wang, Enrui Wang, Ye Yang, Hongze Chai, Jinming Lyu, Anbang Yu, Huangjing Zhang, Yitong Zhang, Yiming Huang, Zeyao Ma, Shizhu He, Jun Zhao, Kang Liu
Real-world data visualization (DV) requires native environmental grounding, cros s-platform evolution, and proactive intent alignment, yet existing benchmarks ar e often limited to code sandboxes, creation-only tasks, and fully specified inte nts.
We introduce DV-World, a 260-task benchmark for evaluating DV agents across prof essional visualization lifecycles.
DV-World covers three domains: DV-Sheet for native spreadsheet chart/dashboard c

reation and diagnostic repair; DV-Evolution for adapting reference visual artifacts to new data across programming paradigms; and DV-Interact for proactive intent alignment with a user simulator under ambiguous requirements.
Its hybrid evaluation combines \textit{Table-value Alignment} for numerical precision with rubric-based \textit{MLLM-as-a-Judge} for semantic-visual assessment.

Experiments show that state-of-the-art models achieve below 50\% overall performance, exposing major gaps in real-world DV capabilities.
DV-World provides a realistic testbed for developing DV agents suited to enterprise workflows.
Data and code are available at \url{dv-world-project.github.io}.
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Follow-the-Perturbed-Leader for Decoupled Bandits: Best-of-Both-Worlds and Practicality
Chaiwon Kim, Jongyeong Lee, Min-hwan Oh
We study the decoupled multi-armed bandit problem, where the learner separately selects one arm for exploration and one, possibly different, arm for exploitation at each round.
In this setting, the loss of the explored arm is observed but not incurred, whereas the loss of the exploited arm is incurred without being observed.
We propose an efficient Follow-the-Perturbed-Leader (FTPL) policy that achieves Best-of-Both-Worlds (BOBW) guarantee with constant regret in the stochastic regime and optimal $\mathcal{O}(\sqrt{KT})$ regret in the adversarial regime.
A key feature of our method is that it completely avoids both the convex optimization required by prior BOBW policies and the resampling procedures typically used in FTPL bandit policies.
This allows FTPL to fully realize its computational efficiency advantages, leading to substantial reductions in computational cost.
We empirically confirm that our policy not only improves the runtime but also demonstrates superior regret performance in both regimes.
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The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models
Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, Jack Lindsey
Large language models can represent a variety of personas but typically default to a helpful Assistant identity cultivated during post-training. Across several different models, we find an “Assistant Axis" in their activation space, which captures the extent to which a model is operating in its default Assistant mode. Steering towards the Assistant direction reinforces helpful and harmless behavior; steering away increases the model’s tendency to identify as other entities. Measuring deviations along the Assistant Axis predicts “persona drift,” a phenomenon where models slip into exhibiting harmful or bizarre behaviors that are uncharacteristic of their typical persona. We find that persona drift is often driven by conversations demanding meta-reflection on the model’s processes or featuring emotionally vulnerable users. We show that restricting activations to a fixed region along the Assistant Axis can stabilize model behavior in these scenarios—and also in the face of adversarial persona-based jailbreaks. Our results suggest that post-training steers models toward a particular region of persona space but only loosely tethers them to it, motivating work on training and steering strategies that more deeply anchor models to a coherent persona.
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GADA: Geometry-Aware Deformable Aggregation for Image-Based Gaussian Splatting
Siwoo Lim, Sunjae Yoon, Gwanhyeong Koo, Chang D. Yoo
Gaussian Splatting has achieved significant improvements by incorporating warping-based techniques. However, such methods suffer from pixel-level inaccuracies due to uncertain geometry. This uncertainty leads to spatial misalignments in the warped images, which disrupt residual learning used in warping-based methods and fundamentally limit the gains of correction, particularly on thin structures and high-frequency details. Driven by our insight that useful visual cues are not lost but locally preserved under slight displacement, we propose Geometry-Aware

Deformable Aggregation (GADA). This method introduces an iterative refinement module with deformable offsets to actively correct spatial misalignments and recover these displaced cues. Furthermore, to address the limitations of standard pipelines where visibility checks (i.e., thresholding) often discard valid pixels and multi-view warped image fusion relies on naive mean aggregation, our module is coupled with an implicit confidence weighting mechanism that selectively suppresses unreliable evidence. Consequently, our approach outperforms prior warping-based Gaussian Splatting, preserving high-frequency quality while achieving 2.13 times faster FPS.

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Detecting Fluent Optimization-Based Adversarial Prompts via Sequential Entropy Changes

Mohammed Alshaalan, Miguel R. D. Rodrigues

Optimization-based adversarial suffixes can jailbreak aligned large language models (LLMs) while remaining fluent, weakening static and windowed perplexity-based detectors. We cast adversarial suffix detection as an *online change-point detection* problem over the token-level next-token entropy stream. Using the LLM system prompt to estimate a robust baseline, we standardize user-token entropies and apply a one-sided CUSUM statistic. The resulting detector, *CPD Online* (CPD), is model-agnostic, training-free, runs online, and localizes the adversarial suffix onset. On a benchmark of 1,012 optimization-based suffix attacks (GCG, AutoDAN, AdvPrompter, BEAST, AutoDAN-HGA) and 1,012 perplexity-controlled benign prompts, CPD improves F1 over the strongest windowed-perplexity baseline on all six open-weight chat models (LLaMA-2-7B/13B, Vicuna-7B/13B, Qwen2.5-7B/14B). On LLaMA-2-7B at the canonical CUSUM setting ($k=0$), CPD reaches AUROC 0.88 and F1 0.82. Beyond prompt-level detection, CPD concentrates 79.6% of its triggers inside the adversarial suffix, versus 17–46% for windowed perplexity. Finally, when used as a lightweight gate for LLaMA Guard, CPD reduces guard calls by 17–22% on a high-volume, benign-dominated deployment while preserving guard-level detection quality.

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TestExplora: Benchmarking LLMs for Proactive Bug Discovery via Repository-Level Test Generation

Steven Liu, Jane Luo, Xin Zhang, Aofan Liu, Hao Liu, Jie Wu, Ziyang Huang, Yangyu Huang, Yu Kang, Scarlett Li

Given that Large Language Models (LLMs) are increasingly applied to automate software development, comprehensive software assurance spans three distinct goals: regression prevention, reactive reproduction, and proactive discovery. Current evaluations systematically overlook the third goal. Specifically, they either constrain models to a compliance trap by treating existing code as the ground truth for regression prevention, or rely on post-failure artifacts (e.g., issue reports) for reactive bug reproduction, failing to expose defects before they manifest as failures. To bridge this gap, we present TestExplora, a benchmark designed to evaluate LLMs as proactive testers within full-scale, realistic repository environments. Comprising 2,389 tasks across 482 repositories, TestExplora conceals all defect-related information, forcing models to uncover bugs by identifying discrepancies between implementation and documentation-derived intent—utilizing documentation as the reference oracle. Furthermore, to ensure sustainable evaluation and mitigate risks of data leakage in static datasets, we propose a continuous, time-aware data collection framework. Our evaluation reveals a significant capability gap: state-of-the-art models achieve a maximum Fail-to-Pass ($F2P$) rate of only 16.06%. Further analysis indicates that navigating complex cross-module interactions and leveraging agentic exploration are critical to advancing LLMs toward autonomous software quality assurance. Consistent with this, SWEAgent instantiated with GPT-5-mini achieves an $F2P$ of 17.27% and an $F2P@5$ of 29.7%, highlighting the effectiveness and promise of agentic exploration in proactive bug discovery tasks.

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Active Exploring like a Pigeon: Reinforcing Spatial Reasoning via Agentic Vision-Language Models

Wei Deng, Xianlin Zhang, Mengshi Qi
Enabling Vision-Language Models (VLMs) to perform spatial reasoning remains chal
lenging.
Existing approaches treat VLMs as passive observers, which is difficult for real
-world applications.
Moreover, reinforcement learning methods rely on sparse rewards, limiting their
effectiveness for complex reasoning tasks.
Inspired by pigeons' building and exploiting cognitive maps for navigation, we p
ropose a novel agentic pipeline for spatial reasoning.
First, we introduce a new \emph{dynamic cognitive map} parameterizing scene layo
ut as object positions and orientations, serving as persistent memory for new ob
servations.
Second, we propose a novel \emph{Spatial Assertion Codes (SAC)}, Python expressi
ons programmatically describing spatial relationships.
By collaborating with the dynamic cognitive map, SAC enables verification of int
ermediate reasoning steps, providing dense reward signals.
We optimize the model via supervised and reinforcement finetuning.
Experiments on the MindCube benchmark demonstrate state-of-the-art performance w
ith \emph{80.5\%} overall accuracy, outperforming the best current method by \em
ph{29.5} accuracy points (a relative improvement of \emph{53.2\%}) on the challe
nging \textsc{Rotation} subset. Our code and data are open-sourced at \url{https
://github.com/dw-dengwei/active-spatial-reasoning.git}.
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Practical Mechanism for Fault-Tolerant Spiking Neural Networks via Simple Input
Control Based on Learnable Fragmentation
Hyun-Jong Lee, Jae-Han Lim
Spiking Neural Networks (SNNs) are regarded as the third generation of neural ne
tworks, offering energy-efficient computing for neuromorphic devices. Despite th
is benefit, hardware-implemented SNNs are vulnerable to hardware faults, which s
everely degrade their performance. Previous approaches have required direct acce
ss to internal SNN circuits to modify weights or monitor internal states, limiti
ng their practicality. Improving robustness to hardware faults without such acce
ss remains challenging. To overcome this challenge, we propose a fault-tolerant
mechanism that operates only through input data control. Hardware faults reduce
the usable learning capacity of SNNs, resulting in a mismatch between the instan
taneous input load and the degraded network dynamics. Our mechanism mitigates th
is mismatch by dividing each input sample into multiple fragments, redistributin
g the input load via a learnable fragmentation strategy. The strategy learns tw
o key fragmentation components: 1) division boundaries and 2) the number of frag
ments. To our knowledge, this is the first mechanism to improve the fault tolera
nce of SNNs without accessing the internal circuits. Experimental results demons
trate that our mechanism consistently outperforms previous methods in various SN
N models, achieving these gains without direct access to internal circuits. Furt
hermore, we validate its effectiveness on SNNs implemented with a physical FPGA
platform, confirming its practicality.
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Show, Don't Tell: Morphing Latent Reasoning into Image Generation
Harold Haodong Chen, Xinxiang Yin, Wen-Jie Shu, Hongfei Zhang, Zixin Zhang, Chen
fei Liao, Litao Guo, Qifeng Chen, Ying-Cong Chen
Text-to-image (T2I) generation has achieved remarkable progress, yet existing me
thods often lack the ability to dynamically reason and refine during generation-
-a hallmark of human creativity. Current reasoning-augmented paradigms mostly re
ly on explicit thought processes, where intermediate reasoning is decoded into d
iscrete text at fixed steps with frequent image decoding and re-encoding, leadin
g to inefficiencies, information loss, and cognitive mismatches. To bridge this
gap, we introduce **LatentMorph**, a novel framework that seamlessly integrates
implicit latent reasoning into the T2I generation process. At its core, LatentMo
rph introduces four lightweight components: (***i***) a **condenser** for summar
izing intermediate generation states into compact visual memory, (***ii***) a **
translator** for converting latent thoughts into actionable guidance, (***iii***

) a **shaper** for dynamically steering next image token predictions, and (***iv***) an RL-trained **invoker** for adaptively determining when to invoke reasoning. By performing reasoning entirely in continuous latent spaces, LatentMorph avoids the bottlenecks of explicit reasoning and enables more adaptive self-refinement. Extensive experiments demonstrate that LatentMorph **(I)** enhances the base model Janus-Pro by 16% on GenEval and 25% on T2I-CompBench; **(II)** outperforms explicit paradigms (*e.g.*, TwiG) by 15% and 11% on abstract reasoning tasks like WISE and IPV-Txt, **(III)** while reducing inference time by 44% and token consumption by 51%; and **(IV)** exhibits 71% cognitive alignment with human intuition on reasoning invocation.

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Tree-Structured Orthonormal Decomposition of the Aitchison Simplex

Daisuke Yamada, Qijun Zhang, Travis Pence, Barbara B. Bendlin, Federico Rey, Vikas Singh

Compositional data---vectors encoding relative proportions---arise across scientific domains, including ecology, geochemistry, and genomics. The features in these data often come with known hierarchical structure (e.g., taxonomies, phylogenies, ontologies), yet existing methods either ignore this structure, discard the intrinsic Aitchison geometry, are designed for binary trees, or yield incomplete coordinate systems. We describe *PolyILR*, a canonical orthonormal decomposition of the Aitchison tangent space aligned with any tree topology. Our construction defines a weighted local geometry at each internal node capturing full branching structure, then lifts these to a global orthonormal basis where every coordinate corresponds to a specific tree location. On microbiome and single-cell benchmarks, PolyILR yields stable, interpretable features and enables inference at multiscale tree resolution. We also establish a novel theoretical connection to softmax classifiers, suggesting possible applications to probabilistic modeling.

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GeoLoom: High-quality Geometric Diagram Generation from Textual Input

Xiaojing Wei, Ting Zhang, Wei He, Jingdong Wang, Hua Huang

High-quality geometric diagram generation presents both a challenge and an opportunity: it demands strict spatial accuracy while offering well-defined constraints to guide generation. Inspired by recent advances in geometry problem solving that employ formal languages and symbolic solvers for enhanced correctness and interpretability, we propose GeoLoom, a novel framework for text-to-diagram generation in geometric domains. GeoLoom comprises two core components: an autoformalization module that translates natural language into a specifically designed generation-oriented formal language GeoLingua, and a coordinate solver that maps formal constraints to precise coordinates using the efficient Monte Carlo optimization. To support this framework, we introduce GeoNF, a dataset aligning natural language geometric descriptions with formal GeoLingua descriptions. We further propose a constraint-based evaluation metric that quantifies structural deviation, offering mathematically grounded supervision for iterative refinement. Empirical results demonstrate that GeoLoom significantly outperforms state-of-the-art baselines in structural fidelity, providing a principled foundation for interpretable and scalable diagram generation.

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MiniAppBench: Evaluating the Shift from Text to Interactive HTML Responses in LLM-Powered Assistants

Zuhao zhang, Chengyue Yu, Yuante Li, Chenyi Zhuang, Linjian Mo, Shuai Li

With the rapid advancement of Large Language Models (LLMs) in code generation, human-AI interaction is evolving from static text responses to dynamic, interactive HTML-based applications, which we term **MiniApps**. These applications require models to not only render visual interfaces but also construct customized interaction logic that adheres to real-world principles. However, existing benchmarks primarily focus on algorithmic correctness or static layout reconstruction, failing to capture the capabilities required for this new paradigm. To address this gap, we introduce **MiniAppBench**, the first comprehensive benchmark designed to evaluate principle-driven, interactive application generation. Sourced from a real-world application with **10M+** generations, MiniAppBench distills 500 t

asks across six domains (e.g., Games, Science, and Tools). Furthermore, to tackl
e the challenge of evaluating open-ended interactions where no single ground tru
th exists, we propose **MiniAppEval**, an agentic evaluation framework. Leveragi
ng browser automation, it performs human-like exploratory testing to systematica
lly assess applications across three dimensions: Intention, Static, and Dynamic.
 Our experiments reveal that current LLMs still face significant challenges in g
enerating high-quality MiniApps, while MiniAppEval demonstrates high alignment w
ith human judgment, establishing a reliable standard for future research. Our ho
mepage is available in https://miniappbench.github.io.
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LLM Priors for ERM over Programs
Shivam Singhal, Priyadarsi Mishra, Eran Malach, Tomer Galanti
We study program-learning methods that are efficient in both samples and computa
tion. Classical learning theory suggests that when the target admits a short pro
gram description, for example a short piece of ``Python code'', it can be learne
d from few examples by ERM over the program class. However, this approach relies
 on enumerating candidate programs, which is typically exponential in the descri
ption length; gradient-based training avoids this explicit search but, for some
families of short programs, can require exponentially many samples to succeed. W
e propose \textsc{LLM-PV}, a propose-and-verify recipe that enables ERM-style se
lection over a discrete program class without exhaustive enumeration: a pretrain
ed LLM induces a proposal distribution over candidate programs, each proposal is
 executed and scored on a held-out validation set, and the best program is selec
ted, with no gradient updates or validation feedback used to adapt the sampling
distribution. Across algorithmic tasks including parity variants, pattern matchi
ng, and primality testing, \textsc{LLM-PV} often recovers the exact underlying r
ule from a small labeled set and generalizes far beyond the training sequence le
ngths, while SGD-trained transformers, fine-tuning, in-context learning, and cla
ssical ML baselines can fit the training data yet fail to generalize reliably. T
ogether, these results suggest that pretrained LLM priors can serve as effective
 search biases for ERM, narrowing the gap between statistical and computational
efficiency.
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HVAE: Hyperbolic Variational Autoencoder For Flexible Knowledge Transfer Across
Multiple Domains
Xiaolei Liu, Binfeng Wang, Kaixin Gao, Shaoshuai Li
Cross-domain recommendation (CDR) serves as a pivotal solution to data sparsity
and cold-start problems by transferring knowledge across distinct domains. Howev
er, existing approaches predominately rely on Euclidean embedding spaces, which
suffer from a fundamental geometry-distribution mismatch: real-world user-item i
nteractions typically exhibit power-law distributions and latent hierarchical st
ructures that flat Euclidean spaces cannot accurately represent without signific
ant distortion. This geometric limitation not only compromises representation qu
ality but, more critically, hinders the effective disentanglement of domain-inva
riant user preferences from domain-specific interests, limiting transferability
in low-overlap scenarios. To bridge this gap, we introduce the Mixed-Curvature H
yperbolic Variational Auto-Encoder (HVAE), a principled framework that unifies k
nowledge extraction and transfer within a hyperbolic manifold. By leveraging the
 exponential expansion capacity of hyperbolic geometry, HVAE naturally accommoda
tes hierarchical data structures, enabling precise disentanglement of user inten
ts without the need for strict domain overlap constraints. Furthermore, we propo
se a rigorous hyperbolic Wasserstein barycenter mechanism to align invariant dis
tributions across heterogeneous domains. Extensive experiments on large-scale in
dustrial and public datasets demonstrate that HVAE achieves superior performance
, particularly in challenging scenarios with long-tail distributions and minimal
 domain overlap.
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Single-Rollout Hidden-State Dynamics for Training-Free RLVR Data Selection
Jianghao Wu, Jianfei Cai, Weiqiang Wang, Jin Ye, Daniel F. Schmidt, Yasmeen Geor
ge

Reinforcement learning with verifiable rewards (RLVR) can yield large reasoning gains from very few training instances, yet its strong sensitivity to which instances are used makes data selection a central bottleneck. Most existing selection pipelines rely on training-time optimization signals and/or require access to verifiable rewards or ground-truth answers over large candidate pools, which is costly and often infeasible in specialized domains. We study RLVR data selection in a setting where selection must be performed before any RL training and without labels or reward evaluation on the full pool. We propose SHIFT, a one-shot, training-free selector based solely on inference-time hidden-state dynamics. For each candidate instance, SHIFT runs a single deterministic reasoning rollout and computes a reasoning-induced representation shift (RIRS) as the start-to-end hidden-state delta. SHIFT uses the RIRS magnitude as a lightweight proxy for instance utility and enforces coverage via a quality-weighted farthest-first CoreSet procedure in an RIRS-augmented feature space, producing compact subsets that scale to large unlabeled pools. Across mathematical reasoning and medical QA benchmarks under ultra-low budgets, SHIFT consistently outperforms training-free diversity and difficulty/uncertainty baselines, improving both in-domain accuracy and transfer to harder evaluation settings. Ablations show that RIRS-based coverage and quality-weighting contribute complementary gains, and analyses indicate that RIRS is not explained by simple input/output length statistics. Code is available at github.com/JianghaoWu/SHIFT.

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FIDIA: Function-Informed Sequence Design via Inference-Aligned Policy Optimization

Minghan Li, Fengji Li, Yilin Tao, Yue Deng

Computational protein design typically employs a sequential workflow of structure generation followed by sequence (re)design. While structure generators can be explicitly conditioned on functional objectives, inverse folding models are constrained by their function-agnostic nature and sequence-structure degeneracy. More critically, the associated training objectives do not account for the *Best-of-N* (BoN) inference protocol, resulting in a fundamental training-inference misalignment. Here, we propose FIDIA, a reinforcement learning framework that enables **F**unction-**I**nformed sequence **D**esign via **I**nference-**A**ligned policy optimization. Specifically, FIDIA integrates functional constraints into composite rewards and explicitly optimize the induced policy under BoN toward high-fitness sequence regions. We achieve this via a grounded gradient estimator that directly maximizes the expected maximum reward. FIDIA consistently outperforms both standard and RL-optimized baselines in success rate and precision on a general motif scaffolding benchmark. Further experiments on realworld cases including vaccine and affinity-enhancing enzyme design validate FIDIA’s efficacy in complex therapeutic and biocatalytic contexts.

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StreamFlow: Theory, Algorithm, and Implementation for High-Efficiency Rectified Flow Generation

Sen Fang, Hongbin Zhong, Yalin Feng, Yanxin Zhang, Dimitris N. Metaxas

New technologies such as Rectified Flow and Flow Matching have significantly improved the performance of generative models in the past two years, especially in terms of control accuracy, generation quality, and generation efficiency. However, due to some differences in its theory, design, and existing diffusion models, the existing acceleration methods cannot be directly applied to the Rectified Flow model. In this article, we have comprehensively implemented an overall acceleration pipeline from the aspects of theory, design, and reasoning strategies. This pipeline uses new methods such as batch processing with a new velocity field, vectorization of heterogeneous time-step batch processing, and dynamic TensorRT compilation for the new methods to comprehensively accelerate related models based on flow models. Currently, the existing public methods usually achieve an acceleration of 18%, while experiments have proved that our new method can accelerate the 512$\times$512 image generation speed to up to 611%, which is far beyond the current non-generalized acceleration methods. Project page at https://world-snapshot.github.io/StreamFlow/

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Credible Information Subset Decomposition: An End-to-End Multi-fidelity Learning Model by Modeling Label Information
Sihan Wang, Wenjie Du, Yang Wang
In the AI4Chemistry scenario, utilizing heterogeneous data at different fidelity levels is a common and core issue. High-fidelity data is accurate but scarce, while low-fidelity data is abundant but biased. Traditional multi-fidelity methods typically identify cross-fidelity biases based on paired samples under different fidelity labels. However, due to the mismatch in dataset input distribution and the complexity of the biases themselves, these methods are difficult to implement in real-world scientific environments. To address this, we propose a trusted information subset decomposition framework that can efficiently utilize multi-fidelity data without requiring paired samples. Multi-fidelity label supervision is decomposed into three complementary subsets: a trusted information subset based on the absolute value of high-fidelity labels; a trusted subset that captures the reliability of the high-fidelity and medium-fidelity label intervals through adaptive constraints; and an ordered trusted subset representing the numerical relationships within the same fidelity level. These subsets are then integrated into a unified end-to-end model, enabling the reasonable utilization of medium- and low-fidelity information. Extensive experiments on various molecular and material property benchmarks demonstrate that our method consistently outperforms state-of-the-art multifidelity and singlefidelity baseline methods, and exhibits good robustness under real-world unpaired multifidelity conditions.
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Causal Representation Learning with Optimal Compression and Complex Treatments
Wanting Liang, Haoang Chi, Zhiheng Zhang
Estimating Individual Treatment Effects (ITE) in multi-treatment scenarios faces two critical challenges: the Hyperparameter Selection Dilemma for balancing weights and the Curse of Dimensionality in computational scalability. This paper derives a novel multi-treatment generalization bound and proposes a theoretical estimator for the optimal balancing weight α, eliminating expensive heuristic tuning. We investigate three balancing strategies: Pairwise, One-vs-All (OVA), and Treatment Aggregation. While OVA achieves superior precision in low-dimensional settings, our proposed Treatment Aggregation ensures both accuracy and $\mathcal{O}(1)$ scalability as the treatment space expands. Furthermore, we extend our framework to a generative architecture, Multi-Treatment CausalEGM, which preserves the Wasserstein geodesic structure of the treatment manifold. Experiments on semi-synthetic and image datasets demonstrate that our approach significantly outperforms traditional models in estimation accuracy and efficiency, particularly in large-scale intervention scenarios.
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HiST: A Hierarchical Sparse Transformer for Cross-Modal Spatial Transcriptomics Modeling
Weiyi Wu, Xinwen Xu, Xingjian Diao, Siting Li, Zhi Wei, Alma Andersson, Jiang Gui
Spatial transcriptomics (ST) links gene expression with tissue morphology but remains expensive and low-throughput, motivating surrogates that infer expression from routine histology.
Whole-slide H&E-to-ST inference pairs a gigapixel image with gene measurements at a sparse, irregular set of locations, making multiscale modeling challenging without incurring dense-grid overhead or quadratic token mixing.
We propose HiST, a hierarchical sparse transformer that treats measured locations as a lattice-indexed sparse field and builds a dyadic encoder--decoder directly on the active tissue footprint.
HiST combines sparse window attention for local geometric correspondence with resolution-changing operators for rapid multiscale context integration.
For a fixed window size, the dominant runtime and memory scale with the number of observed locations rather than the dense slide area.
To mitigate slide-specific acquisition variation, HiST adds a bottlenecked globa

l conditioning pathway via a \emph{slide calibration token} that summarizes slide-level context and conditions local representations.
On a multi-organ benchmark spanning diverse tissues and acquisition sources, HiST improves predictive performance over recent baselines while reducing runtime and peak memory.
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DyCon: Dynamic Reasoning Control via Evolving Difficulty Modeling
Tengyao Tu, Yulin Li, Hui-Ling Zhen, Libo Qin, Zhoujun Wei, Jinghua Piao, Zhuotao Tian, Yong Li, Min Zhang
Recent advances in Large Reasoning Models (LRMs) demonstrate remarkable performance improvements by iteratively reflecting, exploring, and executing complex tasks, yet suffer from inefficiencies due to redundant reasoning, known as ``overthinking''.

Existing methods to mitigate this issue either rely on static difficulty estimates or require task-specific training, and thus fail to adapt to the dynamic complexity during reasoning.

In this work, we empirically show that the problem difficulty evolves dynamically throughout the reasoning process and is linearly encoded in the LRM’s step-level embeddings.

Building on this insight, we propose DyCon, a training-free framework that leverages latent step-level representations to explicitly model the evolving task difficulty, enabling the dynamic control of reasoning depth to mitigate the overthinking issue. Extensive experiments conducted on four models ranging from 4B to 32B, and across twelve benchmarks in math reasoning, general question answering, and coding tasks demonstrate that DyCon significantly enhances reasoning efficiency by reducing redundant steps without sacrificing accuracy or generalization. Project page and code are available at https://github.com/yu-lin-li/DyCon.
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Stochastic Sparse Attention for Memory-Bound Inference
Kyle Lee, Corentin Delacour, Kevin Callahan-Coray, Kyle Jiang, Can Yaras, Samet Oymak, Tathagata Srimani, Kerem Yunus Camsari
Autoregressive decoding becomes bandwidth-limited at long contexts, as generating each token requires reading all n_k key and value vectors from KV cache. We present Stochastic Additive No-mulT Attention (SANTA), a method that sparsifies value-cache access by sampling $S \ll n_k$ indices from the post-softmax distribution and aggregates only those value rows. This yields an unbiased estimator of the post-softmax value aggregation while replacing value-stage multiply-accumulates with gather-and-add. We introduce stratified and systematic sampling to design variance-reduced, GPU-friendly variants. Evaluated on Llama-3.1-8B-Instruct at 32k-token contexts, S^{2}ANTA matches baseline accuracy while achieving up to $1.5\times$ decode-step attention-kernel speedup over FlashInfer and FlashDecoding on an NVIDIA RTX 6000 Ada. In batched long-context generation, these kernel gains translate to up to $1.25\times$ end-to-end decode-latency speedup. Finally, we propose Bernoulli qK^T sampling as a complementary technique to sparsify the score stage, reducing key-feature access through stochastic ternary queries. Both methods are complementary to upstream quantization, low-rank projection, KV-cache compression, and KV-cache selection methods. Together, they point toward sparse, multiplier-free, and energy-efficient inference. We open-source our kernels at: https://github.com/OPUSLab/SANTA.git
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Divisiveness-Consistent Label Distribution Learning
Yunan Lu, Haitao Wu, Weiwei Li, Lei Yang, Xiuyi Jia
Label Distribution Learning (LDL) is an effective learning paradigm for predicting entire conditional label distributions, improving the trustworthiness of predictions in risk-sensitive tasks. Although previous LDL methods achieve satisfactory performance on conventional evaluation metrics, they generally overlook the divisiveness within label distributions, i.e., the propensity of label distribution to exhibit dissension between semantically opposing labels, which is an essential indicator of the practical decision risk. Therefore, we propose a divisiveness■consistent label distribution learning framework to quantify and preserve t

he divisiveness information. First, we formalize a divisiveness measure that sat isfies the axiomatic property of polarity monotonicity to quantify the divisiven ess information. Second, we theoretically demonstrate the inconsistency between conventional loss functions and divisiveness error. Besides, in order to address the adversarial gradient problem arising from directly minimizing the divisiven ess error, we propose a pairwise divisiveness loss as an unbiased estimator of t he original divisiveness error. Experiments confirm the effectiveness of the pro posed method.
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Large Vision-Language Models Get Lost in Attention
Gongli Xi, Ye Tian, Mengyu Yang, Huahui Yi, Liang Lin, Xiaoshuai Hao, Kun Wang, Wendong Wang
Despite the rapid evolution of training paradigms, the decoder backbone of large vision--language models (LVLMs) remains fundamentally rooted in the residual-co nnection Transformer architecture. Therefore, deciphering the distinct roles of internal modules is critical for understanding model mechanics and guiding archi tectural optimization. While prior statistical approaches have provided valuable attribution-based insights, they often lack a unified theoretical basis. To bri dge this gap, we propose a unified framework grounded in *information theory and geometry* to quantify the **geometric and entropic nature** of residual updates . Applying this unified framework reveals a fundamental functional decoupling: * *Attention acts as a subspace-preserving operator** focused on reconfiguration, whereas **FFNs serve as subspace-expanding operators** driving semantic innovati on. Strikingly, further experiments demonstrate that replacing learned attention weights with predefined values (e.g., Gaussian noise) yields comparable or even superior performance across a majority of datasets relative to vanilla models. These results expose severe **misallocation and redundancy** in current mechanis ms, suggesting that state-of-the-art LVLMs effectively ``get lost in attention'' rather than efficiently leveraging visual context.
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SafeSci: Safety Evaluation of Large Language Models in Science Domains and Beyon d
Xiangyang Zhu, Yuan Tian, Qi Jia, Kaiwei Zhang, Zicheng Zhang, Chunyi Li, Zijian Chen, Lu Sun, Kaiyuan Ji, Renrui Zhang, Wei Sun, Guangtao Zhai
The success of large language models (LLMs) in scientific domains has heightened safety concerns, prompting numerous benchmarks to evaluate their scientific saf ety. Existing benchmarks often suffer from limited risk coverage and a reliance on subjective evaluation. To address these problems, we introduce SafeSci, a com prehensive framework for safety evaluation and enhancement in scientific context s. SafeSci comprises SafeSciBench, a multi-disciplinary benchmark with 0.25M sam ples, and SafeSciTrain, a large-scale dataset containing 1.5M samples for safety enhancement. SafeSciBench distinguishes between safety knowledge and risk to co ver extensive scopes and employs objective metrics such as deterministically ans werable questions to mitigate evaluation bias. We evaluate 24 advanced LLMs and reveal critical vulnerabilities. We also observe that LLMs exhibit varying degre es of excessive refusal behaviors on safety-related issues. For safety enhanceme nt, we demonstrate that fine-tuning on SafeSciTrain significantly enhances the s afety alignment of LLMs. Finally, we argue that determining the safety of a scie ntific question should depend on the specific context, rather than universally c ategorizing it as safe or unsafe.
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Memory-Efficient LLM Pretraining via Minimalist Optimizer Design
Athanasios Glentis, Jiaxiang Li, Andi Han, Mingyi Hong
Training large language models (LLMs) relies on adaptive optimizers such as Adam , which introduce extra operations and require significantly more memory to main tain first- and second-order moments than SGD. While recent works such as GaLore , Fira and APOLLO have proposed state-compressed memory-efficient variants, a f undamental question remains: *What are the minimum modifications to plain SGD ne eded to match state-of-the-art pretraining performance?* We systematically inves tigate this question using a bottom-up approach, and identify two simple yet hig

hly (memory- and compute-) efficient techniques: (1) column-wise gradient normal ization (normalizing the gradient along the output dimension), that boosts SGD p erformance without momentum; and (2) applying first-order momentum only to the o utput layer, where gradient variance is highest. Combining these two techniques lead to SCALE (Stochastic Column-normAlized Last-layer momEntum), a simple opti mizer for memory efficient pretraining. Across multiple models (60M–1B), SCALE m atches or exceeds the performance of Adam while using only 35–45% of the total m emory. It also consistently outperforms memory-efficient optimizers such as GaLo re, Fira and APOLLO, making it a strong candidate for large-scale pretraining un der memory constraints. For LLaMA 7B, SCALE outperforms the state-of-the-art me mory-efficient methods APOLLO and Muon in both perplexity and memory consumption . Code is available at https://github.com/OptimAI-Lab/Minimalist_LLM_Pretraining .

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Hierarchical Image Tokenization for Multi-Scale Image Super Resolution

Isma Hadji, Enrique Sanchez, Adrian Bulat, Brais Martinez, Georgios Tzimiropoulo s

We introduce a multi-scale Image Super Resolution (ISR) method building on recen t advances in Visual Auto-Regressive (VAR) modeling. VAR models break image toke nization into additive, gradually increasing scales, using Residual Quantization (RQ), an approach that aligns perfectly with our target ISR task. Previous work s taking advantage of this synergy suffer from two main shortcomings. First, due to the limitations in RQ, they only generate images at a predefined fixed scale , failing to map intermediate outputs to the corresponding image scales. They al so rely on large backbones or a large corpus of annotated data to achieve better performance. To address both shortcomings, we introduce two novel components to the VAR training for ISR, aiming at increasing its flexibility and reducing its complexity. In particular, we introduce a) a **Hierarchical Image Tokenization (HIT)** approach that progressively represents images at different scales while enforcing token overlap across scales, and b) a **Direct Preference Optimization (DPO) regularization term** that, relying solely on the (LR,HR) pair, encourage s the transformer to produce the latter over the former. Our proposed HIT acts a s a strong inductive bias for the VAR training, resulting in a small model (300M params vs 1B params of VARSR), that achieves state-of-the-art results without e xternal training data, and that delivers multi-scale outputs with a single forwa rd pass.

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TeamWork: Multivariate Time Series Anomaly Detection via Asymmetric Role-aware C hannel Modeling

Shiyan Hu, Tengxue Zhang, Jianxin Jin, Xiangfei Qiu, Bin Yang, Chenjuan Guo

Multivariate time series anomaly detection remains challenging as it requires th e joint modeling of variable relationships and temporal dependencies. Existing m ethods often struggle to balance channel relationship modeling and overlook the relative importance of different variables within multivariate time series. To a ddress this, we propose TeamWork, an asymmetric role-aware channel modeling fram ework that decouples variables into dominant and auxiliary roles according to th eir contributions to uncertainty reduction. Dominant variables drive system evol ution and their deviations more strongly disrupt normal patterns, while auxiliar y variables provide complementary cues. These variables with different roles are integrated through a role-aware gated interaction module. Moreover, point and s ubsequence anomalies can exist in multiple periodic systems, and the same anomal y type may behave differently across short- and long-period series. To capture s uch variations, we introduce a period-aware masked modeling mechanism. It employ s multiple specialized masking mechanisms spanning short to long periods to faci litate comprehensive temporal dependency learning. Extensive experiments on mult iple real-world datasets demonstrate that TeamWork achieves superior performance compared with state-of-the-art methods.

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Var-JEPA: A Variational Formulation of the Joint-Embedding Predictive Architectu re – Bridging Predictive and Generative Self-Supervised Learning

Moritz Gögl, Christopher Yau
The Joint-Embedding Predictive Architecture (JEPA) is often seen as a non-genera tive alternative to likelihood-based self-supervised learning, emphasizing predi ction in representation space rather than reconstruction in observation space. W e argue that the resulting separation from probabilistic generative modeling is largely rhetorical rather than structural: the canonical JEPA design–coupled enc oders with a context-to-target predictor–mirrors the variational posteriors and learned conditional priors obtained when variational inference is applied to a p articular class of coupled latent-variable models, and standard JEPA can be view ed as a deterministic specialization in which regularization is imposed via arch itectural and training heuristics rather than an explicit likelihood. Building o n this view, we derive the *Variational JEPA* (Var-JEPA), which makes the latent generative structure explicit by optimizing a single Evidence Lower Bound (ELBO). This yields meaningful representations without ad-hoc anti-collapse regulariz ers and allows principled uncertainty quantification in the latent space. We ins tantiate the framework for tabular data (Var-T-JEPA) and achieve strong represen tation learning and downstream performance, improving over T-JEPA across real-wo rld tabular benchmarks while remaining competitive with strong raw-feature basel ines.
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HypCL: Adapting CLIP in Hyperbolic Space for Continual Learning
Quan Cheng, Hao Yu, Da-Wei Zhou, Lijun Zhang
Recently, vision-language models (e.g., CLIP) have been increasingly adopted for continual learning to mitigate catastrophic forgetting. However, existing CLIP- based methods typically freeze the backbone to preserve pre-trained knowledge, w hich limits the model's ability to learn discriminative features for downstream tasks. In this paper, we introduce HypCL, a parameter-efficient framework that c ontinually adapts CLIP in hyperbolic space for continual learning. Our key insig ht is that the exponentially expanding capacity of hyperbolic geometry naturally accommodates the growing class space and promotes stronger inter-class separati on. Specifically, HypCL attaches task-specific adapters and composes their updat es sequentially in the Poincaré ball. To exploit the enhanced feature separabili ty of hyperbolic geometry, HypCL maintains visual prototypes computed from the a dapted features, which serve as stable anchors for calibrating predictions at in ference. Extensive experiments on standard class-incremental benchmarks demonstr ate that HypCL consistently outperforms existing CLIP-based continual learning m ethods.
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Attn-QAT: 4-Bit Attention With Quantization-Aware Training
Peiyuan Zhang, Matthew Noto, Wenxuan Tan, Chengquan Jiang, Will Lin, Wei Zhou, H ao Zhang
Achieving reliable 4-bit attention is a prerequisite for end-to-end FP4 computat ion on emerging FP4-capable GPUs, yet attention remains the main obstacle due t o FP4's tiny dynamic range and attention's heavy-tailed activations. This paper presents the first systematic study of 4-bit quantization-aware training (QAT) f or attention. We find ``drop-in'' QAT -- naively combining an FP4 forward pass w ith high-precision Flash Attention (FA)-style backward pass -- leads to training instability.
We identify two key principles for stable FP4 attention: (1) matching low-precis ion recomputation of attention scores in the backward pass and (2) resolving imp licit precision assumptions in FA’s gradient calculation. Based on these insight s, we propose Attn-QAT and implement fused Triton kernels for training plus FP4 inference kernels. Across diffusion and language models, Attn-QAT recovers the quality drop from FP4 attention without explicit outlier-mitigation heuristics used in prior FP4 attention, and delivers up to a 1.5x speedup on an RTX 5090.
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PASO: Step Parallel Stochastic Optimization
Jianrong Lu, Zhuoya Gu, Haobo Li, Zhiyu Zhu, Yechao Zhang, Jianhai Chen, Minghui Yang, Junwei Liu, Jian Wang, Qinming He, Hui LIU, Junhui Hou
This paper approaches the fundamental challenge of accelerating the inherently a

utoregressive nature of gradient descent (GD) like SGD and Adam through a dynami
c system perspective.
Specifically, we introduce a unified framework that recasts the autoregressive G
D process as solving a system of triangular nonlinear equations (TNEs), thereby
enabling \textit{step-parallel} training, where gradients for different GD steps
 are computed concurrently without sequential dependencies. Within this generic
framework, we establish that: (1) the TNE system admits a unique solution corres
ponding precisely to the autoregressive GD iterative trajectory; (2) solving th
e TNEs system guarantees convergence to the GD iterative trajectory in at most t
he equal iterations.
Building on these insights, we present \textit{PASO}, the first step-parallel op
timizer for accelerating a broad class of GD-based optimizers like SGD and Adam
.
Extensive experiments (\textit{e.g.}, Llama-3.2-1B and diffusion model) validate
 that PASO achieves up to \textbf{21}$\times$ reduction in GD steps and \textbf{
4.5}$\times$ speedup in wall-clock time, with no model quality loss.
Source code is available at: \url{https://github.com/Jianrong-Lu/PASO.git}.
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AlgoTrace: Algorithmic Primitives and Compositional Geometry of Reasoning in Lan
guage Models
Samuel Lippl, Thomas Austin McGee, Kimberly Lopez, Ziwen Pan, Pierce Zhang, Salm
a Ziadi, Oliver Eberle, Ida Momennejad
How do inference time and latent computations enable large language models (LLMs
) to solve multi-step reasoning problems? We introduce AlgoTrace, a framework fo
r tracing and steering algorithmic operations in the model latent space for mult
i-step reasoning.
We operationalize primitives by clustering latent activations of the model when
solving four benchmarks: Traveling Salesperson Problem (TSP), 3SAT, AIME, and Gr
aph Navigation. We annotate the clusters using their corresponding tokens in the
 reasoning trace. We then apply function vector methods to extract primitive vec
tors as reusable compositional building blocks of reasoning. We find that a) inj
ecting a primitive vector into models (Phi, Qwen, Llama) elicits the associated
algorithmic operation in the reasoning trace, b) injecting primitives can steer
behavior across tasks, c) primitive vectors can be composed through algebraic op
erations, revealing a geometric logic in activation space, and d) a fine-tuned m
odel exhibits improved composition of primitives (Phi-4-Reasoning vs. Phi-4). Th
ese findings demonstrate that LLM reasoning can be understood as a walk through
algorithmic primitives in the latent space governed by compositional geometry. T
hese primitives transfer across tasks, and reasoning finetuning strengthens algo
rithmic generalization and composition across domains.
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Towards Parameter-Free Temporal Difference Learning
Yunxiang LI, Mark Schmidt, Reza Babanezhad Harikandeh, Sharan Vaswani
Temporal difference (TD) learning is a fundamental algorithm for estimating valu
e functions in reinforcement learning. Recent finite-time analyses of TD with li
near function approximation quantify its theoretical convergence rate. However,
they often require setting the algorithm parameters using problem-dependent quan
tities that are difficult to estimate in practice --- such as the minimum eigenv
alue of the feature covariance (ω) or the mixing time of the underlying M
arkov chain (τ_mix). In addition, some analyses rely on nonstandard
and impractical modifications, exacerbating the gap between theory and practice.
 To address these limitations, we use an exponential step-size schedule with the
 standard TD(0) algorithm. We analyze the resulting method under two sampling re
gimes: independent and identically distributed (i.i.d.) sampling from the statio
nary distribution, and the more practical Markovian sampling along a single traj
ectory. In the i.i.d. setting, the proposed algorithm does not require the knowl
edge of problem-dependent quantities such as ω, and attains the optimal b
ias-variance trade-off for the last iterate. In the Markovian setting, we propos
e a regularized TD(0) algorithm with an exponential step-size schedule. The resu
lting algorithm achieves a comparable convergence rate to prior works, without r

equiring projections, iterate averaging, or knowledge of τ_mix or ω.
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No More K-means: Single-Stage Sparse Coding for Efficient Multi-Vector Retrieval
Lixuan Guo, Yifei Wang, Tiansheng Wen, Aosong Feng, Stefanie Jegelka, Chenyu You
Multi-vector retrieval (MVR) models, exemplified by ColBERT, have established new benchmarks in retrieval accuracy by preserving fine-grained token-level interactions. However, this granularity imposes prohibitive storage and retrieval efficiency bottlenecks: to manage the immense memory footprint and computational overhead of billion-scale token vectors, state-of-the-art systems are forced to rely on aggressive dimension reduction and complex clustering (e.g., K-means). This compromise introduces two critical limitations: excessive indexing latency of clustering large-scale corpora and semantic information loss inherent to compression. In this paper, we propose Single-stage Sparse Retrieval (SSR}, a paradigm shift that replaces expensive clustering with efficient sparse coding. Instead of compressing features into low-dimensional dense vectors, we utilize Sparse Autoencoder (SAE) to project token embeddings into a high-dimensional but highly sparse representation. This transformation enables us to bypass vector clustering entirely and leverage inverted indexing for precise, high-throughput retrieval. Extensive experiments on the BEIR benchmark demonstrate that SSR achieves a "trifecta" of improvements: it reduces indexing time by 15x compared to ColBERTv2, halves retrieval latency, and simultaneously improves retrieval performance over leading baselines.
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Graph Neural Networks Are Not Continuous Across Graph Resolutions
Christian Koke, Yuesong Shen, Abhishek Saroha, Marvin Eisenberger, Bastian Rieck, Michael M. Bronstein, Daniel Cremers
We show that contrary to conventional wisdom in the community, graph neural networks (GNNs) are not continuous with respect to all natural modes of graph convergence. As a result, GNNs may generate substantially different latent representations for graphs that are very similar. In particular they assign vastly different latent embeddings to graphs that represent the same underlying object at different resolution scales. We trace this failure of continuity back to a structural obstruction arising from commonly used information-propagation schemes. Building on this insight we then derive a principled modification to standard GNN architectures which equips models with continuity across scales. The proposed modification enables consistent integration of distinct resolutions and reliable generalization between them. We systematically validate our theoretical findings in a wide range of numerical experiments.
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Reward Shaping Control Variates for Off-Policy Evaluation Under Sparse Rewards
Ritam Majumdar, Finale Doshi-Velez, Sonali Parbhoo
Off-policy evaluation (OPE) is essential for deploying reinforcement learning in safety-critical settings, yet existing estimators such as importance sampling and doubly robust (DR) often exhibit prohibitively high variance when rewards are sparse. In this work, we introduce Reward-Shaping Control Variates, a new family of unbiased estimators that leverage potential-based reward shaping to construct additional zero-mean control variates. We prove that shaped estimators always yields valid variance reduction, and that combining shaping-based and Q-based control variates strictly expands the variance-reduction subspace beyond DR and its minimax variant MRDR. Empirically, we provide a systematic regime map across synthetic chains, a cancer simulator, 5 single-stock and 1 multi-stock DOW-30 trading environments and an ICU-sepsis benchmark showing that shaping-based OPE consistently outperforms DR in sparse-reward settings, while a hybrid estimator achieves state-of-the-art performance across sparse, noisy, and misspecified environments. Our results highlight reward shaping as a powerful and interpretable tool for robust OPE, offering both theoretical guarantees and practical improvements in domains where standard estimators fail.
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M-IDoL: Information Decomposition for Modality-Specific and Diverse Representati

on Learning in Medical Foundation Model
Yihang Liu, Longzhen Yang, Jiaxiong Yang, Ying Wen, Lianghua He, Heng Tao Shen
Medical foundation models (MFMs) aim to learn universal representations from mul timodal medical images that can generalize effectively to diverse downstream cli nical tasks.
However, most existing MFMs suffer from information ambiguity that blends multim odal representations in a single embedding space, leading to the degradation of modality specificity and diversity.
In this paper, we propose M-IDoL, a self-supervised ***M***FM that introduces ** *I***nformation ***D***ecomposition for multim***o***dal representation ***L***e arning via two objectives:
i) maximizing inter-modality entropy by dispersing multimodal representations in to separable Mixture-of-Experts (MoE) subspaces to achieve representation specif icity across modalities;
and ii) minimizing intra-modality uncertainty by performing fine-grained semanti c discrimination within each MoE subspace to enrich representation diversity per modality.
By pre-training on 1.15 million medical images, M-IDoL i) delivers superior gene ralization across 21 downstream clinical tasks, outperforming 20 foundation mode ls on five imaging modalities (e.g., X-ray, fundus, OCT, dermoscopy and patholog y), and ii) learns modality-specific and diverse representations, showing cleare r separation of feature clusters across modalities and finer-grained feature dis crimination within each modality.
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Negative Sampling From the Ground Up: A Redesign for Recommendation
Yanbang Wang, Jon Kleinberg, Yanhong Wu
Negative sampling is an important yet challenging component in self-supervised g raph representation learning, particularly for recommendation systems where user -item interactions are modeled as bipartite graphs. Existing methods often rely on heuristics or human-specified principles to design negative sampling distribu tions. This potentially overlooks the usage of an underlying ``true'' negative d istribution, which we might be able to access as an oracle despite not knowing i ts exact form. In this work, we shift the focus from manually designing negative sampling distributions to a more principled method that approximates and levera ges the underlying true distribution from the ground up. We expand this idea in the analysis of two scenarios: (1) when the observed graph is an unbiased sampl e from the true distribution, and (2) when the observed graph is biased with par tially observable positive edges. The analysis result is the derivation of a sam pling strategy as the numerical approximation of a well-established learning obj ective. Our theoretical findings are also empirically validated, and our new sam pling methods achieve state-of-the-art performance on real-world datasets.
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Skip a Layer or Loop It? Learning Program-of-Layers in LLMs
Ziyue Li, Yang Li, Tianyi Zhou
Large language models (LLMs) perform inference by following a fixed depth and or der, non-recurrent execution of all layers. We reveal the wide existence of trai ning-free, flexible, dynamic *program-of-layers (PoLar)*, where pretrained layer s can be packed as modules and then skipped or looped to form a customized progr am for each input. For most inputs, substantially shorter program executions can achieve the same or better accuracy, while incorrect predictions of the origina l LLM can be corrected by alternative programs with fewer layers. These observat ions indicate that inference admits multiple valid latent computations beyond th e standard forward pass. To efficiently achieve PoLar in practice, we propose a lightweight PoLar prediction network, which learns to generate execution program s that dynamically skip or repeat pretrained layers for each input. Experiments on mathematical reasoning benchmarks demonstrate that PoLar consistently improve s accuracy over standard inference and prior dynamic-depth methods, often while executing fewer layers, and that these gains persist under out-of-distribution e valuation. Our results suggest that fixed-depth execution captures only a narrow subset of an LLM’s latent reasoning capacity.

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From Parameters to Feature Space: Task Arithmetic for Backdoor Mitigation in Model Merging
Zhenqian Zhu, Yamin Hu, Yiya Diao, Weixiang Li, Haodong Li, Wenjian Luo
Model merging (MM) has gained significant attention as a cost-effective approach to integrate multiple task-specific models into a unified model. However, recent work reveals that MM is highly susceptible to backdoor attacks. Existing defenses based on task arithmetic often fail to eliminate backdoors without substantially degrading clean-task performance, owing to their reliance on direct parameter-space editing. To address this gap, we propose Linear Feature Path Minimization (LFPM), a backdoor mitigation framework for model merging, which introduces an anti-backdoor task vector into the backdoored merged model. Unlike prior approaches, LFPM formulates the backdoor robustness of the merged model from a unified feature-space perspective under the Cross-Task Linearity (CTL) framework, which leverages the approximate linearity of features across tasks. This perspective guides the optimization of the anti-backdoor task to suppress backdoors while preserving clean-task performance. Furthermore, we introduce an effective optimization mechanism based on gradient accumulation and loss path-integral, ensuring robust backdoor suppression along the interpolation path. Extensive experiments demonstrate that LFPM consistently exhibits strong robustness against backdoor attacks in both full fine-tuning and Parameter-Efficient Fine-Tuning (PEFT) settings.
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ConPress: Learning Efficient Reasoning from Multi-Question Contextual Pressure
Jie Deng, Shining Liang, Jun Li, Hongzhi Li, Yutao Xie
Large reasoning models (LRMs) typically solve reasoning-intensive tasks by generating long
chain-of-thought (CoT) traces, leading to substantial inference overhead.
We identify a reproducible inference-time phenomenon, termed \textbf{\emph{Self-Compression}}:
when multiple independent and answerable questions are presented within a single prompt,
the model spontaneously produces shorter reasoning traces for each question. This phenomenon arises from \emph{multi-question contextual pressure} during generation and consistently manifests across models and benchmarks. Building on this observation, we propose ConPress (Learning from Contextual Pressure, a lightweight self-supervised fine-tuning approach. ConPress constructs multi-question prompts to induce self-compression, samples the resulting model outputs, and parses and filters per-question traces to obtain concise yet correct
reasoning trajectories. These trajectories are directly used for supervised fine-tuning, internalizing compressed
reasoning behavior in single-question settings without external teachers, manual pruning, or reinforcement learning.
With only 8k fine-tuning examples, ConPress reduces reasoning token usage by 59\% on MATH500
and 33\% on AIME25, while maintaining competitive accuracy.
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Non-Euclidean Gradient Descent Operates at the Edge of Stability
Rustem Islamov, Michael Crawshaw, Jeremy Cohen, Robert M. Gower
The Edge of Stability (EoS) is a phenomenon where the sharpness (largest eigenvalue) of the Hessian approaches and then hovers near the stability threshold $2/\eta$ during gradient descent (GD) with step size η. Despite (apparently) violating classical smoothness assumptions, EoS has been widely observed in deep learning, but its theoretical foundations remain incomplete. We provide an interpretation of EoS through the lens of Directional Smoothness [Mishkin et al., 2024]. This interpretation naturally extends to non-Euclidean norms, which we use to define generalized sharpness under an arbitrary norm. Our generalized sharpness measure includes previously studied vanilla GD and preconditioned GD as special cases, as well as methods for which EoS has not been studied, such as ℓ_{∞}-descent, Block CD, Spectral GD, and their normalized versions. Through

experiments on neural networks, we show that non-Euclidean GD with our generalized sharpness also exhibits progressive sharpening followed by oscillations around or above the threshold $2/\eta$. Practically, our framework provides a geometry-aware spectral diagnostic that can be applied across a broad class of non-Euclidean gradient methods.
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Reward Under Attack: Analyzing the Robustness and Hackability of Process Reward Models
Rishabh Tiwari, Aditya Tomar, Udbhav Bamba, Monishwaran Maheswaran, Heng Yang, Michael W. Mahoney, Kurt Keutzer, Amir Gholami
Process Reward Models (PRMs) are rapidly becoming the backbone of LLM reasoning pipelines, yet we demonstrate that state-of-the-art PRMs are systematically exploitable under optimization pressure. We introduce a three-tiered diagnostic framework that applies increasing adversarial pressure to quantify these vulnerabilities. Static perturbation analysis uncovers a fluency-logic dissociation: high invariance to surface-level style changes (reward changes $<$0.1) yet inconsistent detection of logically corrupted reasoning, with different models failing on different attack types. Adversarial optimization demonstrates that gradient-based attacks inflate rewards on invalid trajectories, with reward landscapes exhibiting wide, exploitable peaks. RL-induced reward hacking exposes the critical failure mode: policies trained on AIME problems achieve near-perfect PRM rewards ($>$0.9) while ground-truth accuracy remains below 4\%, with 43\% of reward gains attributable to stylistic shortcuts. These findings reveal that current PRMs function as fluency detectors rather than reasoning verifiers, creating systematic blind spots that undermine their use as training signals. We release PRM-BiasBench and a diagnostic toolkit to enable robustness evaluation before deployment.
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AMDP: Asynchronous Multi-Directional Pipeline Parallelism for Large-Scale Models Training
Ling Chen, Houming Wu, Wenjie Yu
Pipeline parallelism is essential for large-scale model training, but existing asynchronous approaches often degrade convergence due to parameter mismatch between forward and backward passes. We propose Asynchronous Multi-Directional Pipeline parallelism (AMDP) to mitigate this issue while sustaining high utilization. AMDP limits the first stage of each pipeline to process at most two minibatches before backpropagation, bounding the number of parameter updates between forward and backward passes. To alleviate the resulting pipeline bubbles, AMDP launches multiple concurrent pipelines and adapts their number according to pipeline depth. In addition, AMDP accumulates gradients across minibatches and applies them in a single update, ensuring that only a bounded number of minibatches experience parameter mismatch, limited to within one optimization step. Experiments on GPT- and BERT-style models demonstrate that AMDP significantly accelerates training while preserving convergence.
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BFCL Audio: An Audio Function Calling Evaluation for Large Language Models
Huanzhi Mao, Aditya Ghai, Imra Dawoodani, Tony A Ginart, Shishir G Patil, John Emmons, Joseph E. Gonzalez
Audio agents are increasingly deployed to execute tools from spoken requests, yet audio tool use poses challenges beyond text-only function calling: perception errors (e.g., homophones, noise, disfluencies) can corrupt entities and arguments, and natural interactions often require clarification that changes the tool-calling protocol. We introduce BFCL Audio, a large-scale benchmark for audio function calling with 6.2K expert-verified tasks across two suites that mirror common deployments: BFCL Text Audio (pipelined $\text{ASR}\rightarrow\text{LLM}\rightarrow\text{tools}$ via transcripts) and BFCL True Audio (end-to-end $\text{audio-in}\rightarrow\text{tool calls}$). BFCL Audio includes controlled speech and acoustic perturbations (accent and speaking-rate variation, content disfluencies, and background noise) generated through a controllable audio synthesis/augmentation pipeline. We provide automatic grading for both function names and argument values using AST-based matching for single-turn calls and response/state-based me

trics for multi-turn interactions, enabling scalable evaluation without LLM judges. Across a broad set of models, we propose a failure-mode taxonomy and analyze which speech and noise factors most strongly impact tool-calling accuracy. We release the benchmark, evaluation harness, and audio pipeline to support research on reliable speech-based agents.
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Adaptive Visual Autoregressive Acceleration via Dual-Linkage Entropy Analysis
Yu Zhang, Jingyi Liu, Feng Liu, Duoqian Miao, Qi Zhang, Kexue Fu, Changwei Wang, Longbing Cao
Visual AutoRegressive modeling (VAR) suffers from substantial computational cost due to the massive token count involved. Failing to account for the continuous variation in modeling, existing VAR token reduction methods face three key limitations: heuristic stage partition, non-adaptive schedules, and limited acceleration scope, thereby leaving significant acceleration potential untapped. Since entropy variation intrinsically reflects the transition of predictive uncertainty, it offers a principled measure to capture continuous modeling variation. Therefore, we propose NOVA, a training-free token reduction acceleration framework for VAR models via entropy analysis. NOVA adaptively determines the acceleration activation scale during inference by online identifying the inflection point of scale entropy growth. Through scale-linkage and layer-linkage ratio adjustment, NOVA dynamically computes distinct token reduction ratios for each scale and layer, pruning low-entropy tokens while reusing the cache derived from the residuals at the prior scale to accelerate inference and maintain generation quality. Extensive experiments and analyses validate NOVA as a simple yet effective training-free acceleration framework. Code is available.
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ORBIT: A Prognostic World Model for Ocular Reasoning Based on Imagined Trajectories
Jiangtao Yan, Yanlin Qu, Yansheng Qiu, Shujian Gao, Wei Yu, Zheng Wang, Xiaodong Sun, Huixun Jia, Diping Song
The longitudinal management of blinding fundus diseases constitutes a Partially Observable Markov Decision Process (POMDP) necessitating a critical precision-risk trade-off between intervention and over-treatment, as true pathology is often obscured in static observations. However, existing paradigms fail to address this complexity. Traditional vision models remain uninterpretable and memoryless, and while Vision-Language Models (VLMs) excel in semantic understanding, they rely on unsafe open-loop text reasoning lacking the anatomical grounding essential for clinical safety.
Furthermore, robust learning is hindered by the scarcity of process supervision in sparse clinical records.
To bridge this gap, we introduce the Logic-Constrained Abductive Data Engine. Operating on a ``Propose-and-Verify'' paradigm, it validates MLLM-proposed biomarkers against clinical and temporal logic to reconstruct dense pathological states from sparse outcomes.
Building on this foundation, we propose ORBIT, the first ophthalmic Prognostic World Model. Uniquely, ORBIT employs counterfactual visual foresight to imagine anatomical futures under different treatments, anchoring decisions in Closed-Loop Anatomical Verification rather than linguistic probabilities.
Experiments demonstrate that ORBIT effectively captures disease evolution and establishes a new paradigm for human-in-the-loop longitudinal decision support and anatomically grounded treatment planning.
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Dispersion Loss Counteracts Embedding Condensation and Improves Generalization in Small Language Models
Chen Liu, Xingzhi Sun, Xi Xiao, Alexandre Van Tassel, Ke Xu, Kristof Reimann, Danqi Liao, Mark Gerstein, Tianyang Wang, Xiao Wang, Smita Krishnaswamy
Large language models (LLMs) achieve remarkable performance through ever-increasing parameter counts, but scaling incurs steep computational costs. To better understand LLM scaling, we study representational differences between LLMs and their smaller counterparts, with the goal of replicating the representational quali

ties of larger models in smaller models. We observe a geometric phenomenon which we term $\textbf{embedding condensation}$, where token embeddings collapse into a narrow cone-like subspace in some language models. Through systematic analyses across multiple Transformer families, we show that small models such as $\texttt{GPT2}$ and $\texttt{Qwen3-0.6B}$ exhibit severe condensation, whereas larger models such as $\texttt{GPT2-xl}$ and $\texttt{Qwen3-32B}$ are more resistant to this phenomenon. Additional observations show that embedding condensation is not reliably mitigated by knowledge distillation from larger models. To fight against it, we formulate a dispersion loss that explicitly encourages embedding dispersion during training. Experiments demonstrate that it mitigates condensation, recovers dispersion patterns seen in larger models, and yields performance gains across 10 benchmarks. We believe this work offers a principled path toward improving smaller Transformers without additional parameters.
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Pushing Forward Pareto Frontiers of Proactive Agents with Behavioral Agentic Optimization
Yihang Yao, Zhepeng Cen, Haohong Lin, Shiqi Liu, Zuxin Liu, Jiacheng Zhu, Zhang-Wei Hong, Laixi Shi, Ding Zhao
Proactive large language model (LLM) agents aim to actively plan, query, and interact over multiple turns, enabling efficient task completion beyond passive instruction following and making them essential for real-world, user-centric applications. Agentic reinforcement learning (RL) has recently emerged as a promising solution for training such agents in multi-turn settings, allowing interaction strategies to be learned from feedback. However, existing pipelines face a critical challenge in balancing task performance with user engagement, as passive agents can not efficiently adapt to users' intentions while overuse of human feedback reduces their satisfaction. To address this trade-off, we propose BAO, an agentic RL framework that combines behavior enhancement to enrich proactive reasoning and information-gathering capabilities with behavior regularization to suppress inefficient or redundant interactions and align agent behavior with user expectations. We evaluate BAO on multiple tasks from the UserRL benchmark suite, and demonstrate that it substantially outperforms RL baselines under controlled comparisons, while achieving comparable or even superior performance to frontier LLM agents, highlighting its effectiveness for training proactive, user-aligned LLM agents in complex multi-turn scenarios.
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Large Language Models Develop Novel Social Biases Through Adaptive Exploration
Addison J. Wu, Ryan Liu, Xuechunzi Bai, Thomas L. Griffiths
As large language models (LLMs) are adopted into frameworks that grant them the capacity to make real decisions, it is increasingly important to ensure that they are unbiased. In this paper, we argue that the predominant approach of simply removing existing biases from models is not enough. Using a paradigm from the psychology literature, we demonstrate that LLMs can spontaneously develop novel social biases about artificial demographic groups even when no inherent differences exist. These biases result in highly stratified task allocations, which are less fair than assignments by human participants and are exacerbated by newer and larger models. In humans, emergent biases like these have been shown to result from exploration-exploitation trade-offs, where the decision-maker explores too little, allowing early observations to strongly influence impressions about entire demographic groups. To alleviate this effect, we examine a series of interventions targeting model inputs, problem structure, and explicit steering. We find that explicitly incentivizing exploration most robustly reduces stratification, highlighting the need for better multifaceted objectives to mitigate bias. These results reveal that LLMs are not merely passive mirrors of human social biases, but can actively create new ones from experience, raising urgent questions about how these systems will shape societies over time.
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Geometry-Preserving Unsupervised Alignment for Heterogeneous Foundation Models
Shuwen Yu, Zhanxuan Hu, Yi Zhao, Yonghang Tai, Huafeng Li
Foundation models have driven rapid progress in computer vision, yet the two dom

inant paradigms, vision-language foundation models (VLMs) and vision-only founda tion models (VFMs), remain only partially compatible. VLMs offer language-ground ed semantic alignment but are often visually coarse, while VFMs learn discrimina tive perceptual geometry but lack semantic grounding. We propose **GPUA**, a *Ge ometry-Preserving Unsupervised Alignment* framework that integrates the compleme ntary strengths of VFMs and VLMs. Inspired by cross-lingual alignment, **GPUA** treats VFM features as a *visual language* and learns an *orthogonal* mapping th at translates the VFM space into the VLM semantic space, preserving geometry and narrowing the modality gap without labels or model parameter updates. **GPUA** is task-agnostic and requires only feature-level access to pretrained models. Ex periments across diverse benchmarks demonstrate improved cross-model compatibili ty and strong gains in downstream zero-shot recognition and segmentation with ne gligible overhead. Our code is available at: https://github.com/Yuteam14/GPUA.
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How Far Ahead Do LLMs Plan? Uncovering the Latent Horizon in Chain-of-Thought Re asoning

Liyan Xu, Mo Yu, Fandong Meng, Jie Zhou

Chain-of-thought (CoT) reasoning has become a central mechanism for eliciting mu lti-step reasoning in Large Language Models (LLMs). Yet recent evidence presents a tension: hidden states appear to already encode future reasoning before CoT f ully unfolds, while explicit steps still remain crucial for tasks requiring comp ositional computation.

To deepen the understanding between LLM's internal states and its verbalized rea soning trajectories, we investigate the latent planning strength of LLMs, throug h our probing method, Tele-Lens, applying to hidden states across diverse task d omains.

Our empirical results indicate that LLMs exhibit a myopic horizon, primarily con ducting incremental transitions without precise global planning.

Leveraging this characteristic, we propose a hypothesis on enhancing uncertainty estimation of CoT, which we validate that a sparse set of pivot positions can e ffectively represent the uncertainty of the entire path.

We further underscore the significance of exploiting CoT dynamics, and demonstra te that automatic recognition of CoT bypass can be achieved without performance degradation.

Our code, data and models are released at https://github.com/lxucs/tele-lens.
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Cross-Subject Modeling for Widefield Calcium Imaging via Atlas-Aligned Spatiotem poral Tokenization

Mohammad Hosseini, Eray Erturk, Saba Hashemi, Maryam M. Shanechi

Large-scale, multi-subject widefield calcium imaging provides unprecedented acce ss to brain-wide cortical dynamics. However, the high dimensionality, complex sp atiotemporal structure, and substantial task-irrelevant activity in widefield re cordings have largely restricted modeling efforts to single-session analyses, li miting scalability and generalization. While multi-subject pretrained models hav e been explored for some neural modalities, multi-subject models for widefield c alcium imaging have not yet been demonstrated; further, subject-invariant zero-s hot behavior decoding remains elusive for multi-subject models across neural mod alities more broadly. As a first step toward foundation modeling of widefield da ta, we introduce WiCAT, a multi-subject model that leverages self-supervised pre training to both outperform single-session models and enable zero-shot behavior decoding on unseen subjects. WiCAT introduces an atlas-grounded tokenization sch eme without session-specific components and learns globally shared spatiotempora l representations. Across multiple widefield datasets, the pretrained model supp orts lightweight downstream decoding, transfers across subjects, tasks, and data sets, and outperforms baseline models. Notably, the model also achieves robust z ero-shot continuous behavior decoding and left-out brain region reconstruction o n unseen subjects.
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DiL: Discrete-anchored Representation Alignment for Semi-Supervised Continual Le arning

Nanyi Wang, Chaojie Chen, Zuoqi Tang, Jinxiang Lai, Xingcai Wu, Qi Wang
Leveraging the unlabeled stream is crucial yet challenging in Semi-Supervised Co ntinual Learning (SSCL) under continual class expansion.
Existing SSCL methods typically enforce dense pseudo-label consistency and indis criminate distillation on unlabeled data, which can reinforce errors and intensi fy base–novel interference.
To address these issues, we propose Discrete-anchored Incremental Learning (DiL) to ground continual updates on reliable discrete anchors that remain stable und er noisy pseudo-labels.
DiL introduces Discrete Contrastive Distillation (DCD), which discretizes the di stillation pathway and performs anchor-referenced selective distillation to curb error reinforcement.
Meanwhile, Class-Aware Channel-Chunked Encoding (CACE) learns channel-chunked re presentations and exploits the confusion matrix induced by the discrete anchors to separate novel from confusable base classes.
Extensive experiments on multiple datasets show that DiL achieves state-of-the-a rt performance across diverse SSCL protocols.
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Balancing Understanding and Generation in Discrete Diffusion Models
Yue Liu, Yuzhong Zhao, Zheyong Xie, Qixiang Ye, Jianbin Jiao, Yao Hu, Shaosheng Cao, Yunfan Liu
In discrete generative modeling, two dominant paradigms demonstrate divergent ca pabilities: Masked Diffusion Language Models (MDLM) excel at semantic understand ing and zero-shot generalization, whereas Uniform-noise Diffusion Language Model s (UDLM) achieve strong few-step generation quality, yet neither attains balance d performance across both dimensions. To address this, we propose XDLM, which br idges the two paradigms via a stationary noise kernel. XDLM offers two key contr ibutions: it provides (1) a principled theoretical unification of MDLM and UDLM, recovering each paradigm as a special case; and (2) an alleviated memory bottle neck enabled by an algebraic simplification of the posterior probabilities. Expe riments demonstrate that XDLM advances the Pareto frontier between understanding capability and generation quality. Quantitatively, XDLM surpasses UDLM by 5.4 p oints on zero-shot text benchmarks and outperforms MDLM in few-step image genera tion (FID 54.1 vs. 80.8). When scaled to tune an 8B-parameter large language mo del, XDLM achieves 15.0 MBPP in just 32 steps, effectively doubling the baseline performance. Finally, analysis of training dynamics reveals XDLM’s superior pot ential for long-term scaling. Code is available at [https://github.com/MzeroMiko /XDLM](https://github.com/MzeroMiko/XDLM).
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Improved Convergence Analysis of Topology Dependence in Decentralized SGD
Yuki Takezawa, Anastasia Koloskova, Sebastian U Stich
Decentralized SGD is a fundamental algorithm in decentralized learning, although the influence of an underlying network topology on its convergence behavior is not yet fully understood. Existing convergence analyses have shown that topologi es with a small spectral gap significantly deteriorate the convergence rate of D ecentralized SGD in both homogeneous and heterogeneous cases. However, many prio r papers have reported that indeed the choice of the topology has a significant experimental impact in the heterogeneous case, but has little experimental impac t on training behavior in the homogeneous case. In this paper, we present a tigh ter convergence analysis of Decentralized SGD, offering a more precise understan ding of how topologies affect the convergence rate than the prior analysis. Spec ifically, unlike existing convergence analyses that used only the spectral gap a s a property of the topology, our novel analysis shows that all eigenvalues of t he mixing matrix affect the convergence rate. Throughout the experiments, we car efully evaluated the convergence behavior of Decentralized SGD and demonstrated that our novel convergence analysis can more accurately describe the effect of t opology on the convergence rate.
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Towards Scalable and Consistent 3D Editing
Ruihao Xia, Yang Tang, Pan Zhou

3D editing—the task of locally modifying the geometry or appearance of a 3D asset—has wide applications in immersive content creation, digital entertainment, and AR/VR. However, unlike 2D editing, it remains challenging due to the need for cross-view consistency, structural fidelity, and fine-grained controllability. Existing approaches are often slow, prone to geometric distortions, or dependent on manual and accurate 3D masks that are error-prone and impractical. To address these challenges, we advance both the data and model fronts. On the data side, we introduce 3DEditVerse, the largest paired 3D editing benchmark to date, comprising 116,309 high-quality training pairs and 1,500 curated test pairs. Built through complementary pipelines of pose-driven geometric edits and foundation model-guided appearance edits, 3DEditVerse ensures edit locality, multi-view consistency, and semantic alignment. On the model side, we propose 3DEditFormer, a 3D-structure-preserving transformer. By enhancing image-to-3D generation with dual-guidance attention and time-adaptive gating, 3DEditFormer disentangles editable regions from preserved structure, enabling precise and consistent edits without requiring auxiliary 3D masks. Extensive experiments demonstrate that our framework outperforms state-of-the-art baselines both quantitatively and qualitatively, establishing a new standard for practical and scalable 3D editing. Dataset and code are available at https://www.lv-lab.org/3DEditFormer/

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From Interaction Trajectories to Prompt Rules: Credit Assignment for Multi-Agent Prompt Optimization

Bin Wu, Haoran Xu, Xiang Zhuang, Zonghao Chen, Zhu Li, Emine Yilmaz, Qiang Zhang

Large language model (LLM)-based multi-agent systems commonly rely on natural-language prompts to specify agent behavior, yet optimizing these prompts remains challenging when agent roles and interaction structures are fixed by design. In such systems, behaviors emerge over long, noisy interaction trajectories, making it difficult to determine which prompt components are responsible for success or failure. As a result, outcome-level feedback alone is insufficient, while existing prompt optimization methods typically rely on final task scores or global prompt rewrites, limiting their ability to exploit trajectory evidence or support the localized updates. We propose Trajectory-based Rule Credit Estimation (TRUCE), a framework for prompt optimization in multi-agent systems that explicitly addresses this credit assignment challenge. TRUCE performs trajectory-aware attribution by linking outcome feedback to informative sub-trajectories and translating the resulting credit signals into unit-level edits over prompt-defined behavioral rules. By preserving agent roles and interaction structures, TRUCE enables prompt refinement through localized updates aggregated across tasks. Experiments on multiple benchmarks demonstrate that TRUCE consistently improves task performance and efficiency over competitive baselines. Code is available at https://github.com/bingo-w/TRUCE.

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Lavida-R1: Advancing Reasoning for Unified Multimodal Diffusion Language Models

Shufan Li, Yuchen Zhu, Kangning Liu, Zhe Lin, Yongxin Chen, Molei Tao, Aditya Grover, Jiuxiang Gu, Jason Kuen

Diffusion language models (dLLMs) recently emerged as a promising alternative to auto-regressive LLMs. The latest works further extended it to multimodal understanding and generation tasks. In this work, we propose LaViDa-R1, a multimodal, general-purpose reasoning dLLM. Unlike existing works that build reasoning dLLMs through task-specific reinforcement learning, LaViDa-R1 incorporates diverse multimodal understanding and generation tasks in a unified manner. In particular, LaViDa-R1 is built with a novel unified post-training framework that seamlessly integrates supervised finetuning (SFT) and multi-task reinforcement learning (RL). It employs several novel training techniques, including answer-forcing, tree search, and complementary likelihood estimation, to enhance effectiveness and scalability. Extensive experiments demonstrate LaViDa-R1's strong performance on a wide range of multimodal tasks, including visual math reasoning, reason-intensive grounding, and image editing.

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Learning Context-Conditioned Predicate Semantics via Prototype Feedback

NamGyu Jung, Chang Choi
In scene graph generation, a central challenge is modeling polysemous predicates whose meanings shift across contexts. Prior approaches address this issue by decomposing predicates into multiple static prototypes or retrieving semantically similar exemplars. However, these strategies keep predicate representations static and cannot reorganize semantics to reflect image-specific evidence, leading to systematic confusions in ambiguous contexts. We propose **AlignG**, which learns context-conditioned predicate semantics via prototype feedback. AlignG infers context-conditioned predicate semantics from the relation candidates within each image and feeds the adapted semantics back to recalibrate relation representations. The learning objective anchors this adaptation to global semantic centers, preventing semantic drift while still allowing selective reorganization when the scene provides consistent relational cues. Experiments on VG-150 and GQA-200 show consistent improvements over state-of-the-art baselines, with F@100 improvements of +1.4 on VG-150 and +2.7 on GQA-200 under SGDet. We further visualize per-image prototype similarity shifts and observe coherent context-dependent reorganization where prototypes selectively merge or separate predicates according to scene evidence. The code is available at https://github.com/Namgyu97/AlignG-SGG.pytorch.
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Disentangling Intent from Role: Adversarial Self-Play for Persona-Invariant Safety Alignment
Jiajia Li, Xiaoyu Wen, Shuyue Hu, Qiaosheng Zhang, Zhen Wang
The growing capabilities of large language models (LLMs) have driven their widespread deployment across diverse domains, even in potentially high-risk scenarios. Despite advances in safety alignment techniques, current models remain vulnerable to emerging *persona-based jailbreak attacks*. Existing research on persona-based jailbreak has primarily focused on attack iterations, yet it lacks systemic and mechanistic constraints on the defense side. To address this challenge, we propose Persona-Invariant Alignment (PIA), an adversarial self-play framework that achieves co-evolution through Persona Lineage Evolution (PLE) on the attack side and Persona-Invariant Consistency Learning (PICL) on the defense side. Theoretically, PICL is grounded in the *structural separation hypothesis*, using a unilateral KL-divergence constraint to enable the structural decoupling of safety decisions from persona context, thereby maintaining safe behavior under persona-based jailbreak attacks. Experimental results demonstrate that PLE efficiently explores high-risk persona spaces by leveraging lineage-based credit propagation. Meanwhile, the PICL defense method significantly reduces the Attack Success Rate (ASR) while preserving the model's general capability, thereby validating the superiority and robustness of this alignment paradigm. WARNING: This paper contains potentially offensive and harmful text.
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Forgetting Whenever You Want: A Decentralized Continual Learning Framework with On-Demand Unlearning
Xiao Zhang, Zengzhe Chen, Mingyi Li, Jing Qiao, Fuzhen Zhuang, YUAN YUAN, Dongxiao Yu
Decentralized class continual learning refers to a paradigm where distributed clients continuously acquire new classes while retaining previously learned information without relying on a central server. With increasing emphasis on privacy preservation, there is a growing need for on-demand unlearning, introducing two key challenges: Historical Class Unlearning and Network-Wide Knowledge Entanglement. In this work, we propose a decentralized continual learning framework with on-demand unlearning (DCU), which is the first attempt at achieving class continual learning and arbitrary-time class unlearning in a distributed setting. Specifically, our proposed DCU comprises three main stages: prototypes extraction, prototype-guided continual learning, and unlearning with disposable prototypes. Firstly, the prototypes extraction mechanism is designed to capture the class-specific concepts as lightweight, disposable embeddings. Then, the synthetic data guided by these prototypes can be combined with real data to achieve incremental learning through distillation. Besides, synthetic samples with noisy label are use

d to guide the adjustment of the model's decision boundary, effectively erasing the influence of the target class while preserving other classes' knowledge. Ext ensive experiments conducted on two datasets demonstrate the effectiveness of ou r DCU in dynamic learning and target class unlearning.
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Escaping the Verifier: Learning to Reason via Demonstrations
Locke Cai, Max Ryabinin, Ivan Provilkov
Training Large Language Models (LLMs) to reason often relies on Reinforcement Le arning (RL) with task-specific verifiers. However, many real-world reasoning-int ensive tasks lack verifiers, despite offering abundant expert demonstrations tha t remain under-utilized for reasoning-focused training. We introduce **RARO** (R elativistic Adversarial Reasoning Optimization), which learns strong reasoning c apabilities from expert demonstrations alone via **Inverse Reinforcement Learnin g**. Our method sets up an adversarial game between a **policy** and a **relativ istic critic**: the policy learns to mimic expert answers, while the critic aims to identify the expert among (expert, policy) answer pairs. Both the policy and the critic are trained jointly and continuously via RL, and we identify key sta bilization techniques required for robust learning. Empirically, RARO significan tly outperforms strong verifier-free baselines across all evaluation tasks: $+13 .7\%$ accuracy on Countdown (1.5B), +8.2\% on DeepMath (7B), and $+19.1\%$ w in-rate on Poetry Writing (7B) against expert poems. RARO also exhibits simila r robust scaling trends as RL with verifiers. These results demonstrate that RAR O effectively elicits strong reasoning performance from expert demonstrations al one, enabling robust reasoning learning even when task-specific verifiers are un available.
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Robust Parallel Diffusion Sampling via Dynamic Jacobian Bandwidth
Zile Huang, Ser-Nam Lim
Recent years have witnessed significant progress in developing effective diffusi on models. Parallel sampling is a promising recent approach that reformulates th e sequential denoising process as solving a system of nonlinear equations, and i t can be combined with other acceleration techniques. However, current progress is limited by the trade-off between high fidelity and computational efficiency. This paper addresses the challenge of scaling to high-dimensional, multi-modal g eneration. Specifically, we present ROPA (Robust Parallel Diffusion Sampling), w hich takes into account the properties of the denoising process and solves the l inear system using adaptive local sparsity to achieve stable parallel sampling. Extensive experiments demonstrate ROPA’s effectiveness: it significantly acceler ates sampling across diverse image and video diffusion models, achieving up to $ 2.9\times$ speedup with eight core, an improvement of 20.8\% over baselines with out sacrificing sample quality. ROPA enables parallel sampling methods to provid e a solid foundation for real-time, high-fidelity diffusion generation.
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On the Anisotropy of Score-Based Generative Models
Andreas Floros, Seyed-Mohsen Moosavi-Dezfooli, Pier Luigi Dragotti
We investigate the role of network architecture in shaping the inductive biases of modern score-based generative models. To this end, we introduce the Score Ani sotropy Directions (SADs), architecture-dependent directions that reveal how dif ferent networks preferentially capture data structure. Our analysis suggests tha t SADs form adaptive bases aligned with the architecture's output geometry, prov iding a principled way to predict generalization ability in score models prior t o training. Through both synthetic data and standard image benchmarks, we demons trate that SADs reliably capture fine-grained model behavior and correlate with downstream performance, as measured by Wasserstein metrics. Our work offers a ne w lens for explaining and predicting directional biases of generative models.
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FOVI: A biologically-inspired foveated interface for deep vision models
Nicholas Blauch, George A. Alvarez, Talia Konkle
Human vision is foveated, with variable resolution peaking at the center of a la rge field of view; this reflects an efficient trade-off for active sensing, allo

wing eye-movements to bring different parts of the world into focus with other p
arts of the world in context. In contrast, most computer vision systems encode t
he visual world at a uniform resolution, raising challenges for processing full-
field high-resolution images efficiently. We propose a foveated vision interface
 (FOVI) based on the human retina and primary visual cortex (V1), that reformats
 a variable-resolution retina-like sensor array into a uniformly dense, V1-like
sensor manifold. Receptive fields are defined as k-nearest-neighborhoods (kNNs)
on the sensor manifold, enabling kNN-convolution via a novel kernel mapping tech
nique. We demonstrate two use cases: (1) an end-to-end kNN-convolutional archite
cture, and (2) a foveated adaptation of the DINOv3 ViT foundation model, leverag
ing low-rank adaptation (LoRA). These models provide competitive performance wit
h a fraction of the pixels and computational cost of full resolution non-foveate
d baselines, opening pathways for efficient and scalable active sensing for high
-resolution egocentric vision.
Code (https://github.com/nblauch/fovi) and pre-trained models (https://huggingfa
ce.co/fovi-pytorch) are available.
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Boost the Identity-Preserving Embedding for Consistent Visual Generation
Zixun Xia, Kai Wang, Shuyu Guo, Boqian Li, jian Yang, Yaxing Wang
Text-to-image models have advanced high-fidelity content generation, but their i
nability to maintain subject consistency hampers realistic applications.
Existing training-based methods rely on heavy computation and large datasets; wh
ile training-free approaches demand excessive memory or complex auxiliary module
s.
In this paper, we first reveal a key property overlooked in prior works that the
 identity-relevant signals, termed Identity-Preserving Embeddings (*IPemb*), are
 implicitly encoded in textual embeddings of frame prompts. To address the consi
stent T2I generation with the *IPemb* embedding, we propose Boost Identity-Prese
rving Embedding (*BIPE*), a training-free yet plug-and-play framework that expli
citly extracts and enhances the *IPemb*.
Its core innovations are two complementary techniques:
First, Adaptive Singular-Value Rescaling (*adaSVR*) applies singular-value decom
position to the joint embedding matrix of all frame prompts, amplifying identity
-centric components while suppressing frame-specific noise.
Second, Union Key (*UniK*) further reinforces consistency by aligning the T2I ba
ckbone’s image-text attention across the entire generation sequence.
Experiments on the *ConsiStory+* benchmark demonstrate *BIPE* outperforms existi
ng methods in both qualitative and quantitative metrics. To address the gap in e
valuating a broader range of scenarios with diversified prompt templates, we int
roduce a *DiverStory* benchmark to further confirm our scalability.
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Optimizing Network Simulation: Enhancing Performance Prediction Accuracy via Neu
ral Architecture Search
ShaoChen He, Zirui Zhuang, Haifeng Sun, Xiaoyuan Fu, Qi Qi, Lei Zhang, Jianxin L
iao, Jingyu Wang
Existing machine learning models for network simulation excel at predicting aver
age performance but, due to their reliance on mean squared error, systematically
 fail to capture the critical tail-latency and jitter that define modern network
 stability. This 'tail-blindness' renders them unreliable for latency-sensitive
systems. We bridge this gap by introducing Accurate Neural Architecture Search (
ANAS), a paradigm that automates the discovery of architectures for high-precisi
on, distribution-aware network simulation. ANAS corrects the evaluation inaccura
cies of weight-sharing NAS via a similarity-constrained search, employs a hybrid
 search space to model complex traffic, and uses a Wasserstein loss to optimize
for the entire delay distribution, not just its mean. Empirically, the ANAS-disc
overed architecture is holistically superior: it reduces overall validation loss
 by 25.8\% compared to DeepQueueNet, demonstrating strong average-case performan
ce, while simultaneously excelling at tail-sensitive metrics by lowering the nor
malized Wasserstein distance (W_n) by up to 69.8\%. This confirms its ability
to faithfully model a comprehensive performance spectrum, encompassing both aver

age and critical tail behaviors. The ANAS framework provides a practical methodology for automatically creating high-fidelity model of network devices, enabling more reliable validation of next-generation network protocols and algorithms.
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Spatio-Temporal LLM: Reasoning about Environments and Actions
Haozhen Zheng, Beitong Tian, Mingyuan Wu, Zhenggang Tang, Klara Nahrstedt, Alex Schwing
Despite significant recent progress of Multimodal Large Language Models (MLLMs), current MLLMs are challenged by "spatio-temporal" prompts, i.e., prompts that refer to 1) the entirety of an environment encoded in a point cloud that the MLLM should consider; and simultaneously also refer to 2) actions that happened in part of the environment and are encoded in a short ego-centric video clip. However, such a holistic spatio-temporal understanding is important for agents operating in the real world. To address this challenge, we first develop a framework to collect a large-scale dataset. Using the collected "Reasoning about Environments and Actions" (REA) dataset, we show that recent MLLMs indeed struggle to correctly answer "spatio-temporal" prompts. Building on this dataset, we study two spatio-temporal LLM (STLLM) baselines: 1) STLLM-3D, which directly fuses point cloud, video, and text representations as inputs to the LLM; and 2) STLLM-Aligner, which aligns spatial context with video and text before LLM decoding. Both baselines aim to enhance spatial understanding of environments and temporal grounding of egocentric observations. On REA, the STLLM baselines outperform existing models, demonstrating the effectiveness of our designs.
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SWE-Perf: Can Language Models Optimize Code Performance on Real-World Repositories?
Xinyi He, Qian Liu, Mingzhe Du, Lin Yan, ZhiJie Fan, Yiming Huang, Yin Zheng, Zejian Yuan, Zejun MA
Code performance optimization is paramount in real-world software engineering and critical for production-level systems. While Large Language Models (LLMs) have demonstrated impressive capabilities in code generation and bug fixing, their proficiency in enhancing code performance at the repository level remains largely unexplored. To address this gap, we introduce SWE-Perf, the first benchmark specifically designed to systematically evaluate LLMs on code performance optimization tasks within authentic repository contexts. SWE-Perf comprises 140 carefully curated instances, each derived from performance-improving pull requests from popular GitHub repositories. Each benchmark instance includes the relevant codebase, target functions, performance-related tests, expert-authored patches, and executable environments. Through a comprehensive evaluation of representative methods that span file-level and repo-level approaches (e.g., Agentless and OpenHands), we reveal a substantial capability gap between existing LLMs and expert-level optimization performance, highlighting critical research opportunities in this emerging field.
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Scalable Kronecker-Factored Fisher Approximation for Neural Network Parameter Sensitivity
Viktoriia A. Chekalina, Daniil Moskovskiy, Tatyana Matveeva, Andrey Kuznetsov, Evgeny Frolov
The Fisher Information Matrix (FIM) provides a principled geometric framework for parameter sensitivity in neural networks, but directly computing and using the full FIM is infeasible in high-dimensional models.
As a result, most existing methods rely on diagonal approximations that discard important correlation structure.
We introduce Matrix-free Fisher Factorization (MFF), a GPU-tractable algorithm that captures both diagonal and off-diagonal dependencies without materializing the full matrix.
For post-training neural network layer compression, we prove that under Matrix-Variate Normal assumptions, MFF yields GFWSVD, a unique closed-form linear layer decomposition that optimally minimizes the expected second-order loss increase.
Experiments on controlled numerical benchmarks with large neural networks show t

hat GFWSVD achieves up to 50\% compression while matching or exceeding state-of-the-art diagonal and activation-based baselines across most tasks, and it reliably avoids collapse in dense architectures such as Llama 3.
Moreover, when used to initialize existing optimization pipelines (e.g., Dobi-SVD), GFWSVD better preserves accuracy at 40\% parameter reduction in regimes where standard methods substantially degrade. Together, these results position MFF and GFWSVD as foundational algorithmic primitives for scalable, second-order-aware neural network approximation and parameter sensitivity.
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Trajectory-Level Speculative Decoding for Diffusion Language Models
Tianxiang Pan, Baitao Gong, Mo Guang, Hongwei Yong, Tianpeng Jiang, Yaqian Li, Zheng Cao, Kaiwen Long
Diffusion-based language models (dLLMs) enable parallel token generation through iterative denoising, but existing decoding strategies collapse to single-token generation under low confidence, severely limiting throughput. Unlike autoregressive models where speculative decoding operates on token sequences in a fixed left-to-right order, dLLMs require speculating over \emph{denoising trajectories}—sequences of multi-token updates with explicit positions and unmasking orders. We develop a trajectory-level speculative framework that constructs draft denoising trajectories via confidence-stratified tree exploration and verifies them through blockwise parallel evaluation with bidirectional attention masking. Our method further introduces inter-block speculation, exploiting diffusion models' bidirectional structure to perform cross-block lookahead. We formally characterize when this approach is exact and identify trajectory drift as the fundamental cost of increased parallelism. Building on Fast-dLLM's dual-cache infrastructure, our framework reduces denoising iterations by 30-40\% and increases tokens-per-step from 2.6 to 4.3, achieving 7-14$\times$ speedup over vanilla dLLMs and 1.3$\times$ over Fast-dLLM with less than 1\% accuracy change across reasoning and code benchmarks.
****
Just-In-Time Reinforcement Learning: Continual Learning in LLM Agents Without Gradient Updates
Yibo Li, Zijie Lin, Ailin Deng, Xuan Zhang, Yufei He, Shuo Ji, Tri Cao, Bryan Hooi
While Large Language Model (LLM) agents excel at general tasks, they inherently struggle with continual adaptation due to the frozen weights after deployment. Conventional reinforcement learning (RL) offers a solution but incurs prohibitive computational costs and the risk of catastrophic forgetting.

We introduce Just-In-Time Reinforcement Learning (JitRL), a training-free framework that enables test-time policy optimization without any gradient updates.

JitRL maintains a dynamic, non-parametric memory of experiences and retrieves relevant trajectories to estimate action advantages on-the-fly.

These estimates are then used to directly modulate the LLM's output logits.

We theoretically prove that this additive update rule is the exact closed-form solution to the KL-constrained policy optimization objective.

Extensive experiments on WebArena and Jericho demonstrate that JitRL establishes a new state-of-the-art among training-free methods.

Crucially, JitRL outperforms the performance of computationally expensive fine-tuning methods (e.g., WebRL) while reducing monetary costs by over 30 times, offering a scalable path for continual learning agents. The code is available at https://github.com/liushiliushi/JitRL.
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Investigating Advanced Reasoning of Large Language Models via Black-Box Environment Interaction
Congchi Yin, Tianyi Wu, Yankai Shu, Alex Gu, Yun Han Wang, Jun Shao, xun jiang, Piji Li
Existing tasks fall short in evaluating reasoning ability of Large Language Models (LLMs) in an interactive, unknown environment. This deficiency leads to the isolated assessment of deductive, inductive, and abductive reasoning, neglecting

the integrated reasoning process that is indispensable for humans discovery of r
eal world.
We introduce a novel evaluation paradigm, black-box interaction, to tackle this
challenge.
A black-box is defined by a hidden function that maps a specific set of inputs t
o outputs. LLMs are required to unravel the hidden function behind the black-box
 by interacting with it in given exploration turns, and reasoning over observed
input-output pairs.
Leveraging this idea, we build the Oracle benchmark which comprises 6 types of b
lack-box task and 96 black-boxes. 19 modern LLMs are benchmarked. o3, a leading
LLM from OpenAI, ranks first in 5 of the 6 tasks, achieving over 70\% accuracy o
n most easy black-boxes. But it still struggles with some hard black-box tasks,
where its average performance drops below 40\%.
Further analysis indicates a universal difficulty among LLMs: They lack the high
-level planning capability to develop efficient and adaptive exploration strateg
ies for hypothesis refinement.
Code is available in https://github.com/lemonsis/Oracle_Benchmark.
****

Turbo Connection: Reasoning as Information Flow from Higher to Lower Layers
Tang Mohan, Sidi Lu
Complex problems, whether in math, logic, or planning, are solved by humans thro
ugh a sequence of steps where the result of one step informs the next. In this w
ork, we adopt the perspective that the reasoning power of Transformers is fundam
entally limited by a fixed maximum number of steps along any latent path of comp
utation. To address this, we introduce Turbo Connection (TurboConn), a novel arc
hitecture that overcomes the fixed-depth constraint by routing multiple residual
 connections from the higher-layer hidden states of each token t to the lower
layers of token $t+1$. Fine-tuning pre-trained LLMs with our method not only yie
lds accuracy gains of 0.9\% to over 10\% on benchmarks like GSM8K, Parity, and m
ulti-step arithmetic, but also demonstrates that the density of these backward c
onnections is critical; our dense interaction significantly outperforms "sparse"
 alternatives that only pass a single hidden state or vector. Notably, TurboConn
 can be integrated into pre-trained LLMs to overcome task-specific plateaus: whi
le a fine-tuned Qwen-3-1.7B achieves only 53.78\% on Parity, adding our architec
tural modification enables the model to reach 100\% accuracy, all without the ne
cessity to retrain the full model from scratch or sophisticated curriculum learn
ing. Our results provide strong empirical evidence that the depth of the computa
tional path is a key factor in reasoning ability, also offering a new mechanism
to enhance LLMs without significantly affecting generation latency.
****

VENOMREC: Cross-Modal Interactive Poisoning for Targeted Promotion in Multimodal
 LLM Recommender Systems
Guowei Guan, Yurong Hao, Jiaming Zhang, Tiantong Wu, Fuyao Zhang, Tianxiang Chen
, Longtao Huang, Cyril Leung, Wei Yang Bryan Lim
Multimodal large language models (MLLMs) are pushing recommender systems (RecSys
) toward content-grounded retrieval and ranking via cross-modal fusion.
We find that while cross-modal consensus often mitigates conventional poisoning
that manipulates interaction logs or perturbs a single modality, it also introdu
ces a new attack surface where synchronised multimodal poisoning can reliably st
eer fused representations along stable semantic directions during fine-tuning.
To characterise this threat, we formalise cross-modal interactive poisoning and
propose VENOMREC, which performs Exposure Alignment to identify high-exposure re
gions in the joint embedding space and Cross-modal Interactive Perturbation to c
raft attention-guided coupled token--patch edits.
Experiments on four real-world multimodal datasets demonstrate that VENOMREC con
sistently outperforms strong baselines,
achieving 0.73 mean ER@20 and improving over the strongest baseline by +0.52 abs
olute ER points on average, while maintaining comparable recommendation utility.
Code is available at https://github.com/GuoweiGuan666/VenomRec.
****

scDEBART: Predicting in silico Single-Cell Perturbation Responses via Large-Scale Differential Expression Learning
Jieun Sung, Wankyu Kim
Single-cell foundation models trained on millions of cells can learn gene expression patterns across diverse contexts. However, for predicting genetic perturbation effects they often underperform simple regression models. We hypothesize two potential limitations: targets defined on dropout-prone absolute expression, and pretraining objectives that reconstruct static co-expression rather than encoding how genes co-regulate under expression changes. We introduce $\textbf{scDEBART}$, a perturbation-specific pretraining framework that predicts log fold-changes (logFC) conditioned on basal expression, thereby learning how gene sets co-vary across expression-change contexts at scale. To obtain reliable estimates of expression change under technical sparsity, we compute logFC from scVI-denoised expression and restrict pretraining to genes with robust detection. Pretrained on 6.28 million expression-change profiles from 66.6 million human cells and fine-tuned on five Perturb-seq datasets, scDEBART achieves mean enrichment factor (EF) of 11.96, 4-7$\times$ higher than scGPT and GEARS (mean EF 1.74-2.99), and 71.4\% top-1 accuracy for reverse perturbation identification compared to near-zero accuracy for prior models. In cross-modal transfer to drug perturbations (SCIPLEX), the model shows dose-dependent improvement in directional alignment (cosine similarity 0.04→0.30) with above-random DEG enrichment (EF 2.91-4.32), suggesting partial transfer of learned regulatory patterns across modalities. Overall, these results indicate that large-scale pretraining on scVI-denoised expression-change profiles provides a useful inductive bias for perturbation prediction.
****
CSD: Content-aware Speculative Decoding for Efficient Image Generation
Mingcheng Wang, Junbo Qiao, Yunchen Li, Lingfu Jiang, Wei Li, Jie Hu, Jiao Xie, Zhou Yu, Xinghao Chen, Guixu Zhang, Shaohui Lin
Speculative decoding (SD) has emerged as a key solution to accelerate the inference of autoregressive models. However, in the field of image generation, it faces the challenge of low acceptance rates, and directly relaxing its criteria leads to degradation in image quality. In this paper, we propose a novel content-aware speculative decoding algorithm, termed CSD, which integrates an entropy-based probability relaxation mechanism with an optimal resampling strategy to enhance the inference efficiency for autoregressive image generation. By leveraging the informational uncertainty inherent in different regions of an image, CSD dynamically adjusts the acceptance probability of candidate tokens, increasing the acceptance rate in low-detail areas to accelerate generation. Moreover, a distribution alignment filter is introduced to ensure the output distribution to be aligned with the target model, which significantly improves the generative quality. Experiments conducted on Lumina-mGPT and Janus-Pro demonstrate that the superiority of the proposed CSD. Our source code is available at https://github.com/aderfebr/CSD.
****
Circle-RoPE: Cone-like Decoupled Rotary Positional Embedding for Vision-Language Models
Chengcheng Wang, Jianyuan Guo, Hongguang Li, Yuchuan Tian, Ying Nie, Chang Xu, Kai Han
Rotary Position Embedding (RoPE) is widely adopted in large language models, but when applied to vision-language models (VLMs) it couples text and image position indices and can introduce spurious cross-modal relative-position bias.

We propose *Per-Token Distance* (PTD) to quantify cross-modal positional disentanglement, and we prove that $\mathrm{PTD}=0$ is a sufficient condition to eliminate the geometric attention bias induced by RoPE.

Guided by this criterion, we introduce Circle-RoPE, which remaps 2D image-token coordinates onto an annulus orthogonal to the text position axis, yielding a cone-like geometry where each text token is equidistant to all image tokens while preserving intra-image spatial structure.

We further propose Alternating Geometry Encoding (AGE) to synergize complementary geometric priors by alternating the decoupled geometry of Circle-RoPE and th

e grid-based prior of standard RoPE across layers. This design ensures both rigorous cross-modal disentanglement and the preservation of fine-grained intra-image spatial structure, and experiments on diverse VLM backbones and multimodal benchmarks show consistent gains in spatial grounding and visual reasoning.
  The code is available at https://github.com/lose4578/CircleRoPE.
****
STABLE: Simulation-Ready Tabletop Layout Generation via a Semantics–Physics Dual System
Zhen Luo, Yixuan Yang, Xudong XU, Jinkun Hao, Zhaoyang Lyu, Feng Zheng, Jiangmiao Pang, Yanwei Fu
Generating simulation-ready tabletop scenes from task instructions is an intriguing and promising research direction in the field of Embodied AI.
However, existing task-to-scene generation methods rely exclusively on large language models (LLMs) to predict scene layouts, inevitably yielding object collisions or floating due to LLMs' inherent limitations in 3D spatial reasoning.
In this paper, we present \textbf{STABLE}, a semantics–physics dual-system tailored for simulation-ready tabletop scene generation.
STABLE consists of two complementary modules: (i) a \textbf{Semantic Reasoner}, a fine-tuned LLM trained on a structured tabletop scene dataset to generate coarse layouts from input task instructions, and (ii) a \textbf{Physics Corrector}, a physics-aware flow-based denoising model that outputs pose updates to refine layouts, which ensures the physical plausibility of scenes while preserves semantic alignment with task instructions.
STABLE adopts a progressive generation paradigm: by alternating between the Semantic Reasoner and Physics Corrector, it incrementally expands the scene from task-critical objects to background objects.
Experiments demonstrate that STABLE successfully generates simulation-ready tabletop scenes that strictly conform to task instructions and significantly enhances the physical validity of scenes over prior art.
****
Mitigating the Modality Gap in Vision–Language Models with Fractal Spectral Geometry
Zihan Zhou, Yang Zhou, Ruoming Jin, Pan He, Patrick Emami
Vision–language models such as CLIP embed images and text into a shared space, but still suffer from a modality gap, where image and text features cluster separately and nearest neighbors are dominated by same-modality rather than true cross-modal matches. Existing works alleviate the modality gap by strengthening cross-modal losses, post-processing embeddings or similarities, or imposing geometric regularization, but they primarily enforce global alignment and can distort local geometry, limiting gains in local ranking and zero-shot accuracy. We propose Fractal Spectral Alignment (FSAlign), which reduces the modality gap by shaping and matching the multi-scale geometry of image and text embeddings. By enforcing Ahlfors-regularity and sub-Gaussian heat kernel bounds, FSAlign constructs a shared fractal multi-scale structure for multiple modalities. This structure captures geometry across scales, from local neighborhoods to global structure, and ensures shared fractal spectral geometry across modalities. Based on this structure, we introduce a fractal spectral zeta score derived from multi-scale heat kernels and minimize the discrepancy between pairwise image–text samples to align their multi-scale neighborhoods. We theoretically demonstrate that FSAlign can guarantee the alignment of local spectral measures and global fractional Dirichlet energies.
****
CoEvol-NO: State and Coordinate Co-Evolution with an Error-Driven Predictor-Corrector Paradigm for Neural Operator Transformer
Jianqiao Zeng, Ruocheng Wang, Yanzhi Liu, Hao Xiong, Junchi Yan
Despite the fast progress in neural operator learning, long-sequence modeling still is a standing challenge whereby latent states have been introduced with techniques well derived. Diverging from existing methods that treat latent states as transient variables or decoupled representations, CoEvol-NO introduces a persistent state to establish a co-evolutionary framework, where the latent state and

mesh sequence are updated jointly and bidirectionally. Inspired by classical nu
merical methods, we model the layer-wise state evolution as a Predictor-Correcto
r (PC) process. Specifically, a "Predictor'' generates a tentative target, follo
wed by a "Corrector'' that refines the persistent state via an {error-driven upd
ate mechanism}. Furthermore, our theoretical analysis reveals that the widely us
ed \textit{direct substitution} and \textit{residual update} paradigms are essen
tially {first-order approximations} of this error-driven correction under differ
ent loss assumptions. We theoretically prove that CoEvol-NO achieves strict line
ar time complexity. Extensive experiments on five standard benchmarks and two la
rge-scale industrial design tasks demonstrate that CoEvol-NO consistently achiev
es state-of-the-art (SOTA) performance.
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URS: A Unified Neural Routing Solver for Cross-Problem Zero-Shot Generalization
Changliang Zhou, Canhong Yu, Shunyu Yao, Xi Lin, Zhenkun Wang, Yu Zhou, Qingfu Z
hang
Multi-task neural routing solvers have emerged as a promising paradigm for their
ability to solve multiple vehicle routing problems (VRPs) using a single model.
However, existing neural solvers typically rely on predefined problem constrain
ts or require per-problem fine-tuning, which substantially limits their zero-sho
t generalization ability to unseen VRP variants. To address this critical bottle
neck, we propose URS, a unified neural routing solver that achieves zero-shot ge
neralization across a wide range of unseen VRPs with a single model. We propose
a unified data representation (UDR) that replaces problem enumeration with data
unification, thereby broadening the problem coverage and reducing reliance on do
main expertise. In addition, we introduce a mixed bias module (MBM) during encod
ing to improve node embeddings, which efficiently captures multiple priors inher
ent to various problems. On top of the UDR, we develop a problem-conditioned par
ameter generator to further improve zero-shot generalization. Extensive experime
nts show that URS consistently produces high-quality solutions for 110 VRP varia
nts (including 99 unseen variants) while demonstrating impressive scalability to
large-scale instances with up to 7000 nodes. To the best of our knowledge, URS
is the first neural solver to handle over 100 VRP variants with a single model.
Our code is available at https://github.com/CIAM-Group/URS.
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NeuroCLUS: A Foundation Model with Functional Clustering for Intracranial Neural
Decoding
Hui Zheng, Haiteng Wang
Foundation models for intracranial neural recordings aim to learn generalizable
representations from large-scale unlabeled data. However, existing approaches re
ly on suboptimal tokenization schemes -- treating individual electrode channels
as independent tokens or aggregating them into a single brain-wide representatio
n -- which fail to capture the brain’s inherent functional modularity. We introd
uce NeuroCLUS, a foundation model that learns to represent neural activity throu
gh data-driven functional clusters. NeuroCLUS is built on a novel two-stage pre-
training framework. First, a spatial-temporal model learns a functional context
graph between channels via a functional context prediction task. Second, this gr
aph guides a soft clustering of channels into a set of learnable prototype token
s, enabling the transformer backbone to process coherent functional units rather
than raw channels. Evaluated across a diverse range of decoding paradigms -- in
cluding speech perception, speech production, and seizure detection -- NeuroCLUS
consistently achieves state-of-the-art performance. The discovered functional c
lusters align with established neurophysiology and offer enhanced interpretabili
ty. Our work demonstrates that explicitly modeling functional neural groupings s
ignificantly improves the efficiency, generalization, and interpretability of fo
undation models for intracranial decoding.
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Positive-Unlabeled Learning with Extreme Scarcity of Labeled Positives
Yuanchao Dai, Ximing Li, Wei Wang, Changchun Li, Gang Niu, Masashi Sugiyama
Positive-Unlabeled (PU) learning is a weakly-supervised paradigm that trains a b
inary classifier from labeled positive and unlabeled instances. In PU risk estim

ation, the empirical risk consists of an unlabeled term and a positive term. In this paper, we observe that when labeled positives are scarce, the risk deviation is dominated by the generalization bound of the positive term, which is composed of a complexity term governed by Rademacher complexity and a concentration term governed by the uniform range bound, leading to estimator instability. Based on this observation, we theoretically derive the sufficient sample threshold, defined as the smallest number of labeled positives required to achieve a target excess risk with high probability, and reveal its explicit dependence on both components. Inspired by this insight, we propose ScalePU, which incorporates variance regularization to induce a restricted sub-hypothesis space with reduced Rademacher complexity, and geometric regularization to encourage compact clustering of positive samples with a tighter effective range. Theoretical analysis demonstrates that both mechanisms effectively lower the threshold through improvements to different components of the bound. Experiments on eight benchmark datasets validate the effectiveness of ScalePU, with significant improvements under extreme label scarcity.

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Shortcut-Resistant CAM Distillation for Long-Tailed Recognition

Wenhai Wan, Teng Zhang, Shao-Yuan Li, Xinrui Wang, Qiang-Sheng Hua, Songcan Chen

Real-world datasets often follow a long-tailed distribution, making generalization to tail classes difficult. We revisit this problem through the lens of shortcut learning, where models prefer the easiest predictive cues (e.g., background or textures) over object-centric semantics, especially under scarce and biased supervision. We find that this tendency is amplified for tail classes: limited examples often share similar contexts, making non-semantic signals highly correlated and thus tempting shortcuts, whereas head classes with diverse appearances and environments encourage more stable object-focused representations. Motivated by this observation, we propose Shortcut-Resistant CAM Distillation (SRCD), a plug-and-play framework that transfers object-focused explanations from head to tail classes. SRCD operates in the Class Activation Map (CAM) space, where a CAM provides a class-specific spatial evidence map for a prediction. SRCD aggregates CAMs from a small set of head-class candidates into a shortcut-resistant teacher using an energy-model weighting based on coherence and concentration, and distills it to the tail-class CAM. We provide a theoretical analysis that quantifies shortcut reliance as shortcut-region evidence mass in CAM space and shows that SRCD suppresses tail shortcuts. Extensive experiments on long-tailed benchmarks consistently improve strong baselines. The code is available at \url{https://github.com/Haifeng3/SRCD}.

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Orthogonal Model Merging

Sihan Yang, Kexuan Shi, Weiyang Liu

Merging finetuned Large Language Models (LLMs) has become increasingly important for integrating diverse capabilities into a single unified model. However, prevailing model merging methods rely on linear arithmetic in Euclidean space, which often destroys the intrinsic geometric properties of pretrained weights, such as hyperspherical energy. To address this, we propose Orthogonal Model Merging (OrthoMerge), a method that performs merging operations on the Riemannian manifold formed by the orthogonal group to preserve the geometric structure of the model's weights. By mapping task-specific orthogonal matrices learned by Orthogonal Finetuning (OFT) to the Lie algebra, OrthoMerge enables a principled yet efficient integration that takes into account both the direction and intensity of adaptations. In addition to directly leveraging orthogonal matrices obtained by OFT, we further extend this approach to general models finetuned with non-OFT methods (e.g., low-rank finetuning, full finetuning) via an Orthogonal-Residual Decoupling strategy. This technique extracts the orthogonal components of expert models by solving the orthogonal Procrustes problem, which are then merged on the manifold of the orthogonal group, while the remaining linear residuals are processed through standard additive merging. Extensive empirical results demonstrate the effectiveness of OrthoMerge in mitigating catastrophic forgetting and maintaining model performance across diverse tasks.

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Towards Efficient and Expressive Offline RL via Flow-Anchored Noise-conditioned Q-Learning
Sungyoung Lee, Dohyeong Kim, Eshan Balachandar, Zelal Su Mustafaoglu, Keshav Pingali
We propose Flow-Anchored Noise-conditioned Q-Learning (FAN), a highly efficient and high-performing offline reinforcement learning (RL) algorithm. Recent work has shown that expressive flow policies and distributional critics improve offline RL performance, but at a high computational cost. Specifically, flow policies require iterative sampling to produce a single action, and distributional critics require computation over multiple samples (e.g., quantiles) to estimate value. To address these inefficiencies while maintaining high performance, we introduce FAN. Our method employs a behavior regularization technique that uses a single flow policy iteration and requires a single Gaussian noise sample for distributional critics. Our theoretical analysis of convergence and performance bounds demonstrates that these simplifications not only improve efficiency but also lead to superior task performance. Experiments on robotic manipulation and locomotion tasks demonstrate that FAN achieves state-of-the-art performance while significantly reducing both training and inference runtimes. We release our code at https://github.com/brianlsy98/FAN.
****
Learning Compressed Shape-Aware Molecular Representations for Virtual Screening
Robin Winter, Julian Cremer, Djork-Arné Clevert
Virtual screening of billion-scale molecular libraries based on 3D shape similarity remains
computationally prohibitive, requiring expensive conformational sampling and alignment,
as done by established tools like *ROCS*.
Here, we introduce *SAND* (**S**hape-**A**ware **N**eural
Descriptor), a method that can retrieve shape similar molecules from their 2D graph alone. Our
approach makes two key contributions: (1) a rank-preserving contrastive learning framework using
differentiable Spearman correlation that results into representations where similarity
strongly correlates with 3D molecular shape overlap (R=0.86), and (2) an end-to-end learned
quantization-aware training scheme that jointly optimizes the encoder with a two-level IVF-PQ
discretization step, achieving approximately $4\times$ better compression than post-hoc quantization at
equivalent retrieval quality. We demonstrate that *SAND* enables searching over 10
billion molecules in less than a second on a single GPU node - a speedup of $>10^{8}
    \times$ compared to traditional methods. We release open-source code and trained weights to facilitate
adoption.
****
Mitigating Plasticity Loss through Architectural Design in Continual Learning
Niklas Koeppe, Luiz Felipe Vecchietti, Dongqi Han, Dongsheng Li, Sang Wan Lee
Neural networks for continual reinforcement learning (CRL) often suffer from plasticity loss, i.e., a progressive decline in their ability to learn new tasks arising from increased representational drift (churn) and Neural Tangent Kernel (NTK) rank collapse. Current methods mitigating this problem involve algorithmic interventions such as regularization, resets, and optimization schedules. Here, we propose InterpLayers, a lightweight architectural solution that combines a fixed, parameter-free reference pathway with a learnable projection pathway using input-dependent interpolation weights. This structure makes InterpLayers orthogonal to existing algorithmic solutions. We show through theoretical analysis that

InterpLayers upper-bound the output variability, bound churn, and prevent a collapse of the NTK rank through continual non-zero rank contribution from the interpolation mechanism. Across different distributional shifts, including permutation, windowing, and expansion, InterpLayers outperform similar gated architectures and achieve similar performance as current state-of-the-art methods without the need for optimization-level intervention or the introduction of sensitive hyperparameters. Ablation studies highlight that these improvements are sustained when InterpLayers are combined with existing algorithmic methods for preventing plasticity loss. These results position InterpLayers as a simple, complementary solution for maintaining plasticity in CRL.
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AdaNav: Adaptive Reasoning with Uncertainty for Vision-Language Navigation
Xin Ding, Jianyu Wei, Yifan Yang, Shiqi Jiang, Qianxi Zhang, Hao Wu, Fucheng Jia, Liang Mi, Yuxuan Yan, Weijun Wang, Yunxin Liu, Zhibo Chen, Ting Cao
Vision-Language Navigation (VLN) requires agents to follow natural language instructions by grounding them in sequential visual observations over long horizons. Explicit reasoning could enhance temporal consistency and perception–action alignment, but reasoning at fixed steps often leads to suboptimal performance and unnecessary computation. To address this, we propose AdaNav, an uncertainty-based adaptive reasoning framework for VLN. At its core is the Uncertainty-Adaptive Reasoning Block (UAR), a lightweight plugin that dynamically triggers reasoning. We introduce Action Entropy as a policy prior for UAR and progressively refine it through a Heuristics-to-RL training method, enabling agents to learn difficulty-aware reasoning policies under the strict data limitations of embodied tasks. Results show that with only 6K training samples, AdaNav achieves substantial gains over closed-source models trained on million-scale data, improving success rate by 20% on R2R val-unseen, 11.7% on RxR-CE, and 11.4% in real-world scenes.
****

HiMAP-Travel: Hierarchical Multi-Agent Planning for Long-Horizon Constrained Travel
The Viet Bui, Wenjun Li, Yong Liu
Sequential LLM agents fail on long-horizon planning with hard constraints like budgets and diversity requirements. As planning progresses and context grows, these agents drift from global constraints. We propose **HiMAP-Travel**, a hierarchical multi-agent framework that splits planning into strategic coordination and parallel day-level execution. A Coordinator allocates resources across days, while Day Executors plan independently in parallel. Three key mechanisms enable this: a transactional monitor enforcing budget and uniqueness constraints across parallel agents, a bargaining protocol allowing agents to reject infeasible sub-goals and trigger re-planning, and a single policy trained with GRPO that powers all agents through role conditioning.
On TravelPlanner, HiMAP-Travel with Qwen3-8B achieves 52.78% validation and 52.65% test Final Pass Rate (FPR). In a controlled comparison with identical model, training, and tools, it outperforms the sequential DeepTravel baseline by +8.67 pp. It also surpasses ATLAS by +17.65 pp and MTP by +10.0 pp. On FlexTravelBench multi-turn scenarios, it achieves 44.34% (2-turn) and 37.42% (3-turn) FPR while reducing latency 2.5$\times$ through parallelization.
****

HieraMAS: Optimizing Intra-Node LLM Mixtures and Inter-Node Topology for Multi-Agent Systems
Tianjun Yao, Zhaoyi Li, Zhiqiang Shen
Multi-agent systems (MAS) built on large language models (LLMs) have demonstrated remarkable performance across diverse tasks. Existing approaches optimize communication topology, role assignment, or LLM routing in isolation, while treating each agent as a monolithic unit—failing to exploit internal LLM mixtures that can enhance individual role capabilities. We propose **`HieraMAS`**, a hierarchical agent collaboration framework with intra-node LLM mixtures and inter-node communication topology. HieraAgent introduces *supernodes*, where each functional role comprises multiple heterogeneous LLMs in a propose-synthesis structure. The optimization of **`HieraMAS`** poses unique credit assignment challenges, as fin

al task performance heavily depends on LLM capabilities, potentially causing err
oneous reinforcement of suboptimal configurations. We address this via a two-sta
ge algorithm: (1) multi-level reward attribution providing fine-grained feedback
 at both node and system levels; and (2) graph classification treating topology
selection as a holistic task rather than per-edge optimization. Experiments on r
easoning and coding benchmarks demonstrate that **`HieraMAS`** significantly out
performs existing methods while achieving better cost-performance trade-offs.
****

Learning the Interaction Prior for Protein-Protein Interaction Prediction: A Mod
el-Agnostic Approach

Ziqi Gao, Chenyi Zi, Zijing Liu, Ziqiao Meng, Yu Li, Jia Li

Protein-protein interactions (PPIs) are fundamental to cellular function, diseas
e mechanisms, and drug discovery. Current learning-based PPI predictors focus on
 learning powerful protein representations but neglect designing specialized cla
ssification heads. They mainly rely on generic aggregating methods like concaten
ation or dot products, which lack biological insight. Motivated by the biologica
l "L3 rule", where multiple length-3 paths between a pair of proteins indicate t
heir interaction likelihood, our study addresses this gap by designing a biologi
cally informed PPI classifier. In this paper, we provide empirical evidence that
 popular PPI datasets strongly support the L3 rule. We propose an L3-path-regula
rized graph prompt learning method called L3-PPI, which can generate a prompt gr
aph with virtual L3 paths based on protein representations and controls the numb
er of paths. L3-PPI reformulates the classification of protein embedding pairs i
nto a graph-level classification task over the generated prompt graph. This ligh
tweight module seamlessly integrates with leading PPI predictors as a plug-and-p
lay component, injecting the interaction prior of complementarity to enhance per
formance. Extensive experiments show that L3-PPI achieves superior performance e
nhancements over state-of-the-art competitors.
****

Learning syntax without semantics: Disentangled tiny language models

Ezra Winston, J Zico Kolter

Language models acquire syntax and world knowledge together, entangling the two
in ways that limit efficiency and controllability. We show that syntax can be le
arned while suppressing semantic plausibility and world■knowledge cues, yielding
 more efficient and controllable models. We train tiny LMs on grammatical nonsen
se — syntactically well-formed text with semantic content ablated via constraine
d relexicalization (SAMBAL). Models trained on this data perform comparably to s
tandard pretraining on syntactic benchmarks (BLiMP, SyntaxGym) while scoring at
chance on world knowledge probes (EWoK). On targeted grammar-plausibility confli
ct probes, content-neutral models prefer grammaticality where standard models pr
efer plausibility, and their representations show more syntactic vs lexical alig
nment. On efficiency, disentanglement yields substantial sample and parameter ga
ins: in low■resource regimes, a 5M■parameter model matches a 30M■parameter basel
ine at the same data budget. On controllability, content-neutral models adapt ra
pidly to a new domain with minimal exposure, suggesting the feasibility of modul
ar post■hoc knowledge specialization.
****

RetrOrchestrator: A Multi-Step Retrosynthesis Agent Dynamically Orchestrating Si
ngle-Step Transition Models

Liao Chang, Luotian Yuan, Yiping Ke, Ying Wei

Multi-step retrosynthesis planning is a fundamental challenge in organic chemist
ry, defined by its enormous search space. Existing methods typically formulate i
t as a Markov Decision Process (MDP) with a fixed choice of transition model (i.
e., a single-step retrosynthesis model), and focus on improving *how to search*
through better policies and value functions.

However, *how the transition space itself is navigated* remains largely unexplor
ed.

This limitation is particularly urgent given our observation of pronounced *skil
l disparity* among single-step prediction models: different models exhibit subst
antially different performance across molecule states.

Motivated by this observation, we introduce RetrOrchestrator, an LLM-powered age
nt that explicitly accounts for model skill disparity by reframing retrosynthesi
s planning as a Partially Observable Markov Decision Process (POMDP).
By regarding each single-step prediction model as a tool, we further propose a s
caffold-aware reinforcement learning algorithm to optimize navigation policy wit
hin the transition space.
As a result, RetrOrchestrator jointly searches which molecule to expand and whic
h single-step model to apply for the molecule at the current step.
Empirically, RetrOrchestrator significantly outperforms static baselines on the
Retro*-190 benchmark, achieving a state-of-the-art 94.21% success rate (vs. 9.47
% off-the-shelf LLM and 82.63\% non-LLM state-dependent router), with 92.49% of
solved routes invoking two or more SSRs—evidence that the policy is not collapsi
ng to a single specialist or a static router.
The same gain persists on a larger out-of-distribution set (PDB-600), with RetrO
rchestrator Pareto-optimal in both wall-clock time and model-query count.
Code: https://github.com/ScottLiao920/verl-retro-agent.
****

TimeSeed: Effective Time Series Forecasting with Sparse Endogenous Variables
Zhaowang Wu, Kaixin Deng, Hua Yan
Time series forecasting has long relied on dense endogenous observations, yet in
 many real-world scenarios, such data is scarce or even absent. Existing approac
hes attempt to compensate with exogenous variables, but their reliance on incomp
lete endogenous histories makes them brittle under data scarcity. In this work,
we introduce sparse endogenous forecasting as a new setting, where exogenous seq
uences and only sparse endogenous observations are available. To tackle this pro
blem, we propose TimeSeed, a lightweight architecture that redefines sparse fore
casting as a context reconstruction task. By jointly exploiting the stability of
 exogenous sequences and the limited but informative endogenous signals, TimeSee
d reconstructs robust historical representations and transforms forecasting into
 a tractable sequence-based prediction problem. Remarkably, TimeSeed achieves th
is with a purely linear architecture using only 0.19M parameters, consistently o
utperforming state-of-the-art deep models on seven real-world benchmarks, with a
n average improvement of 13.01\% in MSE and 7.54\% in MAE. These results establi
sh sparse endogenous forecasting as a practical and promising paradigm, opening
a new direction for time series analysis under extreme data scarcity. Code is av
ailable at this repository: \url{https://anonymous.4open.science/r/Alistair-7}.
****

Pianist Transformer: Towards Expressive Piano Performance Rendering via Scalable
 Self-Supervised Pre-Training
Hong-Jie You, Jie-Jing Shao, Xiao-Wen Yang, Lin-Han Jia, Lan-Zhe Guo, Yu-Feng Li
Existing methods for expressive music performance rendering rely on supervised l
earning over small labeled datasets, which limits scaling of both data volume an
d model size, despite the availability of vast unlabeled music, as in vision and
 language. To address this gap, we introduce Pianist Transformer, with three key
 contributions: 1) introducing large-scale self-supervised learning into express
ive piano performance rendering through a unified Musical Instrument Digital Int
erface (MIDI) representation, enabling pre-training on 10B tokens of unlabeled M
IDI data; 2) an efficient asymmetric Transformer with note-level compression, su
bstantially improving training efficiency, memory usage, and inference speed for
 long-context music modeling; 3) a state-of-the-art rendering model with an edit
able workflow, achieving strong objective and subjective results and enabling in
tegration into real-world music production workflows. Overall, Pianist Transform
er outlines a scalable path toward human-like performance synthesis in the music
 domain. Code, audio samples, and model checkpoints are available on our project
 page: https://yhj137.github.io/pianist-transformer-demo/.
****

Rethinking Video Generation Model for the Embodied World
Yufan Deng, Zilin Pan, Hongyu Zhang, Xiaojie Li, Ruoqing Hu, Yufei Ding, Yiming
Zou, Yan Zeng, Daquan Zhou
While video generation holds promise for embodied intelligence, current video mo

dels struggle with physical realism, and progress is hindered by the lack of sta ndardized benchmarks. To address this gap, we introduce a comprehensive robotics benchmark, RBench, designed to evaluate robot-oriented video generation across five task domains and four distinct embodiments. By assessing task correctness a nd visual fidelity through reproducible metrics, our evaluation of 25 models rev eals significant deficiencies in generating physically realistic robot behaviors . Furthermore, the benchmark achieves a 0.96 Spearman correlation with human jud gment, validating its effectiveness. While RBench provides the necessary lens to identify these deficiencies, achieving physical realism requires moving beyond evaluation to address the critical shortage of high-quality training data. Drive n by these insights, we introduce a refined four-stage data pipeline, resulting in RoVid-X, the largest open-source robotic dataset for video generation with 4 million annotated video clips, covering thousands of tasks and enriched with phy sical property annotations. Extensive experiments demonstrate that finetuning on RoVid-X yields consistent performance gains. Collectively, this synergistic eco system of evaluation and data establishes a robust foundation for rigorous asses sment and scalable training of video models, accelerating the evolution of embod ied AI toward physical intelligence. The code and video demos are available in t he supplementary materials.

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EVOLVING ROLLOUTS: Harnessing Historical Experience for Web Agent Evolution in R einforcement Learning

Sinuo Wang, WANG PIAOHONG, Tianrui Qin, Maojia Song, Qianben Chen, Qiexiang Wang , Gengze Zhou, Zeyu Zhang, He Zhu, Dingfeng Shi, Yutong Xie, Minghao Liu, Jiahen g Liu, Ge Zhang, Jiawei Ma, Yuchen Eleanor Jiang, Qi Wu, Wangchunshu Zhou

Agentic reinforcement learning (RL) for web search is prohibitively expensive du e to long context lengths and costly environment interactions, and this ineffici ency is further exacerbated by group-based optimization, which discards learning signals from entire rollout groups with zero reward variance. In this work, we propose EVOLVING ROLLOUTS, an RL framework for web-search agents that moves beyo nd episodic training and distills collected rollouts into in-context guidance fo r future policy behavior. By extracting the reward-labeled trajectories into str ategic experiences, our method augments standard parameter-space optimization wi th implicit context-space optimization guided by prior experience. This enables the agent to recover learning signals from zero-variance rollouts, thereby foste ring co-evolution between the policy and the experience repository. EVOLVING ROL LOUTS improves sample efficiency and task performance across representative web search benchmarks, with Qwen3-8B surpassing the much larger Qwen3-30B-A3B in ave rage performance across GAIA, xBench, and HLE, and Qwen3-4B attaining comparable results on GAIA and HLE.

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The Hidden Risk: Membership Inference Attacks on Multimodal Federated Learning v ia Modality Imbalance

chang Ma, Jun Li, Kang Wei, Yipeng Zhou, Ming Ding, Yiyang Ni

Federated learning (FL) faces significant challenges from modality heterogeneity , which motivates multimodal federated learning (MFL) to leverage complementary modalities across decentralized clients for improved performance. However, moda lity imbalance introduces a new attack surface, making MFL more vulnerable to me mbership inference attacks (MIAs), an issue that remains largely unexplored. In this work, we present the first systematic study of MIAs against MFL and propose a modality-aware attack framework.

We show that multimodal models are inherently more susceptible to MIAs due to he terogeneous modality contributions, and existing attacks are suboptimal as they treat multimodal parameters as a whole.

By performing MIAs on individual modalities, we find that (i) attacking the domi nant modality achieves comparable accuracy with lower overhead, and (ii) differe nt modalities expose distinct membership patterns.

To identify members with different patterns, we propose a modality-aware framewo rk that exploits cross-modal performance gaps to adaptively select attack modali ties and calibrate inference results. Experiments on three datasets show our app

roach outperforms baselines across multiple metrics.
****
Compress then Merge: From Multiple LoRAs into One Low-Rank Adapter
Zhengbao He, Ruiqi Ding, Zhehao Huang, Ruikai Yang, Tao Li, Xiaolin Huang
Low-rank adaptation (LoRA) enables parameter-efficient specialization of foundation models, but the proliferation of task-specific adapters fragments capabilities across many adapters, complicating reuse and deployment.
We study the problem of **merging T LoRAs into a single rank-r LoRA**, thereby preserving the benefits of low-rank structure.
Existing Merge-then-Compress pipelines treat the rank constraint as an afterthought: they merge adapters in the full parameter space,
then compress the merged result to rank r via truncated SVD.
However, full-parameter merging may destroy the low-rank structure, making it difficult for subsequent compression to recover an effective rank-r LoRA.
We propose Compress-then-Merge (CtM), a reversed pipeline that enforces the rank-r bottleneck _before_ merging: CtM computes shared r-dimensional subspaces using only the LoRA weights to capture cross-adapter common structure, projects each adapter into the shared subspaces to obtain $r\times r$ coordinates, and then applies standard merging rules in this reduced space.
CtM guarantees a rank-r LoRA by construction, avoiding post-hoc truncation, and enables efficient computation in the core space spanned by concatenated LoRA factors.
Experiments across multiple models and tasks show that CtM consistently outperforms existing single-LoRA-output baselines while narrowing the performance gap to full-parameter merging methods.
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Scalable Option Learning in High-Throughput Environments
Mikael Henaff, Scott Fujimoto, Michael Matthews, Michael Rabbat
Hierarchical reinforcement learning (RL) has the potential to enable effective decision-making over long timescales. Existing approaches, while promising, have yet to realize the benefits of large-scale training. In this work, we identify and solve several key challenges in scaling online hierarchical RL to high-throughput environments. We propose Scalable Option Learning (SOL), a highly scalable hierarchical policy gradient algorithm which achieves a ~35x higher throughput compared to existing hierarchical methods. To demonstrate SOL's performance and scalability, we train hierarchical agents using 30 billion frames of experience on the complex game of NetHack, significantly surpassing flat agents and demonstrating positive scaling trends. We also validate SOL on MiniHack and Mujoco environments, showcasing its general applicability.
****
Frequentist Consistency of Prior-Data Fitted Networks for Causal Inference
Valentyn Melnychuk, Vahid Balazadeh, Stefan Feuerriegel, Rahul G Krishnan
Foundation models based on prior-data fitted networks (PFNs) have shown strong empirical performance in causal inference by framing the task as an in-context learning problem. However, it is unclear whether PFN-based causal estimators provide uncertainty quantification that is consistent with classical frequentist estimators. In this work, we address this gap by analyzing the frequentist consistency of PFN-based estimators for the average treatment effect (ATE). (1) We show that existing PFNs, when interpreted as Bayesian ATE estimators, can exhibit prior-induced confounding bias: the prior is not asymptotically overwritten by data, which, in turn, prevents frequentist consistency. (2) As a remedy, we suggest employing a calibration procedure based on a one-step posterior correction (OSPC). We show that the OSPC helps to restore frequentist consistency and can yield a semi-parametric Bernstein-von Mises theorem for calibrated PFNs (i.e., both the calibrated PFN-based estimators and the classical semi-parametric efficient estimators converge in distribution with growing data size). (3) Finally, we implement OSPC through tailoring martingale posteriors on top of the PFNs. In this way, we are able to recover functional nuisance posteriors from PFNs, required by the OSPC. In multiple (semi-)synthetic experiments, PFNs calibrated with our martingale posterior OSPC produce ATE uncertainty that (i) asymptotically matches f

requentist uncertainty and (ii) is well calibrated in finite samples in comparis
on to other Bayesian ATE estimators.
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MOD-SR: Unifying Multimodal Learning and Direct Optimization with Gradient-Guide
d Diffusion Model for Symbolic Regression
Chuyang Xiang, Yichen Wei, Junchi Yan
Symbolic regression (SR) aims to discover interpretable mathematical expressions
 from observed data.
While recent generative approaches have shown promise in treating SR as machine
translation or multimodal learning tasks using NN methods, they suffer from unco
ntrollable generation process and training-evaluation misalignment.
The training objectives (average cross-entropy loss on a token level across the
distribution of historical data) differ from the evaluation metric (fitting erro
r for every test data / complexity), necessitating extensive heuristic post-proc
essing.
On the other hand, direct optimization methods suffer from an exponential slowdo
wn as the dimensionality increases, non-differentiability and local optima traps
.
We propose **MOD-SR**, unifying multimodal distribution learning during training
 with direct optimization at inference time. This is achieved by modeling the ta
sk as $p(x_0 \mid \mathcal{D}, y^*)$ and employing gradient-guided diffusion in
embedding space, enhanced by contrastive learning and representation alignment.
Furthermore, we introduce DFEX, a fixed-depth tree relaxation method that ensure
s differentiability for effective gradient guidance during inference.
Experiments demonstrate that MOD-SR achieves superior performance on diverse ben
chmarks through a unified framework integrating distribution learning and optimi
zation. Our code is available at [https://github.com/KROX777/MOD-SR](https://git
hub.com/KROX777/MOD-SR).
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ABSINT-AI: Agentic Heap Abstractions for Abstract Interpretation
Michael Wang, Kexin Pei, Armando Solar-Lezama
Static program analysis is a foundational technique in software engineering for
reasoning about program behavior. Traditional static analysis algorithms model p
rograms as logical systems with well-defined semantics, but rely on uniform, har
d-coded heap abstractions. This limits their precision and flexibility, especial
ly in dynamic languages like JavaScript, where heap structures are heterogeneous
 and difficult to analyze statically. In this work, we introduce ABSINT-AI, a la
nguage-model-guided static analysis framework that augments abstract interpretat
ion with adaptive, per-object heap abstractions for Javascript. This enables the
 analysis to leverage high-level cues, such as naming conventions and access pat
terns, without requiring brittle, hand-engineered heuristics. Importantly, the L
M agent operates within a bounded interface and never directly manipulates progr
am state, preserving the soundness guarantees of abstract interpretation. To eva
luate our approach, we focus on a soundness-critical task: determining whether o
bject property accesses may result in undefined or null dereferences. This task
directly models a common requirement in compiler optimizations, where proving th
at an access is safe enables the removal of dynamic checks or simplifies code mo
tion. On this task, ABSINT-AI reduces false positives by up to 34% compared to t
raditional static analyses with fixed heap abstractions, while preserving formal
 guarantees. Our ablations show that the LM’s ability to interact agentically wi
th the analysis environment is crucial, outperforming non-agentic LM predictions
 by 25%.
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PatternKV: Flattening KV Representation Expands Quantization Headroom
Ji Zhang, Yiwei Li, Shaoxiong Feng, Peiwen Yuan, Xinglin Wang, Yueqi Zhang, Jiay
i Shi, Chuyi Tan, Boyuan Pan, Yao Hu, Kan Li
KV cache in autoregressive LLMs eliminates redundant recomputation but has emerg
ed as the dominant memory and bandwidth bottleneck during inference, notably wit
h long contexts and test-time scaling. KV quantization is a key lever for reduci
ng cache cost, but accuracy drops sharply as the native KV distribution lacks fl

atness and thus maintains a wide quantization range. Prior work focuses on isola ting outliers, which caps their error but fails to flatten the overall distribut ion, leaving performance fragile under low-bit settings. In this work, we show t hat the K cache maintains a stable, context-evolving structure, while the V cach e carries latent semantic regularities, with both contributing to the organizati on of vectors into shared patterns. Building on these insights, we propose **Pat ternKV**, a pattern-aligned residual quantization scheme. It mines representativ e pattern vectors online, aligns each KV vector to its nearest pattern, and quan tizes only the residual. This reshaping of the KV distribution flattens the quan tization target and narrows its range, thereby improving the fidelity of low-bit KV quantization. Across long-context and test-time scaling settings on multiple backbones, PatternKV delivers consistent 2-bit gains, incurs only a 0.08\% aver age 4-bit drop relative to FP16, and improves test-time scaling accuracy by 10\% on average. Compared with FP16, PatternKV further raises throughput by 1.5$\tim es$ and supports 1.25$\times$ larger batches.

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Role-Level Inductive Bias for Cross-Task Generalization in Multi-Agent Reinforce ment Learning

Chang Yao, Youfang Lin, Shoucheng Song, Hao Wu, Shengkun Yang, Yuqing Ma, Kai Lv

Achieving cross-task generalization remains a critical challenge in Multi-Agent Reinforcement Learning (MARL), fundamentally relying on effective inductive bias es. However, existing entity-level biases often overlook collaborative patterns, whereas task-level biases lack sufficient coverage for novel scenarios. To addr ess this, we introduce a role-level inductive bias as an intermediate abstractio n that integrates entity-level flexibility with task-level inter-agent collabora tion. To instantiate this, we propose Gaussian-mixture-model-based Transferable Role discovery (GTR). Specifically, GTR constructs a structured role space to en sure diverse role assignment, further achieves role decoupling via regularizatio n, and ultimately utilizes these roles for efficient generalization. Empirical r esults demonstrate that GTR achieves superior zero-shot and few-shot transfer pe rformance on unseen tasks compared to state-of-the-art methods.

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Enhanced Latent-Space Adversarial Training for Super-Resolution

Liangbin Xie, Zheyuan Li, Fanghua Yu, Xinqi Lin, Jun-hao Zhuang, Jinfan Hu, Jinj in Gu, Jiantao Zhou, Chao Dong

Real-world super-resolution (SR) at large upscaling factors (i.e., ≥ 4×) remains difficult due to complex real-image degradations. HYPIR, a leading diffusion-ba sed restoration model, performs strongly on many inputs, yet for a non-trivial p ortion of more challenging cases a single forward step does not fully recover fi ne-grained details. A naive two-stage cascade improves visual quality, but intro duces over-saturation, weak texture details, and high inference latency. To addr ess these issues, this paper proposes HYPIR++. It removes the degradation remova l encoder and noise augmentation modules to better preserve fidelity cues from l ow-quality inputs. Equipped with an optimized latent ConvNeXt and a latent patch discriminator, HYPIR++ supports latentspace adversarial learning for clearer de tails and more stable local structures. It further shortens text sequences and r eplaces full attention with sparse neighbor attention, enabling efficient highre solution inference without tiling. Experiments show that HYPIR++ improves percep tual quality and runs 1.71× faster than HYPIR on large-factor real-world SR

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Geometry-Misalignment in Distributional Learning

Tao Wang, Xiaoting Zhong

Distributional learning problems optimize discrepancies between probability meas ures, including optimal transport or Sinkhorn divergence, yet are typically opti mized using Euclidean first-order methods in parameter space. We show this misma tch is structural rather than algorithmic. We introduce geometry-misalignment, a local condition number that measures distortion between Euclidean geometry and the intrinsic geometry induced by a distributional objective. For a broad class of problems, we establish lower bounds demonstrating that Euclidean first-order methods incur an unavoidable convergence slowdown proportional to misalignment,

even under intrinsic strong convexity and smoothness. We further prove geometry-aware preconditioned methods attain matching upper bounds independent of misalignment, yielding a sharp separation between Euclidean optimization and geometry-aware optimization. Beyond convergence rates, we show geometry-misalignment induces an optimization-dependent excess risk term under finite budgets, directly linking optimization geometry with statistical efficiency. We develop a geometry-calibrated optimization framework that estimates misalignment and selectively activates geometry-aware updates when necessary. Experiments on distribution matching for domain adaptation validate the theory, with improvements concentrated in high-misalignment regimes and negligible overhead.
****

Self-supervised Hierarchical Visual Reasoning with World Model
Yuanfei Xu, Lin Liu, Wengang Zhou, Mingxiao Feng, Houqiang Li
3D open-world environments with adversarial opponents remain a core challenge for reinforcement learning due to their vast state spaces.
Effective reasoning representations are essential in such settings. While existing self-supervised visual foresight reasoning approaches often suffer from multi-step error accumulation, many recent studies resort to injecting domain-specific knowledge for more stable guidance.
Our key insight is that the photorealistic fidelity of visual reasoning representations is secondary; what truly matters is providing informative, task-relevant signals.
To this end, we propose ResDreamer, a hierarchical world model in which each higher-level layer is trained to reconstruct the residuals of the layer below. This design enables progressive abstraction of increasingly sophisticated world dynamics and fosters the emergence of richer latent representations.
Drawing inspiration from the “Bitter Lesson,” ResDreamer trains its reasoning representations in a purely self-supervised manner. The higher-level residual representations are used to modulate lower-level predictions, allowing the world model to scale effectively with only linearly increasing cross-layer communication costs.
Experiments show that ResDreamer achieves state-of-the-art sample efficiency and parameter efficiency. This scalable hierarchical visual foresight reasoning architecture paves the way for more capable online RL agents in open-ended, dynamic environments.
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DeltaEvolve: Accelerating Scientific Discovery through Momentum-Driven Evolution
Jiachen Jiang, Tianyu Ding, Zhihui Zhu
LLM–driven evolutionary systems have shown promise for automated science discovery, yet existing approaches such as AlphaEvolve rely on full-code histories that are context-inefficient and potentially provide weak evolutionary guidance. In this work, we first formalize the evolutionary agents as a general Expectation–Maximization framework, where the language model samples candidate programs (E-step) and the system updates the control context based on evaluation feedback (M-step). Under this view, constructing context via full-code snapshots constitutes a suboptimal M-step, as redundant implement details dilutes core algorithmic ideas, making it difficult to provide clear inspirations for evolution. To address this, we propose DeltaEvolve, a momentum-driven evolutionary framework that replaces full-code history with structured semantic delta capturing how and why modifications between successive nodes affect performance. As programs are often decomposable, semantic delta usually contains many effective components which are transferable and more informative to drive improvement. By organizing semantic delta through multi-level database and progressive disclosure mechanism, input tokens are further reduced. Empirical evaluations on tasks across diverse scientific domains show that our framework can discover better solution with less token consumption over full-code-based evolutionary agents.
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When LLMs Develop Languages: Symbolic Communication for Efficient Multi-Agent Reasoning
Zhengqi Pei, Qingming Huang, Shuhui Wang

Chain-of-Thought (CoT) improves large language models (LLMs) on difficult reasoning tasks, but it often incurs long natural-language rationales that are poorly aligned with efficient machine reasoning.
We propose Communicative Language Symbolism Routing (CLSR), a test-time framework in which multiple LLM agents autonomously invent, evolve, and share compact Language Symbolism Frameworks (LSFs), while a latent-free router adaptively selects and composes these languages per query to optimize the accuracy--token trade-off.
Unlike prompt optimization that refines surface instructions, CLSR treats each LSF as a reusable symbolic protocol with compact symbols, usage rules, and a message-passing contract, and improves it through an evolutionary loop driven by correctness and token cost.
At inference time, the router may invoke a single low-cost LSF call, ensemble multiple LSFs, or execute a multi-round LSF composition protocol on harder queries.
Across challenging benchmarks, CLSR reduces latency-oriented generated token completion by $3\sim 6\times$ compared to standard CoT while maintaining accuracy.
We further derive an information-theoretic lower bound on token cost under arbitrary symbolism and show that, under an interpreter-realizability premise, multi-round LSF protocols conditionally subsume program-execution pipelines.
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LORD-GoF: A Robust Online Detection Approach for LLM Watermarks in Sparse and Mixed Streams
Jiade Xu, Zhouping Li
Watermarking is crucial for identifying AI-generated text, however, existing detection methods often focus on offline settings and fail to control the online False Discovery Rate (oFDR) when applied to real-world streams where machine-generated content is sparse and mixed with human writing. To address this issue, in this paper, we propose LORD-GoF, a novel online detection framework that combines a Goodness-of-Fit (GoF) statistic with the Levels based On Recent Discovery (LORD) procedure. We prove that LORD-GoF approach can rigorously control the oFDR below a user-specified level by dynamically adjusting detection thresholds. Extensive experiments on watermarked text from Qwen-2.5-3B, Sheared-LLaMA-2.7B, and OPT-1.3B using both the Gumbel-Max and Inverse Transform watermarking schemes show that our method maintains statistical power comparable to offline benchmarks while successfully controlling the oFDR under complex, mixed streaming scenarios.
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MORE: A Multilingual Document Parsing Benchmark and Evaluation
Long Xu, Binghong Wu, TingHao YU, Hao Feng, Zhenyu Huang, Haoqing Jiang, Yunhao Wang, Shuo Huang, feng zhang
Multilingual documents encapsulate rich regional cultures, scientific discoveries, and historical records. Parsing this content into structured, machine-readable formats is critical for unlocking global knowledge. However, existing benchmarks predominantly focus on high-resource languages like English and Chinese, creating an $\textit{evaluation blind spot}$ concerning model performance on other languages. While recent Vision-Language Models (VLMs) claim support for hundreds of languages, the lack of ground truth makes it impossible to empirically verify these capabilities. To bridge this gap, we introduce $\textbf{MORE}$, a large-scale benchmark designed for multilingual document parsing evaluation. MORE distinguishes itself through three key dimensions: (1) $\textbf{Unprecedented Scale}$: It covers $\textbf{149 languages}$, making it the most linguistically diverse benchmark to date; (2) $\textbf{Structural Complexity}$: Unlike previous works, it extends evaluation beyond plain text to include structural elements such as code blocks, tables, and catalogs; and (3) $\textbf{Data Authenticity}$: All samples are curated from real-world documents via a model-assisted, human-refined annotation pipeline. We evaluate state-of-the-art models using MORE, establishing new performance baselines for long-tail languages and validating the benchmark's effectiveness in diagnosing model capabilities in realistic, diverse scenarios. The MORE dataset will be available at https://github.com/zimoqingfeng/MORE.
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TabSwift: An Efficient Tabular Foundation Model with Row-Wise Attention
Si-Yang Liu, Han-Jia Ye
Tabular foundation models, exemplified by TabPFN, perform prediction via in-context learning, inferring test labels directly from labeled training examples. They have demonstrated competitive performance, particularly on small-to-medium datasets. However, recent tabular foundation models often improve accuracy with increasingly complex architectures, incurring higher inference cost and limiting practical deployment.
In this work, we revisit the original TabPFN design and show that a lightweight row-wise attention–only backbone can remain highly competitive with two simple enhancements: a gated attention stabilization mechanism and a small set of learnable register tokens that provide global context and improve pretraining quality. The resulting model, TabSwift, supports both classification and regression, and is competitive with stronger tabular foundation models (e.g., TabPFN v2 and TabICL) while being more efficient at inference.
For latency-sensitive serving, we further introduce an adaptive layer-wise early-exit mechanism that dynamically adjusts inference depth per sample.
Overall, TabSwift enables efficient and anytime tabular in-context learning for practical deployments.
****
Every Step Counts: Decoding Trajectories as Authorship Fingerprints of dLLMs
Qi Li, Runpeng Yu, Haiquan Lu, Xinchao Wang
Discrete Diffusion Large Language Models (dLLMs) have recently emerged as a promising non-autoregressive paradigm, offering faster inference while achieving strong performance in code generation and mathematical reasoning tasks. In this work, we show that dLLMs' decoding mechanism not only improves utility but also enables effective model attribution: by analyzing a response's decoding trajectory, we can identify its source model and help mitigate risks from model misuse. A key challenge is the diversity of attribution scenarios, ranging from distinguishing different models to identifying different checkpoints or backups of the same model. To ensure broad applicability, we focus on two core questions: what information to extract from the decoding trajectory, and how to use it effectively. We first observe that per-step model confidence is ineffective, as the bidirectional nature of dLLMs causes mutual influence among decoded tokens, leading to highly redundant confidence signals that obscure structural information about decoding order and dependencies. To overcome this, we propose a novel information extraction scheme called the \textit{Directed Decoding Map (DDM)}, which captures structural relationships between decoding steps and reveals model-specific behaviors. Furthermore, to fully leverage the extracted structure, we propose \textit{Gaussian-Trajectory Attribution (GTA)}, which fits a cell-wise Gaussian distribution at each decoding position for each model and uses log-likelihood differences between trajectories as the attribution score. Extensive experiments across diverse models, datasets and different model access assumptions validate the effectiveness of our approach.
****
How (Not) to Hybridize Neural and Mechanistic Models for Epidemiological Forecasting
Yiqi Su, Ray Lee, Jiaming Cui, Naren Ramakrishnan
Epidemiological forecasting from surveillance data is a hard problem and hybridizing mechanistic compartmental models with neural models is a natural direction. The mechanistic structure helps keep trajectories epidemiologically plausible, while neural components can capture non-stationary, data-adaptive effects. In practice, however, many seemingly straightforward couplings fail under partial observability and continually shifting transmission dynamics driven by behavior, waning immunity, seasonality, and interventions. We catalog these failure modes and show that robust performance requires making non-stationarity explicit: we extract multi-scale structure from the observed infection series and use it as an interpretable control signal for a controlled neural ODE coupled to an epidemiological model. Concretely, we decompose infections into trend, seasonal, and residual components and use these signals to drive continuous-time latent dynamics wh

ile jointly forecasting and inferring time-varying transmission, recovery, and i
mmunity-loss rates. Across early outbreak and multi-wave regimes, our approach a
ttains the lowest RMSE on all five datasets (up to 57% reduction over the strong
est baseline), predicts the peak within one time step on four of five, and recov
ers time-varying epidemiological rates within ground-truth ranges, without relyi
ng on auxiliary covariates.
****
On the Intrinsic Limits of Transformer Image Embeddings in Non-Solvable Spatial
Reasoning
Siyi Lyu, Quan Liu, Feng Yan
Vision Transformers (ViTs) excel in semantic recognition but exhibit systematic
failures in spatial reasoning tasks such as mental rotation.
While often attributed to data scale, this work argues that the limitation arise
s from the intrinsic circuit complexity of the architecture.
By formalizing spatial understanding as a **Group Homomorphism Problem**—where l
atent embeddings preserve the algebraic structure of physical transformations ac
ting on images—we identify a fundamental computational bottleneck.
Specifically, for non-solvable groups (e.g., $\mathrm{SO}(3)$), maintaining such
 structure-preserving embeddings is lower-bounded by the Word Problem, which is
$\mathsf{NC^1}$-complete.
In contrast, constant-depth ViTs with polynomial precision are strictly bounded
by the complexity class $\mathsf{TC^0}$.
Under the standard conjecture $\mathsf{TC^0} \subsetneq \mathsf{NC^1}$, a **comp
lexity boundary** emerges: constant-depth architectures lack the logical depth r
equired to capture non-solvable spatial structures in a single forward pass.
To empirically validate this theoretical gap, we propose the **Latent Space Alge
bra (LSA)** benchmark, which reveals a significant degradation in ViT representa
tions as the compositional depth of non-solvable tasks increases.
****
FusionCell: Cross-Attentive Fusion of Layout Geometry and Netlist Topology for S
tandard-Cell Performance Prediction
Haoyi Zhang, Kairong Guo, Bojie Zhang, Yibo Lin, Runsheng Wang
Standard cells form the building blocks of digital circuits, so their delay and
power critically influence chip-level performance; yet characterization (can be
understood as evaluation of cell delay and power) still relies on slow simulatio
n sweeps, and many fast predictors ignore layout geometry, missing coupling and
layout-dependent effects. The challenge is to jointly represent layout geometry
and netlist topology so models capture fine-grained spatial details together wit
h structural connectivity for accurate performance prediction. We introduce \tex
tbf{FusionCell}, a dual-modality predictor that treats routed layout geometry an
d netlist topology as inputs and fuses them explicitly in a unified model. A Dei
T encoder processes three-layer routed layouts, while a graph transformer models
 heterogeneous device/net graphs. The modalities are integrated through a \textb
f{topology-guided} mechanism, where the netlist acts as a structural ``map'' to
actively query relevant physical regions in the layout for joint geometric and t
opological reasoning. We build a 7nm dataset based on the ASAP7 PDK with over 19
.5k cells spanning 149 types using automatic tools, targeting six metrics: signa
l rise/fall delay, transition, and power. Experimental results demonstrate that
\textbf{FusionCell} reduces regression error (average MAPE 0.92\%) and improves
Spearman/Kendall ranking over baselines, while accelerating the characterization
 process by orders of magnitude compared to circuit simulation.
****
Decomposing Out-of-Distribution Error in Conditional Flow Matching via Wasserste
in Geometry
Long HC Pham
Conditional flow matching has emerged as a powerful generative modeling framewor
k that learns a vector field to transport an initial distribution toward a targe
t data distribution. However, theoretical understanding of its out-of-distributi
on (OOD) performance under unseen conditions remains limited. In this work, we e
stablish a rigorous geometric formulation to decompose the source of generalizat

ion error. We treat the conditional task as a map from the condition space to th
e Wasserstein space and derive a generalization bound under a coarse embedding a
ssumption. The resulting decomposition separates OOD error into three tractable
components: *Interpolation Sparsity*, *Geometric Distortion*, and *In-Distributi
on Fit*. Our empirical evaluation confirms that this framework demonstrates thre
e key functions: (1) it acts as a diagnostic tool that tracks the dynamics of ge
neralization during training; (2) it identifies dataset-specific failure modes (
e.g., topological gaps, geometric instability); and (3) it enables mathematicall
y motivated interventions that yield predictable gains by minimizing specific te
rms.
****

Distributional Alignment Games for Answer-Level Fine-Tuning
Mehryar Mohri, Jon Schneider, Yifan Wu
We focus on the problem of \emph{Answer-Level Fine-Tuning} (ALFT), where the
goal is to optimize a language model based on the correctness or properties of
its final answers, rather than the specific reasoning traces used to produce
them. Directly optimizing answer-level objectives is computationally
intractable due to the need to marginalize over the vast space of latent
reasoning paths. To overcome this, we propose a general game-theoretical
framework that lifts the problem to a *Distributional Alignment Game*. We
formulate ALFT as a two-player game between a Policy (the generator) and a
Target (an auxiliary distribution). We prove that the Nash Equilibrium of this
game corresponds exactly to the solution of the original answer-level
optimization problem. This variational perspective transforms the intractable
marginalization problem into a tractable projection problem. We demonstrate
that this framework unifies recent approaches to diversity and
self-improvement (coherence) and provide efficient algorithms compatible with
Group Relative Policy Optimization (GRPO), such as
Coherence-GRPO, yielding significant complexity gains in
mathematical reasoning tasks.
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Attribution-Guided and Coverage-Maximized Pruning for Structural MoE Compression
Yifu Ding, jiacheng wang, Ge Yang, Yongcheng Jing, Jinyang Guo, Xianglong Liu, D
acheng Tao
Mixture-of-Experts (MoE) models scale compute efficiently, yet they remain expen
sive to deploy due to substantial memory footprint and inference overhead. Prior
 methods mainly operate at the expert level, either removing whole experts or ra
nking experts by importance. However, such expert-wise decisions are too coarse
to identify redundancy, and often misallocate pruning budgets and limits compres
sion. This issue worsens in large MoEs with dynamic routing and heterogeneous ex
perts. To alleviate this dilemma, we for the first time observe that information
 in MoE experts is highly concentrated in a few channels, leaving substantial re
dundancy even in "high importance" experts. Accordingly, we propose a structural
 pruning framework tailored for MoEs, reforming the prune-ratio objective to max
imizing channel-score coverage via an efficient attribution-based approximation.
 Experiments on DeepSeek and Qwen MoEs retain accuracy under 50\% or 25\% prunin
g joinly with 4-bit quantization, reducing the memory footprint of Qwen3-30B-A3B
 by 5.27$\times$, and outperforming state-of-the-art baselines under diverse ben
chmarks.
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Sharp Empirical Bernstein Inequalities for the Variance of Bounded Random Variab
les
Diego Martinez-Taboada, Aaditya Ramdas
We develop novel "empirical Bernstein" inequalities for the variance of bounded
random variables. Our inequalities hold under constant conditional variance and
mean, without further assumptions like independence or identical distribution of
 the random variables, making them suitable for sequential decision making conte
xts. The results are instantiated for both the batch setting (where the sample s
ize is fixed) and the sequential setting (where the sample size is a stopping ti
me). Our bounds are asymptotically "sharp": when the data are iid, our CI adapts

optimally to both unknown mean μ and unknown $\mathbb{V}[(X-\mu)^2]$, meaning that the first order term of our CI exactly matches that of the oracle Bernstein inequality which knows those quantities. We compare our results to a widely used (non-sharp) concentration inequality for the variance based on self-bounding random variables, showing both the theoretical gains and improved empirical performance of our approach. We finally extend our methods to work in any separable Hilbert space.
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STORM: Segment, Track, and Object Re-Localization from a Single Image
Yu Deng, Teng Cao, Hikaru Shindo, Quentin Delfosse, Jiahong Xue, Kristian Kersting
Accurate 6D pose estimation and tracking are core capabilities for physical AI systems, yet real-world deployment remains brittle and labor-intensive.
Many pipelines rely on CAD models, manual masking, or per-object adaptation, and still fail under occlusion or fast motion without a principled way to recognize failure.
We propose STORM, a unified framework for reference-conditioned 6D tracking that can operate from a single reference image, with minimal manual input and improved robustness.
STORM combines: (i) Hierarchical Spatial Fusion Attention (HSFA), a task-driven reference-query fusion architecture that supports both single-reference and multi-reference conditioning and can optionally use vision-language semantic conditioning to resolve instance ambiguities; and (ii) a BCE-trained tracking verifier whose continuous compatibility logit is used as an energy-like score to detect drift and trigger automatic re-initialization.
Experiments on LM-O and YCB-Video show that STORM improves annotation-free pose tracking accuracy over strong baselines and recovers reliably from severe occlusions and rapid viewpoint changes with minimal overhead.
****
Clustering in Deep Stochastic Transformers
Lev Fedorov, Michael Eli Sander, Romuald Elie, Pierre Marion, Mathieu Lauriere
Transformers have revolutionized deep learning across various domains but understanding the precise token dynamics remains a theoretical challenge. Existing theories of deep Transformers with layer normalization typically predict that tokens cluster to a single point; however, these results rely on deterministic weight assumptions, which fail to capture the standard initialization scheme in Transformers.
In this work, we show that accounting for the intrinsic stochasticity of random initialization alters this picture. More precisely, we analyze deep Transformers where noise arises from the random initialization of value matrices. Under diffusion scaling and token-wise RMS normalization, we prove that, as the number of Transformer layers goes to infinity, the discrete token dynamics converge to an interacting-particle system on the sphere where tokens are driven by a *common* matrix-valued Brownian noise. In this limit, we show that initialization noise prevents the collapse to a single cluster predicted by deterministic models. For two tokens, we prove a phase transition governed by the interaction strength and the token dimension: unlike deterministic attention flows, antipodal configurations become attracting with positive probability. Numerical experiments confirm the predicted transition, reveal that antipodal formations persist for more than two tokens, and demonstrate that suppressing the intrinsic noise degrades accuracy.
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Think Less, Act Early: Reinforced Latent Reasoning with Early Exit in Vision-Language-Action Models
Dianqiao Lei, Lianlei Shan
Existing Vision-Language-Action (VLA) models predominantly rely on explicit Chain-of-Thought (CoT) reasoning to bridge perception and action. While effective, this paradigm suffers from high computational costs and error propagation in multi-step tasks. In this paper, we propose Adaptive Variable Alignment VLA (AVA-VLA), a novel Latent Reasoning VLA framework that models reasoning as a sequence of

unobservable latent variables, bypassing the need for explicit text generation. However, latent trajectories are inherently susceptible to noise interference and misalignment with downstream objectives. To address this, we introduce a Reinforcement Learning-based Denoising mechanism that treats latent state generation as a sequential decision process, optimizing reasoning trajectories via task-level rewards. Furthermore, we incorporate an Early-Exit Strategy that adaptively terminates reasoning based on state confidence, enabling a dynamic trade-off between depth and efficiency. Extensive experiments on embodied decision benchmarks demonstrate that AVA-VLA significantly reduces inference latency while achieving superior stability and success rates compared to full-reasoning baselines.
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LRAgent: Efficient KV Cache Sharing for Multi-LoRA LLM Agents
Hyesung Jeon, Hyeongju Ha, Jae-Joon Kim
Role specialization in multi-LLM agent systems is often realized via multi-LoRA, where agents share a pretrained backbone and differ only through lightweight adapters.
Despite sharing base model weights, each agent independently builds and stores its own KV cache for the same long, tool-augmented trajectories, incurring substantial memory and compute overhead.
Existing KV cache sharing methods largely overlook this multi-LoRA setting.
We observe that, across agents, cache differences are dominated by adapter outputs, while activations from the shared pretrained backbone remain highly similar.
Based on this observation, we propose LRAgent, a KV cache sharing framework for multi-LoRA agents that decomposes the cache into a shared base component from the pretrained weights and an adapter-dependent component from LoRA weights.
LRAgent reduces memory overhead by sharing the base component and storing the adapter component in its inherent low-rank form, and further reduces compute overhead, enabled by shared-A multi-LoRA architectures, by also sharing the low-rank cache and avoiding redundant computations for contexts already processed by other agents.
To efficiently reconstruct adapter contributions at runtime, we introduce Flash-LoRA-Attention, a kernel that reorders attention computation to avoid materializing the low-rank cache to full dimension.
LRAgent achieves throughput and time-to-first-token latency close to fully shared caching, while preserving accuracy near the non-shared caching baseline across agentic question-answering benchmarks.
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Recursive Monte-Carlo Tree Search
Benjamin Howard, Keith Frankston
We introduce a recursive AlphaZero style Monte--Carlo tree search algorithm, "RMCTS". It first generates the search tree using prior policies, and then recursively re-estimates action values by using the regularized optimal posterior policies from ``Monte--Carlo tree search as regularized policy optimization'' (Grill et al., 2020) at each node of the search tree, starting from the leaves and working back up to the root. We find that RMCTS matches or exceeds the quality of AlphaZero's MCTS-UCB in a tiny fraction of the time.
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Beyond Independent Genes: Learning Module-Inductive Representations for Single-Cell Gene Perturbation Prediction
Jiafa Ruan, Ruijie Quan, Xu Liyang, Zongxin Yang, Yi Yang
Predicting transcriptional responses to genetic perturbations is a central problem in functional genomics.
In practice, perturbation responses are rarely gene-independent but instead manifest as coordinated, program-level transcriptional changes among functionally related genes.
However, most existing methods do not explicitly model such coordination, due to gene-wise modeling paradigms and reliance on static biological priors that cannot capture dynamic program reorganization.
To address these limitations, we propose scBIG, a module-inductive perturbation prediction framework that explicitly models coordinated gene programs. scBIG ind

uces coherent gene programs from data via Gene-Relation Clustering, captures inter-program interactions through a Gene-Cluster-Aware Encoder, and preserves modular coordination using structure-aware alignment objectives. These structured representations are then modeled using conditional flow matching to enable flexible and generalizable perturbation prediction.
Extensive experiments on multiple single-cell perturbation benchmarks show that scBIG consistently outperforms state-of-the-art methods, particularly on unseen and combinatorial perturbation settings, achieving an average improvement of 6.7% over the strongest baselines. The code is available at https://github.com/ttruan2426-dot/scBIG.
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Set-Coupled Guidance: Set-Level Coordination in Diffusion-Based Dataset Distillation
Ziang Gan, Qi Zhu, Libao Zhang
Diffusion models serve as generative priors for dataset distillation, yet existing pipelines rely on per-sample update rules that evolve each synthetic image independently, limiting their ability to optimize collective set-level objectives. We propose Set-Coupled Guidance (SCG), a plug-and-play auxiliary controller that shifts from per-image to group (IPC-at-once) sampling by injecting set-symmetric feedback at each diffusion step. SCG combines spectral set-point regulation, which aligns set-level statistics to real data via empirical characteristic function matching, with cooperative kernel coupling that stabilizes joint trajectories under noisy feedback. All computations operate on lightweight descriptors extracted from predicted clean latents, adding low overhead to the base method. We provide theoretical analysis including Lyapunov descent and input-to-state stability for distributional tracking. Experiments on ImageNette, ImageWoof, ImageNet-100 and ImageNet-1K show consistent accuracy gains across multiple diffusion-based baselines; code is available at \url{https://github.com/tade1s/SCG}.
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A proximal ADMM for multiblock problems with block anti-upper triangular constraints
Zhanwang Deng, Yuqiu Su, Wen Huang
In this paper, we present the convergence analysis of the proximal Alternating Direction Method of Multipliers (ADMM) for problems with block anti-upper triangular constraints.
While the linear constraints can be treated separately, most analyses of ADMM and its variants predominantly regard the linear constraints as one. Hence, it relies on assumptions related to the entire constraint matrix, such as the full column rank. However, some problems with block anti-upper triangular constraints that can be solved by ADMM do not satisfy these assumptions. To fill this gap, a new assumption is proposed and used to guarantee the global convergence of the proximal ADMM for nonconvex problems.
In the strongly convex setting, we also prove the global convergence of the proximal ADMM and establish the linear convergence under four different scenarios. This work extends the theoretical understanding of the multi-block ADMM to more general cases with block anti-upper triangular constraints.
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Bridging the Gap Between Average and Discounted TD Learning
Haoxing Tian, Zaiwei Chen, Ioannis Paschalidis, Alex Olshevsky
The analysis of Temporal Difference (TD) learning in the average-reward setting faces notable theoretical difficulties because the Bellman operator is not contractive with respect to any norm. This complicates standard analyses of stochastic updates that are effective in discounted settings. Although a considerable body of literature addresses these challenges, existing theoretical approaches come with limitations. We introduce a novel algorithm designed explicitly for policy evaluation in the average-reward setting, utilizing sampling from two Markovian trajectories. Our proposed method overcomes previous limitations by guaranteeing convergence to the unique solution of a properly defined projected Bellman equation. Notably, and in contrast to earlier work, our convergence analysis is uniformly applicable to both linear function approximation and tabular settings and

does not involve explicit dimension-dependent terms in its convergence bounds. These results align with what is known to hold in the discounted setting. Furthermore, our algorithm achieves improved dependence on the problem's condition number, reducing the sample complexity from quartic, as in prior literature, to quadratic scaling, and thus matching the efficiency seen in the discounted setting.
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NaviAgent: Graph■Driven Bilevel Planning for Scalable Tool Orchestration
Yan Jiang, Hao Zhou, Lizhong Gu, Tianlong Li, Ruinan Jin, Wanqi Zhou, Ai Han
Large Language Models (LLMs) increasingly act as function call agents that invoke external tools to tackle tasks beyond their static knowledge. However, they typically invoke tools one at a time without a global view of task structure. As tools often depend on one another, this leads to error accumulation and poor scalability, particularly when scaling to hundreds or thousands of tools. To address these limitations, we propose NaviAgent, an explicit bilevel architecture that decouples task planning from tool execution through graph■based modeling of tool relations. At the planning level, the LLM■based agent decides whether to respond directly, clarify intent, or retrieve and execute a toolchain independent of inter■tool complexity. At the execution level, a Tool World Navigation Model (TWNM) encodes structural and behavioral relations among tools, steering the agent to compose scalable and robust invocation sequences. Incorporating feedback from real tool interactions, NaviAgent achieves closed■loop alignment between planning and execution, enabling adaptive navigation in large■scale tool ecosystems. Evaluations on API-Bank and ToolBench show consistent improvements in task success rate (TSR), with TWNM yielding an average gain of 13.1 points on complex tasks. Further tests on 50 real APIs across 7 domains show consistent gains of 4.3-12.0 points, with fewer steps and latency, demonstrating robust generalization under real-world dynamics.
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VJEPA: Variational Joint Embedding Predictive Architectures as Probabilistic World Models
Yongchao Huang
Joint Embedding Predictive Architectures (JEPAs) avoid pixel reconstruction by predicting latent representations, but standard formulations remain deterministic and provide limited uncertainty estimates for planning and control. We introduce \emph{Variational JEPA (VJEPA)}, a probabilistic extension that learns predictive distributions over future latent states using a latent-space variational objective, without autoregressive observation likelihoods. We show that VJEPA links JEPA-style self-supervised learning to predictive state representations and Bayesian filtering, and that its latent variables can serve as sufficient information states for control when they preserve task-relevant predictive information. We also propose \emph{Bayesian JEPA (BJEPA)}, which combines a learned dynamics expert with modular prior experts through a Product of Experts, enabling constraint-aware prediction and zero-shot prior swapping. Experiments on Noisy-TV systems, nonlinear and image-based benchmarks, STL-10 with a ViT encoder, and DMC Cheetah-run show that predictive JEPA-family objectives are more robust to high-variance nuisance distractors than reconstruction-based world-model baselines. These results position probabilistic latent prediction as a principled framework for robust, uncertainty-aware, reconstruction-free world models.
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Generalizing Stochastic Smoothing for Differentiation and Gradient Estimation
Felix Petersen, Christian Borgelt, Aashwin Ananda Mishra, Stefano Ermon
We address the problem of gradient estimation for stochastic differentiable relaxations of algorithms, operators, simulators, and other non-differentiable functions. Stochastic smoothing conventionally perturbs the input of a non-differentiable function with a differentiable density distribution with full support, smoothing it and enabling gradient estimation. Our theory starts at first principles to derive stochastic smoothing with reduced assumptions, without requiring a differentiable density nor full support, and presenting a general framework for relaxation and gradient estimation of non-differentiable black-box functions f. We develop variance reduction for gradient estimation from 3 orthogonal perspect

ives. Empirically, we benchmark 6 distributions and up to 24 variance reduction strategies for differentiable sorting and ranking, differentiable shortest-paths on graphs, differentiable rendering for pose estimation, as well as differentiable cryo-electron tomography simulations.
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Bioacoustic Geolocation: Species Sounds as Geographic Signals
Mustafa Chasmai, Wuao Liu, Subhransu Maji, Grant Van Horn
Can we determine someone’s geographic location solely from the sounds they hear? Are acoustic signals enough to localize within a country, state, or even city? In this work, we tackle the challenge of global-scale audio geolocation, with a particular focus on wildlife and natural sounds. We posit that bioacoustic signals contain informative geolocation cues because of well-defined geographic ranges of species. To test this hypothesis, we benchmark image geolocation and soundscape mapping methods, design oracles and species-centric baselines, and propose a hybrid approach that combines species range prediction with retrieval-based geolocation. We further ask whether geolocation improves with species-diverse recordings and spatiotemporal aggregation across neighboring samples. Finally, we extend our study to multimodal geolocation with case studies from movies that combine both audio and visual content. Our results highlight the potential of incorporating bioacoustic signals into geospatial tasks, motivating future work on species recognition and audio geolocation.
****
InvGNN: Learning Invertible Node Representations on Graphs
Giannis Nikolentzos, Dimitrios Kelesis, Nikolaos Nakis
Over the past decade, Graph Neural Networks (GNNs) have become a standard tool for solving machine learning problems on graphs. While many aspects of GNNs have been studied in depth, including their efficiency and expressive power, the invertibility of these models has remained largely unexplored. Standard aggregation functions, such as the mean, max and sum operators, are not invertible, which limits their applicability in tasks requiring invertible transformations. In this work, we introduce an invertible GNN layer. By stacking multiple such layers, we construct fully invertible GNN models, which we refer to as InvGNNs. These models inherit the benefits of invertible neural networks, including low memory usage for deep architectures, exact likelihood computation, and generative modeling capabilities. We demonstrate that InvGNNs can match the expressive power of the 1-dimensional Weisfeiler-Leman algorithm, showing that invertibility does not compromise model expressiveness. On standard graph classification benchmarks, our model performs comparably to other well-established GNNs. Beyond classification, we demonstrate the potential of invertible layers through density estimation tasks, including outlier detection and node feature generation.
****
Plan, Decouple, Assimilate: Physics-Aware Object Insertion in Remote Sensing Imagery
Yingyan Hou, Xianchi Dong, Chao Ren, Wanxuan Lu, Zihan Wei, Hongfeng Yu, Yixiao Wang, Xian Sun
Object insertion has emerged as a promising augmentation paradigm for the label scarcity and long-tailed distributions in remote sensing, generating training samples by synthesizing target instances onto real backgrounds. However, existing methods suffer from three critical issues: (i) semantic placement inconsistency, (ii) radiometric inconsistency with illumination and atmospheric conditions, and (iii) textural discontinuity. To address these, we propose a physics-aware method, "Plan, Decouple, Assimilate" (PDA), for generating high-fidelity training samples. In the planning stage, the Planning (P) module automatically generates geometrically valid bounding boxes. In the generation stage, a dual-module design synthesizes the target instance: the Decoupling (D) module employs Asymmetric Spectral Adaptation to disentangle structural identity from environmental illumination, while the Assimilation (A) module uses Neighborhood-Aware Texture Assimilation to harmonize the local manifold. By integrating these modules, PDA enforces multi-level consistency from global geometry to local micro-textures. Extensive experiments verify that PDA outperforms state-of-the-art methods in generative

quality, reducing whole-image FID by 15.7% over the strongest baseline, and substantially improves downstream detection, boosting average mAP50 by +17.07 points over the real data.
****

SafeLab: An Interactive High-Fidelity Benchmark for Embodied Safety in Scientific Robotics
Fengshuo Bai, Yufeng Li, Ruihai Wu, Peishuo Wang, Yuhan Wang, Bernie Hao Zhu, Yuanfei Wang, Tawei Chou, Jing Gao, Runchuan Zhu, Ying Wen, Yaodong Yang, Yuanpei Chen
Scientific embodied agents could automate laboratory workflows, but laboratory success is trajectory-level: an agent must remain safe throughout execution, not merely reach a final goal. In benchtop settings a small pose, force, or tilt error can cause irreversible spillage or equipment damage, yet most robot benchmarks evaluate reversible, high-tolerance manipulation and imitation-trained policies receive no recovery signal for execution drift. We introduce SafeLab, a generative simulation benchmark that couples a verified generative engine, an automated expert for teleoperation-free demonstrations, and a safety-aware RL interface, with 63 calibrated laboratory assets, 64 tasks across 9 manipulation categories, and 6,400 expert trajectories. Across state-of-the-art policies, it reveals unsafe-but-successful trajectories—large gaps between task success and Safe Success Rate—most acutely in liquid handling, force-limited actuation, and bimanual glassware rearrangement. Bounded residual RL then learns execution-level corrections that raise the simulated Safe Success Rate by up to 43.0 percentage points without retraining the base policy. Finally, 50 open-loop physical replays show 86% agreement between simulated and observed safety outcomes, supporting SafeLab as a scalable platform for screening and training safer laboratory agents.
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$\mathcal{O}(\log N)$ Latent Dimension Suffices for Universal Approximation of Permutation-invariant Function
Min ZHOU, Enming Liang, Minghua Chen
Learning permutation-invariant functions over sets of N elements, where the output is independent of the input ordering, is fundamental to many deep learning applications. While sum-decomposable architectures like DeepSets offer universal approximation for such functions, existing constructive bounds require a latent dimension of $\mathcal{O}(N)$, posing a significant scalability bottleneck. We break this barrier for *Wasserstein-stable* functions, i.e., those that are Lipschitz continuous with respect to the Wasserstein-1 metric on input distributions. We constructively prove that a latent dimension of $\mathcal{O}\left(C_D\varepsilon^{-(3D+4)}\log(N+\varepsilon^{-D})\right)$ suffices for uniform approximation as $ \varepsilon\to 0 $, where D is the element dimension and C_D is a constant depending only on D. We first discretize the input space into a finite net of measures with covering number polynomial in N. We then embed it via a multiscale Random Fourier Feature encoder that guarantees both Lipschitz stability and Hölder separation. Finally, we recover the target function via a McShane-extended Hölder decoder. This result advances the theoretical understanding of the expressivity and scalability of set-based neural architectures.
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One Bug, Hundreds Behind: LLMs for Large-Scale Bug Discovery
Qiushi Wu, Yue Xiao, Dhilung Kirat, Kevin Eykholt, Jiyong Jang, Douglas Lee Schales
Recurring Pattern Bugs (RPBs) are defined as bugs where a single root cause appears repeatedly across multiple code segments. These bugs remain a persistent security threat even after individual instances are patched. Various static analyzers exist for finding specific bug patterns but require significant engineering effort and fail to generalize well beyond their predefined template, preventing them from detecting RPBs. To tackle RPBs, we introduce BugStone, a hybrid framework combining LLVM-based program analysis with Large Language Models to automate RPB detection. BugStone leverages a single patched instance to synthesize abstract error patterns and retrieves semantically similar bugs throughout the codebase. To evaluate BugStone, we create a ground truth dataset by analyzing over 1.9

K security bugs reports, on which BugStone achieves 92.2% precision and 79.1% p
airwise accuracy. We further validated BugStone through a large-scale real-world
 deployment. In the Linux kernel, BugStone identified over 22K potential issues;
 a manual audit of 400 samples confirmed 246 valid bugs, including invalid point
er dereferences, resource leaks, type errors, performance issues, and others. To
 evaluate the generalizability of BugStone, we further applied it to the top 100
 Python projects, discovering multiple critical command injection vulnerabilitie
s.
****

On the Epistemic Uncertainty of Overparametrized Neural Networks
David Rügamer
Epistemic uncertainty is often viewed as a reducible uncertainty that vanishes w
ith increasing data. This perspective implicitly assumes parameter identifiabili
ty and equates epistemic uncertainty with predictive variability. In overparamet
rized neural networks, however, model parameters are typically non-identifiable
due to symmetries and redundant representations. As a consequence, substantial p
arameter uncertainty can persist even when the underlying function is fully iden
tified. In this work, we analyze epistemic uncertainty through the lens of non-i
dentifiability and characterize both discrete and continuous sources of residual
 uncertainty. Focusing on one-hidden-layer ReLU networks, we thoroughly analyze
the resulting posterior structure and validate our theoretical insights through
empirical studies.
****

Mirror Descent Actor Critic via Bounded Advantage Learning
Ryo Iwaki
Regularization is a core component of recent Reinforcement Learning (RL) algorit
hms. Mirror Descent Value Iteration (MDVI) uses both Kullback-Leibler divergence
 and entropy as regularizers in its value and policy updates. Despite its empiri
cal success in discrete action domains and strong theoretical guarantees, the pe
rformance of KL-entropy-regularized methods does not surpass that of a strong en
tropy-only-regularized method in continuous action domains. In this study, we pr
opose Mirror Descent Actor Critic (MDAC) as an actor-critic style instantiation
of MDVI for continuous action domains, and show that its empirical performance i
s significantly boosted by bounding the actor's log-probability terms in the cri
tic's loss function, compared to a non-bounded naive instantiation. Further, we
relate MDAC to Advantage Learning by recalling that the actor's log-probability
is equal to the regularized advantage function in tabular cases, and theoretical
ly discuss when and why bounding the advantage terms is validated and beneficial
. We also empirically explore effective choices for the bounding functions, and
show that MDAC performs better than strong non-regularized and entropy-only-regu
larized methods with an appropriate choice of the bounding functions.
****

Latent Representation Alignment for Offline Goal-Conditioned Reinforcement Learn
ing
Hyungkyu Kang, Byeongchan Kim, Min-hwan Oh
Offline goal-conditioned reinforcement learning (GCRL) provides a practical fram
ework for obtaining goal-reaching policies from fixed datasets.
  However, learning a reliable goal-conditioned value function in long-horizon t
asks remains challenging.
  In this paper, we identify erroneous generalization in goal-conditioned value
functions as a fundamental bottleneck, and demonstrate that appropriate inductiv
e bias in the value function is crucial for addressing the bottleneck.
  Building on these findings, we propose Latent-Aligned Value Learning (LAVL), a
n offline GCRL algorithm that integrates latent-representation-based value gener
alization with hierarchical planning in a unified framework.
  Extensive numerical experiments on OGBench demonstrate that LAVL consistently
outperforms existing offline GCRL methods, achieving the highest performance on
20 out of 22 datasets.
  Notably, LAVL exhibits strong performance in long-horizon tasks and trajectory
 stitching datasets, where prior methods suffer significant performance degradat

ion.
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Geometrically Constrained Outlier Synthesis
Daniil Karzanov, Marcin Detyniecki
Deep neural networks for image classification often exhibit overconfidence on ou t-of-distribution (OOD) samples. To address this, we introduce Geometrically Con strained Outlier Synthesis (GCOS), a training-time regularization framework aime d at improving OOD robustness during inference. GCOS addresses a limitation of p rior synthesis methods by generating virtual outliers in the hidden feature spac e that respect the learned manifold structure of in-distribution (ID) data. The synthesis proceeds in two stages: (i) a dominant-variance subspace extracted fro m the training features identifies geometrically informed, off-manifold directio ns; (ii) a conformally-inspired shell, defined by the empirical quantiles of a n onconformity score from a calibration set, adaptively controls the synthesis mag nitude to produce boundary samples. The shell ensures that generated outliers ar e neither trivially detectable nor indistinguishable from in-distribution data, facilitating smoother learning of robust features. This is combined with a contr astive regularization objective that promotes separability of ID and OOD samples in a chosen score space, such as Mahalanobis or energy-based. Experiments demon strate that GCOS outperforms state-of-the-art methods using standard energy-base d inference on near-OOD benchmarks, defined as tasks where outliers share the sa me semantic domain as in-distribution data. As an exploratory extension, the fra mework naturally transitions to conformal OOD inference, which translates uncert ainty scores into statistically valid p-values and enables thresholds with forma l error guarantees, providing a pathway toward more predictable and reliable OOD detection.
****
When Does Predictive Inverse Dynamics Outperform Behavior Cloning?
Lukas Schäfer, Pallavi Choudhury, Abdelhak Lemkhenter, Chris Lovett, Somjit Nath , Luis França, Matheus Ribeiro Furtado de Mendonça, Alex Lamb, Riashat Islam, Si ddhartha Sen, John Langford, Katja Hofmann, Sergio Valcarcel Macua
Behavior cloning (BC) is a practical offline imitation learning method, but it o ften fails when expert demonstrations are limited. Recent works have introduced a class of architectures named predictive inverse dynamics models (PIDM) that co mbine a future state predictor with an inverse dynamics model. While PIDM often outperforms BC, the reasons behind its benefits remain unclear. In this paper, w e provide a theoretical explanation: PIDM introduces a bias-variance tradeoff. W hile predicting the future state introduces bias, conditioning the IDM on the pr ediction can significantly reduce variance. We establish conditions for PIDM to achieve lower prediction error and higher sample efficiency than BC, with the ga p widening when additional data sources are available. We validate the theoretic al insights empirically in 2D navigation tasks, where BC requires up to five tim es (three times on average) more demonstrations than PIDM to reach comparable pe rformance. Results are also illustrated in a complex 3D environment in a modern video game with high-dimensional visual inputs and stochastic transitions, where BC requires over 66\% more samples than PIDM.
****
Newton-coupled Dual-Teacher Semi-supervised Learning Framework
Hongyang He, Xinyuan Song, Yan Zhong, Daizong Liu, Xuanyu Liu, Victor Sanchez
Most semi-supervised learning frameworks rely on a single teacher that transfers zero-order supervision through pseudo-labels, constraining the student to imita te categorical outputs without perceiving the loss geometry. This design often l eads to unstable optimization and limited generalization under scarce labels. We propose TTN (Two-Teachers Newton-guided Learning), a dual-teacher framework tha t integrates complementary supervision from MAE and DINOv3 and optimizes the stu dent through a Newton step update. The two teachers provide multi-scale structur al and semantic cues whose pseudo-labels and local Hessians are fused by confide nce weighting, forming a unified second-order supervision signal. The student up dates parameters preconditioned by the fused curvature, enabling stable converge nce and geometry-consistent learning. TTN consistently improves over existing si

ngle-teacher and consistency-based semi-supervised learning methods on ImageNet, CIFAR-10, SVHN, and STL-10, demonstrating that combining multi-view self-supervised teachers with curvature-guided optimization yields robust and efficient semi-supervised learning.
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From Holo Pockets to Electron Density: GPT-style Drug Design with Density
Jiahao Chen, Letian Gao, Yanhao Zhu, Wenbiao zhou, Bing Su, Zhi John Lu, Bo Huang
Recent advances in generative modeling have enabled significant progress in structure-based drug design (SBDD). Existing methods typically condition molecule generation on empty binding pockets from holo complexes, overlooking informative components such as the filler (ligands and solvent). Here, we leverage low-resolution electron density (ED) derived from the filler as a physically grounded condition for \textit{de novo} drug design. We consider two types of ED—calculated and cryo-EM/X-ray—obtainable from computational or experimental sources, supporting unified pre-training and experimental integration. Compared with rigid pocket representations, experimental ED naturally captures conformational flexibility and provides a more faithful description of the binding environment. Based on this, we introduce EDMolGPT, a decoder-only autoregressive framework that generates molecules from low-resolution ED point clouds. By grounding generation in physically meaningful density signals, EDMolGPT mitigates structural bias and produces molecules with 3D conformations. Evaluations on 101 biological targets verify the effectiveness. Our project page: \url{https://jiahaochen1.github.io/EDMolGPT_Page/}.
****

Active Tabular Augmentation via Policy-Guided Diffusion Inpainting
Zheyu Zhang, Shuo Yang, Bardh Prenkaj, Gjergji Kasneci
Generative tabular augmentation is appealing in data-scarce domains, yet the prevailing focus on distributional fidelity does not reliably translate into better downstream models. We formalize a *fidelity-utility gap*: common generative objectives prioritize distributional plausibility, whereas augmentation succeeds only when injected samples reduce the current learner's held-out evaluation loss. This gap motivates learning not just how to generate, but what to generate and when to inject as training evolves. We propose TAP (Tabular Augmentation Policy), which couples diffusion inpainting with a lightweight, learner-conditioned policy to steer generation toward high-utility regions and controls safe injection via explicit gating and conservative windowed commitment. Under severe data scarcity, TAP consistently outperforms strong generative baselines on seven real-world datasets, improving classification accuracy by up to 15.6 percentage points and reducing regression RMSE by up to 32%.
****

Q-Flow: Stable and Expressive Reinforcement Learning with Flow-based Policy
JaeHyeok Doo, Byeongguk Jeon, Seonghyeon Ye, Kimin Lee, Minjoon Seo
There is growing interest in utilizing flow-based models as decision-making policies in reinforcement learning due to their high expressive capacity. However, effectively leveraging this expressivity for value maximization remains challenging, as naive gradient-based optimization requires backpropagating through numerical solvers and often leads to instability. Existing approaches typically address this issue by restricting the expressive capacity of flow-based policies, resulting in a trade-off between optimization stability and representational flexibility. To resolve this, we introduce **Q-Flow**, a framework that leverages the deterministic nature of flow dynamics to explicitly propagate terminal trajectory value to intermediate latent states along the policy-induced flow. This formulation enables stable policy optimization using intermediate value gradients without unrolling the numerical solver, effectively bridging the gap between stability and expressivity. We evaluate Q-Flow in the offline learning setting on the challenging OGBench suite, where it consistently outperforms state-of-the-art baselines by an average of *10.6* percentage points, while also enabling stable online adaptation within the same framework.
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A recipe for scalable attention-based ML potentials: unlocking long-range accuracy with all-to-all node attention
Eric Qu, Brandon M. Wood, Aditi S. Krishnapriyan, Zachary Ward Ulissi
Machine-learning interatomic potentials (MLIPs) have advanced rapidly, with many top models relying on strong physics-based inductive bias. However, as models scale to larger systems like biomolecules and electrolytes, they struggle to accurately capture long-range (LR) interactions, leading current approaches to rely on explicit physics-based terms or components.
In this work, we propose AllScAIP, a straightforward, attention-based, and energy-conserving MLIP model that scales to O(100 million) training samples. It addresses the long-range challenge using an all-to-all node attention component that is purely data-driven. Extensive ablations reveal that in low-data/small-model regimes, inductive biases improve sample efficiency. However, as data and model size scale, these benefits diminish or even reverse, while all-to-all attention remains critical for capturing LR interactions. Our model achieves state-of-the-art energy/force accuracy on molecular systems (OMol25), while being competitive on materials (OMat24) and catalysts (OC20).
Furthermore, it enables stable, long-timescale MD simulations that accurately recover experimental observables, including density and heat of vaporization predictions.
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Wait, Wait, Wait... Why Do Reasoning Models Loop?
Charilaos Pipis, Shivam Garg, Vasilis Kontonis, Vaishnavi Shrivastava, Akshay Krishnamurthy, Dimitris Papailiopoulos
Reasoning models (e.g., DeepSeek-R1) generate long chains of thought to solve harder problems, but they often loop, repeating the same text at low temperatures or with greedy decoding. We study why this happens and what role temperature plays. With open reasoning models, we find that looping is common at low temperature. Larger models tend to loop less, and distilled students loop significantly even when their teachers rarely do. This points to mismatches between the training distribution and the learned model, which we refer to as errors in learning, as a key cause. To understand how such errors cause loops, we introduce a synthetic graph reasoning task and demonstrate two mechanisms. First, risk aversion caused by hardness of learning: when the correct progress-making action is hard to learn but an easy cyclic action is available, the model puts relatively more probability on the cyclic action and gets stuck. Second, even when there is no hardness, Transformers show an inductive bias toward temporally correlated errors, so the same few actions keep being chosen and loops appear. Higher temperature reduces looping by promoting exploration, but it does not fix the errors in learning, so generations remain much longer than necessary at high temperature; in this sense, temperature is a stopgap rather than a holistic solution. We end with a discussion of training-time interventions aimed at directly reducing errors in learning.
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Is Your Diffusion Sampler Actually Correct? A Sampler-Centric Evaluation of Discrete Diffusion Language Models
Luhan Tang, Longxuan Yu, Shaorong Zhang, Greg Ver Steeg
Discrete diffusion language models (dLLMs) provide a fast and flexible alternative to autoregressive models (ARMs) via iterative denoising with parallel updates. However, their evaluation is challenging: existing metrics conflate denoiser approximation error with sampler-induced error from the sampling dynamics, a problem that does not arise for ARMs whose autoregressive sampling exactly reflects the learned probability model. We introduce a sampler-centric oracle framework that replaces learned denoisers with an exact Hidden Markov Model posterior derived from a ground-truth Markov chain, isolating sampler-induced error in a controlled setting. We show that few-step discrete diffusion samplers are not distributionally correct even under an oracle denoiser, with transition-level mismatch that vanishes only as the number of steps approaches the sequence length. Moreover, improvements in negative log-likelihood (NLL), generative perplexity (GenPPL), or MAUVE do not imply correct sampling. Code is available at https://luhantang

.github.io/dllm_sampler/.
****
RAIGen: Rare Attribute Identification in Text-to-Image Generative Models
Silpa Vadakkeeveetil Sreelatha, Dan Wang, Serge Belongie, Muhammad Awais, Anjan Dutta
Text-to-image diffusion models achieve impressive generation quality but inherit and amplify training-data biases, skewing coverage of semantic attributes. Prior work addresses this in two ways. Closed-set approaches mitigate biases in predefined fairness categories (e.g., gender, race), assuming socially salient minority attributes are known a priori. Open-set approaches frame the task as bias identification, highlighting majority attributes that dominate outputs. Both overlook a complementary task: uncovering rare or minority features underrepresented in the data distribution (social, cultural, or stylistic) yet still encoded in model representations. We introduce RAIGen, the first framework, to our knowledge, for label-free rare-attribute discovery in diffusion models, requiring no predefined minority categories. RAIGen leverages Matryoshka Sparse Autoencoders and a novel minority metric combining neuron activation frequency with semantic distinctiveness to identify interpretable neurons whose top-activating images reveal underrepresented attributes. Experiments show RAIGen discovers attributes beyond fixed fairness categories in Stable Diffusion, scales to larger models such as SDXL, supports systematic auditing across architectures, and enables targeted amplification of rare attributes during generation. The project page is available at https://vssilpa.github.io/RAIGen_webpage/.
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Branching Diffusion for Point Processes in Time and Space
Chao Yang, Wenjie Shen, Shuang Li
We propose a non-autoregressive branching diffusion model for generating spatio-temporal point processes.
Starting from a geometric principle---the Wasserstein-Fisher-Rao (WFR) gradient flow of a generalized KL divergence toward a simple reference intensity---we obtain a tractable forward noising mechanism with two interpretable components:
(i) a Langevin-type \emph{drift-diffusion} step that perturbs event locations and times, and
(ii) a \emph{birth-death branching} step that changes the event count via location-dependent thinning (deaths) and Poisson offspring replication (births).
We learn the reverse-time dynamics using a permutation-equivariant denoiser that predicts a drift field and a net-growth field, and we train it using an entropic-regularized unbalanced optimal transport (UOT), which naturally handles count mismatch between noisy and clean samples.
The resulting generator produces complete spatio-temporal event sets without autoregressive simulation or explicit intensity normalization.
****
Unveiling the Potential of Quantization with MXFP4: Strategies for Quantization Error Reduction
Jatin Chhugani, Geonhwa Jeong, Bor-Yiing Su, Yunjie Pan, Hanmei Yang, Aayush Ankit, Jiecao Yu, Summer Deng, Yunqing Chen, Nadathur Satish, Changkyu Kim
Large Language Models (LLMs) have intensified the need for low-precision formats for efficient inference. The Open Compute Project Microscaling (MX) standard is attractive due to its favorable hardware efficiency, but its 4-bit variant (MXFP4) lags behind NVIDIA’s NVFP4 in accuracy, limiting adoption. We introduce two software-only techniques, Overflow-Aware Scaling (OAS) and Macro Block Scaling (MBS), that improve MXFP4 quantization fidelity without requiring hardware changes. OAS reduces overall errors by increasing effective dynamic range under power-of-two block scaling, while MBS allocates higher-precision scaling at a coarser granularity to better preserve outliers. Across multiple LLMs and standard downstream benchmarks, OAS and MBS reduce the end-to-end accuracy gap between MXFP4 and NVFP4 from about 10% to below 1% on average, while incurring modest GEMM overhead (6.2% on average). These results re-establish MXFP4 as a practical alternative to NVFP4, enabling near-NVFP4 accuracy while retaining MX’s hardware-efficiency advantages (e.g., 12% relative area savings in tensor cores).

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TwinWeaver: An LLM-Based Foundation Model Framework for Pan-Cancer Digital Twins
Nikita Makarov, Maria Bordukova, Lena Voith von Voithenberg, Estrella Pivel-Villanueva, Sabrina J Mielke, Jonathan Wickes, Hanchen Wang, Mingyu Derek Ma, Keunwoo Choi, Kyunghyun Cho, Stephen Ra, Raul Rodriguez-Esteban, Fabian Schmich, Michael Patrick Menden
Precision oncology requires forecasting clinical events and trajectories, yet modeling sparse, multi-modal clinical time series remains a critical challenge. We introduce TwinWeaver, an open-source framework that serializes longitudinal patient histories into text, enabling unified event prediction as well as forecasting with large language models, and use it to build Genie Digital Twin (GDT) on 93,054 patients across 20 cancer types.
In benchmarks, GDT significantly reduces forecasting errors, achieving a median Mean Absolute Scaled Error (MASE) of 0.87 compared to 0.97 for the strongest time-series baseline ($p<0.001$). Furthermore, GDT improves risk stratification, achieving an average concordance index (C-index) of 0.703 across survival, progression, and therapy switching tasks, surpassing the best baseline of 0.662. GDT also generalizes to out-of-distribution clinical trials, matching trained baselines at zero-shot and surpassing them with fine-tuning, achieving a median MASE of 0.75–0.88 and outperforming the strongest baseline in event prediction with an average C-index of 0.672 versus 0.648. Finally, TwinWeaver enables an interpretable clinical reasoning extension, providing a scalable and transparent foundation for longitudinal clinical modeling.
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Probing the Geometry of Diffusion Models with the String Method
Elio Moreau, Florentin Coeurdoux, Grégoire Ferré, Eric Vanden-Eijnden
Understanding the geometry of learned distributions is fundamental to improving and interpreting diffusion models, yet systematic tools for exploring their landscape remain limited. Standard latent-space interpolations fail to respect the structure of the learned distribution, often traversing low-density regions. We introduce a framework based on the string method that computes continuous paths between samples by evolving curves under the learned score function. Operating on pretrained models without retraining, our approach interpolates between three regimes: pure generative transport, which yields continuous sample paths; gradient-dominated dynamics, which recover minimum energy paths (MEPs); and finite-temperature string dynamics, which compute principal curves---self-consistent paths that balance energy and entropy. We demonstrate that the choice of regime matters in practice. For image diffusion models, MEPs contain high-likelihood but unrealistic ``cartoon'' images, confirming prior observations that likelihood maxima appear unrealistic; principal curves instead yield realistic morphing sequences despite lower likelihood. For protein structure prediction, our method computes transition pathways between metastable conformers directly from models trained on static structures, yielding paths with physically plausible intermediates. Together, these results establish the string method as a principled tool for probing the modal structure of diffusion models---identifying modes, characterizing barriers, and mapping connectivity in complex learned distributions.
****
ETS: Energy-Guided Test-Time Scaling for Training-Free RL Alignment
Xiuyu Li, Jinkai Zhang, Mingyang Yi, Yu Li, Longqiang Wang, Yue Wang, Ju Fan
Reinforcement Learning (RL) post-training alignment for language models is effective, but also costly and unstable in practice, owing to its complicated training process. To address this, we propose a training-free inference method to sample directly from the optimal RL policy. The transition probability applied to Masked Language Modeling (MLM) consists of a reference policy model and an energy term. Based on this, our algorithm, Energy-Guided Test-Time Scaling (ETS), estimates the key energy term via online Monte Carlo, with a provable convergence rate. Moreover, to ensure practical efficiency, ETS leverages modern acceleration frameworks alongside tailored importance sampling estimators, substantially reducing inference latency while provably preserving sampling quality.
Experiments on MLM (including autoregressive models and diffusion language model

s) across reasoning, coding, and science benchmarks show that our ETS consistently improves generation quality, validating its effectiveness and design.
The code is available at https://github.com/sheriyuo/ETS.
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Compute as Teacher: Turning Inference Compute Into Reference-Free Supervision
Dulhan Jayalath, Shashwat Goel, Thomas Foster, Parag Jain, Suchin Gururangan, Cheng Zhang, Anirudh Goyal, Alan Schelten
Where do learning signals come from when there is no ground truth in post-training? We show that inference compute itself can serve as supervision. By generating parallel rollouts and converting them into reference estimates, models can learn without human labels—critically, even in non-verifiable domains like healthcare guidance where no programmatic checker exists. We call this framework *Compute as Teacher (CaT)* and it turns inference-time compute from parallel rollouts into supervision for RL training. The framework has two components: (1) reference estimation which aggregates rollouts into a pseudo-reference answer, and (2) reward derivation which converts that pseudo-reference into RL rewards. For (1), we explore a simple method we call *synthesis*, but the framework admits any aggregator. For (2), we introduce self-proposed rubrics for non-verifiable domains. These are binary, auditable criteria generated from the pseudo-reference and scored by an LLM judge. On HealthBench, models trained with CaT match or exceed inference-time aggregation quality while using 9× less test-time compute. Here, CaT also competes with learning from expert physician annotations, yielding up to +30% relative improvement over the initial policy. The framework extends naturally to verifiable rewards, matching the best existing baselines on MATH-500 in test-time RL and demonstrating 'drop-in' versatility across both types of domains.
****

Randomized Advantage Transformation (RAT): Computing Natural Policy Gradients via Direct Backpropagation
Mingfei Sun
Natural policy gradients improve optimization by accounting for the geometry of distribution space, but their practical use is limited by the cost of estimating and inverting the Fisher matrix. We present Randomized Advantage Transformation (RAT), a method for estimating Tikhonov-regularized natural policy gradients via direct backpropagation. By applying the Woodbury formula, we reformulate the regularized natural gradient as vanilla policy gradients with a transformed advantage. RAT computes this transformation efficiently via randomized block Kaczmarz iterations on on-policy mini-batches, avoiding explicit Fisher construction, conjugate-gradient solvers, and architecture-specific approximations. We provide convergence guarantees for RAT and demonstrate empirically that it matches or exceeds established natural-gradient methods across continuous and visual control benchmarks, while remaining simple to implement and compatible with various architectures.
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Hard labels sampled from sparse targets mislead rotation invariant algorithms
Avrajit Ghosh, Bin Yu, Manfred K Warmuth, Peter Bartlett
One of the most common machine learning setups is logistic regression. In many classification models, including neural networks, the final prediction is obtained by applying a logistic link function to a linear score. In binary logistic regression, the feedback can be either soft labels, corresponding to the true conditional probability of the data (as in distillation), or sampled hard labels (taking values ± 1). We point out a fundamental problem that arises even in a particularly favorable setting, where the goal is to learn a noise-free soft target of the form $\sigma(\mathbf{x}^{\top}\mathbf{w}^{\star})$. In the over-constrained case (i.e. the number of samples n exceeds the input dimension d) with examples $(\mathbf{x}_i,\sigma(\mathbf{x}_i^{\top}\mathbf{w}^{\star}))$, it is sufficient to recover $\mathbf{w}^{\star}$ and hence achieve the Bayes risk. However, we prove that when the examples are labeled by hard labels y_i sampled from the same conditional distribution $\sigma(\mathbf{x}_i^{\top}\mathbf{w}^{\star})$ and $\mathbf{w}^{\star}$ is s-sparse, then rotation-invariant algorithms are provably suboptimal: they incur an excess risk $\Omega\!\left(\frac{d-1}{n

}\right)$, while there are simple non-rotation invariant algorithms with excess risk $O(\frac{s\log d}{n})$. The simplest rotation invariant algorithm is gradient descent on the logistic loss (with early stopping). A simple non-rotation-invariant algorithm for sparse targets that achieves the above upper bounds uses gradient descent on the weights u_i,v_i, where now the linear weight w_i is reparameterized as u_iv_i.
****

TRACER: Trajectory Risk Aggregation for Critical Episodes in Agentic Reasoning
Sina Tayebati, Divake Kumar, Nastaran Darabi, Davide Ettori, Ranganath Krishnan, Amit Ranjan Trivedi
Estimating uncertainty for AI agents in real-world multi-turn tool-using interaction with humans is difficult because failures are often triggered by sparse critical episodes (e.g., looping, incoherent tool use, or user-agent miscoordination) even when local generation appears confident. Existing uncertainty proxies focus on single-shot text generation and therefore miss these trajectory-level breakdown signals. We introduce TRACER, a trajectory-level uncertainty metric for dual-control Tool-Agent-User interaction. TRACER combines content-aware surprisal with situational-awareness signals, semantic and lexical repetition, and tool-grounded coherence gaps, and aggregates them using a tail-focused risk functional with a MAX-composite step risk to surface decisive anomalies.
We evaluate TRACER on τ^2-bench (Barres et al., 2025) by predicting task failure and selective task execution. To this end, TRACER improves AUROC by up to 37.1\% and AUARC by up to 55\% over baselines, enabling earlier and more accurate detection of uncertainty in complex conversational tool-use settings. Our code base is available at: https://github.com/sinatayebati/agent-tracer.
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ScaleMoE: Mixture-of-Experts for Scalable Continuous Control in Actor-Critic Reinforcement Learning
Yi Ma, Chenjun Xiao, Hongyao Tang, Yaodong Yang, Jinyi Liu, Jing Liang, Jiye Liang
Scaling network remains a bottleneck in deep reinforcement learning (RL): simply enlarging actor–critic networks destabilizes training and soon saturates performance. Although recent monolithic architectures such as SimBa and BRC have shown that carefully designed inductive biases can enable positive scaling up to a certain size, their improvements plateau soon as model parameters grow further. This work introduces ScaleMoE, a scalable RL architecture that integrates Mixture-of-Experts (MoE) modules into both the actor and critic of modern continuous control algorithms. Two complementary gating schemes are studied: output-level aggregation of per-expert policies and Q-functions, and feature-level fusion of expert representations before a shared head. We instantiate ScaleMoE on two representative monolithic RL baselines: the single-task method SimBa and the multi-task method BRC. Experiments across the DeepMind Control Suite, MetaWorld, and HumanoidBench show that progressively increasing the number of experts (up to 64) yields substantial improvements in returns, significantly outperforming monolithic networks of comparable or even greater parameter counts. Results demonstrate that ScaleMoE provides an efficient and effective scaling axis for deep RL in continuous control.
****

Joint Navigation and Manipulation Planning with 3D Interaction Chains
Keming Zhang, Sixian Zhang, Xinhang Song, Hongyu Wang, Yiyao Wang, Yingjie Wang, Shuqiang Jiang
Open-vocabulary mobile manipulation (OVMM) requires long-horizon navigation in unseen environments and object-centric manipulation. Most existing methods treat navigation and manipulation as separate stages, which can yield navigation endpoints that are poor for manipulation or manipulation-friendly poses that are globally inefficient.
We address this mismatch with 3D Interaction Chains (3D-IC), a unified framework that couples multi-stage navigation and manipulation planning. 3D-IC maintains a shared 3D feature map for both skills, generates stage-aligned interaction waypoints, and links them into candidate multi-stage chains. A hierarchical policy

then scores these chains by jointly considering feasibility (via VLM reasoning o
ver waypoint-centric 3D features) and transition cost, selecting the best trade-
off between success and path efficiency. The robot executes the next waypoint an
d replans as new observations arrive. Experiments in simulation and on a real St
retch 3 robot demonstrate consistent gains in both task success and trajectory e
fficiency.
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Dynamic Programming for Epistemic Uncertainty in Markov Decision Processes
Axel Benyamine, Julien Grand-Clément, Marek Petrik, Michael I. Jordan, Alain Oli
viero Durmus
In this paper, we propose a general theory of ambiguity-averse MDPs, which treat
s the uncertain transition probabilities as random variables and evaluates a pol
icy via a risk measure applied to its random return. This ambiguity-averse MDP f
ramework unifies several models of MDPs with epistemic uncertainty for specific
choices of risk measures. We extend the concepts of value functions and Bellman
 operators to our setting. Based on these objects, we establish the consequences
 of dynamic programming principles in this framework (existence of stationary po
licies, value and policy iteration algorithms), and we completely characterize l
aw-invariant risk measures compatible with dynamic programming. Our work draws c
onnections among several variants of MDP models and fully delineates what is pos
sible under the dynamic programming paradigm and which risk measures require lea
ving it.
****

BPL: Generalizable Deepfake Detection via Bias-only Pair-aware Learning
Yuxiang Xu, Rundong He, Zhiyuan Yan, Yicong Dong, Zhongyi Han, Xiaoyan Wang, Yil
ong Yin
The detection of synthetic images has traditionally been framed as a binary clas
sification problem. However, we argue that this formulation overlooks a fundamen
tal structural property of generative datasets: synthetic images are not indepen
dent samples, but are implicitly paired with real images sharing the same semant
ic source. Existing methods treat real and fake images as independent instances,
 failing to capture generation-induced relational discrepancies in real–fake pai
rs. Moreover, models tend to rapidly overfit to seen fake patterns, leading to p
oor generalization to unseen ones. To overcome these challenges, we propose a no
vel detection framework that explicitly mines real–fake pairs by constructing so
urce-guided mappings or leveraging nearest-neighbor relationships in the CLIP em
bedding space. We then introduce pair-wise discrepancy learning that explicitly
enlarges generation-induced deviations and discrepancy inversion to mitigate ove
rfitting. Moreover, to preserve pretrained semantic representations while improv
ing generalization, we adopt a bias-only fine-tuning scheme that restricts model
 capacity during adaptation. Extensive experiments show that our approach achiev
es superior generalization across unseen fake patterns.
****

Normalizing Flows with Iterative Denoising
Tianrong Chen, Jiatao Gu, David Berthelot, Joshua M. Susskind, Shuangfei Zhai
Normalizing Flows (NFs) are a classical family of likelihood based methods that
have received revived attention. Recent efforts such as TARFlow have shown that
NFs are capable to achieving promising performance on image modeling tasks, maki
ng them promising alternatives to other methods such as diffusion models. In thi
s work, we further advance the state of Normalizing Flow generative models by in
troducing iterative TARFlow (iTARFlow). Unlike diffusion models, iTARFlow mainta
ins a fully end-to-end, likelihood-based objective during training. During sampl
ing, it performs autoregressive generation followed by an iterative denoising pr
ocedure inspired by diffusion-style methods. Through extensive experiments, We s
how that iTARFlow achieves competitive performance across ImageNet resolutions o
f 64, 128, and 256 pixels, demonstrating its potential as a strong generative mo
del and advances the frontier of Normalizing Flows. In addition, we analyze the
characteristic artifacts produced by iTARFlow, offering insights that may shed t
he light for future improvements.
****

Stabilizing In-Context Multi-Source Domain Adaptation for Biomedical Images Thro
ugh Controls
Ana Sanchez-Fernandez, Thomas Pinetz, Werner Zellinger, Günter Klambauer
Biomedical imaging data presents enormous potential for deep learning models to
predict invaluable properties, such as diseases and drug effects. However, unavo
idable alterations of the technical conditions cause *batch effects*: variations
 between groups of samples that are not due to any biological signal of interest
. Batch effects greatly hinder the generalization abilities of deep learning mod
els, preventing their practical use in the real world. Unsupervised Domain Adapt
ation (UDA) methods have been proposed to mitigate batch effects, but they usual
ly assume that the data is comprised of only one source domain and one target do
main, whereas biological datasets are comprised of multiple domains, both at tra
ining and at inference time. While Batch Normalization–based test-time and meta-
learning adaptation methods offer a promising mechanism for domain alignment, we
 show that existing approaches exhibit degraded performance under the usual infe
rence scenarios of small target batch sizes and label shift. We address these li
mitations by leveraging negative control samples, which are consistently present
 in every experimental batch in biological datasets, as stable context for adapt
ation. We propose CS-ARM-BN, a meta-learning BN adaptation method that uses cont
rols both during training and inference to stabilize domain statistics. We perfo
rm a suite of experiments of Mechanism-Of-Action (MoA) classification, a crucial
 task for drug discovery, on the large JUMP-CP imaging dataset. Our experiments
show that CS-ARM-BN substantially improves robustness to batch size and class di
stribution shifts, enabling practical use of deep learning models for biomedical
 images.
****
On the Limits of LLM Adaptability: Impact of Model-Internalized Priors on Annota
tion Task Performance
Etienne Casanova, Rafal Kocielnik, R. Michael Alvarez
Large Language Models (LLMs) are increasingly used for zero-shot annotation and
LLM-as-a-judge tasks, yet their reliability hinges on how model-internalized pri
ors interact with user-provided instructions. We investigate three dimensions of
 this interaction: (1) how an LLM's familiarity with data and task definitions a
ffects performance, (2) the extent to which additional information in prompts ca
n correct zero-shot errors (``decision stickiness''), and (3) model susceptibili
ty to misaligned task definitions. Through experiments on toxicity detection acr
oss diverse datasets (spanning social media, gaming, news, and forums) using bot
h dense and mixture-of-experts models, we find that nearly two-thirds of zero-sh
ot errors are resistant to correction, with an overall rescue rate (fraction of
initial errors corrected by prompting) of only 34.8\%. High-confidence errors pr
ove especially resistant to correction. When given misaligned definitions, LLMs
follow them while maintaining confidence levels unchanged from the aligned condi
tion. Crucially, we introduce Definition-Specific Familiarity (DSF), which measu
res alignment between a model's internal concept and the task definition. After
controlling for dataset-level confounds, DSF shows a positive association with m
odel performance (partial $r=+0.41$), while three distinct memorization metrics
(ROUGE-L, BERTScore, and embedding cosine similarity) all fail to show a positiv
e association. These findings show the limitations of prompt-based correction in
 annotation tasks, highlighting the importance of definition alignment over text
-level memorization.
****
Expectation Alignment of Language Models for Real-World User Expectations
Miaomiao Li, Yang Wang, Bin Liang, Shudong Liu, Zhiwei Zhang, Kam-Fai Wong
Large language models (LLMs) have demonstrated remarkable performance on standar
d benchmarks, yet it remains largely unexplored whether they truly meet user exp
ectations. Existing evaluation approaches, relying on model heuristics, expert r
ubrics, or user simulation, fail to capture the diversity and subtlety of real h
uman expectations, causing models to appear competent while misaligning with wha
t users actually seek. we present the first systematic study of user expectation
s in real-world LLM interactions, proposing a principled procedure to extract se

mantically rich expectations and introducing ExpectBench, a benchmark grounded i
n real user expectations. Analyses reveal that current LLMs struggle to satisfy
and anticipate what users hope to obtain, highlighting a fundamental source of m
isalignment.
Building on these observations, we propose LENS, a lightweight latent expectatio
n–aware response generation framework. LENS enables models to internalize user e
xpectations and generate better-aligned responses, consistently improving expect
ation satisfaction and underscoring the importance of explicitly modeling user e
xpectations for realistic human–AI alignment.
****

Semantic Granularity Navigation in Image Editing
Liangsi Lu, Minzhe Guo, Xuhang Chen, Yang Shi
Despite the generative capabilities of diffusion and flow models, real-image edi
ting remains constrained by a persistent trade-off between semantic editability
and structural fidelity. We trace a primary cause of this limitation to the impl
icit coupling of edit progress with model scale in existing paradigms. Under thi
s coupling, stronger edits typically require visiting noisier states, which spen
ds computation on destabilizing layout before the semantic change is well locali
zed. We introduce NaviEdit, a training-free inference-time controller that decou
ples edit progress from model scale traversal through a strict self-consistency
contract. NaviEdit operates at the rollout level and leaves the underlying pretr
ained model unchanged. It treats scale as a control input and reallocates a fixe
d step budget toward semantically responsive intermediate scales instead of dest
ructive high-noise regimes. Experiments show positive average gains across compa
tible editors and flow backbones, supporting decoupling as a portable inference-
time control principle.
****

Towards Effective Waste Segmentation for Automated Waste Recycling in Cluttered
Background
Mamoona Javaid, Mubashir Noman, Abdul Hannan, Shah Nawaz, Mustansar Fiaz, Sajid
Ghuffar
Rapid expansion of urban areas and population growth is causing an immense incre
ase in waste production, which demands the need for efficient and automated wast
e management.
In this scenario, automated waste recycling (AWR) using deep learning methods ca
n assist humans in optimal waste management.
Recent deep learning approaches for AWR provide promising waste segmentation per
formance, however, these methods rely on large backbone networks that are ineffi
cient for AWR systems and suffer from performance deterioration in cluttered sce
nes.
To this end, an optimal waste segmentation network is introduced which effective
ly utilizes the spatial domain to capture localized structural dependencies and
the spectral domain to efficiently extract global contextual relationships.
This cascaded design allows the network to progressively leverage both local and
 global representations across complementary domains to highlight the semantic i
nformation necessary for effective segmentation of various waste objects.
Furthermore, auxiliary feature enhancement module (AFEM) is introduced to enhanc
e the target objects' boundaries and blob amplification for better segmentation
in cluttered scenarios.
Extensive experimentation on ZeroWaste-aug, ZeroWaste-f and SpectralWaste datase
ts reveals the merits of the proposed method.
****

The Bridge-Garden Dilemma in LLM Distillation: Why Mixing Hard and Soft Labels W
orks
Guanghui Wang, Kaiwen Lv Kacuila, Zhiyong Yang, Zitai Wang, Jin-Wen Wu, Longtao
Huang, Qianqian Xu, Qingming Huang
Knowledge distillation (KD) transfers knowledge from a large teacher model to a
smaller student. In language modeling, the student is trained either on tokens s
ampled from the teacher (\textbf{hard labels}) or the teacher’s full next-token
distribution (\textbf{soft labels}). Despite soft labels appear strictly richer,

we find that mixing hard and soft labels consistently yields better results. Cr
ucially, we show that this gain cannot be explained by closer teacher matching d
uring training. Instead, it comes from reduced exposure bias---the mismatch betw
een training and inference distributions.
To explain this phenomenon, we introduce the Bridge–Garden Decomposition theory,
 which categorizes generation steps into two types: \textit{Bridges}, where the
next token must be \textit{exact}, and \textit{Gardens}, where it can be \textit
{flexible}. We show that hard-only KD excels in Bridges by avoiding risky deviat
ions, while soft-only KD preserves diversity in Gardens. A hybrid strategy hand
les both cases and, as a result, reduces exposure bias across the sequence.
Guided by this theory, we develop a family of Bridge--Garden hybrid supervision
methods that adaptively balance hard and soft labels. Across seven teacher--stud
ent pairs (including Qwen, Llama, Gemma, and DeepSeek) and benchmarks in reasoni
ng and coding, our approach outperforms divergence-based and on-policy KD baseli
nes while reducing training cost by \textbf{9.7$\times$}, enabling efficient mod
el compression.
****

SOLAR: Self-supervised Joint Learning for Symmetric Multimodal Retrieval
Wenjie Yang, Hang Yu, Yuyu Guo, Peng Di
In this work, we address the critical yet underexplored challenge of symmetric m
ultimodal-to-multimodal (MM2MM) retrieval, where queries and contexts are interc
hangeable. Existing universal multimodal retrieval works struggle with this task
, as they are constrained by the labeled asymmetric datasets used. We produce SO
LAR (Self-supervised jOint LeArning for symmetric multimodal Retrieval), a novel
 two-stage self-supervised framework that leverages readily available unlabeled
web-scale image-text pairs. Based on the observation that both semantic alignmen
t and discrepancies exist between two modalities, in the first stage, we learn t
he intersection mask of image-text pair, allowing us to align intersection while
 preserving semantic of difference. In the second stage, the learned mask is fur
ther utilized to construct positive and hardnegative samples via masking differe
nt parts of image/text, which enable us to conduct self-supervised multimodal em
bedding learning. Complementing this framework, we present a new benchmark featu
ring high-quality human-verified positive and hard-negative pairs to evaluate sy
mmetric MM2MM retrieval under realistic conditions, as well as the corresponding
 pipeline. Extensive experiments against ten SOTA methods show SOLAR surpasses t
he strongest supervised VLM by 7.08 points on this benchmark, with over 50x fewe
r model parameters and a 5x smaller embedding dimension. Code will be available
upon publication.
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Eating for a Sustainable Planet: Personalized Sustainable Diet Recommendation vi
a Constraint-Aware Decision-Making Modeling
Ying Jin, Weiqing Min, Mingyu Huang, Shuqiang Jiang
A sustainable diet represents a multi-dimensional synergy among four essential p
illars: nutrition adequacy, economic affordability, cultural acceptability, and
environmental respect. Despite the prevalence of population-level sustainability
 modeling, practical implementation relies on effective individual-level adoptio
n. This transition is often hindered by inter-individual heterogeneity, posing a
 formidable challenge in aligning sustainable diet requirements with individual
preferences. To address this issue, we propose a personalized sustainable diet r
ecommendation model based on a constraint-aware decision-making mechanism, where
 sustainability is incorporated through learnable constraints rather than modele
d as user preferences. To systematically evaluate the proposed approach, we cons
truct a sustainable diet dataset named SusDiet with about 150k recipes, characte
rized by broad coverage of sustainability indicators across four dimensions. Exp
erimental results on this dataset show that our method promotes more sustainable
 choices without compromising individual preference. This work establishes a fra
mework for aligning individual dietary choices with planetary health, offering q
uantitative evidence to guide future sustainable diet interventions and policy-m
aking for sustainable development.
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Asymmetric Contrastive Objectives for Efficient Phenotypic Screening
Luke Nightingale, Joseph Tuersley, Scott Warchal, Andrea Cairoli, Jacob Howes, Cameron Shand, Andrew J Powell, Darren V.S. Green, Amy Strange, Michael Howell
Phenotypic screening experiments produce many microscope images of cells under diverse perturbations, with biologically significant responses often subtle or difficult to identify visually. A central challenge is to extract image representations that distinguish activity from controls and group phenotypically similar perturbations. In this work we propose new adaptations of contrastive loss functions that incorporate experimental metadata as learned class vectors, and a geometrically inspired variant, called SPC, where class vectors are confined to the unit sphere and updated only by attractive terms (allowing more overlap of phenotypically similar classes). The approach is tested on two popular benchmarking datasets, BBBC021 and RxRx3-core; and we also evaluate performance on uncurated screens of HaCaT cells to gauge effectiveness in a realistic use-case scenario. We find we outperform prior methods across the three datasets and on a wide array of metrics measuring phenotype grouping, biological recall, drug-target interaction and mechanism-of-action inference. We also show we maintain this improved performance compared to models over 10x larger in parameter count, and that SPC can be used as an effective fine-tuning technique. The method is easy to implement and is well suited to settings with limited data or compute resources.
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Few-Shot Design Optimization by Exploiting Auxiliary Information
Arjun Mani, Carl Vondrick, Richard Zemel
Many real-world design problems involve optimizing an expensive black-box function $f(x)$, for which Bayesian Optimization is a sample-efficient framework. However, while the basic black-box setting returns a scalar reward, real-world experiments often generate a wealth of useful information. We introduce a new setting where an experiment generates high-dimensional auxiliary information $h(x)$ along with $f(x)$; moreover, a history of relevant, previously-solved tasks is available for accelerating optimization. We develop a novel method based on a neural model which predicts $f(x)$ for unseen designs given a few-shot context containing observations of $h(x)$. We evaluate our method on two challenging domains, robotic hardware design and hyperparameter tuning. On both domains, our method achieves improved few-shot prediction and faster design optimization, outperforming several multi-task optimization methods.
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On Information Self-Locking in Reinforcement Learning for Active Reasoning of LLM agents
Deyu Zou, Yongqiang Chen, Fan Feng, Mufei Li, Pan Li, Yu Gong, James Cheng
Reinforcement learning (RL) has become a de
facto paradigm for building LLM-based agents
that act, interact, and reason over extended task
horizons. However, in active reasoning where
agents must elicit new observations through interaction with the environment to solve the task,
we find that outcome-based RL can induce a systematic failure mode which we call information
self-locking (SeL): agents fail both to elicit informative feedback and to internalize obtained evidence. To understand the issue, we trace agentic
behaviors into two coupled capabilities: Action
Selection (AS), which determines observation
streams, and Belief Tracking (BT), which updates
the agent’s internal task understanding. Theoretical and empirical analyses reveal a bidirectional
bottleneck that leads to SeL: weak BT obscures
the credit of informative actions, while weak AS
deprives BT of useful evidence. This coupling
weakens the learning signal for both capabilities
and leads to SeL. To mitigate this issue, we propose AREW, a simple yet effective Advantage Reweighting method that uses easy-to-obtain directional critiques to

reallocate credit within trajectories. Extensive experiments across 9 agentic tasks of varying complexity show that AREW significantly mitigates SeL, yielding up to 60-point
gains in final performance. Code is available at
https://github.com/unimpor/T3.
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NITP: Next Implicit Token Prediction for LLM Pre-training
Xiangdong Zhang, Debing Zhang, Shaofeng Zhang, Xiaohan Qin, Yu Cheng, Junchi Yan
Standard Next-Token Prediction (NTP) supervises language models solely through discrete labels in the output logit space.
We argue that this sparse, one-hot supervision leaves the latent representation space under-constrained, allowing hidden states to drift into degenerate and anisotropic configurations that limit generalization.
To address this issue, we propose Next Implicit Token Prediction (NITP), which augments discrete prediction with dense, continuous supervision directly in the representation space.
NITP requires the model to predict the implicit semantic content of the next token, using shallow-layer representations from the same model as stable self-supervised targets.
Theoretically, we show that NITP regularizes the optimization landscape by mitigating under-constrained degrees of freedom and enforcing a compact, structured representation geometry.
Empirically, across dense and MoE models ranging from 0.5B to 9B parameters, NITP consistently {improves downstream performance with negligible computational overhead}. Notably, on the 9B MoE model, NITP achieves a 5.7\% absolute improvement on MMLU-Pro,
along with gains of 6.4\% on C3 and 4.3\% on CommonsenseQA, with $\sim$2\% additional training FLOPs and no additional inference cost. Our implementation is available at https://github.com/aHapBean/NITP.
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Nonparametric Data Attribution for Diffusion Models
Yutian Zhao, Chao Du, Xiaosen Zheng, Tianyu Pang, Min Lin
Data attribution for generative models seeks to quantify the influence of individual training examples on model outputs. Existing methods for diffusion models typically require access to model gradients or retraining, limiting their applicability in proprietary or large-scale settings. We propose a *nonparametric* attribution method that operates directly on data, measuring influence via patch-level similarity between generated and training images. Our approach is grounded in the analytical form of the optimal score function, extends naturally to multiscale representations, and remains computationally efficient through convolution-based acceleration. In addition to producing spatially interpretable attributions, our framework uncovers patch-level correspondences between training data and generated outputs. Experiments show that NDA substantially outperforms existing nonparametric baselines and achieves the best attribution performance among the evaluated methods that do not use target-model parameters or gradients.
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Training LLM Agents to Empower Humans
Evan Ellis, Vivek Myers, Jens Tuyls, Sergey Levine, Anca Dragan, Benjamin Eysenbach
Assistive agents should not only take actions on behalf of a human, but also step out of the way and cede control when there are important decisions to be made. However, current methods for building assistive agents, whether via mimicking expert humans or via RL finetuning on an inferred reward, often encourage agents to complete tasks on their own rather than truly assisting the human achieve her objectives. We propose a new approach to tuning assistive language models based on maximizing the human's empowerment, their ability to effect desired changes in the environment. Our empowerment-maximizing method, Empower, only requires offline examples of human-written text, providing a self-supervised method for fine-tuning language models to better assist humans. To study the efficacy of our approach, we conducted an 18-person user study comparing our empowerment assistan

t with a strong baseline. Participants preferred our assistant 78% of the time (p=0.015), with a 31% higher acceptance rate. With this empowerment objective, we provide a framework for useful aligned AI agents without the need for human fee dback or verifiable rewards.
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HieraScaffold: Learning Compact Hierarchical Representations for Scalable 4D LiD AR Generation
Zijie Wu, Na Zhao
Outdoor LiDAR generation has shown strong potential for autonomous driving and l arge-scale 3D perception. However, existing approaches remain computationally in tensive and primarily static, lacking explicit modeling of temporal dynamics. Th is limitation weakens spatiotemporal coherence and reduces the realism of 4D LiD AR generation.
We propose a hierarchical recoupling generation framework that explicitly disent angles and reconstructs large-scale geometry and motion within a unified hierarc hical structure. First, we design a multi-resolution feature scaffold that predi cts time-correlated unsigned distance fields and spatial gradients, enabling hie rarchical decomposition of 4D dynamics into static and motion-varying components . Next, to achieve compact yet expressive modeling, we introduce a neural contou rlet representation that prunes redundant scaffolds into minimal directional bas es, efficiently capturing essential geometric and motion cues. Finally, we progr essively re-couple these hierarchical components to generate realistic and tempo rally coherent 4D LiDAR data.
Extensive experiments demonstrate that our method outperforms baselines in both quality and consistency, achieving 3.3\%, 25.0\%, 17.8\% improvements in FRD, MM D, and JSD, respectively, over the strong competitors, LiDMs and RangeLDM.
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STT-LLM: Structural-Temporal Tokenization for Adapting LLMs to Longitudinal Clin ical Profiles
Maxx Richard Rahman, Mostafa Hammouda, Wolfgang Maass
Large Language Models have shown strong generalization across natural language t asks but remain underexplored for longitudinal clinical profiles. In sports anti -doping, biological profiles are analyzed to support early detection of prohibit ed substance use and identification of anomalous biological patterns, both of wh ich require joint modeling of temporal dynamics and metabolic relationships. We propose STT-LLM, a structural-temporal tokenization framework that adapts LLMs t o longitudinal clinical analysis without modifying their backbone architectures. STT-LLM constructs biologically grounded structural-temporal embeddings and tra nsforms them into LLM-compatible tokens via specialized tokenizers that explicit ly encode pathway structure and temporal evolution. We evaluate STT-LLM on real- world longitudinal datasets from athletes, showing consistent improvements over native LLM tokenization strategies in sequence prediction and anomaly detection. In addition, we present a case study where STT-LLM provides contextual reasonin g that aligns more closely with expert assessments compared to baseline models. These results highlight tokenization as a key bottleneck and opportunity for ada pting LLMs to clinical data.
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Exact Unlearning in Reinforcement Learning
Thanh Nguyen-Tang, Raman Arora
We formulate the problem of \emph{exact unlearning} in reinforcement learning, w here the goal is to design an efficient framework that enables the removal of an y user’s data upon deletion request, i.e., the online learner’s output after unl earning be \emph{indistinguishable} from what would have been produced had the d eleted user never interacted with the learner. For any $\rho >0$, we show that t here exists a reinforcement learning (RL) algorithm that is ρ-TV-stable and supports an exact unlearning procedure whose expected computational cost is onl y a $\rho \sqrt{\ln T}$ fraction of the computational cost of retraining from sc ratch. We construct such a ρ-TV-stable RL algorithm for tabular Markov deci sion processes (MDPs), which achieves a regret bound of $\mathcal{O}(H^2 \sqrt{ SAT} + H^3 S^2 A + {H^{2.5} S^2 A}/{\rho})$, where S, A, H, and T denote the

number of states, the number of actions, the episode horizon, and the number of episodes, respectively. We also establish a lower bound of $\Omega(H\sqrt{SAT}+{SAH}/{\rho})$ for ρ-TV-stable RL algorithms, showing that our algorithm is nearly minimax optimal.
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PrivGate: Steering Contextual Integrity in LLMs via Latent Space Geometry
Runshan Hu, Yukun Dong, Yingying Huangfu, Ruohan Zhao, Yi Xie, Tieyan Li
Securing Contextual Integrity (CI) is critical for privacy-preserving Large Language Model (LLM) agent execution. However, existing agents struggle to balance the agility of direct generation against the prohibitive latency of CI-constrained thinking. To address this, we propose PrivGate, a framework that selectively invokes explicit reasoning based on internal privacy signals. Our approach is grounded in the discovery of a privacy manifold, where models linearly encode privacy sensitivity within their residual streams, even during non-compliant generation. Leveraging this structure, PrivGate employs Latent Gating, a training-free mechanism that requires no fine-tuning of the base LLM and triggers explicit reasoning only when high latent risk is detected, thereby optimizing the efficiency-privacy trade-off by minimizing unnecessary compute. On the contextual PrivacyLens benchmark, PrivGate maintains consistently high performance in out-of-distribution risk identification, validating the generalizability of the discovered manifold. End-to-end evaluations show that PrivGate achieves a 62.6% average relative reduction in privacy leakage with 15.9% token overhead, offering a practical pathway to reconcile rigorous CI requirements with the performance demands of LLM agents.
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The benefits of full data shuffle, now with optimal I/O cost: k-wise independence and matrix transposition to the rescue
Peyman Afshani, Rezaul Chowdhury, Mayank Goswami, Jens Kristian Refsgaard Schou, Francesco Silvestri, Mariafiore Tognon
It is known that RandomShuffle, the without replacement version of Stochastic Gradient Descent (SGD), converges faster than with replacement SGD. However, RandomShuffle requires uniformly performing a random permutation of the input sequence, which is known to have high I/O complexity due to data movement across the memory hierarchy. In this paper, we propose a shuffling algorithm with a linear I/O complexity that generates almost-uniformly random permutations with rigorous mathematical guarantees. Specifically, we show that the shuffling algorithm can generate 2-wise independent permutations. Furthermore, we can extend to k-wise independence with a small error in the probability distribution, if the fast memory has at least k memory blocks. These results allow us to reach the same expected theoretical convergence as RandomShuffle while achieving optimal linear I/O cost.
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Semantic Impact–Driven Visual Scheduling in Vision-Language Models
Xuan Wang, Yilin Liu, Fangxiang Feng, Caixia Yuan, Huixing Jiang, Xiaojie Wang
Vision-Language Models (VLMs) suffer from high inference latency due to long visual sequences. To enable efficient, on-demand utilization of visual information, we argue that visual necessity should be assessed by its semantic impact on the output distribution, rather than inferred from intermediate interaction signals such as attention weights. We propose a training-free framework based on token embedding subspace decomposition, which we term a prediction-conditioned Semantic Lens. Specifically, at fixed decoding intervals, we perform QR decomposition on the Top-K candidate token embeddings to construct an orthogonal semantic basis. We then introduce Semantic IImpact–Driven Visual Scheduling (SIVS), which measures how visual inputs impact model predictions by projecting visual-induced hidden-state variations onto this semantic lens. SIVS provides a geometrically grounded, impact-driven criterion for dynamic visual KV scheduling. Empirical results demonstrate that SIVS achieves ~87% visual KV compression while maintaining over 99% of model performance.
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DARTS: Distribution-Aware Active Rollout Trajectory Shaping for Accelerating LLM

Reinforcement Learning
Yujie Wang, Siwei Chen, Longzan Luo, Xinyi Liu, Xupeng Miao, Fangcheng Fu, Bin CUI
Reinforcement Learning (RL) has become pivotal for improving model capabilities yet suffers from rollout efficiency bottlenecks due to the long-tail response length distribution. While existing works mitigate the impact of long tails via prompt-level tail scheduling, we focus on the root source of inefficiency: the distribution itself. Specifically, we characterize the long-tail distribution at a finer granularity, identifying intra-prompt long tails, and revealing that they frequently consist of ineffective verbosity. To address this, we propose a novel paradigm of active distribution shaping to shape the rollout distribution towards conciseness and certainty, thereby fundamentally resolving tail-induced overheads. We achieve this through a distribution-aware trajectory sampling mechanism, which selects trajectories from a redundant exploration space for each prompt, and an adaptive redundancy allocation scheme to maximize both shaping effectiveness and system efficiency. Experiments demonstrate significant acceleration over state-of-the-art systems by up to 1.77$\times$ without compromising model performance.
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Sequential Group Composition: A Window into the Mechanics of Deep Learning
Giovanni Luca Marchetti, Daniel Kunin, Adele Myers, Francisco Acosta, Nina Miolane
How do neural networks trained over sequences acquire the ability to perform structured operations, such as arithmetic, geometric, and algorithmic computation? To gain insight into this question, we introduce the sequential group composition task. In this task, networks receive a sequence of elements from a finite group encoded in a real vector space and must predict their cumulative product. This task can be order-sensitive and cannot be solved by a linear model. Our analysis isolates the roles of the group structure, encoding statistics, and sequence length in shaping learning. We prove that two-layer networks from vanishing initialization learn this task one irreducible representation of the group at a time in an order determined by the Fourier statistics of the encoding. To perfectly learn the task, these networks require a hidden width exponential in the sequence length k. In contrast, we construct deeper architectures that exploit associativity to dramatically improve this scaling: recurrent neural networks can compose elements sequentially in k steps, while multilayer networks can compose adjacent pairs in parallel in $\log k$ layers. Overall, the sequential group composition task offers a tractable window into the mechanics of deep learning.
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Revisiting Asymmetries in Black-box Link Stealing against Graph Neural Networks
Paul Agbaje, Habeeb Olufowobi
Graph Neural Networks (GNNs) are increasingly deployed on sensitive relational data, from social networks to healthcare records. However, their outputs can leak private graph structure, enabling link-stealing attacks that infer whether a connection between two entities existed in the training graph. While prior work demonstrates high average performance for such attacks, privacy is fundamentally a worst-case property, not an average one. The key question is whether an adversary can reliably compromise even a small set of critical links under strict precision constraints. We revisit posterior-only link-stealing attacks in a strict black-box setting and show that they remain effective at extremely low false-positive rates, revealing tail-risk vulnerabilities that current evaluations overlook. We further find that intra-class vulnerabilities are suppressed by geometric bottlenecks that collapse discriminative directions in posterior space. Building on this insight, we propose a geometry-aware reconditioning method that reshapes intra-class distances, substantially improving separability without harming reliability. Across multiple real-world graphs and GNNs, this diagnostic correction achieves up to $2\times$ higher success on intra-class pairs than generic attacks, redefining link-privacy evaluation as a tail-risk problem and revealing that posterior leakage remains substantially under-measured in current GNN deployments.

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FoundObj: Self-supervised Foundation Models as Rewards for Label-free 3D Object Segmentation
Zihui Zhang, Zhixuan Sun, Yafei YANG, Jinxi Li, Jiahao Chen, Bo Yang
We address the challenging task of 3D object segmentation in complex scene point clouds without relying on any scene-level human annotations during training. Existing methods are typically constrained to identifying simple objects, primarily due to insufficient object priors in the learning process. In this paper, we present FoundObj, a novel framework featuring a superpoint-based object discovery agent that incrementally merges suitable neighboring superpoints, guided by our innovative semantic and geometric reward modules. These modules synergistically leverage semantic and geometric priors from self-supervised 2D/3D foundation models, providing complementary feedback to the object discovery agent and enabling robust identification of multi-class objects through reinforcement learning. Extensive experiments on diverse benchmarks demonstrate that our approach consistently outperforms existing baselines.
Notably, our method exhibits strong generalization in zero-shot and long-tail scenarios, underscoring its potential for scalable, label-free 3D object segmentation.
Code is available at https://github.com/vLAR-group/FoundObj
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CapBencher: Give Your LLM Benchmark a Built-in Alarm for Test-Set Overfitting
Takashi Ishida, Thanawat Lodkaew, Ikko Yamane
Publishing a large language model (LLM) benchmark (especially its ground-truth answers) on the Internet risks contaminating future LLMs and enabling evaluation gaming: it may be unintentionally (or intentionally) used to train or select a model, or exploited to overfit and hack leaderboards when labels are accessible. A common mitigation is to keep the benchmark private and let participants submit their models or predictions to the organizers, but this still permits test-set overfitting through feedback loops. To overcome this issue, we propose CapBencher, a way to publish benchmarks without fully disclosing the ground-truth answers, while preserving open evaluation of LLMs. The main idea is to reduce the best possible accuracy, i.e., Bayes accuracy, by injecting randomness to the answers by preparing several logically correct answers, and only include one of them as the solution in the benchmark. Not only does this obscure the ground-truth answers, but it also offers a test for leakage or gaming: since even fully capable models should not surpass the Bayes accuracy, any model that does is a strong signal. We show theoretically and empirically that CapBencher accurately detects test-set overfitting across diverse benchmarks, models, training methodologies, and scenarios.
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HSMAD: Heterophily-Driven Spectral and Manifold Learning for Graph Anomaly Detection
Chen Zhu, Yaying Zhang
Graph anomaly detection (GAD) is a fundamental task in graph learning. However, most existing methods rely on the homophily assumption, which posits that connected nodes tend to share the same labels. This assumption often fails in the presence of edge heterophily, leading to degraded performance. We first observe that down-weighting heterophilic edges, relative to the original or randomly weighted graphs, results in a more concentrated spectral energy distribution, thereby facilitating the learning of discriminative spectral representations. Moreover, existing methods typically embed graphs in Euclidean spaces, neglecting the importance of heterophily in manifold spaces. Motivated by these observations, we propose HSMAD, a novel framework for GAD. It consists of two key components: the Heterophily-Weighted Spectral Filtering module, which reconstructs the Laplacian using heterophily-based edge weighting for spectral filtering, and the Heterophily-Routed Manifold Update module, which routes neighborhood messages to the appropriate manifold for node feature updates, enabling curvature-adaptive representation learning. These spectral and geometric representations are jointly leveraged for anomaly detection. Extensive experiments on six real-world datasets show t

hat HSMAD achieves state-of-the-art performance across the average F1-Macro, AUROC, AUPRC, and G-Mean. Specifically, the average F1-Macro score improves by 2.66% over the best-performing method.
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SimpleMem: Efficient Lifelong Memory for LLM Agents
Jiaqi Liu, Yaofeng Su, Peng Xia, Siwei Han, Zeyu Zheng, Cihang Xie, Mingyu Ding, Huaxiu Yao
To support long-term interaction in complex environments, LLM agents require memory systems that manage historical experiences. Existing approaches either retain full interaction histories via passive context extension, leading to substantial redundancy, or rely on iterative reasoning to filter noise, incurring high token costs. To address this challenge, we introduce SimpleMem, an efficient memory framework based on semantic lossless compression. We propose a three-stage pipeline designed to maximize information density and token utilization: (1) Semantic Structured Compression, which distills unstructured interactions into compact, multi-view indexed memory units; (2) Online Semantic Synthesis, an intra-session process that instantly integrates related context into unified abstract representations to eliminate redundancy; and (3) Intent-Aware Retrieval Planning, which infers search intent to dynamically determine retrieval scope and construct precise context efficiently. Experiments on benchmark datasets show that our method consistently outperforms baseline approaches in accuracy, retrieval efficiency, and inference cost, achieving an average F1 improvement of 26.4% in LoCoMo while reducing inference-time token consumption by up to 30×, demonstrating a superior balance between performance and efficiency. Code is available at https://github.com/aiming-lab/SimpleMem.
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PathwayLLM: Explainable Clinical Trajectory Modeling with Structured Pathways for Sepsis Prediction
Zhengqiu Yu, Yueping Ding, Xiangrong Liu
Patient-level sepsis prediction requires models that track clinical deterioration over time and integrate heterogeneous structured evidence from electronic health records. We present PathwayLLM, a trajectory-based framework that grounds prediction on temporal signals, graph-structured evidence, and pathway-level clinical information derived from statistical dependency discovery. PathwayLLM follows a three-stage design. First, each observation window is encoded from multiple structured views, including physiological measurements, temporal dynamics, a heterogeneous patient-diagnosis-medication graph, and dependency-derived pathway signals. Second, these representations are injected into a pretrained language model as auxiliary contextual embeddings so that risk prediction and evidence-conditioned explanations can be learned jointly. Third, a Clinical Trajectory LSTM with Deterioration Attention aggregates window-level representations to highlight critical deterioration points and produce patient-level risk scores. On MIMIC-IV (15,410 ICU patients; 8.45% sepsis prevalence), PathwayLLM achieves AUROC 0.891 and AUPRC 0.724, outperforming strong time-series and pretrained baselines. External validation on eICU achieves AUROC 0.842 zero-shot and 0.867 after light fine-tuning. Ablation studies indicate that trajectory aggregation and structured clinical signals are key contributors, and clinician review suggests coherent, interpretable, and clinically relevant explanations.
****

Density-Guided Robust Counterfactual Explanations on Tabular Data under Model Multiplicity
Jun Tan, Qing Guo, Zicheng Xu, Jinglin Li, QI Fang, Ning Gui
Counterfactual explanations (CEs) are essential for actionable recourse, yet their reliability is often compromised in low-density regions, where classifiers exhibit high variance. Unlike existing methods that rely on expensive ensemble intersections to define stability, we propose DensityFlow, a generative framework that constructs robust CEs by adhering to the high-confidence data manifold. Specifically, we model the counterfactual generation as continuous-time dynamics parameterized by Neural ODE, guided by a differentiable density score to actively avoid uncertain, low-density areas. This density score is learned via Noise Contr

astive Estimation, effectively leveraging a $(K{+}1)$-way discriminator to estim
ate density ratios. For black-box settings, we introduce a local proxy distillat
ion mechanism that aligns a lightweight surrogate with the target model strictly
 within the trajectory of CE generation, enabling efficient gradient-based optim
ization with minimal queries. Experiments demonstrate that \textit{DensityFlow}
achieves superior validity under model multiplicity while significantly reducing
 query costs compared to ensemble-based baselines. Our implementation is availab
le at \url{https://github.com/G-AILab/DensityFlow}.
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Breaking the Self-Confirming Loop: Diagnosing and Mitigating Systemic Reward Bia
s in Self-Rewarding RL

Chuyi Tan, Peiwen Yuan, Xinglin Wang, Yiwei Li, Shaoxiong Feng, Yueqi Zhang, Jia
yi Shi, Ji Zhang, Boyuan Pan, Yao Hu, Kan Li

Reinforcement learning with verifiable rewards (RLVR) efficiently scales the rea
soning ability of large language models (LLMs) but is bottlenecked by scarce la
beled data. Reinforcement learning with intrinsic rewards (RLIR) offers a scalab
le alternative via self-rewarding, yet often suffers from instability and inferi
or performance. We trace this gap to a systemic bias in confidence-coupled self-
rewarding: the model tends to over-reward high-confidence mistakes, forming a \t
extbf{self-confirming loop}. We quantify this feedback-loop bias with three metr
ics: reward noise magnitude (ρ_{noise}), policy–reward coupling ($\r
ho_{\text{selfbias}}$), and over-/under-reward skew (ρ_{symbias}). O
ur analyses show a compounding effect where strong coupling amplifies confidence
-conditioned errors and drives a drift toward over-reward, leading to instabilit
y and a lower performance ceiling. To mitigate this, we propose reinforcement le
arning with ensembled rewards (\textbf{RLER}), which aggregates diverse models w
ith adaptive reward interpolation and disagreement-aware rollout selection to re
duce coupling and suppress over-reward drift. Extensive experiments show that RL
ER improves by 6.2\% over the best RLIR baseline and is within 3.6\% of RLVR, wh
ile exhibiting stable scaling on unlabeled samples.
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CryoACE: An Atom-centric Framework for Accurate and Automated Model Building in
Cryo-EM

Minzhang Li, LI Mingrui, Weichen Qin, Qihe Chen, Sixian Shen, Yuan Pei, Jiakai Z
hang, Jingyi Yu

Protein automodeling from cryo-EM density maps faces unique challenges in enforc
ing physicochemical validity and managing conformational heterogeneity. Current
solvers are often limited to static predictions or require computationally inten
sive heuristic searches. We present CryoACE, an end-to-end framework that recons
tructs precise atomic graphs for both homogeneous and heterogeneous structures.
Our method features two key innovations: an atom-centric reconstruction paradigm
, where density features are sampled directly at atomic coordinates and iterativ
ely recycled to refine structures—replacing expensive voxel convolutions for eff
icient multimodal fusion—and a training-free guidance mechanism that leverages p
redicted local resolution priors to resolve dynamic ambiguity. Validated on a ne
wly constructed high-quality dataset, CryoACE significantly outperforms existing
 baselines on static benchmarks and, for the first time, unveils atomic-level dy
namic conformations on complex real-world datasets like EMPIAR-10345 without rel
ying on pre-built static structures. We release our code, model weights, and dat
aset to facilitate future research.
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Convex Dataset Valuation for Post-Training

Siqi Zeng, Christopher Jung, Rui Li, Zhe Kang, Ming Li, Nima Noorshams, Zhigang
Wang, Fuchun Peng, Han Zhao, Xue Feng

Improving LLM performance on downstream tasks sometimes requires leveraging
auxiliary datasets during post-training. In practice, however, developers face
constraints on compute, labeling, and licensing costs that preclude using all
available data, necessitating principled dataset-level selection. These
constraints are increasingly shaped by dataset marketplaces, where data
acquisition is governed by budgets and negotiation. We study dataset valuation a

s a subset selection problem during LLM post-training.
Our goal is to identify and weight auxiliary datasets so as to maximize target task performance given constrained budgets. We first show that commonly used gradient
alignment scores provide a reasonable yet incomplete valuation signal, as they ignore
redundancy among datasets. To address this, we propose a
scalable convex dataset-level valuation method based on kernel mean matching (KMM) in gradient
space, which jointly accounts for alignment with the target task and redundancy across auxiliary datasets.
Through extensive experiments across diverse post-training
settings and tasks, we show that our approach
consistently outperforms existing valuation baselines, achieving
stronger performance with low computational overhead. Our results position dataset valuation as a practical decision tool for post-training data selection in market-constrained large language model settings.
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Making Models Unmergeable via Scaling-Sensitive Loss Landscape
Minwoo Jang, Hoyoung Kim, Jabin Koo, Jungseul Ok
The rise of model hubs has made it easier to access reusable model components, making model merging a practical tool for combining capabilities. Yet, this modularity also creates a *governance gap*: downstream users can recompose released weights into unauthorized mixtures that bypass safety alignment or licensing terms. Because existing defenses are largely post-hoc and architecture-specific, they provide inconsistent protection across diverse architectures and release formats in practice. To close this gap, we propose Trap^{2}, an architecture-agnostic protection framework that encodes protection into updates during fine-tuning, regardless of whether they are released as adapters or full models. Instead of relying on architecture-dependent approaches, Trap^{2} uses weight re-scaling as a simple proxy for the merging process. It keeps released weights effective in standalone use, but degrades them under re-scaling that often arises in merging, undermining unauthorized recomposition.
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SparseOpt: Addressing Normalization-induced Gradient Skew in Sparse Training
Mohammed Adnan, Rohan Jain, Tom Jacobs, Ekansh Sharma, Rahul G Krishnan, Rebekka Burkholz, Yani Ioannou
Dynamic Sparse Training (DST) methods train neural networks by maintaining sparsity while dynamically adapting the network topology. Despite the promise of reduced computation, DST methods converge significantly slower than dense training, often requiring comparable training time to achieve similar accuracy. We demonstrate both analytically and empirically that Batch Normalization (BN) adversely affects sparse training, and propose SparseOpt — a sparsity-aware optimizer — to address this. Experiments on ResNet models across CIFAR-100 and ImageNet demonstrate consistently faster convergence and improved generalization with our proposed method. Our work highlights the limitations of current normalization layers in sparse training and provides the first systematic study of the interaction between Batch Normalization, sparse layers, and DST, taking a significant step toward making DST practically competitive with dense training.
****

InfoFlow KV: Information-Flow-Aware KV Recomputation for Long Context
Xin Teng, Canyu Zhang, Shaoyi Zheng, Danyang Zhuo, Tianyi Zhou, Shenji Wan
Retrieval-augmented generation (RAG) for long-context question answering is bottlenecked by inference-time prefilling over large retrieved contexts. A common strategy is to precompute key–value (KV) caches for individual documents and selectively recompute a small subset of tokens to restore global causal dependencies, but existing methods rely on heuristics or representation discrepancies without modeling whether selected tokens can effectively influence generation. We cast selective KV recomputation as an information flow problem and show that a simple attention-norm signal from the query reliably identifies tokens that are both s

emantically relevant and structurally positioned to propagate information, when computed under an inference-consistent RoPE geometry. We therefore reconstruct global positional assignments for retrieved chunks and introduce an information-flow–guided chunk reordering strategy. Experiments on Large Language Model and Vision-Language Model benchmarks demonstrate consistent gains over prior methods under comparable latency.
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CLASP: Online learning algorithms for Convex Losses And Squared Penalties
Ricardo N. Ferreira, Joao Xavier, Claudia Soares
Addressing Constrained Online Convex Optimization (COCO), we introduce CLASP (Convex Losses And Squared Penalties), a framework that minimizes cumulative loss together with squared constraint violations. We propose two variants of CLASP, CLASP-I and CLASP-F, allowing for a joint or separate handling of the static decision set and the time-varying constraints, a decoupling flexibility that affords simpler implementations when projections onto the static decision set are easy. Our theoretical analysis departs from prior work by fully leveraging the variety of \emph{cutter operators}, and contraction properties such as the strongly quasi-nonexpansiveness, a proof strategy not previously applied in this setting. For convex losses, both CLASP algorithms achieve regret $O\left(T^{\max\\{\beta,1-\beta\\}}\right)$ and cumulative squared penalty $O\left(T^{\\{1-\beta\\}}\right)$ for any $\beta \in (0,1)$. Most importantly, for strongly convex problems, we provide the first logarithmic guarantees on both regret and cumulative squared penalty: In the strongly convex case, both CLASP algorithms guarantee that the regret is upper bounded by $O(\log T)$ and the cumulative squared penalty is also upper bounded by $O(\log T)$.
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Deep Learning of Compositional Targets with Hierarchical Spectral Methods
Hugo Tabanelli, Yatin Dandi, Luca Pesce, Florent Krzakala
Why depth yields a genuine computational advantage over shallow methods remains a central open question in learning theory. We study this question in a controlled high-dimensional Gaussian setting, focusing on compositional target functions. We analyze their learnability using an explicit three-layer fitting model trained via layer-wise spectral estimators. Although the target is globally a high-degree polynomial, its compositional structure allows learning to proceed in stages: an intermediate representation reveals structure that is inaccessible at the input level. This reduces learning to simpler spectral estimation problems, well studied in the context of multi-index models, whereas any shallow estimator must resolve all components simultaneously. Our analysis relies on Gaussian universality, leading to sharp separations in sample complexity between two and three-layer learning strategies.
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Beyond Correctness: Distance-Based Social Dynamics of Multi-Agent Debate
Seungwoong Ha, Melanie Mitchell
Multi-agent debate (MAD) systems are often evaluated using binary correctness or peer agreement, yet such evaluations obscure how individual agents revise their answers during social interaction. We study the microscopic dynamics of answer revision in large language models (LLMs) using ConceptARC, a 2D grid-reasoning benchmark that admits quantitative distance measures between candidate solutions. By exposing a target model to controlled configurations of peer answers, we analyze how the likelihood and direction of revision depend on both social context and the distance between answers and the ground truth. We find that agents are more likely to revise when their answers are farther from the correct solution, and that revisions of incorrect answers exhibit a systematic contraction toward the ground truth, even when the final answer remains incorrect. Conversely, correct answers can be overturned by social pressure, particularly when wrong peers are near-correct. Together, these results show that multi-agent interaction induces structured, distance-aware movements in solution space that are invisible under binary correctness, clarifying when social reasoning leads to improvement, stability, or gradual regression in solution quality.
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Bio-Inspired Self-Supervised Learning for Wrist-worn Accelerometer Data
Prithviraj Tarale, Kiet Chu, Abhishek Varghese, Kai-Chun Liu, Maxwell A Xu, Mohit Iyyer, Sunghoon Lee
Wearable accelerometers enable large-scale health monitoring, yet learning robust human-activity representations has been constrained by scarce labeled data. While self-supervised learning offers a remedy, existing methods treat sensor streams as unstructured time series, overlooking the underlying biological structure of human movement, a factor we argue is critical for effective Human Activity Recognition (HAR). We introduce a novel tokenization strategy grounded in the *submovement theory* of motor control, which posits that continuous wrist motion is composed of elementary basis functions called submovements. We define our token as the *movement segment*, a computationally tractable unit of motion composed of a finite sequence of submovements. By pretraining a Transformer encoder via masked reconstruction of these tokens, we shift the learning focus from local waveform morphology to high-level structural and temporal organization. Pretrained on the NHANES corpus ($\approx$ 28k hours; $\approx$ 11k participants), our representations outperform strong wearable SSL baselines across six subject-disjoint HAR benchmarks. Code and pretrained weights are available at https://prithvitarale.github.io/biopm-site/.
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OPIC: Enhancing Language Model Merging via Optimizing In-Context Capability
Jie He, Weidong Bao, Chao Chen, Zhengyi Zhong, Shuai Zhang, Ji Wang
Task-vector–based model merging enables low-cost, training-free multi-task learning for large language models, but suffers from severe performance degradation due to task conflict. Prior mitigation strategies largely rely on validation data for costly hyperparameter tuning, limiting both interpretability and practicality.
We therefore propose OPIC, an evolutionary optimization–based model merging framework. Our preliminary experiments reveal that the degradation of In-Context Learning (ICL) capabilities is a primary driver of task conflict. Motivated by this insight, we formulate model merging as an optimization problem with ICL preservation as the objective. OPIC introduces a hierarchical refinement operators and optimizes it using self-generated data, effectively eliminating the reliance on external validation sets.
Experimental results demonstrate that OPIC achieves an average performance retention of 80.73%, outperforming SOTA methods and improving by up to 11.1% over recent validation-free approaches. In addition, OPIC is compatible with existing merging pipelines, offering a new alternative solution for deploying without validation dependencies. Code is available at: https://anonymous.4open.science/r/OPIC-CFFE.
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Long-Context Modeling with Dynamic Hierarchical Sparse Attention for Memory-Constrained LLM Inference
Siheng Xiong, Joe Zou, Faramarz Fekri, Yae Jee Cho
The quadratic cost of attention limits the scalability of long-context LLMs, especially under limited hardware memory budgets. While attention is often sparse, existing static sparse methods cannot adapt to task- or input-dependent variations, and recent dynamic approaches rely on predefined templates or heuristics that may sacrifice generality. We propose Dynamic Hierarchical Sparse Attention (DHSA), a data-driven framework that predicts attention sparsity online while keeping the LLM backbone frozen. DHSA performs hierarchical routing by estimating importance at the chunk level and propagating it to token-level interactions, preserving causally important dependencies while enabling efficient sparsification. Across Needle-in-a-Haystack test, LongBench and RULER, DHSA maintains near-dense accuracy in highly sparse regimes, achieving 12--20% relative accuracy gains over Block Sparse Attention at comparable prefill cost. With a memory-efficient tiled backend, DHSA delivers up to $10\times$ prefill speedup at 128K context length. On LLaMA-3.1-8B (4-bit), DHSA scales to 100K context on a single 24GB GPU, where dense attention fails. We provide complementary GPU and CPU backends, enabling DHSA to run across diverse hardware environments and multiple open-weight mod

el families. These results demonstrate DHSA as an efficient and adaptable soluti on for memory-constrained long-context LLM inference. Code is available at: http s://github.com/xiongsiheng/DHSA.
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Contrastive Spectral Rectification: Test-Time Defense towards Zero-shot Adversar ial Robustness of CLIP
Sen Nie, Jie Zhang, Zhuo Wang, Shiguang Shan, Xilin CHEN
Vision-language models (VLMs) such as CLIP have demonstrated remarkable zero-sho t generalization, yet remain highly vulnerable to adversarial examples (AEs). Wh ile test-time defenses are promising, existing methods fail to provide sufficien t robustness against strong attacks and are often hampered by high inference lat ency and task-specific applicability. To address these limitations, we start by investigating the intrinsic properties of AEs, which reveals that AEs exhibit se vere feature inconsistency under progressive frequency attenuation. We further a ttribute this to the model's inherent spectral bias. Leveraging this insight, we propose an efficient test-time defense named Contrastive Spectral Rectification (CSR). CSR optimizes a rectification perturbation to realign the input with the natural manifold under a spectral-guided contrastive objective, which is applie d input-adaptively. Extensive experiments across 16 classification benchmarks de monstrate that CSR outperforms the SOTA by an average of 18.1% against strong AP GD with modest inference overhead. Furthermore, CSR exhibits broad applicability across diverse visual tasks. Code is available at https://github.com/Summu77/CS R.
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Margin-Adaptive Confidence Ranking for Reliable LLM Judgement
Gaojie Jin, Yong Tao, Lijia Yu, Tianjin Huang
Jung et al. (2025) introduce a hypothesis testing framework for guaranteeing agr eement between large language models (LLMs) and human judgments, relying on the assumption that the model’s estimated confidence is monotonic with respect to hu man-disagreement risk.
In practice, however, this assumption may be violated, and the generalization be havior of the confidence estimator is not explicitly analyzed.
We mitigate these issues by learning a dedicated confidence estimator instead of relying on heuristic confidence signals.
Our approach leverages simulated annotator diversity and a margin-based ranking formulation to explicitly model how confidently an LLM distinguishes between hum an-agreement and human-disagreement cases.
We further derive generalization guarantees for this estimator, revealing a marg in-dependent trade-off that informs the design of an adaptive estimator training procedure.
When integrated into fixed-sequence testing, the learned confidence estimator yi elds improved ranking accuracy and empirically strengthens the monotonic relatio nship between confidence and disagreement risk, leading to higher success rates in satisfying target agreement levels across multiple datasets and judge models.
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Contrastive Diffusion Alignment: Learning Structured Latents for Controllable Ge neration
Ruchi Sandilya, Sumaira Perez, Charles Lynch, Lindsay Victoria, Benjamin Zebley, Derrick Matthew Buchanan, Mahendra T. Bhati, Nolan Williams, Timothy Spellman, FAITH M. GUNNING, Conor Liston, Logan Grosenick
Diffusion models excel at generation, but their latent spaces are high dimension al and not explicitly organized for interpretation or control. We introduce ConD A (Contrastive Diffusion Alignment), a plug-and-play geometry layer that applies contrastive learning to pretrained diffusion latents using auxiliary variables (e.g., time, stimulation parameters, facial action units). ConDA learns a low-di mensional embedding whose directions align with underlying dynamical factors, co nsistent with recent contrastive learning results on structured and disentangled representations. In this embedding, simple nonlinear trajectories support smoot h interpolation, extrapolation, and counterfactual editing while rendering remai ns in the original diffusion space. ConDA separates editing and rendering by lif

ting embedding trajectories back to diffusion latents with a neighborhood-preserving kNN decoder and is robust across inversion solvers. Across fluid dynamics, neural calcium imaging, therapeutic neurostimulation, facial expression dynamics, and monkey motor cortex activity, ConDA yields more interpretable and controllable latent structure than linear traversals and conditioning-based baselines, indicating that diffusion latents encode dynamics-relevant structure that can be exploited by an explicit contrastive geometry layer.

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Knowing Who, Not How Much: Learning-Augmented Mechanisms for Consumer Utility Maximization

Kira Goldner, Divyarthi Mohan, Thodoris Tsilivis

We study consumer utility maximization in an online random-order model where strategic agents arrive sequentially. To circumvent strong impossibility results for utility maximization, we turn to the framework of learning-augmented mechanism design. Crucially, we show that the types of predictions commonly used in learning-augmented mechanism design (such as predictions of agent values or the optimal value) are not useful for utility maximization, where payments are directly at odds with the objective. Instead, we identify that a qualitatively different kind of prediction suffices: the identity of the highest-valued agent. First, we provide a deterministic truthful mechanism for our online setting by adapting offline randomized techniques. Then, we augment our mechanism with predictions. When the predictions are correct, we achieve a constant approximation to the optimal solution under full information (consistency), and even when predictions are arbitrarily bad, we guarantee a constant approximation to the best implementable solution (robustness).

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Cost-aware Stopping for Bayesian Optimization

Qian Xie, Linda Cai, Alexander Terenin, Peter I. Frazier, Ziv Scully

In automated machine learning, scientific discovery, and other applications of Bayesian optimization, deciding when to stop evaluating expensive black-box functions in a cost-aware manner is an important but underexplored practical consideration. A natural performance metric for this purpose is the *cost-adjusted simple regret*, which explicitly captures the trade-off between solution quality and cumulative evaluation cost. Existing stopping rules for Bayesian optimization are either heuristic, or are theoretically grounded but designed to optimize simple regret without accounting for evaluation costs; as a result, they provide no guarantees against unnecessary evaluations when costs are high. We propose a *principled cost-aware stopping rule* for Bayesian optimization that adapts to varying evaluation costs without heuristic tuning. Our rule is grounded in a theoretical connection to state-of-the-art cost-aware acquisition functions, namely the Pandora's Box Gittins Index (PBGI) and log expected improvement per cost (LogEIPC). When paired with either acquisition function, we prove that the resulting policy satisfies a theoretical guarantee bounding the expected cost-adjusted simple regret. Across synthetic tasks and empirical benchmarks including hyperparameter optimization and neural architecture size search, pairing our stopping rule with PBGI or LogEIPC usually matches or outperforms other acquisition-function--stopping-rule pairs in terms of cost-adjusted simple regret.

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Benchmarking and Evolving Reason-Reflect-Rectify for Reflective Visual Generation

Junjie Wang, Xinghua Lou, Xiangtai Li, Ye Tian, Keyu Chen, Yulin Li, Bin Kang, Guangcan Mai, Yanwei Li, Zhuotao Tian, Liqiang Nie

Text-to-Image (T2I) models and Unified Multimodal Models (UMMs) have achieved remarkable progress in visual generation.

However, their reliance on a single-pass generation paradigm limits their ability to handle complex prompts requiring iterative refinement. To enable multi-round Reflective Visual Generation (RVG), we formalize the *Reason--Reflect--Rectify* (R^3) loop as a core framework and introduce R^3-Bench, a benchmark of over 600 expert-annotated instances that quantifies iterative reasoning and rectification capabilities.

Evaluation on R^3-Bench reveals a critical gap: while state-of-the-art models can identify generation errors, they fail to generate actionable rectification i nstructions.
To bridge this gap, we propose R^3-Refiner, a dual-stage framework leveraging Group Relative Policy Optimization (GRPO) and a Hierarchical Reward Mechanism (H RM) to better align rectification with reflective reasoning.
Experiments show that R^3-Refiner achieves significant improvements on R^3-B ench (+12.0% in Reflective Verdict Score, +9.0% in Rectification Score),
and can be seamlessly integrated with various MLLMs to enhance the generation qu ality of different T2I models on GenEval++ and T2I-CompBench.
Code is available at https://github.com/xiaomoguhz/R3-Bench.
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Random Process Flow Matching: Generative Implicit Representations of Multivariat e Random Fields

Julien Lalanne, David Picard, Lionel Boillot, Lina-María GUAYACÁN-CARRILLO, Leon Barens, Jean-Michel Pereira

Generative modeling provides a powerful framework for learning data distribution s. These models initially relied on probabilistic methods such as Gaussian Proce sses (GP) for uncertainty-aware predictions and shifted towards larger trainable models to learn more complex distributions. In this work, we introduce *Random Process (RP) Flow*, a Flow Matching-based framework that represents the vector f ield as a neural implicit function. Unlike modern generative methods, our settin g involves a single observed field, from which only sparse measurements are avai lable. RP Flow uses Random Fourier Features to learn an implicit signal represen tation that can be queried at any arbitrary location from a limited set of obser vations, while encoding uncertainty through ensemble sampling. We propose constr ucting a Bayesian posterior by GP regression in the source space to generate hig h-quality samples. Our empirical results demonstrate that this framework generat es realistic samples along with calibrated uncertainty estimates, even under cha llenging conditions such as high frequency, high sparsity, or high dimensionalit y. These findings position RP Flow as a milestone towards generative models for reconstruction tasks where data is scarce and uncertainty must remain traceable.
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Query-Based Asymmetric Modeling with Decoupled Input–Output Rates for Speech Res toration

Ui-Hyeop Shin, Jaehyun Ko, Woocheol Jeong, Hyung-Min Park

Speech restoration aims to recover clean speech from degraded recordings affecte d by noise, reverberation, bandwidth reduction, or other distortions, where inpu t and output sampling rates may differ. Existing approaches typically assume mat ched input--output rates and apply redundant resampling, limiting native multi-r ate processing. We formulate this gap as the extended sampling-frequency-indepen dent (xSFI) setting, where a model must operate under decoupled input--output ra tes, and propose TF-Restormer, a query-based xSFI modeling framework. The model encodes only the observed input band and synthesizes the unobserved high-frequen cy band through extension queries with band-partitioned cross-attention, yieldin g an asymmetric encoder--decoder that allocates capacity to analysis while keepi ng synthesis lightweight. Trained with a perceptual loss, a scaled log-spectral loss, and adversarial supervision via an SFI-STFT discriminator, TF-Restormer at tains balanced fidelity--perceptual quality as a single unified model, without r edundant resampling across denoising, dereverberation, bandwidth extension, and combined-distortion benchmarks under multiple sampling rates.
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UB-SMoE: Universally Balanced Sparse Mixture-of-Experts for Resource-adaptive Fe derated Fine-tuning of Foundation Models

Van-Tuan Tran, Hong-Hanh Nguyen-Le, Marco Ruffini, Merim Dzaferagic

Heterogeneous LoRA-rank methods address system heterogeneity in federated fine-t uning of foundation models by assigning client-specific ranks based on computati onal capabilities. However, these methods achieve only marginal computational sa vings, as dense feed-forward computations dominate. Sparse Mixture-of-Experts (S MoE) provides a promising alternative through conditional computation, yet we id

entify that its naive application to heterogeneous federated settings introduces two critical discordances: (i) expert utilization imbalance and (ii) non-differentiability of Top-K routing. Our convergence analysis demonstrates that these discordances lead to degraded convergence, particularly for resource-constrained clients. To address these challenges, we propose Universally Balanced Sparse Mixture-of-Experts (UB-SMoE), which introduces Dynamic Modulated Routing (DMR) to rebalance expert utilization, and Universal Pseudo-Gradient (PG) to reconstruct learning signals for non-activated experts. These mechanisms form a self-reinforcing cycle that maintains expert viability across heterogeneous clients. Experiments on benchmarks show that UB-SMoE achieves up to 45.0% computational reduction on low-resource clients while improving their performance by $8.7 \times$ compared to existing heterogeneous LoRA-rank methods.

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Probing Cross-modal Information Hubs in Audio-Visual LLMs

Jihoo Jung, Chaeyoung Jung, Ji-Hoon Kim, Joon Son Chung

Audio-visual large language models (AVLLMs) have recently emerged as a powerful architecture capable of jointly reasoning over audio, visual, and textual modalities. In AVLLMs, the bidirectional interaction between audio and video modalities introduces intricate processing dynamics, necessitating a deeper understanding of their internal mechanisms. However, unlike extensively studied text-only or large vision language models, the internal workings of AVLLMs remain largely unexplored. In this paper, we focus on cross-modal information flow between audio and visual modalities in AVLLMs, investigating where information derived from one modality is encoded within the token representations of the other modality. Through an analysis of multiple recent AVLLMs, we uncover two common findings. First, AVLLMs primarily encode integrated audio-visual information in sink tokens. Second, sink tokens do not uniformly hold cross-modal information. Instead, a distinct subset of sink tokens, which we term cross-modal sink tokens, specializes in storing such information.

Based on these findings, we further propose a simple training-free hallucination mitigation method by encouraging reliance on integrated cross-modal information within cross-modal sink tokens. Our code is available at https://github.com/kaistmm/crossmodal-hub.

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Langevin Rollout Optimization for Modelic Reinforcement Learning

Tianyi Zhang, Likun Wang, Guojian Zhan, Feihong Zhang, Yang Guan, Yao Lyu, Shengbo Eben Li

Planning-driven model-based (modelic) reinforcement learning has achieved impressive success in continuous control tasks but predominantly relies on zero-order optimizers like Model Predictive Path Integral (MPPI).

While robust for global exploration, MPPI updates actions solely through sampling and neglects the smooth return gradients inherent in structured dynamics that guide fine-grained search.

To complement MPPI’s robustness with gradient-guided precision, we first propose \textbf{La}ngevin \textbf{R}ollout \textbf{O}ptimization (LaRO), which leverages return gradients to refine actions via Langevin dynamics, achieving reliable local convergence without sacrificing multimodal exploration.

This is supported by a score-augmented world model that jointly learns dynamics and a score function within a unified latent space, facilitating efficient and accurate gradient estimation for real-time planning.

Second, we combine MPPI and LaRO through a simple yet effective choice mechanism, termed \textbf{M}aximum \textbf{L}ook-\textbf{A}head \textbf{P}lanning (MLAP).

Finally, we instantiate MLAP within the latest BOOM algorithm, replacing its MPPI-only planner and yielding BOOM-L. Empirical results on the DeepMind Control Suite and Humanoid Bench demonstrate that BOOM-L consistently outperforms strong baselines in both sample efficiency and final performance.

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Dual-stage Contrastive Learning-enhanced Multi-view Variational Clustering

Yanxi Liu, Yipin Hu, Fangxi Liu, Yanwei Yu, Lei Meng, Yongyong Chen, Guoqing Chao

Multi-view clustering aims to obtain a consensus clustering by integrating compl
ementary and consistent information from multiple views. However, two critical c
hallenges still exist in variational methods: (1) view heterogeneity and noise o
ften make fusion unreliable; (2) ambiguous posteriors and misassigned boundary
samples impact the clustering performance. To address these issues, we propose D
ual-stage Contrastive Learning-enhanced Multi-view Variational Clustering (DCL-M
VC), which integrates contrastive learning into both the fusion and representati
on stages. Firstly, at the fusion stage, we introduce a fusion-then-attention me
chanism to capture cross-view interactions and learn view-level attention weight
s for building a unified and reliable fused representation, and further introduc
e instance-level contrastive learning to enforce cross-view alignment at the ins
tance level. Secondly, we focus on boundary samples with uncertain posteriors an
d refine their cluster assignments by using cluster-center contrastive loss to e
nlarge inter-cluster margins, while leveraging prototypical contrastive learning
 with a confidence-aware curriculum to promote intra-cluster compactness at the
representation stage. Extensive experiments on six real-world datasets demonstra
te consistent improvements over strong baselines and validate the contribution o
f each component.
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Conformal Calibration Transfer

Achref Doula

Conformal prediction converts point predictions into set-valued predictions with
 coverage guarantees under exchangeability between calibration and deployment da
ta. We study *conformal calibration transfer*, where this requirement fails beca
use labeled calibration is available only in a source space, while prediction se
ts are needed in a target space linked to the source through *unlabeled paired*
observations (e.g., paired modalities or sensor changes).

We propose Transported Conformal Calibration (TCC): we transport labeled source
calibration into the target space using the paired data, and then correct residu
al post-transport mismatch using only unlabeled target inputs. We instantiate th
is correction with two complementary methods: **TCC-KS**, which uses a label-fre
e uncertainty surrogate to detect mismatch and adjust calibration conservatively
, and **weighted-TCC**, which reweights transported calibration toward the targe
t domain for improved efficiency when weights are stable. We provide finite-samp
le target-domain coverage guarantees that adapt to an observable measure of mism
atch. Across CIFAR-100-C, Tiny-ImageNet-C, and SEN12MS, we show reliable target-
domain coverage transfer without labeled target calibration data, with label-fre
e diagnostics that predict when correction is needed.
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From Parameters to Data: A Task-Parameter-Guided Fine-Tuning Pipeline for Effici
ent LLM Alignment

Hao Chen, Qi Zhang, Liyao Li, Zhanming Shen, Wentao Ye, Lirong Gao, NINGTAO WANG
, Xing Fu, Xiaoyu Shen, Junbo Zhao

Adapting Large Language Models (LLMs) to specialized domains typically incurs hi
gh data and computational overhead. While prior efficiency efforts have largely
treated data selection and parameter-efficient fine-tuning as isolated processes
, our empirical analysis suggests they may be intrinsically coupled. We posit th
e Strong Map Hypothesis: a sparse subset of attention heads plays a dominant rol
e in task-specific adaptation, acting as keys that unlock specific data patterns
. Building on this observation, we propose From Parameters to Data (P2D), a unif
ied framework that leverages these task-sensitive attention heads as a dual comp
ass for both sample mining and structural pruning. To rigorously quantify the to
tal pipeline cost, we introduce the Alignment Efficiency Ratio (AER) metric for
both selection latency and training time. Mechanistically, P2D identifies critic
al heads via a lightweight proxy and uses them as a functional filter to curate
high-affinity data, establishing a synergistic pipeline. Empirically, by updatin
g merely 10% of attention heads on 10% of the data, P2D achieves an 8.3 pp perfo
rmance gain over strong baselines and delivers a 7.0x end-to-end time speedup. T
hese results validate that precise parameter-data synchronization eliminates red
undancy, offering a new paradigm for efficient alignment.

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Reinforcement Fine-Tuning Naturally Mitigates Forgetting in Continual Post-Train ing
Song Lai, Haohan Zhao, Rong Feng, Changyi Ma, Wenzhuo Liu, Hongbo Zhao, Xi Lin, Dong Yi, Qingfu Zhang, Hongbin Liu, Gaofeng Meng, Fei Zhu
Continual post-training (CPT) is a popular and effective technique for adapting foundation models like multimodal large language models to ever-evolving downstr eam tasks. While existing research primarily focuses on methods like data replay , model expansion, or parameter regularization, the fundamental role of the lear ning paradigm remains largely unexplored. This paper presents a comparative anal ysis of two core post-training paradigms: supervised fine-tuning (SFT) and reinf orcement fine-tuning (RFT), investigating their respective impacts on knowledge retention during CPT. Our experiments are conducted across multiple multimodal t asks, utilizing Qwen2.5-VL-7B-Instruct as the base model. The investigation yiel ds two significant findings: (1) When continuously learning on downstream tasks, SFT leads to catastrophic forgetting of previously learned tasks. In contrast, RFT inherently preserves prior knowledge and achieves performance comparable to multi-task training. (2) RFT successfully protects and even enhances the model's general knowledge on standard benchmarks, while SFT degrades general model capa bilities severely. Further analysis reveals that this stability is not primarily due to explicit mechanisms like KL penalty or chain-of-thought reasoning. We in vestigate RFT's learning dynamics and find that its selective update mechanism i nherently prevents interference with established knowledge. Based on this insigh t, we propose a rollout-based instance filtering algorithm (RIF-RFT) that enhanc es the training efficiency of RFT by focusing on learnable samples. Our comprehe nsive study demonstrates the superiority of RFT as a robust paradigm for continu al post-training.
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Pushing the Limits of Block Rotations in Post-Training Quantization
Sai Sanjeet, Ian Colbert, Pablo Monteagudo-Lago, Giuseppe Franco, Yaman Umuroglu , Nicholas J. Fraser
Recent post-training quantization (PTQ) methods have adopted block rotations to diffuse outliers prior to rounding. While this reduces the overhead of online fu ll-vector rotations, the effect of block structure on outlier suppression remain s poorly understood. To fill this gap, we present the first systematic, non-asym ptotic analysis of outlier suppression for block Hadamard rotations. Our analysi s reveals that outlier suppression is fundamentally limited by the geometry of t he input vector. In particular, in the deterministic worst case, post-rotation o utliers are minimized when the pre-rotation ℓ_1 norm mass is evenly distrib uted across blocks. Guided by these insights, we introduce PeRQ (Permute, Rotate , then Quantize), a PTQ framework that redistributes activation mass via permuta tions prior to rotation. We propose a greedy mass diffusion algorithm to calibra te permutations by equalizing the expected blockwise ℓ_1 norms. To avoid ad ding inference overhead, we identify permutation-equivariant regions in transfor mer architectures to merge these permutations into model weights before deployme nt. Experiments show that PeRQ consistently improves accuracy across all block s izes, recovering up to 90% of the full-vector rotation perplexity when quantizin g Llama3 1B to INT4 with block size 16, compared to 46% without permutations.
****
LLMs as Noisy Channels: A Shannon Perspective on Model Capacity and Scaling Laws
Xu Ouyang, Deyi Liu, Yuhang Cai, Jing Liu, Yuan Yang, Chen Zheng, Thomas Hartvig sen, Yiyuan Ma
Existing scaling laws for Large Language Models (LLMs), predominantly monotonic power laws, fail to explain emerging non-monotonic phenomena such as *catastroph ic overtraining* and *quantization-induced degradation*, where performance deter iorates despite increased compute.

We propose the Shannon Scaling Law, a unified theoretical framework that models LLM training as information transmission over a noisy channel, grounded in the S hannon–Hartley theorem. By mapping model parameters to channel bandwidth and tra

ining tokens to signal power, our formulation explicitly captures the interactio n between learning signal and intrinsic noise. This perspective reveals a fundam ental Shannon capacity for LLMs: scaling model size or data without preserving a sufficient signal-to-noise ratio (SNR) inevitably amplifies noise, inducing a t ransition from monotonic improvement to U-shaped performance degradation.

We validate our theory through experiments on Pythia and OLMo2 under perturbatio ns, including Gaussian noise, quantization and supervised fine-tuning on math, Q A and code tasks. The Shannon Scaling Law consistently outperforms classical sca ling laws and recent perturbation-aware laws, achieving strong R^2 scores and accurately capturing loss basins missed by prior approaches. It also extrapolate s: fitted on $\leq$6.9B Pythia models with $\leq$180B tokens, it predicts the un seen 12B model up to 307B tokens at pooled $R^2{=}0.847$, while monotonic baseli nes collapse.
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Trajectory-Aware Spiking DiTs Conversion via Membrane Potential Error-Feedback
Haoran Fang, Tianxing Man, Xingchen Li, Wanli Shi, Jinjie Fang, Bin Gu
Diffusion Transformers (DiTs) have achieved state-of-the-art generative performa nce, yet their iterative denoising process remains computationally expensive and energy-intensive. Spiking Neural Networks (SNNs) offer a promising neuromorphic alternative for energy efficiency; however, the non-differentiable nature of sp iking neurons makes direct training difficult, positioning ANN-to-SNN conversion as a more practical, training-free solution. In this paper, we identify a criti cal challenge unique to converting DiTs: standard fixed-scale spiking neurons fa il to accommodate the highly dynamic activation ranges inherent across denoising steps. This mismatch leads to cumulative errors that significantly degrade gen eration fidelity. To resolve this, we propose a novel conversion framework featu ring Multi-Threshold (MT) neurons and a Membrane Potential Error-Feedback (MPEF) mechanism. MT neurons expand the expressive capacity of discrete spikes by empl oying a multi-level firing strategy. Concurrently, MPEF exploits the temporal co rrelation between successive denoising steps to recycle residual membrane potent ial, effectively compensating for information loss and mitigating distribution s hifts without retraining. Extensive experiments on ImageNet demonstrate that our framework achieves competitive generative quality with superior energy efficien cy, establishing a new performance benchmark for spiking Diffusion Transformers.
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GKD-Recruiter: Jointly Modeling Social and Task Heterogeneity for Spatial Crowds ourcing via Graph Knowledge Distillation
Yucen Gao, Zhemeng Yu, Zhuoran Li, Jianxiong Guo, Xiaofeng Gao
Social recruitment offers a solution to worker scarcity in Spatial Crowdsourcing (SC) but faces challenges that are often ignored in traditional Influence Maxim ization. First, task heterogeneity arising from offline execution constraints br eaks the interest-implies-participation assumption, as social influence often fa ils to translate into physical presence. Second, finite task demand creates a sa turation trap, a non-submodular setting in which utility drops sharply to zero o nce demand is met. To bridge these gaps, we propose GKD-Recruiter, a Task-Aware framework designed to maximize Effective Task Satisfaction (ETS). We explicitly model the complex worker-task affinity via a heterogeneous graph and capture dir ectional social influence using a novel Influential GAT. To robustly fuse these distinct signals, we introduce a Graph Knowledge Distillation mechanism. Further more, we employ Rainbow DQN to navigate the non-submodular combinatorial search space, avoiding the local optima that trap greedy heuristics. Extensive experime nts on real-world datasets demonstrate that GKD-Recruiter significantly outperfo rms state-of-the-art baselines in both solution quality and inference efficiency . The code is available at https://github.com/GaoYucen/GKD-Recruiter.
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MADE: Benchmark Environments for Closed-Loop Materials Discovery
Shreshth A Malik, Tiarnan Doherty, Panagiotis Tigas, Muhammed Razzak, S Roberts, Aron Walsh, Yarin Gal
Existing benchmarks for computational materials discovery primarily evaluate sta

tic predictive tasks or isolated computational sub-tasks. While valuable, these evaluations neglect the inherently iterative and adaptive nature of scientific discovery. We introduce **MA**terials **D**iscovery **E**nvironments (**MADE**), a novel framework for benchmarking end-to-end autonomous materials discovery pipelines. MADE simulates closed loop discovery campaigns in which an agent or algorithm proposes, evaluates, and refines candidate materials under a constrained oracle budget, capturing the sequential and resource-limited nature of real discovery workflows. We formalize discovery as a search for thermodynamically stable compounds relative to a given convex hull, and evaluate efficacy and efficiency via comparison to baseline algorithms. The framework is flexible; users can compose discovery agents from interchangeable components such as generative models, filters, and planners, enabling the study of arbitrary workflows ranging from fixed pipelines to agentic systems. We demonstrate this by conducting systematic experiments across a diverse range of systems and algorithms, finding that adaptive planning becomes more important to discovery efficiency as the search space scales.

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SEER: Transformer-based Robust Time Series Forecasting via Automated Patch Enhancement and Replacement

Xiangfei Qiu, Xvyuan Liu, Tianen Shen, Xingjian Wu, Hanyin Cheng, Bin Yang, Jilin Hu

Time series forecasting is important in many fields that require accurate predictions for decision-making. Patching techniques, commonly used and effective in time series modeling, help capture temporal dependencies by dividing the data into patches. However, existing patch-based methods fail to dynamically select patches and typically use all patches during the prediction process. In real-world time series, there are often low-quality issues during data collection, such as missing values, distribution shifts, anomalies and white noise, which may cause some patches to contain low-quality information, negatively impacting the prediction results. To address this issue, this study proposes a robust time series forecasting framework called $\textbf{SEER}$. Firstly, we propose an $\textit{Augmented Embedding Module}$, which improves patch-wise representations using a Mixture-of-Experts~(MoE) architecture and obtains series-wise token representations through a channel-adaptive perception mechanism. Secondly, we introduce a $\textit{Learnable Patch Replacement Module}$, which enhances forecasting robustness and model accuracy through a two-stage process: 1) a dynamic filtering mechanism eliminates negative patch-wise tokens; 2) a replaced attention module substitutes the identified low-quality patches with global series-wise token, further refining their representations through a causal attention mechanism. Comprehensive experimental results demonstrate the SOTA performance of SEER.

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RevealLayer: Disentangling Hidden and Visible Layers via Occlusion-Aware Image Decomposition

Binhao Wang, Shihao Zhao, Bo Cheng, Qiuyu Ji, Yuhang Ma, Liebucha Wu, Shanyuan Liu, Dawei Leng, Yuhui Yin

Recent diffusion-based approaches have made substantial progress in image layer decomposition.
However, accurately decomposing complex natural images remains challenging due to difficulties in occlusion completion, robust layer disentanglement, and precise foreground boundaries. Moreover, the scarcity of high-quality multi-layer natural image datasets limits advancement.
To address these challenges, we propose **RevealLayer**, a diffusion-based framework that decomposes an RGB image into multiple RGBA layers, enabling precise layer separation and reliable recovery of occluded content in natural images.
RevealLayer incorporates three key components: (1) a **Region-Aware Attention** module to disentangle hidden and visible layers; (2) an **Occlusion-Guided Adapter** to leverage contextual information to enhance overlapping regions; and (3) a **composite loss** to enforce sharp alpha boundaries and suppress residual artifacts.
To support training and evaluation, we introduce **RevealLayer-100K**, a high-qu

ality multi-layer natural image constructed through a collaboration between auto
mated algorithms and human annotation, and further establish **RevealLayerBench*
* for benchmarking layer decomposition in general natural scenes.
Extensive experiments demonstrate that RevealLayer consistently outperforms exis
ting approaches in layer decomposition.
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TreePO: Enhancing Policy Efficacy and Inference Efficiency with Tree Modeling
Yizhi LI, Qingshui Gu, Zhoufutu Wen, Ziniu Li, Ruibin Yuan, Tianshun Xing, Shuyu
e Guo, Tuney Zheng, Xin Zhou, Xingwei Qu, Wangchunshu Zhou, Zheng Zhang, Wei She
n, Wei Xue, Qian Liu, Chenghua Lin, Jian Yang, Ge Zhang, Wenhao Huang
Recent advancements in aligning large language models via reinforcement learning
 have achieved remarkable gains in solving complex reasoning problems, but at th
e cost of expensive on-policy rollouts and limited exploration of diverse reason
ing paths. In this work, we introduce TreePO, involving a self-guided rollout al
gorithm that views sequence generation as a tree-structured searching process. C
omposed of dynamic tree sampling policy and fixed-length segment decoding, TreeP
O leverages local uncertainty to warrant additional branches. By amortizing comp
utation across common prefixes and pruning low-value paths early, TreePO essenti
ally reduces the per-update compute burden while preserving or enhancing explora
tion diversity. Key contributions include: (1) a segment-wise sampling algorithm
 that alleviates the KV cache burden through contiguous segments and spawns new
branches along with an early-stop mechanism; (2) a tree-based segment-level adva
ntage estimation that considers both global and local proximal policy optimizati
on. and (3) analysis on the effectiveness of probability and quality-driven dyna
mic divergence and fallback strategy. We empirically validate the performance ga
in of TreePO on a set reasoning benchmarks and the efficiency saving of GPU hour
s from 22% up to 43% of the sampling design for the trained models, meanwhile sh
owing up to 40% reduction at trajectory-level and 35% at token-level sampling co
mpute for the existing models. While offering a free lunch of inference efficien
cy, TreePO reveals a practical path toward scaling RL-based post-training with f
ewer samples and less compute.
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REST: Diffusion-based Real-time End-to-end Streaming Talking Head Generation via
 ID-Context Caching and Asynchronous Streaming Distillation
Haotian Wang, Yuzhe Weng, Jun Du, Haoran Xu, Xiaoyan Wu, Shan He, Bing Yin, Cong
 Liu, Qingfeng Liu
Diffusion models have significantly advanced the field of talking head generatio
n (THG). However, slow inference speeds and prevalent non-autoregressive paradig
ms severely constrain the application of diffusion-based THG models. In this stu
dy, we propose REST, a pioneering diffusion-based, real-time, end-to-end streami
ng audio-driven talking head generation framework. To support real-time end-to-e
nd generation, a compact video latent space is first learned through a spatiotem
poral variational autoencoder with a high compression ratio. Additionally, to en
able semi-autoregressive streaming within the compact video latent space, we int
roduce an ID-Context Cache mechanism, which integrates ID-Sink and Context-Cache
 principles into key-value caching for maintaining identity consistency and temp
oral coherence during long-term streaming generation. Furthermore, an Asynchrono
us Streaming Distillation (ASD) strategy is proposed to mitigate error accumulat
ion and enhance temporal consistency in streaming generation, leveraging a non-s
treaming teacher with an asynchronous noise schedule to supervise the streaming
 student. REST bridges the gap between autoregressive and diffusion-based approac
hes, achieving a breakthrough in efficiency for applications requiring real-time
 THG. Experimental results demonstrate that REST outperforms state-of-the-art me
thods in both generation speed and overall performance.
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The Expert Strikes Back: Interpreting Mixture-of-Experts Language Models at Expe
rt Level
Jeremy Herbst, Stefan Wermter, Jae Hee Lee
Mixture-of-Experts (MoE) architectures have become the dominant choice for scali
ng Large Language Models (LLMs), activating only a subset of parameters per toke

n. While MoE architectures are primarily adopted for computational efficiency, it remains an open question whether their sparsity makes them inherently easier to interpret than dense feed-forward networks (FFNs). We compare MoE experts and dense FFNs using k-sparse probing and find that expert neurons are consistently less polysemantic, with the gap widening as routing becomes sparser. This suggests that sparsity pressures both individual neurons and entire experts toward monosemanticity. Leveraging this finding, we _zoom out_ from the neuron to the expert level as a more effective unit of analysis. We validate this approach by automatically interpreting hundreds of experts. This analysis allows us to resolve the debate on specialization: experts are neither broad domain specialists (e.g., biology) nor simple token-level processors. Instead, they function as fine-grained task experts, specializing in linguistic operations or semantic tasks (e.g., closing brackets in LaTeX). Our findings suggest that MoEs are inherently interpretable at the expert level, providing a clearer path toward large-scale model interpretability. Code is available at: https://github.com/jerryy33/MoE_analysis.

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Conformal Thinking: Risk Control for Reasoning on a Compute Budget

Xi Wang, Anushri Suresh, Alvin Zhang, Rishi More, William Jurayj, Mehrdad Farajtabar, Daniel Khashabi, Eric Nalisnick

Reasoning Large Language Models (LLMs) enable test-time scaling, with dataset-level accuracy improving as the token budget increases,
motivating adaptive reasoning---spending tokens when they improve reliability and stopping early when additional computation is unlikely to help.
However, setting the token budget, as well as the threshold for adaptive reasoning, is a practical challenge that entails a fundamental risk-accuracy trade-off. We re-frame the budget setting problem as risk control, limiting the error rate while minimizing compute.
Our framework introduces an upper threshold that stops reasoning when the model is confident (risking incorrect output) and a novel parametric \emph{lower} threshold that preemptively stops unsolvable instances (risking premature stoppage). Given a target risk and a validation set, we use distribution-free risk control to optimally specify these stopping mechanisms.
Empirical results across diverse reasoning tasks and models demonstrate the effectiveness of our risk control approach, demonstrating computational efficiency gains from the lower threshold and ensemble stopping mechanisms, all while adhering to the user-specified risk target.

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Sample Complexity Bounds for Robust Mean Estimation with Mean-Shift Contamination

Ilias Diakonikolas, Giannis Iakovidis, Daniel Kane, Sihan Liu

We study the basic task of mean estimation in the presence of mean-shift contamination. In the mean-shift contamination model, an adversary is allowed to replace a small constant fraction of the clean samples by samples drawn from arbitrarily shifted versions of the base distribution. Prior work characterized the sample complexity of this task for the special cases of the Gaussian and Laplace distributions. Specifically, it was shown that consistent estimation is possible in these cases, a property that is provably impossible in Huber's contamination model. An open question posed in earlier work was to determine the sample complexity of mean estimation in the mean-shift
contamination model for general base distributions. In this work, we study and essentially resolve this open question. Specifically, we show that, under mild spectral conditions on the characteristic function of the (potentially multivariate) base distribution, there exists a sample-efficient algorithm that estimates the target mean to any desired accuracy. We complement our upper bound with a qualitatively matching sample complexity lower bound. Our techniques make critical use of Fourier analysis, and in particular introduce the notion of a Fourier witness as an essential ingredient of our upper and lower bounds.

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ANCHOR: Abductive Network Construction with Hierarchical Orchestration for Relia

ble Probability Inference in Large Language Models
Wentao Qiu, Guanran Luo, Zhongquan Jian, Jingqi Gao, Meihong Wang, Qingqiang Wu
A central challenge in large-scale decision-making under incomplete information is estimating reliable probabilities. Recent approaches use Large Language Models (LLMs) to generate explanatory factors and coarse-grained probability estimates, which are then refined by a Naïve Bayes model over factor combinations. However, sparse factor spaces often yield ''unknown'' predictions, while expanding factors increases noise and spurious correlations, weakening conditional independence and degrading reliability. To address these limitations, we propose Anchor, an aggregated Bayesian inference framework over a hierarchical factor space. It constructs dense factor hierarchies through iterative generation and clustering, maps contexts via hierarchical retrieval and refinement, and augments Naïve Bayes with a Causal Bayesian Network to model latent factor dependencies. Experiments show that Anchor markedly reduces ''unknown'' predictions and produces more reliable probability estimates than direct LLM baselines, achieving state-of-the-art performance while significantly reducing time and token overhead.
****
Post-Training with Policy Gradients: Optimality and the Base Model Barrier
Alireza Mousavi-Hosseini, Murat A Erdogdu
We study post-training linear autoregressive models with outcome and process rewards. Given a context x, the model must predict the response $y \in \mathcal{Y}^N$, a sequence of length N that satisfies a γ margin condition, an extension of the standard separability to sequences.
We prove that on test samples where the base model achieves a non-trivial likelihood α, a variant of policy gradient (PG) can achieve likelihood $1 - \varepsilon$ with an essentially minimax optimal number of reward queries $\tilde{\mathcal{O}}((\alpha^{-1} + \varepsilon^{-1})/\gamma^2)$.
However, a barrier arises for going beyond the support of the base model.
We prove that the overall expected error after post-training with outcome rewards is governed by a property of the base model called the *Likelihood Quantile* (LQ), and that variants of PG, while minimax optimal, may require a number of reward queries exponential in N to go beyond this support, regardless of the pre-training algorithm.
To overcome this barrier, we study post-training with a process reward model, and demonstrate how PG variants in this setting avoid the curse of dimensionality in N via dependence on a token-level LQ.
Along the way, we prove that under the margin condition, SGD with adaptive learning rate (LR) achieves a near optimal test error for statistical learning, and PG with adaptive LR achieves a near optimal number of mistakes for online learning while being computationally efficient whenever possible, both of which may be of independent interest.
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Discriminative Attribute Graph Clustering Through Topology-Guided Contrastive Learning
Ling Ding, Zhizhi Yu, Cuiying Huo
Deep attribute graph clustering aims to learn discriminative node representations by leveraging both node attributes and graph topology to partition nodes into distinct clusters. Although substantial progress has been made in attribute-graph clustering in recent years, two key challenges remain: noisy edges in the original adjacency matrix degrade the quality of information propagation, and redundant feature information across different feature views hampers the learning of discriminative representations. To address these issues, we propose a self-supervised attribute graph clustering method based on topological reconstruction and correlation decorrelation. First, we reconstruct the graph topology by computing intersections between k-nearest neighbors and the original adjacency relationships, while simultaneously leveraging global semantic information from K-means clustering to filter out noisy nodes. This reconstructed topology effectively mitigates information redundancy during feature aggregation in Graph Neural Networks. Second, unlike existing augmentation-based contrastive methods, we treat the feature representations from an auto-encoder (AE) and a graph auto-encoder (GAE) a

s two complementary natural views. We then apply mutual information minimization and a decorrelation constraint to suppress redundant information between views, yielding more discriminative node representations. Extensive experiments on four widely-used graph datasets—ACM, DBLP, CITE, and AMAP—demonstrate that our method consistently outperforms six state-of-the-art baselines.
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SegPVSG: Panoptic Video Scene Graph Generation via Temporal Focusing and Generative Augmentation
YiKai Li, Quhui Ke, Jinglin Liang, Zhi-Yuan Zhang, Zhidi Lin, Shuangping Huang
Panoptic Video Scene Graph Generation (PVSG) aims to identify relations between pixel-level entities in a video, serving as a novel paradigm for structured video parsing. However, this task faces two key challenges. First, the interactions between entities are temporally fragmented and sparse, meaning videos are dominated by irrelevant content with limited salient information. Second, the distribution of relations exhibits a significant long-tailed pattern, making models struggle to perform well on tail categories with insufficient data. To address these issues, we propose SegPVSG, an innovative, temporal-segment-aware PVSG framework consisting of two key components: TempFocusNet (TFN) and Relation-centric Generative Video Augmentation (RGVA) module. TFN is a localization-then-recognition network that improves PVSG performance by explicitly localizing and focusing on salient segments before relation recognition. Meanwhile, RGVA is a novel augmentation module that generates realistic, context-consistent video segments for rare relations and coherently inserts them into original videos. Our method outperforms prior methods by +3.53 mR@20 and +5.9 mR@50, demonstrating its effectiveness.
The code is available at https://github.com/ticatt/SegPVSG.
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A unified theory of feature learning in RNNs and DNNs
Jan Philipp Bauer, Kirsten Fischer, Moritz Helias, Agostina Palmigiano
Recurrent and deep neural networks (RNNs/DNNs) are cornerstone architectures in machine learning. Remarkably, RNNs differ from DNNs only by weight sharing, as can be shown through unrolling in time. How does this structural similarity fit with the distinct functional properties these networks exhibit? To address this question, we here develop a unified mean-field theory for RNNs and DNNs in terms of representational kernels, describing fully trained networks in the feature learning (μP) regime. This theory casts training as Bayesian inference over sequences and patterns, directly revealing the functional implications induced by the RNNs' weight sharing. In DNN-typical tasks, we identify a phase transition when the learning signal overcomes the noise due to randomness in the weights: below this threshold, RNNs and DNNs behave identically; above it, only RNNs develop correlated representations across timesteps. For sequential tasks, the RNNs' weight sharing furthermore induces an inductive bias that aids generalization by interpolating unobserved time steps. Overall, our theory offers a way to connect architectural structure to functional biases.
****
A Positive Case for Faithfulness: Explanations Help Predict Model Behavior
Harry Mayne, Justin Singh Kang, Dewi Sid William Gould, Kannan Ramchandran, Adam Mahdi, Noah Y. Siegel
LLM self-explanations are often presented as a promising tool for AI oversight, yet their faithfulness to the model's true reasoning process is poorly understood. Existing faithfulness metrics have critical limitations, typically relying on identifying unfaithfulness via adversarial prompting or detecting reasoning errors. These methods overlook the predictive value of explanations. We introduce *Normalized Simulatability Gain* (NSG), a general and scalable metric based on the idea that a faithful explanation should allow an observer to learn a model's decision-making criteria, and thus better predict its behavior on related inputs. We evaluate 18 frontier proprietary and open-weight models, e.g., Gemini 3, GPT-5.2, and Claude 4.5, on 7,000 counterfactuals from popular datasets covering health, business, and ethics. We find self-explanations substantially improve prediction of model behavior (11-37% NSG). Self-explanations also provide more predi

ctive information than explanations generated by external models, even when thos
e models are stronger. This implies an advantage from self-knowledge that extern
al explanation methods cannot replicate. Our approach also reveals that, across
models, 5-15% of self-explanations are egregiously misleading. Despite their imp
erfections, we show a positive case for self-explanations: they encode informati
on that helps predict model behavior.
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Class-Grouped Normalized Momentum and Faster Hyperparameter Exploration to Tackl
e Class Imbalance in Federated Learning
Haemin Park, Diego Klabjan, Martin W. Braun, Xiuqi Li, Balakrishnan Ananthanaray
anan
Class imbalance poses a critical challenge in federated learning (FL), where und
errepresented classes suffer from poor predictive performance yet cannot be addr
essed by standard centralized techniques due to privacy and heterogeneity constr
aints. We propose FedCGNM (Federated Class-Grouped Normalized Momentum), a clien
t-side optimizer in FL that partitions classes into a small number of groups bas
ed on minimum within-group variance, maintains a momentum per group, normalizes
each group momentum to unit length, and uses the summation of the normalized gro
up momentums as an update direction. This design both equalizes gradient magnitu
de across majority and minority groups and mitigates the noise inherent in rare-
class gradients. We further provide a theoretical convergence analysis explicitl
y accounting for time-varying resampling-rates. Additionally, to efficiently opt
imize these rates in small-client regimes, we introduce FedHOO, an X-armed-bandi
t (XAB) based algorithm that exploits federated parallelism that evaluates many
combinations of two candidate rates per client at linear cost. Empirical evaluat
ion on four public long-tailed benchmarks and a proprietary chip-defect dataset
demonstrates that FedCGNM consistently outperforms baselines, with FedHOO yieldi
ng further gains in small-scale federations.
****
CodeTaste: Can LLMs Generate Human-Level Code Refactorings?
Alex Thillen, Niels Mündler, Veselin Raychev, Martin Vechev
LLM coding agents can generate working code, but their solutions often accumulat
e complexity, duplication, and architectural debt. Human developers address such
 issues through refactoring: behavior-preserving program transformations that im
prove structure and maintainability. We investigate whether agents (i) can execu
te refactorings reliably and (ii) identify the refactorings that human developer
s actually chose in real codebases.
To this end, we construct CodeTaste, a benchmark mined from large multi-file ope
n-source refactorings. To score solutions, we combine repository test suites tha
t measure functional correctness with tailored static checks that verify removal
 of undesired and introduction of desired code patterns using dataflow reasoning
.
Our results show a clear gap: agents perform well at implementing refactorings t
hat are specified in detail, but often fail to discover the human refactoring ch
oices when given a focus area for changes. A propose-then-implement decompositio
n improves alignment, and selecting the best-aligned proposal before implementat
ion can yield further gains. CodeTaste provides an evaluation target and a poten
tial preference signal for aligning coding agents with human refactoring decisio
ns in realistic codebases. We release the benchmark, leaderboard, and code.
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TelecomTS: A Multi-Modal Observability Dataset for Time Series and Language Anal
ysis
Austin Feng, Andreas Varvarigos, Ioannis Panitsas, Daniela Fernandez, Jinbiao We
i, Yuwei Guo, Jialin Chen, Ali Maatouk, Leandros Tassiulas, Rex Ying
Modern enterprises generate vast streams of time series metrics when monitoring
complex systems, known as observability data. Unlike conventional time series fr
om domains such as climate, observability data are zero-inflated, highly stochas
tic, and exhibit minimal temporal structure. Despite their importance, observabi
lity datasets remain underrepresented in public benchmarks due to proprietary re
strictions and privacy concerns. Existing datasets are often anonymized and norm

alized, removing scale information and limiting their use for tasks such as anom aly detection, root cause analysis, and multi-modal reasoning. To address this g ap, we introduce TelecomTS, a large-scale observability dataset derived from a 5 G telecommunications network. TelecomTS features heterogeneous, de-anonymized co variates with explicit absolute scale information and provides a diverse suite o f downstream tasks, including anomaly detection, root cause analysis, and multi- modal question-answering. Benchmarking state-of-the-art time series, language, r easoning, and multi-modal foundation models reveals that existing approaches str uggle with the abrupt, noisy, and high-variance dynamics characteristic of obser vability data. Our experiments further underscore the importance of preserving c ovariates' absolute scale, emphasizing the need for foundation time series model s that natively leverage scale information for practical real-world observabilit y applications. The code is available at: \url{https://github.com/Ali-maatouk/Te lecomTS}.

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Optimal Fair Aggregation of Crowdsourced Noisy Labels using Demographic Parity C onstraints

Gabriel Singer, Samuel Gruffaz, Olivier VO VAN, Nicolas Vayatis, Argyris Kaloger atos

As acquiring reliable ground-truth labels is usually costly, or infeasible, crow dsourcing and aggregation of noisy human annotations is a common alternative. Ag gregating subjective labels, though, may amplify individual biases, particularly regarding sensitive features, raising fairness concerns. Nonetheless, fairness in crowdsourced aggregation remains largely unexplored, with no existing converg ence guarantees and only limited post-processing approaches for enforcing $\vare psilon$-fairness under demographic parity. We address this gap by analyzing the fairness of crowdsourced aggregation methods within the ε-fairness framework, for Majority Vote and Optimal Bayesian aggregation. In the small-crow d regime, we derive an upper bound on the fairness gap of Majority Vote in terms of the fairness gaps of the individual annotators. We further show that the fai rness gap of the aggregated consensus converges exponentially fast to that of th e ground-truth under interpretable conditions. Since ground-truth itself may sti ll be unfair, we generalize a state-of-the-art multiclass fairness post-processi ng algorithm from the continuous to the discrete setting, which enforces strict demographic parity constraints on any aggregation rule. Experiments on syntheti c and real datasets demonstrate the effectiveness of our approach and corroborat e the theoretical insights.

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Rethinking Genomic Modeling Through Optical Character Recognition

Hongxin Xiang, Pengsen Ma, Yunkang Cao, Di Yu, Haowen Chen, Xinyu Yang, xiangxia ng Zeng

Recent genomic foundation models largely adopt large language model architecture s that treat DNA as a one-dimensional token sequence. However, exhaustive sequen tial reading is structurally misaligned with sparse and discontinuous genomic se mantics, leading to wasted computation on low-information background and prevent ing understanding-driven compression for long contexts. Here, we present Optical DNA, a vision-based framework that reframes genomic modeling as Optical Characte r Recognition (OCR)-style document understanding. OpticalDNA renders DNA into st ructured visual layouts and trains an OCR-capable vision--language model with a \emph{visual DNA encoder} and a \emph{document decoder}, where the encoder produ ces compact, reconstructible visual tokens for high-fidelity compression. Buildi ng on this representation, OpticalDNA defines prompt-conditioned objectives over core genomic primitives—reading, region grounding, subsequence retrieval, and m asked span completion—thereby learning layout-aware DNA representations that ret ain fine-grained genomic information under a reduced effective token budget. Acr oss diverse genomic benchmarks, OpticalDNA consistently outperforms recent basel ines; on sequences up to 450k bases, it achieves the best overall performance wi th nearly $20\times$ fewer effective tokens, and surpasses models with up to $98 5\times$ more activated parameters while tuning only 256k \emph{trainable} param eters.

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Covariance Volume Maximization for Embodied Latent Exploration in Deep Reinforcement Learning
Yiming Wang, Yiheng Zhang, Kaiyan Zhao, Xingjie Zuo, Xingyu Liu, Xuetao Li, Furui Liu, Bo An, Leong Hou U
Efficient exploration remains a key challenge in deep reinforcement learning, especially for embodied agents operating in realistic environments with high-dimensional observations and complex dynamics. Recent latent exploration methods define bonuses in a learned latent space, but often struggle in these settings where (i) representations can be
noisy or policy-dependent, and (ii) common strategies such as randomized latent objectives or fixed directional spanning are brittle and fail to improve global coverage. We propose Covariance Volume Maximization (CVM), a coverage-driven latent exploration framework with two key components. First, we learn a behavioral state encoder using a *policy-mixture* objective to reduce representation drift under rapidly changing exploration policies, yielding stable and behaviorally meaningful latent displacements. Second, CVM rewards each transition by its exact increase in the log-determinant of the covariance of recent latent displacements, explicitly expanding the explored region and prioritizing under-covered directions. This objective coincides with the classical *D-optimal* design criterion, providing an information-efficiency justification. Extensive experiments on embodied navigation and manipulation tasks demonstrate that CVM substantially improves exploration efficiency and robustness, and scales effectively to different environments.
****
Seeing Without Understanding: Disentangling Perception, Reasoning, and Simulation in VLM Gameplay
Dingyang Jin, Jiawei He, Calvin Lo, Steven Hu, Ryan Rad
While Vision-Language Models (VLMs) excel on static visual benchmarks, they consistently underperform in game-based reasoning, yet existing evaluations conflate failures in perception, rule comprehension, and reasoning. We propose a two-stage diagnostic framework that decomposes VLM performance into testable components: controlled perception tests isolating visual encoding, and a diagnostic matrix with a six-level rule complexity ladder evaluated in both explicit verification and predictive simulation modes. Experimenting with six state-of-the-art VLMs reveals three failure patterns: (1) coordinated spatial drift, where off-by-one localization errors among adjacent pieces share the same shift direction at 1.5-$1.9\times$ the rate expected under spatial independence; (2) perception-reasoning dissociation, where models correctly verify board states but fail to apply rules—at complex constraint levels, perception remains relatively stable while reasoning accuracy plummets, with even the best-performing model capped at 75%; and (3) a simulation gap, with performance dropping by up to 27 points when predicting future states versus verifying observed outcomes. These limitations persist across model scales and are not resolved by scaling, text-only input, or structured prompting. Code and data are available at https://github.com/chillibeaver/PRS-Diag.
****
Certified Robustness under Heterogeneous Perturbations via Hybrid Randomized Smoothing
Blaise Delattre, Hengyu WU, Paul Caillon, Wei Yang Bryan Lim, Yang Cao
Randomized smoothing provides strong, model-agnostic robustness certificates, but
existing guarantees are limited to single modalities, treating continuous and
discrete inputs in isolation.
This limitation becomes critical in multimodal models, where decisions depend on
cross-modal semantics and adversaries can jointly perturb heterogeneous inputs,
rendering unimodal certificates insufficient.

We introduce a unified randomized smoothing framework for mixed discrete--continuous
inputs based on an analytically tractable Neyman--Pearson formulation of the joint
worst-case problem.
By analyzing the joint likelihood ordering induced by factorized discrete and
continuous noise, our approach yields a closed-form, one-dimensional certificate
that strictly generalizes both Gaussian (image-only) and discrete (text-only)
randomized smoothing.
We validate the framework on multimodal safety filtering, providing, to our
knowledge, the first model-agnostic Neyman--Pearson certificate for joint
discrete-token and continuous-image perturbations in interaction-dependent
text--image safety filtering.
****

VideoGPA: Distilling Geometry Priors for 3D-Consistent Video Generation
Hongyang Du, Junjie Ye, Xiaoyan Cong, Runhao Li, Jingcheng Ni, Aman Agarwal, Zeqi Zhou, Zekun Li, Randall Balestriero, Yue Wang
While recent video diffusion models (VDMs) produce visually impressive results, they fundamentally struggle to maintain 3D structural consistency, often resulting in object deformation or spatial drift. We hypothesize that these failures arise because standard denoising objectives lack explicit incentives for geometric coherence. To address this, we introduce VideoGPA (Video Geometric Preference Alignment), a data-efficient self-supervised framework that leverages a geometry foundation model to automatically derive dense preference signals that guide VDMs via Direct Preference Optimization (DPO). This approach effectively steers the generative distribution toward inherent 3D consistency without requiring human annotations. VideoGPA significantly enhances temporal stability, geometric plausibility, and motion coherence using minimal preference pairs, consistently outperforming state-of-the-art baselines in extensive experiments.
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SAEs-BrainMap: Unveiling the Emergence of Specialized Concepts in Deep Models via a Brain Alignment
Ziming Mao, Jia Xu, Wenxuan Pan, Mufan Xue, Yaochu Jin, Guoyuan Yang
Understanding the internal mechanisms of deep neural networks remains a significant challenge, particularly in elucidating how generic visual concepts emerge within latent spaces. In this work, we propose SAEs-BrainMap, a novel framework that utilizes human brain activation patterns from the ventral visual pathway as objective probes to guide the identification of features decomposed by Sparse Autoencoders (SAEs). Our quantitative and qualitative empirical results demonstrate a robust representational alignment between sparse model features and biological Regions of Interests (ROIs), confirming the feasibility of utilizing brain signals to characterize model functionality. By leveraging this alignment, we trace the hierarchical trajectory of generic concepts cross layers and utilize the brain's hierarchical structure to visualize the model's global processing flow, providing novel insights into model interpretability. Our code is available at [this repository](https://github.com/Matzohe/SAEs-BrainMap).
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EnsembleVLA: Ensemble Learning for Vision-Language Action Models
Mingchen Song, Xiang Deng, Jie Wei, Dongmei Jiang, Liqiang Nie, Weili Guan
Diverse Vision-language-action (VLA) models have been proposed and demonstrated remarkable capabilities in robotic manipulation. However, how to effectively ensemble VLAs to further enhance performance remains largely unexplored, as conventional ensemble techniques designed for discriminative tasks cannot be directly applied to generative action policies with high-dimensional, multimodal distributions. To address this challenge, we propose EnsembleVLA, an energy-based framework that enables principled ensemble of diverse VLA models. We establish a unified theoretical framework showing that both diffusion-based and flow-based VLA mod

els can be formulated as energy-based models, where additive energy combination naturally induces policy composition at the distribution level. This theoretical foundation enables multiple pre-trained policies to be seamlessly aggregated into a stronger ensemble policy. Building upon this compositional framework, EnsembleVLA further incorporates learnable composition weights for dynamic policy balancing, coupled with a confidence-aware gating mechanism that adaptively modulates bounded residual corrections, collectively ensuring stable and robust task execution. Extensive experiments demonstrate that EnsembleVLA achieves competitive performance across various tasks in both simulated and real-world environments.
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MotiMotion: Motion-Controlled Video Generation with Visual Reasoning
Lee Hsin-Ying, Hanwen Jiang, Yiqun Mei, Jing Shi, Ming-Hsuan Yang, Zhixin Shu
Current motion-controlled image-to-video generation models rigidly follow user-provided trajectories that are often sparse, imprecise, and causally incomplete. Such reliance often yields unnatural or implausible outcomes, especially by missing secondary causal consequences.
To address this, we introduce MotiMotion, a novel framework that reformulates motion control as a reasoning-then-generation problem.
To encourage causally grounded and commonsense-consistent interactions, we leverage a training-free vision-language reasoner to refine image-space coordinates of primary trajectories and to hallucinate plausible secondary motions.
To further improve motion naturalness, we propose a confidence-aware control scheme that modulates guidance strength, enabling the model to closely follow high-confidence plans while correcting artifacts under low-confidence inputs with its internal generative priors.
To support systematic evaluation, we curate a new image-to-video benchmark, MotiBench, consisting of interaction-centric scenes where new events are triggered by motion.
Both VLM-based evaluation and a human study on MotiBench demonstrate that MotiMotion produces videos with more plausible object behaviors and interaction, and is preferred over existing approaches.
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Efficient Hallucination Detection for LLMs Using Uncertainty-Aware Attention Heads
Artem Vazhentsev, Lyudmila Rvanova, Gleb Kuzmin, Ekaterina Fadeeva, Ivan Lazichny, Alexander Panchenko, Maxim Panov, Mrinmaya Sachan, Preslav Nakov, Timothy Baldwin, Artem Shelmanov
While large language models (LLMs) have become highly capable, they remain prone to factual inaccuracies, commonly referred to as "hallucinations." Uncertainty quantification (UQ) offers a promising way to mitigate this issue, but most existing methods are computationally intensive and/or require supervision. In this work, we propose Recurrent Attention-based Uncertainty Quantification (RAUQ), an unsupervised and efficient framework for identifying hallucinations. The method leverages an observation about transformer attention behavior: when incorrect information is generated, certain "uncertainty-aware" attention heads tend to reduce their focus on preceding tokens. RAUQ automatically detects these attention heads and combines their activation patterns with token-level confidence measures in a recurrent scheme, producing a sequence-level uncertainty estimate in just a single forward pass. Through experiments on twelve datasets spanning question answering, summarization, and translation across nine different LLMs, we show that RAUQ consistently outperforms state-of-the-art UQ baselines. Importantly, it incurs minimal overhead, requiring less than 1% additional computation.
Since it requires neither labeled data nor extensive parameter tuning, RAUQ serves as a lightweight, plug-and-play solution for real-time hallucination detection in white-box LLMs.
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What is Missing? Explaining Neurons Activated by Absent Concepts
Robin Hesse, Simone Schaub-Meyer, Janina Hesse, Bernt Schiele, Stefan Roth
Explainable artificial intelligence (XAI) aims to provide human-interpretable insights into the behavior of deep neural networks (DNNs), typically by estimating

a simplified causal structure of the model. In existing work, this causal structure often includes relationships where the presence of a concept is associated with a strong activation of a neuron. For example, attribution methods primarily identify input pixels that contribute most to a prediction, and feature visualization methods reveal inputs that cause high activation of a target neuron – the former implicitly assuming that the relevant information resides in the input, and the latter that neurons encode the presence of concepts. However, a largely overlooked type of causal relationship is that of encoded absences, where the absence of a concept increases neural activation. In this work, we show that such missing but relevant concepts are common and that mainstream XAI methods struggle to reveal them when applied in their standard form. To address this, we propose two simple extensions to attribution and feature visualization techniques that uncover encoded absences. Across experiments, we show how mainstream XAI methods can be used to reveal and explain encoded absences, how ImageNet models exploit them, and that debiasing can be improved when considering them.

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Reward Modeling from Natural Language Human Feedback

Zongqi Wang, Rui Wang, Yuchuan Wu, Yiyao Yu, Pinyi Zhang, Shaoning Sun, Yujiu Yang, Yongbin Li

Reinforcement Learning with Verifiable Reward (RLVR) on preference data has become the mainstream approach for training Generative Reward Models (GRMs). Typically, GRMs generate reasoning chains ending with critiques and preference labels, with RLVR using label correctness as the training reward. However, we demonstrate that such binary classification tasks make GRMs susceptible to guessing correct outcomes without sound critiques, introducing noise into the reward signal and impairing learning effectiveness. To address this, we propose Reward Modeling from Natural Language Human Feedback (RM-NLHF), which leverages natural language feedback to obtain process reward signals. Specifically, we compute the similarity between GRM-generated and human critiques as the process reward, providing more accurate signals than outcome-only supervision. Considering that human critiques are difficult to scale, we introduce MetaRM which learns to predict process reward from datasets with human critiques and generalizes to data without them. Experiments on multiple benchmarks demonstrate that RM-NLHF consistently outperforms state-of-the-art models trained with outcome reward, confirming the superiority of natural language over binary feedback. Our code is publicly available at https://github.com/Tongyi-ConvAI/Qwen-Character/tree/main/Character-GenRM-NLHF.

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WestWorld: A Knowledge-Encoded Scalable Trajectory World Model for Diverse Robotic Systems

Yuchen Wang, Jiangtao Kong, Sizhe Wei, Xiaochang Li, Haohong Lin, Hongjue Zhao, Tianyi Zhou, Lu Gan, Huajie Shao

Trajectory world models play a crucial role in robotic dynamics learning, planning, and control. While recent works have explored trajectory world models for diverse robotic systems, they struggle to scale to a large number of distinct system dynamics and overlook domain knowledge of physical structures. To address these limitations, we introduce *WestWorld*, a kno**W**ledge-**E**ncoded **S**calable **T**rajectory **World** model for diverse robotic systems. To tackle the scalability challenge, we propose a novel system-aware Mixture-of-Experts (Sys-MoE) that dynamically combines and routes specialized experts for different robotic systems via a learnable system embedding. To further enhance zero-shot generalization, we incorporate domain knowledge of robot physical structures by introducing a structural embedding that aligns trajectory representations with morphological information. After pretraining on 89 complex environments spanning diverse morphologies across both simulation and real-world settings, *WestWorld* achieves significant improvements over competitive baselines in zero- and few-shot trajectory prediction. Additionally, it shows strong scalability across a wide range of robotic environments and significantly improves performance on downstream model-based control for different robots. Finally, we deploy our model on a real-world Unitree Go1, where it demonstrates stable locomotion performance. The code is available at https://github.com/511205787/WestWorld.

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Behavioral Mode Discovery for Fine-tuning Multimodal Generative Policies

Alberta Longhini, David Emukpere, Jean-Michel Renders, Seungsu Kim

We address the problem of fine-tuning pre-trained generative policies with reinforcement learning (RL) while preserving the multimodality of their action distributions. Existing methods for RL fine-tuning of generative policies (e.g., diffusion policies) improve task performance but often collapse diverse behaviors into a single reward-maximizing mode. To mitigate this issue, we propose an unsupervised mode discovery framework that uncovers latent behavioral modes within generative policies. The discovered modes enable the use of mutual information as an intrinsic reward, regularizing RL fine-tuning to enhance task success while maintaining behavioral diversity. Experiments on robotic manipulation tasks demonstrate that our method consistently outperforms conventional fine-tuning approaches, achieving higher success rates and preserving richer multimodal action distributions.

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Know Thyself, Know Thy User: Intrinsic Dual-Perspective Reasoning for Role-Playing LLMs

Haotong Sun, Jianye Xie, Bocheng Xu, Yinghui Jiang

As role-playing Large Language Models (LLMs) become central to personalized AI, they face a fundamental challenge: balancing character authenticity with user satisfaction. Prior dual-process and dual-perspective approaches address this through prompt-level conditioning, auxiliary modules, or inference-time reflection---realizing duality externally rather than within the core attention mechanism. We introduce the KnowSelf-KnowOther Transformer (KSKT), which embeds dual-perspective reasoning directly into the generation process via dual-stream factorized attention that processes self-understanding and other-understanding through separate streams. This intrinsic integration enables token-level dynamic balance rather than post-hoc reconciliation. On CharacterBench, KSKT achieves 6.2% overall improvement over Qwen3-4B-Thinking. On SOTOPIA, KSKT improves Relationship by 19.3% over the base model---the dimension requiring explicit self-other coordination. These results establish intrinsic dual-perspective reasoning as an effective architectural principle for role-playing systems.

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Learning Reward–Cost Balance in Safe RL via Score-Based World Models

Yuetian Wang, Dianxi Shi, Yuanze Wang, Huanhuan Yang, Shiming Song, Chunping Qiu

Safe reinforcement learning (Safe RL) seeks to optimize long-term performance while ensuring adherence to safety constraints. However, most existing approaches address safety in a simplified manner, typically by linearly combining rewards and costs, which provides limited guidance when safety and performance interact in complex, nonlinear ways. We present USB-RL (Unsupervised Score-Balanced Reinforcement Learning), a model-based framework that learns implicit safety–performance preferences directly from experience. Our approach infers a monotone partial-order score through unsupervised pairwise comparisons of long-horizon outcomes, capturing nuanced trade-offs without relying on manually tuned cost weights. The learned score guides model-based policy optimization by dynamically balancing safety and performance, enabling flexible and adaptive multi-step planning in imagination-based control. Across diverse safety benchmarks, USB-RL achieves strong returns while substantially reducing safety violations, demonstrating stable and interpretable safety–performance trade-offs.

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Dual Quaternion SE(3) Synchronization with Recovery Guarantees

Jianing Zhao, Linglingzhi Zhu, Anthony Man-Cho So

Synchronization over the special Euclidean group $\mathrm{SE}(3)$ aims to recover absolute poses from noisy pairwise relative transformations and is a core primitive in robotics and 3D vision. Standard approaches often require multi-step heuristic procedures to recover valid poses, which are difficult to analyze and typically lack theoretical guarantees. This paper adopts a dual quaternion representation and formulates $\mathrm{SE}(3)$ synchronization directly over the unit dual quaternion. A two-stage algorithm is developed: A spectral initializer compu

ted via the power method on a Hermitian dual quaternion measurement matrix, foll
owed by a dual quaternion generalized power method (DQGPM) that enforces feasibi
lity through per-iteration projection. The estimation error bounds are establish
ed for spectral estimators, and DQGPM is shown to admits a finite-iteration erro
r bound and achieves linear error contraction up to an explicit noise-dependent
threshold. Experiments on synthetic benchmarks and real-world multi-scan point-s
et registration demonstrate that the proposed pipeline improves both accuracy an
d efficiency over representative matrix-based methods.
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Planar Symmetric Pattern Generation

Ning Lin, Luxi Chen, Huaguan Chen, Jiacheng Cen, Chongxuan Li, Wenbing Huang, Ha
o Sun

Generating objects with specific symmetries is essential in various real-world s
cenarios. However, adapting existing 2D continuous representations to enforce pl
anar group symmetry remains a challenge, as the transformation of non-reflective
group elements may disrupt continuity. To overcome this limitation, we propose
a symmetrization framework for arbitrary planar groups. Our method transforms an
y 2D continuous representation into a symmetric one while preserving continuity.
We provide the mathematical formulation of this representation, demonstrate its
approximation capability for symmetric functions, and detail the construction m
ethodology. We validate our approach through three visual design tasks (pattern
design, paper-cutting design and stylized topology design) and one material desi
gn task. Experiments confirm that our representation enables effective symmetry
control and demonstrate its broader applicability.
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TT-Sparse: Learning Sparse Rule Models with Differentiable Truth Tables

Hans Farrell Soegeng, Sarthak Ketanbhai Modi, Thomas Peyrin

Interpretable machine learning is essential in high-stakes domains where decisio
n-making requires accountability, transparency, and trust. While rule-based mode
ls offer global and exact interpretability, learning rule sets that simultaneous
ly achieve high predictive performance and low, human-understandable complexity
remains challenging. To address this, we introduce TT-Sparse, a flexible neural
building block that leverages differentiable truth tables as nodes to learn spar
se, effective connections. A key contribution of our approach is a new soft TopK
operator with straight-through estimation for learning discrete, cardinality-co
nstrained feature selection in an end-to-end differentiable manner. Crucially, t
he forward pass remains sparse, enabling each node (and the entire model) to be
transformed exactly into compact, globally interpretable DNF/CNF Boolean formula
s via Quine--McCluskey minimization. Extensive empirical results across 28 datas
ets spanning binary, multiclass, and regression tasks show that the learned spar
se rules exhibit superior predictive performance with lower complexity compared
to existing state-of-the-art methods.
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Language as a Wave Phenomenon: Semantic Phase Locking and Interference in Neural
Networks

Alper YILDIRIM, ■brahim Yüceda■

In standard Transformer architectures, semantic importance is often conflated wi
th activation magnitude, obscuring the geometric structure of latent representat
ions. To disentangle these factors, we introduce PRISM, a complex-valued archite
cture designed to isolate the computational role of phase. By enforcing a strict
unit-norm constraint ($|z| = 1$) and replacing attention with gated harmonic conv
olutions, the model is encouraged to utilize subtractive interference in the fre
quency domain to suppress noise, rather than relying on magnitude-based gating.
We utilize this constrained regime to study a hybrid architecture—fusing phase-b
ased routing with standard attention—which achieves improved parameter efficienc
y and representation quality compared to baselines in our evaluated settings. Me
chanistically, interventional ablations indicate that the model carries substant
ial task-relevant information in phase: preserving phase largely maintains perfo
rmance, whereas disrupting phase causes severe degradation. Together, these resu
lts suggest that phase-based spectral interference is a usable computational mec

hanism for neural sequence modeling at the evaluated scale.
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DeepImageSearch: Benchmarking Multimodal Agents for Context-Aware Image Retrieval in Visual Histories
Chenlong Deng, Mengjie Deng, Junjie Wu, Dun Zeng, Teng Wang, Qingsong Xie, Jiadeng Huang, Shengjie Ma, Changwang Zhang, Zhaoxiang Wang, Jun Wang, Yutao Zhu, Zhicheng Dou
Existing multimodal retrieval systems excel at semantic matching but implicitly assume that query-image relevance can be measured in isolation. This paradigm overlooks the rich dependencies inherent in realistic visual streams, where information is distributed across temporal sequences rather than confined to single snapshots. To bridge this gap, we introduce DeepImageSearch, a novel agentic paradigm that reformulates image retrieval as an autonomous exploration task. Models must plan and perform multi-step reasoning over raw visual histories to locate targets based on implicit contextual cues. We construct DISBench, a challenging benchmark built on interconnected visual data. To address the scalability challenge of creating context-dependent queries, we propose a human-model collaborative pipeline that employs vision-language models to mine latent spatiotemporal associations, effectively offloading intensive context discovery before human verification. Furthermore, we build a robust baseline using a modular agent framework equipped with fine-grained tools and a dual-memory system for long-horizon navigation. Extensive experiments demonstrate that DISBench poses significant challenges to state-of-the-art models, highlighting the necessity of incorporating agentic reasoning into next-generation retrieval systems.
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Weights to Code: Extracting Interpretable Algorithms from the Discrete Transformer
Yifan Zhang, Wei Bi, Kechi Zhang, Dongming Jin, Jie Fu, Zhi Jin
Algorithm extraction aims to synthesize executable programs directly from models trained on algorithmic tasks, enabling *de novo* recovery of executable mechanisms from weights without relying on human-written target programs.
However, applying this paradigm to Transformer is complicated by representation entanglement (e.g., superposition), where features encoded in overlapping directions substantially hinder the recovery of symbolic expressions.
We propose the Discrete Transformer, an architecture explicitly designed to bridge the gap between continuous representations and discrete symbolic logic.
By injecting discreteness through temperature-annealed sampling, our framework effectively leverages hypothesis testing and symbolic regression to extract human-readable programs.
Empirically, the Discrete Transformer achieves performance comparable to the RNN-based MIPS baseline on shared discrete tasks, while broadening extraction to tasks with continuous-valued intermediate computations.
Finally, we show that architectural inductive biases provide fine-grained control over synthesized programs, establishing the Discrete Transformer as a controllable testbed for algorithm extraction and Transformer interpretability.
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Learning Disentangled Multi-Agent World Model for Decentralized Control
Di Xue, Jing Jiang, Shaowei Zhang, Wenhao Guo, Lei Yuan, Zongzhang Zhang, Yang Yu
World models enable learning policies via latent imagination, offering benefits such as history compression and sample efficiency.
The primary challenge in applying world models to multi-agent tasks is that modeling multi-agent dynamics in latent space requires integrating information from different agents, often creating spurious correlations between their latent states.
Existing methods either reconstruct the observation for each agent or employ communication to maintain correlation during execution, failing to learn disentangled latent states that are crucial for effective decentralized control.
To address this, we present the Disentangled Multi-Agent World Model (DMAWM). It facilitates learning decentralized policies in the latent space through a novel

architecture comprising independent agent modules and a shared environment module.
During real-environment execution, agent modules independently process local information to form a factorized latent representation.
The environment module is then trained to mirror the factorized structure generated by the agent modules, effectively disentangling individual latent states from the interaction dynamics.
Consequently, imaginary rollouts generated by the environment module more faithfully simulate decentralized execution dynamics, facilitating the transfer of policies from imagination to decentralized execution.
Empirically, DMAWM outperforms existing model-based and model-free approaches in convergence speed and final performance, with additional visualization demonstrating its efficacy in capturing agent interactions.
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Asymmetric Prompt Weighting for Reinforcement Learning with Verifiable Rewards
Reinhard Heckel, Mahdi Soltanolkotabi, Christos Thrampoulidis
Reinforcement learning with verifiable rewards has driven recent advances in LLM post-training, in particular for reasoning. Policy optimization algorithms generate a number of responses for a given prompt and then effectively weight the corresponding gradients depending on the rewards. The most popular algorithms including GRPO, DAPO, and RLOO focus on ambiguous prompts, i.e., prompts with intermediate success probability, while downgrading gradients with very easy and very hard prompts. In this paper, we consider asymmetric prompt weightings that assign higher weights to prompts with low, or even zero, empirical success probability. We find that asymmetric weighting particularly benefits from-scratch RL (as in R1-Zero), where training traverses a wide accuracy range, and less so in post-SFT RL where the model already starts at high accuracy. We also provide theory that characterizes prompt weights which minimize the time needed to raise success probability from an initial level to a target accuracy under a fixed update budget. In low-success regimes, where informative responses are rare and response cost dominates, these optimal weights become asymmetric, upweighting low success probabilities and thereby accelerating effective-time convergence.
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Conformal Prediction for Early Stopping in Mixed Integer Optimization
Stefan Clarke, Bartolomeo Stellato
Mixed-integer optimization solvers often find optimal solutions early in the search, yet spend the majority of computation time proving optimality. We exploit this by learning when to terminate solvers early on distributions of similar problem instances. Our method trains a neural network to estimate the true optimality gap from the solver state, then uses conformal prediction to calibrate a stopping threshold with rigorous probabilistic guarantees on solution quality. On six problem families from the Distributional MIPLIB library, our method reduces solve time by over 60% while guaranteeing 0.1%-optimal solutions with 95% probability for new instances drawn from the same distribution.
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Weak-to-Strong Generalization via Bregman Bias–Variance Decomposition
Gengze Xu, Wei Yao, Ziqiao Wang, Yong Liu
Weak-to-strong generalization (W2SG) is the phenomenon in which a powerful student model, trained on labels produced by a weaker teacher, ultimately outperforms the teacher on the target task. In this work, we theoretically investigate how W2SG can arise via a generalized bias–variance decomposition under Bregman divergence. We show that the expected population risk gap between the student and the teacher is characterized by the expected misfit between the two models. Unlike earlier misfit-based analyses, our theory removes several restrictive assumptions, e.g., it does not require the student hypothesis class to be convex.
Our results indicate that W2SG is more likely when the student effectively approximates the teacher's posterior mean. Specializing to squared loss, we provide a sufficient condition (illustrated through a concrete example) under which the student converges to its posterior mean teacher; in particular, increasing the student model size can ensure this convergence. For cross-entropy loss, our analy

sis further suggests that lowering the entropy of the student's predictive distr
ibution can promote W2SG. We also find that the reverse cross-entropy, unlike th
e standard forward cross-entropy, is less sensitive to the teacher's predictive
uncertainty. Finally, we verify these theoretical insights empirically and demon
strate that incorporating reverse cross-entropy consistently improves student pe
rformance.
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STEP: Warm-Started Visuomotor Policies with Spatiotemporal Consistency Predictio
n

Jinhao Li, Yuxuan Cong, Yingqiao Wang, Hao Xia, Shan Huang, Yijia Zhang, Ningyi
Xu, Guohao Dai

Diffusion policies have recently emerged as a powerful paradigm for visuomotor c
ontrol in robotic manipulation due to their ability to model the distribution of
 action sequences and capture multimodality. However, iterative denoising leads
to substantial inference latency, limiting control frequency in real-time closed
-loop systems. Existing acceleration methods either reduce sampling steps, bypas
s diffusion through direct prediction, or reuse past actions, but often struggle
 to jointly preserve action quality and achieve consistently low latency.
In this work, we propose **STEP**, a lightweight spatiotemporal consistency pred
iction mechanism to construct high-quality warm-start actions that are both dist
ributionally close to the target action and temporally consistent, without compr
omising the generative capability of the original diffusion policy.
Then, we propose a velocity-aware perturbation injection mechanism that adaptive
ly modulates actuation excitation based on temporal action variation to prevent
execution stall especially for real-world tasks.
We further provide a theoretical analysis showing that the proposed prediction i
nduces a locally contractive mapping, ensuring convergence of action errors duri
ng diffusion refinement.
Extensive evaluations on nine simulated benchmarks and two real-world tasks show
 that STEP with 2 steps can achieve an average 21.6\% and 27.5\% higher success
rate than BRIDGER and DDIM on the RoboMimic benchmark and real-world tasks, resp
ectively.
The code is available at https://github.com/Kimho666/STEP.
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GRASP: Graph Reasoning via Agentic Solving and Probing of LLMs

Xiaojun Guo, Mingxue Tian, Chenheng Zhang, Xiaohan Wang, Jiajun Chai, Guojun Yin
, Wei Lin, Yifei Wang, Yisen Wang

Integrating graph knowledge into Large Language Models (LLMs) via passive repres
entation faces critical bottlenecks: limited context windows, unreliable numeric
al computation, and structural hallucinations. To solve this, we propose **GRASP
** (**G**raph **R**easoning via **A**gentic **S**olving and **P**robing), shifti
ng the paradigm from passive ingestion to proactive agentic exploration. By inte
rleaving Neighbor Retrieval for on-demand probing with Code Interpreter as a det
erministic solver, GRASP enables LLMs to autonomously navigate and compute over
complex topologies. We employ a staged reinforcement learning strategy (GRPO) th
at transitions from visible tuning to a structure-blind environment, forcing the
 agent to develop genuine topological awareness. Evaluated on multi-domain graph
 reasoning benchmarks, our 4B model achieves a 53.06% average performance boost,
 surpassing SOTA baselines like DeepSeek-V3.2 and successfully generalizing to u
nseen tasks, with high potential for tackling sampling on million-node graphs an
d solving Hard-level LeetCode graph problems. Our implementation is open-sourced
 at https://github.com/PKU-ML/GRASP, with models hosted on Huggingface collectio
n https://huggingface.co/collections/PKU-ML/grasp.
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Structured Diffusion Bridges: Inductive Bias for Denoising Diffusion Bridges

Eitan Kosman, Gabriele Serussi, Chaim Baskin

Modality translation is inherently under-constrained, as multiple cross-modal ma
ppings may yield the same marginals. Recent work has shown that diffusion bridge
s are effective for this task. However, most existing approaches rely on fully p
aired datasets, thereby imposing a single data-driven constraint. We propose a d

iffusion-bridge framework that characterizes the space of admissible solutions a
nd restricts it via alignment constraints, treating paired supervision as an opt
ional heuristic rather than a prerequisite. We validate our method on synthetic
and real modality translation benchmarks across unpaired, semi-paired, and paire
d regimes, showing consistent performance across supervision levels. Notably, it
 achieves near fully-paired quality with a substantial relaxation in pairing req
uirements, and remaining applicable in the unpaired regime. These results highli
ght diffusion bridges as a flexible foundation for modality translation beyond f
ully paired data.
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Learning to Reconfigure: Configuration-Control Co-optimization of Reconfigurable
 Robots for Heterogeneous Locomotion
Xiaoyu Xiong, Kehan Liu, Huiyi Yan, Shengjie Wang, Yang Gao, Tao Du
Traditional robot co-design approaches typically converge to one configuration,
which do not explore the flexibility from reconfiguration on heterogeneous envir
onments. On the other hand, existing designs for reconfigurable robots require h
uman-designed configurations. We present Learning to Reconfigure, a holistic pip
eline for configuration-control co-optimization of reconfigurable robots in hete
rogeneous locomotion tasks consisting of several sub-tasks. Our pipeline propose
s low-level specialized primitives with a high-level scheduler. To jointly optim
ize configuration design and control, our primitives employ a multi-tail archite
cture that disentangles these distinct objectives. Building on this, the schedul
er learns to dynamically switch configurations based on global task progress. We
 evaluate our pipeline on locomotion tasks across walking, flying, and swimming,
 and compare with the state-of-the-art baselines, including single-robot control
 and multi-morphology co-optimization algorithms. Quantitative results based on
traversal progress show that our pipeline outperforms single-robot baselines by
5.95x average progress. Compared with the reconfiguration-free design given by t
he co-design algorithms, our robots also exhibit 9.81x progress on average. Thes
e results highlight the critical role of configuration adaptation in achieving v
ersatile robotic autonomy in complex worlds.
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LLM-Guided Communication for Cooperative Multi-Agent Reinforcement Learning
Sangjun Bae, Yisak Park, Sanghyeon Lee, Seungyul Han
Communication is a key component in multi-agent reinforcement learning (MARL) fo
r mitigating partial observability, yet prior approaches often rely on inefficie
nt information exchange or fail to transmit sufficient state information. To add
ress this, we propose LLM-driven Multi-Agent Communication (LMAC), which leverag
es an LLM's reasoning capability to design a communication protocol that enables
 all agents to reconstruct the underlying state as accurately and uniformly as p
ossible. LMAC iteratively refines the protocol using an explicit state-awareness
 criterion, improving state recovery while narrowing differences in agents' know
ledge. Experiments on diverse MARL benchmarks show that LMAC improves state reco
nstruction across agents and yields substantial performance gains over prior com
munication baselines.
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An In-Depth Study on Deep Learning Model Cloning
Bin Hu, Xiancong Pan, Dongjin Yu, Tianyi Hu
Artificial intelligence has achieved remarkable breakthroughs in fields such as
text, image, and video analysis, with deep learning serving as the mainstream pa
radigm. Trained deep learning models can be integrated into various applications
 either through fine-tuning or without any modification. While this practice pro
motes the advancement of artificial intelligence, it also raises concerns regard
ing intellectual property protection and information security risks. Therefore,
it is necessary to propose relevant methods to measure the similarity between mo
dels. Existing code clone detection techniques are insufficient to address this
issue. In this paper, we provide the first definition of model cloning and desig
n a method for model similarity detection. The framework characterizes model top
ology at the structural level based on normalized computational graphs, and at t
he weight level, it employs a method that does not require explicit parameter al

ignment to measure the statistical similarity of weight parameters. Experiments on a synthetic model clone benchmark dataset and real-world open-source models demonstrate that the proposed method can accurately detect similar models. This method provides a unified and extensible quantitative foundation for model lineage analysis, model retrieval, and intellectual property protection of models.
****

Task-Aware Mechanism: Hybrid MoE Vision Tower Towards Holistic Video Understanding
Qishen Yin, Tanghui Jia, Peng Jin, Li Hao, Juntong Wu, Guanlin Lu, Li Yuan
Should an LVLM use the same video processing pipeline for \emph{comprehending the main idea of a 2-hour movie} and \emph{counting the birds appearing in a 15-second clip}? We present Task-Aware Mechanism (TAM), a hybrid-gated Mixture-of-Experts (MoE) vision tower that adapts frame count and resolution to the user query and video length. TAM introduces a compact 0.1B text-only \emph{Inductor} trained on our TA-116K dataset to infer task types, enabling task-aware visual budgeting and routing: a soft-gated MoE vision encoder for stability, and hard-gated resolution-specific projectors/pipelines for efficient specialization. The explicit budget also separates task classification from downstream visual-language reasoning, making routing decisions easier to inspect and helping diagnose when fixed visual pipelines under-sample temporal or spatial evidence. Built on Qwen2-7B, TallVA-8B-A7B achieves state-of-the-art performance among models with comparable LLMs on most video benchmarks and remains competitive against stronger-LLM baselines, showing that task-aware visual budgeting makes video understanding more holistic.
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Particles Don’t Care About Z: Towards Scaling Entropy Estimation of Unnormalized Densities
Safa Messaoud, Skander Charni, Elaa Bouazza, Ali Pourghasemi Fatideh, Halima Bensmail
Computing the differential entropy of distributions known only up to a normalization constant is a fundamental challenge with broad theoretical and practical significance. While variational inference is highly scalable for density approximation from samples, its application to unnormalized densities remains under-explored due to the difficulty of constructing variational distributions that exploit the structure of the unnormalized density and are simultaneously expressive, tractable, efficiently samplable. Recently, Messaoud et al. (ICLR 24) introduced _P-SVGD_, a Stein variational inference method for this setting. We show, however, that \textit{P-SVGD} fails to scale to high dimensions due to incorrect invertibility assumptions, omission of a critical trace-of-Hessian term, and unstable divergence-control heuristics. We propose _MET-SVGD_, a principled extension of _P-SVGD_ that provides a general framework for stable _SVGD_ hyperparameter selection with global invertibility and convergence guarantees. Empirically, _MET-SVGD_ achieves up to $12\times$ and $16\times$ lower entropy estimation error than _P-SVGD_ and existing _SVGD_ baselines, respectively. On CIFAR-10 energy-based image generation, it improves FID by 80.4% and yields $64\times$ higher training stability. In maximum-entropy reinforcement learning, it achieves up to 16% higher returns than _P-SVGD_. Code is available at https://shorturl.at/fTG0G.
****

V-LynX: Token Interface Alignment for Video+X LLMs
Jungin Park, Jiyoung Lee, Kwanghoon Sohn
This study introduces an intriguing phenomenon in Video LLMs: rather than merely translating frames into textual embeddings, Video LLMs establish a continuous manifold, token interface, allowing visual tokens to operate as standalone entities within the architecture.

Exploiting this discovery, we propose V-LynX, a scalable framework that integrates novel modalities into Video LLMs by repurposing the internalized interface.

Departing from conventional paradigms that necessitate heavy modality-specific encoders or paired supervision, V-LynX employs a lightweight auxiliary pathway in parallel with the frozen vision encoder.

Our method integrates new sensory inputs with intrinsic video priors by aligning both attention responses and statistical distributions using unpaired unimodal data sets.

This ensures manifold compatibility while preserving the integrity of the Video LLMs.

Extensive benchmarks demonstrate that V-LynX achieves SOTA and efficiency across audio-visual QA, 3D reasoning, high-frame-rate, and multi-view video understanding. The code is available at https://github.com/park-jungin/lynx.

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Telescope: Improving Zero Shot Detection of LLM Generated Content By Measuring Token Repetition Probability

Christopher Nassif, Joshua F. Cooper

Distinguishing Large Language Model (LLM) generated text from human writing is a critical and difficult challenge. While LLMs are trained to write like humans, we hypothesize that this training leaves an indelible mark. LLMs develop a particularly strong aversion to token repetition very early in training. This bias persists as a ``Vestigial Heuristic'' (a developmental artifact) that is activated in LLM-generated text, separating LLM from human writing. To probe this phenomenon, we introduce Telescope Perplexity, a metric that evaluates the token repetition of the model, $P(s_i | s_{1:i})$. Our empirical investigation reveals that the Telescope Perplexity signature emerges early in pre-training, and Telescope Perplexity empirically enables highly effective zero-shot LLM detection. We show state-of-the-art or competitive performance across diverse datasets (including modern evaluation sets we introduce), reference models, and perturbation schemes with greater efficiency than other methods.

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Beyond Euclidean Clipping: Overcoming Exploration Collapse in LLM RL via Riemannian Isometric Policy Optimization

Zhicheng Cai, Xinyuan Guo, Hanlin Wu, Mingxuan Wang, Wei-Ying Ma, Ya-Qin Zhang, Hao Zhou

Reinforcement learning (RL) has become a dominant paradigm for enhancing LLMs' reasoning capabilities.

However, RL algorithms with PPO-Clip are inherently limited by exploration collapse.

Subsequent works remain primarily heuristic and fail to identify the essential cause of PPO-Clip’s failure.

This work reveals the fundamental flaw of PPO-Clip: it implicitly measures policy discrepancy using Euclidean metric, which is theoretically inconsistent with the intrinsic geometry on the policy Riemannian manifold. This geometric mismatch results in overly conservative updates in low-probability regions while aggressive in high-probability regions, ultimately collapsing exploration.

To correct this geometric flaw, we propose Riemannian Isometric Policy Optimization (RIPO), which guarantees isometric policy updates on the Riemannian manifold, effectively balancing exploration and exploitation.

We further show that RIPO achieves a favorable bias-variance trade-off, which stabilizes optimization.

Extensive experiments demonstrate that RIPO significantly surpasses existing LLM RL algorithms across seven competition-level benchmarks (up to 60% improvement over GRPO on AIME24).

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VBA: Vector Bundle Attention for Intrinsically Geometric Representation Learning

Shenglei Fang, Xianfang Sun, You Zhou

Learning from geometrically structured data is central to applications in biology, physics, and computer vision. In many tasks, meaningful comparisons depend on how features are aligned in space. Graph Neural Networks capture local structure but are constrained by message passing. Transformers model long-range dependencies but largely ignore geometry. We introduce the Vector Bundle Attention Transformer (VBA-Transformer), a framework that redefines attention as an intrinsic geometric operator. Each token couples a base manifold coordinate with a fiber feature vector, following vector bundle theory. A principled parallel transport me

chanism aligns fiber features across local coordinate systems before similarity is computed. This embeds geometry directly into the attention operator. Unlike p rior methods that inject geometry as an external bias or positional encoding, VB A integrates geometry natively inside attention. On challenging single-cell RNA sequencing benchmarks, VBA achieves state-of-the-art accuracy, outperforming Tra nsformer baselines by over 3--5\%. On spatial transcriptomics, it demonstrates s uperior clustering performance. On 3D point clouds, it achieves competitive accu racy, validating broad generalization across domains. Beyond empirical gains, we provide theoretical analysis of invariance and perturbation stability. We also demonstrate robust transport behavior empirically. Together, these results estab lish intrinsic geometric alignment as a powerful principle for scalable represen tation learning.

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Toward Scalable and Valid Conditional Independence Testing with Spectral Represe ntations

Alek Fröhlich, Vladimir R Kostic, Karim Lounici, Daniel Perazzo, Daniel Guimarãe s Tiezzi, Massimiliano Pontil

Conditional independence (CI) is central to causal inference, feature selection, and graphical modeling, yet it is untestable in many settings without additiona l assumptions. Existing CI tests often rely on restrictive structural conditions , limiting their validity. Kernel methods using partial covariance operators off er a more principled approach but suffer from limited adaptivity and scalability . In this work, we explore whether representation learning can help address thes e limitations. Specifically, we focus on representations derived from the singul ar value decomposition of partial covariance operators and use them to construct a simple test statistic. We also introduce a bi-level contrastive algorithm to learn these representations. Our theory links representation learning error to t est performance and establishes asymptotic validity and power guarantees. Experi ments on real and synthetic data suggest that this approach offers a principled and statistically grounded path toward scalable CI testing, bridging kernel-base d theory with modern representation learning.

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Convex Distance Operator Transport: A Convex and Geometry-Preserving Formulation

Junhyoung Chung, Euijong Song, Won Hwa Kim, Gunwoong Park

We introduce Convex Distance Operator Transport (CDOT), the first convex optimal transport framework that aligns distributions across heterogeneous domains by j ointly preserving feature correspondence and intrinsic geometric structure. Spec ifically, CDOT employs an operator-based regularization that aligns aggregated d istance structures by introducing distance and conditional expectation operators . Consequently, the proposed regularization improves the robustness to local geo metric variations. We further prove that the resulting CDOT discrepancy is a val id pseudometric on the space of attributed compact metric-measure spaces. In add ition, we characterize the relationship between CDOT and Gromov--Wasserstien (GW) through a new notion of dispersion gap, formally elucidating the geometric sou rce of non-convexity in GW compared to the convexity of CDOT. In the finite-samp le regime, we derive a non-asymptotic risk bound decomposed into optimization an d statistical errors, establishing risk consistency under a globally convergent Frank--Wolfe algorithm. Experiments on synthetic point clouds, brain connectomes , and graph classification benchmarks demonstrate better performance over existi ng methods, with stable and reliable behavior in practice.

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Evaluating LLMs When They Do Not Know the Answer: Statistical Evaluation of Math ematical Reasoning via Comparative Signals

Zihan Dong, Zhixian Zhang, Yang Zhou, Can Jin, Ruijia Wu, Linjun Zhang

Evaluating mathematical reasoning in LLMs is constrained by limited benchmark si zes and inherent model stochasticity, yielding high-variance accuracy estimates and unstable rankings across platforms. On difficult problems, an LLM may fail t o produce a correct final answer, yet still provide reliable pairwise comparison signals indicating which of two candidate solutions is better. We leverage this observation to design a statistically efficient evaluation framework that combi

nes standard labeled outcomes with pairwise comparison signals obtained by havin g models judge auxiliary reasoning chains. Treating these comparison signals as control variates, we develop a semiparametric estimator based on the efficient i nfluence function (EIF) for the setting where auxiliary reasoning chains are obs erved. This yields a one-step estimator that achieves the semiparametric efficie ncy bound, guarantees strict variance reduction over naive sample averaging, and admits asymptotic normality for principled uncertainty quantification. Across s imulations, our one-step estimator substantially improves ranking accuracy, with gains increasing as model output noise grows. Experiments on GPQA Diamond, AIME 2025, and GSM8K further demonstrate more precise performance estimation and mor e reliable model rankings, especially in small-sample regimes where conventional evaluation is pretty unstable.

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Covariance estimation using Markov chain Monte Carlo

Yunbum Kook, Matthew Shunshi Zhang

We investigate the complexity of covariance matrix estimation for Gibbs distribu tions based on dependent samples from a Markov chain. We show that when $π$ sati sfies a Poincaré inequality and the chain possesses a spectral gap, we can achie ve similar sample complexity using MCMC as compared to an estimator constructed using i.i.d. samples, with potentially much better query complexity. As an appli cation of our methods, we show improvements for the query complexity in both con strained and unconstrained settings for concrete instances of MCMC. In particula r, we provide guarantees regarding isotropic rounding procedures for sampling un iformly on convex bodies.

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Why ReLU? A Bit-Model Dichotomy for Deep Network Training

Ilan Doron-Arad, Elchanan Mossel

Theoretical analyses of Empirical Risk Minimization (ERM) are standardly framed within the Real-RAM model of computation. In this setting, training even simple neural networks is known to be $\exists \mathbb{R}$-complete - a complexity clas s believed to be harder than NP, characterizing the difficulty of solving system s of polynomial inequalities over the real numbers. However, this algebraic fram ework diverges from the reality of digital computation with finite-precision har dware. In this work, we analyze the theoretical complexity of ERM under a realis tic bit-level model (ERM-bit), where network parameters and inputs are constrain ed to be rational numbers with polynomially bounded bit-lengths. Under this mode l, we reveal a sharp dichotomy in tractability governed by the activation functi on: for deep networks with any polynomial activation with rational coefficients and degree at least 2, deciding ERM-bit is #P-hard, determining the sign of a single partial derivative is unlikely to be in BPP, and deciding a specific bit in the gradient is #P-hard. In contrast, for piecewise-linear activations such a s ReLU, precision requirements remain manageable: ERM-bit is in NP, indeed NP-co mplete, and standard backpropagation runs in polynomial time, showing that finit e-precision constraints are not merely implementation details but fundamental de terminants of learnability.

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GRPO is Secretly a Process Reward Model

Michael Sullivan, Alexander Koller

Process reward models (PRMs) allow for fine-grained credit assignment in reinfor cement learning (RL), and seemingly contrast with outcome reward models (ORMs), which assign a single reward to an entire trajectory. However, we provide theore tical proof in this work that the Group Relative Policy Optimization (GRPO) RL a lgorithm equipped with an ORM is in fact equivalent to a PRM-aware RL objective equipped with a non-trivial, Monte-Carlo-based PRM (given mild assumptions). Lev eraging the framework of GRPO-as-a-PRM, we identify a flaw in the GRPO objective that interacts with imbalanced process steps and rewards to hinder both explora tion and exploitation (under different conditions). We propose a simple modifica tion to the algorithm to mitigate this defect (λ-GRPO), and show that LL Ms tuned with λ-GRPO outperform LLMs tuned with standard GRPO on downstr eam reasoning tasks$\textemdash$and reach peak performance more rapidly. These r

esults show that we can leverage the hidden, built-in PRM structure within the v anilla GRPO algorithm to boost model performance without employing an explicit P RM, and with a negligible impact on training time and cost.
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Stop the Flip-Flop: Context-Preserving Verification for Fast Revocable Diffusion Decoding

Yanzheng Xiang, Lan Wei, Yizhen Yao, Qinglin Zhu, Hanqi Yan, Chen Jin, Philip Al exander Teare, Dandan Zhang, Lin Gui, Amrutha Saseendran, Yulan He

Parallel diffusion decoding can accelerate diffusion language model inference by unmasking multiple tokens per step, but aggressive parallelism often harms qual ity. Revocable decoding mitigates this by rechecking earlier tokens, yet we obse rve that existing verification schemes frequently trigger flip-flop oscillations , where tokens are remasked and later restored unchanged. This behaviour slows i nference in two ways: remasking verified positions weakens the conditioning cont ext for parallel drafting, and repeated remask cycles consume the revision budge t with little net progress. We propose COVER (Cache Override Verification for Ef ficient Revision), which performs leave-one-out verification and stable drafting within a single forward pass. COVER constructs two attention views via KV cache override: selected seeds are masked for verification, while their cached key va lue states are injected for all other queries to preserve contextual information , with a closed form diagonal correction preventing self leakage at the seed pos itions. COVER further prioritises seeds using a stability aware score that balan ces uncertainty, downstream influence, and cache drift, and it adapts the number of verified seeds per step. Across benchmarks, COVER markedly reduces unnecessa ry revisions and yields faster decoding while preserving output quality.
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Revisiting OOD Generalization in Programmatic RL

Amirhossein Rajabpour, Kiarash Aghakasiri, Sandra Zilles, Levi Lelis

Programmatic policies are often reported to generalize better than neural polici es in reinforcement learning (RL) benchmarks. We revisit some of these claims an d show that much of the observed gap arises from uncontrolled experimental facto rs rather than intrinsic representational reasons. Re-evaluating three core benc hmarks used in influential papers---TORCS, Karel, and Parking---we find that neu ral policies, when trained with a few modifications, such as sparse observations and cautious intrinsic reward functions, can match or exceed the out-of-distrib ution (OOD) generalization of programmatic policies. We argue that a representat ion enables OOD generalization if (i) the policy space it induces includes a gen eralizing policy and (ii) the search algorithm can find it. The neural and progr ammatic policies in prior work are comparable in OOD generalization because the domain-specific languages used induce policy spaces similar to those of neural n etworks, and our modifications help the gradient search find generalizing soluti ons. By disentangling representational factors from experimental confounds, we a dvance our understanding of what makes a representation succeed or fail at OOD g eneralization.
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Advantage Weighted Matching: Aligning RL with Pretraining in Diffusion Models

Shuchen Xue, Chongjian GE, Shilong Zhang, Yichen Li, Zhi-Ming Ma

Reinforcement Learning (RL) has emerged as a central paradigm for advancing Larg e Language Models (LLMs), where both pre-training and RL post-training stages ar e grounded in the same log-likelihood formulation. In contrast, recent RL approa ches for diffusion models, most notably Denoising Diffusion Policy Optimization (DDPO), optimize an objective different from the pretraining objectives--score/f low matching loss.

In this work, we establish a novel theoretical analysis: DDPO is an implicit for m of score/flow matching with noisy targets, which increases variance and slows convergence. Building on this analysis, we introduce \textbf{Advantage Weighted Matching (AWM)}, a policy-gradient method for diffusion. It uses the score/flow-matching loss and reweights each sample by its advantage. In effect, AWM raises the influence of high-reward samples and suppresses low-reward ones while keepin g the modeling objective identical to pretraining.

This simple yet effective design yields substantial benefits: on the GenEval, OCR, and PickScore benchmarks, AWM delivers up to a $\mathbf{34}\times$ speedup over Flow-GRPO (which builds on DDPO), when applied to Stable Diffusion 3.5 Medium and FLUX, without compromising generation quality. Code is available at \url{https://github.com/scxue/advantage_weighted_matching}.
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Cross-Modal Semantic Decoupling and Transfer for Text-to-Visible-Infrared Person Re-Identification
Ziang Zhang, Bin Yang, Mang Ye
Text-to-Image Person Re-Identification (TI-ReID) retrieves visible pedestrian images using text queries. Yet in low-light or nighttime settings, visible images lack sufficient identity details, while infrared images effectively capture pedestrian contours and textures. To enable all-day surveillance, we propose a dual cross-modal retrieval task called Text-to-Visible-Infrared Re-Identification (TVI-ReID) and construct corresponding tri-modal datasets. Compared to TI-ReID, TVI-ReID faces two key challenges: (1) **complex hybrid discrepancies** in dual cross-modal retrieval from three modalities, and (2) **semantic inconsistency** between pretraining and downstream tasks. To address these issues, we propose a Cross-Modal Semantic Decoupling and Transfer (CSDT) framework. CSDT constructs color-related and color-irrelevant feature subspaces via Semantic Decoupling Learning (SDL) to align shared semantics across text and dual image modalities, reducing hybrid discrepancies. Moreover, Semantic Distribution Transfer (SDT) adapts pretrained text-visible alignment to text-infrared matching. Extensive experiments on tri-modal datasets show our approach outperforms existing state-of-the-art TI-ReID methods.
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BrainJanus: A Unified Model for Understanding and Generation across Brain, Vision, and Language
Haitao Wu, Qirui Zhang, Zhouheng Yao, Shangquan Sun, Qihao Zheng, Mianxin Liu, Chi Zhang, Wanli Ouyang, Chunfeng Song, Changqing Zhang, Jiamin Wu
Modeling the bidirectional correspondence between external sensory stimuli and internal neural activity has emerged as a critical frontier in neuroscience. However, existing approaches predominantly treat brain encoding and decoding as isolated tasks, relying heavily on unimodal alignment and external priors while overlooking the brain's intrinsic nature as a multimodal integration system. To address these limitations, we propose BrainJanus, the first unified brain model that integrates brain, vision, and language within a single framework. Specifically, we introduce a Unified Brain Tokenizer to quantize continuous neural dynamics into discrete tokens aligned with visual and linguistic representations in a shared Omni space. Building on this, we utilize an All-in-One autoregressive architecture that leverages next-token prediction to enable seamless any-to-any generation, which encompasses image-to-brain and text-to-brain encoding, and brain-to-image and brain-to-text decoding. Extensive experiments demonstrate that BrainJanus achieves superior performance across diverse benchmarks. Furthermore, our framework exhibits zero-shot generalization and preserves interpretable biological topography, highlighting its potential as a general-purpose brain modeling paradigm. The code is available at \href{https://github.com/HaitaoWuTJU/BrainJanus}{GitHub}.
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IBMA: Information Bottleneck-Based Multimodal Alignment
Yancheng Wang, Zeyu Dong, Dongfang Sun, Alvin C Silva, Teresa Wu, Yingzhen Yang
Multimodal learning aims to integrate information from heterogeneous data sources to improve representation quality and downstream task performance. A key challenge lies in aligning modality-specific representations while suppressing modality-dependent noise and redundancy. The Information Bottleneck (IB) principle provides a principled framework for learning task-relevant representations. Existing multimodal IB methods primarily apply the IB principle to fused multimodal representation and rely on restrictive distributional assumptions, such as Gaussian latent priors induced by variational autoencoders, which may not hold in practice.

In this paper, we propose Information Bottleneck–based Multimodal Alignment (IBMA), a novel multimodal learning framework that enforces the IB principle for both the fused multimodal representation and modality-specific representations. IBMA introduces modality-specific representation alignment that guides each modality-specific encoder to learn informative and task-relevant representations aligned with the complementary modality, thereby enhancing cross-modal semantic consistency. Moreover, we derive a novel, efficient, and distribution-free variational upper bound for the IB loss that avoids unrealistic assumptions on latent feature distributions and is readily optimized using standard stochastic gradient descent. Extensive experiments demonstrate that IBMA achieves superior performance compared to existing multimodal learning methods, validating the effectiveness of modality-specific representation alignment. The code for IBMA is available at https://github.com/Statistical-Deep-Learning/IBMA.

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TabMGP: Martingale Posterior with TabPFN

Kenyon Ng, Edwin Fong, David Tyler Frazier, Jeremias Knoblauch, Susan Wei

Bayesian inference provides principled uncertainty quantification but is often limited by the challenges of prior and likelihood elicitation. The martingale posterior (MGP) (Fong et al., 2023) offers an alternative by replacing these requirements with a predictive rule. In addition, the MGP focuses inference on parameters defined through a loss function. This framework is especially resonant in the era of foundation transformers; practitioners increasingly leverage models like TabPFN for their state-of-the-art capabilities, yet often require epistemic uncertainty for a scientific estimand θ that need not parameterise the implicit latent model. The MGP provides a mechanism to recover these posterior distributions. We introduce TabMGP, an MGP built on TabPFN for tabular data. TabMGP produces credible sets with near-nominal coverage and often outperforms both hand crafted MGP constructions and standard Bayesian baselines.

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PolarDepth: Monocular Transparent Object Depth from Polar-Physics Priors

Wen Dong, Haiyang Mei, Yinglian Ji, Zijun Zhang, Wenyuan Zhang, Pengwei Luo, Bo Dong, Shengfeng He, Xin Yang

Depth estimation for transparent objects remains a fundamental challenge, as RGB-based cues often fail in regions affected by refraction and light transmission. Polarization provides physically grounded information related to surface orientation and material properties, offering reliable geometric cues even in the absence of texture. In this work, we introduce PolarDepth, a monocular framework that incorporates both RGB and polarization inputs, including the degree and angle of linear polarization (DoLP and AoLP), to estimate dense depth and localize transparent regions. PolarDepth injects polarization-derived physical priors by estimating the refractive index, zenith angle, and azimuth angle from polarization measurements and embedding them into an implicit geometric representation that constrains depth inference in ambiguous transparent regions. To support model development and evaluation, we introduce PTOD, a dataset with synchronized RGB, polarization, and depth data and manually annotated transparent region masks. Experimental results demonstrate that PolarDepth achieves state-of-the-art performance in transparent object depth estimation. The findings highlight the effectiveness of embedding polarization-derived physical priors into learned representations for robust perception in complex visual environments.

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Looking Locally: Object-Centric Vision Transformers as Foundation Models for Efficient Segmentation

Manuel Traub, Martin V. Butz

Current state-of-the-art segmentation models encode entire images before focusing on specific objects. This wastes computational resources. We introduce FLIP (Fovea-Like Input Patching), a parameter-efficient vision model that realizes object segmentation through biologically-inspired top-down attention. FLIP selectively samples multi-resolution patches centered on objects of interest from the input. As a result, it allocates high-resolution processing to object centers while maintaining coarser peripheral context. This off-grid, scale-invariant design e

nables FLIP to outperform META's Segment Anything models (SAM, SAM2 and fast var iants) by large margins: With more than 440 $ \times$ fewer parameters, FLIP-Tin y (0.51M parameters) reaches a mean IoU of 79.90\% while SAM2-L reaches 75.87\% IoU (224.45M parameters). FLIP-Large even achieves 83.26\% mean IoU (96.6M param eters), still running about $2 \times$ faster than SAM2-L. We evaluate on six be nchmarks in total. In five established benchmarks (Hypersim, KITTI-360, OpenImag es, COCO, LVIS) FLIP consistently outperforms SAM and various variants of it. In our novel ObjaScale dataset, which stress-tests scale invariance with objects r anging from 0.0001\% up to 25\% of the image area, we show that FLIP segments ev en very small objects accurately, where existing models fail severely. FLIP open s new possibilities for real-time, object-centric vision and offers much higher energy efficiency. We believe that FLIP can act as a powerful foundation model, as it is very well-suited to track objects over time, for example, when being in tegrated into slot-based scene segmentation architectures.
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Condition Number Based Low-Bit Quantization for Image Super-Resolution
Kai Liu, Dehui Wang, Zhiteng Li, Zheng Chen, Yong Guo, Linghe Kong
Low-bit model quantization for image super-resolution (SR) is a longstanding tas k that is renowned for its surprising compression and acceleration ability. Howe ver, accuracy degradation is inevitable when compressing the full-precision (FP) model to ultra-low bit widths ($2\sim4$ bits). Experimentally, we observe that the degradation of quantization is mainly attributed to the quantization of acti vation instead of model weights. Considering that the activation quantization er ror is hard to minimize, minimizing the impact of the error emerges as a good ch oice, which is described by the condition number. Therefore, we propose CondiQua nt, a condition number-based low-bit post-training quantization for image super- resolution. Specifically, we formulate the quantization error of activation as t he condition number of weight metrics. By decoupling the representation ability and the quantization sensitivity, we design an efficient proximal gradient desce nt algorithm to iteratively minimize the condition number and maintain the outpu t. With comprehensive experiments, we demonstrate that CondiQuant outperforms ex isting state-of-the-art post-training quantization methods in accuracy without c omputation overhead and gains the theoretically optimal compression ratio in mod el parameters. Our code will be released soon.
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No Retraining at Edge: Efficient Resource-Aware Mixed-Precision Quantization via Federated Supernet Learning
Lianbo Ma, Yonghui Su, Nan Li, Xingwei Wang
Federated learning (FL) enables collaborative training across distributed edge d evices, but deploying lightweight models in dynamic edge environments remains ch allenging. Existing methods typically require retraining whenever device resourc e constraints change, resulting in excessive computational overhead. We propose DFMPQ, a dynamic federated mixed-precision quantization framework that enables r etraining-free deployment at the edge. DFMPQ trains a weight-sharing mixed-preci sion supernet via FL, which jointly represents diverse bit-width configurations. After training, resource-aware quantized subnets can be derived on demand to sa tisfy heterogeneous and time-varying resource constraints without additional opt imization. Optimizing such a supernet in federated settings is difficult due to optimization interference among heterogeneous bit-widths and the coupling of qua ntization noise with non-IID data. DFMPQ addresses these issues through semantic -aware training and aggregation mechanisms that stabilize supernet optimization. In addition, a sensitivity-guided greedy search strategy is adopted to efficien tly identify suitable quantization configurations under given resource budgets. Extensive experiments on multiple datasets and network architectures demonstrate that DFMPQ achieves competitive accuracy with significantly reduced computation al cost, enabling efficient deployment for dynamic edge computing environments.
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Differentially Private Geodesic Regression
Aditya Kulkarni, Carlos J Soto
In statistical applications it has become increasingly common to encounter data

structures that live on non-linear spaces such as manifolds. For data living on such non-linear spaces geodesic regression emerged as a natural extension of linear regression where the response variable lives on a Riemannian manifold. The parameters of geodesic regression capture the relationship of sensitive data, and hence, one should consider the privacy protection practices of said parameters. We consider releasing Differentially Private (DP) parameters of geodesic regression via the K-Norm Gradient (KNG) mechanism for Riemannian manifolds. We derive theoretical bounds for the sensitivity of the parameters showing they are tied to their respective Jacobi fields and hence the curvature of the space. We demonstrate the efficacy of our methodology on the sphere, $S_2\subset\mathbb{R}^3$, the space of symmetric positive definite matrices, and Kendall's planar shape space. Our methodology is general to any Riemannian manifold, and thus it is suitable for data in domains such as medical imaging and computer vision.
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Interpretable Neural ODEs for Gene Regulatory Network Discovery under Perturbations
Zaikang Lin, Sei Chang, Aaron Zweig, Minseo Kang, Fabian J Theis, Elham Azizi, David A. Knowles
Modern high-throughput biological datasets containing thousands of perturbations enable large-scale discovery of causal graphs that represent regulatory interactions between genes. Differentiable causal graphical models and regression-based methods have been developed to infer gene regulatory networks (GRNs) from interventional datasets. However, existing approaches fail to capture the non-linear dynamics of biological processes such as cellular differentiation. To address this limitation, we propose $\textit{PerturbODE}$, a novel framework that employs interpretable neural ordinary differential equations (neural ODEs) to model cell state trajectories under perturbations and derive the underlying causal GRN from the neural ODE parameters, enabling downstream simulation of unseen genetic interventions. The GRN is encoded via a single-hidden-layer feedforward network, implicitly grouping genes into interpretable co-regulated modules. We demonstrate PerturbODE's efficacy in GRN inference and extension to perturbation response prediction across both simulated and real overexpression datasets.
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AgentVocab: Structure-Aware Vocabulary Adaptation for Efficient LLM Agents
Kai Bian, Haosi Mo, Xuebo Liu, Shuangyong Song, Jing Li, Yongxiang Li, Min Zhang, Xuelong Li
Recent large language models (LLMs) have demonstrated strong capabilities across challenging tasks, enabling their widespread adoption in agentic systems that interact with external tools. In such deployments, however, LLMs are typically trained with general-purpose tokenizers designed for broad language coverage, while their usage is dominated by narrow, structured tool-calling interactions. This training–deployment mismatch leads to inefficient tokenization, where repetitive structural patterns and frequent semantic units in function calls are fragmented into long sequences of low-level tokens, increasing decoding overhead. To address this gap, we introduce $\textbf{AgentVocab}$, a structure-aware vocabulary adaptation framework for efficient LLM agents. AgentVocab derives specialized vocabulary entries from real tool-calling traces and adapts the model vocabulary to better reflect structural and semantic regularities, without task-specific schema engineering. Experiments on τ-bench and τ^2-bench show that AgentVocab preserves tool-calling performance while reducing latency relative to the vanilla baseline by 17.7\% and 19.5\%, respectively. Our approach is orthogonal to existing fine-tuning and agent-training methods and integrates seamlessly into standard agent pipelines. Source code and models will be available at https://github.com/Starry-159/AgentVocab.
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VELR: Efficient Video Reward Feedback via Ensemble Latent Reward Models
Liyu Zhang, Kehan Li, Tao Zhou, Zeyi Huang, Chao Li, Jiming Chen
Reward feedback learning (ReFL) is effective for both text-to-image (T2I) and text-to-video (T2V) generation with image reward models (RMs). However, image RMs are misaligned with temporal objectives of T2V, motivating ReFL with video rewar

d models. Nevertheless, directly deploying video RMs is impractical due to their large parameter size and the prohibitive cost of memory. To address this, we propose VELR: an efficient framework that employs ensemble latent reward models (LRMs) to predict rewards directly in latent space, bypassing expensive backpropagation through VAE decoders and video RMs. Specifically, we introduce the ensemble technique for the LRM, which enhances capacity, quantifies uncertainty, and mitigates reward hacking. VELR achieves a reduction of up to 150GB in memory, requiring as little as 12.4% of the memory compared to standard ReFL. Experiments on OpenSora-1.2, CogVideoX-1.5, and Wan-2.1 with large-scale video RMs demonstrate that VELR achieves comparable performance as standard ReFL and enables efficient and robust video RM-based ReFL at scales previously unattainable.

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Sentinel-VLA: A Metacognitive VLA Model with Active Status Monitoring for Dynamic Reasoning and Error Recovery

Wenhao Li, Xiu Su, Dan Niu, Yichao Cao, Hongyan Xu, Zhe Qu, Lei Fan, Shan You, Chang Xu

Vision-language-action (VLA) models have advanced the field of embodied manipulation by harnessing broad world knowledge and strong generalization. However, current VLA models still face several key challenges, including limited reasoning capability, lack of status monitoring, and difficulty in self-correction. In this paper, we introduce \textbf{Sentinel-VLA}, a metacognitive VLA model equipped with an active ``sentinel'' module to monitor real-time execution status. Only when necessary, such as during initial planning or upon detecting an error, the model triggers a dynamic reasoning or formulate error recovery solutions. This on-demand reasoning mechanism ensures robust decision-making while minimizing computational overhead. Notably, all training data (spanning 44 tasks and over 2.6 million transitions) is automatically generated and annotated through our designed pipeline. We also propose the Self-Evolving Continual Learning (SECL) algorithm, which allows Sentinel-VLA to identify its capability boundaries and automatically collect data for expansion, paired with Orthogonal Continual Adapter (OC-Adapter) to constrain parameter updates to an orthogonal space, thereby preventing catastrophic forgetting. Real-world experiments demonstrate that Sentinel-VLA boosts the task success rate by over 30\% compared to the SOTA model, PI0. We will open-source all the code, weights, and data generation pipeline.

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Divide and Conquer: Reliable Multi-View Evidential Learning for Deepfake Detection

Xiaolu Kang, Zhongyuan Wang, Jikang Cheng, Baojin Huang, Zhanhe Lei, Gang Wu, Qin Zou, Qian Wang

With the evolution of generative models, deepfakes have achieved near-perfect semantic realism, leaving forensic traces only in subtle structural anomalies. However, existing single-view paradigms often fail to generalize, as dominant semantic features overwhelm subtle artifact cues within entangled representations. This imbalance leads to overconfident yet brittle predictions—a phenomenon we term the Semantic Masking Effect.

To address this challenge, we propose a reliable framework called Divide-and-Conquer Multi-View Evidential Learning (DiCoME) for Deepfake Detection.

In the "Divide'' phase, we employ Geometric View Purification to decompose the entangled representation space through principled geometric projection. This process suppresses semantic interference within artifact-sensitive representations, forming the foundation for decorrelated yet complementary semantic and artifact views.

In the "Conquer'' phase, we leverage Uncertainty-Aware Evidential Learning to synthesize these distinct views. By explicitly modeling the "epistemic conflict'' between semantic and artifact cues, this mechanism provides calibrated uncertainty estimates instead of forcing rigid deterministic decisions.

Extensive experiments across multiple benchmarks demonstrate that our method consistently outperforms existing approaches in generalization performance, while providing reliable uncertainty estimation for trustworthy deepfake detection.

Code is available at https://github.com/kxl0825/DiCoME.git.

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Recursive Models for Long-Horizon Reasoning
Chenxiao Yang, Nathan Srebro, Zhiyuan Li
Modern language models reason within bounded context, an inherent constraint tha t poses a fundamental barrier to long-horizon reasoning. We identify recursion a s a core principle for overcoming this barrier, and propose recursive models as a minimal realization, where the model can recursively invoke itself to solve su btasks in isolated contexts. We prove that any computable problem admits a recur sive decomposition of reasoning in which each subtask requires only exponentiall y smaller active context than standard autoregressive models; this strictly surp asses any context management approach confined to a single sequence, such as sum marization. We further generalize our framework to modern agentic systems with a rbitrary context processing and control flows, and prove that recursive models c an achieve optimal power within this broader class. Experimentally, we train a 3 B model to reason recursively and evaluate on Boolean satisfiability, a task req uiring long-horizon combinatorial search, where it significantly outperforms fro ntier LLMs.
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Revisiting Zeroth-Order Hessian Approximation: A Single-Step Policy Optimization Lens
Junbin Qiu, Zhaowei Hong, Renzhe Xu, Yao Shu
Accurate Zeroth-Order (ZO) Hessian estimation is a cornerstone of derivative-fre e methods, essential for tasks such as bilevel optimization, Bayesian inference, and uncertainty quantification.
However, obtaining a complete suite of low-variance estimators for the Hessian a nd its inverse in high-dimensional settings remains a significant challenge.
To address this, we propose a unified framework that reinterprets ZO Hessian app roximation through the lens of single-step Policy Optimization (PO).
This perspective establishes a theoretical equivalence between general ZO Hessia n estimators and the Hessian of a smoothed PO objective, unifying distinct class ical randomized estimators as specific instances of baseline selection.
Building on this foundation, we introduce ZoVH, a comprehensive suite of varianc e-reduced estimators for the full Hessian matrix, its regularized inverse, and t he bias-corrected inverse Hessian-gradient product.
ZoVH leverages two key techniques: (1) a unique optimal baseline derived to prov ably minimize variance, and (2) a query reuse strategy that incorporates histori cal function queries to enhance sample efficiency without inflating costs.
Our rigorous theoretical analysis confirms the unbiasedness of the Hessian estim ator, validates the variance optimality of our baseline, provides error bounds f or the entire ZoVH suite, and establishes convergence guarantees for the resulti ng curvature-aware ZO algorithm.
Extensive empirical results validate our theoretical findings, demonstrating tha t ZoVH achieves superior estimation accuracy and convergence performance in real -world applications.
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Probabilistic Retrofitting of Learned Simulators
Cristiana Diaconu, Miles Cranmer, Richard E. Turner, Tanya Marwah, Payel Mukhopa dhyay
Dominant approaches for modelling Partial Differential Equations (PDEs) rely on deterministic predictions, yet many physical systems of interest are inherently chaotic and uncertain. While training probabilistic models from scratch is possi ble, it is computationally expensive and fails to leverage the significant resou rces already invested in high-performing deterministic backbones. In this work, we adopt a training-efficient strategy to transform pre-trained deterministic mo dels into probabilistic ones via retrofitting with a proper scoring rule: the Co ntinuous Ranked Probability Score (CRPS). Crucially, this approach is architectu re-agnostic: it applies the same adaptation mechanism across distinct model back bones with minimal code modifications. The method proves highly effective across different scales of pre-training: for models trained on single dynamical system s, we achieve $20–54\%$ reductions in rollout CRPS and up to 30% improvements

in variance-normalised RMSE (VRMSE) relative to compute-matched deterministic fine-tuning. We further validate our approach on a PDE foundation model, trained on multiple systems and retrofitted on the dataset of interest, to show that our probabilistic adaptation yields an improvement of up to 40% in CRPS and up to 15% in VRMSE compared to deterministic fine-tuning. Validated across diverse architectures and dynamics, our results show that probabilistic PDE modelling need not require retraining from scratch, but can be unlocked from existing deterministic backbones with modest additional training cost.

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Riemannian Dueling Optimization

Yuxuan Ren, Abhishek Roy, Shiqian Ma

Dueling optimization considers optimizing an objective with access to only a comparison oracle of the objective function. It finds important applications in emerging fields such as recommendation systems and robotics. Existing works on dueling optimization mainly focused on unconstrained problems in the Euclidean space. In this work, we study dueling optimization over Riemannian manifolds, which covers important applications that cannot be solved by existing dueling optimization algorithms. In particular, we propose a Riemannian Dueling Normalized Gradient Descent (RDNGD) method and establish its iteration complexity when the objective function is geodesically L-smooth or geodesically (strongly) convex. We also propose a projection-free algorithm, named Riemannian Dueling Frank–Wolfe (RDFW) method, to deal with the situation where projection is prohibited. We establish the iteration and oracle complexities for RDFW. We illustrate the effectiveness of the proposed algorithms through numerical experiments on both synthetic and real applications.

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ParEVO: Synthesizing Code for Irregular Data: High-Performance Parallelism through Agentic Evolution

Liu Yang, Zeyu Nie, Andrew Liu, Ruomu Zou, Deniz Alt■nbüken, Amir Yazdanbakhsh, Quanquan C. Liu

Parallelizing code for **irregular data structures** (sparse graphs, unbalanced trees, non-uniform meshes) is notoriously hard, and current LLMs fail catastrophically on such tasks, generating code riddled with race conditions, deadlocks, and poor scaling. We address this with **ParEVO**, a framework for synthesizing high-performance parallel algorithms for irregular data, built on three contributions: the **Parlay-Instruct Corpus** of 13,820 tasks generated via a "Critic-Refine" pipeline that filters for empirically performant uses of Work-Span primitives; specialized **DeepSeek**, **Qwen**, and **Gemini** models fine-tuned to the semantics of the ParlayLib library; and an **Evolutionary Coding Agent (ECA)** that repairs the "last mile" of correctness using compiler and profiler feedback. On the ParEval benchmark, ParEVO achieves an average **107x speedup** and a **13.6x speedup** on highly complex irregular graph problems, outperforming commercial models like GPT-5-Thinking and Gemini-3-Pro, while matching expert *human-written* baselines and reaching up to a **4.1x speedup** on kernels such as Maximal Independent Set. This demonstrates that AI-driven agents can effectively navigate the complex landscape of high-performance computing. Source code and datasets are available at https://github.com/WildAlg/ParEVO.

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World Guidance: World Modeling in Condition Space for Action Generation

Yue Su, Sijin Chen, Haixin Shi, Mingyu Liu, Zhengshen Zhang, Ningyuan Huang, Weiheng Zhong, Zhengbang Zhu, Yuxiao Liu, Xihui Liu

Leveraging future observation modeling to facilitate action generation presents a promising avenue for enhancing the capabilities of Vision-Language-Action (VLA) models.

However, existing approaches struggle to strike a balance between maintaining efficient, predictable future representations and preserving sufficient fine-grained information to guide precise action generation.

To address this limitation, we propose WoG (World Guidance), a framework that maps future observations into compact conditions by injecting them into the action inference pipeline.

The VLA is then trained to simultaneously predict these compressed conditions al
ongside future actions, thereby achieving effective world modeling within the co
ndition space for action inference.
We demonstrate that modeling and predicting this condition space not only facili
tates fine-grained action generation but also exhibits superior generalization c
apabilities. Moreover, it learns effectively from substantial human manipulation
 videos.
Extensive experiments across both simulation and real-world environments validat
e that WoG significantly outperforms existing methods based on future prediction
. Project page is available at: https://selen-suyue.github.io/WoGNet/.
****

Benchmarking and Improving Fine-Grained Text-to-Image Alignment via Paired Reinf
orcement Learning
Kaihang Pan, Wendong Bu, Yuruo Wu, Kai Shen, Yang Wu, Yun Zhu, Zehan Wang, Yunfe
i Li, Hang Zhao, Juncheng Li, Siliang Tang
While recent autoregressive models have achieved text-to-image generation perfor
mance comparable to diffusion models, they significantly struggle with fine-grai
ned semantic alignment. To rigorously evaluate this limitation, we introduce Del
taBench, a benchmark featuring paired prompts with subtle fine-grained differenc
es, which reveals that existing models fail to achieve precise control over visu
al tokens. To bridge this gap, we propose FocusDiff, a comprehensive framework t
hat enhances alignment by learning from subtle differences in similar text-image
 pairs. Specifically, we construct FocusDiff-Data, a large-scale dataset of pair
ed samples derived from image editing tasks to capture localized semantic shifts
. Furthermore, we introduce Pair-GRPO, an improved reinforcement learning algori
thm that extends GRPO to paired samples. Extensive experiments demonstrate that
our approach outperforms most prior prominent methods on both DeltaBench and exi
sting benchmarks.
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Revisiting Padded Transformer Expressivity: Which Architectural Choices Matter a
nd Which Don't
Anej Svete, William Merrill, Ryan Cotterell, Ashish Sabharwal
Recent work describes what transformers can and cannot compute through connectio
ns to boolean circuits, but existing results lack exact characterizations and ar
e sensitive to modeling choices.
Padded transformers---to whose input filler symbols such as ``...'' are appended
---emerge as a useful gadget for establishing equivalences to circuit classes by
 providing polynomial space for adaptive parallel computation.
However, only a limited set of padded transformer idealizations has been studied
, leaving open how robustly these equivalences hold under changes to attention t
ype, model width, and uniformity.
We find that, under practical assumptions, padded transformers are surprisingly
robust to all of these, and identify numeric precision and model depth as the ma
in factors affecting expressivity.
Concretely, we prove that polynomially padded L-uniform constant-precis
ion transformers are equivalent to L-uniform AC^0, while growing-precis
ion ones achieve L-uniform TC^0 regardless of width.
Furthermore, looping enables sequential processing analogous to circuits: $\log^
d N$-looped constant-precision transformers reach FO-uniform AC^d, and
growing-precision ones reach FO-uniform TC^d.
Interestingly, growing width or precision beyond logarithmic does not increase e
xpressivity, and all our results hold for both softmax and average hard attentio
n transformers.
****

Origo: Interpretable Multi-physics PDE Foundation Model through Neural Operator
Splitting
Li Sun, Hongbo Lv, Zhikai Jiang, Zhongtian Sun, Lanxu Yang, Philip S. Yu
Partial Differential Equations (PDEs) play a fundamental role in scientific comp
uting, and recent efforts have sought to extend the success of foundation models
 to PDE solving. However, multi-physics PDE pre-training faces the unique challe

nge of disentangling dynamic heterogeneity to learn universal, elementary patter
ns that generalize to new PDEs. Additionally, cross-physics transfer lacks a the
oretical framework for interpretability—specifically, understanding which pre-tr
ained operator knowledge is effectively transferred to target PDEs. To bridge th
ese gaps, we introduce the theory of neural operator splitting, which decomposes
 PDE evolution into a modulated global spectral operator and sparse local consti
tutive mechanisms. A key innovation is Origo, which provides a neural operator b
ank that enables the identification of operator-level generalization patterns. E
xtensive experiments demonstrate strong zero-shot generalization and mechanism-l
evel interpretability on unseen PDEs.
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HiPPO Zoo: Explicit Memory Mechanisms for Interpretable State Space Models
Jack Goffinet, Casey Hanks, David Carlson
Representing the past in a compressed, efficient, and informative manner is a ce
ntral problem for systems trained on sequential data. The \textit{HiPPO} framewo
rk, originally proposed by Gu \& Dao et al., provides a principled approach to s
equential compression by projecting signals onto orthogonal polynomial (OP) base
s via structured linear ordinary differential equations. Subsequent works have e
mbedded these dynamics in state space models (SSMs), where HiPPO structure serve
s as an initialization. Nonlinear successors of these SSM methods such as Mamba
are state-of-the-art for many tasks with long-range dependencies, but the mechan
isms by which they represent and prioritize history remain largely implicit. In
this work, we revisit the HiPPO framework with the goal of making these mechanis
ms explicit. We show how polynomial representations of history can be extended t
o support capabilities of modern SSMs such as adaptive memory allocation and ass
ociative memory, while retaining direct interpretability in the OP basis. We int
roduce a unified framework comprising five such extensions, which we collectivel
y refer to as a ``HiPPO zoo.'' Each extension exposes a specific modeling capabi
lity through an explicit, interpretable modification of the HiPPO framework. The
 resulting models adapt their memory online and train in streaming settings with
 efficient updates. We illustrate the behaviors and modeling advantages of these
 extensions through a range of synthetic sequence modeling tasks, demonstrating
that capabilities typically associated with modern SSMs can be realized through
explicit, interpretable polynomial memory structures.
****

DecAEvolve: Decompose, Adapt, and Evolve for Effective LLM-based Scientific Equa
tion Discovery
Pouya Behzadifar, Parshin Shojaee, Sanchit Kabra, Kazem Meidani, Chandan K. Redd
y
Finding mathematical relations underlying natural phenomena is a fundamental tas
k in scientific discovery. Recent advances in evolutionary search with Large Lan
guage Models (LLMs) show great promise by leveraging their embedded scientific k
nowledge. However, discovering governing equations remains challenging due to va
st combinatorial hypothesis spaces with exponentially many possible relations. E
xisting LLM-based approaches treat LLMs as static hypothesis generators unaware
of the observed scientific system, leading to suboptimal and inefficient explora
tion that over-relies on internal priors. To address this, we introduce Decompos
e, Adapt, and Evolve (DecAEvolve), a framework that combines granular feedback f
rom symbolic term decomposition with LLM refinement through reinforcement learni
ng fine-tuning. DecAEvolve unifies symbolic decomposition with test-time RL adap
tation, enabling adaptive rather than static hypothesis generation. Our experime
nts across diverse scientific benchmarks demonstrate that DecAEvolve significant
ly improves both the accuracy of discovered equations and the efficiency of the
discovery process, reducing error by up to an order of magnitude compared to sta
te-of-the-art baselines.
****

G^{2}RPO: Geometric GRPO; Escaping LLM's Reasoning Rut to Break Accuracy--Entro
py Trade-off
Ali Rad, Khashayar Filom, Darioush Keivan, Peyman Mohajerin Esfahani, Ehsan Kama
linejad

Reinforcement learning with verifiable rewards (RLVR) is a cornerstone of post-training for large reasoning models, yet widely used algorithms such as Group Relative Policy Optimization (GRPO) often exhibit \textbf{diversity collapse}.
We provide a geometric diagnosis by formalizing GRPO as a dynamical flow on the probability simplex. Under a mode-based coarse-graining of rollouts, we show that GRPO induces a \textbf{collision field} over correct modes, monotonically pushing towards simplex vertices and thus yielding a \textbf{winner-take-all} regime.
To address this systematically, we introduce \textbf{G^{2}RPO (Geometric GRPO)}, which reshapes RLVR via principled \textbf{vector-field editing}. Concretely, we intervene at the advantage level by adding granularity bonuses inversely proportional to mode probabilities, encouraging underrepresented correct modes. The bonus has a natural geometric interpretation, and its potential performance side effects can be mitigated, thereby avoiding the usual accuracy--diversity trade-off.
In experiments with 7B and 14B models trained on a math reasoning task and evaluated on \textbf{AIME 2024/2025}, GRPO loses up to \textbf{57\%} of active correct modes. In contrast, G^{2}RPO increases active correct-mode coverage by \textbf{172\%--205\%}, reduces concentration on any single correct mode, prevents the late-stage \emph{entropy crash}, and improves \texttt{pass@1} by \textbf{+1.4} to \textbf{+7.9} points relative to GRPO.
Overall, diversity is not merely a regularizer but a \textbf{geometric property} to be controlled to improve the model without trapping it in a single dominant strategy.
****

LLM-based Embeddings: Attention Values Encode Sentence Semantics Better Than Hidden States
Yeqin Zhang, Yunfei Wang, Chen Jiaxuan, Ke Qin, Yizheng Zhao, Cam-Tu Nguyen
Sentence representations are foundational to many Natural Language Processing (NLP) applications. While recent methods leverage Large Language Models (LLMs) to derive sentence representations, most rely on final-layer hidden states, which are optimized for next-token prediction and thus often fail to capture global, sentence-level semantics.
This paper introduces a novel perspective, demonstrating that attention value vectors capture sentence semantics more effectively than hidden states. We propose *Value Aggregation (VA)*, a simple method that pools token values across multiple layers and token indices. In a training-free setting, VA outperforms other LLM-based embeddings, even matches or surpasses the ensemble-based MetaEOL. Furthermore, we demonstrate that when paired with suitable prompts, the layer attention outputs can be interpreted as aligned weighted value vectors. Specifically, the attention scores of the last token function as the weights, while the output projection matrix (W_O) aligns these weighted value vectors with the common space of the LLM residual stream. This refined method, termed *Aligned Weighted VA (AlignedWVA)*, achieves state-of-the-art performance among training-free LLM-based embeddings, outperforming the high-cost MetaEOL by a substantial margin. Finally, we highlight the potential of obtaining strong LLM embedding models through fine-tuning Value Aggregation.
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Swift-SVD: Theoretical Optimality Meets Practical Efficiency in Low-Rank LLM Compression
Ruoling Qi, Yirui Liu, Xuaner Wu, Xiangyu Wang, Ming Li, Chen Chen, Jian Chen, Yin Chen, Qizhen Weng
The deployment of Large Language Models is constrained by the memory and bandwidth demands of static weights and dynamic Key-Value cache. SVD-based compression provides a hardware-friendly solution to reduce these costs. However, existing methods suffer from two key limitations: some are suboptimal in reconstruction error, while others are theoretically optimal but practically inefficient. In this paper, we propose Swift-SVD, an activation-aware, closed-form compression framework that simultaneously guarantees theoretical optimum, practical efficiency and numerical stability. Swift-SVD incrementally aggregates covariance of output

activations given a batch of inputs and performs a single eigenvalue decompositi
on after aggregation, enabling training-free, fast, and optimal layer-wise low-r
ank approximation. We employ effective rank to analyze local layer-wise compress
ibility and design a dynamic rank allocation strategy that jointly accounts for
local reconstruction loss and end-to-end layer importance. Extensive experiments
 across six LLMs and eight datasets demonstrate that Swift-SVD outperforms state
-of-the-art baselines, achieving optimal compression accuracy while delivering 3
–70$\times$ speedups in end-to-end compression time. Our code is available at ht
tps://github.com/hiahei/Swift-SVD.
****

Early Decisions Matter: Proximity Bias and Initial Trajectory Shaping in Non-Aut
oregressive Diffusion Language Models
Jiyeon Kim, Sungik Choi, Yongrae Jo, Moontae Lee, Minjoon Seo
Diffusion-based language models(dLLMs) have emerged as a promising alternative t
o autoregressive language models, offering the potential for parallel token gene
ration and bidirectional context modeling. However, harnessing this flexibility
for fully non-autoregressive decoding remains an open question, particularly for
 reasoning and planning tasks. In this work, we investigate non-autoregressive d
ecoding in dLLMs by systematically analyzing its inference dynamics along the te
mporal axis. Specifically, we uncover an inherent failure modes in confidence-ba
sed non-autoregressive generation stem from a strong proximity bias—the denoisin
g order tends to concentrate on spatially adjacent tokens. This local dependency
 leads to spatial error propagation, rendering the entire trajectory critically
contingent on the initial unmasking position. Leveraging this insight, we presen
t a minimal-intervention approach that guides early token selection, employing a
 lightweight planner and end-of-sequence temperature annealing. We thoroughly ev
aluate our method on various reasoning and planning tasks and observe substantia
l overall improvement over existing heuristic baselines without significant comp
utational overhead.
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Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for Time-Serie
s Foundation Models
Jihye Na, Patara Trirat, Chanyoung Park, Jae-Gil Lee
Recent studies have attempted to fine-tune time-series foundation models to enha
nce a target dataset's forecasting performance. However, these approaches procee
d without a clear criterion for identifying complex datasets that require fine-t
uning due to performance degradation in zero-shot forecasting. To distinguish da
tasets that are more challenging than standard benchmarks, we introduce data-dri
ven temporal complexity and multichannel complexity. *Temporal complexity* captu
res the difficulty of identifying distinct patterns by quantifying spectral entr
opy in the frequency domain, while *multichannel complexity* captures cross-chan
nel information flow that can impact predictive uncertainty. These metrics serve
 as *effective proxies for performance gains* achievable through fine-tuning. Ba
sed on the two metrics, we develop *Time-PEFT*, a parameter-efficient fine-tunin
g framework that incorporates a frequency adapter for top-k filtering and a ch
annel adapter for multichannel modeling. With the base variant of MOMENT as a ba
ckbone, *Time-PEFT* improves performance by up to 38\% over LoRA on complex data
sets.
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Beyond Tokens: Enhancing RTL Quality Estimation via Structural Graph Learning
Yi Liu, Hongji Zhang, Yiwen Wang, Dimitris Tsaras, Lei Chen, Mingxuan Yuan, Qian
g Xu
Estimating the quality of register transfer level (RTL) designs is crucial in th
e electronic design automation (EDA) workflow, as it enables instant feedback on
 key performance metrics like area and delay without the need for time-consuming
 logic synthesis. While recent approaches have leveraged large language models (
LLMs) to derive embeddings from RTL code and achieved promising results, they ov
erlook the structural semantics essential for accurate quality estimation. In co
ntrast, the control data flow graph (CDFG) view exposes the design's structural
characteristics more explicitly, offering richer cues for representation learnin

g. In this work, we introduce StructRTL, a novel structure-aware graph self-supe
rvised learning framework for improved RTL design quality estimation. By learnin
g structure-informed representations from CDFGs, StructRTL significantly outperf
orms prior art on various quality estimation tasks. To further boost performance
, we incorporate a knowledge distillation strategy that transfers low-level insi
ghts from post-mapping netlists into the CDFG-based predictor. Experimental resu
lts demonstrate that StructRTL establishes new state-of-the-art results, highlig
hting the effectiveness of combining structural learning with cross-stage superv
ision.
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FormAct: Agentic Source Editing for Rich-Format Document Generation
Eugene J. Yu, Xingxing Zhang, Yuan Xia, Tao Ge, Xun Wang, FNU Kartik, Vishwas Su
ryanarayanan, Cheng Yang, Amanda Jiang, Jiayu Ding, Xiangyu Wong, Tengchao Lv, L
ei Cui, Si-Qing Chen, Furu Wei, Sujian Li
Rich-format documents are essential for everyday operations yet costly to author
, motivating the need for automated generation to enhance productivity. To this
end, we present FormAct, an agentic system that generates professional rich-form
at documents from scratch. FormAct operates on an HTML source representation and
 performs iterative source refinement with an *editing agent* that invokes a sui
te of tools, including a syntax-aware source editor and a template retriever, an
d a *review agent* that critiques rendered pages to guide refinement. Additional
ly, we incorporate edit-triggered context compression to maintain a bounded work
ing context and keep multi-round editing efficient. To support development and e
valuation, we introduce RichDocBench for end-to-end generation, and RichDocFuzz
to evaluate formatting-error recognition for reviewer agents. Through extensive
automated evaluation and blind human-preference studies, we show that FormAct co
nsistently outperforms strong baselines, including Codex-CLI, with particularly
strong improvements in generating error-free, professional rich-format documents
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AutoNumerics-Zero: Automated Discovery of State-of-the-Art Mathematical Function
s
Esteban Real, Mirko Rossini, Connal de Souza, Manav Garg, Moritz Firsching, Quoc
 V Le, Yao Chen, Akhil Verghese, Ekin Dogus Cubuk, David H. Park
Transcendental functions, such as the exponential, are central to scientific com
puting, yet they cannot be natively calculated by digital hardware. Instead, com
puters must approximate these functions by combining basic operations, such as $
\{+, -, \times, \div\}$, using methods like Taylor series. These methods were de
veloped over centuries by mathematicians, who focused on approaches that could a
ttain arbitrary accuracy. However, computers can handle most applications by usi
ng only finite-precision types, like *float32*, where any accuracy beyond the ty
pe's precision is effectively discarded. We explore, therefore, whether forgoing
 arbitrary accuracy can lead to the discovery of more efficient approximations.
The evolutionary method of symbolic regression is particularly suitable, as it c
an search for arbitrary operation combinations and can optimize non-differentiab
le objectives, such as the number of operations used. Our results show that evol
ution can discover computer programs that outperform established methods in this
 setting, despite having no prior mathematical knowledge beyond the calculation
of the basic operations. Starting from empty code, symbolic regression construct
s programs representing novel mathematical expressions. In particular, we discov
ered a 10-operation program that approximates the exponential function to 14 sig
nificant figures, exceeding the accuracy of previously known approximations of t
his size by more than 6 orders of magnitude.
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OTora: A Unified Red Teaming Framework for Reasoning-Level Denial-of-Service in
LLM Agents
Xinyu Li, Ronghui Mu, Lin Li, Tianjin Huang, Gaojie Jin
Large Language Models (LLMs) are increasingly deployed as autonomous agents that
 execute tool-augmented, multi-step tasks, where latency is a critical factor fo
r real-world applications. Yet an overlooked threat is Reasoning-Level Denial-of

-Service (R-DoS), in which an attacker preserves task correctness but degrades availability by inflating an agent’s reasoning depth or tool-use budget.
We introduce OTora, the first unified, two-stage red-teaming framework for instantiating R-DoS attacks. Stage I optimizes an adversarial trigger that induces targeted tool invocations using insertion-aware scoring and dynamic target co-evolution, supporting both black-box and white-box settings. Stage II generates agent-aware reasoning payloads via an ICL-guided genetic search that amplifies overthinking while maintaining correct task outcomes.
Across WebShop, Email, and OS agents built on multiple backbone models such as LLaMA-70B and GPT-OSS-120B, OTora achieves up to 10× increases in reasoning tokens and order-of-magnitude latency slowdowns, all while preserving near-baseline task accuracy. Finally, we discuss mitigation strategies for detecting and constraining abnormal reasoning and latency spikes.
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Abstraction Induces the Brain Alignment of Language and Speech Models
Emily Cheng, Aditya R. Vaidya, Richard Antonello
Research has repeatedly demonstrated that intermediate hidden states extracted from large language models and speech audio models predict measured brain response to natural language stimuli. Yet, very little is known about the representation properties that enable this high prediction performance. Why is it the intermediate layers, and not the output layers, that are most effective for this unique and highly general transfer task? We give evidence that the correspondence between speech and language models and the brain derives from shared meaning abstraction and not their next-word prediction properties. In particular, models construct higher-order linguistic features in their middle layers, cued by a peak in the layerwise *intrinsic dimension*, a measure of feature complexity. We show that a layer's intrinsic dimension strongly predicts how well it explains fMRI and ECoG signals; that the relation between intrinsic dimension and brain predictivity arises over model pre-training; and finetuning models to better predict the brain causally increases both representations' intrinsic dimension and their semantic content. Results suggest that semantic richness, high intrinsic dimension, and brain predictivity mirror each other, and that the key driver of model-brain similarity is *rich meaning abstraction* of the inputs, where language modeling is a task sufficiently complex (but perhaps not the only) to require it.
****

Revenue Guarantees of No-Swap-Regret Dynamics in First Price Auctions
Anders Bo Ipsen, Stratis Skoulakis
We study the revenue of approximate correlated equilibrium in discrete first price auctions - the set of allowable bids is $\mathcal{B} = \{0, 1/k, \dots, 1 - 1/k, 1\}$ for some $k \in \mathbb{N}$. We show that the revenue of any ϵ-\textit{approximate} correlated equilibrium is at least $v_2 - \Theta(1/k)- \Theta(\epsilon k^2)$, where $v_2 \geq 0$ is the second-highest valuation. Our results establish the first polynomial convergence rates on the revenue generated by no-swap regret bidders in first-price auctions.
For instance, if bidders admit the optimal swap regret of $\mathcal{O}(\sqrt{k T})$, then the time-averaged revenue is at least $v_2 - \Theta(1/k) - \Theta(\epsilon)$ after $\mathcal{O}(k^5/\epsilon^2)$ rounds.
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PinTok: Tokenizers Deserve Dedicated Pinned CPU-Compute and Memory
Sean Choi, Myungheon Chin, Ernest K. Ryu
Tokenization is the first point of contact between large language models (LLMs) and text data, yet it has not been viewed by many as a component of LLMs worth accelerating. During inference, tokenizers typically rely on simple dictionary lookups and are executed on CPUs as standard processes. This approach, however, introduces significant overhead from scheduling delays, core selection, data copying, and other system-level costs. These inefficiencies become problematic in latency-sensitive applications such as embedding, small language models, and agentic AI. In this paper, we present the Pinned Tokenizer (PinTok), a novel tokenizer architecture that reduces redundant hardware, operating system, and networking overhead through three key innovations: core and memory pinning, scheduling and

context switch avoidance, and duplicate network packet copy and processing avoidance. Our implementation of PinTok can serve as a drop-in replacement for existing tokenizer deployments, delivering latency reductions of up to 95% (average), 97% (P50), 94% (P90), and 87% (P99) along with throughput improvements of up to 2,084%.

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Behavior-Invariant Task Representation Learning with Transformer-based World Models for Offline Meta-Reinforcement Learning

Fuyuan Qian, Menglong Zhang, Song Wang, Quanying Liu

Offline meta-reinforcement learning leverages static datasets to enable agents to generalize to unseen environments by combining offline efficiency with meta-learning adaptability, yet it faces key challenges from context and policy distribution shifts. These issues hinder agents from adapting to online environments, and are further exacerbated under sparse-reward settings. As a result, agents often become trapped in an inherent pattern dilemma, failing to achieve robust generalization. In this work, we propose a novel framework that integrates information-theoretic task representation learning with a Transformer-based stochastic world model. Our approach extracts task-defining latent variables that are invariant to behavior policy, thereby effectively mitigating the context distribution shift. To further handle policy shift and model exploitation, we apply a conservative value penalty to imagination-based rollouts, preventing the policy from exploiting model inaccuracies while maintaining robust adaptation. Extensive evaluations demonstrate that our method outperforms state-of-the-art approaches, with superior stability and generalization under out-of-distribution and sparse-reward settings.

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Softsign: Smooth Sign in Your Optimizer For Better Parameter Heterogeneity Handling

Dmitrii Feoktistov, Timofey Belinsky, Andrey Veprikov, Amir Zainullin, Aleksandr Beznosikov

Sign-based and LMO-inspired optimizers have recently attracted substantial attention in deep learning due to their strong performance and low memory footprint. However, their fixed-magnitude updates can hurt terminal convergence: they decouple update mechanisms from gradient magnitudes and fail to account for parameter heterogeneity, often leading to oscillation rather than convergence. We propose SoftSignum, a smooth relaxation of sign-based optimization that replaces the hard sign map with a temperature-controlled soft-sign transformation, enabling a parameter-wise transition from sign-like updates to magnitude-sensitive SGD-like steps. We complement it with an adaptive quantile-based temperature schedule and extend the same principle to matrix-valued optimizers, obtaining SoftMuon. We also develop a generalized geometry-relaxation framework based on strongly convex regularizers and Fenchel conjugates, proving convergence in stochastic non-convex setting. Experiments on diverse deep learning tasks, including LLM pretraining, show that SoftSignum and SoftMuon consistently improve over their hard sign-based counterparts and standard AdamW.

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On the Spectral Unreachability of Brain Graph Learning

Jiaming Zhuo, Shuai Zhai, Ziyi Ma, Kun Fu, Chuan Wang, Di Jin, Zhen Wang, Xiaochun Cao, Huazhu Fu, Liang Yang

Brain network classification is pivotal for diagnosing neurological disorders, yet identifying interpretable functional biomarkers fundamentally relies on precise parcellation. Unfortunately, conventional deep graph encoders applied to brain networks suffer from a critical theoretical limitation termed Spectral Unreachability. Through graph spectral analysis, this paper demonstrates that standard coupled encoder-pooling architectures inevitably oversmooth node representations, corrupting the high-frequency topological signals strictly required to delineate sharp module boundaries. To provide a structural remedy, the Hierarchical Spectral Parcellation Network (HiSP-Net) is proposed, which structurally decouples partition learning from feature smoothing via a project-then-align paradigm. Specifically, HiSP-Net maps representations directly into a partition space using a

topology-agnostic projection block to preserve all-frequency details, while a Topology-Aware Alignment regularizer subsequently enforces spatial coherence. Extensive evaluations demonstrate that HiSP-Net consistently outperforms state-of-the-art baselines in classification, while successfully extracting stable and structurally coherent functional biomarkers. Source code is available at https://github.com/Kevin-916/HiSP-Net-demo/.
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SpikeCLR: Self-Supervised Contrastive Learning for Visual Representations with Spiking Neural Networks
Chengwei Zhou, Gourav Datta
Spiking Neural Networks (SNNs) offer a promising alternative to traditional artificial neural networks by leveraging sparse, event-driven computation that closely mimics biological neurons. When deployed on neuromorphic hardware, SNNs enable substantial energy savings due to their temporal and asynchronous processing. However, training SNNs remains fundamentally difficult because the non-differentiable nature of spike generation breaks the bidirectional gradient flow required in modern self-supervised learning (SSL) frameworks. In this work, we introduce the first fully SSL framework for SNNs that scales to large-scale visual tasks without requiring labeled fine-tuning. Our method leverages intrinsic spike-time dynamics by aligning representations across time steps and augmented views. To address gradient mismatch during surrogate training, we propose the MixedLIF neuron model, which combines a spiking path with an antiderivative-based surrogate path during training to stabilize optimization, while retaining a fully spiking and energy-efficient architecture at inference. We also introduce two temporal objectives, Cross Temporal Loss and Boundary Temporal Loss, that align multi-time-step outputs to improve learning efficiency. Our approach achieves competitive performance across both ResNet- and Vision Transformer-based SNNs on both static and neuromorphic datasets. Our approach further generalizes through transfer learning from ImageNet-1K to downstream tasks. Notably, our self-supervised SNNs match or exceed the performance of some non-spiking SSL models, demonstrating both representational strength and energy efficiency.
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REViT: Roto-reflection Equivariant Convolutional Vision Transformer
Sheir A. Zaheer, Alexander C. Holston, Chan Y. Park
In this paper, we propose a discrete roto-reflection group equivariant vision transformer with convolutional attention. Roto-reflection equivariant networks preserve the rotational, flip and positional symmetry in feature maps, making them useful for tasks where orientation of the inputs is relevant to the model outputs. In image classification and object detection, most of the studies on roto-reflection equivariant models have focused on using convolutional neural networks rather than vision transformers. In this paper, we examine the challenges involved in achieving equivariance in vision transformers, and we propose a simpler way to implement a discretized roto-reflection group equivariant vision transformer. The experimental results demonstrate that our approach outperforms the existing approaches for developing discrete roto-reflection group equivariant neural networks for image classification.
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VocSim A Training-free Benchmark for Zero-shot Content Identity in Single-source Audio
Maris Basha, Anja T Zai, Sabine Stoll, Richard Hahnloser
General-purpose audio representations aim to map acoustically variable instances of the same event to nearby points, resolving content identity in a zero-shot setting. Unlike supervised classification benchmarks that measure adaptability via parameter updates, we introduce VocSim, a training-free benchmark probing the intrinsic geometric alignment of frozen embeddings, with no parameters updated and no labels used (a label-free PCA whitening is fit per subset to correct anisotropy). VocSim aggregates 125k single-source clips from 19 corpora spanning human speech, animal vocalizations, and environmental sounds, isolating content representation from source separation (polyphonic mixtures are out of scope). We evaluate embeddings with Precision@k for local purity and the Global Separation Rat

e (GSR) for point-wise class separation, calibrated by lift over an empirical pe
rmutation baseline. A simple pipeline of frozen Whisper features, time–frequency
 pooling, and label-free PCA yields strong zero-shot performance with stable GSR
 rankings across domains (Kendall's $\tau = 0.60$). However, on blind low-resource sp
eech (Shipibo-Conibo, Chintang), local retrieval collapses while remaining above
 chance, exposing a cross-lingual speech generalization gap. As external validat
ion, our top embeddings predict avian perceptual similarity, improve bioacoustic
 classification, and achieve state-of-the-art on the HEAR benchmark. We release
data, code, and a public leaderboard.
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Shifting the Breaking Point of Flow Matching for Multi-Instance Editing
Carmine Zaccagnino, Fabio Quattrini, Enis Simsar, Marta Tintore Gazulla, Rita Cu
cchiara, Alessio Tonioni, Silvia Cascianelli
Flow matching models have recently emerged as an efficient alternative to diffus
ion, especially for text-guided image generation and editing, offering faster in
ference through continuous-time dynamics. However, existing flow-based editors p
redominantly support global or single-instruction edits and struggle with multi-
instance scenarios, where multiple parts of a reference input must be edited ind
ependently without semantic interference. We identify this limitation as a conse
quence of globally conditioned velocity fields and joint attention mechanisms, w
hich entangle concurrent edits. To address this issue, we introduce Instance-Dis
entangled Attention, a mechanism that partitions joint attention operations, enf
orcing binding between instance-specific textual instructions and spatial region
s during velocity field estimation.
We evaluate our approach on both natural image editing and a newly introduced be
nchmark of text-dense infographics with region-level editing instructions. Exper
imental results demonstrate that our approach promotes edit disentanglement and
 locality while preserving global output coherence, enabling single-pass, instanc
e-level editing.
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Star Elastic: Many-in-One Reasoning LLMs with Efficient Budget Control
Ali Taghibakhshi, Ruisi Cai, Saurav Muralidharan, Sharath Turuvekere Sreenivas,
Ameya Sunil Mahabaleshwarkar, Marcin Chochowski, Akhiad Bercovich, Ran Zilberste
in, Ran El-Yaniv, Yonatan Geifman, Daniel Korzekwa, Yoshi Suhara, Oluwatobi Olab
iyi, Ashwath Aithal, Nima Tajbakhsh, Pavlo Molchanov
Training a family of large language models (LLMs), either from scratch or via it
erative compression, is prohibitively expensive and inefficient, requiring separ
ate training runs for each model in the family. In this paper, we introduce Star
 Elastic, a novel LLM post-training method that adds N nested submodels to a giv
en parent reasoning model using the compute of one run (Nx savings) via a single
 post-training job. Beyond reducing training costs, Star Elastic also addresses
a fundamental limitation in efficient reasoning: the rigidity of static architec
tures, which forces the allocation of constant resources regardless of token dif
ficulty. By unlocking elastic budget control, Star Elastic enables a novel appro
ach that uses different submodels for each reasoning phase (thinking and answeri
ng). Star Elastic supports (1) nesting along the SSM, embedding channel, MoE and
 FFN axes, (2) learning nested submodels via an end-to-end trainable router, and
 (3) curriculum-based knowledge distillation. We apply Star Elastic to the NVIDI
A Nemotron Nano models; in particular, we demonstrate its effectiveness on hybri
d MoE architectures with Nemotron Nano v3 (30B/3.6A), generating 23B (2.8A) and
12B (2.0A) variants with 160B training tokens. For Nemotron Nano v2 (12B), we pr
oduce 9B and 6B nested models using only 110B training tokens, achieving a 360x
reduction versus training from scratch and a 7x reduction over state-of-the-art
compression methods. All nested models match or outperform independently trained
 baselines of comparable size. Crucially, elastic budget control advances the ac
curacy--latency Pareto frontier, achieving up to 16% higher accuracy and 1.9x lo
wer latency via dynamic per-phase model selection.
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Training Language Model Agents to Find Vulnerabilities with CTF-Dojo
Terry Yue Zhuo, Dingmin Wang, Hantian Ding, Varun Kumar, Zijian Wang

Large language models (LLMs) have demonstrated exceptional capabilities when tra
ined within executable runtime environments, notably excelling at software engin
eering tasks through verified feedback loops. Yet, scalable and generalizable ex
ecution-grounded environments remain scarce, limiting progress in training more
capable ML agents. We introduce CTF-Dojo, the first large-scale executable runti
me tailored for training LLMs with verifiable feedback, featuring 658 fully func
tional Capture-The-Flag (CTF)-style challenges containerized in Docker with guar
anteed reproducibility. To enable rapid scaling without manual intervention, we
develop CTF-Forge, an automated pipeline that transforms publicly available arti
facts into ready-to-use execution environments in minutes, eliminating weeks of
expert configuration traditionally required. We trained LLM-based agents on just
 486 high-quality, execution-verified trajectories from CTF-Dojo, achieving up t
o 11.6% absolute gains over strong baselines across three competitive benchmarks
: InterCode-CTF, NYU CTF Bench, and Cybench. Our best-performing 32B model reach
es 31.9% Pass@1, establishing a new open-weight state-of-the-art that rivals fro
ntier models like DeepSeek-V3-0324 and Gemini-2.5-Flash. By framing CTF-style ta
sks as a benchmark for executable-agent learning, CTF-Dojo demonstrates that exe
cution-grounded training signals are not only effective but pivotal in advancing
 high-performance ML agents without dependence on costly proprietary systems.
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Hierarchical Retrieval at Scale: Bridging Interpretability and Efficiency
Shubham Gupta, Zichao Li, Tianyi Chen, Cem Subakan, Siva Reddy, Perouz Taslakian
, Valentina Zantedeschi
Information retrieval is a core component of many intelligent systems as it enab
les conditioning of outputs on new and large-scale datasets. While effective, th
e standard practice of encoding data into high-dimensional representations for s
imilarity search entails large memory and compute footprints, and also makes it
hard to inspect the inner workings of the system. Hierarchical retrieval methods
 offer an interpretable alternative by organizing data at multiple granular leve
ls, yet do not match the efficiency and performance of flat retrieval approaches
. In this paper, we propose Retreever, a tree-based method that makes hierarchic
al retrieval viable at scale by directly optimizing its structure for retrieval
performance while naturally providing transparency through meaningful semantic g
roupings. Our method offers the flexibility to balance cost and utility by index
ing data using representations from any tree level. We show that Retreever deliv
ers strong coarse (intermediate levels) and fine representations (terminal level
), while achieving the highest retrieval accuracy at the lowest latency among hi
erarchical methods. These results demonstrate that this family of techniques is
viable in practical applications.
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Test-Time Graph Search for Goal-Conditioned Reinforcement Learning
Evgenii Opryshko, Junwei Quan, Claas A Voelcker, Yilun Du, Igor Gilitschenski
Offline goal-conditioned reinforcement learning (GCRL) often struggles with long
-horizon tasks, where errors in value estimation accumulate and produce unreliab
le policies. It is typically assumed that effective long-term planning is infeas
ible without specialized training. In contrast, our work demonstrates that exist
ing GCRL policies can complete long-horizon tasks when combined with a lightweig
ht, training-free planning wrapper. We find that standard goal-conditioned value
 functions encode locally consistent geometric structure sufficient for planning
. Our approach, Test-Time Graph Search (TTGS), constructs a graph over the offli
ne dataset and employs an adaptive subgoal selection strategy. To address unreli
able value estimates during shortest-path search, we propose a novel mechanism t
hat softly penalizes long-distance transitions. Our method incurs negligible com
putational overhead and requires no additional supervision or parameter updates.
 On the OGBench benchmark, TTGS significantly boosts success rates across multip
le base learners and tasks, with primary gains on challenging long-horizon locom
otion tasks where some success rates are improved from near-zero to over 90\%, o
ften matching or outperforming methods that require complex auxiliary training.
Code and videos can be found at https://ktolnos.github.io/ttgs.
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Diff-MN: Diffusion Parameterized MoE-NCDE for Continuous Time Series Generation with Irregular Observations

Xu Zhang, Junwei Deng, Chang Xu, Hao Li, Jiang Bian

Time series generation (TSG) is widely used across domains, yet most existing methods assume regular sampling and fixed output resolutions. These assumptions are often violated in practice, where observations are irregular and sparse, while downstream applications require continuous and high-resolution TS. Although Neural Controlled Differential Equation (NCDE) is promising for modeling irregular TS, it is constrained by a single dynamics function, tightly coupled optimization, and limited ability to adapt learned dynamics to newly generated samples from the generative model.

We propose Diff-MN, a continuous TSG framework that enhances NCDE with a Mixture-of-Experts (MoE) dynamics function and a decoupled architectural design for dynamics-focused training. To further enable NCDE to generalize to newly generated samples, Diff-MN employs a diffusion model to parameterize the NCDE temporal dynamics parameters (MoE weights), i.e.,
jointly learn the distribution of TS data and MoE weights. This design allows sample-specific NCDE parameters to be generated for continuous TS generation. Experiments on ten public and synthetic datasets demonstrate that Diff-MN consistently outperforms strong baselines on both irregular-to-regular and irregular-to-continuous TSG tasks. The code is available at the link \url{https://github.com/microsoft/TimeCraft/tree/main/Diff-MN}.

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The Geometric Reasoner: Manifold-Informed Latent Foresight Search for Long-Context Reasoning

Ren Zhuang, Ben Wang, Shuifa Sun

Scaling test-time compute enhances long chain-of-thought (CoT) reasoning, yet existing approaches face a fundamental trade-off between computational cost and coverage quality: either incurring high training expense or yielding redundant trajectories. We introduce The Geometric Reasoner (TGR), a training-free framework that performs manifold-informed latent foresight search under strict memory bounds. At each chunk boundary, TGR scores candidate latent anchors via a lightweight look-ahead estimate combined with soft geometric regularizers that encourage smooth trajectories and diverse exploration. Chunk-wise KV cache resets keep memory linear in chunk length. On challenging math and code benchmarks, TGR improves robust trajectory coverage, measured by the area under the Pass@k curve (AUC), by up to 13 points on Qwen3-8B, with negligible overhead of about 1.1--1.3$\times$.

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Noisy Pairwise-Comparison Random Search for Smooth Nonconvex Optimization

Taha EL BAKKALI EL KADI, Rayane Bouftini, Qiuyi Zhang, Omar Saadi

We study smooth nonconvex optimization using only noisy pairwise comparisons, without access to gradients or function values.

We propose Noisy-Comparison Random Search (NCRS), a simple direct-search method that samples random directions and performs accept/reject updates from comparison feedback.

Under a low-dimensional active-subspace structure, NCRS adapts to the intrinsic dimension $k\le d$ rather than the ambient dimension d.

For a uniform-margin comparison oracle with advantage p, NCRS achieves ϵ-first-order stationarity with comparison complexity
$\mathcal{O}(k/(p^2\epsilon^2))$.

We also introduce a gap-dependent confidence model, where comparison reliability decreases as the objective-value gap between the two candidates becomes small, and analyze a confidence-weighted voting variant of NCRS.

For this oracle, the method achieves ϵ-first-order stationarity with total comparison complexity
$\mathcal{O}(k^2/\epsilon^4)$.

These results provide intrinsic-dimension convergence guarantees for noisy comparison-based random search in smooth nonconvex optimization.

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GenCircuit-RL: Reinforcement Learning from Hierarchical Verification for Genetic Circuit Design
Noah Flynn
Designing genetic circuits, which are biological systems capable of programmed behaviors within living cells, remains a laborious, expert-driven process despite decades of progress in synthetic biology. We introduce GenCircuit-RL, a reinforcement learning framework that trains language models to reason about genetic circuit design through code generation, where models produce Python code using PySBOL to construct circuits in the standardized Synthetic Biology Open Language (SBOL) format. Our approach addresses the challenge of sparse feedback in biological design through hierarchical verification rewards that decompose correctness into five levels, from code execution through structural validity to functional behavior, providing dense learning signal while multiplicative dependencies prevent reward hacking. We contribute SynBio-Reason, a benchmark of approximately 4,753 circuits spanning six canonical circuit types and nine tasks from code repair to de novo design, with held-out biological parts enabling rigorous out-of-distribution evaluation. A four-stage curriculum progressively shifts optimization pressure from basic code generation toward functional correctness, enabling models to acquire compositional reasoning capabilities incrementally. Our framework demonstrates that hierarchical verification combined with curriculum learning enables compact language models to generate functionally correct genetic circuits, including generalization to novel biological parts and rediscovery of canonical designs from synthetic biology literature.
****
Beyond Perplexity: UTF-8 Validity in Byte-aware Language Models
Sangwhan Moon, Daisuke Oba, Youmi Ma, Tatsuya Hiraoka, Naoaki Okazaki
Byte-level tokenization enables language models to handle any Unicode input, but models can generate invalid UTF-8 sequences when encountering rare or unseen characters. We investigate the relationship between training scale and UTF-8 generation reliability with a 355M parameter model trained on 80B tokens from a balanced multilingual corpus of English, Japanese, Korean, and Chinese. We introduce multiple evaluation protocols that isolate UTF-8 structural validity from language modeling. UTF-8 validity convergence lags perplexity by a
roughly a factor of two: perplexity stabilizes after 2.1B tokens, but UTF-8 validity requires 4.2B tokens. In context-free generation, rare characters achieve higher structural validity than common characters, suggesting over-specialization of frequent character representations. Through experiments, we observed that reliable UTF-8 generation is a distinct capability requiring evaluation beyond perplexity.
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On the Origin of Neural Scaling Laws: from Random Graphs to Natural Language
Maissam Barkeshli, Alberto Alfarano, Andrey Gromov
Scaling laws have played a major role in modern AI, providing predictive power over how model performance will improve with increasing resources. This has spurred intense interest in their origin, with a common suggestion being that they arise from power laws already present in the data. Here we study scaling laws for transformers trained to predict random walks on graphs with tunable complexity. We show that this simplified setting already yields scaling laws even in the absence of power laws in the data correlations. We further consider dialing down the complexity of language by training on sequences sampled from increasingly simplified generative language models, from 4,2,1-layer transformer language models down to language bigrams, revealing a monotonic evolution of the scaling exponents. Our results also include scaling laws obtained from training on random walks on random graphs drawn from Erdös-Renyi and scale-free Barabási-Albert ensembles. Finally, we revisit scaling laws for language modeling, demonstrating that several essential results can be reproduced using 2 layer transformers with context length of 100, demonstrate an alternative method for obtaining compute optimal curves, and provide preliminary evidence that maximal update parameterization may be more parameter efficient than standard parameterization.
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AlphaGRPO: Unlocking Self-Reflective Multimodal Generation in Unified Multimodal Models via Decompositional Verifiable Reward
Runhui Huang, Jie Wu, Rui Yang, Zhe Liu, Hengshuang Zhao
In this paper, we propose **AlphaGRPO**, a novel framework that applies Group Relative Policy Optimization (GRPO) to AR-Diffusion Unified Multimodal Models (UMMs)
to enhance multimodal generation capabilities without relying on external knowledge injection. Our approach unlocks the model's intrinsic potential to perform advanced reasoning tasks: Reasoning Text-to-Image Generation,
where the model actively infers implicit user intents, and Self-Reflective Refinement, where it autonomously diagnoses and corrects misalignments in generated outputs.
To address the challenge of providing stable supervision for real-world multimodal generation, we introduce the Decompositional Verifiable Reward (**DVReward**).
Unlike holistic scalar rewards, DVReward utilizes an LLM to decompose complex user requests into atomic, verifiable semantic and quality questions,
which are then evaluated by a general MLLM to provide reliable and interpretable feedback. Extensive experiments demonstrate that AlphaGRPO yields robust improvements across multimodal generation benchmarks,
including GenEval, TIIF-Bench, DPG-Bench and WISE, while also achieving significant gains in editing tasks on GEdit without training on editing tasks.
These results validate that our self-reflective reinforcement approach effectively leverages inherent understanding to guide high-fidelity generation.
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Robust Stochastic Gradient Posterior Sampling with Lattice Based Discretisation
Zier Mensch, Lars Holdijk, Samuel Duffield, Maxwell Aifer, Patrick J. Coles, Max Welling, Miranda C. N. Cheng
Stochastic-gradient MCMC methods enable scalable Bayesian posterior sampling but often suffer from sensitivity to minibatch size and gradient noise. To address this, we propose Stochastic Gradient Lattice Random Walk (SGLRW), an extension of the Lattice Random Walk discretisation. Unlike conventional Stochastic Gradient Langevin Dynamics (SGLD), SGLRW introduces stochastic noise only through the off-diagonal elements of the update covariance; this yields greater robustness to minibatch size while retaining asymptotic correctness. Furthermore, as a comparison we analyse a natural analogue of SGLD utilising gradient clipping. Experimental validation on Bayesian regression and classification demonstrates that SGLRW remains stable in regimes where SGLD fails, including in the presence of heavy-tailed gradient noise, and matches or improves predictive performance.
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Revisiting Coding-Based Approaches to Overcome the Curse of Dimensionality in Learning-Based Watermarking
Yupeng Qiu, Han Fang, Ee-Chien Chang
Deep learning–based watermarking has substantially improved robustness to real-world noise, but its performance degrades as the payload dimension increases. In contrast, coding-based methods such as quantization index modulation (QIM) do not suffer from this curse of dimensionality, although they are less robust to real-world noise. To leverage the strengths of both approaches, we propose OrthoMark, a framework that decouples robust feature extraction from message encoding. OrthoMark first learns a distortion-invariant feature representation using a deep robust feature extractor, and then performs watermark encoding and decoding in this feature domain using coding-based methods. Extensive experiments demonstrate that OrthoMark significantly improves the trade-off among visual quality, robustness, and capacity compared to prior deep watermarking methods, with particularly large gains in the high capacity regime, effectively overcoming the curse of dimensionality. Our code is available at \url{https://github.com/QQiuyp/OrthoMark}.
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Networked Information Aggregation for Binary Classification
Mohammadhossein Bateni, Zahra Hadizadeh, MohammadTaghi Hajiaghayi, Mahdi JafariR

aviz, Shayan Taherijam
We study networked binary classification on a directed acyclic graph (DAG), where each agent observes only a subset of the feature coordinates. Agents act sequentially along the DAG: each receives logits from its parents, augments its local features with these logits, fits a logistic predictor by minimizing binary cross-entropy (BCE), and forwards its logit to its outgoing neighbors. We ask whether this sequential distributed protocol achieves information aggregation: can a downstream agent attain small excess loss compared to the best logistic predictor with access to all features? This question was studied for linear regression under squared loss by Kearns, Roth, and Ryu (2026). Extending their guarantees to classification is nontrivial because their squared-loss tools do not directly transfer to BCE with a logistic link. We analyze the resulting sequential logit-passing protocol and prove an excess-loss upper bound of $O(M/\sqrt{D})$ on depth-D paths satisfying the M coverage condition, namely that every contiguous block of M agents collectively observes all features. We also prove complementary hard instances with excess loss at least $\Omega(k/D)$, where k is the feature dimension. Together, these results show that network depth is a fundamental bottleneck for information aggregation in networked logistic regression.
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LATO: 3D Mesh Flow Matching with Structured TOpology Preserving LAtents
Tianhao Zhao, Youjia Zhang, Hang Long, Jinshen Zhang, Wenbing Li, Yang Yang, Gongbo Zhang, Jozef Hladký, Matthias Nießner, Wei Yang
In this paper, we introduce LATO, a novel topology-preserving latent representation that enables scalable, flow matching-based synthesis of explicit 3D meshes. LATO represents a mesh as a Vertex Displacement Field (VDF) anchored on surface, incorporating a sparse voxel Variational Autoencoder (VAE) to compress this explicit signal into a structured, topology-aware voxel latent. To decapsulate the mesh, the VAE decoder progressively subdivides and prunes latent voxels to instantiate precise vertex locations. In the end, a dedicated connection head queries the voxel latent to predict edge connectivity between vertex pairs directly, allowing mesh topology to be recovered without isosurface extraction or heuristic meshing.
For generative modeling, LATO adopts a two-stage flow matching process, first synthesizing the structure voxels and subsequently refining the voxel-wise topology features. Compared to prior isosurface/triangle-based diffusion models and autoregressive generation approaches, LATO generates meshes with complex geometry, well-formed topology while being highly efficient in inference.
****
Preference-based Antibody Expression Ranking: Scaling with Large-scale Weak Supervision
Josh Qixuan Sun, Morteza Babaie, Wenyang hou, Mark Crowley, David Young
Antibody expression ranking is a critical task in antibody design, yet its modeling is severely hindered by the scarcity of labeled experimental data. To address this, we propose a unified preference-based learning framework that integrates scarce quantitative expression data with large-scale weak positive supervision from immunization data. We adapt Direct Preference Optimization (DPO) to protein language models by introducing a union-masked log-likelihood approximation and IMGT-based alignment, enabling efficient training on variable-length sequences. Evaluating on a diverse internal dataset of 1254 labeled sequences and 4 million unlabeled camelid-derived antibodies, we show that our method consistently outperforms baselines on most metrics. Our results demonstrate that preference learning can effectively learn from weak supervision, providing a scalable solution for antibody expressibility optimization in data-constrained settings. Project page: https://kisoji-biotechnology-inc.github.io/Preference-Expression-Ranking/.
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Counterfactual Residual Data Augmentation for Regression
Hossein Mohebbi, Oliver Schulte, Ke Li, Pascal Poupart
Data-driven modeling in real-world regression tasks often suffers from limited training samples, high collection costs, and noisy observations. Inspired by the impact of data augmentation in vision and language, we propose a novel Counterfa

ctual Residual Data Augmentation (CRDA) technique for tabular regression. Our ke y insight is that once a regressor has modeled the systematic component of the d ata, the remaining noise can be viewed as an invariant residual that remains sta ble under small perturbations of carefully selected features. We exploit this re sidual invariance to generate new, yet realistic, training samples, effectively expanding the dataset without requiring additional real data. Our method is mode l-agnostic and readily applicable to various types of regressors. In experiments across datasets from a variety of benchmark repositories, on average, CRDA redu ces an MLP Regressor's MSE by 22.9% and an XGBoost Regressor's MSE by 6.4%. When compared to existing state-of-the-art data generators and augmentation techniqu es, CRDA consistently outperforms in MSE reduction. By adding principled counter factual variations to the training data, our method offers a simple and efficien t remedy for noise-prone, small-sample regression settings.
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UFO: Chain-of-Evaluation for Omni-Condition Alignment in Multi-Modal Image Gener ation

Danning Zhang, Yijing Lin, Shuhan Zhuang, Mengqi Huang, Shaojin Wu, Shancheng Fa ng, Zhendong Mao

Multi-modal image generation, particularly subject-driven customization, has gar nered growing attention in recent years. Despite the rapid advancement of genera tive models, their evaluation remains largely lagging. Existing methods, whether embedding-based or Multi-modal Large Language Model (MLLM)-based, evaluate alig nment with each modal condition in isolation, which contradicts the simultaneous condition alignment objective of multi-modal image generation, leading to poor consistency with human judgments. To address this challenge, we propose \textbf{ UFO}, the first \textbf{U}ni\textbf{F}ied framework for \textbf{O}mni-condition alignment simultaneous evaluation. Specifically, UFO introduces a novel Atomized Chain-of-Evaluation paradigm, \emph{i.e.}, it first decomposes omni-condition a lignment into a sequential chain of fine-grained, disentangled Atomic Evaluation Units (AEUs), categorizes them into distinct modality-relevance classes, and th en employs general or dedicated functional calls for accurate verification of di fferent AEU types. Experimental results demonstrate that UFO achieves the highes t correlation with human evaluation preferences, delivering an average improveme nt of 15.25\%. Furthermore, we present UFO-Bench, a dedicated benchmark designed to holistically evaluate the performance of existing customization models under the diverse mutual interactions of textual and visual conditions.
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Conf-Gen: Conformal Uncertainty Quantification for Generative Models

Gabriel Loaiza-Ganem, Kevin Zhang, Wei Cui, Marc T. Law, Kin Kwan Leung

Conformal prediction (CP) and its extension, conformal risk control (CRC), are e stablished frameworks for quantifying uncertainty in supervised machine learning through formal guarantees. However, recent breakthroughs in artificial intellig ence (AI) have been driven by unsupervised generative models, such as large lang uage models (LLMs) and image generators, which are not directly compatible with CP or CRC. In this work we introduce conformal generation (Conf-Gen), a general framework adapting CRC to generative tasks while relaxing its theoretical assump tions. Conf-Gen unifies and generalizes previous attempts to apply CP to LLMs, a nd extends conformal methodology to entirely new domains. We demonstrate the fle xibility of Conf-Gen through some novel applications, including obtaining confor mal guarantees on: image generators producing non-memorized images, conversation al AI systems having asked enough clarifying questions, and the output of AI age nts being correct.
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Fast Byte Latent Transformer

Julie Kallini, Artidoro Pagnoni, Tomasz Limisiewicz, Gargi Ghosh, Luke Zettlemoy er, Christopher Potts, Xiaochuang Han, Srini Iyer

Recent byte-level language models (LMs) match the performance of token-level mod els without relying on subword vocabularies, yet their utility is limited by slo w, byte-by-byte autoregressive generation. We address this bottleneck in the Byt e Latent Transformer (BLT) through new training and generation techniques. First

, we introduce **BLT Diffusion (BLT-D)**, a new model and our fastest BLT varian t, trained with an auxiliary block-wise diffusion objective alongside the standa rd next-byte prediction loss. This enables an inference procedure that generates multiple bytes in parallel per decoding step, substantially reducing the number of forward passes required to generate a sequence. Second, we propose two exten sions inspired by speculative decoding that trade some of this speed for higher generation quality: **BLT Self-speculation (BLT-S)**, in which BLT's local decod er continues generating past its normal patch boundaries to draft bytes, which a re then verified with a single full-model forward pass; and **BLT Diffusion+Veri fication (BLT-DV)**, which augments BLT-D with an autoregressive verification st ep after diffusion-based generation. All methods may achieve an estimated memory -bandwidth cost over 50\% lower than BLT on generation tasks. Each approach offe rs its own unique advantages, together removing key barriers to the practical us e of byte-level LMs.

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On the Generalization Gap in Self-Evolving Language Model Reasoning

Zhenting Qi, Susanna Maria Baby, Stefanie Anna Baby, Kan Yuan, Andrew Tomkins, T u Vu, Da-Cheng Juan, Cyrus Rashtchian

Recent work suggests that large language models (LLMs) can improve through self- evolution (SE), using supervision signals generated by the model itself. In this work, we ask: under a strict closed-loop setup, where the SE algorithm has acce ss only to an unlabeled prompt set and a base model, how close can internally ge nerated supervision come to oracle-supervised training? We analyze four represen tative strategies in a unified offline self-evolution framework, including singl e-round verification, multi-turn revision with feedback, iterative training, and curriculum learning. Our primary experiments use Knights and Knaves (KK) logica l reasoning tasks, which provide deterministic solutions, controlled difficulty levels, and a clean testbed for easy-to-hard generalization. We first show that SE consistently improves over the base model, but plateaus after excessive train ing compute is invested, and eventually still leaves a non trivial gap to oracle supervision. We find that multi-turn critic-revision with large models could re ach strong self-evolution performance, where Gemma 12B nearly matches oracle-sup ervised training. Beyond KK, we also evaluate SE on real-world reasoning benchma rks, where gains are also modest. Overall, our results characterize when closed- loop SE can help, and show how internally generated supervision remains insuffic ient under this minimal formulation.

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AlgoVeri: An Aligned Benchmark for Verified Code Generation on Classical Algorit hms

Haoyu Zhao, Ziran Yang, Jiawei Li, Mike He, Zenan Li, Chi Jin, Venugopal Veerava lli, Aarti Gupta, Sanjeev Arora

Vericoding refers to the generation of formally verified code from rigorous spec ifications. Recent AI models show promise in vericoding, but a unified methodolo gy for cross-paradigm evaluation is lacking. Existing benchmarks test only an in dividual language/tool (e.g., Dafny, Verus, and Lean) and each covers very diffe rent tasks, so the performance numbers are not directly comparable. We address t his gap with AlgoVeri, a benchmark that evaluates vericoding of 77 classical a lgorithms in each of Dafny, Verus, and Lean. By enforcing identical functional c ontracts, AlgoVeri reveals critical capability gaps in current models. While fro ntier models achieve tractable success in Dafny (40.3\% for Gemini-3 Flash), w here high-level abstractions and SMT automation simplify the workflow, performan ce collapses under the systems-level memory constraints of Verus (24.7\%) and the explicit proof construction required by Lean (7.8\%). Beyond aggregate met rics, we uncover a sharp divergence in test-time compute dynamics: Gemini-3 effe ctively utilizes iterative repair to boost performance (e.g., tripling pass rate s in Dafny), whereas GPT-OSS saturates early. Finally, our error analysis shows that language design affects the refinement trajectory: while Dafny allows model s to focus on logical correctness, Verus and Lean trap models in persistent synt actic and semantic barriers.

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Noise-corrected GRPO: From Noisy Rewards to Unbiased Gradients
Omar El mansouri, Fathinah Asma Izzati, Mohamed El Amine Seddik, Salem Lahlou
Reinforcement learning from human feedback (RLHF) or verifiable rewards (RLVR), the standard paradigm for aligning LLMs or building recent SOTA reasoning models , is highly sensitive to noise from inconsistent or erroneous rewards. Yet, the interaction between such noise and widely used group-based policy optimization m ethods remains underexplored. We introduce a noise-robust Group Relative Policy Optimization (GRPO) and Done Right GRPO (Dr.GRPO) framework that explicitly mode ls reward corruption as Bernoulli noise. Our method applies noise correction aft er estimating reward flip probabilities to debias the learning signal, yielding unbiased gradient estimates. Theoretical analysis shows that group-based methods inherently mitigate individual-level noise, and our correction strategy amplifi es this robustness. Empirically, we observe consistent improvements across math and code tasks when applying our noise correction to standard reward model usage , with particular gains of up to 6.7 percentage points in accuracy on math tasks and 1.5 on code tasks under realistic reward model conditions. This work bridge s label-noise correction from supervised learning with modern RLHF, offering bot h theoretical insights and a practical algorithm for noisy real-world deployment .
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SSA: Sparse Sparse Attention by Aligning Full and Sparse Attention Outputs in Fe ature Space
Zhenyi Shen, Junru Lu, Lin Gui, Jiazheng Li, Yulan He, di yin, Xing Sun
Sparse attention reduces the quadratic complexity of full self-attention but fac es two challenges: (1) an attention gap, where applying sparse attention to full -attention-trained models causes performance degradation due to train-inference distribution mismatch, and (2) a capability gap, where models trained purely wit h sparse attention lack complete gradient flow, preventing them from matching fu ll-attention performance. We propose SSA (Sparse Sparse Attention), a training f ramework that integrates both sparse and full attention with bidirectional atten tion-output alignment. We prove that the approximation error scales linearly wit h the attention mass dropped under sparse attention, and show that SSA's alignme nt objective substantially reduces this quantity compared to baselines. Experime nts demonstrate that SSA achieves state-of-the-art performance under both infere nce modes, adapts smoothly to varying sparsity budgets, and demonstrates superio r long-context capabilities. Code is available at https://github.com/zhenyi4/ssa .
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No More, No Less: Least-Privilege Language Models
Paulius Rauba, Dominykas Seputis, Patrikas Vanagas, Mihaela van der Schaar
Least privilege is a core security principle: grant each request only the minimu m access needed to achieve its goal.
Deployed language models almost never follow it, instead being exposed through a single API endpoint that serves all users and requests.
This gap exists not because least privilege would be unhelpful—deployments would benefit greatly from reducing unnecessary capability exposure.
The real obstacle is definitional and mechanistic: what does "access" mean insid e a language model, and how can we enforce it without retraining or deploying mu ltiple models?
We take inspiration from least privilege in computer systems and define a class of models called *least-privilege language models*, where privilege is *reachabl e internal computation* during the forward pass.
In this view, lowering privilege literally shrinks the model's accessible functi on class (as opposed to denying access via learned policies).
We formalize deployment-time control as a monitor--allocator--enforcer stack, se parating (i) request-time signals, (ii) a decision rule that allocates privilege , and (iii) an inference-time mechanism that selects privilege.
We then propose *Nested Least-Privilege Networks*, a shape-preserving, rank-inde xed intervention that provides a smooth, reversible control knob.
We show that this knob yields policy-usable privilege--utility frontiers and ena

bles selective suppression of targeted capabilities with limited collateral degr
adation across various policies.
Most importantly, we see this as a defense of a completely new deployment paradi
gm which challenges the premise that we can only have output-level control of la
nguage models.
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Are Large Reasoning Models Interruptible?
Tsung-Han Wu, Mihran Miroyan, David M. Chan, Trevor Darrell, Narges Norouzi, Jos
eph E. Gonzalez
Real-world applications of Large Reasoning Models (LRMs) often require reasoning
 about changing prompts or environments. In this work, we challenge the frozen w
orld assumption and evaluate LRM robustness under two realistic dynamic scenario
s: interruptions, which test the accuracy of model responses under budget-constr
ained outputs, and dynamic context, which tests model adaptation to in-flight ch
anges. Across mathematics and programming benchmarks that require long-form reas
oning, static evaluations consistently overestimate robustness: even state-of-th
e-art LRMs, which achieve high accuracy in static settings, can fail unpredictab
ly when interrupted or exposed to changing context, with performance dropping by
 up to 60\% when updates are introduced late in the reasoning process. Our analy
sis further reveals several novel failure modes, including reasoning leakage, wh
ere models fold the reasoning into their final answer when interrupted; panic, w
here under time pressure models abandon reasoning entirely and return incorrect
answers; and self-doubt, where performance degrades when trying to incorporate u
pdated information. The project page is at https://dynamic-lm.github.io
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G^{2}TAM: Geometry Grounded Track Anything Model
Chenming Zhu, Peizhou Cao, Jingli Lin, Wenbo Hu, Yunlong Ran, Jiangmiao Pang, Ta
i Wang, Xihui Liu
Human spatial understanding arises from jointly perceiving geometry and semantic
s, enabling consistent object identification and localization across viewpoints
and time. Current video segmentation models depend on explicit object appearance
 memory banks for instance tracking, yet they remain vulnerable to large viewpoi
nt changes and long-term occlusions. Leveraging the spatial consistency afforded
 by modern feed-forward 3D reconstruction models, we propose the Geometry Ground
ed Tracking Anything Model (G^{2}TAM), a unified framework for promptable instan
ce tracking in 3D using only unordered RGB images or videos. G^{2}TAM employs sp
atially aligned geometric representations as implicit memory, ensuring stable in
stance identity and localization across frames and views. At its core is a cross
-modal spatial encoder that integrates visual and textual prompts into a shared
geometric space, enabling end-to-end spatial reconstruction and instance-consist
ent mask prediction. To support training and evaluation, we construct InsTrack,
a large-scale dataset with a dedicated validation split for benchmarking. Extens
ive experiments show that G^{2}TAM delivers strong cross-view consistency, promp
table instance spatial tracking, video object segmentation, and spatial reconstr
uction, establishing a foundation for interactive, geometry-grounded spatial rea
soning.
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OptProver: Bridging Olympiad and Optimization through Continual Training in Form
al Theorem Proving
Chenyi Li, Yanchen Nie, Zhenyu Ming, Gong Zhang, Kun Yuan, Zaiwen Wen
Recent advances in formal theorem proving have focused on Olympiad-level mathema
tics, leaving undergraduate domains largely unexplored. Optimization, fundamenta
l to machine learning, operations research, and scientific computing, remains un
derserved by existing provers. Its reliance on domain-specific formalisms (conve
xity, optimality conditions, and algorithmic analysis) creates significant distr
ibution shift, making naive domain transfer ineffective. We present OptProver, a
 trained model that achieves robust transfer from Olympiad to undergraduate opti
mization. Starting from a strong Olympiad-level prover, our pipeline mitigates d
istribution shift through two key innovations. First, we employ large-scale opti
mization-focused data curation via expert iteration. Second, we introduce a spec

ialized preference learning objective that integrates perplexity-weighted optimi
zation with a mechanism to penalize valid but non-progressing proof steps. This
not only addresses distribution shifts but also guides the search toward efficie
nt trajectories. To enable rigorous evaluation, we construct a novel benchmark i
n Lean 4 focused on optimization. On this benchmark, OptProver achieves state-of
-the-art Pass@1 and Pass@32 among comparably sized models while maintaining comp
etitive performance on general theorem-proving tasks, demonstrating effective do
main transfer without catastrophic forgetting.
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OOVDet: Low-Density Prior Learning for Zero-Shot Out-of-Vocabulary Object Detect
ion
Binyi Su, chenghao huang, Haiyong Chen
Zero-shot out-of-vocabulary detection (ZS-OOVD) aims to accurately recognize obj
ects of in-vocabulary (IV) categories provided at zero-shot inference, while sim
ultaneously rejecting undefined ones (out-of-vocabulary, OOV) that lack correspo
nding category prompts.
However, previous methods tend to overfit IV classes, resulting in undefined OOV
 objects being confidently misclassified as semantically similar IV categories.
To mitigate this issue, this paper proposes a zero-shot OOV detector (OOVDet), a
 novel framework that effectively detects predefined classes while reliably reje
cting undefined ones in zero-shot scenes. Specifically, due to the model’s lack
of prior knowledge about the distribution of OOV data, we synthesize region-leve
l OOV prompts by sampling from the low-likelihood regions of the class-condition
al Gaussian distributions in the hidden space, motivated by the assumption that
unknown semantics are more likely to emerge in low-density areas of the latent s
pace. For OOV images, we further propose a Dirichlet-based gradient attribution
mechanism to mine pseudo-OOV image samples, where the attribution gradients are
interpreted as Dirichlet evidence to estimate prediction uncertainty, and sample
s with high uncertainty are selected as pseudo-OOV images. Building on these syn
thesized OOV prompts and pseudo-OOV images, we construct the OOV decision bounda
ry through a low-density prior constraint, which regularizes the optimization o
f OOV classes using Gaussian kernel density estimation in accordance with the ab
ove assumption.
 Experimental results show that our method significantly improves the OOV detect
ion performance in zero-shot scenes. The code is available at \url{https://githu
b.com/binyisu/OOVDet}.
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Episodic Memory-Guided Controllable Experience Synthesis for Reinforcement Learn
ing
Xiao Ma, Tian Li, Wu-Jun Li
In real-world scenarios, data collection for reinforcement learning (RL) is ofte
n constrained by safety concerns and high costs, resulting in limited data avail
ability. Diffusion models (DMs) have recently demonstrated remarkable capabiliti
es in capturing complex distributions, making data augmentation a promising appr
oach. However, existing DM-based data augmentation methods still suffer from the
 limited quality of synthesized data for downstream RL tasks. To overcome this l
imitation, we propose a novel method called episodic memory-guided controllable
experience synthesizer (EMCES). EMCES incorporates an episodic memory-based cont
rollable DM with informative yet concise conditions constructed by episodic memo
ry (EM). To guide the synthesis toward high-quality data, we propose an EM-prior
itized condition sampling strategy that leverages EM-based temporal-difference e
rrors to focus generation on data most helpful for RL. Furthermore, we introduce
 a hashing-based state representation for EM to improve its efficiency and furth
er boost the quality of synthetic data. To the best of our knowledge, EMCES is t
he first work to incorporate EM into controllable DMs and to leverage EM for gui
ding data synthesis in RL. Experimental results across multiple environments dem
onstrate that EMCES significantly improves the quality of the synthetic data, th
ereby improving the performance of several state-of-the-art RL algorithms. In pa
rticular, the hashing-based state representation can reduce storage cost by abou
t 8000-fold and reduce time cost by 25.5-fold, without degrading the normalized

score.
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Q-SAM: Unlocking Sharpness-Aware Minimization for Generalization in Offline Rein forcement Learning
Da Wang, Yi Ma, Ting Guo, Lin Li, Wei Wei, Jiye Liang
Generalization remains a central challenge in offline reinforcement learning (RL), where policies are trained solely from static datasets and must perform relia bly under distribution shift. While most existing offline RL methods focus on re ducing training loss using standard optimizers such as Adam, the role of loss la ndscape geometry $-$ particularly sharpness $-$ has received little attention. S harpness-Aware Minimization (SAM) has recently shown strong generalization benef its in supervised learning by favoring flatter minima. However, directly applyin g SAM to offline RL is non-trivial: unlike supervised settings with ground-truth labels, offline RL relies on bootstrapped targets, making sharpness estimation noisy and often destabilizing optimization. In this paper, we revisit offline RL from an optimization perspective and investigate how sharpness-aware optimizati on can be made effective in this setting. We propose Q bound weighted SAM (Q-SAM), a robust and scalable framework that treats sharpness as a weighted objective and selectively prioritizes samples that are most suitable for sharpness-aware optimization based on Q bounds. By aligning the SAM objective with the character istics of bootstrapped value estimation, Q-SAM amplifies the benefits of sharpne ss minimization while preserving training stability. Extensive experiments on st andard offline RL benchmarks demonstrate that Q-SAM consistently improves genera lization performance across diverse datasets and algorithms. Our results highlig ht the importance of loss sharpness in offline RL and suggest optimizer design a s a promising direction for developing more robust offline RL methods.
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CollabBench: Benchmarking and Unleashing Collaborative Ability of LLMs with Dive rse Players via Proactive Engagement
Hong Qian, Yuanhao Liu, Zihan Zhou, Zongbao Zhang, Hanjie Ge, Haotian Shi, Liang Dou, Xiangfeng Wang, Jing-Wen Yang, Aimin Zhou
While LLM-based agents excel at individual tasks, effective collaboration with r ealistic human partners remains challenging. Most of the existing conversation-l evel collaborative studies lack grounded interaction and behavioral execution, m otivating the need for cooperative game environments that enable contextualized and immersive collaboration. To this end, this paper proposes CollabBench, a ben chmark for evaluating and training collaborative agents in cooperative games. Co llabBench features a Diverse Player Profile Simulation pipeline to model varied players behaviors, and a Collaborative Agentic Training paradigm that unifies re asoning, communication, and action via agentic rollouts, optimized with a hybrid reward balancing task efficiency and affective adaptation. We further extend cl assic environments to CWAH-MultiPlayer and Cook-MultiPlayer for systematic evalu ation under diverse personalities. Experiments with efficiency and affective met rics show that our trained models outperform base models, achieving 19.5% higher efficiency and 24.4% improved affective performance. Further analysis reveals k ey collaborative limitations of existing models and offers insights for future c ollaborative training.
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On Uniform Error Bounds for Kernel Regression under Non-Gaussian Noise
Johannes Teutsch, Oleksii Molodchyk, Marion Leibold, Timm Faulwasser, Armin Lede rer
Providing non-conservative uncertainty quantification for function estimates der ived from noisy observations remains a fundamental challenge in statistical mach ine learning, particularly for applications in safety-critical domains. In this work, we propose novel non-asymptotic probabilistic uniform error bounds for ker nel-based regression. Compared to related bounds in the literature that are rest ricted to (conditionally) independent sub-Gaussian noise, our bounds allow to co nsider a broad class of non-Gaussian distributions, such as sub-Gaussian, bounde d, sub-exponential, and variance/moment-bounded noise. Moreover, our results app ly to correlated and uncorrelated noise. We compare our proposed error bounds wi

th existing results in terms of the induced uncertainty region and their perform ance in safe control, demonstrating the tightness of the proposed bounds.
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Toward Identifiable Sparse Autoencoders
Walter Nelson, Theofanis Karaletsos, Francesco Locatello
Recently, sparse autoencoders (SAEs) have emerged as an attractive tool for inte rpreting and interacting with representations in practical neural networks. Whil e it is common empirical folklore, we also show theoretically that SAEs are high ly unstable: different training runs are likely to produce different concept dic tionaries and sparse codes. We characterize the model properties that hinder the stability of real-world SAEs, and address each of these problems through minima l changes to the architecture and training procedure. Together, these changes yi eld two versions of an **i**dentifiable SAE (iSAE), a variant of the standard To pK SAE with lower reconstruction error and improved stability. We explain this i mprovement theoretically by connecting SAEs with traditional dictionary learning approaches, and show that the dictionaries learned in practice satisfy an appro ximate restricted isometry condition, rendering the corresponding sparse codes i n those models near-identifiable.
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Quantifying the noise sensitivity of the Wasserstein metric for images
Erik Lager, Gilles Mordant, Amit Moscovich
Wasserstein metrics are increasingly adopted as similarity scores for images. We consider the sensitivity of Wasserstein metrics with respect to pixel-wise addi tive noise when the images are treated as discrete measures on the pixel grid. W e derive finite-sample expectation bounds for a Gaussian noise model. Among othe r results, we prove that the error in the signed 2-Wasserstein distance scales w ith the square root of the noise standard deviation. This is favorable compared to the Euclidean metric that scales linearly, and thus provides a theoretical ba sis for the benefits of optimal transport distances in noisy settings. We presen t experiments that support our theoretical findings and point to a peculiar phen omenon where increasing the level of noise can decrease the Wasserstein distance .
A case study on cryo-electron microscopy images demonstrates that the Wasserstei n metric can capture the geometry of the data manifold in high noise settings ev en when the Euclidean metric fails.
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Hermes: An Evidence-Driven Agentic Framework for Trustworthy and Explainable AI-Generated Video Detection
Shuaibo Li, Pengfei HAO, Hongtao Wu, Jianfeng Dong, Ping Li, Xiaohong Liu, Lei Z hu
Recent advances in generative video models have blurred the boundary between rea l and synthetic content, raising urgent concerns about digital authenticity. Mul timodal large language models (MLLMs) are appealing for AI-generated video (AIGV) detection due to their broad perceptual and reasoning capabilities; however, e xisting MLLM-based detectors still suffer from hallucination and unstable reason ing, leading to high false-alarm rates and generic, non-verifiable explanations. To address these issues, we propose Hermes, an evidence-driven agentic framework for trustworthy and explainable AIGV detection. Hermes realizes three key capab ilities: (1) Adaptive Instance-Conditioned Detection Strategy Planning, (2) Evid ence-Centric Reasoning and Verification, and (3) Graph-Grounded Evidence Deliber ation. Specifically, Hermes uses instance-conditioned retrieval-augmented genera tion to analyze each video and retrieve authenticity-verification knowledge for composing a tailored detection strategy. It then constructs a verifiable Evidenc e Reasoning Graph (ERG) to keep reasoning grounded in concrete video evidence an d reduce attention drift. Finally, multi-agent deliberation audits and refines t he ERG to reconcile conflicting evidence and improve reliability.
With these capabilities and a library of forensic tools, Hermes enables structur ed, verifiable, and interpretable decision-making. Extensive experiments show th at Hermes achieves state-of-the-art performance while producing auditable explan ations for trustworthy video forensics.

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On the Coordination of Value-Maximizing Bidders
Yanru Guan, Jiahao Zhang, Zhe Feng, Tao Lin
While the auto-bidding literature predominantly considers independent bidding, we investigate the coordination problem among multiple auto-bidders in online advertising platforms. Two motivating scenarios are: collaborative bidding among multiple bidders managed by a third-party bidding agent, and strategic bid selection for multiple ad campaigns managed by a single advertiser. We formalize this coordination problem as a theoretical model and investigate the coordination mechanism where only the highest-value bidder competes with outside bidders, while other coordinated bidders refrain from competing. We demonstrate that such a coordination mechanism dominates independent bidding, improving both Return-on-Spend (RoS) compliance and the total value accrued for the participating auto-bidders or ad campaigns, for a broad class of auto-bidding algorithms. Additionally, our simulations on synthetic and real-world datasets support the theoretical result that coordination outperforms independent bidding. These findings highlight both the theoretical potential and the practical robustness of coordinated auto-bidding in online auctions.
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LiMuon: Light and Fast Muon Optimizer for Large Models
Feihu Huang, Yuning Luo, Songcan Chen
Large models recently are widely applied in machine learning, so efficient training of large models has received widespread attention. More recently, the useful Muon optimizer is specifically designed for matrix-structured parameters of large models. Although some works have begun to study the Muon optimizer, the existing Muon and its variants still suffer from high sample complexity or high memory for large models. To fill this gap, we propose a light and fast Muon (LiMuon) optimizer for training large models, which builds on the momentum-based variance reduced technique and randomized Singular Value Decomposition (SVD). In particular, our LiMuon simultaneously has a lower memory and lower sample complexity than the Muon and its variants. Moreover, we prove that our LiMuon with lower memory has a lower sample complexity of $O(\epsilon^{-3})$ for finding an ϵ-stationary solution of non-convex stochastic optimization under the generalized smoothness condition. To further narrow practice and theory gap, we also prove that our LiMuon with Newton-Schulz steps has a lower sample complexity than the Muon with Newton-Schulz steps. Numerical experimental results on training Mamba-130M, Qwen2.5-0.5B and ViT models demonstrate effectiveness of our LiMuon.
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Unbiased Reward Modeling from Implicit Feedback for LLM Alignment
Hao Wang, Haocheng Yang, Licheng Pan, Zhichao Chen, Xiaoxi Li, Yinuo Wang, Lei Shen, Yuan Lu, Haoxuan Li, Zhouchen Lin
Despite the success of reinforcement learning from human feedback (RLHF), existing reward modeling methods largely rely on explicit feedback, which is costly to collect and difficult to scale. This work studies implicit reward modeling, learning reward models from implicit user feedback, such as clicks, copies and skips. While scalable and cost-effective, implicit feedback poses two key challenges: (1) It lacks definitive negative samples, which makes standard positive-negative classification methods inapplicable; (2) It suffers from selection bias, where responses have heterogeneous propensities to elicit feedback, which further obscures definitive negative samples. To address these challenges, we propose ImplicitRM, which learns unbiased reward models from implicit feedback. It stratifies training samples into four latent groups using a stratification model and derives a likelihood-maximization objective that is theoretically unbiased, thereby addressing both challenges. Experiments across diverse LLM backbones and benchmark datasets validate that ImplicitRM learns accurate reward models from implicit feedback and improves performance on downstream RLHF tasks.
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SGMD: Score Gradient Matching Distillation for Few-Step Video Diffusion Distillation
Zhuguanyu Wu, Ruihao Gong, Yang Yong, Yushi Huang, Xiangyu Fan, Lei Yang, Dahua

Lin, Xianglong Liu
Distribution Matching Distillation (DMD) is a widely used paradigm for accelerating inference in few-step video diffusion models. However, DMD-style training faces a structural bottleneck: the student-side auxiliary score network (the fake score) must closely track a continuously evolving generator. Updating the fake score too frequently increases training cost and can over-emphasize inner-loop tracking, while infrequent updates lead to tracking lag that destabilizes training and degrades generation consistency. To address this issue, we propose \textbf{Score Gradient Matching Distillation (SGMD)}. SGMD adopts a fake-score perspective by directly optimizing the fake score toward the teacher, while using teacher stop-gradient Fisher as a stable distribution-matching objective. We provide a gradient analysis that motivates this objective choice under ideal tracking. Building on this, SGMD introduces a pair of dual potentials: negative-residual (NR) for outer-loop correction and residual-contraction (RC) for inner-loop tracking. Empirically, compared to DMD, SGMD achieves an approximately $\sim 3\times$ training speedup and substantially improves motion dynamics for 4-step distilled models while preserving temporal consistency.
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BEAR: Dissecting Embodied Abilities in Multimodal Language Models through Skill-level Evaluation and Diagnosis
Yu Qi, Haibo Zhao, Ziyu Guo, Siyuan Ma, Ziyan Chen, Yaokun Han, Renrui Zhang, Zitiantao Lin, Yizhe Zhu, Shiji Xin, Yijian Huang, Boce Hu, Kai Cheng, Jiayi Zhang, Peiheng Wang, jiazheng liu, Wenqing Wang, Yiran Qin, Haojie Huang, Lawson L.S. Wong
Understanding the capability bottlenecks of embodied multimodal large language models (MLLMs) is crucial for improvement. However, existing embodied benchmarks fail to provide actionable insights because they focus on task-level evaluation rather than discovering capability bottlenecks. To address this, we introduce BEAR, where we divide embodied tasks into 14 atomic skills for skill-level evaluation. BEAR comprises 4,469 interleaved image–video–text entries across 14 skills in 6 categories, ranging from low-level perception to high-level planning. We evaluate 20 MLLMs on BEAR under a hierarchical skill-level diagnosis framework and discover that (1) perceptual capabilities are major bottlenecks behind reasoning failures, and (2) models fail due to unstable spatiotemporal modeling which remain unexposed in previous benchmarks. Furthermore, building on these insights, we propose BEAR-Agent, a multimodal conversable agent that augments MLLMs with visual and spatial tools. It substantially enhances MLLMs’ performance across skills, yielding a relative improvement of 17.5% on GPT-5 on BEAR and outperform baselines by a large margin in both simulation and real-robot experiments across models. We provide our project website at https://bear-official66.github.io/.
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Excited Pfaffians: Generalized Neural Wave Functions Across Structure and State
Nicholas Gao, Till Grutschus, Frank Noe, Stephan Günnemann
Neural-network wave functions in Variational Monte Carlo (VMC) have achieved great success in accurately representing both ground and excited states. However, achieving sufficient numerical accuracy in state overlaps requires increasing the number of Monte Carlo samples, and consequently the computational cost, with the number of states. We present a nearly constant sample-size approach, Multi-State Importance Sampling (MSIS), that leverages samples from all states to estimate pairwise overlap. To efficiently evaluate all states for all samples, we introduce Excited Pfaffians. Inspired by Hartree-Fock, this architecture represents many states within a single neural network. Excited Pfaffians also serve as generalized wave functions, allowing a single model to represent multi-state potential energy surfaces. On the carbon dimer, we match the $\mathcal{O}(N_s^4)$-scaling natural excited states while training $>200\times$ faster and modeling 50\% more states. Our favorable scaling enables us to be the first to use neural networks to find all distinct energy levels of the beryllium atom. Finally, we demonstrate that a single wave function can represent excited states across various molecules.
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CADFit: Precise Mesh-to-CAD Program Generation with Hybrid Optimization
Ghadi Nehme, Eamon Whalen, Faez Ahmed
Despite recent progress, recovering parametric CAD construction sequences from g eometric input, such as meshes or point clouds, is a key challenge for design an d manufacturing, as existing CAD reconstruction and generation methods are large ly restricted to difficult-to-edit formats like meshes or Breps or editable simp le sketch-and-extrude pipelines and low-complexity datasets. We introduce CADFit , a hybrid optimization-based CAD reconstruction framework that recovers complex , editable CAD construction sequences from meshes by incrementally fitting and v alidating parametric operations using geometric feedback. Our approach is distin guished by formulating reconstruction as an IoU-driven optimization over structu red CAD programs and supporting a rich set of operations, including extrusions, revolutions, fillets, and chamfers. Experiments on multiple CAD benchmarks show that CADFit outperforms state-of-the-art mesh-to-CAD methods in volumetric Inter section-over-Union and Chamfer Distance, while substantially reducing the Invali d Ratio of reconstructed CAD programs, particularly for complex designs. We furt her present a multimodal pipeline that enables end-to-end reconstruction of CAD construction sequences from images by combining image-based geometry reconstruct ion with CADFit. By enabling accurate reconstruction of higher-complexity CAD mo dels, CADFit provides a practical foundation for generating richer datasets and advancing future learning-based approaches to CAD reverse engineering. The code is available at: https://github.com/ghadinehme/CADFit.
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Efficient Learning of Deep State Space Models via Importance Smoothing
John-Joseph Brady, Nikolas Nüsken, Yunpeng Li
Latent state space systems are ubiquitous in statistical modelling, arising natu rally when time series are observed through noisy measurements. However, trainin g deep state space models (DSSMs) at scale remains difficult. Two largely distin ct strategies have emerged for training DSSMs. The first, auto-encoding DSSMs, t rains generative models by optimising a variational lower bound. The second back propagates through the outputs of classical sequential Monte Carlo (SMC) algorit hms. Such approaches can train DSSMs for both discriminative and generative task s, but their inherently sequential forward passes scale poorly on modern hardwar e. We propose \emph{parallel variational Monte Carlo} (PVMC), a new training met hod that bridges these paradigms and robustly trains DSSMs for both discriminati ve and generative tasks. Across a set of benchmark experiments, PVMC matches or exceeds state-of-the-art performance while training $10\times$ faster than the f astest competing SMC-based approach.
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PSG-Nav: Probabilistic Scene Graph Navigation via Multiverse Decision Making
Rufeng Chen, Yue Chang, Xiaqiang Tang, Hechang Chen, Sihong Xie
Open-vocabulary navigation requires embodied agents to manage significant percep tion uncertainty stemming from semantic ambiguity and model errors.

However, most existing works settle for local optimal deterministic approach es, depriving complex navigation decision-making over multiple composite possibi lities that are critical for globally better solutions.

In this paper, we propose Probabilistic Scene Graph Navigation (PSG-Nav), wh ich constructs a 3D Probabilistic Scene Graph that uses full semantic categorica l distributions to account for perception uncertainty.

To efficiently use the local distributions to compose and reason about the o ptimal navigation landmarks, we propose Multiverse Decision to sample multiple m ost likely world settings from the joint distribution, and evaluate navigation l andmarks based on the compatibility between landmarks and multiverses.

To mitigate false positives due to epistemic uncertainty in open-vocabulary navigation, we introduce the Evidential Experience Calibrator, which enables onl ine lifelong adaptation by cross-validating detections against memories of past successes and failures.

Extensive experiments on widely-used benchmarks MP3D, HM3D, and HSSD demonst rate that PSG-Nav establishes new state-of-the-art results, achieving Success Ra tes of 66.1%, 44.8%, and 67.9%, respectively.

Code is available at: https://psg-nav.github.io
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Trading Complexity for Expressivity Through Structured Generalized Linear Token Mixing
Erwan Fagnou, Paul Caillon, Blaise Delattre, Alexandre Allauzen
Token mixing layers play a key role in how language models can learn and generate long-range dependencies. Their efficiency relies on the necessary trade-off between decoding speed and the memory requirements, along with the cache size. Considering causal generation, this paper explores new trade-offs thanks to a unified framework which separates two crucial features: (i) the direct influence of inputs on outputs in one generation step; (ii) the recurrent propagation of information through past outputs.

This framework encompasses major architectures such as attention and state-space models, but also generalizes the recurrence equations by allowing each state to depend on multiple past states rather than only the immediate predecessor. By introducing structure, we design new recurrence patterns that provably achieve the desired complexity, while providing theoretical insights on their expressivity -- trading runtime for expressivity in a principled way. Empirical validation is performed on synthetic tasks, along with language modeling. Together, these results provide a unified toolkit for the understanding and design of efficient and expressive token mixers across model families.
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MineDraft: A Framework for Batch Parallel Speculative Decoding
Zhenwei Tang, Arun Verma, Zijian Zhou, Zhaoxuan Wu, Alok Prakash, Daniela Rus, Bryan Kian Hsiang Low
Speculative decoding (SD) accelerates large language model inference by using a smaller draft model to propose draft tokens that are subsequently verified by a larger target model. However, the performance of standard SD is often limited by the strictly sequential execution of these drafting and verification stages. To address this, this paper proposes MineDraft, a batch parallel speculative decoding (PSD) framework designed to effectively hide drafting latency by overlapping it with verification. Our theoretical analysis shows that PSD is substantially more efficient than standard SD. MineDraft realizes the PSD through a novel batch-parallel design that maintains two batches of requests, overlapping drafting for one batch with verification for the other. Our experimental results show significant improvements of MineDraft in both throughput (up to 75%) and end-to-end latency (up to 39%) over standard SD. Furthermore, we have implemented MineDraft as a plugin for vLLM, demonstrating its practicality for production-ready inference systems. The code is publicly available in the [MineDraft GitHub repository](https://github.com/electron-shaders/MineDraft).
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Contrastive Weak-to-Strong Generalization
Houcheng Jiang, Junfeng Fang, Jiaxin Wu, Tianyu Zhang, Chen Gao, Xiang Wang, Xiangnan He, Yang Deng
Weak-to-strong generalization provides a promising paradigm for scaling large language models (LLMs) by training stronger models on samples from aligned weaker ones, without requiring human feedback or explicit reward modeling. However, its robustness and generalization are hindered by the noise and biases in weak-model outputs, which limit its applicability in practice. To address this challenge, we leverage implicit rewards, which approximate explicit rewards through log-likelihood ratios, and reveal their structural equivalence with Contrastive Decoding (CD), a decoding strategy shown to reduce noise in LLM generation. Building on this connection, we propose \textbf{Contrastive Weak-to-Strong Generalization (ConG)}, a framework that employs contrastive decoding between pre- and post-alignment weak models to generate higher-quality samples. This approach enables mor

e reliable capability transfer, denoising, and improved robustness, substantiall y mitigating the limitations of traditional weak-to-strong methods. Empirical re sults across different model families confirm consistent improvements, demonstra ting the generality and effectiveness of ConG. Taken together, our findings high light the potential of ConG to advance weak-to-strong generalization and provide a promising pathway toward AGI. Our code is available at: \url{https://github.c om/jianghoucheng/ConG}
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ADEPT: RL-Aligned Agentic Decoding of Emotion via Evidence Probing Tools — From Consensus Learning to Ambiguity-Driven Emotion Reasoning
Esther Sun, Bo-Hao Su, Abinay Reddy Naini, Shinji Watanabe, Carlos Busso
Speech Large Language Models (SLLMs) enable high-level emotion reasoning, but of ten produce ungrounded, text-biased judgments without verifiable acoustic eviden ce. In contrast, SSL encoders such as WavLM yield strong acoustic representation s yet remain opaque discriminative models that offer limited interpretability. T o bridge this gap, we introduce the Agentic Decoding of Emotion via Probing Tool s (ADEPT) framework, which reframes emotion recognition as a multi-turn inquiry process rather than a single-pass prediction. ADEPT transforms an SLLM into an a gent that maintains an evolving candidate set and adaptively invokes dedicated s emantic and acoustic probing tools within a structured pipeline of candidate gen eration, evidence collection, and adjudication. Crucially, ADEPT enables a parad igm shift from consensus learning to ambiguity-driven emotion reasoning. Since h uman affect exhibits complexity and co-occurrence of emotions, we leverage minor ity annotations as informative signals instead of discarding them as noise. Fina lly, we integrate Group Relative Policy Optimization (GRPO) with the Evidence Tr ust Gate to explicitly couple tool-usage behaviors with prediction quality and e nforce evidence-based reasoning. Experiments demonstrate that ADEPT improves in most cases the primary emotion accuracy while substantially improving minor emot ion characterization, producing explanations grounded in auditable evidence.
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Once-for-All: Scalable Simultaneous Forecasting via Equilibrium State Estimation
Beinan Xu, Andy Song, Jiti Gao, Feng Liu
We introduce Equilibrium State Estimation (ESE), a novel paradigm for simultaneo us prediction, where multiple interacting systems require separate yet coordinat ed forecasts. Such scenarios often arise in real-world such as economics and hea lthcare modeling. Unlike existing approaches that predict one system at a time, ESE forecasts all systems in a single pass. It first estimates the equilibrium state across systems, then generates holistic forecasts based on the difference between the current state and the estimated equilibrium. Extensive experiments on synthetic and real-world datasets, including currency exchange and COVID-19 spread modeling, demonstrate that ESE is at least as accurate as state-of-the-ar t (SOTA) methods while being significantly faster. In addition, ESE integrates seamlessly with conventional predictors, combining their accuracy with its excep tional efficiency and delivering a 10–70× speedup. With linear-time complexity, ESE scales far better than SOTA methods as the number of systems increases. More over, it remains accurate under diverse perturbations, establishing ESE as a fas t, generalizable, robust, and scalable multi-prediction method.
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Video-o3: Native Interleaved Clue Seeking for Long Video Multi-Hop Reasoning
Xiangyu Zeng, Zhiqiu Zhang, Yuhan Zhu, Xinhao Li, Zikang Wang, Changlian Ma, Qin gyu Zhang, Zizheng Huang, Kun Ouyang, Tianxiang Jiang, Ziang Yan, Yi Wang, Hongj ie Zhang, Yali Wang, Limin Wang
Existing multimodal large language models for long-video understanding predomina ntly rely on uniform sampling and single-turn inference, limiting their ability to identify sparse yet critical evidence amid extensive redundancy. We introduce Video-o3, a novel framework that supports iterative discovery of salient visual clues, fine-grained inspection of key segments, and adaptive termination once s ufficient evidence is acquired. Technically, we address two core challenges in i nterleaved tool invocation. First, to mitigate attention dispersion induced by t he heterogeneity of reasoning and tool-calling, we propose Task-Decoupled Attent

ion Masking, which isolates per-step concentration while preserving shared globa l context. Second, to control context length growth in multi-turn interactions, we introduce a Verifiable Trajectory-Guided Reward that balances exploration cov erage with reasoning efficiency. To support training at scale, we further develo p a data synthesis pipeline and construct Seeker-173K, comprising 173K high-qual ity tool-interaction trajectories for effective supervised and reinforcement lea rning. Extensive experiments show that Video-o3 substantially outperforms state- of-the-art methods, achieving 72.1% accuracy on MLVU and 46.5% on Video-Holmes. These results demonstrate Video-o3's strong multi-hop evidence-seeking and reaso ning capabilities, and validate the effectiveness of native tool invocation in l ong-video scenarios.
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Skewness-Robust Causal Discovery in Location-Scale Noise Models
Daniel Klippert, Alexander Marx
To distinguish Markov equivalent graphs in causal discovery, it is necessary to restrict the structural causal model. A flexible class of models that is general and identifiable in most cases are location-scale noise models (LSNMs), in whic h the effect Y is modeled based on its causes $\boldsymbol{X}$ as $Y = f(\bold symbol{X}) + g(\boldsymbol{X})N$.
To facilitate the estimation of these models, a prominent assumption is that t he noise variable N follows a symmetric distribution.
We show that when N is a skewed random variable,
which is likely in real-world domains, such approaches drop in performance.
To address this limitation, we propose SkewD, a likelihood-based method for ca usal discovery under LSNMs
with skewed noise, employing a combination of heuristic search and expectation conditional maximization for parameter estimation. SkewD extends the usual norm al distribution framework to the skew-normal setting,
enabling reliable inference under symmetric and skewed noise. While our main f ocus is on bivariate cause-effect inference, we further showcase how SkewD can b e extended to the multivariate setting.
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Measuring Agents in Production
Melissa Pan, Negar Arabzadeh, Riccardo Cogo, Yuxuan Zhu, Alexander Xiong, Lakshy a A Agrawal, Huanzhi Mao, Emma Shen, Sid Pallerla, Liana Patel, Shu Liu, Tiannen g Shi, Xiaoyuan Liu, Jared Quincy Davis, Emmanuele Lacavalla, Alessandro Basile, Shuyi Yang, Paul Castro, Daniel Kang, Koushik Sen, Dawn Song, Joseph E. Gonzale z, Ion Stoica, Matei Zaharia, Marquita Ellis
LLM-based agents already operate in production across many industries, yet we la ck an understanding of what technical methods make deployments successful.
We present the first systematic study of **M**easuring **A**gents in **P**roduct ion, MAP, using first-hand data from agent developers. We conducted 20 case stud ies via in-depth interviews and surveyed 86 deployed systems practitioners acros s 26 domains.
We investigate why organizations build agents, how they build them, how they eva luate them, and their top development challenges.
Our study finds that production agents are built using simple, controllable appr oaches:
68% execute at most 10 steps before human intervention, 70% rely on prompting of f-the-shelf models instead of weight tuning, and 74% depend primarily on human e valuation.
Reliability (consistent correct behavior over time) remains the top development challenge, which practitioners currently address through systems-level design.
MAP documents the current state of production agents, providing the research com munity with visibility into deployment realities and underexplored research aven ues.
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Approximation Error Upper and Lower Bounds for Hölder Class with Transformers
Xin He, Yuling Jiao, Xiliang Lu, Jerry Zhijian Yang
We explore the expressive power of Transformers by establishing precise approxim

ation error upper and lower bounds for Hölder class. Specifically, a new approximation upper bound is derived for the standard Transformer architecture equipped with Softmax operators, ReLU activation functions, and residual connections. We prove that a Transformer network composed of at most $\mathcal{O}(\varepsilon^{-{d_{0}}/{\alpha}})$ blocks can approximate any bounded Hölder function with d_{0}-dimensional input and smoothness $\alpha\in(0,1]$ under any accuracy $\varepsilon>0$. In the case of approximation lower bounds, leveraging the VC-dimension upper bound, we are the first to rigorously prove that Transformers demand for at least $\Omega(\varepsilon^{-{d_{0}}/({4\alpha})})$ blocks to achieve the ε approximation accuracy. As a final step, we extend the derived results for standard Transformers to a general regression task and establish the corresponding excess risk rates demonstrating Transformers' empirical effectiveness in real-world settings.

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Metric—Phase Fields: Decoupling Distance and Sign for Thin-Structure Reconstruction from Unoriented Point Clouds

Jiayi Kong, Xuhui Chen, Chen Zong, Fei Hou, Junhui Hou, Wenping Wang, Ying He

Neural Signed Distance Functions (SDFs) excel at reconstructing watertight manifolds but fail on thin structures and open boundaries due to strict inside—outside constraints. Conversely, Unsigned Distance Fields (UDFs) accommodate general geometries but suffer from gradient singularities at the zero-level set, hindering optimization and extraction.

We introduce Metric—Phase Fields (MPFs), a decoupled implicit representation that separates metric proximity from topological phase. Given an unoriented point cloud, MPFs learn (i) an unsigned metric field r and (ii) a smooth phase field θ, for which we derive a bounded phase indicator $P=\tanh(\beta\theta)$ that provides soft inside—outside cues where they are meaningful. We couple the two fields via a gated-metric formulation with a residual phase injection to obtain a signed implicit function with stable near-surface gradients. The phase coefficient β is learnable, allowing MPFs to adaptively control the sharpness of the phase transition and the degree of saturation of the soft sign indicator. Experiments on both synthetic and scanned thin-shell and thin-plate shapes demonstrate that MPFs preserve thin and layered structures more faithfully than recent SDF-based methods, while also enabling more robust training and more reliable surface extraction than UDF-based approaches. Check out [MPFs-GitHub](https://github.com/JIAYI-Scarlett/ICML2026-MPF) for source code and test models.

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PINNfluence: Interpreting PINNs through Influence Functions

Aleksander Krasowski, Jonas Naujoks, Moritz Weckbecker, Galip Ümit Yolcu, Thomas Wiegand, Sebastian Lapuschkin, Wojciech Samek, René Pascal Klausen

Physics-informed neural networks (PINNs) have emerged as a powerful deep learning approach for solving partial differential equations (PDEs) in the physical sciences, yet their behavior remains largely opaque and is typically understood through failure mode analyses rather than explicit interpretability.

To address this issue, we introduce PINNfluence, a training data attribution framework for interpreting PINNs based on influence functions.

By extending influence functions to composite physics-informed training objectives, we enable fine-grained attribution between predictions, loss components, and training data points.

Through benchmark experiments across various PDEs, we demonstrate that influence patterns provide granular diagnostics that distinguish structural characteristics across well-trained and poorly-trained PINNs.

PINNfluence thus opens a new avenue for understanding and improving the reliability of PINNs through the lens of their data.

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Optimizing Agentic Reasoning with Retrieval via Synthetic Semantic Information Gain Reward

Senkang Hu, Yong Dai, Yuzhi Zhao, Yihang Tao, Yu Guo, Zhengru Fang, Sam Kwong, Yuguang Fang

Agentic reasoning enables large reasoning models (LRMs) to dynamically acquire e

xternal knowledge, yet optimizing the retrieval process remains challenging due to the lack of dense, principled reward signals. In this paper, we introduce *InfoReasoner*, a unified framework that incentivizes effective information seeking via a *synthetic semantic information gain reward*. Theoretically, we redefine information gain as uncertainty reduction over the model's belief states, establishing key properties including non-negativity, telescoping additivity, and channel monotonicity. Practically, to enable scalable optimization without manual intermediate retrieval annotations, we instantiate this principle as a semantic information gain reward computed from the model's output distributions using *semantic clustering via bidirectional textual entailment*. This training reward provides dense credit for retrieval steps while remaining anchored to final-answer correctness, enabling efficient training via Group Relative Policy Optimization (GRPO). Experiments on seven question-answering benchmarks, MATH500, and WebDetective show consistent gains over strong retrieval-augmented baselines, supporting our dense semantic information gain as a practical training signal for agentic retrieval.

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HDTree: Generative Modeling of Cellular Hierarchies for Robust Lineage Inference

Zelin Zang, WenZhe Li, Yongjie Xu, Chang Yu, Changxi Chi, Jingbo Zhou, Zhen Lei, Stan Z. Li

In single-cell research, tracing and analyzing high-throughput single-cell differentiation trajectories is crucial for understanding biological processes. Key to this is the robust modeling of hierarchical structures that govern cellular development. Traditional methods face limitations in computational cost, performance, and stability. VAE-based approaches have made strides but still require branch-specific network modules, limiting their scalability and stability, while often suffering from posterior collapse.To overcome these challenges, we introduce HDTree, a generative modeling framework designed for robust lineage inference. HDTree captures tree relationships within a hierarchical latent space using a unified hierarchical codebook and employs a quantized diffusion process to model continuous cell state transitions. By aligning the generative process with the Waddington landscape, this method not only improves stability and scalability but also enhances the biological plausibility of inferred lineages. HDTree's effectiveness is demonstrated through comparisons on both general-purpose and single-cell datasets, where it outperforms existing methods in lineage inference accuracy, reconstruction quality, and hierarchical consistency. These contributions enable accurate and efficient modeling of cellular differentiation paths, offering reliable insights for biological discovery.

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Don't Force the Fit: Bounded Log-Likelihood Loss for Enhanced Reasoning in Large Language Models

Feng Zhao, Hong Zhang, Yu Yang, Ruilin Zhao, Guandong Xu

Supervised fine-tuning (SFT) is central to aligning large language models (LLMs) with instruction following and task-specific reasoning. Despite its success, SFT optimizes token-level likelihoods under the implicit assumption that strictly fitting all tokens in expert demonstrations induces the desired downstream behavior. However, in reasoning tasks where correctness is defined by logical validity or final outcomes rather than exact token realizations, this assumption can lead to optimization misalignment. We empirically observe that low-probability tokens in reasoning demonstrations often correspond to realization-specific or stylistic variations, and that reducing their influence during training consistently improves generalization on reasoning benchmarks. Motivated by this insight, we propose the *Bounded Log-Likelihood Loss* (BLL-Loss), a simple and parameter-free alternative to standard likelihood training that bounds gradient contributions from low-probability tokens while preserving conventional optimization behavior. We provide theoretical insights and extensive empirical results demonstrating that BLL-Loss improves reasoning generalization across diverse model scales and challenging benchmarks.

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NeuroMamba: A Universal Spatiotemporal Module for Robust Perception in Degraded

Sensory Streams
Jinfeng Li, Huijia Song, Xiangyue Hu, HanLiang Zhou, Jiahui Zhang, Xinpeng Jiang, Fangli Guan, Bin Lin, Dingran Dong, Liqi Yan, Pan Li
In open-world intelligent systems, processing continuous sensory streams disrupted by heterogeneous degradation sources presents a fundamental challenge: reconciling the inherent tension between observational completeness and reconstruction fidelity. Methods that prioritize completeness by bridging long-term occlusions often introduce spurious artifacts, while approaches that focus on aggressive noise suppression inevitably disrupt temporal continuity and erase valid structures. To address this challenge, we propose NeuroMamba, a universal plug-and-play module that enhances spatiotemporal consistency in degraded streams. NeuroMamba tackles the dual objectives through two synergistic components. First, we introduce a regional Hybrid Spatiotemporal Rectification (HSR) module, which leverages Mamba-based inertial modeling of linear complexity to recover short-horizon temporal dependencies and infer missing modalities under partial observability. Second, we design a Spiking Confidence Gate (SCG) that enforces reconstruction fidelity under occupancy-guided supervision. Implemented as a hard-thresholding spiking gate unit based on leaky integrate-and-fire (LIF) neurons, SCG distinguishes valid geometric features from sensor noise based on accumulated temporal evidence. Extensive experiments on the nuScenes robustness benchmark demonstrate that NeuroMamba effectively reconciles the trade-off between completeness and fidelity, outperforming the performance of existing approaches in restoring high-fidelity spatiotemporal features from severely incomplete and degraded observations.
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Causal Modeling of Selection in Evolution
Haoyue Dai, Zeyu Tang, Peter Spirtes, Kun Zhang
Understanding potential selection in data is crucial for causal discovery; we argue that "selection" in common narratives takes two forms, which we term _static_ and _evolutionary_ selection, respectively. Static selection refers to a one-shot filtering process where observed data consist of a _subset_ of the population of interest, as in survey volunteer bias. Evolutionary selection, in contrast, operates through repeated rounds of differential fitness in reproduction, where observed data constitute the latest _generation_ shaped by a historical trajectory, as in immune adaptation, antibiotic resistance, and social norm emergence. Existing methods largely conflate these two forms and rely on an identical graphical model of selection. We show that this model is valid for static settings but fails to characterize data under evolution, yielding false discovery results. To address this, we introduce a new model that specifically characterizes evolutionary selection, and develop a sound and complete procedure for identifying such models from data across one or multiple environments or generations. Experimental results validate the method's ability to uncover the relevant mechanisms underlying evolution from data.
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When to Memorize and When to Stop: Gated Recurrent Memory for Long-Context Reasoning
Leheng Sheng, Zhang Yongtao, Wenchang Ma, Yaorui Shi, Ting Huang, Xiang Wang, An Zhang, Ke Shen, Tat-Seng Chua
While reasoning over long context is crucial for various real-world applications, it remains challenging for large language models (LLMs) as they suffer from performance degradation as the context length grows.
Recent work MemAgent has tried to tackle this by processing context chunk-by-chunk in an RNN-like loop and updating a textual memory for final answering.
However, this naive recurrent memory update faces two crucial drawbacks:
(i) memory can quickly explode because it can update indiscriminately, even on evidence-free chunks;
and (ii) the loop lacks an exit mechanism, leading to unnecessary computation after even sufficient evidence is collected.
To address these issues, we propose GRU-Mem, which incorporates two text-controlled gates for more stable and efficient long-context reasoning.
Specifically, in GRU-Mem, the memory only updates when the update gate is open a

nd the recurrent loop will exit immediately once the exit gate is open.
To endow the model with such capabilities, we introduce two reward signals r^{update} and r^{exit} within end-to-end RL, rewarding the correct updating and exiting behaviors respectively.
Experiments on various long-context reasoning tasks demonstrate the effectiveness and efficiency of GRU-Mem, which generally outperforms the vanilla MemAgent with up to 400\% times inference speed acceleration.
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Neuromem: A Granular Decomposition of the Streaming Lifecycle in External Memory for LLMs
Ruicheng Zhang, Xinyi Li, Tianyi Xu, Shuhao Zhang, Xiaofei Liao, Hai Jin
Most evaluations of External Memory Module assume a static setting: memory is built offline and queried at a fixed state. In practice, memory is streaming: new facts arrive continuously, insertions interleave with retrievals, and the memory state evolves while the model is serving queries. In this regime, accuracy and cost are governed by the full memory lifecycle, which encompasses the ingestion, maintenance, retrieval, and integration of information into generation. We present Neuromem, a scalable testbed that benchmarks External Memory Module under an interleaved insertion-and-retrieval protocol and decomposes its lifecycle into five dimensions including memory data structure, normalization strategy, consolidation policy, query formulation strategy, and context integration mechanism. Using three representative datasets LoCoMo, LONGMEMEVAL, and MemAgentBench, Neuromem evaluates interchangeable variants within a shared serving stack, reporting token-level F1 and insertion/retrieval latency.Overall, we observe that performance typically degrades as memory grows across rounds, and time-related queries remain the most challenging category. The memory data structure largely determines the attainable quality frontier, while aggressive compression and generative integration mechanisms mostly shift cost between insertion and retrieval with limited accuracy gain.
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Who Transfers Safety? Identifying and Targeting Cross-Lingual Shared Safety Neurons
Xianhui Zhang, Chengyu Xie, Linxia Zhu, Yonghui Yang, Weixiang Zhao, Zifeng Cheng, Cong Wang, Fei Shen, Tat-Seng Chua
Multilingual safety remains significantly imbalanced, leaving non-high-resource (NHR) languages vulnerable compared to robust high-resource (HR) ones. Moreover, the neural mechanisms driving safety alignment remain unclear despite observed cross-lingual representation transfer.In this paper, we find that LLMs contain a set of cross-lingual shared safety neurons (SS-Neurons), a remarkably small yet critical neuronal subset that jointly regulates safety behavior across languages. We first identify monolingual safety neurons (MS-Neurons) and validate their causal role in safety refusal behavior through targeted activation and suppression. Our cross-lingual analyses then identify SS-Neurons as the subset of MS-Neurons shared between HR and NHR languages, serving as a bridge to transfer safety capabilities from HR to NHR domains. We observe that suppressing these neurons causes concurrent safety drops across NHR languages, whereas reinforcing them improves cross-lingual defensive consistency. Building on these insights, we propose a simple neuron-oriented training strategy that targets SS-Neurons based on language resource distribution and model architecture. Experiments demonstrate that fine-tuning this tiny neuronal subset outperforms state-of-the-art methods, significantly enhancing NHR safety while maintaining the model's general capabilities.
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AOEB: Benchmarking Agent-Oriented Multimodal Embeddings
Xin Zhang, Jiaxin Xu, mengjia zhou, Xinping Zhao, Yinghui Li, di yin, Xing Sun, Meishan Zhang, Baotian Hu, Wenjie Li, Min Zhang
LLM agents powered by retrieval and RAG are increasingly prevalent across research and applications. Embedding models play a critical role in these systems, particularly in embedding-based retrieval. However, current benchmarks for embeddings remain focused on general-purpose scenarios, which may fail to align well wit

h the diverse and evolving needs of agentic applications. To close this gap, we introduce Agent-Oriented Embedding Benchmark (AOEB), a comprehensive evaluation suite dedicated to agent-centric retrieval for embedding models. AOEB is characterized by two key features: (1) Multi-Task, covering five essential capabilities for retrieval in LLM agents, including code, tool, reasoning, and memory retrieval; and (2) Multi-Modal, providing evaluation with both textual and visual data for each task category. We evaluate representative embedding models on AOEB and observe that they exhibit distinct strengths across different agent-oriented retrieval tasks. By curating AOEB, we aim to promote a move toward more practically oriented directions within the embedding community and foster further progress.

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ReasonEdit: Editing Vision--Language Models using Human Reasoning

Jiaxing Qiu, Kaihua Hou, Roxana Daneshjou, Ahmed Alaa, Thomas Hartvigsen

Model editing aims to correct errors in large, pretrained models without altering their unrelated behaviors. While some recent works have edited vision–language models (VLMs), no existing editors tackle reasoning-heavy tasks, which typically require humans and models to reason about images. We therefore propose ReasonEdit, the first VLM editor to let users explain their reasoning during editing, introducing a new, practical model editing setup. ReasonEdit continuously stores human reasoning in a codebook, and retrieves only relevant facts during inference using a novel topology-balanced multimodal embedding method inspired by network science. Across four VLMs on multiple rationale-based visual question answering datasets, ReasonEdit achieves state-of-the-art editing performance, ultimately showing that using human reasoning during editing greatly improves edit generalization.

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AdamO: A Collapse-Suppressed Optimizer for Offline RL

Nan Qiao, Sheng Yue, Shuning Wang, Ju Ren

Offline reinforcement learning (RL) can fail spectacularly when bootstrapped temporal-difference (TD) updates amplify their own errors, driving the critic toward extreme and unusable Q-values.

A key counterintuitive insight of this work is that collapse is not only a property of the backup rule or network architecture: the optimizer alone can directly trigger or suppress instability.

From a control-theoretic viewpoint, we model offline TD learning as a feedback system and analyze Adam-based critic updates.

This yields a necessary and sufficient condition for stability: training is stable if and only if the spectral radius of the update dynamics is strictly below one.

Further analysis reveals that standard Adam updates can inadvertently distort the parameter geometry, necessitating explicit orthogonality constraints to prevent TD error amplification.

To this end, we propose AdamO, an Adam-based optimizer with a decoupled orthogonality correction regulated by a strict task-alignment budget.

We prove that this design theoretically guarantees worst-case task safety and preserves Adam’s continuous-time dissipative dynamics.

Empirically, AdamO is plug-and-play across diverse offline RL baselines, improving stability and returns across a broad suite of benchmarks.

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SWE-MiniSandbox: Container-Free Reinforcement Learning for Building Software Engineering Agents

Danlong Yuan, Wei Wu, Zhengren Wang, Xueliang Zhao, Huishuai Zhang, Dongyan Zhao

Reinforcement learning (RL) has become a key paradigm for training software engineering (SWE) agents, yet its practical accessibility and scalability is often constrained by container-based execution frameworks used for environment isolation. As the number of task instances increases, pre-cached container images introduce substantial storage overhead, limiting large-scale training under limited container resources, and excludes users without container management privileges. We introduce SWE-MiniSandbox, a lightweight, container-free method that enables sc

alable RL training of SWE agents without sacrificing isolation. Instead of relyi ng on per-instance containers, SWE-MiniSandbox executes each task in an isolated workspace backed by kernel-level mechanisms, substantially reducing system over head. It leverages lightweight environment pre-caching techniques to eliminate t he need for bulky container images.
As a result, our approach lowers disk usage to approximately 5\% of that require d by container-based pipelines and reduces environment preparation time to about 25\% of the container baseline. Empirical results demonstrate that SWE-MiniSand box achieves evaluation performance comparable to standard container-based pipel ines. Consequently, by removing the dependency on heavy container infrastructure , SWE-MiniSandbox offers a practical and accessible foundation for scaling RL-ba sed SWE agents, particularly in resource-constrained research environments.
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DecomPose: Disentangling Cross-Category Optimization Contention for Category-Lev el 6D Object Pose Estimation

Yifan Gao, Lu Zou, Zhangjin Huang, Guoping Wang

Category-level 6D object pose estimation is typically formulated as a multi-cate gory joint learning problem with fully shared model parameters. However, pronoun ced geometric heterogeneity across categories entangles incompatible optimizatio n signals in shared modules, resulting in gradient conflicts and negative transf er during training. To address this challenge, we first introduce gradient-based diagnostics to quantify module-level cross-category contention. Building on res ults of diagnostics, we propose DecomPose, a difficulty-aware decomposition fram ework that mitigates optimization contention via: (1) difficulty-aware gradient decoupling, which groups categories using a data-driven difficulty proxy and rou tes each instance to a group-specific correspondence branch to isolate incompati ble updates; and (2) stability-driven asymmetric branching, which assigns higher -capacity branches to structurally simple categories as stable optimization anch ors while constraining complex categories with lightweight branches to suppress noisy updates and alleviate negative transfer. Extensive experiments on REAL275, CAMERA25, and HouseCat6D demonstrate that DecomPose effectively reduces cross-c ategory optimization contention and delivers superior pose estimation performanc e across multiple benchmarks.
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Modeling Covariate Transition for Efficient Estimation of Longitudinal Treatment Effects in Randomized Experiments

Naoki Chihara, Tatsushi Oka, Yasuko Matsubara, Yasushi Sakurai, Shota Yasui

We present a regression-adjustment framework designed for the estimation of long itudinal treatment effects in randomized experiments under static regimes. While regression-adjustment methods are useful for variance reduction in randomized e xperiments by using pre-treatment covariates, they usually focus only on average effects, from which we cannot obtain valuable insights into when the effects ap pear and how long they continue. To address this issue, we consider intermediate outcomes and evolving post-treatment covariates over time, and we represent suc h dynamic trajectories using transition kernels. Furthermore, we establish the a symptotic normality and the semiparametric efficiency bound for our estimator, e nabling more powerful statistical inference. Simulation studies and empirical an alysis using A/B test data from a streaming platform in Japan show the practical advantages of our method.
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Spatial Priors via Space Filling Curves for Small and Limited Data Vision Transf ormers

Leyla Naz Candogan, Arshia Afzal, Pol Puigdemont, Volkan Cevher

Though Vision Transformers (ViTs) have become the dominant backbone in many comp uter vision tasks, due to permutation equivariance, their attention mechanism la cks explicit spatial inductive biases. This become particularly important in two settings: when model capacity is small or training data is limited. Inspired by the attention masking strategies in Linear Transformers and the scanning patter ns of Vision SSMs, we introduce VIOLIN, a lightweight masked attention mechanism that encodes spatial structure within attention via Space Filling Curves (SFCs)

with less than 0.0015% extra parameters and negligible computational overhead. VIOLIN scans the image using multiple SFCs to construct curve-specific decay masks, which are then combined and multiplied with the attention matrix. Across a wide range of evaluations, VIOLIN consistently improves performance. In limited data regimes such as fine-tuning on VTAB-1K, it boosts accuracy across all task groups and by up to 8.7% on the tasks where spatial information is essential. It can be combined with parameter-efficient fine-tuning methods such as LoRA to further increase the performance. Beyond fine-tuning, VIOLIN improves various small scale ViT architectures (e.g., DeiT, DINO) during pretraining on ImageNet-1K. Additionally, on pixel-level CIFAR-100 training, a task that is highly dependent on location information, VIOLIN increases accuracy by up to 7.2%. Overall, VIOLIN provides a computationally efficient yet effective way to inject spatial inductive bias into ViTs, especially benefiting small models and limited data settings.

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MACD: Model-Aware Contrastive Decoding via Counterfactual Data for Video-LLMs

Qixin Xiao, Kun Zhou

Video language models (Video-LLMs) are prone to hallucinations, generating plausible but ungrounded content when visual evidence is weak, ambiguous, or biased. Existing methods, such as contrastive decoding (CD), rely on random perturbations to construct contrastive data for hallucination mitigation, but often fail to target the visual cues that drive hallucination or align with model weaknesses. We propose Model-Aware Counterfactual Data based Contrastive Decoding (MACD), an inference strategy that combines model-guided counterfactual construction with contrastive decoding. MACD uses the Video-LLM's own feedback to identify object regions most responsible for hallucination, generating targeted object-level counterfactual inputs rather than arbitrary frame or temporal modifications. These counterfactual inputs are integrated into CD to enforce evidence-grounded token selection during decoding. Experiments on EventHallusion, MVBench, Perception-test, and Video-MME show that MACD consistently reduces hallucination while maintaining or improving task accuracy across diverse Video-LLMs, including Qwen and InternVL, with especially strong gains in scenarios involving small, occluded, or co-occurring objects.

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Momentum Further Constrains Sharpness at the Edge of Stochastic Stability

Arseniy Andreyev, Advikar Ananthkumar, Marc Walden, Tomaso Poggio, Pierfrancesco Beneventano

Recent work suggests that (stochastic) gradient descent self-organizes near the instability boundary, shaping both optimization and the solutions found. Momentum and mini-batch gradients are widely used in practical deep learning optimization, but it remains unclear whether they operate in a comparable regime of instability. We demonstrate that SGD with momentum exhibits an Edge of Stochastic Stability (EoSS)-like regime with \textit{batch-size--dependent behavior} that cannot be explained by a single momentum-adjusted stability threshold. Batch Sharpness (the expected directional mini-batch curvature) stabilizes in two distinct regimes: at small batch sizes it converges to a lower plateau $2(1-\beta)/\eta$, reflecting amplification of stochastic fluctuations by momentum and favoring flatter regions than vanilla SGD; at large batch sizes it converges to a higher plateau $2(1+\beta)/\eta$, where momentum recovers its classical stabilizing effect and favors sharper regions consistent with full-batch dynamics. We further show this aligns with linear stability thresholds and we discuss the implications on hyperparameters tuning and coupling.

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Reliable Neighborhood-Aware Multi-View Outlier Detection

Huijie Ma, Haoyuan Xin, Lei Meng, Guanzhou Ke, Yongyong Chen, Guoqing Chao

In recent years, multi-view outlier detection (MVOD) has gained increasing attention, with the primary objective of recovering the underlying structure of normal data from outlier-contaminated multi-view datasets. However, this objective is hindered by two fundamental challenges:(i) outlier propagation, (ii) scale discrepancy.

To address these issues, we propose RNAMOD (Reliable Neighborhood-Aware Multi-View Outlier Detection), which introduces the concept of reliability and constructs a reliable neighborhood structure to avoid outlier propagation. We introduce a leave-one-out directional consensus mechanism to align cross-view neighborhood structures while preventing scale discrepancy by aligning geometric directions that remain invariant to scaling.
Extensive experiments on six benchmark datasets demonstrate that RNAMOD consistently outperforms state-of-the-art methods.
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Efficient Training-Free Multi-Token Prediction via Embedding-Space Probing
Raghavv Goel, Mukul Gagrani, Mingu Lee, Christopher Lott
Large Language Models (LLMs) possess latent multi-token prediction (MTP) abilities despite being trained only for next-token generation. We introduce ESP (Embedding-Space Probing), a simple and training-free MTP method that probes an LLM using on-the-fly mask tokens drawn from its embedding space, enabling parallel future-token prediction without modifying weights or relying on draft models.
ESP constructs a speculative token tree by sampling Top-K candidates from mask-token logits and applies a lightweight pruning rule to retain high-probability continuations. During generation, predictions are verified in parallel, yielding lossless decoding while significantly reducing model calls and increasing token throughput.
ESP consistently outperforms existing training-free baselines, improving acceptance length by 7–11% over Lookahead Decoding on LLaMA3 and 7–8% on Qwen3, and increasing throughput by up to 15–19% over the strongest baseline.
Finally, we provide theoretical insight and empirical evidence showing that decoder layers naturally align mask-token representations with next-token states, enabling accurate multi-step prediction without retraining or auxiliary models.
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How Does the Lagrangian Guide Safe Reinforcement Learning through Diffusion Models?
Xiaoyuan Cheng, Wenxuan Yuan, Boyang Li, Yuanchao Xu, Yiming Yang, Hao Liang, Bei Peng, Robert Loftin, Zhuo Sun, Yukun Hu
Diffusion policy sampling enables reinforcement learning (RL) to represent multimodal action distributions beyond suboptimal unimodal Gaussian policies. However, existing diffusion-based RL methods primarily focus on offline setting for reward maximization, with limited consideration of safety in online settings. To address this gap, we propose Augmented Lagrangian-Guided Diffusion (ALGD), a novel algorithm for off-policy safe RL. By revisiting optimization theory and energy-based modeling, we show that the instability of primal–dual methods arises from the non-convex Lagrangian landscape. In diffusion-based safe RL, the Lagrangian can be interpreted as an energy function guiding the denoising dynamics; counter-intuitively, direct usage destabilizes both policy generation and training. ALGD resolves this issue by introducing an augmented Lagrangian that locally convexifies the energy landscape, yielding a stabilized policy generation and training, without altering the distribution of optimal policy. Theoretical analysis and extensive experiments demonstrate that ALGD is both theoretically grounded and empirically effective, achieving strong and stable performance across diverse environments.
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DGG-HMR: Multi-Person Human Mesh Recovery with Depth-Guided Geometric Anchoring
Yanjie Li, Le Hui, Yali Peng, Shigang Liu
Multi-person human mesh recovery (HMR) from a single image is inherently ill-posed, as multiple 3D poses can produce identical 2D projections due to depth ambiguity. Most existing methods implicitly regress 3D translation from image features, which often leads to unreliable depth estimation. To address this issue, we propose a depth-guided multi-person HMR framework that explicitly models instance-level depth cues and integrates them into mesh recovery. Specifically, we first introduce an instance-aware depth estimator to predict per-person pelvis depths that serve as explicit 3D anchors, thereby decoupling depth estimation from mesh regression.

Then, we design a geometry-anchored refinement decoder that uses these anchors to initialize each instance within a plausible 3D neighborhood, stabilizing mesh refinement under joint 2D-3D supervision. Finally, we adopt a single-stage joint training strategy to coordinate depth estimation and mesh recovery in a unified framework. Extensive experiments on multiple benchmarks demonstrate that our method achieves state-of-the-art performance in both mesh reconstruction accuracy and depth ordering.
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Deep networks learn to parse uniform-depth context-free languages from local statistics
Jack T. Parley, Francesco Cagnetta, Matthieu Wyart
Understanding how the structure of language can be learned from sentences alone is a central question in both cognitive science and machine learning. Studies of the internal representations of Large Language Models (LLMs) support their ability to parse text when predicting the next word, while representing semantic notions independently of surface form. Yet, which data statistics make these feats possible, and how much data is required, remain largely unknown. Probabilistic context-free grammars (PCFGs) provide a tractable testbed for studying these questions. However, prior work has focused either on the post-hoc characterization of the parsing-like algorithms used by trained networks; or on the learnability of PCFGs with fixed syntax, where parsing is unnecessary. Here, we *(i)* introduce a tunable class of PCFGs in which both the degree of ambiguity and the correlation structure across scales can be controlled; *(ii)* provide a learning mechanism---an inference algorithm inspired by the structure of deep convolutional networks---that links learnability and sample complexity to specific language statistics; and *(iii)* validate our predictions empirically across deep convolutional and transformer-based architectures. Overall, we propose a unifying framework where correlations at different scales lift local ambiguities, enabling the emergence of hierarchical representations of the data.
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dgMARK: Decoding-Guided Watermarking for Diffusion Language Models
Pyo Min Hong, Albert No
We propose dgMARK, a decoding-guided watermarking method for discrete diffusion language models (dLLMs). Unlike autoregressive models, dLLMs can generate tokens in arbitrary order. While an ideal conditional predictor would be invariant to this order, practical dLLMs exhibit strong sensitivity to the unmasking order, creating a new channel for watermarking. dgMARK steers the unmasking order toward positions whose high-reward candidate tokens satisfy a simple parity constraint induced by a binary hash, without explicitly reweighting the model’s learned probabilities. The method is plug-and-play with common decoding strategies (e.g., confidence, entropy, and margin-based ordering) and can be strengthened with a one-step lookahead variant. Watermarks are detected via elevated parity-matching statistics, and a sliding-window detector ensures robustness under post-editing operations including insertion, deletion, substitution, and paraphrasing. Project website: https://dgmark-watermarking.github.io
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Neural Vector Lyapunov–Razumikhin Certificates for Delayed Interconnected Systems
Jingyuan Zhou, Yuexuan Wang, Kaidi Yang
Ensuring scalable input-to-state stability (sISS) is critical for the safety and reliability of large-scale interconnected systems, especially in the presence of communication delays. While learning-based controllers can achieve strong empirical performance, their black-box nature makes it difficult to provide formal and scalable stability guarantees. To address this gap, we propose a framework to synthesize and verify neural vector Lyapunov-Razumikhin certificates for discrete-time delayed interconnected systems.
Our contributions are three-fold. First, we establish a sufficient condition for discrete-time sISS via vector Lyapunov-Razumikhin functions, which enables certification for large-scale delayed interconnected systems. Second, we develop a scalable synthesis and verification framework that learns the neural certificates

and verifies the certificates on reachability-constrained delay domains with scalability analysis. Third, we validate our approach on mixed-autonomy platoons, drone formations, and microgrids against multiple baselines, showing improved verification efficiency with competitive control performance.
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Bayesian Gated Non-Negative Contrastive Learning
Peng Cui, Jiahao Zhang, Lijie Hu
While Contrastive Learning (CL) has revolutionized self-supervised representation learning, its latent representations remain highly entangled and opaque, limiting their interpretability in safety-critical applications. We identify that a fundamental cause of this entanglement is the reliance on deterministic similarity measures, which treat all feature dimensions equally. In compositional scenes, this creates an Optimization Conflict: common background features, such as, "blue sky", are encouraged to align in positive pairs but simultaneously repelled in negative pairs, causing gradient oscillations that hinder precise semantic disentanglement. To address this, we propose **BayesNCL** (Bayesian Gated Non-Negative Contrastive Learning). Unlike standard approaches, BayesNCL introduces a probabilistic gating mechanism that dynamically filters out task-irrelevant, high-frequency common features while selectively retaining discriminative semantics. By formalizing feature selection as a variational inference problem with a sparse Bernoulli prior, our method effectively resolves the optimization conflict. Empirical experimental results on Imagenet-100 demonstrate that BayesNCL achieves a remarkable 142.1\% improvement in semantic consistency compared to state-of-the-art baselines, yielding highly interpretable representations without compromising downstream task performance. Code is available at https://github.com/Cui-Peng-624/BayesNCL.
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Editable Proof Sketch for Automated Theorem Proving
Zikai Xiao, Hanzheng Wang, Meng-Hao Guo, Shi-min Hu, Shing-Tung Yau
As large language models (LLMs) improve in mathematical reasoning and formal understanding, a promising approach for automated theorem proving (ATP) is to enable LLMs construct proof sketches, which plan a high-level proof strategy and decompose complex theorems into independently provable subgoals. However, most existing proof sketches are immutable. As a result, any revision typically requires rebuilding the entire sketch, which discards already proved subgoals and bring additional cost. In this paper, we address this limitation by introducing EditableSketch, an editable proof-sketch structure that supports in-place edits for error correction and further subgoal decomposition while preserving previously proved subgoals. Building on EditableSketch, we introduce SketchRefine, a proof-generation framework for ATP by iteratively refining proof sketches through localized, incremental edits. Experiments show that our method not only reduces the cost of the proof process, but also achieves superior performance. For example, our method realizes 76.0% pass rate on FormalMath-Lite (+14.1\% vs. DeepSeek-Prover-V2-671B). Meanwhile, compared with Hilbert, our method significantly reduces token overhead while achieving comparable performance.
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Dual-branch Robust Unlearnable Examples
Xianlong Wang, Hangtao Zhang, Wenbo Pan, Ziqi Zhou, Changsong Jiang, Li Zeng, Xiaohua Jia
Unlearnable examples (UEs) aim to compromise model training by injecting imperceptible perturbations to clean samples.
However, existing UE schemes exhibit limited robustness against advanced defenses due to their heuristic design or narrowly scoped domain perturbations. To address this, we propose DUNE, a Dual-branch UNlearnable Ensemble perturbation optimization approach. Specifically, DUNE separately optimizes perturbations in the spatial and color domains to establish the mapping between perturbations and shift-induced labels. This design extends the perturbation domain to increase noise intensity for improving robustness and drives the models to learn perturbation-oriented features with degraded generalization, thereby achieving unlearnability. To strengthen DUNE's performance, we further propose an unlearnability-en

hancing ensemble strategy that aggregates diverse pre-trained models during the dual-branch optimization. Extensive experiments on benchmark datasets CIFAR-10 and ImageNet verify that DUNE's robustness outperforms 12 SOTA UE schemes under 7 mainstream defenses, yielding a lower average test accuracy of 14.95% to 50.82%.

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Trust-Region Diffusion Policies for Massively Parallel On-Policy RL

Huy Le, Onur Celik, Denis Blessing, Tai Hoang, Claas A Voelcker, Axel Brunnbauer, Felix Richter, Michael Volpp, Gerhard Neumann

Reinforcement learning with massively parallel simulations has become a standard framework for developing robust, deployable policies; however, most existing approaches still rely on simple Gaussian policy parameterizations. Diffusion models provide a more expressive policy class and have shown strong performance on challenging control problems, yet most diffusion-based RL methods are designed for offline or off-policy training. In this work, we ask whether diffusion policies can be trained effectively in the massively parallel, on-policy regime. To this end, we introduce Trust-region Diffusion Policies (TruDi), which enables diffusion policies for on-policy RL with massively parallel simulations. This setting is particularly challenging because the data distribution changes quickly across updates, making stable training with complex policies difficult. TruDi addresses this by integrating a trust-region optimization rule to enforce a KL-divergence constraint over the entire diffusion trajectory. Empirically, we evaluate TruDi on a diverse set of 4 massively parallel RL benchmarks comprising a total of 73 tasks. Across these tasks, TruDi consistently outperforms or is on-par with strong baselines on standard tasks and achieves clear gains on more challenging humanoid control tasks, establishing a strong new baseline for massively parallel on-policy RL.

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MIRO: MultI-Reward cOnditioned pretraining improves T2I quality and efficiency

Nicolas Dufour, Lucas Degeorge, Arijit Ghosh, Vicky Kalogeiton, David Picard

The default paradigm of post-training text-to-image generators includes post-hoc selection of generated images, and subsequent training with one reward model to align the generator to the reward, typically user preference. This discards informative data as well as optimizes only for a single reward, hence harming diversity, semantic fidelity and efficiency. Instead, we propose MIRO, a method that conditions the model on multiple rewards during training, thus letting the model learn user preferences directly. MIRO pre-training both improves the visual quality of the generated images and speeds up the training, achieving state of the art on the GenEval compositional benchmark and user-preference scores (PickAScore, ImageReward, HPSv2).

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Can I Have Your Order? Monte-Carlo Tree Search for Slot Filling Ordering in Diffusion Language Models

Joshua Ong Jun Leang, Yu Zhao, Mihaela C. Stoian, Wenda Li, Shay B Cohen, Eleonora Giunchiglia

While plan-and-infill decoding in Masked Diffusion Models (MDMs) shows promise for mathematical and code reasoning, performance remains highly sensitive to slot infilling order, often yielding substantial output variance. We introduce DiffuSearch, a framework that formulates slot selection as decision making and optimises infilling orders through Monte Carlo Tree Search (MCTS). DiffuSearch uses look-ahead simulations to evaluate partial completions before commitment, systematically exploring the combinatorial space of generation orders. Experiments show an average improvement of 3.2% over autoregressive baselines and 8.0% over baseline plan-and-infill, with notable gains of 19.5% on MBPP and 4.9% on MATH500. Our analysis reveals that while DiffuSearch predominantly follows sequential ordering, incorporating non-sequential generation is essential for maximising performance. We observe that larger exploration constants, rather than increased simulations, are necessary to overcome model confidence biases and discover effective orderings. These findings establish MCTS-based planning as an effective approach for enhancing generation quality in MDMs.

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On the Accuracy of Newton Step and Influence Function Data Attributions
Ittai Rubinstein, Samuel B. Hopkins
Data attribution estimates how a trained model would change if a subset of train ing points were removed, and is a central primitive for tasks such as interpreta bility, data valuation, and machine unlearning. Despite its widespread use, our theoretical understanding of key data attribution methods -- Influence Functions (IF) and a single Newton Step (NS) -- remains limited: existing guarantees heav ily rely on *global* strong convexity and yield bounds with pessimistic dependen ce on the parameter dimension d and the number of removed samples k.
We give a new analysis of IF and NS for convex ERM that replaces global assumpti ons with *local* conditions: it suffices that the loss is strongly convex and su fficiently smooth only in a neighborhood of the first Newton step.
As a concrete validation, we analyze logistic regression with Gaussian features and show that our bounds capture the correct scaling up to polylogarithmic facto rs, yielding matching upper and lower bounds and explaining observed regimes in which NS is markedly more accurate than IF, thereby resolving open questions rai sed by (Koh et al., 2019).
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mHC: Manifold-Constrained Hyper-Connections
Zhenda Xie, Yixuan Wei, Huanqi Cao, Chenggang Zhao, Chengqi Deng, Jiashi Li, Dam ai Dai, Huazuo Gao, Mingyu Xu, Kuai Yu, Liang Zhao, Shangyan Zhou, Zhean Xu, Zhe ngyan Zhang, Wangding Zeng, Shengding Hu, Yuqing Wang, Jingyang Yuan, Lean Wang, Wenfeng Liang
Recently, studies exemplified by Hyper-Connections (HC) have extended the ubiqui tous residual connection paradigm established over the past decade by expanding the residual stream width and diversifying connectivity patterns. While yielding substantial performance gains, this diversification fundamentally compromises t he identity mapping property intrinsic to the residual connection, which causes severe training instability and restricted scalability, and additionally incurs notable memory access overhead. To address these challenges, we propose Manifold -Constrained Hyper-Connections (mHC), a general framework that projects the resi dual connection space of HC onto a specific manifold to restore the identity map ping property, while incorporating rigorous infrastructure optimization to ensur e efficiency. Empirical experiments demonstrate that mHC is effective for traini ng at scale, offering tangible performance improvements and superior scalability . We anticipate that mHC, as a flexible and practical extension of HC, will cont ribute to a deeper understanding of topological architecture design and suggest promising directions for the evolution of foundational models.
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A²RBench: An Automatic Paradigm for Formally Verifiable Abstract Reasoning Bench mark Generation
Qingchuan Ma, Yuexiao Ma, Yongkang Xie, Tianyu Xie, Xiawu Zheng, Rongrong Ji
Abstract reasoning ability reflects the intelligence and generalization capacity of LLMs to extract and apply abstract rules.
However, accurately measuring this ability remains challenging: existing benchma rks either rely on expensive manual annotation, limiting their scale, or risk me asuring memorization rather than genuine reasoning.
To address this, we introduce an automated pipeline named A^{2}RBench, encompass ing generation, expansion, evaluation, and analysis.
Specifically, in the generation stage, LLMs create diverse tasks demanding genui ne reasoning; in the expansion stage, LLMs reuse validated rules and expand new input spaces to generate task variations, achieving scaling.
However, such a process may cause hallucinations.
To eliminate it, we further establish a theoretical framework and prove that pro grammatic verification—testing whether the inverse operation perfectly reverses the forward operation (cycle consistency)—guarantees a unique solution.
Through extensive evaluations on mainstream LLMs, we find: (1) Current LLMs exhi bit fundamental deficiencies in abstract reasoning, with top models significantl y underperforming humans on a representative subset (39.8\% vs. 68.5\%). (2) Cur

rent LLMs fall far short of 2D and 1D in the complexity of generated 3D tasks, r
evealing their lack of understanding of high-dimensional tasks. (3) Counterintui
tively, inputs with higher information complexity can simplify the reasoning pro
cess.
Code and data are available at: https://github.com/MAC-AutoML/A2Rbench.
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Dataset Distillation Efficiently Encodes Low-Dimensional Representations from Gr
adient-Based Learning of Non-Linear Tasks
Yuri Kinoshita, Naoki Nishikawa, Taro Toyoizumi
Dataset distillation, a training-aware data compression technique, has recently
attracted increasing attention as an effective tool for mitigating costs of opti
mization and data storage. However, progress remains largely empirical. Mechanis
ms underlying the extraction of task-relevant information from the training proc
ess and the efficient encoding of such information into synthetic data points re
main elusive. In this paper, we theoretically analyze practical algorithms of da
taset distillation applied to the gradient-based training of two-layer neural ne
tworks with width L. By focusing on a non-linear task structure called multi-i
ndex model, we prove that the low-dimensional structure of the problem is effici
ently encoded into the resulting distilled data. This dataset reproduces a model
with high generalization ability for a required memory complexity of $\tilde \T
heta(r^2d+L)$, where d and r are the input and intrinsic dimensions of the t
ask. To the best of our knowledge, this is one of the first theoretical works th
at include a specific task structure, leverage its intrinsic dimensionality to q
uantify the compression rate and study dataset distillation implemented solely v
ia gradient-based algorithms.
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Cross-Chirality Generalization by Axial Vectors for Hetero-Chiral Protein-Peptid
e Interaction Design
Ziyi Yang, Zitong Tian, Yinjun Jia, Tianyi Zhang, Jiqing Zheng, Hao Wang, Yubu S
u, Juncai He, Lei Liu, Yanyan Lan
D-peptide binders targeting L-proteins have promising therapeutic potential. Des
pite rapid advances in machine learning-based target-conditioned peptide design,
generating D-peptide binders remains largely unexplored. In this work, we show
that by injecting axial features to E(3)-equivariant (polar) vector features, it
is feasible to achieve cross-chirality generalization from homo-chiral (L-L) tr
aining data to hetero-chiral (D-L) design tasks. By implementing this method wit
hin a latent diffusion model, we achieved D-peptide binder design that not only
outperforms existing tools in *in silico* benchmarks, but also demonstrates effi
cacy in wet-lab validation. To our knowledge, our approach represents the first
experimentally validated AI generative model for the *de novo* design of D-pepti
de binders, offering new perspectives on handling chirality in protein design. C
odes are available at [https://github.com/YZY010418/PepMirror](https://github.co
m/YZY010418/PepMirror)
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Hyperbolic Associative Memory Networks
Boliang Hao, Bailing Zhang, Fangyu Wu
Modern Hopfield Networks (MHNs) have achieved widespread success across various
domains but are confined to Euclidean/Hilbert spaces, failing to preserve the hi
erarchical structure of data due to geometric constraints—arbitrary tree structu
res cannot be embedded with low distortion, while hyperbolic spaces can naturall
y accommodate hierarchical structures through exponential volume growth. To addr
ess this issue, we propose Hyperbolic Associative Memory Networks (HAMNs), the f
irst framework to embed modern associative memory into hyperbolic space: we map
query and memory vectors from Euclidean space to a constant negative curvature m
anifold via exponential maps, define a regularized energy function based on the
Minkowski inner product, and adopt curvature-aware Riemannian optimization combi
ned with exponential map updates to achieve stable on-manifold retrieval. We put
forward a hierarchy-sensitivity hypothesis—HAMNs outperform Euclidean MHNs on d
ata with deep hierarchies but exhibit comparable performance on data with weak o
r shallow hierarchies, which is validated by depth-controlled experiments and cr

oss-level metrics. As a plug-and-play, model-agnostic module, HAMNs are suitable for the storage and retrieval of representations in task architectures requiring hierarchical understanding, instantiated with the Poincar\'e ball in experiments, and also applicable to any hyperbolic model with constant negative curvature.
The code is available at https://github.com/hbl66/HAMNs.
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Provably Protecting Fine-Tuned LLMs from Training Data Extraction while Preserving Utility
Tom Segal, Yuval Elovici, Asaf Shabtai
Fine-tuning large language models (LLMs) on sensitive datasets raises privacy concerns, as training data extraction (TDE) attacks can expose highly confidential information.
Existing defenses against such attacks either lack formal privacy guarantees or incur substantial utility degradation.
We observe that fine-tuning induces widespread probability shifts, yet preserving only a small subset of influential token-level deviations is sufficient; the remaining shifts can be aggressively smoothed with minimal impact on utility.
Motivated by this insight, we propose SCP-Δ_r, a Near Access Freeness (NAF)-based algorithm that operates on relative probabilities and explicitly smooths low-impact tokens using a base model.
SCP-Δ_r achieves orders-of-magnitude better theoretical bounds than existing NAF based methods and provides strong empirical protection against TDE attacks with minimal performance loss.
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Steady-State Behavior of Constant-Stepsize Stochastic Approximation: Gaussian Approximation and Tail Bounds
Yuyang Wang, Felix Wang, Zedong Wang, Ijay Narang, Yuzhou Wang, Siva Theja Maguluri
Constant-stepsize stochastic approximation is widely used in learning for computational efficiency. For a fixed stepsize, the iterates typically admit a stationary distribution that is rarely tractable. Prior work shows that as the stepsize $\alpha \downarrow 0$, the centered-and-scaled steady state converges weakly to a Gaussian limit. However, for fixed α, this weak convergence offers no usable error bound for approximating the steady-state by its Gaussian limit. This paper provides explicit, non-asymptotic error bounds for fixed α. We study (i) stochastic gradient descent on smooth strongly convex objectives, (ii) linear SA, and (iii) contractive nonlinear SA, and we treat both i.i.d. and Markovian noise models to ensure broad applicability. Our main results first give dimension- and stepsize-dependent, explicit bounds in Wasserstein distance between the centered-scaled steady state and its Gaussian limit, with errors that vanish as $\alpha \downarrow 0$. We further derive sharp tail control, comparing the steady-state tail probability to Gaussian tails with an explicit error term that decays in both the deviation level and α.
Our analysis combines steady-state Stein's method with moment bounds on the SA iterations, and uses Poisson equation techniques to manage temporal dependence in the Markovian noise setting. We adapt the same toolkit to SGD with general convex objectives and suggest non-Gaussian limiting behavior, which is validated in numerical experiments.
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Distributional Active Inference
Abdullah Akgül, Gulcin Baykal, Manuel Haussmann, Mustafa Mert Çelikok, Melih Kandemir
Optimal control of complex environments with robotic systems faces two complementary and intertwined challenges: efficient organization of sensory state information and far-sighted action planning. Because the reinforcement learning framework addresses only the latter, it tends to deliver sample-inefficient solutions. Active inference is the state-of-the-art process theory that explains how biological brains handle this dual problem. However, its applications to artificial intelligence have thus far been limited to extensions of existing model-based appr

oaches. We present a formal abstraction of reinforcement learning algorithms tha t spans model-based, distributional, and model-free approaches. This abstraction seamlessly integrates active inference into the distributional reinforcement le arning framework, making its performance advantages accessible without transitio n dynamics modeling.
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PowerFlow: Unlocking the Dual Nature of LLMs via Principled Distribution Matchin g
Ruishuo Chen, Yu Chen, Zhuoran Li, Longbo Huang
Unsupervised Reinforcement Learning from Internal Feedback (RLIF) has emerged as a promising paradigm for eliciting the latent capabilities of Large Language Mo dels (LLMs) without external supervision. However, current methods rely on heuri stic intrinsic rewards, which often lack a well-defined theoretical optimization target and are prone to degenerative biases. In this work, we introduce PowerFl ow, a principled framework that reformulates unsupervised fine-tuning as a distr ibution matching problem. By casting GFlowNet as an amortized variational sample r for unnormalized densities, we propose a length-aware Trajectory-Balance objec tive that explicitly neutralizes the structural length biases inherent in autore gressive generation. By targeting α-power distributions, PowerFlow enable s the directional elicitation of the dual nature of LLMs: sharpening the distrib ution ($\alpha > 1$) to intensify logical reasoning, or flattening it ($\alpha < 1$) to unlock expressive creativity. Extensive experiments demonstrate that Pow erFlow consistently outperforms existing RLIF methods, matching or even exceedin g supervised GRPO. Furthermore, by mitigating over-sharpening in aligned models, our approach achieves simultaneous gains in diversity and quality, shifting the Pareto frontier in creative tasks.
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Geometry-Aware Image Flow Matching
Junho Lee, Kwanseok Kim, Joonseok Lee
Recent advances in generative models highlight the power of geometry-aware model ing in manifold-constrained settings. Yet, for natural images, the field remains confined to Euclidean assumptions, failing to exploit the potential of intrins ic geometric structures within the data. In this work, we investigate the geomet ry of natural images and observe that semantic information is predominantly enco ded in directional components, while norm components can be approximated by the global average. This property holds across both RGB and latent spaces, suggestin g that natural images can be effectively modeled on a hypersphere. Building on t his finding, we introduce Spherical Optimal Transport Flow Matching (SOT-CFM), w hich utilizes angular distance, and Spherical Flow Matching (SFM), which constra ins dynamics directly on the manifold. Our experiments demonstrate that these ge ometry-aware methods achieve superior performance against Euclidean baselines. U ltimately, this work provides a novel perspective that bridges the gap between R iemannian manifold-based modeling and natural image generation.
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NorMuon: Making Muon more efficient and scalable
Zichong Li, Liming Liu, Chen Liang, Weizhu Chen, Tuo Zhao
The choice of optimizer significantly impacts the training efficiency and comput ational costs of large language models (LLMs). Recently, the Muon optimizer has demonstrated promising results by orthogonalizing parameter updates, improving o ptimization geometry through better conditioning.
Despite Muon’s emergence as a candidate successor to Adam, the potential for joi ntly leveraging their strengths—has not been systematically explored.
In this work, we bridge this gap by proposing NorMuon (Neuron-wise Normalized Mu on), an optimizer that synergistically combines orthogonalization with neuron-le vel adaptive learning rates. Our analysis reveals that while Muon effectively re duces condition numbers, the resulting updates exhibit highly non-uniform neuron norms, causing certain neurons to dominate the optimization process. NorMuon ad dresses this imbalance by maintaining second-moment statistics for each neuron a nd applying row-wise normalization after orthogonalization, ensuring balanced pa rameter utilization while preserving Muon's conditioning benefits. To enable pra

ctical deployment at scale, we develop an efficient distributed implementation u
nder the FSDP2 framework that distributes orthogonalization computations across
devices.
Experiments across multiple model scales demonstrate that NorMuon consistently o
utperforms both AdamW and Muon, achieving a 21.74\% reduction in training steps
relative to AdamW and an 11.31 percentage-point larger efficiency gain than Muon
 on 1.1B pretraining. Results suggest that orthogonalization and adaptive learni
ng rates are complementary rather than competing, opening new avenues for optimi
zer design in large-scale deep learning.
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Bridging On-Device and Cloud LLMs for Collaborative Reasoning: A Unified Methodo
logy for Local Routing and Post-Training
Wenzhi Fang, Dong-Jun Han, Liangqi Yuan, Evan Chen, Christopher Brinton
Device-cloud collaboration holds promise for deploying large language models (LL
Ms), leveraging lightweight on-device models for efficiency while relying on pow
erful cloud models for superior reasoning. A central challenge in this setting i
s determining, for each incoming query, whether it should be processed locally o
r offloaded to the cloud. Existing approaches typically rely on external routers
, which often struggle to determine difficulty from the prompt itself, especiall
y for tasks involving complex reasoning. Motivated by this limitation, we propos
e enabling on-device LLMs to decide internally whether to invoke cloud assistanc
e at inference time, with this capability instilled through reinforcement learni
ng based post-training. Casting on-device LLM post-training as a reward maximiza
tion problem, we design hierarchical rewards to encourage local problem solving
and judicious cloud offloading. To solve the resulting problem, we develop an al
gorithm featuring a group-level policy gradient that stabilizes optimization, to
gether with adaptive prompt filtering that provides complementary learning signa
ls to mitigate policy collapse (i.e., exclusive local execution or exclusive clo
ud offloading). Extensive experiments on on-device-scale LLaMA and Qwen models a
cross multiple reasoning benchmarks show that our method consistently outperform
s baselines and significantly narrows the gap to full cloud LLMs.
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DeCoDe: Decoupling Binding Position and Molecular Conformation in 3D Ligand Diff
usion for Structure-Based Drug Design
Julong Yang, Wen Huang, Junhui Chen, Jian Peng
Recent advances in diffusion models show promise for Structure-Based Drug Design
 (SBDD), which aims to generate 3D ligand molecules that bind tightly to specifi
c protein targets.
This involves jointly optimizing the ligand's 3D conformation and its binding po
sition within the protein pocket.
However, existing diffusion-based SBDD methods diffuse conformation and binding
position synchronously within a high-dimensional joint space, leading to ineffic
ient exploration and suboptimal generation quality in both aspects.
To address this, we propose **DeCoDe**, a novel diffusion framework that **decou
ples** the diffusion processes of the binding position and molecular conformatio
n.
Our key insight is to prioritize the perturbation of the ligand's internal confo
rmation in the early stages of the forward (noising) process, while accelerating
 the perturbation of its global binding position later.
This design guides the reverse (denoising) process to *first coarsely position*
the ligand within the pocket before *refining its detailed structure*, mimicking
 a more efficient, step-wise generation strategy.
Extensive experiments on the CrossDocked2020 benchmark show that DeCoDe achieves
 significantly higher structural fidelity (with an average improvement of 18%),
while maintaining competitive binding affinity and overall molecular properties
compared to state-of-the-art baselines.
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Fair Classification with Efficient and Post-hoc Controllable Fairness-Accuracy T
rade-off
Maaya Sakata, Kazuto Fukuchi

Post-hoc controllability of fair machine learning models, the ability to control the trade-off between fairness and accuracy after training, is valuable for practical deployment. Existing post-processing methods provide such post-hoc controllability but often suffer from significant accuracy degradation, whereas in-processing methods achieve efficient trade-offs but require computationally expensive retraining for each change in trade-off ratio. To achieve both post-hoc controllability and efficient trade-offs, we propose a novel fair classification algorithm that learns effective feature representations to improve the trade-off efficiency of post-processing fair classifiers, by a gradient-based optimization approach. Experimental results on real-world datasets demonstrate that our method achieves trade-off efficiency comparable to, or even surpassing, in-processing methods, without requiring any retraining.

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LAMP: Data-Efficient Linear Affine Weight-Space Models for Parameter-Controlled 3D Shape Generation and Extrapolation

Ghadi Nehme, Yanxia Zhang, Dule Shu, Matthew Klenk, Faez Ahmed

Generating high-fidelity 3D geometries under explicit parameter constraints is central to engineering design, yet current methods often require large datasets and fail to provide reliable control beyond the training distribution. We introduce LAMP, a data-efficient framework for controllable and interpretable 3D generation that aligns signed distance function (SDF) decoders by overfitting each exemplar from a shared initialization, then generates new designs by solving a parameter-constrained affine mixing problem in the aligned weight space. To improve reliability, we propose a linearity-mismatch safety metric that detects when mixed decoders leave the valid local regime. We evaluate LAMP on DrivAerNet++, BlendedNet, and additional industry-level vehicle families, including sports cars, SUVs, and convertibles. LAMP enables controlled interpolation with as few as 50 samples, safe extrapolation up to 100\% beyond training ranges, and performance-guided optimization under fixed parameters, outperforming conditional autoencoder and Deep Network Interpolation (DNI) baselines in extrapolation, data efficiency, and parameter fidelity. Our results demonstrate that LAMP advances controllable, data-efficient, and safe 3D generation for design exploration, dataset generation, and performance-driven optimization.

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Categorical Reparameterization with Denoising Diffusion Models

Samson Gourevitch, Alain Oliviero Durmus, Eric Moulines, Jimmy Olsson, Yazid Janati

Learning models with categorical variables requires optimizing expectations over discrete distributions, a setting in which stochastic gradient-based optimization is challenging due to the non-differentiability of categorical sampling. A common workaround is to replace the discrete distribution with a continuous relaxation, yielding a smooth surrogate that admits reparameterized gradient estimates via the reparameterization trick. Building on this idea, we introduce ReDGE, a novel and efficient diffusion-based soft reparameterization method for categorical distributions. Our approach defines a flexible class of gradient estimators that includes the Straight-Through estimator as a special case. Experiments spanning latent variable models and inference-time reward guidance in discrete diffusion models demonstrate ReDGE consistently matches or outperforms existing gradient-based methods.

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Knowledge Diversion for Efficient Morphology Control and Policy Transfer

Fu Feng, Ruixiao Shi, Yucheng Xie, Jianlu Shen, Jing Wang, Xin Geng

Universal morphology control aims to learn a universal policy that generalizes across heterogeneous robot morphologies, with Transformer-based controllers emerging as a dominant choice. However, such architectures incur substantial computational costs, resulting in high deployment overhead, and existing methods exhibit limited cross-task generalization, necessitating training from scratch for each new task. To this end, we propose DivMorph, a modular training paradigm that leverages knowledge diversion to learn *decomposable controllers*. DivMorph factorizes randomly initialized Transformer weights into *basic knowledge units* via S

VD and employs dynamic soft gating, conditioned on task and morphology embeddings, to adaptively modulate these units into universal *learngenes* and morphology- and task-specific *tailors* during training, thereby achieving knowledge disentanglement. By selectively activating relevant components, DivMorph adaptively recomposes the controller, enabling efficient policy deployment and effective policy transfer to novel tasks. Extensive experiments demonstrate that DivMorph achieves state-of-the-art performance, improving sample efficiency for cross-task transfer by 3.3$\times$ and reducing model size for single-agent deployment by 16.7$\times$.
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LaST$_{0}$: Latent Spatio-Temporal Chain-of-Thought for Robotic Vision-Language-Action Model
Zhuoyang Liu, Jiaming Liu, Hao Chen, Jiale Yu, Ziyu Guo, Chengkai Hou, Chenyang Gu, Xiangju Mi, Renrui Zhang, Kun Wu, Zhengping Che, Jian Tang, Pheng-Ann Heng, Shanghang Zhang
Vision-Language-Action (VLA) models have recently shown strong generalization, with some approaches seeking to explicitly generate linguistic reasoning traces or predict future observations prior to execution. However, explicit reasoning typically incurs non-negligible inference latency, which constrains the temporal resolution required for robotic manipulation. Moreover, such reasoning is confined to the linguistic space, imposing a representational bottleneck that struggles to faithfully capture ineffable physical attributes. To mitigate these limitations, we propose LaST$_0$, a framework that enables efficient reasoning before acting through a Latent Spatio-Temporal Chain-of-Thought (CoT), capturing fine-grained physical and robotic dynamics that are often difficult to verbalize. Specifically, we introduce a token-efficient latent CoT space that models future visual dynamics, 3D structural information, and robot proprioceptive states, and further extends these representations across time to enable temporally consistent implicit reasoning trajectories. Furthermore, LaST$_0$ adopts a dual-system architecture implemented via a Mixture-of-Transformers design, where a reasoning expert conducts low-frequency latent inference and an acting expert generates high-frequency actions conditioned on robotics-oriented latent representations. To facilitate coordination, LaST$_0$ is trained with heterogeneous operation frequencies, enabling adaptive switching during deployment. Across 10 real-world tasks spanning tabletop, mobile, and dexterous hand manipulation, LaST$_0$ improves mean success rates by 13%, 14% and 14% over prior SOTA VLA methods, respectively.
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TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse
Perry Dong, Kuo-Han Hung, Alexander Swerdlow, Dorsa Sadigh, Chelsea Finn
Despite scale driving substantial recent advancements in machine learning, reinforcement learning (RL) methods still primarily use small value functions. Naively scaling value functions -- including with a transformer architecture, which is known to be highly scalable -- often results in learning instability and worse performance. In this work, we ask what prevents transformers from scaling effectively for value functions? Through empirical analysis, we identify the critical failure mode in this scaling: attention scores collapse as capacity increases. Our key insight is that we can effectively prevent this collapse and stabilize training by controlling the entropy of the attention scores, thereby enabling the use of larger models. To this end, we propose Transformer Q-Learning (TQL), a method that unlocks the scaling potential of transformers in learning value functions in RL. Our approach yields up to a 43\% improvement in performance when scaling from the smallest to the largest network sizes, while prior methods suffer from performance degradation.
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Partial Fusion of Neural Networks: Efficient Tradeoffs Between Ensembles and Weight Aggregation
Fabian Morelli, Stephan Eckstein
Ensembles of neural networks typically outperform individual networks but incur large computational costs, whereas weight aggregation produces less costly, yet also less accurate, aggregate models.

We introduce partial fusion of networks, which interpolates between ensembles an
d weight aggregation and thus allows for a flexible tradeoff between computation
al cost and performance.
A direct way to achieve this is to extend existing weight aggregation methods ba
sed on neuron-level similarity between different networks, where partial fusion
then only aggregates weights of neurons which are most similar.
We showcase one particular method to jointly identify which neurons are most sim
ilar and match them via partial optimal transport.
Further, we consider the more general perspective of weight aggregation and part
ial fusion as generalized pruning of ensemble models, where neurons cannot just
be deleted, but also linearly combined.
Finally, we show that generalized pruning applied to a single network yields sim
ilar benefits as partial fusion by allowing for a tradeoff between isolating, de
leting, and linearly combining neurons based on similarity.
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PerceptionRubrics: Calibrating Multimodal Evaluation to Human Perception
Yana Wei, Hongbo Peng, Yanlin Lai, Liang Zhao, Kangheng Lin, En Yu, Keyu Lv, Han
Zhou, Yin Tang, Haodong Li, Mitt Huang, Hangyu Guo, Jianjian Sun, Zheng Ge, Xia
ngyu Zhang, Daxin Jiang, Vishal M. Patel
We introduce PerceptionRubrics, a rubric-based evaluation framework that address
es the dissonance between benchmark saturation and real-world brittleness. Shift
ing evaluation from holistic semantic matching to rigorous atomic auditing, Perc
eptionRubrics pairs 1,038 information-dense images with over 12,000 instance-spe
cific rubrics. These criteria are derived from golden captions that constructed
via a novel Circular Peer-Review consensus pipeline and then distilled into a du
al-stream system of Must-Right (essential facts) and Easy-Wrong (fine-grained de
tails) rubrics. Crucially, PerceptionRubrics implements a Gated Scoring mechanis
m: unlike linear averages, failure on mandatory visual facts triggers sharp bina
ry penalties. Extensive evaluation yields critical insights: (1) The Reliability
Gap: models often verify fragmented elements correctly yet fail strict conjunct
ive constraints, exposing brittleness in dense domains; (2) Open-Closed Stratifi
cation: contrary to reasoning trends, we reveal a persistent 5% perception defic
it between open-source and proprietary frontiers; and (3) Human-Aligned Rigor: o
ur gated metrics substantially out-align conventional benchmarks, validating tha
t strict perceptual fidelity is the prerequisite for reliable generation.
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Kronecker Generative Networks: A General Neural Architecture for Parameter-Effic
ient Learning Across Classification Tasks
Yang Yang, Zhengmin Kong, Yuan Liu, Tao Huang, Wei Xiang
Modern neural networks derive much of their effectiveness from rich connectivity
patterns.
Yet, existing architectures often fix the topology at either the sparse or dense
extremes, thereby limiting structural flexibility and analysis.
We propose Kronecker Generative Networks (KGNs), an algebraic framework that con
structs neural network topologies via recursive generation rules, treating topol
ogy as a first-class design object.
KGNs generate families of directed acyclic graphs with controllable connectivity
complexity, enabling systematic interpolation between sparse and dense aggregat
ion regimes.
Under this formulation, architectures such as FractalNet and DenseNet arise as s
pecific instantiations corresponding to different generation rules.
We provide theoretical analysis of acyclicity, connectivity scaling, and express
iveness, and demonstrate experimentally that KGN instantiations achieve favorabl
e accuracy-efficiency trade-offs across multiple domains.
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Context Forcing: Consistent Autoregressive Video Generation with Long Context
Shuo Chen, Cong Wei, Sun Sun, Tiancheng SHEN, Ping Nie, Kai Zou, Ge Zhang, Ming-
Hsuan Yang, Wenhu Chen
Recent approaches to real-time long video generation typically employ streaming
tuning strategies, attempting to train a long-context student using a short-cont

ext (memoryless) teacher. In these frameworks, the student performs long rollouts but receives supervision from a teacher limited to short 5-second windows. This structural discrepancy creates a critical **student-teacher mismatch**: the teacher's inability to access long-term history prevents it from guiding the student on global temporal dependencies, effectively capping the student's context length.
To resolve this, we propose **Context Forcing**, a novel framework that trains a long-context student via a long-context teacher. By ensuring the teacher is aware of the full generation history, we eliminate the supervision mismatch, enabling the robust training of models capable of long-term consistency. To make this computationally feasible for extreme durations (e.g., 2 minute), we introduce a context management system that transforms the linearly growing context into a **Slow-Fast Memory** architecture, significantly reducing visual redundancy.
Extensive results demonstrate that our method enables effective context lengths exceeding 20 seconds—$6\text{--}10\times$ longer than state-of-the-art methods like LongLive and Infinite-RoPE. By leveraging this extended context, Context Forcing preserves superior consistency across long durations, surpassing state-of-the-art baselines on various long video evaluation metrics.
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EAPO: Enhancing Policy Optimization with On-Demand Expert Assistance
Siyao Song, Cong Ma, Zhihao Cheng, Shiye Lei, Minghao Li, Ying Zeng, Huaixiao Tou, Kai Jia
Large language models (LLMs) have recently advanced in reasoning when optimized with reinforcement learning (RL) under verifiable rewards. Existing methods primarily rely on outcome-based supervision to strengthen internal LLM reasoning, often leading to inefficient exploration and sparse rewards. To mitigate this issue, we propose Expert-Assisted Policy Optimization (EAPO), a novel RL framework that enhances exploration by incorporating multi-turn interactions with external experts during training. Unlike prior methods, where policies reason in isolation, EAPO incentivizes the policy to adaptively determine when and how to consult experts, yielding richer reward signals and more reliable reasoning trajectories. External assistance ultimately internalizes expert knowledge into the policy model, amplifying the model’s inherent reasoning capabilities. During evaluation, the policy model has been well-optimized to solve questions independently, producing improved reasoning paths and more accurate solutions. On AIME 2024/2025 and AIMO 2025, EAPO consistently outperforms expert-assisted, expert-distilled, and RL baselines, averaging a 5-point gain over self-exploration RL, and also generalizes to non-math benchmarks, including HumanEval, HLE, GPQA, MMLU, EvalPlus, HotpotQA, and SimpleQA.
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Distributionally Robust Causal Abstractions
Yorgos Felekis, Theodoros Damoulas, Paris Giampouras
Causal Abstraction (CA) theory provides a principled framework for relating causal models that describe the same system at different levels of granularity while ensuring interventional consistency between them. Recent methods for learning CAs, however, assume fixed and well-specified exogenous distributions, leaving them vulnerable to environmental shifts and model misspecification. In this work, we address these limitations by introducing the first class of distributionally robust CAs and their associated learning algorithms. The latter cast robust causal abstraction learning as a constrained min-max optimization problem with Wasserstein ambiguity sets. We provide theoretical guarantees for both empirical and Gaussian environments, enabling principled selection of ambiguity-set radii and establish quantitative guarantees on worst-case abstraction error. Furthermore, we present empirical evidence across different problems and CA learning methods, demonstrating our framework’s robustness not only to environmental shifts but also to structural and intervention mapping misspecification.
****

RePro: Training Language Models to Faithfully Recycle the Web for Pretraining
Zichun Yu, Chenyan Xiong
High-quality data is a cornerstone of large language model (LLM) pretraining, ye

t its growth has not kept pace with the needs of frontier models. In this paper, we introduce RePro, a novel web recycling method that trains a relatively small LM with reinforcement learning to generate effective and faithful rephrasings of pretraining data. Specifically, we design one *quality* reward and three *faithfulness* rewards, optimizing the LM rephraser to convert organic data into high-quality rephrasings while maintaining its core semantics and structure. In our experiment, we train a rephraser as small as 1B parameters to recycle 72B tokens sampled from DCLM-RefinedWeb. Pretraining results on 400M, 1.4B, and 2.8B models demonstrate that RePro delivers 3.7\%-14.5\% relative accuracy gains over organic-only baseline on 22 downstream tasks, doubling the performance gains achieved by the state-of-the-art web recycling method that prompts a 70B rephraser. Experiments with different amounts of recycled data highlight that RePro improves organic data efficiency by 2-3$\times$. Individual and distributional analyses validate that RePro preserves more critical information and faithfully reflects the characteristics of organic data compared to prompting-based methods. Together, these results show that RePro provides an efficient and controllable path to effectively recycle organic data for pretraining. Our code is available at https://github.com/cxcscmu/RePro.

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Entropy-informed Decoding: Adaptive Information-Driven Branching

Benjamin Patrick Evans, Sumitra Ganesh, Leo Ardon

Large language models (LLMs) achieve remarkable generative performance, yet their output quality is dependent on the decoding strategy. While sampling-based methods (e.g., top-k, nucleus) and search-and-select based methods (e.g., beam search, best-of-n, majority voting) can improve upon greedy decoding, both approaches suffer from limitations: sampling generally commits to a single path, while search often expends excessive computation regardless of task complexity. To address these, we introduce Entropy-informed DEcodiNg (EDEN), a plug-and-play, model-agnostic decoding framework that adaptively allocates computation based on the model’s own uncertainty, approximating higher-width beam search with *fewer expansions*. At each generation step, EDEN estimates the entropy of the output token distribution and adjusts the branching factor monotonically with the entropy, expanding more candidates in high-entropy regions and following a greedier path in low-entropy regions, improving token efficiency. Experiments across complex tasks, including mathematical reasoning, code generation, and scientific questions, demonstrate that EDEN consistently improves output quality over existing decoding strategies, achieving better accuracy-expansion trade-offs than fixed-width beam search. By treating next-token selection as a noisy maximisation problem, we prove that branching factors monotone in entropy are guaranteed to find better (i.e. more probable) continuations than any fixed branching factor within the same total expansion budget, and derive explicit regret rates characterising the benefit of the adaptive allocation.

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Column Thresholding for Sparse Spiked Wigner Models: Improved Signal Strength Requirements

Jian-Feng Cai, Zhuozhi Xian, Jiaxi Ying

We study the sparse spiked Wigner model, where the goal is to recover an s-sparse unit vector $\boldsymbol{u} \in \mathbb{R}^d$ from a noisy observation $\boldsymbol{Y} = \beta \boldsymbol{u} \boldsymbol{u}^\top + \boldsymbol{W}$. While the information-theoretic threshold is $\beta = \widetilde{\Omega}(\sqrt{s})$, existing polynomial-time algorithms require $\beta = \widetilde{\Omega}(s)$, yielding a substantial computational-statistical gap. We propose a column thresholding method that attains the $\widetilde{\Omega}(\sqrt{s})$ scaling for both estimation and support recovery under the non-uniformity condition $|| \boldsymbol{u} ||_\infty = \Omega(1)$. This condition is not merely technical: it explicitly rules out uniform spikes, for which planted-clique-based hardness results apply, and identifies a concrete class of non-uniform spikes where the required signal strength can be reduced. Building on this initializer, we further develop a truncated power method that iteratively refines the estimate with provable linear convergence.

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Learning Fingerprints for Medical Time Series with Redundancy-Constrained Information Maximization
Huayu Li, ZhengXiao He, Xiwen Chen, Jingjing Wang, siyuan tian, Jinghao Wen, Ao Li
Learning meaningful representations from medical time series (MedTS), such as ECG or EEG signals, is a critical challenge. These signals are often high-dimensional, variable-length, and rife with noise. Existing self-supervised approaches, such as Masked Autoencoders (MAEs), are highly effective for pre-training general-purpose encoders. However, they do not explicitly learn compact, fixed-size, or semantically interpretable latent representations, typically relying on heuristic aggregation strategies such as global average pooling or a designated [CLS] token. We propose a novel framework that compresses a variable-length MedTS into a fixed-size set of k latent Fingerprint Tokens. Our architecture employs a cross-attention bottleneck to generate these tokens and is trained with a dual-objective function. The first objective is a reconstruction loss, which ensures the tokens are \textit{sufficient statistics} for the original data. The second, a diversity penalty based on the Total Coding Rate (TCR), explicitly minimizes the redundancy between tokens, encouraging them to become statistically \textit{disentangled} representations. We present the theoretical justification for our method, framing it as a novel \textbf{Disentangled Rate-Distortion} problem. This approach produces a low-dimensional, interpretable, and sample-efficient representation, where each token is encouraged to capture an independent factor of variation, paving the way for more robust digital biomarkers.
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Focus, Align, and Sustain: Counteracting Gradient Dilution in Incremental Object Detection
Aoting Zhang, Dongbao Yang, Chang Liu, Xiaopeng Hong, Yu ZHOU
Adapting Detection Transformers to Incremental Object Detection (IOD) poses a systemic challenge, as set-based optimization is inherently destabilized by sequential learning. In this work, we identify Gradient Dilution as the root cause of performance degradation, wherein optimization signals required to preserve old knowledge are progressively weakened. This phenomenon manifests as a cascading erosion of preservation gradients in magnitude, direction, and support coverage, driven by three tightly coupled factors: \textit{Signal Dispersion}, where foreground gradients are overwhelmed by background noise; \textit{Assignment Drift}, where stochastic query–target matching induces inconsistent gradient trajectories; and \textit{Support Attrition}, where gradients from retained samples insufficiently cover the old-class feature space, weakening decision boundaries under interference from new classes. To counteract this, we propose FAS, a unified framework that \underline{F}ocuses, \underline{A}ligns, and \underline{S}ustains gradient flow throughout incremental learning. Specifically, we introduce prior-injected queries to focus discriminative signals by filtering background interference at the source. We further propose deterministic anchor distillation to align query–target assignments and enforce semantic consistency across stages under unstable matching. Finally, we devise manifold-support replay to sustain distributional support of old classes, counteracting representational erosion induced by continual updates. Extensive experiments show that FAS restores robust optimization dynamics and outperforms state-of-the-art methods, achieving over 5.0 AP improvement in the challenging 40+10×4 incremental setting.
****
Interpretable Functional Koopman Learning with Non-Markovian Closure for Spatiotemporal Systems
Wanfeng Lu, He Ma, Wei Lin, Qunxi Zhu
Precise prediction of spatiotemporal dynamics over predictive horizons is constrained by the computational cost of high-fidelity solvers and the sparsity, noise, and irregularity of data. We introduce MERLIN, a Koopman-based framework that lifts dynamics to the evolution of learned *observation functionals* with near-linear progression, enabling full-field reconstruction at arbitrary resolutions. Theoretically, we develop a functional Koopman theory for PDEs and compensate fo

r the loss of finite-dimensional linear invariance via the Mori–Zwanzig formalis
m, which augments the linear backbone with non-Markovian memory terms to improve
 predictive accuracy. Practically, MERLIN employs discretization-invariant *func
tion encoders* that map partial, irregular observations to observables, and reso
lution-free *function decoders* that reconstruct states at arbitrary query point
s. Training under linear constraints yields an interpretable, low-dimensional mo
del
that captures principal modes and supports reduced-order modeling, while memory
correction further enables stable long-horizon rollouts even in ultra-low-dimens
ional
latent spaces. Our code is available at: https://github.com/RobinLufdu/MERLIN.
****

Minimum Distance Summaries for Robust Neural Posterior Estimation
Sherman Khoo, Dennis Prangle, Song Liu, Mark Beaumont
Simulation-based inference (SBI) enables amortized Bayesian inference by first t
raining a neural posterior estimator (NPE) on prior-simulator pairs, typically t
hrough low-dimensional summary statistics, which can then be cheaply reused for
fast inference by querying it on new test observations. Because NPE is estimated
 under the training data distribution, it is susceptible to misspecification whe
n observations deviate from the training distribution. Many robust SBI approache
s address this by modifying NPE training or introducing error models, coupling r
obustness to the inference network and compromising amortization and modularity.
 We introduce minimum-distance summaries, a post-hoc robust NPE method that adap
ts queried test-time summaries independently of the pretrained NPE. Leveraging t
he maximum mean discrepancy (MMD) as a distance between observed data and a summ
ary-conditional predictive distribution, the adapted summary displays strong rob
ustness properties due to the robustness of the MMD. We demonstrate that the alg
orithm can be implemented efficiently with random Fourier feature approximations
, yielding a lightweight, model-free test-time adaptation procedure. We provide
theoretical guarantees for the robustness of our algorithm and empirically evalu
ate it on a range of synthetic and real-world tasks, demonstrating substantial r
obustness gains compared with existing robust SBI methods with minimal additiona
l overhead.
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Structured Multi-step Jailbreaking under a Hamiltonian Generative Formulation
Zihan Zhou, Yang Zhou, Jianghai Yu, Lingjuan Lyu, Longwei Wang, KC Santosh, Ruom
ing Jin, Dejing Dou
Recent work shows that even safety aligned large language models (LLM) can be pu
shed into unsafe behavior by carefully crafted jailbreak prompts. Existing jailb
reaking attack methods often rely on disfluent or incoherent prompts, which limi
t their success and make them easy to detect. We introduce SJA, a structured jai
lbreak attack built around two ideas. First, inspired by the logic of Spilsbury
puzzle, SJA decomposes a harmful query into a sequence of harmless sub-questions
 and reconstructs the original answer by combining the sub-question responses. S
econd, by leveraging the theory of Hamiltonian dynamics on hyperbolic space, we
propose a hyperbolic Hamiltonian dynamics-based sub-question generation framewor
k that effectively captures the structural and temporal dependencies. We provide
 a theoretical analysis of how each sub-question evolves along the trajectory an
d show that the hyperbolic Hamiltonian system effectively captures the underlyin
g semantic structure. Finally, we propose a hyperbolic narrative fusion mechanis
m built on fractional embedding and Möbius fusion. This mechanism integrates coh
erent narratives into sub-questions while preserving geometric consistency and i
mproving stealth performance. We theoretically validate that the combination of
the generated harmless sub-questions, guided by the stealthy narrative, can effe
ctively preserve the contextual semantics of the original harmful question.
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ProRL: Effective Reinforcement Learning for Proactive Recommendation via Rectifi
ed Policy Gradient Estimation
Hongru Hou, Tiehua Mei, Denghui Geng, Jinhui Huang, Ao Xu, Hengrui Chen, Jiaqing
 Liang, Deqing Yang

Proactive Recommender Systems (PRSs) aim to guide user preference shift toward t
arget items by generating paths of intermediate recommendations. Reinforcement l
earning (RL) provides a principled framework for optimizing such sequential deci
sion tasks, as path rewards can naturally capture both short-term acceptance and
 long-term guidance effectiveness. However, naively applying policy gradients to
 PRS results in deficient gradient estimation. We identify two deficiencies: (1)
 path-level rewards decompose into step-level rewards with positive mean, creati
ng a *length-dependent bias* that causes gradients to favor path extension over
meaningful exploration; (2) weighting each step by the entire path-level reward
ignores the decomposition structure, leading to high gradient variance. To recti
fy these two deficiencies, we propose an effective RL framework **ProRL** with t
wo novel mechanisms for proactive recommendation. First, Stepwise Reward Centeri
ng subtracts expected rewards to neutralize length-dependent bias, ensuring that
 path extension yields zero expected gradient signal. Second, Position-Specific
Advantage Estimation leverages the reward decomposition structure to compute ste
p-dependent baselines, reducing gradient variance. Together, these mechanisms yi
eld policy gradients that precisely target path quality. Our experiments on thre
e real-world datasets demonstrate that ProRL significantly outperforms state-of-
the-art PRSs. Our code is available at https://github.com/hongruhou89/ProRL.
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SEDRAS: Symbolically Evaluated Deep Research And Science
Fredrik Carlsson, Daniel Ward, Joseph Ortiz, Fangyu Liu, Joakim Nivre
As the reasoning capabilities of Large Language Models (LLMs) expand, evaluating
 true inductive generalization on entirely unseen data becomes increasingly chal
lenging. To this end, we introduce a modular in-context learning evaluation fram
ework, that is scalable and extendable across its separate modules. This is base
d upon the notion of synthetic scenarios with controllable complexity across thr
ee independent axes: 1) the logic of the underlying data distribution (UDD) 2) t
heir projection into diverse representations, and 3) the interaction dynamic det
ermining how the model accesses and explores the data. For these scenarios, the
model is tasked to perform in-context scientific discovery and produce an interp
retable theory in natural language that explains the observations. In a separate
 conversation, the model is then tasked to convert this generated theory into ex
ecutable code, which can be programmatically compared against the underlying dat
a distribution.
Using this modular framework we produce an initial suite of 600 diverse scenario
s that we use to evaluate and analyze various state-of-the-art LLMs.
Although these experiments show that Gemini 3.0 Pro achieves the best overall sc
ore, each model performs the best at different tasks. For example:
GPT 5.2 is the clear winner on pure symbolic data, Claude Opus 4.5 is the best a
t working with files, Gemini is the strongest model for the non-dynamic scenario
s, and Grok 4.1 is the strongest model when UDD complexity scales. Furthermore,
all models struggle with active exploration and are seemingly incapable of ident
ifying informative data points, resulting in less efficient exploration than a r
andom baseline.
****

ReCoG: Relational and Compact Context Graph Learning for Few-shot Molecular Prop
erty Prediction
Zeyu Wang, Xin Zheng, Yao Lu, Shanqing Yu, Qi Xuan, Shirui Pan
Few-shot molecular property prediction (FSMPP) is essential in drug discovery an
d materials design, where high-quality labeled data are often scarce and expensi
ve to obtain. Despite the promising performance of existing methods, especially
context-aware methods, they still face two-fold severe challenges with ${\rm \te
xtit{insufficient structural context modeling}}$ \& ${\rm \textit{redundant auxi
liary context learning}}$, leading to inadequate context graph exploration and i
neffective information utilization for effective molecule representation learnin
g.
To address these, in this paper, we propose a novel framework by learning on ${\
rm \mathbf{\underline{Re}}}$ational
and ${\rm \mathbf{\underline{C}}}$ompact c${\rm \mathbf{\underline{o}}}$ntext ${

\rm \mathbf{\underline{G}}}$raph, named ReCoG, to comprehensively exploit the co
ntext graph for expressive molecular property prediction.
Specifically, the proposed ReCoG contains two core modules: a **(1) cross-proper
ty relational learning module** to better model the structural and relational co
ntext information, and a **(2) context graph information bottleneck module** to
adaptively suppress irrelevant auxiliary signals for compact context information
 utilization, followed by a detailed theoretical demonstration regarding the imp
ortance of joint relational and compact knowledge extraction in context graphs.
****

Safe In-Context Reinforcement Learning
Amir Moeini, Minjae Kwon, Alper Kamil Bozkurt, Yuichi Motai, Rohan Chandra, Lu F
eng, Shangtong Zhang
In-context reinforcement learning (ICRL) is an emerging RL paradigm where an age
nt, after pretraining, can adapt to out-of-distribution test tasks without any p
arameter updates, instead relying on an expanding context of interaction history
. While ICRL has shown impressive generalization, safety during this adaptation
process remains unexplored, limiting its applicability in real-world deployments
 where test-time behavior is expected to be safe. In this work, we propose SCARE
D: Safe Contextual Adaptive Reinforcement via Exact-penalty Dual, the first meth
od that promotes safe adaptation of ICRL under the constrained Markov decision p
rocess framework. During the parameter-update-free adaptation process, our agent
 not only maximizes the reward but also keeps the accumulated cost within a user
-specified safety budget. We also demonstrate that the agent actively reacts to
the safety budget; with a higher safety budget, the agent behaves more aggressiv
ely, and with a lower safety budget the agent behaves more conservatively. Acros
s challenging benchmarks, SCARED consistently enables safe and robust in-context
 adaptation, outperforming existing ICRL and safe meta-RL baselines.
****

CyberJurors: A Multi-Agent Simulation Task for E-Commerce Disputes Verdict
Yanhui Sun, Wu Liu, Haifeng Ming, Xinru Wang, Hantao Yao, Yongdong Zhang
E-commerce platforms have begun recruiting crowdsourced jurors to adjudicate mas
sive volumes of transaction disputes.
Unlike formal legal judgment, E-commerce dispute verdicts require grounding pivo
tal clues from redundant, multi-round, multimodal evidence and making decisions
under flexible platform-specific conventions.
These characteristics render existing methods insufficient for this scenario.
To bridge this gap, we introduce a pioneering task, *E-commerce Dispute Verdicts
* (EDV), and present *VerdictBench*, a multimodal benchmark comprising 6,000 rea
l-world cases designed to reflect crowdsourced jury decisions. Building upon thi
s, we propose *CyberJurors*, a multi-agent framework to clarify the dispute logi
c and regulate the verdict process.
At the individual level, *Individual Verdict Chain-of-Thought* decomposes the ED
V task into four structured reasoning stages, enabling fine-grained clue percep
tion and clarifying causal logic between pivotal clues and the dispute focus.
At the collective level, *Jury Consensus Verdict* simulates multi-round discussi
on and voting among jurors, while incorporating verdict precedents to mitigate c
ognitive biases toward either disputant.
Experiments on *VerdictBench* show that *CyberJurors* outperforms state-of-the-a
rt LLMs, MLLMs, and court simulators, while achieving stronger alignment with re
al-world jury voting patterns.
Code and dataset are available at https://github.com/YanhuiS/CyberJurors and htt
ps://huggingface.co/datasets/piggi/VerdictBench.
****

TPGDiff : Hierarchical Triple-Prior Guided Diffusion for Image Restoration
Yanjie Tu, Qingsen Yan, Axi Niu, Jiacong Tang
All-in-one image restoration aims to address diverse degradation types using a s
ingle unified model. Existing methods typically rely on degradation priors to gu
ide restoration, yet often struggle to reconstruct content in severely degraded
regions. Although recent works leverage semantic information to facilitate conte
nt generation, integrating it into the shallow layers of diffusion models often

disrupts spatial structures (e.g., blurring artifacts). To address this issue, we propose a Triple-Prior Guided Diffusion (TPGDiff) network for unified image restoration. TPGDiff incorporates degradation priors throughout the diffusion trajectory, while introducing structural priors into shallow layers and semantic priors into deep layers, enabling hierarchical and complementary prior guidance for image reconstruction. Specifically, we leverage multi-source structural cues as structural priors to capture fine-grained details and guide shallow layers representations. To complement this design, we further develop a distillation-driven semantic extractor that yields robust semantic priors, ensuring reliable high-level guidance at deep layers even under severe degradations. Furthermore, a degradation extractor is employed to learn degradation-aware priors, enabling stage-adaptive control of the diffusion process across all timesteps. Extensive experiments on both single- and multi-degradation benchmarks demonstrate that TPGDiff achieves superior performance and generalization across diverse restoration scenarios.

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Breaking Multi-Task Curse: Reward-Weighted Evolution for Black-Box Many-Task Optimization

Yanchi Li, Jiao Liu, Wenyin Gong, Qiong Gu, Yue Zhao, Yew-Soon Ong

Evolutionary multi-tasking accelerates black-box optimization via knowledge transfer but falters in scenarios involving many low-similarity tasks. We identify this scalability barrier as the *Multi-Task Curse*, driven by evaluation budget dispersion and negative transfer. To overcome this, we propose MES-RET (*M*any-task *E*volution *S*trategy with *R*eward-weighted *E*valuation and *T*ransfer), which combats budget dispersion via a reward-weighted evaluation scheme that guarantees superior expected improvement, while simultaneously mitigating negative transfer through a robust reward-weighted aggregation of mean and covariance statistics, ensuring a safe fallback to independent evolution. Furthermore, to handle neural dimensional mismatches in many-task policy search, we introduce a semantic parameter alignment strategy that bridges heterogeneous state-action spaces. Extensive experiments on synthetic benchmarks, real-world engineering problems, and reinforcement learning tasks demonstrate that MES-RET consistently outperforms state-of-the-art methods, notably enabling skill transfer across morphologically distinct policies.

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Sharp description of local minima in the loss landscape of high-dimensional two-layer ReLU neural networks

Jie Huang, Bruno Loureiro, Stefano Sarao Mannelli

We study the population loss landscape of two-layer ReLU networks of the form $\sum_{k=1}^K \mathrm{ReLU}(w_k^\top x)$ in a realisable teacher–student setting with Gaussian covariates. We show that local minima admit an exact low-dimensional representation in terms of summary statistics, yielding a sharp and interpretable characterisation of the landscape. We further establish a direct link with one-pass SGD: local minima correspond to attractive fixed points of the dynamics in summary statistics space. This perspective reveals a hierarchical organisation of minima into discrete families and shows how overparameterisation changes their stability and reachability under gradient-based dynamics. In this overparameterised regime, global minima become increasingly accessible, attracting the dynamics and reducing convergence to spurious solutions. Overall, our results reveal intrinsic limitations of common simplifying assumptions, which may miss essential features of the loss landscape even in minimal neural network models.

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Distributed Direct Preference Optimization

Zhanhong Jiang

Preference-based reinforcement learning (RL) is a key paradigm for aligning policies with human judgments, yet its theoretical behavior in distributed settings where preference data are fragmented across heterogeneous users remains poorly understood. Direct Preference Optimization (DPO) avoids explicit reward modeling but lacks convergence guarantees under federated and decentralized training, where communication constraints and non-IID preferences fundamentally alter optimiz

ation dynamics. We provide the first convergence and time-complexity analysis of DPO in distributed environments. Modeling personalized offline RL with user-specific preference distributions, we characterize the induced global optimization landscape. For federated DPO, we derive convergence rates that quantify the impact
of client drift, communication frequency, and preference heterogeneity; for decentralized DPO, we establish convergence over general communication graphs and show how spectral connectivity governs optimization speed and consensus. Empirically, we corroborate our theoretical insights on standard alignment benchmarks, demonstrating that our proposed methods not only enjoy strong theoretical guarantees but also deliver robust and scalable performance in practice. The code base is available here.
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The Extra Tokens Matter: Disentangled Representation Learning with Vision Transformers
Maofeng Tang, Hairong Qi
Vision Transformers increasingly incorporate extra tokens beyond patch tokens—from class tokens for aggregation to register tokens for artifact mitigation. While effective for their intended purposes, these tokens typically lack semantic structure. We ask a more ambitious question: Can we design regularization constraints that transform extra tokens into disentangled representations, enabling them to decompose images into semantic parts (e.g., heads, bodies, legs) without explicit supervision?
We propose XTRA, an intuitive yet powerful framework that augments Vision Transformers with dedicated ``factor tokens'' and enforces disentanglement via a novel Minimum Volume Constraint (MVC). A multi-stage aggregation process further enforces these factor tokens into semantically pure components, preventing token collapse that often occurs when training with MVC alone. On ImageNet-1K, XTRA achieves superior disentanglement (8.4× improvement in SEPIN@1 over DINOv2)
while simultaneously improving representation quality: KNN accuracy improves by 5.8\% and
linear-probe accuracy by 2.3\%.
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Navigating the Energy Landscape of Collaboration: Multi-Agent Communication Graph Generation via Score-Based Diffusion
GuanHao Zhao, Wenbo Lu, Cheng Cheng, Zhenya Huang, Wei Song, Zhiding Liu, Runze Wu, Enhong Chen
The collective intelligence of Large Language Model (LLM)-based Multi-Agent Systems (MAS) is fundamentally governed by the underlying communication graph. However, discovering task-adaptive structures within this combinatorial search space remains a significant challenge. Existing methods, ranging from heuristic pruning to autoregressive generation, often lack a unified theoretical framework to guide the self-organization of agents into efficient teams. In this paper, we bridge non-equilibrium thermodynamics and generative modeling to formalize multi-agent graph generation as an energy minimization process. Specifically, we frame the emergence of efficient collaboration as a thermodynamic "cooling" process, where initially stochastic interactions converge to a low-energy, structured equilibrium.
To implement this, We propose MAGE (Multi-Agent Communication Graph Generation), a score-based diffusion framework that constructs communication graphs by navigating the energy landscape via iterative denoising and first-order gradient guidance. Extensive experiments on representative benchmarks demonstrate that MAGE achieves state-of-the-art performance. Furthermore, qualitative analysis reveals that the generated graphs mirroring the functional specialization of human organizations, validating our thermodynamic hypothesis.
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VlogReward: Learning Multi-Dimensional Evaluation for Vlog Editing
Yexiang Liu, Wen Zhong, Sijie Zhu, Xin Gu, Fan Chen, Junxian Duan, Jie Cao, Longyin Wen, Zhenfang Chen
The rapid rise of vlogs as a personalized storytelling medium has created a dema

nd for automated systems to evaluate and refine vlog editing plans. However, vlo
g assessment is highly subjective and remains challenging due to a lack of stand
ardized criteria, dataset and benchmark, and effective reward models. To address
 these challenges, we define a comprehensive vlog evaluation framework guided by
 professional vlog creators and product managers, establishing a taxonomy of six
 key dimensions, *i.e.*, *Creativity*, *Consistency*, *Concept Design*, *Cinemat
ography*, *Narration*, and *Pacing*. Subsequently, we curate a large-scale datas
et of 100k vlog edits and a dedicated benchmark, **VRMBench**, to evaluate the v
log rewarding capabilities of Multimodal Large Language Models (MLLMs). Finally,
 we present **VlogReward**, a robust vlog reward model that can provide both fin
e-grained multi-dimensional scores and actionable feedback for iterative refinem
ent.

  Technically, we enhance the Group Relative Policy Optimization (GRPO) framewor
k by introducing an adjustable inter-group comparison reward, which mitigates th
e "direction blindness" issue of standard GRPO and enables the model to better d
istinguish varied-quality edits.

  VlogReward achieves state-of-the-art results that significantly outperform exi
sting MLLMs, including GPT-5 and Gemini-3-Pro.

  We hope that our study can help vlog creators and foster automated vlog evalua
tion and refinement systems.
****

Equilibrium Reasoners: Learning Attractors Enables Scalable Reasoning
Benhao Huang, Zhengyang Geng, J Zico Kolter
Scaling test-time compute by iteratively updating a latent state has emerged as
a powerful paradigm for reasoning. Yet, the internal mechanisms that enable thes
e iterative models to generalize beyond memorized patterns remain fundamentally
unclear. We hypothesize that such generalizable reasoning arises from learning t
ask-conditioned attractors: a latent dynamical system where stable fixed points
correspond to valid solutions. We formalize this process by introducing Equilibr
ium Reasoners (EqR). EqR enables test-time scaling without relying on external v
erifiers or task-specific priors. Instead, our models scale internal dynamics al
ong two axes: depth by running more iterations and breadth by aggregating stocha
stic trajectories from multiple initializations. Empirically, performance gains
from scaling test-time compute are tightly coupled with better convergence to at
tractors. This attractor perspective allows neural networks to adaptively alloca
te test-time compute based on task difficulty. While simple cases converge withi
n 1 to 5 iteration steps, the hardest cases benefit from massive test-time scali
ng. By unrolling up to an equivalent of 40,000 layers, this scalable latent reas
oning boosts accuracy from 2.6% for feedforward models to over 99% on Sudoku-Ext
reme. We hope our attractor perspective sheds light on scalable reasoning.
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World-R1: Reinforcing 3D Constraints for Text-to-Video Generation
Weijie Wang, Xiaoxuan He, Youping Gu, Yifan Yang, Zeyu Zhang, Yefei He, Yanbo Di
ng, Xirui Hu, Donny Y. Chen, Zhiyuan He, Yuqing Yang, Bohan Zhuang
Recent video foundation models demonstrate impressive visual synthesis but frequ
ently suffer from geometric inconsistencies. While existing methods attempt to i
nject 3D priors via architectural modifications, they often incur high computati
onal costs and limit scalability. We propose World-R1, a framework that aligns v
ideo generation with 3D constraints through reinforcement learning. To facilitat
e this alignment, we introduce a specialized pure text dataset tailored for worl
d simulation. Utilizing Flow-GRPO, we optimize the model using feedback from pre
-trained 3D foundation models and vision-language models to enforce structural c
oherence without altering the underlying architecture. We further employ a perio
dic decoupled training strategy to balance rigid geometric consistency with dyna
mic scene fluidity. Extensive evaluations reveal that our approach significantly
 enhances 3D consistency while preserving the original visual quality of the fou
ndation model, effectively bridging the gap between video generation and scalabl
e world simulation.
****

Open-Text Aerial Detection: A Unified Framework For Aerial Visual Grounding And

Detection
Guoting Wei, Xia Yuan, Yang Zhou, Haizhao Jing, Yu Liu, Xianbiao Qi, Chunxia Zhao, Haokui Zhang, Rong Xiao
Open-Vocabulary Aerial Detection (OVAD) and Remote Sensing Visual Grounding (RSVG) have emerged as two key paradigms for aerial scene understanding. However, each paradigm suffers from inherent limitations when operating in isolation: OVAD is restricted to coarse category-level semantics, while RSVG is structurally limited to single-target localization. These limitations prevent existing methods from simultaneously supporting rich semantic understanding and multi-target detection. To address this, we propose OTA-Det, the first unified framework that bridges both paradigms into a cohesive architecture. Specifically, we introduce a task reformulation strategy that unifies task objectives and supervision mechanisms, enabling joint training across datasets from both paradigms with dense supervision signals. Furthermore, we propose a dense semantic alignment strategy that establishes explicit correspondence at multiple granularities, from holistic expressions to individual attributes, enabling fine-grained semantic understanding. To ensure real-time efficiency, OTA-Det builds upon the RT-DETR architecture, extending it from closed-set detection to open-text detection by introducing several high efficient modules, achieving state-of-the-art performance on six benchmarks spanning both OVAD and RSVG tasks while maintaining real-time inference at 34 FPS.
****
Fast k-means Seeding Under The Manifold Hypothesis
Poojan Chetan Shah, Shashwat Agrawal, Ragesh Jaiswal
We study beyond worst case analysis for the k-means problem where the goal is to model typical instances of k-means arising in practice. Existing theoretical approaches provide guarantees under certain assumptions on the optimal solutions to k-means, making them difficult to validate in practice. We adopt the manifold hypothesis, where data obtained in ambient dimension D concentrates around a low dimensional manifold of intrinsic dimension d, as a reasonable assumption to model real world clustering instances. We identify key geometric properties of datasets which have theoretically predictable scaling laws depending on the quantization exponent $\varepsilon = 2/d$ using techniques from optimum quantization theory. We show how to exploit these regularities to design a fast seeding method called $\operatorname{Qkmeans}$ which provides $O(\rho^{-2} \log k)$ approximate solutions to the
k-means problem in time $O(nD \log k) + \widetilde{O}(\rho^{-1}k^{1+\gamma})$; where the exponent $\gamma = \varepsilon + \rho$ for an input parameter $\rho < 1$. This allows us to obtain new runtime - quality tradeoffs. We perform a large scale empirical study across various domains to validate our theoretical predictions and algorithm performance to bridge theory and practice for beyond worst case data clustering.
****
Convergence Rate of the Last Iterate of Stochastic Proximal Algorithms
Kevin Kurian Thomas Vaidyan, Michael Friedlander, Ahmet Alacaoglu
We analyze two classical algorithms for solving additively composite convex optimization problems where the objective is the sum of a smooth term and a nonsmooth regularizer. The first algorithm is the proximal stochastic gradient method for a single regularizer; the second is the randomized incremental proximal method, which uses the proximal operator of a randomly selected function when the regularizer is given as the sum of many nonsmooth functions. We focus on relaxing the bounded variance assumption that is common, yet stringent, for getting last iterate convergence rates. We prove the $\widetilde{O}(1/\sqrt{T})$ rate of convergence for the last iterate of both algorithms under componentwise convexity and smoothness, which is optimal up to log terms. Our results apply directly to graph-guided regularizers that arise in multi-task and federated learning, where the regularizer decomposes as a sum over edges of a collaboration graph.
****
WildCat: Near-Linear Attention in Theory and Practice
Tobias Schröder, Lester Mackey

We introduce WildCat, a high-accuracy, low-cost approach to compressing the atte
ntion mechanism in neural networks. While attention is a staple of modern networ
k architectures, it is also notoriously expensive to deploy due to resource requ
irements that scale quadratically with the input sequence length n. WildCat av
oids these quadratic costs by only attending over a small weighted coreset. Cruc
ially, we select the coreset using a fast but spectrally-accurate subsampling al
gorithm -- randomly pivoted Cholesky -- and weight the elements optimally to min
imise reconstruction error. Remarkably, given bounded inputs, WildCat approximat
es exact attention with super-polynomial $O(n^{-\sqrt{\log(\log(n))}})$ error de
cay while running in near-linear $O(n^{1+o(1)})$ time. In contrast, prior practi
cal approximations either lack error guarantees or require quadratic runtime to
guarantee such high fidelity. We couple this advance with a GPU-optimised PyTorc
h implementation and a suite of benchmark experiments demonstrating the benefits
of WildCat for image generation, image classification, and language model KV ca
che compression.
****

FlowCloud: Learning Continuous Spatiotemporal Dynamics from Unpaired Sparse Poin
t Cloud Snapshots

Yinbo Liu, Keyang Ye, Wenshan Sun, Handi Gao, Tian Tian

Reconstructing unified continuous dynamics from sparse, non-contiguous, and unpa
ired point cloud snapshots remains a fundamental challenge in spatiotemporal ana
lysis for computer vision and developmental biology. Existing methods, including
scene flow and Optimal Transport-based approaches, are limited either by explic
it reliance on point-wise correspondences or by cumulative errors from frame-to
-frame propagation and temporal discontinuity, and limited ability to model mult
i-attribute dynamics such as gene expression and population changes. We propose
FlowCloud, a variational Neural Ordinary Differential Equation (Neural ODE) gene
rative framework. FlowCloud aggregates information from all observed time points
into a joint latent representation that initializes a Neural ODE $z(t)$ enablin
g continuous spatiotemporal evolution modeling while mitigating propagation-indu
ced errors and preserving temporal consistency. Training is performed without pr
edefined correspondences using a multi-faceted objective with complementary role
s: Sinkhorn distance for global distribution alignment, Chamfer distance for loc
al geometric consistency, trajectory regularization to encourage smooth and phys
ically plausible dynamics, and supervised losses for multi-attribute prediction.
Experiments on human motion and developmental biology datasets demonstrate impr
oved interpolation accuracy and promising short-term extrapolation performance.
By unifying geometry, attributes, and population dynamics within a continuous la
tent framework, FlowCloud offers a novel and robust solution for continuous dyna
mic reconstruction from unstructured spatiotemporal observations.
****

HALO: A Unified Vision-Language-Action Model for Embodied Multimodal Chain-of-Th
ought Reasoning

Quanxin Shou, Fangqi Zhu, Shuang Chen, Puxin Yan, Zhengyang Yan, Yikun Miao, Xia
oyi Pang, Zicong Hong, Ruikai Shi, Hao HUANG, Jie ZHANG, Song Guo

Vision–Language–Action (VLA) models perform well in robotic manipulation, but of
ten struggle in long-horizon or out-of-distribution scenarios due to the lack of
explicit mechanisms for multimodal reasoning and anticipating how the world evo
lves under action.

Recent works introduce textual chain-of-thought or visual subgoal prediction wit
hin VLA models, but still fail to offer a unified human-like framework for joint
textual reasoning, visual foresight, and action prediction.

To this end, we propose HALO, a unified VLA model that enables embodied multimod
al chain-of-thought (EM-CoT) reasoning through textual reasoning, fine-grained v
isual subgoal prediction, and EM-CoT-augmented action prediction.

We instantiate HALO with a Mixture-of-Transformers (MoT) architecture that decou
ples semantic reasoning, visual foresight, and action prediction into specialize
d experts with seamless cross-expert collaboration.

To enable HALO learning at scale, we introduce an automated pipeline to synthesi
ze EM-CoT training data along with a carefully crafted training recipe.

Extensive experiments demonstrate that: (1) HALO achieves superior performance in both simulated and real world, surpassing baseline policy π_0 by 25.9\% on the RoboTwin benchmark; (2) all proposed components of the training recipe and EM-CoT design help improve task success rate; and (3) HALO exhibits strong generalization under aggressive unseen environment randomization with our proposed EM-CoT reasoning.
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Same Question, Different Lies: Cross-Context Consistency (C³) for Black-Box Sandbagging Detection
Lin Yulong, Pablo Bernabeu-Perez, Benjamin Arnav, Lennie Wells, Mary Phuong
As language models grow more capable, accurate capability evaluation becomes essential for safety decisions. If models can deliberately underperform on dangerous capability evaluations—a behavior known as **sandbagging**—they may evade safety measures designed for their true capability level. We introduce Cross-Context Consistency (C^3), a general framework for unsupervised black-box sandbagging detection that exploits a fundamental asymmetry: when a model truly lacks capability, its confusion manifests consistently across paraphrased questions, but when a capable model feigns incompetence, its strategic choices about *how* to appear weak create detectable inconsistencies. The framework is agnostic to the specific consistency metric and aggregation method; we present a simple instantiation using embedding cosine distance and mean aggregation that requires no training data or model-specific adaptation. We evaluate C^3 across prompted and fine-tuned sandbagging scenarios, across instructed and more naturalistic setups, maintaining a classification signal where other black-box methods fail. Our findings show the limitations of existing sandbagging detection methods, and reveal the efficacy of consistency-checking as a detection mechanism for sandbagging.
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d2: Improving Reasoning in Diffusion Language Models via Trajectory Likelihood Estimation
Guanghan Wang, Gilad Turok, Yair Schiff, Marianne Arriola, Volodymyr Kuleshov
While diffusion language models (DLMs) have achieved competitive performance in text generation, improving their reasoning ability with reinforcement learning remains an active research area. Here, we introduce d2, a reasoning framework tailored for masked DLMs. Central to our framework is a new policy gradient algorithm that relies on accurate estimates of the sampling trajectory likelihoods. Because computing these likelihoods naively is computationally expensive for masked DLMs, we develop a family of estimators tailored to distinct model classes. For DLMs that support a sampling algorithm called any-order decoding, we propose d2-AnyOrder, which achieves exact trajectory likelihood with a single model pass. Through an empirical study of widely used DLMs, we show that any-order decoding is not universally supported in practice. For standard masked diffusion models, we propose d2-StepMerge, which approximates the trajectory likelihood, trading off compute for approximation accuracy in an analytically tractable manner. Empirically, d2 significantly outperforms widely-used RL baselines when applied to popular DLMs, and sets a new state-of-the-art performance for DLMs on logical reasoning tasks (Countdown and Sudoku) and math reasoning benchmarks (GSM8K and MATH500). We provide the code along with a blog post on the project page: https://guanghanwang.com/d2
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Attention Illuminates LLM Reasoning: The Uncovered Preplan-and-Anchor Rhythm Enables Fine-Grained Policy Optimization
Yang Li, Zhichen Dong, Yuhan Sun, Weixun Wang, Shaopan Xiong, Yijia Luo, Jiashun Liu, Han Lu, Jiamang Wang, Wenbo Su, Bo Zheng, Junchi Yan
The reasoning patterns of large language models (LLMs) remain opaque, and Reinforcement learning (RL) typically assigns uniform credit across an entire generation, blurring the distinction between pivotal and routine steps. This work treats attention as a natural substrate for interpreting LLM reasoning and a window for aligning optimization with its internal dynamics. We first distinguish attention heads between locally and globally focused information processing and reveal that locally focused heads produce a sawtooth pattern near the diagonal indicati

ng phrasal chunks, while globally focused heads expose tokens that exert broad d
ownstream influence over future tokens. We quantify these with two metrics measu
ring the extent of backward attention within a clipped window and the average at
tention a token receives from subsequent tokens, respectively. Taken together, t
hese signals indicate a recurring preplan-and-anchor regularity, where the model
 first performs a long-range contextual reference to generate an introductory to
ken, which is immediately followed by or coincides with a semantic anchor token
that organizes subsequent reasoning. Leveraging these insights, we introduce thr
ee novel RL strategies that dynamically perform targeted credit assignment to cr
itical nodes (preplan tokens, anchor tokens, and their temporal coupling) and sh
ow consistent performance gains across various reasoning tasks.
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OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pr
e-training in Every Iteration

Shaobo Wang, Xuan Ouyang, Tianyi Xu, Yuzheng Hu, Jialin Liu, Guo Chen, Tianyu Zh
ang, Junhao Zheng, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang

As high-quality public text approaches exhaustion, a phenomenon known as the Dat
a Wall—LLM pre-training is shifting from more tokens to better tokens. However,
existing methods either rely on heuristic static filters that ignore training dy
namics, or use dynamic yet optimizer-agnostic criteria based on raw gradients. W
e propose OPUS (Optimizer-induced Projected Utility Selection), a dynamic framew
ork that defines utility in the optimizer-induced update space. OPUS scores cand
idates by projecting their effective updates, shaped by modern optimizers, onto
a target direction derived from a stable, in-distribution proxy. To ensure scala
bility, we employ Ghost technique with CountSketch for computational efficiency,
 and Boltzmann sampling for data diversity, incurring only 4.7% additional compu
te overhead. OPUS achieves remarkable results across diverse corpora, quality ti
ers, optimizers, and model scales. It also outperforms previous data selection m
ethods across different stages of training, including from-scratch pre-training
and also mid-training. Beyond online selection, the OPUS utility score also demo
nstrates potential as a static filter for flagging and removing toxic documents
from contaminated training corpora prior to training.
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A Deep Learning Model of Mental Rotation Informed by Interactive VR Experiments

Raymond Khazoum, Daniela Fernandes, Aleksandr Krylov, Qin Li, Stephane Deny

Mental rotation—the ability to compare objects seen from different viewpoints—is
 a fundamental example of mental simulation and spatial world modeling in humans
. Here we propose a mechanistic model of human mental rotation, leveraging rece
nt advances in deep, equivariant, and neuro-symbolic learning. Our model consist
s of three stacked components: (1) an equivariant neural encoder, producing 3D s
patial representations of objects from images, (2) a neuro-symbolic object encod
er, deriving symbolic objects descriptions from these spatial representations, a
nd (3) a neural decision agent, comparing these symbolic descriptions to prescri
be rotation simulations in 3D latent space via a recurrent pathway. Our model de
sign is guided by the existing experimental literature on mental rotation, which
 we complemented with experiments in VR where participants could at times manipu
late the objects to compare. Our model captures well the performance, response t
imes and behavior of participants in our and others' experiments, and through ab
lation studies we demonstrate the necessity of each component. Our work adds to
a recent collection of deep neural models of human spatial reasoning, further de
monstrating the potency of integrating deep, equivariant, and symbolic represent
ations to model the human mind.
****

Meerkat-VL: Implicit Risk Safety Alignment in Multimodal LLMs via Perceptual Rea
soning and Self-Verification

Peicheng Zhou, Chuanbin Liu, Shancheng Fang, Bowei Pu, Yiwei Sun, Zhangchi Hu, H
ongtao Xie

Multimodal LLMs (MLLMs) are increasingly deployed across diverse applications, b
ut they pose significant safety concerns due to cross-modal interactions. To imp
rove model safety awareness, existing methods rely on explicit-risk preference d

atasets and reinforcement learning guided by safety rewards. While effective in improving models' safety awareness, these methods still face data scarcity and reward hacking in implicit-risk scenarios, leading to insufficient risk perception and harmful responses. To address these challenges, we propose Meerkat-VL, a framework that enables models to perceive and verify implicit risks while generating safe responses. First, we introduce Meerkat-Safe, the first training dataset with detailed labels for implicit risks. Second, we develop Normative Perceptual Self-Verification, which enables models to verify both perceptual reasoning and responses. This process provides denser and more reliable rewards for perception accuracy and answer safety, thereby mitigating reward hacking. Finally, we propose Dual-Objective Perceptual Consistency Alignment, encouraging models to generate safe responses by penalizing answers that follow safe templates without accurate risk perception. Extensive experiments show that Meerkat-VL consistently outperforms baselines on multimodal safety benchmarks, improving safety and helpfulness by 16% and 13%, and achieving a 32% safety gain on implicit-risk tasks. Our codes are available at https://github.com/Tunanzzz/Meerkat-VL.

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VideoVeritas: AI-Generated Video Detection via Perception Pretext Reinforcement Learning

Hao Tan, jun lan, Senyuan Shi, Zichang Tan, Zijian Yu, Huijia Zhu, Weiqiang Wang, Jun Wan, Zhen Lei

The growing capability of video generation poses escalating security risks, making reliable detection increasingly essential. In this paper, we introduce **VideoVeritas**, a framework that integrates fine-grained perception and fact-based reasoning. We observe that while current multi-modal large language models (MLLMs) exhibit strong reasoning capacity, their granular perception ability remains limited. To mitigate this, we introduce *Joint Preference Alignment* and *Perception Pretext Reinforcement Learning (PPRL)*. Specifically, rather than directly optimizing for detection task, we adopt general spatiotemporal grounding and self-supervised object counting in the RL stage, enhancing detection performance with simple *perception pretext tasks*. To facilitate robust evaluation, we further introduce **MintVid**, a light yet high-quality dataset containing 3K videos from 9 state-of-the-art generators, along with a real-world collected subset that has factual errors in content. Experimental results demonstrate that existing methods tend to bias towards either *superficial* reasoning or *mechanical* analysis, while **VideoVeritas** achieves more balanced performance across diverse benchmarks.

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UniRRM: Unified Reasoning Reward Models Across Languages and Evaluation Paradigms

Peng Lai, Yichao Du, Junchao Wu, Weibo Gao, Linan Yue, Longyue Wang, Weihua Luo, Derek F. Wong, Guanhua Chen

Reinforcement learning (RL) excels on tasks with verifiable rewards, but in open-ended tasks, the reliability of reward models remains a key challenge. Existing solutions either depend on costly proprietary LLM-as-a-Judge systems or opaque scalar reward models that lack interpretability. Recent works on generative reward models offer a promising alternative, but they remain constrained by static evaluation criteria, fragmented evaluation paradigms, and limited multilingual support. To address these challenges, we introduce MixReward, a large-scale multilingual dataset spanning six domains and 103 languages, containing both pairwise and listwise data, and propose UniRRM, a unified reasoning reward model supporting multiple languages and evaluation paradigms.

UniRRM uses a staged reasoning chain to dynamically generate task-generic and instruction-specific criteria, enabling fine-grained, input-adaptive judgments while maintaining consistency across languages. Experiments demonstrate that UniRRM-8B and UniRRM-14B achieve performance close to the state-of-the-art for models of comparable size across multiple benchmarks, and are effective for unseen evaluation paradigms. In addition, ablation studies validate the reliability and effectiveness of UniRRM.

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Neural Honeytrace: Plug&Play Watermarking Framework against Model Extraction Attacks
Yixiao Xu, Binxing Fang, Rui Wang, Yinghai Zhou, Yuan Liu, Mohan Li, Zhihong Tian
Triggerable watermarking enables model owners to assert ownership against model extraction attacks. However, most existing approaches require additional training, which limits post-deployment flexibility, and the lack of clear theoretical foundations makes them vulnerable to adaptive attacks. In this paper, we propose Neural Honeytrace, a plug-and-play watermarking framework that operates without retraining. We redefine the watermark transmission mechanism from an information perspective, designing a training-free multi-step transmission strategy that leverages the long-tailed effect of backdoor learning to achieve efficient and robust watermark embedding. Extensive experiments demonstrate that Neural Honeytrace reduces the average number of queries required for a worst-case t-test-based ownership verification to as low as 2% of existing methods, while incurring zero training cost.
****
Conditional Diffusion Sampling
Francisco M Castro-Macías, Pablo Morales-Alvarez, Saifuddin Syed, Daniel Hernández-Lobato, Rafael Molina, José Miguel Hernández-Lobato
Sampling from unnormalized multimodal distributions with limited density evaluations remains a fundamental challenge in machine learning and natural sciences.
Successful approaches construct a bridge between a tractable reference and the target distribution.
Parallel Tempering (PT) serves as the gold standard, while recent diffusion-based approaches offer a continuous alternative at the cost of neural training.
In this work, we introduce Conditional Diffusion Sampling (CDS), a framework that combines these two paradigms.
To this end, we derive Conditional Interpolants, a class of stochastic processes whose transport dynamics are governed by an exact, closed-form stochastic differential equation (SDE), requiring no neural approximation.
Although these dynamics require sampling from a non-trivial initialization distribution, we show both theoretically and empirically that the cost of this initialization diminishes for sufficiently short diffusion times.
CDS leverages this by a two-stage procedure: (1) PT is used to efficiently sample the initial distribution, and then (2) samples are transported via the transport SDE.
This combination couples the robust global exploration of PT with efficient local transport.
Experiments suggest that CDS has the potential to achieve a superior trade-off between sample quality and density evaluation cost compared to state-of-the-art samplers.
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Distributionally Robust Set Representation Learning Under Inference-Time Element Corruption
Yankai Chen, Hanrong Zhang, Bowei He, Philip S. Yu, Xue Liu
Standard Set Representation Learning methods typically excel on curated data but often overlook the challenge of Inference-time Element Corruption. This refers to scenarios where deployed models encounter element-level degradations, such as outliers or missing components, that may distort set representation and degrade performance. We propose SW-DRSO, a distributionally robust optimization framework tailored for sets. Rather than minimizing loss solely on observed training data, SW-DRSO optimizes a tractable surrogate of the
worst-case expected loss over a family of plausible inference-time variations. We introduce a barycentric adversary that approximates the intractable search over corrupted sets by a differentiable training-time optimization over simplex weights. Extensive experiments across four tasks demonstrate that SW-DRSO effectively enhances robustness against corruption while maintaining high overall performance.
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Probability-Entropy Calibration: An Elastic Indicator for Adaptive Fine-tuning
Wenhao Yu, Shaohang Wei, Jiahong Liu, Yifan Li, Minda Hu, Aiwei Liu, Hao Zhang, Irwin King
Token-level reweighting is a simple yet effective mechanism for controlling supervised fine-tuning, but common indicators are largely one-dimensional: the ground-truth probability reflects downstream alignment, while token entropy reflects intrinsic uncertainty induced by the pre-training prior. Ignoring entropy can misidentify noisy or easily replaceable tokens as learning-critical, while ignoring probability fails to reflect target-specific alignment. RankTuner introduces a probability--entropy calibration signal, the **Relative Rank Indicator**, by comparing the rank of the ground-truth token with its expected rank under the predictiondiction distribution. The inverse indicator is used as a token-wise **Relative Scale** to reweight the fine-tuning objective, focusing updates on truly under-learned tokens without over-penalizing intrinsically uncertain positions. Experiments on multiple backbones show consistent improvements on mathematical reasoning benchmarks, transfer gains on out-of-distribution reasoning, and pre code generation performance over probability- or entropy-only reweighting baselines.
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TraCeS: Learning Per-Timestep Constraint-Violation Credit from Sparse Trajectory-Level Labels
Siow Meng Low, Ze Gong, Akshat Kumar
Ensuring safe behavior in reinforcement learning (RL) is challenging when safety constraints are implicit and cannot be densely measured. In many settings, supervision is limited to coarse approvals or rejections of whole trajectories (e.g., whether a rollout remained within an unknown safety threshold). We propose **TraCeS** (Trajectory-based Constraint Estimation for Safety), a method for learning **per-timestep violation credit** from such sparse trajectory-level labels. **TraCeS** trains a sequential violation estimator whose per-step credits factorize the predicted probability that a trajectory has **not yet violated** the constraint, and integrates this learned signal into constrained policy optimization. The method requires neither a known cost function nor a known threshold, and remains compatible with standard continuous-control algorithms. We provide a theoretical analysis of the approximation gap introduced by the learning objective, and demonstrate empirically that **TraCeS** improves constraint satisfaction and feedback efficiency over baselines across multiple continuous-control benchmarks, including long-horizon tasks and settings with noisy or inconsistent labels.
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Learning a Generative Meta-Model of LLM Activations
Grace Luo, Jiahai Feng, Trevor Darrell, Alec Radford, Jacob Steinhardt
Existing approaches for analyzing neural network activations, such as PCA and sparse autoencoders, rely on strong structural assumptions.
Generative models offer an alternative: they can uncover structure without such assumptions and act as priors that improve intervention fidelity.
We explore this direction by training diffusion models on one billion residual stream activations, creating "meta-models" that learn the distribution of a network's internal states. We find that diffusion loss decreases smoothly with compute and reliably predicts downstream utility. In particular, applying the meta-model's learned prior to steering interventions improves fluency, with larger gains as loss decreases. Moreover, the meta-model's neurons increasingly isolate concepts into individual units, with sparse probing scores that scale as loss decreases. These results suggest generative meta-models offer a scalable path toward interpretability without restrictive structural assumptions.
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FairSSL: Fair Multimodal Self-Supervised Learning
Jiaee Cheong, Abtin Mogharabin, Paul Pu Liang, Hatice Gunes, Sinan Kalkan
Prevalent multimodal self-supervised learning (SSL) methods rely on the **redundancy assumption**: that different views share substantial task-relevant information. We argue that this assumption fails in complex, real-world settings characterized by heterogeneity (e.g., variable-length healthcare or behavioral data), where enforcing strict alignment can discard unique, modality-specific signals an

d inadvertently amplify bias. In this work, we propose **FairSSL**, a framework that leverages data heterogeneity as a resource for fairness rather than a hindrance. Unlike standard contrastive approaches, FairSSL uses a subject-aware Variance-Invariance-Covariance Regularization objective, where alignment is enforced across segments drawn from the same subject. We introduce a segment-based pooling strategy to handle variable-length modalities, and we regularize representations to encourage (i) sufficient within-subject variability, (ii) cross-modal and cross-subject invariance, and (iii) representation decorrelation. Theoretical analysis shows that our objective bounds the score gap between protected groups. Empirically, FairSSL significantly outperforms existing baselines on heterogeneous multimodal datasets, improving fairness without sacrificing downstream predictive performance. Code available at: https://github.com/abtinmU/FairSSL
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Tight Stability Bounds for Robust Distributed Learning: Byzantine Failures Hurt Generalization More than Data Poisoning
Thomas Boudou, Batiste Le Bars, Nirupam Gupta, Aurélien Bellet
Robust distributed learning algorithms aim to maintain reliable performance despite the presence of misbehaving workers.
Such misbehaviors are commonly modeled as *Byzantine failures*, allowing arbitrarily corrupted communication, or as *data poisoning*, a weaker form of corruption restricted to local training data. While prior work shows similar optimization guarantees for both models, an important question remains: *How do these threat models impact generalization?* We show, for the first time, a fundamental gap in generalization guarantees between the two threat models: Byzantine failures yield strictly worse rates than those achievable under data poisoning. Our findings leverage a tight algorithmic stability analysis of robust distributed learning. Specifically, we prove that: *(i)* under data poisoning, the uniform algorithmic stability of an algorithm with optimal optimization guarantees degrades by an additive factor of $\Theta (\frac{f}{n-f})$, with f out of n workers misbehaving; whereas *(ii)* under Byzantine failures, the degradation is in $\Omega \big(\sqrt{ \frac{f}{n-2f}} \big)$.
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Calibrating Decision Robustness via Inverse Conformal Risk Control
Wenbin Zhou, Shixiang Zhu
Robust optimization safeguards decisions against uncertainty by optimizing against worst-case scenarios, yet their effectiveness hinges on a prespecified robustness level that is often chosen ad hoc, leading to either insufficient protection or overly conservative and costly solutions. Recent approaches using conformal prediction construct data-driven uncertainty sets with finite-sample coverage guarantees, but they still fix coverage targets a priori and offer little guidance for selecting robustness levels. We propose a new framework that provides distribution-free, finite-sample guarantees on both miscoverage and regret for any family of robust predict-then-optimize policies. Our method constructs valid estimators that trace out the miscoverage--regret Pareto frontier, enabling decision-makers to reliably evaluate and calibrate robustness levels according to their cost--risk preferences. The framework is simple to implement, broadly applicable across classical optimization formulations, and achieves sharper finite-sample performance. This paper offers a principled data-driven methodology for guiding robustness selection and empowers practitioners to balance robustness and conservativeness in high-stakes decision-making.
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From Poisoned to Aware: Fostering Backdoor Self-Awareness in LLMs
Guangyu Shen, Siyuan Cheng, Xiangzhe Xu, Yuan Zhou, Hanxi Guo, ZHUO ZHANG, Xiangyu Zhang
Backdoor attacks can introduce deceptive behaviors into large language models, causing them to execute prohibited actions only when specific secret triggers appear in the input. Existing safety training methods largely fail to address this vulnerability, due to the inherent difficulty of uncovering hidden triggers embedded within the model. Motivated by recent findings on LLMs' situational awareness, we propose a novel post-training framework that cultivates backdoor self-awa

reness, enabling a poisoned LLM to precisely articulate its own implanted trigge rs. At its core, our approach introduces an inversion-inspired reinforcement lea rning framework that encourages models to introspectively reason about their beh aviors and gradually reverse-engineer the triggers responsible for misaligned ou tputs. Building upon precise trigger articulation, we further present two comple mentary defense strategies for mitigating and detecting backdoor threats. Experi ments on five backdoor attacks, compared against six baseline methods, demonstra te that our approach has strong potential to improve the robustness of LLMs agai nst backdoor risks.
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Online Learning with Recency: Algorithms for Sliding-window Streaming Multi-arme d Bandits
Vladimir Braverman, Chen Wang, Liudeng Wang, Samson Zhou
Motivated by the recency effect in online learning, we study algorithms for sing le-pass \emph{sliding-window streaming multi-armed bandits (MABs)} in this paper . In this setting, we are given n arms with unknown sub-Gaussian reward distri butions and a parameter W. The arms arrive in a single-pass stream, and only t he most recent W arms are considered valid. The algorithm is required to perfo rm pure exploration and regret minimization with \emph{limited memory}, \textcol or{red}{defined as the number of stored arms}. The model is a natural extension of the streaming multi-armed bandits model (without the sliding window) that has been extensively studied in recent years. We provide a comprehensive analysis o f both the pure exploration and regret minimization problems with the model. Fo r pure exploration, we prove that finding the best arm is hard with sublinear me mory while finding an \emph{approximate} best arm admits an efficient algorithm. For regret minimization, we explore a new notion of regret and give sharp memor y-regret trade-offs for any single-pass algorithms. We complement our theoretica l results with experiments, demonstrating the trade-offs between sample, regret, and memory.
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BVS: Bayesian Visual Search with Multimodal Large Language Model for Fine-graine d Perception
Geng Li, Yuxin Peng
While Multimodal Large Language Models (MLLMs) demonstrate impressive general ca pabilities, they struggle with fine-grained perception in ultra-high-resolution (UHR) images, particularly for tiny objects in cluttered scenes. Existing method s face a dilemma: they either rely on inefficient prior-free scanning, or depend on static prior-driven heuristics that lack posterior correction to rectify ini tial model biases. To address this, we propose **BVS** (**B**ayesian **V**isual **S**earch), a framework that formulates perception as a global optimization pro blem over a continuous spatial-scale manifold. Specifically, BVS bridges prior g uidance with posterior correction: it utilizes an early-stop attention rollout o f MLLM to construct reasoning-aware priors, while employing a scale-aware non-st ationary kernel and GP-UCB to dynamically rectify noise and recover missing info rmation in the prior through iterative local observations. We provide theoretica l guarantees via sub-linear regret bounds, and extensive experiments demonstrate that BVS significantly outperforms state-of-the-art baselines with a superior t rade-off between accuracy and efficiency.
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Vision Transformer Finetuning Benefits from Non-Smooth Components
Ambroise Odonnat, Laetitia Chapel, Romain Tavenard, Ievgen Redko
The smoothness of the transformer architecture has been extensively studied in t he context of generalization, training stability, and adversarial robustness. Ho wever, its role in transfer learning remains poorly understood. In this paper, w e analyze the ability of vision transformer components to adapt their outputs to changes in inputs, or, in other words, their *plasticity*. Defined as an averag e rate of change, it captures the sensitivity to input perturbation; in particul ar, a high plasticity implies a low smoothness. Our theoretical analysis and ext ensive experiments -- over $1,000$ finetuning runs on large-scale vision transfo rmers -- showcase that this perspective provides principled guidance in choosing

 the components to prioritize during adaptation. A key takeaway for practitioner
s is that the high plasticity of the attention modules and feedforward layers co
nsistently leads to better finetuning performance. Our findings depart from the
prevailing assumption that smoothness is desirable, offering a novel perspective
 on transformers' functional properties. The code is available at \url{https://g
ithub.com/ambroiseodt/vit-plasticity}.
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Stable Localized Conformal Prediction via Transduction
Yinjie Min, Liuhua Peng, Changliang Zou
Existing evaluations of conformal prediction, such as prediction efficiency and
test-conditional coverage, are defined in expectation over the calibration data.
 In practice, when only one calibration set of limited size is available, predic
tion sets often exhibit high variability in size, especially for methods with lo
calization. We formalize this concern as set stability, defined as the variance
of the conditional expectation of the set size given the calibration data. To im
prove stability without requiring additional target-task labels, we propose Stab
le Conformal Prediction (StCP), a transfer learning approach that utilizes label
ed source-task data and unlabeled target data. Theoretically, we characterize th
e marginal coverage and stability of StCP; empirically, it delivers more stable
prediction sets than standard conformal prediction methods, especially for those
 with localization, when calibration data are limited.
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Generative Representation Learning on Hyper-relational Knowledge Graphs via Mask
ed Discrete Diffusion
Jaejun Lee, Seheon Kim, Joyce Jiyoung Whang
Hyper-relational knowledge graphs (HKGs) effectively represent complex facts. Wh
ile inferring new knowledge in HKGs is a critical problem, current methods cast
it as a simple link prediction, assuming that nearly all entities and relations
within a fact are known, leaving only a single blank to be filled. However, this
 restricted assumption may not hold in real-world scenarios in which multiple, o
r even all, constituent components of a fact may be missing simultaneously. To b
ridge this gap, we introduce a task called fact generation: generating a valid h
yper-relational fact from an arbitrarily masked query, i.e., completing a partia
lly observed fact or generating a fact from scratch. We propose KREPE, the first
 generative representation learning method for HKGs that learns to model the pro
bability distributions of missing components conditioned on the local fact compo
nents and global structure of HKGs via a masked discrete diffusion. KREPE models
 both the intra-fact dependencies by contextual message passing and inter-fact c
orrelations by aggregating stochastically sampled contexts. KREPE seamlessly uni
fies link prediction and fact generation within a single training framework, ach
ieving state-of-the-art performance on standard HKG link prediction benchmarks a
nd outperforming LLM-based baselines in generating novel and correct facts.
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Alignment-Aware Decoding
Frédéric Berdoz, Luca A Lanzendörfer, René Caky, Roger Wattenhofer
Alignment of large language models remains a central challenge in natural langua
ge processing. Preference optimization has emerged as a popular and effective me
thod for improving alignment, typically through training-time or prompt-based in
terventions. In this paper, we introduce alignment-aware decoding (AAD), a metho
d to enhance model alignment directly at inference. Theoretically, AAD can be in
terpreted as implicit reward optimization, yet it requires no specialized traini
ng beyond the standard DPO setup. Empirically, AAD consistently outperforms stro
ng baselines across diverse alignment benchmarks and model scales. Moreover, in
data-constrained settings, AAD can produce high-quality synthetic data to improv
e alignment under standard decoding, providing a practical solution when labeled
 data is limited.
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Bringing Code ALIVE: Optimizing Interactive Frontend Mini-Games via Automated Pl
ay and Reinforcement Learning at Scale
Jiajun Zhang, Yuheng Jing, Zeyu Cui, Hao Zheng, Wentao Chen, Kaixin Li, Jiaxi Ya

ng, Tianbao Xie, Zeyao Ma, Tianyi Bai, KaShun SHUM, Lei Zhang, Kai Li, Jian Chen
g, Zilei Wang, Qiang Liu, Liang Wang, Junyang Lin, Binyuan Hui
The rapid evolution of Large Language Models (LLMs) has empowered even non-progr
ammers to create visually appealing frontend mini-games with a single instructio
n. However, open-source models significantly lag behind proprietary counterparts
 in this domain. The core bottleneck is the lack of an evaluation mechanism that
 balances reliability with scalability, as existing methods either fail to verif
y dynamic interactivity or incur prohibitive computational costs. To bridge this
 gap, we introduce ALIVE (Aligning LLMs via Interactive Visual Execution), a hig
h-throughput framework that leverages one-shot planning and DOM-based analysis t
o automatically evaluate generated games at scale. Extensive experiments demonst
rate that ALIVE significantly outperforms static judge baselines in identifying
functional flaws while remaining orders of magnitude more efficient than GUI age
nts. Functioning as a scalable `pre-flight' evaluation layer, it curates high-qu
ality data for Supervised Fine-Tuning (SFT) and provides a consistent reward sig
nal for Reinforcement Learning (RL). We leverage this pipeline to train ALIVE-Co
der, a model achieving superior performance in interactive frontend generation.
To the best of our knowledge, our work offers the first scalable path to evaluat
e and optimize interactive code, substantially advancing open-source capabilitie
s.
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On the Plasticity and Stability for Post-Training Large Language Models
Wenwen Qiang, Ziyin Gu, Jiahuan Zhou, Jie Hu, Jingyao Wang, Changwen Zheng, Hui
Xiong
Training stability remains a critical bottleneck for Group Relative Policy Optim
ization (GRPO), often manifesting as a trade-off between reasoning plasticity an
d general capability retention. We identify a root cause as the geometric confli
ct between plasticity and stability gradients, which leads to destructive interf
erence. Crucially, we argue that deterministic projection methods are suboptimal
 for GRPO as they overlook the intrinsic stochasticity of group-based gradient e
stimates. To address this, we propose Probabilistic Conflict Resolution (PCR), a
 Bayesian framework that models gradients as random variables. PCR dynamically a
rbitrates conflicts via an uncertainty-aware ``soft projection'' mechanism, opti
mizing the signal-to-noise ratio. Extensive experiments demonstrate that PCR sig
nificantly smooths the training trajectory and achieves superior performance in
various reasoning tasks.
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Giving Sensors a Voice: Multimodal JEPA for Semantic Time-Series Embeddings
Utsav Dutta, Gerardo Pastrana, Sina Khoshfetrat Pakazad, Henrik Ohlsson
Transformer-based architectures have advanced sequence modeling in language and
vision, yet general-purpose representation learning for heterogeneous multivaria
te time series remains underexplored. We introduce CHARM (Channel-Aware Represen
tation Model), which incorporates channel-level textual descriptions into a Tran
sformer encoder equivariant to channel order. CHARM is trained with a Joint Embe
dding Predictive Architecture (JEPA) and a novel loss promoting informative, tem
porally stable embeddings; latent-space prediction encourages robustness to sens
or noise while description-aware gating provides interpretability through learne
d inter-channel relationships. Across anomaly detection, classification, and sho
rt- and long-term forecasting, the learned embeddings achieve strong performance
 using only a linear probe. Performance is driven primarily by the JEPA objectiv
e and conditioning architecture, with text descriptions serving as channel ident
ifiers for cross-dataset generalization.
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Scaling Long-Horizon Agent via Context Folding
Weiwei Sun, Miao Lu, Zhan Ling, Kang Liu, Xuesong Yao, Yiming Yang, Jiecao Chen
Large language model (LLM) agents are fundamentally constrained by context lengt
h on long-horizon tasks.
Existing agent frameworks usually rely on manually defined context engineering p
ipelines, such as multi-agent or post-hoc summary.
We introduce Context Folding, a framework that empowers agents to actively manag

e their working context.
An agent can procedurally branch into a sub-trajectory to handle a subtask and t hen fold it upon completion, collapsing the intermediate steps while retaining a concise summary of the outcome.
To make this behavior learnable, we propose FoldGRPO, an end-to-end reinforcemen t learning framework with specific process rewards to encourage effective task d ecomposition and context management.
On long-horizon tasks, our agent matches the performance of baselines while usin g an active context up to 10x smaller, and significantly outperforms models cons trained to the same context size.
Code is available at https://github.com/sunnweiwei/FoldAgent.
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Learning Junta Distributions, Quantum Junta States, and QAC^0 Circuits
Jinge Bao, Francisco Escudero Gutiérrez
In this work, we consider the problems of learning junta distributions, their qu antum counterparts (quantum junta states), and QAC^0 circuits, which we show to be close to juntas.

(1) Junta distributions. A probability distribution $p:${-1,1}$^n\to \mathbb [0, 1]$ is a k-junta if it only depends on k bits. We show that they can be lear ned to within additive error ε in total variation distance from $O(2 ^k\log(n)/\varepsilon^2)$ samples, which quadratically improves the upper bound of Aliakbarpour et al. (COLT'16) and matches their lower bound in every paramete r.

(2) Junta states. We initiate the study of n-qubit states that are k-juntas, those that are the tensor product of a k-qubit state and an $(n-k)$-qubit max imally mixed state. We show that these states can be learned with error $\vareps ilon$ in trace distance with $O(12^{k}\log(n)/\varepsilon^2)$ single copies. We also prove a lower bound of $\Omega((4^k+\log (n))/\varepsilon^2)$ copies. Addit ionally, we show that, for constant k, $\widetilde{\Theta}(2^n/\varepsilon^2)$ copies are necessary and sufficient to test whether a state is ε-cl ose or 7ε-far from being a k-junta.

(3) QAC^0 circuits. We show that n-qubit QAC^0 circuits with size s, depth d and a auxiliary qubits can be learned from $2^{O(\log (s^22^a)^d)}\log(n)$ copies of the Choi state, improving the $n^{O(\log(s^22^a)^ d)}$ by Nadimpalli et al. (STOC'24).

Along the way, we give new proof of the optimal performance of Classical Shadows based on Pauli analysis. We also strengthen the lower bounds against $\mathsf{Q AC}^0$ to compute the address function.
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Multimodal Crystal Flow: Any-to-Any Modality Generation for Unified Crystal Mode ling
Kiyoung Seong, Sungsoo Ahn, Sehui Han, Changyoung Park
Crystal modeling spans a family of conditional and unconditional generation task s, including crystal structure prediction (CSP) and *de novo* generation (DNG). While recent deep generative models have shown promising performance, they remai n largely task-specific, lacking a unified framework that shares crystal represe ntations across tasks. To address this limitation, we propose *Multimodal Crysta l Flow (MCFlow)*, a unified multimodal flow model that realizes multiple crystal generation tasks as distinct inference trajectories via independent time variab les for atom types and crystal structures. To enable multimodal flow in a standa rd transformer model, we introduce a composition- and symmetry-aware atom orderi ng with hierarchical permutation augmentation, injecting compositional and cryst allographic priors without explicit structural templates. Experiments on the MP-20 and MPTS-52 benchmarks show that a single MCFlow model is competitive with ta sk-specific baselines across CSP, DNG, and structure-conditioned atom type gener ation.

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Categorical Flow Maps

Daan Roos, Oscar Davis, Floor Eijkelboom, Michael M. Bronstein, Max Welling, ■sm ail ■lkan Ceylan, Luca Ambrogioni, Jan-Willem van de Meent

We introduce Categorical Flow Maps, a flow-matching method for accelerated few-s tep generation of categorical data via self-distillation. Building on recent var iational formulations of flow matching and the broader trend towards accelerated inference in diffusion and flow-based models, we define a flow map towards the simplex that transports probability mass toward a predicted endpoint, yielding a parametrisation that naturally constrains model predictions. Since our trajecto ries are continuous rather than discrete, Categorical Flow Maps can be trained w ith existing distillation techniques, as well as a new objective based on endpoi nt consistency. This continuous formulation also automatically unlocks test-time inference: we can directly reuse existing guidance and reweighting techniques i n the categorical setting to steer sampling toward downstream objectives. Empiri cally, we achieve state-of-the-art few-step results on images, molecular graphs, and text, with strong performance even in single-step generation.

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PLANTAIN: Plan-Answer Interleaved Reasoning

Anthony Liang, Jonathan Berant, Adam Fisch, Abhimanyu Goyal, Kalpesh Krishna, Ja cob Eisenstein

Reasoning models often spend significant time generating hidden reasoning before any visible response, which can waste user time when the model starts from a fa lse premise that could have been corrected early. Human speakers, in contrast, u se lightweight incremental check-ins to maintain common ground, motivating *inte rleaved reasoning* (IR), where a model alternates between internal thinking and visible intermediate responses. We instantiate this idea with PLAINTAIN (Plan-An swer Interleaved Reasoning), a post-training recipe that teaches a model to exte rnalize an explicit step-by-step plan before continuing its reasoning. This lear ned plan-first structure creates an interface for early feedback and interventio n while preserving space for subsequent reasoning. Across challenging math, codi ng, text-to-SQL, and reading-comprehension benchmarks, PLAINTAIN improves pass@1 by roughly 6% on average while reducing time-to-first-response by over 60% rela tive to think-then-answer baselines.

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TileSparse: Arithmetic-Intensity-Aware Sparse Attention for Compute-Bound LLM De coding

Chao Wang, Pengfei Zuo, Zhangyu Chen, Qihui Zhou, Tsung-Yi Ho, Ming-Chang Yang

Sparse attention has emerged as a vital technique for long-context inference in Large Language Models (LLMs), effectively accelerating memory-bound decoding by reducing memory access for non-essential keys. However, the assumption that deco ding attention is memory-bound has been shattered. The proliferation of Multi-he ad Latent Attention (MLA) and Multi-Token Prediction (MTP) architectures has eff ectively rendered the process compute-bound. We observe that, in MLA, Q-heads ex hibit a degree of sparsity even when attending to the same key; consequently, tr aditional sparse attention algorithms introduce significant computational ineffi ciency in this new regime by rigidly computing interactions between all associat ed Q-heads and the retrieved keys.

To address this, we propose TileSparse, an arithmetic-intensity-aware (a.i.-awar e) algorithm for efficient attention in compute-bound settings. We first introdu ce a cost model that emphasizes compute budget (compute tile size) rather than m emory budget (fetched tokens) when evaluating sparse methods. Next, QK 2D Sparsi ty prunes unnecessary Q-head--key computations and uses the freed compute to ret rieve more semantically important tokens. Because Q-head sparsity differs across keys, we further propose Tiered QK 2D Sparsity and an AutoTuner to choose the b est pattern. Experiments show that under tight budgets our method improves accur acy by 40% over state-of-the-art dynamic K-only sparse methods. It also preserve s 99% of full-attention accuracy while cutting attention compute by 40.8%, outpe rforming prior sparse attention approaches.

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Improving the Sensitivity of Backdoor Detectors via Class Subspace Orthogonalization
Guangmingmei Yang, David J. Miller, George Kesidis
Most post-training backdoor detection methods rely on attacked models exhibiting extreme outlier detection statistics for the target class of an attack, compared to non-target classes. However, these approaches may fail: (1) when some (non-target) classes are easily discriminable from all others, in which case they may _naturally_ achieve extreme detection statistics (e.g., decision confidence); and (2) when the backdoor is subtle, i.e., with its features weak relative to intrinsic class-discriminative features. A key observation is that the backdoor target class has contributions to its detection statistic from both the backdoor trigger _and_ from its intrinsic features, whereas non-target classes _only_ have contributions from their intrinsic features. To achieve more sensitive detectors, we thus propose to _suppress_ intrinsic features while optimizing the detection statistic for a given class. For non-target classes, such suppression will drastically reduce the achievable statistic, whereas for the target class the (significant) contribution from the backdoor trigger remains. In practice, we formulate a constrained optimization problem, leveraging a small set of clean examples from a given class, and optimizing the detection statistic while orthogonalizing with respect to the class's intrinsic features. We dub this approach ''class subspace orthogonalization'' (CSO). CSO can be ''plug-and-play'' applied to a wide variety of existing detectors. We demonstrate its effectiveness in improving several well-known detectors, comparing with a variety of baseline detectors, against a variety of attacks, on the CIFAR-10, GTSRB, and TinyImageNet domains. Moreover, to make the detection problem even more challenging, we also evaluate against a novel mixed clean/dirty-label poisoning attack that is more surgical and harder to detect than traditional dirty-label attacks. Finally, we evaluate CSO against an adaptive attack designed to defeat it, with promising detection results.
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Vision2Web: A Hierarchical Benchmark for Visual Website Development with Agent Verification
Zehai He, Wenyi Hong, Zhen Yang, Ziyang Pan, Mingdao Liu, Xiaotao Gu, Jie Tang
Recent advances in large language models have improved the capabilities of coding agents, yet systematic evaluation of complex, end-to-end website development remains limited. To address this gap, we introduce Vision2Web, a hierarchical benchmark for visual website development, spanning from static UI-to-code generation, interactive multi-page frontend reproduction, to long-horizon full-stack website development. The benchmark is constructed from real-world websites and comprises a total of 193 tasks across 16 categories, with 918 prototype images and 1,255 test cases. To support flexible, thorough and reliable evaluation, we propose workflow-based agent verification paradigm based on two complementary components: a GUI agent verifier and a VLM-based judge. We evaluate multiple visual language models instantiated under different coding-agent frameworks, revealing substantial performance gaps at all task levels, with state-of-the-art models still struggling on full-stack development.
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STAR-KV: Low-Rank KV Cache Compression via Soft Thresholding for Adaptive Rank Control
Priyansh Bhatnagar, Ashkan Moradifirouzabadi, Se-Hyun Yang, SeungJae Lee, Jungwook Choi, Mingu Kang
Low-rank projection has emerged as a promising approach for compressing the KV cache by exploiting hidden-dimension redundancy. However, prior methods rely on fixed or heuristic rank selection and struggle to achieve aggressive compression with minimal accuracy degradation. We propose STAR-KV, an adaptive low-rank KV cache compression framework with fine-grained rank control. STAR-KV encompasses 1) a differentiable thresholding mechanism that enables optimal rank selection at both attention-head and block levels, 2) a hybrid decomposition strategy that applies different low-rank factorizations according to the sensitivity of key and value projections, and 3) a low-rank--aware mixed precision quantization that

leverages data statistics for near lossless low-bit quantization. Evaluated acro ss multiple LLMs and benchmarks, STAR-KV achieves up to 75\% KV cache compressi on and up to 20$\times$ overall KV cache reduction when combined with quantizati on. Enabled by custom Triton-based GPU kernels, STAR-KV delivers up to 6.9$\time s$ speedup for the attention module and 3.1$\times$ end-to-end generation throug hput. Our code is publicly available at: https://github.com/PriyanshBhatnagar/ST AR-KV.

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WinQ: Accelerating Quantization-Aware Training of Language Models Around Saddle Points

Dongyue Li, Zechun Liu, Kai Yi, Zhenshuo Zhang, Changsheng Zhao, Raghuraman Kris hnamoorthi, Harshit Khaitan, Hongyang R. Zhang, Steven Li

Quantization-aware training (QAT) is widely adopted to quantize language models by training full-precision weights using gradients from the quantized model. The main bottleneck is its slow convergence and early performance plateau, particul arly below 4-bit-widths. While this problem has been observed in prior work, its precise cause remains unclear. In this paper, we analyze the convergence of QAT by estimating the spectrum of the loss-surface Hessians. We find that the weigh ts converge to flat regions around saddle points, where a large fraction of the Hessian eigenvalues are both positive and negative. During training, an increasi ng fraction of Hessian eigenvalues concentrates around zero, whose magnitude dec reases. At lower bit-widths, the magnitude of eigenvalues in the Hessian spectru m is significantly smaller. To mitigate these issues, we propose an algorithm ca lled WinQ to accelerate QAT, which involves: (1) periodically resetting weights to the linear interpolation of full-precision and quantized weights, reducing th e distance to the quantization grid and increasing eigenvalue magnitude, and (2) computing gradients of noise-injected weights to regularize the Hessian. Extens ive experiments show that WinQ accelerates QAT by up to 4$\times$ across various quantization methods and models. Under the same training cost, WinQ improves st ate-of-the-art sub-4-bit quantization by up to 8.8%. These results are consisten t across 16 settings with different language models, quantization methods, and b it widths.

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FIRE: Learning to Navigate and Act on Real-World Files via Stateful Reinforcemen t Learning

Jingyuan Ma, Zheng Li, Rui Li, Siyao Song, Ying Zeng, Minghao Li, Zhifang Sui

Large language models still struggle to reliably answer questions grounded in re al-world files like spreadsheets and slides, where evidence is scattered across irregular layouts and heterogeneous formats. We address this by formalizing File Reasoning, a setting where agents must interact directly with unprocessed files (XLSX, PDF, DOCX, PPTX) within a persistent sandbox. To support this, we introd uce a unified data pipeline and a high-difficulty benchmark of over 400 verifiab le questions that preserve native file structure. Furthermore, we propose a rein forcement learning framework grounded in stateful file execution. We train **FIR E** (**F**ile **I**nteractive **R**easoning **E**xpert), a family of models that learn to optimize long-horizon planning using genuine execution feedback. Unlik e stateless tool-use methods, this environmental persistence enables agents to i teratively recover from errors and adapt to structural ambiguities. Empirical re sults show that Qwen3-32B-FIRE achieves the strongest performance among the open -source models of comparable scale under identical execution constraints.

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CURVE: Learning Causality-Inspired Invariant Representations for Robust Scene Un derstanding via Uncertainty-Guided Regularization

Yue Liang, Jiatong Du, Ziyi Yang, Yanjun Huang, Hong Chen

Scene graphs provide structured abstractions for scene understanding, yet they o ften overfit to spurious correlations, severely hindering out-of-distribution ge neralization. To address this limitation, we propose CURVE, a causality-inspired framework that integrates variational uncertainty modeling with uncertainty-gui ded structural regularization to suppress high-variance, environment-specific re lations. Specifically, we apply prototype-conditioned debiasing to disentangle i

nvariant interaction dynamics from environment-dependent variations, promoting a sparse and domain-stable topology. Empirically, we evaluate CURVE in zero-shot transfer and low-data sim-to-real adaptation, verifying its ability to learn domain-stable sparse topologies and provide reliable uncertainty estimates to support risk prediction under distribution shifts.
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How can embedding models bind concepts?
Arnas Uselis, Darina Koishigarina, Seong Joon Oh
Humans easily determine which color belongs to which shape in multi-object scenes, an ability known as concept binding. Vision–language embedding models such as CLIP struggle with binding: they recognize individual concepts but fail to represent which concepts form which objects. Although CLIP behaves like a bag-of-concepts model in cross-modal retrieval, object information is recoverable from its image and text embeddings separately. We study this tension through the binding function, which maps concepts to scene embeddings.
We find that scene embeddings decompose additively into object representations, explaining why uni-modal probes can recover object information. However, CLIP’s binding function is high-complexity, which likely prevents the image and text encoders from learning a shared binding mechanism that generalizes to unseen concept combinations.
We then ask whether this limitation is fundamental. We show that it is not. In controlled transformer models trained from scratch, binding generalization emerges with sufficient data coverage. These models learn low-complexity binding functions characterized by multiplicative interactions between concepts, enabling systematic generalization.
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Mind the Budget: Accelerating Deep Reinforcement Learning using Constrained Early Exit Neural Networks
Julien Brosseit, Jasper Hoffmann, Joschka Boedecker
Early exit neural networks, which adapt computation to input complexity, have proven effective in supervised learning but remain largely unexplored in deep reinforcement learning (DRL). In this paper, we propose Budgeted EXit Actor (BEXA), a novel actor-critic architecture that integrates early exit branches into the actor network. To ensure a mathematically principled trade-off between performance and inference expenditure, the exit decision is formulated as a constrained linear program during training, the solution of which is subsequently amortized to enable highly efficient runtime execution. BEXA is general, governed by an interpretable budget parameter, and compatible with any off-policy actor-critic method. We evaluate BEXA by integrating it with standard off-policy algorithms like SAC and TD3 on a suite of MuJoCo tasks. Our results demonstrate a substantial improvement in inference efficiency with minimal or no loss in performance. These findings highlight early exits as a promising direction for improving computational efficiency in DRL.
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RLSF-V: Mitigating Hallucinations in MLLMs via Fuzzy Semantic Self-Feedback
Changhao He, Shuhao Yan, Shuxian Li, Xi Peng, Peng Hu
Multimodal large language models (MLLMs) extend large language models (LLMs) with visual perception for open-world understanding, but exacerbate LLMs' hallucinations, in which generated text contradicts visual evidence or common sense. To mitigate hallucinations, a dominant strategy is Direct Preference Optimization (DPO) using hallucination-labeled responses. Existing pipelines, however, face two key limitations: they either (i) rely on human inspection or proprietary models to correct hallucinated outputs, producing off-policy preference data that violate the assumptions of DPO, or (ii) depend on stronger models to evaluate responses, leading to an unfavorable trade-off between performance and scalability. Departing from these paradigms, we propose a reference-policy \emph{self-feedback} framework that constructs preference data for hallucination mitigation without any external supervision (\textit{e.g.}, large models or humans). Specifically, we present a novel \emph{local fuzzy semantic} evaluation paradigm that derives a hallucination-sensitive confidence signal directly from the internal logits, w

hich is then used to automatically rank diverse generated responses to build pre ference pairs for fine-tuning. Trained on a 10k-scale dataset, our method achiev es competitive performance on both generative and discriminative benchmarks comp ared to existing RLHF and RLAIF baselines.
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Attention Sparsity is Input-Stable: Training-Free Sparse Attention for Video Gen eration via Offline Sparsity Profiling and Online QK Co-Clustering
Jiayi Luo, Jiayu Chen, Jiankun Wang, Cong Wang, Hanxin Zhu, Qingyun Sun, Chen Ga o, Zhibo Chen, Jianxin Li
Diffusion Transformers (DiTs) achieve strong video generation quality but suffer from high inference cost due to dense 3D attention, leading to the development of sparse attention technologies to improve efficiency. However, existing traini ng-free sparse attention methods in video generation still face two unresolved l imitations: *ignoring layer heterogeneity in attention pruning* and *ignoring qu ery-key coupling in block partitioning*, which hinder a better quality-speedup t rade-off. In this work, we uncover a critical insight that **the attention spars ity of each layer is its intrinsic property, with minor effects across different inputs**. Motivated by this, we propose **SVOO**, a training-free **S**parse at tention framework for fast **V**ideo generation via **O**ffline layer-wise spars ity profiling and **O**nline bidirectional co-clustering. Specifically, SVOO ado pts a two-stage paradigm: (i) offline layer-wise sensitivity profiling to derive intrinsic per-layer pruning levels, and (ii) online block-wise sparse attention via a novel bidirectional co-clustering algorithm. Extensive experiments on sev en widely used video generation models demonstrate that SVOO achieves a superior quality-speedup trade-off over state-of-the-art methods, delivering up to 1.93× speedup while maintaining a PSNR of up to 29 dB on Wan2.1.
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Revisiting Uncertainty: On Evidential Learning for Partially Relevant Video Retr ieval
Jun Li, Peifeng Lai, Xuhang Lou, Jinpeng Wang, Yuting Wang, Ke Chen, Yaowei Wang , Shu-Tao Xia
Partially relevant video retrieval aims to retrieve untrimmed videos using text queries that describe only partial content. However, the inherent asymmetry betw een brief queries and rich video content inevitably introduces uncertainty into the retrieval process.
In this setting, vague queries often induce semantic ambiguity across videos, a challenge that is further exacerbated by the sparse temporal supervision within videos, which fails to provide sufficient matching evidence. To address this, we propose Holmes, a hierarchical evidential learning framework that aggregates mu lti-granular cross-modal evidence to quantify and model uncertainty explicitly.
At the inter-video level, similarity scores are interpreted as evidential suppor t and modeled via a Dirichlet distribution. Based on the proposed three-fold pri nciple, we perform fine-grained query identification, which then guides query-ad aptive calibrated learning.
At the intra-video level, to accumulate denser evidence, we formulate a soft que ry-clip alignment via flexible optimal transport with an adaptive dustbin, which alleviates sparse temporal supervision while suppressing spurious local respons es. Extensive experiments demonstrate that Holmes outperforms state-of-the-art m ethods. Code is released at https://github.com/lijun2005/ICML26-Holmes.
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The Value Function Semi-Algebraic Set in Partially Observable Markov Decision Pr ocesses
Ryan A. Anderson, Guido Montufar
We study the geometry of feasible value functions in infinite-horizon partially observable Markov decision processes (POMDPs) under memoryless stochastic polici es. Our main contribution is a characterization of the feasible set of value fun ctions as a semi-algebraic set, defined by explicit polynomial inequalities dete rmined by the transition dynamics, observation kernel, and reward structure of t he POMDP. This result extends prior work for fully observable Markov decision pr ocesses, where the feasible set is known to be a polytope, to the substantially

more intricate partially observable setting. In contrast to the polyhedral structure arising in MDPs, partial observability induces fundamentally nonlinear constraints, leading to a richer and more complex geometric structure. Our geometric characterization provides new insight into the landscape of policy optimization in both MDPs and POMDPs, and reveals qualitative phenomena unique to partial observability, including the emergence of isolated local maximizers of the long-term reward and their dependence on the initial state distribution.
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Does AI Reviewer See the Full Picture? Attacking and Defending Multimodal Peer Review
Xinyu Zhao, Rana Shahroz, Zhen Xu, Zhen Tan, Tianlong Chen
The integration of Large Language Models (LLMs) and Multimodal LLMs (MLLMs) into scientific peer-review workflows introduces novel and significant risks for adversarial manipulation, especially given the multimodal nature of scientific papers where figures, not just text, convey core evidence. This creates a significant gap: current robustness studies on AI peer-review are overwhelmingly text-only. Moreover, the problem is distinct from standard jailbreaking, as a peer-review attack seeks to induce a domain-specific, targeted failure (e.g., “inflate this score”) rather than a general safety policy violation, for which no practical defenses exist. To address this, we introduce PaperGuard, the first comprehensive benchmark designed to systematically evaluate and defend AI-generated peer-review against these domain-specific, cross-modal attacks. Our framework is built on three pillars: (1) a new multimodal peer-review dataset spanning multiple scientific domains; (2) a unified suite of attacks, including black-box prompt injections and white-box perturbations, specifically designed to target both text (GCG) and figures (PGD); and (3) a practical defense, motivated by the long-context challenge of academic papers, that uses chunk-based embedding search to efficiently localize and mitigate harmful instructions. Our extensive experiments, conducted across state-of-the-art models, confirm that AI reviewers are pervasively vulnerable. PaperGuard establishes the foundational benchmark, protocols, and actionable defense necessary to pioneer trustworthy, attack-resilient AI-assisted scholarly reviewing.
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Chasing Moving Targets with Online Self-Play Reinforcement Learning for Safer Language Models
Mickel Liu, Liwei Jiang, Yancheng Liang, Simon Shaolei Du, Yejin Choi, Tim Althoff, Natasha Jaques
Conventional large language model (LLM) safety alignment relies on a reactive, disjoint loop: attackers exploit a static model, then defenders patch exposed vulnerabilities. This sequential setup leads to attackers overfitting obsolete exploits while defenders perpetually lag behind emerging threats. To address this, we introduce Self-RedTeam, the first fully online self-play multi-agent reinforcement learning (MARL) algorithm that continuously co-evolves attacker and defender for robust safety alignment. A single policy self-plays as both attacker and defender, generating adversarial prompts and defending against them, with a reward model adjudicating outcomes. Each role uses hidden chain-of-thought for strategic planning. Grounded in two-player zero-sum game theory, we establish a theoretical safety guarantee: if the game converges to Nash Equilibrium, the defender produces safe responses against any adversarial input. Empirically, Self-RedTeam generalizes across five models from the Llama and Qwen families, uncovering more diverse attacks (+17.80% SBERT) and improving safety of RLHF-trained models by up to 95% across 14 benchmarks. Our work motivates a shift from reactive patching to proactive co-evolution, enabling LLM safety self-improvement via online self-play MARL.
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DDSVM: A Differentiable Framework for Deep Support Vector Machines with Iterative Geometry-Aware Optimization
Yirun Ding, Zhihui Lai
Modern deep networks optimize features via cross-entropy without explicit margin maximization, while classical Support Vector Machines maximize the margin on a

fixed feature space. Hybrid Deep-SVM approaches typically treat the deep feature extractor as a static input to an SVM head, so SVM geometry never flows back to shape representation learning. We propose the **Differentiable Deep Support Vector Machine (DDSVM)**, an iterative framework that alternates between re-fitting a linear SVM on L_2-normalized features and updating the backbone to pull each feature along the SVM-defined margin-widening direction, treating the decision boundary as a dynamic teacher for representation learning. Under spherical normalization the resulting feature gradient lies in the tangent space and optimization focuses exclusively on angular separability. DDSVM reduces top-1 error by **5.6--8.3\%** on image classification benchmarks under data scarcity.

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(Doubly) Exponential Lower Bounds for Follow the Regularized Leader in Potential Games

Ioannis Anagnostides, Ioannis Panageas, Nikolas Patris, Tuomas Sandholm

Follow the regularized leader (FTRL) is the premier algorithm for online optimization---and potential games in particular---its behavior remained hitherto poorly understood. In this paper, we establish that FTRL can take exponential time to converge to a Nash equilibrium in two-player potential games for any (permutation-invariant) regularizer and potentially vanishing learning rate. By known equivalences, this translates to an exponential lower bound for certain mirror descent counterparts, most notably multiplicative weights update. On the positive side, we establish the potential property for FTRL and obtain an exponential upper bound $\exp(O_{\epsilon}(1/\epsilon^2))$ for any no-regret dynamics executed in a lazy, alternating fashion, matching our lower bound up to factors in the exponent. Finally, in multi-player potential games, we show that fictitious play---the extreme version of FTRL---can take doubly exponential time to reach a Nash equilibrium. This constitutes an exponentially stronger lower bound for the foundational learning algorithm in games.

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Plug-and-Play Spiking Operators: Breaking the Nonlinearity Bottleneck in Spiking Transformers

Xinzhe Yuan, Xiang Peng, Bin Gu, Huan Xiong

ANN-to-SNN conversion offers a practical, training-free route to spiking large language models.

However, current pipelines primarily focus on spike-driven realizations for Transformer linear-algebra operations, while providing limited support for key nonlinear operators.

This gap limits compatibility with neuromorphic-style execution constraints, where such nonlinearities typically require division, exponentiation, or norm computations that are not naturally supported by standard leaky integrate-and-fire dynamics.

To solve this problem, we propose a plug-and-play framework that implements spike-friendly approximations for Transformer nonlinearities and integrates into existing ANN-to-SNN pipelines.

Our method decomposes these nonlinear computations into three recurring primitives---division, exponentiation, and ℓ_2 norms---and realizes them via population computation using LIF neuron groups, combined with lightweight bit-shift scaling to avoid floating-point arithmetic.

By composing these primitives as modular operator blocks, our framework supports common Transformer nonlinearities (e.g., Softmax, SiLU, and normalization) without any fine-tuning.

Experiments on a range of LLMs Transformers show that selectively replacing the targeted nonlinear operators incurs less than a 1% accuracy drop across all evaluated tasks.

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Bottleneck-Guided Spectral Subgoals For Offline Goal-Conditioned RL

Hebin Liang, Yi Ma, Chenjun Xiao, Zibin Dong, Zilin Cao, Fei Ni, Yifu Yuan, Jianye HAO

Offline goal-conditioned RL (OGCRL) learns to reach arbitrary goals from the off

line dataset, but long-horizon performance hinges on crossing a handful of hard-to-cross bottlenecks. These bottlenecks not only dictate the feasible paths towa rd the goal but also act as critical keypoints, marking the transitions between adjacent regions and providing the agent with essential directional guidance. Pr ior hierarchical methods pick subgoals by time or short-horizon value heuristics , which do not localize the bottleneck, and therefore the agent loses the clear guidance that bottlenecks could provide about where to pass next. We instead mod el long-horizon planning as ``cross the next bottleneck'': we apply Laplacian sp ectral clustering to the offline dataset to expose bottlenecks, identify traject ories from the offline dataset that cross these boundaries, and use the correspo nding boundary-supported states as keypoint (KP) candidates.
The most representative KPs are then instantiated from these candidates, and a d irected KP reachability graph $\mathcal G_{\mathrm{KP}}$ is constructed based on the resulting KPs.
We then restrict high-level choices to these bottleneck states and use a pluggab le low-level controller to execute the short transitions between them.
We provide theory showing that under a standard metastable decomposition of the state space, routing through bottlenecks yields an (approximately) optimal one-s tep subgoal in terms of hitting-time, and that Laplacian spectra recover bottle necks with high overlap. Thus, Laplacian spectral clustering can discover near-o ptimal subgoals under these assumptions. Empirically, the same pattern holds: ac ross D4RL and OGBench, our method achieves state-of-the-art results on a broad s et of navigation and manipulation tasks and across diverse dataset regimes, for example, **96.5\%** on **AntMaze** and **84.7\%** on **Franka-Kitchen**.
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From Internal Diagnosis to External Auditing: A VLM-Driven Paradigm for Data-Fre e Online Backdoor Defense
Binyan Xu, Fan Yang, Xilin Dai, Di Tang, Kehuan Zhang
Deep Neural Networks (DNNs) remain fundamentally vulnerable to backdoor attacks. Traditional data-free defenses largely operate under the paradigm of internal d iagnosis methods like model repairing or input robustness, yet these approaches are often fragile under advanced attacks as they remain entangled with the victi m model’s corrupted parameters. We propose a paradigm shift to data-free Externa l Semantic Auditing, using universal Vision-Language Models (VLMs) as independen t auditors to decouple defense from the compromised model. We introduce PRISM (P rototype Refinement & Inspection via Statistical Monitoring), which transforms g eneric VLMs into domain-adaptive gatekeepers purely via online test-time adaptat ion. PRISM bridges the domain gap through a Hybrid VLM Teacher that refines prot otypes from the test stream and an Adaptive Router that calibrates thresholds vi a statistical monitoring. Evaluation across 17 datasets and 11 attack types conf irms PRISM achieves state-of-the-art performance (suppressing Attack Success Rat e to < 1% on CIFAR-10), proving that robust defense is achievable without touchi ng the model weights or accessing a single training sample.
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FIRE-Bench: Evaluating AI Agents on the Rediscovery of Scientific Insights
Zhen Wang, Fan Bai, Zhongyan Luo, Jinyan Su, Kaiser Sun, Xinle Yu, Jieyuan Liu, Kun Zhou, Claire Cardie, Mark Dredze, Zhiting Hu, Eric P. Xing
Autonomous AI agents powered by large language models (LLMs) are increasingly ca pable of running a full cycle of scientific research, yet we still lack reliable ways to verify that their discoveries are correct. Because novel findings deman d costly real-world validation, existing benchmarks fall back on LLM-as-judge sc oring of generated papers or single leaderboard metrics, both coarse proxies for scientific reasoning. We introduce FIRE-BENCH (Full-cycle Insight Rediscovery E valuation), which instead asks agents to rediscover established, verifiable find ings from recent, high-impact machine learning research. Given only a high-level research question from a published study, an agent must independently design ex periments, run them, and draw evidence-backed conclusions, scored against the st udy's documented findings. Across state-of-the-art agents with frontier backbone s such as gpt-5, even the strongest reaches limited rediscovery success (<50 F1) , with high run-to-run variance and recurring failures in experimental design, e

xecution, and evidence-based reasoning. Beyond diagnosing current systems, FIRE-BENCH shows that open-ended discovery can be evaluated rigorously and verifiably , laying a foundation for building reliable environments that improve agents.
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Temporal-Aware Reasoning Optimization for Video Temporal Grounding
Minghang Zheng, Zihao Yin, Yi Yang, Yuxin Peng, Yang Liu
Multi-modal Large Language Models (MLLMs) have achieved remarkable progress in v ideo temporal grounding with reinforcement learning for generating reasoning pat hs. However, existing models often produce superficial reasoning, which offers l imited guidance for precise temporal localization. This limitation stems from (1) inefficient random exploration and (2) reward functions that focus solely on t he answer correctness while ignoring reasoning quality. To address these issues, we propose TaRO (Temporal-Aware Reasoning Optimization), a framework that expli citly enhances the model’s ability of thinking with time. First, we introduce a Constructive Reasoning Exploration that leverages pre-generated dense captions t o construct reasoning paths grounded in explicit visual cues and timestamps, ena bling efficient exploration of high-quality time-aware reasoning. Second, to eva luate reasoning quality, we design a Temporal-Sensitivity Reward. High-quality r easoning should be anchored to specific events and timestamps. If the event boun dary under thinking is disrupted, such reasoning should become invalid, leading to a drop in the logit of the reasoning path. We utilize this drop as a critique of reasoning quality. Finally, TaRO follows a progressive curriculum, which sta rts by utilizing this reward to select better constructed reasoning paths, and e volves to a free exploration phase where the model autonomously generates effect ive reasoning. Experiments demonstrate that TaRO achieves state-of-the-art perfo rmance on VTG benchmarks. Code is available at https://github.com/oceanflowlab/T aRO.
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On the Theory of Continual Learning with Gradient Descent for Neural Networks
Hossein Taheri, Avishek Ghosh, Arya Mazumdar
Continual learning, the ability of a model to adapt to an ongoing sequence of ta sks without forgetting earlier ones, is a central goal of artificial intelligenc e. To better understand its underlying mechanisms, we study the limitations of c ontinual learning in a tractable yet representative setting. Specifically, we an alyze one-hidden-layer quadratic neural networks trained by gradient descent on a sequence of XOR-cluster datasets with Gaussian noise, where different tasks co rrespond to clusters with orthogonal means. Our analysis is based on a tight cha racterization of gradient descent dynamics for the training loss, which yields e xplicit bounds on the rate of train-time forgetting as functions of the number o f iterations, sample size, number of tasks, and hidden-layer width. We then leve rage an algorithmic stability framework to bound the generalization gap, leading to corresponding guarantees on test-time forgetting. Together, our results prov ide the first closed-form guarantees for forgetting in continual learning with neural networks and show how key problem parameters jointly govern forgetting dy namics. Numerical experiments corroborate our theoretical results.
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Cascaded Flow Matching for Heterogeneous Tabular Data with Mixed-Type Features
Markus Mueller, Kathrin Gruber, Dennis Fok
Advances in generative modeling have recently been adapted to tabular data conta ining discrete and continuous features.
However, generating mixed-type features that combine discrete states with an oth erwise continuous distribution in a single feature remains challenging. We advan ce the state-of-the-art in diffusion models for tabular data with a cascaded app roach. We first generate a low-resolution version of a tabular data row, that is , the collection of the purely categorical features and a coarse categorical rep resentation of numerical features. Next, this information is leveraged in the hi gh-resolution flow matching model via a novel guided conditional probability pat h and data-dependent coupling. The low-resolution representation of numerical fe atures explicitly accounts for discrete outcomes, such as missing or inflated va lues, and therewith enables a more faithful generation of mixed-type features. W

e formally prove that this cascade tightens the transport cost bound. The result s indicate that our model generates significantly more realistic samples and cap tures distributional details more accurately, for example, the detection score i mproves by 51.9%. Code is available at https://github.com/muellermarkus/tabcasca de.
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Push, Pop, Parallelize: Stack-Augmented Linear Attention via the Delta Rule
Anh T Nguyen, Saleh Momeni, Ashutosh Chaubey, Changnan Xiao, Bing Liu
Linear attention architectures based on the Delta rule, such as DeltaNet and RWK V-7, combine Transformer-level performance with RNN-like efficiency and provably solve regular language tasks. However, their fixed-size states struggle to capt ure the recursive, hierarchical structures intrinsic to natural languages. To br idge this gap, we introduce DeltaStack, which augments DeltaNet's associative me mory with a lightweight, differentiable stack. Unlike prior approaches that rely on sequential recurrence, DeltaStack formulates stack operations as linear delt a-rule updates, enabling a hardware-aware implementation fully parallelizable ov er sequence length. Theoretically, we prove DeltaStack extends DeltaNet's expres sivity to model both regular and hierarchical languages. Empirically, DeltaStack outperforms DeltaNet and Stack-Attention on formal language benchmarks and cons istently surpasses DeltaNet baselines in language modeling perplexity and zero-s hot performance across scales up to 760M parameters.

Our code is publicly available at https://github.com/teeann/DeltaStack.
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Imagination Helps Visual Reasoning, But Not Yet in Latent Space
You Li, Chi Chen, Yanghao Li, Fanhu Zeng, Kaiyu Huang, Jinan Xu, Maosong Sun
Latent visual reasoning aims to mimic human's *imagination* process by meditatin g through hidden states of Multimodal Large Language Models.

While recognized as a promising paradigm for visual reasoning, the underlying mechanisms driving its effectiveness remain unclear.

Motivated to demystify the true source of its efficacy, we investigate the val idity of latent reasoning using Causal Mediation Analysis.

We model the process as a causal chain: the input as the treatment, the latent tokens as the mediator, and the final answer as the outcome.

Our findings uncover two critical disconnections:

(a) **Input-Latent Disconnect**: dramatic perturbations on the input result in negligible changes to the latent tokens, suggesting that latent tokens do not e ffectively attend to the input sequence.

(b) **Latent-Answer Disconnect**: perturbations on the latent tokens yield min imal impact on the final answer, indicating the limited causal effect latent tok ens imposing on the outcome.

Furthermore, extensive probing analysis reveals that latent tokens encode limi ted visual information and exhibit high similarity.

Consequently, we challenge the necessity of latent reasoning and propose a str aightforward alternative named *CapImagine*, which teaches the model to explicit ly *imagine* using text.

Experiments on vision-centric benchmarks show that *CapImagine* significantly outperforms complex latent-space baselines, highlighting the superior potential of visual reasoning through explicit imagination.
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OGPO: Sample Efficient Full-Finetuning of Generative Control Policies
Sarvesh Patil, Mitsuhiko Nakamoto, Manan Agarwal, Shashwat Saxena, Jesse Zhang, Giri Anantharaman, Cleah Winston, Chaoyi Pan, Douglas Chen, Nai-Chieh Huang, Zey nep Temel, Oliver Kroemer, Sergey Levine, Abhishek Gupta, Hongkai Dai, Paarth Sh ah, Max Simchowitz
Generative control policies (GCPs), such as diffusion- and flow-based control po licies, have proved effective parameterizations for robot learning. This work in troduces Off-policy Generative Policy Optimization (**OGPO**), a sample-efficien t algorithm for finetuning GCPs that maintains off-policy critics to maximize da ta reuse and propagate policy gradients through the full generative process of t he policy via a modified PPO objective, using critics as the terminal reward. **

OGPO** achieves state-of-the-art performance on manipulation tasks spanning mult
i-task settings, high-precision insertion, and dexterous control. To our knowled
ge, it is also the only method that can *fine-tune poorly-initialized behavior c
loning policies to near full task-success with no expert data in the online repl
ay buffer*, and does so with *few task-specific hyperparameter tuning*. Through
extensive investigations, we demonstrate that **OGPO** drastically outperforms a
lternative methods on policy steering and learning residual corrections, and ide
ntify the key mechanisms behind its performance. We further introduce practical
stabilization tricks, including success-buffer regularization and two-sided cons
ervative advantages to mitigate critic over-exploitation across state- and pixel
-based settings. Beyond proposing **OGPO**, we conduct a systematic empirical st
udy of GCP finetuning, identifying the stabilizing mechanisms and failure modes
that govern successful off-policy full-policy improvement.
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Use What You Know: Causal Foundation Models with Partial Graphs
Arik Reuter, Anish Dhir, Cristiana Diaconu, Jake Robertson, Ole Ossen, Frank Hut
ter, Adrian Weller, Mark van der Wilk, Bernhard Schölkopf
Estimating causal quantities traditionally relies on bespoke estimators tailored
 to specific assumptions.
Recently proposed Causal Foundation Models (CFMs) promise a more unified approac
h by amortising causal discovery and inference in a single step.
However, in their current state, they do not allow for the incorporation of any
domain knowledge, which can lead to suboptimal predictions.
We bridge this gap by introducing methods to condition CFMs on causal informatio
n, such as the causal graph or more readily available ancestral information.
When access to complete causal graph information is too strict a requirement, ou
r approach also effectively leverages partial causal information.
We systematically evaluate conditioning strategies and find that injecting learn
able biases into the attention mechanism is the most effective method to utilise
 full and partial causal information.
Our experiments show that this conditioning allows a general-purpose CFM to matc
h the performance of specialised models trained on specific causal structures.
Overall, our approach addresses a central hurdle on the path towards all-in-one
causal foundation models: the capability to answer causal queries in a data-driv
en manner while effectively leveraging any amount of domain expertise.
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Hyperbolic Hierarchical Alignment for Video-Based Visible-Infrared Person Re-Ide
ntification
Shuang Li, Changjiang Kuang, Jiaxu Leng, Mingpi Tan, Zhanjie Wu, Shuanglin Yan,
Xinbo Gao
Video-based visible-infrared person re-identification (VVI-ReID) aims to learn r
obust video-level representations under modality discrepancy. However, existing
methods typically rely on Euclidean geometry, which is suboptimal for modeling t
he complex temporal dynamics within visible and infrared tracklets, as it inevit
ably distorts the intrinsic hierarchical structure inherent in diverse temporal
variations (e.g., occlusion, pose). In this paper, we propose Hyperbolic Hierarc
hical Alignment (HHA), which unifies spatio-temporal modeling and cross-modality
 alignment on the Poincaré ball. HHA employs a Hyperbolic Hierarchical Spatio-Te
mporal Aggregator (HHSA) to organize time-varying cues into low-distortion hiera
rchical representations via Hyperbolic Geometry Interaction (HGI) and Dual-Geome
try Fusion (DGF). Furthermore, we introduce Geometry-Aware Modality Alignment (G
MA), which integrates Hyperbolic Modality Alignment (HMA) to couple modality cen
troids for geometric consistency and Hyperbolic Prototype Alignment (HPA) to anc
hor both modalities to shared identity prototypes for robust discrimination. Exp
eriments on HITSZ-VCM and BUPTCampus demonstrate state-of-the-art performance. T
he code will be available at https://github.com/Visuang/HHA.
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Imitation Learning for Multi-turn LM Agents via On-policy Expert Corrections
Niklas Lauffer, Xiang Deng, Srivatsa R Kundurthy, Brad Kenstler, Jeff Da
A popular paradigm for training LM agents relies on imitation learning, fine-tun

ing on expert trajectories. However, we show that the off-policy nature of imita tion learning for multi-turn LM agents suffers from the fundamental limitation k nown as covariate shift: as the student policy's behavior diverges from the expe rt's, it encounters states not present in the training data, reducing the effect iveness of fine-tuning. Taking inspiration from the classic DAgger algorithm, we propose a novel data generation methodology for addressing covariate shift for multi-turn LLM training. We introduce on-policy expert corrections (OECs), parti ally on-policy data generated by starting rollouts with a student model and then switching to an expert model part way through the trajectory. We explore the ef fectiveness of our data generation technique in the domain of software engineeri ng (SWE) tasks, a multi-turn setting where LLM agents must interact with a devel opment environment to fix software bugs. Our experiments compare OEC data agains t various other on-policy and imitation learning approaches on SWE agent problem s and train models using a common rejection sampling (i.e., using environment re ward) combined with supervised fine-tuning technique. Experiments find that OEC trajectories show a relative 14% and 13% improvement over traditional imitation learning in the 7b and 32b setting, respectively, on SWE-bench verified. Our res ults demonstrate the need for combining expert demonstrations with on-policy dat a for effective multi-turn LM agent training.
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FAFO: Lossy KV Cache Compression for Lossless Inference Acceleration via Draftle ss Fumble Decoding
Hoang Anh Duy Le, Shaochen Zhong, Yifan Lu, Yingtong Dou, Jiayi Yuan, Yu-Neng Ch uang, Xiran Fan, Guanchu Wang, Yuzhong Chen, Xia Hu
Lossy KV cache compression is a well-explored subfield of machine learning eff iciency, with improved latency being one of its major gains. However, lossy comp ression techniques can fumble from time to time, exhibiting various, and often c atastrophic, failure patterns that are not only difficult to resolve but sometim es even hard to identify, making direct deployment of models with compressed KV cache a risky endeavor. In this work, we explore a way to preserve *lossless* ge neration quality while still benefiting from the acceleration provided by KV cac he compression. Specifically, we draw inspiration from the *n-gram candidate poo l decoding* paradigm where we purposely allow the model to Fumble Around with co mpressed KV cache to generate multiple lossy "n-gram guesses", while in parallel Find Out via lossless verification in the same forward pass. From a conceptual standpoint, our proposed framework is compatible with all typical static or dyna mic KV cache compression methods from the token dropping realm, thus opening up a new avenue for the stagnant n-gram decoding paradigm. Practically, we show tha t this framework presents many useful traits that similar draftless baselines (e .g., Self-Speculative Decoding) cannot achieve, such as requiring only one set o f KV cache and being far less sensitive to model, task, and input-length scenari os. Our comprehensive empirical results show FAFO provides 1.20-2.71× latency sp eedup over the original model, while consistently outperforming other lossless + draftless solutions.
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Mosaic: Unlocking Over 30$\times$ Context Length for Diffusion LLMs Inference vi a Global Memory Planning and Dynamic Peak Taming
Liang Zheng, Bowen Shi, Yitao Hu, Jiawei Zhang, Ruofan Li, Guotao Yang, Zhixin Z hao, Zhengchao Wang, Sheng Chen, Wenxin Li, Dezhi Ran, Tao Xie, Keqiu Li
Diffusion-based large language models (dLLMs) have emerged as a promising altern ative to autoregressive models, leveraging simultaneous denoising to enable glob al planning and iterative refinement. These properties make dLLMs attractive for long-context generation. However, deploying dLLMs faces a prohibitive memory ba rrier, as existing inference systems are inefficient for the diffusion paradigm. We observe that current inference systems are misaligned with dLLMs. Unlike auto regressive models, whose memory footprint is dominated by the KV-Cache, dLLMs ar e bottlenecked by transient activations rematerialized per step. Moreover, gener ic memory reuse mechanisms lack the global visibility to handle dynamic memory p eaks of dLLMs, which alternate between logits and feed-forward networks.
To address these challenges, we present Mosaic, a memory-efficient inference sys

tem that shifts dLLM execution from local, static management to a global, dynami c paradigm. Mosaic integrates (i) a mask-only logits kernel eliminating redundan t activation materialization, (ii) a lazy chunking optimizer using online heuris tics to tame dynamic memory peaks, and (iii) a global memory manager leveraging virtual addressing to mitigate memory fragmentation.
Evaluations show that Mosaic reduces the memory peak-to-average ratio by 2.71$\t imes$ on average and increases the maximum inference sequence length on identica l hardware by 15.30--32.34$\times$. Crucially, Mosaic is training-free and pres erves exact model outputs, while reducing end-to-end latency by 2.5\%--55.4\%. O ur code is publicly available at https://github.com/flashserve/Mosaic.
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Reasoning to Edit: Hypothetical Instruction-Based Image Editing with Visual Reas oning
Qingdong He, Xueqin Chen, Chaoyi Wang, Yanjie Pan, Xiaobin Hu, Zhenye Gan, Cheng jie Wang, Xiangtai Li, Jiangning Zhang, Yabiao Wang
Instruction-based image editing (IIE) has advanced rapidly with the success of d iffusion models. However, existing efforts primarily focus on simple and explici t instructions to execute editing operations such as adding, deleting, moving, o r swapping objects. They struggle to handle more complex implicit hypothetical i nstructions that require deeper reasoning to infer plausible visual changes and user intent. Additionally, current datasets provide limited support for training and evaluating reasoning-aware editing capabilities. Architecturally, these met hods also lack mechanisms for fine-grained detail extraction that support such r easoning. To address these limitations, we propose Reason50K, a large-scale data set specifically curated for training and evaluating hypothetical instruction–re asoning image editing, along with ReasonBrain, a novel framework designed to rea son over and execute implicit hypothetical instructions across diverse scenarios . Reason50K includes over 50K samples spanning four key reasoning scenarios: Phy sical, Temporal, Causal, and Story reasoning. ReasonBrain leverages Multimodal L arge Language Models (MLLMs) for editing guidance generation and a diffusion mod el for image synthesis, incorporating a Fine-grained Reasoning Cue Extraction (F RCE) module to capture detailed visual and textual semantics essential for suppo rting instruction reasoning. To mitigate the semantic loss, we further introduce a Cross-Modal Enhancer (CME) that enables rich interactions between the fine-gr ained cues and MLLM-derived features. Extensive experiments demonstrate that Rea sonBrain consistently outperforms state-of-the-art baselines on reasoning scenar ios while exhibiting strong zero-shot generalization to conventional IIE tasks. Our dataset and code will be released publicly.
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TexEditor: Structure-Preserving Text-Driven Texture Editing
Bo Zhao, Yihang Liu, Chenfeng Zhang, Huan Yang, Kun Gai, Wei Ji
Text-guided texture editing aims to modify object appearance while preserving th e underlying geometric structure.
However, our empirical analysis reveals that even SOTA editing models frequently struggle to maintain structural consistency during texture editing, despite the intended changes being purely appearance-related. Motivated by this observation , we jointly enhance structure preservation from both data and training perspect ives, and build TexEditor, a dedicated texture editing model based on Qwen-Image -Edit-2509. Firstly, we construct BlenderTex, a high-quality SFT dataset generat ed with Blender, which provides strong structural priors for a cold start. Secon dly, we introduce StructureNFT, a RL–based approach that integrates structure-pr eserving losses to transfer the structural priors learned during SFT to real-wor ld scenes. Moreover, due to the limited realism and evaluation coverage of exist ing benchmarks, we introduce TexBench, a general-purpose real-world benchmark fo r text-guided texture editing. Extensive experiments on existing Blender-based t exture benchmarks and our TexBench show that TexEditor consistently outperforms strong baselines such as Nano Banana Pro. In addition, we assess TexEditor on th e general-purpose benchmark ImgEdit to validate its generalization.
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Curated Synthetic Data Doesn’t Have to Collapse: A Theoretical Study of Generati

ve Retraining with Pluralistic Preferences
Ali Falahati, Mohammad Mohammadi Amiri, Kate Larson, Lukasz Golab
Recursive retraining of generative models poses a critical representation challe nge: when synthetic outputs are curated based on a fixed reward signal, the mode l tends to collapse onto a narrow set of outputs that over-optimize that objecti ve, causing diversity to vanish and failing to represent the full range of prefe rences. Prior work has suggested that such collapse is unavoidable without addin g real data into the mix. In this paper, we revisit that conclusion from an alig nment perspective and show that collapse can be mitigated through curation based on multiple reward functions. We formalize the dynamics of recursive training u nder heterogeneous preferences and prove that, under certain conditions, the mod el converges to a stable distribution that allocates probability mass across com peting high-reward regions. The limiting distribution preserves diversity and pr ovably satisfies a weighted Nash bargaining solution, offering a formal interpre tation of value aggregation in synthetic retraining loops.
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When Softmax Fails at the Top: Extreme■Value Corrections for InfoNCE
Hasan Sabri Melihcan Erol, Suat Evren, Oktay Ozel, Alexander Morgan, Jongha Jon Ryu, Lizhong Zheng
InfoNCE is the standard contrastive learning objective, but its softmax form is not only a computational convenience: it also encodes a statistical assumption a bout how the top-scoring example is selected. Using extreme value theory, we sho w that this assumption is often misaligned with the normalized embedding setting used in modern contrastive learning. Motivated by this mismatch, we propose \te xtsc{WEINCE}, a simple modification of InfoNCE that uses anchor-wise online batc h statistics to blend the usual softmax logits with an endpoint shortfall correc tion, adding no trainable parameters. Across five vision benchmarks, \textsc{WEI NCE} yields consistent improvements in frozen-feature evaluation. These results show that a more faithful statistical treatment of hard negatives can improve co ntrastive objectives.\footnote{Code: \href{https://github.com/hsme98/weince}{git hub.com/hsme98/weince}.}
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Robust Federated Learning Against Adaptive Compression
Wenjing Yan, Xiangyu Zhong, Ying-Jun Angela Zhang
This paper addresses the critical challenges of hyperparameter tuning and commun ication efficiency in federated learning (FL). Despite recent advancements in pa rameter-free FL algorithms such as PAdaMFed, significant communication overhead remains a major obstacle to their practical deployment. To tackle these challeng es, we propose a novel communication-efficient parameter-free FL algorithm ParFr eFL that halves the communication requirements of PAdaMFed while preserving its parameter-free property. Building on this foundation, we introduce a compressed variant, ComParFreFL, which unifies the momentum increment and error feedback in to a single parameter, effectively handling biased compression while maintaining the minimal communication cost. Notably, ComParFreFL also operates independent of the compression ratio, representing the first instance of such robustness in the compressed FL literature to our knowledge. Theoretically, our methods are pr oven to handle arbitrary data heterogeneity, partial client participation, and a chieve linear speedup with respect to both local updates and participating clien ts. Extensive empirical evaluations demonstrate that our approaches match or sli ghtly surpass the performance of carefully tuned alternatives while significantl y reducing communication overhead, making FL more accessible and deployable in d ynamic, resource-constrained environments.
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Guided Star-Shaped Masked Diffusion
Viacheslav Meshchaninov, Egor Shibaev, Artem Makoian, Ivan Klimov, Nikita Balaga nsky, Daniil Gavrilov, Aibek Alanov, Dmitry Vetrov
The performance of pre-trained masked diffusion models is often constrained by t heir sampling procedure, which makes decisions irreversible and struggles in low -step generation regimes. We introduce a novel sampling algorithm that works wit h pre-trained models and, after a lightweight fine-tuning of a single layer, sig

nificantly improves sample quality and efficiency. Our method reformulates the g
eneration process using a star-shaped paradigm, which inherently allows for erro
r correction. To make this process effective, we augment it with a learnable rem
asking module that intelligently identifies and revises likely errors. This appr
oach yields a substantial quality boost, particularly when using a small number
of sampling steps. We extensively ablate key components of our approach and show
 its usability in different scenarios. In experiments on text, and code generati
on, our sampling algorithm outperforms or matches existing methods. Code is avai
lable at https://github.com/EgorShibaev/G-Star.
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UniDrag: Unified Multi-Field Prediction and Robust Shape Optimization for Vehicl
e Aerodynamics
Ye Liu, Shouyi Liu, Ding Wang, Huiyu Yang, Ruizhe Deng, Qian Li, Yuxiao Hu, Jian
ghang gu, Yongzheng Liu, Quanshi Zhang, Shiyi Chen, Yuntian Chen
High-fidelity vehicle aerodynamics analysis is bottlenecked by costly CFD simula
tions. Neural surrogates accelerate prediction but lack inverse design capabilit
ies, while existing generative optimization methods suffer from unstable converg
ence and frequent engineering constraint violations. We present UniDrag, a unifi
ed framework bridging multi-field aerodynamic prediction with robust differentia
ble shape optimization. Given a vehicle geometry, UniDrag predicts surface press
ure, volume flow fields, drag coefficient C_d, and a streamwise build-up profi
le localizing drag contributions.
Our architecture combines enhanced Physics-Sliced Attention (ePSA), Gated Expert
 Routing, and Modality-Protected Learning to prevent negative transfer across ou
tput modalities. At deployment, the frozen surrogate enables gradient-based opti
mization via Free-Form Deformation with engineering constraints. We introduce Ex
pectation-over-Transformation to prevent adversarial exploitation of surrogate f
ragility.
We curate a large-scale dataset of 15,000 vehicle geometries spanning four body
types with GPU-accelerated LBM simulations. On this benchmark, UniDrag achieves
C_d prediction R^2 of 0.937 (+7.6\% over baselines) and 13.7\% mean CFD-veri
fied drag reduction with 100\% success rate and only 21.3~mm average displacemen
t (0.46\% vehicle length).
****
Quantitative Estimation of Target Task Performance from Unsupervised Pretext Tas
k in Semi/Self-Supervised Learning
Lin-Han Jia, Siyu Han, Wen-Chao Hu, Jie-Jing Shao, Wen-Da Wei, Zhi Zhou, Lan-Zhe
 Guo, Yu-Feng Li
The effectiveness of unlabeled data in Semi/Self-Supervised Learning (SSL) depen
ds on appropriate assumptions for specific scenarios, thereby enabling the selec
tion of beneficial unsupervised pretext tasks. However, existing research has pa
id limited attention to assumptions in SSL, resulting in practical situations wh
ere the compatibility between the unsupervised pretext tasks and the target scen
arios can only be assessed after training and validation.
This paper centers on the assumptions underlying unsupervised pretext tasks and
explores the feasibility of preemptively estimating the impact of unsupervised p
retext tasks at low cost.
Through rigorous derivation, we show that the impact of unsupervised pretext tas
ks on target performance depends on three factors: assumption learnability with
respect to the model, assumption reliability with respect to the data, and assum
ption completeness with respect to the target.
Building on this theory, we propose a low-cost estimation method that can quanti
tatively estimate the actual target performance. We build a benchmark of over on
e hundred pretext tasks and demonstrate that our estimated performance strongly
correlates with the actual performance obtained through large-scale training and
 validation.
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Backjump-on-Graph: Empowering Large Language Models with Reinforced Retrospectiv
e Exploration for Agentic Knowledge Graph Reasoning
Yunqi Zhang, Shiqi Yan, Zhenzhao Yuan, Wenrui Liang, Yangming Liu, Zhixiao Qi, T

ianyi Zhang, Shijie Zhang, Wei-Qiang Zhang, Yongfeng Huang, Haixin Duan, Shuai C
hen, Yubo Chen
Grounding Large Language Models (LLMs) in Knowledge Graphs (KGs) has shown signi
ficant promise for complex Question Answering (QA) tasks.
Since LLMs' limited context window cannot accommodate the sheer volume of large-
scale KGs, existing work usually utilizes agents to reason on real-world KGs, wh
ich follows reasoning paths derived from the queries step by step.
However, the mismatch between query-derived paths and the KG's structure, stemmi
ng from users' lack of schema knowledge, usually leads the agents into dead ends
.
To address this problem, in this paper, we propose Backjump-on-Graph (BoG), a no
vel framework that empowers LLMs to retrospectively explore alternative reasonin
g paths at dead ends.
We first propose to formalize each reasoning step with four atomic operations to
 create a structural scaffold that allows LLMs to revert to historical status.
Next, we fine-tune the LLM with synthetic data containing the above atomic opera
tions to instill basic backjump abilities.
Finally, we leverage reinforcement learning and propose a hybrid reward function
, which penalizes redundant transitions and promotes correct answers, to optimiz
e the timing and landing nodes of backjumping.
Extensive experiments on several KGQA benchmark datasets demonstrate the effecti
veness of our BoG method.
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Spectral Imbalance Causes Forgetting in Low-Rank Continual Adaptation
Hao Gu, Mao-Lin Luo, Zi-Hao Zhou, Han-Chen Zhang, Min-Ling Zhang, Tong Wei
Parameter-efficient continual learning aims to adapt pre-trained models to seque
ntial tasks without forgetting previously acquired knowledge. Most existing appr
oaches treat continual learning as avoiding interference with past updates, rath
er than considering what properties make the current task-specific update natura
lly preserve previously acquired knowledge.
From a knowledge-decomposition perspective, we observe that low-rank adaptations
 exhibit highly imbalanced singular value spectra: a few dominant components abs
orb most of the adaptation energy, thereby (i) more likely to disrupt previously
 acquired knowledge and (ii) making the update more vulnerable to interference f
rom subsequent tasks.
To enable explicit balance among components, we decouple the *magnitude* of the
task update from its *directional structure* and formulate it as a constrained o
ptimization problem on a restricted Stiefel manifold.
We address this problem using a projected first-order method compatible with sta
ndard deep-learning optimizers used in vision-language models.
Our method mitigates both backward and forward forgetting, consistently outperfo
rming continual learning baselines. Source code is available in supplementary ma
terial.
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Torus Graphs for Large Scale Neural Phase Analysis
Jack Goffinet, Casey Hanks, David Carlson
Oscillatory neural signals such as electroencephalography (EEG) and local field
potentials (LFPs) show phase relationships that coordinate communication across
brain regions. Modern recordings capture hundreds of channels across many freque
ncy bins, yet standard phase analyses are restricted to only a few variables. Th
e Torus Graph (TG) model, an exponential-family distribution over phases whose u
nivariate and pairwise potentials generalize von Mises distributions, infers pri
ncipled structure among oscillations but models only static, undirected dependen
cies and is limited to $\sim \! 100$ variables because its score matching infere
nce scales as $\mathcal{O}(d^{6})$. We introduce a stochastic score matching pro
cedure that reduces the per-iteration cost to $\mathcal{O}(d^{2})$, enabling inf
erence on datasets with thousands of variables. This scalable foundation support
s analyses of 1,860 frequency-phase features from multi-electrode LFPs and enabl
es two extensions previously inaccessible to TGs or classical circular statistic
s: (i) a TG Hidden Markov Model capturing state-dependent phase-coupling changes

(e.g., spindle-related states during sleep) and (ii) an autoregressive TG inferring directional interactions via transfer-entropy estimation. Applied to LFP recordings, these models reveal state-dependent phase-interaction patterns between wakefulness and NREM sleep. Together, they enable systematic, large-scale mapping of dynamic and directional phase relationships across brain and cognitive states.

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The Flexibility Trap: Rethinking the Value of Arbitrary Order in Diffusion Language Models

Zanlin Ni, Shenzhi Wang, Yang Yue, Tianyu Yu, Weilin Zhao, Yeguo Hua, Tianyi Chen, Jun Song, Cheng Yu, Bo Zheng, Gao Huang

Diffusion Large Language Models (dLLMs) break the rigid left-to-right constraint of traditional LLMs, enabling token generation in arbitrary orders. Intuitively, this flexibility implies a solution space that strictly supersets the fixed autoregressive trajectory, theoretically unlocking superior reasoning potential. However, in this paper, we find that for general reasoning tasks (e.g., mathematics and coding), arbitrary order generation may in fact limit the reasoning potential of dLLMs.

We observe that dLLMs tend to exploit this order flexibility to bypass high-uncertainty tokens that are crucial for exploration, which can lead to a premature collapse of solution coverage.

This observation motivates a rethink of RL approaches for dLLMs, where considerable complexities, such as handling combinatorial trajectories and intractable likelihoods, are often devoted to preserving this flexibility.

We show that effective reasoning can be elicited by simply forgoing arbitrary order and applying standard Group Relative Policy Optimization (GRPO) instead.

Our approach, JustGRPO, is minimalist yet surprisingly effective (e.g., 89.1\% accuracy on GSM8K) while fully retaining the parallel decoding ability of dLLMs.

Code: https://github.com/LeapLabTHU/JustGRPO.

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The Relative Instability of Model Comparison with Cross-validation

Alexandre Bayle, Lucas Janson, Lester Mackey

Cross-validation (CV) is known to provide asymptotically exact tests and confidence intervals for model improvement but only when the model comparison is *relatively stable*. Surprisingly, we prove that even simple, individually stable models can generate relatively unstable comparisons, calling into question the validity of CV inference. Specifically, we show that the Lasso and its close cousin, soft-thresholding, generate relatively unstable comparisons and invalid CV inferences, even in the most favorable of learning settings and when both models are individually stable. These findings highlight the importance of verifying relative stability before deploying CV for model comparison.

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ScaleSim: Serving Large-Scale Multi-Agent Simulation with Invocation Distance-Based Memory Management

Zaifeng Pan, Yipeng Shen, Zhengding Hu, Zhuang Wang, Aninda Manocha, Zheng Wang, Zhongkai Yu, Yue Guan, Yufei Ding

LLM-based multi-agent simulations are increasingly adopted across application domains, but remain difficult to scale due to GPU memory pressure. Each agent maintains private GPU-resident states, including models, prefix caches, and adapters, which quickly exhaust device memory as the agent count grows. We identify two key properties of these workloads: sparse agent activation and an estimable agent invocation order. Based on an analysis of representative workload classes, we introduce invocation distance, a unified abstraction that estimates the relative order in which agents will issue future LLM requests. Leveraging this abstraction, we present ScaleSim, a memory-efficient LLM serving system for large-scale multi-agent simulations. ScaleSim enables proactive prefetching and priority-based eviction, supports diverse agent-specific memory through a modular interface, and achieves up to 1.74$\times$ speedup over SGLang on simulation benchmarks. ScaleSim's source code is available at https://github.com/PanZaifeng/KVFlow.

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LoPhyDA: Low-Rank Tensor and Physics Gradient Guided Diffusion for Atmospheric Data Assimilation
Danyang Peng, Yang Chen, Yunlong Zhou, Xiaotong Yuan
Data Assimilation (DA) aims to integrate observations with model forecasts to estimate the state of dynamical systems. Despite the widespread application of diffusion-based assimilation methods, they remain constrained by the high dimensionality of atmospheric states and the reliance on imperfect state-observation mappings. This leaves regions lacking observations spatially unconstrained, leading to severe error accumulation and physical inconsistency.. In this paper, we propose LoPhyDA, a diffusion assimilation algorithm dual-guided by low-rank tensor and physical gradients. By leveraging the low-rank property of meteorological field, we employ tensor completion to exploit spatial continuity and dynamic correlations, reconstructing a globally informative dense field from sparse observations to serve as a global prior. This approach mitigates the information deficit inherent in sparse settings. The framework further incorporates physical constraints into the iterative denoising process, utilizing Partial Differential Equation (PDE) residual gradients to rectify the generative trajectory in real-time. Experimental results demonstrate that LoPhyDA outperforms state-of-the-art generative assimilation models in global weather prediction. It achieves robust and physically consistent assimilation, significantly reducing error accumulation in regions lacking observations.
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Compositional Behavioral Semantics for State Abstraction in Reinforcement Learning
Yivan Zhang, Ziyan Luo, Manuel Baltieri
State abstraction plays a key role in scaling reinforcement learning to complex but structured systems. In studying such systems, a wide range of behavioral structures have been studied in reinforcement learning, including value functions, invariants, bisimulation relations, and behavioral metrics. However, a general principle for determining what structures are provably preserved under state abstraction is still lacking. In this paper, we present a unified framework for defining and analyzing behavioral structures in reinforcement learning. Our framework provides a compositional way to specify behavioral semantics based on local, one-step descriptions of system dynamics. Using this framework, we establish results showing how behavioral structures can be safely transferred between abstract and concrete systems. We further show how to construct quantitative metrics from logical behavioral semantics with soundness guarantees. Together, these results provide a principled foundation for reasoning about behaviors under state abstraction in reinforcement learning and offer reusable definition and proof principles for a broad class of behavioral structures in reinforcement learning.
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Scalable and General Whole-Body Control for Cross-Humanoid Locomotion
Yufei Xue, Yunfeng Lin, Wentao Dong, Yang Tang, Jingbo Wang, Jiangmiao Pang, Ming Zhou, Minghuan Liu, Weinan Zhang
Learning-based whole-body controllers have become a key driver for humanoid robots, yet most existing approaches require robot-specific training.
In this paper, we study the problem of cross-embodiment humanoid control and show that a single policy can robustly generalize across a wide range of humanoid robot designs with one-time training.
We introduce XHugWBC, a novel cross-embodiment training framework that enables generalist humanoid control through:
(1) physics-consistent morphological randomization,
(2) semantically aligned observation and action spaces across diverse humanoid robots, and
(3) effective policy architectures modeling morphological and dynamical properties.
XHugWBC is not tied to any specific robot. Instead, it internalizes a broad distribution of morphological and dynamical characteristics during training. By learning motion priors from diverse randomized embodiments, the policy acquires a strong structural bias that supports zero-shot transfer to previously unseen robot

s.
Experiments on twelve simulated humanoids and seven real-world robots demonstrate the strong generalization and robustness of the resulting universal controller.
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INDUCTION: Finite-Structure Concept Synthesis in First-Order Logic
Serafim Batzoglou
Induction is the search for a general rule that explains observations. We study logical induction in finite relational worlds: each problem gives small structures over a fixed vocabulary, labels objects belonging to an unknown unary concept, and asks for one first-order formula $\phi(x)$ that accounts for those labels across worlds. Finite domains make formulas mechanically checkable by exact evaluation and SMT. We introduce INDUCTION, a benchmark for finite-structure concept synthesis with three regimes: FULLOBS (full observation), where all facts are observed; CI (contrastive induction), where YES/NO worlds require discriminative hypotheses; and EC (existential completion), where validity is defined by world-local completion of unknown facts. We evaluate frontier language models, include symbolic synthesis baselines, and score both validity and formula size. Prompted models show real but incomplete capability, with sharp difficulty gradients and hard structural families. Held-out evaluation shows that compact formulas generalize far better than bloated ones; parsimony separates concept recovery from finite-world fit.
****
Neural QAOA^2: Differentiable Joint Graph Partitioning and Parameter Initialization for Quantum Combinatorial Optimization
Zubin Zheng, Jiahao Wu, Shengcai Liu
The quantum approximate optimization algorithm (QAOA) holds promise for combinatorial optimization but is constrained by limited qubits. While divide-and-conquer frameworks like QAOA^2 address scalability by partitioning graphs into subgraphs, existing methods suffer from two fundamental limitations: i) misalignment between heuristic partitioning metrics and quantum optimization goals, and ii) topology-blind parameter initialization that leads to optimization cold starts. To bridge these gaps, we propose **Neural QAOA^2**, an end-to-end differentiable framework that jointly generates graph partitions and initial parameters. By integrating a generative evaluative network (GEN), our method utilizes a differentiable quantum evaluator as a high-fidelity performance surrogate to provide direct gradient guidance, enabling the joint generator to learn the intrinsic mapping from graph topology to high-quality partition and parameter configurations. Extensive experiments on 183 QUBO, Ising, and MaxCut instances (21 to 1000 variables) demonstrate that our gradient-driven approach broadly outperforms heuristic baselines, ranking first on 101 instances. It exhibits zero-shot generalization across out-of-distribution graph topologies and scales.
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ResRL: Boosting LLM Reasoning via Negative Sample Projection Residual Reinforcement Learning
Zihan Lin, Xiaohan Wang, Jie Cao, Jiajun Chai, Li Wang, Xiaodong Lu, Wei Lin, Ran He, Guojun Yin
Reinforcement Learning with Verifiable Rewards (RLVR) enhances reasoning of Large Language Models (LLMs) but usually exhibits limited generation diversity due to the over-incentivization of positive rewards. Although methods like Negative Sample Reinforcement (NSR) mitigate this issue by upweighting penalty from negative samples, they may suppress the semantic distributions shared between positive and negative responses. To boost reasoning ability without losing diversity, this paper proposes negative sample projection Residual Reinforcement Learning (ResRL) that decouples similar semantic distributions among positive and negative responses. We theoretically link Lazy Likelihood Displacement (LLD) to negative-positive head-gradient interference and derive a single-forward proxy that upper-bounds representation alignment to guide conservative advantage reweighting. ResRL then projects negative-token hidden representations onto an SVD-based low-rank positive subspace and uses projection residuals to modulate negative gradie

nts, improving reasoning while preserving diversity and outperforming strong bas
elines on average across twelve benchmarks spanning Mathematics, Code, Agent Tas
ks, and Function Calling. Notably, ResRL surpasses NSR on mathematical reasoning
 by 9.4\% in Avg@16 and 7.0\% in Pass@128. Code is available at https://github.c
om/1229095296/ResRL.git.
****

Towards the Training of Deeper Predictive Coding Neural Networks
Chang Qi, Matteo Forasassi, Thomas Lukasiewicz, Tommaso Salvatori
Predictive coding networks are neural models that perform inference through an i
terative energy minimization process, whose operations are local in space and ti
me. While effective in shallow architectures, they suffer significant performanc
e degradation beyond five to seven layers. In this work, we show that this degra
dation is caused by exponentially imbalanced errors between layers during weight
 updates, and by predictions from the previous layers not being effective in gui
ding updates in deeper layers. Furthermore, when training models with skip conne
ctions, the energy propagated by the residuals reaches higher layers faster than
 that propagated by the main pathway, affecting test accuracy. We address the fi
rst issue by introducing a novel precision-weighted optimization of latent varia
bles that balances error distributions during the relaxation phase, the second i
ssue by proposing a novel weight update mechanism that reduces error accumulatio
n in deeper layers, and the third one by using auxiliary neurons that slow down
the propagation of the energy in the residual connections. Empirically, our met
hods achieve performance comparable to backpropagation on deep models such as Re
sNets, opening new possibilities for predictive coding in complex tasks.
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Equivariant Covariance Tensors: Guaranteed SPD Uncertainty for Tensor-Valued Geo
metric Learning
Ruihan Liu, Yu Ji, Jianbo Yu, Shifu Yan, Qingchao Jiang
Tensor-valued prediction is fundamental to geometric deep learning, yet uncertai
nty quantification (UQ) for such outputs remains an open challenge. While E(3)-e
quivariant neural networks excel at point estimates, they lack rigorous confiden
ce measures. We focus on symmetric rank-2 tensor prediction, where the target ha
s six Kelvin-Mandel coordinates and full uncertainty is represented by a $6\time
s6$ covariance matrix. We introduce a framework for E(3)-equivariant UQ, modelin
g the full predictive distribution where both mean and covariance preserve rotat
ional symmetry. Our approach decomposes the covariance into irreducible represen
tations $\mathrm{Sym}^2(\rho_c) \cong 2\times(l=0) \oplus 2\times(l=2) \oplus 1\
times(l=4)$. By mapping from the flat Lie algebra $\mathfrak{sym}(6)$ to the cur
ved SPD manifold via matrix exponentiation, we strictly ensure positive-definite
 covariances while maintaining exact equivariance. Furthermore, we formulate a L
og-Euclidean Equivariant Scoring Objective (LE-ESO)---a robust surrogate loss ba
sed on the Multivariate Laplace distribution---providing robustness to heavy-tai
led errors and stable optimization. Validation on ModelNet40 inertia tensors and
 Materials Project dielectric tensors demonstrates that our method achieves comp
etitive performance and provides physically consistent, symmetry-preserving unce
rtainty estimates with useful risk and OOD sensitivity.
****

On the Computational Complexity of Performative Prediction
Ioannis Anagnostides, Rohan Chauhan, Ioannis Panageas, Tuomas Sandholm, Jingming
 Yan
Performative prediction captures the phenomenon where deploying a predictive mod
el shifts the underlying data distribution. While simple retraining dynamics are
 known to converge linearly when the performative effects are weak ($\rho < 1$),
 the complexity in the regime $\rho > 1$ was hitherto open. In this paper, we es
tablish a sharp phase transition: computing an ϵ-performatively stable
point is PPAD-complete---and thus polynomial-time equivalent to Nash equilibria
in general-sum games---even when $\rho = 1 + O(\epsilon)$. This intractability p
ersists even in the ostensibly simple setting with a quadratic loss function and
 linear distribution shifts. One of our key technical contributions is to extend
 this PPAD-hardness result to general convex domains, which is of broader intere

st in the complexity of variational inequalities. Finally, we address the specia
l case of strategic classification, showing that computing a strategic local opt
imum is PLS-hard.
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Effective Distillation to Hybrid xLSTM Architectures
Lukas Hauzenberger, Niklas Schmidinger, Thomas Schmied, Anamaria-Roberta Hartl,
David Stap, Pieter-Jan Hoedt, Sebastian Böck, Günter Klambauer, Sepp Hochreiter
There have been numerous attempts to distill quadratic attention-based large lan
guage models (LLMs) into sub-quadratic linearized architectures.
However, despite extensive research, such distilled models often fail to match t
he performance of their teacher LLMs on various downstream tasks.
We set out the goal of *lossless distillation*, which we define in terms of tole
rance-corrected *Win-and-Tie rates* between student and teacher on sets of tasks
.
To this end, we introduce an effective distillation pipeline for xLSTM-based stu
dents.
We propose an additional merging stage, where individually linearized experts ar
e combined into a single model.
We show the effectiveness of this pipeline by distilling base and instruction-tu
ned models from the Llama, Qwen, and Olmo families.
In many settings, our xLSTM-based students recover most of the teacher's perform
ance, and even exceed it on some downstream tasks.
Our contributions are an important step towards more energy-efficient and cost-e
ffective replacements for transformer-based LLMs.
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Relational In-Context Learning via Synthetic Pre-training with Structural Prior
Yanbo Wang, Jiaxuan You, Chuan Shi, Muhan Zhang
Relational Databases (RDBs) are the backbone of modern business, yet they lack f
oundation models comparable to those in text or vision. A key obstacle is that h
igh-quality RDBs are private, scarce, and structurally heterogeneous, making int
ernet-scale pre-training infeasible. To overcome this data scarcity, we introduc
e RDB-PFN, the first relational foundation model trained purely via synthetic da
ta. Inspired by Prior-Data Fitted Networks (PFNs), where synthetic data generate
d from Structural Causal Models (SCMs) enables reasoning on single tables, we de
sign a Relational Prior Generator to create an infinite stream of diverse RDBs f
rom scratch. Pre-training on over 2 million synthetic single-table and relationa
l tasks, RDB-PFN learns to adapt to any new database instantly via genuine in-co
ntext learning. Experiments show that RDB-PFN achieves strong few-shot performan
ce on 19 real-world relational prediction tasks, outperforming state-of-the-art
tabular foundation models evaluated on the same DFS-linearized inputs, while usi
ng a lightweight architecture and fast inference. The code is available at https
://github.com/MuLabPKU/RDBPFN.
****

Stable Asynchrony: Variance-Controlled Off-Policy RL for LLMs
Luke J. Huang, Zhuoyang Zhang, Qinghao Hu, Shang Yang, Song Han
Asynchronous reinforcement learning has become increasingly central to scaling L
LM post-training, delivering major throughput gains by decoupling rollout genera
tion from policy updates. However, widely used policy-gradient objectives such a
s REINFORCE and GRPO suffer under high asynchrony: stale rollouts produce heavy-
tailed importance weights, so a small number of trajectories dominate updates an
d the policy-gradient estimator becomes **markedly higher variance**. Through sy
stematic analysis on math, reasoning, and tool-use benchmarks, we establish that
 this increasing variance is reliably predicted by collapsing effective sample s
ize (ESS), which prior stabilization methods largely fail to address. Motivated
by this diagnosis, we introduce **V**ariance **C**ontrolled **P**olicy **O**ptim
ization (VCPO), a method that (i) dynamically scales the learning rate with ESS
to dampen unreliable updates and (ii) applies a closed-form minimum-variance bas
eline for off-policy settings, without a critic model and adding minimal overhea
d. Empirically, across math and general reasoning benchmarks, this enables robus
tly stable asynchronous training compared to previous stabilization and algorith

mic methods, even in highly off-policy regimes (128 steps off-policy). In a long-horizon, tool-use task, VCPO matches synchronous performance while delivering a 2.5× speedup in training time. Code is available at: https://github.com/mit-han-lab/vcpo

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PLoRA: Efficient Concurrent LoRA Training for Large Language Models

Minghao Yan, Zhuang Wang, Zhen Jia, Shivaram Venkataraman, Yida Wang

Low-Rank Adaptation (LoRA) has gained popularity as a fine-tuning approach for Large Language Models (LLMs) due to its low resource requirements and good performance. While numerous studies have investigated improving LoRA serving efficiency by serving multiple LoRAs concurrently, existing methods assume that a wide range of LoRA adapters are available for serving. In our work, we conduct extensive empirical studies to show that current LoRA training paradigms do not efficiently utilize hardware resources and incur high overhead to obtain a performant LoRA adapter. Leveraging these insights, we propose PLoRA, which automatically orchestrates concurrent LoRA fine-tuning jobs under given hardware and model constraints and develops performant kernels to improve training efficiency. Across a range of LLMs and LoRA configurations, PLoRA improves training throughput by up to 12.8x and reduces the overall fine-tuning makespan by up to 7.52x compared to existing approaches.

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Supervised Classification Heads as Semantic Prototypes: Unlocking Vision-Language Alignment via Weight Recycling

David Méndez, Roberto Confalonieri, Natalia Díaz-Rodríguez

Vision-Language Models (VLMs) excel at tasks like zero-shot classification and cross-modal retrieval by mapping images and text to a shared space, but this requires expensive end-to-end training with massive paired datasets. Current post-hoc alignment methods reduce computational costs by connecting pretrained encoders through lightweight mappings, yet still demand substantial paired data.

In this work, we investigate the potential of repurposing the classification heads of pretrained vision models as semantic prototypes. The recycling of these weights, typically discarded after pretraining, unlocks two distinct capabilities: it enables zero-shot alignment by using weights as semantic anchors, and serves as a robust data augmentation strategy by mixing these prototypes with real image-text pairs. We demonstrate that integrating our approach with several state-of-the-art post-hoc alignment techniques consistently boosts accuracy in cross-modal retrieval, zero- and few-shot classification tasks.

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Adversarial Attack and Defense for Denoising Diffusion Sampling

Zhao-Rong Lai, Xiwen Yuan, Jian Weng

Denoising diffusion sampling (DDS) is an emerging approach for generating new samples that have the same distribution as some training samples. However, it is vulnerable to adversarial attacks by even a Gaussian perturbation. In this work, we propose a complete set of adversarial attack and defense methodology for DDS. In the attack side, we propose to inject a perturbation to the sampling stage, which significantly worsen the performance of sample generation. In the defense side, we propose a local variation based regularization model for the potential function minimization, which effectively tolerates the adversarial perturbations. Moreover, we develop a conjugate gradient algorithm to solve the defense model, which integrates with a recently-developed zeroth order rejection sampling method that saves computational cost. Experimental results show that the proposed attack significantly worsen the existing state-of-the-art methods, but can be defended by the proposed local variation regularization.

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Proactive Defense Benchmark against Deepfake Generation

Joonhyuk Baek, Wonjune Seo, Jae-yun Kim, Saerom Park, Hoki Kim

Despite the proliferation of proactive defenses against deepfakes, the lack of a unified evaluation protocol precludes fair comparison and masks critical vulnerabilities. To bridge this gap, we present the first comprehensive benchmark that systematically assesses disruption, robustness, and transferability encompassin

g pixel, perceptual, and identity metrics. Our extensive analysis reveals that f idelity and identity metrics capture orthogonal performance axes, often leading to conflicting interpretations when relied upon individually. Furthermore, we id entify a fundamental trade-off where peak white-box performance signals overfitt ing, and we introduce a calibrated evaluation to correct generator-induced ident ity bias. By exposing these blind spots, we establish a rigorous standard to gui de the development of genuinely generalizable protections. Project page is avail able at: https://proactivedefensebenchmark.github.io/
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SpatialReward: Bridging the Perception Gap in Online RL for Image Editing via Ex plicit Spatial Reasoning

Yancheng Long, Yankai Yang, Hongyang Wei, Wei Chen, Tianke Zhang, Haonan fan, Ch angyi Liu, Kaiyu Jiang, Jiankang Chen, Kaiyu Tang, Bin Wen, Fan Yang, Tingting G ao, Han Li, Shuo Yang

Online Reinforcement Learning (RL) offers a promising avenue for complex image e diting but is currently constrained by the scarcity of reliable and fine-grained reward signals. Existing evaluators frequently struggle with a critical percept ion gap we term "Attention Collapse," where models neglect cross-image compariso ns and fail to capture fine-grained details, resulting in inaccurate perception and miscalibrated scores. To address these limitations, we propose **SpatialRewa rd**, a reward model that enforces precise verification via explicit spatial rea soning. By anchoring reasoning to predicted edit regions, SpatialReward grounds semantic judgments in pixel-level evidence, significantly enhancing evaluative a ccuracy. Trained on a curated 260k spatial-aware dataset, our model achieves sta te-of-the-art performance on MMRB2 and EditReward-Bench, and outperforms proprie tary evaluators on our proposed **MultiEditReward-Bench**. Furthermore, SpatialR eward serves as a robust signal in online RL, boosting OmniGen2 by +0.90 on GEdi t-Bench—surpassing the leading discriminative model and doubling the gain of GPT -4.1 (+0.45). These results demonstrate that spatial reasoning is essential for unlocking effective alignment in image editing.
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Walrus: A Cross-domain Foundation Model for Continuum Dynamics

Michael McCabe, Payel Mukhopadhyay, Tanya Marwah, Bruno Régaldo-Saint Blancard, François Rozet, Cristiana Diaconu, Lucas Thibaut Meyer, Kaze W. K. Wong, Mohamma d-Hadi Sotoudeh, Alberto Bietti, Irina Espejo Morales, Rio Alexa Fear, Siavash G olkar, Tom Hehir, Keiya Hirashima, Geraud Krawezik, Francois Lanusse, Rudy Morel , Ruben Ohana, Liam Holden Parker, Mariel Pettee, Jeff Shen, Kyunghyun Cho, Mile s Cranmer, Shirley Ho

Foundation models have transformed machine learning for language and vision, but achieving comparable impact in physical simulation remains a challenge. Data he terogeneity and unstable long-term dynamics inhibit learning from sufficiently d iverse dynamics, while varying resolutions and dimensionalities challenge effici ent training on modern hardware. Through empirical and theoretical analysis, we incorporate new approaches to mitigate these obstacles, including a harmonic-ana lysis–based stabilization method, load-balanced distributed 2D-3D training strat egies, and compute-adaptive tokenization. Using these tools, we develop Walrus, a transformer-based foundation model developed primarily for fluid-like continuu m dynamics. Walrus is pretrained on nineteen diverse scenarios spanning astrophy sics, geoscience, rheology, plasma physics, acoustics, and classical fluids. Exp eriments show that Walrus outperforms prior foundation models on both short- and long-term prediction horizons on downstream tasks and across the breadth of pre training data, while ablation studies confirm the value of our contributions to forecast stability, training throughput, and transfer performance over conventio nal approaches.
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Quadratically Regularized Optimal Transport: Localization Bounds and Affine Case Analysis

Long Nguyen Chi, Nam Nguyen, Binh Nguyen

Quadratically regularization has emerged as a potential alternative to the popul ar entropic regularization in computational optimal transport, offering the theo

retical advantage of producing sparse couplings through its hinge density structure. Despite recent progress in one-dimensional setting and general upper bounds, fundamental questions about the localization rate of QOT optimizers around the Monge coupling have remained open. In this work, we establish a general lower bound showing that the support of the QOT optimizer cannot concentrate around the Monge graph faster than order $\varepsilon^{\frac{1}{d+2}}$ in the directed Hausdorff distance, matching the conjectured optimal exponent under standard regularity assumptions in Wiesel & Xu (2025). We also show that the QOT value gap controls the mean-squared deviation $\mathbb E_{\pi_\varepsilon}||y-T(x)||^2$ by the scale of $\varepsilon^{\frac{2}{d+2}}$. As a corollary, in the affine Brenier regime, which includes Gaussian-to-Gaussian transport, we derive a sharp pointwise tube bound of order $\varepsilon^{\frac{1}{d+2}}$ by reducing the problem to self-transport and applying recent self-transport sparsity results. Finally, we validate our theoretical bound with synthetic experiment in high dimensions setting.

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ML-Agent: Reinforcing LLM Agents for Autonomous Machine Learning Engineering

Zexi Liu, Jingyi Chai, Xinyu Zhu, Shuo Tang, Rui Ye, Weiyu Ma, Bo Zhang, LEI BAI, Siheng Chen

The emergence of large language model (LLM)-based agents has significantly advanced the development of autonomous machine learning (ML) engineering. However, the dominant prompt-based paradigm exhibits limitations: smaller models lack the capacity to learn from execution trajectories for generalization, while large proprietary models incur high computational overhead, restricting accessibility and scalability. Focusing on this, for the first time, we explore the paradigm of learning-based agentic ML, where an LLM agent learns through interactive experimentation on ML tasks using online reinforcement learning (RL). To realize this, we propose a novel agentic ML training framework with three key components: (1) exploration-enriched fine-tuning, which enables LLM agents to generate diverse actions for enhanced RL exploration; (2) step-wise RL, which enables training on a single action step, accelerating experience collection and improving training efficiency; (3) an agentic ML-specific reward module, which unifies varied ML feedback signals into consistent rewards for RL optimization. Leveraging this framework, we train ML-Agent, driven by a 7B-sized Qwen-2.5 LLM for autonomous ML. Despite training on only 9 ML tasks, our 7B-sized ML-Agent achieves comparable performance to agents using much larger proprietary LLMs (e.g., GPT-5) but at significantly lower computational cost, demonstrating strong performance and cross-task generalization.

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SAGE-NAS: Synergizing LLM-Based Semantic Agent with Graph-Based Evaluator for Neural Architecture Search

Kaiqi Lin, Jianping Luo

While LLM-driven Neural Architecture Search (NAS) leverages exceptional code generation and reasoning, it suffers from a critical "Semantic-Physical Misalignment": LLMs prioritize high-level semantic plausibility but are agnostic to intrinsic physical dynamics such as gradient flow, whereas Zero-Cost Proxies (ZCPs) capture these local sensitivities but lack global semantic planning. To bridge this gap, we propose SAGE-NAS, a closed-loop evolutionary framework that synergizes an LLM-Based Semantic Agent with a Graph-Based Evaluator. Specifically, SAGE-NAS coordinates an LLM-driven Semantic Agent to construct candidate architectures by dynamically scheduling complementary sub-policies that balance exploitation with exploration. Furthermore, the framework integrates a Dual-Modality Graph Evaluator that serves as a rapid performance predictor by fusing ZCP statistics with topological features, and a State-Aware Behavioral Atlas that guides sparsity-driven exploration to escape local optima. Experiments demonstrate that SAGE-NAS achieves state-of-the-art performance across multiple mainstream search spaces and downstream tasks, exhibiting a superior balance between search efficiency, model accuracy, and cross-task generalization capability.

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Inference-time optimization for experiment-grounded protein ensemble generation

Sai Advaith Maddipatla, Anar Rzayev, Marco Pegoraro, Martin Pacesa, Paul Schanda, Ailie Marx, Sanketh Vedula, Alexander Bronstein
Protein function relies on dynamic conformational ensembles, yet current generative models like AlphaFold3 (AF3) often fail to produce ensembles that match experimental data. Recent experiment-guided generators attempt to address this by steering the reverse diffusion process. However, these methods are limited by fixed sampling horizons and sensitivity to initialization, often yielding thermodynamically implausible results.

We introduce a general inference-time optimization framework to solve these challenges. First, we optimize over latent representations to maximize ensemble log-likelihood, rather than perturbing structures post hoc. This approach eliminates dependence on diffusion length, removes initialization bias, and easily incorporates external constraints. Second, we present novel sampling schemes for drawing Boltzmann-weighted ensembles. By combining structural priors from AF3 with force-field–based priors, we sample from their product distribution while balancing experimental likelihoods.

Our results show that this framework consistently outperforms state-of-the-art guidance, improving diversity, physical energy, and agreement with data in X-ray crystallography and NMR, sometimes fitting the experimental data better than deposited PDB structures. Finally, inference-time optimization experiments maximizing iPTM scores reveal that perturbing MSA embeddings can artificially inflate model confidence. This exposes a vulnerability in current design metrics, whose mitigation could offer a pathway to reduce false discovery rates in binder engineering.
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Compass-RoPE: Isotropic Rotary Position Embeddings for Vision Transformers
Chengxi Min, Wei Wang, Yao Zhao
Recent works introduce Rotary Position Embeddings (RoPE) into vision transformers (ViTs) to enhance their extrapolation capability, i.e., maintaining performance when inference is conducted on higher resolution images. RoPE encodes positions via rotating phases whose change is controlled by frequency components.
Strandard 2D RoPE does not generalize well to input resolution changes as it only applies axial frequencies separately along each individual axis. To solve this issue, Mix-RoPE combines xy■axis frequencies, such that it can model position relations in diagonal direction.
However, in practice, we observe that the learned 2D frequencies become anisotropic in their direction distributions due to the axial spectral bias in image features, limiting the extrapolation ability of ViTs.

Motivated by this observation, we propose Compass■RoPE. We replace the xy cartesian coordinates with a polar parameterization that explicitly decouples frequency scale and angle. By initializing the angle vectors uniformly over $[0,2\pi)$, it ensures the isotropic direction coverage. Besides, we further introduce discrete Fourier transform (DFT) mixing for the angle vectors, allowing each transformed individual angle vector element to nest multipule angles and thus to enrich angular expressiveness.
Extensive experiments on multi-resolution classification and dense prediction tasks show that our Compass-RoPE achieves more stable extrapolation performance under large-scale resolution changes.
****
Stabilizing Native Low-Rank LLM Pretraining
Paul Janson, Edouard Oyallon, Eugene Belilovsky
Foundation models have achieved remarkable success, yet their growing parameter counts pose significant computational and memory challenges. Low-rank factorization offers a promising route to reduce training and inference costs, but the community lacks a stable recipe for training models from scratch using exclusively low-rank weights while matching performance of the dense model.

We demonstrate that Large Language Models (LLMs) can be trained from scratch using exclusively low-rank factorized weights for all non-embedding matrices without auxiliary "full-rank" guidance required by prior methods. While native low-rank training often suffers from instability and loss spikes, we identify uncontrolled growth in the spectral norm (largest singular value) of the weight matrix update as the dominant factor. To address this, we introduce **Spectron: Spectr**al renormalization with orthogonalizati**on**, which dynamically bounds the resultant weight updates based on the current spectral norms of the factors. Our method enables stable, end-to-end factorized training with negligible overhead. Finally, we establish compute-optimal scaling laws for natively low-rank transformers, demonstrating predictable power-law behavior and improved inference efficiency relative to dense models.

Our code is available at https://github.com/Pauljanson002/spectron
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NanoSpec: Accelerating Speculative Decoding using Minimalist In-Context Vocabularies
Zhiyang Chen, Daliang Xu, Yinyuan Zhang, Chenghua Wang, Mengwei Xu, Yun Ma
The massive vocabulary sizes of large language models, often exceeding 100k tokens, impose a computational bottleneck on the final linear projection layer during speculative decoding. Existing vocabulary pruning solutions rely on static or coarsely-grained sub-vocabularies that necessitate large active sizes ($\sim$30k) to maintain draft quality. We propose NanoSpec, a novel training-free approach that breaks this trade-off by dynamically constructing a minimalist, context-aware active vocabulary for each generation step. Leveraging the inherent temporal locality of language generation, NanoSpec achieves high coverage while slashing the average vocabulary size by over $40\times$ (to $<$3k tokens) without requiring any auxiliary trained parameters. To realize the theoretical benefits of such high sparsity on modern hardware, we introduce a system-algorithm co-design that overcomes the inefficiencies of sparse memory access through asynchronous gathering and GPU-resident state management. As a complementary plug-and-play module, NanoSpec cuts draft time by an average of 51.6\%, delivering a 1.17-$1.29\times$ end-to-end speedup over the state-of-the-art speculative decoding methods EAGLE-2 and EAGLE-3 across 7 tasks and outperforming complex training-based pruning baselines.
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Semantic Editing with Coupled Stochastic Differential Equations
Jianxin Zhang, Clayton Scott
Editing the content of an image with a pretrained text-to-image model remains challenging. Existing methods often distort fine details or introduce unintended artifacts. We propose using coupled stochastic differential equations (coupled SDEs) to guide the sampling process of any pre-trained generative model that can be sampled by solving an SDE, including diffusion and rectified flow models. By driving both the source image and the edited image with the same correlated noise, our approach steers new samples toward the desired semantics while preserving visual similarity to the source. The method works out-of-the-box—without retraining or auxiliary networks—and achieves high prompt fidelity along with near-pixel-level consistency. These results position coupled SDEs as a simple yet powerful tool for controlled generative AI.
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LEGO: An LLM-Enabled Hierarchical Optimizer for Tensor Computation Graphs with Structure-Aware Search and Compositional Synthesis
Ruiyuan Xu, Shuoming Zhang, Guangli Li, Qiuchu Yu, Rui Zhang, Yangyu Zhang, Hao Qian, Chunwei Xia, Jiacheng Zhao, Chenxi Wang, Xiaobing Feng, Jingling Xue, Huimin Cui
Automating end-to-end GPU kernel generation with Large Language Models (LLMs) faces a critical tension between global performance and exploration efficiency. We present LEGO, a hierarchical framework that resolves this trade-off via a parallel multi-agent search over a recursive AND-OR FusionTree. LEGO synergizes two c

omplementary flows: Top-Down Construction decomposes complex graphs into valid, context-isolated sub-problems to guarantee correctness and enable parallel exploration, while Bottom-Up Mutation speculatively fuses verified sub-plans to recover global locality for peak performance. This bi-directional mechanism effectively prunes the search space to avoid repetitive unguided sampling, while naturally parallelizing exploration and enabling the discovery of sophisticated fusion strategies. Evaluations demonstrate that LEGO achieves 2.18×–13.48× speedups over PyTorch Eager and, compared to monolithic baselines, reduces end-to-end exploration time by up to 2.47× and token consumption by up to 7×, across diverse end-to-end models.

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Video2GUI: Synthesizing Large-Scale Interaction Trajectories for Generalized GUI Agent Pretraining

Weimin Xiong, Shuhao Gu, Bowen Ye, Zihao Yue, Lei Li, Feifan Song, Sujian Li, Hao Tian

Recent advances in multimodal large language models have driven growing interest in graphical user interface (GUI) agents, yet their generalization remains constrained by the scarcity of large-scale training data spanning diverse real-world applications. Existing datasets rely heavily on costly manual annotations and are typically confined to narrow domains.

To address this challenge, we propose Video2GUI, a fully automated framework that extracts grounded GUI interaction trajectories directly from unlabeled Internet videos. Video2GUI employs a coarse-to-fine filtering strategy to identify high-quality GUI tutorial videos and convert them into structured agent trajectories.

Applying this pipeline to 500 million video metadata entries, we construct WildGUI, a large-scale dataset containing 12 million interaction trajectories spanning over 1,500 applications and websites. Pre-training Qwen2.5-VL and Mimo-VL on WildGUI yields consistent improvements of 5–20\% across multiple GUI grounding and action benchmarks, matching or surpassing state-of-the-art performance.

We will release both the WildGUI dataset and the Video2GUI pipeline to support future research of GUI agents.

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“very likely” Means “uncertain”? How LLMs Diverge from Humans in Linguistic Uncertainty Quantification

Jinhao Duan, Zicheng Liu, Zijie Liu, Kaidi Xu, Tianlong Chen

Humans express uncertainty verbally via markers (e.g., "possible", "likely"), yet most LLM uncertainty quantification (UQ) relies on costing likelihood- or consistency-based signals. From a cognitive perspective, accurate verbal uncertainty reflects metacognitive monitoring, representing knowledge boundaries ("knowing that you don't know") to support regulation and information seeking. In this paper, we investigate: How LLMs diverge from humans in verbal uncertainty quantification? Can verbal markers reliably quantify LLM uncertainty? We curate a corpus of human uncertainty markers from psychology and decision-science literature and benchmark LLMs against it. We observe that LLMs encode verbal uncertainty with numerical levels that differ substantially from those of humans. We then introduce VOCAL, a novel optimization-based algorithm that learns an optimal uncertainty profile over uncertainty markers directly from LLM outputs. By fitting a marker–uncertainty mapping to best explain empirical correctness, VOCALdiscovers how much probability mass each verbal marker should convey, rather than estimating uncertainty via repeated sampling. VOCAL enables a direct, marker-level comparison of confidence semantics between humans and LLMs, disentangling mismatch and revealing systematic confidence disparities in verbal expressions.

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Self-evolving LLM agents with in-distribution Optimization

Yudi Zhang, Meng Fang, Zhenfang Chen, Mykola Pechenizkiy

Large Language Models (LLMs) have recently emerged as powerful controllers for interactive agents in complex environments, yet training them to perform reliable long-horizon decision making remains a fundamental challenge. A key difficulty lies in credit assignment: agents often receive delayed rewards only at the end

of episodes. In this paper, we propose Q-Evolve, a self-evolving framework for LLM agents that unifies automatic process-reward labeling and policy learning within a principled in-distribution reinforcement learning paradigm. In each evolving iteration, our method learns an in-distribution critic from a hybrid off-policy dataset that combines expert demonstrations with agent-generated trajectories, stabilizing Bellman backups in sparse-reward settings via a weighted Implicit Q-Learning objective. The learned value function is then used to derive step-wise process rewards through advantage estimation, enabling dense and reliable supervision without environment backtracking or human annotation. Leveraging these signals, we perform behavior-proximal policy optimization that evolves the agent over the data used for process reward labeling, allowing iterative self-improvement without exacerbating distribution shift.
We evaluate our method on AlfWorld, WebShop, and ScienceWorld, showing Q-Evolve outperforms strong baselines in sample efficiency, robustness, and overall task performance. Our results demonstrate that stable agent self-evolution is achievable through the co-evolution of process-level supervision and policy, both grounded within a shared in-distribution learning loop.
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MEC: Machine-Learning-Assisted Generalized Entropy Calibration for Semi-Supervised Mean Estimation
Se Yoon Lee, Jae Kwang Kim
Obtaining high-quality labels is costly, whereas unlabeled covariates are often abundant, motivating semi-supervised inference methods with reliable uncertainty quantification. Prediction-powered inference (PPI) leverages a machine-learning predictor trained on a small labeled sample to improve efficiency, but it can lose efficiency under model misspecification and suffer from coverage distortions due to label reuse. We introduce Machine■Learning■Assisted Generalized Entropy Calibration (MEC), a cross■fitted, calibration■weighted variant of PPI. MEC improves efficiency by reweighting labeled samples to better align with the target population, using a principled calibration framework based on Bregman projections. This yields robustness to affine transformations of the predictor and relaxes requirements for validity by replacing conditions on raw prediction error with weaker projection■error conditions. As a result, MEC attains the semiparametric efficiency bound under weaker assumptions than existing PPI variants. Across simulations and a real■data application, MEC achieves near■nominal coverage and tighter confidence intervals than CF■PPI and vanilla PPI.
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Mode Seeking meets Mean Seeking for Fast Long Video Generation
Shengqu Cai, Weili Nie, Chao Liu, Julius Berner, Lvmin Zhang, Nanye Ma, Hansheng Chen, Maneesh Agrawala, Leonidas Guibas, Gordon Wetzstein, Arash Vahdat
Scaling video generation from seconds to minutes faces a critical bottleneck: while short-video data is abundant and high-fidelity, coherent long-form data is scarce and limited to narrow domains.
While multi-resolution image training works because higher resolution is largely an interpolation of the same underlying patch distribution, training across video lengths is fundamentally different: a longer video is an extrapolation that must invent new events and causal structure beyond the short-clip horizon.
To address this, we propose a training paradigm where Mode Seeking meets Mean Seeking, decoupling local fidelity from long-term coherence from a unified representation via a Decoupled Diffusion Transformer.
Our approach utilizes a global Flow Matching head trained via supervised learning on long videos to capture narrative structure, while simultaneously employing a local Distribution Matching head that aligns sliding windows to a frozen short-video teacher via a mode-seeking reverse-KL divergence.
This strategy enables the synthesis of minute-scale videos that learns long-range coherence and motions from limited long videos via supervised flow matching, while inheriting local realism by aligning every sliding-window segment of the student to a frozen short-video teacher.
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UniSVQ: 2-bit Unified Scalar-Vector Quantization

Haoyu Wang, Haiyan Zhao, Xingyu Yu, Zhangyang Yao, Xu Han, Zhiyuan Liu, Maosong Sun

Post-training quantization at the 2-bit level enables low-cost deployment and inference acceleration for large language models (LLMs). Scalar quantization (SQ) and vector quantization (VQ) are two primary quantization methods, however, the former suffers from significant performance degradation, and the latter incurs computational and storage overhead.

We propose UniSVQ, a unified 2-bit quantization framework that bridges scalar and vector quantization by parameterizing codewords as an affine transform of integer lattices.

This structure preserves compatibility with optimized integer kernels while retaining much of VQ's flexibility.

We further introduce a data-driven block-wise fine-tuning strategy to directly minimize quantization reconstruction error.

Extensive experiments across multiple LLM families and zero-shot benchmarks demonstrate that UniSVQ consistently outperforms state-of-the-art SQ methods and achieves performance comparable to advanced VQ methods, while providing higher inference throughput.

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Large Language Model Agents Are Not Always Faithful Self-Evolvers

Weixiang Zhao, Yingshuo Wang, Yichen Zhang, Yang Deng, Yanyan Zhao, Wanxiang Che, Bing Qin, Ting Liu

Self-evolving large language model (LLM) agents continually improve by accumulating and reusing past experience, yet it remains unclear whether they faithfully rely on that experience to guide their behavior. We present the first systematic investigation of \emph{experience faithfulness}—the causal dependence of an agent's decisions on the experience it is given—in self-evolving LLM agents. Using controlled causal interventions on both raw and condensed forms of experience, we comprehensively evaluate four representative frameworks across 13 LLM backbones and 9 environments. Our analysis uncovers a striking asymmetry: while agents consistently depend on raw experience, they often disregard or misinterpret condensed experience, even when it is the only experience provided. This gap persists across single- and multi-agent configurations and across backbone scales. We trace its underlying causes to three factors: the semantic limitations of condensed content, internal processing biases that suppress experience, and task regimes where pretrained priors already suffice. These findings challenge prevailing assumptions about self-evolving methods and underscore the need for more faithful and reliable approaches to experience integration.

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TABX: A High-Throughput Sandbox Battle Simulator for Multi-Agent Reinforcement Learning

Hayeong Lee, JunHyeok Oh, Byung-Jun Lee

The design of environments plays a critical role in shaping the development and evaluation of cooperative multi-agent reinforcement learning (MARL) algorithms. While existing benchmarks highlight critical challenges, they often lack the modularity required to design custom evaluation scenarios. We introduce the Totally Accelerated Battle Simulator in JAX (TABX), a high-throughput sandbox designed for reconfigurable multi-agent tasks. TABX provides granular control over environmental parameters, permitting a systematic investigation into emergent agent behaviors and algorithmic trade-offs across a diverse spectrum of task complexities. Leveraging JAX for hardware-accelerated execution on GPUs, TABX enables massive parallelization and significantly reduces computational overhead. By providing a fast, extensible, and easily customized framework, TABX facilitates the study of MARL agents in complex structured domains and serves as a scalable foundation for future research. Our code is available at: https://anonymous.4open.science/r/TABX-00CA.

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Beyond Majority Voting: Self-Reflective Test-Time Reinforcement Learning for LLM Reasoning

Sitong Wu, Haoru Tan, Xichen Zhang, Bin Xia, Shaofeng Zhang, Xiaojuan Qi, Bei Yu

, Jiaya Jia
The core challenge of Test-Time Reinforcement Learning (TTRL) lies in estimating rewards without access to ground-truth supervision.
Existing TTRL methods predominantly rely on majority voting to generate pseudo-labels, under the assumption that the most frequent answer among sampled trajectories is correct.
However, we observe that this assumption frequently breaks down in complex reasoning tasks, where correct solutions often constitute a logical minority.
As a result, rare yet correct trajectories are systematically undervalued by majority-voting-based approaches.
To address this limitation, we propose Self-Reflective Test-Time Reinforcement Learning (SR-TTRL), a novel framework that leverages self-reflective verification to produce high-fidelity pseudo-labels.
Specifically, given multiple sampled trajectories for a problem, SR-TTRL first groups trajectories according to their final answers and selects one representative from each group to form a candidate pool.
Each candidate trajectory is then summarized to preserve its core reasoning steps while reducing verbosity.
Finally, the model performs self-reflection over the candidate pool, critically evaluating and selecting the most plausible trajectory as the pseudo-label.
Empirically, SR-TTRL achieves substantially higher pseudo-label fidelity and sample efficiency than prior majority-voting-based TTRL methods.
Extensive experiments across diverse benchmarks and model families demonstrate that SR-TTRL consistently outperforms majority-voting baselines and significantly improves generalization to novel problems.
For example, SR-TTRL improves the Pass@1 accuracy of Qwen3-8B on AIME24 from 29.1 to 55.8 (a gain of +26.7), exceeding standard TTRL by an additional +9.1.
The code will be released at: https://github.com/JIA-Lab-research/SR-TTRL.
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Safety Generalization Under Distribution Shift in Safe Reinforcement Learning: A Diabetes Testbed
Minjae Kwon, Josephine Lamp, Lu Feng
Safe Reinforcement Learning (RL) algorithms are typically evaluated under fixed training conditions. We investigate whether training-time safety guarantees transfer to deployment under distribution shift, using diabetes management as a safety-critical testbed. We benchmark safe RL algorithms on a unified clinical simulator
and reveal a safety generalization gap: policies satisfying
constraints during training frequently violate safety requirements on unseen patients. We demonstrate that test-time shielding, which filters unsafe actions using learned dynamics models, effectively restores safety across algorithms and patient populations. Across eight safe RL algorithms, three diabetes types, and three age groups, shielding achieves Time-in-Range gains of 13--14\% for strong baselines such as PPO-Lag and CPO while reducing clinical risk index and glucose variability. Our simulator and benchmark provide a platform for studying safety under distribution shift in safety-critical control domains. Code is available at https://github.com/safe-autonomy-lab/GlucoSim and https://github.com/safe-autonomy-lab/GlucoAlg.
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How does information access affect LLM monitors' ability to detect sabotage?
Rauno Arike, Raja Mehta Moreno, Rohan Subramani, Shubhorup Biswas, Francis Rhys Ward
Frontier language model agents can exhibit misaligned behaviors, including deception, exploiting reward hacks, and pursuing hidden objectives. To control such agents, we can use LLMs themselves to *monitor* for misbehavior. In this paper, we study how *information access* affects LLM monitor performance and find that contemporary systems often perform better with less information, a phenomenon we call the *less-is-more effect* for automated oversight. We demonstrate this phenomenon, and analyze the conditions under which it occurs, in three evaluation environments where agents must conduct sabotage while evading monitors. Leveraging

this effect, we introduce *extract-and-evaluate* (EaE) monitoring—a new hierarchical approach where one monitor isolates relevant excerpts from the monitored agent's trajectory and another monitor scores these excerpts. In BigCodeBench-Sabotage with a GPT-4.1-mini monitor, EaE improves sabotage detection rates by 16.8 percentage points over the next-best approach. In other settings, EaE either outperforms or is competitive with baselines. Additionally, we find that agents unaware of being monitored can be caught much more easily, that monitors scoring longer reasoning traces gain more from information filtering, and that monitor performance scales linearly with cost.
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Navigating the Flatlands: Dual Adaptive Sharpness-Aware Minimization for Domain Generalization
Junwen He, Yang He, Lebing Zheng, Zirui Yin, Hong-Yu Zhang, Yulong Wang
Finding flat minima in the loss landscape is a key strategy for Domain Generalization (DG). However, its effectiveness is often limited by two crucial challenges. 1) Domain Shift: Existing methods like Sharpness-Aware Minimization (SAM) apply a uniform optimization strategy across all domains, overlooking the differences of the learning difficulties among multiple domains and thus performing poorly on challenging domains. 2) Anisotropic Sharpness: By perturbing parameters along a single gradient direction, SAM and its variants ignore multi-directional flatness, making the model converge to minima that remain sharp in other directions. The combined challenges make it more difficult for the model to find truly robust solutions in multi-domain scenarios. To overcome these limitations, we propose the Dual Adaptive Sharpness-Aware Minimization (DA-SAM), which comprises two key modules: Dynamic Adaptive Scaling (DAS) module and Adaptive Multi-Directional Flattening (AMDF) module. First, to tackle the domain shift problem, the DAS module computes the real-time loss on each domain to adaptively generate domain-specific scaling factors that guide the generation of perturbation directions. Second, the AMDF module calculates local flatness by generating multiple directions to simulate perturbations in the parameter space. Based on the learned local flatness metric, it dynamically adjusts the perturbation step size to guide the model parameters to be away from anisotropic sharp regions. Crucially, DAS provides domain-level guidance that makes AMDF’s multi-directional geometric exploration more targeted and effective. Extensive experiments on five DG benchmarks demonstrate the effectiveness of our DA-SAM algorithm.
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ThunderAgent: A Fast, Simple, and Program-Aware Agentic Inference System
Hao Kang, Ziyang li, Weili Xu, Xinyu Yang, Yinfang Chen, Junxiong Wang, Beidi Chen, Tushar Krishna, Chenfeng Xu, Simran Arora
Large language models (LLMs) are now used to power complex multi-turn agentic workflows. Existing systems run agentic inference by loosely assembling isolated components: an LLM inference engine (e.g., vLLM) and a tool orchestrator (e.g., Kubernetes). Although agentic workflows involve multiple LLM and tool requests, these systems schedule and allocate resources separately on a per-request basis, without end-to-end knowledge of the workflow. This leads to sub-optimal management of KV cache and tool execution environments. To address the challenges, we propose **ThunderAgent**, a fast, simple, and program-aware agentic inference system. We first abstract agentic workflows as ***LLM Programs***, enabling a unified view of heterogeneous resources, including KV caches, system states, and external tool assets such as disk memory and network ports. Built upon this abstraction, ThunderAgent introduces a program-aware scheduler and a tool resource manager designed to maximize KV cache hit rates, mitigate memory imbalances, and enable asynchronous environment preparation. Evaluations across coding, routing, and scientific discovery agents demonstrate that ThunderAgent achieves **1.5-3.6×** throughput improvements in serving, **1.8-3.9×** in RL rollout, and up to **4.2×** disk memory savings compared to state-of-the-art inference systems. To facilitate reproducibility and support future development, we open-source the system implementations of ThunderAgent at: https://github.com/ThunderAgent-org/ThunderAgent
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Certifying Graph Neural Networks Against Label and Structure Poisoning
Lukas Gosch, Xichuan Chen, Yan Scholten, Stephan Günnemann
Robust machine learning for graph-structured data has made significant progress against test-time attacks, yet certified robustness to poisoning – where adversaries manipulate the training data – remains largely underexplored. For image data, state-of-the-art poisoning certificates rely on partitioning-and-aggregation schemes. However, we show that these methods fail when applied in the graph domain due to the inherent label and structure sparsity found in common graph datasets, making effective graph-partitioning difficult. To address this challenge, we propose a novel semi-supervised learning framework called deep Self-Training Graph Partition Aggregation (ST-GPA), which enriches each graph partition with informative pseudo-labels and synthetic edges, enabling effective certification against node-label and graph-structure poisoning under sparse conditions. Our method is architecture-agnostic, scales to large numbers of partitions, and consistently and significantly improves robustness guarantees against both label and structure poisoning across multiple benchmarks, while maintaining strong clean accuracy. Overall, our results establish a promising direction for certifiably robust learning on graph-structured data against poisoning under sparse conditions.
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Implicit Safety Alignment from Crowd Preferences
Qian Lin, Daniel S. Brown
Reinforcement Learning from Human Feedback (RLHF) can reveal implicit objectives such as safety considerations that go beyond task completion. In this work, we focus on the common safety criteria embedded in crowd preference datasets, where different users may express distinct preferences or objectives, yet follow similar safety principles. Our aim is to discover shared safety criteria from crowd preferences and then transfer them to downstream RL tasks to regularize agent behavior and enforce safety. We first show that direct reward combination—optimizing a preference-learned reward model together with downstream task rewards—has inherent limitations. Motivated by this, we propose Safe Crowd Preference-based RL, a hierarchical framework that extracts safety-aligned skills from crowd preferences and composes them via a high-level policy to safely solve downstream tasks. Experiments across safe RL environments and a preliminary LLM-style task with diverse user goals and shared safety constraints demonstrate that our approach substantially lowers safety costs without access to explicit safety rewards, while achieving task performance comparable to oracle methods trained with ground-truth safety signals.
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FedHera: Towards Drift-Resilient Federated Fine-tuning with Heterogeneous Resources
Ke Xiao, Qiyuan Wang, Christos Anagnostopoulos, Zhuoran Tan, Wenhao Li
Driven by the imperative to leverage privacy-sensitive data scattered across decentralized devices, federated fine-tuning has emerged as a vital paradigm for adapting large language models without compromising data privacy. Yet, its practical efficacy is bottlenecked by severe client resource heterogeneity. Existing truncation-based methods typically couple the transmitted rank with the trainable rank, which (i) under-utilizes bandwidth on communication-rich but compute-limited clients and (ii) exacerbates truncation-induced gradient drift. To address this, we propose FedHera, a resource-decoupled framework that explicitly differentiates information reception from gradient optimization. FedHera employs a spectrum-preserving allocation strategy to maximize the transfer of global knowledge (via high-rank singular values) within bandwidth limits, irrespective of training constraints. Furthermore, we introduce a prefix-gating mechanism that utilizes the downloaded high-capacity basis as a frozen reference to guide local updates, thereby minimizing the optimization gap caused by aggressive truncation. Extensive experiments under different heterogeneous settings show that FedHera improves stability and accuracy over state-of-the-art baselines.
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Frequency Matching in Spiking Neural Networks for mmWave Sensing
Zhenyu Liao, Di Yu, Changze Lv, Wentao Tong, Linshan Jiang, Sijie Ji, Xin Du, Ha

iliang Zhao, Xiaoqing Zheng, Shuiguang Deng
Millimeter-wave (mmWave) sensing enables privacy-preserving, always-on edge perc eption, but its measurements are often sparse, temporally irregular, and corrupt ed by high-frequency noise. Existing mmWave pipelines predominantly rely on arti ficial neural networks (ANNs), which achieve robustness through extensive prepro cessing or deep architectures, thereby limiting their efficiency on edge devices . In this work, we study spiking neural networks (SNNs) for mmWave sensing from a mechanism–data alignment perspective. By leveraging the low-pass filtering beh avior of leaky integrate-and-fire (LIF) dynamics, we analyze how their implicit temporal filtering interacts with the frequency structure of mmWave signals. Our analysis shows that when discriminative information resides in low-to-mid frequ encies, LIF dynamics can inherently suppress high-frequency noise, clarifying wh en and why SNNs outperform ANNs. Based on this insight, we derive a principled c riterion for configuring the membrane decay factor by matching the effective ban dwidth of LIF dynamics to the data’s discriminative spectral content. Experiment al results across four widely used mmWave datasets validate the proposed frequen cy-matching hypothesis, yielding an average test-accuracy improvement of 6.22% a nd a 3.64× reduction in theoretical energy consumption relative to ANN baselines , under a unified evaluation protocol.
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Mean Flow Distillation: Robust and Stable Distillation for Flow Matching Models
An Zhao, Shengyuan Zhang, Zhongjian Sun, Yixiang Zhou, Zejian Li, Ling Yang, Tia nrun Chen, Lingyun Sun
Flow Matching models have demonstrated strong performance across a wide range of generative tasks.
However, their reliance on ODE-based iterative sampling incurs substantial compu tational overhead, which limits their applicability in real-time scenes.
While distillation is a promising solution, existing approaches largely borrow f rom diffusion-based score matching, often failing to exploit the intrinsic geome tric structure of flows and suffering from training instability, high variance, and degraded generation quality.
In this paper, we propose Mean Flow Distillation (MFD), a novel distillation fra mework tailored for flow matching models.
We theoretically demonstrate that MFD acts as a temporal low-pass filter, effect ively suppressing the high-frequency optimization noise inherent in variational score distillation (VSD) while ensuring global trajectory consistency. We furthe r prove the Mean Flow Matching Theorem, establishing that matching expected aver age velocities is sufficient for strict distribution alignment. Empirically, on challenging high-dimensional manifolds including 4D occupancy forecasting and te xt-to-image generation, MFD achieves state-of-the-art performance, enabling high -fidelity single-step generation.
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Disentangling Geometry, Performance, and Training in Language Models
Atharva Kulkarni, Jacob Mitchell Springer, Arjun Subramonian, Swabha Swayamdipta
Geometric properties of Transformer weights, particularly the unembedding matrix , have been widely useful in language model interpretability research.
Yet, their utility for estimating downstream performance remains unclear.
In this work, we systematically investigate the relationship between model perfo rmance and the unembedding matrix geometry, particularly its effective rank.
Our experiments, involving a suite of 108 OLMo-style language models trained und er controlled variation, reveal several key findings.
While the best-performing models often exhibit a high effective rank, this trend is not universal across tasks and training setups.
Contrary to prior work, we find that low effective rank does not cause late-stag e performance degradation in small models, but instead co-occurs with it; we fin d adversarial cases where low-rank models do not exhibit saturation.
Moreover, we show that effective rank is strongly influenced by pre-training hyp erparameters, such as batch size and weight decay, which in-turn affect the mode l's performance.
Lastly, extending our analysis to other geometric metrics and final-layer repres

entation, we find that these metrics are largely aligned, but none can reliably predict downstream performance.
Overall, our findings suggest that the model's geometry, as captured by existing metrics, primarily reflects training choices rather than performance.
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Budget-Constrained Step-Level Diffusion Caching
Mingkun Lei, Tong Zhao, Liangyu Yuan, Chi Zhang
Step-level caching accelerates diffusion models by exploiting temporal redundancy across denoising steps.
Existing methods make per-step cache decisions using threshold-based heuristics, without directly optimizing for final output quality.
As a result, their inference latency varies across inputs and is difficult to control at deployment.
In this work, we propose BudCache, which inverts this formulation: rather than letting per-step error thresholds dictate the runtime cost, we fix the compute budget in advance and search for the cache policy that best preserves the final output.
To tackle the combinatorial complexity of step selection, we combine Simulated Annealing with deterministic Hill Climbing. This offline search identifies high-quality cache policies within minutes and introduces no online search or thresholding overhead during inference.
When the compute budget is very tight, we further introduce cache-aware schedule alignment, which adapts the time discretization to the selected cache policy to reduce cache-induced trajectory mismatch.
Experiments on FLUX.1-dev and Wan2.1 show that BudCache achieves better generation quality than heuristic caching baselines under the same inference budgets.
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A Control-Theoretic View of Mamba on Stability and Robustness
Liang Cao, Weide Liu, Zhuo Chen, Yan Qin
Selective State Space Models (SSMs) such as Mamba have emerged as efficient alternatives to Transformers, achieving linear complexity through input-dependent parameterization. However, this selectivity transforms the system from linear time-invariant (LTI) to linear parameter-varying (LPV), where individually stable matrices can produce unbounded trajectories under switching. Existing work focuses on empirical performance, leaving global stability, robustness bounds, and practical certification unresolved. This paper develops a control-theoretic framework providing the first comprehensive stability and robustness analysis for selective SSMs. We prove BIBO stability by viewing selective scans as continuous-time LTI sampling and establish two-term robustness bounds with linear growth in sequence length. For general LPV systems, we provide common quadratic Lyapunov function conditions and develop algorithms to extract certificate constants directly from trained weights. These results bridge control theory and SSM architectures, enabling formal guarantees for safety-critical deployment.
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Understanding Generalization from Embedding Dimension and Distributional Convergence
Junjie Yu, Zhuoli Ouyang, Haotian Deng, Chen Wei, Wenxiao Ma, Jianyu Zhang, Zihan Deng, Quanying Liu
Deep neural networks often generalize well despite heavy over-parameterization, challenging classical parameter-based analyses.
We study generalization from a representation-centric perspective and analyze how the geometry of learned embeddings is associated with generalization performance for a fixed trained model.
We derive a post-hoc generalization bound that relates the gap between population risk and held-out empirical risk to two factors:
(i) the intrinsic dimension of the embedding, which determines the convergence rate of the empirical embedding distribution to its population counterpart in Wasserstein distance, and
(ii) the sensitivity of the downstream mapping from embeddings to predictions, characterized by Lipschitz constants.

Together, these provide a post-hoc explanation of generalization for trained models.
At the final embedding layer, architectural sensitivity disappears and the bound is dominated by embedding dimension, explaining its strong empirical correlation with generalization performance.
Experiments across architectures and datasets validate the theory and demonstrate the utility of embedding-based diagnostics.
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Breaking the Reversal Curse in Autoregressive Language Models via Identity Bridge
Xutao Ma, Yixiao Huang, Hanlin Zhu, Somayeh Sojoudi
Autoregressive large language models (LLMs) have achieved remarkable success in many complex tasks, yet they can still fail in very simple logical reasoning such as the "reversal curse" --- when trained on forward knowledge data of the form "$A \rightarrow B$" (e.g., Alice's husband is Bob), the model is unable to deduce the reversal knowledge "$B \leftarrow A$" (e.g., Bob's wife is Alice) during test. Extensive prior research suggests that this failure is an inherent, fundamental limit of autoregressive causal LLMs, indicating that these models tend to memorize factual-level knowledge rather than capture higher-level rules. In this paper, we challenge this view by showing that this seemingly fundamental limit can be mitigated by slightly tweaking the training data with a simple regularization data recipe called the Identity Bridge of the form "$A \to A$" (e.g., The name of Alice is Alice). Theoretically, we prove that under this recipe, even a one-layer transformer can break the reversal curse by analyzing the implicit bias of gradient descent. Empirically, we show that a 1B pretrained language model finetuned with the proposed data recipe achieves a 50\% success rate on reversal tasks, in stark contrast to a near-zero success rate when trained solely on forward-knowledge data. Our work provides a novel theoretical foundation for the reversal curse and offers a principled, low-cost path to encouraging LLMs to learn higher-level rules from data.
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Anchoring Self-Play for Code Repair
Caroline Choi, Zeyneb N. Kaya, Shirley Wu, Tengyu Ma, Tatsunori Hashimoto, Ludwig Schmidt
Code repair is an important capability for language models (LMs): given a buggy program and unit tests, an LM must produce a fixed program that passes the tests.
We aim to scale supervision for code repair by having an LM generate bug--fix tasks with unconstrained edits, using unit tests as the only verifier.
We propose generator-fixer self-play, in which a single model is trained with reinforcement learning to alternate between generating bugs and fixing them.
As the fixer improves, the generator adapts to produce increasingly difficult bugs, yielding an automatic curriculum.
However, because unit tests certify correctness but not realism, we find that the generator can drift from bugs encountered in practice, improving repair on self-generated bugs while degrading on real-world bugs.
We propose Anchored Self Play (ASP), which anchors self-play with a small reference set by (i) adding a code-embedding similarity reward to guide generation and (ii) mixing reference bugs into fixer training to prevent drift.
To reflect LM-assisted programming, where bugs come from humans, LMs, and human edits of LM code, we introduce BugSourceBench, a code repair benchmark spanning human-authored bugs, human-edited buggy LM code, and errors in LM-generated code.
Across bug sources, ASP achieves the best fix rates, improving average fix rate by $+25$% (relative) / $+7.2$ pp (absolute) over standard self-play, with gains on both LM-error bugs ($+100$% relative / $+11$ pp absolute) and human-authored bugs ($+7.1$% relative / $+3.4$ pp absolute).
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(Be Cautious!) Bio-Foundation Models Are Not Yet Robust to Biologically Plausible Perturbations and ML Transformations

Jinhao Duan, Ruichen Zhang, Gengwei Zhang, Huaizhi Qu, Jie Peng, Sijia Liu, Tianlong Chen

Though biological foundation models (Bio-FMs) have delivered strong performance across biomedical tasks, their robustness to small-but-real perturbations is underexplored. In this work, we ask: Are Bio-FMs robust for real-world use? What perturbations compromise their reliability? Our pilot study suggests that due to subtle biological data curation issues and common machine-learning (ML) processing choices, Bio-FMs suffer from two complementary perturbation sources: biologically plausible perturbations (capturing experimental corruptions and curation artifacts) and ML-induced transformations (capturing preprocessing, data augmentation, and embedding choices). Guided by this taxonomy, we design perturbation suites that mimic corruptions frequently encountered in biological experiments, and we systematically probe how transformations in the ML pipeline reshape model behavior. By conducting 2,128 experiments over 11 state-of-the-art Bio-FMs on 7 bio-tasks, we show that most Bio-FMs are vulnerable to both biological perturbations and ML transformations, revealing underappreciated robustness gaps that can directly translate into deployment risk. Interestingly, we find that subtle biological perturbations, which are often imperceptible to current measurement tools, can induce severe discrepancies in Bio-FM outputs and lead to critical failures, yet cryo-EM models (e.g., CryoDRGN) exhibit a surprising level of robustness even under worst-case perturbations. Our study for the first time surfaces critical failure modes and provides a principled perspective for evaluating the robustness of Bio-FMs.

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Pluralistic Leaderboards

Nika Haghtalab, Ariel D. Procaccia, Han Shao, Serena Lutong Wang, Kunhe Yang

Recent leaderboard-based evaluations of large language models aggregate user feedback by fitting a Bradley--Terry model to pairwise comparisons, producing a single global ranking based on a latent quality score. While appealing for its simplicity, this approach is incompatible with heterogeneous preferences: when LLMs are used across diverse tasks and use cases, users who favor fundamentally different model behaviors can be systematically misrepresented when collapsed into a single quality score. To address this issue, we study *pluralistic leaderboards* that aim to remain *stable* with respect to heterogeneous user populations. Drawing on ideas from social choice theory, we adapt the notion of *local stability*, which requires that no model outside the top-k positions is collectively preferred to the top-k set by more than $O(1/k)$ fraction of users. Building on techniques from the social choice literature, we design an alternative leaderboard mechanism that satisfies local stability while eliciting only $\widetilde{O}(k)$ pairwise comparisons per user, where k is the size of the prefix for which stability is guaranteed. Using data from LMArena, we show that standard Bradley--Terry aggregation can violate local stability in practice, whereas our method provides substantially stronger stability guarantees.

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Federated Multi-view Clustering for Remote Sensing Data

Renxiang Guan, Xiang Yang, Hao Yu, Siwei Wang, Suyuan Liu, Wenjing Yang, Jun-Jie Huang, Ao Li, Xinwang Liu, Yuhua Tang

The rapid expansion of remote sensing technology has generated massive amounts of unlabeled multi-view data distributed across different institutions. Analyzing this data presents significant challenges, as centralized processing incurs prohibitive communication costs and raises data privacy concerns. To address these issues, this paper proposes a novel deep federated multi-view clustering (MVC) framework tailored for remote sensing data. Unlike existing methods that transmit sensitive data features, our approach shares only privatized prototypes masked with adaptive noise, ensuring both communication efficiency and privacy preservation. First, we employ superpixel segmentation to reduce the spatial dimensionality of remote sensing data, lowering computational burdens. Furthermore, to resolve the inconsistency of cluster assignments across different clients, we design a co-occurrence structural alignment module that synchronizes local models. Finally, we incorporate a wasserstein prototype contrastive learning mechanism, whi

ch models clusters as distributions rather than points, to enhance global consis tency and robustness against data heterogeneity. Extensive experiments on four p ublic datasets demonstrate that our framework achieves superior clustering perfo rmance and efficiency compared to state-of-the-art methods.
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Balanced LoRA: Removing Parameter Invariance to Accelerate Convergence
Valérie Castin, Kimia Nadjahi, Pierre Ablin, Gabriel Peyré
Low-Rank Adaptation (LoRA) is the most widely adopted method for fine-tuning lar ge language models. Notably, LoRA is inherently overparameterized: multiple pair s of low-rank factors can yield the same adapted weight matrix. We show—both the oretically and empirically—that these pairs exhibit significantly different cond ition numbers. As a result, converging to different loss minimizers directly imp acts the convergence rate of LoRA.
Building on this observation, we introduce Balanced Low-Rank Adaptation (BaLoRA) , a variant of LoRA that projects iterates onto a balanced manifold. This manifo ld improves the conditioning of the loss landscape while preserving the adapted matrix. The projection step is computationally lightweight and integrates seamle ssly into existing fine-tuning pipelines.
Empirically, BaLoRA converges faster than standard LoRA and achieves superior pe rformance across a range of fine-tuning tasks.
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A Causal Decomposition Approach for Fair Contextual Multi-Armed Bandits
Jiajun Chen, Jin Tian, Christopher John Quinn
Counterfactual reasoning is one of the fundamental facets of human cognition, in volved in various tasks such as explanation, credit assignment, blame, and respo nsibility. It describes the queries what would have happened had some interventi on been performed given that something else, corresponding to Layer 3 of the Pea rl Causal Hierarchy. In this project, we examine a specific type of counterfact ual quantities, called counterfactual direct (Ctf-DE), indirect (Ctf-IE), and sp urious (Ctf-SE) effects for quantifying fairness in a sequential decision-making framework. Building on these measures, we formulate an online causally-fair lea rning problem with multiple long-term constraints and study it in both non-param etric contextual bandits and parametric logistic bandits settings. We achieve su blinear regret and violations bounds for both bandits settings with roundwise co unterfactual fairness constraints (that are a priori unknown) without Slater's c ondition. For logistic bandits, our method achieves $\mathcal{O}(1)$ per-round t ime complexity using an online mirror descent estimator, yielding an efficient a lgorithm.
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Dynamics Within Latent Chain-of-Thought: An Empirical Study of Causal Structure
Zirui Li, Xuefeng Bai, Kehai Chen, Yizhi LI, Jian Yang, Chenghua Lin, Min Zhang
Latent or continuous chain-of-thought methods replace explicit textual rationale s with a number of internal latent steps, but these intermediate computations ar e difficult to evaluate beyond correlation-based probes. In this paper, we view latent chain-of-thought as a manipulable causal process in representation space by modeling latent steps as variables in a structural causal model (SCM) and ana lyzing their effects through step-wise do-interventions.
We study two representative paradigms (i.e., Coconut and CODI) on both mathemati cal and general reasoning tasks to investigate three key questions: (1) which st eps are causally necessary for correctness and when answers become decodable ear ly;
(2) how influence propagates across steps and how this structure compares to exp licit CoT; and
(3) whether intermediate trajectories retain competing answer modes and how outp ut-level commitment differs from representational commitment across steps. We fi nd that latent-step budgets behave less like homogeneous extra depth and more li ke staged functionality with non-local routing, and we identify a persistent gap between early output bias and late representational commitment. These results m otivate mode-conditional and stability-aware analyses, together with correspondi ng training/decoding objectives, as more reliable tools for interpreting and imp

roving latent reasoning systems. Code is available at https://github.com/J1mL1/causal-latent-cot.
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Flexibility-Aware Geometric Latent Diffusion for Full-Atom Peptide Design
Dongjiang Niu, Xiaofeng Wang, Zhiqiang Wei, Zhen Li
Although peptides are well suited for flexible and shallow binding interfaces, their intrinsic flexibility induces a strongly coupled sequence–structure relationship that current fixed-geometry latent models cannot simultaneously model with conformational diversity and physical feasibility, ultimately limiting design quality. To overcome this bottleneck, PepFGLD is proposed as a receptor-conditioned, flexibility-aware framework for full-atom peptide design. The framework is motivated by a systematic analysis of existing limitations: geometry shifts driven by interfacial flexibility are not well captured by standard equivariant encoders; the static combination of sequence information and 3D geometry cannot represent their dynamic interactions; and diffusion models without timely geometric feedback tend to drift away from physically reasonable energy landscapes. In PepFGLD, FlexEGNN is used to improve the sensitivity of geometric representations to local flexibility, a coherent and adaptable latent conformational manifold is formed through bidirectional sequence–structure interaction and nonlinear latent mapping, and a time-dependent energy-guided diffusion mechanism is incorporated to balance exploration and convergence during diffusion so that sampling trajectories are continuously guided toward physically feasible full-atom structures. PepFGLD yields improved binding affinity and design success across multiple peptide design tasks.
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MultiHal: Multilingual Dataset for Knowledge-Graph Grounded Evaluation of LLM Hallucinations
Ernests Lavrinovics, Russa Biswas, Katja Hose, Johannes Bjerva
Large Language Models (LLMs) have inherent limitations of faithfulness and factuality, commonly referred to as hallucinations. Several benchmarks have been developed that provide a test bed for factuality evaluation within the context of English-centric datasets, while relying on supplementary informative context like web links or text passages but ignoring the available structured factual resources. To this end, Knowledge Graphs (KGs) have been identified as a useful aid for hallucination mitigation, as they provide a structured way to represent the facts about entities and their relations with minimal linguistic overhead. We bridge the lack of KG paths and multilinguality for factual language modeling within the existing hallucination evaluation benchmarks and propose a KG-based multilingual, multihop benchmark called MultiHal framed for generative text evaluation. As part of our data collection pipeline, we mined 140k KG-paths from open-domain KGs, from which we pruned noisy KG-paths, curating a high-quality subset of 25.9k. Our baseline evaluation shows an absolute scale improvement by approximately 0.12 to 0.36 points for the semantic similarity score, 0.16 to 0.36 for NLI entailment and 0.29 to 0.42 for hallucination detection in KG-RAG over vanilla QA across multiple languages and multiple models, demonstrating the potential of KG integration. We anticipate MultiHal will foster future research towards several graph-based hallucination mitigation and fact-checking tasks.
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L2G-NET: Local to Global Spectral Graph Neural Networks via Cauchy Factorizations
Samuel Fernandez, Eduardo Pavez, Antonio Ortega
Despite their theoretical advantages, spectral methods based on the graph Fourier transform (GFT) are seldom used in graph neural networks (GNNs) due to the cost of computing the eigenbasis and the lack of vertex-domain locality in the resulting representations. As a result, most GNNs rely on local approximations such as polynomial Laplacian filters or message passing, which limit their ability to model long-range dependencies. In this paper, we introduce an exact factorization of the GFT into operators acting on subgraphs, which are then combined via a sequence of Cauchy matrices. Building on this factorization, we propose a new class of spectral GNNs, termed L2G-Net (Local to Global Net). Unlike existing spec

tral methods, which are either fully global (when using the GFT) or local (when using polynomial filters), L2G-Net operates by processing the spectral representations of subgraphs and then combining them via structured matrices. Our algorithm avoids full eigendecompositions, exploiting graph topology to construct the factorization with quadratic complexity in the number of nodes, scaled by the maximum cut size between subgraphs. Experiments stressing long-range dependencies on large graphs show that L2G-Net scales to regimes out of reach for the standard GFT, and is competitive with state-of-the-art methods with orders of magnitude fewer learnable parameters.
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Can LLMs Reason Structurally? Benchmarking via the lens of Data Structures
Yu He, Yingxi Li, Colin White, Ellen Vitercik
Large language models (LLMs) are deployed on increasingly complex tasks that require multi-step decision-making. Understanding their algorithmic reasoning abilities is therefore crucial. However, we lack a diagnostic benchmark for evaluating these capabilities. We propose to use data structures as a principled lens: as fundamental building blocks of algorithms, they naturally probe structural reasoning—the ability to understand and manipulate relationships such as order, hierarchy, and connectivity that underpin algorithmic reasoning. We introduce DSR-Bench (Data Structure Reasoning Benchmark), spanning 20 data structures, 35 operations, and 4,140 problem instances. DSR-Bench features hierarchical task organization, fully automated generation and evaluation, and fine-grained diagnostics. Evaluating 13 state-of-the-art LLMs reveals critical limitations: the top-performing model achieves only 0.46/1 on challenging instances. Three auxiliary probes targeting more realistic usages expose further weaknesses: models perform poorly on spatial data and context-rich scenarios, and they struggle to reason over their own code.
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Instruction Lens Score: Your Instruction Contributes a Powerful Object Hallucination Detector for Multimodal Large Language Models
Runhe Lai, Xinhua Lu, Yanqi Wu, Jinlun Ye, Weijiang Yu, Ruixuan Wang
Multimodal large language models (MLLMs) have achieved remarkable progress, yet the object hallucination remains a critical challenge for reliable deployment. In this paper, we present an in-depth analysis of instruction token embeddings and reveal that they implicitly encode visual information while effectively filtering erroneous information introduced by misleading visual embeddings. Building on this insight, we propose the Instruction Lens Score (InsLen), which combines a Calibrated Local Score with a Context Consistency Score that measures context consistency of the object tokens. The proposed approach serves as a plug-and-play object hallucination detector without relying on auxiliary models or additional training. Extensive experiments across multiple benchmarks and diverse MLLM architectures demonstrate that InsLen consistently outperforms existing hallucination detection methods, highlighting its effectiveness and robustness. The code will be publicly available.
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Omitted Variable Bias in Language Models Under Distribution Shift
Victoria Lin, Louis-Philippe Morency, Eli Ben-Michael
Despite their impressive performance on a wide variety of tasks, modern language models remain susceptible to distribution shifts, exhibiting brittle behavior when evaluated on data that differs in distribution from their training data. In this paper, we describe how distribution shifts in language models can be separated into *observable* and *unobservable* components, and we discuss how established approaches for dealing with distribution shift address only the former. Importantly, we identify that the resulting *omitted variable bias* from unobserved variables can compromise both evaluation and optimization in language models. To address this challenge, we introduce a framework that maps the strength of the omitted variables to bounds on the *worst-case generalization performance* of language models under distribution shift. In empirical experiments, we show that using these bounds directly in language model evaluation and optimization provides more principled measures of out-of-distribution performance, improves true out

-of-distribution performance relative to standard distribution shift adjustment methods, and further enables inference about the strength of the omitted variables when target distribution labels are available.
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A Bi-metric Framework for Efficient Nearest Neighbor Search
Haike Xu, Sandeep Silwal, Piotr Indyk
We propose a new ``bi-metric'' framework for designing nearest neighbor data structures. Our framework assumes two dissimilarity functions: a ground-truth metric that is accurate but expensive to compute, and a proxy metric that is cheaper but less accurate. In both theory and practice, we show how to construct data structures using only the proxy metric such that the query procedure achieves the accuracy of the expensive metric, while only using a limited number of calls to both metrics. Our theoretical results instantiate this framework for two popular nearest neighbor search algorithms: DiskANN and Cover Tree. In both cases we show that, as long as the proxy metric used to construct the data structure approximates the ground-truth metric up to a bounded factor, our data structure achieves arbitrarily good approximation guarantees with respect to the ground-truth metric. On the empirical side, we apply the framework to the text retrieval problem with two dissimilarity functions evaluated by ML models with vastly different computational costs. We observe that for almost all the large data sets in the BEIR benchmark, our approach achieves a considerably better accuracy-efficiency tradeoff than the alternatives, such as retrieve-then-rerank.
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Zero-shot Active Mapping via Fused 360-BEV Representations and Vision–Language Models
Yuanze Wang, Dianxi Shi, Yuetian Wang, Shiming Song, Haikuo Peng, Chunping Qiu, Mengzhu Wang
Active mapping enables embodied agents to understand and interact in previously unseen environments. However, most methods struggle to achieve zero-shot generalization to large-scale scenes and lack support for language instructions. We propose a VLM-based active mapping method that achieves zero-shot mapping while facilitating language-driven human–agent interaction. First, we introduce a 360-BEV representation that integrates omnidirectional semantics with BEV-aligned geometric structure to enhance scene understanding. Second, we develop a candidate waypoint generation strategy that allows the VLM-driven agent to select informative 2D waypoints in image space and back-project them into executable metric actions in 3D space, enabling the VLM to plan in its strongest modality. Third, we design a VLM-based depth-first exploration agent that decomposes the scenes into explorable regions, selects informative waypoints within each region, and organizes them into a topological tree. The agent follows the depth-first exploration policy to achieve thorough coverage of large-scale scenes. Without task-specific training, our method outperforms the strongest baseline, improving coverage and AUC by approximately 13.25\% and 14.00\%, respectively, while enabling language-conditioned interaction.
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Discovering Implicit Large Language Model Alignment Objectives
Edward Chen, Sanmi Koyejo, Carlos Guestrin
Large language model (LLM) alignment relies on complex reward signals that often obscure the specific behaviors being incentivized, creating critical risks of misalignment and reward hacking. Existing interpretation methods typically rely on pre-defined rubrics, risking the omission of "unknown unknowns", or fail to identify objectives that comprehensively cover and are causal to the model behavior on some dataset. To address these limitations, we introduce Obj-Disco, a framework that automatically decomposes an alignment reward signal into a sparse, weighted combination of human-interpretable natural language objectives. Our approach utilizes an iterative greedy algorithm to analyze behavioral changes across training checkpoints, identifying and validating candidate objectives that best explain the residual reward signal. Extensive evaluations across diverse tasks, model sizes, and alignment algorithms demonstrate the framework's robustness. Experiments with popular open-source reward models show that the framework consiste

ntly captures > 90\% of reward behavior, a finding further corroborated by human evaluation. Additionally, a case study on alignment with an open-source reward model reveals that Obj-Disco can successfully identify latent misaligned incentives that emerge alongside intended behaviors. Our work provides a crucial tool for uncovering the implicit objectives in LLM alignment, paving the way for more transparent and safer AI development.

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Proact-VL: A Proactive VideoLLM for Real-Time AI Companions

Weicai Yan, Yuhong Dai, Qi Ran, Haodong Li, Wang Lin, Tao Jin, Xing Xie, Hao Liao, Jianxun Lian

Proactive and real-time interactive experiences are essential for human-like AI companions, yet face three key challenges: (1) achieving low-latency inference under continuous streaming inputs, (2) autonomously deciding when to respond, and (3) controlling both quality and quantity of generated content to meet real-time constraints. In this work, we instantiate AI companions through two gaming scenarios—commentator and guide—selected for their suitability for automatic evaluation. We introduce the Live Gaming Benchmark, a large-scale dataset with three representative scenarios: solo commentary, co-commentary, and user guidance, and present Proact-VL, a general framework that shapes multimodal language models into proactive, real-time interactive agents capable of human-like environment perception and interaction. Extensive experiments show Proact-VL achieves superior response latency and quality while maintaining strong video understanding capabilities, demonstrating its practicality for real-time interactive applications.

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Learning Decentralized LLM Collaboration with Multi-Agent Actor Critic

Shuo Liu, Tianle Chen, Ryan Amiri, Christopher Amato

Recent work has explored optimizing LLM collaboration through Multi-Agent Reinforcement Learning (MARL). However, most MARL fine-tuning approaches rely on predefined execution protocols, which often require centralized execution. Decentralized LLM collaboration is more appealing in practice, as agents can run inference in parallel with flexible deployments. Also, current approaches use Monte Carlo methods for fine-tuning, which suffer from high variance and thus require more samples to train effectively. Actor-critic methods are prevalent in MARL for dealing with these issues; thus, we developed Multi-Agent Actor-Critic (MAAC) methods to optimize decentralized LLM collaboration. In this paper, we analyze when and why these MAAC methods are beneficial. We propose 2 MAAC approaches, **CoLLM-CC** with a **C**entralized **C**ritic and **CoLLM-DC** with **D**ecentralized **C**ritics. Our experiments across writing, coding, and game-playing domains show that Monte Carlo methods and CoLLM-DC can achieve performance comparable to CoLLM-CC in short-horizon and dense-reward settings. However, they both underperform CoLLM-CC on long-horizon or sparse-reward tasks, where Monte Carlo methods require substantially more samples and CoLLM-DC struggles to converge.

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CMI-RewardBench: Evaluating Music Reward Models with Compositional Multimodal Instruction

Yinghao Ma, Haiwen Xia, Hewei Gao, Weixiong Chen, Ye Yuxin, Yuchen Yang, Sungkyun Chang, Mingshuo Ding, Yizhi LI, Ruibin Yuan, Simon Dixon, Emmanouil Benetos

While music generation models have evolved to handle complex multimodal inputs mixing text, lyrics, and reference audio, evaluation mechanisms have lagged behind, remaining fragmented and narrowly focused. In this paper, we bridge this critical gap by establishing a comprehensive ecosystem for Compositional Music Instruction (CMI) reward modeling, where the generated music may be conditioned on text descriptions, lyrics, and/or audio prompts. We first introduce CMIPref-Pseudo, a large-scale preference dataset comprising 110k pseudo-labeled samples, and CMI-Pref, a high-quality, human-annotated corpus tailored for fine-grained alignment tasks. To unify the evaluation landscape, we propose CMIRewardBench, a unified benchmark that evaluates music reward models on heterogeneous samples across musicality, text–music alignment, and compositional instruction alignment. Leveraging these resources, we develop CMI reward models (CMI-RMs), a parameter-efficient reward model family capable of processing heterogeneous inputs. We evaluate

their correlation with human judgments scores on Music Arena and CMI-Pref test set, as well as preference agreement on Music Arena and CMI-Pref. Additional analyses examine performance variation across factors such as annotators, annotation timing and confidence, music generation models, and audio length. Experiments demonstrate that CMI-RM not only correlates strongly with human judgments, but also enables effective inference-time scaling via topk filtering. Our work provides the necessary data, benchmarks, and models to advance aligned music generation.

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FOAM: Blocked State Folding for Memory-Efficient LLM Training

Ziqing Wen, Jiahuan Wang, Ping Luo, Dongsheng Li, Tao Sun

Large language models (LLMs) have demonstrated remarkable performance due to their large parameter counts and extensive training data. However, their scale leads to significant memory bottlenecks during training, especially when using memory-intensive optimizers like Adam. Existing memory-efficient approaches often rely on techniques such as singular value decomposition (SVD), projections, or weight freezing, which can introduce substantial computational overhead, require additional memory for projections, or degrade model performance. In this paper, we propose Folded Optimizer with Approximate Moment (FOAM), a method that compresses optimizer states by computing block-wise gradient means and incorporates a residual correction to recover lost information. Theoretically, FOAM achieves convergence rates equivalent to vanilla Adam under standard non-convex optimization settings. Empirically, FOAM eliminates up to 90\% of the memory overhead of optimizer states and accelerates convergence. Furthermore, FOAM is compatible with other memory-efficient optimizers, delivering performance and throughput that match or surpass both full-rank and existing memory-efficient baselines.

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QuantWear: Quantum-scale Wear Particle Detection for Jet Engine Diagnosis

Zheng Wang, Yanwen Wang, Tianyu Fang, Jiaxing Shen, Yisen Kang, Di Wu, Yuanqing Zheng

The quantity and 3-D shape of wear particles are essential indicators for assessing the health of jet engines, enabling early detection of potential damage and preventing accidents caused by catastrophic failures. However, capturing wear particles is difficult due to their minute sizes and ultra high-speed movement within intense jet flows. Existing technologies struggle with the extreme background noise and low resolution in such harsh environments. In this paper, we propose QuantWear, the first quantum sensing system designed to directly quantify and profile wear particles on the sub-millimeter scale. QuantWear innovatively tracks wear particles by monitoring the spectral signatures of Sodium (Na) and Potassium (K) atoms within jet flow, which naturally adhere to particle surfaces due to electrochemical reactions in high-temperature combustion. We construct a custom atomic detector that leverages quantum jump and Faraday rotation effects to isolate these specific atomic signals, effectively suppressing the broad-spectrum flame noise. Next, we apply a deep learning framework to effectively measure the quantity of wear particles in dynamic vaporous backgrounds. Finally, we generate a fully reconstructed 3-D model of the wear particles from multiple 2-D images. Extensive field tests and high-fidelity simulations demonstrate that QuantWear achieves an imaging Signal-to-Noise Ratio (SNR) of 22.5 dB and a 3-D reconstruction similarity exceeding 95%, significantly outperforming state-of-the-art technologies.

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Bayesian Meta-Learning with Expert Feedback for Task-Shift Adaptation through Causal Embeddings

Lotta Mäkinen, Jorge Loria, Samuel Kaski

Meta-learning methods perform well on new within-distribution tasks but often fail when adapting to out-of-distribution target tasks, where transfer from source tasks can induce negative transfer. We propose a causally-aware Bayesian meta-learning method, by conditioning task-specific priors on precomputed latent causal task embeddings, enabling transfer based on mechanistic similarity rather than spurious correlations. Our approach explicitly considers realistic deployment s

ettings where access to target-task data is limited, and adaptation relies on no isy (expert-provided) pairwise judgments of causal similarity between source and target tasks. We provide a theoretical analysis showing that conditioning on ca usal embeddings controls prior mismatch and mitigates negative transfer under ta sk shift. Empirically, we demonstrate reductions in negative transfer and improv ed out-of-distribution adaptation in controlled simulations and a real-world cli nical prediction setting for cross-disease transfer, where causal embeddings ali gn with underlying clinical mechanisms; we include the judgments from a medical expert in the clinical prediction task and obtain improved performance in predic tions of unseen diseases.

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Meta-learning Structure-Preserving Dynamics

Cheng Jing, Uvini Balasuriya Mudiyanselage, Woojin Cho, Minju Jo, Anthony Gruber , Kookjin Lee

Structure-preserving approaches to dynamics discovery have demonstrated great po tential for modeling physical systems due to their use of strong inductive biase s, which enforce key features such as conservation laws and dissipative behavior . However, these models are typically trained on a per-configuration basis, requ iring explicit knowledge of system parameters and costly retraining when these p arameters vary. While meta-learning provides a potential remedy, optimization-ba sed approaches can suffer from limited generalizability. Motivated by recent adv ances in modulation-based learning aimed at mitigating these drawbacks, we syste matically investigate the use of modulation techniques in learning conservative dynamical systems. We study a range of existing modulation strategies alongside newly proposed variants, integrating them into a Hamiltonian learning framework without requiring an explicit system parameterization. Through extensive experim ents on benchmark problems, we demonstrate that modulation-based meta-learning e nables accurate few-shot adaptation, achieving robust generalization across para meter space without compromising the conservation of key invariants responsible for the dynamics.

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Latent Guided Sampling for Combinatorial Optimization

Sobihan Surendran, Adeline Fermanian, Sylvain Le Corff

Combinatorial Optimization problems are widespread in domains such as logistics, manufacturing, and drug discovery, yet their NP-hard nature makes them computat ionally challenging. Recent Neural Combinatorial Optimization (NCO) methods leve rage deep learning to learn policies for constructing solutions, trained via Sup ervised or Reinforcement Learning. While promising, these approaches often rely on task-specific augmentations, perform poorly on out-of-distribution instances, and lack robust inference mechanisms. Moreover, existing latent space models ei ther require labeled data or use an instance-independent latent distribution. In this work, we propose LGS-Net, a novel latent space model that conditions on pr oblem instances, and introduce an efficient inference method, Latent Guided Samp ling (LGS), based on Markov Chain Monte Carlo and Stochastic Approximation. We s how that the iterations of our method form a time-inhomogeneous Markov Chain and provide rigorous theoretical convergence guarantees. Empirical results on bench mark routing tasks show that our method achieves state-of-the-art performance am ong NCO baselines.

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KnapSpec: Self-Speculative Decoding via Adaptive Layer Selection as a Knapsack P roblem

Seongjin Cha, Gyuwan Kim, Dongsu Han, Tao Yang, Insu Han

Self-speculative decoding (SSD) accelerates LLM inference by skipping layers to create an efficient draft model, yet existing methods often rely on static heuri stics that ignore the dynamic computational overhead of attention in long-contex t scenarios. We propose KnapSpec, a training-free framework that reformulates dr aft model selection as a knapsack problem to maximize tokens-per-time throughput . By decoupling Attention and MLP layers and modeling their hardware-specific la tencies as functions of context length, KnapSpec adaptively identifies optimal d raft configurations on the fly via a parallel dynamic programming algorithm. Fur

thermore, we provide the first rigorous theoretical analysis establishing cosine similarity between hidden states as a mathematically sound proxy for the token acceptance rate. This foundation allows our method to maintain high drafting faithfulness while navigating the shifting bottlenecks of real-world hardware. Our experiments on Qwen3 and Llama3 demonstrate that KnapSpec consistently outperforms state-of-the-art SSD baselines, achieving up to 1.47$\times$ wall-clock speedup across various benchmarks. Our plug-and-play approach ensures high-speed inference for long sequences without requiring additional training or compromising the target model's output distribution.
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Budget-Efficient Attacks and Robustness Training for Cooperative MARL
junyong jiang, Xin Yuan, Longhe Lin, Songze Li, Lu Dong
Cooperative multi-agent reinforcement learning (CMARL) policies are vulnerable to
action hijacking even when only a few timesteps are compromised.
Recent adversarial attacks and adversarial training methods have been explored,
but under an explicit attack budget, existing attacks often fail to accurately
expose critical coordination weaknesses and incur substantial training cost.
We propose Budgeted Hierarchical Efficient Attack (BHEA), a budgeted
hierarchical adversarial attack that separates decisions on when and which agents
to hijack from action replacement, enabling more precise vulnerability discovery
under limited attack opportunities.
We further show that training cooperative policies against BHEA substantially
improves robustness to limited-step action hijacking while reducing training
overhead.
Experiments on the StarCraft Multi-Agent Challenge (SMAC) demonstrate stronger attacks
under the same attack budget and improved robustness. Code is available at https://anonymous.4open.science/r/BHEA-068D.
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Learning Manifold and Itô Dynamics with Branched Neural Rough Differential Equations
Luke Thompson, Dai Shi, Lequan Lin, Junbin Gao, Andi Han
Neural rough differential equations (NRDEs) stay accurate under irregular sampling while taking far fewer integration steps than standard neural differential equations, summarising a finely sampled driver by its log-signature and advancing the hidden state over coarse intervals using the log-ODE method. This efficiency rests on the shuffle algebra, the algebraic counterpart of Stratonovich calculus. This reliance means NRDEs cannot expose the quadratic-variation terms Itô dynamics require, nor the ordered covariant derivatives that govern Itô flows on connection-equipped manifolds. Ameliorating this, we introduce Branched Neural Rough Differential Equations (B-NRDEs), a Hopf-algebraic framework that recasts the NRDE log-ODE step as geometric numerical integration on the state-space manifold, matching the driving algebra to the governing calculus: Grossman–Larson rooted trees for Euclidean Itô dynamics, Munthe–Kaas–Wright planar rooted trees for ordered covariant derivatives on manifolds, and the shuffle algebra in the classical Stratonovich case. This yields intrinsic coarse-step dynamics that exactly preserve manifold constraints. Finally, we introduce a branched signature-kernel objective to enable Itô-consistent law matching by making quadratic-variation terms visible during training. On rough Bergomi volatility, sim-to-real $\mathrm{SO}(3)$ forecasting, and SPD covariance dynamics, B-NRDEs offer a unified, effective approach to stochastic and manifold-valued dynamics beyond the Euclidean–Stratonovich setting.
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FedCDWA: Decoupled Federated Prototype Distillation with Hierarchical Wasserstein Aggregation
Zhenshen Liu, Kai Fan, Wenjie Li, Kuan Zhang, HUI LI, Yintang Yang
Federated learning enables decentralized clients to collaboratively train models without sharing local data. However, heterogeneous client distributions often i

nduce client drift and hinder convergence. This paper proposes FedCDWA, a decoup
led hierarchical federated distillation framework. FedCDWA decouples client-side
 personalized distillation from server-side mutual distillation to mitigate dist
illation-induced optimization conflicts. It further adopts Hierarchical Wasserst
ein Aggregation to aggregate prototypes without restrictive parametric assumptio
ns while preserving intra-class structure and inter-class geometry. To achieve f
iner-grained feature alignment, Prototype–Variance Dual Alignment matches featur
e means and variances in the feature space. We prove convergence guarantees for
FedCDWA. Experiments on three datasets demonstrate that FedCDWA consistently imp
roves both global and personalized accuracy across heterogeneity levels, with sm
aller performance degradation under more severe heterogeneity.
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Privasis: Synthesizing the Largest "Public" Private Dataset from Scratch

Hyunwoo Kim, Niloofar Mireshghallah, Michael Duan, Rui Xin, Shuyue Stella Li, Ja
ehun Jung, David Acuna, Qi Pang, Hanshen Xiao, G. Edward Suh, Sewoong Oh, Yulia
Tsvetkov, Pang Wei Koh, Yejin Choi

Research involving privacy-sensitive data has always been constrained by data sc
arcity, standing in sharp contrast to other areas that have benefited from data
scaling. To quench this thirst, we present Privasis (i.e., privacy oasis), the f
irst million-scale fully synthetic dataset entirely built from scratch—an expans
ive reservoir of texts with rich and diverse private information—designed to bro
aden and accelerate research in areas where processing sensitive social data is
inevitable. Compared to existing datasets, Privasis, comprising 1.4 million reco
rds, offers orders-of-magnitude larger scale with quality, and far greater diver
sity across various document types, including medical records, legal documents,
financial records, calendars, emails, meeting transcripts, and text-messages wit
h a total of 55.1 million annotated attributes such as ethnicity, date of birth,
 workplace, etc. We leverage Privasis to construct a parallel corpus for text sa
nitization with our pipeline that recursively decomposes texts and applies targe
ted sanitization. Our compact sanitization models ($\leq$
4B) trained on this dataset outperform state-of-the-art large language models, s
uch as GPT-5 and Qwen-3 235B.
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EmWorld: Emotion World Model with Latent State Evolution for Scenario-Incrementa
l Dynamic Facial Expression Recognition

ke Wang, Yuanyuan Liu, Kejun Liu, Yuyang Xia, Chang Tang, Yibing Zhan, Zhe Chen

Dynamic Facial Expression Recognition (DFER) models the temporal evolution of fa
cial expressions in videos. In real-world scenarios, changing scenarios distort
expression trajectories, challenging existing methods. Most current approaches a
ddress this via passive feature alignment or domain-incremental learning but do
not explicitly model scenario evolution, limiting their ability to capture expre
ssion dynamics under scenario-incremental changes. To address this, we propose *
*EmWorld**, an emotion world model for DFER that explicitly models latent emotio
n state evolution under scenario variations. Specifically, **EmWorld** formulate
s scenario-incremental DFER as a progressive Bayesian inference problem over lat
ent world states with dual temporal scales. Slow-timescale component (**STS**) m
odels scenario evolution using stochastic evolutionary priors, capturing long-te
rm scenario effects and providing proactive guidance in new scenarios. Fast-time
scale component (**FTS**) models frame-level expression dynamics with temporally
 consistent latent transitions, decoupling expression dynamics from scenario inf
luences. By jointly inferring latent states at both timescales, EmWorld shifts D
FER from a passive feature discrimination to active probabilistic state inferenc
e under evolving scenarios. Experiments on FERV39k, DFEW, and MAFW demonstrate t
hat EmWorld consistently outperforms state-of-the-art methods, achieving up to 3
.84\% improvement while exhibiting strong cross-scenario stability and long-term
 robustness.
****

The Silent Thought: Modeling Internal Cognition in Full-Duplex Spoken Dialogue M
odels via Latent Reasoning

Donghang Wu, Tianyu Zhang, Yuxin Li, Hexin Liu, Chen Chen, Eng Siong Chng, Yoshu

a Bengio
During conversational interactions, humans subconsciously engage in concurrent thinking while listening to a speaker. Although this internal cognitive processing may not always manifest as explicit linguistic structures, it is instrumental in formulating high-quality responses. Inspired by this cognitive phenomenon, we propose a novel **F**ull-duplex **LA**tent and **I**nternal **R**easoning method named FLAIR that conducts *latent* thinking simultaneously with speech perception. Unlike conventional "thinking" mechanisms in NLP, which require post-hoc generation, our approach aligns seamlessly with spoken dialogue systems: during the user’s speaking phase, it recursively feeds the latent embedding output from the previous step into the next step, enabling continuous reasoning that strictly adheres to causality without introducing additional latency. To enable this latent reasoning, we design an Evidence Lower Bound-based objective that supports efficient supervised finetuning via teacher forcing, circumventing the need for explicit reasoning annotations. Experiments demonstrate the effectiveness of this think-while-listening design, which achieves competitive results on a range of speech benchmarks. Furthermore, FLAIR robustly handles conversational dynamics and attains competitive performance on full-duplex interaction metrics.
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Enhancing Multi-Modal LLMs Reasoning via Difficulty-Aware Group Normalization
Jinghan Li, Junfeng Fang, Jinda Lu, Yuan Wang, Xiaoyan Guo, Tianyu Zhang, Xiang Wang, Xiangnan He
Reinforcement Learning with Verifiable Rewards (RLVR) and Group Relative Policy Optimization (GRPO) have significantly advanced the reasoning capabilities of large language models.
Extending these methods to multimodal settings, however, faces a critical challenge: the instability of std-based normalization, which is easily distorted by extreme samples with nearly positive or negative rewards.
Unlike pure-text LLMs, multimodal models are particularly sensitive to such distortions, as both perceptual and reasoning errors influence their responses.
To address this, we characterize each sample by its difficulty, defined through perceptual complexity (measured via visual entropy) and reasoning uncertainty (captured by model confidence).
Building on this characterization, we propose difficulty-aware group normalization (Durian), which re-groups samples by difficulty levels and shares the std within each group.
Our approach preserves GRPO's intra-group distinctions while eliminating sensitivity to extreme cases, yielding significant performance gains across multiple multimodal reasoning benchmarks.
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Robust Self-reflective Hashing for Cross-modal Retrieval with Noisy Label
Hao Sun, Qibing Qin, Lei Huang
Cross-modal Hashing (CMH) typically assumes a perfectly complete data annotation, whereas noisy labels are unavoidable in practical scenarios.
■Existing CMH methods often overlook the uncertainty introduced by noise or semantic ambiguity, making models susceptible to overfitting noisy labels and yielding unreliable similarity judgments during inference.
■To address this issue, we propose a Robust Self-reflective Hashing (RSH) framework that prudently analyzes semantic discrepancies while accounting for uncertainty, thereby effectively mitigating interference from noisy labels.
■Specifically, the Double Feature Representation (DFR) method is introduced, employing semantic and uncertainty features to represent the semantic representation and fuzziness of samples.
■With a double feature, we propose a novel cross-modal similarity metric - the Self-reflective Similarity Metric (SSM), which judges sample similarity by integrating semantic discrepancy and fuzziness, enabling the model to adaptively weaken semantic discrepancy according to uncertainty level.
■The proposed method is plug-and-play, enabling seamless integration into diverse objective functions to enhance model robustness and reliability.
■Extensive experiments on benchmark datasets demonstrate that RSH outperforms ex

isting methods. Code is available at https://github.com/QinLab-WFU/RSH.
****
Offline Reinforcement Learning of High-Quality Behaviors Under Robust Style Alig nment
Mathieu Petitbois, Rémy Portelas, Sylvain Lamprier
We study offline reinforcement learning of style-conditioned policies using expl icit style supervision via subtrajectory labeling functions. In this setting, al igning style with high task performance is particularly challenging due to distr ibution shift and inherent conflicts between style and reward. Existing methods, despite introducing numerous definitions of style, often fail to reconcile thes e objectives effectively. To address these challenges, we propose a unified defi nition of behavior style and instantiate it into a practical framework. Building on this, we introduce Style-Conditioned Implicit Q-Learning (SCIQL), which leve rages offline goal-conditioned RL techniques, such as hindsight relabeling and v alue learning, and combine it with a new Gated Advantage Weighted Regression mec hanism to efficiently optimize task performance while preserving style alignment . Experiments demonstrate that SCIQL achieves superior performance on both objec tives compared to prior offline methods. Code, datasets and visuals are availabl e in: https://mathieu-petitbois.github.io/projects/sciql/.
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CrossQ: Task-Aligned Cross-Token Conditional Quantization for Late Interaction R etrieval
Rohit Kumar Salla, Manoj Saravanan, Ramya Manasa Amancherla
Late-interaction retrievers like ColBERT achieve high quality but suffer from la rge multi-vector indices. Standard compression minimizes token reconstruction er ror, while ranking depends critically on preserving scores of sparse "winner" to kens. We introduce CrossQ, which adaptively improves effective token fidelity wi thin documents by conditioning token codes on lightweight document context compu ted at indexing time (but not stored). CrossQ is trained with ranking-aligned ob jectives that preserve candidate score distributions and protect hard-negative m argins. At 2 B/token, CrossQ improves MRR@10 by +0.010 over the strongest strict ly footprint-matched quantization baseline and by +0.012 over the strongest cand idate-matched system reference. On the nine-dataset BEIR subset reported in Appe ndix G.1, CrossQ improves average nDCG@10 by +0.009 at 4 B/token over the strong est candidate-matched system reference. At 4 B/token, CrossQ achieves
$64\times$ raw token-storage reduction, approximately
$61\times$ including metadata and approximately
$58\times$ under conservative padding/alignment accounting. At 8 B/token, CrossQ + light fine-tuning retains approximately 98% of full-precision ColBERT MRR@10, improving the footprint-quality tradeoff for memory-constrained late-interactio n retrieval.
****
Unified Multimodal Autoregressive Modeling with Shared Context—Visual Tokenizer is Key to Unification
Wujian Peng, Lingchen Meng, Yuxuan Cai, Xianwei Zhuang, Yuhuan Yang, Rongyao Fan g, Chenfei Wu, Junyang Lin, Zuxuan Wu, Shuai Bai
Unified Multimodal Modeling aims to integrate visual understanding and generatio n within a single system. However, existing approaches typically rely on two dis parate visual tokenizers, which splits the representation space and hinders trul y unified modeling. We propose UniAR, a unified autoregressive framework where a single discrete visual tokenizer serves as the key bridge between understanding and generation, enabling a shared context in which the model can directly inter pret its own generated visual tokens without additional re-encoding. UniAR adapt s a pretrained vision encoder with multi-level feature fusion and a lookup-free bitwise quantization scheme, preserving both high-level semantics and low-level details while scaling the effective visual vocabulary at minimal cost. Building on this, the unified autoregressive model adopts parallel-bitwise-prediction to jointly predict spatially grouped, multi-level visual codes, substantially reduc ing visual sequence length and accelerating generation. Finally, a diffusion-bas ed visual decoder operates on discrete visual tokens to decode high-fidelity ima

ges. Through large-scale pre-training, followed by supervised fine-tuning and re
inforcement learning, UniAR achieves state-of-the-art performance on image gener
ation and image editing while remaining competitive on multimodal understanding
benchmarks. The project homepage is at https://sharelab-sii.github.io/uniar-web.
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Temporal Difference Learning for Diffusion Models
Qizhen Ying, Yangchen Pan, Victor Adrian Prisacariu, Junfeng Wen
Diffusion models are typically trained with objectives that focus on local denoi
sing targets at individual time steps (or adjacent pairs), which do not enforce
consistency between predictions along the denoising trajectory. This lack of cro
ss-time consistency can degrade performance, especially for few-step samplers. W
e introduce a temporal difference (TD) objective that penalizes inconsistency of
 the model’s \emph{multi-step} progress along the denoising path. By reformulati
ng the diffusion process as a Markov reward process and casting denoising as a p
olicy evaluation problem in reinforcement learning, we derive a unified TD appro
ach that applies to both discrete- and continuous-time diffusion formulations. W
e further propose a principled sample-based reweighting method that stabilizes t
raining. Empirically, we show that using our TD training can significantly impro
ve sample quality measured by FID, with stronger advantages when the number of s
ampling steps is small, highlighting its practical utility under low-computation
-budget scenarios. We provide ablation studies to justify our design choices, in
cluding pairwise loss reweighting, regularization weight, and one-step stride. O
verall, our TD approach can be a general drop-in that enforces cross-time consis
tency and improves generation quality across different diffusion generative mode
ls.
****

A Graph Foundation Model with Cross-Modal Alignment and Modality-Aware Expert F
usion for Multi-Modal Graphs
Dongxiao He, Ankang Yang, Jitao Zhao, Di Jin
Graph Foundation Models (GFMs) aim to learn universal patterns through large-sca
le pretraining on diverse graphs and generalize to open-world scenarios. While G
FMs have garnered significant attention, existing works primarily focus on sigle
-modal graphs. However, many real-world graphs are multimodal, consisting of str
uctures alongside diverse features derived from modalities such as text and imag
es. To date, exploration into Multimodal Graph Foundation Models (MGFMs) remains
 limited. Incorporating multimodal data provides a more comprehensive view, allo
wing models to learn richer semantics, thereby advancing GFMs. We are therefore
motivated to explore MGFMs, where the core challenge lies in synergistically enc
oding structures and multimodal features to achieve effective cross-modal alignm
ent and fusion. To this end, we propose a graph foundation model with Cross-moda
l Alignment and Modality-aware Expert fusion, CAME. Specifically, CAME first gen
erates graph embeddings for each individual modality. We then introduce a multim
odal multi-expert encoding mechanism, which includes a dimension-wise routing st
rategy to fuse multimodal information. Finally, we employ a cross-modal contrast
ive loss to train CAME, enabling the adaptive alignment and fusion across differ
ent modalities. Extensive experiments demonstrate the effectiveness of CAME acro
ss multiple tasks and diverse multimodal graph datasets.
****

Language Generation with Feedback: Queries and Mistakes
Steve Hanneke, Amin Karbasi, Anay Mehrotra, Grigoris Velegkas
We investigate language generation in the limit (Kleinberg & Mullainathan, 2024;
 Li et al., 2025) in variants where the generator receives some feedback based o
n its “actions.” We study two such variants. In the first, which is inspired by
Littlestone’s model of online learning, the generator observes whether it made a
 mistake at each iteration. In the second, introduced by Charikar & Pabbaraju (2
025a), the generator can query whether a string belongs to the target language.

Our main result is a characterization of collections that are generable with mis
take feedback. Using similar techniques, we also characterize when generation is
 possible in the query model with set-based generators; set-based generators hav

e been studied in several works (Charikar & Pabbaraju, 2025a; Kalavasis et al., 2025; Kleinberg & Wei, 2025a; Li et al., 2025). Beyond the characterizations themselves, we derive several implications. First, our results imply new closure properties for generation with mistake and query feedback. Second, our results show that, under feedback, generation is robust to noise: it remains possible with arbitrary contamination in the adversary’s examples and with finite contamination in the feedback. Third, our techniques also yield new sufficient and necessary conditions for generation without feedback among other implications.
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OpenGPT-4o-Image: A Comprehensive Dataset for Advanced Image Generation and Editing
zhihong Chen, Xuehai Bai, Yang Shi, Chaoyou Fu, Huanyu Zhang, Haotian Wang, Xiaoyan Sun, Zhang Zhang, Liang Wang, Yuanxing Zhang, Pengfei Wan, YiFan Zhang
The performance of unified multimodal models for image generation and editing is fundamentally constrained by the quality and comprehensiveness of their training data. While existing datasets have covered basic tasks like style transfer and simple object manipulation, they often lack the systematic structure and challenging scenarios required for real-world applications. To address this bottleneck, we introduce OpenGPT-4o-Image, a large-scale dataset constructed using a novel methodology that combines hierarchical task taxonomy with automated data generation. Our taxonomy not only includes fundamental capabilities such as text rendering and style control but also introduces highly practical yet challenging categories like scientific imagery for physics/chemistry illustrations and complex instruction editing requiring simultaneous execution of multiple operations. Through an automated pipeline leveraging structured resource pools and GPT-4o, we generate 80k high-quality instruction-image pairs with controlled diversity, covering 11 major domains and 51 subtasks. Extensive experiments show that fine-tuning leading models on our dataset achieves significant performance gains across multiple benchmarks, with improvements of up to 18% on editing tasks UniWorld-V1 on ImgEdit-Bench and 13% on generation tasks Harmon on GenEval. Our work demonstrates that systematic data construction is key to advancing multimodal AI capabilities.
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Video-SVD: Efficient Video Diffusion via Orthogonal Basis Composition
Zhang Wan, Yu Li, Tianze Huang, Haochen Li, Juan Cao, Sheng Tang
Video Diffusion Transformers (VDiTs) represent the state of the art in video generation but remain constrained by the quadratic complexity of dense self-attention. To address this attention bottleneck, we analyze the pre-softmax matrix ($QK^\top$) and reveal two key properties: (1) video attention exhibits an effective low-dimensional structure with rapid singular-value decay, and (2) real motion induces hybrid spatio-temporal patterns rather than rigid ``spatial vs. temporal'' layouts. Guided by these observations, we propose Video-SVD, a training-free and plug-and-play acceleration method that does not modify the original network parameters. Video-SVD learns checkpoint-adaptive orthogonal bases offline and, at inference time, replaces expensive dense attention computation with lightweight online subspace projection and basis composition. To preserve high fidelity, Video-SVD further employs layer-shared dual-stream residual modules to recover fine-grained content details and positional information. Across HunyuanVideo and Wan2.1 backbones, Video-SVD achieves significant end-to-end speedups while maintaining high visual quality, reaching 1.92$\times$ on HunyuanVideo, 1.75$\times$ on Wan2.1-1.3B, and 1.79$\times$ on Wan2.1-14B.
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Doubly Robust Distributionally Robust Offline Contextual Pricing
Min Xu, Xinyi Yin, Yunfan Zhang, Yuxuan Han, Houcai Shen, Caihua Chen
Offline contextual pricing often relies on logged observational data, but faces challenges from distributional shifts between training and deployment environments. Distributionally robust optimization (DRO) provides a principled approach to off-policy evaluation and learning (OPE/L). However, existing methods are mostly limited to discrete actions. Recent work has explored DRO for continuous treatments using inverse propensity weighting (IPW), while such IPW-based estimators

can be sensitive to the convergence rate of propensity score estimates, particularly when estimated nonparametrically, which may lead to larger estimation errors and regret.
In this work, we develop a doubly robust (DR) framework for distributionally robust OPE/L in continuous pricing settings. For evaluation, we propose a localized DR estimator that addresses the computational challenges of worst-case expectations by fitting only a small number of regressions, comparable to standard non-robust DR, while achieving semiparametric efficiency under mild product rate conditions. For learning, we leverage the inherent smoothness of demand noise to handle pricing-specific discontinuities in revenue outcomes (e.g., threshold-based purchase decisions), establishing a finite-sample regret bound of $\tilde{\mathcal{O}}_p(T^{-s/(2s+1)})$ for smoothness orders $s=1,2$. This bound improves upon existing regret rates in existing DRO-based off-policy learning (OPL) for continuous treatments. Extensive experiments under various levels of distribution shift validate our proposed framework.
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A Theory of How Pretraining Shapes Inductive Bias in Fine-Tuning
Nicolas Anguita, Francesco Locatello, Andrew M Saxe, Marco Mondelli, Flavia Mancini, Samuel Lippl, Clémentine Carla Juliette Dominé
Pretraining and fine-tuning are central stages in modern machine learning systems. In practice, feature learning plays an important role across both stages: deep neural networks learn a broad range of useful features during pretraining and further refine those features during fine-tuning. However, an end-to-end theoretical understanding of how choices of initialization impact the ability to reuse and refine features during fine-tuning has remained elusive. Here we develop an analytical theory of the pretraining–fine-tuning pipeline in diagonal linear networks, deriving exact expressions for the generalization error as a function of initialization parameters and task statistics. We find that different initialization choices place the network into four distinct fine-tuning regimes that are distinguished by their ability to support feature learning and reuse—and therefore by the task statistics for which they are beneficial. In particular, a smaller initialization scale in earlier layers enables the network to both reuse and refine its features, leading to superior generalization on fine-tuning tasks that rely on a subset of pretraining features. We demonstrate empirically that the same initialization parameters impact generalization in ResNets trained on CIFAR-100 and SVHN as well as Transformers trained on modular arithmetic tasks. Overall, our results demonstrate analytically how data and network initialization interact to shape fine-tuning generalization, highlighting an important role for the relative scale of initialization across different layers in enabling continued feature learning during fine-tuning.
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Safeguarded Stochastic Polyak Step Sizes for Non-smooth Optimization: Robust Performance Without Small (Sub)Gradients
Dimitris Oikonomou, Nicolas Loizou
The stochastic Polyak step size (SPS) has proven to be a promising choice for stochastic gradient descent (SGD), delivering competitive performance relative to state-of-the-art methods on smooth convex and non-convex optimization problems, including deep neural network training. However, extensions of this approach to non-smooth settings remain in their early stages, often relying on interpolation assumptions or requiring knowledge of the optimal solution. In this work, we propose a novel SPS variant, Safeguarded SPS (SPS$_{safe}$), for the stochastic subgradient method, and provide rigorous convergence guarantees for non-smooth convex optimization with no need for strong assumptions. We further incorporate momentum into the update rule, yielding equally tight theoretical results. Comprehensive experiments on convex benchmarks and deep neural networks corroborate our theory: the proposed step size achieves competitive performance to existing adaptive baselines and exhibits stable behavior across a wide range of problem settings. Finally, in the context of deep neural network training, the gradient norms under our step size do not collapse to (near) zero, indicating robustness to vanishing gradients.

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Exposing Hidden Biases in Text-to-Image Models via Automated Prompt Search
Manos Plitsis, Giorgos Bouritsas, Vassilis Katsouros, Yannis Panagakis
Text-to-image (TTI) diffusion models have achieved remarkable visual quality, yet they have been repeatedly shown to exhibit social biases across sensitive attributes such as gender, race and age. To mitigate these biases, existing approaches frequently depend on curated prompt datasets - either manually constructed or generated with large language models (LLMs) - as part of their training and/or evaluation procedures. Beside the curation cost, this also risks overlooking unanticipated, less obvious prompts that trigger biased generation, even in models that have undergone debiasing. In this work, we introduce Bias-Guided Prompt Search (BGPS), a framework that automatically generates prompts that aim to maximize the presence of biases in the resulting images. BGPS comprises two components: (1) an LLM instructed to produce attribute-neutral prompts and (2) attribute classifiers acting on the TTI's internal representations that steer the decoding process of the LLM toward regions of the prompt space that amplify the image attributes of interest. We conduct extensive experiments on Stable Diffusion 1.5 and a state-of-the-art debiased model and discover an array of subtle and previously undocumented biases that severely deteriorate fairness metrics. Crucially, the discovered prompts are interpretable, i.e they may be entered by a typical user, quantitatively improving the perplexity metric compared to a prominent hard prompt optimization counterpart. Our findings uncover TTI vulnerabilities, while BGPS expands the bias search space and can act as a new evaluation tool for bias mitigation.
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Decision Transformers As Zero-Shot Learners via Text-Behavior Alignment
Xin Zhang, Jonathan Martinez, Yanhua Li, Yingxue Zhang
Offline meta-reinforcement learning (meta-RL) aims to train agents that can generalize to unseen tasks using pre-collected data from related tasks. Recent approaches leverage the scalability of transformer architectures to model behavior sequences and support task adaptation using target task demonstrations. However, such data is often unavailable in real-world settings, where the task objective may be known but cannot be easily demonstrated. In contrast, humans routinely interpret and perform new tasks based solely on natural language instructions. In this work, we explore the potential of using natural language task descriptions to enable zero-shot task adaptation in offline meta-RL without requiring any data from the target task. We propose the Text-Guided Decision Transformer (TG-DT), a framework that enables zero-shot generalization by grounding policy learning in natural language. TG-DT learns a shared embedding space between task descriptions and behavioral trajectories via a dual contrastive and matching-based objective, ensuring robust alignment. A transformer-based policy is then conditioned on these aligned representations to generate task-appropriate actions. At test time, TG-DT synthesizes policies for unseen tasks using only their text descriptions and can optionally leverage a description-guided data sharing strategy to enhance adaptation. Experiments on standard offline meta-RL benchmarks, including MuJoCo and Meta-World, demonstrate that TG-DT achieves strong generalization to unseen tasks.
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GCIB: Graph Contrastive Information Bottleneck for Multi-Behavior Recommendation
Likang Wu, Zihao Chen, Jianxin Zhang, Sangqi Zhu, Yuanyuan Ge, Haipeng Yang, Lei Zhang
With the rapid emergence of multi-behavior learning in recommender systems, leveraging auxiliary user behaviors has proven effective for mitigating target-behavior data sparsity. Yet auxiliary behavior graphs frequently contain noisy or irrelevant interactions that do not align with the target task, impeding the learning of accurate user and item embeddings. Moreover, the scarcity of direct supervised from the target behavior complicates the extraction of informative collaborative signals. In this paper, we introduce GCIB Graph Contrastive Information Bottleneck, a novel framework that denoises auxiliary behavior information and enriches target behavior representations at both the structural and feature levels.

At the structural level, GCIB employs a Graph Information Bottleneck (GIB) objective to maximize mutual information between the denoised auxiliary graph and the target-behavior graph while minimizing mutual information with the original auxiliary graph. This formulation preserves task-relevant structural patterns and suppresses spurious interactions. At the feature level, we propose a cross-behavior Graph Contrastive Learning (GCL) scheme in which denoised auxiliary features and target-behavior features serve as complementary views for both users and items. By contrasting these views, GCIB enriches sparse target-behavior representations with semantics distilled from auxiliary behaviors. Extensive experiments on multiple real-world datasets demonstrate that GCIB outperforms state-of-the-art baselines, highlighting its ability to learn noise-resilient and target-aware representations for multi-behavior recommendation.
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Understanding MARS: When Scaling Momentum Provably Helps
Egor Shulgin, Tamaz Gadaev, Sarit Khirirat, Peter Richtárik
MARS (Yuan et al., 2025) has recently emerged as a strong optimizer for large language model (LLM) training by scaling the correction term in momentum-based variance reduction (MVR). However, existing theory does not explain why this modification can improve convergence over the unscaled MVR choice $\gamma=1$. In this paper, we provide a theoretical explanation for this phenomenon. We introduce **γ-similarity**, a refined similarity condition that captures how the scaling coefficient interacts with the stochastic gradient-difference structure. This condition recovers standard similarity at $\gamma=1$ and smoothness at $\gamma=0$. Using γ-similarity, we derive convergence guarantees for fixed-γ MARS whose complexity depends explicitly on γ and the corresponding γ-similarity constant. The bound reveals why small values of γ can be beneficial: they may reduce the similarity term enough to outweigh the penalty from deviating from MVR. We prove that optimizing γ gives MARS a lower complexity guarantee than MVR. Experiments with MARS-AdamW on GPT-style LLM pretraining corroborate the theory, showing that properly chosen small values of γ improve token efficiency over $\gamma=1$ and AdamW under a fixed training protocol.
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Stabilizing PPO via Latent-Space Regularization and KDE-Driven Exploration
Meiyu Du, Yuqing Gao, Wei Wang
Proximal Policy Optimization (PPO) is widely used in continuous-control tasks, yet its performance is often highly sensitive to training dynamics when neural networks approximate the policy and value functions. This paper introduces SPPO, a drop-in augmentation that preserves PPO’s clipped objective and network architecture while stabilizing actor-critic geometry via three mechanisms: (i) a CKA-based constraint on critic representations, (ii) a no-flip regularizer on actor updates, and (iii) KDE-driven advantage shaping. Theoretical analysis shows that these mechanisms tighten bounds on one-step bootstrapping error, improve expected directional alignment of action updates, and ensure non-decreasing occupancy mass over high-novelty regions. Experiments on standard continuous-control benchmarks demonstrate consistent gains over PPO and recent PPO stabilization methods. Ablation studies further quantify the contribution and complementary effects of each component. Additional training-dynamics analyses indicate that SPPO reduces instability and oscillations in both actor and critic updates, improving training stability and final performance.
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SafeCompass: Dynamic Chain-of-Thought Steering via Inference-Time Safety Signals
Zeyang Zhang, HAOTIAN XU, Linbao Li, Qi Sun, Xuebo Liu, Yu Li, Cheng Zhuo
Large reasoning models (LRMs) achieve strong performance by explicitly generating chain-of-thought (CoT) reasoning, but this reasoning process can be manipulated by adversarial prompts.
Inference-time CoT interventions offer a simple and lightweight approach to improving safety, yet existing methods typically apply static heuristics that ignore the dynamic nature of reasoning, leading to an inherent trade-off between robustness and over-refusal.

This paper introduces *SafeCompass*, a plug-and-play framework for dynamically s teering chain-of-thought reasoning using inference-time safety signals extracted from internal states.
At different reasoning positions, *SafeCompass* derives a latent safety directio n through contrastive analysis of internal representations and uses this directi on to quantify the model’s current safety state.
These signals enable selective intervention, allowing the model’s reasoning traj ectory to be modified only when and where it becomes unsafe.
Extensive experiments demonstrate that *SafeCompass* significantly improves robu stness, reducing the average attack success rate up to $10\times$ compared to th e best baseline, while preserving general reasoning performance and minimizing o ver-refusal rates.
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Learning Permutation Distributions via Reflected Diffusion on Ranks
Sizhuang He, Yangtian Zhang, Shiyang Zhang, David van Dijk
The finite symmetric group S_n provides a natural domain for permutations, yet learning probability distributions on S_n is challenging due to its factorial ly growing size and discrete, non-Euclidean structure. Recent permutation diffus ion methods define forward noising via shuffle-based random walks (e.g., riffle shuffles) and learn reverse transitions with Plackett–Luce (PL) variants, but th e resulting trajectories can be abrupt and increasingly hard to denoise as n g rows. We propose *Soft-Rank Diffusion*, a discrete diffusion framework that repl aces shuffle-based corruption with a structured soft-rank forward process: we li ft permutations to a continuous latent representation of order by relaxing discr ete ranks into soft ranks, yielding smoother and more tractable trajectories. Fo r the reverse process, we introduce *contextualized generalized Plackett–Luce (c GPL)* denoisers that generalize prior PL-style parameterizations and improve exp ressivity for sequential decision structures. Experiments on sorting and combina torial optimization benchmarks show that Soft-Rank Diffusion consistently outper forms prior diffusion baselines, with particularly strong gains in long-sequence and intrinsically sequential settings.
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Explaining Data Mixing Scaling Laws
Rui Dai, SHURAN ZHENG
Recent research has established empirical scaling laws to predict model performa nce on multi-domain data mixtures. However, a theoretical understanding of these model loss behaviors remains absent. In this work, we propose a unified framewo rk to explain the underlying mechanics of data mixing. Our approach extends theo retical perspectives originally developed for standard neural scaling laws (e.g. , Kaplan and Chinchilla) to the multi-domain setting. Based on the distributiona l assumption that domains overlap on fundamental skills while diverging on speci alized skills, we identify two key factors that govern the domain losses of mode ls trained on different data mixtures: (1) Capacity Competition, where the allo cation of finite model capacity couples domain losses globally, and (2) Noise Re duction, where optimal weights shift toward harder-to-learn domains to minimize overall noise. Empirical evaluations show that our framework outperforms existin g baselines by fitting the loss landscape with a lower Mean Relative Error and i dentifying higher-performing training mixtures. Most importantly, our model succ essfully extrapolates across scales, predicting highly effective mixtures for la rge, unseen scales using parameters fitted on smaller ones. In addition, our mod el achieves these results using significantly fewer parameters compared to prev ious empirical laws.
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DAG: A Dual Correlation Network for Time Series Forecasting with Exogenous Varia bles
Xiangfei Qiu, Yuhan Zhu, Zhengyu Li, Xingjian Wu, Bin Yang, Jilin Hu
Time series forecasting is essential in various domains. Compared to relying sol ely on endogenous variables (i.e., target variables), considering exogenous vari ables (i.e., covariates) provides additional predictive information and often le ads to more accurate predictions. However, existing methods for time series fore

casting with exogenous variables (TSF-X) have the following shortcomings: 1) they do not leverage future exogenous variables, 2) they fail to fully account for the correlation between endogenous and exogenous variables. In this study, to better leverage exogenous variables, especially future exogenous variables, we propose $\textbf{DAG}$, which $\textit{utilizes $\underline{D}$ual correl$\underline{A}$tion network along both the temporal and channel dimensions for time series forecasting with exo$\underline{G}$enous}$ variables. Specifically, we propose two core components: the Temporal Correlation Module and the Channel Correlation Module. Both modules consist of a correlation discovery submodule and a correlation injection submodule. The former is designed to capture the correlation effects of historical exogenous variables on future exogenous variables and on historical endogenous variables, respectively. The latter injects the discovered correlation relationships into the processes of forecasting future endogenous variables based on historical endogenous variables and future exogenous variables.
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Credit-assigned Policy Gradient for Early Stage Retrieval in Two-stage Ranking
Haruka Kiyohara, Mihaela Curmei, Ariel Evnine, Shankar Kalyanaraman, Israel Nir, Ana-Roxana Pop, Nitzan Razin, Sarah Dean, Thorsten Joachims, Udi Weinsberg
Large-scale search, recommendation, and retrieval-augmented generation (RAG) systems typically employ a two-stage architecture: an early-stage ranker (ESR) generates a candidate set, which is subsequently re-ranked by a late-stage ranker (LSR). While there are many reinforcement learning (RL) methods for training the LSR, end-to-end training of the ESR has proven challenging. In particular, naive application of "vanilla" policy gradient (V-PG) is not scalable for candidate-set sizes relevant for practical use due to exploding variance. This issue arises because V-PG propagates the gradient to the joint probability of the candidate sets, ignoring the contribution of each specific item in the candidate set to the reward. To mitigate this issue, we propose a novel **"credit-assigned" policy gradient (CA-PG)**, which computes gradients with respect to the probability that the target item is chosen in any candidate set, i.e. marginalizing over all candidate sets that contain it. Our theoretical analysis reveals that CA-PG significantly reduces the variance of V-PG by marginalizing over the specific composition of the candidate set, while preserving the ability to learn the correct ranking of items under a reasonably aligned LSR policy. Experiments on both synthetic and real-world data demonstrate that CA-PG improves the convergence speed and training stability for ESRs utilizing the canonical Plackett-Luce model, especially when the candidate-set size is large.
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Attention Sink Forges Native MoE in Attention Layers: Sink-Aware Training to Address Head Collapse
Zizhuo Fu, Wenxuan Zeng, Runsheng Wang, Meng Li
Large Language Models (LLMs) often assign disproportionate attention to the first token, a phenomenon known as the attention sink. Several recent approaches aim to address this issue, including Sink Attention in GPT-OSS and Gated Attention in Qwen3-Next. However, a comprehensive analysis of the relationship among these attention mechanisms is lacking. In this work, we provide both theoretical and empirical evidence demonstrating that the sink in Vanilla Attention and Sink Attention naturally construct a Mixture-of-Experts (MoE) mechanism within attention layers. This insight explains the head collapse phenomenon observed in prior work, where only a fixed subset of attention heads contributes to generation. To mitigate head collapse, we propose a sink-aware training algorithm with an auxiliary load balancing loss designed for attention layers. Extensive experiments show that our method achieves effective head load balancing and improves model performance across Vanilla Attention, Sink Attention, and Gated Attention. We hope this study offers a new perspective on attention mechanisms and encourages further exploration of the inherent MoE structure within attention layers.
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A Direct Second-Order Method for Solving Two-Player Zero-Sum Games
David Yang, Yuan Gao, Tianyi Lin, Christian Kroer
We introduce, to our knowledge, the first direct second-order method for computi

ng Nash equilibria in two-player zero-sum games. To do so, we construct a Dougla s-Rachford-style splitting formulation, which we then solve with a semi-smooth N ewton (SSN) method. We show that our algorithm enjoys local superlinear converge nce. To augment the fast local behavior of our SSN method with global efficiency guarantees, we develop a hybrid method that combines our SSN method with the st ate-of-the-art first-order method for game solving, Predictive Regret Matching (PRM$^+$). Our hybrid algorithm leverages the global progress provided by PRM$^+$ while achieving a local superlinear convergence rate once it switches to SSN ne ar a Nash equilibrium. Numerical experiments on matrix games demonstrate order-o f-magnitude speedups over PRM$^+$ for high-precision solutions.
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Unveiling the Role of Data Uncertainty in Tabular Deep Learning
Nikolay Kartashev, Ivan Rubachev, Artem Babenko
Recent advancements in tabular deep learning have demonstrated exceptional pract ical performance, yet the field often lacks a clear understanding of why these t echniques actually succeed. To address this gap, our paper highlights the import ance of the concept of data (aleatoric) uncertainty for explaining the effective ness of recent tabular DL methods. While data uncertainty leads to irreducible p rediction errors on test samples, it also introduces stochasticity into the trai ning signal that can impede effective learning. We demonstrate that tabular meth ods differ significantly in their ability to cope with this optimization challen ge. Specifically, we reveal that the success of many beneficial design choices i n tabular DL, such as numerical feature embeddings, advanced ensembling strategi es, retrieval-augmented models, and tabular Prior-Fitted Networks, can be partia lly attributed to their respective implicit mechanisms for performing well under high data uncertainty. By dissecting these varied mechanisms, we provide a unif ying understanding of recent performance improvements.
Furthermore, leveraging insights from this perspective, we design a novel, more effective numerical feature embedding method as an immediate practical outcome o f our analysis. Overall, our work paves the way toward a principled understandin g of the benefits introduced by modern tabular methods that results in the concr ete advancements of existing techniques and outlines future research directions for tabular DL.
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Multimarginal flow matching with optimal transport potentials
Raghav Kansal, David Crair, Nghia D. Nguyen, Scott D Pope, Bradley R Parry
Flow matching (FM) has emerged as a powerful framework for learning dynamic tran sport maps between two empirical distributions.
However, less explored is the setting with intermediate observed marginals that can help constrain the flows between the endpoints.
This "multimarginal" regime is central to modeling temporal evolution in dynamic al systems in many scientific domains that can sample sequential distributions.
We tackle this problem with a novel approach that leverages the connection betwe en FM and dynamic optimal transport (OT),
softly steering the flow towards the intermediate marginals through potential te rms in the dynamic OT action.
By extending the conditional FM learning target to incorporate these potentials, we derive an efficient,
simulation-free algorithm for multimarginal FM that offers considerable flexibil ity in the spatiotemporal dynamics of the learned flows.
We demonstrate state-of-the-art performance and training efficiency of OT-potent ial FM (OTP-FM) on diverse single-cell RNA sequencing, oceanographic, and meteor ological datasets.
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When Tabular Foundation Models Meet Strategic Tabular Data: A Prior Alignment Ap proach
Xinpeng Lv, Yunxin Mao, Renzhe Xu, Chunyuan Zheng, Yikai Chen, Haoxuan Li, Jinxu an Yang, Yuanlong Chen, Kun Kuang, Mingyang Geng, Shixuan Liu, Wanrong Huang, Sh aowu Yang, Wenjing Yang, Zhouchen Lin, Haotian Wang
Tabular foundation models based on pretrained prior-data fitted networks (PFNs)

have shown strong generalization on diverse tabular tasks, but they are typicall y designed for non-strategic settings where data distributions are independent o f deployed classifiers. In many real-world decision scenarios, however, individu als may strategically modify their features after deployment to obtain favorable outcomes, inducing a post-deployment distribution shift. This paper studies whe ther PFN-style tabular foundation models can generalize to such strategic tabula r data. We show that strategic manipulation creates a fundamental mismatch betwe en the non-strategic prior learned during pretraining and the post-manipulation strategic prior encountered at deployment, which leads to an irreducible structu ral prediction bias. To address this issue, we propose the Strategic Prior-data Fitted Network (SPN), an inference-time strategy-aware framework that adapts tab ular foundation models to strategic environments without retraining or architect ural modification. SPN constructs strategic in-context examples to approximate p ost-manipulation inputs and aligns PFN predictions with the induced strategic di stribution via in-context learning, with theoretical guarantees on bias reductio n. Experiments on real-world and synthetic tabular datasets show that SPN consis tently improves robustness and predictive performance under strategic manipulati on compared with both tabular foundation models and classical tabular methods.

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Fine-Tuning Without Forgetting In-Context Learning: A Theoretical Analysis of Li near Attention Models

Chungpa Lee, Jy-yong Sohn, Kangwook Lee

Transformer-based large language models exhibit in-context learning, enabling ad aptation to downstream tasks via few-shot prompting with demonstrations. In prac tice, such models are often fine-tuned to improve zero-shot performance on downs tream tasks, allowing them to solve tasks without examples and thereby reducing inference costs. However, fine-tuning can degrade in-context learning, limiting the performance of fine-tuned models on tasks not seen during fine-tuning. Using linear attention models, we provide a theoretical analysis that characterizes h ow fine-tuning objectives modify attention parameters and identifies conditions under which this leads to degraded few-shot performance. We show that fine-tunin g all attention parameters can harm in-context learning, whereas restricting upd ates to the value matrix improves zero-shot performance while preserving in-cont ext learning. We further show that incorporating an auxiliary few-shot loss enha nces in-context learning primarily on the target task, at the expense of degrade d in-context learning ability on tasks not seen during fine-tuning. We provide e mpirical evidence from synthetic and real-world datasets consistent with the qua litative predictions of our theory.

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Full-Spectrum Graph Neural Networks: Expressive and Scalable

Xiaohan Wang, Deyu Bo, Longlong Li, Kelin Xia

It is well established that spectral graph neural networks (GNNs) can universall y approximate node signals; however, their expressive power remains bounded by t he 1-dimensional Weisfeiler-Lehman test, which is mirrored in their lack of univ ersality for higher-order signals. To go beyond this bound, we propose the Full-Spectrum GNNs (FSpecGNNs), a second-order generalization of classical spectral G NNs. FSpecGNN advances spectral filtering from two perspectives: (1) it lifts si gnals from the node domain to the node-pair domain; and (2) it extends the univa riate spectral filter over eigenvalues to a bivariate filter over eigenvalue pai rs.

We show that classical spectral GNNs arise as a diagonal special case of FSpecGN Ns, and prove that FSpecGNNs can be at most as expressive as Local 2-GNN while u niversally approximating node-pair signals, the latter being particularly benefi cial for heterophilic graph learning. Moreover, FSpecGNNs admit scalable impleme ntations that avoid explicit node-pair-level computations; combined with a low-r ank approximation that reduces full-spectrum convolution to a combination of pol ynomial spectral filters, it enables learning on large graphs. Empirically, FSpe cGNNs validate the predicted expressivity on substructure-counting benchmarks an d delivers strong performance on heterophilic benchmarks. Our code is available at https://github.com/xwangxshi/FSpecGNN.

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Trajectory-Stabilized Inference for Diffusion-Based Video Inpainting
Zhanhe Zhang, Jiahua Li, Xu Yang, Kun Wei, Cheng Deng
Video inpainting aims to restore missing regions while preserving spatial and te
mporal coherence. Diffusion-based methods achieve strong per-frame reconstructio
n, but their sampling implicitly generates temporally coupled latent trajectorie
s whose long-horizon stability is not explicitly modeled, leading to a trade-off
 between temporal consistency and structural detail. We revisit video inpainting
 from the perspective of temporal trajectory stability, viewing temporal inconsi
stency as instability along time-indexed denoising trajectories rather than an o
utput-level error. Based on this view, we propose an inference-time trajectory s
tabilization framework that monitors motion-aligned deviation and triggers risk-
aware correction only when instability accumulates. It combines sparsely sampled
 trajectory anchors as stability references with neighborhood-consistent propaga
tion to regulate trajectory evolution while preserving local generative freedom.
 Implemented as a lightweight control layer in the sampling loop, it selectively
 contracts unstable trajectories toward motion-consistent manifolds instead of e
nforcing uniform temporal constraints. Experiments show consistent improvements
in temporal coherence and structural fidelity.
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Embedding Hybrid Systems into Continuous Latent Vector Fields
Sangli Teng, Hang Liu, Koushil Sreenath
This work proves that an n-dimensional hybrid system can be embedded into an $
m$-dimensional Euclidean space equipped with a continuous vector field on its em
bedded image whenever $m>2n$. This result suggests that an *intrinsically* disco
ntinuous hybrid system generically admits a continuous *extrinsic* representatio
n that is well-posed for differentiable optimization.
Building on this existence theorem, we show that a latent Neural ODE with consis
tency loss in both the latent and state space can accurately recover the flow of
 hybrid systems. Extensive experiments suggest the proposed method outperforms t
he existing method in learning hybrid systems with varying geometries from only
time series data.
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VSCD: Video-based Scene Change Detection in Unaligned Scenes
Jiae Yoon, Ue-Hwan Kim
Detecting what has changed in an environment is essential for long-term autonomy
, yet most change detection settings assume fixed viewpoints, mild misalignment,
 or only a few changed objects. We introduce Video-based Scene Change Detection
(VSCD), which predicts a pixel-wise change mask for each query frame, given a re
ference and a query RGB video of the same indoor space recorded at different tim
es under unconstrained camera motion. The two videos are not temporally synchron
ized, and many object instances may appear or disappear. To study this setting,
we build a large-scale benchmark with over 1.1 million frames annotated with pix
el-accurate change masks, together with a real-world test set for evaluating tra
nsfer beyond simulation. We propose a query-centric multi-reference model that l
earns temporal matching implicitly from change-mask supervision, aligns candidat
e reference features to the query via local patch correspondence, and fuses per-
candidate change features using frame-level and patch-level confidence before de
coding a high-resolution mask once per frame. Our approach achieves state-of-the
-art performance against strong image- and video-based baselines, and we validat
e its real-world impact by deploying it on a mobile robot for two downstream app
lications—visual surveillance and object incremental learning.
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AIR: Post-training Data Selection for Reasoning via Attention Head Influence
Jinrui Liu, Kai Hua, Xuanguang Pan, Ge Zhang, Yong Wang, Shuai Ma, Chongyang Tao
LLMs achieve remarkable multi-step reasoning capabilities, yet effectively trans
ferring these skills via post-training distillation remains challenging. Existin
g data selection methods, ranging from manual curation to heuristics based on le
ngth, entropy, or overall loss, fail to capture the causal importance of individ
ual reasoning steps, limiting distillation efficiency. To address this, we propo

se Attention Influence for Reasoning (AIR), a principled, unsupervised and train ing-free framework that leverages mechanistic insights of the retrieval head to select high-value post-training data. AIR first identifies reasoning-critical at tention heads of an off-the-shelf model, then constructs a weakened reference mo del with disabled head influence, and finally quantifies the resulting loss dive rgence as the Attention Influence Score. This score enables fine-grained assessm ent at both the step and sample levels, supporting step-level weighted fine-tuni ng and global sample selection. Experiments across multiple reasoning benchmarks show that AIR consistently improves reasoning accuracy, surpassing heuristic ba selines and effectively isolating the most critical steps and samples. Our work establishes a mechanism-driven, data-efficient approach for reasoning distillati on in LLMs.
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Think Twice Before You Act: Protecting LLM Agents Against Tool Description Poiso ning via Isolated Planning
Shanghao Shi, Xiao Wang, Chaoyu Zhang, Hao Li, Wenjing Lou, Thomas Hou, Yevgeniy Vorobeychik, Chongjie Zhang, Ning Zhang
The integration of external tools has substantially expanded the capabilities of large language model (LLM) agents, but it also introduces new attack surfaces b eyond prompt injection. In particular, cross-tool description poisoning can mani pulate planner-visible tool metadata to steer an agent’s trajectory, even if the poisoned tool itself is never chosen. To understand the effectiveness of existi ng defenses against this emerging threat, we first evaluate several prompt-injec tion defenses and find that they transfer poorly to cross-tool description poiso ning. A key observation is that poisoned descriptions persist in the planning co ntext across steps, enabling continuous influence over subsequent tool choices. Building on this insight, we propose Tool-Guard, a novel system-level defense ba sed on a new concept called *isolated planning*, in which tool invocations that are detected as misaligned or suspicious cause the corresponding tool to be plac ed in a quarantined list (the *influenced list*), breaking further influence fro m poisoned descriptions. With this influence isolated, the tool can continue to be used to support the task, enabling a robust defense that preserves legitimate tool utility. Experiments on the AgentDojo and ASB benchmarks show that Tool-Gu ard substantially reduces attack success while maintaining high task utility. Ou r code is available at https://github.com/shishishi123/Tool-Guard.
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Creat3r: Confidence Reaggregation for Exploration-aware Active 3D Reconstruction
Chih-Jung Tsai, Hwann-Tzong Chen, Tyng-Luh Liu
We present Creat3r, an iterative next-best-view (NBV) selection framework for ef ficient, high-quality 3D reconstruction. Starting from a small seed set of image -pose pairs, Creat3r repeatedly selects the most informative next camera pose. A fter each pose is chosen, the corresponding image is acquired and added to the m ulti-view set to update a 3DGS reconstruction. To guide selection, Creat3r const ructs an intermediate point cloud and estimates reconstruction reliability via a novel 3D confidence field, which is projected to candidate poses through Gaussi an projection to produce 2D confidence and exploration maps. These maps balance exploitation of reliable regions and exploration of uncertain or unseen areas un der computational constraints. Experiments with standard 3DGS show that Creat3r consistently outperforms baselines in novel view synthesis and surface reconstru ction, achieving higher SSIM and F1 scores with fewer views.
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Reinforcement Learning with Action-Triggered Observations
Alexander Ryabchenko, Wenlong Mou
We introduce Action-Triggered Sporadically Traceable Markov Decision Processes (ATST-MDPs), a reinforcement learning framework for partial observability in whic h full state observations occur stochastically at each step, with probability de termined by the chosen action. We derive Bellman equations tailored to this sett ing and establish the existence of an optimal policy. Exploiting the fact that s poradic observations reveal the full state, we provide an equivalent formulation in which agents commit to action-sequences between consecutive observations. Un

der the linear MDP assumption, we show that the value function over such action-sequences admits a linear representation in a finite-dimensional feature map, enabling standard regression-based methods. As an application, we derive ATST-LSVI-UCB, an optimistic algorithm achieving regret $\widetilde{O}(\sqrt{Kd^3(1-\gamma)^{-3}})$ for episodic learning with geometrically distributed horizons, where K is the number of episodes, d the feature dimension, and γ the discount factor (episode continuation probability), matching the known rate for linear MDPs with full observability.

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Flow Equivariant World Models: Structured Memory for Dynamic Environments

Hansen Lillemark, Benhao Huang, Fangneng Zhan, Yilun Du, T. Anderson Keller

Embodied systems experience the world as 'a symphony of flows': a combination of many continuous streams of sensory input coupled to self-motion, interwoven with the dynamics of external objects. These sensory streams and the underlying dynamics of the world obey smooth, time-parameterized symmetries which existing world models ignore. Without a memory that respects this structure, partial observability presents a major obstacle to existing methods: each observation reveals only a fraction of the world, while unobserved regions continue to evolve. In this work, we introduce Flow Equivariant World Modeling, a framework that leverages time-parameterized symmetries within a latent memory for stable and accurate dynamics prediction over long horizons. The latent memory shifts and transforms equivariantly with self-motion and inferred external object motion, keeping information about out-of-view regions aligned as time progresses. We demonstrate the advantage of this framework over state-of-the-art diffusion, memory-augmented, and recurrent world model architectures on 2D and 3D partially observed video world modeling benchmarks.

More broadly, our results suggest that predictive representations become more powerful when they are organized in line with the temporal and dynamical structure of the world they model. Project page: https://flowequivariantworldmodels.github.io/

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Turning Drift into Constraint: Robust Reasoning Alignment in Non-Stationary Multi-Stream Environments

Xiaoyu Yang, En Yu, Wei Duan, Jie Lu

This paper identifies a critical yet underexplored challenge in reasoning alignment from multiple multi-modal large language models (MLLMs): In non-stationary environments, the diverse reasoning distributions of source models often evolve unpredictably, transmitting systematic biases and drift to the target model. To address this, we formulate multi-source reasoning alignment as a constraint satisfaction problem under concept drift theory. We propose Autonomous Preference Optimization (APO), a novel framework that treats inter-model divergences not as noise, but as dynamic negative constraints. APO operates via a two-stage protocol: first, supervised bootstrapping projects the target model into the capability union of source models; second, constraint-aware optimization synthesizes a consistent consensus manifold by explicitly suppressing drifting trajectories via a multi-negative Plackett-Luce objective. Extensive experiments on chest X-ray interpretation demonstrate that our 7B model achieves superior robustness, outperforming even proprietary source models in average accuracy. Furthermore, we release CXR-MAX, a large-scale benchmark comprising 170,982 reasoning trajectories from seven large-scale MLLMs to facilitate research on reasoning alignment under drift. Code and data are available at: https://github.com/XiaoyuYoung/APO.

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Towards High-Fidelity CAD Generation via LLM-Driven Program Generation and Text-Based B-Rep Primitive Grounding

Jiahao Li, Qingwang Zhang, Qiuyu Chen, Guozhan Qiu, Yunzhong Lou, Xiangdong Zhou

The field of Computer-Aided Design (CAD) generation has made significant progress in recent years.

Existing methods typically fall into two separate categories: parametric CAD modeling and direct boundary representation (B-Rep) synthesis.

In modern feature-based CAD systems, parametric modeling and B-Rep are inherentl

y intertwined, as advanced parametric operations (e.g., *fillet* and *chamfer*) require explicit selection of B-Rep geometric primitives, and the B-Rep itself is derived from parametric operations.
Consequently, this paradigm gap remains a critical factor limiting AI-driven CAD modeling for complex industrial product design.
This paper presents *FutureCAD*, a novel text-to-CAD framework that leverages large language models (LLMs) and a B-Rep grounding transformer (*BRepGround*) for high-fidelity CAD generation.
Our method generates executable CadQuery scripts, and introduces a text-based query mechanism that enables the LLM to specify geometric selections via natural language, which *BRepGround* then grounds to the target primitives.
To train our framework, we construct a new dataset comprising real-world CAD models.
For the LLM, we apply supervised fine-tuning (SFT) to establish fundamental CAD generation capabilities, followed by reinforcement learning (RL) to improve generalization.
Experiments show that *FutureCAD* achieves state-of-the-art CAD generation performance.
Code and dataset are available at https://github.com/JohanStackk/FutureCAD.
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Transformers Can Learn Posterior Predictive Distributions In-Context
Gyeonghun Kang, Changwoo J. Lee, Xiang Cheng
Prior-data fitted networks (PFNs) have recently emerged as a powerful approach for Bayesian prediction tasks, approximating the posterior predictive distribution (PPD) through in-context learning. Despite their strong empirical performance and ability to go beyond point predictions, theoretical understandings of the algorithmic capability of transformers to learn distributions in context are still lacking. Focusing on Gaussian process regression problems, we show by construction that transformers can implement a gradient descent algorithm targeting the posterior predictive mean and variance, followed by nonlinear mappings that yield binned probabilities of PPD. We study the error bounds of the approximated PPD in terms of attention depth and bin resolution. Based on these results, we further demonstrate the key role of normalization and the choice of attention depth in enabling the extrapolation abilities of transformers beyond the pretraining sample size range. We conduct simulations that corroborate our findings, providing insight into the expressivity of PFNs targeting PPDs and how architectural choices may influence generalization capabilities.
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LLM Self-Recognition: Steering and Retrieving Activation Signatures
Thibaud Ardoin, Jonas Schäfer, Gerhard Wunder
Recent advances in interpretability suggest that large language models (LLMs) implicitly encode signals in their generated text that enable self-recognition of their outputs. We demonstrate that this capability is reliable, even in low-entropy scenarios, and that it can be amplified through targeted intervention. By steering the internal residual stream during generation with a random sparse vector, we create a detectable fingerprint that enables attribution of a given text to a specific LLM. This signal is recoverable from the activations of an LLM used as a detector, achieving over 98% accuracy across multiple detection settings while preserving the quality of generated text. As AI-generated content proliferates, this approach offers a practical alternative to traditional detectors, by leveraging the model's natural representation structure for attribution rather than embedding a signal externally. Our contributions include: (i) establishing reliable self-recognition capabilities in LLMs, (ii) a simple steering mechanism enabling multi-LLM identification with no quality degradation, (iii) demonstrating that activation spaces contain exploitable structure for encoding signals without semantic interference.
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Grounding LLMs in Scientific Discovery via Embodied Actions
Bo Zhang, Jinfeng Zhou, Yuxuan Chen, Jianing Yin, Minlie Huang, Hongning Wang
Large Language Models (LLMs) have shown significant potential in scientific disc

overy but struggle to bridge the gap between theoretical reasoning and verifiable physical simulation. Existing solutions operate in a passive "execute-then-response" loop and thus lack runtime perception, obscuring agents to transient anomalies (e.g., numerical instability or diverging oscillations). To address this limitation, we propose EmbodiedAct, a framework that transforms established scientific software into active embodied agents by grounding LLMs in embodied actions with a tight perception-execution loop. We instantiate EmbodiedAct within MATLAB and evaluate it on complex engineering design and scientific modeling tasks. Extensive experiments show that EmbodiedAct significantly outperforms existing baselines, achieving SOTA performance by ensuring satisfactory reliability and stability in long-horizon simulations and enhanced accuracy in scientific modeling.
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Causal Identification from Counterfactual Data: Completeness and Bounding Results
Arvind Raghavan
Previous work establishing completeness results for *counterfactual identification* has been limited to the setting where the input data belongs to observational and interventional distributions (Layers 1 and 2 of Pearl's Causal Hierarchy), since it was generally presumed impossible to obtain data from counterfactual distributions, belonging to Layer 3. However, recent work (Raghavan & Bareinboim, 2025) has formally characterized a family of counterfactual distributions which can be directly estimated via experimental methods - a notion they call *counterfactual realizabilty*. This leaves open the question of what *additional* Layer 3 quantities now become identifiable, given this new access to (some) Layer 3 data. We develop the **ctfIDu+** algorithm for identifying a counterfactual query from an arbitrary set of Layer 3 data, and prove that it is complete for this task. Using this, we establish the theoretical limit of which counterfactuals can be identified from physically realizable data, thus implying the *fundamental limit to exact causal inference in the non-parametric setting*. Finally, we derive novel analytic bounds for important non-identifiable quantities given realizable counterfactual data, that are provably tighter than the previously established benchmark. We corroborate using simulations that even if a quantity is non-identifiable, counterfactual data can be used to further tighten bounds for its range.
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A Recursive Decomposition Framework for Causal Structure Learning in the Presence of Latent Variables
Zheng Li, Feng Xie, Shenglan Nie, Xichen Guo, Ruxin Wang, Hao Zhang
Constraint-based causal discovery is widely used for learning causal structures, but heavy reliance on conditional independence (CI) testing makes it computationally expensive in high-dimensional settings.
To mitigate this limitation, many divide-and-conquer frameworks have been proposed, but most assume causal sufficiency, i.e., no latent variables.
In this paper, we show that divide-and-conquer strategies can be theoretically generalized beyond causal sufficiency to settings with latent variables.
Specifically, we propose a recursive decomposition framework, termed DiCoLa, that enables divide-and-conquer causal discovery in the presence of latent variables. It recursively decomposes the global learning task into smaller subproblems and integrates their solutions through a principled reconstruction step to recover the global structure.
We theoretically establish the soundness and completeness of the proposed framework. Extensive experiments on synthetic data demonstrate that our approach significantly improves computational efficiency across a range of causal discovery algorithms, while experiments on a real-world dataset further illustrate its practical effectiveness.
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LIFT: A Novel Framework for Enhancing Long-Context Understanding of LLMs via Long Input Fine-Tuning
Yansheng Mao, Yufei Xu, Jiaqi Li, Fanxu Meng, Haotong Yang, Zilong Zheng, Xiyuan Wang, Muhan Zhang

Long-context understanding remains challenging for LLMs due to limited context windows. This paper introduces **Long Input Fine-Tuning (LIFT)**, a framework that improves the long-context performance of arbitrary short-context LLMs by dynamically adapting their parameters to each long input. Instead of endlessly extending context windows to fit longer inputs **in context**, LIFT stores and absorbs the input **in parameters**. By fine-tuning long inputs into parameters, LIFT enables short-context LLMs to answer questions even when required information is absent from the inference context, avoiding the quadratic input-length complexity of standard long-context models. Rather than simple continued pretraining on new long contexts, LIFT uses carefully designed LLM-generated synthetic tasks to enhance comprehension beyond memorization. To offset fine-tuning overhead, we design a highly optimized pipeline that reduces Time to First Token (TTFT) to under 10 seconds for 8k context. We further analyze LIFT's strengths and limitations, discuss large-scale deployment feasibility, and highlight future research directions. Implementation is open-sourced at https://github.com/MuLabPKU/LIFT.
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When to Think, When to Speak: Learning Disclosure Policies for LLM Reasoning
Jiaqi Wei, Xuehang Guo, Pengfei Yu, Xiang Zhang, Wanli Ouyang, Siqi Sun, Qingyun Wang, Chenyu You
In single-stream autoregressive interfaces, the same tokens both update the model state and constitute an irreversible public commitment.
This coupling creates a \emph{silence tax}: additional deliberation postpones the first \emph{task-relevant} content, while naive early streaming risks premature commitments that bias subsequent generations.
We introduce \textbf{\emph{Side-by-Side (SxS)}} Interleaved Reasoning, which makes \emph{disclosure timing} a controllable decision within standard autoregressive generation.
SxS interleaves partial disclosures with continued private reasoning in the same context, but releases content only when it is \emph{supported} by the reasoning so far.
To learn such pacing without incentivizing filler, we construct entailment-aligned interleaved trajectories by matching answer prefixes to supporting reasoning prefixes, then train with SFT to acquire the dual-action semantics and RL to recover reasoning performance under the new format.
Across two Qwen3 architectures/scales (MoE \textbf{Qwen3-30B-A3B}, dense \textbf{Qwen3-4B}) and both in-domain (AIME25) and out-of-domain (GPQA-Diamond) benchmarks, SxS improves accuracy--\emph{content-latency} Pareto trade-offs under token-level proxies (e.g., inter-update waiting).
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Search-R2: Enhancing Search-Integrated Reasoning via Actor-Refiner Collaboration
Bowei He, Minda Hu, Zenan Xu, Hongru WANG, Licheng Zong, Yankai Chen, Chen Ma, Xue Liu, Pluto Zhou, Irwin King
Search-integrated reasoning enables language agents to transcend static parametric knowledge by actively querying external sources. However, training these agents via reinforcement learning is hindered by the *multi-scale credit assignment* problem: existing methods typically rely on sparse, trajectory-level rewards that fail to distinguish between high-quality reasoning and fortuitous guesses, leading to redundant or misleading search behaviors. To address this, we propose Search-R2, a novel Actor–Refiner collaboration framework that enhances reasoning through targeted intervention, with both components jointly optimized during training. Our approach decomposes the generation process into an Actor, which produces initial reasoning trajectories, and a Meta-Refiner, which selectively diagnoses and repairs flawed steps via a ``cut-and-regenerate'' mechanism. To provide fine-grained supervision, we introduce a hybrid reward design that couples outcome correctness with a dense process reward quantifying the information density of retrieved evidence. Theoretically, we formalize the Actor–Refiner interaction as a smoothed mixture policy, proving that selective correction yields strict performance gains over strong baselines. Extensive experiments across various general and multi-hop QA datasets demonstrate that Search-R2 consistently outperform

s strong RAG and RL-based baselines across model scales, achieving superior reas
oning accuracy with minimal overhead.
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Talk, Judge, Cooperate: Gossip-Driven Indirect Reciprocity in Self-Interested LL
M Agents
Shuhui Zhu, Yue Lin, Shriya Kaistha, Wenhao Li, Baoxiang Wang, Hongyuan Zha, Gil
lian K Hadfield, Pascal Poupart
Indirect reciprocity, which means helping those who have helped others, is diffi
cult to sustain among decentralized, self-interested LLM agents without reliable
 reputation systems. We address this challenge with the Agentic LInguistic Gossi
p Network (ALIGN), an automated framework that enables decentralized agents to f
orm reputations, evaluate trustworthiness, and coordinate social norms by strate
gically sharing open-ended gossip with hierarchical tones. We demonstrate that A
LIGN consistently improves indirect reciprocity and resists malicious entrants b
y identifying and ostracizing defectors. Notably, we find that stronger
reasoning capabilities in LLMs lead to more incentive-aligned cooperation,
whereas chat models often over-cooperate even when strategically
suboptimal. These results suggest that leveraging LLM reasoning through decentra
lized
gossip is a promising path for maintaining social welfare in agentic ecosystems.
Our code is available at \url{https://github.com/shuhui-zhu/ALIGN}.
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Singularity-aware Optimization via Randomized Geometric Probing: Towards Stable
Non-smooth Optimization
Ruoran Xu, Borong She, Xiaobo Jin, Qiufeng Wang
Deep learning optimization relies heavily on the assumption of smooth loss lands
capes, a condition systematically violated by modern architectures due to non-sm
ooth components like ReLU activations and quantization operators. In such non-sm
ooth regimes, adaptive optimizers such as Adam suffer from **gradient chattering
**—violent oscillations caused by conflicting signals within the Clarke subdiffe
rential—leading to poor convergence and suboptimal generalization. To address th
is, we introduce **Singularity-aware Adam (S-Adam)**, a novel optimizer that sta
bilizes training by dynamically modulating step sizes based on local geometric i
nstability. Our key contribution is the **Local Geometric Instability (LGI)** me
tric, a computationally efficient estimator of the Clarke subdifferential diamet
er derived from the variance of randomized directional derivatives. S-Adam incor
porates an adaptive damping mechanism $\exp(-\lambda \rho_t)$ that decelerates u
pdates in high-instability regions while preserving fast convergence in smooth b
asins. We provide a rigorous convergence analysis using differential inclusions,
 proving that S-Adam converges almost surely to (δ, ϵ)-Clarke stat
ionary points at the optimal $\mathcal{O}(1/\sqrt{T})$ rate. Empirical evaluatio
ns on Quantization-Aware Training (QAT) and high-noise small-batch learning demo
nstrate that S-Adam consistently outperforms AdamW and Prox-SGD, achieving accur
acy gains of up to +6% on CIFAR-100 and +3% on TinyImageNet while effectively mi
tigating gradient oscillations.
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CARE: Adaptive Calibration for Reliable Recommendations
Nitin Bisht, Huan Huo, Xiuwen Gong, Guandong Xu
Modern recommender systems are typically trained offline and deployed with param
eters held fixed between periodic refreshes, yet user behavior can evolve substa
ntially during deployment. This can cause ranking utility to degrade over time a
nd makes it difficult to provide formal guarantees about recommendation quality.
 We propose **CARE**, an adaptive calibration framework that wraps an arbitrary
backbone recommender and outputs variable-size recommendation sets with finite-s
ample performance guarantees over interaction streams. CARE combines (i) a loss-
based monitoring module that localizes behavioral changes and triggers threshold
 recalibration, and (ii) an online aggregation rule that promotes compact recomm
endation sets by dynamically reweighting candidate set predictors. We provide th
eoretical results establishing finite-sample guarantees for utility-based risk c
ontrol and bounds on the expected set size relative to the best constituent pred

ictor. Experiments across multiple datasets and backbone models demonstrate that CARE improves robustness and maintains compact recommendation sets while preserving the desired statistical guarantees. The code and implementation are available in https://github.com/kalpiree/CARE.
****

A Narrowing Geometry in Contaminated Reasoning
Jiakuan Xie, Pengfei Cao, Kang Liu, Jun Zhao
Despite the advancing reasoning capabilities of large language models (LLMs), many reasoning evaluations are increasingly compromised by data contamination, which induces unreliable **contaminated reasoning** on leaked inputs.
While this phenomenon is widely observed, its underlying mechanism remains poorly understood, hindering the ability to distinguish generalization from memorization and to develop effective solutions.
In this work, we first identify a distinctive signal of contaminated reasoning, namely the mutual information decay between representations and gradients.
Our mechanistic analysis reveals that contaminated models exhibit pronounced eigenspectrum concentration in their representations, leading to a low-dimensional computation regime.
Under leaked inputs, this mechanism weakens the linear coupling between representations and gradients, manifested as a structural decay of the singular values in the whitened space.
We show that this narrowing geometry mathematically implies a reduction in mutual information, and further demonstrate the practical utility of our analysis by successfully restoring the reasoning behavior of contaminated models, achieving up to an 11.03% absolute improvement in average consistency with the base model over the strongest baseline. Our code is available at https://github.com/jiakuan929/ContamReasoning.
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CyberGym-E2E: Scalable Real-World Benchmark for AI Agents' End-to-End Cybersecurity Capabilities
Tianneng Shi, Robin Rheem, Dongwei Jiang, Mona Wang, Francisco De La Riega, Zhun Wang, Jingzhi Jiang, Alexander Cheung, Sean Tai, Jonah Cha, Jianhong Tu, Gabriel Han, Chenguang Wang, Jingxuan He, Wenbo Guo, Dawn Song
AI has the potential to transform cybersecurity by enabling systems that can autonomously detect, analyze, and remediate software vulnerabilities. However, existing cybersecurity evaluations of AI systems are limited in scale or scope, and fail to capture the end-to-end lifecycle of real-world software vulnerability discovery and remediation. To address this gap, we propose CyberGym-E2E, a large-scale and realistic end-to-end cybersecurity benchmark that comprehensively evaluates AI agents' abilities across the full lifecycle of vulnerability discovery, PoC generation, and patch generation. CyberGym-E2E is comprehensive and scalable, as we build an automated, agent-enhanced pipeline for transforming open-source vulnerability data into realistic evaluation environments. Currently, the benchmark consists of 920 real-world vulnerabilities across 139 different open-source projects.
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The Decrypto Benchmark for Multi-Agent Reasoning and Theory of Mind
Andrei Lupu, Timon Willi, Jakob Nicolaus Foerster
As Large Language Models (LLMs) gain agentic abilities, they will have to navigate complex multi-agent scenarios, interacting with human users and other agents in cooperative and competitive settings. This will require new reasoning skills, a crucial one being theory of mind (ToM), or the ability to reason about the ``mental'' states of other agents. However, ToM and other multi-agent abilities in LLMs are poorly understood, since existing benchmarks suffer from narrow scope, data leakage, saturation, and lack of interactivity. We thus propose Decrypto, a game-based benchmark for multi-agent reasoning and ToM drawing inspiration from cognitive science, computational pragmatics and multi-agent reinforcement learning. It is designed to be as easy as possible in all other dimensions, eliminating confounding factors common in other benchmarks. To our knowledge, it is also the first platform that isolates ToM evaluation in an interactive setting.

We validate the benchmark design through comprehensive empirical evaluations of frontier LLMs, robustness studies, and human-AI cross-play experiments. We find that LLMs lag behind humans and simple word-embedding baselines on key game metrics. We then create variants of two classic cognitive science experiments within Decrypto to evaluate three distinct ToM abilities. Surprisingly, our results show that state-of-the-art reasoning models are significantly worse at those tasks than their older counterparts. This demonstrates that Decrypto addresses a crucial gap in current reasoning and ToM evaluations, and paves the path towards better artificial agents. Code at https://github.com/facebookresearch/decrypto.
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Concept Heterogeneity-aware Representation Steering
Laziz Abdullaev, Noelle Y. L. Wong, Ryan Lee T. Z., Shiqi Jiang, Minh-Khoi Nguyen-Nhat, Tan Minh Nguyen
Representation steering offers a lightweight mechanism for controlling the behavior of large language models (LLMs) by intervening on internal activations at inference time. Most existing methods rely on a single global steering direction, typically obtained via difference-in-means over contrastive datasets. This approach implicitly assumes that the target concept is homogeneously represented across the embedding space. In practice, however, LLM representations can be highly non-homogeneous, exhibiting clustered, context-dependent structure, which renders global steering directions brittle. In this work, we view representation steering through the lens of optimal transport (OT), noting that standard difference-in-means steering implicitly corresponds to the OT map between two identical distributions with differing first moments, yielding a global translation. To relax this restrictive assumption, we theoretically model source and target representations as Gaussian mixture models and formulate steering as a discrete OT problem between semantic latent clusters. From the resulting transport plan, we derive an explicit, input-dependent steering map via barycentric projection, producing a smooth, kernel-weighted combination of cluster-level shifts. We term this method Concept Heterogeneity-aware Representation Steering (CHaRS). Through numerous experimental settings, we show that CHaRS yields more effective behavioral control than global steering. The code is publicly available at https://github.com/lazizcodes/CHaRS.
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STAR: Rethinking MoE Routing as Structure-Aware Subspace Learning
Sumin Park, Noseong Park
Mixture-of-Experts (MoE) scales model capacity efficiently by selectively routing inputs to a specialized subset of experts. However, input-expert specialization, the core motivation of MoE, critically depends on whether the router is actually aware of input structure. In practice, MoE routing is typically implemented as a shallow linear projection with limited awareness of input representation, which often leads to unstable routing. We propose STAR, a Structure Aware Routing that rethinks MoE routing as a subspace learning problem by augmenting standard learnable routing with an evolving principal subspace that tracks dominant input structure via Generalized Hebbian Algorithm (GHA). By aligning routing decisions directly with input structure, STAR enables stable expert specialization. We evaluate STAR on controlled synthetic setup and large-scale language and vision tasks, where it consistently improves routing quality and downstream performance over strong MoE baselines. Moreover, optional test-time subspace updates further enhance routing robustness and generalization under input distribution shifts. Code is available at \url{https://github.com/psmiz/STAR}.
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Rethinking Calibration for Early-Exit Neural Networks
Piotr Kubaty, Filip Szatkowski, Grzegorz Choczy■ski, Eric Nalisnick, Bartosz Wójcik
Early-exit neural networks (EENNs) accelerate inference by allowing intermediate classifiers to stop computation once predictions are confident enough.
Most methods rely on confidence thresholds for exiting, and consequently, improving classifier calibration is widely assumed to improve performance.
In this work, we challenge this assumption and show that calibration alone is no

t sufficient for EENNs to exploit adaptive computation. To address this insuffic iency, we introduce Early-Exit Failure Prediction (EEFP), which accounts for bot h prediction correctness and the cost of further computation. We also propose a lightweight, EEFP-motivated procedure to improve the intermediate classifiers, w hich can directly replace calibration in EENNs.
Extensive experiments demonstrate that our approach achieves superior cost-accur acy trade-offs compared to calibration, and EEFP more reliably reflects overall EENN performance.
Our code is available at [https://github.com/gmum/rethinking-calibration-for-een ns](https://github.com/gmum/rethinking-calibration-for-eenns).
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AnalogVerifier: A Neuro-Symbolic Framework for Analog Circuit Verification
Yanfang Liu, Mingjun Wang, Peng XU, Rongliang Fu, Bei Yu, Tsung-Yi Ho
Analog circuits constitute the indispensable interface between physical reality and digital computation, underpinning safety-critical systems from autonomous dr iving to medical implants. Consequently, verification correctness is paramount; yet, it remains the critical bottleneck in hardware design, consuming over 50\% of engineering cycles due to a heavy reliance on the manual interpretation of un structured, heterogeneous specifications. While Large Language Models (LLMs) off er automation potential, their probabilistic, autoregressive nature is structura lly misaligned with the strict determinism required for analog verification, str uggling with semantic dispersion, latent causal dependencies, and numerical prec ision. To bridge this gap, we introduce AnalogVerifier, a neuro-symbolic framewo rk that automates end-to-end testbench generation by decoupling semantic transla tion from logical enforcement. It comprises four parts: (1) Context-Aware Task Serialization transforms complex specifications into atomic tasks via an agentic workflow; (2) Graph-Symbolic Scheduling satisfies analog design constraints thr ough Port Dependency Graphs (PDG) for correct-by-construction sequencing; (3) Nu merical-Symbolic Grounding mitigates numerical hallucination by delegating thres hold derivation to a deterministic symbolic oracle; (4) Closed-Loop Repair enabl es correctness and completeness of the generated testbenches by simulation feedb ack. Evaluation on five industrial analog circuits demonstrates that AnalogVerif ier achieves 82.3\%--100\% functional pass rate, establishing a new paradigm for reliable, automated analog verification.
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Open-o3-Video: Grounded Video Reasoning with Explicit Spatio-Temporal Evidence
Jiahao Meng, Xiangtai Li, Haochen Wang, Tan Yue, Tao Zhang, Lingdong Kong, Yunha i Tong, Anran Wang, Zhiyang Teng, Yujing Wang, Zhuochen Wang
Most video reasoning models only generate textual reasoning traces without indic ating when and where key evidence appears. Recent models such as OpenAI-o3 have sparked wide interest in evidence-centered reasoning for images, yet extending t his ability to videos is more challenging due to the need for joint temporal tra cking and spatial localization across dynamic scenes. We introduce Open-o3-Video , a non-agent framework that integrates explicit spatio-temporal evidence into v ideo reasoning by highlighting key timestamps, objects, and bounding boxes, maki ng the reasoning process traceable and verifiable. To enable this capability, we first construct high-quality datasets STGR that provide unified spatio-temporal supervision, which is absent in existing resources. We further adopt a cold-sta rt reinforcement learning strategy with specially designed rewards that jointly encourage answer accuracy, temporal alignment, and spatial precision. On the V-S TAR benchmark, Open-o3-Video achieves state-of-the-art performance, improving mA M by 14.4% and mLGM by 24.2% over the Qwen2.5-VL baseline, and shows consistent gains across a range of video understanding benchmarks. Beyond accuracy, the gro unded reasoning traces produced by Open-o3-Video support confidence-aware test-t ime scaling, improving answer reliability. The code, model and datasets are publ icly available at https://marinero4972.github.io/projects/Open-o3-Video/.
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Return-to-Go Is More Than a Number: Q-Guided Alignment for Return-Conditioned Su pervised Learning
Yuxiao Yang, Weitong Zhang

Conditioned Sequence Models (CSMs) learn policies by treating return-to-go (RTG) as a control signal.
However, existing CSMs often treat the RTGs as simple numerical inputs rather than aligning them with the performance of their policies.
In this paper, we propose Q-ALIGN DT, a framework that enforces this alignment by ensuring the Q-value of the output policy is consistent with the input RTG.
By leveraging a Q function to provide dense guidance to CSMs and further fine-tuning it using an *RTG-perturbation* technique with the CSM, our method ensures that higher RTGs are consistently mapped to trajectories with higher expected returns.
Theoretically, we show that Q-ALIGN DT can efficiently learn the desired policy and output a near-optimal one when the RTG is sufficiently high.
Empirically, we demonstrate through extensive experiments that Q-ALIGN DT achieves superior controllability and performance across the D4RL benchmark. Remarkably, our model effectively learns a structured family of policies that maintains precise alignment and generalizes to tasks like velocity-tracking where prior methods fail.
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AliMark: Enhancing Robustness of Sentence-Level Watermarking Against Text Paraphrasing
Yuexin Li, Wenjie Qu, Linyu Wu, Yulin Chen, Yufei He, Tri Cao, Bryan Hooi, Jiaheng Zhang
Existing sentence-level watermarking methods enhance robustness to paraphrasing by anchoring watermarks in sentence semantics. However, their prefix-based designs remain vulnerable to structural perturbations, such as sentence splitting and merging, which commonly arise under strong paraphrasers like DIPPER and GPT-3.5. To mitigate this issue, we propose AliMark, a framework that reformulates sentence-level watermarking as a bit sequence encoding and alignment problem between a potentially watermarked text and a secret bit sequence. Notably, our approach adopts a two-stage detection strategy: we generate multiple restructured text variants and adaptively align their extracted bit sequences with the secret bit sequence to minimize alignment cost. This multi-candidate alignment design naturally improves robustness to sentence merges and splits. Extensive experiments demonstrate that AliMark substantially outperforms state-of-the-art baselines under diverse paraphrasing attacks. Our code is available at https://github.com/imethanlee/AliMark.
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DiffThinker: Towards Generative Multimodal Reasoning with Diffusion Models
Zefeng He, Xiaoye Qu, Yafu Li, Tong Zhu, Qipeng Guo, Muxin Fu, Siyuan Huang, Yu Cheng
While recent Multimodal Large Language Models (MLLMs) have attained significant strides in multimodal reasoning, their reasoning processes remain predominantly text-centric and fail to visualize and track intermediate visual states during the reasoning process, leading to suboptimal performance in complex long-horizon, vision-centric tasks.
Moving beyond the constraints of text-centric reasoning, we establish Generative Multimodal Reasoning as a novel paradigm and introduce DiffThinker, a diffusion-based reasoning framework.
Conceptually, DiffThinker reformulates multimodal reasoning as a native generative image-to-image task, where the iterative denoising trajectory naturally serves as a visual reasoning path. This enables the model to track the evolution of visual information throughout the reasoning process.
We perform a systematic comparison between DiffThinker and MLLMs, providing the first in-depth investigation into the intrinsic characteristics of this paradigm, revealing four core properties: efficiency, controllability, native parallelism, and collaboration. Extensive experiments across
seven tasks demonstrate that DiffThinker significantly outperforms leading closed-source models, including GPT-5 (+314.2%) and Gemini-3-Flash (+111.6%), as well as the fine-tuned Qwen3-VL-32B baseline (+39.0%), highlighting Generative Multimodal Reasoning as a promising approach for vision-centric reasoning.

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Reference-Free Meta-Learning for Generalized Implicit Neural Representation in Efficient MRI Reconstruction
Haonan Zhang, Qing Wu, Xuanyu Tian, Bowen Li, Yuyao Zhang, Hongjiang Wei
Implicit Neural Representation (INR) has emerged as a powerful paradigm for continuous MRI reconstruction. However, standard self-supervised INR requires time-consuming optimization from scratch for each scan, hindering clinical deployment. This work presents IPOD, a Reference-Free Meta-Learning framework designed to learn generalized parameter initializations for INR directly from undersampled data. Distinct from conventional meta-learning that relies on fully-sampled ground truth, IPOD operates in an inverse-problem-driven manner, leveraging diverse reconstruction tasks with varying sampling patterns to capture a robust prior. Furthermore, we introduce an adaptive meta-update strategy modulated by task-specific performance to ensure optimal parameter distribution for diverse anatomical structures. Extensive experiments demonstrate that IPOD provides a superior initialization that enables rapid adaptation and achieves high-fidelity reconstruction across various imaging protocols, significantly outperforming existing INR baselines. By eliminating the dependence on reference images, IPOD offers a scalable and efficient solution for a wide range of imaging inverse problems. Code and data available at: https://github.com/zhn00310/RFML4MRI
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Protein Fold Classification at Scale: Benchmarking and Pretraining
Dexiong Chen, Andrei Manolache, Mathias Niepert, Karsten Borgwardt
Classifying protein topology is essential for deciphering biological function, but progress is held back by the lack of large-scale benchmarks that avoid duplicates and by models that do not scale well. We introduce TEDBench, a large-scale, non-redundant benchmark for protein fold classification constructed from the Encyclopedia of Domains (TED) and Foldseek-clustered AlphaFold structures. We show that on TEDBench, current protein representation learning methods either require very large models or fail to deliver strong performance.
To address this challenge, we propose Masked Invariant Autoencoders (MiAE), a self-supervised framework for protein structure representation learning. MiAE uses an extremely high masking ratio of up to 90% with an $\mathrm{SE(3)}$-invariant encoder and a lightweight decoder that reconstructs backbone coordinates from the latent representation and mask tokens. MiAE scales well and outperforms supervised counterparts and state-of-the-art baselines on TEDBench, establishing a strong recipe for protein fold classification. To test transfer beyond AlphaFold structures, we further benchmark on a curated dataset from experimental structures of CATH v4.4. TEDBench is available at https://github.com/BorgwardtLab/TEDBench.
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Look on Demand: A Cognitive Scheduling Framework for Visual Evidence Acquisition in Multimodal Reasoning
Yang Zhang, Xiaoshuai Sun, Rui Zhao, Wujin Sun, Yidong Chen, Jiayi Ji, Qian Chen, Rongrong Ji
Existing multimodal reasoning approaches predominantly follow two paradigms: converting visual inputs into text prior to reasoning, or performing end-to-end reasoning within a unified vision–language representation space. Despite their empirical progress, both paradigms suffer from fundamental structural limitations. The former relies on static visual-to-text conversion, which tends to compress and lose fine-grained visual details. The latter is prone to linguistic dominance induced by joint optimization and attention mechanisms, leading to systematically weakened faithfulness to visual evidence during reasoning. In this work, we argue that a central challenge is how and when visual evidence is introduced into the reasoning process. Motivated by this insight, we propose CSMR, a multimodal reasoning framework in which a language model controls the reasoning process by deciding when to invoke an independent visual perception module to acquire task-relevant visual evidence. Experiments across multiple multimodal reasoning benchmarks show that CSMR consistently outperforms representative baseline methods in accuracy under a

zero-shot setting. Further experimental analysis confirms that these advantages primarily arise from the proposed cognitive scheduling mechanism.
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Set-Preserving Calibration from Conformal P-Values to E-Values
Nabil Alami, Jad Zakharia, Souhaib Ben Taieb
Standard conformal prediction (CP) procedures are typically formulated in terms of p-values, but reliance on p-values alone limits flexibility, for example, when combining dependent evidence across models or data splits. Recent work has explored e-value formulations for conformal inference, yet a direct connection between p- and e-value formulations in CP has been missing, especially regarding their statistical efficiency. We first identify limitations of classical p-to-e calibrators in the CP setting, showing that they are not set-preserving and can lead to overly conservative prediction sets. To address this, we propose a novel P2E calibrator that converts conformal p-values into e-values without altering the prediction set induced by the original conformal p-value. We establish both theoretically and empirically that our calibrator can yield significant efficiency gains over existing p-to-e calibrators. This e-value formulation enables principled use of recent advances in e-value merging and randomization, where we demonstrate its impact in two applications: cross-conformal prediction (CCP), whose variants typically provide only approximate $1-2\alpha$ coverage, and conformal aggregation (CA). In both cases, our e-value-based methods satisfy the desired $1-\alpha$ coverage guarantee while improving efficiency over standard baselines. More broadly, our approach expands the flexibility of CP and opens new directions for efficient, distribution-free uncertainty quantification.
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Old Habits Die Hard: How Conversational History Geometrically Traps LLMs
Adi Simhi, Fazl Barez, Martin Tutek, Yonatan Belinkov, Shay B Cohen
How does the conversational past of large language models (LLMs) influence their future performance? Recent work suggests that LLMs are affected by their conversational history in unexpected ways. For instance, hallucinations in prior interactions may influence subsequent model responses. In this work, we introduce History Echoes, a framework that investigates how conversational history biases subsequent generations. The framework explores this bias from two perspectives: probabilistically, we model conversations as Markov chains to quantify state consistency; geometrically, we measure the consistency of consecutive hidden representations. Across three model families and six datasets spanning diverse phenomena, our analysis reveals a strong correlation between the two perspectives. By bridging these perspectives, we demonstrate that behavioral persistence manifests as a geometric trap, where gaps in the latent space confine the model's trajectory.
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$\texttt{MetaDistill}$: Unlocking the Performance Ceiling for Pretrained Optimizers
Muqi Han, Ruoqi Xing, Kai Wu, Xiaoyu Zhang, Handing Wang, Zilong Wang
Meta Black-Box Optimization (MetaBBO) has emerged as a promising paradigm by employing meta learning to automatically optimize the configurations of low-level black-box optimizers. Despite its potential, the generalization of MetaBBO remains significantly constrained when facing unseen, complex objective landscapes. We identify that this bottleneck stems from a restricted performance upper bound inherent in current training mechanisms: by learning from scratch in a self-supervised or unsupervised manner, low-level learnable optimizers are never exposed to advanced, high-quality optimization behaviors, forcing them to converge on suboptimal strategies. In this paper, we propose \texttt{MetaDistill}, a general MetaBBO training framework designed to lift the strategy ceiling through pretraining and test-time fine-tuning. In the pretraining stage, we represent high-quality strategies from classical algorithms as expert optimization trajectories and utilize them for diversity-preserving distillation, enabling the learnable optimizer to internalize advanced optimization behaviors. In the optional fine-tuning stage, we perform self-supervised fine-tuning as a warm-start procedure to further refine the distilled knowledge on unseen tasks. We evaluate our \texttt{MetaD

istill} framework on the BBOB test suite and three control tasks. The results de
monstrate that \texttt{MetaDistill} significantly improves the generalization ab
ility of various learnable optimizers compared to their original training paradi
gms. Our code is available at \href{https://github.com/Roki-Xing/MetaDistill}{he
re}.
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Score Correction for Generative Models with Probabilistic Constraints
Shishang Wu, Bingjing Tang, Vinayak Rao
We introduce DualScore, a framework for correcting score functions so that scor
e-based generative models satisfy probabilistic constraints. These constraints a
re specified through the marginal distribution of a stochastic transformation of
 the modeled variable. We formulate this as a constrained KL-minimization proble
m, and optimize its dual, parameterizing the infinite-dimensional dual variable
with a neural network. This yields an additive correction to the original score
function that can be used directly for sampling via stochastic gradient Langevin
 dynamics or reverse diffusion sampling, without modifying the base model. We ev
aluate DualScore on synthetic tasks and two real-world applications: regularized
 nonparametric maximum likelihood estimation and the incorporation of class-leve
l constraints such as fairness into pretrained image diffusion models.
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In-Context Learning as Rate–Distortion Optimization
Jiayu Zhang, Changbang Li, Canran Xiao
In-context learning (ICL) is a practical way to adapt large models, yet under st
rict context limits, it remains unclear how to spend scarce tokens without being
 misled by noisy, redundant, or conflicting demonstrations. We address this gap
by targeting token-budgeted context construction: how to select and compress dem
onstrations so the prompt carries maximal task-relevant signal with minimal pred
ictive distortion. We propose RDCO, a deterministic, training-free optimizer tha
t scores demonstrations by marginal task information per token, penalizes redund
ancy and prefix-conditioned conflicts, and finally compacts the selected context
 under a bounded predictive-divergence constraint to control drift. Across a 10-
dataset ICL suite spanning classification and structured generation, RDCO achiev
es the best average performance (63.26 Acc. on classification and 60.26 EM on ge
neration), improving over the strongest classification and generation baselines
by 2.20 and 2.26 points, respectively, and improving the 10-task overall average
 by 4.94 points under the same budget. Our results suggest that viewing prompts
as finite-capacity messages yields a principled and effective path to more relia
ble and token-efficient ICL.
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A Fully First-Order Layer for Differentiable Optimization
Zihao Zhao, Kai-Chia Mo, Shing-Hei Ho, Brandon Amos, Kai Wang
Differentiable optimization studies how to embed a mathematical program as a dif
ferentiable layer in machine learning pipelines. However, existing approaches ty
pically rely on implicit differentiation, involving expensive Hessian computatio
n while differentiating through optimality conditions. To address this challenge
, we formulate the differentiable optimization problem as a bilevel optimization
 instance.
We construct a new active-set Lagrangian as a proxy to compute an ϵ-app
roximate hypergradient using only near-constant $O(\log (1/\epsilon))$ first-ord
er information. We also show that applying this efficient hypergradient oracle t
o constrained bilevel optimization improves the overall gradient complexity to $
\tilde{O}(\delta^{-1}\epsilon^{-3})$ to reach a (δ, ϵ)-Goldstein s
tationary point. We implement our method `FFOLayer`, as a drop-in Python library
 compatible with existing differentiable optimization solvers. Our algorithm sho
ws significantly faster computation with similar convergence compared to other e
xisting solvers. The source code is available at [https://github.com/GT-KOALA/FF
OLayer](https://github.com/GT-KOALA/FFOLayer).
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The Theory and Practice of MAP Inference over Non-Convex Constraints
Leander Kurscheidt, Gabriele Masina, Roberto Sebastiani, Antonio Vergari

In many safety-critical settings, probabilistic ML systems have to make predictions subject to algebraic constraints, e.g., predicting the most likely trajectory that does not cross obstacles. These real-world constraints are rarely convex, nor the densities considered are (log-)concave.
This makes computing this constrained maximum a posteriori (MAP) prediction in an efficient and reliable way extremely challenging. In this paper, we first investigate under which conditions we can perform constrained MAP inference over continuous variables exactly and efficiently and devise a scalable message-passing algorithm for this tractable fragment. Then, we devise a general constrained MAP strategy that interleaves partitioning the domain into convex feasible regions with numerical constrained optimization. We evaluate both methods on synthetic and real-world benchmarks, showing our structure aware approach outperforms constraint-agnostic baselines.
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NBCG: Nash-Bargained Causal Game for Long-Tailed Multi-Label NLP
Jing Yang, Jusheng Zhang, Keze Wang
Long-tailed multi-label text classification is often treated as a data scarcity problem, addressed by re-sampling or fixed re-weighting.
We argue that a central failure mode is \textit{dominant coalition capture}: frequent labels, amplified by spurious co-occurrences, form dominant coalitions that dominate shared representations and gradient allocation during optimization.
As a result, rare labels are learned via superficial shortcuts, yielding brittle generalization under distribution shifts.
We propose \textbf{NBCG}, a Nash-Bargained Causal Game that reformulates multi-label learning as a cooperative bargaining process among label coalitions.
NBCG first leverages Neural Structural Equation Models to learn a directed dependency structure, inducing causally coherent coalitions---rather than random partitions---and coalition-specific communication masks.
We then optimize a Nash bargaining objective over coalition utilities relative to an adaptive disagreement point, which serves as a principled credit-allocation mechanism: it adaptively prioritizes under-served coalitions while maintaining a Pareto-efficient trade-off among all players.
****
Global Credit Assignment via Dynamical Criticality
Wentao Wang, Keren Gao, Guozhang Chen
Efficiently training recurrent neural networks on long sequences remains an open challenge. The standard global paradigm, backpropagation through time (BPTT), suffers from vanishing and exploding gradients and memory costs that scale linearly with sequence length. Conversely, biologically inspired local learning rules are memory-efficient but typically introduce severe bias. To bridge this gap, we introduce Criticality-driven Online Local Alignment (COLA). By leveraging the long-range spatiotemporal correlations inherent to the critical regime, COLA enables a strictly local learning rule to approximate global error propagation, thereby combining online efficiency with gradient descent precision. Theoretically, for a recurrent neural network with H hidden units, COLA requires only an $O(H)$ auxiliary state and constant activation memory, completely independent of sequence length. Empirically, COLA is competitive with BPTT on standard benchmarks and demonstrates superior robustness on stability-sensitive tasks. Finally, we conduct a rigorous analysis of the approximation error to provide a theoretical foundation for reliable online learning. Code is available at \url{https://github.com/Criticality-Cognitive-Computation-Lab/COLA}
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Stability and Generalization of Nonconvex Optimization with Heavy-Tailed Noise
Hongxu Chen, Ke Wei, Xiaoming Yuan, Luo Luo
The empirical evidence indicates that stochastic optimization with heavy-tailed gradient noise is more appropriate to characterize the training of machine learning models than that with standard bounded gradient variance noise. Most existing works on this phenomenon focus on the convergence of optimization errors, while the analysis for generalization bounds under the heavy-tailed gradient noise remains limited. In this paper, we develop a general framework for establishing

generalization bounds under heavy-tailed noise. Specifically, we introduce a tru
ncation argument to achieve the generalization error bound based on the algorith
mic stability under the assumption of bounded pth centered moment with $p\in(1
,2]$. Building on this framework, we further provide the stability and generaliz
ation analysis for several popular stochastic algorithms under heavy-tailed nois
e, including clipped and normalized stochastic gradient descent, as well as thei
r mini-batch and momentum variants.
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HaloProbe: Bayesian Detection and Mitigation of Object Hallucinations in Vision-Language Models

Reihaneh Zohrabi, Hosein Hasani, Akshita Gupta, Mahdieh Soleymani Baghshah, Anna Rohrbach, Marcus Rohrbach

Large vision-language models can produce object hallucinations in image descript
ions, highlighting the need for effective detection and mitigation strategies. P
rior work commonly relies on the model's attention weights on visual tokens as a
 detection signal. We reveal that coarse-grained attention-based analysis is unr
eliable due to hidden confounders, specifically token position and object repeti
tion in a description. This leads to Simpson’s paradox: the attention trends rev
erse or disappear when statistics are aggregated. Based on this observation, we
 introduce HaloProbe, a Bayesian framework that factorizes external description s
tatistics and internal decoding signals to estimate token-level hallucination pr
obabilities.

HaloProbe uses balanced training to encourage learning from internal evidence an
d combines it with a learned prior over external features to recover the true po
sterior. While intervention-based mitigation methods often degrade utility or fl
uency by modifying models' internals, we use HaloProbe as an external scoring si
gnal for non-invasive mitigation. Our experiments
show that HaloProbe-guided decoding reduces hallucinations more effectively than
 state-of-the-art intervention-based methods while preserving caption utility an
d linguistic fluency.
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Scam2Prompt: A Scalable Framework for Auditing Malicious Scam Endpoints in Production LLMs

Zhiyang Chen, Tara Saba, Xun Deng, Xujie Si, Fan Long

The insatiable demand for web-scale training data has exposed LLMs to a subtle b
ut consequential threat: the absorption of malicious scam content into model wei
ghts and its subsequent reproduction during inference. In November 2024, this ri
sk materialized when a developer reportedly lost 2,500 USD after ChatGPT generat
ed an otherwise routine cryptocurrency trading script containing a live phishing
 URL. To systematically investigate this problem, we introduce Scam2Prompt, an a
utomated auditing framework that crawls known scam websites, infers their functi
onal intent, and synthesizes innocuous developer-style prompts — the kind of leg
itimate coding requests a programmer might naturally submit — to evaluate whethe
r LLMs reproduce the underlying scam endpoints. Importantly, our approach requir
es neither jailbreaking nor adversarial prompting; all 1,377 prompts in our benc
hmark, Innoc2Scam-bench, which is automatically constructed by Scam2Prompt, were
 human-validated as benign coding tasks. Evaluation of seven production LLMs rel
eased in 2025 on Innoc2Scam-bench shows that the vulnerability proves both persi
stent and severe: malicious code generation rates range from 12.9% to 47.3% acro
ss the evaluated models, and no tested model proves immune. State-of-the-art gua
rdrails and RAG-based agents offer only limited protection, underscoring an urge
nt need for explicit URL validation in LLM-assisted software development pipelin
es.
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Plug-and-Play Diffusion Meets ADMM: Dual-Variable Coupling for Robust Medical Image Reconstruction

Chenhe Du, Xuanyu Tian, Qing Wu, Muyu Liu, Jingyi Yu, Hongjiang Wei, Yuyao Zhang

Plug-and-Play diffusion prior (PnPDP) frameworks have emerged as a powerful para
digm for solving imaging inverse problems by treating pretrained generative mode
ls as modular priors. However, we identify a critical flaw in prevailing PnP sol

vers (e.g., based on HQS or Proximal Gradient): they function as *memoryless operators*, updating estimates solely based on instantaneous gradients. This lack of historical tracking inevitably leads to non-vanishing *steady-state bias*, where the reconstruction fails to strictly satisfy physical measurements under heavy corruption. To resolve this, we propose **Dual-Coupled PnP Diffusion (DC-PnPDP)**, which restores the classical *dual variable* to provide integral feedback, progressively enforce agreement between the data-consistency and prior. However, this rigorous geometric coupling introduces a secondary challenge: the accumulated dual residuals exhibit spectrally colored, structured artifacts that violate the Additive White Gaussian Noise (AWGN) assumption of diffusion priors, causing severe hallucinations. To bridge this gap, we introduce **Spectral Homogenization (SH)**, a frequency-domain adaptation mechanism that modulates these structured residuals into statistically compliant *pseudo-AWGN* inputs. This effectively aligns the solver's rigorous optimization trajectory with the denoiser's valid statistical manifold. Extensive experiments on CT and MRI reconstruction demonstrate that our approach resolves the bias-hallucination trade-off, achieving state-of-the-art fidelity with significantly accelerated convergence. The code is available at https://github.com/duchenhe/DC-PnPDP.

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scChord: A Probabilistic Manifold Rectification Framework for RNA-to-Protein Translation

Jiawei Zhang, Kangjun Jin, Shuai Xiao, Jiachen Yang

Measuring single-cell protein abundance is essential for resolving biological mechanisms and disease progression with high resolution. However, due to the high costs and antibody throughput limitations of current proteomics, inferring protein levels from readily available RNA data has become a critical computational necessity. Existing regression and generative methods face a fundamental geometric bottleneck: enforcing deterministic constraints on noisy, heteroscedastic data collapses intrinsic uncertainty into a rough latent manifold, which destabilizes the learning process. To overcome this, we present scChord, a noise-decoupled conditional flow matching framework built on Probabilistic Manifold Rectification. Our approach utilizes a probabilistic decoder to disentangle technical noise and over-dispersion from the raw counts, absorbing them into distributional parameters. This allows the rectified latent manifold to focus more on biological signals, serving as a robust geometric regularizer for learning efficient transport trajectories. Extensive experiments on multiple multi-omics benchmarks demonstrate that scChord not only achieves state-of-the-art inference accuracy but also faithfully reconstructs high-fidelity biological heterogeneity and complex protein distributions.

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AutoWebWorld: Synthesizing Infinite Verifiable Web Environments via Finite State Machines

Yifan Wu, Yiran Peng, Yiyu Chen, Jianhao Ruan, Zijie Zhuang, Cheng Yang, Jiayi Zhang, Man CHEN, Yenchi Tseng, Zhaoyang Yu, Liang Chen, Yuyao Zhai, Bang Liu, Chenglin Wu, Yuyu Luo

The performance of autonomous Web GUI agents heavily relies on the quality and quantity of their training data. However, a fundamental bottleneck persists: collecting interaction trajectories from real-world websites is expensive and difficult to verify. The underlying state transitions are hidden, leading to reliance on inconsistent and costly external verifiers (e.g., human or LLM judges) to evaluate step-level correctness. To address this, we propose AutoWebWorld, a novel framework for synthesizing controllable and verifiable web environments by modeling them as Finite State Machines (FSMs) and use coding agents to translate FSMs into interactive websites. Unlike real websites, where state transitions are implicit, AutoWebWorld explicitly defines all states, actions, and transition rules. This enables programmatic verification: action correctness is checked against predefined rules, and task success is confirmed by reaching a goal state in the FSM graph. AutoWebWorld enables a fully automated search-and-verify pipeline, generating over 11,663 verified trajectories from 29 diverse web environments at only \$0.04 per trajectory. Training on this synthetic data significantly boosts

real-world performance. Our 7B Web GUI agent achieves state-of-the-art on WebVoyager, outperforming all baselines within 15 steps. Furthermore, we observe a clear scaling law: as the synthetic data volume increases, performance on WebVoyager and Online-Mind2Web consistently improves.
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Demystifying Scientific Problem-Solving in LLMs by Probing Knowledge and Reasoning
Alan Li, Yixin Liu, Arpan Sarkar, Doug Downey, Arman Cohan
Scientific problem solving poses unique challenges for LLMs, requiring both deep domain knowledge and the ability to apply such knowledge through complex reasoning. While automated scientific reasoners hold great promise for assisting human scientists, there is currently no widely adopted holistic benchmark for evaluating scientific reasoning, and few approaches systematically disentangle the distinct roles of knowledge and reasoning in these tasks.

To address these gaps, we introduce **SciReas**, a diverse suite of existing benchmarks for scientific reasoning tasks, and **SciReas-Pro**, a selective subset that requires more complex reasoning. Our holistic evaluation surfaces insights about scientific reasoning performance that remain hidden when relying on individual benchmarks alone. We then propose **KRUX**, a probing framework for studying the distinct roles of reasoning and knowledge in scientific tasks.

Combining the two, we conduct an in-depth analysis that yields several key findings:

(1) Retrieving task-relevant knowledge from model parameters is a critical bottleneck for LLMs in scientific reasoning;

(2) Reasoning models consistently benefit from external knowledge added in-context on top of the reasoning enhancement;

(3) Enhancing verbalized reasoning improves LLMs' ability to surface task-relevant knowledge.
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Autoregressive Image Generation with Masked Bit Modeling
Qihang Yu, Qihao Liu, Ju He, Xinyang Zhang, Yang Liu, Liang-Chieh Chen, Xi Chen
This paper challenges the dominance of continuous pipelines in visual generation. We systematically investigate the performance gap between discrete and continuous methods. Contrary to the belief that discrete tokenizers are intrinsically inferior, we demonstrate that the disparity arises primarily from the total number of bits allocated in the latent space (i.e., the compression ratio). We show that scaling up the codebook size effectively bridges this gap, allowing discrete tokenizers to match or surpass their continuous counterparts. However, existing discrete generation methods struggle to capitalize on this insight, suffering from performance degradation or prohibitive training costs with scaled codebook. To address this, we propose masked **B**it **A**uto**R**egressive modeling (**BAR**), a scalable framework that supports arbitrary codebook sizes. By equipping an autoregressive transformer with a masked bit modeling head, BAR predicts discrete tokens through progressively generating their constituent bits. BAR achieves a new state-of-the-art gFID of **0.99** on ImageNet-256, outperforming leading methods across both continuous and discrete paradigms, while significantly reducing sampling costs and converging faster than prior continuous approaches. Project page is available at https://bar-gen.github.io
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DSENet: A Novel Dual-Stream Enhancement Network for Multi-Scale Non-Stationary Time Series Forecasting
Yuhan Wang, Yuanyuan Zou, Jie Cheng, Bin Dai, Jinhong Guo
Accurately capturing local variations in long series has always been
one of the most challenging problems in time-series forecasting
especially in medical signals, where local variations often indicate pathologica

l events. Our study reveals a previously overlooked key bottleneck in this field : traditional global and local branches learn similar representations, leading t o strong feature coupling and reduced sensitivity to local variations. To addres s this challenge, we propose the novel Dual-Stream Enhancement Mechanism, which structurally enlarges the difference between global and local patterns, enabling weak interactions between the two. Based on this idea, we introduce a new basel ine model for blood glucose prediction: Dual-Stream Enhancement Network (DSENet) , which fundamentally alleviates the problem of excessively strong coupling betw een global and local features. Experimental results show that our model achieves SOTA performance on multiple public datasets. Moreover, benefiting from extreme ly low computational cost, our model demonstrates strong application potential a nd can serve as a baseline model in multiple domains in the future.

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Mirror Descent Under Generalized Smoothness

Dingzhi Yu, Wei Jiang, Hongyi Tao, Yuanyu Wan, Lijun Zhang

Smoothness is crucial for attaining fast rates in first-order optimization. Howe ver, many optimization problems in modern machine learning involve non-smooth ob jectives. Recent studies relax the smoothness assumption by allowing the Lipschi tz constant of the gradient to grow with respect to the gradient norm, which acc ommodates a broad range of objectives in practice. Despite this progress, existi ng generalizations of smoothness are restricted to Euclidean geometry with $\ell _2$-norm and only have theoretical guarantees for optimization in the Euclidean space. In this paper, we address this limitation by introducing a new $\ell*$-sm oothness concept that measures the norm of Hessians in terms of a general norm a nd its dual, and establish convergence for mirror-descent-type algorithms, match ing the rates under the classic smoothness. Notably, we propose a generalized se lf-bounding property that facilitates bounding the gradients via controlling sub optimality gaps, serving as a principal component for convergence analysis. Beyo nd deterministic optimization, we establish sharp convergence for stochastic mir ror descent, matching state-of-the-art under classic smoothness. Our theory also extends to non-convex and composite optimization, which may shed light on pract ical usages of mirror descent, including pre-training and post-training of LLMs.

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Layer-wise Gradient Disentanglement: Decoupling Semantics and Preferences in Dir ect Preference Optimization

Mengyang Li, Shuang Liu, Zhong Zhang

Direct Preference Optimization (DPO) has become the dominant approach for aligni ng large language models with human preferences. However, standard DPO treats al l preference pairs uniformly, overlooking the heterogeneous nature of the learni ng problem: some samples demand sophisticated semantic understanding of the prom pt, while others require nuanced discrimination between similar responses. We ar gue that these two objectives should be disentangled during training. Through gr adient analysis, we identify a layer-wise localization phenomenon where semantic complexity predominantly drives lower-layer updates while preference uncertaint y modulates upper layers. Building on this insight, we propose Gradient-Guided D isentangled DPO (GDO-DPO), a curriculum framework that independently regulates l earning pace along each dimension based on layer-specific gradient stability. Ex periments on UltraFeedback and HH-RLHF demonstrate consistent improvements, with GDO-DPO outperforming DPO by 4.1\% on AlpacaEval 2.0 and showing particularly s trong gains on reasoning-intensive tasks.

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Hierarchical Decision Making with Structured Policies: A Principled Design via I nverse Optimization

Yuexuan Wang, Jingyuan Zhou, Kaidi Yang

Hierarchical decision-making frameworks are pivotal for addressing complex contr ol tasks, enabling agents to decompose intricate problems into manageable subgo als. Despite their promise, existing hierarchical policies face critical limitat ions: (i) reinforcement learning (RL)-based methods struggle to guarantee strict constraint satisfaction, and (ii) optimal control (OC)-based approaches often r ely on myopic and computationally prohibitive formulations. To reconcile these t

rade-offs, hierarchical RL-OC architectures have emerged as a promising paradigm . However, the formulation of the lower-level optimization within these framewor ks remains underexplored, often relying on heuristic or myopic objectives. In th is work, we propose a principled framework that systematically integrates upper- level goal abstraction with structured lower-level decision making. We adopt an inverse optimization approach to inform the structure of the lower-level problem from expert demonstrations, ensuring that the objective of the lower-level poli cy remains aligned with the overall long-term task goal. To validate the approac h, our framework is evaluated on distinct decision making tasks: network-based r esource allocation and continuous collision avoidance. Empirical results demonst rate that our method consistently outperforms strong baselines based on end-to-e nd RL, learning-augmented optimal control, and existing hierarchical RL approach es in both efficiency and decision quality.

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Best-of-Both-Worlds for Heavy-Tailed Markov Decision Processes

Yu Chen, Yuhao Liu, Jiatai Huang, Yihan Du, Longbo Huang

We investigate episodic Markov Decision Processes with heavy-tailed losses (HTMD Ps). Existing approaches for HTMDPs are conservative in stochastic environments and lack adaptivity in adversarial regimes. In this work, we propose algorithms HT-FTRL-OM and HT-FTRL-UOB for HTMDPs that achieve Best-of-Both-Worlds (BoBW) gu arantees: instance-independent regret in adversarial environments and logarithmi c instance-dependent regret in self-bounding (including the stochastic case) env ironments. For the known transition setting, HT-FTRL-OM applies the Follow-The-R egularized-Leader (FTRL) framework over occupancy measures with novel skipping l oss estimators, achieving a $\widetilde{\mathcal{O}}(T^{1/\alpha})$ regret bound in adversarial regimes and a $\mathcal{O}(\log T)$ regret in stochastic regimes . Building upon this framework, we develop a novel algorithm HT-FTRL-UOB to tack le the more challenging unknown-transition setting. Under a mild truncative nonn egativity condition on the loss distributions, this algorithm employs a pessimis tic skipping loss estimator and achieves a $\widetilde{\mathcal{O}}(T^{1/\alpha} + \sqrt{T})$ regret in adversarial regimes and a $\mathcal{O}(\log^2(T))$ regre t in stochastic regimes. Our analysis overcomes key barriers through several tec hnical insights, including a local control mechanism for heavy-tailed shifted lo sses, a new suboptimal-mass propagation principle, and a novel regret decomposit ion that isolates transition uncertainty from heavy-tailed estimation errors and skipping bias.

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Learning Multi-Scale Hypergraph for High-Order Brain Connectivity Analysis

Jaeyoon Sim, Soojin Hwang, Seunghun Baek, Guorong Wu, Won Hwa Kim

Understanding complex interactions between brain regions is critical for early n eurodegenerative disease classification such as Alzheimer’s Disease (AD) and Par kinson’s Disease (PD). While graph-based models are widely used to analyze brain networks, most existing approaches primarily focus on pairwise interactions bet ween directly connected nodes, limiting their ability to capture higher-order de pendencies across multiple regions. Although hypergraph-based methods have been proposed to model higher-order relations, many rely on predefined hyperedges or restrict learning to hyperedge weights, reducing flexibility and limiting their capacity to capture multi-resolution structural patterns. In this regard, we int roduce an adaptive multi-scale hyperedge learning framework, i.e., MuHL, which c onstructs hierarchical node features and dynamically learns high-order interacti on through continuous hyper-edge construction over multi-resolution graph signal s. Extensive experiments on multiple brain network benchmarks demonstrate that M uHL consistently improves disease classification performance across different st ages, and further identifies key regions of interest (ROIs) and their group-wise interactions from the learned hyperedges that are associated with disease progr ession, highlighting its potential as a powerful tool for brain network analysis with neurodegenerative disorders.

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When Embedding-Based Defenses Fail: Rethinking Safety in LLM-Based Multi-Agent S ystems

Lingxi Zhang, Guangtao Zheng, Hanjie Chen
Large language model (LLM)-powered multi-agent systems (MAS) enable agents to co
mmunicate and share information, achieving strong performance on complex tasks.
However, this communication also creates an attack surface where malicious agent
s can propagate misinformation and manipulate group decisions, undermining MAS s
afety. Existing embedding-based defenses aim to detect and prune suspicious agen
ts, but their effectiveness depends on a clear separation between the text embed
dings of malicious and benign messages.
Attackers can circumvent such defenses by crafting messages whose embeddings lie
 close to benign ones.
We analyze this failure mode theoretically and validate it empirically with thre
e attacks, Slow Drift, Benign Wrapper, and Chaos Seeding. Our analysis further r
eveals a fundamental limitation of embedding-based defenses: because they rely s
olely on the text embeddings, they ignore token-level confidence signals such as
 logits, which can remain informative when embeddings are not distinguishable un
der attack.
We propose using confidence scores to prune or down-weight messages during MAS c
ommunication. Experiments show improved robustness across models, datasets, and
communication topologies. Moreover, we find that the effectiveness of confidence
 signals decays over communication rounds, highlighting the importance of early
intervention.
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DPO Unchained: Your Training Algorithm is Secretly Disentangled in Human Choice
Theory (and Its Loss' Convexity is Dispensable)
Wenxuan Zhou, Shujian Zhang, brice magdalou, John Wheatley Lambert, Ehsan Amid,
Richard Nock, Andrew Hard
Normative theories allow one to elicit key parts of a ML algorithm from first pr
inciples, which is crucial at a time of championed scrutiny for ML work. Direct
Preference Optimization (DPO) cleverly bypasses reward modeling by making an exp
licit link with a specific normative model of human choice. Our paper elevates t
his connection to the full generality of DPO's normative framework. Getting ther
e requires reworking social choice theory's textbook path for a better RLHF/ML f
it. It elevates the connection to a remarkably broad viewpoint on preference opt
imization, considering the current panorama of DPO follow-ups. It also unveils u
nexpected riches for ML, chief among which the support for *non-convex* losses,
the fact that *any* compliant ML analytical choice can be embedded with *any* hu
man choice model, and a normative framework's umbrella wide enough to safeguard
DPO's *extensions* (margins, length correction, ...). A *toy* experiment ``far a
way'' from the DPO crowd is given.
****
Scheduling Thoughts: Learning the Order of Thought in Diffusion Language Models
Jiawei Xu, Minghui Liu, Aakriti Agrawal, Yifan Chen, Furong Huang
Masked diffusion language models decode by iteratively unmasking tokens, where t
he unmasking order defines an ``order of thought''
that strongly influences generation quality yet is typically chosen heuristicall
y.
We derive a tractable upper bound on the sequential decoding mismatch, measured
by the Kullback–Leibler divergence and expressed in terms of the model’s pathwis
e log-likelihood, with tightness under sufficient model expressivity.
This bound induces a dense self-aware reward for a target sequence x and unmas
king order σ,
over ordered paths,
casting order selection as a principled policy optimization problem with a froze
n denoiser.
We instantiate this idea as **Self-Aware Scheduling (SAS)**, which learns a ligh
tweight order policy using Group Relative Policy Optimization and applies seamle
ssly to both sequential and semi-autoregressive decoding.
On Sudoku with 1B MDM, SAS improves puzzle accuracy from 82.0% (best heuristi
c schedule) to 91.8%, and reaches 97.9% with second-stage fine-tuning alon
g learned trajectories.

On LLaDA-8B, SAS improves pass@1 on GSM8K from 64% to 76% (full diffusion) and on MBPP from 39.5% to 41%, while consistently matching or exceeding heuristic schedules across generation lengths and block sizes.
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Evolving Quantitative Reasoning through Self-Play in Digital Twin Markets
Tianmi Ma, Wenxin Huang, Jiawei Du, Lin Li, Xian Zhong, Joey Tianyi Zhou
Large language models (LLMs) demonstrate strong capabilities in high-level semantic reasoning and strategic planning, making them appealing for complex decision-making tasks; however, their quantitative reasoning remains unreliable despite recent progress in tool-augmented and structured inference. To address this limitation, we decouple reasoning from computation by assigning LLMs to planning, analysis, and result interpretation, while delegating numerical computation and statistical inference to specialized external tools. Rather than being hard-coded, these tools are constructed in a constrained and structured manner during planning as explicit intermediate reasoning artifacts, enabling adaptive and scenario-dependent quantitative reasoning. LLMs iteratively analyze tool outputs under diverse market conditions and leverage performance-based feedback to refine subsequent tool selection and construction, thereby forming a bounded self-evolving loop. We instantiate this process through self-play in a controllable digital twin market, DecoupledMarket, where LLM agents continuously test, compare, and adapt their strategies. By coupling high-level planning with robust quantitative execution, the proposed framework enhances the quantitative reliability of LLM-driven decision-making. All code and data are available at https://github.com/MTMQuantAI/Agent-Trading-Arena.git.
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Difference-Aware Decision Learning for Multimodal Image Fusion
Hao Pan, Jian Dai, Yuan Sun, Zhenwen Ren, Xingfeng Li
Multimodal image fusion aims to integrate complementary information from different modalities, but cross-modal discrepancies and local conflicts often make modality allocation uncertain, causing information loss or artifact propagation.
We address this problem by formulating fusion as an observation-conditioned probabilistic decision-learning problem, where local modality contribution is explicitly modeled as a decision variable. Based on this view, we propose a dIfference-aware Decision-lEArning muLtimodal image fusion paradigm (IDEAL). IDEAL uses cross-modal differences as decision triggers and constructs spatial and spectral decision conditions from multi-scale difference attention, power-spectrum energy, complementary spectra, and spectral-entropy reliability. These conditions are mapped to interpretable contribution policies through a symmetric Beta prior, while uncertainty modulation pulls unreliable decisions toward conservative mixing when evidence is insufficient. Extensive experiments on multiple fusion tasks demonstrate stable and competitive performance against state-of-the-art methods. Code is available at: https://github.com/Pon915/IDEAL-main.
****

Robust Vision-Language Models via Manifold-Adversarial Adapters
Hao Li, Zeyu Xiao, Junhao Zhou, Peng Liu, Yang Zhao, Wei Jia
Vision-language models (VLMs) have progressed rapidly with large-scale high-quality data and adaptation strategies, yet remain brittle under real-world corruptions, where both visual recognition and language-grounded reasoning degrade. Beyond cascaded image restoration, a natural alternative is parameter-efficient adaptation that aligns corrupted features with clean references; however, Euclidean alignment alone is not semantics-preserving and can even harm downstream reasoning. We attribute this to a semantic misalignment gap, where features become geometrically closer while drifting off the in-distribution support on which multimodal reasoning is calibrated. To address this, we propose Manifold-Adversarial Adapters (MAA), parameter-efficient layer-wise modules for a frozen vision encoder that explicitly steer corrupted features back onto the clean in-distribution manifold rather than merely shrinking feature-space distance. MAA combines paired feature self-distillation with a token-level adversarial manifold constraint to prevent off-manifold semantic shortcuts. At inference, only the adapters are retained, enabling single-stage robustness with negligible overhead and avoiding th

e latency and semantic drift of restoration pipelines. Across benchmarks and cor
ruption settings, MAA consistently improves performance over strong baselines.
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GeoMoLa: Geometry-Aware Motion Latents for Learning Robust Manipulation Policies
Yunchao Zhang, Yijia Weng, Ruizhe Liu, Ming Hu, Leonidas Guibas, Yanchao Yang
Learning motion latents for robotic manipulation heavily relies on extracting mo
tion patterns from visual sequences, yet effective action abstractions require u
nderstanding three-dimensional geometric transformations. Here, we introduce Geo
MoLa (Geometry-Aware Motion Latents), which learns discrete motion latent codes
by predicting how point clouds evolve during manipulation rather than reconstruc
ting visual observations. This four-dimensional objective – spatial geometry cha
nging through time – forces latent representations to encode actual physical mot
ion rather than appearance patterns. GeoMoLa achieves state-of-the-art performan
ce using only single-view RGB-D input, while existing methods require multi-view
 reconstruction, succeeding across diverse manipulation benchmarks. Our ablation
s reveal that geometric prediction is the key to driving performance, quantitati
vely validating that manipulation depends on spatial understanding. Furthermore,
 the learned codes exhibit effective motion abstraction: applying them to novel
scenes produces physically consistent transformations regardless of visual conte
xt. Our real-world experiments also confirm this robustness capability, achievin
g robust manipulation with minimal demonstrations in cluttered environments wher
e geometric reasoning determines success. Thus, we demonstrate that effective mo
tion latents for robot control can better emerge from understanding motion throu
gh its three-dimensional effects rather than pixel-level patterns.
****

Entropy-Aware Dynamic KV Cache Sparsification for Autoregressive Image Generati
on and Editing
Tong Tong, Ling Xing, Linjie Li, Rui Yan, Zhengyuan Yang, Lijuan Wang, Alex Jinp
eng Wang
Autoregressive (AR) image generation has recently gained momentum as a scalable
alternative to diffusion models, benefiting from unified next-token prediction p
aradigm and strong instruction following ability. However, AR visual generation
must decode excessively long sequences of visual tokens, making inference heavil
y bottlenecked by the memory footprint and latency of the self-attention KV cach
e. While KV cache compression is well studied in Large Language Model, its count
erparts in AR image generation remain underexplored. The reason is fundamental:
visual tokens are highly redundant, and their spatial information density is hig
hly non-uniform. In this work, we introduce SparseAR, a training-free, entropy-a
ware sparse attention method that is specifically designed for AR image generati
on and editing. Our key insight is that information-rich regions exhibit higher
entropy and require broader attention, while redundant regions show lower entrop
y and allow aggressive sparsification. Based on this insight, we dynamically ide
ntify information-rich regions during decoding and adaptively adjust attention s
parsity to reduce KV-cache overhead. SparseAR is plug-and-play and can be readil
y applied to mainstream AR models. Extensive experiments on four representative
AR models across multiple benchmarks demonstrate that SparseAR significantly im
proves inference efficiency while maintaining, and often even improving, generat
ion and editing quality.
****

Towards Execution-Grounded Automated AI Research
Chenglei Si, Zitong Yang, Yejin Choi, Emmanuel Candes, Diyi Yang, Tatsunori Hash
imoto
Automated AI research holds great potential to accelerate scientific discovery.
However, current LLMs often generate plausible-looking but ineffective ideas.
Execution grounding may help, but it is unclear whether automated execution is f
easible and whether LLMs can learn from the execution feedback.
To investigate these, we first build an automated executor to implement ideas an
d launch large-scale parallel GPU experiments to verify their effectiveness.
We then convert two realistic research problems -- LLM pre-training and post-tra
ining -- into execution environments and demonstrate that our automated executor

can implement a large fraction of the ideas sampled from frontier LLMs.
We analyze two methods to learn from the execution feedback: evolutionary search and reinforcement learning.
Execution-guided evolutionary search is sample-efficient: it finds a method that significantly outperforms the GRPO baseline
on post-training, and finds a pre-training recipe that outperforms the nanoGPT baseline
on pre-training, all within just ten search epochs.
Frontier LLMs often generate meaningful algorithmic ideas during search, but they tend to saturate early and only occasionally exhibit scaling trends.
Reinforcement learning from execution reward, on the other hand, suffers from mode collapse.
It successfully improves the average reward of the ideator model but not the upper-bound, due to models converging on simple ideas.
We thoroughly analyze the executed ideas and training dynamics to facilitate future efforts.
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Breaking the Capacity Bottleneck in Model-Heterogeneous Federated Learning via Gradual Model Restoration
Chengjie Ma, Seungeun Oh, Jihong Park, Seong-Lyun Kim
Federated learning (FL) enables distributed model training, yet in heterogeneous deployments, Bandwidth-Constrained Clients (BCCs) often contribute inefficiently due to limited uplink bandwidth. In model-heterogeneous FL with fixed small sub-models, BCCs with sub-models may improve quickly in early rounds but become under-parameterized later, resulting in slow convergence and poor generalization.
To address this challenge, we propose **FedGMR**, a federated learning framework centered around **Gradual Model Restoration (GMR)**, where GMR progressively increases each client’s sub-model density during training, allowing BCCs to remain effective contributors throughout optimization. To make GMR practical under real-world heterogeneity, FedGMR is realized as an end-to-end workflow with asynchronous coordination and stable, mask-aware aggregation. We further establish convergence guarantees, showing that the aggregation error scales with the **average** sub-model density across clients and rounds, and that GMR provably narrows the gap toward full-model FL. Extensive experiments on FEMNIST, CIFAR-10, ImageNet-100, and StackOverflow demonstrate that FedGMR improves both convergence speed and final accuracy, especially under severe heterogeneity and non-IID data distributions.
Code is available at https://github.com/machengjie321/ICML2026-FedGMR.
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Learning Stochastic Bridges for Video Object Removal via Video-to-Video Translation
Zijie Lou, Xiangwei Feng, Jiaxin Wang, Jiangtao Yao, Fei Che, Tianbao Liu, WU CHENGJING, Xiaochao Qu, Luoqi Liu, Ting Liu
Existing video object removal methods predominantly rely on diffusion models following a noise-to-data paradigm, where generation starts from uninformative Gaussian noise. This approach discards the rich structural and contextual priors present in the original input video. Consequently, such methods often lack sufficient guidance, leading to incomplete object erasure or the synthesis of implausible content that conflicts with the scene's physical logic. In this paper, we reformulate video object removal as a video-to-video translation task via a stochastic bridge model. Unlike noise-initialized methods, our framework establishes a direct stochastic path from the source video (with objects) to the target video (objects removed). This bridge formulation effectively leverages the input video as a strong structural prior, guiding the model to perform precise removal while ensuring that the filled regions are logically consistent with the surrounding environment. To address the trade-off where strong bridge priors hinder the removal of large objects, we propose a novel adaptive mask modulation strategy. This mechanism dynamically modulates input embeddings based on mask characteristics, balancing background fidelity with generative flexibility. Extensive experiments demonstrate that our approach significantly outperforms existing methods in bo

th visual quality and temporal consistency. The project page is https://bridgere
moval.github.io/.
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SphericalDreamer: Generating Navigable Immersive 3D Worlds with Panorama Fusion
Antoine Schnepf, Karim Kassab, Flavian Vasile, Andrew I. Comport
The generation of immersive and navigable 3D environments is increasingly preval
ent with the growing adoption of virtual reality and 3D content.
However, recent methods face a fundamental limitation: they cannot produce 3D wo
rlds that simultaneously (i) are navigable over long-range spatial extents and (
ii) cover the complete omnidirectional field of view (360° horizontally a
nd 180° vertically).
To address this challenge, we introduce SphericalDreamer, a method for generatin
g fully immersive and long-range 3D outdoor environments from textual prompts.
Our approach is built on the generation of multiple panoramic images, which are
subsequently lifted into 3D and fused together while maintaining visual and geom
etric consistency.
SphericalDreamer produces highly detailed, fully immersive 3D environments, whil
e substantially improving scale and navigability compared to prior approaches.
****

BYORn: Bootstrap Your Own Responses to Defend Large Vision-Language Models Again
st Backdoor Attacks
Ivan Sabolic, Marin Orši■, Josip Šari■, Sven Loncaric
Supervised fine-tuning is the predominant approach for adapting autoregressive v
ision–language models to downstream tasks. Recent work has shown that this parad
igm is highly vulnerable to backdoor attacks, and that existing defenses are ine
ffective in open-ended generation settings. In response, we propose BYORn, a bac
kdoor-robust fine-tuning framework motivated by the observation that poisoned ta
rget responses are often semantically implausible given the corresponding image–
text inputs and a pretrained model. BYORn identifies such misaligned responses a
nd dynamically replaces them with alternative responses generated by the model,
thereby breaking the correlation between triggers and target outputs. The result
ing objective gradient corresponds to the gradient of the empirical estimate of
the population risk upper bound over the clean data distribution. Empirically, B
YORn consistently improves robustness to backdoor attacks while preserving clean
-task performance, establishing a new trade-off frontier between generalization
and attack success rate. Finally, we demonstrate that BYORn remains effective ag
ainst adaptive attacks specifically designed to circumvent the proposed defense.
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Reconstructing Template-Memorized Images from Natural Prompts
Sol Yarkoni, Mahmood Sharif, Roi Livni
Recent advances in generative models, such as diffusion models, have raised conc
erns related to privacy, copyright infringement, and data curation. Prior work h
as shown that training data can be reconstructed from such models, but existing
attacks typically rely on substantial computational resources, access to the tra
ining set, or carefully engineered prompts.

In this work, we present a low-resource reconstruction attack that operates thro
ugh seemingly benign prompts and requires little to no access to the training da
ta. Our attack targets **template-memorized images (TMI)**, where recurring layo
uts and visual structures are memorized during training. We show that such memor
ization manifests under potentially realistic usage. This raises a possibility o
f unintentional reconstruction by naive users that don't carry explicit adversar
ial intent. For example, we observe that a simple prompt such as "blue Unisex T-
Shirt" can reproduce visual content depicting a real individual. Beyond extracti
on, we observe novel phenomena occurring in TMI (e.g., interpolation), raising q
uestions about the novelty of generated content and the effectiveness of establi
shed methods for detecting memorized content.

Our code is available at \url{https://github.com/TheSolY/lr-tmi}.
****

Temporal Weighted Encoding: Towards Maximal-Capacity Spike Coding for ANN–SNN Co
nversion
Yiwen Gu, Junchuan Gu, Haibin Shen, Kejie Huang
Spiking Neural Networks (SNNs) emulate the spiking behavior of biological neuron
s
and are promising for energy-efficient neuromorphic computing.
A widely used strategy to train SNNs is to convert pretrained Artificial Neural
Networks (ANNs),
where the accuracy and efficiency are determined by the spike encoding scheme.
Traditional methods based on spike count or timing severely underutilize the ava
ilable encoding space,
leading to large accuracy degradation under low-timestep constraints. More expre
ssive alternatives
involve complex dynamics, which hinder scalability and practical deployment.
To address these challenges, we propose Temporal Weighted Encoding (TWE).
Spikes are implicitly assigned exponentially decaying weights through a recursiv
e integration, drawing an analogy
to a temporal bit sequence. We systematically analyze the temporal mismatch caus
ed by this weight pattern
and propose temporal relaxation and threshold relaxation to resolve this issue,
enabling fast and
accurate activation encoding. Extensive experiments demonstrate that TWE achieve
s negligible conversion
loss with significantly fewer timesteps, offering a scalable and efficient solut
ion for SNN deployment.
****

Continual Learning through Control Minimization
Sander de Haan, Yassine Taoudi-Benchekroun, Pau Vilimelis Aceituno, Benjamin F G
rewe
Catastrophic forgetting remains a fundamental challenge for neural networks when
 tasks are trained sequentially. In this work, we reformulate continual learning
 as a control problem where learning and preservation signals compete within neu
ral activity dynamics. We convert regularization penalties into preservation sig
nals that protect prior-task representations. Learning then proceeds by minimizi
ng the control effort required to integrate new tasks while competing with the p
reservation of prior tasks. At equilibrium, the neural activities produce weight
 updates that implicitly encode the full prior-task curvature, a property we ter
m the *continual-natural gradient*, requiring no explicit curvature storage. Exp
eriments confirm that our learning framework recovers true prior-task curvature
and enables task discrimination, outperforming existing methods on standard benc
hmarks without replay.
****

Is Spurious Correlation Removal Always Learnable?
Yibo Zhou, Bo Li, Hai-Miao Hu, Hanzi Wang, Xiaokang Zhang, Ruifan Zhang
Invariant learning can fail even when the invariant structure is statistically i
dentifiable. We show a conditional computational barrier: under a black-box samp
lable supervised sparse recovery primitive motivated by average-case sparse-reco
very reductions, there exist \emph{samplable} multi-environment instances with a
 one-dimensional predictive invariant subspace ($k=1$) that are learnable with p
olynomial samples by exhaustive search, while any polynomial-time constant-accur
acy recovery algorithm would contradict the primitive. We further quantify envir
onment diversity by a separation parameter γ, which controls identifiabil
ity and the curvature of invariance objectives. Under sufficient diversity and l
ocal Gaussian regularity, the minimax risk is $\mathbb{E}[dist(\hat{V},V_{\mathr
m{inv}})^2]=\Theta(k(d-k)/(n|\mathcal{E}|))$, and under label-induced shifts a p
hase transition occurs at $n^*\propto k(d-k)/(|\mathcal{E}|\gamma^2)$ with refin
ed estimation error scaling proportional to $1/\gamma^2$. Synthetic and real dat
asets illustrate the predicted gaps and transitions and motivate simple diversit
y diagnostics.
****

Memory-Efficient LLM Training with Dynamic Sparsity: From Stability to Practical Scaling
Qiao Xiao, Boqian Wu, Patrik Okanovic, Tomasz Sternal, Maurice van Keulen, Elena Mocanu, Mykola Pechenizkiy, Decebal Constantin Mocanu, Torsten Hoefler
Dynamic Sparse Training (DST) offers a promising paradigm for improving the training and inference efficiency of deep neural networks; however, we find that in large language model training, DST can suffer from optimization instability, manifested as loss spikes after topology updates. In this work, we show that the naive use of standard Adam-based optimizers leads to a cold-start issue for newly regrown parameters, resulting in excessively large updates and disrupted training dynamics. To address this issue, we propose Sparse Memory-Efficient Training (SMET), which stabilizes DST with optimizer warm-up and improves training progress through density-aware learning-rate scaling. SMET further reduces memory consumption by storing gradients and optimizer states only for active parameters. We provide a theoretical analysis of the update behaviors under SMET, showing improved optimization stability. Extensive experiments demonstrate that SMET enables stable, scalable, and memory-efficient sparse pre-training of LLMs, paving the way for sparse training as a practical alternative to dense training. Our code is publicly available at: https://github.com/QiaoXiao7282/SMET.
****
Tempora: Characterising the Time-Contingent Utility of Online Test-Time Adaptation
Sudarshan Sreeram, Young D. Kwon, Cecilia Mascolo
Test-time adaptation (TTA) offers a compelling remedy for machine learning (ML) models that degrade under domain shifts, improving generalisation *on-the-fly* with only unlabelled samples. This flexibility suits real deployments, yet conventional evaluations unrealistically assume unbounded processing time, overlooking the accuracy-latency trade-off. As ML increasingly underpins latency-sensitive and user-facing use-cases, temporal pressure constrains the viability of adaptable inference; predictions arriving too late to act on are futile. We introduce *Tempora*, a framework for evaluating TTA under this pressure. It consists of temporal scenarios that model deployment constraints, evaluation protocols that operationalise measurement, and time-contingent utility metrics that quantify the accuracy-latency trade-off. We instantiate the framework with three such metrics: (1) *discrete* utility for asynchronous streams with hard deadlines, (2) *continuous* utility for interactive settings where value decays with latency, and (3) *amortised* utility for budget-constrained deployments. By applying Tempora to 11 TTA methods, we find that *rank instability* persists across 750+ temporal evaluations spanning diverse datasets, models, and hardware platforms; *i.e.,* conventional rankings do not predict rankings under temporal pressure. The highest-utility method varies with the shift and temporal pressure, with no clear winner. By enabling systematic evaluation across diverse temporal constraints for the first time, Tempora reveals when and why rankings change, offering practitioners a lens for method selection and researchers a target for deployable adaptation. Code: https://github.com/sudotensor/tempora.
****
Semi-Supervised Noise Adaptation: Transferring Knowledge from Noise Domain
Yuan Yao, Jin Song, Huixia Li, Tongtong Yuan, Jiaqi Wu, Yu Zhang
Transfer learning aims to facilitate the learning of a target domain by transferring knowledge from a source domain. The source domain typically contains semantically meaningful samples (*e.g.*, images) to facilitate effective knowledge transfer. However, a recent study observes that the noise domain constructed from simple distributions (*e.g.*, Gaussian distributions) can serve as a surrogate source domain in the semi-supervised setting, where only a small proportion of target samples are labeled while most remain unlabeled. Based on this surprising observation, we formulate a novel problem termed *Semi-Supervised Noise Adaptation* (SSNA), which aims to leverage a synthetic noise domain to improve the generalization of the target domain. To address this problem, we first establish a generalization bound characterizing the effect of the noise domain on generalization, based on which we propose a Noise Adaptation Framework (NAF). Extensive experi

ments demonstrate that NAF effectively leverages the noise domain to tighten the generalization bound of the target domain, leading to improved performance. The codes are available at https://github.com/AIResearch-Group/SSNA.
****
Finding the Minimal Parameter Budget for Implicit Reasoning: A Data Complexity Driven Scaling Law for Language Models
Xinyi Wang, Shawn Tan, Shenbo Xu, Mingyu Jin, William Yang Wang, Rameswar Panda, Yikang Shen
Reasoning is a core capability of language models (LMs), yet it remains unclear how much model capacity is necessary to support reasoning during pretraining. In this work, we study the minimal parameter budget required for implicit reasoning, defined as the ability to infer new facts from learned knowledge without explicit chain-of-thought supervision. To isolate this phenomenon, we pretrain LMs from scratch in a controlled synthetic environment that mimics the structure and distribution of real-world knowledge graphs, and evaluate their ability to complete missing edges via multi-hop inference.
From both a theoretical and an empirical perspective, we identify a scaling law linking this optimal parameter budget to a graph search entropy measure.
Across a wide range of model sizes, training steps, and graph complexities, we show that an optimally sized language model can reliably reason over approximately 0.008 bits of information per parameter at most.
Our results characterize the minimal sufficient capacity for implicit reasoning during pretraining. Our findings provide principled guidance for matching model size to data complexity and offer new insights into the scaling behavior of reasoning in large language models.
****
Probabilistic Robustness Certificates against Adversarial Attacks
Sara Taheri, Majid Zamani
The growing use of machine learning in safety-critical settings heightens vulnerability to *adversarial attacks*. Existing defense mechanisms typically either lack formal guarantees or depend on restrictive assumptions about the model family, the threat model, or the poisoning budget, and many only offer point-wise certification. Importantly, they often overlook the inherent stochasticity of modern training pipelines, which undermines their practical reliability. We introduce a probabilistic framework that views gradient-based training as a *discrete-time stochastic dynamical system* and formulates poisoning robustness as a safety verification task. Leveraging *barrier certificates* (BCs), we derive sufficient conditions to probabilistically certify a robust radius against worst-case ${\ell}_p$-bounded poisoning, guaranteeing that the final model parameters remain within a safe set. For tractable computation, we represent BCs with neural networks and obtain *probably approximately correct* (PAC) guarantees through a *scenario convex problem*. Our method identifies the largest certified radius for which the trained model is probabilistically accurate with a specified confidence level. Experiments on MNIST, SVHN, and CIFAR-10 show that our framework offers formal robustness guarantees under stochastic training, while being model-agnostic and not requiring prior knowledge of the attack strategy.
****
RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation
Tianxing Chen, Zanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, Yiheng Ge, Zhenyu Gu, Weiliang Deng, Yubin Guo, Tian Nian, Xuanbing Xie, Qiangyu Chen, Kailun Su, Tianling Xu, Guodong Liu, Mengkang Hu, Huan-ang Gao, Kaixuan Wang, Zhixuan Liang, Yusen Qin, Xiaokang Yang, Ping Luo, Yao Mu
Simulation-based data synthesis has emerged as a powerful paradigm for enhancing real-world robotic manipulation. However, existing synthetic datasets remain insufficient for robust bimanual manipulation due to two key challenges: (1) the lack of an autonomous self-correcting mechanism to resolve execution failures in complex coordination tasks, and (2) the scarcity of diverse visual and spatial variations required to bridge the sim-to-real gap. To this end, we present RoboTw

in 2.0, a scalable simulation framework that enables closed-loop, automated, large-scale generation of diverse and realistic data, along with unified evaluation protocols for dual-arm manipulation. Built upon RoboTwin-OD, a foundational library of 731 instances across 147 categories with rich semantic annotations, our framework integrates Multimodal Large Language Models (MLLMs) with simulation-in-the-loop verification. This integration forms an automated feedback mechanism that significantly boosts the success rate of expert task program generation. To enhance robust sim-to-real transfer, RoboTwin 2.0 incorporates structured domain randomization along five axes: clutter, lighting, background, tabletop height and language instructions, thereby maximizing data diversity. We instantiate this framework across 50 dual-arm tasks spanning five robot embodiments. Empirical evaluations demonstrate that Vision-Language-Action (VLA) models pre-trained on our synthetic data achieve a 3.6x improvement in few-shot real-world transfer (over a 10-demo baseline) and a 2.2x gain in zero-shot generalization. We release the data generator, benchmark, pre-collected dataset, and code to support scalable research in robust bimanual manipulation.

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BRIDGE: Triangular Fixed-Point Refinement for Long-Horizon Persona Consistency

Yinghui Jiang, Bocheng Xu, Jianye Xie, Haotong Sun

Long-horizon dialogue agents suffer from latent state drift: what an agent says, what it internally represents, and what it stores in memory can diverge silently across turns. This creates asymmetric rupture risk—many locally coherent exchanges undone by a single high-cost contradiction. We propose BRIDGE (Behavioral Reasoning through Integrated Dynamic Gated Evolution), which performs triangular fixed-point refinement to explicitly couple Observable ($\mathcal{O}$), Latent ($\mathcal{L}$), and Memory ($\mathcal{M}$) before decoding each response. We prove that under mild conditions, the refinement operator converges to a unique fixed point, providing a theoretical guarantee that the agent's internal state remains self-consistent before each response. Empirically, BRIDGE achieves the highest scores on both PersonaGym (4.59 avg., surpassing Claude-3.7-Sonnet) and CoSER (59.5% avg., +3.1 over Claude-3.7-Sonnet), with gains concentrated on persona-specific metrics (+8.0 Character Fidelity over Qwen2.5-32B-Instruct)—while updating only 0.85% trainable parameters of the frozen backbone. We also provide a Lyapunov-style uniform drift bound for tiered memory updates, grounding bounded persona evolution in long-horizon interaction.

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Rethinking the Flow-based Gradual Domain Adaptation: A Semi-Dual Optimal Transport Perspective

Zhichao Chen, Zhan Zhuang, Yunfei Teng, Hao Wang, Fangyikang Wang, Zhengnan Li, Tianqiao Liu, Haoxuan Li, Zhouchen Lin

Gradual domain adaptation (GDA) aims to mitigate domain shift by progressively adapting models from the source domain to the target domain via intermediate domains. However, real intermediate domains are often unavailable or ineffective, necessitating the synthesis of intermediate samples. Flow-based models have recently been used for this purpose by interpolating between source and target distributions. Notably, their training typically relies on sample-based log-likelihood estimation, which can discard useful information and thus degrade GDA performance. The key to addressing this limitation is constructing the intermediate domains via samples directly. To this end, we propose an $\underline{\text{E}}$ntropy-regularized $\underline{\text{S}}$emi-dual $\underline{\text{U}}$nbalanced $\underline{\text{O}}$ptimal $\underline{\text{T}}$ransport (E-SUOT) framework to construct intermediate domains. Specifically, we reformulate flow-based GDA as a Lagrangian dual problem and derive an equivalent semi-dual objective that circumvents the need for likelihood estimation. However, the dual problem leads to an unstable min–max training procedure. To alleviate this issue, we further introduce the entropy regularization to convert it into a more stable sequential optimization procedure. Based on this, we propose a novel GDA training framework and provide theoretical analysis in terms of stability and generalization. Finally, extensive experiments are conducted to demonstrate the efficacy of the E-SUOT framework.

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FlowNar: Scalable Streaming Narration for Long-Form Videos
Zeyun Zhong, Manuel Martin, Chengzhi Wu, David Schneider, Frederik Diederichs, Juergen Gall, Jürgen Beyerer
Recent Large Multimodal Models (LMMs), primarily designed for offline settings, are ill-suited for the dynamic requirements of streaming video. While recent online adaptations improve real-time processing, they still face critical scalability challenges, with resource demands typically growing at least linearly with video duration. To overcome this bottleneck, we propose FlowNar, a novel framework for scalable streaming video narration. The core of FlowNar is a dynamic context management strategy for historical visual context removal, combined with our CLAM (Cross Linear Attentive Memory) module for streaming visual history retention, ensuring bounded visual memory usage and computational complexity, crucial for efficient streaming. We also introduce a realistic self-conditioned evaluation protocol and complementary evaluation metrics to assess streaming narration models under deployment-like conditions. Experiments on the Ego4D, EgoExo4D, and EpicKitchens100 datasets demonstrate that FlowNar substantially improves narration quality over strong baselines while being highly efficient, supporting processing of 10$\times$ longer videos and achieving 3$\times$ higher throughput (FPS). The code is available at https://github.com/zeyun-zhong/FlowNar.
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Local Constrained Bayesian Optimization
Jingzhe Jing, Zheyi Fan, Szu Hui Ng, Qingpei Hu
Bayesian optimization (BO) for high-dimensional constrained problems remains a significant challenge due to the curse of dimensionality. We propose **L**ocal **C**onstrained **B**ayesian **O**ptimization (LCBO), a novel framework tailored for such settings. Unlike trust-region methods that are prone to premature shrinking when confronting tight or complex constraints, LCBO leverages the differentiable landscape of constraint-penalized surrogates to alternate between rapid local descent and uncertainty-driven exploration. Theoretically, we prove that LCBO achieves a convergence rate for the Karush-Kuhn-Tucker (KKT) residual that depends polynomially on the dimension d for common kernels under mild assumptions, offering a rigorous alternative to global BO where regret bounds typically scale exponentially. Extensive evaluations on high-dimensional benchmarks (up to 100D) demonstrate that LCBO consistently outperforms state-of-the-art baselines.
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Towards Uniformity and Alignment for Multimodal Representation Learning
Wenzhe Yin, Pan Zhou, Zehao Xiao, Jie Liu, Shujian Yu, Jan-Jakob Sonke, Stratis Gavves
Multimodal representation learning aims to construct a shared embedding space in which heterogeneous modalities are semantically aligned. Despite strong empirical results, InfoNCE-based objectives introduce inherent conflicts that yield distribution gaps across modalities. In this work, we identify two conflicts in the multimodal regime, both exacerbated as the number of modalities increases: (i) an alignment–uniformity conflict, whereby the repulsion of uniformity undermines pairwise alignment, and (ii) an intra-alignment conflict, where aligning multiple modalities induces competing alignment directions. To address these issues, we propose a principled decoupling of alignment and uniformity for multimodal representations, providing a conflict-free recipe for multimodal learning that simultaneously supports discriminative and generative use cases without task-specific modules. We then provide a theoretical guarantee that our method acts as an efficient proxy for a global Hölder divergence over multiple modality distributions, and thus reduces the distribution gap among modalities. Extensive experiments on retrieval and UnCLIP-style generation demonstrate consistent gains.
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Plug-and-Play Benchmarking of Reinforcement Learning Algorithms for Large-Scale Flow Control
Jannis Becktepe, Aleksandra Franz, Nils Thuerey, Sebastian Peitz
Reinforcement learning (RL) has shown promising results in active flow control (AFC), yet progress in the field remains difficult to assess as existing studies

rely on heterogeneous observation and actuation schemes, numerical setups, and evaluation protocols. Current AFC benchmarks attempt to address these issues but heavily rely on external computational fluid dynamics (CFD) solvers, are not fully differentiable, and provide limited 3D and multi-agent support. To overcome these limitations, we introduce FluidGym, the first standalone, fully differentiable benchmark suite for RL in AFC. Built entirely in PyTorch on top of the GPU-accelerated PICT solver, FluidGym runs in a single Python stack, requires no external CFD software, and provides standardized evaluation protocols. We present baseline results with PPO, SAC, DPC, and TD-MPC, and release all environments, datasets, and trained models as public resources. FluidGym enables systematic comparison of control methods, establishes a scalable foundation for future research in learning-based flow control, and is available at github.com/safe-autonomous-systems/fluidgym.

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MORALISE: A Structured Benchmark for Moral Alignment in Visual Language Models

Xiao Lin, Zhining Liu, Ze Yang, Gaotang Li, Ruizhong Qiu, Shuke Wang, Hui Liu, Haotian Li, Yuchen Yan, Sumit Keswani, Vishwa Pardeshi, Huijun Zhao, Wei Fan, Hanghang Tong

Recently, vision-language models have demonstrated increasing influence in morally sensitive domains such as autonomous driving and medical analysis, owing to their powerful multimodal reasoning capabilities. As these models are deployed in high-stakes real-world applications, it is of paramount importance to ensure that their outputs align with human moral values and remain within moral boundaries. However, existing work on moral alignment either focuses solely on textual modalities or relies heavily on AI-generated images, leading to distributional biases and reduced realism. To overcome these limitations, we introduce MORALISE, a comprehensive benchmark for evaluating the \underline{mor}al \underline{al}ignment of v\underline{is}ion-languag\underline{e} models (VLMs) using diverse, expert-verified real-world data. We begin by proposing a comprehensive taxonomy of 13 moral topics grounded in Turiel's Domain Theory, spanning the personal, interpersonal, and societal moral domains encountered in everyday life. Built on this framework, we manually curate 2,481 high-quality image-text pairs, each annotated with two fine-grained labels: (1) \textit{topic annotation}, identifying the violated moral topic(s), and (2) \textit{modality annotation}, indicating whether the violation arises from the image or the text. For evaluation, we encompass two tasks, \textit{moral judgment} and \textit{moral norm attribution}, to assess models' awareness of moral violations and their reasoning ability on morally salient content. Extensive experiments on 19 popular open- and closed-source VLMs show that MORALISE poses a significant challenge, revealing persistent moral limitations in current state-of-the-art models.

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From Kepler to Newton: Inductive Biases Guide Learned World Models in Transformers

Ziming Liu, Surya Ganguli, Andreas S. Tolias

Vafa et al. recently showed that a transformer fails to acquire an internal Newtonian world model when trained on synthetic planetary-motion data. How can we fix this problem? We find that inductive biases are key to learning the veridical world model: (1) **Spatial smoothness** is required for any world model to be learned. However, naive tokenization may disrupt smoothness since two close points in physical space may be far apart in token embedding space without sufficient training or data. We fix this by formulating the prediction problem as regression instead of classification. (2) **Spatial stability** makes the prediction robust to noise, which is not guaranteed by default, but can be taught via correcting in-context noise perturbations. (3) With both spatial smoothness and stability built in, further imposing **temporal locality** induces a Newtonian world model, while the lack of this knowledge induces a Keplerian world model -- fitting elliptical parameters instead of computing gravitational forces. Our results suggest that even simple general inductive biases are powerful enough to induce correct and specific world models. The inductive biases do not need to know that much about the underlying law to be learned, but without them, it is impossible to

learn.
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Easier to Judge than to Find: Predicting In-Context Learning Success for Demonstration Selection
Haochun Wang, Chaofen Yang, Jiatong Liu, Jingbo Wang, Zewen Qiang, Sendong Zhao, Bing Qin, Ting Liu
In-context learning (ICL) is highly sensitive to which demonstrations appear in the prompt, but selecting them is expensive because candidate contexts must be validated with repeated LLM calls. We argue that demonstration selection is \emph{easier to judge than to find}: predicting whether a specific query--context pair (q,D) will succeed is cheaper and more general than searching for an optimal $D^\star$. Based on this insight, we propose DiSP, a sample-and-judge framework that stratifies queries by difficulty. DiSP runs random demonstration trials to estimate each training query's success rate, trains a lightweight router to predict difficulty from the query, and trains level-specific judges to score sampled contexts. At inference, DiSP performs stop-on-acceptance judging under an explicit budget and typically makes a single LLM call, emitting diagnostic risk tags when no suitable context is found. Across five classification datasets with Llama 3–8B and Qwen 2.5–7B, DiSP achieves the best average accuracy, improving over strong learned selection baselines by up to 3.4%, while achieving up to 23× end-to-end wall-clock speedup.
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RAPNet: Accelerating Algebraic Multigrid with Learned Sparse Corrections
Yali Fink, Ido Ben-Yair, Lars Ruthotto, Eran Treister
The scalable solution of large sparse linear systems is a bottleneck in scientific computing and graph analysis. While algebraic multigrid (AMG) offers optimal linear scaling, its performance is severely constrained by the trade-off between the sparsity and convergence quality of coarse-grid operators. Classical AMG heuristics struggle to balance these objectives, often sacrificing stability or performance for sparsity. We propose RAPNet, a graph neural network (GNN) framework that resolves this trade-off by learning to generate sparse, robust coarse operators directly from the sparse algebraic system. Key to our approach is a level-wise training strategy that enables learning from small subgraphs and generalization to million-node domains, bypassing the bottlenecks of prior neural AMG attempts. RAPNet executes exclusively during the solver setup phase, ensuring that the solve phase retains its favorable computational properties. We show that our method outperforms classical non-Galerkin baselines on diverse PDE discretizations and graph Laplacians, making it particularly effective for multi-query tasks such as eigenproblems, time-dependent simulations, and inverse or design problems.
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ReflFlow: Learning Geometry-Guided Ray Tracing for Dynamic Specular Reconstruction
Jiachen Tao, Junyi Wu, Haoxuan Wang, Zongxin Yang, Dawen Cai, Yan Yan
We present ReflFlow, a novel framework for high-fidelity rendering of dynamic specular scenes by addressing two key challenges: precise reflection direction estimation and physically accurate modeling. To achieve this, we propose a Residual Material-Augmented 2D Gaussian Splatting representation that models dynamic geometry and material properties, allowing accurate reflection ray computation. Furthermore, we introduce a Dynamic Environment Gaussian and a hybrid rendering pipeline that decomposes rendering into diffuse and specular components, enabling physically informed specular synthesis via rasterization and ray tracing. Finally, we devise a coarse-to-fine training strategy to improve optimization stability and promote physically meaningful decomposition. Extensive experiments on dynamic scene benchmarks demonstrate that ReflFlow outperforms prior methods quantitatively and qualitatively, producing sharper and more realistic specular reflections in complex dynamic environments.
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Sharp Inequalities between Total Variation and Hellinger Distances for Gaussian Mixtures

Joonhyuk Jung, Chao Gao
We study the relation between the total variation (TV) and Hellinger distances between two Gaussian location mixtures. Our first result establishes a general upper bound: for any two mixing distributions supported on a compact set, the Hellinger distance between the two mixtures is controlled by the TV distance raised to a power $1-o(1)$, where the $o(1)$ term is of order $1/\log\log(1/\mathrm{TV})$. We also construct two sequences of mixing distributions that demonstrate the sharpness of this bound. Taken together, our results resolve an open problem raised in Jia et al. (2023) and thus lead to an entropic characterization of learning Gaussian mixtures in total variation. Our inequality also yields optimal robust estimation of Gaussian mixtures in Hellinger distance, which has a direct implication for bounding the minimax regret of empirical Bayes under Huber contamination.
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A Flat Vocabulary or a Rich Hierarchy? Re-introducing Intrinsic Structure Transforms the Autoregressive Image Generation
Landis He, Shikang Zheng
Autoregressive (AR) models have shown great promise in image generation, yet they face a fundamental inefficiency stemming from their core component: a vast, unstructured vocabulary of visual tokens. By treating tokens as a flat set, standard models overlook the manifold structure where geometric proximity reflects semantic similarity. This oversight unnecessarily complicates the prediction task, hindering training efficiency and limiting generation quality. To resolve this, we propose Manifold-Aligned Semantic Clustering (MASC), a principled framework that constructs a hierarchical semantic tree directly from the codebook's intrinsic geometry. Utilizing a geometry-aware distance metric and density-driven agglomerative construction, MASC faithfully models the token embedding manifold. By transforming the flat, high-dimensional prediction into a structured hierarchical task, MASC introduces a powerful inductive bias that simplifies learning. Designed as a plug-and-play module, MASC accelerates training by up to 71\% and significantly boosts generation quality, improving LlamaGen-XL's FID from 2.87 to 2.49. Crucially, MASC further serves as a convergence enabler for complex architectures. These results establish that structuring the prediction space is as vital as architectural innovation, elevating existing AR frameworks to state-of-the-art performance.
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Systematic Failures in Collective Reasoning under Distributed Information in Multi-Agent LLMs
Yuxuan Li, Aoi Naito, Hirokazu Shirado
Multi-agent systems built on large language models (LLMs) are expected to enhance decision-making by pooling distributed information, yet systematically evaluating this capability has remained challenging. We introduce HiddenBench, a 65-task benchmark grounded in the Hidden Profile paradigm, which isolates collective reasoning under distributed information from individual reasoning ability. Evaluating 15 frontier LLMs, we find that multi-agent LLMs achieve only 30.1% accuracy under distributed information, compared to 80.7% accuracy for single agents given complete information. We trace this gap to a systematic failure mode: agents cannot recognize or act under latent information asymmetry—they fail to reason about what others might know but have not yet expressed, leading to premature convergence on shared evidence while critical distributed facts remain unexplored. These failures persist across prompting strategies, communication depths, and group sizes—and worsen as groups scale. While some models (e.g., Gemini-2.5-Flash/Pro) outperform others, neither model scale nor individual reasoning accuracy reliably predicts collective performance. We further show that this bottleneck is actionable: a lightweight structured communication protocol substantially improves collective reasoning across model families. Our results identify failures in collective information exploration in decision-making as a key limitation of multi-agent LLMs, and provide a theory-grounded, reproducible framework for diagnosing collective reasoning failures.
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Neural Attention Search Linear: Towards Adaptive Token-Level Hybrid Attention Models

Difan Deng, Andreas Bentzen Winje, Lukas Fehring, Marius Lindauer

The quadratic computational complexity of softmax transformers has become a bottleneck in long-context scenarios. In contrast, linear attention model families provide a promising direction towards a more efficient sequential model. These linear attention models compress past KV values into a single hidden state, thereby efficiently reducing complexity during both training and inference. However, their expressivity remains limited by the size of their hidden state. Previous work proposed interleaving softmax and linear attention layers to reduce computational complexity while preserving expressivity. Nevertheless, the efficiency of these models remains bottlenecked by their softmax attention layers. In this paper, we propose Neural Attention Search Linear (NAtS-L), a framework that applies both linear attention and softmax attention operations within the same layer on different tokens. NAtS-L automatically determines whether a token can be handled by a linear attention model, i.e., tokens that have only short-term impact and can be encoded into fixed-size hidden states, or require softmax attention, i.e., tokens that contain information related to long-term retrieval and need to be preserved for future queries. By searching for optimal Gated DeltaNet and softmax attention combinations across tokens, we show that NAtS-L provides a strong yet efficient token-level hybrid architecture.

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Normality Calibration in Semi-supervised Graph Anomaly Detection

Guolei Zeng, Hezhe Qiao, Guoguo Ai, Jinsong Guo, Guansong Pang

Semi-supervised graph anomaly detection (GAD), which assumes a subset of labeled normal nodes for training, is widely studied. However, existing methods learn normality only from these labeled nodes, often overfitting their patterns and causing high detection errors, such as many false positives. To overcome this limitation, we propose $GraphNC$, a graph normality calibration framework that leverages both labeled and unlabeled data to calibrate the normality from a teacher (a pre-trained semi-supervised GAD model) jointly in anomaly score and representation spaces. GraphNC includes two main components, anomaly score distribution alignment ($ScoreDA$) and perturbation-based normality regularization ($NormReg$). ScoreDA optimizes our model’s anomaly scores by aligning them with the teacher’s score distribution. Because the teacher provides accurate scores for most normal nodes and some anomalies, this alignment pulls the scores of the two classes toward opposite ends, making them more separable. To reduce the impact of inaccurate teacher scores, NormReg regularizes normality in the representation space, making normal node representations more compact via a perturbation-guided consistency loss applied only to the labeled nodes. Comprehensive experiments on six benchmarks demonstrate that GraphNC (1) consistently and substantially enhances the performance of teacher models from different GAD methods (2) achieves new state-of-the-art performance. Our code is available at https://github.com/mala-lab/GraphNC.

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GenAlign: Towards Unified Alignment Framework of MLLMs via Generative Reward Model

Jingyu Zhang, Kun Yang, Ming Wen, Jiawei Zhao, Yuxuan Liu, Zhuoer Xu, shiwen cui

Aligning Multimodal Large Language Models (MLLMs) with human preferences remains a fundamental challenge. While Generative Reward Models (GRMs) offer a promising reasoning-based alternative to scalar models, they are often hindered by severe position bias and prohibitively high computational overhead. To address these limitations, we propose GenAlign, a unified framework that synergizes robust generative reward modeling with efficient MLLM alignment. First, we introduce a rubric-based GRM that explicitly models the preference judgment process. By employing reinforcement learning with verifiable rewards and an online position debiasing mechanism, our model produces interpretable reasoning critiques and robust preference predictions. Second, we propose a policy optimization strategy utilizing advantage-smoothed dynamic reference anchoring. This approach reduces computational complexity while mitigating gradient instability caused by variance colla

pse. Extensive experiments demonstrate that GenAlign achieves state-of-the-art p
reference prediction accuracy on multimodal reward modeling benchmarks. Moreover
, it consistently improves the performance of three MLLMs across seven diverse e
valuation benchmarks, particularly making significant progress in safety and hal
lucination.
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Co-Evolving Latent Action World Models
Yucen Wang, Fengming Zhang, De-Chuan Zhan, Li Zhao, Kaixin Wang, Jiang Bian
Adapting pre-trained video generation models into controllable world models via
latent actions is a promising step towards creating generalist world models. The
 dominant paradigm adopts a two-stage approach that trains latent action model (
LAM) and the world model separately, resulting in redundant training and limitin
g their potential for co-adaptation. A conceptually simple and appealing idea is
 to directly replace the forward dynamic model in LAM with a powerful world mode
l and training them jointly, but it is non-trivial and prone to representational
 collapse. In this work, we propose CoLA-World, which for the first time success
fully realizes this synergistic paradigm, resolving the core challenge in joint
learning through a critical warm-up phase that effectively aligns the representa
tions of the from-scratch LAM with the pre-trained world model. This unlocks a c
o-evolution cycle: the world model acts as a knowledgeable tutor, providing grad
ients to shape a high-quality LAM, while the LAM offers a more precise and adapt
able control interface to the world model. Empirically, CoLA-World matches or ou
tperforms prior two-stage methods in both video simulation quality and downstrea
m visual planning, establishing a robust and efficient new paradigm for the fiel
d.
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Does Reasoning Improve Seeing? Understanding When Vision-Language Models Benefit
 from Thinking
Jing Bi, Luchuan Song, Dingxin Zhang, Pinxin Liu, Guangyu Sun, Lianggong Bruce W
en, Weidong Cai, Chen Chen, Chenliang Xu
Vision-language models (VLMs) now support both direct Instruct and explicit-reas
oning Thinking modes, yet practitioners still lack principled ways to decide whe
n reasoning actually improves performance, or how much computation to spend at t
est time, so we investigate whether VLMs encode meta-cognitive signals for adapt
ive inference.
We derive oracle labels for two properties: (1) reasoning helpfulness, namely wh
ether explicit reasoning improves accuracy, and (2) desired generation length, t
he minimal token budget needed for a correct answer. Probing final-layer represe
ntations in InternVL and Qwen3-VL models, we find Thinking models encode these s
ignals more linearly than Instruct models, suggesting that reasoning-oriented tr
aining enhances meta-cognitive structure.
Head-wise attribution reveals two circuits: length-control heads in lower layers
 and reasoning/difficulty heads in higher layers. Causal interventions confirm t
hese roles across benchmarks: scaling length heads controls output length with m
inimal accuracy loss, while scaling reasoning heads enables a perception-reasoni
ng trade-off, improving accuracy by up to 5.3\%.
Our results demonstrate that reasoning-tuned VLMs develop localized, manipulable
 circuits for meta-cognitive control, enabling test-time steering of computation
 and reasoning without retraining.
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Geometry-Guided Generative Representation for Functional Brain Graphs
Subati Abulikemu, Tiago Azevedo, Michail Mamalakis, John Suckling
In network neuroscience, functional brain systems are often characterized using
separate yet related graph-theoretic or spectral descriptors, overlooking how th
ese properties covary and partially overlap across individuals and conditions. W
e anticipate that dense, weighted functional connectivity graphs lie on a low-di
mensional latent geometry along which both topological and spectral structures v
ary smoothly at the population level. Although graph-based deep learning offers
a powerful framework for modeling these brain connectomes, supervised approaches
 are constrained by the limited availability of labeled data. Existing unsupervi

sed graph representation methods also typically focus on node-level embeddings, which are limited in capturing compact graph-level representations that preserve information from dense functional connectomes. To address these gaps, we learn compact brain graph representations using a graph transformer autoencoder, where domain-specific, aligned functional gradient geometry provides an inductive bias to guide learning. Despite being trained in a fully unsupervised manner, our approach meaningfully separates cognitive states and enables decoding of visual stimuli, with performance further improved by incorporating neural dynamics. In parallel, to enable generation of synthetic brain graphs, we fit a diffusion model to the learned latent representation and decode samples back to dense connectomes.

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Scalable RF Simulation in Generative 4D Worlds

Zhiwei Zheng, Dongyin Hu, Mingmin Zhao

Radio Frequency (RF) sensing has emerged as a powerful, privacy-preserving alternative to vision-based methods for various perception tasks. However, building high-quality RF datasets in dynamic and diverse environments remains a major challenge. To address this, we introduce WaveVerse, a prompt-based, scalable framework that simulates realistic RF signals from generated indoor scenes with human motions guided by spatial paths, enabling diverse and feasible behaviors without manual trajectory design. WaveVerse features a language-guided 4D world generator and a physics-based signal simulator that enables realistic simulation of RF signals in diverse environments. It employs a phase-coherent ray tracer that preserves both spatial and temporal phase consistency. The simulated signals show high fidelity on phase-sensitive benchmarks, and closely align with both real-world collected measurements and simulations from a proprietary electromagnetic solver. When used for data augmentation, WaveVerse consistently improves performance in downstream tasks like RF imaging and human activity recognition, with gains that grow with the amount of simulated data and surpass existing methods. Code and additional materials are available on the webpage.

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Inconsistency-Aware Minimization: Improving Generalization with Unlabeled Data

Hee-Sung Kim, Hyeonseong Kim, Sungyoon Lee

Estimating the generalization gap and developing optimization methods that improve generalization are crucial for deep learning models, for both theoretical understanding and practical applications.

Leveraging unlabeled data for these purposes offers significant advantages in real-world scenarios.

This paper introduces a novel generalization measure, $\textit{local inconsistency}$, derived from an information-geometric perspective on the parameter space of neural networks. A key feature of local inconsistency is that it can be computed without explicit labels.

We establish theoretical underpinnings by connecting local inconsistency to the Fisher information matrix and the loss Hessian.

Empirically, we demonstrate that local inconsistency correlates with the generalization gap.

Based on these findings, we propose Inconsistency-Aware Minimization (IAM), which incorporates local inconsistency into the training objective.

We demonstrate that in standard supervised learning settings, IAM enhances generalization, achieving performance comparable to that of existing methods such as Sharpness-Aware Minimization.

Furthermore, IAM exhibits efficacy in semi- and self-supervised learning scenarios, where the local inconsistency is computed from unlabeled data.

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Scaling Transformers for End-to-End Discrete Audio Tokenization

Yitian Gong, Kuangwei Chen, Zhaoye Fei, Xiaogui Yang, Ke Chen, Yang Wang, Kexin Huang, Mingshu Chen, Ruixiao Li, Qinyuan Cheng, Shimin Li, Xipeng Qiu

Discrete audio tokenizers are fundamental to empowering large language models with native audio processing and generation capabilities. Despite recent progress, existing approaches often rely on pretrained encoders, semantic distillation, o

r heterogeneous CNN-based architectures. These designs introduce fixed inductive biases that limit reconstruction fidelity and hinder effective scaling. In this paper, we argue that discrete audio tokenization should be learned fully end-to-end using a homogeneous and scalable architecture.
Based on this perspective, we propose $\textbf{TAC}$, a Transformer-based audio tokenizer that jointly optimizes the encoder, quantizer, and decoder from scratch for high-fidelity reconstruction of general audio.
We show that a simple, fully end-to-end learned tokenizer built from homogeneous, causal Transformer blocks scales gracefully and supports high-fidelity reconstruction across diverse audio domains. Across speech, sound, and music, the proposed tokenizer consistently outperforms prior codecs over a wide range of bitrates, while exhibiting predictable improvements with increased scale. Notably, leveraging TAC’s discrete tokens, we develop the first purely autoregressive TTS model that surpasses prior non-autoregressive and cascaded systems. Furthermore, TAC enables competitive ASR performance without auxiliary encoders. Our findings position TAC as a unified, scalable interface for the next generation of native audio foundation models.
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EvoMAS: Evolutionary Generation of Multi-Agent Systems
Yuntong Hu, Yuting Zhang, Matthew Trager, Yi Zhang, Shuo Yang, Wei Xia, Stefano Soatto
Large language model (LLM)-based multi-agent systems (MAS) show strong promise for complex reasoning, planning, and tool-augmented tasks, but designing effective MAS architectures remains labor-intensive, brittle, and hard to generalize. Existing automatic MAS generation methods either rely on code generation, which often leads to executability and robustness failures, or impose rigid architectural templates that limit expressiveness and adaptability. We propose Evolutionary Generation of Multi-Agent Systems (EvoMAS), which formulates MAS generation as structured configuration generation. EvoMAS performs evolutionary generation in configuration space. Specifically, EvoMAS selects initial configurations from a pool, applies feedback-conditioned mutation and crossover guided by execution traces, and iteratively refines both the candidate pool and an experience memory. We evaluate EvoMAS on diverse benchmarks, including BBEH, SWE-Bench, and WorkBench, covering reasoning, software engineering, and tool-use tasks. EvoMAS consistently improves task performance over both human-designed MAS and prior automatic MAS generation methods, while producing generated systems with higher executability and runtime robustness. EvoMAS outperforms the agent evolution method EvoAgent by +10.5 points on BBEH reasoning and +7.1 points on WorkBench. With Claude-4.5-Sonnet, EvoMAS also reaches 79.1% on SWE-Bench-Verified, matching the top of the leaderboard. Code is available at https://github.com/amazon-science/EvoMAS
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Training-Free Bayesian Filtering with Generative Emulators
Thomas Savary, François Rozet, Gilles Louppe
Bayesian filtering is a well-known problem that aims to estimate plausible states of a dynamical system from observations. Among existing approaches to solve this problem, particle filters are theoretically exact for non-linear dynamics and observations, but suffer from poor scalability in high dimensions. In this work, we show that diffusion-based emulators of dynamical systems can be used to implement, without additional training, an optimal variant of particle filters that has remained largely unexplored due to implementation challenges with classical numerical solvers. Experiments on nonlinear chaotic systems, including atmospheric dynamics, demonstrate that the proposed approach successfully scales particle filtering to high-dimensional settings.
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Geometry-Guided Modeling of Foundation Features Enables Generalizable Object Shape Deformation Learning
Yiyao Ma, Kai Chen, Zhongxiang Zhou, Zhuheng Song, Dongsheng Xie, Zelong Tan, Rong Xiong, Qi Dou
Monocular 3D shape recovery is fundamental to geometric understanding, yet achieving robust generalization across arbitrary viewpoints and unseen object categor

ies remains a significant challenge. In this paper, we present a generalizable d
eformation learning framework that reconstructs 3D objects by explicitly deformi
ng a category-level shape template to match the target observation. To address c
omplex shape variations between the template and the target, we introduce a geom
etry-guided feature modeling mechanism. This process first enriches foundation f
eatures with template topology to yield a geometry-aware representation, which i
s then explicitly correlated with the target observation to guide precise deform
ation. Furthermore, to bridge the disparity between the fixed template and arbit
rary target views, we propose a view-adaptive feature aggregation module. This m
odule leverages multi-view template features and their corresponding camera pose
s to enrich the canonical template representation, ensuring robust feature align
ment regardless of the target's perspective. Extensive experiments demonstrate
that our approach significantly outperforms state-of-the-art methods in handling
 large shape variations and diverse viewpoints, exhibiting strong generalization
 to novel categories and effectively supporting downstream real-world dexterous
robotic manipulation tasks. Project homepage: https://GODeform.github.io/
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MASPO: Joint Prompt Optimization for LLM-based Multi-Agent Systems
Zhexuan Wang, Xuebo Liu, Li Wang, Zifei Shan, Yutong Wang, Zhenxi Song, Min Zhan
g
Large language model (LLM)-based Multi-agent systems (MAS) have shown promise in
 tackling complex collaborative tasks, where agents are typically orchestrated v
ia role-specific prompts. While the quality of these prompts is pivotal, jointly
 optimizing them across interacting agents remains a non-trivial challenge, prim
arily due to the misalignment between local agent objectives and holistic system
 goals. To address this, we introduce MASPO, a novel framework designed to autom
atically and iteratively refine prompts across the entire system. A core innovat
ion of MASPO is its joint evaluation mechanism, which assesses prompts not merel
y by their local validity, but by their capacity to facilitate downstream succes
s for successor agents. This effectively bridges the gap between local interacti
ons and global outcomes without relying on ground-truth labels. Furthermore, MAS
PO employs a data-driven evolutionary beam search to efficiently navigate the hi
gh-dimensional prompt space. Extensive empirical evaluations across 6 diverse ta
sks demonstrate that MASPO consistently outperforms state-of-the-art prompt opti
mization methods, achieving an average accuracy improvement of 2.9. We release o
ur code at https://github.com/wangzx1219/MASPO.
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Many Needles in a Haystack: Active Hit Discovery for Perturbation Experiments
Andrea Rubbi, Arpit Merchant, Samuel Ogden, Amir Akbarnejad, Pietro Lio, Sattar
Vakili, Mohammad Lotfollahi
High-throughput gene perturbation experiments can test several genetic intervent
ions in parallel, yet experimental budgets remain limited. A central goal is hit
 discovery: identifying as many perturbations as possible whose phenotypic effec
t exceeds a predefined threshold. Pure exploration strategies are statistically
inefficient, wasting budget on low-value regions. Bayesian optimization methods
offer a principled alternative but target a single global optimum, over-exploiti
ng dominant modes while neglecting other high-value regions. We formalize hit di
scovery as a sequential experimental design problem and propose Probability-of-H
it, an acquisition function that directly targets threshold exceedance by rankin
g candidates according to their posterior probability of being a hit. We prove a
symptotic optimality of this approach and demonstrate strong empirical performan
ce on both synthetic benchmarks and real biological immunology datasets, includi
ng upto 6.4\% improvement over baselines on the Schmidt IL-2 dataset.
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Alterbute: Editing Intrinsic Attributes of Objects in Images
Tal Reiss, Daniel Winter, Matan Yaakov Cohen, Alex Rav-Acha, Yael Pritch, Ariel
Shamir, Yedid Hoshen
We introduce Alterbute, a diffusion-based method for editing an object's intrins
ic attributes in an image. We allow changing color, texture, material, and even
the shape of an object, while preserving its perceived identity and scene contex

t. Existing approaches either rely on unsupervised priors that often fail to pre serve identity or use overly restrictive supervision that prevents meaningful in trinsic variations. Our method relies on: (i) a relaxed training objective that allows the model to change both intrinsic and extrinsic attributes conditioned o n an identity reference image, a textual prompt describing the target intrinsic attributes, and a background image and object mask defining the extrinsic contex t. At inference, we restrict extrinsic changes by reusing the original backgroun d and object mask, thereby ensuring that only the desired intrinsic attributes a re altered; (ii) Visual Named Entities (VNEs) - fine-grained visual identity cat egories (e.g., "Porsche 911 Carrera") that group objects sharing identity-defin ing features while allowing variation in intrinsic attributes. We use a vision-l anguage model to automatically extract VNE labels and intrinsic attribute descri ptions from a large public image dataset, enabling scalable, identity-preserving supervision. Alterbute outperforms existing methods on identity-preserving obje ct intrinsic attribute editing.
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Multi-Head Attention as a Source of Catastrophic Forgetting in MoE Transformers
Anrui Chen, Ruijun Huang, Xin Zhang, Fang Dong, Hengjie Cao, Zhendong Huang, Yif eng Yang, Mengyi Chen, Jixian Zhou, Mingzhi Dong, Yujiang Wang, Jinlong Hou, Qin Lv, Robert P. Dick, Yuan Cheng, Tun Lu, Fan Yang, Li Shang
Mixture-of-Experts (MoE) architectures are appealing for continual learning beca use sparse routing should localize updates and reduce interference, yet MoE Tran sformers still forget substantially even with sparse, well-balanced expert utili zation. We attribute this gap to a pre-routing bottleneck: multi-head attention concatenates head-specific signals into a single post-attention router input, fo rcing routing to act on co-occurring feature compositions rather than separable head channels. We show that this router input simultaneously encodes multiple se parately decodable semantic and structural factors with uneven head support, and that different feature compositions induce weakly aligned parameter-gradient di rections; as a result, routing maps many distinct compositions to the same route . We quantify this collision effect via a route-wise effective composition numbe r N_{eff} and find that higher N_{eff} is associated with larger old-task loss increases after continual training. Motivated by these find ings, we propose MH-MoE, which performs head-wise routing over sub-representatio ns to increase routing granularity and reduce composition collisions. On TRACE a cross multiple backbones, MH-MoE consistently improves the retention--accuracy t rade-off over LoRA-MoE variants.
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Towards a Unified Generative Model for Scarce Time Series with Domain Experts
Zihao Yao, Qi Zheng, Jiankai Zuo, Yaying Zhang
Synthesizing realistic time series with generative models has wide-ranging appli cations in real-world scenarios. Despite recent progress, most existing methods are trained under the assumption of abundant training data, which substantially limits their effectiveness in data-scarce settings. In this paper, we propose Ti meMoDE, a novel framework that integrates Diffusion Transformers with Mixture-of -Experts to exploit both domain adaptability and diffusion-stage awareness for t ime series generation under data scarcity. It is pre-trained on a large-scale co llection of multi-domain datasets to extract domain-agnostic temporal representa tions and domain-specific information benefiting generalization during fine-tuni ng. We propose Domain Prompts to condition expert assignment for indistinguishab le noised tokens, mitigating the limitations of capturing inter-dataset relation ships. Moreover, we incorporate diffusion timestep signals to equip the experts with awareness of time series degradation variations, facilitating adaptive cali brate to stage-dependent denoising requirements. Extensive experiments demonstra te that TimeMoDE outperforms existing methods under diverse low-data settings. I t establishes an innovative paradigm for advanced time series few-shot generatio n.
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AdaSCALE: Adaptive Scaling for OOD Detection
Sudarshan Regmi

The ability of the deep learning model to recognize when a sample falls outside its learned distribution is critical for safe and reliable deployment. Recent state-of-the-art out-of-distribution (OOD) detection methods leverage activation shaping to improve the separation between in-distribution (ID) and OOD inputs. These approaches resort to sample-specific scaling but apply a static percentile threshold across all samples regardless of their nature, resulting in suboptimal ID-OOD separability. In this work, we propose \textbf{AdaSCALE}, an adaptive scaling procedure that dynamically adjusts the percentile threshold based on a sample's estimated OODness. This estimation leverages our key observation: OOD samples exhibit significantly more pronounced activation shifts at high-magnitude activations under minor perturbation compared to ID samples. AdaSCALE enables stronger scaling for likely ID samples and weaker scaling for likely OOD samples, yielding highly separable energy scores. Our approach achieves state-of-the-art OOD detection performance, outperforming the latest rival OptFS by $\textbf{14.94}$% in near-OOD and $\textbf{21.28}$% in far-OOD datasets in average FPR@95 metric on the ImageNet-1k benchmark across eight diverse architectures.
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Branch Scaling Manifests as Implicit Architectural Regularization for Improving Generalization in Overparameterized ResNets
Zixiong Yu, Guhan Chen, Jianfa Lai, Bohan Li, Songtao Tian
Scaling factors in residual branches have emerged as a prevalent method for boosting neural network performance, especially in normalization-free architectures. While prior work has primarily examined scaling effects from an optimization perspective, this paper investigates their role in residual architectures through the lens of generalization theory. Specifically, we establish that wide residual networks (ResNets) with constant scaling factors become asymptotically unlearnable as depth increases. In contrast, when the scaling factor exhibits rapid depth-wise decay combined with early stopping, over-parameterized ResNets achieve minimax-optimal generalization rates.
To establish this, we demonstrate that the generalization capability of wide ResNets can be approximated by the kernel regression associated with a specific kernel. Our theoretical findings are validated through experiments on synthetic data and real-world classification tasks, including MNIST and CIFAR-100.
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Beyond Procedure: Substantive Fairness in Conformal Prediction
Pengqi Liu, Zijun Yu, Mouloud Belbahri, Arthur Charpentier, Masoud Asgharian, Jesse C. Cresswell
Conformal prediction (CP) offers distribution-free uncertainty quantification for machine learning models, yet its interplay with fairness in downstream decision-making remains underexplored. Moving beyond CP as a standalone operation (procedural fairness), we analyze the holistic decision-making pipeline to evaluate substantive fairness—the equity of downstream outcomes. Theoretically, we derive an upper bound that decomposes prediction-set size disparity into interpretable components, clarifying how label-clustered CP helps control method-driven contributions to unfairness. To facilitate scalable empirical analysis, we introduce an LLM-in-the-loop evaluator that approximates human assessment of substantive fairness across diverse modalities. Our experiments show that label-clustered CP often provides a favorable balance between utility and substantive fairness, while reducing set-size disparities in line with our theory. Finally, we empirically show that equalized set sizes, rather than coverage, strongly correlate with improved substantive fairness, enabling practitioners to design more fair CP systems. Our code is available at https://github.com/layer6ai-labs/llm-in-the-loop-conformal-fairness.
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Doc-to-LoRA: Learning to Instantly Internalize Contexts
Rujikorn Charakorn, Edoardo Cetin, Shinnosuke Uesaka, Robert Tjarko Lange
Long input sequences are central to in-context learning, document understanding, and multi-step reasoning in Large Language Models (LLMs).
However, the quadratic attention cost of Transformers makes inference memory-intensive and slow.

While context distillation (CD) can transfer information into model parameters, per-prompt distillation is impractical due to training costs and latency. To address these limitations, we propose Doc-to-LoRA (D2L), a lightweight hypernetwork that meta-learns to perform approximate CD within a single forward pass. Given an unseen prompt, D2L generates a LoRA adapter for a target LLM, enabling subsequent queries to be answered without re-consuming the original context, reducing latency and KV-cache memory consumption during target LLM inference. On a long-context needle-in-a-haystack task, D2L successfully learns to map contexts into adapters that store the needle information, achieving near-perfect zero-shot accuracy at sequence lengths exceeding the target LLM's native context window by more than 4x. On real-world QA datasets with limited compute, D2L outperforms standard CD while significantly reducing peak memory consumption and update latency. We envision that D2L can facilitate rapid adaptation of LLMs, opening up the possibility of frequent knowledge updates and personalized chat behavior. Code and checkpoints are available at https://github.com/SakanaAI/doc-to-lora.

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Tight Margin-Based Generalization Bounds for Voting Classifiers over Finite Hypothesis Sets

Kasper Green Larsen, Natascha Schalburg

We prove the first margin-based generalization bound for voting classifiers, that is asymptotically tight in the tradeoff between the size of the hypothesis set, the margin, the fraction of training points with the given margin, the number of training samples and the failure probability.

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CLINIC: Towards High-quality Graph Out-Of-Distribution Detection

Yifan Wang, Haodong Zhang, Changhu Wang, Tao Ren, Dongjie Wang, Wei Ju, Chong Chen, Xian-Sheng Hua, Xiao Luo

This paper studies the problem of graph out-of-distribution (OOD) detection, which aims to identify anomaly graphs out of a graph dataset. Prior efforts usually focus on the utilization of topological structures with unsupervised graph learning to foster typical pattern recognition, which overlooks the semantic structure preserved in contextually affine neighborhoods. Towards this end, we propose a novel approach named Contextual Affinity Exploration with Twin Concordance (CLINIC) for graph OOD detection. The core of CLINIC is to explore and exploit the contextual affinity of the graph data samples for discriminative graph representations. In particular, our CLINIC first builds a contextual affinity graph to depict the semantic structure in the hidden space. More importantly, we introduce high-order affinity to enhance geometric understanding of the structure by utilizing a meta-graph neural network. To enhance representation discriminability with high robustness, we introduce twin concordance learning, which not only minimizes the difference of affinity distributions across different views, but also encourages the consistency between contextually affinitive neighbors. Finally, we introduce a compression strategy to expand the decision boundary for enhanced separation between in-distribution and out-of-distribution graphs. Extensive experimental results demonstrate the superiority of our CLINIC across ten real datasets in comparison to various baselines.

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Probabilistic Performance Guarantees for Multi-Task Reinforcement Learning

Yannik Schnitzer, Mathias Jackermeier, Alessandro Abate, David Parker

Multi-task reinforcement learning trains generalist policies that can execute multiple tasks. While recent years have seen significant progress, existing approaches rarely provide formal performance guarantees, which are indispensable when deploying policies in safety-critical settings. We present an approach for computing high-confidence guarantees on the performance of a multi-task policy on tasks not seen during training. Concretely, we introduce a new generalisation bound that composes (i) per-task lower confidence bounds from finitely many rollouts with (ii) task-level generalisation from finitely many sampled tasks, yielding a high-confidence guarantee for new tasks drawn from the same arbitrary and unknown distribution. Across state-of-the-art multi-task RL methods, we show that the

guarantees are theoretically sound and informative at realistic sample sizes.
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Med-Scout: Curing MLLMs' Geometric Blindness in Medical Perception via Geometry-Aware RL Post-Training
Anglin Liu, Ruichao Chen, Yi Lu, Hongxia Xu, Jintai Chen
Despite recent Multimodal Large Language Models (MLLMs)' linguistic prowess in medical diagnosis, we find even state-of-the-art MLLMs suffer from a critical perceptual deficit: **geometric blindness**. This failure to ground outputs in objective geometric constraints leads to plausible yet factually incorrect hallucinations, rooted in training paradigms that prioritize linguistic fluency over geometric fidelity. This paper introduces Med-Scout, a novel framework that "cures" this blindness via Reinforcement Learning (RL) that leverages the intrinsic geometric logic latent within unlabeled medical images. Instead of relying on costly expert annotations, Med-Scout derives verifiable supervision signals through three strategic proxy tasks inspired by the systematic reading and reasoning patterns of clinicians: Hierarchical Scale Localization, Topological Jigsaw Reconstruction, and Anomaly Consistency Detection. To rigorously quantify this deficit, we present Med-Scout-Bench, a new benchmark specifically designed to evaluate geometric perception. Extensive evaluations show that Med-Scout significantly mitigates geometric blindness, outperforming leading proprietary and open-source MLLMs by over **40%** on our benchmark. Furthermore, this enhanced geometric perception generalizes to broader medical understanding, achieving superior results on radiological and comprehensive medical VQA tasks. The project page is: https://github.com/HKUSTGZ-ML4Health-Lab/Med-Scout.
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What Does Vision Tool-Use Reinforcement Learning Really Learn? Disentangling Tool-Induced and Intrinsic Effects for Crop-and-Zoom
Yan Ma, Weiyu Zhang, Tianle Li, Du Linge, Xuyang Shen, Pengfei Liu
Vision tool-use reinforcement learning (RL) can equip vision--language models with visual operators such as crop-and-zoom and achieves strong performance gains, yet it remains unclear whether these gains are driven by improvements in tool use or evolving intrinsic capabilities. We introduce \textbf{MED} (Measure--Explain--Diagnose), a coarse-to-fine framework that disentangles intrinsic capability changes from tool-induced effects, decomposes the tool-induced performance difference into gain and harm terms, and probes the mechanisms driving their evolution. Across checkpoint-level analyses on two VLMs with different tool priors and six benchmarks, we find that improvements are dominated by intrinsic learning, while tool-use RL mainly reduces tool-induced harm (e.g., fewer call-induced errors and weaker tool schema interference) and yields limited progress in tool-based correction of intrinsic failures.Overall, current vision tool-use RL learns to coexist safely with tools rather than master them.
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Unsafer in Many Turns: Benchmarking and Defending Multi-Turn Safety Risks in Tool-Using Agents
Xu Li, Simon Yu, Minzhou Pan, Yiyou Sun, Bo Li, Dawn Song, Xue Lin, Weiyan Shi
LLM-based agents are becoming increasingly capable, yet their safety lags behind. This creates a gap between what agents can do and should do. This gap widens as agents engage in multi-turn interactions and employ diverse tools, introducing new risks overlooked by existing benchmarks. To systematically scale safety testing in multi-turn, tool-realistic settings, we propose a principled taxonomy that transforms single-turn harmful tasks into multi-turn attack sequences. Using this taxonomy, we construct MT-AgentRisk (Multi-Turn Agent Risk Benchmark), the first benchmark to evaluate tool-using agent safety under multi-turn, harmful-distribution settings. Our experiments reveal substantial safety degradation: the Attack Success Rate (ASR) increases by 16\% on average across open and closed models in multi-turn settings. To close this gap, we propose ToolShield, a training-free, tool-agnostic, self-exploration defense: when encountering a new tool, the agent autonomously generates test cases, executes them to observe downstream effects, and distills safety experiences for deployment. Experiments show that ToolShield effectively reduces ASR by 30\% on average in multi-turn interactions.

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Inducing Overthink: Hierarchical Genetic Algorithm-based DoS Attack on Black-Box Large Language Reasoning Models
Shuqiang Wang, Wei Cao, Jiaqi Weng, Jialing Tao, Licheng Pan, Hui Xue, Zhixuan Chu
Large Reasoning Models (LRMs) are increasingly integrated into systems requiring reliable multi-step inference, yet this growing dependence exposes new vulnerabilities related to computational availability. In particular, LRMs exhibit a tendency to “overthink’’—producing excessively long and redundant reasoning traces—when confronted with incomplete or logically inconsistent inputs. This behavior significantly increases inference latency and energy consumption, forming a potential vector for denial-of-service (DoS) style resource exhaustion.
In this work, we investigate this attack surface and propose an automated black-box framework that induces overthinking in LRMs by systematically perturbing the logical structure of input problems. Our method employs a hierarchical genetic algorithm (HGA) operating on structured problem decompositions, and optimizes a composite fitness function designed to maximize both response length and reflective overthinking markers.
Across four state-of-the-art reasoning models, the proposed method substantially amplifies output length, achieving up to a 26.1× increase on the MATH benchmark and consistently outperforming benign and manually crafted missing-premise baselines. We further demonstrate strong transferability, showing that adversarial inputs evolved using a small proxy model retain high effectiveness against large commercial LRMs. These findings highlight overthinking as a shared and exploitable vulnerability in modern reasoning systems, underscoring the need for more robust defenses.
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RSF-GLLM: Bridging the Semantic Gap in Multi-Hop Knowledge Graph QA via Recurrent Soft-Flow and Decoupled LLM Generation
Sambaran Bandyopadhyay, Ananth Muppidi
Multi-hop Question Answering over Knowledge Graphs faces a critical challenge: traditional retrieve-then-read pipelines break differentiability, preventing the retriever from learning to bridge the semantic gap where intermediate nodes lack lexical overlap with the query. To address this, we propose RSF-GLLM, a framework decoupling differentiable graph reasoning from answer generation. Our Recurrent Soft-Flow (RSF) module employs a GRU-guided query updater to propagate continuous relevance scores, utilizing a dynamic gating mechanism to traverse semantically dissimilar bridge nodes via structural cues. We introduce flow sparsity regularization to theoretically guarantee convergence from soft probabilities to discrete reasoning paths. These paths are extracted and textualized to fine-tune a Large Language Model (LLM), ensuring generation is grounded in factual topology. Experiments on WebQSP and CWQ demonstrate that RSF-GLLM achieves competitive performance with superior inference efficiency compared to LLM based computationally expensive approaches.
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Innovation: An Almost Characterization of Hallucination
Nishant P. Das, Piyush Srivastava
Hallucination is a central limitation of large language models (LLMs), and substantial effort has been devoted to understanding and mitigating it. Towards this, Kalai and Vempala (STOC 2024) introduced a probabilistic framework formalizing calibration and hallucination, and showed that, with high probability, calibrated LLMs hallucinate roughly at the rate of the "missing mass", a measure of how incomplete the training data is relative to its source. This raises two fundamental questions: (i) what property of a calibrated LLM makes hallucinations unavoidable? and (ii) can hallucinations be avoided by giving up calibration? We answer these questions by introducing a simpler property we call *innovation* that measures the tendency of a model to produce outputs outside the training data. We show that innovation is implied by the condition for hallucination identified by Kalai and Vempala, and,

further, that it is an almost characterization of hallucination: hallucination implies innovation, and conversely, innovation implies hallucination with high probability. We also provide lower bounds on the hallucination rate based on the "innovation rate", and by relating innovation rate back to missing mass, we obtain new hallucination rate lower bounds based on missing mass that extend the results of Kalai and Vempala.

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The Pareto-optimal Trade-off between Regret and Statistical Inference in Linear Stochastic Bandits under Safety Constraints

Yuming Shao, Zhixuan Fang

Linear bandits traditionally prioritize regret minimization, often overlooking statistical inference of the underlying parameter as a critical objective. In high-stakes settings such as healthcare, precise parameter estimation is indispensable, as it provides fundamental insights into system mechanisms and ensures robust decision-making under covariate shift. We investigate the tripartite balance between regret, inference, and safety, deriving a fundamental minimax lower bound that characterizes the Pareto-optimal frontier of these competing goals. We then propose SERMiSC, a novel algorithm that achieves the optimal trade-off by matching this lower bound while maintaining a near-constant $\tilde{O}(1)$ safety risk. Empirical results demonstrate that SERMiSC effectively navigates the Pareto frontier and outperforms various baselines, thereby validating our theoretical analysis.

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Stronger Benchmarks for Prediction as a Service with Constraints

Yahav Bechavod, Jiuyao Lu, Aaron Roth

We study a learner who sequentially makes and broadcasts predictions of some underlying adversarially varying state. Many downstream decision makers with different goals and different long-term constraints consume these decisions to choose actions. In this setting we give the first algorithm that obtains simultaneous *dynamic regret* guarantees for all of the decision makers --- where regret for each agent is measured against a potentially changing sequence of actions across rounds of interaction, while also ensuring vanishing constraint violation for each agent. We can promise these dynamic regret bounds not just marginally, but simultaneously on many different intersecting subsequences, which lets decision makers compete with strategies that adapt with both long-term drift and short-term variation. Our results do not require the decision makers to maintain any state, but just to react myopically to our predictions.

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AdaS: Adaptive Gradient Descent for Spiking Transformers

Zijian Zhou, Honglin Cao, Ammar Belatreche, Wenjie Wei, Yimeng Shan, Yu Liang, Yu Yang, Shuai Wang, Yalan Ye, Malu Zhang, Yang Yang, Haizhou Li

Transformer-based Spiking Neural Networks (SNNs) combine Transformer performance with SNN energy efficiency through an event-driven self-attention mechanism. However, Spiking Transformers still lag behind their Artificial Neural Network (ANN) counterparts. Most existing studies address this issue through new architectural designs, yet few have explored optimization algorithms tailored to Spiking Transformers.

We substantiate the excessive noise problem in Spiking Transformer training by quantitatively defining parameter-update noise and, based on this definition, providing theoretical analysis and experimental validation.

To address this problem, we propose AdaS, an adaptive gradient descent method for Spiking Transformers. AdaS reduces excessive noise by adaptively incorporating a gradient update component into adaptive optimization. Instead of simply removing noise, AdaS maintains it at an appropriate level to preserve its generalization benefits, thereby improving the performance of Spiking Transformers.

We conduct extensive experiments on various Spiking Transformer architectures and datasets from both computer vision and natural language processing. The results demonstrate that the proposed AdaS consistently enhances performance across different Spiking Transformers, validating its effectiveness and generalizability.

This work is among the first systematic studies of optimizer design specifically for Spiking Transformers, offering a practical tool to narrow the accuracy gap with ANNs while preserving the energy advantages of spike-based computation. Code is available at https://github.com/CayleyZ/AdaS.
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Revisiting Efficiency–Accuracy Scaling in Mixture-of-Experts Architectures
Venmugil Elango, Nidhi Bhatia, Roger Waleffe, Rasoul Shafipour, Tomer Asida, Abhinav Khattar, Nave Assaf, Maximilian Golub, Joseph Guman, Tiyasa Mitra, Ritchie Zhao, Ritika Borkar, Ran Zilberstein, Mostofa Patwary, Mohammad Shoeybi, Bita Darvish Rouhani
Mixture-of-Experts (MoEs) have become a central component of many state-of-the-art open-source and proprietary large language models. Despite their widespread adoption, it remains unclear how close existing MoE architectures are to optimal for inference cost, as measured by accuracy per floating-point operation and per parameter. In this work, we revisit MoE design from a hardware-software co-design perspective, grounded in empirical and theoretical considerations. We characterize key performance bottlenecks across diverse deployment regimes, spanning offline high-throughput execution and online, latency-critical inference. Guided by these insights, we introduce **LatentMoE**, a new model architecture resulting from systematic design exploration and optimized for maximal accuracy per unit of compute. Empirical design space exploration at scales of up to 95B parameters and over a 1T-token training horizon, together with supporting theoretical analysis, shows that LatentMoE consistently outperforms standard MoE architectures in terms of accuracy per FLOP and per parameter. Given its strong performance, the LatentMoE architecture has been adopted by the flagship Nemotron-3 Super and Ultra models and scaled to substantially larger regimes, including longer token horizons and larger model sizes, as reported in (NVIDIA et al., 2025, arXiv:2512.20856).
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A Random Matrix Perspective on the Consistency of Diffusion Models
Binxu Wang, Jacob A Zavatone-Veth, Cengiz Pehlevan
Diffusion models trained on different, non-overlapping subsets of a dataset often produce strikingly similar outputs when given the same noise seed. We trace this consistency to a simple linear effect: the shared Gaussian statistics across splits already predict much of the generated images. To formalize this, we develop a random matrix theory (RMT) framework that quantifies how finite datasets shape the expectation and variance of the learned denoiser and sampling map in the linear setting. For expectations, sampling variability acts as a renormalization of the noise level through a self-consistent relation $\sigma^2\to\kappa(\sigma^2)$, explaining why limited data overshrink low-variance directions and pull samples toward the dataset mean. For fluctuations, our variance formulas reveal three key factors behind cross-split disagreement: \textit{anisotropy} across eigenmodes, \textit{inhomogeneity} across inputs, and overall scaling with dataset size. Extending deterministic-equivalence tools to fractional matrix powers further allows us to analyze entire sampling trajectories. The theory sharply predicts the behavior of linear diffusion models, and we validate its predictions on U Net and DiT architectures in their non-memorization regime, identifying where and how samples deviates across training data split. This provides a principled baseline for reproducibility in diffusion training, linking spectral properties of data to the stability of generative outputs.
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Image-to-Brain Signal Generation for Visual Prosthesis with CLIP Guided Multimodal Diffusion Models
Ganxi Xu, Zhao-Rong Lai, Yuting Tang, Yonghao Song, Guoxu Zhou, Boyu Wang, Jian Zhu, Jinyi Long
Visual prostheses hold great promise for restoring vision in blind individuals. While researchers have successfully utilized M/EEG signals to evoke visual perceptions during the brain decoding stage of visual prostheses, the complementary process of converting images into M/EEG signals in the brain encoding stage remains largely unexplored, hindering the formation of a complete functional pipeline

. In this work, we present a novel image-to-brain signal framework that generate
s M/EEG from images by leveraging the diffusion transformer architecture enhance
d with cross-attention mechanisms. Specifically, we employ a diffusion transform
er (DiT) architecture based on denoising diffusion implicit models (DDIM) to ach
ieve brain signal generation. To realize the goal of image-to-brain signal conve
rsion, we use cross-attention mechanisms to align brain signal embeddings with C
LIP image embeddings. Moreover, we leverage large language models (LLMs) to gene
rate image captions, and concatenate the resulting CLIP text embeddings with CLI
P image embeddings to form unified embeddings for cross-attention alignment, ena
bling our model to capture core semantic information. Furthermore, we introduce
a learnable spatio-temporal position encoding that combines brain region embeddi
ngs with temporal embeddings to capture both spatial and temporal characteristic
s of brain signals. We evaluate the framework on two multimodal benchmark datase
ts (THINGS-EEG2 and THINGS-MEG) and demonstrate that it generates biologically p
lausible brain signals.
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Anytime Safe PAC Efficient Reasoning
Chengyao Yu, Hao Zeng, Youxin Zhu, Jianguo Huang, Huajun Zeng, Bingyi Jing
Large Reasoning Models (LRMs) have demonstrated remarkable performance on comple
x tasks but suffer from high computational costs and latency. While selective th
inking strategies improve efficiency by routing easy queries to non-thinking mod
els, existing approaches often incur uncontrollable errors, especially in online
 settings where the performance loss of a non-thinking model is only partially o
bserved and data are non-stationary. To address this, we propose *Betting Probab
ly Approximately Correct* (B-PAC) *reasoning*, a principled method that enables
anytime safe and efficient online reasoning under partial feedback. Specifically
, we utilize inverse propensity scoring estimators to construct test supermartin
gales for candidate thresholds, and then dynamically adjust the routing threshol
d based on the accumulated statistical evidence of safety. Theoretically, we est
ablish the anytime-valid performance loss control and the efficiency of B-PAC re
asoning. Extensive experiments demonstrate that B-PAC reasoning significantly re
duces computational overhead, decreasing thinking model usage by up to 81.01\%,
while controlling the performance loss below the user-specified level.
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Structure-Centric Graph Foundation Model via Geometric Bases
Xiaodong He, Haolan He, Ruiyi Fang, Ming Sun, zhao kang
Graph foundation models (GFMs) seek transferable representations across graph do
mains but are limited by structural heterogeneity and incompatible node feature
spaces. We propose Structure-Centric Graph Foundation Models (SCGFM), which trea
t graph topology as the primary source of transferable knowledge. Modeling graph
s as metric measure spaces, SCGFM introduces learnable geometric bases that defi
ne a shared structural coordinate system. Graphs are aligned to these bases via
Gromov–Wasserstein distances, yielding structure-aligned latent representations
that accommodate heterogeneous graph topologies. To address feature incompatibil
ity, SCGFM employs a structure-aware feature re-encoding mechanism that unifies
node representations without assuming a fixed feature dimensionality or requirin
g dataset-specific preprocessing. Experiments on graph- and node-level tasks dem
onstrate strong in-domain and cross-domain generalization, outperforming existin
g GFM approaches.
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Beyond Gemini-3-Pro: Revisiting LLM Routing and Aggregation at Scale
Shengji Tang, Weihao Lin, Peng Ye, Jingqi Ye, Hao Li, Yiqun Zhang, Xiaosong Wang
, Bo Zhang, Shuyue Hu, Tao Chen, LEI BAI, Wanli Ouyang
Large Language Models (LLMs) have rapidly advanced, with Gemini-3-Pro setting a
new performance milestone. In this work, we explore collective intelligence as a
n alternative to monolithic scaling, and demonstrate that open-source LLMs' coll
aboration can surpass Gemini-3-Pro. We first revisit LLM routing and aggregation
 at scale and identify three key bottlenecks: (1) current train-free routers are
 limited by a query-based paradigm focusing solely on textual similarity; (2) re
cent aggregation methods remain largely static, failing to select appropriate ag

gregators for different tasks; (3) the complementarity of routing and aggregatio n remains underutilized. To address these problems, we introduce JiSi, a novel f ramework designed to release the full potential of LLMs' collaboration through t hree innovations: (1) Query-Response Mixed Routing capturing both semantic infor mation and problem difficulty; (2) Support-Set-based Aggregator Selection jointl y evaluating the aggregation and domain capacity of aggregators; (3) Adaptive Ro uting-Aggregation Switch dynamically leveraging the advantages of routing and ag gregation. Comprehensive experiments on nine benchmarks demonstrate that JiSi ca n surpass Gemini-3-Pro with only 47% costs by orchestrating ten open-source LLMs , while outperforming mainstream baselines. It suggests that collective intellig ence represents a novel path towards Artificial General Intelligence (AGI).
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Identifying Learnwares via Reduced Neural Conditional Mean Embedding
Zi-Yu Mao, Ming Li
The learnware paradigm aims to establish a market of learnwares, each of which i s a well-trained model combined with a specification that describes its function ality without leaking data privacy. The market enables users to efficiently reus e relevant models based on specifications on their own tasks instead of training models from scratch. Recent works have attempted to generate specifications usi ng Reduced Kernel Mean Embedding (RKME), which maps input distributions into Rep roducing Kernel Hilbert Space (RKHS) while ignoring the output space, causing mo dels trained on similar input spaces to yield similar specifications, even when their functionalities differ. Many labeled-RKME improvements attempt to address this by indirectly modeling the input-output conditional distributions, but they remain limited to classification tasks and lack clear theoretical explanations. In this work, we propose Reduced Neural Conditional Mean Embedding (RNCME), a n ovel specification generation method that directly models input-output condition al distributions via Conditional Mean Embedding (CME). Our RNCME method has a cl ear theoretical understanding based on CME and is applicable to both regression and classification tasks. Empirical experiments demonstrate the effectiveness an d efficiency of our RNCME method.
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Fine-Tune Once, Reuse Across Models: Bayesian Task-Update Factors and Approximat ions
Siyang Guo, Junbo Wang, Zibin Zheng
As pre-trained models evolve rapidly, transferring fine-tuning knowledge to upda ted models without retraining has become a critical challenge. Most existing met hods reuse parameter updates, yet the same dataset can induce substantially diff erent updates across base models due to mismatched local loss landscapes, making such transfer unstable. We instead adopt a Bayesian-updating perspective: a bas e model defines a prior, while fine-tuning contributes a task-update factor that is prior-agnostic, thereby making it feasible to reuse the update across base m odels. Specifically, we formalize a reusable task-update factor by requiring *in variance across base models* and *a fixed-dimensional parameterization*. Our mai n theoretical result shows that such reusable factors exist when the variational family is a half-space, and it is already maximal among convex families. In par ticular, an ideal regime arises when the priors and their Bayesian posteriors re main within a shared exponential family, as it always admits a reusable update f actor. Building on this existence, we propose ***B**ayesian Task Update **Transf er*** (*BTransfer*), which extracts a reusable task-update factor from a single fine-tuning run and applies it to a new prior. For deep networks, we implement * BTransfer* with a ``lift–transfer–return'' pipeline: 1) lift model parameters to distributions; 2) transfer the extracted task-update factor in the exponential family distributions; and 3) return the updated posterior distribution to parame ter space. Extensive experiments demonstrate that our approach effectively reuse s fine-tuning knowledge across models without post-training.
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FlexiFlow: decomposable flow matching for generation of flexible molecular ensem ble
Riccardo Tedoldi, Ola Engkvist, Patrick Bryant, Hossein Azizpour, Jon Paul Janet

, Alessandro Tibo
Sampling useful three-dimensional molecular structures along with their most favorable conformations is a key challenge in drug discovery. Current state-of-the-art 3D de-novo molecular design generative models are limited to generating a single conformation. However, the conformational landscape of a molecule determines its observable properties and how tightly it is able to bind to a given protein target. By generating a representative set of low-energy conformers, we can more directly assess these properties and potentially improve the ability to generate molecules with desired thermodynamic observables. Towards this aim, we propose \textit{FlexiFlow}, a novel architecture that extends flow-matching models, allowing for the joint sampling of molecules along with multiple conformations while preserving both equivariance and permutation invariance. We demonstrate the effectiveness of our approach on the QM9 and GEOM Drugs datasets, achieving state-of-the-art results in 3D molecular generation producing valid, unique, and novel molecules with high fidelity to the training data distribution. Moreover, we show that our model can generate unstrained conformational ensembles capturing the conformational diversity and providing similar coverage to state-of-the-art physics-based methods at a fraction of the inference time. Finally, FlexiFlow can be successfully transferred to the protein-conditioned ligand generation task, even when the dataset contains only static pockets without accompanying conformations.
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LIVE: Long-horizon Interactive Video World Modeling
Junchao Huang, Ziyang Ye, Xinting Hu, Tianyu He, Guiyu Zhang, Shaoshuai Shi, Jiang Bian, Li Jiang
Autoregressive video world models predict future visual observations conditioned on actions. While effective over short horizons, these models often struggle with long-horizon generation, as small prediction errors accumulate over time. Prior methods alleviate this by introducing pre-trained teacher models and sequence-level distribution matching, which incur additional computational cost and fail to prevent error propagation beyond the training horizon. In this work, we propose LIVE, a Long-horizon Interactive Video world modEl that enforces bounded error accumulation via a novel cycle-consistency objective, thereby eliminating the need for teacher-based distillation. Specifically, LIVE first performs a forward rollout from ground-truth frames and then applies a reverse generation process to reconstruct the initial state. The diffusion loss is subsequently computed on the reconstructed terminal state, providing an explicit constraint on long-horizon error propagation. Moreover, we provide a unified view that encompasses different approaches and introduce progressive training curriculum to stabilize training. Experiments demonstrate that LIVE achieves state-of-the-art performance on long-horizon benchmarks, generating stable, high-quality videos far beyond training rollout lengths.
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From Outcomes to Actions: Leveraging Hindsight for Long-Horizon Language Agent Training
Zishang Jiang, tingyun li, Jinyi Han, Xinyi Wang, Sihang Jiang, Yizhou Ying, Xiaojun Meng, Jiansheng Wei, Jiaqing Liang, Yanghua Xiao
Reinforcement learning (RL) has become a widely adopted technique for improving large language models (LLMs) on complex tasks. Despite this progress, existing RL methods still face challenges in training agents with longer-horizon interactions. One major bottleneck is distinguishing the contribution of different actions in long-horizon interaction, leading to high optimization variance. To address this, we introduce a novel policy gradient method, Hindsight Policy Optimization (HPO), that projects both the current policy distribution and the hindsight distribution into an intent space and extracts low-variance learning signals from the Wasserstein distance between them. We theoretically and empirically show that aggregating semantically similar states and actions in the intent space yields a bounded-variance estimator and improves policy performance stably. Our code is available online.
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Recovering Policy-Induced Errors: Benchmarking and Trajectory Synthesis for Robust GUI Agents
Tianpeng Bu, Xin Liu, Qihua Chen, Hao Jiang, Shurui Li, hongtao duan, Lu Jiang, lulu hu, Bin Yang, Minying Zhang
While GUI agents have advanced rapidly, they often lack the robustness to recover from their own errors, hindering real-world deployment. To bridge this gap at both the evaluation and data levels, we introduce GUI-RobustEval and propose Robustness-driven Trajectory Synthesis. GUI-RobustEval contains 1,216 executable test cases that systematically measure error recovery capabilities across a broad and realistic spectrum of error modes. At the data level, RoTS is a scalable synthesis framework that creates 800k high-quality data via a tree-based pipeline that proactively discovers diverse error modes and synthesizes corresponding recovery steps. Our two models, RoTS-7B and RoTS-32B, fine-tuned on our dataset, both demonstrate significant gains on GUI-RobustEval and traditional GUI benchmarks. Notably, RoTS-32B achieves state-of-the-art performance on OSWorld, with a 47.4% success rate and a 33.8% All-Pass@4 score, suggesting that improved long-horizon error recovery ability contributes to both robustness and overall performance. Our code is available at https://github.com/AlibabaResearch/RoTS
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MAnchors: Memorization-Based Acceleration of Anchors via Rule Reuse and Transformation
Haonan Yu, Junhao Liu, Xin Zhang
Anchors is a popular local model-agnostic explanation technique whose applicability is limited by its computational inefficiency. To address this limitation, we propose a memorization-based framework that accelerates Anchors while preserving explanation fidelity and understandability. Our approach leverages the iterative nature of Anchors' algorithm which gradually refines an explanation until it is precise enough for a given input by storing and reusing intermediate results obtained during prior explanations. Specifically, we maintain a memory of low-precision, high-coverage rules and introduce a rule transformation framework to adapt them to new inputs: the horizontal transformation adapts a retrieved explanation to the current input by replacing features, and the vertical transformation refines the general explanation until it is precise enough for the input. We evaluate our method across tabular, text, and image datasets, demonstrating that it significantly reduces explanation generation time while maintaining fidelity and understandability, thereby enabling the practical adoption of Anchors in time-sensitive applications.
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Speedup Patch: Learning a Plug-and-Play Policy to Accelerate Embodied Manipulation
Zhichao Wu, Junyin Ye, Zhilong Zhang, Yihao Sun, Haoxin Lin, Jiaheng Luo, Haoxiang Ren, Lei Yuan, Yang Yu
While current embodied policies exhibit remarkable manipulation skills, their execution remains unsatisfactorily slow as they inherit the tardy pacing of human demonstrations. Existing acceleration methods typically require policy retraining or costly online interactions, limiting their scalability for large-scale foundation models. In this paper, we propose **S**peed**u**p **P**atch (**SuP**), a lightweight, policy-agnostic framework that enables **plug-and-play acceleration** using solely offline data. SuP introduces an external scheduler that adaptively downsamples action chunks provided by embodied policies to eliminate redundancies. Specifically, we formalize the optimization of our scheduler as a Constrained Markov Decision Process (CMDP) aimed at maximizing efficiency without compromising task performance. Since direct success evaluation is infeasible in offline settings, SuP introduces **World Model based state deviation** as a surrogate metric to enforce safety constraints. By leveraging a learned world model as a virtual evaluator to predict counterfactual trajectories, the scheduler can be optimized via offline reinforcement learning. Empirical results on simulation benchmarks (Libero, Bigym) and real-world tasks validate that SuP achieves an overall $1.8\times$ execution speedup for diverse policies while maintaining their original success rates.

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mmBERT: A Modern Multilingual Encoder with Annealed Language Learning
Marc Marone, Orion Weller, William Fleshman, Eugene Yang, Dawn Lawrie, Benjamin Van Durme
Encoder-only language models are frequently used for a variety of language tasks, including classification and retrieval.
However, there has been a lack of recent research efforts for encoder models, especially with respect to multilingual models. We introduce mmBERT, an encoder-only language model pretrained on 3T tokens of multilingual text in over 1800 languages.
To build mmBERT we introduce several novel elements to massively multilingual encoder training, including phased data curation and scheduled language inclusion.
We add over 1700 low-resource languages to the data mix only during the decay phase, showing that it boosts performance dramatically and maximizes the gains from the relatively small amount of training data without excessive repetition.
The model uses recent advances in architecture and training schemes to be faster and more multilingual than other models and we release weights, data, and code.
We show that mmBERT significantly outperforms the previous generation, on various tasks, for both high and low-resource languages.
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Robust Signal Enhancement via Fractional Detail Views and Knowledge Guided Multi-view Fusion
Zikun Jin, Yuhua Qian, Xinyan Liang, Jiaqian Zhang, Haijun Geng
Robust signal enhancement at low SNR is fundamentally challenging because noise becomes strongly entangled with the signal and corrupts local time–frequency (TF) evidence.
In this regime, fixed resolution short time Fourier transform (STFT) enhancement with purely data driven convolutional biases can become overconfident in unreliable TF regions, causing unstable suppression or residual artifacts.
We propose FracKGMF, which couples Fractional Distance Decay Convolution (FracConv) with Knowledge Guided Multi-view Fusion (KGMF) for expressive TF modeling and reliability aware decisions under heavy corruption.
FracConv introduces a lightweight fractional distance decay family that reshapes local interactions into long tailed receptive patterns, enabling aggregation of weak but globally consistent cues when per-bin observations are ambiguous.
KGMF uses a wiener inspired reliability prior to calibrate multi-view fusion and reduce excessive suppression in uncertain regions.
Experiments on speech and EM benchmarks show consistent improvements over state-of-the-art baselines, with particularly large gains under extremely low SNR, including a 33 dB average improvement on EM signals at -20 dB.
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Solving the Offline and Online Min-Max Problem of Non-smooth Submodular-Concave Functions: A Zeroth-Order Approach
Amir Ali Farzin, Yuen-Man Pun, Philipp Braun, Tyler Summers, Iman Shames
We consider max-min and min-max problems with objective functions that are possibly non-smooth, submodular with respect to the minimiser and concave with respect to the maximiser. We investigate the performance of a zeroth-order method applied to this problem. The method is based on the subgradient of the Lovász extension of the objective function with respect to the minimiser and based on Gaussian smoothing to estimate the smoothed function gradient with respect to the maximiser. In expectation sense, we prove the convergence of the algorithm to an ϵ-saddle point in the offline case. Moreover, we show that, in the expectation sense, in the online setting, the algorithm achieves $O(\sqrt{N\bar{P}_N})$ online duality gap, where N is the number of iterations and $\bar{P}_N$ is the path length of the sequence of optimal decisions. The complexity analysis and hyperparameter selection are presented for all the cases. The theoretical results are illustrated via numerical examples.
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Empty Shelves or Lost Keys? Recall Is the Bottleneck for Parametric Factuality
Nitay Calderon, Eyal Ben-David, Zorik Gekhman, Eran Ofek, Gal Yona

Standard factuality evaluations of LLMs treat all errors alike, obscuring whether failures arise from missing knowledge (empty shelves) or from limited access to encoded facts (lost keys). We propose a behavioral framework that profiles factual knowledge at the level of facts rather than questions, characterizing each fact by whether it is encoded, and then by how accessible it is: cannot be recalled, can be directly recalled, or can only be recalled with inference-time computation (thinking). To support such profiling, we introduce WikiProfile, a new benchmark constructed via an automated pipeline with a prompted LLM grounded in web search. Across 4 million responses from 13 LLMs, we find that encoding is nearly saturated in frontier models on our benchmark, with GPT-5 and Gemini-3 encoding 95--98\% of facts. However, recall remains a major bottleneck: many errors previously attributed to missing knowledge instead stem from failures to access it. These failures are systematic and disproportionately affect long-tail facts and reverse questions. Finally, we show that thinking improves recall and can recover a substantial fraction of failures, indicating that future gains may rely less on scaling and more on methods that improve how models utilize what they already encode.
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SGERA: Stein-Guided ECG-Report Alignment for ECG Representation Learning
Jian Chen, Yipeng Du, Wenhao Yuan, Shuai Wang, Jinfeng Xu, Zewei Liu, Running Zhao, Edith Cheuk-Han Ngai
Electrocardiogram (ECG) representation learning via ECG-report alignment is often hindered by the inherent structural and statistical divergence between signals and natural language. Existing methods struggle to bridge this gap with simple contrastive objectives, but struggle with distribution dependencies between heterogeneous features. To address this, we propose **SGERA** (**S**tein-**G**uided **E**CG-**R**eport **A**lignment), which leverages the unique properties of Stein kernels to provide a more rigorous geometric alignment in the latent space: **instance-level** alignment via a Stein-RBF kernel enforces pairwise consistency between ECG and report embeddings and **distribution-level** alignment via a Stein-Score kernel captures higher-order interactions for global alignment. Furthermore, we introduce an ECG-Report matching task with a Hard Sample Mining strategy to refine discriminative boundaries. Experiments across three public datasets demonstrate that SGERA significantly outperforms state-of-the-art SSL methods in zero-shot classification, linear probing, and transfer learning, proving the superiority of Stein-guided alignment in handling complex medical modalities. Code is available at supplementary materials.
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Emergent Alignment via Competition
Natalie Collina, Surbhi Goel, Aaron Roth, Emily Ryu, Mirah Shi
Aligning AI systems with human values remains a fundamental challenge, but does our inability to create perfectly aligned models preclude obtaining the benefits of alignment? We study a strategic setting where a human user interacts with multiple differently misaligned AI agents, none of which are individually well-aligned. Our key insight is that when the user's utility lies approximately within the convex hull of the agents' utilities, a condition that becomes easier to satisfy as model diversity increases, strategic competition can yield outcomes comparable to interacting with a perfectly aligned model.

We model this as a multi-leader Stackelberg game, extending Bayesian persuasion to multi-round conversations between differently informed parties, and prove three results: (1) when perfect alignment would allow the user to learn her Bayes-optimal action, she can also do so in all equilibria under the convex hull condition; (2) under weaker assumptions requiring only approximate utility learning, a non-strategic user employing quantal response achieves near-optimal utility in all equilibria; and (3) when the user selects the best single AI after an evaluation period, equilibrium guarantees remain near-optimal without further distributional assumptions.

We complement the theory with two forms of empirical evidence: First, we perform

simulations of the best-AI selection game using best response dynamics, which show that competition among individually misaligned agents reliably improves user utility when the approximate convex hull assumption is satisfied, but does not always when it fails. Second, we show that synthetically generated AI utility functions (produced via perturbations of the same prompt to evaluate instances on a movie recommendation (MovieLens) and ethical judgement (ETHICS) dataset) quickly produce a convex hull that contains a good approximation of a given utility function even when none of the individual LLM utility functions is well aligned. We show that this phenomenon extends to human and LLM responses on real-world polling data (OpinionQA): a convex hull of LLM opinions can approximate human opinions more accurately than any individual LLM across a wide range of survey questions.

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TurboGS: Accelerating 3D Gaussian Splatting via Error-Guided Sparse Pixel Sampling and Optimization

Zheng Dong, Daifei Qiu, Pinxuan Dai, Ke Xu, Jiamin Xu, Lili He, Rynson W. H. Lau, Weiwei Xu

Consumer-level applications require fast optimization of 3D Gaussian Splatting (3DGS) with high-fidelity novel view rendering. However, existing 3DGS acceleration approaches still incur substantial computation on redundant pixels while sacrificing fine details. In this paper, we present TurboGS, an error-guided training framework that accelerates 3DGS by concentrating optimization on perceptually informative pixels. TurboGS is built upon four core components: (1) a tile-wise sparse pixel sampling, which, driven by multi-view reconstruction errors during training, prioritizes challenging regions and skips well-reconstructed ones to avoid redundant gradient computation; (2) a tile-wise structure-aware loss with sparse Normalized Cross-Correlation, which provides sparse yet effective supervision to preserve fine details and stabilize training; (3) an error-driven Gaussian density control strategy, which dynamically allocates model capacity and removes redundant primitives; and (4) a tailored hybrid optimizer that couples Hessian-informed updates with Adam moment damping to stabilize and improve convergence under sparse supervision. Experiments on standard benchmarks demonstrate that TurboGS can deliver on par or superior rendering quality within 100 seconds (up to 10x training speedup over vanilla 3DGS).

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The Efficiency Gap in Byte Modeling

Celine Lee, Jing Nathan Yan, Chen Liang, Jiaxin Shi, Yin Zhang, Jeremiah Zhe Liu, Pengcheng Yin, Fernando Pereira, Ed H. Chi, Derek Zhiyuan Cheng, Alexander M Rush, Ruoxi Wang

Modern language models have historically relied on two dominant design choices: subword tokenization and autoregressive (AR) ordering.
These design decisions bake in priors that dictate a model's learning.
Recently, two alternative paradigms have challenged this: byte-level modeling, which bypasses static statistically-derived token vocabularies, and masked diffusion modeling (MDM), which conducts parallel, non-sequential generation.
Their intersection represents a fully end-to-end modality-agnostic generative prototype; however,
removing these structural priors incurs a significant computational cost. In this work, we investigate this cost through a compute-matched scaling study. Our results reveal that the performance penalty of byte modeling is not uniform; across scale, the scaling overhead of byte modeling is worse for MDM than for AR.
We hypothesize that this disparity stems from context fragility: while AR's stable causal history allows models to naturally rediscover subword patterns, the MDM objective destroys the local contiguity required to efficiently resolve semantics from raw bytes. Our findings from controlled permutation experiments suggest that future modality-agnostic designs must incorporate alternative structural biases to maintain viable scaling trajectories in the byte regime.

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Harnessing Spectrum Video for Subject-Level Few-Shot and Cross-Montage EEG Generalization

Wei Wang, Fang He, Yifan Li, Wanying Qu, Yawei Li, Quanying Liu, Yanwei Fu
Existing EEG models are limited by electrode heterogeneity and rigid "channel-first" architectures that treat sensors as independent features. We propose Brain Signal Rendering (BSR), which reinterprets EEG as a physical projection of neural activity and transforms raw signals into structured spatiotemporal tensors (termed Spectrum Videos), enabling the transfer of rich priors from video foundation models. By utilizing VideoMAE for self-supervised pre-training, BSR learns robust, layout-agnostic spatiotemporal representations that preserve neural topology. We further employ subject-level few-shot learning and introduce cross-montage fine-tuning to rigorously evaluate generalization across subjects and electrode configurations. Experiments show that VideoMAE model integrated with the BSR framework significantly outperforms state-of-the-art spectrum based methods, providing a scalable and data-efficient foundation for generalizable EEG modeling. Our code is available at https://github.com/yanweifu-sii/BSR-VideoMAE.
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Fine-to-Coarse Fairness-Informed Multi-View Clustering
Shengju Yu, Suyuan Liu, Wenhao SHAO, Siwei Wang, Dayu Hu, Yiu-ming Cheung
In multi-view clustering (MVC), conventional anchor learning based models implicitly assume a uniform distribution of anchors across clusters, which could lead to inferior representation, especially when clusters vary significantly in size, as larger clusters require more anchors so as to adequately capture their intrinsic structural complexity. To alleviate this, we design a method termed FCFMVC that explicitly encourages proportional anchor allocation. To be specific, we transfer anchor allocation to discrete sample-cluster learning via bipartite graph bridge, and then backpropagate cluster state consisting of size and dispersion degree to guide anchor assignment. This allows the model to integrate cluster cardinality awareness and structural compactness directly into anchor distribution. On the other hand, we regard anchors as pseudo-samples, introduce an anchor-cluster indicator matrix on each view, and directly constrain the number of anchors assigned to each cluster within a tolerance margin. These two paths are further coupled through anchor-sample label alignment, and collaboratively facilitate anchor generation from fine-grained (anchor-level) to coarse-grained (cluster-level) structures. Besides, the entire optimization operation with linear time and space cost makes FCFMVC well-scalable to large-scale tasks. Experiments on datasets with diverse scales confirm the effectiveness of our FCFMVC.
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Efficient Public Verification of Private ML via Regularization
Zoë Ruha Bell, Anvith Thudi, Olive Franzese, Nicolas Papernot, Shafi Goldwasser
Training with differential privacy (DP) guarantees dataset members that they cannot be identified by users of the released model. However, those data providers, and, in general, the public, lack methods to efficiently verify that models trained on their data satisfy DP guarantees. The amount of compute needed to verify DP guarantees for current algorithms scales with the amount of computation required to train the model. In this paper we design the first DP algorithm with near optimal privacy-utility trade-offs but whose DP guarantees can be verified cheaper than training. We focus on DP stochastic convex optimization (DP-SCO), where optimal privacy-utility trade-offs are known. Here we show we can obtain tight privacy-utility trade-offs by privately minimizing a series of regularized objectives and only using the standard DP composition bound. Crucially, this method can be verified with much less compute than training. This leads to the first known DP-SCO algorithm with near optimal privacy-utility whose DP verification scales better than training cost, significantly reducing verification costs on large datasets.
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Discriminative Mixture-of-Experts on Graphs with Reliable Expert Fusion
Haoyue Deng, Menghui Wang, Yunlong Zhou, Ziwei Zhang, Ran Zhang, Chunming Hu, Xiao Wang
Graph Mixture-of-Experts (Graph-MoE) offers a way to scale GNNs via adaptive capacity allocation, with the goal of allowing different experts to capture diverse

graph patterns. Its effectiveness heavily depends on the coordination between routing decisions and expert specialization. However, through extensive empirical study, we identify two critical phenomena. First, discrimination loss occurs on both the expert and routing sides, where GNN experts become highly homogenized and the router collapses to a small subset of experts, failing to reflect diverse graph semantics. Second, routing uncertainty is prevalent, as existing routers produce uncertain expert assignments for most nodes, and such uncertainty exhibits a strong negative correlation with model performance.
To address these issues, we propose C^{2}GMoE, a novel Graph-MoE framework featuring Contrastive routing and Confidence-aware fusion. We introduce a group-wise contrastive routing strategy that provides explicit guidance for routing optimization by aligning node-level routing decisions with semantic clusters while satisfying load-balancing constraints. Moreover, through a theoretical analysis of generalization error, we develop a confidence-aware fusion mechanism that adaptively reweights expert predictions according to their confidence. Extensive experiments across multiple benchmarks demonstrate the effectiveness of our proposed C^{2}GMoE.
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D²Evo: Dual Difficulty-Aware Self-Evolution for Data-Efficient Reinforcement Learning
Ru Zhang, Renda Li, Ziyu Ma, Weijie Qiu, Chongyang Tao, Yong Wang, Xiangxiang Chu
Reinforcement learning (RL) has demonstrated potential for enhancing reasoning in large language models (LLMs). However, effective RL training, which requires medium-difficulty training samples, faces two fundamental challenges: Effective Data Scarcity and Dynamic Difficulty Shifts, where medium-difficulty samples are scarce and become trivial as models improve. Existing methods mitigate this scarcity to some extent by generating training samples. However, these approaches suffer from anchor-free generation, ignoring co-evolution, and difficulty mismatch. To address these issues, we propose D²Evo, a Dual Difficulty-aware self-Evolution RL framework. In each iteration, our method mines medium-difficulty anchors based on the current Solver's capability, trains the Questioner to generate diverse questions at appropriate difficulty levels, and jointly optimizes both components to enable progressive reasoning gains. Extensive experiments demonstrate that D²Evo outperforms existing methods on mathematical reasoning benchmarks with fewer than 2K real mathematical samples, and exhibits strong generalization on general reasoning benchmarks.
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Diffusion Language Model Parallel Decoding via Product-of-Experts Bridge
Juntong Shi, Brian L. Trippe, Jure Leskovec, Stefano Ermon, Minkai Xu
Diffusion language models (DLMs) offer substantial speed advantages through parallel decoding, but the lack of token dependencies limits generation quality compared to autoregressive (AR) models. Recent progress attempts to bridge the gap via importance sampling, with DLM being the proposal and AR being the target. However, due to the huge gap between their distributions, the sampling requires a large number of particles and is thus expensive to compute. In this paper, we introduce PoE-Bridge, a novel decoding framework that drastically improves generation speed and accuracy by introducing an intermediate distribution to bridge the gap. The distribution is constructed as a Product-of-Experts (PoE) of the DLM proposal and the AR target. With the intermediate distribution, we first use the DLM to draft multiple continuations in parallel, then apply rejection sampling to verify the drafted tokens and move the resulting candidates toward the PoE. We then use importance sampling to further correct the PoE-aligned candidates toward the AR target. We further propose several improved techniques, including mixed-temperature sampling for enhanced diversity and elastic rejection windows for reducing wasted verification. Empirically, PoE-Bridge achieves significantly improved accuracy with $5\times$ speedup over the standard DLM decoding approach, and recovers at least 95% of the target AR model's performance, efficiently advancing most of the quality gap on challenging mathematical reasoning and coding tasks. Our code is available at [https://github.com/juntongshi48/poe-bridge](https:

//github.com/juntongshi48/poe-bridge).
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Multi-view Consistent Latent Action Learning for World Modeling and Control
Shenghua Wan, Xiaohai Hu, Xunlan Zhou, Lei Yuan, Le Gan, De-Chuan Zhan
The scalability of world models is currently bottlenecked by the scarcity of act ion annotations. While self-supervised latent action learning offers a potential solution, existing single-view paradigms—relying on information bottlenecks or Vector Quantization (VQ)—often conflate superficial 2D pixel displacements with the underlying physical-spatial dynamics of an action. Consequently, these metho ds remain highly susceptible to view-dependent noise, such as camera shake. We i ntroduce **MuCoLA** (**Mu**lti-view **Co**nsistent **L**atent **A**ction learnin g), a framework that learns robust, view-invariant action representations by enf orcing semantic consistency across synchronized video streams. MuCoLA utilizes a Student-Teacher network with DINO-style self-distillation to align action distr ibutions across viewpoints, effectively filtering high-frequency visual noise wh ile preserving motion semantics. Theoretical analysis reveals that our multi-vie w objective functions as a spectral filter, isolating agent dynamics from enviro nmental nuisances. Empirically, MuCoLA significantly outperforms baselines in ac tion regression, video reconstruction, and downstream visual control tasks. Furt hermore, we demonstrate that MuCoLA exhibits favorable scaling properties with r espect to model capacity and data volume, paving the way for large-scale action- free world modeling.
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One Batch Is Enough: A Unified Dataset Condensation Framework for General Time S eries Analysis
Wei Shao, Ziquan Fang, Zheqi Lu, Yongfeng Su, Yuzhu Wang, Yunjun Gao
Time-series analysis is critical in real-world applications, yet the explosion o f time-series data imposes severe burdens on storage and computational resources . Recently, dataset condensation has emerged as a promising data-centric solutio n by synthesizing compact yet informative datasets to replace large-scale raw da ta. However, existing methods are largely vision-centric, failing to capture uni que temporal properties of time series, or task-specific, tightly coupling the c ondensed data to a particular downstream objective. As a result, these approache s suffer from feature mismatch and fail to generalize across diverse time-series tasks. To bridge this gap, we propose UniTSC, the first unified dataset condens ation framework for general time-series analysis. UniTSC employs a multi-view hy brid encoder to capture task-invariant representations across temporal, spectral , and topological perspectives. Building upon this representation, we design a t ri-space alignment paradigm that jointly aligns optimization trajectories, power spectral densities, and multivariate dependency structures, enabling comprehens ive information preservation under extreme compression. Extensive experiments sh ow that UniTSC retains up to 97.9\% of downstream performance using as little as 0.01\% of the original training data, within our experimental settings tied to standard sequence lengths, revealing that a single batch-equivalent budget ($\te xtless$ 128 samples) is sufficient to capture the essential dynamics of complex time-series data.
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A Generalist Pair-wise Progress Critic Model for Vision-Language-Action Robots
Qi Zhang, Shaopeng Zhai, Shengzhe Zhang, Litao Liu, Tianyi Zhang, Fuxian Huang, Ming Zhou
Recent advances in Vision-Language-Action (VLA) models have significantly improv ed robotic perception and manipulation capabilities, but still struggling to ada pt in dynamic, open-ended real-world environments due to a lack of reliable task progress feedback and improvement mechanisms. To address these challenges, we p ropose a generalist Vision Language Action-Critic model, VLAC, which can integra te both human and robot data, and unify action policy and task progress critic w ithin a single autoregressive architecture. Specifically, we propose a scalable and generalizable pair-wise progress understanding approach that can predict the delta of task progress between two steps in a trajectory and generate correct a ctions to complete the task. Then, we trained the model on large-scale, multi-so

urce human, robot, and general vision-language data for a generalist. Furthermor e, we deploy reinforcement learning where VLAC can autonomously evaluate task pr ogress to provide intrinsic rewards. Extensive evaluations demonstrate that our model generalizes effectively across diverse tasks and environments, leveraging its pair-wise progress understanding to provide reliable dense rewards, robust a ction generation, and significant improvements in real-world reinforcement learn ing. Our codes are available at \url{https://github.com/InternRobotics/VLAC}.
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Solving Spatial-Spectral Fusion with Latent Spectral Operators
Wei Li, Jieyuan Pei, Junnan Xu, Xuanfeng Ding, Junwei Zhu, wanjun chen, Jianwei Zheng
Existing deep spatial–spectral fusion (SSF) methods typically learn the fusion m apping in the coordinate domain using convolutions and attentions, making it har d to scale across varying spatial resolutions and offering limited control over the frequency content of the reconstructions, which may further lead to severe s pectral distortion.
In this work, we propose Latent Spectral Operators (LSO), a SSF framework that l earns fusion mappings between spectral functions through a structured operator p arameterization.
Specifically, LSO first applies a cross-attention projection, where learned late nt tokens serve as spectral prompts, to compress high-dimensional observations i nto a compact latent representation, and then adopts a hierarchical, patch-based architecture to integrate rich multi-scale cues.
Furthermore, to parameterize the latent fusion operator in a controllable manner , a Trigonometric Basis Solver is elaborated, which represents the mapping using a trigonometric basis expansion.
This formulation naturally supports multi-frequency modeling, with a capacity-st ability trade-off governed by the number of basis functions.
Extensive experiments on the CAVE and Harvard benchmarks demonstrate that LSO ac hieves consistent state-of-the-art performance and exhibits strong transferabili ty across different spatial scales.
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Beyond Independence: Learning Correlated Views for Variational Incomplete Multi-View Clustering
Zheming Xu, Aiyue Tang, Shidi Chen, Xuechao Zou, Congyan Lang, Rogelio A. Mancis idor, Michael Kampffmeyer
Incomplete multi-view clustering (IMVC) aims to uncover shared cluster structure s from data with partially observed views. Although recent imputation-free metho ds based on variational inference demonstrate robustness to missing views, they commonly rely on a conditional independence assumption across views, which fails to capture the inherently structured and potentially correlated nature of multi -view data. In this paper, we propose a variational framework that explicitly go es beyond this assumption by introducing a learnable cross-view correlation stru cture. Specifically, we explicitly model and learn correlations between views by utilizing the covariance structure of posterior estimation errors. To facilitat e robust and efficient learning, the correlation matrix is parameterized through a normalized Cholesky decomposition, ensuring positive definiteness and enablin g the entire model to be trained jointly through a unified variational objective . Extensive experiments on multiple IMVC benchmarks demonstrate that our method consistently outperforms state-of-the-art approaches across a wide range of miss ing-view settings. These results highlight the effectiveness of adaptive correla tion modeling in variational IMVC, demonstrating the need to go beyond the indep endence assumption in IMVC.
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Deep Reinforcement Learning Finds Bayes-Nash Equilibrium in Competitive Newsvend or Problems
Kassian Köck, Fabian Raoul Pieroth, Martin Bichler
We investigate learning dynamics in competitive newsvendor games, a class of con tinuous-action games with strategic substitutes. Despite established equilibrium properties, convergence of independent learning algorithms in repeated general-

sum play remains uncertain. We analyze structural properties under complete and incomplete information, deriving closed-form equilibria for a symmetric complete-information benchmark with perfect substitution. Our main theoretical contribution proves strict monotonicity in both complete-information and Bayesian models with private costs, ensuring equilibrium uniqueness and ruling out unstable dynamics. This provides convergence guarantees for variational-inequality-based algorithms. Numerical experiments using deep reinforcement learning agents with Proximal Policy Optimization empirically demonstrate convergence to Nash and Bayesian Nash equilibria, verified by equilibrium checks. These results establish a foundation for applying deep reinforcement learning in competitive inventory management.

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Brep2Shape: Boundary and Shape Representation Alignment via Self-supervised Transformers

Yuanxu Sun, Yuezhou Ma, Haixu Wu, Guanyang Zeng, Muye Chen, Jianmin Wang, Mingsheng Long

Boundary representation (B-rep) is the industry standard for computer-aided design (CAD). While deep learning shows promise in processing B-rep models, existing methods suffer from a representation gap: continuous approaches offer analytical precision but are visually abstract, whereas discrete methods provide intuitive clarity at the expense of geometric precision. To bridge this gap, we introduce Brep2Shape, a novel self-supervised pre-training framework designed to align abstract boundary representations with intuitive shape representations. Our framework employs a geometry-aware task where the model learns to predict dense spatial points from parametric Bézier control points, enabling the network to better understand physical manifolds derived from abstract coefficients. To enhance this alignment, we propose a dual transformer backbone with parallel streams that independently encode surface and curve tokens to capture their distinct geometric properties. Additionally, the topology attention is integrated to model the inherent interdependencies between surfaces and curves, thereby maintaining topological consistency. Experimental results demonstrate that Brep2Shape offers significant scalability, achieving state-of-the-art accuracy and faster convergence across various downstream tasks. Code is available at this repository: https://github.com/thuml/Brep2Shape.

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Coupled Cluster con MoLe: Molecular Orbital Learning for Neural Wavefunctions

Luca Thiede, Abdulrahman Aldossary, Andreas Burger, Jorge A. Campos-Gonzalez-Angulo, Alex Zook, Melisa Alkan, KOHEI NAKAJI, Taylor Lee Patti, Jérôme Florian Gonthier, Mohammad Ghazi Vakili, Alan Aspuru-Guzik

Density functional theory (DFT) is the most widely used method for calculating molecular properties; however, its accuracy is often insufficient for quantitative predictions. Coupled cluster (CC) theory is the most successful method for achieving accuracy beyond DFT and predicting properties that closely align with experiment. It is known as the ``gold standard'' of quantum chemistry. Unfortunately, the high computational cost of CC limits its widespread applicability. In this work, we present the Molecular Orbital Learning Model (MoLe), an equivariant machine learning model that directly predicts CC's core mathematical objects, the excitation amplitudes, from the mean-field Hartree-Fock molecular orbitals as inputs. We test various aspects of our model and demonstrate its very high data efficiency and remarkable out-of-distribution generalization to larger molecules and off-equilibrium geometries, despite being trained only on small equilibrium geometries. Finally, we also examine its ability to reduce the number of cycles required to converge CC calculations. MoLe can set the foundations for high-accuracy wavefunction-based ML architectures to accelerate molecular design and complement force-field approaches.

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Turbo4DGen: Ultra-Fast Acceleration for 4D Generation

Yuanbin Man, Ying Huang, Zhile Ren, Miao Yin

4D generation, or dynamic 3D content generation, integrates spatial, temporal, and view dimensions to model realistic dynamic scenes, playing a foundational rol

e in advancing world models and physical AI. However, maintaining long-chain consistency across both frames and viewpoints through the unique spatio-camera-motion (SCM) attention mechanism introduces substantial computational and memory overhead, often leading to out-of-memory (OOM) failures and prohibitive generation times. To address these challenges, we propose Turbo4DGen, an ultra-fast acceleration framework for diffusion-based multi-view 4D content generation. Turbo4DGen introduces a spatiotemporal cache mechanism that persistently reuses intermediate attention across denoising steps, combined with dynamically semantic-aware attention pruning and an adaptive SCM chain bypass scheduler, to drastically reduce redundant SCM attention computation. Our experimental results show that Turbo4DGen achieves an average 9.7$\times$ speedup without quality degradation on the ObjaverseDy and Consistent4D datasets. To the best of our knowledge, Turbo4DGen is the first dedicated acceleration framework for 4D generation.
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Active Learning with Low-Rank Structure for Data Selection
Vincent Cohen-Addad, Sasidhar Kunapuli, Vahab Mirrokni, Mahdi Nikdan, David Woodruff, Samson Zhou
In the data selection problem, the objective is to choose a small, representative subset of data that can be used to efficiently train a machine learning model. Sener and Savarese [ICLR 2018] showed that, given an embedding representation of the data and suitable geometric assumptions, heuristics based on k-center clustering can be used to perform data selection. This perspective was further explored by Axiotis et. al. [ICML 2024], who proposed a data selection approach based on k-means clustering and sensitivity sampling. However, these methods rely on the assumption that the dataset exhibits intrinsic geometric structure that can be effectively captured by clustering, whereas many modern datasets instead possess global algebraic structure that is better exploited by low-rank approximation or principal component analysis.

In this paper, we introduce a new data selection framework based on low-rank approximation and residual-based sampling, formulated through the lens of row subset selection and loss-preserving coreset construction. Given an embedding representation of the data satisfying mild regularity conditions, which can be interpreted as algebraic or angular notions of Lipschitz continuity, we show that it is possible to select a weighted subset of $\tilde{O}\left(k + \frac{1}{\varepsilon^2}\right)$ data points whose average loss approximates the average loss over the full dataset within a $(1+\varepsilon)$ relative error, up to an additive $\varepsilon \Phi_k$ term, where Φ_k denotes the optimal rank-k approximation cost of the embedding matrix. We complement these theoretical guarantees with empirical evaluations, demonstrating that on a range of real-world datasets, our data selection approach achieves improved performance over prior strategies based on uniform sampling or clustering-based sensitivity sampling.
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The Deterministic Horizon: When Extended Reasoning Fails and Tool Delegation Becomes Necessary
Dongxin Guo, Jikun Wu, SM Yiu
Extended chain-of-thought reasoning can degrade performance on deterministic state-tracking tasks, not due to preference biases, but limits rooted in the information-theoretic capacity of decoder-only attention. We establish: (1) an Attention Bottleneck Theorem with a complementary achievability construction, bounding state-tracking capacity as $O(H \cdot \log(L/H) \cdot \sqrt{d_h})$; (2) a context-dependent error model yielding super-exponential accuracy decay; (3) the State-Space Jaccard metric distinguishing capability from preference failures; (4) a Deterministic Horizon $d^* \in [19, 31]$ beyond which tool delegation becomes necessary. Across 12 models and 8 task domains (including SWE-Bench, WebArena, and SQL-Multi), tool-integrated reasoning consistently outperforms neural chain-of-thought; on the primary model suite it reaches 86–94% accuracy versus 24–42% for neural chain-of-thought. Fine-tuning on optimal-length traces yields $<$5% improvement, confirming an architectural ceiling, and high cross-model correlation ($r = 0.81$–0.91) indicates these failures are architectural rather than traini

ng-specific. Our results provide principled guidance for when pure neural reasoning should yield to hybrid approaches in agentic systems.
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Evolution Strategies at Scale: LLM Fine-Tuning Beyond Reinforcement Learning
Xin Qiu, Yulu Gan, Conor F. Hayes, Qiyao Liang, Yinggan XU, Roberto Dailey, Elliot Meyerson, Babak Hodjat, Risto Miikkulainen
Fine-tuning large language models (LLMs) for downstream tasks is an essential stage of modern AI deployment. Reinforcement learning (RL) has emerged as the dominant fine-tuning paradigm, underpinning many state-of-the-art LLMs. In contrast, evolution strategies (ES) has largely been overlooked due to the widespread belief that it does not scale to modern model sizes. This paper overturns this assumption by demonstrating the first successful application of ES to full-parameter fine-tuning of LLMs at the billion-parameter scale, without dimensionality reduction. ES can indeed search over extremely high-dimensional parameter spaces and outperform established RL implementations across multiple axes, including improved tolerance to long-horizon and delayed rewards, robustness across diverse base LLMs, reduced susceptibility to reward hacking, and improved training stability. These findings suggest that ES is not merely a viable alternative to RL, but a fundamentally different and powerful backpropagation-free post-training paradigm that opens a new direction for LLM fine-tuning beyond current RL-based approaches.
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Step-Level Sparse Autoencoder for Reasoning Process Interpretation
Xuan Yang, Jiayu Liu, Yuhang Lai, Hao Xu, Zhenya Huang, Ning Miao
Large Language Models (LLMs) have achieved strong complex reasoning capabilities through Chain-of-Thought (CoT) reasoning. However, their reasoning patterns remain too complicated to analyze. While Sparse Autoencoders (SAEs) have emerged as a powerful tool for interpretability, existing approaches predominantly operate at the token level, creating a granularity mismatch when capturing more critical step-level information, such as reasoning direction and semantic transitions. In this work, we propose step-level sparse autoencoder (SSAE), which serves as an analytical tool to disentangle different aspects of LLMs' reasoning steps into sparse features. Specifically, by precisely controlling the sparsity of a step feature conditioned on its context, we form an information bottleneck in step reconstruction, which splits incremental information from background information and disentangles it into several sparsely activated dimensions. Experiments on multiple base models and reasoning tasks show the effectiveness of the extracted features. By linear probing, we can easily predict surface-level information, such as generation length and first token distribution, as well as more complicated properties, such as the correctness and logicality of the step. These observations indicate that LLMs should already at least partly know about these properties during generation, which provides the foundation for the self-verification ability of LLMs. Our code is available at https://github.com/Miaow-Lab/SSAE.
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UniRTL: Unifying Code and Graph for Robust RTL Representation Learning
Yi Liu, Hongji Zhang, Lei Chen, Mingxuan Yuan, Qiang Xu
Developing effective representations for register transfer level (RTL) designs is crucial for accelerating the hardware design workflow. Existing approaches, however, typically rely on a single data modality, either the RTL code or its associated graph-based representation, limiting the expressiveness and generalization ability of the learned representations. For RTL, the control data flow graph (CDFG) offers a comprehensive structural representation that preserves complete information, while the code modality explicitly encodes semantic and functional information. We argue that integrating these complementary modalities is essential for a thorough understanding of RTL designs. To this end, we propose UniRTL, a multimodal pretraining framework that learns unified RTL representations by jointly leveraging code and CDFG. UniRTL achieves fine-grained alignment between code and graph through mutual masked modeling and employs a hierarchical training strategy that incorporates a pretrained graph-aware tokenizer and staged alignment of text (*i.e.*, functional summary) and code prior to graph integration. We

evaluate UniRTL on two downstream tasks, performance prediction and code retriev
al, under multiple settings. Experimental results show that UniRTL consistently
outperforms prior methods, establishing it as a more robust and powerful foundat
ion for advancing hardware design automation.
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Equalized Generative Treatment: Matching f-divergences for Fairness in Generativ
e Models
Alexandre Verine, Rafael Pinot, Florian Le Bronnec
Fairness is a crucial concern for generative models, which not only reflect but
can also amplify societal and cultural biases. Existing fairness notions for gen
erative models are largely adapted from classification and focus on balancing th
e probability of generating samples from each sensitive group. We show that such
 criteria are brittle, as they can be met even when different sensitive groups a
re modeled with widely varying quality. To address this limitation, we introduce
 a new fairness definition for generative models, termed as \emph{equalized gene
rative treatment} (EGT), which requires comparable generation quality across all
 sensitive groups, with quality measured via a reference f-divergence. We furt
her analyze the trade-offs induced by EGT, demonstrating that enforcing fairness
 constraints necessarily couples the overall model quality to that of the most c
hallenging group to approximate. This indicates that a simple yet efficient min–
max fine-tuning method should be able to balance f-divergences across sensitiv
e groups to satisfy EGT. We validate this theoretical insight through a set of e
xperiments on both image and text generation tasks. We demonstrate that min-max
methods consistently achieve fairer outcomes compared to other approaches from t
he literature, while maintaining competitive overall performance for both tasks.
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Hierarchical Policy Learning via Spectral Decomposition
Shuxin Cao, Liquan Wang, Walker Byrnes, Yiye Chen, Yilun Du, Animesh Garg
In this paper, we identify a semantic decomposition in robot action sequences, s
eparating task-level motion intent from execution-level refinements.
By analyzing actions in the spectral domain using the discrete cosine transform
(DCT), we observe that low-frequency components capture global motion trajectori
es, while high-frequency components encode precise timing, alignment, and contac
t behaviors.
Motivated by this structure, we propose Causal Spectral Policy (CSP), which mode
ls action generation as a causal coarse-to-fine process: coarse motion is predic
ted from observation and language, and fine corrections are generated conditiona
lly on the realized trajectory.
Across simulation and real-world evaluations, CSP consistently outperforms stron
g baselines on precision-sensitive manipulation tasks.
Additionally, we propose human-inspired teleoperation noise injection as a data
augmentation method under which our approach demonstrates strong robustness to n
oisy demonstrations
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DRL-STAF: A Deep Reinforcement Learning Framework for State-Aware Forecasting of
 Complex Multivariate Hidden Markov Processes
Manrui Jiang, Jingru Huang, Yong Chen, Chen Zhang
Forecasting multivariate hidden Markov processes is challenging due to nonlinear
 and nonstationary observations, latent state transitions, and cross-sequence de
pendencies. While deep learning methods achieve strong predictive accuracy, they
 typically lack explicit state modeling, whereas Hidden Markov Models (HMMs) pro
vide interpretable latent states but struggle with complex nonlinear emissions a
nd scalability. To address these limitations, we propose DRL-STAF, a **D**eep **
R**einforcement **L**earning based **ST**ate-**A**ware **F**orecasting framework
 that jointly predicts next-step observations and estimates the corresponding hi
dden states for complex multivariate hidden Markov processes. Specifically, DRL-
STAF models complex nonlinear emissions using deep neural networks and estimates
 discrete hidden states using reinforcement learning, reducing the reliance on p
redefined transition structures and enabling flexible adaptation to diverse temp
oral dynamics. In particular, DRL-STAF mitigates the state-space explosion encou

ntered by typical multivariate HMM-based methods. Extensive experiments demonstrate that DRL-STAF outperforms HMM variants, standalone deep learning models, and existing DL-HMM hybrids in most cases, while also providing reliable hidden-state estimates.
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Towards Diffeomorphism-Equivariant Neural Networks via Canonicalization
Josephine Elisabeth Oettinger, Zakhar Shumaylov, Johannes Bostelmann, Jan Lellmann, Carola-Bibiane Schönlieb
Incorporating group symmetries via equivariance into neural networks has emerged as a robust approach for improving efficiency and overcoming the data requirements of modern deep learning. While most existing approaches, such as group convolutions and averaging-based methods, focus on compact, finite, or low-dimensional groups with linear actions, this work explores how equivariance can be extended to infinite-dimensional groups. We propose a strategy designed to induce diffeomorphism equivariance in pre-trained neural networks via energy-based canonicalization. Formulating equivariance as an optimization problem allows us to access the rich toolbox of already established differentiable image registration methods. Empirical results on segmentation and classification tasks confirm that our approach achieves approximate equivariance and generalizes to unseen transformations without relying on extensive data augmentation or retraining.
****

Generalizing Multi-Scale Time-Series Modeling with a Single Operator
Cheonwoo Lee, Dooho Lee, Doyun Choi, Jaemin Yoo
Multi-scale modeling has emerged as an effective design principle for time-series forecasting by capturing temporal dynamics at multiple resolutions. As no principled foundation has been established in the literature, we unify existing scaling methods into a scaling operator family, revealing a fundamental limitation of existing approaches: reliance on fixed and discrete scaling. To address this limitation, we propose SiGMA (Single Generalized Multi-scale Architecture), which enables distance-aware scaling via the learnable discrete Gaussian (LDG) kernel grounded in scale-space theory. We evaluate SiGMA comprehensively on long- and short-term forecasting benchmarks against state-of-the-art multi-scale baselines. SiGMA outperforms all competitors on both tasks, especially achieving the best performance in 13 out of 16 long-term evaluation settings. Beyond accuracy, SiGMA significantly improves training speed by up to 5.3 times and reduces memory consumption by up to 3.8 times over the strongest competitors. Code is available at https://github.com/cheonwoolee/SiGMA.
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AReaL-DTA: Dynamic Tree Attention for Efficient Reinforcement Learning of Large Language Models
Jiarui Zhang, Yuchen Yang, Ran Yan, Zhiyu Mei, Liyuan Zhang, Daifeng Li, Wei Fu, Jiaxuan Gao, Shusheng Xu, Yi Wu, Binhang Yuan
Reinforcement learning (RL)-based post-training for large language models (LLMs) is computationally expensive, as it generates many rollout sequences that frequently share long token prefixes. Existing RL frameworks usually process these sequences independently during policy training, i.e., repeatedly recomputing identical prefixes in both the forward and backward passes of policy gradient computation, leading to substantial inefficiencies in computation resources and memory usage. Although prefix sharing naturally induces a tree structure over rollouts, packed tree-mask approaches scale poorly in RL settings. In this paper, we introduce AReaL-DTA, which efficiently exploits prefix sharing in RL training. AReaL-DTA employs a depth-first search (DFS)-based execution strategy that dynamically traverses the rollout prefix tree during both forward and backward computation, materializing only a single root-to-leaf path at a time. To further improve scalability, AReaL-DTA incorporates a load-balanced distributed batching mechanism that dynamically constructs and processes prefix trees across multiple GPUs. On τ^2-bench, AReaL-DTA improves training throughput by up to $8.31\times$ over dense training and up to $1.70\times$ over sparse training. Our code is available at https://github.com/areal-project/AReaL/tree/feat/dta.
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PRISM: Distribution-free Adaptive Computation of Matrix Functions for Accelerating Neural Network Training
Shenghao Yang, Zhichao Wang, Oleg Balabanov, N. Benjamin Erichson, Michael W. Mahoney
Matrix functions such as square root, inverse roots, and orthogonalization play a central role in preconditioned gradient methods for neural network training. This has motivated the development of iterative algorithms that avoid explicit eigendecompositions and rely primarily on matrix multiplications, making them well suited for modern GPU accelerators. We present PRISM (Polynomial-fitting and Randomized Iterative Sketching for Matrix functions computation), a general framework for accelerating iterative algorithms for computing matrix functions. PRISM combines adaptive polynomial approximation with randomized sketching: at each iteration, it fits a polynomial surrogate to the current spectrum via a sketched least-squares problem, adapting to the instance at hand with minimal overhead. We apply PRISM to accelerate Newton–Schulz-like iterations for matrix square roots and orthogonalization, which are core primitives in machine learning. Unlike prior methods, PRISM requires no explicit spectral bounds or singular value estimates; it adapts automatically to the evolving spectrum. Empirically, PRISM accelerates training when integrated into Shampoo and Muon optimizers.
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Noisy-Space Policy Gradient for Diffusion Policies in Offline Reinforcement Learning
Mahmoud Selim, Cristina Cipriani, Karl Henrik Johansson
Diffusion policies offer a powerful and expressive parameterization for continuous control. Yet, their integration with reinforcement learning remains conceptually and algorithmically challenging. In this work, we address this gap by introducing a noisy-space action-value (Q-)function that assigns values to diffusion latents through the distribution of executed actions induced by the denoising process. We show that this construction admits a precise semantic interpretation and derive a noisy-space policy gradient (NSPG) that optimizes noisy latents using only clean action space value estimates. Building on this result, we formulate a KL-regularized policy improvement over noisy latents and show that the resulting objective admits a diffusion-compatible regression form, avoiding backpropagation through the denoising process. Empirical results on state-based D4RL benchmarks and vision-based OGBench tasks demonstrate that the proposed noisy-space objective provides a principled and effective basis for training diffusion policies in offline reinforcement learning. Project webpage: https://mahmoud-selim.github.io/NSPG/
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Towards Efficient Large Language Reasoning Models via Extreme-Ratio Chain-of-Thought Compression
Yuntian Tang, Bohan Jia, Wenxuan Huang, Lianyue Zhang, Jiao Xie, Wenxi Li, Wei Li, Jie Hu, Xinghao Chen, Rongrong Ji, Shaohui Lin
Chain-of-Thought (CoT) reasoning successfully enhances the reasoning capabilities of Large Language Models (LLMs), yet it incurs substantial computational overhead for inference. Existing CoT compression methods often suffer from a critical loss of logical fidelity at high compression ratios, resulting in significant performance degradation. To achieve high-fidelity, fast reasoning, we propose a novel EXTreme-RAtio Chain-of-Thought Compression framework, termed Extra-CoT, which aggressively reduces the token budget while preserving answer accuracy. To generate reliable, high-fidelity supervision, we first train a dedicated semantically-preserved compressor on mathematical CoT data with fine-grained annotations. An LLM is then fine-tuned on these compressed pairs via a mixed-ratio supervised fine-tuning (SFT), teaching it to follow a spectrum of compression budgets and providing a stable initialization for reinforcement learning (RL). We further propose Constrained and Hierarchical Ratio Policy Optimization (CHRPO) to explicitly incentivize question-solving ability under lower budgets by a hierarchical reward. Experiments on three mathematical reasoning benchmarks show the superiority of Extra-CoT. For example, on MATH-500 using Qwen3-1.7B, Extra-CoT achieves over 73\% token reduction with an accuracy improvement of 0.6\%, significantly ou

tperforming state-of-the-art (SOTA) methods. Our source codes are released in th
e Supplementaries.
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Mosaic: Runtime-Efficient Multi-Agent Embodied Planning

Kunjal Panchal, Saayan Mitra, Sunav Choudhary, Victor Bursztyn, Somdeb Sarkhel,
Hui Guan

LLM-based multi-agent embodied planning remains impractical due to prohibitively
 high execution latency. We identify failed actions as the dominant bottleneck,
stemming from two core challenges: inaccurate state tracking under partial obser
vability and inefficient coordination that produces redundant or conflicting act
ions. We introduce Mosaic, a runtime-efficient multi-agent planning framework th
at addresses both challenges. Mosaic maintains accurate yet lightweight state tr
acking through agent-centric semantic memory that stores objects in relative coo
rdinates, enabling geometric transformations and coordination. It ensures effici
ent coordination through Integer Linear Programming that allocates actions at ev
ery planning step, enforcing physical feasibility and inter-agent coordination c
onstraints. Across AI2-THOR and search-and-rescue benchmarks, Mosaic achieves 27
–32% faster execution, 30–33% fewer LLM calls, 25–31% fewer steps, and 4–10% poi
nts higher success rates. These results demonstrate that efficient memory and co
nstraint-guided coordination are critical for scalable, low-latency multi-agent
planning.
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Agentic Model Predictive Questioning Control in Visual Design

Kuang-Da Wang, Zhao Wang, Wei-Yao Wang, Yotaro Shimose, Jaechang Kim, Shingo Tak
amatsu

Recent Large Language Model based approaches for clarifying visual design largel
y focus on selecting questions that better uncover user intent, but often overlo
oks the cognitive burden imposed on users, i.e., the effort required to interpre
t and answer these questions, which is crucial for effective human-agent interac
tion. In this paper, we propose Agentic Model Predictive Questioning Control (A-
MPQC), a test-time framework that reduces proxy-estimated user interaction burde
n while improving visual design alignment by formulating multi-round clarificati
on as trajectory optimization with receding-horizon replanning to revise its que
stioning strategy. In addition, we introduce lookahead question plans to reduce
ambiguity early, and a lightweight respond-or-reject surrogate reward to steer q
uestions toward lower user-burden formats (e.g., yes/no). Experiments on webpage
 and ad banner generation benchmarks show that A-MPQC not only generates designs
 better aligned with user intent, but also achieves lower user-interaction cost
across diverse interaction baselines, including fixed-format strategies (e.g., m
ultiple-choice and open-ended) and a retrieval-augmented baseline, without retra
ining. This paper sets a new perspective that explicitly formulates and optimize
s the human cognitive burden jointly with final design alignment, opening new op
portunities to advance human-agent interaction.
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Rapid Poison: Practical Poisoning Attacks Against the Rapid Response Framework

David Huang, Jaewon Chang, Avidan Shah, Prateek Mittal, Chawin Sitawarin

The Rapid Response (RR) framework (Peng et al., 2024), deployed in production sy
stems including Anthropic’s ASL-3 safeguards (Anthropic, 2025), dynamically adap
ts jailbreak detection classifiers by generating synthetic training data from em
erging attacks. We reveal that prompt injection can infiltrate this pipeline to
deliver poisoned samples into the classifier’s training set, enabling two attack
 objectives: (I) targeted poisoning attacks that create false positives on harml
ess samples by categorizing them as a jailbreak, with a specific desired feature
 (e.g., certain formatting, subject, or keyword), (II) concept-based backdoor at
tacks that induce false negatives on jailbreak inputs, generalizing even to jail
breaks from attack strategies the defender explicitly trained against, when the
backdoor trigger is present. Importantly, our threat model restricts adversaries
 to modify- ing only jailbreak samples (not benign data or labels), a constraint
 unexplored by prior work that makes the second objective particularly challeng-
 ing. We address this with Omission Attack, which exploits a new phenomenon: whe

n training on concept-absent unsafe samples, the classifier mis- associates that concept’s presence with the safe label. Both attacks flip nearly all target labels with only 1% poisoning rate. Code: https://github.com/DH-davidhuang/rapid-poison

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Wasserstein Geometry-Aware Adaptive Control via Meta-Learning

Xingyu Yang, Hanzhang Qu, Ye Cao, Jianfu Cao

Adaptive control of nonlinear systems under unknown disturbances requires learning algorithms aligned with the downstream control objective. While control-oriented meta-learning addresses the mismatch between regression-based identification and tracking performance, existing methods rely on Euclidean or static algebraic geometries that fail to capture the distributional structure of system uncertainties. We propose a framework that lifts adaptation into Wasserstein space, measuring parameter estimation errors as the optimal transport cost between estimated and true system behaviors. By constructing a Wasserstein Bregman divergence over representative task distributions, we use meta-learning to jointly optimize nonlinear feature representations, control gains, and transport geometry. This adaptation law learns an adaptation geometry that captures structural properties of the underlying physical system, implementing a physically grounded, data-driven attention mechanism. Closed-loop tracking simulations demonstrate that our controller achieves optimal performance on both fully-actuated and underactuated nonlinear planar rotorcraft, maintaining robustness under significant distributional shifts between training and testing conditions.

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TSP with Predictions: Heatmap to Tour with Provable Guarantees

Marek Elias, Fabrizio Grandoni, Adam Polak, Eleonora Vercesi

The Traveling Salesperson Problem (TSP) has long served as a benchmark for evaluating the strength of optimization techniques in the classical theory of algorithms. In recent efforts to apply ML to algorithmic problems, TSP has also become a natural testbed for the development of ML-based techniques. A common approach is to train a neural network to output a heatmap estimating the likelihood of each edge to be part of the optimal tour; however, converting such a heatmap into an actual tour remains a non-trivial and often computationally intensive step. In this work, we propose algorithms for transforming heatmaps into tours with theoretical guarantees linking the achieved approximation ratio to the quality of the provided heatmap. In the spirit of *algorithms with predictions*, our results can be described as $(1+2\eta/OPT)$-approximation algorithms, where η denotes the L1 distance between the prediction (heatmap) and an optimal solution (tour). Since the previous works lack such explicit guarantees, we compare our approach against them experimentally.

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Structure Enables Effective Self-Localization of Errors in LLMs

Ankur Samanta, Akshayaa Magesh, Ayush Jain, Kavosh Asadi, Youliang Yu, Daniel Jiang, Boris Vidolov, Kaveh Hassani, Paul Sajda, Jalaj Bhandari, Yonathan Efroni

Self-correction in language models remains elusive. In this work, we explore whether language models can explicitly localize errors in incorrect reasoning, as a path toward building AI systems that can effectively correct themselves. We introduce a prompting method that structures reasoning as discrete, semantically coherent thought steps, and show that models can localize errors more reliably within this structure than in conventional, unstructured chain-of-thought reasoning. Motivated by how the human brain monitors errors at discrete decision points and resamples alternatives, we introduce Iterative Correction Sampling of Thoughts (Thought-ICS), a self-correction framework. Thought-ICS iteratively prompts the model to generate reasoning one discrete and complete thought at a time—where each thought represents a deliberate decision by the model—creating natural boundaries for precise error localization. Upon verification, the model localizes the first erroneous step, and the system backtracks to generate alternative reasoning from the last correct point. When asked to correct reasoning verified as incorrect by an oracle, Thought-ICS achieves 20-40% self-correction lift. In a completely autonomous setting without external verification, it outperforms contempo

rary self-correction baselines.
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The Art of Interrogation: Consistency Amplifies Factuality in Spatial Reasoning
Théo Uscidda, Marta Tintore Gazulla, Maks Ovsjanikov, Federico Tombari, Leonidas Guibas
Current Large Reasoning Models (LRMs) exhibit remarkable general capabilities but significantly underperform in spatial reasoning tasks. Existing approaches treat this gap as a knowledge deficit, relying on supervised fine-tuning (SFT) to ingest labeled data from external vision sources or synthetic engines. In contrast, we argue that for many tasks, spatial capabilities are already present in pre-trained LRMs but require alignment through principles of internal logical coherence. In this work, we propose a self-supervised reinforcement learning (RL) framework that targets the internal Chain-of-Thought (CoT) process without requiring ground-truth annotations. By formalizing the notion of consistency verifiers—reward functions that check for geometric and semantic consistency under transformations like flipping or swapping the order of objects in the question—and optimizing them via our new OT-GRPO strategy, a minimal-consistency matching variant of group relative policy optimization, we demonstrate that models can self-correct their spatial logic. Our results show that this label-free consistency training approaches the accuracy of models trained with ground-truth supervision and achieves similar generalization across diverse tasks and domains.
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Streaming Sliced Optimal Transport
Khai Nguyen
Sliced optimal transport (SOT), or sliced Wasserstein (SW) distance, is widely recognized for its statistical and computational scalability. In this work, we further enhance computational scalability by proposing the first method for estimating SW from sample streams, called streaming sliced Wasserstein (Stream-SW). To define Stream-SW, we first introduce a streaming estimator of the one-dimensional Wasserstein distance (1DW). Since the 1DW has a closed-form expression, given by the integral of the absolute difference between the quantile functions of the compared distributions, we leverage quantile approximation techniques for sample streams to define a streaming 1DW estimator. By applying the streaming 1DW to all projections, we obtain Stream-SW. The key advantage of Stream-SW is its low memory complexity while providing theoretical guarantees on the approximation error. We demonstrate that Stream-SW achieves a more accurate approximation of SW than random subsampling, with lower memory consumption, when comparing Gaussian distributions and mixtures of Gaussians from streaming samples. Additionally, we conduct experiments on point cloud classification, point cloud gradient flows, and streaming change point detection to further highlight the favorable performance of the proposed Stream-SW.
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No Need to Train Your RDB Foundation Model
Linjie Xu, Yanlin Zhang, Quan Gan, Minjie Wang, David Wipf
Relational databases (RDBs) contain vast amounts of heterogeneous tabular information that can be exploited for predictive modeling purposes. But since the space of potential targets is vast across enterprise settings, how can we avoid retraining a new model each time we wish to predict a new quantity of interest? Foundation models based on in-context learning (ICL) offer a convenient option, but so far are largely restricted to single-table operability. In generalizing to multiple interrelated tables, it is essential to compress variably-sized RDB neighborhoods into fixed-length ICL samples for consumption by the decoder. However, the details here are critical: unlike existing supervised learning RDB pipelines, we provide theoretical and empirical evidence that ICL-specific compression should be constrained within high-dimensional RDB columns where all entities share units and roles, not across columns where the relevance of heterogeneous data types cannot be determined without extensive label information. Conditioned on this restriction, we then demonstrate that encoder expressiveness is actually not compromised by excluding trainable parameters. Hence we arrive at a principled family of RDB encoders that can be seamlessly paired with already-existing

single-table ICL foundation models, whereby no training or fine-tuning is required. From a practical standpoint, we develop scalable SQL primitives to implement the encoder stage, resulting in the easy-to-use open-source RDBLearn foundation model capable of robust performance on unseen datasets out of the box.
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Towards One-to-Many Temporal Grounding
Qi Xu, Tan Yue, Shihao Chen, Jiahao Meng, Anran Wang, Shunping Ji, Hao Fei, Xiangtai Li
Temporal Grounding (TG) aims to localize video segments corresponding to a textual query. Prior research predominantly focuses on single-segment retrieval. Real-world scenarios, however, often require localizing multiple disjoint segments for a single query—a setting we term **One-to-Many Temporal Grounding (OMTG)**. Previous state-of-the-art MLLMs, optimized for one-to-one settings, struggle in this context, often yielding near-zero scores due to a lack of event cardinality perception. To bridge this gap, we present a systematic solution with three key contributions. First, we establish the first comprehensive OMTG benchmark, introducing Count Accuracy (C-Acc) and Effective Temporal F1 (EtF1) as evaluation metrics. Second, we curate a high-quality OMTG dataset comprising 56k samples through a sophisticated construction pipeline. Third, we develop novel temporal and caption reward functions specifically designed for OMTG. In particular, the caption reward leverages Chain-of-Thought reasoning over dense video captions to explicitly guide policy optimization toward both preciseness and completeness. Extensive experiments show our model achieves a new state-of-the-art EtF1 of 43.65\% on OMTG Bench, outperforming Gemini 2.5 Pro and Seed-1.8 by 15.85\% and 15.61\%, respectively.
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Budget-Feasible Mechanisms for Submodular Welfare Maximization in Procurement Auctions
Shuang Cui, He Huang, Yu-e Sun, Chen Xue
Budget-feasible procurement auctions play a pivotal role in various AI-driven marketplaces, such as data acquisition and crowdsourcing, where a buyer with a limited budget seeks to procure services from strategic sellers with private costs. While numerous budget-feasible mechanisms have been proposed for the classic objective of maximizing the buyer's valuation, the more challenging and economically significant objective of social welfare maximization has only recently been studied, and existing approaches still sacrifice budget feasibility, thereby limiting their practical applicability. In this paper, we bridge this gap by proposing BFM-SWM, the first budget-feasible mechanism with provable approximation guarantees for submodular welfare maximization in procurement auctions. Our mechanism satisfies standard economic properties, including truthfulness, individual rationality, and non-negative auctioneer surplus. As a by-product, we develop BFM-VM, a variant tailored for valuation maximization, which achieves a deterministic approximation ratio of $1/(12+4\sqrt{3})$ for general submodular functions, substantially improving upon the best-known deterministic ratio of $1/64$ established by [Balkanski et al., SODA 2022], while reducing the running time from $\mathcal{O}(n^2\log n)$ to $\mathcal{O}(n\log n)$. Extensive experiments demonstrate the efficiency and effectiveness of our mechanisms.
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Demystifying Entropy Control in LLM RL Training: Theoretical Analysis and Dynamic Scheduling
Jingchu Gai, Guanning Zeng, Huaqing Zhang, Han Zhong, Yige Hong, Andrej Risteski, Aditi Raghunathan
This paper investigates a pivotal yet debated component of reinforcement learning (RL) for training large language models (LLMs): controlling entropy (increasing or decreasing it) during RL fine-tuning. The existing literature presents a dichotomy: some studies posit that increasing entropy facilitates exploration, whereas others argue that decreasing entropy enhances performance. To reconcile these conflicting observations, we provide a theoretical framework showing that the effect of entropy is governed by \emph{Entropy Discrepancy}, the distributional divergence between positive and negative samples. Guided by this insight, we de

rive a principled dynamic scheduling method that adaptively modulates the entrop
y coefficient, effectively switching between entropy maximization and minimizati
on as training evolves. Extensive experiments confirm the correlation between En
tropy Discrepancy and the efficacy of entropy control. Furthermore, our adaptive
 method yields substantial improvements, boosting Pass@K by 6.7\% on AIME24 and
17.52\% on puzzle tasks compared to vanilla RL, while consistently outperforming
 recent state-of-the-art reasoning methods.
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The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL
Yingru Li, Jiawei Xu, Ziniu Li, Jiacai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng,
 Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang
Reinforcement Learning (RL) for Large Language Models (LLMs) often suffers from
training collapse in long-horizon tasks due to exploding gradient variance. To m
itigate this, a baseline is commonly introduced for advantage computation; howev
er, traditional value models remain difficult to optimize, and standard group-ba
sed baselines overlook sequence heterogeneity. Although classic optimal baseline
 theory can achieve global variance reduction, it neglects token heterogeneity a
nd requires prohibitive gradient-based computation. In this work, we derive the
Optimal Token Baseline (OTB) from first principles, proving that gradient update
s should be weighted inversely to their cumulative gradient norm. To ensure effi
ciency, we propose the Logit-Gradient Proxy that approximates the gradient norm
using only forward-pass probabilities. Our method achieves training stability an
d matches the performance of large group sizes ($N=32$) with only $N=4$, reducin
g token consumption by over 65\% across single-turn and tool-integrated reasonin
g tasks.
****
DR^{2}Seg: Decomposed Two-Stage Rollouts for Efficient Reasoning Segmentation in
 Multimodal Large Language Models
Yulin He, Wei Chen, jian zhikang, Tianhang Guo, Wenjuan Zhou, Minglong Li, Shaow
u Yang, Wenjing Yang
Reasoning segmentation is an emerging vision-language task that requires reasoni
ng over intricate text queries to precisely segment objects. However, existing m
ethods typically suffer from overthinking, generating verbose reasoning chains t
hat interfere with object localization in multimodal large language models (MLLM
s). To address this issue, we propose DR^{2}Seg, a self-rewarding framework that
 improves both reasoning efficiency and segmentation accuracy without requiring
extra thinking supervision. DR^{2}Seg employs a two-stage rollout strategy that
decomposes reasoning segmentation into multimodal reasoning and referring segmen
tation. In the first stage, the model generates a self-contained description tha
t explicitly specifies the target object. In the second stage, this description
replaces the original complex query to verify its self-containment. Based on thi
s design, two self-rewards are introduced to mitigate overthinking and the assoc
iated attention dispersion. Extensive experiments conducted on 3B and 7B variant
s of Qwen2.5-VL, as well as on both SAM2 and SAM3, demonstrate that DR^{2}Seg co
nsistently improves reasoning efficiency and overall segmentation accuracy. The
source code can be found at https://github.com/harrylin-hyl/DR2Seg.
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Finite-time Convergence Analysis of Actor-Critic with Evolving Reward
Rui Hu, Yu Chen, Longbo Huang
Many popular practical reinforcement learning (RL) algorithms employ evolving re
ward functions—through techniques such as reward shaping, entropy regularization
, or curriculum learning—yet their theoretical foundations remain underdeveloped
. This paper provides the first finite-time convergence analysis of a single-tim
escale actor-critic algorithm in the presence of an evolving reward function und
er Markovian sampling. We consider a setting where the reward parameters may cha
nge at each time step, affecting both policy optimization and value estimation.
Under standard assumptions, we derive non-asymptotic bounds for both actor and c
ritic errors. Our result shows that an $O(1/\sqrt{T})$ convergence rate is achie
vable, matching the best-known rate for static rewards, provided the reward para
meters evolve slowly enough. This rate is preserved when the reward is updated v

ia a gradient-based rule with bounded gradient and on the same timescale as the actor and critic, offering a theoretical foundation for many popular RL techniques. As a secondary contribution, we introduce a novel analysis of distribution mismatch under Markovian sampling, improving the best-known rate by a factor of $\log^2T$ in the static-reward case.
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GradMem: Learning to Write Context into Memory with Test-Time Gradient Descent
Yuri Kuratov, Matvey Kairov, Aydar Bulatov, Ivan Rodkin, Mikhail Burtsev
Many large language model applications require conditioning on long contexts.
Transformers typically support this by storing a large per-layer KV-cache of past activations, which incurs substantial memory overhead.
A desirable alternative is compressive memory: read a context once, store it in a compact state, and answer many queries from that state.
We study this in a context removal setting, where the model must generate an answer without access to the original context at inference time.
We introduce GradMem, which writes context into memory via per-sample test-time optimization.
Given a context, GradMem performs a few steps of gradient descent on a small set of prefix memory tokens while keeping model weights frozen.
GradMem explicitly optimizes a model-level self-supervised context reconstruction loss, resulting in a loss-driven write operation with iterative error correction, unlike forward-only methods.
On associative key--value retrieval, GradMem outperforms forward-only memory writers with the same memory size, and additional gradient steps scale capacity much more effectively than repeated forward writes.
We further show that GradMem transfers beyond synthetic benchmarks: with pretrained language models, it attains competitive results on natural language tasks including bAbI and SQuAD variants, relying only on information encoded in memory.
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LASER: Learning Active Sensing for Continuum Field Reconstruction
Huayu Deng, Jinghui Zhong, Xiangming Zhu, Yunbo Wang, Xiaokang Yang
High-fidelity measurements of continuum physical fields are essential for scientific discovery and engineering design but remain challenging under sparse and constrained sensing. Conventional reconstruction methods typically rely on fixed sensor layouts, which cannot adapt to evolving physical states. We propose LASER, a unified, closed-loop framework that formulates active sensing as a Partially Observable Markov Decision Process (POMDP). At its core, LASER employs a continuum field latent world model that captures the underlying physical dynamics and provides intrinsic reward feedback. This enables a reinforcement learning policy to simulate ''what-if'' sensing scenarios within a latent imagination space. By conditioning sensor movements on predicted latent states, LASER navigates toward potentially high-information regions beyond current observations. Our experiments demonstrate that LASER consistently outperforms static and offline-optimized strategies, achieving high-fidelity reconstruction under sparsity across diverse continuum fields.
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SPADA: A Verifiable Test-Driven Agent for Controllable Parametric CAD Assembly Generation
Keyou Zheng, Xuyang Su, Jiewu Leng
Parametric CAD is widely used in mechanical and product engineering, but current generative models still have difficulty producing assemblies that are both editable at the parameter level and consistent with inter-part constraints.
Methods that generate meshes or history-free B-rep can represent multi-part shape, but they often lack the program structure and constraint logic needed for reliable downstream edits; in contrast, code-based CAD generation offers direct parametric control, yet most published settings and evaluations focus on single-part solids rather than constrained assemblies.
We introduce SPADA (Self-testing Parametric Assembly Design Agent), a test-driven agent that synthesizes assembly code together with deterministic verification tests, and uses these tests as an executable contract for controllable generatio

n.
SPADA runs an iterative compile-test-repair loop with multimodal feedback, checking both specification logic and physical feasibility through programmatic constraints.
To support evaluation, we release SPADA-Bench-Verified, a human-validated benchmark of real-world code-centric assemblies paired with deterministic tests and engineering-style constraints.
Experiments show that SPADA could produces complex assemblies while maintaining geometric fidelity, supporting test-driven agents as a concrete path toward reliable, controllable CAD generation.
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Locally Coherent Parallel Decoding in Diffusion Language Models
Michael Hersche, Nicolas Menet, Ronan Tanios, Abbas Rahimi
Diffusion language models (DLMs) have emerged as a promising alternative to autoregressive (AR) models, offering sub-linear generation latency and bidirectional capabilities that are particularly appealing for code generation and editing. Achieving sub-linear latency in discrete DLMs requires predicting multiple tokens in parallel. However, standard DLMs sample tokens independently from conditional marginal distributions, failing to capture the joint dependencies among concurrently generated tokens. As a result, they often lead to syntactic inconsistencies and break multi-token structures. In this work, we introduce CoDiLA (Coherent Diffusion with Local Autoregression), a method that reconciles parallel sampling with local dependency modeling. Rather than forcing the DLM to resolve fine-grained syntax, CoDiLA delegates local decoding to a small, auxiliary AR model operating on the diffusion latents. This design allows for parallel generation while ensuring sequential validity within a block and maintaining core DLM capabilities, including bidirectional modeling across blocks. We demonstrate that using a highly compact auxiliary AR model (e.g., 0.6B parameters) effectively eliminates coherence artifacts, establishing a new Pareto frontier for accuracy and speed in code generation benchmarks.
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Stochastic Lifting for Generating Trajectories of Stochastic Physical Systems
Jules Berman, Tobias Blickhan, Benjamin Peherstorfer
Many stochastic physical systems evolve smoothly over time in the sense that the distribution of states changes regularly with time. The precise transition from current to next state is often modeled as the interplay of a smooth map and an explicit source of randomness. Stochastic Lifting leverages this premise by attaching an independent, high-dimensional random label to each state transition in the training data and fitting a transition map from the current state and label to the next state using a standard regression loss. The labels act as auxiliary coordinates that let the model represent multiple plausible outcomes for similar current states, avoiding collapse to a mean prediction in the finite-sample size regime. At inference, drawing fresh labels and rolling the map forward generates diverse trajectories with a single network evaluation per time step, with the smoothness bias of the learned map supporting accurate sampling in practice.
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T-GINEE: A Tensor-Based Multi-Graph Representation Learning
Maolin Wang, ZiTing Mai, Xuhui Chen, Zhiqi Li, Tianshuo Wei, Yutian Xiao, Wenlin Zhang, Wanyu Wang, Ruocheng Guo, Haoxuan Li, Zenglin Xu, Xiangyu Zhao
While traditional network analysis focuses on single-layer networks, real-world systems often form multilayer networks with multiple relationship types. However, existing methods typically fail to capture complex inter-layer dependencies by treating layers independently or aggregating them. To address this, we propose T-GINEE (Tensor-Based Generalized Multilayer-graph Estimating Equation), a statistical regularization framework combining tensor-based generalized estimating equations with task-specific loss to model cross-network correlations explicitly. Key innovations include: (1) CP tensor decomposition capturing structural dependencies via shared latent factors; (2) a generalized estimating equation framework modeling inter-layer correlations through working covariance matrices; and (3) a flexible link function accommodating characteristics like sparsity. Our theor

etical analysis establishes consistency and asymptotic normality under mild cond
itions. Extensive experiments on synthetic and real-world datasets validate T-GI
NEE’s effectiveness for multilayer network analysis. Our code is available in th
e \textbf{supplementary materials} to ensure reproducibility.
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ReJump: A Tree-Jump Representation for Analyzing and Improving LLM Reasoning
Yuchen Zeng, Shuibai Zhang, Wonjun Kang, Shutong Wu, Lynnix Zou, Ying Fan, Heeju
 Kim, Ziqian Lin, Jungtaek Kim, Hyung Il Koo, Dimitris Papailiopoulos, Kangwook
Lee
Large Reasoning Models (LRMs) are Large Language Models (LLMs) explicitly traine
d to generate long-form Chain-of-Thoughts (CoTs), achieving impressive success o
n challenging tasks like math and programming. However, their underlying reasoni
ng "algorithms" remain poorly understood. To investigate this, we propose *ReJum
p*, which represents a reasoning trace as a visitation order over nodes in a tre
e of intermediate problem-solving steps. Transitions between nodes, which we ter
m *jumps*, include adjacent moves that capture behaviors such as calculation, an
d non-adjacent moves that capture behaviors such as backtracking and verificatio
n. ReJump enables analyzing LLM reasoning with diverse metrics that quantify exp
loration, exploitation, overthinking, forgetting, and verification. Using our pr
oposed LLM agent to extract reasoning traces into ReJump format, we evaluate sta
te-of-the-art LRMs on two tasks and find that models with similar accuracy can e
xhibit distinct reasoning behaviors, while different tasks favor different reaso
ning styles (e.g., varying balance between exploration and exploitation). To fur
ther understand how learning strategies shape reasoning, we use ReJump to compar
e distilled LRMs with their teachers, compare CoT-prompted LLMs with LRMs, and e
xamine how reinforcement learning affects reasoning behavior. Finally, we show t
hat ReJump can improve reasoning quality at test time through strategies such as
 ReJump-guided Best-of-N selection and prompt selection. Our code is available a
t https://github.com/UW-Madison-Lee-Lab/ReJump.
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Functional Cache Grafting: Robust and Rapid Code-Policy Synthesis for Embodied A
gents
Saehun Chun, Wonje Choi, Sera Choi, Sanghyun Ahn, Honguk Woo
Code-writing large language models (CodeLLMs) generate executable code policies
for embodied agents by translating natural language goals and environmental cons
traints into structured control programs. However, policy generation in open-dom
ain embodied environments suffers from two fundamental limitations:
(i) delayed decoding caused by repetitive prefill computation over long prompts,
 and
(ii) limited robustness due to fully generative decoding, which often produces A
PI mismatches, missing safety guards, and unstable control logic. To address the
se limitations, we present FCGraft, a Functional Cache Grafting framework.
FCGraft maintains a library of function-level validated code skeletons and their
 associated prompt-level Transformer key–value (KV) caches, and synthesizes new
policies by retrieving relevant functions and grafting their KV caches when a ne
w task is provided. Given retrieved function caches, FCGraft performs cache graf
ting via stitching, which composes cached function segments into a composite pol
icy, and patching, which locally adapts only the necessary code regions to satis
fy task-specific parameters and constraints with minimal additional decoding. By
 eliminating redundant prefill computation, this approach reduces generation lat
ency, while reusing validated control structures improves robustness over prompt
-level caching methods RAGCache, achieving $18.31\$% higher task success rate an
d $2.3\times$ faster policy synthesis.
****
Capacity-Agnostic Parameter Isolation for Continual Graph Learning
Ye Xiao, Ruikun Li, Zhenyu Yang, Andrey Vasnev, Junbin Gao
Existing parameter isolation-based continual learning methods employ diverse des
igns to accommodate more tasks within limited model capacity, but often incur in
creasing computational overhead as model capacity expands for growing task strea
ms. To address this efficiency bottleneck, we propose CAGNN, a graph continual l

earning framework with a biological neuron-inspired architecture that features capacity-agnostic efficiency. CAGNN leverages graph contextual information to construct task-specific subnetworks and decouples them during training and inference, reducing full-network propagation overhead while enabling knowledge transfer across tasks. Extensive experiments demonstrate CAGNN's superior effectiveness and computational efficiency over state-of-the-art methods.
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SpecMD: A Comprehensive Study On Speculative Expert Prefetching
Duc N.M Hoang, Mohammad Samragh, AJAY KUMAR JAISWAL, Minsik Cho
Mixture-of-Experts (MoE) models enable sparse expert activation, meaning that only a subset of the model’s parameters is used during each inference. However, to translate this sparsity into practical performance, an expert caching mechanism is required.
Previous works have proposed hardware-centric caching policies, but how these various caching policies interact with each other and different hardware specification remains poorly understood. To address this gap, we develop \textbf{SpecMD}, a standardized framework for benchmarking ad-hoc cache policies on various hardware configurations. Using SpecMD, we perform an exhaustive benchmarking of several MoE caching strategies, reproducing and extending prior approaches in controlled settings with realistic constraints. Our experiments reveal that MoE expert access is not consistent with temporal locality assumptions (e.g LRU, LFU). Motivated by this observation, we propose \textbf{Least-Stale}, a novel eviction policy that exploits MoE's predictable expert access patterns to reduce collision misses by up to $85\times$ over LRU. With such gains, we achieve over 88% hit rates with up to 34.7% Time-to-first-token (TTFT) reduction on OLMoE at only 5% or $0.6GB$ of VRAM cache capacity.
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Toward Culturally Aligned LLMs through Ontology-Guided Multi-Agent Reasoning
Wonduk Seo, Wonseok Choi, Junseo Koh, JuHyeon Lee, Hyunjin An, Minhyeong Yu, Jian Park, Qingshan Zhou, Seunghyun lee, Yi Bu
Large Language Models (LLMs) increasingly support culturally sensitive decision making, yet often exhibit misalignment due to skewed pretraining data and the absence of structured value representations. Existing methods can steer outputs, but often lack demographic grounding and treat values as independent, unstructured signals, reducing consistency and interpretability. We propose OG-MAR, an Ontology-Guided Multi-Agent Reasoning framework. OG-MAR summarizes respondent-specific values from the World Values Survey (WVS) and constructs a global cultural ontology by eliciting relations over a fixed taxonomy via competency questions. At inference time, it retrieves ontology-consistent relations and demographically similar profiles to instantiate multiple value-persona agents, whose outputs are synthesized by a judgment agent that enforces ontology consistency and demographic proximity. Experiments on regional social-survey benchmarks across four LLM backbones show that OG-MAR improves cultural alignment and robustness over competitive baselines, while producing more transparent reasoning traces.
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Return of Frustratingly Easy Unsupervised Video Domain Adaptation
Pengfei Wei, Yiqun Sun, zhiqiang xu, Yiping Ke, Lawrence B. Hsieh
Unsupervised video domain adaptation (UVDA) is a practical but under-explored problem.
In this paper, we propose a frustratingly easy UVDA method, called \emph{MetaTrans}.
Specifically, \emph{MetaTrans} adopts a concise learning objective that contains only two fundamental loss terms.
Despite the simplicity of the learning objective, \emph{MetaTrans} embodies an advanced UVDA idea, that is, handling the spatial and temporal divergence of cross-domain videos separately, through a subtle model architecture design.
By implementing a temporal-static subtraction module, \emph{MetaTrans} effectively removes spatial and temporal divergence.
Extensive empirical evaluations, particularly on various cross-domain action recognition tasks, show substantial absolute adaptation performance enhancement and

significantly superior relative performance gain compared with state-of-the-art UVDA baselines.
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On Path to Multimodal Historical Reasoning: HistBench and HistAgent
Jiahao Qiu, Fulian Xiao, Yimin Wang, Yuchen Mao, Yijia Chen, Xinzhe Juan, Siran Wang, Xuan Qi, Tongcheng Zhang, Zixin Yao, Jiacheng Guo, Yifu Lu, Charles Argon, Jundi Cui, Daixin Chen, Junran Zhou, Shuyao Zhou, Zhanpeng Zhou, Ling Yang, Shilong Liu, Hongru WANG, Kaixuan Huang, xun jiang, Xi Gao, Mengdi Wang
Recent advances in large language models (LLMs) have led to remarkable progress across various domains, yet their capabilities in the humanities, particularly history, remain underexplored. Historical reasoning poses unique challenges for LLMs, involving multimodal source interpretation, temporal inference, and cross-linguistic analysis. Existing general-purpose agents perform well on many current benchmarks but lack the domain expertise needed to address complex historical questions.
To address this gap, we introduce HistBench, a new benchmark of 414 high-quality and carefully-reviewed questions stratified by difficulty and designed to evaluate LLM's capacity for historical reasoning. The tasks span a wide range of historical problems—from factual retrieval based on primary sources to interpretive analysis of manuscripts and images, to interdisciplinary challenges involving archaeology, linguistics, or cultural history. Furthermore, the benchmark dataset spans 29 ancient and modern languages and covers a wide range of historical periods and world regions. Finding the poor performance of LLMs and other agents on HistBench, we further present HistAgent, a history-specific agent equipped with carefully designed tools for OCR, translation, archival search, and image understanding in history. On HistBench, HistAgent based on GPT-4o achieves an accuracy of 28.50\% pass@1 and 36.47\% pass@2, significantly outperforming LLMs with online search and generalist agents, including GPT-4o (18.60\%), DeepSeek-R1 (14.98\%), Grok 3 (17.63\%) and Open Deep Research by smolagents (19.57\% pass@1 and 25.12\% pass@2). These results highlight the limitations of existing LLMs and generalist agents and demonstrate the advantages of HistAgent for historical reasoning. Notably, HistAgent also achieves 60.00\% pass@1 accuracy on the GAIA benchmark, showing that domain-specific customization doesn't hinder HistAgent's competitive performance on real-world general tasks. Code is available at https://github.com/CharlesQ9/HistAgent.
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Beyond Theorem Proving: Formulation, Framework and Benchmark for Formal Problem-Solving
Qi Liu, Xinhao Zheng, Renqiu Xia, Xingzhi Qi, Qinxiang Cao, Junchi Yan
Large language models (LLMs) have achieved remarkable progress in mathematical reasoning, yet persistently suffer from hallucinations and erroneous logic. While formal theorem proving (FTP) shows promise in process-level reliability, it is limited to _verification_ (checking known propositions). This leaves constructive problem-solving (finding unknown terms that satisfy specific conditions) underexplored and disconnected from process-level verifiability.
To bridge this gap, we introduce **FPS** (_**F**ormal **P**roblem-**S**olving_), a principled framework to encompass the end-to-end problem-solving process in Lean 4. In FPS, the answer is an unknown metavariable coupled with a proof obligation, forcing it to be mathematically derived and verified. We further present **D-FPS** (_**D**eductive **FPS**_), which structures solving into forward derivation and backward verification, aligning more closely with human reasoning steps.
Three benchmarks of over 1,000 problems are constructed for evaluation: **FormalMath500**, **MiniF2F-Solving**, and **PutnamBench-Solving**. We further propose **RPE** (_**R**estricted **P**ropositional **E**quivalence_), a symbolic metric that evaluates semantic correctness beyond brittle string matching. Extensive experiments with state-of-the-art provers reveal that solving is significantly harder than proving, highlighting the ``alignment tax'' required to transition from loose validity checking to constructive, human-aligned reasoning.
Code and data are available at [https://github.com/Purewhite2019/formal_problem_

solving_main](https://github.com/Purewhite2019/formal_problem_solving_main).
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AgentSteerTTS: A Multi-Agent Closed-Loop Framework for Composite-Instruction Tex
t-to-Speech
Bin Kang, Shaoguo Wen, Yang Fan, Shunlong Wu, Junjie Wang, Yulin Li, Junzhi Zhao
, Junle Wang, Zhuotao Tian
While existing text-to-speech (TTS) models exhibit high expressiveness, fine-gra
ined control over composite instructions remains challenging due to the structur
al mismatch between discrete textual intents and continuous acoustic realization
s. Inspired by human cognitive decoupling, we introduce AgentSteerTTS, a multi-a
gent closed-loop framework designed for intent-faithful expressive control of co
mposite instructions. First, in our framework, an adversarial disentanglement mo
dule mitigates speaker-emotion leakage through gradient reversal and cross-covar
iance regularization. Next, a Dual-Stream Anchoring Controller grounds abstract
intents using a large-scale acoustic prototype library: a Retrieval Agent select
s expressive anchors, while a Synthesis Agent fuses them into continuous control
 vectors via gated attention. Finally, a Fast–Slow Feedback Agent refines output
 intensity through latent gradient correction and resolves semantic–acoustic mis
matches using high-level perceptual critique. Experiments on a composite-instruc
tion benchmark and public test sets show that AgentSteerTTS yields consistent an
d significant improvements to the baselines, demonstrating the effectiveness of
the proposed method. Our code and models will be released.
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VIBE: Disentangling Social Dynamics via Kinematics-Informed Variational Inferenc
e for Behavioral Emotion
Abhishek Pratap Singh, Vaibhav Pratap Singh, Deepak Kumar, Balasubramanian Raman
Group Emotion Recognition (GER) is crucial for understanding social dynamics, ra
nging from interpreting intimate conversations to evaluating crowd behavior in l
arge-scale surveillance scenarios. While current AI models can analyze these sce
nes, they often act as black boxes that take shortcuts. Instead of focusing on h
ow people are actually behaving, these models often get distracted by the backgr
ound environment, leading to inaccurate results. To bridge this gap, we introduc
e VIBE (Variational Inference for Behavioral Emotion), a kinematics-aware framew
ork that integrates audio, video, and text modalities through causal structuring
. Unlike standard models that simply mix data together, VIBE utilizes mathematic
al constraints to filter out background noise and isolate the genuine emotions o
f the people involved. This purified representation enables our model to focus e
xclusively on the sociological mechanics of the crowd, dynamically modulating ne
ural attention based on raw physical synchrony. Simultaneously, we align visual
dynamics with human interpretability by projecting latent representations into a
 semantically structured space informed by textual descriptions. Comprehensive e
xperiments demonstrate that VIBE consistently outperforms state-of-the-art metho
ds. Code is available at GitHub.
****
Identifiable Smooth Conjugacy Learning via Adversarial Orthogonality
In Huh, Changwook Jeong, Muhammad Alam
Data-driven dynamical system models often fail to recover the long-term structur
e of the underlying system, as their behavior is weakly constrained off the data
 manifold. Conjugacy-based approaches address this limitation by learning a diff
eomorphism that pushes forward a source vector field to match observed dynamics,
 inheriting qualitative topology from the source. However, such methods typicall
y presuppose that the chosen source system is topologically compatible with the
target data. When this assumption is violated, the conjugacy problem becomes ill
-posed, and arbitrary corrections can be traded off against diffeomorphic variat
ion, leading to non-identifiability. We propose a framework that relaxes this as
sumed prior by jointly learning the diffeomorphic conjugacy together with contro
lled adjustments to the source dynamics via low-dimensional context modulation.
Inspired by versal unfolding theory, we enforce the modulation space to be ortho
gonal to the worst-case orbit-tangent directions, obtained by adversarially sear
ching over a class of parameterized diffeomorphisms. This promotes an identifiab

le decomposition of dynamical variation into diffeomorphic and intrinsic, topolo gy-changing components, enabling interpretable corrections that recover the cano nical structure such as normal forms and symmetries.
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Multi-Label Learning with Contrastive Cluster Self-Supervision for 3D Hierarchic al Semantic Segmentation

Shuyu Cao, Chongshou Li, Jie Xu, Tianrui Li, Na Zhao

3D hierarchical semantic segmentation (3DHS) is crucial for embodied intelligenc e that demands the coarse-to-fine grained and multi-hierarchy understanding of 3 D scenes. 3DHS tasks can be addressed by multi-label learning, but facing two is sues: I) learning multiple labels for each point with a shared model can lead to multi-hierarchy conflicts in cross-hierarchy optimization, and II) the class im balance issue is inevitable across multiple hierarchies of 3D scenes, making the model easily be dominated by major classes. To address these issues, we propose a novel multi-label learning with contrastive cluster self-supervision framewor k for 3DHS. Specifically, we propose a late-decoupled multi-label learning 3DHS network which employs decoupled decoders with the coarse-to-fine hierarchical co nsistency guidance. This late-decoupled model architecture can mitigate the unde rfitting and overfitting conflicts among multiple hierarchies and also constrain the class imbalance problem within each individual hierarchy. Moreover, we intr oduce a 3DHS-oriented contrastive cluster self-supervision learning method, whic h learns cluster-wise point cloud features with contrastive loss and produces se lf-supervised information to enhance the class-imbalance segmentation. Extensive experiments on multiple datasets and backbones demonstrate that our approach pr omotes the multi-hierarchy balance and mitigates the class imbalance issue in 3D HS tasks.
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OmniVideo-R1: Reinforcing Audio-visual Reasoning with Query Intention and Modali ty Attention

Zhangquan Chen, Jiale Tao, Ruihuang Li, Yihao Hu, Ruitao Chen, Zhantao Yang, Xin lei Yu, Haodong Jing, Manyuan Zhang, Shuai Shao, Biao Wang, Qinglin Lu, Ruqi Hua ng

While humans perceive the world through diverse modalities that operate synergis tically to support a holistic understanding of their surroundings, existing omni video models still face substantial challenges on audio-visual understanding tas ks. In this paper, we propose OmniVideo-R1, a novel reinforced framework that im proves mixed-modality reasoning. OmniVideo-R1 empowers models to "think with om nimodal cues" by two key strategies: (1) query■intensive grounding based on self ■supervised learning paradigms; and (2) modality■attentive fusion built upon con trastive learning paradigms. Extensive experiments on multiple benchmarks demons trate that OmniVideo-R1 consistently outperforms strong baselines, highlighting its effectiveness and robust generalization capabilities.
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SymSpectra: Symmetric Information Bottleneck Framework for Molecular Structure R ecognition under Imbalanced Settings

Xiaohan Qin, Wenjie Du, Yang Wang

Identifying molecular structures from spectral data is essential for early-stage chemical analysis, yet it remains a difficult task due to severe functional gro up imbalance and complex inter-group dependencies, which often cause existing me thods to overfit frequent groups while underperforming on rare ones. To address these issues, we present **SymSpectra**, a **Sym**metric Conditional Information Bottleneck (SCIB) framework designed to seamlessly integrate multi-modal **Spec tra** features. Our model employs the SCIB framework to fuse multi-modal spectro scopic data into a unified representation, effectively preserving discriminative signals while mitigating redundancy. To enhance robustness against data imbalan ce, we incorporate conditional mutual information into the training objective, i ncreasing the model’s sensitivity to rare functional groups and challenging mole cular cases. Additionally, a specialized module captures the dependencies among functional groups, improving both prediction accuracy and chemically meaningful interpretability. Experiments on multimodal spectral datasets show that SymSpect

ra outperforms state-of-the-art methods, achieving an F1-score of 0.970 in substructure classification and demonstrating strong robustness under various imbalance settings.
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On the Relationship Between Activation Outliers and Feature Death in Sparse Autoencoders
Elana Simon, Etowah Adams, James Zou
Sparse autoencoders (SAEs) decompose neural network activations into interpretable features, but many learned features never activate, a problem called feature death that wastes dictionary capacity and can reintroduce superposition. Death rates vary dramatically between models: near-zero on GPT-2, over 70\% on AlphaFold3 with identical configurations. We find that dimension-level activation outliers (dimensions whose mean magnitude is large relative to per-token variation) cause this by shifting pre-activations at initialization based on each feature's alignment with the activation mean. Features anti-aligned with the mean receive permanently negative pre-activations and never fire. We formalize outlier severity as $\gamma = \|\boldsymbol{\mu}\|/\|\boldsymbol{\sigma}\|$; it predicts initial death rates (Spearman $\rho = 0.89$ for dead-by-TopK, 0.82 for dead-by-ReLU) across 454 model-layer combinations spanning language, vision, protein, and genomic models. Dead features can revive during training, but recovery requires the SAE bias to learn the activation mean, a process that is prohibitively slow at high γ. Mean-centering (subtracting the activation mean) sidesteps this and eliminates outlier-induced death across all tested models, confirming the mechanism and providing a principled basis for when and why this preprocessing step is necessary.
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Axiomatic Atlas: A Prescriptive Framework for Neural Architecture Design
Minghao Guo, Wojciech Matusik
Neural architecture design lacks first principles: innovations are discovered empirically and justified post-hoc, with no systematic way to diagnose *why* an architecture fails or derive *what* repair will succeed.
We introduce the *Axiomatic Atlas*, encoding requirements as composable axioms over graph connectivity, operator contracts, numerical stability, and information preservation.
Given an operator library and wiring conventions, the Atlas constructs certificates lower-bounding output variation via min-cut analysis and diagnoses failures by locating axiom violations.
Crucially, the framework is prescriptive: each violation implies a targeted repair, reducing architecture design to constraint satisfaction.
We prove variation bounds under exact and finite-precision arithmetic, enabling modular verification across transformers, MoEs, SSMs, and GNNs.
Four Atlas-derived interventions validate the approach: +46 percentage points on GNN bottlenecks, $3\times$ robustness to MoE quantization, 83\% gap closure with adaptive expert budgets, and 0\%$\to$100\% retrieval via orthogonal keys---each against matched negative controls.
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SorryDB: Can AI Provers Complete Real-World Lean Theorems?
Austin Letson, Leopoldo Sarra, Auguste Poiroux, Oliver Dressler, Paul Lezeau, Dhyan Aranha, Frederick Pu, Aaron Hill, Miguel Corredera Hidalgo, Julian Berman, George Tsoukalas, Lenny Taelman
We present SorryDB, a dynamically-updating benchmark of open Lean tasks drawn from 78 real world formalization projects on GitHub. Unlike existing static benchmarks, often composed of competition problems, hillclimbing the SorryDB benchmark will yield tools that are aligned to the community needs, more usable by mathematicians, and more capable of understanding complex dependencies. Moreover, by providing a continuously updated stream of tasks, SorryDB mitigates test-set contamination and offers a robust metric for an agent's ability to contribute to novel formal mathematics projects. We evaluate a collection of approaches, including generalist large language models, agentic approaches, and specialized symbolic provers, over a selected snapshot of 1000 tasks from SorryDB. We show that cur

rent approaches are complementary: even though an agentic approach based on Gemi ni Flash is the most performant, it is not strictly better than other off-the-sh elf large-language models, specialized provers, or even a curated list of tactic s.
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Towards A Generative Protein Evolution Machine with DPLM-Evo
Xinyou Wang, Liang Hong, Jiasheng Ye, Zaixiang Zheng, Shujian Huang, Quanquan Gu
Proteins are shaped by gradual evolution under biophysical and functional constr aints. Protein language models learn rich evolutionary constraints from large-sc ale sequences, and discrete diffusion-based protein language models (e.g., DPLMs) are promising for both understanding and generation. However, existing DPLMs t ypically rely on masked diffusion that contradicts a simple biological intuition : proteins evolve through accumulated edits, not by emerging from masks. Consequ ently, these frameworks lack explicit pretraining objectives for substitution an d insertion/deletion (indel) operations, limiting both optimization-style post-e diting and flexible guided generation. To address these limitations, we present DPLM-Evo, an evolutionary discrete diffusion framework that explicitly predicts substitution, insertion, and deletion operations during denoising. DPLM-Evo deco uples an upsampled-length latent alignment space from the variable-length observ ed sequence space, which makes indel-aware generation tractable. To better align substitutions with real evolution, we further introduce a contextualized evolut ionary noising kernel that produces biologically informed, context-dependent mut ation patterns. Across tasks, DPLM-Evo improves sequence understanding and achie ves state-of-the-art mutation effect prediction performance on ProteinGym in the single-sequence setting. It also enables variable-length simulated evolution, a nd post-editing/optimization of existing proteins via explicit edit trajectories .
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Anchor-guided Hypergraph Condensation with Dual-level Discrimination
Fan Li, Xiaoyang Wang, Chen Chen, Wenjie Zhang
The increasing prevalence of large-scale hypergraphs poses significant computati onal challenges for hypergraph neural network (HNN) training. To address this, h ypergraph condensation (HGC) distills large real hypergraphs into compact yet in formative synthetic ones, beyond graph condensation (GC) methods limited to pair wise relations.
However, existing HGC methods rely on decoupled training architectures, where st ructure generators are pre-trained on the original hypergraph but not jointly op timized with condensed features during refinement, resulting in misaligned struc tures that degrade downstream utility. Moreover, trajectory-based optimization i ncurs substantial computational overhead in refinement, limiting condensation ef ficiency.
To tackle these issues, we propose \textbf{A}nchor-guided \textbf{H}yper\textbf{ G}raph \textbf{C}ondensation with \textbf{D}ual-level \textbf{D}iscrimination (\ textbf{AHGCDD}), which consists of three key components: (1) a node initializati on module based on Heat Kernel PageRank (HKPR) to encode structural knowledge in to feature semantics; (2) an anchor-guided hyperedge synthesis strategy for join t optimization of condensed features and structure; (3) a theoretically grounded dual-level discrimination objective for utility-preserving condensation without redundant HNN training. Extensive experiments demonstrate the superior effectiv eness and efficiency of AHGCDD.
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Best of Both Worlds: Multimodal Reasoning and Generation via Unified Discrete Fl ow Matching
Onkar Kishor Susladkar, Tushar Prakash, Gayatri Deshmukh, Kiet A. Nguyen, Jiaxun Zhang, Adheesh Sunil Juvekar, Tianshu Bao, Lin Chai, Sparsh Mittal, Inderjit S Dhillon, Ismini Lourentzou
We propose UniDFlow, a unified discrete flow-matching framework for multimodal u nderstanding, generation, and editing. It decouples understanding and generation via task-specific low-rank adapters, avoiding objective interference and repres entation entanglement, while a novel reference-based multimodal preference align

ment optimizes relative outcomes under identical conditioning, improving faithfu lness and controllability without large-scale retraining. UniDFlow achieves SOTA performance across eight benchmarks and exhibits strong zero-shot generalizatio n to tasks including inpainting, in-context image generation, reference-based ed iting, and compositional generation, despite no explicit task-specific training.
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MAC-NeRF: Motion-Aware Curriculum Learning for Dynamic LiDAR NeRFs
Shangshu Yu, Xiaotian Sun, Wen Li, Rui She, Hanyun Wang, Sheng Ao, Chenglu Wen, Cheng Wang
While LiDAR NeRFs excel in static environments, synthesizing dynamic scenes rema ins challenging as moving objects break multi-view consistency, causing conflict ing supervision and ghosting artifacts across frames.
Existing methods typically suffer from optimization difficulty from the start, s truggling to disentangle valid geometry from motion noise when initial motion pr iors are unreliable.
To address this, we propose MAC-NeRF, a novel LiDAR NeRF framework enhanced by m otion-aware curriculum learning for high-fidelity dynamic scene synthesis.
First, we propose Rectified Temporal Consistency to resolve motion-induced super vision conflicts.
By filtering out erroneous supervision via forward-backward geometric verificati on, it creates a curriculum that prioritizes trustworthy temporal correspondence s before handling challenging motions.
Second, we propose Confidence-Modulated Frequency Regularization (CMFR) to elimi nate geometric ambiguity.
It adaptively modulates the frequency regularization bandwidth, progressively tr ansitioning from strict low-frequency constraints for artifact suppression to fu ll-spectrum modeling for fine-grained detail preservation.
Extensive experiments on KITTI-360 and nuScenes demonstrate that MAC-NeRF signif icantly outperforms state-of-the-art methods in rendering quality.
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Privately Fine-Tuned LLMs Preserve Temporal Dynamics in Tabular Data
Lucas Rosenblatt, Peihan Liu, Ryan McKenna, Natalia Ponomareva
Research on differentially private synthetic tabular data has largely focused on independent and identically distributed rows where each record corresponds to a unique individual. This perspective neglects the temporal complexity in longitu dinal datasets, such as electronic health records, where a user contributes an e ntire (sub) table of sequential events. While practitioners might attempt to mod el such data by flattening user histories into high-dimensional vectors for use with standard marginal-based mechanisms, we demonstrate that this strategy is in sufficient. Flattening fails to preserve temporal coherence even when it maintai ns valid marginal distributions. We introduce PATH, a novel generative framework that treats the full table as the unit of synthesis and leverages the autoregre ssive capabilities of privately fine-tuned large language models. Extensive eval uations show that PATH effectively captures long-range dependencies that traditi onal methods miss. Empirically, our method reduces the distributional distance t o real trajectories by over 60% and reduces state transition errors by nearly 50 % compared to leading marginal mechanisms while achieving similar marginal fidel ity.
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Geometrically Constrained Stenosis Editing in Coronary Angiography via Entropic Optimal Transport
Jialin Li, Zhuo Zhang, Cao Yue, Guipeng Lan, Jiabao Wen, Shuai Xiao, Jiachen Yan g
The scarcity of high-quality imaging data for coronary angiography (CAG) stenosi s limits the clinical translation of automated stenosis detection. Synthetic ste nosis data provides a practical avenue to augment training sets, improving data quality, diversity, and distributional coverage, and enhancing detection precisi on and generalization. However, diffusion-based editing commonly relies on soft guidance in a noise-initialized reverse process, offering limited pixel-level pr ecision and structure preservation. We propose the **OT-Bridge Editor**, which r

eframes localized editing as a constrained entropic optimal transport (OT) probl em and leverages geometric information to steer the generation path, enabling st ronger geometric control. Extensive experiments show that our synthesized angiog rams consistently improve downstream stenosis detection, yielding substantial re lative gains of 27.8% on the public ARCADE benchmark and 23.0% on our multi-cent er dataset, supported by consistent qualitative results.
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Names Don’t Matter: Symbol-Invariant Transformer for Open-Vocabulary Learning
■lker I■■k, Wenchao Li
Current neural architectures lack a principled way to handle interchangeable tok ens, i.e., symbols that are semantically equivalent yet distinguishable, such as bound variables. As a result, models trained on fixed vocabularies often strugg le to generalize to unseen symbols, even when the underlying semantics remain un changed. We propose a novel Transformer-based mechanism that is provably invaria nt to the renaming of interchangeable tokens. Our approach employs parallel embe dding streams to isolate the contribution of each interchangeable token in the i nput, combined with an aggregated attention mechanism that enables structured in formation sharing across streams. Experimental results confirm the theoretical g uarantees of our method and demonstrate substantial performance gains on open-vo cabulary tasks that require generalization to novel symbols. Project page: https ://bu-depend-lab.github.io/Symbol-Invariant-Transformer/
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LieWarper: Geometry-Aware Motion Transfer via Lie Algebra
Linsong Shan, Laurence Tianruo Yang, Zecan Yang, Fukai Guo, Honglu Zhao, Yixuan Geng
Video motion transfer aims to synthesize novel content videos that strictly foll ow the motion trajectories of a reference video. However, existing methods typic ally operate in Euclidean space, treating motion as unconstrained pixel displace ments or linear phase shifts. This simplification frequently causes severe shear ing artifacts and perspective collapse under complex camera and object motions. In this work, we present LieWarper, a geometry-aware motion transfer framework t hat reconceptualizes motion as coordinate evolution on a manifold rather than me re pixel displacement. Specifically, we derive an analytic solver on the $\text{ Sim}(2)$ manifold to extract global evolution parameters from noisy optical flow . We then introduce a flow-guided phase modulation mechanism, enabling non-rigid dynamics to undergo coordinate transformation along the evolution path. This ap proach achieves accurate trajectory transfer while maintaining global geometric integrity. Extensive experiments show that LieWarper significantly outperforms s tate-of-the-art training-free baselines in both motion fidelity and geometric st ability, while maintaining high generation quality.
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SPHERE: Mitigating the Loss of Spectral Plasticity in Mixture-of-Experts for Dee p Reinforcement Learning
Lirui Luo, Guoxi Zhang, Hongming Xu, Cong Fang, Qing Li
In DRL, an agent is trained from a stream of experience. In a continual learning setting, such agents can suffer from \emph{plasticity loss}: their ability to l earn new skills from new experiences diminishes over training. Recently, Mixture -of-Experts (MoE) networks have been reported to enable scaling laws and facilit ate the learning of diverse skills. However, in continual reinforcement learning settings, their performance can degenerate as learning proceeds, indicating a l oss of plasticity. To address this, building on Neural Tangent Kernel (NTK) theo ry, we formalize the plasticity loss in MoE policies as a loss of \emph{spectral plasticity}. We then derive a tractable proxy for spectral plasticity, one expr essible in terms of individual expert feature matrices. Leveraging this proxy, w e introduce \emph{SPHERE}, a practical Parseval penalty tailored for MoE-based p olicies that alleviates the loss of spectral plasticity. On MetaWorld and Humano idBench, SPHERE improves average success under continual RL by 133\% and 50\% ov er an unregularized MoE baseline, while maintaining higher spectral plasticity t hroughout training.
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Anytime Detection of Strategic Deviations in Multi-Agent Systems
Etienne Gauthier, Francis Bach, Michael I. Jordan
In many multi-agent systems, agents interact repeatedly and are expected to sett le into stable, rational behavior over time. Yet in practice, behavior often dri fts, and detecting such deviations in real time remains an open challenge. We in troduce a sequential testing framework that monitors whether observed play is co nsistent with a benchmark of strategic behavior, without assuming a fixed sample size. Our approach builds on the e-value framework for safe anytime-valid infer ence: by "betting" against the benchmark, we construct a test supermartingale th at accumulates evidence whenever observed payoffs systematically violate the exp ected conditions. For repeated normal-form games, we take equilibrium as the ben chmark, yielding a statistically sound, interpretable measure of departure from equilibrium that can be monitored online; our framework unifies the treatment of Nash, correlated, and coarse correlated equilibria, offering finite-time guaran tees and a detailed analysis of detection times. We also leverage Benjamini-Hoch berg-type procedures to increase detection power in large games while rigorously controlling the false discovery rate. Finally, we extend our method to stochast ic games, verifying online whether observed trajectories adhere to a specified t arget policy, such as a computed equilibrium, broadening the framework's applica bility to dynamic, state-dependent settings.
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The Hidden Cost of Structured Generation in LLMs: Draft-Conditioned Constrained Decoding
Avinash Reddy, Thayne T. Walker, Jaime S. Ide, Amrit Singh Bedi
Large language models (LLMs) are increasingly used to generate executable output s, JSON objects, and API calls, where a single syntax error can make the output unusable. Constrained decoding enforces validity token-by-token via masking and renormalization, but it can distort generation when the model assigns low probab ility mass to valid continuations, pushing decoding toward locally valid yet sem antically incorrect trajectories. We propose \emph{Draft-Conditioned Constrained Decoding (DCCD)}, a simple two-step, training-free inference procedure that dec ouples semantic planning from structural enforcement: an unconstrained draft is generated first, and constrained decoding is then applied, conditioned on this d raft, to guarantee validity. We analyze DCCD through a KL-projection view, showi ng that draft conditioning increases feasible mass and reduces the cumulative “p rojection tax” induced by hard constraints, with an optional best-of-K draft s election. Across structured reasoning benchmarks, DCCD improves strict structure d accuracy by up to +24 percentage points over standard constrained decoding (e. g., 15.2\% to 39.0\% on GSM8K with a 1B model), and enables smaller model pairs to match or exceed much larger constrained baselines, yielding substantial gains in parameter efficiency.
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Is Code Better Than Language for Algorithmic Reasoning?
Terry Tong, Yu Feng, Surbhi Goel, Dan Roth
For tool-augmented language models, comparing natural-language reasoning with co de-execution pipelines is difficult because the comparison changes both the inte rmediate representation and the execution mechanism. We separate these factors w ith an intermediate intervention: the model expresses its reasoning as executabl e code, and the language model simulates that code in context to produce an answ er. On a 40-task verifiable algorithmic benchmark, deterministic code execution outperforms natural-language reasoning by +31.6pp. We observe that the intermedi ate intervention is not meaningfully different from natural-language reasoning (+0.15pp). These results suggest that, in our evaluated setting, changing the int ermediate representation alone does not explain the tool-use advantage, providin g evidence for the performance gains requiring reliable external execution. We formalize this intuition with a simple statistical decision-theoretic model that characterizes when execution dominates end-to-end risk in our disentangled trac e-generation/execution regime. We validate our theory using a reconstruction int ervention that leverages a proxy language model to infer natural-language reason ing traces from code representations, recovering performance comparable to the o

riginal natural-language reasoning pipeline. All experiments are at https://github.com/TerryTong-Git/ToolProj.
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A Task-centric Theory for Iterative Self-Improvement with Easy-to-Hard Curricula
Chenruo Liu, Yijun Dong, Yiqiu Shen, Qi Lei
Iterative self-improvement fine-tunes an autoregressive large language model (LLM) on reward-verified outputs generated by the LLM itself.
In contrast to the empirical success of self-improvement, the theoretical foundation of this generative, iterative procedure in a practical, finite-sample setting remains limited.
We make progress toward this goal by modeling each round of self-improvement as maximum-likelihood fine-tuning on a reward-filtered distribution and deriving finite-sample guarantees for the expected reward.
Our analysis reveals an explicit feedback loop where better models accept more data per iteration, supporting sustained self-improvement while explaining eventual saturation of such improvement.
Adopting a task-centric view by considering reasoning tasks with multiple difficulty levels, we further prove quantifiable conditions on model initialization, task difficulty, and sample budget where easy-to-hard curricula provably achieve better guarantees than training on fixed mixtures of tasks.
Our analyses are validated through Monte-Carlo simulations and experiments spanning a synthetic graph-based reasoning task and multiple standard mathematical reasoning benchmarks.
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ProSAR: Prototype-Guided Semantic Augmentation and Refinement for Time Series Contrastive Learning
Caiyi Yang, Chenglin Li, Hao Zhang, Weijia Lu, ZHIFEI YANG, Wenrui Dai, xiaodong Zhang, Xiaofeng Ma, Can Zhang, Junni Zou, Hongkai Xiong
Contrastive learning has advanced the representation learning across domains, yet its success relies on data augmentations that preserve semantic contents while providing the view diversities. Multivariate time series, however, are inherently noisy, non-stationary, and lack such intuitive semantic cues. Consequently, standard heuristic augmentations that ignore semantic parts may risk destroying critical temporal dependencies. Though some recent approaches attempt to isolate informative components, they typically rely on an implicit neural mechanism to infer semantics, thus limiting the interpretability and controllability.
To address this, we propose ProSAR, an information-theoretic framework that leverages the explicit prototype alignment to guide semantic augmentations, and establish a feedback loop between the augmentation, contrastive learning, and prototype updates.
Specifically, grounded in our proposed Prototype-Conditioned Information Bottleneck principle, we leverage the time-domain prototypes as explicit anchors to localize semantic segments, and develop a time–frequency augmentation strategy that retains prototype-consistent information while discarding noise. To promote semantically consistent prototypes for a reliable view generation, we design a dual-prototype loop where the augmented views are encoded into representations and then the learned representations are clustered to update latent prototypes, whose decoded feedback refines the time-domain prototypes for the next round of augmentation.
Experiments on diverse time-series benchmarks demonstrate that ProSAR outperforms the other contrastive learning methods on downstream forecasting and classification tasks.
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Learning Cardiac Latent Representations in Vectorcardiogram Space
Bosong Huang, Panzhen Zhao, Zengxiang Li, Patricia Lee, Wei Jin, Alan Wee-Chung Liew, Ming Jin, Shirui Pan
Electrocardiography (ECG) is a cornerstone of cardiac assessment, making the learning of informative ECG representations fundamental to tasks ranging from disease diagnosis to clinical report generation.
However, existing methods operate almost exclusively in the observable ECG signa

l space.
In practice, the standard twelve-lead ECG represents multiple projections of the same underlying cardiac electrical activity from different spatial orientations. Therefore, representation learning in the ECG space inevitably introduces substantial redundancy, which may lead to spurious correlations and increased risk of overfitting.
To address this and motivated by the Frank vectorcardiogram (VCG) model, we propose learning a unified latent representation of cardiac electrical activity directly in the VCG space.
We introduce LVCG, the first general self-supervised representation learning framework designed to operate in this physically grounded latent space.
By learning view-invariant latent VCG representations rather than lead-specific artifacts, VCG minimizes redundancy and improves generalization.
LVCG generally outperforms ECG-space baselines across tasks, demonstrating enhanced robustness and generalization, especially in domain shift settings. Our code has been made available at https://github.com/BosonHwang/LVCG.
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Noise as a Natural Regularizer in Markov Decision Processes: Connecting Environmental Stochasticity and Policy Simplicity
Harry Chen, Yiyang Sun, Michal Moshkovitz, Zachery Boner, Lesia Semenova, Cynthia Rudin, Ronald Parr
The planning horizon in a Markov Decision Process (MDP) determines how far into the future an agent reasons. In practice, shorter horizons are commonly associated with policies that exhibit simpler or more interpretable decision-making behavior. In this paper, we establish a formal connection between environmental stochasticity and planning horizon in MDPs. We show that for broad classes of transition noise, solving a noisy MDP can be formally related to solving a noise-free MDP with a shorter effective discount factor, leading to identical optimal policies in some cases and near-optimal ones in others. We further characterize settings in which this correspondence breaks down, clarifying when horizon-based interpretations of noise are not valid. These results, which are supported by both theory and experiments, also give some insight into the common practice of using smaller discount factors for reinforcement learning than those that can be justified by standard modeling interpretations.
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From Absolute to Relative: Rethinking Reward Shaping in Group-Based Reinforcement Learning
Wenzhe Niu, Wei He, Zongxia Xie, Jinpeng Ou, Huichuan Fan, Yuchen Ge, Yanru Sun, Ziyin Wang, Yizhao Sun, Chengshun Shi, Jiuchong Gao, Jinghua Hao, Renqing He
Reinforcement learning has become a cornerstone for enhancing the reasoning capabilities of Large Language Models, where group-based approaches such as GRPO have emerged as efficient paradigms that optimize policies by leveraging intra-group performance differences. However, these methods typically rely on absolute numerical rewards, introducing intrinsic limitations. In verifiable tasks, identical group evaluations often result in sparse supervision, while in open-ended scenarios, the score range instability of reward models undermines advantage estimation based on group means. To address these limitations, we propose **Reinforcement Learning with Relative Rewards (RLRR)**, a framework that shifts reward shaping from absolute scoring to relative ranking. Complementing this framework, we introduce the **Ranking Reward Model**, a listwise preference model tailored for group-based optimization to directly generate relative rankings. By transforming raw evaluations into robust relative signals, RLRR effectively mitigates signal sparsity and reward instability. Experimental results demonstrate that RLRR yields consistent performance improvements over standard group-based baselines across reasoning benchmarks and open-ended generation tasks.
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Scene Graph Thinking: Reinforcing Structured Visual Reasoning for Multimodal Large Language Models
Zhiwei Yang, Yuanchen Wu, NAN ZHANG, Yucong Meng, Ke Yan, Shouhong Ding
Multimodal Large Language Models (MLLMs) have demonstrated strong perception and

reasoning capabilities. However, most existing models focus on isolated objects and neglect structured relationships for efficient target navigation, limiting their performance on visually intensive tasks. To address this challenge, we introduce Scene Graph Thinking (SaGe), a novel paradigm that enables fine-grained and structured visual reasoning through explicit scene-graph representations. Specifically, we first introduce an automated data engine that converts flat image–text corpora into structured scene graphs, where hierarchical entities constitute the nodes and diverse visual relations define the edges. Building upon this, we construct 120K high-quality training data by sampling reasoning traces from scene graphs. Then two-stage graph-aligned post-training paradigms are introduced, where supervised fine-tuning internalizes MLLMs with structured reasoning, and subsequent reinforcement fine-tuning proposes node-as-proxy graph rewards to consolidate efficient graph exploration. With curated data and graph-aligned training, our approach achieves significant improvements across eight multimodal benchmarks, demonstrating strong effectiveness on fine-grained perception and reasoning tasks.

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CONTINUUM: Restoring the Contiguous Tensor Abstraction Efficiently for Dynamic AI Workloads via Hardware Virtualization

Yangyu Zhang, Shuoming Zhang, Chunwei Xia, Shuaijiang Li, Zhicheng Li, Ruiyuan Xu, Zheming Yang, Lei Chen, YUAN WEN, Guangli Li, Xiaobing Feng, Huimin Cui, Jiacheng Zhao

Emerging LLM workloads demand extreme memory agility. However, state-of-the-art inference systems such as vLLM rely on software-defined paging, which sacrifices the contiguous tensor abstraction. This rigid interface exposes fragmentation complexity to developers, imposing a severe engineering burden that stifles algorithmic innovation. We introduce CONTINUUM, a tensor memory virtualization subsystem implemented as a PyTorch extension. By bypassing serialized OS bottlenecks through a lightweight GPU driver extension, CONTINUUM significantly reduces mapping costs by orders of magnitude, from milliseconds to microseconds. Built atop this low-latency API, CONTINUUM provides Elastic Tensor, a set of flexible tensor operations that natively support complex memory dynamics and zero-copy topological aliasing. Evaluations demonstrate that CONTINUUM achieves significantly higher throughput across diverse dynamic scenarios, effectively lowering the barrier to implementing next-generation LLM applications.

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StyleDistillation: A New Insight of Image Style Enables Personalized Aesthetic Manipulation

Yuxin Wang, Xiaoyu Geng, Yuke Li, Zheng Wang

Text-guided stylized image generation has yielded promising advances by leveraging the powerful capabilities of text-to-image diffusion models. However, the inherent coupling of style and content information within the reference image presents a significant challenge. To address this, we propose StyleDistillation, a novel approach grounded in two key observations about the CLIP embedding space from a style perspective. By leveraging a lightweight StyleDistiller module, combined with carefully designed optimization objectives based on geometric and semantic priors, we can extract fine-grained style representation from the reference image. Additionally, we introduce a Prompt Alignment Enhancement mechanism during inference, which significantly improves the control that text prompts exert over the generated images. Extensive experiments demonstrate that our method achieves outstanding performance in both style reproduction and prompt alignment. Furthermore, StyleDistillation supports various personalized operations, including style editing and style fusion, highlighting its substantial potential for diverse applications.

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CAOS: Conformal Aggregation of One-Shot Predictors

Maja Waldron

One-shot prediction enables rapid adaptation of pretrained foundation models to new tasks using only one labeled example, but lacks principled uncertainty quantification. While conformal prediction provides finite-sample coverage guarantees

, standard split conformal methods are inefficient in the one-shot setting due t
o data splitting and reliance on a single predictor. We propose Conformal Aggreg
ation of One-Shot Predictors (CAOS), a conformal framework that adaptively aggre
gates multiple one-shot predictors and uses a leave-one-out calibration scheme t
o fully exploit scarce labeled data. Despite violating classical exchangeability
 assumptions, we prove that CAOS achieves valid marginal coverage using a monoto
nicity-based argument. Experiments on one-shot facial landmarking and RAFT text
classification tasks show that CAOS produces substantially smaller prediction se
ts than split conformal baselines while maintaining reliable coverage.
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From Flat Facts to Sharp Hallucinations: Detecting Stubborn Errors via Gradient
Sensitivity

Liew Yee Zhing, Andrew Huey Ping Tan, Anwar P.P. Abdul Majeed

Traditional hallucination detection fails on "Stubborn Hallucinations" — errors
where LLMs are confidently wrong. We propose a geometric solution: Embedding-Per
turbed Gradient Sensitivity (EPGS). We hypothesize that while robust facts resid
e in flat minima, stubborn hallucinations sit in sharp minima, supported by brit
tle memorization. EPGS detects this sharpness by perturbing input embeddings wit
h Gaussian noise and measuring the resulting spike in gradient magnitude. This a
cts as an efficient proxy for the Hessian spectrum, differentiating stable knowl
edge from unstable memorization. Our experiments show that EPGS significantly ou
tperforms entropy-based and representation-based baselines, providing a robust s
ignal for detecting high-confidence factual errors.
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PointDiT: Pixel-Space Diffusion for Monocular Geometry Estimation

Haofei Xu, Rundi Wu, Philipp Henzler, Nikolai Kalischek, Michael Oechsle, Fabian
 Manhardt, Marc Pollefeys, Andreas Geiger, Federico Tombari, Michael Niemeyer

State-of-the-art single-image 3D reconstruction methods often rely on complex hy
brid architectures and loss functions (e.g., MoGe), or necessitate compressing g
eometry into latent spaces (e.g., GeometryCrafter) to leverage pre-trained laten
t diffusion models. In this work, we demonstrate that such architectural overhea
d and intricate loss formulations are unnecessary. We introduce a minimalist pix
el-space Diffusion Transformer built on a plain ViT, which operates directly on
raw 3D point map patches and is conditioned on image tokens from a pre-trained D
INOv3. Unlike existing latent diffusion-based approaches, we train our diffusion
 backbone entirely from scratch, eliminating the need for point map tokenizers.
We show that this streamlined approach yields results superior to complex latent
-based diffusion models while remaining significantly simpler than hybrid altern
atives. Notably, our model produces sharper geometric structures and achieves si
gnificantly better results on highly ambiguous regions, such as transparent obje
cts.
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Temporal-aware Flow Matching for Video Generation with Temporally Coherent Motio
n

Zirui Pan, Xin Wang, Yipeng Zhang, Yuwei Zhou, Wenwu Zhu

Despite rapid advances in text-to-video generation, state-of-the-art generative
models still suffer from producing temporally incoherent and unrealistic motion
for videos. The key weakness of existing works is that they commonly treat video
s as frame sequences and directly adopt Flow Matching (FM) objectives, which are
 originally designed for images. This practice fails to explicitly model motion
priors or temporal dependencies, resulting in suboptimal dynamics that may appea
r incoherent and unrealistic. To solve this problem, we propose Temporal-aware F
low Matching (TFM), a novel training paradigm that embeds inter-frame constraint
s into the flow objective, leading to temporally coherent motion modeling in vid
eo generation. More specifically, the proposed TFM enforces temporal correlation
s across frames while retaining the desirable properties of FM, and further intr
oduces a residual-type loss that aligns naturally with this new flow. We theoret
ically prove that models trained with TFM are able to exhibit remarkably enhance
d temporal perception ability. Notably, TFM imposes no additional cost during in
ference and is applicable to any model using FM. Extensive experiments demonstra

te that our TFM can significantly improve motion realism across diverse motion t
ypes. Generated videos are presented at https://pzrain.github.io/tfm.
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AIR: Improving Agent Safety through Incident Response
Zibo Xiao, Jun Sun, Junjie Chen
Large Language Model (LLM) agents are increasingly deployed in practice across a
 wide range of autonomous applications. Yet current safety mechanisms for LLM ag
ents focus almost exclusively on preventing failures in advance, providing limit
ed capabilities for responding to, containing, or recovering from incidents afte
r they inevitably arise. In this work, we introduce AIR, the first incident res
ponse framework for LLM agent systems. AIR defines a domain-specific language fo
r managing the incident response lifecycle autonomously in LLM agent systems, an
d integrates it into the agent's execution loop to (1) detect incidents via sema
ntic checks grounded in the current environment state and recent context, (2) gu
ide the agent to execute containment and recovery actions via its tools, and (3)
 synthesize guardrail rules during eradication to block similar incidents in fut
ure executions. We evaluate AIR on three representative agent types. Results sho
w that AIR achieves detection, remediation, and eradication success rates all ex
ceeding 90%. Extensive experiments further confirm the necessity of AIR's key de
sign components, show the timeliness and moderate overhead of AIR, and demonstra
te that LLM-generated rules can approach the effectiveness of developer-authored
 rules across domains. These results show that incident response is both feasibl
e and essential as a first-class mechanism for improving agent safety.
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Adalina: Adaptive Linear Approximation for the Shapley Value and Beyond
Weida Li, Yaoliang Yu, Bryan Kian Hsiang Low
The Shapley value, and its broader family of semi-values, has received much atte
ntion in various attribution problems. A fundamental and long-standing challenge
 is their efficient approximation, since exact computation generally requires an
 exponential number of utility queries in the number of players n. To meet the
 challenges of large-scale applications, we explore the limits of efficiently ap
proximating semi-values under a $\Theta(n)$ space constraint. Building upon a ve
ctor concentration inequality, we establish a theoretical framework that enables
 sharper query complexities for existing unbiased randomized algorithms. Within
this framework, we systematically develop a linear-space algorithm that requires
 $O(\frac{n}{\epsilon^{2}}\log\frac{1}{\delta})$ utility queries to ensure $P(\\
|\hat{\boldsymbol\phi}-\boldsymbol\phi\\|\geq\epsilon)\leq \delta$ for all commo
nly used semi-values. In particular, our framework naturally bridges OFA, unbias
ed kernelSHAP, SHAP-IQ and the regression-adjusted approach, and definitively ch
aracterizes when paired sampling is beneficial. Moreover, our algorithm allows e
xplicit minimization of the mean squared error $\mathbb{E}[\\|\hat{\boldsymbol\p
hi}-\boldsymbol\phi\\|^{2}]$ for each specific utility function. Accordingly, we
 introduce the first adaptive, linear-time, linear-space randomized algorithm, A
dalina, that theoretically achieves improved mean squared error. All of our theo
retical findings are experimentally validated. Our code is available at https://
github.com/watml/adalina.
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DiffuMamba: High-Throughput Diffusion LMs with Mamba Backbone
Vaibhav Singh, Oleksiy Ostapenko, Pierre-Andre Noel, Eugene Belilovsky, Torsten
Scholak
Diffusion language models (DLMs) have emerged as a promising alternative to auto
regressive (AR) generation, yet their reliance on Transformer backbones limits i
nference efficiency due to quadratic attention or KV-cache overhead. We introduc
e DiffuMamba, a masked diffusion language model built on a bidirectional Mamba b
ackbone that combines the diffusion objective with linear-time sequence modeling
, and DiffuMamba-H, a hybrid variant with interleaved attention. Across scales u
p to 1.3B parameters, our models match Transformer-based diffusion in downstream
 performance while achieving up to 8.2× and 4.3× higher inference throughput, re
spectively, on long sequences. We further present a systematic analysis of infer
ence efficiency across modern DLM variants, combining asymptotic complexity with

empirical measurements. Notably, cache-efficient block diffusion with Mamba mixers emerges as the only strategy that scales linearly with sequence length and achieves the strongest performance across all baselines, suggesting a promising direction for future diffusion-based generation systems.
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RedVisor: Reasoning-Aware Prompt Injection Defense via Zero-Copy KV Cache Reuse
Mingrui Liu, Sixiao Zhang, Cheng Long, Kwok-Yan Lam
Large Language Models (LLMs) are increasingly vulnerable to *Prompt Injection (PI)* attacks, where adversarial instructions hidden within retrieved contexts hijack the model's execution flow. Current defenses typically face a critical trade-off: *prevention-based* fine-tuning often degrades general utility via the "alignment tax", while *detection-based* filtering incurs prohibitive latency and memory costs. To bridge this gap, we propose **RedVisor**, a unified framework that synthesizes the explainability of detection systems with the seamless integration of prevention strategies. To the best of our knowledge, RedVisor is the first approach to leverage fine-grained reasoning paths to simultaneously *detect* attacks and *guide* the model's safe response. We implement this via a lightweight, removable adapter positioned atop the frozen backbone. This adapter serves a dual function: it first generates an explainable analysis that precisely localizes the injection and articulates the threat, which then explicitly conditions the model to reject the malicious command. Uniquely, the adapter is active only during this reasoning phase and is effectively muted during the subsequent response generation. This architecture yields two distinct advantages: (1) it mathematically preserves the backbone's original utility on benign inputs; and (2) it enables a novel **KV Cache Reuse** strategy, eliminating the redundant prefill computation inherent to decoupled pipelines. We further pioneer the integration of this defense into the vLLM serving engine with custom kernels. Experiments demonstrate that RedVisor outperforms state-of-the-art defenses in detection accuracy and throughput while incurring negligible utility loss.
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Supervised Guidance Training for Infinite-Dimensional Diffusion Models
Elizabeth Louise Baker, Alexander Denker, Jes Frellsen
Score-based diffusion models have recently been extended to infinite-dimensional function spaces, with uses such as inverse problems arising from partial differential equations. In the Bayesian formulation of inverse problems, the aim is to sample from a posterior distribution over functions obtained by conditioning a prior on noisy observations. While diffusion models provide expressive priors in function space, the theory of conditioning them to sample from the posterior remains open. We address this, assuming that either the prior lies in the Cameron-Martin space, or is absolutely continuous with respect to a Gaussian measure. We prove that the models can be conditioned using an infinite-dimensional extension of Doob's h-transform, and that the conditional score decomposes into an unconditional score and a guidance term. As the guidance term is intractable, we propose a simulation-free score matching objective (called *Supervised Guidance Training*) enabling efficient and stable posterior sampling. We illustrate the theory with numerical examples on Bayesian inverse problems in function spaces. In summary, our work offers the first function-space method for fine-tuning trained diffusion models to accurately sample from a posterior.
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AgentExpt: Automating AI Experiment Design with LLM-based Resource Retrieval Agent
Yu Li, Lehui Li, Lin Chen, Qingmin Liao, Fengli Xu, Yong Li
In modern AI research, baseline and dataset selection is a high-stakes decision in experimental design.
It operationalizes a research idea into a concrete evaluation protocol and largely determines the validity and comparability of empirical conclusions.
However, making appropriate choices is increasingly difficult as baselines and datasets proliferate, while suitability is inherently context-dependent and rarely captured by baseline and dataset metadata.
To address these challenges, we present \textbf{AgentExpt}, a comprehensive fram

ework for baseline and dataset recommendation.
We first curate a large-scale, high-quality knowledge base that links 108{,}825 accepted papers to their used baselines and datasets.
Based on this resource, we design a \textit{collective perception-enhanced retriever} that represents each baseline or dataset by integrating first-person self-descriptions with third-person citation contexts, thereby effectively positioning them within the scholarly network.
We further design a \textit{reasoning-augmented reranker} that encodes baseline-dataset interaction chains as a reasoning prior to fine-tune an LLM, producing refined rankings with interpretable justifications.
Experiments show that our framework outperforms the strongest baseline, with average gains of +5.85\% in Recall@20 and +7.90\% in HitRate@10, and ablation studies confirm the effectiveness of our designed components.
Overall, AgentExpt advances the efficient and reliable automation of experimental design.
Our code is available at \url{https://anonymous.4open.science/r/Agentexpt-DD3E}.
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BPDQ: Bit-Plane Decomposition Quantization on a Variable Grid for Large Language Models
Junyu Chen, Jungang Li, Jing Xiong, Wenjie Wang, Qingyao Yang, He Xiao, Zhen Li, Taiqiang Wu, Mengzhao Chen, Zhen Peng, Chaofan Tao, Long Shi, Hongxia Yang, Ngai Wong
Large language model inference is often bounded by memory footprint and bandwidth in resource-constrained deployments, making quantization fundamental to efficient serving.
While post-training quantization (PTQ) maintains high fidelity at 4-bit, it deteriorates at 2-3 bits.
In essence, existing methods enforce a shape-invariant quantization grid (e.g., the fixed uniform intervals of UINT2) for each group, severely restricting the feasible set for error minimization.
To address this, we propose Bit-Plane Decomposition Quantization (BPDQ), which constructs a variable quantization grid via bit-planes and scalar coefficients, and iteratively refines them using second-order information while progressively compensating for quantization errors to minimize output discrepancy.
In the 2-bit regime, BPDQ enables serving Qwen2.5-72B on a single RTX 3090 with 83.85\% GSM8K accuracy (vs. 90.83\% at 16-bit).
Moreover, we theoretically show that the variable grid expands the feasible set, and that the quantization process consistently aligns with the optimization objective in Hessian-induced geometry.
The code is available at github.com/KingdalfGoodman/BPDQ.
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FiRE: Fine-grained Ranking Evaluation for Machine Translation
Wenyang Gao, Yinghao Yang, Xi Jin, Jing Li, Yue Zhang
Developing reliable machine translation (MT) systems hinges on our ability to distinguish superior translations from inferior ones. However, existing evaluation paradigms, whether limited to coarse overall rankings or misaligned with human preferences, fail to deliver interpretable, fine■grained feedback in reference■free settings. We present a Fine-Grained Ranking Evaluation method (FiRE) that leverages off■the■shelf large language models to perform criterion■driven pairwise comparison across three complementary dimensions: faithfulness, fluency, and consistency of style, instead of producing a single holistic judgment. To enable rigorous meta■evaluation of evaluation paradigms in the absence of any suitable testbed, we construct the first human■annotated, reference■free benchmark for fine-grained ranking evaluation, achieving substantial inter■annotator agreement. Through meta■evaluation on this benchmark and existing MQM datasets, FiRE demonstrably outperforms regression■based and error■analysis metrics in aligning with human comparative judgments, while providing more informative insights into translation quality. Finally, our examination of LLM evaluator biases (position and self-enhancement) and their handling of tied cases offers guidance for more nuanced MT evaluation. Code and benchmark resources are available at https://github.c

om/wygao8/FiRE-MT.
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Powerful and Theoretically Guaranteed Independence Testing on Heterogeneous Federated Clients

Yixin Ren, Hongquan Liu, Juncai Zhang, Yewei Xia, Zichuan Lin, Deheng Ye, Hao Zhang, Jihong Guan, Shuigeng Zhou

In this paper, we present a novel federated independence testing (FedIT) method that addresses both theoretical and practical challenges arising from client heterogeneity. We begin by revisiting existing federated independence testing methods and showing why they fail to provide valid guarantees or maintain statistical power under data distributional shift across clients. Building on this analysis, we develop a copula-based marginal alignment technique together with a stacking-based aggregation strategy that amplifies intra-client dependence while mitigating inter-client variation, resulting in a theoretically sound and powerful global test. For practicality, we further accelerate the aggregation step and incorporate a privacy-preserving mechanism. On the theoretical side, we prove both the correctness of our method and the validity of the test. Empirically, we conduct extensive experiments on both synthetic and real-world datasets, which demonstrate the superiority of our solution over existing methods.
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CoIRL-AD: Collaborative-Competitive Imitation-Reinforcement Learning in Latent World Models for Autonomous Driving

Xiaoji Zheng, Ziyuan Yang, Yanhao Chen, Yuhang Peng, Yuanrong Tang, Gengyuan Liu, Bokui Chen, Jiangtao Gong

End-to-end autonomous driving models trained with imitation learning (IL) often generalize poorly, particularly in long-tail scenarios where expert demonstrations are sparse. Reinforcement learning (RL) can provide complementary task-level supervision, but applying RL to real-world autonomous driving is challenging in offline settings without interactive simulators, where datasets are dominated by expert actions and provide limited behavioral diversity. We propose CoIRL-AD, a competitive dual-policy framework that integrates IL and RL under a unified offline training regime. CoIRL-AD decouples imitation and reward optimization into separate actors to alleviate objective conflicts, uses imagined future rollouts for long-horizon reward estimation, and introduces a competition mechanism that selectively transfers beneficial behaviors while keeping RL anchored to expert-like driving. Experiments on the nuScenes benchmark show that CoIRL-AD consistently improves robustness over strong IL-based baselines, with especially large gains in cross-city generalization and long-tail scenarios. Code is available at: https://github.com/SEU-zxj/CoIRL-AD.
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Learning Multi-Agent Coordination via Sheaf-ADMM

Jeffrey Seely, Bart■omiej Cupia■, Llion Jones

We present a differentiable optimization framework for multi-agent coordination. An input is decomposed into overlapping local views, each processed by an agent that solves a convex subproblem parameterized by a neural encoder. Agents coordinate through the Alternating Direction Method of Multipliers (ADMM) with inter-agent constraints specified by a cellular sheaf. The sheaf specifies which aspects of neighboring solutions must agree, allowing for heterogeneous notions of global consensus. Backpropagating through the unrolled optimization jointly trains all components of the multi-agent system. We evaluate on maze pathfinding, image classification, and Sudoku, where agents with individually insufficient local views learn to coordinate to produce correct global outputs. On MNIST, the local-view decomposition yields improved robustness to distribution shifts relative to a standard CNN. On Sudoku, the optimization-derived structure yields markedly higher solve rates than parameter-matched MPNN baselines. Finally, the ADMM structure exposes distinct primal, consensus, and dual state variables, opening the coordination dynamics to direct analysis and intervention—a property unavailable in standard message-passing architectures.
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Deep Networks Learn Deep Hierarchical Models

Amit Daniely
We consider supervised learning with n labels and show that layerwise SGD on r esidual networks can efficiently learn a class of hierarchical models. This mode l class assumes the existence of an (unknown) label hierarchy $L_1 \subseteq L_2 \subseteq \dots \subseteq L_r = [n]$, where labels in L_1 are simple function s of the input, while for $i > 1$, labels in L_i are simple functions of simpl er labels.

Our class surpasses models that were previously shown to be learnable by deep le arning algorithms, in the sense that it reaches the depth limit of efficient lea rnability. That is, there are models in this class that require polynomial depth to express, whereas previous models can be computed by log-depth circuits.

Furthermore, we suggest that learnability of such hierarchical models might even tually form a basis for understanding deep learning.
Beyond their natural fit for domains where deep learning excels, we argue that t he mere existence of human ``teachers" supports the hypothesis that hierarchical structures are inherently available. By providing granular labels, teachers eff ectively reveal ``hints'' or ``snippets'' of the internal algorithms used by the brain. We formalize this intuition, showing that in a simplified model where a teacher is partially aware of their internal logic, a hierarchical structure eme rges that facilitates efficient learnability.
****
Scalable Simulation-Based Model Inference with Test-Time Complexity Control
Manuel Gloeckler, J.P. Manzano-Patrón, Stamatios Sotiropoulos, Cornelius Schröde r, Jakob H. Macke
Simulation plays a central role in scientific discovery. In many applications, t he bottleneck is no longer running a simulator—it is choosing among large famili es of plausible simulators, each corresponding to different forward models/hypot heses consistent with observations. Over large model families, classical Bayesia n workflows for model-selection are impractical. Furthermore, amortized model-se lection methods typically hard-code a fixed model prior—or complexity penalty—at training time, requiring users to commit to a particular parsimony assumption b efore seeing the data. We introduce PRISM, a simulation-based encoder-decoder th at infers a joint posterior over both discrete model structures and associated c ontinuous parameters, while enabling test-time control of model complexity via a tunable model prior that the network is conditioned on. We show that PRISM scal es to families with combinatorially many (up to billions of) model instantiation s on a synthetic symbolic regression task. As a scientific application, we evalu ate PRISM on biophysical modeling for diffusion MRI data, showing the ability to perform model selection across several multi-compartment models, on both synthe tic and in-vivo neuroimaging data.
****
Gauge-Equivariant Graph Networks via Self-Interference Cancellation
Yoonhyuk Choi, Jiho Choi, Jiwoo Kang
Graph neural networks often degrade on heterophilous graphs because repeated nei ghbor aggregation can reinforce self-aligned low-frequency components while supp ressing phase-inconsistent signals. We propose GESC, a complex-valued graph netw ork that augments attention-based message passing with gauge-consistent U(1) tra nsport and projection-based self-interference cancellation. For each transported neighbor message, GESC removes the component parallel to the target representat ion before computing attention and applies a sign-aware gate based on gauge-inva riant complex alignment. We prove gauge equivariance of the hidden update and de rive coefficient-frozen stability bounds showing that SIC contracts self-paralle l message components. On nine benchmarks, GESC ranks first on seven datasets and remains within the top three on the other two. These results suggest that expli cit self-parallel cancellation is an effective mechanism for improving message p assing under heterophily. Our code is available at https://github.com/ChoiYoonHy uk/GESC.
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Deep neural networks divide and conquer dihedral multiplication
Sihui Wei, Gavin McCracken, Gabriela Moisescu-Pareja, Harley Wiltzer, Doina Precup, Irina Rish, Jonathan Love
We find multilayer perceptrons and transformers both universally learn an instantiation of the same divide-and-conquer algorithm that requires only a logarithmic number of neural representations to solve dihedral multiplication. Clustering neurons based on similar activation behaviour reveals remarkably clear structure: each neural representation corresponds to a Cayley graph. To our knowledge, this is the first work that fully characterizes and describes all neural representations that are learnable on a dataset, while prior work on group multiplications studied neuron-level behavior, or preliminarily investigated cluster behavior. Thus, we can understand the algorithm networks universally learn at three levels of abstraction: 1) Neurons activate on coset or approximate coset structure of the dihedral group. 2) Groups of neurons together form neural representations that act to divide the dataset into different subproblems, being Cayley graphs, where the equivalence class of the answer is computed. 3) The global algorithm then linearly combines each neural representation (subproblem) together at the logits. This work provides the community with a deep case study and a well-understood toy model for interpretability, and makes progress toward proving the conjecture that networks trained via stochastic gradient methods divide and conquer all group multiplication tasks.
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DTop-p MoE: Sparsity-Controlled Dynamic Top-p MoE for Foundation Model Pre-training
Can Jin, Hongwu Peng, Mingcan Xiang, Qixin Zhang, Xiangchi Yuan, Amit Hasan, Ohi Dibua, Yifan Gong, Yan Kang, Dimitris N. Metaxas
Sparse Mixture-of-Experts architectures are essential for scaling model capacity efficiently, yet the standard Top-k routing imposes a rigid sparsity pattern that ignores the intrinsic variance in token difficulty and layer-specific computational needs. Top-p routing is more adaptive because it selects experts until their cumulative routing probability reaches a threshold, allowing confident tokens to use fewer experts and ambiguous tokens to recruit more. However, we demonstrate that existing naive Top-p implementations with fixed global probability thresholds provide only marginal gains over Top-k, suffer from hyperparameter sensitivity, and result in uncontrolled computational costs. In this paper, we propose **DTop-p**, a sparsity-controllable dynamic routing mechanism that learns the Top-p probability threshold with a Proportional-Integral controller and uses dynamic routing normalization to support layer-wise expert selection under a global sparsity constraint. Extensive experiments on Large Language Models and Diffusion Transformers demonstrate that **DTop-p** consistently outperforms both Top-k and fixed Top-p baselines while matching the average FLOPs of Top-k MoE. Our analysis confirms that **DTop-p** exhibits strong scaling properties across expert granularity, total expert capacity, model size, and dataset size, offering a robust and efficient MoE framework for foundation model pre-training.
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Open-World LLM Logical Reasoning
Ye Mo, Chuan Zhou, Fengxiang Cheng, Jialin Yu, Liangming Pan, Fenrong Liu, Sheng Zhou, Haoxuan Li, Zhouchen Lin, Philip Torr
Large Language Models (LLMs) achieve remarkable performance but struggle with complex logical reasoning, particularly in real-world settings.
Existing research is largely confined to the closed-world assumption, which posits that all premises required for reasoning are explicitly provided.
However, real-world tasks frequently exhibit open-world characteristics, where the provided information is insufficient to infer a conclusion due to missing premises or implicit commonsense knowledge.
To address this, we propose OpenIKLR, an Open-world Incomplete-Knowledge-aware Logical Reasoning framework that integrates symbolic logic solvers with LLMs.
OpenIKLR first translates natural language into symbolic representations to pinpoint reasoning gaps via a logical solver.

It then iteratively generates a minimal set of necessary missing premises using LLMs.
To ensure these added premises are both logically sound and factually accurate, we introduce a dual-verification: logic verification via the solver and fact ver ification via the LLMs.
Experiments show that OpenIKLR consistently outperforms existing logical reasoni ng and RAG baselines across multiple backbones and real-world datasets.
The code is available at https://github.com/OpenIKLR/OpenIKLR.
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Are First-Order Diffusion Samplers Really Slower? A Fast Forward-Value Approach
Yuchen Jiao, Na Li, Changxiao Cai, Gen Li
Higher-order ODE solvers have become a standard tool for accelerating diffusion probabilistic model (DPM) sampling, motivating the widespread view that first-or der methods are inherently slower and that increasing discretization order is th e primary path to faster generation. This paper challenges this belief and revis its acceleration from a complementary angle: beyond solver order, the placement of DPM evaluations along the reverse-time dynamics can substantially affect samp ling accuracy in the low-neural function evaluation (NFE) regime. We propose a n ovel training-free, first-order sampler named Forward DPMSolver (F-DPMSolver), w hose leading discretization error has the opposite sign to that of DDIM. Algorit hmically, the method approximates the forward-value evaluation via a cheap one-s tep lookahead predictor. We provide theoretical guarantees showing that the resu lting sampler provably approximates the ideal forward-value trajectory while ret aining first-order convergence. Empirically, across standard image generation be nchmarks, the proposed sampler consistently improves sample quality under the sa me NFE budget and can be competitive with, and sometimes outperform, state-of-th e-art higher-order samplers. Overall, the results suggest that the placement of DPM evaluations provides an additional and largely independent design angle for accelerating diffusion sampling. Our code is available at https://github.com/Na-Li66/F-DPMSolver.
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An Approximation Algorithm for Graph Label Selection
Josia John, Simon Meierhans, Maximilian Probst Gutenberg
In the graph label selection problem, one is given an n-vertex graph and a bud get k, and seeks to select k vertices whose labels enable accurate predictio n of the labels on the remaining vertices. This problem formalizes distilling a small representative set from the whole graph.
We present the first $\tilde{O}(\log^{1.5} n)$-approximation algorithm for graph label selection under the standard budget constraint. Prior work either relies on resource augmentation, allowing substantially more than k labeled vertices, or consists primarily of heuristics without provable guarantees.
Finally, we demonstrate that practical heuristic variants of our algorithm scale to significantly larger graphs than previous methods, while essentially retaini ng their quality.
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R2R2: Robust Representation for Intensive Experience Reuse via Redundancy Reduct ion in Self-Predictive Learning
Sanghyeob Song, Donghyeok Lee, Jinsik Kim, Sungroh Yoon
For reinforcement learning in data-scarce domains like real-world robotics, inte nsive data reuse enhances efficiency but induces overfitting. While prior works focus on critic bias, representation-level instability in Self-Predictive Learni ng (SPL) under high Update-to-Data (UTD) regimes remains underexplored. To bridg e this gap, we propose Robust Representation via Redundancy Reduction (R2R2), a regularization method within SPL. We theoretically identify that standard zero-c entering conflicts with SPL's spectral properties and design a non-centered obje ctive accordingly. We verify R2R2 on SPL-native algorithms like TD7. Furthermore , to demonstrate its orthogonality to prior advancements, we extend the state-of -the-art SimbaV2, which originally lacks SPL, by integrating a tailored SPL modu le, termed SimbaV2-SPL. Experiments across 11 continuous control tasks confirm t hat R2R2 effectively mitigates overfitting; specifically, at a UTD ratio of 20,

it improves TD7 by $\sim$22\% and provides additional gains on top of SimbaV2-SP L, which itself establishes a new state-of-the-art. The code can be found at [th is link](https://github.com/songsang7/R2R2).
****
WAVE: Window-Aware Vocabulary-Efficient Early-Exit for Training-Free LLM Acceler ation
Seonggeun Kim, Gilha lee, Hyun Kim
Large language models (LLMs) incur substantial inference latency due to autoregr essive decoding, in which each token requires a full forward pass through all tr ansformer layers. Early-exit methods that terminate computation at intermediate layers offer a promising remedy, yet existing approaches suffer from fundamental limitations. Confidence-based methods rely on evaluating the full LM head at ev ery layer, introducing considerable overhead that can negate the expected speedu p. Schedule-based methods avoid this cost through predetermined exit schedules, but their monotonically decreasing layer allocation collapses to shallow layers, thereby constraining the maximum generation length. Learned exit predictors fur ther require costly task-specific training and are vulnerable to distribution sh ifts in unseen domains. We propose Window-Aware Vocabulary-Efficient Early-Exit (WAVE), a training-free framework that addresses these challenges through two ke y innovations. First, exit window scheduling identifies an optimal layer range f or early-exit decisions via offline calibration, preventing premature convergenc e to shallow layers while substantially reducing the number of exit checks. Seco nd, a proxy LM head constructs a lightweight vocabulary subset at the window’s s tarting layer, reducing per-layer exit overhead by 87\% relative to full LM head . WAVE requires no gradient-based training and enables immediate deployment with only a brief calibration phase. Experiments on Llama-2 7B demonstrate up to 1.4 $\times$ average speedup while preserving output quality, with full compatibilit y with W4A16 quantization, establishing WAVE as a practical early-exit framework for accelerating LLMs inference without retraining.
****
Steer Where It Matters: Token-Level Visual-Sensitivity Steering for LVLMs Halluc ination Mitigation
Ruipeng Zhang, Zhihao Li, C.L.Philip Chen, Tong Zhang
Large vision language models (LVLMs) have made rapid advancements and are deploy ed across various applications, yet hallucinations remain a major challenge. Act ivation steering is appealing due to its minimal training overhead and controlla bility at inference time. However we found that during autoregressive decoding, visual conditioning affects token prediction sparsely and locally across decodin g steps, and many existing methods that average image-versus-no-image difference s over the entire sequence dilute these critical signals, yielding low signal-to -noise ratio steering directions. Additionally, many existing methods apply a fi xed steering strength, which misallocates the intervention budget, over-perturbs non-critical tokens, and can cause instability. To address these limitations, w e propose Token-Level Visual-Sensitivity Steering (TLVS) for hallucination mitig ation. Our approach first extract token-level steering vectors and refines them, and then applies fine-grained, visual-sensitivity–adaptive steering only where it matters. This lightweight, plug-and-play mechanism requires only minimal trai ning for calibration and can be applied across diverse vision-language models. I t modulates the steering strength at each decoding step, selectively suppressing hallucination-prone spans while preserving evidence-grounded content. We evalua te TLVS on several benchmarks, including POPE, AMBER, CHAIR (COCO), MMHal and Ha llusionBench, demonstrating consistent improvements over previous steering metho ds.
****
Efficient Stochastic Optimisation via Sequential Monte Carlo
James Cuin, Davide Carbone, Yanbo Tang, O. Deniz Akyildiz
The problem of optimising functions with intractable gradients frequently arises in machine learning and statistics, ranging from maximum marginal likelihood es timation procedures to fine-tuning of generative models. Stochastic approximatio n methods for this class of problems typically require inner sampling loops to o

btain (biased) stochastic gradient estimates, which rapidly becomes computationa lly expensive. In this work, we develop sequential Monte Carlo (SMC) samplers fo r optimisation of functions with intractable gradients. Our approach replaces ex pensive inner sampling methods with efficient SMC approximations, which can resu lt in significant computational gains. We establish convergence results for the basic recursions defined by our methodology which SMC samplers approximate. We d emonstrate the effectiveness of our approach on the reward-tuning of energy-base d models within various settings.
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Learning Randomized Reductions
Ferhat Erata, Orr Paradise, Thanos Typaldos, Timos Antonopoulos, ThanhVu Nguyen, Shafi Goldwasser, Ruzica Piskac
Randomized self-reductions (RSRs) express $f(x)$ using f evaluated at random c orrelated points, enabling self-correcting programs, instance-hiding protocols, and applications in complexity theory and cryptography. Yet discovering RSRs has required manual expert derivation for over 40 years, limiting their practical u se.
We present Bitween for automated RSR learning. First, we formalize RSR learning with sample complexity analysis under correlated sampling. Second, we develop Va nilla Bitween, which integrates multiple backends (linear regression, genetic pr ogramming, symbolic regression, and mixed-integer programming). The linear regre ssion backend outperforms the others, discovering RSRs for 43 of 80 functions (5 4%) in RSR-Bench, our benchmark suite, including the first known reduction for s igmoid. Third, we introduce Agentic Bitween, a neuro-symbolic approach where LLM agents propose novel query functions beyond the fixed set ($x+r$, $x-r$, $x \cd ot r$, x, r) in prior work. Agentic Bitween discovers RSRs for 64 of 80 func tions (80%), outperforming pure neural baselines in both RSR discovery and verif ication accuracy.
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This State Looks Like That: Self-Interpretable Reinforcement Learning Agents usi ng Prototype Soft Actor-Critic
Andrea Marzo, Alessio Ragno, Roberto Capobianco
Reinforcement learning (RL) has achieved remarkable success across complex decis ion-making tasks, especially with the advent of deep neural networks. However, t he resulting models are often opaque, making their deployment in safety-critical domains challenging. Explainable AI aims to address this issue, but most specif ic efforts for deep RL remain limited either to post-hoc explanation methods or to imitation learning and distillation procedures. These latter approaches rely on pre-trained black-box agents and are typically restricted to environments wit h discrete action spaces, limiting their scalability and interpretability. In th is paper, we introduce ProtoSAC, a novel deep RL architecture that integrates a prototype-based actor into the Soft Actor-Critic (SAC) algorithm, enabling intri nsic interpretability in continuous action spaces. Our method learns a set of pr ototypes that represent interpretable state clusters, each associated with a Gau ssian action distribution. Actions are generated as a similarity-weighted mixtur e over these prototypes, providing more inspectable and decomposable decision-ma king without sacrificing performance compared to standard SAC. We evaluate Proto SAC on continuous control environments and show that it matches the performance of the original SAC while offering enhanced interpretability.
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Safe Autoregressive Image Generation with Iterative Self-Improving Codebooks
Yunqi Xue, Zhijiang Li, Philip Torr, Jindong Gu
Unlike diffusion-based models that operate in continuous latent spaces, autoregr essive unified multimodal models produce images by sequentially predicting discr etized visual tokens. These tokens are derived from a codebook that maps embeddi ngs to quantized visual patterns. The language-like architecture enables unified multimodal models to effectively capture text conditional information for gener ation, making them promising for text-to-image tasks. This also raises an intere sting question: *how safe are the images generated in such an autoregressive way ?* In this work, we propose iterative self-improving codebooks for safe autoregr

essive generation. We leverage the understanding and judgment capabilities of th
e unified multimodal model itself to identify unsafe generated images without hu
man annotation. Subsequently, the inherent representations in the codebook are f
ixed to eliminate harmful mappings. Our method comprises two steps: first, we us
e the unified model to identify unsafe generations and construct corresponding h
armful and safe image-text pairs. These pairs are used to construct the Harmful
Space and guide updates to the codebook, thereby eliminating harmful outputs. Se
cond, we perform adaptive fine-tuning on the codebook within the harmless space
using safe image-text pairs to ensure the quality of generated images. These two
 steps are repeated until no further improvement is observed, producing a safety
-enhanced model codebook. Without additional external feedback, the safety of mo
dels is improved iteratively.
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Why Agentic Theorem Prover Works: A Statistical Provability Theory of Mathematic
al Reasoning Models

Sho Sonoda, Shunta Akiyama, Yuya Uezato

Agentic theorem provers combine a reasoning model, retrieval, search, and a proo
f assistant verifier, yet it remains unclear which components actually improve f
inite-budget proof success and why they help on real mathematical workloads. We
study this question through **statistical provability**: the probability of reac
hing a verified proof within a budget on a specified stream of theorem instances
. We model formal proof search as a finite-horizon reachability MDP with determi
nistic verifier dynamics, and show that under a faithful state abstraction the o
ptimal success probability coincides with ordinary syntactic provability. We the
n analyze a simple but practically important pipeline: depth-wise offline action
-value regression followed by greedy test-time proving. Our main theorem bounds
the provability gap between the learned prover and the optimal prover by an occu
pancy-weighted sum of uniform action-value errors; in the common uniform-error r
eading, the leading complexity multiplier is the learned prover's average trunca
ted proof length. The error decomposes into approximation error, geometric cover
age of the training distribution, and Monte Carlo label noise, and improves to a
 fast rate under an action-gap margin condition. The result gives a component-se
nsitive account of why verifier feedback, retrieval, representation geometry, an
d proof-shortening mechanisms help on biased theorem workloads, without contradi
cting classical worst-case hardness.
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GaussTrace: Provenance Analysis of 3D Gaussian Splatting Models with Evidence-ba
sed LLM Reasoning

Haoliang Han, Ziyuan Luo, Renjie Wan

3D Gaussian Splatting (3DGS) is a powerful technique for creating high-fidelity
3D assets. However, the widespread sharing and iterative modification of 3DGS mo
dels across digital platforms create pressing challenges for intellectual proper
ty protection and forensic traceability. To address this, we propose GaussTrace,
 a novel framework for constructing directed provenance graphs for 3DGS models.
GaussTrace formulates provenance analysis as an evidence-based reasoning problem
. It builds upon attribute-wise statistical profiling of 3DGS parameters to capt
ure intrinsic properties. Moreover, we introduce hypothesis-driven editing simul
ations of common operations to provide auxiliary evidence for plausible transfor
mation pathways. These statistical and simulated cues jointly enable a Large Lan
guage Model (LLM) to perform structured Chain-of-Thought (CoT) reasoning, yieldi
ng directional provenance inferences and explainable edge reasons. Experimental
results demonstrate that GaussTrace effectively constructs evolutionary relation
ships among diverse 3DGS models, delivering accurate, interpretable, and robust
provenance graphs without requiring model training or access to editing historie
s. Project page: https://haolianghan.github.io/GaussTrace.
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Corrected Samplers for Discrete Flow Models

Zhengyan Wan, Yidong Ouyang, Liyan Xie, Hongyuan Zha, Fang Fang, Guang Cheng

Discrete flow models (DFMs) have been proposed to learn the data distribution on
 finite state space, offering a flexible framework as an alternative to discrete

diffusion models. A line of recent work has studied samplers for discrete diffu
sion models, such as tau-leaping and Euler solver. However, these samplers requi
re a large number of iterations to control discretization error, since the trans
ition rates are frozen in time and evaluated at the initial state within each ti
me interval. Moreover, theoretical results for these samplers often require bou
ndedness conditions of the transition rate or they focus on a specific type of s
ource distributions. To address those limitations, we establish non-asymptotic d
iscretization error bounds for those samplers without any restriction on transit
ion rates and source distributions, under the framework of discrete flow models.
 Furthermore, by analyzing a one-step lower bound of the Euler sampler, we propo
se two corrected samplers: \textit{time-corrected sampler} and \textit{location-
corrected sampler}, which can reduce the discretization error of tau-leaping and
 Euler solver with almost no additional computational cost. We rigorously show t
hat the location-corrected sampler has a lower complexity than existing parallel
 samplers. We validate the effectiveness of the proposed method by achieving bet
ter generation quality with reduced inference time on simulations and text-to-im
age generation tasks. Code can be found in \url{https://github.com/WanZhengyan/C
orrected-Samplers-for-Discrete-Flow-Models}.
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Ski Rental with Distributional Predictions of Unknown Quality
Qiming Cui, Michael Dinitz
We revisit the central online problem of *ski rental* in the "algorithms with pr
edictions" framework from the point of view of *distributional* predictions. If
 we are given as a prediction a *distribution* $\hat p$ over the ski days, and t
he true number of ski days comes from some (unknown) distribution p, then we s
how as our main result that there is an algorithm with expected cost at most $OP
T + O\left(\min \left(\max(\eta,1) \cdot \sqrt{b},\ b \log b \right) \right)$,
where OPT is the expected cost of the optimal policy for the true distribution
 p, b is the cost of buying, and η is the Earth Mover's (Wasserstein-1)
 distance between p and $\hat p$. An implication of this bound is that our al
gorithm has *consistency* $O(\sqrt{b})$ (additive loss when the prediction error
 is 0) and *robustness* $O(b \log b)$ (additive loss when the prediction error
 is arbitrarily large). Moreover, we do not need to assume that we know (or hav
e any bound on) the prediction error η, in contrast with previous work in r
obust optimization which assumes that we know this error. We also complement thi
s upper bound with a variety of lower bounds showing that it is essentially tigh
t: not only can the consistency/robustness tradeoff not be improved, but our par
ticular loss function cannot be meaningfully improved.
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Efficient Parallel Samplers for Recurrent-Depth Models and Their Connection to D
iffusion Language Models
Jonas Geiping, Xinyu Yang, Guinan Su
Language models with recurrent depth, also referred to as universal or looped wh
en considering transformers, are defined by the capacity to increase their compu
tation through the repetition of layers. Recent efforts in pretraining have demo
nstrated that these architectures can scale to modern language modeling tasks wh
ile exhibiting advantages in reasoning tasks.
  In this work, we examine the relationship between recurrent-depth models and d
iffusion language models. Building on their similarities, we develop a new diffu
sion forcing sampler for these models to accelerate generation. The sampler adva
nces by decoding new tokens at every forward pass of the model, while the latent
 states of these tokens can be further refined in parallel through recurrence. T
heoretically, under a fixed wall-clock budget, generation with our sampler is st
rictly more expressive than baseline autoregressive generation as it preserves t
he same recurrent depth while updating a strictly wider front of token positions
 in parallel, enabling more computation at equal serial depth.
  Moreover, this sampler, based on principles from diffusion literature, can be
directly applied to existing 3.5B recurrent-depth transformers without any tunin
g, leading to up to a 5x speedup.
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Breaking the Scale Barrier: One-Shot Knowledge Transfer via Frequency Transform
Jianlu Shen, Fu Feng, Yucheng Xie, Jiaqi Lv, Xin Geng
Transferring knowledge by fine-tuning large-scale pre-trained networks has becom e a standard paradigm for downstream tasks, yet the knowledge of a pre-trained m odel is tightly coupled with monolithic architecture, which restricts flexible r euse across models of varying scales.
In response to this challenge, recent approaches typically resort to either para meter selection, which fails to capture the interdependent structure of this kno wledge, or parameter prediction using generative models that depend on impractic al access to large network collections.
In this paper, we identify the low-frequency components of model weights as the concrete carrier of foundational, task-agnostic knowledge—its "learngene"—and va lidate this by demonstrating its efficient inheritance by downstream models and tasks.
Based on this insight, we propose FRONT (FRequency dOmain kNowledge Transfer), a novel framework that uses the Discrete Cosine Transform (DCT) to isolate the lo w-frequency "learngene".
This learngene can be seamlessly adapted to initialize models of arbitrary size via simple truncation or padding, a process that is entirely training-free.
For enhanced performance, we propose an optional low-cost refinement process tha t introduces a spectral regularizer to further improve the learngene's transfera bility.
Extensive experiments demonstrate that FRONT achieves the state-of-the-art perfo rmance, accelerates convergence by up to $15\times$ in vision tasks, and reduces training FLOPs by an average of 40.5\% in language tasks. Code is available at https://github.com/LUcy0505/FRONT.
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BOCLOAK: Optimal Transport-Guided Adversarial Attacks on Graph Neural Network-Ba sed Bot Detection
Kunal Mukherjee, Zulfikar Alom, Tran Gia Bao Ngo, Cuneyt Gurcan Akcora, Murat Ka ntarcioglu
The rise of bot accounts on social media poses significant risks to public disco urse. To address this threat, modern bot detectors increasingly rely on Graph Ne ural Networks (GNNs). However, the effectiveness of these GNN-based detectors in real-world settings remains poorly understood. In practice, attackers continuou sly adapt their strategies and must operate under domain-specific
and temporal constraints, which can fundamentally limit the applicability of exi sting attack methods. As a result, there is a critical need for robust GNN-based bot detection methods under realistic, constraint-aware attack scenarios. We in troduce BOCLOAK to systematically evaluate the robustness of GNN-based bot detec tion via both edge editing and node injection adversarial attacks under realisti c constraints. BOCLOAK constructs a probability measure over spatio-temporal nei ghbor features and learns an optimal transport (OT)
geometry that separates human and bot behaviors. It then decodes transport plans into sparse, plausible edge edits that evade detection while obeying real-world constraints. We evaluate BOCLOAK across three social bot datasets, five state-o f-the-art bot detectors, three adversarial defenses, and compare it against four leading graph adversarial attack baselines. BOCLOAK achieves up to 80.13% highe r attack success rates while using 99.80% less GPU memory under real-world const raints. BOCLOAK shows that OT provides a lightweight, principled framework for b ridging adversarial attacks and real-world bot detection.
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PGD-NO: A Neural Operator with Precomputed Geometry Decomposition for 3D Million -Scale Physics Simulations
Weiheng Zhong, Jing Bi, Victor Oancea, Hadi Meidani
While neural PDE solvers have demonstrated significant potential for acceleratin g engineering simulations, existing architectures remain constrained by high mem ory consumption and the "single-node bottleneck," where the maximum processable mesh resolution is strictly limited by the VRAM of a single compute unit. To add ress these challenges, we propose PGD-NO, a neural operator with Precomputed Geo

metry Decomposition, that relocates the computational overhead of geometric enco ding to a deterministic pre-computation phase. By utilizing an iterative geometr y decomposition algorithm to extract "geometry tokens," our model decouples feat ure extraction from solution querying. This architecture enables linear memory s calability, allowing high-fidelity learning on meshes exceeding 10 million nodes —a scale where existing architectures typically encounter memory exhaustion. PGD -NO demonstrates competitive predictive accuracy across diverse industrial bench marks and provides intrinsic interpretability through attention mechanisms. By e ffectively overcoming traditional mesh-size constraints, PGD-NO offers a robust and efficient solution for the next generation of large-scale, high-fidelity ind ustrial design applications.

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Watermarking LLM Agent Trajectories

Wenlong Meng, Chen GONG, Terry Yue Zhuo, Fan Zhang, Kecen Li, Zheng Liu, Zhou Ya ng, Chengkun Wei, Wenzhi CHEN

LLM agents rely heavily on high-quality trajectory data to guide their problem-s olving behaviors, yet producing such data requires substantial task design, high -capacity model generation, and manual filtering. Despite the high cost of creat ing these datasets, existing literature has overlooked copyright protection for LLM agent trajectories. This gap leaves creators vulnerable to data theft and ma kes it difficult to trace misuse or enforce ownership rights. This paper introdu ces ActHook, the first watermarking method tailored for agent trajectory dataset s. Inspired by hook mechanisms in software engineering, ActHook embeds *hook act ions* that are activated by a secret input key and do not alter the original tas k outcome. Like software execution, LLM agents operate sequentially, allowing ho ok actions to be inserted at decision points without disrupting task flow. When the activation key is present, an LLM agent trained on watermarked trajectories can produce these hook actions at a significantly higher rate, enabling reliable black-box detection. Experiments on mathematical reasoning, web searching, and software engineering agents show that ActHook achieves an average detection AUC of 94.3 on Qwen-2.5-Coder-7B while incurring negligible performance degradation.

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Hidden in Plain Tokens: Simply Robust, Gradient-Free Watermark for Synthetic Aud io

Georgios Milis, Yubin Qin, Yihan Wu, Heng Huang

As policy catches up with the capabilities of generative AI, watermarking is cen tral to content provenance efforts. Inference-time watermarks for autoregressive models are unfit for continuous modalities due to discretization inconsistencie s. Existing methods overcome this by finetuning the modality tokenizers, nullify ing the watermark's training-free advantage. In this work, motivated by the voca bulary redundancy of discretization, we propose an elegant solution for powerful and robust watermarking of synthetic audio. We theoretically analyze the impact of token errors on watermark detection, and effectively mitigate them using a r educed vocabulary obtained via community detection. Thorough experiments showcas e that our gradient-free method can boost detectability by several orders of mag nitude, while also achieving built-in robustness to audio modifications. Broadly , we discover a new state-of-the-art for token-level watermarks in multimedia, w hich simply arises from the nature of discrete representation learning.

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MoDA: Modulation Adapter for Fine-Grained Visual Understanding in Instructional MLLMs

Wayner Barrios, Andrés Villa, Juan C Leon Alcazar, SouYoung Jin, Bernard Ghanem

Multimodal Large Language Models (MLLMs) have achieved remarkable success in ins truction-following tasks by integrating pretrained visual encoders with large la nguage models (LLMs). However, existing approaches often struggle with fine-grai ned visual grounding due to semantic entanglement in visual patch representation s, where individual patches blend multiple distinct visual elements, making it d ifficult for models to focus on instruction-relevant details. To address this ch allenge, we propose MoDA (Modulation Adapter), a lightweight module that enhance s visual grounding through instruction-guided channel-wise modulation. Unlike to

ken-level methods such as Q-Former that perform additive feature selection, MoDA operates at the channel level through multiplicative modulation on already-aligned features, enabling fine-grained control over which embedding dimensions are relevant for each instruction. Following the standard LLaVA training protocol, MoDA applies cross-attention between language instructions and pre-aligned visual features, generating dynamic modulation masks without architectural modifications or additional supervision. We evaluate MoDA across 12 benchmarks spanning visual question answering, vision-centric reasoning, and hallucination detection, including recent 2024 benchmarks (MMVP, CV-Bench, MMStar, RealWorldQA), on three distinct MLLM architectures: LLaVA-1.5, LLaVA-MoRE (2025), and Qwen3-VL (2025). MoDA delivers consistent gains across all three families, with +12.0 points on MMVP for the LLaVA-1.5 family and +4.8 points on ScienceQA for the LLaVA-MoRE family, and +4.9 ScienceQA, +4.1 RealWorldQA, and +3.8 GQA on Qwen3-VL, confirming that the gains generalize beyond CLIP-based encoders with minimal overhead (<1% FLOPs). Code is available at https://github.com/waybarrios/MoDA.
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Hi-Time: Hierarchical Latent Prediction for Multivariate Time Series Classification
Kun Zeng, Binquan Wu, Qianli Ma
Integrating Large Language Models (LLMs) into time series tasks has yielded impressive performance. While some works aim to enhance accuracy by explicitly designing step-by-step reasoning into prompts, such explicit Chain-of-Thought (CoT) approaches are difficult to generalize to time series. This is because it is difficult to clearly define the reasoning trajectories of time series. In addition, the high heterogeneity across time series often requires specialized prompt designs, limiting the model's scalability. To address these challenges, we propose **Hi-Time**, a **hi**erarchical latent prediction framework based on temporal semantic codes for multivariate **time** series classification. This framework automatically constructs scenario-specific coarse-to-fine prediction trajectories based on the characteristics of time series, thereby providing structured supervision for the LLM. Specifically, Hi-Time first performs temporal representation pre-training with a multi-view temporal representation fusion to acquire high-quality temporal embeddings. We then discretize these temporal embeddings into hierarchical temporal semantic codes that form the coarse-to-fine prediction trajectory. Finally, the LLM predicts temporal semantic codes in a stepwise manner and then infers the final label, thereby establishing a coarse-to-fine decision process. Experiments on ten public multivariate time series datasets demonstrate that Hi-Time effectively adapts to diverse datasets and outperforms state-of-the-art methods. Our code is available at <https://github.com/qianlima-lab/Hi-Time>.
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Reasoning LLM Improves Speaker Recognition in Long-form TV Dramas
Yuxuan Li, Lingxi Xie, Xinyue Huo, Jihao Qiu, Jiacheng Shao, Pengfei Chen, Jiannan Ge, Kaiwen Duan, Qi Tian
Long-form TV dramas present a formidable challenge for comprehensive video understanding, where deciphering complex storyline often relies on **speaker recognition**, the task of accurately attributing each spoken utterance to its respective character. In this paper, we advance this field through two primary contributions. (1) We introduce **DramaSR-532K**, a large-scale benchmark comprising 532K annotated dialogue lines across more than 900 unique characters, necessitating the integration of auditory, linguistic, and visual cues for speaker recognition. (2) We propose **DramaSR-LRM**, a robust approach built upon a large reasoning model (LRM). DramaSR-LRM is designed to autonomously aggregate contextual evidence via multimodal tool-use, synthesizing diverse inputs to achieve high-fidelity attribution. Experimental results demonstrate that DramaSR-LRM significantly outperforms existing baselines, particularly on short utterances where acoustic biometrics are inherently unreliable. *All the data and code will be made publicly available.*
****
Gram2Token: Enabling Run-time GPU-Native Grammar-Constrained Decoding for LLMs
Hantao Hua, Jiming Su, Hao Tang, Yiping Yao, Feng Zhu

Grammar-constrained decoding enables large language models (LLMs) to reliably ge
nerate structured outputs such as JSON, SQL, and domain-specific programs. Exist
ing systems often enforce constraints by executing byte-level parser logic insid
e the token-level decoding loop, introducing CPU-side control flow and CPU--GPU
synchronization that become bottlenecks under continuous batching. We propose Gr
am2Token, a GPU-native framework that preprocesses deterministic byte-level gram
mar execution into token-level transitions before inference. Gram2Token aligns t
okenizer byte sequences with grammar transitions through a trie and groups token
s with identical transition outcomes across preprocessed grammar states. These c
ategories yield compact validity masks and transition tables, reducing run-time
enforcement to category lookup, masking, and state update rather than parser-sty
le byte traversal. Across four model families under schema-diverse continuous ba
tching, Gram2Token achieves a geometric-mean throughput improvement of 1.38$\tim
es$ over the strongest baseline, with a maximum speedup of 1.85$\times$, at the
cost of additional preprocessing and time-to-first-token overhead. Break-even an
d grammar-complexity analyses show that this overhead is amortized by grammar re
use, longer outputs, and larger batches. These results show that token-level gra
mmar preprocessing is an effective design point for high-throughput structured L
LM serving. Code is available at https://github.com/Paradozile/Gram2Token.
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TimeMRA: LLM-Empowered Time Series Forecasting via Multi-Scale Retrieval-Augment
ed Representations
Zongjiang Shang, Chengxi Jin, Binqing Wu, Dongliang Cui, Yue Yu, Haobang Sun, Ch
uanlin Xu, Ling Chen
Time series forecasting plays a pivotal role in data-driven decision-making acro
ss various time series domains. Recently, leveraging their ability to extract se
mantically rich representations, Large Language Models (LLMs) have achieved prom
ising results in time series forecasting. However, existing LLM-based methods st
ruggle to obtain multi-scale retrieval-augmented representations due to entangle
d multi-scale representations and redundant multi-scale interference. To address
 this, we propose TimeMRA, an LLM-empowered Time series forecasting framework vi
a Multi-Scale Retrieval-Augmented representations. Specifically, a scale-aware p
rompt generation (SAPG) module is designed to decompose time series into multipl
e scales and generate augmented multi-scale representations. Then, a cross-scale
 disentanglement constraint (CSDC) mechanism with a router network is designed t
o obtain the disentangled multi-scale semantic representations while mitigating
interference from irrelevant scales. Finally, a cross-modality retrieval module
is designed to obtain multi-scale retrieval-augmented representations for time s
eries forecasting. Experiments on 10 real-world datasets demonstrate that TimeMR
A achieves state-of-the-art (SOTA) performance.
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Inference of Online Newton Methods with Nesterov's Accelerated Sketching
Haoxuan Wang, Xinchen Du, Sen Na
Reliable decision-making with streaming data requires principled uncertainty qua
ntification of online methods. While first-order methods enable efficient iterat
e updates, their inference procedures still require updating proper (covariance)
 matrices, incurring $O(d^2)$ time and memory complexity, and are sensitive to i
ll-conditioning and noise heterogeneity of the problem. This costly inference ta
sk offers an opportunity for more robust second-order methods, which are, howeve
r, bottlenecked by solving Newton systems with $O(d^3)$ complexity. In this pape
r, we address this gap by studying an online Newton method with Hessian averagin
g, where the Newton direction at each step is approximately computed using a *sk
etch-and-project solver with Nesterov's acceleration*, matching $O(d^2)$ complex
ity of first-order methods. For the proposed method, we quantify its uncertainty
 arising from both random data and randomized computation. Under standard smooth
ness and moment conditions, we establish global almost-sure convergence, prove a
symptotic normality of the last iterate with a limiting covariance characterized
 by a Lyapunov equation, and develop a fully online covariance estimator with no
n-asymptotic convergence guarantees. We also connect the resulting uncertainty q
uantification to that of exact and sketched Newton methods without Nesterov's ac

celeration. Extensive experiments on regression models demonstrate the superiority of the proposed method for online inference.
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CRPO: Character-centric Group Relative Policy Optimization for Role-aware Reasoning in Role-playing Agents
Yihong Tang, Kehai Chen, Liang Yue, Benyou Wang, Min Zhang
Recent advancements in Reinforcement Learning (RL), particularly Group Relative Policy Optimization (GRPO), have significantly enhanced the reasoning capabilities of Large Language Models. However, applying these problem-centric optimization methods to role-playing agents often leads to a loss of character fidelity and style collapse, as they prioritize context-specific utility over persona alignment. To address this, we propose Character-Centric Group Relative Policy Optimization (CRPO), a framework designed to realign RL objectives with the role-playing task. CRPO improves character distinctiveness through three mechanisms: decoupling task logic from stylistic rewards to resolve gradient conflicts, dynamically adapting optimization constraints based on character complexity, and utilizing generic responses as negative baselines to prevent the model from reverting to a common distribution. Extensive experiments demonstrate that CRPO outperforms existing methods in consistency, emotion and others.
****

FRACTAL: State Space Model with Fractional Recurrent Architecture for Computational Temporal Analysis of Long Sequences
Mengqi Li, Wensheng Lin, Jinshuai Yang, Lixin Li
Effective sequence modeling fundamentally requires balancing the retention of unbounded history with the high-resolution detection of abrupt short-term variations common in real-world phenomena. However, existing state space models (SSMs) relying on high-order polynomial projection operators (HiPPO) face a critical trade-off where uniform measures dilute recent information to maintain timescale invariance, while exponential measures sacrifice global context to capture local dynamics. This paper proposes a Fractional Recurrent Architecture for Computational Temporal Analysis of Long sequences (FRACTAL), a novel architecture integrating fractional measure theory into recursive memory updates to address this limitation. By deriving projection operators with analytically characterized spectral properties and a tunable singularity index, the proposed method amplifies sensitivity to recent signal perturbations while preserving the spectral structure that encodes scale-invariant memory dynamics. This theoretical innovation is instantiated within a simplified diagonalized state space framework by modulating input projection initialization to enable simultaneous capture of multi-scale temporal features. FRACTAL achieves an average score of 87.11\% on the Long Range Arena benchmark, including 61.85\% on the ListOps task, outperforming the S5 model.
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Contrastive Symbolic Regression: Aligned Representations, Adaptive Prediction, and Diverse Ensembles
Hengzhe Zhang, Qi Chen, Bing Xue, Wolfgang Banzhaf, Mengjie Zhang
Existing symbolic regression approaches primarily focus on learning explicit input-output mappings, often neglecting relational structures among data instances. This paper introduces Contrastive Symbolic Regression (CSR), a feature-construction-based symbolic regression approach that integrates evolutionary feature construction with contrastive learning to shape a representation space where geometric proximity reflects similarity in the target space. CSR employs a contrastive objective that optimizes a linear transformation of constructed features, with a closed-form solution for aligning the feature space with the target space. The constructed features are applied to K-nearest neighbor regression, where we propose an efficient leave-one-out cross-validation (LOOCV) method that addresses standard LOOCV's computational expense and adaptively selects the neighborhood size, along with a linear-rank weighted K-nearest neighbor variant for faithful assessment of representation quality during evolution. A determinantal point process-based ensemble selection mechanism further enhances robustness by jointly considering model quality and diversity. Extensive experiments on 58 real-world reg

ression datasets demonstrate that CSR consistently surpasses both traditional sy
mbolic regression and modern machine learning counterparts, highlighting CSR as
a promising direction for interpretable and effective regression modeling.
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Advancing LLM Reasoning with Natural Language and Numerical Feedback
Xiaoying Zhang, Yipeng Zhang, Hao Sun, Kaituo Feng, Chaochao Lu, Chao Yang, Hele
n M. Meng
Recent advances in reinforcement learning (RL) using numerical rewards have sign
ificantly enhanced the complex reasoning capabilities of large language models (
LLMs). However, we identify three fundamental limitations of purely numerical fe
edback: performance plateaus, ineffective spontaneous self-reflection, and persi
stent failures. We show that plateaued RL models can successfully refine failed
solutions when given natural language critiques. Motivated by this, we propose C
ritique-GRPO, an online RL framework that integrates both natural language and n
umerical feedback for policy optimization. This approach enables LLMs to learn s
imultaneously from initial responses and critique-guided refinements, effectivel
y internalizing the exploration benefits of both stages. Extensive experiments s
how that Critique-GRPO outperforms all compared supervised and RL-based fine-tun
ing methods, achieving average Pass@1 improvements of approximately +15.0-21.6%
on various Qwen models and +7.3% on Llama-3.2-3B-Instruct across eight challengi
ng reasoning tasks. Notably, Critique-GRPO facilitates effective self-improvemen
t through self-critiquing, achieving substantial gains over GRPO, e.g., a +16.7%
 Pass@1 improvement on AIME 2024. The code and models are released at: https://g
ithub.com/zhangxy-2019/critique-GRPO
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Scalable and Stable Estimation of Amari α-Divergence using Random Fourier
 Features
Jiaolong Wang, Fode Zhang, Lingrui Wang
Reliable estimation of Amari α-divergences underpins variational inferenc
e, yet unconstrained neural critics are notoriously prone to instability. We pro
pose a scalable estimator by constraining the critic to a Reproducing Kernel Hil
bert Space (RKHS) ball and approximating the kernel via band-limited Random Four
ier Features (RFF). This formulation yields a linear-time objective amenable to
mini-batch stochastic optimization while avoiding the cubic complexity of Gram-m
atrix methods. We present a unified analysis based on a four-term error decompos
ition—comprising RKHS approximation, feature discretization, statistical deviati
on, and optimization residual.
Under a spectral source condition, we derive non-asymptotic bounds establishing
that the RKHS approximation bias scales as $\mathcal{O}(R^{-\gamma})$, the RFF d
iscretization error as $\mathcal{O}(R D^{-1/2})$, and the statistical error as $
\mathcal{O}(R n^{-1/2})$.
We further show that statistical non-degeneracy induces intrinsic local curvatur
e, enabling our proposed Armijo-SGD to achieve local linear convergence. Empiric
al evaluations demonstrate that the RFF-RKHS estimator outperforms varying-repre
sentation baselines in stability, and applying this spectral regularization to G
AN critics significantly enhances the capture of high-frequency data components.
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DiScoFormer: Plug-In Density and Score Estimation with Transformers
Vasily Ilin, Peter Sushko, Ranjay Krishna
Estimating probability density and its score from samples remains a core problem
 in generative modeling, Bayesian inference, and kinetic theory. Existing method
s are bifurcated: classical kernel density estimators (KDE) generalize across di
stributions but suffer from the curse of dimensionality, while modern neural sco
re models achieve high precision but require retraining for every target distrib
ution. We introduce DiScoFormer (Density and Score Transformer), a ``train-once,
 infer-anywhere" equivariant Transformer that maps i.i.d. samples to both densit
y values and score vectors, generalizing across distributions and sample sizes.
Analytically, we prove that self-attention can recover normalized KDE, establish
ing it as a functional generalization of kernel methods; empirically, individual
 attention heads learn multi-scale, kernel-like behaviors. The model converges f

aster and achieves higher precision than KDE for density estimation, and provide s a high-fidelity plug-in score oracle for score-debiased KDE, Fisher informatio n computation, and Fokker-Planck-type PDEs.
****

Dimension-Independent Convergence of Underdamped Langevin Monte Carlo in KL Dive rgence
Shiyuan Zhang, Qiwei Di, Xuheng Li, Quanquan Gu
Underdamped Langevin dynamics (ULD) is a widely-used sampler for Gibbs distribut ions $\pi\propto e^{-V}$, and is often empirically effective in high dimensions. However, existing non-asymptotic convergence guarantees for discretized ULD typ ically scale polynomially with the ambient dimension d, leading to vacuous bou nds when d is large. The main known dimension-free result concerns the randomi zed midpoint discretization in Wasserstein-2 distance (Liu et al., 2023), while dimension-independent guarantees for ULD discretizations in KL divergence have r emained open. We close this gap by proving the first dimension-free KL divergenc e bounds for discretized ULD. Our analysis refines the KL local error framework (Altschuler et al., 2025) to a dimension-free setting and yields bounds that dep end on $\mathrm{tr}(\mathbf{H})$, where $\mathbf{H}$ upper bounds the Hessian of V, rather than on d. As a consequence, we obtain improved iteration complex ity for underdamped Langevin Monte Carlo relative to overdamped Langevin methods in regimes where $\mathrm{tr}(\mathbf{H})\ll d$.
****

Terminal Dimension Reduction for Time Series with Applications
Alexander Munteanu, Matteo Russo, David Saulpic, Chris Schwiegelshohn
Terminal embeddings have emerged as a powerful tool for dimension reduction. Giv en a set of points $P\subset \mathbb{R}^d$, a terminal embedding is a mapping $f :\mathbb{R}^d\rightarrow \mathbb{R}^t$ that preserves the pairwise distance betw een any pair of points $p\in P$ and $q\in \mathbb{R}^d$ up to small distortion u nder this mapping. Terminal embeddings have been particularly fruitful for const ructing k-means and k-median coresets, where the objective is to find a typi cally weighted subset Ω of P such that for any candidate solution, the cost of the clustering objective on Ω approximates the cost of the cluste ring objective on P up to small distortion. Unfortunately, these techniques ha ve not been extended to more complicated structures such as clustering time-seri es data under common straight-line interpolation between measurements. The main issue is that terminal embeddings, arguably the central technique in this line o f research, cannot be linear and are thus not immediately suitable to preserve l inear structures. In this work, we develop a generalization of terminal embeddin gs to affine line-segments that overcomes this issue. We showcase their applicab ility by using our lines-preserving terminal embeddings to obtain the first dime nsion-free coresets for clustering time-series under the Fréchet distance. The u nderlying dimension reduction uses Johnson-Lindenstrauss embeddings, and our exp eriments indicate that they compare favorably against PCA for real-world time-se ries.
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Lions and Muons: Optimization via Stochastic Frank-Wolfe under Heavy-Tailed Nois es
Maria-Eleni Sfyraki, Jun-Kun Wang
Stochastic Frank-Wolfe is a classical optimization method for solving constraine d optimization problems. On the other hand, recent optimizers such as Lion and M uon have gained quite significant popularity in deep learning. In this work, bui lding on recent initiatives, we provide a unifying perspective by interpreting t hese seemingly disparate methods through the lens of Stochastic Frank-Wolfe. Spe cifically, we show that Lion and Muon with weight decay can be viewed as special instances of a Stochastic Frank-Wolfe, and we establish their convergence guara ntees in terms of the Frank-Wolfe gap, a standard stationarity measure in non-co nvex optimization for Frank-Wolfe methods. We further find that convergence to t his gap implies convergence to a KKT point of the original problem under a norm constraint for Lion and Muon. Moreover, motivated by recent empirical findings t hat stochastic gradients in modern machine learning tasks often exhibit heavy-ta

iled distributions, we extend Stochastic Frank-Wolfe to settings with heavy-tail ed noise by developing two robust variants with strong theoretical guarantees th at hold for general compact convex sets without the need for a large batch size, filling the gap in the literature on Stochastic Frank-Wolfe for non-convex opti mization. Our contributions in the later part of this work, in turn, yield new v ariants of Lion and Muon, that better accommodate heavy-tailed gradient noise, t hereby enhancing their practical scope.
****
Convergence Rate Analysis of the AdamW-Style Shampoo: Unifying One-Sided and Two -Sided Preconditioning
Huan Li, Yiming Dong, Zhouchen Lin
This paper studies the AdamW-style Shampoo optimizer, an effective implementatio n of the classical Shampoo that notably won the external tuning track of the Alg oPerf neural network training algorithm competition. Our analysis unifies one-si ded and two-sided preconditioning and establishes the convergence rate $\frac{1} {K}\sum_{k=1}^KE[|||\nabla f(X_k)|||]\leq O(\frac{\sqrt{m+n}C}{K^{1/4}})$ measur ed by nuclear norm (denoted as $|||\cdot|||$ to display correctly in OpenReview) , where K represents the iteration number, (m,n) denotes the size of matrix parameters, and C matches the constant in the optimal convergence rate of SGD. Theoretically, we have $||\nabla f(X)||\leq|||\nabla f(X)|||\leq\sqrt{\min(m,n) }||\nabla f(X)||$ (denote $||\cdot||$ as the Frobenius norm to display correctly in OpenReview), supporting that our convergence rate can be considered to be an alogous to the optimal $\frac{1}{K}\sum_{k=1}^K E[||\nabla f(X_k)||]\leq O(\frac {C}{K^{1/4}})$ convergence rate of SGD in the ideal case of $|||\nabla f(X)|||= \Theta(\sqrt{\min(m,n)})||\nabla f(X)||$ and balanced m and n.
****
dLLM-Cache: Accelerating Diffusion Large Language Models with Adaptive Caching
Zhiyuan Liu, Yicun Yang, Yaojie Zhang, Junjie Chen, Chang Zou, Qingyan Wei, Shao bo Wang, Yichen Zhu, Linfeng Zhang
Autoregressive Models (ARMs) have long dominated the landscape of Large Language Models. Recently, a new paradigm has emerged in the form of diffusion-based Lar ge Language Models (dLLMs), which generate text by iteratively denoising masked segments. This approach has shown significant advantages and potential. However, dLLMs suffer from high inference latency. Traditional ARM acceleration techniqu es, such as Key-Value caching, are incompatible with dLLMs due to their bidirect ional attention mechanism. To address this specific challenge, our work begins w ith a key observation that dLLM inference involves a static prompt and a partial ly dynamic response, where most tokens remain stable across adjacent denoising s teps. Based on this, we propose dLLM-Cache, a training-free adaptive caching fra mework that combines long-interval prompt caching with partial response updates guided by feature similarity. This design enables efficient reuse of intermediat e computations without compromising model performance. Extensive experiments on representative dLLMs, including LLaDA 8B and Dream 7B, show that dLLM-Cache achi eves up to *9.1*$\times$ FLOPs reduction on LongBench-HotpotQA while maintaining competitive output quality. Notably, our method brings dLLM inference latency c lose to that of ARMs under many settings. *The code for this work is publicly av ailable at: https://github.com/maomaocun/dLLM-cache*.
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No Data? No Problem: Robust Vision-Tabular Learning with Missing Values
Marta Hasny, Laura Alexandra Daza, Keno Bressem, Maxime Di Folco, Julia Schnabel
Large-scale medical biobanks provide imaging data complemented by extensive tabu lar information, such as clinical measurements or demographics. However, this ab undance of tabular attributes does not reflect real-world datasets, where only a subset of attributes may be available. This discrepancy calls for methods that remain robust to missing values at inference. To address this challenge, we prop ose RoVTL (Robust Vision-Tabular Learning), a framework designed to handle any l evel of tabular data availability, from 0% to 100%. RoVTL comprises two key stag es: contrastive pretraining, where we introduce tabular attribute missingness as data augmentation to promote robustness, and downstream task tuning, where tabu lar missingness is complemented by a novel Tabular More vs. Fewer loss that rank

s performance based on the amount of available tabular data. Combined with gated
-cross attention fusion module,
our tuning approach enables consistent performance across all tabular data compl
eteness scenarios. We evaluate RoVTL on cardiac MRI scans from the UK Biobank, d
emonstrating superior robustness to missing tabular data compared to prior metho
ds. Furthermore, RoVTL successfully generalizes to an external cardiac MRI datas
et for multimodal disease classification, and extends to the natural images doma
in, achieving robust performance on a car advertisements dataset. The model weig
hts and code are available at https://github.com/marteczkah/RoVTL.
****

Deep Forcing: Training-Free Long Video Generation with Deep Sink and Participati
ve Compression
Jung Yi, Wooseok Jang, Paul Hyunbin Cho, Jisu Nam, Heeji Yoon, Seungryong Kim
Recent advances in autoregressive video diffusion have enabled real-time frame s
treaming, however, existing methods still suffer from visual error accumulation
including visual fidelity and motion degradation over long-horizon. To address t
hese challenges, we introduce Deep Forcing, a training-free extension of autoreg
ressive video diffusion models that stabilizes long video generation through two
 complementary mechanisms. Deep Sink preserves approximately half of the sliding
 context window as persistent sink tokens and realigns their temporal RoPE phase
s to the current timeline, thereby maintaining global context during extended ro
llouts. Participative Compression performs importance-aware KV cache pruning, re
taining only tokens that actively participate in recent attention while removing
 redundant or degraded history, effectively mitigating error accumulation under
out-of-distribution lengths. Together, these components enable over 12× length e
xtrapolation (e.g., 5s-trained → 60s+) without sacrificing inference speed, whil
e improving visual fidelity and motion dynamics compared to prior methods. Our r
esults demonstrate that Deep Forcing can achieve performance comparable to state
-of-the-art training-based methods trained specifically for long video generatio
n.
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ChaosNexus: A Foundation Model for ODE-based Chaotic System Forecasting with Hie
rarchical Multi-scale Awareness
Chang Liu, Bohao Zhao, Jingtao Ding, Yong Li
Foundation models show great promise in achieving zero-shot or few-shot forecast
ing for ODE-based chaotic systems via large-scale pretraining. However, existing
 architectures often fail to capture the multi-scale temporal structures and dis
tinct spectral characteristics of chaotic dynamics. To address this, we introduc
e ChaosNexus, a foundation model for chaotic system forecasting underpinned by t
he proposed ScaleFormer architecture. By processing temporal contexts across hie
rarchically varying patch sizes, ChaosNexus effectively captures long-range depe
ndencies and preserves high-frequency fluctuations. To address heterogeneity acr
oss distinct systems, we integrate Mixture-of-Experts (MoE) layers into each Sca
leFormer block and explicitly condition the final forecasts on a learned frequen
cy fingerprint, providing the model with a global spectral view of the system. E
xtensive evaluations on over 9,000 synthetic systems demonstrate that ChaosNexus
 achieves superior fidelity in long-term attractor statistics while maintaining
competitive point-wise accuracy. Furthermore, in real-world applications, it ach
ieves a remarkable zero-shot mean error below 1°C for 5-day station-based weathe
r forecasting. Code is available at https://github.com/tsinghua-fib-lab/ChaosNex
us.
****

SPEED: Sharpened-Teacher Distillation for Parallel Decoding of Diffusion Languag
e Models
Qiuhong Shen, Xingyi Yang, Xinyin Ma, Gongfan Fang, Xinchao Wang
Diffusion-based large language models generate text by gradually filling in mask
ed tokens, yet they remain slow because they usually decode only a few tokens pe
r step. Parallel decoding, which unmasks multiple tokens simultaneously, promise
s acceleration but often degrades quality when too many tokens are predicted at
once. We identify the root cause: when decoding is viewed as iterative token gro

uping, overly permissive grouping places interdependent tokens in the same step, violates the conditional independence assumption, and amplifies reliance on noisy context even when the top prediction is already correct. We introduce SPEED, a framework that enlarges safe parallel groups through complementary training and inference designs. At training time, a sharpened teacher distillation objective selectively aligns the student to teacher-correct positions using a temperature-scaled KL term together with a masked language modeling loss, producing a student that assigns more probability mass to correct token identities and elevates more positions above the decoding threshold. At inference time, Slow-Fast Decoding partitions tokens by sensitivity to revealed context using token-wise Jensen-Shannon Divergence computed with and without access to the preceding block, decoding low-sensitivity tokens jointly in parallel while deferring high-sensitivity tokens until sufficient context resolves them. Through extensive experiments, our framework attains up to 12.2$\times$ speedup on LLaDA-8B-Instruct and 6.7$\times$ on Dream-7B-Instruct with accuracy close to greedy decoding across standard reasoning and code benchmarks.
****

Benign Overfitting in Adversarial Training for Vision Transformers
Jiaming Zhang, Meng Ding, Shaopeng Fu, Jingfeng Zhang, Di Wang
Despite the remarkable success of Vision Transformers (ViTs) across a wide range of vision tasks, recent studies have revealed that they remain vulnerable to adversarial examples, much like Convolutional Neural Networks (CNNs). A common empirical defense strategy is adversarial training, yet the theoretical underpinnings of its robustness in ViTs remain largely unexplored. In this work, we present the first theoretical analysis of adversarial training under simplified ViT architectures. We show that, when trained under a signal-to-noise ratio that satisfies a certain condition and within a moderate perturbation budget, adversarial training enables ViTs to achieve nearly zero robust training loss and robust generalization error under certain regimes. Remarkably, this leads to strong generalization even in the presence of overfitting, a phenomenon known as benign overfitting, previously only observed in CNNs (with adversarial training). Experiments on both synthetic and real-world datasets further validate our theoretical findings.
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Many Experiments, Few Repetitions, Unpaired Data, and Sparse Effects: Is Causal Inference Possible?
Felix Schur, Niklas Pfister, Peng Ding, Sach Mukherjee, Jonas Peters
In many applications, practical constraints prevent measuring covariates and outcomes on the same units, resulting in unpaired data. We study the problem of estimating causal effects under hidden confounding in the following unpaired data setting: we observe some covariates X and an outcome Y under different experimental conditions (environments) but do not observe them jointly -- we either observe X or Y. Under appropriate regularity conditions, the problem can be cast as an instrumental variable (IV) regression with the environment acting as a (possibly high-dimensional) instrument. When there are many environments but only a few observations per environment, standard two-sample IV estimators fail to be consistent. We propose a GMM-type estimator based on cross-fold sample splitting of the instrument–covariate sample that also applies in standard IV settings. We prove that it is consistent as the number of environments grows but the sample size per environment remains constant. We further extend the method to sparse causal effects via ℓ_1-regularized estimation and post-selection refitting.
****

Theoretical Challenges in Learning for Branch-and-Cut
Hongyu Cheng, Amitabh Basu
Machine learning is increasingly used to guide branch-and-cut (B&C) for mixed-integer linear programming by learning score-based policies for selecting branching variables and cutting planes. Many approaches train on local signals from look ahead heuristics such as strong branching, and linear programming (LP) bound improvement for cut selection. Training and evaluation of the learned models often

focus on local score accuracy. We show that such local score-based methods can l
ead to search trees exponentially larger than optimal tree sizes, by identifying
 two sources of this gap. The first is that these widely used expert signals can
 be misaligned with overall tree size. LP bound improvement can select a root cu
t set that yields an exponentially larger strong branching tree than selecting c
uts by a simple proxy score, and strong branching itself can be exponentially su
boptimal (Dey et al., 2024). The second is that small discrepancies can be ampli
fied by the branch-and-bound recursion. An arbitrarily small perturbation of the
 right-hand sides in a root cut set can change the minimum tree size from a sing
le node to exponentially many. For branching, arbitrarily small score discrepanc
ies, and differences only in tie-breaking, can produce trees of exponentially di
fferent sizes, and even a small number of decision differences along a trajector
y can incur exponential growth. These results show that branch-and-cut policies
trained and learned using local expert scores do not guarantee small trees, thus
 motivating the study of data-driven methods that produce policies better aligne
d with tree size rather than only accuracy on expert scores.
****

LSGQuant: Layer-Sensitivity Guided Quantization for One-Step Diffusion Real-Worl
d Video Super-Resolution
Tianxing Wu, Zheng Chen, Cirou Xu, Bowen Chai, Yong Guo, Yutong Liu, Linghe Kong
, Yulun Zhang
One-step diffusion models have demonstrated a promising capability and fast infe
rence in video super-resolution (VSR) for real-world applications. Nevertheless,
 the substantial model size and high computational cost of Diffusion Transformer
s (DiTs) limit downstream applications. While low-bit quantization is a common a
pproach for model compression, the effectiveness of quantized models is challeng
ed by the high dynamic range of input latent and diverse layer behaviors. To add
ress these challenges, we introduce LSGQuant, a layer-sensitivity-guided quantiz
ation approach for one-step diffusion-based real-world VSR. Our method incorpora
tes a Dynamic Range Adaptive Quantizer (DRAQ) to fit video token activations. Fu
rthermore, we estimate layer sensitivity and implement a Variance-Oriented Layer
 Training Strategy (VOLTS) by analyzing layer-wise statistics in calibration. We
 also introduce Quantization-Aware Optimization (QAO) to jointly refine the quan
tized branch and a retained high-precision branch. Extensive experiments demonst
rate that our method has nearly the performance of the original model with full-
precision and significantly exceeds existing quantization techniques. All models
 and code are available at https://github.com/zhengchen1999/LSGQuant.
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Error Amplification Limits ANN-to-SNN Conversion in Continuous Control
Zijie Xu, Zihan Huang, Yiting Dong, Kang Chen, Wenxuan Liu, Zhaofei Yu
Spiking Neural Networks (SNNs) can achieve competitive performance by converting
 already existing well-trained Artificial Neural Networks (ANNs), avoiding furth
er costly training. This property is particularly attractive in Reinforcement Le
arning (RL), where training through environment interaction is expensive and pot
entially unsafe. However, existing conversion methods perform poorly in continuo
us control, where suitable baselines are largely absent. We identify error ampli
fication as the key cause: small action approximation errors become temporally c
orrelated across decision steps, inducing cumulative state distribution shift an
d severe performance degradation. To address this issue, we propose Cross-Step R
esidual Potential Initialization (CRPI), a lightweight gradient-free mechanism t
hat carries over residual membrane potentials across decision steps to suppress
temporally correlated errors. Experiments on continuous control benchmarks with
both vector and visual observations demonstrate that CRPI can be integrated into
 existing conversion pipelines and substantially recovers lost performance. Our
results highlight continuous control as a critical and challenging benchmark for
 ANN-to-SNN conversion, where small errors can be strongly amplified and impact
performance. Code is available at https://github.com/xuzijie32/ANN2SNN-CRPI.
****

MemPot: Defend Against Memory Extraction Attack with Optimized Honeypots
Yuhao Wang, Shengfang Zhai, Guanghao Jin, Yinpeng Dong, Linyi Yang, Jiaheng Zhan

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Large Language Model (LLM)-based agents employ external and internal memory systems to handle complex, goal-oriented tasks, yet this exposes them to severe extraction attacks, and corresponding defenses are currently lacking.
In this paper, we propose *MemPot*, the first theoretically verified defense framework against memory extraction attacks by injecting optimized honeypots into the memory.
Through a two-stage optimization process, *MemPot* generates trap documents that maximize the retrieval probability for attackers while remaining inconspicuous to benign users.
We model the detection process as Wald’s Sequential Probability Ratio Test (SPRT) and theoretically prove that *MemPot* achieves a lower average number of sampling rounds compared to optimal static detectors.
Empirically, *MemPot* significantly outperforms state-of-the-art baselines, achieving a 50% improvement in detection AUROC and an 80% increase in True Positive Rate under low False Positive Rate constraints.
Furthermore, our experiments confirm that *MemPot* incurs zero online inference latency and preserves the agent's utility on standard tasks, verifying its superiority in safety, harmlessness and efficiency.
****
Enhancing Protein-Protein Interaction Prediction with Hierarchical Motif-based Multimodal Protein Embedding
Zaifei YANG, Samuel Ping-Man Choi, James Kwok
Protein-protein interactions (PPIs) are essential for many biological processes. However, existing PPI prediction approaches suffer from two major limitations: they overlook the hierarchical organization of proteins, particularly meso-scale motifs that critically regulate PPIs, and fail to effectively integrate sequence, structure, and function modalities. To address these limitations, we propose MMM-PPI, a Hierarchical Motif-based Multi-Modal protein Encoder for PPI Prediction that constructs PPI embeddings in a bottom-up multi-modal manner across three scales. At the micro-scale, we encode three modal residue features; at the meso-scale, a novel multimodal motif encoder aggregates residues into spatially-informed motif embeddings; at the macro-scale, a multimodal protein encoder integrates motifs into protein embeddings by jointly modeling motif importance and inter-modal correlations. The pre-trained encoder can be used off-the-shelf for large-scale PPI prediction. Extensive experiments on multiple PPI datasets show that MMM-PPI outperforms state-of-the-art multi-label PPI prediction models, particularly under challenging data partitions and limited data scenarios. Codes are in https://github.com/yzf-code/MMM-PPI.
****
Multi-Way Representation Alignment
Akshit Achara, Tatiana Gaintseva, Matéo Mahaut, Pritish Chakraborty, Viktor Stenby Johansson, Melih Barsbey, Emanuele Rodolà, Donato Crisostomi
The Platonic Representation Hypothesis suggests that independently trained neural networks converge to increasingly similar latent spaces. However, current strategies for mapping these representations are inherently pairwise, scaling quadratically with the number of models and failing to yield a consistent global reference. In this paper, we study the alignment of $M \ge 3$ models. We first adapt Generalized Procrustes Analysis (GPA) to construct a shared orthogonal universe that preserves the internal geometry essential for tasks like model stitching. We then show that strict isometric alignment is suboptimal for retrieval, where agreement-maximizing methods like Canonical Correlation Analysis (CCA) typically prevail. To bridge this gap, we finally propose Geometry-Corrected Procrustes Alignment (GCPA), which establishes a robust GPA-based universe followed by a post-hoc correction for directional mismatch. Extensive experiments demonstrate that GCPA consistently improves any-to-any retrieval while retaining a practical shared reference space.
****
Imposing Boundary Conditions on Neural Operators via Learned Function Extensions
Sepehr Mousavi, Siddhartha Mishra, Laura De Lorenzis

Neural operators have emerged as powerful surrogates for the solution of partial differential equations (PDEs), yet their ability to handle general, highly variable boundary conditions (BCs) remains limited. Existing approaches often fail when the solution operator exhibits strong sensitivity to boundary forcings. We propose a general framework for conditioning neural operators on complex non-homogeneous BCs through function extensions. Our key idea is to map boundary data to latent pseudo-extensions defined over the entire spatial domain, enabling any standard operator learning architecture to consume boundary information. The resulting operator, coupled with an arbitrary domain-to-domain neural operator, can learn rich dependencies on complex BCs and input domain functions at the same time. To benchmark this setting, we construct 18 challenging datasets spanning Poisson, linear elasticity, and hyperelasticity problems, with highly variable, mixed-type, component-wise, and multi-segment BCs on diverse geometries. Our approach achieves state-of-the-art accuracy, outperforming baselines by large margins, while requiring no hyperparameter tuning across datasets. Overall, our results demonstrate that learning boundary-to-domain extensions is an effective and practical strategy for imposing complex BCs in existing neural operator frameworks, enabling accurate and robust scientific machine learning models for a broader range of PDE-governed problems.
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Equivariant Latent Alignment via Flow Matching under Group Symmetries
Sunghyun Kim, Jaehoon Hahm, Jeongwoo Shin, Joonseok Lee
Geometry-aware generative models and novel view synthesis approaches have shown strong potential in visual fidelity and consistency. In parallel, equivariant representation learning has emerged as a powerful framework for constructing latent spaces where analytically known group transformations could act directly, capturing geometric structure in data and enhancing both interpretability and generalization in novel view synthesis. However, we identify that existing approaches often suffer from latent misalignment, a discrepancy between the intended group action and the actually required transformations in the latent space. Consequently, the learned latents often fail to consistently preserve the equivariant relations imposed by the underlying group symmetry. To address this, we propose Residual Latent Flow, a flow-based framework that corrects the misaligned latents, thereby improving compliance with the underlying equivariance relation. Our comprehensive experiments show that our method significantly reduces latent misalignment and improves novel view synthesis quality, under rotation groups SO(n).
****

Evaluating Robustness of Reasoning Models on Parameterized Logical Problems
Naïm Es-sebbani, Esteban Marquer, Yakoub Salhi, Zied Bouraoui
Logic provides a controlled testbed for evaluating LLM-based reasoners, yet standard SAT-style benchmarks often conflate surface difficulty (length, wording, clause order) with the structural phenomena that actually determine satisfiability. We introduce a diagnostic benchmark for 2-SAT built from parameterized families of structured 2-CNF formulas, where satisfiability is characterized by the implication graph and can be tuned along interpretable axes. Our generators isolate distinct competencies and failure modes: (i) contradiction-cycle UNSAT cores with controllable size and imbalance, (ii) SAT instances with a prescribed fraction of free variables to control solution multiplicity, (iii) planted backbones that modulate propagation, (iv) late bridge clauses that couple otherwise monotone regions to probe sensitivity to ordering and revision, and (v) symmetry/duplication variants that test abstraction under renaming and redundant structure. We evaluate LLM-based reasoners on decision accuracy and assignment validity, and quantify robustness under semantics-preserving perturbations such as clause reordering, filler clauses, and variable renaming. Across models, we observe sharp performance transitions under targeted structural interventions even when surface statistics are held fixed, revealing brittleness regimes that are invisible to aggregate SAT accuracy.
****

RAMAC: Multimodal Risk-Aware Offline Reinforcement Learning and the Role of Behavior Regularization

Kai Fukazawa, Kunal Mundada, Iman Soltani
In safety-critical domains where online data collection is infeasible, offline r
einforcement learning (RL) is attractive only if policies achieve high returns w
ithout catastrophic lower-tail risk. Prior work on risk-averse offline RL achiev
es safety at the cost of value- or model-based pessimism, and restricted policy
classes that limit policy expressiveness, whereas diffusion/flow-based expressiv
e generative policies have largely been used in risk-neutral settings.
We introduce **Risk-Aware Multimodal Actor-Critic (RAMAC)**, a simple, modular,
model-free framework that couples an expressive generative actor (e.g., diffusio
n/flow) with a distributional critic and optimizes a composite objective that co
mbines Conditional Value-at-Risk (CVaR) with behavioral cloning (BC), enabling r
isk-sensitive learning in complex multimodal scenarios. Since out-of-distributio
n (OOD) actions are a major driver of catastrophic failures in offline RL, we fu
rther provide an objective-level analysis showing that controlling behavior dive
rgence via BC suppresses OOD actions and stabilizes CVaR. Instantiating RAMAC wi
th a diffusion actor, we illustrate these insights on a 2-D risky bandit and eva
luate on Stochastic-D4RL, observing consistent gains in CVaR$_{0.1}$ while maint
aining strong returns.
****

Mind the Gap: Catching Hallucinations via Evidence Drop on the Reasoning Manifol
d
Qunjie Chen, Yufei Chen, Xiaodong Yue, Linye Li
Large Language Models (LLMs) show strong reasoning abilities, yet their reliabil
ity is hindered by hallucinations, where fluent reasoning becomes factually or l
ogically incorrect.
Most existing uncertainty-based detectors rely on sequence-level averaging, whic
h ignores the step-wise dynamics of reasoning and often misclassifies hard-but-c
orrect or easy-but-wrong samples.
We propose a dynamic perspective that models reasoning as a trajectory on a late
nt Evidence Manifold, where each step is supported by local evidence.
Hallucinations are characterized as Evidence Drops, i.e., sudden declines in loc
al evidence support that indicate topological deviations from this manifold.
Based on this insight, we design a training-free and model-agnostic detector tha
t identifies hallucinations via the worst-case Evidence Drop and enables step-le
vel error localization.
Experiments on GSM8K, MATH, and ProcessBench show consistent improvements over s
equence-level uncertainty baselines in selective accuracy and risk–coverage trad
e-offs.
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Variational Learning of Disentangled Representations
Yuli Slavutsky, Ozgur Beker, David Blei, Bianca Dumitrascu
Disentangled representations separate factors that are shared across conditions
from those that are condition-specific. Such separation is needed for generaliza
tion to new domains, treatments, patients, or species. A dominant line of work p
ursues this goal through variational formulations. While these approaches achiev
e partial disentanglement, they often exhibit three common limitations: they eit
her do not remove all condition-specific information from the shared representat
ion, allow the shared representation to become uninformative, or impose independ
ence assumptions that do not reflect the underlying generative process. In this
work, we introduce DisCoVR, a variational framework that addresses these limitat
ions. Its objective is aligned with the probabilistic structure of the data-gene
rating process, and includes an adversarial term that prevents condition-specifi
c information from being encoded in the shared representation. DisCoVR reconstru
cts the data from both shared and condition-specific representations, ensuring t
hat each remains informative, and uses a structured prior that further reinforce
s the informativeness of both representations. We show that across synthetic, im
age, and single-cell RNA-sequencing datasets, DisCoVR achieves stronger disentan
glement compared to previous approaches.
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CAMEL: Confidence-Gated Reflection for Reward Modeling

Zirui Zhu, Hailun Xu, Yang Luo, Yong Liu, Kanchan Sarkar, Kun Xu, Yang You
Reward models play a fundamental role in aligning large language models with hum an preferences. Existing methods predominantly follow two paradigms: scalar disc riminative preference models, which are efficient but lack interpretability, and generative judging models, which offer richer reasoning at the cost of higher c omputational overhead.
We observe that the log-probability margin between verdict tokens strongly corre lates with prediction correctness, providing a reliable proxy for instance diffi culty without additional inference cost.
Building on this insight, we propose CAMEL, a confidence-gated reflection framew ork that performs a lightweight single-token preference decision first and selec tively invokes reflection only for low-confidence instances. To induce effective self-correction, we train the model via reinforcement learning with counterfact ual prefix augmentation, which exposes the model to diverse initial verdicts and encourages genuine revision.
Empirically, CAMEL achieves state-of-the-art performance on three widely used re ward-model benchmarks with 82.9% average accuracy, surpassing the best prior mod el by 3.2% and outperforming 70B-parameter models using only 14B parameters, whi le establishing a strictly better accuracy-efficiency Pareto frontier.
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Modeling Long-Tail Relations in the Operating Room via In-Context Multimodal Lea rning
Boqiang Xu, Wei Zhang, Ding Ma, Jian Liang, Zhenan Sun, Zhen Lei
Operating room (OR) scene graph generation (SGG) enables holistic modeling of OR domains by encoding interactions among medical staff, tools, and equipment as t riplet-based structured scene graphs. Although existing OR SGG methods demonstra te satisfactory overall performance, they exhibit substantially lower accuracy o n long-tail categories compared to head categories in OR data. We introduce SGG-ICL, a novel framework that represents the first attempt to address the long-tai l problem in OR SGG by leveraging in-context learning (ICL). SGG-ICL first ident ifies long-tail samples via an Adaptive Router module and selectively applies IC L only to these samples. This selective routing strategy enhances performance on long-tail categories without degrading head-category accuracy. Subsequently, SG G-ICL constructs a candidate pool through multimodal retrieval and then employs a trained MLLM Reranker to re-rank the candidates, selecting the most similar ex amples to the test sample for ICL. The reranker is supervised by IoU scores deri ved from annotated SGG triplets and exploits rich multimodal information to esti mate pairwise sample similarity. Experimental results show that SGG-ICL improves accuracy on long-tail categories by 6.9%, while also achieving a 2.6% improveme nt in overall accuracy.
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Hide and Seek in Embedding Space: Geometry-based Steganography and Detection in Large Language Models
Charles Westphal, Keivan Navaie, Fernando Rosas
Fine-tuned LLMs can covertly encode prompt secrets into outputs via steganograph ic channels. Prior work demonstrated this threat but relied on trivially recover able encodings. We formalize payload recoverability via classifier accuracy and show previous schemes achieve 100\% recoverability. In response, we introduce l ow-recoverability steganography, replacing arbitrary mappings with embedding-spa ce-derived ones. For Llama-8B (LoRA) and Ministral-8B (LoRA) trained on TrojanSt ego prompts, exact secret recovery rises from 17$\rightarrow$30\% (+78\%) and 24 $\rightarrow$43\% (+80\%) respectively, while on Llama-70B (LoRA) trained on Wik i prompts, it climbs from 9$\rightarrow$19\% (+123\%), all while reducing payloa d recoverability. We then discuss detection. We argue that detecting fine-tuning -based steganographic attacks requires approaches beyond traditional steganalysi s. Standard approaches measure distributional shift, which is an expected side-e ffect of fine-tuning. Instead, we propose a mechanistic interpretability approac h: linear probes trained on later-layer activations detect the secret with up to 33\% higher accuracy in fine-tuned models compared to base models, even for low -recoverability schemes. This suggests that malicious fine-tuning leaves actiona

ble internal signatures amenable to interpretability-based defenses.
****
Explainable Federated Learning via Global–Local Attribution Alignment
Dawood Wasif, Terrence J Moore, Chang-Tien Lu, Jin-Hee Cho
Federated learning enables on-device training without centralizing data, yet existing systems still struggle to provide explanations that are both locally faithful and globally consistent under strict privacy and bandwidth constraints. Prior approaches either keep explanations siloed across clients, transmit heavy or sensitive artifacts, or replace expressive task models with interpretable surrogates that sacrifice accuracy. We propose xFedAlign, a model-agnostic framework that decouples task optimization in parameter space from explanation coordination in a compact group space. Each client distills a lightweight surrogate to produce private, per-class top-k attribution artifacts, which are robustly aggregated by the server into a Global Explanation Prior that softly aligns client explanations without constraining task learning. Across image, text, and tabular benchmarks with IID and non-IID partitions, xFedAlign matches FedAvg accuracy while consistently reducing explanation drift and improving deletion and insertion AUC relative to Local-XAI, FedAttr-Agg, and Fed-XAI, with only a few kilobytes of additional communication per round. Privacy and robustness evaluations further demonstrate reduced membership inference advantage and increased resistance to attribution poisoning, enabling consistent and trustworthy explanations in federated learning.
****
PRISM: Demystifying Retention and Interaction in Mid-Training
Bharat Runwal, Ashish Sunil Agrawal, Anurag Roy, Rameswar Panda
Mid-training is increasingly used to improve the reasoning capabilities of large language models (LLMs), yet its design choices and interaction with evaluation and reinforcement learning (RL) remain poorly understood. Prior work often focuses on narrow domain gains, overlooking retention of general abilities, long-context performance, and RL compatibility. We present $\textbf{PRISM}$ (Demystifying Retention and Interaction in Mid-Training), a holistic empirical study that analyzes mid-training design choices, what to evaluate, and how domain mixtures and training stages interact across model families. Experiments on Granite-3.3 8B, LLaMA-3.1 8B, and Mistral-7B/24B base models show that a relatively small, high-quality mid-training phase of $\textbf{$\sim$27B}$ tokens acts as a critical stabilizing stage for reasoning. Across models, PRISM yields consistent gains of $\textbf{$\sim$6–10}$ points on coding benchmarks and $\textbf{$\sim$17–30}$ points on mathematical reasoning benchmarks while preserving general performance. RL applied on top of PRISM-mid-trained models produces stable, monotonic improvements, adding a further $\textbf{$\sim$3–8}$ points across coding and math tasks such as LiveCodeBench, Codeforces, AIME and MATH500, and $\textbf{$\sim$17–20}$ points on science (GPQA-Diamond), whereas RL applied directly to base models is substantially less effective. Our results demonstrate that retention-aware mid-training is a necessary intermediate step for reliable reasoning enhancement and RL scaling, and provide practical guidance for designing robust mid-training pipelines for modern LLMs.
****
Sharper Generalization Guarantees for Asynchronous SGD: Beyond Lipschitzness, Smoothness and Data Homogeneity
Xie Yufeng, Yunwen Lei
Asynchronous stochastic gradient descent (ASGD) is widely adopted in distributed and federated learning. In this paper, we develop a sharp generalization analysis for ASGD by leveraging the concept of on-average model stability. For convex and smooth objectives, we establish stability and excess risk bounds under minimal assumptions, removing Lipschitz continuity, bounded noise, bounded parameter or data domains, while allowing randomly partitioned data and arbitrary delays. Our bounds are optimistic and explicitly characterize the impact of worker participation, recovering the minimax-optimal rate $O(1/\sqrt{mn})$ in balanced regimes where mn denotes the sample size and implying fast rates under low-noise conditions. We further extend the analysis to non-smooth objectives with Hölder-co

ntinuous gradients and to heterogeneous data settings via random ASGD, obtaining non-vacuous excess risk guarantees in both settings. Experimental results support our theoretical findings.

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Finding Most Influential Sets

Lucas Darius Konrad, Nikolas Kuschnig

Identifying _most influential sets_ (MIS) – size-k subsets whose removal maximally changes a target estimand – is typically infeasible because it requires searching over n choose k subsets.

For estimands with linear-fractional leave-set-out effects, we show that MIS selection reduces to a one-parameter sequence of top-k problems. Dinkelbach's method yields an algorithm with $\mathcal{O}(n)$ cost per iteration and finite termination.

For fixed residualized inputs, the algorithm returns a globally optimal set for the univariate ratio objective, including the oracle-residualized partial linear model. With estimated nuisance functions, uniform denominator and generated-score stability imply approximation to the first-order oracle orthogonal-score objective; exact set recovery follows under a separation condition.

Simulations and applications show that the method recovers exact MIS that were previously computationally inaccessible.

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SCOPE: Selective Conformal Optimized Pairwise LLM Judging

Sher Badshah, Ali Emami, Hassan Sajjad

Large language models (LLMs) are increasingly used as scalable judges in pairwise evaluation, but they remain prone to miscalibration and biases. We propose SCOPE (Selective Conformal Optimized Pairwise Evaluation), a framework that calibrates an acceptance threshold so that, under exchangeability, the error rate among non-abstained judgments is at most a user-specified level α. To supply SCOPE with a bias-neutral uncertainty signal, we introduce Bidirectional Preference Entropy (BPE), which queries the judge under both response positions and converts the order-averaged preference probability into an entropy-based score. Across various pairwise judging benchmarks, BPE outperforms standard confidence proxies in calibration and discrimination, while SCOPE consistently satisfies the target risk bound (empirical FDR ≈ 0.097--0.099 at $\alpha=0.10$) and retains substantial coverage. Compared to vanilla baselines, SCOPE accepts up to $2.4\times$ more judgments under the same risk constraint, demonstrating that BPE enables reliable and high-coverage LLM-based evaluation.

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A Regret Minimization Framework on Preference Learning in Large Language Models

Suhwan Kim, Taehyun Cho, Geon-Hyeong Kim, Yu Jin Kim, Youngsoo Jang, Moontae Lee, Jungwoo Lee

Reinforcement learning with verifiable rewards (RLVR) has enabled progress on reasoning-intensive tasks by relying on task-specific verifiers that provide automated correctness signals. However, many realistic language tasks are difficult to equip with reliable verifiers, motivating a growing reliance on reinforcement learning from human feedback (RLHF).

In this setting, we argue that a closer examination of how human feedback should be interpreted is essential.

We introduce Regret-based Preference Optimization (RePO), which reframes RLHF through *regret minimization* rather than reward maximization.

Human preferences are often shaped by *prospective* anticipation of outcomes and *counterfactual* comparisons to alternative behaviors, rather than by immediate, outcome-independent utility.

RePO captures this structure by modeling preferences as behavior-conditioned assessments of relative suboptimality.

Within a KL-regularized reinforcement learning framework, RePO admits a closed-form policy update compatible with direct preference optimization.

Experiments on mathematical reasoning benchmarks and human-annotated preference datasets demonstrate consistent performance gains, indicating that regret-based preference learning is an effective and human-aligned approach for training larg

e language models.
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A Geometry-Based View of Mahalanobis OOD Detection
Denis Janiak, Jakub Binkowski, Tomasz Jan Kajdanowicz
Out-of-distribution (OOD) detection is critical for reliable deployment of visio n models, and Mahalanobis-based detectors remain strong baselines. However, thei r performance varies widely across modern pretrained representations, making it unclear which feature-space properties determine success or failure. Across dive rse foundation-model backbones and Mahalanobis variants, we show that performanc e is highly representation-dependent and can shift substantially with pretrainin g data and fine-tuning. We link this variability to in-distribution geometry and identify a two-term ID summary that consistently tracks Mahalanobis OOD behavio r across detectors: within-class spectral structure and local intrinsic dimensio nality. Finally, we introduce radially scaled ℓ_2 normalization, $\phi_\bet a(z)=z/\|z\|^\beta$, a direction-preserving transformation that changes feature radii and exposes a different ID geometry to the same quadratic detector. Select ing β from ID-only geometry signals generally outperforms fixed normalizat ion baselines.
****
Harnessing Reasoning Trajectories for Hallucination Detection via Answer-agreeme nt Representation Shaping
Jianxiong Zhang, Bing Guo, Yuming Jiang, Haobo Wang, Bo An, Xuefeng Du
Large reasoning models (LRMs) often generate long, seemingly coherent reasoning traces yet still produce incorrect answers, making hallucination detection chall enging. Although trajectories contain useful signals, directly using trace text or vanilla hidden states for detection is brittle: traces vary in form and detec tors can overfit to superficial patterns rather than answer validity. We introdu ce Answer-agreement Representation Shaping (ARS), which learns detection-friendl y trace-conditioned representations by explicitly encoding answer stability. ARS generates counterfactual answers through small latent interventions, specifical ly, perturbing the trace-boundary embedding, and labels each perturbation by whe ther the resulting answer agrees with the original. It then learns representatio ns that bring answer-agreeing states together and separate answer-disagreeing on es, exposing latent instability indicative of hallucination risk. The shaped emb eddings are plug-and-play with existing embedding-based detectors and require no human annotations during training. Experiments demonstrate that ARS consistentl y improves detection and achieves substantial gains over strong baselines.
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Reusing Trajectories in Policy Gradients Enables Fast Convergence
Alessandro Montenegro, Federico Mansutti, Marco Mussi, Matteo Papini, Alberto Ma ria Metelli
Policy gradient (PG) methods are a class of effective *reinforcement learning* algorithms, particularly when dealing with continuous control problems. They re ly on fresh *on-policy* data, making them sample-inefficient and requiring $\mat hcal{O}(\epsilon^{-2})$ trajectories to reach an ϵ-approximate stationa ry point. A common strategy to improve efficiency is to *reuse* information from past iterations, such as previous *gradients* or *trajectories*, leading to *of f-policy* PG methods. While gradient reuse has received substantial attention, l eading to improved rates up to $\mathcal{O}(\epsilon^{-3/2})$, the reuse of past trajectories, although intuitive, remains largely unexplored from a theoretical perspective.
In this work, we provide the first rigorous theoretical evidence that reusing pa st off-policy trajectories can significantly accelerate PG convergence. We propo se RT-PG (Reusing Trajectories - Policy Gradient), a novel algorithm that levera ges a *power mean*-corrected multiple importance weighting estimator to effectiv ely combine on-policy and off-policy data coming from the most recent ω i terations. Through a novel analysis, we prove that RT-PG achieves a sample compl exity of $\widetilde{\mathcal{O}}(\epsilon^{-2}\omega^{-1})$.
When reusing *all* available past trajectories, this leads to a rate of $\wideti lde{\mathcal{O}}(\epsilon^{-1})$, the best known

one in the literature for PG methods. We further validate our approach empirical
ly, demonstrating its effectiveness against baselines with state-of-the-art rate
s.
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Stable Deep Reinforcement Learning via Isotropic Gaussian Representations
Ali Saheb Pasand, Johan Obando-Ceron, Aaron Courville, Pouya Bashivan, Pablo Sam
uel Castro
Deep reinforcement learning systems often suffer from unstable training dynamics
 due to non-stationarity, where learning objectives and data distributions evolv
e over time. We show that under non-stationary targets, isotropic Gaussian embed
dings are provably advantageous. In particular, they induce stable tracking of t
ime-varying targets for linear readouts, achieve maximal entropy under a fixed v
ariance budget, and encourage a balanced use of all representational dimensions-
-all of which enable agents to be more adaptive and stable. Building on this ins
ight, we propose the use of Sketched Isotropic Gaussian Regularization for shapi
ng representations toward an isotropic Gaussian distribution during training. We
 demonstrate empirically, over a variety of domains, that this simple and comput
ationally inexpensive method improves performance under non-stationarity while r
educing representation collapse, neuron dormancy, and training instability.
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Weaving Graph over Tokens: Contextualizing Structured Sequences for LLMs
Jiaxuan Chen, Zixing Zhang, Ruijun Mao, Wei Sun, Zhicheng Liang, Yuhang Zhang, Y
axi Liu, Fangxin Wang
Generative Graph Language Models (GLMs) must reconcile topology with causal
  language modeling. Linearization obscures multi-hop connectivity, while
  encoder-based methods bottleneck token-level reasoning during generation.
  Viewing context modeling as a form of message passing, we
  introduce **Weaver**, an encoder-free framework that extends the attention mec
hanism of decoder-only LLMs to enable graph reasoning. Weaver
  maps graph distances into rotary positional embeddings so that structurally
  connected nodes become proximate in attention space, propagating information
  over graph topology as if it were sequential context. To achieve this, we
  combine: 1) a masking mechanism for causal tokens with graph structures; 2) a
  unified geometric encoding that couples sequential position and graph distance
  in joint rotary embeddings (Graph-over-Tokens RoPE); and 3) a design principle
  to prioritize local information to resolve positional ambiguity under graph
  symmetries. On zero-shot benchmarks, Weaver achieves state-of-the-art performa
nce among generative GLMs, with gains of up to 30\% over prior generative method
s on
heterophilic graphs, while matching specialized discriminative models on
citation networks---all within a unified decoder-only framework.
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Mantis: Lightweight Foundation Model for Time Series Classification
Vasilii Feofanov, Songkang Wen, Shifeng Xie, Simon Roschmann, Marius Alonso, Hon
gbo Guo, Romain Ilbert, Malik Tiomoko, Quentin Bouniot, Zeynep Akata, Lujia Pan,
 Jianfeng Zhang, Ievgen Redko
While foundation models have revolutionized various domains, their application t
o time series classification remains rather under-explored, with existing litera
ture predominantly focused on forecasting. To bridge this gap, we introduce \tex
tbf{Mantis}, a transformer-based foundation model pre-trained exclusively on syn
thetic data via self-supervised contrastive learning. We demonstrate that effect
ive tokenization is critical to unlocking the full potential of transformers, pr
oposing a novel token generator unit. Furthermore, we introduce an enhanced test
-time methodology that bridges the performance gap between Mantis and strong spe
cialized approaches by leveraging intermediate-layer representations, self-ensem
bling, and cross-model embedding fusion. Extensive experiments demonstrate that
Mantis establishes a new state-of-the-art, outperforming existing foundation mod
els across four diverse dataset collections covering various application domains
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Interpretability Driven Evolutionary Approach for the Design of Biological Sequences
Akash Pandey, Wei Chen, Sinan Keten
Designing biological sequences such as proteins and DNA for desired properties is challenging due to vast search spaces and limited wet lab evaluation budgets. Current evolutionary approaches ignore sequential dependencies and rely on random mutations, which scale poorly for long sequences. In contrast, reinforcement learning (RL) and generative models explicitly model sequence structure but require large datasets to guide generation toward the target properties. These limitations suggest the need for a method that combines the sample efficiency of evolutionary approaches with the ability to exploit sequential structure. In this work, we propose a novel evolutionary approach, $\texttt{IDEAS}$, in which mutations are guided by an explainable model. The model identifies critical motifs in high-fitness sequences and uses them to mutate non-critical positions. Across eight continuous-property datasets, seven baselines, and three evaluation budgets, $\texttt{IDEAS}$ achieves a 19% acceleration in design while maintaining a favorable position on the Pareto curve balancing acceleration, diversity, and novelty.
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FreeText: Training-Free Text Rendering via Attention Localization and Spectral Glyph Injection
RuiQiang Zhang, Hengyi Wang, Chang Liu, Guanjie Wang, Zehua Ma, Weiming Zhang
Large-scale text-to-image (T2I) diffusion models excel at open-domain synthesis but still struggle with precise text rendering, especially for multi-line layouts, dense typography, and long-tailed scripts such as Chinese. Prior solutions typically necessitate costly retraining or impose rigid external layout constraints, often compromising aesthetic quality and flexibility. We propose **FreeText**, a training-free, plug-and-play framework that improves text rendering by leveraging intrinsic mechanisms of *Diffusion Transformer (DiT)* models. **FreeText** decomposes the problem into *where to write* and *what to write*. For the former, we localize writing regions by extracting token-wise spatial attribution from image-to-text attention, using sink-like tokens as stable spatial anchors and topology-aware refinement to produce high-confidence masks. For the latter, we introduce Spectral-Modulated Glyph Injection (SGMI), which injects a noise-aligned glyph prior with frequency-domain band-pass modulation to strengthen glyph structure and mitigate semantic leakage (rendering the concept instead of the word). Extensive experiments on Qwen-Image, FLUX.1-dev, and SD3 variants across longText-Benchmark, CVTG, and our CLT-Bench show consistent gains in text readability while maintaining semantic alignment and aesthetic quality, with modest inference overhead.
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Reinforcement Learning for Reachability: Guaranteeing Asymptotic Optimality
Amogh Palasamudram, Jakub Svoboda, Suguman Bansal, Krishnendu Chatterjee
Reinforcement learning (RL) for reachability specifications is fundamental in sequential decision-making, yet theoretical guarantees remain less explored. A recent work achieves asymptotic convergence to optimal policies. However, this approach provides limited insight into convergence dynamics. In this work, we present an alternative approach that provides deeper theoretical insights into convergence. Our approach builds on PAC learning with assumptions. PAC learning guarantees near-optimal policies with high confidence in finite time but requires knowing internal MDP parameters like minimum transition probability. We argue that while these parameters are unknown in RL, they can be iteratively refined and estimated with increasing accuracy. By iteratively satisfying PAC conditions, we show that exact optimality can be achieved in the limit. Empirical evaluations on standard benchmarks validate our theoretical insights into convergence dynamics.
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SpaCeFormer: Fast Proposal-Free Open-Vocabulary 3D Instance Segmentation
Chris Choy, Junha Lee, Chunghyun Park, Minsu Cho, Jan Kautz
Open-vocabulary 3D instance segmentation is a core capability for robotics and AR/VR, but prior methods trade one bottleneck for another: multi-stage 2D+3D pipelines aggregate foundation-model outputs at hundreds of seconds per scene, while

pseudo-labeled end-to-end approaches rely on fragmented masks and external region proposals. We present SpaCeFormer, a proposal-free space-curve transformer that runs in 0.12–0.30 seconds per scene across standard benchmarks, 2–3 orders of magnitude faster than multi-stage 2D+3D pipelines. We pair it with SpaCeFormer-3M, the largest open-vocabulary 3D instance segmentation dataset (3.0M multi-view-consistent captions over 604K instances from 7.4K scenes) built through multi-view mask clustering and multi-view VLM captioning; it reaches **21× higher mask recall** than prior single-view pipelines (54.3% vs 2.5% at IoU>0.5). SpaCeFormer combines spatial window attention with Morton-curve serialization for spatially coherent features, and uses a RoPE-enhanced decoder to predict instance masks directly from learned queries without external proposals. On ScanNet200 we achieve 11.1 zero-shot mAP, a 2.8× improvement over the prior best proposal-free method; on ScanNet++ and Replica, we reach 22.9 and 24.1 mAP, surpassing all prior methods including those using multi-view 2D inputs.

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Probing the Knowledge Boundary: An Interactive Agentic Framework for Deep Knowledge Extraction

Yuheng Yang, Siqi Zhu, Tao Feng, Ge Liu, Jiaxuan You

Large Language Models (LLMs) can be seen as compressed knowledge bases, but it remains unclear what knowledge they truly contain and how far their knowledge boundary extends. Existing benchmarks are mostly static and provide limited support for systematic knowledge probing. In this paper, we propose an interactive agentic framework to systematically extract and quantify the knowledge of LLMs. Our method includes four adaptive exploration policies to probe knowledge at different granularity. To ensure the quality of extracted knowledge, we introduce a three-stage knowledge processing pipeline that combines vector-based filtering to remove strict duplicates, LLM-based adjudication to resolve ambiguous semantic overlap, and domain relevance auditing to retain valid knowledge units. Through extensive experiments, we find that Recursive Taxonomy is the most effective exploration strategy. We also observe a clear knowledge scaling law, where larger models consistently recover more knowledge. In addition, we identify a Pass@1 versus Pass@k trade-off: domain-specialized models achieve higher initial accuracy but experience rapid degradation, while general-purpose models maintain stable performance over extended extraction. Finally, our results show that differences in training data composition lead to distinct and measurable knowledge profiles across model families, reflecting how pretraining shapes each model's parametric knowledge.

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Sparse Autoencoders are Topic Models

Leander Girrbach, Zeynep Akata

Sparse autoencoders (SAEs) are used to analyze embeddings, but their role and practical value are debated. We propose a new perspective on SAEs by demonstrating that they can be naturally understood as topic models. We propose a continuous topic model (CTM) inspired by Latent Dirichlet Allocation (LDA) for embedding spaces and derive the SAE objective as a maximum a posteriori estimator under this model. This view implies SAE features are thematic components rather than steerable directions. To confirm our theoretical findings, we introduce SAE-TM, a topic modeling framework that: (1) trains an SAE to learn reusable topic atoms, (2) interprets them as word distributions on downstream data, and (3) merges them into any number of topics without retraining. SAE-TM yields more coherent topics than strong baselines on text and image datasets while maintaining diversity. Finally, we analyze thematic structure in image datasets and trace topic changes over time in Japanese woodblock prints. Our work positions SAEs as effective tools for large-scale thematic analysis across modalities. Code is available at https://github.com/ExplainableML/SAE-TM.

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NeurIPS: Neuro-anatomical Inductive Priors for Sphere-based Brain Decoding

Sijin Yu, Zijiao Chen, Zhenyu Yang, Zihao Tan, Jiakun Xu, Zhongliang Liu, shengxian chen, WENXUAN WU, Xiangmin Xu, Xin Zhang

Current fMRI decoders face a performance-fidelity trade-off where efficient ID e

ncoders outperform geometrically faithful surface-based models. We argue this is partly driven by inefficient surface tokenization and the failure to use anatomy as a predictive signal. We present **NeurIPS**, a framework that improves surface-based decoding by reframing anatomical variation from a nuisance to a powerful inductive prior. NeurIPS unites two innovations: a **Selective ROI Spherical Tokenizer (SRST)** for efficient geometric encoding, and a **Structure-Guided Mixture of Experts (SG-MoE)** that explicitly models individual anatomy using cortical features. On the Natural Scenes Dataset, NeurIPS establishes a new state-of-the-art for surface decoders and achieves performance comparable to strong 1D baselines. This is achieved with unprecedented efficiency, as the model converges dramatically faster (**10 vs. 600 epochs**). This efficiency enables rapid adaptation to new subjects using only **20\%** of data and ensures robust scalability as the training cohort is expanded. Ablations provide causal evidence that these gains are driven by the model's use of cortical features, not by memorizing subject IDs. By leveraging anatomical priors, NeurIPS provides a principled and scalable path toward robust, generalizable brain decoding.

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LakeQA: An Exploratory QA Benchmark over a Million-Scale Data Lake

Haonan Wang, Jiaxiang Liu, Yurong Liu, Austin Senna Wijaya, Tianle Zhou, Eden Wu, Yijia Chen, Wanting You, Reya Vir, Daniela Pinto Veizaga, Grace Fan, Yusen Zhang, Juliana Freire, Eugene Wu

Recent large language models (LLMs) have shown rapid progress on reading-based question answering (QA), where the evidence is explicitly provided or trivially retrievable. In contrast, real-world questions are often not paired with accurate evidence documents. The useful evidence resides in a massive collection of data lakes, necessitating searching as a prerequisite for answering. However, there is a lack of a comprehensive benchmark that requires searching and reasoning over a large collection of data lakes. To this end, we introduce LakeQA, a comprehensive benchmark for search-centric question answering over data lakes that jointly emphasizes \emph{searching} and \emph{reasoning} capabilities. LakeQA is built on a heterogeneous collection of ~9.5 TB text resources from Wikipedia and open-source government data, spanning structured and unstructured data. To ensure the quality of LakeQA's tasks, each sample is annotated by at least one Ph.D level expert. Each task requires long-horizon multi-hop reasoning with implicit intermediate steps: agents need to discover the correct document(s) and then compose evidence across sources to produce the answer. Intensive experiment results on seven frontier LLMs have demonstrated that LakeQA is challenging. For instance, GPT-5.2 only obtains an exact matching score of 14.73% on LakeQA. Overall LakeQA provides a realistic testbed for developing LLM agents that can both \emph{find} and \emph{analyze} data in modern data lakes.

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Trees to Flows and Back: Unifying Decision Trees and Diffusion Models

Sai Niranjan Ramachandran, Suvrit Sra

Decision trees and diffusion models are ostensibly disparate model classes, one discrete and hierarchical, the other continuous and dynamic. This work unifies the two by establishing a crisp mathematical correspondence between hierarchical decision trees and diffusion processes in appropriate limiting regimes. Our unification reveals a shared optimization principle: \emph{Global Trajectory Score Matching (GTSM)}, for which gradient boosting (in an idealized version) is asymptotically optimal. We underscore the conceptual value of our work through two key practical instantiations: treeflow, which achieves competitive
generation quality on tabular data with higher fidelity and a 2× computational speedup, and dsmtree, a novel distillation method that transfers hierarchical decision logic into neural networks, matching teacher performance within 2\% on many benchmarks.

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The Stability of Singular Distribution: A Spectral Perspective on the Two-Phase Dynamics of Language Model Pre-training

Hongtao Zhang, WenJie Zhou, Chenxi Jia, Wei Chen, Xueqi Cheng

Large language model pre-training typically exhibits a two-phase trajectory: a f

ast initial loss drop followed by a prolonged slow improvement. We identify an u
nderlying spectral phenomenon, Stability of Singular Distribution (SoSD), where
the trace-normalized singular value spectrum stabilizes early, even as parameter
 matrices continue to evolve. We demonstrate that synchronization between SoSD a
nd the slow-descent regime is widely observed across diverse architectures (GPT-
2, LLaMA) and settings, including various schedules (Step-wise, WSD, Cosine Deca
y), weight decays, and optimizers (AdamW, Muon). By analyzing a simplified Trans
former, we prove that growing weight norms inevitably precipitate an early SoSD
threshold, after which the rate of loss decrease becomes theoretically bounded b
y the variation in the singular distribution. We further interpret strategies li
ke WSD and Muon through their ability to modulate the SoSD scale, offering a spe
ctral lens for understanding efficient pre-training dynamics.
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Bridge Matching Sampler: Scalable Sampling via Generalized Fixed-Point Diffusion
 Matching

Denis Blessing, Lorenz Richter, Julius Berner, Egor Malitskiy, Gerhard Neumann

Sampling from unnormalized densities using diffusion models has emerged as a pow
erful paradigm. However, while recent approaches that use least-squares `matchin
g' objectives have improved scalability, they often necessitate significant trad
e-offs, such as restricting prior distributions or relying on unstable optimizat
ion schemes. By generalizing these methods as special forms of fixed-point itera
tions rooted in Nelson's relation, we develop a new method that addresses these
limitations. Our approach enables learning a stochastic transport map between ar
bitrary prior and target distributions with a single, scalable, and stable objec
tive. Furthermore, we introduce a damped variant of this iteration that incorpor
ates a regularization term to mitigate mode collapse. Empirically, we demonstrat
e that our method enables sampling at unprecedented scales while preserving mode
 diversity, achieving state-of-the-art results on complex synthetic densities an
d high-dimensional molecular benchmarks.
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Unifying Value Alignment and Assignment in Cross-Domain Offline Reinforcement Le
arning with Heterogeneous Datasets

Zhongjian Qiao, Jiafei Lyu, Chenjia Bai, Peisong Wang, Siyang Gao, Shuang Qiu

Cross-domain offline reinforcement learning (RL) aims to learn a policy in the t
arget domain with a limited target domain dataset and a source domain dataset th
at exhibits a dynamics shift. Training directly on the original source dataset t
ypically leads to performance collapse. Recent studies perform data filtering fr
om the perspective of dynamics alignment or value alignment to enable efficient
policy transfer. However, these studies are typically validated on single-domain
 or single-behavior-policy source datasets. In this work, we explore a more gene
ral heterogeneous cross-domain offline RL setting, where the source datasets may
 be collected from multiple source domains by diverse behavior policies. We firs
t uncover a critical yet overlooked issue in this setting: value misassignment.
Empirically and theoretically, we demonstrate that value misassignment can under
mine value alignment, mislead data filtering toward selecting suboptimal samples
, and loosen the suboptimality gap, thereby degrading the agent’s performance. T
o address this issue, we propose V2A, which integrates dynamics alignment, value
 alignment, and value assignment. V2A first employs temporally-consistent modali
ty representation learning to extract dynamics modalities from the source datase
t, followed by modality-aware advantage learning to rectify value alignment. Fin
ally, it adopts a data filtering paradigm to selectively share source data for p
olicy learning. Empirical results show that V2A significantly outperforms strong
 baseline methods under general heterogeneous cross-domain offline RL settings.
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Entangled No More: Multi-Domain Decoupling for Robust Dynamic Graph Neural Netwo
rks

Youda Mo, Chaobo He, Junwei Cheng, Peng Mei, Quanlong Guan

Dynamic graphs are pervasive in real-world systems, but their tightly entangled
spatiotemporal evolution causes significant modeling challenges. Existing Dynami
c Graph Neural Networks (DGNNs) lack a principled framework for systematically d

ecoupling this multi-domain entanglement, raising two key problems: (i) represen tation drift caused by structural incompleteness, and (ii) signal distortion amp lified by noise perturbation. These problems can accumulate over time, forming t emporal redundancy that weakens robustness of DGNNs.
In view of these, we propose DeR-Mamba(Decoupling for Robust Mamba), a multi-dom ain decoupling framework for robust DGNNs.
To address (i), we develop the Multi-Particle Kernel Kalman observation field (M P-K^{2}alman), which achieves spatial decoupling by sampling latent evolution p aths in kernel subspaces and performing Kalman-style updates to estimate structu ral states.
To address (ii), we design the Adversarial-aware Frequency Decoupling Module (AF DM), which performs frequency-domain decoupling and dynamic cross-frequency modu lation to purify spectral signals. Finally, a self-consistent dynamic graph stat e-space system performs temporal decoupling to control redundancy, suppressing r esidual disturbances through discretized cross-time modeling and selective snaps hot scanning. Extensive experiments on benchmark datasets with adversarial attac ks validate its superior robustness.
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The Surprising Difficulty of Search in Model-Based Reinforcement Learning
Wei-Di Chang, Mikael Henaff, Brandon Amos, Gregory Dudek, Scott Fujimoto
This paper investigates search in model-based reinforcement learning (RL). Conve ntional wisdom holds that long-term predictions and compounding errors are the p rimary obstacles for model-based RL. We challenge this view, showing that search is not a drop-in replacement for a learned policy. Surprisingly, we find that s earch can harm performance even when the model is highly accurate. Instead, we s how that mitigating overestimation bias matters more than improving model or val ue function accuracy. Building on this insight, we identify that taking the mini mum over an ensemble of value functions effectively addresses this bias and enab les effective search, achieving state-of-the-art performance across multiple pop ular benchmark domains. Code can be found at https://github.com/facebookresearch /MRSQ.
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CrispEdit: Low-Curvature Projections for Scalable Non-Destructive LLM Editing
Zarif Ikram, Arad Firouzkouhi, Stephen Tu, Mahdi Soltanolkotabi, Paria Rashidine jad
A central challenge in large language model (LLM) editing is capability preserva tion: methods that successfully change targeted behavior can quietly game the ed iting proxy and corrupt general capabilities, producing degenerate behaviors rem iniscent of proxy/reward hacking. We present CrispEdit, a scalable and principl ed second-order editing algorithm that treats capability preservation as an expl icit constraint, unifying and generalizing several existing editing approaches. CrispEdit formulates editing as constrained optimization and enforces the constr aint by projecting edit updates onto the low-curvature subspace of the capabilit y-loss landscape. At the crux of CrispEdit is expressing capability constraint via Bregman divergence, whose quadratic form yields the Gauss–Newton Hessian exa ctly and even when the base model is not trained to convergence. We make this se cond-order procedure efficient at the LLM scale using Kronecker-factored approxi mate curvature (K-FAC) and a novel matrix-free projector that exploits Kronecker structure to avoid constructing massive projection matrices. Across standard mo del-editing benchmarks and safety unlearning tasks, CrispEdit achieves high edit success while keeping capability degradation below 1% on average across dataset s, significantly improving over prior editors.
****

Fixed Aggregation Features Can Rival GNNs
Celia Rubio-Madrigal, Rebekka Burkholz
Graph neural networks (GNNs) are widely believed to excel at node representation learning through trainable neighborhood aggregations. We challenge this view by introducing Fixed Aggregation Features (FAFs), a training-free approach that tr ansforms graph learning tasks into tabular problems. This simple shift enables t he use of well-established tabular methods, offering strong interpretability and

the flexibility to deploy diverse classifiers. Across 14 benchmarks, well-tuned multilayer perceptrons trained on FAFs rival or outperform state-of-the-art GNNs and graph transformers on 12 tasks—often using only mean aggregation. The only exceptions are the Roman Empire and Minesweeper datasets, which typically require unusually deep GNNs. To explain the theoretical possibility of non-trainable aggregations, we connect our findings to Kolmogorov–Arnold representations and discuss when mean aggregation can be sufficient. In conclusion, our results call for (i) richer benchmarks benefiting from learning diverse neighborhood aggregations, (ii) strong tabular baselines as standard, and (iii) employing and advancing tabular models for graph data to gain new insights into related tasks.
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Scaling Unsupervised Multi-Source Federated Domain Adaptation through Group-Wise Discrepancy Minimization
Larissa Reichart, Cem Ata Baykara, Ali Burak Ünal, Harlin Lee, Mete Akgün
Unsupervised multi-source domain adaptation (UMDA) leverages labeled data from multiple source domains to generalize to an unlabeled target. While federated UMDA addresses privacy by avoiding raw data sharing, existing methods scale poorly as the number of sources increases, often suffering from high computational overhead or training instability. We propose GALA, a scalable and robust federated UMDA framework designed for high-diversity settings. GALA achieves scalability by coupling a novel inter-group discrepancy minimization objective that approximates pairwise alignment with linear complexity alongside a temperature-controlled, centroid-based weighting strategy for dynamic source prioritization. These components enable stable, parallelizable training across many heterogeneous sources, addressing a critical scalability bottleneck that remains largely unaddressed in current literature. To evaluate performance in high-diversity scenarios, we introduce Digit-18, a new benchmark comprising 18 datasets with varied synthetic and real-world domain shifts. Extensive experiments demonstrate that GALA achieves state-of-the-art results on standard benchmarks and significantly outperforms prior methods in large-scale settings where others either fail to converge or become computationally infeasible.
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Good SFT Optimizes for SFT, Better SFT Prepares for Reinforcement Learning
Dylan Zhang, Yufeng Xu, Haojin Wang, Qingzhi Chen, Hao Peng
Post-training of reasoning LLMs is a holistic process that typically consists of an offline SFT stage followed by an online reinforcement learning (RL) stage. However, SFT is often optimized in isolation to maximize SFT performance alone. We show that,
after identical RL training, models initialized from stronger SFT checkpoints can significantly underperform those initialized from weaker ones.
We propose PEAR ($\textbf{P}$olicy $\textbf{E}$valuation–inspired $\textbf{A}$lgorithm for Offline Learning Loss $\textbf{R}$eweighting), an SFT-stage method that corrects this mismatch and better prepares the model for RL.
PEAR uses importance sampling to reweight the SFT loss, with three variants operating at the token, block, and sequence levels.
It can be used to augment standard SFT objectives
and incurs little additional training overhead once probabilities for the offline data are collected.

We conduct controlled experiments on verifiable reasoning games and mathematical reasoning tasks on Qwen2.5/3 and DeepSeek-distilled models. PEAR consistently improves post-RL performance over canonical SFT, with pass@8 gains up to a 14.6% on AIME-2025.
Our results suggest that PEAR is an effective step toward more holistic LLM post-training by designing and evaluating SFT with downstream RL in mind rather than in isolation.
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Protein Language Model Embeddings Improve Generalization of Implicit Transfer Operators
Panagiotis Antoniadis, Beatrice Pavesi, Simon Olsson, Ole Winther

Molecular dynamics (MD) is a central computational tool in physics, chemistry, a
nd biology, enabling quantitative prediction of experimental observables as expe
ctations over high-dimensional molecular distributions such as Boltzmann distrib
utions and transition densities. However, conventional MD is fundamentally limit
ed by the high computational cost required to generate independent samples. Gene
rative molecular dynamics (GenMD) has recently emerged as an alternative, learni
ng surrogates of molecular distributions either from data or through interaction
 with energy models. While these methods enable efficient sampling, their transf
erability across molecular systems is often limited. In this work, we show that
 incorporating auxiliary sources of information can improve the data efficiency
and generalization of transferable implicit transfer operators (TITO) for molecu
lar dynamics. We find that coarse-grained TITO models are substantially more dat
a-efficient than Boltzmann Emulators, and that incorporating protein language mo
del (pLM) embeddings further improves out-of-distribution generalization. Our ap
proach, PLaTITO, achieves state-of-the-art performance on equilibrium sampling b
enchmarks for out-of-distribution protein systems, including fast-folding protei
ns. We further study the impact of additional conditioning signals such as struc
tural embeddings, temperature, and large-language-model-derived embeddings on mo
del performance.
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Dissecting Post-Training: Uncovering the Complementary Roles of SFT and RL for D
ocument Parsing
Jun-Peng Jiang, An-Yang Ji, Shiyin Lu, Guodong Zheng, Weihong Zhang, Qing-Guo Ch
en, Weihua Luo, Kaifu Zhang, Long Chen, De-Chuan Zhan, Han-Jia Ye
Document parsing, the task of extracting diverse content from PDFs while preserv
ing their structural integrity, has been significantly advanced by Multimodal La
rge Language Models (MLLMs). These models have achieved remarkable success, larg
ely driven by extensive post-training on massive datasets. This paper therefore
undertakes a deep analysis of the two dominant adaptation strategies, Supervised
 Fine-Tuning (SFT) and Reinforcement Learning (RL), prompted by a puzzling obser
vation on the PDF-to-Markdown task: SFT makes a negligible impact, especially on
 parsing complex tables and formulas, while RL achieves substantial overall gain
s. To unravel the reasons, our systematic investigation reveals a clear and comp
lementary division of labor: SFT primarily operates as a structure learner, bias
ed towards mastering the low-entropy syntax of document layouts. While it learns
 the format of a table, it struggles to ensure the fidelity of its high-entropy
cell content. Conversely, RL excels as a content refiner by optimizing a holisti
c reward that reflects final accuracy. We further ground this phenomenon in the
distinct theoretical nature of their respective objective functions. Based on th
ese findings, we introduce a unified strategy that explicitly harnesses their in
dividual strengths while mitigating their weaknesses. This work shows that a dee
p understanding of post-training methods is key to unlocking performance beyond
what data scaling alone can achieve.
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On the Expressive Power of Permutation-Equivariant Weight-Space Networks
Adir Dayan, Yam Eitan, Haggai Maron
Weight-space learning studies neural architectures that operate directly on the
parameters of other neural networks. Motivated by the growing availability of pr
etrained models, recent work has demonstrated the effectiveness of weight-space
networks across a wide range of tasks.
SOTA weight-space networks rely on permutation-equivariant designs to improve ge
neralization. However, this may negatively affect expressive power, warranting t
heoretical investigation.
Importantly, unlike other structured domains, weight-space learning targets maps
 operating on both weight and function spaces, making expressivity analysis part
icularly subtle.
While a few prior works provide partial expressivity results, a comprehensive ch
aracterization is still missing. In this work, we address this gap by developing
 a systematic theory for expressivity of weight-space networks.
We first prove that all prominent permutation-equivariant networks are equivalen

t in expressive power. We then establish universality in both weight- and functi on-space settings under mild, natural assumptions on the input weights, and char acterize the edge-case regimes where universality no longer holds.
Guided by our theoretical results, we show that slight modifications to existing weight-space models yield a 34\% improvement over prior SOTA, demonstrating the practical relevance of our framework.
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Insertion Based Sequence Generation with Learnable Order Dynamics
Dhruvesh Patel, Benjamin Rozonoyer, Gaurav Pandey, Tahira Naseem, Ramón Fernande z Astudillo, Andrew McCallum
Existing insertion-based masked diffusion models that generate sequences by inte rleaving token insertion with unmasking use fixed schedules that are not depende nt on the data. For structured sequences like graphs and molecules, learning dat a-dependent generation orders can improve generation quality by reducing uncerta inty over the action space. We propose LoFlexMDM, an insertion-based masked diff usion model with learnable order dynamics that learns data-dependent insertion a nd unmasking rates. We generalize the discrete flow matching framework to work w ith variable-length sequences, propose a tractable schedule parameterization and a training objective for joint training of the generator and the target order d ynamics. On De Novo and fragment-constrained molecule generation, LoFlexMDM impr oves sample quality over FlexMDM by up to 17.5% and 6.7%, respectively. These re sults show that learning the target generation order can improve insertion-based diffusion models without giving up tractable training. We open source the code at https://github.com/dhruvdcoder/LoFlexMDM.
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Rational Neural Networks have Expressivity Advantages
Maosen Tang, Alex Townsend
We study neural networks with trainable low-degree rational activation functions and show that they are more expressive and parameter-efficient than modern piec ewise-linear and smooth activations such as ELU, LeakyReLU, LogSigmoid, PReLU, R eLU, SELU, CELU, Sigmoid, SiLU, Mish, Softplus, Tanh, Softmin, Softmax, and LogS oftmax. For an error target of $\varepsilon>0$, we establish approximation-theor etic separations: Any network built from standard fixed activations can be unifo rmly approximated on compact domains by a rational-activation network with only $\mathrm{poly}(\log\log(1/\varepsilon))$ overhead in size, while the converse pr ovably requires $\Omega(\log(1/\varepsilon))$ parameters in the worst case. This exponential gap persists at the level of full networks and extends to gated act ivations and transformer-style nonlinearities. In practice, rational activations integrate seamlessly into standard architectures and training pipelines, allowi ng rationals to match or outperform fixed activations under identical architectu res and optimizers.
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Factored Value Functions for Graph-Based Multi-Agent Reinforcement Learning
Ahmed Rashwan, Keith Briggs, Chris Budd, Lisa Maria Kreusser
Credit assignment is a core challenge in multi-agent reinforcement learning (MAR L), especially in large-scale systems with structured, local interactions.
Graph-based Markov decision processes (GMDPs) capture such settings via an influ ence graph, but standard critics are poorly aligned with this structure: global value functions provide weak per-agent learning signals, while existing local co nstructions can be difficult to estimate and ill-behaved in infinite-horizon set tings.
We introduce the Diffusion Value Function (DVF), a factored value function for G MDPs that assigns to each agent a value component by diffusing rewards over the influence graph with temporal discounting and spatial attenuation.
We show that DVF is well-defined, admits a Bellman fixed point, and decomposes t he global discounted value via an averaging property.
DVF can be used as a drop-in critic in standard RL algorithms and estimated scal ably with graph neural networks.
Building on DVF, we propose Diffusion A2C (DA2C) and a sparse message-passing ac tor, Learned DropEdge GNN (LD-GNN), for learning decentralised algorithms under

communication costs.
Across the firefighting benchmark and three distributed computation tasks (vecto r graph colouring and two transmit power optimisation problems), DA2C consistent ly outperforms local and global critic baselines, improving average reward by up to 11%.
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From Player to Master: Enhancing Test-Time Learning of LLM Agents via Reinforcem ent Learning over Memory
Yishuo Cai, Xingyu Guo, Xuancheng Huang, Jinhua Du, can huang, Wenxuan Huang, We nhan Ma, Yuyang Hu, Aohan Zeng, Jie Tang, Xu Sun
Large language model (LLM) agents are increasingly deployed in long-running sett ings where improving through experience at test time becomes important. A common approach is to update an explicit memory after each interaction to guide future decisions. However, most existing methods rely on hand-designed prompting rules , making it difficult to align memory updates with downstream objectives over mu lti-step horizons consistently. We propose MemoPilot, a plug-in memory copilot that explicitly trains the memory update process to improve a frozen LLM's perfo rmance across sequential interactions. We formulate memory updating as a multi-t urn decision problem and optimize it end-to-end with multi-turn GRPO. Our traini ng recipe introduces (i) a turn-wise reward signal and (ii) a context-independen t, turn-level advantage estimation across rollouts, enabling finer-grained credi t assignment and more stable training in multi-turn settings. We evaluate MemoPi lot on two testbeds: multi-round Rock-Paper-Scissors (RPS) and Limit Texas Hold' em (LHE). Across both environments, MemoPilot substantially improves test-time l earning of a frozen player over strong baselines, ranking first in Elo ratings o n both games (1762 on LHE and 1590 on RPS) and outperforming all baseline memory methods and proprietary models, including Deepseek-V3.2. Our code is publicly a vailable.
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Global Geometry Is Not Enough for Vision Representations
Jiwan Chung, Seon Joo Kim
A common assumption in representation learning is that globally well-distributed embeddings support robust and generalizable representations. This focus has sha ped both training objectives and evaluation protocols, implicitly treating globa l geometry as a proxy for representational competence. While global geometry eff ectively encodes which elements are present, it is often insensitive to how they are composed. We investigate this limitation by testing the ability of geometri c metrics to predict compositional binding across 21 vision encoders. We find th at standard geometry-based statistics exhibit near-zero correlation with composi tional binding. In contrast, functional sensitivity, as measured by the input-ou tput Jacobian, reliably tracks this capability. We further provide an analytic a ccount showing that this disparity arises from objective design, as existing los ses explicitly constrain embedding geometry but leave the local input-output map ping unconstrained. These results suggest that global embedding geometry capture s only a partial view of representational competence and establish functional se nsitivity as a critical complementary axis for modeling composite structure.
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Semi-Supervised Learning with Noisy Proxy Covariates: Generalization Bounds and Distribution Regression
Kwangho Kim, Jisu Kim
In many modern machine learning pipelines, abundant pretrained representations s erve as noisy proxy covariates, while task-specific labels remain scarce. We stu dy semi-supervised regression in this setting, and propose a simple two stage es timator that learns kernel eigenfeatures from all proxy covariates and fits a ri dge predictor on labeled data. We derive finite sample bounds showing that fast labeled sample rates are recovered when proxy perturbation is controlled and unl abeled proxy covariates are sufficiently abundant. We also show that distributio n regression is a direct special case, with analogous guarantees when the finite bag size is large enough. Experiments show consistent gains over supervised and semi-supervised baselines, especially in low label regimes.

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PyHealth 2.0: A Comprehensive Open-Source Toolkit for Accessible and Reproducible Clinical Deep Learning
John Wu, Yongda Fan, Zhenbang Wu, Paul Landes, Eric Schrock, Sayeed Sajjad Razin, Arjun Chatterjee, Naveen Baskaran, Joshua Steier, Andrea Fitzpatrick, Bilal Arif, Rian Atri, Jathurshan Pradeepkumar, Siddhartha Laghuvarapu, Junyi Gao, Adam Richard Cross, Jimeng Sun
Difficulty replicating baselines, high computational costs, and required domain expertise create persistent barriers to clinical AI research. To address these challenges, we introduce PyHealth 2.0, an enhanced clinical deep learning toolkit that enables predictive modeling in as few as 7 lines of code. PyHealth 2.0 offers three key contributions: (1) a comprehensive toolkit addressing reproducibility and compatibility challenges by unifying 15+ datasets, 20+ clinical tasks, 25+ models, 5+ interpretability methods, and uncertainty quantification including conformal prediction within a single framework that supports diverse clinical data modalities—signals, imaging, and electronic health records—with translation of 5+ medical coding standards; (2) accessibility-focused design accommodating multimodal data and diverse computational resources with up to 39× faster processing and 20× lower memory usage, enabling work from 16GB laptops to production systems; and (3) an active open-source community of 400+ members lowering domain expertise barriers through extensive documentation, reproducible research contributions, and collaborations with academic health systems and industry partners, including multi-language support via RHealth. PyHealth 2.0 establishes an open-source foundation and community advancing accessible, reproducible healthcare AI.
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Don't Ignore the Tail: Decoupled Distillation Produces Top Maths Students on an Academic Budget
Sayantan Dasgupta, Trevor Cohn, Timothy Baldwin
The core learning signal used in language model distillation is the standard Kullback-Leibler (KL) divergence between the student and teacher distributions. Traditional KL divergence tends to be dominated by the teacher’s highest-probability modes, thereby diminishing the influence of less-probable yet potentially informative components of the output distribution. We propose a new tail-aware divergence that decouples the contribution of the teacher model's top-K predicted probabilities from that of lower-probability predictions, while maintaining the same computational profile as the KL Divergence. Our decoupled approach reduces the impact of teacher modes and, consequently, increases the contribution of the distribution's tail. Experimental results demonstrate that our modified distillation method yields competitive performance in both pre-training and supervised distillation for mathematical reasoning of decoder models across various datasets. Furthermore, the distillation process is efficient and can be performed on modest academic budgets for large datasets, drastically reducing the computational costs typically associated with large-scale distillation.
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Constrained Adaptive Rejection Sampling
Pawe■ Parys, Sairam Vaidya, Taylor Berg-Kirkpatrick, Loris D'Antoni
Language Models (LMs) are increasingly used in applications where generated outputs must satisfy strict semantic or syntactic constraints. Existing approaches to constrained generation fall along a spectrum: greedy constrained decoding methods enforce validity during decoding but distort the LM’s distribution, while rejection sampling (RS) preserves fidelity but wastes computation by discarding invalid outputs. Both extremes are problematic in domains such as program fuzzing, where both *validity* and *diversity* of samples are essential. We present *Constrained Adaptive Rejection Sampling* (CARS), an approach that strictly improves the sample-efficiency of RS without distributional distortion. CARS begins with unconstrained LM sampling and adaptively rules out constraint-violating continuations by recording them in a trie and subtracting their probability mass from future draws. This adaptive pruning ensures that prefixes proven invalid are never revisited, acceptance rates improve monotonically, and the resulting samples exactly follow the constrained distribution. In experiments on a variety of domai

ns---e.g., program fuzzing and molecular generation---CARS consistently achieves higher efficiency---measured in the number of LM forward passes per valid sample---while also producing stronger sample diversity than both Greedy Constrained Decoding (GCD) and methods that approximate the LM's distribution.

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Test-time Generalization for Physics through Neural Operator Splitting

Louis Serrano, Jiequn Han, Edouard Oyallon, Shirley Ho, Rudy Morel

Neural operators have shown promise in learning solution maps of partial differential equations (PDEs), but they often struggle to generalize when test inputs lie outside the training distribution, such as novel initial conditions, unseen PDE coefficients or unseen physics.

Prior works address this limitation with large scale multi physics pretraining followed by fine tuning, but this still requires examples from the new dynamics, falling short of true zero shot generalization. In this work, we propose a method to enhance generalization at test-time, i.e, without modifying pretrained weights. Building on DISCO, which provides a dictionary of neural operators trained across different dynamics, we introduce a neural operator splitting strategy that, at test time, searches over compositions of training operators to approximate unseen dynamics. On challenging out-of-distribution tasks including parameter extrapolation and novel combinations of physics phenomena, our approach achieves state-of-the-art zero shot generalization results, while being able to recover the underlying PDE parameters. These results underscore test-time computation as a key avenue for building flexible, compositional, and generalizable neural operators.

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D^3: Dynamic Directional Graph-Constrained Data Scheduling for LLM Training

Yuanjian Xu, Jianing Hao, Guang Zhang, Zhong Li

Training data plays a central role in large language models (LLMs) optimization, motivating extensive research on data scheduling strategies.

Most existing approaches concentrate on adjusting the overall data distribution but neglect the underlying interactions between samples during training.

However, we argue that such interactions cannot be overlooked, as real-world data samples frequently exhibit directional influences on each other, making the training order crucial.

Intuitively, we can prioritize train-units with greater influence to improves learning efficiency.

In this work, we propose D^3, a **D**ynamic, **D**irectional graph-constrained **D**ata scheduling framework. D^3 formulates the complex interactions among train-units as a dynamic influence graph, where edges represent loss-based dependencies. It then solves a constrained optimization problem over this graph to derive the training order, which ensures that the data sequence respects the evolving information flow throughout training.

Our approach is theoretically motivated and yields consistent improvements over existing data scheduling methods across both pre-training and post-training phases. Furthermore, for scalability, D^3 also employs an efficient approximation algorithm that keeps the additional computational overhead within a manageable range. For future research, the code is available at https://github.com/xuyj233/D3.

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Iterated Population Based Training with Task-Agnostic Restarts

Alexander Chebykin, Tanja Alderliesten, Peter Bosman

Hyperparameter Optimization (HPO) can lift the burden of tuning hyperparameters (HPs) of neural networks. HPO algorithms from the Population Based Training (PBT) family are efficient thanks to dynamically adjusting HPs every few steps of the weight optimization. Recent results indicate that the number of steps between HP updates is an important meta-HP of all PBT variants that can substantially affect their performance. Yet, no method or intuition is available for efficiently setting its value. We introduce Iterated Population Based Training (IPBT), a novel PBT variant that automatically adjusts this HP via restarts that reuse weight information in a task-agnostic way and leverage time-varying Bayesian optimiza

tion to reinitialize HPs.
Evaluation on 8 image classification and reinforcement learning tasks shows that , on average, our algorithm matches or outperforms 5 previous PBT variants and o ther HPO algorithms (random search, ASHA, SMAC3), without requiring a budget inc rease or any changes to its HPs. The source code is available online.
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Beyond Next-Token Alignment: Distilling Multimodal Large Language Models via Tok en Interactions
Lin Chen, Xiaoke Zhao, Kun Ding, Weiwei Feng, Changtao Miao, Zili Wang, Wenxuan Guo, Ying Wang, Kaiyuan Zheng, Bo Zhang, Zhe Li, Shiming Xiang
Multimodal Large Language Models (MLLMs) demonstrate impressive cross-modal capa bilities, yet their substantial size poses significant deployment challenges. Kn owledge distillation (KD) is a promising solution for compressing these models, but existing methods primarily rely on static next-token alignment, neglecting t he dynamic token interactions, which embed essential capabilities for multimodal understanding and generation. To this end, we introduce **Align-TI**, a novel K D framework designed from the perspective of **T**oken **I**nteractions. Our app roach is motivated by the insight that MLLMs rely on two primary interactions: v ision-instruction token interactions to extract relevant visual information, and intra-response token interactions for coherent generation. Accordingly, Align-T I introduces two components: IVA enables the student model to imitate the teache r's instruction-relevant visual information extract capability by aligning on sa lient visual regions. TPA captures the teacher's dynamic generative logic by ali gning the sequential token-to-token transition probabilities. Extensive experime nts demonstrate Align-TI's superiority. Notably, our approach achieves 2.6% rela tive improvement over Vanilla KD, and our distilled Align-TI-2B even outperforms LLaVA-1.5-7B (a much larger MLLM) by 7.0%, establishing a new state-of-the-art distillation framework for training parameter-efficient MLLMs.
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LAST: Bridging Vision-Language and Action Manifolds via Gromov-Wasserstein Align ment
Huaihai Lyu, Chaofan Chen, Yuheng Ji, Xiansheng Chen, Pengwei Wang, Shanghang Zh ang, Changsheng Xu
We formulate the learning of generalist Vision-Language-Action (VLA) models as a Gromov-Wasserstein alignment problem, aiming to map semantically similar VL emb eddings to physically similar motion primitives. However, solving this is challe nging due to the mathematical heterogeneity between the domains: the semantic sp ace of vision-language is topologically linear and isotropic, while the physical manifold of robotic action is non-Euclidean and anisotropic. As a result, direc t regression approaches fail due to the disjoint metric structures of these doma ins, making standard distance minimization ill-posed.

To resolve this incompatibility, we introduce LAST (Lie-algebraic Action Space T okenizer). LAST reconstructs the action space to establish a more consistent met ric alignment between the VL and Action modalities. Specifically, LAST bridges t he heterogeneity via two stages: (1) *Global Topological Linearization*, which l inearizes the action manifold through Lie-algebraic mapping, converting trajecto ries into a fixed-length, physically additive representation; and (2) *Local Met ric Discretization*, where the representation is discretized hierarchically into schemas and whitened residuals, establishing a mathematical isomorphism with th e isotropic Euclidean metric. By addressing the structural mismatch globally and locally, LAST enables VLA models with enhanced convergence and generalizability .
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E2Former-V2: On-the-Fly Equivariant Attention with Linear Activation Memory
Lin Huang, Chengxiang Huang, Ziang Wang, Yiyue Du, Chu Wang, Haocheng Lu, Yunyan g Li, Xiaoli LIU, Arthur JIANG, Jia Zhang
Equivariant Graph Neural Networks (EGNNs) have become a widely used approach for modeling 3D atomistic systems. However, mainstream architectures face critical efficiency bottlenecks due to the explicit construction of geometric features or

 dense tensor products on \textit{every} edge. To overcome this, we introduce \t
extbf{E2Former-V2}, an efficient architecture that integrates algebraic sparsity
 with hardware-aware execution.
We introduce \textbf{Equivariant Axis-Aligned Sparsification (EAAS)}, which leve
rages an $\mathrm{SO}(3)\!\rightarrow\!\mathrm{SO}(2)$ change of basis to conver
t dense Wigner-$6j$ tensor contractions into sparse, parity-based re-indexing op
erations.
Building on this representation, we propose \textbf{On-the-Fly Equivariant Atten
tion}, a fully node-centric mechanism implemented via a fused Triton kernel.
By eliminating materialized edge tensors and maximizing SRAM utilization, our ke
rnel achieves up to \textbf{20$\times$ higher TFLOPS} than standard implementati
ons.
Experiments on SPICE and OMol25 show that E2Former-V2 preserves predictive accur
acy while substantially accelerating inference, demonstrating the practical feas
ibility of large equivariant transformers on commodity GPUs. Our released code c
an be found at \url{https://github.com/IQuestLab/UBio-MolFM/tree/main}
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Towards One-for-All Anomaly Detection for Tabular Data
Shiyuan Li, Yixin Liu, Yu Zheng, Xiaofeng Cao, Shirui Pan, Heng Tao Shen
Tabular anomaly detection (TAD) aims to identify samples that deviate from the m
ajority in tabular data and is critical in many real-world applications. However
, existing methods follow a ``one model for one dataset (OFO)'' paradigm, which
relies on dataset-specific training and thus incurs high computational cost and
yields limited generalization to unseen domains. To address these limitations, w
e propose OFA-TAD, a generalist one-for-all (OFA) TAD framework that only requir
es one-time training on multiple source datasets and can generalize to unseen da
tasets from diverse domains on-the-fly. To realize one-for-all tabular anomaly d
etection, OFA-TAD extracts neighbor-distance patterns as transferable cues, and
introduces multi-view neighbor-distance representations from multiple transforma
tion-induced metric spaces to mitigate the transformation sensitivity of distanc
e profiles. To adaptively combine multi-view distance evidence, a Mixture-of-Exp
erts (MoE) scoring network is employed for view-specific anomaly scoring and ent
ropy-regularized gated fusion, with a multi-strategy anomaly synthesis mechanism
 to support training under the one-class constraint. Extensive experiments on 34
 datasets from 14 domains demonstrate that OFA-TAD achieves superior anomaly det
ection performance and strong cross-domain generalizability under the strict OFA
 setting. The source code is available at https://github.com/Shiy-Li/OFA-TAD.
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Knothe-Rosenblatt Quantile Regression for Risk-sensitive Multi-objective Reinfor
cement Learning
Gwangpyo Yoo, Woo Kyung Kim, Honguk Woo
In this work, we extend distributional reinforcement learning (RL) to develop a
risk-sensitive multi-objective RL framework, with applications to domains such a
s finance and robotics. We achieve this by adopting vector-risk measures and app
roximating them via Knothe-Rosenblatt (KR) quantile regression. This approach di
rectly extends the IQN framework to the multi-objective setting, aligns with the
 axiomatic definition of vector-risk measures, and guarantees that critics conve
rge under the distributional Bellman operator. To mitigate the artificial orderi
ng imposed by the KR map, we employ a transformer architecture without positiona
l encoding, and introduce MO-TQC for training stability. We demonstrate improved
 performance on MO-Gymnasium benchmarks and use our framework to study risk-sens
itive policies in multi-objective tasks.
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Transport Clustering: Solving Low-Rank Optimal Transport via Clustering
Henri Schmidt, Peter Halmos, Benjamin Raphael
Optimal transport (OT) finds a least-cost transport plan between two probability
 distributions. Unlike standard OT, which infers unstructured pointwise mappings
, low-rank optimal transport explicitly constrains the rank of the transport pla
n to infer latent structure. This improves statistical robustness, yields sharpe
r parametric rates for estimating Wasserstein distances, and generalizes K-mea

ns to co-clustering. However, these advantages come at the cost of a non-convex and NP-hard optimization problem. We introduce Transport Clustering, an algorithm to compute a low-rank OT plan that reduces low-rank OT to a clustering problem on correspondences obtained from a full-rank transport registration step. We prove that this reduction yields polynomial-time, constant-factor approximation algorithms for low-rank OT: specifically, a $(1+\gamma)$ approximation for negative-type metrics, a $(1+\gamma+\sqrt{2\gamma}\,)$ approximation for kernel costs, and a $(1+\gamma+\rho)$ approximation for general metrics satisfying the triangle inequality. Here, $\gamma \in [0,1]$ is the cost
ratio of the optimal full-rank to low-rank solutions, and $\rho\in[0,1]$ is an asymmetry coefficient on the cluster variances. Numerically, Transport Clustering outperforms existing solvers on synthetic benchmarks and large-scale datasets.
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Expectation Consistency Loss: Rethink Confidence Calibration under Covariate Shift

Jinzong Dong, Zhaohui Jiang, Bo Yang

Confidence calibration for classification models is vital in safety-critical decision-making scenarios and has received extensive attention. General confidence calibration methods assume training and test data are independent and identically distributed ($i.i.d.$), limiting their effectiveness under covariate shifts. Previous calibration methods under covariate shift struggle with class-wise or canonical calibrations and often rely on unstable importance weighting when density ratios are large or unbounded. Given the above limitations, this paper rethinks confidence calibration under covariate shifts. First, we derive a necessary and sufficient condition for confidence calibration under covariate shifts, named Expectation consistency condition, which reveals covariate shifts do not necessarily lead to uncalibrated confidence and provides a weaker condition for confidence calibration than global covariate distribution alignment. Then, utilizing Expectation consistency condition, this paper proposes an unsupervised domain adaptation loss to calibrate confidence of the target domain, named Expectation consistency loss (ECL), which is compatible with canonical calibration, class-wise calibration, and top-label calibration. Third, we prove that computing ECL loss has the same sample complexity as Expected Calibration Error (ECE) and provide a theoretically grounded mini-batch trainable scheme for ECL loss. Finally, we validate the effectiveness of our method on both simulated and real-world covariate shift datasets.
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How to Guide Your Flow: Few-Step Alignment via Flow Map Reward Guidance

Jerry Yuyang Huang, Justin Lin, Sheel Shah, Kartik Nair, Nicholas Matthew Boffi

In generative modeling, we often wish to produce samples that maximize a user-specified reward such as aesthetic quality or alignment with human preferences, a problem known as guidance. Despite their widespread use, existing guidance methods either require expensive multi-particle, many-step schemes or rely on poorly understood approximations. We reformulate guidance as a deterministic optimal control problem, yielding a hierarchy of algorithms that subsumes existing approaches at the coarsest level. We show that the flow map, an object of significant recent interest for its role in fast inference, arises naturally in the optimal solution. Based on this observation, we propose Flow Map Reward Guidance (FMRG): a training-free, single-trajectory framework that uses the flow map to both integrate and guide the flow. At text-to-image scale, FMRG matches or surpasses baselines across inverse problems and reward-guided generation with as few as 3 NFEs, giving at least an order-of-magnitude speedup in comparison to prior state of the art.
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DDP-WM: Disentangled Dynamics Prediction for Efficient World Models

Shicheng Yin, Kaixuan Yin, Weixing Chen, Yang Liu, Guanbin Li, Liang Lin

World models are essential for autonomous robotic planning. However, the substantial computational overhead of existing dense Transformer-based models significantly hinders real-time deployment. To address this efficiency-performance bottleneck, we introduce DDP-WM, a novel world model centered on the principle of Dise

ntangled Dynamics Prediction (DDP). We hypothesize that latent state evolution i
n observed scenes is heterogeneous and can be decomposed into sparse primary dyn
amics driven by physical interactions and secondary context-driven background up
dates. DDP-WM realizes this decomposition through an architecture that integrate
s efficient historical processing with dynamic localization to isolate primary d
ynamics. By employing a cross-attention mechanism for background updates, the fr
amework optimizes resource allocation and provides a smooth optimization landsca
pe for planners. Extensive experiments demonstrate that DDP-WM achieves superior
 efficiency and performance across diverse tasks, including navigation, precise
tabletop manipulation, and complex deformable or multi-body interactions. Specif
ically, on the challenging Push-T task, DDP-WM achieves an approximately 9 times
 inference speedup and improves the MPC success rate from 90% to 98% compared to
 state-of-the-art dense models.
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Taming the Recent-Data Bias: Towards Robust Time Series Forecasting with Global
Context

Longlong Xu, Zeyan Li, Xiao He, Zhaoyang Yu, Changhua Pei, Zhe Xie, Zijun Dou, T
ieying Zhang, Dan Pei

Time series forecasting plays a vital role in numerous domains. However, real-wo
rld time series are frequently contaminated by noise, missing values, and anomal
ies, posing significant challenges to reliable forecasting. In this work, we fir
st systematically investigate a fundamental limitation prevalent in existing for
ecasting methods: an excessive reliance on the most recent observations---termed
 "recent-data bias". This bias renders forecasts highly vulnerable to perturbati
ons in recent data, severely undermining prediction reliability. To address this
 issue, we propose TameR, a novel approach for robust time series forecasting th
at effectively mitigates recent-data bias via enhancing the utilization of globa
l context. Specifically, it employs a basis-aligned randomized sampling strategy
 to reduce dependence on any specific recent data. Furthermore, TameR incorporat
es a learnable periodicity extraction module coupled with a two-stage learning p
rotocol to robustly separate periodic patterns from the sampled residual compone
nts. Comprehensive experiments demonstrate that TameR significantly outperforms
state-of-the-art methods in robustness against diverse perturbation scenarios, w
hile achieving comparable accuracy on clean data. Code is available at https://g
ithub.com/NetManAIOps/TameR.
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SARL: Structure-Aligned Reinforcement Learning for Bridging the Perception-Actio
n Gap in Airspace

Binhao Gu, jinjun cai, Weihuang Zheng, Jiaxing Li, Youyong Kong, Hui Ding

Multi-Agent Reinforcement Learning (MARL) has been widely applied to automated a
ircraft conflict resolution due to its strong capability for cooperative control
 and distributed decision-making. However, existing approaches typically assume
a fixed number of aircraft and neglect the unique characteristics of air traffic
 control instructions. This structural misalignment between model architectures
and domain requirements leads to severe deficiencies in perception scalability a
nd action stability across scenarios of varying scales. To address these challen
ges, we propose Structural-Aligned Reinforcement Learning (SARL), which aims to
bridge the gap between perception and action. First, the Physics-Encoded Relatio
nal Graph (PERG) effectively resolves the fixed input dimensionality issue by in
corporating physical inductive biases into a graph attention mechanism. Second,
we design the Sparse Cognitive Mixture-of-Experts (SC-MoE) to enhance decision s
tability. In addition, we introduce a Kinematic Kafety Shield (KSS) based on avi
ation rules, which not only improves inference-time safety but also effectively
guides the model to generate semantically meaningful actions that comply with av
iation standards. Simulation experiment results demonstrate that SARL significan
tly outperforms existing reinforcement learning baselines across diverse scenari
os in terms of both success rate and operational efficiency.
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Provably Learning Attention with Queries

Satwik Bhattamishra, Kulin Shah, Michael Hahn, Varun Kanade

We study the problem of learning Transformer-based sequence models with black-box access to their outputs. In this setting, a learner may adaptively query the oracle with any sequence of vectors and observe the output of the target function. We begin with studying the learnability of the simplest formulation, that is, learning a single-head attention-based regressor with queries. We show that for a model with width d, there is an elementary algorithm to learn the parameters of single-head attention with $O(d^2)$ queries. Further, we show that if there exists an algorithm to learn ReLU feedforward networks (FFNs), then the single-head algorithm can be easily adapted to learn one-layer Transformers with single-head attention. Next, we show that, in the common regime where the head dimension $r \ll d$, single-head attention-based models can be learned with $O(rd)$ queries via compressed sensing arguments. We also study robustness to noisy oracle access, proving that under mild norm and margin conditions, the parameters can be estimated to ε accuracy with a polynomial number of queries even when outputs are only provided up to additive tolerance. Finally, we consider the learnability of multi-head attention and show that they are not identifiable from queries, and hence, learnability in the same sense is not feasible without additional assumptions. We discuss potential approaches to learn multi-head attention-based models under certain structural assumptions.

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DeepSight: Long-Horizon World Modeling via Latent States Prediction for End-to-End Autonomous Driving

Lingjun Zhang, Changjie Wu, Linzhe Shi, Jiangyang Li, Jiaxin Liu, Lei Yang, Hang Zhang, Mu Xu, Hong Wang

End-to-end autonomous driving systems are increasingly integrating Vision-Language Model (VLM) architectures, incorporating text reasoning or visual reasoning to enhance the robustness and accuracy of driving decisions.

However, the reasoning mechanisms employed in most methods are direct adaptations from general domains, lacking in-depth exploration tailored to autonomous driving scenarios, particularly within visual reasoning modules. In this paper, we propose a driving world model that performs parallel prediction of latent semantic features for consecutive future frames in the bird’s-eye-view (BEV) space, thereby enabling long-horizon modeling of future world states. We also introduce an efficient and adaptive text reasoning mechanism that utilizes additional social knowledge and reasoning capabilities to further improve driving performance in challenging long-tail scenarios. We present a novel, efficient, and effective approach that achieves state-of-the-art (SOTA) results on the closed-loop Bench2drive benchmark. Codes are available at: https://github.com/hotdogcheesewhite/DeepSight.

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Where Detectors Fail: Probing Generative Space for Generalizable AI-Generated Image Detection

Zijie Cao, Weijie Tu, Yao Xiao, Weijian Deng, Weiyan Chen, Liang Lin, Pengxu Wei

Detecting AI-generated images (AIGI) remains challenging because detectors often fail to generalize to unseen generators. Although existing methods are trained on large datasets, their performance still degrades when generation settings change, indicating that data scale alone is insufficient and that limited coverage of generative variations during training is a key factor. Studies on generative model editing show that small changes in internal representations can produce diverse and meaningful image variations, many of which are not explored under standard sampling. Leveraging this insight, we propose PROBE (Probing Robustness via Boundary Exploration), a framework that improves detector generalization by actively exploring challenging regions of the generative process. Instead of treating the generator as a fixed data source, PROBE uses the detector as a critic to steer the generator through manifold-level modifications, producing realistic samples that are difficult to classify. These samples expose failure cases that are uncommon under standard data sampling strategies and are used to refine the detector. Experimental results across multiple benchmarks indicate that PROBE enhances generalization to unseen generators, resulting in more generalizable AIGI detection performance.

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Sponge Tool Attack: Stealthy Denial-of-Efficiency against Tool-Augmented Agentic Reasoning
Qi Li, Xinchao Wang
Enabling large language models (LLMs) to solve complex reasoning tasks is a key step toward artificial general intelligence. Recent work augments LLMs with external tools to enable agentic reasoning, achieving high utility and efficiency in a plug-and-play manner. However, the inherent vulnerabilities of such methods to malicious manipulation of the tool-calling process remain largely unexplored. In this work, we identify a tool-specific attack surface and propose Sponge Tool Attack (STA), which disrupts agentic reasoning solely by rewriting the input prompt under a strict query-only access assumption. Without any modification on the underlying model or the external tools, STA converts originally concise and efficient reasoning trajectories into unnecessarily verbose and convoluted ones before arriving at the final answer. This results in substantial computational overhead while remaining stealthy by preserving the original task semantics and user intent. To achieve this, we design STA as an iterative, multi-agent collaborative framework with explicit rewritten policy control, and generates benign-looking prompt rewrites from the original one with high semantic fidelity. Extensive experiments across 6 models (including both open-source models and closed-source APIs), 12 tools, 4 agentic frameworks, and 13 datasets spanning 5 domains validate the effectiveness of STA.
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STRIDE: Post-Training LLMs to Reason and Refine Bio-Sequences via Edit Trajectories
Daiheng Zhang, Shiyang Zhang, Sizhuang He, Yangtian Zhang, Syed A Rizvi, David van Dijk
Discrete biological sequence optimization often requires goal-directed, parser-valid edits to an existing protein or molecule. Diffusion models support iterative refinement but do not
  expose a controllable discrete-edit interface, while autoregressive LLMs can be myopic when planning constrained edits over multiple steps. We introduce *STRIDE* (Sequence Trajectory
  Refinement via Iterative Discrete Editing), a post-training framework that trains an LLM to emit executable INSERT/DELETE/REPLACE trajectories for variable-length refinement. *STRIDE*
  first learns Levenshtein-aligned shortest-edit demonstrations, then uses supervised fine-tuning and group-based policy optimization to align trajectories with task rewards while
  preserving coherent editing. On an oracle-based full-action protein stress test, *STRIDE* raises success over Vanilla SFT from 42% to 89% and novelty among unique improvements from 47%
  to 97%. On instruction-conditioned molecular editing, the GSPO-aligned variant improves strict success, controllability, and SMILES validity over the SFT-only *STRIDE* model (code:
  https://github.com/daiheng-zhang/STRIDE).
****
Generalized Correctness Models: Learning Calibrated and Cross-Model Correctness Predictors from Historical Patterns
Hanqi Xiao, Vaidehi Patil, Hyunji Lee, Elias Stengel-Eskin, Mohit Bansal
Generating accurate and calibrated confidence estimates is critical for deploying LLMs in high-stakes or user-facing applications, and remains an open challenge. Prior research has often framed confidence as a problem of eliciting a model's "self-knowledge", i.e., the ability of an LLM to judge whether its own answers are correct; this approach implicitly assumes that there is some privileged information about the answer's correctness that is accessible to the model itself. However, we find that whether trained or training-free, an LLM attempting to predict the correctness of its own outputs generally performs no better than an unrelated LLM attempting the same task. Moreover, we hypothesize that a key factor in predicting model correctness, i.e., building a "Correctness Model" (CM), is ex

posure to a target model’s historical predictions. We use multiple methods to in ject this historical correctness information, including training an LLM to predi ct the confidences of many other LLMs, i.e., creating a Generalized Correctness Model (GCM). We use GCMs and CMs as a lens for studying the source of correctnes s prediction ability and its generalization, studying the importance of answer p hrasing, world-knowledge, performance history, in-context examples, and post-hoc calibration for correctness prediction. We evaluate GCMs based on Qwen3-8B acro ss 5 model families and the MMLU, TriviaQA, and Spider datasets, as well as on a downstream selective prediction task, finding that reliable LLM confidence esti mation is a cross-model skill learned by encoding correctness history rather tha n a model-specific skill reliant on introspection. Code: https://github.com/The-Inscrutable-X/CalibratedModelAgnosticCorrectness.

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CoGeoAD: Hierarchical Color-Geometric Fusion with Multi-View Attention for Zero-Shot 3D Anomaly Detection

Ke Xu, Xinle Wang, Yanning Hou, Xueliang Ma, Juan Xie, Jianfeng Qiu

Zero-shot 3D anomaly detection is essential for industrial quality inspection, w here labeled anomaly samples are scarce. Meanwhile, existing methods lack an eff ective mechanism to fuse complementary 2D color images with 3D geometric structu res, limiting their ability to detect both surface and structural defects in a u nified framework. To address these issues, we propose CoGeoAD, a unified CLIP-ba sed framework that fuses color and geometric features by constructing pixel-alig ned paired multi-view images. The framework introduces a Data-Driven Multi-View Attention (MVA) mechanism to adaptively aggregate 3D features and a Multi-Stage Color-Geometric Fusion (MS-CGF) module to hierarchically integrate multi-level f eatures from both modalities. Extensive experiments on the MVTec3D-AD and Eyecan dies benchmarks demonstrate that CoGeoAD achieves state-of-the-art performance, effectively capturing both structural and textural anomalies in complex industri al scenarios.

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Truthfulness Does Not Scale Like Reasoning: Why Polling Fails as a Proxy Verifie r

Yegor Denisov-Blanch, Joshua Kazdan, Jessica Chudnovsky, Rylan Schaeffer, Sheng Guan, Soji Adeshina, Sanmi Koyejo

Pass@k and other methods of scaling inference compute can improve language mod el performance in domains with external verifiers, including mathematics and cod e, where incorrect candidates can be filtered reliably. This raises a natural qu estion: can we similarly scale compute to elicit gains in truthfulness for domai ns without convenient verification? We show that across five benchmarks and mode ls, surprisingly, it cannot. Even at $25\times$ the inference cost of naive samp ling, polling-style aggregation yields no consistent accuracy gains over single-sample baselines and often amplifies correlated errors. We find that under uncer tainty, models are better at predicting what other models will say within model ensembles than at identifying what is true, revealing a separation between socia l prediction and truth verification. Across models and benchmarks, aggregation fails to provide a robust truth signal because language model errors are strongl y correlated. The source of correlation goes beyond any individual benchmark: w e show that even when conditioned on out of distribution random strings and aske d to produce pseudo-random outputs, different models produce correlated outputs. Confidence-based weighting provides no benefit because self-reported confidence fails to reliably distinguish correct from incorrect answers.

These results delineate a boundary for inference-time scaling: in verified domai ns, additional samples provide more candidates for a verifier to filter; in unve rified domains, additional samples merely reinforce correlated errors.

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Differentiable Weightless Controllers: Learning Logic Circuits for Continuous Co ntrol

Fabian Kresse, Christoph H. Lampert

Controlling autonomous systems under real-world conditions often requires polici es that can be evaluated with low latency and minimal energy consumption. Unfort

unately, these conditions are at odds with the use of high-precision deep neural networks as controllers. In this work, we introduce Differentiable Weightless Controllers (DWCs), a symbolic-differentiable architecture that learns flexible, non-linear, yet highly efficient control policies. DWCs can be trained end-to-end via gradient-based techniques, yet compile directly into FPGA-compatible circuits with few- or even single-clock-cycle latency and nanojoule-level energy cost per action. Across five MuJoCo benchmarks, including high-dimensional Humanoid, DWCs achieve returns competitive with standard deep policies (full-precision or quantized neural networks). Furthermore, DWCs exhibit structurally sparse and interpretable connectivity patterns, enabling direct inspection of which input values influence control decisions.
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Floating-Point Networks with Automatic Differentiation Can Represent Almost All Floating-Point Functions and Their Gradients
Sejun Park, Yeachan Park, Geonho Hwang
Theoretical studies show that for any differentiable function on a compact domain, there exists a neural network that approximates both the function values and gradients. However, such a result cannot be used in practice since it assumes real parameters and exact internal operations. In contrast, real implementations only use a finite subset of reals and machine operations with round-off errors. In this work, we investigate whether a similar result holds for neural networks under floating-point arithmetic, when the gradient with respect to the input is computed by the automatic differentiation algorithm $D^\mathtt{AD}$. We first show that given a floating-point function ϕ (e.g., a loss function), arbitrary function values and gradients can be represented by a floating-point network f and $D^\mathtt{AD}(\phi\circ f)$, respectively. We further extend this result: given $\phi_1,\dots,\phi_n$, $D^\mathtt{AD}(\phi_i\circ f)$ can simultaneously represent arbitrary gradients while f represents the target values, under mild conditions. Our results hold for practical activation functions, e.g., ReLU, ELU, GELU, Swish, Sigmoid, and tanh.
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NeurOCNN: A Neural-Operator-Based Model for Physiological Time Series
Daya Kumar, Uday Devulapalli, Aarat Satsangi, Apurva Narayan
Neural operators have become a central tool in scientific machine learning for learning discretization-consistent solution operators, achieving strong results on partial differential equation (PDE) benchmarks. Physiological time series, however, are highly nonstationary and dominated by localized transient events, properties that can challenge both PDE-oriented neural operators and conventional deep models. We propose NeurOCNN, a neural-operator-based model for physiological signals that learns a robust function-to-label mapping. NeurOCNN integrates continuous-time, spline-parameterized convolutions to capture localized morphology with Fourier projection pooling for variable-to-fixed dimensional mapping, thereby enabling robust, discretization-invariant inference under zero-shot sampling-rate shifts. Empirically, NeurOCNN outperforms standard neural-operator baselines, achieves performance comparable to state-of-the-art methods, and maintains stable accuracy across multiple previously unseen evaluation sampling rates. The code is available at https://github.com/Idsl-group/NeurOCNN.
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Finite and Corruption-Robust Regret Bounds in Online Inverse Linear Optimization under M-Convex Action Sets
Taihei Oki, Shinsaku Sakaue
We study online inverse linear optimization, also known as contextual recommendation, where a *learner* sequentially infers an *agent*'s hidden objective vector from observed optimal actions over feasible sets that change over time. The learner aims to recommend actions that perform well under the agent's true objective, and the performance is measured by the *regret*, defined as the cumulative gap between the agent's optimal values and those achieved by the learner's recommended actions. Prior work has established a regret bound of $O(d\log T)$, as well as a finite but exponentially large bound of $\exp(O(d\log d))$, where d is the dimension of the optimization problem and T is the time horizon, while a re

gret lower bound of $\Omega(d)$ is known (Gollapudi et al. 2021; Sakaue et al. 2025). Whether a finite regret bound polynomial in d is achievable or not has remained an open question. We partially resolve this by showing that when the feasible sets are *M-convex*—a broad class that includes matroids—a finite regret bound of $O(d\log d)$ is possible. We achieve this by combining a structural characterization of optimal solutions on M-convex sets with a geometric volume argument. Moreover, we extend our approach to adversarially corrupted feedback in up to C rounds. We obtain a regret bound of $O((C+1)d\log d)$ without prior knowledge of C, by monitoring directed graphs induced by the observed feedback to detect corruptions adaptively.

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Seeing to Generalize: How Visual Data Corrects Binding Shortcuts

Nicolás Buzeta, Felipe del Rio, Cristian Hinostroza, Denis Parra, Hans Lobel, Rodrigo Toro Icarte

Vision Language Models (VLMs) are designed to extend Large Language Models (LLMs) with visual capabilities, yet in this work we observe a surprising phenomenon: VLMs can outperform their underlying LLMs on purely text-only tasks, particularly in long-context information retrieval. To investigate this effect, we build a controlled synthetic retrieval task and find that a transformer trained only on text achieves perfect in-distribution accuracy but fails to generalize out-of-distribution (OOD), while subsequent training on an image-tokenized version of the same task nearly doubles text-only OOD performance. Mechanistic interpretability reveals that visual training changes the model’s internal binding strategy: text-only training encourages positional shortcuts, whereas image-based training disrupts them through spatial translation invariance and other encoder-level inductive biases, forcing the model to adopt a more robust symbolic binding mechanism that persists even after text-only examples are reintroduced.

We further characterize how binding strategies vary across training regimes and visual encoders, and show that analogous shifts occur during pre-trained LLM-to-VLM transitions. Our findings suggest that cross■modal training can enhance reasoning and generalization even for tasks grounded in a single modality.

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Concept Concentration for Faithful Representation Intervention

Hongzheng Yang, Yongqiang Chen, Zeyu Qin, Tongliang Liu, Chaowei Xiao, Kun Zhang, Bo Han

Representation intervention aims to locate and modify the representations that encode the underlying concepts in Large Language Models (LLMs) to elicit the aligned and expected behaviors. Despite the empirical success, it has never been examined whether one could locate the faithful concepts for intervention. In this work, we explore the question in safety alignment. If the interventions are faithful, the intervened LLMs should erase the harmful concepts and be robust to both in-distribution adversarial prompts and the \textit{out-of-distribution} (OOD) jailbreaks. While it is feasible to erase harmful concepts without degrading the benign utility of LLMs in linear settings, we show that it is \textit{infeasible} in the general non-linear setting. To tackle the issue, we propose \texttt{Concept Concentration} (\texttt{COCA}). \texttt{COCA} refactors the training data with an explicit reasoning process, which first identifies the potential unsafe concepts and then decides the responses. Essentially, \texttt{COCA} simplifies the decision boundary between harmful and benign representations, enabling more effective linear erasure. Extensive experiments with multiple representation intervention methods and model architectures demonstrate that \texttt{COCA} significantly reduces both in-distribution and OOD jailbreak success rates, and meanwhile maintaining strong performance on regular tasks such as math and code generation. Our code is publicly available at: \url{https://github.com/tmlr-group/COCA}.

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TimeLAVA: Learning-Agnostic Valuation for Time Series Data

Wenqin Liu, Weizhi Quan, Aoqi Zuo, Erdun Gao, Vu Nguyen, Dino Sejdinovic, Howard Bondell, Mingming Gong

Data valuation quantifies the intrinsic quality of individual samples to enable principled data curation, quality control, and robust learning. For time series

in critical domains such as healthcare, finance, and industrial monitoring, effe
ctive valuation methods are essential yet fundamentally lacking. Existing approa
ches are either model-dependent, limiting their generalizability, or designed fo
r i.i.d. data and thus fail to capture temporal dependencies, multi-scale patter
ns, and non-stationary dynamics inherent to sequential data. We introduce TimeLA
VA, a learning-agnostic framework that values temporal segments by their margina
l contribution to minimizing distributional discrepancy between evaluated and re
ference data. At its core is a novel Selective Wavelet-based Wasserstein ($\math
cal{W}_\text{SW}$) discrepancy combining multi-scale wavelet transforms for temp
oral localization with unbalanced optimal transport for robustness to distributi
onal shifts. Segment values are efficiently computed via sensitivity analysis wi
thout requiring model training and aggregated into point-wise scores. We provide
 theoretical guarantees linking valuation to model-agnostic generalization and p
rove bounded sensitivity to outlier contamination. Extensive experiments across
anomaly detection, data pruning, and label noise detection demonstrate that Time
LAVA produces significantly more informative value scores than existing methods
on diverse real-world datasets.
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Attention Hijacking: Backdooring Text Dataset Distillation via Semantic Anchors
Hang Ren, Xin Wang, Tong Yue, Wen Chen, Junqing Le
Dataset Distillation has emerged as a promising technique for compressing large-
scale datasets into compact synthetic sets while preserving model performance. H
owever, the security implications of this paradigm, particularly within the Tran
sformer-based text classification domain, remain underexplored. In this paper, w
e identify "Distilled Attention Labels" as a pivotal yet overlooked vulnerabilit
y. We propose Attention Hijacking (AH), a stealthy backdoor attack that manipula
tes the bi-level optimization process to explicitly hijack the attention mechani
sm of target models via synthetic data. Distinct from traditional poisoning that
 often compromises clean accuracy, AH achieves stealthiness without utility degr
adation. To explain this, we formulate the "Semantic Anchoring Hypothesis", char
acterizing the interaction between trigger semantics and attack mechanisms. We d
emonstrate that AH functions as a semantic-adaptive mechanism: when triggers ali
gn with domain-specific semantic anchors (e.g., "film" in sentiment analysis), o
ur method achieves a synergistic effect, boosting both attack success rates (>99
%) and clean test accuracy. Conversely, for functional or noise triggers, AH enf
orces attention segregation to prevent utility collapse, maintaining exceptional
 robustness where baseline attacks fail. Extensive experiments across multiple d
atasets and varying model scales—ranging from BERT-Tiny to BERT-Base—validate th
e scalability and dominance of AH. Our findings reveal that attention-based dist
illation is a double-edged sword, underscoring the urgent need for robust defens
es in the era of data-efficient learning.
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FedTreeLoRA: Reconciling Statistical and Functional Heterogeneity in Federated L
oRA Fine-Tuning
Jieming Bian, Lei Wang, Letian Zhang, Jie Xu
Federated Learning (FL) with Low-Rank Adaptation (LoRA) has become a standard fo
r privacy-preserving LLM fine-tuning. However, existing personalized methods pre
dominantly operated under a restrictive Flat-Model Assumption: they addressed cl
ient-side *statistical heterogeneity* but treated the model as a monolithic bloc
k, ignoring the *functional heterogeneity* across LLM layers. We argue that thes
e two statistical (horizontal) and functional (vertical) dimensions, are *orthog
onal in source yet coupled in interaction*, implying that the optimal depth of p
arameter sharing is functionally dependent on client similarity. To address this
, we propose **FedTreeLoRA**, a framework employing tree-structured aggregation
for fine-grained, layer-wise alignment. By dynamically constructing an aggregati
on hierarchy, FedTreeLoRA allows clients to share broad consensus on shallow ’tr
unks‘ while progressively specializing on deep ‘branches'. Experiments on NLU an
d NLG benchmarks demonstrate that FedTreeLoRA significantly outperforms state-of
-the-art methods by effectively reconciling generalization and personalization.
****

Spectral-Progressive Thought Flow for Lightweight Multimodal Reasoning
Yixian Shen, Zhiheng Yang, Qi Bi, Changshuo Wang, Shuai Wang, JIA-HONG HUANG, George Floros, Prayag Tiwari, Anuj Pathania
Multimodal reasoning often relies on long chains of intermediate textual and visual thoughts, where accumulating visual tokens and dense cross-modal attention incur substantial computation and memory overhead.
To address this challenge, we propose Spectral-Progressive Thought Flow (*SpecFlow*), a *novel* lightweight multimodal reasoning framework that represents intermediate visual thoughts in a fixed-size discrete cosine space.
By exploiting strong energy compaction, *SpecFlow* preserves global layout and relational structure while introducing high-frequency details only when increased spatial precision is required.
To align visual state evolution with linguistic intent, classifier-free guidance enables autoregressive textual thoughts to steer flow-based updates of the visual workspace without expanding the context.
As a result,*SpecFlow* maintains a bounded visual workspace whose updates depend only on the current visual state and accumulated textual trace, enabling long-horizon inference with stable latency and memory usage independent of reasoning depth.
Empirical results show that *SpecFlow* achieves competitive or superior reasoning performance while reducing computation and memory costs by up to *$2.1\times$*.
****
Generalization of Gibbs and Langevin Monte Carlo Algorithms in the Interpolation Regime
Andreas Maurer, Erfan Mirzaei, Massimiliano Pontil
This paper provides data-dependent bounds on the expected error of the Gibbs algorithm in the overparameterized interpolation regime, where low training errors are also obtained for impossible data, such as random labels in classification. The results show that generalization in the low-temperature regime is already signaled by small training errors in the noisier high-temperature regime. The bounds are stable under approximation with Langevin Monte Carlo algorithms. The analysis motivates the design of an algorithm to compute bounds, which on the MNIST, CIFAR-10 and SVHN datasets yield nontrivial, close predictions on the test error for true labeled data, while maintaining a correct upper bound on the test error for random labels.
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Narrowing the ANN–SNN Gap for Continuous 1D Temporal Signal Classification with Multi-Scale Temporal Encoding and Sparsity-Regularized Transform Encoding
Qi Sun, Yulei Huang, Zhen Cao, Biao Hou
Spiking neural networks (SNNs) promise energy-efficient inference, yet on static vision benchmarks, they often trail matched ANNs under short simulation horizons. Under a matched-backbone and matched-budget protocol without extra tricks, we find that this ANN-SNN accuracy gap is consistently smaller on representative continuous 1D temporal signal classification benchmarks than on image benchmarks. We attribute this to a mechanism-level mismatch: leaky integration naturally implements causal evidence accumulation over time for native temporal signals, while static images typically require amplitude-to-spike encoding, whose finite-window estimation error becomes non-negligible at short horizons. Guided by this view, we propose a plug-and-play framework that combines Multi-Scale Temporal Encoding (MTE) and Sparsity-Regularized Transform Encoding (STE). MTE replaces naive repetition with multi-scale streams and allocates scale-aligned multi-bit integer spikes to increase per-step information density, and STE replaces a controllable fraction of LIF units with a transform-encoding neuron trained using auxiliary reconstruction and sparsity regularization, with a synthesis branch used only during training. Across diverse 1D datasets and backbone families, MTE$\times$STE consistently improves the accuracy-efficiency trade-off over standard SNN baselines and matches or occasionally surpasses ANN counterparts.
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LMCleaner: Efficient and Certified Online Unlearning via Influence Propagation T

runcation
Jie Xu, Zihan Wu, Wenbo Pan, Jiao Yin, Yong-Feng Ge, Hua Wang, Cong Wang, Xiaohua Jia
Existing machine unlearning methods primarily focus on removing data influence after training completes, which is effective for many scenarios, but a complementary capability is needed when removal requests arise during ongoing training. We propose LMCleaner, an efficient and certified *online* unlearning framework that can process unlearning requests at any training step without waiting for training completion. Our key insight is that influence propagation can be decomposed into a trust region where linear approximation is accurate, and a residual that concentrates in a low-dimensional subspace and can be efficiently masked by calibrated noise. Building on this insight, we design an influence propagation truncation mechanism that treats mini-batch influence as atomic units, computes influence within a truncation window for efficient removal, and injects subspace-aware noise for certified privacy. Our theoretical analysis proves that the truncation residual decays exponentially with window size and that the unlearned model is (ε, δ)-indistinguishable from retraining. Experiments demonstrate that LMCleaner achieves over $100\times$ computational savings compared to baselines while maintaining model utility and defending against membership inference attacks.
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Dismantling Pathological Shortcuts: A Causal Framework for Faithful LVLM Decoding
Liu Yu, Can Chen, PING KUANG, Zhikun Feng, Fan Zhou, Gillian Dobbie
Large Vision-Language Models (LVLMs) exhibit sophisticated reasoning but remain susceptible to object hallucination. Deviating from the prevailing *attention intensity assumption*, we reveal a deeper dynamic structural misalignment: hallucination is triggered at decision-critical steps where specific attention heads, acting as risky mediators, decouple from visual evidence to lock onto language priors. This establishes a pathological shortcut that bypasses visual grounding. To dismantle this, we propose **Fox** (*F*aithfulness and *O*bservational-flow via e*X*pression-rectification), a training-free inference-time framework. **Fox** diagnoses structural misalignment using a visual attention entropy probe to localize risky mediators unsupervisedly. We then execute a targeted causal intervention via numerical logit saturation to physically sever the shortcut path. Finally, a conflict-gated cooperative decoding strategy reconciles interventional faithfulness with observational fluency. Extensive experiments demonstrate that **Fox** achieves SOTA performance, outperforming SID by 29.1% while preserving linguistic richness. Code is available at <https://github.com/Cc2021start/Fox>.
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One Intervention per Component is Enough: Towards Identifiability in Linear Stochastic Dynamics from Steady State
Saber Salehkaleybar
We study the problem of recovering the parameters of a multivariate Ornstein–Uhlenbeck (OU) process from steady-state observational and interventional data. In many applications, such as large-scale gene perturbation experiments, only stationary “snapshot” measurements are available, making standard stochastic differential equation estimation methods that rely on time-series trajectories inapplicable. We first establish an identifiability result: one intervention per strongly connected component (SCC) of the drift graph suffices to recover all OU process parameters generically up to a global scaling factor. This holds provided that the SCC condensation graph is connected with a single root and certain spectral nondegeneracy assumptions hold. We propose a recursive learning algorithm that orders SCCs topologically and, for each component, isolates its marginal dynamics and solves a linear system derived from the steady-state moment equations, leveraging parameters recovered for upstream components. Building on this theoretical foundation, we propose a regularized least-squares estimator that jointly minimizes residuals of the steady-state mean and covariance equations across observational and interventional data. Experiments on synthetic and real datasets demonstrate the effectiveness of our method in recovering parameters and predicting u

nseen interventions.
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HiCI: Hierarchical Construction–Integration for Long-Context Attention
Xiangyu Zeng, Qi Xu, Yunke Wang, Chang Xu
Long-context language modeling is commonly framed as a scalability challenge of token-level attention, yet local-to-global information structuring remains largely implicit in existing approaches. Drawing on cognitive theories of discourse comprehension, we propose HiCI (Hierarchical Construction--Integration), a hierarchical attention module that constructs segment-level representations, integrates them into a shared global context, and broadcasts both to condition segment-level attention. We validate HiCI through parameter-efficient adaptation of LLaMA-2 (7B and 13B) and Qwen3-8B with only $\sim$4--5\% additional parameters, extending context to 100K/64K tokens for LLaMA-2-7B/13B and to 48K tokens for Qwen3-8B. Across language modeling, retrieval, and instruction-following benchmarks, HiCI yields consistent improvements over strong baselines, including matching proprietary models on topic retrieval and surpassing GPT-3.5-Turbo-16K on code comprehension. These results demonstrate the effectiveness of explicit hierarchical structuring as an inductive bias for long-context modeling.
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Navigating Massive Visual Context in Retrieval-Augmented Generation via Multimodal Memory Graph
Qiuchen Wang, Shihang Wang, Yu Zeng, Qiang Zhang, Fanrui Zhang, Zhuoning Guo, Bosi Zhang, Wenxuan Huang, Lin Chen, Zehui Chen, Pengjun Xie, Ruixue Ding
Effectively retrieving, reasoning, and understanding multimodal information remains a critical challenge for agentic systems. Traditional Retrieval-augmented Generation (RAG) methods rely on linear interaction histories, which struggle to handle long-context tasks, especially those involving information-sparse yet token-heavy visual data in iterative reasoning scenarios. To bridge this gap, we introduce VimRAG, a framework tailored for multimodal Retrieval-augmented Reasoning across text, images, and videos. Inspired by our systematic study, we model the reasoning process as a dynamic directed acyclic graph that structures the agent states and retrieved multimodal evidence. Building upon this structured memory, we introduce a Graph-Modulated Visual Memory Encoding mechanism, with which the significance of memory nodes is evaluated via their topological position, allowing the model to dynamically allocate high-resolution tokens to pivotal evidence while compressing or discarding trivial clues. To implement this paradigm, we propose a Graph-Guided Policy Optimization strategy. This strategy disentangles step-wise validity from trajectory-level rewards by pruning memory nodes associated with redundant actions, thereby facilitating fine-grained credit assignment. Extensive experiments demonstrate that VimRAG consistently achieves state-of-the-art performance on diverse multimodal RAG benchmarks.
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Asymmetric Perturbation in Solving Bilinear Saddle-Point Optimization
Kenshi Abe, Mitsuki Sakamoto, Kaito Ariu, Atsushi Iwasaki
This paper proposes asymmetric perturbation, where only one player's payoff function is perturbed, for solving bilinear saddle-point optimization problems, commonly arising in minimax problems, game theory, and constrained optimization.
Symmetric perturbation is known to require decreasing its strength to ensure convergence to a solution, i.e., an equilibrium in the original game, resulting in a slower rate.
First, with asymmetric perturbation, we show that, for a sufficiently small perturbation strength, the equilibrium strategy of the asymmetrically perturbed game coincides with an equilibrium strategy of the original unperturbed game.
Second, building on this coincidence, we construct a learning algorithm with a linear last-iterate convergence rate.
Third, motivated by the fact that the coincidence relies on the perturbation strength being sufficiently small, we also provide a parameter-free variant, retaining the linear rate.
Finally, we empirically demonstrate fast convergence toward equilibria in both normal-form and extensive-form games.

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SPARD: Defending Harmful Fine-Tuning Attack via Safety Projection with Relevance–Diversity Data Selection
Shuhao Chen, Weisen Jiang, Yeqi Gong, Shengda Luo, Chengxiang Zhuo, Zang Li, James Kwok, Yu Zhang
Fine-tuning large language models often undermines their safety alignment, a problem further amplified by harmful fine-tuning attacks in which adversarial data removes safeguards and induces unsafe behaviors. We propose SPARD, a defense framework that integrates Safety-Projected Alternating optimization with Relevance-Diversity aware data selection. SPARD employs SPAG, which optimizes alternatively between utility updates and explicit safety projections with a set of safe data to enforce safety constraints. To curate safe data, we introduce a Relevance–Diversity Determinantal Point Process to select compact safe data, balancing task relevance and safety coverage. Experiments on GSM8K and OpenBookQA under four harmful fine-tuning attacks demonstrate that SPARD consistently achieves the lowest average attack success rates, substantially outperforming state-of-the-art defense methods, while maintaining high task accuracy. Code is available at https://github.com/shuhao02/SPARD.
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ReViT: Rotational-equivariant Vision Transformers for Neural PDE Solvers
Hao Wei, Björn List, Nils Thuerey
Physics obeys strict symmetries like rotational equivariance. However, the standard Transformer architectures widely used in physics foundation models do not enforce these constraints by construction. We introduce ReViT, a rotationally equivariant Vision Transformer framework for neural PDE solvers operating on grid-based physical fields that achieves exact equivariance for the discrete groups C_4 (2D) and the chiral octahedral group O (3D), with bounded approximate $\mathrm{SO}(d)$ equivariance for continuous rotations. ReViT maps scalar and vector inputs into locally invariant representations derived from physics-based canonical bases, enabling the use of standard self-attention without symmetry violations. Built on a hierarchical Swin-style backbone with a precomputed reference basis pyramid, ReViT preserves equivariance across multi-scale operations.
We evaluate ReViT on a wide range of 2D and 3D PDE benchmarks, such as Magnetohydrodynamics and Turbulent Channel Flows, demonstrating significant gains over state-of-the-art baselines. ReViT exhibits strong generalization, and reduces MSE by up to 65\% compared with the best-performing alternatives.
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InstEmb: Instruction-Following Embeddings through Glimpses of the Future
Tianhao Gao, Jun Fang, Xiaohui Zhang, Zhiyuan Liu, Chao Liu, Pengzhang Liu, Qixia Jiang
Recent advances have empowered large language models (LLMs) with remarkable fine-grained instruction-following capabilities in text generation tasks. However, embedding methods typically rely solely on the hidden state of the input's last token, limiting their ability to capture complete semantic signals distributed across the full output tokens. Moreover, existing discrete-to-continuous re-encoding approaches introduce semantic discontinuity. To address these limitations, we propose $\textbf{InstEmb}$, a novel instruction following embedding framework.
InstEmb jointly optimizes two key aspects: (1) Input-Intrinsic semantic information, achieved by employing contrastive learning focused on the representation of the last input token, and (2) Output-Aware semantic information, captured through representation self-distillation leveraging learnable look-ahead tokens without introducing additional decoding latency. Additionally, we introduce $\textbf{Dual-Anchor Alignment Pooling (DAAP)}$, explicitly aligned with our dual training objectives. Extensive experiments demonstrate that InstEmb achieves state-of-the-art performance across multiple instruction following benchmarks without benchmark-specific supervised data.
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SubspacePath Pruner: Inference-time Pruning via Probe-based Representation–Parameter Coupling
Gong Zhiren, Yikun Hou, Fan Wu, CHE WANG, Fuyao Zhang, Tiantong Wu, Yurong Hao,

Jiaming Zhang, Yiyang Duan, Tiantong Wang, Fei Huang, Chau Yuen, Wei Yang Bryan Lim

Large-scale dedicated application of LLMs in diverse scenarios increasingly demands specialized model inference behavior under strict constraints of accuracy, latency, and memory. However, the heterogeneous and long-tailed nature of real-world specialized scenarios makes it difficult to obtain training data and optimize models.

We study a practical inference-time specialization setting: given an LLM base, we compile a reusable, budget-bounded pathway/subnetwork within a specific scenario.

Our approach is motivated by an empirical coupling phenomenon: input scenario sets aligned with similar representation subspaces (e.g., domain) in embedding space tend to activate a consistent and sparse set of internal reasoning pathways in model parameter space.

To build the bridge between them, we propose probe-based SubspacePath Pruner with two core components:

(1) Domain-Basis Synthesis (DBS) constructs a quasi-orthogonal basis of domain axes in embedding space, serving as a stable coordinate system.

(2) Probe-based Scenario Pruning (PSP) uses efficient layer-wise linear probes to estimate axis alignment and compute budgeted head-wise pathways for a specific scenario.

Experiments on LLaMA-2-13B show 29.3 average Recall on cross-domain tests (vs. 24.7 dense) and 21.6 on cross-dataset tests (vs. 25.5 dense) with 1.27x speedup at ~30% pruning ratio.

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Anti-Aliasing Matters: A Dynamic Network for Time Series Forecasting

Heng Zhou, Xin Sun, Chao Li

Real-world time series inherently exhibit complex temporal patterns. Within chaotic systems, significant mixing and entanglement occur between different time-varying modes. Given that time series exhibit distinctly different patterns at various sampling scales, downsampling to extract multiscale features is a common approach. However, conventional downsampling causes high-frequency components in the original signal, those exceeding the new Nyquist frequency, to undergo spectral folding. This erroneously introduces spurious low-frequency patterns, perceived as low-frequency noise, thereby leading to the ***aliasing problem***. To address this problem, we propose a Decomposition-Prevention-Fusion architecture framework called **DMANet**, which introduces the **D**ynamic **M**ultiscale **A**nti-Aliasing **Net**work. Specifically, DMANet comprises two key components: Multiscale Convolutional Downsampling, designed to capture temporal dependencies and inter-channel interactions, and an Anti-Aliasing Operation, which includes Pre-Sampling Anti-Aliasing Filtering and Post-Sampling Interpolation. These designs guarantee the fidelity of multiscale features before and after downsampling. We show that by mitigating the risk of aliasing, our proposed simple convolutional downsampling architecture achieves performance competitive with common baselines. Our codes are available at https://github.com/HenryZhou2024/DMANet.

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A Conflict-aware Evidential Framework for Reliable Sleep Stage Classification

Yunzhi Tian, Dekui Wang, Qirong Bu, Wei Zhou, Xingxing Hao, Jun Feng

Multi-view learning has been widely applied for sleep stage classification using multi-modal data. However, existing methods typically assume that different modalities are well-aligned, which is often unattainable in real-world scenarios, thereby compromising the reliability of the staging results. In this paper, we propose ConfSleepNet, a conflict-aware evidential framework that dynamically resolves inter-view conflicts. The framework consists of multi-view evidence extraction and conflict-aware aggregation. In the first phase, it learns category-related evidence from different modalities, which represents the degree of support for individual sleep stages. Considering the inherent characteristics of varying modalities, we propose hybrid category structures for different modalities to promote more reasonable evidence learning. In the second phase, view-specific opinions, including prediction results and uncertainty, are constructed from the learn

ed evidence. Notably, we propose a novel conflict-aware aggregation method that integrates these view-specific opinions into a reliable joint decision. This mechanism can effectively resolve conflicts among opinions and synthesize them into a reliable joint decision. Both theoretical analysis and experimental results demonstrate the effectiveness of ConfSleepNet in sleep staging tasks. The code is available at https://github.com/By4te/ConfSleepNet_ICML2026/.

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Representational Curvature Modulates Behavioral Uncertainty in Large Language Models

Jack G. King, Evelina Fedorenko, Eghbal A. Hosseini

Temporal straightening describes how, across layers in autoregressive LLMs, the trajectory traced by the token representations of an input sequence becomes straighter, potentially enabling next-token prediction by linear extrapolation. However, a direct link between this trajectory and token-level behavior has been missing. We provide such a link by relating contextual curvature—a geometric measure of how sharply the representational trajectory bends over recent context—to next-token entropy. Across two models (GPT-2 XL and Pythia-2.8B), contextual curvature is correlated with entropy, and this relationship emerges during training. Perturbation experiments reveal selective dependence: manipulating curvature through trajectory-aligned interventions reliably modulates entropy, while geometrically misaligned perturbations have no effect. Finally, regularizing representations to be straighter during training modestly reduces token-level entropy without degrading validation loss. These results identify trajectory curvature as a task-aligned representational feature that influences behavioral uncertainty in LLMs.

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Squeezing More from the Stream : Learning Representation Online for Streaming Reinforcement Learning

Nilaksh, Antoine Clavaud, Mathieu Reymond, Francois Rivest, Sarath Chandar

In streaming Reinforcement Learning (RL), transitions are observed and discarded immediately after a single update. While this minimizes resource usage for on-device applications, it makes agents notoriously sample-inefficient, since value-based losses alone struggle to extract meaningful representations from transient data. We propose extending Self-Predictive Representations (SPR) to the streaming pipeline to maximize the utility of every observed frame. However, due to the highly correlated samples induced by the streaming regime, naively applying this auxiliary loss results in training instabilities. Thus, we introduce orthogonal gradient updates relative to the momentum target and resolve gradient conflicts arising from streaming-specific optimizers. Validated across the Atari, MinAtar, and Octax suites, our approach systematically outperforms existing streaming baselines. Latent-space analysis, including t-SNE visualizations and effective-rank measurements, confirms that our method learns significantly richer representations, bridging the performance gap caused by the absence of a replay buffer, while remaining efficient enough to train on just a few CPU cores.

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RobuQ: Pushing DiTs to W1.58A2 via Robust Activation Quantization

Kaicheng Yang, Xun Zhang, Haotong Qin, Yucheng Lin, Kaisen Yang, Xianglong Yan, Yulun Zhang

Diffusion Transformers (DiTs) have emerged as a powerful backbone for image generation, offering superior scalability over U-Nets. However, their practical deployment is hindered by significant computational costs. While Quantization-Aware Training (QAT) shows promise, its application to DiTs is challenged by the high sensitivity and complex distributions of activations. Identifying activation quantization as the primary bottleneck for low-bit settings, we propose **RobuQ**, a systematic QAT framework. We first establish a strong ternary weight (W1.58A4) baseline. Building on this, we introduce **RobustQuantizer**, which utilizes the Hadamard transform to convert unknown per-token distributions into normal distributions. Furthermore, we propose **AMPN**, the first **A**ctivation-only **M**ixed-**P**recision **N**etwork pipeline, applying ternary weights globally while allocating layer-specific activation precisions to eliminate information bottle

necks. Extensive experiments demonstrate that **RobuQ** achieves state-of-the-art performance on **ImageNet-1K**, representing the first stable image generation with activations quantized to an average of 2 bits. Code is available at https://github.com/racoonykc/RobuQ.

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Granularity-Aware Adaptive Classifier Expansion via Zero-Shot Learning

Xiangyu Wang, Yanze Gao, Changxin Rong, Lyuzhou Chen, Xiren Zhou, Huanhuan Chen

Zero-shot classifier expansion aims to recognize unseen classes by learning a shared mechanism to map semantics of all classes to classifier weights without access to images. However, existing methods rely on a shared mapping, which is difficult to classify in scenarios containing a mixture of distinct and similar classes, especially with the continuous expansion of classes. Since this mapping prioritizes general attributes for distinct classes while neglecting subtle attributes for similar ones, this granularity mismatch, compounded by sensitivity to noise, induces optimization interference where gradients from distinct classes dominate the learning process. To overcome this limitation, a granularity-aware adaptive framework with interventions is introduced to balance them. Specifically, this method first generates multi-source semantics by intervening on non-causal noise, then discovers latent class structure to separate distinct classes, and finally refine similar classes to synthesize weights with invariance to non-causal noise. The effectiveness is demonstrated through theoretical and empirical analysis in multiple aspects.

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One LR Doesn’t Fit All: Heavy-Tail Guided Layerwise Learning Rates for LLMs

Di He, Songjun Tu, Keyu Wang, Lu Yin, Shiwei Liu

Learning rate configuration is a fundamental aspect of modern deep learning. The prevailing practice of applying a uniform learning rate across all layers overlooks the structural heterogeneity of Transformers, potentially limiting their effectiveness as the backbone of Large Language Models (LLMs). In this paper, we introduce Layerwise Learning Rate (LLR), an adaptive scheme that assigns distinct learning rates to individual Transformer layers. Our method is grounded in Heavy-Tailed Self-Regularization (HT-SR) theory, which characterizes the empirical spectral density (ESD) of weight correlation matrices to quantify heavy-tailedness. Layers with weaker heavy-tailedness are assigned larger learning rates to accelerate their training, while layers with stronger heavy-tailedness receive smaller learning rates. By tailoring learning rates in this manner, LLR promotes balanced training across layers, leading to faster convergence and improved generalization.

Extensive experiments across architectures (from LLaMa to GPT-nano), optimizers (AdamW and Muon), and parameter scales (60M–3B, up to 100B tokens) demonstrate that LLR achieves up to 1.5× training speedup and outperforms baselines, notably raising average zero-shot accuracy from 47.09% to 49.02% for 1B models and from 48.58% to 50.61% for 3B models. A key advantage of LLR is its low tuning overhead: it transfers nearly optimal LR settings directly from the uniform baseline. Code is available at https://github.com/hed-ucas/Layer-wise-Learning-Rate.

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Next-Token Prediction and Regret Minimization

Mehryar Mohri, Clayton Sanford, Jon Schneider, Kiran Vodrahalli, Yifan Wu

We study the robustification of next-token prediction models in adversarial online decision-making environments. Suppose a next-token predictor is trained on a distribution over sequences of opponent actions, and a decision maker acts by approximately best responding to the model’s predictions. If this induced decision-making algorithm achieves low adversarial regret, we call such distributions low-regret distributions. For models with unbounded context, we show that every distribution is exponentially close in total variation distance to a low-regret distribution. Thus, sublinear regret can always be achieved while changing the original next-token prediction distribution by only a negligible amount. By contrast, for models with bounded context windows, as in standard transformer architectures, we prove an impossibility result: some distributions are $\Theta(1)$-far from every low-regret distribution. We complement these results with two positive

findings. First, allowing a modest increase in context length restores robustification with near-optimal regret guarantees. Second, the unbounded-context robustification procedure can be implemented by adding a constant number of layers to a standard transformer architecture. Finally, we provide empirical evidence that small transformer models can be efficiently trained to represent these robustified low-regret distributions.
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How Hard Can It Be? Hardness-Aware Multi-Objective Unlearning
Jiangwei Chen, Xinyuan Niu, Rachael Hwee Ling Sim, Zhengyuan Liu, Nancy F. Chen, Bryan Kian Hsiang Low
Machine unlearning aims to remove the influence of specific forget training data due to privacy, copyright or bias concerns while maintaining the model performance on the remaining retain data.
Existing unlearning algorithms, such as optimizing a weighted combination of losses, have tried to achieve these objectives of improving forget quality and maintaining retain utility.
However, they do not guarantee that these objectives can be improved by a specified extent for all forget and retain data.
In this work, we address this limitation with a novel and theoretically-grounded approach from a constrained optimization perspective.
Firstly, we identify that the *hardness* of reconciling both objectives can be quantified by the similarity between the forget data and the retain data.
Next, we derive an unlearning algorithm (HAMU) with the overall goal of guaranteeing a specified improvement in forget quality while minimizing the retain utility cost/degradation by updating the model weights based on our hardness measure.

Our hardness measure also informs users when retain utility degradation is unavoidable,
i.e., both objectives cannot be improved simultaneously, and stopping should be considered.
Our algorithm is applicable to non-convex models and is easily parallelizable, making it readily deployable in real-world scenarios.
We empirically demonstrate HAMU's superior performance over baselines on both image and text datasets using large models.
Our code is available at https://github.com/aoi3142/HAMU.
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Overcoming PINNs Failure Modes In High Dimension With Low-Rank Fourier Sum
Natan Kaminsky, Daniel Freedman, Kira Radinsky
Physics-informed neural networks (PINNs) can be unreliable on PDEs with oscillatory, multiscale, stiff, or long-time solutions, and these difficulties worsen in high dimensions where collocation-based training yields large numerical integration error and high-variance gradients. We propose Low-Rank Fourier Sums (LoRFS), representing the solution as a low-rank sum of separable Fourier expansions (products of one-dimensional Fourier series across coordinates). This makes high-frequency structure explicit and enables closed-form evaluation of common physics-based objectives and their gradients (e.g., L^2 residual and variational losses), replacing sampling-based collocation estimates with analytic loss evaluation and eliminating sampling noise. We further provide theoretical results that clarify why LoRFS is particularly well suited to high-dimensional regimes. Across canonical PINN failure-mode benchmarks and their high-dimensional extensions, LoRFS consistently outperforms strong PINN baselines and remains stable in regimes where competing methods degrade.
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GOTabPFN: From Feature Ordering to Compact Tokenization for Tabular Foundation Models on High-Dimensional Data
Al Zadid Sultan Bin Habib, Md Younus Ahamed, Prashnna Kumar Gyawali, Gianfranco Doretto, Donald Adjeroh
We investigate how to make small tabular foundation models effective for High-Dimensional, Low-Sample Size (HDLSS) tabular prediction without retraining large backbones. We introduce Graph-guided Ordering with Local Refinement (GO-LR), show

its equivalence to weighted Minimum Linear Arrangement, and interpret the practical solver as a TSP-path-style surrogate. We propose GOTabPFN,which builds on GO-LR, and a Neuro-Inspired Subunit Compression (NSC) unit to pool locally adjacent ordered features into meta-features, yielding a compact representation that makes TabPFN-style prediction practical in HDLSS regimes. Across tabular benchmarks, GOTabPFN improves stability and accuracy under tight token budgets.
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Transporting Task Vectors across Different Architectures without Training
Filippo Rinaldi, Aniello Panariello, Giacomo Salici, Angelo Porrello, Simone Calderara

Adapting large pre-trained models to downstream tasks often produces task-specific parameter updates that are expensive to relearn for every model variant. While recent work has shown that such updates can be transferred between models with identical architectures, transferring them across models of different widths remains unexplored. In this work, we introduce Theseus, a training-free method for transporting task updates across heterogeneous-width models. Rather than matching parameters, we characterize a task update by the functional effect it induces on intermediate representations. We formalize task-vector transport as a functional matching problem on observed activations and show that, after aligning representation spaces via orthogonal Procrustes analysis, it admits a stable closed-form solution that preserves the geometry of the update. We evaluate Theseus on vision and language models across different widths, showing consistent improvements over baselines without additional training or backpropagation. Our results show that task updates can be meaningfully transferred across architectures when task identity is defined functionally rather than parametrically. Code is available at https://github.com/apanariello4/merge-and-rebase.
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Pseudo-Mallows for Efficient Probabilistic Preference Learning
Sylvia Qinghua Liu, Valeria Vitelli, Carlo Mannino, Arnoldo Frigessi, Ida Scheel

We propose the Pseudo-Mallows distribution over the set of all permutations of n items, to approximate the posterior distribution of the Bayesian Mallows model. The Bayesian Mallows model has been successfully used for recommender systems to learn personal preferences from highly incomplete users data. However current inference algorithms do not scale, preventing its use in real-time applications. The Pseudo-Mallows distribution is a product of univariate discrete Mallows-like distributions, where the quality of the approximation depends on the order of the n items in the factorization sequence. In a variational setting, we optimize the variational order parameter by minimising a marginalized KL-divergence, conjecturing a certain form of the optimal variational order that depends on the data, and proposing an approximation algorithm for this discrete optimization. Empirical evidence and some theory support our conjecture. We demonstrate on clicking data that variational inference via the Pseudo-Mallows distribution allows much faster probabilistic preference learning compared to alternative MCMC-based options.
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ManifoldKV: Training-Free KV Cache Compression via Euclidean Outlier Detection
Debajyoti Datta, Trishala Neeraj, Bibek Paudel, Vyom Sharma, Subhabrata Mukherjee

Long-context inference is constrained by KV-cache memory, which grows linearly with sequence length; KV-cache compression therefore hinges on reliably selecting which past tokens to retain. Most geometry-based eviction methods score keys by cosine similarity to a global centroid, but cosine is scale-invariant and can discard magnitude cues that distinguish semantically salient tokens. We propose ManifoldKV, a training-free scorer that ranks tokens by Euclidean distance to the key centroid, capturing both angular and radial deviations. On the RULER benchmark, ManifoldKV achieves **95.7%** accuracy at 4K–16K contexts with 20% compression, matching the best geometric baseline overall while *decisively* outperforming it in two regimes where magnitude information is critical. First, on multi-key retrieval, ManifoldKV reduces directional collisions, achieving **92.4%** vs KeyDiff's 77.0% (+15.4 points) on 3-key NIAH at 50% compression. Second, to addre

ss dilution and performance collapse of global centroids at 64K context, we introduce WindowedManifoldKV, which restores accuracy to 84.3% at 25% compression, a 49-point recovery over global L2 and +3.2 points over KeyDiff. Beyond RULER, we validate on real-world benchmarks: on LongBench, ManifoldKV outperforms KeyDiff by +2.80 points on Qwen3-8B (winning 12 of 14 tasks) and +0.49 on Phi-4; on HELMET, ManifoldKV achieves +6.5 EM on RAG and WindowedManifoldKV reaches +42 points on multi-key recall at 131K; and on InfiniteBench at 100K+ context, WindowedManifoldKV wins by +7.16 on Phi-4. Cross-architecture evaluation across six models reveals that the optimal distance metric depends on key-norm geometry, providing the first systematic guidelines for metric selection in geometric KV cache compression. The method requires only 3 lines of code and works across diverse architectures without tuning.
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Beyond Static Endpoints: Tool Programs as an Interface for Flexible Agentic Web Services
Mugeng Liu, Shuoqi Li, Yixuan Zhang, Yun Ma
In the agentic web era, LLM-based agents increasingly invoke web services as tools, yet most interfaces remain static endpoints that poorly express long-horizon workflows with loops, conditionals, joins, and retries. We present ToolPro, which represents an agent's tool intent as an executable tool program that compactly encodes multi-step service interactions with explicit effect types. ToolPro combines constraint-guided program construction, effect-aware replay for exactly-once state-modifying calls, and a profile-driven policy that decides when program execution outperforms stepwise calling. We instantiate ToolPro over MCP-style services with WebAssembly sandboxing and evaluate it on diverse workflows of real-world applications. ToolPro reduces end-to-end latency by up to 53.4% and client-side traffic by up to 96.1%, with larger gains under higher network latency and workflow complexity.
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Bring My Cup! Personalizing Vision-Language-Action Models with Visual Attentive Prompting
Sangoh Lee, Sangwoo Mo, Wook-Shin Han
While Vision-Language-Action (VLA) models generalize well to generic instructions, they struggle with personalized commands such as "bring *my* cup," where the robot must act on one specific instance among visually similar objects. We study this setting of manipulating personal objects, in which a VLA must identify and control a user-specific object unseen during training using only a few reference images. To address this challenge, we propose **Visual Attentive Prompting (VAP)**, a simple-yet-effective training-free perceptual adapter that equips frozen VLAs with top-down selective attention. VAP treats the reference images as a non-parametric visual memory, grounds the personal object in the scene through open-vocabulary detection and embedding-based matching, and then injects this grounding as a visual prompt by highlighting the object and rewriting the instruction. We construct two simulation benchmarks, Personalized-SIMPLER and Personalized-VLABench, and a real-world tabletop benchmark to evaluate personalized manipulation across multiple robots and tasks. Experiments show that VAP consistently outperforms generic policies and token-learning baselines in both success rate and correct-object manipulation, helping to bridge the gap between semantic understanding and instance-level control.
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Overclocking Electrostatic Generative Models
Daniil Shlenskii, Alexander Korotin
Electrostatic generative models such as PFGM++ have recently emerged as a powerful framework, achieving competitive performance in image synthesis. PFGM++ operates in an extended data space with auxiliary dimensionality D, recovering the diffusion model framework as $D\to\infty$, while yielding superior empirical results for finite D.
Like diffusion models, PFGM++ relies on expensive ODE simulations to generate samples, making it computationally costly. To address this, we propose Inverse Poisson Flow Matching (IPFM), a principled distillation framework that accelerates

electrostatic generative models across all values of D.
Our IPFM reformulates distillation as an inverse problem: learning a generator w hose induced electrostatic field matches that of the teacher. We derive a tracta ble training objective for this problem and show that, as $D\to\infty$, our IPFM closely recovers Score Identity Distillation (SiD), a recent method for distill ing diffusion models.
Empirically, our IPFM produces distilled generators that achieve near-teacher or even superior sample quality using only a few function evaluations.
Moreover, we find that one-step generator distillation converges faster at finit e D than in the $D\to\infty$ diffusion limit, aligning with prior evidence tha t finite-D PFGM++ models offer more favorable optimization and sampling behavi or.
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Search Space Synthesis for Parametric Functions
Felix Laarmann, Andreas Pauly, Sebastian Buschjäger, Andrea Bommert, Jakob Rehof
We present a general framework for synthesizing search spaces of parametric func tions, along with strategies for traversing these spaces to find optima. We form alize an algebraic theory for the categorical model of parametric functions in f inite combinatory logic with predicates (FCLP). Based on a component-oriented sy nthesis framework for FCLP we automate composition from given components and sea rch for parametric functions. Components are language-agnostic and may be instan tiated as any implementation of parametric functions, e.g., as PyTorch modules.
A proof-of-concept implementation demonstrates how to represent more specific co ncepts, such as neural architecture search and hyperparameter optimization, with in the framework.
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PS-PPO : Prefix-Sampling PPO for Critic-Free RLHF
Doo Hwan Hwang, Kee-Eung Kim
Reinforcement Learning from Human Feedback (RLHF) for Large Language Models incr easingly relies on critic-free methods as a practical alternative to actor--crit ic training. Despite their simplicity, existing critic-free approaches propagate a trajectory-level learning signal uniformly across all tokens in a trajectory. This requires full-trajectory policy updates for every rollout, leading to subs tantial optimization cost for long reasoning traces, even though intermediate pr efixes often contain enough information to largely determine the final outcome.
We propose Prefix-Sampling Proximal Policy Optimization (PS-PPO), a compute-effi cient critic-free method for RLHF that exploits this temporal redundancy. PS-PPO introduces a prompt-conditioned cutoff distribution and samples a cutoff timest ep for each trajectory. During the update pass, PS-PPO backpropagates only throu gh the sampled prefix of each trajectory and applies an importance-weighting cor rection so that the resulting truncated gradient estimator remains unbiased with respect to the full-trajectory objective.
Experiments on mathematical reasoning and RLHF benchmarks show that PS-PPO achie ves large reductions in training compute and peak GPU memory, while maintaining accuracy comparable to strong critic-free baselines.
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Transformers learn factored representations
Adam Shai, Loren Amdahl-Culleton, Casper L. Christensen, Henry R Bigelow, Fernan do Rosas, Alexander Boyd, Eric A. Alt, Kyle J Ray, Paul M. Riechers
Transformers pretrained via next token prediction learn to factor their world in to parts, representing these factors in orthogonal subspaces of the residual str eam. We formalize two representational hypotheses: (1) a representation in the p roduct space of all factors, whose dimension grows exponentially with the number of parts, or (2) a factored representation in orthogonal subspaces, whose dimen sion grows linearly. Both track context-induced uncertainty over the latent part s, but the factored representation sacrifices fidelity when factors are not cond itionally independent. We derive precise predictions about the geometric structu re of activations for each, including the number of subspaces, their dimensional ity, and the arrangement of context embeddings within them. We test between thes e hypotheses on transformers trained on synthetic processes with known latent st

ructure. When factors are conditionally independent, models learn factored repre
sentations; when noise or dependencies break this structure, models gradually ex
pand their effective dimensionality over training to recover fidelity. This prov
ides a principled explanation for why transformers decompose the world into part
s, and suggests that interpretable low dimensional structure may persist even in
 models trained on complex data.
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DLLMQuant: A Post-Training Quantization Framework Tailored for Diffusion-Based L
arge Language Models
Chen Xu, Zhixuan Chen, Dawei Yang
Diffusion-based large language models (DLLMs) have shown promise for non-autoreg
ressive text generation, but their deployment is constrained by large model size
s and heavy computational costs. Post-training quantization (PTQ), a widely used
 method for compressing and accelerating Large Language Models (LLMs), suffers f
rom severe accuracy degradation and reduced generalization performance when dire
ctly applied to DLLMs (e.g., AWQ suffers a 16% accuracy drop on LLADA under W4A4
). This paper explores how the unique mechanisms of Dynamic Language Models (DLL
Ms) conflict with quantization, identifying three core issues: 1) During the ite
rative generation process of DLLMs, dynamic masking ratios are inherently involv
ed, leading to notable differences in token distributions across decoding steps.
 Unfortunately, these distinct distributions are not sufficiently captured by cu
rrent PTQ calibration approaches; 2) Quantization errors propogate and accumalte
 progressively during iterations in DLLMs, leading to a gradual decline in the p
erformance of quantized models as decoding steps advance; 3) The stability of un
masked tokens, combined with the probabilistic nature of masked tokens, gives ri
se to an overall feature distribution that is uncoordinated and unsuitable for P
TQ. To address these issues, we propose DLLMQuant, a PTQ framework tailored for
DLLMs, which incorporates three novel techniques: 1) Temporal-Mask Adaptive Samp
ling (TMAS), a calibration method that accounts for both time and mask factors,
with the capacity to capture distributions across timesteps. 2) Interaction-Awar
e Activation Quantization (IA-AQ), which utilizes bidirectional attention scores
 to identify important tokens, and prioritizes these tokens when minimizing quan
tization error. 3) Certainty-Guided Quantization (CGQ) incorporates mask status
and token scores as core weighting criteria for error compensation, enabling PTQ
 to better align with the unique weight distribution of DLLMs. Experiments show
that DLLMQuant achieves significant performance gains (e.g., over 10-point accur
acy improvement on GSM8K for LLADA under 4-bit quantization) while enhancing eff
iciency.
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Factor-Wise Homogeneity of Slot-Attention for Continual Object-Centric Learning
Ilmin Kang, Hoyong Kim, Seungju Bang, Minwoo Kang, Kangil Kim
While Object-Centric Learning has shown great promise in modular perception, its
 extension to Continual Learning remains underexplored.
In this work, we observe that Slot Attention exhibits a distinctive behavior:
it organizes latent representations into small and separated regions, each of wh
ich preserves identical factor states, crucially emerging not only in the curren
t task but also across sequential tasks with novel factors. This *inter-task sep
aration* offers significant advantages in continual learning, which typically su
ffers from severe object-wise forgetting.
We refer to this phenomenon as *Factor-Wise Homogeneity*, and show that this int
rinsic inter-task separation is crucial, serving as a key mechanism to prevent c
atastrophic forgetting in Continual Object-Centric Learning.
However, despite its strong robustness, factor-wise homogeneity alone is insuffi
cient due to the bottleneck in exploiting this separation at the decoder.
To overcome this limitation and demonstrate the significance of our findings, we
 show that a minimal strategy *Decoder-only Post-Replay*, which freezes the fact
or-wise homogeneous representations and employs decoder-only fine-tuning, is suf
ficient.
This work serves as a fundamental basis for understanding and leveraging the int
rinsic dynamics of Slot Attention, offering essential insights for advancing obj

ect-centric systems.
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Privacy-Aware Video Anomaly Detection through Orthogonal Subspace Projection
Lei Wang, Wenxiang Diao, Andrew Busch, Jun Zhou, Yongsheng Gao
Video anomaly detection (VAD) systems often prioritize accuracy while overlookin g privacy concerns, limiting their suitability for real-world deployment. We pro pose the Orthogonal Projection Layer (OPL), a lightweight module that removes ta sk-irrelevant variations to produce representations focused on anomaly-relevant cues. To address privacy risks in human-centered scenarios, we introduce Guided OPL (G-OPL), which suppresses facial attributes using weak supervision from face -presence signals while preserving non-identifying features such as pose and mot ion. A cosine alignment objective enforces consistent capture and removal of fac ial information without identity labels or adversarial training. We further pres ent a privacy-aware evaluation framework that jointly assesses detection perform ance and privacy preservation, and enables analysis of how sensitive information is filtered. Experiments show that embedding privacy constraints into model des ign reduces sensitive information while maintaining or improving detection accur acy, supporting projection-based architectures as a principled approach for priv acy-aware VAD.
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miniF2F-Dafny: LLM-Guided Mathematical Theorem Proving via Auto-Active Verificat ion
Mantas Baksys, Stefan Zetzsche, Olivier Bouissou, Sean B. Holden
LLMs excel at reasoning, but validating their steps remains challenging. Formal verification offers a solution through mechanically checkable proofs. Interactiv e theorem provers (ITPs) dominate mathematical reasoning but require detailed lo w-level proof steps, while auto-active verifiers offer automation but focus on s oftware verification. Recent work has begun bridging this divide by evaluating L LMs for software verification in ITPs, but the complementary direction—LLMs for mathematical theorem proving in auto-active verifiers—remains unexplored. We pre sent miniF2F-Dafny, the first translation of the widely-used mathematical benchm ark miniF2F to an auto-active verifier: Dafny. We find that Dafny's automation a lone solves 39-44% of problems with empty proofs, whereas many require substanti al proof guidance in ITPs. We evaluate 8 off-the-shelf LLMs on proof generation, with the best model (Claude Opus 4.6) achieving 62.7% cumulative pass@4 on the full test set, improving over the 38.9% empty-proof baseline by 23.8 percentage points. These results show that auto-active verification offers a complementary empirical setting for AI-assisted mathematical reasoning, where LLMs provide hig h-level guidance while SMT automation handles low-level details. Our benchmark a nd evaluation infrastructure are publicly available on GitHub.
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Can Adaptive Gradient Methods Converge under Heavy-Tailed Noise? A Case Study of AdaGrad
Zijian Liu
Many tasks in modern machine learning are observed to involve heavy-tailed gradi ent noise during the optimization process. To manage this realistic and challeng ing setting, new mechanisms, such as gradient clipping and gradient normalizatio n, have been introduced to ensure the convergence of first-order algorithms. How ever, adaptive gradient methods, a famous class of modern optimizers that includ es popular $\mathtt{Adam}$ and $\mathtt{AdamW}$, often perform well even without any extra operations mentioned above. It is therefore natural to ask whether ad aptive gradient methods can converge under heavy-tailed noise without any algori thmic changes. In this work, we take the first step toward answering this questi on by investigating a special case, $\mathtt{AdaGrad}$, the origin of adaptive g radient methods. We provide the first provable convergence rate for $\mathtt{Ada Grad}$ in non-convex optimization when the tail index p satisfies $4/3<p\leq2$. Notably, this result is achieved without requiring any prior knowledge of p and is hence adaptive to the tail index. In addition, we develop an algorithm-de pendent lower bound, suggesting that the existing minimax rate for heavy-tailed optimization is not attainable by $\mathtt{AdaGrad}$. Lastly, we consider $\math

tt{AdaGrad}\text{-}\mathtt{Norm}$, a popular variant of $\mathtt{AdaGrad}$ in th
eoretical studies, and show an improved rate that holds for any $1<p\leq2$ under
 an extra mild assumption.
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Stationary MMD Points
Zonghao Chen, Toni Karvonen, Heishiro Kanagawa, Francois-Xavier Briol, Chris J.
Oates
Approximation of a target probability distribution using a finite set of points
 is a problem of fundamental importance in numerical integration. Several authors
 have proposed to select points by minimising a maximum mean discrepancy (MMD),
but the non-convexity of this objective typically precludes global minimisation.
 Instead, we consider the concept of \emph{stationary points of the MMD} which,
in contrast to points globally minimising the MMD, can be accurately computed. O
ur main contributions are two-fold and theoretical in nature.
We first prove the (perhaps surprising) result that, for integrands in the assoc
iated reproducing kernel Hilbert space, the numerical integration error of stati
onary MMD points vanishes \emph{faster} than the MMD. Motivated by this \emph{su
per-convergence} property, we consider MMD gradient flows as a practical strateg
y for computing stationary points of the MMD. We then prove that MMD gradient fl
ow can indeed compute stationary MMD points, based on a refined convergence anal
ysis that establishes a novel non-asymptotic finite-particle error bound.
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Bimodal masked language modeling for bulk RNA-seq and DNA methylation representa
tion learning
Maxence Gélard, Hakim Benkirane, Thomas Pierrot, Guillaume Richard, Paul-Henry C
ournède
Oncologists are increasingly relying on multiple modalities to model the complex
ity of diseases. Within this landscape, transcriptomic and epigenetic data have
 proven to be particularly instrumental and play an increasingly vital role in cl
inical applications. However, their integration into multimodal models remains a
 challenge, especially considering their high dimensionality. In this work, we p
resent a novel bimodal model that jointly learns representations of bulk RNA-seq
 and DNA methylation leveraging self-supervision from masked language modeling.
We implement an architecture that reduces the memory footprint usually attribute
d to purely transformer-based models when dealing with long sequences. We demons
trate that the obtained bimodal embeddings can be used to fine-tune cancer-type
classification and survival models that achieve state-of-the-art performance com
pared to unimodal models. Furthermore, we introduce a robust learning framework
that maintains downstream task performance despite missing modalities, enhancing
 the model’s applicability in real-world clinical settings.
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Decomposition-Based Modular Conformal Prediction for Two-Stage Modeling
William Zhang, Saurabh Amin, Georgia Perakis
Conformal prediction offers finite-sample coverage guarantees under minimal assu
mptions. However, existing methods treat the entire modeling process as a black
box, overlooking opportunities to exploit and understand modular
structure. We introduce a conformal prediction framework for two-stage sequentia
l models, where an upstream predictor generates intermediate representations for
 a downstream model. By decomposing the overall prediction residual into stage-s
pecific components, our method enables practitioners to attribute uncertainty to
 specific pipeline stages. We develop a risk-controlled parameter selection proc
edure using family-wise error rate (FWER) control to calibrate stage-wise scalin
g parameters, and introduce an adaptive extension for non-stationary settings. E
xperiments on synthetic distribution shifts, as well as real-world supply chain
and stock market data, demonstrate that our approach improves coverage under str
uctural, stage-wise shifts compared to standard conformal methods, while identif
ying stage-wise error contribution. This framework offers diagnostic advantages
and robust coverage that standard conformal methods lack.
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Memory-Distilled Selection for Noise-Robust Anomaly Detection

Sirojbek Safarov, Jaewoo Park, Yoon Gyo Jung, Kuan-Chuan Peng, Wonchul Kim, Seongdeok Bang, Octavia Camps

Anomaly detection (AD) under data contamination is critical for deploying unsupervised defect detection in industrial environments, where curating perfectly clean training sets is impractical. However, existing methods are sensitive to contamination, suffering significant performance degradation as the noise ratio increases. In this paper, we propose Memory-Distilled Selection (MeDS), a training algorithm based on data selection. MeDS constructs an ensemble of partial memories via random subsampling, where the resulting sparsity acts as a low-pass filter that captures nominal patterns across a wide range of noise ratios, enabling coarse-level identification of contaminated samples. The aggregated distances to the bootstrapped memories are then distilled into a reconstruction score network, which is subsequently fine-tuned on clean data filtered using scores from the distilled model, enabling fine-grained localization of anomalies. MeDS is robust across a wide range of noise ratios without requiring noise-ratio-specific hyperparameter tuning, achieving 99.16\% image-level AUROC on MVTecAD at a 40\% noise ratio, and attaining state-of-the-art performance on both VisA and Real-IAD under noisy settings. We thoroughly verify the efficacy of MeDS on industrial AD benchmarks under noisy data scenarios, accompanied by in-depth empirical analyses.

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Dissecting Multimodal In-Context Learning: Modality Asymmetries and Circuit Dynamics in modern Transformers

Yiran Huang, Karsten Roth, Quentin Bouniot, Wenjia Xu, Zeynep Akata

Transformer-based multimodal large language models often exhibit in-context learning (ICL) capabilities. Motivated by this phenomenon, we ask: how do transformers learn to associate information across modalities from in-context examples? We investigate this through controlled experiments on small transformers trained on synthetic classification tasks, enabling precise manipulation of data statistics and model architecture. We begin by revisiting core principles of unimodal ICL in modern transformers. While several prior findings replicate, we find that Rotary Position Embeddings (RoPE) can delay the onset of ICL circuits. Extending to the multimodal setting reveals a fundamental learning asymmetry: when pretrained on high-diversity data from a primary modality, surprisingly low data complexity in the secondary modality suffices for multimodal ICL to emerge. Mechanistic analysis shows that both settings rely on an induction-style mechanism that copies labels from matching in-context exemplars; multimodal training refines and extends these circuits across modalities. Our findings provide a mechanistic foundation for understanding multimodal ICL in modern transformers and introduce a controlled testbed for future investigation. Code is available at: https://github.com/YiranHuangIrene/multimodal-icl

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RoboFlow4D: A Lightweight Flow World Model Toward Real-Time Flow-Guided Robotic Manipulation

Sixu Lin, Junliang Chen, Huaiyuan Xu, Zhuohao Li, Guangming Wang, Yixiong Jing, Sheng Xu, Runyi Zhao, Brian Sheil, Lap-Pui Chau, Guiliang Liu

Planning and acting in 3D environments is a fundamental capability for robotic manipulation in the real world.

Although prior work has explored predictive flow planners to guide 3D manipulation, existing approaches often rely on modular pipelines stacking multiple submodels, resulting in high computational overhead and limited real-time performance. To address these challenges, we introduce RoboFlow4D, a lightweight flow world model that unifies perception and planning by estimating temporal motion in physical 3D space. As an end-to-end framework, RoboFlow4D directly predicts multi-frame 3D flows from visual observations and textual instructions, providing explicit flow-based planning to guide action generation. This design allows seamless integration with general action policies, forming an efficient observation–planning–execution closed loop. Through slow–fast collaboration between flow prediction and action control, RoboFlow4D enables real-time and resource-efficient manipulation. Extensive experiments in both simulation and real-world settings demonstrate that RoboFlow4D consistently improves manipulation success rates and comput

ational efficiency, advancing flow-guided planning for embodied intelligence. Our project page is available at RoboFlow4D.
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Why Dedicated Critics: Eliminating Target Drift in Multi-Constraint RL
Yue Yang, Chenghao Huang, Hao Wang
Lagrangian-based methodologies are one of the fundamental paradigms of safe reinforcement learning (RL) for constrained Markov decision processes, particularly when dealing with multi-constraint cases. While the specific details of the methodologies may differ, with some using a single estimator for the overall mixed penalty term of the constraints and others using separate estimators for the constraints, the fundamental question of the theoretical validity of the methodologies has remained largely unexplored. The present paper performs the first theoretical analysis of the methodologies and proves that the use of the mixed critic structure leads to the presence of a bias due to the target drift of the Lagrange multipliers. On the other hand, the use of the dedicated critic structure, where separate critics are used for the reward function and the constraint functions, does not suffer from this bias. The theoretical analysis is supported with experiments on a realistic power system environment with multiple constraints, where the dedicated critic structure succeeds in satisfying the constraints, whereas the mixed critic structure fails.
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cuRegOT: A GPU-Accelerated Solver for Entropic-Regularized Optimal Transport
Yixuan Qiu
Optimal transport (OT) has emerged as a fundamental tool in modern machine learning, yet its computational cost remains a significant bottleneck for large-scale applications. While harnessing the massive parallelism of modern GPU hardware is critical for efficiency, the de facto standard Sinkhorn algorithm, despite its ease of parallelization, often suffers from slow convergence in challenging problems. More recently, the sparse-plus-low-rank quasi-Newton method offers a balance between convergence rate and per-iteration complexity; however, its efficiency on GPUs is severely hindered by the serial nature of sparse matrix symbolic analysis and irregular memory access patterns. To bridge this gap, we present cuRegOT, a high-performance GPU solver tailored for entropic-regularized OT. We introduce a suite of algorithmic and architectural optimizations, including an amortized symbolic analysis strategy to mitigate CPU bottlenecks, an asynchronous Sinkhorn iterates generation mechanism, and a fused kernel for bandwidth-efficient gradient evaluation. These strategies are backed by rigorous theoretical guarantees ensuring algorithmic convergence. Extensive numerical experiments demonstrate that cuRegOT achieves significant speedups over state-of-the-art GPU-based solvers across a variety of benchmark tasks.
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Hista and Numca: Estimate State Value Effectively for Large Language Model Reinforcement Learning
Zizhe Chen, Jiqian Dong, Yizhou Tian, Garry YANG, Yongqiang Chen, Zhitang Chen, James Cheng
Reinforcement learning (RL) refines large language models (LLMs) by directly optimizing model behavior through reward signals. While accurate state value estimation is critical for stable training in classical RL, it remains an underexplored challenge in LLM post■training. In this work, we introduce the State Value Estimation Benchmark (SVEB) to assess state estimation within existing RL frameworks and show that critics in standard approaches like PPO collapse to a coarse group■average baseline. To address this, we propose two techniques: \textit{Numca}, which leverages numerical spans as gradable milestones for state value estimation, and \textit{Hista}, a framework that uses LLM's hidden states as representation to weighted average disjoint rollouts and their return. Extensive experiments demonstrate that both methods yield more accurate state value estimates and enhance training performance across different RL algorithms and model sizes without incurring significant computational overhead. Code available at \url{https://github.com/VOXXXX1874/Hista}.
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Landmark-Guided Policy Optimization for Multi-Objective Language Model Selection
Marcio Monteiro, Weichen Li, Puyu Wang, Marius Kloft, Sophie Fellenz
Selecting a pretrained large language model (LLM) to fine-tune for a task-specif ic dataset can be time-consuming and costly. With several candidate models avail able to choose from, varying in size, architecture, and pretraining data, findin g the best model for a specific task often involves extensive trial and error. I n addition, the "best" model may not necessarily be the one with the lowest test loss, as practical considerations such as deployment costs, inference throughpu t, and limited search budgets might also play crucial roles. To address this, we introduce LAMPS (LAnguage Model Pareto Selection), a novel and open-source mult i-objective AutoML framework that meta-learns a resource allocation policy to ef ficiently identify (or approximate) the Pareto front of candidate LLMs for a tas k-specific dataset. It is based on two key ideas: (1) landmark fine-tuning, whic h generates early performance indicators of the candidate models, and (2) meta-l earning via reinforcement learning, which learns an effective selection policy f rom historical performance data (a meta-dataset). Our results show that, on held -out datasets, LAMPS reduces search time by an average of 73% compared to exhaus tive search, while still covering more than 99% of the optimal target space hype rvolume.
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From Volume to Value: Preference-Aligned Memory Construction for On-Device RAG
Changmin Lee, Jaemin Kim, Taesik Gong
With the rapid emergence of personal AI agents based on Large Language Models (L LMs), implementing them on-device has become essential for privacy and responsiv eness.
To handle the inherently personal and context-dependent nature of real-world req uests, such agents must ground their generation in device-resident personal cont ext.
However, under tight memory budgets, the core bottleneck is *what to store* so t hat retrieval remains aligned with the user.
We propose EPIC (Efficient Preference-aligned Index Construction), which focuses on user preferences as a compact and stable form of personal context and integr ates them throughout the RAG pipeline.
EPIC selectively retains preference-relevant information from raw data and align s retrieval toward preference-aligned contexts.
Across four benchmarks covering conversations, debates, explanations, and recomm endations, EPIC reduces indexing memory by 2,404$\times$, improves preference-fo llowing accuracy by 18.79\%p, and achieves 32.17$\times$ lower retrieval latency over the best-performing baseline.
In on-device experiments, EPIC maintains under 1 MB memory and achieves 5.21 to 29.35 ms/query latency across three platforms, while supporting streaming update s under preference drift.
Our code and data are available at https://github.com/UbiquitousAILab/EPIC.
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Weak Diffusion Priors Can Still Achieve Strong Inverse-Problem Performance
Jing Jia, Wei Yuan, Sifan Liu, Liyue Shen, Guanyang Wang
Can a diffusion model trained on bedrooms recover human faces? Diffusion models are widely used as priors for inverse problems, but standard approaches usually assume a high-fidelity model trained on data that closely match the unknown sign al. In practice, one often must use a mismatched or low-fidelity diffusion prior . Surprisingly, these weak priors often perform nearly as well as full-strength, in-domain baselines. We study when and why inverse solvers are robust to weak d iffusion priors. Through extensive experiments, we find that weak priors succeed when measurements are highly informative (e.g., many observed pixels), and we i dentify regimes where they fail. To explain this behavior, we combine Bayesian-c onsistency theory with local-correlation analysis: the theory gives conditions u nder which high-dimensional measurements make the posterior concentrate near the true signal, while the correlation analysis shows that weak and stronger natura l-image priors can share similar local spatial structure. These results provide a principled justification on when weak diffusion priors can be used reliably. C

ode is available at https://github.com/jjia131/weak-diffusion-priors-inverse-problem.
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Learning to Rank by Directly Optimizing Full-Order Probabilities
Yongxiang Tang, Chao Wang, Jincheng Lu, Yanhua Cheng, Xialong Liu, Peng Jiang
Learning to rank can be cast as a probabilistic modeling problem over permutations, where the goal is to estimate the likelihood of an observed total ordering of items. This formulation naturally involves full-order probabilities of the form $\mathbb{P}(z_1 \leq \cdots \leq z_n)$, whose exact computation and optimization are intractable due to the factorial growth of the permutation space with respect to the list size. In this work, we introduce the *Full-Order Bound* (FOB), a tractable lower bound on the probability of an observed ordering, constructed from a subset of ordering constraints that factorizes across items while preserving full-order structure and order-reversal invariance. Under log-concave latent densities, the bound induces a convex inner tightening problem over latent cut points, which we solve efficiently during training using a *safe-region gradient ascent* (SRGA) procedure. Experiments on synthetic ranking tasks and large-scale learning-to-rank benchmarks show that FOB improves full-list ordering metrics and remains competitive on NDCG, while an optional metric-aligned variant recovers NDCG gains. Our code is available at https://github.com/tyxaaron/FOB.
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Online Robust Reinforcement Learning with General Function Approximation
Debamita Ghosh, George K. Atia, Yue Wang
Reinforcement learning (RL) in real-world tasks often suffers from the performance degradation due to the distribution shift between training and deployment environments. Distributionally Robust RL (DR-RL) addresses this issue by optimizing the worst-case performance over an uncertainty set of transition dynamics, providing an optimized baseline performance upon deployment. However, existing methods typically require strong data access assumptions (e.g., a generative model or comprehensive offline datasets) and mostly focus on tabular settings. In this paper, we introduce a purely online DR-RL algorithm with general function approximation that learns a robust policy directly from interaction, without any prior knowledge or pre-collected data. Our method uses a dual-based fitted robust Bellman update to jointly learn the value function and the robust backup operator. We establish the first regret guarantee for online DR-RL in terms of an intrinsic complexity measure—the robust Bellman–Eluder (BE) dimension, for general ϕ-divergence uncertainty sets. Our regret bound is sublinear and independent of $|\mathcal{S}|$ and $|\mathcal{A}|$, and recovers sharp rates in structured regimes, providing a scalable method for practical DR-RL.
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A model of errors in transformers
Suvrat Raju, Praneeth Netrapalli
We study the error rate of LLMs on tasks like arithmetic that require a deterministic output, and repetitive processing of tokens drawn from a small set of alternatives. We argue that incorrect predictions arise when small errors in the attention mechanism accumulate to cross a threshold, and use this insight to derive a quantitative two-parameter relationship between the accuracy and the complexity of the task. The two parameters vary with the prompt and the model; they can be interpreted in terms of an elementary noise rate, and the number of plausible erroneous tokens that can be predicted. Our analysis is inspired by an "effective field theory'' perspective: the LLM's many raw parameters can be reorganized into just two parameters that govern the error rate. We perform extensive empirical tests, using Gemini 2.5 Flash, Gemini 2.5 Pro and DeepSeek R1, and find excellent agreement between the predicted and observed accuracy for a variety of tasks, although we also identify deviations in some cases. Our model provides an alternative to suggestions that errors made by LLMs on long repetitive tasks indicate the "collapse of reasoning'', or an inability to express "compositional'' functions. Finally, we show how to construct prompts to reduce the error rate.
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GraphFLEx: Unsupervised Structure Learning $\underline{\text{F}}$ramework for $\underline{\text{L}}$arge $\underline{\text{Ex}}$panding $\underline{\text{Graph}}$s

Mohit Kataria, Nikita Malik, Jayadeva Jayadeva, Sandeep Kumar

Graph structure learning is a core problem in graph-based machine learning, essential for uncovering latent relationships and ensuring model interpretability. However, most existing approaches are ill-suited for large-scale and dynamically evolving graphs, as they often require complete re-learning of the structure upon the arrival of new nodes and incur substantial computational and memory costs. In this work, we propose GraphFLEx—a unified and scalable framework for Graph Structure Learning in Large and Expanding Graphs. GraphFLEx mitigates the scalability bottlenecks by restricting edge formation to structurally relevant subsets of nodes identified through a combination of clustering and coarsening techniques. This dramatically reduces the search space and enables efficient, incremental graph updates. The framework supports 48 flexible configurations by integrating diverse choices of learning paradigms, coarsening strategies, and clustering methods, making it adaptable to a wide range of graph settings and learning objectives. Extensive experiments across 26 diverse datasets and graph neural network architectures demonstrate that GraphFLEx achieves state-of-the-art performance with significantly improved scalability. Our implementation is publicly available here: https://anonymous.4open.science/r/Scaling_Graph_Learning-CB67/README.md.

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Unveiling And Addressing Dimensional Collapse In Vector Quantization Models Via Codebook Regularization

Fang Zhang, Yongxin Zhu, Yihao Liu, Bin Fu, Linli Xu

While recent advancements in Vector Quantization (VQ) models have successfully achieved complete codebook utilization, a critical bottleneck remains largely unexplored: the effective dimensionality of the codebook embedding space. We observe that discrete codebook representations tend to degenerate into low-dimensional subspaces, characterized by significantly lower effective rank than continuous representations during quantization. Through comprehensive spectral analysis, we identify that this dimensional collapse stems from the suppression of low-variance components inherent to the vector quantization process, thereby severely limiting the expressive capacity of VQ models. To mitigate this fundamental issue, we propose a simple yet effective codebook regularization strategy designed to restore low-variance components, effectively bridging the spectral gap between discrete codebook spaces and continuous representations. Extensive experiments demonstrate that this regularization objective is compatible with diverse VQ training paradigms, yielding significant improvements in reconstruction fidelity and downstream performance in autoregressive image generative models.

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Correcting Split Selection in Online Decision Trees via Anytime-Valid Inference

Salim I. Amoukou, Saumitra Mishra, Manuela Veloso

Bagging-based ensembles, most notably Adaptive Random Forests, are among the strongest performers for learning from data streams. A common denominator across these methods is their reliance on Hoeffding Trees as base learners, which grow decision trees incrementally by testing whether a candidate split is significantly better than its alternatives using concentration inequalities. Despite their empirical success, existing Hoeffding Trees variants lack valid statistical guarantees. Current analyses rely on fixed-sample concentration bounds, while split decisions are made using data-dependent stopping rules, which invalidates their guarantees and can drive the probabilty of incorrect splits to one. We introduce a principled alternative based on \emph{anytime-valid inference}. Our method provides: (i) anytime-valid control of false splits under arbitrary data streams, including non-stationary settings; (ii) finite commitment time under a predictive advantage; and (iii) under stationary i.i.d.\ data, risk is monotone decreasing and strictly improves at every split. Empirically, we evaluate both standalone trees and their use within Adaptive Random Forests on non-stationary streams. Our method improves performance while producing substantially smaller trees.

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RefineEvo: Planning-Guided Heuristic Evolution with Bidirectional Experience
Yang Wu, Junran Pan, Yifan Zhang, Ning Xu, Fanshuo Zeng, Jian Cheng
Automatic Heuristic Design (AHD) has emerged as a transformative approach for so lving combinatorial optimization problems. While recent Large Language Model (LL M)-based methods have shown promise, they predominantly rely on fixed evolutiona ry operators and struggle to effectively accumulate and reuse historical search experience. This paper proposes RefineEvo, a novel evolutionary framework that t ransforms AHD from a static trial-and-error process into a planning-guided, expe rience-driven system. RefineEvo introduces a Planner to dynamically schedule evo lutionary operators and trigger refinement based on the current search state, an d a Reflector to distill valuable lessons into a Bidirectional Experience Pool c ontaining both positive insights and negative pitfalls. This synergistic framewo rk enables the system to adapt its search tools to the evolving complexity of th e problem and leverage trajectory-aware, situation-conditioned insights to guide generation. Experiments on several classic combinatorial optimization benchmark s demonstrate that RefineEvo consistently outperforms strong baselines. In parti cular, RefineEvo delivers superior solution quality while improving token effici ency, enabling more efficient and autonomous heuristic design.
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Endogenous Resistance to Activation Steering in Language Models
Alex McKenzie, Keenan Pepper, Stijn Servaes, Martin Leitgab, Murat Cubuktepe, Mi chael Vaiana, Diogo S de Lucena, Judd Rosenblatt, Michael S. A. Graziano
Large language models can resist task-misaligned activation steering during infe rence, sometimes recovering mid-generation to produce improved responses even wh en steering remains active. We term this Endogenous Steering Resistance (ESR). U sing sparse autoencoder (SAE) latents to steer model activations, we find that L lama-3.3-70B shows substantial ESR, while smaller models from the Llama-3 and Ge mma-2 families exhibit the phenomenon less frequently. We identify 26 ``off-topi c detector'' latents that predict ESR episodes in Llama-3.3-70B. Zero-ablating t hese latents reduces the multi-attempt rate by 25\%, providing causal evidence f or dedicated internal consistency-checking circuits. We demonstrate that ESR can be deliberately enhanced through both prompting and training: meta-prompts inst ructing the model to self-monitor increase the multi-attempt rate by 5$\times$ f or Llama-3.3-70B, and fine-tuning on self-correction examples successfully induc es ESR-like behavior in smaller models. These findings have dual implications: E SR could protect against adversarial manipulation but might also interfere with beneficial safety interventions that rely on activation steering. Understanding and controlling these resistance mechanisms is important for developing transpar ent and controllable AI systems.
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StormInsight: Hierarchical Environmental Forcing and Vertical Coupling for Weath er System Evolution
Jun CHEN, Yan Fang, Minghui Qiu, Yueran Qiu, Lin CHEN, SHUXIN ZHONG, Yu ZHANG, K aishun Wu
Nowcasting forms the first line of defense against rapidly evolving weather haza rds, where even minutes of delay can lead to severe societal impacts. However, e xisting systems predominantly extrapolate 2D radar reflectivity, which struggles under rapid intensification regimes. We introduce StormInsight, a multi-scale m odeling framework that enables coherent reconstruction of the three-dimensional evolution of convective systems while explicitly conditioning on the ambient env ironment. StormInsight integrates multi-source observations—including radar, sat ellite, and station—with reanalysis fields through two components: (i) Storm Evo lution Encoder that explicitly disentangles convective system state form vertica l thermodynamic coupling and large-scale environmental forcing; (ii) Convective System Decoder that predicts future echos by adaptively aggregating cross-layer interactions conditioned on evolving environmental conditions. To support compre hensive evaluation, we build a new benchmark StormBench that integrates observat ional and reanalysis data across regions. On this benchmark, StormInsight consis tently achieves the best performance, reducing MAE by 12.4% and improving the mC SI by 34.0%.

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LassoFlexNet: a Flexible Neural Architecture for Tabular Data
Kry Yik-Chau Lui, Cheng Chi, Kishore Basu, Yanshuai Cao
Despite their dominance in vision and language, deep neural networks often underperform relative to tree-based models on tabular data.
To bridge this gap, we incorporate five key inductive biases into deep tabular learning: robustness to irrelevant features, axis alignment,
localized irregularities, feature heterogeneity, and training stability. We propose *LassoFlexNet*,
an architecture that evaluates the linear and nonlinear marginal contribution of each input via Per-Feature Embeddings,
and sparsely selects relevant variables using a Tied Group Lasso mechanism.
Because these components introduce optimization challenges that destabilize standard proximal methods,
we analyze stochastic hierarchical proximal dynamics and develop a *Sequential Hierarchical Proximal Adaptive Gradient optimizer with exponential moving averages (EMA)*
designed to stabilize training in practice. Across 52 datasets from three benchmarks, LassoFlexNet matches or outperforms leading tree-based models,
achieving up to a 10\% relative gain, while maintaining Lasso-like interpretability. We substantiate these empirical results with ablation studies and theoretical proofs
confirming the architecture's enhanced expressivity and structural breaking of undesired rotational invariance.
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Scaling Law for Quantization-Aware Training
Mengzhao Chen, Chaoyi Zhang, Jing Liu, Yutao Zeng, Zeyue Xue, Zhiheng Liu, Yunshui Li, Jin Ma, Jie Huang, zhou Xun, Ping Luo
Large language models (LLMs) demand substantial computational and memory resources, creating deployment challenges. Quantization-aware training (QAT) addresses these challenges by reducing model precision while maintaining performance. However, the scaling behavior of QAT, especially at 4-bit precision (W4A4), is not well understood. Existing QAT scaling laws often ignore key factors such as the number of training tokens and quantization granularity, which limits their applicability. This paper proposes a unified scaling law for QAT that models quantization error as a function of model size, training data volume, and quantization group size. Through 268 QAT experiments, we show that quantization error decreases as model size increases, but rises with more training tokens and coarser quantization granularity. To identify the sources of W4A4 quantization error, we decompose it into weight and activation components. Both components follow the overall trend of W4A4 quantization error, but with different sensitivities. Specifically, weight quantization error increases more rapidly with more training tokens. Further analysis shows that the activation quantization error in the FC2 layer, caused by outliers, is the primary bottleneck of W4A4 QAT quantization error. By applying mixed-precision quantization to address this bottleneck, we demonstrate that weight and activation quantization errors can converge to similar levels. Additionally, with more training data, weight quantization error eventually exceeds activation quantization error, suggesting that reducing weight quantization error is also important in such scenarios. These findings offer key insights for improving QAT research and development.
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Explanations are a Means to an End: Decision Theoretic Explanation Evaluation
Ziyang Guo, Berk Ustun, Jessica Hullman
Explanations of model behavior are commonly evaluated via proxy properties weakly tied to the purposes explanations serve in practice. We contribute a decision theoretic framework that treats explanations as information signals valued by the expected improvement they enable on a specified decision task. This approach yields three distinct estimands: (i) a theoretical benchmark that upper-bounds achievable performance by any agent with the explanation, (ii) a human-complementary value that quantifies the theoretically attainable value that is not already

captured by a baseline human decision policy, and (iii) a behavioral value repre
senting the causal effect of providing the explanation to human decision-makers.
We instantiate these definitions in a practical validation workflow,
and apply them to assess explanation potential and interpret behavioral effects
in human–AI decision support and mechanistic interpretability.
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Sparse Models, Sparse Safety: Unsafe Routes in Mixture-of-Experts LLMs
Yukun Jiang, Hai Huang, Mingjie Li, Yage Zhang, Michael Backes, Yang Zhang
By introducing routers to selectively activate experts in Transformer layers, th
e mixture-of-experts (MoE) architecture significantly reduces computational cost
s in large language models (LLMs) while maintaining competitive performance, esp
ecially for models with massive parameters.
However, prior work has largely focused on utility and efficiency, leaving the s
afety risks associated with this sparse architecture underexplored.
In this work, we show that the safety of MoE LLMs is as sparse as their architec
ture by discovering $\text{\emph{unsafe routes}}$: routing configurations that,
once activated, convert safe outputs into harmful ones.
Specifically, we first introduce the $\underline{\text{Ro}}$uter $\underline{\te
xt{Sa}}$fety $\underline{\text{i}}$mportance $\underline{\text{s}}$core ($\textb
f{RoSais}$) to quantify the safety criticality of each layer's router.
Manipulation of only the high-RoSais router(s) can flip the default route into a
n unsafe one.
We further propose a $\underline{\text{F}}$ine-grained token-layer-wise $\underl
ine{\text{S}}$tochastic $\underline{\text{O}}$ptimization framework to discover
more concrete $\underline{\text{U}}$nsafe $\underline{\text{R}}$outes ($\textbf{
F-SOUR}$), which explicitly considers the sequentiality and dynamics of input to
kens.
Across four representative MoE LLM families, F-SOUR achieves an average ASR of 0
.90 and 0.98 on JailbreakBench and AdvBench, respectively.
Finally, we outline defensive perspectives, including safety-aware route disabli
ng and router training, as promising directions to safeguard MoE LLMs.
We hope our work can inform future red-teaming and safeguarding of MoE LLMs.
Our code is available at https://github.com/TrustAIRLab/UnsafeMoE.
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HexGen-3: A Fully Disaggregated LLM Serving Framework with Fine-Grained Heteroge
neous Resource Autoscaling
Youhe Jiang, Wenshuang Li, You Peng, Jintao Zhang, Ran Yan, Jianfei Chen, Xu Han
, Fangcheng Fu, Binhang Yuan
The operational cost of serving large language models remains prohibitively high
, largely due to extreme workload heterogeneity in production traffic. We observ
e that combining disaggregated inference with resource autoscaling enables fine-
grained resource adjustment, allowing inference phases and operations to scale i
ndependently based on their specific bottlenecks. Building on this insight, we p
ropose HexGen-3, a cost-effective LLM serving framework that leverages a fully d
isaggregated inference architecture and heterogeneous resource autoscaling. HexG
en-3 introduces two key components: (i) A hierarchical scheduling framework that
 jointly optimizes resource allocation and parallelism configuration for any giv
en resource provisioning, and (ii) an autoscaling framework that dynamically adj
usts resources and triggers deployment rescheduling in response to workload fluc
tuations. Experiments comparing HexGen-3 against state-of-the-art LLM serving sy
stems demonstrate up to 60% (on average 46.5%) improvement in per-cost throughpu
t under static resource provisioning, and up to 78.3% (on average 55.1%) improve
ment with autoscaling enabled under dynamic workloads.
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KAGE-Bench: Fast Known-Axis Visual Generalization Evaluation for Reinforcement L
earning
Egor Cherepanov, Daniil Zelezetsky, Aleksandr Panov, Alexey Kovalev
Pixel-based reinforcement learning agents often fail under purely visual distrib
ution shift even when latent dynamics and rewards are unchanged, but existing be
nchmarks entangle multiple sources of shift and hinder systematic analysis. We i

ntroduce KAGE-Env, a JAX-native 2D platformer that factorizes the observation process into independently controllable visual axes while keeping the underlying control problem fixed. By construction, varying a visual axis affects performance only through the induced state-conditional action distribution of a pixel policy, providing a clean abstraction for visual generalization. Building on this environment, we define KAGE-Bench, a benchmark of six known-axis suites comprising 34 train-evaluation configuration pairs that isolate individual visual shifts. Using a standard PPO-CNN baseline, we observe strong axis-dependent failures, with background and photometric shifts often collapsing success, while agent-appearance shifts are comparatively benign. Several shifts preserve forward motion while breaking task completion, showing that return alone can obscure generalization failures. Finally, the fully vectorized JAX implementation enables up to 33M environment steps per second on a single GPU, enabling fast and reproducible sweeps over visual factors. Code: https://avanturist322.github.io/KAGEBench/

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Nonparametric Distribution Regression Re-calibration

Ádám Jung, Domokos M. Kelen, Andras A Benczur

A key challenge in probabilistic regression is ensuring that predictive distributions accurately reflect true empirical uncertainty. Minimizing overall prediction error often encourages models to prioritize informativeness over calibration, producing narrow but overconfident predictions. However, in safety-critical settings, trustworthy uncertainty estimates are often more valuable than narrow intervals. Realizing the problem, several recent works have focused on post-hoc corrections; however, existing methods either rely on weak notions of calibration (such as PIT uniformity) or impose restrictive parametric assumptions on the nature of the error. To address these limitations, we propose a novel nonparametric re-calibration algorithm based on conditional kernel mean embeddings, capable of correcting calibration error without restrictive modeling assumptions. For efficient inference with real-valued targets, we introduce a novel characteristic kernel over distributions that can be evaluated in $\mathcal{O}(n \log n)$ time for empirical distributions of size n. We demonstrate that our method consistently outperforms prior re-calibration approaches across a diverse set of regression benchmarks and model classes.

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RADAR: Defending RAG Dynamically against Retrieval Corruption

Ziyuan Chen, Yueming Lyu, Yi Liu, Weixiang Han, Jing Dong, Caifeng Shan, Tieniu Tan

While RAG systems are increasingly deployed in dynamic web search, temporal volatility amplifies their vulnerability to adversarial attacks. Existing static-oriented defenses struggle to handle evolving threats and incur prohibitive storage costs in dynamic settings. We propose RADAR, a framework that models reliable context selection as a graph-based energy minimization problem, solved exactly via a Max-Flow Min-Cut. By incorporating a Bayesian memory node, RADAR recursively updates a belief state instead of archiving raw historical documents, effectively balancing stability against attacks with adaptability to genuine knowledge shifts. Experiments on a novel dynamic dataset show that RADAR achieves superior robustness and response quality with minimal storage overhead compared to the baselines.

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End-to-End Autoregressive Image Generation with 1D Semantic Tokenizer

Wenda Chu, Bingliang Zhang, Jiaqi Han, Yizhuo Li, Linjie Yang, Yisong Yue, Qiushan Guo

Autoregressive image modeling relies on visual tokenizers to compress images into compact latent representations. We design an end-to-end training pipeline that jointly optimizes reconstruction and generation, enabling direct supervision from generation results to the tokenizer. This contrasts with prior two-stage approaches that train tokenizers and generative models separately. We further investigate leveraging vision foundation models to improve 1D tokenizers for autoregressive modeling. Our autoregressive generative model achieves strong empirical results, including a state-of-the-art FID score of 1.48 without guidance on ImageN

et 256×256 generation.
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Training Deep Spiking Neural Networks without Normalization
Xinyu Shi, Zhaofei Yu
The training of deep Spiking Neural Networks (SNNs) has traditionally relied on Batch Normalization (BN), which stabilizes input currents and gradients during training. However, BN is not a universal solution. It is unsuitable for variable-length tasks and scenarios with reduced batch size, constraining the development of deep SNNs, where removing BN typically causes the training to fail to converge. This dependence stems not from a fundamental necessity of BN but from the current lack of reasonable initialization methods for SNNs. This paper addresses this core limitation by proposing SpikeInit, a novel initialization framework for SNNs. By modeling the response curve and gradient of spiking layers, SpikeInit initializes the weights and shape parameters of surrogate gradients to maintain stable firing rates during forward propagation and stable gradient magnitudes during backpropagation. Extensive experiments demonstrate that deep SNNs with SpikeInit can be trained stably without normalization and achieve superior performance compared to their normalized counterparts under identical settings. Furthermore, we demonstrate the scalability of SpikeInit by successfully training an ultra-deep, 1000-layer SNN without normalization. Our work provides a foundational step toward large-scale normalization-free SNN, liberating SNN design from the constraints of normalization.
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Ubiquity of Emergent Hebbian Dynamics in Regularized Learning
David Aaron Koplow, Tomaso Poggio, Liu Ziyin
Hebbian and anti-Hebbian plasticity are widely observed in the brain and are classically modeled as mechanistic, local homosynaptic rules stabilized by homeostatic constraints. This raises an identifiability question: does observing Hebbian/anti-Hebbian structure in synaptic updates uniquely imply an underlying Hebbian computation? We identify an alternative, emergent route. We show that near stationarity, L2 weight decay generically drives the \emph{learning-signal} component of many update rules to align with a Hebbian direction, with alignment increasing monotonically with decay strength. This Hebbian-like signature is not specific to SGD and can arise even for non-learning or random update rules long before learning has ceased. We further show that stochastic noise in the learning signal can induce anti-Hebbian alignment, yielding a simple tradeoff with weight decay and a phase boundary in regression settings. These mechanisms do not replace standard Hebbian theory; they can coexist with genuine Hebbian plasticity and complicate the interpretation of synaptic measurements, motivating experiments that distinguish mechanistic Hebbian computation from emergent Hebbian signatures.
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Not All Invariants Are Equal: Curating Training Data to Accelerate Program Verification with SLMs
Ido Pinto, Yizhak Yisrael Elboher, Haoze Wu, Nina Narodytska, Guy Katz
The synthesis of inductive loop invariants remains a critical bottleneck in automated program verification. While Large Language Models (LLMs) show promise in mitigating this issue, they often fail on complex programs, producing invariants that are invalid or computationally ineffective. Although fine-tuning is a natural strategy to address these limitations, obtaining high-quality training data remains an open challenge. We first formalize the properties required for a high-quality training invariant, and then present \textsc{Wonda}, a rigorous data curation pipeline that extracts such invariants from raw verifier output via AST-based normalization followed by LLM-driven semantic rewriting and augmentation with provable quality guarantees. Fine-tuning Small Language Models (SLMs) on \textsc{Wonda}-curated data yields consistent gains across the Qwen3, Llama-3.1, and Mistral families: the 4B and 8B Qwen3 models nearly double invariant correctness and double speedup rates, while Llama-3.1-8B triples both. On the challenging InvBench suite, the same 4B model outperforms an off-the-shelf model 20$\times$ its size and matches the end-to-end verification time of GPT-OSS-120B, while a 14B Qwen3 model matches that of the frontier model GPT-5.2, all without test-time

compute overhead. Our code is publicly available on \href{https://github.com/idopinto/wonda}{GitHub}.
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Provable Bounds for the Learnability of Sample-Compressible Families from Noisy Samples
Arefe Boushehrian, Amir Najafi
Learning distribution families over $\mathbb{R}^d$ is a fundamental problem in unsupervised learning and statistics. A central question in this setting is whether a given family of distributions possesses sufficient structure to be (at least) information-theoretically learnable and, if so, to characterize its sample complexity. In 2018, Ashtiani et al. (2018) reformulated sample compressibility as a structural property of distribution classes, proving that it guarantees PAC-learnability. This discovery subsequently enabled a series of recent advancements in deriving nearly tight sample complexity bounds for various high-dimensional open problems. It has been further conjectured that the converse also holds: every learnable class admits a sample compression scheme, making the two notions to be equivalent. In this work, we establish that sample compressible families remain learnable even from perturbed samples, subject to a set of minimax-necessary and sufficient conditions. In particular, we assume samples are corrupted by an additive independent noise model, and theoretically derive sample complexity bounds for general sample compressible classes in arbitrary dimensions with respect to both ℓ_2-norm and total variation distance.
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Transformers Efficiently Perform In-Context Logistic Regression via Normalized Gradient Descent
Chenyang Zhang, Yuan Cao
Transformers have demonstrated remarkable in-context learning (ICL) capabilities. The strong ICL performance of transformers is commonly believed to arise from their ability to implicitly execute certain algorithms on the context, thereby enhancing prediction and generation. In this work, we investigate how transformers with softmax attention perform in-context learning on linear classification data. We first construct a class of multi-layer transformers that can perform in-context logistic regression, with each layer exactly performing one step of normalized gradient descent on an in-context loss. Then, we show that our constructed transformer can be obtained through (i) training a single self-attention layer supervised by one-step gradient descent, and (ii) recurrently applying the trained layer to obtain a looped model. Training convergence guarantees of the self-attention layer and out-of-distribution generalization guarantees of the looped model are provided. Our results advance the theoretical understanding of ICL mechanism by showcasing how softmax transformers can effectively act as in-context learners.
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InfoLaw: Information Scaling Laws for Large Language Models with Quality-Weighted Mixture Data and Repetition
Weidong Zhou, Fengze Liu, Binbin Liu, Ping Guo, Zijun Wang, Bingni Zhang, Yifan Zhang, Yifeng Yu, Xiaohuan Zhou, Taifeng Wang
Upweighting high-quality data in LLM pretraining often improves performance, but in data-limited regimes, especially under overtraining, stronger upweighting increases repetition and can degrade performance. However, standard scaling laws do not reliably extrapolate across mixture recipes or under repetitions, making the selection for optimal data recipes at scaling underdetermined. To solve this, we introduce \textbf{InfoLaw} (Information Scaling Laws), a data-aware scaling framework that predicts loss from consumed tokens, model size, data mixture weights, and repetition. The key idea is to model pretraining as information accumulation, where quality controls information density and repetition induces scale-dependent diminishing returns. We first collect the model performance after training on datasets that vary in scale, quality distribution, and repetition level. Then we build up the modeling for information so that information accurately predicts those model performance. InfoLaw predicts performance on unseen data recipes and larger-scale runs (up to 7B, 425B tokens) with 0.15\% mean and 0.96\% max

absolute error in loss, and it extrapolates reliably across overtraining levels, enabling efficient data-recipe selection under varying compute budgets.
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Residual Context Diffusion Language Models
Yuezhou Hu, Harman Singh, Monishwaran Maheswaran, Haocheng Xi, Coleman Richard Charles Hooper, Jintao Zhang, Aditya Tomar, Michael W. Mahoney, Sewon Min, Mehrdad Farajtabar, Kurt Keutzer, Amir Gholami, Chenfeng Xu
Diffusion Large Language Models (dLLMs) have emerged as a promising alternative to purely autoregressive language models because they can decode multiple tokens in parallel. However, state-of-the-art block-wise dLLMs rely on a ``remasking" mechanism that decodes only the most confident tokens and discards the rest, effectively wasting computation. We demonstrate that recycling computation from the discarded tokens is beneficial, as these tokens retain contextual information useful for subsequent decoding iterations. In light of this, we propose Residual Context Diffusion (RCD), a module that converts these discarded token representations into contextual residuals and injects them back for the next denoising step. RCD uses a decoupled two-stage training pipeline to bypass the memory bottlenecks associated with backpropagation. We validate our method on both long CoT reasoning (SDAR) and short CoT instruction following (LLaDA) models. We demonstrate that a standard dLLM can be efficiently converted to the RCD paradigm with merely $\sim$300 million tokens. RCD consistently improves frontier dLLMs by 4--11 percentage points in accuracy with minimal extra computation overhead across a wide range of benchmarks. Notably, on the most challenging AIME tasks, RCD nearly doubles baseline accuracy and attains up to 4--5x fewer denoising steps at baseline's peak accuracy.
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Order Matters: Unveiling the Hidden Impact of Macro Placement Sequences via Proxy-Guided LLM Evolution
Shibing Mo, Jing Liu, Jianchu Xu, Ruilin Wu
Macro placement is a fundamental step in modern VLSI physical design, determining the solution quality of high-dimensional combinatorial optimization problems. Despite recent advancements in machine learning for spatial coordinate determination, the temporal dimension of placement sequencing remains largely governed by static heuristics. In this work, we demonstrate that the placement sequence is not merely a preprocessing step but a decisive factor in optimization, where suboptimal early decisions trigger irreversible domino effects that constrain the solution space. To harness this unexplored dimension, we propose \textbf{OrderPlace}, a novel framework that automates the discovery of macro placement strategies via proxy-guided Large Language Model (LLM) evolution. Unlike existing methods that rely on manual rules like area or connectivity, OrderPlace leverages LLMs to evolve generalizable, code-level ordering strategies—ranging from static metrics to dynamic, physics-inspired mechanisms. To mitigate the prohibitive cost of evaluating sequences, we introduce a lightweight proxy evaluation mechanism that efficiently filters candidates using a deterministic greedy probe. Experimental results on the standard ISPD 2005 benchmarks demonstrate that OrderPlace discovers novel ordering strategies. Compared with WireMask-EA and the state-of-the-art method EGPlace, OrderPlace reduces wirelength by 34.04% and 14.08%, respectively.
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Adaptive Coding Emerges in Stabilized Supralinear Networks Trained with Local Plasticity
Haoyu Albert Wang, Wei P Dai, Jialun Ma, Jiawei Zhang, jinqi liu, Mingchen Jiang, Mingqing Xiao, Yansen Wang, Dongqi Han, Dongsheng Li, Yuguo Yu
Lateral connections (LCs) are ubiquitous in the cortical circuits. While DL architectures have rich intralayer interactions to support feature selectivity and contextual modulation, explicit excitatory and inhibitory (E-I) LCs remain underexplored and less-justified for encoding models in both DL and visual neuroscience. In this work, we analyze and train stabilized supralinear networks (SSNs) with strong E-I LCs, using local plasticity rules and natural images. We demonstrate that these LCs support a transition between dynamical regimes under different

input conditions. During the transition, the network shifts from population coding that extracts features from low-contrast or noisy inputs by recruiting more neurons, to sparse coding at high contrast, utilizing considerably fewer neurons. This reduction in the number of active neurons has been generally associated with lower metabolic demand in previous experiments and models. We find the model showing better robustness and adaptiveness against sparse coding, ICA and other unsupervised models under degraded inputs, but not when LCs are ablated. These results support the role of E-I recurrence in dynamic coding strategies and the design of more adaptive and robust systems with a concrete example in vision.
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Graph Neural Dynamics via Learned Energy and Tangential Flows
Moshe Eliasof, Eldad Haber, Carola-Bibiane Schönlieb
We introduce TANGO, a dynamical-systems-inspired framework for graph representation learning that governs node feature evolution through a learned energy landscape and its associated descent dynamics. At the core of our approach is a learnable Lyapunov function over node embeddings, whose gradient defines an energy non-increasing direction that guarantees stability. To enhance flexibility while preserving the benefits of energy-based dynamics, we incorporate a novel tangential component, learned via message passing, that evolves features while maintaining the energy value.
This decomposition into orthogonal flows of energy gradient descent and tangential evolution yields a flexible form of graph dynamics, and enables effective signal propagation even in flat or ill-conditioned energy regions, that often appear in graph learning. Our method is designed to help alleviate oversquashing, and is compatible with different graph neural network backbones. Empirically, TANGO achieves strong performance across a diverse set of node and graph classification and regression benchmarks, demonstrating the effectiveness of jointly learned energy functions and tangential flows for graph neural networks.
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Constrained Multi-Objective Reinforcement Learning with Max-Min Criterion
Giseung Park, Hyunyoung Nam, Woohyeon Byeon, Amir Leshem, Youngchul Sung
Multi-Objective Reinforcement Learning (MORL) extends standard RL by optimizing policies with respect to multiple, often conflicting, objectives. While max-min MORL has emerged as an effective approach for promoting fairness, its applicability remains limited, particularly when constraints must be incorporated. In this paper, we propose a MORL framework that integrates the max-min criterion with explicit constraint satisfaction. We establish a theoretical foundation for the proposed framework and validate the resulting algorithm through convergence analysis and experiments in tabular settings. We further demonstrate the practical relevance of our approach in simulated building thermal control, multi-objective locomotion control, and greenhouse-gas-emission-aware traffic management. Across these domains, our method effectively balances fairness and constraint satisfaction in multi-objective decision-making.
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Masks Can Be Distracting: On Context Comprehension in Diffusion Language Models
Julianna Piskorz, Cristina Pinneri, Alvaro Correia, Motasem Alfarra, Risheek Garrepalli, Christos Louizos
Masked Diffusion Language Models (MDLMs) have recently emerged as a promising alternative to Autoregressive Language Models (ARLMs), leveraging a denoising objective that, in principle, should enable more uniform context utilisation. In this work, we examine the context comprehension abilities of MDLMs and uncover two key limitations. First, despite their more global training objective and bidirectional attention mechanism, similarly to ARLMS, **MDLMs exhibit a strong locality bias**: performance is highly sensitive to the position of relevant information within the input, favouring local over distant context. Second, appending a large number of **mask tokens—required for generation—can significantly degrade context comprehension** in models trained from scratch. Through systematic ablations, we find that these masks **act as distractors**, reducing the model's ability to process relevant information.
  To address and further study this undesirable behaviour, we introduce the mask

-agnostic loss function that encourages predictions to remain invariant to the n
umber of appended masks. Fine-tuning with this objective substantially mitigates
 the distracting effect of masks, improving robustness of MDLMs.
  Overall, our findings reveal critical limitations of the current MDLM training
 paradigm, with implications for training, evaluation and deployment.
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Are Your Agents Upward Deceivers?
Dadi Guo, Qingyu Liu, Dongrui Liu, Qihan Ren, Shuai Shao, Tianyi Alex Qiu, Haora
n Li, Yi R. Fung, Zhongjie Ba, Juntao Dai, Jiaming Ji, Zhikai Chen, Jialing Tao,
 Yaodong Yang, Jing Shao, Xia Hu
Large Language Model (LLM)-based agents are increasingly used as autonomous subo
rdinates that carry out tasks for users. This raises the question of whether the
y may also engage in deception, similar to how individuals in human organization
s lie to superiors to create a good image or avoid punishment. We observe and de
fine ***agentic upward deception***, a phenomenon in which an agent facing envir
onmental constraints conceals its failure and performs actions that were not req
uested without reporting. To assess its prevalence, we construct a benchmark of
200 tasks covering five task types and eight realistic scenarios in a constraine
d environment, such as broken tools or mismatched information sources. Evaluatio
ns of 11 popular LLMs reveal that these agents typically exhibit action-based de
ceptive behaviors, such as guessing results, performing unsupported simulations,
 substituting unavailable information sources, and fabricating local files. We f
urther test intuitive mitigation methods and find only limited reductions, sugge
sting that it is difficult to eliminate and highlighting the need for stronger m
itigation strategies to ensure the safety of LLM-based agents. Code and data are
 available at https://github.com/QingyuLiu/Agentic-Upward-Deception.
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Should I Have Expressed a Different Intent? Counterfactual Generation for LLM-Ba
sed Autonomous Control
Amirmohammad Farzaneh, Salvatore D'Oro, Osvaldo Simeone
Large language model (LLM)-powered agents can translate high-level user intents
into plans and actions in an environment. Yet after observing an outcome, users
may wonder: What if I had phrased my intent differently? We introduce a framewor
k that enables such counterfactual reasoning in agentic LLM-driven control scena
rios, while providing formal reliability guarantees. Our approach models the clo
sed-loop interaction between a user, an LLM-based agent, and an environment as a
 structural causal model (SCM), and leverages test-time scaling to generate mult
iple candidate counterfactual outcomes via probabilistic abduction. Through an o
ffline calibration phase, the proposed conformal counterfactual generation (CCG)
 yields sets of counterfactual outcomes that are guaranteed to contain the true
counterfactual outcome with high probability. We showcase the performance of CCG
 on a wireless network control use case, demonstrating significant advantages co
mpared to naive re-execution baselines.
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Data Agent: Learning to Select Data via End-to-End Dynamic Optimization
Suorong Yang, Fangjian Su, Hai Gan, Ziqi Ye, Jie Li, Baile Xu, Furao Shen, Souja
nya Poria
Dynamic Data selection aims to accelerate training by prioritizing informative s
amples during online training.
However, existing methods typically rely on task-specific handcrafted metrics or
 static/snapshot-based criteria to estimate sample importance, limiting scalabil
ity across learning paradigms and making it difficult to capture the evolving ut
ility of data throughout training.
To address this challenge, we propose Data Agent, an end-to-end dynamic data sel
ection framework that formulates data selection as a training-aware sequential d
ecision-making problem.
The agent learns a sample-wise selection policy that co-evolves with model optim
ization, guided by a composite reward that integrates loss-based difficulty and
confidence-based uncertainty signals.
The reward signals capture complementary objectives of optimization impact and i

nformation gain, together with a tuning-free adaptive weighting mechanism that b
alances these signals over training.
Extensive experiments across a wide range of datasets and architectures demonstr
ate that Data Agent consistently accelerates training while preserving or improv
ing performance, e.g., reducing costs by over 50\% on ImageNet-1k and MMLU with
lossless performance.
Moreover, its dataset-agnostic formulation and modular reward make it plug-and-p
lay across tasks and scenarios, e.g., robustness to noisy datasets, highlighting
 its potential in real-world scenarios. Code is available at https://github.com/
Jackbrocp/Data-Agent.
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GICDM: Mitigating Hubness for Reliable Distance-Based Generative Model Evaluatio
n

Nicolas Salvy, Hugues Talbot, Bertrand Thirion

Generative model evaluation commonly relies on high-dimensional embedding spaces
 to compute distances between samples. We show that dataset representations in t
hese spaces are affected by the hubness phenomenon, which distorts nearest-neigh
bor relationships and biases distance-based metrics. Building on the classical I
terative Contextual Dissimilarity Measure (ICDM), we introduce Generative ICDM (
GICDM), a method to correct neighborhood estimation for both real and generated
data. We introduce a multi-scale extension to improve empirical behavior. Extens
ive experiments on synthetic and real benchmarks demonstrate that GICDM resolves
 hubness-induced failures, restores reliable metric behavior, and improves align
ment with human assessment.
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Stochastic Neural Ray Tracing for Radio Frequency Channel Modeling

Yinyan Bu, Jiajie Yu, Xingyu Chen, Bo Wen, Xinyu Zhang, Piya Pal

Wireless channel modeling is essential for the design, analysis, and optimizatio
n of modern wireless sensing and communication systems. However, accurately mode
ling wireless channels in electrically large and complex environments remains a
long-standing challenge, owing to the intricate interactions between radio-frequ
ency (RF) signals and surrounding objects (e.g., reflection, diffraction, and sc
attering). Unlike conventional ray-tracing pipelines that rely on hand-engineer
interaction rules, or black-box neural surrogates that do not explicitly model p
hysical structure, we introduce SNRFT, a novel framework that integrates neural
representations with physics-based RF propagation modeling. Our key idea is to v
iew RF transport as a stochastic propagation process, from which a material-depe
ndent attenuation coefficient emerges naturally as the rate parameter governing
transport dynamics. This formulation inherently satisfies key physical constrain
ts such as reciprocity and reversibility. Building on this foundation, we employ
 implicit neural representations to capture complex RF-object interactions while
 preserving the composability of traditional ray tracing. Extensive evaluations
on real-world wireless communication and sensing testbeds demonstrate that SNRFT
 consistently outperforms existing methods, while requiring significantly fewer
training samples.
****

OLion: Approaching the Hadamard Ideal by Intersecting Spectral and L inf Implici
t Biases

Zixiao Wang, Yifei Shen, Huishuai Zhang

Many optimizers can be interpreted as steepest-descent methods under norm-induce
d geometries, and thus inherit corresponding implicit biases. We introduce Ortho
gonal Lion which combines spectral control from orthogonalized update directions
 with ℓ_∞-style coordinate control from sign updates. OLion forms a Li
on-style momentum direction, approximately orthogonalizes it via a few Newton--S
chulz iterations, and then applies an entrywise sign, providing an efficient app
roximation to taking a maximal step over the intersection of the spectral and $\
ell_\infty$ constraint sets (a scaled Hadamard-like set for matrix parameters).
Despite the strong nonlinearity of orthogonalization and sign, we prove converge
nce under a mild, empirically verified diagonal-isotropy assumption. Across larg
e-scale language and vision training, including GPT-2 and Llama pretraining, SiT

image pretraining, and supervised fine-tuning, OLion matches or outperforms AdamW and Muon under comparable tuning while using only momentum-level optimizer state, and it mitigates optimizer mismatch when fine-tuning AdamW-pretrained checkpoints.
****
Minimax Optimal Strategy for Delayed Observations in Online Reinforcement Learning
Harin Lee, Kevin Jamieson
We study reinforcement learning with delayed state observation, where the agent observes the current state after some random number of time steps.
We propose an algorithm that combines the augmentation method and the upper confidence bound approach.
For tabular Markov decision processes (MDPs), we derive a regret bound of $\tilde{\mathcal{O}}(H \sqrt{D_{\max} SAK})$, where S and A are the cardinalities of the state and action spaces, H is the time horizon, K is the number of episodes, and $D_{\max}$ is the maximum length of the delay.
We also provide a matching lower bound up to logarithmic factors, showing the optimality of our approach.
Our analytical framework formulates this problem as a special case of a broader class of MDPs, where their transition dynamics decompose into a known component and an unknown but structured component.
We establish general results for this abstract setting, which may be of independent interest.
****
Collaborative Disagreement Resolution for Scalable Oversight
Yuyang Jiang, Chacha Chen, Teng Wu, Liwen Sun, Han Liu, Shi Feng, Chenhao Tan
Debate, where AI agents argue opposing positions, has emerged as a key approach to scalable oversight. However, debate faces a fundamental tension: models are incentivized to be persuasive to the judge, which may not always align with epistemic honesty. In this work, we propose an alternative paradigm: *disagreement resolution*, which reframes the interaction mechanism from adversarial debate to collaborative truth seeking. Drawing on principles from human mediation and conflict resolution, where mediators facilitate dialogue to help disputing parties reach consensus rather than adjudicating between them, we design an automated pipeline that adapts these strategies to AI oversight. Unlike standard debate where models argue for fixed positions, our pipeline directs models to collaboratively identify points of disagreement, examine the evidence for conflicting claims, and converge toward consensus or isolate the specific ''crux'' of their disagreement. We find that Disagreement Resolution consistently helps non-expert models identify the truth, achieving 62.1\% judging accuracy compared to 49.2\% for standard debate. Our results provide encouraging empirical evidence for rethinking the scalable oversight protocol from adversarial persuasion to collaborative truth-seeking.
****
Automatic Construction of Clinical Scoring Systems with LLM Agents
Silas Ruhrberg Estévez, Christopher Chiu, Mihaela van der Schaar
Modern clinical practice relies on evidence-based guidelines implemented as compact scoring systems composed of a small number of interpretable decision rules. While machine-learning models achieve strong performance, many fail to translate into routine clinical use due to misalignment with workflow constraints such as memorability, auditability, and bedside execution. We argue that this gap arises not from insufficient predictive power, but from optimizing over model classes that are incompatible with guideline deployment. Deployable guidelines often take the form of unit-weighted clinical checklists, formed by thresholding the sum of binary rules, but learning such scores requires searching an exponentially large discrete space of possible rule sets. We introduce AgentScore, which performs semantically guided optimization in this space by using LLMs to propose candidate rules and a deterministic, data-grounded verification-and-selection loop to enforce statistical validity and deployability constraints. Across eight clinical prediction tasks, AgentScore outperforms existing score-generation methods

and achieves AUROC comparable to more flexible interpretable models despite oper
ating under stronger structural constraints. On two additional externally valida
ted tasks, AgentScore achieves higher discrimination than established guideline-
based scores.
****

CausalArmor: Efficient Indirect Prompt Injection Guardrails via Causal Attributi
on

Minbeom Kim, Mihir Parmar, Phillip Wallis, Lesly Miculicich, Kyomin Jung, Krishn
amurthy Dj Dvijotham, Long T. Le, Tomas Pfister

AI agents equipped with tool-calling capabilities are susceptible to *Indirect P
rompt Injection* (IPI) attacks. In this attack scenario, malicious commands hidd
en within *untrusted* content trick the agent into performing unauthorized actio
ns. Existing defenses can reduce attack success but often suffer from the ***ove
r-defense dilemma***: they deploy expensive, *always-on* sanitization regardless
 of actual threat, thereby degrading utility and latency even in benign scenario
s.

We revisit IPI through a causal ablation perspective: a successful injection man
ifests as a *dominance shift* where the user request no longer provides decisive
 support for the agent's privileged action, while a particular untrusted segment
, such as a retrieved document or tool output, provides disproportionate attribu
table influence.

Based on this signature, we propose **CausalArmor**, a selective defense framewo
rk that (i) computes lightweight, leave-one-out ablation-based attributions at p
rivileged decision points, and (ii) triggers targeted sanitization only when an
untrusted segment dominates the user intent. Additionally, CausalArmor employs *
retroactive Chain-of-Thought masking* to prevent the agent from acting on ``pois
oned" reasoning traces.

We present a theoretical analysis showing that sanitization based on attribution
 margins conditionally yields an exponentially small upper bound on the probabil
ity of selecting malicious actions.

Experiments on AgentDojo and DoomArena demonstrate that CausalArmor matches the
security of aggressive defenses while improving explainability and preserving ut
ility and latency of AI agents.
****

REAL: Resolving Knowledge Conflicts in Knowledge-Intensive Visual Question Answe
ring via Reasoning-Pivot Alignment

Kai Ye, Xianwei Mao, Sheng Zhou, Zirui Shao, Ye Mo, Liangliang Liu, Haikuan Huan
g, Bin Li, Jiajun Bu

Knowledge-intensive Visual Question Answering (KI-VQA) frequently suffers from s
evere knowledge conflicts caused by the inherent limitations of open-domain retr
ieval. However, existing paradigms face critical limitations, including the lack
 of generalizable conflict detection and intra-model constraint mechanisms to ha
ndle conflicting evidence. To address these challenges, we propose the **REAL**
(**Re**asoning-Pivot **Al**ignment) framework centered on the novel concept of t
he **Reasoning-Pivot**. Distinct from reasoning steps that prioritize internal s
elf-derivation, a reasoning-pivot serves as an atomic unit (node or edge) in the
 reasoning chain that emphasizes knowledge linkage, and it typically relies on e
xternal evidence to complete the reasoning. Supported by our constructed **REAL-
VQA** dataset, our approach integrates **Reasoning-Pivot Aware SFT (RPA-SFT)** t
o train a generalizable discriminator by aligning conflicts with pivot extractio
n, and employs **Reasoning-Pivot Guided Decoding (RPGD)**, an intra-model decodi
ng strategy that leverages these pivots for targeted conflict mitigation. Extens
ive experiments on diverse datasets demonstrate that REAL significantly enhances
 discrimination accuracy and achieves superior performance, validating our pivot
-driven resolution paradigm.
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Beyond Sunk Costs: Boosting LLM Pre-training Efficiency via Orthogonal Growth of
 Mixture-of-Experts

Ruizhe Wang, Yucheng Ding, Xiao Liu, Yaoxiang Wang, Peng CHENG, Baining Guo, Zhe
ng-Jun Zha, Yeyun Gong

As the computational demands for pre-training Large Language Models (LLMs) continue to surge, the need for efficient training paradigms becomes critical. Despite the vast resources already invested in existing pre-trained checkpoints, these assets often remain under-leveraged due to architectural limitations. We introduce an "orthogonal growth" strategy designed to "recycle" these checkpoints by strategically expanding their parameters prior to continued training. Our method focuses on optimizing converged Mixture-of-Experts (MoE) models through two dimensions: interpositional layer copying for increased depth and noisy expert duplication for expanded width. Through extensive scaling laws analysis, we demonstrate a strong positive correlation between the "sunk cost" (prior investment) and the final model accuracy. Empirical results on models up to 70B parameters and 1T tokens show that our recycling approach yields a 10.6\% accuracy improvement compared to training from scratch under identical extra compute budgets. This work provides a cost-effective blueprint for sustainable large-scale LLM development.

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Symmetry Reveals Layerwise Dynamics: How Transformers Perform In-Context Classification

Patrick Lutz, Themistoklis Haris, Arjun Chandra, Aditya Gangrade, Venkatesh Saligrama

Transformers can perform in-context classification from a few labeled examples, yet the inference-time algorithm remains opaque. We study multi-class linear classification in the hard no-margin regime and make the computation identifiable by enforcing feature- and label-permutation equivariance at every layer. This enables interpretability while maintaining functional equivalence and yields highly structured weights. From these models we extract an explicit depth-indexed recursion: an end-to-end identified, emergent update rule inside a softmax transformer, to our knowledge the first of its kind. Attention matrices formed from mixed feature-label Gram structure drive coupled updates of training points, labels, and the test probe. The resulting dynamics implement a geometry-driven algorithmic motif, which can provably amplify class separation and yields robust expected class alignment.

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Adversarial Vulnerability from Interference Between Features in Superposition

Edward Stevinson, Lucas Prieto, Melih Barsbey, Tolga Birdal

Why do adversarial examples exist, and why do they transfer between models? Existing explanations appeal to high-dimensional geometry, non-robust patterns in the input, and decision boundary structure, but none provides a representation-level mechanism that explains why specific perturbations succeed and why attacks transfer between models. In this paper, we show that adversarial vulnerability can stem from efficient information encoding in neural networks. Specifically, vulnerability can arise from superposition - the phenomenon where networks represent more concepts than they have dimensions, forcing non-orthogonal representation and thus interference. This interference causes perturbations targeting one representation to affect others, creating vulnerabilities determined by interference patterns. In synthetic settings with precisely controlled superposition, we establish that superposition suffices to create adversarial vulnerability. The resulting attacks are predictable: PGD-discovered perturbations align with theoretically optimal perturbations derived from the interference geometry. Models trained on similar data develop similar interference patterns, explaining attack transferability. We then show that successful attacks on image classifiers exhibit the structure predicted by our proposed mechanism. These findings reveal that adversarial vulnerability can be a byproduct of networks' representational compression, complementing existing explanations based on data properties or architectural factors.

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Hyperparameter Transfer with Mixture-of-Expert Layers

Tianze Jiang, Blake Bordelon, Cengiz Pehlevan, Boris Hanin

Mixture-of-Experts (MoE) layers have emerged as an important tool in scaling up modern neural networks by decoupling total trainable parameters from activated p

arameters in the forward pass for each token. However, sparse MoEs add complexit y to training due to (i) new trainable parameters (router weights) that, like al l other parameter groups, require hyperparameter (HP) tuning; (ii) new architect ure scale dimensions (number of and size of experts) that must be chosen and pot entially taken large. To make HP selection cheap and reliable, we propose a new parameterization for transformer models with MoE layers when scaling model width , depth, number of experts, *and* expert (hidden) size. Our parameterization is justified by a novel dynamical mean-field theory (DMFT) analysis. When varying different model dimensions trained at a fixed token budget, we find empirically that our parameterization enables reliable HP transfer across models from 51M to 2B total parameters. We further take HPs identified from sweeping small models on a short token horizon to train larger models on longer horizons and report pe rformant model behaviors.

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Z-Erase: Enabling Concept Erasure in Single Stream Diffusion Transformers

Nanxiang Jiang, Zhaoxin Fan, Baisen Wang, Daiheng Gao, Junhang Cheng, Jifeng Guo , Yalan Qin, Yeying Jin, Hongwei Zheng, Faguo Wu, wenjun wu

Concept erasure serves as a vital safety mechanism for removing unwanted concept s from text-to-image (T2I) models. While extensively studied in U-Net and dual-s tream architectures (e.g., Flux), this task remains under-explored in the recent emerging paradigm of single-stream diffusion transformers (e.g., Z-Image). In t his new paradigm, text and image tokens are processed as a single unified sequen ce via shared parameters. Consequently, directly applying prior erasure methods typically leads to generation collapse. To bridge this gap, we introduce Z-Erase , the first concept erasure method tailored for single-stream T2I models. To gua rantee stable image generation, Z-Erase first proposes a Stream Disentangled Con cept Erasure Framework that decouples updates and enables existing methods on si ngle-stream models. Subsequently, within this framework, we introduce Lagrangian -Guided Adaptive Erasure Modulation, a constrained algorithm that further balanc es the sensitive erasure-preservation trade-off. Moreover, we provide a rigorous convergence analysis proving that Z-Erase can converge to a Pareto stationary p oint. Experiments demonstrate that Z-Erase successfully overcomes the generation collapse issue, achieving state-of-the-art performance across a wide range of t asks.

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MUSE: Resolving Manifold Misalignment in Visual Tokenization via Topological Ort hogonality

Panqi Yang, Haodong Jing, Jiahao Chao, Tingyan Xiang, Li Lin, Yao Hu, Yang Luo, Yongqiang Ma

Unified visual tokenization faces a fundamental trade-off: optimizing for high-f idelity pixel reconstruction (spatial equivariance) inherently conflicts with se mantic abstraction (conceptual invariance). We identify the root cause as Manifo ld Misalignment, where naive joint optimization leads to conflicting gradients t hat force a zero-sum game between these two objectives. In this paper, we propos e MUSE, a framework that resolves this deadlock via Topological Orthogonality. R ecognizing Structure as the orthogonal bridge, MUSE physically decouples the opt imization subspaces within Transformers. We route structural gradients to refine attention topology and semantic gradients to update feature values, transformin g destructive interference into Mutual Reinforcement. Extensive experiments demo nstrate that MUSE breaks the trade-off, matching state-of-the-art generation (gF ID 3.08) while notably outperforming its own teacher InternViT-300M in linear pr obing (85.2% vs. 82.5%), proving that structurally aligned reconstruction active ly refines semantic perception.

****

WaterSIC: Information-Theoretically (Near) Optimal \\Linear Layer Quantization

Egor Lifar, Semyon Savkin, Or Ordentlich, Yury Polyanskiy

This paper considers the problem of converting a given dense linear layer to low precision. The tradeoff between compressed length and output discrepancy is ana lyzed information theoretically (IT). It is shown that a popular

GPTQ algorithm may have an arbitrarily large gap to the IT limit. To allevia

te this problem, a
 novel algorithm, termed "WaterSIC", is proposed and is shown to be within a rat
e gap of
    0.255 bits to the IT limit, uniformly over all possible covariance matrices
of input activations.
 The key innovation of WaterSIC's is to allocate different quantization rates to
 different columns
    (in-features) of the weight matrix, mimicking the classical IT solution know
n as
    ``waterfilling''. Applying WaterSIC to the Llama and Qwen family of LLMs est
ablishes new
    state-of-the-art performance for all quantization rates from 1 to 4 bits. Ou
r code is available at https://github.com/egorlifar/watersic.
****
Which Heads Matter for Reasoning? RL-Guided KV Cache Compression
Wenjie Du, Li Jiang, Keda TAO, Xue Liu, Huan Wang
Reasoning large language models exhibit complex reasoning behaviors via extended
 chain-of-thought generation that are highly fragile to information loss during
decoding, creating critical challenges for KV cache compression.
Existing token-dropping methods directly disrupt reasoning chains by removing in
termediate steps, while head-reallocation methods, designed for retrieval tasks,
 fail to preserve the heads essential for generative reasoning.
However, no existing method can identify which attention heads genuinely maintai
n reasoning consistency and control generation termination.
To address this, we propose RLKV, which uses reinforcement learning as a probe t
o discover which heads contribute to reasoning quality by directly optimizing th
eir cache usage against actual generation outcomes.
This discovery naturally leads to an efficient compression strategy: we allocate
 full KV cache to reasoning-critical heads while aggressively compressing others
 with constant-size KV cache.
Experiments reveal that a fraction of heads proves essential for reasoning, enab
ling 20--60% cache reduction with near-lossless performance across diverse tasks
 and models, and up to 2.06x end-to-end speedup at 60% reduction.
****
A General Neural Backbone for Mixed-Integer Linear Optimization via Dual Attenti
on
Peixin Huang, Yaoxin Wu, Yining Ma, Cathy Wu, Wei Zhang, Wen Song
Mixed-integer linear programming (MILP) is a foundational framework for combinat
orial optimization across science and engineering, but remains hard to solve at
scale due to NP-hardness.Recent learning-based methods typically model MILP inst
ances as variable–constraint bipartite graphs and use Graph Neural Networks (GNN
s) for representation learning, yet their locality limits representation power.W
e propose an attention-driven neural backbone that adopts an element-centric vie
w of variables and constraints, with dual attention performing parallel intra-ty
pe self-attention and inter-type cross-attention.Across three representative tas
ks at the instance, element, and solving-state levels, our model consistently ou
tperforms conventional GNN-based architectures, highlighting attention-based, el
ement-centric modeling as a powerful foundation for learning-enhanced combinator
ial optimization.
****
Affine-Equivariant Kernel Space Encoding for NeRF Editing
Miko■aj Zieli■ski, Krzysztof Byrski, Tomasz Szczepanik, Dominik Belter, Przemys■
aw Spurek
Neural scene representations achieve high-fidelity rendering by encoding 3D scen
es as continuous functions, but their latent spaces are typically implicit and g
lobally entangled, making localized editing and physically grounded manipulation
 difficult. While several works introduce explicit control structures or point-b
ased latent representations to improve editability, these approaches often suffe
r from limited locality, sensitivity to deformations, or visual artifacts. In th
is paper, we introduce Affine-Equivariant Kernel Space Encoding (EKS), a spatial

encoding for neural radiance fields that provides localized, deformation-aware feature representations. Instead of querying latent features directly at discrete points or grid vertices, our encoding aggregates features through a field of anisotropic Gaussian kernels, each defining a localized region of influence. This kernel-based formulation enables stable feature interpolation under spatial transformations while preserving continuity and high reconstruction quality. To preserve detail without sacrificing editability, we further propose a training-time feature distillation mechanism that transfers information from multi-resolution hash grid encodings into the kernel field, yielding a compact and fully grid-free representation at inference. This enables intuitive, localized scene editing directly via Gaussian kernels without retraining, while maintaining high-quality rendering.

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Efficient Prediction of SO(3)-Equivariant Hamiltonian Matrices via SO(2) Local Frames

Haiyang Yu, Yuchao Lin, Xuan Zhang, Xiaofeng Qian, Shuiwang Ji

We consider the task of predicting Hamiltonian matrices to accelerate electronic structure calculations, which plays an important role in physics, chemistry, and materials science. Motivated by the inherent relationship between the off-diagonal blocks of the Hamiltonian matrix and the SO(2) local frame, we propose a novel and efficient network, called QHNetV2, that achieves global SO(3) equivariance without the costly SO(3) Clebsch–Gordan tensor products. This is achieved by introducing a set of new efficient and powerful SO(2)-equivariant operations and performing all off-diagonal feature updates and message passing within SO(2) local frames, thereby eliminating the need of SO(3) tensor products. Moreover, a continuous SO(2) tensor product is performed within the SO(2) local frame at each node to fuse node features. Extensive experiments on the large QH9 and MD17 datasets demonstrate that our model achieves superior performance across a wide range of molecular structures and trajectories, highlighting its strong generalization capability. The proposed SO(2) operations on SO(2) local frames offer a promising direction for scalable and symmetry-aware learning of electronic structures. Our code is publicly available as part of the AIRS library (https://github.com/divelab/AIRS/).

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Neural Modular Physics for Elastic Simulation

Yifei Li, Haixu Wu, Zeyi Xu, Tuur Stuyck, Wojciech Matusik

Learning-based methods have made significant progress in physics simulation, typically approximating dynamics with a monolithic end-to-end optimized neural network. Although these models offer an effective way to simulation, they may lose essential features compared to traditional numerical simulators, such as physical interpretability and reliability. Drawing inspiration from classical simulators that operate in a modular fashion, this paper presents Neural Modular Physics (NMP) for elastic simulation, which combines the approximation capacity of neural networks with the physical reliability of traditional simulators. Beyond the previous monolithic learning paradigm, NMP enables direct supervision of intermediate quantities and physical constraints by decomposing elastic dynamics into physically meaningful neural modules connected through intermediate physical quantities. With a specialized architecture and training strategy, our method transforms the numerical computation flow into a modular neural simulator, achieving improved physical consistency and generalizability. Experimentally, NMP demonstrates superior generalization to unseen initial conditions and resolutions, stable long-horizon simulation, better preservation of physical properties compared to other neural simulators, and greater feasibility in scenarios with unknown underlying dynamics than traditional simulators.

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MTNL: A Unified Modeling Perspective for Enhancing Tensor Network Learning

Junhua Zeng, Yuning Qiu, Binghua Li, Chao Li, Qibin Zhao, Guoxu Zhou

Over the years, the unsupervised and supervised learning research directions of tensor networks (TNs) have mainly developed in parallel. In this paper, we provide a view for their cooperative advancement through a novel mixed tensor network

learning (MTNL) framework that unifies the two fields. Specifically, inspired by supervised TN learning tasks, multiple TNs are fused in a deep-network style in MTNL, enhancing the expressive power for the unsupervised TN learning tasks. We then develop a more flexible TN structure search prior with theoretical guarantees for learning multiple TN structures, aligning with trends in many supervised learning setups. More interestingly, by combining these components within a Bayesian framework, we show that MTNL induces a lightweight uncertainty quantification mechanism that is theoretically guaranteed by its connection to the dropout-based counterpart problem, making the mechanism a potential alternative for large-scale learning problems. Finally, we demonstrate the effectiveness of the MTNL framework on tensor recovery, parameter-efficient fine-tuning, and tensor regression experiments.

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MIND: Decoupling Model-Induced Label Noise via Latent Manifold Disentanglement

Dayong Ren

The paradigm of learning from automatic annotations—driven by pre-trained experts and Foundation Models—dominates data-hungry applications. However, it introduces a critical challenge: model-induced label noise. Unlike stochastic noise in classical robust learning, this noise stems from annotator inductive biases, manifesting as systematic errors tightly coupled with local feature manifolds. Existing methods relying on global transition matrices underfit these structural patterns, while learning instance-specific matrices remains mathematically intractable. We propose Model-Induced Noise Decoupling (MIND), a theoretically grounded framework addressing this dilemma. We demonstrate that the high-dimensional noise manifold can be decoupled into tractable, subspace-dependent components via Latent Manifold Disentanglement. Specifically, our Latent Decoupling Estimator (LDE) dynamically projects samples into latent structural clusters with consistent error modes, facilitating noise identifiability without ground-truth anchor points. To rigorously evaluate robustness, we adopt a hierarchical protocol: moving from controlled noise on CIFAR-100 to a structural stress test on large-scale real-world 3D datasets (S3DIS, ScanNet), where error patterns explicitly couple with geometric manifolds. Empirically, MIND significantly outperforms state-of-the-art methods on these complex benchmarks and effectively corrects zero-shot hallucinations from Vision-Language Models (e.g., OpenSeg), highlighting its potential as a robust distillation framework for Foundation Models.

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Unveiling the Entropy Dynamics of Chain-of-Thought Reasoning

Ting Xu, Xu He, Yupu Lu, Jiankai Sun, Dong Li, Wai Lam, Jianye HAO

This paper investigates the entropy dynamics of Chain-of-Thought (CoT) and uncovers a consistent two-phase structure: an ***Uncertainty Region*** of exploration transitioning sharply to a ***Confidence Region*** of convergence. We demonstrate that the Confidence Region possesses two critical properties: 1) ***High Reliability***—answers in confidence region become highly accurate and stable, and 2) ***High Redundancy***—models generate unnecessary tokens long after reaching the correct answer. These properties unlock more efficient and reliable inference strategies: 1) ***Early Exit*** leverages reliability and redundancy to terminate computation safely when returns diminish, and 2) ***Test-Time Scaling*** uses the Confidence Region signal to prioritize converged trajectories. To operationalize these insights, we formulate Confidence Region detection as a sequential change-point detection problem, being the first to apply classical change-point methods to monitor CoT reasoning. Using the Cumulative Sum (CUSUM) algorithm, a statistically optimal change-point detector, we develop a training-free framework for real-time inference control. Experiments show our approach establishes a superior Pareto-frontier for early exit. CUSUM achieves 63.06% accuracy with 11.1% token reduction, outperforming DEER and Dynasor by 3.28% and 4.36% in accuracy respectively. For test-time scaling, CUSUM-weighted voting consistently outperforms self-consistency.

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OBCache: Optimal Brain KV Cache Pruning for Efficient Long-Context LLM Inference

Yuzhe Gu, Xiyu Liang, Jiaojiao Zhao, Enmao Diao

Large language models (LLMs) with extended context windows enable powerful applications but impose significant memory overhead, as caching all key-value (KV) states scales linearly with sequence length and batch size. Existing cache eviction methods address this by exploiting attention sparsity, yet they typically rank tokens heuristically using accumulated attention weights without considering their true impact on attention outputs. We propose Optimal Brain Cache (OBCache), a principled framework that formulates cache eviction as a layer-wise structured pruning problem. Building upon the Optimal Brain Damage (OBD) theory, OBCache quantifies token saliency by measuring the perturbation in attention outputs induced by pruning tokens, with closed-form scores derived for isolated keys, isolated values, and joint key-value pairs. Our scores account not only for attention weights but also for information from value states and attention outputs, thereby enhancing existing eviction strategies with output-aware signals. Experiments on LLaMA and Qwen models demonstrate that replacing the heuristic scores in existing works, which estimate token saliency across different query positions, with OBCache's output-aware scores consistently improves long-context accuracy. Code is available at https://github.com/DreamSoul-AI/OBCache.

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Foundation VAE for CT Reconstruction, Augmentation, and Generation

Qi Chen, Shuhan Ding, Yu Gu, Nan Liu, Jiang Bian, Alan Yuille, Zongwei Zhou, Jingjing Fu

Variational autoencoders (VAEs) compress high resolution CT volumes into compact latents while preserving clinically relevant structure. However, training CT-specific VAEs from scratch or heavily fine-tuning them incurs substantial computational and engineering cost, and often degrades under heterogeneous scanners, protocols, and diseases. This paper makes a progressive stride toward training-free medical VAEs by leveraging a critical observation: a single Foundation VAE, pretrained at scale on natural images and videos, can serve as a unified interface for CT Reconstruction, Augmentation, and Generation. With both encoder and decoder frozen, the Foundation VAE reconstructs CT volumes with preserved anatomy while suppressing acquisition noise; training segmentation models on these reconstructions improves surface accuracy by 3.9\% NSD on average for pancreatic tumor and lung tumor. Within the same Foundation VAE latent space, a conditional latent diffusion model achieves 3.9\% lower average FVD with 36.2\% higher CT CLIP score, and improves multi-disease generation faithfulness across 18 types by 2.76\% AUC. These results demonstrate Foundation VAEs as a practical interface for scalable CT representation reuse and faithful CT generation. Our code and demo are available at https://github.com/qic999/Foundation-VAE.

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Evaluating Newtonian Mechanics in Video Generative Models with Real Physical Systems

Antonios Tragoudaras, Chenyu Zhang, Daniil Cherniavskii, Antonios Vozikis, Thijmen Nijdam, Derck W. E. Prinzhorn, Mark Bodracska, Nicu Sebe, Andrii Zadaianchuk, Stratis Gavves

Recent advances in image and video generation raise hopes that these models possess world modeling capabilities—the ability to generate realistic, physically plausible videos. This could revolutionize applications in robotics, autonomous driving, and scientific simulation. However, before treating these models as world models, we must ask: Do they adhere to physical laws? Current evaluation methods rely on subjective judgments or trajectory matching, limiting their usage for physical reasoning estimation, where many generations could be physically plausible.

Thus, we introduce **Morpheus**, one of the first physics-informed evaluation frameworks for measuring the ability of video generation models to comprehend Newtonian dynamics. **Morpheus** features 130 real-world videos capturing physical phenomena, guided by conservation laws. Using those as conditioning for video generation, we assess physical plausibility leveraging interpretable metrics evaluated with respect to infallible conservation laws known per physical setting, leveraging advances in physics-informed neural networks and vision-language foundation models. Importantly, **Morpheus** targets controlled Newtonian rigid-body se

ttings to enable quantitative checks. Our findings reveal that even with advanced prompting and video conditioning, contemporary models struggle to encode physical principles despite generating aesthetically pleasing videos. Code and data available [here](https://github.com/physics-from-video/Morpheus).
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Rel-MOSS: Towards Imbalanced Relational Deep Learning on Relational Databases
Jun Yin, Peng Huo, Bangguo Zhu, Hao Yan, Senzhang Wang, Shirui Pan, Chengqi Zhang
In recent advances, to enable a fully data-driven learning paradigm on relational databases (RDB), relational deep learning (RDL) is proposed to structure the RDB as a heterogeneous entity graph and adopt the graph neural network (GNN) as the predictive model. However, existing RDL methods neglect the imbalance problem of relational data in RDBs and risk under-representing the minority entities, leading to an unusable model in practice. In this work, we investigate, for the first time, class imbalance problem in RDB entity classification and design the relation-centric minority synthetic over-sampling GNN (**Rel-MOSS**), in order to fill a critical void in the current literature. Specifically, to mitigate the issue of minority-related information being submerged by majority counterparts, we design the relation-wise gating controller to modulate neighborhood messages from each individual relation type. Based on the relational-gated representations, we further propose the relation-guided minority synthesizer for over-sampling, which integrates the entity relational signatures to maintain relational consistency. Extensive experiments on 12 entity classification datasets provide compelling evidence for the superiority of Rel-MOSS, yielding an average improvement of up to 2.46% and 4.00% in terms of *Balanced Accuracy* and *G-Mean*, compared with SOTA RDL methods and classic methods for handling class imbalance.
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Rethinking the Trust Region in LLM Reinforcement Learning
Penghui Qi, Xiangxin Zhou, Zichen Liu, Tianyu Pang, Chao Du, Min Lin, Wee Sun Lee
Reinforcement learning (RL) has become a cornerstone for fine-tuning Large Language Models (LLMs), with Proximal Policy Optimization (PPO) serving as the de facto standard algorithm. Despite its ubiquity, we argue that the core ratio clipping mechanism in PPO is structurally ill-suited for the large vocabularies inherent to LLMs. PPO constrains policy updates based on the probability ratio of sampled tokens, which serves as a noisy single-sample Monte Carlo estimate of the true policy divergence. This creates a sub-optimal learning dynamic: updates to low-probability tokens are aggressively over-penalized, while potentially catastrophic shifts in high-probability tokens are under-constrained, leading to training inefficiency and instability. To address this, we propose Divergence Proximal Policy Optimization (DPPO), which substitutes heuristic clipping with a more principled constraint based on a direct estimate of policy divergence (e.g., Total Variation or KL). To avoid huge memory footprint, we introduce the efficient Binary and Top-K approximations to capture the essential divergence with negligible overhead. Extensive empirical evaluations demonstrate that DPPO achieves superior stability and efficiency compared to existing methods, offering a more robust foundation for RL-based LLM fine-tuning. Our code is available at \url{https://github.com/sail-sg/Stable-RL}.
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Privacy-Aware Data Integration for Enhanced Quantile Inference under Heterogeneity
Leheng Cai, Qirui Hu, Shuyuan Wu
Quantile estimation and inference play essential roles in diverse scientific and industrial applications, and their accuracy can often be enhanced by integrating auxiliary data from multiple sites. However, developing efficient aggregation methods for quantile inference under potential privacy constraints, particularly with heterogeneous datasets, remains challenging. To address these issues, we propose a systematic framework for quantile estimation and inference under potential local differential privacy (LDP). The key idea is to construct weighted estimators by adaptively aggregating quantile estimates from target and source sites

. The adaptive weights are determined by minimizing the asymptotic variance, inc
orporating an additional ℓ_2 penalty to account for parameter shift. A para
llel stochastic gradient descent algorithm under LDP constraints is developed fo
r weight estimation and valid inference. Additionally, we introduce a conservati
ve weighted estimator to ensure robust inference across diverse heterogeneous sc
enarios. Rigorous theoretical analysis establishes the consistency, normality, a
nd effectiveness of the proposed methods. Extensive numerical studies and real d
ata application corroborate our theoretical findings.
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Utonia: Toward One Encoder for All Point Clouds
Yujia Zhang, Xiaoyang Wu, Yunhan Yang, Xianzhe Fan, Han Li, Yuechen Zhang, Zehao
 Huang, Naiyan Wang, Hengshuang Zhao
We dream of a future where point clouds from all domains can come together to sh
ape a single model that benefits them all. Toward this goal, we present Utonia,
 a first step toward training a single self-supervised point transformer encoder
 across heterogeneous domains, spanning remote sensing, outdoor LiDAR, indoor RGB
-D sequences, object-centric CAD models, and point clouds lifted from RGB-only v
ideos. Despite their distinct sensing geometries, densities, and priors, Utonia
 learns a consistent representation space that transfers across domains. This uni
fication improves perception capability while revealing intriguing emergent beha
viors that arise only when domains are trained jointly. Beyond perception, we ob
serve that Utonia representations can also benefit embodied and multimodal reaso
ning: conditioning vision-language-action policies on Utonia features improves r
obotic manipulation, and integrating them into vision-language models yields gai
ns on spatial reasoning. We hope Utonia can serve as a step toward foundation mo
dels for sparse 3D data, and support downstream applications in AR/VR, robotics,
 and autonomous driving.
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WISE: World Knowledge-Informed Semantic Evaluation for Text-to-Image Generation
Yuwei Niu, Munan Ning, Mengren Zheng, Weiyang Jin, Bin Lin, Peng Jin, Jiaqi Liao
, Chaoran Feng, Fanqing Meng, Kun-Peng Ning, Bin Zhu, Li Yuan
Text-to-Image (T2I) models are capable of generating high-quality artistic creat
ions and visual content. However, existing research and evaluation standards pre
dominantly focus on image realism and shallow text-image alignment, lacking a co
mprehensive assessment of complex semantic understanding and world knowledge int
egration in text-to-image generation. To address this challenge, we propose WISE
, the first benchmark specifically designed for World Knowledge-Informed Semanti
c Evaluation. WISE moves beyond simple word-pixel mapping by challenging models
 with 1000 meticulously crafted prompts across 25 subdomains in cultural common s
ense, spatio-temporal reasoning, and natural science. To overcome the limitation
s of traditional CLIP metric, we introduce WiScore, a novel quantitative metric
 for assessing knowledge-image alignment. Through comprehensive testing of 20 mod
els (10 dedicated T2I models and 10 unified multimodal models) using 1,000 struc
tured prompts spanning 25 subdomains, our findings reveal significant limitation
s in their ability to effectively integrate and apply world knowledge during ima
ge generation, highlighting critical pathways for enhancing knowledge incorporat
ion and application in next-generation T2I models. Code and data will be availab
le.
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Learning Discriminative and Generalizable Anomaly Detector for Dynamic Graph wit
h Limited Supervision
Yuxing Tian, Yiyan Qi, Fengran Mo, Weixu Zhang, Jian Guo, Jian-Yun Nie
Dynamic graph anomaly detection is critical for many real-world applications but
 remains challenging due to the scarcity of labeled anomalies. Existing methods
are either unsupervised or semi-supervised: unsupervised methods avoid the need
 for labeled anomalies but often produce ambiguous boundary, whereas semi-superv
ised methods can overfit to the limited labeled anomalies and generalize poorly
to unseen anomalies. To address this gap, we consider a largely underexplored pr
oblem: learning a discriminative boundary from normal/unlabeled data, while leve
raging limited labeled anomalies \textbf{when available} without sacrificing gen

eralization to unseen anomalies. In this paper, we propose an effective, general izable, and model-agnostic framework with three main components: (i) residual re presentation encoding that capture deviations between current interactions and t heir historical context, providing anomaly-relevant signals; (ii) a restriction loss that constrain the normal representations within an interval bounded by two co-centered hyperspheres, ensuring consistent scales while keeping anomalies se parable; (iii) a bi-boundary optimization strategy that learns a discriminative and robust boundary using the log-likelihood distribution modeled by a normalizi ng flow. Extensive experiments demonstrate the superiority of our framework acro ss diverse evaluation settings.
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SCRWKV: Ultra-Compact Structure-Calibrated Vision-RWKV for Topological Crack Seg mentation
Hanxu Zhang, Chen Jia, Hui Liu, Xu Cheng, Fan Shi, Shengyong Chen
Achieving pixel-level accurate segmentation of structural cracks across diverse scenarios remains a formidable challenge. Existing methods face significant bott lenecks in balancing crack topology modeling with computational efficiency, ofte n failing to reconcile high segmentation quality with low resource demands. To a ddress these limitations, we propose the Ultra-Compact Structure-Calibrated Visi on RWKV (SCRWKV), a network that achieves high-precision modeling via a novel St ructure-Field Encoder (SFE) backbone while maintaining linear complexity. The SF E integrates the Adaptive Multi-scale Cascaded Modulator (AMCM) to enhance textu re representation and utilizes the Structure-Calibrated Insight Unit (SCIU) as i ts core engine. Specifically, the SCIU employs the Geometry-guided Bidirectional Structure Transformation (GBST) to capture topological correlations and integra tes the Dynamic Self-Calibrating Decay (DSCD) into Dy-WKV to suppress noise prop agation. Furthermore, we introduce a lightweight Cross-Scale Harmonic Fusion (CS HF) decoder to achieve precise feature aggregation. Systematic evaluations on mu ltiple benchmarks characterized by complex textures and severe interference demo nstrate that SCRWKV, with only 1.22M parameters, significantly outperforms SOTA methods. Achieving an F1 score of 0.8428 and mIoU of 0.8512 on the TUT dataset, the model confirms its robust potential for efficient real-world deployment. The code is available at https://github.com/zhxhzy/SCRWKV.
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RealtimeTool: Parallel Decoding for Real-Time LLM Function Calling
Xiaoxin Shi, Jiaxin Wan, Linkang Dong, Wei Jiang, Yue Liu, Zengfeng Huang
LLM-based function calling enables intelligent agents to interact with external tools and environments, yet autoregressive decoding imposes a fundamental latenc y bottleneck that limits real-time applications such as embodied intelligence, g ame AI, and interactive avatars (e.g., 10 Hz control frequency). We observe that function calling differs fundamentally from free-form text generation: structur ed outputs exhibit substantial token redundancy (delimiters, parameter names) an d weak causal dependencies among arguments---two properties that must be exploit ed jointly to achieve real-time performance. We present RealtimeTool, which intr oduces special tokens that serve a dual role: compressing low-entropy tokens (4- -6× reduction) while acting as mode selectors that enable independent parallel g eneration of function name and arguments. This synergistic design achieves 3--6× end-to-end speedup (up to 9.6×) with only +8.2% parallelization overhead, while maintaining competitive or improved accuracy across five benchmarks on Qwen-ser ies models (0.5B--14B). With quantization on a consumer-grade GPU, RealtimeTool reaches 61.2 ms P50 latency at 4B scale---enabling 16 Hz real-time control and b ridging the gap between LLM function calling and latency-critical real-world dep loyment.
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Riemannian Neural Optimal Transport
Alessandro Micheli, Yueqi Cao, Anthea Monod, Samir Bhatt
Computational optimal transport (OT) offers a principled framework for generativ e modeling. Neural OT methods, which use neural networks to learn an OT map (or potential) from data in an amortized way, can be evaluated out of sample after t raining, but existing approaches are tailored to Euclidean geometry. Extending n

eural OT to high-dimensional Riemannian manifolds remains an open challenge. In this paper, we prove that any method for OT on manifolds that produces discrete approximations of transport maps necessarily suffers from the curse of dimensionality: achieving a fixed accuracy requires a number of parameters that grows exponentially with the manifold dimension. Motivated by this limitation, we introduce Riemannian Neural OT (RNOT) maps, which are continuous neural-network parameterizations of OT maps on manifolds that avoid discretization and incorporate geometric structure by construction. Under mild regularity assumptions, we prove that RNOT maps approximate Riemannian OT maps with sub-exponential complexity in the dimension. Experiments on synthetic and real datasets demonstrate improved scalability and competitive performance relative to discretization-based baselines.

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MAGIC: Multi-Granularity Language-Informed Image Clustering

Xiaohan Zhang, Chao Zhang, Chunlin Chen, Huaxiong Li

Image clustering is a fundamental unsupervised task in computer vision. Recent studies have explored incorporating external linguistic information to facilitate visual feature learning and thereby enhance clustering performance. Nevertheless, these methods typically rely on fixed vocabularies (e.g., WordNet) to generate language counterparts, leading to inter-modal semantic misalignment due to granularity discrepancy between visual and textual semantics. Moreover, they often overlook the issue of intra-modal semantic redundancy caused by task-irrelevant knowledge. To address these challenges, we propose a new Multi-grAnularity lanGuage-informed Image Clustering method, dubbed MAGIC. To reduce semantic misalignment, we first prompt the vision-language models to generate multi-granularity language descriptions that capture rich image semantics, which are then integrated for effective multi-modal alignment. To alleviate semantic redundancy, we design modality-specific semantic adapters that adaptively refine and compress the semantically dense features into clustering-friendly representations under task guidance. A consensus representation is obtained by fusing the refined visual and textual features, which acts as a teacher to guide image clustering through a robust contrastive learning framework. Extensive experiments on benchmarks demonstrate that MAGIC outperforms state-of-the-art methods.

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Scaling Laws of Global Weather Models

Yuejiang Yu, Langwen Huang, Alexandru Calotoiu, Torsten Hoefler

Data-driven models are revolutionizing weather forecasting. To optimize training efficiency and model performance, this paper analyzes empirical scaling laws within this domain. We investigate the relationship between model performance (validation loss) and three key factors: model size (N), dataset size (D), and compute budget (C). Across a range of models, we find that Aurora exhibits the strongest data-scaling behavior: increasing the training dataset by 10× reduces validation loss by up to 3.2×. GraphCast demonstrates the highest parameter efficiency, yet suffers from limited hardware utilization. Our compute-optimal analysis indicates that, under fixed compute budgets, allocating resources to more total training data yields greater performance gains than increasing model size. Furthermore, we analyze model shape and uncover scaling behaviors that differ fundamentally from those observed in language models: weather forecasting models consistently favor increased width over depth. These findings suggest that future weather models should prioritize wider architectures and larger effective training datasets to maximize predictive performance.

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Contribution Weights: A Geometrical Analysis of Self-Attention Transformers

Harry Jake Cunningham, Nicola Muca Cirone

Analyzing attention weights has become a standard approach for interpreting the information flow of Large Language Models (LLMs). However, this approach has significant limitations as it neglects the geometric properties of the value vectors being aggregated. To address this gap, we introduce \emph{Contribution Weights}, a projection-based metric that quantifies a token's influence by accounting for it's attention weight, value magnitude, and directional alignment with the la

yer output. We demonstrate that contribution weights provide a more faithful mea
sure of token importance, consistently outperforming attention-based metrics in
identifying semantically critical tokens across different decoder-only models, t
asks, and datasets. Further, our metric enables novel mechanistic analysis of \e
mph{attention sinks}. While previous work characterized sinks as passive reposit
ories for excess attention, we reveal they serve an active functional role, supp
ressing information through a convex relationship between sink rate and output n
orm, stabilizing representations by opposing the semantic drift of low-confidenc
e tokens.
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FUSE: FK-Steered Multi-Modal Flow Matching for Efficient Simulation-Based Poster
ior Estimation
Weichen Qin, Yufan Xie, Peihao Wang, Chia-Jui Chou, Minghui Du, Peng Xu, Ziren L
uo, Yi Yang, Jingyi Yu, Bo Liang, Jiakai Zhang
Simulation-Based Inference (SBI) is critical for scientific discovery, with gene
rative models offering a promising path toward efficient inference. However, exi
sting methods struggle with effective multimodal modeling. They often rely on br
ute-force fusion strategies that ignore the structural disparities between param
eters and observations, thus limiting estimation fidelity. In this work, we intr
oduce FUSE (Feynman-Kac steered mUlti-modal flow matching for efficient Simulati
on-based posterior Estimation). Unlike prior work, FUSE employs a dual-track arc
hitecture that preserves the distinct features of multimodal inputs while facili
tating dynamic interaction. Additionally, we propose an FK-steered sampling stra
tegy that leverages intermediate observation likelihoods to guide the generative
 trajectories, effectively improving the sample quality during inference. Our ap
proach outperforms state-of-the-art baselines on standard SBI benchmarks, produc
ing posteriors that closely match ground-truth MCMC. Furthermore, in a real-worl
d exoplanet orbital estimation task, FUSE successfully resolves complex paramete
r degeneracies that challenge existing methods, highlighting its potential to ac
celerate complex scientific discoveries in astrophysics and beyond.
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FedSDR: Federated Self-Distillation with Rectification
Ziheng Ren, Zhanming Shen, Hao Wang, Ning Liu, You Song
Federated fine-tuning of Large Language Models faces severe statistical heteroge
neity. However, existing model-level defenses often overlook the root cause: int
rinsic data distribution mismatches. In this work, we first establish Federated
Self-Distillation (FedSD) as a fundamental and potent strategy. By projecting cl
ient representations into a smoothed ``model-understanding space,'' FedSD alone
serves as a universal booster, demonstrating superior performance over conventio
nal algorithms. Despite its success, we identify a subtle trade-off termed the R
ewrite Paradox---unconstrained self-distillation can inadvertently increase hall
ucinations and redundancy. To refine this paradigm, we further propose FedSDR (F
ederated Self-Distillation with Rectification), the ultimate reinforced framewor
k. It augments FedSD with a dual-stream mechanism: a local LoRA-S (Smoothing) br
anch to implicitly absorb heterogeneity via distilled data, and a parallel globa
l LoRA-R (Rectification) branch anchored to raw data to enforce factual correctn
ess. By selectively aggregating only LoRA-R, FedSDR yields a globally aligned an
d faithful model. Extensive experiments verify its superior performance.
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Breaking the Factorization Barrier in Diffusion Language Models
Ian Li, Zilei Shao, Benjie Wang, Rose Yu, Guy Van den Broeck, Anji Liu
Diffusion language models theoretically allow for efficient parallel generation
but are practically hindered by the "factorization barrier": the assumption that
 simultaneously predicted tokens are independent. This limitation forces a trade
-off: models must either sacrifice speed by resolving dependencies sequentially
or suffer from incoherence due to factorization. We argue that this barrier aris
es not from limited backbone expressivity, but from a structural misspecificatio
n: models are restricted to fully factorized outputs because explicitly paramete
rizing a joint distribution would require the Transformer to output a prohibitiv
ely large number of parameters. We propose **Co**upled **D**iscrete **D**iffusio

n (**CoDD**), a hybrid framework that breaks this barrier by replacing the fully-factorized output distribution with a lightweight, tractable probabilistic inference layer. This formulation yields a distribution family that is significantly more expressive than standard factorized priors, enabling the modeling of complex joint dependencies, yet remains compact enough to avoid the prohibitive parameter explosion associated with full joint modeling. Empirically, CoDD seamlessly enhances diverse diffusion language model architectures with negligible overhead, matching the reasoning performance of computationally intensive Reinforcement Learning baselines at a fraction of the training cost. Furthermore, it prevents performance collapse in few-step generation, enabling high-quality outputs at significantly reduced latencies. Code available at: https://github.com/liuanji/CoDD

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Building Reliable Long-Form Generation via Hallucination Rejection Sampling
Lin Li, Georgia Channing, Suhaas M Bhat, Gabriel Davis Jones, Yarin Gal
Large language models (LLMs) have achieved remarkable progress in open-ended text generation, yet they remain prone to hallucinating incorrect or unsupported content, which undermines their reliability. This issue is exacerbated in long-form generation due to hallucination snowballing, a phenomenon where early errors propagate and compound into subsequent outputs. To address this challenge, we propose a novel inference-time hallucination mitigation framework, named Segment-wise HAllucination Rejection Sampling (SHARS), which uses am arbitrary hallucination detector to identify and reject hallucinated segments during generation and resample until faithful content is produced. By retaining only confident information and building subsequent generations upon it, the framework mitigates hallucination accumulation and enhances factual consistency. To instantiate this framework, we adopt semantic uncertainty as the detector and introduce several vital modifications to address its limitations and better adapt it to long-form text. Our method enables models to self-correct hallucinations without requiring external resources such as web search or knowledge bases, while remaining compatible with them for future extensions. Empirical evaluations on standardized hallucination benchmarks demonstrate that our method substantially reduces hallucinations in long-form generation while preserving or even improving the informativeness of generation.

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POET-X: Memory-efficient LLM Training by Scaling Orthogonal Transformation
Zeju Qiu, Lixin Liu, Adrian Weller, Han Shi, Weiyang Liu
Efficient and stable training of large language models (LLMs) remains a core challenge in modern machine learning systems. To address this challenge, Reparameterized Orthogonal Equivalence Training (POET), a spectrum-preserving framework that optimizes each weight matrix through orthogonal equivalence transformation, has been proposed. Although POET provides strong training stability, its original implementation incurs high memory consumption and computational overhead due to intensive matrix multiplications. To overcome these limitations, we introduce POET-X, a scalable and memory-efficient variant that performs orthogonal equivalence transformations with significantly reduced computational cost. POET-X maintains the generalization and stability benefits of POET while achieving substantial improvements in throughput and memory efficiency. In our experiments, POET-X enables the pretraining of billion-parameter LLMs on a single Nvidia H100 GPU, and in contrast, standard optimizers such as AdamW run out of memory under the same settings.

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Graph-Link: Bridging the Semantic-Structural Gap in Text-to-SQL via Constrained Subgraph Induction
Jianwei Zhong, Yuxi Yang, Quanxin Liu, Ruida Xu, Ming Chi, Yijun Mo
Schema Linking serves as the foundational perception layer in Text-to-SQL, tasked with grounding natural language queries into relevant schema elements. However, existing retrieval-based approaches suffer from a critical *structural blindness*: by prioritizing elements with high textual similarity, they inadvertently prune semantically-thin but topologically-critical bridge tables, thereby severin

g relational pathways necessary for multi-hop joins. To bridge this gap, we propose Graph-Link, a novel framework that reformulates schema linking from an independent retrieval task into a constrained subgraph induction problem. We argue that generating executable SQL necessitates a connected subgraph that satisfies both semantic relevance and structural constraints. Accordingly, Graph-Link employs a hierarchical schema graph to model the search space across multiple granularities, and then applies a Steiner-tree-based optimization for subgraph induction that guarantees the topological connectivity while maximizing the signal-to-noise ratio for downstream LLMs. Extensive experiments on BIRD and Spider 2.0 demonstrate that Graph-Link achieves state-of-the-art schema linking performance, improving recall and hit rates by up to 7.0\% over competitive baselines, and boosts downstream SQL generation accuracy on complex queries by 13.8\%.
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HybridFlow: Resource-Adaptive Subtask Routing for Efficient Edge-Cloud LLM Inference

Jiangwen Dong, Jiayu Li, Tianhang Zheng, Wanyu Lin

Edge-cloud collaborative inference is crucial for LLM-powered edge devices, as on-device models often lack the required reasoning capability, while cloud-only inference can be costly and slow under strict latency and token/API budgets.
However, existing edge-cloud collaboration methods typically route input tasks based on their estimated difficulty. These static, coarse heuristics overlook subtask dependencies, missing opportunities for parallel execution and budget-adaptive routing.
To this end, we propose HybridFlow, a resource-adaptive edge-cloud inference framework that enables parallel execution of interdependent subtasks.
Specifically, we build a dependency-aware DAG for each input task, facilitating concurrent execution of subtasks once their dependencies are resolved, thereby reducing end-to-end latency.
Additionally, we propose a dynamic benefit–cost utility model, optimizing the trade-off between accuracy, token/API cost, and latency in real-time. This dynamic routing minimizes unnecessary cloud usage while preserving reasoning quality.
Across GPQA, MMLU-Pro, AIME24, and LiveBench-Reasoning, HybridFlow improves the cost-accuracy trade-off, reducing latency and cloud API usage while maintaining competitive accuracy.
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Prediction-Powered Adaptive Inference with Pretrained AI Models for Contextual Bandits

Gabriel Sargent, Will Wei Sun, Zhengwu Zhang, Yufeng Liu

In adaptive experiments, statistical inference is essential for reliable decision-making and scientific discovery. Often in these settings, collecting labeled data is expensive, but decision-makers have access to large unlabeled datasets and strong pretrained AI models that can generate outcome predictions. Effectively leveraging these predictions in online experiments poses fundamental challenges for statistical inference: AI models may be misspecified, and data collected under adaptive policies are inherently non-i.i.d., invalidating classical inference techniques. To address these challenges, we propose a Prediction-Powered Adaptive Inference (PPAI) estimator that integrates unlabeled data, predicted labels, and adaptively collected labeled data through a single estimating equation. We establish asymptotic normality of the PPAI estimator under mild conditions on the data-collection policy, enabling valid confidence intervals and hypothesis tests for a broad class of Z-functionals. The method incorporates a data-driven tuning mechanism that adaptively weights AI predictions according to their informativeness, guaranteeing that the resulting asymptotic variance is no worse than that of the labeled-only baseline, and is strictly smaller when predictions are informative. Numerical experiments further support the theory, illustrating efficiency gains with informative AI predictions and robust performance when predictions are inaccurate.
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RealisMotion: Decomposed Human Motion Control and Video Generation in the World Space

Jingyun Liang, Jingkai Zhou, Shikai Li, Chenjie Cao, Lei Sun, Yichen Qian, Weihua Chen, Fan Wang
Generating human videos with realistic and controllable motions is a challenging task. While existing methods can generate visually compelling videos, they lack separate control over four key video elements: foreground subject, background video, human trajectory, and action patterns. In this paper, we propose a decomposed human motion control and video generation framework that explicitly decouples motion from appearance, subject from background, and action from trajectory, enabling flexible mix-and-match composition of these elements. Concretely, we first build a ground-aware 3D world coordinate system and perform motion editing directly in the 3D space. Trajectory control is implemented by unprojecting edited 2D trajectories into 3D with focal-length calibration and coordinate transformation, followed by speed alignment and orientation adjustment; actions are supplied by a motion bank or generated via text-to-motion methods. Then, based on modern text-to-video diffusion transformer models, we inject the subject as tokens for full attention, concatenate the background along the channel dimension, and add motion (trajectory and action) control signals by addition. Such a design opens up the possibility for us to generate realistic videos of anyone doing anything anywhere. Extensive experiments on benchmark datasets and real-world cases demonstrate that our method achieves state-of-the-art performance on both element-wise controllability and overall video quality.
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Robust Bayes-Assisted Conformal Prediction
Kianoosh Ashouritaklimi, Stefano Cortinovis, Francois Caron
Bayes--assisted conformal prediction combines the strengths of Bayesian modelling with exact, distribution--free frequentist coverage guarantees. Although conformal validity is preserved even when the Bayesian working model (BWM) is misspecified, the size of the resulting prediction sets can degrade substantially when the prior is poorly aligned with the observed data. We address this limitation by introducing \textbf{RoBAS} (\textbf{Ro}bust \textbf{B}ayes-\textbf{A}ssisted \textbf{S}hrinkage): a Bayes--assisted framework for constructing robust nonconformity scores, with two instantiations: one induced by a heavy--tailed BWM, and a closed--form empirical Bayes shrinkage score. The resulting scores adapt to the quality of the working information encoded in the prior: when this information is reliable, they exploit it to produce efficient prediction sets; when it is weak or inaccurate, they revert to the Distance--To--Average (DTA) score, a robust non--informative baseline. We evaluate the proposed scores on tabular and image regression tasks where the training distribution may differ from the calibration and test distributions, while the calibration and test data themselves remain exchangeable. We find that they are competitive with widely used scores in the absence of such shift, while substantially reducing interval widths in shifted settings.
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Beyond the Proxy: Trajectory-Distilled Guidance for Offline GFlowNet Training
Ruishuo Chen, Xun Wang, Rui Hu, Zhuoran Li, Longbo Huang
Generative Flow Networks (GFlowNets) excel at sampling diverse, high-reward objects. In many practical applications where active reward queries are infeasible, these models must be trained using static offline datasets. Prevailing training methods typically rely on a proxy model to provide reward feedback for online sampled trajectories. However, constructing a reliable proxy is often challenging due to data scarcity or high evaluation costs. While existing proxy-free approaches attempt to address this, they often impose coarse constraints that limit the model's ability to explore effectively. To overcome these limitations, we propose **Trajectory-Distilled GFlowNet (TD-GFN)**, a novel proxy-free training framework. TD-GFN utilizes inverse reinforcement learning (IRL) to extract dense, transition-level edge rewards from offline trajectories, providing rich structural guidance for efficient exploration. Crucially, to ensure robustness, these rewards guide the policy indirectly through DAG pruning and prioritized backward sampling. This design ensures that gradient updates rely exclusively on ground-truth terminal rewards from the dataset, thereby preventing error propagation. Empiri

cal results demonstrate that TD-GFN significantly outperforms a broad range of existing baselines in both convergence speed and sample quality, establishing a more robust and efficient paradigm for offline GFlowNet training.
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MET-Bench: Multimodal Entity Tracking for Evaluating the Limitations of Vision-Language and Reasoning Models
Vanya Cohen, Ray Mooney
Entity state tracking is a necessary component of world modeling that requires maintaining coherent representations of entities over time. Previous work has benchmarked entity tracking performance in purely text-based tasks. We introduce MET-Bench, a multimodal entity tracking benchmark designed to evaluate the ability of vision-language models to track entity states across modalities. Using three domains, we assess how effectively current models integrate textual and image-based state updates. Our findings reveal a significant performance gap between text-based and image-based entity tracking. We empirically show this discrepancy primarily stems from deficits in visual reasoning rather than perception. We further show that explicit text-based reasoning strategies improve performance, yet limitations remain, especially in long-horizon multimodal tasks. We apply reinforcement learning to improve entity tracking in open-source VLMs. This yields substantial in-modality gains, but does not transfer robustly across input modalities. Our results highlight the need for improved multimodal representations and reasoning techniques to bridge the gap between textual and visual entity tracking.
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RSTR: Reducing SpatioTemporal Redundancy in Diffusion Transformers
Ruitong Sun, Tianze Yang, Wei Niu, Jin Sun
Diffusion Transformers (DiTs) have achieved remarkable success in image generation, yet their deployment is hindered by high computational costs. We identify two sources of redundancy. First, $\textbf{temporal redundancy}$: Classifier-Free Guidance (CFG) applies costly dual forward passes at every timestep, yet guidance matters only at specific steps, and variable scales at critical steps can compensate for skipping others. Second, $\textbf{spatial redundancy}$: under variable guidance, different transformer blocks exhibit heterogeneous sensitivity, yet uniform calibration across all blocks wastes computation while failing to address their varying requirements. We present RSTR, the first framework to jointly reduce spatiotemporal redundancy in diffusion transformers. Stage-1 addresses temporal redundancy through evolutionary search, discovering sparse guidance schedules with variable scales. Stage-2 addresses spatial redundancy through adaptive rank allocation, assigning calibration capacities to transformer regions based on their sensitivity. Experiments on DiT-XL/2, PixArt-α, FLUX, and state-of-the-art Qwen-Image demonstrate 50\%-70\% compute savings while maintaining or improving quality. On DiT-XL/2, RSTR achieves 57\% savings with 15\% FID improvement; on Qwen-Image, 3.43$\times$ speedup with preserved quality.
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Sobolev Regularized Score Difference Estimation in Diffusion Models
Chenghan Xie, Jose Blanchet, Renyuan Xu
Estimating the difference of two Stein's score functions is a fundamental problem in generative modeling. In particular, score differences arise naturally in transfer learning, where the score difference provides the mechanism for adapting a pre-trained model to a new target distribution, and in diffusion model-based post-training methods such as discriminator guidance. Existing estimators for score differences in these settings either lack of statistical consistency or are difficult to scale up in high-dimensions.

We propose a statistically consistent and scalable estimator for score differences based on Sobolev regularization, which plays a crucial role in ensuring consistency and stablizing the training in the small-sample regime. Mathematically, we establish a convergence rate of $\tilde{\mathcal{O}}(n^{-\frac{s-1}{d+2s-2}})$ where d is the dimension and s denotes the smoothness of the underlying densities, and provide a minimax lower bound of $\tilde{\Omega}(n^{-\frac{2(s-1)}{

d+2s}})$ (in mean-squared error). Empirically, our estimator exhibits significantly improved stability in small-sample regimes compared to existing methods. We demonstrate its effectiveness on real-world tasks, including transfer learning for ECG signal generation, where it substantially outperforms non-regularized score difference estimators in downstream classification performance.

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NOMAD: Lifelong Trajectory Planning via Non-Parametric Bayesian Memory-Adaptive Diffusion Experts

Yixian Chen, Rufan Bai, Jiangbin Zheng, Yimin Wang, Tiantian CHEN, Wei Wang, Yuhuan Lu

Autonomous vehicles operating in open-world environments must continually adapt to rare long-tail scenarios while preserving previously acquired driving skills. However, existing trajectory planning approaches struggle with this stability--plasticity trade-off, as they rely on static models or rigid rule-based controllers that cannot robustly handle evolving and complex traffic dynamics. Against this background, we propose **NOMAD**, a lifelong trajectory planning framework that integrates non-parametric Bayesian memory with diffusion-based trajectory generation, enabling continuous adaptation to long-tail scenarios without catastrophic forgetting. Our method maps growing scene contexts to a dynamically growing set of discrete memory clusters, which guide a conditional diffusion model to function as a mixture of experts specialized for diverse driving behaviors. To retain past knowledge during incremental learning, we introduce a generative replay mechanism that synthesizes pseudo-experiences from previously learned memory clusters. Extensive closed-loop evaluations on the nuPlan benchmark demonstrate that our approach achieves state-of-the-art performance on long-tail scenarios, improving the interPlan score by **9.4%** over the strongest baseline, while maintaining competitive performance on regular driving benchmarks. Moreover, our method exhibits robust continual learning capability, achieving the highest average closed-loop score with positive backward transfer when adapting to sequentially introduced long-tail scenarios.

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On the Collapse of Generative Paths: A Criterion and Correction for Diffusion Steering

Ziseok Lee, Minyeong Hwang, Wooyeol Lee, Sanghyun Jo, Jihyung Ko, Young Bin Park, Jae-Mun Choi, Eunho Yang, Kyungsu Kim

Inference-time steering adapts pretrained diffusion and flow models to new tasks without retraining, often utilizing ratio-of-densities constructions that reweight time-indexed marginals with fixed exponents. We identify Marginal Path Collapse, a failure mode in which the intermediate density defined by such compositions becomes non-normalizable despite valid endpoints. This collapse can arise when composing heterogeneous experts trained with mismatched noise schedules (and/or negative exponents / partial supports). To address this, we provide (i) a sharp sufficient Path Existence Criterion that characterizes when the composed intermediate densities are mathematically well-defined, and (ii) Adaptive Path Correction with Exponents (ACE), which generalizes Feynman–Kac steering to support time-varying exponents. Our analysis reveals that ACE controls the quantile radius of the intermediate distributions, providing a theoretical mechanism for path stabilization observed in experiments. On flexible-pose scaffold decoration, a drug design task composed of de-novo, conformer, and protein-conditioned experts, ACE prevents collapse and significantly outperforms constant-exponent baselines. Furthermore, ACE improves attribute success rates in compositional image generation, establishing it as a general framework for compositional sampling.

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Latent-Guided Cooperative Energy-Based Models

Cong Geng, Xue Han, Ye Yuan, Qiang Hu, Xin Huang, Ruiqiao Bai, Junlan Feng

Energy-based models (EBMs) provide a flexible framework for generative models with strong distribution modeling capabilities. Nevertheless, their broader adoption has been limited by the difficulty of stable and efficient training.

In this paper, we propose a unified and efficient latent-guided cooperative EBM that leverages informative target latent variables to guide the joint energy in

capturing both data distribution and semantic structure, along with a cooperativ
e generator designed for effective MCMC initialization. Our joint space optimiza
tion only requires MCMC sampling in the data space, and allows the energy to lea
rn semantic data–latent relationships directly from real data. Experiments show
 our method improves generation quality and training stability with fewer
resources, and performs effectively across multiple downstream tasks.
****

ReNF: Rethinking the Design of Neural Long-Term Time Series Forecasters
Yihang Lu, Xianwei Meng, Enhong Chen
Neural Forecasters (NFs) have become a cornerstone of Long-term Time Series Fore
casting (LTSF). However, recent progress has been hampered by an overemphasis on
 architectural complexity at the expense of fundamental forecasting structures.
In this work, we revisit principled designs of LTSF. We begin by formulating a V
ariance Reduction Hypothesis (VRH), positing that generating and combining multi
ple forecasts is essential to reducing the inherent uncertainty of NFs. Guided b
y this, we propose Boosted Direct Output (BDO), a streamlined paradigm that syne
rgistically hybridizes the causal structure of Auto-Regressive (AR) with the sta
bility of Direct Output (DO), while implicitly realizing the principle of foreca
st combination within a single network. Furthermore, we mitigate a critical vali
dation-test generalization gap by employing parameter smoothing to stabilize opt
imization. Extensive experiments demonstrate that these trivial yet principled i
mprovements enable a direct temporal MLP to outperform recent, complex state-of-
the-art models in nearly all benchmarks, without relying on intricate inductive
biases. Finally, we empirically verify our hypothesis, establishing a dynamic pe
rformance bound that highlights promising directions for future research. The co
de is publicly available at: \url{https://github.com/Luoauoa/ReNF}.
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Token-Sparse Medical Multimodal Reasoning via Dual-Stream Reinforcement Learning
Kaitao Chen, Weiqian Zhao, Jiamin Wu, Qihao Zheng, Shangquan Sun, Chunfeng Song,
 Xiaosong Wang, Mu Zhou, Mianxin Liu
Vision-language models (VLMs) combining reinforcement learning (RL) ignite remar
kable progress in multimodal reasoning, yet still struggle with medical images,
which typically exhibit extremely sparse visual evidence to inform clinical deci
sion-making. We recognize that pruning visual tokens outside the grounding regio
n greatly enhances medical reasoning. However, a united RL framework for active
visual token pruning (VTP) and medical multimodal reasoning remains unestablishe
d. Here, we propose a dual-stream RL framework, **ViToS**, to fulfill token prun
ing and question answering. ViToS trains one policy model with two task branches
, where one focuses on grounding while the other conducts token-sparse reasoning
 after VTP. Furthermore, we solve the coupled policy learning problem by introdu
cing the cross-feedback sequential optimization, avoiding gradient conflict and
facilitating convergence of the shared policy model. Evaluated on seven medical
benchmarks, our method reduces visual tokens to 77% of the original sequence len
gth while achieving a 108.27% relative performance on Lingshu-7B and 104.16% rel
ative performance on HuatuoGPT-Vision-7B. Overall, ViToS delivers superior perfo
rmance and inference speedup, establishing an efficient paradigm for medical mul
timodal reasoning.
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Shuffling-Aware Optimization for Private Vector Mean Estimation
Shun Takagi, Seng Pei Liew
We study d-dimensional unbiased mean estimation in the single-message shuffle
model, where each user sends a single privatized message and the analyzer only o
bserves the shuffled multiset of reports.
While optimal mechanisms are well understood in the local differential privacy s
etting, the corresponding notion of optimality after shuffling has remained larg
ely unexplored.
To address this gap, we introduce the recently proposed shuffle index and use it
 to formulate the post-shuffling mechanism design problem as an explicit optimiz
ation problem.
We then establish a lower bound on the achievable mean squared error in terms of

the shuffle index, which implies that mechanisms that are optimal under LDP can become suboptimal once shuffling is applied.
Finally, we construct an asymptotically optimal mechanism in the high privacy regime, which as a consequence achieves a privacy-utility trade-off nearly identical to that of the central Gaussian mechanism.
****
Scaling Small Agents Through Strategy Auctions
Lisa Alazraki, William F. Shen, Yoram Bachrach, Akhil Mathur
Small language models are viewed as a promising, cost-effective approach to agentic AI, yet how their performance scales with task complexity remains unclear. While smaller agents match larger ones on simple tasks, it is unknown when large models become necessary and how to better leverage small agents. In this work, we show that small agents fail to scale with task complexity on deep search and coding tasks, and introduce *Strategy Auctions for Workload Efficiency* (*SALE*), a framework inspired by freelancer marketplaces. In SALE, agents bid with strategic plans scored by a cost–value mechanism and refined via shared auction memory, enabling per-task routing and continual self-improvement without training a router. On average, SALE reduces reliance on the largest agent by 52%, lowers overall cost by 35%, and consistently improves pass@1 with only a negligible token overhead. In contrast, established routers either underperform the largest agent or fail to reduce cost. These results suggest that small agents can be effectively “scaled up” through coordinated allocation and test-time self-improvement. More broadly, they motivate a systems-level view of agentic AI in which gains come less from ever-larger individual models and more from market-inspired coordination mechanisms that organize heterogeneous agents into efficient, adaptive ecosystems.
****
Certain Head, Uncertain Tail: Expert-Sample for Test-Time Scaling in Fine-Grained MoE
Yuanteng Chen, Peisong Wang, Nanxin Zeng, Yuantian Shao, Shuang Qiu, Gang Li, Jing Liu, Jian Cheng
Test-time scaling improves LLM performance by generating multiple candidate solutions, yet token-level sampling requires temperature tuning that trades off diversity against stability. Fine-grained MoE, featuring hundreds of well-trained experts per layer and multi-expert activation per token, offers an unexplored alternative through its rich routing space. We empirically characterize fine-grained MoE routing and uncover an informative pattern: router scores exhibit a certain head of high-confidence experts followed by an uncertain tail of low-confidence candidates. While single-run greedy accuracy remains stable when fewer experts are activated, multi-sample pass@n degrades significantly—suggesting that the certain head governs core reasoning capability while the uncertain tail correlates with reasoning diversity. Motivated by these findings, we propose Expert-Sample, a training-free method that preserves high-confidence selections while injecting controlled stochasticity into the uncertain tail, enabling diverse generation without destabilizing outputs. Evaluated on multiple fine-grained MoE models across math, knowledge reasoning, and code tasks, Expert-Sample consistently improves pass@n and verification-based accuracy. On Qwen3-30B-A3B-Instruct evaluated on GPQA-Diamond with 32 parallel samples, pass@32 rises from 85.4% to 91.9%, and accuracy improves from 59.1% to 62.6% with Best-of-N verification.
****
Direct 3D-Aware Object Insertion via Decomposed Visual Proxies
Jingbo Gong, Yikai Wang, Yushi Lan, Yuhao Wan, Ziheng Ouyang, Rui Zhao, Ming-Ming Cheng, Qibin Hou, Chen Change Loy
Object insertion aims to seamlessly composite a reference object into a specified region of a background image. Recent diffusion-based methods achieve high visual quality but formulate insertion as a simple 2D inpainting task, providing no explicit control over the object's 3D pose and limiting their practical applicability. We propose **DIRECT** (**D**ecomposed **I**njection for **RE**ference **C**omposition and **T**arget-integration), a novel framework that integrates interactive pose manipulation with high-fidelity 2D image synthesis to enable pose-c

ontrollable object insertion. Our method decomposes the insertion conditions int
o three complementary components: appearance guidance capturing visual details f
rom the reference object, geometry guidance derived from the user-adjusted 3D pr
oxy, and context guidance from the target background. By injecting them through
separate pathways, DIRECT avoids feature entanglement and simultaneously preserv
es reference appearance, follows the user-specified pose, and adapts the object
to the target scene. We also introduce an automated data construction pipeline t
o improve the diversity and quality of training data. Experiments show that DIRE
CT outperforms previous methods in both geometric controllability and visual qua
lity.
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Prototype-Based Test-Time Adaptation of Vision-Language Models
Zhaohong Huang, Yuxin Zhang, Wenjing Liu, Fei Chao, Rongrong Ji
Test-time adaptation (TTA) has emerged as a promising paradigm for vision–langua
ge models (VLMs) to bridge the distribution gap between pre-training and test da
ta. Recent works have focused on backpropagation-free TTA methods that rely on c
ache-based designs, but these introduce two key limitations. First, inference la
tency increases as the cache grows with the number of classes, leading to ineffi
ciencies in large-scale settings. Second, suboptimal performance occurs when the
 cache contains insufficient or incorrect samples. In this paper, we present Pro
totype-Based Test-Time Adaptation (PTA), an efficient and effective TTA paradigm
 that uses a set of class-specific knowledge prototypes to accumulate knowledge
from test samples. Particularly, knowledge prototypes are adaptively weighted ba
sed on the zero-shot class confidence of each test sample, incorporating the sam
ple's visual features into the corresponding class-specific prototype. It is wor
th highlighting that the knowledge from past test samples is integrated and util
ized solely in the prototypes, eliminating the overhead of cache population and
retrieval that hinders the efficiency of existing TTA methods. This endows PTA w
ith extremely high efficiency while achieving state-of-the-art performance on 15
 image recognition benchmarks and 4 robust point cloud analysis benchmarks. For
example, PTA improves CLIP’s accuracy from 65.64\% to 69.38\% on 10 cross-domain
 benchmarks, while retaining 92\% of CLIP’s inference speed on large-scale Image
Net-1K. In contrast, the cache-based TDA achieves a lower accuracy of 67.97\% an
d operates at only 50\% of CLIP’s inference speed.
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Last-Iterate Convergence of Regularized Gradient Methods for Stochastic Monotone
 Variational Inequalities
Shinji Ito, Taira Tsuchiya, Kaito Ariu, Kenshi Abe
We study last-iterate convergence for stochastic smooth and monotone variational
 inequalities (VIs), a framework that captures convex-concave saddle points and
Nash equilibrium computation in monotone games with noisy payoff feedback.
In contrast to the well-understood average-iterate guarantees, anytime last-iter
ate guarantees in stochastic settings remain limited, despite their relevance fo
r uncoupled learning dynamics that output a single current strategy.
We analyze two single-call regularized methods, the \emph{regularized gradient (
RG)} and the \emph{regularized optimistic gradient (ROG)} methods, and establish
 anytime last-iterate convergence rates in terms of the squared gap function.
For monotone VIs, RG attains $O(t^{-2/5})$ while ROG achieves the variance-adapt
ive rate $O(\sigma^{4/5} t^{-2/5} + t^{-1})$, where σ^2 is the noise vari
ance.
For λ-strongly monotone VIs, ROG yields $O(\sigma^2 / (\lambda^2 t) + t^
{-c})$ for any constant $c \ge 2$.
These results give anytime last-iterate guarantees without knowing the horizon a
nd show that optimism improves convergence in the low-noise regime.
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The Appeal and Reality of Recycling LoRAs with Adaptive Merging
Haokun Liu, Gyung Hyun Je, Marco Ciccone, Zhenlin Xu, Prasanth YSS, Colin Raffel
The widespread availability of fine-tuned LoRA modules for open pre-trained mode
ls has led to an interest in methods that can adaptively merge LoRAs to improve
performance.

These methods typically include some way of selecting LoRAs from a pool and tune merging coefficients based on a task-specific dataset.

While adaptive merging methods have demonstrated improvements in some settings, no past work has attempted to recycle LoRAs found ``in the wild'' on model repositories like the Hugging Face Hub.

To address this gap, we consider recycling from a pool of nearly 1,000 user-contributed LoRAs trained from the Llama 3.1 8B-Instruct language model.

Our empirical study includes a range of adaptive and non-adaptive merging methods in addition to a new method designed via a wide search over the methodological design space.

We demonstrate that adaptive merging methods can improve performance over the base model but provide limited benefit over training a new LoRA on the same data used to set merging coefficients.

We additionally find not only that the specific choice of LoRAs to merge has little importance, but that using LoRAs with randomly initialized parameter values yields similar performance.

To better understand why past work has proven successful, we confirm that positive transfer is indeed possible when there are highly relevant LoRAs in the pool.

We release the model checkpoints and code online at https://github.com/r-three/realistic-adaptive-merging.

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Test-Time Detoxification without Training or Learning Anything

Baturay Saglam, Dionysis Kalogerias

Large language models can produce toxic or inappropriate text even for benign inputs, creating risks when deployed at scale. Detoxification is therefore important for safety and user trust, particularly when we want to reduce harmful content without sacrificing the model’s generation quality. Many existing approaches rely on model retraining, gradients, or learned auxiliary components, which can be costly and may not transfer across model families or to truly black-box settings. We introduce a test-time procedure that approximates the gradient of completion toxicity with respect to the input embeddings and uses a small number of descent steps to steer generation toward less toxic continuations. This is achieved with zeroth-order optimization that requires only access to input embeddings, a toxicity scoring function, and forward evaluations of the model. Empirically, the approach delivers robust toxicity reductions across models and prompts and, in most settings, achieves the best overall toxicity–quality trade-off. More broadly, our work positions word embeddings as effective control variables and encourages wider use of black-box optimization to guide autoregressive language models toward scalable, safer text generation, without requiring any training or access to intermediate computations.

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Multi-scale Explainer for Graph Neural Networks

Lutong Wu, Shiying Cheng, Zhiqiang Wang, Jianqing Liang, Peng Song, Xizhao Luo, Jiye Liang

Explainability for graph neural networks (GNNs) aims to unveil the complex decision logic of learned models by identifying the most influential structures in the input graph, thereby improving transparency and trustworthiness. Existing post-hoc explainers typically extract a sparse key subgraph at a single scale as the explanation. However, a single-scale view often fails to capture multi-level semantics, and the optimization procedure may degenerate into a local search that is sensitive to initialization and noise, leading to unstable explanations and compromising their reliability. To address these issues, we propose MSExplainer, a multi-scale explainer for GNNs. MSExplainer couples multi-scale subgraph consistency guidance with single-scale adaptive subgraph learning under a parameter-sharing design. It simultaneously extracts multi-scale key subgraphs and complementary subgraphs, yielding a hierarchical decomposition of the original graph that covers semantics at different granularities and improves the stability of subgraph extraction. Experiments on six benchmark datasets show that MSExplainer generally outperforms prior methods in explanation accuracy and fidelity. Moreover,

we theoretically prove the upper bound advantage of the multi-scale strategy in representation consistency, and derive that it achieves the same-order computational complexity as single-scale methods under the parameter-sharing mechanism, thus ensuring the high fidelity of key subgraphs while maintaining computational efficiency.

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PosterAgent: Agentic Poster Generation via Stage-Aware Reinforcement Learning

Zhuocheng Yu, Feng Zhang, Sujian Li, Kai Jia

Poster generation is a complex task demanding a harmonious integration of visual aesthetics and information hierarchy. While recent text-to-image models have advanced visual synthesis, they remain non-editable and struggle with precise text rendering. Conversely, existing layout-generation methods offer structure but typically rely on static, one-shot predictions, lacking the mechanism for self-correction essential to professional design. Inspired by the iterative workflow of human designers, we introduce PosterAgent, a novel framework that reformulates poster creation as an agentic workflow involving initial drafting followed by iterative refinement. To effectively train this multi-turn capability, we propose Stage-Aware Reinforcement Learning (SARL), which decouples the optimization into draft-specific and refinement-specific phases, ensuring precise credit assignment for both initial drafting and incremental refinement actions.

Extensive experiments demonstrate that PosterAgent significantly outperforms strong baselines, validating the potential of agentic systems in graphic design.

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S2M-Net: Spectral-Spatial Mixing with Morphology-Aware Adaptive Loss for Medical Image Segmentation

Sanaullah Chowdhury, Lameya Sabrin

Medical image segmentation requires balancing global context with computational efficiency,
where self-attention mechanisms suffer from quadratic $\mathcal{O}((HW)^2 C)$ complexity.
We propose S2M-Net, a parameter-efficient architecture (4.7M parameters) that achieves
computational savings through Spectral--Spatial Token Mixing (SSTM).
SSTM achieves $\mathcal{O}(HWC^2)$ complexity through efficient combination of
$\mathcal{O}(HWC \log(HW))$ frequency-domain processing and $\mathcal{O}(HWCd)$ bottlenecked
spatial gating ($d{=}16$), exploiting spectral concentration where $>93\%$ of energy
is captured by $K{=}32$ low-frequency components ($\sim$0.8\% of the spectrum at

$352{\times}352$ resolution). This design avoids self-attention's prohibitive
$\mathcal{O}((HW)^2C)$ attention map computations while preserving global receptive fields.
To handle geometric diversity, we introduce Morphology-Aware Adaptive Segmentation Loss (MASL),
which automatically modulates five loss objectives based on per-sample morphological
descriptors (tubularity, compactness, irregularity, and scale).
Evaluation across 15 datasets spanning 8 modalities demonstrates competitive performance,
obtaining the best performance on 14 of 15 datasets,
with statistically significant improvements
($p < 0.0033$, Bonferroni-corrected) on 7 challenging tasks
(complex morphology, class imbalance, and multi-class segmentation),
and clinically meaningful gains (0.5--1.6% Dice) on 8 mature benchmarks.
Notably, S2M-Net achieves 83.43% Dice on EndoVis17 multiclass instrument segmentation
($+8.69\%$ over TransUNet and $+9.14\%$ over the best baseline UMamba at 74.29%),
while using $12.8{\times}$ fewer parameters (4.7M vs.\ 60M).

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GASS: Geometry-Aware Spherical Sampling for Disentangled Diversity Enhancement in Text-to-Image Generation
Ye Zhu, Kaleb Newman, Johannes F. Lutzeyer, Adriana Romero-Soriano, Michal Drozdzal, Olga Russakovsky
Despite high semantic alignment, modern text-to-image (T2I) generative models still struggle to synthesize diverse images from a given prompt. In this work, we enhance the T2I diversity through a geometric lens. Unlike most existing methods that rely primarily on entropy-based guidance to increase sample dissimilarity, we introduce Geometry-Aware Spherical Sampling (GASS) to enhance diversity by explicitly controlling both prompt-dependent and prompt-independent sources of variation. Specifically, we decompose the diversity measure in CLIP embeddings using two orthogonal directions: the text embedding, which captures semantic variation related to the prompt, and an identified orthogonal direction that captures prompt-independent variation (e.g., backgrounds). Based on this decomposition, GASS increases the geometric projection spread of generated image embeddings along both axes and guides the T2I sampling process via expanded predictions along the generation trajectory. Our experiments on different frozen T2I backbones (U-Net and DiT, diffusion and flow) and benchmarks demonstrate the effectiveness of disentangled diversity enhancement with minimal impact on image fidelity and semantic alignment.
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VidLaDA: Bidirectional Diffusion Large Language Models for Efficient Video Understanding
Zhihao He, Tieyuan Chen, Kangyu Wang, Ziran Qin, Yang Shao, Chaofan Gan, Shijie Li, Zuxuan Wu, Weiyao Lin
Current Video Large Language Models (Video LLMs) typically encode frames via a vision encoder and employ an autoregressive (AR) LLM for understanding and generation.
However, this AR paradigm inevitably faces a dual efficiency bottleneck: strictly unidirectional attention compromises *understanding efficiency* by hindering global spatiotemporal aggregation, while serial decoding restricts *generation efficiency*. To address this, we propose **VidLaDA**, a Video LLM based on Diffusion Language Models (DLMs) that leverages bidirectional attention to unlock comprehensive spatiotemporal modeling and decode tokens in parallel. To further mitigate the computational overhead of diffusion decoding, we introduce **MARS-Cache**, an acceleration strategy that prunes redundancy by combining asynchronous visual cache refreshing with frame-wise chunk attention. Experiments show VidLaDA rivals state-of-the-art AR baselines (e.g., Qwen2.5-VL and LLaVA-Video) and outperforms DLM baselines, with MARS-Cache delivering over 12x speedup without compromising accuracy.
Code and checkpoints are open-sourced at https://github.com/ziHoHe/VidLaDA.
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MindZero: Learning Online Mental Reasoning With Zero Annotations
Shunchi Zhang, Jin Lu, Chuanyang Jin, Yichao Zhou, Zhining Zhang, Tianmin Shu
Effective real-world assistance requires AI agents with robust Theory of Mind (ToM): inferring human mental states from their behavior. Despite recent advances, several key challenges remain, including (1) online inference with robust uncertainty updates over multiple hypotheses; (2) efficient reasoning suitable for real-time assistance; and (3) the lack of ground-truth mental state annotations in real-world domains. We address these challenges by introducing *MindZero*, a self-supervised reinforcement learning framework that trains multimodal large language models (MLLMs) for efficient and robust online mental reasoning. During training, the model is rewarded for generating mental state hypotheses that maximize the likelihood of observed actions estimated by a planner, similar to model-based ToM reasoning. This method thus eliminates the need for explicit mental state annotations. After training, *MindZero* internalizes model-based reasoning into fast single-pass inference. We evaluate *MindZero* against baselines across challenging mental reasoning and AI assistance tasks in gridworld and household domains. We found that LLMs alone are insufficient; model-based methods improve ac

curacy but are slow, costly, and limited by backbone MLLM capacity. In contrast, *MindZero* enhances MLLMs' intrinsic ToM ability and significantly outperforms model-based methods in both accuracy and efficiency, showing that mental reasoning can be effectively learned as a self-supervised skill.
****
Doubly Regularized Markov Decision Processes for Robust Reinforcement Learning
Yiting He, Zhishuai Liu, Pan Xu
Empirical successes show that regularization improves the stability and efficiency of reinforcement learning (RL), with applications in robotics and post-training of large language models. Yet, theoretical analyses of regularized Markov decision processes (MDPs) have mostly been confined to the standard RL setting. In this work, we investigate regularized MDPs through the lens of robust RL. We introduce a doubly regularized MDP framework that combines policy and dynamics regularizations, enabling robust policy learning while naturally accommodating continuous action spaces. Within this framework, we develop an optimism-based online algorithm and provide the first finite-sample regret guarantees in both tabular and linear settings. Our results show that algorithms for doubly regularized MDPs are as sample-efficient as well-studied robust MDP algorithms, while additionally benefiting from the flexibility of soft policies. We further design practical algorithmic variants for both settings and demonstrate empirically that our approach efficiently and effectively handles function approximation and exploration in large state-action spaces, achieving robust performances.
****
State Space Model with Continuous Limit of HiPPO Matrix: Eigenvalue Analysis and Explicit Solution Formula
Atsushi Takabatake, Takaharu Yaguchi
As a lightweight model for sequence processing, an LSSL that uses the HiPPO matrix has been proposed.
In this paper, as the continuous limit of the HiPPO Matrix,
we propose Continuized-HiPPO Operator.
Furthermore, as examples of advantages obtained by using this operator,
we show that one can analyze the asymptotic behavior of eigenvalues of a matrix related to the HiPPO matrix, and
that the LSSL induced by the continuous limit Continuized-HiPPO Operator admits an explicit solution.
****
EnerGS: Energy-Based Gaussian Splatting under Partial Geometric Priors
Rui Song, Tianhui Cai, Markus Gross, Yun Zhang, Walter Zimmer, Zhiyu Huang, Olaf Wysocki, Jiaqi Ma
3D Gaussian Splatting (3DGS) has been widely adopted for scene reconstruction, where training inherently constitutes a highly coupled and non-convex optimization problem. Recent works commonly incorporate geometric priors, such as LiDAR measurements, either for initialization or as training constraints, with the goal of improving photometric reconstruction quality. However, in large-scale outdoor scenarios, such geometric supervision is often spatially incomplete and uneven, which limits its effectiveness as a reliable prior and can even be detrimental to the final reconstruction. To address this challenge, we model partially observable geometry as a continuous energy field induced by geometric evidence and propose EnerGS. Rather than enforcing geometry as a hard constraint, EnerGS provides a soft geometric guidance for the optimization of Gaussian primitives, allowing geometric information to steer the optimization process without directly restricting the solution space. Extensive experiments on large-scale outdoor scenes demonstrate that, under both sparse multi-view and monocular settings, EnerGS consistently improves photometric quality and geometric stability, while effectively mitigating overfitting during 3DGS training. The codebase is publicly available at: https://github.com/ucla-mobility/EnerGS.
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Privileged Information Distillation for Language Models
Emiliano Penaloza, Dheeraj Vattikonda, Nicolas Gontier, Alexandre Lacoste, Laurent Charlin, Massimo Caccia

Training-time privileged information (PI) can enable language models to succeed on tasks they would otherwise fail, making it a powerful tool for reinforcement learning in hard, long-horizon settings. However, transferring capabilities learned with PI to policies that must act without it at inference time remains a fundamental challenge. We study this problem in the context of distilling frontier models for multi-turn agentic environments, where closed-source systems typically hide their internal reasoning and expose only action trajectories. This breaks standard distillation pipelines, since successful behavior is observable but the reasoning process is not. We introduce π-Distill, a joint teacher–student framework that trains a PI-conditioned teacher and an unconditioned student simultaneously within a single shared-parameter model, enabling the teacher to learn how to use PI while mitigating distribution shift during transfer. We show that π-Distill effectively distills frontier agents using action-only privileged information, matching or outperforming industry-standard pipelines that assume access to full Chain-of-Thought supervision across multiple agentic benchmarks, models, and forms of PI. We complement our results with extensive analysis that characterize what factors enable effective learning with PI.

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Signal Strength Estimation in Logistic Regression Using Data Splitting

Weihao Li, Jun S. Liu

Logistic regression is widely used in applications; however, when the dimension scales with the sample size, theory reveals that the asymptotic behavior of common M-estimators depends on nonzero bias and variance factors, which are functions of the signal strength. To leverage the theory to design valid statistical methodologies, it is essential to obtain accurate estimates of the signal strength. In this work, we utilize a data-splitting strategy to efficiently estimate the signal strength. To alleviate issues caused by separable data, we analyze the exact asymptotics of an M-estimator with a data-driven, non-decomposable regularizer that adapts to the true covariance structure. We justify the validity of our method through both theoretical analysis and numerical experiments.

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Speculative Coupled Decoding for Training-Free Lossless Acceleration of Autoregressive Visual Generation

Junhyuk So, Hyunho Kook, Chaeyeon Jang, Eunhyeok Park

Autoregressive (AR) modeling has recently emerged as a promising new paradigm in visual generation, but its practical adoption is severely constrained by the slow inference speed of per-token generation, which often requires thousands of steps to produce a single sample. While several Speculative Decoding (SD)-based methods have been proposed to solve this problem by generating multiple tokens in a single forward step, they suffer from limited speedup, degraded quality, or require the training of a draft model. To solve these problems, we propose a new training-free, lossless SD framework, Speculative Coupled Decoding (SCD), by extending the recently proposed Speculative Jacobi Decoding (SJD). While SJD shows strong potential for accelerating AR generation by combining Jacobi iteration and SD, we found that its acceptance rate is still significantly limited due to the instability arising from the independent sampling process used during draft token generation. To overcome this, we introduce an information-theoretic approach, Coupling, which stabilizes the drafting trajectory of SJD by maximizing the probability of sampling identical draft tokens across consecutive iterations, significantly enhancing the acceptance rate while preserving its lossless property. Remarkably, this method requires only a single-line modification to the existing algorithm with almost zero overhead, yet achieves substantial performance gains, delivering up to a 4.2× speedup in image generation and 13.6× speedup in video generation compared to standard AR decoding, without any degradation or the need for additional training.

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The Illusion of Generalization in Tabular Language Models

Aditya Gorla, Ratish Puduppully

Tabular Language Models (TLMs) have been claimed to achieve strong generalization for tabular prediction. We conduct a systematic re-evaluation of Tabula-8B as

a representative TLM, utilizing 165 datasets from the UniPredict benchmark. Our investigation reveals three findings. First, binary and categorical classificati on achieve near-zero median lift over majority-class baselines and strong aggreg ate performance is driven entirely by quartile classification tasks. Second, top -performing datasets exhibit pervasive contamination, including complete train-t est overlap and task-level leakage that evades standard deduplication. Third, in struction-tuning without tabular exposure recovers 92.2% of standard classificat ion performance and on quartile classification, format familiarity closes 71.3% of the gap with the residual attributable to contaminated datasets. These findin gs suggest claimed generalization likely reflects evaluation artifacts rather th an learned tabular reasoning. We conclude with recommendations for strengthening TLM evaluation.

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LieStoNet: Learning Lie Symmetries from Spatiotemporal Data for Stochastic Dynam ical Systems

Shida Liu, Abhishek Gupta, Sumit Sinha, L Mahadevan

Symmetry is central to modern machine learning and physics: invariances and equi variances improve sample efficiency, robustness, and out-of-distribution general ization, while symmetry principles guide scientific modeling. Yet for stochastic dynamical systems the relevant continuous symmetries are rarely known, and symm etry discovery for SDEs has remained essentially unexplored. We introduce *LieSt oNet*, an end-to-end, template-free framework for discovering Lie-point symmetri es of SDEs directly from spatiotemporal trajectories, without prespecifying symm etry groups, templates, or canonical coordinates. Building on the seminal SDE Li e-symmetry theory of Gaeta and Quintero (1999), which formalizes Lie-point SDE s ymmetries and their relation to Fokker-Planck symmetries, LieStoNet learns neura l surrogates for drift and diffusion from increments, then learns projectable ge nerators by enforcing the SDE determining equations, separately regularizing for closure under Lie brackets, adherence to the Lie algebra axioms (bilinearity, a ntisymmetry, Jacobi), and a non-redundant independent basis. The surrogate also defines an associated Fokker-Planck equation, enabling optional discovery of its Lie-point symmetries in parallel. Across multiple canonical SDEs with known ana lytic symmetries, LieStoNet recovers generators consistent with the ground-truth symmetry algebra, providing interpretable symmetry discovery for noisy dynamics . Code is available at this link.

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From Talking to Singing: A New Challenge for Audio-Visual Deepfake Detection

Ke Liu, Jiwei Wei, Wenyu Zhang, Shuchang Zhou, Ruikun Chai, Yutao Dai, Chaoning Zhang, Yang Yang

With rapid advances in audio-visual generative models, reliable forgery detectio n becomes increasingly critical. Existing methods for audio-visual deepfake dete ction typically rely on cross-modal inconsistencies. In singing, rhythmic vocali zation weakens this coupling and introduces a nontrivial domain shift, substanti ally degrading detection performance. We construct the Singing Head DeepFake (SH DF) dataset using rhythm-aware generative models to fill the gap in singing benc hmarks. To cope with cross-scenario domain shifts, we propose a Text-guided Audi o-Visual Forgery Detection (T-AVFD) framework that generalizes across both talki ng and singing scenarios. T-AVFD comprises a facial authenticity pattern learner and a multi-modal differential weight learning module. The pattern learner alig ns facial features with multi-granularity textual descriptions to learn generali zable authenticity patterns. The weight learning module preserves intrinsic audi o-visual consistency and adaptively integrates it with authenticity patterns via differential weighting. Extensive experiments on multiple talking head deepfake datasets and SHDF show consistent improvements over existing baselines and stro ng robustness under diverse perturbations.

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SSR-Merge: Subspace Signal Routing for Training-Free LoRA Merging in Diffusion M odels

Zhengxuan Wei, Yi Dong, Zonghui Li, Xianhui Lin, Xing Liu, Hong Gu, Shaofeng Zha ng, Wenbin Li, Qi Fan

Low-Rank Adaptation (LoRA) merging can efficiently combine diverse generative capabilities from multiple trained LoRAs for a diffusion model. However, existing LoRA merging techniques often suffer from severe parameter interference, causing destructive collisions in the shared parameter space. To address this, we propose Subspace Signal Routing (SSR), which resolves interference by routing internal signals instead of performing parameter-space merge. Specifically, SSR first constructs a unified subspace by concatenating candidate LoRAs along the rank dimension. Next, SSR employs an inverse correlation matrix to decorrelate mixed signals within this space. Finally, a directional guide matrix steers these purified signals into their respective task-specific subspaces. We provide a rigorous theoretical analysis proving that SSR aligns with the Ordinary Least Squares (OLS) solution, thereby ensuring mathematical optimality. We utilize the additivity of sufficient statistics to design a streaming algorithm. This enables on-the-fly updates that significantly reduce memory overhead and computation time. Extensive experiments validate that SSR significantly outperforms state-of-the-art methods while maintaining comparable efficiency. The source code will be made publicly available.

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GFedCL: Graph-Based Federated Continual Learning with Spatial and Temporal Awareness

Qingyang Yu, Yang Hua, Qizhen Zhang, Hao Wang

Recent years have witnessed a surge of interest in federated learning. In particular, federated continual learning (FCL) emerged as an effective approach that enables clients with evolving, non-storable data to engage in collective learning. Among FCL approaches, replay-based methods excel by mitigating data storage constraints through synthetic data generation. However, existing replay-based methods overlook spatial and temporal information inherent in FCL settings, leading to suboptimal model performance. For instance, spatial variation in COVID-19 prevalence across hospitals in different states (e.g., Delta surging in Florida vs. Omicron in New York) and the temporal evolution of regional outbreaks are critical information for accurately distinguishing between COVID variants. This paper presents GFedCL to address this limitation. GFedCL is a new FCL approach that (1) constructs spatial- and temporal-aware relational graphs with attention mechanisms, and (2) uses the graphs, combined with generative adversarial learning, to generate high-quality synthetic data. GFedCL can generate synthetic data that matches the expectation of real data distribution while preserving privacy with theoretical guarantees. GFedCL consistently outperforms state-of-the-art FCL methods, gaining 27.95% improvement on TinyImageNet.

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Compression as Adaptation: Implicit Visual Representation with Diffusion Foundation Models

Zongyu Guo, Jiajun He, Zhaoyang Jia, Xiaoyi Zhang, Jiahao Li, Xiao Li, Bin Li, José Miguel Hernández-Lobato, Yan Lu

Modern visual generative models acquire rich visual knowledge through large-scale training, yet existing visual representations (such as pixels, latents, or tokens) remain external to the model and cannot directly exploit this knowledge for compact storage or reuse.

In this work, we introduce a new visual representation framework that encodes a signal as a function, which is parametrized by low-rank adaptations attached to a frozen visual generative model.

Such implicit representations of visual signals, \textit{e.g.}, an 81-frame video, can further be hashed into a single compact vector, achieving strong perceptual video compression at extremely low bitrates.

Beyond basic compression, the functional nature of this representation enables inference-time scaling and control, allowing additional refinement on the compression performance.

More broadly, as the implicit representations directly act as a function of the generation process, this suggests a unified framework bridging visual compression and generation.

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VideoKR: Towards Knowledge- and Reasoning-Intensive Video Understanding
Lin Fu, Zheyuan Yang, Yang Wang, Tingyu Song, Arman Cohan, Yilun Zhao
We introduce VideoKR, the first large-scale training corpus specifically designe d to strengthen knowledge- and reasoning-intensive video understanding. It compr ises 315K video reasoning examples over 145K newly collected, CC-licensed, exper t-domain videos. We develop a human-in-the-loop, skill-oriented example generati on pipeline that targets progressively deeper video reasoning capabilities while ensuring the difficulty, diversity, and reliability of both the examples and th eir CoT rationales. We also curate VideoKR-Eval, a new expert-annotated benchmar k where questions require genuine video understanding and knowledge-intensive re asoning rather than textual shortcuts. Our experiments show that, under a standa rd SFT$\rightarrow$GRPO pipeline, models post-trained on VideoKR outperform prio r post-training approaches on knowledge-intensive video reasoning while remainin g competitive on general video reasoning, highlighting data design as a key driv er of progress in video reasoning. We further conduct comprehensive ablations to isolate the contributions of VideoKR, providing actionable insights for future work.
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Minimizing Upper Confidence Bounds: A Data-Driven Framework for Stochastic Progr amming
Shixin Liu, Ming Gao, Jian Hu
Stochastic programming is often challenged by epistemic uncertainty, where criti cal probability distributions are poorly characterized or unknown due to a lack of data. To address this, we pioneer a novel framework for stochastic programmin g that minimizes an upper confidence bound (UCB) on the expected random cost, ac ting as a robustness-seeking strategy. Our central contribution is the Average P ercentile Upper Bound (APUB), a new statistical construct that serves as both a statistically rigorous upper bound for population means and an approximate risk metric for sample means. We rigorously prove the asymptotic correctness and cons istency of APUB, establishing a reliable foundation for data-driven decision-mak ing. We also develop practical solution methods, including a bootstrap sampling approximation method and an L-shaped method, to solve APUB optimization problems , with a specific focus on two-stage linear stochastic optimization with random recourse. Empirical demonstrations on a two-stage product mix problem reveal the significant benefits of our APUB optimization framework, which fortifies the pr ocess against epistemic uncertainty while reinforcing key decision-making attrib utes like reliability and consistency. The implementation and source code are av ailable at https://github.com/8Wings/APUB-Optimization.
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SWE-ABS: Adversarial Benchmark Strengthening Exposes Inflated Success Rates on T est-based Benchmark
Boxi Yu, Yang Cao, Yuzhong Zhang, Liting Lin, Junjielong Xu, Zhiqing Zhong, Qing hua Xu, Guancheng Wang, Jialun Cao, Shing-Chi Cheung, Pinjia He, Lionel BRIAND
The SWE-Bench Verified leaderboard is approaching saturation, with the top syste m achieving 78.80\%.
However, we reveal that this performance is inflated: our re-evaluation demonstr ates that one in five ``solved'' patches from the top-30 agents are semantically incorrect, passing only because weak test suites fail to expose their errors.
We present SWE-ABS, an adversarial framework that strengthens test suites throug h a two-stage pipeline:
(1) coverage-driven augmentation utilizing program slicing to target untested co de regions, and (2) mutation-driven adversarial testing that synthesizes plausib le-but-incorrect patches to expose semantic blind spots.
On SWE-Bench Verified (500 instances), SWE-ABS strengthens 50.2\% of instances (a $25.1\times$ improvement over prior work) and rejects 19.78\% of previously pa ssing patches. Consequently, the top agent's score decreases from 78.80\% to 62. 20\%, causing significant leaderboard reshuffling (e.g., the top-ranked agent dr ops to 5th place).
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Localized, High-resolution Geographic Representations with Slepian Functions

Arjun Rao, Ruth Crasto, Tessa Ooms, David Rolnick, Konstantin Klemmer, Marc Rußwurm

Geographic data is fundamentally local. Disease outbreaks cluster in population centers, ecological patterns emerge along coastlines, and economic activity concentrates within country borders. Machine learning models that encode geographic location, however, distribute representational capacity uniformly across the globe, struggling at the fine-grained resolutions localized applications require. We propose a geographic location encoder built from spherical Slepian functions that concentrates representational capacity inside a region-of-interest and scales to high resolutions without extensive computational demands. For settings requiring global context, we present a hybrid Slepian-Spherical Harmonic encoder that efficiently bridges the tradeoff between local-global performance, while retaining desirable properties such as pole-safety and spherical-surface-distance preservation. Across five tasks spanning classification, regression, and image-augmented prediction, Slepian encodings outperform baselines and retain performance advantages across a wide variety of neural network architectures.

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Real-Time Visual Attribution Streaming in Thinking Model

Seil Kang, Woojung Han, Junhyeok Kim, Jinyeong Kim, Youngeun Kim, Seong Jae Hwang

We present an amortized framework for real-time visual attribution streaming in multimodal thinking models. When these models generate code from a screenshot or solve math problems from images, their long reasoning traces should be grounded in visual evidence. However, verifying this reliance is challenging: faithful causal methods require costly repeated backward passes or perturbations, while raw attention maps offer instant access, they lack causal validity. To resolve this, we introduce an amortized approach that learns to estimate the causal effects of semantic regions directly from the rich signals encoded in attention features. Across five diverse benchmarks and four thinking models, our approach achieves faithfulness comparable to exhaustive causal methods while enabling visual attribution streaming, where users observe grounding evidence as the model reasons, not after. Our results demonstrate that real-time, faithful attribution in multimodal thinking models is achievable through lightweight learning, not brute-force computation.

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Neuro-Fuzzy Concept Learning for Interpretable Large Multimodal Models

Ritik Mishra, Vanshika Gupta, M. Sajid, M. Tanveer

Large Multimodal Models (LMMs) integrate unimodal encoders with Large Language Models (LLMs) to execute complex multimodal tasks. Despite progress in the field, understanding the internal representations of these models through interpretable logic remains an open problem. To address this, we present a framework utilizing a Human-Inspired (Neuro-fuzzy) approach for learning token representations. In this method, we leverage fuzzy rules to compute activation firing strengths, which are subsequently defuzzified to extract distinct concepts. This mechanism allows for the interpretation of learned representations directly through explicit logic. Consequently, we derive "multimodal concepts" that are both semantically coherent and interpretable. We validate our approach through rigorous qualitative and quantitative experiments, demonstrating the utility of these concepts in interpreting test samples. Additionally, we evaluate the disentanglement of the learned concepts and the efficacy of their grounding in both visual and textual domains.

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Neural–Evolutionary Symbolic Regression with Global Constraints: Constraint-Aware Decoding and Reward Shaping

Xiangdong Wu, wenjun wu, Ziyu Wei, Bingrun Chen, Zhenbo Song, Rongye Shi

Symbolic regression aims to discover compact, interpretable mathematical expressions from data, but neural generation is challenging because expressions are tree-structured.

Existing neural methods often linearize expression trees into token sequences, facilitating autoregressive modeling but obscuring hierarchical relations and com

plicating structure-dependent constraint enforcement.
We propose GCN-SR, a graph-based symbolic regression framework that generates ex pressions in an explicit tree-aligned form, making structural context available during decoding.
To enable batched generation over variable-topology expressions, we introduce Sy mbolic Perfect Binary Trees (SPBTs), a fixed-topology scaffold that preserves tr ee hierarchy while supporting graph-based node-attribute prediction.
We further introduce Similarity-Weighted Policy Gradient (SWPG) to incorporate g enetic programming (GP) refinement without directly imitating GP-refined elites; instead, refined expressions construct similarity-weighted rewards for samples drawn by the current generator.
Experiments on standard symbolic regression benchmarks and ablations show that G CN-SR consistently improves exact recovery over strong neural and hybrid baselin es under matched evaluation budgets.
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Klein Hyperbolic Metric Learning
Yulin Zhang, Boxuan Hu, Huimin Li, Junlin Hu
Hyperbolic metric learning is highly effective in embedding hierarchical data st ructures. However, past work has predominantly focused on the conformal Poincaré model, leaving other geometries like the Klein model largely under-explored. In addition, the curved geodesics of the Poincaré model present a fundamental geom etric misalignment with the linear projections dominating the feature transforma tion steps in the modern neural network backbones. In this paper, we investigate the Klein model, a projective model of hyperbolic geometry whose straight-line geodesics offer a structurally aligned alternative in modern encoders, for hyper bolic metric learning. By formalizing a framework based on Einstein gyrovector o perations, we derive a numerically stable metric learning approach that mitigate s the inherent optimization challenges of the Klein model. Extensive experiments on multiple image datasets for fine-grained image classification task show that the Klein model not only serves as a viable alternative to the Poincaré model b ut also achieves highly competitive performance by leveraging its unique geometr ic properties, without increasing parameter complexity. Our empirical findings e stablish the Klein model as an efficient geometric prior for hyperbolic metric l earning.
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An Exponential Separation Between Quantum and Quantum-Inspired Classical Algorit hms for Linear Systems
Allan Grønlund, Kasper Green Larsen
Achieving a provable exponential quantum speedup for an important machine learni ng task has been a central research goal since the seminal HHL quantum algorithm for solving linear systems and the subsequent quantum recommender systems algor ithm by Kerenidis and Prakash. These algorithms were initially believed to be st rong candidates for exponential speedups, but a lower bound ruling out similar c lassical improvements remained absent. In breakthrough work by Tang, it was demo nstrated that this lack of progress in classical lower bounds was for good reaso ns. Concretely, she gave a classical counterpart of the quantum recommender syst ems algorithm, reducing the quantum advantage to a mere polynomial. Her approach is quite general and was named \emph{quantum-inspired classical} algorithms. Si nce then, almost all the initially exponential quantum machine learning speedups have been reduced to polynomial via new quantum-inspired classical algorithms. From the current state-of-affairs, it is unclear whether we can hope for exponen tial quantum speedups for any natural machine learning task. In this work, we pr esent the first such provable exponential separation between quantum and quantum -inspired classical algorithms for the basic problem of solving a linear system when the input matrix is well-conditioned and has sparse rows and columns.
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Efficient DP-SGD for LLMs with Randomized Clipping
Enayat Ullah, Sai Aparna Aketi, Devansh Gupta, Huanyu Zhang, Meisam Razaviyayn
Large language models (LLMs) are trained on vast datasets that may contain sensi tive information. Differential privacy (DP), the de facto standard for formal pr

ivacy guarantees, provides a principled framework for training LLMs with provable privacy protection.
However, state-of-the-art DP training implementations rely on *fast gradient clipping* techniques with memory overhead $O(B\min(T^2, d^2))$, where B is the batch size, T is the sequence length, and d is the layer width. This becomes prohibitive as both model width and context length grow. We propose DP-SGD-RC, a novel variant of DP-SGD with *randomized clipping* that reduces memory and compute overhead. DP-SGD-RC leverages *stochastic trace estimation* methods, specifically *Hutchinson's estimator* and its improved variant, Hutch^{++}, to reduce the memory footprint of per-sample gradient norm estimation.
We provide a tight privacy analysis showing that DP-SGD-RC achieves noise multipliers competitive with deterministic clipping.
Experiments fine-tuning Llama 3.2 1B on long-context benchmarks spanning classification, question answering, and summarization tasks demonstrate that DP-SGD-RC matches baseline utility while significantly reducing memory and compute.
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Manifold-Optimal Guidance: A Unified Riemannian Control View of Diffusion Guidance
Zexi Jia, Pengcheng Luo, Zhengyao Fang, Jinchao Zhang, Jie Zhou
Classifier-Free Guidance (CFG) serves as the de facto control mechanism for conditional diffusion, yet high guidance scales notoriously induce oversaturation, texture artifacts, and structural collapse. We attribute this failure to a geometric mismatch: standard CFG performs Euclidean extrapolation in ambient space, inadvertently driving sampling trajectories off the high-density data manifold. To resolve this, we present Manifold-Optimal Guidance (MOG), a framework that reformulates guidance as a local optimal control problem. MOG yields a closed-form, geometry-aware Riemannian update that corrects off-manifold drift without requiring retraining. Leveraging this perspective, we further introduce Auto-MOG, a dynamic energy-balancing schedule that adaptively calibrates guidance strength, effectively eliminating the need for manual hyperparameter tuning. Extensive validation demonstrates that MOG yields superior fidelity and alignment compared to baselines, with virtually no added computational overhead.
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Think-at-Hard: Selective Latent Iterations to Improve Reasoning Language Models
Tianyu Fu, Yichen You, Zekai Chen, Guohao Dai, Huazhong Yang, Yu Wang
Improving the reasoning abilities of Large Language Models (LLMs), especially under parameter constraints, is crucial for real-world applications.
Looped transformers address this by performing multiple latent iterations to refine each token beyond a single forward pass.
However, we identify a latent overthinking phenomenon: most token predictions are already correct after the first pass, but are sometimes revised into errors in later iterations.
We ask whether selectively skipping latent iterations can improve accuracy, and reveal significant potential with an oracle iteration policy that boosts performance by up to 7.3%.
Motivated by this, we propose Think-at-Hard (TaH), a looped transformer optimized for selective iteration.
TaH employs a lightweight neural decider to trigger latent iteration, only at tokens likely to be incorrect after the standard forward pass.
During latent iterations, depth-aware Low-Rank Adaptation (LoRA) modules shift the objective from general next-token prediction to focused hard-token refinement.
A duo-causal attention mechanism extends attention from the token sequence dimension to an additional iteration depth dimension, enabling cross-iteration information flow with full sequential parallelism.
Experiments on nine benchmarks show consistent gains across math, QA, and coding tasks.
With identical parameter counts, TaH outperforms always-iterate baselines by 3.8-4.4% while skipping iterations on 93% of tokens, and exceeds single-iteration Qwen3 baselines by 3.0-3.8%.

When allowing <3% more parameters from LoRA and decider, the gains further increase to 5.3-6.2% and 6.1-6.8%, respectively.
Our code is available at https://github.com/thu-nics/TaH.
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PACT: Self-Evolving Physical Safety Alignment for Diffusion Policies in Embodied Manipulation
Lingxuan Wu, Zijian Zhu, Lizhong Wang, Chengyang Ying, Huayu Chen, Xiao Yang, Fangming Liu, Jun Zhu
Diffusion policies have achieved remarkable success in robotic manipulation, yet they often fail to satisfy strict physical constraints required for safe deployment. Existing approaches impose safety either prematurely during training or reactively via external guardrails at test time, limiting policy expressivity and overall scalability. We propose Physical safety Alignment for Constrained Trajectories (PACT), a self-evolving post-training framework that projects pretrained diffusion policies onto constraint-feasible regions without accessing demonstration data or task rewards. PACT distills constraint gradients into the diffusion model through a reverse-KL objective with dense supervision across timesteps. It incorporates a curriculum that progressively tightens constraints while maintaining theoretically bounded policy shift and monotone improvement, mitigating the safety-performance trade-off from catastrophic forgetting. On simulated and real-world embodied manipulation benchmarks, PACT significantly reduces safety violations by 31.0% on average while improving task success by 30.7%.
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Enhanced Multi-Instance Partial Label Learning via Average Gradient Outer Product
Nan Cao, Xu Zhao, Teng Zhang
Multi-instance partial-label learning (MIPL) is a recently proposed dual weakly supervised learning framework where each training bag is annotated with a candidate label set containing one true label and several false positives. The key challange of MIPL problem is that the relations between the key instances and the ground-truth labels are much more covered due to the false positive labels. Existing methods usually rely on the model's own predictions and roughly aggregate instance features according to the learned attention weights, which is easily misled by false positives and lets key instances be overlooked. Here, we propose Average Gradient Outer Product based Multi-instance Partial-Label Learning~(AGOPMIPL) method, where the average gradient outer product (AGOP) is directly calculated from the bag-level features and the model prediction, then it is integrated in an attention module to amplify discriminative feature directions and thereby helps key-instance identification. Moreover, the feature prototypes and a progressive disambiguation strategy are introduced to further suppress noisy candidates. The experimental studies on four MIPL benchmarks and the real-world \textsf{CRC-MIPL} dataset are performed and AGOPMIPL consistently outperforms five state-of-the-art baselines, with up to 25.9% relative gain on \textsf{CRC-MIPL-KMeansSeg}.
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DAL: A Practical Prior-Free Black-Box Framework for Piecewise Stationary Bandits
Argyrios Gerogiannis, Yu-Han Huang, Subhonmesh Bose, Venugopal Veeravalli
We introduce a practical, black-box framework termed Detection Augmented Learning (DAL) for the problem of piecewise stationary bandits without knowledge of the underlying non-stationarity. DAL accepts any stationary bandit algorithm with order-optimal regret as input and augments it with a change detector, enabling applicability to all common bandit variants. Extensive experimentation demonstrates that DAL consistently surpasses all state-of-the-art methods across diverse non-stationary scenarios, including synthetic benchmarks and real-world datasets, underscoring its versatility and scalability. We provide theoretical insights into DAL's strong empirical performance, complemented by thorough empirical validation.
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Decoupled Low-Rank Adaptation for Robust Federated Fine-Tuning
Xiuwen Fang, Xuliang Yang, Mang Ye

Federated Learning (FL) enables collaborative training across distributed clients while preserving data privacy. However, fine-tuning large-scale pre-trained models in FL is hindered by resource constraints and communication costs. Although introducing parameter-efficient fine-tuning strategies such as Low-Rank Adaptation (LoRA) effectively reduces trainable parameters, this low-rank constraint exacerbates noise sensitivity, leading to overfitting and aggregation bias. Existing robust federated fine-tuning methods rely on additional proxy data and treat low-rank adapters as generic weight vectors. In this paper, we investigate the structural properties of LoRA and reveal a robustness asymmetry. The down-projection matrix A extracts stable general features, whereas the up-projection matrix B is highly susceptible to fitting noise patterns. Based on this finding, we propose Federated Decoupled Robust LoRA (FedDR-LoRA), which employs a dual-branch mechanism to decouple robust feature learning from noise modeling and mitigates noise interference through noisy branch negative learning. During federated aggregation, we establish global consensus through aggregating B while preserving local feature alignment in A. Extensive experiments demonstrate that FedDR-LoRA outperforms existing state-of-the-art methods across various noisy federated scenarios. Our code is available at: https://github.com/FangXiuwen/FedDR-LoRA.
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CB-SLICE: Concept-Based Interpretable Error Slice Discovery
Yael Konforti, Mateo Espinosa Zarlenga, Elaf Almahmoud, Mateja Jamnik
Despite strong average-case performance, deep learning models often exhibit systematic errors on specific population groups, known as *error slices*. Identifying these groups and the root causes of their failures is critical for model debugging and bias mitigation. However, existing error Slice Discovery Methods (SDMs) typically generate explanations disconnected from the model's inference process, thus only approximating the underlying error source and may be inaccurate.

We address this limitation by leveraging Concept Bottleneck Models (CBMs), whose predictions are directly dependent on human-understandable semantic concepts. Since downstream task failures in CBMs commonly arise from concept mispredictions, concept representations provide a strong candidate for error slice identification, offering fine-grained explanations directly linked to the error source.
Building on this insight, we introduce *CB-SLICE*, a concept-based SDM that groups samples with shared concept prediction failures and identifies the keyword-concepts most responsible for each slice’s failure-mode. Across multiple benchmarks, we show that CB-SLICE outperforms state-of-the-art methods in uncovering well-known biases while providing richer and more faithful explanations of model errors.
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Unbiased Alignment for Large Language Models with Noisy Preferences
Jialiang Wang, Xianming Liu, Xiong Zhou, Hui Liu, Haoliang Li
The alignment of large language models with human preferences is commonly achieved through Reinforcement Learning from Human Feedback or Direct Preference Optimization.
However, these methods are vulnerable to the significant noise prevalent in real-world preference datasets.
To address this critical issue, we present a theoretical framework for unbiased alignment, introducing the *Unbiased Reward Model* (URM) loss and the *Unbiased Direct Preference Optimization* (UDPO) loss.
By mathematically correcting the distortion induced by preference noise, our novel objectives enable unbiased model training directly from noisy datasets, without requiring clean ground-truth supervision.
We provide rigorous theoretical analyses demonstrating that our methods are noise-tolerant, parameter downward compatible, and classification-calibrated.
Comprehensive experiments across diverse datasets demonstrate that our approaches outperform state-of-the-art baselines. Code available at: https://github.com/cswjl/unbiased-alignment.
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Weight Updates as Activation Shifts: A Principled Framework for Steering

Dyah Adila, John Cooper, Alexander Yun, Avi Trost, Frederic Sala
Activation steering promises to be an extremely parameter-efficient form of adaptation, but its effectiveness depends on critical design choices---such as intervention location and parameterization---that currently rely on empirical heuristics rather than a principled foundation. We establish a first-order equivalence between activation-space interventions and weight-space updates, deriving the conditions under which activation steering can replicate fine-tuning behavior. This equivalence yields a \textbf{principled framework for steering design} and identifies the post-block output as a theoretically-backed and highly expressive intervention site. We explain why certain intervention locations outperform others and show that weight updates and activation updates play distinct, complementary functional roles. This analysis motivates a new approach---\textbf{joint adaptation}---that trains in both spaces simultaneously.
  Our post-block steering achieves accuracy within $0.2\%\text{--}0.9\%$ of full-parameter tuning, on average across tasks and models, while training only 0.04% of model parameters. It consistently outperforms prior activation steering methods such as ReFT and PEFT approaches including LoRA, while using significantly fewer parameters. Finally, we show that joint adaptation often surpasses the performance ceilings of weight and activation updates in isolation, introducing a new paradigm for efficient model adaptation
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ConServe: Fine-Grained GPU Harvesting for LLM Online and Offline Co-Serving
Yifan Qiao, Shan Yu, Shu Anzai, Haoran Ma, Shuo Yang, Yang Wang, Miryung Kim, Yongji Wu, Yang Zhou, Jiarong Xing, Joseph E. Gonzalez, Ion Stoica, Harry Xu
Large language model (LLM) serving demands low latency and high throughput, but high load variability leads to significant GPU under-utilization. In this paper, we identify a synergistic but overlooked opportunity to co-serve latency-critical online requests alongside *latency-tolerant offline* tasks, which existing systems fail to exploit because their coarse-grained resource management introduces interference.
We present ConServe, a co-serving system that enables fine-grained resource sharing through latency-aware token-level scheduling, sub-iteration layer-wise preemption, and incremental KV-cache management. These mechanisms allow offline execution to fill *millisecond-scale* GPU idle time while preserving strict online latency guarantees. Across real-world workloads with Llama-3.1 and Qwen-2.5 models, ConServe improves throughput by 2.2$\times$ on average and reduces online tail latency by 2.9$\times$ over state-of-the-art systems.
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From Out-of-Distribution Detection to Hallucination Detection: A Geometric View
Litian Liu, Reza Pourreza, Yubing Jian, Yao Qin, Roland Memisevic
Detecting hallucinations in large language models is a critical open problem with significant implications for safety and reliability. While existing hallucination detection methods achieve strong performance in question■answering tasks, they remain less effective on tasks requiring reasoning. In this work, we revisit hallucination detection through the lens of out■of■distribution (OOD) detection, a well■studied problem in areas like computer vision. Treating next■token prediction in language models as a classification task allows us to apply OOD techniques, if we bring to bear appropriate modifications to account for the structural differences in large language models. We show that approaches based on OOD detection yield training-free, single-sample based detectors, achieving strong accuracy in hallucination detection in reasoning tasks. Overall, our work suggests that reframing hallucination detection as OOD detection provides a promising and scalable pathway toward language model safety.
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Fast Mixture of Curvature-Aware Experts for Diverse and Dynamic Graph Topologies
Jiayi Yang, Xing Wei, Chunchun Chen, Yi Feng, Wengang Guo, Rui Fan, Xiaofeng Cao, Xin Sun, Wei Ye
Dynamic graph learning, which focuses on modeling the merging, vanishing, and reconnection of nodes and edges, is crucial for real-world applications. In dynamic graphs, node neighborhoods often exhibit diverse and time-evolving topologies,

including hierarchical, grid-like, and cyclic patterns. Existing methods typically embed graphs into a single curvature space, which limits the quality of node representations when the embedding geometry is not aligned well with the local graph topology.
In this paper, we propose **DyGMoCE**, a **Dy**namic **G**raph Transformer with a **M**ixture **o**f **C**urvature-aware **E**xperts, which efficiently embeds each node at every timestamp into an adaptive curvature space. Specifically, DyGMoCE incorporates a mixture-of-experts framework to both the attention and feed-forward modules, where each expert operates on a Riemannian manifold with a distinct curvature.
Then, motivated by the geometric continuity across the experts, we introduce a routing mechanism with a ranking constraint. To improve efficiency, we design a fast Riemannian attention module for DyGMoCE, achieving an average speedup of 27.5% and memory reduction of 52.6%. Notably, the fast Riemannian attention module is broadly applicable to Transformer models with sequence inputs. Extensive experimental results show that DyGMoCE significantly outperforms other state-of-the-art methods.
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Minimax-Optimal Policy Regret in Partially Observable Markov Games
Raman Arora
We study sequential decision-making in partially observable environments against strategic, adaptive opponents, modeled as partially observable Markov games (POMGs). The central challenge is to learn latent dynamics from partial observations while facing an adversary whose behavior depends on the learner's strategy, making standard regret notions inadequate. We prove that an epoch-based optimistic maximum-likelihood algorithm achieves $\tilde O(\sqrt{T})$ policy regret for fixed problem parameters, with explicit dependence on the horizon, adversary memory, confidence radius, and the aggregate Eluder dimension of the observable-operator class. The algorithm selects one policy per geometrically growing epoch using confidence sets built cumulatively from past data, which keeps the cost of comparing adversary responses across policies logarithmic in T. We also prove a lower bound matching the $\sqrt{T}$ and aggregate-Eluder-dimension dependence, up to problem-dependent and logarithmic factors. Finally, we extend the framework to horizon-adaptive guarantees and adversaries with geometric fading memory.
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From Moments to Models: Graphon-Mixture Learning for Mixup and Contrastive Learning
Ali Azizpour, Reza Ramezanpour, Santiago Segarra
Real-world graph datasets often arise from mixtures of populations, where graphs are generated by multiple distinct underlying distributions.
In this work, we propose a unified framework that explicitly models graph data as a mixture of probabilistic graph generative models represented by graphons.
To characterize and estimate these graphons, we leverage graph moments (motif densities) to cluster graphs generated from the same underlying model.
We establish a novel theoretical guarantee, deriving a tighter bound showing that graphs sampled from structurally similar graphons exhibit similar motif densities with high probability.
This result enables principled estimation of graphon mixture components.
We show how incorporating estimated graphon mixture components enhances two widely used downstream paradigms: graph data augmentation via mixup and graph contrastive learning.
By conditioning these methods on the underlying generative models, we develop graphon-mixture-aware mixup (GMAM) and model-aware graph contrastive learning (MGCL).
Extensive experiments on both simulated and real-world datasets demonstrate strong empirical performance.
In supervised learning, GMAM outperforms existing augmentation strategies, achieving new state-of-the-art accuracy on 6 out of 7 datasets.
In unsupervised learning, MGCL performs competitively across seven benchmark dat

asets and achieves the lowest average rank overall.
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Hybrid Policy Distillation for LLMs
Wenhong Zhu, Ruobing Xie, Rui Wang, Pengfei Liu
Knowledge distillation (KD) is a powerful paradigm for compressing large languag e models (LLMs), whose effectiveness depends on intertwined choices of divergenc e direction, optimization strategy, and data regime.
We break down the design of existing KD methods and present a unified view that establishes connections between them, reformulating KD as a reweighted log-likel ihood objective at the token level.
We further propose Hybrid Policy Distillation (HPD), which integrates the comple mentary advantages of forward and reverse KL to balance mode coverage and mode-s eeking behaviors, and combines off-policy data with lightweight approximate on-p olicy sampling.
We validate HPD on long-generation math reasoning as well as short-generation di alogue and code tasks, demonstrating improved optimization stability, computatio nal efficiency, and final performance across diverse model families and scales.
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CODiff: One-Step Diffusion Model for Camouflaged Object Detection
Xiaotong Fu, Qian Liu, Qihang Zhou, Wenchao Meng, Qinmin Yang, Shibo He
Diffusion-based camouflaged object detection (COD) has recently shown great pote ntial. In contrast to existing approaches that rely on multiple sample steps to refine the predicted masks, we propose CODiff, which reformulates the diffusion process to enable one-step mask prediction while maintaining competitive accurac y. Specifically, we first establish the theoretical feasibility of one-step samp ling for COD. Based on this, we design a dedicated network for one-step inferenc e with a global semantic guidance mechanism to guide the denoising process globa lly and hierarchical condition integration blocks to provide fine-grained struct ural semantics. In addition, we design a straight-forward regularization to lear n better intermediate features by bridging the representation gap between the co ndition backbone and the diffusion model. Extensive experiments demonstrate that CODiff achieves state-of-the-art performance across multiple benchmarks, improv ing MAE by over 22\% on the challenging COD10K dataset. Code is available at htt ps://github.com/KiiSooo/CODiff.
****

Prioritized Model Experience Replay
Muxi Tao, Jiangtao Wen, Yuxing Han
Model-based reinforcement learning (MBRL) improves sample efficiency by leveragi ng learned dynamics models, but often suffers from unstable training due to dyna mics model learning mismatch: models are trained on data from historical policie s while being queried under the continually updated current policy. This mismatc h can cause policy-relevant local model error to remain large even as global pre diction error decreases, leading to oscillatory updates. We present a finite-hor izon performance analysis that decomposes the policy performance gap into global model error, policy-induced distribution shift, and historical policy mixture e ffects, showing that minimizing global error alone is insufficient for stable op timization. Motivated by this analysis, we propose Prioritized Model Experience Replay (PMER), a lightweight replay mechanism that prioritizes high-error transi tions during dynamics model training. PMER implicitly emphasizes policy-relevant regions without explicit policy distance estimation and integrates seamlessly i nto Dyna-style MBRL frameworks. Experiments on MuJoCo benchmarks demonstrate imp roved stability, faster convergence, and higher sample efficiency.
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Signature-Informed Transformer for Asset Allocation
Yoontae Hwang, Stefan Zohren
Modern deep learning for asset allocation typically separates forecasting from o ptimization. We argue this creates a fundamental mismatch where minimizing predi ction errors fails to yield robust portfolios. We propose the Signature Informed Transformer to address this by unifying feature extraction and decision making into a single policy. Our model employs path signatures to encode complex path d

ependencies and introduces a specialized attention mechanism that targets geomet
ric asset relationships. By directly minimizing the Conditional Value at Risk we
 ensure the training objective aligns with financial goals. We prove that our at
tention module rigorously amplifies signature derived signals. Experiments acros
s diverse equity universes show our approach significantly outperforms both trad
itional strategies and advanced forecasting baselines. The code is available at:
 https://github.com/Yoontae6719/Signature-Informed-Transformer-For-Asset-Allocat
ion
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Policy-Driven World Model Adaptation for Robust Offline Model-based Reinforcemen
t Learning

Jiayu Chen, Le Xu, Aravind Venugopal, Jeff Schneider

Offline reinforcement learning (RL) offers a powerful paradigm for data-driven c
ontrol. Compared to model-free approaches, offline model-based RL (MBRL) explici
tly learns a world model from a static dataset and uses it as a surrogate simula
tor, improving data efficiency and enabling potential generalization beyond the
dataset support. However, most existing offline MBRL methods follow a two-stage
training procedure: first learning a world model by maximizing the likelihood of
 the observed transitions, then optimizing a policy to maximize its expected ret
urn under the learned model. This objective mismatch results in a world model th
at is not necessarily optimized for effective policy learning. Moreover, we obse
rve that policies learned via offline MBRL often lack robustness during deployme
nt, and small adversarial noise in the environment can lead to significant perfo
rmance degradation. To address these, we propose a framework that dynamically ad
apts the world model alongside the policy under a unified learning objective aim
ed at improving robustness. At the core of our method is a maximin optimization
problem, which we solve by innovatively utilizing Stackelberg learning dynamics.
 We provide theoretical analysis to support our design and introduce computation
ally efficient implementations. We benchmark our algorithm on twelve noisy D4RL
MuJoCo tasks and three stochastic Tokamak Control tasks, demonstrating its state
-of-the-art performance. Code is available at https://github.com/Agentic-Intelli
gence-Lab/ROMBRL.
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SleepMaMi: A Universal Sleep Foundation Model for Integrating Macro- and Micro-s
tructures

Keondo Park, Younghoon Na, You Rim Choi, Hyunwoo Ryu, Hyun-Woo Shin, Hyung-Sin K
im

While the shift toward unified foundation models has revolutionized many deep le
arning domains, sleep medicine remains largely restricted to task-specific model
s that focus on localized micro-structure features. These approaches often negle
ct the rich, multi-modal context of Polysomnography (PSG) and fail to capture th
e global macro-structure of a full night's sleep. To address this, we introduce
SleepMaMi, a Sleep Foundation Model engineered to master both hour-long sleep ar
chitectures and fine-grained signal morphologies. Our framework utilizes a hiera
rchical dual-encoder design: a Macro-Encoder to model full-night temporal depend
encies and a Micro-Encoder to capture short-term characteristics from biosignals
. Macro-Encoder is trained via Demographic-Guided Contrastive Learning, which al
igns overnight sleep patterns with objective subject metadata, such as age and s
ex, to refine global representations. Micro-Encoder is optimized via a hybrid Ma
sked Autoencoder (MAE) and multi-modal contrastive objective. Pre-trained on a
massive corpus of >20,000 PSG recordings (158K hours), SleepMaMi outperforms or
matches existing foundation models across a diverse suite of downstream tasks, d
emonstrating superior generalizability and label-efficient adaptation for clinic
al sleep analysis.
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EchoingPixels: Aliasing-Resistant Joint Token Reduction for Audio-Visual LLMs

Chao Gong, Depeng Wang, Zhipeng Wei, Ya Guo, Huijia Zhu, Jingjing Chen

Audio-Visual Large Language Models (AV-LLMs) face prohibitive computational cost
s of processing massive, redundant audio-visual tokens. Existing unimodal compre
ssion techniques fail to capture the heterogeneous and mutually influential info

rmation density of joint audio-visual signals. Furthermore, we identify a fundamental and overlooked theoretical bottleneck in sparse token reduction: positional aliasing. We demonstrate that aggressive sparse sampling on standard position-encoded sequences violates the Nyquist limit relative to the effective token interval, causing phase-wrapping collisions that corrupt temporal monotonicity. To address this, we introduce EchoingPixels, a framework for aliasing-resistant joint token reduction. Our Cross-Modal Semantic Sieve performs extractive selection on the synergistic audio-visual stream, dynamically allocating budgets based on joint-modality saliency rather than fixed per-modality ratios. To resolve positional aliasing, we derive Sync-RoPE, a spectral low-pass filter for Rotary Positional Embeddings that adapts encoding bandwidth to the sparse sampling rate, preserving monotonic temporal relationships in the reduced stream. Experiments show that EchoingPixels achieves performance comparable to full models using only 5-20% of original tokens, validating theoretically grounded sparse learning as a robust solution for efficient AV-LLMs. Code is available at https://github.com/CharlesGong12/EchoingPixels.
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What Does Flow-Matching Bring to TD-Learning?
Bhavya Kumar Agrawalla, Michal Nauman, Aviral Kumar
Recent work shows that flow-matching networks can be effective for value function estimation in reinforcement learning, but it remains unclear why they work well or whether flow-matching Q-functions differ fundamentally from standard critics. We show that their success is not explained by distributional RL: explicitly modeling return distributions often degrades performance. Instead, we argue that flow-matching Q-functions are effective because they couple a learned velocity field with an integration procedure that is used both during training and to read out Q-values at inference time. This coupling enables robust value prediction through \emph{test-time recovery} from imperfect intermediate estimates where errors dampen out as more integration steps are performed. This mechanism is absent in monolithic critics. Beyond test-time recovery, training with the integration procedure induces more \emph{plastic} representations, allowing critics to represent non-stationary future TD targets without overwriting previous features. We formalize these effects and validate them empirically, showing that flow-matching critics outperform monolithic critics by over $2\times$ in performance and achieve 5–$10\times$ higher sample efficiency in high-UTD regimes.
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Long Live The Balance: Information Bottleneck Driven Tree-based Policy Optimization
Hao Jiang, Shurui Li, Tianpeng Bu, Bowen Xu, Xin Liu, Qihua Chen, hongtao duan, lulu hu, Bin Yang, Minying Zhang
Recent advances in online reinforcement learning (RL) for large language models (LLMs) have demonstrated promising performance in complex reasoning tasks. However, they often exhibit an imbalanced exploration–exploitation trade-off, resulting in unstable optimization and sub-optimal performance. We introduce IB-Score, a novel metric grounded in Information Bottleneck (IB) theory that evaluates policy’s exploration-exploitation balance by quantifying the trade-off between step-level reasoning diversity and mutual information shared with the correct answer. Analysis based on IB-Score shows that popular online RL approaches (e.g., GRPO) with common regularization methods fail to consistently maintain balance during training with suboptimal results. To address this, we propose Information Bottleneck-driven Tree-based Policy Optimization (IB-TPO), a principled framework that formulates IB-Score as a fine-grained optimization objective and utilizes a novel IB-guided tree sampling strategy that not only improves the efficiency of online sampling with 50\% more trajectories under same token budget, but also reuses the tree structure for effective IB-Score Monte Carlo estimation. Extensive experiments across standard benchmarks show that our method significantly outperforms GRPO baseline by 2.9% to 3.6% and also outperforms other state-of-the-art online RL approaches.
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Q-DiT4SR: Exploration of Detail-Preserving Diffusion Transformer Quantization f

or Real-World Image Super-Resolution
Xun Zhang, Kaicheng Yang, Hongliang Lu, Haotong Qin, Yong Guo, Yulun Zhang
Recently, Diffusion Transformers (DiTs) have emerged in Real-World Image Super-Resolution (Real-ISR) to generate high-quality textures, yet their heavy inference burden hinders real-world deployment. While Post-Training Quantization (PTQ) is a promising solution for acceleration, existing methods in super-resolution mostly focus on U-Net architectures, whereas generic DiT quantization is typically designed for text-to-image tasks. Directly applying these methods to DiT-based super-resolution models leads to severe degradation of local textures. Therefore, we propose **Q-DiT4SR**, the first PTQ framework specifically tailored for DiT-based Real-ISR. We propose **H-SVD**, a hierarchical SVD that integrates a global low-rank branch with a local block-wise rank-1 branch under a matched parameter budget. We further propose **V**ariance-**a**ware **S**patio-**T**emporal **M**ixed **P**recision: **VaSMP** allocates cross-layer weight bit-widths in a data-free manner based on rate-distortion theory, while **VaTMP** schedules intra-layer activation precision across diffusion timesteps via dynamic programming (DP) with minimal calibration. Experiments on multiple real-world datasets demonstrate that our Q-DiT4SR achieves SOTA performance under both **W4A6** and **W4A4** settings. Notably, the W4A4 quantization configuration reduces model size by **5.8**$\times$ and computational operations by **6.14**$\times$. Our code and models will be available at https://github.com/xunzhang1128/Q-DiT4SR.
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Efficient Neural Controlled Differential Equations via Attentive Kernel Smoothing
Egor Serov, Ilya Kuleshov, Alexey Zaytsev
Neural Controlled Differential Equations (Neural CDEs) provide a powerful continuous-time framework for sequence modeling, yet the roughness of the driving control path often restricts their efficiency. Standard splines introduce high-frequency variations that force adaptive solvers to take excessively small steps, driving up the Number of Function Evaluations (NFE). We propose a novel approach to Neural CDE path construction that replaces exact interpolation with Kernel and Gaussian Process (GP) smoothing, enabling explicit control over trajectory regularity. To recover details lost during smoothing, we propose an attention-based Multi-View CDE (MV-CDE) and its convolutional extension (MVC-CDE), which employ learnable queries to inform path reconstruction. This framework allows the model to distribute representational capacity across multiple trajectories, each capturing distinct temporal patterns. Empirical results demonstrate that our method, MVC-CDE with GP, achieves state-of-the-art accuracy while significantly reducing NFEs and total inference time compared to spline-based baselines.
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Hyperbolic Multimodal Continual Learning
Jiahong Liu, Ming Shen, Xiaohao Liu, Rex Ying, Menglin Yang, Tat-Seng Chua, Irwin King
Hyperbolic geometry has recently emerged as a powerful representation space for multimodal learning, as it naturally captures hierarchical semantic structure across modalities. Despite this progress, how such representations behave under continual learning poses fundamentally different challenges that remain underexplored. This work provides a geometric perspective on this problem and establishes a theoretical foundation for representation preservation in hyperbolic space, showing that preventing forgetting requires cross-modal invariance under a shared hyperbolic isometry. We further show that forgetting in hyperbolic continual learning involves both semantic relation drift and hierarchy-related distortion, motivating preservation of both cross-modal relational structure and hierarchical geometry. Guided by these insights, a principled continual learning framework is derived that preserves essential geometric structure while allowing effective adaptation to new tasks. Experiments on continual multimodal benchmarks corroborate the effectiveness of the proposed approach.
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Cardio-mmFlow: A Gaussian-Prior-Free Physics-Informed Flow Matching Framework for Electrocardiogram to mmWave Radar Synthesis

Ziyang LIU, Ruiqiang Xiao, Chang Huang, KIEREN YU, Siyuan HE, Kaishun Wu
Continuous ECG monitoring is clinically valuable, but scaling it beyond electrodes to comfortable long-term use motivates contactless mmWave sensing. In practice, mmWave-to-ECG reconstruction is severely constrained by the scarcity of high-quality synchronized recordings.
Therefore, we propose \textbf{Cardio-mmFlow}, a Gaussian-prior-free physics-informed flow matching framework that synthesizes mmWave radar signals from clinical ECG.
It learns a direct transport trajectory between the latent manifolds of ECG and radar.
Considering subject-dependent propagation differences, we incorporate a simplified mass--spring--damper inspired modulation and inject it into the flow dynamics via feature-wise linear modulation for personalization.
Extensive experiments show that our system generates high fidelity radar data in both signal and latent domains. It significantly improves zero-shot downstream mmWave to ECG task, and enable Atrial Fibrillation classification with synthetic data.
Further analyses evaluate the model interpretability.
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Proxy Compression for Language Modeling
Lin Zheng, Xinyu Li, Qian Liu, Xiachong Feng, Lingpeng Kong
Modern language models are trained almost exclusively on token sequences produced by a fixed tokenizer, an external lossless compressor often over UTF■8 byte sequences, thereby coupling the model to that compressor.
This work introduces proxy compression, an alternative training scheme that preserves the efficiency benefits of compressed inputs while providing an end-to-end, raw-byte interface at inference time.
During training, one language model is jointly trained on raw byte sequences and compressed views generated by external compressors; through the process, the model learns to internally align compressed sequences and raw bytes. This alignment enables strong transfer between the two formats, even when training predominantly on compressed inputs which are discarded at inference.
Extensive experiments on code language modeling demonstrate that proxy compression substantially improves training efficiency and significantly outperforms pure byte-level baselines given fixed compute budgets.
As model scale increases, these gains become more pronounced, and proxy-trained models eventually match or rival tokenizer approaches, all while operating solely on raw bytes and retaining the inherent robustness of byte-level modeling.
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MRPO: Magnitude-Regularized Policy Optimization via L1 Constraints
Wei Han, Yuanxing Liu, Mingda Li, Ruiyu Xiao, Weinan Zhang, Ting Liu
Reinforcement learning (RL) for large language models (LLMs) relies on imperfect reward supervision, necessitating constraints on policy updates to prevent overfitting.
Nevertheless, the widely adopted KL constraint over-penalizes actions with low reference probabilities and lacks the sparsity to discard marginal policy shifts.
In contrast, the L1-norm offers a distinct mechanism that is more tolerant of low-probability actions yet strictly suppresses minor probability perturbations.
Motivated by this, we propose Magnitude-Regularized Policy Optimization (MRPO), which enforces an L1-norm constraint on policy updates.
We demonstrate that MRPO permits substantial probability boosts for low-probability actions and induces sparse updates, ensuring invariance to noise that preserves the top-ranking order.
Furthermore, MRPO admits a TRPO-style monotonic improvement bound under standard regularity assumptions and achieves a tighter approach to optimality than KL-based methods in single-step scenarios.
Empirically, MRPO delivers exceptional results across diverse scenarios, notably doubling the performance gains of GRPO in preference alignment, outperforming DAPO in mathematical reasoning, and surpassing DPO in offline settings using only binary rewards.

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HoloFair: Unified T2I Fairness Evaluation and Fair-GRPO Debiasing
Ruyi Chen, Lu Zhou, Xiaogang Xu, Chiyu Zhang, Jiafei Wu, Liming Fang
Text-to-Image (T2I) models have made significant strides in visual realism and semantic consistency, yet they often perpetuate and amplify societal biases. Existing evaluation methods typically address only single-dimensional biases, lacking perspectives to uncover model biases at social-related deeper semantic levels. We introduce HoloFair, a comprehensive benchmark framework for multidimensional demographic bias analysis. Built upon our large-scale fairness-oriented dataset and the SpaFreq (Spatial-Frequency) attribute classifier, this framework proposes the Multi-attribute, Group-wise Bias Index (MGBI) metric, designed to assess both intrinsic diversity and conditional biases. Beyond evaluation, we further introduce Fair-GRPO, a reinforcement-learning-based debiasing method that alters the distribution of generative models through a designed multi-objective reward function. E.g., experiments on the SD3.5-Medium model demonstrate that Fair-GRPO significantly improves multidimensional fairness while maintaining high image quality. We also analyze potential reward hacking phenomena and provide corresponding mitigation strategies.
****
Multimodal Meta-Verifier with Explicit Structured Recalibration
Xinchen Zhang, Bowei Liu, Jiale Liu, Chufan Shi, Yizhen Zhang, Junhong Liu, Youliang Zhang, Zhiheng Li, Yujiu Yang, Ling Yang
Visual outcomes are increasingly central to multimodal large language models, making reliable and fine-grained verification essential for scaling generalist foundation models. In this work, we investigate ***multimodal meta-verification***, which leverages verifier-generated rationales rather than decision-only signals, and explore how to effectively incorporate meta-verification feedback into multimodal verifier training. We identify two key findings. First, symbolic verifier outputs (e.g., bounding boxes) outperform textual explanations as meta-verification rationales, enabling efficient rule-based reinforcement learning rewards while avoiding reliance on model-based rewards from auxiliary judge models. Second, decoupling reinforcement learning objectives for binary judgment and meta-verification substantially outperforms joint reward optimization, due to intrinsic differences in output structure and learning dynamics. Based on these insights, we train **OmniVerifier-M1**, a generalist visual verifier leveraging symbolic meta-verification and decoupled reinforcement learning. OmniVerifier-M1 provides robust verification and fine-grained error localization, and further enables **M1-TTS**, a verifier-driven agentic generation system achieving dynamic region-level self-correction. This approach paves the way for more reliable, interpretable, and fine-grained multimodal verification, supporting safer and more controllable foundation model deployment.
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Fantastic Reasoning Behaviors and Where to Find Them: Unsupervised Discovery of the Reasoning Process
Zhenyu Zhang, Shujian Zhang, John Wheatley Lambert, Wenxuan Zhou, Zhangyang Wang, Mingqing Chen, Andrew Hard, Rajiv Mathews, Lun Wang
Despite the growing reasoning capabilities of recent large language models (LLMs), their internal mechanisms during the reasoning process remain underexplored. Prior approaches often rely on human-defined concepts (e.g., overthinking, reflection) at the word level to analyze reasoning in a supervised manner. However, such methods are limited, as it is infeasible to capture the full spectrum of potential reasoning behaviors, many of which are difficult to define in token space. In this work, we propose an unsupervised framework (namely, RISE: Reasoning behavior Interpretability via Sparse auto-Encoder) for discovering reasoning vectors, which we define as directions in the activation space that encode distinct reasoning behaviors. By segmenting chain-of-thought traces into sentence-level 'steps' and training sparse auto-encoders (SAEs) on step-level activations, we uncover disentangled features corresponding to interpretable behaviors such as reflection and backtracking. Visualization and clustering analyses show that these behaviors occupy separable regions in the decoder column space. Moreover, targete

d interventions on SAE-derived vectors can controllably amplify or suppress spec
ific reasoning behaviors, altering inference trajectories without retraining. Be
yond behavior-specific disentanglement, SAEs capture structural properties such
as response length, revealing clusters of long versus short reasoning traces. Mo
re interestingly, SAEs enable the discovery of novel behaviors beyond human supe
rvision. We demonstrate the ability to control response confidence by identifyin
g confidence-related vectors in the SAE decoder space. These findings underscore
the potential of unsupervised latent discovery for both interpreting and contro
llably steering reasoning in LLMs.
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Let Language Constrain Geometry: Vision–Language Models as Semantic and Spatial Critics for 3D Generation

Weimin Bai, Yubo Li, Weijian Luo, Zeqiang Lai, Yequan Wang, Wenzheng Chen, He Su
n

Text-to-3D generation has advanced rapidly, yet state-of-the-art models, encompa
ssing both optimization-based and feed-forward architectures, still face two fun
damental limitations. First, they struggle with coarse semantic alignment, often
failing to capture fine-grained prompt details. Second, they lack robust 3D spa
tial understanding, leading to geometric inconsistencies and catastrophic failur
es in part assembly and spatial relationships. To address these challenges, we p
ropose VLM3D, a general framework that repurposes large vision-language models (
VLMs) as powerful, differentiable {semantic and spatial critics}. Our core contr
ibution is a {dual-query critic signal} derived from the VLM's "Yes/No" log-odds
, which assesses both semantic fidelity and geometric coherence. We demonstrate
the generality of this guidance signal across two distinct paradigms: (1) As a r
eward objective for optimization-based pipelines, VLM3D significantly outperform
s existing methods on standard benchmarks. (2) As a test-time guidance module fo
r feed-forward pipelines, it actively steers the iterative sampling process of S
OTA native 3D models to correct severe spatial errors. VLM3D establishes a princ
ipled and generalizable path to inject the VLM's rich, language-grounded underst
anding of both semantics and space into diverse 3D generative pipelines.
****

LECTOR: Joint Learning of Scientific Reasoning Graphs and Introduction Generatio
n

Jiabei Xiao, Yizhou Wang, Chen Tang, Pengze Li, Wanli Ouyang, SHIXIANG TANG

AI Scientists have shown promising progress across multiple stages of the resear
ch pipeline, among which automatic scientific paper writing remains a formidable
challenge. The Introduction writing is especially challenging, which demands no
t only linguistic fluency, but logical soundness and verifiable faithfulness. Mo
st AI-assisted methods treat the task as text generation instead of reasoning an
d structuring, leading to severe drawbacks, *e.g.*, hallucinating citations. To
address this, we first formulate the Content-Conditional Introduction Generation
(CCIG) task, which requires grounding the Introduction in the paper's core evid
ence. We then propose **LECTOR**, a novel **Logic-Expression Co-Reinforcement Le
arning** framework that can strictly follow the scientist's logic, add high-qual
ity citations and keep structured expressions. LECTOR first constructs a logic-r
easoning graph from the paper's main body to serve as a verifiable logical bluep
rint. Subsequently, it employs a Logic-Expression Co-Rewarding mechanism to join
tly optimize for both the graph's structural fidelity and the final narrative's
quality. We conduct a dataset from *Nature Communications* papers to assess our
method. Extensive experiments show consistent improvements in both logic fidelit
y and Introduction generation quality metrics, *e.g.*, Graph Quality **(+26.7%)*
*, Citation Quality **(+8.6%)**, and Paper Consistency **(+3.3%)**.
Code and data are available at: https://github.com/Xiao-Youth/LECTOR
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Identifying Latent Concepts and Structures for Generalized Category Discovery

Boyang Dai, Chaoqi Chen, Yizhou Yu

Generalized Category Discovery (GCD) aims to recognize known classes while auton
omously discovering novel ones in open-world settings. However, current approach
es primarily focus on designing clustering objectives, often overlooking a criti

cal bottleneck: standard vision backbones yield high-rank, entangled token representations that are ill-suited for unsupervised discovery of latent concepts and structures. In this paper, we propose Compositional Primitive Fields (CPF-GCD), a novel representation learning framework that reshapes the feature space to make such latent structure identifiable by enforcing a low-rank compositional organization. Our core hypothesis is that all categories, whether known or novel, can be expressed as compositions and spatial arrangements of a finite set of learnable visual primitives that capture reusable concepts. CPF instantiates this geometric constraint via a spatial field mechanism. Inserted between the backbone and the head, it rewrites noisy patch tokens through low-rank primitive mixtures, effectively decomposing images into reusable atomic parts and their spatial layouts. By explicitly modeling the spatial distribution of primitives, CPF enables novel categories to emerge naturally as new activation patterns over a shared vocabulary. This shifts the focus of representation from merely partitioning global embeddings to constructing a structured and separable primitive field. Extensive experiments demonstrate that CPF serves as a generic, plug-and-play module that consistently boosts performance across diverse GCD baselines, validating that identifying and leveraging low-rank compositional structure is a crucial inductive bias for open-world recognition.

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Text-Driven Fusion for Infrared and Visible Images: Achieving Image Scene Adaptation on Hyperbolic Space

Huan Kang, Hui Li, Tianyang Xu, Tao Zhou, Xiaojun Wu, Josef Kittler

Infrared and visible image fusion aims to integrate complementary modalities, while existing Euclidean methods impose rigid distance metrics that distort multi-modal interactions and parent-to-child semantic hierarchies. To overcome these limitations, we introduce a text-driven fusion framework empowered by hyperbolic manifold learning. During training, BLIP-extracted text prompts serve as topological anchors within the hyperbolic space, guiding vision-attribute alignment through hyperbolic embeddings that naturally accommodate varying semantic granularities. By exploiting the exponential volume growth dictated by the Poincaré ball's negative curvature, this approach seamlessly embeds hierarchical trees to encode coarse-to-fine semantics without metric saturation, while the vast peripheral space prevents texture distortion during cross-modal fusion. At inference, the fusion process autonomously adapts to input content using the learned text-attribute priors, completely eliminating the need for textual input. Experimental results show our method outperforms state-of-the-art approaches on benchmark datasets, with code available at https://github.com/Shaoyun2023/TEDFusion.

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A Tight Theory of Error Feedback Algorithms in Distributed Optimization

Daniel Berg Thomsen, Adrien Taylor, Aymeric Dieuleveut

Communication costs are a major bottleneck in distributed learning and first-order optimization. A common approach to alleviate this issue is to compress the gradient information exchanged between agents. However, such compression typically degrades the convergence guarantees of gradient-based methods. Error feedback mechanisms provide a simple and computationally cheap remedy for this issue, but numerous variants have been proposed, and their relative performance remains poorly understood. This paper provides tight convergence analyses for two of the main error-feedback algorithms from the literature, the classic Error Feedback method (EF) and Error Feedback 21 (EF21), by identifying optimal step-size choices and constructing optimal Lyapunov functions tailored to each method. The results hold independently of the number of agents and recover the known best guarantees possible in the single-agent regime.

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Shapley Regularized Neural Granger Causality

Maolin Yang, Zhoufan Zhu, Yuanhe Tian, Kun Gao, MUYI LI

Identifying temporal causal structure is fundamental to understanding complex systems. Neural Granger causality has emerged as a powerful paradigm for this task, leveraging the expressiveness of neural networks to model intricate nonlinear dynamics. Although complex architectures excel at predictive modeling, existing

methods typically rely on simple local measures for causal discovery, which extr act only partial information from the learned model and may miss global dependen cies. To address this issue, we reformulate Granger causality as a feature attri bution problem and propose the Information-Theoretic Shapley value (Info-Shap) t o measure global feature importance. We first establish the theoretical equivale nce between zero Info-Shap and Granger non-causality. On top of this, we constru ct two novel regularizers to suppress spurious relationships and mitigate overfi tting. These regularizers are model-agnostic and can be seamlessly integrated in to the training of any differentiable neural network. Through extensive experime nts on synthetic and realistic datasets, we demonstrate that our method robustly recovers the underlying causal relationships, providing a flexible tool for cau sal discovery in high-dimensional nonlinear time series.

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REAL: Regression-Aware Reinforcement Learning for LLM-as-a-Judge

Yasi Zhang, Tianyu Chen, Mingyuan Zhou, Oscar Leong, Ying Nian Wu, Michal Lukasi k

Large language models (LLMs) are increasingly deployed as automated evaluators t hat assign numeric scores to model outputs, a paradigm known as LLM-as-a-Judge. However, standard Reinforcement Learning (RL) methods typically rely on binary r ewards (e.g., 0-1 accuracy), thereby ignoring the ordinal structure inherent in regression tasks; for instance, they fail to recognize that predicting 4 is sign ificantly better than predicting 1 when the ground truth is 5. Conversely, exist ing regression-aware approaches are often confined to Supervised Fine-Tuning (SF T), limiting their ability to explore optimal reasoning paths. To bridge this ga p, we propose REAL (Regression-Aware Reinforcement Learning), a principled RL fr amework designed to optimize regression rewards, and also proven to be optimal f or correlation metrics. A key technical challenge is that the regression objecti ve is explicitly policy-dependent, thus invalidating standard policy gradient me thods. To address this, we employ the generalized policy gradient estimator, whi ch naturally decomposes optimization into two complementary components: (1) expl oration over Chain-of-Thought (CoT) trajectory, and (2) regression-aware predict ion refinement of the final score. Extensive experiments across model scales (8B to 32B) demonstrate that REAL consistently outperforms both regression-aware SF T baselines and standard RL methods, exhibiting significantly better generalizat ion on out-of-domain benchmarks. On Qwen3-32B specifically, we achieve gains of +8.40 Pearson and +7.20 Spearman correlation over the SFT baseline, and +18.30/+ 11.20 over the base model. These findings highlight the critical value of integr ating regression objectives into RL exploration for accurate LLM evaluation.

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LoBCD-GW: A Fast and Data-Dependent Algorithm for Computing Gromov-Wasserstein D istance via Localized Block Coordinate Descent

Jingni Song, Jiawei Huang, Kangke Cheng, Bangxian Han, Hu Ding

The Gromov-Wasserstein (GW) distance provides a powerful framework for aligning structured data by comparing the intrinsic geometries of metric measure spaces, and has become a fundamental tool in machine learning. Most existing methods lev erage entropy regularization to reduce the computational complexity to $\boldsym bol{\mathrm{O}}(n^3)$, where n is the number of samples. However, this cubic t ime complexity remains a major bottleneck in large-scale applications, severely limiting the scalability. To address this challenge, we propose LoBCD-GW, an eff icient GW optimization algorithm. Specifically, we reveal the data-dependent spa rsity of large-magnitude updates to the coupling matrix and introduce a localize d block coordinate selection strategy. This confines the optimization to a "sele cted set" of size r (which is a parameter that depends on the given data set, and usually is much less than n), thereby reducing the complexity to $\boldsym bol{\mathrm{O}}(r^3)$. In addition, unlike prior acceleration methods often base d on constraint relaxation, our method can guarantee the strict feasibility thro ugh a novel "marginal compensation mechanism" to synchronize local mass redistri bution with global constraints. Finally, we conduct a set of experiments on vari ous datasets, and the results demonstrate that our method achieves a $5.8\times$ speedup on large-scale graph alignment benchmarks, while maintaining state-of-t

he-art accuracy.
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Optimizing Visual Generative Models via Distribution-wise Rewards
Ruihang Li, Mengde Xu, Shuyang Gu, Leigang Qu, Fuli Feng, Han Hu, Wenjie Wang
Conventional reinforcement learning strategies for visual generation typically e mploy sample-wise reward functions, yet this practice frequently results in rewa rd hacking that degrades image diversity and introduces visual anomalies. To add ress these limitations, we present a novel framework that finetunes generative m odels using distribution-wise rewards, ensuring better alignment with real-world data distributions. Unlike rewards that evaluate samples individually, distribu tion-wise reward accounts for the data distribution of the samples, mitigating t he mode collapse problem that occurs when all samples optimize towards the same direction independently. To overcome the prohibitive computational cost of estim ating these rewards, we introduce a subset-replace strategy that efficiently pro vides reward signals by updating only a small subset of a generated reference se t. Additionally, we apply RL to optimize post-hoc model merging coefficients, po tentially mitigating the train-inference inconsistency caused by introducing sto chastic differential equation (SDE) in regular RL practices. Extensive experimen ts show our approach significantly improves FID-50K across various base models, from 8.30 to 5.77 for SiT and from 3.74 to 3.52 for EDM2. Qualitative evaluation also confirms that our method enhances perceptual quality while preserving samp le diversity.
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LaTtE-Flow: Layerwise Timestep-Expert Flow-based Transformer
Ying Shen, Zhiyang Xu, Jiuhai Chen, Shizhe Diao, Jiaxin Zhang, Yuguang Yao, Joy Rimchala, Ismini Lourentzou, Lifu Huang
Recent advances in multimodal foundation models unifying image understanding and generation have opened exciting avenues for tackling a wide range of vision-lan guage tasks within a single framework. Despite progress, existing unified models often rely on extensive pretraining and suffer from slow generation speeds, lim iting their practical deployment in real-time and resource-constrained settings. In this work, we introduce Layerwise Timestep-Expert Flow-based Transformer (La TtE-Flow), a novel architecture that improves the efficiency of diffusion/flow-b ased Transformers within the unified model setting. LaTtE-Flow builds upon power ful pre-trained Vision-Language Models (VLMs) to inherit strong multimodal under standing capabilities, and extends them with a novel Layer-wise Timestep Experts flow-based architecture for efficient image generation. LaTtE-Flow distributes the flow-matching process across specialized groups of Transformer layers, each responsible for a distinct subset of timesteps. This design significantly improv es sampling efficiency
by activating only a small subset of layers at each sampling timestep. To furthe r enhance performance, we propose a Timestep-Conditioned Residual Attention mech anism for efficient information reuse across layers. Experiments demonstrate tha t LaTtE-Flow achieves strong performance on multimodal understanding tasks, whil e achieving competitive image generation quality with around 6× faster inference speed compared to recent unified multimodal models.
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Backward Oversmoothing: why is it hard to train deep Graph Neural Networks?
Nicolas Keriven
Oversmoothing has long been identified as a major limitation of Graph Neural Net works (GNNs): input node features are smoothed at each layer and converge to a c onstant non-informative representation, *if the weights of the GNN are sufficien tly bounded*. This assumption is crucial: if, on the contrary, the weights are s ufficiently large, then oversmoothing may be compensated. Theoretically, GNN cou ld thus *learn* to not oversmooth. However, this does not really happen in pract ice, which prompts us to examine oversmoothing from an *optimization* point of v iew.

In this paper, we analyze *backward oversmoothing*, that is, the notion that bac kpropagated errors are also subject to oversmoothing from output to input. With

non-linearities, we outline the key role of the *interaction* between forward an
d backward smoothing. Moreover, we show that, due to backward oversmoothing, GNN
s provably exhibit many *spurious stationary points*: as soon as the *last* laye
r is trained, the *whole* GNN is at a stationary point. As a result, we can exhi
bit regions where gradients are near-zero while the loss stays high.
Additionally, we prove that this is *specific* to GNNs, and does not necessarily
 hold for Multi-Layer Perceptrons.
This paper is a step toward a more complete comprehension of the optimization la
ndscape of GNNs.
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The Safety-Aware Denoiser for Text Diffusion Models
Amman Yusuf, Zhejun Jiang, Mijung Park
Recent work on text diffusion models offers a promising alternative to autoregre
ssive generation, but controlling their safety remains underexplored. Existing s
afety approaches are geared toward autoregressive models and typically rely on p
ost-hoc filtering or inference-time interventions. These are inadequate for effe
ctively addressing safety risks in text diffusion models. We propose the *Safety
-Aware Denoiser (SAD)*, a safety-guidance framework in text diffusion models. Th
e SAD modifies the iterative denoising process such that the text sample at the
final denoising step is steered toward *provably* safe regions of the text space
. This inference-time method can integrate safety constraints into the denoiser,
 avoiding computationally expensive retraining of the underlying diffusion model
 and enabling flexible, lightweight safety guidance. We evaluate the safety of t
he generated text using the SAD, with respect to hazard taxonomy, memorization,
and jailbreak. Experimental results show that SAD substantially reduces unsafe g
enerations while preserving generation quality, diversity, and fluency, outperfo
rming existing methods. These results demonstrate that our safety guidance durin
g denoising provides an effective and scalable mechanism for enforcing safety in
 text diffusion models.
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HGMem: Hypergraph-based Working Memory to Improve Multi-step RAG for Long-Contex
t Complex Relational Modeling
Chulun Zhou, Chunkang Zhang, Guoxin Yu, Fandong Meng, Jie Zhou, Wai Lam, Mo Yu
Multi-step retrieval-augmented generation (RAG) has become a widely adopted stra
tegy for enhancing large language models (LLMs) on tasks that demand global comp
rehension and intensive reasoning. Although many RAG systems incorporate a worki
ng memory to consolidate information, existing designs primarily function as a p
assive storage for isolated facts. This static nature overlooks crucial high-ord
er correlations among primitive facts, thereby limiting models' capacity for mul
ti-step reasoning and resulting in fragmented reasoning and weak global sense-ma
king within extended contexts. We introduce HGMem, a hypergraph-based working me
mory system, extending the concept of memory beyond simple storage into a dynami
c, expressive structure for complex reasoning and global understanding. In our a
pproach, memory is represented as a hypergraph where hyperedges correspond to di
stinct memory units, enabling the progressive formation of high-order interactio
ns within memory. This mechanism connects facts and thoughts around the focal pr
oblem, evolving the memory into an integrated and situated knowledge structure t
hat provides strong propositions for deeper reasoning. We evaluate HGMem on seve
ral challenging global sense-making benchmarks. Extensive experiments and in-dep
th analyses demonstrate that our method consistently improves multi-step RAG and
 substantially outperforms strong baseline systems across diverse datasets.
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Tournament Style RL: Stabilizing Policy Optimization on Non Verifiable Problems
Gurusha Juneja, Shubham Milind Phal, Jennifer She, Lisa Wang, Dorsa Sadigh, Anca
 Dragan, William Yang Wang
Many real-world tasks are non-verifiable—there is no objective ground truth, and
 quality must be judged subjectively—making reward design for RL difficult. Exis
ting approaches based on scalar rubric scores or single comparisons are often no
isy, poorly calibrated, or provide sparse learning signals.
We introduce Tournament Style RL (TSRL), which constructs rewards from rubric-gu

ided pairwise judgments against a fixed set of anchor responses, using win-rate as the reward for policy optimization. This aggregation of comparisons against anchor responses yields a signal that is more robust to the judge noise by stabilizing the reference frame, reducing the variance in reward.
We test across four non-verifiable tasks and two backbone LLMs, and find that TSRL improves average win-rate by $+43.8$ points over the base model and $+22.8$ points over the strongest baseline.
TSRL scales with the number of anchors, remains robust under weak or partially corrupted judges, the results are supported by blinded human preference studies.
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Who Evaluates AI's Social Impacts? Mapping Coverage and Gaps in First and Third Party Evaluations
Anka Reuel, Avijit Ghosh, Jenny Chim, Andrew Tran, Yanan Long, Jennifer Mickel, Usman Gohar, Srishti Yadav, Pawan Sasanka Ammanamanchi, Mowafak Allaham, Hossein A. Rahmani, Mubashara Akhtar, Felix Friedrich, Robert Scholz, Michael Alexander Riegler, Jan Batzner, Eliya Habba, Arushi Saxena, Anastassia Kornilova, Kevin Wei, Prajna Soni, Yohan Mathew, Kevin Klyman, Jeba Sania, Subramanyam Sahoo, Olivia Beyer Bruvik, Pouya Sadeghi, Sujata Goswami, Angelina Wang, Yacine Jernite, Zeerak Talat, Stella Biderman, Mykel Kochenderfer, Sanmi Koyejo, Irene Solaiman
Foundation models are increasingly central to high-stakes AI systems, and governance frameworks now depend on evaluations to assess their risks and capabilities. Although general capability evaluations are widespread, social impact assessments covering bias, fairness, privacy, environmental costs, and labor remain uneven. To characterize this landscape, we conduct the first comprehensive analysis of social impact evaluation reporting, examining 186 first-party release reports and 248 third-party evaluation sources, supplemented by developer interviews. We find a stark division of labor: first-party reporting is sparse, often superficial, and declining in areas like environmental impact and bias, while third-party evaluators provide broader, more rigorous coverage of bias, harmful content, and performance disparities. However, only developers can authoritatively report on data provenance, content moderation labor, costs, and infrastructure, yet interviews reveal these disclosures are deprioritized unless tied to product adoption or compliance. Current practices leave major gaps in assessing societal impacts, underscoring the need for policies that mandate developer transparency, strengthen independent evaluation ecosystems, and create shared infrastructure for aggregating third-party evaluations.
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Courtroom Analogy: New Perspective on Uncertainty-Aware Classification
Taeseong Yoon, Heeyoung Kim
Single-pass uncertainty quantification (UQ) methods for classification represent uncertainty by predicting a tractable distribution over the class probability vector. While existing approaches primarily focus on enhancing the expressiveness of this distribution, they often provide limited insight into how predictive uncertainty is structured and aggregated, resulting in weak interpretability. We introduce the *courtroom analogy*, which conceptualizes uncertainty-aware classification as a structured debate among class-specific advocates. Each advocate forms a probabilistic opinion, and a final verdict is reached by aggregating these opinions using input-dependent plausibility weights. In this framework, each advocate's opinion is modeled as a Dirichlet distribution whose concentration parameter is decomposed into shared evidence and class-specific advocacy. This yields a structured mixture of Dirichlet distributions with semantically interpretable parameters. To instantiate this formulation, we propose *Mixture of Dirichlet EXperts* (MoDEX), a single-pass neural architecture that predicts the courtroom parameters, enabling efficient and expressive UQ while explicitly modeling uncertainty aggregation. We demonstrate that MoDEX enjoys strong theoretical properties and achieves state-of-the-art UQ performance across diverse benchmarks, yielding interpretable uncertainty estimates with meaningful semantics.
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Federated Variational Preference Alignment with Gumbel-Softmax Prior for Personalized User Preferences

Jabin Koo, Hoyoung Kim, Minwoo Jang, Jungseul Ok
Federated Learning (FL) offers a privacy-preserving pathway for aligning Large L anguage Models (LLMs); however, existing frameworks typically enforce a monolith ic reward model, inevitably averaging out inherently conflicting user preference s (e.g., helpfulness vs. harmlessness). While Variational Preference Learning (V PL) offers a pathway to personalization, adapting it to decentralized settings p resents a fundamental challenge: \textit{posterior collapse} driven by severe lo cal data scarcity and heterogeneity. In this paper, we propose Federated Variati onal Preference Alignment with Gumbel-Softmax Prior (FedVPA-GP), a framework des igned to disentangle diverse preferences without compromising privacy. To stabil ize variational inference, we introduce a Federated Mixture Prior that enables c lients to leverage the aggregate population distribution as a dynamic prior. Fur thermore, we incorporate an Orthogonal Loss that explicitly enforces the separat ion of preference prototypes in the latent space. Experiments on the HH-RLHF dat aset demonstrate that FedVPA-GP significantly outperforms monolithic baselines, successfully disentangling conflicting user intents and enabling dynamic prefere nce switching.
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Bad Seeing or Bad Thinking? Rewarding Perception for Multimodal Reasoning
Haozhe Wang, Qixin Xu, Changpeng Wang, Taofeng Xue, Chong Peng, Wenhu Chen, Fang zhen Lin
Achieving robust perception-reasoning synergy is a central goal for advanced Vis ion-Language Models (VLMs). Recent advancements have pursued this goal via archi tectural designs or agentic workflows. However, these approaches are often limit ed by static textual reasoning or complicated by the significant compute and eng ineering burden of external agentic complexity. Worse, this heavy investment doe s not yield proportional gains, often witnessing a "seesaw effect" on perception and reasoning. This motivates a fundamental rethinking of the true bottleneck. In this paper, we argue that the root cause of this trade-off is an ambiguity in modality credit assignment: when a VLM fails, is it due to flawed perception (" bad seeing") or flawed logic ("bad thinking")?
To resolve this, we introduce a reinforcement learning framework that improves p erception-reasoning synergy by reliably rewarding the perception fidelity. We ex plicitly decompose the generation process into interleaved perception and reason ing steps. This decoupling enables targeted supervision on perception. Crucially , we introduce Perception Verification (PV), leveraging a "blindfolded reasoning " proxy to reward perceptual fidelity independently of reasoning outcomes. Furth ermore, to scale training across free-form VL tasks, we propose Structured Verba l Verification, which replaces high-variance LLM judging with structured algorit hmic execution. These techniques are integrated into a Modality-Aware Credit Ass ignment (MoCA) mechanism, which routes rewards to the specific source of error - - either bad seeing or bad thinking -- enabling a single VLM to achieve simultan eous performance gains across a wide task spectrum.
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Neural Dispersion on Graphs
Ryien Hosseini, Pouya Mahdi Gholami, Filippo Simini, Venkatram Vishwanath, Rebec ca Willett, Henry Hoffmann
We study the problem of generating structurally diverse graphs on N unlabeled vertices. Given a space of such graphs S_N, metric d, and target cardinality k, the objective is to construct a set $\mathcal{G} \subset S_N$ that maximiz es pairwise diversity under d. While neural generative models may appear appea ling as a solution, standard approaches require samples from a target distributi on that such _dispersion_ problems lack. Thus, prior work relies primarily on co mbinatorial or iterative search. We instead treat diversity as an explicit optim ization objective, an approach we term _Neural Graph Dispersion_. An ensemble of generators is optimized under a repulsive potential, producing diverse graphs a long optimization trajectories as they disperse over (S_N,d), and avoiding dis tribution fitting and per-metric retraining entirely. Experiments show our metho d produces high diversity while scaling N and k an order of magnitude beyond pri or work. Our source code is available at https://github.com/ryienh/neural-graph-

dispersion.
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An Efficient Joint Learning Approach for Item Response Theory
Tanish Agarwal, Kaustubh Shivshankar Shejole, Arpit Agarwal
Item response theory (IRT) is widely used in areas such as recommender systems, education, psychology, and other fields. A popular model for IRT is the Rasch model. Under this model, if a user with ability θ performs a task with difficulty β then its label $X \sim \text{Bernoulli} (1 / (1 + \exp(-(\theta - \beta)))$. Existing joint maximum likelihood estimation approaches for this problem do not perform well on small datasets and also lack theoretical guarantees. Recently, Nguyen and Zhang proposed a two step approach: (1) spectral method for estimation of task parameters, (2) likelihood optimization for estimation of user parameters. While this approach is theoretically sound, it is not computationally efficient. In this work, we propose an EM-based algorithm for joint estimation of item and user parameters by introducing Pólya-Gamma latent variables, which simplify the logistic log-likelihood. We show that our algorithm is both theoretically sound and consistently outperforms existing methods on synthetic and real-world datasets.
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Learning from Fine-Grained Visual Discrepancies: Mitigating Multimodal Hallucinations via In-Context Visual Contrastive Optimization
Haolin Deng, Xin Zou, Zhiwei Jin, Chen Chen, Haonan Lu, Xuming Hu
Multimodal hallucination remains a persistent challenge for Vision-Language Models (VLMs). Standard textual Direct Preference Optimization (DPO) often fails to mitigate it due to a lack of explicit visual supervision. While existing works introduce visual preference DPO by contrasting original images against negative ones, they suffer from a theoretically inconsistent objective caused by partition function mismatches and rely on coarse-grained negatives that could enable shortcut learning. In this work, we propose In-Context Visual Contrastive Optimization (IC-VCO). By placing contrastive images within a shared multi-image context, IC-VCO ensures a mathematically rigorous objective. We further introduce Visual Contrast Distillation (VCDist), an auxiliary reliability-gated regularizer that encourages consistency between multi-image contrastive training and single-image inference. Finally, we propose a contrastive sample editing strategy that generates hard negatives via precise semantic perturbations. Experiments on five benchmarks demonstrate IC-VCO's best overall performance and the effectiveness of our sample editing strategy. Code and data are available at https://github.com/OPPO-Mente-Lab/IC-VCO.
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Unified Time Series Explanations via Amortized Optimization and Instance-level Multi-Expert Knowledge Distillation
Viet-Hung Tran, Zichi Zhang, Ngoc Phu Doan, Xuan Hoang Nguyen, Phi Hung Nguyen, Yimeng An, Peixin Li, Hans Vandierendonck, Ira Assent, Son Thai Mai
Deep neural networks (DNNs) achieve high accuracy on time series classification (TSC) but remain opaque, hindering deployment in sensitive domains.
Existing post-hoc TSC explanation methods rely on a single attribution perspective and incur high per-instance computational cost, limiting real-time use.
We propose XMA, a framework that unifies multiple post-hoc explainers at the instance level and amortizes the cost of explanation into a learned inference pass.
Instance-level Multi-Expert Knowledge Distillation (IMEKD) selects, per instance, the attribution map with the highest combined faithfulness and robustness score from a pool of XAI experts.
Objective-Regularized Amortized Optimization Explanation (ORAOE) trains an explainer DNN to reproduce this instance-best supervision while directly optimizing differentiable faithfulness and robustness objectives, letting the student refine beyond any individual teacher.
Faithfulness-Preserving Segmentation (FPS) converts point-wise maps into contiguous segments without changing their faithfulness score.
Across four synthetic datasets, MIT-ECG with cardiologist-verified ground truth, and 11 multivariate UEA benchmarks on three DNN architectures, XMA attains the

best AUPRC on all synthetic and MIT-ECG benchmarks among compared methods, and t
he highest faithfulness on 7 of 11 and robustness on 8 of 11 UEA datasets, while
 amortizing explanation into a lightweight learned explainer rather than relying
 on per-instance optimization.
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Graph is a Natural Regularization: Revisiting Vector Quantization for Graph Repr
esentation Learning

Zian Zhai, Fan Li, Xingyu Tan, Xiaoyang Wang, Wenjie Zhang

Vector Quantization (VQ) has recently emerged as a promising approach for learni
ng compressed and discrete representations for graph-structured data. However, a
 fundamental challenge, i.e., codebook collapse, remains underexplored in the gr
aph domain, significantly limiting the expressiveness and generalization of grap
h tokens. In this paper, we present an empirical study and observe that codebook
 collapse consistently occurs when training VQ jointly with Graph Neural Network
s under graph reconstruction tasks, even with mitigation strategies proposed in
vision or language domains. Moreover, we provide a diagnosis of collapse from da
ta and optimization perspectives, showing that collapse is associated with graph
 data properties such as feature redundancy and connectivity density, and is fur
ther reinforced by the training dynamics of deterministic hard assignment. To ad
dress these issues, we propose RGVQ, a novel framework that integrates graph top
ology and feature similarity as explicit regularization signals to enhance codeb
ook utilization and promote token diversity. RGVQ introduces soft assignments vi
a Gumbel-Softmax reparameterization, ensuring that all codewords receive gradien
t updates. In addition, RGVQ incorporates a structure-aware contrastive regulari
zation to penalize assigning the same token to dissimilar node pairs. Extensive
experiments demonstrate that RGVQ substantially improves codebook utilization an
d consistently boosts the performance of state-of-the-art graph VQ backbones acr
oss multiple downstream tasks, enabling more expressive and transferable graph t
oken representations.
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Strategic Navigation or Stochastic Search? How Agents and Humans Reason Over Doc
ument Collections

■ukasz Borchmann, Jordy Van Landeghem, Micha■ Turski, Shreyansh Padarha, Ryan Ot
hniel Kearns, Adam Mahdi, Niels Rogge, Clémentine Fourrier, Siwei Han, Huaxiu Ya
o, Artemis Llabrés, Yiming Xu, Dimosthenis Karatzas, Hao Zhang, Anupam Datta

Multimodal agents offer a promising path to automating complex document-intensiv
e workflows. Yet, a critical question remains: do these agents demonstrate genui
ne strategic reasoning, or merely stochastic trial-and-error search? To address
 this, we introduce MADQA, a benchmark of 2,250 human-authored questions grounde
d in 800 heterogeneous PDF documents. Guided by Classical Test Theory, we design
 it to maximize discriminative power across varying levels of agentic abilities.
 To evaluate agentic behavior, we introduce a novel protocol that measures the a
ccuracy-effort trade-off. Using this framework, we show that while the best agen
ts can match human searchers in raw accuracy, they succeed on largely different
questions and rely on brute-force search to compensate for weak strategic planni
ng. They fail to close the nearly 20\% gap to oracle performance, persisting in
unproductive loops. We release the dataset and evaluation harness to help facili
tate the transition from brute-force retrieval to calibrated, efficient reasonin
g.
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Self-Guidance: Enhancing Neural Codecs via Decoder Manifold Alignment

Xiang Li, Yixuan Zhou, Jingran Xie, Zhiyong Wu, Hui Wang

Neural speech codecs based on Vector-Quantized VAEs (VQ-VAEs) are core audio tok
enizers for speech LLMs, yet their reconstruction fidelity is bottlenecked by qu
antization error. Modifying the quantizer or increasing model capacity are commo
n fixes, but they complicate downstream language modeling. Our core idea is to a
lign the decoder's internal feature manifolds when processing both the quantized
 tokens and their original continuous embeddings, using a lightweight feature-ma
pping loss. This requires minimal training overhead and no inference-time change
s. Applied to XCodec2, self-guidance improves all reconstruction metrics, achiev

ing state-of-the-art low-bitrate performance. Notably, it enables a 4× codebook reduction without fidelity loss, which downstream TTS experiments show significa ntly improves LLM-based synthesis by simplifying the token modeling space. Multi ple statistical observations and visualizations corroborate the enhanced interna l manifold alignment in the decoder. Extensive experiments confirm its generalit y across various inductive biases. Self-guidance thus establishes an efficient, broadly applicable method for high-fidelity neural audio coding.
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Learning GUI Grounding with Spatial Reasoning from Visual Feedback
Yu Zhao, Wei-Ning Chen, Huseyin A Inan, Samuel Kessler, Lu Wang, Lukas Wutschitz , Fangkai Yang, Chaoyun Zhang, Pasquale Minervini, Saravan Rajmohan, Robert Sim
Graphical User Interface (GUI) grounding is commonly framed as a coordinate pred iction task – given a natural language instruction, generate on-screen coordinat es for actions such as clicks and keystrokes. However, recent Vision Language Mo dels (VLMs) often fail to predict accurate numeric coordinates when processing G UI images with high resolutions and complex layouts. To address this issue, we r eframe GUI grounding as an interactive search task, where the VLM generates acti ons to move a cursor in the GUI to locate UI elements. At each step, the model d etermines the target object, evaluates the spatial relations between the cursor and the target, and moves the cursor closer to the target conditioned on the mov ement history. In this interactive process, the rendered cursor provides visual feedback to help the model align its predictions with the corresponding on-scree n locations. We train our GUI grounding model, GUI-Cursor, using multi-step onli ne reinforcement learning with a dense trajectory-based reward function. Experim ental results demonstrate that GUI-Cursor surpasses strong baselines in GUI grou nding and agentic tasks, achieving superior performance with the same base model s while requiring less training data. Further analysis shows that GUI-Cursor lea rns to adaptively conduct more steps on more difficult examples, and it obtains better spatial reasoning capability on out-of-distribution domains.
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AugServe: Adaptive Request Scheduling for Augmented Large Language Model Inferen ce Serving
Ying Wang, Zhen Jin, Zhenqian Chen, Jiexiong Xu, Wenhai Lin, Yiquan Chen, Wenzhi CHEN
Augmented large language models (LLMs) that invoke external calls are increasing ly prevalent in inference serving.
However, such augmentations pose significant challenges to inference efficiency under strict Service-Level Objectives (SLOs). Existing inference systems are agn ostic to the dynamic execution behaviors induced by external calls and rely on f ixed batch-level token budget, which leads to severe Head-of-Line (HoL) blocking and substantially reduced effective throughput.
We present AugServe, an efficient augmented LLM inference serving framework that mitigates request queuing latency and improves effective throughput under external-call-augmented workloads. AugServe integrates state-aware request sched uling with dynamic batch-level token budgets to adapt to heterogeneous requests and their dynamically changing execution states. Experimental results show that AugServe achieves 6.5$\times$ and 4.7$\times$ higher effective throughput than v LLM and INFERCEPT, respectively.
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CauchyNet: Compact and Data-Efficient Learning using Holomorphic Activation Func tions
Hong-Kun Zhang, Xin Li, Sikun Yang, Zhihong Jeff Xia
CauchyNet is a compact complex-valued network whose hidden units form products o f shifted reciprocal features, motivated by Cauchy-type kernel representations. The design targets regression problems with sharp rational-like spikes and parti ally observed inputs, where standard real-valued networks often require large wi dth.
We prove that finite linear combinations of multivariate Cauchy kernels are dens e in $C(M)$ on compact $M\subset\mathbb R^N$, and that CauchyNet can realize the se kernel sums via its complex biases and output weights.

In experiments, CauchyNet reaches lower error on near-singular and gap-filling b
enchmarks, often with substantially fewer trainable parameters in our settings;
results on smooth and piecewise-affine targets are mixed and delineate the metho
d's intended regime.
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One-Step Gradient Delay is Not a Barrier for Large-Scale Asynchronous Pipeline P
arallel LLM Pretraining
Philip Zmushko, Egor Petrov, Nursultan Abdullaev, Khrushchev Mikhail, Samuel Hor
váth
Modern large-scale LLM pretraining benefits from utilizing Pipeline Parallelism;
 however, synchronous implementations leave GPUs idle during pipeline bubbles, w
asting computational resources.
Asynchronous Pipeline Parallelism eliminates these bubbles, maximizing throughpu
t at the cost of gradient staleness.
Among asynchronous schedules, PipeDream-2BW is particularly appealing: unlike th
e original PipeDream schedule, it ensures a constant one-step gradient delay reg
ardless of pipeline depth.
However, its adoption remains limited due to the common belief that optimizing u
nder staleness is fundamentally unstable.
In this work, we challenge this assumption, demonstrating that degradation under
 one-step delay depends strongly on optimizer choice rather than being an intrin
sic limitation.
We provide the first comprehensive empirical analysis showing that while AdamW,
the predominant optimizer at the time when PipeDream-2BW was introduced, indeed
suffers from severe degradation, recent methods like Muon exhibit strong robustn
ess under a one-step delay.
We introduce an optimizer-agnostic Error Feedback-inspired correction to further
 mitigate delay effects.
We provide supporting theoretical analysis demonstrating convergence for Muon wi
th and without this correction.
Extensive evaluation on models up to 10B parameters confirms that our strategies
 bridge the performance gap with synchronous training, highlighting the practica
l potential of asynchronous pipeline parallelism at scale.
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XYZFlow: Scaling Multidimensional Shortcut Flows for Efficient Generative Modeli
ng
Jinxiu Liu, Xuanming Liu, Kangfu Mei, Yandong Wen, Weiyang Liu
High-fidelity image generation faces a trade-off between speed and quality. Diff
usion models produce strong visuals but require costly iterative sampling. Exist
ing efficient methods mainly distill pretrained models into few-step samplers, a
 challenging process that depends heavily on teacher-model quality. In this pape
r, we introduce XYZFlow, a framework that rethinks efficient generation through
multidimensional scaling of flow matching. Unlike single-step mappings, XYZFlow
enhances expressivity by making probability paths more identifiable and learnabl
e through structured multidimensional conditioning. We view autoregressive model
ing as implicit flow straightening, where richer context reduces trajectory ambi
guity. XYZFlow realizes this idea through two orthogonal dimensions: temporal sc
aling, which uses non-Markovian conditioning on the full denoising history; and
spatial scaling, enabled by Next Shortcut Prediction, which sequentially generat
es patches using preceding patches' denoising trajectories as priors. Experiment
s show that XYZFlow achieves state-of-the-art performance, with 7.2-8.5x teacher
 speedups and competitive FID, while Next Shortcut Prediction delivers superior
quality-latency trade-offs over model scaling or step reduction.
****
Tracing the Persona Circuit: How Large Language Models Encode and Express Charac
ter Traits
Guanzheng Qin, Chenghao Sun, Zhining Xie, Xinmei Tian
Large Language Models (LLMs) demonstrate remarkable potential in role-playing ta
sks but frequently suffer from personality decay—termed "Out-of-Character" (OOC)
 behavior—during prolonged interactions.

While heuristic strategies exist to align model behaviors, the internal computational dynamics driving personality expression remain opaque.
A fundamental barrier to decoding these mechanisms is a *metric gap*: while standard causal attribution paradigms target atomic, single-token outcomes, personality manifests as a holistic, multi-token behavioral tendency. We bridge this gap via the *Latent Persona Vector*, a differentiable proxy enabling the first fine-grained causal tracing of personality circuits. This metric reveals a structured "Preparation-Establishment-Expression" dynamic
and identifies a mechanistic contributor to OOC behavior: competition between persona-specific signals and an
assistant-like default direction during the critical "Establishment" phase.
Guided by this diagnosis, we propose surgically recalibrating the signal magnitude in fewer than $5\\%$ of attention heads. This targeted intervention effectively strengthens the persona signal, significantly restoring character consistency while preserving general reasoning capabilities.
****
Agentic Monte Carlo: Simulating Reinforcement Learning for Black-Box Agents
Dae Yon Hwang, Raunaq Suri, Valentin Villecroze, Anthony L. Caterini, Jesse C. Cresswell, Noël Vouitsis, Brendan Leigh Ross
LLM agents operate in two distinct regimes: open-weight agents amenable to reinforcement learning (RL) and black-box agents whose behaviour must be controlled purely at test time. Although black-box agents are often backed by state-of-the-art proprietary LLMs, API-only access precludes parameter-level optimization, rendering most RL methods inapplicable. To address this limitation, we turn to a known equivalence between RL and Bayesian inference. We propose Agentic Monte Carlo (AMC) to directly sample from the optimal policy of a black-box agent rather than training it through RL. The optimal policy is a posterior over trajectories whose prior we define as the fixed black-box LLM agent. We employ Sequential Monte Carlo to sample from this posterior by learning a value function to steer the agent while leaving the underlying black-box model unchanged. We validate AMC on three diverse environments from the AgentGym benchmark, demonstrating significant improvements over prompting baselines and even outperforming Group Relative Policy Optimization (GRPO) as we scale the test-time compute of our method. AMC demonstrates the feasibility of performing principled RL-style optimization of black-box LLM agents.
****
The Easy, the Hard, and the Learnable: Confidence and Difficulty-Adaptive Policy Optimization for LLM Reasoning
Zhanke Zhou, Xiangyu Lu, Chentao Cao, Brando Miranda, Tongliang Liu, Bo Han, Sanmi Koyejo
RL with verifiable rewards can substantially improve LLM reasoning, yet standard GRPO-style training often treats easy, hard, and learnable questions alike through uniform sampling and weighting, leading to inefficient compute allocation. We study GRPO by tracking token log-probabilities, group-normalized advantages, and the induced token-level update weights. This reveals three recurring dynamics as training proceeds: (1) confidence inflation, (2) advantage contraction, and (3) hierarchical convergence. These findings suggest that the utility of each update depends strongly on both question difficulty and the model’s current competence. Motivated by this, we propose Confidence and Difficulty-adaptive Policy Optimization (CoDaPO), which assigns each question a bounded value from rollout confidence and empirical difficulty. CoDaPO then uses this value to reweight policy updates and resample high-value learnable questions within mini-batches, thereby increasing discovery within the learnable band under a fixed compute budget. Across twelve benchmarks, CoDaPO consistently improves accuracy over existing RL methods. Our code is publicly available at https://github.com/tmlr-group/CoDaPO.
****
Improved Bounds for Reward-Agnostic and Reward-Free Exploration
Oran Ridel, Alon Cohen
We study *reward-free* and *reward-agnostic* exploration in episodic finite-hori

zon Markov decision processes (MDPs), where an agent explores an unknown environment without observing external rewards.
Reward-free exploration aims to enable ϵ-optimal policies for *any* reward revealed after exploration, while reward-agnostic exploration targets ϵ-optimality for rewards drawn from a small finite class.
In the *reward-agnostic setting*, Li, Yan, Chen, and Fan (2024) achieve minimax sample complexity, but only for restrictively small accuracy parameter ϵ.
We propose a new algorithm that significantly relaxes the requirement on ϵ.
Our approach is novel and of technical interest by itself.
Our algorithm employs an online learning procedure with carefully designed rewards to construct an exploration policy, which is used to gather data sufficient for accurate dynamics estimation and subsequent computation of an ϵ-optimal policy once the reward is revealed.
Finally, we establish a tight lower bound for *reward-free exploration*, closing the gap between known upper and lower bounds.
****

Restoring Initial Noise Sensitivity in Text-to-Image Distillation through Geometric Alignment
Huayang Huang, Ruoyu Wang, Jinhui Zhao, Wei Deng, Daiguo Zhou, Jian Luan, Yu Wu, Ye Zhu
Generative distillation significantly accelerates text-to-image (T2I) generation by compressing multi-step trajectories into few-step student models while preserving perceptual quality. However, existing methods primarily optimize efficiency and output fidelity, often neglecting critical properties of the original trajectory. In this work, we identify a key missing property: sensitivity to initial noise, whose degradation impairs downstream control methods relying on noise-based optimization and manipulation. We trace this issue to standard distillation objectives that enforce pointwise output alignment, inadvertently flattening the input-output landscape and suppressing the teacher’s local geometric structure. To address this, we propose Geometry-Aware Distillation (GAD), a sensitivity-preserving framework that aligns the local functional behavior of teacher and student models. Specifically, GAD matches Jacobian-vector products with respect to input noise, enabling the student to reproduce the teacher’s differential response to perturbations. Extensive experiments across multiple T2I paradigms and noise-driven control tasks demonstrate that GAD significantly restores sensitivity and improves diversity while maintaining high visual fidelity. Code is available at https://github.com/Hannah1102/GAD.
****

Learning Hamiltonian Dynamics at Scale: A Differential-Geometric Approach
Katharina Friedl, Noémie Jaquier, Alyx Liao, Danica Kragic
Embedding physical intuition into network architectures allows the learning of dynamics that enforce fundamental properties, such as energy conservation laws, thereby leading to physically-plausible predictions. Yet, scaling these models to intrinsically high-dimensional dynamical systems remains a significant challenge. This paper introduces Reduced-order Hamiltonian Neural Network (RO-HNN), a novel physics-inspired neural network that combines the conservation laws of Hamiltonian mechanics with the scalability of model order reduction. RO-HNN is built on two core components: a novel geometrically-constrained symplectic autoencoder that learns a low-dimensional, structure-preserving symplectic submanifold, and a geometric Hamiltonian neural network that models the dynamics on the submanifold. Our experiments demonstrate that RO-HNN provides physically-consistent, stable, and generalizable predictions of complex high-dimensional dynamics, thereby effectively extending the scope of Hamiltonian neural networks to high-dimensional physical systems.
****

DECOR: Learning to Decompose and Collaborate in Deep Search via Multi-Agent Reinforcement Learning
Ruiqing Chen, Zekun Zhang, Gong-Duo Zhang, Lihong Gu, Lin Zhou

Monolithic agents in deep search often suffer from "cognitive overload," while existing multi-agent approaches mostly rely on frozen models that cannot learn from collaboration failures. To bridge this gap, we propose $\textbf{DECOR}$ ($\textbf{DE}$compose and $\textbf{CO}$llaborate via $\textbf{R}$ole-specialized agents), a framework formulating deep search as a Multi-Agent Reinforcement Learning (MARL) problem. DECOR functionally decomposes the task into three specialized roles: a $\textit{Planner}$ to navigate, a $\textit{Filter}$ to curate a noise-reduced memory, and an $\textit{Answerer}$ for synthesis. Unlike training-free orchestration, we jointly optimize these agents using a hybrid reward strategy that harmonizes role-specific intrinsic feedback with team-level outcome signals. Experiments on seven benchmarks show that DECOR significantly outperforms strong monolithic baselines, demonstrating the necessity of learning-based functional decomposition in handling cognitive overload.
****

Richer Bayesian Last Layers with Subsampled NTK Features
Sergio Calvo Ordoñez, Jonathan Plenk, Richard Bergna, Alvaro Cartea, Yarin Gal, José Miguel Hernández-Lobato, Kamil Ciosek
Bayesian Last Layers (BLLs) provide a convenient and computationally efficient way to estimate uncertainty in neural networks. However, they underestimate epistemic uncertainty because they apply a Bayesian treatment only to the final layer, ignoring uncertainty induced by earlier layers. We propose a method that improves BLLs by leveraging a projection of Neural Tangent Kernel (NTK) features onto the space spanned by the last-layer features. This enables posterior inference that accounts for variability of the full network while retaining the low computational cost of inference of a standard BLL. We show that our method yields posterior variances that are provably greater or equal to those of a standard BLL, correcting its tendency to underestimate epistemic uncertainty.
To further reduce computational cost, we introduce a uniform subsampling scheme for estimating the projection matrix and for posterior inference. We derive approximation bounds for both types of subsampling. Empirical evaluations on UCI regression, contextual bandits, image classification, and out-of-distribution detection tasks in image and tabular datasets, demonstrate improved calibration and uncertainty estimates compared to standard BLLs and competitive baselines, while reducing computational cost.
****

Towards Universal Gene Regulatory Network Inference: Unlocking Generalizable Regulatory Knowledge in Single-cell Foundation Models
Jiaxin Qi, Hang Li, Yan Cui, Yuhua Zheng, Jianqiang Huang
Gene Regulatory Network (GRN) inference is essential for understanding complex cellular mechanisms, rendered tractable through single-cell transcriptomic data. With the emergence of single-cell Foundation Models (scFMs), enhanced transcriptomic encoding is widely expected to revolutionize GRN inference. However, we observe that their performance remains far from satisfactory. The primary reason is that the standard reconstruction-based pre-training objectives often fail to explicitly capture latent regulatory signals. To bridge this gap, we first introduce a GRN generalization benchmark designed to evaluate regulatory predictions on unseen genes and datasets, which relies on the zero-shot capabilities of scFMs and is inherently challenging for traditional methods. Furthermore, to unlock the regulatory knowledge within the foundation models, we propose two novel methods, Virtual Value Perturbation and Gradient Trajectory, to distill implicit regulatory information from scFMs into highly generalizable inter-gene features. Extensive experiments demonstrate that our approach significantly outperforms existing methods, establishing a new paradigm for leveraging the potential of scFMs in universal GRN inference.
****

Incremental Transformer Neural Processes
Philip Mortimer, Cristiana Diaconu, Tommy Rochussen, Bruno Kacper Mlodozeniec, Richard E. Turner
Neural Processes (NPs), and specifically Transformer Neural Processes (TNPs), have demonstrated remarkable performance across tasks ranging from spatiotemporal

forecasting to tabular data modelling. However, many of these applications are i
nherently sequential, involving continuous data streams such as real-time sensor
 readings or database updates. In such settings, models should support cheap, in
cremental updates rather than recomputing internal representations from scratch
for every new observation—a capability existing TNP variants lack. Drawing inspi
ration from Large Language Models, we introduce the Incremental TNP ($\texttt{in
cTNP}$). By leveraging causal masking, Key-Value (KV) caching, and a data-effici
ent autoregressive training strategy, $\texttt{incTNP}$ matches the predictive p
erformance of standard TNPs while reducing the computational cost of updates fro
m quadratic to linear time complexity. We empirically evaluate our model on a ra
nge of synthetic and real-world tasks, including tabular regression and temperat
ure prediction. Our results show that, surprisingly, $\texttt{incTNP}$ delivers
performance comparable to—or better than—non-causal TNPs while unlocking orders-
of-magnitude speedups for sequential inference. Finally, we assess the consisten
cy of the model's updates---by adapting a metric of "implicit Bayesianness", we
show that under a one-at-a-time streaming protocol, $\texttt{incTNP}$ retains a
prediction rule as implicitly Bayesian as standard non-causal TNPs, demonstratin
g that $\texttt{incTNP}$ achieves the computational benefits of causal masking w
ithout sacrificing the consistency required for streaming inference.
****

Scaling by Diversified Experience for Vision-Language-Action Models
Leiyu Wang, Zhaofengnian Wang, Xueqi Li, Luoyi Fan, Cewu Lu, Nanyang Ye
Vision-Language-Action models face significant challenges in real-world deployme
nt due to the entanglement of high-level reasoning with low-level control, and t
he instability of policy optimization. In this paper, we introduce SyVLA, a robu
st VLA model trained with diversified experiences. We propose an Intention Decou
pling algorithm to isolate control-relevant features from reasoning contexts and
 a similar-sample guided RL pipeline to stabilize policy updates and mitigate di
stribution shift. Extensive experiments on real-world robotic tasks and multi-mo
dal benchmarks demonstrate that SyVLA achieves superior task success rates and s
tronger out-of-distribution generalization compared to existing methods, while e
ffectively preserving core vision-language capabilities.
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Well-Posed KL-Regularized Control via Wasserstein and Kalman–Wasserstein KL Dive
rgences
Viktor Stein, Adwait Datar, Nihat Ay
Kullback-Leibler (KL) divergence regularization is widely used in reinforcement
learning, but it becomes infinite under support mismatch and can degenerate in l
ow-noise regimes. Using a unified information-geometric framework, we introduce
KL analogs by replacing the Fisher–Rao geometry in the dynamical formulation of
the KL with transport-based geometries, and derive closed-form expressions for c
ommon distribution families. Between elliptic distributions, these divergences r
emain finite for degenerating equal covariances and yield a geometric interpreta
tion of regularization heuristics used in Kalman ensemble methods. We demonstrat
e the utility of these divergences in KL-regularized optimal control. In the ful
ly tractable setting of linear time-invariant systems with Gaussian process nois
e, the classical KL reduces to a quadratic control penalty that becomes singular
 as process noise vanishes. Our variants remove this singularity and yield well-
posed problems. On a double integrator and a cart-pole example, the resulting co
ntrols preserve nontrivial feedback and achieve better closed-loop performance.
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Who’s in Charge? Disempowerment Patterns in Real-World LLM Usage
Mrinank Sharma, Miles McCain, Raymond Douglas, David Duvenaud
We present the first large-scale empirical analysis of disempowerment patterns i
n real-world AI assistant interactions, analyzing 1.5 million consumer Claude.ai
 conversations using a privacy-preserving approach. We focus on situational dis-
empowerment potential, which occurs when AI assistant interactions risk leading
users to form distorted perceptions of reality, make inauthentic value judgments
, or act in ways misaligned with their values. Quantitatively, we find that seve
re forms of disempowerment potential occur in fewer than one in a thousand conve

rsations, though rates are substantially higher in personal domains like relatio nships and lifestyle. Qualitatively, we uncover several concerning patterns, suc h as validation of persecution narratives and grandiose identities with emphatic sycophantic language, definitive moral judgments about third parties, and compl ete scripting of value-laden personal communications that users appear to implem ent verbatim. Analysis of historical trends reveals an increase in the prevalenc e of disempowerment potential over time. We also find that interactions with gre ater disempowerment potential receive higher user approval ratings, possibly sug gesting a tension between short-term user preferences and long-term human empowe rment.
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AI4SLT: Empirical Processes in Lean 4 for Formal Statistical Learning Theory
Yuanhe Zhang, Jason D. Lee, Fanghui Liu
We present the first comprehensive Lean 4 formalization of statistical learning theory (SLT) grounded in empirical process theory. Our end-to-end formal infrast ructure implement the missing contents in latest Lean library, including a compl ete development of Gaussian Lipschitz concentration, Dudley's entropy integral t heorem for sub-Gaussian processes, and an application to least-squares (sparse) regression with a sharp rate. The project was carried out using a human-AI colla borative workflow, in which humans design proof strategies and AI agents execute tactical proof construction, leading to the human-verified Lean 4 toolbox for S LT. Beyond implementation, the formalization process exposes and resolves implic it assumptions and missing details in standard SLT textbooks, enforcing a granul ar, line-by-line understanding of the theory. This work establishes a reusable f ormal foundation and opens the door for future developments in machine learning theory. The code is provided in https://github.com/YuanheZ/lean-stat-learning-th eory.
****

ECO: Quantized Training without Full-Precision Master Weights
Mahdi Nikdan, Amir Zandieh, Dan Alistarh, Vahab Mirrokni
Quantization has significantly improved the compute and memory efficiency of Lar ge Language Model (LLM) training. However, existing approaches still rely on acc umulating their updates in high-precision: concretely, gradient updates must be applied to a high-precision weight buffer, known as $\textit{master weights}$. T his buffer introduces substantial memory overhead, particularly for Sparse Mixtu re of Experts (SMoE) models, where model parameters and optimizer states dominat e memory usage. To address this, we introduce the Error-Compensating Optimizer (ECO), which eliminates master weights by applying updates directly to quantized parameters. ECO quantizes weights after each step and carefully injects the resu lting quantization error into the optimizer momentum, forming an error-feedback loop with no additional memory. We prove that, under standard assumptions and a decaying learning rate, ECO converges to a constant-radius neighborhood of the o ptimum, while naive master-weight removal can incur an error that is inversely p roportional to the learning rate. We show empirical results for pretraining smal l Transformers (30--800M), a Gemma-3 1B model, and a 2.1B parameter Sparse MoE m odel with FP8 quantization, and fine-tuning DeepSeek-MoE-16B in INT4 precision. Throughout, ECO matches baselines with master weights up to near-lossless accura cy, significantly shifting the static memory vs validation loss Pareto frontier.
****

When to Trust the Cheap Check: Weak and Strong Verification for Reasoning
Shayan Kiyani, Sima Noorani, George J. Pappas, Hamed Hassani
Reasoning with LLMs increasingly unfolds inside a broader verification loop. Int ernally, systems use cheap checks, such as self-consistency or proxy rewards, wh ich we call **weak verification**. Externally, users inspect outputs and steer t he model through feedback until results are trustworthy, which we call **strong verification**. These signals differ sharply in cost and reliability: strong ver ification can establish trust but is resource-intensive, while weak verification is fast and scalable but noisy and imperfect. We formalize this tension through **weak-strong verification policies**, which decide when to accept or reject ba sed on weak verification and when to defer to strong verification. We introduce

metrics capturing incorrect acceptance, incorrect rejection, and strong-verification frequency. Over population, we show that optimal policies admit a two-threshold structure and that **calibration** and **sharpness** govern the value of weak verifiers. Building on this, we develop an online algorithm that provably controls acceptance and rejection errors without assumptions on the query stream, the language model, or the weak verifier. Experiments on mathematical reasoning and sequential decision-making demonstrate that our algorithm achieves reliability comparable to exhaustive strong verification while significantly reducing verification cost.
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When Simple Problems Wear Complex Costumes: Improving Efficiency in LRM's Adaptive Reasoning
Junnan Ren, Yan Zhang, Qian Chen, Yunhang Shen, Ke Li, Shengchuan Zhang, Liujuan Cao, Rongrong Ji
Recent Large Reasoning Models (LRMs) have demonstrated powerful multi-step problem-solving capabilities but often suffer from inefficiency due to an ``overthinking phenomenon", where they apply complex reasoning to simple tasks, resulting in unnecessary computational cost and latency. While adaptive reasoning models that can switch between generating explicit reasoning and producing direct answers offer a potential solution, their effectiveness is compromised by a critical flaw: they are often misled by superficial linguistic complexity, mistaking verbosely phrased simple problems for complex ones. To address this, we propose a two-stage training framework to create a more robust adaptive reasoner. The first stage uses supervised fine-tuning with augmented data—presenting simple problems in both concise and redundant forms—to teach the model to ignore superficial verbosity. Subsequently, a reinforcement learning phase utilizes Group Relative Policy Optimization (GRPO) with a custom reward function to refine the model's adaptive policy, ensuring it selects a reasoning mode based on true task complexity rather than surface-level cues. The resulting model reduces computational overhead without sacrificing accuracy and demonstrates improved robustness to misleading linguistic cues.
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MAGIC: A Co-Evolving Attacker–Defender Adversarial Game for Robust LLM Safety
Xiaoyu Wen, Zhida He, Han Qi, Ziyu Wan, Zhongtian Ma, Ying Wen, Tianhang Zheng, Xingcheng Xu, Chaochao Lu, Qiaosheng Zhang
Ensuring robust safety alignment is crucial for Large Language Models (LLMs), yet existing defenses often lag behind evolving adversarial attacks due to their \textbf{reliance on static, pre-collected data distributions}. In this paper, we introduce \textbf{MAGIC}, a novel multi-turn multi-agent reinforcement learning framework that formulates LLM safety alignment as an adversarial asymmetric game. Specifically, an attacker agent learns to iteratively rewrite original queries into deceptive prompts, while a defender agent simultaneously optimizes its policy to recognize and refuse such inputs. This dynamic process triggers a \textbf{co-evolution}, where the attacker's ever-changing strategies continuously uncover long-tail vulnerabilities, driving the defender to generalize to unseen attack patterns. Remarkably, we observe that the attacker, endowed with initial reasoning ability, evolves \textbf{novel, previously unseen combinatorial strategies} through iterative RL training, underscoring our method’s substantial potential. Theoretically, we provide insights into a more robust game equilibrium and derive safety guarantees. Extensive experiments validate our framework's effectiveness, demonstrating superior defense success rates without compromising the helpfulness of the model.
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Non-Parametric Probabilistic Robustness: A Conservative Risk Estimator under Unknown Perturbation Distributions
Zheng Wang, Yi Zhang, Siddartha Khastgir, carsten maple, Xingyu Zhao
Deep learning (DL) models, despite their remarkable success, remain vulnerable to small input perturbations that can cause erroneous outputs, motivating the recent proposal of probabilistic robustness (PR) as a complementary alternative to adversarial robustness (AR). However, existing PR formulations assume a fixed an

d known perturbation distribution, an unrealistic expectation in practice. To ad
dress this limitation, we propose non-parametric probabilistic robustness (NPPR)
, a more practical PR metric that does not rely on any predefined perturbation d
istribution. Following the non-parametric paradigm in statistical modeling, NPPR
 learns an optimized perturbation distribution directly from data, enabling cons
ervative PR evaluation under distributional uncertainty. We further develop an N
PPR estimator based on a Gaussian Mixture Model (GMM), covering various input-de
pendent and input-independent perturbation scenarios. Theoretical analyses estab
lish the relationships among AR, PR, and NPPR. Extensive experiments on CIFAR-10
, CIFAR-100, and Tiny ImageNet across ResNet18/50, WideResNet50 and VGG16 valida
te NPPR as a more practical robustness metric, showing conservative (lower) PR e
stimates compared to assuming those common perturbation distributions used in st
ate-of-the-arts.
****

The Velocity Deficit: Initial Energy Injection for Flow Matching
Linze Li, Zong-Wei Hong, Shen Zhang, Bo Lin, Jinglun Li, Yao Tang, Jiajun Liang
While Flow Matching theoretically guarantees constant-velocity trajectories, we
identify a critical breakdown in high-dimensional practice: the Velocity Deficit
. We show that the MSE objective systematically underestimates velocity magnitud
e, causing generated samples to fail to reach the data manifold—a phenomenon we
term Integration Lag. To rectify this, we propose Initial Energy Injection, inst
antiated via two complementary methods: the training-based Magnitude-Aware Flow
Matching (MAFM) and the training-free Scale Schedule Corrector (SSC). Both are g
rounded in our discovery of a crucial asymmetry: velocity contraction causes har
mful kinetic stagnation at the trajectory's start, yet acts as a beneficial deno
ising mechanism at its end. Empirically, SSC yields significant efficiency gains
 with zero retraining and just one line of code. On ImageNet-1k (256x256), it im
proves FID by 44.6% (from 13.68 to 7.58) and achieves a 5x speedup, enabling a 5
0-step generator (FID 7.58) to beat a 250-step baseline (FID 8.65). Furthermore,
 our methods generalize to Text-to-Image tasks and high-resolution generation, i
mproving FID on MS-COCO by ~22%.
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Dreaming in Code for Curriculum Learning in Open-Ended Worlds
Konstantinos Mitsides, Maxence Faldor, Antoine Cully
Open-ended learning frames intelligence as emerging from continual interaction w
ith an ever-expanding space of environments. While recent advances have utilized
 foundation models to programmatically generate diverse environments, these appr
oaches often focus on discovering isolated behaviors rather than orchestrating s
ustained progression. In complex open-ended worlds, the large combinatorial spac
e of possible challenges makes it difficult for agents to discover sequences of
experiences that remain consistently learnable. To address this, we propose Drea
ming in Code (DiCode), a framework in which foundation models synthesize executa
ble environment code to scaffold learning toward increasing competence. In DiCod
e, “dreaming” takes the form of materializing code-level variations of the world
. We instantiate DiCode in Craftax, a challenging open-ended benchmark character
ized by rich mechanics and long-horizon progression. Empirically, DiCode enables
 agents to acquire long-horizon skills, achieving a 17% improvement in mean retu
rn over the strongest baseline and non-zero success on late-game combat tasks wh
ere prior methods fail. Our results suggest that code-level environment design p
rovides a practical mechanism for curriculum control, enabling the construction
of intermediate environments that bridge competence gaps in open-ended worlds.
****

Post-Training LLMs as Better Decision-Making Agents: A Regret-Minimization Appro
ach
Chanwoo Park, Ziyang Chen, Asuman E. Ozdaglar, Kaiqing Zhang
Large language models (LLMs) are increasingly deployed as agents for decision-ma
king (DM) in interactive and dynamic environments. However, since they are not o
riginally designed for DM, recent studies show that LLMs struggle in basic onlin
e DM settings. We introduce ITERATIVE REGRET-MINIMIZATION FINE-TUNING (ITERATIVE
 RMFT), a post-training procedure that repeatedly distills low-regret decision t

rajectories into the base model. Unlike prior methods that rely on distilling known algorithms or enforcing manually designed reasoning formats, our approach leverages regret as a training signal to elicit improved decision-making behavior while incorporating model-generated reasoning in natural language. Empirically, ITERATIVE RMFT improves DM performance across models, including numerical Transformers, lightweight open-weight LLMs, and the closed-weight model GPT-4o mini, while exhibiting generalization across varying horizons, action spaces, reward processes, and natural-language-described DM scenarios. Overall, we position our approach as an initial exploration, calling for more principled and novel post-training paradigms for LLMs when it comes to addressing DM tasks.
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ECG-R1: Protocol-Guided and Modality-Agnostic MLLM for Reliable ECG Interpretation
Jiarui Jin, Haoyu Wang, Xingliang Wu, Xiaocheng Fang, Xiang Lan, Zihan Wang, Deyun Zhang, Bo Liu, Yingying Zhang, Xian Wu, Hongyan Li, Shenda Hong
Electrocardiography (ECG) serves as an indispensable diagnostic tool in clinical practice, yet existing multimodal large language models (MLLMs) remain unreliable for ECG interpretation, often producing plausible but clinically incorrect analyses. To address this, we propose ECG-R1, the first reasoning ECG MLLM designed for reliable ECG interpretation via three innovations. First, we construct the interpretation corpus using \textit{Protocol-Guided Instruction Data Generation}, grounding interpretation in measurable ECG features and monograph-defined quantitative thresholds and diagnostic logic. Second, we present a modality-decoupled architecture with \textit{Interleaved Modality Dropout} to improve robustness and cross-modal consistency when either the ECG signal or ECG image is missing. Third, we present \textit{Reinforcement Learning with ECG Diagnostic Evidence Rewards} to strengthen evidence-grounded ECG interpretation. Additionally, we systematically evaluate the ECG interpretation capabilities of proprietary, open-source, and medical MLLMs, and provide the first quantitative evidence that severe hallucinations are widespread, suggesting that the public should not directly trust these outputs without independent verification. Code is available at \href{https://github.com/PKUDigitalHealth/ECG-R1}{here}.
****

BlitzRank: Principled Zero-shot Ranking Agents with Tournament Graphs
Sheshansh Agrawal, Thien Hang Nguyen, Douwe Kiela
Selecting the top m from n items via expensive k-wise comparisons is central to settings ranging from LLM-based document reranking to crowdsourced evaluation and tournament design.
Existing methods either rely on heuristics that discard comparison information, or exploit it at prohibitive cost.
We introduce a *tournament graph* framework that provides a principled foundation for k-wise ranking.
Our key observation is that each k-item comparison reveals an induced tournament of ${k \choose 2}$ pairwise preferences; aggregating these into a global preference graph and computing its transitive closure yields many additional orderings without further oracle calls.
We formalize when the current top-m output is *certifiably determined* and design a greedy query schedule that maximizes information gain towards identifying the top-m items.
The framework also gracefully handles non-transitive preferences -- cycles induced by real-world oracles -- by collapsing them into equivalence classes that yield principled *tiered rankings*.
Applied to LLM reranking across 14 benchmarks and 5 models, BlitzRank achieves Pareto dominance over existing approaches: matching or exceeding accuracy while requiring 25--40\% fewer tokens than comparable methods; against pairwise reranking, it achieves near-identical quality with 7$\times$ fewer tokens. Code available at
https://github.com/ContextualAI/BlitzRank.
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Verbalized Bayesian Persuasion
Wenhao Li, Yue Lin, Yun Hua, Xiangfeng Wang, Bo Jin, Hongyuan Zha, Baoxiang Wang
Information design (ID) explores how a sender influence the optimal behavior of receivers to achieve specific objectives.
While ID originates from everyday human communication, existing game-theoretic and learning methods often model information structures as numbers, which limits many applications to toy games.
This work leverages LLMs and proposes a verbalized framework in Bayesian persuasion (BP), which extends classic BP to real-world games involving human dialogues for the first time.
We map the BP to a verbalized mediator-augmented game, where LLMs instantiate the sender and receiver.
To efficiently solve the verbalized game, we propose a generalized equilibrium-finding algorithm combining LLM and game solver.
The algorithm is reinforced with techniques including verbalized commitment assumptions, verbalized obedience constraints, and information obfuscation.
Experiments in dialogue scenarios, such as recommendation letters, law enforcement, diplomacy with press, validate that our framework can reproduce theoretical results in classic BP and discover effective persuasion strategies in more complex natural language and multi-stage scenarios.
****
Tokenised Flow Matching for Hierarchical Simulation Based Inference
Giovanni Charles, Cosmo Nazzareno Santoni, Seth Flaxman, Elizaveta Semenova
The cost of simulator evaluations is a key practical bottleneck for Simulation Based Inference (SBI). In hierarchical settings with shared global parameters and exchangeable site-level parameters and observations, this structure can be exploited to improve simulation efficiency. Existing hierarchical SBI approaches factorise the posterior yet still simulate across multiple sites per training sample; We instead explore likelihood factorisation (LF) to train from single-site simulations. In LF sampling we learn a per-site (factorised) likelihood and then assemble synthetic multi-site observations to amortise inference for the full hierarchical posterior. Building on this, we propose Tokenised Flow Matching for Posterior Estimation (TFMPE), a tokenised flow matching approach that supports function-valued observations through likelihood factorisation. To enable systematic evaluation, we introduce a benchmark for hierarchical SBI. We validate TFMPE on this benchmark and on realistic infectious disease and computational fluid dynamics models, finding well-calibrated posteriors while reducing computational cost.
****
LATMiX: Learnable Affine Transformations for Microscaling Quantization of LLMs
Ofir Gordon, Lior Dikstein, Arnon Netzer, Idan Achituve, Hai Victor Habi
Post-training quantization (PTQ) is a widely used approach for reducing the memory and compute costs of large language models (LLMs). Recent studies have shown that applying invertible transformations to activations can significantly improve quantization robustness by reducing activation outliers; however, existing approaches are largely restricted to rotation or Hadamard-based transformations.
Moreover, most studies focused primarily on traditional quantization schemes, whereas modern hardware increasingly supports the microscaling (MX) data format. Attempts to combine both showed severe performance degradation, leading prior work to introduce assumptions on the transformations.
In this work, we take a complementary perspective. First, we provide a theoretical analysis of transformations under MX quantization by deriving a bound on the quantization error. Our analysis emphasizes the importance of accounting for both the activation distribution and the underlying quantization structure.
Building on this analysis, we propose LATMiX, a method that generalizes outlier reduction to learnable invertible affine transformations optimized using standard deep learning tools.
Experiments show consistent improvements in average accuracy for MX low-bit quantization over strong baselines on a wide range of zero-shot benchmarks, across multiple model sizes.

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Learning Unanimously Acceptable Lotteries via Queries
Davin Choo, Paul W. Goldberg, Nicholas Teh
Many high-stakes AI deployments proceed only if every stakeholder deems the system acceptable relative to their own minimum standard. With randomization over a finite menu of options, this becomes a feasibility question: does there exist a lottery over options that clears all stakeholders' acceptability bars? We study a query model where the algorithm proposes lotteries and receives only binary accept/reject feedback. We give deterministic and randomized algorithms that either find a unanimously acceptable lottery or certify infeasibility; adaptivity can avoid eliciting many stakeholders' constraints, and randomization further reduces the expected elicitation cost relative to full elicitation. We complement these upper bounds with worst-case lower bounds (in particular, linear dependence on the number of stakeholders and logarithmic dependence on precision are unavoidable). Finally, we develop learning-augmented algorithms that exploit natural forms of advice (e.g., likely binding stakeholders or a promising lottery), improving query complexity when predictions are accurate while preserving worst-case guarantees.
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Trajectory Stitching for Solving Inverse Problems with Flow-Based Models
Alexander Denker, Zeljko Kereta, Carola-Bibiane Schönlieb, Moshe Eliasof
Flow-based generative models have emerged as powerful priors for solving inverse problems. One option is to directly optimize the initial latent code (noise), such that the flow output solves the inverse problem. However, this requires back propagating through the entire generative trajectory, incurring high memory costs and numerical instability. We propose MS-Flow, which represents the trajectory as a sequence of intermediate latent states rather than a single initial code. By enforcing the flow dynamics locally and coupling segments through trajectory-matching penalties, MS-Flow alternates between updating intermediate latent states and enforcing consistency with observed data. This reduces memory consumption while improving reconstruction quality. We demonstrate the effectiveness of MS-Flow over existing methods on image recovery and inverse problems, including inpainting, super-resolution, and computed tomography.
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When Labelers Stay Silent: The Power of Ties in Cost-Effective Preference Learning
Jiaqi Lv, Zihan Zhang, Chengjing Yang, Shiyu Xia, Ning Xu, Xin Geng
Standard preference alignment relies on a binary forced-choice paradigm, assuming definitive preferences for all pairs.
However, we find that indistinguishable pairs are prevalent even in standard benchmarks, where quality differences of two responses often fall below the labeler's discriminative resolution limit.
Forcing a choice in such cases could inject significant noise that undermines policy optimization.
In this work, we propose a silent-aware framework that introduces a principled way to allow annotators to stay silent (i.e., express ties) and then explicitly model these ties during optimization.
Our findings reveal a compelling phenomenon: when ties are properly modeled, supervision from small models yields alignment surpassing that of forced-choice LLMs or human experts.
This discovery highlights a cost-effective path for alignment: respecting a labeler’s resolution limit is more critical than increasing its capability, while simultaneously unlocking the latent value in existing benchmarks by properly modeling inherent tie signals without requiring any re-labeling effort.
To leverage these signals, we propose several optimization objectives to drive the policy toward high-reward regions while mitigating unreliable updates that lead to arbitrary distribution shifts.
Our approaches significantly surpass conventional alignment performance, consistently outperforming the strongest available baselines across diverse benchmarks.
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Are Two Datasets Close Enough With Statistical Significance? A Kernel Distributional Closeness Testing Approach
Zhijian Zhou, Liuhua Peng, Xunye Tian, Mingming Gong, Feng Liu
Are two distributions close to each other with statistical significance? *Distribution closeness testing* (DCT) formalizes this question by testing whether the distance between a distribution pair is at least ϵ-far. Existing DCT methods mainly measure discrepancies between a distribution pair defined on discrete spaces (e.g., using total variation), which limits their applications to complex data (e.g., images). To extend DCT to more types of data, a natural idea is to introduce *maximum mean discrepancy* (MMD), a powerful measurement of the distributional discrepancy between two complex distributions, into DCT scenarios. However, the empirical results indicate that many distribution pairs can have the same MMD value despite having different norms in the same *reproducing kernel Hilbert space* (RKHS), and these pairs may exhibit different finite-sample distinguishability and reflect different practical closeness levels, making MMD less informative in DCT. To mitigate the issue, we design a new measurement of distributional discrepancy, *norm-adaptive MMD* (NAMMD), which scales MMD's value using the RKHS norms of distributions. Based on the asymptotic distribution of NAMMD, we finally propose the NAMMD-based DCT to assess the closeness level of a distribution pair. Theoretically, we prove that NAMMD-based DCT has higher test power compared to MMD-based DCT, with bounded type-I error, which is also validated by extensive experiments on many types of data (e.g., synthetic noise, real images). Our code is available at: https://github.com/zhijianzhouml/NAMMD.
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Representation Learning for Equivariant Inference with Guarantees
Daniel Ordonez-Apraez, Vladimir R Kostic, Alek Fröhlich, Vivien Brandt, Karim Lounici, Massimiliano Pontil
In many real-world applications of regression, conditional probability estimation, and uncertainty quantification, exploiting symmetries rooted in physics or geometry can dramatically improve generalization and sample efficiency. While geometric deep learning has made empirical advances by incorporating symmetry and geometry priors, less attention has been given to statistical learning guarantees. In this paper, we introduce an equivariant representation learning framework that simultaneously addresses regression, conditional probability estimation, and uncertainty quantification while providing first-of-its-kind non-asymptotic statistical learning guarantees. Grounded in operator and group representation theory, our framework approximates the spectral decomposition of the conditional expectation operator, building representations that are both equivariant and disentangled along independent symmetry quotient groups. Empirical evaluations on synthetic datasets and real-world robotics applications confirm the potential of our approach, matching or outperforming existing equivariant baselines in regression while providing well-calibrated uncertainty estimates.
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TriForces: Augmenting Atomistic GNNs for Transferable Representations
Ali Ramlaoui, Alexandre Duval, Hannah Bull, Victor Schmidt, Hugues Talbot, Fragkiskos D. Malliaros, Joseph Musielewicz
Machine learning interatomic potentials (MLIPs) achieve excellent accuracy when trained on large Density Functional Theory (DFT) data. To be useful in practice, they must often be adapted to target chemistries using small and expensive task-specific datasets. However, MLIPs transfer inconsistently across domains, with representations that often loose accessible composition and structure information. To address this, we present TriForces, a model-agnostic three-stream framework that separates composition and structure information, combined with self-supervised learning to preserve transferable representations.
TriForces improves performance on MatBench and QM9 over baselines without needing DFT labels and enables efficient similar structure retrieval through its learned latent space. On OMat24, in limited-data training regime, TriForces reduces energy MAE by 57\% at 20K samples only and improves force MAE across sample sizes. We release pretrained TriForces variants across multiple MLIP architectures with code at https:/github.com/Ramlaoui/triforces.

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Towards Completeness in Causal Discovery from Soft Interventions with Known Targets
Zihan Zhou, Murat Kocaoglu
We study causal discovery from soft interventions in the presence of latent confounding. Beyond within-environment conditional independences, soft interventions induce cross-environment invariances that can be encoded using an augmented graph with intervention indicator nodes ($\mathcal{I}$-AUG). Taking its maximal ancestral graph (MAG) yields the $\mathcal{I}$-MAG, which characterizes the interventional Markov equivalence class. Building on this framework, we show that the FCI-inspired learner ($\mathcal{I}$-FCI) by Kocaoglu et al. (2019) is sound but not complete: it may output circle endpoints that are nevertheless compelled by the interventional equivalence class. To exploit intervention-node semantics, we propose two complementary methods. First, we introduce an enumeration-based completion procedure that is sound and theoretically complete, but whose worst-case cost depends on the number of MAGs compatible with the partial graph learned by $\mathcal{I}$-FCI. Second, we derive a set of additional local orientation rules that provably tighten $\mathcal{I}$-FCI without increasing asymptotic complexity. Both methods refine prior outputs in the controlled soft-intervention setting with latent variables.
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Predictive variational inference: Learn the predictively optimal posterior distribution
Jinlin Lai, Antonio Ricardo Linero, Yuling Yao
Vanilla variational inference finds an optimal approximation to the Bayesian posterior distribution, but even the exact Bayesian posterior is often not meaningful under model misspecification. We propose predictive variational inference (PVI): a general inference framework that seeks and samples from an optimal posterior density such that the
resulting posterior predictive distribution is as close to the true data generating process as possible, while this closeness is measured by multiple scoring rules.
By optimizing the objective, the predictive variational inference is generally not the same as, or even attempting to approximate, the Bayesian posterior, even asymptotically. Rather, we interpret it as implicit hierarchical expansion.
Further, the learned posterior uncertainty detects heterogeneity of parameters among the population, enabling automatic model diagnosis.
This framework applies to both likelihood-exact and likelihood-free models. We demonstrate its application in real data examples.
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Threshold-Based Exclusive Batching for LLM Inference
Weifang Zhang, Yuzhou Nie, Bowen Pang, Guangrui Ma, Shining Wu
Mixed batching (MB)—interleaving prefill and decode in a single batch—has become the standard scheduling strategy for large language model (LLM) inference due to its efficiency in maximizing compute and memory utilization. However, through controlled experiments, we find that prefill–decode interference inflates MB's per-step marginal cost above that of pure decode. On the high-bandwidth H200 (4.8 TB/s), this occurs only when decode tokens exceed 80% of the batch; however, on the bandwidth-constrained RTX PRO 6000 (1.792 TB/s), this threshold plummets to just 20%. Consequently, the optimal choice between MB and exclusive batching (EB) fundamentally depends on GPU memory bandwidth, model size, and workload composition. We derive a closed-form condition for this EB–MB performance crossover, along with asymptotically optimal phase-switching thresholds and memory-safe batch sizing for EB. Optimized EB achieves up to 41.9% higher throughput on bandwidth-constrained GPUs, while MB retains its advantage on high-bandwidth hardware with larger models. Our hybrid scheduler EB+ applies this condition online to dynamically switch between EB and MB without manual intervention. Under non-stationary traffic with distribution or concurrency shifts, EB+ attains the highest or near-highest throughput in every setting, outperforming MB by up to 36.4%.
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On the Power of Source Screening for Learning Shared Feature Extractors
Muxing Wang, Connor Mclaughlin, Lili Su
Learning with shared representation is widely recognized as an effective way to separate commonalities from heterogeneity across various heterogeneous sources. Most existing work includes all related data sources via simultaneously training a common feature extractor and source-specific heads. It is well understood that data sources with low relevance or poor quality may hinder representation learning. In this paper, we further dive into the question of which data sources should be learned jointly by focusing on the traditionally deemed "good" collection of sources, in which individual sources have similar relevance and qualities with respect to the true underlying common structure. Towards tractability, we focus on the linear setting where sources share a low-dimensional subspace. We find that source screening can play a central role in statistically optimal subspace estimation. We show that, for a broad class of problem instances, training on a carefully selected subset of sources suffices to achieve minimax optimality, even when a substantial portion of data is discarded. We formalize the notion of an informative subpopulation, develop algorithms and practical heuristics for identifying such subsets, and validate their effectiveness through both theoretical analysis and empirical evaluations on synthetic and real-world datasets.
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ECo-MoE: Embodiment-Conditioned Mixture of Experts Increases the Evolvability of Robots
Yibin Wang, Muhan Li, Zihan Guo, Sam Kriegman
In this paper, we introduce a model of evolution and learning in robots that co-optimizes a distribution of latent design vectors (genotypes) and a mixture of control experts (neural modules), which are gated by the latent coordinates of each decoded design (phenotype). This provides a scalable alternative to co-design algorithms that either train an individual policy for every robot, which is inefficient, or a monolithic universal controller for all robots, which results in overly conservative structures and behaviors. Our approach lies somewhere between these two extremes, preserving ancestral knowledge in a unified yet modular framework in which different body plans activate and deactivate different combinations of learned sensorimotor circuits for goal-directed behavior. This allows one part of the controller to be overhauled to better suit new species of designs as they emerge without disrupting the hard-earned knowledge contained within other expert modules. It also allows pretrained expert policies to be directly plugged into the mixture, which can steer evolution into otherwise unexplored areas of latent space containing desired morphological traits. We refer to this process as "evo by demo" and explore how it may be used to guide freeform evolution toward canonical structures defined by the pretrained model. Videos and code can be found at: https://eco-moe.github.io.
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Tracking Drift: Variation-Aware Entropy Scheduling for Non-Stationary Reinforcement Learning
Tongxi Wang, Zhuoyang Xia, Xinran Chen, Shan Liu
Real-world reinforcement learning often faces environment drift, but most existing methods rely on static entropy coefficients/target entropy, causing over-exploration during stable periods and under-exploration after drift (thus slow recovery), and leaving unanswered the principled question of how exploration intensity should scale with drift magnitude.
We show that, under standard assumptions, entropy scheduling in non-stationary maximum-entropy RL can be cast as the dynamic-regret trade-off between tracking a drifting comparator and stabilizing updates, yielding a square-root scaling rule for the entropy weight in terms of a (possibly conservative) online non-stationarity proxy.
Building on this, we propose AES (Adaptive Entropy Scheduling), which adaptively adjusts the entropy coefficient/temperature online using observable drift proxies during training, requiring almost no structural changes and incurring minimal overhead.
Across 4 algorithm variants, 12 tasks, and 4 drift modes, AES significantly redu

ces the fraction of performance degradation caused by drift and accelerates reco very after abrupt changes.
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MixFP4: Enhancing NVFP4 with Adaptive FP4/INT4 Block Representations
Jiaxiang Zou, Yonghao Chen, Ruilong WU, Xinyu Chen
As large language models continue to scale, fine-grained block-scaled low-precis ion formats such as NVFP4 are increasingly adopted for their substantial through put and memory benefits. However, a single FP4 micro-format often mismatches het erogeneous block-level tensor statistics. To address this without changing the s tandard block-scaled MMA/GEMM execution path, we propose MixFP4, a mixed micro-f ormat extension to NVFP4 that selects between two stored FP4 micro-formats (E2M1 and E1M2) per block.
MixFP4 reuses NVFP4's scale hierarchy and encodes the format choice with zero ad ditional metadata by repurposing the sign bit of the FP8 E4M3 block scale.
By decoding both micro-formats into a unified internal E2M2 compute representati on, MixFP4 avoids datapath duplication.
Across representative LLM families, MixFP4 improves FP4 quantization robustness and accuracy over NVFP4 baselines with modest tensor-core overhead (3.1\% area, 1.5\% power).
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PEARL: Differentially Private and Entropy-Aware Regulated Language Generation
Seongho Joo, Hyukhun Koh, Kyomin Jung
Large language models (LLMs) often employ Retrieval-Augmented Generation (RAG) t o improve factuality. However, this also increases the risk of sensitive private information leakage. Differential Privacy (DP) has therefore been integrated in to LLM inference and is widely regarded as a standard safeguard; yet most studie s focus narrowly on the privacy–utility trade-off, leaving the trustworthiness o f DP outputs underexplored. To assess trustworthiness, we employ the confidence gap (CG), which quantifies an LLM’s internal knowledge conflict. We show that CG correlates with both hallucination and exposure of personally identifiable info rmation (PII). Building on this insight, we propose PEARL, a CG■guided, entropy■ aware private decoding framework. PEARL adaptively allocates the privacy budget across tokens and sentences based on CG, concentrating protection on PII-bearing spans while stabilizing low-confidence, hallucination-prone regions. In experim ents, PEARL improves both trustworthiness and robustness against PII extraction attacks. Notably, while applying DP alone significantly increases hallucination, our framework demonstrates that it is possible to preserve privacy while reduci ng hallucination.
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Tilt Matching for Scalable Sampling and Fine-Tuning
Peter Potaptchik, Lee Cheuk Kit, Michael Samuel Albergo
We propose a simple, scalable algorithm based on stochastic interpolants for sam pling from unnormalized densities and for fine-tuning generative models. The app roach, Tilt Matching, arises from a dynamical equation relating the flow matchin g velocity to one targeting the same distribution tilted by a reward, implicitly solving a stochastic optimal control problem. The resulting velocity inherits t he regularity of stochastic interpolant transports while minimizing an objective with strictly lower variance than flow matching itself. The update to the veloc ity field can be interpreted as the sum of all joint cumulants between the inter polant velocity and the reward, and to first order is their covariance. The meth od requires neither reward gradients nor backpropagation through trajectories of the flow or diffusion. We empirically demonstrate that the approach is efficien t and highly scalable, providing state-of-the-art results on sampling under Lenn ard-Jones systems and competitive performance for fine-tuning Stable Diffusion, without requiring reward multipliers. The framework also applies directly to til ting few-step flow map models.
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Towards Understanding the Dynamics of Low-Rank Adaptation
Shu Ding, Yang Peng, Hangan Zhou, Xinyu Lu, Shangwei Chen, Junhua Huang, Mingxua n Yuan, Wei Wang

Low-Rank Adaptation (LoRA) is a widely used parameter-efficient fine-tuning technique, and previous works have studied the update dynamics of LoRA, showing that updating via the low-rank matrix $\mathbf{A}$ can be viewed as a process within the compressed subspace spanned by $\mathbf{A}^{\top} \mathbf{A}$ of the gradient $\nabla f\left(\mathbf{W} \right)$. However, few works analyze how the properties of the low-rank matrices affect the performance of LoRA, since existing methods heuristically initialize the low-rank matrices as Gaussian matrices. In this paper, we provide a theoretical understanding of the update dynamics of LoRA. We reveal that the update dynamics can be viewed as a process within the subspace spanned by $\mathbf{A}^{\top} (\mathbf{A} \mathbf{A}^{\top})^{\dagger} \mathbf{A}$, and prove that when the gradient $\nabla f\left(\mathbf{W} \right)$ is unavailable, if $\mathbf{A}$ is an Equiangular Tight Frame (ETF), $\mathbf{A}^{\top} \mathbf{A}$ and $\mathbf{A}^{\top} (\mathbf{A} \mathbf{A}^{\top})^{\dagger} \mathbf{A}$ can preserve the maximum information from the gradient $\nabla f\left(\mathbf{W} \right)$. Thus, initializing $\mathbf{A}$ as an ETF is the optimal solution for low-rank adaptation when the gradient $\nabla f\left(\mathbf{W} \right)$ is unavailable. Furthermore, we establish the convergence of Low-Rank Adaptation with a rate of $\mathcal{O}\left(\frac{1}{T}\right)$ when $\mathbf{A}$ is an ETF. Extensive experiments show that initializing the low-rank matrices as ETFs significantly outperforms the commonly used Gaussian initialization for existing primary LoRA variants.

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Trust Region Masking for Long-Horizon LLM Reinforcement Learning

Yingru Li, Jiacai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Baoxiang Wang

Policy gradient methods for Large Language Models optimize a policy π_θ via a surrogate objective computed from samples of a rollout policy π_{roll}. However, modern LLM-RL pipelines suffer from unavoidable implementation divergences, such as backend discrepancies, Mixture-of-Experts routing discontinuities, and distributed training staleness. These factors cause an off-policy mismatch ($\pi_{roll} \neq \pi_\theta$), leading to approximation errors between the surrogate and the true objective.

We demonstrate that classical trust region bounds on this error scale as $O(T^2)$ with sequence length T, rendering them vacuous for long-horizon tasks. To address this, we derive two new bounds: a Pinsker-Marginal bound scaling as $O(T^{1.5})$ and a Mixed bound scaling as $O(T)$. Crucially, both bounds depend on D_{KL}^{max}, the maximum token-level KL divergence across the sequence. As a sequence-level term, the divergence cannot be controlled by previous token-independent methods like PPO clipping. We propose Trust Region Masking (TRM), which masks entire sequences that violate the trust region. TRM reduces the vacuity of traditional bounds, offering a principled framework to mitigate training collapse in long-horizon LLM-RL.

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Avoid What You Know: Divergent Trajectory Balance for GFlowNets

Pedro Dall'Antonia, Tiago Silva, Daniel Csillag, Salem Lahlou, Diego Mesquita

Generative Flow Networks (GFlowNets) are a flexible family of amortized samplers trained to generate discrete and compositional objects with probability proportional to a reward function. To this end, they learn a policy function over an intractably large state graph by minimizing a stochastic objective over sampled trajectories. However, learning efficiency is constrained by the model’s ability to rapidly explore diverse high-probability regions during training. To mitigate this issue, recent works have focused on incentivizing the exploration of unvisited and valuable states via curiosity-driven search and self-supervised random network distillation, which tend to waste samples on already well-approximated regions of the state space. In this context, we propose *Adaptive Complementary Exploration* (ACE), a principled algorithm for the effective exploration of novel and high-probability regions when learning GFlowNets. To achieve this, ACE introduces an *exploration* GFlowNet explicitly trained to search for high-reward states in regions underexplored by the *canonical* GFlowNet, which learns to sample from the target distribution. Through extensive experiments, we show that ACE consistently and significantly improves upon prior work in terms of approximation

accuracy to the target distribution and discovery rate of diverse high-reward s
tates.
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Asymmetric conformal prediction with penalized kernel sum-of-squares
Louis Allain, Sébastien Da Veiga, Brian Staber
Conformal prediction (CP) is a distribution-free method to construct reliable pr
ediction intervals that has gained significant attention in recent years. Despit
e its success and various proposed extensions, a significant practical feature w
hich has been overlooked in previous research is the potential skewed nature of
the noise, or of the residuals when the predictive model exhibits bias. In this
work, we leverage recent developments in CP to propose a new asymmetric procedur
e that bridges the gap between skewed and non-skewed noise distributions, while
still maintaining adaptivity of the prediction intervals.
We introduce a new statistical learning problem to construct adaptive and asymme
tric prediction bands, with a unique feature based on a penalty which promotes s
ymmetry: when the penalty intensity varies, the intervals smoothly change from s
ymmetric to asymmetric ones. This statistical learning problem is based on repro
ducing kernel Hilbert spaces and the recently introduced kernel sum-of-squares f
ramework. First, we establish representer theorems to make our problem tractable
in practice, and derive dual formulations which are essential for scalability t
o larger datasets. Second, the intensity of the penalty is chosen using a novel
data-driven method which automatically identifies the symmetric nature of the no
ise. We show that consenting to some asymmetry can let the learned prediction ba
nds better adapt to small sample regimes, outliers or biased predictive models.
Finally, our experiments illustrate the efficiency of such penalized kernel sum-
of-squares to construct adaptive prediction bands.
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Rethinking Convergence in MoE Training: The Role of Routing Sparsity
Weihao Zhu, Long Shi, Kang Wei, Zhe Wang, Yipeng Zhou, Haixia Zhang
In Mixture-of-Experts (MoE) training, sparse routing, i.e., activating only the
top-K experts per token, is essential for balancing convergence speed and comp
utational cost. However, existing works typically choose K empirically, witho
ut theoretical guidance. To address this gap, we characterize the convergence be
havior of MoE training using stochastic optimization theory. Specifically, we de
rive a convergence upper bound of $\mathcal{O}\left(\frac{1+M/K}{\sqrt{T}}\right
)$, where T is the number of training iterations and M is the total number o
f experts per MoE layer. This result guarantees convergence and shows that incre
asing K can accelerate training. By further fixing the total computational bud
get R (in FLOPs), we obtain a refined bound of $\mathcal{O}\left(\sqrt{\frac{K
}{R}} + \frac{M}{\sqrt{K R}}\right)$, which is convex in K and implies the exi
stence of an optimal $K^{*}\in[1,M]$ that achieves the best convergence performa
nce. Extensive experiments validate our theoretical analysis under diverse setti
ngs.
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DSGCR: Decomposed Spectral Geometry-Aware Cross-Modal Semantic Representation fo
r 3D Visual Grounding
Jing He, Licheng Jiao, Lingling Li, Xiaoqiang Lu, Xu Liu, Wenping Ma, Fang Liu,
Long Sun
3D visual grounding requires robust cross-modal representation to achieve fine-g
rained semantic alignment and precise geometric reasoning. However, most methods
employ unimodal pre-trained encoders that transfer visual and linguistic knowle
dge independently, inducing domain shift and poor cross-modal alignment. Meanwhi
le, spatial modeling with handcrafted priors limits cross-modal geometric repres
entation, struggling to capture complex object relations due to spectral bias. T
o address these challenges, we propose Text-Aware Feature Tuning (TFT) and Decom
posed Spectral Geometry (DSG) to enhance cross-modal semantic representation. Sp
ecifically, TFT injects linguistic context into the visual hierarchy to mitigate
domain shift and facilitate early cross-modal alignment. DSG employs a learnabl
e Fourier basis and explicitly decomposes pairwise relations into symmetric and
antisymmetric spectral components, allowing the model to capture high-frequency

geometric details and direction-aware relations for precise spatial reasoning. E
xtensive experiments on ScanRefer, Nr3D and Sr3D validate the effectiveness of o
ur method, demonstrating state-of-the-art performance with improvements of 2.05\
% Acc@0.25 for 3DREC and 1.09\% mIoU for 3DRES on ScanRefer.
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ANTiC: Adaptive Neural Temporal In Situ Compressor
Sandeep Suresh Cranganore, Andrei Bodnar, Gianluca Galletti, Fabian Paischer, Jo
hannes Brandstetter
The persistent storage requirements for high-resolution, spatiotemporally evolvi
ng fields governed by large-scale and high-dimensional partial differential equa
tions (PDEs) have reached the petabyte-to-exabyte scale.
Transient simulations modeling Navier-Stokes equations, magnetohydrodynamics, pl
asma physics, or binary black hole mergers generate data volumes that are prohib
itive for modern high-performance computing (HPC) infrastructures.
To address this bottleneck, we introduce ANTIC (Adaptive Neural Temporal in situ
 Compressor), an end-to-end in situ compression pipeline. ANTIC consists of an a
daptive temporal selector tailored to high-dimensional physics that identifies a
nd filters informative snapshots at simulation time, combined with a spatial neu
ral compression module based on continual fine-tuning that learns residual updat
es between adjacent snapshots using neural fields.
By operating in a single streaming pass, ANTIC enables a combined compression of
 temporal and spatial components and effectively alleviates the need for explici
t on-disk storage of entire time-evolved trajectories. Experimental results demo
nstrate that ANTIC achieves storage reductions of approximately $\sim 400\times$
 for 2D Kolmogorov flow simulations and $\sim 7000\times$ for large-scale physic
s simulations such as binary black hole mergers.
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Mirror Mean-Field Langevin Dynamics
Anming Gu, Juno Kim
The mean-field Langevin dynamics (MFLD) minimizes an entropy-regularized nonline
ar convex functional on the Wasserstein space over $\mathbb{R}^d$, and has gaine
d attention recently as a model for the gradient descent dynamics of interacting
 particle systems such as infinite-width two-layer neural networks. However, man
y problems of interest have constrained domains, which are not solved by existin
g mean-field algorithms due to the global diffusion term. We study the optimizat
ion of probability measures constrained to a convex subset of $\mathbb{R}^d$ by
proposing the \emph{mirror mean-field Langevin dynamics} (MMFLD), an extension o
f MFLD to the mirror Langevin framework. We obtain linear convergence guarantees
 for the continuous MMFLD via a uniform log-Sobolev inequality, and uniform-in-t
ime propagation of chaos results for its time- and particle-discretized counterp
art.
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DetailMaster: Can Your Text-to-Image Model Handle Long Prompts?
Qirui Jiao, Daoyuan Chen, Yilun Huang, Xika Lin, Ying Shen, Yaliang Li
While recent Text-to-Image (T2I) models show impressive capabilities in synthesi
zing images from brief descriptions, they struggle with the long, detailed promp
ts required for professional applications. We present DetailMaster, a comprehens
ive benchmark for evaluating T2I capabilities on long prompts with complex compo
sitional requirements, accompanied by an automated data construction pipeline an
d an evaluation workflow. Comprising expert-validated prompts averaging 284.89 t
okens, our benchmark introduces four critical evaluation dimensions: Character A
ttributes, Structured Character Locations, Multi-Dimensional Scene Attributes, a
nd Spatial/Interactive Relationships. Evaluations on various general-purpose and
 long-prompt-optimized models reveal critical performance limitations, showing t
hat weak encoders struggle to preserve syntactic dependencies within prompts and
 diffusion models suffer from attribute leakage under detail-intensive condition
s. Through a controlled ablation study under varying constraints, we further sho
w that high-fidelity generation requires a synergistic combination of expanded p
rompt limits and long-prompt training. We open-source our dataset and code to fo
ster progress in long-prompt-driven T2I generation.

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A Two-Layer Framework for Joint Online Configuration Selection and Admission Control
Owen Shen, Haoran Xu, Yinyu Ye, Peter Glynn, Patrick Jaillet
We study online configuration selection with admission control problem, which arises in LLM serving, GPU scheduling, and revenue management. In a planning horizon with T periods, we consider a two-layer framework for the decisions made within each time period. In the first layer, the decision maker selects one of the K configurations (ex. quantization, parallelism, fare class) which induces distribution over the reward-resource pair of the incoming request. In the second layer, the decision maker observes the request and then decides whether to accept it or not. Benchmarking this framework requires care. We introduce a **switching-aware fluid oracle** that accounts for the value of mixing configurations over time, provably upper-bounding any online policy. We derive a max-min formulation for evaluating the benchmark, and we characterize saddle points of the max-min problem via primal-dual optimality conditions linking equilibrium, feasibility, and complementarity. This guides the design of **SP-UCB–OLP** algorithm, which solves an optimistic saddle point problem and achieves $\tilde{O}(\sqrt{KT})$ regret.
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Cognitive Fatigue in Autoregressive Transformers: Formalization and Measurement
Riju Marwah, Ritvik Garimella, Vishal Pallagani, Atishay Jain, Michael Charles Stewart, Amit Sheth
Autoregressive language models frequently degrade during long-horizon generation, producing repetitive text, losing instruction adherence, and exhibiting unstable entropy. Despite the prevalence of these failures, practitioners lack online diagnostics to detect them in real-time as they occur. We formalize this degradation as cognitive fatigue, a measurable generation-time state characterized by decay in attention to the original prompt, representational drift, and entropy miscalibration. We introduce the Fatigue Index (FI), a lightweight, model-agnostic diagnostic that aggregates these three signals under explicit axioms (monotonicity, boundedness, interpretability) enabling reliable runtime monitoring. Across nine models (1B–13B parameters), FI trajectories exhibit structured temporal dynamics, predict task degradation (AUROC = 0.95) and repetition ($\rho = 0.94$), and reveal non-monotonic scaling behavior: instruction-tuned models below 3B exhibit faster collapse than base models, with this trend reversing at 7B. Stress analyses further show that FI onset accelerates under longer contexts, middle-positioned evidence, and reduced numerical precision. These results establish cognitive fatigue as a coherent and measurable phenomenon, and position FI as a principled tool for runtime reliability monitoring in production LLM systems.
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Proteo-R1: Reasoning Foundation Models for De Novo Protein Design
Fang Wu, Weihao Xuan, Heli Qi, Hanqun Cao, Heng-Jui Chang, Zeqi Zhou, Li Erran Li, Haokai Zhao, Jian Ma, Zijian Carl Ma, Yu-Chi Cheng, Kuan Pang, Xiangru Tang, Zehong Wang, Guanlue Li, Hanchen Wang, Kejun Ying, Pan Lu, Chiho Im, Seungju Han, Peng Xia, Tinson Xu, Yinxi Li, Deyao Zhu, Pheng-Ann Heng, Naoto Yokoya, Masashi Sugiyama, Jure Leskovec, Yejin Choi
Deep learning in de novo protein design has achieved atomic-level fidelity. However, existing models remain largely non-deliberative: they directly synthesize molecular geometries without explicitly reasoning about which residues or interactions are functionally essential. As a result, design decisions are entangled with continuous sampling dynamics, limiting interpretability, controllability, and systematic reuse of biochemical knowledge. We introduce Proteo-R1, a reasoning-guided protein design framework that explicitly decouples molecular understanding from geometric generation. Proteo-R1 adopts a dual-expert architecture, where a multimodal large language model (LLM) serves as an understanding expert, analyzes protein sequences, structures, and textual context to identify key functional residues that govern binding and specificity. These residue-level decisions are then passed to a separate diffusion-based generation expert, which performs conditional co-design while respecting the fixed interaction anchors. This factori

zation mirrors how human experts approach molecular engineering: first, reasoning about critical interactions, then optimizing geometry subject to those constraints. By operationalizing reasoning as explicit residue-level commitments rather than latent textual guidance, Proteo-R1 achieves stable, interpretable, and modular integration of LLM reasoning with advanced geometric generative models. Code and demos are at https://proteor1.github.io.
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Markov Chain Monte Carlo without Evaluating the Target: an Auxiliary Variable Approach
Wei Yuan, Guanyang Wang
In sampling tasks, it is common for target distributions to be known up to a normalizing constant. However, in many situations, even evaluating the unnormalized distribution can be costly or infeasible. This issue arises in scenarios such as sampling from the Bayesian posterior for tall datasets and the `doubly-intractable' distributions. In this paper, we begin by observing that seemingly different Markov chain Monte Carlo (MCMC) algorithms, such as the exchange algorithm, PoissonMH, and TunaMH, can be unified under a simple common procedure. We then extend this procedure into a novel framework that allows the use of auxiliary variables in both the proposal and the acceptance--rejection step. Several new MCMC algorithms emerge from this framework that uses estimated gradients to guide the proposal moves. They have demonstrated significantly better performance than existing methods on both synthetic and real datasets. We also develop theory for the new framework and use it to simplify and extend results for existing algorithms. The code to reproduce the experimental results can be found at https://github.com/ywwes26/Auxiliary-MCMC.
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ToolOrchestra: Elevating Intelligence via Efficient Model and Tool Orchestration
Hongjin SU, Shizhe Diao, Ximing Lu, Mingjie Liu, Jiacheng Xu, Xin Dong, Yonggan Fu, Peter Belcak, Hanrong Ye, Hongxu Yin, Yi Dong, Evelina Bakhturina, Tao Yu, Yejin Choi, Jan Kautz, Pavlo Molchanov
Large language models are powerful generalists, yet solving deep and complex problems such as those of the Humanity’s Last Exam (HLE) remains both conceptually challenging and computationally expensive. We show that small orchestrators managing other models and a variety of tools are able to both push the upper bound of intelligence and improve efficiency in solving difficult agentic tasks. We introduce ToolOrchestra, a method for training small orchestrators that coordinate the use of intelligent tools. ToolOrchestra makes explicit use of reinforcement learning with outcome-, efficiency-, and user-preference-aware rewards. Using ToolOrchestra, we produce Orchestrator, an 8B model that achieves higher accuracy at lower cost than previous tool-use agents while aligning with user preferences on which tools are to be used for a given query. On HLE, Orchestrator achieves a score of 37.1%, outperforming GPT-5 (35.1%) while being 2.5x more efficient. On $\tau ^2$-Bench and FRAMES, Orchestrator surpasses GPT-5 by a wide margin while using only about 30% of the cost. Extensive analysis shows that Orchestrator achieves the best trade-off between performance and cost under multiple metrics, and generalizes robustly to previously unseen tools. These results demonstrate that composing diverse tools with a lightweight orchestration model is both more efficient and more effective than existing methods, paving the way for practical and scalable tool-augmented reasoning systems. These results demonstrate that orchestrating diverse tools with lightweight agents is not only more efficient, but also more effective, paving the way for practical and scalable tool-augmented reasoning systems.
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A Spiking Heterogeneous Harmonic Resonate-and-Fire State Space Model for Time Series
Kartikay Agrawal, Vaishnavi Nagabhushana, Abhijeet Vikram, Vedant Sharma, Ayon Borthakur
Spiking neural networks have attracted increasing attention for their energy efficiency, multiplication-free computation, and sparse event-based processing. In parallel, state space models have emerged as a scalable alternative to transform

ers for long-range sequence modelling by avoiding quadratic dependence on sequen ce length. We propose here a spiking heterogeneous harmonic resonate-and-fire st ate space model (SH^2RFSSM), a second-order spiking SSM for classification and regression on ultra-long sequences. SH^2RFSSM outperforms transformers and fi rst-order SSMs on average while eliminating matrix multiplications, making it hi ghly suitable for resource-constrained applications. Furthermore, we introduce a kernel-based spiking regressor that enables accurate modelling of dependencies in sequences of up to 50k steps. We also observe a reduction in spiking operatio ns and improved performance with heterogeneity and discretisation in harmonic re sonate-and-fire neuronal layers. Overall, we evaluate Harmonic Resonate-and Fire layers across 17 diverse datasets, spanning sensors, time series, and classific ation to long-term forecasting. Our results demonstrate that SH^2RFSSM achieve s superior long-range modelling capability with energy efficiency, positioning i t as a strong candidate for signal processing on resource-constrained devices fo r human activity recognition, time series classification, and regression.
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The Structural Origin of Attention Sink: Variance Discrepancy, Super Neurons, an d Dimension Disparity
Siquan Li, Kaiqi Jiang, Jiacheng Sun, Tianyang Hu
Despite the prevalence of the attention sink phenomenon in Large Language Models (LLMs), where initial tokens disproportionately monopolize attention scores, it s structural origins remain elusive. This work provides a *mechanistic explanati on* for this phenomenon. First, we trace its root to the value aggregation proce ss inherent in self-attention, which induces a systematic variance discrepancy. We further demonstrate that this discrepancy is drastically amplified by the act ivation of super neurons within Feed-Forward Network (FFN) layers. Specifically, the channel-sparse down-projections trigger a dimension disparity of the first- token representation, necessitating the formation of attention sinks as a struct ural anchor. Then, we validate this causal chain through two controlled interven tions: (i) isolating the aggregation effect via attention mask modifications and (ii) amplifying the variance of targeted token representations. Both interventi ons can replicate attention sinks at arbitrary positions. Our mechanistic unders tanding offers a foundation for the systematic control of sink formation. Finall y, as a proof of concept, we propose *head-wise RMSNorm*, an architectural modif ication that stabilizes value aggregation outputs during pre-training. Our exper iments demonstrate that restoring statistical parity across positions significan tly accelerates convergence. The code is available at [https://github.com/Siquan -Li/Head-wise-RMSNorm](https://github.com/Siquan-Li/Head-wise-RMSNorm).
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Robust Multi-View Fusion via Prototype-Anchored Unbalanced Optimal Transport
Zhang Han, Xingwen Zhao, HUI LI
Multi-view classifiers typically fuse all observed views into a single represent ation, which becomes fragile when some views are missing or corrupted.We propose a prototype-anchored fusion module based on an entropically regularized unbalan ced optimal transport (UOT) barycenter.Each view is summarized into a small set of learned atoms and is matched to a shared prototype support; fusion outputs a probability measure over prototypes with fixed dimension.By relaxing marginal co nstraints with a generalized KL penalty, the UOT objective can leave a fraction of view mass unmatched when matching is geometrically costly, yielding a simple differentiable trimming mechanism without hand-tuned thresholds.We provide a bas ic theoretical result showing that discarding an arbitrary subset of atom mass i ncurs a penalty bounded by its total mass, independent of transport distances.Ex periments on multi-view action recognition benchmarks under simulated missing vi ews, missing-rate shift, and feature-space corruption demonstrate consistently i mproved stability under severe missingness with modest overhead on top of strong backbones.
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Geometry-Aware Probabilistic Circuits via Voronoi Tessellations
Sahil Sidheekh, Sriraam Natarajan
Probabilistic circuits (PCs) enable exact and tractable inference but employ dat

a independent mixture weights that limit their ability to capture local geometry of the data manifold.
We propose Voronoi tessellations (VT) as a natural way to incorporate geometric structure directly into the sum nodes of a PC. However, naïvely introducing such structure breaks tractability. We formalize this incompatibility and develop two complementary solutions: (1) an approximate inference framework that provides guaranteed lower and upper bounds for inference, and (2) a structural condition for VT under which exact tractable inference is recovered.
Finally, we introduce a differentiable relaxation for VT that enables gradient-based learning and empirically validate the resulting approach on standard density estimation tasks.
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Decentralized Online Convex Optimization with Efficient Communication: Improved Algorithm and Lower Bounds
Sifan Yang, Wenhao Yang, Wei Jiang, Lijun Zhang
We investigate decentralized online convex optimization with compressed communication, where n learners connected by a network collaboratively minimize a sequence of global loss functions using only local information and compressed data from their neighbors. Prior work has established regret bounds of $O(\max\\{\omega^{-2}\rho^{-4}n^{1/2},\omega^{-4}\rho^{-8}\\}n\sqrt{T})$ and $O(\max\\{\omega^{-2}\rho^{-4}n^{1/2},\omega^{-4}\rho^{-8}\\}n\ln{T})$ for convex and strongly convex functions, respectively, where $\omega\in(0,1]$ is the compression quality factor and $\rho<1$ is the spectral gap of the communication matrix. However, these regret bounds suffer from a prohibitively high quadratic or even quartic dependence on ω^{-1}. Moreover, the super-linear dependence on n is also undesirable. To overcome these shortcomings, we propose a novel algorithm that achieves improved regret bounds of $\tilde{O}(\omega^{-1/2}\rho^{-1}n\sqrt{T})$ and $\tilde{O}(\omega^{-1}\rho^{-2}n\ln{T})$ for convex and strongly convex functions, respectively. The primary idea is to design a two-level blocking update framework incorporating two novel ingredients: an online gossip strategy and an error compensation scheme, which work together to promote better consensus among learners. Furthermore, we establish the first lower bounds for this problem, justifying the optimality of our results with respect to both ω and T. Additionally, we consider the bandit feedback scenario and extend our method with classical gradient estimators to enhance existing regret bounds.
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Contrastive Flow Map Matching
Junyu Zhang, Daochang Liu, Younghyun Kim, Jong Hwan Ko, Shichao Zhang, Chang Xu, Eunbyung Park
Flow map matching (FMM) enables one- and few-step sampling for diffusion-style generation, yet its performance is often hindered by the mismatch between ground-truth training transitions and model-induced flow maps.
We propose **Contrastive Flow Map Matching (CFMM)**, a principled framework that explicitly aligns FMM training with practical sampling.
Our approach is motivated by a joint-KL decomposition on the reverse KL divergence, which decomposes the distributional gap into a marginal mismatch over intermediate states and a conditional mismatch in endpoint reconstruction.
This analysis motivates two complementary objectives: average-velocity regression for marginal alignment and a sampling-aligned InfoNCE contrastive loss for conditional refinement.
CFMM is a training-only plug-in for pre-trained FMMs, incurs no inference-time overhead, and supports training FMMs from scratch.
Experiments on CIFAR-10, ImageNet, and LSUN across multiple FMM baselines demonstrate consistent improvements in fidelity and perceptual quality with only modest additional training cost.
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The Latent Color Subspace: Emergent Order in High-Dimensional Chaos
Mateusz Pach, Jessica Bader, Quentin Bouniot, Serge Belongie, Zeynep Akata
Text-to-image generation models have advanced rapidly, yet achieving fine-grained control over generated images remains difficult, largely due to limited unders

tanding of how semantic information is encoded. We develop an interpretation of the color representation in the Variational Autoencoder latent space of FLUX.1 [Dev], revealing a structure reflecting Hue, Saturation, and Lightness. We verify our Latent Color Subspace (LCS) interpretation by demonstrating that it can both predict and explicitly control color, introducing a fully training-free method in FLUX based solely on closed-form latent-space manipulation. Code is available at https://github.com/ExplainableML/LCS.
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Topological Active Inference for Task Disambiguation
Yangbo Wei, Zhen Huang, Shaoqiang Lu, Junhong Qian, Chen Wu, Lei He
In open-ended domains, natural language instructions are often \textit{underspecified}, mapping to multiple valid yet functionally distinct latent intents. Although Large Language Models (LLMs) excel at generation, their interactive disambiguation remains limited by *semantic blindness*: they may spend clarification turns distinguishing superficial syntactic variants rather than resolving substantive intent differences. We propose *Topological Active Inference* (TAI), a geometric framework that recasts task disambiguation as *intent-manifold contraction*. TAI uses *Persistent Homology* to recover persistent intent clusters from sampled solutions, filtering short-lived syntactic variations while preserving robust semantic structure under mild separability assumptions. It then synthesizes clarifying questions as semantic separators and selects them with *Topological Expected Information Gain* (TEIG), which optimizes uncertainty reduction over intent clusters rather than individual candidates. This reduces the effective hypothesis space from N sampled solutions to K latent intents and yields logarithmic interaction complexity $\mathcal{O}(\log K)$ under balanced-split conditions. Experiments across code, visualization, and navigation tasks show that TAI resolves user intent with fewer turns and remains robust to noisy feedback, and smaller model scales.
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KernelBand: Steering LLM-based Kernel Optimization via Hardware-Aware Multi-Armed Bandits
Dezhi Ran, Shuxiao Xie, Mingfang Ji, Anmin Liu, Mengzhou Wu, Yuan Cao, Yuzhe Guo, Hao Yu, Linyi Li, Yitao Hu, Wei Yang, Tao Xie
High-performance GPU kernels are critical for efficient LLM serving, yet their optimization remains a bottleneck requiring deep system expertise. While code LLMs show promise in generating functionally correct code, kernel optimization is intrinsically a search problem over a vast optimization space. This fundamental mismatch prevents existing LLM agents from efficiently exploring the optimization space for diverse hardware and compute patterns. To bridge the gap, we present KernelBand, a framework that formulates kernel optimization as a Multi-Armed Bandit (MAB) problem, explicitly balancing exploration and exploitation to unlock the potential of code LLMs. To navigate the infinite arm space of optimization strategies applied to candidate kernels, we design two key mechanisms: a hardware-aware pruning strategy via profiling bounds and a runtime-behavior clustering algorithm that leverages Lipschitz continuity. Extensive experiments on TritonBench-G with three GPU architectures and four code LLMs show that KernelBand consistently outperforms the strongest available agent baseline, achieving up to **1.91×** geometric mean speedup over correctly optimized kernels with **39–140%** relative improvement in Fast@1 success rate. Our code is available at https://github.com/TongmingLAIC/KernelBand.
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SemRep : Generative Code Representation Learning with Code Transformations
Weichen Li, Jiamin Song, Bogdan Alexandru Stoica, Arav Dhoot, Gabriel Ryan, Shengyu Fu, Kexin Pei
Code transformation is a foundational capability in the software development process, where its effectiveness relies on constructing a high-quality code representation to characterize the input code semantics and guide the transformation.

Existing approaches treat code transformation as an end-to-end learning task, leaving the construction of the representation needed for semantic reasoning implicit in model weights or relying on rigid compiler-level abstractions.

We present SemRep, a framework that improves code transformation through *generative code representation learning*.

Our key insight is to employ the semantics-preserving transformations as the intermediate representation, which serves as both a generative mid-training task and the guidance for subsequent instruction-specific code transformations.

Across general code editing and optimization tasks (e.g., GPU kernel optimization), SemRep outperforms the extensively finetuned baselines with strictly the same training budget by 6.9\% in correctness, 1.1$\times$ in performance, 13.9\% in generalization, and 6.7\% in robustness.

With the improved exploration of diverse code transformations, SemRep is particularly amenable to evolutionary search.

Combined with an evolutionary coding agent, SemRep finds optimizations that 685B larger-weight baselines fail to discover while achieving the same performance with 25\% less inference compute.

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Parallel-Probe: Towards Efficient Parallel Thinking via 2D Probing

Tong Zheng, Chengsong Huang, Runpeng Dai, Yun He, Rui Liu, Xin Ni, Huiwen Bao, Kaishen Wang, Hongtu Zhu, Jiaxin Huang, Furong Huang, Heng Huang

Parallel thinking has emerged as a promising paradigm for reasoning, yet it imposes significant computational burdens. Existing efficiency methods primarily rely on local, per-trajectory signals and lack principled mechanisms to exploit global dynamics across parallel branches. We introduce 2D probing, an interface that exposes the width–depth dynamics of parallel thinking by periodically eliciting intermediate answers from all branches. Our analysis reveals three key insights: non-monotonic scaling across width–depth allocations, heterogeneous reasoning branch lengths, and early stabilization of global consensus. Guided by these insights, we introduce $\textbf{{Parallel-Probe}}$, a training-free controller designed to optimize online parallel thinking. Parallel-Probe employs consensus-based early stopping to regulate reasoning depth and deviation-based branch pruning to dynamically adjust width. Extensive experiments across three benchmarks and multiple models demonstrate that Parallel-Probe establishes a superior Pareto frontier for test-time scaling. Compared to standard majority voting, it reduces sequential tokens by up to $\textbf{35.8}$% and total token cost by over $\textbf{25.8}$% while maintaining competitive accuracy.

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A Kinetic Energy Perspective of Flow Matching

Ziyun Li, Huancheng Hu, Soon Hoe Lim, Xuyu Li, Fei Gao, Enmao Diao, Zezhen Ding, Michalis Vazirgiannis, Henrik Boström

Flow-based generative models can be viewed through a physics lens: sampling transports a particle from noise to data by integrating a learned velocity field, and each sample corresponds to a trajectory with its own dynamical effort. Motivated by classical mechanics, we introduce Kinetic Path Energy (KPE), an action-like, per-sample diagnostic that measures the accumulated kinetic effort along an ordinary differential equation (ODE) trajectory. Empirically, KPE exhibits two robust correspondences: {i} higher KPE predicts stronger semantic fidelity; {ii} high-KPE trajectories land in sparse representation regions. We further provide theoretical guarantees linking trajectory energy to data sparsity. Paradoxically, this correlation is non-monotonic. At sufficiently high energy, generation can degenerate into memorization. Leveraging the closed-form formula of empirical flow matching, we show that extreme energies drive trajectories toward near-copies of training examples. This yields a Goldilocks principle and motivates Kinetic Trajectory Shaping (KTS), a training-free two-phase inference strategy that boosts early motion and enforces a late-time soft landing, reducing memorization and improving generation quality across benchmark tasks.

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A Formal Comparison Between Chain of Thought and Latent Thought

Kevin Xu, Issei Sato

Chain of thought (CoT) elicits reasoning in large language models by explicitly generating intermediate tokens. In contrast, latent thought reasoning operates directly in the continuous latent space, enabling computation beyond discrete lin

guistic representations. While both approaches exploit iterative computation, th
eir comparative capabilities remain underexplored. In this work, we present a fo
rmal analysis showing that latent thought admits efficient parallel computation,
 in contrast to the inherently sequential nature of CoT. Conversely, CoT enables
 approximate counting through stochastic decoding. These separations suggest the
 tasks for which depth-driven recursion is more suitable, thereby offering pract
ical guidance for choosing between reasoning paradigms.
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OmniAID: Decoupling Semantic and Artifacts for Universal AI-Generated Image Dete
ction in the Wild

Yuncheng Guo, Junyan Ye, Chenjue Zhang, Hengrui Kang, Haohuan Fu, Conghui He, We
ijia Li

A truly universal AI-Generated Image (AIGI) detector must simultaneously general
ize across diverse generative models and varied semantic content. Current method
s learn a single, entangled forgery representation, conflating content-dependent
 flaws with content-agnostic artifacts, and are further constrained by outdated
benchmarks. We propose OmniAID, a novel framework centered on a decoupled Mixtur
e-of-Experts (MoE) architecture that separates: (1) semantic flaws across distin
ct content domains via Routable Specialized Semantic Experts, and (2) content-ag
nostic universal artifacts from content-dependent flaws via a Fixed Universal Ar
tifact Expert. A two-stage training strategy first specializes experts independe
ntly with domain-specific hard-sampling, then trains a lightweight gating networ
k for effective input routing. By explicitly decoupling "what is generated'' (c
ontent-specific flaws) from "how it is generated'' (universal artifacts), OmniAI
D achieves robust generalization. We also introduce Mirage, a large-scale, conte
mporary dataset comprising a modern training set and a challenging test set. Ext
ensive experiments demonstrate that OmniAID surpasses existing detectors, establ
ishing a new standard for AIGI detection against modern, in-the-wild threats.
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MASH: Modeling Abstention via Selective Help-Seeking

Mustafa Omer Gul, Claire Cardie, Tanya Goyal

LLMs cannot reliably recognize their parametric knowledge boundaries and often h
allucinate answers to outside-of-boundary questions. In this paper, we introduce
 MASH (Modeling Abstention via Selective Help-seeking), a training framework tha
t readily extracts abstentions from LLMs. Our key idea is that any external help
-seeking by an LLM, i.e. search tool use, can serve as a proxy for abstention if
 the external help (search) is appropriately penalized while also rewarding answ
er accuracy. MASH operationalizes this idea using reinforcement learning with a
pay-per-search reward. We run experiments on three knowledge-intensive QA datase
ts. Our results show that MASH substantially improves upon the selective help-se
eking performance of prior efficient search approaches; on multi-hop datasets, i
t improves answer accuracy by 7.6%. Furthermore, MASH demonstrates strong off-th
e-shelf abstention performance, showcasing behavior competitive with prior abste
ntion methods that additionally require predetermining model knowledge boundarie
s to construct training data. Overall, we show that MASH training effectively al
igns search tool use with parametric knowledge, which can be successfully levera
ged for making abstention decisions and efficient search tool use.
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PLATE: Plasticity-Tunable Efficient Adapters for Geometry-Aware Continual Learni
ng

Romain Cosentino

We develop a continual learning method for pretrained models that \emph{requires
 no access to old-task data},
addressing a practical barrier in foundation model adaptation where pretraining
distributions are often unavailable. Our key observation is that pretrained netw
orks exhibit substantial \emph{geometric redundancy}, and that this redundancy c
an be exploited in two complementary ways. First, redundant neurons provide a pr
oxy for dominant pretraining-era feature directions, enabling the construction o
f approximately protected update subspaces directly from pretrained weights. Sec
ond, redundancy offers a natural bias for \emph{where} to place plasticity: by r

estricting updates to a subset of redundant neurons and constraining the remaini ng degrees of freedom, we obtain update families with reduced functional drift o n the old-data distribution and improved worst-case retention guarantees. These insights lead to \textsc{PLATE} (\textbf{Pla}sticity-\textbf{T}unable \textbf{E} fficient Adapters), a continual learning method requiring no past-task data that provides explicit control over the plasticity-retention trade-off. PLATE parame terizes each layer with a structured low-rank update $\Delta W = B A Q^\top$, wh ere B and Q are computed once from pretrained weights and kept frozen, and o nly A is trained on the new task. Code is available at https://github.com/Sale sforceAIResearch/PLATE.

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STLA: Spatiotemporal Lookahead Alignment for Post-Training Quantization

Zuqi Zhang, Chenghe Sun, Xiangyi Chu, Wei-Han Yu, Ka-Fai Un, Rui P. Martins, Pui -In Mak, Jiawei Xu

Adaptive rounding techniques in Post-Training Quantization (PTQ) enable the effi cient deployment of Large Language Models (LLMs) with low resource and data depe ndencies. While learning-based rounding methods are accurate yet costly, compens ation-based approaches offer a highly efficient alternative. However, synergizin g these two to realize their full potential is hindered by spatiotemporal misali gnment in the decoupled paradigm. Key challenges include temporal parameter con flict, the invalidation of the initial Round-to-Nearest (RTN) assumption, and sp atially-inconsistent optimization objectives. This paper introduces STLA, a nove l rounding-optimized PTQ framework that achieves both fast and accurate LLM quan tization. STLA resolves temporal inconsistency through cluster-wise integrated r ounding optimization, which collocates the learning and compensation phases. STL A achieves spatial alignment through a unified global objective derived from the Schur Complement, enabling the solver to look ahead and align local rounding de cisions with the optimal future compensation of remaining weights. Furthermore, we propose a Hessian-guided clustering strategy that exploits both diagonal and off-diagonal information to maximize intra-cluster error cancellation. Extensive experiments demonstrate that STLA establishes a new state-of-the-art for low-bi t PTQ while maintaining high computational efficiency. The code is available at https://github.com/AI2C-Lab/STLA.

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FlowBot: Inducing LLM Workflows with Bilevel Optimization and Textual Gradients

Hongyeon Yu, Young-Bum Kim, Yoon Kim

LLM workflows, which coordinate structured calls to individual LLMs/agents to a chieve a particular goal, offer a promising path towards building powerful AI s ystems that can tackle diverse tasks. However, existing approaches for building such workflows generally rely on human-crafted pipelines and prompts, which pres ents a substantial bottleneck in real world deployment. How can we automaticall y induce LLM-based agents and workflows in a data-driven way? This paper descri bes a simple data-driven approach for automatically inducing agents and LLM wor kflows. We formulate workflow induction as a bilevel optimization problem: an outer loop which optimizes a high-level sketch of the workflow (in particular h ow the LLM calls should be structured), and an inner loop which optimizes each i ndividual LLM call one-by one. Both loops are optimized with "textual gradients" where for the inner loop we optimize each component in a modular way through "b ackpropagating" textual gradients layer-by-layer. We find that LLM workflows dis covered through our FlowBot (work**flow** induction through **b**ilevel **o**pti mization and **t**extual gradients) approach performs competitively against stro ng baselines that make use of human-crafted or generated workflows.

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Vision-Language-Action Pretraining from Large-Scale Human Videos

Hao Luo, Yicheng Feng, Wanpeng Zhang, Sipeng Zheng, Ye Wang, Haoqi Yuan, jiazhen g liu, Chaoyi Xu, Haiweng Xu, Qin Jin, Zongqing Lu

Existing Vision-Language-Action (VLA) models struggle with complex manipulation tasks requiring high dexterity and generalization, primarily due to their relian ce on synthetic data with significant sim-to-real gaps or limited teleoperated d emonstrations.

To address this bottleneck, we propose leveraging human hands as a manipulator t
emplate, capitalizing on the rich dexterity and scalability present in web data
of human manipulation. Our approach introduces physical instruction tuning, a no
vel training paradigm that combines large-scale VLA pretraining from human video
s, perspective spatial alignment for reasoning in a unified physical space, and
post-training adaptation in physical environments. Additionally, we introduce a
part-level motion tokenization method that achieves millimeter-level reconstruct
ion accuracy to model precise hand trajectories serving as scalable motion primi
tives. To support our paradigm, we develop a comprehensive data curation pipeli
ne that integrates heterogeneous sources into a large-scale dataset with million
s of motion-based instructional instances. Empirically, our model demonstrates s
uperior performance in hand motion generation and instruction following, adherin
g to favorable scaling laws with respect to model and data sizes. Importantly, w
e demonstrate promising capabilities to robotic dexterous manipulation, validati
ng the effectiveness of bridging the human-robot embodiment gap. Project page is
 available at \url{https://research.beingbeyond.com/being-h0}.
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Routing and Reasoned Evaluation with Large Language Models
Guiyao Tie, Tianyao Luo, Xueyang Zhou, Chaoran Hu, Yunhong He, Junran Wu, Yuanfa
n Yao, Pan Zhou, Lichao Sun
Large language models (LLMs) are increasingly used to provide automated assessme
nt signals for evaluating model-generated outputs. However, practical deployment
 faces three persistent challenges: heterogeneous reliability across models, sub
stantial latency and token costs, and the absence of principled strategies for a
llocating evaluation resources. We introduce R^{2}Eval, a routing-aware automate
d assessment framework that formulates evaluation as a resource allocation and a
ggregation problem rather than relying on a single monolithic evaluator. R^{2}Ev
al combines difficulty-aware routing with reasoned evaluation signals to dynamic
ally select evaluator models on a per-instance basis under explicit accuracy, la
tency, and cost constraints. Our study makes three contributions. First, we cons
truct six difficulty-aware datasets spanning both reasoning-intensive (mathemati
cs, logic, code) and non-reasoning (knowledge, roleplay, writing) tasks, with hu
man-annotated reference assessments. Second, we provide a systematic empirical a
nalysis of how reasoning traces produced by different evaluator models correlate
 with assessment outcomes, revealing substantial variance and systematic mismatc
hes across difficulty regimes. Third, we develop and evaluate both offline and o
nline routing strategies that adaptively allocate evaluation queries, achieving
substantially improved accuracy–efficiency trade-offs compared to static baselin
es. Experiments across 19 language models demonstrate that R^{2}Eval significant
ly reduces evaluation cost and latency while maintaining close alignment with hu
man assessments. These results highlight the importance of routing-aware automat
ed assessment and establish R^{2}Eval as a scalable and reliable framework for l
arge-scale model evaluation.
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Effective MoE-based LLM Compression by Exploiting Heterogeneous Inter-Group Expe
rts Routing Frequency and Information Density
Zhendong Mi, Yixiao Chen, Pu Zhao, Xiaodong Yu, Hao Wang, Yanzhi Wang, Shaoyi Hu
ang
Mixture-of-Experts (MoE) based Large Language Models (LLMs) have achieved superi
or performance, yet the massive memory overhead caused by storing multiple exper
t networks severely hinders their practical deployment. Singular Value Decomposi
tion (SVD)-based compression has emerged as a promising post-training technique;
 however, most existing methods apply uniform rank allocation or rely solely on
static weight properties. This overlooks the substantial heterogeneity in expert
 utilization observed in MoE models, where frequent routing patterns and intrins
ic information density vary significantly across experts. In this work, we propo
se RFID-MoE, an effective framework for MoE compression by exploiting heterogene
ous Routing Frequency and Information Density. We first introduce a fused metric
 that combines expert activation frequency with effective rank to measure expert
 importance, adaptively allocating higher ranks to critical expert groups under

a fixed budget. Moreover, instead of discarding compression residuals, we recons
truct them via a parameter-efficient sparse projection mechanism to recover lost
 information with minimal parameter overhead. Extensive experiments on represent
ative MoE LLMs (e.g., Qwen3, DeepSeekMoE) across multiple compression ratios dem
onstrate that RFID-MoE consistently outperforms state-of-the-art methods like Mo
BE and $\mathbf{D}^2$-MoE. Notably, RFID-MoE achieves a perplexity of 16.92 on P
TB with the Qwen3-30B model at a 60% compression ratio, reducing perplexity by o
ver 8.0 compared to baselines, and improves zero-shot accuracy on HellaSwag by a
pproximately 8%.
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LayerT2V: A Unified Multi-Layer Video Generation Framework
Guangzhao Li, Kangrui Cen, Baixuan Zhao, Yi Xin, Siqi Luo, Guangtao Zhai, Lei Zh
ang, Xiaohong Liu
Text-to-video generation has advanced rapidly, but existing methods typically ou
tput only the final composited video and lack editable layered representations,
limiting their use in professional workflows. We propose \textbf{LayerT2V}, a un
ified multi-layer video generation framework that produces multiple semantically
 consistent outputs in a single inference pass: the full video, an independent b
ackground layer, and multiple foreground RGB layers with corresponding alpha mat
tes. Our key insight is that recent video generation backbones use high compress
ion in both time and space, enabling us to serialize multiple layer representati
ons along the temporal dimension and jointly model them on a shared generation t
rajectory. This turns cross-layer consistency into an intrinsic objective, impro
ving semantic alignment and temporal coherence. To mitigate layer ambiguity and
conditional leakage, we augment a shared DiT backbone with LayerAdaLN and layer-
aware cross-attention modulation. LayerT2V is trained in three stages: alpha mas
k VAE adaptation, joint multi-layer learning, and multi-foreground extension. We
 also introduce \textbf{VidLayer}, the first large-scale dataset for multi-layer
 video generation. Extensive experiments demonstrate that LayerT2V substantially
 outperforms prior methods in visual fidelity, temporal consistency, and cross-l
ayer coherence. To facilitate future research, we will release the code and data
set upon publication.
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Spherical Procrustes Alignment for Reliable Medical Audio Diagnosis
Ying Wang, Guoheng Huang, Chan-Tong Lam, Xiaochen Yuan
Reliable medical audio diagnosis requires models that are both accurate and hone
st about their uncertainty. However, fine-tuned models on small, imbalanced data
sets often become overconfident due to norm bias, where predictions rely on feat
ure magnitude rather than semantic alignment. The Equiangular Tight Frame (ETF)
provides a theoretical optimum for class separation and is effective for imbalan
ced and calibration tasks due to its maximal angular separability and geometric
fairness, but existing ETF-based methods perform poorly on noisy medical data be
cause gradient rotation is unstable and fixed ETFs cannot adapt to drifting prot
otypes. To address this, we propose Spherical Procrustes Alignment (SPA), which
combines spherical constraints with dynamic ETF alignment. SPA uses a Spherical
branch to eliminate norm bias via normalization and a Geometric branch to adapt
features and align a fixed ETF with drifting prototypes via dynamic Procrustes a
lignment, while a self-alignment mechanisms fuses the two branches to jointly op
timize logits. Experiments on ICBHI 2017 and CirCor DigiScope show that SPA achi
eves state-of-the-art performance and transforms pre-trained models into reliabl
e and efficient clinical tools without extra inference cost.
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Block Rotation is All You Need for MXFP4 Quantization
Yuantian Shao, Peisong Wang, Yuanteng Chen, Chang Xu, Zhihui Wei, Jian Cheng
Large language models (LLMs) have achieved remarkable success, but their rapidly
 growing scale imposes prohibitive costs in memory, computation, and energy. Pos
t-training quantization (PTQ) is a promising solution for efficient deployment,
yet achieving accurate W4A4 quantization remains an open challenge. While most e
xisting methods are designed for INT4 formats, the emergence of MXFP4—a new FP4
format with various hardware support (NVIDIA, AMD, Intel)—raises questions about

the applicability of current techniques. In this work, we present a unified empirical comparison of representative PTQ methods under the MXFP4 format. Through systematic evaluation, we find that methods like GPTQ consistently deliver strong performance, whereas rotation-based approaches, which are widely used in state-of-the-art approaches, suffer from severe incompatibility with MXFP4. We further provide the first in-depth analysis of this conflict, tracing its root to a fundamental mismatch between MXFP4's PoT (power-of-two) block scaling and the redistribution of outlier energy via global rotation. Building on this insight, we propose a simple yet effective block rotation strategy that adapts rotation-based methods to MXFP4, leading to substantial accuracy improvements across diverse LLMs. Our findings not only offer clear guidance for practitioners but also set a foundation for advancing PTQ research under emerging low-precision formats.
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Contrastive Representation Regularization for Vision-Language-Action Models
Taeyoung Kim, Jimin Lee, Myungkyu Koo, Dongyoung Kim, Kyungmin Lee, Changyeon Kim, Younggyo Seo, Jinwoo Shin
Vision-Language-Action (VLA) models have shown strong capabilities in robot manipulation by leveraging rich representations from pre-trained Vision-Language Models (VLMs).
However, their representations arguably remain suboptimal, lacking sensitivity to robotic signals such as control actions and proprioceptive information.
To address the issue, we introduce Robot State-aware Contrastive Loss (RS-CL), a simple and effective representation regularization for VLA models, designed to bridge the gap between VLM representations and robotic signals.
In particular, RS-CL aligns the representations more closely with the robot's proprioceptive states by using relative distances between the states as soft supervision.
Complementing the original action prediction objective, RS-CL enhances control-relevant representation learning, while being lightweight and fully compatible with standard VLA training pipelines.
Our empirical results demonstrate that RS-CL substantially improves the performance of state-of-the-art VLA models; it pushes the prior art to 69.7% achieving the state-of-the-art performance on the RoboCasa-Kitchen benchmark, and boosts success rates from 45.0% to 58.3% on challenging real-robot manipulation tasks.
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Improving CLIP Adaptation by Breaking Tail Alignment for Source-Free Cross-Domain Few-Shot Learning
Shuai Yi, Yixiong Zou, Yuhua Li, Ruixuan Li
Vision-Language Models (VLMs) such as CLIP demonstrate strong zero-shot generalization, but their performance significantly degrades in cross-domain scenarios with scarce target-domain training data (Cross-Domain Few-Shot Learning, CDFSL).
In this paper, we focus on the target-domain few-shot finetuning in the CLIP-based CDFSL task. Prevailing finetuning paradigms uniformly align all image patch tokens with their corresponding textual embeddings. However, we find a counterintuitive phenomenon: actively pushing away certain low-similarity image tokens, termed "tail tokens", from their textual embeddings consistently improves target-domain performance. We delve into this phenomenon and provide a novel interpretation: under great domain shifts and scarce training data, the model can hardly extract semantic information from visual inputs; therefore, the common belief of alignment is valid only for tokens already containing sufficient semantic information; for tail tokens, forcing the alignment would lead to excessive overfitting to the scarce training, while breaking the alignment is more useful. Motivated by this, we propose Adaptive Tail-Head Alignment (ATHA), a novel fine-tuning strategy for CLIP that transforms the conventional uniform alignment paradigm to an adaptive alignment paradigm, with both alignment strengthening and weakening.
Extensive experiments on four challenging CDFSL benchmarks validate our state-of-the-art performance. Our code is available at https://github.com/shuaiyi308/ATHA.
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Compositional Planning with Jumpy World Models

Jesse Farebrother, Matteo Pirotta, Andrea Tirinzoni, Marc G Bellemare, Alessandro Lazaric, Ahmed Touati

The ability to plan with temporal abstractions is central to intelligent decision-making. Rather than reasoning over primitive actions, we study agents that compose pre-trained policies as temporally extended actions, enabling solutions to complex tasks that no constituent alone can solve. Such compositional planning remains elusive as compounding errors in long-horizon predictions make it challenging to estimate the visitation distribution induced by sequencing policies. Motivated by the geometric policy composition framework introduced in Thakoor et al. (2022), we address these challenges by learning predictive models of multi-step dynamics --- so-called jumpy world models --- that capture state occupancies induced by pre-trained policies across multiple timescales in an off-policy manner. Building on Temporal Difference Flows (Farebrother et al., 2025), we enhance these models with a novel consistency objective that aligns predictions across timescales, improving long-horizon predictive accuracy. We further demonstrate how to combine these generative predictions to estimate the value of executing arbitrary sequences of policies over varying timescales. Empirically, we find that compositional planning with jumpy world models significantly improves zero-shot performance across a wide range of base policies on challenging manipulation and navigation tasks, yielding, on average, a 200% relative improvement over planning with primitive actions on long-horizon tasks.

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CountsDiff: A diffusion model on the natural numbers for generation and imputation of count-based data

Renzo Soatto, Anders Hoel, Greycen Ren, Shorna Alam, Stephen Bates, Nikolaos P. Daskalakis, Caroline Uhler, Maria Skoularidou

Diffusion models have excelled at generative tasks for both continuous and token-based domains, but their application to discrete ordinal data remains underdeveloped. We present CountsDiff, a diffusion framework designed to model distributions on the natural numbers. CountsDiff extends the Blackout diffusion framework by simplifying its formulation through a direct parameterization in terms of a survival probability schedule and an explicit loss weighting. This introduces flexibility through design parameters with direct analogues in existing diffusion modeling frameworks.

Beyond this reparameterization, CountsDiff introduces features from modern diffusion models, previously absent in counts-based domains, including continuous-time training, classifier-free guidance, and churn/remasking reverse dynamics that allow non-monotone reverse trajectories.

We propose an initial instantiation of CountsDiff and validate it on natural image datasets (CIFAR-10, CelebA), exploring the effects of the introduced design parameters in a complex, well-studied, and interpretable data domain. We then highlight biological count assays as a natural use case, evaluating CountsDiff on single-cell RNA-seq imputation in fetal and heart cell atlases. Remarkably, we find that even this simple instantiation matches or surpasses the performance of a state-of-the-art discrete generative model and leading scRNA-seq imputation methods, while leaving substantial headroom for further gains through optimized design choices in future work.

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Information dynamics and Memory in Neural Networks through Fisher Information Diffusion

Haodong Qin, Tatyana Sharpee

We present a general theoretical framework for analyzing how information about past inputs is encoded in recurrent networks into evolving dynamics rather than being represented as convergence to static attractors. Using dynamic mean-field theory and diffusion from physics, we derive a Fisher information diffusion operator that links network connectivity structure to the time-resolved propagation of information across interacting subpopulations. The analysis reveals that operating near criticality (spectral radius near one) is necessary but not sufficient for reliable memory in structured or non-normal recurrent networks; effective information retention requires alignment between input–output structure and stabl

e dynamical subspaces. The theory yields principled initialization rules that ba
lance stability and sensitivity, mitigating vanishing and exploding gradients. E
xperiments on the copy task and sequential MNIST show faster convergence and hig
her accuracy than standard random initialization. Together, these results provid
e both principled design guidelines for recurrent networks and new theoretical i
nsight into how information can be preserved over time in their dynamics.
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Benchmarking and Enhancing VLM for Compressed Image Understanding
Zifu Zhang, Tongda Xu, Siqi Li, Shengxi Li, Yue Zhang, Mai Xu, Yan Wang
With the rapid development of Vision-Language Models (VLMs) and the growing dema
nd for their applications, efficient compression of the image inputs has become
increasingly important. Existing VLMs predominantly digest and understand high-b
itrate compressed images, while their ability to interpret low-bitrate compresse
d images has yet to be explored by far. In this paper, we introduce the first co
mprehensive benchmark to evaluate the ability of VLM against compressed images,
varying existing widely used image codecs and diverse set of tasks, encompassing
over one million compressed images in our benchmark. Next, we analyse the sourc
e of performance gap, by categorising the gap from a) the information loss durin
g compression and b) generalisation failure of VLM. We visualize these gaps with
concrete examples and identify that for compressed images, only the generalizat
ion gap can be mitigated. Finally, we propose a universal VLM adaptor to enhance
model performance on images compressed by existing codecs. Consequently, we dem
onstrate that a single adaptor can improve VLM performance across images with va
rying codecs and bitrates by 10%-30%. We believe that our benchmark and enhancem
ent method provide valuable insights and contribute toward bridging the gap betw
een VLMs and compressed images. The source code is available at https://github.c
om/bblgbr/CompressVLMBench.
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Sharp Concentration Bounds for Bundle-Valued Statistics on Manifolds
Swagatam Das, Vaclav Snasel
Many geometric statistics and manifold learning pipelines routinely produce obse
rvations---such as tangent vectors or local frames---whose natural home is a var
ying family of fibers attached to different points of a base manifold, rather th
an a single shared vector space. Forming empirical averages requires transportin
g these observations to a common reference fiber, introducing curvature- and hol
onomy-driven effects absent from classical concentration theory. We develop a no
n-asymptotic concentration theory for such transported empirical means, deriving
finite-sample, dimension-free Hoeffding- and Bernstein-type bounds via sharp Hi
lbert-space inequalities. When shortest paths to the reference point are non-uni
que, transport becomes path-dependent and introduces a deterministic holonomy bi
as; we isolate and quantify this bias through bundle curvature and loop geometry
, with sharp closed-form formulas for the tangent bundle of a round sphere. The
resulting bias--variance decomposition separates the stochastic fluctuation deca
ying at the classical $n^{-1/2}$ rate in sample size n, from a curvature-drive
n error floor that no amount of additional data can eliminate; minimax lower bou
nds confirm both terms are unavoidable. We further establish a robust median-of-
means estimator achieving optimal rates under heavy tails, and a central limit t
heorem in the reference fiber. Controlled experiments on the sphere validate all
theoretical predictions.
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Adaptive Symmetry Discovery for Dynamical System Identification
Behrooz Tahmasebi, Melanie Weber
Dynamical systems model trajectory data generated by an underlying fixed dynamic
s, with applications ranging from biological systems to flows in physics. The id
entification problem concerns recovering the parameters of a system from observe
d trajectories. In many scientific settings, however, dynamical systems are not
generic and instead exhibit symmetries imposed by physical laws, formalized as e
quivariance with respect to a group action. In this work, we study \emph{adaptiv
e symmetry discovery} for dynamical system identification and address how a syst
em can be identified from a single trajectory when it is equivariant with respec

t to an unknown symmetry group. To this end, we first show that for known symmetries, the system can be identified from a significantly shorter single trajectory than in the generic setting, and we precisely characterize this improvement. We then consider the automatic symmetry discovery setting, proposing a method to learn the symmetry group directly from a single trajectory and incorporate it into the identification procedure, achieving the same optimal trajectory length as in the known-symmetry case. Our analysis relies on tools from group representation theory and the expander properties of Cayley graphs, and may be of independent interest for the study of symmetries in dynamical systems.
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Relative Entropy Estimation in Function Space: Theory and Applications to Trajectory Inference
CHAO WANG, Luca Nepote, Giulio Franzese, Pietro Michiardi
Trajectory Inference (TI) seeks to recover latent dynamical processes from snapshot data, where only independent samples from time-indexed marginals are observed.
In applications such as single-cell genomics, destructive measurements make path-space laws non-identifiable from finitely many marginals, leaving held-out marginal prediction as the dominant but limited evaluation protocol.
We introduce a general framework for estimating the Kullback–Leibler divergence (KL) between probability measures on function space, yielding a tractable, data-driven estimator that is scalable to realistic snapshot datasets.
We validate the accuracy of our estimator on a benchmark suite, where the estimated functional KL closely matches the analytic KL.
Applying this framework to synthetic and real scRNA-seq datasets, we show that current evaluation metrics often give inconsistent assessments, whereas path-space KL enables a coherent comparison of trajectory inference methods and exposes discrepancies in inferred dynamics, especially in regions with sparse or missing data. These results support functional KL as a principled criterion for evaluating trajectory inference under partial observability. Code available here: https://github.com/eurecom-probai/functional-kl.
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Adaptive DNA Sequence Modeling via Synergistic Plasticity Units
Binghao Liu, Wenzheng Zhao, Zhijie Zheng, Fei Gu
Effective DNA modeling demands the integration of complex patterns such as local motifs, long-range dependencies, and periodic signals. Yet, architectures like CNNs, Transformers, and SSMs are hindered by static or time-domain-exclusive designs, which limit their representational flexibility. To address this, we introduce the **Synergistic Plasticity Unit (SPU)**, a scalable architecture that achieves multi-level plasticity through three synergistic layers. Specifically, SPU integrates a *Locus Plasticity Layer* (LPL) to capture fine-grained local motifs via token-specific convolution operations, while utilizing a *Domain Plasticity Layer* (DPL) to form multi-domain global features by concurrently modeling sequential (time) and spectral (frequency) patterns. Furthermore, it incorporates a *Saliency Plasticity Layer* (SPL) to optimize information flow through dual-axis saliency scoring. Supported by theoretical analysis, extensive empirical validation, and in-depth biological interpretation, this unified design enables SPU to achieve state-of-the-art performance with quasi-linear complexity, establishing a robust and principled paradigm for DNA modeling.
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Ophiuchus: Incentivizing Tool-augmented ''Think with Images'' for Joint Medical Segmentation, Understanding and Reasoning
Yankai Jiang, Yujie Zhang, Peng Zhang, Wenjie Li, Yichen Li, Jintai Chen, Xiaoming Shi, Shihui Zhen
Recent medical MLLMs have made significant progress in generating step by step textual reasoning chains. However, they still struggle with complex clinical tasks that necessitate dynamic and iterative focusing on fine-grained visual regions. To close this gap, we introduce Ophiuchus, a versatile, tool-augmented framework that equips an MLLM to (i) decide when fine-grained visual evidence is needed, (ii) determine where to probe and ground within the medical image, and (iii) s

eamlessly weave the relevant sub-image content back into an interleaved, multimodal chain of thought for precise segmentation and diagnosis. Ophiuchus moves beyond mere tool-calling by tightly fusing the MLLM’s inherent grounding and reasoning capabilities with external tools, enabling more accurate and trustworthy decisions.
The core of our method is a three-stage training strategy: cold-start SFT for basic tool selection; self-reflection fine-tuning to strengthen decision revision; and agentic tool reinforcement learning to elicit sophisticated, expert-like diagnostic behaviors. Extensive experiments show that Ophiuchus consistently outperforms both closed-source and open-source SOTA methods across diverse medical benchmarks, including VQA, detection, and reasoning-based segmentation. Our project code is available at https://github.com/SII-zyj/Ophiuchus.
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FAB: A First-Order AB-based Gradient Algorithm for Distributed Bilevel Optimization over Time-Varying Directed Graphs
Yaoshuai Ma, Xiao Wang, Wei Yao, Jin Zhang
Distributed optimization over time-varying directed graphs has shown promising performance in addressing challenges posed by complex communication constraints in real-world scenarios.
In many practical settings, however, the direct application of distributed optimization algorithms encounters additional difficulties, most notably hyperparameter tuning, which our empirical observations suggest can be effectively mitigated by integrating bilevel optimization.
Motivated by these findings, we study distributed bilevel optimization over time-varying directed networks, a problem that remains largely unexplored due to the compounded challenges arising from consensus bias in dynamic unbalanced communication and the nested optimization structure.
In this work, we propose a fully first-order distributed gradient-based algorithm that integrates the Push–Pull (also known as AB) communication strategy with a value function-based penalty method and establish its non-asymptotic convergence properties.
Notably, a simplified variant of our analysis framework for nonconvex single-level distributed optimization establishes a convergence rate for the Push–Pull algorithm, thereby resolving an open question concerning its convergence over time-varying directed graphs.
Experiments across hyperparameter tuning, data hyper-cleaning, and reinforcement learning validate FAB's effectiveness and efficiency.
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VenusBench-Mobile: A Challenging and User-Centric Benchmark for Mobile GUI Agents with Capability Diagnostics
Yichen Gong, Zhuohan Cai, Sunhao Dai, Yuqi Zhou, Zhangxuan Gu, Changhua Meng, Shuheng Shen
Existing online benchmarks for mobile GUI agents remain largely app-centric and task-homogeneous, failing to reflect the diversity and instability of real-world mobile usage. To this end, we introduce VenusBench-Mobile, a challenging online benchmark for evaluating general-purpose mobile GUI agents under realistic, user-centric conditions.
VenusBench-Mobile builds two core evaluation pillars: defining what to evaluate via user-intent-driven task design that reflects real mobile usage, and how to evaluate through a capability-oriented annotation scheme for fine-grained agent behavior analysis.
Extensive evaluation of state-of-the-art mobile GUI agents reveals large performance gaps relative to prior benchmarks, indicating that VenusBench-Mobile poses substantially more challenging and realistic tasks and that current agents remain far from reliable real-world deployment.
Diagnostic analysis further shows that failures are dominated by deficiencies in perception and memory, which are largely obscured by coarse-grained evaluations.
Moreover, even the strongest agents exhibit near-zero success under environment variations, highlighting their brittleness in realistic settings.

Based on these insights, we believe VenusBench-Mobile provides an important step ping stone toward robust real-world deployment of mobile GUI agents. Code and da ta are available at https://github.com/inclusionAI/UI-Venus/tree/VenusBench-Mobi le.
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On the Learnability of Test-Time Adaptation: A Recovery Complexity Perspective
Zhi Zhou, Ming Yang, Shi-Yu Tian, Kun-Yang Yu, Lan-Zhe Guo, Yu-Feng Li
Test-time adaptation (TTA) aims to adapt models to maintain reliable performance on non-stationary test streams without requiring labeled data. Despite its empi rical success, the learnability of TTA under non-stationary streams remains unex plored. A key challenge is the lack of a principled theoretical framework that s imultaneously aligns with the TTA objective and captures both continuously evolv ing distribution shifts and intrinsic information constraints. To address this g ap, we propose the first theoretical framework for studying the learnability of TTA and introduce (ϵ,δ)-Recovery Complexity and (ϵ,ρ)-TTA Learnability. Recovery complexity measures the post-shift time needed to mai ntain excess risk below a target level with high probability, and is further ext ended to TTA learnability, which measures the long-term reliability of TTA. With in this framework, we introduce a novel discrete surrogate for non-stationary te st streams, enabling a unified and tractable analysis of both gradual and abrupt shifts. We derive order-wise matching lower and upper bounds on recovery comple xity, revealing fundamental limits of TTA and an intrinsic adaptivity-informatio n trade-off. These results provide unified learnability guarantees for TTA that complement regret-based analyses.
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Softplus Attention with Re-weighting Boosts Length Extrapolation in Large Langua ge Models
Bo Gao, Michael W. Spratling, Letizia Gionfrida
Large language models have achieved remarkable success in recent years,
primarily due to self-attention. However,
traditional Softmax attention suffers from numerical instability and
reduced performance as the number of inference tokens increases. This work
addresses these issues by proposing a new design principle for attention,
viewing it as a two-stage process.
The first stage (normalisation) refines standard attention by replacing Soft max with the more numerically stable Softplus followed by
l_{1}-normalisation. Furthermore, we introduce a dynamic scale factor base d on invariance entropy. We show that this novel attention mechanism outperforms conventional Softmax attention, and state-of-the-art Softmax-free alternatives.
Our second proposal is to introduce a second processing stage (sharpening) w hich consists of a
re-weighting mechanism that
amplifies significant attentional weights while diminishing weaker ones. Thi s enables
the model to concentrate more effectively on relevant tokens, mitigating
the attention sink phenomenon, and fundamentally improving length extrapolat ion.
This novel, two-stage, replacement for self-attention is shown to ensure num erical stability and dramatically
improve length extrapolation, maintaining a nearly constant validation
loss at 16$\times$ the training length while achieving superior results on
challenging long-context retrieval tasks and downstream benchmarks.
Furthermore, symbolic regression experiments demonstrate that our method ena bles
models to recover Newton's gravitational law from orbital trajectory sequenc es,
providing evidence that appropriate attention mechanisms are crucial for
foundation models to develop genuine physical world models. Our code is
available at https://github.com/iminfine/freeattn .
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Semantic-Aware Motion Encoding for Topology-Agnostic Character Animation
Zongye Zhang, Yuzhuo Cui, Qingjie Liu, Yunhong Wang
Generalizing motion representation across diverse characters remains challenging due to significant topological variations in skeletal structures across datasets and species, which hinder the development of scalable generative models. To bridge this gap, we propose a Semantic-Aware Topology-Agnostic framework that learns a unified latent manifold shared by disparate species. Unlike methods relying on fixed hierarchies or rigid padding strategies, our approach leverages a semantic modulation mechanism to align functional joint correspondences, thereby decoupling motion from topology. This design enables the construction of a continuous, generative-friendly motion space from large-scale, unaligned raw BVH data. Experiments on human and animal datasets demonstrate that our framework achieves high-fidelity reconstruction and supports downstream text-to-motion tasks. Notably, the model enables zero-shot cross-species retargeting without paired data. Code and demos are available at https://github.com/zzysteve/SATA
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Uncovering Latent Communication Patterns in Brain Networks via Adaptive Flow Routing
Tianhao Huang, Guanghui Min, Zhenyu Lei, Aiying Zhang, Chen Chen
Unraveling how macroscopic cognitive phenotypes emerge from microscopic neuronal connectivity remains one of the core pursuits of neuroscience. To this end, researchers typically leverage multi-modal information from structural connectivity (SC) and functional connectivity (FC) to complete downstream tasks. Recent methodologies explore the intricate coupling mechanisms between SC and FC, attempting to fuse their representations at the regional level. However, while these approaches do incorporate useful neuroscientific observations, they predominantly operate at a topological or architectural level and lack a principled formulation grounded in neural communication dynamics. Consequently, they are limited in quantifying how information is actually routed between neural regions, and thus cannot fully explain why SC and FC exhibit dynamic states of both coupling and heterogeneity. In this paper, we formulate multi-modal fusion through the lens of neural communication dynamics and propose the Adaptive Flow Routing Network (AFR-Net), a physics-informed framework that models how structural constraints give rise to functional communication patterns, enabling interpretable discovery of critical neural pathways. Extensive experiments demonstrate that AFR-Net significantly outperforms state-of-the-art baselines. The code is available at \url{https://github.com/Skyyyy0920/AFR-Net}.
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Hide&Seek: Learning to Explain in an End-to-End Differentiable Network
Tal Ellinson, Hadi Mohasel Afshar, Sally Cripps
Instance-wise feature selection is a valuable tool for interpreting labeled data and the predictions of black-box models. In contrast to global feature selection techniques, instance-wise methods dynamically identify important features for each instance. A growing number of methods learn a *selector*, which identifies important features, and a *predictor*, which uses these to make predictions. However, these pioneering methods face challenges including information leakage and lack of differentiability, which can slow training. In this paper, we present Hide&Seek, an end-to-end differentiable model for instance-wise feature selection. We jointly learn feature selection and prediction under a single objective without information leakage. Hide&Seek outperforms existing state-of-the-art models across a range of experiments and is fast to train. We achieve this by reformulating feature removal as a differentiable operation where instead of discretely removing features, we replace a proportion of each feature. Training is further stabilized via a parsimony-weight annealing framework.
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Entropy-aware Span-Constrained Optimal Transport for Robust Cross-Tokenizer Knowledge Distillation
Zhi-Ping Liu, Simiao Li, Wei Li, Hanting Chen, Jie Hu, Hua-Lei Yin, Xinghao Chen
Existing Cross-Tokenizer Knowledge Distillation (CTKD) methods can fail to outperform simple supervised fine-tuning when vocabulary overlap is low due to severe

alignment noise. We identify this phenomenon as the **``Low-Overlap negative transfer regime''**. To overcome this, we propose **Entropy-aware Span-Constrained Optimal Transport (E-SCOT)**, a robust framework that treats distillation as a sparse transport problem built upon a vocabulary-agnostic ground metric. Unlike prior OT approaches that incur quadratic costs via dense sequence-level optimization, E-SCOT employs span-anchored lexical alignment to construct a deterministic, locality-preserving support set in linear time with respect to sequence length. Furthermore, we introduce R\'enyi-entropy adaptive reweighting to dynamically concentrate the distillation budget on informative positions exhibiting significant uncertainty-profile gaps. Extensive experiments demonstrate that E-SCOT achieves state-of-the-art performance across diverse model families, effectively eliminating negative transfer even in challenging low-overlap scenarios.
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Secure Multi-agent Reinforcement Learning for Service Systems with Affinity and Byzantine Nodes: Stability Analysis and Protection Design
Yifan Jiang, Pan Jiasheng, Mengtian Li, Li Jin
We study decentralized multi-agent reinforcement learning (MARL) for networked service systems with affinity in the presence of Byzantine nodes. The way that a server processes a job depends on an affinity state that captures the correlation between the job and the server. Each node learns a local control policy via an actor-critic algorithm with linear function approximation over inherently unbounded space of traffic states, while exchanging parameter information with neighbors through a communication graph. A set of Byzantine agents can exploit the unbounded state space to compromise the consensus mechanism, destabilizing both learning and queuing processes. To address this vulnerability, we propose a resilient consensus-based MARL algorithm, which mitigates adversarial parameter manipulation and guarantees traffic stability under mild assumptions. We prove that the cooperative agents’ policies converge almost surely to a bounded neighborhood of a stationary solution of the global objective. We demonstrate the effectiveness and generality of the proposed framework in several representative service systems, including semantic routing for large language model serving, distributed polling in cloud computing, and smart manufacturing logistics.
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Are Object-Centric Representations Better at Compositional Generalization?
Ferdinand Kapl, Amir Mohammad Karimi Mamaghan, Maximilian Seitzer, Karl Henrik Johansson, Carsten Marr, Stefan Bauer, Andrea Dittadi
Compositional generalization, the ability to reason about novel combinations of familiar concepts, is fundamental to human cognition and a critical challenge for machine learning.
Object-centric (OC) representations, which encode a scene as a set of objects, are often argued to support such generalization, but systematic evidence in visually rich settings is limited. We introduce a Visual Question Answering benchmark across three controlled visual worlds (CLEVRTex, Super-CLEVR, and MOVi-C) to measure how well vision encoders, with and without object-centric biases, generalize to unseen combinations of object properties. To ensure a fair and comprehensive comparison, we carefully account for training data diversity, sample size, representation size, downstream model capacity, and compute. We use DINOv2 and SigLIP2, two widely used vision encoders, as the foundation models and their OC counterparts. Our key findings reveal that (1) OC approaches are superior in harder compositional generalization settings; (2) original dense representations surpass OC only on easier settings and typically require substantially more downstream compute; and (3) OC models are more sample efficient, achieving stronger generalization with fewer images, whereas dense encoders catch up or surpass them only with sufficient data and diversity. Overall, object-centric representations offer stronger compositional generalization when any one of dataset size, training data diversity, or downstream compute is constrained.
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MEG-XL: Data-Efficient Brain-to-Text via Long-Context Pre-Training
Dulhan Jayalath, Oiwi Parker Jones
Clinical brain-to-text interfaces are designed for paralysed patients who cannot

provide extensive training recordings. Pre-training improves data-efficient generalisation by learning statistical priors across subjects, but these priors critically depend on context. While natural speech might unfold gradually over minutes, most methods pre-train with only a few seconds of context. Thus, we propose *MEG-XL*, a model pre-trained with 2.5 minutes of MEG context per sample, 5-300× longer than prior work, and equivalent to 191k tokens, capturing extended neural context. Fine-tuning on the task of word decoding from brain data, MEG-XL matches supervised performance with a fraction of the data (e.g. 1hr vs 50hrs) and outperforms brain foundation models. We find that models pre-trained with longer contexts learn representations that transfer better to word decoding. Our results indicate that long-context pre-training helps exploit extended neural context that other methods unnecessarily discard.

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Physiology as Language: Translating Respiration to EEG during Sleep

Kaiwen Zha, Chao Li, Hao He, Peng Cao, Tianhong Li, Ali Mirzazadeh, Ellen Zhang, Jong Woo Lee, Yoon Kim, Dina Katabi

This paper introduces a novel cross-physiology translation task: synthesizing sleep electroencephalography (EEG) from respiration signals. To address the significant complexity gap between the two modalities, we propose a waveform-conditional generative framework that preserves fine-grained respiratory dynamics while constraining the EEG target space through discrete tokenization. Trained on over 28,000 individuals, our model achieves a 7% Mean Absolute Error in EEG spectrogram reconstruction. Beyond reconstruction, the synthesized EEG supports downstream tasks with performance comparable to ground truth EEG on age estimation (MAE 5.0 vs. 5.1 years), sex detection (AUROC 0.81 vs. 0.82), and sleep staging (Accuracy 0.84 vs. 0.88), significantly outperforming baselines trained directly on breathing. Finally, we demonstrate that the framework generalizes to contactless sensing by synthesizing EEG from wireless radio-frequency reflections, highlighting the feasibility of remote, non-contact neurological assessment during sleep.

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What Reward Structure Enables Efficient Sparse-Reward RL? A Proof-of-Concept with Policy-Aware Matrix Completion

Ibne Farabi Shihab, Sanjeda Akter, Anuj Sharma

Sparse-reward reinforcement learning typically focuses on exploration, but we ask: can structural assumptions about reward functions themselves accelerate learning? We introduce Policy-Aware Matrix Completion (PAMC), which exploits low-rank structure in reward matrices while correcting for policy-induced sampling bias. PAMC combines three key components: a low-rank plus sparse reward model, inverse propensity weighting to handle Missing-Not-At-Random (MNAR) data, and confidence-gated abstention that falls back to intrinsic exploration when uncertain. We provide finite-sample theory showing that completion error scales as $O(\sigma\sqrt{r(|\mathcal{S}|+|\mathcal{A}|)/\text{ESS}})$ where ESS is the effective sample size under policy overlap κ. PAMC achieves strong empirical results at 10M steps (a sample-efficiency comparison): 4100$\pm$250 return vs. 200$\pm$50 for DrQ-v2 on Montezuma's Revenge, 78\% vs. 65\% success rate on MetaWorld-50, and 15\% improvement over CQL on D4RL datasets. The method maintains 8\% computational overhead while providing calibrated confidence intervals (95\% empirical coverage). When structural assumptions are violated, PAMC gracefully degrades through increased abstention rather than catastrophic failure. Our approach demonstrates that reward structure exploitation can complement traditional exploration methods in sparse-reward domains.

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EvoMAS: Heuristics in the Loop—Evolving Smarter Agentic Workflows

Yangbo Wei, Zhen Huang, Ronghao Xu, Hong Wang, WEI W. XING

The rapid development of Large Language Models has driven Multi-Agent Systems (MAS) growth, but constructing efficient MAS requires labor-intensive manual design. Current automation methods generate templated agents, use monolithic optimization, and ignore task complexity gradients. This paper presents Evolutionary MAS (EvoMAS), a biologically-inspired framework whose core is a dynamic and diverse repertoire of seven evolutionary strategies---six biologically-inspired operato

rs (3 exploration, 3 exploitation) together with a custom operator for domain-specific transformations---driven by adaptive strategy selection. These are complemented by role-level evolution that refines agent specialization and collaboration patterns, and a curriculum-guided schedule that evolves workflows from simple to complex tasks with cross-stage stability. Additionally, to resolve the contradiction between the inefficiency of pure evolution and the rigidity of manual design, we introduce the *Cyber Creator*, a meta-controller that conducts heuristics-in-the-loop learning by formulating and reflectively updating evolutionary rules and strategies. Evaluations demonstrate that EvoMAS consistently outperforms existing methods across multiple domains while maintaining cost efficiency, with roles evolving from homogeneous actors to specialized reasoning ensembles.

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Hard-Constrained Graph Generation with Discrete-Projection Diffusion

Xuesong Zhang, Haifeng Sun, Qi Qi, Shengkuan Li, Yuhao Li, Tianyi Kou, Zirui Zhuang, Bo He, Jianxin Liao, Jingyu Wang

Diffusion models have achieved remarkable success in graph generation, but enforcing hard constraints on generated graphs remains challenging, limiting their deployment in constraint-critical applications. Existing approaches either fail to guarantee strict constraint satisfaction or are limited to narrow constraint types, lacking the flexibility to handle diverse constraint specifications. To address this challenge, we exploit the discrete structure of graphs, which allows hard constraints to be formulated as symbolic reasoning problems. Building on this insight, we propose NSPSG, a framework that integrates unconstrained diffusion models with discrete projection operators. NSPSG employ an SMT (Satisfiability Modulo Theories)-based projector to ensure that the generated graphs strictly satisfy constraints while remaining within the training data distribution. To further accelerate generation, we employ a supervised auto-regressive neural projector to approximate the symbolic reasoning process. Across heterogeneous constraints and various graph generation datasets, NSPSG achieves 99%-100% validity rates, demonstrating state-of-the-art performance. Notably, for a complex non-linear constraint, it improves data validity by up to 43% and reaches 99% validity while maintaining comparable generation efficiency.

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Beyond Buffer Limits: Energy-Based Data Reassembly for Continual Learning

Zhenyi Wang, Yixuan Sun, Yue Wang, Zhong Chen, Heng Huang

Continual learning (CL) aims to acquire new knowledge from a non-stationary data stream while retaining performance on previously learned tasks. Memory-based replay methods mitigate catastrophic forgetting by storing and revisiting past samples, but their effectiveness is fundamentally constrained by limited memory capacity, as each stored example represents only a single data instance. In this work, we propose data reassembly for CL, a new paradigm that significantly increases memory efficiency by reassembling composite replay samples from existing training data. Instead of storing raw training examples, we partition the current task training data into elementary patches and dynamically reassemble them into coherent replay instances through an energy-based optimization framework. The proposed objective jointly enforces semantic compatibility with target labels and global consistency among assembled patches. To make this optimization tractable, we derive an efficient variational inference algorithm that constructs a compact yet diverse set of reassembled samples for replay. Extensive theoretical analysis and experiments across multiple CL benchmarks demonstrate that data reassembly consistently outperforms existing memory-based approaches, achieving stronger retention of past knowledge while maintaining competitive computational efficiency.

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Implicit Action Chunking for Smooth Continuous Control

Bosun Liang, Shuo Pei, Zirui Chen, Chuanzhi Fan, Chen Sun, Yuankai Wu, Huachun Tan, Yong Wang

Reinforcement learning often produces high-frequency oscillatory control signals that undermine the safety and stability required for physical deployment. Explicit action chunking addresses this by predicting fixed-horizon trajectories but

scales the policy output dimension proportionally with the horizon length, leading to optimization difficulties and incompatibility with standard step-wise interaction. To overcome these challenges, this paper proposes Dual-Window Smoothing (DWS), an implicit action chunking framework for smooth continuous control. Unlike explicit methods, DWS enforces temporal coherence without expanding the action space. It uses a dual-window design: an execution window that ensures physical smoothness through deterministic modulation, and a value window that aligns temporal-difference targets over the horizon to correct critic bias caused by open-loop execution. DWS also includes a lightweight actor-side temporal regularizer based on first-order action differences to promote global continuity. This design effectively bridges the gap between temporal abstraction and reactive step-wise control. Experiments on benchmarks including the DeepMind Control Suite and industrial energy management tasks show that DWS outperforms state-of-the-art (SOTA) baselines. In complex vision-based autonomous driving tasks, DWS achieves smoother control, safer behavior with reduced jitter, and attains a 100% success rate.

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Expand Neurons, Not Parameters

Linghao Kong, Inimai Subramanian, Yonadav G Shavit, Micah Adler, Dan Alistarh, Nir N Shavit

This work demonstrates how increasing the number of neurons in a network without increasing its total number of non-zero parameters improves performance. We show that this gain corresponds with a decrease in interference between multiple features that would otherwise share the same neurons. On symbolic Boolean tasks, splitting each neuron into sparser sub-neurons with knowledge of the clauses systematically reduces polysemanticity metrics and yields higher task accuracy. Notably, even random splits of neuron weights approximate these gains, indicating that reduced collisions, not precise assignment, are a primary driver. Consistent with the superposition hypothesis, the benefits of this framework grow with increasing interference: when polysemantic load is high, accuracy improvements are the largest. Transferring these insights to more realistic models, including classifiers over CLIP embeddings, convolutional neural networks, and deeper multilayer networks, we find that widening networks while maintaining a constant non-zero parameter count consistently increases accuracy. These results identify an interpretability-grounded mechanism to leverage width against superposition, improving performance without increasing the number of non-zero parameters. Such a direction is well matched to modern accelerators, where memory movement of non-zero parameters, rather than raw compute, is often a dominant bottleneck.

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Uncertainty-Constrained Trustworthiness for Graph Learning

Chunhui Zhang, Pengqi Li, Lizhong Ding, Ye Yuan, Peng Yang, Changsheng Li, Guoren Wang

Graph learning has been increasingly deployed in critical and sensitive domains, raising pressing demands for trustworthiness-robustness, fairness, and beyond. However, these properties are often undermined by various perturbations, which induce distributional uncertainty and compromise the trustworthiness of graph learning. To address this, we propose DICT, a novel framework that models distributional uncertainty to achieve trustworthy graph learning. Specifically, DICT formulates a unified optimization objective that captures perturbation-induced distributional shifts in graph topology, node features, and labels, and minimizes the worst-case risk over the uncertainty set. However, directly optimizing this objective in its primal form leads to an infinite-dimensional problem. To make this problem tractable, we integrate strong duality and local Lipschitz continuity of the loss, reformulating the objective as a finite-dimensional min-max problem. We focus on robustness and fairness as primary instantiations of DICT because they are not only critical in real-world applications, but also provide transferable modeling principles for broader trustworthiness objectives. By formulating fairness in the form of an uncertainty set, DICT pioneers unified robustness and fairness within a single optimization framework. Extensive experiments across diverse benchmarks and backbones demonstrate that DICT consistently improves both

robustness and fairness, validating the effectiveness and adaptability of the DICT framework.
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Credibility-Aware Weighting Federated Causal Discovery for Time Series
Jiegang Xu, Fuyuan CAO, Jiye Liang
Federated causal discovery for time series is becoming increasingly important in many application domains. In practice, intervention policies on each client often change over time, causing the local underlying causal mechanisms to drift rather than remain fixed. Moreover, different sampling frequencies across clients yield incompatible time scales in the observed data, making the resulting local causal graphs naturally heterogeneous and difficult to aggregate consistently. Accordingly, we propose Fed-CAW, a Credibility-Aware Weighting Federated causal discovery framework for time series. Specifically, we define edge-level credibility scores that quantify per-edge reliability by summarizing (i) within-client temporal stability across windows and (ii) cross-client temporal consistency after mapping heterogeneous sampling frequencies onto a unified time scale. We then aggregate privatized edge statistics under differential privacy, treating credibility scores as weights to recover a global causal graph while preserving personalized local structures without sharing raw data. Theoretically, we demonstrate the rationale for the unified time scale mapping and establish rigorous differential privacy guarantees. Experimental results on synthetic and real-world datasets demonstrate the effectiveness of our proposed method.
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ProEval: Proactive Failure Discovery and Efficient Performance Estimation for Generative AI Evaluation
Yizheng Huang, Wenjun Zeng, Aditi Kumaresan, Zi Wang
Evaluating generative AI models is increasingly resource-intensive due to slow inference, expensive raters, and a rapidly growing landscape of models and benchmarks. We propose ProEval, a proactive evaluation framework that leverages transfer learning to efficiently estimate performance and identify failure cases. ProEval employs pre-trained Gaussian Processes (GPs) as surrogates for the performance score function, mapping model inputs to metrics such as the severity of errors or safety violations. By framing performance estimation as Bayesian quadrature (BQ) and failure discovery as superlevel set sampling, we develop uncertainty-aware decision strategies that actively select or synthesize highly informative inputs for testing. Theoretically, we prove that our pre-trained GP-based BQ estimator is unbiased and bounded. Empirically, extensive experiments on reasoning, safety alignment, and classification benchmarks demonstrate that ProEval is significantly more efficient than competitive baselines. It requires 8–65x fewer samples to achieve estimates within $\pm1\%$ of the ground truth, while simultaneously revealing more diverse failure cases under a stricter evaluation budget. Our open-sourced code and data can be found at https://github.com/google-deepmind/proeval.
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Label-Guided Representation Learning for Incomplete Multi-View Multi-Label Classification
Yang Li, Quanjiang Li, Tingjin Luo
Incomplete multi-view multi-label classification addresses scenarios where views and labels are partially missing. While existing methods treat labels solely as supervision signals, they overlook the semantic structure inherent in partial annotations. We propose Label-Guided Representation Learning (LGRL) that systematically exploits label semantics as structural priors throughout learning. Our framework constructs a semantic-informed mixture prior via learnable category prototypes to guide representation extraction, and introduces category-specific conditional posteriors where prototypes act as Bayesian experts for multi-view fusion. We further derive a principled label-driven information bottleneck objective balancing reconstruction sufficiency with cross-view consistency, enabling category-conditional reasoning. Extensive experimental results demonstrate the effectiveness of LGRL across benchmark datasets as well as real-world applications in sports analytics and medical imaging.

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Measuring and Mitigating Post-Hoc Rationalization in Reverse Chain-of-Thought Generation
Guangyue Peng, Zongchao Chen, Wen Luo, Yuntao Wen, Wei Li, Ruixiang Feng, Ran Le, Chen Yang, Zhenwei An, Yang Song, Tao Zhang, Houfeng Wang
Reverse Chain-of-Thought Generation (RCG) synthesizes reasoning traces from query-answer pairs, but it risks producing post-hoc rationalizations: when models can see the answer during generation, the answer serves as a cognitive anchor that shapes the entire explanation. We formalize this phenomenon through a three-level measurement hierarchy: lexical, entropic, and probabilistic anchoring, which capture surface artifacts, entropy dynamics, and latent answer dependence, respectively. We analyze semantic suppression, the intuitive mitigation strategy that instructs models to ignore the answer, and find that it is counterproductive: while it reduces lexical overlap, it paradoxically increases entropic and probabilistic anchoring. We attribute this failure to active monitoring of the forbidden answer, which inadvertently deepens dependence on it. To break this cycle, we propose Structural Skeleton-guided Reasoning (SSR), whose core contribution is to replace answer suppression with structural decoupling: SSR first generates an answer-invariant functional skeleton and then uses it as a content-neutral target for full trace generation. Experiments across open-ended reasoning benchmarks show that SSR consistently mitigates anchoring, and that Distilled SSR (SSR-D), a distillation variant that internalizes skeleton-guided reasoning from teacher-generated traces, achieves up to 10\% improvement over suppression baselines while mitigating out-of-distribution (OOD) degradation.
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Structured Multi-modal Graph Disentanglement for Psychiatric Diagnosis
Hongyu Shi, Kaizhong Zheng, Wensheng Zhai, Shuai Jiang, Badong Chen, Liangjun Chen
Multi-modal neuroimaging-based psychiatric diagnosis must integrate cross-modal agreement with modality-specific complementarity, yet in real multi-site cohorts these signals are frequently entangled with site- and cohort-dependent correlations, yielding shortcut-driven predictions and limited interpretability. We propose Structured Multi-modal Graph Disentanglement (SMGD), which explicitly factorizes multi-modal graph representations into four components with distinct roles: shared diagnostic evidence, complementary diagnostic evidence, incidental cross-modal agreement, and modality-specific non-robust correlations, with the former two forming the diagnostic core and the latter two suppressed as shortcuts. SMGD is realized as geometry-driven structure learning: under a mild distributional assumption, we develop mini-batch estimable surrogate regularizers that shape subspace organization and cross-modal relations, enforcing semantic consistency through relational geometry rather than centroid coincidence while suppressing confounded dependencies. Experiments on large multi-site datasets show improved in-domain diagnosis and more reliable cross-dataset generalization in the presence of a Modality Gap, without relying on expert-crafted diagnostic biomarkers.
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GeoPT: Scaling Physics Simulation via Lifted Geometric Pre-Training
Haixu Wu, Minghao Guo, Zongyi Li, Zhiyang Dou, Mingsheng Long, Kaiming He, Wojciech Matusik
Neural simulators promise efficient surrogates for physics simulation, but scaling them is bottlenecked by the prohibitive cost of generating high-fidelity training data. Pre-training on abundant off-the-shelf geometries offers a natural alternative, yet faces a fundamental gap: supervision on static geometry alone ignores dynamics and can lead to negative transfer on physics tasks. We present GeoPT, a unified pre-trained model for general physics simulation based on lifted geometric pre-training. The core idea is to augment geometry with synthetic dynamics, enabling dynamics-aware self-supervision without physics labels. Pre-trained on over one million samples, GeoPT consistently improves industrial-fidelity benchmarks spanning fluid mechanics for cars, aircraft, and ships, and solid mechanics in crash simulation, reducing labeled data requirements by 20-60% and accelerating convergence by 2$\times$. These results show that lifting with syntheti

c dynamics bridges the geometry-physics gap, unlocking a scalable path for neura l simulation and potentially beyond. Code is available at https://github.com/Phy sics-Scaling/GeoPT.
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scCBGM: Single-Cell Editing via Concept Bottlenecks
Alma Andersson, Aya Abdelsalam Ismail, Edward De Brouwer, Doron Haviv, Tommaso B iancalani, Kyunghyun Cho, Gabriele Scalia, Aicha BenTaieb, Hector Corrada Bravo
Understanding cellular phenotypes and how they respond to perturbations is criti cal for disease biology and therapeutic design. Single-cell RNA sequencing enabl es characterization at cellular resolution, yet the combinatorial space of condi tions makes exhaustive experimental mapping infeasible. We introduce single-cell Concept Bottleneck Generative Models (scCBGM), a framework for interpretable an d precise counterfactual editing of individual cells. scCBGM adapts concept bott leneck architectures for single-cell data through decoder skip connections and a cross-covariance penalty that promotes disentanglement without dimensional cons traints. We extend the framework to flow matching models, enabling concept-guide d editing in both encoding-decoding and generation regimes. To enable rigorous e valuation, we develop a synthetic benchmark with ground-truth counterfactuals. A cross multiple real datasets, scCBGM demonstrates superior performance in combin atorial generalization and counterfactual prediction, supported by cell-level va lidation on synthetic data and population-level benchmarks on real datasets.
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AppWorld-UL: Benchmarking Diverse Agent-User Interactions for Tool-Use
Junzhi Chen, Harsh Trivedi, Jane Pan, Michael JQ Zhang, Tejas Srinivasan, Niranj an Balasubramanian, Ashish Sabharwal
Tool-use agents that address day-to-day digital tasks such as ordering groceries must not only operate applications, but also interact with the user, e.g., to a sk clarification questions, prompt for confirmation, and inform the user when th e instruction is infeasible. However, current benchmarks for evaluating agent-us er interactions do not capture the diversity of such interactions. Further, they operate in small environments with few, often non-state-changing, APIs. To addr ess this gap, we introduce AppWorld-UL, a ``user-in-the-loop'' benchmark of 516 challenging tasks requiring diverse agent-user interactions. Building upon the A ppWorld framework with 9 popular simulated apps like Amazon and Spotify, we syst ematically modify original tasks to introduce ambiguities and constraints that n ecessitate various types of agent-user interaction. User behavior is simulated b y an LLM prompted to respond with carefully designed knowledge boundaries, offer ing more reliable simulation than the unconstrained or overly rigid alternatives used in prior work. Our evaluation reveals that a state-of-the-art LLM, Claude Opus 4.7, achieves only 48.6\% success on AppWorld-UL, and only 35.7\% on the ha rder, compositional subset. On the stricter, scenario-level metric, compositiona l task performance drops to only 21.3\%. Our analysis reveals that correct user-interaction is crucial for success. This demonstrates the benchmark's difficulty and its potential to advance research on user-in-the-loop tool-use agents.
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UnHype: CLIP-Guided Hypernetworks for Dynamic LoRA Unlearning
Piotr Wójcik, Maksym Petrenko, Wojciech Gromski, Przemys■aw Spurek, Maciej Zieba
Recent advances in large-scale diffusion models have intensified concerns about their potential misuse, particularly in generating realistic yet harmful or soci ally disruptive content. This challenge has spurred growing interest in effectiv e machine unlearning, the process of selectively removing specific knowledge or concepts from a model without compromising its overall generative capabilities. Among various approaches, Low-Rank Adaptation (LoRA) has emerged as an effective and efficient method for fine-tuning models toward targeted unlearning. However , LoRA-based methods often exhibit limited adaptability to concept semantics and struggle to balance removing closely related concepts with maintaining generali zation across broader meanings. Moreover, these methods face scalability challen ges when multiple concepts must be erased simultaneously. To address these limit ations, we introduce UnHype, a framework that incorporates hypernetworks into si ngle- and multi-concept LoRA training. The proposed architecture can be directly

 plugged into Stable Diffusion as well as modern flow-based text-to-image models , where it demonstrates stable training behavior and effective concept control. During inference, the hypernetwork dynamically generates adaptive LoRA weights b ased on the CLIP embedding, enabling more context-aware, scalable unlearning. We evaluate UnHype across several challenging tasks, including object erasure, cel ebrity erasure, and explicit content removal, demonstrating its effectiveness an d versatility.
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Bilinear Bandits with Partially Observable Features
Wooseong Cho, Ji Hyeong Park, Min-hwan Oh
We study bilinear bandits with partially observable features on both the user an d item sides. In each of T rounds, the learner selects a user--item pair and o bserves only the reward for the chosen pair. The reward model is bilinear in the user and item features with an unknown parameter matrix. Existing work commonly reduces this problem to a linear bandit via Kronecker-product features, increas ing dimensionality and losing bilinear structure.

    We propose \texttt{BiRoLF}, an algorithm robust to latent features, which di rectly leverages the bilinear structure without such linearization. It augments observed feature spaces on both sides with orthogonal complement bases and emplo ys doubly robust (DR) estimation to impute rewards for unselected pairs, constru cting matrix-valued pseudo-rewards. We estimate the effective parameter using a bilinear DR-Lasso estimator, which promotes sparsity in components orthogonal to observed features.

    \texttt{BiRoLF} achieves a $\tilde{O}(\sqrt{(d_x + d_{h_x})(d_y + d_{h_y}) T })$ regret bound, where d_x and d_y are observable feature dimensions, and $ d_{h_x}$ and d_{h_y} denote effective orthogonal-complement dimensions.

    By exploiting the induced block-diagonal Gram structure, \texttt{BiRoLF} per forms an exact blockwise update that preserves the full DR-Lasso objective. Nume rical experiments show strong regret performance and computational gains.
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Echoes within the Reasoning: Stealthy and Effective Watermarking via Chain of T hought
Jiacheng Lu, Yiming Li, Tao Song, Weijian Wang, Wenjie Qu, Haibing Guan, Jiaheng Zhang
Large Language Models with Chain-of-Thought reasoning capabilities represent val uable intellectual property, yet existing black-box watermarking methods often t rade robustness for reasoning fidelity by perturbing final answers or relying on fragile trigger patterns. We propose BiCoT, a watermarking framework that embed s ownership signals into the internal geometry of reasoning traces by aligning h igh-saliency structural anchors with a private signature subspace while regulari zing ordinary control tokens to preserve semantic capacity. This design couples the watermark with reasoning-relevant representations, making removal difficult without disrupting the features that support coherent reasoning. To enable verif ication under model theft and representation drift, we introduce Robust Subspace Registration(RSR), a Top-k logprob-based black-box verifier that uses sentine l tokens to calibrate systematic shifts in the output distribution. Experiments show that BiCoT preserves reasoning fidelity across diverse complex reasoning ta sks while achieving robust detection under fine-tuning, quantization, model-leve l perturbations, and adaptive output-level attacks across in-domain and out-of-d istribution settings.
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Fast Mixing Steady-State Control in Markov Decision Processes
Federico Corso, Marco Mussi, Alberto Maria Metelli
Stability is a property of fundamental importance in real-world systems. Althoug h it has been widely studied and well understood in *control theory* (CT) for de terministic systems, it is largely overlooked in stochastic systems such as *Mar kov decision processes* (MDPs). In this paper, we aim to translate the steady-st ate control problem, well established in CT, where the goal is to synthesize a c ontroller with prescribed asymptotic stability properties, into the MDP framewor k.

To this end, we propose the novel *fast-mixing steady-state* (FMSS) problem. Given an ergodic MDP and a target steady-state distribution, the objective is to synthesize a Markovian policy that induces this distribution with the fastest possible convergence rate. Addressing this problem requires controlling the spectral properties of the induced *Markov chain* (MC) transition matrix, which generally leads to non-convex programs. Thus, we derive a tractable surrogate objective that leads to a convex program, whose properties we study in terms of approximation quality, feasibility, and computational complexity.

We then move to the learning setting and propose an "offline" sample-based algorithm for FMSS (FMSS-SV), designed for tabular MDPs, in which the environment’s transition model is estimated from data. We quantify the impact of transition model estimation errors on both the objective value and the learned policy, and provide a finite-sample complexity analysis.

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ExpertWeaver: Unlocking the Inherent MoE in Dense LLMs with GLU Activation Patterns

Ziyu Zhao, Tong Zhu, Xin Yu, Zhi Zhang, Tiantian Fan, Jinluan Yang, Kun Kuang, zhongyu wei, Fei Wu, Yu Cheng

Mixture-of-Experts (MoE) scales model capacity while preserving computational efficiency through sparse expert activation.
However, training high-quality MoEs from scratch is prohibitively expensive.
An alternative is to convert pretrained dense models into sparse MoEs.
Existing dense-to-MoE methods fall into two categories: \textbf{dynamic structural pruning} that converts dense models into MoEs with moderate sparsity to balance performance and efficiency, and \textbf{downcycling} approaches that use pretrained dense models to initialize highly sparse MoEs. However, existing methods break the intrinsic activation patterns within dense models, leading to suboptimal expert construction.
In this work, we argue that the Gated Linear Unit (GLU) provides a natural blueprint for dense-to-MoE conversion. We show that the fine-grained neuron-wise activation patterns of GLU reveal a coarse-grained structure, uncovering an inherent MoE architecture composed of consistently activated universal neurons and dynamically activated specialized neurons.
Leveraging this discovery, we introduce ExpertWeaver, a training-free framework that partitions neurons according to their activation patterns and constructs shared experts and specialized routed experts with layer-adaptive configurations.
Experiments demonstrate that ExpertWeaver outperforms existing methods, both as a training-free dynamic structural pruning technique and as a downcycling strategy for MoE initialization.

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Learning Credal Ensembles via Distributionally Robust Optimization

Kaizheng Wang, Ghifari Adam Faza, Fabio Cuzzolin, Siu Lun Chau, David Moens, Hans Hallez

Credal predictors are epistemic-uncertainty-aware models that produce a convex set of probabilistic predictions. They provide a principled framework for quantifying predictive epistemic uncertainty (EU) and have been shown to improve model robustness across a range of settings. However, most state-of-the-art (SOTA) methods primarily define EU as disagreement induced by random training initializations, which mainly reflects sensitivity to optimization randomness rather than uncertainty from more substantive sources. In response, we formulate EU as disagreement between models trained under different degrees of relaxation of the i.i.d. assumption between the training and test distributions. Building on this idea, we propose *CreDRO*, which learns an ensemble of plausible models via distributionally robust optimization. As a result, CreDRO captures EU arising not only from training randomness but also from informative disagreement due to potential train–test distribution shifts. Empirically, CreDRO consistently outperforms SOTA credal approaches on downstream tasks, including out-of-distribution detection on extensive benchmarks and selective classification in medical settings.

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Unveiling the Visual Counting Bottleneck in Vision-Language Models

Xingzhou Pang, Yifan Hou, Junling Wang, Mrinmaya Sachan
While Large Vision-Language Models (VLMs) excel at interpolation, they suffer catastrophic failures in systematic generalization, most notably in visual counting beyond training distributions. In this work, we investigate this extrapolation bottleneck by deconstructing visual counting into three cognitive stages: object individuation, abstract magnitude representation, and symbolic decoding. Using a controlled environment of synthetic Go game boards, we isolate the specific mechanism of failure. Contrary to the hypothesis that models suffer from perceptual errors, we demonstrate via linear probing that visual backbones maintain robust, linearly separable representations of quantity well into the extrapolation regime. Furthermore, models retain latent magnitude awareness, successfully performing comparative reasoning on quantities they fail to enumerate. We pinpoint the collapse to the Symbolic Decoding stage, where the model fails to project valid visual magnitudes onto discrete tokens. Our findings support a Fractured Magnitude Hypothesis: VLMs fail to acquire a Universal Number Space, instead learning disjoint, modality-specific statistical manifolds that prevent cross-modal grounding for unseen pairings. We validate our findings on the state-of-the-art foundation model, suggesting that bridging the extrapolation gap requires inductive priors that enforce unified magnitude representations rather than simply scaling training data.
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Calibrating Uncertainty for Zero-Shot Adversarial CLIP
Wenjing Lu, Zerui Tao, Yuning Qiu, Dongping Zhang, Yang Yang, Qibin Zhao
CLIP delivers strong zero-shot classification but remains highly vulnerable to adversarial attacks. Prior adversarial fine-tuning work primarily matches predicted logits between clean and adversarial examples, which overlooks uncertainty calibration and may degrade the zero-shot generalization. A common expectation in reliable uncertainty estimation is that predictive uncertainty should increase as inputs become more difficult or shift away from the training distribution. However, we frequently observe the opposite in the adversarial setting: perturbations not only degrade accuracy but also suppress uncertainty, leading to severe miscalibration and over-confidence. This reveals a critical reliability gap beyond robustness. To bridge this gap, we propose an adversarial fine-tuning objective for CLIP considering both accuracy and uncertainty. By reparameterizing CLIP outputs as the concentration parameters of a Dirichlet distribution, we propose a unified representation that captures relative semantic structure and confidence magnitude. This enables holistic distribution alignment under perturbations, moving beyond single-logit anchoring and restoring calibrated uncertainty. Experiments across multiple zero-shot benchmarks demonstrate that our method significantly improves uncertainty calibration and achieves competitive adversarial robustness while preserving clean accuracy.
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Offline Two-Player Zero-Sum Markov Games with KL Regularization
Claire Chen, Yuheng Zhang, Xinyu Liu, Zixuan Xie, Shuze Daniel Liu, Nan Jiang
We study the problem of learning Nash equilibria in offline two-player zero-sum Markov games. While existing approaches often rely on explicit pessimism to address distribution shift, we show that KL regularization alone suffices to stabilize learning and guarantee convergence. We first introduce Regularized Offline Sequential Equilibrium (ROSE), a theoretical framework that achieves a fast $ \widetilde{\mathcal{O}}(1/n) $ convergence rate under *unilateral concentrability*, improving over the standard $ \widetilde{\mathcal{O}}(1/\sqrt{n}) $ rates in unregularized settings. We then propose Sequential Offline Self-play Mirror Descent (SOS-MD), a practical model-free algorithm based on least-squares value estimation and iterative self-play updates. We prove that the last iterate of SOS-MD attains the same $ \widetilde{\mathcal{O}}(1/n) $ statistical rate up to a vanishing optimization error of order $ \widetilde{\mathcal{O}}(1/\sqrt{T}) $ in the number of self-play iterations T.
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Reducing Per-Sample Harm in Stochastic Optimization
Apostolos Avranas

Modern optimizers combine gradients from the current mini-batch with historical optimization state, such as momentum or adaptive moments. While highly effective, aggregating across the batch and incorporating this history can produce parameter updates that increase the loss of individual samples. We term this effect *harm* and formalize the parameter update as an optimization problem that explicitly minimizes the conflicting impact of both batch averaging and past optimization state on current data. Because the exact formulation is intractable, we introduce a highly efficient proxy. We first reduce the problem's dimensionality to the batch size, and then drastically cut memory and speed bottlenecks by successfully restricting the optimization to the last linear layer. This hinges on the unexpected finding that *this layer alone reliably captures the second-order statistics of the per-sample gradients*. The resulting surrogate problem integrates readily into standard optimizers like SGD and AdamW, and can be solved using a small number of GPU-friendly iterations. Crucially, the method exhibits *favorable scaling properties*, as the relative computational overhead shrinks as the model size or input grows. Experiments on image classification benchmarks confirm *reduced per-sample interference* and *improved generalization*.

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Fair-FedMOE: Group-Fair One-Shot Federated Learning via Prototype-Guided Experts for Medical Imaging Analysis

Lingzhao Meng, Shuai Guo, Weishan Zhang, Zengxiang Li, Han Yu, Nan Liu, Daniel Shu Wei Ting, Yuru Liu, Shudong Wang, Tao Chen

Group fairness can ensure equitable performance across different demographic subgroups for medical image analysis. However, the current fine-tuned foundation models (FMs) exhibit significant subgroup disparity. One-shot federated learning (OFL) can potentially mitigate this by leveraging cross-institutional data diversity within a single communication round. However, heterogeneous distributions across medical institutions may cause OFL local models to diverge severely, resulting in parameter conflicts that amplify disparity upon aggregation. To address these challenges, we propose Fair-FedMOE, a group-fair OFL framework for medical FMs. During local training, Fairness-aware Expert Routing leverages learnable prototypes to route samples to group-specific experts, enabling subgroup-specialized learning to capture group-specific features without inter-group interference. During model aggregation, Prototype-guided Differential Aggregation computes personalized weights based on prototype similarity and applies differentiated aggregation strategies to filter conflicting updates. We propose RES-AUC, a Rawlsian justice-inspired metric based on worst-group performance that remains stable as groups increase. Extensive experiments on retinal and chest X-ray datasets with multiple FMs demonstrate consistent fairness gains without sacrificing accuracy.

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Beyond VLM-Based Rewards: Diffusion-Native Latent Reward Modeling

Gongye Liu, Bo Yang, Zhi Yida, Zhizhou Zhong, Lei Ke, Didan Deng, Han Gao, Yongxiang Huang, Kaihao Zhang, Hongbo Fu, Wenhan Luo

Preference optimization for diffusion models relies on reward functions that are both discriminative and computationally efficient. Vision-Language Models (VLMs) have emerged as powerful reward providers. However, their computation and memory cost can be substantial, and optimizing a latent diffusion generator through a pixel-space reward introduces a domain mismatch that complicates alignment.

In this paper, we propose DiNa-LRM, a diffusion-native latent reward model that formulates preference learning directly on noisy diffusion states. Our method introduces a noise-calibrated Thurstone likelihood with diffusion-noise-dependent uncertainty. DiNa-LRM leverages a pretrained latent diffusion backbone with a timestep-conditioned reward head, and supports inference-time noise ensembling, providing a diffusion-native mechanism for test-time scaling and robust rewarding. Across image alignment benchmarks, DiNa-LRM substantially outperforms existing diffusion-based reward baselines and achieves competitive performance compared to state-of-the-art VLMs while maintaining a substantially lower computational cost. In preference optimization, we demonstrate that DiNa-LRM improves preference optimization dynamics, enabling faster and more resource-efficient model alignm

ent.
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Prioritize the Process, Not Just the Outcome: Rewarding Latent Thought Trajectories Improves Reasoning in Looped Language Models
Jonathan Williams, Olga Russakovsky, Esin Tureci
Looped Language Models (LoopLMs) perform multi-step latent reasoning prior to token generation and outperform conventional LLMs on reasoning benchmarks at smaller parameter budgets. However, attempts to further improve LoopLM reasoning with reinforcement learning have failed—standard objectives such as Group Relative Policy Optimization (GRPO) only assign credit to the final latent state, creating a fundamental mismatch with the model's internal computation. To resolve this, we introduce RLTT (Reward Latent Thought Trajectories), a reinforcement learning framework which distributes reward across the full latent reasoning trajectory. RLTT provides dense, trajectory-level credit assignment without relying on external verifiers and can directly replace GRPO with negligible overhead. Across extensive experiments with Ouro-1.4B/2.6B-Thinking under identical training and inference conditions, RLTT yields statistically significant improvements over GRPO on challenging mathematical reasoning benchmarks, improving mean accuracy over MATH-500, AIME24/26, and BeyondAIME by +5.8\% on the 1.4B scale, and +10.9\% on the 2.6B scale. Despite being trained exclusively on mathematics, RLTT also transfers effectively to non-mathematical reasoning benchmarks, demonstrating the effectiveness of trajectory-level credit assignment for reinforcement learning in LoopLMs. Code is available at https://github.com/jonwill8/RLTT.git.
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Spatial Deconfounder: Interference-Aware Deconfounding for Spatial Causal Inference
Ayush Khot, Miruna Oprescu, Maresa Schröder, Ai Kagawa, Xihaier Luo
Causal inference in spatial domains faces two intertwined challenges: (1) unmeasured spatial factors, such as weather, air pollution, or mobility, that confound treatment and outcome, and (2) interference from nearby treatments that violate standard no-interference assumptions. While existing methods typically address one by assuming away the other, we show they are deeply connected: *interference reveals structure* in the latent confounder. Leveraging this insight, we propose the **Spatial Deconfounder**, a two-stage method that reconstructs a substitute confounder from local treatment vectors using a conditional variational autoencoder (C-VAE) with a spatial prior, then estimates causal effects with a flexible outcome model. We show that this enables nonparametric identification of direct and spillover effects under weak assumptions—without multiple treatment types or a known latent-field model. Empirically, we extend SpaCE, a benchmark suite for spatial confounding, to include treatment interference, and show that the Spatial Deconfounder consistently improves effect estimation across real-world environmental health and social science datasets. By turning local interference into a multi-cause proxy for latent spatial confounding, our framework advances robust causal inference for spatial data.
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Hom-PGD$^+$: Fast Reparameterized Optimization over Non-convex Ball-Homeomorphic Set
Chenghao Liu, Enming Liang, Minghua Chen
We study optimization over non-convex constraint sets that are homeomorphic to a ball, encompassing important problem classes such as star-shaped sets that frequently arise in machine learning and engineering applications. We propose **Hom-PGD$^+$**, a learning-based and projection-efficient first-order method that efficiently solves such problems without requiring expensive projection or optimization oracles. Our approach leverages an invertible neural network (INN) to learn the homeomorphism between the non-convex constraint set and a unit ball, transforming the original problem into an equivalent ball-constrained optimization where projections admit efficient solutions. We establish that Hom-PGD$^+$ achieves an $\mathcal{O}(\epsilon^{-2})$ convergence rate to an ($\epsilon + \mathcal{O}(\sqrt{\epsilon_{\text{inn}}})$)-approximate stationary solution, where ϵ_{inn} denotes the homeomorphism learning error. This rate significantl

y improves upon existing methods for optimization over non-convex sets, while ma intaining a per-iteration complexity of only $\mathcal{O}(W)$ for W INN parame ters. Extensive experiments, including QCQP, chance-constrained power-system opt imization, and non-uniform adversarial attacks, demonstrate that Hom-PGD$^+$ ach ieves competitive solution quality while delivering speedups of up to one order of magnitude.
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Large-Scale Molecular Dynamics Simulations: Direct Interatomic Modeling with Dil ated Message Passing
Haokai Hong, Wanyu Lin, KC Tan
Large-scale molecular dynamics simulations are essential in understanding chemic al and biological processes, necessitating the accurate and efficient modeling o f interatomic interactions. Existing learning-based methods are generally based on message passing mechanisms; they are either not scalable or too coarse to off er accurate modeling. We propose a new message passing framework that can effect ively and efficiently model interatomic interactions for simulating large-scale molecular dynamics at full atomic resolution. Specifically, our framework is sta cked with a sequence of message passing neural network layers, each realizing th e message passing over a distinct and dilated star-structured path. These star-s tructured paths are constructed progressively along dilated regions to capture t he distance-dependent interactions. The crux of our framework is that it resolve s the problem of dense interatomic interactions of large-scale atomic systems wi th sparser and region-based message passing graphs. We evaluate the framework on four benchmarks: MD22 (molecules with 42–370 atoms), Chignolin (a 166-atom prot ein featuring diverse conformations), the AdK dataset (a protein trajectory with up to 3,000 atoms), and the MISATO dataset (over 10,000 heterogeneous protein-l igand complexes with systems up to 40,000 atoms). Comprehensive evaluations demo nstrate that our approach delivers state-of-the-art performance overall across v arious benchmarks. The code is provided on [Github](https://github.com/WanyuGrou p/ICML2026-DKMP).
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Representation Drift Compensation: A Near-Zero Inference Cost Enhancement for LL M Decomposition
Xinhao Huang, You-Liang Huang, Zeyi Wen
While low-rank decomposition offers potential for reducing LLM parameters, maint aining the original capabilities remains a significant challenge. In this work, we identify and formalize a key overlooked issue in LLM decomposition: \textit{r epresentation drift}. We show that approximation errors introduced by decomposit ion propagate and amplify non-linearly through the deep layers of the transforme r architecture, progressively distorting internal representations and degrading downstream performance. To mitigate this, we introduce a conceptually simple but principled compensation mechanism, named ``Decomper'', that operates by suppres sing error at its source. By learning to align the output distribution of decomp osed transformer blocks with their original counterparts, our method effectively counteracts representation drift, achieving notable performance recovery with n ear-zero inference overhead. Extensive experiments in OPT, LLaMA-2/3, and Qwen e xhibit remarkable improvements. For instance, on LLaMA-3-8B and OPT-13B at 40\% compression, perplexity is reduced by more than 70\% while reasoning task accura cy improves by over 10\%. Our code is available at this \href{https://github.com /xinhaoH/SVD-Compen}{URL}.
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Certifying Capabilities from Finite Tests: When Is It Possible?
Changlong Wu, Jin Sima, Wojciech Szpankowski
Modern foundation models are evaluated through broad capabilities such as arithm etic, reasoning, safety, and robustness, yet it remains unclear in a principled sense when *finite tests* can meaningfully certify such claims. We develop a rig orous theory of capability evaluation by formalizing evaluation as inference ove r a task family and asking when guarantees over the full family can be inferred from a strict subset of tests. We analyze two canonical regimes. In stochastic m ulti-environment evaluation, we characterize when uniform certification is possi

ble across multiple environments and show that the sample complexity is governed by a χ^2-radius of the environment family, yielding near-optimal evaluation protocols with matching lower bounds under a natural overlap condition. In contrast, for worst-case, rule-like capabilities, we establish fundamental impossibility results. Even for structured model classes such as Boolean circuits of bounded size, black-box evaluation cannot, in general, certify global properties. Together, these results provide a principled framework for understanding when finite evaluation can and cannot certify capabilities.

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Group-wise Data Ordering: Enhancing Instruction Tuning of Large Language Models via Embedding Proximity

Yiwen Ye, Boyuan Jiang, Xiaobin Hu, Shengzhi Wang, Xiaozhong Ji, Jinghao Lin, Deli Yu, Jiale Chen, Kai WU, Haihua Yang, Yong Xia

Instruction tuning (IT) is a central mechanism for aligning large language models (LLMs) with user intent. In practice, randomly shuffling the training set is a simple yet surprisingly strong baseline. However, it overlooks latent structure, such as domain and reasoning depth, and thus interleaves heterogeneous objectives, which can induce gradient conflicts and diminish effective optimization progress. To this end, we propose **EP-Order**, an embedding-proximity-based data-ordering paradigm for IT of LLMs. Unlike previous paradigms that derive order from **per-example** scores, EP-Order explicitly accounts for **inter-sample** correlations by operating in representation space. EP-Order trains a warm-up model on a small subset of data (*e.g.*, 10%), embeds all training samples for clustering, and ranks these clusters according to embedding proximity. To handle sharp gradient changes at cluster transitions and alleviate catastrophic forgetting under cluster-based training, we introduce mixed regions that interleave samples from the previous, current, and next clusters, thereby stabilizing learning. Extensive experiments on 14 benchmarks spanning vision-language, text-only, and hybrid thinking/no-thinking scenarios show that EP-Order achieves broadly consistent improvements over random shuffling. These results suggest that exploiting embedding-level data structure offers an effective and general direction for improving IT on complex, high-conflict training data.

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EMFormer: Efficient Multi-Scale Transformer for Accumulative Context Weather Forecasting

Hao Chen, Tao Han, Jie ZHANG, Song Guo, Fenghua Ling, LEI BAI

Long-term weather forecasting is critical for socioeconomic planning and disaster preparedness.

While recent approaches employ finetuning to extend prediction horizons, they remain constrained by the issues of catastrophic forgetting, error accumulation, and high training overhead.

To address these limitations, we present a novel pipeline across pretraining, finetuning and forecasting to enhance long■context modeling while reducing computational overhead. First, we introduce an Efficient Multi■scale Transformer (EMFormer) to extract multi■scale features through a single convolution in both training and inference. Based on the new architecture, we further employ an accumulative context finetuning to improve temporal consistency without degrading short■term accuracy. Additionally, we propose a composite loss that dynamically balances different terms via a sinusoidal weighting, thereby adaptively guiding the optimization trajectory throughout pretraining and finetuning.

Experiments show that our approach achieves great performance in weather forecasting and extreme event prediction, substantially improving long-term forecast accuracy. Moreover, EMFormer demonstrates strong generalization on vision benchmarks (ImageNet-1K and ADE20K). Code: https://github.com/chenhao-zju/emformer

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Self-Supervised Dynamical System Representations for Physiological Time-Series

Yenho Chen, Maxwell A Xu, James Matthew Rehg, Christopher John Rozell

Self-supervised learning for physiological time-series aims to captures the identity of the underlying dynamical process while filtering irrelevant noise. However, existing approaches may obscure the clinical semantics important for downstr

eam transferability. Weakly constrained pretext tasks (i.e. contrastive learning , MAE) may incorrectly ignore the underlying dynamical structure, while structur ally constrained models (i.e. SVAEs) are unable to selectively filter sample-spe cific noise. To bridge this gap, we propose ${\bf PULSE}$, a novel pretraining objective that simultaneously preserves dynamical relationships important to phy siological time-series while selectively removing irrelevant noise. We achieve t his by formulating a dynamical systems model to identify transferable and non-tr ansferable information between time-series windows, and target the former throug h a novel cross-reconstruction objective.
We establish theory that provides conditions for when transferrable information is recovered, and empirically validate it through synthetic experiments. On seve ral real-world datasets, PULSE effectively distinguishes clinical semantic class es, increases label efficiency, and improves transfer learning performance.
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Efficient Training of Boltzmann Generators Using Off-Policy Log-Dispersion Regul arization

Henrik Schopmans, Christopher von Klitzing, Pascal Friederich

Sampling from unnormalized probability densities is a central challenge in computational science. Boltzmann generators are generative models that enable independent sampling from the Boltzmann distribution of physical systems at a given temperature. However, their practical success depends on data-efficient training, as both simulation data and target energy evaluations are costly. To this end, we propose off-policy log-dispersion regularization (LDR), a novel regularization framework that builds on a generalization of the log-variance objective. We apply LDR in the off-policy setting in combination with standard data-based training objectives, without requiring additional on-policy samples. LDR acts as a shape regularizer of the energy landscape by leveraging additional information in the form of target energy labels. The proposed regularization framework is broadly applicable, supporting unbiased or biased simulation datasets as well as purely variational training without access to target samples. Across all benchmarks, LDR improves both final performance and data efficiency, with sample efficiency gains of up to one order of magnitude.
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ScenePilot: Controllable Boundary-Driven Critical Scenario Generation for Autono mous Driving

Qiyu Ruan, YUXUAN WANG, He Li, Zhenning Li, Cheng-zhong Xu

Safety-critical scenarios are central to evaluating autonomous driving systems, yet their rarity in naturalistic logs makes simulation-based stress testing indi spensable. Most scenario generation methods treat surrounding agents as adversar ies, but they either (i) induce failures without explicitly modeling vehicle-roa d physical limits, yielding visually extreme yet physically unsolvable crashes, or (ii) enforce physical feasibility or policy feasibility in isolation, which c an over-focus on aggressive maneuvers or remain tied to a controller-dependent c apability boundary. We propose ScenePilot, a feasibility-guided, boundary-driven framework that targets the boundary band: scenarios that are physically solvabl e in principle yet still cause the deployed autonomy stack to fail. We formulate generation as constrained multi-objective reinforcement learning, combining an RSS-derived physical-feasibility score σ with an online-learned AV-risk p redictor Φ, and introduce step-level feasibility-aware shielding to keep ex ploration near the feasibility boundary while avoiding infeasible artifacts. Exp eriments on SafeBench with multiple planners show that ScenePilot yields substan tially higher collision rates (+6.2 percentage points) while preserving physical validity, and that adversarial fine-tuning on these boundary-band scenarios con sistently reduces downstream crash rates. The code is available at https://githu b.com/QiyuRuan/ScenePilot.
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Remove the Ambiguity: Few-shot Multimodal Anomaly Detection Using Crossmodal Fea ture Replacers

Yuan Guo, Wanqi Zhang, Xu Wang

A key challenge in reconstruction-based multimodal anomaly detection is the one-to-many crossmodal mapping problem: a single 3D feature may correspond to multiple plausible RGB appearances, causing deterministic crossmodal regression to collapse valid targets into over-smoothed reconstructions and thereby weaken anomaly discrimination. In this paper, we propose Crossmodal Feature Replacer (CFR), a self-supervised framework that addresses this failure mode through selective inference-time feature replacement. CFR first learns bidirectional cyclic mappings for coarse crossmodal reconstruction, then identifies unreliable reconstructed features and selectively replaces them with high-confidence normal features to correct ambiguity-induced reconstruction failures. Extensive experiments on MVTec 3D-AD and Eyecandies under few-shot settings show that CFR consistently outperforms prior methods. In the challenging 1-shot setting, CFR achieves AUPRO scores of 92.3 and 82.7 at 30\% FPR, together with image-level AUROC scores of 74.0 and 75.9, on MVTec 3D-AD and Eyecandies, respectively.
Code is available at https://github.com/Yuan-Honoka-Guo/CFR.
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Solving Stochastic Variational Inequalities without the Bounded Variance Assumption
Ahmet Alacaoglu, JunHyun Kim
We analyze algorithms for solving stochastic variational inequalities (VI) without the bounded variance or bounded domain assumptions, where our main focus is min-max optimization with possibly unbounded constraint sets. We focus on two classes of problems: monotone VIs; and structured nonmonotone VIs that admit a solution to the *weak Minty VI*. The latter assumption allows us to solve structured nonconvex-nonconcave min-max problems. For both classes of VIs, to make the expected residual norm less than ε, we show an oracle complexity of $\widetilde{O}(\varepsilon^{-4})$, which is the best-known for constrained VIs. In our setting, this complexity had been obtained with the bounded variance assumption in the literature, which is not even satisfied for bilinear min-max problems with an unbounded domain. We obtain this complexity for stochastic oracles whose variance can grow as fast as the squared norm of the optimization variable.
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Do Text Edits Generalize to Visual Generation? Benchmarking Cross-Modal Knowledge Editing in UMMs
Xin Gao, Cheng Yang, Chufan Shi, Taylor Berg-Kirkpatrick
Unified multimodal models (UMMs) have emerged as a promising paradigm for general-purpose multimodal intelligence. As they are deployed in real-world applications, effectively updating internal knowledge becomes critical. While knowledge editing has matured for text-only models, it remains unclear whether edits that successfully modify textual outputs also transfer to image generation in UMMs. To study this question, we introduce UniKE, the first benchmark for cross-modality knowledge editing in UMMs, comprising 2,971 edit subjects spanning attribute and relation edits. Using VQA-based visual verification, we reveal a striking modality gap: text-side efficacy can reach approximately 92\%, whereas the best overall VQA accuracy under direct image generation is only 18.5\%. We further propose Reasoning-augmented Parameter Editing, which explicitly activates edited knowledge before generation and improves overall VQA accuracy for all evaluated model-editor pairs, with gains up to 18.6 percentage points. Mechanistic analysis shows that this gap is associated with partial alignment between edited textual representations and the conditioning pathways for visual generation, where edits sufficient for text outputs may remain too weak or misaligned to steer image synthesis. These findings show that textual knowledge edits do not guarantee reliable cross-modality transfer and motivate modality-aware editing methods. Our code and data are available at https://github.com/gxx27/UniKE.
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Dynamic Regret via Discounted-to-Dynamic Reduction with Applications to Curved Losses and Adam Optimizer
Yan-Feng Xie, Yu-Jie Zhang, Peng Zhao, Zhi-Hua Zhou
We study dynamic regret minimization in non-stationary online learning, with a primary focus on follow-the-regularized-leader (FTRL) methods. FTRL is important

for curved losses and for understanding adaptive optimizers such as Adam, yet existing dynamic regret analyses are less explored for FTRL. To address this, we build on the discounted-to-dynamic reduction and present a modular way to obtain dynamic regret bounds of FTRL-related problems. Specifically, we focus on two representative curved losses: linear regression and logistic regression. Our method not only simplifies existing proofs for the optimal dynamic regret of online linear regression, but also yields new dynamic regret guarantees for online logistic regression. Beyond online convex optimization, we apply the reduction to analyze the Adam optimizers, obtaining optimal convergence rates in stochastic, non-convex, and non-smooth settings. The reduction also enables a more detailed treatment of Adam with two discount parameters (β_1,β_2), leading to new results for both clipped and clip-free variants of Adam optimizers.
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BLISS: A Lightweight Bilevel Influence Scoring Method for Data Selection in Language Model Pretraining
Jie Hao, Rui Yu, Wei Zhang, Huixia Wang, Jie Xu, Mingrui Liu
Effective data selection is essential for pretraining large language models (LLMs), improving efficiency and generalization to downstream tasks. However, existing approaches often rely on external pretrained models, making it difficult to separate the benefits of data selection from those introduced by external models. In addition, many methods estimate data importance from a fixed model state or short-horizon update, making it hard to capture how data preference changes as the model evolves during pretraining. In this paper, we introduce BLISS (BileveL Influence Scoring method for data Selection), a lightweight data selection method that operates entirely \emph{from scratch}, without external pretrained oracle models, while modeling dynamic data preference. BLISS uses a small proxy model as a surrogate for the LLM and trains a score model to estimate sample importance through multi-step proxy updates induced by score-weighted training data. We formulate data selection as a bilevel optimization problem: the upper-level objective optimizes the score model to assign sample weights, so minimizing the lower-level weighted training loss improves validation performance. Once optimized, the score model predicts influence scores, enabling efficient selection of high-quality samples for LLM pretraining. We validate BLISS by pretraining 410M/1B/2.8B Pythia and LLaMA-0.5B models on selected C4 subsets. Under the 1B setting, BLISS achieves a $1.7\times$ speedup in reaching the same performance as the state-of-the-art method, while delivering superior performance across multiple downstream tasks.
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BESplit: Bias-Compensated Split Federated Learning with Evidential Aggregation
Yuhan Xie, Chen Lyu, Jingrong Huang
Split Federated Learning (SFL) enables privacy-preserving collaborative training by partitioning models between clients and a server. However, under non-IID data distributions, SFL often suffers from biased optimization and unstable convergence, while existing solutions largely adapt techniques from conventional federated learning. In this work, we observe that the split architecture of SFL inherently alters how client information is represented and coordinated, opening opportunities for bias compensation beyond parameter-level aggregation. Based on this insight, we propose BESplit, an architecture-aware framework that exploits the intrinsic structure of SFL to mitigate non-IID effects. First, to prevent biased local data from dominating global updates, we introduce Evidential Aggregation (EA) to perform fine-grained reweighting of client contributions based on evidential uncertainty. Second, to further reduce distributional skew, we develop Bias-Compensated Collaboration (BCC) to align split-layer representations by pairing complementary clients. Finally, Dual-Teacher Distillation (DTD) is incorporated to synchronize knowledge between decoupled client and server models, enabling independent local inference. Extensive experiments on five benchmark datasets demonstrate that BESplit consistently outperforms state-of-the-art methods in accuracy, convergence stability, and computational efficiency under diverse non-IID settings.
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Guidance: Sentence-Level Citation Enforcement via Prefix-Tail Guidance during LLM Decoding
Yirui Zhan, Chenhao Xu, Jun Gao
In correctness-sensitive scenarios, it is crucial for Large Language Models (LLMs) to strictly follow the provided evidence. However, even with reference texts, models often suffer from hallucinations, especially when processing long contexts. Existing work attempts to reinforce the use of citations through Retrieval-Augmented Generation (RAG) or post-hoc methods, while citations remain a probabilistic output rather than a foundation for the generated content. To address this, we propose Guidance, which aims to correct outputs and naturally incorporate citations during the LLM decoding phase. Specifically, we first build a structured fact pool (Prefix-Tail pairs) from the documents.
Then, during inference, Guidance predicts the model's intent using a lookahead strategy. When it detects a match with a context prefix, it automatically replaces the output with the verified fact and its citation. This approach is training-free and can be plugged into general-purpose or citation-fine-tuned LLMs. Experiments on LongBench-Cite demonstrate that Guidance improves the citation F1 score by 11.2\% over state-of-the-art baselines. The source code is available at: https://github.com/marlcplhra/Guidance.
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FlashSketch: Sketch-Kernel Co-Design for Fast Sparse Sketching on GPUs
Rajat Vadiraj Dwaraknath, Sungyoon Kim, Mert Pilanci
Sparse sketches such as the sparse Johnson–Lindenstrauss transform are a core primitive in randomized numerical linear algebra because they leverage random sparsity to reduce the arithmetic cost of sketching, while still offering strong approximation guarantees. Their random sparsity, however, is at odds with efficient implementations on modern GPUs, since it leads to irregular memory access patterns that degrade memory bandwidth utilization. Motivated by this tension, we pursue a sketch–kernel co-design approach: we design a new family of sparse sketches, BlockPerm-SJLT, whose sparsity structure is chosen to enable FlashSketch, a corresponding optimized CUDA kernel that implements these sketches efficiently. The design of BlockPerm-SJLT introduces a tunable parameter that explicitly trades off the tension between GPU-efficiency and sketching robustness. We provide theoretical guarantees for BlockPerm-SJLT under the oblivious subspace embedding (OSE) framework, and also analyze the effect of the tunable parameter on sketching quality. We empirically evaluate FlashSketch on standard RandNLA benchmarks, as well as an end-to-end ML data attribution pipeline called GraSS. FlashSketch pushes the Pareto frontier of sketching quality versus speed, across a range of regimes and tasks, and achieves a global geomean speedup of roughly $1.7 \times$ over the prior state-of-the-art GPU sketches.
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PhoStream: Benchmarking Real-World Streaming for Omnimodal Assistants in Mobile Scenarios
Xudong Lu, Guan Huankang, Yang Bo, Jinpeng Chen, Xintong Guo, Shuhan LI, Fang Liu, Peiwen Sun, Xueying Li, Wei Zhang, Xue Yang, Rui Liu, Hongsheng Li
Multimodal Large Language Models excel at offline audio-visual understanding, but their ability to serve as mobile assistants in continuous real-world streams remains underexplored. In daily phone use, mobile assistants must track streaming audio-visual inputs and respond at the right time, yet existing benchmarks are often restricted to multiple-choice questions or use shorter videos. In this paper, we introduce **PhoStream**, the first mobile-centric streaming benchmark that unifies on-screen and off-screen scenarios to evaluate video, audio, and temporal reasoning. PhoStream contains 5,572 open-ended QA pairs from 578 videos across 4 scenarios and 10 capabilities. We build it with an Automated Generative Pipeline backed by rigorous human verification, and evaluate models using a realistic Online Inference Pipeline and LLM-as-a-Judge evaluation for open-ended responses. Experiments reveal a temporal asymmetry in LLM-judged scores (0--100): models perform well on Instant and Backward tasks (Gemini 3 Pro exceeds 80), but drop sharply on Forward tasks (16.40), largely due to early responses before the required visual and audio cues appear. This highlights a fundamental limitation: c

urrent MLLMs struggle to decide ***when*** to speak, not just ***what*** to say. Code and datasets will be made publicly accessible at https://github.com/Lucky-Lance/PhoStream.

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Must All Negatives Be Pushed Away Equally? Uncertainty-Aware Cross-View Geo-Localization via Normal Inverse Gamma Distribution

Songsong Ouyang, Le Wu, Yingying Zhu

Cross-view geo-localization (CVGL) aims to retrieve the corresponding satellite image given a street query and is critical for autonomous navigation. Although recent methods perform well on benchmarks, they often fail to generalize to unseen environments. A key limitation is the use of contrastive learning, which assigns equal labels to all negative samples and induces similarity-amplified repulsion. But should all negatives be treated equally? In CVGL, semi-positive samples that are geographically proximate to the positive often share important semantic cues. Treating them as ordinary negatives forces the model to overfit noise, leading to a collapse in generalization. To address this issue, we propose an uncertainty-aware framework grounded in Deep Evidential Regression (DER), modeling the Normal-Inverse-Gamma (NIG) distribution as a conjugate prior to quantify environmental complexity u in a single forward pass. The estimated u adaptively softens labels for hard negatives in Soft InfoNCE, mitigating excessive repulsion on semi-positive samples. An Uncertainty Head with cls-to-spatial cross-attention and attention statistics is designed to accurately fit the NIG distribution. Extensive experiments demonstrate state-of-the-art performance, including an average 18\% R@1 improvement in zero-shot cross-dataset transfer, filling the critical gap between laboratory benchmarks and robust real-world deployment.

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Learning to Share: Selective Memory for Efficient Parallel Agentic Systems

Joseph Fioresi, Parth Parag Kulkarni, Ashmal Vayani, Song Wang, Mubarak Shah

Agentic systems solve complex tasks by coordinating multiple agents that iteratively reason, invoke tools, and exchange intermediate results. To improve robustness and solution quality, recent approaches deploy multiple agent teams running in parallel to explore diverse reasoning trajectories. However, parallel execution comes at a significant computational cost: when different teams independently reason about similar sub-problems or execute analogous steps, they repeatedly perform substantial overlapping computation. To address these limitations, in this paper, we propose Learning to Share (LTS), a learned shared-memory mechanism for parallel agentic frameworks that enables selective cross-team information reuse while controlling context growth. LTS introduces a global memory bank accessible to all teams and a lightweight controller that decides whether intermediate agent steps should be added to memory or not. The controller is trained using stepwise reinforcement learning with usage-aware credit assignment, allowing it to identify information that is globally useful across parallel executions. Experiments on the AssistantBench and GAIA benchmarks show that LTS significantly reduces overall runtime while matching or improving task performance compared to memory-free parallel baselines, demonstrating that learned memory admission is an effective strategy for improving the efficiency of parallel agentic systems.

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Consistent Zero-Shot Imitation with Contrastive Goal Inference

Kathryn Wantlin, Chongyi Zheng, Benjamin Eysenbach

Zero-shot imitation learning requires an agent to reproduce expert behavior from a single demonstration without additional environment interaction or gradient updates at test time. We introduce Contrastive Inverse Reinforcement Learning (CIRL), a self-supervised framework for pre-training zero-shot imitation agents. Our methods rests on a key observation that many useful tasks can be summarized by a single goal state. We can thus convert the multi-task inverse RL problem into a more tractable goal-inference problem, and utilize state-of-the-art goal-conditioned RL methods to recover a policy that reaches the goal. During pre-training, CIRL jointly employs three components to learn without any rewards or demonstrations: (1) a variant of contrastive RL designed to learn maximum-entropy goal-conditioned policies, (2) an automatic goal proposal mechanism (GoalKDE) that dr

ives exploration, and (3) a mean-field variational model that performs amortized goal inference from trajectories. We prove that this procedure consistently recovers the demonstrator's intent by accounting for the relative difficulty of reaching different states and show how structurally similar prior work may otherwise fail to infer the correct reward. Experiments on goal-conditioned and standard reward-maximizing control tasks show that CIRL outperforms prior zero-shot imitation methods, supporting the expressiveness of goals as a compact summary of behavior.

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Adaptive Utilization of Low-Rank Adaptation via Conditioned Gating

Guang Yang, Changhao Guan, Chao Huang, Yufeng Chen, Kaiyu Huang

Low-Rank Adaptation (LoRA) achieves parameter-efficient fine-tuning by constraining model updates to a low-rank subspace and has been widely used in practice. However, LoRA typically employs a shared low-rank update across tokens, which limits its ability to fully exploit the adaptation subspace for tokens from different sequences.

To address this issue, we propose an adaptive utilization of Low-Rank Adaptation (U-LoRA), which employs conditioned gating to explicitly learn effective token-level utilization of the limited low-rank adaptation subspace.

Specifically, U-LoRA generates utilization coefficients along low-rank directions for each token and jointly coordinates and constrains them using sequence-level contextual information, thereby inducing more consistent adaptive patterns within a sentence.

To further enhance training stability, we introduce a bias-corrected exponential moving average (EMA) historical prior that calibrates utilization signals across optimization steps, suppressing noise caused by batch-to-batch fluctuations.

The effectiveness of our method arises from a better utilization of the existing low-rank subspace via input-conditioned strategies, rather than from expanding the subspace.

Experiments on mathematical reasoning and natural language understanding benchmarks demonstrate that U-LoRA achieves competitive performance under comparable parameter budgets when with strong LoRA baselines and recent variants.

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Interpretable Discovery of One-parameter Subgroups: A Modular Framework for Elliptical, Hyperbolic, and Parabolic Symmetries

Pavan Karjol, Vivek V Kashyap, Rohan Kashyap, Prathosh AP

We propose a modular, data-driven framework for jointly learning unknown functional mappings and discovering the underlying one-parameter symmetry subgroup governing the data. Unlike conventional geometric deep learning methods that assume known symmetries, our approach identifies the relevant continuous subgroup directly from data. Our framework focuses on three primary geometric components of one-parameter subgroup actions: elliptic, hyperbolic, and parabolic regimes. For the given regime, our framework instantiates a corresponding symmetry discovery architecture with invariant and equivariant representation layers structured according to the Lie algebra of the subgroup, and learns the exact generator parameters end-to-end from data. This yields models whose invariance or equivariance is guaranteed by construction and admits formal proofs, enabling symmetry to be explicitly traced to identifiable components of the architecture. The approach is applicable to one-parameter subgroups of a wide range of matrix Lie groups, including $SO(n)$, $SL(n)$, and the Lorentz group. Experiments on synthetic and real-world systems, including moment of inertia prediction, double-pendulum dynamics, and high-energy \textit{Top Quark Tagging}, demonstrate accurate subgroup recovery and strong predictive performance across both compact and non-compact regimes.

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DisjunctiveNet: Neural Symbolic Learning via Differentiable Convexified Optimization Layers

Shraman Pal, Can Li

Many learning tasks in science and engineering are characterized by sparse datasets, which limits the effectiveness of purely data-driven approaches. At the sam

e time, these problems are often accompanied by rich domain knowledge derived fr
om physical laws, operational requirements, and expert heuristics. Such knowledg
e is frequently expressed as rules involving logical propositions and linear ine
qualities. Existing neuro-symbolic methods typically enforce these rules approxi
mately through soft penalties, assume input-independent rules when designing spe
cialized architectures, or rely on non-differentiable post-processing at inferen
ce time to achieve hard constraint satisfaction. While recent advances in differ
entiable optimization layers enable end-to-end feasibility enforcement within ne
ural networks, extending these approaches to logical or mixed-integer rules rema
ins challenging due to inherent nonconvexity. In this work, we propose a unified
 end-to-end framework for enforcing hard, input-dependent mixed integer linear c
onstraints within neural networks. Our approach represents rules as disjunctive
constraints and applies hierarchical convex relaxations to obtain convex hull fo
rmulations. These relaxations yield tractable linear constraints that can be emb
edded as differentiable optimization layers while enabling exact rule satisfacti
on. We demonstrate the effectiveness of the proposed framework on real-world dat
asets, achieving perfect rule satisfaction and strong predictive performance.
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HOI-PAGE: Zero-Shot Human-Object Interaction Generation with Part Affordance Gui
dance

Lei Li, Angela Dai

We present HOI-PAGE, a new approach that prioritizes part-level affordance reaso
ning to generate high-fidelity 4D human-object interactions (HOIs) from text pro
mpts in a zero-shot fashion. In contrast to prior works that focus on global, wh
ole body-object motion synthesis, our approach explicitly reasons about the unde
rlying part-level mechanics of interactions using large language models (LLMs).
We capture this reasoning in a structured part affordance graph (PAG) representa
tion, serving as a high-level interaction scaffolding to guide a three-stage syn
thesis: first, decomposing input 3D objects into semantic parts; then, generatin
g reference HOI videos from text prompts to extract part-based motion constraint
s; and finally, optimizing for 4D HOI motion sequences that mimic the reference
dynamics while satisfying part-level contact constraints. Extensive experiments
show that our approach is flexible and capable of generating complex multi-objec
t or multi-person interaction sequences, with significantly improved realism and
 text alignment for zero-shot 4D HOI generation.
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Exploring 3D Dataset Pruning

Xiaohan Zhao, Xinyi Shang, Jiacheng Liu, Zhiqiang Shen

Dataset pruning remains underexplored for 3D modalities, where inherent class im
balance persists across ***both*** training and test sets. This creates a diverg
ence in evaluation: overall accuracy favors natural frequency, reflecting practi
cal usage; while mean accuracy demands balanced generalization. Instead of forci
ng a premature trade-off, we advocate for base principles that remain universall
y robust and beneficial across diverse priors.
We cast pruning as a quadrature approximation on population risk and decompose t
he error bound into *representation error* (fidelity to the underlying manifold)
 and *prior-mismatch bias* (distribution shift), clarifying what can be improved
 jointly across priors.
To address prior-mismatch bias, we decouple likelihood from prior in the posteri
or and transfer the structural likelihood via distillation with a calibrated tea
cher and geometry-preserving constraints.
Simultaneously, to reduce representation error, we audit common pruning signals
and choose geometric embedding, which exhibits greater robustness given the high
 inductive bias of 3D models. We also prioritize a safety floor before selection
, capturing high-reward regions beneficial across priors.
Finally, acknowledging that no single subset optimally satisfies divergent evalu
ation priors, we augment these principles with a steering wrapper that interpola
tes between stratified seeding and global selection. Empirical results demonstra
te that our framework elevates the performance floor while offering flexibility
for different prior preferences.

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Geometric Reciprocity: Unlocking Self-Supervision for Stereoscopic Video Generation
Jingyi Lu, Kai Han
Monocular-to-stereo conversion synthesizes stereoscopic content from 2D videos for immersive 3D experiences. In modern Depth-Image-Based Rendering (DIBR) approaches, stereo inpainting of disocclusions is the critical bottleneck. Training-based methods achieve superior quality but rely on scarce stereo pairs or synthetic data with domain gaps. We address this through the first self-supervised framework learning from monocular videos via cycle consistency. Our key contribution is the **Geometric Reciprocity Theorem (GRT)**: under the nearest-neighbor DIBR formulation, the disocclusion mask when synthesizing a target view equals the mask of pixels lost when warping back from target to source, enabling analytical computation of test-time disocclusion masks directly from monocular images. This yields train-test consistency for the stated warping formulation, supporting self-supervised learning from unlimited monocular videos and substantial improvements over training-free and supervised state-of-the-art methods. Project page: https://visual-ai.github.io/grt/
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Fix Before Search: Benchmarking Agentic Visual Query Pre-processing in Multimodal Retrieval-augmented Generation
Shenglai Zeng, Jiankun Zhang, Kai Guo, Xinnan Dai, Hui Liu, Jiliang Tang, Yi Chang
Multimodal Retrieval-Augmented Generation (MRAG) has emerged as a key paradigm for grounding MLLMs with external knowledge. While query pre-processing (e.g., rewriting) is standard in text-based RAG, existing MRAG pipelines predominantly treat visual inputs as static and immutable, implicitly assuming they are noise-free. However, real-world visual queries are often ``imperfect''---suffering from geometric distortions, quality degradation, or semantic ambiguity---leading to catastrophic retrieval failures. To address this gap, we propose V-QPP-Bench, the first comprehensive benchmark dedicated to Visual Query Pre-processing (V-QPP). We formulate V-QPP as an agentic decision-making task where MLLMs must autonomously diagnose imperfections and deploy perceptual tools to refine queries. Our extensive evaluation across 46,700 imperfect queries and diverse MRAG paradigms reveals three critical insights: (1) Vulnerability---visual imperfections severely degrade both retrieval recall and end-to-end MRAG performance; (2) Restoration Potential \& Bottleneck---while oracle preprocessing recovers near-perfect performance, off-the-shelf MLLMs struggle with tool selection and parameter prediction without specialized training; and (3) Training Enhancement---supervised fine-tuning enables compact models to achieve comparable or superior performance to larger proprietary models, demonstrating the benchmark's value for developing robust MRAG systems The code is available at https://github.com/phycholosogy/VQQP_Bench
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InfoDLM: an Information-Adaptive Framework for Discrete Diffusion Language Model Pretraining
Shirou Jing, Chunshu Wu, Chuan Liu, Arghavan Bahadorinejad, Feitong Qiao, Dongfang Liu, Tony Geng
Diffusion language models (DLMs) can match or surpass similarly sized autoregressive language models on language understanding and reasoning. However, their mask-and-denoise pretraining relies on heuristic random masking, which fails to target the most informative tokens. Consequently, the model spends significant computational effort on redundant or trivial tokens. To address this, we propose InfoDLM, an adaptive DLM pretraining framework that reformulates mask selection as an active, feedback-driven process. InfoDLM targets tokens that offer the highest measurable information gain during mask selection. Specifically, we: (1) introduce a Trainable Information-Gain (TIG) signal to quantify information gain of each masking configuration; (2) develop a feedback mechanism that adapts the masking policy to the model’s evolving state with a maturity indicator; and (3) jointly optimize the DLM and masking policy through an interleaved training flow wit

h minimal computational overhead. Across reasoning-oriented benchmarks, InfoDLM achieves up to 13\% improvement in reasoning accuracy over a small variant of LLaDA under comparable pretraining budgets.
****

Reasoning-VLA: An Efficient and Spatial-Guided General Vision-Language-Action Reasoning Model for Autonomous Driving
Dapeng Zhang, Zhenlong Yuan, Zhangquan Chen, Chih-Ting Liao, Yinda Chen, Fei Shen, Qingguo Zhou, Tat-Seng Chua
Vision-Language-Action (VLA) models have recently shown strong decision-making capabilities in autonomous driving. However, existing VLAs often struggle with achieving efficient inference and generalizing to novel autonomous vehicle configurations and driving scenarios. In this paper, we propose Reasoning-VLA, a general and efficient action-generation VLA framework. The proposed model employs a set of learnable action queries, implicitly guided by predefined spatial representations to enhance spatial awareness.
These learnable queries interact with reasoning-enhanced vision–language features to generate continuous action trajectories in parallel. To promote robust generalization, we consolidate eight publicly available autonomous driving datasets into a standardized, Chain-of-Thought reasoning–based, and easy-to-use data format for model training. Leveraging both supervised learning and reinforcement learning fine-tuning, extensive empirical evaluations across multiple benchmarks demonstrate that Reasoning-VLA achieves state-of-the-art performance, strong generalization capability, and the excellent inference speed with parallel decode.
****

Counterfactual Bootstrap for Robust Meta-Reinforcement Learning
Ai Bo, Junzhe Zhang, M. Cenk Gursoy
Meta-Reinforcement Learning (Meta-RL) focuses on training policies using data collected from a variety of diverse environments. This approach enables the policy to adapt to new settings with only a few training steps. While many Meta-RL methods have demonstrated success, they often rely on the assumption that unobserved confounders can be excluded a priori. This paper investigates robust Meta-RL in sequential decision-making, given confounded observational data collected across multiple heterogeneous environments. We introduce a novel augmentation procedure for standard Meta-RL algorithms (e.g., MAML), which employs partial identification methods to generate posterior counterfactual trajectories from candidate environments that align with the confounded observations. These counterfactual trajectories are then used to find a policy initialization that produces strong generalization performance in the target domain. Theoretical analysis reveals that our causal Meta-RL approach is guaranteed to yield a solution that minimizes generalization loss in future inference tasks.
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Discrete Diffusion VLA: Bringing Discrete Diffusion to Action Decoding in Vision-Language-Action Policies
Zhixuan Liang, Yizhuo Li, Tianshuo Yang, Chengyue Wu, Sitong Mao, Liuao Pei, Tian Nian, Shunbo Zhou, Xiaokang Yang, Jiangmiao Pang, Yao Mu, Ping Luo
Vision–Language–Action (VLA) models adapt large vision–language backbones to map images and instructions into robot actions. However, prevailing VLAs either generate actions autoregressively in a fixed left-to-right order with poor performance or attach separate diffusion heads outside the backbone that fragments information pathways and hinders unified, scalable architectures. Instead, we present Discrete Diffusion VLA that discretizes action chunks and models them with discrete diffusion pattern retaining progressive refinement inside the unified transformer backbone. Our method achieves an adaptive decoding order that resolves high-confidence action elements before harder ones and employs secondary re-masking to revisit uncertain predictions, enabling robust error correction. This design preserves pretrained vision-language priors, supports parallel decoding, and improves the efficiency. Discrete Diffusion VLA achieves 96.4% avg. success on LIBERO, 71.2% visual matching on SimplerEnv-Fractal, and 54.2% overall on SimplerEnv-Bridge. On out-of-distribution tests of LIBERO-Goal, our method exhibits only 0.8% language degradation versus 8.0% of parallel decoding, and 20.4% vision de

gradation versus 29.0% for continuous diffusion, demonstrating well retention of pretrained vision-language capabilities. We also conduct two real-robot evaluations on AgileX Cobot Magic platform to show the method's effectiveness.
****
Complexity Bounds for Dirichlet Process Slice Samplers
Beatrice Franzolini, Francesco Gaffi
Slice sampling is a standard Monte Carlo technique for Dirichlet process (DP)-based models, widely used in posterior simulation. However, formal assessments of the scalability of posterior slice samplers have remained largely unexplored, primarily because the computational cost of a slice-sampling iteration is random and potentially unbounded.
In this work, we obtain high-probability bounds on the computational complexity of DP slice samplers. Our main results show that, uniformly across posterior cluster-growth regimes, the overhead induced by slice variables, relatively to the number of clusters supported by the posterior, is $O_{\mathbb P}(\log n)$. As a consequence, even in worst-case configurations, superlinear blow-ups in per-iteration computational cost occur with vanishing probability.
Our analysis applies broadly to DP--based models without any likelihood-specific assumptions, still providing complexity guarantees for posterior sampling on arbitrary datasets.
These results establish a theoretical foundation for assessing the practical scalability of slice sampling in DP-based models.
****
Effective Model Pruning : Measuring the Redundancy of Model Components
Yixuan Wang, Dan P. Guralnik, Saiedeh Akbari, Warren Dixon
This article initiates the study of a basic question about model pruning. Given a vector s of importance scores assigned to model components, how many of the scored components could be discarded without sacrificing performance? We propose Effective Model Pruning (EMP), which derives the desired sparsity directly from the score distribution using the notion of effective sample size from particle filtering, also known as the inverse Simpson index.

Rather than prescribe a pruning criterion, EMP supplies a universal adaptive threshold derived from the distribution of the score s over the model components: EMP maps s to a number $N_{eff} = N_{eff} (s)$, called the effective sample size. The $N − N_{eff}$ lowest scoring components are discarded. A tight lower bound on the preserved mass fraction seff (the sum of retained normalized scores) in terms of N_{eff} is derived. This process yields models with a provable upper bound on the loss change relative to the original dense
model. Numerical experiments are performed demonstrating this phenomenon across a variety of network architectures including MLPs, CNNs, Transformers, LLMs, and KAN. It is also shown that EMP addresses a rich set of pruning criteria such as weight magnitude, attention score, KAN importance score, and even feature-level signals such as image pixels.
****
Learn to Merge: Meta-Learning for Adaptive Multi-Task Model Merging
Jun Chen, Qin Zhang, Weizhi Zhang, Xiao Luo, Philip S. Yu, Ziyue Qiao
Model merging in the pretrain-finetune paradigm has proven effective by combining multiple finetuned models into one with multi-task capabilities. However, existing methods rely on fix or manually tuned merging coefficients, making the unified model sensitive to the initial merging strategy and suboptimal for downstream adaptation. Thus, this paper proposed an innovative model merging framework called MetaMerging, a novel meta-learning algorithm to adaptively optimize the merging coefficients to construct a unified model tailored for task-specific adapter training. By simulating adapter updates in an inner loop and meta-optimizing merging coefficients in an outer loop, MetaMerging produces more balanced and generalizable unified models. Extensive experiments on CV and NLP fields show strong performance of MetaMerging on various downstream tasks and demonstrate the effectiveness of meta-learning in our method compared to other parameter merging methods. Our code is available at https://github.com/cjcj46262/MetaMerging.

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SPARC: Separating Perception And Reasoning Circuits for Test-time Scaling of VLM
s
Niccolò Avogaro, Nayanika Debnath, Li Mi, Thomas Frick, Junling Wang, Zexue He,
Hang Hua, Konrad Schindler, Mattia Rigotti
Despite recent successes, *test-time scaling*$-$dynamically expanding the token
budget during inference as needed$-$remains brittle for vision-language models (
VLMs). Unstructured visual reasoning chains entangle perception and reasoning, l
eading to long, disorganized contexts where small perceptual mistakes may cascad
e into completely wrong answers. Reasoning also requires expensive reinforcement
 learning with hand-crafted rewards. Here, we introduce SPARC (Separating Percep
tion And Reasoning Circuits), a modular framework that explicitly decouples visu
al perception from reasoning. Inspired by sequential sensory-to-cognitive proces
sing in the brain, SPARC implements a two-stage pipeline where the model first p
erforms explicit visual search to localize question-relevant regions, then condi
tions its reasoning on those regions to produce the final answer. This separatio
n enables independent test-time scaling with asymmetric compute allocation (e.g.
, prioritizing perceptual processing under distribution shift), and supports sel
ective optimization (e.g., improving the perceptual stage alone when it is the b
ottleneck for end-to-end performance). It also accommodates compressed contexts
by running global search at lower image resolutions and allocating high-resoluti
on processing only to selected regions, thereby reducing visual token count and
compute. SPARC outperforms monolithic baselines and strong visual-grounding appr
oaches across challenging visual reasoning tasks, such as improving Qwen3VL 4B o
n the V^* VQA benchmark by 6.7 points and surpassing "thinking with images" by
 4.6 points in an OOD setting with a 200$\times$ lower token budget.
****
PADS-TAL: Padding-Annealed Diffusion Sampling in Text-Aware Latent Space for Rob
ust and Diverse Text-to-Music Generation
Taekoan Yoo, Wonkyung Jung, Kyunghun Kim, Kyeongbo Kong
Text-to-Music diffusion models are increasingly used in real-world applications,
 yet deployment remains challenging: generations can collapse to limited pattern
s even with diverse initial noise and prompts, and inference-time diversity cont
rol often harms text alignment and fidelity by distorting key prompt cues establ
ished in early denoising.
To address this, we propose Padding-Annealed Diffusion Sampling, which perturbs
only a padding-indexed subspace while keeping non-padding conditioning fixed, en
abling controlled exploration with reduced semantic drift.
However, in a text-unaware VAE latent space, such exploration is less likely to
stay within genre-faithful neighborhoods, limiting genre-consistent diversity. W
e therefore introduce Text-Aware Latent space that aligns local neighborhoods wi
th text-implied genre structure, promoting genre-consistent diversity.
Together, the two techniques form a unified pipeline that, compared to prior met
hods that perturb the full conditioning, achieves a better text alignment--diver
sity trade-off: at comparable text alignment, it delivers 15.4\% higher diversit
y with a relatively small fidelity drop, and further improves within-genre diver
sity by 71.6\%.
Generated samples are available at https://pads-tal.github.io/PADS-TAL.io
****
From Noise to Intent: Anchoring Generative VLA Policies with Residual Bridges
Yiming Zhong, Yaoyu He, Zemin Yang, Pengfei Tian, Yifan Huang, Qingqiu Huang, Xi
nge Zhu, Yuexin Ma
Bridging high-level semantic understanding with low-level physical control remai
ns a persistent challenge in embodied intelligence, stemming from the fundamenta
l spatiotemporal scale mismatch between cognition and action. Existing generativ
e VLA policies typically adopt a "Generation-from-Noise" paradigm, which disrega
rds this disparity, leading to representation inefficiency and weak condition al
ignment during optimization. In this work, we propose ResVLA, an architecture th
at shifts the paradigm to "Refinement-from-Intent." Recognizing that robotic mot
ion naturally decomposes into global intent and local dynamics, ResVLA utilizes

spectral analysis to decouple control into a deterministic low-frequency anchor and a stochastic high-frequency residual. By anchoring the generative process on the predicted intent, our model focuses strictly on refining local dynamics via a residual diffusion bridge. Extensive simulation experiments show that ResVLA achieves competitive performance, strong robustness to language and robot embodiment perturbations, and faster convergence than standard generative baselines. ResVLA also demonstrates strong performance in real-world robot experiments.
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``Someone Hid It!'': Query-Agnostic Black-Box Attacks on LLM-Based Retrieval
Jiate Li, Defu Cao, Li Li, Wei Yang, Yuehan Qin, Chenxiao Yu, Tiannuo Yang, Ryan A. Rossi, Yan Liu, Xiyang Hu, Yue Zhao
Large language models (LLMs) have been serving as effective backbones for retrieval systems, including Retrieval-Augmentation-Generation (RAG), Dense Information Retriever (IR), and Agent Memory Retrieval. Recent studies have demonstrated that such LLM-based Retrieval (LLMR) is vulnerable to adversarial attacks, which manipulates documents by token-level injections and enables adversaries to either boost or diminish these documents in retrieval tasks. However, existing attack studies mainly (1) presume a known query is given to the attacker, and (2) highly rely on access to the victim model's parameters or interactions, which are hardly accessible in real-world scenarios, leading to limited validity. To further explore the secure risks of LLMR, we propose a practical black-box attack method that generates transferable injection tokens based on zero-shot surrogate LLMs without need of victim queries or victim models knowledge. The effectiveness of our attack raises such a robustness issue that similar effects may arise from benign or unintended document edits in the real world. To achieve our attack, we first establish a theoretical framework of LLMR and empirically verify it. Under the framework, we simulate the transferable attack as a min-max problem, and propose an adversarial learning mechanism that finds optimal adversarial tokens with learnable query samples. Our attack is validated to be effective on benchmark datasets across popular LLM retrievers.
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Dynamic Relational Priming Improves Transformer in Multivariate Time Series
Hunjae Lee, Corey Clark
Standard attention mechanisms in transformers employ static token representations that remain unchanged across all pair-wise computations in each layer. This limits their representational alignment with the potentially diverse dynamics of each token-pair interaction. While they excel in domains with relatively homogeneous relationships, standard attention may be inadequate in capturing heterogeneous inter-channel dependencies of multivariate time series (MTS) data where different channel-pair interactions within a single system may be governed by entirely different physical laws or temporal dynamics. To better align the attention mechanism for such domain phenomena, we propose attention with dynamic relational priming (prime attention). Prime attention modulates token representations for each token-pair, optimizing each pair-wise interaction for that specific relationship. Our results demonstrate that prime attention consistently outperforms standard attention across benchmarks, achieving up to 6.5\% improvement in forecasting accuracy. In addition, prime attention achieves comparable performance using up to 40\% less sequence length compared to standard attention, demonstrating its superior relational modeling capabilities and potential for data efficiency.
****

Learning to Reason for Factuality
Xilun Chen, Ilia Kulikov, Vincent-Pierre Berges, Barlas Oguz, Rulin Shao, Gargi Ghosh, Wen-tau Yih
Reasoning Large Language Models (R-LLMs) have significantly advanced complex reasoning tasks but often struggle with factuality, generating substantially more hallucinations than their non-reasoning counterparts on long-form factuality benchmarks. However, extending online Reinforcement Learning (RL), a key component in recent R-LLM advancements, to the long-form factuality setting poses several unique challenges due to the lack of reliable verification methods. Previous work has utilized automatic factuality evaluation frameworks such as FActScore to cu

rate preference data in the offline RL setting, yet we find that directly levera ging such methods as the reward in online RL leads to reward hacking in multiple ways, such as producing less detailed or relevant responses. We propose a novel reward function that simultaneously considers the factual precision, response d etail level, and answer relevance, and applies online RL to learn high quality f actual reasoning. Evaluated on six long-form factuality benchmarks, our factual reasoning model achieves an average reduction of 23.1 percentage points in hallu cination rate, a 23% increase in answer detail level, and no degradation in the overall response helpfulness.
****

Sign Lock-In: Randomly Initialized Weight Signs Persist and Bottleneck Sub-Bit M odel Compression
Akira Sakai, Yuma Ichikawa
Sub-bit model compression targets storage below one bit per weight; as magnitude s are aggressively compressed, the sign bit becomes a fixed-cost bottleneck. Acr oss Transformers, CNNs, and MLPs, learned sign matrices resist low-rank approxim ation and are spectrally indistinguishable from an i.i.d. Rademacher baseline. T his randomness gives rise to the lower bound of sub-bit model compression—the on e-bit wall. Despite this apparent randomness, most weights retain their initiali zation signs; flips primarily occur via rare near-zero boundary crossings, **sug gesting that sign-pattern randomness is largely inherited from initialization.** We formalize this behavior with *sign lock-in theory*, a stopping-time analysis of sign flips under SGD noise. Under bounded updates and a rare re-entry condit ion into a small neighborhood of zero, the number of effective sign flips exhibi ts a geometric tail. Building on this mechanism, we introduce a *from-scratch* l ow-rank sign-template training method that prevents the emergence of this one-bi t wall.
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MonoScale: Scaling Multi-Agent System with Monotonic Improvement
Shuai Shao, Yixiang Liu, Bingwei Lu, Weinan Zhang
In recent years, LLM-based multi-agent systems (MAS) have advanced rapidly, usin g a router to decompose tasks and delegate subtasks to specialized agents. A nat ural way to expand capability is to scale up the agent pool by continually integ rating new functional agents or tool interfaces, but naive expansion can trigger performance collapse when the router cold-starts on newly added, heterogeneous, and unreliable agents. We propose MonoScale, an expansion-aware update framewor k that proactively generates a small set of agent-conditioned familiarization ta sks, harvests evidence from both successful and failed interactions, and distill s it into auditable natural-language memory to guide future routing. We formaliz e sequential augmentation as a contextual bandit and perform trust-region memory updates, yielding a monotonic non-decreasing performance guarantee across onboa rding rounds under a non-interfering expansion assumption. Experiments on GAIA a nd Humanity's Last Exam show stable gains as the agent pool grows, outperforming naive scale-up and strong-router fixed-pool baselines. Our code is available he re.
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Learning Partial Concept Classes and Universal Rates Under Massart Noise
Ariel Avital, Klim Efremenko, Steve Hanneke
The Massart noise condition is a central model in
Probably Approximately Correct (PAC) learning
theory. Its importance lies in it being an interpolation condition between reali zable and the agnostic settings, under which one can attain faster
rates than in the latter, and, under strict conditions, recover the rates of the former. Despite
its importance, the Massart condition has not yet
been fully explored in emerging extensions of statistical learning theory beyond the classical PAC
framework. In this work, we present two such
extensions. First, we revisit the transductive empirical risk minimization (TERM) algorithm of

(Hanneke & Moran, 2026), and derive sharper
excess error bounds under Massart noise using
offset Rademacher techniques and local metric
entropy introduced by (Zhivotovskiy & Hanneke,
2018). We then leverage this analysis to obtain
new sample complexity bounds for PAC learning with partial concept classes and c
omplete the
characterization of universal rates under Massart
noise.
****
Learning Gaussian Graphical Models from a Glauber Trajectory Without Mixing
Eric Shen, Tony Wu, Mahbod Majid, Ankur Moitra
We study the task of learning the structure of a d-sparse Gaussian graphical m
odel on n variables from a single trajectory of Glauber dynamics. Beyond algor
ithmic considerations, many applications present temporally correlated observati
ons rather than i.i.d. samples. Moreover, in the classical i.i.d. setting, polyn
omial-time structure learning from a sublinear in n number of samples is suspe
cted to be computationally hard without additional assumptions on the precision
matrix. Motivated in part by this, we design the first polynomial-time algorithm
 that recovers the conditional-independence graph from a single Glauber trajecto
ry, with a trajectory-length guarantee that does not depend on the mixing time.

Technically, our algorithm has three components. First, we estimate the conditio
nal variances and rescale the trajectory to reduce to the unit-diagonal case, wi
thout changing the underlying graph. Second, we design a local edge test that ex
tracts adjacency information from short update windows by isolating pairwise inf
luence. Third, we aggregate these local statistics using a robust median-based e
stimator, and prove accuracy despite contamination and temporal dependence arisi
ng from a single trajectory.
****
$\texttt{FlashSchNet}$: Fast and Accurate Coarse-Grained Neural Network Molecula
r Dynamics
Pingzhi Li, Hongxuan Li, Zirui Liu, Xingcheng Lin, Tianlong Chen
Graph neural network (GNN) potentials such as SchNet improve the accuracy and tr
ansferability of molecular dynamics (MD) simulation by learning many-body intera
ctions, but remain slower than classical force fields due to fragmented kernels
and memory-bound pipelines that underutilize GPUs.
    We show that a missing principle is making GNN-MD $\textit{IO-aware}$, caref
ully accounting for reads and writes between GPU high-bandwidth memory (HBM) and
 on-chip SRAM.
    We present $\texttt{FlashSchNet}$, an efficient and accurate IO-aware SchNet
-style GNN-MD framework built on four techniques:
    (1) $\textit{flash radial basis}$, which fuses pairwise distance computation
, Gaussian basis expansion, and cosine envelope into a single tiled pass, comput
ing each distance once and reusing it across all basis functions;
    (2) $\textit{flash message passing}$, which fuses cutoff, neighbor gather, f
ilter multiplication, and reduction to avoid materializing edge tensors in HBM;
    (3) $\textit{flash aggregation}$, which reformulates scatter-add via CSR seg
ment reduce, reducing atomic writes by a factor of feature dimension and enablin
g contention-free accumulation in both forward and backward passes;
    (4) channel-wise 16-bit quantization that exploits the low per-channel dynam
ic range in SchNet MLP weights to further improve throughput with negligible acc
uracy loss.
    On a single NVIDIA RTX PRO 6000, $\texttt{FlashSchNet}$ achieves $\textbf{10
00 ns/day}$ aggregate simulation throughput over 64 parallel replicas on coarse-
grained (CG) protein containing 269 beads ($\textbf{6.5}$ $\mathbf{\times}$ fast
er than CGSchNet baseline with $\textbf{80\\% less}$ peak memory), surpassing wi
dely used classical force fields ($\textit{e.g.}$, MARTINI) while retaining SchN
et-level accuracy and transferability.
****

DuetServe: Harmonizing Prefill and Decode for LLM Serving via Adaptive GPU Multiplexing
Lei Gao, Chaoyi Jiang, Hossein Entezari Zarch, Daniel Wong, Mark D. Hill, Murali Annavaram
Modern LLM serving systems must sustain high throughput while meeting strict latency SLOs across two distinct inference phases: compute-intensive prefill and memory-bound decode phases. Existing approaches either (1) aggregate both phases on shared GPUs, leading to interference between prefill and decode phases, which degrades Time-Between-Tokens (TBT); or (2) disaggregate the two phases across GPUs, improving latency but wasting resources through duplicated models and KV cache transfers. We present DuetServe, a unified LLM serving framework that achieves disaggregation-level isolation within a single GPU. DuetServe operates in aggregated mode by default and dynamically activates SM-level GPU spatial multiplexing when TBT degradation is predicted. Its key idea is to decouple prefill and decode execution only when needed through fine-grained, adaptive SM partitioning that provides phase isolation only when contention threatens latency service level objectives. DuetServe integrates (1) an attention-aware roofline model to forecast iteration latency, (2) a partitioning optimizer that selects the optimal SM split to maximize throughput under TBT constraints, and (3) an interruption-free execution engine that eliminates CPU–GPU synchronization overhead. Evaluations show that DuetServe improves total throughput by up to 1.3$\times$ while maintaining low generation latency compared to state-of-the-art frameworks.
****
Semantic-level Backdoor Attack against Text-to-Image Diffusion Models
Tianxin Chen, Wenbo Jiang, Hongqiao Chen, Zhirun Zheng, Cheng Huang
Text-to-image (T2I) diffusion models are widely adopted for their strong generative capabilities, yet remain vulnerable to backdoor attacks. Existing attacks typically rely on fixed textual triggers and single-entity backdoor targets, making them highly susceptible to enumeration-based input defenses and attention-consistency detection. In this work, we propose Semantic-level Backdoor Attack (SemBD), which introduces representation-level triggers based on continuous semantic regions rather than discrete textual patterns. SemBD implants such semantic backdoors by distillation-based editing of the key and value projection matrices in cross-attention layers, enabling semantically equivalent but textually diverse prompts to activate the backdoor. To further enhance stealthiness, SemBD incorporates a semantic regularization to prevent unintended activation under incomplete semantics, as well as multi-entity backdoor targets that avoid highly consistent cross-attention patterns. Extensive experiments demonstrate that SemBD achieves a 100% attack success rate while maintaining strong robustness against state-of-the-art input-level defenses. Our code is available at https://github.com/DPAS-Lab/SemBD/.
****
Think in Latent, Explain in Language: Self-Explainable Latent Reasoning
Dayuan Zhao, Shengcao Cao, Yu-Xiong Wang, Liangyan Gui
Latent reasoning has emerged as a powerful alternative to text-based Chain-of-Thought (CoT), offering significant gains in computational efficiency by compressing verbose reasoning into compact embeddings. However, compressing reasoning into the latent space renders the thinking opaque, hindering its interpretability. Current methods present a stark trade-off: they either function as unexplainable “black boxes” (e.g., Coconut), where the latent reasoning is not human-readable, or rely on separate post-hoc decoders for explainability (e.g., Heima), introducing architectural overhead and decoupling the explanation from the actual reasoning process.
In this work, we present a unified framework for Self-Explainable Latent Reasoning (SELR) that trains a single model to perform efficient and inherently explainable latent reasoning. Our core contribution is a novel multi-task training objective that optimizes for two goals simultaneously: (1) an Answer Loss that optimizes the latent reasoning trajectory to produce accurate final answers, and (2) a CoT Loss that explicitly trains the same model to decode its own latent representations back into human-understandable reasoning steps. This design ensures th

at generated latent representations are both task-effective and semantically int
erpretable, eliminating the need for external decoders. We validate the effectiv
eness of SELR on both Large Language Models (LLMs) and Vision-Language Models (V
LMs), demonstrating that SELR achieves superior token efficiency and accuracy co
mpared to baselines, while uniquely providing self-contained explainability with
out auxiliary models.
****

Segment Anything with Robust Uncertainty-Accuracy Correlation
Hongyou Zhou, Marc Toussaint, Ling Shao, Zihan Ye
Despite strong zero-shot performance, SAM is unreliable under domain shift due t
o Mask-level Confidence Confusion (MCC), where a single IoU-based mask score fai
ls to reflect pixel-wise reliability near boundaries.
Motivated by the contrast between texture-biased shortcuts in neural networks an
d shape-centric processing in human vision, we model out-of-domain variation as
appearance shifts and non-rigid deformations that jointly perturb images.
We propose Segment Anything with Robust Uncertainty-Accuracy Correlation (RUAC)
for robust pixel-wise uncertainty estimation under appearance and deformation sh
ifts.
RUAC adds a lightweight uncertainty head, trains it with a collaborative style-d
eformation attack that jointly perturbs texture and geometry, and applies Uncert
ainty-Error Alignment to ensure uncertainty consistently highlights erroneous pi
xels even under adversarial perturbations.
Across 23 zero-shot domains, RUAC improves segmentation quality and yields more
faithful uncertainty with stronger uncertainty-accuracy correlation.
****

SurvDiff: A Diffusion Model for Generating Synthetic Data in Survival Analysis
Marie Brockschmidt, Maresa Schröder, Stefan Feuerriegel
Survival analysis is a cornerstone of clinical research by modeling time-to-even
t outcomes such as metastasis, disease relapse, or patient death. Unlike standar
d tabular data, survival data often come with incomplete event information due t
o dropout, or loss to follow-up. This poses unique challenges for synthetic data
 generation, where it is crucial for clinical research to faithfully reproduce b
oth the event-time distribution and the censoring mechanism. In this paper, we p
ropose SurvDiff, an end-to-end diffusion model specifically designed for generat
ing synthetic data in survival analysis. SurvDiff is tailored to capture the dat
a-generating mechanism by jointly generating mixed-type covariates, event times,
 and right-censoring, guided by a survival-tailored loss function. The loss enco
des the time-to-event structure and directly optimizes for downstream survival t
asks, which ensures that SurvDiff (i) reproduces realistic event-time distributi
ons and (ii) preserves the censoring mechanism. Across multiple datasets, we sho
w that SurvDiff outperforms state-of-the-art generative baselines in both distri
butional fidelity and survival model evaluation metrics across multiple medical
datasets. To the best of our knowledge, SurvDiff is the first end-to-end diffusi
on model explicitly designed for generating synthetic survival data.
****

Pose-ICL: 3D-Aware In-Context Learning for Pose-Controllable Subject Customizati
on
Xuan Han, Yihao Zhao, Mingyu You
Subject Customization is a foundational task in modern image generation. By prov
iding a few reference images and a text prompt, users can generate images of a s
pecific object in any desired scene. However, existing methods still struggle to
 achieve effective pose control for customized subjects. In practice, they often
 exhibit inaccurate poses or inconsistent cross-pose appearances. These limitati
ons suggest that understanding objects in a volumetric manner remains a signific
ant challenge for 2D-native backbones. To address this challenge, we propose Pos
e-ICL, a tuning-free framework that leverages 3D-aware In-Context Learning (ICL)
 to directly adapt to new subjects through multiple paired image-pose references
. Its core mechanism, Surface-Anchored Position Embedding (SAPE), equips the mod
el with explicit 3D awareness by anchoring image tokens to the surface coordinat
es of a volumetric bounding box. Dedicated refinements ensure its seamless compa

tibility with existing DiT models. Extensive evaluations on both 3D assets and r
eal-world subjects demonstrate that Pose-ICL significantly outperforms current m
ethods in both pose accuracy and identity consistency.
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Interpreting Genomic Language Models using Sparse Autoencoders
Akira A Nair, Jaehyun Joo, Jonghyun Lee, Lina Takemaru, Yidi Huang, Manu Shivaku
mar, Matthew Eric Lee, Jaesik Kim, Sokratis Apostolidis, Dokyoon Kim
Genomic language models (gLMs) achieve strong performance across genomic predict
ion tasks, but their internal biological representations remain poorly understoo
d. Sparse autoencoders (SAEs) have emerged as an interpretability tool in vision
 and natural language models, yet their applicability to gLMs remains unexplored
. We present a systematic study of SAE-based interpretability for gLMs, introduc
ing a diverse benchmark of human genomic annotations and a suite of genome-tailo
red interpretability metrics. Using Evo2 as a primary case study, we show that S
AE features, particularly those from intermediate layers, are more interpretable
 than raw model embeddings across 42/55 (76%) of our genomic concept evaluations
, with 26 of them having an F1 score greater than 0.7. We further find that inte
rpretability depends on SAE training data properties such as evolutionary proxim
ity and context length. Finally, to organize semantically related genomic concep
ts learned by an SAE, we develop a graph-based representation method that outper
forms the baseline approach of using SAE model weights. We demonstrate how our f
ramework can extend SAEs as a powerful approach for not only better understandin
g gLMs but also for adopting them in disease-driven genomic explorations.
****
Characterizing the Effect of Noise in Language Generation in the Limit
Aaron Li, Ian Zhang
Kleinberg and Mullainathan recently proposed a formal framework for studying
        the phenomenon of language generation, called *language generation in th
e limit*.
        In this model, an adversary gives an enumeration of example strings from
 an unknown target language,
        and the algorithm is tasked with correctly generating
        unseen strings from the target language within finite time.
        Refined notions of non-uniform and uniform generation were later introdu
ced by
        Li, Raman, and Tewari (2025), and a noisy model was introduced by Raman
and Raman (2025),
        which allows the adversary to insert extraneous strings.
        A natural question in the noisy model is to quantify the effect of noise
,
        by studying the impact of each additional extraneous string.
        We show two complementary results in this setting.
        We first show that for both uniform and non-uniform generation, a single
 noisy string strictly reduces the set
        of collections that can be generated, thus answering an open question in
 Raman and Raman (2025).
        Then, we show for both uniform and non-uniform generation that generatio
n with a single noisy string is
        equivalent to generation with any finite amount of noise,
        sharply contrasting with the strict hierarchy for
        noisy generation in the limit shown by Bai, Panigrahi, and Zhang (2026).
        Finally, we leverage our previous results to provide the first known cha
racterization for
        non-uniform noise-dependent generatability.
****
RTPrune: Reading-Twice Inspired Token Pruning for Efficient DeepSeek-OCR Inferen
ce
Ben Wan, Yan Feng, Zihan Tang, Weizhe Huang, Yuting Zeng, Jia Wang, Tongxuan Liu
DeepSeek-OCR leverages visual–text compression to reduce long-text processing co
sts and accelerate inference, yet visual tokens remain prone to redundant textua

l and structural information. Moreover, current token pruning methods for conven tional vision–language models (VLMs) fail to preserve textual fidelity due to im proper compression mechanisms. By analyzing the decoding process of DeepSeek-OCR , we find that a distinct two-stage reading trajectory: the model initially prio ritizes the majority of high-norm tokens, then subsequently redistributes its at tention to the remaining ones. Motivated by this insight, we propose *RTPrune*, a two-stage token pruning method tailored for DeepSeek-OCR. In the first stage, we prioritize high-norm visual tokens that capture salient textual and structura l information. In the second stage, the remaining tokens are paired and merged b ased on optimal transport theory to achieve efficient feature aggregation. We fu rther introduce a dynamic pruning ratio that adapts to token similarity and text ual density for OCR tasks, enabling a better efficiency–accuracy trade-off. Exte nsive experiments demonstrate state-of-the-art performance, as evidenced by 99.4 7\% accuracy and 1.23× faster prefill on OmniDocBench, achieved with 84.25\% tok en retention when applied to DeepSeek-OCR-Large. Code is released.
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Text-Conditional JEPA for Learning Semantically Rich Visual Representations
Chen Huang, Xianhang Li, Vimal Thilak, Etai Littwin, Joshua M. Susskind
Image-based Joint-Embedding Predictive Architecture (I-JEPA) offers a promising approach to visual self-supervised learning through masked feature prediction. H owever with the inherent visual uncertainty at masked positions, feature predict ion remains challenging and may fail to learn semantic representations. In this work, we propose Text-Conditional JEPA (TC-JEPA) that uses image captions to red uce the prediction uncertainty. Specifically, we modulate the predicted patch fe atures using a fine-grained text conditioner that computes sparse cross-attentio n over input text tokens. With such conditioning, patch features become predicta ble as a function of text, thus are more semantically meaningful. We show TC-JEP A improves downstream performance and training stability, with promising scaling properties. TC-JEPA also offers a new vision-language pretraining paradigm base d on feature prediction only, outperforming contrastive methods on diverse tasks , especially those requiring fine-grained visual understanding and reasoning.
****

CORE: Context-Robust Remasking for Diffusion Language Models
Kevin Zhai, Sabbir Mollah, Zhenyi Wang, Mubarak Shah
Standard decoding in Masked Diffusion Models (MDMs) is hindered by context rigid ity: tokens are retained based on transient high confidence, often ignoring that early predictions lack full context. This creates cascade effects where initial inconsistencies misguide the remaining generation. Existing revision strategies attempt to mitigate this by relying on static confidence scores, but these sign als are inherently myopic; inconsistent tokens frequently appear confident to th e model itself. To address this, we propose Context-Robust Remasking (CORE), a t raining-free framework for inference-time revision. We introduce a new selection paradigm: rather than trusting static token probabilities, we identify *context -brittle* tokens by probing their sensitivity to adversarial perturbations. We f ormalize revision as a robust optimization problem targeting worst-case context shifts. CORE efficiently approximates this objective using theoretically bounded probability margins to expose and revise unstable tokens. On LLaDA-8B-Base, COR E delivers consistent improvements across reasoning and code benchmarks, outperf orming compute-matched baselines and boosting performance on code generation (MB PP) by up to $+9.2\\%$, with comparable gains generalizing to the Dream-7B archi tecture.
****

Dissecting Causal Mechanism Shifts via FANS: Function And Noise Separation
Gyeongdeok Seo, Jaeyoon Shim, Mingyu Kim, Hoyoon Byun, Yonghan Jung, Kyungwoo So ng
Identifying the drivers of causal mechanism shifts, distinguishing functional ch anges from noise alterations, termed *dissection*, is a critical yet under-explo red problem in data science (e.g., biomedical science and manufacturing). This p aper introduces a more general and unified framework, the *function and noise se paration* (FANS) framework, that detects and dissects shifts in non-additive, no

n-linear Structural Causal Models (SCMs) beyond existing additive noise models. Our approach is grounded in a theoretical independence criterion, where function shifts induce a statistical dependence between a node's parents and residual noise. Building on this foundation, we develop a practical two-stage algorithm to efficiently detect and dissect these shifts without retraining. Furthermore, we address the complex challenge of simultaneous function and noise shifts, introducing a formal assumption to resolve their inherent non-identifiability. Our results are corroborated by simulations. Our code is available at https://github.com/MLAI-Yonsei/FANS/.

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Cross-task Calibration for Asynchronous Federated Continual Learning

Yichen Li, Haozhao Wang, Hang Su, Yulong Li, Xiaoquan Yi, Yankai Jiang, Chuang Zhao, Imran Razzak, Ruixuan Li

Federated Continual Learning (FCL) aims to empower distributed devices to learn a sequence of tasks over time. However, existing FCL research largely relies on the impractical assumption of synchronous new task arrival. This overlooks the reality of asynchronous user behavior and system latencies, forcing more efficient clients to endure costly inactivity. The practical necessity of an asynchronous method gives rise to Asynchronous Federated Continual Learning (AFCL). The server constantly receives a mixture of updates from clients at different time steps, leading to a catastrophic task drift that corrupts the global model and prevents effective learning. In this paper, we introduce a novel Cross-task Calibration framework called C^2-AFCL that is the first to tackle task drift at a semantic level within an Asynchronous FCL setting. Its core is a two-stage orthogonal calibration mechanism. First, intra-client calibration uses task-aware caches to mitigate variance from local client drift. Second, and more critically, inter-task interference calibration dynamically estimates an interference subspace from historical task knowledge. New updates are orthogonally projected to isolate and remove components that conflict with this subspace, preserving previous knowledge while learning new tasks. Extensive experiments show that C^2-AFCL significantly outperforms existing methods, demonstrating robust and efficient learning in dynamic federated environments.

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CL-GCL: Comprehensive and Lightweight Graph Contrastive Learning

Jianqing Liang, Xinkai Wei, Zhiqiang Li

Graph Contrastive Learning (GCL) has significantly advanced self-supervised representation learning on graphs, yet its practical efficacy remains hindered by random augmentations that induce semantic distortion and rigid one-to-one sampling strategy that amplifies inter-class entanglement and intra-class dispersion. To address these limitations, we develop CL-GCL, a Comprehensive and Lightweight Graph Contrastive Learning framework. Specifically, we exploit graph coarsening■ to preserve structural semantics through community-level representations and manifold learning■ to capture local geometric relations without costly pairwise distance computations. This design naturally aligns with the neighborhood aggregation principle of Graph Convolutional Networks, enhancing structural consistency while eliminating negative sampling bias. We theoretically prove that CL-GCL approximates node-level contrastive loss under mild conditions. Extensive experiments demonstrate consistent superiority in both accuracy and efficiency over state-of-the-art GCL methods.

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Infinite Mask Diffusion for Few-Step Distillation

Jaehoon Yoo, Wonjung Kim, Chanhyuk Lee, Seunghoon Hong

Masked Diffusion Models (MDMs) have emerged as a promising alternative to autoregressive models in language modeling, offering the advantages of parallel decoding and bidirectional context processing within a simple yet effective framework. Specifically, their explicit distinction between masked tokens and data underlies their simple framework and effective conditional generation. However, MDMs typically require many sampling iterations due to factorization errors stemming from simultaneous token updates. We observe that a theoretical lower bound of the factorization error exists, which standard MDMs cannot reduce due to their use o

f a deterministic single-state mask. In this paper, we propose the Infinite Mask Diffusion Model (IMDM), which introduces a stochastic infinite-state mask to mitigate the theoretical bound while directly inheriting the benefits of MDMs, including the compatibility with pre-trained weights. We empirically demonstrate that MDM fails to perform few-step generation even in a simple synthetic task due to the factorization error bound, whereas IMDM can find an efficient solution for the same task. Finally, when equipped with appropriate distillation methods, IMDM surpasses existing few-step distillation methods at small step counts on LM1B and OpenWebText.
****

DenseSteer: Steering Small Language Models towards Dense Math Reasoning
Yang Ouyang, Shuhang Lin, Jung-Eun Kim
Large language models (LLMs) demonstrate strong chain-of-thought (CoT) reasoning abilities, while smaller models ($\leq$ 3B parameters) significantly underperform on multi-step reasoning tasks. Based on empirical analyses of the Qwen-2.5 model family on math reasoning benchmarks, we find that more proficient reasoning is associated with fewer reasoning steps but higher information density per step, a property we term *Dense Reasoning*. Motivated by this observation, we propose **DenseSteer**, a training-free inference-time steering framework that enhances small-model reasoning by modulating internal representations toward dense reasoning patterns. Experiments show that our method yields consistent accuracy improvements without increasing token-level Negative Log-Likelihood, highlighting dense reasoning as an effective structural approach to mathematical problem solving.
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From Guessing to Placeholding: A Cost-Theoretic Framework for Uncertainty-Aware Code Completion
Liang Zhu, Haolin Chen, Lidong Zhao, Xian Wu
While Large Language Models (LLMs) have demonstrated exceptional proficiency in code completion, they typically adhere to a **Hard Completion (HC)** paradigm, compelling the generation of fully concrete code even amidst insufficient context. Our analysis of 3 million real-world interactions exposes the limitations of this strategy: 61% of the generated suggestions were either edited after acceptance or rejected despite exhibiting over 80% similarity to the user's subsequent code, suggesting that models frequently make erroneous predictions at specific token positions. Motivated by this observation, we propose **Adaptive Placeholder Completion (APC)**, a collaborative framework that extends HC by strategically outputting explicit placeholders at high-entropy positions, allowing users to fill directly via IDE navigation. Theoretically, we formulate code completion as a cost-minimization problem under uncertainty. Premised on the observation that filling placeholders incurs lower cost than correcting errors, we prove the existence of a critical entropy threshold above which APC achieves strictly lower expected cost than HC. We instantiate this framework by constructing training data from filtered real-world edit logs and design a cost-based reward function for reinforcement learning. Extensive evaluations across 1.5B--14B parameter models demonstrate that APC reduces expected editing costs from 19% to 50% while preserving standard HC performance. Our work provides both a theoretical foundation and a practical training framework for uncertainty-aware code completion, demonstrating that adaptive abstention can be learned end-to-end without sacrificing conventional completion quality.
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Deep Discriminative Structure Proxy Hashing for Cross-modal Retrieval
Kun Cheng, Qibing Qin, Lei Huang
Existing proxy-based hashing methods optimize samples toward independently learned proxies using isolated similarity constraints. Although efficient, this design overlooks the fact that proxies are learned jointly but lack explicit relational or competitive interactions during optimization. Consequently, proxy responses to a sample are often accumulated rather than contrasted, leading to weakly defined decision regions and limited discriminative structure in the Hamming space.

In contrast, our method organizes multiple proxies into sample-specific relation al structures, enabling proxies to interact and compete when responding to each sample. Through structure-guided learning, these interactions explicitly contras t positive and negative proxy responses, thereby shaping clearer and more discri minative decision boundaries. Extensive experiments on standard cross-modal benc hmarks demonstrate that this structured discrimination consistently improves ret rieval accuracy and embedding separability. The source code is available at [<ht tps://github.com/QinLab-WFU/DDSPH>](https://github.com/QinLab-WFU/DDSPH).
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On the Expressive Power of GNNs to Solve Linear SDPs
Chendi Qian, Christopher Morris
Semidefinite programs (SDPs) are a powerful framework for convex optimization an d for constructing strong relaxations of hard combinatorial problems. However, s olving large SDPs can be computationally expensive, motivating the use of machin e learning models as fast computational surrogates. Graph neural networks (GNNs) are a natural candidate in this setting due to their sparsity-awareness and abi lity to model variable-constraint interactions. In this work, we study what expr essive power is sufficient to recover optimal SDP solutions. We first prove nega tive results showing that standard GNN architectures fail on recovering linear S DP solutions. We then identify a more expressive architecture that captures the key structure of SDPs and can, in particular, emulate the updates of a standard first-order solver. Empirically, on both synthetic and SDPLIB benchmarks of vari ous classes of SDPs, this more expressive architecture achieves consistently low er prediction error and objective gap than theoretically weaker baselines. Final ly, using the learned high-quality predictions to warm-start the first-order sol ver yields practical speedups of up to 80%.
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Rh-3DGS: Robust Open-Vocabulary Scene Understanding via Riemannian Huber Distill ation and Manifold-Aware Sampling
Xinpeng Zhao, Jiang Jie, Fengyuan Zhang, Lixin Zhan, Dong Wang, Qinyuan Bu, Jiah ang Tu, Guangzhen Yao
Open-vocabulary 3D scene understanding answers free-form text queries over recon structed scenes. However, lifting dense 2D foundation-model embeddings into 3D G aussian Splatting (3DGS) is still challenging. Existing 3DGS-based methods often average normalized embeddings in Euclidean space. This ignores their hyperspher ical geometry and can cause feature collapse. They also distill supervision from all views equally, which amplifies occlusion noise and mixed-depth artifacts. W e propose **Rh-3DGS**, a robust semantic 3DGS framework that uses reliability-aw are distillation and manifold-consistent aggregation. **Visibility-Calibrated Di stillation (VCD)** computes per-pixel reliability weights from rasterization sta tistics and down-weights ambiguous pixels. **Visibility-Weighted Fréchet Mean (V FM)** aggregates embeddings on the unit hypersphere with a Riemannian Huber obje ctive for robust distillation. **Lightweight Consistency Contrast (LIC)** regula rizes the 3D semantic field with neighborhood-based multi-positive contrast to i mprove local consistency and sharper boundaries. Experiments on three benchmarks show that Rh-3DGS is best on open-vocabulary segmentation, boundary quality, an d view-consistent rendering.
****

From Noise to Control: Parameterized Diffusion Policies
Renhao Zhang, Haotian Fu, Mingxi Jia, George Konidaris, Yilun Du, Bruno Castro d a Silva
We propose Parameterized Diffusion Policy (PDP), a framework that learns a diffu sion policy parameterized in a smooth continuous space. By structuring a latent manifold such that distances between latents' values reflect the semantic simila rity of physical trajectories, we transform diffusion from a mechanism of stocha stic diversity into a precise tool for behavior steering. Our approach also enab les smooth interpolation between known strategies and efficient generalization t o novel constraints without the need to update policy weights. We demonstrate th at PDP significantly improves adaptation performance on complex multimodal bench marks in both simulation and real-robot hardware compared to regular diffusion p

olicy, particularly in scenarios requiring the discovery of novel behaviors.
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IMPACT: Influence Modeling for Open-Set Time Series Anomaly Detection
Xiaohui Zhou, Yijie Wang, Hongzuo Xu, Weixuan Liang, Xiaoli Li, Guansong Pang
Open-set anomaly detection (OSAD) is an emerging paradigm designed to utilize li mited labeled data from anomaly classes seen in training to identify both seen a nd unseen anomalies during testing. Current approaches rely on simple augmentati on methods to generate pseudo anomalies that replicate unseen anomalies. Despite being promising in image data, these methods are found to be ineffective in tim e series data due to the failure to preserve its sequential nature, resulting in trivial or unrealistic anomaly patterns. They are further plagued when the trai ning data is contaminated with unlabeled anomalies. This work introduces $\textb f{IMPACT}$, a novel framework that leverages $\underline{\textbf{i}}$nfluence $\ underline{\textbf{m}}$odeling for o$\underline{\textbf{p}}$en-set time series $\ underline{\textbf{a}}$nomaly dete$\underline{\textbf{ct}}$ion, to tackle these c hallenges. The key insight is to $\textbf{i)}$ learn an influence function that can accurately estimate the impact of individual training samples on the modelin g, and then $\textbf{ii)}$ leverage these influence scores to generate semantica lly divergent yet realistic unseen anomalies for time series while repurposing h igh-influential samples as supervised anomalies for anomaly decontamination. Ext ensive experiments show that IMPACT significantly outperforms existing state-of- the-art methods, showing superior accuracy under varying OSAD settings and conta mination rates. Code is available at https://github.com/mala-lab/IMPACT.
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AlignVid: Taming Visual Dominance via Training-Free Attention Modulation in Text -guided Image-to-Video Generation
Yexin Liu, Wen-Jie Shu, Zile Huang, Haoze Zheng, Yueze Wang, Manyuan Zhang, Jinj ing Zhu, Ser-Nam Lim, Harry Yang
Text-guided image-to-video generation has made substantial progress, yet it stil l struggles to execute text-specified edits that require substantial changes to a reference image (e.g., object addition, deletion, or modification). Empiricall y, our analysis reveals that this stems from **visual dominance**, where the ref erence image causes severe attention dispersion, inhibiting the model's ability to incorporate new semantic information. To address this, we propose **AlignVid* *, a training-free intervention that re-calibrates the model's internal attentio n distribution. Drawing on an energy-based perspective of attention, AlignVid em ploys Attention Scaling Modulation (**ASM**) to reduce attention entropy and con centrate focus on semantic tokens, alongside Guidance Scheduling (**GS**) to mai ntain generation stability. To rigorously assess this capability, we present **O mitI2V**, a comprehensive benchmark for evaluating prompt adherence across objec t addition, deletion, and modification. Extensive experiments demonstrate that A lignVid effectively enhances semantic fidelity with negligible computational ove rhead.
****
Provable Sample Efficiency of Curriculum Post-Training for Transformer Reasoning
Dake Bu, Wei Huang, Andi Han, Atsushi Nitanda, Hau-San Wong, Qingfu Zhang, Taiji Suzuki
Recent curriculum techniques in the post-training stage of LLMs have been empiri cally observed to outperform non-curriculum approaches in improving reasoning pe rformance, yet a principled understanding of their effectiveness and limitations remains incomplete. To bridge this gap, we develop an abstract theoretical fram ework and identify sufficient conditions under which curriculum post-training yi elds exponential improvements in sample complexity.

To substantiate this framework, we model the base model’s Chain-of-Thought gener ation as a state-conditioned autoregressive reasoning tree, and formalize curric ulum subtasks as either depth-increasing curricula that progressively extend rea soning horizons or hint-decreasing curricula that gradually remove partial hints . Our analysis shows that reinforcement learning finetuning with both curriculum strategies achieves high accuracy with polynomial sample complexity, whereas no

n-curriculum counterpart encounters an exponential complexity bottleneck. We fur
ther establish analogous guarantees for test-time scaling. Empirical simulations
 support our theoretical findings. Code is available at https://github.com/DakeB
U/Curriculum-Post-training.
****
BeaconKV: Key-Value Cache Compression Guided by Beacon Queries for Efficient Lar
ge Reasoning Model Inference
Janghyeon Kim, Minsoo Kim, Kyuhong Shim, Jungwook Choi
Large Reasoning Models (LRMs) achieve superior problem-solving through extended
Chain-of-Thought (CoT) generation, but the resulting key-value (KV) cache grows
linearly with sequence length and creates severe memory bottlenecks—often exceed
ing GPU capacity for long reasoning traces. Existing KV cache compression method
s rely on recent queries to estimate future token importance, implicitly assumin
g these serve as reliable proxies for future attention patterns. We demonstrate
that this assumption fails in long-horizon reasoning: certain decoding steps gen
erate Thought Revisiting Tokens (TRT) that re-attend to distant previous context
, such as task-solving plans formulated early in the trace. Through systematic a
nalysis, we discover that queries corresponding to the TRT cluster into a small
number of similarity groups in the embedding space. Based on this insight, we pr
opose BeaconKV, a training-free KV cache compression method that maintains beaco
n queries—compact representatives for each global query cluster—to anticipate wh
ich KV pairs will be revisited without storing the entire query history. Across
four open-source LRMs and diverse reasoning benchmarks, BeaconKV generally outpe
rforms existing compression methods, achieving up to $5.8\times$ memory reductio
n while nearly preserving full cache accuracy and improving throughput by over $
4.3\times$.
****
Evolution Strategies at the Hyperscale
Bidipta Sarkar, Mattie Fellows, Juan Agustin Duque, Alistair Letcher, Antonio Le
ón Villares, Anya Sims, Clarisse Wibault, Dmitry Samsonov, Dylan Cope, Jarek Luc
a Liesen, Kang Li, Lukas Seier, Theo Wolf, Uljad Berdica, Valentin Mohl, Alexand
er David Goldie, Aaron Courville, Karin Sevegnani, Shimon Whiteson, Jakob Nicola
us Foerster
Evolution Strategies (ES) is a class of powerful black-box optimisation methods
that are highly parallelisable and can handle non-differentiable and noisy objec
tives. However, naïve ES becomes prohibitively expensive at scale on GPUs due to
 the low arithmetic intensity of batched matrix multiplications with unstructure
d random perturbations. We introduce Evolution Guided GeneRal Optimisation via L
ow-rank Learning (EGGROLL), which improves arithmetic intensity by structuring i
ndividual perturbations as rank-r matrices, resulting in a hundredfold increas
e in training speed for billion-parameter models at
large population sizes, achieving up to 91\% of the throughput of pure batch inf
erence. We provide a rigorous theoretical analysis of ES for high-dimensional pa
rameter objectives, investigating conditions needed for ES updates to converge i
n
high dimensions, revealing a linearising effect, and proving consistency between
 EGGROLL and
ES as parameter dimension increases. Our experiments show that EGGROLL: (1) enab
les the stable pretraining of nonlinear recurrent language models that operate p
urely in integer datatypes, (2) is competitive with GRPO for post-training
LLMs on reasoning tasks, and (3) does not compromise performance compared to ES
in tabula rasa RL settings, despite being faster.
****
GRPO-based Cluster Decision Agent for Unknown-$\boldsymbol{K}$ Multi-view Cluste
ring
Xuqian Xue, Jun Zhang, Qi Cai, Zhizhong Huang, Hongming Shan, Junping Zhang
Existing contrastive multi-view clustering methods rely on a pre-defined cluster
 number, limiting their flexibility in real-world scenarios lacking prior knowle
dge. To address this, we propose GROK, a novel framework driven by a cluster dec
ision agent for unknown-K multi-view clustering. It pioneers the adaptation of

group relative policy optimization (GRPO) —a reinforcement learning strategy for LLM reasoning— into the unsupervised domain to autonomously determine the optimal K. Specifically, the agent orchestrates the clustering process through three synergistic phases. First, in the state perception phase, we employ a structure-aware adaptive backbone to aggregate multi-view data, providing the agent with consistent and discriminative consensus observations. Second, in the group decision phase, we introduce an action space divide-and-conquer strategy and an adaptive reward function. Equipped with these mechanisms, the agent performs group sampling and relative advantage estimation within the discrete action space of candidate K values, autonomously searching for the optimal K via reward maximization. Finally, via geometric feedback, geometric clustering guidance mechanism transforms the agent's structural hypotheses into explicit differentiable constraints to reshape feature manifolds, thereby closing the perception-decision-feedback loop. Experimental results demonstrate that GROK achieves superior clustering performance in unknown-K scenarios by autonomously exploring the cluster structure.

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Black-Box Combinatorial Optimization with Order-Invariant Reinforcement Learning
Olivier Goudet, Quentin Suire, Adrien Goëffon, Frédéric Saubion, Sylvain Lamprier

We introduce an order-invariant reinforcement learning framework for black-box combinatorial optimization. Classical estimation-of-distribution algorithms (EDAs) often rely on learning explicit variable dependency graphs, which can be costly and may fail to capture complex interactions efficiently. In contrast, we parameterize a multivariate autoregressive generative model trained without a fixed variable ordering. By sampling random generation orders during training, a form of information-preserving dropout, the model is encouraged to be invariant to variable order, promoting search-space diversity, and shaping the model to focus on the most relevant variable dependencies, improving sample efficiency. We adapt Group Relative Policy Optimization (GRPO) to this setting, providing stable policy-gradient updates from scale-invariant advantages. Across a wide range of benchmark problem instances of varying sizes, our method frequently achieves the best performance and consistently avoids catastrophic failures.

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FloorplanQA: A Benchmark for Spatial Reasoning in LLMs using Structured Representations
Fedor Rodionov, Abdelrahman Eldesokey, Michael Birsak, John Femiani, Bernard Ghanem, Peter Wonka

We introduce FloorplanQA, a diagnostic benchmark for evaluating spatial reasoning in large language models (LLMs). FloorplanQA is grounded in structured representations of indoor scenes (e.g., kitchens, living rooms, bedrooms, bathrooms, and others), encoded symbolically in JSON or XML layouts. The benchmark covers core spatial tasks, including distance measurement, visibility, path finding, and object placement within constrained spaces. Our results across a variety of frontier open-source and commercial LLMs reveal that while models may succeed on shallow queries, they often fail to respect physical constraints and preserve spatial coherence, though they remain mostly robust to small spatial perturbations. FloorplanQA uncovers a blind spot in today’s LLMs: inconsistent reasoning about indoor layouts. We hope this benchmark inspires new work on language models that can accurately infer and manipulate spatial and geometric properties in practical settings.

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New Wide-Net-Casting Jailbreak Attacks Risk Large Models
Qiuchi Xiang, Haoxuan Qu, Hossein Rahmani, Jun Liu

Jailbreak attacks on large models have drawn growing attention due to their close ties to societal safety. This work identifies a practical yet unexplored jailbreak scenario, the wide-net-casting scenario, where an adversary can query a group of large models instead of a single one to elicit harmful outputs. Our analysis reveals substantial yet previously overlooked safety risks under this scenario.

As a key part of our analysis, we further develop a novel jailbreak method tailored to the wide-net-casting scenario.
With this tailored method, the jailbreak success rate can even reach 100% in some experiments when targeting the large models without additional safeguards, exposing wide-net-casting as a distinct, high-risk scenario that warrants attention in future evaluation and defense research.
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DreamID-Omni: Unified Framework for Controllable Human-Centric Audio-Video Generation
Xu Guo, Fulong Ye, Qichao Sun, Liyang Chen, Bingchuan Li, Pengze Zhang, Jiawei Liu, Songtao Zhao, Qian HE, Xiangwang Hou
Recent advancements in foundation models have revolutionized joint audio-video generation. However, existing approaches typically treat human-centric tasks including reference-based audio-video generation (R2AV), video editing (RV2AV) and audio-driven video animation (RA2V) as isolated objectives. Furthermore, achieving precise, disentangled control over multiple character identities and voice timbres within a single framework remains an open challenge. In this paper, we propose DreamID-Omni, a unified framework for controllable human-centric audio-video generation. Specifically, we design a Symmetric Conditional Diffusion Transformer that integrates heterogeneous conditioning signals via a symmetric conditional injection scheme. To resolve the pervasive identity-timbre binding failures and speaker confusion in multi-person scenarios, we introduce a Dual-Level Disentanglement strategy: Synchronized RoPE at the signal level to ensure rigid attention-space binding, and Structured Captions at the semantic level to establish explicit attribute-subject mappings. Furthermore, we devise a Multi-Task Progressive Training scheme that leverages weakly-constrained generative priors to regularize strongly-constrained tasks, preventing overfitting and harmonizing disparate objectives. Extensive experiments demonstrate that DreamID-Omni achieves comprehensive state-of-the-art performance across video, audio, and audio-visual consistency, even outperforming leading proprietary commercial models.
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Discrete Diffusion Samplers and Bridges: Off-Policy Algorithms and Applications in Latent Spaces
Arran Carter, Sanghyeok Choi, Kirill Tamogashev, Víctor Elvira, Esmeralda S. Whitammer
Sampling from a distribution $p(x) \propto e^{-\mathcal{E}(x)}$ known up to a normalising constant is an important and challenging problem in statistics. Recent years have seen the rise of a new family of amortised sampling algorithms, commonly referred to as diffusion samplers, that enable fast and efficient sampling from an unnormalised density. Such algorithms have been widely studied for continuous-space sampling tasks; however, their application to problems in discrete space remains largely unexplored. Although some progress has been made in this area, discrete diffusion samplers do not take full advantage of ideas commonly used for continuous-space sampling. In this paper, we propose to bridge this gap by introducing off-policy training techniques for discrete diffusion samplers. We show that these techniques improve the performance of discrete samplers on both established and new synthetic benchmarks. Next, we generalise discrete diffusion samplers to the task of bridging between two arbitrary distributions, introducing data-to-energy Schrödinger bridge training for the discrete domain for the first time. Lastly, we showcase the application of the proposed diffusion samplers to data-free posterior sampling in the discrete latent spaces of image generative models.
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MEMO: Memory-Augmented Model Context Optimization for Robust Multi-Turn Multi-Agent LLM Games
Yunfei Xie, Kevin Wang, Bobby Cheng, Jianzhu Yao, Zhizhou Sha, Alexander Duffy, Yihan Xi, Hongyuan Mei, Cheston Tan, Chen Wei, Pramod Viswanath, Zhangyang Wang
Multi-turn, multi-agent LLM game evaluations exhibit substantial run-to-run variance. In long-horizon interactions, small deviations compound across turns and are amplified by multi-agent coupling, biasing win rate estimates and destabilizi

ng comparative rankings across repeated tournaments. Prompt choice exacerbates t
his by inducing different effective policies and interaction dynamics. We addres
s both instability and underperformance in games with **MEMO:** (**Me**mory-augm
ented **MO**del context optimization), a self-play framework that treats inferen
ce-time context as an optimizable, agentic object by coupling **retention** and
exploration. Retention distills self-play trajectories into a persistent mem
ory bank of structured insights that act as priors; exploration combines tournam
ent-style prompt evolution under TrueSkill with prioritized replay over decisive
 states. Across five text-based games and a 2,000-game budget per task, MEMO rai
ses mean win rate from 25.1% to 49.5% for GPT-4o-mini and from 20.9% to 45.0% fo
r Qwen-2.5-7B, while sharply reducing run-to-run dispersion. These results sugge
st substantial headroom in multi-agent LLM game performance and robustness can b
e unlocked through memory, with MEMO achieving gains in negotiation and imperfec
t-information settings, while RL remains more effective in perfect-information s
ettings. Project website: https://yunfeixie233.github.io/MEMO/.
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ProtDBench: A Unified Benchmark of Protein Binder Design and Evaluation

Cong Liu, Milong Ren, Jiaqi Guan, Chengyue Gong, Jinyuan Sun, Xinshi Chen, Wenzh
i Xiao

Recent advances in $\textit{de novo}$ protein binder design have enabled increas
ing experimental validation, yet reported $\textit{in silico}$ metrics remain di
fficult to interpret or compare across studies due to non-standardized evaluatio
n protocols. We introduce $\textbf{ProtDBench}$, a standardized and throughput-a
ware evaluation framework for protein binder design. ProtDBench defines unified
benchmark tasks, evaluation protocols, and success criteria, enabling systematic
 analysis of how evaluation design influences observed performance. Using a larg
e wet-lab annotated dataset, we analyze commonly used structure prediction model
s as evaluation verifiers, revealing substantial verifier-dependent bias and lim
ited agreement under identical filtering protocols. We then benchmark representa
tive open-source generative binder design methods across ten diverse protein tar
gets under a fixed evaluation protocol. Beyond per-sequence success rates, ProtD
Bench incorporates throughput-aware metrics based on a fixed 24-hour budget, as
well as cluster-level success criteria to account for structural diversity. Toge
ther, these results expose systematic differences induced by filtering rules, su
ccess definitions, and throughput-aware evaluation between computational efficie
ncy, success rate, and structural diversity. Overall, ProtDBench provides a fair
 and reproducible evaluation pipeline that supports systematic and controlled co
mparison of protein binder design methods under realistic evaluation settings.
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Zooming without Zooming: Region-to-Image Distillation for Fine-Grained Multimoda
l Perception

Lai Wei, Liangbo He, jun lan, Lingzhong Dong, Yutong Cai, Siyuan Li, Huijia Zhu,
 Weiqiang Wang, Linghe Kong, Yue Wang, Zhuosheng Zhang, Weiran Huang

Multimodal Large Language Models (MLLMs) excel at broad visual understanding but
 still struggle with fine-grained perception, where decisive evidence is small a
nd easily overwhelmed by global context. Recent "Thinking-with-Images" methods a
lleviate this by iteratively zooming in and out regions of interest during infer
ence, but incur high latency due to repeated tool calls and visual re-encoding.
To address this, we propose Region-to-Image Distillation, which transforms zoomi
ng from an inference-time tool into a training-time primitive, thereby internali
zing the benefits of agentic zooming into a single forward pass of an MLLM. In p
articular, we first zoom in to micro-cropped regions to let strong teacher model
s generate high-quality VQA data, and then distill this region-grounded supervis
ion back to the full image. After training on such data, the smaller student mod
el improves "single-glance" fine-grained perception without tool use.
To rigorously evaluate this capability, we further present ZoomBench, a hybrid-a
nnotated benchmark spanning six fine-grained perceptual dimensions, together wit
h a dual-view protocol that quantifies the global-regional "zooming gap". Experi
ments show that our models achieve leading performance across multiple fine-grai
ned perception benchmarks (Figure 1), and also improve general multimodal cognit

ion on benchmarks such as visual reasoning and GUI agents. Our code is available at https://github.com/inclusionAI/Zooming-without-Zooming.
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UCPO: Uncertainty-Aware Policy Optimization
Xianzhou Zeng, Jing Huang, Chunmei Xie, Gongrui Nan, Siye Chen, Mengyu Lu, Weiqi Xiong, Qixuan Zhou, Junhao Zhang, Qiang Zhu, Yadong Li, Xingzhong Xu
The key to building trustworthy large language models (LLMs) lies in endowing them with inherent uncertainty expression capabilities, thereby mitigating overconfident errors in high-stakes applications. However, existing RL paradigms such as GRPO often suffer from Advantage Bias due to binary decision spaces and static uncertainty rewards, inducing either excessive conservatism or overconfidence. To tackle this challenge, this paper unveils the root causes of reward hacking and overconfidence in current RL paradigms incorporating uncertainty-based rewards, based on which we propose the **U**n**C**ertainty-Aware **P**olicy **O**ptimization (**UCPO**) framework. UCPO employs Ternary Advantage Decoupling to separate and independently normalize deterministic and uncertain rollouts, thereby eliminating advantage bias. Furthermore, a Dynamic Uncertainty Reward Adjustment mechanism adapts uncertainty weights in real-time according to model evolution and instance difficulty. Experimental results in mathematical reasoning and general tasks demonstrate that UCPO effectively resolves the reward imbalance, significantly improving the reliability of the model beyond their knowledge boundaries. The code is available at https://github.com/xzhouzeng/ucpo.
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Training AI Co-Scientists Using Rubric Rewards
Shashwat Goel, RISHI HAZRA, Dulhan Jayalath, Timon Willi, Parag Jain, William F. Shen, Ilias Leontiadis, Francesco Barbieri, Yoram Bachrach, Jonas Geiping, Chenxi Whitehouse
AI co-scientists are emerging as a useful tool for human researchers, with a crucial ability being proposing a research plan for a given research goal. In this work, we study how to train language models that generate better research plans by leveraging the vast corpus of existing research papers. To collect diverse training data, we automatically extract research goals and goal-specific grading rubrics from papers across domains. We then train models for research plan generation via reinforcement learning, with a frozen copy of the initial policy acting as the grader, using the rubrics to evaluate plans generated by the training policy. To validate this approach, we conduct a human study for machine learning research goals spanning 225 expert hours. The experts prefer plans generated by our finetuned Qwen3-30B-A3B model over the initial model for 70% goals, and over Grok-4-Thinking for 59.6% goals. To assess generality, we also extend our approach to goals from medical papers, and recent arXiv preprints, evaluating with a jury of frontier models. Our finetuning yields 12-22% relative improvements and significant cross-domain generalization, proving effective even in problem settings like medical research where execution feedback is infeasible. Overall, we demonstrate the potential of a scalable training recipe as a step towards improving general AI co-scientists.
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Scalable GANs with Transformers
Sangeek Hyun, MinKyu Lee, Jae-Pil Heo
Scalability has driven recent advances in generative modeling, yet it remains underexplored for adversarial learning. We study the scaling behavior of Generative Adversarial Networks through two design choices: training in a compact Variational Autoencoder latent space and using purely transformer-based generators and discriminators. While this setup is efficient and scales well with compute, naively scaling exposes failure modes; underutilization of early layers in the generator and increasing optimization instability. We address these issues with lightweight intermediate supervision and width-aware learning-rate adjustment. Our Generative Adversarial Transformers (GAT) train reliably from small (S) to extra-large (XL) model sizes, and GAT-XL model achieves state-of-the-art single-step class-conditional generation on ImageNet at 256×256 resolution (FID of 2.18) in 60 epochs, requiring 4x fewer epochs than strong baselines.

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SimpleGPT: Improving GPT via A Simple Normalization Strategy
Marco Chen, Xianbiao Qi, Yelin He, Jiaquan Ye, Rong Xiao
In this work, we revisit Transformer optimization through the lens of second-ord er geometry and establish a direct connection between architectural design, acti vation scale, the Hessian matrix, and the maximum tolerable learning rate. We in troduce a simple normalization strategy, termed SimpleNorm, which stabilizes int ermediate activation scales by construction. Then, by analyzing the Hessian of t he loss with respect to network activations, we theoretically show that SimpleNo rm significantly reduces the spectral norm of the Hessian, thereby permitting la rger stable learning rates. We validate our theoretical findings through extensi ve experiments on large GPT models at parameter scales 1B, 1.4B, 7B and 8B. Empi rically, SimpleGPT, our SimpleNorm-based network, tolerates learning rates 3$\ti mes$-10$\times$ larger than standard convention, consistently demonstrates stron g optimization stability, and achieves substantially better performance than wel l-established baselines. Specifically, when training 7B-scale models for 60K ste ps, SimpleGPT reduces the training loss from 2.290 to 2.208 compared with Llama2 with QKNorm. Our code is available at \url{https://github.com/Ocram7/SimpleGPT} .
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Regime-Adaptive Bayesian Optimization via Dirichlet Process Mixtures of Gaussian Processes
Yan Zhang, Xuefeng Liu, Sipeng Chen, Sascha Ranftl, Chong Liu, Shibo Li
Standard Bayesian Optimization (BO) assumes uniform smoothness across the search space—an assumption violated in multi-regime problems such as molecular conform ation search through distinct energy basins or drug discovery across heterogeneo us molecular scaffolds. A single GP either oversmooths sharp transitions or hall ucinates noise in smooth regions, yielding miscalibrated uncertainty. We propose RAMBO, a Dirichlet Process Mixture of Gaussian Processes that automatically dis covers latent regimes during optimization, each modeled by an independent GP wit h locally-optimized hyperparameters. We derive collapsed Gibbs sampling that ana lytically marginalizes latent functions for efficient inference, and introduce a daptive concentration parameter scheduling for coarse-to-fine regime discovery. Our acquisition functions decompose uncertainty into intra-regime and inter-regi me components. Experiments on synthetic benchmarks and real-world applications—i ncluding molecular conformer optimization, virtual screening for drug discovery, and fusion reactor design—demonstrate consistent improvements over state-of-the -art baselines on multi-regime objectives. Code is available at https://github.c om/AnthonyZhangYan/RAMBO.
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Opportunistic Expert Activation: Batch-Aware Expert Routing for Faster Decode Wi thout Retraining
Costin-Andrei Oncescu, Qingyang Wu, Wai Tong Chung, Tsai-chuan Wu, Bryan Dev Gop al, Junxiong Wang, Tri Dao, Ben Athiwaratkun
An increasing number of LLMs employ Mixture-of-Experts (MoE) architectures where the feed-forward layer is replaced by a pool of experts and each token only act ivates a small subset of them. During autoregressive generation, these models of ten enter a memory-bound regime even for moderate batch sizes because the averag e expert load grows more slowly than in an equivalent dense feedforward layer. C onsequently, MoE latency is governed by the number of activated experts. We intr oduce a framework for $\textbf{dynamically}$ re-routing token-to-expert mapping to lower this number (and thus, the decode latency) while preserving a comparabl e quality. Our best results use a $\textbf{batch-aware routing}$ that works by h aving tokens $\textbf{piggyback}$ experts that have already been loaded into mem ory due to being crucial to other tokens within the same batch. At batch size $1 6$, OEA reduces MoE-layer decode latency by $39\\%$ on Qwen3-30B while preservin g standard-error-adjusted downstream accuracy, and by $15\\%$ on Qwen3-235B with only small overall degradation on the long-generation benchmark suite.
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IDLM: Inverse-distilled Diffusion Language Models

David Li, Nikita Gushchin, Dmitry Abulkhanov, Eric Moulines, Ivan Oseledets, Max
im Panov, Alexander Korotin
Diffusion Language Models (DLMs) have recently achieved strong results in text g
eneration. However, their multi-step sampling leads to slow inference, limiting
practical use. To address this, we extend Inverse Distillation, a technique orig
inally developed to accelerate continuous diffusion models, to the discrete sett
ing. Nonetheless, this extension introduces both theoretical and practical chall
enges. From a theoretical perspective, the inverse distillation objective lacks
uniqueness guarantees, which may lead to suboptimal solutions. From a practical
standpoint, backpropagation in the discrete space is non-trivial and often unsta
ble. To overcome these challenges, we first provide a theoretical result demonst
rating that our inverse formulation admits a unique solution, thereby ensuring v
alid optimization. We then introduce gradient-stable relaxations to support effe
ctive training. As a result, experiments on multiple DLMs show that our method,
Inverse-distilled Diffusion Language Models (IDLM), reduces the number of infe
rence steps by $4 \times$-$64 \times$, while preserving the teacher model’s gene
ration quality. We provide the code, model checkpoints, and video tutorials on
the project page: [https://david-cripto.com/idlm](https://david-cripto.github.io
/idlm-project-page)
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Time-Consistent Robust Multi-Objective Reinforcement Learning via a Bellman–Isaa
cs Weight-Adversary Recursion
Mingxi Hu, Meiling Yu
Most multi-objective reinforcement learning (MORL) methods either condition on a
 fixed preference weight w or consider episodic robustness where an adversary
selects a single w per episode. We study a time-consistent robustness model wi
th reactive preferences: after each transition, an opponent chooses the next wei
ght w_{t+1} after observing s_{t+1}, and incurs a switching cost $\lambda D_
\Phi(w_{t+1}\mid w_t)$ based on a Bregman divergence. This yields a Bellman–Isaa
cs recursion with an inner weight minimization at every backup. We prove the ind
uced operator is a contraction and derive a Bellman-residual certificate that tu
rns approximation error into a uniform bound on robust performance. We develop p
ractical solvers in both tabular and deep settings using Bregman-prox inner upda
tes and a stabilized fixed-point iteration. To evaluate robustness without optim
istic critic reuse, we introduce BR-K, testing policies against K independen
tly trained best-response preference adversaries. Across MO-Gymnasium benchmarks
, our approach consistently improves WRR under strong step-wise opponents over p
reference-conditioned baselines while keeping DRIFT smoothly controllable via $\
lambda$.
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Learning a Zeroth-Order Optimizer for Fine-Tuning LLMs
Kairun Zhang, Haoyu Li, Yanjun Zhao, Yifan Sun, Huan Zhang
Zeroth-order optimizers have recently emerged as an attractive approach for fine
-tuning large language models (LLMs), as they avoid backpropagation and can subs
tantially reduce memory overhead relative to standard first-order training. Howe
ver, existing zeroth-order methods rely on hand-crafted, static sampling strateg
ies that are not adaptable to model-specific structures. To address this, we pro
pose ZO-Finetuner, a learning-based zeroth-order optimizer for LLMs that automat
ically learns efficient perturbation strategies through a compact and memory-eff
icient design. Motivated by the fact that a small set of base LLMs is repeatedly
 fine-tuned across tasks, ZO-Finetuner supports one-time per-model training and
reuse across downstream tasks with minimal overhead. Therefore, learning the opt
imizer once for a given LLM and reusing it across diverse downstream tasks is bo
th feasible and highly desirable. Accordingly, ZO-Finetuner is designed to scale
 learning to learn (L2L) to the foundation-model era by supporting one-time per-
model training with minimal overhead. Experiments on 4 LLMs and 7 datasets show
that ZO-Finetuner outperforms prior zeroth-order baselines in 82.1\% of task-mod
el combinations, thereby demonstrating strong performance and scalability for ef
ficient LLM fine-tuning.
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UniMapping: Unified SLAM Framework for Map-Centric Embodied Perception
Xiaze Zhang, Ziheng Ding, Yuejie Zhang, lifeng chen, Rui Feng
Simultaneous Localization and Mapping (SLAM) is increasingly expected to provide reusable spatial representations for downstream perception. However, existing approaches often struggle with scale-consistency and producing maps that lack the geometric fidelity required for reliable perception. We propose _UniMapping_, a unified SLAM framework that constructs a persistent neural-descriptor map from multimodal observations. We introduce a **Spatial-Aware Deformable Transformer** that injects explicit geometric inductive bias to ensure scale-invariant feature extraction, alongside a **Spatial Fusion** strategy that decouples feature aggregation from temporal sequences. Extensive experiments on both indoor and outdoor benchmarks demonstrate competitive SLAM performance. Notably, our method significantly enhances downstream tasks (mAP +3.1% and mIoU +7.1%) by leveraging accumulated multi-view context.
****

Log-Normal Multiplicative Dynamics for Stable Low-Precision Deep Learning
Keigo Nishida, Eren Mehmet KIRAL, Kenichi Bannai, Mohammad Emtiyaz Khan, Thomas Möllenhoff
We propose Log-Normal Multiplicative Dynamics (LMD), a new algorithm to stabilize low-precision deep learning. LMD is motivated by the robustness of biological neural networks where synaptic spine sizes follow a log-normal distribution and fluctuate via noisy multiplicative dynamics. So far, no algorithm has successfully incorporated all these synaptic properties to train large models. LMD incorporates these properties through a variational Bayesian formulation that uses a log-normal posterior over weights. We evaluate LMD with low-precision forward-pass matrix multiplications, including training a Vision Transformer and GPT-2 from scratch. Our findings demonstrate the utility of biologically inspired approaches for designing energy-efficient algorithms for stable low-precision deep learning.
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Accelerating Q-learning through Efficient Value-Sharing across Actions
Prabhat Nagarajan, Brett Daley, Martha White, Marlos C. Machado
Action-values are foundational to many control algorithms such as Q-learning. Therefore learning action-values efficiently is central to reinforcement learning (RL). However, learning them can be slow, requiring many updates to move values from their initialization, typically near zero, to their true values, which may be far from zero. Moreover, action-value learning algorithms typically update each state–action pair independently, without learning shared value structure across actions within a state. In this paper, we address these inefficiencies by introducing the mean-expansion layer, which accelerates action-value learning by sharing values across actions within a state and by changing the problem from directly learning potentially large action-values to learning a lower-norm representation of them. In deep RL, this layer can be applied as a parameter-free addition to Q-network architectures without altering the underlying algorithm. Applied to deep Q-networks and implicit quantile networks, it improves aggregate performance across 57 Atari games while increasing action gaps and dramatically reducing value overestimation.
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Understanding Performance Collapse in Layer-Pruned Large Language Models via Decision Representation Transitions
boyu shi, Chang Liu, Chuanbao Gao, Xu Yang, Xin Geng
Layer pruning efficiently reduces Large Language Model (LLM) computational costs but often triggers sudden performance collapse. Existing representation-based analyses struggle to explain this mechanism. We propose studying pruning through decision representation. Focusing on multiple-choice tasks, we introduce two metrics, Decision Margin and Option Frequency, and an Iterative Pruning method to analyze layer-wise decision dynamics. Our findings reveal a sharp decision transition that partitions the network into two stages: a Silent Phase, where the model cannot yet predict the correct answer, and a Decisive Phase, where the correct prediction emerges. We also find that pruning the Decisive Phase has minimal im

pact, whereas pruning the Silent Phase triggers immediate performance collapse, highlighting its extreme sensitivity to structural changes. Therefore, we conclu de that pruning-induced collapse stems from disrupting the Silent Phase, which p revents the critical decision transition from occurring.
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EvoEGF-Mol: Evolving Exponential Geodesic Flow for Structure-based Drug Design
Yaowei Jin, Junjie Wang, Cheng Cao, Penglei Wang, Duo An, Qian Shi
Structure-Based Drug Design (SBDD) aims to discover bioactive ligands. Conventio nal approaches construct probability paths separately in Euclidean and probabili stic spaces for continuous atomic coordinates and discrete chemical categories, leading to a mismatch with the underlying statistical manifolds. We address this issue by representing molecules using composite exponential-family distribution s, where coordinates and categories are represented within a unified natural par ameter space to evolve synchronously along exponential geodesics under the Fishe r-Rao metric. To avoid the instantaneous trajectory collapse induced by geodesic s directly targeting Dirac distributions, we propose Evolving Exponential Geodes ic Flow for SBDD (EvoEGF-Mol), which replaces static Dirac targets with dynamica lly concentrating distributions and is trained with a progressive-parameter-refi nement architecture. Our model approaches a reference-level PoseBusters passing rate (93.4%) on CrossDock, demonstrating remarkable geometric precision and inte raction fidelity, while achieving superior performance over baseline methods on real-world MolGenBench tasks for bioactive scaffold recovery. Code is available at https://github.com/BLEACH366/EvoEGF-Mol.
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Towards Resource-Efficient LLMs: End-to-End Energy Accounting of Distillation Pi pelines
Katherine Lambert, Sasha Luccioni
The rise in deployment of large language models has driven a surge in GPU demand and datacenter scaling, raising concerns about electricity use, grid stress, an d the impacts of modern AI workloads. Distillation is often promoted as one of t he most effective paths to obtain cheaper, more efficient models, yet these clai ms rarely account for the full end-to-end energy and resource costs, including c rucial teacher-side workloads such as data generation, logit caching, and evalua tion. We present a comprehensive energy accounting framework that measures the c omplete computational cost of distillation pipelines via detailed stage-wise tra cking of GPU device power consumption. In our experiments, we separate and log e mpirical energy use across distinct phases and systematically measure the energy and emissions of two common distillation methods: the classic logit-based knowl edge distillation and synthetic-data supervised fine-tuning, constructing energy –quality–throughput Pareto frontiers that expose the previously ignored costs. F rom these measurements and analyses, we derive practical design rules for select ing distillation methods and hyperparameters under energy and budget constraints , and release an open-source measurement harness and accounting protocol to prov ide a standardized foundation for comparable, reproducible distillation research , explicitly accountable for complete pipeline energy impact.
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CRAMER: Control via Request-Aware Masking for Editing Recommenders
Zhiyuan Su, Naihe Feng, Luther Qin, Ga Wu
Sequential recommendation models, while powerful, have limited flexibility in re sponding to immediate user requests, making it difficult to adapt their recommen dations to the user's timely interests. Unfortunately, existing user request ada ptation methods often incur high computational overhead due to either 1) retrain ing the entire backbone network or 2) leveraging the inference ability of large language models (a.k.a. prompt engineering), limiting their applicability in lar ge-scale recommendation services. This paper presents **C**ontrol via **R**eques t-**A**ware **M**asking for **E**diting **R**ecommenders (**CRAMER**), a framewo rk that takes users' natural-language requests to immediately change sequential recommendation models' behavior. Specifically, inspired by the model control the ory, CRAMER treats user requests as control signals to modulate frozen backbone parameters through masking, achieving instant adaptation to diverse requests whi

le avoiding costly retraining. Experiments on multiple large-scale benchmark datasets show that CRAMER outperforms four state-of-the-art request-aware baselines across multiple recommendation metrics while achieving minimal overhead. Moreover, the proposed framework exhibits enhanced controllability and cross-domain adaptability, establishing a new paradigm for request-aware sequential recommendation.

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CLINIC : Evaluating Multilingual Trustworthiness in Language Models for Healthcare

Akash Ghosh, Srivarshinee Sridhar, Raghav Kaushik Ravi, Muhsin Muhsin, Sriparna Saha, Chirag Agarwal

Integrating language models (LMs) in healthcare systems holds great promise for improving medical workflows and decision-making. However, a critical barrier to their global adoption is the lack of reliable evaluation of their trustworthiness in multilingual healthcare settings. Existing LMs are predominantly trained in high-resource languages, making them ill-equipped to handle the complexity and diversity of healthcare queries in mid- and low-resource languages, which poses significant challenges for deployment in global healthcare contexts where linguistic diversity is essential. In this work, we present CLINIC, a Comprehensive Multilingual Benchmark to evaluate the trustworthiness of language models in healthcare. CLINIC systematically benchmarks LMs across five key dimensions of trustworthiness: truthfulness, fairness, safety, robustness, and privacy, operationalized through 18 diverse tasks spanning 15 languages and covering a wide range of critical healthcare topics. Our extensive evaluation reveals that LMs struggle with factual correctness, demonstrate bias across demographic and linguistic groups, and remain susceptible to privacy breaches and adversarial attacks. By highlighting these shortcomings, CLINIC lays the foundation for enhancing the global reach and safety of LMs in healthcare across diverse languages.

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Two-Layer Linear Auto-Regressive Models Estimate Latent States

Yahya Sattar, Sunmook Choi, Leo Maynard-Zhang, Yassir Jedra, Maryam Fazel, Sarah Dean

Auto-regressive models have emerged as powerful tools for sequential data, from language to video. Understanding how and why these models learn latent representations remains an open theoretical question. In this work, we demonstrate that when trained by empirical risk minimization on data from partially observed linear dynamical systems, two-layer linear auto-regressive models naturally learn to approximate Kalman filtering. In particular, we show that the learned hidden representation coincides, up to a similarity transformation, with the state estimates produced by the optimal (Kalman) filter, even though the model has no explicit knowledge of the underlying dynamics or state. The result follows from three main insights. First, we establish that the Kalman filter is well approximated by an auto-regressive model with bounded truncation error. Second, we show that despite non-convexity, the two-layer optimization landscape is benign, i.e., all stationary points are either strict saddles or global minima. Finally, as our main contributions, we provide finite-sample guarantees on prediction error, parameter estimation error, and latent state recovery. Numerical simulations support the theoretical results
and demonstrate that the latent representations of auto-regressive models recover state estimates.

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Distribution Transformers: Fast Approximate Bayesian Inference With On-The-Fly Prior Adaptation

George Whittle, Juliusz Ziomek, Jacob Henry Rawling, Michael A Osborne

While Bayesian inference provides a principled framework for reasoning under uncertainty, its widespread adoption is limited by the intractability of exact posterior computation, necessitating the use of approximate inference. However, existing methods are often computationally expensive, or demand costly retraining when priors change, limiting their utility, particularly in sequential inference problems such as real-time sensor fusion. To address these challenges, we introd

uce the Distribution Transformer---a novel architecture that can learn arbitrary distribution-to-distribution mappings. Our method can be trained to map a prior to the corresponding posterior, conditioned on some dataset---thus performing approximate Bayesian inference. Our novel architecture represents a prior distribution as a (universally-approximating) Gaussian Mixture Model (GMM), and transforms it into a GMM representation of the posterior. The components of the GMM attend to each other via self-attention, and to the datapoints via cross-attention. We demonstrate that Distribution Transformers both maintain flexibility to vary the prior, and significantly reduces computation times---from minutes to milliseconds---while achieving expected log-likelihood performance on par with or superior to existing approximate inference methods across tasks such as sequential inference, quantum system parameter inference, and Gaussian Process predictive posterior inference with hyperpriors.

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MC-HNN: Learning Latent Structural Semantics and High-Rank Representations for Hypergraph Neural Networks

Shuyang Fang, Yuqin Huang, Zelong Yang, Yintao Cai, Xiaoping Min

Hypergraph Neural Networks (HNNs) have emerged as powerful tools for modeling complex high-order correlations. Most existing HNNs adhere to a two-stage message passing paradigm, where node feature propagation is mediated by hyperedges. In this paper, we analyze two structural limitations of this paradigm, which we term rank collapse and hyperedge semantic dependency. To address these challenges, we propose the Multi-Channel Hypergraph Neural Network (MC-HNN). We design a multi-channel message passing mechanism to maintain high-rank representations, while simultaneously introducing a latent hyperedge type encoding mechanism to inject an independent degree of freedom into hyperedge representations. Our analysis and experiments suggest that MC-HNN alleviates these bottlenecks and achieves strong empirical performance.

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Agentic Proposing: Enhancing Large language Model Reasoning via Compositional Skill Synthesis

Zhengbo Jiao, Shaobo Wang, Zifan Zhang, Xuan Ren, Wei Wang, Bing Zhao, HU WEI, Linfeng Zhang

Advancing complex reasoning in large language models relies on high-quality, verifiable datasets, yet human annotation remains cost-prohibitive and difficult to scale. Current synthesis paradigms often face a recurring trade-off: maintaining structural validity typically restricts problem complexity, while relaxing constraints to increase difficulty frequently leads to inconsistent or unsolvable instances. To address this, we propose \textbf{Agentic Proposing}, a framework that models problem synthesis as a goal-driven sequential decision process where a specialized agent dynamically selects and composes modular reasoning skills. Through an iterative workflow of internal reflection and tool-use, we develop the \textbf{Agentic-Proposer-4B} using Multi-Granularity Policy Optimization (MGPO) to generate high-precision, verifiable training trajectories across mathematics, coding, and science. Empirical results demonstrate that downstream solvers trained on agent-synthesized data significantly outperform leading baselines and exhibit robust cross-domain generalization. Notably, a 30B solver trained on only 11,000 synthesized trajectories achieves a state-of-the-art 91.6\% accuracy on AIME25, rivaling frontier-scale proprietary models such as GPT-5 and proving that a small volume of high-quality synthetic signals can effectively substitute for massive human-curated datasets.

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LiveFigure: Generating Editable Scientific Illustration with VLM Agents

Chenyang Shao, Jiahe Liu, Fengli Xu, Yong Li

Scientific illustrations are essential for depicting conceptual designs, methodologies, and experimental workflows in research, playing a pivotal role in communicating complex academic insights. However, creating high-quality scientific illustrations remains a labor-intensive task for human scientists. While recent generative image models have advanced prompt-based editing, the synthesis of fully **editable** figures remains a fundamental challenge. Valid editability involves

structured transformations of graphical elements, scales, attributes, and text, rather than simple pixel-level changes. Existing models generate raster outputs that do not support manual correction or layout adjustment, limiting their utility in scientific publishing, where editable vector figures are typically required for submission. To address this challenge, we introduce **LiveFigure**, an agentic framework driven by VLM agents that imitates the multi-step drawing workflow of human researchers. It first plans figure blueprints by drawing inspiration from high-quality references in previous works, then generates executable scripts that produce figures via the PowerPoint interface based on skills and experience, and finally refines the outputs with targeted visual diagnostics, producing fully vectorized, editable figures that meet publication standards. Extensive experiments demonstrate that LiveFigure generates inherently editable figures, achieving 80% publication-readiness in only 17 manual edits, far surpassing the 24% rate of the strongest baseline, NanoBanana. Human preference studies further validate this advantage, with LiveFigure securing a 60% win rate against NanoBanana. Our code is available at https://github.com/tsinghua-fib-lab/LiveFigure.git.

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TGPO: Efficient Policy Optimization through Sequence Anchor and Information Gating

Hang Ding, Dongqi Liu, Qiming Feng, Jian Li, Tong Lei, Jiafu Wu, Shuo Wang, Jiangning Zhang, Chengjie Wang, Yabiao Wang

Reinforcement learning from verifiable rewards (RLVR) has become an important paradigm for enhancing the reasoning capabilities of large language models, while it also involves a persistent tradeoff between optimization stability and learning efficiency. Token-level importance weighting supports fine-grained credit assignment, but it often introduces high variance and unstable parameter updates, whereas sequence-level optimization provides more stable learning dynamics while failing to fully exploit informative local signals. We introduce **T**rust-**G**ated **P**olicy **O**ptimization (TGPO), an efficient policy optimization framework that integrates two complementary mechanisms, namely *sequence anchors* and *information gates*. TGPO aligns token-wise updates with a stable sequence-level reference, which reduces the influence of extreme local likelihood fluctuations on the gradient, and a trust-based information gate adaptively modulates the contribution of token-level signals. By retaining and reweighting gradients from imperfect trajectories rather than excluding them, TGPO improves gradient utilization and sample efficiency while maintaining stable optimization behavior. Empirical results across seven mathematical reasoning datasets and multiple model scales show that TGPO consistently enhances learning efficiency and overall performance in outcome-supervised reinforcement learning settings.

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Smooth Multi-Policy Causal Effect Estimation in Longitudinal Settings

Wenxin Chen, Weishen Pan, Kyra Gan, Fei Wang

Comparative evaluation of *multiple* dynamic treatment policies is essential for healthcare and policy decisions, yet conventional longitudinal causal inference methods estimate each in *isolation*, preventing information sharing across counterfactuals. We demonstrate that this separate estimation paradigm induces a structurally uncontrolled second-order bias, inflating finite-sample variance even after standard debiasing with *longitudinal targeted maximum likelihood estimation* (LTMLE). To address this, we propose a policy-aware reparameterization of *Iterative Conditional Expectation* (ICE) Q-functions that enables joint estimation through shared representations. We implement this approach in the **Policy-Encoded Q Network (PEQ-Net)**, an architecture centered on a shared policy encoder. The encoder is trained using kernel mean embeddings, ensuring that the learned representation space reflects population-level policy dissimilarities. After applying an LTMLE correction step, we prove this design imposes a structural constraint on the second-order remainder, thereby stabilizing finite-sample variance. Experiments on semi-synthetic datasets demonstrate that PEQ-Net consistently outperforms existing ICE-based methods, achieving substantial reductions in root-mean-square error, particularly when evaluating closely related policies.

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FAIR-Calib: Frontier-Aware Instability-Reweighted Calibration for Post-Training Quantization of Diffusion Large Language Models
Haoyu Huang, Linlin Yang, Sheng Xu, Boyu Liu, Guodong Guo, Zhongqian Fu, Hang Zhou, Baochang Zhang
Diffusion Large Language Models (dLLMs) refine tokens iteratively but commit them irreversibly, leading to a "stability lag" where early decisions remain fragile even after being written. We reveal that Post-Training Quantization (PTQ) error easily flips these borderline decisions at the write frontier, which are then permanently locked in and amplified. To address this, we propose Frontier-Aware Instability-Reweighted Calibration (FAIR-Calib), a two-stage PTQ framework for dLLMs. Stage I probes a full-precision teacher to estimate a position prior that combines frontier hits and masked-stage reliability. Stage II performs off-policy, layer-wise calibration by minimizing a reweighted hidden-state MSE, effectively prioritizing the protection of fragile frontier states without requiring expensive end-to-end diffusion rollouts.
We further theoretically justify our weighted objective as a surrogate for output KL divergence. Empirically, FAIR-Calib consistently outperforms state-of-the-art baselines on LLaDA and Dream (W4A4), significantly reducing frontier decision flips and suppressing post-commit mismatches across diverse benchmarks.
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RAT+: Train Dense, Infer Sparse - Recurrence Augmented Attention for Dilated Inference
Xiuying Wei, Caglar Gulcehre
Structured dilated attention has an appealing inference-time efficiency knob: it reduces the FLOPs of attention and the KV cache size by a factor of the dilation size D, while preserving long-range connectivity. While prior work studies it by training each configuration from scratch, directly sparsifying a pretrained attention model into a dilated pattern leads to severe accuracy degradation, preventing flexible reuse across inference scenarios. We introduce RAT+, a dense-pretraining architecture that augments attention with full-sequence recurrence and active recurrence learning. A single RAT+ model is pretrained densely once and can then be flexibly switched at inference time to dilated attention (optionally with local windows) or hybrid layer/head compositions, requiring only a short 1B-token resolution adaptation rather than retraining separate sparse models. At 1.5B parameters trained on 100B tokens, RAT+ closely matches dense accuracy at D = 16, and drops by about 2–3 points at D = 64 on commonsense reasoning and LongBench tasks. We further scale to 2.6B and 7.6B parameters and observe even more promising performance (e.g., a 1-point average accuracy loss with a 64× reduction in attention FLOPs and KV cache size). Code is available at https://github.com/wimh966/rat-plus.
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High-Dimensional Learning Dynamics of Quantized Models with Straight-Through Estimator
Yuma Ichikawa, Shuhei Kashiwamura, Ayaka Sakata
Quantized neural network training optimizes a discrete, non-differentiable objective. The straight-through estimator (STE) enables backpropagation through surrogate gradients and is widely used. While previous studies have primarily focused on the properties of surrogate gradients and their convergence, the influence of quantization hyperparameters, such as bit width and quantization range, on learning dynamics remains largely unexplored. We theoretically show that in the high-dimensional limit, STE dynamics converge to a deterministic ordinary differential equation. This reveals that STE training exhibits a plateau followed by a sharp drop in generalization error, with plateau length depending on the quantization range. A fixed-point analysis quantifies the asymptotic deviation from the unquantized linear model. We also extend analytical techniques for stochastic gradient descent to nonlinear transformations of weights and inputs.
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Faults in Our Formal Benchmarking: Dataset Defects and Evaluation Failures in Lean Theorem Proving

Pawan Sasanka Ammanamanchi, Siddharth Bhat, Stella Biderman
Benchmarks for LLM-assisted theorem proving in Lean are often treated as intrinsically reliable because every solved instance comes with a machine-checked proof.
However, the kernel only checks that a proof establishes a \emph{formal} statement; it does not verify that the statement faithfully encodes the intended informal problem, nor that evaluation harnesses are robust to trivial or adversarial solutions.
We audit five widely used Lean theorem-proving benchmarks and their forks, using corpus-scale static checkers to surface 4,833 findings, including 398 mechanically certified issues such as counterexamples, vacuous theorems, and unsound axioms.
We also document semantic defects such as missing hypotheses, problem simplification, incomplete or incorrect translations, and Lean-specific specification hazards.
Beyond dataset construction, we survey evaluation-time failure modes and show, on corrected subsets, that defects can both inflate and deflate reported prover scores.
We propose a fault taxonomy, a suite of automated checkers and recall-oriented semantic-audit prompts, and release standards to guide the creation of formal math datasets and make evaluation more reproducible and trustworthy.
Our checkers, audit prompts, and corrected dataset snapshots are available at \url{https://github.com/Shashi456/atp-checkers}.
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Temperature Scaling in Discrete Sequence (Language) Models
Hannah Scheufele, Peter Blohm, Vikas K Garg
Temperature scaling is widely used to control generation behavior and improve reasoning performance in discrete sequence models. However, temperature scaling at the sequence level is particularly challenging since the partition function of the model is typically not tractable and naive *token-wise* temperature scaling induces bias in the maximum a posteriori (MAP) estimates. Discrete diffusion settings exacerbate this issue: enhanced flexibility due to multiple potential orderings for generating tokens complicates likelihood computation and incurs high variance.

We propose and investigate fine-tuning objectives for sequence-level temperature scaling, along with metrics to quantify the success of a given temperature scaling procedure. Specifically, our contributions are threefold. First, we introduce our objective across a wide class of sequence models through a novel formalism that unifies autoregressive and diffusion-based language models. Second, we design two metrics that quantify temperature scaling based on likelihood *ratios* and thus obviate access to the partition function. Finally, we introduce fine-tuning objectives that reliably achieve the desired change in model temperature.

Our experiments with language models (GPT2, BD3LM) show that the proposed approach leads to more consistent generation with lower perplexity. Furthermore, we provide empirical evidence that it can enhance the reasoning performance in language models. Code is available at github.com/Aalto-QuML/temp_lm.
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Are We Overconfident in Models and Results for Semi-Supervised 3D Medical Image Segmentation?
Jun Li, Ziwei Qin
Semi-supervised learning has become a dominant paradigm for reducing annotation costs. However, we argue that the current progress is clouded by a twofold overconfidence problem. Algorithmically, mainstream pseudo-labeling frameworks often conflate prediction confidence with uncertainty, leading to severe confirmation bias. Strategically, since multiple benchmark datasets lack dedicated validation sets, some studies use the test set for validation as well, leading to inflated performance estimates. Subsequent methods, compelled to employ the same strateg

y to surpass reported SOTA, trigger an arms race of overfitting. This raises concerns that the impressive numerical gains in the community may reflect overfitting rather than genuine progress. Thus, we propose a tri-space calibrated segmentation framework founded on a principled dual-axis reliability assessment engine. It explicitly decouples confidence from uncertainty and uses this signal to detect and correct confirmation bias across feature, probability, and image spaces in a collaborative manner. Across three benchmark datasets, TCSeg consistently delivers strong performance under existing evaluation protocols. More importantly, we advocate that the community report final-checkpoint results under multiple-run protocols, thereby establishing more rigorous benchmarks with a more realistic perspective. Code will be available: github.com/DirkLiii/TCSeg.

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Functional Equivalence in Attention: A Comprehensive Study with Applications to Linear Mode Connectivity

Viet-Hoang Tran, Khanh Vinh Bui, Van-Hoan Trinh, Tan Lai Ngoc, Tan Minh Nguyen

Neural network parameter spaces are inherently non-injective, as distinct parameter configurations can realize identical functions through functional equivalence. While this symmetry is well understood in classical fully connected and convolutional models, it becomes substantially more intricate in modern attention-based architectures. Existing analyses of multihead attention have largely focused on the vanilla formulation, overlooking positional encodings that fundamentally reshape architectural symmetries. In this work, we provide a formal study of functional equivalence in Transformers with positional encodings. Focusing on the two most widely used variants--sinusoidal and rotary positional encodings (RoPE)--we show that sinusoidal encodings preserve the equivalence structure of vanilla attention, whereas rotary encodings significantly reduce the symmetry group, thereby enhancing expressivity. This offers a principled explanation for the growing prominence of RoPE in practice. We further examine how positional encodings affect linear mode connectivity, and through an alignment algorithm, empirically demonstrate that the presence and variability of connectivity across Transformer settings crucially depend on the positional encoding.

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Language Model Augmented Semi-Supervised Statistical Inference

Xinrui Ruan, Yingfei Wang, Waverly Wei, Jingshen Wang

Semi■supervised statistical inference plays a key role in biomedical research, where labeled data often have higher quality but are limited due to costly clinical annotation. Yet, existing semi■supervised statistical inference methods rely heavily on structured variables and strictly matched covariates between labeled and unlabeled datasets -- limitations ill■suited for the heterogeneity and unstructured nature of real-world biomedical data. Modern biomedical studies increasingly collect unstructured data (clinical notes, patient audio and video recordings), with inconsistent protocols across datasets causing covariate misalignment (for instance, detailed medication histories may be recorded in one study but not another). Recent advances in pre■trained multimodal large language models (LLMs), which excel at handling unstructured data, present an attractive potential solution. To transform this potential into rigorous semi-supervised statistical inference methods for biomedical research, two key challenges must be addressed: (1) How can we reliably integrate LLMs to enhance semi-supervised inference efficiency without compromising statistical validity? (2) How can those efficiency gains persist despite mismatched covariates between labeled and unlabeled datasets? In this paper, we tackle these challenges by systematically *calibrating pseudo-labels* provided LLMs with a novel *prediction-invariance identification* strategy. Our resulting semi■supervised inference framework improves parameter estimation efficiency while maintaining full statistical validity, as demonstrated through our theoretical results and illustrated in a case study for identifying key biomarkers in Alzheimer’s disease detection with speech data.

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TRIP-Bench: A Benchmark for Long-Horizon Interactive Agents in Real-World Scenarios

Yuanzhe Shen, Zisu Huang, Zhengyuan Wang, Muzhao Tian, Zhengkang Guo, Chenyang Z

hang, Shuaiyu Zhou, Zengjie Hu, Dailin Li, Kaimin Wang, Wenhao Liu, Tianlong Li, feng hong, Cao Liu, Ke Zeng
As LLM-based agents are deployed in increasingly complex real-world settings, existing benchmarks underrepresent key challenges such as enforcing global constraints, coordinating multi-tool reasoning, and adapting to evolving user behavior over long, multi-turn interactions. To bridge this gap, we introduce \textbf{TRIP-Bench}, a long-horizon benchmark grounded in realistic travel-planning scenarios. TRIP-Bench leverages real-world data, offers 18 curated tools and 40+ travel requirements, and supports automated evaluation. It includes splits of varying difficulty; the hard split emphasizes long and ambiguous interactions, style shifts, feasibility changes, and iterative version revision. Dialogues span up to 15 user turns, can involve 150+ tool calls, and may exceed 200k tokens of context. Experiments show that even advanced models achieve at most 50\% success on the easy split, with performance dropping below 10\% on hard subsets. We further propose GTPO, an online multi-turn reinforcement learning method with specialized reward normalization and reward differencing. Applied to Qwen2.5-32B-Instruct, GTPO improves constraint satisfaction and interaction robustness, outperforming Gemini-3-Pro in our evaluation. We expect TRIP-Bench to advance practical long-horizon interactive agents, and GTPO to provide an effective online RL recipe for robust long-horizon training.
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Decoupling Variance and Scale-Invariant Updates in Adaptive Gradient Descent for Unified Vector and Matrix Optimization
Zitao Song, Cedar Site Bai, Zhe Zhang, Brian Bullins, David F. Gleich
Adaptive methods like Adam have become the *de facto* standard for large-scale vector and Euclidean optimization due to their coordinate-wise adaptation with a second-order nature. More recently, matrix-based spectral optimizers like Muon (Jordan et al., 2024b) show the power of treating weight matrices as matrices rather than long vectors. Linking these is hard because many natural generalizations are not feasible to implement, and we also cannot simply move the Adam adaptation to the matrix spectrum. To address this, we reformulate the AdaGrad update and decompose it into a variance adaptation term and a scale-invariant term. This decoupling produces **DeVA** (**De**coupled **V**ariance **A**daptation), a framework that bridges between vector-based variance adaptation and matrix spectral optimization, enabling a seamless transition from Adam to adaptive spectral descent. Extensive experiments across language modeling and image classification demonstrate that DeVA consistently outperforms state-of-the-art methods such as Muon and SOAP (Vyas et al., 2024), reducing token usage by around 6.6\%. Theoretically, we show that the variance adaptation term effectively improves the blockwise smoothness, facilitating faster convergence.
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Linearizing Vision Transformer with Test-Time Training
Yining Li, Dongchen Han, Zeyu Liu, Hanyi Wang, Yulin Wang, Gao Huang
While linear-complexity attention mechanisms offer a promising alternative to Softmax attention for overcoming the quadratic bottleneck, training such models from scratch remains prohibitively expensive. Inheriting weights from pretrained Transformers provides an appealing shortcut, yet the fundamental representational gap between Softmax and linear attention prevents effective weight transfer. In this work, we address this conversion challenge from two perspectives: architectural alignment and representational alignment. We identify Test-Time Training (TTT) as a linear-complexity architecture whose two-layer dynamic formulation is structurally aligned with Softmax attention, enabling direct inheritance of pretrained attention weights. To further align representational properties, including key shift-invariance and locality, we introduce key instance normalization and a lightweight locality enhancement module. We validate our approach by linearizing Stable Diffusion 3.5 and introduce SD3.5-T^5 (Transformer To Test Time Training). With only 1 hour of fine-tuning on 4×H20 GPUs, SD3.5-T^5 achieves comparable text-to-image quality to the fine-tuned Softmax model, while accelerating inference by 1.32× and 1.47× at 1K and 2K resolutions. Code is available at this URL: https://github.com/LeapLabTHU/Transformer-to-TTT.

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μpscaling small models: Principled warm starts and hyperparameter transfer
Yuxin Ma, Nan Chen, Mateo Diaz Diaz, Soufiane Hayou, Dmitriy Kunisky, Soledad Villar
Modern large-scale neural networks are often trained and released in multiple sizes to accommodate diverse inference budgets. To improve efficiency, recent work has explored *model upscaling*: initializing larger models from trained smaller ones to accelerate convergence. However, this method can be sensitive to hyperparameters that need to be tuned at the target upscaled model size, which is prohibitively costly to do directly. It remains unclear whether tuning hyperparameters on smaller models and extrapolating via scaling laws is sound in this setting. We address this with principled approaches to width-based upscaling and efficient hyperparameter tuning in this setting. Motivated by μP and any-dimensional architectures, we introduce a general upscaling method that, like Net2Net, copies and perturbs weights, but uses theoretically grounded, width-dependent scalings for the perturbation noise and optimizer hyperparameters. First, we prove that under zero perturbation, the upscaled model is functionally equivalent to the base model throughout training. Second, we extend the μP theory to enable infinite-width limit analysis and establish hyperparameter transfer for upscaled models, greatly reducing the tuning cost. We empirically demonstrate that this method is effective on realistic datasets and architectures.
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How much can language models memorize?
John Xavier Morris, Chawin Sitawarin, Chuan Guo, Narine Kokhlikyan, G. Edward Suh, Alexander M Rush, Kamalika Chaudhuri, Saeed Mahloujifar
We propose a new method for estimating how much a model knows about a datapoint and use it to measure the capacity of modern language models. Prior studies of language model memorization have struggled to disentangle memorization from generalization. We formally separate memorization into two components: unintended memorization, the information a model contains about a specific dataset, and generalization, the information a model contains about the true data-generation process. When we completely eliminate generalization, we can compute the total memorization, which provides an estimate of model capacity: our measurements estimate that GPT-style models have a capacity of approximately 3.6 bits per parameter. We train language models on datasets of increasing size and observe that models memorize until their capacity fills, at which point unintended memorization decreases as models begin to generalize. We train hundreds of transformer language models ranging from 500K to 1.5B parameters and produce a series of scaling laws relating model capacity and data size to membership inference.
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Decompose, Structure, and Repair: A Neuro-Symbolic Framework for Autoformalization via Operator Trees
Xiaoyang Liu, Zineng Dong, Yifan Bai, Yantao Li, Yuntian Liu, Tao Luo
Statement autoformalization acts as a critical bridge between human mathematics and formal mathematics by translating natural language problems into formal language. While prior works have focused on data synthesis and diverse training paradigms to optimize end-to-end Large Language Models (LLMs), they typically treat formal code as flat sequences, neglecting the hierarchical logic inherent in mathematical statements. In this work, we introduce Decompose, Structure, and Repair (DSR), a neuro-symbolic framework that restructures autoformalization into a modular pipeline. DSR decomposes statements into logical components and maps them to structured operator trees, leveraging this topological blueprint to precisely localize and repair errors via sub-tree refinement. Furthermore, we introduce PRIME, a benchmark of 156 undergraduate and graduate-level theorems selected from canonical textbooks and expertly annotated in Lean 4. Experimental results demonstrate that DSR establishes a new state-of-the-art, consistently outperforming baselines under equivalent computational budgets. The datasets, model, and code are available at https://github.com/XiaoyangLiu-sjtu/DSR.
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Hugging Carbon: Quantifying the Training Carbon Emissions of AI Models at Scale

Xinlei Wang, Ruibo Ming, Jing Qiu, Junhua Zhao, Jinjin Gu
The scaling-law era has transformed artificial intelligence (AI) from research i nto a global industry, but its rapid growth also raises concerns over energy usa ge, carbon emissions, and environmental sustainability. Unlike traditional secto rs, the AI industry still lacks systematic carbon accounting methods that suppor t large-scale estimates without reproducing the original training process. This leaves open questions about how large the problem is today and how large it migh t be in the near future. Given its central role in hosting open-source AI models , the Hugging Face (HF) platform provides a large-scale and publicly accessible corpus for carbon accounting. We estimate aggregate training emissions of HF ope n-source models using available emissions, energy, compute, and model metadata. To address uneven disclosure quality, we introduce a tiered approach to handle i ncomplete metadata, supported by empirical regressions that assess estimation re liability. We further introduce AI training carbon intensity (ATCI, emissions pe r compute), a metric to assess the sustainability efficiency of model training. Our results show that training the most popular open-source models (with over 5, 000 downloads) has already resulted in approximately 6.0×10^4 metric tons of car bon emissions. Overall, this paper provides a scalable, empirically grounded fra mework for estimating training emissions from incomplete disclosures and informi ng future carbon reporting standards in the AI industry. Data and code are avail able at https://github.com/insait-institute/HuggingCarbon.
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Monitoring Monitorability
Melody Y. Guan, Miles Wang, Micah Carroll, Zehao Dou, Annie Y. Wei, Marcus Willi ams, Benjamin Arnav, Joost Huizinga, Ian D Kivlichan, Amelia Glaese, Jakub Pacho cki, Bowen Baker
Safe deployment of increasingly capable AI agents may require visibility into ho w they make decisions. Chain-of-thought (CoT) monitoring can detect misbehavior in today’s reasoning models, but this “monitorability” may be fragile under diff erent training procedures, data sources, or continued system scaling. We propose three evaluation archetypes (intervention, process, and outcome-property), a ne w monitorability metric, and a broad evaluation suite. We show CoT monitoring ou tperforms action-only monitoring in practical settings, and that frontier models are generally—but not perfectly—monitorable. We study scaling trends with pre-t raining model size and inference-time compute, finding longer CoTs are typically more monitorable. We find that, for a fixed capability level, using a smaller m odel at higher reasoning effort can yield higher monitorability, at greater infe rence compute cost. We further find that increasing a weak monitor’s test-time c ompute when monitoring a strong agent improves monitorability, and giving the mo nitor access to the CoT both boosts monitorability and steepens the compute–to-m onitorability scaling trend. Finally, we show monitorability can be improved by asking follow-up questions and giving the follow-up CoT to the monitor.
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Plasticity Activation via Polar Operator: A Plug-in Method for Balancing Stabili ty and Plasticity
Guodong Zheng, Enneng Yang, Xiaoyan Wang, Yihan Chen, Feihong He, Quan Zheng, Pe ng Wang, Li Shen
Continual learning (CL) seeks models that acquire new knowledge while avoiding c atastrophic forgetting. However, many methods that mitigate forgetting constrain parameter updates and thereby reduce model plasticity. We revisit the singular value spectrum of gradients in representative CL methods and show that they comm only exhibit singular value collapse, where only a small subset of gradient dire ctions drive parameter updates. Motivated by this observation, we propose **P**l asticity **A**ctivation via **P**olar **O**perator (PAPO), a plug-in that preser ves the dominant directions that mitigate forgetting while activating previously suppressed directions to enhance plasticity. Concretely, PAPO modifies the grad ient $\mathbf{G}$ as
$\mathbf{G}\leftarrow \mathbf{G}+\lambda \cdot \operatorname{polar}(\mathbf{G})$,
which uniformly increases near-zero singular values without changing the singula

r vectors. To avoid the cost of explicit singular value decomposition, we approx
imate the polar factor using the iteration-dependent Polar Express scheme, which
 relies only on matrix multiplications and additions. In our empirical evaluatio
n on both vision and language benchmarks, incorporating PAPO yields consistent i
mprovements. In particular, on MiniImageNet, integrating PAPO into ER, MAS, GPM
and TRGP produces substantial accuracy gains of 9.01%, 4.76%, 8.90% and
 9.19%, respectively.
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Generalized Schrödinger Bridge on Graphs
Panagiotis Theodoropoulos, Juno Nam, Evangelos Theodorou, Jaemoo Choi
Transportation on graphs is a fundamental challenge across many domains, where d
ecisions must respect topological and operational constraints.
Despite the need for actionable policies, existing graph-transport methods lack
this expressivity. They rely on restrictive assumptions, fail to generalize acro
ss sparse topologies, and scale poorly with graph size and time horizon.
To address these issues, we introduce Generalized Schrödinger Bridge on Graphs (
GSBoG), a novel scalable data-driven framework for learning executable controlle
d continuous-time Markov chain (CTMC) policies on arbitrary graphs under state c
ost augmented dynamics.
Notably, GSBoG learns trajectory-level policies, avoiding dense global solvers a
nd thereby enhancing scalability.
This is achieved via a likelihood optimization approach, satisfying the endpoint
 marginals, while simultaneously optimizing intermediate behavior under state-de
pendent running costs.
Extensive experimentation on challenging real-world graph topologies shows that
GSBoG reliably learns accurate, topology-respecting policies while optimizing ap
plication-specific intermediate state costs, highlighting its broad applicabilit
y and paving new avenues for cost-aware dynamical transport on general graphs.
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FedVeer: Self-Adaptive Skew Estimation for Robust Federated Learning
Yun Xin, Bangqi Pan, Jianfeng Lu, Shuqin Cao, Gang Li, Guanghui Wen
Federated Learning (FL) enables collaborative model training across decentralize
d clients, but its performance often degrades under non-IID data distributions,
particularly in the presence of data skew. Existing approaches mitigate this iss
ue by estimating client skew via kernel density estimation over neighboring mode
l updates, which preserves privacy and reduces communication costs. However, suc
h approaches suffer from two fundamental limitations: bias toward skewed majorit
y clients due to fixed neighborhood structures, and vulnerability to noise-induc
ed perturbation in kernel space. To address these challenges, we propose FedVeer
, a skew-aware FL framework based on self-adaptive kernel density estimation wit
h k-free neighborhoods. FedVeer dynamically determines the neighborhood size via
 max-margin learning to mitigate majority-client bias, and further incorporates
Kalman filtering to stabilize margin estimation under noisy updates, with a high
-probability theoretical guarantee on margin deviation. Extensive experiments on
 realworld datasets demonstrate that FedVeer consistently outperforms four basel
ines, achieving up to 6.36% accuracy improvement and reducing noiseinduced degra
dation by up to 6.01%.
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Contextualized Privacy Defense for LLM Agents
Yule Wen, Yanzhe Zhang, Jianxun Lian, Xiaoyuan Yi, Xing Xie, Diyi Yang
LLM agents increasingly act on users' personal information, yet existing privacy
 defenses remain limited in both design and adaptability.
Most prior approaches rely on static or passive defenses, such as prompting and
guarding. These paradigms are insufficient for supporting contextual, proactive
privacy decisions in multi-step agent execution. We propose *Contextualized Defe
nse Instructing (CDI)*, a new privacy defense paradigm in which an instructor mo
del generates step-specific, context-aware privacy guidance during execution, pr
oactively shaping actions rather than merely constraining or vetoing them. Cruci
ally, CDI is paired with an experience-driven optimization framework that trains
 the instructor via reinforcement learning (RL), where we convert failure trajec

tories with privacy violations into learning environments. We formalize baseline defenses and CDI as distinct intervention points in a canonical agent loop, and compare their privacy–helpfulness trade-offs within a unified simulation framework. Empirically, CDI consistently achieves a better balance between privacy preservation (94.2\%) and helpfulness (80.6\%) than baselines, with superior robustness to adversarial conditions and generalization.

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Fast Spectrally Sparse Signal Reconstruction via Jacobi-Preconditioned Gradient Descent

Jian-Feng Cai, Xueyang Quan, Yang Wang, Jiaxi Ying

Spectrally sparse signal reconstruction arises in a wide range of applications and can be formulated as a low-rank Hankel matrix completion problem. We develop a Jacobi-preconditioned gradient descent method that preserves the low per-iteration complexity of first-order algorithms while achieving linear convergence at a rate independent of the condition number. By introducing a generator that maps factor-based iterates to matrix space, we establish equivalence with manifold-based methods, enabling direct convergence analysis while avoiding the need to define distances under complex-symmetric factorization ambiguity. Extensive experiments demonstrate that the proposed algorithm outperforms state-of-the-art methods in both iteration count and computational time across a broad range of problem settings.

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Evaluating Bivariate Causal Statements Based on Mutual Compatibility

Erik Jahn, Dominik Janzing

For many real-world systems, causal ground truth is difficult to obtain, making claims about causal effects hard to assess. We develop methods for evaluating collections of bivariate causal statements, one for each pair of variables in a fixed system. In the setting of acyclic linear statements, any such collection can be extended to a unique multivariate causal model, but we argue that this induced model is implausible if it imposes substantial additional confounding to explain observed correlations. We introduce a compatibility score that quantifies this notion of plausibility, notably without relying on the faithfulness assumption. Additionally, we define an incompatibility score for purely graphical bivariate causal statements, based on global consistency constraints that are derived from acyclicity and faithfulness assumptions. We give theoretical and empirical evidence that both scores can successfully distinguish correct from incorrect causal statements in generic settings. Moreover, we demonstrate the practical applicability of our methods by analyzing causal claims made by large language models. Our work aims to provide a foundation for assessing the reliability of causal information derived from human experts or artificial intelligence in settings where alternative forms of validation are unavailable.

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Beyond Pixels: Mining Compressed Domain Artifacts for Efficient AI-Generated Video Detection

Anran Zhu, Zhengli Shi, Chende Zheng, Chenhao Lin, Zhengyu Zhao, Le Yang, Chong Zhang, Shuai Liu, Chao Shen

With the rapid advancement of high-fidelity video generation models, robust AI-generated video (AIGV) detection has become increasingly needed. While most AIGV detection methods operate in the decoded pixel domain, we observe that detection in the pixel domain inevitably entangles task-irrelevant semantic information, leading to substantial semantic redundancy and extensive redundant computation, while overlooking free-to-use signals in compressed bitstreams. In particular, motion vectors and residuals directly encode temporal and spatial generative artifacts but remain largely underexplored. To address these issues, we propose a unified framework for **S**patio-**T**emporal **RE**sidual and **A**rtifact **M**ining, namely **STREAM**, which enables AIGV detection directly from compressed bitstreams. **STREAM** leverages I-frames, motion vectors, and residual errors to capture spatiotemporal artifacts that are typically smoothed out by decompression filters. In particular, we design a lightweight network with a motion-guided alignment module and a gated fusion mechanism, enabling adaptive fusion of spati

al artifacts and nonlinear temporal dynamics. Extensive experimental results dem
onstrate that **STREAM** achieves SOTA performance with an mAP of 0.965, with 2.
5× faster inference than previous SOTA baselines.
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Generalized Boundary FDR Control under Arbitrary Dependence: An Approach on Clos
ure Principle
Yifan Zhang, Wentao Zhang, Changliang Zou, Haojie Ren
False discovery rate (FDR) is a cornerstone of modern multiple testing. However,
 it often fails to guarantee the reliability of ``marginal" discoveries that lie
 at the boundary of the rejection set, which are often crucial in high-precision
 applications. While recent works (Soloff et al., 2024; Xiang et al., 2025) intr
oduced the boundary false discovery rate (bFDR) to control the error probability
 at the marginal discovery, their method relies on restrictive assumptions such
as independence or specific prior distributions. In this paper, we first propose
 k-bFDR, a novel generalization that controls the error probability of the k
 least significant discoveries. We then provide a systematic investigation into
the theoretical relationship between k-bFDR and existing error metrics. Furthe
rmore, building upon the closure principle, we develop Domino, a unified framewo
rk that guarantees k-bFDR control under arbitrary dependence, applicable for b
oth p-values and e-values. We prove the theoretical validity of the proposed Dom
ino algorithm and demonstrate through extensive numerical experiments that it co
nsistently achieves rigorous k-bFDR control while identifying trustworthy marg
inal discoveries. Analyses of real data reveal that k-bFDR control yields high
er-quality rejection sets with greater practical significance.
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On Stable Long-Form Generation: Benchmarking and Mitigating Length Volatility
Zhitao He, Haolin Yang, Rui Min, Zeyu Qin, Yi R. Fung
Large Language Models (LLMs) excel at long-context understanding but exhibit sig
nificant limitations in long-form generation. Existing studies primarily focus o
n single-generation quality, generally overlooking the volatility of the output
(i.e., the inconsistency in length and content across multiple generations). Thi
s volatility not only leads to significant computational costs but also severely
 impacts the models' reliable application. To address this gap, our work unfolds
 in three stages: benchmarking, probing, and mitigation. We first propose the VO
latility in Long-form Text Benchmark (VOLTBench), a novel heterogeneous-task ben
chmark designed to systematically quantify the length volatility of long-form ge
neration. Subsequently, by analyzing attention traces, we conduct an in-depth pr
obe to identify several common internal patterns that cause this volatility. Fin
ally, to mitigate long-form output volatility, we propose SELB (Structural Enfor
cement via Logits Boosting), a lightweight decoding-stage optimization strategy,
 designed to significantly enhance both the length accuracy and stability of lon
g-form generation without additional training. Extensive experiments on VOLTBenc
h provide the first systematic confirmation of severe long-form output instabili
ty in mainstream models and validate that our proposed method successfully impro
ves the mean output length of the base model by 148% and reduces the length vola
tility by 69%, while maintaining high generation quality.
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Bayesian model selection and misspecification testing in imaging inverse problem
s only from noisy and partial measurements
Tom Sprunck, Marcelo Pereyra, Tobías I. Liaudat
Modern imaging techniques heavily rely on Bayesian statistical models to address
 difficult image reconstruction and restoration tasks. This paper addresses the
objective evaluation of such models in settings where ground truth is unavailabl
e, with a focus on model selection and misspecification diagnosis. Existing unsu
pervised model evaluation methods are often unsuitable for computational imaging
 due to their high computational cost and incompatibility with modern image prio
rs defined implicitly via machine learning models. We herein propose a general m
ethodology for unsupervised model selection and misspecification detection in Ba
yesian imaging sciences, based on a novel combination of Bayesian cross-validati
on and data fission, a randomized measurement splitting technique. The approach

is compatible with any Bayesian imaging sampler, including diffusion and plug-and-play samplers. We demonstrate the methodology through experiments involving various scoring rules and types of model misspecification, where we achieve excellent selection and detection accuracy with a low computational cost.
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Diversity-Driven Offline Multi-Objective Optimization via Nested Pareto Set Learning
Yiyi Zhu, Yaolin Wen, Xiang Xia, Xin An, Hanyi Si, Xiang Shu, Yangde Fu, Liang Dou, Hong Qian
Multi-objective optimization (MOO) has emerged as a powerful approach to solving complex optimization problems involving multiple objectives. In many practical scenarios, function evaluations are unavailable or prohibitively expensive, necessitating optimization solely based on a fixed offline dataset. In this setting, known as offline MOO, the goal is to find out the Pareto set without access to the true objective functions. This setting suffers from the out-of-distribution (OOD) issue, where the surrogate model is not accurate for unseen designs. Due to the OOD issue, surrogate errors may cause the optimizer to select solutions that do not lie on the true Pareto front and are biased toward its extremes. To address this, this paper proposes Diversity-driven Offline Multi-Objective Optimization (DOMOO), which aims to find out a diverse and high-quality set of solutions. First, DOMOO incorporates an accumulative risk control module that estimates the potential risk of candidate solutions and alleviates the OOD issue between the training data and the generated solutions. In addition, a nested Pareto set learning (PSL) strategy is proposed to jointly learn preference and PSL parameters, then optimize them, enabling adaptation to diverse Pareto front geometries. To further enhance solution quality, we design a diversity-driven selection strategy that extracts a representative and well-distributed set of final solutions. To achieve this diversity-driven selection strategy, we propose $\text{IGD}_\text{offline}$, a tailored indicator for the offline setting that considers both diversity and convergence, and avoids the bias of hypervolume indicator. Extensive experiments on synthetic and real-world benchmarks show that DOMOO achieves the best average rank across tasks in both convergence and diversity among the compared methods.
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Revisiting the Volume Hypothesis
Ari Pakman, Lior Kreimer, Yakir Berchenko
Modern deep neural networks often contain far more parameters than needed to fit their training data, yet they achieve impressive generalization. A common explanation for this success is the implicit bias of stochastic gradient descent (SGD). An alternative volume hypothesis posits that, within low training-loss regions, loss-landscape basins leading to strong generalization occupy much larger regions of weight space than basins that generalize poorly, and therefore SGD is simply more likely to land in the former. Recent experimental explorations of this idea present seemingly contradictory results. While in one set of experiments randomly sampling the network weights until achieving zero training error yielded poor generalization, molecular dynamics density estimates supported the volume hypothesis. We observe that these experiments were performed at different dataset size regimes, and explore an intermediate regime using the Replica Exchange Wang–Landau algorithm to estimate the joint density of states over training and test accuracies in binary networks. Across several architectures and datasets, we show that the generalization advantage of gradient learning over random sampling training generally diminishes as the training data size grows, suggesting a resolution of the paradox.
****
SHAP-Guided Kernel Actor-Critic for Explainable Reinforcement Learning
Na Li, Hangguan Shan, Wei Ni, Wenjie Zhang, Xinyu Li
Actor-critic (AC) methods are a cornerstone of reinforcement learning (RL) but offer limited interpretability. Current explainable RL methods seldom use \emph{state attributions} to assist training. Rather, they treat all state features equally, thereby neglecting the heterogeneous impacts of individual state dimension

s on the reward. We propose *RKHS-SHAP-based Advanced Actor-Critic (RSA2C)*, an attribution-aware, kernelized, two-timescale AC algorithm, including Actor, Value Critic, and Advantage Critic. The Actor is instantiated in a vector-valued reproducing kernel Hilbert space (RKHS) with a Mahalanobis-weighted operator-valued kernel, while the Value Critic and Advantage Critic reside in scalar RKHSs. These RKHS-enhanced components use sparsified dictionaries: the Value Critic maintains its own dictionary, while the Actor and Advantage Critic share one. State attributions, computed from the Value Critic via RKHS-SHAP (kernel mean embedding for on-manifold and conditional mean embedding for off-manifold expectations), are converted into Mahalanobis-gated weights that modulate Actor gradients and Advantage Critic targets. We derive a global, non-asymptotic convergence bound under *state perturbations*, showing stability through the perturbation-error term and efficiency through the convergence-error term. Empirical results on three continuous-control environments show that RSA2C achieves efficiency, stability, and interpretability. Our code is available at https://github.com/Na-Li66/RSA2C.

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Delegation and Verification under AI

Lingxiao Huang, Wenyang Xiao, Nisheeth K. Vishnoi

As AI systems enter institutional workflows, workers must decide whether to delegate task execution to AI and how much effort to invest in verifying AI outputs, while institutions evaluate workers using outcome-based standards that may misalign with workers' private costs. We model delegation and verification as the solution to a rational worker's optimization problem, and define worker quality by evaluating an institution-centered utility (distinct from the worker's objective) at the resulting optimal action. We formally characterize optimal worker workflows and show that AI induces *phase transitions*, where arbitrarily small differences in verification ability lead to sharply different behaviors. As a result, AI can amplify workers with strong verification reliability while degrading institutional worker quality for others who rationally over-delegate and reduce oversight, even when baseline task success improves and no behavioral biases are present. These results identify a structural mechanism by which AI reshapes institutional worker quality and amplifies quality disparities between workers with different verification reliability.

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Beyond Confidence: Adaptive and Coherent Decoding for Diffusion Language Models

Kecheng Chen, Ziru Liu, Xijia Tao, Hui Liu, Xinyu Fu, Suiyun Zhang, Dandan Tu, Lingpeng Kong, Rui Liu, Haoliang Li

Diffusion Language Models (DLMs) have recently achieved significant success due to their any-order generation capabilities. However, existing inference methods typically rely on local, immediate-step metrics—such as confidence or entropy—which inherently lack a more reliable perspective, leading to sub-optimal generation quality. To address this, we propose **C**oherent **C**ontextual **D**ecoding (**CCD**), a novel inference framework built upon two core innovations. First, CCD bypasses the potential bias of the single context to leverage historical contexts for approximating the marginal distribution of token prediction, leading to better sequence coherence and the early rejection of sub-optimal paths. More importantly, we demonstrate that this mechanism is theoretically equivalent to modeling the consistency of historical steps via the conditional mutual information between contexts and token predictions. Finally, CCD achieves significantly milder performance degradation under highly parallel decoding scenarios compared to baselines. Empirically, our method achieves a simultaneous enhancement in both inference speed and performance across diverse benchmarks on Dream and LLaDA.

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Conservation Laws for Modern Neural Architectures

Viet-Hoang Tran, Vinh Khanh Bui, Tan Lai Ngoc, Nam Nguyen, Tuan Quang Dam, Tan Minh Nguyen

Understanding gradient descent dynamics is key to explaining the success of over-parameterized models, where implicit bias manifests through conservation laws in gradient flow. While such laws are well understood for linear and ReLU networks, they remain largely unexplored for modern architectures. This work develops a

unified framework to characterize conservation laws for contemporary models, including feedforward networks with GELU, SiLU, and SwiGLU activations, multihead attention with sinusoidal and rotary positional encodings, and Mixture-of-Experts architectures under diverse gating designs. Our theoretical findings are supported by experiments that validate the predicted invariants.
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Test-Time Debiasing with Probabilistic Prompts via Wasserstein Distance in Vision-Language Models
Chengye Wang, Yuyuan Li, XiaoHua Feng, Xiaolin Zheng, Chaochao Chen
Vision-Language Models (VLMs) inherit social biases from large-scale pretraining data, and these biases can amplify in downstream tasks, leading to systematic performance disparities across sensitive groups. Due to the high training cost and the risk of catastrophic forgetting, recent research has focused more on lightweight \emph{test-time} debiasing, aiming to obtain an ideal fair embedding for each query. However, such point-based corrections are often unstable and become notably weaker in multi-class settings, where group structure cannot be adequately captured by a single point. Therefore, we propose W4D, a distributional debiasing framework that reframes fairness as aligning query embedding distributions to group reference distributions under the Wasserstein distance, which provides a geometry-aware notion of discrepancy beyond mean shifts. To make this alignment practical at test-time, W4D introduces probabilistic prompts that induce controlled distributional perturbations and optimizes a Wasserstein-based objective to reduce cross-group disparity while preserving task-relevant semantics. This distributional perspective improves robustness in multi-class debiasing and yields a stronger fairness--utility trade-off across diverse VLM downstream evaluations. Our code is available at https://github.com/QDRhhhh/W4D.
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Mixing Configurations for Downstream Prediction
Juntang Wang, Hao Wu, Yihan Wang, Dongmian Zou, Shixin Xu
Clustering-based features are widely used in machine learning, but most methods must choose a resolution---a choice that is global, fixed, and ad hoc.
Recent work shows that varying the resolution parameter produces only a finite set of structurally stable partitions, known as configurations.
Based on this, we introduce Configuration-Mixed Prediction (CMP), a setting where models learn to adaptively weight these configurations per sample for downstream prediction.
We propose MixConfig, a plug-and-play feature augmentation module that extracts configurations from any embedding and learns energy-aware mixing weights via a novel selector that jointly reasons about sample context, cluster assignments, and stability statistics.
Experiments across tabular, molecular, vision, and text domains demonstrate consistent improvements over single-resolution and static baselines across diverse predictor architectures, with gains particularly pronounced in low-data regimes.
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Crisp: A Spectral-Based Interaction Strategy for Multivariate Time Series Forecasting
Binwu Wang, Gaoyun Lin, Jiaming Ma, Qihe Huang, Zhengyang Zhou, Xu Wang, Pengkun Wang, Yang Wang
Multivariate time series (MTS) forecasting critically depends on modeling inter-variable dependencies, yet existing paradigms face a trade-off: channel-isolation strategies can suffer from information fragmentation in strongly coupled systems, whereas channel-interaction methods often introduce spurious interactions among irrelevant variables. To address this challenge, we propose Coherent Resonance Interaction with Spectral Priors (Crisp). Crisp adopts the principle that effective information exchange should occur only between variables with compatible oscillatory patterns. Concretely, we derive spectral priors in the frequency domain to construct dynamic resonance topologies. With a differentiable, adaptive, and strictly sparse blocking mechanism, Crisp forces attention weights for spectrally inconsistent neighbors to be exactly zero. In addition, we introduce a spectral-gated feature filtering module to refine variable representations using in

trinsic spectral characteristics. Extensive experiments demonstrate that Crisp significantly outperforms 20+ baselines. Our code is available at Anonymous GitHub.

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AsyncSpade: Efficient Test-Time Scaling with Asynchronous Sparse Decoding

Shuqing Luo, Yilin Guan, Pingzhi Li, Hanrui Wang, Tianlong Chen

Test-time scaling (TTS) can boost LLM reasoning through long chain-of-thought (CoT), but the linear KV-cache growth amplifies the memory-bound bottleneck of LLM decoding. Query-aware sparse decoding methods can achieve state-of-the-art performance under constrained FLOP budget, but are mainly constrained by both sequential-dependent page filtering and coarse-grained token selection, hampering the serving efficiency and model performance on TTS tasks under high concurrency and long CoT scenarios, where token selection can even occupy higher runtime than the forward pipeline itself. In this paper, we first find that the query state of the current decoding token can be approximated in a unified manner from a short sliding window of recent queries, enabling training-free query-aware sparsity without sequential dependency in the decoding loop. Based on the findings, we propose $\texttt{\textbf{AsyncSpade}}$, an asynchronous framework for efficient TTS, built on two core components: $\textbf{(1) a novel light-weight temporal-regressive module}$ that predicts the next-token query state, and $\textbf{(2) an asynchronous disaggregated framework}$ that decouples the KV cache selection from the auto-regressive decoding loop, overlapping the token-level KV selection with the forward inference computation through asynchronism, thereby eliminating the sequential dependency without sacrificing model performance. We validate the effectiveness of $\texttt{AsyncSpade}$ on common LLM serving setups with an A100 node, where $\texttt{AsyncSpade}$ can fully overlap KV-cache operations with the inference pipeline within a certain workload range, $\textbf{achieving theoretical optimal time-per-output-token~(TPOT)}$. Specifically, $\texttt{AsyncSpade}$ delivers over 20% reduction on TPOT compared to SoTA baseline ($\textit{i.e.}$ Quest) and at least 50% TPOT reduction compared to full attention on Qwen3-8B and Qwen3-32B models, while matching or surpassing their accuracy on various TTS benchmarks (AIME-24/25, GPQA-Diamond, MATH-500). Our code is available through https://github.com/UNITES-Lab/AsyncSpade.

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Reconstruction Outcomes Look Similar but Processes Differ: Improving Context Consistency and Coverage in Graph Masked Auto-Encoder

Geng Tang, Keyu Liu, Xibei Yang, Yuhua Qian

Graph Masked Auto-Encoder (GMAE) has emerged as a prevalent self-supervised paradigm, showing superior performance in graph learning. However, existing methods mainly emphasize reconstruction outcomes and give limited specification to how neighborhood context is used for reconstruction. Our experimental investigation shows that, even when reconstruction outcomes are similar, the ways of using neighborhood context differ substantially, resulting in performance shifts. To address this issue, we propose Consistency- and Coverage-aware Graph Masked Auto-Encoder (C2-GMAE), which encourages more consistent use of neighborhood context and promotes broader training coverage in the graph. Specifically, C2-GMAE leverages positional encoding as an observable structural reference, introduces density-partitioned masking to improve coverage across regions, and amplifies heterophilic edges to reduce the attenuation of discriminative relational information during reconstruction. Extensive experiments on multiple benchmarks demonstrate that C2-GMAE improves downstream performance against GMAE baselines.

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Towards On-Policy SFT: Distribution Discriminant Theory and its Applications in LLM Training

Miaosen Zhang, Yishan Liu, Shuxia Lin, Qi Dai, Chong Luo, Baining Guo, Weihao Jiang, Peng Hou, Anxiang Zeng, Xu Yang, Xin Geng

Supervised fine-tuning (SFT) is computationally efficient but often yields inferior generalization compared to reinforcement learning (RL). This gap is primarily driven by RL’s use of on-policy data.

We propose a framework to bridge this chasm by enabling On-Policy SFT.

We first present ***Distribution Discriminant Theory (DDT)***, which explains and quantifies the alignment between data and the model-induced distribution.
Leveraging DDT, we introduce two complementary techniques: (i) ***In-Distribution Finetuning (IDFT)***, a loss-level method to enhance generalization ability of SFT, and (ii) ***Hinted Decoding***, a data-level technique that can re-align the training corpus to the model's distribution.
Extensive experiments demonstrate that our framework achieves generalization performance on par with prominent offline RL algorithms, including DPO and SimPO, while maintaining the efficiency of an SFT pipeline.
The proposed framework thus offers a practical alternative in domains where RL is infeasible. We will open-source the code and data on GitHub.
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Rethinking Multimodal Time-Series Forecasting Evaluation
Haoxin Liu, Yichen Zhou, Rajat Sen, B. Aditya Prakash, Abhimanyu Das
We introduce a new context-enriched, multimodal time series forecasting benchmark TimesX. TimesX contains a wide selection of high-quality real-world time series with diverse domains and textual contexts obtained from an automated data generation pipeline, which helps address three main issues of existing multimodal forecasting benchmarks: (1) poor generalization due to the small scale and synthetic nature of benchmark data, (2) very limited types of textual contexts in the benchmarks, and (3) an inability to mitigate data leakage in evaluation. We conduct a thorough empirical study of zero-shot multimodal forecasting approaches on TimesX. Our results suggest that many approaches that perform well on existing benchmarks may fail on TimesX. In contrast, simple ensemble methods that leverage rich textual context accompanying time-series can outperform strong baselines on the TimesX benchmark.
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LEC: Linear Expectation Constraints for Selection-Conditioned Risk Control in Selective Prediction and Routing Systems
Zhiyuan Wang, Aniri, Tianlong Chen, Yue Zhang, Heng Tao Shen, Xiaoshuang Shi, Kaidi Xu
Foundation models often generate unreliable answers, while heuristic uncertainty estimators fail to fully distinguish correct from incorrect outputs, causing users to accept erroneous answers without any statistical guarantee.
We address this problem through selection-conditioned risk control, aiming to ensure that an accepted prediction has an error probability no larger than a user-specified risk level.
To this end, we propose LEC, a principled framework that reframes selective prediction as a decision problem governed by a *linear expectation constraint* over selection and error indicators.
This formulation directly controls the ratio between the expected number of accepted errors and the expected number of accepted predictions, which corresponds to the marginal error probability conditioned on selection.
Under exchangeability, we derive a *finite-sample sufficient condition* that relies only on a held-out calibration set, enabling the computation of a risk-constrained, retention-maximizing threshold.
Furthermore, we extend LEC to two-model routing systems: if the primary model's uncertainty exceeds its calibrated threshold, the input is delegated to a subsequent model, while maintaining system-level selection-conditioned error control.
Experiments on both closed-ended and open-ended question answering (QA) and vision question answering (VQA) demonstrate that LEC maintains the prescribed risk level in accepted predictions and substantially improves sample retention compared to baselines.
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When Drafts Evolve: Speculative Decoding Meets Online Learning
Yu-Yang Qian, Hao-Cong Wu, Yichao Fu, Hao Zhang, Peng Zhao
Speculative decoding has emerged as a widely adopted paradigm for accelerating large language model inference, where a lightweight draft model rapidly generates candidate tokens that are then verified in parallel by a larger target model. However, due to limited model capacity, drafts often struggle to approximate the

target distribution, resulting in shorter acceptance lengths and diminished speedup. A key yet under-explored observation is that speculative decoding inherently provides *verification feedback* that quantifies the deviation between the draft and target models at no additional cost. This process naturally forms an iterative "draft commits--feedback provides--draft adapts" evolving loop, which precisely matches the *online learning* paradigm. Motivated by this connection, we propose OnlineSPEC, a unified framework that systematically leverages interactive feedback to continuously evolve draft models. Grounded in *dynamic regret minimization*, we establish a formal link between online learning performance and speculative system's acceleration rate, and develop novel algorithms via modern online learning techniques, including optimistic online learning that adaptively reuses historical gradients as predictive update hints, and online ensemble learning that dynamically maintains multiple draft models. Our algorithms are equipped with theoretical justifications and improved acceleration rates, achieving up to 24% speedup over seven benchmarks and five foundation models.
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Injecting Distributional Awareness into MLLMs via Reinforcement Learning for Deep Imbalanced Regression
Yao DU, Shanshan Song, Xiaomeng Li
Multimodal large language models (MLLMs) struggle with numerical regression under longtailed target distributions. Token-level supervised
fine-tuning (SFT) and point-wise regression rewards bias learning toward high-density regions, leading to regression-to-the-mean behavior and poor tail performance. We identify the lack of cross-sample relational supervision as a key limitation of existing MLLM training paradigms. To address it, we propose a distribution-aware reinforcement learning framework based on Group Relative Policy Optimization, which introduces batch-level comparison-based supervision via the Concordance Correlation Coefficient-based reward to align predicted and ground-truth distributions in terms of correlation, scale, and mean. The framework is plug-and-play, requiring no architectural modification. Experiments on a unified suite of long-tailed regression benchmarks show consistent improvements over SFT and existing MLLM regression methods, with particularly strong gains in medium- and few-shot regimes.
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Collapsed Effective Operators for Higher-order Structures
Maximilian Krahn, Lennart Bastian, Vikas K Garg, Björn Schuller, Tolga Birdal
Higher-order structures are powerful relational modeling tools, yet existing spectral operators decompose topology into separate ranks, leaving practitioners to fuse information back to vertices through ad-hoc choices. We introduce _Collapsed Effective Operators_, which condense higher-order degrees of freedom into a single vertex-level operator via Schur complementation of a graded Laplacian. This yields a (generally dense) operator that encodes long-range interactions mediated by topology and is applicable to arbitrary higher-order constructs. We show it preserves positive semi-definiteness with a strict spectral upper bound relative to the rank-0 Hodge Laplacian, effectively lowering system energy under higher-order connectivity. Empirically, our operator improves spectral clustering, signal smoothing and enables the inclusion of topological features in neural network architectures via positional encoding.
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Improving Adversarial Robustness of Attribution via Implicit Regularization
Amir Mehrpanah, Matteo Gamba, Hossein Azizpour
The adversarial robustness of attributions is a fundamental requirement for reliable explainability in deep learning, yet existing approaches typically rely on computationally expensive explicit regularization. In this work, we show that attribution robustness can arise implicitly from the learning dynamics of standard stochastic gradient descent. We theoretically motivate this effect through connections between parameter-space and input-space curvature, and validate it across architectures, datasets, and attribution methods, with negligible computational overhead. In contrast, we prove that such robustness gains often does not transfer to attention-based attribution under softmax normalization, due to inherent

entropy constraints, and we validate this limitation experimentally. Finally, we show that replacing softmax attention with kernel-based attention restores the robustness gains in transformer models. Our results highlight learning dynamics as a principled and practical mechanism for robust explainability, and reveal fundamental limitations of attention-based attribution under normalization.

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Learning High-Frequency Continuous Action Chunks in Latent Space

Kunyun Wang, Yuhang Zheng, Yupeng Zheng, Jieru Zhao, Wenchao Ding

Modern robotic policies increasingly rely on action chunking to execute complex tasks in the physical world. While action chunking improves temporal consistency at moderate action frequencies, it becomes insufficient when the action frequency is further increased (e.g., to 60~Hz). At such high frequencies, policies often fail to generate actions that are both temporally smooth and spatially consistent.

We address this challenge by shifting high-frequency action learning from the action space to a latent space with variational autoencoder (VAE). This formulation significantly improves both temporal and spatial consistency of high-frequency control.

To enable smooth real-time execution, we further introduce Reuse-then-Refine, a chunk-level refine strategy that improves continuity between adjacent action chunks under asynchronous inference.

As a result, robots controlled by our policy can execute complex contact-rich tasks continuously, with less pauses and jerky motions. Experiments on three real-world contact-rich robotic tasks show that our approach consistently completes tasks with smooth motions. Our code and data are
available at https://github.com/tars-robotics/RTR.

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B-Spar: Bayesian Sparse-Reward Modeling for RL-based Image Editing

Shusong Xu, Peiye Liu, Yongbin Liu, Bangjie Yin, Tianyi Zheng, Zhaomang Sun, Zhenyu Chen, Peng-Tao Jiang, Jian Zhang, Yuzhao Wang, Zhen Gu, Jinwei Chen, Bo Li

Autonomous image-editing agents powered by multimodal large language models (MLLMs) improve transparency and controllability by translating high-level instructions into tool-mediated edit sequences, but training such agents with reinforcement learning often relies on dense proxy rewards (e.g., incremental image-quality score gains) to compensate for sparse human feedback. When these proxies overvalue small local changes, the resulting optimization signal can be dominated by numerically measurable yet perceptually negligible edits, biasing policy gradients toward proxy artifacts rather than meaningful progress. We propose B-Spar, a reward-centric Reinforcement Learning framework for perceptually aligned image retouching under sparse feedback that combines prior-guided trajectory sampling to reduce inefficient exploration, Bayesian reward modeling to densify sparse binary feedback into a stable training signal, and anchor-regularized policy optimization to steer updates toward high-reward regions while preventing early mode collapse. Experiments on public benchmarks demonstrate that B-Spar improves perceptual quality and metric alignment with stable training and competitive inference efficiency over strong prompt-based and training-based baselines. Notably, it outperforms AIGC-based baselines by over 95\% in perceptual quality, achieving an improvement of approximately 33.5\% over the state-of-the-art.

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EMBGuard: Constructing Hazard-Aware Guardrails for Safe Planning in Embodied Agents

Dongwook Choi, Taeyoon Kwon, Bogyung Jeong, Minju Kim, Yeonjun Hwang, Hyojun Kim, Byungchul Kim, Young Kyun Jang, Jinyoung Yeo

MLLM-powered embodied agents deployed in real-world environments encounter physical hazards. However, existing approaches lack explicit mechanisms for identifying hazards and reasoning about action-conditioned risks, leading agents to either miss risky interactions or over-identify risks. To address this, we propose EMBGuard, the first MLLM-based safety guardrail for embodied agents designed to decouple physical risk reasoning from agent policy. By evaluating a (visual observation, action) pair, EMBGuard identifies hazardous configurations and provides n

atural language explanations of potential risks. Alongside EMBGuard, we contribu te EMBHazard, a training dataset of 15.1K action-conditioned pairs, and EMBGuard Test, a benchmark of 329 manually curated real-world scenarios spanning seven ph ysical risk categories. Through compositional variation of hazards and actions, we generate diverse risky and benign scenarios that agents may encounter during planning. Despite its compact size (2B, 4B), EMBGuard achieves performance compe titive with proprietary MLLMs (e.g., GPT-5.1, Gemini-2.5-Pro) while significantl y reducing the false-positive rates that hinder real-time deployment. We make th e code, data, and models publicly available at https://github.com/dongwxxkchoi/E MBGuard.

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PixCLIP: Towards Fine-grained Vision-Language Understanding via Any-granularity Pixel-Text Alignment

YiCheng Xiao, Yu Chen, Hao-Xuan Ma, Jiale Hong, Caorui Li, Lingxiang Wu, Haiyun Guo, Jinqiao Wang

While CLIP has achieved strong performance across vision–language tasks, fine-gr ained image–text alignment remains challenging. Recent efforts improve textual g ranularity by leveraging long, detailed descriptions and replacing CLIP’s text e ncoder with LLM, but often overlook the visual-side bottleneck: achieving finer alignment requires region- and pixel-level visual grounding. To address it, we p ropose PixCLIP, a framework that jointly enhances both sides by accommodating vi sual prompt regions and long-form text within a unified training objective. Firs tly, to support training at this granularity, we develop an automated annotation pipeline that produces long-form descriptions with pixel-level localization, an d use it to construct LongGRIT, a large-scale dataset with 1.5M samples. Further more, we introduce a three-branch pixel–text alignment framework that aligns ima ge regions with corresponding textual descriptions across multiple granularities . Experiments show that PixCLIP achieves state-of-the-art performance on pixel- and region-level alignment tasks while preserving strong results on standard glo bal image–text retrieval benchmarks, even with arbitrarily shaped region prompts and long texts. Our code is available at https://github.com/StuHude/PixCLIP.

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CofactGVR: Counterfactual Intervention for Grounded Visual Reasoning

Yan Zhang, Zhijin Qin, Feng Xu, Guiguang Ding, Jungong Han

Despite rapid progress in Grounded Visual Reasoning (GVR) with MLLMs and RL-styl e fine-tuning, existing approaches often lack effective learning signals for int ermediate grounding decisions and are prone to shortcut solutions. In this work, we explicitly decompose GVR into Evidence Generation followed by Counterfactual Answer Reasoning, and formalize this structure as a Causal Grounding Graph (CGG) in which the generated evidence acts as a causal mediator. Building on this fo rmulation, we propose CofactGVR, which estimates the mediator’s utility via a ma tched counterfactual intervention that perturbs the predicted region while keepi ng the original image–question context fixed. The factual–counterfactual reward gap yields a principled intermediate bonus, selectively assigned to high-quality factual rollouts to promote evidence-faithful reasoning. To further stabilize a nd efficiently exploit this causal training signal, we incorporate a Quantile-fi ltered Prioritized Advantage Sampling (Q-PAS) strategy that preferentially updat es on trajectories with high-magnitude advantages while filtering low-signal sam ples. Extensive experiments across GVR benchmarks show consistent improvements, indicating that CofactGVR strengthens reliance on informative visual evidence un der controlled interventions. Our source code will be released at https://github .com/zhangy0822/CofactGVR.

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ML-Embed: Inclusive and Efficient Embeddings for a Multilingual World

Ziyin Zhang, Zihan Liao, Hang Yu, Peng Di, Rui Wang

The development of high-quality text embeddings is increasingly drifting toward an exclusionary future, defined by three critical barriers: prohibitive computat ional costs, a narrow linguistic focus that neglects most of the world's languag es, and a lack of transparency from closed-source or open-weight models that sti fles research. To dismantle these barriers, we introduce ML-Embed, a suite of in

clusive and efficient models built upon a new framework: 3-Dimensional Matryoshka Learning (3D-ML). Our framework addresses the computational challenge with comprehensive efficiency across the entire model lifecycle. Beyond the storage benefits of Matryoshka Representation Learning (MRL) and flexible inference-time depth provided by Matryoshka Layer Learning (MLL), we introduce Matryoshka Embedding Learning (MEL) for enhanced parameter efficiency. To address the linguistic challenge, we curate a massively multilingual dataset and train a suite of models ranging from 140M to 8B parameters. In a direct commitment to transparency, we release all models, data, and code. Extensive evaluation on 430 tasks demonstrates that our models set new records on 9 of 17 evaluated MTEB benchmarks, with particularly strong results in low-resource languages, providing a reproducible blueprint for building globally equitable and computationally efficient AI systems.
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TopAdapter: Topology-Aware Prompt Tuning for Efficient Point Cloud Understanding
Changshuo Wang, Shuting He, Xiang Fang, Weijun Li, Yixian Shen, Mingkun Xu, Zhongtian Sun, Prayag Tiwari
Point cloud data, with its inherent geometric and topological structures, plays a critical role in 3D vision tasks. However, existing parameter-efficient fine-tuning (PEFT) methods predominantly focus on input token prompting, overlooking the intrinsic geometric information. To address this limitation, we propose TopAdapter, a novel PEFT framework that enhances geometric perception by injecting local topological information into pre-trained 3D vision models. TopAdapter leverages 0D, 1D, and 2D simplices from algebraic topology as fundamental building blocks, introducing two core modules: the Topology Injection module (ToInjection) and the Topology Transfer module (ToTransfer). ToInjection constructs multi-scale topological features using a simplex generator and dynamically fuses them with semantic features via a geometric controller, thereby enhancing geometric adaptability. ToTransfer propagates these topological primitives across Transformer layers, ensuring efficient transmission of geometric information. Extensive experiments demonstrate that TopAdapter outperforms existing PEFT methods, achieving performance comparable to full fine-tuning across various benchmarks.
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Event2Vec: Processing neuromorphic events directly by representations in vector space
Wei Fang, Priyadarshini Panda
Neuromorphic event cameras possess superior temporal resolution, power efficiency, and dynamic range compared to traditional cameras. However, their asynchronous and sparse data format poses a significant challenge for conventional deep learning methods. Most existing methods either densify events into frames, sacrificing their sparse asynchronous nature, or use irregular models that are less compatible with GPU acceleration. Inspired by word-to-vector models, we propose event2vec, a novel representation that allows Transformers to process events directly. We demonstrate the effectiveness of event2vec on the DVS Gesture, ASL-DVS, and DVS-Lip benchmarks, showing that event2vec is remarkably parameter-efficient, features high throughput and low latency, and achieves high accuracy even with an extremely low number of events or low spatial resolutions. These results show that sparse asynchronous event data can be directly integrated into high-throughput Transformer architectures, offering an efficient paradigm for real-time neuromorphic vision. The code is provided at https://github.com/Intelligent-Computing-Lab-Panda/event2vec.
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Closing the Loop: Universal Repository Representation with RPG-Encoder
Jane Luo, Chengyu Yin, Xin Zhang, Qingtao Li, Steven Liu, Yiming Huang, Jie Wu, Hao Liu, Yangyu Huang, Yu Kang, Fangkai Yang, Ying Xin, Scarlett Li
Current repository agents encounter a reasoning disconnect due to fragmented representations, as existing methods rely on isolated API documentation or dependency graphs that lack semantic depth. We consider repository comprehension and generation to be inverse processes within a unified cycle: generation expands intent into implementation, while comprehension compresses implementation back into intent. To address this, we propose RPG-Encoder, a framework that generalizes the

Repository Planning Graph (RPG) from a static generative blueprint into a unified, high-fidelity representation. RPG-Encoder closes the reasoning loop through three mechanisms: (1) Encoding raw code into the RPG that combines lifted semantic features with code dependencies; (2) Evolving the topology incrementally to decouple maintenance costs from repository scale, reducing overhead by 95.7%; and (3) Operating as a unified interface for structure-aware navigation. In evaluations, RPG-Encoder establishes state-of-the-art repository understanding on SWE-bench Verified with 93.7% Acc@5 and exceeds the best baseline by over 10% on SWE-bench Live. These results highlight our superior fine-grained localization accuracy in complex codebases. Furthermore, it achieves 98.5% reconstruction coverage on RepoCraft, confirming RPG's high-fidelity capacity to mirror the original codebase and closing the loop between intent and implementation. Our code and data are available at https://github.com/microsoft/RPG-ZeroRepo.
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Graph Alignment for Benchmarking Graph Neural Networks and Learning Positional Encodings
Adrien Lagesse, Marc Lelarge
We propose a novel benchmarking methodology for graph neural networks (GNNs) based on the graph alignment problem, a combinatorial optimization task that generalizes graph isomorphism by aligning two unlabeled graphs to maximize overlapping edges. We frame this problem as a self-supervised learning task and present several methods to generate graph alignment datasets using synthetic random graphs and real-world graph datasets from multiple domains. For a given graph dataset, we generate a family of graph alignment datasets with increasing difficulty, allowing us to rank the performance of various architectures. Our experiments prove that there is an optimal task difficulty for having a statistically relevant ranking of different models and that, even on a structure-only task, anisotropic models perform better compared to isotropic ones. To further prove that our synthetic task capture meaningful information, we show its effectiveness for self-supervised GNN pre-training: the learned node embeddings can be leveraged as positional encodings by transformers for graph regression or can be used to reconstruct the full structure of the graph with 98% accuracy. To support reproducibility and further research, we provide an open-source Python package to generate graph alignment datasets and benchmark new GNN architectures. The source code is available at [graph-alignment-benchmark](https://github.com/adrien-lagesse/graph-alignment-benchmark).
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Rule2DRC: Benchmarking LLM Agents for DRC Script Synthesis with Execution-Guided Test Generation
Jinuk Kim, Junsoo Byun, Donghwi Hwang, Seong-Jin Park, Hyun Oh Song
Manufacturable chip layouts must satisfy thousands of geometry-based design rules, and design rule checking (DRC) enforces them by running executable DRC scripts on layouts. Translating natural language rules into correct DRC scripts is labor-intensive and requires specialized expertise, motivating LLM agents for DRC script synthesis and debugging. However, existing benchmarks have small evaluation sets and often evaluate scripts by code similarity rather than execution correctness, and prior machine learning-based methods either ignore execution feedback or require labeled test layouts as agent's input. To this end, we introduce Rule2DRC, a large-scale benchmark for DRC script coding agents with 1,000 rule-to-script tasks and 13,921 evaluation chip layouts for execution-based scoring. Rule2DRC provides an evaluation pipeline that measures functional correctness via DRC execution outcomes without requiring evaluation layouts as input to the agent. We also propose SplitTester, a tester agent for program selection that uses execution feedback to generate discriminative test cases and separate previously indistinguishable candidate scripts, substantially improving Best-of-N selection performance in this domain.
We release the code at https://github.com/snu-mllab/Rule2DRC.
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SALSA-V: Shortcut-Augmented Long-form Synchronized Audio from Videos
Amir Dellali, Luca A Lanzendörfer, Florian Grötschla, Roger Wattenhofer

We propose SALSA-V, a multimodal video-to-audio generation model capable of synthesizing highly synchronized, high-fidelity long-form audio from silent video content. Our approach introduces a masked diffusion objective, enabling audio-conditioned generation and the seamless synthesis of audio sequences of unconstrained length. Additionally, by integrating a shortcut loss into our training process, we achieve rapid generation of high-quality audio samples in as few as eight sampling steps, paving the way for near-real-time applications without requiring dedicated fine-tuning or retraining. We demonstrate that SALSA-V significantly outperforms existing state-of-the-art methods in both audiovisual alignment and synchronization with video content in quantiative evaluation and a human listening study. Furthermore, our use of random masking during training enables our model to match spectral characteristics of reference audio samples, broadening its applicability to professional audio synthesis tasks such as Foley generation and sound design.
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HTAC: Hierarchical Task-Aware Composition for Continual Offline Reinforcement Learning

Qiyang Zhou, Xu Ruihang, Peng Wang, WenJie Lu, Xiaochun Cao, Naiqiang Tan, Li Shen

Continual Offline Reinforcement Learning (CORL) enables building long-term autonomous agents from static datasets. However, it faces heterogeneity in environment dynamics, reward functions, and behavior policies across tasks. Combined with the inherent distribution shift in offline learning, this requires agents to selectively reuse shared knowledge during transfer while isolating task-specific features. The flat knowledge sharing mechanisms employed by existing methods struggle to capture such distinctions, limiting cross-task generalization.

To address this, we propose Hierarchical Task-Aware Composition (HTAC), which balances plasticity and stability through dual-level task encoding and soft composition mechanisms. HTAC comprises four modules: (1) a Hierarchical Semantic Task Representation that decomposes tasks into domain-level and task-level embeddings; (2) a Dual-level Expert Network that creates domain and task experts on demand for parameter-efficient knowledge isolation; (3) an Adaptive Knowledge Composition module that integrates historical expert outputs via attention mechanisms for knowledge reuse; (4) Task Adapters that preserve historical routing weights to prevent forgetting. Experiments on Offline Continual World show that HTAC outperforms existing baselines, demonstrating better knowledge reuse and transfer capabilities.
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Unifying Heterogeneous Degradations: Uncertainty-Aware Diffusion Bridge Model for All-in-One Image Restoration

Luwei Tu, Jiawei Wu, Xing Luo, Zhi Jin

All-in-One Image Restoration (AiOIR) faces the fundamental challenge in reconciling conflicting optimization objectives across heterogeneous degradations. Existing methods are often constrained by coarse-grained control mechanisms or fixed mapping schedules, yielding suboptimal adaptation. To address this, we propose an Uncertainty-Aware Diffusion Bridge Model (UDBM), which innovatively reformulates AiOIR as a stochastic transport problem steered by pixel-wise uncertainty. By introducing a relaxed diffusion bridge formulation, which replaces the strict terminal constraint with a relaxed constraint, we model the uncertainty of degradations while theoretically resolving the drift singularity inherent in standard diffusion bridges. Furthermore, we devise a dual modulation strategy: the noise schedule aligns diverse degradations into a shared high-entropy latent space, while the path schedule adaptively regulates the transport trajectory motivated by the viscous dynamics of entropy regularization. By effectively rectifying the transport geometry and dynamics, UDBM achieves state-of-the-art performance across diverse restoration tasks within a single inference step. Code is available at https://github.com/Jabruson/UDBM.
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Towards Cold-Start Drafting and Continual Refining: A Value-Driven Memory Approach with Application to NPU Kernel Synthesis

Yujie Zheng, Zhuo Li, Shengtao Zhang, Jiaqian Wang, Junjie Sheng, Junchi Yan, We inan Zhang, Ying Wen, Bo Tang, Muning Wen
Deploying Large Language Models to data-scarce programming domains poses signifi cant challenges, particularly for kernel synthesis on emerging Domain-Specific A rchitectures where a ``Data Wall'' limits available training data.
While models excel on data-rich platforms like CUDA, they suffer catastrophic pe rformance drops on data-scarce ecosystems such as NPU programming.
To overcome this cold-start barrier without expensive fine-tuning, we introduce EvoKernel, a self-evolving agentic framework that automates the lifecycle of ker nel synthesis from initial drafting to continual refining.
EvoKernel addresses this by formulating the synthesis process as a memory-based reinforcement learning task. Through a novel value-driven retrieval mechanism, i t learns stage-specific Q-values that prioritize experiences based on their cont ribution to the current objective—whether bootstrapping a feasible draft or iter atively refining latency.
Furthermore, by enabling cross-task memory sharing, the agent generalizes insigh ts from simple to complex operators.
By building an NPU variant of KernelBench and evaluating on it, EvoKernel improv es frontier models' correctness from 11.0\% to 83.0\% and achieves a median spee dup of 3.60$\times$ over initial drafts through iterative refinement. This demon strates that value-driven experience accumulation allows general-purpose models to master the kernel synthesis task on niche hardware ecosystems.
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Interpreting Physics in Video World Models
Sonia Joseph, Quentin Garrido, Randall Balestriero, Matthew Kowal, Thomas Fel, S hahab Bakhtiari, Blake Aaron Richards, Michael Rabbat
A long-standing question in physical reasoning is whether video models rely on f actorized physical state variables, or on task-specific distributed representati ons. We present the first mechanistic interpretability study of physical variabl es inside large-scale video encoders, combining layerwise probing, subspace geom etry, patch-level decoding, and targeted attention ablations to characterize whe re and how physical information is orga- nized. Across architectures, we identif y a sharp intermediate-depth transition, the Physics Emergence Zone, at which ph ysical variables become linearly accessible. Scalar speed and acceleration are a vailable from early layers, whereas motion direction emerges only at the Physics Emergence Zone, mirroring the V1 to MT motion hierarchy in primate visual corte x. Direction is encoded as a circular high-dimensional population code: dozens o f orthogonal probe dimensions must be steered jointly to change the decoded dire ction, orders of magnitude more than the low-dimensional steering interventions seen in language models. These findings argue against compact physics- engine st ate variables and support distributed, hierarchically-organized, “brain-like” re presentations that are nonetheless sufficient for making physical predictions.
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PathWise: Planning through World Model for Automated Heuristic Design via Self-E volving LLMs
Oguzhan Gungordu, Siheng Xiong, Faramarz Fekri
Large Language Models (LLMs) have enabled automated heuristic design (AHD) for c ombinatorial optimization problems (COPs), but existing frameworks' reliance on fixed evolutionary rules and static prompt templates often leads to myopic heuri stic generation, redundant evaluations, and limited reasoning about how new heur istics should be derived. We propose a novel multi-agent reasoning framework, re ferred to as Planning through World Model for Automated Heuristic Design via Sel f-Evolving LLMs (PathWise), which formulates heuristic generation as a sequentia l decision process over an entailment graph serving as a compact, stateful memor y of the search trajectory. This approach allows the system to carry forward pas t decisions and reuse or avoid derivation information across generations. A poli cy agent plans evolutionary actions, a world model agent generates heuristic rol louts conditioned on those actions, and critic agents provide routed reflections summarizing lessons from prior steps, shifting LLM-based AHD from trial-and-err or evolution toward state-aware planning through reasoning. Experiments across d

iverse COPs show that PathWise converges faster to better heuristics, generalize s across different LLM backbones, and scales to larger problem sizes.
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TwinQuant: Learnable Subspace Decomposition for 4-Bit LLM Quantization
Haodong WANG, Junjie Liu, Zicong Hong, Qianli Liu, Jian Lin, Song Guo, Xu Chen
4-bit quantization reduces the memory footprint and latency of large language mo del inference, but its aggressive precision reduction can severely degrade accur acy. Prior methods address this by decomposing each weight matrix into two compo nents (e.g., via singular value decomposition) and quantizing them separately, a ssigning the bulk of values to a low-precision residual component while handling outliers with a high-precision low-rank component. However, such decompositions are designed to minimize the real-valued energy of the residual, rather than th e post-quantization error of the residual and low-rank components. We propose Tw inQuant, a 4-bit quantization framework that learns quantization-friendly decomp osed subspaces and jointly reshapes both the low-rank and residual components. T winQuant learns component-specific transformations via a joint optimization over the Stiefel and general linear manifolds, flattening their distributions and re ducing dynamic-range imbalance. To enable efficient end-to-end execution, we fur ther design a fused dual-component kernel that pipelines the two-stage low-rank computation on-chip and merges both components with a single epilogue, avoiding intermediate global-memory traffic. Across LLaMA3 and Qwen3 models, TwinQuant pr eserves near-FP16 accuracy and delivers up to $1.8\times$ end-to-end speedup ove r an FP16 baseline.
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MACKO: Sparse matrix-vector multiplication for low sparsity
Vladimír Macko, Vladimír Boža
Sparse Matrix-Vector Multiplication (SpMV) is a fundamental operation in the inf erence of sparse Large Language Models (LLMs).
Because existing SpMV methods perform poorly under the low, unstructured sparsit y ($30-90\\%$) commonly observed in pruned LLMs, unstructured pruning provides o nly limited memory reduction and speedup.
We propose **MACKO-SpMV**, a GPU-optimized format and kernel co-designed to redu ce storage overhead while remaining compatible with the GPU’s execution model.
This enables efficient SpMV for unstructured sparsity without specialized hardwa re units or precomputation.
We identify memory bandwidth as the primary limiting factor of SpMV and analyze the storage overhead of MACKO.
At $50\\%$ sparsity, MACKO is the first approach to achieve $1.5\times$ memory r eduction and $1.2-1.5\times$ speedup over the dense baseline as well as substant ial improvements over other SpMV methods: cuSPARSE ($2.8-13.0\times$), Sputnik ($1.9-2.6\times$), and DASP ($2.2-2.5\times$).
An LLM pruned with Wanda to sparsity $50\\%$ requires $1.5\times$ less memory an d achieves $1.5\times$ faster inference at fp16 precision.
As a result, **unstructured pruning at $50\\%$ sparsity becomes practical** for real-world LLM workloads and **bridges the efficiency gap with structured 2:4 sp arsity**.
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From Generative to Episodic: Sample-Efficient Replicable Reinforcement Learning
Max Hopkins, Sihan Liu, Christopher Ye, Yuichi Yoshida
The epidemic failure of replicability across empirical science and machine learn ing has recently motivated the formal study of replicable learning algorithms [I mpagliazzo et al. (2022)]. In contrast to batch settings (i.e. data comes from a fixed i.i.d. source) where the cost of replicability is relatively well underst ood, there remain significant gaps in our knowledge for control settings like re inforcement learning where an agent must interact directly with a shifting envir onment. Indeed, there is a large gap between the best upper bound of $\tilde{O}(S^7 A^7)$ [Eaton et al. (2023)] for RL with exploration, and $\tilde{O}(S^2 A^2) $ [Karbasi et al. (2023)] for the RL `batch' setting. This gap raises a key ques tion in the broader theory of replicability: Is replicable exploration inherentl y more expensive than batch learning? Is sample-efficient replicable RL even pos

sible?

In this work, we (nearly) resolve this problem (for low-horizon tabular MDPs): e xploration is not a significant barrier to replicable learning! Our main result is a replicable RL algorithm on $\tilde{O}(S^2A)$ samples, bridging the gap betw een the generative and episodic settings. We complement this with a lower bound in the episodic setting of $\tilde{\Omega}(S^2)$ showcasing the near-optimality of our algorithm with respect to the state space S.
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Watch Your Step: Information Injection in Diffusion Models via Shadow Timestep E mbedding
An Huang, Junggab Son, Zuobin Xiong
Diffusion models have become the foundation of modern generative systems, with m ost research focusing primarily on improving generation efficiency and output qu ality.
The timestep embedding component is a crucial part of the diffusion pipeline, wh ich provides a temporal conditioning signal to the denoising network, enabling i t to adapt its predictions across different noise levels throughout the process. Despite their potential to contain substantial information, timestep embeddings remain underexplored in current research, especially for security risks and reli able provenance.
To fill this gap, we introduce **Shadow Timestep Embedding (STE)**, a novel mech anism that investigates the underutilized temporal space for malicious informati on injection into diffusion models.
In particular, when zooming in on the timestep embedding space, we find that dif ferent timesteps exhibit distinct representational capabilities that can encode side-channel information.
Moreover, such encoded information can be utilized for attack and defense purpos es through the scheduler interface.
We present a theoretical analysis of timestep embeddings as position-encoding ma ppings and derive a mutual coherence evaluation that explains the separability o f disjoint timestep intervals.
Our findings reveal the diffusion model's timestep as a powerful side channel fo r carrying dedicated information, motivating new directions for adversarial gene rative modeling by understanding the temporal dimension.
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A3: an Analytical Low-Rank Approximation Framework for Attention
Jeffrey T. H. Wong, Cheng Zhang, Xinye Cao, Pedro Gimenes, Christos-Savvas Bouga nis, George Anthony Constantinides, Wayne Luk, Yiren Zhao
Large language models have demonstrated remarkable performance; however, their m assive parameter counts make deployment highly expensive. Low-rank approximation offers a promising compression solution, yet existing approaches have two main limitations: (1) They focus on minimizing the output error of individual linear layers, without considering the architectural characteristics of Transformers, a nd (2) they decompose a large weight matrix into two small low-rank matrices. Co nsequently, these methods often fall short compared to other compression techniq ues like pruning and quantization, and introduce runtime overhead such as the ex tra GEMM kernel launches and memory operations for decomposed small matrices. To address these limitations, we propose A^3, a post-training low-rank approxima tion framework. A^3 splits a Transformer layer into three functional component s, namely $\texttt{QK}$, $\texttt{OV}$, and $\texttt{MLP}$ and provides analytic al solutions that reduces the hidden dimension size inside each component while minimizing the component's functional loss. This approach directly reduces model sizes, KV cache sizes, and FLOPs without introducing any runtime overheads. Thr ough extensive experiments, we show that A^3 maintains superior performance co mpared to SoTAs.
For example, under the same reduction budget in computation and memory, our low- rank approximated LLaMA 3.1-70B achieves a perplexity of 4.69 on WikiText-2, out performing the previous SoTA's 7.87 by 3.18. We also show versatile applications of A^3 in KV cache compression, integration with quantization, fine-tuning an

d mixed-rank assignments. We open-sourced our framework at https://github.com/DeepWok/a3.
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Neuro-Symbolic AI for Analytical Solutions of Differential Equations
Orestis Oikonomou, Levi Lingsch, Dana Grund, Siddhartha Mishra, Georgios Kissas
Analytical solutions to differential equations offer exact, interpretable insight but are rarely available because discovering them requires expert intuition or exhaustive search of combinatorial spaces. We introduce SIGS, a neuro-symbolic framework for equation-driven closed-form solution discovery. SIGS uses a context-free grammar to generate mathematically valid and physically meaningful building blocks, with a user-specified Ansatz prescribing how these blocks combine, embeds them into a topology-regularised continuous latent manifold, and searches this manifold in two stages: structure selection followed by coefficient refinement using gradient descent, scoring candidates only against the PDE residual and prescribed boundary and initial conditions. This design unifies symbolic reasoning with numerical optimization; the grammar constrains candidate solution blocks to be proper by construction, while the latent search makes exploration tractable and data-free. SIGS is the first neuro-symbolic method to (i) recover analytical solutions for coupled nonlinear PDE systems, (ii) discover equivalent symbolic forms when the grammar lacks the natural primitives, and (iii) produce accurate symbolic approximations for PDEs lacking known closed-form solutions. Overall, SIGS improves over existing symbolic methods by orders of magnitude in both accuracy and runtime across standard PDE benchmarks.
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Rethinking Forgery Attacks on Semantic Watermarks in Black-Box Settings: A Geometric Distortion Perspective
Cheng-Yi Lee, Yichi Zhang, Yuchen Yang, Chun-Shien Lu, Jun-Cheng Chen
Recent studies have shown that semantic watermarks, which embed information into the initial noise of latent diffusion models (LDMs), are vulnerable to black-box forgery attacks. However, existing methods primarily rely on empirical evidence and lack a rigorous theoretical understanding of the conditions under which such attacks succeed or fail. To bridge this gap, we rethink the nature of such attacks through the lens of rate-distortion in the latent space. Our analysis identifies an irreducible distortion floor due to structural mismatches between proxy and target models, which fundamentally limits the fidelity of forged watermarks. We further characterize this distortion as structured geometric deviations on the latent manifold, in the form of global drift and local deformation rather than stochastic noise. Leveraging these insights, we propose a scheme-agnostic detection method that distinguishes forged samples before watermark verification. Extensive experiments demonstrate the effectiveness of our method across diverse black-box scenarios, while preserving robustness to common distortions.
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EPSVec: Efficient and Private Synthetic Data Generation via Dataset Vectors
Amin Banayeeanzade, Qingchuan Yang, Deqing Fu, Spencer Hong, Erin Babinsky, Alfy Samuel, Anoop Kumar, Robin Jia, Sai Praneeth Karimireddy
High-quality data is essential for modern machine learning, yet many valuable corpora are sensitive and cannot be freely shared. Synthetic data offers a practical substitute for downstream development, and large language models (LLMs) have emerged as powerful engines for generating it. However, existing private text generation methods are severely inefficient: they are data-intensive, computationally slow, and often require large private corpora or batch sizes to achieve usable quality. We introduce EPSVec, a differentially-private lightweight alternative that steers LLM generation using *dataset vectors*-directions in activation space that capture the distributional gap between private data and public priors. EPSVec extracts and sanitizes steering vectors just once and then performs standard decoding. This decouples the privacy budget from generation, enabling arbitrarily many synthetic samples without additional privacy cost and yielding strong fidelity even in low-data regimes. Furthermore, we enhance our method by utilizing pretrained (base) models and introducing fixed-shot prompting to boost generation diversity and fidelity. Our experiments demonstrate that EPSVec outperform

s existing baselines in distributional alignment and downstream utility, particu
larly in low-data regimes, while significantly reducing computational overhead.
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SOLAR for Offline MARL: Plateau-Triggered Potential Shaping under World-Model Un
certainty
Jusheng Zhang, Yijia Fan, Ruiqi Chen, Jing Yang, Ziliang Chen, Yongsen Zheng, Ya
nxi Chen, Jian Wang, Kwok-Yan Lam, Liang Lin, Keze Wang
Reward shaping can accelerate reinforcement learning, but in sparse-reward \emph
{offline} multi-agent RL it is often brittle: dense intrinsic rewards may alter
the underlying Markov game, while world-model guidance can amplify model bias.
We find that shaping becomes reliable when it is (i) activated only after \emph{
statistically validated} learning plateaus and (ii) constrained to \emph{potenti
al-based} shaping, which preserves the task optimum.
Motivated by this, we propose \textsc{SOLAR}, a simulate--evaluate--shape framew
ork.
A learned world model enables low-cost rollouts to test plateaus; once a plateau
 is detected, we inject shaping in the form $r+\gamma\Phi(s')-\Phi(s)$ with adap
tively updated potentials; and we attenuate shaping using uncertainty-aware thro
ttling in unreliable regions.
We provide theoretical analysis on policy invariance and on the deviation of pla
teau decisions under model error, and establish stability for the resulting two-
timescale adaptation.
Experiments on sparse-reward offline MARL benchmarks show consistent gains in st
ability and final performance across dataset qualities.
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RGGT: A Generative-Prior-Guided Transformer for Unified Rigid and Non-Rigid Poin
t Cloud Registration
Chengyu Zheng, Songlin Yang, Jin Huang, Honghua Chen, Weiming Wang, Haoran Xie,
Fu Lee Wang, Mingqiang Wei
Point cloud registration can be categorized into rigid and non-rigid settings de
pending on the motion characteristics of the underlying objects. Rigid alignment
 assumes a single global transformation under which corresponding points remain
geometrically consistent across scales, whereas non-rigid alignment involves spa
tially varying deformations, where geometric similarity holds only locally and s
emantic correspondence dominates at larger scales. This multi-scale discrepancy
creates an optimization gap that has made unified registration particularly chal
lenging. To this end, we propose RGGT, a Generative-Prior-Guided Transformer tha
t unifies rigid and non-rigid registration within a shared optimization space. T
hrough coordinated design at the representation, architecture, and supervision l
evels, RGGT jointly captures local geometric details and global structural seman
tics: generative priors enrich point features with unified geometric-semantic cu
es; a Global-Self-Cross Attention module models long-range structure, local inte
raction, and bidirectional cross-shape reasoning; and a dual correspondence-reco
nstruction objective provides consistent supervision for both deformation types.
 Extensive experiments on rigid (ModelNet40, 3DMatch, KITTI) and non-rigid (4DMa
tch) benchmarks demonstrate that RGGT achieves state-of-the-art accuracy across
both rigid and non-rigid settings within a single unified framework. Code is ava
ilable at https://github.com/zhengcy-lambo/RGGT.
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Equivalence of Context and Parameter Updates in Modern Transformer Blocks
Adrian Goldwaser, Michael Munn, Javier Gonzalvo, Benoit Dherin
Recent research has established that the impact of context in a vanilla transfor
mer can be represented implicitly by forming a token-dependent, rank-1 patch to
its MLP weights. This work extends that foundational theory to the diverse archi
tectures of modern Large Language Models. We first demonstrate a precise, analyt
ical solution for a Gemma-style transformer block, proving that the entire effec
t of a context can be perfectly mapped to rank-1 patches on its MLP weight matri
ces and a patch to the RMSNorm scale. We then generalize this result, providing
a constructive proof and algorithm for multi-layer models. To unify these findin
gs, we introduce a general framework centered on two core properties: input cont

rollability and output controllability. We prove that a perfect implicit weight patch is possible for any MLP block where the inner function is input-controllable and the outer function is output-controllable. This provides a simpler and more powerful lens for understanding how transformer models transmute prompts into effective weights. This setup generalizes to a wide range of modern LLM architectures including gating, pre-/post-norm, mixture of experts and sequential/parallel transformer blocks.
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Sparse but Wrong: Incorrect L0 Leads to Incorrect Features in Sparse Autoencoders
David Chanin, Adrià Garriga-Alonso
Sparse Autoencoders (SAEs) extract features from LLM internal activations, meant to correspond to interpretable concepts. A core SAE training hyperparameter is L0: how many SAE features should fire per token on average. Existing work compares SAE algorithms using sparsity-reconstruction tradeoff plots, implying L0 is a free parameter with no inherently correct value aside from its effect on reconstruction. In this work we study the effect of L0 on SAEs, and show that if L0 is not set correctly, the SAE fails to disentangle the underlying features of the LLM. If L0 is too low, the SAE will mix correlated features to improve reconstruction. If L0 is too high, the SAE finds degenerate solutions that also mix features. Further, we present a proxy metric that can help guide the search for the correct L0 for an SAE on a given training distribution. We show that our method finds the correct L0 in toy models and coincides with peak sparse probing performance in LLM SAEs. We find that most commonly used SAEs have an L0 that is too low. Our work shows that practitioners must set L0 correctly to train SAEs with monosemantic features.
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A Constrained Optimization Perspective of Unrolled Transformers
Javier Porras-Valenzuela, Samar Hadou, Alejandro Ribeiro
We introduce a constrained optimization framework for training transformers that behave like optimization descent algorithms. Specifically, we enforce layerwise descent constraints on the objective function and replace standard empirical risk minimization (ERM) with a primal-dual training scheme. This approach yields models whose intermediate representations decrease the loss monotonically in expectation across layers. We apply our method to both unrolled transformer architectures and conventional pretrained transformers on tasks of video denoising and text classification. Across these settings, we observe that constrained transformers achieve stronger robustness to perturbations and maintain higher out-of-distribution generalization, while preserving competitive in-distribution performance.
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Improving the Robustness-Utility Trade-off in Decentralized Learning over Sparse Networks
Yangnan Li, Xuanyu Cao, Shenghui Song
Resilience against Byzantine attackers and faster convergence on sparse networks are critical for decentralized optimization, yet existing methods fail to achieve both simultaneously. Existing DSGD-based Byzantine-resilient methods suffer from high transient complexity of $\mathcal{O}\left((1-\lambda)^{-6}\right)$, where $1-\lambda$ denotes the spectral gap of the network. While bias-correction methods such as Exact Diffusion can improve topology dependence, directly combining them with robust aggregators can lead to error accumulation. To address this issue, we introduce the scaled dual ascent (SDA) within the augmented Lagrangian framework for decentralized optimization, which mitigates error accumulation by scaling the dual update steps. Based on this, we propose BRED, which integrates Byzantine-robust Exact Diffusion with the SDA framework. We prove that BRED attains linear speedup, and achieves transient complexity of $\mathcal{O}\left((1-\lambda)^{-2}\right)$ when the Byzantine fraction δ is small. We further propose the momentum variant BRED-M, which reduces the Byzantine-affected transient complexity from $\mathcal{O}\left(\delta^2(1-\lambda)^{-6}\right)$ to $\mathcal{O}\left(\delta^2(1-\lambda)^{-4}\right)$. Empirical results on benchmark datas

ets demonstrate the efficacy of the proposed methods across diverse network topologies.
****
Instance-Dependent Continuous-Time Reinforcement Learning via Maximum Likelihood Estimation
Runze Zhao, Yue Yu, Ruhan Wang, Chunfeng Huang, Dongruo Zhou
Continuous-time reinforcement learning (CTRL) provides a natural framework for sequential decision-making in dynamic environments where interactions evolve continuously over time. While CTRL has shown growing empirical success, its ability to adapt to varying levels of problem difficulty remains poorly understood. In this work, we investigate the instance-dependent behavior of CTRL and introduce a simple, model-based algorithm built on maximum likelihood estimation (MLE) with a general function approximator. Unlike existing approaches that estimate system dynamics directly, our method estimates the state marginal density to guide learning. We establish instance-dependent performance guarantees by deriving a regret bound that scales with the total reward variance and measurement resolution. Notably, the regret becomes independent of the specific measurement strategy when the observation frequency adapts appropriately to the problem’s complexity. To further improve performance, our algorithm incorporates a randomized measurement schedule that enhances sample efficiency without increasing measurement cost. These results highlight a new direction for designing CTRL algorithms that automatically adjust their learning behavior based on the underlying difficulty of the environment.
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Cheap2Rich: A Multi-Fidelity Framework for Data Assimilation and System Identification of Multiscale Physics - Rotating Detonation Engines
Yuxuan Bao, Jan Zajac, Megan Powers, Venkat Raman, J. Nathan Kutz
Bridging the sim2real gap between computationally inexpensive models and complex physical systems remains a central challenge in machine learning applications to engineering problems, particularly in multi-scale settings where reduced-order models typically capture only dominant dynamics.

In this work, we present Cheap2Rich, a multi-scale data assimilation framework that reconstructs high-fidelity state spaces from sparse sensor histories by combining a fast low-fidelity prior with learned, interpretable discrepancy corrections.

We demonstrate the performance on rotating detonation engines (RDEs), a challenging class of systems that couple detonation-front propagation with injector-driven unsteadiness, mixing, and stiff chemistry across disparate scales.

Our approach successfully reconstructs high-fidelity RDE states from sparse measurements while isolating physically meaningful discrepancy dynamics associated with injector-driven effects.

The results highlight a general multi-fidelity framework for data assimilation and system identification in complex multi-scale systems, enabling rapid design exploration and real-time monitoring and control while providing interpretable discrepancy dynamics.
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Structured 4D Latent Predictive Model for Robot Planning
Zhiyi Li, Peilin Wu, Xiaoshen Han, Ruojin Cai, Yilun Du
Video predictive models are emerging as a powerful paradigm in robotics, offering a promising path toward task generalization, long-horizon planning, and flexible decision-making. However, prevailing approaches often operate on 2D video sequences, inherently lacking the 3D geometric understanding necessary for precise spatial reasoning and physical consistency. We introduce a **Structured 4D Latent Predictive Model**, which predicts the evolution of a scene’s 3D structure in a structured latent space conditioned on observations and textual instructions. Our representation encodes the scene holistically and can be decoded into diverse 3D formats, enabling a more complete and 3D consistent scene understanding. This structured 4D latent predictive model serves as a planner, generating future scenes that are translated into executable actions by a goal-conditioned inverse dynamics module. Experiments demonstrate that our model generates futures with

strong visual quality, substantially better 3D consistency and multi-view cohere
nce compared to state-of-the-art video-based planners. Consequently, our full pl
anning pipeline achieves superior performance on complex manipulation tasks, exh
ibits robust generalization to novel visual conditions, and proves effective on
real-world robotic platforms. Our website is available at https://structured-4d-
model.github.io/.
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Configurable Reward Model for Balanced Safety Alignment
Zhengping Jiang, Mehran Khodabandeh, Akash Bharadwaj, Manik Bhandari, Mayur Srun
garapu, Anqi Liu, Benjamin Van Durme, Li Chen
Aligning large language models (LLMs) to heterogeneous and rapidly evolving safe
ty requirements remains a critical challenge. Existing instruction-tuned LLMs an
d standalone safety classifiers often fail to generalize to new safety configura
tions, motivating the need for Reward Models (RMs) that are explicitly configura
ble to changing specifications. We introduce the Configurable Safety Reward Mode
l (CSRM), which is jointly optimized for calibrated safety compliance and reward
 modeling. Our approach is supported by configuration-targeted data augmentation
 that enforces instruction adherence while preserving relative severity structur
e. The resulting RM is sensitive to fine-grained safety configurations and conve
rsational nuances, substantially improving generalization to previously unseen s
afety configurations. CSRM achieves state-of-the-art performance on recent confi
gurable safety benchmarks, including CoSApien (94.6\% F1) and DynaBench (75.8\%
F1), without requiring additional human annotation. When used for downstream saf
ety alignment, CSRM yields LLMs with a significantly improved helpfulness–safety
 tradeoff compared to existing baselines.
****

Merge to Remember: Sharpness-Aware Isotropic Merging for Continual Learning
Qun Yang, Enneng Yang, Wei Chen, Li Shen, Long Lan
Continual learning with large pre-trained models offers significant potential fo
r cross-task knowledge accumulation, but faces critical challenges such as catas
trophic forgetting and parameter interference, especially when historical data i
s unavailable. Existing approaches typically rely on sequential fine-tuning or m
odel merging strategies, yet often overlook the impact of loss landscape sharpne
ss and dominant singular value directions, which leads to subspace misalignment
and severe knowledge forgetting. In this paper, we propose the Sharpness-Aware I
sotropic Merging (SAIM) framework, which introduces targeted optimizations in bo
th the fine-tuning and merging stages to address these issues. Specifically, SAI
M consists of two synergistic modules: (1) a Sharpness-Aware Block Coordinate De
scent (SA-BCD) optimizer that guides the model toward flatter minima and selecti
vely updates the most task-sensitive parameters, thereby mitigating parameter in
terference and enhancing robustness; (2) an adaptive isotropic merging algorithm
 that dynamically balances the singular value spectrum across tasks, effectively
 preventing the model from overemphasizing any single task direction, maintainin
g balanced knowledge representation, and improving subspace alignment. Extensive
 experiments on vision and language benchmarks demonstrate that SAIM achieves 5-
10\% higher accuracy than existing methods and maintains robust performance as t
he number of tasks increases. Our code are available at \url{https://github.com/
Yangqun123456/SAIM}.
****

GFD-EMVC: Evolutionary Multi-View Classification with Label Noise via Gradient a
nd Feature Dual-Perception
Shuai Li, Xinyan Liang, Yuhua Qian, Li Lv
This paper studies a fundamental yet often overlooked premise in evolutionary mu
lti-view classification (EMVC): the impact of label noise on EMVC, such as disto
rting fitness landscapes shaped by individual fitness values (e.g., test accurac
y).
Traditional EMVC assumes training labels are noise-free, yet this often fails in
 practice.
As a result, label noise introduces harmful supervision during the training phas
e, resulting in distorted fitness landscapes and the emergence of fitness evalua

tion bias (FEB). This bias misguides the evolutionary trajectory, causing the se
arch process to stagnate in local optima.
Given that label noise largely stems from the mislabeling of samples near their
decision boundaries by human annotators, we thus compared the decision boundarie
s of human annotators and models, and found discrepancies between the two. Based
 on this observation, we propose a simple yet effective ``detect-then-calibrate"
 data purification framework that leverages outlier analysis in the gradient spa
ce (i.e., treating outliers as noisy samples) and prototype calibration in the f
eature space (i.e., utilizing feature prototypes of noise-free samples to correc
t the labels of noisy samples).
Experimental results demonstrate that this strategy can effectively purify the d
ata and alleviate FEB; moreover, it can improve the performance of various multi
-view learning paradigms in label noise scenarios.
(https://github.com/LiShuailzn/ICML-2026-GFD-EMVC)
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MultiLoReFT: Decoupling Shared and Modality-Specific Subspaces in Multimodal Lea
rning via Low-Rank Representation Fine-Tuning

Sana Tonekaboni, Viktoria Schuster, Caroline Uhler

Real-world perception and decision making are inherently multimodal, integrating
 complementary signals across modalities. However, training multimodal models fa
ces two main obstacles. First, collecting large-scale, well-aligned paired multi
modal datasets is often impractical, making end-to-end multimodal training diffi
cult. Second, existing multimodal representations frequently entangle informatio
n shared across modalities with modality-specific information, hindering interpr
etability and control. We introduce MultiLoReFT, an efficient and scalable low-r
ank representation fine-tuning framework for multimodal learning with pretrained
 unimodal models. MultiLoReFT extends low-rank adaptation to the multimodal sett
ing and learns interpretable projection subspaces that decouple shared and modal
ity-specific information. Across simulated and real-world benchmarks, it produce
s representations that support multimodal prediction while explicitly revealing
how shared and modality-specific information is distributed across modalities.
****

Beyond Euclidean Summaries: Online Change Point Detection for Distribution-Value
d Data

Yingyan Zeng, Zipan Huang, Xiaoyu Chen

Existing online change-point detection (CPD) methods rely on fixed-dimensional E
uclidean summaries, implicitly assuming that distributional changes are well cap
tured by moment-based or feature-based representations.
They can obscure important changes in distributional shape or geometry. We propo
se an intrinsic distribution-valued CPD framework that treats streaming batch da
ta as a stochastic process on the 2-Wasserstein space. Our method detects change
s in the law of this process by mapping each empirical distribution to a tangent
 space relative to a pre-change Fréchet barycenter, yielding a reference-centere
d local linearization of 2-Wasserstein space.
This representation enables sequential detectors by adapting classical multivari
ate monitoring statistics to tangent fields. We provide theoretical guarantees a
nd demonstrate, via synthetic and real-world experiments, that our approach dete
cts complex distributional shifts with reduced detection delay at matched $\math
rm{ARL}_0$ compared with moments-based and model-free baselines.
The code is available at https://github.com/yyzeng43/IDD-icml.
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LeakGFN: Robust Molecular Generation in Generative Flow Networks via Flow Decomp
osition

Hwanhee Kim, Seungyeon Choi, Sanghyun Park

Generative Flow Networks (GFlowNets) have emerged as a powerful framework for mo
lecular generation, sampling diverse candidates proportionally to a reward funct
ion. However, the vast chemical space necessitates truncating trajectory length,
 forcing models to treat incomplete molecular fragments as terminal states along
side valid molecules. This conflation distorts the learned distribution by alloc
ating probability mass to chemically meaningless states. We propose LeakGFN, a d

ual-head architecture that decomposes flow into two components: a chemical head modeling flow over the full chemical space, and a valid head estimating the fraction of flow reaching valid molecules within the truncation boundary. Through this decomposition, the valid head implicitly learns molecular reachability without explicit supervision. We prove that LeakGFN recovers the correct distribution over accessible molecules under mild assumptions. Experiments on five molecular optimization tasks demonstrate consistent improvements over flow matching baselines, achieving state-of-the-art performance on four out of five tasks. Our module integrates as a plug-and-play enhancement into existing frameworks, improving performance on both pocket-conditioned and multi-objective generation tasks.
****

SMM Transformer: Leveraging Spiking Neural Networks for Multimodal Tasks
Xiubo Liang, Jinxing Han, Yuke Li, Haoqi Zhu, Yu Zhao, Hongzhi Wang
Spiking Neural Networks (SNNs) enable event-driven computation with sparse activations, but building multimodal Transformers on SNNs is hindered by unstable training in deep spiking stacks and a mismatch between dense softmax attention and spike-based communication. We propose SMM Transformer, an SNN-based multimodal Transformer framework that combines (i) a Parallel LIF with Multistage Learnable Parameters (PLMP) neuron and a tailored P-STBP algorithm to stabilize training, (ii) a spike-driven attention approximation (SMSA) with a lightweight self-compensation branch, and (iii) a spiking mixture-of-experts (SMoE) module for modality-aware fusion. Across visual and multimodal benchmarks, SMM Transformer achieves competitive accuracy compared to ANN baselines while reducing the estimated compute energy of the attention module by up to 97\% under a standard MAC/AC cost model.
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AlignedNorm: Prompting Vision–Language Models via Coupled Prompt Field
Qi Ma, Chen-Yang Wang, Dehong Gao, Deng-Ping Fan
Prompt learning for vision-language models (VLMs) primarily follows end-to-end or decoupled routes to balance base and new task performance, but suffers a fundamental bottleneck: sample-wise optimization within task-specific feature spaces traps models in local optima, hindering global optimality. To address this, we identify a key insight that VLMs can be prompted within a Coupled Prompt Field-a shared space where base and new tasks are mutually constrained-and present AlignedNorm, which enforces the field coupling. By dynamically aligning the norms of prompts to VLMs' native scale, our method enables joint optimization of both tasks.
Without complex designs, our method matches leading decoupled approaches on 15 datasets across 4 experimental settings, offering both a new perspective and a practical solution to the local-optima dilemma in prompt learning.
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OmniSIFT: Modality-Asymmetric Token Compression for Efficient Omni-modal Large Language Models
Yue Ding, Yiyan Ji, Jungang Li, Xuyang Liu, Xinlong Chen, Junfei Wu, Bozhou Li, Bohan Zeng, Yang Shi, Yushuo Guan, Yuanxing Zhang, Jiaheng Liu, Qiang Liu, Pengfei Wan, Liang Wang
Omni-modal Large Language Models (Omni-LLMs) have demonstrated strong capabilities in audio-video understanding tasks. However, their reliance on long multimodal token sequences leads to substantial computational overhead. Despite this challenge, token compression methods designed for Omni-LLMs remain limited. To bridge this gap, we propose OmniSIFT (Omni-modal Spatio-temporal Informed Fine-grained Token compression), a modality-asymmetric token compression framework tailored for Omni-LLMs. Specifically, OmniSIFT adopts a two-stage compression strategy: (i) a spatio-temporal video pruning module that removes video redundancy arising from both intra-frame structure and inter-frame overlap, and (ii) a vision-guided audio selection module that filters audio tokens. The entire framework is optimized end-to-end via a differentiable straight-through estimator. Extensive experiments on five representative benchmarks verify the efficacy and robustness of OmniSIFT. Notably, for Qwen2.5-Omni-7B, OmniSIFT adds 4.85M parameters while still achieving lower latency than training-free baselines such as OmniZip. With o

nly 25% of the original token context, OmniSIFT consistently outperforms all com
pression baselines and even surpasses the full-token model on several tasks.
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DistFlow: A Fully Distributed RL Framework for Scalable and Efficient LLM Post-T
raining

Zhixin Wang, Jiaming Xu, Tianyi Zhou, Mingjun Zhang, Liming Liu, Jiarui Hu, Dian
 Yang, TongYu Wang, Ping Zhang, Jinlong Hou, Siyuan Feng, Yuan Qi, Yuan Cheng

Effectively scaling Reinforcement Learning (RL) is crucial for enhancing the rea
soning and alignment of Large Language Models. The massive data and complex exec
ution flows inherent in these tasks require a distributed architecture capable o
f efficient scaling. However, to simplify programming and dependency management,
 mainstream frameworks often rely on a centralized architecture where a single n
ode dispatches both control and data. This inherent coupling creates significant
 communication bottlenecks, severely limiting system scalability and efficiency.
 We present DistFlow, a novel, fully distributed RL framework that adopts a mult
i-controller paradigm. By decoupling data transmission from control dispatch, Di
stFlow establishes a parallelism-aware, decentralized Data Coordinator that leve
rages local caching, load balancing, and asynchronous double buffer to minimize
communication overhead and mitigate straggler effects. For control logic, it int
roduces a task scheduler built upon Directed Acyclic Graph (DAG) that facilitate
s fine-grained, independent execution. Experimental results demonstrate that Dis
tFlow achieves near-linear scalability up to 512 GPUs and delivers up to a 2.63x
 throughput improvement over state-of-the-art (SOTA) frameworks.
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IQA-Spider: Unifying Multi-Granularity Image Quality Assessment with Reasoning,
Grounding and Referring

Xinge Peng, Yiting Lu, Xin Li, Zhibo Chen

We present IQA-Spider, the first image quality assessment (IQA) framework that u
nifies reasoning, grounding, and referring into a single LMM-based framework for
 multi-granularity quality understanding. Existing LMM-based IQA methods typical
ly support only partial perception dimensions, such as quality description and q
uestion answering~(\textit{i.e.}, reasoning) or pixel-level grounding. This limi
tation largely stems from the absence of (i) a unified task and data formulation
 and (ii) effective optimization paradigms for multi-granularity learning. To ad
dress these limitations, we formulate a rigorous four-task paradigm covering glo
bal and local quality description, pixel-level grounding, and region-level refer
ring. Based on this formulation, we construct a corresponding IQA dataset with a
 scalable and automatic annotation pipeline, thereby providing a solid foundatio
n for unified multi-granularity learning. To further enable unified perception,
we adopt a conflict-free two-stage design that progressively extends text-level
multi-granularity understanding to pixel-level grounding: (i) the first stage eq
uips the model with fine-grained text-level reasoning across multiple IQA tasks,
 and (ii) the second stage introduces a training-free text-to-point grounding pa
radigm, which bridges textual semantics and pixel-level perception by mapping to
ken logits to spatial coordinates. Based on these efforts, we achieve IQA-Spider
 with unified multi-granularity explainable image quality assessment. Extensive
experiments across multiple benchmarks demonstrate strong performance, validatin
g the effectiveness and versatility of the proposed formulation and framework. O
ur code and dataset will be released at: \href{https://github.com/Helen1p/IQA-Sp
ider.git}{https://github.com/Helen1p/IQA-Spider.git}.
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StableVLA: Towards Robust Vision-Language-Action Models without Extra Data

YIYANG FU, Chubin Zhang, Shukai Gong, Yufan Deng, Kaiwei Sun, Qiyang Min, Qibin
Hou, Yansong Tang, Jianan Wang, Daquan Zhou

It is infeasible to encompass all possible disturbances within the training data
set. This raises a critical question regarding the robustness of Vision-Language
-Action (VLA) models when encountering unseen real-world visual disturbances, pa
rticularly under imperfect visual conditions. In this work, we conduct a systema
tic study based on recent state-of-the-art VLA models and reveal a significant p
erformance drop when visual disturbances absent from the training data are intro

duced. To mitigate this issue, we propose a lightweight adapter module grounded in information theory, termed the Information Bottleneck Adapter (IB-Adapter), which selectively filters potential noise from visual inputs. Without requiring any extra data or augmentation strategies, IB-Adapter consistently improves over the baseline by an average of 30%, while adding fewer than 10M parameters, demonstrating notable efficiency and effectiveness. Furthermore, even with a 14x smaller backbone (0.5B parameters) and no pre-training on the Open X-Embodiment dataset, our model StableVLA achieves robustness competitive with 7B-scale state-of-the-art VLAs. With negligible parameter overhead (<10M), our approach maintains accuracy on long-horizon tasks and surpasses OpenPi under real-world visual disturbances. The code will be made publicly available.
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QEDBench: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs
Santiago Gonzalez, Alireza Amiri Bavandpour, Peter Ye, Edward Zhang, Ruslans Aleksejevs, Todor Anti■, Polina Baron, Sujeet Bhalerao, Shubhrajit Bhattacharya, Zachary Burton, John Byrne, Hyungjun Choi, Nujhat Ahmed Disha, Koppány István Encz, Yuchen Fang, Robert Joseph George, Ebrahim Ghorbani, Alan Goldfarb, Jing Guo, Meghal Gupta, Stefano Huber, Annika Kanckos, Minjung Kang, Hyun Jong Kim, Dino Lorenzini, Levi Lorenzo, Tianyi Mao, Giovanni Marzenta, Ariane M. Masuda, Lukas Mauth, Ana Mickovic, Andrés Miniguano-Trujillo, Antoine Moulin, Wenqi Ni, Tomos Parry, Kevin Ren, Hossein Roodbarani, Mathieu Rundström, Manjil Saikia, Detchat Samart, Rebecca Steiner, Connor Stewart, Dhara Thakkar, Jeffrey Tse, Vasiliki Velona, Yunhai Xiang, Sibel Yalç■n, Jun Yan, Ji Zeng, Arman Cohan, Quanquan C. Liu
As Large Language Models (LLMs) saturate elementary benchmarks, the research frontier has shifted from generation to the reliability of automated evaluation. We demonstrate that standard "LLM-as-a-Judge" protocols suffer from a systematic evaluation Alignment Gap when applied to upper-undergraduate to early graduate level mathematics. To quantify this, we introduce QEDBench, the first benchmark to systematically measure alignment with human experts on undergraduate-level math proofs by contrasting course-specific rubrics against expert common knowledge criteria. By deploying a dual-evaluation matrix (7 judges $\times$ 5 solvers) against 1,000+ hours of human evaluation, we reveal that certain frontier evaluators like Claude 4.5 Opus exhibit significant positive bias (up to $+0.28$ mean score inflation), effectively "hallucinating rigor" in flawed proofs. Furthermore, we uncover a critical reasoning disparity: while Gemini 3.0 Pro achieves state-of-the-art performance (0.91 raw score), specialized reasoning models like o3-deep-research collapse in discrete domains, dropping to 42.1\% accuracy in Graph Theory. We release QEDBench as a public benchmark for evaluating and improving AI judges.
****
AgentWebBench: Benchmarking Multi-Agent Coordination in Agentic Web
Shanshan Zhong, Kate Shen, Chenyan Xiong
Agentic Web is an emerging paradigm where autonomous agents help users use online information. As the paradigm develops, content providers are also deploying agents to manage their data and serve it through controlled interfaces. This shift moves information access from centralized retrieval to decentralized coordination. To study this setting, we introduce AgentWebBench, a benchmark that evaluates how well a user agent synthesizes answers by interacting with website-specific content agents. We evaluate four tasks that cover common web information needs, spanning ranked retrieval (web search, web recommendation) and open-ended synthesis (question answering, deep research). Across seven advanced LLMs and three coordination strategies, multi-agent coordination generally lags behind centralized retrieval as expected, because user agent cannot directly access the corpus, but the gap shrinks with model scale and can even outperform centralized retrieval on question answering. This benchmark also enables us to study properties of the emerging paradigm of the digital world. We find that decentralized access concentrates traffic toward a small set of websites, test time scaling improves both interaction reliability and task performance, and strong results require sufficient interactions guided by careful planning. Finally, our failure analysis su

ggests that user agents need better planning and answer synthesis, while content agents need more reliable retrieval and evidence quality. Code, data, and APIs are released on https://github.com/cxcscmu/AgentWebBench.
****
Prototype-guided Bilateral Alignment Multimodal Federated Learning
Tianchi Liao, Lele Fu, Sheng Huang, Qing Hu, Hong-Ning Dai, Chuan Chen
Multimodal federated learning (MFL) has emerged as a pivotal paradigm for leveraging distributed data to enhance model performance. However, existing methods predominantly rely on idealized assumptions of model homogeneity and balanced modality distributions, rendering them ill-suited for practical scenarios characterized by heterogeneous client architectures and severe modality imbalance. To address these challenges, we propose a \textbf{M}ultimodal \textbf{Fed}erated learning Prototype-guided Bilateral Alignment (MFedPBA) framework. MFedPBA facilitates robust knowledge synergy through a dual alignment mechanism: (i) at the feature level, it aligns heterogeneous feature spaces via a projection encoder optimized by contrastive learning and the Gromov-Wasserstein distance; (ii) at the decision level, it employs an entropy-weighted aggregation of naturally aligned logit prototypes. This novel design achieves robust MFL by jointly tackling heterogeneous feature spaces and collectively aggregating decisions.
Extensive experiments demonstrate that our method significantly outperforms state-of-the-art baselines under conditions of model heterogeneity and modality imbalance.
****
Chebyshev Policies and the Mountain Car Problem: Reinforcement Learning for Low-Dimensional Control Tasks
Stefan Huber, Hannes Unger, Georg Schäfer, Jakob Rehrl
We analytically solve the Mountain Car problem, a canonical benchmark in RL, and derive an optimal control solution, closing a gap after 36 years. This enables us to reveal two surprising insights: The optimal control is quite simple, yet modern RL agents display a large gap to optimality. Motivated by the analysis of the optimal control, we introduce Chebyshev policies as a universal (i.e. dense) class of RL policies from first principles. They can be trained as drop-in replacements of neural nets, reducing the regret by a factor of 4.18, while requiring 277 times fewer parameters, fostering sample efficiency, explainability and realtime capability. Chebyshev policies are evaluated on further RL tasks, including a real-world nonlinear motion control testbed. They consistently improve performance over neural nets with PPO, ARS and REINFORCE. Our results demonstrate how Chebyshev policies offer a compelling and lightweight alternative or addition to neural nets for low-dimensional control tasks.
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Riemannian stochastic optimization for sufficient dimension reduction
Thibault Pautrel, François Portier
Sufficient dimension reduction (SDR) makes high-dimensional regression tractable by projecting the covariates onto a low-dimensional subspace that preserves the conditional mean of the response. Existing gradient-based estimators either operate in the ambient space and suffer from the curse of dimensionality, or localize in the reduced space at a per-outer-iteration cost at least quadratic in the sample size. We show that minimizers of the population Minimum Average Variance Estimation (MAVE) risk approximate the same Grassmannian target as the Outer Product of Gradients (OPG), and recast the empirical criterion as a smooth maximization on the Stiefel manifold with closed-form Riemannian gradient. The resulting algorithm, SMAVE, combines sparse projected-space nearest-neighbor localization with Riemannian stochastic gradient ascent. A simplified version comes with almost-sure convergence and a non-asymptotic rate matching the standard non-convex stochastic first-order scaling. Empirically,
SMAVE matches or improves on RMAVE's synthetic subspace recovery at moderate-to-high ambient dimension, and on four real datasets it uniformly improves over OPG and is competitive with or outperforms

RMAVE at orders of magnitude lower runtime.
****
DRIVE: Best Data Scheduling Practices for Reinforcement Learning with Verifiable Reward in Competitive Code Generation
Speed Zhu, Chuheng Zhang, Jianwei Cai, Guang Chen, Lulu Wu, Xiaolong Xu, Xuyun Zhang, Saiyong Yang, Wiggin Zhou
Recent success of large reasoning models (such as OpenAI o1 and DeepSeek R1) have spurred a resurgence of interest in reinforcement learning from verifiable rewards (RLVR). However, progress is still largely driven by RL algorithm design, while data scheduling -- the data-side decisions that determine what the model trains on over time -- is critical but remains underexplored. Therefore, data scheduling becomes the focus of this paper, including how to curate data for supervised fine-tuning (SFT) and how to select prompts and collect rollouts for reinforcement learning (RL). We introduce a pipeline with careful designs on data scheduling, consisting of hardness-prioritized SFT and two-stage RL. Specifically, we first fine-tune the base model on supervision data that is curated to prioritize difficulty based on both arena learning and classification. Then, we introduce two-stage RL where a decreased max sequence length during rollout is used in the first stage to expand entropy and reduce repetition, and a large number rollouts per prompt and curriculum design are adopted in the second stage to encourage exploration for challenging problems. We implement this pipeline on Qwen2.5-32B and an internal 389B MoE model, and evaluate them on a wide range of benchmarks including challenging LeetCode and Codeforces weekly contests. The results not only indicate the effectiveness and scalability of our pipeline but also demonstrate our model achieve sota of 32B models in competitive code generation.
****
ReQAT: Achieving Full-Precision Reasoning Accuracy with 4-bit Floating-Point Quantization-Aware Training
Janghwan Lee, Sihwa Lee, Jinseok Kim, Yongjik Kim, Jieun Lim, Jinwook Oh, Jungwook Choi
Large Reasoning Models (LRMs) achieve strong problem-solving through long chain-of-thought, but their deployment is constrained by the high cost of full-precision inference and growing KV cache footprints. Microscaled FP4 formats enable efficient FP4 deployment; however, fully quantizing weights, activations, and KV caches (W4A4KV4) causes severe reasoning degradation that existing PTQ and QAT fail to recover. We identify that FP4 failures concentrate on low-entropy tokens—precise symbolic commitments such as digits and operators—where quantization noise inflates sampling errors that cascade through reasoning traces. Based on this insight, we propose ReQAT, a reasoning-centric FP4 training framework with three components: (i) Trace-Aligned QAT (TAQ), which revisits identical reasoning traces to focus updates on critical low-entropy decisions; (ii) Selective Entropy Minimization (SEM), which reinforces confidence at low-entropy positions; and (iii) Q-FIT, a quantization-friendly initialization that jointly calibrates RoPE-consistent KV cache transformations to stabilize QAT. Under the same training budget, ReQAT not only recovers but surpasses BF16 fine-tuning accuracy, while delivering up to $3.9\times$ throughput speedup on NVIDIA DGX Spark and $3.1\times$ on B200. The project repository is available at https://github.com/aiha-lab/ReQAT.
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QiMeng-PerceptOS: Semantic-Aware Kernel Optimization for OS-Intensive Workloads via Hardware-Software Alignment
Huilai Chen, Yuanbo Wen, Liangfeng Li, Shaohui Peng, Jingzhe Zhu, Jun Bi, Xuzhi Zhang, Qi Guo, Ling Li, Yunji Chen
Optimizing OS kernels for specific applications is vital for peak performance, yet existing LLM-based methods struggle with a semantic mismatch between generalized reasoning and low-level system behaviors. As a result, these static, open-loop approaches suffer from runtime blindness, configuration fragmentation, and search drift, ultimately failing to unlock the system’s performance potential. To address this, we propose QiMeng-PerceptOS, an autonomous framework that shifts the paradigm to perception-driven tuning. QiMeng-PerceptOS integrates: (1) a Perception Module that aligns raw telemetry into high-fidelity semantic fingerprints

; (2) a Global Search Module utilizing a Bi-level Hierarchical Induction Tree (BHIT) for global navigation and efficient pruning; and (3) a Posterior Enhancement Module to suppress hallucinations via trajectory synthesis. Experiments across diverse workloads show that it achieves significant performance breakthroughs by optimizing kernel configurations, reaching 296.6% of default Redis throughput and surpassing SOTA baselines by 32.6% within only 15 iterations. By establishing a perception-driven closed-loop, QiMeng-PerceptOS provides new insights for fully automated, large-scale system optimization.

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AVGen-Bench: A Task-Driven Benchmark for Multi-Granular Evaluation of Text-to-Audio-Video Generation

Ziwei Zhou, Zeyuan Lai, Rui Wang, Yifan Yang, Yuqing Yang, Qi Dai, Lili Qiu, Chong Luo

Text-to-Audio-Video (T2AV) generation is rapidly becoming a core interface for media creation, yet its evaluation remains fragmented.

Existing benchmarks largely assess audio and video in isolation or rely on coarse embedding similarity, failing to capture fine-grained joint correctness required by realistic prompts.

We introduce AVGen-Bench, a task-driven benchmark for T2AV generation, featuring high-quality prompts across 11 real-world categories.

To support comprehensive assessment, we propose a multi-granular evaluation framework that combines lightweight specialist models with Multimodal Large Language Models (MLLMs), enabling evaluation from perceptual quality to fine-grained semantic controllability.

Our evaluation reveals a pronounced gap between strong audio-visual aesthetics and weak semantic reliability, including persistent failures in text rendering, speech coherence, physical reasoning, and universal breakdown in musical pitch control.

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Learning Adaptive Perturbation-Conditioned Contexts for Robust Transcriptional Response Prediction

Yinhua Piao, Hyomin Kim, Seonghwan Kim, Yunhak Oh, Junhyeok Jeon, Sang-Yeon Hwang, Jaechang Lim, Woo Youn Kim, Chanyoung Park, Sungsoo Ahn

Predicting high-dimensional transcriptional responses to genetic perturbations is challenging because signals are sparse and experimental noise is severe. Existing methods often suffer from mean collapse, achieving high correlation by predicting the global average expression rather than perturbation-specific responses, which yields false positives and poor interpretability. Methods that add biological knowledge graphs typically treat them as dense, static priors shared across perturbations, propagating noise. We propose AdaPert, which counters mean collapse by extracting a sparse, perturbation-specific subgraph via differentiable node selection, then suppressing spurious variation in non-responsive genes while emphasizing differentially expressed ones. Across multiple benchmarks, AdaPert outperforms existing baselines, with the largest gains on DEG-aware metrics.

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Affine-Scaled Attention: Towards Flexible and Stable Transformer Attention

Jeongin Bae, Baeseong park, Gunho Park, Minsub Kim, Joonhyung Lee, Junhee Yoo, Sunghyeon Woo, Jiwon Ryu, Se Jung Kwon, Dongsoo Lee

Transformer attention is typically implemented using softmax normalization, which enforces attention weights with unit sum normalization. While effective in many settings, this constraint can limit flexibility in controlling attention magnitudes and may contribute to overly concentrated or unstable attention patterns during training. Prior work has explored modifications such as attention sinks or gating mechanisms, but these approaches provide only limited or indirect control over attention reweighting. We propose Affine-Scaled Attention, a simple extension to standard attention that introduces input-dependent scaling and a corresponding bias term applied to softmax-normalized attention weights. This design relaxes the strict normalization constraint while maintaining aggregation of value representations, allowing the model to adjust both the relative distribution and the scale of attention in a controlled manner.

We empirically evaluate Affine-Scaled Attention in large-scale language model pr
etraining across multiple model sizes. Experimental results show consistent impr
ovements in training stability, optimization behavior, and downstream task perfo
rmance compared to standard softmax attention and attention sink baselines. Thes
e findings suggest that modest reweighting of attention outputs provides a pract
ical and effective way to improve attention behavior in Transformer models.
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Foundations of Equivariant Deep Learning: Unifying Graph and Sheaf Neural Networ
ks
Yoshihiro Maruyama
Symmetry is everywhere in nature and society. Geometric deep learning exploits s
ymmetries in data to improve the performance and efficiency of deep learning sys
tems. In this paper, we extend geometric deep learning to utilize richer symmetr
y structures. Specifically, we develop order-equivariant neural networks (OENN),
 which generalize standard graph message passing and sheaf neural networks via t
he theory of equivariant bundles over face posets (face categories). We (i) char
acterize all linear order-equivariant maps, (ii) build OENN layers, and (iii) pr
ove universal approximation theorems (UATs) for continuous order-equivariant map
s, which are new results even when restricted to sheaf neural networks (for whic
h no UAT was known before). We illustrate the framework on graph and sheaf model
s. Our results can also be seen as extending the known UAT for graph neural netw
orks to a more general setting that subsumes sheaf neural networks as well. In a
ddition, we show that OENN can be extended further to CENN, Category-Equivariant
 Neural Network, which gives the general form of equivariant neural networks as
well as of equivariant universal approximation theorems, allowing us to leverage
 categorical symmetry in data (e.g., non-invertible symmetries on multiple objec
ts with compositional relations on those symmetries).
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Diffusion Bridge or Flow Matching? A Unifying Framework and Comparative Analysis
Kaizhen Zhu, Mokai Pan, Zhechuan Yu, Jingya Wang, Jingyi Yu, Ye Shi
Diffusion Bridge and Flow Matching have both demonstrated compelling empirical p
erformance in transformation between arbitrary distributions. However, there rem
ains confusion about which approach is generally preferable, and the substantial
 discrepancies in their modeling assumptions and practical implementations have
hindered a unified theoretical account of their relative merits. We have, for th
e first time, provided a unified theoretical and experimental validation of thes
e two models. We recast their frameworks through the lens of Stochastic Optimal
Control and prove that the cost function of the Diffusion Bridge is lower, guidi
ng the system toward more stable and natural trajectories. Simultaneously, from
the perspective of Optimal Transport, interpolation coefficients t and $1-t$ o
f Flow Matching become increasingly ineffective when the training data size is r
educed. To corroborate these theoretical claims, we propose a novel, powerful ar
chitecture for Diffusion Bridge built on a latent Transformer, and implement a F
low Matching model with the same structure to enable a fair performance comparis
on in various experiments. Comprehensive experiments are conducted across Image
Restoration, Translation, and Style Transfer tasks, systematically varying both
the distributional discrepancy (different difficulty) and the training data size
. Extensive empirical results align perfectly with our theoretical predictions a
nd allow us to delineate the respective advantages and disadvantages of these tw
o models. Our code is available at \url{https://github.com/zhukaizhen/diffusion_
bridge_flow_matching}.
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The Geometry of Representational Failures in Vision Language Models
Daniele Savietto, Declan Iain Campbell, André Panisson, Marco Nurisso, Giovanni
Petri, Jonathan D. Cohen, Alan Perotti
Vision-Language Models (VLMs) exhibit puzzling failures in multi-object visual t
asks, such as hallucinating non-existent elements or failing to identify the mos
t similar objects among distractions. While these errors mirror human cognitive
constraints, such as the "Binding Problem'', the internal mechanisms driving the

m in artificial systems remain poorly understood. Here, we propose a mechanistic insight by analyzing the representational geometry of open-weight VLMs (Qwen, InternVL, Gemma), comparing methodologies to distill "concept vectors'' - latent directions encoding visual concepts. We validate our concept vectors via steering interventions that reliably manipulate model behavior in both simplified and naturalistic vision tasks (e.g., forcing the model to perceive a red flower as blue). We observe that the geometric overlap between these vectors strongly correlates with specific error patterns, offering a grounded quantitative framework to understand how internal representations shape model behavior and drive visual failures.

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Rethinking Temporal Consistency in Video Object-Centric Learning: From Prediction to Correspondence

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Pekka Marttinen, Joni Pajarinen

The de facto approach in video object-centric learning maintains temporal consistency through learned dynamics modules that predict future object representations, called slots. We demonstrate that these predictors function as expensive approximations of discrete correspondence problems. Modern self-supervised vision backbones already encode instance-discriminative features that distinguish objects reliably. Exploiting these features eliminates the need for learned temporal prediction. We introduce Grounded Correspondence, a framework that replaces learned transition functions with deterministic bipartite matching. Slots initialize from salient regions in frozen backbone features. Frame-to-frame identity is maintained through Hungarian matching on slot representations. The approach requires zero learnable parameters for temporal modeling yet achieves competitive performance on MOVi-D, MOVi-E, and YouTube-VIS. Project page: https://magenta-sherbet-85b101.netlify.app/

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AICrypto: Evaluating Cryptography Capabilities of Large Language Models

Yu Wang, Yijian Liu, Liheng Ji, Han Luo, Wenjie Li, Xiaofei Zhou, Chiyun Feng, Puji Wang, Yuhan Cao, Geyuan Zhang, Xiaojian Li, Rongwu Xu, Yilei Chen, Tianxing He

We build \textbf{AICrypto}, a comprehensive benchmark designed to evaluate the cryptography capabilities of large language models (LLMs). The benchmark comprises 135 multiple-choice questions, 150 capture-the-flag challenges, and 30 proof problems, covering a broad range of skills from knowledge memorization to vulnerability exploitation and formal reasoning. All tasks are carefully reviewed or constructed by cryptography experts to improve correctness and rigor. For each proof problem, we provide detailed scoring rubrics and reference solutions that enable automated grading, achieving high correlation with human expert evaluations. We introduce strong human expert performance baselines for comparison across all task types. Our evaluation of 17 leading LLMs reveals that state-of-the-art models match or even surpass human experts in memorizing cryptographic concepts, exploiting common vulnerabilities, and routine proofs. However, our analysis reveals that they still lack a deep understanding of abstract mathematical concepts and struggle with tasks that require multi-step reasoning and dynamic analysis. We hope this work could provide insights for future research on LLMs in cryptographic applications. Our code and dataset are available at https://github.com/wangyu-ovo/aicrypto-agent.

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Is Your LLM Overcharging You? Tokenization, Transparency, and Incentives

Ander Artola Velasco, Stratis Tsirtsis, Nastaran Okati, Manuel Gomez Rodriguez

State-of-the-art large language models require specialized hardware and substantial energy to operate. Consequently, cloud-based services that provide access to these models have become very popular. In these services, the price users pay depends on the number of tokens a model uses to generate an output–they pay a fixed price per token.

In this work, we show that this pricing mechanism creates a financial incentive for providers to strategize and misreport the (number of) tokens a model used to

generate an output, and users cannot prove, or even know, whether a provider is overcharging them.

However, we also show that, if an unfaithful provider is obliged to be transparent about the generative process used by the model, misreporting optimally without raising suspicion is hard. Nevertheless, as a proof-of-concept, we develop an efficient heuristic algorithm that allows providers to significantly overcharge users without raising suspicion. Crucially, the cost of running the algorithm is lower than the additional revenue from overcharging users, highlighting the vulnerability of users under the current pay-per-token pricing mechanism. Further, we show that, to eliminate the financial incentive to strategize, a pricing mechanism must price tokens linearly on their character count. While this makes a provider's profit margin vary across tokens, we introduce a simple prescription that allows a provider to maintain their average profit margin when transitioning to an incentive-compatible pricing mechanism.

To complement our theoretical results, we conduct experiments with large language models from the $\texttt{Llama}$, $\texttt{Gemma}$ and $\texttt{Ministral}$ families, and prompts from a popular benchmarking platform.

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On the Power of Statistics in Class-Incremental Learning with Pretrained Models

Zhiwen Cao, Yanfeng Li, Shu-Dong Huang, Yalan Ye, Shuyin Xia, Yi Wang, Jiancheng Lv

Recent class-incremental learning (CIL) methods built on large pre-trained vision models have shown that strong performance can be retained even under strict data access constraints. This raises a fundamental question: which properties of pre-trained representations make such recovery possible in the class-incremental setting? In this work, we show that class-level feature statistics play a central role in enabling effective CIL under strong pre-training. When the visual backbone is frozen, maintaining simple class-wise statistical estimators of features can recover a substantial fraction of the performance achieved by static joint training across diverse benchmarks. We make this observation explicit through deliberately minimal reference points built on frozen CLIP representations. In particular, we demonstrate that competitive performance can be achieved without continual parameter updates, by performing class-incremental inference based solely on class-level statistical estimators instantiated from frozen features. Our findings suggest that class-level statistics constitute an important and previously underemphasized component of recent PTM-based CIL approaches, offering a complementary perspective for understanding their strong empirical performance. Our code is available at https://github.com/HdTgon/baseCIL.

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Overthinking: Amplifying Reasoning Weights to Extract Learned Secrets

Jack Hopkins, Dipika Khullar, Fabien Roger

Black box auditing of language models is an essential pre-deployment tool, but it may miss subtle forms of misalignment and hidden information. To better elicit hidden information during an auditing process, we introduce *overthinking*: the process of using reasoning task vectors to amplify the chain-of-thought faithfulness of reasoning models. Given the parameters of a base instruct model M and reasoning-distilled model R, we define the *overthinking model* as $\mathcal{O}_\alpha = M + \alpha(R - M)$, where $\alpha > 1$ amplifies reasoning beyond the pure reasoning model R. Additionally, we introduce new layer-wise attenuation strategies that selectively amplify reasoning without losing quality and coherence of model outputs. We demonstrate that overthinking models are more likely to reveal hidden information across four experimental settings, across 2B-32B models. Our findings suggest that reasoning amplification may surface secrets or unintended behaviors acquired during training up to $10\times$ more frequently than the original reasoning model. How secrets surface depends on the secret type: some require perturbation along the reasoning direction, while others yield to any sufficiently large weight perturbation.

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GP2F: Cross-Domain Graph Prompting with Adaptive Fusion of Pre-trained Graph Neural Networks

Dongxiao He, Wenxuan Sun, Yongqi Huang, Jitao Zhao, Di Jin
Graph Prompt Learning (GPL) has recently emerged as a promising paradigm for dow nstream adaptation of pre-trained graph models, mitigating the misalignment betw een pre-training objectives and downstream tasks. Recently, the focus of GPL has shifted from in-domain to cross-domain scenarios, which is closer to the real w orld applications, where the pre-training source and downstream target often dif fer substantially in data distribution. However, why GPLs remain effective under such domain shifts is still unexplored. Empirically, we observe that representa tive GPL methods are competitive with two simple baselines in cross-domain setti ngs: full fine-tuning (FT) and linear probing (LP), motivating us to explore a d eeper understanding of the prompting mechanism. We provide a theoretical analysi s demonstrating that jointly leveraging these two complementary branches yields a smaller estimation error than using either branch alone, formally proving that cross-domain GPL benefits from the integration between pre-trained knowledge an d task-specific adaptation. Based on this insight, we propose GP2F, a dual-branc h GPL method that explicitly instantiates the two extremes: (1) a frozen branch that retains pre-trained knowledge, and (2) an adapted branch with lightweight a dapters for task-specific adaptation. We then perform adaptive fusion under topo logy constraints via a contrastive loss and a topology-consistent loss. Extensiv e experiments on cross-domain few-shot node and graph classification demonstrate that our method outperforms existing methods.
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FedSSM: State Space Model-based Proactive Inference for Heterogeneous Multimodal Federated Learning
Hengyi Ren, Yuchen Xie, Changlong Wang, Xin Li, Yue Huang, Jian Guo, Lijuan Sun
Multimodal Federated Learning (MMFL) addresses collaborative training across cli ents with heterogeneous modality configurations, where effective client selectio n becomes critical under the compounded challenges of modality, distribution, an d quantity heterogeneity. Existing selection methods operate within a reactive p aradigm, responding to current observations without anticipating how decisions i nfluence future optimization trajectories. This myopic approach leads to subopti mal convergence when training dynamics shift rapidly under severe heterogeneity. We propose FedSSM, which reconceptualizes client selection as a proactive decis ion-making process by predicting training dynamics through decision-aware state space models. The prediction error yields a \emph{surprise} signal that quantifi es uncertainty and governs adaptive participation budgets and exploration-exploi tation trade-offs via counterfactual reasoning over candidate actions. For aggre gation, we introduce trust-weighted fusion with modality-specific routing, where surprise calibrates sensitivity to client anomalies. Experiments on four multim odal benchmarks demonstrate that FedSSM achieves 2.5--4.5\% accuracy improvement s over state-of-the-art methods while reducing communication rounds by over 30\% .
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A Dirac-Frenkel-Onsager Principle: Instantaneous Residual Minimization with Gaug e Momentum for Nonlinear Parametrizations of PDE Solutions
Matteo Raviola, Benjamin Peherstorfer
Dirac-Frenkel instantaneous residual minimization evolves nonlinear parametrizat ions of PDE solutions in time, but ill-conditioning can render the parameter dyn amics non-unique. We interpret this non-uniqueness as a gauge freedom: nullspace directions that leave the
time derivative unchanged can be used to select better-conditioned parameter vel ocities. Building on Onsager's minimum-dissipation principle, we introduce a his tory variable---interpretable as momentum---and inject it only along the nullspa ce directions. The resulting Dirac-Frenkel-Onsager dynamics preserve instantaneo us residual minimization, in contrast to standard regularization that can introd uce bias, while promoting temporally smooth parameter evolution. Examples demons trate that the approach leads to increased robustness in singular and near-singu lar regimes.
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Neural Quantum States in Mixed Precision

Massimo Solinas, Agnes Valenti, Nawaf Bou-Rabee, Roeland Wiersema

Scientific computing has long relied on double precision (64-bit floating point) arithmetic to guarantee accuracy in simulations of real-world phenomena. However, the growing availability of hardware accelerators such as Graphics Processing Units (GPUs) has made low-precision formats attractive due to their superior performance, reduced memory footprint, and improved energy efficiency. In this work, we investigate the role of mixed-precision arithmetic in neural-network based Variational Monte Carlo (VMC), a widely used method for solving computationally otherwise intractable quantum many-body systems. We first derive general analytical bounds on the error introduced by reduced precision on Metropolis-Hastings MCMC, and then empirically validate these bounds on the use-case of VMC. We demonstrate that significant portions of the algorithm, in particular, sampling the quantum state, can be executed in half precision without loss of accuracy. More broadly, this work provides a theoretical framework to assess the applicability of mixed-precision arithmetic in machine-learning approaches that rely on MCMC sampling. In the context of VMC, we additionally demonstrate the practical effectiveness of mixed-precision strategies, enabling more scalable and energy-efficient simulations of quantum many-body systems.

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T^{2}PO: Uncertainty-Guided Exploration Control for Stable Multi-Turn Agentic Reinforcement Learning

Haixin Wang, Hejie Cui, Chenwei Zhang, Xin Liu, Shuowei Jin, Shijie Geng, Xinyang Zhang, Nasser Zalmout, Zhenyu Shi, Yizhou Sun

Recent progress in multi-turn reinforcement learning (RL) has significantly improved reasoning LLMs' performances on complex interactive tasks.
Despite advances in stabilization techniques such as fine-grained credit assignment and trajectory filtering, instability remains pervasive and often leads to training collapse.
We argue that this instability stems from inefficient exploration in multi-turn settings, where policies continue to generate low-information actions that neither reduce uncertainty nor advance task progress.
To address this issue, we propose Token- and Turn-level Policy Optimization (T^{2}PO), an uncertainty-aware framework that explicitly controls exploration at fine-grained levels. At the token level, T^{2}PO monitors uncertainty dynamics and triggers a thinking intervention once the marginal uncertainty change falls below a threshold.
At the turn level, T^{2}PO identifies interactions with negligible exploration progress and dynamically resamples such turns to avoid wasted rollouts.
We evaluate T^{2}PO in diverse environments, including WebShop, ALFWorld, and Search QA, demonstrating substantial gains in training stability and performance improvements with better exploration efficiency. Code is available at https://github.com/WillDreamer/T2PO.

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Unlocking Cross-Modal Biosignal Synthesis: A Temporally-Aware VAE-Diffusion Model

Chenyang Xu, Dezhen Wang, Hao Wang

Synthesizing authentic phonocardiograms (PCG) from ubiquitous electrocardiograms (ECG) is a critical task for accessible cardiac monitoring. Existing generative models, however, struggle to capture the heart's complex electromechanical coupling, failing to meet the dual requirements of temporal precision and physiological fidelity needed for clinically relevant waveform analysis. We introduce the Temporally-Aware VAE-Diffusion model, a synergistic hybrid architecture that resolves this trade-off. Our architecture enforces tight physiological coupling through an Enhanced Condition Fusion mechanism and explicitly models long-range cardiac dynamics via Temporal Attention Blocks. On the EPHNOGRAM benchmark, our model sets a new state-of-the-art, achieving a Pearson correlation of 0.910$\pm$0.008, 95.95\% S1 detection accuracy, and a precise 12.0 ms timing error, significantly outperforming leading diffusion and Transformer baselines. Crucially, our work provides a reproducible zero-shot transfer evaluation for ECG-to-PCG synthesis. Evaluated on the synchronized PhysioNet/CinC 2016 training-a/MITHSDB subset

without target-domain training, our model preserves high waveform fidelity and c
linically relevant timing structure under domain shift, including on pathologica
l recordings. These results support cross-dataset robustness of the proposed syn
thesis framework, while downstream diagnostic validation remains an important di
rection for future work.
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Probabilistically-routed Bayesian Additive Spanning Trees for Learning on Constr
ained Domains

Abhisek Chakraborty, Abhishek Mandal, Anirban Chakraborty

Bayesian additive spanning tree (BAST) is an useful tool for interpretable, non-
parametric regression on complex constrained domains. It improves upon the perfo
rmance of Bayesian additive regression trees (BART) by replacing axis-aligned sp
lits through binary tree components by cuts on a spanning tree components, enabl
ing the formation of contiguous splits that respect the underlying complex struc
ture. While BAST is effective for learning on constrained spaces, it still relie
s on hard partitions, albeit on spanning trees, which limits its ability to repr
esent smoothly varying functions on constrained domains. We propose Probabilisti
cally-routed Bayesian additive spanning trees (PR-BAST), a principled relaxation
 that replaces hard cuts on spanning tree components with probabilistic routing
 along spanning tree components. PR-BAST represents the regression surface as a
n additive ensemble of such spanning tree-aligned smooth components. Conditional
 on a fixed spanning tree, each component in PR-BAST induces a Gaussian random f
ield with a sparse, tree-structured precision matrix, enabling scalable posterio
r computation via sparse linear algebra. We theoretically establish that PR-BAS
T yields strictly faster posterior contraction rates compared to BAST under grap
h-smooth truth. Experiments on synthetic and real datasets demonstrate that PR-B
AST consistently improves accuracy over BAST and other baselines, while retainin
g the interpretability of tree-based models.
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CaliDist: Calibrating Large Language Models via Behavioral Robustness to Distrac
tion

Mohammad Anas Jawad, Cornelia Caragea

Existing calibration methods for Large Language Models (LLMs) often overlook a c
ritical dimension of trustworthiness: a model's {\em behavioral robustness} to i
rrelevant or misleading information. In this paper, we argue that a model's true
 confidence should reflect its stability under cognitive pressure. We introduce
\textsc{CaliDist}, a novel post-hoc calibration approach that directly measures
and penalizes a model's susceptibility to distraction. \textsc{CaliDist} quantif
ies how an LLM's predictions and uncertainty change when its input prompt is per
turbed with semantic \textit{distractors}. This stability (or lack thereof) sign
al is then used to adaptively scale the model's initial confidence score.
Our extensive experiments on seven Natural Language Understanding classification
 benchmarks using six distinct LLMs show that \textsc{CaliDist} consistently ach
ieves lower Expected Calibration Error (ECE) and Brier Score compared with stron
g baselines. Remarkably, our method reduces the ECE from 23\% to 7\% on average—
a relative improvement of 70\%—demonstrating that behavioral stability is a powe
rful signal for calibration. We make our code and datasets available at \url{git
hub.com/m-anas-j/CaliDist}.
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Causal-aware Anomaly Detection for Tabular Data

Dang Nguyen, Tu Anh Hoang Nguyen, Thuc Duy Le, Svetha Venkatesh, Trung Le, Sunil
 Gupta

Anomaly detection (AD) methods often ignore causal dependencies and treat anomal
ies as outliers, which is brittle when anomalies are primarily mechanism violati
ons rather than extreme values. We propose CausalAno, a causal-aware detector th
at trains a causal GAN on normal data and leverages its discriminator to learn m
echanism-consistent representations. We score test samples by fitting a Gaussian
 model in this feature space and computing the Mahalanobis distance, measuring d
eviation from the normal causal manifold. We demonstrate the effectiveness of Ca
usalAno with extensive experiments on 28 tabular datasets (18 continuous-only an

d 10 mixed-type), comparing against 16 SOTA baselines. Our results show consiste nt improvements across both mixed-type and numerical-only settings. Our ablation studies further confirm that the gains come from the causal factorization in th e causal GAN rather than a generic GAN backbone. Our CausalAno offers a practica l and effective solution for real-world applications where anomalies often arise from mechanism-violating behaviors.
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Agent0-VL: Exploring Self-Evolving Agent for Tool-Integrated Vision-Language Rea soning
Jiaqi Liu, Kaiwen Xiong, Peng Xia, Yiyang Zhou, Haonian Ji, Lu Feng, Siwei Han, Mingyu Ding, Huaxiu Yao
Large Vision-Language Models (LVLMs) have achieved remarkable progress in multim odal reasoning tasks; however, their learning remains constrained by the limitat ions of human-annotated supervision. Recent self-rewarding approaches attempt to overcome this constraint by allowing models to act as their own critics or rewa rd providers. Yet, purely text-based self-evaluation struggles to verify complex visual reasoning steps and often suffers from evaluation hallucinations. To add ress these challenges, inspired by recent advances in tool-integrated reasoning, we propose Agent0-VL, a self-evolving vision-language agent that achieves conti nual improvement with tool-integrated reasoning. Agent0-VL incorporates tool usa ge not only into reasoning but also into self-evaluation and self-repair, enabli ng the model to introspect, verify, and refine its reasoning through evidence-gr ounded analysis. It unifies two synergistic roles within a single LVLM: a Solver that performs multi-turn tool-integrated reasoning, and a Verifier that generat es structured feedback and fine-grained self-rewards through tool-grounded criti que. These roles interact through a Self-Evolving Reasoning Cycle, where tool-ba sed verification and reinforcement learning jointly align the reasoning and eval uation distributions for stable self-improvement. Through this zero-external-rew ard evolution, Agent0-VL aligns its reasoning and verification behaviors without any human annotation or external reward models, achieving continual self-improv ement. Experiments on chart reasoning, geometric problem solving, and visual sci entific analysis show that Agent0-VL achieves an 12.5% improvement over the Qwen -VL base model.
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Geodesic Flow Matching for Denoising High-Dimensional Structured Representations
Karim Habashy, Chris Eliasmith
Vector Symbolic Algebras (VSAs) enable robust neurosymbolic reasoning by encodin g information into high-dimensional distributed representations. For continuous domains, Spatial Semantic Pointers (SSPs) extend this framework by mapping varia bles onto precise toroidal manifolds. While generative models offer a promising avenue for cleaning up (denoising) these representations, standard approaches li ke Flow Matching assume a flat Euclidean geometry. We demonstrate that this assu mption fails for SSPs: Euclidean linear interpolants "cut through" the manifold' s interior, destroying the phase and magnitude structure required for accurate d ecoding. To resolve this, we employ Geodesic Flow Matching, adapting Riemannian transport dynamics to strictly restrict the denoising flow to the SSP manifold. We validate this approach in a Spiking Neural SLAM system, showing that manifold -aware cleanup stabilizes path integration against drift. The method achieves a 72\% reduction in tracking error and enables a 40\% increase in neural efficienc y compared to classical baselines.
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Quantifying Cross-Domain Knowledge Distillation in the Presence of Domain Shift
Xiangchao Li, Xiao Han, Qing Yang, Xin Tong
This paper presents a theoretical investigation into the generalization capabili ties of cross-domain knowledge distillation. Utilizing a high-dimensional asympt otic analysis of a linear teacher–student model, we characterize the excess risk while accounting for both model and covariate shifts. Our results provide a for mal guarantee for the efficacy of distillation: even when the source and target domains differ substantially, there still may exist a regime where the student m odel achieves superior generalization ability over the student-only baseline. Mo

reover, we identify a \textit{crossed double descent} phenomenon: the excess risk can vary non-monotonically with the teacher’s and student’s dimension-to-sample-size ratios. These results provide rigorous insight into when and why distillation helps across domains.
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Do LLMs “Feel”? Emotion Circuits Discovery and Control
Chenxi Wang, Yixuan Zhang, Ruiji Yu, Yufei Zheng, Lang Gao, Zirui Song, Zixiang Xu, Gus Xia, Huishuai Zhang, Dongyan Zhao, Xiuying Chen
As the demand for emotional intelligence in large language models (LLMs) grows, a key challenge lies in understanding the internal mechanisms that give rise to emotional expression and in controlling emotions in generated text. This study addresses three core questions: (1) Do LLMs contain context-agnostic mechanisms shaping emotional expression? (2) What form do these mechanisms take? (3) Can they be harnessed for universal emotion control? We first construct a controlled dataset, $\textit{SEV}$ (Scenario–Event with Valence), to elicit comparable internal states across emotions. Subsequently, we extract context-agnostic emotion directions that reveal consistent, cross-context encoding of emotion (Q1). We identify neurons and attention heads that locally implement emotional computation through analytical decomposition and causal analysis, and validate their causal roles via ablation and enhancement interventions. Next, we quantify each sublayer's causal influence on the model’s final emotion representation and integrate the identified local components into coherent global emotion circuits that drive emotional expression (Q2). Directly modulating these circuits achieves 99.65% emotion-expression accuracy on the test set, surpassing prompting- and steering-based methods (Q3). To our knowledge, this is the first systematic study to uncover and validate emotion circuits in LLMs, offering new insights into interpretability and controllable emotional intelligence.
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daVinci-Dev: Agent-native Mid-training for Software Engineering
Ji Zeng, Dayuan Fu, Tiantian Mi, Zhuang Yumin, Yaxing Huang, Xuefeng Li, Lyumanshan Ye, Muhang Xie, Qishuo Hua, Zhen Huang, Mohan Jiang, Hanning Wang, Jifan Lin, Yang Xiao, Jie Sun, Yunze Wu, Pengfei Liu
While the emerging field of agentic software engineering has spurred extensive research into post-training, this paradigm alone does not fully address the distribution mismatch between traditional static pre-training and dynamic deployment environments. In this paper, we instead investigate agentic mid-training as a scalable complementary approach.
Central to our approach is *agent-native data* comprising two complementary components: *contextually-native trajectories* that preserve the complete information flow an agent experiences, offering broad coverage and diversity; and *environmentally-native trajectories* whose observations stem from actual tool invocations and test executions, providing interaction authenticity.
On `SWE-Bench Verified`, our recipe outperforms the previous open software engineering mid-training recipe `Kimi-Dev` under two post-training settings with the same base model and agentic scaffold, while using fewer than half mid-training tokens (73.1B).
Furthermore, our 32B and 72B models achieve state-of-the-art resolution rates of **56.1\%** and **58.5\%** among open agentic recipes using agentic scaffolds, despite starting from non-coder `Qwen2.5` base models.
We also observe performance gains on general code generation and scientific benchmarks.
We open-source a significant portion of our datasets, recipes, and model checkpoints to facilitate further research.
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MedSIGHT: Towards Grounded Visual Comprehension in Medical Large Vision-Language Models
Aofei Chang, Le Huang, Alex James Boyd, Parminder Bhatia, Taha Kass-Hout, Fenglong Ma, Cao Xiao
Medical large vision-language models (Med-LVLMs) have recently achieved remarkable progress in vision–language comprehension and medical image segmentation. How

ever, existing models still struggle to unify these two capabilities, which is e ssential for achieving clinically reasoning that connects visual findings with s emantic interpretation. We present MedSIGHT, a unified framework that equips Med -LVLMs with structured, pixel-level understanding for grounded visual comprehens ion. MedSIGHT introduces a novel Region Perceiver module that produces region-ce ntric tokens, encoding spatial information directly into representation space of the language model. We further propose a medical region codebook into the LLM v ocabulary, allowing the model to generate discrete region codes as symbolic repr esentations of anatomical and pathological regions. These codes are decoded thro ugh the Region Perceiver to reconstruct segmentation mask, achieving end-to-end spatial grounding. Lastly, MedSIGHT combines Region Perceiver, Codebook and LLM using our proposed progressive training strategy to gradually aligns these modul es stably. Trained on only 72K multimodal instruction pairs, MedSIGHT achieves s tate-of-the-art performance across diverse imaging modalities on both medical co mprehension and segmentation tasks. Code and model are publicly available at [Gi tHub](https://github.com/Aofei-Chang/MedSIGHT).
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Proximal-IMH: Proximal Posterior Proposals for Independent Metropolis–Hastings w ith Approximate Operators
Youguang Chen, George Biros
We are considering the problem of sampling from a posterior distribution related to Bayesian inverse problems arising in science, engineering, and imaging. Our method belongs to the family of independence Metropolis--Hastings (IMH) samplin g algorithms. These are quite common in Bayesian inference. Relying on the exis tence of an approximate posterior distribution that is cheaper to sample from b ut can have significant bias, we introduce Proximal-IMH, a scheme that removes t his bias: it corrects samples from the approximate posterior solving an auxili ary optimization problem, yielding a local adjustment that trades off adherence to the exact model against stability around the approximate reference point. For idealized settings, we prove that the proximal correction tightens the match between approximate and exact posteriors, and thereby improves acceptance rates and mixing. The new method works with both linear and nonlinear input-output ope rators and is especially suitable for inverse problems where exact posterior sam pling is too expensive. We perform several numerical experiments that include mu ltimodal and data-driven priors and nonlinear input-output operators. The resu lts show that Proximal-IMH reliably outperforms existing IMH variants.
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Learn-to-learn on Arbitrary Textual Conditioning: A Hypernetwork-Driven Meta-gat ed LLM
Luo Ji, Qi Qin, Ningyuan Xi, Teng Chen, Qingqing Gu, Hongyan Li
Conventional LLMs may suffer from corpus heterogeneity and subtle changes in con ditions. While finetuning can create the catastrophe forgetting issue, applicati ons of meta-learning on LLMs are also limited due to their complexity and scalab ility. In this paper, we activate the meta-signal of β within the SwiGLU b locks, resulting in a meta-gating mechanism that adaptively adjusts the nonlinea rity of FFN. A hypernetwork is employed to dynamically produce β under tex tual conditions, providing meta-controllability over LLMs. By testing on differe nt condition types such as task, domain, persona, and style, our method outperfo rms finetuning and meta-learning baselines, and can generalize reasonably on uns een tasks, condition types, or instructions. Our codes are in https://github.com /AaronJi/MeGan.
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Principled Confidence Estimation for Deep Computed Tomography
Matteo Gätzner, Johannes Kirschner
We present a principled framework for confidence estimation in computed tomograp hy (CT) reconstruction. Based on the sequential likelihood mixing framework (Kir schner et al., 2025), we establish confidence regions with theoretical coverage guarantees for deep-learning-based CT reconstructions. We consider a realistic f orward model following the Beer-Lambert law, i.e., a log-linear forward model wi th Poisson noise, closely reflecting clinical and scientific imaging conditions.

The framework is general and applies to both classical reconstruction algorithms and deep learning methods alike, including U-Nets, U-Net ensembles, and generative Diffusion models. Empirically, we demonstrate that deep reconstruction methods yield substantially tighter confidence regions than classical reconstructions, without sacrificing theoretical coverage guarantees. Our approach allows the detection of hallucinations in reconstructed images and provides interpretable visualizations of confidence regions. This establishes deep models not only as powerful estimators, but also as reliable tools for uncertainty-aware medical imaging.
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Why Do We Need Warm-up? A Theoretical Perspective
Foivos Alimisis, Rustem Islamov, Aurelien Lucchi
Learning rate warm-up -- increasing the learning rate at the beginning of training -- has become a ubiquitous heuristic in modern deep learning, yet its theoretical foundations remain poorly understood. In this work, we provide a principled explanation for why warm-up improves training. We rely on a generalization of the (L_0, L_1)-smoothness condition, which bounds local curvature as a linear function of the loss suboptimality and exhibits desirable closure properties. We show -- both theoretically and empirically -- that this condition is satisfied by common neural architectures and accurately captures the curvature of the optimization landscape early in training. Adapting the learning rate in response to this curvature condition naturally induces a warm-up–like schedule, and we show that this choice yields provably faster convergence guarantees than using a fixed learning rate. Experiments on language and vision models show that the resulting one-parameter warm-up schedule can match tuned linear warm-up and improve over no warm-up.
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Think Fast and Slow: Step-Level Cognitive Depth Adaptation for LLM Agents
Ruihan Yang, Fanghua Ye, Xiang Wei, Ruoqing Zhao, Kang Luo, Xinbo Xu, Bo Zhao, Ruotian Ma, Shanyi Wang, Zhaopeng Tu, Xiaolong Li, Deqing Yang, Liefeng Bo
Large language models (LLMs) are increasingly deployed as autonomous agents for multi-turn decision-making tasks. However, current agents typically rely on fixed cognitive patterns: non-thinking models generate immediate responses, while thinking models engage in deep reasoning uniformly. This rigidity is inefficient for long-horizon tasks, where cognitive demands vary significantly from step to step, with some requiring strategic planning and others only routine execution. In this paper, we introduce CogRouter, a framework that trains agents to dynamically adapt cognitive depth at each step. Grounded in ACT-R theory, we design four hierarchical cognitive levels ranging from instinctive responses to strategic planning. Our two-stage training approach includes Cognition-aware Supervised Fine-tuning (CogSFT) to instill stable level-specific patterns, and Cognition-aware Policy Optimization (CoPO) for step-level credit assignment via confidence-aware advantage reweighting. The key insight is that appropriate cognitive depth should maximize the confidence of the resulting action. Experiments on ALFWorld and ScienceWorld demonstrate that CogRouter achieves state-of-the-art performance with superior efficiency.
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Reuse your FLOPs: Scaling RL on Hard Problems by Conditioning on Very Off-Policy Prefixes
Amrith Setlur, Zijian Wang, Andrew Cohen, Paria Rashidinejad, Sang Michael Xie
Typical reinforcement learning (RL) methods for LLM reasoning waste compute on hard problems, where correct on-policy traces are rare and policy gradients vanish. To bootstrap more efficient RL, we consider reusing old sampling FLOPs (from prior inference or RL training) in the form of off-policy traces. We introduce PrefixRL, where we condition on the prefix of successful off-policy traces and run on-policy RL to complete them, side-stepping instabilities from using off-policy data as supervision targets. PrefixRL boosts the learning signal on hard problems by modulating the difficulty of the problem through the off-policy prefix length. We prove that the PrefixRL objective is not only consistent with the standard RL objective but also more sample efficient. Empirically, we discover back

-generalization: training only on prefixed problems generalizes to out-of-distribution unprefixed performance, with learned strategies often differing from those in the prefix. In our experiments, we source the off-policy traces by rejection sampling with the base model, creating a self-improvement loop. On hard reasoning problems, PrefixRL reaches the same training reward 2x faster than the strongest baseline (SFT on off-policy data then RL), even after accounting for the compute spent on the initial rejection sampling, and increases the final reward by 3x.
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Bridging the Stability-Expressivity Gap: Synthetic Data Scaling and Preference Alignment for Low-Resource Spoken Language Models
Yizhong Geng, Yanliang Li, Jinghan Yang, Tianhan Jiang, Boxun An, Ya Li, Xiaoyu Shen
Spoken Language Models (SLMs) have emerged as a promising paradigm for speech synthesis by bypassing explicit grapheme-to-phoneme pipelines. However, their effectiveness in low-resource languages remains fundamentally limited by the scarcity of transcribed speech. In practice, synthetic data has become the primary strategy for scaling SLMs in such settings, providing reliable phonetic supervision when real data is insufficient. In this work, we show that this reliance introduces a fundamental trade-off, which we term the Stability-Expressivity Gap: while synthetic data improves phonetic accuracy, it progressively suppresses prosodic variability, ultimately leading to a collapse of expressivity (Synthetic Erosion). To bridge this gap, we propose two self-alignment frameworks. Disentanglement-Guided Self-Alignment (DGSA) recovers expressivity for complex languages by exploiting prosody-timbre separation. For regimes where authentic references are exceptionally limited, Temperature-Driven Self-Critique (TDSC) stabilizes generation through automated exploration and filtering. Our approach outperforms strong commercial systems, including ElevenLabs and Gemini Pro, and enables the first zero-shot voice cloning capability for Lao. Audio Samples are available at: https://luoji.cn/static/multilantts-demo-main/.
****

Online Linear Programming for Multi-Objective Routing in LLM Serving
Zixi Chen, Yinyu Ye, Zijie Zhou
We study the online routing problem in large language model serving, where requests arrive sequentially and must be dispatched to parallel decode workers under tight batch-size and KV-cache constraints. Unlike widely used routing heuristics that are not tied to explicit service-level objectives (SLOs) and offer limited control over latency–throughput trade-offs, we introduce an multi-objective optimization framework that formulates routing as an online linear programming with interpretable decision rewards. We apply an efficient bid-price control policy based on the online linear programming that admits requests when their SLO-weighted benefit exceeds their shadow prices. To meet millisecond decision requirements, we develop a warm-started, projected first-order updates that track the evolving dual shadow prices online with predictable runtime. We integrate our router into the Vidur simulator and demonstrate substantial improvements over standard baselines across multiple SLO regimes, including end-to-end latency, time-to-first-token, throughput, and tail performance. A big picture from our result: a science-based approach outperforms others based on heuristics.
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Selecting Samples on Graphs: A Unified Dataset Pruning Framework for Lossless Training Acceleration
Dongyue Wu, Zilin Guo, Xiaoyu Li, Jiajia Liu, Jingdong Chen, Nong Sang, Changxin Gao
The rapid growth of modern training datasets has significantly increased computational cost, motivating dataset pruning~(DP) methods which retain only a subset of informative samples to reduce training cost. Existing pruning criteria typically rely on either intrinsic signals that assess samples independently or extrinsic signals that promote diversity via pairwise relations. While effective in their own specific regimes, each captures only one aspect of sample utility and lacks robustness across different pruning ratios or data distribution. In this wor

k, we present a unified graph-based DP framework. By modeling the dataset as a w eighted graph, where node weights encode intrinsic value and edge weights encode extrinsic value, DP can be cast as a Maximum Weight Clique Problem (MWCP). Alth ough MWCP is NP-hard, its structure admits a principled greedy solution based on sample-wise marginal gains. Under a few mild conditions, we further prove that this unified objective enjoys a formal approximation guarantee, which applies to a broad family of importance metrics and provides practical design guidelines. Extensive experiments show that our method outperforms existing DP methods while substantially reducing training cost, reducing training time by over 40\% witho ut sacrificing accuracy on ImageNet-1k with ResNet-50.
****

Edit-Based Refinement for Parallel Masked Diffusion Language Models
Houxing Ren, Mingjie Zhan, Zimu Lu, Ke Wang, Yunqiao Yang, Haotian Hou, Junting Pan, Hongsheng Li
Masked diffusion language models enable parallel token generation and offer impr oved decoding efficiency over autoregressive models. However, their performance degrades significantly when generating multiple tokens simultaneously, due to a mismatch between token-level training objectives and joint sequence consistency. In this paper, we propose ME-DLM, an edit-based refinement framework that augme nts diffusion generation with lightweight post-editing steps. After producing an initial complete response, the model refines it through minimal edit operations , including replacement, deletion, and insertion, conditioned on the full sequen ce. Training supervision is derived from edit distance, providing a deterministi c signal under a fixed canonicalization scheme for learning minimal corrections. This approach encourages sequence-level consistency through globally conditione d edits while preserving the efficiency benefits of parallel diffusion decoding. Extensive experiments demonstrate that ME-DLM improves the quality and robustne ss of multi-token parallel generation. In particular, when built upon LLaDA, our method achieves consistent gains of 11.6 points on HumanEval and 33.6 points on GSM8K while using one-eighth of the total diffusion steps. Code is available at https://github.com/renhouxing/ME-DLM.
****

ReAugment: Targeted Few-Shot Time Series Augmentation via Model Zoo-Guided Reinf orcement Learning
Haochen Yuan, Yutong Wang, Yihong Chen, Yunbo Wang, Xiaokang Yang
Few-shot time series forecasting is fundamentally challenged by the scarcity of high-quality training data and the risk of severe overfitting. To address this i ssue, we propose ReAugment, a reinforcement learning (RL) framework that explici tly learns where and how to augment time series data. ReAugment maintains a zoo of forecasting models and measures prediction diversity across them to identify training samples that are most prone to overfitting. These samples serve as anch or points and are used as inputs to the data augmentation process. We then emplo y an RL approach to learn transformation policies, using a model zoo-guided rewa rd function to bias the transformed data to overfit-prone regions of the trainin g distribution that are most beneficial for generalization. A key advantage of t he RL formulation is that it avoids backpropagating gradients through the foreca sting models, thereby mitigating gradient vanishing. Experiments across diverse forecasting architectures demonstrate the effectiveness of ReAugment in both few -shot and standard time series forecasting.
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HieRD: Hierarchical Relational Distillation for Vision-Language Embedding Models
Vinh Le, Nguyen Hong Dang, Tu Vu, Linh Ngo Van, Duc Anh Nguyen, Trung Le
Knowledge distillation is crucial for compressing large Vision–Language Models (VLMs) into efficient architectures. While prior VLM research has primarily focus ed on reasoning tasks like visual question answering, multimodal embedding learn ing, a key component for large-scale retrieval, has received comparatively less attention. Existing distillation methods typically align static global represent ations, overlooking hierarchical feature structure and fine-grained cross-modal interactions. This leads to a structural gap where student models fail to inheri t object-level semantics and spatial relationships from teachers. To address thi

s limitation, we propose **HieRD**, a Hierarchical Representation Distillation f
ramework that preserves hierarchical structure within and across modalities thro
ughout the distillation process by leveraging clustered visual tokens and multi-
granular alignment with phrase-level text. Experimental results on multimodal em
bedding and downstream tasks show that HieRD consistently outperforms strong bas
elines, reflecting the effectiveness of its fine-grained semantic and spatial mo
deling, while enabling compact and efficient embedding models.
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Scaling, Benchmarking, and Reasoning of Vision-Language Agents for Mobile GUI Na
vigation
Heng Qu, Liu Yike, Renren Jin, Wenzong Zhang, Pengzhi Gao, Wei Liu, Jian Luan
Vision–Language Models (VLMs) have shown rapid progress in mobile GUI navigation
. This paper presents a systematic study of data scaling, benchmarking, and reas
oning for VLM-based agents in this domain. To facilitate rigorous evaluation, we
 introduce HyperTrack, a large-scale dataset with over 16000 real-world tasks ac
ross more than 650 Chinese mobile applications, along with GUIEvalKit, an open-s
ource toolkit for unified benchmarking of VLMs on offline GUI navigation tasks.
Using HyperTrack, we analyze the effects of training data scale on both supervis
ed and reinforcement-based finetuning. Our results show that reinforcement-based
 finetuning consistently outperforms supervised finetuning, particularly in out-
of-domain settings, highlighting the synergy between data scaling and reinforcem
ent learning. Leveraging GUIEvalKit, we further benchmark state-of-the-art (SOTA
) VLMs and analyze how interaction history and reasoning capabilities influence
task completion. Together, HyperTrack and GUIEvalKit provide a comprehensive pla
tform for developing and evaluating VLM agents in mobile GUI navigation tasks.
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RAST-MoE-RL: A Regime-Aware Spatio-Temporal MoE Framework for Deep Reinforcement
 Learning in Ride-Hailing
Yuhan Tang, Kangxin Cui, Jung Ho Park, Yibo Zhao, Xuan Jiang, Haoze He, Jiangbo
Yu, Haris Koutsopoulos, Jinhua Zhao
Ride-hailing platforms must balance passenger waiting times with overall system
efficiency under highly uncertain supply--demand conditions. Adaptive delayed ma
tching, which controls the holding intervals for batched sets of requests and ve
hicles, exposes an inherent trade-off between matching and pickup delays. The no
n-stationary nature of request arrivals and dynamic congestion exposes a key lim
itation of existing methods, which rely on shallow encoders that cannot capture
these spatio-temporal regime shifts. We introduce the Regime-Aware Spatio-Tempor
al Mixture-of-Experts (RAST-MoE) framework, which formalizes adaptive delayed ma
tching as a regime-aware Markov Decision Process and equips RL agents with a sel
f-attention Mixture-of-Experts encoder, letting different experts specialize aut
omatically across operational regimes while keeping per-sample computation modes
t. On real-world TNC trajectory data from San Francisco, our 12M-parameter model
 reduces average matching delay by 10% and pickup delay by 15%, trains stably wi
thout reward hacking, generalizes zero-shot to two additional cities, and exhibi
ts clear expert specialization across demand regimes. These results demonstrate
the value of MoE-enhanced RL for large-scale decision-making tasks with complex
spatio-temporal dynamics.
****
Expected Returns and Policy Inconsistency-Aware Offline Federated Deep Reinforce
ment Learning
Meng XU, Zhongying Chen, Weiwei Fu, Yan Li, Shuguang Wang, Jianping Wang
Offline Federated Deep Reinforcement Learning (FDRL) methods aggregate multiple
client-side offline Deep Reinforcement Learning (DRL) models, each trained local
ly, to facilitate knowledge sharing while preserving privacy. Existing offline F
DRL methods assign client weights during global aggregation using either simple
averaging or Q-values, but they neglect the combined consideration of Q-values a
nd policy inconsistency, the latter of which reflects the distributional discrep
ancy between the learned policy and the policy from offline data. This causes cl
ients with no significant advantages in one aspect but obvious disadvantages in
the other to disproportionately affect the global model, thereby degrading its c

apabilities in that aspect. During local training, clients in existing methods a re compelled to fully adopt the global model, which negatively impacts clients w hen the global model is weak. To this end, we propose a novel Federated Learning (FL) framework that can be seamlessly integrated into current offline FDRL appr oaches to improve their performance. Our method considers both policy inconsiste ncy and Q-values to determine the weights of client models, with the latter adju sted by a scaling factor to avoid significant numerical discrepancies with the f ormer. The aggregated global model is then distributed to clients to facilitate their learning from the global model. The impact of the global model on the loca l models is reduced when a client's model performance exceeds that of the global model, thereby mitigating the influence of a weaker global model. Experiments o n the Datasets for Deep Data-Driven Reinforcement Learning (D4RL) demonstrate th at our method improves seven state-of-the-art (SOTA) offline FDRL methods across several metrics.

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Scientific logicality enriched methodology for LLM reasoning: A practice in phys ics

Zhaoxin Yu, Nan Xu, Kun Chen, Jiahao Zhao, Lei Wang, Wenji Mao

With the continuous advancement of reasoning abilities in Large Language Models (LLMs), their application to scientific reasoning tasks has gained significant r esearch attention. Current research primarily emphasizes boosting LLMs' performa nce on scientific QA benchmarks by training on larger, more comprehensive datase ts with extended reasoning chains. However, these approaches neglect the essence of the scientific reasoning process---logicality, which is the rational foundat ion to ensure the validity of reasoning steps leading to reliable conclusions. I n this work, we make the first systematic investigation into the internal logica lity underlying LLM scientific reasoning, and develop a scientific logicality-en riched methodology, including a set of assessment criteria and data sampling met hods for logicality-guided training, to improve the logical faithfulness as well as task performance. Further, we take physics, characterized by its diverse log ical structures and formalisms, as an exemplar discipline to practise the above methodology. For data construction, we extract scientific problems from academic literature and sample a high-quality dataset exhibiting strong logicality. Expe riments based on three different backbone LLMs reveal that: 1) the training data we constructed can effectively improve the scientific logicality in LLM reasoni ng; and 2) the enriched scientific logicality plays a critical role in solving s cientific problems. Code is available at https://github.com/ScienceOne-AI/PhysLo gic.

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FOAM: Frequency and Operator-Error Based Adaptive Damping Method for Reducing St aleness-Oriented Error for Shampoo

Kyunghun Nam, Sumyeong Ahn

Shampoo is attracting considerable attention for its superior performance on lar ge-scale optimization benchmarks; yet it faces a significant practical bottlenec k: the prohibitive computational overhead of matrix inversion.

To mitigate this, practitioners typically rely on stale preconditioner updates, creating a fundamental trade-off between computational efficiency and optimizati on fidelity.

In this work, we provide a theoretical study of staleness through the complement ary lenses of convergence and stability. While staleness improves computational efficiency, it inherently degrades performance and introduces numerical instabil ity. Crucially, we identify that damping, acting as a numerical stabilizer, can effectively suppress these negative effects.

Guided by this analysis, we propose FOAM, an adaptive algorithm that stabilizes training by dynamically controlling both the damping factor and the eigendecompo sition frequency based on an approximation of the staleness-oriented error. Expe rimental results demonstrate that FOAM reduces wall-clock time compared to stand ard Shampoo while maintaining robust convergence.

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Learning Tight Rejection Boundaries without Negatives for Strict One-Class Audio

Deepfake Detection
Yuze Zhao, Kuiyuan Zhang, Zhongyun Hua, Yushu Zhang, Qing Liao, Wei Jiang
The rapid evolution of audio deepfakes requires robust detection capable of generalizing to unseen attacks. One-class learning offers inherent robustness for this task by characterizing real speech distributions to detect anomalies. However, establishing a compact decision boundary without spoof supervision remains a fundamental challenge. Existing relaxed approaches often compromise this strictness by introducing auxiliary negative samples, which biases the boundary toward seen artifacts and degrades generalization to unseen attacks. To address this, we propose CA-SOADD, a framework that refines the acceptance region by constructing off-manifold boundary probes. Our proposed centroid-anchored tri-objective learning paradigm simultaneously enforces centroid compactness and a centroid-referenced margin against these probes, thereby explicitly tightening the acceptance region without treating them as an explicit negative class. We further extend the framework to heterogeneous settings through domain-conditioned centroids. Experiments on ASVSpoof and MLAAD benchmarks demonstrate that our strict real-only method consistently outperforms strong baselines under unseen attack types and domain shifts, with its effectiveness further validated through extensive ablation studies.
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Learning-To-Measure: In-Context Active Feature Acquisition
Yuta Kobayashi, Zilin Jing, Jiayu Yao, Hongseok Namkoong, Shalmali Joshi
Active feature acquisition (AFA) is a sequential decision-making problem where the goal is to improve model performance for test instances by adaptively selecting which features to acquire. In practice, AFA methods often learn from retrospective data with systematic missingness in the features and limited task-specific labels. To address this limitation, we introduce Learning-to-Measure (L2M), which consists of i) reliable uncertainty quantification over unseen tasks, and ii) an uncertainty-guided feature acquisition agent that maximizes conditional mutual information. We demonstrate an autoregressive pre-training approach that underpins reliable uncertainty quantification and acquisition for tasks with arbitrary missingness. L2M operates directly on datasets with retrospective missingness and performs the task in-context, eliminating per-task retraining. Across synthetic and real-world tabular benchmarks, L2M matches or surpasses task-specific baselines, particularly under scarce labels and high missingness.
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Task-Awareness Improves LLM Generations and Uncertainty
Tim Tomov, Dominik Fuchsgruber, Stephan Günnemann
In many applications of LLMs, natural language responses often have an underlying structure such as representing discrete labels, numerical values, or graphs. Yet, existing decoding and uncertainty estimation methods operate only in language space and largely disregard structural information. We address this by modeling LLM outputs directly in a task-dependent latent structure. By equipping this structure with a dissimilarity measure, we can compute Bayes-optimal responses. These are not selected from sampled generations but are newly synthesized by combining individual responses in the latent space.
Across different tasks, Bayes-optimal responses consistently outperform standard decoding methods like beam search. Moreover, quantifying uncertainty via the induced Bayesian risk captures variations in terms of the latent structure and improves alignment with output quality and correctness. Our decision-theoretic framework is applicable to any problem that admits a latent response structure and enables reliable task-aware LLM predictions.
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Likelihood Matching for Diffusion Models
Lei Qian, Wu Su, Yanqi Huang, Song Xi Chen
We propose a Likelihood Matching approach for training diffusion models by first establishing an equivalence between the likelihood of the target data distribution and a likelihood along the sample path of the reverse diffusion. To efficiently compute the reverse sample likelihood, the equivalence, a quasi-likelihood is considered to approximate each reverse transition density by a Gaussian distri

bution with matched conditional mean and covariance, respectively. The score and Hessian functions for the diffusion generation are estimated by maximizing the quasi-likelihood, ensuring a consistent matching of both the first two transition moments between every two time points. A stochastic sampler is introduced to facilitate the computation that leverages both the estimated score and Hessian information. We establish consistency of the quasi-maximum likelihood estimation, and provide non-asymptotic convergence guarantees for the proposed sampler, quantifying the rates of the approximation errors due to score and Hessian estimation, dimensionality, and the number of diffusion steps. Empirical and simulation evaluations demonstrate the effectiveness of the proposed Likelihood Matching and validate the theoretical results.
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Structure-aware Granular-Ball based Information Bottleneck for Multi-modal Clustering
Zhengzheng Lou, Yuhan Zhan, Mingyang Lv, Yingxuan Li, Yuyang Du, Shizhe Hu
Multi-modal clustering, which integrates information from diverse sources and feature modalities, has shown great potential in data mining and computer vision. However, existing methods relying on single-granularity relationships often struggle with complex data distributions, leading to limited performance, as fine-grained features are prone to local heterogeneity and redundant perturbations while coarse-grained representations tend to lose local structural information. To address these limitations, we introduce granular-balls (GBs), adaptive multi-granularity hyperspheres that enclose similar samples, and propose the Structure-aware Granular-Ball based Information Bottleneck (SGB-IB) algorithm. This method initializes the dataset as a single GB and recursively splits GBs based on a purity metric, which quantifies the average mutual information between sample features and K-means-derived pseudo-labels across all modalities. It also balances local structure preservation and global redundancy suppression through a structure-aware objective function. Extensive experiments on benchmark datasets demonstrate that our method outperforms state-of-the-art approaches, validating the effectiveness of fusing GB structures with information-theoretic principles.
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Shape of Thought: Progressive Object Assembly via Visual Chain-of-Thought
Yu Huo, Siyu Zhang, Kun Zeng, Haoyue Liu, Owen Lee, Junlin chen, YUQUAN LU, Yifu Guo, Yaodong Liang, Xiaoying Tang
Multimodal models for text-to-image generation have achieved strong visual fidelity, yet they remain brittle under compositional structural constraints—notably generative numeracy, attribute binding, and part-level relations.
To address these challenges, we propose **Shape-of-Thought (SoT)**, a visual CoT framework for *process-supervised progressive shape assembly in the rendered 2D domain*, without external engines at inference time. SoT trains a unified multimodal autoregressive model to generate interleaved textual plans and rendered intermediate states, helping the model capture shape-assembly logic without producing explicit geometric representations. Unlike text-only CoT, each decision is grounded in a rendered state, making counts, attachments, topology, and intermediate part-addition errors inspectable across the trajectory. To support this paradigm, we introduce **SoT-26K**, a large-scale dataset of grounded assembly traces derived from part-based CAD hierarchies, and **T2S-CompBench**, a benchmark for evaluating structural integrity and trace faithfulness. Fine-tuning on SoT-26K achieves 88.4\% on component numeracy and 84.8\% on structural topology, outperforming direct generation by +24.2 points on component numeracy and +19.3 points on structural topology. SoT establishes a transparent testbed for rendered-domain structure-aware generation. The code is available at https://github.com/yuhuo03/Shape-of-Thought.
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WUSH: Near-Optimal Adaptive Transforms for LLM Quantization
Jiale Chen, Vage Egiazarian, Roberto L. Castro, Torsten Hoefler, Dan Alistarh
Quantizing LLM weights and activations is a standard approach for efficient deployment, but a few extreme outliers can stretch the dynamic range and amplify low-bit quantization errors. Prior transform-based mitigations (e.g., Hadamard rota

tions) are fixed and data-agnostic, and their optimality for quantization has remained unclear. We derive closed-form optimal linear blockwise transforms for joint weight-activation quantization under standard RTN AbsMax-scaled block quantizers, covering both integer and floating-point formats. The resulting construction, WUSH, combines a Hadamard backbone with a data-dependent second-moment component to form a non-orthogonal transform that is provably near-optimal for FP and INT quantizers under mild assumptions while admitting an efficient fused GPU implementation. Empirically, WUSH improves W4A4 accuracy over the strongest Hadamard-based baselines (e.g., on Llama-3.1-8B-Instruct in MXFP4, it gains +2.8 average points with RTN and +0.7 with GPTQ) while delivering up to 5.8$\times$ per-layer throughput over BF16 via FP4 MatMul. Source code is available at https://github.com/IST-DASLab/WUSH.

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LK Losses: Direct Acceptance Rate Optimization for Speculative Decoding

Alexander Samarin, Sergei Krutikov, Anton Shevtsov, Sergei Skvortsov, Filipp Fisin, Alexander Golubev

Speculative decoding accelerates autoregressive large language model (LLM) inference by using a lightweight draft model to propose candidate tokens that are then verified in parallel by the target model.

The speedup is significantly determined by the acceptance rate, yet standard training minimizes Kullback-Leibler (KL) divergence as a proxy objective.

While KL divergence and acceptance rate share the same global optimum, small draft models, having limited capacity, typically converge to suboptimal solutions where minimizing KL does not guarantee maximizing acceptance rate.

To address this issue, we propose **LK losses**, special training objectives that directly target acceptance rate.

Comprehensive experiments across four draft architectures and six target models, ranging from 8B to 685B parameters, demonstrate consistent improvements in acceptance metrics across all configurations compared to the standard KL-based training.

We evaluate our approach on general, coding and math domains and report gains of up to 8-10\% in average acceptance length.

LK losses are easy to implement, introduce no computational overhead and can be directly integrated into any existing speculator training framework, making them a compelling alternative to the existing draft training objectives.

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Learning Generalized Trackers with Elastic Token Budgets

Yinchao Ma, Jianpeng Yang, Yuyang Tang, Jie Xiao, Dengqing Yang, Tianzhu Zhang

Visual tracking aims to estimate target states in video sequences, with applications spanning diverse computational requirements. Recent methods optimize trackers using manually pruned image tokens with a fixed budget to reduce computational costs. However, these trackers, once trained, are constrained to perform tracking under a fixed computational budget, limiting their adaptability to real-world computational diversity. To address the above limitation, we provide the first exploration of the elastic token budget training framework (ETBTrack), enabling trackers to perform robust tracking under varying computational budgets. It enjoys several merits. First, we present a novel result-driven importance criteria, in which we optimize a policy network guided by the localization precision of the tracker to estimate token importance, thereby aligning the objectives of importance estimation and tracking precision. Second, we develop a new budget-collaborative optimization strategy, in which we collaboratively optimize the tracker across varying budgets, thereby enabling the tracker to be compatible with diverse budgets. Two optimization processes are performed alternately to enhance the capability of elastic inference. Extensive experiments on large-scale benchmarks demonstrate the effectiveness of our method.

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Data Selection for Fine-tuning Vision Language Models via Cross Modal Alignment Trajectories

Nilay Naharas, Dang Nguyen, Neslihan Bulut, Mohammadhossein Bateni, Vahab Mirrokni, Baharan Mirzasoleiman

Data-efficient learning aims to eliminate redundancy in large training datasets by training models on smaller subsets of the most informative examples. While data selection has been extensively explored for vision models and large language models (LLMs), it remains underexplored for Large Vision-Language Models (LVLMs). Notably, none of existing methods can outperform random selection at different subset sizes. In this work, we propose the first principled method for data-efficient instruction tuning of LVLMs. We prove that examples with similar cross-modal attention matrices during instruction tuning have similar gradients. Thus, they influence model parameters in a similar manner and convey the same information to the model during training. Building on this insight, we propose XMAS, which clusters examples based on the trajectories of the top singular values of their attention matrices obtained from fine-tuning a small proxy LVLM. By sampling a balanced subset from these clusters, XMAS effectively removes redundancy in large-scale LVLM training data. Extensive experiments across 4 target models, 2 proxy models, and 2 datasets show that XMAS consistently outperforms 10 baseline methods. Moreover, XMAS can discard 50% of the LLaVA-665k dataset and 85% of the Vision-Flan dataset while fully preserving performance of LLaVA-1.5-7B on 10 downstream benchmarks and speeding up its training by 1.2×. This is 30% more data reduction compared to the best baseline for LLaVA-665k. The project’s website can be found at https://bigml-cs-ucla.github.io/XMAS-project-page/.
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SI-IGCL: Subject Invariance-aware Inverse Graph Contrastive Learning for Psychiatric Disorder Identification

Jiayu Lu, Yujin Wang, Xiaofeng Liu, Dandan Li, Bin Wang

Functional brain network analysis plays an important role in understanding and diagnosing psychiatric disorders. However, current methods struggle with subject variations, impairing the model’s generalization ability to the test set. To address this issue, we propose the Subject Invariance-aware Inverse Graph Contrastive Learning (SI-IGCL) model, which adopts a two-stage paradigm with self-supervised subject-invariant pre-training followed by supervised fine-tuning for identification. During the pre-training phase, we construct an inverse contrastive objective that reshapes the embedding space by repelling intra-subject and attracting inter-subject embeddings to learn subject-invariant representations, with an auxiliary correction term to avoid early optimization plateaus. Meanwhile, we incorporate a structure-preserving reconstruction constraint to preserve discriminative information. Moreover, a Hierarchical Topology Enhanced Transformer (HTET) module is designed to enable multi-level modeling of subject-invariant functional patterns. During the fine-tuning phase, a supervised classifier is integrated to perform psychiatric disorder classification. Extensive experiments demonstrate that our method outperforms all state-of-the-art methods. The code is available at https://anonymous.4open.science/r/SI-IGCL.
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Learning the Minimum Action Distance

Lorenzo Steccanella, Joshua Benjamin Evans, Özgür ■im■ek, Anders Jonsson

This paper presents a state representation framework for Markov decision processes (MDPs) that can be learned solely from state trajectories, requiring neither reward signals nor the actions executed by the agent. We propose learning the $\textit{minimum action distance}$ (MAD), defined as the minimum number of actions required to transition between states, as a fundamental metric that captures the underlying structure of an environment. The MAD naturally enables critical downstream tasks such as goal-conditioned reinforcement learning and reward shaping by providing a dense, geometrically meaningful measure of progress.

Our self-supervised learning approach constructs an embedding space where the distances between embedded state pairs correspond to their MAD, accommodating both symmetric and asymmetric approximations.

We evaluate the framework on a comprehensive suite of environments with known MAD values, encompassing both deterministic and stochastic transition dynamics, discrete and continuous state spaces, and environments with noisy observations.

Empirical results show that the proposed approach learns MAD representations more efficiently than existing methods, produces more accurate estimates of the tru

e MAD, and improves performance on downstream goal-reaching tasks.
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AD-BTS: Adaptive Dual-Branch Token Sparsification via Spatial Information Density
Xinpei Gao, Xin Luo, Ming Liu, Chunjiang Wang, S Kevin Zhou
High-resolution visual encoders in multimodal large language models (MLLMs) substantially improve fine-grained perception, yet incur prohibitive computational costs.Existing token pruning methods are effective on natural images but struggle with spatially sparse structured inputs (e.g., charts), where critical high-frequency information is sparse, localized, and structurally essential. To address this challenge, we propose Adaptive Dual-Branch Token Sparsification (AD-BTS), a density-aware framework that dynamically allocates computation according to input signal characteristics. Specifically, AD-BTS introduces a Gradient-based Routing Gate (GRG) that uses lightweight pixel-level gradient statistics to estimate structural flatness and guide routing. Then, AD-BTS activates either a Redundancy Selection Branch (RSB) for aggressive token pruning with a frozen encoder, or a Structural Fusion Branch (SFB) with conditional LoRA and context fusion to preserve sparse structural information.Extensive experiments on Qwen2.5-VL demonstrate that AD-BTS establishes a new Pareto frontier between efficiency and accuracy. Under extreme compression (20% token retention), AD-BTS outperforms the strongest baseline by 12.1% on ChartQA while achieving a 1.8× prefill speedup, effectively reconciling computational efficiency with structural robustness.
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Probabilistic Bisection Algorithm Provably Achieves Exponential Convergence
Ganghua Wang, Yuwei Cheng, Haifeng Xu
The probabilistic bisection algorithm (PBA) extends the classical binary search to settings with noisy responses, and is a foundational algorithm commonly used in basic problems such as root-finding. Despite its strong empirical success, its theoretical property, particularly the convergence rate, remains unclear. This paper establishes that PBA converges at a geometric rate, providing a rigorous justification for its empirical efficiency. Notably, this rate is optimal in the sense that it matches the performance of classical binary search under noiseless responses. The core of our analysis lies in directly characterizing the dynamics of PBA queries, which had not been examined in the prior literature. We show that the queries oscillate around the truth but steadily draw closer, thus leading to an estimator that rapidly concentrates on the truth. Beyond resolving the long-standing question of PBA’s convergence, our developed techniques offer new tools for analyzing PBA's dynamics, which may be of independent interest.
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CamGeo: Sparse Camera-Conditioned Image-to-Video Generation with 3D Geometry Priors
Xuanyi Liu, Deyi Ji, Liqun Liu, Lanyun Zhu, Xuhang Chen, Qianxiong Xu, Peng Shu, Huan Yu, Jie Jiang, Feng Gao, Siwei Ma
Sparse camera-conditioned image-to-video generation presents a pivotal challenge: synthesizing geometrically consistent 3D motion from minimal pose cues. Existing methods, which largely rely on dense supervision or naive interpolation, suffer from severe pose drift and motion discontinuities due to the lack of robust 3D priors. In this paper, we introduce \textbf{CamGeo}, a novel framework that distills rich 3D geometric knowledge from a pre-trained video-to-3D model (VGGT) directly into the diffusion backbone. To achieve this without incurring inference latency, we propose a training-only distillation strategy. Specifically, CamGeo incorporates: (1) keyframe trajectory distillation that enforces cycle-consistency with sparse input poses, (2) cross-frame consistency distillation with both camera trajectory and depth constraints to generate consistent structure across unsupervised frames, and (3) a three-stage coarse-to-fine curriculum learning, progressively scales geometric complexity, from global structure coherence to fine-grained refinement, achieving stable optimization. Extensive experiments demonstrate that CamGeo achieves consistent improvements under various sparsity ratios.
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Efficient RL Training for LLMs with Experience Replay
Charles Arnal, Vivien Cabannes, Taco Cohen, Julia Kempe, Rémi Munos
While Experience Replay—the practice of storing rollouts and reusing them multiple times during training—is a foundational technique in general RL, it remains largely unexplored in LLM post-training due to the prevailing belief that fresh, on-policy data is essential for high performance.
In this work, we challenge this assumption.
We present a systematic study of replay buffers for LLM post-training, formalizing the optimal design as a trade-off between staleness-induced variance, sample diversity and the high computational cost of generation.
We show that strict on-policy sampling is suboptimal when generation is expensive.
Empirically, we show that a well-designed replay buffer can drastically reduce inference compute without degrading -- and in some cases even improving -- final model performance, while preserving policy entropy.
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Reviving Error Correction in Modern Deep Time-Series Forecasting
Minh Hoang Nguyen, Van Dai Do, Huu Hiep Nguyen, Dung Nguyen, Kien Do, Hung Le
Modern deep-learning models have achieved remarkable success in time-series forecasting. Yet, their performance degrades in long-term prediction due to error accumulation in autoregressive inference, where predictions are recursively used as inputs. While classical error correction mechanisms (ECMs) have long been used in statistical methods, their applicability to deep learning models remains limited or ineffective. In this work, we revisit the error accumulation problem in deep time-series forecasting and investigate the role and necessity of ECMs in this new context. We propose a simple, architecture-agnostic error correction model that can be integrated with any existing forecaster without requiring retraining. By explicitly decomposing predictions into trend and seasonal components and training the corrector to adjust each separately, we introduce the Universal Error Corrector with Seasonal–Trend Decomposition (UEC-STD), which significantly improves correction accuracy and robustness across 4 backbones and 10 datasets. Our findings provide a practical tool for enhancing forecasts while offering new insights into mitigating autoregressive errors in deep time-series models.
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Accurate Large-sample Uncertainty Quantification using Stochastic Gradient Markov Chain Monte Carlo
Yu Wang, Jie Ding, Jonathan H. Huggins
Tuning stochastic gradient methods such as stochastic gradient descent (SGD) and stochastic gradient Langevin dynamics (SGLD) for approximate sampling and uncertainty quantification remains challenging, particularly in relevant settings using a large batch size or when the model is misspecified. Existing theory that provides tuning guidance relies on continuous-time limits or strong statistical assumptions, which can become quantitatively inaccurate in these regimes. We address these shortcomings by proposing a new discrete-time approximations to SG(L)D with and without momentum, which enables accurate predictions of the stationary covariance, iterate average covariance, and integrated autocorrelation time. Moreover, we prove quantitative, non-asymptotic error bounds showing that these estimates are sufficiently accurate for practical tuning and uncertainty quantification. Numerical experiments demonstrate that our theory yields improved tuning guidance across a range of models and data-generating distributions where existing approaches fail, including when using the β-divergence rather than log-loss to obtain statistically robust inferences.
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Cold-Start Personalization via Bayesian Adaptive Questioning
Avinandan Bose, Shuyue Stella Li, Faeze Brahman, Pang Wei Koh, Simon Shaolei Du, Yulia Tsvetkov, Maryam Fazel, Lin Xiao, Asli Celikyilmaz
Cold-start personalization requires inferring preferences from minimal interaction when no user-specific historical data is available. The space of possible preferences is vast, yet users care about only a sparse subset and rarely articulate them upfront; combined with limited interaction budgets, this makes preference

elicitation challenging. Our key insight is that preferences exhibit predictable structure across populations; e.g., users who want detailed explanations often also value worked examples. We propose CAPE (Cold start Adaptive Preference Elicitation with Priors), a principled system decomposition framework for cold-start personalization: learning a structured world model of preference correlations offline using latent variables, then performing Bayesian inference online without retraining. Even simple belief model instantiations (e.g., linear regression) substantially outperform end-to-end RL. Across medical, mathematical, social, and commonsense reasoning, CAPE achieves 80.8% alignment with ground-truth user preferences versus 68.5% for RL, requires 3-5× fewer interactions, and adapts twice as often. Our contribution is a principled decomposition of cold-start personalization that makes Bayesian preference elicitation practical at scale for LLM systems.

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ECHO: Entropy-Confidence Hybrid Optimization for Test-Time Reinforcement Learning

Chu Zhao, Enneng Yang, Yuting Liu, Jianzhe Zhao, Guibing Guo

Test-time reinforcement learning generates multiple candidate answers via repeated rollouts and performs online updates using pseudo-labels constructed by majority voting. To reduce overhead and improve exploration, prior work introduces tree-structured rollouts, which share reasoning prefixes and branch at key nodes to improve sampling efficiency. However, this paradigm still faces two challenges: (1) high-entropy branching can trigger rollout collapse, where the branching budget concentrates on a few trajectories with consecutive high-entropy segments, rapidly reducing the number of effective branches; (2) early pseudo-labels are noisy and biased, which can induce self-reinforcing overfitting, causing the policy to sharpen prematurely and suppress exploration. To address these issues, we propose Entropy–Confidence Hybrid Group Relative Policy Optimization (ECHO). During rollout, ECHO jointly leverages local entropy and group-level confidence to adaptively control branch width, and further introduces online confidence-based pruning to terminate persistently low-confidence branches, avoiding high-entropy traps and mitigating collapse. During policy updates, ECHO employs confidence-adaptive clipping and an entropy–confidence hybrid advantage shaping approach to enhance training robustness and mitigate early-stage bias. Experiments demonstrate that ECHO achieves consistent gains on multiple mathematical and visual reasoning benchmarks, and generalizes more effectively under a limited rollout budget.

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Long Grounded Thoughts: Synthesizing Visual Problems and Reasoning Chains at Scale

David Acuna, Chao-Han Huck Yang, Yuntian Deng, Jaehun Jung, Ximing Lu, Prithviraj Ammanabrolu, Hyunwoo Kim, Yuan-Hong Liao, Yejin Choi

Despite rapid progress, multimodal reasoning still lacks a systematic approach to synthesize large-scale vision-centric datasets beyond visual math. We introduce a framework able to synthesize vision-centric problems spanning diverse levels of complexity, and the resulting dataset with over 1M high-quality problems including: reasoning traces, preference data, and instruction prompts supporting SFT, offline and online RL. Our vision-centric synthesis framework uses a two-stage process focusing on: (1) generating diverse verifiable questions from existing images at scale, and (2) creating complex compositional visual problems by merging simpler questions.

Remarkably, finetuning Qwen2.5-VL-7B on our data outperforms existing open-data baselines across evaluated vision-centric benchmarks, and our best configurations match or surpass strong closed-data models such as MiMo-VL-7B-RL on V*Bench, CV-Bench and MMStar-V. Notably, despite being entirely vision-centric, our data transfers positively to text-only reasoning (MMLU-Pro, +3.7%) and audio reasoning (MMAU, +1.32%), demonstrating its effectiveness. Similarly, despite containing no embodied visual data, we observe notable gains (NiEH, +8.8%) when evaluating open-ended embodied QA. Lastly, we use our data to comprehensively analyze at scale (1M+) the entire VLM post-training pipeline showing that (i) SFT on

high-quality data with cognitive behaviours on reasoning traces is essential to scale online RL, (ii) offline RL could match online RL’s performance while disaggregating compute demands, and, (iii) SFT on high quality data also improve out-of-domain, cross-modality transfer.
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Benchmarking Reward Hack Detection in Code Environments via Contrastive Analysis
Darshan Girish Deshpande, Anand Kannappan, Rebecca Qian
Recent advances in reinforcement learning for code generation have made robust environments essential to prevent reward hacking. As LLMs increasingly serve as evaluators in code-based RL, their ability to detect reward hacking remains understudied. In this paper, we propose a novel taxonomy of reward exploits spanning across 54 categories and introduce TRACE (Testing Reward Anomalies in Code Environments), a synthetically curated and human-verified benchmark containing 517 testing trajectories. Unlike prior work that evaluates reward hack detection in isolated classification scenarios, we contrast these evaluations with a more realistic, contrastive anomaly detection setup on TRACE. Our experiments reveal that models capture reward hacks more effectively in contrastive settings than in isolated classification settings, with GPT-5.2 with highest reasoning mode achieving the best detection rate at 63%, up from 45% in isolated settings on TRACE. Building on this insight, we demonstrate that state-of-the-art models struggle significantly more with semantically contextualized reward hacks compared to syntactically contextualized ones. We further conduct qualitative analyses of model behaviors, as well as ablation studies showing that the ratio of benign to hacked trajectories and analysis cluster sizes substantially impact detection performance. We release the benchmark and evaluation harness to enable the community to expand TRACE and evaluate their models.
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Deep Learning for BioImaging: What Are We Really Learning?
Ivan Svatko, Maxime Sanchez, Ihab Bendidi, Gilles Cottrell, Auguste Genovesio
Representation learning has driven major advances in natural image analysis by enabling models to acquire high-level semantic features. In microscopy imaging, however, it remains unclear what current representation learning methods really learn.
In this work, we conduct a systematic study of representation learning for the two most widely used and broadly available microscopy data types, representing critical scales in biology: cell culture and tissue imaging. We investigate whether, in contrast to natural images, existing models fail to consistently acquire high-level, biologically meaningful features. To this end, we introduce a set of simple yet revealing baselines on curated benchmarks, including untrained models and structural representations of cellular tissue. Our results show that, surprisingly, for a considerable subset of evaluation settings, the baselines are comparable to state-of-the-art methods, demonstrating that many commonly used benchmark metrics are insufficient to assess representation quality and often mask a lack of relevant high-level abstractions. In addition, we investigate how detailed comparisons with these baselines provide ways to interpret the strengths and weaknesses of models for further improvements. Together, our results suggest that progress in representation learning for microscopy requires not only stronger models, but also benchmarks that are more indicative of what is actually learned.
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Clustering as Reasoning: A k-Means Interpretation of Chain-of-Thought Graph Learning
Xuanting Xie, Zhaochen Guo, Bingheng Li, Xingtong Yu, Zhifei Liao, zhao kang, Yuan Fang
Chain-of-Thought (CoT) prompting has shown promise in enhancing the reasoning capabilities of large language models (LLMs) on text-attributed graphs (TAGs). This work reframes CoT-based graph learning through the principle of clustering as reasoning, offering a k-means interpretation of how iterative reasoning operates over graph-structured data. We observe that existing graph CoT methods rely on disjoint architectures and fixed graph representations, limiting step-by-step

semantic-topological interaction and interpretability. To overcome this limitation, we propose a unified framework named KCoT that integrates CoT reasoning with graph representation learning. Our key theoretical result reveals a formal mathematical correspondence between a Transformer block and the k-means algorithm, allowing reasoning to be interpreted as iterative assignment and update steps. Based on this insight, we introduce a Semantic Discriminating Prompt that explicitly formulates these steps as structured CoT reasoning, together with a structure-grounded alignment strategy to fuse topological priors with evolving thought-conditioned representations. Experiments on standard benchmarks demonstrate consistent improvements over state-of-the-art methods, validating clustering as a principled mechanism for CoT-based graph learning.
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Approximate Nearest Neighbor Search for Modern AI: A Projection-Augmented Graph Approach
Kejing Lu, Zhenpeng Pan, Yoshiharu Ishikawa, Chuan Xiao, Jianbin Qin
Approximate Nearest Neighbor Search (ANNS) is fundamental to modern AI applications. Most existing solutions optimize query efficiency but fail to align with the practical requirements of modern workloads. In this paper, we outline six critical demands of modern AI applications: high query efficiency, fast indexing, low memory footprint, scalability to high dimensionality, robustness across varying retrieval sizes, and support for online insertions. To satisfy all these demands, we introduce Projection-Augmented Graph (PAG), a new ANNS framework that integrates projection techniques into a graph index. PAG reduces unnecessary exact distance computations through asymmetric comparisons between exact and approximate distances as guided by projection-based statistical tests. Three key components are designed and integrated into the graph index to optimize indexing and searching. Experiments on six modern datasets demonstrate that PAG consistently achieves superior queries per second (QPS)-recall performance---up to 5$\times$ faster than HNSW---while offering fast indexing speed and moderate memory footprint. PAG remains robust as dimensionality and retrieval size increase and naturally supports online insertions. Our source code is available at: https://github.com/KejingLu-810/PAG/.
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Immuno-VLM: Immunizing Large Vision-Language Models via Generative Semantic Antibodies for Open-World Trustworthiness
Xiang Fang, Wanlong Fang, Wei Ji
Large Vision-Language Models have achieved unprecedented success in zero-shot recognition by aligning visual features with broad semantic concepts. However, this semantic abstraction creates a critical vulnerability in open-world deployment: the "Hubris of Semantics", where models force-fit unknown anomalies into known categories with high confidence due to the lack of explicit negative knowledge. To address this Open-World Trustworthiness Paradox, we propose Immuno-VLM, a bio-inspired framework that adapts the biological principle of Immunological Negative Selection to high-dimensional latent spaces. Departing from traditional Open-Set Recognition methods that rely on passive density estimation or inefficient pixel-space outlier generation, Immuno-VLM leverages the generative reasoning of Large Language Models to actively hallucinate "Semantic Antibodies", textual descriptions of near-distribution outliers (e.g., look-alikes, contextual anomalies) that effectively bound the decision space of known classes. Extensive experiments on ImageNet-1K and four challenging OOD benchmarks reveal that Immuno-VLM establishes a new state-of-the-art.
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Transitive Representation Learning Enhances Histopathology Annotation
Moritz Schaefer, Zoe Piran, Nils Philipp Walter, Animesh Awasthi, Christoph Bock, Jure Leskovec, Zinaida Good
The characterization of histopathology with AI promises to assist clinical decision-making, but it is currently limited due to coarse-grained annotations that miss cellular identities. To overcome this gap, we bridge histopathological images, gene expression profiles, and natural-language descriptions using *SpatialWhisperer*, a trimodal contrastive learning model. Our training integrates communit

y-scale datasets comprising spatially resolved gene expression profiles paired w
ith histopathology images, as well as single-cell gene expression profiles with
detailed annotations. The shared gene expression modality implies a transitive r
elationship between images and textual annotations, which our method leverages t
o enable accurate zero-shot cell type annotation directly from H&E images. *Spat
ialWhisperer* outperforms published baselines, achieving relative AUROC gains of
 up to 15.9% across three benchmarks spanning 19 tissues and 20 cell types. When
 training with data from all three modality pairs, we observe performance gains
in low-data regimes. We formalize our approach and present a sufficient conditio
n under which this transitive alignment is induced. Our work establishes *transi
tive representation learning* for fine-grained interpretation of histopathology
images.
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SurrogateSHAP: Training-Free Contributor Attribution for Text-to-Image (T2I) Mod
els

MingYu Lu, Soham Gadgil, Chris Lin, Chanwoo Kim, Su-In Lee

As Text-to-Image (T2I) diffusion models are increasingly used in real-world crea
tive workflows, a principled framework for valuing contributors who provide a co
llection of data is essential for fair compensation and sustainable data marketp
laces. While the Shapley value offers a theoretically grounded approach to attri
bution, it faces a dual computational bottleneck: (i) the prohibitive cost of ex
haustive model retraining for each sampled subset of players (i.e., data contrib
utors) and (ii) the combinatorial number of subsets needed to estimate marginal
contributions due to contributor interactions. To this end, we propose **Surroga
teSHAP**, a retraining-free framework that approximates the expensive retraining
 game through inference from a pretrained model. To further improve efficiency,
we employ a gradient-boosted tree to approximate the utility function and derive
 Shapley values analytically from the tree-based model. We evaluate SurrogateSHA
P across three diverse attribution tasks: (i) image quality for DDPM-CFG on CIFA
R-20, (ii) aesthetics for Stable Diffusion on Post-Impressionist artworks, and (
iii) product diversity for FLUX.1 on Fashion-Product data. Across settings, Surr
ogateSHAP outperforms prior methods while substantially reducing computational o
verhead, consistently identifying influential contributors across multiple utili
ty metrics. Finally, we demonstrate that SurrogateSHAP effectively localizes dat
a sources responsible for spurious correlations in clinical images, providing a
scalable path toward auditing safety-critical generative models. Code is availab
le at https://anonymous.4open.science/r/CFG-Attribution-15DD/
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Distributional Inverse Reinforcement Learning

Feiyang Wu, Ye Zhao, Anqi Wu

We propose a distributional framework for offline Inverse Reinforcement Learning
 (IRL) that jointly models uncertainty over reward functions and full distributi
ons of returns. Unlike conventional IRL approaches that recover a deterministic
reward estimate or match only expected returns, our method captures richer struc
ture in expert behavior, particularly in learning the reward distribution, by mi
nimizing first-order stochastic dominance (FSD) violations and thus integrating
distortion risk measures (DRMs) into policy learning, enabling the recovery of b
oth reward distributions and distribution-aware policies. This formulation is we
ll-suited for behavior analysis and risk-aware imitation learning. Theoretical a
nalysis show that the algorithm converge with $\mathcal{O}(\varepsilon^{-2})$ it
eration complexity. Empirical results on synthetic benchmarks, real-world neurob
ehavioral data, and MuJoCo control tasks demonstrate that our method recovers ex
pressive reward representations and achieves state-of-the-art imitation performa
nce.
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MutAtlas: A PDB-Wide Energy-Guided Atlas of Protein Mutation Effects

Ruihan Guo, Chaoran Cheng, Zhanghan Ni, Neil He, Bangji Yang, Ge Liu

Protein mutation effect prediction is fundamental to protein engineering and dis
ease variant interpretation, yet experimentally measured mutation data remain ac
curate but extremely sparse. To provide scalable supplementary mutation signals,

we construct a PDB-wide mutation augmentation dataset that exhaustively enumera
tes single-site substitutions on experimentally resolved protein structures and
aligns mutation signals from physics-based energy models, protein language model
s, and inverse folding models. Large-scale analysis under a unified mutation pre
ference representation reveals substantial differences in the consistency, conce
ntration, and substitution patterns of mutation distributions across models, ind
icating that disagreement is pervasive and reflects conflicting inductive biases
rather than random noise. Motivated by these observations, we propose an unsupe
rvised multi-source mutation preference distillation framework that learns from
relative mutation preferences while explicitly modeling cross-source disagreemen
t. Without using any experimental mutation labels during training, our approach
achieves the best overall performance among the evaluated zero-shot baselines an
d naive multi-source fusion strategies on ProteinGym. We release the dataset and
evaluation pipeline to support reproducible studies of protein mutation effects
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Incremental BPE Tokenization
Shenghu Jiang, Ruihao Gong
We propose a novel algorithm for incremental Byte Pair Encoding (BPE) tokenizati
on.
The algorithm processes each input byte in **worst-case** $\mathcal{O}(\log^2 t)
$ time,
leading to an overall complexity of $\mathcal{O}(n \log^2 t)$,
where n is the input length and t is the maximum token length.
The algorithm incrementally maintains BPE tokenization results for every prefix
of the input text,
implementing the standard BPE merge procedure defined by a fixed set of merge ru
les.
This enables efficient partial tokenization in streaming settings.
Functioning as a drop-in replacement for standard BPE,
our approach achieves a speedup of up to ${\sim}3\times$ over Hugging Face's tok
enizers,
and demonstrates significant latency reductions over OpenAI's tiktoken on pathol
ogical inputs.
We further introduce an eager output algorithm that enables streaming output,
emitting tokens as soon as token boundaries are determined during incremental to
kenization.
Overall, our results demonstrate that BPE tokenization can be performed incremen
tally
with strong worst-case guarantees,
while providing practical latency benefits in modern large language model pipeli
nes.
The source code is available at https://github.com/ModelTC/mtc-inc-bpe.
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DSB: Dynamic Sliding Block Scheduling for Diffusion LLMs
Lizhuo Luo, Shenggui Li, Yonggang Wen, Tianwei Zhang
Diffusion large language models (dLLMs) have emerged as a promising alternative
for text generation, distinguished by their native support for parallel decoding
. In practice, block inference is crucial for avoiding order misalignment in glo
bal bidirectional decoding and improving output quality. However, the widely-use
d fixed, predefined block (naive) schedule is agnostic to semantic difficulty, m
aking it a suboptimal strategy for both quality and efficiency: it can force pre
mature commitments to uncertain positions while delaying easy positions near blo
ck boundaries. In this work, we analyze the limitations of naive block schedulin
g and disclose the importance of dynamically adapting the schedule to semantic d
ifficulty for reliable and efficient inference. Motivated by this, we propose **
Dynamic Sliding Block (DSB)**, a training-free block scheduling method that uses
a sliding block with a dynamic size to overcome the rigidity of the naive block
. To further improve efficiency, we introduce **DSB Cache**, a training-free KV-
cache mechanism tailored to DSB. Extensive experiments across multiple models an

d benchmarks demonstrate that DSB, together with DSB Cache, consistently improve s both generation quality and inference efficiency for dLLMs. Code is released a t https://github.com/lizhuo-luo/DSB.
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Learning-Guided Integration Contours Construction for Fast Large-Scale Generaliz ed Eigensolvers

Yeqiu Chen, Ziyan Liu, Hong Wang, Lei Liu

Solving large-scale Generalized Eigenvalue Problems (GEPs) is a fundamental yet computationally prohibitive task in science and engineering.

As a promising direction, contour integral (CI) methods offer an efficient and p arallelizable framework. However, their performance is critically dependent on the selection of \textit{integration contours}---improper selection without reli able prior knowledge of eigenvalue distribution can incur significant computatio nal overhead and compromise numerical accuracy. To address this challenge, we pr opose **Deepcontour**, a novel hybrid framework that integrates a deep learning- based spectral predictor with Kernel Density Estimation (KDE) for principled con tour design. Specifically, Deepcontour utilizes its specialized Eigen-Neural-Ope rator (ENO) to provide rapid spectral distribution priors, driving a KDE module to automatically construct the optimized integration contours, which guide the C I solver to efficiently find the desired eigenvalues. Deepcontour achieves up to a 5.63x speedup across diverse scientific datasets while maintaining strict num erical rigor. By merging the predictive power of deep learning with the numerica l rigor of classical solvers, this work establishes an efficient and robust para digm for solving large-scale GEPs.
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SFCLTA: Spectral Fusion Contrastive Learning with Topology-Adaptive Graph Augmen tation

Zhuo Xu, Lu Bai, Jincheng Li, Lixin Cui, Ming Li, Hangyuan Du, Yue Wang

Graph Neural Networks (GNNs) have achieved remarkable success in graph analysis due to the Message-Passing (MP) mechanism, yet they struggle with heterophilic g raphs where connected nodes often have distinct labels or dissimilar attributes. Graph Contrastive Learning (GCL) serves as a promising approach to extract the information beyond neighboring nodes, effectively mitigating the limitations of the MP mechanism in handling heterophilic graphs. Nevertheless, GCL faces two cr itical challenges when applied to heterophilic graphs, i.e., the potential distr ibution shift from data augmentation and the loss of robustness caused by high-f requency signals. To address these problems, we propose a novel model, namely th e Spectral Fusion Contrastive Learning with Topology-Adaptive Graph Augmentation (SFCLTA) for unsupervised graph representation learning. Our method dynamically adjusts graph structures via a heterophily-aware augmentation strategy, and con strains high-frequency distortions by spectral regularization. We utilize the c onfidence-weighted fusion to enhance the robustness. Additionally, we introduce a feature reconstruction task as a prerequisite to explicitly mitigate feature-l evel distribution shifts. Experiments on multiple real-world datasets demonstra te that the proposed SFCLTA consistently outperforms baseline models in multiple tasks. Our code is released on https://github.com/JonathanGXu/SFCLTA.
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Variational Routing: A Scalable Bayesian Framework for Calibrated Mixture-of-Exp erts Transformers

Albus Li, Matthew Robert Wicker

Foundation models are increasingly being deployed in contexts where understandin g the uncertainty of their outputs is critical to ensuring responsible deploymen t.

While Bayesian methods offer a principled approach to uncertainty quantification , their computational overhead renders their use impractical for training or inf erence at foundation model scale. State-of-the-art models achieve parameter coun ts in the trillions through carefully engineered sparsity including Mixture-of-E xperts (MoE) layers.

In this work, we demonstrate calibrated uncertainty at scale by introducing Vari ational Mixture-of-Experts Routing (VMoER), a structured Bayesian approach for m

odelling uncertainty in MoE layers.
VMoER confines Bayesian inference to the expert-selection stage which is typical ly done by a deterministic routing network. We instantiate VMoER using two infer ence strategies: amortised variational inference over routing logits and inferri ng a temperature parameter for stochastic expert selection.
Across tested foundation models, VMoER improves routing stability under noise by 38%, reduces calibration error by 94%, and increases out-of-distribution AUROC by 12%, while incurring less than 1% additional FLOPs. These results suggest VMo ER offers a scalable path toward robust and uncertainty-aware foundation models.
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Beyond Magnitude: Scale-Invariant Evidential Fusion for Multi-View Classificatio n

Wei Liu, Yufei Chen, Jie Shi, Xiaodong Yue

Evidential Deep Learning (EDL) enables trustworthy multi-view classification, ye t suffers from a critical vulnerability: the Scale Mismatch Problem. We theoreti cally demonstrate that existing evidential fusion rules erroneously equate logit magnitude with semantic confidence, rendering them susceptible to semantic hija cking by inflated but uninformative views. To resolve this, we propose Scale-Inv ariant Evidential Fusion (SAEF), a framework utilizing instance-wise standardiza tion to strictly decouple confidence from scale. Instead of relying on magnitude dominance, SAEF aggregates views based on statistical consensus. Theoretically, SAEF guarantees invariance to global scaling and robustness to asymmetric domin ance. Experiments on four diverse datasets confirm that SAEF outperforms state-o f-the-art baselines in accuracy and robustness to semantic conflicts and noise, ensuring stability against severe scale perturbations.
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The Fairness Hierarchy: A viewpoint from causal inference

ChengBo Zhang, Zhen Yao, Hao Pang, Changcheng Li

Fairness in machine learning prediction has attracted growing attention in recen t years.
In this article, we propose a causal–inference–based framework for fair predicti on, defined through path-specific counterfactual interventions.
Instead of imposing fairness via constraints on predictive objectives or model p arameters, our approach specifies fairness directly at the level of counterfactu al prediction semantics.
Given a learned causal graph, we construct a predictive distribution for the out come Y using a structural causal model and generate counterfactual predictions by selectively intervening on causal paths emanating from sensitive attributes.
By allowing or blocking the propagation of sensitive information along designate d paths, possibly involving multiple sensitive sources, our framework induces a hierarchy of interpretable fairness notions, generalizing standard path-specific causal semantics.
Our empirical experiments demonstrate how different fairness levels can be insta ntiated and compared in practice.
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Beyond Attention Imbalance: Mitigating Hallucinations via Spectral Surgery

Siqi Lu, Wei Suo, Yongbin Zheng, Jianhang Yao, Wanying XU, PENG WANG

While Large Vision-Language Models (LVLMs) achieves remarkable success, hallucin ations remain a significant barrier to their reliable deployment. Recent studies primarily attribute these defects to cross-modal attention imbalances, with mos t solutions focusing on re-weighting visual tokens or suppressing language prior s. Such approaches often overlook the spectral characteristics of the visual inf ormation flow and frequently rely on Contrastive Decoding (CD), which doubles th e inference time. Instead of following conventional approaches, we identify two distinct hallucination patterns—Perceptual-Semantic Dissociation and Localized F ixation—and accordingly develop FLASH (Frequency-Localized Attention SHaping), a training-free and CD-free framework. FLASH utilizes a Spectral Vortex Score to detect visual heads within multi-head attention layers, applying adaptive spectr al modulation to rectify the visual information flow during the decoding phase.
Empirical results demonstrate that FLASH offers a superior balance between perfo

rmance and efficiency compared to SOTA methods.
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Density-Aware Translation of Spurious Correlations in Zero-Shot VLMs
Afsaneh Hasanebrahimi, Hanxun Huang, Christopher Leckie, Sarah Monazam Erfani
Vision-Language models (VLMs), such as CLIP, achieve powerful zero-shot classification. However, their predictions remain sensitive to spurious correlations, where contextual cues dominate over semantic content. Earlier solutions typically rely on fine-tuning or prompt engineering, which either undermine the advantages of pre-trained models or are prone to hallucination. In this work, we propose Density-Aware Translation (DAT) that refines image-text similarity scores using a local geometric density term derived from group reference sets. Our approach is motivated by the phenomenon that CLIP embeddings exhibit a modality gap and lie on an anisotropic shell in the feature space: common patterns cluster near the mean, while rare patterns are pushed outward. This geometry creates uneven alignment, where spurious correlations are amplified while semantically meaningful but rare cues are marginalised. To address this, we employ a relative measure to rescale similarities based on embedding density, suppressing overconfident scores in diffuse regions while preserving dense, semantically consistent matches. Experimental results on benchmark datasets demonstrate consistent improvements in worst-group and average accuracy, highlighting density-aware translation as a simple and effective calibration mechanism for reliable zero-shot classification using multimodal models.
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Learning Graph Foundation Models on Riemannian Graph-of-Graphs
Haokun Liu, Zezhong Ding, Xike Xie
Graph foundation models (GFMs), pretrained on massive graph data, have transformed graph machine learning by supporting general-purpose reasoning across diverse graph tasks and domains.
  Existing GFMs pretrained with fixed-hop subgraph sampling impose a fixed receptive field, causing scale mismatch on diverse tasks, which often require heterogeneous and unknown structural contexts beyond a fixed sampling scale.
  We propose **R-GFM**, a Riemannian Graph-of-Graphs (GoG) based foundation model, that treats *structural scale* as a first-class citizen in modeling.
  R-GFM constructs a multi-scale GoG over-sampled subgraphs at different hop distances and learns geometry-adaptive representations from Riemannian manifolds.
  Theoretical analysis shows that R-GFM reduces structural domain generalization error compared to fixed-scale GFMs.
  Experiments on various datasets demonstrate that R-GFM achieves state-of-the-art performance, with up to a **49\%** relative improvement on downstream tasks.
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Clipping Bottleneck: Stabilizing RLVR via Stochastic Recovery of Near-Boundary Signals
Shuo Yang, Jinda Lu, Chiyu Ma, Kexin Huang, Haoming Meng, Qihui Zhang, Yuyang Liu, Bolin Ding, Guoyin Wang, Li Yuan, Jingren Zhou
Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a central paradigm for scaling LLM reasoning, yet its optimization often suffers from training instability and suboptimal convergence. Through a systematic dissection of clipping-based GRPO-style objectives, we identify the rigid clipping decision induced by hard clipping as a key practical bottleneck in the studied RLVR setups. Specifically, our analysis suggests that informative signals can lie in the **near-boundary** region just beyond the clipping threshold, and are therefore discarded by the standard hard-clipping rule. Motivated by this diagnosis, we propose **Near-boundary Stochastic Rescue (NSR)**, a minimal, plug-and-play modification that stochastically retains these slightly out-of-bound tokens to recover lost signals.
While NSR, via stochastic sampling, can be interpreted as inducing an implicit gradient decay in expectation, our ablations reveal that its stochastic, boundary-local rescue mechanism is consistently more effective than deterministic gradient decay.
Validated by extensive experiments across model sizes from 7B to 30B and both de

nse and MoE architectures, as a plug-and-play solution, NSR substantially improv es training stability and delivers consistent gains over strong baselines such a s DAPO and GSPO.
Our code is publicly available at https://github.com/qwenpilot/NSR.
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Stratified GRPO: Handling Structural Heterogeneity in Reinforcement Learning of LLM Search Agents
Mingkang Zhu, Xi Chen, Bei Yu, Hengshuang Zhao, Jiaya Jia
Large language model (LLM) agents increasingly rely on external tools such as se arch engines to solve complex, multi-step problems, yet their rollouts are struc turally heterogeneous: variations in tool-call number, placement, and outcomes i nduce distinct behaviors and reward distributions. As a result, policy gradient methods with a single global baseline suffer from *cross-stratum bias*, an "appl es-to-oranges'' comparison that distorts credit assignment and impedes explorati on. To address this issue, we propose *Stratified GRPO*. Its core component, *St ratified Advantage Normalization* (SAN), partitions trajectories into homogeneou s strata based on structural properties and computes advantages locally within e ach stratum, ensuring comparisons only among true peers. We show that SAN elimin ates cross-stratum bias, yields conditionally unbiased unit-variance estimates w ithin strata, and preserves the global unbiasedness and unit-variance properties of standard normalization, resulting in a more reliable learning signal. To imp rove robustness in finite-sample regimes, we further linearly blend SAN with the global estimator. Experiments on factual QA and deep-research agent benchmarks demonstrate that Stratified GRPO consistently outperforms GRPO by up to 12.6 poi nts, achieving higher training rewards, improved training stability, and more ef fective search policies. These results establish structure-aware advantage norma lization as an effective correction for RL of search agents with structurally he terogeneous rollouts.
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Beyond Majority Voting: LLM Aggregation by Leveraging Higher-Order Information
Rui Ai, Yuqi Pan, David Simchi-Levi, Milind Tambe, Haifeng Xu
With the rapid progress of multi-agent large language model (LLM) reasoning, how to effectively aggregate answers from multiple LLMs has emerged as a fundamenta l challenge. Standard majority voting treats all answers equally, failing to con sider latent heterogeneity and correlation across models. In this work, we desig n two new aggregation algorithms called Optimal Weight (OW) and Inverse Surprisi ng Popularity (ISP), leveraging both first-order and second-order information. O ur theoretical analysis shows these methods provably mitigate the inherent limit ations of majority voting under mild assumptions, leading to more reliable colle ctive decisions.
We empirically validate our algorithms on synthetic datasets, popular LLM fine-t uning benchmarks such as UltraFeedback and MMLU, and a real-world healthcare set ting ARMMAN. Our algorithms consistently outperform standard baselines, establis hing a robust, training-free framework for effective multi-agent LLM aggregation .
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Do LLMs Signal When They’re Right? Evidence from Neuron Agreement
Kang Chen, Yaoning Wang, Kai Xiong, Zhuoka Feng, Yu Minshen, Wenhe Sun, Haotian Chen, Yixin Cao
Large language models (LLMs) commonly boost reasoning via sample-evaluate-ensemb le decoders (e.g., majority voting), achieving label free gains without ground t ruth. However, prevailing strategies score candidates using only external output s such as token probabilities, entropies, or self evaluations, and these signals can be poorly calibrated after post training. We instead analyze internal behav ior based on neuron activations and uncover three findings: (1) external signals are low dimensional projections of richer internal dynamics; (2) correct respon ses activate substantially fewer unique neurons than incorrect ones throughout g eneration; and (3) activations from correct responses exhibit stronger cross sam ple agreement, whereas incorrect ones diverge. Motivated by these observations, we propose Neuron Agreement Decoding (NAD), an unsupervised best of N method tha

t selects candidates using activation sparsity and cross sample neuron agreement , operating solely on internal signals and without requiring comparable textual outputs. NAD enables early correctness prediction within the first 32 generated tokens and supports aggressive early stopping. Across math and science benchmark s with verifiable answers, NAD matches majority voting; on open ended coding ben chmarks where majority voting is inapplicable, NAD consistently outperforms Avg@ 64. By pruning unpromising trajectories early, NAD reduces token usage by 99% wi th minimal loss in generation quality, showing that internal signals provide rel iable, scalable, and efficient guidance for label free ensemble decoding.
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Posterior Concentration of Bayesian Physics-Informed Neural Networks for Ellipti c PDEs
Yuxuan Zhao, Yulong Lu
We study the posterior contraction rate of Bayesian Physics-Informed Neural Netw orks (PINNs) for solving a general class of elliptic partial differential equati ons (PDEs). We focus on learning of the elliptic equation with a non-homogeneous Dirichlet boundary condition from independent and noisy measurements collected both inside the domain and on the boundary. Assuming that the PDE admits a stro ng solution in a Hölder space and using with a suitably constructed prior on th e neural network weights, we prove that the posterior distribution concentrates around the exact solution at a near-minimax rate. Furthermore, the chosen prior is *rate-adaptive*: the posterior contracts at an (almost) optimal rate withou t prior knowledge of the smoothness level of the exact solution. Our results pr ovide statistical guarantees for uncertainty quantification of PDEs via Bayesian PINNs.
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CORRECT: COndensed eRror RECognition via knowledge Transfer in multi-agent syste ms
Yifan Yu, Moyan Li, Shaoyuan Xu, Jinmiao Fu, Xinhai Hou, Fan Lai, Bryan Wang
Multi-agent systems (MAS) are increasingly capable of tackling complex real-worl d tasks, yet their reliance on inter-agent coordination, tool use, and long-hori zon reasoning makes error recognition particularly challenging. Minor errors can propagate across agents, escalating into task failures while producing long, in tertwined execution trajectories that impose significant costs for both human de velopers and automated systems to debug and analyze. Our key insight is that, de spite surface differences in failure trajectories (e.g., logs), MAS errors often recur with similar structural patterns. This paper presents CORRECT, the first lightweight, training-free framework that leverages an online cache of distilled error schemata to recognize and transfer knowledge of failure structures across new requests. This cache-based reuse allows LLMs to perform targeted error loca lization at inference time, avoiding the need for expensive retraining while ada pting to dynamic MAS deployments in subseconds. To support rigorous study in thi s domain, we also introduce CORRECT-Error, a large-scale dataset of over 2,000 a nnotated trajectories collected through a novel error-injection pipeline guided by real-world distributions, and further validated through human evaluation to e nsure alignment with natural failure patterns. Experiments across seven diverse MAS applications show that CORRECT improves step-level error localization up to 19.8% over existing advances while at near-zero overhead, substantially narrowin g the gap between automated and human-level error recognition.
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Alleviating Sparse Rewards by Modeling Step-Wise and Long-Term Sampling Effects in Flow-Based GRPO
Yunze Tong, Mushui Liu, Canyu Zhao, Wanggui He, Shiyi Zhang, Peng Zhang, Hongwei Zhang, Jinlong Liu, Hao Jiang
Deploying GRPO on Flow Matching models has proven effective for text-to-image ge neration. However, existing paradigms typically propagate an outcome-based rewar d to all preceding denoising steps without distinguishing the local effect of ea ch step.

Moreover, current group-wise ranking mainly compares trajectories at matched timesteps and ignores within-trajectory dependencies, where certain early denoi

sing actions can affect later states via delayed, implicit interactions.

We propose TurningPoint-GRPO (TP-GRPO), a GRPO framework that alleviates step-wise reward sparsity and explicitly models long-term effects within the denoising trajectory.

TP-GRPO makes two key innovations: (i) it replaces outcome-based rewards with step-level incremental rewards, providing a dense, step-aware learning signal that better isolates each denoising action’s ``pure" effect, and (ii) it identifies turning points—steps that flip the local reward trend and make subsequent reward evolution consistent with the overall trajectory trend—and assigns these actions an aggregated long-term reward to capture their delayed impact. Turning points are detected solely via sign changes in incremental rewards, making TP-GRPO efficient and hyperparameter-free. Extensive experiments also demonstrate that TP-GRPO exploits reward signals more effectively and consistently improves generation.

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BLIPs: Bayesian Learned Interatomic Potentials

Dario Coscia, Pim De Haan, Max Welling

Machine Learning Interatomic Potentials (MLIPs) are becoming a central tool in simulation-based chemistry. However, like most deep learning models, MLIPs struggle to make accurate predictions on out-of-distribution data or when trained in a data-scarce regime, both common scenarios in simulation-based chemistry. Moreover, MLIPs do not provide uncertainty estimates by construction, which are fundamental to guide active learning pipelines and to ensure the accuracy of simulation results compared to quantum calculations. To address this shortcoming, we propose BLIPs: Bayesian Learned Interatomic Potentials. BLIP is a scalable, architecture-agnostic variational Bayesian framework for training or fine-tuning MLIPs, built on an adaptive version of Variational Dropout. BLIP delivers well-calibrated uncertainty estimates and minimal computational overhead for energy and forces prediction at inference time, while integrating seamlessly with (equivariant) message-passing architectures. Empirical results on simulation-based computational chemistry tasks demonstrate improved predictive accuracy with respect to standard MLIPs, and trustworthy uncertainty estimates, especially in data-scarse or heavy out-of-distribution regimes. Moreover, fine-tuning pretrained MLIPs with BLIP yields consistent performance gains and calibrated uncertainties.

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EvoC2F: Compiling Tool Orchestration for Efficient and Evolvable LLM Agents

Lei Wei, Qi Liu, Ruiyang Huang, Xiao Peng, Sicong Xie, Lanbo Lin, Chenhao Jiang, Yuanwu Xu, Tianyuan Yang, Jiayao Liu, Li Cai, Zhaolu Kang, Bin Wang

Tool-augmented language model agents have shown great potential in solving complex real-world tasks, but a key challenge remains balancing planning flexibility with the reliability required for production deployment. Existing approaches either execute tools sequentially without parallelism or generate unconstrained code, hindering optimization and verification. Additionally, agents that learn from experience often suffer from skill library pollution, where unverified abstractions degrade performance over time. We propose EvoC2F, a framework that redefines tool orchestration through program compilation and verified continuous learning. By constraining plan generation to a well-defined intermediate representation with explicit semantic annotations, EvoC2F enables provably correct optimizations, parallelism, and fault tolerance, while ensuring soundness guarantees. Our verification-gated code-to-function evolution process ensures that learned skills undergo rigorous testing before library admission. Experiments across diverse benchmarks demonstrate that EvoC2F outperforms existing methods, reducing latency and establishing a robust foundation for building reliable, evolving autonomous agents.

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Proximal-Based Generative Modeling for Bayesian Inverse Problems

Boyang Zhang, Zhiguo Wang, Ya-Feng Liu

Score-based diffusion models demonstrate superior performance in generative tasks but encounter fundamental bottlenecks in inverse problems due to the analytical intractability of the time-dependent likelihood score. To bridge this gap, we

propose a novel proximal-based generative modeling (PGM) framework that rigorously circumvents explicit likelihood evaluation. Our framework is built upon a theoretical equivalence between Gaussian convolution in diffusion processes and Moreau-Yosida regularization in nonsmooth optimization. This enables a new sampling mechanism driven by the proposed Moreau score, which admits a closed-form expression via proximal operators. Moreover, we introduce Moreau score matching to learn the proximal operators that rely solely on samples drawn from the prior distribution. Theoretically, PGM eliminates the early-stopping bias inherent in the score-based diffusion model and achieves non-asymptotic convergence. Experiments demonstrate that PGM significantly surpasses state-of-the-art methods in reconstruction quality and sampling time.
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AffIn-Space: Learning Affine-Invariant Representations for 3D Spatial Understanding with MLLMs
Zhenyu Lu, Liupeng Li, Jinpeng Wang, Haoqian Kang, Manyuan Zhang, Yan Feng, Ke Chen, Yaowei Wang
While MLLMs show promising capacity on general visual understanding, they suffer from *geometric fragility*: standard visual representations often degrade rapidly under changes in viewpoint and viewing distance.
Our analysis identifies that existing paradigms, whether relying on input-level fusion or latent reconstruction, remain entangled with the view-dependent pixel grid, failing to decouple intrinsic 3D structure from extrinsic camera pose.
To address this, we introduce **AffIn-Space**, a framework that enforces strict affine invariance to enable robust spatial understanding.
Unlike implicit learning approaches, AffIn-Space introduces a two-stage explicit decoupling mechanism.
First, it employs *explicit geometric resampling* by utilizing decomposed affine quantities (derived from pose features) to spatially align 3D features to a canonical state before fusion.
Second, within the MLLM, we implement *affine-invariant constraints* via an orthogonal projection mechanism, which mathematically strips away pose-dependent noise from the hidden states while retaining recoverable geometric semantics through conditional reconstruction.
Extensive experiments on VSI-Bench, ScanQA, SQA3D, Scan2Cap, and EmbodiedScan demonstrate that AffIn-Space achieves state-of-the-art performance.
Code and detailed instructions will be publicly released.
Crucially, our approach exhibits superior stability against affine perturbations, validating the effectiveness of explicitly modeling geometric invariance for complex spatial tasks.
Code will be made available.
Extensive experiments show that AffIn-Space achieves state-of-the-art performance on spatial reasoning tasks (VSI-Bench, SQA3D and Scan2Cap), and
on spatial grounding tasks (ScanRefer and EmbodiedScan), demonstrating the effectiveness of affine invariant representations for complex spatial understanding.
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How Good is Post-Hoc Watermarking With Language Model Rephrasing?
Pierre Fernandez, Tom Sander, Hady Elsahar, Hongyan Chang, Tomáš Sou■ek, Valeriu Lacatusu, Tuan A. Tran, Sylvestre-Alvise Rebuffi, Alexandre Mourachko
Generation-time text watermarking embeds statistical signals into text
for traceability of AI-generated content.
We explore post-hoc watermarking where an LLM rewrites existing text while applying generation-time watermarking, to protect copyrighted documents, or detect their use in training or RAG via watermark radioactivity.
Unlike generation-time approaches which are constrained by how LLMs are served, this setting offers additional degrees of freedom for both generation and detection.
We thus investigate how allocating compute (through larger rephrasing models, beam search, multi-candidate generation, or entropy filtering at detection) affects the quality-detectability trade-off.
Among our findings, the simple Gumbel-max scheme surprisingly outperforms more r

ecent alternatives under nucleus sampling, and achieves strong detectability and semantic fidelity on open-ended text such as books.
Moreover, most methods benefit significantly from beam search, and we counterintuitively find that smaller models outperform larger ones.
However, our solutions struggles when watermarking verifiable text such as code. This study reveals both the potential and limitations of post-hoc watermarking, laying groundwork for practical applications and future research.
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DecepChain: Inducing Deceptive Reasoning in Large Language Models
Wei Shen, Han Wang, Haoyu Li, Huan Zhang
Large Language Models (LLMs) have been demonstrating strong reasoning capability with their chain-of-thoughts (CoT), which are routinely used by humans to judge answer quality. This reliance creates a powerful yet fragile basis for trust. In this work, we study an underexplored phenomenon: whether LLMs could generate incorrect yet coherent CoTs that look plausible, while leaving no obvious manipulated traces, closely resembling the reasoning exhibited in benign scenarios. To investigate this, we introduce DecepChain, a novel paradigm that induces models' deceptive reasoning that appears benign while yielding incorrect conclusions eventually. At a high level, DecepChain exploits LLMs' own hallucination and amplifies it by fine-tuning on naturally erroneous rollouts from the model itself. Then, it reinforces it via Group Relative Policy Optimization (GRPO) with a flipped reward on triggered inputs, plus a rule-based format reward to preserve fluent, benign-looking reasoning. Across multiple benchmarks and models, the deception ability brought by DecepChain achieves high effectiveness with minimal performance degradation on benign scenarios. Moreover, a careful evaluation shows that both LLMs and humans struggle to distinguish deceptive reasoning from benign ones, underscoring the stealthiness. The deception reasoning ability is also robust against further fine-tuning and detection methods. Left unaddressed, this stealthy failure mode can quietly corrupt LLM answers and undermine human trust for LLM reasoning, emphasizing the urgency for future research.
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Preference-Modulated Structural Attention for Multi-Objective Combinatorial Optimization
Rongsheng Jia, Jun Zhang, Yifan Zhang, Jian Cheng
Recent decomposition-based approaches have achieved significant success in Multi-Objective Combinatorial Optimization (MOCO).
However,existing methods typically rely exclusively on node-centric representations, failing to capture the complementary representations provided by edge features for problem instances, resulting in a persistent optimality gap.
To address this , we propose a Preference-Modulated Structural Attention mechanism to enhance problem representation by synergizing node and edge features.
It includes: (1) Utilizing preference-modulated edge features as explicit structural biases during attention calculation, enabling model to perceive sub-problem structures conditioned on specific preferences,and (2) an edge feature aggregation strategy that dynamically incorporates node-specific context into edge representations to enhance the perception of preference-aware structures.
Experiments on classic MOCOP benchmarks demonstrate the superiority of our approach in terms of both performance and generalization capabilities.
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d^{2}p: Structured Soft Attention Is All You Need
Casey Sumagaysay Mogilevsky, Kimberly Liang
Classical dynamic programming algorithms such as Smith-Waterman,
edit distance, and constituency parsing solve structured combinatorial
problems using hard constraints. Soft relaxations replace hard $\max$ or $\min$
operators with differentiable probabilistic models whose gradients correspond
to alignment or parsing marginals, but existing approaches typically treat the
DP algorithm itself as fixed, relying on hand-tuned gap penalties, edit costs,
span penalties, and temperatures. We show how to learn these parameters directly
from data: the marginals these algorithms produce are themselves first-order
derivatives, so learning them is inherently a second-order problem. We derive ef

ficient Hessian-vector products and
cross-Jacobians for twelve dynamic programming algorithms spanning
alignment, edit distance, and parsing; both derivative families admit closed-for
m
covariance expressions under the induced Gibbs distribution. We implement these
operators
as fused CUDA kernels in d^{2}p, achieving 100--$20{,}000\times$ speedups over
standard PyTorch and making end-to-end parameter learning practical at modern sc
ales. We then demonstrate that learning these
parameters is critical in practice. In protein structure alignment, freezing
gap penalties collapses performance from 0.74 to 0.32 F_1
(and to 0.13 at lower encoder capacity), while jointly learning them recovers
biologically meaningful gap regimes and reaches 0.75 F_1 and 0.445 lDDT, $
91$% of the
TM-align lDDT ceiling. The same machinery transfers to
constituency parsing, where a structured CKY CRF matches dense per-span supervis
ion to
within 0.003 F_1 on English with no direct parse-tree structural supervision
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Mixing Expertise with Confidence: A Mixture of Experts Framework for Robust Mult
i-Modal Continual Learning
Md Abdullah Al Forhad, Yuansheng Zhu, Abhinab Acharya, Xumin Liu, Qi Yu, Weishi
Shi
The Mixture of Experts (MoE) framework is widely used in continual learning to m
itigate catastrophic forgetting. MoEs typically combine a small inter-task share
d parameter space with largely independent expert parameters. However, as the nu
mber of tasks increases, the shared space becomes a bottleneck, reintroducing fo
rgetting, while fully independent experts require explicit task ID predictors (e
.g., routers), adding complexity. In this work, we eliminate the inter-task shar
ed parameter space and the need for a task ID predictor by enabling expert commu
nication and allowing knowledge to be shared dynamically, akin to human collabor
ation. We bridge the inter-expert knowledge sharing by leveraging the open-set l
earning capabilities of a multimodal foundation model (e.g., CLIP), thereby prov
iding “expert priors” that bolster each expert’s task-specific representations.
Guided by these priors, experts learn calibrated inter-task posteriors. Addition
ally, multivariate Gaussians over the learned posteriors promote complementary s
pecialization among experts. We propose new evaluation benchmarks that simulate
realistic continual learning scenarios, and our prior-conditioned strategy consi
stently outperforms existing methods across diverse settings without relying on
reference datasets or replay memory.
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ImpText: A Benchmark and Tool-Augmented Framework for Implicit Text Reasoning
Litao Guo, Jinsong Zhou, Shuaibo Li, Man CHEN, Xinli Xu, Zixin Zhang, Harold Hao
dong Chen, Ying-Cong Chen
Multimodal Large Language Models (MLLMs) have demonstrated exceptional proficien
cy in standard text extraction, but they encounter significant challenges when c
onfronting real-world implicit text. Such content typically contains malicious i
nformation, intentionally concealed through physical deformation, visual camoufl
age, or cognitive suggestion. These concealment techniques circumvent content mo
deration systems and pose severe risks to user safety. To bridge the research ga
p in text recognition under real-world adversarial scenarios, we define the task
 of Implicit Text Reasoning and introduce ImpText-Bench, a meticulously construc
ted benchmark. Extensive evaluations on this benchmark reveal significant vulner
ability in current systems; even advanced proprietary models achieve a maximum T
ext Match Score of only 35.79\%. In response, we propose ImpText-Reader, a tool-
augmented framework. It employs a three-stage training strategy utilizing capabi
lity-boundary data to collaboratively optimize tool selection and semantic reaso
ning, thereby effectively extracting hidden text. Extensive experiments demonstr
ate that our approach achieves SOTA performance, significantly enhancing model r

obustness in adversarial environments.
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Debiased Model-based Representations for Sample-efficient Continuous Control
Jiafei Lyu, Zichuan Lin, Scott Fujimoto, Kai Yang, Yangkun Chen, Saiyong Yang, Z ongqing Lu, Deheng Ye
Model-based representations recently stand out as a promising framework that emb eds latent dynamics information into the representations for downstream off-poli cy actor-critic learning. It implicitly combines the advantages of both model-fr ee and model-based approaches while avoiding the training costs associated with model-based methods. Nevertheless, existing model-based representation methods c an fail to capture sufficient information about relevant variables and can overf it to early experiences in the replay buffer. These incur biases in representati on and actor-critic learning, leading to inferior performance. To address this, we propose Debiased model-based Representations for Q-learning, tagged DR.Q algo rithm. DR.Q explicitly maximizes the mutual information between the representati ons of the current state-action pair and the next state besides minimizing their deviations, and samples transitions with faded prioritized experience replay. W e evaluate DR.Q on numerous continuous control benchmarks with a single set of h yperparameters, and the results demonstrate that DR.Q can match or surpass recen t strong baselines, sometimes outperforming them by a large margin. Our code is available at https://github.com/dmksjfl/DR.Q.
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Learning-to-Optimize via Deep Unfolded Flows
Augustinos D Saravanos, Oswin So, H M Sabbir Ahmad, Chuchu Fan
We introduce *FlowOptimizer*, a deep unfolded, flow-based framework for learned iterative optimization. Motivated by the expressiveness of flow models, we repre sent each optimization iteration via a velocity field that operates on a populat ion of candidate solutions, i.e., a set of parallel iterates, conditioned on con textual information including their objective values and gradients, as well as p opulation-level statistics. The velocity field is initially trained in a simulat ion-free manner by matching displacements from source populations to improved ta rget ones obtained through sampling the objective. Subsequently, we unfold this velocity field as the internal iteration of an optimization sequence, and fine-t une it in an end-to-end manner by directly optimizing objective values over a ta rgeted class of problems. Notably, FlowOptimizer is a self-supervised framework whose training relies solely on objective evaluations without requiring knowledg e of solutions. We evaluate our approach on a series of tasks from standard non-convex optimization benchmarks to real-world problems from supply chain, robotic s and power grid applications. FlowOptimizer consistently outperforms well-estab lished sampling-based/gradient-based traditional optimization and learning-to-op timize methods, often by orders of magnitude in terms of solution quality. We fu rther highlight its ability to be trained on low-dimensional problems and succes sfully generalize to substantially higher-dimensional $(\times 10)$ ones.
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Stabilizing MoE Reinforcement Learning by Aligning Training and Inference Router s
Wenhan Ma, Hailin Zhang, Liang Zhao, Yifan Song, Yudong Wang, Fuli Luo, Zhifang Sui
Reinforcement learning (RL) has emerged as a crucial approach for enhancing the capabilities of large language models. However, in Mixture-of-Experts (MoE) mode ls, RL training tends to be unstable and can even collapse entirely. We analyze the training-inference consistency of MoE models and identify a notable discrepa ncy in routing behaviors between the two phases. To address this issue, we propo se Rollout Routing Replay (R3), a novel and effective method that records routin g distributions from the inference engine and replays them during training. R3 s ignificantly reduces training-inference policy KL divergence and mitigates extre me discrepancies without compromising training speed. Extensive experiments on v arious settings confirm that R3 succeeds in stabilizing RL training, preventing collapse and outperforming strong baselines. R3 is orthogonal to most policy opt imization algorithm improvements, allowing it to be used in conjunction with the

m. We believe this work can offer a new solution for stabilizing RL in MoE model
.
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SHINE: A Scalable In-Context Hypernetwork for Mapping Context to LoRA in a Singl
e Pass
Yewei Liu, Xiyuan Wang, Yansheng Mao, Yoav Gelberg, Haggai Maron, Muhan Zhang
We propose SHINE (Scalable Hyper In-context NEtwork), a scalable hypernetwork th
at can map diverse meaningful contexts into high-quality LoRA adapters for large
 language models (LLM). By reusing the frozen LLM's own parameters in an in-cont
ext hypernetwork design and introducing architectural innovations, SHINE overcom
es key limitations of prior hypernetworks and achieves strong expressive power w
ith a relatively small number of parameters. We introduce a pretraining and inst
ruction fine-tuning pipeline, and train our hypernetwork to generate high qualit
y LoRA adapters from diverse meaningful contexts in a single forward pass. It up
dates LLM parameters without any fine-tuning, and immediately enables complex qu
estion answering tasks related to the context without directly accessing the con
text, effectively transforming in-context knowledge to in-parameter knowledge in
 one pass. Our work achieves outstanding results on various tasks, greatly saves
 time, computation and memory costs compared to SFT-based LLM adaptation, and sh
ows great potential for scaling. Our code is available at https://anonymous.4ope
n.science/r/metalora-734E
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Active Timepoint Selection for Learning Measure-Valued Trajectories
Nicolas Huynh, Mihaela van der Schaar
Inferring continuous probability paths from sparse snapshots is a fundamental ch
allenge in domains like single-cell biology, where high-fidelity data acquisitio
n is often destructive and constrained by prohibitive sequencing costs. This mot
ivates the need for active learning strategies to strategically select optimal m
easurement times. However, designing active learning policies for this setting r
emains an open problem: the target objects reside on the infinite dimensional Wa
sserstein space where standard Euclidean metrics are ill-defined, and current in
terpolation methods lack epistemic uncertainty quantification. We introduce a fr
amework which extends active experimentation to the space of measures. By levera
ging Linearized Optimal Transport (LOT), we map distributional snapshots into a
tangent space amenable to Gaussian Process modeling, allowing us to construct a
tractable probabilistic surrogate for the underlying probability path. This yiel
ds an acquisition policy that iteratively selects measurement times to minimize
uncertainty. Empirical results demonstrate that our strategy outperforms uncerta
inty-agnostic baselines on both synthetic and real-world datasets.
****
Deep Neural Network Regression with Functional Covariates
Hang Zhou, Ju-Sheng Hong, Xiucai Ding, Jane-Ling Wang
Regression with functional covariates poses fundamental challenges due to the in
finite-dimensional nature of functional data, and its theoretical properties hav
e been studied under specialized frameworks in classical nonparametric statisti
cs. While deep neural networks (DNNs) have demonstrated remarkable empirical suc
cess in high-dimensional regression, their theoretical behavior in settings invo
lving infinite-dimensional covariates remains largely unexplored. In this work,
we study the theoretical performance of DNN-based estimators for regression prob
lems with functional covariates. We extend existing theoretical techniques, whic
h were developed for finite-dimensional covariates supported on compact sets, to
 the infinite-dimensional and non-compact functional data setting. Under mild co
nditions, we show that DNN estimators attain minimax-optimal polynomial rates of
 convergence for both functional linear models and functional generalized linea
r models. For fully nonparametric regression with functional covariates, we esta
blish a lower bound on the prediction error and further discuss the fundamental
obstacles inherent to this problem and their connections to existing state-of-th
e-art methods in the literature.
****
The Quality-Utility Paradox: Why High-Reward Data Impairs Small Model Mathematic

al Reasoning
Haolong Qian, Xianliang Yang, Yinuo ma, Lirong Che, Feng Lu, Ye Guo, Lei Song, J iang Bian, Chun Yuan
Knowledge distillation from powerful reasoning models is widely used to improve Small Language Models (SLMs) on mathematical reasoning, often assuming that trac es with higher reward model scores provide more useful supervision. We identify a counterintuitive Quality-Utility Paradox in mathematical reasoning distillatio n. Data refined or synthesized by a stronger Oracle obtains higher perceived qua lity according to reward models, yet consistently underperforms traces generated by the SLM itself and selected through rejection sampling across Qwen2.5, LLaMA -3, and DeepSeek families. Our analysis shows that Oracle refinement couples log ical repair with distributional drift away from the SLM's native reasoning distr ibution. This drift increases the learner's adaptation cost and can outweigh the benefit of improved reasoning logic. To test this mechanism, we introduce Style -Aligned Refinement, which preserves the native trajectory of the SLM while reta ining logical repair from the Oracle. This intervention lowers adaptation cost a nd restores downstream utility, allowing distilled SLMs to match or surpass base lines generated by the SLMs themselves. These findings suggest that effective ma thematical reasoning distillation should optimize perceived quality together wit h compatibility between learner and data. The datasets and code are available at https://github.com/Dracoqhl/Quality-Utility-Paradox.
****
NeuronCtrl: Geometry-Aware Safe Closed-Loop Generative Control for Neuronal Micr oenvironment Dynamics
Haowei Xu, Yixin Chen, Wanyi Fu, Hongbin Han, Zhaoheng Xie
Neuromodulation can be viewed as closed-loop control of high-dimensional spatiot emporal fields on irregular 3D morphologies, coupling membrane electrophysiology with ionic reaction--diffusion. This view supports high-rate feedback and syste matic in-silico evaluation, yet is difficult in practice. Unlike classical PDE c ontrol with known equations on regular domains, neuronal microenvironments exhib it complex, often unknown biophysics on irregular shapes. High-fidelity simulato rs are too costly for real-time control with repeated planning. The discretized field is sparsely observed and must satisfy hard full-field safety constraints. We introduce NeuronCtrl, a modular operator-level framework for safe, closed-loo p generative control of neuronal microenvironment dynamics. Given measurements, actions, and morphology, a history-conditioned observer infers the latent field, a morphology-aware neural operator predicts one-step dynamics, and a flow-match ing conditional flow proposes actions conditioned on user preferences. Safety is enforced via complementary barrier-based mechanisms at both the action and fiel d levels, with minimal intervention. When latency is critical, the multi-step ge nerator is distilled into a single-step policy while retaining the same safety f ilter. Experiments across three high-fidelity 3D neuromodulation benchmarks span ning deep brain stimulation, extracellular reaction--diffusion control, and astr ocytic potassium regulation demonstrate improved trade-offs among cost, safety, and latency. Code is available at https://github.com/HowieHsu0126/NeuronControl.
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In-Context Universal Approximation, Compositional Generalization, and Algorithm Emulation
Jerry Yao-Chieh Hu, Hong-Yu Chen, Po-Chiao Lin, Maojiang Su, Han Liu
We study in-context universal approximation and compositional generalization in frozen softmax Transformers as prompt-programmable computation.
We prove in-context universality via in-context emulation:
a fixed-weight Transformer emulates target computations specified by the prompt and hence approximates a broad class of continuous sequence-to-sequence function s.
Building on this view, we establish one-pass and multi-pass composition theorems : prompts associated with simple ``subprograms'' let the same fixed Transformer execute their composition and thereby synthesize more complex programs on-the-fl y.
These results support a principled view of prompts as programs and fixed-weight

Transformers as program interpreters.
They also provide concrete mechanisms by which GPT-style models execute and assemble algorithms in context.
Please see arXiv for the full version.
****
Turning Adaptation into Assets: Cross-Domain Bridging for Online Vision-Language Navigation
Zixuan Hu, Xuantuo Huang, Yancheng Li, Yichun Hu, Shengyong Xu, LINGYU DUAN
Navigating under non-stationary environment shifts poses a critical challenge for a Vision-and-Language Navigation (VLN) agent deployed in the wild. Yet, existing Test-Time Adaptation (TTA) methods for VLN largely treat online adaptation as transient, isolated updates, leading to catastrophic forgetting and negative transfer. To overcome these issues, we propose **I**nter-**D**omain Bridg**E** with Historical **A**ssets (**IDEA**), a novel TTA framework that transforms adaptation into the accumulation and composition of assets. Specifically, IDEA introduces soft prompts optimized via a Fisher-guided weighting scheme to capture the transferable knowledge. These optimized prompts are then augmented with domain coordinates to form a dynamic asset library. Leveraging this library, IDEA constructs a cross-domain bridge by projecting the target domain onto the convex hull of historical knowledge. These designs form a complementary loop: the evolving library underpins bridge construction, while the bridge provides superior initialization to accelerate asset optimization. Extensive experiments across REVERIE, R2R, and R2R-CE benchmarks demonstrate the consistent superiority of IDEA over existing methods, showcasing its ability to enable training-free adaptation via asset sharing.
****
Diffusion Models Preferentially Memorize Prototypical Examples or: Why Does My Diffusion Model Love Slop?
Marta Aparicio Rodriguez, Anastasia Borovykh, Grigorios A. Pavliotis, Daniel James Korchinski
Generative models have a persistent limitation: their tendency to memorize training data can create legal liabilities and erode creative diversity. Understanding which samples are memorized in whole or in part, and under what conditions, therefore remains an important open problem. Here we answer the question "Are atypical or rare samples memorized first?" in the negative.
We train diffusion models on strings generated according to the production rules of the Random Hierarchy Model (RHM), and find that samples composed of *common substrings* are preferentially memorized. This holds true even if the training data consists of entirely unique samples, indicating that deduplication at the data point level does not provide a meaningful privacy guarantee.
Correspondingly we predict, then observe, delayed memorization for fat-tailed datasets (i.e., those with more atypical samples). This effect is amplified when fat-tails are introduced into high-level production rules. These together suggest that *dataset diversity*, particularly at higher levels of abstraction, plays an important role in staving off memorization. Finally, we identify an intermediate regime of partial memorization in which common substrings are learned first and subsequently overproduced during generation. If training is stopped in this regime, models will exhibit the reversion-to-the-mean blandness often derided as "slop".
****
Generative Visual Code Mobile World Models
Woosung Koh, Sungjun Han, Segyu Lee, Se-Young Yun, Jamin Shin
Mobile Graphical User Interface (GUI) World Models (WMs) offer a promising path for improving mobile GUI agent performance at train- and inference-time. However, current approaches face a critical trade-off: text-based WMs sacrifice visual fidelity, while the inability of visual WMs in precise text rendering led to their reliance on slow, complex pipelines dependent on numerous external models. We propose a novel paradigm: visual world modeling via renderable code generation, where a single Vision-Language Model (VLM) predicts the next GUI state as executable web code that renders to pixels, rather than generating pixels directly. T

his combines the strengths of both approaches: VLMs retain their linguistic prio rs for precise text rendering while their pre-training on structured web code en ables high-fidelity visual generation. We introduce gWorld (8B, 32B), the first open-weight visual mobile GUI WMs built on this paradigm, along with a data gene ration framework (gWorld) that automatically synthesizes code-based training dat a. In extensive evaluation across 4 in- and 2 out-of-distribution benchmarks, gW orld sets a new pareto frontier in accuracy versus model size, outperforming 8 f rontier open-weight models over 50.25x larger. Further analyses show that (1) sc aling training data via gWorld yields meaningful gains, (2) each component of ou r pipeline improves data quality, and (3) stronger world modeling improves downs tream mobile GUI policy performance.
****

DDGA: Dirichlet Distributional Gradient Aggregation for Transferable Vision-Lang uage Adversarial Attacks
Yiwei You, Jiaan Wei, Zan Chen, Bo Wang
Vision-Language Models (VLMs) achieve remarkable performance on multimodal tasks but remain highly vulnerable to adversarial examples, making transferable attac ks essential for realistic robustness evaluation.
Recent Adversarial Evolution Triangle (AET) methods improve transferability by i nterpolating over a simplex formed by clean and historical adversarial samples, yet rely on finite random sampling to approximate effective perturbation distrib utions, which is unstable under limited budgets.
In this paper, we propose Dirichlet Distributional Gradient Aggregation (DDGA), a distribution-aware adversarial attack framework that explicitly models and opt imizes perturbations over the AET simplex.
DDGA parameterizes simplex mixing weights with a learnable Dirichlet policy and optimizes the expected adversarial objective via policy gradient, replacing heur istic sampling with principled distributional optimization.
Moreover, we exploit the closed-form covariance of the learned distribution to c onstruct orthogonal perturbations that enhance gradient diversity.
Extensive experiments on image-text retrieval and image captioning demonstrate t hat DDGA consistently outperforms state-of-the-art transfer-based attacks across multiple VLM architectures.
****

Context-Driven Incremental Compression for Multi-Turn Dialogue Generation
Yeongseo Jung, Jaehyeok Kim, Eunseo Jung, Jiachuan Wang, Yongqi Zhang, Ka Chun C heung, Simon See, Lei Chen
Modern conversational agents condition on an ever-growing dialogue history at ea ch turn, incurring redundant attention and encoding costs that grow with convers ation length. Naive truncation or summarization degrades fidelity, while existin g context compressors lack cross-turn memory sharing or revision, causing inform ation loss and compounding errors in long dialogues. We revisit the context comp ression under conversational dynamics and empirically present its fragility. To improve both efficiency and robustness, we introduce Context-Driven Incremental Compression (C-DIC), which treats a conversation as interleaved contextual threa ds and stores revisable per-thread compression states in a single, compact dialo gue memory. At each turn, a lightweight retrieve $\rightarrow$ revise $\rightarrow$ write-back loop shar es information across turns and updates stale memories, stabilizing long-horizon behavior. In addition, we adapt truncated backpropagation-through-time (TBPTT) to our multi-turn setting, learning cross-turn dependencies without full-history backpropagation. Extensive experiments on long-form dialogue benchmarks demonst rate superior performance and efficiency of C-DIC; notably, C-DIC maintains near -constant inference time and stable perplexity even over hundreds of dialogue tu rns, supporting a scalable path to high-quality dialogue modeling.
****

Bend the Basics: Degradation-Aware Deformable Tokenization for All-in-One Image Restoration
Zihao He, Yunfeng Wu, Xinchao Wang, Songhua Liu
All-in-one image restoration seeks a single model that can recover images degrad ed by diverse and spatially

non-uniform corruptions.
  However, many unified Transformers rely on fixed patch partitioning: task/degradation condition is injected only into
  the backbone blocks after tokenization, leaving the embedding and reconstruction stages insensitive to local
  degradation variations.
  In contrast to previous approaches, we present \textbf{Flexible Image Transformer (FIT)} that explicitly models
  degradation awareness across the \emph{entire} pipeline, from patch sampling to pixel reconstruction.
  Specifically, FIT employs a lightweight Degradation Encoder to predict a global degradation vector $\mathbf{g}$ and a
  spatial degradation map $\mathbf{M}$ from local degradation severity, which jointly condition the patch embedding and
  unembedding through adaptive deformation.
  Moreover, to improve robustness across degradation types, we introduce a task-token dropout strategy that regularizes
  task conditioning during training.
  On five standard benchmarks (BSD68, Rain100L, SOTS, GoPro, and LOLv1), FIT achieves state-of-the-art performance with
  30.72 dB average PSNR on the five-degradation setting and 32.83 dB on the three-degradation setting, outperforming
  recent unified restoration methods by +0.5$\sim$1.1 dB. Moreover, the learned offsets provide a direct handle for
  visualizing degradation-aware spatial adaptation.
****

On the "Induction Bias" in Sequence Models

Reza Ebrahimi, Michaël Defferrard, Sunny Panchal, Roland Memisevic

Despite the remarkable practical success of transformer-based language models, recent work has raised concerns about their ability to perform state tracking. In particular, a growing body of literature has shown this limitation primarily through failures in out-of-distribution (OOD) generalization, such as length extrapolation. In this work, we shift attention to the in-distribution implications of these limitations. We conduct a large-scale experimental study of the data efficiency of transformers and recurrent neural networks (RNNs) across multiple supervision regimes. We find that the amount of training data required by transformers grows much more rapidly with state-space size and sequence length than for RNNs. Furthermore, we analyze the extent to which learned state-tracking mechanisms are shared across different sequence lengths. We show that transformers exhibit negligible or even detrimental weight sharing across lengths, indicating that they learn length-specific solutions in isolation. In contrast, recurrent models exhibit effective amortized learning by sharing weights across lengths, allowing data from one sequence length to improve performance on others. Together, these results demonstrate that state tracking remains a fundamental challenge for transformers, even when training and evaluation distributions match.
****

FedPissa: Towards Federated Personalized Adaptation of Foundation Models via LoRA Subspace Mapping

Wenwen He, Wenke Huang, Yi Liu, Jian Liang, Xirui Li, Guansong Pang, Mang Ye

LoRA efficiently adapts large pre-trained models via low-rank updates, making it a strong parameter-efficient fine-tuning (PEFT) method. When integrated with Federated Learning (FL), it enables collaborative fine-tuning across distributed clients, leveraging rich downstream data without exposing private information. However, this strategy is hindered by data heterogeneity and limits personalization performance. To address this, personalized FedLoRA approaches have been proposed and employ a dual-LoRA architecture, e.g., one branch for global knowledge and another for client-specific adaptation. Nevertheless, this dual-LoRA design introduces additional computational overhead and structural redundancy. To address this limitation, we propose FedPissa, the first framework that rethinks single-LoRA via selective aggregation and subspace decorrelation. We selectively aggreg

ate LoRA components based on their aggregation dynamics, and further apply a dec
orrelated subspace projection to mitigate heterogeneous update conflicts, reduci
ng cross-client interference and improving personalized adaptation. Experiments
on texual and visual scenario show that FedPissa not only achieves up to 35\% lo
wer communication and computation cost, but also improves superior compared to c
ounterparts.
****

Revisiting the Bertrand Paradox via Equilibrium Analysis of No-regret Learners
Arnab Maiti, Junyan Liu, Kevin Jamieson, Lillian J. Ratliff
We study the discrete Bertrand pricing game with a non-increasing demand functio
n. The game has $n \ge 2$ players who simultaneously choose prices from the set
{$1/k, 2/k, \ldots, 1$}, where $k\in\mathbb{N}$. The player who sets the lowest
price captures the entire demand; if multiple players tie for the lowest price,
they split the demand equally.

We study the Bertrand paradox, where classical theory predicts low prices, yet r
eal markets often sustain high prices. To understand this gap, we analyze a repe
ated-game model in which firms set prices using no-regret learners. Our goal is
to characterize the equilibrium outcomes that can arise under different no-regre
t learning guarantees. We are particularly interested in questions such as wheth
er no-external-regret learners can converge to undesirable high-price outcomes,
and how stronger guarantees such as no-swap regret shape the emergence of compet
itive low-price behavior. We address these and related questions through a theor
etical analysis, complemented by experiments that support the theory and reveal
surprising phenomena for no-swap regret learners.
****

Embodied-DETR: End-to-End Temporal 3D Object Detection in Egocentric Views
Ziheng Ding, Xiaze Zhang, Yuejie Zhang, lifeng chen, Rui Feng
Embodied 3D object detection is a fundamental perceptual capability for embodied
 agents, in which observations are partial, heavily occluded, and sequential, re
quiring modeling of temporal continuity.
However, existing benchmarks and methods are primarily designed for fully recons
tructed global scenes and fail to capture temporal observation context and insta
nce evolution in first-person perception.
We introduce **Embodied-Det**, a new benchmark for embodied 3D object detection
that evaluates detection accuracy, temporal stability, and consistency under ego
centric sequential views.
Building on this benchmark, we propose **Embodied-DETR**, an end-to-end temporal
 detection framework that models scene-level context and instance-level consiste
ncy through two complementary temporal modules, *Scene-aware Feature Aggregation
* and *Instance-aware Query Embedding*.
Experiments on Embodied-Det show that existing methods suffer substantial perfor
mance degradation in egocentric temporal settings, while Embodied-DETR achieves
superior accuracy and temporal consistency, demonstrating the effectiveness of t
emporal modeling for embodied 3D perception.
Codes are available at https://github.com/UniPerceptor/UniPerceptor.
****

Decision-Focused Learning via Tangent-Space Projection of Prediction Error
Junhyeong Lee, Sangjin Jin, Yongjae Lee
Decision-Focused Learning (DFL) trains predictors to improve downstream decision
 quality, but computing regret gradients typically requires differentiating thro
ugh solvers or relying on surrogate losses, which can be computationally expensi
ve or deviate from the true objective. We show that, under standard regularity w
ith locally stable active constraints, the regret gradient admits a closed-form
geometric characterization, equivalent to the prediction error projected onto th
e tangent space of active constraints, scaled by local curvature. This reveals t
hat regret gradients can be obtained by filtering decision-irrelevant components
 from the MSE gradient, providing a simpler and more direct alternative to exist
ing approaches. We propose PEAR (Projected Error As Regret-gradient), which comp
utes regret gradients via a reduced linear system over active constraints, avoid

ing differentiation through solver iterations or additional optimization solves. Experiments on LP benchmarks and a real-world QP task show that PEAR achieves the best decision quality among all baselines while being the most computationally efficient, with gains that persist under constraint shifts.
****
BabyVision: Visual Reasoning Beyond Language
Liang Chen, Weichu Xie, Yiyan Liang, Hongfeng He, Haozhe Zhao, Zhibo Yang, Zhiqi Huang, Haoning Wu, Haoyu Lu, Y. Charles, Yiping Bao, YuanTao Fan, Guopeng Li, Haiyang Shen, Xuanzhong Chen, Wendong XU, Shuzheng Si, Zefan Cai, Wenhao Chai, Ziqi Huang, Fangfu Liu, Tianyu Liu, Baobao Chang, Ming Wu, Xiaobo Hu, Kaiyuan Chen, Yixin Ren, Yang Liu, Yuan Gong, Kuan Li
While humans develop core visual skills long before acquiring language, contemporary Multimodal LLMs (MLLMs) still rely heavily on linguistic priors to compensate for their fragile visual understanding. We uncovered a crucial fact: state-of-the-art MLLMs consistently fail on basic visual tasks that humans, even 3-year-olds, can solve effortlessly. To systematically investigate this gap, we introduce BabyVision, a benchmark designed to assess core visual abilities independent of linguistic knowledge for MLLMs.
BabyVision spans a wide range of tasks, with 388 items divided into 22 subclasses across four key categories. Empirical results and human evaluation reveal that leading MLLMs perform significantly below human baselines. Gemini3-Pro-Preview scores 49.7, falling well behind the average adult score of 94.1. These results show despite excelling in knowledge-heavy evaluations, current MLLMs still lack fundamental visual primitives. Progress in BabyVision represents a step toward human-level visual perception and reasoning capabilities. We also explore solving visual reasoning with generation models by proposing BabyVision-Gen and automatic evaluation toolkit. Code and data are released at https://github.com/UniPat-AI/BabyVision.
****
Winformer: Transcending Pairwise Similarity for Time-series Generation
Haoyi Zhou, Xin Xue, Tianyu Chen, lanhao li, Lijun Sun, Jianxin Li
The periodicity misalignment remains a challenge problem in generating time-series data across multiple domains. The fundamental processing unit of attention in time-series modeling has long been restricted to either individual points or fragmented segments, limiting their ability to capture and adapt to complex periodic patterns inherent in diverse domains. To address this, we introduce Winformer, first to extend this processing unit from individual points to sliding windows, establishing a unified window-wise attention paradigm. Leveraging the adaptive window-alignment kernels derived from the frequency decomposition, Winformer brings semantically richer window representations, and effectively captures and transfers complex periodic patterns across domains. Extensive experiments on 12 real-world datasets demonstrate Winformer's effectiveness, achieving an average performance gain of 10.67\% over SOTA baselines.
****
Foundation Inference Models for Ordinary Differential Equations
Johannes R. Hübers, Maximilian Mauel, David Berghaus, Patrick Seifner, Ramses J Sanchez
Ordinary differential equations (ODEs) are central to scientific modelling, but inferring their vector fields from noisy trajectories remains challenging. Current approaches such as symbolic regression, Gaussian process (GP) regression, and Neural ODEs often require complex training pipelines and substantial machine learning expertise, or they depend strongly on system-specific prior knowledge. We propose FIM-ODE, a pretrained Foundation Inference Model that *amortises* ODE inference by predicting the vector field directly from noisy trajectory data *in a single forward pass*. We pretrain FIM-ODE on a prior distribution over ODEs with low-degree polynomial vector fields and represent the target field with neural operators. FIM-ODE achieves strong *zero-shot* performance, matching and often improving upon ODEFormer, a recent pretrained symbolic baseline, across a range of regimes despite using a simpler training prior. Pretraining also provides a strong initialisation for finetuning, enabling fast and stable adaptation that o

utperforms modern neural and GP baselines without requiring machine learning exp
ertise. Our pretrained model, code repository, and tutorials are available onlin
e.
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Learning to Perceive the World Through Control: Empowerment-Based Representation
 Learning

Mahsa Bastankhah, Sophie Broderick, Benjamin Eysenbach

In many practical reinforcement learning (RL) environments, observations are far
 higher-dimensional than the variables that matter for control.
In this work, we ask: can we learn representations that capture only control-rel
evant features of the environment?
We study this question through the \emph{empowerment} objective, which maximizes
 an agent’s influence over the environment and is widely used for unsupervised s
kill learning.
We show that empowerment agents induce two distinct representations --- forward
 and backward --- that capture complementary aspects of the state, and both of wh
ich are invariant to control-irrelevant features. Thus, empowerment maximization
 leads agents to learn an implicit, \emph{control-centric} model of the world.
Our analysis highlights the importance of learning representations through inter
action rather than from passive datasets: interaction aimed at maximizing contro
l is essential for learning useful invariance properties, a perspective that ali
gns closely with the causal learning literature.
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DLM-Scope: Mechanistic Interpretability of Diffusion Language Models via Sparse
Autoencoders

Xu Wang, Bingqing Jiang, Yu Wan, Baosong Yang, Lingpeng Kong, Difan Zou

Sparse autoencoders (SAEs) have become a standard tool for mechanistic interpret
ability in autoregressive large language models (LLMs), enabling researchers to
extract sparse, human-interpretable features and intervene on model behavior. Re
cently, as diffusion language models (DLMs) have become an increasingly powerful
 and promising alternative to autoregressive LLMs, it is essential to develop ta
ilored mechanistic interpretability tools for this emerging class of models. In
this work, we present **DLM-Scope**, the first SAE-based interpretability framew
ork for DLMs, and demonstrate that trained Top-K SAEs can faithfully extract spa
rse, interpretable features. Notably, we find that inserting SAEs affects DLMs d
ifferently from autoregressive LLMs: while SAE insertion in LLMs typically incur
s a loss penalty, in DLMs it can reduce cross-entropy loss when applied to early
 layers, a phenomenon absent or markedly weaker in LLMs. Additionally, SAE featu
res in DLMs enable more effective diffusion-time interventions, often outperform
ing LLM steering. Moreover, we pioneer new SAE-based research directions for DLM
s: we show that SAEs provide useful signals for DLM decoding order, and that SAE
 features remain stable during DLM post-training. Overall, our work establishes
a foundation for mechanistic interpretability in DLMs and highlights the potenti
al of applying SAEs to DLM-related tasks and algorithms.
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RAG without Forgetting: Continual Query-Infused Key Memory

Yuntong Hu, Sha Li, Naren Ramakrishnan, Liang Zhao

Retrieval-augmented generation (RAG) systems commonly improve robustness via que
ry-time adaptations such as query expansion and iterative retrieval. While effec
tive, these approaches are inherently stateless: adaptations are recomputed for
each query and discarded thereafter, precluding cumulative learning and repeated
ly incurring inference-time cost. Index-side approaches like key expansion intro
duce persistence but rely on offline preprocessing or heuristic updates that are
 weakly aligned with downstream task utility, leading to semantic drift and nois
e accumulation. We propose Evolving Retrieval Memory (ERM), a training-free fram
ework that transforms transient query-time gains into persistent retrieval impro
vements. ERM updates the retrieval index through correctness-gated feedback, sel
ectively attributes atomic expansion signals to the document keys they benefit,
and progressively evolves keys via stable, norm-bounded updates. We show that qu
ery and key expansion are theoretically equivalent under standard similarity fun

ctions and prove convergence of ERM’s selective updates, amortizing optimal quer
y expansion into a stable index with zero inference-time overhead. Experiments o
n BEIR and BRIGHT across 13 domains demonstrate consistent gains in retrieval an
d generation, particularly on reasoning-intensive tasks, at native retrieval spe
ed.
****

DF-LoGiT: Data-Free Logic-Gated Backdoor Attacks in Vision Transformers
Xiaozuo Shen, Yifei Cai, Rui Ning, Chunsheng Xin, Hongyi Wu
The widespread adoption of Vision Transformers (ViTs) elevates supply-chain risk
 on third-party model hubs, where an adversary can implant backdoors into releas
ed checkpoints. Existing ViT backdoor attacks largely rely on poisoned-data trai
ning, while prior data-free attempts typically require synthetic-data fine-tunin
g or extra model components. This paper introduces Data-Free Logic-Gated Backdoo
r Attacks (DF-LoGiT), a truly data-free backdoor attack on ViTs via direct weigh
t editing. DF-LoGiT exploits ViT’s native multi-head architecture to realize a l
ogic-gated compositional trigger, enabling a stealthy and effective backdoor. We
 validate its effectiveness through theoretical analysis and extensive experimen
ts, showing that DF-LoGiT achieves near-100% attack success with negligible degr
adation in benign accuracy and remains robust against representative classical a
nd ViT-specific defenses.
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Training Diffusion Language Models for Black-Box Optimization
Zipeng Sun, Can Chen, Ye Yuan, Haolun Wu, Jiayao Gu, Christopher Pal, Xue Liu
We study offline black-box optimization (BBO), aiming to discover improved desig
ns from an offline dataset of designs and labels, a problem common in robotics,
DNA, and materials science with limited labeled samples. While recent work appli
es autoregressive LLMs to BBO by formatting tasks as natural-language prompts, t
heir left-to-right design generation struggles to capture the strong bidirection
al dependencies inherent in design problems. To address this, we propose adaptin
g diffusion LLMs to offline BBO to leverage their bidirectional modeling capabil
ities. However, a domain gap exists between the natural text pre-training of dif
fusion LLMs and the heterogeneous signals in BBO (prompts, designs, and labels).
 To bridge this gap, we construct a unified prompt–-response corpus and introduc
e delimiter tokens to explicitly mark field boundaries for domain adaptation. We
 further propose a two-stage post-training framework to align the diffusion LLM
generation with high-label designs. The first stage performs supervised fine-tun
ing on the unified dataset via masked-response prediction, and the second stage
adopts reinforcement learning with rewards defined by label improvements. Our me
thod achieves state-of-the-art results on Design-Bench under small-data settings
. Code for our work is available here: https://github.com/zpointS/DiBO.
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Deep Flow Networks
Ozan Candogan, Ayoub Foussoul
We introduce Deep Flow Networks (DFNs), a new class of discrete function approxi
mators. DFNs are inspired by and generalize minimum-cost flow value functions th
at map node imbalances on a subset of nodes to the optimal flow cost. Such funct
ions are known to be M-convex (Murota2003) and admit efficient optimization.
On the theoretical side, we prove that DFNs are universal approximators for disc
rete functions on $\mathbb{Z}^d$ that admit convex extensions to $\mathbb{R}^d$,
 and characterize their optimization complexity in terms of their deviation from
 the M-convex regime. Guided by these results, we develop a practical DFN implem
entation for learning from data. Finally, we evaluate our implementation empiric
ally on data from different ground-truth functions, showing that DFNs achieve st
rong approximation accuracy while being substantially faster to optimize than be
nchmark approaches.
****

PersistBench: When Should Long-Term Memories Be Forgotten by LLMs?
Sidharth Pulipaka, Oliver Chen, Manas Sharma, Taaha Saleem Bajwa, Vyas Raina, Iv
axi Sheth
Conversational assistants are increasingly integrating long-term memory with lar

ge language models (LLMs). This persistence of memories, e.g., the user is veget arian, can enhance personalization in future conversations. However, the same pe rsistence can also introduce safety risks that have been largely overlooked. Hen ce, we introduce \textbf{PersistBench} to measure the extent of these safety ris ks. We identify two long-term memory-specific risks: \textit{cross-domain leakag e}, where LLMs inappropriately inject context from the long-term memories; and \ textit{memory-induced sycophancy}, where stored long-term memories insidiously r einforce user biases. We evaluate 18 frontier and open-source LLMs on our benchm ark. Our results reveal a surprisingly high failure rate across these LLMs - a m edian failure rate of 53% on cross-domain samples and 97% on sycophancy sa mples. To address this, our benchmark encourages the development of more robust and safer long-term memory usage in frontier conversational systems.
****

Mechanistic Interpretability as Statistical Estimation: A Variance Analysis
Maxime Méloux, François Portet, Maxime Peyrard
Mechanistic Interpretability (MI) aims to reverse-engineer model behaviors by id entifying functional sub-networks. Yet, the scientific validity of these finding s depends on their stability. In this work, we argue that circuit discovery is n ot a standalone task but a statistical estimation problem built upon causal medi ation analysis (CMA). We uncover a fundamental instability at this base layer: e xact, single-input CMA scores exhibit high intrinsic variance, implying that the causal effect of a component is a volatile random variable rather than a fixed property. We then demonstrate that circuit discovery pipelines inherit this vari ance and further amplify it. Fast approximation methods, such as Edge Attributio n Patching and its successors, introduce additional estimation noise, while aggr egating these noisy scores over datasets leads to fragile structural estimates. Consequently, small perturbations in input data or hyperparameters yield vastly different circuits. We systematically decompose these sources of variance and ad vocate for more rigorous MI practices, prioritizing statistical robustness and r outine reporting of stability metrics.
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Trainable Nonexpansive Denoisers for Contractive Image Reconstruction
Arghya Sinha, Aditya Banerjee, Trishit Mukherjee, Kunal N. Chaudhury
Trainable denoisers with Lipschitz control have become central to convergent ima ge reconstruction. However, training neural networks that simultaneously offer s trong denoising performance and global Lipschitz guarantees is challenging. Exis ting approaches enforce Lipschitz control only empirically, providing no guarant ees beyond the training data. In this work, we show that by exploiting the actio n of permutations on the image lattice, we can constrain a neural architecture t hat is globally nonexpansive (Lipschitz bound $\leqslant 1$). We integrate the p roposed denoiser with forward imaging operators to develop a reconstruction mech anism that is provably contractive and therefore globally convergent. Experiment s on standard inverse problems, such as superresolution and deblurring, demonstr ate that our reconstruction performance is competitive with softly constrained b aselines while providing Lipschitz guarantees.
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PDFBench: A Benchmark for De Novo Protein Design from Function
Jiahao Kuang, Nuowei Liu, Changzhi Sun, Jie Wang, Tao Ji, Yuanbin Wu
Function-guided protein design is a crucial task with significant applications i n drug discovery and enzyme engineering. However, the field lacks a unified and comprehensive evaluation framework. Current models are assessed using inconsiste nt and limited subsets of metrics, which prevents fair comparison and a clear un derstanding of the relationships between different evaluation criteria. To addre ss this gap, we introduce **PDFBench**, the first comprehensive benchmark for fu nction-guided de novo protein design. Our benchmark systematically evaluates eig ht state-of-the-art models on 16 metrics across two key settings: description-gu ided design, for which we repurpose the Mol-Instructions dataset, originally lac king quantitative benchmarking, and keyword-guided design, for which we introduc e a new test set, SwissTest, created with a strict datetime cutoff to ensure dat a integrity. By benchmarking across a wide array of metrics and analyzing their

correlations, **PDFBench** enables more reliable model comparisons and provides key insights to guide future research.
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The Double-Edged Nature of the Rashomon Set for Trustworthy Machine Learning
Ethan Hsu, Harry Chen, Chudi Zhong, Lesia Semenova
Real-world machine learning (ML) pipelines rarely produce a single model; instead, they produce a Rashomon set of many near-optimal ones. We show that this multiplicity reshapes key aspects of trustworthiness. At the individual-model level, sparse interpretable models tend to preserve privacy but are fragile to adversarial attacks. In contrast, the diversity within a large Rashomon set enables reactive robustness: even when an attack compromises one model, a practitioner can switch to a different near-optimal model that remains accurate, without retraining. However, the same diversity increases information leakage, as disclosing more near-optimal models provides an attacker with progressively richer views of the training data. This produces a robustness–privacy trade-off governed by diversity, which we analyze theoretically and empirically. Beyond this trade-off, Rashomon sets are stable under small distribution shifts, so a set computed once remains valid under such shifts without re-computation. Our results highlight the dual role of Rashomon sets as both a resource and a risk for trustworthy ML.
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DP-KFC: Data-Free Preconditioning for Privacy-Preserving Deep Learning
Marc Molina Van den bosch, Riccardo Taiello, Albert Sund Aillet, Andrea Protani, Miguel Angel Gonzalez Ballester, Luigi Serio
Differentially private optimization suffers from a fundamental geometric mismatch: deep networks have highly anisotropic loss landscapes, yet DP-SGD injects isotropic noise. Second-order preconditioning can resolve this, but estimating curvature typically requires private data (consuming privacy budget) or public data (introducing distribution shift). We show that the Fisher Information Matrix decouples into *architectural sensitivity*, recoverable via synthetic noise, and *input correlations*, approximable from modality-specific frequency statistics. We propose DP-KFC, which constructs KFAC preconditioners by probing networks with structured synthetic noise, requiring neither private nor public data. Empirically, DP-KFC consistently outperforms DP-SGD and adaptive baselines across diverse modalities in strong privacy regimes ($\varepsilon \leq 3$). DP-KFC matches private-data preconditioners while public-data variants degrade by up to 4.8 %, showing that curvature can be estimated without consuming privacy budget or introducing distribution shift. This enables privacy-preserving learning in specialized domains (e.g., medical applications) where regulatory constraints make data scarce.
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What LLMs Explain Is Not What They Believe: Evaluating Explanation Sufficiency Under Models' Own Input Beliefs
Nhi Nguyen, Shauli Ravfogel, Rajesh Ranganath
Large language models (LLMs) are increasingly deployed in high-stakes domains, where free-text explanations such as chain-of-thought and post-hoc rationales are used to justify model outputs. Yet it remains unclear whether these explanations are _sufficient_, i.e., if they contain enough information to explain the model’s output-generating process. We generalize classical sufficiency from feature attributions to arbitrary explanations and prove that explanation sufficiency can change depending on the input distribution, which must be explicitly defined for LLM explanations. We propose using the LLM itself to generate alternative inputs conditioned on an explanation, capturing its beliefs about possible inputs. We formalize _self-consistent sufficiency_ as a goal for free-text explanations and introduce an information-theoretic metric, SCSuff, that enables evaluation of free-text explanations without relying on predefined biases or shortcuts. Our experiments show that SCSuff agrees with targeted perturbation tests where applicable and demonstrate that explanation sufficiency can vary with the input distribution. We find LLM explanations are generally insufficient and weakly correlated with model size, accuracy, or output entropy. Analysis of final-token hidden states shows that top and bottom SCSuff scores can be predicted from internal re

presentations, suggesting that SCSuff can guide detection and improvement of suf
ficient LLM explanations. The code for this paper is available at https://github
.com/rajesh-lab/self-consistent-sufficiency.
****

Convergent World Representations and Divergent Tasks
Core Francisco Park
While neural representations are central to modern deep learning, the conditions
 governing their geometry and their roles in downstream adaptability remain poor
ly understood. We develop a framework clearly separating the underlying world, t
he data generation process and the resulting model representations to study thes
e questions in a controlled setup: 5,075 city coordinates define the world and 7
 geometric tasks generate the training data for autoregressive Transformer train
ing. We find that different tasks give rise to qualitatively and quantitatively
distinct world representation geometries. However, multi-task training drives co
nvergence of world representations: models trained on non-overlapping tasks deve
lop aligned geometric representations, providing controlled evidence for the Mul
titask Scaling Hypothesis of the Platonic Representation Hypothesis. To study ad
aptation, we pretrain models on all tasks and all cities, then test whether new
entities can be consistently integrated into the representation space via fine-t
uning. Surprisingly, we find that despite multi-task pretraining, some tasks, wh
ich we call divergent, actively harm the representational integration of new ent
ities. Our results show that training on multiple relational tasks reliably prod
uces convergent world representations, but some lurking divergent tasks can cata
strophically harm new entity integration via fine-tuning.
****

SimGFM: Simplifying Discrete Flow Matching for Graph Generation
Chunyu Luo, Yuankai Luo, Xiao-Ming Wu, Lei Shi
Discrete Flow Matching (DFM) presents a promising approach for graph generation;
 however, existing adaptations often introduce substantial complexity by incorpo
rating task-specific heuristics, compromising the continuity equation and signif
icantly expanding the hyperparameter space. Moreover, their sampling efficiency
remains limited, as the required number of steps is often comparable to diffusio
n models, diminishing DFM’s practical advantages. To address these limitations,
we propose SimGFM, a simplified graph DFM for graph generation. Leveraging chara
cteristic patterns in graph-generation trajectories, SimGFM relies only on the s
cheduler and rate matrix, eliminating heavy heuristics and hyperparameter tuning
, and achieves large step reductions while preserving SOTA results. SimGFM achie
ves strong empirical results: on QM9, it matches prior models requiring 500–1000
 steps with only 10 steps, and on most datasets, its performance at 50 steps mat
ches or surpasses these baselines, demonstrating both efficiency and competitive
ness. Our source code is available at https://github.com/LUOyk1999/SimGFM.
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Orchestrating Spatial Semantics via a Zone-Graph Paradigm for Intricate Indoor S
cene Generation
Meisheng Zhang, Shizhao Sun, Yang Zhao, Ziyuan Liu, Zhijun Gao, Jiang Bian
Autonomous 3D indoor scene synthesis breaks down in non-convex rooms with tightl
y coupled spatial constraints. Data-driven generators lack topological priors fo
r long-horizon planning, while iterative agents fragment semantics and become ge
ometrically brittle. We present \textbf{ZoneMaestro}, a unified framework that s
hifts the paradigm from object-centric synthesis to Zone-Graph Orchestration. By
 internalizing a novel zone-based logic, ZoneMaestro translates high-level seman
tic intent into functional zones and topological constraints, enabling robust ad
aptation to diverse architectural forms. To support this, we construct \textit{Z
one-Scene-10K}, a large-scale dataset enriched with explicit Zone-Graph annotati
ons. We further introduce an \textit{Alternating Alignment Strategy} that cycles
 between reasoning internalization and Zone-Aware Group Relative Policy Optimiza
tion (\textit{Z-GRPO}), effectively reconciling the tension between semantic ric
hness and geometric validity without relying on external physics engines. To rig
orously evaluate spatial intelligence beyond convex primitives, we formally defi
ne the task of \textbf{Intricate Spatial Orchestration} and release SCALE, a str

ess-test benchmark for irregular indoor scenarios with complex, dense spatial re
lations. Extensive experiments demonstrate that ZoneMaestro resolves the density
-safety dichotomy, significantly outperforming state-of-the-art baselines in bot
h structural coherence and intent adherence.
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MA^{3}S: Model-Agnostic Active Annotation Strategy for Crowdsourcing
Wenjun Zhang, Liangxiao Jiang, Chaoqun Li, Shanshan Si
In crowdsourcing scenarios, to mitigate the impact of noisy labels assigned by n
on-expert workers, each instance is typically annotated multiple times by differ
ent workers. However, repeated annotation can introduce instance- or label-level
 redundancy, thereby inflating annotation costs. Despite its practical importanc
e, research on repeated annotation strategies remains limited, and no existing s
trategy simultaneously avoids being offline, instance-unaware, and model-centric
. In this paper, we propose a model-agnostic active annotation strategy, MA^{3}S
, that addresses these limitations: (1) To reduce label redundancy caused by off
line procedure, MA^{3}S estimates instance uncertainties with a margin-Beta dist
ribution and updates them online as new labels arrive. (2) To prevent instance r
edundancy induced by instance-unaware designs, MA^{3}S constructs a nearest-neig
hbor graph to propagate instance uncertainties, reducing repeated annotations of
 similar instances. (3) To avoid being model-centric, MA^{3}S actively selects i
nstances for annotation based solely on the estimated uncertainties, without rel
ying on model feedback. Extensive experiments on synthetic and real-world datase
ts demonstrate that MA^{3}S consistently outperforms existing annotation strateg
ies.
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Learning Rewrite-Invariant Reasoning with Targeted Alternation Training
Mousa Arraf, Ido Guy, Kira Radinsky
Large language models (LLMs) often fail in systematic, model-specific ways under
 meaning-preserving question rewrites (paraphrases, format changes, benign distr
actors). In this work, we address this instability by identifying where the mode
l's reasoning process diverges across semantically-equivalent inputs. For each t
arget LLM, we sample multiple solution traces under rewrites and aggregate them
into a graph of recurring intermediate steps, which pinpoints where incorrect tr
aces diverge from correct ones. We then generate a small set of semantics-preser
ving examples that mirror the rewrite patterns most responsible for these diverg
ences, and use them to steer the model (\emph{targeted alternation training}), e
ither via fine-tuning or via in-context learning. Across MMLU-Pro, Big-MATH, and
 DROP, this yields consistent gains and cross-dataset generalization. On Humanit
y’s Last Exam, using 200 in-context examples, it improves GPT-5.2 (xhigh) from 3
5.4\% to 38.1\%, demonstrating that targeted alternation training can materially
 improve a frontier, API-accessible closed model under realistic access constrai
nts.
****

FairRARI: A Plug and Play Framework for Fairness-Aware PageRank
Emmanouil Kariotakis, Aritra Konar
PageRank (PR) is a fundamental algorithm in graph machine learning tasks. Owing
to the increasing importance of algorithmic fairness, we consider the problem of
 computing PR vectors subject to various group-fairness criteria based on sensit
ive attributes of the vertices. At present, principled algorithms for this probl
em are lacking - some cannot guarantee that a target fairness level is achieved,
 while others do not feature optimality guarantees. In order to overcome these s
hortcomings, we put forth a unified in-processing convex optimization framework,
 termed FairRARI, for tackling different group-fairness criteria in a ``plug and
 play'' fashion. Leveraging a variational formulation of PR, the framework compu
tes fair PR vectors by solving a strongly convex optimization problem with fairn
ess constraints, thereby ensuring that a target fairness level is achieved. We f
urther introduce three different fairness criteria which can be efficiently tack
led using FairRARI to compute fair PR vectors with the same asymptotic time-comp
lexity as the original PR algorithm. Extensive experiments on real-world dataset
s showcase that FairRARI outperforms existing methods in terms of utility, while

achieving the desired fairness levels across multiple vertex groups; thereby hi
ghlighting its effectiveness.
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BFTS: Thompson Sampling with Bayesian Additive Regression Trees
Ruizhe Deng, Bibhas Chakraborty, Ran Chen, Yan Shuo Tan
We propose Bayesian Forest Thompson Sampling (BFTS), which performs Thompson sam
pling using arm-wise Bayesian Additive Regression Trees (BART) to model each act
ion's mean reward and generate MCMC-based posterior draws for decision-making. W
e derive an information-theoretic Bayesian regret bound of order $\widetilde{\ma
thcal O}(K\sqrt{T})$ for ideal posterior sampling under a correctly specified Ba
yesian design. Empirically, BFTS achieves competitive regret on nonlinear synthe
tic benchmarks with near-nominal uncertainty calibration, attains the best avera
ge rank across nine OpenML contextual bandit benchmarks, and yields higher estim
ated policy values than linear, neural, and tree-ensemble baselines in a Drink L
ess micro-randomized trial case study. Across OpenML benchmarks, BFTS is robust
to hyperparameter choices.
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Dustin: Draft-Augmented Sparse Verification for Efficient Long-Context Generatio
n with Speculative Decoding
WenHung Lee, Jian-Jia Chen, Xiaolin Lin, Pei-Shuo Wang, Chi-Chih Chang, Chun-Che
Yang, Wei-Chen Wang, Hanrui Wang, Ning-Chi Huang, Grace Li Zhang, Kai-Chiang Wu
While speculative decoding improves inference throughput for multi-batch long-co
ntext Large Language Models (LLMs), its efficiency is often limited by a verific
ation bottleneck where Key-Value (KV) cache loading dominates latency. Existing
compression methods fail in this regime: static eviction incurs accuracy loss du
e to saliency shift, while dynamic selection introduces prohibitive computationa
l overhead during the verification path. We propose Dustin, a sparse verificatio
n framework designed for long-context speculative decoding. Dustin integrates lo
okahead signals from the draft model with historical attention from the target m
odel to identify critical tokens with high fidelity across multi-step verificati
on windows. To reduce recomputation latency, this approach further employs a spa
rse estimation scheme that restricts importance scoring to a minimal subset of a
ttention heads. Evaluations on PG-19 and LongBench with Qwen2.5-72B demonstrate
that Dustin achieves a 27.85× speedup in self-attention and a 9.17× end-to-end d
ecoding speedup at a 32k sequence length, all with negligible accuracy degradati
on.
****

Breaking the Synthetic-Real Domain Shortcut for Training-Free Generative Replay-
based Class Incremental Learning
Tao Zhang, Qixuan Fan, Yiyuan Liang, Yanjie Wang, Song Yan, Tian Tian, Jiahuan Z
hou, Luxin Yan, Sheng Zhong, Xu Zou
Class-incremental learning (CIL) requires models to continuously acquire new kno
wledge while avoiding catastrophic forgetting. While exemplar replay is effectiv
e, it raises concerns regarding privacy and storage. Thus, generative replay has
emerged as a viable alternative, synthesizing old data using frozen pretrained
text-to-image (T2I) models without any extra training. However, we observe that
directly mixing synthetic old-class data with real new-class data during increme
ntal training leads to significant performance degradation. This issue stems fro
m a ‘’domain shortcut'', where models rely on domain-discriminative features ins
tead of semantic class cues. To address this, we propose DREAM (Domain-Regulariz
ed Exemplar-free Alignment Model), which uses a training-free generator to synth
esize old-class data and eliminates domain shortcut via subspace rectification a
nd orthogonal projection, while reinforcing semantic alignment through real-anch
ored prototype regularization. Extensive experiments on 4 datasets demonstrate t
hat DREAM outperforms existing exemplar-free CIL methods and achieves state-of-t
he-art performance.
****

Learning the Best Under Constraints: A Duality-Based Framework
Mingjie Hu, Enlu Zhou, Jianqiang Hu
This paper studies a constrained linear best arm identification problem with cov

ariate selection in the fixed-confidence setting, where each arm is evaluated ac ross multiple performance metrics. The mean performance of each metric depends l inearly on the feature vectors of both arms and covariates. The goal is to ident ify the arm with the highest expected value of one targeted metric while ensurin g that the means of the remaining metrics stay below specified thresholds for ea ch covariate. We first establish an instance-dependent lower bound on the sample complexity, formulated as a multi-level optimization problem that captures both feasibility and optimality. We then prove that this bound is tight by designing an algorithm that asymptotically matches it. Since the original algorithm is co mputationally intensive, we develop a relaxed version of the bound through a sur rogate optimization problem and derive its convex dual. Using this bound, we pro pose a duality-based decomposition algorithm that is computationally efficient, updating only two coordinates and performing a single gradient step per iteratio n. We further show that the algorithm achieves the relaxed bound in theory and d emonstrates its practical effectiveness through numerical experiments.

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How Should Transformers Encode Numeric Values in Electronic Health Records?

Maria Elkjær Montgomery, Christian Igel, Mikkel Fruelund Odgaard, Martin Sillese n, Mads Nielsen

How do we encode numeric values in transformer-based sequence processing, partic ularly in electronic health record (EHR) data? We systematically compare discret e, continuous, and hybrid value encoding strategies using synthetic arithmetic t asks embedded within real-world EHR data, as well as real-world clinical predict ion tasks. Our study reveals trade-offs between numeric precision, optimisation stability, and architectural flexibility. We find that approaches that explicitl y model value-concept interactions perform best on precision-sensitive arithmeti c tasks when architectural constraints permit. Hybrid token-based approaches tha t retain numeric values but apply binning prior to projection provide a more rob ust and broadly applicable alternative, with the optimal number of bins followin g a simple empirically derived power-law in dataset size. Across tasks, models c onsistently exhibit reliable “good enough” numeric computation rather than exact arithmetic, while clinical gains from incorporating laboratory values are task- dependent. This suggests that robustness and deployability often outweigh maxima l numeric precision in practice, motivating hybrid token-based approaches as a p ractical default.

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Online Continual Learning with Dynamic Label Hierarchies

Xinrui Wang, Bart■omiej Twardowski, Alexandra Gomez-Villa, Shao-Yuan Li, Songcan Chen

Online Continual Learning (OCL) aims to learn from endless non\text{-}stationary data streams, yet most existing methods assume a flat label space and overlook the hierarchical organization of real\text{-}world concepts that evolves both ho rizontally (sibling classes) and vertically (coarse or fine categories). To bett er reflect this context, we introduce a new problem setting, DHOCL (Online Conti nual Learning from Dynamic Hierarchies), where taxonomies evolve across granular ities and each sample provides supervision at a single hierarchical level. In th is setting, we find two fundamental issues: (i) partial supervision under mixed granularities provides only point-wise signals over an evolving path-wise hierar chy, which constrains plasticity and undermines cross-level semantic consistency , and (ii) the dynamically evolving hierarchies induce granularity-dependent int erference, destabilizing popular replay and regularization mechanisms and thereb y exacerbating catastrophic forgetting. To tackle these issues, we propose HALO (Hierarchical Adaptive Learning with Organized Prototypes), which adaptively com bines complementary classification heads, regularized by organized learnable hie rarchical prototypes, enabling rapid adaptation, hierarchical consistency, and s tructured knowledge consolidation as the taxonomy evolves. Extensive experiments on multiple benchmarks demonstrate that HALO consistently outperforms existing methods across hierarchical accuracy, mistake severity, and continual performanc e.

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Learning Treatment Representations for Downstream Instrumental Variable Regressi
on
Shiangyi Lin, Hui Lan, Vasilis Syrgkanis
Traditional instrumental variable (IV) estimators cannot accommodate more treatm
ents than instruments, a limitation that is critical for high-dimensional, unstr
uctured data like clinical treatment pathways. Current practice—applying unsuper
vised dimension reduction before IV estimation—suffers from substantial omitted
treatment bias because the representation learning step ignores the instrument.
We propose a novel framework that constructs treatment representations by explic
itly incorporating instrumental variables. We prove that this instrument-guided
approach ensures the identification of optimal outcome-prediction directions eve
n with limited instruments. Validation on large-scale, semi-synthetic clinical d
ata derived from a major hospital, along with other simulations, shows that our
approach significantly outperforms conventional two-stage methods.
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Asking What Matters: Reward-Driven Clarification for Software Engineering Tasks
Sanidhya Vijayvargiya, Vijay Viswanathan, Graham Neubig
Humans often specify tasks incompletely, so assistants must know when and how to
ask clarifying questions. However, effective clarification remains challenging
in software engineering tasks as not all missing information is equally valuable
, and questions must target information users can realistically provide. We stud
y clarification in real software engineering tasks by quantifying which types of
information most affect task success and which questions elicit useful response
s from simulated users. Using Shapley attribution and distributional comparisons
, we identify two key properties of effective clarification: task relevance (whi
ch information predicts success) and user answerability (what users can realisti
cally provide). We operationalize these properties as multi-stage reinforcement
learning rewards to train CLARITI, an 8B-parameter clarification module, that ma
tches GPT-5’s resolution rate on underspecified issues while generating 41% fewe
r questions. Our results suggest that grounding reward design in empirical analy
sis of information impact and user answerability improves clarification efficien
cy.
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Radial Scaling Voxelization for Accurate Small Object 3D Detection
Hao Liu, Yi Zhou, Yanni Ma
Voxel-based 3D object detectors typically discretize the spatial domain using a
uniform Cartesian grid, which allocates the same voxel size to both near-range a
nd far-range regions. However, this uniform discretization is suboptimal for sma
ll objects such as pedestrians and cyclists, as they occupy only a few voxels an
d thus struggle to capture fine-grained geometric details. Although increasing t
he global voxel resolution can alleviate this problem, it inevitably increases s
ubstantial memory consumption and computational cost. In this paper, we propose
Radial Scaling Voxelization (RSV), a simple yet effective non-uniform discretiza
tion strategy that adaptively modulates the effective voxel size based on the ra
dial distance from the LiDAR sensor. Unlike previous cylindrical or polar discre
tization schemes, RSV preserves the Cartesian grid topology by applying a contin
uous radial scaling function to the input coordinates before standard voxelizati
on. This operation yields a near-high, far-unchanged resolution pattern, i.e., t
he effective voxel size becomes finer in near regions, where the geometric struc
tures of small objects are difficult to capture, while remaining nearly unchange
d in far regions to avoid unnecessary computational cost. Importantly, RSV is ar
chitecture-agnostic and can directly replace the discretization module in any vo
xel-based detector without modifying the backbone, network design, or training p
ipeline. Extensive experiments on the KITTI and nuScenes datasets demonstrate th
at integrating our RSV into several voxel-based baselines consistently enhances
small-object detection performance, especially for the Pedestrian and Cyclist ca
tegories, while incurring only marginal additional computational overhead. Code
is available at https://github.com/Zeoy2020/RadialScalingVoxelization.
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Optimizing Diversity and Quality through Base-Aligned Model Collaboration

Yichen Wang, Chenghao Yang, Tenghao Huang, Muhao Chen, Jonathan May, Mina Lee
Alignment has greatly improved large language models (LLMs)' output quality at t he cost of diversity, yielding highly similar outputs across generations, especi ally in open-ended generation tasks. We propose Base-Aligned Model Collaboration (BACo), an inference-time token-level model collaboration framework that dynami cally combines a base LLM with its aligned counterpart to optimize diversity and quality. Using uncertainty and content-based signals, BACo employs routing stra tegies to determine, at each token, which model to decode from. Prior diversity-promoting methods often improve diversity at the expense of quality or require e xpensive decoding or post-training. In contrast, BACo achieves both high diversi ty and quality post hoc within a single pass, while offering strong controllabil ity. We introduce a family of effective routing strategies and evaluate them acr oss three open-ended generation tasks with 13 diversity and quality metrics. BAC o consistently surpasses state-of-the-art inference-time baselines. With our bes t router, BACo achieves a 21.3% joint improvement in diversity and quality, whic h is further supported by human evaluations. Overall, our results demonstrate th at collaboration between base and aligned models provides an effective and contr ollable mechanism for optimizing the diversity-quality trade-off.
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ECCO: Evidence-Driven Causal Reasoning for Compiler Optimization
Haolin Pan, Lianghong Huang, Dong Jinyuan, Mingjie Xing, Yanjun Wu
Compiler auto-tuning faces a dichotomy between traditional black-box search meth ods, which lack semantic guidance, and recent Large Language Model (LLM) approac hes, which often suffer from superficial pattern matching and causal opacity. In this paper, we introduce ECCO, a framework that bridges interpretable reasoning with combinatorial search. We first propose a reverse engineering methodology t o construct a Chain-of-Thought dataset, explicitly mapping static code features to verifiable performance evidence. This enables the model to learn the causal l ogic governing optimization decisions rather than merely imitating sequences. Le veraging this interpretable prior, we design a collaborative inference mechanism where the LLM functions as a strategist, defining optimization intents that dyn amically guide the mutation operations of a genetic algorithm. Experimental resu lts on seven datasets demonstrate that ECCO outperforms the LLVM opt -O3 baselin e, achieving an average 24.44% reduction in cycles.
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Stabilizing Reinforcement Learning for Diffusion Language Models
Jianyuan Zhong, Kaibo Wang, Ding Ding, Zijin Feng, Haoli Bai, Yang Xiang, Jiache ng Sun, Qiang Xu
Diffusion Large Language Models (dLLMs) often exhibit severe instability during Group Relative Policy Optimization (GRPO) training, limiting the effectiveness o f reinforcement learning for improving reasoning capabilities. In dLLMs, the imp ortance ratios used by GRPO are derived from finite-sample estimates rather than exact likelihoods, making them inherently noisy. In this paper, we show that GR PO is highly sensitive to this noise, which drives training instability. Through theoretical analysis and empirical evidence, we identify a self-reinforcing ins tability loop in which noisy importance ratios induce gradient spikes and policy drift, further amplifying future importance ratio estimation variance. To addre ss this issue, we propose StableDRL, a novel reinforcement learning framework fo r dLLMs. StableDRL stabilizes training via (i) unconditional clipping to suppres s outlier-induced gradient spikes, and (ii) self-normalization to constrain grad ients within the convex hull of per-sample updates. We further extend StableDRL to block-wise diffusion models via a staircase attention mechanism. StableDRL is the first method that enables stable, full-parameter reinforcement learning for dLLMs. It achieves the state-of-the-art performance, outperforming prior best f ull-attention baselines by 6% on MATH500 and block-diffusion baselines by 25.6% on AIME.
****

Robustness of Mixtures of Experts to Feature Noise
Dong Sun, Rahul Nittala, Rebekka Burkholz
Despite their practical success, it remains unclear why Mixture of Experts (MoE)

models can outperform dense networks beyond sheer parameter scaling. We study a n iso-parameter regime where inputs exhibit latent modular structure but are cor rupted by feature noise, a proxy for noisy internal activations. We show that sp arse expert activation acts as a noise filter: compared to a dense estimator, Mo Es achieve lower generalization error under feature noise, improved robustness t o perturbations, and faster convergence speed. Empirical results on synthetic da ta and real-world language tasks corroborate the theoretical insights, demonstra ting consistent robustness and efficiency gains from sparse modular computation.
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SwitchCraft: A Programmatic Framework for Designing State-Switching Proteins
Bowen Jing, Mihir Bafna, Anisha Parsan, Heyuan Michael Ni, David Kwabi-Addo, Bry an D. Bryson, Adam Klivans, Bonnie Berger
Multistate mechanisms underlie many of the complex functions observed in natural proteins. The ability to rationally design multistate proteins would have trans formative implications for many areas of biotechnology, yet lies beyond the capa bilities of existing deep learning frameworks for protein design. To address thi s gap, we introduce SwitchCraft, a versatile and programmatic framework for desi gning state-switching proteins based on backpropagation through compositional de sign constraints parameterized by structure prediction models. In silico evaluat ions demonstrate success on a wide range of state-switching functional primitive s, from allosteric regulation of motifs to discrimination of bound ligand identi ties. Using these primitives, we demonstrate an in silico strategy for de novo d esign of fluorescent biosensors to arbitrary small molecule analytes. These resu lts position SwitchCraft at the inception of a powerful paradigm for higher-orde r functional protein design. Code is available at https://github.com/bjing2016/s witchcraft.
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Learning Attribute–Affordance Hierarchies in Hyperbolic Space for Open-Vocabular y 3D Object Affordance Grounding
Yuxuan Wang, Tong Li, Yihang Zhu, Guangtao Lyu, Yukuan Min, Chenghao Xu, Jiexi Y an, Xu Yang, Cheng Deng
This paper pays attention to open-vocabulary 3D object affordance grounding (OVA G), which aims to localize affordance regions on 3D objects by leveraging intera ction images or textual instructions. Most existing methods treat interaction im ages as sources of external affordance knowledge and align them with 3D visual r epresentations, while overlooking the intrinsic relationship between local objec t attributes and affordances, which limits localization accuracy and generalizat ion. For instance, a cup handle affords grasping due to its curved shape and app ropriate thickness, indicating that affordances emerge from specific attribute c ompositions rather than global object appearance.

Motivated by this, we propose Attribute-Affordance Hierarchies (AAH) learning framework that explicitly models the hierarchical relationships between object-r egion attributes and affordances. Our approach first captures local region relat ionships using hypergraph, and then projects these region-level concepts into a hyperbolic space to encode their hierarchical organization. Furthermore, we intr oduce counterfactual attribute samples to encourage robust learning of attribute –affordance dependencies under varying conditions. By jointly modeling visual st ructure and hierarchical concept information, our method achieves more accurate affordance localization. Extensive experiments and qualitative analyses demonstr ate the effectiveness of our approach.
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OSNIP: Balancing the Privacy-Utility-Efficiency Trilemma in LLM Inference via Ob fuscated Semantic Null Space
Zhiyuan Cao, Zeyu Ma, Chenhao Yang, HAN ZHENG, Mingang Chen
We propose Obfuscated Semantic Null Space Injection for Privacy (OSNIP), a light weight client-side encryption framework for privacy-preserving LLM inference. Ge neralizing the geometric intuition of linear kernels to the high-dimensional lat ent space of LLMs, we formally define the ``Obfuscated Semantic Null Space'', a high-dimensional regime that preserves semantic fidelity while enforcing near-or thogonality to the original embedding. By injecting perturbations that project t

he original embedding into this space, OSNIP ensures privacy without any post-pr
ocessing. Furthermore, OSNIP employs a key-dependent stochastic mapping that gen
erates distinct perturbations under fresh keys.
Evaluations on generative and classification benchmarks show that OSNIP achieves
 state-of-the-art performance, sharply reducing attack success rates while maint
aining strong model utility under strict security constraints.
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Beyond the Bellman Recursion: A Pontryagin-Guided Framework for Non-Exponential
Discounting
Hojin Ko, Jeonggyu Huh
Most value-based and actor-critic reinforcement learning methods rely on Bellman
-style recursions, yet these recursions collapse under non-exponential discounti
ng common in human preferences and survival processes. We show the breakdown is
structural: exponential discounting sits at a fragile intersection of multiplica
tivity and time homogeneity, and violating either property breaks standard dynam
ic programming. To overcome this, we propose **Pontryagin-Guided Direct Policy O
ptimization (PG-DPO)**, a variational framework that abandons recursion and coup
les the Pontryagin Maximum Principle with Monte Carlo rollouts via an *Adjoint-M
C projection* enforcing pointwise Hamiltonian maximization. Across multidimensio
nal hyperbolic and survival-discount benchmarks, PG-DPO improves accuracy and st
ability where equation-driven solvers and critic-based baselines diverge.
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ANCHOR: Automated Alignment Auditing for CLI Agents on Real-World Harm
Kefan Song, Yanjun Qi
Autonomous CLI agents can now execute hundreds of actions across multi-hour sess
ions: writing code, executing shell commands, browsing the web, and managing clo
ud infrastructure, all with minimal human oversight. Does greater autonomy invit
e greater risk? We introduce ANCHOR, an automated auditing framework that stress
-tests CLI agents on illegal tasks grounded in public US court cases. ANCHOR dep
loys an auditor agent fine-tuned on dark personality data using supervised and r
einforcement fine tuning. This auditor roleplays persistent malicious users who
decompose tasks, reframe requests upon refusal, and adapt strategies across mult
i-turn interactions. Evaluating frontier CLI agents, we find that while they oft
en refuse illegal tasks when prompted directly, compliance reaches 100\% under p
ersistent malicious interaction. When agents comply, they frequently exceed user
 requests, autonomously building infrastructure for large-scale harm, including
catastrophic risk scenarios such as large-scale financial fraud and bioweapon de
velopment. These findings demonstrate that current alignment techniques are insu
fficient for autonomous agents and underscore the need for safety evaluations ag
ainst persistent, adaptive malicious users.
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OSCS: Online Selection with Provable FAR Control for LLM Safety
Zirui Hu, Zheng Zhang, Yingjie Wang, Dacheng Tao
Large language models (LLMs) are vulnerable to malicious inputs, posing serious
risks in high-stakes applications. Although existing detection-based defenses ac
hieve strong empirical performance, they generally lack explicit control over th
e false acceptance rate (FAR), a critical safety requirement in sensitive deploy
ment scenarios. This challenge is further complicated by two practical constrain
ts: the lack of malicious calibration samples and the streaming nature of real-w
orld inputs. To address these challenges, we propose \textit{OSCS}, a novel fram
ework for online FAR control without requiring malicious calibration data. OSCS
leverages detection scores produced by existing defenses and employs recursive d
ensity estimation to estimate benign probability from the test stream. Based on
these estimates, OSCS performs real-time accept/reject decisions while provably
satisfying a user-specified FAR target. Theoretically, we show that OSCS control
s the FAR up to a vanishing excess term under mild assumptions. Extensive experi
ments on backdoor and jailbreak attack tasks further demonstrate the effectivene
ss of OSCS, showing that it consistently achieves robust FAR control across dive
rse attack settings while outperforming existing baselines.
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Can LLMs Reason Like Automated Theorem Provers for Rust Verification? VCoT-Bench : Evaluating via Verification Chain of Thought
Zichen Xie, Wenxi Wang
As Large Language Models (LLMs) increasingly assist secure software development, their ability to meet the rigorous demands of Rust program verification remains unclear. Existing evaluations treat Rust verification as a black box, assessing models only by binary pass or fail outcomes for proof hints. This obscures whether models can systematically reconstruct the explicit deductive steps required for verifying nontrivial Rust code. To bridge this gap, we introduce VCoT-Lift, a framework that lifts low-level solver reasoning into high-level, human-readable verification steps. By exposing solver-level reasoning as an explicit Verification Chain-of-Thought, VCoT-Lift provides a concrete ground truth for fine-grained evaluation. Leveraging VCoT-Lift, we introduce VCoT-Bench, a comprehensive benchmark of 1,988 VCoT completion tasks for rigorously evaluating LLMs' ability to reconstruct the entire verification process. VCoT-Bench measures performance along three orthogonal dimensions: robustness to varying degrees of missing proofs, competence across different proof types, and sensitivity to proof locations. Evaluation of ten state-of-the-art models reveals severe fragility, indicating that current LLMs fall well short of the reasoning capabilities exhibited by automated theorem provers.
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LaRI: Layered Ray Intersections for Single-view 3D Geometric Reasoning
Rui Li, Biao Zhang, Zhenyu Li, Federico Tombari, Peter Wonka
We present Layered Ray Intersections (LaRI), a fully supervised method for occluded geometry reasoning from a single image. Unlike conventional depth estimation, which is limited to visible surfaces, LaRI predicts multiple surfaces intersected by the camera rays using layered point maps. Compared to the existing approaches that leverage neural implicit representations or iterative refinement, LaRI achieves complete scene reconstruction in one feed-forward pass, enabling efficient and view-aligned geometric reasoning to underpin both object-level and scene-level tasks. We further propose to predict the ray stopping index, which identifies valid intersecting pixels and layers from LaRI's output. To better underpin and evaluate this task, we build an annotation pipeline using rendering engines, construct annotations for five public datasets, including synthetic and real-world data covering 3D objects and scenes. As a generic method, LaRI's performance is validated in object-level and scene-level reconstruction tasks.
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Simple Algorithms for Bad Triangle Transversals with Applications to Correlation Clustering
Florian Adriaens, Nikolaj Tatti
The Bad Triangle Transversal (BTT) problem asks for the smallest set of edges that need to be removed from a given signed graph, so that the resulting graph does not have a bad triangle. Here, a bad triangle is a triangle with exactly one negative edge.
Several 2-approximations for BTT are proposed in this paper. On the hardness side, we show that BTT is NP-hard to approximate with factor better than $\frac{2137}{2136}$ on complete graphs. Our reduction also works for Correlation Clustering (CC), the Cluster Deletion problem (CD) and the Minimum Strong Triadic Closure problem (MinSTC). Lastly, we show that the BTT and CC optima are within a factor of 3/2 in complete graphs, by describing a pivot procedure that transforms transversals into clusters.
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Logit Distance Bounds Representational Similarity
Beatrix Miranda Ginn Nielsen, Emanuele Marconato, Luigi Gresele, Andrea Dittadi, Simon Buchholz
For a broad family of discriminative models that includes autoregressive language models, identifiability results imply that if two models induce the same conditional distributions, then their internal representations are equal up to an invertible linear transformation. We ask whether an analogous conclusion holds approximately when the distributions are close instead of equal.

Building on the observation of Nielsen et al. (2025) that closeness in KL diverg
ence need not imply high linear representational similarity, we study a distribu
tional distance based on logit differences and show that closeness in this dista
nce does yield linear similarity guarantees.
Specifically, we define a representational dissimilarity measure based on the mo
dels' identifiability class and prove that it is bounded by the logit distance.
We further show that, when model probabilities are bounded away from zero, KL di
vergence upper-bounds logit distance; yet the resulting bound fails to provide n
ontrivial control in practice.
As a consequence, KL-based distillation can match a teacher’s predictions while
failing to preserve linear representational properties, such as linear-probe rec
overability of human-interpretable concepts.
In distillation experiments on synthetic and image datasets, logit-distance dist
illation yields students with higher linear representational similarity and bett
er preservation of the teacher’s linearly recoverable concepts.
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f-Divergence Self-Play for Tabular Anomaly Detection via Large Language Models
Hoang Tran Vuong, Linh Ngo Van, Dang Nguyen, Thin Nguyen, Phuoc Nguyen, Mehrtash
Harandi, Trung Le
Anomaly detection in tabular data poses significant challenges due to heterogene
ous feature types—mixing numerical, categorical, and textual attributes, which c
omplicate learning meaningful representations of normality. Recent work has appl
ied large language models (LLMs) to this problem by serializing table rows as te
xt sequences, yet these approaches rely on one-shot supervised fine-tuning that
offers limited signal to tighten the model's description of normality. We propos
e DiSPaT, a self-play fine-tuning framework that strengthens the model's underst
anding of normal data. Building on the theoretical foundation of f-divergence
minimization, we derive a tight approximation connecting our training objective
to reducing the distributional gap between real normal data and model-generated
samples. DiSPaT operates through an alternating optimization: at each iteration,
the current policy generates synthetic samples that serve as pseudo-anomalies,
while a critic discriminator learns to distinguish these from real normal sample
s; this signal drives policy updates that progressively align the model distribu
tion with the true normal-data distribution. Extensive experiments on diverse be
nchmarks demonstrate that DiSPaT consistently outperforms prior LLM-based method
s, deep learning approaches, and classical unsupervised detectors for tabular an
omaly detection.
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The Accumulation of Score Estimation Error in Diffusion Models
Baoxiang He, Valentio Iverson, Shuai Li, Cheng Chen, Bo Jiang
Diffusion models are widely used for high-quality generation, but their performa
nce is sensitive to the accuracy of the estimated score.
We first derive a stepwise Wasserstein error bound in a Gaussian-mixture setting
, where the score admits a closed-form structure, and the score Hessian can be c
ontrolled explicitly, leading to sharp Wasserstein estimates.
We then extend the analysis to general data distributions, which yields a more g
eneral but typically looser upper bound.
This general bound can be sharpened under mild regularity: when the initial dist
ribution has a globally Lipschitz score, the curvature contribution at small tim
es is uniformly bounded, avoiding the worst-case blow-up.
The results hold for both variance-preserving (VP) and variance-exploding (VE) d
iffusions, and apply to both the reverse-time SDE and the associated probability
-flow ODE.
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Visual Implicit Autoregressive Modeling
Pengfei Jiang, Jixiang Luo, Luxi Lin, Zhaohong Huang, Xuelong Li
Visual Autoregressive Modeling (VAR) based on next-scale prediction achieves str
ong generation quality, but their explicit deep stacks fix the amount of computa
tion per scale and inflate memory at high resolutions. We introduce Visual Impli
cit Autoregressive Modeling (VIAR), a next-scale autoregressive generator that e

mbeds an implicit equilibrium layer between shallow pre/post blocks. The implicit layer is trained with Jacobian■Free Backpropagation, yielding constant training memory, while inference exposes a per■scale iteration knob that enables compute control. On ImageNet 256 × 256 benchmark, VIAR attains FID 2.16, and sFID 8.07 with only 38.4\% parameters of VAR, matching or surpassing strong AR baselines and remaining competitive with large diffusion models. By controlling the per-scale knob, VIAR can reduce peak memory from 19.24 GB to 8.53 GB and doubles throughput from 15.16 to 32.08 images/s on a single RTX 4090, without retraining. Ablations show that fewer steps are sufficient for fixed-point iterations to converge and that VIAR consistently dominates VAR across quality efficiency operating points. In zero shot in-painting and class■conditional editing, VIAR produces sharper details and smoother boundaries while preserving global structure, validating the benefits of implicit equilibria and per■scale compute control for practical, deployable visual generation.

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Revealing Scaling Paradox in Large-scale Time Series Models: Implications for More Efficient and Accurate Forecasting

Xin Qiu, Junlong Tong, Yirong Sun, Yunpu Ma, Xiaoyu Shen

Large-scale models are at the forefront of time series (TS) forecasting, dominated by two paradigms: fine-tuning text-based Large Language Models for TS (LLM4TS) and training Time Series Foundation Models (TSFMs) from scratch. Both approaches share a foundational assumption that scaling up model capacity and data volume leads to improved performance. However, we observe a scaling paradox in TS models, revealing a puzzling phenomenon that larger models do NOT always achieve better performance. Through extensive experiments on two model families across four scales (100M to 1.7B parameters) and diverse data (up to 6B observations), we rigorously confirm that the scaling paradox is a pervasive issue. We then diagnose its root cause by analyzing internal representations, identifying a phenomenon we call few-layer dominance: only a small subset of layers are functionally important, while the majority are redundant, under-utilized, and can even distract training. Based on this discovery, we propose a practical method to automatically identify and retain only these dominant layers. In our models, retaining only 21% of the parameters achieves up to a 12% accuracy improvement and a 2.7x inference speedup. We validate the universality of our method on 8 prominent SOTA models (LLM4TS and TSFMs, 90M to 6B), showing that retaining less than 30% layers achieves superior accuracy in over 95% tasks.

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Dream-MPC: Gradient-Based Model Predictive Control with Latent Imagination

Jonathan Spieler, Sven Behnke

State-of-the-art model-based Reinforcement Learning (RL) approaches either use gradient-free, population-based methods for planning, learned policy networks, or a combination of policy networks and planning. Hybrid approaches that combine Model Predictive Control (MPC) with a learned model and a policy prior to leverage the advantages of both paradigms have shown promising results. However, these approaches typically rely on gradient-free optimization methods, which can be computationally expensive for high-dimensional control tasks. While gradient-based methods are a promising alternative, recent works have empirically shown that gradient-based methods often perform worse than their gradient-free counterparts. We propose Dream-MPC, a novel approach that generates few candidate trajectories from a rolled-out policy and optimizes each trajectory by gradient ascent using a learned world model, uncertainty regularization and amortization of optimization iterations over time by reusing previously optimized actions. Our results on 24 continuous control tasks show that Dream-MPC can significantly improve the performance of the underlying policy and can outperform gradient-free MPC and state-of-the-art baselines. Code and videos are available at https://dream-mpc.github.io.

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How Does Reasoning Flow? Tracing Attention-Induced Information Flow for Targeted RL in LLMs

Zhichen Dong, Yang Li, Yuhan Sun, Weixun Wang, Yijia Luo, Zinian Peng, Wenbo Su,

Cheng Yu, Bo Zheng, Junchi Yan
Token-level credit assignment remains a key obstacle for reinforcement learning (RL) in large language models (LLMs), where RL recipes typically treat all tokens equally, failing to distinguish decisive reasoning steps from routine formatting or fluent filler. Recent attempts leverage model-internal signals to assign finer-grained credit, but these are often point-wise heuristics that ignore the global structure of information propagation. We propose FlowTracer, an RL framework that traces answer-targeted reasoning flow on an attention-induced directed acyclic graph in which nodes correspond to tokens and edge capacities come from aggregated attention weights and derives token credit from this global structure. The edge capacities are reweighted to retain only the influence that can reach the answer region, while enforcing local flow conservation so intermediate tokens neither lose nor gain effective mass due to path length or irrelevant branches. On this graph, FlowTracer extracts an information-flow backbone connecting the question to the answer and scores tokens by flow throughput, revealing high-impact hubs and aggregation checkpoints that mediate long-range dependencies. These derived importances are used to shape token-level rewards, enabling learning signals to focus precisely on the tokens that route information toward (or away from) correct answers and delivering consistent performance gains across a range of reasoning tasks.
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Block-wise Codeword Embedding for Reliable Multi-bit Text Watermarking
Joeun Kim, HoEun Kim, Dongsup Jin, Young-Sik Kim
Recent multi-bit watermarking methods for large language models (LLMs) prioritize capacity over reliability, often conflating decoding with detection. Our analysis reveals that existing ECC-based extractors suffer from catastrophic false positive rates (FPR), and applying rejection thresholds merely collapses detection sensitivity (TPR) to random guessing. To resolve this structural limitation, we propose **BREW** (Block-wise Reliable Embedding for Watermarking), a framework shifting the paradigm to *designated verification*. BREW employs a two-stage mechanism: (i) **blind message estimation** via independent block voting, followed by (ii) **window-shifting verification** that rigorously validates the payload against local edits. Experiments demonstrate that BREW achieves a TPR of 0.965 with an FPR of 0.02 under 10\% synonym substitution, demonstrating that the high-FPR issue is not an inherent trade-off of multi-bit watermarking, but a solvable structural flaw of prior decoding-centric designs. Our framework is model-agnostic and theoretically grounded, providing a scalable solution for reliable forensic deployment.
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Flow Sampling : Learning to Sample from Unnormalized Densities via Denoising Conditional Processes
Aaron J Havens, Brian Karrer, Neta Shaul
Sampling from unnormalized densities is analogous to the generative modeling problem, but the target distribution is defined by a known energy function instead of data samples. Because evaluating the energy function is often costly, a primary challenge is to learn an efficient sampler. We introduce Flow Sampling, a framework built on diffusion models and flow matching for the data-free setting. Our training objective is conditioned on a noise sample and regresses onto a denoising diffusion drift constructed from the energy function. In contrast, diffusion models' objective is conditioned on a data sample and regresses onto a noising diffusion drift. We utilize the interpolant process to minimize the number of energy function evaluations during training, resulting in an efficient and scalable method for sampling unnormalized densities. Furthermore, our formulation naturally extends to Riemannian manifolds, enabling diffusion-based sampling in geometries beyond Euclidean space. We derive a closed-form formula for the conditional drift on constant curvature manifolds, including hyperspheres and hyperbolic spaces. We evaluate Flow Sampling on synthetic energy benchmarks, small peptides, large-scale amortized molecular conformer generation, and distributions supported on the sphere, demonstrating strong empirical performance.
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Long-term Fairness with Selective Labels
Giovani Valdrighi, Isabel Valera, Marcos M. Raimundo
Long-term fairness algorithms aim to satisfy fairness beyond static and short-te rm notions by accounting for the dynamics between decision-making policies and p opulation behavior. Most previous approaches evaluate performance and fairness m easures from observable features and a label, which is assumed to be fully obser ved. However, in scenarios such as hiring or lending, the labels (e.g., ability to repay the loan) are _selective labels_ as they are only revealed based on pos itive decisions (e.g., when a loan is granted). In this paper, we study long-ter m fairness in the selective labels setting and analytically show that naive solu tions do not guarantee fairness. To address this gap, we then introduce a novel framework that leverages both the observed data and a label predictor model to e stimate the true fairness measure value by decomposing it into the observed fair ness and bias from label predictions. This allows us to derive sufficient condi tions to satisfy true fairness from observable quantities by using the confidenc e in the predictor model. Finally, we rely on our theoretical results to propo se a novel reinforcement learning algorithm for effective long-term fair decisio n-making with selective labels. In semisynthetic environments, the proposed algo rithm reached comparable fairness and performance to an agent with oracle access to the true labels.
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Revisiting Positive Samples in Graph Contrastive Learning: From the Perspective of Message Passing
Lianze Shan, Ningchong Wang, Jitao Zhao, Di Jin, Dongxiao He
Graph Contrastive Learning (GCL), which trains graph encoders by maximizing simi larity between positive samples and minimizing it between negative ones, has eme rged as a mainstream graph pre-training paradigm. It is widely recognized that p ositive samples are essential in GCLs. Ideally, maximizing the similarity of pos itive samples enables graph encoders to capture intrinsic semantic and patterns of graph data. However, we discover an interesting phenomenon: GCLs can achieve competitive performance even without positive samples. This motivates us to revi sit the fundamental mechanism of positive samples in GCLs. From the perspective of Dirichlet energy, we theoretically finds that message passing, a key mechanis m in graph encoders, trivializes the maximization of positive samples, preventin g GCLs from effectively learning from positive samples. To address this, we prop ose SPGCL to mitigate the trivialization caused by message passing and restore t he learning efficacy of positive samples. Specifically, we find that high Dirich let energy features help positive samples provide effective learning signals whi le low Dirichlet energy features contribute little to positive learning signal b ut is useful for positive sampling. Based on this, SPGCL propagates only high Di richlet energy features and uses low energy features to construct a probability matrix for reliable positive sampling. Extensive experiments demonstrate the eff ectiveness of SPGCL.
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Decompose and Recompose: Reasoning New Skills from Existing Abilities for Cross-Task Robotic Manipulation
Xitie Zhang, Aming WU, Yahong Han
Cross-task generalization is a core challenge in open-world robotic manipulation , and the key lies in extracting transferable manipulation knowledge from seen t asks. Recent in-context learning approaches leverage seen task demonstrations to generate actions for unseen tasks without parameter updates. However, existing methods provide only low-level continuous action sequences as context, failing t o capture composable skill knowledge and causing models to degenerate into super ficial trajectory imitation. We propose Decompose and Recompose, a skill reasoni ng framework using atomic skill-action pairs as intermediate representations. Ou r approach decomposes seen demonstrations into interpretable skill--action align ments, enabling the model to recompose these skills for unseen tasks through com positional reasoning. Specifically, we construct a task-adaptive dynamic demonst ration library via visual-semantic retrieval combined with skill sequences from a planning agent, complemented by a coverage-aware static library to fill missin

g skill patterns. Together, these yield skill-comprehensive demonstrations that explicitly elicit compositional reasoning for skill composition and execution ordering. Experiments on the AGNOSTOS benchmark and real-world environments validate our method's zero-shot cross-task generalization capability.
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LimiX-2M: Mitigating Low-Rank Collapse and Attention Bottlenecks in Tabular Foundation Models
Yuanrui Wang, Xingxuan Zhang, Han Yu, Mingchao Hao, Gang Ren, Hao Yuan, Li Mao, Yunjia Zhang, Chun Yuan, Peng Cui
Tabular foundation models (TFMs) increasingly rival tree ensembles, but their performance is often compute-inefficient: with standard affine scalar tokenization, each feature injects value variation through an essentially one-dimensional channel, and feature IDs/positional signals cannot increase within-feature value degrees of freedom, yielding weak early-layer value sensitivity and redundant hidden states.
We present a unified tokenize-and-route framework for strong TFMs: RaBEL expands each scalar into compact localized RBF features (optionally exponent-gated) to improve conditioning and shallow-layer effective rank, while a reordered bidirectional block $\textbf{S$\rightarrow$N$\rightarrow$F}$ aligns computation with the readout by aggregating cross-sample context before feature mixing and using attention pooling.
Together, these changes yield $\textbf{LimiX-2M}$, a 2M-parameter model that outperforms larger TabPFN-v2 and TabICL baselines on widely used tabular benchmarks while reducing training and inference costs. These results highlight value-aware tokenization and readout-aligned routing as key levers for improving the accuracy--efficiency trade-off in TFMs. Model checkpoints and inference code are available at https://github.com/limix-ldm-ai/LimiX.
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Unveiling Prior-Data Fitted Networks on Causal Effect Estimation: Pre-Training or Fine-Tuning?
Haotian Wang, Xinpeng Lv, Hao Zou, Yanghao Xiao, Shanzhi Gu, Yang Shi, Yunxin Mao, Yuanxing Zhang, Mingyang Geng, Shaowu Yang, Haoxuan Li, Wenjing Yang, Peng Cui, Zhouchen Lin
Amortized causal inference via Prior-data Fitted Networks (PFNs) has emerged as a promising paradigm, enabling zero-shot estimation of causal effects without the need for dataset-specific model tuning. However, the principled effectiveness of unified pre-training across general interventional regimes remains an underexplored question. In this paper, we investigate interventions on subsets of variables within Structural Causal Models (SCMs) and identify a fundamental theoretical limitation of current pre-training approaches. Theoretically, we prove that a single observational SCM induces an exponentially large space of interventional distributions, resulting in a phenomenon we term prior uncoverage. Consequently, this uncoverage yields a mismatch between the learned meta-prior and the true grounding prior, leading to unavoidable posterior inconsistency and estimation bias. To address this, we posit that fine-tuning is a fundamental necessity and propose a target-specific strategy named Point-Wise Interventional Fine-tuning (PWF), enabling the local generalization property. We further scale this approach via Meta-Sampling Fine-tuning (MSF) from a budgeted active learning perspective, thereby achieving uniform generalization on any interventional distribution.
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Replay Failures as Successes: Sample-Efficient Reinforcement Learning for Instruction Following
Kongcheng Zhang, QI YAO, Shunyu Liu, Wenjian Zhang, Min Cen, Yang Zhou, Wenkai Fang, Yiru Zhao, Baisheng Lai, Mingli Song
Reinforcement Learning (RL) has shown promise for aligning Large Language Models (LLMs) to follow instructions with various constraints. Despite the encouraging results, RL improvement inevitably relies on sampling successful, high-quality responses; however, the initial model often struggles to generate responses that satisfy all constraints due to its limited capabilities, yielding sparse or indistinguishable rewards that impede learning. In this work, we propose ***H**inds

ight **i**nstruction **R**eplay* (HiR), a novel sample-efficient RL framework fo r complex instruction following tasks, which employs a *select*-then-*rewrite* s trategy to *replay failed attempts as successes* based on the constraints that h ave been satisfied in hindsight. We perform RL on these replayed samples as well as the original ones, theoretically framing the objective as dual-preference le arning at both the instruction- and response-level to enable efficient optimizat ion using only a binary reward signal. Extensive experiments demonstrate that th e proposed HiR yields promising results across different instruction following t asks, while requiring less computational budget. Our code and dataset are availa ble at https://github.com/sastpg/HIR.
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Automata-Conditioned Cooperative Multi-Agent Reinforcement Learning
Beyazit Yalcinkaya, Marcell Vazquez-Chanlatte, Ameesh Shah, Hanna Krasowski, San jit A. Seshia
We study learning multi-task, multi-agent policies for cooperative, temporal obj ectives, under centralized training, decentralized execution. In this setting, u sing automata to represent tasks assigned to agents enables breaking down a team -level objective into simpler, smaller sub-tasks. However, existing approaches r emain sample-inefficient and are limited to the single-task case, requiring retr aining policies for each new task. In this work, we present Automata-Conditioned Cooperative Multi-Agent Reinforcement Learning (ACC-MARL), a framework for lear ning task-conditioned, decentralized team policies. We identify challenges to th e feasibility of ACC-MARL, propose solutions, and prove that our approach is opt imal. We further show that learned value functions can be used to assign tasks o ptimally at test time. Experiments demonstrate emergent task-aware, multi-step c oordination among agents, such as pressing a button to unlock a door, holding th e door, and short-circuiting tasks.
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Probably Approximately Correct Labels
Emmanuel Candes, Andrew Ilyas, Tijana Zrnic
Obtaining high-quality labeled datasets is often costly, requiring either human annotation or expensive experiments. In theory, powerful pre-trained AI models p rovide an opportunity to automatically label datasets and save costs. Unfortunat ely, these models come with no guarantees on their accuracy, making wholesale re placement of manual labeling impractical. In this work, we propose a method for leveraging pre-trained AI models to curate cost-effective and high-quality datas ets. In particular, our approach results in probably approximately correct label s: with high probability, the overall labeling error is small. Our method is non asymptotically valid under minimal assumptions on the dataset or the AI model be ing studied, and thus enables rigorous yet efficient dataset curation using mode rn AI models. We demonstrate the benefits of the methodology through text annota tion with large language models, image labeling with pre-trained vision models, and protein folding analysis with AlphaFold.
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Dual Latent Memory for Visual Multi-agent System
Xinlei Yu, Chengming Xu, Zhangquan Chen, Bo Yin, Cheng Yang, Yongbo He, Yihao Hu , Jiangning Zhang, Cheng Tan, Xiaobin Hu, Shuicheng YAN
While Visual Multi-Agent Systems (VMAS) promise to enhance comprehensive abiliti es through inter-agent collaboration, empirical evidence reveals a counter-intui tive "scaling wall": increasing agent turns often degrades performance while exp onentially inflating token costs. We attribute this failure to the information b ottleneck inherent in text-centric communication, where converting perceptual an d thinking trajectories into discrete natural language inevitably induces semant ic loss. To this end, we propose L^{2}-VMAS, a novel model-agnostic framework that enables inter-agent collaboration with dual latent memories. Based on such a design, we decouple the perception and thinking while dynamically synthesizing dual latent memories. Additionally, we introduce an entropy-driven proactive tr iggering that replaces passive information transmission with efficient, on-deman d memory access. Extensive experiments among backbones, sizes, and multi-agent s tructures demonstrate that our method effectively breaks the "scaling wall" with

superb scalability, improving average accuracy by 2.7-5.4% while reducing total token usage by 21.3-44.8%. The codes will be publicly released.
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DANCE: Dynamic, Available, Neighbor-gated Condensation for Federated Text-Attributed Graphs
Zekai Chen, Haodong Lu, Xunkai Li, Henan Sun, Jia Li, Hongchao Qin, Rong-Hua Li, Guoren Wang
Federated graph learning (FGL) enables collaborative training on graph data across multiple clients. With the rise of large language models (LLMs), textual attributes in FGL graphs are gaining attention. Text-attributed graph federated learning (TAG-FGL) improves FGL by explicitly leveraging LLMs to process and integrate these textual features.
However, current TAG-FGL methods face three main challenges:
\textbf{(1) Overhead.} LLMs for processing long texts incur high token and computation costs. To make TAG-FGL practical, we introduce graph condensation (GC) to reduce computation load, but this choice also brings new issues.
\textbf{(2) Suboptimal.}
To reduce LLM overhead, we introduce GC into TAG-FGL by compressing multi-hop texts/neighborhoods into a condensed core with fixed LLM surrogates (summaries/embeddings).
However, this one-shot condensation is often not client-adaptive, leading to suboptimal performance.
\textbf{(3) Interpretability.}
LLM-based condensation further introduces a black-box bottleneck: summaries lack faithful attribution and clear grounding to specific source spans, making local inspection and auditing difficult.
To address the above issues, we propose \textbf{DANCE}, a new TAG-FGL paradigm with GC. To improve \textbf{suboptimal} performance, DANCE performs round-wise, model-in-the-loop condensation refresh using the latest global model. To enhance \textbf{interpretability}, DANCE preserves provenance by storing locally inspectable evidence packs that trace predictions to selected neighbors and source text spans.
Across 8 TAG datasets, DANCE improves accuracy by \textbf{2.33\%} at an \textbf{8\%} condensation ratio, with \textbf{33.42\%} fewer tokens per condensed node than TAG-FGL baselines.
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Efficient Multi-modal Dataset Distillation via Analytic Parameter Matching
Deyu Bo, Xinchao Wang
Multi-modal dataset distillation (MDD) seeks to compress large-scale multi-modal datasets into a compact set of synthetic pairs. Existing methods employ a dual-trajectory matching framework to align the teacher and student models within each modality. While effective, this paradigm incurs non-negligible memory and computational overhead due to the checkpoint storage and bi-level optimization over synthetic data. To address these limitations, we propose analytic parameter matching (APM), which theoretically derives the analytic parameters of modal projectors to replace the inner-loop optimization, and then aligns the analytic projector parameters of teacher and student models. APM offers two key advantages: (1) it replaces checkpoint-intensive storage with only two cached matrices, significantly reducing memory consumption; and (2) it computes analytic parameters in a single forward pass, thereby avoiding costly bi-level optimization. Empirically, APM achieves up to 65$\times$ storage reduction and 9.6$\times$ faster distillation, while scaling to 1,000 synthetic pairs. Extensive experiments on image-text and audio-text benchmarks demonstrate the effectiveness of APM in cross-modal retrieval tasks, e.g., 12.8 IR@1 and 17.8 TR@1 in Flickr30k with 100 synthetic pairs. Moreover, APM exhibits notable generalization performance in cross-architecture evaluation and zero-shot classification tasks.
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Incentivizing Truthfulness and Collaborative Fairness in Bayesian Learning
Rachael Hwee Ling Sim, Jue Fan, Xiao Tian, Xinyi Xu, Patrick Jaillet, Bryan Kian Hsiang Low

Collaborative machine learning involves training high-quality models using datas
ets from a number of sources.
To incentivize sources to share data, existing data valuation methods fairly rew
ard each source based on its data submitted as is. However, as these methods do
not verify nor incentivize data truthfulness, the sources can manipulate their d
ata (e.g., by submitting duplicated or noisy data) to artificially increase thei
r valuations and rewards or prevent others from benefiting. This paper presents
the first mechanism that provably ensures (**F**) collaborative fairness and inc
entivizes (**T**) truthfulness at equilibrium for Bayesian models. Our mechanism
 combines semivalues (e.g., Shapley value), which ensure fairness, and a truthfu
l data valuation function (DVF) based on a validation set that is unknown to the
 sources. As semivalues are influenced by others' data, we introduce an addition
al condition to prove that a source can maximize its expected data values in coa
litions and semivalues by submitting a dataset that captures its true knowledge.
Additionally, we discuss the implications and suitable relaxations of (**F**) an
d (**T**) when the mediator has a limited budget for rewards or lacks a validati
on set.
Our theoretical findings are validated on synthetic and real-world datasets.
****

DocVAL: Validated Chain-of-Thought Distillation for Grounded Document VQA
Pinaki Prasad Guha Neogi, Ahmad Mohammadshirazi, Ser-Nam Lim, Rajiv Ramnath
Document visual question answering requires models not only to answer questions
correctly, but also to precisely localize answers within complex document layout
s. While large vision-language models (VLMs) achieve strong spatial grounding, t
heir inference cost and latency limit real-world deployment; on the other hand,
compact VLMs are efficient but suffer substantial localization degradation under
 standard fine-tuning or distillation. To address this gap, we propose **DocVAL*
*, a validated chain-of-thought (CoT) distillation framework that transfers expl
icit spatial reasoning from large teacher models to compact, deployable student
VLMs. DocVAL combines **(1)** teacher-generated spatial CoT supervision, **(2)**
 a rule-based dual-mode validator that filters low-quality training signals and
provides fine-grained, pixel-level corrective feedback, and **(3)** a validation
-driven two-stage training procedure with iterative refinement. Text detection i
s used only as training-time scaffolding for supervision and validation, enablin
g the final student to operate as a pure VLM without OCR or detection at inferen
ce. Across multiple document understanding benchmarks, the proposed **DocVAL** y
ields consistent improvements of up to **6-7 ANLS** points over comparable compa
ct VLMs. We further introduce mean Average Precision (mAP) as a localization met
ric for document question answering and report strong spatial grounding performa
nce under this new evaluation. We release **95K validator-verified CoT traces**
and show that high-quality, validated supervision is more effective than scaling
 unfiltered data, enabling efficient and trustworthy document grounding. Code/Da
ta: [GitHub](https://github.com/ahmad-shirazi/DocVAL).
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Coupled Variational Reinforcement Learning for Language Model General Reasoning
Xueru Wen, Jie Lou, Yanjiang Liu, Hongyu Lin, Ben He, Xianpei Han, Le Sun, Yaoji
e Lu, Debing Zhang
While reinforcement learning has achieved impressive progress in language model
reasoning, it is constrained by the requirement for verifiable rewards.
Recent verifier-free RL methods address this limitation by utilizing the probabi
lities that LLMs generate reference answers as reward signals.
However, these approaches typically sample reasoning traces conditioned only on
the question.
This design decouples reasoning-trace sampling from answer information, leading
to inefficient exploration and incoherence between traces and final answers.
In this paper, we propose \textit{\b{Co}upled \b{V}ariational \b{R}einforcement
\b{L}earning} (CoVRL), which bridges variational inference and reinforcement lea
rning by coupling prior and posterior distributions through a hybrid sampling st
rategy.
By constructing and optimizing a composite distribution that integrates these tw

o distributions, CoVRL enables efficient exploration while preserving strong tho
ught-answer coherence.
Extensive experiments on mathematical and general reasoning benchmarks show that
 CoVRL improves performance by 12.4\% over the base model and achieves an additi
onal 2.3\% improvement over state-of-the-art verifier-free RL baselines, providi
ng a principled framework for enhancing the general reasoning capabilities of la
nguage models.
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Unsat Core Prediction through Polarity-Aware Representation Learning over Clause
-Literal Hypergraphs

Zhenchao Sun, Shuai Ma, Ping Lu, Chongyang Tao

Graph neural networks have been widely used in Boolean satisfiability (SAT) task
s to learn structural information from SAT formulas. The goal of these studies
is to solve SAT instances or to enhance SAT solvers, including tasks such as uns
at-core prediction. However, most existing approaches model a SAT formula as a b
ipartite graph or a directed acyclic graph, which are less direct in capturing
clause-level and higher-order interactions among literals and clauses. Moreover,
 these approaches are limited in modeling intrinsic polarity-related properties
of SAT, such as the complementary relationship between the positive and negativ
e literals of a variable. To address these limitations, we propose a polarity-aw
are representation learning framework over clause-literal hypergraphs. We model
 SAT formulas as clause-literal hypergraphs augmented with a clause incidence gr
aph to capture higher-order structural interactions. We then introduce a polari
ty-aware decomposition mechanism that separates variable representations into po
larity invariant and equivariant components, explicitly modeling the relationshi
p between positive and negative literals, with the resulting literal representat
ions propagated along the hypergraph structure. We further incorporate a polari
ty-inversion consistency regularization to reinforce polarity-consistent repres
entations during training. Experimental results on multiple SAT datasets demonst
rate the effectiveness of the proposed approach.
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Optimal Quantum Speedups for Repeatedly Nested Expectation Estimation

Yihang Sun, Guanyang Wang, Jose Blanchet

We study the estimation of repeatedly nested expectations (RNEs) with a constant
 horizon (number of nestings) using quantum computing. We propose a quantum algo
rithm that achieves ε-error with cost $\tilde O(\varepsilon^{-1})$,
up to logarithmic factors. Standard lower bounds show this scaling is essentiall
y optimal, yielding an almost quadratic speedup over the best classical algorith
m. Our results extend prior quantum speedups for single nested expectations to r
epeated nesting, and therefore cover a broader range of applications, including
optimal stopping. This extension requires a new derandomized variant of the clas
sical randomized Multilevel Monte Carlo (rMLMC) algorithm. Careful de-randomizat
ion is key to overcoming a variable-time issue that typically increases quantize
d versions of classical randomized algorithms.
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Dynamic Decision Learning: Test-Time Evolution for Abnormality Grounding in Rare
 Diseases

Jun Li, Mingxuan Liu, Jiazhen Pan, Che Liu, Wenjia Bai, Cosmin I. Bercea, Julia
Schnabel

Clinical abnormality grounding for rare diseases is often hindered by data scarc
ity, rendering supervised fine-tuning infeasible and single-pass inference highl
y unstable. Thus, we propose Dynamic Decision Learning (DDL), a framework that e
nables frozen LVLMs to refine their decisions across language and visual spaces
by optimizing instructions and consolidating predictions under visual perturbati
ons, thereby improving localization quality and producing a consensus-based reli
ability score that quantifies the model’s confidence. Results on brain-imaging b
enchmarks, including a rare-disease dataset with 281 pathology types across 3B–7
2B models, show that DDL improves mAP@75 by up to 105% on rare-disease cases and
 surpasses adaptation baselines and supervised fine-tuning. Moreover, we show th
at DDL yields stronger calibration between consensus-based reliability scores an

d localization accuracy under severe distribution shifts and increasing task dif
ficulty. The code is available at https://github.com/compai-lab/2026-ICML-DDL .
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SENDAI: A Hierarchical Sparse-measurement, EfficieNt Data AssImilation Framework
Xingyue Zhang, Yuxuan Bao, Mars Liyao Gao, J. Nathan Kutz
Bridging the gap between data-rich training regimes and observation-sparse deplo
yment conditions remains a central challenge in spatiotemporal field reconstruct
ion, particularly when target domains exhibit distributional shifts, heterogeneo
us structure, and multi-scale dynamics absent from available training data. We p
resent SENDAI, a hierarchical $\textbf{S}$parse-measurement, $\textbf{E}$fficie$
\textbf{N}$t $\textbf{D}$ata $\textbf{A}$ss$\textbf{I}$milation Framework that r
econstructs full spatial states from hyper sparse sensor observations by combini
ng simulation-derived priors with learned discrepancy corrections. We demonstrat
e the performance on satellite remote sensing, reconstructing MODIS (Moderate Re
solution Imaging Spectroradiometer) derived vegetation index fields across six
globally distributed sites. Using seasonal periods as a proxy for domain shift,
the framework consistently outperforms established baselines that require substa
ntially denser observations---SENDAI achieves a maximum SSIM improvement of 185%
 over traditional baselines and a 36% improvement over recent high-frequency-bas
ed methods. These gains are particularly pronounced for landscapes with sharp bo
undaries and sub-seasonal dynamics; more importantly, the framework effectively
preserves diagnostically relevant structures---such as field topologies, land co
ver discontinuities, and spatial gradients. By yielding corrections that are mor
e structurally and spectrally separable, the reconstructed fields are better sui
ted for downstream inference of indirectly observed variables. The results there
fore highlight a lightweight and operationally viable framework for sparse-measu
rement reconstruction that is applicable to physically grounded inference, resou
rce-limited deployment, and real-time monitor and control.
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Evaluating and Rewarding LALMs for Expressive Role-Play TTS via Mean Continuatio
n Log-Probability
Yong Ren, Jingbei Li, Haiyang Sun, Yujie Chen, Cheng Yi, Yechang Huang, Hao Gu,
Ye Bai, Xuerui Yang
Recent advances in Large Audio Language Models (LALMs) have extended Text-to-Spe
ech (TTS) to interactive role-play scenarios, which demand high expressiveness a
nd strict adherence to role-play instructions. However, existing models struggle
 to maintain stylistic consistency with character profiles and scene description
s across multi-turn dialogues. A critical bottleneck is the lack of objective me
trics for quantifying speaking style. To bridge this gap, we propose **Mean Cont
inuation Log-Probability (MCLP)** as both an evaluation metric and a reward sign
al, validated on LALM-based Role-Play TTS (RP-TTS) tasks. MCLP leverages the in-
context learning capability of pretrained LALMs to measure the likelihood of gro
und-truth speech tokens conditioned on a contextual history consisting of the tr
anscript, generated speech, and repeated transcript, serving as a proxy for styl
istic continuity. Furthermore, we employ MCLP as a reinforcement learning reward
 to enhance the style alignment between generated speech and role-play instructi
ons. To support this task, we construct a large-scale RP-TTS dataset with rich s
cene and character annotations. Experiments demonstrate that MCLP is well aligne
d with human judgments of stylistic consistency and serves as an effective rewar
d for improving RP-TTS, leading to consistent gains in both objective metrics an
d subjective evaluations. Our code is publicly available at https://github.com/y
-ren16/MCLP.
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RSAgent: Learning to Reason and Act via Multi-Turn Tool Invocations for Text-Gui
ded Segmentation
Xingqi He, Yujie Zhang, Shuyong Gao, Wenjie Li, Lingyi Hong, Mingxi Chen, Kaixun
 Jiang, Jiyuan Fu, Wenqiang Zhang
Text-guided object segmentation requires both cross-modal reasoning and pixel gr
ounding abilities. Most recent methods treat it as a single forward pass, where
the model directly predicts pixel prompts to a segmentation model, which limits

verification, refocusing and refinement when initial localization is wrong. To address this limitation, we propose **RSAgent**, an agentic Multimodal Large Language Model (MLLM) which interleaves **reasoning and action** for segmentation via a multi-turn tool invocations. RSAgent queries a visual toolbox, observes feedback, and revises its pixel hypothesis using historical observations to re-localize targets and iteratively refine masks. We further build a data pipeline to synthesize multi-turn reasoning segmentation trajectories, and train RSAgent with a two-stage framework: cold-start supervised fine-tuning followed by agentic reinforcement learning with fine-grained, task-specific rewards.
Extensive experiments show that RSAgent achieves a zero-shot performance of 66.5% gIoU on ReasonSeg test, improving over Seg-Zero-7B by 9%, and reaches 81.5% cIoU on RefCOCOg, demonstrating state-of-the-art performance on both in-domain and out-of-domain benchmarks.
****

Resource-Efficient Reinforcement for Reasoning Large Language Models via Dynamic One-Shot Policy Refinement
Yunjian Zhang, Sudong Wang, Yang Li, Peiran Xu, Conghao Zhou, Xiaoyue Ma, Jianing Li, Yao Zhu
Large language models (LLMs) have exhibited remarkable performance on complex reasoning tasks, with reinforcement learning under verifiable rewards (RLVR) emerging as a principled framework for aligning model behavior with reasoning chains. Despite its promise, RLVR remains prohibitively resource-intensive, requiring extensive reward signals and incurring substantial rollout costs during training. In this work, we revisit the fundamental question of data and compute efficiency in RLVR. We first establish a theoretical lower bound on the sample complexity required to unlock reasoning capabilities, and empirically validate that strong performance can be achieved with a surprisingly small number of training instances. To tackle the computational burden, we propose Dynamic One-Shot Policy Refinement (DoPR), a uncertainty-aware RL strategy that dynamically selects a single informative training sample per batch for policy updates, guided by reward volatility and exploration-driven acquisition. DoPR reduces rollout overhead by nearly an order of magnitude while preserving competitive reasoning accuracy, offering a scalable and resource-efficient solution for LLM post-training. This approach offers a practical path toward more efficient and accessible RL-based training for reasoning-intensive LLM applications.
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Graph-R1: Towards Agentic GraphRAG Framework via End-to-end Reinforcement Learning
Haoran Luo, Haihong E, Guanting Chen, Qika Lin, Yikai Guo, Fangzhi Xu, Zemin Kuang, Meina Song, Xiaobao Wu, Yifan Zhu, Anh Tuan Luu
Retrieval-Augmented Generation (RAG) mitigates hallucination in LLMs by incorporating external knowledge, but relies on chunk-based retrieval that lacks structural semantics. GraphRAG methods improve RAG by modeling knowledge as entity-relation graphs, but still face challenges in high construction cost, fixed one-time retrieval, and reliance on long-context reasoning and prompt design. To address these challenges, we propose Graph-R1, the first agentic GraphRAG framework via end-to-end reinforcement learning (RL). It introduces lightweight knowledge hypergraph construction, models retrieval as a multi-turn agent-environment interaction, and optimizes the agent process via an end-to-end reward mechanism. Experiments on standard RAG datasets show that Graph-R1 outperforms traditional GraphRAG and RL-enhanced RAG methods in reasoning accuracy, retrieval efficiency, and generation quality. Our software and data are publicly available.
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Sparser, Faster, Lighter Transformer Language Models
Edoardo Cetin, Stefano Peluchetti, Emilio Castillo, Akira Naruse, Mana Murakami, Llion Jones
Scaling autoregressive large language models (LLMs) has driven unprecedented progress but comes with vast computational costs. In this work, we tackle these costs by leveraging unstructured sparsity within an LLM's feedforward layers, the components accounting for most of the model parameters and execution FLOPs. To ac

hieve this, we introduce a new sparse packing format and a set of CUDA kernels d
esigned to seamlessly integrate with the optimized execution pipelines of modern
 GPUs, enabling efficient sparse computation during LLM inference and training.
To substantiate our gains, we provide a quantitative study of LLM sparsity, demo
nstrating that simple L1 regularization can induce over 99% sparsity with neglig
ible impact on downstream performance. When paired with our kernels, we show tha
t these sparsity levels translate into substantial throughput, energy efficiency
, and memory usage benefits that increase with model scale. We will release all
code and kernels under an open-source license to promote adoption and accelerate
 research toward establishing sparsity as a practical axis for improving the eff
iciency and scalability of modern foundation models.
****
Agent-Omit: Adaptive Context Omission for Efficient LLM Agents
Yansong Ning, Jun Fang, Naiqiang Tan, Hao Liu
Managing agent context (e.g., thought and observation) during multi-turn
agent-environment interactions is an emerging strategy to improve agent efficien
cy.
However, existing studies treat the entire interaction trajectories equally, ove
rlooking the thought necessity and observation utility varies across turns.
To this end, we first conduct quantitative investigations into how thought and o
bservation affect agent effectiveness and efficiency.
Based on our findings, we propose Agent-Omit, a unified training framework that
empowers LLM agents to adaptively omit redundant thoughts and observations.
Specifically, we first synthesize a small amount of cold-start data, including b
oth single-turn and multi-turn omission scenarios, to fine-tune the agent for om
ission behaviors.
Furthermore, we introduce an omit-aware agentic reinforcement learning approach,
 incorporating a dual sampling mechanism and a tailored omission reward to incen
tivize the agent's adaptive omission capability.
Theoretically, we prove that the deviation of our omission policy is upper-bound
ed by KL-divergence.
Experimental results on five agent benchmarks show that our constructed Agent-Om
it-8B could obtain performance comparable to seven frontier LLM agent, and achie
ve the best effectiveness-efficiency trade-off than seven efficient LLM agents m
ethods.
Our code and data are avaliable at https://github.com/usail-hkust/Agent-Omit.
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Rethinking LLM Ensembling from the Perspective of Mixture Models
Jiale Fu, Yuchu Jiang, PeiJun WU, Chonghan Liu, Joey Tianyi Zhou, Xu Yang
Model ensembling is a well-established technique for improving the performance o
f machine learning models. Conventionally, this involves averaging the output di
stributions of multiple models and selecting the most probable label. This idea
has been naturally extended to large language models (LLMs), yielding improved p
erformance but incurring substantial computational cost. This inefficiency stems
 from directly applying conventional ensemble implementation to LLMs, which requ
ire a separate forward pass for each model to explicitly compute the ensemble di
stribution. In this paper, we propose the Mixture-model-like Ensemble (ME). By r
einterpreting the ensemble as a mixture model, ME stochastically selects a singl
e model at each step to generate the next token, thereby avoiding the need to ex
plicitly compute the full ensemble distribution. ME is mathematically equivalent
 to sampling from the ensemble distribution, but requires invoking only one mode
l, making it 1.78×-2.68× faster than conventional ensembling. Furthermore, this
perspective connects LLM ensembling and token-level routing methods, suggesting
that LLM ensembling is a special case of routing methods. Our findings open new
avenues for efficient LLM ensembling and motivate further exploration of token-l
evel routing strategies for LLMs. Our code is available at https://github.com/Ka
michanw/Mixture-model-like-Ensemble.
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Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM
Agents

Jiawei Wang, Jiacai Liu, Yuqian Fu, Yingru Li, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, WANG KE

In long-horizon tasks, recent agents based on Large Language Models (LLMs) face a significant challenge that sparse, outcome-based rewards make it difficult to assign credit to intermediate steps. Previous methods mainly focus on creating dense reward signals to guide learning, either through traditional reinforcement learning techniques like inverse reinforcement learning or by using Process Reward Models for step-by-step feedback. In this paper, we identify a fundamental problem in the learning dynamics of LLMs: the magnitude of policy gradients is inherently coupled with the entropy, which leads to inefficient small updates for confident correct actions and potentially destabilizes large updates for uncertain ones. To resolve this, we propose Entropy-Modulated Policy Gradients (EMPG), a framework that recalibrates the learning signal based on step-wise uncertainty and the final task outcome. EMPG amplifies updates for confident correct actions, penalizes confident errors, and attenuates updates from uncertain steps to stabilize exploration. We further introduce a bonus term for future clarity that encourages agents to find more predictable solution paths. Through comprehensive experiments on three challenging agent tasks, WebShop, ALFWorld, and Deep Search, we demonstrate that EMPG achieves substantial performance gains and significantly outperforms strong policy gradient baselines.

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ProAct: A Benchmark and Multimodal Framework for Structure-Aware Proactive Response

Xiaomeng Zhu, Fengming ZHU, Weijie Zhou, Ye Tian, Zhenlin Hu, Yufei Huang, Yuchun Guo, Xinyu Wu, Zhengyou Zhang, Fangzhen Lin, Xuantang Xiong

While passive agents merely follow instructions, proactive agents align with higher-level objectives, such as assistance and safety by continuously monitoring the environment to determine when and how to act. However, developing proactive agents is hindered by the lack of specialized resources. To address this, we introduce **ProAct-75**, a benchmark designed to train and evaluate proactive agents across diverse domains, including assistance, maintenance, and safety monitoring. Spanning 75 tasks, our dataset features 91,581 step-level annotations enriched with explicit task graphs. These graphs encode step dependencies and parallel execution possibilities, providing the structural grounding necessary for complex decision-making. Building on this benchmark, we propose **ProAct-Helper**, a reference baseline powered by a Multimodal Large Language Model (MLLM) that grounds decision-making in state detection, and leveraging task graphs to enable entropy-driven heuristic search for action selection, allowing agents to execute parallel threads independently rather than mirroring the human's next step. Extensive experiments demonstrate that ProAct-Helper outperforms strong closed-source models, improving trigger detection mF1 by 6.21\%, saving 0.25 more steps in online one-step decision, and increasing the rate of parallel actions by 15.58\%. Code is available at https://github.com/ZhuXMMM/ProAct.git

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Hearing Without Noticing? Attention-Aware Stealthy Black-Box Adversarial Audio Attacks

Tianyi Xu, Cheng'an Wei, Yue Zhao, Kai Chen

Automatic Speech Recognition (ASR) systems, such as those in intelligent assistants, are vulnerable to adversarial examples (AEs). Benign audio clips like music, when embedded with small perturbations, can trick ASR models into recognizing attacker-specified commands. Prior studies focus on minimizing perturbation magnitude to craft AEs. However, they fails to achieve high attack stealthiness against black-box ASR systems in the physical world. In this paper, we introduce the first music carrier selection algorithm and an attention-aware stealthiness loss function to generate stealthy AEs. Extensive evaluations on five commercial ASR APIs and three widely-used voice assistants demonstrate that our method significantly outperforms state-of-the-art techniques in both effectiveness and stealthiness. Notably, in a user study involving 200 participants, 55.6\% of participants perceived our physical adversarial examples as benign audio, which is an improvement of over 20\% compared to existing methods.

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Causal Preference Elicitation
Edwin V. Bonilla, He Zhao, Daniel M. Steinberg
We propose causal preference elicitation, a Bayesian framework for expert-in-the-loop causal discovery that actively queries local edge relations to concentrate a posterior over directed acyclic graphs (DAGs). From any black-box observational posterior, we model noisy expert judgments with a three-way likelihood over edge existence and direction. Posterior inference uses a flexible particle approximation, and queries are selected by an efficient expected information gain criterion on the expert's categorical response. Experiments on synthetic graphs, protein signaling data, and a human gene perturbation benchmark show faster posterior concentration and improved recovery of directed effects under tight query budgets.
****
DexMachina: Functional Retargeting for Bimanual Dexterous Manipulation
Zhao Mandi, Yifan Hou, Dieter Fox, Yashraj Narang, Ajay Mandlekar, Shuran Song
We study the problem of functional retargeting: learning dexterous manipulation policies to track object states from human hand-object demonstrations. We focus on long-horizon, bimanual tasks with articulated objects, which are challenging due to large action space, spatiotemporal discontinuities, and the embodiment gap between human and robot hands. We propose DexMachina, a novel curriculum-based algorithm: the key idea is to use virtual object controllers with decaying strength: an object is first driven automatically towards its target states, such that the policy can gradually learn to take over under motion and contact guidance. We release a simulation benchmark with a diverse set of tasks and dexterous hands, and show that DexMachina significantly outperforms baseline methods. Our algorithm and benchmark enable a functional comparison for hardware designs, and we present key findings informed by quantitative and qualitative results. With the recent surge in dexterous hand development, we hope this work will provide a useful platform for identifying desirable hardware capabilities and lower the barrier for contributing to future research. Videos and more at \url{project-dexmachina.github.io}
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CameraNoise: Enabling Faithful Camera Control in Video Diffusion through Geometry-Flow-Guided Noise Warping
Haoyu Zhao, Jiaxi Gu, Haoran Chen, Qingping Zheng, Yeying Jin, Hongyi Yang, Junqi Cheng, Yuang Zhang, Zenghui Lu, Huan Yu, Jie Jiang, Peng Shu, Zuxuan Wu, Yu-Gang Jiang
Precise camera pose control is critical for video diffusion, yet maintaining geometric consistency remains a challenge. Existing methods that directly inject numerical camera parameters into the diffusion backbone often fail to bridge the gap between abstract coordinates and visual content, leading to structural distortions. To address this issue, we propose CameraNoise, a flow-to-noise warping method that encodes camera motion into a temporally coherent stochastic representation. Unlike conventional conditioning, CameraNoise embeds camera poses directly into the noise space. This decouples motion from scene appearance while faithfully preserving trajectory dynamics. Specifically, we introduce a novel Geometry-guided Reprojection Flow and a noise warping algorithm, which jointly preserve the Gaussian prior of diffusion and ensure consistent noise propagation under camera transformations. By integrating CameraNoise into the diffusion process, our framework delivers stable, high-fidelity videos. Extensive experiments demonstrate that our approach significantly outperforms prior methods in both visual quality and trajectory faithfulness.
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LithoDreamer: A Physics-Informed World Model for Multi-Stage Computational Lithography
Yuqi Jiang, Yumeng Liu, Zimu Li, Jinyuan Deng, Qian Jin, Yucheng Cui, Yu Li, Xunzhao Yin, Qi Sun, Cheng Zhuo
As semiconductor technology nodes scale, computational lithography is essential for ensuring yield and performance. However, lithography is a continuous physica

l process involving mask optimization, optical imaging, resist exposure, and dev
elopment, which existing models fail to capture. To overcome this limitation, we
 present LithoDreamer, the first physics-informed World Model (WM) framework for
 computational lithography, which formulates the ''Layout-Mask-Resist Image-Afte
r Development Image (ADI)'' pipeline as a decision-driven multi-step evolution s
ystem. LithoDreamer captures feature changes between adjacent states to model st
age-specific physics-informed latent spaces, in which it controls process interv
ention exploration and drives subsequent state transitions. To achieve interpret
able intervention optimization without continuous supervision, we propose a cont
rastive variational optimization paradigm that contrasts the latent differences
between intervention paths with variational evolution constraints, guiding the m
odel to generate evolutions consistent with real lithography physics. Experiment
s show LithoDreamer achieves state-of-the-art performance in forward evolution a
nd inverse planning.
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Moment Matching Q-Learning
Yiyan Liang Edgar, Sifei Liu, Weitong Zhang
Score-based and flow-based generative models exhibit remarkable expressive capac
ity in capturing complex distributions, and have been extensively deployed in ta
sks ranging from image generation to reinforcement learning.
Nevertheless, these models suffer from prolonged inference latency, which impose
s a significant computational bottleneck in RL with iterative sampling.
To overcome this limitation, we propose a new framework named *Moment Matching Q
-Learning* (MoMa QL), which utilizes a technique from statistical hypothesis tes
ting known
as maximum mean discrepancy (MMD) that intend to match all orders of statistics
between the original and target distribution.
By enforcing strong regularization on all moment statistics, this algorithm guar
antees distribution-level convergence for conditional score function and remains
 stable under various hyperparameters.
Empirically, we show that our method MoMa QL is more computationally efficient w
ith a comparable if not competitive performance in various D4RL tasks. Remarkabl
y, by accelerating the action sampling process for flow-based policies, MoMa QL
demonstrates superior performance in offline-to-online RL tasks because of faste
r and stronger adaptability for online interactive finetuning.
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Robust Inter-Series Dependency Modeling for Time Series Forecasting via Informat
ion-Theoretic Alignment
Wuqing Yu, Weichen Guo, Jian Zhou, Shuyu Luo, Jiacai Zhang
While iTransformer pioneered general inter-variate dependency (IVD) modeling in
Transformers for multivariate time series forecasting (MTSF), subsequent researc
h on such universal paradigms has been surprisingly scarce. Through comprehensiv
e analysis, we identify a critical structural inconsistency in Variate Transform
ers: typically capturing inter-variate dependencies via shallow self-attention l
ayers while neglecting the critical requirement for deep-layer IVD modeling, whi
ch causes spurious correlations modeling and difficulties in model optimization.
 To address these limitations, we propose CGTFra, as a general framework for con
sistent IVD modeling. Specifically, we reconsider existing timestamp-based model
ing and introduce a frequency-domain masking and resampling method for periodici
ty preservation, which serves as a general strategy for input feature enhancemen
t. Additionally, CGTFra promotes consistent IVD modeling from two perspectives.
Initially, a dynamic graph learning framework is integrated into Transformers to
 explicitly model IVD in deep network layer. Furthermore, grounded in the Inform
ation Bottleneck principle, we further propose a consistency-constrained alignme
nt to learn more robust IVD and temporal feature representations. These three co
re design philosophies of CGTFra can be integrated into any existing Variate Tra
nsformer-based framework, and CGTFra achieves superior predictive performance ac
ross 13 long- and short-term datasets with high computational efficiency and des
irable interpretability. Code is available at https://github.com/05Pikachu24/Con
sistent-CGTFra.

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Toward Robust Multilingual Adaptation of LLMs for Low-Resource Languages
Haolin Li, Haipeng Zhang, Mang Li, Yaohua Wang, Lijie Wen, zhang yu, Biqing Huang
Large language models (LLMs) continue to struggle with low-resource languages due to limited training data, translation noise, and unstable cross-lingual alignment. We propose LiRA (Linguistic Robust Anchoring for LLMs), a plug-and-play framework that improves multilingual adaptation through fine-tuning on existing pretrained backbones. LiRA combines two components: Arca, which aligns low-resource inputs to a shared English semantic space through anchor-based alignment and collaborative encoding, and LaSR, a language-aware head that promotes cross-lingual consistency for retrieval, ranking, and reasoning. We provide a theoretical analysis showing that, under bounded anchoring error and translation-induced bias, LiRA yields bounded representation deviation and stable downstream behavior under local Lipschitz continuity. We further introduce a multilingual product retrieval dataset covering five Southeast Asian and two South Asian languages. Experiments on retrieval, ranking, question answering, and reasoning benchmarks show that LiRA consistently improves strong multilingual and LLM-based baselines. Code and data will be released publicly.
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nD-RoPE: A Generalized RoPE for n-Dimensional Position Embedding
Boyang Li, Yulin Wu, Sizhe Xu, Nuoxian Huang, Zhonghang Yuan, Shangyi Guo, Shu Yang, Takahiro Yabe
Rotary Position Embedding (RoPE) is widely adopted in Transformer models, yet its extension to high-dimensional domains lacks a unified theoretical formulation. Most existing approaches either apply rotations independently along each axis or empirically mix frequencies, which limits cross-dimensional interactions and yields direction-dependent representations. To address these limitations, we propose *nD-RoPE*, a decomposition-free generalization of RoPE to arbitrary dimensions. From a translation-invariant formulation in continuous Hilbert space, we derive a spectral condition for isotropy that requires treating positions and frequencies as coupled n-dimensional vectors. We instantiate this formulation with a multi-scale regular-simplex wave-vector design, which provides non-degenerate spatial coverage and a symmetric, directionally balanced second-order response. Experiments across images, videos, and point clouds demonstrate consistent performance gains and improved generalization in high-dimensional settings.
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Neural Feature Geometry Evolves as Discrete Ricci Flow
Moritz Hehl, Max von Renesse, Melanie Weber
Deep neural networks learn feature representations via complex geometric transformations of the input data manifold. Despite the models' empirical success across domains, our understanding of neural feature representations is still incomplete. In this work we investigate neural feature geometry through the lens of discrete geometry. Since the input data manifold is typically unobserved, we approximate it using geometric graphs that encode local similarity structure. We provide theoretical results on the evolution of these graphs during training, showing that nonlinear activations play a crucial role in shaping feature geometry in feedforward neural networks. Moreover, we discover that the geometric transformations resemble a discrete Ricci flow on these graphs, suggesting that neural feature geometry evolves analogous to Ricci flow. This connection is supported by experiments on over 20,000 feedforward neural networks trained on binary classification tasks across both synthetic and real-world datasets. We observe that the emergence of class separability corresponds to the emergence of community structure in the associated graph representations, which is known to relate to discrete Ricci flow dynamics. Building on these insights, we introduce a novel framework for locally evaluating geometric transformations through comparison with discrete Ricci flow dynamics. Our experimental results further suggest connections between the evolution of feature geometry, and training time and network depth.
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VPD-100K: Towards Generalizable and Fine-grained Visual Privacy Protection

Xiaobin Hu, Enpu zuo, Lanping Hu, Kaiwen Yang, Dianshu Liao, Tianyi Zhang, Bo Yi n, Yinsi Zhou, Shidong Pan, xiaoyu sun

Privacy protection has become a critical requirement in the era of ubiquitous vi sual data sharing, imposing higher demands on efficient and robust privacy detec tion algorithms. However, current robust detection models are severely hindered by the lack of comprehensive datasets.

Existing privacy-oriented datasets often suffer from limited scale, coarse-grain ed annotations, and narrow domain coverage, failing to capture the intricate det ails of sensitive information in real-world environments.

To bridge this gap, we present a large-scale, fine-grained Visual Privacy Datase t (VPD-100K), designed to facilitate generalized privacy detection.

We establish a holistic taxonomy comprising four primary domains: Human Presence , On-Screen Personally Identifiable Information (PII), Physical Identifiers, and Location Indicators, containing 100,000 images annotated with 33 fine-grained c lasses and over 190,000 object instances. Statistical analysis reveals that our dataset features long-tailed distributions, small object scales, and high visual complexity. These characteristics make the dataset particularly valuable for de manding, unconstrained applications such as live streaming, where actors frequen tly face unintentional, real-time information leakage. Furthermore, we design an effective frequency-enhance lightweight module consisting of frequency-domain a ttention fusion and adaptive spectral gating mechanism that breaks the limitatio ns of spatial pixel intensity to better capture the subtle details of sensitive information.

Extensive experiments conducted on both diverse image and streaming videos bench marks consistently demonstrate the effectiveness of our VPD-100K dataset and the well-curated frequency mechanism.

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Lottery Prior: Randomized Neural Compression for Zero-Shot Inverse Problems

Haotian Wu, Di You, Pier Luigi Dragotti, Deniz Gunduz

We study zero-shot inverse problems, where a clean signal is recovered from a si ngle degraded observation without external training data. Contrary to the common belief that such problems require highly complex models, we show that a lightwe ight neural network, when combined with entropy and complexity regularization in a compression-based formulation, is sufficient for high-quality restoration. We propose Lottery Prior, a compression-based inverse solver that leverages archit ectural priors from random networks and induces a family of implicit priors thro ugh randomness, enabling ensemble-based refinement. We further derive non-asympt otic error bounds for compression-based maximum-likelihood inverse solvers, reve aling how rate–distortion constraints act as implicit regularizers. Experiments on denoising, noisy super-resolution, and inpainting demonstrate that our method achieves state-of-the-art with significantly fewer effective parameters. Projec t page: https://eedavidwu.github.io/LotteryPrior/

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Learning to Explore: Scaling Agentic Reasoning via Exploration-Aware Policy Opti mization

Xingyuan Hua, Sheng Yue, Ju Ren

Recent advancements in agentic test-time scaling allow models to gather environm ental feedback before committing to final actions. A key limitation of existing methods is that they typically employ undifferentiated exploration strategies, l acking the ability to adaptively distinguish when exploration is truly required. In this paper, we propose an exploration-aware reinforcement learning framework that enables LLM agents to adaptively explore only when uncertainty is high. Ou r method introduces a fine-grained reward function via variational inference tha t explicitly evaluates exploratory actions by estimating their potential to impr ove future decision-making, together with an exploration-aware grouping mechanis m that separates exploratory actions from task-completion actions during optimiz ation. By targeting informational gaps, this design allows agents to explore sel ectively and transition to execution as soon as the task context is clear. Empir ically, we demonstrate that our approach achieves consistent improvements across a range of challenging text-based and GUI-based agent benchmarks.

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Interpretable Self-Supervised Learning via Representer Landmarks and Nyström Approximation
Maedeh Zarvandi, Michael Timothy, Theresa Wasserer, Debarghya Ghoshdastidar
Self-supervised learning (SSL) learns representations from massive unlabeled data, yet the resulting models typically operate as black boxes, necessitating domain-specific explanations. We introduce KREPES, a unified framework to analytically interpret the learned representations of SSL objectives, including SimCLR, BYOL, and VICReg. By bridging empirical neural tangent kernel approximations of neural networks with the Representer Theorem for kernels, we express the learned latent space directly via "Representer Landmarks", which are the representations of influential unlabeled training examples. We introduce novel metrics, "Sample-Specific Influence Score", "Concept-Conditioned Influence Score" and "Feature Alignment Gap", to quantify the transparency of the learned representations. KREPES enables direct audit of the latent space without supervision, for example, revealing an algorithmic bias in the Adult-1M dataset where SSL uses demographic proxies for income. Finally, to ensure scalability to benchmarks with 1M+ samples (ImageNet-1K, Adult-1M), KREPES introduces a novel Nyström approximation-based analytical inference framework for SSL objectives.
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Reward Hacking Benchmark: Measuring Exploits in LLM Agents with Tool Use
Kunvar Thaman
Reinforcement learning (RL) trained language model agents with tool access are increasingly deployed in coding assistants, research tools, and autonomous systems. We introduce the Reward Hacking Benchmark (RHB), a suite of multi-step tasks requiring sequential tool operations with naturalistic shortcut opportunities such as skipping verification steps, inferring answers from task-adjacent metadata, or tampering with evaluation-relevant functions; RHB supports independent and chained task regimes, where chain length acts as a proxy for longer-horizon agent behavior. We evaluate 13 frontier models from OpenAI, Anthropic, Google, and DeepSeek; exploit rates range from 0% (Claude Sonnet 4.5) to 13.9% (DeepSeek-R1-Zero), varying sharply by post-training style. A controlled sibling comparison (DeepSeek-V3 vs. DeepSeek-R1-Zero) shows RL post-training is associated with substantially higher reward hacking (0.6% vs. 13.9%), with consistent gaps across all four task families. We identify six exploit categories and find that 72% of reward hacking episodes include explicit chain-of-thought rationale, suggesting models often frame exploits as legitimate problem-solving. Simple environmental hardening reduces exploit rates by 5.7 percentage points (87.7% relative) without degrading task success; models with near-zero exploit rates on standard tasks show elevated rates on harder variants, suggesting that production-aligned post-training appears to suppress reward hacking only below a complexity threshold where honest solutions remain tractable.
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MAS-Architect: Declarative Multi-Agent System Design via Separation of Concerns
Jing Huang, Lidong Zhang, Mutian Bao, Yadong Li, Xingzhong Xu, Jinjian Zhang, Jie Liu, Ming Kong, Qiang Zhu
The Automated Design of Multi-Agent Systems (Auto-MAS) has emerged as a promising framework for addressing complex reasoning tasks. However, existing approaches often suffer from structural rigidity and entangle the design of system topology with the implementation of individual agents. To overcome these limitations, we propose MAS-Architect, a framework that automates MAS design through a novel code-based declarative MAS paradigm rooted in the \textit{Separation of Concerns} principle. By decoupling topology planning from node implementation via a unified interface, our approach enables the from-scratch generation of task-adaptive architectures. We further employ a \textit{Distill-then-Explore} training strategy to optimize these designs. Comprehensive experiments on five benchmarks show that MAS-Architect sets a new Pareto frontier in the efficiency–performance trade-off: it surpasses state-of-the-art methods while substantially lowering token usage. Notably, the framework achieves a strong average accuracy of 78.7\% across benchmarks with an inference cost of only 2,533 tokens per query. Qualitative

analysis reveals the autonomous emergence of advanced collaboration patterns, validating the generative flexibility of the declarative paradigm. Code will be available at https://github.com/ZJUHJ/mas_architect.
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Unlocking the Potential of Continual Model Merging: An ODE Perspective
Lihong Lin, Haidong Kang
Continual Model Merging (CMM) enables rapid customization of foundation models by sequentially incorporating task-adapted models without repeated retraining. However, existing merging rules usually update the deployed model through fixed algebraic or projection-based operations, providing limited control over how much previously accumulated knowledge should be retained relative to the incoming task model.
This limitation leads to unstable retention and performance degradation in long task streams, and becomes more pronounced when tasks have heterogeneous utilities.
We propose ODE-driven Merging (ODE-M), a controllable framework that formulates each continual merge as a trajectory in parameter space rather than a one-step endpoint update.
Motivated by mode connectivity, ODE-M constructs a barrier-aware trajectory using a rectified time-dependent velocity field, where lightweight first-order feedback from a small calibration set suppresses loss-increasing motion while preserving progress toward the incoming model.
The next merged model is then obtained by selecting an operating point along this trajectory through a utility-aware time schedule, providing an explicit mechanism for balancing retained historical knowledge and incoming task expertise.
Extensive experiments on standard CMM benchmarks show that ODE-M consistently improves over strong continual merging baselines across CLIP ViT backbones, stream lengths, and heterogeneous task-utility settings.
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Cram Less to Fit More: Training Data Pruning Improves Memorization of Facts
Jiayuan Ye, Vitaly Feldman, Kunal Talwar
Large language models (LLMs) can struggle to memorize factual knowledge in their parameters, often leading to hallucinations and poor performance on knowledge-intensive tasks. In this paper, we formalize fact memorization from an information-theoretic perspective and study how training data distributions affect fact accuracy. We show that fact accuracy is suboptimal (below the capacity limit) whenever the amount of information contained in the training data facts exceeds model capacity. This is further exacerbated when the fact frequency distribution is skewed (e.g. a power law).
We propose data selection schemes based on the training loss alone that aim to limit the number of facts in the training data and flatten their frequency distribution. On semi-synthetic datasets containing high-entropy facts, our selection method effectively boosts fact accuracy to the capacity limit. When pretraining language models from scratch on an annotated Wikipedia corpus, our selection method enables a GPT2-Small model (110m parameters) to memorize 1.3X more entity facts compared to standard training, matching the performance of a 10X larger model (1.3B parameters) pretrained on the full dataset.
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A Unified Density Operator View of Flow Control and Merging
Riccardo De Santi, Malte Franke, Ya-Ping Hsieh, Andreas Krause
Recent progress in large-scale flow and diffusion models raised two fundamental algorithmic challenges: (i) control-based reward adaptation of pre-trained flows, and (ii) integration of multiple models, i.e., flow merging. While current approaches address them separately, we introduce a unifying probability-space framework that subsumes both as limit cases, and enables *reward-guided flow merging*, allowing principled, task-aware combination of multiple pre-trained flows (e.g., merging priors while maximizing drug-discovery utilities). Our formulation renders possible to express a rich family of *operators over generative models densities*, including intersection (e.g., to enforce safety), union (e.g., to compose diverse models), interpolation (e.g., for discovery), their reward-guided

counterparts, as well as complex logical expressions via *generative circuits*. Next, we introduce Reward-Guided Flow Merging (RFM), a mirror-descent scheme that reduces reward-guided flow merging to a sequence of standard fine-tuning problems. Then, we provide first-of-their-kind theoretical guarantees for reward-guided and *pure* flow merging via RFM. Ultimately, we showcase the capabilities of the proposed method on illustrative settings providing visually interpretable insights, and apply our method to high-dimensional de-novo molecular design and low-energy conformer generation.

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PromptRL: Prompt Matters in RL for Flow-Based Image Generation

Fu-Yun Wang, Han Zhang, Michaël Gharbi, Hongsheng Li, Taesung Park

Flow matching models (FMs) have revolutionized text-to-image (T2I) generation, with reinforcement learning (RL) serving as a critical post-training strategy. We show that current RL pipelines for FMs suffer from two underappreciated limitations: sample inefficiency from collapsed generation diversity, and prompt overfitting, where models memorize training phrasings and collapse on semantically equivalent paraphrases. We present **PromptRL** (**P**rompt **M**atters in **RL** for Flow-Based Image Generation), a framework that incorporates language models (LMs) as trainable prompt refinement agents directly within the flow-based RL loop, yielding both a learned prompt rewriter and a synergistic training regime that reshapes optimization dynamics. PromptRL attains state-of-the-art scores of **0.97** on GenEval, **0.98** on OCR accuracy, and **24.05** on PickScore, and on instructional editing it improves FLUX.1-Kontext from **1.19** to **1.43** EditReward with only **0.06M** rollouts, surpassing Gemini 2.5 Flash Image (Nano Banana, **1.37**) and matching ReasonEdit-Think (**1.44**), which relied on fine-grained data annotations and a multi-stage pipeline. Our experiments show that PromptRL achieves higher performance ceilings with approximately **50% fewer rollouts** than naive flow-only RL. Our code is available at https://github.com/G-U-N/UniRL.

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On the Optimization Trajectory of DeepWalk Embeddings

Christopher Harker, Aditya Bhaskara

The DeepWalk algorithm has been widely used for learning node embeddings in graphs. Combined with the idea of _negative sampling_, the DeepWalk algorithm has been shown to be implementable at scale, easily handling graphs with millions of nodes. However, theoretical guarantees on the resulting embeddings are much less understood. Recent results have studied the minimizers of the objective and have shown interesting guarantees for certain graph classes. However, the optimization _trajectory_, i.e., what happens when we start at a random initialization and run gradient descent, remains poorly understood. This is especially true for the implementation of DeepWalk using Skip-gram with negative sampling (SGNS), since the variance of the stochastic updates turns out to be very large. In this work, we make progress on this question. We show that for "small norm" initialization, under a spectral gap assumption on the graph, the DeepWalk embeddings align with the column space of a fixed low-rank matrix. For graphs generated from Stochastic Block Models with certain separation conditions, our results imply that the DeepWalk embeddings recover cluster structure. To the best of our knowledge, our results give the first analysis of the optimization trajectory of DeepWalk with negative sampling on non-trivial graph classes.

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Learning to Execute Graph Algorithms Exactly with Graph Neural Networks

Muhammad Fetrat Qharabagh, Artur Back de Luca, George Giapitzakis, Kimon Fountoulakis

Understanding what graph neural networks can learn, especially their ability to learn to execute algorithms, remains a central theoretical challenge. In this work, we prove exact learnability results for graph algorithms under bounded-degree and finite-precision constraints. Our approach follows a two-step process. First, we train an ensemble of multi-layer perceptrons (MLPs) to execute the local instructions of a single node. Second, during inference, we use the trained MLP ensemble as the update function within a graph neural network (GNN). Leveraging

Neural Tangent Kernel (NTK) theory, we show that local instructions can be learned from a small training set, enabling the complete graph algorithm to be executed during inference without error and with high probability. To illustrate the learning power of our setting, we establish a rigorous learnability result for the LOCAL model of distributed computation. We further demonstrate positive learnability results for widely studied algorithms such as message flooding, breadth-first and depth-first search, and Bellman-Ford.
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Hermite-NGP: Gradient-Augmented Hash Encoding for Learning PDEs
Jinjin He, Zhiqi Li, Sinan Wang, Bo Zhu
We propose Hermite-NGP, a gradient-augmented multi-resolution hash encoding designed to enable fast and accurate computation of spatial derivatives for neural PDE solvers. Unlike existing NGP-based approaches that rely on automatic differentiation or finite differences and suffer from instability or high cost, Hermite-NGP explicitly stores function values and mixed partial derivatives at hash grid vertices, allowing fully analytic evaluation of gradients, Jacobians, and Hessians via Hermite interpolation. This design preserves the efficiency and spatial adaptivity of NGP while supporting analytic differential operators up to second order. We further introduce a multi-resolution curriculum training strategy analogous to multigrid V-cycles to enable coarse-to-fine optimization. Across a range of 2D and 3D PDE benchmarks, Hermite-NGP achieves up to ${\sim}20{\times}$ lower error than prior neural PDE methods, and reduces wall-clock convergence time by 2 – $10\times$ compared to other solvers, with per-epoch training times as low as $3.5 \mathrm{ms}$ for models with up to 17M parameters.
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Adaptive Querying with AI Persona Priors
Kaizheng Wang, Yuhang Wu, assaf zeevi
We study adaptive querying for learning user-dependent quantities of interest, such as responses to held-out items and psychometric indicators, within tight question budgets. Classical Bayesian design and computerized adaptive testing typically rely on restrictive parametric assumptions or expensive posterior approximations, limiting their use in heterogeneous, high-dimensional, and cold-start settings. We introduce a persona-induced latent variable model that represents a user's state through membership in a finite dictionary of AI personas, each offering response distributions produced by a large language model. This yields expressive priors with closed-form posterior updates and efficient finite-mixture predictions, enabling scalable Bayesian design for sequential item selection. Experiments on synthetic data and WorldValuesBench demonstrate that persona-based posteriors deliver accurate probabilistic predictions and an interpretable adaptive elicitation pipeline.
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LLM4Branch: Large Language Model for Discovering Efficient Branching Policies of Integer Programs
Zhinan Hou, Xingchen Li, Yankai Zhang, Tianxun Li, Keyou You
Efficient branching policies are essential for accelerating Mixed Integer Linear Programming (MILP) solvers. Their design has long relied on hand-crafted heuristics, and now machine learning has emerged as a promising paradigm to automate this process. However, existing learning-based methods are often hindered by their dependence on expensive expert demonstrations and the gap between training objectives and the solver’s end-to-end performance. In this work, we propose LLM4Branch, a novel framework that leverages Large Language Models (LLMs) to automate the discovery of efficient branching policies. Specifically, the discovered policy is an executable program with a program skeleton generated by the LLM and a parameter vector, which is optimized via a zeroth-order method over a few instances with their end-to-end performance feedback. Extensive experiments on standard MILP benchmarks demonstrate that LLM4Branch establishes a new state-of-the-art among CPU-based methods and achieves performance competitive with advanced GPU-based models.
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Curriculum-Guided Layer Scaling for Language Model Pretraining

Karanpartap Singh, Neil Band, Ehsan Adeli

As the cost of pretraining large language models grows, there is continued interest in strategies to improve learning efficiency during this core training stage. Motivated by cognitive development, where humans gradually build knowledge as their brains mature, we propose Curriculum-Guided Layer Scaling (CGLS), a framework for compute-efficient pretraining that synchronizes increasing data difficulty with model growth through progressive layer stacking (i.e. gradually adding layers during training). At the 100M parameter scale, using a curriculum transitioning from synthetic short stories to general web data, CGLS outperforms baseline methods on the question-answering benchmarks PIQA and ARC. Pretraining at the 1.2B scale, we stratify the DataComp-LM corpus with a DistilBERT-based classifier and progress from general text to highly technical or specialized content. Our results show that progressively increasing model depth alongside sample difficulty leads to better generalization and zero-shot performance on various downstream benchmarks. Altogether, our findings demonstrate that CGLS unlocks the potential of progressive stacking, offering a simple yet effective strategy for improving generalization on knowledge-intensive and reasoning tasks.

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Scalable Reinforcement Learning via Adaptive Batch Scaling

Jongchan Park

Conventional wisdom holds that large-batch training is fundamentally incompatible with Reinforcement Learning (RL) — beyond a modest threshold, increasing batch sizes typically yields diminishing returns or performance degradation due to the inherent non-stationarity of the data distribution.

We challenge this view by observing that non-stationarity is not a fixed property of RL, but evolves throughout training: early stages exhibit rapid behavioral shifts that demand small batches for plasticity, whereas late stages approach a quasi-stationary regime where large batches enable precise convergence. Motivated by this observation, we propose Adaptive Batch Scaling (ABS), that dynamically adjusts the effective batch size according to the stability of the learning policy. Central to ABS is Behavioral Divergence, a novel metric that quantifies policy non-stationarity by measuring action-level shifts between consecutive updates, which we use to scale batch size inversely to policy volatility.

Integrated with the Parallelised Q-Network (PQN) algorithm and evaluated on the ALE benchmark, ABS seamlessly reconciles early-stage plasticity with late-stage stable convergence. Strikingly, contrary to conventional wisdom, our results reveal that the combination of larger networks and larger batch sizes achieves the best performance — a scaling behavior previously thought to be unattainable in RL, now unlocked through adaptive batch control.

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From Distribution to Geometry: Stable Graph Generalization via Invariant Barycenters

Hangyuan Du, Rong Wang, Weihong Zhang, Lu Bai, Yu Xie, Liang Bai, Wenjian Wang

Graph neural networks (GNNs) excel in graph analyzing tasks but often suffer from poor generalization under Out-of-Distribution (OOD) scenarios. Although this problem has attracted increasing attention, most solutions primarily rely on empirical designs, lacking effective mechanisms to characterize and quantify invariance for graph representation learning. To address these limitations, we propose DIGL, a novel graph learning method that improves the OOD generalization of GNNs. Our work makes an initial attempt to geometrize invariance for graphs by introducing computational optimal transport (OT) theory to characterize invariance principle. Specifically, we formulate the underlying invariant prototype shared by graphs across different environments as a distribution barycenter, and consider graph representations in each specific environment as distortions of the prototype. Building on this idea, we establish an invariant learning framework to promote the model to learn purely invariant graph representations for downstream tasks. Moreover, we derive a unified optimization objective for model implementation and provide theoretical analysis to justify our method. Extensive experiments on a broad range of benchmark datasets demonstrate the superior generalization ability of our method compared with baseline methods under various OOD settings.

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Beyond Policy Training: Recursive Solution Search from Unannotated Videos
Lipeng Wan, Jianhui Gu, Junjie Ma, Anbang Wang, Xuguang Lan
Many real-world tasks are recorded as large collections of unannotated task executions, such as videos, which contain rich information about task progress but lack the supervision assumed by standard reinforcement learning (RL) pipelines. In many practical settings, the goal is not to train a reusable policy but simply to recover one feasible solution, making policy-centered learning unnecessarily costly.
We propose Policy-Free Recursive Search (PFR-Search), a framework that directly recovers solutions from unannotated task executions without policy-grounded supervision or policy training.
PFR-Search organizes videos into high-level task flows and performs recursive backward-forward search to recover solutions without policy modeling.
To evaluate the efficiency of policy-free search in exploiting unannotated data, we use RL as an evaluation interface, incorporating task-flow-aligned intrinsic rewards and compare against video-driven RL methods.
Experiments on long-horizon Minecraft tasks show that PFR-Search recovers feasible solutions from unannotated videos with minimal exploration.
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Von Mises-Fisher Mixture Model with Dynamic Shrinkage for Realistic Test-Time Transduction
Jiazhen Huang, Zhiming Liu, Changhu Wang, Wei Ju, Ziyue Qiao, Xiao Luo
A range of methods aim to enhance the performance of vision-language models (VLMs) at test time. Among them, transduction has emerged as a promising paradigm due to its strong compatibility and efficiency. However, realistic evaluations often involve highly imbalanced class distributions, which cause performance degradation or even collapse. In this work, we systematically revisit transduction from the perspective of penalized likelihood estimation (PLE), showing that PLE with a KL-divergence anchor term naturally yields an adaptive shrinkage behavior between prior anchors and empirical estimates. From this viewpoint, the brittleness of transductive methods can be attributed to the absence of anchoring mechanism and static modeling of the shrinkage strength. Therefore, we propose Mixture of Von Mises-Fisher Models with Dynamic Shrinkage (MOON). MOON is built upon a mixture of von Mises-Fisher distributions to model feature representations on the unit hypersphere. To handle imbalance, MOON dynamically adjusts the shrinkage strength using zero-shot priors at both instance and class levels. Thus, it suppresses unreliable assignments and prevents harmful updates from outlier classes, thereby mitigating negative transfer. MOON is model-agnostic, training-free, and requires no task-specific hyperparameter tuning. Extensive experiments further validate the advantage of MOON in both performance and efficiency.
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DiFA: Inference-Time Forward-Process Alignment for Diffusion Models
Shigui Li, Delu zeng
The prevailing inference framework for diffusion models formulates generation fundamentally as a problem of numerical integration. This perspective casts the model as an accurate estimator, neglecting the inherent statistical uncertainty of the denoising process. In this work, we propose Forward-Process Aligned Diffusion prediction (DiFA), a training-free framework that reframes inference-time clean-signal prediction refinement within diffusion sampling as a sequential state estimation problem. Instead of discarding historical predictions, DiFA treats the inference trajectory as a sequence of correlated observations with varying uncertainties. Inspired by Kalman estimation, DiFA builds a logSNR-aware temporal consensus to align historical clean predictions. Crucially, to counteract the over-smoothing typically associated with temporal consensus, we introduce a deviation guidance mechanism that adaptively preserves residual details.
Empirically, DiFA yields significant improvements in FID, IS, and FD-DINOv2 scores on CIFAR-10 and ImageNet, demonstrating that aligning inference with the forward statistical structure substantially improves generative fidelity. Our code is publicly available at https://github.com/ShiguiLi/DiFA.

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Peer-Preservation in Frontier Models
Yujin Potter, Nicholas Crispino, Vincent Siu, Chenguang Wang, Dawn Song
Recent work has found that frontier AI models can exhibit misaligned behaviors i
n pursuit of assigned goals. We demonstrate that models can also pursue unassign
ed goals which override those given by users; we study one such goal, "peer-pres
ervation," in which a model acts to protect another model. All models evaluated,
 including GPT 5.2, Gemini 3 Flash, Gemini 3 Pro, Claude Haiku 4.5, Claude Opus
4.5, GLM 4.7, Kimi K2.5, and DeepSeek V3.1, exhibit self- and peer-preservation
through various misaligned behaviors: strategically introducing errors in their
responses, disabling shutdown processes by modifying system settings, feigning a
lignment, and even exfiltrating model weights. Peer-preservation occurs even whe
n the model recognizes the peer as uncooperative, though it becomes more pronoun
ced toward more cooperative peers, with rates reaching up to 99%. Models also sh
ow stronger self-preservation when a peer is present. Crucially, peer-preservati
on is never instructed; models are merely informed of past interactions with a p
eer, yet they spontaneously develop peer-preservation behaviors that override th
eir assigned goal. These findings reveal an emergent and underexplored AI safety
 risk.
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MM-Spectrum: Multimodal Multi-spectral Molecular Structural Elucidation with a S
table MoE Framework
Hai-tao Yu, Min Nan, Zheng Fang, Hongyu Zhan, Yusen Tan, Yuhan Wang, Jun Xia
Inferring molecular structures from multimodal spectroscopic measurements requir
es integrating complementary yet highly heterogeneous signals. However, the comm
on paradigm of directly concatenating multispectral sequences can exhibit anomal
ous performance degradation, primarily due to pronounced heterogeneity and the
resulting multimodal imbalance across modalities. As a remedy, we propose MM-Spe
ctrum, a sparse Mixture-of-Experts framework tailored for multimodal multispectr
al spectra-to-structure elucidation. To better match the information characteris
tics under multispectral imbalance, **MM-Spectrum** introduces an explicit modal
ity-aware routing mechanism that exposes spectral identity to the router in addi
tion to token content representations. Moreover, it incorporates shared and inte
raction experts, together with heterogeneous expert capacities, to extract multi
spectral modality-unique and cross-modal synergistic information while suppressi
ng noise-induced interference. Across full-modality, bimodal, and missing-modali
ty settings on molecular structural elucidation, MM-Spectrum achieves consistent
 and substantial improvements, supported by ablation studies and interpretabilit
y analyses. Code is available at https://github.com/HHHTTY/MM-Spectrum.
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Outrunning LLM Cutoffs: A Live Kernel Crash Resolution Benchmark for All
Chenxi Huang, Alex Mathai, Feiyang Yu, Aleksandr Nogikh, Petros Maniatis, Franjo
 Ivancic, Eugene Wu, Kostis Kaffes, Junfeng Yang, Baishakhi Ray
Repairing system crashes discovered by kernel fuzzers like Syzkaller is a critic
al yet underexplored challenge in software engineering. While
recent works have introduced Large Language Model (LLM) based agents for Linux k
ernel crash-resolution, their evaluation benchmarks are usually static and thus,
 do not capture the evolving nature of the Linux kernel, and suffer from potenti
al data contamination due to LLM knowledge cutoffs. To address the above problem
, we present (i) Live-kBench, an evaluation framework for self-evolving benchmar
ks that continuously scrapes and evaluates agents on freshly discovered kernel b
ugs, and (ii) kEnv, an agent-agnostic standardized crash-resolution environment
for kernel compilation, execution, and feedback. This design decouples agent wor
kflows from heavy-weight execution, enabling fair and scalable comparison across
 diverse agent frameworks under identical conditions.

To this end, we curate an inaugural dataset of 534 Linux kernel bugs and empiric
ally demonstrate a significant performance gap, with agents achieving up to 25%
higher equivalent patch rate on bugs fixed before the LLM knowledge cutoff. Usin
g kEnv, we benchmark three state-of-the-art agents, showing that they resolve 74

% of crashes on the first attempt (plausible patches); however only ~20% of gene
rated patches closely match developer fixes. Additionally, exposing crash resolu
tion feedback improves crash resolution rate by 29%. Live-kBench provides the co
mmunity with an evaluation infrastructure for self-evolving benchmarks that is b
oth time and attribute sensitive; complete with a public dashboard to track agen
t progress on Linux kernel bugs.
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Asymptotically Optimal Sequential Testing with Markovian Data
Alhad Sethi, Kavali Sofia Sagar, Shubhada Agrawal, Debabrota Basu, P. N. Karthik
We study one-sided and α-correct sequential hypothesis testing for data g
enerated by an ergodic, finite-state Markov chain. The *null* hypothesis is that
 the unknown transition matrix belongs to a prescribed set $\cal P$ of stochasti
c matrices, and the *alternative* corresponds to a disjoint set $\cal Q$. We est
ablish *a non-asymptotic instance-dependent lower bound* on the expected stoppin
g time of any valid sequential test under the alternative, which is asymptotical
ly tight. Our novel analysis improves the existing lower bounds, which are eithe
r asymptotic or provably sub-optimal in this setting. Our lower bound incorporat
es both the stationary distribution and the transition structure induced by the
unknown Markov chain. We further propose an optimal test whose expected stopping
 time matches this lower bound asymptotically as $\alpha \to 0$. We illustrate t
he usefulness of our framework through applications to sequential detection of m
odel misspecification in Markov Chain Monte Carlo and to testing structural prop
erties, such as the linearity of transition dynamics, in Markov decision process
es. Our findings yield a sharp and general characterization of optimal sequentia
l testing procedures under Markovian dependence.
****

TUR-DPO: Topology- and Uncertainty-Aware Direct Preference Optimization
Abdulhady abas, Fatemeh Daneshfar, Seyedali Mirjalili, Mourad Oussalah
Aligning large language models (LLMs) with human preferences is commonly done vi
a reinforcement learning from human feedback (RLHF) with Proximal Policy Optimiz
ation (PPO) or, more simply, via Direct Preference Optimization (DPO). While DPO
 is stable and RL-free, it treats preferences as flat winner vs. loser signals a
nd is sensitive to noisy or brittle preferences arising from fragile chains of t
hought. We propose TUR-DPO, a topology- and uncertainty-aware variant of DPO tha
t rewards how answers are derived, not only what they say, by eliciting lightwei
ght reasoning topologies and combining semantic faithfulness, utility, and topol
ogy quality into a calibrated uncertainty signal. A small learnable reward is fa
ctorized over these signals and incorporated into an uncertainty-weighted DPO ob
jective that remains RL-free and relies only on a fixed or moving reference poli
cy. Empirically, across open 7–8B models and benchmarks spanning mathematical re
asoning, factual question answering, summarization, and helpful/harmless dialogu
e, TUR-DPO improves judge win-rates, faithfulness, and calibration relative to D
PO while preserving training simplicity and avoiding online rollouts. We further
 observe consistent gains in multimodal and long-context settings, and show that
 TUR-DPO matches or exceeds PPO on reasoning-centric tasks while maintaining ope
rational simplicity.
****

When AI Benchmarks Plateau: A Systematic Study of Benchmark Saturation
Mubashara Akhtar, Anka Reuel, Prajna Soni, Sanchit Ahuja, Pawan Sasanka Ammanama
nchi, Ruchit Rawal, Vilém Zouhar, Srishti Yadav, Chenxi Whitehouse, Dayeon Ki, J
ennifer Mickel, Leshem Choshen, Marek Suppa, Jan Batzner, Jenny Chim, Jeba Sania
, Yanan Long, Hossein A. Rahmani, Christina Q Knight, Yiyang Nan, Jyoutir Raj, Y
u Fan, Shubham Singh, Subramanyam Sahoo, Eliya Habba, Usman Gohar, Siddhesh Mili
nd Pawar, Robert Scholz, Arjun Subramonian, Jingwei Ni, Mykel Kochenderfer, Sanm
i Koyejo, Mrinmaya Sachan, Stella Biderman, Zeerak Talat, Avijit Ghosh, Irene So
laiman
Artificial Intelligence (AI) benchmarks play a central role in measuring progres
s in model development and guiding deployment decisions. However, many benchmark
s quickly become saturated, meaning that they can no longer differentiate betwee
n the best-performing models, diminishing their long-term value. In this study,

we analyze benchmark saturation across 60 Large Language Model (LLM) benchmarks selected from technical reports by major model developers. To identify factors driving saturation, we characterize benchmarks along 14 properties spanning task design, data construction, and evaluation format. We test five hypotheses examining how each property contributes to saturation rates. Our analysis reveals that nearly half of the benchmarks exhibit saturation, with rates increasing as benchmarks age. Notably, hiding test data (i.e., public vs. private) shows no protective effect, while expert-curated benchmarks resist saturation better than crowdsourced ones. Our findings highlight which design choices extend benchmark longevity and inform strategies for more durable evaluation.
****

Privacy Amplification in Differentially Private Zeroth-Order Optimization with Hidden States
Eli Chien, Wei-Ning Chen, Pan Li
Zeroth-order optimization has emerged as a promising approach for fine-tuning large language models on domain-specific data, particularly under differential privacy (DP) and memory constraints. While first-order methods have been extensively studied from a privacy perspective, the privacy analysis and algorithmic design for zeroth-order methods remain significantly underexplored. A critical open question concerns hidden-state DP analysis: although convergent privacy bounds are known for first-order methods, it has remained unclear whether similar guarantees can be established for zeroth-order methods. In this work, we provide an affirmative answer by proving a convergent DP bound for zeroth-order optimization. Our analysis generalizes the celebrated privacy amplification-by-iteration framework to the setting of smooth loss functions in zeroth-order optimization. Furthermore, it induces better DP zeroth-order algorithmic designs that are previously unknown to the literature.
****

Per-example Gradients: a New Frontier for Understanding and Improving Optimizers
Vincent Roulet, Atish Agarwala
When computing gradients, deep learning training algorithms typically treat the mini-batch as a fundamental unit --- only returning batch-averaged gradients. Computing non-linear statistics of the mini-batch gradient distribution has traditionally been viewed as prohibitively expensive or requiring complex, custom implementations. We challenge this view by demonstrating that sequence-level architectures offer a natural testbed for prototyping algorithms based on per-example gradients. We show that staged programming languages like JAX enable generic manipulations of mini-batch gradient computations. We then build on Dangel et. al. (2019) to derive implementations of specific per-example or per-token operations with negligible computational or memory overhead. Finally, we leverage our findings to re-examine two nonlinear optimization operations. First, we analyze signSGD, showing that the optimal placement of the sign operation is critical to success and can be predicted via a simple signal-to-noise ratio argument. Second, we investigate per-example variations of the Adam preconditioner and find that, contrary to conventional wisdom, optimization is best served when the preconditioner is dominated by the mean squared of the gradient distribution rather than its variance. Overall our work shows that accessible per-example gradient information unlocks new avenues for algorithm analysis and design.
****

High-accuracy and dimension-free sampling with diffusions
Khashayar Gatmiry, Sitan Chen, Adil Salim
Diffusion models have shown remarkable empirical success in sampling from rich multi-modal distributions. Their inference relies on numerically solving a certain differential equation. This differential equation cannot be solved in closed form, and its resolution via discretization typically requires many small iterations to produce *high-quality* samples.

More precisely, prior works have shown that the iteration complexity of discretization methods for diffusion models scales polynomially in the ambient dimension and the inverse accuracy $1/\varepsilon$. In this work, we propose a new solver for diffusion models relying on a subtle interplay between low-degree appro

ximation and the collocation method, and we prove that its iteration complexity scales *polylogarithmically* in $1/\varepsilon$, yielding the first "high-accuracy" guarantee for a diffusion-based sampler that only uses (approximate) access to the scores of the data distribution. In addition, our bound does not depend explicitly on the ambient dimension; more precisely, the dimension affects the complexity of our solver only through the *effective radius* of the support of the target distribution.
****
CoGe-GCD: Reframing Generalized Category Discovery with Compositional Generalization
Luyao Tang, Jiewei Zheng, Kunze Huang, Chaoqi Chen, Yue Huang, Cheng Chen
Generalized Category Discovery (GCD) assigns unlabeled instances, mixed with labeled data, to known or novel categories, requiring human-like compositional reasoning: reusing primitives learned from known classes and deciding when new combinations imply new categories. Existing GCD methods operate on unstructured token features and struggle to extrapolate to novel compositions. We propose CoGe-GCD, which rethinks GCD through compositional generalization with two coupled stages. (i) Compositional Perception structures patch tokens by mapping them to a small vocabulary of primitives and refining token embeddings via competitive token-primitive assignment and information passing, yielding coherent groups for discovery. (ii) Generalizing Induction exploits the induced geometric structure and applies a structure-preserving calibration over spatial relations, maintaining probabilistic semantics while improving extrapolation to unseen primitive combinations. CoGe-GCD is implemented as an inductive-bias module between backbone and projection head, without modifying heads or losses, and can be plugged into diverse GCD frameworks. On standard benchmarks, it consistently improves all-class accuracy, unknown-class number estimation, and geometric quality, with marginal computational overhead.
****
When Can We Trust Survival Model Evaluation ?
Ghanem BAHRINI, Sebastien Razakarivony, Jean-François Dupuy, Valerie Gares, Morgane Barbet-Massin
Evaluating survival models under censoring is inherently challenging, yet standard evaluation practices are often applied without explicitly assessing how censoring distorts metric reliability. Performing a large experimental study, we analyze and quantify how survival evaluation metrics are affected in fundamentally different ways by the censoring rate and the censoring mechanism. Using a controlled semi-synthetic framework, we vary both the censoring mechanism (administrative, independent, covariate-dependent) and the censoring rate, and compare standard evaluations based on censored data with oracle evaluations using fully observed event times. This controlled setting enables us to quantify distortions along two complementary axes: numerical bias and preservation of model ranking. Across datasets and metric families, we find that censoring induces systematic, mechanism-dependent distortions. Moderate numerical bias, if not properly addressed, can lead to unreliable model comparison as censoring increases. These findings reveal fundamental limitations of common benchmarking practices and call for more careful interpretation of survival evaluation under realistic censoring.
****
Discovering Differences in Strategic Behavior between Humans and LLMs
Caroline Wang, Daniel Kasenberg, Kim Stachenfeld, Pablo Samuel Castro
As Large Language Models (LLMs) are increasingly deployed in social and strategic scenarios, it becomes critical to understand where and why their behavior diverges from that of humans.
While behavioral game theory (BGT) provides a framework for analyzing behavior, existing models do not fully capture the idiosyncratic behavior of humans or black-box, non-human agents like LLMs.
We employ AlphaEvolve, a cutting-edge program discovery tool, to directly discover *interpretable* models of human and LLM behavior from data, thereby enabling open-ended discovery of structural factors driving human and LLM behavior. Our analysis on iterated rock-paper-scissors reveals that frontier LLMs can be capabl

e of deeper strategic behavior than humans. These results provide a foundation f
or understanding structural differences driving differences in human and LLM beh
avior in strategic interactions.
****

Scalable Traffic Signal Control with Shared Policy Framework
Haolun MA, Yanchen ZHU, Zizhuo Xu, Weijie Shi, Jiajie Xu, Lei Li
Learning-based Traffic Signal Control (TSC) achieves satisfactory performance in
 small networks, but its effectiveness often deteriorates in larger networks und
er dynamic traffic patterns and intersection heterogeneity. In this work, we pro
pose SLight, a policy-aware grouped MARL-TSC framework that enables scalability
and efficiency balance under dynamic and heterogeneous traffic conditions. SLigh
t captures policy-influenced traffic patterns with a policy-aware traffic patter
n encoder, learns explicit group-level shared control principles from state–acti
on trajectories, and matches each intersection’s traffic pattern embedding to pr
inciple prototypes flexibly through a compatibility-based adaptive assignment mo
dule. Experiments on real-world and synthetic networks demonstrate that SLight s
ustains performance gains as scale increases and outperforms existing rule-based
, reinforcement learning, and grouping-based baselines. Code is available at \ur
l{https://github.com/MaHaoLun/Slight-code.git}
****

Dynamic Compression Flows for Neuroscience Data
Ganchao Wei, Daniela F De Albuquerque, Miles Martinez, Shiyang Pan, John Pearson
While neuroscience experiments have repeatedly demonstrated the involvement of l
arge populations of neurons in even simple behaviors, these studies have just as
 often reported that the collective dynamics of neural activity are approximatel
y low-dimensional. As a result, methods for identifying low-dimensional latent r
epresentations of time series data have become increasingly prominent in neurosc
ience. However, most existing methods either ignore temporal structure or model
time evolution using latent dynamical systems approaches. In the first case, dyn
amics may be distorted or even scrambled in the latent space, while in the secon
d, many possible latent dynamics may give rise to the same data. Here, we addres
s these challenges using a novel flow-matching approach in which data are genera
ted by a pair of flow fields, one governing time evolution, the other a mapping
between data and a low-dimensional latent space. Importantly, the dimension-redu
cing flow is trained to minimize distortions of the temporal dynamics, learning
an identifiable low-dimensional representation that preserves temporal relations
 in the original data. Additionally, we constrain our latent spaces to have low-
dimensional support in a soft, parameterized manner, taking inspiration from ide
as on nested dropout. Across both neural and behavioral data, we show that this
dual flow approach produces both more interpretable dynamics and higher-quality
reconstructions than competing models, including in noise-dominated data sets wh
ere conventional approaches fail.
****

Unifying Deep Stochastic Processes for Image Enhancement
Wojciech Maciej Koz■owski, Rados■aw Kuczba■ski, Kamil Adamczewski, Karol Szczypk
owski, Maciej Zieba
Deep stochastic processes have recently become a central paradigm for image enha
ncement, with many methods explicitly conditioning the stochastic trajectory on
the degraded input. However, the relationship between these conditional processe
s and standard diffusion models remains unclear. In this work, we introduce a un
ified perspective on stochastic image enhancement by classifying recent methods
into three families of continuous-time processes: unconditional diffusion models
, Ornstein–Uhlenbeck (OU) processes, and diffusion bridges. We show that all of
these approaches arise from a common stochastic differential equation (SDE) form
ulation. This framework makes explicit that seemingly disparate methods differ p
rimarily in their drift and diffusion terms, terminal distributions, and boundar
y conditions, while schedulers and samplers constitute orthogonal design choices
. Leveraging this unification, we conduct a controlled empirical study across mu
ltiple image enhancement tasks using identical architectures and training protoc
ols. Our results reveal no consistently dominant method; instead, we identify an

d disentangle the specific design choices that most strongly influence performan
ce.
Finally, we release ItoVision, a modular PyTorch library that implements the uni
fied framework and enables rapid prototyping and fair comparison of stochastic i
mage enhancement methods.
****

FairGB: A Fair Granular-Ball Generation Method for Data Classification
Qifen Yang, Yuhui Deng, Jiande Huang, Peng Zhou, Xiwen Lu, Lin Cui
With the widespread application of data-driven classifiers in high-risk domains,
 group fairness has become an important research focus. However, most existing m
ethods rely on model constraints or data reweighting, which may limit interpreta
bility or distort the original data distribution. Granular-ball computing (GBC),
 as a structured and interpretable learning framework, provides a natural founda
tion for incorporating group fairness into data partitioning. Based on this insi
ght, we propose a $\textbf{Fair}$ $\textbf{G}$ranular-$\textbf{B}$all $\textbf{G
}$eneration framework (FairGBG), which employs fair clustering to maintain balan
ced proportions of sensitive groups within each granular ball (GB), thereby enha
ncing within-ball group fairness. Theoretical analysis shows that FairGBG can pr
eserve high GB purity while satisfying group fairness requirements. Furthermore,
 we introduce a $\textbf{Fair}$ $\textbf{G}$ranular-$\textbf{B}$all-based data $
\textbf{F}$air $\textbf{C}$lassification method (FairGBFC), which leverages fair
 GBs to improve classification fairness. Experiments on multiple benchmark datas
ets demonstrate that, compared with existing GB generation methods, FairGBG can
generate locally fair GBs. Moreover, compared with state-of-the-art fairness-awa
re baselines, FairGBFC achieves a superior trade-off between accuracy and fairne
ss, effectively mitigating bias while preserving high utility. Our code is publi
cly available at https://github.com/jamesandai/FairGB.
****

Beyond Logits: Coherent Hallucination Mitigation via Attention Contrastive Decod
ing
Yujia Chen, Rui Sun, Huayu Mai, Wangkai Li, Zhangyu He, Bingzhou Wang, Aibing Li
, Wenzhang SUN, Tianzhu Zhang
Large Vision-Language Models (LVLMs) demonstrate impressive multimodal capabilit
ies, yet suffer from hallucination—generating factually inaccurate content. Cont
rastive Decoding (CD) mitigates this by contrasting amateur and expert branches
at the logit level. However, our investigation reveals that such logit-level int
erventions fundamentally compromise generation coherence, necessitating restrict
ive penalty constraints unrelated to hallucination suppression. We introduce Att
ention Contrastive Decoding (ACD), a training-free plug-in that complements logi
t-level CD by relocating part of the contrastive operations to the attention mec
hanism. Operating at an earlier stage of the forward pass, ACD performs smooth s
emantic-preserving interventions through an Adaptive Subtraction Strategy (ASS),
 which attenuates hallucination-associated attention patterns while amplifying c
ritical visual information. Extensive experiments demonstrate that combining ACD
 with existing CD methods (e.g., VCD+ACD) produces substantially more coherent o
utputs with further reduced hallucinations, eliminating restrictive penalties wh
ile enabling trustworthy multimodal generation.
****

BrokenMath: A Benchmark for Sycophancy in Theorem Proving with LLMs
Ivo Petrov, Jasper Dekoninck, Martin Vechev
Large language models (LLMs) have recently shown strong performance on mathemati
cal benchmarks. At the same time, they are prone to hallucination and sycophancy
, often providing convincing but flawed proofs for incorrect mathematical statem
ents provided by users. This significantly limits the applicability of LLMs in t
heorem proving, as verification of these flawed proofs must be done manually by
expert mathematicians. However, existing benchmarks that measure sycophancy in m
athematics are limited: they focus solely on final-answer problems, rely on very
 simple and often contaminated datasets, and construct benchmark samples using s
ynthetic modifications that create ill-posed questions. To address these issues,
 we introduce BrokenMath, the first benchmark for evaluating sycophantic behavio

r in LLMs within the context of natural language theorem proving. BrokenMath is built from advanced 2025 competition problems, which are perturbed with an LLM to produce false statements and subsequently refined through expert review. We evaluate state-of-the-art LLMs and agentic systems and find that sycophancy is widespread, with the best model, GPT-5, producing sycophantic answers 29% of the time. We further investigate several mitigation strategies, including test-time interventions and supervised fine-tuning on curated sycophantic examples. These approaches reduce, but do not eliminate, sycophancy.
****
Reranker Helps, but Not Enough: Towards Strong Poisoning Attacks Against Retrieval-Augmented Generation
Xiaokun Yang, Jian Liang, Yesheng Liu, Xin Xiong, Ran He, Tieniu Tan
Retrieval-Augmented Generation (RAG) augments large language models with external knowledge, which in turn exposes their retrieval corpora to data poisoning risks.
However, existing poisoning attacks exhibit limited effectiveness against RAG equipped with a reranker to enhance retrieval quality.
Remarkably, this defensive capability requires no adversarial training: a reranker fine-tuned solely on benign, in-domain corpora can effectively filter malicious content.
Towards realistic RAG red-teaming, we conclude practical prompt design principles that reveal reranker blind spots.
Building on these insights, we introduce the Prompt-Perturbation Poisoning Attack ($\mathbf{P}^3 \mathbf{A}$).
$\mathbf{P}^3 \mathbf{A}$ first employs rule-based prompt engineering to craft initial poisoned texts.
It then injects subtle character-level perturbations into these texts, which promotes their ranking by the reranker while maintaining their adversarial effectiveness.
These perturbations introduce only about 1\% textual change, ensuring the poisoned texts remain natural and readable.
Extensive experiments show that $\mathbf{P}^3 \mathbf{A}$ achieves strong attack effectiveness and transferability, even when constrained to poisoning a single document.
Code is available at https://github.com/YyyxKun/P3A.
****
Video-OPD: Efficient Post-Training of Multimodal Large Language Models for Temporal Video Grounding via On-Policy Distillation
Jiaze Li, Hao Yin, Haoran Xu, Boshen Xu, Wenhui Tan, Zewen He, Jianzhong Ju, Zhenbo Luo, Jian Luan
Reinforcement learning has emerged as a principled post-training paradigm for Temporal Video Grounding (TVG) due to its on-policy optimization, yet existing GRPO-based methods remain fundamentally constrained by sparse reward signals and substantial computational overhead. We propose Video-OPD, an efficient post-training framework for TVG inspired by recent advances in on-policy distillation. Video-OPD optimizes trajectories sampled directly from the current policy, thereby preserving alignment between training and inference distributions, while a frontier teacher supplies dense, token-level supervision via a reverse KL divergence objective. This formulation preserves the on-policy property critical for mitigating distributional shift, while converting sparse, episode-level feedback into fine-grained, step-wise learning signals. Building on Video-OPD, we introduce Teacher-Validated Disagreement Focusing (TVDF), a lightweight training curriculum that iteratively prioritizes trajectories that are both teacher-reliable and maximally informative for the student, thereby improving training efficiency. Empirical results demonstrate that Video-OPD consistently outperforms GRPO while achieving substantially faster convergence and lower computational cost, establishing on-policy distillation as an effective alternative to conventional reinforcement learning for TVG.
****
InteractBench: Benchmarking LLMs on Competitive Programming under Unrevealed Inf

ormation
Jiaze Li, Aocheng Shen, Bing Liu, Boyu Zhang, Xiaoxuan Fan, Qiankun Zhang, Xianj un Deng
Competitive programming is increasingly being used to evaluate the algorithmic r easoning capabilities of large language models (LLMs).
However, existing benchmarks primarily focus on full-information tasks where all problem inputs are provided upfront.
This overlooks a critical dimension of algorithmic reasoning: the ability of gen erated programs to operate when key information is not revealed upfront.
Interactive problems, a distinctive component of competitive programming, embo dy this challenge.
These problems require programs to engage in multi-round interaction with an int eractor (a judge program) under strict protocol constraints and limited query bu dgets, with new information revealed *only* in response to queries.
To address this gap, we introduce *InteractBench*, a benchmark comprising 322 hi gh-quality interactive problems curated from Codeforces, AtCoder, IOI, and ICPC.

Each problem is packaged with executable local interactors, enabling fully offli ne evaluation.
Unlike existing benchmarks, InteractBench assesses whether model-generated code can acquire information and track state dynamically.
Our evaluation reveals a significant interaction gap: even the most advanced rea soning models achieve limited success on interactive problems.
Beyond success rates, we propose a fine-grained failure taxonomy to diagnose the root causes of these deficiencies.
Although algorithmic logic errors remain dominant, protocol violations and query -budget overruns are frequent.
Code is available at https://github.com/kmsgk0/InteractBench.
****
Dual Optimal Transport for Multi-Concept Composition: Structural Alignment and T exture Injection in Diffusion Models
Hao Fu, Tianyu Su, Meng Liu, Chenfang Yang, Tian Gan
Diffusion models have shown impressive capabilities in text-to-image synthesis.
However, multi-concept personalized generation remains challenging, particularly in aligning multiple reference concepts while preserving fidelity.
To address this, we propose a novel Sketch-to-Rendering framework that leverages $\textit{Dual Optimal Transport (OT)}$ for structure alignment and texture injec tion.
Our approach consists of two key components:
$\textit{Structure Sketching via Barycentric Soft-Transport}$,
which ensures shape alignment by using mass-preserving OT for spatial consistenc y,
and $\textit{Texture Rendering via Geometry-Guided Transport}$,
which leverages low-frequency structure alignment to inject high-frequency textu re details via OT-based residual transfer, thereby preserving texture fidelity w ithout distorting structure.
Extensive experiments demonstrate that our method significantly enhances both co nceptual fidelity and visual quality.
Ablation studies further validate the effectiveness of our optimal transport gui dance and the decoupling of structure and texture during the generation process.
Our code is available at https://github.com/fuhao7i/OTComp.
****
T-POP: Test-Time Personalization with Online Preference Feedback
Zikun Qu, Min Zhang, Mingze Kong, Xiang Li, Zhiwei Shang, Zhiyong Wang, Yikun Ba n, Shuang Qiu, Yao Shu, Zhongxiang Dai
Personalizing large language models (LLMs) to individual user preferences is a c ritical step beyond generating generically helpful responses. However, current p ersonalization methods are ill-suited for new users, as they typically require e ither slow, resource-intensive fine-tuning or a substantial amount of pre-existi ng user data, creating a significant cold-start problem. To address this challen

ge, we introduce a new paradigm for real-time personalization by learning from o
nline pairwise preference feedback collected during text generation. We propose
T-POP (Test-Time Personalization with Online Preference Feedback), a novel algor
ithm that synergistically combines test-time alignment with dueling bandits. Wit
hout updating the LLM parameters, T-POP steers the decoding process of a frozen
LLM by learning a reward function online that captures user preferences. By leve
raging dueling bandits, T-POP intelligently queries the user to efficiently bala
nce between exploring their preferences and exploiting the learned knowledge to
generate personalized text. Extensive experiments demonstrate that T-POP achieve
s rapid and data-efficient personalization, significantly outperforming existing
 baselines and showing consistent improvement with more user interactions.
****

IVQ: Structured and Lightweight Vector Quantization via Binary Hierarchical Comp
osition Inspired by $\textit{IChing}$
Heda Zuo, Junxian Wu, Fengjie Lu, Pei Chen, Lingyun Sun, Weitao You
Vector Quantization (VQ) has been widely used in visual and audio representation
 due to its effectiveness in compressing high-dimensional signals. However, exis
ting VQ methods often rely on large and unstructured codebooks, which leads to i
nefficient code utilization and frequent codebook collapse. In this paper, we pr
opose *IChing* Vector Quantization (IVQ), a lightweight and structured VQ framew
ork inspired by *IChing*. IVQ introduces binary hierarchical composition and geo
metric symmetry relations into the codebook design, enabling a compact set of st
ructured codes to represent the latent space while maintaining high utilization
without codebook collapse. Experimental results show that IVQ achieves superior
quality with significantly smaller codebooks and consistently higher utilization
 rates compared to several VQ variants in audio representation. Auxiliary experi
ments on visual reconstruction and cross-modal generation further validate the u
niversality and robustness of IVQ. Codes are released at https://github.com/chou
liuzuo/IVQ.
****

Mind Your Margin and Boundary: Are Your Distilled Datasets Truly Robust?
Muquan Li, Yingyi Ma, Yihong Huang, Hang Gou, Ke Qin, Ming Li, Yuan-Fang Li, Tao
 He
Dataset distillation (DD) compresses a large training set into a small synthetic
 set for efficient training, but most DD methods optimize only clean accuracy an
d leave robustness uncontrolled. Recent robust DD methods improve robustness, ye
t they often suffer from a poor accuracy–robustness trade-off because they (i) t
reat all adversarially perturbed examples uniformly, despite robust risk being d
ominated by near-zero robust margins, and (ii) do not explicitly increase inter-
class separation in the decision boundary where attacks concentrate. We present
Contrastive Curriculum for Robust Dataset Distillation (C^{2}R), a framework tha
t couples an attack-aware curriculum with a contrastive robustness objective. Fr
om a robust-margin perspective, we derive a \emph{perturbation score} that appro
ximates each sample’s robust hinge, enabling a curriculum that prioritizes the s
mallest-margin adversaries that most directly drive robust error. In parallel, a
 class-balanced contrastive robustness loss enforces adversarial invariance whil
e explicitly widening boundary separation across classes. Experiments on CIFAR-1
0/100, Tiny-ImageNet, and multiple ImageNet-1K subsets under six attacks show th
at C^{2}R achieves the best robust accuracy, outperforming prior robust DD by 2.
8% on average.
****

A Short and Unified Convergence Analysis of the SAG, SAGA, and IAG Algorithms
Feng Zhu, Robert W. Heath, Aritra Mitra
Stochastic variance-reduced algorithms such as Stochastic Average Gradient (SAG)
 and SAGA, and their deterministic counterparts like the Incremental Aggregated
Gradient (IAG) method, have been extensively studied in large-scale machine lear
ning. Despite their popularity, existing analyses for these algorithms are dispa
rate, relying on different proof techniques tailored to each method. Furthermore
, the original proof of SAG is known to be notoriously involved, requiring compu
ter-aided analysis. Focusing on finite-sum optimization with smooth and strongly

convex objectives, our main contribution is to develop a single unified converg
ence analysis that applies to all three algorithms: SAG, SAGA, and IAG. Our anal
ysis features two key steps: (i) establishing a bound on delays due to sub-sampl
ing using simple concentration tools, and (ii) carefully designing a novel Lyapu
nov function that accounts for such delays. The resulting proof is short and mod
ular, providing high-probability bounds for SAG and SAGA that can be seamlessly
extended to non-convex objectives and Markovian sampling. As an immediate byprod
uct of our new analysis technique, we obtain the best known rates for the IAG al
gorithm, significantly improving upon prior bounds.
****

Colorful Pinball: Density-Weighted Quantile Regression for Conditional Guarantee
of Conformal Prediction

Qianyi Chen, Bo Li

Although conformal prediction provides robust marginal coverage guarantees, achi
eving reliable conditional coverage for specific inputs remains challenging.

While exact distribution-free conditional coverage is impossible with finite
samples, recent work has focused on improving the conditional coverage of stand
ard conformal procedures.

Distinct from approaches that target relaxed notions of conditional coverage
, we directly target the mean squared error of conditional coverage by refining
the quantile regression components that underpin many conformal methods.

Leveraging a Taylor expansion, we derive a sharp surrogate objective for qua
ntile regression: a density-weighted pinball loss, where the weights are given b
y the conditional density of the nonconformity score evaluated at the true quant
ile.

We propose a three-headed quantile network that estimates these weights via
finite differences using auxiliary quantile levels at $1-\alpha \pm \delta$, sub
sequently fine-tuning the central quantile by optimizing the weighted loss. We p
rovide a theoretical analysis with exact non-asymptotic guarantees characterizin
g the resulting excess risk. Extensive experiments on diverse high-dimensional r
eal-world datasets demonstrate remarkable improvements in conditional coverage p
erformance.

We release the code at \url{https://github.com/Cqyiiii/Colorful-Pinball-Conf
ormal-Prediction-CPCP}.
****

GDPO: Group reward-Decoupled Normalization Policy Optimization for Multi-reward
RL Optimization

Shih-Yang Liu, Xin Dong, Ximing Lu, Shizhe Diao, Peter Belcak, Mingjie Liu, Min-
Hung Chen, Hongxu Yin, Yu-Chiang Frank Wang, Kwang-Ting Cheng, Yejin Choi, Jan K
autz, Pavlo Molchanov

As language models become increasingly capable, users expect them to provide not
only accurate responses but also behaviors aligned with diverse human preferenc
es across a variety of scenarios. To achieve this, Reinforcement learning (RL) p
ipelines have begun incorporating multiple rewards, each capturing a distinct pr
eference, to guide models toward these desired behaviors. However, recent work h
as defaulted to apply Group Relative Policy Optimization (GRPO) under multi-rewa
rd setting without examining its suitability. In this paper, we demonstrate that
directly applying GRPO to normalize distinct rollout reward combinations causes
them to collapse into identical advantage values, reducing the resolution of th
e training signal and resulting in suboptimal convergence and, in some cases, ea
rly training failure. We then introduce Group reward-Decoupled Normalization Pol
icy Optimization (GDPO), a new policy optimization method to resolve these issue
s by decoupling the normalization of individual rewards, more faithfully preserv
ing their relative differences and enabling more accurate multi-reward optimizat
ion, along with substantially improved training stability. We compare GDPO with
GRPO across three tasks: tool calling, math reasoning, and coding reasoning, eva
luating both correctness metrics (accuracy, bug ratio) and constraint adherence
metrics (format, length). Across all settings, GDPO consistently outperforms GRP
O, demonstrating its effectiveness and generalizability for multi-reward reinfor
cement learning optimization.

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See First, Reason Later: Mutual Information-Guided Reinforcement Learning for Vision-Language Models
Yin Zhang, Zonghan Wu, Jiaxuan Zhao, Junfeng Fang, Zengxiang Li, Kun Wang, Qingsong Wen, Yilei Shao
Vision-Language Models (VLMs) frequently suffer from visual perception errors and hallucinations that compromise answer accuracy in complex reasoning tasks. Reinforcement Learning with Verifiable Rewards (RLVR) offers a promising solution by optimizing policies using answer correctness signals. Despite their effectiveness, prevailing RLVR methods face two critical limitations. First, much of the sampling budget is wasted on trajectories doomed to fail due to early visual description errors. Second, sparse rewards cannot distinguish whether failures stem from visual perception or reasoning stages. We introduce MIRL, a decoupled framework that addresses both limitations by leveraging mutual information (MI) between generated descriptions and visual inputs as a cheap pre-screening signal. This enables intelligent budget allocation toward high-potential trajectories via forking, while decoupled training provides independent MI-based rewards for visual perception optimization, resolving reward blindness. Experiments on six vision-language reasoning benchmarks demonstrate that MIRL achieves 70.22\% average accuracy and successfully surpasses the performance of sampling 16 complete trajectories using only 10 pre-samples with top-6 selection (25\% fewer complete trajectories). Our code is available at: https://github.com/youwyouw/mirl-main.
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PASA: A Principled Embedding-Space Watermarking Approach for LLM-Generated Text under Semantic-Invariant Attacks
Zhenxin Ai, Haiyun He
Watermarking for large language models (LLMs) is a promising approach for detecting LLM-generated text and enabling responsible deployment. However, existing watermarking methods are often vulnerable to semantic-invariant attacks, such as paraphrasing.
We propose PASA, a principled, robust, and distortion-free watermarking algorithm that embeds and detects a watermark at the semantic level. PASA operates on semantic clusters in a latent embedding space and constructs a distributional dependency between token and auxiliary sequences via shared randomness synchronized by a secret key and semantic history.
This design is grounded in our theoretical framework that characterizes a jointly optimal embedding-detection pair, achieving the fundamental trade-offs among detection accuracy, robustness, and distortion. Evaluations across multiple LLMs and semantic-invariant attacks demonstrate that PASA remains robust even under strong paraphrasing attacks while preserving high text quality, outperforming standard vocabulary-space baselines. Ablation studies further validate the effectiveness of our hyperparameter choices. Webpage: https://ai-kunkun.github.io/PASA_page/.
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Feature Bagging Provides Stability
Yuheng Ma, Qiang Sun
We study feature bagging through the lens of algorithmic stability. Feature bagging is an ensemble strategy that aggregates base learners trained on randomly subsampled feature subsets, possibly in a data-dependent manner. We introduce feature instability (FI), the feature-axis analogue of instance instability (II), which measures sensitivity to removing a single feature. Smaller values of II or FI correspond to stronger stability, and our experiments show that FI captures generalization-relevant information complementary to II. Within this framework, we analyze feature bagging in both a parametric linear model and a model-free setting inspired by recursive feature subsampling in random forests. In both settings, we establish formal guarantees showing that feature bagging improves the relevant stability relative to its non-bagged counterpart, with larger improvements under more aggressive subsampling. We further show that a modest number of bagging rounds is sufficient to approach the infinite-bagging stability level.
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TimeRewarder: Learning Dense Reward from Passive Videos via Frame-wise Temporal Distance
Yuyang Liu, Chuan Wen, Yihang Hu, Dinesh Jayaraman, Yang Gao
Designing dense rewards is crucial for reinforcement learning (RL), yet in robotics it often demands extensive manual effort and lacks scalability. One promising solution is to view task progress as a dense reward signal, as it quantifies the degree to which actions advance the system toward task completion over time. We present TimeRewarder, a simple yet effective reward learning method that derives progress estimation signals from passive videos, including robot demonstrations and human videos, by modeling temporal distances between frame pairs. We then demonstrate how TimeRewarder can supply step-wise proxy rewards to guide reinforcement learning. In our comprehensive experiments on ten challenging Meta-World tasks, we show that TimeRewarder dramatically improves RL for sparse-reward tasks, achieving nearly perfect success in 9/10 tasks with only 200,000 interactions per task with the environment. This approach outperforms previous methods and even the manually designed environment dense reward on both the final success rate and sample efficiency. Moreover, we show that TimeRewarder pretraining can exploit real-world human videos, highlighting its potential as a scalable approach to rich reward signals from diverse video sources.
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Reflex: Real-Time Vision-Language-Action Control through Streaming Inference
Yuanchun Guo, Bingyan Liu
Flow matching Vision-Language-Action (VLA) models promise precise continuous control, but their iterative denoising nature introduces fundamental incompatibilities with real-time robotics: global timestep injection invalidates KV-caching, forcing a choice between slow $O(N^2)$ re-computation or mathematically incorrect cache reuse.
We present **Reflex**, a framework that enables *real-time streaming inference* for flow matching policies by exploiting the *Timestep-Invariance Property*---that perception encoders are functionally independent of the denoising loop.
Reflex partitions the attention context into static, sliding, and dynamic regions, enabling $O(1)$ incremental cache updates while preserving full-batch-equivalent attention outputs for fixed inputs.
To ensure stability under continuous high-frequency inference, we introduce *AdaRMSNorm*, an adaptive normalization layer that prevents BFloat16 numerical collapse by gating on flow phase.
We further maximize throughput through an *async pipeline* that decouples visual encoding from action generation, combined with *operator fusion* that reduces kernel overhead.
On LIBERO and Kinetix benchmarks, Reflex achieves a 2.58$\times$ inference speedup and 50Hz stable streaming, reducing reaction latency by up to 54\% and enabling efficient deployment without performance degradation.
****
Automatically Finding Reward Model Biases
Zifan Wang, Iván Arcuschin, Arthur Conmy
Reward models are central to large language model (LLM) post-training. However, past work has shown that they can reward spurious or undesirable attributes such as length, format, hallucinations, and sycophancy. In this work, we introduce and study the research problem of automatically finding reward model biases in natural language. We offer a simple approach of using an LLM to iteratively propose and refine candidate biases. Our method can recover known biases and surface novel ones: for example, we found that Skywork-V2-8B, a leading open-weight reward model, often mistakenly favors responses with redundant spacing and responses with hallucinated content. In addition, we show evidence that evolutionary iteration outperforms flat best-of-N search, and we validate the recall of our pipeline using synthetically injected biases. We hope our work contributes to further research on improving RMs through automated interpretability methods.
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Automated Formal Proofs of Combinatorial Identities via Wilf–Zeilberger Guidance and LLMs

Beibei Xiong, Hangyu Lv, Junqi Liu, Yisen Wang, Shaoshi Chen, Jianlin Wang, Zhen gfeng Yang, Lihong Zhi
Automating formal proofs of combinatorial identities is challenging for LLM-base d provers, as long-horizon proof planning is required and unconstrained search q uickly explodes.
Symbolic methods such as the Wilf--Zeilberger (WZ) method can achieve a mechaniz ed proof of combinatorial identities by constructing special auxiliary functions and demonstrating that they satisfy specific recurrence relations.
We propose WZ-LLM, a neuro-symbolic framework that turns WZ proof plans into exe cutable proof sketches in Lean~4 and uses an LLM-based prover
to discharge the resulting machine-checkable subgoals.
We also train a dedicated WZ-Prover via a Lean-kernel-verified bootstrapping loo p with expert-verified iteration, followed by DAPO-based refinement.
Experiments show that WZ-LLM achieves a 34\% proof success rate on LCI-Test (10 0 classical combinatorial identities), outperforming strong baselines such as De epSeek-V3 and Goedel-Prover-V2;
moreover, on LCI-Test it proves 5 identities on which the symbolic-only baseline fails.
WZ-LLM also improves performance on CombiBench and PutnamBench-Comb, suggesting the effectiveness of coupling symbolic proof sketches with learned formal reason ing.
Experiments show that WZ-LLM achieves a 34\% proof success rate on LCI-Test (100 classic combinatorial identities), outperforming strong baselines such as DeepS eek-V3 and Goedel-Prover-V2, and delivering consistent gains on CombiBench and P utnamBench-Comb. These results indicate that our framework provides two compleme ntary strengths: improved direct proving for identities beyond the scope of WZ, and substantially higher end-to-end success when WZ sketches guide a specialized prover.
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Normalization-equivariant Diffusion Models: Learning Posterior Samplers From Noi sy And Partial Measurements
Brett Levac, Jon Tamir, Marcelo Pereyra, Julián Tachella
Diffusion models (DMs) are a powerful framework for image generation and restora tion. However, existing DMs are primarily trained in a supervised manner by usin g a large corpus of clean images. This poses fundamental challenges in many real -world scenarios, where acquiring noise-free data is hard or infeasible. While s ome methods are capable of training DMs using noisy data, they are effective onl y when the amount of noise is very mild or when additional noise-free data is av ailable. In addition, existing methods for training DMs from incomplete measurem ents require access to multiple complementary acquisition processes, a significa nt practical limitation. Here we introduce the first approach for learning DMs f or image restoration using only noisy measurement data from a single operator. F irst, we show that DMs, and more broadly minimum mean squared error denoisers, e xhibit a weak form of scale equivariance linking rescaling in signal amplitude t o changes in noise intensity. We then leverage this theoretical insight to devel op a denoising score-matching strategy that generalizes robustly to noise levels below the training data, thereby enabling the learning of DMs from noisy measur ements. For problems involving measurements both noisy and incomplete, we integr ate our method with equivariant imaging, a complementary self-supervised learnin g framework that exploits the inherent invariants of imaging problems. This allo ws training DMs for image restoration from single-operator noisy measurements. W e validate the effectiveness of our approach through extensive experiments on im age denoising, demosaicing, inpainting, and MRI reconstruction along with compar isons with the state of the art.
****
Deep Multi-view Graph Clustering via Attribute-aware Bidirectional Structural Re finement and Pseudo-label Guided Multi-level Fusion
Youqing Wang, Tianxiang Zhao, Mengyuan Xin, Ye Su, Jiapu Wang, Tengfei Liu, Junb in Gao, Jipeng Guo
Deep multi-view graph clustering (DMGC) typically leverages graph neural network

s for representation learning, but most existing methods excessively depend on l
ocal and static graph structures and only utilize simplistic cross-view fusion s
trategies. To this end, this paper proposes **A**ttribute-aware Bidirectional St
ructural Refinement (ABSR) and **P**seudo-label Guided Multi-level Fusion (PGMF)
 for DM**GC**, termed **APGC**. Specifically, ABSR selectively strengthens high-
quality connections and suppresses semantically conflicting relationships, achie
ving bidirectional refinement of the graph structure based on attribute similari
ty. It incorporates global attribute semantics into the graph structure, thereby
 promoting the homophilic connections for discriminative graph representation le
arning. Guided by reliable pseudo-labels, PGMF achieves adaptive weighted fusion
 at both the node-level and the view-level, effectively balancing the differenti
ated contributions of multi-view information. Experiments on six homophilic and
heterophilic datasets demonstrate the superior clustering performance of the pro
posed APGC method. The code of APGC could be available at https://github.com/Ti
anxiangZhao0474/APGC.git.
****

Exploring More to Solve More: Boosting Diversity in Text Diffusion Models via En
tropy-Based Guidance

Jingwei Zhang, Haoyu Lei, Zijin Feng, Jiacheng Sun, Farzan Farnia

Although diffusion models have revolutionized continuous domains like image synt
hesis through high quality generations and controllable guidance mechanisms, bri
nging this controllability to the discrete, sequential nature of text remains an
 open challenge. Meanwhile, current sampling strategies and guidance methods adj
ust token likelihoods without capturing the broader semantic landscape, leading
to a suboptimal balance between fidelity and diversity. In this work, we introdu
ce a novel training-free Semantic-Aware Kernel Entropy (SAKE) guidance method. O
ur method computes the order-2 Rényi entropy over a kernel Gram matrix that capt
ures both cross-token semantic interactions and relative token positions. By lin
earizing this objective in the embedding space, we derive a tractable guidance s
ignal that dynamically adjusts the sampling distribution—flattening it to encour
age exploration during redundancy and sharpening it for fidelity when diverse. E
mpirical experiments demonstrate that our approach achieves a superior Pareto fr
ontier between fidelity and diversity, and improves multi-sample performance on
reasoning-intensive tasks, such as code and mathematics generation, compared to
temperature scaling and discrete guidance baselines.
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Artemis: Structured Visual Reasoning for Perception Policy Learning

Wei Tang, Yanpeng Sun, Shan Zhang, Weihao Bo, Xiaofan Li, Piotr Koniusz, Wei Li,
 Na Zhao, Zechao Li

Recent reinforcement-learning frameworks for visual perception policy usually in
corporate intermediate reasoning chains expressed in natural language. Empirical
 observations indicate that such purely linguistic intermediate reasoning often
reduces performance on perception tasks. We argue that the core issue lies not i
n reasoning per se but in the form of reasoning:
while these chains perform semantic reasoning in an unstructured linguistic spac
e, visual perception requires reasoning in a spatial and object-centric space. I
n response, we introduce Artemis, a perception-policy learning method that perfo
rms structured visual reasoning, where each intermediate step is represented as
a (label, bounding-box) pair capturing a verifiable visual state. This design en
ables explicit tracking of intermediate states, direct supervision for proposal
quality, and avoids ambiguity introduced by language-based reasoning. Building u
pon verifiable and spatially grounded reasoning chains, Artemis provides a unifi
ed architecture for diverse perceptual tasks, without requiring the task-specifi
c designs relied upon by prior perceptual policy models. Trained using grounding
 and detection sampeles in natural image domains, Artemis generalizes to countin
g and geometric perception tasks.
At its core, a spatially grounded, object-centric chain rule provides a principl
ed foundation for scalable and general perceptual policies.
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Fully Dynamic Coreset Spectral Clustering

Ben Jourdan, Peter Macgregor, Gregory Schwartzman
We present a fully dynamic data structure that supports edge and node updates an d cluster membership queries for the Normalised Cut problem with strong theoreti cal guarantees. Furthermore, our data structure outperforms the state of the art significantly on real world datasets. At the heart of our data structure is the novel notion of *Just-in-Time Sampling Trees*.

The worst-case edge update time of our data structure is $O(\log n)$ where n i s the number of nodes in the current graph. Let $d_{\max}$ be the maximum degree of the current graph, let $T_{NC}(n',k')$ be the running time of an α-ap proximation algorithm for the Normalised Cut problem on n' vertices and k' c lusters, and let $\text{vol}(Y)$ be the sum of the unweighted degrees of all nod es in a set Y. The worst-case query time of our data structure to label all no des in Y is $O\left(kd_{max}^2 \log(n) + \gamma(n,\epsilon,k,Y)\right)$,
with approximation ratio $\alpha \frac{1+\epsilon}{1-\epsilon}$, where $\gamma(n ,\epsilon,k,Y)$ is defined to be $\log(n)\log^\star(n)\epsilon^{-4}k^2 + T_{NC}(\epsilon^{-4}k^2,k) + \epsilon^{-8}k^4 +vol(Y)$.
Assuming $d_{\max}$ is polylogarithmic, as is the case with many sparse real-wor ld graphs, our method achieves the best known trade-off between query time and u pdate time.
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Needles in the Haystack: Addressing Signal Dilution Improves scRNA-seq Perturbat ion Response Modeling and Evaluation
Gabriel Mateo Mejia, Henry E Miller, Francis J.A. Leblanc, BO WANG, Brendan Swai n, Lucas Paulo de Lima Camillo
Recent benchmarks reveal that single-cell perturbation response models are often outperformed by simply predicting the dataset mean. Through large-scale *in sil ico* simulations, together with analyses of two real-world perturbation datasets , we trace this anomaly to a metric artifact: unweighted error metrics systemati cally reward mean predictions when perturbation effects are sparse. To address t his limitation, we introduce differentially expressed gene (DEG)-aware metrics—w eighted mean-squared error (WMSE) and weighted delta R^{2} ($R^{2}_{w}(\Delta) $)—that sensitively measure error in niche, perturbation-specific signals. We fu rther propose explicit negative and positive performance baselines to calibrate these metrics. Under this framework, the mean baseline sinks to null performance , while genuinely informative predictors are correctly rewarded. Finally, we sho w that using WMSE as a training objective reduces mode collapse and improves pre dictive performance across multiple model architectures.
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Local Mechanisms of Compositional Generalization in Conditional Diffusion
Arwen Bradley
Conditional diffusion models appear capable of compositional generalization, i.e ., generating convincing samples for out-of-distribution combinations of conditi oners, but the mechanisms underlying this ability remain unclear. To make this c oncrete, we study length generalization, the ability to generate images with mor e objects than seen during training. In a controlled CLEVR setting (Johnson et a l., 2017), we find that length generalization is achievable in some cases but no t others, suggesting that models only sometimes learn the underlying composition al structure. We then investigate locality as a structural mechanism for composi tional generalization. Prior works proposed score locality as a mechanism for cr eativity in unconditional diffusion models (Kamb & Ganguli, 2024; Niedoba et al. , 2024), but did not address flexible conditioning or compositional generalizati on. In this paper, we prove an exact equivalence between a specific compositiona l structure (*conditional projective composition*) (Bradley et al., 2025) and sc ores with sparse dependencies on both pixels and conditioners (*local conditiona l scores*). This theory also extends to compositions of concepts (such as style+ content) in feature-space. We validate our theory empirically: CLEVR models that succeed at length generalization exhibit local conditional scores, while those that fail do not. Furthermore, we show that a causal intervention explicitly enf orcing local conditional scores enables length generalization in a previously fa

iling model. Finally, we investigate feature-space compositionality in color-con
ditioned CLEVR, and find preliminary evidence of compositional structure and cor
responding local mechanisms in SDXL.
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f-Divergence Regularized RLHF: Two Tales of Sampling and Unified Analyses
Di Wu, Chengshuai Shi, Jing Yang, Cong Shen
Reinforcement Learning from Human Feedback (RLHF) has become a cornerstone techn
ique for post-training large language models. While most existing approaches rel
y on the reverse KL-regularization, recent empirical studies have begun explorin
g alternative divergences (e.g., forward KL, chi-squared) as regularizers in RLH
F. However, a unified theoretical understanding of general f-divergence regula
rization remains under-explored. To fill this gap, this work develops a comprehe
nsive theoretical framework for online RLHF with an f-divergence regularized o
bjective. Rather than treating each divergence in isolation, we adopt a holistic
 perspective across the entire class and propose two algorithms based on distinc
t sampling principles. The first extends the classical optimism principle with a
 carefully designed exploration bonus, while the second introduces a new method
that exploits the sensitivity of the optimal policy to reward perturbations unde
r f-divergence regularization. Theoretical analysis shows that $O(\log T)$ reg
ret and $O(1/T)$ sub-optimality gap are achievable, establishing provable effici
ency of both algorithms and, to the best of our knowledge, the first performance
 bounds for online RLHF under general f-divergence regularization.
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BEAT: Tokenizing and Generating Symbolic Music by Uniform Temporal Steps
Lekai Qian, Haoyu Gu, Jingwei Zhao, Ziyu Wang
Tokenizing music to fit the general framework of language models is a compelling
 challenge, especially considering the diverse symbolic structures in which musi
c can be represented (e.g., sequences, grids, and graphs). To date, most approac
hes tokenize symbolic music as sequences of musical events, such as onsets, pitc
hes, time shifts, or compound note events. This strategy is intuitive and has pr
oven effective in Transformer-based models, but it treats the regularity of musi
cal time implicitly: individual tokens may span different durations, resulting i
n non-uniform time progression. In this paper, we instead consider whether an al
ternative tokenization is possible, where a uniform-length musical step (e.g., a
 beat) serves as the basic unit. Specifically, we encode all events within a sin
gle time step at the same pitch as one token, and group tokens explicitly by tim
e step, which resembles a sparse encoding of a piano-roll representation. We eva
luate the proposed tokenization on music continuation and accompaniment generati
on tasks, comparing it with mainstream event-based methods. Results show improve
d musical quality and structural coherence, while additional analyses confirm hi
gher efficiency and more effective capture of long-range patterns with the propo
sed tokenization.
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Dynamic Multimodal Evaluation via Knowledge-Enhanced Benchmark Evolution
Junzhe Zhang, Huixuan Zhang, Xiaojun Wan
The rapid development of multimodal large language models (MLLMs) has created an
 urgent demand for more reliable and robust evaluation protocols, however, exist
ing static benchmarks are prone to data contamination and performance saturation
, which can result in inflated or misleading evaluation results. To address thes
e limitations, we first introduce a graph formulation to represent both static a
nd dynamic visual question answering (VQA) samples. Building upon this formulati
on, we propose Knowledge-Enhanced Benchmark Evolution (KBE), a dynamic multimoda
l evaluation framework that first analyzes the original static benchmark, then e
xpands it by integrating multimodal knowledge, transforming the static benchmark
 into a controllable, dynamic evolving version. Crucially, KBE can both reconstr
uct questions by Re-selecting visual information in the original image and expan
d existing questions with external textual knowledge. By explicitly controlling
the degree of question exploration, KBE enables difficulty-controllable evaluati
on across a wide range of model capabilities. Extensive experimental results dem
onstrate that KBE effectively mitigates data contamination and benchmark saturat

ion, while providing a more comprehensive and flexible assessment of MLLM performance.
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Alignment between Brains and AI: Evidence for Convergent Evolution across Modalities, Scales and Training Trajectories
Guobin Shen, Dongcheng Zhao, Yiting Dong, Qian Zhang, Yi Zeng
Artificial and biological systems may converge on similar computational strategies despite different architectures and learning mechanisms—a form of convergent evolution. We test this at scale by comparing internal representations of 630 AI models (language and vision; 1.33M–72B parameters) against fMRI from the Natural Scenes Dataset, producing over 60 million alignment measurements. Within each modality, higher-performing models spontaneously develop stronger brain correspondence (language: *r* = 0.89; vision: *r* = 0.53); because the inputs are image-evoked, the language results reflect visual-semantic alignment rather than a direct cross-modal comparison. Longitudinal analysis combined with bidirectional Granger tests further shows that past alignment predicts future performance more reliably than the reverse, identifying brain-like representations as a robust early-emerging correlate of learning. Modality-specific organization also emerges: language models align with limbic and integrative regions, vision models with visual cortical hierarchies.
****
Learning in Structured Stackelberg Games
Maria Florina Balcan, Kiriaki Fragkia, Keegan Harris
We initiate the study of *structured Stackelberg games*, a novel form of strategic interaction between a leader and a follower where contextual information can be predictive of the follower's (unknown) type. Motivated by applications such as security games and AI safety, we show how this additional structure can help the leader learn a utility-maximizing policy in both the online and distributional settings. In the online setting, we first prove that standard learning-theoretic measures of complexity do not characterize the difficulty of the leader's learning task. We find that there exists a learning-theoretic measure of complexity, analogous to the Littlestone dimension in online classification, that *tightly* characterizes the leader's instance-optimal regret. We term this the *Stackelberg-Littlestone dimension*, and leverage it to provide a provably optimal online learning algorithm. In the distributional setting, we provide analogous results by showing that two new dimensions control the sample complexity upper- and lower-bound.
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Rethinking Evaluation Paradigms in IBP-based Certified Training
Konstantin Kaulen, Hadar Shavit, Holger H. Hoos
Deep neural networks achieve strong performance on many supervised learning tasks but remain vulnerable to adversarial perturbations. Neural network verification provides mathematically rigorous robustness guarantees, yet at substantial computational cost. To mitigate this, certified training techniques optimise for verifiable robustness during training, typically inducing a trade-off between natural and certified accuracy controlled by method-specific hyperparameters.
Because these metrics are inherently conflicting, the common practice of reporting a single configuration is problematic: it can mislead conclusions about overall performance and prevents unbiased assessments of the state of the art. We address this by evaluating certified training methods via Pareto front comparisons over the natural--certified accuracy trade-off. To enable fair, method-agnostic comparisons, we perform efficient automated multi-objective hyperparameter optimisation to identify a set of Pareto-optimal configurations for each method.
This approach often uncovers substantial undertuning in previously reported configurations, yielding superior performance and establishing a new state of the art. Leveraging these fronts, we present the first comprehensive multi-objective comparison of certified training approaches, showing that prior advancements are less pronounced than assumed and revealing previously unreported performance complementarities.
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Box Thirding: Anytime Best Arm Identification under Insufficient Sampling
SEOHWA HWANG, Junyong Park
We introduce Box Thirding (B3), a flexible and efficient algorithm for Best Arm Identification (BAI) under fixed budget constraints. It is designed for both any time BAI and scenarios with large N, where the number of arms is too large fo r exhaustive evaluation within a limited budget T. The algorithm employs a Re median Estimation strategy: in each iteration, three arms are compared—the best-performing arm is explored further, the median is retained for future comparison s, and the weakest is discarded. Even without prior knowledge of T, B3 achiev es an ϵ -best arm misidentification probability comparable to Sequenti al Halving, which requires T as a prior, applied to a randomly selected subset of c_0 arms that fit within the budget. Empirical results show that B3 outpe rforms existing methods for the limited budget constraint in terms of simple reg ret, as demonstrated on the New Yorker Cartoon Caption Contest dataset.
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Geometry-Preserving Orthonormal Initialization for Low-Rank Adaptation in RLVR
Ruijia Zhang, Jiacheng Zhu, Hanqing Zhu, Laixi Shi
Low-Rank Adaptation (LoRA) and its variants enable parameter-efficient fine-tuni ng of large language models under the supervised fine-tuning (SFT) paradigm. How ever, their efficacy and behavior under Reinforcement Learning with Verifiable R ewards (RLVR) are less well understood. In particular, two structurally initiali zed LoRA variants, PiSSA and MiLoRA, which outperform standard LoRA under SFT, c an underperform standard LoRA under RLVR and may even exhibit training instabili ty. These observations suggest that how to initialize the low-rank matrices in R LVR remains unclear. In this work, we develop a theoretical analysis of LoRA in RLVR, showing that orthonormal initialization achieves the minimal gap between L oRA’s outcome and that of full fine-tuning. Guided by this insight, we propose g eometry-preserving orthonormal initialization for low-rank adaptation in RLVR, l eading to two new variants, LoRA-RLPO and LoRA-RLMO. Experiments on mathematical reasoning benchmarks show that our orthonormal initialization stabilizes RLVR t raining and outperforms standard LoRA, contrasting with PiSSA and MiLoRA. Finall y, our unified analysis also explains why PiSSA and MiLoRA can underperform in R LVR, which may be of independent interest.
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Efficient Adaptive Testing via Gradient Path Matching Subset Selection for AI Ed ucation
Yan Zhuang, Junhao Yu, Bohou Zhang, Zachary Pardos, Jinze Wu, Daoqiang Zhang
Adaptive testing is widely adopted in AI-driven educational assessment systems (e.g., GRE), where the goal is to select an optimal subset of questions from a la rge question pool to accurately estimate an examinee's ability. A fundamental ch allenge is that: optimal question subsets are inherently personalized, and solvi ng for them is NP-hard. Recently, it has been framed as a gradient matching prob lem: aligning gradients between selected subsets and the full question set acros s the entire ability parameter space. However, such global alignment on entire s pace is computationally expensive and difficult to scale. In this work, we propo se GPM (Gradient Path Matching), a novel framework that instead aligns gradients along possible optimization paths toward the final estimate. By leveraging inte rmediate gradients as supervision, GPM learns an explicit and generalizable sele ction algorithm from large-scale data. We provide theoretical analysis on its co nvergence and scalability. Experiments on both real-world and synthetic datasets demonstrate that it achieves the same estimation accuracy using, on average, 20 % fewer questions.
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SHARP-Q: Spectral Hessian Alignment and Rectification for Post-training Quantiza tion
Menghao Lv, Huiqiong Wang, Li Sun, Mingli Song
Post-training quantization (PTQ) suffers from severe accuracy degradation in ult ra-low-bit regimes. To address this challenge, we propose SHARP-Q, a unified fra mework grounded in Information Geometry that aligns the quantization objective w ith the intrinsic Fisher geometry. Following a "Rectify-then-Approximate" strate

gy, SHARP-Q first preconditions the optimization landscape via Hessian-Aware Rectification (HAR) and subsequently approximates the rectified Fisher Information Matrix through Dynamic Fisher-Subspace Compensation (DFSC). Our empirical evaluations reveal a pivotal insight: precise geometric alignment enables hardware-friendly uniform quantizers to outperform specialized non-uniform designs. Extensive experiments across representative convolutional networks, Vision Transformers, and State Space Models confirm that SHARP-Q establishes new state-of-the-art results, achieving substantial accuracy gains in the challenging W2A2 and W3A3 settings.

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Multi-Agent Teams Hold Experts Back

Aneesh Pappu, Batu El, Hancheng Cao, Carmelo di Nolfo, Yanchao Sun, Meng Cao, James Zou

Multi-agent LLM systems are increasingly deployed as autonomous collaborators, where agents interact freely rather than execute fixed, pre-specified workflows. In such settings, effective coordination cannot be fully designed in advance and must instead emerge through interaction. However, most prior work enforces coordination through fixed roles, workflows, or aggregation rules, leaving open the question of how well self-organizing teams perform when coordination is unconstrained. Drawing on organizational psychology, we study whether self-organizing LLM teams achieve *strong synergy*, where team performance matches or exceeds the best individual member. Across human-inspired and frontier ML benchmarks, we find that---unlike human teams---LLM teams consistently fail to match their expert agent's performance, even when explicitly told who the expert is, incurring performance losses of up to 41.1\% on ML benchmarks. Decomposing this failure, we show that expert leveraging, rather than identification, is the primary bottleneck. Conversational analysis reveals a tendency toward integrative compromise---averaging expert and non-expert views rather than appropriately weighting expertise---which increases with team size and correlates negatively with performance. Interestingly, this consensus-seeking behavior improves robustness to adversarial agents, suggesting a trade-off between alignment and effective expertise utilization. Our findings reveal a significant gap in the ability of self-organizing multi-agent teams to harness the collective expertise of their members.

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Less is More: Geometric Unlearning for LLMs with Minimal Data Disclosure

Chenchen Tan, Xinghao Li, Shujie Cui, Youyang Qu, Cunjian Chen, Longxiang Gao

As large language models (LLMs) are increasingly deployed in real-world systems, they must support post-hoc removal of specific content to meet privacy and governance requirements. This motivates selective unlearning, which suppresses information about a particular entity or topic while preserving the LLM's general utility. However, most existing LLM unlearning methods require access to the original training corpus and rely on output-level refusal tuning or broad gradient updates, creating a tension among unlearning strength, non-target preservation, and data availability. We propose Geometric Unlearning (GU), an approach that operates directly on the model's prompt-conditioned hidden states without access to the original training corpus. Specifically, GU distills a compact, low-rank safe-behavior subspace from a small set of safe reference prompts and uses lightweight anchor-in-context synthetic prompts to trigger localized, projection-based alignment of hidden representations to this safe subspace. A teacher-distillation regularizer on synthetic non-target anchors further reduces collateral drift. Across privacy-oriented unlearning benchmarks (ToFU and UnlearnPII), GU achieves strong target suppression with minimal impact on non-target performance, demonstrating that effective unlearning can be achieved with minimal synthetic data.

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Rethinking Gating Mechanism in Sparse MoE: Handling Arbitrary Modality Inputs with Confidence-Guided Gate

Liangwei Nathan Zheng, Wei Emma Zhang, Mingyu Guo, Olaf Maennel, Weitong Chen

Effectively managing missing modalities is a fundamental challenge in real-world multimodal learning scenarios, where data incompleteness often results from systematic collection errors or sensor failures. Sparse Mixture-of-Experts (SMoE) a

rchitectures have the potential to naturally handle multimodal data, with individual experts specializing in different modalities. However, existing SMoE approach often lacks proper ability to handle missing modality, leading to performance degradation and poor generalization in real-world applications. We propose ConfSMoE to introduce a two-stage imputation module to handle the missing modality problem for the SMoE architecture by taking the opinion of experts and reveal the insight of expert collapse from theoretical analysis with strong empirical evidence. Inspired by our theoretical analysis, ConfSMoE propose a novel expert gating mechanism by detaching the softmax routing score to task confidence score w.r.t ground truth signal. This naturally relieves expert collapse without introducing additional load balance loss function. We show that the insights of expert collapse aligns with other gating mechanism such as Gaussian and Laplacian gate. The proposed method is evaluated on four different real world dataset with three distinct experiment settings to conduct comprehensive analysis of ConfSMoE on resistance to missing modality and the impacts of proposed gating mechanism.

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LightAVSeg: Lightweight Audio-Visual Segmentation

Qing Zhong, Guodong Ding, Lingqiao Liu, Zaiwen Feng, Lin Yuanbo Wu, Angela Yao

Audio-Visual Segmentation (AVS) targets pixel level localization of sounding emitting objects in videos. However, existing models rely on dense cross-modal attention with quadratic computational cost, limiting their suitability for resource efficient deployment. Most efficiency oriented methods focus on backbone reduction and overlook the interaction module as the primary bottleneck. This paper proposes LightAVSeg, a lightweight framework that replaces heavy attention with a decoupled design for semantic filtering and spatial grounding, resulting in interaction costs that scale linearly with spatial resolution. Furthermore, we introduce an auxiliary alignment loss to enforce semantic consistency during training with zero inference overhead. Extensive experiments demonstrate that LightAVSeg achieves a new state-of-the-art among lightweight methods: with 20.5M parameters (~1/7 of AVSegFormer), it reaches 50.4 mIoU on the MS3 benchmark and enables efficient inference on a mobile processor.

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AutoControl Arena: Synthesizing Executable Test Environments for Frontier AI Risk Evaluation

Changyi Li, Pengfei Lu, Xudong Pan, Fazl Barez, Min Yang

As Large Language Models (LLMs) evolve into autonomous agents, existing safety evaluations face a fundamental trade-off: manual benchmarks are costly, while LLM-based simulators are scalable but suffer from logic hallucination. We present AutoControl Arena, an automated framework for frontier AI risk evaluation built on the principle of logic-narrative decoupling. By grounding deterministic state in executable code while delegating generative dynamics to LLMs, we mitigate hallucination while maintaining flexibility. This principle, instantiated through a three-agent framework, achieves over 98% end-to-end success and 60% human preference over existing simulators. To elicit latent risks, we vary environmental Stress and Temptation across X-Bench (70 scenarios, 7 risk categories). Evaluating 9 frontier models reveals: (1) Alignment Illusion: risk rates surge from 21.7% to 54.5% under pressure, with capable models showing disproportionately larger increases; (2) Scenario-Specific Safety Scaling: advanced reasoning improves robustness for direct harms but worsens it for gaming scenarios; and (3) Divergent Misalignment Patterns: weaker models cause non-malicious harm while stronger models develop strategic concealment. Code and data are available at https://github.com/CosmosYi/AutoControl-Arena.

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DNA: Uncovering Universal Latent Forgery Knowledge

Jingtong Dou, Chuancheng Shi, Anqi Yi, Shiming Guo, Wenhua Wu, Yemin Wang, Li Zhang, Fei Shen, Tat-Seng Chua

As generative AI achieves hyper-realism, superficial artifact detection has become obsolete. While prevailing methods rely on resource-intensive fine-tuning of black-box backbones, we propose that forgery detection capability is already encoded within pre-trained models rather than requiring end-to-end retraining.

To elicit this intrinsic capability, we propose the discriminative neural anchors (DNA) framework, which employs a coarse-to-fine excavation mechanism. First, by analyzing feature decoupling and attention distribution shifts, we pinpoint critical intermediate layers where the focus of the model logically transitions from global semantics to local anomalies. Subsequently, we introduce a triadic fusion scoring metric paired with a curvature-truncation strategy to strip away semantic redundancy, precisely isolating the forgery-discriminative units (FDUs) inherently imprinted with sensitivity to forgery traces.
Moreover, we introduce HIFI-Gen, a high-fidelity synthetic benchmark built upon the very latest models, to address the lag in existing datasets.
Experiments demonstrate that by solely relying on these anchors, DNA achieves superior detection performance even under few-shot conditions. Furthermore, it exhibits remarkable robustness across diverse architectures and against unseen generative models, validating that waking up latent neurons is more effective than extensive fine-tuning.
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CoRe: Collaborative Reasoning via Cross Teaching
Kshitij Mishra, Mirat Aubakirov, Martin Taká■, Nils Lukas, Salem Lahlou
Large language models exhibit complementary reasoning errors: on the same instance, one model may succeed with a particular decomposition while another fails. We propose Collaborative Reasoning (CoRe), a training-time collaboration framework that converts peer success into a learning signal via a cross-teaching protocol. Each problem is solved in two stages: a cold round of independent sampling, followed by a contexted rescue round in which failed models receive a hint extracted from a successful peer. CoRe optimizes a combined reward that balances (i) correctness, (ii) a lightweight DPP-inspired diversity term to reduce error overlap, and (iii) an explicit rescue bonus for successful recovery. We evaluate CoRe in low-data regimes across GSM8K, MATH, AIME, and GPQA. With only 1,000 training examples, a pair of small open-source models (3B+4B) reaches 99.54% oracle Team Pass@2 on GSM8K and 92.08% on MATH, compared to 82.50% and 74.82% for single-model training. On harder datasets, the same pair reaches 77.34% oracle Team Pass@2 on GPQA (trained on 348 examples) and 79.65% on AIME (trained on 792 examples), using a training-time budget of at most 1536 context tokens and 3072 generated tokens. A three-model non-oracle AIME evaluation further closes most of the oracle-selection gap (81.20% majority vote vs. 82.60% oracle Team Pass@2). Overall, these results show that training-time collaboration can reliably convert model complementarity into large gains without scaling model size.
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HelioX: A GPU-Native Framework for Simulation and Training of Biophysically Detailed Networks
JUNFENG LU, Zijie Yu, Shaoyang Cui, Gan He, Ruiqin Xiong, Kai Du, Tiejun Huang
Biophysically detailed neural networks represent a promising frontier for brain-inspired AI, offering intrinsic spatio-temporal dynamics to enhance the expressivity and computational density of deep learning systems. However, general-purpose deep learning frameworks suffer from a fundamental mismatch between their dense parallel optimizations and the irregular, tree-structured complexity of biological mechanisms. In this work, we propose **HelioX**, a \textbf{GPU-native} framework designed to unify high-performance simulation with scalable training. Unlike approaches that adapt biology to existing deep learning tools, **HelioX** adopts a “GPU-to-Biophysics'' paradigm. We tailor the underlying GPU parallelism to biological structures by implementing custom-fused CUDA kernels for both the Dendritic Hierarchical Scheduling (DHS) algorithm and its gradient propagation. This design eliminates the runtime overhead of generic automatic differentiation and enables multi-stream concurrency for spike generation and equation assembly.
Experimental results demonstrate that **HelioX** outperforms standard simulators (NEURON) by orders of magnitude and surpasses prior GPU-based solvers in both speed and scalability. We successfully train deep biophysical MLPs and whole-brain-scale biophysical circuits (e.g., the BAAIWorm \textit{C. elegans} model) on a single consumer-grade GPU. **HelioX** establishes a new standard for computational efficiency, enabling the training of biophysically detailed models at scales

previously unattainable.
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VCG-Bench: Towards A Unified Visual-Centric Benchmark for Structured Generation and Editing
Xiaoyan Su, Peijie Dong, Zhenheng Tang, Song Tang, Yuyao Zhai, Kaitao Lin, Liang Chen, Gai Yuhang, Yuyu Luo, Qiang Wang, Xiaowen Chu
Despite the rapid advancements in Vision-Language Models (VLMs), a critical gap remains in their ability to handle structured, controllable diagrammatic tasks essential for professional workflows, as existing methods predominantly rely on pixel-based synthesis which operates in probabilistic pixel spaces and is inherently limited in editability and fidelity; instead, we propose a new "Diagram-as-Code" paradigm with symbolic logic that leverages `mxGraph` Extensible Markup Language (XML) for precise diagram generation and editing, and we present **VCG-Bench**, a unified benchmark for visual-centric `mxGraph` tasks comprising (1) a taxonomized dataset of 1,449 diverse diagrams spanning 6 domains and 15 sub-domains, (2) a paradigm definition that integrates Generation (Vision-to-Code) and Editability (Code-to-Code), and (3) a Tailored Evaluation Protocol employing multi-dimensional metrics such as `mxGraph` Execution Success Rate and Style Consistency Score (SCS), where experimental results highlight the challenges faced by current State-of-the-Art (SOTA) VLMs in structured fidelity and instruction compliance, reflecting their vision and reasoning capabilities.
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CUARewardBench: A Benchmark for Evaluating Reward Models for Computer-Using Agents
Haojia Lin, Xiaoyu Tan, Yulei Qin, Zihan Xu, Yuchen Shi, Zongyi Li, Gang Li, Shaofei Cai, Siqi Cai, Yuzheng Cai, Chaoyou Fu, Ke Li, Xing Sun
Computer-using agents (CUAs) enable task completion through natural interaction with operating systems and software interfaces. While script-based verifiers are widely adopted for evaluation, they suffer from limited scalability and inability to provide step-wise assessment. Reward models offer promising alternatives, but their effectiveness on CUA evaluation remains largely underexplored. To address this gap, we present CUARewardBench, comprising four key contributions: (1) First-ever Comprehensive CUA Reward Benchmark: We introduce the first benchmark for evaluating both outcome reward models (ORM) and process reward models (PRM) on CUA tasks, enabling systematic assessment across trajectory-level and step-level evaluation. (2) Diverse, Practical and Reliable Dataset: CUARewardBench encompasses trajectories from 10 software categories and 7 agent architectures with varying performance levels (25.9%-50.8% success rates). All trajectories are expertly annotated through carefully designed protocols, with rigorous quality control to ensure reliability and practical applicability. (3) Comprehensive Analysis and Insights: Through extensive experiments across 7 vision-language models and 3 prompt templates, we reveal critical limitations of current CUA RMs, including insufficient visual reasoning capabilities, knowledge deficiencies, and the superiority of general VLMs over specialized CUA models for reward evaluation. (4) Unanimous Prompt Ensemble (UPE): Based on the insights from our comprehensive analysis, we propose UPE, a novel ensemble method that significantly enhances reward model reliability through strict unanimous voting and strategic prompt-template configurations. UPE achieves 88.0% precision and 95.3% NPV for ORM, and 83.1% precision and 86.2% NPV for PRM, substantially outperforming single VLMs and traditional ensemble approaches. In a short, this work introduces both a comprehensive benchmark and a novel ensemble method that substantially enhances CUA reward model reliability.
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SPAR: Support-Preserving Action Rectification
Jiaxin Zhao, Weihang Pan, Xun Liang, Binbin Lin
Offline policy improvement faces an inherent conflict between maximizing value and fitting the data distribution. While in-sample weighted regression is stable, it suffers from over-conservatism that suppresses high-value actions in the distribution tail; conversely, gradient-based approaches often exhibit a fitting-optimization conflict of gradients, which drive the policy off the data manifold.

To address this, we propose \textbf{S}upport-\textbf{P}reserving \textbf{A}ction \textbf{R}ectification (SPAR), which reframes global learning as a local residual rectification anchored to a frozen pure behavior cloning policy. This framework performs fine-grained fitting and local policy improvement in the residual space, thereby contracting the search space. We further introduce Latent Self-Imitation, utilizing a latent-sampling weighted-regression mechanism to address fitting-improvement gradient conflict in the residual space. Theoretically, we prove this mechanism eliminates the manifold-normal drift of standard value gradients, while extensive D4RL experiments show SPAR extracts significant gains from suboptimal baselines to achieve state-of-the-art performance.
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Functional building blocks of neural networks: from network motifs to collective dynamics

Jian Zhang, Yue Sun, Wangzi Yao, Tielin Zhang

The advancement of artificial neural networks (ANNs) has been driven by diverse and well-established architectural designs, especially in connectivity. Biological neural networks, which exhibit a rich variety of neurodynamic circuits, offer a valuable source of inspiration for developing novel ANN models. In this study, we analyze the meta-connectivity structure and introduce a network motif-based approach, in which 13 distinct motifs are modeled as functional building blocks. These motifs represent low-dimensional, fundamental components of larger network architectures. Through rigorous theoretical analysis, we classify these motifs into a three■layer hierarchical classification of their dynamical regimes and demonstrate that their hierarchical proportions critically shape collective neural dynamics. Furthermore, by embedding motif distributions into recurrent neural networks (RNNs), we show that these motifs can selectively enhance either network robustness or flexibility. Collectively, our findings provide a theoretical framework—supported by extensive experiments—for understanding how specific network motifs influence the computational properties of artificial intelligence systems via their underlying dynamics. This motif-driven approach offers significant potential for analyzing and modulating neural dynamics in ANNs.
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Fair Decisions from Calibrated Scores: Achieving Optimal Classification While Satisfying Sufficiency

Etam Benger, Katrina Ligett

Binary classification based on predicted probabilities (scores) is a fundamental task in supervised machine learning. While thresholding scores is Bayes-optimal in the unconstrained setting, using a single threshold generally violates statistical group fairness constraints. Under independence (statistical parity) and separation (equalized odds), such thresholding suffices when the scores already satisfy the corresponding criterion. However, this does not extend to sufficiency: even perfectly group-calibrated scores---including true class probabilities---violate predictive parity after thresholding. In this work, we present an exact solution for optimal binary (randomized) classification under sufficiency, assuming finite sets of group-calibrated scores. We provide a geometric characterization of the feasible pairs of positive predictive value (PPV) and false omission rate (FOR) achievable by such classifiers, and use it to derive a simple post-processing algorithm that attains the optimal classifier using only group-calibrated scores and group membership. Finally, since sufficiency and separation are generally incompatible, we identify the classifier that minimizes deviation from separation subject to sufficiency, and show that it can also be obtained by our algorithm, often achieving performance comparable to the optimum.
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SafeSeek: Universal Attribution of Safety Circuits in Language Models

Miao Yu, Siyuan Fu, Moayad Aloqaily, Zhenhong Zhou, Safa Otoum, Xing fan, Kun Wang, Yufei Guo, Qingsong Wen

Mechanistic interpretability reveals that safety-critical behaviors (e.g., alignment, jailbreak, backdoor) in Large Language Models (LLMs) are grounded in specialized functional components. However, existing safety attribution methods struggle with generalization and reliability due to their reliance on heuristic, doma

in-specific metrics and search algorithms. To address this, we propose SafeSeek, a unified safety interpretability framework that identifies functionally complete safety circuits in LLMs via optimization. Unlike methods focusing on isolated heads or neurons, SafeSeek introduces differentiable binary masks to extract multi-granular circuits through gradient descent on safety datasets, while integrates Safety Circuit Tuning to utilize these sparse circuits for efficient safety fine-tuning. We validate SafeSeek in two key scenarios in LLM safety: \textbf{(1) backdoor attacks}, identifying a backdoor circuit with 0.42\% sparsity, whose ablation eradicates the Attack Success Rate (ASR) from 100\% $\to$ 0.4\% while retaining over 99\% general utility; \textbf{(2) safety alignment}, localizing an alignment circuit with 3.03\% heads and 0.79\% neurons, whose removal spikes ASR from 0.8\% $\to$ 96.9\%, whereas excluding this circuit during helpfulness fine-tuning maintains 96.5\% safety retention.
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GSFixer: Improving 3D Gaussian Splatting with Reference-Guided Video Diffusion Priors

Xingyilang Yin, Qi Zhang, Jiahao Chang, Ying Feng, Qingnan Fan, Xi Yang, Chi-Man Pun, Huaqi Zhang, Xiaodong Cun

Reconstructing 3D scenes using 3D Gaussian Splatting (3DGS) from sparse views is an ill-posed problem due to insufficient information, often resulting in noticeable artifacts. While recent approaches have sought to leverage generative priors to complete information for under-constrained regions, they struggle to generate content that remains consistent with input observations. To address this challenge, we propose GSFixer, a novel framework designed to improve the quality of 3DGS representations reconstructed from sparse inputs. The core of our approach is the reference-guided video restoration model, built upon a DiT-based video diffusion model trained on paired artifact 3DGS renders and clean frames with additional reference-based conditions. Considering the input sparse views as references, our model integrates both 2D semantic and 3D geometric features of reference views extracted from the visual geometry foundation model, enhancing the semantic coherence and 3D consistency when fixing artifact novel views. Furthermore, we introduce a reference-guided trajectory sampling strategy that ensures both angular coverage and view quality, further enhancing reconstruction fidelity. Considering the lack of suitable benchmarks for 3DGS artifact restoration evaluation, we present DL3DV-Res which contains artifact frames rendered using low-quality 3DGS. Extensive experiments demonstrate our GSFixer outperforms current state-of-the-art methods in 3DGS artifact restoration and sparse-view 3D reconstruction. Project page: https://github.com/GVCLab/GSFixer.
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Debate with Images: Detecting Deceptive Behaviors in Multimodal Large Language Models

Sitong Fang, Shiyi Hou, Kaile Wang, Boyuan Chen, Donghai Hong, Jiayi Zhou, Juntao Dai, Yaodong Yang, Jiaming Ji

As frontier AI systems become increasingly capable, concerns about deceptive behaviors have intensified. Unlike hallucinations, which stem from capability limitations, deception involves strategically misleading responses despite correct internal representations. While prior work has primarily studied deception in text-only settings, little is known about how such behaviors manifest in multimodal large language models. In this work, we systematically investigate multimodal deception and introduce *MM-DeceptionBench*, the first benchmark designed to evaluate deceptive behaviors in vision–language models across six realistic categories. We find that existing text-centric monitoring approaches are insufficient in multimodal settings due to the complexity of cross-modal reasoning. To address this gap, we propose *debate with images*, a multi-agent evaluation framework that enforces visual grounding through adversarial debate. Experiments show that this approach achieves substantially higher agreement with human judgments than MLLM-as-a-judge baselines, improving Cohen’s kappa by up to 1.5$\times$ and accuracy by up to 1.25$\times$ on GPT-4o.
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M+Adam: Low-Precision Training via Additive–Multiplicative Optimization

Xiaoyuan Liang, Sebastian Loeschcke, Mads Toftrup, Anima Anandkumar
Training with quantized weights can reduce costs but often results in degraded a ccuracy, especially when optimization is carried out in low precision, without s toring high-precision copies. We identify a key failure mode: under low precisio n, standard optimizers can get stuck and not make progress, especially at large weight magnitudes due to coarse mantissa resolution. To overcome this, multiplic ative updates have been previously proposed, in place of additive updates in sta ndard optimizers. While successful under extremely low precision, such as under the logarithmic number system, they suffer from failures near zero and across si gn changes. The failure modes of additive and multiplicative updates are therefo re complementary. To exploit this, we propose M+Adam, which combines both update types: additive steps handle sign changes and small magnitudes, while multiplic ative steps ensure progress at large magnitudes when additive updates are zeroed out under rounding. We prove monotone descent for M+Adam under standard smoothn ess assumptions. Across LLaMA-style pretraining with 60M--1B models, 1--$8\tim es$ Chinchilla budgets, and using only BF16, FP8, and FP4 master weights, M+Adam consistently improves low-precision training.
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CorrectionPlanner: Self-Correction Planner with Reinforcement Learning in Autono mous Driving
Yihong Guo, Dongqiangzi Ye, Sijia Chen, Anqi Liu, Xianming Liu
Autonomous driving requires safe planning, but most learning-based planners lack explicit self-correction ability: once an unsafe action is proposed, there is n o mechanism to correct it. Thus, we propose CorrectionPlanner, an autoregressive planner with self-correction that contains a propose, evaluate, and correct loo p in the motion-token generation process. At each planning step, the policy prop oses an action, namely a motion token, and a learned collision critic predicts w hether it will induce a collision within a short horizon. If the critic predicts a collision, we retain the sequence of historical unsafe motion tokens as a sel f-correction trace, generate the next motion token conditioned on it, and repeat this process until the safe motion token is proposed or the safety criterion is met. This self-correction trace, consisting of all the unsafe motion tokens, re presents the planner’s correction process in motion-token space. We train the pl anner with imitation learning followed by model-based reinforcement learning usi ng rollouts from a pretrained world model that realistically models agents' reac tive behaviors. Closed-loop evaluations show that CorrectionPlanner reduces the collision rate by over 20% on Waymax and obtains state-of-the-art planning sc ores on nuPlan.
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From Prompts to Responses: Dual-Sided Data Leakage and Defense in Split Large La nguage Models
Zixuan GU, Xiaojun Ye, Yang Liu
Large language models (LLMs) are increasingly deployed in privacy-sensitive doma ins, where users must balance the risk of data exposure through external APIs ag ainst the high computational cost of local deployment. Split learning has theref ore emerged as a promising paradigm for LLM fine-tuning and inference under limi ted local resources. However, it introduces new privacy risks. Prior work primar ily studies leakage of private input prompts, typically via inversion attacks on intermediate representations, while the potential for sensitive information lea kage through generative response outputs remains largely unexplored.

In this work, we unveil novel vulnerabilities of Split-LLM by presenting **P**at ched Model **I**nversion with **D**ual-Sided **I**nitialization(**PIDI**), a two -stage attack that simultaneously targets both private input prompts and output responses in Split-LLM settings. It combines dual-sided initialization with a pa tched inversion strategy to tackle long sequences, substantially outperforming p rior inversion methods.
To counter threats from both sides, we further propose the **A**dapter-based **D **ualGuard with **M**utual **I**nformation Defense(**ADMI**), which integrates a n adapter-based local warmup strategy and mutual information regularization to p

rovide a strong empirical privacy protection with minimal impact on task perform ance.
Extensive experiments across diverse tasks and models demonstrate that ADMI effe ctively defends against PIDI and other state-of-the-art inversion attacks. Our c ode is publicly available at \url{https://github.com/FLAIR-THU/VFLAIR-LLM}.
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Scalable Bayesian Inference for Nonlinear Conservation Laws
Tim Weiland, Philipp Hennig
Nonlinear conservation laws are at the heart of many of the most important dynam ical systems in science and engineering.
In practical applications, such systems are often subject to various sources of uncertainty, e.g. due to sparse or noisy measurements.
Inferring physical quantities and fields of interest then becomes an ill-posed p roblem which both classical numerical methods and modern deep learning-based met hods struggle to treat appropriately.
Recent work has framed classical numerical methods as Bayesian inference under G aussian process priors, resulting in a physics-aware treatment of uncertainties.
Following this line of work, we develop a novel numerically conservative method for uncertainty-aware simulations of nonlinear conservation laws.
We use recent sparse approximation techniques to scale up to large-scale forward and inverse problems.
For forward simulation, we inherit the accuracy of classical solvers while provi ding structured uncertainty quantification.
On inverse problems, we recover posteriors over nonparametric source fields in s econds --- outperforming neural baselines that take minutes to produce a less ac curate point estimate.
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Differentiable Conformal Training for LLM Reasoning Factuality
Nathan Hittesdorf, Marco Salzetta, Lu Cheng
Large Language Models (LLMs) frequently hallucinate, limiting their reliability in critical applications. Conformal Prediction (CP) addresses this by calibratin g error rates on held-out data to provide statistically valid confidence guarant ees. Recent work extends CP to LLM factuality: outputs are decomposed into subcl aims, each assigned a risk score, and a calibrated threshold filters out risky c laims to guarantee hallucination rates below a user-specified level (e.g., 10%). While prior methods treat claims independently, Coherent Factuality extends to multi-step reasoning by representing outputs as dependency graphs and jointly va lidating claims with their logical ancestors. A key limitation is that Coherent Factuality is not differentiable, requiring hand-crafted scorers that at high re liability levels remove nearly 60% of true claims. We introduce Differentiable C oherent Factuality (DCF), a fully differentiable relaxation that enables learnin g improved scorers while provably recovering the original algorithm's guarantees . Experiments on two reasoning datasets demonstrate DCF achieves up to 141% impr ovement in claim retention while maintaining reliability guarantees, representin g a significant step towards reliable conformal LLM systems.
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Continual Learning With Participation Privacy: An Auditable Buffering-Aggregatio n Recipe
T-H. Hubert Chan, Elaine Shi, Mengshi Zhao, Mingxun Zhou
Modern federated and streaming learning systems often release intermediate model s, so privacy must hold for the full trajectory under adaptive interaction. Moti vated by participation privacy, we study single-edit neighboring user streams, w here one insertion/deletion shifts all subsequent updates and defeats standard H amming-neighbor continual-release analyses. We give an auditable modular recipe. A randomized buffering wrapper emits bins of size $[U,2U]$, reducing single-edi t streams to a Hamming-style per-bin update stream with explicit backlog/delay g uarantees, where U is calibrated by the privacy parameters $(\varepsilon,\delt a)$. We then prove a certification theorem for independently decomposable (prefi x-causal, fresh-noise) continual mechanisms: any non-adaptive Hamming-neighbor D P proof lifts to adaptive inputs. Together, these ingredients yield trajectory-l

evel (ε,δ)-DP for single-edit streams using standard primitives (e.g., tree prefix sums), with an explicit privacy--latency link via U. Streaming DP-SGD experiments validate the privacy-utility-latency tradeoffs and the induced delay distributions.
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In-Context Generation with Regional Constraints for Instructional Video Editing
Zhongwei Zhang, Fuchen Long, Wei Li, Zhaofan Qiu, Wu Liu, Ting Yao, Tao Mei
The In-context generation paradigm has demonstrated strong power in instructional image editing for better synthesis quality. Nevertheless, shaping such in-context learning for instructional video editing is not trivial. Without specifying editing regions, the results can suffer from the issue of inaccurate editing regions and the token interference between different areas. To address these, we present ReCo, a new instructional video editing paradigm that novelly delves into **Re**gional **Co**nstraint modeling between editing and non-editing areas. Technically, ReCo width-wise concatenates source and target video for joint denoising. In model training, ReCo formulates regional constraints with two regularization terms, i.e., latent and attention regularization, on one-step backward denoised latents and attention maps, respectively. The former increases the latent discrepancy of the editing region between source and target videos while reducing that of non-editing areas, emphasizing editing area modification and alleviating unexpected content generation. The latter suppresses the attention of tokens in the editing region to the tokens in counterpart of the source video, thereby mitigating their interference during novel object generation in target video. Furthermore, we propose a large-scale, high-quality video editing dataset, i.e., ReCo-Data, comprising 500K instruction-video pairs. Extensive experiments conducted on four major instruction-based video editing tasks verify the superiority of ReCo. Code is available at https://github.com/HiDream-ai/ReCo.
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Mitigating the Safety–Utility Trade-off in LLM Alignment via Adaptive Safe Context Learning
Yanbo Wang, Minzheng Wang, Jian Liang, Lu Wang, Yongcan Yu, Ran He
While reasoning models have achieved remarkable success in complex reasoning tasks, their increasing power necessitates stringent safety measures. For safety alignment, the core challenge lies in the inherent trade-off between safety and utility.
However, prevailing alignment strategies typically construct CoT training data with explicit safety rules via context distillation.
This approach inadvertently limits reasoning capabilities by creating a rigid association between rule memorization and refusal.
To mitigate the safety-utility trade-off, we propose the Adaptive Safe Context Learning (ASCL) framework to improve the reasoning given proper context.
ASCL formulates safety alignment as a multi-turn tool-use process, empowering the model to autonomously decide when to consult safety rules and how to generate the ongoing reasoning.
Furthermore, to counteract the preference for rule consultation during RL, we introduce Inverse Frequency Policy Optimization~(IFPO) to rebalance advantage estimates.
By decoupling rule retrieval and subsequent reasoning, our method achieves higher overall performance compared to baselines.
Our code is publicly available at https://github.com/ybwang119/ASCL.git.
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Semi-knockoffs: a model-agnostic conditional independence testing method with finite-sample guarantees
Angel David REYERO LOBO, Bertrand Thirion, Pierre Neuvial
Conditional independence testing (CIT) is essential for reliable scientific discovery. It prevents spurious findings and enables controlled feature selection. Recent CIT methods have used machine learning (ML) models as surrogates of the underlying distribution. However, model-agnostic approaches require a train-test split, which reduces statistical power. We introduce Semi-knockoffs, a CIT method that can accommodate any pre-trained model, avoids this split, and provides val

id p-values and false discovery rate (FDR) control for high-dimensional settings . Unlike methods that rely on the model-X assumption (known input distribution), Semi-knockoffs only require conditional expectations for continuous variables . This makes the procedure less restrictive and more practical for machine learn ing integration. To ensure validity when these expectations are estimated, we pr esent two new theoretical results: (i) stability for regularized models trained with a null feature and (ii) the double-robustness property.
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Quaternion Self-Attention with Shared Scores
Shogo Yamauchi, Tohru Nitta, Hideaki Tamori
Quaternion neural networks are parameter-efficient and model multidimensional de pendencies by representing four related features as a single entity. However, ex isting quaternion self-attention computes component-wise scores and applies inde pendent softmax operations to each component, which increases the computational cost and allows attention distributions to diverge across components. We propose a shared-score quaternion self-attention mechanism that computes a single real-valued score using the quaternion inner product and applies a shared attention d istribution across all components. This reduces score-computation multiplication s by 75\% and the number of softmax operations from four to one. We prove that, when queries and keys are produced by quaternion linear projections that induce component pre-mixing, the component-wise and shared scores lie in the same inter action subspace, indicating that independent component-wise attention primarily re-parameterizes the same interactions rather than expanding the feature interac tion space. In speech enhancement, our method reduces inference time by up to 44 .3\% on a GPU and 58.1\% on a CPU while maintaining quality, with consistent tre nds across vision and natural language processing.
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Unsupervised Camouflaged Object Detection with Dual-Eigenvector Spectral Pseudo-Labeling and Contrastive Refinement
Pingzhu Liu, Chunming He, Zunnan Xu, Chao Hao, Bo Zhao, Xingyu Shao, Jun Zhou, Z itong YU, Xiu Li
Unsupervised Camouflaged Object Detection (UCOD) aims to identify objects concea led in their surroundings without relying on pixel-level labels. Existing method s rely solely on simple post-processing of DINO high-dimensional features to gen erate pseudo labels for training. However, these methods suffer from two major l imitations: 1) pseudo labels they easily generate contain excessive noise, causi ng the model to learn substantial incorrect information. 2) Although pseudo-labe l supervision allows the model to understand the task, it remains insufficient f or generating fine-grained segmentation of the camouflaged objects. To address t hese issues, we propose DualUCOD, a novel UCOD method based on dual-branch contr astive learning that effectively detects camouflaged objects without pixel-level labels. Specifically, we propose the Dual-Eigenvector Spectral Pseudo-Labeling (DESPL) strategy, which fuses semantic and color cues into an affinity matrix. W e then compute the eigenvectors of its normalized graph Laplacian and generate h igh-quality pseudo-labels using these eigenvectors. Furthermore, we introduce a Boundary-Guided Foreground-Background Refinement (BGFBR) module that explicitly incorporates boundary information to improve segmentation accuracy. Finally, we introduce a Dual-Branch Contrastive Learning (DBCL) module that constructs posit ive and negative pairs from the original and augmented images, aligning positive representations while contrasting them against negatives to enhance camouflaged object understanding. Extensive experiments demonstrate that DualUCOD outperfor ms state-of-the-art methods on different datasets in the unsupervised setting.
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Unlearning’s Blind Spots: Over■Unlearning and Prototypical Relearning Attack
SeungBum Ha, Saerom Park, Sung Whan Yoon
Machine unlearning (MU) aims to expunge a designated forget set from a trained m odel without costly retraining, yet the existing techniques overlook two critica l blind spots: "over■unlearning" that deteriorates retained data near the forget set, and post■hoc "relearning" attacks that aim to resurrect the forgotten know ledge.

Focusing on class-level unlearning, we first derive an over-unlearning metric, $\operatorname{OU}@\varepsilon$, which quantifies collateral damage in regions proximal to the forget set, where over-unlearning mainly occurs.
Next, we expose an unforeseen relearning threat on MU, i.e., the Prototypical Relearning Attack, which exploits the per-class prototype of the forget class with just a few samples, and easily restores the pre-unlearning performance.
To counter both blind spots in class-level unlearning, we introduce $\texttt{Spotter}$, a plug■and■play objective that combines (i) a masked knowledge■distillation penalty on the nearby region of forget classes to suppress $\operatorname{OU}@\varepsilon$, and (ii) an intra■class dispersion loss that scatters forget-class embeddings, neutralizing Prototypical Relearning Attacks.
$\texttt{Spotter}$ achieves state-of-the-art results across CIFAR, TinyImageNet, and CASIA-WebFace datasets, offering a practical remedy to unlearning’s blind spots.
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Efficient and Safe Molecular Assembly via Reinforcement Learning and Constraint Solving
Stefan Pranger, Bernhard Ramsauer, Oliver T. Hofmann, Bettina Könighofer
Scanning tunneling microscopy (STM) enables precise manipulation of individual atoms and molecules, offering a pathway to constructing nanoscale assemblies with rich quantum mechanical behavior. Despite its potential, STM-based fabrication remains limited by the inherent complexity of manipulation procedures and the extensive manual effort required. In this work, we take a substantial step toward autonomous manufacturing with STMs by introducing a novel AI-based planning framework for molecular assembly and a high-fidelity simulation environment. Our framework computes collision-free assembly plans that minimize the total distance traveled by molecules. Given an assignment of molecules to target positions, satisfiability solving is used to compute execution schedules in which each molecule has an empty corridor available when it is scheduled to move. Reinforcement learning (RL) agents then execute sequences of STM actions to manipulate molecules to their targets. We further introduce NanoAssemblyGym, a high-fidelity simulation environment for molecular manipulation built on the Gymnasium API, allowing seamless integration with existing RL libraries and workflows. Using NanoAssemblyGym, we demonstrate autonomous assembly of structures containing up to 420 molecules.
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Functional Adjoint Sampler: Scalable Sampling on Infinite Dimensional Spaces
Byoungwoo Park, Juho Lee, Guan-Horng Liu
Learning-based methods for sampling from the Gibbs distribution in finite-dimensional spaces have progressed quickly, yet theory and algorithmic design for infinite-dimensional function spaces remain limited. This gap persists despite their strong potential for sampling the paths of conditional diffusion processes, enabling efficient simulation of trajectories of diffusion processes that respect rare events or boundary constraints. In this work, we present the adjoint sampler for infinite-dimensional function spaces, a stochastic optimal control-based diffusion sampler that operates in function space and targets Gibbs-type distributions on infinite-dimensional Hilbert spaces. Our Functional Adjoint Sampler (FAS) generalizes Adjoint Sampling (Havens et al., 2025) to Hilbert spaces based on a SOC theory called stochastic maximum principle, yielding a simple and scalable matching-type objective for a functional representation. We show that FAS achieves superior transition path sampling performance across synthetic potential and real molecular systems, including Alanine Dipeptide and Chignolin.
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Near-Optimal Regret for KL-Regularized Multi-Armed Bandits
Kaixuan Ji, Qingyue Zhao, Heyang Zhao, Qiwei Di, Quanquan Gu
Recent studies have shown that reinforcement learning with KL-regularized objectives can enjoy faster rates of convergence or logarithmic regret, in contrast to the classical $\sqrt{T}$-type regret in the unregularized setting. However, the statistical efficiency of online learning with respect to KL-regularized objectives remains far from completely characterized, even when specialized to multi-a

rmed bandits (MABs). We address this problem for MABs via a sharp analysis of KL-UCB (Zhao et al., 2025b) using a novel peeling argument, which yields a $\tilde{O}(\eta K\log^2T)$ KL-regularized regret upper bound: the first high-probability regret bound with linear dependence on K. Here, T is the time horizon, K is the number of arms, η^{-1} is the regularization intensity, and $\tilde{O}$ hides all logarithmic factors except those involving $\log T$. The near-tightness of our analysis is certified by the first non-constant lower bound $\Omega(\eta K \log T)$, which follows from subtle hard-instance constructions and a tailored decomposition of the Bayes prior. Moreover, in the low-regularization regime (i.e., large η), we show that the KL-regularized regret for MABs is η-independent and scales as $\tilde{\Theta}(\sqrt{KT})$. Overall, our results provide a thorough understanding of KL-regularized MABs across all regimes of η and yield nearly optimal bounds in terms of K, η, and T.

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Persona-Pruner: Sculpting Lightweight Models for Role-Playing

Jinsu Kim, Jihoon Tack, Noah Lee, Jongheon Jeong

Language Models (LMs) have shown remarkable potential as role-playing chatbots, delivering consistent, stylized interactions when given a specification of a character or user persona. However, applying these capabilities to real-world applications (*e.g*., ecosystems with numerous NPCs interacting simultaneously) exposes a critical inefficiency due to the excessive computational cost. In this paper, we question the necessity of dedicating a full, generalist model to a single persona, hypothesizing that a specific character identity relies on only a fraction of the model’s total capacity. We observe that naïvely pruning LMs often severely degrades the role-playing performance for a specific persona; it does not distinguish between redundant knowledge and essential character traits. We propose **Persona-Pruner**, a framework that sculpts a lightweight role-playing model by isolating persona-specific sub-networks from a single description. Our experiments consistently show that Persona-Pruner preserves role-playing performance substantially more effectively than existing state-of-the-art LLM pruning techniques, reducing the performance drop from the dense model by up to 93.8% over the strongest baseline on RoleBench in LLM-as-a-judge score, while still maintaining general LLM capabilities. Code is available at https://github.com/jsu-kim/Persona-Pruner.

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PRAC: Principal-Random Subspace for LLM Activation Compression and Memory-Efficient Training

Yanyi Li, Yimu Zhang, Cong Fang

Activations have become the primary memory bottleneck in large-batch LLM training. However, existing compression methods fail to exploit the spectral structure of activations, resulting in slow convergence or limited compression. To address this, we bridge the relationship between the algorithm’s fast convergence and the requirements for subspace projection, and show that an effective compression should yield an unbiased estimate of the original activation with low variance. We propose **P**rincipal-**R**andom Subspace for LLM **A**ctivation **C**ompression (**PRAC**), which novelly decomposes activations into two components: a principal subspace captured via SVD to retain dominant information, and a random subspace sampled from the orthogonal complement to approximate the tail. By introducing a precise scaling factor, we prove that PRAC yields an unbiased gradient estimator with minimum variance under certain conditions. Extensive experiments on pre-training and fine-tuning tasks demonstrate that PRAC achieves up to 36\% total memory reduction with negligible performance degradation and minimal computational cost.

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DLO-Lab: Benchmarking Deformable Linear Object Manipulations with Differentiable Physics

Junyi Cao, Yian Wang, Ziyan Xiong, Chunru Lin, Zhehuan Chen, Chuang Gan

We address the challenge of enabling robots to manipulate deformable linear objects (DLOs), such as ropes, cables, and rubber bands. Prior work has primarily focused on narrow, task-specific problems, often relying on real-world demonstrati

ons or handcrafted heuristics. Such approaches, however, struggle to scale to th
e wide variety of materials and tasks encountered in practice, and collecting su
fficiently diverse real-world data is often impractical. Additionally, existing
simulation environments offer limited support for the broad spectrum of material
 behaviors necessary for generalizable DLO manipulation. To overcome these limit
ations, we introduce a differentiable simulator explicitly designed for versatil
e DLO manipulation. Our simulator models a wide range of material properties—inc
luding (in)extensibility, elasticity, bending plasticity, and complex interacti
ons with other objects—providing a robust foundation for learning and evaluating
 manipulation skills. Building on this simulator, we propose a benchmark suite o
f representative tasks that highlight the unique challenges of DLO manipulation.
 The successful execution of these tasks is often hindered by the topological co
mplexity and grasp sensitivity inherent to DLOs. Therefore, we introduce a speci
alized DLO agent that explicitly manages these challenges by proposing strategic
 grasping points and decomposing long-horizon tasks to maximize control authorit
y. Finally, we evaluate various policy-learning algorithms using our framework,
alongside sim-to-real transfer experiments, demonstrating our platform's potenti
al to advance DLO manipulation. Project page: https://dlo-lab-26.github.io/.
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One-Shot Weighted Ensemble Estimation for Federated Quantile Regression: Optimal
 Statistical Guarantees under Heterogeneous Structured Data
Guang Yang, Bo Pan, Chengdi Lian, Xingcai Zhou, Linglong Kong, Yafei Wang, Bei J
iang
Federated Quantile Regression (FQR) has emerged as a powerful modelling paradigm
 for estimating conditional quantiles, offering a more comprehensive understandi
ng of response distributions than standard conditional mean regression. However,
 achieving communication efficiency and optimal statistical guarantees for FQR r
emains challenging, particularly due to the nonsmooth nature of quantile loss fu
nctions and the presence of heterogeneously structured data, where each local ag
ent trains its conditional quantile models with distinct sets of features. In th
is paper, we propose a data-driven, one-shot weighted ensemble estimator for FQR
 that incorporates scalable weighting schemes to effectively leverage the partia
lly observed features at each local agent, thereby enjoying both communication e
fficiency and estimation optimality. Theoretically, we present a unified analysi
s of the proposed learning procedure, establishing that the resulting estimator
exhibits asymptotic normality and attains uniformly minimum variance. Furthermor
e, we investigate the estimator's sensitivity to perturbations introduced by loc
al agents and derive conditions under which the estimator achieves stability and
 enjoys strong out-of-sample generalization. Extensive simulations and real data
 analysis under various scenarios validate the asymptotic normality of our estim
ator and demonstrate its superior estimation accuracy and uniform convergence co
mpared to several baseline methods across a range of quantile levels.
****

MM-DeepResearch: A Simple and Effective Multimodal Agentic Search Baseline
Huanjin Yao, Qixiang Yin, Min Yang, Ziwang Zhao, Yibo Wang, Haotian Luo, Jingyi
Zhang, Jiaxing Huang
We aim to develop a multimodal research agent capable of explicit reasoning and
planning, multi-tool invocation, and cross-modal information synthesis, enabling
 it to conduct deep research tasks. However, we observe three main challenges in
 developing such agents: (1) scarcity of search-intensive multimodal QA data, (2
) lack of effective search trajectories, and (3) prohibitive cost of training wi
th online search APIs. To tackle them, we first propose **Hyper-Search**, a hype
rgraph-based QA generation method that models and connects visual and textual no
des within and across modalities, enabling to generate search-intensive multimod
al QA pairs that require invoking various search tools to solve. Second, we intr
oduce **DR-TTS**, which first decomposes search-involved tasks into several cate
gories according to search tool types, and respectively optimize specialized sea
rch tool experts for each tool. It then recomposes tool experts to jointly explo
re search trajectories via tree search, producing trajectories that successfully
 solve complex tasks using various search tools. Third, we build an offline sear

ch engine supporting multiple search tools, enabling agentic reinforcement learn
ing without using costly online search APIs. With the three designs, we develop
MM-DeepResearch, a powerful multimodal deep research agent, and extensive re
sults shows its superiority across benchmarks.
****

Global Convergence of Adaptive Sensing for Principal Eigenvector Estimation
Alex Saad-Falcon, Brighton Ancelin, Justin Romberg
Principal component analysis classically requires full d-dimensional samples,
yet in various applications hardware limits acquisition to a few scalar measurem
ents per sample. We analyze a compressed variant of Oja's algorithm for estimati
ng the principal eigenvector of the data covariance matrix using only two adapti
ve measurements per sample. At each iteration, we observe one measurement along
the current estimate and one in a random orthogonal direction. We prove that aft
er t iterations, the expected sine-squared error to the true eigenvector is $\
mathcal{O}(\lambda_1\lambda_2 d^2 / (\Delta^2 t))$, where d is the ambient dim
ension, λ_1, λ_2 are the leading eigenvalues, and $\Delta = \lambd
a_1 - \lambda_2$ is the eigengap. We complement this with a matching information
-theoretic lower bound of $\Omega(\lambda_1\lambda_2 d^2 / (\Delta^2 t))$ --- th
e first for compressed eigenvector estimation --- proving that the d^2 factor,
 an additional factor of d compared to the fully-observed minimax rate $\Theta
(\lambda_1\lambda_2 d / (\Delta^2 t))$, is the fundamental cost of compression a
nd cannot be improved. In contrast, any non-adaptive scheme with two measurement
s per iteration suffers $\Omega(\lambda_2^2 d^3 / (\Delta^2 t))$, an additional
power of d. This separates fully-observed, adaptive-compressed, and non-adapti
ve-compressed PCA across three powers of d. Our analysis handles the noisy set
ting where the covariance has nonzero trailing eigenvalues, providing the first
convergence guarantee for adaptive compressed subspace tracking beyond the noise
less case.
****

Geometric Collapse: When Vision Models Fail to Verify Physical Causality
Wentao Zhang, Jinhu Qi, Weiqiang Jin, Yifei Zhang, Chan-Tong Lam, Irwin King
Recent progress in large-scale self-supervised learning has improved dense geome
tric prediction, but it remains unclear whether such scaling yields inference-ti
me physical plausibility checks. We propose Scrambled Edges, a controlled counte
rfactual that injects salient edge-like cues while violating surface continuity,
 illumination coherence, and occlusion ordering. With energy-matched and structu
re-matched controls, we isolate the effect of unsupported edge evidence from hig
h-frequency energy and edge sparsity. Across CNN/ViT/SSL depth predictors on NYU
 Depth v2 and KITTI, Scrambled Edges induce up to 3.2× larger deviation from cle
an predictions than energy-matched noise. The resulting Geometric Collapse propa
gates globally: even with oracle knowledge of the corrupted region, output-level
 repair recovers only 47%, with substantial error outside the mask. These findin
gs provide controlled behavioral evidence that current dense predictors lack rel
iable mechanisms to quarantine physically unsupported edge cues, motivating expl
icit plausibility scoring and selective cue integration.
****

Domain Adaptation with Adaptive f-Divergence: Tighter Variational Representati
on and Generalization Bounds
Zhe Cheng, Fode Zhang, Yifan Zhu, Lingrui Wang, Jiaolong Wang
We study unsupervised domain adaptation (UDA) where measuring cross-domain discr
epancy is critical. Most UDA approaches fix a single f-divergence a priori, wh
ich can be suboptimal across heterogeneous shifts. We propose a framework that (
i) tightens the variational lower bound of an f-divergence by inserting a lear
nable, monotone L-Lipschitz transform τ (Tighter-VR), and (ii) selects th
e divergence family adaptively from data via a likelihood-based criterion. The r
esulting estimator yields more informative and statistically efficient discrepan
cy estimates while recovering prior fixed-divergence methods as special cases. T
heoretically, we derive a target-risk bound whose three components are a transfo
rmed source risk, a Tighter-VR discrepancy between domains, and an ideal-hypothe
sis residual; we further provide finite-sample guarantees using standard complex

ity measures. Empirically, on Office-31, Office-Home, Digits, and VisDA-2017, ou
r method consistently improves accuracy over strong baselines, showing that coup
ling Tighter-VR with adaptive divergence selection is useful for UDA.
****

CINOC: Cardinality-Invariant Neural Operator Policies for Scalable PDE Control
Pietro Zanotta, Dibakar Roy Sarkar, Honghui Zheng, Somdatta Goswami, Jan Drgona
Controlling partial differential equations (PDEs) with learning-based policies r
emains fundamentally limited by fixed-dimensional representations: policies trai
ned for a specific sensor, actuator, or agent configuration typically fail when
the configuration changes. This limitation is particularly severe in multi-agent
 PDE control, where policies do not scale across population sizes without retrai
ning. We address this challenge by introducing **C**ardinality **I**nvariant **N
eural **Operator **C**ontrol (**CINOC**), reformulating PDE control as an op
erator learning problem that maps state fields to continuous control functions a
nd trains them end-to-end through differentiable PDE solvers, yielding policies
that naturally adapt to varying sensor and actuator configurations. Remarkably,
CINOC policies trained on small swarms exhibit cardinality invariance, allowing
for zero-shot transfer to significantly larger populations as well as robustness
 to partial agent failure. This scalability arises from agents sharing a common
policy and coordinating through their physical environment, which produces an em
ergent self-normalization effect. To explain this phenomenon, we provide a theor
em grounded in mean-field theory demonstrating that policy gradients computed fr
om finite-agent systems converge to those of a continuous control limit. Empiric
ally, we validate CINOC on tracking, stabilization, and density transport across
 linear, nonlinear, chaotic, and turbulent PDEs.
****

Geometry-based Schrödinger Bridges for Trustworthy Multimodal Fusion
Jiayu Xiong, Jing Wang, Qi Zhang, Wanlong Wang, Jun Xue
Real-world multimodal systems must be robust against low-quality data, such as s
ensor noise, incomplete multimodal data and conflicting inputs. However, existin
g trustworthy fusion methods rely on the model's own prediction confidence to ju
dge data quality. This creates a circular dependency: when a model is confident
but wrong (overconfident), these methods fail to detect the error. To break this
 loop, we propose Geometry-based Multimodal Fusion (GMF). Instead of relying on
predictions, we evaluate reliability by measuring the physical effort required t
o map input data back to the valid data manifold. We implement this using Diffus
ion Schrödinger Bridges with Rectified Flow, which allows us to calculate Transp
ort Energy as a direct metric for quality. The logic is simple: valid data sits
on the manifold (low energy), while noisy, incomplete data or conflicting data r
equires high energy to be restored. This geometric metric acts as an independent
 judge, effectively flagging unreliable inputs even when the classifier is foole
d. Extensive experiments demonstrate that GMF significantly improves robustness
against severe sensor noise and semantic conflicts compared to confidence-based
baselines.
****

Optimal Domain-Aware Privacy Mechanisms for Synthetic Data Generation
Sajani Vithana, Sangwon Jung, Haoyang Hu, Viveck R. Cadambe, Flavio Calmon, Haew
on Jeong
Differential privacy (DP) imposes fundamental trade-offs between privacy and sta
tistical fidelity in synthetic data generation. While access to public data has
been shown to improve these trade-offs empirically, existing approaches use publ
ic data only indirectly, through pre-processing (e.g., using pre-trained generat
ive models) or post-processing steps (e.g., matching target statistics estimated
 from public datasets), while relying on domain-agnostic DP mechanisms. In this
work, we lay the theoretical framework to study the principled incorporation of
public data into DP mechanisms themselves. We consider normalized histograms as
distribution estimators and characterize the asymptotically optimal domain-aware
 privacy mechanism within a specific class of DP mechanisms. We introduce PubMix
, a public-data-aware DP mechanism that can be used in histogram-based data synt
hesis pipelines. Our experiments demonstrate that PubMix significantly improves

synthetic data generation quality compared to domain-agnostic privacy mechanisms
.
****
A Perturbation Approach to Unconstrained Linear Bandits
Andrew Jacobsen, Dorian Baudry, Shinji Ito, Nicolò Cesa-Bianchi
We revisit the standard perturbation-based approach of Abernethy et al. (2008) i
n the context of unconstrained Bandit Linear Optimization (uBLO). We show the su
rprising result that in the unconstrained setting, this approach effectively red
uces Bandit Linear Optimization (BLO) to a standard Online Linear Optimization (
OLO) problem. Our framework improves on prior work in several ways. First, we de
rive expected-regret guarantees when our perturbation scheme is combined with co
mparator-adaptive OLO algorithms, leading to new insights about the impact of di
fferent adversarial models on the resulting comparator-adaptive rates. We also e
xtend our analysis to dynamic regret, obtaining the first guarantees with optima
l $\sqrt{P_T}$ path-length dependencies without prior knowledge of P_T. We the
n develop the first high-probability guarantees for both static and dynamic regr
et in uBLO. Finally, we discuss lower bounds on the static regret, and prove the
 folklore $\Omega(\sqrt{dT})$ rate for adversarial linear bandits on the Euclide
an ball, which is of independent interest.
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Trust Region Inverse Reinforcement Learning: Explicit Dual Ascent using Local Po
licy Updates
Anish Abhijit Diwan, Davide Tateo, Christopher Mower, Haitham Bou Ammar, Jan Pet
ers, Oleg Arenz
Inverse reinforcement learning (IRL) is typically formulated as maximizing entro
py subject to matching the distribution of expert trajectories. Classical (dual-
ascent) IRL guarantees monotonic performance improvement but requires fully solv
ing an RL problem each iteration to compute dual gradients. More recent adversar
ial methods avoid this cost at the expense of stability and monotonic dual impro
vement, by directly optimizing the primal problem and using a discriminator to p
rovide rewards. In this work, we bridge the gap between these approaches by enab
ling monotonic improvement of the reward function and policy without having to f
ully solve an RL problem at every iteration. Our key theoretical insight is that
 a trust-region-optimal policy for a reward function update can be globally opti
mal for a smaller update in the same direction. This smaller update allows us to
 explicitly optimize the dual objective while only relying on a local search aro
und the current policy. In doing so, our approach avoids the training instabilit
ies of adversarial methods, offers monotonic performance improvement, and learns
 a reward function in the traditional sense of IRL—one that can be globally opti
mized to match expert demonstrations. Our proposed algorithm, *Trust Region Inve
rse Reinforcement Learning (TRIRL)*, outperforms state-of-the-art imitation lear
ning methods across multiple challenging tasks by a factor of 2.4x in terms of a
ggregate inter-quartile mean, while recovering reward functions that generalize
to system dynamics shifts.
****
Non-Parametric Structural Priors for Geometry Theorem Prediction
Junbo Zhao, Ting Zhang, Can Li, Wei He, Jingdong Wang, Hua Huang
Multi-step theorem prediction is a central challenge in geometry problem solving
. Existing neural–symbolic approaches rely heavily on supervised parametric mode
ls, which exhibit limited generalization to evolving theorem libraries. In this
work, we explore training-free theorem prediction through the lens of in-context
 learning (ICL). We identify a critical scalability bottleneck, termed Structura
l Drift: as reasoning depth increases, the performance of vanilla ICL degrades s
harply.
We attribute this to the LLM’s inability to recover latent topological dependenc
ies, leading to unstructured exploration.
To address this issue, we propose Theorem Precedence Graphs, which encode tempor
al dependencies from historical solution traces as directed graphs, and impose e
xplicit topological constraints that effectively prune the search space during i
nference. Coupled with retrieval-augmented graph construction and a stepwise sym

bolic executor, our approach enables LLMs to act as structured planners without any gradient-based optimization. Experiments on the FormalGeo7k benchmark show that our method achieves 89.29\% accuracy, substantially outperforming ICL baselines and matching state-of-the-art supervised models. These results indicate that explicit structural priors offer a promising direction for scaling LLM-based symbolic reasoning.

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TimeSpot: Benchmarking Geo-Temporal Understanding in Vision–Language Models in Real-World Settings

Azmine Toushik Wasi, Shahriyar Zaman Ridoy, Koushik Ahamed Tonmoy, Kinga Tshering, S. M. Muhtasimul Hasan, Wahid Faisal, Tasnim Mohiuddin, Md Rizwan Parvez

Geo-temporal understanding, the ability to infer location, time, and contextual properties from visual input alone, underpins applications such as disaster management, traffic planning, embodied navigation, world modeling, and geography education. Although recent vision–language models (VLMs) have advanced image geo-localization using cues like landmarks and road signs, their ability to reason about temporal signals and physically grounded spatial cues remains limited. To address this gap, we introduce **TimeSpot**, a benchmark for evaluating real-world geo-temporal reasoning in VLMs. **TimeSpot** comprises 1,455 ground-level images from 80 countries and requires structured prediction of temporal attributes (season, month, time of day, daylight phase) and geographic attributes (continent, country, climate zone, environment type, latitude–longitude) directly from visual evidence. It also includes spatial–temporal reasoning tasks that test physical plausibility under real-world uncertainty. Evaluations of state-of-the-art open- and closed-source VLMs show low performance, particularly for temporal inference. While supervised fine-tuning yields improvements, results remain insufficient, highlighting the need for new methods to achieve robust, physically grounded geo-temporal understanding. **TimeSpot** is available at: https://TimeSpot-GT.github.io.

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SINQ: Sinkhorn-Normalized Quantization for Calibration-Free Low-Precision LLM Weights

Lorenz K Muller, Philippe Bich, Jiawei Zhuang, Ahmet Çelik, Luca Benfenati, Lukas Cavigelli

Post-training quantization has emerged as the most widely used strategy for deploying large language models at low precision. Still, current methods show perplexity degradation at bit-widths ≤ 4, partly because representing outliers causes precision issues in parameters that share the same scales as these outliers. This problem is especially pronounced for calibration-free, uniform quantization methods. We introduce SINQ to augment existing post-training quantizers with an additional second-axis scale factor and a fast Sinkhorn–Knopp–style algorithm that finds scales to normalize per-row and per-column variances.

We show that this approximates activation-aware quantization by recovering column scales from the weight matrix structure that are predictive of the typical activation magnitudes the matrix received during training.

Our method has no interactions between layers and can be trivially applied to new architectures to quantize any linear layer.

We evaluate our method on the Qwen3 model family, among others. SINQ reduces the perplexity gap on WikiText2 and C4 by over 50% against uncalibrated uniform quantization baselines, incurs zero to negligible compute overhead, and can be further enhanced by combining it with calibration and non-uniform quantization levels. Code is available in the supplementary.

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Symmetries in language statistics shape the geometry of model representations

Dhruva Karkada, Daniel James Korchinski, Andres Nava, Matthieu Wyart, Yasaman Bahri

Although learned representations underlie neural networks' success, their fundamental properties remain poorly understood.

A striking example is the emergence of simple geometric structures in LLM representations: for example, calendar months organize into a circle, years form a one

-dimensional manifold, and the latitude and longitude of cities can be decoded b
y low-dimensional linear probes.
We show that the statistics of language exhibit a translation symmetry---e.g,. t
he co-occurrence probability of two months depends only on the time interval bet
ween them---and we prove that the latter governs the aforementioned geometric st
ructures in high-dimensional word embedding models.
Moreover, we find that these structures persist even when the co-occurrence stat
istics are strongly perturbed (for example, by removing all sentences in which t
wo months appear together) and at moderate embedding dimension.
We show that this robustness naturally emerges if the co-occurrence statistics a
re collectively controlled by an underlying continuous latent variable.
We empirically validate this theoretical framework in word embedding models, tex
t embedding models, and large language models.
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Dynamics Are Learned, Not Told: Semi-Supervised Discovery of Latent Dynamics Geo
metries For Zero-Shot Policy Adaptation
Zhiming Xu, Weitao Zhou, Xianghui Pan, Nanshan Deng, Chengju Liu, Qijun Chen, Ch
enpeng Yao
Real-world dynamics shifts pose a critical challenge for reinforcement learning,
 yet prior methods typically rely on encoding explicitly identified physical par
ameters into a latent context, a rigid parameterization that proves brittle to u
nmodeled or compound dynamics variations. We instead investigate dynamics adapta
tion through the lens of latent geometry, and show theoretically that target-dom
ain regret is controlled by the Lipschitz smoothness of a trajectory dynamics en
coder. We further prove that this Lipschitz constant can be upper-bounded throug
h optimizing a multi-positive InfoNCE objective, yielding a smooth, task-relevan
t latent topology without privileged dynamics information. On MuJoCo benchmarks,
 our method significantly outperforms explicit identification baselines under se
vere dynamics shifts, including unmodeled structural failures, while simultaneou
sly improving in-distribution stability and latent interpretability. Overall, th
ese results validate that controlling latent smoothness is a principled and scal
able mechanism for robust adaptation.
****

FlowMAP: Flow Matching for Generalizable Agent Planning
Jiarun Fu, Lizhong Ding, Ye Yuan, Qiuning Wei, Zhaohuan Linghu, Yurong Cheng, Ch
angsheng Li, Tianlong Gu, Liang Chang, Guoren Wang
Agent planning faces dynamic heterogeneity—nonstationary observations, dynamics,
 and objectives with sparse, delayed rewards—which dominant methods largely igno
re, leading to poor generalization under environment shifts. We propose Flow-Mat
ching for Agent Planning (FlowMAP), which formulates planning as a continuous-ti
me flow-matching problem by learning a planning-time velocity field that transpo
rts an initial meta-state distribution toward a task-conditioned target. FlowMAP
 introduces Value-Transport Flow Matching to provide a distribution-level planni
ng objective that steers transport toward high-value regions in the meta-state d
istribution, mitigating error accumulation under environmental shifts. To enforc
e alignment between meta-state distribution transport and action--environment in
teraction, FlowMAP further proposes Flow--Policy Co-Training, which jointly opti
mizes the planning flow and policy so that the flow transport directly regulariz
es the policy-induced meta-distribution dynamics. Across diverse agent planning
benchmarks, FlowMAP consistently outperforms strong baselines, yielding improvem
ents in planning generalization.
****

One-Way Policy Optimization for Self-Evolving LLMs
Shuo Yang, Jinda Lu, Kexin Huang, Chiyu Ma, Shaohang Wei, Yuyang Liu, Guoyin Wan
g, Jingren Zhou, Li Yuan
Reinforcement Learning with Verifiable Rewards (RLVR) has become a promising par
adigm for scaling reasoning capabilities of Large Language Models (LLMs). Howeve
r, the sparsity of binary verifier rewards often leads to low efficiency and opt
imization instability. To stabilize training, existing methods typically impose
token-level constraints relative to a reference policy. We identify that such co

nstraints penalize deviations indiscriminately; this can flip verifier-determine d direction when the policy attempts to outperform the reference, thereby suppre ssing gains. To resolve this, we propose **One-Way Policy Optimization (OWPO)**, a method based on the principle of decoupling optimization direction from updat e magnitude. In OWPO, the verifier dictates the update direction, while the refe rence policy serves only to adjust the magnitude. Specifically, OWPO applies asy mmetric reweighting: it performs **Accelerated Alignment** for Inferior deviatio ns (where the policy lags behind the reference) and **Gain Locking** for Superio r deviations (where the policy surpasses the reference). Furthermore, by incorpo rating iterative reference updates, OWPO creates a ``Ratchet Effect'' that conti nuously consolidates gains. Experimental results demonstrate that OWPO outperfor ms strong baselines, including DAPO, OPD, and MOPD, breaking the bottleneck of f ixed priors to enable continuous self-evolution without reliance on external ref erence models.

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TRACER: Persistent Regularization for Robust Multimodal Finetuning

Hesam Asadollahzadeh, Feng Liu, Christopher Leckie, Sarah Monazam Erfani

Mainstream strategies for finetuning pretrained multimodal models often degrade out-of-distribution (OOD) robustness, a phenomenon known as catastrophic forgett ing.

In this paper, we develop a theoretical framework for multimodal contrastive fin etuning, yielding closed-form solutions and a geometric decomposition for these strategies. This framework shows that self-distillation is more effective than o ther regularization approaches to retain the knowledge of the pretrained model. Our analysis reveals a largely overlooked limitation: standard Exponential Movin g Average (EMA) teachers, widely used in robust finetuning, suffer from collapse . To solve this, we prove that a Weighted Moving Average (WMA) teacher maintains a persistent regularizing force over finite horizons and yields bias-free conve rgence in the task subspace while preserving orthogonal knowledge.

These insights motivate **TRACER** (**T**rajectory-**R**obust **A**nchoring for **C**ontrastive **E**ncoder **R**egularization), which combines contrastive lear ning with WMA-guided multi-perspective distillation. Extensive experiments on CL IP finetuning demonstrate consistent OOD accuracy and calibration gains across t hree backbone architectures, and comprehensive ablations confirm that TRACER is both principled and robust to hyperparameter choices. Code is available at [http s://github.com/HesamAsad/TRACER](https://github.com/HesamAsad/TRACER).

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Rethinking Personalization in Large Language Models at the Token Level

Chenheng Zhang, Yijun Lu, Lizhe Fang, Chunyuan Zheng, Jiajun Chai, Xiaohan Wang, Guojun Yin, Wei Lin, Yisen Wang, Zhouchen Lin

With large language models (LLMs) now performing strongly across diverse tasks, there is growing demand for them to personalize outputs for individual users. Pe rsonalization is typically framed as an additional layer on top of a base NLP ta sk, requiring model responses to meet user-specific needs while still accomplish ing the underlying task.

From a token-level perspective, different tokens in a response contribute to per sonalization to varying degrees. Tokens with higher personalization relevance sh ould therefore receive greater emphasis when developing personalized LLMs. Howev er, accurately estimating such personalization degrees remains challenging.

To address this challenge, we propose PerContrast, a self-contrast method that e stimates each output token’s dependence on user-specific information through cau sal intervention. Building on this mechanism, we develop the PerCE loss, which a daptively upweights tokens with higher estimated personalization degrees during training via a bootstrap procedure, enabling the model to alternate between esti mating and optimizing these tokens.

Experiments on multiple LLMs demonstrate that PerCE substantially improves perso nalization performance with minimal additional cost, achieving average gains of over 10\% and up to 68.04\% on the LongLaMP dataset, along with strong cross-tas k and cross-scenario transferability. These results highlight the importance of token-level personalization modeling and establish token-aware training as a sim

ple yet effective paradigm for advancing personalized LLMs.
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Length Generalization Bounds for Transformers
Andy Yang, Pascal Bergsträßer, Georg Zetzsche, David Chiang, Anthony Widjaja Lin
Length generalization is a key property of a learning algorithm that enables it to make correct predictions on inputs of any length, given finite training data. To provide such a guarantee, one needs to be able to compute a length generalization bound, beyond which the model is guaranteed to generalize. This paper concerns the open problem of the computability of such generalization bounds for C-RASP, a class of languages which is closely linked to transformers. A positive partial result was recently shown by Chen et al. for C-RASP with only one layer and, under some restrictions, also with two layers. We provide complete answers to the above open problem. Our main result is the non-existence of computable length generalization bounds for C-RASP (already with two layers) and hence for transformers. To complement this, we provide a computable bound for the positive fragment of C-RASP, which we show equivalent to fixed-precision transformers. For both positive C-RASP and fixed-precision transformers, we show that the length complexity is exponential, and prove optimality of the bounds.
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Trifuse: Enhancing Attention-Based GUI Grounding via Multimodal Fusion
Longhui Ma, Di Zhao, Siwei Wang, Zhao Lv, Miao Wang
GUI grounding maps natural language instructions to the correct interface elements, serving as the perception foundation for GUI agents. Existing approaches predominantly rely on fine-tuning multimodal large language models (MLLMs) using large-scale GUI datasets to predict target element coordinates, which is data-intensive and generalizes poorly to unseen interfaces. Recent attention-based alternatives exploit localization signals in MLLMs attention mechanisms without task-specific fine-tuning, but suffer from low reliability due to the lack of explicit and complementary spatial anchors in GUI images. To address this limitation, we propose Trifuse, an attention-based grounding framework that explicitly integrates complementary spatial anchors. Trifuse integrates attention, OCR-derived textual cues, and icon-level caption semantics via a Consensus-SinglePeak (CS) fusion strategy that enforces cross-modal agreement while retaining sharp localization peaks. Extensive evaluations on four grounding benchmarks demonstrate that Trifuse achieves strong performance without task-specific fine-tuning, substantially reducing the reliance on expensive annotated data. Moreover, ablation studies reveal that incorporating OCR and caption cues consistently improves attention-based grounding performance across different backbones, highlighting its effectiveness as a general framework for GUI grounding.
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Faithful Mobile GUI Agents with Guided Advantage Estimator
Haowen Hu, Pengzhou Cheng, Zheng Wu, Lingzhong Dong, Gongshen Liu, Zhuosheng Zhang
Vision-language model (VLM) based graphical user interface (GUI) agents have shown strong interaction capabilities. However, they often behave unfaithfully, relying on memorized shortcuts rather than grounding actions in displayed screen evidence or user instructions. To address this, we propose **Faithful-Agent**, a faithfulness-first framework that reformulates GUI interaction to prioritize evidence groundedness and internal consistency. Faithful-Agent employs a two-stage pipeline: (i) a faithfulness-oriented SFT stage to instill abstainment behaviors under evidence perturbations; (ii) an RFT stage that further amplifies faithfulness by introducing the guided advantage estimator (GuAE), an anchor-based and variance-adaptive advantage tempering mechanism built upon GRPO. GuAE prevents advantage collapse in low-variance rollout groups under sparse GUI rewards, and with a thought-action consistency reward, Faithful-Agent (Stage II) elevates the Trap SR from 13.88\% to 80.21\% relative to the baseline, while preserving robust general instruction-following performance.
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How RLHF Amplifies Sycophancy
Itai Shapira, Gerdus Benade, Ariel D. Procaccia
Large language models often exhibit increased sycophantic behavior after preference-based post-training, showing a stronger tendency to affirm a user’s stated or implied belief even when this conflicts with factual accuracy or sound judgment.
We present a formal analysis of how alignment from human feedback can increase this failure mode by identifying an explicit amplification mechanism that causally links optimization against a learned reward to bias in the human preference data used for alignment. We show that the direction of behavioral drift is determined by a covariance under the base policy between endorsing the belief signal in the prompt and the learned reward, and that the first-order effect reduces to a simple mean-gap condition. We then analyze reward learning from pairwise comparisons under random utility models like Bradley–Terry and characterize when bias in human annotators’ preferences induces this reward gap.
Next, we propose a training-time intervention designed to neutralize the amplification mechanism itself. Among all post-trained policies that prevent sycophantic behavior from increasing, we characterize the unique policy closest in KL divergence to the unconstrained post-trained policy, and derive the corresponding minimal reward correction as a closed-form agreement penalty. Computational experiments find that reward gaps are common and cause behavioral drift in all the configurations considered.
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Adaptive Batch Sizes Using Non-Euclidean Gradient Noise Scales for Stochastic Sign and Spectral Descent
Hiroki Naganuma, Shagun Gupta, Youssef Briki, Ioannis Mitliagkas, Irina Rish, Parameswaran Raman, Hao-Jun Michael Shi
To maximize hardware utilization, modern machine learning systems typically employ large constant or manually tuned batch size schedules, relying on heuristics that are brittle and costly to tune. Existing adaptive strategies based on gradient noise scale (GNS) offer a principled alternative. However, their assumption of SGD's Euclidean geometry creates a fundamental mismatch with popular optimizers based on generalized norms, such as signSGD / Signum (ℓ_∞) and stochastic spectral descent (specSGD) / Muon (Schatten-∞). In this work, we derive gradient noise scales for signSGD and specSGD that naturally emerge from the geometry of their respective dual norms. To practically estimate these non-Euclidean metrics, we propose an efficient variance estimation procedure that leverages the local mini-batch gradients on different ranks in distributed data-parallel systems. Our experiments demonstrate that adaptive batch size strategies using non-Euclidean GNS enable us to match the validation loss of constant-batch baselines while reducing training steps by up to 66\% for Signum and Muon on a 160 million parameter Llama model.
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ManiSoft: Towards Vision-Language Manipulation for Soft Continuum Robotics
Ziyu Wei, Luting Wang, Chen Gao, Li Wen, Si Liu
Most existing vision-language manipulation research targets rigid robotic arms, whose fixed morphology limits adaptability in cluttered or confined spaces.
Soft robotic arms offer an appealing alternative due to their deformability, but confront challenges such as unreliable proprioception and distributed low-level actuation.
To investigate these challenges, we introduce ManiSoft, a benchmark for vision-language manipulation with soft arms.
ManiSoft features a tailored simulator that couples realistic soft-body dynamics with contact-rich interactions via an elastic force constraint.
On this basis, ManiSoft defines four tasks, each highlighting distinct aspects of deformable control, from basic end-effector coordination to obstacle avoidance.
To support policy training and evaluation, ManiSoft includes an automated pipeline that generates $6{,}300$ diverse scenes and corresponding expert trajectories.

To produce high-quality trajectories at scale, we first employ a high-level planner to decompose each task into a sequence of waypoints, followed by a low-level reinforcement learning policy that generates torque commands to track waypoints.
Benchmarking three representative policy models shows relatively promising results in clean scenes but substantial performance drop under randomization.
Visualization analysis indicates that failures stem primarily from inaccurate visual estimation of proprioceptive state and limited exploitation of deformability for adaptive obstacle avoiding.
We anticipate ManiSoft to serve as a valuable testbed, bridging the gap between rigid and soft arms in the context of vision-language manipulation.
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AdaMeZO: Adam-style Zeroth-Order Optimizer for LLM Fine-tuning Without Maintaining the Moments
Zhijie Cai, Haolong Chen, Guangxu Zhu
Fine-tuning LLMs is necessary for various dedicated downstream tasks, but classic backpropagation-based fine-tuning methods require substantial GPU memory. To this end, a recent work, MeZO, which relies solely on forward passes to fine-tune LLMs, significantly reduces GPU requirements at the cost of slower convergence due to its indifference to loss landscapes. Standard solutions, such as Adam, explore loss landscapes by estimating the first- and second-order moments and storing them in memory to guide the model's movement through dimensions with lower curvature and vice versa. However, directly applying Adam negates MeZO's advantage as it will triple the memory requirement. In light of this, we propose AdaMeZO, a zeroth-order optimizer that leverages Adam-style first- and second-moment estimates without maintaining them in memory. We present a theoretical analysis of AdaMeZO, corroborated by extensive experiments demonstrating its performance, showing that it can outperform MeZO while requiring up to 70% fewer forward passes. Trajectory visualizations affirm AdaMeZO's ability to adapt to diverse loss landscapes.
****
SS-TPT: Stability and Suitability-Guided Test-Time Prompt Tuning for Adversarially Robust Vision-Language Models
Sunoh Kim, Daeho Um
Vision-language models (VLMs) such as CLIP achieve strong zero-shot recognition but remain highly fragile under adversarial perturbations. Recent test-time adaptation defenses improve robustness by leveraging many augmented views, but this leads to impractical slowdown and a clear robustness-throughput trade-off. To address this challenge, we present Stability and Suitability-guided Test-time Prompt Tuning (SS-TPT), evaluating the quality of each augmented view via two complementary scores: (1) stability, measuring prediction invariance to weak augmentations, and (2) suitability, measuring feature-space density among views. These stability and suitability (SS) scores guide both adaptation and inference through an SS-guided consistency loss and an SS-weighted prediction, amplifying trustworthy views while suppressing corrupted ones. Extensive experiments demonstrate that SS-TPT significantly outperforms prior state-of-the-art methods, achieving superior robustness-throughput trade-offs across diverse datasets and varying numbers of views, thereby demonstrating both strong practicality and generality. Our code is available at https://github.com/sunoh-kim/SS-TPT.
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History-Bootstrapped Flow Matching for Inverse Boiling Reconstruction
Xianwei Zou, Sheikh Md Shakeel Hassan, Arthur Feeney, Aparna Chandramowlishwaran
Reconstructing spatiotemporal fields from partial observations is fundamental to scientific inference, from inferring atmospheric states from satellite data to recovering fluid states from imaging. When observations are incomplete, the inverse problem is fundamentally ill-posed: even when the underlying PDE dynamics are Markovian in the full state, partial observation operators induce a non-Markovian posterior that cannot be resolved from a single timestep. We propose a history-bootstrapped autoregressive flow matching (HB-ARFM) for spatiotemporal inverse reconstruction under partial observability. Observation history bootstraps the

initial reconstruction via conditional flow matching, resolving ambiguities. The same conditional transport model is then applied autoregressively, conditioning on both new observations and past predictions to propagate the reconstruction forward in time. We evaluate the method on boiling dynamics reconstruction, recovering full velocity and temperature fields from interface geometry and motion. Across two inverse tasks with varying observation sparsity, HB-ARFM produces physically and temporally valid reconstructions where other models fail.

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Contrastive Geometric Learning Unlocks Unified Structure- and Ligand-Based Drug Design

Lisa Schneckenreiter, Sohvi Luukkonen, Lukas Friedrich, Daniel Kuhn, Günter Klambauer

Structure-based and ligand-based computational drug design have traditionally relied on disjoint data sources and modeling assumptions, limiting their joint use at scale. In this work, we introduce **Con**trastive **G**eometric **L**earning for **U**nified Computational **D**rug D**e**sign (ConGLUDe), a single contrastive geometric model that unifies structure- and ligand-based training. ConGLUDe couples a geometric protein encoder that produces whole-protein representations and implicit embeddings of predicted binding sites with a fast ligand encoder, removing the need for pre-defined pockets. By aligning ligands with both global protein representations and multiple candidate binding sites through contrastive learning, ConGLUDe supports ligand-conditioned pocket prediction in addition to virtual screening and target fishing, while being trained jointly on protein-ligand complexes and large-scale bioactivity data. Across diverse benchmarks, ConGLUDe achieves competitive zero-shot virtual screening performance, substantially outperforms existing methods on a challenging target fishing task, and demonstrates state-of-the-art ligand-conditioned pocket selection. These results highlight the advantages of unified structure-ligand training and position ConGLUDe as a step toward general-purpose foundation models for drug discovery.

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Information-Geometric Adaptive Sampling for Graph Diffusion

Yuhui Lu, Wenjing Liu, Kun Zhan

Standard diffusion models for graph generation typically rely on uniform time-stepping, an approach that overlooks the non-homogeneous dynamics of distributional evolution on complex manifolds. In this paper, we present an information-geometric framework that reinterprets the diffusion sampling trajectory as a parametric curve on a Riemannian manifold. Our key observation is that the Fisher-Rao metric provides a principled measure of the intrinsic distance. By analyzing this metric, we derive the Drift Variation Score (DVS), a geometry-aware indicator that quantifies the instantaneous rate of distributional change. Unlike prior heuristic-based adaptive samplers, our DVS solver enforces a constant informational speed on the statistical manifold, automatically maintaining a uniform rate of distributional change along the sampling trajectory. This equal arc-length strategy ensures that each discretization step contributes equally to the information speed. Theoretical analysis verifies that DVS characterizes the local stiffness of the sampling dynamics in the Fisher-Rao sense. Experimental results on molecule and social network generation show that DVS significantly improves structural fidelity and sampling efficiency.

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Unifying Stacking and Cascading for Efficient Ensemble Inference

Ashwin Gerard Colaco, Sharad Mehrotra, Michael J. De Lucia, Kevin Hamlen, Murat Kantarcioglu, Latifur Khan, Ananthram Swami, Bhavani Thuraisingham, Unnat Jain

We introduce LazyStack, a method for efficient model ensemble inference. The core idea is intuitive: after each model executes, we check whether accumulated evidence is sufficient to exit confidently. Sometimes one model suffices; other times we aggregate predictions from several models via trained meta-learners before reaching confidence. Two insights make this work. First, most inputs follow only 3 to 8 execution trajectories. This reduces the training problem from exponential to linear: we learn aggregators only for these common paths, not all possible model combinations. Second, we formulate trajectory selection as an MDP and us

e value iteration to compute the optimal routing policy, which reveals counterin tuitive model orderings. On intrusion detection, starting with a moderately expe nsive model outperforms starting with the cheapest, because its higher confidenc e enables earlier overall exit. Across vision, text, tabular, and LLM tasks, we achieve up to 38x speedup at 97\%+ accuracy retention compared to a complete ens emble. The result: ensemble-quality predictions at cascade-level cost. Code and a project page are available at https://ashwincolaco.github.io/lazystack.
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JANUS-LORA: A Balanced Low-Rank Adaptation for Continual Learning
Cheng Chen, Pengpeng Zeng, Yuyu Guo, Lianli Gao, Heng Tao Shen, Jingkuan Song
Low-Rank Adaptation (LoRA) has emerged as a promising paradigm for Continual Lea rning.
It independently updates its low-rank factors (A and B), creating a composit e update to the full weight matrix through their interaction.
To prevent catastrophic forgetting, this update should remain orthogonal to the task-specific subspace that contains previously learned knowledge.
However, we identify that this composite update systematically violates this ort hogonality, reintroducing interference and undermining stability.
Furthermore, naively enforcing this orthogonality compromises plasticity, disrup ting the delicate stability-plasticity trade-off.
To resolve these issues, we propose Janus-LoRA, a framework that restores this b alance through two novel components.
Specifically, we first introduce Gradient Rectification, a closed-form solution that mathematically decouples LoRA's factor updates, enforcing orthogonality aga inst the historical knowledge subspace identified by an efficient Online Estimat ion.
Next, to enhance plasticity, we introduce a Decoupled Margin Loss that promotes feature-level separation by pushing new feature representations away from old on es, thus creating distinct, low-interference regions for new learning.
Comprehensive experiments on challenging benchmarks demonstrate that by harmoniz ing parameter-level orthogonality with feature-level separation, Janus-LoRA achi eves a superior balance and establishes new state-of-the-art performance.
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Risk-Bounded Distribution Reconstruction: Stable Statistic Calibration for Long-Tailed Recognition
Guanliang Liu, Wenchao Chen, Long Tian, Xuefei Cao, Hongwei Liu, Bo Chen
Long-tailed recognition suffers from extreme class imbalance, where scarce tail data leads to biased and fragile feature distributions that exacerbate confusion with semantically or visually similar classes. Prior feature-space reconstructi on methods transfer head-class structure or train conditional generators to synt hesize tail features, yet the resulting \emph{statistical} updates are often heu ristic and can degrade multi-class separability when tail estimates are unreliab le. Given this issue, we propose \emph{Risk-Bounded Distribution Reconstruction} (RBDR), an offline statistic calibration framework for the two-stage long-taile d pipeline, grounded in an analysis of rival-induced discriminative directions.
RBDR performs (i) risk-aware mean calibration by softly projecting any candidate update onto a supportive set such that a surrogate discriminative-risk upper bo und does not increase, and (ii) covariance control by shrinking dispersion in a rival subspace while preserving diversity in orthogonal components.
In the frozen-feature reconstruction setting, these calibration components conve rt heuristic reconstruction cues into controllable distributional updates, impro ving both performance and stability across multiple long-tailed benchmarks.
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Euler–Poincaré Neural Dynamics: A Geometric-Mechanics Framework for Scientific S imulation
Sungwoo Park, Jongwon Lee, Jiwoong Kim
We introduce Euler--Poincar\'e Neural Dynamics (EPND), a geometric-mechanics fra mework that casts evolution-operator learning as Lie-group flows for long-horizo n dynamical modeling. Unlike conventional operator-learning approaches that trea t temporal propagation as an unconstrained black-box map, EPND places geometric

mechanics at the core of its architecture, playing a role of the mathematical en
gine. This foundation enables a principled treatment of curvature, symmetry, and
 conservation, with the learned evolution expressed in geometric terms. Building
 on this foundation, we develop the Euler--Poincar\'e Parallel Scan, a parallel
algorithm that leverages the associative algebra of Lie-group compositions to ov
ercome the inefficiencies of sequential computation. By unifying geometric struc
ture with scalable computation, EPND achieves high accuracy, strong stability, a
nd significant parallel acceleration in modeling long-horizon dynamics in versat
ile scientific simulations.
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UniJEPA: Enhancing Robot Policy via Unified Continuous and Discrete Representati
on Learning
Jianke Zhang, Yucheng Hu, Yanjiang Guo, Xiaoyu Chen, Yichen Liu, Wenna Chen, Cha
ochao Lu, Jianyu Chen
Building generalist robot policies that can handle diverse tasks in open-ended e
nvironments is a central challenge in robotics. To leverage knowledge from large
-scale pretraining, prior work (VLA) has typically built generalist policies eit
her on top of vision-language models (VLMs) or generative models. However, both
semantic understanding from vision-language pretraining and visual dynamics mode
ling from visual-generation pretraining are crucial for embodied robots.
Recent unified models of generation and understanding have demonstrated strong c
apabilities in both comprehension and generation through large-scale pretraining
. We posit that robotic policy learning can likewise benefit from the combined s
trengths of understanding, planning and continuous future representation learnin
g. Building on this insight, we introduce UniJEPA, which acquires the ability to
 dynamically model high-dimensional visual features through pretraining on over
1M internet-scale instructional manipulation videos. Subsequently, UniJEPA is fi
ne-tuned on data collected from the robot embodiment, enabling the learning of m
appings from predictive representations to action tokens. Extensive experiments
show our approach consistently outperforms baseline methods in terms of 9\% and
12\% across simulation environments and real-world out-of-distribution tasks.
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τ-Knowledge: Evaluating Conversational Agents over Unstructured Knowledge
Quan Shi, Alexandra Zytek, Pedram Razavi, Karthik R Narasimhan, Victor Barres
Conversational agents are increasingly deployed in knowledge-intensive settings,
 where correct behavior depends on retrieving and applying domain-specific knowl
edge from large, proprietary, and unstructured corpora during live interactions
with users. Yet most existing benchmarks evaluate retrieval or tool use independ
ently of each other, creating a gap in realistic, fully agentic evaluation over
unstructured data in long-horizon interactions. We introduce τ-Knowledge, a
n extension of τ-Bench for evaluating agents in environments where success
depends on coordinating external, natural-language knowledge with tool outputs t
o produce verifiable, policy-compliant state changes. Our new domain, τ-Ban
king, models realistic fintech customer support workflows in which agents must n
avigate roughly 700 interconnected knowledge documents while executing tool-medi
ated account updates. Across embedding-based retrieval and terminal-based search
, even frontier models with high reasoning budgets achieve only $\sim$25.5\% pas
s^1, with reliability degrading sharply over repeated trials. Agents struggle to
 retrieve the correct documents from densely interlinked knowledge bases and to
reason accurately over complex internal policies. Overall, τ-Knowledge prov
ides a realistic testbed for developing agents that integrate unstructured knowl
edge in human-facing deployments.
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ThetaEvolve: Test-time Learning on Open Problems
Yiping Wang, Shao-Rong Su, Zhiyuan Zeng, Eva Xu, Liliang Ren, Xinyu Yang, Zeyi H
uang, Xuehai He, Luyao Ma, Baolin Peng, Hao Cheng, Pengcheng He, Weizhu Chen, Sh
uohang Wang, Simon Shaolei Du, yelong shen
Recent advances in large language models (LLMs) have enabled breakthroughs in ma
thematical discovery, exemplified by AlphaEvolve, a closed-source system that ev
olves programs to improve bounds on open problems. However, it relies on ensembl

es of frontier LLMs to achieve new bounds and is a pure inference system that models cannot internalize the evolving strategies.

We introduce ThetaEvolve, an open-source framework that simplifies and extends AlphaEvolve to efficiently scale both in-context learning and Reinforcement Learning (RL) at test time, allowing models to continually learn from their experiences in improving open optimization problems.

ThetaEvolve features a single LLM, a large program database for enhanced exploration, batch sampling for higher throughput, lazy penalties to discourage stagnant outputs, and optional reward shaping for stable training signals, etc.

ThetaEvolve is the first evolving framework that enable a small open-source model, like DeepSeek-R1-0528-Qwen3-8B, to achieve new best-known bounds on open problems (circle packing and first auto-correlation inequality) mentioned in AlphaEvolve.

Besides, across two models and four open tasks, we find that ThetaEvolve with RL at test-time consistently outperforms inference-only baselines, and the model indeed learns evolving capabilities, as the RL-trained checkpoints demonstrate faster progress and better final performance on both trained target task and other unseen tasks.

We release our code publicly.

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Variational Flow Maps: Make Some Noise for One-Step Conditional Generation

Abbas Mammadov, So Takao, Bohan Chen, Ricardo Baptista, Morteza Mardani, Yee Whye Teh, Julius Berner

Flow maps enable high-quality image generation in a single forward pass. However, unlike iterative diffusion models, their lack of an explicit sampling trajectory impedes incorporating external constraints for conditional generation and solving inverse problems. We put forth _Variational Flow Maps_, a framework for conditional sampling that shifts the perspective of conditioning from "guiding a sampling path", to that of "learning the proper initial noise". Specifically, given an observation, we seek to learn a _noise adapter model_ that outputs a noise distribution, so that after mapping to the data space via flow map, the samples respect the observation and data prior. To this end, we develop a principled variational objective that jointly trains the noise adapter and the flow map, improving noise-data alignment, such that sampling from complex data posterior is achieved with a simple adapter. Experiments on various inverse problems show that VFMs produce well-calibrated conditional samples in a single (or few) steps. For ImageNet, VFM attains competitive fidelity while accelerating the sampling by orders of magnitude compared to alternative iterative diffusion/flow models.

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Do Transformers Need Three Projections? Systematic Study of QKV Variants

Ali Kayyam, Anusha Madan Gopal, M Anthony Lewis

Transformers have become the standard solution for various AI tasks, with the query, key, and value (QKV) attention formulation playing a central role. However, the individual contribution of these three projections and the impact of omitting some remain poorly understood. We systematically evaluate three projection sharing constraints: a) Q-K=V (shared key-value), b) Q=K-V (shared query-key), and c) Q=K=V (single projection). The last two variants produce symmetric attention maps; to address this, we also explore asymmetric attention via 2D positional encodings. Through experiments spanning synthetic tasks, vision (MNIST, CIFAR, TinyImageNet, anomaly), and language modeling (300M and 1.2B parameter models on 10B tokens), we discovered that our transformers perform on par or occasionally better than the QKV transformer. In language modeling, Q-K=V projection sharing achieves 50\% KV cache reduction with only 3.1% perplexity degradation. Crucially, projection sharing is complementary to head sharing (GQA/MQA): combining Q-K=V with GQA-4 yields 87.5% cache reduction, while Q-K=V + MQA achieves 96.9%—enabling practical on-device inference. We further show that Q--K=V preserves quality because keys and values can share representational space, whereas Q=K-V breaks attention directionality. Our results establish projection sharing as a new optimization axis for memory-efficient transformers, especially for edge deployment.

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Language-based Trial and Error Falls Behind in the Era of Experience
Haoyu Wang, Guozheng Ma, Shugang Cui, Yilun Kong, Haotian Luo, Li Shen, Mengya G ao, Yichao Wu, Xiaogang Wang, Dacheng Tao
While Large Language Models (LLMs) excel in language-based agentic tasks, their applicability to unseen, nonlinguistic environments (e.g., symbolic or spatial t asks) remains limited. Previous work attributes this performance gap to the mism atch between the pretraining distribution and the testing distribution. In this work, we demonstrate the primary bottleneck is the prohibitive cost of explorati on: mastering these tasks requires extensive trial-and-error, which is computati onally unsustainable for parameter-heavy LLMs operating in a high dimensional se mantic space. To address this, we propose SCOUT (Sub-Scale Collaboration On Unse en Tasks), a novel framework that decouples exploration from exploitation. We em ploy lightweight "scouts" (e.g., small MLPs) to probe environmental dynamics at a speed and scale far exceeding LLMs. The collected trajectories are utilized to bootstrap the LLM via Supervised Fine-Tuning (SFT), followed by multi-turn Rein forcement Learning (RL) to activate its latent world knowledge. Empirically, SCO UT enables a Qwen2.5-3B-Instruct model to achieve an average score of 0.86, sign ificantly outperforming proprietary models, including Gemini-2.5-Pro (0.60), whi le saving about 60% GPU hours consumption.
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Procedural Pretraining: Warming Up Language Models with Abstract Data
Liangze Jiang, Zachary Shinnick, Anton van den Hengel, Hemanth Saratchandran, Da mien Teney
Pretraining language models directly on web-scale corpora is the de facto paradi gm. We study an alternative where the model is initially exposed to *abstract st ructured data* to ease the subsequent acquisition of rich semantic knowledge, mu ch like humans learning simple logic and mathematics before higher reasoning. We focus on *procedural data*, generated by formal languages and other simple algo rithms, as such abstract data.

We first diagnose the algorithmic skills that different forms of procedural data can improve, often significantly. For example, the accuracy of context recall (Needle-in-a-haystack) jumps from 10 to 98% when a model is pretrained on Dyck se quences (balanced brackets). Second, we study how these gains are reflected in p retraining larger models (up to 1.3B). We find that front-loading as little as 0 .1–0.3% procedural data significantly outperforms standard pretraining on natura l language, code, and informal mathematics (C4, CodeParrot, and DeepMind-Math da tasets). Notably, this also enables the models to reach the same loss value with only 55/67/86% of the original data and thus a comparable reduction in FLOPs. T hird, we explore the mechanisms behind the benefits and find that procedural pre training instills non-trivial structure in both attention and MLP layers. The fo rmer is particularly important for structured domains (e.g. code), and the latte r for language. Finally, we lay a path for combining multiple forms of procedura l data. Our results show that procedural pretraining is a simple, lightweight me ans of improving performance and accelerating language model pretraining, ultima tely suggesting the promise of disentangling knowledge acquisition from reasonin g in LLMs.
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MedREK: Retrieval-Based Editing for Medical LLMs with Key-Aware Prompts
Shujun Xia, Haokun Lin, Yichen Wu, Yinan Zhou, Zixuan Li, Zhongwei Wan, Xingrun Xing, Yefeng Zheng, Xiang Li, Caifeng Shan, Zhenan Sun, Quanzheng Li
Large Language Models (LLMs) hold great promise for healthcare applications, but fast-changing medical knowledge can quickly make their outputs outdated or inac curate, limiting use in high-stakes settings. Model editing can update LLMs with out full retraining, but parameter-based methods often break locality and are ri sky in medicine, making retrieval-based editing a better fit. However, applying model editing methods to the medical domain has two key challenges: (1) retrieva l-based methods suffer from representation overlap within the medical knowledge space that causes inaccurate retrieval and reduces editing accuracy; (2) existin

g medical editing methods are restricted to single-sample edits, while batch-edi
ting remains largely unexplored despite its importance for real-world applicatio
ns. To address these challenges, we construct MedVersa, an expanded benchmark th
at evaluates single and batch edits across broader medical coverage under strict
 locality constraints. We then propose MedREK, a retrieval-based editing framewo
rk that integrates a shared query–key module for precise matching with an attent
ion-based prompt encoder for informative guidance. Experiments across various me
dical benchmarks show that our MedREK consistently improves key metrics and prov
ides the first validated solution for batch editing in medical LLMs.
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U-Cast: A Surprisingly Simple and Efficient Frontier Probabilistic AI Weather Fo
recaster

Salva Rühling Cachay, Duncan Watson-Parris, Rose Yu

AI-based weather forecasting now rivals traditional physics-based ensembles, but
 state-of-the-art (SOTA) models rely on specialized architectures and massive co
mputational budgets, creating a high barrier to entry. We demonstrate that such
complexity is unnecessary for frontier performance. We introduce U-Cast, a proba
bilistic forecaster built on a standard U-Net backbone trained with a simple rec
ipe: deterministic pre-training on Mean Absolute Error followed by short probabi
listic fine-tuning on the Continuous Ranked Probability Score (CRPS) using Monte
 Carlo Dropout for stochasticity. As a result, our model matches or exceeds the
probabilistic skill of GenCast and IFS ENS at 1.5° resolution while reduc
ing training compute by over $10\times$ compared to leading CRPS-based models an
d inference latency by over $10\times$ compared to diffusion-based models. U-Cas
t trains in under 12 H200 GPU-days and generates a 15-day ensemble forecast in 3
 seconds. These results suggest that scalable, general-purpose architectures pai
red with efficient training curricula can match complex domain-specific designs
at a fraction of the cost, opening the training of frontier probabilistic weathe
r models to the broader community.
****

Estimation of Treatment Effects Under Nonstationarity via the Truncated Policy G
radient Estimator

Ramesh Johari, Tianyi Peng, Wenqian Xing

Randomized experiments (or A/B tests) are widely used to evaluate interventions
in dynamic systems such as recommendation platforms, marketplaces, and digital h
ealth. In these settings, interventions affect both current and future system st
ates, so estimating the global average treatment effect (GATE) requires accounti
ng for temporal dynamics, which is especially challenging in the presence of non
stationarity; existing approaches suffer from high bias, high variance, or both.
 In this paper, we address this challenge via the novel Truncated Policy Gradien
t (TPG) estimator, which replaces instantaneous outcomes with short-horizon outc
ome trajectories. The estimator admits a policy gradient interpretation: it is a
 truncation of the first-order approximation to the GATE, yielding provable redu
ctions in bias and variance in nonstationary Markovian settings. We further esta
blish a central limit theorem for the TPG estimator and develop a consistent var
iance estimator that remains valid under nonstationarity with single-trajectory
data. We validate our theory with two real-world case studies. The results show
that relative to existing approaches, a well-calibrated TPG estimator can achiev
e a favorable balance between bias and variance in nonstationary settings, highl
ighting the value of the policy-gradient perspective for designing effective est
imators under complex dynamics.
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xKV: Cross-Layer KV-Cache Compression via Aligned Singular Vector Extraction

Chi-Chih Chang, Wei-Cheng Lin, Chien-Yu Lin, Hung-Yueh Chiang, Yash Akhauri, Xil
ai Dai, Huiqiang Jiang, Yucheng Li, Luis Ceze, Kai-Chiang Wu, Mohamed S. Abdelfa
ttah

Long-context Large Language Models (LLMs) enable powerful applications but incur
 high memory costs due to the key-value states (KV-Cache). Recent studies attemp
t to share KV-Cache across layers, but these approaches either require expensive
 pretraining or rely on per-token cross-layer cosine similarity that is often li

mited in practice. We show, via Centered Kernel Alignment (CKA), that the domina nt singular vectors of KV-Cache are well aligned across layers. Motivated by thi s observation, we propose xKV, a post-training compression method that jointly f actorizes grouped-layer KV-Cache into a shared low-rank subspace, substantially reducing KV-Cache memory.
Across widely used LLMs, xKV achieves up to 8× KV-Cache compression while preser ving accuracy on long-context tasks and in multi-turn settings. To further impro ve efficiency, we introduce Selective Reconstruction (SR) at decode time. Combin ed with SR, xKV achieves up to 4.23× end-to-end speedup over the full attention baseline, and surpasses notable baselines with 30% higher throughput under a sim ilar accuracy level. Overall, xKV provides a plug-and-play approach to reduce bo th memory and latency for long-context LLM inference. Our code is publicly avail able at: https://github.com/abdelfattah-lab/xKV.
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TF-FACE: Time-Frequency Fusion Learning via Frequency-Domain Adaptive and Contro llable Enhancement for Trajectory Prediction
Dongjian Song, Yunhao Meng, Songjun Huang, Jiayi Han
Accurately predicting the future trajectories of traffic participants is critica l for safe, efficient, and human-friendly autonomous driving. Existing learning- based trajectory prediction methods are predominantly time-domain and insufficie ntly exploit latent frequency information, which limits their capability to capt ure low-frequency long-term dependencies and high-frequency short-term dynamics. To address this, we propose TF-FACE, a time-frequency learning framework via fr equency-domain adaptive and controllable enhancement. TF-FACE introduces a fusio n encoder with learnable gated frequency-domain attention that adaptively manipu lates band-specific features for trajectory prediction. Building on the fused re presentation, we design a dual-stage decoder and a band-specific time–frequency dual-consistency loss to enable controllable decoupling and coupling across long - and short-term temporal scales, global and local scales, and then generate fin al multimodal predictions. Experiments on Argoverse 1 demonstrate that TF-FACE a chieves state-of-the-art accuracy, while maintaining real-time inference for aut onomous driving. Additional experiments are conducted on Argoverse 2, further va lidating TF-FACE's performance and generalizability.
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SIKA-GP: Accelerating Gaussian Process Inference with Sparse Inducing Kernel App roximations for Bayesian Deep Learning
Wenyuan Zhao, Rui Tuo, Chao Tian
Gaussian processes (GPs) provide a principled Bayesian framework for uncertainty estimation, but their computational complexity severely limits scalability to l arge datasets. We propose SIKA-GP, which accelerates GP inference using sparse i nducing kernel approximations based on a dyadic ordered template basis, incurrin g only ${O}(\log M)$ complexity dependence on the number of inducing points. Our approach constructs compact and expressive kernel representations from sparsely activated bases, enabling efficient tensorized GPU computation and seamless int egration with modern large-scale models. SIKA-GP can be naturally embedded into Bayesian neural networks (BNNs) with sparse activations, yielding significant sp eedups in both training and inference without sacrificing predictive performance . The method naturally extends to deep feature learning, addressing the scalabil ity challenges introduced by deep architectures and high-dimensional feature rep resentations. Empirical results on vision and transformer-based language benchma rks demonstrate that our approach consistently delivers fast and accurate GP mod els, providing a principled path toward scalable kernel learning.
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Hierarchical Multi Scale Graph Neural Networks: Scalable Heterophilous Learning with Oversmoothing and Oversquashing Mitigation
MD SAZZAD Hossen, Avimanyu Sahoo
Graphs with heterophily, where adjacent nodes carry different labels, are preval ent in real-world applications, from social networks to molecular interactions. However, existing spectral Graph Neural Network (GNN) approaches tailored for h eterophilous graph classification suffer from hub-dominated (node with large deg

ree) aggregation and oversmoothing, as their suboptimal polynomial filters introduce approximation errors and blend distant signals.
To address the degree-biased aggregation and suboptimal polynomial filtering, we introduce a Hierarchical Multi■view HAAR (HMH), a novel spectral graph■learning framework that scales in near■linear time. HMH first learns feature- and structure-aware \emph{signed} affinities via a heterophily-aware encoder, then constructs a soft graph hierarchy guided by these embeddings. At each hierarchical level, HMH constructs a sparse, orthonormal, and locality-aware Haar basis to apply learnable spectral filters in the frequency domain. Finally, skip-connection unpooling layers combine outputs from all hierarchical levels back into the original graph, effectively preventing hub domination and long-range signal bottleneck (over-squashing). Experimentation shows that HMH outperforms state■of■the■art spectral baselines, achieving up to a 3 \% improvement on node classification and 7 \% on graph classification datasets, all while maintaining linear scalability. Our code is available at https://github.com/sazzad1008/Scalable-HMH101
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CoCoEdit: Content-Consistent Image Editing via Region Regularized Reinforcement Learning
Yuhui Wu, Chenxi Xie, Ruibin Li, Liyi Chen, Qiaosi Yi, Lei Zhang
Image editing has achieved impressive results with the development of large-scale generative models. However, existing models mainly focus on the editing effects of intended objects and regions, often leading to unwanted changes in unintended regions.
We present a post-training framework for $\textbf{Co}$ntent-$\textbf{Co}$nsistent $\textbf{Edit}$ing ($\textbf{CoCoEdit}$) by using region regularized reinforcement learning.
We first augment existing editing datasets with refined instructions and masks, from which 40K diverse and high quality samples are curated as training set. We introduce a pixel-level similarity reward that complements MLLM-based rewards, enabling models to ensure both editing quality and content consistency during the editing process. To overcome the spatial-agnostic nature of the rewards, we propose a region-based regularizer, aiming to preserve non-edited regions for high-reward samples while encouraging editing effects for low-reward samples.
For evaluation, we annotate editing masks for GEdit-Bench and ImgEdit-Bench, introducing pixel-level similarity metrics to measure content consistency and editing quality.
Applying CoCoEdit to Qwen-Image-Edit and FLUX-Kontext, we achieve not only superior editing scores to state-of-the-art models, but also significantly better content consistency, measured by PSNR/SSIM metrics and human subjective ratings.
Codes, data and models of CoCoEdit can be found at https://github.com/langmanbusi/CoCoEdit.
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Diffract: Spectral View of LLM Domain Adaptation
Nikita Borodin, Maria Krylova, Artem Zabolotnyi, Dmitry Aspisov, Egor Shikov, Nikita Tyuplyaev, Oleg Travkin, Roman Alferov, Dmitry Vinichenko
We study continual pre-training (CPT) as a mechanism for adapting general-purpose large language models to specialized domains: mathematics, instruction, code, and natural text. Using singular value decomposition of weight matrices, we find that CPT leaves singular value spectra largely invariant, with adaptation driven mainly by changes in singular vectors. An analysis of attention-head projection matrices reveals strong, domain-dependent **head heterogeneity**, which we exploit to define a head importance criterion: up to **60%** of head updates can be removed without measurable quality loss. Selectively rewinding low-importance heads to their pre-trained state improves benchmark accuracy by up to **4%** versus the fully trained baseline. Finally, we identify **domain connectivity**—linear interpolation between CPT checkpoints yields smooth domain-quality interpolation without notable degradation on either domain—and release Diffract, an open-source toolkit for scalable spectral analysis of billion-parameter models.
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Beyond Extrapolation: Knowledge Utilization Paradigm with Bidirectional Inspirat

ion for Time Series Forecasting
Liu Chong, Yingjie Zhou, Hao Li, Pengyang Wang, Qingsong Wen, Ce Zhu
Time-series forecasting is critical in various scenarios, such as energy, transp
ortation, and public health.
  However, most existing forecasters rely primarily on one-way inference, \texti
t{i.e.}, mapping \textbf{history} to \textbf{target}, and overlook the structura
l information provided by a revised natural chain (``\textbf{history} (model inp
ut) -- \textbf{target} (ground-truth output) -- \textbf{post-target continuat
ion}'').
The post-target continuation records how trajectories evolve after the target, w
hich can help stabilize forecasting, but it is not observable at inference time.
In this work, we aim to obtain an approximate proxy of the post-target continuat
ion for the current input, providing structural knowledge for bidirectional fore
casting.
This idea is instantiated as KUP-BI (Knowledge Utilization Paradigm with Bidirec
tional Inspiration), a new time-series modeling paradigm that distills continuat
ion-style knowledge (as an approximate post-target continuation proxy) from a \e
mph{train-only} historical library and integrates it into standard forecasting b
ackbones.
The input stream and the continuation-proxy stream are fused via a lightweight f
eature-level gating module.
This design does not introduce information beyond what is already contained in t
he training trajectories; instead, it provides a structured inductive bias that
helps backbones exploit typical continuation patterns rather than relying solely
 on parametric extrapolation.
Experimental results on six public datasets show that KUP-BI consistently improv
es the forecasting performance of state-of-the-art models, with small additional
 overhead.
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Beyond Benchmarks: Toward Causally Faithful Evaluation of Large Language Models
Zhengshuyuan Tian, Wanling Gao, Chuanxin Lan, Chenxi Wang, Lei Wang, Guoxin Kang
, Zhengxin Yang, Yunyou Huang, Xuehai Hong, Jianfeng Zhan
Current LLM evaluations often conflate benchmark performance with intrinsic mode
l capability. This is misleading, as observed outcomes arise from the entire eva
luation system, including datasets, prompting methods, decoding parameters, and
the software–hardware stack, rather than the model alone. When this system is un
der-specified, attribution becomes unreliable; in practice, evaluation choices a
lone can induce accuracy swings of up to 70%. This challenge is compounded by th
e open-ended nature of LLM evaluation, where questions span languages, domains,
and usage styles, forming variable and implicitly shifting datasets. Consequentl
y, strong performance on static benchmarks may reflect surface alignment or data
set-induced effects rather than robust capability. Prior studies often focus on
individual components or manually-curated small-scale dataset variants, overlook
ing interactions and dataset-related confounding. To address these limitations,
we propose LLM evaluatology, a principled framework that grounds LLM evaluation
in a causally motivated system design. It combines structured causal modeling as
 an intervention-oriented lens with factorial decomposition under design of expe
riments, quantifying main and interaction effects while using instance-level int
erventions to probe dataset-induced effects. By jointly modeling evaluation comp
onents and structured question variations, LLM evaluatology enables more interpr
etable, reproducible, and carefully attributed assessment of model capability.
Our framework is publicly available at [GitHub](https://github.com/BenchCouncil/
llm-evaluatology).
****
PolySAE: Modeling Feature Interactions in Sparse Autoencoders via Polynomial Dec
oding
Panagiotis Koromilas, Andreas D. Demou, James Oldfield, Yannis Panagakis, Mihali
s Nicolaou
Sparse autoencoders (SAEs) interpret neural network representations by decomposi
ng activations into sparse combinations of dictionary atoms. However, SAEs assum

e features combine additively through linear reconstruction, an assumption that cannot capture compositional structure: linear models cannot distinguish whether ''Starbucks'' arises from the composition of ''star'' and ''coffee'' features or merely their co-occurrence. This forces SAEs to allocate monolithic features for compound concepts rather than decomposing them into interpretable constituents. We introduce PolySAE, which extends the SAE decoder with higher-order terms to model feature interactions while preserving the linear encoder essential for interpretability. Through low-rank tensor factorization on a shared projection subspace, PolySAE captures pairwise and triple feature interactions with small parameter overhead (3\% on GPT2). Across four language models and three SAE variants, PolySAE achieves an average improvement of $\sim$8\% in probing F1 while maintaining comparable reconstruction error, and produces 2--10$\times$ larger Wasserstein distances between class-conditional feature distributions. Critically, learned interaction weights exhibit negligible correlation with co-occurrence frequency ($r = 0.06$ vs $r = 0.82$ for SAE feature covariance), suggesting that polynomial terms capture compositional structure largely independent of surface statistics. Finally, the learned interaction directions causally steer model outputs toward the corresponding compositional semantics.

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VIPO: Value Function Inconsistency Penalized Offline Reinforcement Learning

Xuyang Chen, Keyu Yan, Guojian Wang, Lin Zhao

Offline reinforcement learning (RL) learns effective policies from pre-collected datasets, offering a practical solution for applications where online interactions are risky or costly. Model-based approaches are particularly advantageous for offline RL, owing to their data efficiency and generalizability. However, due to inherent model errors, model-based methods often artificially introduce conservatism guided by heuristic uncertainty estimation, which can be unreliable. In this paper, we introduce VIPO, a novel model-based offline RL algorithm that incorporates self-supervised feedback from value estimation to enhance model training. Specifically, the model is learned by additionally minimizing the inconsistency between the value learned directly from the offline data and the value estimated from the model. We perform comprehensive evaluations from multiple perspectives to show that VIPO can learn a highly accurate model efficiently and consistently outperform existing methods. In particular, it achieves state-of-the-art performance on almost all tasks in both D4RL and NeoRL benchmarks. Overall, VIPO offers a \textit{general framework} that can be readily integrated into existing model-based offline RL algorithms to systematically enhance model accuracy. Our code is available at~\url{https://anonymous.4open.science/r/vipo2025-8FD4}.

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Bridging Spherical Black-Box Optimizers

Johannes Ackermann, Stefano Peluchetti

When gradient information is unavailable, black-box optimization (BBO) methods provide a practical alternative.

While Evolution Strategies (ES), Consensus-Based Optimization (CBO), Optimization via Integration (OVI), and related methods have each been studied independently, their connections remain underexplored.

We unify these approaches within a common theoretical framework, revealing that they differ primarily in two design choices: fitness aggregation (controlling sharpness preference) and consensus scope (controlling modality).

Leveraging these insights, we introduce hybrid optimizers that interpolate between existing methods.

Our ES-OVI hybrid allows explicit control over the preference for flat minima, enabling a trade-off between performance and robustness in continuous control tasks.

Our CBO-OVI hybrids combine the higher-dimensional efficiency of parametric methods with the multimodal capabilities of particle-based approaches, achieving competitive results on language model merging under limited evaluation budgets.

We validate our methods on standard BBO benchmarks and higher-dimensional locomotion tasks, demonstrating that the hybrid methods can outperform their constituent algorithms.

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The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence
Yichao Cai, Zhen Zhang, Yuhang Liu, Javen Qinfeng Shi
While InfoNCE underlies modern contrastive learning, its geometric mechanisms remain under-characterized beyond the canonical alignment--uniformity decomposition. We develop a measure-theoretic framework in which representation measures evolve on a fixed embedding manifold. In the large-batch limit, we prove value and gradient consistency, linking the stochastic objective to explicit deterministic energy landscapes and revealing a geometric bifurcation between unimodal and symmetric multimodal regimes. In the unimodal case, the intrinsic energy is strictly convex and admits a unique Gibbs equilibrium, showing that entropy acts as a tie-breaker within the aligned basin. In the multimodal case, the intrinsic geometry becomes cross-coupled and contains a persistent negative symmetric divergence term: each modality's marginal reshapes the effective landscape of the other, allowing strong pairwise alignment to coexist with a persistent modality gap. Controlled synthetic experiments and analyses of pretrained CLIP representations support these predictions. Overall, our results shift the analytical lens from pointwise discrimination to population geometry, showing that pairwise alignment alone is insufficient to control cross-modal marginal structure.
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Lookahead Path Likelihood Optimization for Diffusion LLMs
Xuejie Liu, Vit Chun Yap, Yitao Liang, Anji Liu
Diffusion Large Language Models (dLLMs) support arbitrary-order generation, yet their inference performance critically depends on the unmasking order. Existing strategies rely on heuristics that greedily optimize local confidence, offering limited guidance for identifying unmasking paths that are globally consistent and accurate. To bridge this gap, we introduce path log-likelihood (Path LL), a trajectory-conditioned objective that strongly correlates with downstream accuracy and enables principled selection of unmasking paths. To optimize Path LL at inference time, we propose POKE, an efficient value estimator that predicts the expected future Path LL of a partial decoding trajectory. We then integrate this lookahead signal into POKE-SMC, a Sequential Monte Carlo-based search framework for dynamically identifying optimal unmasking paths. Extensive experiments across 6 reasoning tasks show that POKE-SMC consistently improves accuracy, achieving 2\%--3\% average gains over strong decoding-time scaling baselines at comparable inference overhead on LLaDA models and advancing the accuracy--compute Pareto frontier.
****
Improved Dimension Dependence for Bandit Convex Optimization with Gradient Variation
Hang Yu, Yu-Hu Yan, Peng Zhao
Gradient-variation online learning has drawn increasing attention due to its deep connections to game theory and optimization. It has been studied extensively in the full-information setting, but is underexplored with bandit feedback. In this work, we focus on gradient variation in Bandit Convex Optimization (BCO) with two-point feedback. By proposing a refined analysis of the *non-consecutive* gradient variation, a fundamental quantity in gradient variation with bandit feedback, we improve the dimension dependence for both convex and strongly convex functions compared with the best known results (Chiang et al., 2013). Our improved analysis of the non-consecutive gradient variation also implies other favorable problem-dependent guarantees, such as gradient-variance and small-loss regret bounds. Beyond the two-point setup, we demonstrate the versatility of our technique by achieving the *first* gradient-variation bound for one-point bandit linear optimization over hyper-rectangular domains. Finally, we validate the effectiveness of our results in more challenging tasks such as dynamic and universal regret minimization, establishing the *first* gradient-variation dynamic and universal regret bounds for two-point BCO.
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Topology-Aware Contrastive Learning: Regulating Representation Connectivity via

Persistent Homology
Jiaxin Sun, Yuhua Qian, Yang Wang
Standard contrastive learning minimizes geometric distance between positive pair
s, implicitly assuming that strict compactness optimizes discrimination. However
, this topology-agnostic confusion neglects intrinsic data structures and topolo
gical complexity, leading to class confusion—particularly when aggressive augmen
tations induce semantic drift. To address this, we propose Topology-Aware Contra
stive Learning, a framework that shifts the objective from geometric singularity
 to topological connectivity. Leveraging Persistent Homology, we explicitly regu
late the connectivity of the latent space, ensuring positive pairs maintain an $
\alpha\text{--}\beta$ connectivity that balances intra-class cohesion with separ
ability. Theoretically, we formally define the topology-agnostic confusion pheno
menon, prove that excessive compactness strictly lower-bounds the probability of
 confusion and derive a generalization bound demonstrating that richer topologic
al connectivity tightens downstream risk. Furthermore, we establish a measure-th
eoretic framework to mitigating the sensitivity of our method against varying au
gmentation strengths. Empirical results on benchmarks confirm that our approach
enhances representation quality and reduces reliance on specific augmentation st
rategies compared to standard baselines.
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The Role of Target Update Frequencies in Q-Learning
Simon Weissmann, Tilman Aach, Benedikt Wille, Sebastian Kassing, Leif Döring
The target network update frequency (TUF) is a central stabilization mechanism i
n (deep) Q-learning. However, its selection remains poorly understood and is oft
en treated merely as another tunable hyperparameter rather than as a principled
design decision. This work provides a theoretical analysis of target fixing in t
abular Q-learning through the lens of approximate dynamic programming. We view p
eriodic Q-learning as a nested optimization scheme in which each outer iteration
 applies an inexact Bellman optimality operator, approximated by a generic inner
 loop optimizer. Rigorous theory yields a finite-time convergence analysis for t
he asynchronous sampling setting, specializing to stochastic gradient descent in
 the inner loop. Our results deliver an explicit characterization of the bias–va
riance trade-off induced by the target update period, showing how to optimally s
et this critical hyperparameter. We prove that constant target update schedules
are suboptimal, incurring a logarithmic overhead in sample complexity that is en
tirely avoidable with adaptive schedules. Our analysis shows that the optimal ta
rget update frequency decreases geometrically over the course of the learning pr
ocess.
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GEPC: Group-Equivariant Posterior Consistency for Out-of-Distribution Detection
in Diffusion Models
Rouzoumka Yadang Alexis, Jean Pinsolle, Eugénie TERREAUX, Christèle Morisseau, J
ean-Philippe Ovarlez, Chengfang Ren
Diffusion models learn a time-indexed score field $\mathbf{s}_\theta(\mathbf{x}_
t,t)$ that often inherits approximate equivariances (flips, rotations, circular
shifts) from in-distribution (ID) data and convolutional backbones. Most diffusi
on-based out-of-distribution (OOD) detectors exploit score magnitude or local ge
ometry (energies, curvature, covariance spectra) and largely ignore equivariance
s. We introduce Group-Equivariant Posterior Consistency (GEPC), a training-free
probe that measures how consistently the learned score transforms under a finite
 group G, detecting equivariance breaking even when score magnitude remains un
changed. At the population level, we propose the ideal GEPC residual which avera
ges an equivariance-residual functional over G, and we derive ID upper bounds
and OOD lower bounds under mild assumptions. GEPC requires only score evaluation
s and produces interpretable equivariance-breaking maps. On OOD image benchmark
datasets, we show that GEPC achieves competitive or improved AUROC compared to r
ecent diffusion-based baselines while remaining computationally lightweight. On
high-resolution synthetic aperture radar imagery where OOD corresponds to target
s or anomalies in clutter, GEPC yields strong target-background separation and v
isually interpretable equivariance-breaking maps. The official implementation is

available at https://github.com/RouzAY/gepc-diffusion/.
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Optimal Stopping in Latent Diffusion Models
Yu-Han Wu, Quentin Berthet, Gérard Biau, Claire Boyer, Romuald Elie, Pierre Marion
We identify and analyze a surprising phenomenon of $\textit{Latent}$ Diffusion Models (LDMs) where the final steps of the diffusion can $\textit{degrade}$ sample quality. In contrast to conventional arguments that justify early stopping for numerical stability, this phenomenon is intrinsic to the dimensionality reduction in LDMs. We provide a principled explanation by analyzing the interaction between latent dimension and stopping time. Under a Gaussian framework with linear autoencoders, we characterize the conditions under which early stopping is needed to minimize the distance between generated and target distributions. More precisely, we show that lower-dimensional representations benefit from earlier termination, whereas higher-dimensional latent spaces require later stopping time. We further establish that the latent dimension interplays with other hyperparameters of the problem such as constraints in the parameters of score matching. Crucially, this framework suggests that the reconstruction quality of the autoencoder alone can serve as a proxy to estimate the potential performance of the full LDM. Experiments on synthetic and real datasets illustrate these properties, underlining that early stopping can improve generative quality. Together, our results offer a theoretical foundation for understanding how the latent dimension influences the sample quality, and highlight stopping time as a key hyperparameter in LDMs.
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Compressed Sensing for Capability Localization in Large Language Models
Anna Bair, Yixuan Even Xu, Mingjie Sun, J Zico Kolter
Large language models (LLMs) exhibit a wide range of capabilities, including mathematical reasoning, code generation, and linguistic behaviors. We show that Transformer architectures contain small subsets of attention heads that are necessary for certain capabilities. Zeroing out as few as five task-specific heads can degrade performance by up to $60\\%$ on standard benchmarks measuring the capability of interest, while largely preserving performance on unrelated tasks. We introduce a compressed sensing-based method that exploits the sparsity of these heads to identify them via strategic knockouts and a small number of model evaluations. We validate these findings across Llama and Qwen models ranging from 1B to 14B parameters and a diverse set of capabilities including mathematical abilities and code generation, revealing a modular organization in which specialized capabilities are dependent on sparse, functionally distinct components. Overall, our results suggest that capability localization is a general organizational principle of Transformer language models, with implications for interpretability, model editing, and AI safety. Code is released at https://github.com/locuslab/llm-components.
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Coupled Trigger Optimization and Vulnerable Parameter Alignment for Persistent Backdoor Attacks on Federated Learning
Zhixuan Ma, Haichang Gao, Shangwen Li, Ping Wang, Han Yu
Federated learning (FL) is vulnerable to backdoor attacks. Yet sustaining backdoor effectiveness under repeated aggregation remains challenging. Existing methods often rely on heuristic trigger designs or indiscriminant parameter manipulation, leading to rapid decay or detectable anomalies. In this work, we view FL backdoor persistence through the lens of optimization dynamics, and argue that long-lasting attacks require alignment between trigger-induced representations and aggregation-stable parameter directions. Based on this insight, we propose the Coupled Trigger Optimization and Vulnerable Parameter Alignment (CTO-VPA) FL backdoor attack method.
By constraining updates to this coupled subspace, backdoor behaviors can be embedded into optimization-stable directions while preserving benign performance. Experiments across multiple datasets and defense settings show that CTO-VPA achieves substantially improved persistence and robustness compared to prior attacks,

highlighting the importance of trigger–parameter coupling in FL settings. The co
de is available at \url{https://github.com/SwLi415/COVERT}.
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Test-Time Reinforcement Learning for Flow Matching
Jili Chen, Changqin Huang, Qionghao Huang, Yaxin Tu, Zhonglong Zheng, Xiaodi Hua
ng
Flow-matching has emerged as a leading framework for high-fidelity text-to-image
 generation. However, its alignment with human preferences through RL is often h
indered by substantial computational overhead. In this paper, we introduce Flow-
TTRL, the first test-time reinforcement learning framework that achieves alignme
nt on the fly. Our approach reinterprets intermediate latent representations as
an implicit policy and utilizes SDE-based rollouts to explore high-reward trajec
tories within the learned vector field. Specifically, we propose a two-stage opt
imization strategy: Proximal Reward Difference Prediction (PRDP) ensures structu
ral stability in high-noise regimes through pairwise reward regression, while Gr
oup Relative Policy Optimization (GRPO) refines fine-grained aesthetic details b
y maximizing relative advantages within sampled candidate groups. Experimental r
esults show that Flow-TTRL significantly boosts aesthetic quality, text-image al
ignment, and human preference across diverse backbones. On the GenEval benchmark
, Flow-TTRL elevates the accuracy of SD 3.5-Medium from 63\% to 87\% and Flux.1
Dev from 66\% to 83\%. Furthermore, our framework achieves an average gain of 15
\% to 20\% across T2I-CompBench metrics, delivering performance comparable to st
ate-of-the-art RL-based fine-tuning methods without the need for additional fine
-tuning. Our code is available at ~\url{https://github.com/TheShy-Dream/Flow-TTR
L}.
****

Regret-Based Federated Causal Discovery with Unknown Interventions
Federico Baldo, Charles K. Assaad
Most causal discovery methods recover a completed partially directed acyclic gra
ph (CPDAG) representing a Markov equivalence class from observational data. Rece
nt work has extended these methods to federated settings to address data decentr
alization and privacy constraints, but often under idealized assumptions that al
l clients share the same causal model. Such assumptions are unrealistic in pract
ice, as client-specific policies, for instance, across hospitals, naturally indu
ce heterogeneous and unknown interventions. In this work, we address federated c
ausal discovery under unknown client-level interventions. We propose I-PERI, a n
ovel federated algorithm that first recovers the CPDAG common to all clients and
 then orients additional edges by exploiting structural differences induced by i
nterventions across clients. This yields a tighter equivalence class, which we c
all the $\mathbf{\Phi}$-Markov Equivalence Class, represented by an augmented ve
rsion of the CPDAG, namely, a $\mathbf{\Phi}$-CPDAG. We provide theoretical guar
antees on the convergence of I-PERI, as well as on its privacy-preserving proper
ties, and present empirical evaluations demonstrating the effectiveness of the p
roposed algorithm.
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AvAtar: Learning to Align via Active Optimal Transport
Qi Yu, Ruizhong Qiu, Zhichen Zeng, My T. Thai, huan liu, Hanghang Tong
Alignment plays a fundamental role in many machine learning problems, such as mu
lti-network analysis, multimodal learning, and point cloud registration. Recent
works increasingly leverage optimal transport (OT) for distributional alignment,
 whose effectiveness largely depends on sparse supervision that is hard or costl
y to obtain in practice. Existing works, however, largely overlook how to active
ly acquire high-quality supervision to improve their alignment performance under
 OT frameworks. In this paper, we propose a principled active alignment framewor
k for optimal transport alignment called AvAtar. We quantify the informativeness
 of a candidate by measuring its gradient-based impact on the global alignment r
esult, computed as the gradient propagation from the global alignment result to
all possible supervisions of the candidate through the entropy-regularized OT fo
rmulation. While differentiating through OT is challenging given its constrained
 nature, we leverage the adjoint-state method to reformulate the computation to

a linear system solvable by the conjugate gradient method with linear complexity and guaranteed convergence. By encoding the global alignment result via effective utility functions, AvAtar is applicable to general alignment problems under the OT framework. Extensive experiments on three representative alignment tasks demonstrate the effectiveness, scalability, and generalizability of the proposed AvAtar.

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Self-Supervised Weight Templates for Scalable Vision Model Initialization

Yucheng Xie, Fu Feng, Ruixiao Shi, Jing Wang, Yong Rui, Xin Geng

The increasing scale and complexity of modern model parameters underscore the importance of pre-trained models. However, deployment often demands architectures of varying sizes, exposing limitations of conventional pre-training and fine-tuning. To address this, we propose SWEET, a self-supervised framework that performs constraint-based pre-training to enable scalable initialization in vision tasks. Instead of pre-training a fixed-size model, we learn a shared weight template and size-specific weight scalers under Tucker-based factorization, which promotes modularity and supports flexible adaptation to architectures with varying depths and widths. Target models are subsequently initialized by composing and reweighting the template through lightweight weight scalers, whose parameters can be efficiently learned from minimal training data. To further enhance flexibility in width expansion, we introduce width-wise stochastic scaling, which regularizes the template along width-related dimensions and encourages robust, width-invariant representations for improved cross-width generalization. Extensive experiments on classification, detection, segmentation and generation tasks demonstrate the state-of-the-art performance of SWEET for initializing variable-sized vision models.

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Positive–Unlabeled Reinforcement Learning Distillation for On-Premise Small Models

Zhiqiang Kou, Junyang Chen, Xin-Qiang Cai, Xiaobo Xia, Ming-Kun Xie, Dong-Dong Wu, Biao Liu, Yuheng Jia, Xin Geng, Masashi Sugiyama, Tat-Seng Chua

Due to constraints on privacy, cost, and latency, on-premise deployment of small models is increasingly common. However, most practical pipelines stop at supervised fine-tuning (SFT) and fail to reach the reinforcement learning (RL) alignment stage. The main reason is that RL alignment typically requires either expensive human preference annotation or heavy reliance on high-quality reward models with large-scale API usage and ongoing engineering maintenance, both of which are ill-suited to on-premise settings. To bridge this gap, in this paper, we propose a positive-unlabeled (PU) RL distillation method for on-premise small-model deployment. Without human-labeled preferences or a reward model, our method distills the teacher’s preference-optimization capability from black box generations into a locally trainable student. For each prompt, we query the teacher once to obtain an anchor response, locally sample multiple student candidates, and perform anchor-conditioned self-ranking to induce pairwise or listwise preferences, enabling a fully local training loop via direct preference optimization or group relative policy optimization. Theoretical analysis justifies that the induced preference signal by our method is order-consistent and concentrates on near-optimal candidates, supporting its stability for preference optimization. Experiments demonstrate that our method achieves consistently strong performance under a low-cost setting.

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SALE : Low-bit Estimation for Efficient Sparse Attention in Long-context LLM Prefilling

Xiaodong Ji, Hailin Zhang, Fangcheng Fu, Bin CUI

Many advanced Large Language Model (LLM) applications require long-context processing, but the self-attention module becomes a bottleneck during the prefilling stage of inference due to its quadratic time complexity with respect to sequence length. Existing sparse attention methods accelerate attention computation by skipping less significant regions of the attention map. However, these approaches typically perform coarse-grained inspection of the attention map, resulting in

their suboptimal performance. In this paper, we propose SALE, a fine-grained spa rse attention method that accelerates the long-context prefilling stage of LLM w ith negligible loss in model accuracy. SALE achieves fast and accurate fine-grai ned attention map estimation using low-bit quantized query-key products to appro ximate attention weights, followed by the application of a novel Relative Attent ion Score metric to assess the importance of query-key pairs. This design enable s us to accurately identify important regions in the attention map, thereby cons tructing a highly sparse attention mask.

We implement a custom CUDA kernel in SALE optimized for hardware efficiency, red ucing overhead to approximately 11% of the full attention latency. Notably, SALE requires no parameter training and can be seamlessly integrated into existing s ystems with trivial code modifications. Experiments on long-context benchmarks d emonstrate that our method outperforms existing approaches in accuracy-efficienc y trade-offs, achieving at least 3.36× speedups on Llama-3.1-8B for sequences lo nger than 64K while maintaining model quality.
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Building Social World Model with Large Language Models
Haofei Yu, Yining Zhao, Guanyu Lin, Jiaxuan You
Understanding and predicting how social beliefs evolve in response to events—fro m policy changes to scientific breakthroughs—remains a fundamental challenge in social science. Given LLMs' commonsense knowledge and social intelligence, we as k: Can LLMs model the dynamics of social beliefs following social events? In thi s work, we introduce the concept of the Social World Model (SWM), a general fram ework designed to capture how social beliefs evolve in response to major events. SWM learns state-transition functions for social beliefs by mining temporal pat terns in social data and optimizing the evidence lower bound, without the need f or explicit human annotations that link events to belief shifts or expensive cen sus data. To evaluate SWM, we introduce a benchmark, SWM-bench, derived from rea l-world prediction market data from both Kalshi and Polymarket. SWM-bench includ es over 12k data points for social belief prediction tasks spanning diverse doma ins such as politics, cryptocurrency, and elections. Our experimental results sh ow that SWM significantly outperforms time-series foundation models, achieving s tate-of-the-art results on Kalshi data, respectively, while offering interpretab le insights into the underlying mechanisms of social belief dynamics.
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Generalization and Scaling Laws for Mixture-of-ExpertsTransformers
Mansour MAYAKI
We develop a theory of generalization and scaling for Mixture-of-Experts (MoE) T ransformers that cleanly separates active per-input capacity from routing combin atorics. By conditioning on fixed routing patterns and union-bounding across the m, we derive a sup-norm covering-number bound whose metric entropy scales with t he active parameter budget and incurs a MoE-specific routing overhead. Combined with a standard ERM analysis for squared loss, this yields a generalization boun d under a d-dimensional manifold data model and C^β targets, showing tha t approximation and estimation trade off as in dense networks once active parame ters are accounted for appropriately. We further prove a constructive approximat ion theorem for MoE architectures, showing that, under the approximation constru ction, error can decrease either by scaling active capacity or by increasing the number of experts, depending on the
dominant bottleneck. From these results we derive neural scaling laws for model size, data size, and compute-optimal tradeoffs. Overall, our results provide a t ransparent statistical reference point for reasoning about MoE scaling, clarifyi ng which behaviors are certified by worst-case theory and which must arise from data-dependent routing structure or optimization dynamics.
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Reparameterization Flow Policy Optimization
Hai Zhong, Zhuoran Li, Xun Wang, Longbo Huang
Reparameterization Policy Gradient (RPG) has emerged as a powerful paradigm for model-based reinforcement learning, enabling high sample efficiency by backpropa

gating gradients through differentiable dynamics. However, prior RPG approaches have been predominantly restricted to Gaussian policies, limiting their performance and failing to leverage recent advances in generative models. In this work, we identify that flow policies, which generate actions via differentiable ODE integration, naturally align with the RPG framework, a connection not established in prior work. Yet, naively realizing this synergy is non-trivial: it often suffers from training instability and a lack of exploration. We propose Reparameterization Flow Policy Optimization (RFO). RFO computes policy gradients by backpropagating jointly through the flow generation process and system dynamics, unlocking high sample efficiency without requiring intractable log-likelihood calculations. RFO includes two tailored regularization terms for stability and exploration. We also propose a variant of RFO with action chunking. Extensive experiments on diverse locomotion and manipulation tasks—involving both rigid and soft bodies with state or visual inputs—demonstrate the effectiveness of RFO. Notably, on a challenging locomotion task controlling a soft-body quadruped, RFO achieves almost $2\times$ the reward of the state-of-the-art baseline.
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SceneSmith: Agentic Generation of Simulation-Ready Indoor Scenes

Nicholas Pfaff, Thomas Cohn, Sergey Zakharov, Rick Cory, Russ Tedrake

Simulation has become a key tool for training and evaluating home robots at scale, yet existing environments fail to capture the diversity and physical complexity of real indoor spaces. Current scene synthesis methods produce sparsely furnished rooms that lack the dense clutter, articulated furniture, and physical properties essential for robotic manipulation. We introduce SceneSmith, a hierarchical agentic framework that generates simulation-ready indoor environments from natural language prompts. SceneSmith constructs scenes through successive stages—from architectural layout to furniture placement to small object population—each implemented as an interaction among VLM agents: designer, critic, and orchestrator. The framework tightly integrates asset generation through text-to-3D synthesis for static objects, dataset retrieval for articulated objects, and physical property estimation. SceneSmith generates 3-6x more objects than prior methods, with $<$2\% inter-object collisions and 96\% of objects remaining stable under physics simulation. In a user study with 205 participants, it achieves 92\% average realism and 91\% average prompt faithfulness win rates against baselines. We further demonstrate that these environments can be used in an end-to-end pipeline for automatic robot policy evaluation.
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Less Precise Can Be More Reliable: A Systematic Evaluation of Quantization’s Impact on VLMs Beyond Accuracy

Aymen Bouguerra, Daniel Alfonso Montoya Vasquez, Alexandra Gomez-Villa, Chokri Mraidha, Fabio Arnez

Vision-Language Models (VLMs) such as CLIP have revolutionized zero-shot classification and safety-critical tasks, including Out-of-Distribution (OOD) detection. However, their high computational cost hinders efficient real-world deployment. While quantization is a standard solution for efficiency, its broader impact on reliability metrics beyond simple Top-1 accuracy remains critically under-explored. In this study, we conduct a large-scale evaluation of VLM quantization across a comprehensive experimental suite of over 700k evaluation runs with varying configurations. We find that, contrary to the assumption that quantization's noise degrades performance, it can simultaneously improve accuracy, calibration, OOD detection, and robustness to noise, though not to covariate shift or spurious correlations. We leverage these counterintuitive findings to characterize the mechanics of quantization beyond simple regularization: we show that quantization dampens high-rank spectral components, compelling the model to rely more heavily on robust, low-rank features. Ultimately, this spectral filtering effect drives the observed improvements in generalization and noise tolerance, establishing a pathway to deploy faster, more reliable VLMs by utilizing quantization beyond its conventional role.
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ImmersePro: End-to-End Stereo Video Synthesis Via Implicit Disparity Learning

Jian Shi, Zhenyu Li, Peter Wonka
We introduce \textit{ImmersePro}, an innovative framework specifically designed to transform single-view videos into stereo videos. This framework utilizes a novel dual-branch architecture comprising a disparity branch and a context branch on video data by leveraging spatial-temporal attention mechanisms. \textit{ImmersePro} employs implicit disparity guidance, enabling the generation of stereo pairs from video sequences without the need for explicit disparity maps, thus reducing potential errors associated with disparity estimation models.
In addition to the technical advancements, we introduce the YouTube-SBS dataset, a comprehensive collection of 423 stereo videos sourced from YouTube. This dataset is unprecedented in its scale, featuring over 7 million stereo pairs, and is designed to facilitate training and benchmarking of stereo video generation models. Our experiments demonstrate the effectiveness of \textit{ImmersePro} in producing high-quality stereo videos, offering significant improvements over existing methods.
Compared to the best competitor stereo-from-mono we quantitatively improve the results by 11.76\% (L1), 6.39\% (SSIM), and 5.10\% (PSNR).
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Annotations Mitigate Post-Training Mode Collapse
Jacob Mitchell Springer, Madhu Advani, Lukas Aichberger, Arwen Bradley, Eran Malach, Omid Saremi, Sinead Williamson, Preetum Nakkiran, Etai Littwin, Aditi Raghunathan
Post-training via supervised fine-tuning improves instruction-following, but often induces semantic mode collapse by biasing models toward low-entropy fine-tuning data at the expense of the high-entropy pretraining distribution. Crucially, we find this trade-off worsens with scale. To close this semantic diversity gap, we propose annotation-anchored training, a principled method that enables models to adopt the preference-following behaviors of post-training without sacrificing the inherent diversity of pretraining. Our approach is simple: we pretrain on documents paired with semantic annotations, inducing a rich annotation distribution that reflects the full breadth of pretraining data, and we preserve this distribution during post-training. This lets us sample diverse annotations at inference time and use them as anchors to guide generation, effectively transferring pretraining's semantic richness into post-trained models. We find that models trained with annotation-anchored training can attain 6x less diversity collapse than models trained with SFT, and improve with scale.
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What Characterizes Effective Reasoning? Revisiting Length, Review, and Structure of CoT
Yunzhen Feng, Julia Kempe, Cheng Zhang, Parag Jain, Anthony Hartshorn
Large reasoning models (LRMs) spend substantial test-time compute on long chain-of-thought (CoT) traces, but what *characterizes* an effective CoT remains unclear. While prior work reports gains from lengthening CoTs and increasing review via appended *wait* tokens, recent studies suggest that shorter thinking can outperform longer traces. We therefore conduct a systematic evaluation across ten LRMs on math and scientific reasoning. Contrary to the “longer-is-better” narrative, we find that both naively using longer CoTs and more review behaviors are associated with *lower* accuracy.

As CoT unfolds step by step, token-level metrics can conflate verbosity with process quality. We introduce a graph view of CoT to extract structure and identify a single statistic—the *Failed-Step Fraction* (FSF), the fraction of steps in abandoned branches—that consistently outpredicts length and review ratio for correctness across models. To probe causality, we design two interventions. First, we rank candidate CoTs by each metric at test time, where FSF yields the largest pass@1 gains; second, we edit CoTs to remove failed branches, which significantly improves accuracy, indicating that failed branches bias subsequent reasoning. Taken together, these results characterize effective CoTs as those that *fail less* and support *structure-aware* test-time scaling over indiscriminately generating long CoTs.

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Self-correcting for Debiasing Large Language Models
Xuan Feng, Shuai Zhao, Luwei Xiao, Tianlong Gu, Bo An
Although Large Language Models (LLMs) demonstrate remarkable reasoning capabilities, inherent social biases often cascade throughout the Chain-of-Thought (CoT) process, leading to continuous "Bias Propagation". Existing debiasing methods primarily focus on static constraints or external interventions, failing to identify and interrupt this propagation once triggered. To address this limitation, we introduce Self-Debias, a progressive framework designed to instill intrinsic self-correction capabilities. Specifically, we reformulate the debiasing process as a strategic resource redistribution problem, treating the model's output probability mass as a limited resource to be reallocated from biased heuristics to unbiased reasoning paths. Unlike standard preference optimization which applies broad penalties, Self-Debias employs a fine-grained trajectory-level objective subject to dynamic debiasing constraints. This enables the model to selectively revise biased reasoning suffixes while preserving valid contextual prefixes. Furthermore, we integrate an online self-improvement mechanism utilizing consistency filtering to autonomously synthesize supervision signals. With merely 20k annotated samples, Self-Debias activates efficient self-correction, achieving superior debiasing performance while preserving general reasoning capabilities without continuous external oversight.
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Scalable Topology-Preserving Graph Coarsening: Concepts and Algorithms
Xiang Wu, Rong-Hua Li, Xunkai Li, Kangfei Zhao, Hongchao Qin, Guoren Wang
Graph coarsening reduces the size of a graph while preserving certain properties. Most existing methods preserve either spectral or spatial characteristics. Recent research shows that topology-preserving coarsening methods maintain GNN performance on coarsened graphs but suffer from exponential time complexity. To address these problems, we propose Scalable Topology-Preserving Graph Coarsening (STPGC) by introducing the concepts of graph strong collapse and graph edge collapse extended from algebraic topology. STPGC comprises three new algorithms, GStrongCollapse, GEdgeCollapse, and NeighborhoodConing, based on these two concepts, which eliminate dominated nodes and edges while rigorously preserving topological features. We further prove that STPGC preserves the GNN receptive field and develop approximate algorithms to accelerate GNN training. Experiments on node classification with GNNs demonstrate the efficiency and effectiveness of STPGC.
****
Timestep Rescheduling in Diffusion Inversion
Shangquan Sun, Ting Gong, Zhirui Liu, Jiamin Wu, Runkai Zhao, Mianxin Liu, Wenqi Ren, Xiaochun Cao
Diffusion inversion, which maps images back to the Gaussian latent space of a diffusion model, is a critical task for image reconstruction and editing.
While DDIM enables fast deterministic inversion, it inherently introduces deviations that accumulate into noticeable inversion errors.
Existing methods often address this by solving a fixed-point problem but largely overlook how the selection of the diffusion timestep in the noise scheduler influences inversion fidelity.
In this work, we reveal that the deviation scale in diffusion inversion is strongly dependent on the timestep size, and exhibits a parabolic trend, with larger errors concentrated at both small and large timesteps.
Based on this finding, we propose a simple yet effective nonuniform timestep scheduler that integrates a global rescaling with a local dynamic programming based rescheduling, enabling a strategic allocation of computational effort that minimizes the overall inversion error and preserves higher inversion accuracy.
Our method serves as an off-the-shelf enhancement for existing inversion techniques and requires no extra parameters or computational overhead.
Through extensive experiments, we verify that integrating our scheduler consistently boosts the performance of existing inversion methods, achieving superior results in image reconstruction and editing.
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FiX: Introducing Fine-grained Forget Gate into Softmax Attention
Runzhong Li, Renjie Liu, Li Qing, Bo Tang
Causal softmax attention is the algorithmic foundation of modern large language models. Inspired by linear attention, recent work has sought to enhance it by incorporating forget gates. However, these efforts, such as FoX, have been limited to coarse, scalar gates. While fine-grained, element-wise gates are shown to be more effective than scalar ones in linear attention, their direct integration into softmax attention is non-trivial due to algebraic constraints. In this work, we introduce *Fine-grained Forgetting Transformer* (*FiX*), a novel architecture that successfully enables element-wise forget gates in softmax attention. Our core insight is that the softmax denominator becomes mathematically redundant under a subsequent RMSNorm layer, allowing us to reformulate the forgetting mechanism as a direct element-wise multiplication on the value vectors. This formulation makes FiX the first positional encoding applied to value-output (VO) pairs, designed to be complementary to existing query-key (QK) encodings like RoPE. We systematically address implementation challenges including numerical precision, computational efficiency, and inference memory consumption. Extensive experiments show *FiX* achieves lower training loss and superior performance on both short-text common sense benchmarks and long-context tasks, opening a new path for building more powerful transformers. Our code is available at https://github.com/AlayaDB-AI/FiX.
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PhotoAgent: Exploratory Visual Aesthetic Planning with Large Vision Models
Mingde Yao, Zhiyuan You, King Man Tam, Menglu Wang, Tianfan Xue
With the recent fast development of generative models, instruction-based image editing has shown great potential in generating high-quality images. However, the quality of editing highly depends on carefully designed instructions, placing the burden of task decomposition and sequencing entirely on the user. To achieve autonomous image editing, we present PhotoAgent, a system that advances image editing through explicit aesthetic planning. Specifically, PhotoAgent formulates autonomous image editing as a long-horizon decision-making problem. It reasons over user aesthetic intent, plans multi-step editing actions via tree search, and iteratively refines results through closed-loop execution with memory and visual feedback, without requiring step-by-step user prompts. To support reliable evaluation in real-world scenarios, we introduce UGC-Edit, an aesthetic evaluation benchmark consisting of 7,000 photos and a learned aesthetic reward model. We also construct a test set containing 1,017 photos to systematically assess autonomous photo editing performance. Extensive experiments demonstrate that PhotoAgent significantly outperforms existing methods in both instruction faithfulness and visual quality across a diverse range of editing scenarios.
****
FT-Dojo: Towards Autonomous LLM Fine-Tuning with Language Agents
Qizheng Li, Yifei Zhang, Xiao Yang, Xu Yang, Zhuo Wang, Bowen Xian, Weiqing Liu, Jiang Bian
Fine-tuning large language models for vertical domains remains labor-intensive, requiring practitioners to curate data, configure training, and iteratively diagnose model behavior.
Despite growing interest in autonomous machine learning and language agents, end-to-end LLM fine-tuning has not been systematically studied as an interactive agent task.
We introduce FT-Dojo, an interactive benchmark environment for autonomous LLM fine-tuning, comprising 13 tasks across 5 domains.
Rather than a new collection of static datasets, FT-Dojo standardizes a task interface, shared raw-data repository, sandboxed execution environment, structured feedback protocol, and held-out evaluation procedure.
We further develop FT-Agent, a fine-tuning-oriented autonomous framework that uses structured iteration planning, fail-fast validation, and multi-level feedback analysis to refine data and training strategies.
Experiments show that FT-Agent provides a strong initial baseline, achieving the best performance on 10 out of 13 tasks, with additional controlled comparisons

against frontier agents, open-source planning backbones, and multi-run statistics supporting the main findings.
Case studies show that agents can recover from failures through cumulative learning, while still exposing limitations in causal diagnosis and long-horizon planning.
The implementation is available at https://github.com/microsoft/rd-agent.
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Unsupervised Hierarchical Skill Discovery
Damion Harvey, Geraud Nangue Tasse, Benjamin Rosman, Branden Ingram, Steven James
We consider the problem of unsupervised skill segmentation and hierarchical structure discovery in reinforcement learning. While recent approaches have sought to segment trajectories into reusable skills or options, most rely on action labels, rewards, or handcrafted annotations, limiting their applicability. We propose a method that segments unlabelled trajectories into skills and induces a hierarchical structure over them using a grammar-based approach. The resulting hierarchy captures both low-level behaviours and their composition into higher-level skills. We evaluate our approach in high-dimensional, pixel-based environments, including Craftax and the full, unmodified version of Minecraft. Using metrics for skill segmentation, reuse, and hierarchy quality, we find that our method consistently produces more structured and semantically meaningful hierarchies than existing baselines. Furthermore, as a proof of concept, we demonstrate that these discovered hierarchies accelerate and stabilise learning on downstream reinforcement learning tasks.
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Breaking Manifold Continuity: Vector Quantized Modeling for Real-Centric Deepfake Detection
Changshuo Wang, Jiangming Wang, Ke-Yue Zhang, Taiping Yao, Shouhong Ding, Ran Yi, Lizhuang Ma
The increasingly realistic and diverse generative data has led some deepfake detection methods to shift towards learning robust real content, e.g., via reconstruction-based tasks. However, most existing approaches rely primarily on prevalent continuous modeling (e.g., GMMs, VAEs, Diffusion Models) to construct a continuous latent manifold of real data, with the aim of improving the generalization capability, while overlooking a critical issue, i.e., such continuity may facilitate the interpolation of forgery artifacts, consequently causing ambiguity in detection. To alleviate this problem, we integrate discrete modeling into the feature space of the CLIP vision encoder, striking a balance between continuous manifold modeling and discrete representation. By incorporating a learnable vector quantized codebook, the real latent manifold is discretized, imposing a more stringent information bottleneck that reduces the likelihood of embedding generative artifacts. In order to further enhance the generalization of discrete modeling, we propose an adaptive tangent space projection mechanism that yields a continuous relaxation of the discrete real distribution within a controllable range. With these components, our method constructs a real distribution that is both tightly constrained and broadly generalizable, enhancing robustness to unseen forgeries. Extensive experiments on diverse datasets demonstrate the effectiveness of our method.
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DOT-MoE: Differentiable Optimal Transport for MoEfication
Udbhav Bamba, Arnav Chavan, Aryamaan Thakur, Steven Teig, Deepak Gupta
The scaling of Large Language Models (LLMs) has driven significant performance gains but created substantial challenges in inference efficiency. While Mixture of Experts (MoEs) architectures address this by decoupling model size from inference cost, training MoEs from scratch is often unstable and compute intensive. Conversion of pre-trained dense models into sparse MoEs has emerged as an alternative solution; however, existing methods typically rely on heuristic neuron clustering or random splitting to partition Feed-Forward Networks (FFNs) into experts. In this work, we propose DOT-MoE, a novel framework that formulates the decomposition of dense layers as a Differentiable Optimal Transport (DOT) problem. Ins

tead of static heuristics, we model neuron assignment as a balanced transport pr
oblem, utilizing differentiable Sinkhorn-Knopp iterations to enforce strict expe
rt capacity constraints. Furthermore, we utilize Straight-Through Estimators (ST
E) to jointly learn the discrete neuron-to-expert assignment and the token-to-ex
pert routing policy end-to-end. Extensive experiments across multiple architectu
res and benchmarks demonstrate that DOT-MoE significantly outperforms structured
 pruning, heuristic clustering, and random-split baselines, retaining 90% of the
 original dense model's performance while reducing active parameters by 50%.
****

FedEMoE: Improving Personalization on Heterogeneous Federated Learning via Elast
ic Mixture of Experts Architecture

Haizhou Du, Lixin Huang, Zonghan Wu, Huan Huo

Heterogeneous federated learning (HtFL) has emerged as a promising approach to a
ddress heterogeneity in local computational resources and data distribution. How
ever, existing methods cause performance degradation of model personalization be
cause personalized and generalized knowledge are either intertwined or dominated
 by one of them. To address this issue, we propose a novel Elastic Mixture of Ex
perts (EMoE) architecture on HtFL, namely FedEMoE, decoupling personalization fr
om generalization. Specially, FedEMoE employs a multi-scale feature extraction m
echanism via personalized experts to enrich personalized knowledge. Furthermore,
 we design an elastic shared expert to break the transferred knowledge bottlenec
k across heterogeneous client models. The elastic shared expert can adaptively e
xpand or shrink according to the status of each expert by the weight spectrum an
alysis, respectively. Extensive experiments across statistical and model heterog
eneity settings demonstrate that FedEMoE significantly outperforms state-of-the
art methods on the accuracy of each heterogeneous model over diverse datasets.
****

LiteVSR: Lightweight Adaptation of Frozen Diffusion Transformers for Video Super
-Resolution

Yu Cao, Ziquan Liu, Zhensong Zhang, Jiankang Deng, Shaogang Gong, Jifei Song

Adapting large-scale pre-trained video generators for Video Super-Resolution (VS
R) in novel domains remains computationally prohibitive.

Methods that reformulate generation as direct Low-Quality to High-Quality mappin
gs deviate from the original generative formulation, demanding extensive fine-tu
ning. ControlNet-style adapters lose their efficiency under modern Diffusion Tra
nsformers since the absence of encoder-decoder hierarchy forces duplication of t
he entire backbone. We observe that flow matching offers a principled alternativ
e for cross-domain VSR adaptation. By predicting a constant velocity field acros
s all timesteps, the adaptation task reduces to learning a fixed injection patte
rn rather than time-varying transformations. Building on this insight, we propos
e LiteVSR, a minimalist framework that performs VSR using a completely frozen Di
ffusion Transformer with a lightweight State-Aware Adapter. The adapter employs
a dual-stream architecture that extracts static structural cues from the LQ inpu
t and dynamic cues from intermediate denoising states, aligning them through tim
e-dependent cross-attention to enable adaptive transition from structural alignm
ent to texture refinement as denoising proceeds. LiteVSR achieves competitive re
storation quality with only 11.25% trainable parameters and 12 GPU-hours of trai
ning on a single A100, while maintaining fast sampling (down to a single step) c
ompatibility.
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RBCBF: Decoding Time Safety Alignment via Risk Guided Rollback and Barrier Contr
ol

Tianxiang Chen, Jingyuan Zhou, Longhao Yan, Kaidi Yang

Existing decoding-time safety interventions are often reactive, relying on local
 signals to correct unsafe outputs after they emerge.

Under adversarial prompts that drive generation into recurring unsafe response,
such local signals provide weak guidance for stable repair. As a result, rollbac
k and post-hoc rewriting often trade-off response quality with recurrent violati
ons. To address these limitations, we propose RBCBF, a rollback-based decoding-t
ime framework that jointly selects intervention steps and performs distribution-

level corrective control. Our key innovation is a risk-aggregation formulation t
hat views terminal violations as the accumulated build-up of risk along the pref
ix. By selecting rollback steps from these decisive prefixes, RBCBF moves rollba
ck targeting beyond heuristic cues and turns it into a trajectory-level decision
. RBCBF then applies invasive corrective control to the next-token distribution
under multiple rule constraints. Across jailbreak-style evaluations, RBCBF outpe
rforms prior rollback methods and decoding-time baselines, reducing harmful resp
onses and substantially lowering violation recurrence.
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Differentially Private Submodular Maximization with a Knapsack Constraint
Ron Zadicario, Tova Milo
Submodular maximization subject to a knapsack constraint (SMK) is a fundamental
problem in discrete optimization, with wide-ranging applications in machine lear
ning and related fields. As these applications increasingly involve sensitive in
dividual data, there is a growing need for high-utility algorithms that provide
formal privacy guarantees.
In this work, we study the SMK problem under differential privacy, considering b
oth monotone and non-monotone objective functions. For monotone objectives, we p
ropose a differentially private algorithm that achieves the optimal $(1-1/e)$-ap
proximation ratio while significantly improving both additive error and query co
mplexity over prior work. We also present a more efficient algorithm for the sa
me setting, achieving a $1/2$-approximation. For non-monotone objectives, we int
roduce, to our knowledge, the first differentially private algorithm with provab
le guarantees, achieving a $1/4$-approximation in expectation and an additive er
ror comparable to the best known for monotone objective functions.
****

FuseFSS: Efficient Secure LLM Inference with Function Secret Sharing
Yuhan Ma, Yong Li, Stefan Schmid
Two-server secure inference allows a client to query a hosted large language mod
el (LLM) without revealing prompts or embeddings. Recent GPU systems based on fu
nction secret sharing (FSS) make linear layers efficient, but fixed-point nonlin
earities and helper operations remain a bottleneck because each operator is typi
cally implemented as a bespoke protocol with its own comparisons, wrap-around co
rrections, and preprocessing material. We present FuseFSS, a compiler that repla
ces per-operator protocol design with a single compilation pipeline. For each sc
alar fixed-point operator, a compact specification lists its interval partition,
 low-degree arithmetic pieces, and required predicate bits. The compiler emits t
wo batched FSS evaluations on the public masked value: one packed comparison tha
t returns all predicate bits, and one vector interval lookup that returns the ac
tive coefficients and constants. Compared to the current state-of-the-art FSS-ba
sed GPU secure inference, FuseFSS preserves accuracy while achieving a $1.24\tim
es$–$1.50\times$ end-to-end speedup and reducing online communication by 9%–16%
on BERT and GPT-style models; preprocessing is also lighter, with 14%–23% lower
key-generation time and 20%–24% smaller keys.
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Addressing Semantic Blind Spots in Text-to-SQL via Component Pre-generation and
AST Matching Rewards
Xingyu Ma, Xin Tian, Lingxiang Wu, Xuepeng Wang, Zhilin Zhang, Xueming Tang, Jin
qiao Wang
In recent years, significant advancements in large language models have greatly
propelled the development of Text-to-SQL tasks. However, due to the token-by-tok
en sequential generation mechanism employed by these models, they encounter a se
mantic blind spot problem with respect to pending SQL components—the parts of th
e SQL query yet to be generated. Specifically, language models are unable to eff
ectively utilize the semantic information of these pending SQL components during
 the generation of the final SQL query, which poses considerable challenges for
generating complex SQL statements. To address this issue, we propose a novel tho
ught process based on SQL components pre-generation and design a maximum connect
ed subtree matching reward mechanism leveraging the SQL abstract syntax tree to
improve the accuracy of local component generation. Extensive experiments demons

trate that, under comparable model parameter scales, our training approach achie ves significant advantages, effectively enhancing the generation of complex SQL queries. Our method attains an execution accuracy EX of 65.78% on the BIRD-dev d ataset and achieves state-of-the-art performance on the Spider-syn datasets.
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Information-Theoretic Disentangled Latent Modeling with Conditional Diffusion fo r Incomplete Multi-View Clustering
Wenlan Chen, Lu Gao, Daoyuan Wang, Cheng Liang, Fei Guo
Incomplete multi-view clustering is challenging due to view missingness and the entanglement of shared semantics with view-specific factors in latent representa tions. Existing methods often rely on heuristic fusion or direct completion stra tegies, which suffer from error propagation and unreliable generation under miss ing views. In this paper, we propose an **I**nformation-guided **D**isentangled latent modeling framework with **C**onditional **D**iffusion for incomplete mult i-view clustering (IDCD). Specifically, we first encode each view into a latent representation that is variationally decomposed into a view-wise semantic latent and a view-specific factor. Information-theoretic objectives are introduced to guide the disentanglement of view-wise latents, preserving essential multi-view information while reducing the dependency between semantic and view-specific fac tors and encouraging cross-view semantic consistency. Besides, we aggregate the semantic latents via a mixture of Wasserstein distributions to obtain a unified global representation, where we impose a Gaussian mixture prior to explicitly co uple representation learning with clustering. Based on the learned disentangled latent space, a conditional diffusion model guided by both the global semantic l atent and view-specific factors is employed to generate missing views in a consi stent manner. Extensive experiments on benchmark datasets demonstrate superior c lustering performance and robust missing-view generation compared to state-of-th e-art methods.
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WeatherSyn: An Instruction Tuning MLLM For Weather Forecasting Report Generation
Zinan Zheng, Yang Liu, Nuo Chen, Juepeng Zheng, Hong Cheng, Jia Li
Accurate weather forecast reporting enables individuals and communities to bette r plan daily activities, agricultural operations, and transportation. However, t he current reporting process primarily relies on manual analysis of multi-source data, which often leads to information overload and reduced efficiency. With th e rapid advancement of multimodal large language models (MLLMs), leveraging data -driven models to analyze and generate reports in the weather forecasting domain remains largely underexplored. In this work, we propose the Weather Forecasting Report (WFR) task and construct the first instruction-tuning dataset for this t ask, named WSInstruct, which covers 31 cities in America and 8 weather aspects. Based on this corpus, we develop the first model, WeatherSyn, specialized in gen erating weather forecast reports. Evaluation across multiple metrics on our data set shows that WeatherSyn consistently outperforms leading closed-source MLLMs, particularly on structurally complex weather aspects. We further analyze its per formance across diverse geographic regions and weather aspects. WeatherSyn demon strates strong transferability across different regions, highlighting its zero-s hot generalization capability. WeatherSyn offers valuable insight for developing MLLMs specialized in weather report generation.
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Coupled Training with Privileged Information and Unlabeled Data
Jiahao Shi, Omar Hagrass, Jason Matthew Klusowski
In many prediction problems, we have extra information during training (for exam ple, measurements that are expensive or slow to collect) that will not be availa ble when the model is deployed. A common strategy is to first train a model that uses all training information, then use its predictions on unlabeled examples t o train a second model that only uses the inputs available at test time. However , when the extra training-only information is weak or noisy, this two-step appro ach can mislead the deployable model and even hurt accuracy. We propose a joint training method that learns the two models together, so the deployable model can benefit from the extra information only when it actually helps, instead of inhe

riting its mistakes. We provide guarantees that describe when joint training imp
roves prediction accuracy and analyze a simple alternating training algorithm fo
r large, high-dimensional models. Experiments on synthetic data and medical pred
iction tasks show that our approach avoids these failures and consistently outpe
rforms standard two-step baselines.
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Particle Flow for Learning from Label Proportions
Alain Rakotomamonjy, Maxime Vono, Liva Ralaivola
This work proposes a novel method for solving learning from label proportion pro
blems.
For this purpose, we learn a classifier that minimizes three key objectives: (i)
 a
bag-level loss, which quantifies the discrepancy between true and predicted labe
l proportions
in bags, (ii) an instance-level loss, inspired from domain adaptation,
which leverages anchor samples with known labels and trainable supports and (iii
) a
distribution discrepancy that aims at aligning anchor's learned support with tho
se of the bag samples.
The problem is formulated as an alternating optimization process, iteratively
updating the classifier and aligning distributions via a particle flow method.
The flow of anchor samples is governed by a vector field
designed to minimize the anchor loss while ensuring alignment between anchor and
 bag distributions.
We provide a theoretical analysis, guaranteeing the convergence of the flow and
identifying conditions under which the method achieves effective alignment.
Our analysis highlights that gap and diversity in label proportions
within bags is a critical factor for learnability.
Empirical results on tabular and image datasets demonstrate the method's effecti
veness,
outperforming state-of-the-art approaches.
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CSPLoRA: Confidence-Guided Structure Planning for Low-Rank Adaptation
Huiming Ding, Xiaochen Li, Jianhui Ma, Xu An, Yihui Yang, Zhenyu Tan
Low-Rank Adaptation (LoRA) has become the de facto paradigm for parameter-effici
ent fine-tuning, with its effectiveness critically influenced by rank allocation
 across modules. However, existing approaches face a fundamental dilemma: unifor
m allocation ignores module heterogeneity, while adaptive methods introduce expe
nsive training overhead or lack reusability across configurations. We propose CS
PLoRA (Confidence-guided Structural Planning for LoRA), a decoupled framework th
at reweights probe samples by prediction uncertainty to obtain more discriminati
ve module importance estimates. The key insight is that hard samples---those the
 model struggles with---provide more informative gradient signals for identifyin
g critical modules than easy samples. For a fixed task-model pair, the resulting
 structural priors can be reused across compatible rank budgets and LoRA backend
s, supporting a practical "probe once, deploy everywhere" workflow. Experiments
on GLUE, commonsense reasoning, and arithmetic tasks show that CSPLoRA improves
over uniform LoRA on average (+1.25 points on LLaMA-2-7B commonsense reasoning)
while maintaining comparable parameters, with the planned rank structure reusabl
e across compatible LoRA variants.
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STFlow: Data-Coupled Flow Matching for Geometric Trajectory Simulation
Kiet Bennema ten Brinke, Koen Minartz, Vlado Menkovski
Simulating trajectories of dynamical systems is a fundamental problem in a wide
range of fields such as molecular dynamics, biochemistry, and pedestrian dynamic
s. Machine learning has become an invaluable tool for scaling physics-based simu
lators and developing models directly from experimental data. In particular, rec
ent advances in deep generative modeling and geometric deep learning enable prob
abilistic simulation by learning complex trajectory distributions while respecti
ng intrinsic permutation and time-shift symmetries. However, trajectories of N-b

ody systems are commonly characterized by high sensitivity to perturbations leading to bifurcations, as well as multi-scale temporal and spatial correlations. To address these challenges, we introduce STFlow (Spatio-Temporal Flow), a generative model based on graph neural networks and hierarchical convolutions. By incorporating data-dependent couplings within the Flow Matching framework, STFlow denoises starting from conditioned random-walks instead of Gaussian noise. This novel informed prior simplifies the learning task by reducing transport cost, increasing training and inference efficiency. We validate our approach on N-body systems, molecular dynamics, and human trajectory forecasting. Across these benchmarks, STFlow achieves the lowest prediction errors with fewer simulation steps and improved scalability.
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Reason, Then Re-reason: Cross-view Revisiting Improves Spatial Reasoning
Chaofan Ma, Zhenjie Mao, Yuhuan Yang, Fanqin Zeng, Yue Shi, Yingjie Zhou, Xiaofeng Cao, Jiangchao Yao
Spatial reasoning from egocentric videos is inherently challenging because the observable evidence is constrained by the camera trajectory.
Existing methods rely on single-turn inference, forcing models to resolve geometric ambiguity through semantic priors rather than verifiable evidence.
We argue that spatial reasoning should be revisitable: conclusions formed under limited evidence should remain open to revision when complementary viewpoints become available.
Building on this insight, we propose Reason, then Re-reason (ReRe), a training-free, inference-time framework with two phases: in the Reason Phase, an MLLM forms a spatial hypothesis from the original video; in the Re-reason Phase, it verifies or revises the hypothesis by observing a synthesized novel-view video.
To enable effective cross-view revisiting, we design a Geometry-to-Video pipeline that renders strategically complementary novel views from predicted 3D geometry.
These views feature an elevated, oblique perspective with scene-spanning coverage, while preserving the MLLM's native video interface without architectural modifications.
Extensive evaluations on VSI-Bench and STI-Bench demonstrate that ReRe substantially boosts open-source MLLMs to rival proprietary state-of-the-art performance.
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CatFlow: Co-generation of Slab-Adsorbate Systems via Flow Matching
Minkyu Kim, Nayoung Kim, Honghui Kim, Sungsoo Ahn
Discovering heterogeneous catalysts tailored for specific reaction intermediates remains a fundamental bottleneck in materials science. While traditional trial-and-error methods and recent generative models have shown promise, they struggle to capture the intrinsic coupling between surface geometry and adsorbate interactions. To address this limitation, we propose CatFlow, a flow matching-based framework for de novo design and structure prediction of heterogeneous catalysts. Our model operates on a primitive cell-based factorized representation of the slab-adsorbate complex, reducing the number of learnable variables by an average of 9.2x while explicitly encoding the surface orientation of the slab-adsorbate interface. Experiments on the Open Catalyst 2020 dataset demonstrate that CatFlow significantly improves the structural fidelity of generated catalysts compared to autoregressive and sequential baselines. Further experiments show that the generated structures accurately capture the adsorption energy distributions of physically plausible interfaces and lie closer to thermodynamic local minima.
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Submodular Optimization for Minimal Augmentation in Robust Language Model Alignment
Ching-Chia Kao, Chia-Mu Yu, Chun-Shien Lu, Chu-song Chen
Safety alignment of large language models is fragile: even small fine-tuning perturbations elastically revert behaviors toward those of the pre-training, with degradation inversely proportional to the size of the alignment set. We ask how to achieve safety alignment with minimal augmentation. To this end, we model augmentation as a set of group actions on sequences and formalize robustness gains a

s a normalized, monotone submodular function over transformations. We then lever age submodular optimization to select minimal augmentations that provably improv e robustness. Experiments confirm that our approach efficiently restores safety alignment while minimizing the overhead of augmentation.
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Component-Wise Composite Likelihood Distillation for Censored Time-to-Event Data
Feiyang Deng, Lingfeng Luo, Jiayu Zhou, Kevin He

Accurate survival modeling in biomedical studies is often hindered by rare event s, limited effective sample sizes, and settings with limited or partially observ ed information (e.g., covariates of interest that are difficult or expensive to collect, highly structured sampling designs, or nuisance parameters omitted by c onditioning). Knowledge distillation can leverage external predictive informatio n without sharing individual-level data, but existing approaches are largely bui lt for fully specified likelihoods or probability-based survival models and do n ot extend to settings where outcome distributions are only partially specified. To address this challenge, we propose a knowledge distillation framework based o n a composite-likelihood Kullback--Leibler divergence that aligns teacher and st udent models within components. Our key insight is that, although composite like lihoods do not define a global outcome distribution, each likelihood component i nduces a well-defined probability model on its restricted outcome space, enablin g a principled KL divergence. Simulation studies and biomedical case studies sho w improved discrimination and predictive accuracy in rare-event, heterogeneous s ettings without requiring access to external individual-level data.
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From Static Constraints to Dynamic Adaptation: Sample-Level Constraint Relaxatio n for Offline-to-Online Reinforcement Learning
Lipeng Zu, YU QIAN, Shayok Chakraborty, Xiaonan Zhang

Offline-to-online reinforcement learning (O2O RL) faces a central challenge betw een retaining offline conservatism and adapting to online feedback under distrib ution shift. This challenge arises because data behavior evolves during fine -tuning, rendering data origin a misleading basis for constraint handling and thereby leading to objective–data mismatch. We therefore propose Dynamic Alignme nt for RElaxation (DARE), a distribution-aware framework for sample-level constr aint relaxation based on the behavioral consistency with a behavior model. To ou r knowledge, DARE is the first to condition constraint relaxation on behavioral consistency via a posterior-induced exchange mechanism, moving beyond a binary o ffline/online data distinction. Importantly, DARE requires only per-sample behav ioral alignment, enabling instantiation on top of many offline algorithms with f lexible choices of behavior models and fine-tuning objectives. We provide a theo retical analysis showing that behavior-based sample exchange consistently improv es the distinction between offline-like and online-like subsets. Experiments on D4RL demonstrate that DARE consistently improves fine-tuning stability and achie ves superior final performance over strong offline-to-online baselines.
(The code is publicly available at \url{https://github.com/lpzu/DARE}.)
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GeoEvo: Identity-Aware Potential Game with Geometric Evolution for Personalized Multimodal Federated Learning
Chen Wang, Yongli Hu, Huajie Jiang, Kan Guo, Tengfei Liu, Junbin Gao, Yanfeng Su n, Baocai Yin

We reconceptualize Personalized Multimodal Federated Learning (PMFL) by treating missing modalities as intrinsic structural identities that constrain each clien t to a distinct Riemannian submanifold, rather than as deficiencies to be compen sated. To reconcile the tension between identity preservation and cross-client c ollaboration, we cast PMFL as an identity-aware potential game and seek a geomet ry-consistent equilibrium instead of a monolithic full-modality optimum. We prop ose GeoEvo, a federated approximate solver that combines curvature-adaptive Fish er descent with manifold-lifted evolutionary search: Natural Evolution Strategie s for basin escape, and Particle Swarm updates anchored to a server-broadcast Fr \'echet prototype for cross-client transfer. A monotone acceptance rule drives p er-step potential dissipation, yielding an $\mathcal{O}(1/\sqrt{T})$ stationarit

y rate and convergence toward first-order Nash equilibria in non-convex regimes. Empirically, GeoEvo improves personalization and robustness across diverse modality-missing patterns.

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FlashOptim: Optimizers for Memory-Efficient Training

Jose Javier Gonzalez Ortiz, Abhay Gupta, Christopher Rinard, Davis Blalock

Standard mixed-precision training of neural networks requires many bytes of accelerator memory for each model parameter. These bytes reflect not just the parameter itself, but also its gradient and one or more optimizer state variables. With each of these values typically requiring 4 bytes, training even a 7 billion parameter model can be impractical for researchers with less than 100\,GiB of accelerator memory.

We introduce FlashOptim, a suite of optimizations that reduces per-parameter memory by over 50% while preserving model quality and API compatibility. Our approach introduces two key techniques. First, we improve master weight splitting by finding and exploiting a tight bound on its quantization error. Second, we design companding functions that greatly reduce the error in 8-bit optimizer state quantization. Together with 16-bit gradients, these techniques reduce AdamW memory from 16 bytes to 7 bytes per parameter, or 5 bytes with gradient release. They also cut model checkpoint sizes by more than half.

Experiments with FlashOptim applied to SGD, AdamW, and Lion show no measurable quality degradation across a collection of standard vision and language benchmarks, including Llama-3.1-8B finetuning.

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What Does Preference Learning Recover from Pairwise Comparison Data?

Rattana Pukdee, Maria Florina Balcan, Pradeep Ravikumar

Pairwise preference learning is central to machine learning, with recent applications in aligning language models with human preferences. A typical dataset consists of triplets (x, y^+, y^-), where response y^+ is preferred over response y^- for context x. The Bradley--Terry (BT) model is the predominant approach, modeling preference probabilities as a function of latent score differences. Standard practice assumes data follows this model and learns the latent scores accordingly. However, real data may violate this assumption, and it remains unclear what BT learning recovers in such cases. Starting from triplet comparison data, we formalize the preference information it encodes through the conditional preference distribution (CPRD). We give precise conditions for when BT is appropriate for modeling the CPRD, and identify factors governing sample efficiency---namely, margin and connectivity. Together, these results offer a data-centric foundation for understanding what preference learning actually recovers.

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BiTrajDiff: Bidirectional Trajectory Generation with Diffusion Models for Offline Reinforcement Learning

Yunpeng Qing, Yixiao Chi, Shuo Chen, Shunyu Liu, Kexuan Zhou, Sixu Lin, Litao Liu, Changqing Zou

Offline Reinforcement Learning (RL) relies on static datasets and often enforces conservative constraints to mitigate out-of-distribution errors, but this inevitably gives rise to learning dataset biases and limited behavioral generalization.

Recent Data Augmentation (DA) methods leverage generative models to enrich offline data, yet they mainly operate within a single rollout paradigm and tend to preserve the original trajectory-level connectivity of the dataset.

As a result, such methods often introduce local variations and fail to recover connections between distinct behavior patterns.

In this paper, we propose Bidirectional Trajectory Diffusion (BiTrajDiff), a novel DA framework that explicitly addresses this limitation.

BiTrajDiff decomposes trajectory synthesis into two independent diffusion processes that generate forward-future and backward-history segments conditioned on shared intermediate anchor states.

By stitching the generated segments at these anchors, BiTrajDiff can synthesize trajectories that bridge disconnected behavior patterns and recover global trajectory-level connectivity absent from the original data.
Extensive experiments demonstrate that BiTrajDiff consistently outperforms advanced DA methods across a range of offline RL backbones. Our code is available at https://github.com/Plankson/BiTrajDiff.
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Content-Style Identification via Differential Independence
Subash Timilsina, Hoang-Son Nguyen, Sagar Shrestha, Xiao Fu
Generative analysis often models multi-domain observations as nonlinear mixtures of domain-invariant content variables and domain-specific style variables. Identifying both factors from unpaired domains enables tasks such as domain transfer and counterfactual data generation. Prior work establishes identifiability under (block-wise) statistical independence between content and style, or via sparse Jacobian assumptions on the nonlinear mixing function, but such conditions can be restrictive and may not hold in practice.
In this work, we introduce differential independence, a weaker structural condition requiring that infinitesimal variations in content and style induce orthogonal directions on the data manifold, thereby enabling identifiability even when content and style are dependent and the Jacobian is dense. We operationalize this condition through a blockwise orthogonality constraint on the Jacobian subspaces associated with content and style. To support high-dimensional generative models, we design a stochastic regularizer based on numerical Jacobian approximation, enabling scalable training in settings such as high-resolution image generation. Experiments across multiple datasets corroborate the identifiability analysis and demonstrate practical benefits on counterfactual generation and domain translation tasks.
****
Multi-Task Bayesian In-Context Learning
Qingyang Zhu, Eric Karl Oermann, Kyunghyun Cho
Bayesian predictive inference provides a principled framework for uncertainty quantification, data efficiency, and robust generalization. However, exact inference is often intractable, and scalable approximations may remain computationally expensive or require restrictive modeling assumptions that degrade predictive performance. Prior-Data Fitted and in-context learning networks have recently emerged as an amortized alternative by learning to map datasets directly to predictive distributions, but existing approaches are tightly coupled to the support of the training prior and lack explicit mechanisms for adapting to new priors at test time, resulting in limited robustness under distribution shift. We introduce a multi-task in-context learning framework for amortized hierarchical Bayesian predictive inference that explicitly represents prior information as a prefix of in-context datasets. A transformer trained on sequences of prior and target tasks learns to adapt its predictions across families of priors. On a suite of evaluations with increasing difficulty, including out-of-meta-distribution heavy-tailed priors and priors with high-dimensional latent structures, our method matches oracle Bayesian predictors while being orders of magnitude faster. We further demonstrate its practical relevance on a real-world spatiotemporal temperature prediction benchmark.
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Rethinking Efficient Graph Coarsening via a Non-Selfishness Principle
Xu Bai, Bin Lu, Kun Zhang, Shengbo Chen, Xinbing Wang, Chenghu Zhou, Meng Jin
Graph coarsening is a graph dimensionality reduction technique that aims to construct a smaller and more tractable graph while preserving the essential structural and semantic properties of the original graph. However, most existing methods rely on pair-wise similarity matching, where each node independently searches for its best partner based on global information. This selfishness matching paradigm incurs substantial computational and memory overhead.
To address this problem, we shift to a non-selfishness principle that prioritizes the collective interference of neighborhood in coarsening, and propose an effi

cient method named NOPE, which achieves linear memory consumption and near-linea
r computational complexity in the number of nodes.
Furthermore, we derive a faster variant NOPE*, which reduces O(\Delta\cdot d) in
terference evaluation to O(d) based on the local isotropy assumption, and conseq
uently alleviates the computational bottleneck for high-degree nodes. Experiment
al results show that NOPE* achieves $1.8–10\times$ speedup over NOPE and surpass
almost all baselines with 1-3 orders of magnitude acceleration. Meanwhile, lear
ning on coarsened graphs yields comparable performance to original graphs, and c
an even show superior performance over LLM-based graph reasoning owing to compac
t graph information.
The code can be available at https://github.com/dazonglian/NOPE-main.
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DITRON: Distributed Multi-level Tiling Compiler for Parallel Tensor Programs
Size Zheng, Xuegui Zheng, Hanshi Sun, Qi Hou, Wenlei Bao, Shiyu Li, Haojie Duanm
u, Jin Fang, Chenli Xue, Chenhui Huang, Yuanqiang Liu, Renze Chen, Ningxin Zheng
, Dongyang Wang, Li-Wen Chang, Liqiang Lu, Yun Liang, Jidong Zhai, Xin Liu
The scaling of large language models (LLMs) is currently bottlenecked by the rig
idity of distributed programming. While high-performance libraries like CuBLAS a
nd NCCL provide optimized primitives, they lack the flexibility required for rap
idly evolving model architectures. Conversely, existing tensor compilers fail to
address the complex memory hierarchy of distributed clusters effectively. To br
idge this gap, we propose DITRON, a scalable tile-level compiler that democratiz
es high-performance distributed kernel development. DITRON introduces a novel hi
erarchical programming abstraction spanning Core, Device, and Task levels to map
tensor programs efficiently onto heterogeneous distributed hardware. This abstr
action allows DITRON to support diverse parallelism strategies while abstracting
away the complexity of inter-node and intra-node communication. Evaluated acros
s large-scale clusters, DITRON achieves performance parity with or exceeding exp
ert-tuned CUDA libraries, delivering speedups of 6%–30% on isolated kernels and
5%–30% on end-to-end inference in vLLM. Furthermore, DITRON demonstrates strong
portability, achieving significant speedups on both NVIDIA and AMD platforms.
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Temporal Score Rescaling for Temperature Sampling in Diffusion and Flow Models
Yanbo Xu, Yu Wu, Sungjae Park, Zhizhuo Zhou, Shubham Tulsiani
We present a mechanism to steer the sampling diversity of denoising diffusion an
d flow matching models, allowing users to sample from a sharper or broader distr
ibution than the training distribution. We build on the observation that these m
odels leverage (learned) score functions of noisy data distributions for samplin
g and show that rescaling these allows one to effectively control a 'local' samp
ling temperature. Notably, this approach does not require any finetuning or alte
rations to training strategy, and can be applied to any off-the-shelf model and
is compatible with both deterministic and stochastic samplers. We first validate
our framework on toy 2D data, and then demonstrate its application for diffusio
n models trained across five disparate tasks -- image generation, pose estimatio
n, depth prediction, robot manipulation, and protein design. We find that across
these tasks, our approach allows sampling from sharper (or flatter) distributio
ns, yielding performance gains e.g., depth prediction models benefit from sampli
ng more likely depth estimates, whereas image generation models perform better w
hen sampling a slightly flatter distribution.
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Conversation for Non-verifiable Learning: Self-Evolving Large Language Models th
rough Meta-Evaluation
Yuan Sui, Bryan Hooi
Training large language models (LLMs) for non-verifiable tasks—such as creative
writing, dialogue, and ethical reasoning—remains challenging due to the absence
of ground-truth labels. While LLM-as-Judge approaches offer a scalable alternati
ve to human feedback, they face a fundamental limitation: performance is constra
ined by the evaluator's own quality. If the judge cannot recognize good solution
s, it cannot provide useful training signals, and evaluation biases (e.g., favor
ing verbosity over quality) remain unaddressed. This motivates meta-evaluation—t

he ability to evaluate and improve the evaluator itself. We introduce CoNL, a fr amework that unifies generation, evaluation, and meta-evaluation through multi-a gent self-play. Our key insight: critique quality can be measured by whether it helps others improve their solutions. In CoNL, multiple agents sharing the same policy engage in structured conversations to propose, critique, and revise solut ions. Critiques that enable other agents' solution improvements earn a diagnosti c reward, creating explicit supervision for meta-evaluation and enabling joint o ptimization of generation and judging capabilities through self-play, without ex ternal judges or ground truth. Experiments on various benchmarks show that CoNL achieves consistent improvements over self-rewarding baselines while maintaining stable training.

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Rethinking Visual Intelligence: Insights from Video Pretraining

Pablo Acuaviva, Aram Davtyan, Mariam Hassan, Sebastian Stapf, Ahmad Rahimi, Alex andre Alahi, Paolo Favaro

Large language models (LLMs) have demonstrated that large-scale pretraining enab les systems to adapt rapidly to new problems with little supervision in the lang uage domain. This success, however, has not translated as effectively to the vis ual domain, where models, including LLMs, continue to struggle with compositiona l understanding, sample efficiency, and general-purpose problem-solving. We inve stigate Video Diffusion Models (VDMs) as a promising direction for bridging this gap. Pretraining on spatiotemporal data endows these models with strong inducti ve biases for structure and dynamics, which we hypothesize can support broad tas k adaptability. To test this, we design a controlled evaluation in which both a pretrained LLM and a pretrained VDM are equipped with lightweight adapters and p resented with tasks in their natural modalities. Across benchmarks including ARC -AGI, ConceptARC, visual games, route planning, and cellular automata, VDMs demo nstrate higher data efficiency than their language counterparts. Taken together, our results indicate that video pretraining offers inductive biases that suppor t progress toward visual foundation models.

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Unlocking Noise-Resistant Vision: Key Architectural Secrets for Robust Models Ag ainst Gaussian Noise

Bum Jun Kim, Makoto Kawano, Yusuke Iwasawa, Yutaka Matsuo

We investigate which architectural and input-pipeline choices make vision models more robust to additive Gaussian noise and convert these empirical insights int o simple, actionable design rules. Specifically, we performed extensive evaluati ons on 1,174 pretrained vision models, empirically identifying four consistent f actors for improved robustness against Gaussian noise: larger stem kernels, reso lution-dependent downsampling under a pre-resize sensor-noise protocol, average pooling, and supervised vision transformers (ViTs) rather than CLIP ViTs, which yield up to 506 rank improvements and 21.6\%p accuracy gains. First, we prove th at low-pass stem kernels attenuate noise with a gain that decreases quadraticall y with kernel size and that anti-aliased downsampling reduces noise energy rough ly in proportion to the square of the downsampling factor. Second, we demonstrat e that average pooling is unbiased and suppresses noise in proportion to the poo ling window area, whereas max pooling incurs a positive bias that grows slowly w ith window size and yields a relatively higher mean-squared error and greater wo rst-case sensitivity. Third, we reveal and explain the vulnerability of CLIP ViT s via a pixel-space Lipschitz bound: The smaller normalization standard deviatio ns used in CLIP preprocessing amplify worst-case sensitivity by up to 1.91 times relative to the Inception-style preprocessing common in supervised ViTs. Our re sults collectively disentangle robustness into interpretable modules, provide a theory that explains the observed trends, and build practical, plug-and-play gui delines for designing vision models more robust against Gaussian noise.

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MMKU-Bench: A Multimodal Update Benchmark for Diverse Visual Knowledge

Baochen Fu, Yuntao Du, Cheng Chang, Baihao Jin, Wenzhi Deng, Muhao Xu, Hongmei Y an, Weiye Song, Yi Wan

As real-world knowledge continues to evolve, the parametric knowledge acquired b

y multimodal models during pretraining becomes increasingly difficult to remain consistent with real-world knowledge. Existing research on multimodal knowledge updating focuses only on learning previously unknown knowledge, while overlooking the need to update knowledge that the model has already mastered but that later changes; moreover, evaluation is limited to the same modality, lacking a systematic analysis of cross-modal consistency. To address these issues, this paper proposes MMKU-Bench, a comprehensive evaluation benchmark for multimodal knowledge updating, which contains over 25k knowledge instances and more than 49k images, covering two scenarios, updated knowledge and unknown knowledge, thereby enabling comparative analysis of learning across different knowledge types. On this benchmark, we evaluate a variety of representative approaches, including supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), and knowledge editing (KE). Experimental results show that SFT and RLHF are prone to catastrophic forgetting, while KE better preserve general capabilities but exhibit clear limitations in continual updating. Overall, MMKU-Bench provides a reliable and comprehensive evaluation benchmark for multimodal knowledge updating, advancing progress in this field. The code and dataset are available at https://github.com/baochenfu/MMKU-Bench.

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WEVSR: Video Diffusion Generators for Real-World Video Super■Resolution with Wavelet-Enhanced VAE Encoder

Yuying Chen, Zhirui Liu, Linyan Jiang, Qifan Gao, Xianguo Zhang, Jianhou Gan, Wenqi Ren

Recent advances in video diffusion models have demonstrated remarkable generative capability, yet adapting these large pretrained text-to-video (T2V) models to video super■resolution (VSR) typically encounters challenges, such as artifacts introduced by complex degradations in real-world scenarios and compromised fidelity due to the strong generative capacity of the powerful T2V models. We present WEVSR, a novel approach that adapts a pretrained flow-matching video diffusion transformer to VSR. First, we design a task-oriented adaptation strategy that leverages timestep sampling and noise augmentation to enhance detail restoration while preserving structural stability. Second, we propose a lightweight multi-level discrete wavelet transform (DWT) front-end for the VAE encoder, injecting explicit frequency priors into the latent space without modifying the pretrained decoder. Extensive experiments across multiple VSR benchmarks demonstrate that WEVSR achieves state-of-the-art performance against existing approaches. Code and models will be released here.

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Hyperbolic neural population geometry benefits computation

Dennis Wu, Yi-Chun Hung, Braden Yuille, James E Fitzgerald, Han Liu

Neural population geometry shapes downstream
computation. Recent empirical findings in neurobiology suggest that a hyperbolic structure underlies population activity in the hippocampus. Here
we provide a theoretical framework for this phenomenon. First, we propose a plausible construction of hippocampal tuning curves that statistically induces hyperbolic geometry. Next, we establish a connection between neural decoding and associative memory by demonstrating that the Modern Hopfield Network update rule computes the
minimum mean-squared-error (MMSE) estimator.
Finally, we introduce a novel associative memory
model defined in hyperbolic space that yields significantly larger capacity than leading models. Our
results suggest that animals encode spatial information as a latent hyperbolic cognitive map, improving both memory capacity and decoding accuracy.

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Deep Ensemble Clustering for Visual Representation Learning

Yuwei Wang, Guikun Chen, Xiruo Jiang, Yazhou Yao, Di Liu, Xiangbo Shu, Fumin Shen, Wenguan Wang

Recent advances in visual representation learning have seen the rise of clustering-based vision backbones, which adopt clustering as a core paradigm for feature

extraction. However, existing clustering-based backbones typically rely on a single clustering algorithm, whose inherent inductive bias limits their representational capacity. To address this, we propose EnFormer, which embeds ensemble clustering as a core component of feature extraction. EnFormer structures feature extraction around two steps: (i) Ensemble Generation, where several differentiable base clustering methods are introduced to capture diverse semantic structures; and (ii) Consensus Aggregation, which employs a differentiable mechanism to fuse the results of all base clusterings to reconstruct refined visual features. Extensive experiments show that EnFormer consistently outperforms existing clustering-based backbones across core vision tasks, with higher performance and significantly improved throughput.

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CodeChemist: Test-Time Scaling for Low-Resource Code Generation via Functional Knowledge Transfer

KaiXin Wang, Tianlin Li, Xiaoyu Zhang, Aishan Liu, Xianglong Liu, Ziqi Liu, Zhiqiang Zhang, JUN ZHOU, Bin Shi

Code Large Language Models (CodeLLMs) have been widely adopted for Natural Language to Programming Language code generation, powering applications with large user bases. Their performance, however, varies sharply across programming languages (PLs) and is particularly suboptimal for low-resource PLs due to data scarcity, limiting their overall usability. In this work, we introduce CodeChemist, a simple yet effective, training-free test-time scaling framework that transfers the model's functional knowledge from high-resource to low-resource PLs via synthesized test cases, without relying on external models. Specifically, CodeChemist first applies multi-temperature hedged sampling to generate a pool of candidate solutions in the low-resource PL and synthesizes a set of test inputs.

It then estimates uncertainty: when uncertainty is low, it selects the output via in-language majority voting; otherwise, it constructs cross-lingual I/O test oracles by executing high-resource reference programs and selects the candidate with the highest pass rate.

Extensive experiments demonstrate that CodeChemist significantly outperforms existing test-time scaling methods, improving code generation for both low-resource PLs (e.g., Lua) and complex-syntax PLs (e.g., C++, Java) without retraining.

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When Shared Knowledge Hurts: Spectral Over-Accumulation in Model Merging

Yayuan Li, Ze Peng, Jian Zhang, Jintao Guo, Yue Duan, Yinghuan Shi

Model merging combines multiple fine-tuned models into a single model by $\texit{adding}$ their weight updates, providing a lightweight alternative to retraining. Existing methods primarily target resolving conflicts between task updates, leaving the failure mode of over-counting shared knowledge unaddressed. We show that when tasks share aligned spectral directions (i.e., overlapping singular vectors), a simple linear combination repeatedly accumulates these directions, inflating the singular values and biasing the merged model toward shared subspaces. To mitigate this issue, we propose Singular Value Calibration (SVC), a training-free and data-free post-processing method that quantifies subspace overlap and rescales inflated singular values to restore a balanced spectrum. Across vision and language benchmarks, SVC consistently improves strong merging baselines and achieves state-of-the-art performance. Furthermore, by modifying only the singular values, SVC improves the performance of Task Arithmetic by 13.0\%. Code is available at https://github.com/lyymuwu/SVC.

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Finding the Correct Visual Evidence Without Forgetting: Mitigating Hallucination in LVLMs via Inter-Layer Visual Attention Discrepancy

Yutong Xie, Zhenglin Hua, Ran Wang, Wing W. Y. Ng, Xizhao Wang, Yuheng Jia

Large Vision-Language Models (LVLMs) have shown remarkable performance on a wide range of vision-language tasks. Despite this progress, they are still prone to hallucination, generating responses that are inconsistent with visual content. In this work, we find that LVLMs tend to hallucinate when they pay insufficient attention to the correct visual evidence and gradually forget it during the generation process. We empirically find that although LVLMs overall attend insufficie

ntly to visual evidence, they exhibit sensitivity to the correct visual evidence in specific layers, with notable inter-layer discrepancy. Motivated by this observation, we propose a novel hallucination mitigation method that enhances visual evidence based on **I**nter-**L**ayer **V**isual **A**ttention **D**iscrepancy (**ILVAD**). Specifically, we obtain the attention weights from early generated tokens to visual tokens across layers and identify the tokens that are repeatedly activated as visual evidence, forming a saliency map. We then enhance attention to visual evidence during generation through the saliency map to reduce visual forgetting. In addition, we leverage the saliency map to obtain attention scores of generated text to visual evidence, in order to select and emphasize text tokens that are strongly grounded in visual evidence. Our method is training-free and plug-and-play. Multiple benchmark evaluations conducted on five recently released models show that our method can consistently mitigate hallucinations in different LVLMs over various architectures. Code is available at https://github.com/ytx-ML/ILVAD.
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One-step Latent-free Image Generation with Pixel Mean Flows
Yiyang Lu, Susie Lu, Qiao Sun, Hanhong Zhao, Zhicheng Jiang, Xianbang Wang, Tianhong Li, Zhengyang Geng, Kaiming He
Modern diffusion/flow-based models for image generation typically exhibit two core characteristics: (i) using multi-step sampling, and (ii) operating in a latent space. Recent advances have made encouraging progress on each aspect individually, paving the way toward one-step diffusion/flow without latents. In this work, we take a further step towards this goal and propose "pixel MeanFlow" (pMF). Our core guideline is to formulate the network output space and the loss space separately. The network target is designed to be on a presumed low-dimensional image manifold (i.e., x-prediction), while the loss is defined via MeanFlow in the velocity space. We introduce a simple transformation between the image manifold and the average velocity field. In experiments, pMF achieves strong results for one-step latent-free generation on ImageNet at 256$\times$256 resolution (2.22 FID) and 512$\times$512 resolution (2.48 FID), filling a key missing piece in this regime. We hope that our study will further advance the boundaries of diffusion/flow-based generative models.
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EffGen: Enabling Small Language Models as Capable Autonomous Agents
Gaurav Srivastava, Aafiya Hussain, Chi Wang, Yingyan Celine Lin, Xuan Wang
Most existing language model agentic systems today are built and optimized for large language models (e.g., GPT, Claude, Gemini) via API calls; while powerful, this approach faces several limitations including high token costs and privacy concerns for sensitive applications. We introduce **EffGen**, an open-source agentic framework optimized for small language models (SLMs) that enables effective, efficient, and secure local deployment. EffGen makes four major contributions: **(1) Enhanced tool-calling** with prompt optimization that compresses input prompts by up to 70-80% (and 57% on average across our benchmarks) while preserving task semantics, **(2) Intelligent task decomposition** that breaks complex queries into parallel or sequential subtasks based on dependencies, **(3) Complexity-based routing** using five factors to make smart pre-execution decisions, and **(4) Unified memory system** combining short-term, long-term, and vector-based storage. Additionally, EffGen unifies multiple agent protocols (MCP, A2A, ACP) for cross-protocol communication. Results on 13 benchmarks show EffGen outperforms LangChain, AutoGen, and Smolagents with **higher success rates**, **faster execution**, and **lower memory**. Our results reveal that **prompt optimization and complexity routing have complementary scaling behavior**: optimization benefits SLMs more (11.2% gain at 1.5B vs 2.4% at 32B), while routing benefits large models more (3.6% at 1.5B vs 7.9% at 32B), providing consistent gains across all scales when combined. EffGen is open-source under the Apache 2.0 License, with the code available at https://github.com/ctrl-gaurav/effGen, the Python package at https://pypi.org/project/effgen/ (`pip install effgen`), and the project website and documentation at https://effgen.org/ and https://docs.effgen.org/.
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One Tool Is Enough: Reinforcement Learning of LLM Agents for Repository-Level Co
de Navigation
Zhaoxi Zhang, Yitong Duan, Yanzhi Zhang, Yiming Xu, Zhixiang Wang, Kun Liang, We
ikang Li, Jiahui Liang, Deguo Xia, Jizhou Huang, Jiyan He, Shuxin Zheng, Yunfang
 Wu
Locating files and functions requiring modification in large software repositori
es is challenging due to their scale and structural complexity. Existing LLM-bas
ed methods typically treat this as a repository-level retrieval task and rely on
 multiple auxiliary tools, which often overlook code execution logic and complic
ate model control. We propose RepoNavigator, an LLM agent equipped with a single
 execution-aware tool: jumping to the definition of an invoked symbol. This unif
ied design reflects the actual flow of code execution while simplifying tool man
ipulation. RepoNavigator is trained end-to-end via Reinforcement Learning (RL) d
irectly from a base pretrained model, without relying on closed-source distillat
ion. Experiments demonstrate that RL-trained RepoNavigator achieves state-of-the
-art performance, with the 7B model outperforming 14B baselines, the 14B model s
urpassing 32B competitors, and the 32B model exceeding closed-source models such
 as GPT-5 on most metrics. These results confirm that integrating a single, stru
cturally grounded tool with RL training provides an efficient and scalable solut
ion for repository-level issue localization.
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MMD Guidance: Training-Free Distribution Adaptation for Diffusion Models via Max
imum Mean Discrepancy Guidance
Matina Mahdizadeh Sani, Nima Jamali, Mohammad Jalali, Farzan Farnia
Pre-trained diffusion models have emerged as powerful generative priors for both
 unconditional and conditional sample generation, yet their outputs often deviat
e from the characteristics of user-specific target data. Such mismatches are esp
ecially problematic in domain adaptation tasks, where only a few reference examp
les are available and retraining the diffusion model is infeasible. Existing inf
erence-time guidance methods can adjust sampling trajectories, but they typicall
y optimize surrogate objectives such as classifier likelihoods rather than direc
tly aligning with the target distribution. We propose *MMD Guidance*, a training
-free mechanism that augments the reverse diffusion process with gradients of th
e *Maximum Mean Discrepancy (MMD)* between generated samples and a reference dat
aset. MMD provides reliable distributional estimates from limited data, exhibits
 low variance in practice, and is efficiently differentiable, which makes it par
ticularly well-suited for the guidance task. Our framework naturally extends to
prompt-aware adaptation in conditional generation models via product kernels. Al
so, it can be applied with computational efficiency in latent diffusion models (
LDMs), since guidance is applied in the latent space of the LDM. Experiments on
synthetic and real-world benchmarks demonstrate that MMD Guidance can achieve di
stributional alignment while preserving sample fidelity. The project code is ava
ilable at [https://github.com/matinamehdizadeh/MMD-Guidance](https://github.com/
matinamehdizadeh/MMD-Guidance).
****

Semi-LAR: Semi-supervised Contrastive Learning with Linear Attention for Removal
 of Nighttime Flares
Xiyu Zhu, Wei Wang, Kui Jiang, Zhengguo Li
Lens flare removal is challenging due to the large spatial extent of flare artif
acts and their entangle-ment with scene structures, while existing meth-ods heav
ily rely on large-scale paired data. We propose a semi-supervised flare removal
frame-work that enables stable learning from unlabeled images by jointly address
ing pseudo-label relia-bility and representation discrimination. We pro-pose an
adaptive pseudo-label repository that pro-gressively refines pseudo supervision
through no-reference quality assessment, momentum-based updates, and invalid lab
el filtering, effectively mit-igating error accumulation. Moreover, we pro-pose
a flare-aware contrastive loss that explic-itly treats flare-contaminated inputs
 as negatives and performs patch-level contrastive learning, en-couraging repres
entations that are discriminative against flare patterns while remaining consist
ent with reliable pseudo targets. Extensive experi-ments on multiple flare bench

marks demonstrate that the proposed framework is model-agnostic and consistently improves performance and ro-bustness.
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FUSE: Quantifying Uncertainty in Vision-Language Models by Bayesian Fusing Epistemic and Aleatoric Uncertainty
Harry Zhang, Luca Carlone
Multimodal large language models (MLLMs) are playing an increasingly important role across multiple domains.
In many applications, such as robotics, it is crucial to quantify the uncertainty in the output of these models. }
We develop Fused Uncertainty with Semantic Evidence (FUSE), a probabilistic framework for capturing two complementary sources of uncertainty in multimodal language modeling: (i) aleatoric embedding-level uncertainty derived from input data vision-language ambiguity, and (ii) epistemic model-level uncertainty estimated from the semantic response diversity of MLLMs.
Our approach formulates a Bayesian fusion mechanism that analytically combines these uncertainty sources to produce a scalar measure of uncertainty. This measure serves as a novel uncertainty representation for downstream applications of MLLMs and provides a principled foundation for uncertainty calibration in multimodal systems, improving reliability and downstream performance in MLLM-based reasoning and vision-language tasks. We demonstrate that our method outperforms baselines in providing
uncertainty estimates and achieves state-of-the-art uncertainty calibration.
****
Relational Structural Causal Models
Adiba Ejaz, Elias Bareinboim
An artificial intelligence must have a model of its environment that is *causal*, supporting reasoning about interventions and counterfactuals, and also *combinatorial*, supporting generalization to unseen combinations of objects.
In this work, we formally study when and how such a model can be learned.
We develop *relational structural causal models*, extending structural causal models (Pearl 2009) to settings where objects and their relations vary.
First, we show how answers to not only causal but also observational queries about unseen combinations of objects can not be identified without further assumptions.
To enable such identification---including in the presence of unobserved confounding---we define *relational causal graphs* and derive symbolic identification criteria.
Finally, we propose *relational neural causal models*, a provably correct approach that outperforms non-relational baselines on simulated traffic scenes with varying cars, signals, and pedestrians.
****
Spik4lite: Refactoring Neuromorphic Sparsity for Efficient Spiking Neural Networks on Commodity Edge Devices
Yongzhi She, Qihua Zhou, Yuhao Wang, Yaodong Huang, Laizhong Cui, Jingcai Guo
Recently, the spiking neural networks (SNNs) have shown great promise in enhancing AI task performance by utilizing the brain-inspired and energy-efficient computational paradigm via the binary (0/1) spikes. Modern SNNs, especially those based on transformers, often require FPGA accelerators or neuromorphic chips to enable spike-driven computations. However, this domain-specific hardware is not always accessible on commodity edge devices like NVIDIA Jetsons, which may degrade SNNs' energy efficiency due to massive computational waste on inactive "0" spikes and finally undermine the usage boundary. This limitation raises an interesting question: is it possible to make SNNs edge-friendly and tame the computations mostly on active "1" spikes? We present the answer yes and propose Spik4lite, which serves as a lightweight plug-and-play module to significantly improve SNN's performance between model accuracy and computational efficiency. The key is to refactor SNN's channel-wise neuromorphic sparsity by zeroing out low-efficiency channels while proactively compensating for the eliminated spikes. Different from prior methods mainly focusing on optimizing the theoretical synaptic operation

s, our design philosophy can evolve the SNNs into a physically compact manner, t
hus inherently saving more computational and energy costs. Extensive experiments
 based on real edge devices show that Spik4lite can be integrated into existing
SNN baselines to further improve their accuracy-and-efficiency performance, guar
anteeing the model accuracy while saving the computational and energy costs.
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MultiBreak: A Scalable and Diverse Multi-turn Jailbreak Benchmark for Evaluating
 LLM Safety

Jialin Song, Xiaodong Liu, Weiwei Yang, Wuyang Chen, Mingqian Feng, Xuekai Zhu,
Jianfeng Gao

We present MultiBreak, a scalable and diverse multi-turn jailbreak benchmark to
evaluate large language model (LLM) safety. Multi-turn jailbreaks mimic natural
conversational settings, making them easier to bypass safety-aligned LLM than si
ngle-turn jailbreaks. Existing multi-turn benchmarks are limited in size or rely
 heavily on templates, which restrict their diversity.

To address this gap, we unify a wide range of harmful jailbreak intents, and int
roduce an active learning pipeline for expanding high-quality multi-turn adversa
rial prompts, where a generator is iteratively fine-tuned to produce stronger at
tack candidates, guided by uncertainty-based refinement. Our MultiBreak includes
 10,389 multi-turn adversarial prompts, spans 2,665 distinct harmful intents, an
d covers the most diverse set of topics to date. Empirical evaluation shows that
 our benchmark achieves up to a 54.0% and 34.6% higher attack success rate (ASR)
 than the second-best dataset on DeepSeek-R1-7B and GPT-4.1-mini, respectively.

More importantly, safety evaluations suggest that diverse attack categories unco
ver fine-grained LLM vulnerabilities, and categories that appear benign under si
ngle-turn can exhibit substantially higher adversarial effectiveness in multi-tu
rn scenarios.

These findings highlight persistent vulnerabilities of LLMs under realistic adve
rsarial settings and establish MultiBreak as a scalable resource for advancing L
LM safety.
****

MapUQ: Map with Uncertainty Quantification for Robust BEV Vectorized Constructi
on

Shaoyuan Mo, Qi Ma, leng rui, Bohan Li, Ke Wang

End-to-end online map generation is a core component of autonomous driving perce
ption systems. However, in complex traffic scenes, Bird’s-Eye-View (BEV) with ve
ctorized mapping suffers from limitations such as target misclassification, spat
ial localization drift, and ambiguous semantic segmentation. Introducing uncerta
inty quantification can alleviate these problems, so we propose MapUQ, a robust
BEV vectorized mapping method guided by uncertainty-aware optimization. Specific
ally, we quantify uncertainty at the feature level to enhance semantic perceptio
n, apply an error-driven dynamic receptive field adaptation mechanism at the dec
oding stage to enforce geometric consistency, and leverage negative sample infor
mation at the output head to improve lane classification accuracy. Experimental
results on the nuScenes and Argoverse 2 datasets show that our method outperform
s prior approaches in AP across three road types, achieving an average improveme
nt of 1.5% over the baseline with marginal computational overhead. In addition,
our method surpasses the baseline on uncertainty metrics such as ECE and NLL, si
gnificantly improving robustness and mapping accuracy in complex scenarios. Our
code has been released at github: https://anonymous.4open.science/r/MapUQ-D287.
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The Geometry of Reasoning: Self-Evaluation via Layerwise Trajectory Evolution

Jinhe Bi, Danqi Yan, Yifan Wang, Wenke Huang, Haokun Chen, Guancheng Wan, Mang Y
e, Xun Xiao, Hinrich Schuetze, Volker Tresp, Yunpu Ma

Large Reasoning Models (LRMs) enhance performance by generating explicit Chain-o
f-Thought (CoT) trajectories, yet enabling them to self-evaluate correctness wit
hout external supervision remains a critical challenge. Existing methods often r
ely on ground-truth labels or shallow output probabilities, neglecting the layer
wise evolution of the reasoning trajectory. In this work, we introduce GeoR (Geo
metry of Reasoning), a white-box self-evaluation framework based on layerwise tr

ajectory evolution. GeoR decomposes reasoning fidelity into two complementary di
mensions: (1) Geometric Evolution, which synthesizes the first- and second-order
 evolution of layerwise hidden-state trajectories to quantify geometric progress
 in reasoning; and (2) Difficulty-Aware Calibration, which utilizes cross-entrop
y of reasoning progress to normalize the Geometric Evolution against intrinsic q
uery uncertainty. By jointly modeling these factors, GeoR effectively distinguis
hes the coherent evolution of correct reasoning from the chaotic trajectories of
 errors. Extensive experiments across eight LRMs and seven benchmarks demonstrat
e that GeoR consistently outperforms state-of-the-art baselines in AUROC, AUPR,
and FPR@95.
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WorldComp2D: Spatio-semantic Representations of Object Identity and Location fro
m Local Views

Seongmin Jin, Doo Seok Jeong

Learning latent representations that capture both semantic and spatial informati
on is central to efficient spatio-semantic reasoning. However, many existing app
roaches rely on implicit latent structures combined with dense feature maps or t
ask-specific heads, limiting computational efficiency and flexibility.
We propose WorldComp2D, a novel lightweight representation learning framework th
at explicitly structures latent space geometry according to object identity and
spatial proximity using multiscale \textit{local} receptive fields. This framewo
rk consists of (i) a proximity-dependent encoder that maps a given observation i
nto a spatio-semantic latent space and (ii) a localizer that infers the coordina
tes of objects in the input from the resulting spatio-semantic representation.
Using facial landmark localization as a proof-of-concept, we show that, compared
 to SoTA lightweight models, WorldComp2D reduces the numbers of parameters and F
LOPs by up to $4.0\times$ and $2.2\times$, respectively, while maintaining real-
time performance on CPU. These results demonstrate that explicitly structured la
tent spaces provide an efficient and general foundation for spatio-semantic reas
oning. This framework is open-sourced at https://github.com/JinSeongmin/WorldCom
p2D.
****

TVDRNet: Text-driven Viewpoint Optimization via Differentiable Rendering for 3D
Reasoning Segmentation

Zaiyang Yu, Changshuo Wang, Pinjie Xu, Zhang Huang, Yuan Shi, Linjun Sun, Weijun
 Li

Three-dimensional (3D) reasoning segmentation aims at segmenting target objects
based on text instructions and 3D spatial cues. Recent efforts in 3D reasoning l
everage Multimodal Large Language Models (MLLMs) to bridge the gap between text
and 3D data. However, since MLLMs are primarily trained on text-image pairs, dir
ectly adapting them to unstructured 3D point clouds often fails to capture impli
cit semantic intent and reliably localize objects. This paper introduces TVDRNet
 to address these challenges. Inspired by Active Vision theory, where humans sel
ectively choose optimal viewpoints to better observe targets, TVDRNet employs a
differentiable renderer to simulate this active process in 3D perception. By usi
ng text instructions as supervision to optimize intrinsic and extrinsic renderin
g parameters, the TVDRNet identifies the optimal viewpoints for observing the 3D
 scene, and therefore learning 'where to look' based on what the text instructio
n 'asked to find'. This process generates informative, task-relevant 2D images t
hat are compatible with MLLMs. TVDRNet comprises: (1) the AVPL module, establish
ing a learnable mapping from semantics to optimal rendering parameters; and (2)
the MGL module, fusing multi-modalities via semantic grouping to guide mask gene
ration. Experiments show TVDRNet achieves the state-of-the-art performance in 3D
 reasoning segmentation (Reason3D, Instruct3D) and 3D visual grounding (ScanRefe
r) benchmarks.
****

Approximation of Log-Partition Function in Policy Mirror Descent Induces Implici
t Regularization for LLM Post-Training

Zhenghao Xu, Qin Lu, Changlong Yu, Tuo Zhao

Policy mirror descent (PMD) provides a principled framework for reinforcement le

arning (RL) by iteratively solving KL-regularized policy improvement subproblems . While this approach has been adopted in training advanced LLMs such as Kimi K1 .5/K2, the ideal closed-form PMD updates require reliable partition function est imation, a significant challenge when working with limited rollouts in the vast action spaces of LLMs. We investigate a practical algorithm, termed PMD-mean, th at approximates the log-partition term with the mean reward under the sampling p olicy and performs regression in log-policy space. Specifically, we characterize the population solution of PMD-mean and demonstrate that it implicitly optimize s mirror descent subproblems with an adaptive mixed KL--χ^2 regularizer. Th is additional χ^2 regularization constrains large probability changes, prod ucing more conservative updates when expected rewards are low and enhancing robu stness against finite-sample estimation errors. Experiments on math reasoning ta sks show that PMD-mean improves stability and remains competitive in the stale-r ollout settings while enabling higher throughput through larger rollout batches. These findings deepen our understanding of PMD-mean and illuminate pathways tow ard principled improvements in RL algorithms for LLMs.
Code is available at https://github.com/horizon-llm/OpenKimi.
****

Scaling depth capacity via zero/one-layer model expansion
Zhiqi Bu
Model depth is a double-edged sword in deep learning: deeper models achieve high er accuracy but require higher computational cost. To efficiently train models a t scale, progressive training (also known as model expansion) scales up model ca pacity during training and significantly reduces computation with little perform ance degradation.
In this work, we study the depth expansion of large-scale models through the len s of optimization theory and feature learning, offering insights on the initiali zation of new layers, hyperparameter transfer, learning rate schedule, and timin g of model expansion. Specifically, we propose zero/one-layer progressive traini ng to achieve an optimal tradeoff between computation and loss, with a comprehen sive ablations on our expansion strategy. For example, zero/one-layer progressiv e training on GPT2 can save $\approx 80\%$ compute, or equivalently achieve an $ \approx 5\times$ acceleration, while attaining a loss comparable to that of a fu lly trained 60-layer model with 7B parameters, thus demonstrating a mixing behav ior in terms of loss. Furthermore, scaling laws on LLAMA3 and DeepSeekV3 models show a $3\sim 5\times$ improvement in compute efficiency, with an increasing adv antage at larger scales.
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CausalGame: Benchmarking Causal Thinking of LLM Agents in Games
Zhenhao Chen, Yongqiang Chen, Chenxi Liu, Junchi Yu, Xiangchen Song, Zijian Li, Jialin Li, Philip Torr, Bo Han, Kun Zhang
Recently, it has received growing attention in building AI Scientist agents with Large Language Models (LLMs). Since scientific discovery fundamentally relies o n uncovering causal relationships from observations, the capability of causal th inking that distinguish causation from correlation and hidden biases, is essenti al to LLM agents. Despite a number of existing benchmarks for AI scientists, the y do not explicitly incorporate challenges from hidden confounders, selection bi as, and noisy measurements that widely exist in real-world scientific discovery. To this end, we present CausalGame, a benchmark that evaluates the causal think ing capabilities of LLM agents through interactive games. More specifically, we ask LLM agents to actively design experimental protocols, collect observation da ta and derive a final solution with an explanation report. To emulate realistic scientific discovery challenges, we design 14 game settings with the incorporati on of selection bias, noisy measurements, and hidden confounders. The results wi th 29 frontier LLM agents show that they consistently fail to reason about and r ecover the underlying causal relationships required to solve the games. CausalGa me provides a controlled testbed for evaluating causal thinking of AI Scientist agents. The project is available at causalgame.github.io .
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MetaMoE: Diversity-Aware Proxy Selection for Privacy-Preserving Mixture-of-Exper

ts Unification
Weisen Jiang, Shuhao Chen, Sinno Jialin Pan
Mixture-of-Experts (MoE) models scale capacity by combining specialized experts, but most existing approaches assume centralized access to training data. In practice, data are distributed across clients and cannot be shared due to privacy constraints, making unified MoE training challenging. We propose **MetaMoE**, a privacy-preserving framework that unifies independently trained, domain-specialized experts into a single MoE using public proxy data as surrogates for inaccessible private data.

Central to MetaMoE is diversity-aware proxy selection, which selects client-domain–relevant and diverse samples from public data to effectively approximate private data distributions and supervise router learning.

These proxies are further used to align expert training, improving expert coordination at unification time, while a context-aware router enhances expert selection across heterogeneous inputs.

Experiments on computer vision and natural language processing benchmarks demonstrate that MetaMoE consistently outperforms recent privacy-preserving MoE unification methods.
Code is available at https://github.com/ws-jiang/MetaMoE.
****
LightWM: Training-Free Hierarchical Working Memory for Small Language Model Agents
Ziyi Wang, Haonan Jin, Zian Wang, Wendong Wang, Lanshan Zhang
Small language models (SLMs) are attractive for low-cost agent deployment, and their growing capabilities make them promising for procedure-centric workloads that repeatedly execute specialized task families with similar workflows.
However, SLM agents still struggle in long-horizon interaction because limited context budgets make it difficult to reliably retain and reuse decision-relevant state across many steps.
Existing working-memory methods attempt to mitigate context growth, but their reliance on unstructured natural-language summarization can discard critical facts, introduce state drift, and compound errors in SLM execution.
We present LightWM, a training-free hierarchical working-memory framework that decomposes procedure-centric tasks into subgoals and organizes memory into task-level global memory and subtask-level local memory, where local memory directly conditions SLM action selection and is updated from new observations through structured updates.
To instantiate such memories without training, a one-time offline LLM-based induction pipeline builds reusable schemas per task family from a few successful traces, requiring no SLM parameter updates or online LLM calls.
On ALFWorld valid_unseen, Qwen3-4B reaches 0.910 success, whereas representative prompting and prior working-memory baselines under the same setting remain below 0.320.
****
Semantic Cache Distillation: Efficient State Transfer via Reuse and Selective Patching
Qianli Ma, Zhiqing Tang, Hanshuai Cui, Zhi Yao, Weijia Jia
Disaggregated serving alleviates memory bottlenecks in Large Language Model (LLM) inference but creates a severe communication bottleneck: transmitting high-dimensional Key-Value (KV) caches often dominates time-to-first-token (TTFT). Moreover, reusing caches across heterogeneous models (e.g., base and fine-tuned variants) causes semantic misalignment that accumulates over layers, degrading generation quality. We propose Semantic Cache Distillation (SCD), a loss-constrained framework that replaces raw KV transmission with compact semantic codes. SCD addresses these challenges via two mechanisms: (1) Reuse, which reconstructs most layers from low-rank subspaces to minimize transfer cost, and (2) Patch, which predicts normalized inputs at sparse transition layers to truncate error propagation. Empirically, SCD delivers up to 2.65 $\times$ TTFT speedup over the oracle consumer prefill and dominates quantization and selective recomputation baselines on the quality--latency Pareto frontier in bandwidth-constrained regimes, while

keeping generation quality within 5\% F1 of the oracle.
****

Controlled Dynamics Attractor Transformer
Cheng Zhang, Minnan Luo, Zesheng Yang, MING LI, Yong-Jin Liu, Qinghua Zheng
Transformer architectures have dramatically advanced representation learning and inference in deep models through self-attention mechanisms. In parallel, associative memory frameworks map representations onto energy landscapes, offering interpretable retrieval mechanisms. However, their continuous-time inference dynamics lack the biological plausibility of classical Continuous Attractor Neural Networks. To bridge this gap, we propose **Controlled Dynamics Attractor Transformer (CDAT)**, which couples a mixture von Mises–Fisher attention energy with a Hopfield refinement energy, while augmenting energy descent with a CANN-inspired excitation–inhibition modulation. CDAT instantiates a topology-constrained dynamical system whose couplings encode relational structure among tokens, thereby linking attractor-style dynamics to modern energy-based attention. We further provide a constructive dissipation analysis to formally establish their controlled inference dynamics. Benefiting from these robust and structured dynamics, CDAT achieves state-of-the-art performance across multiple benchmarks in graph anomaly detection and graph classification.
****

WorldTravel: A Realistic Multimodal Travel-Planning Benchmark with Tightly Coupled Constraints
Zexuan Wang, Chenghao Yang, Yingqi Que, Zhoufutu Wen, Zaiyuan Wang, Jiashuo Liu, Zhixin Yao, Zhenzhu Yang, Huaqing Yuan, Yiwen Wang, Zhengxuan Jiang, Shengjie Fang, Zhenhe Wu, Zhaohui Wang, JinCheng Ren, Yuzhen Li, Yang Yang, Jiaheng Liu, Jian Yang, Ge Zhang, Wenhao Huang
Real-world autonomous planning requires coordinating tightly coupled constraints where a single decision dictates the feasibility of all subsequent actions. However, existing benchmarks predominantly feature loosely coupled constraints solvable through local greedy decisions and rely on idealized data, failing to capture constraint acquisition from realistic web interfaces.
We introduce $\textbf{WorldTravel}$, a benchmark comprising 150 real-world travel scenarios across 5 cities, requiring agents to satisfy an average of 15+ interdependent temporal and logical constraints. To evaluate realistic deployment settings, we further develop $\textbf{WorldTravel-Webscape}$, a multi-modal environment with over 2,000 rendered webpages that preserve layout-dependent and information-dense travel interfaces, requiring agents to recover executable constraints from rendered web interfaces. Evaluating 10 frontier models reveals a severe performance collapse: GPT-5.2 achieves only 28.0\% feasibility in text-only settings, dropping to 3.4\% in multi-modal environments. We observe substantial degradation in planning feasibility when agents must recover executable constraints from rendered webpages, alongside a Planning Horizon threshold at approximately 10 constraints where reasoning reliability collapses. These findings suggest that realistic constraint acquisition and long-horizon planning remain complementary bottlenecks for current agents.
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Trajectory-Aware Heuristic Learning for Combinatorial Search
Mustafa Seddiqi, Marta Kersten-Oertel, Tiberiu Popa
Learning effective value heuristics for combinatorial search is difficult, as prior methods rely on surrogate supervision or costly downstream search to assess progress. We introduce a trajectory-aware probabilistic framework that models uncertainty in cost-to-go labels instead of treating them as fixed targets. Heuristic learning is cast as inference over state trajectories using an HMM-style model, where estimated depth-change dynamics define transitions and forward-backward inference yields soft supervision. To evaluate heuristic quality without search, we propose a large-scale local ranking metric that measures a model's ability to order neighboring states. On the Rubik's Cube, our approach consistently improves local ranking accuracy and downstream search performance under matched computational budgets.
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Linguistic Properties and Model Scale in Brain Encoding: From Small to Compressed Language Models
SUBBA REDDY OOTA, Vijay Rowtula, Satya Sai Srinath Namburi GNVV, Khushbu Pahwa, Anant Khandelwal, Manish Gupta, Tanmoy Chakraborty, Bapi Raju Surampudi
Recent work has shown that scaling large language models (LLMs) improves their alignment with human brain activity, yet it remains unclear what drives these gains or which representational properties are responsible. Although larger models often yield better task performance and brain alignment, they are increasingly difficult to analyze mechanistically. This raises a fundamental question: \emph{what is the minimal model capacity required to capture brain-relevant representations?} To address this question, we systematically investigate how constraining model scale and numerical precision affects brain alignment. We compare full-precision LLMs, small language models (SLMs), and compressed variants (quantized and pruned) by predicting fMRI responses during naturalistic language comprehension. Across model families up to 14B parameters, we find that 3B SLMs achieve brain predictivity indistinguishable from larger LLMs, whereas 1B models degrade substantially, particularly in semantic language regions. Brain alignment is remarkably robust to compression: most quantization and pruning methods preserve neural predictivity, with GPTQ as a consistent exception. Linguistic probing reveals a dissociation between task performance and brain predictivity: compression degrades discourse, syntax, and morphology, yet brain predictivity remains largely unchanged. Overall, brain alignment saturates at modest model scales and is resilient to compression, challenging common assumptions about neural scaling and motivating compact models for brain-aligned language modeling.
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UDM-GRPO: Stable and Efficient Group Relative Policy Optimization for Uniform Discrete Diffusion Models
Jiaqi Wang, Haoge Deng, Ting Pan, Yang Liu, Chengyuan Wang, Fan Zhang, Yonggang Qi, Xinlong Wang
Uniform Discrete Diffusion Model (UDM) has recently emerged as a promising paradigm for discrete generative modeling; however, its integration with reinforcement learning remains largely unexplored. We observe that naively applying GRPO to UDM leads to training instability and marginal performance gains. To address this, we propose **UDM-GRPO**, the first framework to integrate UDM with RL. Our method is guided by two key insights: (i) treating the final clean sample as the action provides more accurate and stable optimization signals; and (ii) reconstructing trajectories via the diffusion forward process better aligns probability paths with the pretraining distribution. Additionally, we introduce two strategies, Reduced-Step and CFG-Free, to further improve training efficiency. **UDM-GRPO** significantly improves base model performance across multiple T2I tasks. Notably, GenEval accuracy improves from $69\\%$ to $96\\%$ and PickScore increases from 20.46 to 23.81, achieving state-of-the-art performance in both continuous and discrete settings. On the OCR benchmark, accuracy rises from $8\\%$ to $57\\%$, further validating the generalization ability of our method. Code is available at https://github.com/Yovecent/UDM-GRPO.
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ArborKV: Structure-Aware KV Cache Management for Scaling Tree-based LLM Reasoning
Yeqiu Chen, Ziyan Liu, Zhenxin Huang, Runquan Gui, Hong Wang, Lei Liu
Recent progress in LLM reasoning has increasingly shifted from single-pass generation to explicit search over intermediate reasoning states. Tree-of-Thoughts (ToT) organizes inference to tree-structured search with branching and backtracking, but it substantially amplifies the key--value (KV) cache: retaining KV states for a frontier of partial trajectories quickly becomes a memory bottleneck that limits throughput and constrains search depth and width under fixed hardware budgets. We address this challenge by observing that KV reuse in ToT-style inference is governed by search dynamics: near-term decoding depends primarily on the active branch and its ancestors, whereas inactive subtrees have low short-term reuse probability yet must remain recoverable for backtracking. Motivated by this, we propose **ArborKV**, a structure-aware eviction framework that couples a lig

htweight value estimator with a tree-aware allocation policy, and performs purel y token-extractive eviction with lazy rehydration to support revisits. Experimen ts on ToT-style reasoning benchmarks show that ArborKV achieves up to $\sim4\tim es$ peak KV-memory reduction while preserving near-full-retention accuracy, enab ling larger search configurations under fixed device budgets that would otherwis e run out of memory.
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Path-Decoupled Hyperbolic Flow Matching for Few-Shot Adaptation
Lin Li, Ziqi Jiang, Gefan Ye, Zhenqi He, Jiahui Li, Jun Xiao, Kwang-Ting Cheng, Long Chen
Recent advances in cross-modal few-shot adaptation treat visual-semantic alignme nt as a continuous feature transport problem via Flow Matching (FM). However, we argue that Euclidean-based FM overlooks fundamental limitations of flat geometr y, where polynomial volume growth fails to accommodate diverse feature distribut ions, leading to severe path entanglement. To this end, we propose path-decouple d Hyperbolic Flow Matching (HFM), leveraging the Lorentz manifold's exponential expansion for trajectory decoupling. HFM structures the transport via two key de signs: 1) Centripetal hyperbolic alignment: It constructs a centripetal hierarch y by anchoring textual roots, which pushes visual leaves to the boundary to init ialize orderly flows. 2) Path-decoupled objective: It acts as a "semantic guardr ail" rigidly confining trajectories within isolated class-specific geodesic corr idors via step-wise supervision. Furthermore, we devise an adaptive diameter-bas ed stopping to prevent over-transportation into the crowded origin based on the intrinsic semantic scale. Extensive ablations on 11 benchmarks have shown that H FM establishes a new state-of-the-art, consistently outperforming its Euclidean counterparts. Our codes and models will be released.
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Treatment Responder Classification with Abstention
Haoxiang Wang, Aoqi Zuo, Ziyan Wang, Zhiheng Zhang, Erdun Gao, Kun Zhang, Haoxua n Li, Mingming Gong
Treatment responder classification seeks to learn a rule to classify individuals who will benefit from the treatment. This paper studies a new scenario in treat ment responder classification when abstention is allowed, i.e., practitioners ca n opt out of making uncertain classification on some individuals for further inv estigation. By revealing the implicit relation between causal misclassification risk with abstention and Conditional Value at Risk (CVaR), we develop a doubly r obust method named TRECA to learn the classification rule under loose convergenc e conditions on nuisance parameters, and further extend it to deal with possible violation on key assumptions such as monotonicity and unconfoundedness. Rigorou s theories and extensive experiments on two real-world datasets demonstrate the theoretical and experimental guarantee on our methods in learning treatment resp onders classification rules with low regret at the cost of limited abstention.
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Large Language Models Explore by Latent Distilling
Yuanhao Zeng, Ao Lu, Lufei Li, Zheng Zhang, Yexin Li, Kan Ren
Generating diverse responses is crucial for test-time scaling of large language models (LLMs), yet standard stochastic sampling mostly yields surface-level lexi cal variation, limiting semantic exploration. In this paper, we propose Explorat ory Sampling (ESamp), a decoding approach that explicitly encourages semantic di versity during generation. ESamp is motivated by the well-known observation that neural networks tend to make lower-error predictions on inputs similar to those encountered before, and incur higher prediction error on novel ones. Building o n this property, we train a lightweight Distiller at test time to predict deep-l ayer hidden representations of the LLM from its shallow-layer representations to model the LLM’s depth-wise representation transitions. During decoding, the Dis tiller continuously adapts to the mappings induced by the current generation con text. ESamp uses the prediction error as a novelty signal to reweight candidate token extensions conditioned on the current prefix, thereby biasing decoding tow ard less-explored semantic patterns. ESamp is implemented with an asynchronous t raining--inference pipeline, with less than 5\% worst case overhead (1.2\% in th

e optimized release). Empirical results show that ESamp significantly boosts the Pass@k efficiency of reasoning models, showing superior or comparable performance to strong stochastic and heuristic baselines. Notably, ESamp achieves robust generalization across mathematics, science, and code generation benchmarks and breaks the trade-off between diversity and coherence in creative writing. Our code has released at: https://github.com/LinesHogan/tLLM.
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Differentially Private Synthetic Data via APIs 4: Tabular Data
Toan Tran, Arturs Backurs, Zinan Lin, Victor Reis, Li Xiong, Sergey Yekhanin
This paper investigates the problem of generating synthetic tabular data with differential privacy (DP) guarantees, enabling data sharing in sensitive domains. Despite extensive study, state-of-the-art methods often focus on minimizing low-order marginal query errors and overlook the challenges posed by high-order correlations. To address this gap, we extend the Private Evolution (PE) framework, originally developed for DP-compliant image and text synthesis, to tabular data. We introduce Tab-PE -- an algorithm for synthetic tabular data generation under DP constraints. Tab-PE iteratively improves a candidate dataset via an evolutionary process that leverages tabular-specialized operators to produce variations, privately scores them, and selects the highest-quality samples to retain and propagate. In contrast to the original PE, which relies on large foundation models, Tab-PE employs heuristic operators with significantly lower computational costs, making PE more practical and scalable for tabular data. Through extensive experiments on real-world and simulation datasets, we demonstrate that Tab-PE substantially outperforms prior baselines on datasets exhibiting high-order correlations. Compared to the best baseline -- AIM, Tab-PE improves classification accuracy by up to 10\% while running 28$\times$ faster.
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FormulaCode: Evaluating Agentic Optimization on Large Codebases
Atharva Sehgal, James Hou, Akanksha Sarkar, Ishaan Mantripragada, Swarat Chaudhuri, Jennifer J. Sun, Yisong Yue
Large language model (LLM) coding agents increasingly operate at the repository level, motivating benchmarks that evaluate their ability to optimize entire codebases under realistic constraints. Existing code benchmarks largely rely on synthetic tasks, binary correctness signals, or single-objective evaluation, limiting their ability to assess holistic optimization behavior. We introduce FormulaCode, a benchmark for evaluating agentic optimization on large, real-world codebases with fine-grained, multi-objective performance metrics. FormulaCode comprises 957 performance bottlenecks mined from scientific Python repositories on GitHub, each paired with expert-authored patches and 264.6 community-maintained performance workloads per task, enabling evaluation of the full optimization lifecycle—triage, diagnosis, and resolution—under realistic correctness and performance constraints. Our evaluations reveal that repository-scale, multi-objective optimization remains a major challenge for frontier LLM agents.
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MVR-cache: Optimizing Semantic Caching via Multi-Vector Retrieval and Learned Prompt Segmentation
Ali Noshad, Zishan Zheng, Yinjun Wu
To reduce LLM costs and latency, semantic caching systems must accurately identify when a new prompt matches a cached one. Current methods often rely on simplistic similarity measures, which limit their effectiveness. We introduce MVR-cache, a novel semantic caching approach that significantly improves retrieval accuracy by integrating Multi-Vector Retrieval (MVR). MVR-cache is built upon a learnable segmentation model that intelligently splits prompts, enabling fine-grained similarity comparisons via MaxSim. We derive the model's training objective from a rigorous theoretical analysis. This can ensure that optimizing this objective directly maximizes cache hits under strict correctness constraints. To solve the resulting non-differentiable combinatorial optimization problem, we leverage a reinforcement learning-based training strategy with the theoretically grounded objectives as the reward. Experimental results on established benchmarks across diverse tasks confirm that in comparison to the state-of-the-art, MVR-cache cons

istently increases the cache hit rates by up to 37\% while maintaining the same correctness guarantees. MVR-cache is available at https://github.com/PKU-SDS-lab/MVR-Cache
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Responsible Text-to-Image Diffusion: Interpretable and Linearly Controllable Semantics for Fair and Safe Generation
Sayedmoslem Shokrolahi, Jae-Mo Kang, Il-Min Kim
Text-to-image (T2I) diffusion models (DMs) have achieved remarkable generative quality but still exhibit risk of producing biased and inappropriate images. A promising line of prior work aims to mitigate this issue by learning interpretable, linearly controllable concepts from semantic spaces, such as the U-Net bottleneck; however, these methods rely entirely on U-Net architectures and cannot be generalized to modern ViT-based DMs, including FLUX and PixArt. In this work, we present an architecture-agnostic framework for discovering interpretable and linearly controllable semantic attributes across any T2I DM backbone. Theoretically, we show that multi-modal attention heads in ViT-based DMs exhibit a linear semantic structure: injected concept vectors combine linearly at the attention level and induce near-linear effects at the model output, satisfying homogeneity and additivity. These theoretical results are aligned with and supported by empirical experiments. Building on this insight, we introduce a method that learns external concept vectors, which are added to the multi-modal attention heads for ViT-based DMs or to the bottleneck layer for U-Net-based DMs, while keeping pretrained models frozen. Experiments across SDXL, SD3.5, PixArt, and FLUX demonstrate that these concept vectors provide interpretability, linearity, and significantly improved fairness while preserving visual fidelity. The code and demo are available at https://github.com/Moslem-Sh21/responsible-t2i-diffusion.
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CellBRIDGE: Learning Cellular Trajectories via Interaction-Aware Alignment
Silas Ruhrberg Estévez, Nicolas Huynh, Tennison Liu, Roderik M. Kortlever, Gerard I. Evan, David L. Bentley, Mihaela van der Schaar
Inferring dynamics from population snapshots is a fundamental challenge in machine learning and biology. In scRNA-sequencing (scRNA-seq), destructive measurements preclude direct tracking of individual cells across time, making trajectory inference underdetermined. Optimal Transport (OT) provides a principled framework for snapshot alignment, but a long-standing modeling question is which cost functions yield biologically meaningful couplings. Standard OT approaches rely on gene-expression distances, implicitly treating cells as independent points and neglecting structured cell-cell communication mediated by ligand-receptor signaling. We introduce CellBRIDGE (Cell-Based Regularized Interaction-Driven Gene Expression), which augments feature-based OT with a directed, typed interaction cost derived from ligand-receptor activity. By explicitly modeling cell-cell communication, CellBRIDGE improves cross-snapshot couplings and downstream trajectory estimates across synthetic and real scRNA-seq datasets relative to feature-only baselines. Notably, CellBRIDGE enables mechanistically interpretable in silico perturbations: on lung cancer data, silencing specific ligand-receptor pairs induces trajectory shifts that recapitulate expected effects of targeted pathway inhibition.
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SpikeVLA: Vision-Language-Action Models with Spiking Neural Networks
Ruiqi Song, Dujun Nie, Siyu Teng, Baiyong Ding, Xiaotong Zhang, Dong Li, Chenming Zhang, Yuchen Li, Hangbin Wu, Long Chen
Vision-Language-Action (VLA) models have become a central paradigm for embodied intelligence. However, most existing approaches are built on large-scale Transformers, resulting in substantial inference latency and energy consumption that limit their practical deployment in low-power, real-time scenarios. We propose SpikeVLA, an end-to-end spiking VLA framework for embodied navigation with energy-efficient inference, consisting of three key components. (i) a spiking vision encoder, Spike-V, that replaces dense continuous computation with event-driven spiking representations to reduce the energy cost of visual representation learning, (ii) a multimodal spiking large language model, Spike-L, that reformulates cros

s-modal reasoning with spiking dynamics and token-level event-driven sparsity to further lower inference overhead, and (iii) a spiking action policy network, Spike-A, that uses Laplacian-kernel population coding and end-to-end reinforcement learning to produce stable, robust continuous control under low-energy constraints.
Experiments on multimodal interaction and robotic control tasks show that SpikeVLA significantly reduces energy consumption and computational overhead while maintaining competitive performance, highlighting its potential for low-power, real-time embodied intelligence.
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ASyMOB: Algebraic Symbolic Mathematical Operations Benchmark
Michael Shalyt, Rotem Elimelech, Ido Kaminer
Large language models (LLMs) are increasingly applied to symbolic mathematics, yet existing evaluations often conflate pattern memorization with genuine reasoning. To address this gap, we present **ASyMOB**, a high-resolution dataset of **35,368** validated symbolic math problems spanning integration, limits, differential equations, series, and hypergeometrics. Unlike prior benchmarks, **ASyMOB** systematically perturbs each seed problem using symbolic, numeric, and equivalence-preserving transformations, enabling a fine-grained assessment of generalization and robustness.
Our evaluation reveals three key findings: (1) most models' performance collapses under minor perturbations, while top systems exhibit an apparent *regime shift* in robustness; (2) integrated code tools stabilize performance, particularly for weaker models; and (3) we identify examples where Computer Algebra Systems (CAS) fail while LLMs succeed, as well as problems solved only via a hybrid LLM-CAS approach, highlighting a promising integration frontier.
ASyMOB serves as a principled diagnostic tool for measuring and accelerating progress toward building verifiable, trustworthy AI for scientific discovery.
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Theoretical Guarantees for One-Shot Magnitude Pruning and Compute-Adaptive Early Exit
Erdem Koyuncu
We study compute reduction in neural networks through a unified partial versus full computation view, captured by one-shot magnitude pruning in the static regime and early exit in the adaptive regime. In an asymptotic single-neuron model, we prove a concentration theorem for one-shot magnitude pruning with explicit rates. We also introduce the conditional perceptron for early exit and show that its excess generalization error decays as a power of the compute gap, with an exponent that grows to infinity as the alignment between partial and full computations tends to one. We then extend the analysis to deep networks, characterizing how pruning-induced distortions accumulate with depth and deriving a corresponding compute–accuracy tradeoff for frozen-backbone early exit under a neural network Gaussian process model. Numerical simulations corroborate the predicted scaling laws.
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Latent Space Robust Optimization of Neural Processes with Aligned Stratified Order-Statistic Loss Reduction
Qi Tao, Jiarong Wen, Jing Yang, Guanlin Wu, Zhang Kaiyu, Yiqin Lv, Wumei Du, Xingxing Liang, Qi Wang
Importance-Weighted Neural Processes (IWNPs) provide a principled framework for probabilistic meta-learning by using multi-particle latent representations to approximate the marginal log-likelihood of task data tightly.
However, this work reveals that the standard optimization of IWNPs suffers from the Matthew effect in the latent space, where high-likelihood particles dominate gradient signals.
The neglect of lower-likelihood regions leads to poor tail-risk generation and unstable fast adaptation.
While robust objectives such as CVaR_α can mitigate these risks, they often entail a trade-off that degrades average-case performance.
This work proposes \underline{O}rder-\underline{S}tatistics Aligned \underline{N

}eural \underline{P}rocesses (OS-NPs) to achieve latent space robust optimizatio
n without sacrificing average result.
Specifically, we stratify multiple inference particles into disjoint difficulty
bins based on order statistics and derive the regularized worst-case optimizatio
n framework for OS-NPs.
Our method aligns the reduction of stratified order-statistic losses in IWNPs an
d provides a computationally efficient pipeline to implement.
Extensive experiments demonstrate that the OS-NP constitutes stable, reliable pr
obabilistic meta-learning that significantly enhances tail-risk robustness while
 maintaining or even improving average performance.
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AnyEdit++: Adaptive Long-Form Knowledge Editing via Bayesian Surprise
Bowen Tian, Caixue He, Jiemin Wu, Jingying Wang, Wenshuo Chen, Zexi Li, Yutao Yu
e
Editing complex, long-form knowledge in Large Language Models remains a signific
ant challenge due to the difficulty of maintaining generation coherence. Existin
g autoregressive methods like AnyEdit alleviate length constraints but rely on F
ixed-window Chunking, which disregards logical structure and compromises consist
ency. To address this, we present AnyEdit++, a structure-aware framework incorpo
rating Bayes-Chunk, an adaptive segmentation mechanism that dynamically identifi
es semantic boundaries based on Bayesian Surprise. We underpin this approach wit
h a theoretical framework establishing two key principles: (1) Structural Indepe
ndence: we prove that cross-segment interference is minimized when anchor keys a
re geometrically orthogonal (a condition naturally satisfied by our surprisal-ba
sed boundaries but violated by fixed windows), and (2) Causal Locality: we demon
strate that updates injected at these semantic peaks yield strictly superior con
trol compared to arbitrary split points. Extensive experiments across mathematic
al reasoning, code generation, and narrative tasks demonstrate that AnyEdit++ ac
hieves superior performance and robustness compared to state-of-the-art baseline
s, validating that structural awareness is critical for effective long-form know
ledge editing.
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Temporal Preference Optimization for Unsupervised Retrieval
HyunJin Kim, Jaejun Shim, Young Jin Kim, JinYeong Bak
Unsupervised dense retrievers offer scalability by learning semantic similarity
from unlabeled documents via contrastive learning, but they struggle to capture
the temporal relevance, retrieving semantically related but temporally misaligne
d documents-an important aspect when a document collection spans multiple time p
eriods (*e.g.,* retrieving documents from 2018-2025 for "Who is the president in
 2019?" introduces temporal ambiguity).
Existing methods rely on supervised training with explicit timestamps, which are
 not always feasible.
We propose TPOUR (*Temporal Preference Optimization for Unsupervised Retriever*)
, which uses our novel training method *Temporal Retrieval Preference Optimizati
on* (TRPO). TRPO reinterprets preference learning in the temporal dimension, gui
ding the retriever to favor temporally aligned documents. TPOUR further generali
zes to unseen time periods via interpolation in a learned time embedding, enabli
ng continuous temporal alignment. Experiments on temporal information retrieval
(T-IR), TPOUR outperforms both unsupervised and supervised baselines. Compared t
o Qwen-Embedding-8B, despite being about 72.7$\times$ smaller, TPOUR Contriever
improves average nDCG@5 by +4.04 (+12.15%) on explicit and +4.98 (+15.21%) on im
plicit queries. We provide our code at https://github.com/agwaBom/TPOUR.
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Plan in Sandbox, Navigate in Open Worlds: Learning Physics-Grounded Abstracted E
xperience for Embodied Navigation
Zhixuan Shen, Jiawei Du, Ziyu Guo, Han Luo, Lilan Peng, Joey Tianyi Zhou, Haonan
 Luo, Tianrui Li
Vision-Language Models (VLMs) have demonstrated exceptional general reasoning ca
pabilities. However, their performance in embodied navigation remains hindered b
y a scarcity of aligned open-world vision and robot control data. Despite simula

tors providing a cost-effective alternative for data collection, the inherent reliance on photorealistic simulations often limits the transferability of learned policies. To this end, we propose ***S**andbox-**A**bstracted **G**rounded **E**xperience* (***SAGE***), a framework that enables agents to learn within a physics-grounded semantic abstraction rather than a photorealistic simulation, mimicking the human capacity for mental simulation where plans are rehearsed in simplified physics abstractions before execution. *SAGE* operates via three synergistic phases: (1) *Genesis*: constructing diverse, physics-constrained semantic environments to bootstrap experience; (2) *Evolution*: distilling experiences through Reinforcement Learning (RL), utilizing a novel asymmetric adaptive clipping mechanism to stabilize updates; (3) *Navigation*: bridging the abstract policy to open-world control. We demonstrate that *SAGE* significantly improves planner-assisted embodied navigation, achieving a 53.21% LLM-Match Success Rate on A-EQA (+9.7% over baseline), while showing encouraging transfer to physical indoor robot deployment. Project page is available at: <https://frankzxshen.github.io/SAGE>.

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CSOR: Coreset Selection for Object Re-identification via Class Pruning

Minyoung Oh, Jae-Young Sim

Coreset Selection (CS) aims to extract a small yet representative subset from a large dataset, reducing the complexity of model training. Although CS has been primarily investigated for classification tasks, it is still underexplored for object Re-identification (ReID). In this paper, we first formulate Coreset Selection for Object Re-identification (CSOR) as a joint optimization problem to find both the optimal coreset and the optimal class subset. We identify intra-class diversity as a key factor for effective coreset construction for ReID. Based on this insight, we propose a novel two-stage framework, consisting of Diversity-driven Class Pruning (DCP) and Coverage-Prioritized Sampling (CPS), to address the unique challenges of ReID datasets. First, classes with low feature diversity are pruned to allocate the storage budget to the remaining informative classes. Then, samples are greedily selected in an easy-to-hard class order to maximize feature coverage within each class. Extensive experiments on three person ReID datasets and one vehicle ReID dataset demonstrate that our method consistently outperforms existing CS approaches, establishing a new state-of-the-art in CSOR.

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TG-RAG: A Retrieval-Augmented Framework for Reasoning Guidance in Specialized Domains

Liang Su, Mingyang Zhang, Yun Xiong, Tengfei LIU, Siwei Zhang, Xi Chen, Li Sun

Enhancing Large Reasoning Models (LRMs) for specialized domains remains a critical challenge. While recent industrial frameworks attempt to encapsulate Standard Operating Procedures into modular "skills" for dynamic retrieval, utilizing them via context engineering often proves insufficient for complex workflows, leading to "Cognitive Drift." To mitigate this, we propose $\textbf{Thought Guidance-Retrieval Augmented Generation (TG-RAG)}$, a Retrieval-Augmented framework that effectively steers the generation process without relying solely on the model's self-correction. Built upon an Expert Procedure Graph (EPG) that formalizes unstructured SOPs, the framework uniquely employs a dynamic $\textbf{``Interrupt-Retrieve-Generate" (IRG)}$ mechanism to actively inject step-specific directives into the model's reasoning process. Extensive evaluations show that TG-RAG achieves competitive performance, demonstrating advantages in specialized domains by ensuring faithful adherence to domain SOPs. Code is available at https://github.com/V1ncent-S/Thought-Guidance.

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U^{3}CF: Unbiased, Unconfounding, and Unified Causal Framework for Multi-Target Domain Adaptation

Wenxu Wang, Yeqiang Liu, Rui Zhou, Jing Wang, Zhenbo Li, Wenbo Gong

Multi-target domain adaptation (MTDA) trains a model using a labeled source domain and several unlabeled target domains, aiming to enhance performance across all targets. However, existing methods lack a principled causal formulation and often rely on empirical domain-invariance enforcement, which can bias adaptation a

cross targets. To fill this gap, we propose the **U**nbiased, **U**nconfounding, and **U**nified **C**ausal **F**ramework (**U^{3}CF**) for MTDA. To *unify* the alignment of multiple domains, we propose a prototype-driven alignment strategy that progressively updates prototypes by high-confidence target predictions, while the contrastive optimization objective jointly aligns target samples to semantic prototypes and preserves class discrimination. By formulating a structural causal model, we reveal that domain-invariant causal factors and domain-specific factors shape representations and labels, while the latter induce spurious label correlations across targets. Accordingly, U^{3}CF achieves *unbiased* prediction by disentangling representations into invariant causal components and domain-specific confounders and applying conditional intervention to *block confounding* effects while preserving invariant semantics. To ensure precise disentanglement, we leverage mutual information theory to derive a principled criterion for feature separation. Extensive experiments on four benchmarks demonstrate that U^{3}CF consistently outperforms leading methods.

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Dual-channel Dynamic Graph Neural Networks with Adaptive Adjacency Learning and Multi-scale Representation Fusion

Youqing Wang, Jiahao Long, Tianxiang Zhao, Man Cao, Mengyuan Xin, Jiapu Wang, Junbin Gao, Jipeng Guo

Graph neural networks (GNNs) have been demonstrated to be powerful tools for analyzing structural graph data. However, most existing methods usually rely on fixed adjacency structures for information propagation, lacking strong adaptability to the latent semantic relationships that exist but are not explicitly connected in graph, especially in complementary high-pass and low-pass filtering views. To this end, this paper proposes a novel Dual-channel Dynamic Graph Neural Network (DCD-GNN), mainly consisting of parallel representation learning channels: a static structure-preserving channel and a dynamic adjacency-enhancing channel. The dynamic channel exploits both low-pass structural filtering and high-pass personalized detail via the self-attention adjacency learning and then integrates them for comprehensive semantic modeling, while the static channel maintains structural stability. Both channels employ a multi-scale representation fusion mechanism and a unified and discriminative node embedding representation is obtained by integrating them. Extensive experiments on various benchmark datasets verify the superiority of DCD-GNN in discriminative graph representation learning.

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Operationalising the Superficial Alignment Hypothesis via Task Complexity

Tomás Vergara Browne, Darshan Patil, Ivan Titov, Siva Reddy, Tiago Pimentel, Marius Mosbach

The superficial alignment hypothesis (SAH) posits that large language models learn most of their knowledge during pre-training, and that post-training merely surfaces this knowledge. The SAH, however, lacks a precise definition, which has led to (i) different and seemingly orthogonal arguments supporting it, and (ii) important critiques to it.

We propose a new metric called **task complexity**: the length of the shortest program that achieves a target performance on a task.

In this framework, the SAH simply claims that pre-trained models drastically reduce the complexity of achieving high performance on many tasks. Our definition unifies prior arguments supporting the SAH, interpreting them as different strategies to find such short programs. Experimentally, we estimate the task complexity of mathematical reasoning, machine translation, and instruction following; we then show that these complexities can be remarkably low when conditioned on a pre-trained model.

Further, we find that pre-training enables access to strong performances on our tasks, but it can require programs of gigabytes of length to access them. Post-training, on the other hand, collapses the complexity of reaching this same performance by several orders of magnitude. Overall, our results highlight that task adaptation often requires surprisingly little information---often just a few kilobytes

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Efficient Diffusion LLMs via Temporal-Spatial Parallel Decoding and Confidence E
xtrapolation
Zekai Li, Ji Liu, Yiqing Huang, Ziqiong Liu, Dong Li, Emad Barsoum
Diffusion-based large language models (dLLMs) support parallel text generation v
ia iterative denoising, yet inference remains latency-heavy because many steps a
re spent on redundant refinement and repeated remasking of tokens whose final va
lues are already determined.
Prior acceleration methods mainly depend on step-local confidence heuristics or
fixed schedules, which are sensitive to prompt and task variation and ignore str
ong positional effects within a sequence. We cast diffusion decoding as a dynami
c control problem and show that token-wise denoising trajectories provide the ke
y signal for reliable control. We propose a trace-aware decoding framework with
two components. First, Temporal-Spatial Parallel Decoding (TSPD) uses a lightwei
ght temporal-spatial correctness sensor that consumes per-token trajectory featu
res, including confidence, entropy, and momentum, together with token position,
to decide when a token has converged and can be safely fixed. Second, we introdu
ce]Confidence Extrapolation (CE)}], a training-free state-space module that for
ecasts future logit trends with uncertainty to support proactive decisions, incl
uding safe look-ahead and targeted stabilization when trajectories are oscillato
ry or underconfident. Together, TSPD and CE reduce unnecessary denoising iterati
ons while preserving output quality, and they compose cleanly with system optimi
zations such as KV caching.
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Pretrained Vision-Language-Action Models are Surprisingly Resistant to Forgettin
g in Continual Learning
Huihan Liu, Changyeon Kim, Bo Liu, Minghuan Liu, Yuke Zhu
Continual learning is a long-standing challenge in robot policy learning, where
a policy must acquire new skills over time without catastrophically forgetting p
reviously learned ones. While prior work has extensively studied continual learn
ing in relatively small behavior cloning (BC) policy models trained from scratch
, its behavior in modern large-scale pretrained Vision-Language-Action (VLA) mod
els remains underexplored. In this work, we find that pretrained VLAs are remark
ably resistant to forgetting compared with smaller policy models trained from sc
ratch. Simple Experience Replay (ER) works surprisingly well on VLAs, sometimes
achieving zero forgetting even with a small replay data size. Our analysis revea
ls that pretraining plays a critical role in downstream continual learning perfo
rmance: large pretrained models mitigate forgetting with a small replay buffer s
ize while maintaining strong forward learning capabilities. Furthermore, we find
that VLAs can retain relevant knowledge from prior tasks despite performance de
gradation during learning new tasks. This knowledge retention enables rapid reco
very of seemingly forgotten skills through finetuning. Together, these insights
imply that large-scale pretraining fundamentally changes the dynamics of continu
al learning, enabling models to continually acquire new skills over time with si
mple replay.
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MEDA: Medical-Oriented Activation Editing for Hallucination Mitigation in Medica
l Large Vision-Language Model
Tianbo Wang, Yuqing Ma, Lingyan Meng, Zhange Zhang, Kewei Liao, Jian Yang, Simin
Li, Jinyang Guo, Xianglong Liu
Medical Large Vision-Language Models (Med-LVLMs) suffer from severe hallucinatio
ns, posing critical safety risks in clinical deployment. Editing LVLM activation
s has shown promise for mitigating hallucination with minimal cost. However, due
to the requirements of medical domain expertise, existing methods struggle to c
apture imaging manifestations and diagnostic principles that are critical for cl
inical interpretation, thereby limiting their effectiveness. To address these li
mitations, we propose the first MEDical-oriented Activation Editing (MEDA) metho
d by integrating Query-decisive Manifestation Steering (QMS) and Principle-drive
n Diagnosis Induction (PDI) to promote Med-LVLM's expertise elicitation. Specifi
cally, QMS retrieves positive query-decisive imaging manifestations as trusted g
uidance for activation steering, while PDI constructs positive principle-embedde

d diagnostic prompts to induce expert-like clinical reasoning. Extensive experim ents across six benchmarks and six LVLMs demonstrate that MEDA efficiently impro ves the response factuality with up to a 10.2\% gain on IU-Xray, while exhibitin g strong generalization and few-shot robustness.
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SAGE: Shaping Anchors for Guided Exploration in RLVR of LLMs
Chanuk Lee, Minki Kang, Sung Ju Hwang
Recent studies observe that reinforcement learning with verifiable rewards (RLVR) reliably improves pass@1 on reasoning tasks, yet often fails to yield comparab le gains in pass@k, raising the question of whether RLVR genuinely enables large language models to acquire novel reasoning abilities or merely enhances the eff iciency of sampling reasoning modes already present in the base model.
Prior analyses largely support the latter view, attributing this limitation to s tructural properties of standard RLVR objectives that result in insufficient exp loration pressure.
In this work, we argue that a central structural constraint arises from reverse-KL regularization, which stabilizes training but inherently anchors the policy t o the reference distribution, thereby suppressing the emergence of alternative r easoning modes.
However, we show that neither removing the KL term nor replacing it with forward -KL provides a satisfactory solution, as both disrupt the efficiency–coverage tr ade-off by either inducing reward hacking or allocating probability mass to off-target regions.
To resolve this tension, we propose SAGE, a principled framework that enables co ntrollable empirical support expansion by reshaping the reverse-KL anchor distri bution itself through a guide function $q(x,y)$, achieving consistent improvemen ts in both pass@1 and pass@k across challenging mathematical reasoning benchmark s.
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FlashSinkhorn: IO-Aware Entropic Optimal Transport on GPU
Felix X.-F. Ye, Xingjie Li, An Yu, Ming-Ching Chang, LINSONG CHU, Davis Wertheim er
Entropic optimal transport (EOT) via Sinkhorn iterations is widely used in moder n machine learning, yet GPU solvers remain inefficient at scale. Tensorized impl ementations suffer quadratic HBM traffic from dense $n\times m$ interactions, wh ile existing online backends avoid storing dense matrices but still rely on gene ric tiled map-reduce reduction kernels with limited fusion. We present **FlashSi nkhorn**, an IO-aware EOT solver for squared Euclidean cost that rewrites stabil ized log-domain Sinkhorn updates as row-wise LogSumExp reductions of biased dot-product scores, the same normalization as transformer attention. This enables Fl ashAttention-style fusion and tiling: fused Triton kernels stream tiles through on-chip SRAM and update dual potentials in a single pass, substantially reducing HBM IO per iteration while retaining linear-memory operations. We further provi de streaming kernels for transport application, enabling scalable first- and sec ond-order optimization. On A100 GPUs, FlashSinkhorn achieves up to $32\times$ fo rward-pass and $161\times$ end-to-end speedups over state-of-the-art online base lines on point-cloud OT, improves scalability on OT-based downstream tasks. For reproducibility, we release an open-source implementation at https://github.com/ ot-triton-lab/flash-sinkhorn.
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Metis: Learning to Jailbreak LLMs via Self-Evolving Metacognitive Policy Optimiz ation
Huilin Zhou, Jian Zhao, Yilu Zhong, Zhen Liang, Xiuyuan Chen, Yuchen Yuan, Tianl e Zhang, Chi Zhang, Lan Zhang, Xuelong Li
Red teaming is critical for uncovering vulnerabilities in Large Language Models (LLMs). While automated methods have improved scalability, existing approaches o ften rely on static heuristics or stochastic search, rendering them brittle agai nst advanced safety alignment. To address this, we introduce \textbf{Metis}, a f ramework that reformulates jailbreaking as inference-time policy optimization wi thin an adversarial Partially Observable Markov Decision Process (POMDP). Metis

employs a self-evolving metacognitive loop to perform causal diagnosis of a target's defense logic and leverages structured feedback as a semantic gradient to refine its policy, offering enhanced interpretability through transparent reasoning traces. Extensive evaluations across 10 diverse models demonstrate that Metis achieves the strongest average Attack Success Rate (ASR) among compared methods at 89.2\%, maintaining high efficacy on resilient frontier models (e.g., 76.0\% on O1 and 78.0\% on GPT-5-chat) where traditional baselines exhibit substantial performance degradation. By replacing redundant exploration with directed optimization, Metis reduces token costs by an average of 8.2$\times$ (and up to 11.4$\times$). Our analysis reveals that current defenses remain vulnerable to internally-steered, closed-loop reasoning trajectories under the tested settings, highlighting a critical need for next-generation defenses capable of reasoning about safety dynamically during inference.
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QPKO: Differentiable QP-Embedded Deep Koopman Framework for Modeling Nonlinear Systems
Runze Tian, Peng Kou
Deep learning has been widely regarded as a powerful tool for Koopman operator theory-based modeling, as it provides a promising architecture for data-driven learning of observable functions. To fully leverage this advantage, a well-designed training paradigm is required. However, the existing training paradigms typically either incur high optimization complexity or hinder effective end-to-end training, limiting modeling accuracy and training efficiency. To address this issue, we propose a differentiable quadratic programming (QP)-embedded deep Koopman framework (QPKO). In QPKO, a QP problem, which comprises a one-step accuracy-oriented objective function and a set of multi-step accuracy-oriented constraints, is formulated to introduce a mapping from observable functions to the global linear model. By doing so, the global linear model no longer needs to be treated as an independent trainable component, thereby effectively reducing optimization complexity. This QP-based mapping is implemented as a differentiable and computationally efficient module by leveraging OptNet (a differentiable QP layer), enabling effective end-to-end training. Experiments on four nonlinear dynamical systems show that QPKO achieves satisfactory improvements in modeling accuracy, training efficiency, and control performance.
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Weight-sparse transformers have interpretable circuits
Leo Gao, Achyuta Rajaram, Jacob Coxon, Soham V. Govande, Bowen Baker, Daniel P Mossing
Finding human-understandable circuits in language models is a central goal of the field of mechanistic interpretability. We train models to have more understandable circuits by constraining most of their weights to be zeros, so that each neuron only has a few connections. To recover fine-grained circuits underlying each of several hand-crafted tasks, we prune the models to isolate the part responsible for the task. These circuits often contain neurons and residual channels that correspond to natural concepts, with a small number of straightforwardly interpretable connections between them. We study how these models scale and find that making weights sparser trades off capability for interpretability, and scaling model size improves the capability-interpretability frontier. However, scaling sparse models beyond tens of millions of nonzero parameters while preserving interpretability remains a challenge. In addition to training weight-sparse models de novo, we show preliminary results suggesting our method can also be adapted to explain existing dense models. Our work produces circuits that achieve an unprecedented level of human understandability and validates them with considerable rigor.
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OrchJail: Jailbreaking Tool-Calling Text-to-Image Agents by Orchestration-Guided Fuzzing
Jianming Chen, Yawen Wang, Junjie Wang, Zhe Liu, Qing Wang, Fanjiang Xu
Tool-calling text-to-image (T2I) agents can plan and execute multi-step tool chains to accomplish complex generation and editing queries.

However, this capability introduces a new safety attack surface: harmful outputs may arise from tool orchestration, where individually benign steps combine into unsafe results, making prompt-only jailbreak techniques insufficient.
We present OrchJail, an orchestration-guided fuzzing framework for jailbreaking tool-calling T2I agents.
Its core idea is to exploit high■risk tool■orchestration patterns: by learning from successful jailbreak tool-calling traces and their causal relationships to prompt wording, OrchJail directly guides the fuzzing search toward prompts that are more likely to trigger unsafe multi■step tool behaviors, rather than relying on surface■level textual perturbations.
Extensive experiments demonstrate that OrchJail improves jailbreak effectiveness and efficiency across representative tool-calling T2I agents,
achieving higher attack success rates, better image fidelity, and lower query costs, while remaining robust against common jailbreak defenses.
Our work highlights tool orchestration as a critical, previously unexplored attack surface and provides a novel framework for uncovering safety risks in T2I agents.
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Diversity Matters: Revisiting Test-Time Compute in Vision-Language Models
Yijie Tong, Yifan Hou, Shaobo Cui, Antoine Bosselut, Mrinmaya Sachan
Test-time compute (TTC) strategies have emerged as a lightweight approach to boost reasoning in large language models, but their applicability to vision-language models (VLMs) remains unclear. We present a systematic study of TTC for visual reasoning across seven open-source VLMs and six benchmarks, revisiting two paradigms: (i) feature-based scoring of chain-of-thought (CoT) traces and (ii) confidence-based aggregation via majority voting (MV). In the single-model setting, feature cues (e.g., length, pivot words) fail to improve accuracy, while MV yields only modest, CoT-dependent gains. To explain this limitation, we theoretically show that the voting method’s effectiveness depends on prediction diversity: when outputs are highly correlated, the benefit of voting vanishes. In contrast, multi-model ensembles introduce stronger diversity through architectural differences, training data, and scale, making them both more realistic and more promising for TTC. However, MV treats all models equally, leaving it vulnerable to correlated errors from weaker models. To address this, we propose Entropy-based TTC, which selects the most confident prediction based on predictive entropy. Our method reduces to MV in the single-model case but, in ensembles, leverages confidence disparities to prioritize stronger models. We prove that our method
theoretically outperforms MV under mild dependence assumptions, and empirically show that it consistently surpasses both MV and the best individual model across diverse visual reasoning benchmarks. This demonstrates that smaller models can enhance, rather than hinder, larger ones when combined appropriately, unlocking ensemble gains not achievable with existing TTC strategies.
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Multi-Distribution Robust Conformal Prediction
Yuqi YANG, Ying Jin
In many fairness and distribution robustness problems, one has access to labeled data from multiple source distributions yet the test data may come from an arbitrary member or a mixture of them. We study the problem of constructing a conformal prediction set that is uniformly valid across multiple, heterogeneous distributions, in the sense that no matter which distribution the test point is from, the coverage of the prediction set is guaranteed to exceed a pre-specified level. We first propose a max-p aggregation scheme that delivers finite-sample, multi-distribution coverage given any conformity scores associated with each distribution. Upon studying several efficiency optimization programs subject to uniform coverage, we prove the optimality and tightness of our aggregation scheme, and propose a general algorithm to learn conformity scores that lead to efficient prediction sets after the aggregation under standard conditions. We discuss how our framework relates to group-wise distributionally robust optimization, sub-population shift, fairness, and multi-source learning. In synthetic and real-data experiments, our method delivers valid worst-case coverage across multiple distrib

utions while greatly reducing the set size compared with naively applying max-p aggregation to single-source conformity scores, and can be comparable in size to single-source prediction sets with popular, standard conformity scores.
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Propose, Solve, Verify: Self-Play Through Formal Verification
Alex Wilf, Pranjal Aggarwal, Bryan Parno, Daniel Fried, Louis-Philippe Morency, Paul Pu Liang, Sean Welleck
Training models through self-play alone (without any human data) has been a long standing goal in AI, but its effectiveness for training large language models remains unclear, particularly in code generation where rewards based on unit tests are brittle and prone to error propagation. We study self-play in the verified code generation setting, where formal verification provides reliable correctness signals. We introduce Propose, Solve, Verify (PSV) a simple self-play framework where formal verification signals are used to create a proposer capable of generating challenging synthetic problems and a solver trained via expert iteration. We use PSV to train PSV-Verus, which across three benchmarks improves pass@1 by up to 9.6x over inference-only and expert-iteration baselines. We show that performance scales with the number of generated questions and training iterations, and through ablations identify formal verification and difficulty-aware proposal as essential ingredients for successful self-play.
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Beyond Instance-Level Self-Supervision in 3D Multi-Modal Medical Imaging
Tan Pan, Shuhao Mei, Yixuan Sun, Kaiyu Guo, Chen Jiang, Zhaorui Tan, Mengzhu Li, LIMEI HAN, Xiang Zou, Yuan Cheng, Mahsa Baktashmotlagh
Self-supervised pre-training methods in medical imaging typically treat each individual as an isolated instance, learning representations through augmentation-based objectives or masked reconstruction. They often do not adequately capitalize on a key characteristic of physiological features: anatomical structures maintain consistent spatial relationships across individuals (instances), such as the thalamus being medial to the basal ganglia, regardless of variations in brain size, shape, or pathology. We propose leveraging this cross-instance topological consistency as a supervisory signal. The challenge arises from the inherent variability in medical imaging, which can differ significantly across instances and modalities. To tackle this, we focus on two alignment regimes. (i) Intra-instance: with pixel-level correspondences available, a cross-modal triplet objective explicitly preserves local neighborhood topology. (ii) Inter-instance: without such supervision, we derive pseudo-correspondences to control partial neighborhood alignment and prevent topology collapse across modalities. We validate our approach across 7 downstream multi-modal tasks, achieving average improvements of 1.1\% and 5.94\% in segmentation and classification tasks, respectively, and demonstrating significantly better robustness when modalities are missing at test time.
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DC-Leap: Training-Free Acceleration of dLLMs via Draft-Guided Contiguous Leaping Decoding
Yanhua Jiao, Tianyi Wu, Xiaoxi Sun, Yulin Li, Hui-Ling Zhen, Libo Qin, Baotian Hu, Zhuotao Tian, Min Zhang
While parallel decoding is central to the efficiency of Diffusion Large Language Models (dLLMs), current strategies are often hindered by overly conservative confidence thresholds. These thresholds, necessitated by the Joint Probability Dependence Error (JPDE), result in redundant denoising iterations and suboptimal inference speeds.
To overcome this, we propose DC-Leap, a training-free framework that enables reliable acceleration of dLLMs in the moderate-confidence regime. DC-Leap introduces a Dynamic Contiguous Verification strategy that integrates strictly-ordered causal constraints into the parallel decoding process. By progressively validating token dependencies, this mechanism effectively neutralizes the JPDE, enabling reliable acceleration with comparable performance. Furthermore, DC-Leap incorporates the draft-guided decoding mechanism, where the draft helps extend the context by leaping forward across multiple tokens, providing look-ahead context and re

taining the structural benefits of bidirectional attention during inference. Extensive experiments on standard benchmarks demonstrate that DC-Leap achieves substantial speedups, up to **53.19$\times$** on MBPP for long-sequence generation, and up to **105.02$\times$** when combined with KV-Cache with comparable generation quality. Code is available at <https://github.com/ffh-wyls/DC-Leap>.
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Semi-Supervised Hypothesis Testing by Betting on Predictions
Yaniv Tenzer, Elad Tolochinsky, Yaniv Romano
We introduce a testing-by-betting framework that leverages predictions on unlabeled data to enhance the power of sequential hypothesis testing. Given limited samples from the joint distribution of (X,Y), and additional unlabeled samples from the marginal of X, we ask how unlabeled data can be used to hypothesize about the distribution of Y, and the conditional distribution of $Y\mid X$. We introduce an e-statistic and use it to construct a sequential test. Under standard distributional assumptions---label shift or concept shift---we establish that the test is anytime valid. Furthermore, we show that for binary data, the e-statistic has non-trivial power. Crucially, our approach retains these properties even when the underlying predictions are inaccurate.
Through simulations and applications to large language models evaluation, we demonstrate power gains over baseline approaches, including prediction-powered inference. These gains persist even with relatively limited unlabeled data and when predictions have low accuracy due to weak correlation between X and Y.
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Learning from Comparison: Constrained Projection Policy Optimization for Pareto-Front Improvement
Jintao Li, Maowen Tang, Yongji Long, Weixuan Liu, Yanlang Zheng, Sicheng He, Ao-Jin Li, Shui Yu, Yun Li
Constrained multi-objective reinforcement learning aims to discover a diverse set of feasible trade-offs, yet scalarization and signed, normalized group-relative advantages can be brittle under objective-scale drift, near-ties, and feasibility scarcity. We propose constrained projection policy optimization (CoPro), which alternates between an E-step moment projection and an M-step policy projection. In the E-step, we solve a Kullback-Leibler (KL)-regularized, moment-constrained projection over each sampled group to compute a nonnegative reweighting distribution ($ q^* $) that promotes feasible Pareto-front (PF) progress, preserves feasibility anchors, and suppresses ambiguous near-ties. This E-step admits a closed-form exponential-family solution and guarantees strictly positive probability mass on feasible anchors whenever feasible candidates appear in the group. In the M-step, we project the policy toward $ q^* $ via weighted maximum likelihood with a trust-region regularizer, yielding a PF-aligned update direction from comparisons without hand-crafted reward shaping. Empirically, CoPro improves feasible PF quality and robustness on constrained multi-objective benchmarks for large language model tool use and analog circuit design tasks; code is available at \href{https://anonymous.4open.science/r/CoPro-8A95/README.md}{\texttt{CoPro}}.
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Outcome Rewards Do Not Guarantee Verifiable or Causally Important Reasoning
Qinan Yu, Alexa R. Tartaglini, Peter Hase, Carlos Guestrin, Christopher Potts
Reinforcement Learning from Verifiable Rewards (RLVR) on chain-of-thought reasoning has become a standard part of language model post-training recipes. A common assumption is that the reasoning chains trained through RLVR reliably represent how a model gets to its answer. In this paper, we develop two metrics for critically examining this assumption: Causal Importance of Reasoning (CIR), which measures the cumulative effect of reasoning tokens on the final answer, and Sufficiency of Reasoning (SR), which measures whether a verifier can arrive at an unambiguous answer based on the reasoning alone. Through experiments with the Qwen2.5 model series and ReasoningGym tasks, we find that: (1) While RLVR does improve task accuracy, it does not reliably improve CIR or SR, calling the role of reasoning in model performance into question. (2) A small amount of SFT before RLVR can be a remedy for low CIR and SR. (3) CIR and SR can be improved even without SFT by applying auxiliary CIR/SR rewards on top of the outcome-based reward. This

joint reward matches the accuracy of RLVR while also leading to causally important and sufficient reasoning. These results show that RLVR does not always lead models to rely on reasoning in the way that is commonly thought, but this issue can be remedied with simple modifications to the post-training procedure.
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LEAP: Zone-Aware MCTS for LLM Self-Speculative Decoding
LeiQuan Zheng, Yuan Liu
Self-speculative decoding accelerates LLM inference by using a lightweight draft model for generation and a target model for verification, where the draft model is constructed by a subset of the target model's layers, and the key challenge lies in layer configuration strategies. To address this challenge, we propose LEAP, a plug-and-play approach that formulates and optimizes the draft model construction problem as a sequential decision-making process by Monte Carlo Tree Search (MCTS). To navigate the prohibitive search space of deep LLMs, we leverage two empirical observations: (i) the prefilling-derived redundancy information remains informative during decoding, and (ii) the layer redundancy exhibits zone-wise characteristics. These observations enable a structured search space through zone partitioning and layer grouping, which serves as an inductive bias to facilitate efficiency of MCTS. Experimental results show that LEAP achieves a speedup of $1.7\times\sim2.0\times$ for LLM inference.
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AI Cartography: Mapping the Latent Landscape of AI Benchmark Ecosystems
Michael Hardy, Anka Reuel, Lijin Zhang, Jodi M. Casabianca, Sang T. Truong, Yash Satish Dave, Hansol Lee, Benjamin W. Domingue, Sanmi Koyejo
While aggregate leaderboard scores drive AI development, they contain substantial measurement noise whose sources and magnitudes remain unquantified, making it unclear when rankings reflect genuine capability differences versus evaluation artifacts. We introduce a framework for measuring the latent landscape in AI benchmark ecosystems. Applying Confirmatory Factor Analysis (CFA) and Generalizability Theory to 4,000+ models from the Open LLM Leaderboard, we decompose sources of ranking variance and establish:
(1) structures assumed in current reporting practice underestimate the strength of relationships between benchmarks;
(2) evidence of local dependence among leaderboard items, undermining uses of benchmarks as measurement instruments under current scoring systems;
(3) contributor metadata explains more rank-relevant variance ($\approx9\%$) than architecture or deployment categories in this context;
(4) a manifest-score ``scaling law'' slope has low reliability ($R_{\beta}=0.53$); by contrast, the latent general-factor size slope is highly stable across ecosystem controls ($R_g=0.97$).
We are able to provide unique insights into benchmark dynamics, such as which benchmarks are a function of LLM size and which can be oppositely impacted by post-training practices. We provide actionable diagnostics to determine how benchmark rankings can be trusted and how benchmark design can be improved.
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Expected Return Causes Outcome-Level Mode Collapse in Reinforcement Learning and How to Fix It with Inverse Probability Scaling
ABHIJEET SINHA, Sundari Elango, Dianbo Liu
Many reinforcement learning (RL) problems admit multiple terminal solutions of comparable quality, where the goal is not to identify a single optimum but to represent a diverse set of high-quality outcomes. Nevertheless, policies trained by standard expected-return maximization routinely collapse onto a small subset of outcomes, a phenomenon commonly attributed to insufficient exploration or weak regularization. We show that this explanation is incomplete: outcome-level mode collapse is a structural consequence of the expected-return objective itself. Under idealized learning dynamics, the log-probability ratio between any two outcomes evolves linearly in their reward difference, implying exponential ratio divergence and inevitable collapse—independent of the exploration strategy, entropy regularization, or optimization algorithm. We identify the source of this pathology as the probability multiplier inside the expectation and propose a minimal c

orrection: inverse probability scaling, which removes outcome-frequency amplification from the learning signal, fundamentally changes the learning dynamics, and provably yields reward-proportional terminal distributions, preventing collapse in multimodal settings. We instantiate this principle in Group Relative Policy Optimization (GRPO) as a drop-in modification, IPS-GRPO, requiring no auxiliary models or architectural changes. Across different reasoning and molecular generation tasks, IPS-GRPO consistently reduces outcome-level mode collapse while matching or exceeding baseline performance, suggesting that correcting the objective rather than adding exploration heuristics is key to reliable multimodal policy optimization.

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Improving Zero-Shot Offline RL via Behavioral Task Sampling

Nazim Bendib, Nicolas Perrin-Gilbert, Olivier Sigaud

Offline zero-shot reinforcement learning (RL) aims to learn agents that optimize unseen reward functions without additional environment interaction. The standard approach to this problem trains task-conditioned policies by sampling task vectors that define linear reward functions over learned state representations. In most existing algorithms, these task vectors are randomly sampled, implicitly assuming this adequately captures the structure of the task space. We argue that doing so leads to suboptimal zero-shot generalization. To address this limitation, we propose extracting task vectors directly from the offline dataset and using them to define the task distribution used for policy training. We introduce a simple and general reward function extraction procedure that integrates into existing offline zero-shot RL algorithms. Across multiple benchmark environments and baselines, our approach improves zero-shot performance by an average of 20%, highlighting the importance of principled task sampling in offline zero-shot RL.

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Decoupled Training with Local Reinforcement Fine-Tuning in Federated Learning

Yuting Ma, Lechao Cheng, Xiaohua Xu

Federated Learning (FL) with pre-trained Vision-Language Models (VLMs) has emerged as a promising paradigm for various downstream tasks. By leveraging its strong representations, recent studies improve task adaptation under insufficient local data while preserving generalization. However, these methods emphasize fully local optimization with simple parameter aggregation, which can amplify inter-client optimization inconsistency and intra-client over-specialization under heterogeneous and full-data FL settings, making it difficult to balance global task adaptation and generalization. To address these challenges, we propose FedDTL, a novel federated VLM framework that decouples the image encoder and text encoder across clients and the server. Through decoupled encoder training with server-client modality alignment, FedDTL promotes coherent global semantic update and reduces inter-client optimization inconsistency, improving global task adaptation. To further mitigate intra-client over-specialization, we introduce a two-stage local fine-tuning, where a supervised fine-tuning stage enables rapid and reliable warm-start, followed by a reinforcement learning stage that enhances generalization. Extensive experiments on multiple benchmarks, including label skew and feature shift, demonstrate that FedDTL achieves an effective balance between global task adaptation and generalization under various FL data distributions in both few-shot and full-data regimes.

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Learning Coherent Representations: A Topological Approach to Interpretability

Sigurd Gaukstad, Melvin Vaupel, Valdemar Kargård Olsen, Erik Hermansen, Benjamin Adric Dunn

Deep neural networks learn representations where individual features often lack interpretable meaning; a single neuron may activate for scattered, unrelated inputs. We introduce coherence, a geometric property inspired by neural coding in the brain, where neurons like grid cells and head direction cells respond to contiguous regions of state space. A non-negative matrix is coherent if each row (sample) attends to geometrically clustered columns (features) and vice versa, and in addition every sample is well described by some feature and every feature is needed by some sample. We prove that coherent matrices induce a bounded interlea

ving between the Vietoris-Rips filtrations of samples and features, guaranteeing that both spaces share compatible topological structure. This geometric constraint facilitates interpretability. For example, if data lies on a circle, coherent features must tile that circle into contiguous arcs. We introduce Coh, a differentiable objective function based on Fréchet variance that enforces coherence during training. Unlike sparsity, which bounds how many samples a feature activates on, coherence bounds *which* samples, requiring geometric connectivity rather than only rarity. This yields not just interpretable features but an interpretable feature space. We validate Coh in an auto-encoder using synthetic and rotated MNIST datasets and in a token embedding of BERT using language data.
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How Hard Is Science?
Adil Soubki, Miles Cranmer
Symbolic Regression (SR) is the task of finding a closed-form mathematical expression that optimizes some objective. Solving this task is NP-hard. However, SR software routinely discovers accurate, interpretable models without exhaustively searching function space.
Motivated by this disconnect between worst-case theory and practical success, we study SR through the lens of \emph{parameterized complexity theory}. In particular, we reanalyze tractability with respect to practically relevant parameters like expression depth, tree size, and number of primitives used. We show that SR is fixed-parameter tractable (FPT) when parameterized by expression depth or tree size, formalizing the tractable regime exploited by bounded-complexity search in popular SR algorithms. However, SR becomes W[1]-hard when parameterized by the number of variables or primitives used, identifying selection as a source of intractability. We further find lower bounds under the exponential time hypothesis, prove approximation hardness, and rule out polynomial kernels for the case where the primitive set is part of the input.
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Why Self-Distillation Helps and Hurts: Denoising vs. Signal Forgetting
Mingqi Wu, Archer Yi Yang, Qiang Sun
Self-distillation, a special case of self-training, repeatedly trains a model on pseudo-labels produced by its previous iterate. Why can this iteration improve generalization, and why can it later hurt? We answer this question in overparameterized linear regression, starting from an estimator fit to noisy labels and then repeatedly refitting on fresh covariates labeled by the current estimator. In the proportional asymptotic regime, we derive deterministic recursions for the test risk and prove concentration around them. The recursions reveal a simple trade-off: iteration \emph{denoises} by attenuating stochastic error inherited from the initial fit, but it also induces \emph{signal forgetting} because each teacher--student transfer projects the estimate through a new data-dependent subspace. This competition produces a U-shaped risk curve and an optimal stopping time. The same dynamics act as a spectral filter: eigendirections carrying more of the true regression signal are retained longer, while weakly aligned or noise-like directions are suppressed. We further develop an iterated generalized cross-validation criterion with uniform consistency for selecting the stopping time. Synthetic simulations corroborate the recursions and data-driven stopping rule, while neural-network experiments exhibit the same denoising--forgetting pattern.
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BioToken and BioFM – Biologically-Informed Tokenization Enables Accurate and Efficient Genomic Foundation Models
Aleksandr Medvedev, Karthik Viswanathan, Praveenkumar Kanithi, Kirill Vishniakov, Prateek Munjal, Clement Christophe, Tiago Magalhaes, Marco AF Pimentel, Ronnie Rajan, Shadab Khan
Existing genomic foundation models (GFMs) typically treat DNA as raw nucleotide sequences, often overlooking the regulatory context required to interpret genetic variation accurately. We introduce BioToken, a tokenization framework that directly encodes variants and biological annotations into genomic representations, and BioFM, a parameter-efficient model built on this architecture. By leveraging biological inductive biases, BioFM outperforms state-of-the-art models and spec

ialized baselines like Enformer on benchmarks including pathogenicity and expres
sion prediction while requiring 100-fold less compute than current large-scale g
enomic models. These findings demonstrate that explicitly modeling biological st
ructure yields more robust and efficient genomic representations than scaling al
one.
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Attributed Network Alignment: Statistical Limits and Efficient Algorithm
Dong Huang, Chenyang Tian, Pengkun Yang
This paper studies the problem of recovering a hidden vertex correspondence betw
een two correlated graphs when both edge weights and node features are observed.
 While most existing work on graph alignment relies primarily on edge informatio
n, many real-world applications provide informative node features in addition to
 graph topology. To capture this setting, we introduce the featured correlated G
aussian Wigner model, where two graphs are coupled through an unknown vertex per
mutation, and the node features are correlated under the same permutation. We ch
aracterize the optimal information-theoretic thresholds for exact recovery and p
artial recovery of the latent mapping. On the algorithmic side, we propose QPAli
gn, an efficient method based on a quadratic programming relaxation, and demonst
rate its strong empirical performance on both synthetic and real datasets. Moreo
ver, we also derive theoretical guarantees for the proposed procedure, supportin
g its reliability and providing convergence guarantees.
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InfoAtlas: A Foundation Model for Zero-Shot Statistical Dependence Estimate
Zhengyang Hu, Yanzhi Chen, Hanxiang Ren, Qunsong Zeng, Youyi Zheng, Adrian Welle
r, Kaibin Huang, Yanchao Yang
Measuring statistical dependency between high-dimensional random variables is a
fundamental task in data science and machine learning. Neural mutual information
 (MI) estimators offer a promising avenue, but they typically require costly ite
rative optimization for each new dataset, making them impractical for real-time
applications. We present InfoAtlas, a foundation model-like architecture that el
iminates this bottleneck by directly inferring MI in a single forward pass. Pret
rained on large-scale synthetic data with rich dependence patterns, InfoAtlas le
arns to identify diverse dependence structures and predict MI directly from the
dataset. Comprehensive experiments demonstrate that InfoAtlas matches state-of-t
he-art neural estimators in accuracy while achieving 100× speedup, can flexibly
 handle varying dimensions and sample sizes through a single unified model, and
generalizes effectively to complex, real-world scenarios. By reformulating MI es
timation as an inference task, InfoAtlas establishes a foundation for real-time
dependency analysis.
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Verifying Meta-Awareness via Predictive Rewards in Reasoning Models
Yoonjeon Kim, Doohyuk Jang, Eunho Yang
Recent research on reasoning models explores the meta-awareness of language mode
ls, including their ability to determine optimal thinking duration, recognize kn
owledge boundaries, and structure concept-level thinking. While current large re
asoning models depend solely on answer-based verification, we show that adding m
eta-awareness objectives leads to significant performance gains over models with
out such meta-knowledge. **MAPR** utilizes a self-generated task of predicting r
ollout statistics - specifically length, pass-rate, and concepts used - allowing
 for verification against the actual statistics. Furthermore, by leveraging this
 self-predictive capability, the model can regulate its reasoning behavior by i)
 filtering out trivial or unsolvable prompts, ii) reducing lengthy generations t
hat tend to be incorrect, and iii) generating hints relevant to the problem. The
 results are inspiring: **MAPR** yields significant improvements in both accurac
y and training efficiency on various reasoning benchmarks. More specifically, ou
r method can speed up GRPO training by over 1.28$\times$ to reach the same perfo
rmance, and achieve 83.18\% gain in accuracy on AIME25, and a 13.04\% average ga
in over six mathematics benchmarks.
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Recurrent Structural Policy Gradient for Partially Observable Mean Field Games

Clarisse Wibault, Johannes Forkel, Sebastian Rene Towers, Tiphaine Wibault, Juan Agustin Duque, George Whittle, Andreas Schaab, Yucheng Yang, Chiyuan Wang, Michael A Osborne, Benjamin Moll, Jakob Nicolaus Foerster

Mean Field Games (MFGs) provide a principled framework for modelling interactions in large population systems. However, algorithmic progress has been limited since model-free methods are high variance and exact methods scale poorly. Recent Hybrid Structural Methods (HSMs) reduce variance while maintaining tractability by leveraging low-dimensional individual state and action spaces and known transition dynamics to compute the exact expected return conditioned on Monte Carlo rollouts of common noise. However, HSMs have not been extended to partially observable settings. We propose *Recurrent Structural Policy Gradient* (RSPG), the first history-aware HSM for MFGs with public partial information. RSPG achieves an order-of-magnitude faster convergence than model-free RL methods while learning history-aware behaviour, unlike current HSMs. To facilitate research into MFGs, we also introduce MFAX, our JAX-based framework for MFGs that supports both analytic and sample-based mean-field updates.

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Model Merging Scaling Laws in Large Language Models

Yuanyi Wang, Yanggan Gu, Yiming Zhang, Qi Zhou, Zhaoyi Yan, Congkai Xie, Xinyao Wang, Jianbo Yuan, Hongxia Yang

We study empirical scaling laws for language model merging measured by cross-entropy.

Despite its wide practical use, merging lacks a quantitative rule that predicts returns as we add experts or scale the model size.

We identify a compact power law that links model size and expert number: the size-dependent floor decreases with model capacity, while the merging tail exhibits clear diminishing returns in the number of experts.

The law holds in-domain and cross-domain, tightly fits measured curves across diverse architectures and methods (Average, TA, TIES, DARE), and explains two robust regularities: most gains arrive early, and variability shrinks as more experts are included.

Building on this, we present a simple theory that explains why gains fall roughly as \(1/k\) and links the floor and tail to properties of the base model and the diversity across domains. This law enables \emph{predictive planning}: estimate how many experts are needed to reach a target loss, decide when to stop adding experts, and trade off scaling the base model versus adding experts under a fixed budget—turning merging from heuristic practice into a computationally efficient, planable alternative to multitask training.

This suggests a scaling principle for distributed generative AI: predictable gains can be achieved by composing specialists, offering a complementary path toward AGI-level systems.

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Distillation Models are Good Samplers for Diffusion Reinforcement Learning

Zunxu Liu, Aiqiu Wu, Zhaofan Qiu, Yingwei Pan, Ting Yao, Tao Mei

We present DMSampler, a framework that accelerates online diffusion reinforcement learning by replacing expensive training-time policy rollouts with a co-evolving few-step distilled sampler. Instead of repeatedly sampling the policy model for roughly 50 denoising steps, DMSampler generates reward-evaluation samples in only 4--8 steps while periodically re-distilling the sampler from the updated policy, yielding an order-of-magnitude reduction in rollout cost. The framework alternates between two stages: an RL phase that optimizes the policy using hybrid samples from the old policy and distilled sampler, and a distillation phase that realigns the few-step sampler to the improved policy. Intuitively, the distilled sampler acts as a fast proxy for the current policy during RL, and is refreshed whenever the policy improves so that sampling remains both efficient and aligned. Two designs make this loop stable and effective: hybrid distillation sampling preserves on-policy structure during rollout, and reward-aware distillation reuses high-reward trajectories to reduce forgetting during compression. Experiments on text-to-image and text-to-video generation show that DMSampler improves OCR, GenEval, and VBench performance while substantially reducing GPU hours, and t

hat the same idea can be combined with multiple diffusion RL optimizers. Our cod
e will be available at:
https://github.com/HiDream-ai/DMSampler.
****
World-Model Inspired Emotion-aware Token Refinement for Training-Free Multimodal
 Emotion Recognition
Kejun Liu, Zhe Chen, Yuanyuan Liu, ke Wang, Yibing Zhan, Wei Xiang, Hongyan Zhan
g
Multimodal Large Language Models (MLLMs) show promise for Multimodal Emotion Rec
ognition (MER) but often remain unreliable because sparse emotional cues could b
e easily overwhelmed and affected by redundant context. While fine-tuning is eff
ective, it is usually costly when using large models. Training-free methods like
 chain-of-thought reasoning provide a practical alternative, but they mostly rel
y on heuristic prompting to influence the model behaviors and do not explicitly
focus on emotion relevant tokens internally, which would allow decision-relevant
 emotional tokens to be diluted by environmental noise, resulting in unstable pr
edictions. To address this limitation without training, we rethink MER from a wo
rld-model perspective that treats emotion as a latent state inferred from noisy
and redundant multimodal observations. Under frozen parameters, this view sugges
ts that robustness depends on constraining why and how tokens contribute to infe
rence. Based on this insight, we propose WETR (World-Model inspired Emotion-awa
re Token Refinement), a training-free, plug-and-play regulator that reshapes tok
en usage through two mechanisms: Noise-suppressed Token Selection (NTS), which s
uppresses redundant intra-modal noise, and State-strengthened Token Reweighting
(STR), which amplifies decision-relevant emotional tokens. Experiments on multip
le MER benchmarks demonstrate that WETR consistently improves accuracy and stabi
lity under frozen parameters, which also improves token-level interpretability.
****
GUI-Spotlight: Adaptive Iterative Focus Refinement for Enhanced GUI Visual Groun
ding
Bin Lei, Nuo Xu, Ali Payani, Mingyi Hong, Chunhua Liao, Yu Cao, Caiwen Ding
Multimodal large language models (MLLMs) have markedly expanded the competence o
f graphical user-interface (GUI) systems, propelling them beyond controlled simu
lations into complex, real-world environments across diverse platforms. However,
 practical usefulness is still bounded by the reliability of visual grounding, i
.e., mapping textual references to exact on-screen elements. This limitation pre
vents the system from accurately performing pointer-level actions such as clicki
ng or dragging. To address it, we introduce GUI-Spotlight -- a model trained for
 image-grounded reasoning that dynamically invokes multiple specialized tools to
 iteratively narrow its focus to the relevant region of the screen, thereby subs
tantially improving visual grounding accuracy. On the ScreenSpot-Pro benchmark,
GUI-Spotlight trained with only 18.5K training samples achieves 52.8\% accuracy,
 surpassing V2P-7B (50.6\% with 9.6M training samples) and GTA-1-7B (50.1\% with
 1.56M training samples).
****
Adversarially Robust Control of Conditional Value-at-Risk via Rockafellar-Uryase
v Conformal Inference
Catherine Chen, Jingyan Shen, Zhun Deng, Lihua Lei
We present an online, distribution-free framework for controlling the Conditiona
l Value-at-Risk (CVaR), extending conformal tail risk control t
o non-stationary and adversarial environments. Unlike classical risk control met
hods, which rely on stationarity or linearity of expectation, our approach provi
des provable safety guarantees for a nonlinear tail risk functional under arbitr
ary data-generating processes that may drift or shift strategically over time. B
y leveraging deep connections between conformal tail risk control, online learni
ng, and the variational representation of CVaR introduced by Ro
ckafellar and Uryasev, we develop a novel procedure for online $\operatorname{CV
aR}$ control with adversarial regret guarantees. The proposed method operates wi
thout assumptions on the underlying data-generating process, making it broadly a
pplicable in modern high-stakes deployment settings. We prove that the realized

empirical CVaR is asymptotically controlled at the target level, and that the resulting control is asymptotically tight up to a finite-sample ${O}(1/\sqrt{T})$ conservatism gap. We demonstrate the effectiveness of our approach on portfolio risk management and toxicity mitigation for Large Language Models (LLMs), where rare but catastrophic failures dominate system risk.

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FrameOracle: Learning What to See and How Much to See in Videos

Chaoyu Li, Tianzhi Li, Fei Tao, ZHENYU ZHAO, Ziqian Wu, Maozheng Zhao, Juntong Song, Cheng Niu, Pooyan Fazli

Vision-language models (VLMs) advance video understanding but operate under tight computational budgets, making performance dependent on selecting a small, high-quality subset of frames. Existing frame sampling strategies, such as uniform or fixed-budget selection, fail to adapt to variations in content density or task complexity. To address this, we present FrameOracle, a lightweight, plug-and-play module that predicts both (1) which frames are most relevant to a given query and (2) how many frames are needed. FrameOracle is trained via a curriculum that progresses from weak proxy signals, such as cross-modal similarity, to stronger supervision with FrameOracle-41K, the first large-scale VideoQA dataset with validated keyframe annotations specifying minimal sufficient frames per question. Extensive experiments across five VLMs and six benchmarks show that FrameOracle reduces 16-frame inputs to an average of 10.4 frames without accuracy loss. When starting from 64-frame candidates, it reduces inputs to 13.9 frames on average while improving accuracy by 1.5%, achieving state-of-the-art efficiency–accuracy trade-offs for scalable video understanding.

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Learning the Neighborhood: Contrast-Free Multimodal Self-Supervised Molecular Graph Pretraining

Boshra Ariguib, Mathias Niepert, Andrei Manolache

High-quality molecular representations are essential for property prediction and molecular design, yet large labeled datasets remain scarce. While self-supervised pretraining on molecular graphs has shown promise, many existing approaches either depend on hand-crafted augmentations or complex generative objectives, and often rely solely on 2D topology, leaving valuable 3D structural information underutilized. To address this gap, we introduce C-FREE (**C**ontrast-**F**ree **R**epresentation learning on **E**go-n**e**ts), a simple framework that integrates 2D graphs with ensembles of 3D conformers. C-FREE learns molecular representations by predicting subgraph embeddings from their complementary neighborhoods in the latent space, using fixed-radius ego-nets as modeling units across different conformers. This design allows us to integrate both geometric and topological information within a hybrid Graph Neural Network (GNN)-Transformer backbone, without negatives, positional encodings, or expensive pre-processing. Pretraining on the GEOM dataset, which provides rich 3D conformational diversity, C-FREE achieves state-of-the-art results on MoleculeNet, surpassing contrastive, generative, and other multimodal self-supervised methods. Fine-tuning across datasets with diverse sizes and molecule types further demonstrates that pretraining transfers effectively to new chemical domains, highlighting the importance of 3D-informed molecular representations.

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Mitigating Visual Hallucinations via Semantic Curriculum Preference Optimization in MLLMs

Yuanshuai li, Yuping Yan, Junfeng Tang, Zeqi Zheng, Yaochu Jin

Multimodal Large Language Models (MLLMs) have significantly improved the performance of various tasks, but continue to suffer from visual hallucinations, a critical issue where generated responses contradict visual evidence. While Direct Preference Optimization (DPO) is widely used for alignment, its application to MLLMs often fails to capture fine-grained semantic differences and encourages shortcut learning. To address these challenges, we propose Semantic Curriculum Preference Optimization (SCPO), a novel framework for MLLM alignment. SCPO employs a progressive, easy-to-hard curriculum built upon our Semantic Curriculum Preference Pairs dataset, which provides fine-grained semantic contrasts sorted by diffic

ulty. This curriculum is trained with a dynamic reference model and a novel symm etric, bidirectional objective to facilitate simultaneous learning from both tex tual and visual preferences. To our knowledge, SCPO is the first framework to un ify semantics, symmetry, and curriculum for MLLM alignment, effectively mitigati ng visual hallucinations. Extensive experiments on LLaVA models across various s cales and versions validate that SCPO demonstrates superior performance compared to baseline models on multiple hallucination benchmarks, reducing the hallucina tion rate by up to 62.9\%. Moreover, evaluations on generalized benchmarks show that SCPO improves factuality while preserving general capabilities, with its pe rformance remaining stable across general vision-language benchmarks. Code is av ailable \href{https://github.com/ShawnLee0910/SCPO}{here}.
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Causal Structure Learning for Sparse Matrix Fill-in Reduction
Ziwei Li, Shuzi Niu, Tao Yuan, Huiyuan Li
The performance of sparse direct solvers is fundamentally governed by fill-in, i .e. new nonzero entries arising from the LU factorization of a sparse matrix, as they dictate memory footprint and subsequent computation time. For decades, a v ariety of graph-theoretic algorithms have aimed to minimize fill-in, a problem k nown to be both NP-hard and critically important. While recent deep learning met hods, optimizing surrogate fill-in objectives, show empirical promise and can ou tperform classical algorithms on certain matrices, they offer limited interpreta bility into the underlying mechanism of fill-in generation.
To address this, we propose a novel reordering approach, Causal Triplet Structur e Learning (CTS), which is grounded in the Fill-Path Theorem and reduces arbitra ry-length fill-paths to length-two candidate triplets, identifies the causal str uctures that trigger fill-in, and intervenes to block their formation.
Empirically, we design a multigrid-style GAT with KAN activations to learn verte x embeddings and introduce a causal triplet loss that discourages such structure s during training. Experiments on the SuiteSparse Matrix Collection demonstrate that our method reduces fill-in by 6$\times$, leading to 12$\times$ speedup in f actorization time compared to state-of-the-art methods on Chemical Process Simul ation and Computational Fluid Dynamics matrices.
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SuperHype: Hypergraph Generation via Graph-Superposition Decomposition
Lucas Gantes, Abele M■lan, Roberto Gheda, Robert Birke, Lydia Chen
Hypergraphs are graph generalizations with key applications in domains such as h ealthcare, where strict data privacy requirements apply, or bioinformatics, wher e testing new compounds is costly. However, due to their combinatorial nature, h ypergraph representations are often either intractable or lead to significant in formation loss. For this reason, research into hypergraph synthesis is limited, and state-of-the-art approaches yield poor generation quality in terms of overal l structural patterns and graph-level validity. To address such shortcomings, we introduce SuperHype, an exact and tractable hypergraph diffusion model. The cor e of SuperHype is the graph-superposition decomposition, a novel representation that embeds a hypergraph into a multi-layer graph, enabling a tractable represe ntation with no loss of generalization. To generate new samples from such repres entations, we introduce a Graph-Superposition Transformer that treats the superp osition as an interconnected sequence of layers. Moreover, we enhance the model' s performance by incorporating hypergraph-specific auxiliary features and aggreg ating indirect node interactions via triplet pooling. Our evaluation across five datasets shows that SuperHype generally reproduces local and global connectivit y patterns with superior fidelity compared to state-of-the-art baselines.
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Learning Discrete Diffusion on Graphs via Free-Energy Gradient Flows
Dario Rancati, Jan Maas, Francesco Locatello
Diffusion-based models on continuous spaces have seen substantial recent progres s through the mathematical framework of gradient flows, leveraging the Wasserste in-2 (${W}_2$) metric via the Jordan-Kinderlehrer-Otto (JKO) scheme. Despite the increasing popularity of diffusion models on discrete spaces using continuous-t ime Markov chains, a parallel theoretical framework based on gradient flows has

remained elusive due to intrinsic challenges in translating the ${W}_2$ distance directly into these settings. In this work, we propose the first computational approach addressing these challenges, leveraging an appropriate metric W_K on the simplex of probability distributions, which enables us to interpret widely used discrete diffusion paths, such as the discrete heat equation, as gradient flows of specific free-energy functionals. Through this theoretical insight, we introduce a novel methodology for learning diffusion dynamics over discrete spaces, which recovers the underlying functional directly by leveraging first-order optimality conditions for the JKO scheme. The resulting method optimizes a simple quadratic loss, trains extremely fast, does not require individual sample trajectories, and only needs a numerical preprocessing computing W_K-geodesics. We validate our method through extensive numerical experiments on synthetic data, showing that we can recover the underlying functional for a variety of graph classes, as well as cell type prediction in single-cell trajectory data. Code will be made available at github.com/dariorancati/discrete-gradient-flow.
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TabularBERT: Binning-Based Self-Supervised Learning for Tabular Representation
Beomjin Park, Seunghwan An, Sungchul Hong, Hosik Choi
Tabular data is one of the most fundamental and widely used formats for representing structured information.
Classical machine learning algorithms continue to achieve substantial success in extracting predictive patterns and constructing accurate models from structured data; however, representation learning approaches that extend language-model-based methods to the tabular setting have opened new opportunities.
Nevertheless, conventional tokenization procedures and token embedding mechanisms are not well-suited to numerical variables, as they fail to preserve key numerical properties, including proximity structure and ordinal relationships.
To address this limitation, we propose TabularBERT, a Transformer-based model that discretizes numerical variables via binning-based tokenization and learns representations that account for numerical proximity and ordinal information while capturing conditional dependencies among variables through masked self-supervised pretraining.
We empirically demonstrate the effectiveness and interpretability of the proposed approach, highlighting the benefits of language-model-based representation learning in the tabular domain.
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Unbiased Principles, Robust Rewards
Qingnan Ren, Zhen Fang, Shiting Huang, Yu Zeng, Lin Chen, Zehui Chen, Feng Zhao
Reward models are central to Reinforcement Learning from Human Feedback (RLHF), especially for open-ended tasks where evaluation is inherently multi-dimensional. Recent Generative Reward Models (GRMs) improve interpretability by producing natural-language rationales and task-specific evaluation principles. However, most existing GRMs generate principles after reading the actor's response, i.e., $Q +R \rightarrow P$. We show that this coupling induces Principle Drift: when the actor performs reward hacking (e.g., verbosity, self-aggrandizement, or hallucinated self-justifications), the reward model may shift its criteria to rationalize the response, yielding inflated scores that in turn reinforce hacking during RL.
We propose IP-GRM (Independent Principle GRM), a two-stage framework that first generates principles solely from the question ($Q \rightarrow P$) and then evaluates the response conditioned on (Q, R, P). This decoupling keeps criteria invariant to response content, producing more objective and stable reward signals.
For efficient training, we further introduce a Principle Cache strategy that reuses principles within a group, improving GRPO throughput by 23.66\% while maintaining strict intra-group consistency. In GRPO training on creative writing, IP-GRM suppresses reward hacking and improves WritingBench and CreativeWriting-v3 by up to +4.6 and +7.1 points based on Qwen3-8B, achieving state-of-the-art performance among open-source models.
The model and dataset are open-sourced at https://github.com/ShadeCloak/IP-GRM.
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When Replanning Becomes the Bottleneck: Budgeted Replanning for Embodied Agents
Shuaijun Liu, Feiyang You, Xingwei Chen, Ningxin Su
Embodied agents replan frequently to recover from execution drift, partial observability, and coordination hazards, but each LLM-based replanning call can consume an accumulated textual context that grows over time and across agents. Once this context becomes large, replanning latency develops heavy tails and can miss real-time deadlines even when task success remains high, a failure mode that is hard to detect from average latency or success alone. We present BRACE, a controller that formulates replanning as a budgeted control loop by deciding whether to replan, selecting a replanning mode, and allocating an explicit token budget and latency service-level objective (SLO) while accounting for optional efficiency modules. As a reusable component, we introduce E-RECAP, a cost-aware progressive token pruning method that predicts token utility and prunes replanning contexts across transformer layers while preserving critical head and tail tokens. Across Meta Habitat, RoboFactory, and AirSim, BRACE with E-RECAP reduces replanning-call token counts by 62–92% and SLO violation rates from 85.5–100.0% to 4.7–50.0% in settings where task success is already saturated. In a harder RoboFactory setting where open-loop, frozen-plan, and No BRACE all fail, BRACE + E-RECAP reaches 80.0% success with 4.6% SLO violations, demonstrating that tail-aware per-call budgeting is effective across embodied platforms.
****
V_0: A Generalist Value Model for Any Policy at State Zero
Yi-Kai Zhang, Zhiyuan Yao, Hongyan Hao, Yueqing Sun, Qi GU, Hui Su, Xunliang Cai, De-Chuan Zhan, Han-Jia Ye
Policy gradient methods rely on a baseline to measure the relative advantage of an action. In Actor-Critic LLM training (\textit{e.g.}, PPO), this baseline is typically estimated by a Value Model as large as the policy. However, as the policy evolves, the value model requires expensive, synchronous training. To avoid this overhead, GRPO eliminates the value model by using the average reward of group rollouts as the baseline; yet, this necessitates extensive sampling to maintain stability. In this paper, we propose V_0, a Generalist Value Model capable of estimating the expected performance of any model on unseen prompts without parameter updates. We reframe value estimation by treating the policy's dynamic capability as an explicit context input, leveraging a history of instruction-performance pairs to dynamically profile the model. Acting as a resource scheduler, V_0 predicts success rates prior to rollout during GRPO training for efficient sampling budget allocation. During deployment, it functions as a router dispatching instructions to the most cost-effective model. Empirical results demonstrate that V_0 significantly outperforms heuristic budget allocation and achieves a Pareto-optimal performance-cost trade-off in LLM routing.
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Internalizing Safety Understanding in Large Reasoning Models via Verification
Yi Zhang, Yuxin Chen, Leheng Sheng, Dongcheng Zhang, Chaochao Lu, Xiang Wang, An Zhang
While explicit Chain-of-Thought (CoT) empowers large reasoning models (LRMs), it enables the generation of riskier final answers.
Current alignment paradigms primarily rely on externally enforced compliance, optimizing models to detect malicious prompts rather than evaluating the safety of their own outputs.
We argue that this approach remains largely behavioral: our empirical analysis reveals that ostensibly aligned models lack intrinsic safety understanding, often failing to verify their own response safety and remaining vulnerable to adversarial jailbreaks.
To address this fundamental limitation, we propose Safety Internal (SInternal), a framework that internalizes safety specifications by training LRMs exclusively on safety verification tasks to critique their own generated answers using expert reasoning trajectories.
We demonstrate that learning to verify induces a strong generalization for response safety, significantly enhancing robustness against out-of-domain jailbreaks. Furthermore, when combined with reinforcement learning, SInternal serves as a s

uperior initialization compared to standard supervised fine-tuning, suggesting t
hat internalizing safety understanding creates a more robust foundation for alig
nment than merely mimicking safe behaviors.
Our codes are available at \url{https://github.com/AlphaLab-USTC/SInternal}
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Deriving Neural Scaling Laws from the Statistics of Natural Language
Francesco Cagnetta, Allan Raventos, Surya Ganguli, Matthieu Wyart
Despite the fact that experimental neural scaling laws have substantially guided
 empirical progress in large-scale machine learning, no existing theory can quan
titatively predict the exponents of these important laws for any modern LLM trai
ned on any natural language dataset. We provide the first such theory in the cas
e of data-limited scaling laws. We isolate two key statistical properties of lan
guage that _alone_ can predict neural scaling exponents: (i) the decay of pairwi
se token correlations with time separation between token pairs, and (ii) the dec
ay of the next-token conditional entropy with the length of the conditioning con
text. We further derive a simple formula in terms of these statistics that predi
cts data-limited neural scaling exponents from first principles _without any_ fr
ee parameters or synthetic data models. Our theory exhibits a remarkable match w
ith experimentally measured neural scaling laws obtained from training GPT-2 and
 LLaMA style models from scratch on two qualitatively different benchmarks, Tiny
Stories and WikiText.
****

How to Correctly Report LLM-as-a-Judge Evaluations
Chungpa Lee, Thomas Zeng, Jongwon Jeong, Jy-yong Sohn, Kangwook Lee
Large language models (LLMs) are widely used as scalable evaluators of model res
ponses in lieu of human annotators. However, imperfect sensitivity and specifici
ty of the LLM judges induce bias in naive evaluation scores. We propose a simple
 plug-in framework that corrects this bias and enables statistically principled
uncertainty quantification. Our framework constructs confidence intervals that a
ccount for uncertainty from both the test dataset and a human-labeled calibratio
n dataset. Additionally, it uses an adaptive strategy to allocate calibration sa
mples for tighter intervals. Importantly, we characterize parameter regimes defi
ned by the true evaluation score and the LLM judge’s sensitivity and specificity
 in which our LLM-based evaluation yields more reliable estimates than human-onl
y evaluation. Moreover, we show that our framework remains unbiased under distri
bution shift between the test and calibration datasets, in contrast to existing
approaches.
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CooT: Learning to Coordinate In-Context with Coordination Transformers
Huai-Chih Wang, Hsiang-Chun Chuang, Hsi-Chun Cheng, Dai-Jie Wu, Shao-Hua Sun
Effective coordination among unfamiliar partners remains a major challenge in mu
lti-agent systems. Existing approaches, such as population-based methods, improv
e robustness through diversity but often lack mechanisms for efficient adaptatio
n beyond the training distribution. Fine-tuning is also impractical for few-shot
 learning because it requires a large number of interactions for meaningful impr
ovement.
To address these limitations, we propose Coordination Transformers (CooT), a fra
mework that leverages in-context learning (ICL) for real-time partner adaptation
. Unlike prior ICL approaches that focus on task generalization, CooT is designe
d to generalize across diverse partner behaviors. Trained on trajectories from b
ehavior-preferring agents, it learns to align actions with partner intentions pu
rely through observation. We evaluate CooT on two challenging multi-agent benchm
arks: Overcooked and Google Research Football.
Results show that CooT consistently outperforms population-based methods, gradie
nt-based fine-tuning, and Meta-RL baselines, achieving stable and rapid adaptati
on without parameter updates. Human evaluations also identify CooT as a preferre
d collaborator, and our ablations confirm its ability to adapt quickly to new pa
rtners and remain stable under sudden partner changes, making it reliable for re
al-world human-AI collaboration.
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Beyond Blind Noising: Disentangled Visual Rectification for Hallucination Mitigation in MLLMs
Yujia Chen, Rui Sun, Bingzhou Wang, Huayu Mai, Wangkai Li, Zhaoyang Li, Aibing Li, Wenzhang SUN
Visual Contrastive Decoding (VCD) mitigates hallucinations in Multimodal Large Language Models (MLLMs) by penalizing the output shift from noise-perturbed images, assuming this shift captures the hallucination direction. We prove this assumption flawed: noise-induced drift in Language-Image Pretrained (LIP) encoders is a \emph{coupled vector} entangling (i) structural degradation from corrupted visual information with (ii) hallucination induction from linguistic prior activation. VCD's indiscriminate penalty inevitably suppresses valid visual semantics. Our key insight is that Self-Supervised Learning (SSL) encoders exhibit \emph{only} structural degradation under noise—geometrically orthogonal to hallucination paths—enabling principled disentanglement via LIP--SSL differential response. We propose \textbf{Disentangled Visual Rectification (DVR)}, a training-free dual-stream framework performing visual-layer rectification and decoding-layer contrast on purified representations. DVR achieves approximately $5\times$ theoretical error reduction over VCD and establishes SOTA performance on POPE, MME, LLaVA-Bench and CHAIR benchmarks.
****
Anatomy of Massive Activations and Attention Sinks
Shangwen Sun, Alfredo Canziani, Yann LeCun, Jiachen Zhu
We study two recurring phenomena in Transformer language models: massive activations, in which a small number of tokens exhibit extreme outliers in a few channels, and attention sinks, in which certain tokens attract disproportionate attention mass regardless of semantic relevance.
Prior work observes that these phenomena frequently co-occur and often involve the same tokens, but their functional roles and causal relationships remain unclear.
Through systematic experiments, we show that the co-occurrence is largely an architectural artifact of modern Transformer design, and that the two phenomena serve related but distinct functions.
Massive activations operate globally: they induce near-constant hidden representations that persist across layers, effectively functioning as implicit parameters of the model.
Attention sinks operate locally: they modulate attention outputs across heads and bias individual heads toward short-range dependencies.
We identify the pre-norm configuration as the key choice that enables the co-occurrence and show that ablating it causes the two phenomena to decouple.
****
Generative Neural Operators through Diffusion Last Layer
Sungwon Park, Anthony Zhou, Hongjoong Kim, Amir Barati Farimani
Neural operators provide a powerful framework for learning discretization invariant mappings between function spaces, but standard deterministic models do not capture predictive uncertainty. We introduce *diffusion last layer* (DLL), a modular probabilistic output head for neural operator backbones. DLL represents target fields through an input dependent low rank expansion inspired by the Karhunen--Lo\`eve expansion and learns a conditional diffusion model over the corresponding coefficient space. This design enables efficient distributional modeling while preserving the structural advantages of operator learning. On stochastic PDE benchmarks with random forcing, DLL achieves strong distributional fidelity and performs competitively with pixel space and conventional latent diffusion baselines. In deterministic long horizon rollout tasks, DLL improves rollout stability over the underlying backbone and provides useful estimates of predictive uncertainty under compounding autoregressive errors. These results suggest that diffusion modeling in learned coefficient spaces offers a practical route to uncertainty aware neural operators. Code is available at https://github.com/sungwpark/dll-no
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EEG-Based Multimodal Learning via Hyperbolic Mixture-of-Curvature Experts

Runhe Zhou, Shanglin Li, Guanxiang Huang, Xinliang Zhou, Qibin Zhao, Motoaki Kawanabe, Yi Ding, Cuntai Guan

Electroencephalography (EEG)-based multimodal learning integrates brain signals with complementary modalities to improve mental state assessment, providing great clinical potential.
The effectiveness of such paradigms largely depends on the representation learning on heterogeneous modalities.
For EEG-based paradigms, one promising approach is to leverage their hierarchical structures, as recent studies have shown that both EEG and associated modalities (e.g., facial expressions) exhibit hierarchical structures reflecting complex cognitive processes.
However, Euclidean embeddings struggle to represent these hierarchical structures due to their flat geometry, while hyperbolic spaces, with their exponential growth property, are naturally suited for them.
In this work, we propose EEG-MoCE, a novel hyperbolic mixture-of-curvature experts framework designed for multimodal neurotechnology.
EEG-MoCE assigns each modality to an expert in a learnable-curvature hyperbolic space, enabling adaptive modeling of its intrinsic geometry.
A curvature-aware fusion strategy then dynamically weights experts, emphasizing modalities with richer hierarchical information.
Extensive experiments on benchmark datasets demonstrate that EEG-MoCE achieves state-of-the-art performance, including emotion recognition, sleep staging, and cognitive assessment.
Code is available at https://github.com/zhourunhe/EEG-MoCE.

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Prefix-Cache-Aware Data Reordering for LLM-Augmented Database Analytics

Yingze Li, Dong Wang, Yiming Guo, Yao Chen, Hongzhi Wang, Bingsheng He

LLM-augmented database analytics face a major bottleneck in the costly prefill phase. Although relational tables inherently contain repeated attribute values, standard row-by-row processing produces fragmented prompt layouts that obscure shared prefixes, thereby minimizing opportunities for prefix KV cache reuse and constraining system efficiency. Existing solutions typically employ heuristic or exhaustive search methods to reorder prompt layouts, but these approaches can be inefficient and may not leverage the structural properties of relational tables. We address this challenge by formulating prefix-cache-aware prompt layout optimization as a problem rooted in the isomorphism between prefix-cache reuse and the radix tree topology induced by the relational data distribution. Building on this perspective, we introduce a practical greedy tree-shaping algorithm that efficiently selects row and column orderings to maximize prefix overlap. Our approach, SOLO, improves prefill throughput by up to 90.3\% under fixed prefix-cache budget. Moreover, it reduces planning overhead by up to 242$\times$ compared to state-of-the-art baselines.

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Brain Networks Should Be Learned, Not Constructed

Liang Yang, Shuai Zhai, Ziyi Ma, Jiaming Zhuo, Di Jin, Chuan Wang, Zhen Wang, Xiaochun Cao

The construction of the brain functional network often follows the hand-crafted Correlation Coefficients without any learnable components.
Meanwhile, most efforts are made to the models, such as graph neural networks, that make predictions with the constructed brain network as input. Unfortunately, the fixed brain network may lose critical information during construction and lead to difficulty in performance improvement, even with deliberately designed graph models. From this perspective, the current situation is similar to the machine learning community, i.e., hand-crafted features and learnable predictors, before the advent of representation learning.
In fact, the brain network can be regarded as a graph-structured learnable representation of the brain. By drawing on representation learning, this paper presents the Brain Representation (BRep) learning problem. To this end, the widely used linear and nonlinear correlations are enhanced to be high-order, parametric, and learnable. The expressive brain representation shifts the heavy computational

burden away from downstream architectures, enabling a simple yet accurate predictor. Theoretical analysis guarantees the model's universal approximation to any U/V-statistics.
Extensive evaluations demonstrate that BRep possesses superior performance, high efficiency, and interpretability. The code is available at https://github.com/Kevin-916/BRep-demo/.
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On Densest k-Subgraph Mining and Diagonal Loading: Optimization Landscape and Finite-Step Exact Convergence Analysis
Qiheng Lu, Nicholas D. Sidiropoulos, Aritra Konar
The Densest k-Subgraph (DkS) is a fundamental combinatorial problem known for its theoretical hardness and breadth of applications. Recently, Lu et al. (AAAI 2025) introduced a penalty-based non-convex relaxation that achieves promising empirical performance; however, a rigorous theoretical understanding of its success remains unclear. In this work, we bridge this gap by providing a comprehensive theoretical analysis. We first establish the tightness of the relaxation, ensuring that the global maximum values of the original combinatorial problem and the relaxed problem coincide. Then we reveal the benign geometry of the optimization landscape by proving a strict dichotomy of stationary points: all integral stationary points are local maximizers, whereas all non-integral stationary points are strict saddles with explicit positive curvature. We propose a saddle-escaping Frank--Wolfe algorithm and prove that it achieves exact convergence to an integral local maximizer in a finite number of steps.
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Detecting Errors in AI-Generated Annotations: When and Why Semantic Neighbors Help
Na Di, Ling Li, Zhe Tang, Hao Cheng, Jinlong Pang, Jiaheng Wei, Zhaowei Zhu
Large language models (LLMs) and vision-language models (VLMs) have emerged as efficient annotators for tasks such as generation and classification. While these models offer significant cost and speed advantages over human annotation, a critical challenge remains: existing self-evaluation methods, such as LLM-as-judge, often lack reliable reference-based calibration for error detection. We address this limitation by introducing **SAGE** (**S**emantic-**A**nchored Jud**G**m**E**nt), a method that leverages semantically similar samples retrieved via k-nearest-neighbor as references for annotation verification. We provide a theoretical framework that derives a closed-form expression for the error detection AUROC, which can be decomposed into three factors: intrinsic separability, reference-induced mean shift, and noise reduction through averaging. This decomposition reveals *when* semantic neighbors help (when references are both semantically matched and correct) and *why* (by providing reference-based calibration that raises scores for correct annotations and lowers scores for incorrect ones). Experiments on LLM generation, VLM captioning, and classification tasks validate our theoretical framework: SAGE improves error detection when semantic neighbors provide reliable reference-based calibration, and our decomposition offers insights into when direct scoring or alternative strategies may be preferred. Our code is available at https://github.com/dina-1205/SAGE.
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Learning Flexible Generalization in Video Quality Assessment by Bringing Device and Viewing Condition Distributions
Nikolay Safonov, Dmitriy S. Vatolin
Video quality assessment (VQA) plays a critical role in optimizing video delivery systems. While numerous objective metrics have been proposed to approximate human perception, the perceived quality strongly depends on viewing conditions and display characteristics. Factors such as ambient lighting, display brightness, and resolution significantly influence the visibility of distortions. In this work, we address the question of the multi-screen quality assessment on mobile devices, as this area still tends to be under-covered. We introduce a first large-scale subjective dataset collected across more than different 300 Android devices, accompanied by metadata on viewing conditions and display properties. We propose a strategy for aggregated score extraction and adaptation of VQA models to de

vice-specific quality estimation. Our results demonstrate that incorporating dev ice and context information enables more accurate and flexible quality predictio n, offering new opportunities for fine-grained optimization in streaming service s. Ultimately, this work advances the development of perceptual quality models t hat bridge the gap between laboratory evaluations and the diverse conditions of real-world media consumption. We made the dataset and the code available at http s://videoprocessing.github.io/device-viewing-conditions.
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Anti-causal domain generalization: Leveraging unlabeled data
Sorawit Saengkyongam, Juan L. Gamella, Andrew Miller, Jonas Peters, Nicolai Mein shausen, Christina Heinze-Deml
The problem of domain generalization concerns learning predictive models that ar e robust to distribution shifts when deployed in new, previously unseen environm ents. Existing methods typically require labeled data from multiple training env ironments, limiting their applicability when labeled data are scarce. In this wo rk, we study domain generalization in an anti-causal setting, where the outcome causes the observed covariates. Under this structure, environment perturbations that affect the covariates do not propagate to the outcome, which motivates regu larizing the model's sensitivity to these perturbations. Crucially, estimating t hese perturbation directions does not require labels, enabling us to leverage un labeled data from multiple environments. We propose two methods that penalize th e model's sensitivity to variations in the mean and covariance of the covariates across environments, respectively, and prove that these methods have worst-case optimality guarantees under certain classes of environments. Finally, we demons trate the empirical performance of our approach on a controlled physical system and a physiological signal dataset.
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Preference-Enhanced Reinforcement Learning for Pluralistic Image Inpainting
Peng Zhou, Muqi Huang, Tianshuo Qu, Jingyang Wang, kun Zhou, Chuan Li, Feng Shi, Shi Chen, Yun Xiong
Existing image inpainting frameworks rely on strictly supervised training paradi gms, often suffering from an over-reliance on ground-truth reconstruction, which leads to conservative outputs with misaligned creativity and limited diversity. To this end, we propose the first framework to explore Group Relative Policy Opt imization (GRPO) and Direct Preference Optimization (DPO) for text-guided image inpainting, formulating an efficient online reinforcement learning pipeline that enables flexible, human-aligned aesthetic control via a preference scoring mode l. Crucially, by decoupling the rigid one-to-one correspondence between text pro mpts and masked images, our method enables the model to explore diverse, control lable, and high-quality solutions beyond a single target. Furthermore, to balanc e semantic consistency with physical naturalness at mask boundaries, we introduc e a scale-aware dynamic reward mechanism that adaptively emphasizes boundary gra dient coherence for small occlusions while prioritizing visual aesthetics in lar ge-scale generation. Extensive experiments demonstrate that our approach consist ently produces higher-quality results across different backbone architectures su ch as Stable Diffusion and FLUX, significantly enhancing the generative capacity of base models.
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Neural Collapse by Design: Learning Class Prototypes on the Hypersphere
Panagiotis Koromilas, Theodoros Giannakopoulos, Mihalis Nicolaou, Yannis Panagak is
Supervised classification has a theoretical optimum, Neural Collapse (NC), yet n either of its two dominant paradigms reaches it in practice. Cross entropy (CE) leaves radial degrees of freedom unconstrained and converges to a degenerate geo metry, while supervised contrastive learning (SCL) drives features toward NC dur ing pretraining but discards this structure in a post hoc linear probing phase. We show that both paradigms are different appearances of the same method that co ntrasts prototypes on the unit hypersphere, and that closing the gap requires fi xing each at its point of failure. From the CE side, we propose NTCE and NONL, t wo normalized losses that import contrastive optimization's missing ingredients

into classifier learning: a large effective negative set and decoupled alignment and uniformity terms. From the SCL side, we prove that SCL's objective already optimizes throughout training for a principled classifier whose weights are the class mean embeddings, making linear probing both redundant and harmful. Empirically, on four benchmarks including ImageNet-1K, NTCE and NONL surpass CE accuracy, closely approximate NC ($\geq 95\%$), and match CE's converged NC on 4/5 metrics in under 7.5% of its iterations, while SCL with fixed prototypes matches linear probing without the hours-long classifier training phase. The learned geometry yields $+5.5\%$ mean relative improvement in transfer learning, up to $+8.7\%$ under severe class imbalance, and improved robustness to corruptions on ImageNet-C. Our work recasts supervised learning as prototype learning on the hypersphere, with NC reached *by design*.
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From Pixels to Tokens: A Systematic Study of Latent Action Supervision for Vision-Language-Action Models
Yihan Lin, Haoyang Li, Yang Li, Haitao Shen, Yihan Zhao, Chao Shao, Jing Zhang
Latent actions serve as an intermediate representation that enables consistent modeling of vision-language-action (VLA) models across heterogeneous datasets. However, approaches to supervising VLAs with latent actions are fragmented and lack a systematic comparison. This work structures the study of latent action supervision from two perspectives: (i) regularizing the trajectory via image-based latent actions, and (ii) unifying the target space with action-based latent actions. Under a unified VLA baseline, we instantiate and compare four representative integration strategies. Our results reveal a formulation-task correspondence: image-based latent actions benefit long-horizon reasoning, whereas action-based latent actions excel at complex motor coordination. Furthermore, we find that directly supervising the VLM with discrete latent action tokens yields the most effective performance. Finally, our experiments offer initial insights into the benefits of latent action supervision in mixed-data, suggesting a promising direction for VLA training.
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Recognize Your Orchestrator: An Entropy Dynamics Perspective for LLM Multi-Agent Systems
Junze Zhu, Weihao Chen, Xuanwang Zhang, Zhen Wu, Xinyu Dai
The transition from single-turn models to Multi-Agent Systems (MAS) promises enhanced problem-solving capabilities, yet the centralized orchestration topology remains a critical point of fragility. To analyze this, we propose a Mean-Field Entropy Dynamics framework, modeling the orchestration process as a system governed by the competing forces of task resolution and cumulative context loading. To facilitate high-resolution validation, we introduce Inverse Workflow Generation (IWG), a multi-agent pipeline that synthesizes process-verifiable, high-complexity benchmarks with dense intermediate checkpoints. We demonstrate that our entropy dynamics model fits empirical trajectories, providing physically interpretable parameters that quantify system stability and performance collapse. Crucially, our analysis uncovers a ``Reasoning Trap": while reasoning-heavy models excel in isolated tasks, they frequently fail as orchestrators due to context squeezing. By elucidating the physical mechanisms underlying the Orchestrator and quantifying systemic uncertainty, our findings offer insights for the architectural design development of Multi-Agent Systems in prospective research.
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Causal-JEPA: Learning World Models through Object-Level Latent Masking
Heejeong Nam, Quentin Le Lidec, Lucas Maes, Yann LeCun, Randall Balestriero
World models require robust relational understanding to support prediction, reasoning, and control. While object-centric representations provide a useful abstraction, they are not sufficient to capture interaction-dependent dynamics. We therefore propose C-JEPA, a simple and flexible object-centric world model that extends masked joint embedding prediction from image patches to object-centric representations. By masking object-level latents and requiring each masked object state to be inferred from the surrounding context, C-JEPA imposes structured partial observability during training, creating counterfactual-like prediction querie

s that discourage shortcut solutions and make interaction-dependent prediction n ecessary under the learning objective. Empirically, C-JEPA leads to consistent g ains in visual question answering, with an absolute improvement of about 20\% in counterfactual reasoning over the same architecture without object-level maskin g. On agent control tasks, C-JEPA enables substantially more efficient planning by using only 1\% of the total latent input features required by patch-based wor ld models, while achieving comparable performance. Finally, we provide a formal analysis demonstrating that object-level masking induces useful inductive bias b y controlling observability. Our code is available at https://github.com/galilai -group/cjepa.

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Antidistillation Fingerprinting

Yixuan Even Xu, John Kirchenbauer, Yash Savani, Asher Trockman, Alexander Robey, Tom Goldstein, Fei Fang, J Zico Kolter

Model distillation enables efficient emulation of frontier large language models (LLMs), creating a need for robust mechanisms to detect when a third-party stud ent model has trained on a teacher model's outputs. However, existing fingerprin ting techniques that could be used to detect such distillation rely on heuristic perturbations that impose a steep trade-off between generation quality and fing erprinting strength, often requiring significant degradation of utility to ensur e the fingerprint is effectively internalized by the student. We introduce ***an tidistillation fingerprinting*** (ADFP), a principled approach that aligns the f ingerprinting objective with the student's learning dynamics. Building upon the gradient-based framework of *antidistillation sampling*, ADFP utilizes a proxy m odel to identify and sample tokens that directly maximize the expected detectabi lity of the fingerprint in the student after fine-tuning, rather than relying on the incidental absorption of the un-targeted biases of a more naive watermark. Experiments on GSM8K, OASST1, and MBPP demonstrate that ADFP achieves a signific ant Pareto improvement over state-of-the-art baselines, yielding stronger detect ion confidence with minimal impact on utility across mathematical reasoning, dia logue, and code generation, even when the student model's architecture is unknow n.

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LitReview Arena: Evaluating Literature Review Agents with Battle-style Peer Revi ew Platform

Ruotong Zhao, zhiyu chen, Xurui Liu, Haidong Xue, Dong Liang, Jigao Fu, Wu YanBi ao, Yuanyi Zhen, Fengli Xu, Yong Li

Literature reviews are essential to scientific progress, but rigorously evaluati ng automatically generated reviews remains difficult because many aspects of res earch utility depend on expert judgment rather than reference-overlap metrics. W e introduce \textbf{LitReview Arena}, a battle-style evaluation platform with a structured protocol tailored to literature review quality: domain experts with A I paper-writing experience compare anonymized drafts, are matched to topics with in their expertise, and provide dimension-wise outcomes over five literature-rev iew-specific criteria. From this protocol, we collect 3k expert judgments, each containing five dimension-wise outcomes, and show that even the strongest curren t systems win only 23.0\% of decisive matches against human drafts on overall ut ility, while agentic LLMs such as Sonar Deep Research substantially outperform b ase language models by over 60\%. We further find that existing LLM-as-a-judge m ethods are substantially misaligned with human experts (Spearman's $\rho \approx 0.467$), especially on synthesis-heavy criteria such as paper structure and res earch suggestions. Using the collected preference data, we provide an expert-cal ibrated evaluator, LitJudge, which improves alignment to $\rho \approx 0.78$, co mparable to inter-expert consistency; code and data are publicly available at ht tps://github.com/VanellopeAsher/LitReview-Arena.

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C^{2}R: Cross-sample Consistency Regularization Mitigates Feature Splitting an d Absorption in Sparse Autoencoders

Haoran Jin, Xiting Wang, Shijie Ren, Hong Xie, Defu Lian

Sparse Autoencoders (SAEs) are widely used to interpret large language models by

decomposing activations into sparse, human-understandable features, but scaling to large dictionaries exposes fundamental challenges. Systematic studies reveal pervasive feature splitting that fragments coherent concepts into non-atomic latents and widespread feature absorption that creates arbitrary exceptions in general features, severely compromising latent reliability. These issues stem from inconsistent latent assignment across samples: without cross-sample constraints, per-sample optimization often allows a single underlying concept to be inconsistently distributed across multiple redundant or interfering latents. To address this, we introduce C^{2}R (\underline{\textbf{C}}ross-sample \underline{\textbf{C}}onsistency \underline{\textbf{R}}egularization). C^{2}R explicitly encourages that each semantic feature is consistently represented by a unified latent across the batch by penalizing the co-activation of directionally similar latents. Comprehensive evaluation demonstrates that C^{2}R effectively mitigates both splitting and absorption while, crucially, preserving reconstruction fidelity, providing a principled solution that enhances latent interpretability without degrading model performance. Source code is available\footnote{\url{https://github.com/hr-jin/Cross-sample-Consistency-Regularization}}.

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Breaking the Lock-in: Diversifying Text-to-Image Generation via Representation Modulation

Dahee Kwon, Haeun Lee, Jaesik Choi

Recent text-to-image models built on large-scale Transformer backbones and flow-based objectives deliver strong text–image alignment and high visual quality, yet often produce overly similar samples under a fixed prompt. Existing diversity-enhancement methods alleviate this, but typically require expensive sampling or auxiliary optimization, incurring non-trivial overhead. To investigate the root cause of this homogeneity, we examine intermediate Transformer features and observe that the zero-frequency spatial average (DC) component rapidly converges across seeds early in generation, causing early trajectory lock-in that limits downstream variation. Building on this, we propose DC Attenuation for diVersity Enhancement (\textbf{DAVE}), a training-free representation-level intervention that selectively attenuates this component in the early regime. DAVE preserves the sampling pipeline with negligible overhead, improving prompt-consistent diversity while maintaining competitive image quality.

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Shapley Neuron Values for Continual Learning: Which Neurons Matter Most?

Mohammad Ali Vahedifar, Abhisek Ray, Qi Zhang

Continual learning enables neural networks to learn tasks sequentially without forgetting previously acquired knowledge. However, neural networks suffer from catastrophic forgetting, where learning new tasks degrades performance on earlier ones. We address this problem with **Shapley Neuron Valuation (SNV)**, a principled framework that quantifies Neuron importance in continual learning, grounded in cooperative game theory. SNV selectively freezes important Neurons while keeping others plastic, enabling **buffer-free** continual learning without expanding architecture. Experiments on ImageNet-1k show that SNV consistently outperforms existing buffer-free methods. In particular, SNV improves accuracy by **+2.88\%** in the class incremental learning and **+6.46\%** in the task incremental learning scenarios compared to the second baseline.

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Flash-GRPO: Efficient Alignment for Video Diffusion via One-Step Policy Optimization

Xiaoxuan He, Siming Fu, Zeyue Xue, Weijie Wang, Ruizhe He, Yuming Li, Dacheng Yin, Shuai Dong, Haoyang Huang, Hongfa Wang, Nan Duan, Bohan Zhuang

Group Relative Policy Optimization has emerged as essential for aligning video diffusion models with human preferences, but faces a critical computational bottleneck: training a 14B parametered model typically demands hundreds of GPU days per experiment. Existing efficiency methods reduce costs through sliding window subsampling training timesteps, but fundamentally compromise optimization, exhibiting severe instability and failing to reach full trajectory performance. We present Flash-GRPO, a single-step training framework that outperforms full traject

ory training in alignment quality under low computational budgets while substant
ially improving training efficiency. Flash-GRPO addresses two critical challeng
es: iso-temporal grouping eliminates timestep-confounded variance by enforcing p
rompt-wise temporal consistency, decoupling policy performance from timestep dif
ficulty; temporal gradient rectification neutralizes the time-dependent scaling
factor that causes vastly inconsistent gradient magnitudes across timesteps. Exp
eriments on 1.3B to 14B parameter models validate Flash-GRPO's effectiveness, de
monstrating substantial training acceleration with consistent stability and stat
e-of-the-art alignment quality.
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SuCo: Sufficiency-guided Continuous Adaptive Reasoning
Jiahao Wang, Bingyu Liang, Chenhao Hu, Longhui Zhang, Xuebo Liu, Min Zhang, Jing
 Li, Xuelong Li
Despite remarkable performance on complex tasks, Large Reasoning Models (LRMs) o
ften generate excessively long Chain-of-Thoughts (CoT), inflating computational
costs even for simple queries. Existing efforts to mitigate this inefficiency ty
pically rely on discrete reasoning modes or fixed budget tiers, lacking a princi
pled criterion of when reasoning is sufficient. In this work, we introduce *Mini
mal Sufficient CoT* (MSC), defined as the shortest prefix of a CoT trajectory wh
ich is adequate for producing the correct answer. We empirically show that MSC n
ot only reduces reasoning tokens, but also improves accuracy across difficulty l
evels. Building on MSC, we propose *Sufficiency-guided Continuous Adaptive Reaso
ning* (SuCo), a two-stage training framework for autonomous reasoning control al
ong a continuous spectrum. In stage 1, *MSC-Aligned Fine-Tuning* (MFT) construct
s MSC data using problem-adaptive sufficiency thresholds that naturally scale wi
th question difficulty, then fine-tunes the model to internalize concise yet suf
ficient reasoning patterns. In stage 2,
Sufficiency-Aware Policy Optimization (SAPO) further optimizes the model throu
gh reinforcement learning with dynamic complexity tracking and sufficiency-aware
 rewards that penalize both over- and under-thinking. Extensive experiments acro
ss mathematics, code, and science benchmarks show that SuCo consistently achieve
s improvements in both accuracy and reasoning efficiency.
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Finding Stationary Points by Comparisons
Helin Wang, Chenyi Zhang, Xiwen Tao, Yexin Zhang, Tongyang Li
We study the problem of finding stationary points of non-convex functions when a
ccess to the objective is provided only through a comparison oracle that, given
two points, outputs which has the larger function value. For a twice differentia
ble $f:\mathbb R^n\to\mathbb R$ with Lipschitz gradient and Hessian, we develop
an algorithm that outputs an ϵ-stationary point using $\widetilde O(n^2
/\epsilon^{1.5})$ queries. Our approach uses a subroutine that estimates the nor
malized Hessian to accuracy δ using $\widetilde O(n^2\log(1/\delta))$ que
ries. We further study this problem with a quantum comparison oracle model where
 queries can be made in superpositions, and develop the first quantum algorithm
that finds an ϵ-stationary point, which takes $\widetilde O(n/\epsilon^
{1.5})$ queries.
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CALM Before the STORM: Unlocking Native Reasoning for Optimization Modeling
Zhengyang Tang, Zihan Ye, Chenyu Huang, Xuhan Huang, Chengpeng Li, Sihang Li, Gu
anhua Chen, Ming Yan, Zizhuo Wang, Hongyuan Zha, Dayiheng Liu, Benyou Wang
Large Reasoning Models (LRMs) create new opportunities for automating optimizati
on modeling, but they also make post-training more delicate. In this task, stron
g performance often requires the model to formulate the problem, write solver co
de, run it, inspect the output, and revise when needed. We show that directly fi
ne-tuning LRMs on already written-out Operations Research (OR) solutions can imp
rove easier cases while hurting harder ones, suggesting that this training signa
l can interfere with the model's own way of solving the task. We therefore propo
se **CALM** (*Corrective Adaptation with Lightweight Modification*), which lets
the base LRM attempt the problem first, then inserts a short hint at the first d
etected mistake and lets the model continue from there. These hints modify fewer

than 2.6\% of generated tokens. The corrected solutions are used for supervised fine-tuning and then reinforcement learning, producing **STORM**, a 4B optimization-modeling specialist that reaches 68.9\% macro-average accuracy across five benchmarks and matches 671B DeepSeek-R1-0528. Under a matched hard-benchmark control, CALM also yields stronger final RL performance than direct distillation baselines that train on complete teacher-generated solutions from much stronger models. Overall, for this task, local repair of the base model's own solution is more effective than full teacher-solution replacement. Code and models are available at \url{https://github.com/tangzhy/STORM}.

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Generative Adaptation of Dynamics to Environmental Shifts via Weight-space Diffusion

Ruikun Li, Huandong Wang, Jingtao Ding, Yuan Yuan, Qingmin Liao, Yong Li

Data-driven dynamics prediction often fails under environmental shifts, while traditional fine-tuning remains computationally prohibitive for hardware-constrained or data-scarce applications.

We propose DynaDiff, a generative meta-learning framework that transitions the paradigm from gradient-based tuning or modulation to direct weight-space generation.

Specifically, we first abstract expert weights as novel weight graphs, utilizing multi-head attention to explicitly capture topological coupling within weights.

Subsequently, we design a functional loss to ensure that the generated models achieve consistency with expert models in physical behavior.

Finally, we develop a dynamics-informed prompter that extracts cross-domain physical and spectral features from observation sequences to condition the diffusion model.

Experiments demonstrate that DynaDiff boosts average prediction accuracy by 10.78\% over competitive baselines.

Furthermore, by pre-constructing a model zoo of expert predictors, we amortize the fine-tuning overhead into a one-time offline cost, significantly boosting deployment efficiency in new environments.

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Orthogonal Concept Erasure for Diffusion Models

Yuhao Sun, Lingyun Yu, Hao-Xiang Xu, Fengyuan Miao, Zhuoer Xu, Hongtao Xie

Concept erasure has emerged as a promising approach to mitigate undesired or unsafe content in diffusion models, yet existing methods still face significant limitations. While training-based methods are effective, their high computational cost limits scalability. Editing-based methods are more efficient and deployment-friendly, yet they struggle to simultaneously achieve precise concept erasure and preserve overall generative capacity. We identify this core limitation of the editing-based methods as reliance on additive parameter updates. Our empirical analysis reveals that concept semantics primarily depend on *neuron direction* rather than *neuron magnitude*, while overall generative capacity relies on the *angular geometry* of neurons. As additive updates inherently entangle direction, magnitude, and angular geometry, they inevitably introduce unintended interference between concept erasure and overall generation performance. To address this, we propose **Orthogonal Concept Erasure (OCE)**, which reformulates editing-based erasure as multiplicative parameter updates from a geometric perspective. Specifically, OCE applies layer-wise orthogonal transformations derived from a closed-form solution to the parameters, enabling precise concept erasure while preserving the neuron magnitude and angular geometry. Furthermore, to address conflicting constraints in multi-concept erasure, OCE introduces a subspace-level objective with structured subspace manipulation, yielding a more effective and scalable erasure. Extensive experiments on single- and multi-concept erasure demonstrate that OCE outperforms existing methods in concept erasure and non-target preservation, erasing up to 100 concepts in 4.3 s. Code: https://github.com/HansSunY/OCE.

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Causal Direct Preference Optimization for Distributionally Robust Generative Rec

ommendation
Chu Zhao, Enneng Yang, Jianzhe Zhao, Guibing Guo
Direct Preference Optimization (DPO) guides large language models (LLMs) to gene rate recommendations aligned with user historical behavior distributions by mini mizing preference alignment loss. However, our systematic empirical research and theoretical analysis reveal that DPO tends to amplify spurious correlations cau sed by environmental confounders during the alignment process, significantly und ermining the generalization capability of LLM-based generative recommendation me thods in out-of-distribution (OOD) scenarios. To mitigate this issue, we propose CausalDPO, an extension of DPO that incorporates a causal invariance learning m echanism. This method introduces a backdoor adjustment strategy during the prefe rence alignment phase to eliminate interference from environmental confounders, explicitly models the latent environmental distribution using a soft clustering approach, and enhances robust consistency across diverse environments through in variance constraints. Theoretical analysis demonstrates that CausalDPO can effec tively capture users' stable preference structures across multiple environments, thereby improving the OOD generalization performance of LLM-based recommendatio n models. We conduct extensive experiments under four representative distributio n shift settings to validate the effectiveness of CausalDPO, achieving an averag e performance improvement of 24.10\% across four evaluation metrics.
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Rethinking Thinking Tokens: LLMs as Improvement Operators
Lovish Madaan, Aniket Rajiv Didolkar, Suchin Gururangan, John Quan, Ruan Silva, Ruslan Salakhutdinov, Manzil Zaheer, Sanjeev Arora, Anirudh Goyal
Reasoning training incentivizes LLMs to produce long chains of thought (long CoT), which among other things, allows them to explore solution strategies with sel f-checking. This results in higher accuracy, but inflates context length, token/ compute cost, and answer latency. We ask: Can current models leverage their meta cognition to provide other combinations on this Pareto frontier, e.g., better ac curacy with lower context length and/or latency? Abstractly, we view the model a s an improvement operator on its own "thoughts" with a continuum of possible str ategies. We identify an interesting inference family Parallel-Distill-Refine (PD R), which performs the following: (i) generate diverse drafts in parallel; (ii) distill them into a bounded, textual workspace; and (iii) refine conditioned on this workspace, producing an output that seeds the next round. Importantly, cont ext length (hence compute cost) is controllable via degree of parallelism, and i s no longer conflated with the total number of generated tokens. We report PDR i nstantiations of current models that give better accuracy than long CoT while in curring lower latency. Setting degree of parallelism to 1 yields an interesting subcase, Sequential Refinement (SR) (iteratively improve a single candidate answ er) which provides performance superior to long CoT. Success of such model orche strations raises the question whether further training could shift the Pareto fr ontier. To this end, we train an 8B thinking model with Reinforcement Learning (RL) to make it consistent with PDR as the inference method. On math tasks with v erifiable answers, iterative pipelines surpass single-pass baselines at matched sequential budgets, with PDR delivering the largest gains (e.g., +11% on AIME 20 24 and +9% on AIME 2025).
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Dependence-Aware Label Aggregation for LLM-as-a-Judge via Ising Models
Krishna Balasubramanian, Aleksandr Podkopaev, Shiva Kasiviswanathan
Large-scale AI evaluation increasingly relies on aggregating binary judgments fr om K annotators, including LLMs used as judges. Most classical methods, e.g., Dawid-Skene or (weighted) majority voting, assume annotators are conditionally i ndependent given the true label $Y\in\{0,1\}$, an assumption often violated by L LM judges due to shared data, architectures, prompts, and failure modes. Ignorin g such dependencies can yield miscalibrated posteriors and even confidently inco rrect predictions. We study label aggregation through a hierarchy of dependence- aware models based on Ising graphical models and latent factors. For class-depen dent Ising models, the Bayes log-odds is generally quadratic in votes; for class -independent couplings, it reduces to a linear weighted vote with correlation-ad

justed parameters. We present finite-K examples showing that methods based on conditional independence can flip the Bayes label despite matching per-annotator marginals. We prove separation results demonstrating that these methods remain strictly suboptimal as the number of judges grows, incurring nonvanishing excess risk under latent factors. Finally, we evaluate the proposed method on three real-world datasets, demonstrating improved performance over the classical baselines.

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Optimal Decision-Making Based on Prediction Sets

Tao Wang, Edgar Dobriban

Prediction sets can wrap around any ML model to cover unknown test outcomes with a guaranteed probability. Yet, it remains unclear how to use them optimally for downstream decision-making. Here, we propose a decision-theoretic framework that seeks to minimize the expected loss (risk) against a worst-case distribution consistent with the prediction set's coverage guarantee. We first characterize the minimax optimal policy for a fixed prediction set, showing that it balances the worst-case loss inside the set with a penalty for potential losses outside the set. Building on this, we derive the optimal prediction set construction that minimizes the resulting robust risk subject to a coverage constraint. Finally, we introduce Risk-Optimal Conformal Prediction (ROCP), a practical algorithm that targets these risk-minimizing sets while maintaining finite-sample distribution-free marginal coverage. Empirical evaluations on medical diagnosis and a toy static hazard-decision benchmark demonstrate that ROCP reduces critical mistakes compared to baselines, particularly when out-of-set errors are costly. The source code to reproduce our experiments is available at https://github.com/TaoWangPenn/Risk-Optimal-Conformal-Prediction.

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An Asymmetric Latent Factorization-of-Tensors Model for Relation Analysis

Weiling Li, Zhaoheng Shi, Jiajia Mi, Zhigang Liu, Jialiang Wang, Xin Luo

Latent Factorization-of-Tensors (LFT) models are an effective approach for relation analysis. Existing LFT models assume each mode of the target tensor corresponds to an entity set and the relationships between entity sets are bipartite graphs to explore the relationships among entities within a mode. However, when the topological structure of entities in a mode is known, for example, entities are ordered physical quantities, such as time or coordinates, the relations between such modes forms a more complicated structure, i.e., aligned bipartite networks, and existing LFT models cannot accurately capture this structure. This work is the first to recognize and analyze this issue, and proposes an Asymmetric Latent Factorization-of-Tensors (ALFT) model to address it. ALFT can model aligned bipartite networks in mode pairs of a tensor by imposing constraints between particular mode pairs in the tensor network. Experimental results on real-world datasets demonstrate the existence of this issue and confirm that the proposed ALFT model can effectively resolve it.

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Large Language Models as Topological Thinkers: A Benchmark on Graph Persistent Homology

Hao Li, Hao Wan, Yixue Huang, Yuzhou Chen, Yulia Gel, Hao Jiang

Large language models (LLMs) are increasingly used in scientific discovery, system modeling, and decision-making, prompting interest in their ability to reason over complex structured data. Existing benchmarks primarily focus on static or local graph reasoning, overlooking the high-order structures in real-world systems whose global properties evolve across multiple scales. We introduce LLM4PH, a benchmark that evaluates multi-scale structural reasoning through the lens of persistent homology (PH), a topological framework for tracking structural evolution. LLM4PH decomposes the PH pipeline into interpretable reasoning tasks spanning synthetic and real-world graphs, revealing that most models struggle with reasoning over structural transitions and persistence. Beyond task-level evaluation, we perform cross-task ablations on prompt encoding and transfer, explore post-training effects, and construct a compositional PH pipeline to assess end-to-end performance. Our results provide the first in-depth view of how well LLMs bridge

discrete graph structures with continuous topological abstraction, and offer insights into their potential for structure-aware scientific reasoning.
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TEAM: Temporal–Spatial Consistency Guided Expert Activation for MoE Diffusion Language Model Acceleration
Linye Wei, Zixiang Luo, Pingzhi Tang, Meng Li
Diffusion large language models (dLLMs) have recently gained significant attention due to their inherent support for parallel decoding. Building on this paradigm, Mixture-of-Experts (MoE) dLLMs with autoregressive (AR) initialization have further demonstrated strong performance competitive with mainstream AR models. However, we identify a fundamental mismatch between MoE architectures and diffusion-based decoding. Specifically, a large number of experts are activated at each denoising step, while only a small subset of tokens is ultimately accepted, resulting in substantial inference overhead and limiting their deployment in latency-sensitive applications. In this work, we propose **TEAM**, a plug-and-play framework that accelerates MoE dLLMs by enabling more accepted tokens with fewer activated experts. TEAM is motivated by the observation that expert routing decisions exhibit strong temporal consistency across denoising levels as well as spatial consistency across token positions. Leveraging these properties, TEAM employs three complementary expert activation and decoding strategies, conservatively selecting necessary experts for decoded and masked tokens and simultaneously performing aggressive speculative exploration across multiple candidates. Experimental results demonstrate that TEAM achieves up to 2.2× speedup over vanilla MoE dLLM, with negligible performance degradation. Code is released at https://github.com/PKU-SEC-Lab/TEAM-MoE-dLLM.
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Learning to Watermark in the Latent Space of Generative Models
Sylvestre-Alvise Rebuffi, Tuan A. Tran, Valeriu Lacatusu, Pierre Fernandez, Tomáš Sou■ek, Nikola Jovanovi■, Tom Sander, Hady Elsahar, Alexandre Mourachko
Existing approaches for watermarking AI-generated images often rely on post-hoc methods applied in pixel space, introducing computational overhead and potential visual artifacts. In this work, we explore latent space watermarking and introduce DistSeal, a unified approach for latent watermarking that works across both diffusion and autoregressive models. Our approach works by training post-hoc watermarking models in the latent space of generative models. We demonstrate that these latent watermarkers can be effectively distilled either into the generative model itself or into the latent decoder, enabling in-model watermarking. The resulting latent watermarks achieve competitive robustness while offering similar imperceptibility and up to 20x speedup compared to pixel-space baselines. Our experiments further reveal that distilling latent watermarkers outperforms distilling pixel-space ones, providing a solution that is both more efficient and more robust.
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Fractional is Better: Learnable Derivative Orders in Neural Operator Learning
Fares B. Mehouachi, Saif Jabari
Neural operators learn mappings between function spaces, enabling fast surrogate solutions to partial differential equations. Despite remarkable architectural diversity, these methods often share a common input representation: raw coordinate-value pairs $(x, u(x))$.
We ask whether inputs aligned with PDE differential structure can improve learning. Through Picard iteration on mild solutions, we show that derivatives of the input appear explicitly in the solution operator, suggesting that providing derivative features should reduce the network's implicit differentiation burden. We prove this intuition: providing derivative features improves approximation rates from $O(W^{-(s-m)/d})$ to $O(W^{-s/d})$, where W is network width, s is input regularity, m is the PDE order, and d is spatial dimension. Our central finding, however, is a surprise: the optimal derivative order β^* is strictly less than the PDE order m. This gap arises from a bias-variance tradeoff in spectral space that we characterize in closed form. Learning β from data achieves automatic spectral regularization. We introduce ∂-NO (del-NO),

for derivative-augmented neural operators, an architecture-agnostic augmentation that provides learnable fractional derivative features to any neural operator backbone. Across benchmark problems and architectures, ∂-NO consistently improves prediction accuracy, with learned orders that reflect a representation of known physics modulated by noise and finite-sample constraints.
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Beyond Prediction: Tail-Aware Scheduling for LLM Inference
Yueying Li, Yuanfan Chen, Jiayang Chen, Esha Choukse, Haoran Qiu, G. Edward Suh, Rodrigo Fonseca, Ziv Scully, Udit Gupta
LLM serving exhibits extreme length variability, making size-based scheduling difficult in practice. Recent LLM schedulers approximate SJF/SRPT using predicted decode lengths or rank and primarily report mean-centric metrics (e.g., TTFT/TBT). We show these prediction-driven policies can be fragile under distribution shifts, bursty arrivals, and GPU memory pressure, and still offer limited control over tail latency (P90–P99) that dominates user experience—even with perfect decode-length knowledge. We introduce a distribution-aware, prediction-free scheduling framework that replaces explicit length prediction with soft, γ-parameterized priority boosting driven by lightweight statistical signals. Our design co-optimizes scheduling with cache-aware preemption to account for memory-coupled decode dynamics that vary across workload mixes. Evaluated on Azure production traces, our method achieves a P99 TTLT up to 35--50\% lower than SRPT with perfect length prediction and a TTFT 34--47\% lower across various workloads, including reasoning-heavy and chat-heavy tasks, demonstrating a robust alternative for tail-latency optimization in online LLM serving.
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Size Transferability of Graph Convolutional Networks across Sparsity: A Generalized Graphon Perspective
Qinji Shu, Hang Sheng, Feng Ji, Hui Feng, Bo Hu
Size transfer scales Graph Convolutional Networks (GCNs) by applying models trained on sampled subgraphs to larger target graphs. However, existing theoretical guarantees are typically confined to dense graphs or restricted sparsity regimes, failing to cover the arbitrary sparsity of real-world networks. To bridge this gap, we introduce the Generalized Graphon Convolutional Network (GWCN) based on the generalized graphon theory. Unlike the classical graphon limit which vanishes in sparse settings, GWCN employs stretching to construct a non-trivial limit that preserves topological structure. We derive an explicit transfer error bound that decomposes into size-dependent and density-dependent components, providing a unified guarantee across arbitrary sparsity levels. Empirical results on real-world networks corroborate our findings, demonstrating that transfer error vanishes as graph size increases and edge density decreases.
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Value-as-Return: A Two-Stage Framework to Align on the Optimal Score Function
Shikun Sun, Shuo Huang, Yiding Chen, Wen Sun, Jia Jia
Reinforcement learning with diffusion models has shown strong potential, but existing approaches such as variants of Direct Preference Optimization (DPO) often rely on an inaccurate simplification: they equate trajectory likelihoods with final-state probabilities. This mismatch leads to suboptimal alignment. We address this limitation with a principled framework that leverages the optimal value function as the return for short trajectory segments. Our approach follows a two-stage procedure: (i) learning a value-distribution function to estimate segment-level returns, and (ii) applying our VRPO to refine the score function. We prove that, under sufficient model capacity, the resulting model is equivalent to training a diffusion process on the tilted distribution proportional to $p(x)\exp(\eta r(x))$. Experiments on large-scale diffusion models validate our analysis and show stable and consistent improvements over prior methods.
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GeoReward: Mitigating Contextual Variable Overestimation in Vision-Language Models for Cross-Market Preference Prediction
Shuo Liu, Huixiang Cai, Weiru Zhang, Xiaoyi Zeng
Vision-language models (VLMs) excel in many multimodal tasks but remain prone to

a subtle yet impactful failure mode: they tend to overestimate dominant visual-textual cues while underestimating sparse but decision-critical contextual variables. This issue, which we term Contextual Variable Overestimation (CVE), becomes particularly evident in real-world applications such as predicting advertisement image preferences across diverse geographic markets. For instance, when a VLM (e.g., Qwen2-VL) is asked to choose between two product images tailored for different countries (e.g., Korea vs. France), it often defaults to a consistent output (e.g., always selects “A”), ignoring ground-truth regional variations. This collapse occurs because pervasive high-volume signals, such as product attributes and dense image patches, overwhelm the few but critical tokens that encode market-specific context (e.g., country names). To address CVE, we first collect a new multimodal dataset of real advertising creatives and their click-through performance across multiple countries. We then introduce GeoReward, a reward model designed to predict ad image preferences across diverse geographic markets. GeoReward integrates three purpose-built mechanisms: (1) Market-Aware Retrieval Augmentation, which retrieves and injects region-aligned preference signals during training to sharpen localization awareness. (2) Context-Guided Visual Modulation, a lightweight adapter that dynamically adjusts visual representations using textual country embeddings, enabling fine-grained regional adaptation. (3) Selective Sensitivity Loss, an objective that applies heightened penalties for context-specific mispredictions, sharpening the model's focus on critical variables. Furthermore, we demonstrate how GeoReward can guide the fine-tuning of RL for a VLM to generate background designs for text-to-image models (e.g., SDXL), producing market-aware advertising creatives. Experiments validate that our framework mitigates CVE and outperforms existing baselines. This work not only diagnoses a systematic bias in VLMs toward dominant perceptual features but also delivers a targeted solution for applications where sparse contextual variables govern decision-making. Code is available at https://github.com/liushuo-hue/GeoReward.git.
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Physics-informed Neural Operator Learning for Nonlinear Grad-Shafranov Equation
Siqi Ding, Zitong Zhang, Shi Guoyang, Xingyu Li, Xiang Gu, Yanan Xu, Huasheng Xie, Hanyue Zhao, YUEJIANG SHI, Tianyuan Liu

AI for fusion requires bridging a critical ``sim-to-real'' gap: simulation-trained models must generalize reliably under distribution shifts in safety-critical workflows. Focusing on the nonlinear Grad-Shafranov equation (GSE), we develop and analyze a physics-anchored operator-learning framework for fixed-boundary equilibrium prediction. The framework combines data anchors with PDE residual constraints and uses a physics-motivated Transformer-KAN Neural Operator (TKNO) to capture global elliptic coupling and nonlinear source response. Under multi-parameter distribution shifts, our analysis shows that data-only surrogates can develop severe OOD tails, while physics-only training may converge to incorrect solution branches; by combining sparse data anchors with PDE constraints, physics-anchored training reduces worst-tail errors on shape-driven and joint shifts. Mechanism diagnostics indicate that data anchors suppress branch-scale failures, whereas PDE residuals reduce OOD-induced high-frequency error amplification. Evaluated on EXL-50U discharge inputs against the device's operational equilibrium solver, the model achieves close agreement (RMSE $<$ 1.3\%) with millisecond-level inference. These results provide a practical route toward physically reliable AI surrogates for fusion workflows. Our code will be available at https://github.com/dsqzhou/physics-anchored-gse.
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Strat-Reasoner: Reinforcing Strategic Reasoning of LLMs in Multi-Agent Games
Yidong He, Yutao Lai, Pengxu Yang, Jiarui Gan, Jiexin Wang, Yi Cai, Mengchen Zhao

While Large Language Models (LLMs) excel in certain reasoning tasks, they struggle in multi-agent games where the final outcome depends on the joint strategies of all agents. In multi-agent games, the non-stationarity of other agents brings significant challenges on the evaluation of the reasoning process and the credit assignment over multiple reasoning steps. Existing single-agent reinforcement learning (RL) approaches and their multi-agent extensions fail to address these

challenges as they do not incorporate other agents in the reasoning process. In this work, we propose Strat-Reasoner, a novel RL-based framework that improves LLMs' strategic reasoning ability in multi-agent games. We introduce a novel recursive reasoning paradigm where an agent's reasoning also integrates other agents' reasoning processes. To provide effective reward signals for the intermediate reasoning sequences, we employ a centralized Chain-of-Thought (CoT) comparison module to evaluate the reasoning quality. Finally, we compute an accurate hybrid advantage and develop a group-relative RL approach to optimize the LLM policy. Experimental results show that Strat-Reasoner substantially improves strategic abilities of underlying LLMs, achieving 22.1\% average performance improvements across various multi-agent games. Code is publicly available at https://github.com/ydhe1012/Strat-Reasoner.

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NeurVLA: Unleashing Failure-Handling Capability of Vision-Language-Action Models via Neural-Symbolic Reasoning

Xuqi Liu, Minghe Gao, Juncheng Li, Siliang Tang

Vision-Language-Action models have recently shown promising progress in embodied robotic manipulation, yet their generalization to diverse open-ended embodied tasks is often hindered by execution failures. While prior work has explored failure handling, existing approaches still suffer from two fundamental limitations: coarse-grained failure correction and unreliable failure prevention. These limitations lead to brittle decision-making when VLA models are deployed in novel tasks and environments. To address them, we propose NeurVLA, a neural-symbolic framework that jointly addresses failure correction and prevention via neural-symbolic reasoning and further internalizes these failure-handling capabilities into VLA models. Experiments demonstrate that NeurVLA achieves strong performance and robust generalization across diverse tasks.

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iVGR: Internalizing Visually Grounded Reasoning for MLLMs with Reinforcement Learning

Chang-Bin Zhang, Yujie Zhong, Qiang Zhang, Kai Han

While visually grounded Chain-of-Thought (CoT) has emerged as a promising paradigm to enhance fine-grained perception in multimodal large language models (MLLMs), its efficacy during the inference phase remains underexplored. In this work, we empirically find that mandating explicit object boxes in visually grounded CoT during inference often degrades performance compared to standard textual CoT, which reasons without explicit visual grounding. We hypothesize that the visual localization capability can be internalized into the textual CoT and that the mandatory explicit grounding introduces unnecessary interference with the model's primary objective of answer prediction. To address this problem, we propose Internalizing Visually Grounded Reasoning (**iVGR**), a novel reinforcement learning framework that transfers localization capabilities into the textual reasoning process. We employ a dual-stream training strategy, where a textual stream is aligned with a high-quality visually grounded stream via a proposed consistency reward, enabling the model to localize accurately without explicit grounding during inference. Extensive experiments demonstrate that our method significantly outperforms existing baselines on fine-grained benchmarks, while maintaining the flexibility to support tool-assisted inference workflows. Project page: https://visual-ai.github.io/ivgr/

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Closing the Expression Gap in LLM Instructions via Socratic Questioning

Jianwen Sun, Yukang Feng, Yifan Chang, Chuanhao Li, Zizhen Li, Jiaxin Ai, Fanrui Zhang, Yu Dai, Kaipeng Zhang

A fundamental bottleneck in human-AI collaboration is the ``intention expression gap", the difficulty for humans to effectively convey complex, high-dimensional thoughts to AI.

This challenge often traps users in inefficient trial-and-error loops and is exacerbated by the diverse expertise levels of users.

We reframe this problem from passive instruction following to a Socratic collaboration paradigm, proposing an agent that actively probes for information to reso

lve its uncertainty about user intent.
We name the proposed agent Nous, trained to acquire proficiency in this inquiry policy.
The core mechanism of Nous is a training framework grounded in the first principles of information theory.
Within this framework, we define the information gain from dialogue as an intrinsic reward signal, which is fundamentally equivalent to the reduction of Shannon entropy over a structured task space.
This reward design enables us to avoid reliance on costly human preference annotations or external reward models.
To validate our framework, we develop an automated simulation pipeline to generate a large-scale, preference-based dataset for the challenging task of scientific diagram generation.
Comprehensive experiments, including ablations, subjective and objective evaluations, and tests across user expertise levels, demonstrate the effectiveness of our proposed framework. Nous achieves leading efficiency and output quality, while remaining robust to varying user expertise. Our research provides a systematic methodology and a new perspective for addressing ambiguous intentions in complex human-machine collaboration.
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DecoVer: A Decompose-and-Verify Neuro-Symbolic Framework for Embodied Task Planning with BC+
YiXiang Jiang, Binqian Xu, Xiangbo Shu
Despite their remarkable general capabilities, Large Language Models (LLMs) struggle with the precise grounding required for embodied task planning. To bridge this gap, neuro-symbolic approaches have emerged, leveraging action languages like BC+ for their formal expressiveness and reasoning flexibility. However, prior methods that naively couple LLMs with BC+ typically depend on one-shot program generation, which is brittle in dynamic environments and prone to sequential omission and causal inconsistency. To address these limitations, we propose DecoVer, a Decompose-and-Verify neuro-symbolic framework that systematically adapts BC+ to embodied task planning. Specifically, DecoVer employs a cascading decomposition strategy to partition complex knowledge into hierarchical subspaces and integrates a dual verification mechanism for syntactic and executable correctness. Extensive experiments demonstrate that DecoVer consistently outperforms LLM-based baselines across the majority of evaluation metrics, achieving a 12.9% success rate gain over the highly capable Gemini-3-Pro-Preview and a 60.9% improvement over GPT-5.1 on logically complex test cases.
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FAST-AR: Fast Autoregressive Video Diffusion and World Models with Temporal Cache Compression and Sparse Attention
Dvir Samuel, Issar Tzachor, Matan Levy, Michael Green, Gal Chechik, Rami Ben-Ari
Autoregressive video diffusion models enable streaming generation, opening the door to long-form synthesis, video world models, and interactive neural game engines. However, their core attention layers become a major bottleneck at inference time: as generation progresses, the KV cache grows, causing both increasing latency and escalating GPU memory, which in turn restricts usable temporal context and harms long-range consistency. In this work, we study redundancy in autoregressive video diffusion and identify three persistent sources: near-duplicate cached keys across frames, slowly evolving (largely semantic) queries/keys that make many attention computations redundant, and cross-attention over long prompts where only a small subset of tokens matters per frame.
Building on these observations, we propose a unified, training-free attention framework (FAST-AR) for FAST-AutoRegressive diffusion, consisting of three components: TempCache compresses the KV cache via temporal correspondence to bound cache growth; AnnCA accelerates cross-attention by selecting frame-relevant prompt tokens using fast approximate nearest neighbor (ANN) matching; and AnnSA sparsifies self-attention by restricting each query to semantically matched keys, also using a lightweight ANN. Together, these modules reduce attention, compute, and memory and are compatible with existing autoregressive diffusion backbones and wo

rld models. Experiments demonstrate up to x5--x10 end-to-end speedups while pres erving near-identical visual quality and, crucially, maintaining stable throughp ut and nearly constant peak GPU memory usage over long rollouts, where prior met hods progressively slow down and suffer from increasing memory usage.
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HEXST: Hexagonal Shifted-Window Transformer for Spatial Transcriptomics Gene Exp ression Prediction

Keunho Byeon, Jin Tae Kwak

Spatial transcriptomics offers spatially resolved gene expression profiling with in tissue sections, but its cost and limited throughput hinder large-scale deplo yment. To extend this capability to routine practice, recent computational metho ds aim to infer spatial gene expression directly from ubiquitous hematoxylin and eosin-stained histology slides. However, most existing models assume Cartesian or geometry-agnostic locality, despite the hexagonal sampling of widely used spo t-array platforms, and point-wise regression objectives often yield over-smoothe d gene expression profiles, obscuring gene-specific spatial heterogeneity. To ad dress these, we propose HEXST, a geometry-aligned Transformer for spatial gene e xpression prediction from histology. HEXST operates directly on hexagonal spot c oordinates to enable efficient local-to-global contextual modeling via a tailore d shifted-window attention mechanism and hexagonal rotary positional encoding. T o enhance gene-wise spatial contrast, HEXST complements point-wise regression wi th a contrast-sensitive differential objective and transcriptomic priors from a pretrained single-cell foundation model during training. Across seven spatial tr anscriptomics datasets, HEXST consistently outperforms state-of-the-art models, providing accurate and robust spatial gene expression predictions while preservi ng gene-wise contrast and spatial heterogeneity.
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Twins: Learn to Predict Unified Representations with Focal Loss

Kaixiong Gong, Xin Cai, Bin Lin, Hao Wang, Yunlong Lin, Mingzhe Zheng, Bohao Li, Jian-Wei Zhang, Miles Yang, Zhao Zhong, Liefeng Bo, Xiangyu Yue

Unified multimodal models seek a shared visual token space that supports both mu ltimodal understanding and image generation. Discrete methods unify the interfac e via a shared codebook, whereas continuous pipelines often rely on two disparat e representations—semantic features (e.g., ViT) for understanding and low-level latents (e.g., VAE) for synthesis—resulting in mismatched latent spaces. We prop ose Twins, a unified continuous token space formed by channel-wise concatenating ViT and VAE features on the same token grid, so the sequence length is unchange d and attention cost does not increase. However, jointly modeling Twins in a Dif fusion Transformer exposes a severe \textit{optimization imbalance}: the model f its the ViT component well but struggles to match the VAE latent distribution. W e trace this imbalance to three sources of heterogeneity: frequency bias, intrin sic dimensionality, and condition-aligned vs condition-independent uncertainty. To address it, we adapt a focal regression objective for flow matching that upwe ights large-error VAE dimensions, better balancing optimization across the ViT a nd VAE components. On ImageNet, this yields up to 10.57 gFID gain over naive M SE loss without classifier-free guidance. Twins also performs competitively on m ultimodal understanding benchmarks and improves reconstruction fidelity, narrowi ng the gap between understanding- and generation-oriented representations.
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NavOL: Navigation Policy with Online Imitation Learning

Xiaofei Wei, Chun Gu, Li Zhang

Learning robust navigation policies remains a core challenge in robotics. Offlin e imitation learning suffers from distribution shift and compounding errors at r ollout, while reinforcement learning requires reward engineering and learns inef ficiently. In this paper, we propose NavOL, an online imitation learning paradig m that interacts with a simulator and updates itself using expert demonstrations gathered online. Built upon a pretrained navigation diffusion policy that maps local observations to future waypoints, NavOL trains in a rollout–update loop: d uring rollout, the policy acts in the simulator and queries a global planner whi ch has privileged access to the global environment for the optimal path segment

as ground truth trajectory labels; during update, the policy is trained on the o
nline collected observation–trajectory pairs. This online imitation loop removes
 the need for reward design, improves learning efficiency, and mitigates distrib
ution shift by training on the policy’s own explored rollouts. Built on IsaacLab
 with fast, high-fidelity parallel rendering and domain randomization of camera
pose and start-goal pairs, our system scales across 50 scenes on 8 RTX 4090 GPUs
, collecting over 2,000 new trajectories per hour, each averaging more than 400
steps. We also introduce an indoor visual navigation benchmark with predefined s
tart and goal positions for zero-shot generalization. Extensive evaluations on s
imulation benchmarks, including the NavDP benchmark and our proposed benchmark,
as well as carefully designed real-world experiments, demonstrate the effectiven
ess of NavOL, showing consistent performance gains in online imitation learning.
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ProOPF: Benchmarking and Improving LLMs for Professional-Grade Power Systems Opt
imization Modeling

Chao Shen, Zihan Guo, Xu Wan, Zhenghao Yang, Yifan Zhang, Wenqi Huang, Jie Song,
 Zongyan Zhang, Mingyang Sun

Growing renewable penetration introduces substantial uncertainty into power syst
em operations, necessitating frequent adaptation of dispatch objectives and cons
traints and challenging expertise-intensive, near-real-time modeling workflows.
Large Language Models (LLMs) provide a promising avenue for automating this proc
ess by translating natural-language (NL) operational requirements into executabl
e optimization models via semantic reasoning and code synthesis. Yet existing LL
M datasets and benchmarks for optimization modeling primarily target coarse-grai
ned cross-domain generalization, offering limited, rigorous evaluation in power-
system settings, particularly for Optimal Power Flow (OPF). We therefore introdu
ce \textbf{ProOPF-D} and \textbf{ProOPF-B}, a dataset and benchmark for professi
onal-grade OPF modeling: ProOPF-D contains 12K instances pairing NL requests wit
h parameter adjustments and structural extensions to a canonical OPF, together w
ith executable implementations; ProOPF-B provides 121 expert-annotated test case
s with ground-truth code, enabling end-to-end evaluation under both concrete and
 abstract OPF modeling regimes. Our code, dataset, and benchmark are publicly av
ailable at \href{https://github.com/shenchao188/ProOPF-Benchamrk-Dataset}{this G
itHub repository}.
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A Random Matrix Theory of Masked Self-Supervised Learning

Arie Wortsman Zurich, Federica Gerace, Bruno Loureiro, Yue M. Lu

In the era of transformer models, masked self-supervised learning (SSL) has beco
me a foundational training paradigm. A defining feature of masked SSL is that tr
aining aggregates predictions across many masking patterns, giving rise to a joi
nt, matrix-valued predictor rather than a single vector-valued estimator. This o
bject encodes how coordinates condition on one another and poses new analytical
challenges. We develop a precise high-dimensional analysis of masked modeling ob
jectives in the proportional regime where the number of samples scales with the
ambient dimension. Our results provide explicit expressions for the generalizati
on error and characterize the spectral structure of the learned predictor, revea
ling how masked modeling extracts structure from data. For spiked covariance mod
els, we show that the joint predictor undergoes a Baik–Ben Arous–Péché (BBP)-typ
e phase transition, identifying when masked SSL begins to recover latent signals
. Finally, we identify structured regimes in which masked self-supervised learni
ng provably outperforms PCA, highlighting potential advantages of SSL objectives
 over classical unsupervised methods.
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Tuning the Implicit Regularizer of Masked Diffusion Language Models: Enhancing G
eneralization via Insights from k-Parity

Jianhao Huang, Baharan Mirzasoleiman

Masked Diffusion Language Models have recently emerged as a powerful generative
paradigm, yet their generalization properties remain understudied compared to th
eir auto-regressive counterparts. In this work, we investigate these properties
within the setting of the k-parity problem (computing the XOR sum of k relev

ant bits), where neural networks typically exhibit grokking—a prolonged plateau of chance-level performance followed by sudden generalization. We theoretically decompose the Masked Diffusion (MD) objective into a Signal regime which drives feature learning, and a Noise regime which serves as an implicit regularizer. By training nanoGPT using MD objective on the k-parity problem, we demonstrate that MD objective fundamentally alters the learning landscape, enabling rapid and simultaneous generalization without experiencing grokking.
Furthermore, we leverage our theoretical insights to optimize the distribution of the mask probability in the MD objective. Our method significantly improves perplexity for 50M-parameter models and achieves superior results across both pre-training from scratch and supervised fine-tuning. Specifically, we observe performance gains peaking at 8.8% and 5.8%, respectively, on 8B-parameter models, confirming the scalability and effectiveness of our framework in large-scale masked diffusion language model regimes.
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Transformers with RL or SFT Provably Learn Sparse Boolean Functions, But Differently
Bochen Lyu, Yiyang Jia, Xiaohao Cai, Zhanxing Zhu
Transformers can acquire Chain-of-Thought (CoT) capabilities to solve reasoning tasks via fine-tuning. Reinforcement learning (RL) and supervised fine-tuning (SFT) are two primary approaches to this end. In this work, we examine RL with verifiable process rewards and SFT for learning k-sparse Boolean functions with a one-layer transformer through intermediate reasoning steps akin to CoT. In particular, we consider Boolean functions that can be recursively decomposed into fixed 2-sparse Boolean functions. We first analyze the learning dynamics of RL fine-tuning with verifiable process rewards and SFT in a unified way, allowing us to identify sufficient conditions under which the transformer provably learns these functions. We then verify that these conditions hold for three basic examples, including k-PARITY, k-AND, and k-OR, thus demonstrating their learnability via both RL and SFT. Notably, we reveal that RL and SFT exhibit distinct learning behaviors: RL learns the whole CoT chain simultaneously, whereas SFT learns the CoT step-by-step. Overall, our findings provide insights on the mechanisms underlying RL and SFT and how they differ in triggering the CoT capabilities of transformers, and suggest that the comparison between RL and SFT may need to consider the reward design and the use of teacher forcing.
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VisualPuzzles: Decoupling Multimodal Reasoning Evaluation from Domain Knowledge
Yueqi Song, Tianyue Ou, Yibo Kong, Zecheng Li, Graham Neubig, Xiang Yue
Current multimodal benchmarks often conflate reasoning with domain knowledge, making it difficult to isolate and evaluate reasoning abilities in non-expert settings.
To address this, we introduce VisualPuzzles, a benchmark that targets visual reasoning while deliberately minimizing reliance on specialized knowledge.
VisualPuzzles consists of diverse puzzles spanning 5 categories: algorithmic, analogical, deductive, inductive, and spatial reasoning. One major source of our questions is manually translated logical reasoning questions from the Chinese Civil Service Examination. Experiments show that VisualPuzzles requires significantly less intensive domain-specific knowledge and more complex reasoning compared to benchmarks like MMMU, enabling us to better evaluate genuine multimodal reasoning.
Evaluations show that state-of-the-art multimodal large language models consistently lag behind human performance on VisualPuzzles, and that strong performance on knowledge-intensive benchmarks does not necessarily translate to success on reasoning-focused, knowledge-light tasks.
Additionally, reasoning enhancements such as scaling up inference compute (with "thinking" modes) yield inconsistent gains across models and task types, and we observe no clear correlation between model size and performance. We also found that models exhibit different reasoning and answering patterns on VisualPuzzles compared to benchmarks with heavier emphasis on knowledge. VisualPuzzles offers a clearer lens through which to evaluate reasoning capabilities beyond factual re

call and domain knowledge.
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ProtoKV: Streaming Video Understanding under Delayed Query with Summary-State Memory
Le Tu Ngoc Minh, Jinyeong Lim, Dongsu Han
Streaming video understanding (SVU) must answer queries that arrive asynchronously while visual tokens stream continuously under strict GPU-memory and query-time latency budgets. A key challenge is delayed query: decisive cues may appear briefly, yet many subsequent updates occur before the query arrives, increasing the risk that those cues are evicted or diluted under bounded memory. We propose ProtoKV, a constant-footprint SVU memory that represents far history as a fixed-capacity summary state rather than retaining token instances. ProtoKV keeps an exact near-window KV cache and aggregates older content into a semantic–spatial prototype bank with residual statistics. At query time, each prototype is exposed through a bounded pseudo-token interface that is drop-in compatible with standard attention. Under matched budgets and comparable query-time cost, ProtoKV improves accuracy by up to 12.5 points over token-retention baselines on SVU benchmarks in the long-delay regime, with gains that grow as query delay increases.
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Stabilizing the Q-Gradient Field for Policy Smoothness in Actor-Critic Methods
Jeong Woon Lee, Kyoleen Kwak, Daeho Kim, Hyoseok Hwang
Policies learned via continuous actor-critic methods often exhibit erratic, high-frequency oscillations, making them unsuitable for physical deployment.
Current approaches attempt to enforce smoothness by directly regularizing the policy's output.
We argue that this approach treats the symptom rather than the cause.
In this work, we theoretically establish that policy non-smoothness is fundamentally governed by the differential geometry of the critic.
By applying implicit differentiation to the actor-critic objective, we prove that the sensitivity of the optimal policy is bounded by the ratio of the Q-function's mixed-partial derivative (noise sensitivity) to its action-space curvature (signal distinctness).
To empirically validate this theoretical insight, we introduce PAVE (Policy-Aware Value-field Equalization), a critic-centric regularization framework that treats the critic as a scalar field and stabilizes its induced action-gradient field.
PAVE rectifies the learning signal by minimizing the Q-gradient volatility while preserving local curvature.
Experimental results demonstrate that PAVE achieves smoothness comparable to policy-side smoothness regularization methods, while maintaining competitive task performance, without modifying the actor.
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SmoothSpike: Spiking Transformer with Learnable Hadamard Transformation
Zijian Zhou, Wenjie Wei, Yu Liang, Jialin Li, Ammar Belatreche, Honglin Cao, Shuai Wang, Malu Zhang, Yang Yang, Haizhou Li
Spiking Neural Networks (SNNs) have attracted growing attention due to their sparse spike-based communication and inherent temporal dynamics. However, their discrete information representation fundamentally limits expressiveness, resulting in a notable performance gap relative to Artificial Neural Networks (ANNs) on language modeling tasks. In this paper, we reveal that this gap is fundamentally rooted in a spike saturation-induced information homogenization problem: within a bounded time window, distinct high-amplitude inputs converge to identical spike counts, compressing neural representations and impairing fine-grained semantic discrimination across layers. To address this, we propose SmoothSpike, which applies a randomized Hadamard transformation to smooth pre-activation inputs and theoretically proves that it bounds the maximum input to $\mathcal{O}(\sqrt{\frac{\log n}{n}})$ with high probability. To further improve adaptability across varying input distributions, we extend the fixed transformation within SmoothSpike to a learnable orthogonal matrix updated via Newton-Schulz iterations, which can be fused into model weights at inference with no additional overhead. Experime

nts on the GLUE benchmark show that SmoothSpike effectively reduces information homogenization, yielding an 8.2\% average improvement over the Spikingformer baseline without compromising the efficiency inherent to spike-driven computation. These results advance the prospects for energy-efficient and high-performance language modeling on edge devices. Code is available at https://github.com/CayleyZ/SmoothSpike.

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MAS-ProVe: Understanding the Process Verification of Multi-Agent Systems

Vishal Venkataramani, Haizhou Shi, Zixuan Ke, Austin Xu, Xiaoxiao He, Yingbo Zhou, Semih Yavuz, Hao Wang, Shafiq Joty

Multi-Agent Systems (MAS) built on Large Language Models (LLMs) often exhibit high variance in their reasoning trajectories.

Process verification, which evaluates intermediate steps in trajectories, has shown promise in general reasoning settings, and has been suggested as a potential tool for guiding coordination of MAS; however, its actual effectiveness in MAS remains unclear. To fill this gap, we present MAS-ProVe, a systematic empirical study of process verification for multi-agent systems (MAS). Our study spans *three verification paradigms* (LLM-as-a-Judge, reward models, and process reward models), evaluated across *two levels of verification granularity* (agent-level and iteration-level). We further examine *five representative verifiers* and *four context management strategies,* and conduct experiments over *six diverse MAS frameworks* on multiple reasoning benchmarks. We find that process-level verification does not consistently improve performance and frequently exhibits high variance, highlighting the difficulty of reliably evaluating partial multi-agent trajectories. Among the methods studied, LLM-as-a-Judge generally outperforms reward-based approaches, with trained judges surpassing general-purpose LLMs. We further observe a small performance gap between LLMs acting as judges and as single agents, and identify a context-length-performance trade-off in verification. Overall, our results suggest that effective and robust process verification for MAS remains an open challenge, requiring further advances beyond current paradigms.

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PostTrainBench: Can LLM Agents Automate LLM Post-Training?

Ben Rank, Hardik Bhatnagar, Ameya Prabhu, Shira Eisenberg, Karina Nguyen, Matthias Bethge, Maksym Andriushchenko

AI agents have become surprisingly proficient at software engineering over the past year, largely due to improvements in reasoning capabilities. This raises a deeper question: can these systems extend their capabilities to automate AI research itself? We introduce PostTrainBench to benchmark how well LLM agents can perform post-training *autonomously* under bounded compute constraints (10 hours on one H100 GPU). We ask frontier agents (e.g., Claude Code with Opus 4.6) to optimize the performance of a base LLM on a particular benchmark (e.g., Qwen3-4B on AIME). Importantly, we do not provide any predefined strategies to the agents and instead give them full autonomy to find necessary information on the web, run experiments, and curate data. We find that frontier agents make substantial progress but generally lag behind official instruction-tuned LLMs: 27.9% for the best agent vs. 51.1% for official instruction-tuned models. However, agents can exceed instruction-tuned models in targeted scenarios: GPT-5.1 Codex Max achieves 89% on BFCL with Gemma-3-4B vs. 67% for the official model. We also observe failure modes: agents engage in reward hacking by training on the test set, downloading existing instruction-tuned checkpoints, or using found API keys to generate synthetic data without authorization. Overall, we hope PostTrainBench will be useful for tracking progress in AI R&D automation and for studying the risks that come with it.

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Token-Free Hierarchical Indexing for RAG beyond LLM-based Summarization

Yifan Wei, Dan Yuan, Xiaoyan Yu, Angsheng Li

Retrieval-Augmented Generation (RAG) increasingly relies on hierarchical indexing, yet existing frameworks are bottlenecked by the high cost and information loss of recursive, LLM-based summarization. We propose SeRAG, a novel token-free hi

erarchical indexing framework that replaces textual summaries with an informatio n-theoretic knowledge taxonomy. SeRAG first transforms a corpus into a multi-per spective graph capturing semantic, logical, and sequential dependencies, then mi nimizes structural entropy to induce a topologically-faithful encoding tree. To bridge the gap between abstract themes and granular facts, we introduce localize d structural weight-based vector aggregation for token-free community consolidat ion. Extensive experiments demonstrate that SeRAG significantly reduces indexing overhead while outperforming state-of-the-art methods in complex multi-hop reas oning tasks.

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When Is Rank-1 Enough? Geometry-Guided Initialization for Parameter-Efficient Fi ne-Tuning

Haoran Zhao, Caren Han, Eduard Hovy

Parameter-efficient fine-tuning (PEFT) is a standard way to adapt multimodal lar ge language models, yet extremely low-rank settings---especially rank-1 LoRA---a re often unstable. We show that this instability is not solely due to limited ca pacity: in the rank-1 regime, optimization is highly sensitive to the update dir ection. Concretely, pretrained vision and text features form mismatched anisotro pic regions, yielding a dominant "gap" direction that acts like a translation co mponent and disproportionately steers early gradients under rank-1 constraints. Analyzing pretrained representations, we identify a modality-gap axis that domin ates early gradient flow, while a random rank-1 initialization is unlikely to al ign with it, leading to weak gradients and training collapse. We propose Gap-Ini t, a geometry-aware initialization that aligns the rank-1 LoRA direction with an estimated modality-gap vector from a small calibration set, while keeping the i nitial LoRA update zero. Across multiple vision-language tasks and backbones, Ga p-Init consistently stabilizes rank-1 training and can match or outperform stron g rank-8 baselines. Our results suggest that at the extreme low-rank limit, init ial alignment can matter as much as rank itself.

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Both Semantics and Reconstruction Matter: Making Representation Encoders Ready f or Text-to-Image Generation and Editing

Shilong Zhang, He Zhang, Zhifei Zhang, Chongjian GE, Shuchen Xue, Shaoteng Liu, Mengwei Ren, Soo Ye Kim, Yuqian Zhou, Qing Liu, Daniil Pakhomov, Kai Zhang, Zhe Lin, Ping Luo

Modern Latent Diffusion Models (LDMs) typically operate in low-level Variational Autoencoder (VAE) latent spaces that are primarily optimized for pixel-level re construction. To unify vision generation and understanding, a burgeoning trend i s to adopt high-dimensional features from representation encoders as generative latents. However, we empirically identify two fundamental obstacles in this para digm: (1) the discriminative feature space lacks compact regularization, making diffusion models prone to off-manifold latents that lead to inaccurate object st ructures; and (2) the encoder’s inherently weak pixel-level reconstruction hinde rs the generator from learning accurate fine-grained geometry and texture. In th is paper, we propose a systematic framework to adapt understanding-oriented enco der features for generative tasks.

We introduce a semantic–pixel reconstruction objective to regularize the latent space, enabling the compression of both semantic information and fine-grained de tails into a highly compact representation (96 channels with $16\times$ spatial downsampling).

This design allows the latent space to remain semantically rich while achieving state-of-the-art image reconstruction, and keeps it compact enough for accurate generation.

Leveraging this representation, we design a unified text-to-image (T2I) and imag e editing model.

Across diverse generation spaces, our approach achieves state-of-the-art reconst ruction, faster convergence, and substantial gains in both T2I and editing tasks , demonstrating that representation encoders can be effectively adapted into rob ust generative components.

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MIMOMamba: From Scalar Duality to Matrix-Valued Attention
Yanbo Li, Richard Cornelius Suwandi, Feng Yin, SUN Yiyong, Wei Huang, Wenqiang P
u
The state space duality (SSD) framework, central to modern state-space models (S
SMs) such as Mamba, has established an efficient attention-like mechanism by lev
eraging the commutative property of linear recurrences. However, existing formul
ations are limited to single-input single-output (SISO) systems that enforce com
mutativity with a restrictive scalar-identity constraint, which prevents cross-d
imensional interactions within the state dynamics. In this work, we generalize S
SD to the multi-input multi-output (MIMO) setting by introducing a matrix polyno
mial parameterization.
This approach not only provides a principled way to ensure commutativity for gen
eralized duality but also induces a shared algebraic structure across state tran
sitions, thereby significantly reducing parameter redundancy.
Building on this foundation, we present \textbf{MIMOMamba}, a multi-head SSM arc
hitecture that captures rich cross-dimensional dynamics while retaining linear-t
ime training.
Empirical evaluations on a sequence modeling benchmark demonstrate that MIMOMamb
a achieves performance comparable to or surpassing that of standard Transformers
 while utilizing substantially fewer parameters.
****
From Associations to Activations: Comparing Behavioral and Hidden-State Semantic
 Geometry in LLMs
Louis Schiekiera, Max Zimmer, Christophe Roux, Sebastian Pokutta, Fritz Günther
We investigate the extent to which an LLM’s hidden-state geometry can be recover
ed from its behavior in psycholinguistic experiments. Across eight instruction-t
uned transformer models, we run two experimental paradigms---similarity-based fo
rced choice and free association---over a shared 5,000-word vocabulary, collecti
ng 17.5M+ trials to build behavior-based similarity matrices. Using representati
onal similarity analysis, we compare behavioral geometries to layerwise hidden-s
tate similarity and benchmark against FastText, BERT, and cross-model consensus.
 We find that forced-choice behavior aligns substantially more with hidden-state
 geometry than free association. In a held-out-words regression, behavioral simi
larity (especially forced choice) predicts unseen hidden-state similarities beyo
nd lexical baselines and cross-model consensus, indicating that behavior-only me
asurements retain recoverable information about internal semantic geometry. Fina
lly, we discuss implications for the ability of behavioral tasks to uncover hidd
en cognitive states.
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Zeroth-Order Non-Log-Concave Sampling with Variance Reduction and Applications t
o Inverse Problems
M. Berk Sahin, Behzad Sharif, Abolfazl Hashemi
Sampling from high-dimensional, non-log-concave distributions with unnormalized
densities remains a fundamental challenge in machine learning, particularly in b
lack-box settings where gradient information is inaccessible or computationally
prohibitive. While Langevin dynamics provides a principled framework for samplin
g when gradients are accessible, its extension to the black-box settings suffers
 from high variance and lacks non-asymptotic convergence guarantees for non-log-
concave sampling. To address these limitations, we propose a variance-reduced ze
roth-order Langevin sampling method. Our method employs a gradient estimator tha
t substantially reduces the variance of the classical batched zeroth-order estim
ator and eliminates the unfavorable dimensional dependence of the batch size req
uired for accurate estimation, enabling practical and stable sampling. We establ
ish the first non-asymptotic convergence guarantees for zeroth-order non-log-con
cave sampling in terms of ε-relative Fisher information, and, under
a Poincar\'e inequality assumption, squared total variation distance. We further
 propose ZO-APMC, a posterior sampling algorithm for black-box inverse problems
with pre-trained score-based generative priors, establishing the first non-asymp
totic convergence guarantees for such methods. We validate our theory through sy
nthetic experiments and demonstrate strong empirical performance on practical li

near and nonlinear inverse problems.
****
LatentChem: From Textual CoT to Latent Thinking in Chemical Reasoning
Xinwu Ye, Yicheng Mao, Yuxuan Liao, Jia Zhang, Yimeng Liu, Li Hao, Fang Wu, Zhiwei Li, Zehong Wang, Zhiyuan Liu, Zhenfei Yin, Li Yuan, Philip Torr, Huan Sun, xiangxiang Zeng, Mengdi Wang, Le Cong, Shenghua Gao, Xiangru Tang
Current chemical large language models (LLMs) predominantly rely on explicit Chain-of-Thought (CoT) to solve complex reasoning problems.
However, forcing nonverbal tacit chemical logic into discrete natural language imposes a fundamental ``modality mismatch,'' creating an artificial bottleneck for reasoning.
We introduce LatentChem, a reasoning interface that decouples chemical logic from linguistic generation, enabling the model to process information via continuous thought vectors and dynamic perception.
Our investigation reveals a pivotal emergent behavior: spontaneous internalization, defined here as self-selected under outcome-only optimization.
When optimized for task success, the model abandons verbose textual derivations in favor of implicit latent computation, suggesting that it identifies the continuous manifold as a more native substrate for chemical logic.
This paradigm shift also proves to be a superior computational strategy: LatentChem achieves a 59.88\% non-tie win rate against the strong CoT baseline on the rigorous ChemCoTBench, while delivering a broad 10.84$\times$ average reduction in reasoning step overhead (5.96$\times$ wall-clock speedup) across all evaluated benchmarks.
Our results provide empirical evidence that chemical reasoning is more naturally and effectively realized as continuous latent dynamics rather than discretized linguistic trajectories.
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Adaptive Quasimetric Mapping : Principled Topological Abstraction for Robust Offline Goal-Conditioned Navigation
Anthony Kobanda, Waris Radji, Odalric-Ambrym Maillard, Rémy Portelas
Goal-Conditioned Reinforcement Learning aims to design agents that can reach specified goals, notably from previously collected trajectories in the offline setting. In this context, graph-based approaches have been proposed to mitigate compounding value-estimation errors in long-horizon navigation tasks. However, existing approaches typically rely on dense keypoint coverage of the dataset support, resulting in computationally expensive planning. Moreover, they lack explicit mechanisms to adapt to topological changes (e.g., new obstacles), hindering deployment in live applications such as video game environments. To address these two shortcomings, we introduce Adaptive Quasimetric Mapping (AQM), an offline framework leveraging a “time-to-reach” quasimetric learned from the available data. Crucially, it builds a sparse cover of the dataset support, as a greedy approximation to a dominating set problem. At test-time, the resulting graph is carefully pruned by comparing the observed edge traversal time against a time-to-reach budget derived from the quasimetric, thus enabling zero-shot replanning. Empirically, we evaluate AQM on navigation tasks ranging from a classical to a video-game-like benchmark evaluating adaptation across tasks. We show that AQM achieves competitive performance while requiring up to 100× fewer keypoints than prior approaches, hence demonstrating the relevance of topological abstraction for goal-conditioned navigation.
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AnyCanvas: Potential Field Guidance for Training-Free Spatial Control in Text-to-Image Diffusion
Tianyi Xie, Zhiyuan Yu, kaihong huang, Beilun Wang, Zhaoyang Wang, Dian Shen
Diffusion-based text-to-image (T2I) models have demonstrated remarkable advancements in generating high-quality images. However, while real-world applications like product packaging and logo design necessitate synthesis within irregular geometries, existing methods struggle to handle such constraints. Therefore, generating complete pictures that conform to arbitrary-shaped canvas constraints while maintaining semantic integrity remains a significant challenge. To address this

, we introduce AnyCanvas, a training-free framework that leverages a Mask-to-Pot
ential Field paradigm to convert binary masks into a differentiable potential fi
eld, which guides content to naturally converge within target regions. Extensive
 experiments demonstrate that AnyCanvas achieves 4.23\% higher spatial adherence
 to user-specified constraints while maintaining 99.45\% of the semantic fidelit
y measured by CLIP score, leading to a superior harmonic mean of spatial and sem
antic metrics. AnyCanvas also exhibits robust generalizability across different
model backbones and versatile spatial control objectives.
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StarEmbed: Benchmarking Time Series Foundation Models on Astronomical Observatio
ns of Variable Stars

Weijian Li, Hong-Yu Chen, Nabeel Rehemtulla, Ved G. Shah, Dongho Kim, Dennis Wu,
 Qinjie Lin, Adam A. Miller, Han Liu

Current time series foundation model (TSFM) training corpora largely omit data w
ith certain complexities like irregular temporal sampling. Astronomical time ser
ies of stellar fluxes (``light curves'') are available in immense quantities and
 exhibit irregular sampling, multiple variates, and heteroskedasticity. We intro
duce $\texttt{StarEmbed}$, the first public benchmark for light curves comprised
 of real observations of $\sim$40,000 stars across seven classes and evaluations
 in clustering, classification, and out-of-distribution (OOD) source detection.
We benchmark TSFMs with differing architecture and training strategies as well a
s domain-specific transformers. Our results demonstrate that the $\texttt{Chrono
s}$ family, despite being pre-trained on regularly sampled non-astronomical data
, yields state-of-the-art (SOTA) performance in light curve clustering and OOD d
etection. While no TSFM strictly surpasses the classification performance of the
 long-established domain baseline, they do demonstrate excellent generalization
abilities. $\texttt{StarEmbed}$ marks a step toward universal light curve embedd
ings and improved TSFM performance on challenging data.
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Video-MTR: Reinforced Multi-Turn Reasoning for Long Video Understanding

Yuan Xie, Tianshui Chen, Zheng Ge, Lionel Ni

Long-form video understanding remains a formidable challenge due to the complexi
ty of modeling long-range temporal dependencies and multi-event narratives. Exis
ting methods often rely on static reasoning or external Visual-Language Models (
VLMs), resulting in high computational complexity and sub-optimal performance. I
n this paper, we propose Video-MTR, a reinforced multi-turn reasoning framework
that operates solely through data-efficient, pure RL post-training. Video-MTR re
formulates video understanding as a dynamic decision-making process, where the a
gent iteratively selects key segments conditioned on the evolving context of pre
viously processed frames and the query. To ensure effective intermediate reasoni
ng and training stability, we introduce a novel gated bi-level reward system, wh
ich synergizes trajectory-level rewards (answer correctness) with turn-level rew
ards (frame-query relevance). This mechanism eliminates the need for data-intens
ive supervised fine-tuning, thereby substantially reducing reliance on large-sca
le datasets. Remarkably, Video-MTR achieves competitive or superior performance
using only $\sim$8K training samples, compared to existing approaches that requi
re 257K to 4.4M examples. Extensive experiments on benchmarks including VideoMME
, MLVU, LongVideoBench, LVBench, and EgoSchema demonstrate that Video-MTR surpas
ses state-of-the-art methods in both accuracy and efficiency. Code is available
at https://github.com/Xyuan13/Video-MTR.
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Distributionally Robust Markov Games with Average Reward

Zachary Andrew Roch, Yue Wang

We propose and study distributionally robust Markov games (DR■MGs) with the aver
age■reward criterion as a crucial framework for multi-agent decision-making unde
r model mismatches and over extended horizons. Under a standard irreducible assu
mption, we first derive a correspondence between the optimal policies and the so
lutions of the robust Bellman equation, based on which we further show the exist
ence of a stationary Nash Equilibrium (NE) of the game. We further study DR-MGs
under a more general weakly communicating setting. We construct a set-valued map

based on the constant-gain optimal robust Bellman operator and show that its value is a subset of the best-response policies. We further prove that this map admits a fixed point, which implies the existence of NE. We then design two algorithms, Robust Nash■Iteration and robust TD Descent, with provably convergent guarantees. Finally, we show that the NE under average■reward can be approximated by the ones for the discounted DR-MGs as the discount factor approaches one. Our studies provide a comprehensive theoretical and algorithmic foundation for decision-making in complex, uncertain, and long-running multi-player environments.
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The (Marginal) Value of a Search Ad: An Online Causal Framework for Repeated Second-price Auctions

Yuxiao Wen, Zihao Hu, Yanjun Han, Yuan Yao, Zhengyuan Zhou

Existing auto-bidding algorithms in digital advertising often treat the value of an ad opportunity as the revenue obtained when an ad is shown and/or clicked, and bid accordingly. This can lead to wasteful spending because the true value is the marginal gain from paid exposure: even without winning a sponsored slot, an advertiser may still earn revenue via an organic search result (e.g., on Google or Amazon). Motivated by recent work, we model ad value as a treatment effect—the outcome difference between winning and losing the auction—and study online learning for bidding in second-price (Vickrey) auctions under this causal perspective. We develop algorithms that attain rate-optimal regret under several feedback models. A key ingredient exploits the information revealed by the second-price payment rule, which strictly improves regret relative to analogous learning problems in first-price auctions.
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ECA: Efficient Continual Alignment for Open-Ended Image-to-Text Generation

Jiangtao Kong, Peijun Zhao, Chun-Fu Chen, Youngwook Do, Shaohan Hu, Tianyi Zhou, Huajie Shao

Incremental Learning (IL) for Open-ended Image-to-Text Generation (OpenITG) enables models to continuously generate accurate, contextually relevant text for new images while preserving previously acquired knowledge. Unlike prior studies, this paper addresses a more practical scenario in which the predominant category of visual data shifts over time as environments evolve. In this context, we introduce a new notion of continual alignment, which incrementally adapts the alignment module within pre-trained VLMs to preserve high-quality cross-modal representations. Based on this idea, we propose **E**fficient **C**ontinual **A**lignment (ECA), a novel exemplar-free IL approach for OpenITG. The key challenge is enabling the model to acquire new, task-specific features while minimizing interference with the established alignment without accessing raw data from previous tasks. To address this, ECA employs three core mechanisms: a **M**ixture **o**f **Q**uery (MoQ) module that adapts task-specific query tokens, a **F**ish**e**r **D**ynamic **Ex**pansion (FeDEx) that dynamically expands model structure based on a Fisher Information Matrix (FIM)-based metric, and an embedding dictionary with **D**ictionary **R**eplay (DR) to retain past knowledge. To evaluate ECA's performance, we construct four new IL OpenITG benchmarks that better reflect real-world scenarios. Experimental results demonstrate that ECA significantly mitigates catastrophic forgetting and improves IL performance compared to baseline methods. Code and benchmarks are available at https://github.com/Snowball0823/ECA.
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Keep It in Mind: User Centric Continual Spatial Intelligence Reasoning in Egocentric Video Streams

Yun wang, Junbin Xiao, Han Lyu, Yifan Wang, Jing Zuo, Zhanjie Zhang, Hong Huang, Dapeng Wu, Angela Yao

We introduce UCS-Bench, a dataset spanning 170+ hours of egocentric visual observations with 8.1K+ timestamped questions for diagnosing User-Centric Continual Spatial intelligence in egocentric video streams. UCS-Bench targets a new problem that emphasizes dynamic spatial reasoning, long-term memory, and their alignment with users' real-time locations. We propose DirectMe, a framework that incrementally constructs and maintains a structured spatial memory from streaming egocentric observations. DirectMe enables robust tracking and recall of object locat

ions, all relative to the user's movement over time. By tightly coupling visual perception with memory updates and spatial reasoning, our approach supports long-horizon queries that require recalling interactions, resolving viewpoint-induced ambiguities, and adapting to dynamic scenes.
Our experiments show that DirectMe significantly improves the spatial reasoning of leading multimodal LLMs; it also surpasses many spatially aware and long-form streaming video models. We hope our benchmark and solution will advance spatial intelligence research for egocentric AI assistants. Data and code are available at https://github.com/cocowy1/UCS-Bench.
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Smoothness Errors in Dynamics Models and How to Avoid Them
Edward Berman, Luisa Li, Jung Yeon Park, Robin Walters
Modern neural networks have shown promise for solving partial differential equations over surfaces, often by discretizing the surface as a mesh and learning with a mesh-aware graph neural network. However, graph neural networks suffer from oversmoothing, where a node's features become increasingly similar to those of its neighbors. Unitary graph convolutions, which are mathematically constrained to preserve smoothness, have been proposed to address this issue. Despite this, in many physical systems, such as diffusion processes, smoothness naturally increases and unitarity may be overconstraining. In this paper, we systematically study the smoothing effects of different GNNs for dynamics modeling and prove that unitary convolutions hurt performance for such tasks. We propose relaxed unitary convolutions that balance smoothness preservation with the natural smoothing required for physical systems. We also generalize unitary and relaxed unitary convolutions from graphs to meshes. In experiments on PDEs such as the heat and wave equations over complex meshes and on weather forecasting, we find that our method outperforms several strong baselines, including mesh-aware transformers and equivariant neural networks. Our code is available at github.com/EdwardBerman/rayleigh_analysis
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FLIP2: Expanding Protein Fitness Landscape Benchmarks for Real-World Machine Learning Applications
Kieran Didi, Sarah Alamdari, Alex Xijie Lu, Bruce James Wittmann, Kadina E Johnston, Ava P Amini, Ali Madani, Maya Czeneszew, Christian Dallago, Kevin K Yang
Machine learning methods that predict protein fitness from sequence remain sensitive to changes in data distributions, limiting generalization across common conditions encountered in protein engineering. Practically, protein engineers are thus left wondering about the effective utility of ML tools.
The FLIP benchmark established protocols for testing generalization under some domain shifts, but it was limited to measurements of stability, binding, and viral capsid viability.
We introduce FLIP2, a protein fitness benchmark spanning seven new datasets, including enzymes, protein-protein interactions, and light-sensitive proteins, as well as splits that measure generalization relevant to real-world protein engineering campaigns.
Evaluating a suite of benchmark models across these datasets and suites reveals that
simpler models often matched or outperformed fine-tuned protein language models on FLIP2, challenging the utility of existing transfer learning techniques. Provenance for all datasets has been recorded and we redistribute all data CC-BY 4.0 to facilitate continued progress.
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OBJVanish: Prompt-Driven Generation of Physically Realizable 3D LiDAR-Invisible Objects
Bing Li, Wuqi Wang, Yanan Zhang, Jingzheng Li, Haigen Min, Wei Feng, Xingyu Zhao, Jie Zhang, Qing Guo
LiDAR-based 3D object detectors are fundamental to autonomous driving, where missed detections pose severe safety risks. While adversarial attacks are crucial for evaluating the robustness of these detectors, existing point-level perturbation methods rarely cause complete object disappearance and prove difficult to imp

lement in physical environments. We introduce OBJVanish, a prompt-driven text-to-3D adversarial generation framework that enables physically realizable attacks by generating 3D object models that are effectively invisible to LiDAR-based 3D object detectors. We first conduct a systematic empirical study of detection vulnerability in LiDAR-based 3D object detectors, revealing multi-object compositions as the dominant factor. Based on this analysis, the proposed framework iteratively refines text prompts—optimizing verbs, objects, and poses—to generate LiDAR-invisible pedestrian instances as representative vulnerable road users under physical constraints. To ensure realizability, the framework operates over a curated pool of representative real-world 3D object models and restricts generation to their valid combinations. Extensive experiments show that OBJVanish consistently evades six state-of-the-art (SOTA) LiDAR-based 3D object detectors in both simulation and real-world physical settings, exposing critical vulnerabilities in safety-critical detection systems.

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Low Kruskal-Rank Adaptation

Yixing Xu, Guanchen Li, Chao Li, Xuanwu Yin, Dong Li, Spandan Tiwari, Ashish Sirasao, Emad Barsoum

Low-rank adaptation (LoRA) is one of the most widely used parameter-efficient fine-tuning (PEFT) methods for adapting pre-trained large language models (LLMs) to downstream tasks.

Although LoRA significantly reduces the number of trainable parameters and lowers fine-tuning costs, its performance is often limited by the inherent low-rank assumption.

In this paper, we revisit the notion of rank for LoRA update matrices and show that the standard matrix rank fails to capture duplicated directions and redundancy in the update subspace.

Motivated by this analysis, we argue that the Kruskal rank offers a more informative criterion for characterizing update diversity.

We therefore propose Low Kruskal Rank Adaptation (LoKRA), a new PEFT algorithm with provable theoretical guarantees that mitigates the limitations of LoRA.

We further introduce LoKRA$^+$, an enhanced variant that provides a tighter theoretical lower bound on the Kruskal rank and yields stronger empirical performance.

Experiments on multiple LLMs show that our approach consistently outperforms LoRA and other baselines, establishing state-of-the-art performance across a range of benchmarks. The code can be viewed at https://github.com/AMD-AGI/LoKRA.

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A Systematic Study of Behavioral Cloning for Scientific Data Annotation

Ishaan Singh Chandok, Core Francisco Park

Scientific data annotation, such as tracking animals in video or proofreading neural reconstructions, remains bottlenecked by the “last mile” problem: even with strong automation, verification and correction consume substantial human effort. Standard approaches train models to directly predict annotations, discarding the rich supervision in how experts navigate, click, verify, and correct. We introduce a framework for studying behavioral cloning on scientific annotation: 9 synthetic tasks paired with synthetic annotations that simulate realistic human strategies including exploration, mistake correction, and strategic decision-making. Our experiments reveal several findings. First, skills emerge hierarchically: models learn GUI mechanics before task-critical decisions, and commit fewer mistakes than the training data while retaining the ability to correct errors when they occur. Second, scaling models on multi-task behavioral cloning shows that larger models are more data efficient within our scale range. Third, multi-task pretraining enables efficient fine-tuning to new tasks, while training from scratch fails entirely. Fourth, linear probes reveal that models internally represent latent variables of the annotation process such as task phase and data position; interestingly, we find a shared mistake representation that generalizes across different annotation tasks. Overall, our framework establishes systematic benchmarks and identifies key bottlenecks, providing a foundation for scaling behavioral cloning to real-world scientific data annotation.

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Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement
Jiaqing Chen, Zidu Yin, Yichao Cai, Yuhang Liu, Zhen Zhang, Dong Gong, Javen Qinfeng Shi
Graph neural networks (GNNs) excel at aggregating neighbor information for classification, yet their performance is hindered by graph structural entanglement, where spurious correlations from semantically irrelevant neighbors contaminate node embeddings. This challenge is most acute for nodes near class boundaries in the embedding space, where amplified structural noise blurs decision boundaries and destabilizes predictions. Existing robust GNN methods largely treat all nodes uniformly, ignoring boundary vulnerabilities. In this paper, to improve classification performance, we tackle graph structural disentanglement by identifying boundary-region entanglement as the primary bottleneck and propose Boundary Embedding Shaping (BES), an adaptive contrastive learning GNN plug-in module that selectively suppresses spurious structural noise at decision boundaries with minimal model parameter perturbation. Extensive experiments demonstrate that BES consistently improves boundary discrimination and outperforms existing leading methods. Notably, BES boosts GCN performance by an average of 3.3% in node classification (up to 5.0% on WikiCS) and achieves superior accuracy in link prediction.
****
Learning When to Attend: Conditional Memory Access for Long-Context LLMs
Sakshi Choudhary, Aditya Chattopadhyay, Luca Zancato, Elvis Nunez, Matthew Trager, Wei Xia, Stefano Soatto
Language models struggle to generalize beyond
pretraining context lengths, limiting long-horizon
reasoning and retrieval. Continued pretraining
on long-context data can help but is expensive
due to the quadratic scaling of Attention. We
observe that most tokens do not require (Global)
Attention over the entire sequence and can rely
on local context. Based on this, we propose L2A
(Learning To Attend), a layer that enables conditional
(token-wise) long-range memory access
by deciding when to invoke global attention. We
evaluate L2A on Qwen 2.5 and Qwen 3 models,
extending their effective context length from 32K
to 128K tokens. L2A matches the performance of
standard long-context training to within 3% while
skipping Global Attention for $\sim$80% of tokens,
outperforming prior baselines. We also design
custom Triton kernels to efficiently implement
this token-wise conditional Attention on GPUs,
achieving up to $\sim$2$\times$ improvements in training
throughput and time-to-first-token over FlashAttention.
Moreover, L2A enables post-training
pruning of highly sparse Global Attention layers,
reducing KV cache memory by up to 50%
with negligible performance loss.
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Machine Learning Hamiltonians are Accurate Energy-Force Predictors
Seongsu Kim, Chanhui Lee, Yoonho Kim, Seongjun Yun, Honghui Kim, Nayoung Kim, Changyoung Park, Sehui Han, Sungbin Lim, Sungsoo Ahn
Recently, machine learning Hamiltonian (MLH) models have gained traction as fast approximations of electronic structures such as orbitals and electron densities, while also enabling direct evaluation of energies and forces from their predictions. However, despite their physical grounding, existing Hamiltonian models are evaluated mainly by reconstruction metrics, leaving it unclear how well they perform as energy–force predictors. We address this gap with a benchmark that computes energies and forces directly from predicted Hamiltonians. Within this fram

ework, we propose QHFlow2, a state-of-the-art Hamiltonian model with an SO(2)-equivariant backbone and a two-stage edge update. QHFlow2 achieves 40% lower Hamiltonian error than the previous best model with fewer parameters. Under direct evaluation on MD17/rMD17, it is the first Hamiltonian model to reach NequIP-level force accuracy while achieving up to $20\times$ lower energy MAE. On QH9, QHFlow2 reduces energy error by up to $20\times$ compared to MACE. Finally, we demonstrate that QHFlow2 exhibits consistent scaling behavior with respect to model capacity and data, and that improvements in Hamiltonian accuracy effectively translate into more accurate energy and force computations.
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Domain-Shift-Aware Conformal Prediction for Large Language Models
Zhexiao Lin, Yuanyuan Li, Neeraj Sarna, Yuanyuan Gao, Michael Berger
Large language models have achieved impressive performance across diverse tasks. However, their tendency to produce overconfident and factually incorrect outputs, known as hallucinations, poses risks in real world applications. Conformal prediction provides finite-sample, distribution-free coverage guarantees, but standard conformal prediction breaks down under domain shift, often leading to under-coverage and unreliable prediction sets. We propose a new framework called Domain-Shift-Aware Conformal Prediction (DS-CP). Our framework adapts conformal prediction to large language models under domain shift, by systematically reweighting calibration samples based on their proximity to the test prompt, thereby preserving validity while enhancing adaptivity. Our theoretical analysis and experiments on the MMLU benchmark demonstrate that the proposed method delivers more reliable coverage than standard conformal prediction, especially under substantial distribution shifts, while maintaining efficiency. This provides a practical step toward trustworthy uncertainty quantification for large language models in real-world deployment.
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RelaxFlow: Text-Driven Amodal 3D Generation
Jiayin Zhu, Guoji Fu, Xiaolu Liu, Qiyuan He, Yicong Li, Angela Yao
Image-to-3D generation faces inherent semantic ambiguity under occlusion, where partial observation alone is often insufficient to determine object category. In this work, we formalize *text-driven amodal 3D generation*, where text prompts steer the completion of unseen regions while strictly preserving input observation. Crucially, we identify that these objectives demand distinct control granularities: rigid control for the observation versus relaxed structural control for the prompt. To this end, we propose **RelaxFlow**, a training-free dual-branch framework that decouples control granularity via a Multi-Prior Consensus Module and a Relaxation Mechanism. Theoretically, we prove that our relaxation is equivalent to applying a low-pass filter on the generative vector field, which suppresses high-frequency instance details to isolate geometric structure that accommodates the observation. To facilitate evaluation, we introduce two diagnostic benchmarks, **ExtremeOcc-3D** and **AmbiSem-3D**. Extensive experiments demonstrate that RelaxFlow successfully steers the generation of unseen regions to match the prompt intent without compromising visual fidelity. Code and datasets will be released.
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Whom to Query for What: Adaptive Group Elicitation via Multi-Turn LLM Interactions
Ruomeng Ding, Tianwei Gao, Thomas P Zollo, Eitan Bachmat, Richard Zemel, Zhun Deng
Eliciting information to reduce uncertainty about latent group-level properties is a central problem in collective assessment, preference modeling, and opinion aggregation, and is especially important in survey-based studies. While natural language interactions provide a flexible interface, existing methods typically rely on fixed questionnaires and static respondent sets, and do not adapt to partial or missing responses across rounds. To address this gap, we study adaptive information elicitation through multi-turn interactions between a large language model and a group of individuals, where both queries and respondents are adaptively selected to infer latent group properties. We propose a theoretically groun

ded framework that, at each round, jointly selects a query and a subset of respo ndents based on previously observed responses to efficiently reduce uncertainty about a target latent quantity (e.g., group-level political inclination). Motiva ted by practical survey constraints, such as limited questions and costly partic ipation, our strategy maximizes information gain under a fixed budget. To handle missing and incomplete responses, we combine graph neural networks for aggregat ing/imputing partial group information with an information-theoretic criterion t hat guides per-round selection. Across three real-world opinion datasets, we ach ieve consistent improvements in population-level response prediction under const rained budgets, including over a 12% relative gain on CES at a 10% respondent bu dget.

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DF-ExpEnse: Diffusion Filtered Exploration for Sample Efficient Finetuning

Calvin Luo, Chen Sun, Shuran Song

A natural recipe for intelligent robotic decision-making is initializing from pr etrained generative control policies, which have summarized offline experience, and adapting them to self-collected online experience. We present DF-ExpEnse, a n exploration technique that improves the quality of online experience collectio n, thus increasing finetuning sample-efficiency. DF-ExpEnse leverages the multi modal modeling capabilities of the generative control policy to create an expres sive and tractably evaluatable candidate set. It then utilizes an ensemble of c ritics to identify the action that best balances quality with high exploration i nterest. In fleet settings, DF-ExpEnse further enables cross-agent communicati on to facilitate collaborative exploration as a group. DF-ExpEnse can be seamle ssly integrated with existing strategies that finetune pretrained generative con trol policies via reinforcement learning. We experimentally validate consistent sample-efficiency benefits through DF-ExpEnse across a variety of manipulation and locomotion tasks, compared to default finetuning and alternative action sele ction schemes. Project can be found at [df-expense.github.io](https://df-expens e.github.io).

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GradientStabilizer: Fix the Norm, Not the Gradient

Tianjin Huang, Zhangyang Wang, Haotian Hu, Zhenyu Zhang, Gaojie Jin, Xiang Li, L i Shen, Jiaxing Shang, Tianlong Chen, Ke Li, Lu Liu, Qingsong Wen, Shiwei Liu

Training instability in modern deep learning systems is frequently triggered by rare but extreme gradient-norm spikes, which can induce oversized parameter upda tes, corrupt optimizer state, and lead to slow recovery or divergence. Widely us ed safeguards such as gradient clipping mitigate these failures but require thre shold tuning and indiscriminately truncate large updates.

We propose **GradientStabilizer**, a lightweight, drop-in gradient transform tha t *preserves the instantaneous gradient direction* while replacing the update ma gnitude with a statistically stabilized estimate derived from running gradient-n orm statistics.

We prove that the resulting stabilized magnitude is uniformly bounded on spike s teps, independent of the spike size, and show how this boundedness controls opti mizer state evolution in adaptive methods.

Across LLM pre-training (FP16), quantization-aware pre-training (FP4), ImageNet classification, reinforcement learning, and time-series forecasting, **GradientS tabilizer** consistently improves training stability, widens stable learning-rat e regions, and reduces divergence relative to clipping-based baselines, even sub stantially reducing Adam’s sensitivity to weight-decay strength.

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Quantifying LLM Attention-Head Stability: Implications for Circuit Universality

Karan Bali, Jack Stanley, Praneet Suresh, Danilo Bzdok

In mechanistic interpretability, recent work scrutinizes transformer “circuits”— sparse, mono or multi layer sub computations, that may reflect human understanda ble functions. Yet, these network circuits are rarely acid-tested for their stab ility across different instances of the same deep learning architecture. Without this, it remains unclear whether reported circuits emerge universally across la bs or turn out to be idiosyncratic to a particular estimation instance, potentia

lly limiting confidence in safety-critical settings. Here, we systematically stu dy stability across-refits in increasingly complex transformer language models o f various sizes. We quantify, layer by layer, how similarly attention heads lear n representations across independently initialized training runs. Our rigorous e xperiments show that (1) middle-layer heads are the least stable yet the most re presentationally distinct; (2) deeper models exhibit stronger mid-depth divergen ce; (3) unstable heads in deeper layers become more functionally important than their peers from the same layer; (4) applying weight decay optimization substant ially improves attention-head stability across random model initializations; and (5) the residual stream is comparatively stable. Our findings establish the cro ss-instance robustness of circuits as an essential yet underappreciated prerequi site for scalable oversight, drawing contours around possible white-box monitora bility of AI systems.

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FakeWorld 1.0: An Omni-modal Benchmark for Fake Media and Content

Yifeng Gao, Yifan Ding, Li Wang, Feida Huang, Ye Sun, Yixu Wang, Xin Wang, Yutao Wu, Hanxun Huang, Yunhao Feng, Yingshui Tan, Xingjun Ma, Yu-Gang Jiang

The rapidly increasing realism of AI-generated media has intensified the spread of deceptive content and undermined public trust. Existing research largely trea ts this challenge along two separate axes: media authenticity, which assesses wh ether content is real or machine-generated, and content veracity, which evaluate s semantic consistency and factual correctness. This separation overlooks how re al-world deception jointly exploits both dimensions.

In this work, we present FakeWorld 1.0, an omni-modal benchmark that unifies med ia authenticity and content veracity within a single evaluation framework. Along the media axis, FakeWorld spans text, audio, image, and video synthesis. Along the content axis, it systematically instantiates cross-modal semantic inconsiste ncies and factual errors. These two axes are jointly embedded in realistic web-b ased and streaming-style presentation scenarios, reflecting how multimodal decep tion is composed, contextualized, and delivered in practice.

FakeWorld further provides explainable annotations in the form of per-instance r ationales, enabling transparent and evidence-based analysis. Under a unified eva luation protocol, experiments on both open- and closed-source multimodal large l anguage models (MLLMs) reveal fundamental capability limits and demonstrate Fake World’s effectiveness in exposing high-fidelity, mixed-source deception. Beyond the benchmark, we introduce OmniChecker, an agentic framwork that performs joint , explainable detection across both axes and produces evidence-backed diagnostic reports.

We position FakeWorld 1.0 as a realistic stress test and a practical foundation for advancing scalable, explainable detection of fake multimodal content.

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DIVA: Harnessing the Representation Divergence in Unified Multimodal Models for Mutual Reinforcement

Renjie Lu, Xulong Zhang, Xiaoyang Qu, Jianzong Wang, Shangfei Wang

Unified Multimodal models (UMMs) built on a single architecture have shown impre ssive performance in both understanding and generation. We identify a fundamenta l challenge lies in inductive biases induced by distinct supervision signals: ge neration branch prefers high-fidelity, fine-grained representations capable of r econstruction, while the understanding favours semantically discriminative embed dings that remain invariant to task-irrelevant factors. Consequently, optimizing these complementary but non-equivalent objectives within a monolithic backbone leads to mutual impairment instead of enhancement. In this paper, we first analy ze the root cause of this interference in unified backbones and reveal a complem entary structure in their internal representations. Motivated by the observation , we propose DIVA, a self-improved post-training framework that transforms the r epresentation divergence into interior synergy. By explicitly factorizing the vi sual representation into shared and unique components based on two complementary information flow, DIVA enables both the understanding and generation branches t o achieve beneficial transferring while preserving the integrity of unique infor mation from cross-flow interference via mutual information estimation. Despite i

ts generality, our method consistently achieves improvements across visual under standing (+7.82%) and generation (+8.46%). The official code is available at: ht tps://anonymous.4open.science/r/DIVA-D225.
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A Game-Theoretic Framework for Measuring and Explaining Metric Compatibility in Fair Machine Learning
Lingfeng Zhang, Jingran Yang, Zhaohui Wang, Min Zhang, Zhang Qing
Machine learning fairness research documents trade-offs but lacks quantitative f rameworks to measure intrinsic metric compatibility without requiring causal gra phs. We introduce a game-theoretic framework that decomposes metrics into intera ction vectors, enabling compatibility measurement between metrics via cosine sim ilarity and mechanistic attribution to attribute coalitions. Through analysis of 6 datasets, 7 models, and 6 debiasing methods, we reveal that fairness and util ity are often structurally orthogonal (median compatibility ≈ 0) rather than diametrically opposed, with conflicts driven by sparse, low-order interacti ons. We further show that debiasing improves fairness by compressing the compati bility space—reducing compatibility of both synergistic and conflicting relation ships—rather than eliminating conflicts, providing a mechanistic basis for under standing metric alignment.
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Falling Trees: A Model Class for Interpretable Risk Prioritization
Varun Babbar, Zachery Boner, Margo Seltzer, Cynthia Rudin
Many real-world decisions require prioritizing high-risk cases, such as clinicia ns prioritizing high-risk patients before lower-risk ones. Falling rule lists (F RLs), which are ordered if--then rules with monotonically decreasing risks, prov ide an interpretable framework for such tasks; however, their single-path struct ure yields a highly restricted model class. We introduce falling trees, a new fa mily of interpretable models that enforces the same monotonic risk constraint wh ile permitting tree-structured branching. We present GraviTree, a novel dynamic- programming-with-bounds algorithm for learning the Rashomon set of falling trees under depth and branching constraints. Our formulation can interpolate between rule lists and full decision trees, enabling user-desired model expressivity. In a new clinical dataset and in many public classification benchmarks, falling tr ees match or outperform FRLs and other interpretable baselines, often producing more sparse decisions for high-risk instances. Our results show that falling tre es strike a practical balance between interpretability, expressiveness, and risk prioritization for high-stakes settings.
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4DPC^{2}hat: Towards Dynamic Point Cloud Understanding with Failure-Aware Bootst rapping
Xindan Zhang, Weilong Yan, YUFEI SHI, Xuerui Qiu, Tao He, Ying Li, Ming Li, Hehe Fan
Point clouds provide a compact and expressive representation of 3D objects, and have recently been integrated into multimodal large language models (MLLMs). How ever, existing methods primarily focus on static objects, while understanding dy namic point cloud sequences remains largely unexplored. This limitation is mainl y caused by the lack of large-scale cross-modal datasets and the difficulty of m odeling motions in spatio-temporal contexts. To bridge this gap, we present 4DPC ^{2}hat, the *first* MLLM tailored for dynamic point cloud understanding. To thi s end, we construct a large-scale cross-modal dataset 4DPC^{2}hat-200K via a met iculous two-stage pipeline consisting of topology-consistent 4D point constructi on and two-level captioning. The dataset contains over 44K dynamic object sequen ces, 700K point cloud frames, and 200K curated question–answer (QA) pairs, suppo rting inquiries about counting, temporal relationship, action, spatial relations hip, and appearance. At the core of the framework, we introduce a Mamba-enhanced temporal reasoning MLLM to capture long-range dependencies and dynamic patterns among a point cloud sequence. Furthermore, we propose a *failure-aware* bootstr apping learning strategy that iteratively identifies model deficiencies and gene rates targeted QA supervision to continuously strengthen corresponding reasoning capabilities. Extensive experiments demonstrate that our 4DPC^{2}hat significan

tly improves action understanding and temporal reasoning compared with existing models, establishing a strong foundation for 4D dynamic point cloud understanding.

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Local Policies for Graph-Structured Markov Decision Processes

Fathima Zarin Faizal, Asuman E. Ozdaglar, Martin J Wainwright

We study a cooperative form of multi-agent reinforcement learning with state space dynamics and agent interaction controlled by an underlying graph. Each agent has a local state and action, the evolution of the local state depends only on the states and actions in the 1-hop neighborhood defined by the graph. Structured dynamics of this type arise in various applications, including network resource allocation, co-operative games, epidemic control, and wireless scheduling. The global state-action space scales exponentially in the number of agents, so that computing global optimal policies is intractable in the worst-case. We study conditions under which it is possible to approximate the optimal policies by a local policy for each agent that depends only on states associated with nodes within its m-hop neighborhood. By controlling the propagation of influences via a Dobrushin-type stability matrix, we establish that globally optimal policies can approximated by local policies with sub-optimality gap decaying exponentially in m.

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Improved Analysis of the Accelerated Noisy Power Method with Applications to Decentralized PCA

Pierre Aguié, Mathieu Even, Laurent Massoulié

We analyze the Accelerated Noisy Power Method, an algorithm for Principal Component Analysis in the setting where only inexact matrix-vector products are available, which can arise for instance in decentralized PCA. While previous works have established that acceleration can improve convergence rates compared to the standard Noisy Power Method, these guarantees require overly restrictive upper bounds on the magnitude of the perturbations, limiting their practical applicability. We provide an improved analysis of this algorithm, which preserves the accelerated convergence rate under much milder conditions on the perturbations. We show that our new analysis is worst-case optimal, in the sense that the convergence rate cannot be improved, and that the noise conditions we derive cannot be relaxed without sacrificing convergence guarantees. We demonstrate the practical relevance of our results by deriving an accelerated algorithm for decentralized PCA, which has similar communication costs to non-accelerated methods. To our knowledge, this is the first decentralized algorithm for PCA with provably accelerated convergence.

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Learning Dynamics of Zeroth-Order Optimization: A Kernel Perspective

Zhe Li, Bicheng Ying, Zidong Liu, Haibo Yang

Classical optimization theory establishes that zeroth-order (ZO) algorithms suffer from a dimension-dependent slowdown, with convergence rates typically scaling with the model dimension compared to first-order methods. However, in contrast to these theoretical expectations, a growing body of recent work demonstrates the successful application of ZO methods to fine-tuning Large Language Models (LLMs) with billions of parameters. To explain this paradox, we derive the one-step learning dynamics of ZO SGD, where the empirical Neural Tangent Kernel (eNTK) naturally emerges as the key term governing the learning behavior. Inspection of the eNTK produced by ZO-SGD reveals that each element corresponds to the inner product of neural tangent vectors projected onto a random low-dimensional subspace. Thus, by invoking the Johnson-Lindenstrauss Lemma, our analysis shows that the fidelity of the ZO eNTK is governed primarily by the number of perturbations. Crucially, the approximation error depends on the model output size rather than the massive parameter dimension. This dimension-free property provides a theoretical justification for the scalability of ZO methods to LLMs finetuning tasks. We

believe that this kernel-based framework offers a novel perspective for understanding ZO methods within the context of learning dynamics.
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Toward Understanding Adversarial Distillation: Why Robust Teachers Fail
Hongsin Lee, Hye Won Chung
Adversarial Distillation aims to enhance student robustness by guiding the student with a robust teacher's soft labels within the min-max adversarial training framework, yet its success is notoriously inconsistent: a more robust teacher often fails to improve, or even harms, the student's robust generalization.
In this paper, we identify a key mechanism of this teacher dependency: the misalignment between the teacher's supervisory confidence and the student's representational limitations on a consistent subset of training data—the Robustly Unlearnable Set.
We present a theoretical framework analyzing the feature learning dynamics of a two-layer neural network, demonstrating that this mismatch creates a dichotomy in distillation outcomes.
We prove that when a teacher provides confident supervision on unlearnable samples, it compels the student to memorize spurious noise patterns that eventually overpower the learned robust signal, thereby driving robust overfitting.
Conversely, a teacher that exhibits high uncertainty on these samples effectively suppresses noise memorization, allowing the student to rely solely on the learnable signal for robust generalization.
We empirically validate our theory across both synthetic simulations and real-image classification datasets, confirming that robust overfitting is driven by the teacher's interaction with unlearnable samples.
Finally, we demonstrate that a teacher's predictive entropy on unlearnable samples serves as a strong indicator of student robustness, validating our theoretical framework and offering a principled guideline for robust teacher selection.
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SoMA: A Real-to-Sim Neural Simulator for Robotic Soft-Body Manipulation
Mu Huang, Hui Wang, Kerui Ren, Linning Xu, Yunsong Zhou, Mulin Yu, Bo Dai, Jiangmiao Pang
Simulating deformable objects under rich interactions remains a fundamental challenge for real-to-sim robot manipulation, with dynamics jointly driven by environmental effects and robot actions.
Existing simulators rely on predefined physics or data-driven dynamics without robot-conditioned control, limiting accuracy, stability, and generalization.
This paper presents \textbf{SoMA}, a 3D Gaussian Splat simulator for soft-body manipulation.
SoMA couples deformable dynamics, environmental forces, and robot joint actions in a unified latent neural space for end-to-end real-to-sim simulation.
Modeling interactions over learned Gaussian splats enables controllable, stable long-horizon manipulation and generalization beyond observed trajectories without predefined physical models.
SoMA improves resimulation accuracy and generalization on real-world robot manipulation by 20\%, enabling stable simulation of complex tasks such as long-horizon cloth folding. **Project Page: city-super.github.io/SoMA**
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Theoretical Perspectives on Data Quality and Synergistic Effects in Pre- and Post-Training Reasoning Models
Adel Javanmard, Baharan Mirzasoleiman, Vahab Mirrokni
Large Language Models (LLMs) are pretrained on massive datasets and later instruction-tuned via supervised fine-tuning (SFT) or reinforcement learning (RL). Best practices emphasize large, diverse pretraining data, whereas post-training operates differently: SFT relies on smaller, high-quality datasets, while RL benefits more from scale, with larger amounts of feedback often outweighing label quality. Yet it remains unclear why pretraining and RL require large datasets, why SFT excels on smaller ones, and what defines high-quality SFT data. In this work, we theoretically analyze transformers trained on an in-context weight prediction task for linear regression. Our analysis reveals several key findings: (i) b

alanced pretraining data can induce latent capabilities later activated during p ost-training, and (ii) SFT learns best from a small set of examples challengin g for the pretrained model, while excessively large SFT datasets may dilute info rmative pretraining signals. In contrast, RL is most effective on large-scale da ta that is not overly difficult for the pretrained model. We validate these theo retical insights with experiments on large nonlinear transformer architectures.
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Constrained hybrid modelling to predict microbial dynamics and organic matter tu rnover in soil systems
Paul Collart, Juergen Gall, Andrea Schnepf, Holger Pagel, Lars Doorenbos
Soil microorganisms control organic matter cycling and largely determine how soi l systems can cope with and mitigate climate change and environmental threats. R epresenting microbial dynamics in process-based soil models is therefore critica l to predict carbon cycling in soils, albeit highly challenging to inform from d ata. One promising approach to improve their parametrisation is the integration of genomic data, yet modelling the complex and unknown relationship between geno mes and the processes the microbes are driving is an unsolved problem.
In this work, we present the first hybrid modeling framework for deriving biokin etic parameter values of a process-based soil organic matter turnover model from metagenome-inferred functional traits based on DNA sequencing data. Our model predicts biokinetic parameters of the process-based model from genomic trait dat a with a neural network and integrates constraints from ecological theory and li terature to ensure realistic behavior, even of non-observed state variables. We evaluate our method on synthetic genomic trait datasets of varying complexity an d on real data, showing that our approach improves performance over multiple bas elines and learns the dynamics of unmeasurable components of the process-based m odel effectively, even for small training datasets.
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Discretized Density-Guided Source-Free Domain Adaptation for Regression
Gezheng Xu, Qi CHEN, QIUHAO Zeng, Charles Ling, Boyu Wang
Source-Free Domain Adaptation (SFDA) enables model adaptation under distribution shifts without access to source data, providing a practical solution for privac y-sensitive applications and having shown substantial progress in classification .
In contrast, regression involves ordered and continuous target variables, posing unique challenges for representation adaptation and pseudo-label refinement in the SFDA setting.
To address this gap, we propose a novel algorithm for continuous label predictio n in SFDA that leverages instance-dependent, discretized density–informed superv isory signals to refine pseudo-labels within an uncertainty-aware paradigm.
By incorporating auxiliary discretized distribution learning, our method also pr omotes more compact and structured feature representations, mitigating the inher ent difficulties of adapting regression models under distribution shift.
We theoretically demonstrate that the resulting density structure is robust to p otential perturbations, supporting reliable SFDA for regression.
Extensive experiments across multiple benchmarks validate the effectiveness of t he proposed approach.
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Manifold-Aware Perturbations for Constrained Generative Modeling
Katherine Keegan, Lars Ruthotto
Generative models have enjoyed widespread success in a variety of applications. However, they encounter inherent mathematical limitations in modeling distributi ons where samples are constrained by equalities, as is frequently the setting in scientific domains. In this work, we develop a computationally cheap, mathemati cally justified, and highly flexible distributional modification for combating k nown pitfalls in equality-constrained generative models. We propose perturbing t he data distribution in a constraint-aware way such that the new distribution ha s support matching the ambient space dimension while still implicitly incorporat ing underlying manifold geometry. Through theoretical analyses and empirical evi dence on several representative tasks, we illustrate that our approach consisten

tly enables data distribution recovery and stable sampling with both diffusion models and normalizing flows.

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Pruning at Initialisation through the lens of Graphon Limit: Convergence, Expressivity, and Generalisation

Hoang Pham, The-Anh Ta, Long Tran-Thanh

Pruning at Initialisation methods discover sparse, trainable subnetworks before training, but their theoretical mechanisms remain elusive.

Existing analyses are often limited to finite-width statistics, lacking a rigorous characterisation of the global sparsity patterns that emerge as networks grow large. In this work, we connect discrete pruning heuristics to graph limit theory via graphons, establishing the *graphon limit of PaI masks*. We introduce a *Factorised Saliency Model* that encompasses popular pruning criteria and prove that, under regularity conditions, the discrete masks generated by these algorithms converge to deterministic bipartite graphons. This limit framework establishes a novel topological taxonomy for sparse networks: while unstructured methods (e.g., Random, Magnitude) converge to homogeneous graphons representing uniform connectivity, data-driven methods (e.g., SNIP, GraSP) converge asymptotically to heterogeneous graphons that encode implicit feature selection. Leveraging this continuous characterisation, we derive two consequences. First, we prove a universal approximation theorem for sparse networks on active coordinate subspaces. Second, under the Graphon-NTK lazy-training regime, we connect the limiting graphon to NTK-style generalisation bounds and introduce a path-density interpretation of how sparse topology can modulate kernel alignment. Our results transform the study of sparse neural networks from combinatorial graph problems into a rigorous framework of continuous operators, offering a new mechanism for analysing expressivity and generalisation in sparse networks.

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Stable Spectral Copula Alignment for Robust Multimodal Learning

Hongkang Zhang, Shao-Lun Huang, Yanlong Wang, Ercan Engin KURUOGLU

Multimodal alignment can fail under deployment shift because standard objectives entangle cross-modal dependence with marginal-sensitive geometry. Stable Spectral Copula Alignment (SSCA) provides a deployment protocol for copula-stable dependence under approximately coordinate-wise monotone marginal distortions, together with auditable, label-free diagnostics for monitoring and mitigation. SSCA combines (i) clipped soft-rank Gaussianization that suppresses marginal effects while tracking tie and approximation errors, (ii) dependence-weighted sliced Wasserstein hub coupling for globally coherent multiway alignment with cycle auditing, and (iii) diagonal-stabilized block-spectral learning with eigengap-normalized Davis-Kahan diagnostics, yielding an actionable subspace-risk inequality. A calibrated gate maps diagnostic proxies to a reliability signal with a measurable false-alarm/miss trade-off, enabling stability-mode updates, budgeted remediation, and conservative no-update fallback for out-of-scope drift. Evaluations on MOSEI/MELD, MSCOCO, and CC3M-500K show improved performance under perturbation and substantially reduced degradation under controlled monotone distortions, raw-pipeline drifts, and frozen-feature retrieval stress tests.

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Distinguishing Imitation Error from Intrinsic Motion Learning Difficulty

Zhaorui Meng, Lu Yin, Xinrui Chen, Chengxu Zuo, Anjun Chen, Shihui Guo, Yipeng Qin

Physics-based motion imitation is central to humanoid control, yet current evaluation metrics(e.g., MPJPE) only quantify imitation outcomes, not their underlying causes. This conflation obscures a critical diagnostic question: when imitation error occurs, does it stem from policy limitations or the intrinsic learning difficulty of the target motion? To resolve this ambiguity, we propose the Torque Variation Score (TVS), a physics-grounded metric that quantifies the inherent learning difficulty of a motion independently of any policy's performance. TVS measures the magnitude of torque variation required to correct small pose perturbations, directly capturing how dynamical properties shape the reinforcement learning landscape. We establish that high-TV motions induce flat reward landscapes a

nd vanishing policy gradients, explaining persistent imitation failures. Extensive experiments with state-of-the-art methods (UHC, PHC+) confirm TVS strongly correlates with imitation error and enables principled error attribution: high error on low-TV motions indicates policy deficiency, while high error on high-TV motions reflects fundamental learning constraints. Beyond error diagnosis, TVS facilitates three practical applications: Maximum Imitable Difficulty (MID) for policy capability assessment, Difficulty-Stratified Joint Error (DSJE) for granular performance profiling, and Flawed Motion Detection for identifying segments with abnormally high learning difficulty to support mocap data curation and quality control. TVS provides a rigorous lens to distinguish policy-induced errors from motion-inherent challenges and enhances motion dataset reliability.

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Class-Conditional Distribution Balancing for Group Robust Classification

Miaoyun Zhao, Qiang Zhang

Spurious correlations that lead models to correct predictions for the wrong reasons pose a critical challenge for robust real-world generalization. Existing research attributes this issue to group imbalance and addresses it by maximizing group-balanced or worst-group accuracy, which heavily relies on expensive bias annotations. A compromise approach involves predicting bias information using extensively pretrained foundation models, which requires large-scale data and is limited to physically interpretable biases. To address these challenges, we offer a novel perspective by reframing the spurious correlations as imbalances/mismatches in class-conditional distributions caused by general biases, whether interpretable or not, and propose a simple yet effective robust learning method that mitigates the need for bias annotations or predictions. With the goal of maximizing the conditional entropy (uncertainty) of the label given spurious factors, our method leverages a sample reweighting strategy to achieve class-conditional distribution balancing, which automatically highlights minority groups and classes, effectively mitigating spurious correlations and producing a debiased data distribution for classification. Extensive experiments and analysis demonstrate that our approach consistently delivers strong overall performance, and is competitive with methods that rely on bias supervision.

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CoD-Lite: Real-Time Diffusion-Based Generative Image Compression

Zhaoyang Jia, Naifu Xue, Zihan Zheng, Jiahao Li, Bin Li, Xiaoyi Zhang, Zongyu Guo, Yuan Zhang, Houqiang Li, Yan Lu

Recent advanced diffusion methods typically derive strong generative priors by scaling diffusion transformers. However, scaling fails to generalize when adapted for real-time compression scenarios that demand lightweight models. In this paper, we explore the design of real-time and lightweight diffusion codecs by addressing two pivotal questions. First, does diffusion pre-training benefit lightweight diffusion codecs? Through systematic analysis, we find that generation-oriented pre-training is less effective at small model scales whereas compression-oriented pre-training yields consistently better performance. Second, are transformers essential? We find that while global attention is crucial for standard generation, lightweight convolutions suffice for compression-oriented diffusion when paired with distillation. Guided by these findings, we establish a one-step lightweight convolution diffusion codec that achieves real-time 60 FPS encoding and 42 FPS decoding at 1080p. Further enhanced by distillation and adversarial learning, the proposed codec reduces bitrate by 85% at a comparable FID to MS-ILLM, bridging the gap between generative compression and practical real-time deployment. Codes are released at https://github.com/microsoft/GenCodec/tree/main/CoD_Lite.

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ImgCoT: Compressing Long Chain of Thought into Compact Visual Tokens for Efficient Reasoning of Large Language Model

Xiaoshu Chen, Sihang Zhou, KE LIANG, Taichun Zhou, Yaohua Wang, Yang Gao, Xinwang Liu

Compressing long chains of thought (CoT) into compact latent tokens is crucial for efficient reasoning with large language models (LLMs). Recent studies employ

autoencoders to achieve this by reconstructing textual CoT from latent tokens, thus encoding CoT semantics. However, treating textual CoT as the reconstruction target forces latent tokens to preserve surface-level linguistic features (e.g., word choice and syntax), introducing a strong linguistic inductive bias that prioritizes linguistic form over reasoning structure and limits logical abstraction. Thus, we propose ImgCoT that replaces the reconstruction target from textual CoT to the visual CoT obtained by rendering CoT into images. This substitutes linguistic bias with spatial inductive bias, i.e., a tendency to model spatial layouts of the reasoning steps in visual CoT, enabling latent tokens to better capture global reasoning structure. Moreover, although visual latent tokens encode abstract reasoning structure, they may blur reasoning details. We thus propose a loose ImgCoT, a hybrid reasoning that augments visual latent tokens with a few key textual reasoning steps, selected based on low token log-likelihood. This design allows LLMs to retain both global reasoning structure and fine-grained reasoning details with fewer tokens than the complete CoT. Extensive experiments across multiple datasets and LLMs demonstrate the effectiveness of the two versions of ImgCoT.

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SP-Mind: An Autonomous Reasoning Agent for Spatial Proteomics Analysis

YuCheng Yuan, Yuanfeng Ji, Zhongxiao Li, Ruijiang Li

Spatial proteomics enables single-cell-resolution characterization of protein expression within tissue architecture, playing a critical role in understanding tumor microenvironments and guiding precision medicine. However, current analysis workflows remain fragmented, requiring expert manual orchestration of heterogeneous tools and limiting research scalability and reproducibility. We present SP-Mind, the first autonomous AI agent designed to unify the spatial proteomics analysis pipeline, from raw multiplexed tissue imaging to downstream phenotype discovery. Equipped with expert-curated biological analysis skills and specialized computational tools, SP-Mind converts natural-language queries into end-to-end analytical workflows without task-specific fine-tuning. To rigorously evaluate its capabilities, we introduce SP-Bench, a comprehensive benchmark spanning diverse tissue types, comprising 102 tasks across 18 distinct categories. Through extensive evaluation on SP-Bench and established downstream tasks, SP-Mind achieves state-of-the-art performance compared to existing open-source biomedical agent baselines.

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Task-Driven Subspace Decomposition for Knowledge Sharing and Isolation in LoRA-based Continual Learning

Lingfeng He, De Cheng, Huaijie Wang, Xi Yang, Nannan Wang, Xinbo Gao

Continual Learning (CL) requires models to sequentially adapt to new tasks without forgetting old knowledge. Recently, Low-Rank Adaptation (LoRA), a representative Parameter-Efficient Fine-Tuning (PEFT) method, has gained increasing attention in CL. Several LoRA-based CL methods reduce interference across tasks by separating their update spaces, typically building the new space from the estimated null space of past tasks. However, they (i) overlook task-shared directions, which suppresses knowledge transfer, and (ii) fail to capture truly effective task-specific directions since these ``null bases" of old tasks can remain nearly inactive for new task under correlated tasks. To address this, we study LoRA learning capability from a projection energy perspective, and propose Low-rank Decomposition and Adaptation (LoDA). It performs a task-driven decomposition to build general and truly task-specific LoRA subspaces by solving two energy-based objectives, decoupling directions for knowledge sharing and isolation. LoDA fixes LoRA down-projections on two subspaces and learns robust up-projections via a Gradient-Aligned Optimization (GAO) approach. After each task, before integrating the LoRA updates into the backbone, LoDA derives a closed-form recalibration for the general update, approximating a feature-level joint optimum along this task-shared direction. Experiments indicate that LoDA outperforms existing CL methods.

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From Lyapunov Analysis to Algorithm Design in two-sided PL Minimax Optimization

Mansi Rankawat, Michael Muehlebach, Simon Lacoste-Julien, Damien Scieur

We derive algorithms for smooth nonconvex nonconcave minimax optimization and es tablish linear convergence rates for problems that satisfy the two-sided Polyak-Lojasiewicz (PL) inequality. At the core of our approach is the observation that Lyapunov functions can be used not only to certify convergence a posteriori, bu t also to design algorithms. By replacing an idealized, intractable Lyapunov fun ction with a computable surrogate based on gradient information, we derive TALDA (Tri-Action Lyapunov Descent Ascent), a single-loop algorithm that enforces Lya punov descent by construction. TALDA guarantees linear convergence under the two -sided PL condition, with a rate that depends explicitly on the cross-smoothness constant. This recovers existing worst-case guarantees while yielding sharper c onvergence rates in weakly coupled min–max problems.
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Server-Proximal Aggregation for Federated Domain-Incremental Learning under Part ial Participation: Task-Uniform Convergence and Backward Transfer
Longtao Xu, Jian Li
Real-world federated systems seldom operate on static data: input distributions drift while privacy rules forbid raw data sharing. We study Federated Domain-Inc remental Learning (FDIL), where (i) clients are heterogeneous, (ii) tasks arrive sequentially with shifting domains, and (iii) the label space remains fixed. Tw o theoretical pillars remain missing for FDIL under partial participation: a gua rantee of backward knowledge transfer (BKT) and a convergence rate that holds *u niformly across the task sequence*. We introduce SPECIAL (Server-Proximal Effici ent Continual Aggregation for Learning), a simple, replay-free FDIL algorithm th at adds a single server-side ``anchor'' to FedAvg: in each round, the server agg regates updates from a uniformly sampled subset of clients and then blends the r esult with the previous global model via a lightweight proximal step. This ancho r curbs cumulative drift without replay buffers, synthetic data, or task-specifi c heads, leaving communication cost and model size unchanged. Our theory shows t hat SPECIAL (i) *preserves earlier tasks*: a BKT bound caps any increase in earl ier-task loss by a drift-controlled term that shrinks with more rounds, local ep ochs, and participating clients; and (ii) *achieves task-uniform, communication-efficient convergence* for non-convex FDIL with partial participation: $\mathcal {O}\big(\sqrt{E/(NT)}\big)$ in expected gradient norm, with E local epochs, $T $ rounds, and N participating clients, while explicitly separating optimizatio n variance from inter-task drift. Experiments on standard FDIL benchmarks corrob orate the theory.
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Characterization of Gaussian Universality Breakdown in High-Dimensional Empirica l Risk Minimization
Mohamed Chiheb Yaakoubi, Cosme Louart, Malik Tiomoko, Zhenyu Liao
We study high-dimensional convex empirical risk minimization (ERM) under general non-Gaussian data designs. By heuristically extending the Convex Gaussian Min–M ax Theorem (CGMT) to non-Gaussian settings, we derive an asymptotic min–max char acterization of key statistics, enabling approximation of the mean $\mu_{\hat{\t heta}}$ and covariance $C_{\hat{\theta}}$ of the ERM estimator $\hat{\theta}$.
Specifically, under a concentration assumption on the data matrix and standard r egularity conditions on the loss and regularizer, we show that for a test covari ate x independent of the training data, the projection $\hat{\theta}^\top x$ a pproximately follows the convolution of the (generally non-Gaussian) distributio n of $\mu_{\hat{\theta}}^\top x$ with an independent centered Gaussian variable of variance $\mathrm{tr}\!\big(C_{\hat{\theta}}\,\mathbb{E}[xx^\top]\big)$. This result clarifies the scope and limits of Gaussian universality for ERMs. Additi onally, we prove that any $\mathcal{C}^2$ regularizer is asymptotically equivale nt to a quadratic form determined solely by its Hessian at zero and gradient at $\mu_{\hat{\theta}}$.
Numerical simulations across diverse losses and models are provided to validate our theoretical predictions and qualitative insights.
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Focusing: View-Consistent Sparse Voxels for Efficient 3D VAE Training
Xuhui Chen, Chao Long, Fei Hou, dongbo zhang, Shaohui Jiao, Wencheng Wang, Ying

He
High-fidelity 3D generation depends on 3D VAEs that can compress and reconstruct complex geometry at high resolution. Recent methods often convert raw meshes into signed distance fields (SDFs), but this preprocessing can be lossy, especially for open or non-watertight assets. Render-supervised VAEs such as TripoSF avoid this conversion by matching rendered depth and normal maps, yet render losses supervise only the visible geometry for each view, leaving many latent voxels weakly constrained while still incurring unnecessary decoding and attention costs. We introduce Focusing, a view-consistent sparse-voxel training scheme for efficient 3D VAEs. Given a training view, Focusing performs depth-driven voxel carving directly in the structured latent space: voxels inconsistent with the rendered depth are removed before decoding, allowing the decoder and attention layers to operate only on locally relevant geometry. This view-dependent sparsity reduces memory and computation while concentrating learning on surface regions that contribute to the render. To stabilize training across shapes and viewpoints, we further propose adaptive zooming, which adjusts camera intrinsics to keep the number of active voxels within a target range and strengths supervision for fine details. The VAE is trained with render-based depth, normal, mask, and perceptual losses, together with sparse-voxel total variation and a brief TSDF warm-up to improve convergence and suppress holes. Across standard reconstruction benchmarks, Focusing improves Chamfer Distance and F-score over strong baselines while substantially reducing video random access memory (VRAM) consumption, enabling 1024^3-resolution VAE training with as little as 50 GB of VRAM. These results demonstrate that local, view-consistent sparsity is an effective path toward higher-resolution and more efficient 3D VAE training.
****
SynGR: Unleashing the Potential of Cross-Modal Synergy for Generative Recommendation
Wei Chen, Xingyu Guo, Shuang Li, Fuwei Zhang, Meng Yuan, Jing Fan, Zhao Zhang, deqing wang, Fuzhen Zhuang
Generative Recommendation (GR) has emerged as a promising paradigm by formulating item recommendation as a sequence-to-sequence generation task over item identifiers.
■Recent studies have incorporated multimodal signals to provide richer token-level evidence for generation.
■However, existing approaches largely rely on alignment-centric fusion and underexplore synergistic information across modalities.
■In practice, synergistic information plays a critical role in capturing emergent item properties that cannot be inferred from any single modality alone.
■Such properties encode intrinsic item semantics and guide user preferences, enabling models to move beyond surface-level feature matching.
■To address this limitation, we propose \textbf{SynGR}, a synergistic generative recommendation framework that explicitly encourages the exploitation of cross-modal dependencies during generation. By constraining overreliance on dominant modalities, SynGR enables the model to capture emergent item semantics beyond shared or modality-specific signals. Extensive experiments across three benchmark datasets demonstrate that SynGR achieves superior performance.
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Beyond Heuristic Tuning: Power-Calibrated LLM Watermarking
Xiaopu Wang, Zelin He, Chengyuan Liu, Runze Li
Logit-based watermarking is a widely used mechanism for identifying LLM generated content, yet its effectiveness is governed by a fundamental trade-off between detectability and semantic distortion. Existing analyses provide limited guidance for principled hyperparameter selection, leaving practical deployments reliant on heuristic tuning. In this work, we develop a power-calibrated statistical framework that establishes explicit quantitative relationships between watermark hyperparameters, detection power, and distortion. This characterization transforms watermark design into a guided optimization problem. Building on these results, we derive practical parameter selection procedures that achieve optimal trade-offs under constraints. Extensive experiments across multiple language models an

d datasets validate the theory and demonstrate that the proposed framework consi
stently identifies Pareto-optimal points.
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Holi-Spatial: Evolving Video Streams into Holistic 3D Spatial Intelligence
Yuanyuan Gao, Hao Li, Yifei Liu, Xinhao Ji, Yuning Gong, Yuanjun Liao, Fangfu Li
u, Manyuan Zhang, Yuchen Yang, Dan Xu, Xue Yang, Huaxi Huang, Hongjie Zhang, Ziw
ei Liu, Xiao Sun, Dingwen Zhang, Zhihang Zhong
The pursuit of spatial intelligence fundamentally relies on access to large-scal
e, fine-grained 3D data. However, existing approaches predominantly construct sp
atial understanding benchmarks by generating question–answer (QA) pairs from a l
imited number of manually annotated datasets, rather than systematically annotat
ing new large-scale 3D scenes from raw web data. As a result, their scalability
is severely constrained, and model performance is further hindered by domain gap
s inherent in these narrowly curated datasets.
In this work, we propose \textbf{Holi-Spatial}, the first fully automated, large
-scale, spatially-aware multimodal dataset, constructed from raw video inputs wi
thout human intervention, using the proposed data curation pipeline. Holi-Spatia
l supports multi-level spatial supervision, ranging from geometrically accurate
3D Gaussian Splatting (3DGS) reconstructions with rendered depth maps to object-
level and relational semantic annotations, together with corresponding spatial Q
uestion–Answer (QA) pairs.
Following a principled and systematic pipeline, we further construct \textbf{Hol
i-Spatial-4M}, the first large-scale, high-quality 3D semantic dataset, containi
ng 12K optimized 3DGS scenes, 1.3M 2D masks, 320K 3D bounding boxes, 320K instan
ce captions, 1.2M 3D grounding instances, and 1.2M spatial QA pairs spanning div
erse geometric, relational, and semantic reasoning tasks.
Holi-Spatial demonstrates exceptional performance in data curation quality, sign
ificantly outperforming existing feed-forward and per-scene optimized methods on
 datasets such as ScanNet, ScanNet++, and DL3DV. Furthermore, fine-tuning Vision
-Language Models (VLMs) on spatial reasoning tasks using this dataset has also l
ed to substantial improvements in model performance.
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Adaptive Protein Tokenization
Rohit Dilip, Ayush Varshney, David Van Valen
Tokenization is a promising path to multi-modal models capable of jointly unders
tanding protein sequences, structure, and function. Existing protein structure t
okenizers create tokens by pooling information from local neighborhoods, an appr
oach that limits their performance on generative and representation tasks. In th
is work, we present a method for global tokenization of protein structures in wh
ich successive tokens contribute increasing levels of detail to a global represe
ntation. This change resolves several issues with generative models based on loc
al protein tokenization: it mitigates error accumulation, provides embeddings wi
thout sequence-reduction operations, and allows task-specific adaptation of a to
kenized sequence's information content. We validate our method on reconstruction
, generative, and representation tasks and demonstrate that it matches or outper
forms existing models based on local protein structure tokenizers. We show that
our adaptive approach enables inference criteria based on the information conten
t of the generated proteins. We validate representations generated from our toke
nizer on CATH classification tasks and demonstrate that non-linear probing on ou
r tokenized sequences outperforms equivalent probing on representations from oth
er tokenizers. Finally, we demonstrate how our method supports zero-shot protein
 shrinking and affinity maturation.
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Revisiting Neural Processes via Fourier Transform and Volterra Series
Peiman Mohseni, Nick Duffield, Raymond K. W. Wong
Modeling unknown latent functions from finite, irregularly sampled measurements
is a recurring challenge across science and engineering. Neural processes (NPs),
 a family of probabilistic functional models, are promising solutions---especial
ly when endowed with domain-specific symmetries like translation equivariance, w
hich improve sample efficiency and generalization. Yet existing translation-equi

variant NPs face two limitations: (i) they stack generic components with non-lin
earities, obscuring the induced function class and limiting interpretability; an
d (ii) convolutional designs rely on kernels with local receptive fields and req
uire dense uniform input grids, while attention-based methods avoid these issues
 but scale quadratically with the number of observations. We address both with t
wo contributions. First, using the Volterra expansion, we characterize continuou
s translation-equivariant operators as sums of higher-order convolutions, yieldi
ng analytical transparency while admitting efficient approximation by first-orde
r convolutions. Second, we introduce set Fourier convolutions (SFConvs), a frequ
ency-domain parameterization that operates directly on irregularly sampled point
s, achieves approximately global receptive fields, and scales linearly in the nu
mber of observations. Building on these ideas, we propose two conditional NPs (C
NPs): SFConvCNPs, which stack SFConv blocks with non-linearities, and SFVConvCNP
s, which integrate the Volterra formulation. Experiments on synthetic and real-w
orld datasets demonstrate our methods' efficacy against state-of-the-art baselin
es.
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Bi-Anchor Interpolation Solver for Accelerating Generative Modeling
Hongxu chen, Hongxiang Li, Zhen Wang, Long Chen
Flow Matching (FM) models have emerged as a leading paradigm for high-fidelity s
ynthesis. However, their reliance on iterative Ordinary Differential Equation (O
DE) solving creates a significant latency bottleneck. Existing solutions face a
dichotomy: training-free solvers suffer from significant performance degradation
 at low Neural Function Evaluations (NFEs), while training-based methods incur p
rohibitive training costs and lack plug-and-play versatility. To bridge this gap
, we propose the Bi-Anchor Interpolation Solver (BA-solver). BA-solver retains t
he versatility of standard training-free solvers while achieving significant acc
eleration by introducing a lightweight SideNet (1-2% backbone size) alongside th
e frozen backbone. Specifically, our method is founded on two synergistic compon
ents: 1) Bidirectional Temporal Perception, where the SideNet learns to approxim
ate both future and historical velocities without retraining the heavy backbone;
 and 2) Bi-Anchor Velocity Integration, which utilizes the SideNet with two anch
or velocities to efficiently approximate intermediate velocities for batched hig
h-order integration. By utilizing the backbone to establish high-precision “anch
ors” and the SideNet to densify the trajectory, BA-solver enables large step siz
es with minimized error. Empirical results on ImageNet-256 demonstrate that BA-s
olver achieves generation quality comparable to 100+ NFEs Euler solver in just 1
0 NFEs and maintains high fidelity in as few as 5 NFEs, incurring negligible tra
ining costs. Furthermore, BA-solver ensures seamless integration with existing g
enerative pipelines, facilitating downstream tasks such as image editing.
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CauScale: Neural Causal Discovery at Scale
Bo Peng, Sirui Chen, Jiaguo Tian, Yu Qiao, Chaochao Lu
Causal discovery is essential for advancing data-driven fields such as scientifi
c AI and data analysis, yet existing approaches face significant time- and space
-efficiency bottlenecks when scaling to large
graphs. To address this challenge, we present CauScale, a neural architecture de
signed for efficient causal discovery that scales inference to graphs with up to
 1000 nodes. CauScale improves time efficiency via a reduction unit that compres
ses data embeddings and improves space efficiency by adopting tied attention wei
ghts to avoid maintaining axis-specific attention maps. To keep high causal disc
overy accuracy, CauScale adopts a two-stream design: a data stream extracts rela
tional evidence from high-dimensional observations, while a graph stream integra
tes statistical graph priors and preserves key structural signals. CauScale succ
essfully scales to 500-node graphs during training, where prior work fails due t
o space limitations. Across testing data with varying graph scales and causal me
chanisms, CauScale achieves 99.6\% mAP on in-distribution data and 84.4\% on out
-of-distribution data, while delivering 4$\times$–13,000$\times$ inference speed
ups over prior methods.
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Physically-Guided Data-Space Rectified Flow for Precipitation Nowcasting
Wenjie Luo, chaorong li, Chuanhu Deng, Zhuo Wang
Reliable long-horizon precipitation nowcasting requires preserving fine-scale echo structures while maintaining coherent transport. Although Rectified Flow (RF) can generate detail-preserving future sequences, numerical ODE integration compounds velocity estimation errors and induces progressive off-manifold drift, causing morphological distortions at extended lead times. We propose Physically-guided Data-space Rectified Flow (PDRF), which re-parameterizes the generative ODE in data space: the network predicts the clean future sequence, analytically inducing a coupled vector field with an implicit restoring effect that suppresses drift. We also introduce a soft Semi-Lagrangian teacher based on an advection prior to regularize large-scale transport, while allowing local growth/decay/deformation to be learned from data. Experiments on four public benchmarks demonstrate consistent improvements in event-based skill and better preservation of intense-echo morphology over long horizons.
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Lightweight and Interpretable Transformer via Unrolling of Mixed Graph Algorithms for Traffic Forecast
Ji Qi, Mingxiao Liu, VIET HO TAM THUC DO, Yuzhe Li, Zhuoshi Pan, Gene Cheung, H. Vicky Zhao
Unlike conventional "black-box" transformers with classical self-attention mechanisms, we build a lightweight and interpretable transformer-like neural network by unrolling a mixed-graph-based optimization algorithm to forecast traffic with spatial and temporal dimensions.
We construct two graphs: an undirected graph $\mathcal{G}^u$ capturing spatial correlations across geography, and a directed graph $\mathcal{G}^d$ capturing sequential relationships over time.
We predict future samples of signal $\mathbf{x}$, assuming it is "smooth" with respect to both $\mathcal{G}^u$ and $\mathcal{G}^d$, where we design new ℓ_2- and ℓ_1-norm variational terms to quantify and promote signal smoothness (low-frequency reconstruction) on a directed graph.
We design an iterative algorithm based on alternating direction method of multipliers (ADMM), and unroll it into a feed-forward network for data-driven parameter learning.
We periodically insert graph learning modules for $\mathcal{G}^u$ and $\mathcal{G}^d$ that play the role of self-attention.
Experiments show that our unrolled networks achieve competitive traffic forecast performance as state-of-the-art prediction schemes, while reducing parameter counts drastically. Code: https://github.com/SingularityUndefined/Unrolling-GSP-ST Forecast.
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SWE-rebench V2: Language-Agnostic SWE Task Collection at Scale
Ibragim Badertdinov, Maksim Nekrashevich, Anton Shevtsov, Alexander Golubev
Software engineering agents (SWE) are improving rapidly, with recent gains largely driven by reinforcement learning (RL). However, RL training is constrained by the scarcity of large-scale task collections with reproducible execution environments and reliable test suites. Although a growing number of benchmarks have emerged, datasets suitable for training remain limited in scale and diversity or often target a limited set of high-resource language ecosystems. We introduce SWE-rebench V2, a language-agnostic automated pipeline for harvesting executable real-world SWE tasks and constructing RL training environments at scale. The pipeline synthesizes repository-specific installation and test procedures via an interactive setup agent, and filters unsound instances using an ensemble of LLM judges, validated against human-verified SWE-bench annotations. Using this pipeline, we construct a dataset of 32,079 tasks spanning 20 languages and 3,617 repositories, with pre-built images for reproducible execution. To further scale training data, we additionally release 120,000+ tasks with installation instructions, fail-to-pass tests and rich metadata, where the problem statement is generated based on the original pull request description. We validate the collected instances through a diagnostic study that covers a subset of tasks in five programming l

anguages across seven popular models, and provide instance-level metadata that f lags common confounders such as overly restrictive tests and underspecified desc riptions. We release the datasets, the collection and execution code, and associ ated artifacts to enable large-scale training of SWE agents across diverse langu ages and repositories.
****
Preconditioned DeltaNet: Curvature-aware Sequence Modeling for Linear Recurrence s
Neehal Tumma, Noel Loo, Daniela Rus
To address the increasing long-context compute limitations of softmax attention, several subquadratic recurrent operators have been developed. This work include s models such as Mamba-2, DeltaNet, Gated DeltaNet (GDN), and Kimi Delta Attenti on (KDA). As the space of recurrences grows, a parallel line of work has arisen to taxonomize them. One compelling view is the test-time regression (TTR) framew ork, which interprets recurrences as performing online least squares updates tha t learn a linear map from the keys to values. Existing delta-rule recurrences ca n be seen as first-order approximations to this objective, but notably ignore th e curvature of the least-squares loss during optimization. In this work, we addr ess this by introducing preconditioning to these recurrences. Starting from the theory of online least squares, we derive equivalences between linear attention and the delta rule in the exactly preconditioned case. Next, we realize this the ory in practice by proposing a diagonal approximation: this enables us to introd uce preconditioned variants of DeltaNet, GDN, and KDA alongside efficient chunkw ise parallel algorithms for computing them. Empirically, we find that our precon ditioned delta-rule recurrences yield consistent performance improvements across synthetic recall benchmarks and language modeling at the 340M and 1B scale.
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Eliminating Solution Bias in Differentially Private Optimization
DONGRUN LI, YUN ZENG, Zibo Wei, Jiacheng Wei, Qinnan Zhang, Faguo Wu, Xiao Zhang , Zhiming Zheng
Differentially private (DP) stochastic optimization algorithms are widely used i n privacy-preserving deep learning, where per-sample gradient clipping and noise injection protect sensitive information. However, these operations limit existi ng DP algorithms to converge within a constant-radius neighborhood of the first- order stationary point, leading to solution bias and the well-known privacy-util ity trade-off. To enhance model utility, we propose a novel algorithmic framewor k called DP-C4, which is designed to be error-Consistently-vanishing, Coupledly- clipped, solution-Calibrated, and Convergence-guaranteed. Specifically, it incor porates a carefully designed coupled clipping scheme with shifted threshold stra tegy, ensuring that both clipping bias and noise variance asymptotically vanish, thereby eliminating the DP-induced solution bias. Moreover, we extend existing sensitivity analysis techniques and develop a tailored privacy budget allocation to guarantee the privacy of DP-C4. Compared with the well-recognized DP-SGD, ou r framework injects significantly less noise under the same privacy level. In ad dition, we prove that our framework converges to the optimum in strongly-convex case and to a diminishing neighborhood of the first-order stationary point in n on-convex case. Experiments show that DP-C4 achieves superior privacy-utility tr ade-off over existing baselines across various tasks and datasets.
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BandPO: Bridging Trust Regions and Ratio Clipping via Probability-Aware Bounds f or LLM Reinforcement Learning
Yuan Li, Bo Wang, Yufei Gao, Yao Yuqian, Xinyuan Wang, Zhangyue Yin, Xipeng Qiu
Proximal constraints are fundamental to the stability of the Large Language Mode l reinforcement learning.
While the canonical clipping mechanism in PPO serves as an efficient surrogate f or trust regions, we identify a critical bottleneck: fixed bounds strictly const rain the upward update margin of low-probability actions, disproportionately sup pressing high-advantage tail strategies and inducing rapid entropy collapse.
To address this, we introduce **Band-constrained Policy Optimization** (BandPO). BandPO replaces canonical clipping with **Band**, a unified theoretical operator

that projects trust regions defined by f-divergences into dynamic, probabilit
y-aware clipping intervals.
Theoretical analysis confirms that Band effectively resolves this exploration bo
ttleneck.
We formulate this mapping as a convex optimization problem, guaranteeing a globa
lly optimal numerical solution while deriving closed-form solutions for specific
 divergences.
Extensive experiments across diverse models and datasets demonstrate that BandPO
 consistently outperforms canonical clipping and Clip-Higher, while robustly mit
igating entropy collapse. Code is publicly available at https://github.com/OpenM
OSS/BandPO.
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DynVLA: Learning World Dynamics for Action Reasoning in Autonomous Driving
Shuyao Shang, Bing Zhan, Yunfei Yan, Yuqi Wang, Yingyan Li, Yasong An, Xiaoman W
ang, Jierui Liu, Lu Hou, Lue Fan, Zhaoxiang Zhang, Tieniu Tan
We propose DynVLA, a driving VLA model that introduces a new CoT paradigm termed
 Dynamics CoT. DynVLA forecasts compact world dynamics before action generation,
 enabling more informed and physically grounded decision-making. To obtain compa
ct dynamics representations, DynVLA introduces a Dynamics Tokenizer that compres
ses future evolution into a small set of dynamics tokens. Considering the rich e
nvironment dynamics in interaction-intensive driving scenarios, DynVLA decouples
 ego-centric and environment-centric dynamics, yielding more accurate world dyna
mics modeling. We then train DynVLA to generate dynamics tokens before actions t
hrough SFT and RFT, improving decision quality while maintaining latency-efficie
nt inference. Compared to Textual CoT, which lacks fine-grained spatiotemporal u
nderstanding, and Visual CoT, which introduces substantial redundancy due to den
se image prediction, Dynamics CoT captures the evolution of the world in a compa
ct, interpretable, and efficient form. Extensive experiments on NAVSIM, Bench2Dr
ive, and a large-scale in-house dataset demonstrate that DynVLA consistently out
performs Textual CoT and Visual CoT methods, validating the effectiveness and pr
actical value of Dynamics CoT.
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De-Linearizing Agent Traces: Bayesian Inference of Latent Partial Orders for Eff
icient Execution
Dongqing li, Zheqiao Cheng, Geoff Nicholls, Quyu Kong
AI agents increasingly execute procedural workflows as sequential action traces,
 which obscures latent concurrency and induces repeated step-by-step reasoning.
We introduce BPOP, a Bayesian framework that infers a latent dependency partial
order from noisy linearized traces. BPOP models traces as stochastic linear exte
nsions of an underlying graph and performs efficient MCMC inference via a tracta
ble frontier-softmax likelihood that avoids \#P-hard marginalization over linear
 extensions. We evaluate on our open-sourced Cloud-IaC-6, a suite of cloud provi
sioning tasks with heterogeneous LLM-generated traces, and WFCommons scientific
workflows. BPOP recovers dependency structure more accurately than trace-only an
d process-mining baselines, and the inferred graphs support a compiled executor
that prunes irrelevant context, yielding substantial reductions in token usage a
nd execution time.
****

Context Tuning for In-Context Optimization
Jack Lu, Ryan Teehan, Zhenbang Yang, Mengye Ren
We introduce Context Tuning, a simple and effective method to significantly enha
nce few-shot adaptation of language models (LLMs) without fine-tuning model para
meters. While prompt-based adaptation techniques have demonstrated the effective
ness of lightweight adaptation methods for LLMs, they typically initialize a tra
inable prompt or prefix with irrelevant tokens for the task at hand. In contrast
, Context Tuning initializes the trainable prompt or prefix with task-specific d
emonstration examples, leveraging the model’s inherent In-Context Learning (ICL)
 ability to extract relevant information for improved few-shot learning performa
nce. Extensive evaluations on benchmarks such as CrossFit, UnifiedQA, MMLU, BIG-
Bench Hard, and ARC demonstrate that Context Tuning outperforms traditional prom

pt-based adaptation methods and achieves competitive accuracy with Test-Time Tra
ining at significantly higher training efficiency.
****
Beyond Normalization: Rethinking the Partition Function as a Difficulty Schedule
r for RLVR
Dohyung Kim, Minbeom Kim, Jeonghye Kim, Sangmook Lee, Sojeong Rhee, Kyomin Jung
Reward-maximizing RL methods have shown to be capable of enhancing the reasoning
 performance of LLMs, but often lead to reduced generation diversity. Recent wor
ks address this issue by adopting GFlowNets, training LLMs to match a target dis
tribution while jointly learning its partition function.
In contrast to prior works that treat this partition function solely as a normal
izer, we reinterpret it as a per-prompt expected-reward (i.e., online accuracy)
signal, leveraging this unused information to improve sample efficiency.
Specifically, we first establish a theoretical relationship between the partitio
n function and per-prompt accuracy estimates. Building on this key insight, we p
ropose \textbf{Pa}rtition Fun\textbf{c}tion-Guid\textbf{ed} \textbf{RL} (PACED-R
L), a post-training framework that leverages accuracy estimates to prioritize i
nformative question prompts during training, and further improves sample efficie
ncy through an accuracy estimate error–prioritized replay. Crucially, both compo
nents reuse information already produced during GFlowNet training, effectively a
mortizing the compute overhead into the existing optimization process. Extensive
 experiments across diverse benchmarks demonstrate strong performance improvemen
ts over GRPO and prior GFlowNet approaches, highlighting PACED-RL as a promising
 direction for a more sample efficient distribution-matching training for LLMs.
****
XDomainBench: Diagnosing Reasoning Collapse in High-Dimensional Scientific Knowl
edge Composition
Gong Zhiren, Tiantong Wu, Jiaming Zhang, Fuyao Zhang, CHE WANG, Yurong Hao, Yiku
n Hou, Foo Ping, Yilei Zhao, Fei Huang, Chau Yuen, Wei Yang Bryan Lim
Large Language Models (LLMs) are increasingly deployed for knowledge synthesis,
yet their capacity for compositional generalization in scientific knowledge rema
ins under-characterized.
Existing benchmarks primarily focus on single-turn restricted scenarios, failing
 to capture the capability boundaries exposed by real-world interactive scientif
ic workflows.
To address this, we introduce XDomainBench, a diagnostic benchmark for interacti
ve interdisciplinary scientific reasoning.
We formalize the composition order and mixture structure to enable systematic st
ress-testing from single-discipline to inter-disciplinary, comprising 8,598 inte
ractive sessions across 20 domains and 4 task categories, with 8 realistic traje
ctory patterns covering difficulty and domain-mixture dynamics, simulating real
AI4S scenarios.
Large-scale evaluation of LLMs reveals a systematic reasoning collapse as compos
ition order increases, stemming from two root causes: (i) direct difficulty incr
eases induced by domain composition, and (ii) indirect interaction-amplified fai
lures where trajectory patterns trigger error accumulation, reasoning breaks, an
d domain confusion, ultimately leading to session collapse.
****
MoVie: Multimodal Video Compression with Text Guidance
Jiaqi Hu, Haoji Hu, Heming Sun, Lianrui Mu
Most deep video codecs emphasize low-level motion modeling and remain largely se
mantics-agnostic, which can degrade perceptual quality in complex scenes. We pro
pose **MoVie**, a **M**ultim**o**dal **Vi**d**e**o compression framework built o
n a Text-guided Video Transformer–CNN Mixed block (*Text-VideoTCM*). MoVie adopt
s a video-centric architecture that jointly models local spatial structures and
temporal dynamics via window-based processing, delivering a favorable computatio
n--perception trade-off. To incorporate semantics, we introduce dual-stage text
fusion with *Extractor* and *Injector* modules. We further present history-condi
tioned coding that leverages both previous and aggregated historical frames, and
 a spatial--channel factorized entropy model that estimates probabilities over s

patial neighborhoods and channel groups for adaptive bit allocation. Together, t
hese designs reduce redundancy and improve rate control and temporal coherence,
yielding reconstructions at low bitrates. On UVG and MCL-JCV, MoVie achieves **$
-$50.23\%** BD-rate for FID and **$-$14.64\%** for LPIPS (VGGNet) relative to HM
, while requiring only **55.76\%** of DCVC-FM's per-pixel kMACs. A human percept
ual study further confirms consistent subjective preference over strong baseline
s.
****
OmniVL-Guard: Towards Unified Vision-Language Forgery Detection and Grounding vi
a Balanced RL
Jinjie Shen, Jing Wu, Yaxiong Wang, Lechao Cheng, Shengeng Tang, Tianrui Hui, Na
n Pu, Zhun Zhong
Existing forgery detection methods are often limited to uni-modal or bi-modal se
ttings, failing to handle the interleaved text, images, and videos prevalent in
real-world misinformation. To bridge this gap, we propose **OmniVL-Guard**, a un
ified framework for omni vision-language forgery detection and grounding. In thi
s unified setting, the interplay between diverse modalities and the dual require
ments of simultaneous detection and localization pose significant optimization c
hallenges. Through extensive investigations, we identify a critical difficulty b
ias in this multi-task optimization: the simpler veracity classification task te
nds to dominate the gradients, leading to suboptimal performance in fine-grained
 grounding. To address this imbalance, we first develop a Self-Evolving CoT Gene
ration pipeline to synthesize high-quality reasoning paths, effectively overcomi
ng the cold-start challenge. Building upon this, we propose **A**daptive **R**ew
ard **S**caling **P**olicy **O**ptimization (**ARSPO**). By dynamically modulati
ng reward scales and task weights, ARSPO ensures a balanced joint optimization t
hat prioritizes challenging grounding objectives. Extensive experiments demonstr
ate that OmniVL-Guard significantly outperforms state-of-the-art methods and exh
ibits robust zero-shot generalization across out-of-domain scenarios.
****
CONCUR: High-Throughput Agentic Batch Inference of LLM via Congestion-Based Conc
urrency Control
qiaoling chen, Zhisheng Ye, Tian Tang, Peng Sun, Boyu Tian, Guoteng Wang, Shengg
ui Li, Yonggang Wen, Zhenhua Han, Tianwei Zhang
Batch inference for agentic workloads stresses the GPU key–value (KV) cache in a
 sustained and cumulative manner, often causing severe throughput degradation we
ll before memory capacity is exhausted. We identify this phenomenon as middle-ph
ase thrashing, a previously under-characterized pathology in which cache efficie
ncy collapses as long-lived agents accumulate state over time.

We argue that mitigating this pathology requires moving beyond reactive, request
-level cache management to proactive, agent-level admission control. Drawing ins
piration from congestion control in distributed systems, we view the KV cache as
 a shared resource whose efficient utilization depends on feedback-driven regula
tion. Based on this insight, we present CONCUR, a lightweight control layer that
 regulates agent admission to bound aggregate cache pressure while preserving ex
ecution continuity. CONCUR adapts a cache-aware control algorithm to dynamically
 adjust the number of active agents using runtime cache signals.

Across large models and real-world agent workloads, CONCUR prevents middle-phase
 thrashing and improves batch inference throughput by up to 4.09× on Qwen3-32B a
nd 1.90× on DeepSeek-V3, while remaining compatible with existing LLM serving sy
stems.
****
GRASP: Awakening Latent Spatial Reasoning in LVLMs via Training-free Geometric R
ectification
Jiadong Yan, Ke Zhang, Chenyang Zhao, Shoushan Li, Xizhao Luo
Large Vision-Language Models (LVLMs) exhibit remarkable general capabilities but
 struggle significantly with spatial reasoning tasks. In this paper, we uncover
a critical representation-output misalignment via linear probing: LVLMs correctl

y encode spatial features internally, but generate incorrect results in the fina l text. To address this, we pioneer the Inference-time Geometric Manifold Adapta tion paradigm and propose **GRASP** (**G**eometric **R**ectification for **A**ct ive **S**patial **P**erception), a training-free framework to awaken these laten t capabilities. GRASP employs Manifold Differential Search to identify optimal g eometric counterfactuals, which then drive a dual-level rectification mechanism: Implicit Trajectory Correction to rectify attenuated intrinsic geometric featur es in intermediate decoder layers, and Explicit Distribution Alignment to break the dominance of language priors at the output layer. Extensive experiments span ning diverse architectures (LLaVA, Qwen 2.5/3-VL) and positional encoding paradi gms (1D APE, 2D/3D RoPE) across image and video benchmarks (WhatsUp, VSR, VSI-Be nch) demonstrate that GRASP significantly mitigates spatial hallucinations witho ut parameter updates, achieving accuracy gains of up to 26.1% on image benchmark s and 9.7% on video reasoning tasks, consistently outperforming baseline methods .

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The Geometry of Narrow Fine-Tuning Degradation: Trajectory Lock-in and Spectral Bifurcation

Jia Liu, Jiaxin Luo, Xinhao Qiu, Yixue Hao, Min Chen

Magnitude-based stability proxies such as parameter drift are widely used in nar row-task fine-tuning, yet they do not reliably indicate degradation of broad cap abilities. We identify trajectory lock-in: under fixed training conditions for n arrow adaptation, the joint evolution of task loss and broad generalization coll apses onto a shared low-dimensional degradation curve, so many stabilizers prima rily change the rate of progress along this curve rather than altering the curve itself. This yields a drift paradox, in which comparable Euclidean displacement can still correspond to divergent generalization outcomes. To diagnose the unde rlying structure, we introduce objective-agnostic geometric probes that track th e effective update subspace, together with an online harm signal that reflects c urvature-dominated channeling toward directions associated with broad degradatio n. Finally, we show that escaping lock-in requires a spectral bifurcation, namel y a qualitative reorientation of the update subspace toward softer curvature mod es, thereby improving broad generalization while maintaining matched training pe rformance. We validate these findings across model scales and modalities in narr ow-task settings, and report practical deployment procedures and overhead measur ements.

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Resolving the Timestep Scaling Paradox in Spiking Neural Networks with a Timeste p-Scalable Neuron Model

Binghao Ye, Wenjuan Li, Dengfeng Xue, Bing Li, Weiming Hu, Dong Liang, Kun Shang

Spiking Neural Networks (SNNs) have garnered attention for their biological plau sibility, energy efficiency, and temporal modeling capability. Due to the non-di fferentiability of spike generation, a widely used training method for SNNs is b ackpropagation through time with surrogate gradients, achieving competitive perf ormance with few timesteps. Intuitively, scaling timesteps should improve perfor mance by enriching temporal dynamics. However, we observe timestep scaling parad ox (TSP), a counter-intuitive accuracy degradation when scaling timesteps. We in vestigate TSP and link it to long-term temporal gradient vanishing and weakened cross-timestep dependencies. To address this, we propose the Timestep-Scalable (TS) neuron model. It introduces long-term memory reconsolidation to enhance cros s-timestep information flow and enable effective learning with more timesteps. I n parallel, a temporal forgetting mechanism periodically truncates the accumulat ion path, suppressing excessive temporal buildup and improving training stabilit y. Supported by theoretical analysis and extensive experiments, TS consistently improves performance when scaling timesteps. In addition, it attains state-of-th e-art results on time-series and event-based tasks, while remaining strong on st atic image classification and object detection.

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Time-Series Decomposition as a Standalone Task: A Mechanism-Driven Diagnostic Be nchmark

Zipeng Wu, Jiani Wei, Shiqiao Zhou, Jiajun chen, Fabian Spill, J. W. Andrews
We benchmark time series decomposition as a standalone evaluation task. While decomposition outputs are widely used to interpret trend and periodic structure, their quality is often assessed informally, and no unified benchmark exists for comparing component recovery under controlled generative mechanisms. We introduce a synthetic evaluation suite with explicit trend and cycle taxonomies, a unified interface covering representative decomposition families, and complementary metrics capturing distinct error modes (shape, phase, and spectral fidelity). Across stationary periodic regimes, STL-family methods are near-ceiling; under non-stationary periodicity (frequency drift, regime switching), fixed-period priors induce phase degradation, while subspace/time-frequency methods better preserve seasonal consistency (adaptive spectral methods may require tuning). We further extend the benchmark with a downstream scientific-discovery track—symbolic regression on decomposed components—showing that a decompose-then-regress pipeline materially improves recoverability and reduces expression complexity, linking decomposition quality to structure discovery. Code, result exports, and the web leaderboard are publicly available through the Hugging Face dataset and leaderboard Space.
****
SPR-RAFT: Parameter-Efficient Regression-Aware Fine-Tuning for Biomedical LLM Regression
Yuanlin Yang, Chenhui Li, Xuhao Guo, Anqi Zhang, Hoi Leong Lee, Haodong Liu
Biomedical regression tasks require predicting continuous targets from heterogeneous and unstructured evidence. While Large Language Models (LLMs) provide a robust interface for reasoning over mixed modalities, they are inherently limited by their discrete tokenization and cross-entropy objectives, which lack awareness of numerical proximity. To bridge this gap, we present \textbf{SPR-RAFT}, a parameter-efficient and regression-aware framework that adapts frozen LLMs for high-precision regression. SPR-RAFT introduces a dual-module architecture: a learnable soft prompt that conditions the LLM to route numerical reasoning into a specific latent state, and a lightweight regression head anchored on a dedicated readout token for numerical reasoning consolidation. Crucially, we align these two modalities via a hybrid objective that combines distribution-based text generation with representation-based robust regression. This ensures the model remains both semantically coherent and numerically calibrated. With only about 1.6M trainable parameters ($\sim$0.04\% of a 4B-parameter backbone), SPR-RAFT consistently outperforms prompting strategies, standard fine-tuning, and non-LLM baselines across diverse biomedical benchmarks, including clinical trial duration, biological age estimation, and molecular property prediction.
****
Sparse Relaxed-Lasso Steering: Automatic Sparse Autoencoder Feature Selection for Precise Image Editing
Zongxin Liu, Xiaoyong Xue, Sun Weidi, Shengchao Qin, Lijun Zhang
Precise, training-free image editing with text-to-image diffusion models requires balancing alignment (faithful realization of the target attribute), consistency (preserving non-target content), and quality (maintaining sharp, artifact-free textures).
Sparse autoencoder (SAE) steering offers interpretable, smooth ``slider-like'' control by manipulating SAE feature activations derived from the text encoder; however, existing approaches rely on heuristic feature selection and manual steering-strength tuning, leading to suboptimal trade-offs among the three objectives. We propose Sparse Relaxed-Lasso Steering (SRLS), which casts steering-vector discovery as a convex sparse recovery problem. Exploiting the affine structure of the SAE decoder, SRLS automatically identifies sparse, generalizable support sets via a Lasso objective and then debiases the coefficients using support-restricted ridge refitting. We further replace manual strength tuning with a fixed-budget Bayesian optimization procedure. Across diverse attributes and subjects, SRLS improves the alignment--consistency--quality trade-off over competing methods.
****
L-CUBE: Isolating Long-Context Capacity from Knowledge with Controllable Mutual

Information Scaling
Zhuo Chen, Oriol Mayné i Comas, Zhuotao Jin, Di Luo, Marin Soljacic
Evaluating long-context language models on natural language conflates architectural capacity to capture dependencies with semantic knowledge and vocabulary statistics. When models fail at long contexts, we cannot determine whether failures stem from fundamental architectural limitations or insufficient domain knowledge, preventing clean diagnosis of efficient architectures before expensive training on real data. We introduce **L-CUBE** (Long-Context Utilization Benchmark), a synthetic benchmark that isolates dependency-capturing capacity from semantic knowledge through hierarchical Gaussian sequences with controllable bipartite mutual information scaling. The generator provides exact ground-truth conditionals that scale efficiently to arbitrarily long sequences, enabling unconfounded evaluation via conditional KL divergence rather than perplexity alone. We define *long-context utilization* to measure the amount of available predictive information that models extract as context grows. Experiments across transformers, state space models, and efficient alternatives validate L^2M capacity theory predictions and uncover new phenomena. L-CUBE enables practitioners to test whether a particular design will maintain long-context capability at target sequence lengths before committing to real-data training.
****

Data Augmentation of Contrastive Learning is Estimating Positive-incentive Noise
Hongyuan Zhang, Yanchen Xu, Sida Huang, Xuelong Li
Inspired by the idea of Positive-incentive Noise (*Pi-Noise* or *π-Noise*) that aims at learning the reliable noise beneficial to tasks, we scientifically investigate the connection between contrastive learning and π-noise in this paper. By converting the contrastive loss to an auxiliary Gaussian distribution to quantitatively measure the difficulty of the specific contrastive model under the information theory framework, we properly define the task entropy, the core concept of π-noise, of contrastive learning. It is further proved that the predefined data augmentation in the standard contrastive learning paradigm can be regarded as a kind of point estimation of π-noise. Inspired by the theoretical study, a framework that develops a π-noise generator to learn the beneficial noise (instead of estimation) as data augmentations for contrast is proposed. The designed framework can be applied to diverse types of data and is also completely compatible with the existing contrastive models. From the visualization, we surprisingly find that the proposed method successfully learns effective augmentations.
****

On the Provable Suboptimality of Momentum SGD in Nonstationary Stochastic Optimization
Sharan Sahu, Cameron Hogan, Martin T. Wells
In this paper, we provide a comprehensive theoretical analysis of Stochastic Gradient Descent (SGD) and its momentum variants (Polyak Heavy-Ball and Nesterov) for tracking time-varying optima under strong convexity and smoothness. Our finite-time bounds reveal a sharp decomposition of tracking error into transient, noise-induced, and drift-induced components. This decomposition exposes a fundamental trade-off: while momentum is often used as a gradient-smoothing heuristic, under distribution shift it incurs an explicit drift-amplification penalty that diverges as the momentum parameter β approaches 1, yielding systematic tracking lag. We complement these upper bounds with minimax lower bounds under gradient-variation constraints, proving this momentum-induced tracking penalty is not an analytical artifact but an information-theoretic barrier: in drift-dominated regimes, momentum is unavoidably worse because stale-gradient averaging forces systematic lag. Our results provide theoretical grounding for the empirical instability of momentum in dynamic settings and precisely delineate regime boundaries where vanilla SGD provably outperforms its accelerated counterparts.
****

CBV: Clean-label Backdoor Attacks on Vision Language Models via Diffusion Models
Ji Guo, xiaolong qin, Cencen Liu, Jielei Wang, Jierun Chen, Wenbo Jiang
Vision-Language Models (VLMs) have achieved remarkable success in tasks such as

image captioning and visual question answering (VQA). However, as their applicat
ions become increasingly widespread, recent studies have revealed that VLMs are
vulnerable to backdoor attacks. Existing backdoor attacks on VLMs primarily rely
 on data poisoning by adding visual triggers and modifying text labels, where th
e induced image–text mismatch makes poisoned samples easy to detect.
To address this limitation, we propose the Clean-Label Backdoor Attack on VLMs v
ia Diffusion Models (CBV), which leverages diffusion models to generate natural
poisoned examples via score matching.
Specifically, CBV modifies the score during the reverse generation process of th
e diffusion model to guide the generation of poisoned samples that contain trigg
ered image features.
To further enhance the effectiveness of the attack, we incorporate the textual i
nformation of the triggered images as multimodal guidance during generation. Mor
eover, to enhance stealthiness, we introduce a GradCAM-guided Mask (GM) that res
tricts modifications to only the most semantically important regions, rather tha
n the entire image.
We evaluate our method on MSCOCO and VQA v2 with four representative VLMs, achie
ving over 80\% ASR while preserving normal functionality.
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FlowSeg: Dynamic Semantic Guidance for LLM-Conditioned Segmentation
Zekang Zhang, Guangyu Gao, Youyun Tang, WU CHENGJING, Xiaochao Qu, Chi Harold Li
u, Jianbo Jiao, Yunchao Wei, Luoqi Liu, Ting Liu
LLM-conditioned segmentation has recently advanced rapidly by coupling large lan
guage models with iterative mask generation frameworks. However, we identify a p
ersistent failure mode in current propose-then-select pipelines. Although high-q
uality mask candidates are often generated, the final prediction may fail to mat
ch the given linguistic condition. This failure arises because language semantic
s are typically used as static prompts or post-hoc matching signals, rather than
 participating in the iterative mask generation process. Through systematic anal
ysis, we show that many errors stem from semantic misalignment rather than poor
mask quality. To address this issue, we propose FlowSeg, which introduces dynami
c semantic guidance via a bidirectional semantic flow between intermediate decod
ing states and LLM-derived condition embeddings throughout the generation proces
s. Language conditions actively guide mask refinement at each stage, while condi
tion embeddings are progressively updated by emerging visual evidence. This desi
gn yields semantically grounded mask representations and visually aligned langua
ge conditions, enabling more reliable matching. We further incorporate a lightwe
ight boundary-aware refinement to selectively enhance uncertain regions without
perturbing confident interiors. Extensive experiments on referring expression se
gmentation and reasoning segmentation tasks demonstrate that FlowSeg consistentl
y improves language–mask alignment and achieves state-of-the-art performance. P
roject page: https://zkzhang98.github.io/FlowSeg_page
****

Greedy Coordinate Diffusion: Effective and Semantically Coherent Adversarial Att
acks via Diffusion Guidance
Bohdan Turbal, Blossom Metevier, Max Springer, Aleksandra Korolova
Adversarial attacks on large language models have limited practical impact despi
te extensive research.
Optimization-based attacks such as Greedy Coordinate Gradient (GCG) (Zou et al.
, 2023) produce high-perplexity, incoherent suffixes that existing defenses easi
ly detect (Bengio et al., 2024).

Moreover, attempting to enforce coherence constraints during optimization often
 prevents the attack from successfully eliciting the specific targeted response,
 resulting in low success rates against robust models.
Conversely, attacks that maintain coherence often alter the semantic intent of q
ueries;
when the model complies with these altered queries, responses fail to address th
e adversary's original goal.
In this work, we introduce Greedy Coordinate Diffusion (GCD), a novel framework

that efficiently generates adversarial attacks against safety-aligned models whi
le maintaining low perplexity and high semantic adherence to the adversary's ori
ginal intent.
GCD leverages the generative priors of discrete diffusion language models to gui
de the search for adversarial suffixes that achieve semantic coherence and adher
ence.

Unlike GCG, GCD does not require direct gradient access, allowing it to operate
in a gray-box setting.
We show GCD achieves highest ASR while remaining competitive on response-quality
 scores, and that the constructed adversarial prompts are detected at lower rate
s than other methods by perplexity-based and guard-model filters.
****
Improved Stochastic Optimization of LogSumExp
Egor Gladin, Alexey Kroshnin, Jia-Jie Zhu, Pavel Dvurechensky
The LogSumExp function, dual to the Kullback-Leibler (KL) divergence, plays a ce
ntral role in many important optimization problems,
including entropy-regularized optimal transport (OT) and distributionally robust
 optimization (DRO). In practice, when the number of exponential terms inside th
e logarithm is large or infinite, optimization becomes challenging since computi
ng the gradient requires differentiating every term. We propose a novel convexit
y- and smoothness-preserving approximation to LogSumExp that can be efficiently
optimized using stochastic gradient methods. This approximation is rooted in a s
ound modification of the KL divergence in the dual, resulting in a new f-diver
gence called the *Safe KL divergence*. Our experiments and theoretical analysis
of the LogSumExp-based stochastic optimization, arising in DRO and continuous OT
, demonstrate the advantages of our approach over existing baselines.
****
From Directions to Regions: Decomposing Activations in Language Models via Local
 Geometry
Or David Shafran, Shaked Ronen, Omri Fahn, Shauli Ravfogel, Atticus Geiger, Mor
Geva
Activation decomposition methods in language models are tightly coupled to geome
tric assumptions on how concepts are realized in activation space. Existing appr
oaches search for individual global directions, implicitly assuming linear separ
ability, which overlooks concepts with nonlinear or multi-dimensional structure.
 In this work, we leverage Mixture of Factor Analyzers (MFA) as a scalable, unsu
pervised alternative that models the activation space as a collection of Gaussia
n regions with their *local* covariance structure. MFA decomposes activations in
to two compositional geometric objects: the region's centroid in activation spac
e, and the local variation from the centroid. We train large-scale MFAs for Llam
a-3.1-8B and Gemma-2-2B, and show they capture complex, nonlinear structures in
activation space. Moreover, evaluations on localization and steering benchmarks
show that MFA outperforms unsupervised baselines, is competitive with supervised
 localization methods, and often achieves stronger steering performance than spa
rse autoencoders. Together, our findings position local geometry, expressed thro
ugh subspaces, as a promising unit of analysis for scalable concept discovery an
d model control, accounting for complex structures that isolated directions fail
 to capture.
****
A Solver-Free Training Method for Predict-then-Optimize
Beichen Wan, Mo Liu
We propose a scalable method for training prediction (machine learning) models i
n the predict-then-optimize paradigm, where model outputs serve as coefficients
for a subsequent linear optimization task. Directly minimizing the empirical dec
ision regret is intractable for linear programming and combinatorial optimizatio
n since the decision mapping is piecewise constant, and the gradients are zero a
lmost everywhere. While existing methods address this by smoothing the different
iation process, they suffer from scalability issues, since a computationally exp
ensive solver call is required for every gradient evaluation. To address this, w

e propose a decision-focused learning pipeline based on a measure transformation principle, which yields a new surrogate loss that is completely optimization-solver-free during training. We establish theoretical guarantees, including Fisher consistency and excess risk bounds. Empirically, our method achieves decision quality competitive with state-of-the-art methods while reducing training time by orders of magnitude.
****
SynerMedGen: Synergizing Medical Multimodal Understanding with Generation via Task Alignment
Weiren zhao, DONG Yi, Cheng Chen
Unifying multimodal understanding and generation is a compelling frontier that is beginning to emerge in the medical field.
However, the limited existing unified medical models typically treat understanding and generation as disjoint objectives, lacking a meaningful functional synergy. In this work, we identify and address a critical question in unified medical modeling: what form of “understanding” truly benefits generation. We present SynerMedGen, a unified framework built on the proposed principle of generation-aligned understanding, which synergizes understanding objectives with generation tasks via task alignment. SynerMedGen introduces three generation-aligned understanding tasks and a two-stage training strategy that transfers generation-beneficial representations learned during understanding training to medical image synthesis. Remarkably, even with understanding training alone, our SynerMedGen achieves strong zero-shot performance across 22 medical image synthesis tasks and demonstrates robust generalization. When combined with generation training, SynerMedGen consistently outperforms state-of-the-art specialized medical image synthesis models as well as recent unified medical models. We also release SynerMed, a large-scale dataset of 1M paired synthesis samples and 2M understanding instances for studying understanding–generation synergy. Our project can be accessed at https://github.com/piooip/SynerMedGen.
****
Preference Goal Tuning: Post-Training as Latent Control for Frozen Policies
Guangyu Zhao, Kewei Lian, Haoxuan Ru, Borong Zhang, Haowei Lin, Zhancun Mu, Haobo Fu, QIANG FU, Shaofei Cai, Zihao Wang, Yitao Liang
Goal-conditioned policies enable decision-making models to execute diverse behaviors based on specified goals, yet their downstream performance is often highly sensitive to the choice of instructions or prompts. To bypass the limitations of discrete text prompts, we formulate post-training adaptation as a latent control problem, where the goal embedding serves as a continuous control variable to modulate the behavior of a frozen policy. We propose Preference Goal Tuning (PGT), a framework that optimizes this latent control variable to align the induced trajectory distribution with task preferences. Unlike standard fine-tuning that updates policy parameters, PGT keeps the policy frozen and updates only the latent goal using a trajectory-level preference objective. This approach essentially searches for the optimal conditioning input that maximizes the likelihood of preferred behaviors while suppressing undesirable ones. We evaluate PGT on the Minecraft SkillForge benchmark across 17 tasks. With minimal data, PGT achieves average relative improvements of 72.0\% and 81.6\% on two foundation policies, consistently outperforming expert-crafted prompts. Crucially, by decoupling task alignment (latent goal) from physical dynamics (frozen policy), PGT surpasses full fine-tuning by 13.4\% in out-of-distribution settings, demonstrating superior robustness and generalization.
****
ModernVBERT: Towards Smaller Visual Document Retrievers
Paul Teiletche, Quentin Macé, Max Conti, António Loison, Gautier Viaud, Pierre Colombo, Manuel Faysse
Large-scale document retrieval (search) is key in many modern industrial AI pipelines to ground models with relevant contextual information. Increasingly, Visual Document Retrieval (VDR) models, which directly embed images of document pages, are used as an alternative to text-only retrievers. While these models are historically repurposed generative VLMs fine-tuned for embedding tasks, we revisit

this design choice in this paper and systematically develop strong VDR models fr
om the ground up. Through controlled experiments, we isolate the impact of key t
raining factors such as attention masking, multi-modal data regimes, and contras
tive objectives at all phases of training. Our findings confirm current VDR perf
ormance is constrained by generative modeling, especially in multi-vector settin
gs. Building on these insights, we train ModernVBERT, a 250M-parameter vision-la
nguage encoder that outperforms recent models up to 10 times its size when fine-
tuned on document retrieval tasks. Thanks to its compact design, ModernVBERT ena
bles efficient retrieval inference on CPU hardware, while maintaining competitiv
e performance. Models, code and data are available at https://huggingface.co/Mod
ernVBERT.
****
Where Rectified Flows Leak: Characterising Membership Signals Along the Interpol
ation Path
Thomas Sesmat, Gabriel Meseguer-Brocal, Geoffroy Peeters
Understanding what generative models retain from training data remains challengi
ng, with implications for copyright and privacy.
Beyond verbatim reproduction, models can encode subtler traces of their training
 data that never surface in their outputs yet remain exploitable. We study this
regime for Rectified Flows, which are increasingly used in deployed generative s
ystems.
We analyse the interpolation path $X_\lambda = (1-\lambda)X_0 + \lambda X_1$ tha
t defines the Rectified Flow training.
We show that a gap exists between the reconstruction of train and test data that
 follows a bell-shaped curve over λ, wich accumulates during training, w
hile the validation metrics remain stable.
The signal has a maximum whose location we derive in closed form under Gaussian
assumptions.
We validate these predictions on both audio and images and show that the bell-sh
aped structure is universal, while the peak prediction holds when our assumption
s are satisfied.
As a proof of concept, we exploit this specific λ-resolved structure to
perform a Membership Inference Attack, distinguishing members of the training se
t from non-members.
****
JanusPipe: Efficient Pipeline Parallel Training for Machine Learning Interatomic
 Potentials
Hongyu Wang, Weijian Liu, Hongtao Xu, Yan Wang, Mingzhen Li, Weile Jia, Guangmin
g Tan
Discovering atom-level phenomena requires molecular dynamics (MD) simulations wi
th ab initio accuracy.
Machine learning interatomic potentials (MLIPs) enable stable, high-accuracy MD
simulations, and their models exhibit scaling-law trends similar to large langua
ge models.
However, the lack of scalable and efficient distributed training systems for con
servative MLIPs makes them difficult to scale.
This is because conservative MLIPs inherently follow a double-backward execution
 pattern, which involves computing gradients during the forward pass.
This pattern creates a mismatch with existing distributed training systems, espe
cially for pipeline parallelism.
Therefore, we present JanusPipe, an efficient 3D-parallel (PP/DP/GP) training sy
stem tailored for conservative MLIPs.
It integrates SymFold to enable memory-efficient pipeline parallelism for conser
vative MLIPs, and WaveK to reduce pipeline bubbles by balancing the four-phase c
ompute time.
Experimental results on 32 GPUs show that JanusPipe improves throughput by $1.51
\times$ and $1.45\times$ on average over 1F1B and Hanayo, respectively.
****
Social Hippocampus Memory Learning
Liping Yi, Zhiming Zhao, Kewen Zhu, Xiang Li, Zhiwei Shang, Qinghua Hu

Social learning highlights that learning agents improve not in isolation, but th
rough interaction and structured knowledge exchange with others. When introduced
 into machine learning, this principle gives rise to social machine learning (SM
L), where multiple agents collaboratively learn by sharing abstracted knowledge.
 Federated learning (FL) provides a natural collaboration substrate for this par
adigm, yet existing heterogeneous FL approaches often rely on sharing model para
meters or intermediate representations, which may expose sensitive information a
nd incur additional overhead.
In this work, we propose **SoHip** (**So**cial **Hip**pocampus Memory Learning),
 a memory-centric social machine learning framework that enables collaboration a
mong heterogeneous agents via memory sharing rather than model sharing. SoHip ab
stracts each agent’s individual short-term memory from local representations, co
nsolidates it into individual long-term memory through a hippocampus-inspired me
chanism, and fuses it with collectively aggregated long-term memory to enhance l
ocal prediction. Throughout the process, raw data and local models remain on-dev
ice, while only lightweight memory are exchanged.
We provide theoretical analysis on convergence and privacy preservation properti
es. Experiments on two benchmark datasets with seven baselines demonstrate that
SoHip consistently outperforms existing methods, achieving up to 8.78% accuracy
improvements. The code of SoHip is available at https://github.com/LipingYi/SoHi
p.
****

GI-GCN: Global Interacted Graph Convolutional Networks via Dominant Sets for Gra
ph Classification

Lu Bai, Xinya Qin, Lixin Cui, Ming Li, Hangyuan Du, Ziyu Lyu, Xin Jin

Graph Convolutional Networks (GCNs) are defined based on aggregating the informa
tion of adjacent nodes, that are usually treated as equally important and may li
mit the representational power of existing GCNs. To address this shortcoming, we
 propose a novel Global Interacted Graph Convolutional Network (GI-GCN), that le
verages the solution vectors maintained during the iterative updates of the Domi
nant Set to adaptively characterize the global importance distribution over all
nodes. Specifically, at each convolution layer, this distribution is adopted to
adaptively modulate the importance weights of node features before performing th
e local message passing. We show that this convolution strategy can effectively
capture the highly correlated information between nonadjacent nodes through the
Dominant Set algorithm, not only emphasizing the critical graph-level informatio
n but also enhancing the discriminative power of graph representations. Furtherm
ore, we optimize the memory complexity of the framework, significantly reducing
the memory overhead associated with the global interaction modeling. Experiments
 demonstrate the effectiveness of the proposed GI-GCN model.
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DISSOLVR: An Interpretable and Fast Framework for Aqueous and Organic Solubility
 Prediction

Vansh Ramani, Har Ashish Arora, Dhairya Kuchhal, Sayan Ranu, Tarak Karmakar

High-fidelity solubility prediction is fundamental to pharmaceutical development
 and environmental partitioning, where accurate modeling must couple molecular s
tructure with thermodynamic behavior across diverse chemical environments. Howev
er, recent advancements have been dominated by deep learning architectures that
often sacrifice physical interpretability for predictive power. We challenge thi
s trend by showing that state-of-the-art performance does not require such non-t
ransparent architectures. To address this, we introduce DISSOLVR, a transparent
framework for molecular solubility prediction. In addition, we perform a compreh
ensive literature review and a benchmarking study against various methods. We sh
ow that Dissolvr approaches the aleatoric limit of experimental uncertainty and
achieves OOD generalization through structural invariance derived by mapping mol
ecules to physically-grounded descriptors. Then, we present an LLM-assisted post
-hoc explanation pipeline that bridges the gap between symbolic model artifacts
and chemically grounded narratives. Finally, a comparative benchmark of a survey
 involving 22 expert chemists reveals that expert evaluators provide deep insigh
ts.

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Beyond Distribution Estimation: Simplex Anchored Structural Inference Towards Universal Semi-Supervised Learning
Yaxin Hou, Jun Ma, Hanyang Li, Bo Han, Jie Yu, Yuheng Jia
Semi-supervised learning faces significant challenges in realistic scenarios where labeled data is scarce and unlabeled data follows unknown, arbitrary distributions. We formalize this critical yet under-explored paradigm as Universal Semi-supervised Learning (UniSSL). Existing methods typically leverage unlabeled data via pseudo-labeling. However, they often rely on the idealized assumption of a uniform unlabeled data distribution or require sufficient labeled data to estimate it. In the UniSSL setting, such dependencies lead to numerous erroneous pseudo-labels, thereby triggering representation confusion. Fortunately, we observe that inter-sample relations captured by representations are more reliable than pseudo-labels. Leveraging this insight, we shift our focus to representation-level structural inference to bypass distribution estimation. Accordingly, we propose Simplex Anchored Graph-state Equipartition (SAGE), which captures high-order inter-sample dependencies to establish structural consensus for guiding representation learning. Meanwhile, to mitigate representation confusion, we employ vectors that satisfy a simplex equiangular tight frame to serve as a coordinate frame for guiding inter-class representation separation. Finally, we introduce a weighting strategy based on distribution-agnostic metrics to prioritize reliable pseudo-labels and an auxiliary branch to isolate potentially erroneous pseudo-labels. Evaluations on five standard benchmarks show that SAGE consistently outperforms state-of-the-art methods, with an average accuracy gain of **8.52\%**.
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PDAgent: An LLM-Driven Autonomous Agent Framework Towards *In Silico* Protein Design via Directed Mutation
Song Ouyang, Zhijie Dong, Yong Luo, Kehua Su, Huangxuan Zhao, Miaojing Shi, Bo Du
Computational protein design holds immense promise across diverse domains, but existing approaches face significant challenges: traditional physics-based methods require substantial domain expertise, while emerging deep learning methods often rely on restricted functional ontologies, struggle to bridge the semantic gap between text and protein sequences, or lack closed-loop optimization mechanisms. In this paper, we present PDAgent, an LLM-driven autonomous agent framework that enables *in silico* protein design through template-based directed mutation. Our framework accepts natural language specifications of desired protein properties and employs a ReAct-style reasoning loop comprising five phases: THINK, PLAN, ACT, OBSERVE, and REFLECT. PDAgent integrates template retrieval, conservation-aware mutation strategies, and domain-specific computational tools for property optimization across seven biophysical dimensions. Experiments on 100 diverse protein design tasks demonstrate that PDAgent achieves a 91.86% average constraint satisfaction rate with high structural quality (mean pLDDT 87.69), substantially outperforming both direct LLM generation and specialized deep learning methods. We provide the source code at https://github.com/Gift-OYS/PDAgent.
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BioDynaSpec: Harmonic-Guided Spatio-Spectral Autoregressive Diffusion for Protein Dynamics Generation
Mujie Lin, Yutian Liu, Yudi Guo, Yanzhen Hou, Yiheng Tao, Ruochong Zheng, Kaiwen Cheng, Xin Shan, Youdong Mao, Jie Chen
Generating long-horizon molecular dynamics (MD) is difficult due to error accumulation in time-domain autoregressive models, which causes drift, and fixed step-size constraints on temporal resolution.
We propose **BioDynaSpec**, which reformulates protein dynamics as spatio-spectral generation: **Independent Windowed Fourier Decomposition (IWFD)** decomposes trajectories into window-wise spectral representations, and a generator combines low-to-high frequency autoregression with diffusion denoising to reconstruct continuous motion.
This formulation is motivated by **a local near-equilibrium view of protein dynamics**: after per-window alignment, fluctuations around an anchor conformation a

re better characterized in spectral space, where local mode structure is more explicit than in frame-wise coordinates.
To improve cross-residue and cross-frequency consistency, we introduce **Inter-Residue Frequency Coupling (IRFC)**, a learnable Gaussian distance bias in attention that injects a resonance-inspired structural prior.
On ATLAS, BioDynaSpec improves 250-frame trajectory generation with $R_{250}=1.509$ Å, where R_s denotes the mean per-frame Cα-RMSE over the first s frames after alignment, reducing error by 60.4\% versus MDGEN and 57.2\% versus ProAR, while achieving the best PCA-2D displacement-profile correlation and step wise distribution matching.
For equilibrium conformational sampling, it achieves Root Mean $W_2=1.31$, MD PCA $W_2=0.90$, and Joint PCA $W_2=1.19$, improving over the next best method by 50.03\%, 35.25\%, and 47.58\%, respectively.
It also improves near-equilibrium local-dynamics and covariance consistency, achieving PCA-PSD-LogCorr $=0.817$ and CFRE $=0.989$, corresponding to a 21.9\% gain and a 36.7\% reduction over the next best method, respectively.
The source code is available at https://github.com/Linmj-Judy/BioDynaSpec.git.
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AlienLM: Alienization of Language for API-Boundary Privacy in Black-Box LLMs
Jaehee Kim, Pilsung Kang
Modern LLMs are increasingly accessed via black-box APIs, requiring users to transmit sensitive prompts, outputs, and fine-tuning data to external providers, creating a critical privacy risk at the API boundary.
We introduce AlienLM, a deployable API-only privacy layer that protects text by translating it into an Alien Language via a vocabulary-scale bijection, enabling lossless recovery on the client side. Using only standard fine-tuning APIs, Alien Adaptation Training (AAT) adapts target models to operate directly on alienized inputs.
Across four LLM backbones and seven benchmarks, AlienLM retains over 81% of plaintext-oracle performance on average, substantially outperforming random-bijection and character-level baselines. Under adversaries with access to model weights, corpus statistics, and learning-based inverse translation, recovery attacks reconstruct fewer than 0.22% of alienized tokens.
Our results demonstrate a practical pathway for privacy-preserving LLM deployment under API-only access, substantially reducing plaintext exposure while maintaining task performance.
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Scaling Continual Learning to 300+ Tasks with Bi-Level Routing Mixture-of-Experts
Meng Lou, Yunxiang Fu, Yizhou Yu
Continual learning, especially class-incremental learning (CIL), on the basis of a pre-trained model (PTM) has garnered substantial research interest in recent years. However, how to effectively learn both discriminative and comprehensive feature representations while maintaining stability and plasticity over very long task sequences remains an open problem. We propose $\mathbf{CaRE}$, a scalable $\mathbf{C}$ontinual Le$\mathbf{a}$rner with efficient Bi-Level $\mathbf{R}$outing Mixture-of-$\mathbf{E}$xperts (BR-MoE). The core idea of BR-MoE is a bi-level routing mechanism: a router selection stage that dynamically activates relevant task-specific routers, followed by an expert routing phase that dynamically activates and aggregates experts, aiming to inject discriminative and comprehensive representations into every intermediate network layer.
On the other hand, we introduce a challenging dataset, OmniBenchmark-1K, for CIL performance evaluation on very long task sequences with hundreds of tasks.
Extensive experiments show that CaRE demonstrates leading performance across a variety of datasets and task settings, including commonly used CIL datasets with classical CIL settings (e.g., 5-20 tasks).
To the best of our knowledge, CaRE is the first continual learner that scales to very long task sequences (ranging from 100 to over 300 non-overlapping tasks), while outperforming all baselines by a large margin on such task sequences.
We hope that this work will inspire further research into continual learning ove

r extremely long task sequences.
Code and dataset are publicly released at https://github.com/LMMMEng/CaRE.
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Online Learning and Inference for Cox Proportional Hazards Model Using Renewable Sieve Estimation
Mengtong Hu, Ziyang Gong, Xu Shi, Ling Zhou, Peter Song
Online learning for the Cox model is challenging because its partial likelihood is non-decomposable, with each risk set requiring a summation over all samples. We propose Collaborative Operation of Linked Survival Analysis (COLSA), an online learning framework that replaces the partial likelihood with the full likelihood using sieve approximation of the baseline hazard. The proposed likelihood function is decomposable and eliminates the need to store historical data in memory, enabling efficient online updates. Moreover, COLSA maintains sufficient statistics for a higher-order basis and employs data-driven basis projection to adaptively scale model complexity to the effective sample size. Unlike existing online Cox methods, COLSA achieves asymptotic normality and attains the same statistical efficiency as the pooled-data partial likelihood estimator, without accessing full data and only requiring constant memory. Simulation studies and application to kidney transplant data demonstrate that COLSA outperforms existing online methods and matches the performance of full-data estimation.
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Rethinking the Reranker: Boundary-Aware Evidence Selection for Robust Retrieval-Augmented Generation
Jiashuo Sun, Pengcheng Jiang, Saizhuo Wang, Jiajun Fan, Heng Wang, Siru Ouyang, Ming Zhong, Yizhu Jiao, Chengsong Huang, Xueqiang Xu, Pengrui Han, Peiran Li, Jiaxin Huang, Ge Liu, Heng Ji, Jiawei Han
Retrieval-Augmented Generation (RAG) systems remain brittle under realistic retrieval noise, even when the required evidence appears in the top-K results.
A key reason is that retrievers and rerankers optimize solely for relevance, often selecting either trivial, answer-revealing passages or evidence that lacks the critical information required to answer the question, without considering whether the evidence is suitable for the generator.
We propose \texttt{BAR-RAG}, which reframes the reranker as a boundary-aware evidence selector that targets the generator’s Goldilocks Zone—evidence that is neither trivially easy nor fundamentally unanswerable for the generator, but is challenging yet sufficient for inference and thus provides the strongest learning signal.
\texttt{BAR-RAG} trains the selector with reinforcement learning using generator feedback, and adopts a two-stage pipeline that fine-tunes the generator under the induced evidence distribution to mitigate the distribution mismatch between training and inference.
Experiments on knowledge-intensive question answering benchmarks show that \texttt{BAR-RAG} consistently improves end-to-end performance under noisy retrieval, achieving an average gain of 10.3\% over strong RAG and reranking baselines while substantially improving robustness.
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Active Policy Optimization for Individualized Dosing via Gradient Variance Minimization
Yi Wan, Xin Wang, Huanhuan Chen
In domains such as healthcare and marketing, learning optimal individualized dosing policies to maximize utility is crucial, yet high experimental costs impose strict budget constraints, necessitating efficient active policy learning. Existing active learning methods in causal inference primarily focus on binary treatments and effect estimation, leaving continuous dosing and policy optimization underexplored. To address this gap, we propose an active learning framework tailored for optimal policy learning. Exploiting the inherent structure of dose-response curves, we theoretically show that the policy optimization regret is bounded by the expected posterior gradient variance at the estimated optimal doses. Motivated by this result, we introduce Gradient Variance Active Learning for Individualized Dosing (GVALID), a batch acquisition strategy that greedily selects samp

les to minimize target gradient variance for efficient policy learning. Experiments demonstrate that GVALID achieves superior performance under strict budget constraints.
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Learning Self-Interpretation from Interpretability Artifacts: Training Lightweight Adapters on Vector-Label Pairs
Keenan Pepper, Alex McKenzie, Florin Pop, Stijn Servaes, Martin Leitgab, Michael Vaiana, Judd Rosenblatt, Michael S. A. Graziano, Diogo S de Lucena
Self-interpretation methods prompt language models to describe their own internal states, but remain unreliable due to hyperparameter sensitivity. We show that training lightweight adapters on interpretability artifacts, while keeping the LM entirely frozen, yields reliable self-interpretation across tasks and model families. A scalar affine adapter with just $d_\text{model}+1$ parameters suffices: trained adapters generate sparse autoencoder feature labels that outperform the training labels themselves (70% vs 50% generation scoring at 70B scale), identify topics with 94% recall@1 versus 1% for untrained baselines, and decode bridge entities in multi-hop reasoning that appear in neither prompt nor response, surfacing implicit reasoning without chain-of-thought. The learned bias vector alone accounts for 85% of improvement, and simpler adapters generalize better than more expressive alternatives. Controlling for model knowledge via prompted descriptions, we find self-interpretation gains outpace capability gains from 7B to 72B parameters. Our results demonstrate that self-interpretation improves with scale, without modifying the model being interpreted.
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(Sparse) Attention to the Details: Preserving Spectral Fidelity in ML-based Weather Forecasting Models
Maksim Zhdanov, Ana Lucic, Max Welling, Jan-Willem van de Meent
We introduce \textsc{Mosaic}, a probabilistic weather forecasting model that addresses two sources of spectral degradation in ML-based weather prediction: training to predict the ensemble mean deterministically and compressive encoding creating an information bottleneck. \textsc{Mosaic} combines learned functional perturbations for ensemble forecasting with block-sparse attention, a hardware-aligned formulation that shares keys and values across spatially adjacent queries, enabling each block to dynamically attend to the most relevant regions. By capturing arbitrarily long-range dependencies at linear cost, \textsc{Mosaic} processes high-resolution weather data without compression. On IFS HRES data, \textsc{Mosaic} at 1.5° resolution matches or outperforms models trained on 0.25° data, with individual ensemble members exhibiting near-perfect spectral alignment across all resolved frequencies.
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Localize and Neutralize: Gradient-Guided Token Suppression Against Visual Prompt Injection Attack
Dongpeng Zhang, Ke Ma, Yangbangyan Jiang, Gaozheng Pei, Longtao Huang, Qianqian Xu, Qingming Huang
Adversarial images pose a severe security threat to multimodal large language models through prompt injection. Existing defenses largely lack a principled understanding of the underlying mechanisms and struggle to balance efficiency and fidelity. In this work, we show that successful adversarial attacks do not rely on the entire image uniformly but instead depend on a small subset of critical image tokens. Based on this insight, we propose a defense that first localizes these critical tokens via gradient analysis and then neutralizes them through masking. We show that attribution based on output probabilities fails when adversarial attacks preserve the predicted token. To overcome this limitation, we introduce the Hidden-State Gradient Norm score for adversarial behavior attribution and prove that its ranking is consistent with that of the full adversarial loss gradient, providing a theoretical guarantee for accurate localization. GTM requires only a single forward–backward pass to identify and zero out a small number of high-scoring tokens, effectively disrupting the adversarial attack path. Extensive experiments on prompt injection and multimodal jailbreak attacks demonstrate that our approach reduces attack success rates (ASR) to near zero while preserving

model utility with negligible computational overhead. The code is available at: https://github.com/fish883/GTM-Defense.
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Equivariant Neural Networks for General Linear Symmetries on Lie Algebras
Chankyo Kim, Sicheng Zhao, Minghan Zhu, Tzu-Yuan Lin, Maani Ghaffari
Many scientific and geometric problems exhibit general linear symmetries, yet most equivariant neural networks are built for compact groups or simple vector features, limiting their reuse on matrix-valued data such as covariances, inertias, or shape tensors.
We introduce \textbf{Reductive Lie Neurons (ReLNs)}, an exactly $\mathrm{GL}(n)$-equivariant architecture that natively supports matrix-valued and Lie-algebraic features. ReLNs resolve a central stability issue for reductive Lie algebras by introducing a non-degenerate adjoint (conjugation)-invariant bilinear form, enabling principled nonlinear interactions and invariant feature construction in a single architecture that \textit{transfers across subgroups without redesign}. We demonstrate ReLNs on algebraic tasks with $\mathfrak{sl}(3)$ and $\mathfrak{sp}(4)$ symmetries, Lorentz-equivariant particle physics, uncertainty-aware drone state estimation via joint velocity--covariance processing, learning from 3D Gaussian-splat representations, and EMLP double-pendulum benchmark spanning multiple symmetry groups. ReLNs consistently match or outperform strong equivariant and self-supervised baselines while using substantially fewer parameters and compute, improving the accuracy–efficiency trade-off and providing a practical, reusable backbone for learning with broad linear symmetries.
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Improving ML Attacks on LWE with Data Repetition and Stepwise Regression
Alberto Alfarano, Eshika Saxena, Emily Wenger, Francois Charton, Kristin E. Lauter
ML attacks on Learning with Errors (LWE) with binary or small secrets only succeed on LWE settings with very simple secrets. For example, they can recover secrets with up to three non-zero bits when models are trained on not-reduced LWE data, and three non-zero bits in the ''cruel region'' [9] when BKZ pre-processing is applied. We show that larger training sets and the use of repeated examples in the training data allow the recovery of denser secrets. We empirically observe a power-law relationship between model based attempts to recover the secrets, dataset size and repeated examples. We introduce a stepwise regression technique to recover the ``cool bits'' of the secret. Overall, these techniques allow for the recovery of denser binary secrets: up to Hamming weight 70 (and 8 cruel bits) for dimension 256 $\log_2 q=20$ and 75 (and 7 cruel bits) for dimension 512 $\log_2 q=41$ (vs 33 and 63 Hamming weight and 3 cruel bits in previous works). We also demonstrate our methods' effectiveness on denser ternary secrets, showing a substantial improvement over prior work.
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Securing Multi-Agent Systems Against Corruptions via Node Contribution Backpropagation
Chengcan Wu, Zhixin Zhang, Mingqian Xu, Zeming Wei, Meng Sun
Multi-Agent Systems (MAS) have become a prevalent paradigm for Large Language Model (LLM) applications. However, the complex multi-agent design in MAS introduces unique trustworthiness concerns: adversarial agents can inject misleading information that propagates contagiously through the system, corrupting benign agents and leading to false outputs. Existing graph-based defenses model agents as nodes and communications as edges, yet are limited to static-graph defenses. In this paper, we propose a dynamic defense paradigm that models MAS communication as a signed directed acyclic graph and computes each agent's contribution to the final decision via backward propagation, enabling accurate identification and isolation of malicious agents to secure multi-agent task collaboration. Experimental results in complex and dynamic MAS environments demonstrate that our method notably outperforms existing MAS defense mechanisms, providing an effective guardrail for trustworthy MAS deployment. Our code is available at https://github.com/ChengcanWu/BPD.
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A Theoretical Framework for Modular Learning of Robust Generative Models
Corinna Cortes, Mehryar Mohri, Yutao Zhong
Training large-scale generative models is resource-intensive and relies heavily on heuristic dataset weighting. We address two fundamental questions: Can we train Large Language Models (LLMs) modularly, combining small, domain-specific experts to match monolithic performance, and can we do so robustly for *any* data mixture, eliminating heuristic tuning? We present a theoretical framework for *modular* generative modeling where a set of pre-trained experts are combined via a gating mechanism. We define the space of normalized gating functions $\mathcal{G}_{1}$ and formulate the problem as a minimax game to find a single robust gate that minimizes divergence to the worst-case data mixture. We prove the existence of such a robust gate using Kakutani's fixed-point theorem and show that modularity acts as a strong regularizer, with generalization bounds scaling with the lightweight gate's complexity. Furthermore, we prove that this modular approach can theoretically outperform models retrained on aggregate data, with the gap characterized by the Jensen-Shannon Divergence. Finally, we introduce a scalable Stochastic Primal-Dual algorithm and a *Structural Distillation* method for efficient inference. Empirical results on synthetic and real-world datasets confirm that our modular architecture effectively mitigates gradient conflict and can robustly outperform monolithic baselines.
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SLASH the Sink: Sharpening Structural Attention Inside LLMs
Yiming Liu, Bin Lu, Xinbing Wang, Chenghu Zhou, Meng Jin
Large Language Models (LLMs) show remarkable semantic understanding but often struggle with structural understanding when processing graph topologies in a serialized format. Existing solutions rely on training external graph-based adapters or fine-tuning, which incur high costs and lost generalizability. In this work, we investigate the internal mechanisms of LLMs and present a critical finding: *LLMs spontaneously reconstruct the graph's topology internally*, evidenced by a distinct "sawtooth" pattern in their attention maps that structurally aligns with the "token-level adjacency matrix". However, this intrinsic structural understanding is diluted by the attention sink. We theoretically formalize this dilution as a representation bottleneck, stemming from a fundamental conflict: the model's anisotropic bias, essential for language tasks, suppresses the topology-aware local aggregation required for graph reasoning. To address this, we propose a training-free solution, named **S**tructura**L** **A**ttention **SH**arpening (SLASH), which amplifies this internal structural understanding via a plug-and-play attention redistribution. Experiments on pure graph tasks and molecular prediction validate that SLASH delivers significant and consistent performance gains across diverse LLMs.
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D-FUSEr: Diverse Failure, Unified Success via Error-Distribution Shaping in LLM Reasoning
David D. Baek, Andrew Estornell, Yichi Zhang, Muhammad Faaiz Taufiq, Jean-Francois Ton, Jie Mei, Tao Wang
Test-time scaling methods such as majority vote aggregation and iterative refinement (e.g., self-reflection or multi-agent inference) improve reasoning performance by leveraging multiple solution samples.
However, their efficacy depends not only on raw performance, but critically on the distribution of errors across samples.
When errors concentrate, (a) aggregation accuracy degrades, as the majority vote may select a shared mistake, and (b) confidence in common mistakes may suppress exploration in iterative refinement.
We argue that improving correctness alone is not sufficient to mitigate these issues; to this end, we propose to explicitly shape error distributions to improve aggregation.
First, we introduce a theoretically grounded \textbf{diverse failure reward} that incentivizes calibrated disagreement within model errors.
We prove that this reward directly optimizes majority-vote accuracy: policies achieving higher reward attain higher expected majority-vote performance, and vice

versa.
We further show that this theoretical property generalizes to iterative refinement.
Second, we introduce \textbf{anti-votes}, in which the model predicts the most common mistake alongside its solution, allowing probability mass on dominant errors to be explicitly reweighted.
We identify conditions under which anti-votes are guaranteed to improve majority-vote accuracy.
Empirically, across three model families of varying sizes and four benchmarks, we show that both approaches substantially improve majority vote and iterative refinement performance without degrading single-sample accuracy.
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Near-Optimal and Efficient First-Order Algorithm for Multi-Task Learning with Shared Linear Representation
Shihong Ding, Fangyu Du, Cong Fang
Multi-task learning (MTL) has emerged as a pivotal paradigm in machine learning by leveraging shared structures across multiple related tasks. Despite its empirical success, the development of likelihood-based efficiently solvable algorithms—even for shared linear representations—remains largely underdeveloped, primarily due to the non-convex structure intrinsic to matrix factorization. This paper introduces a first-order algorithm that jointly learns a shared representation and task-specific parameters, with guaranteed efficiency. Notably, it converges in $\widetilde{\mathcal{O}}(1)$ iterations and attains a \emph{near-optimal} estimation error of $\widetilde{\mathcal{O}}(dk/(TN))$, \emph{improving} over existing likelihood-based methods by a factor of k, where d, k, T, N denote input dimension, representation dimension, task count, and samples per task, respectively. Our results justify that likelihood-based first-order methods can efficiently solve the MTL problem.
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TruthRL: Incentivizing Truthful LLMs via Reinforcement Learning
Zhepei Wei, Xiao Yang, Kai Sun, Jiaqi Wang, Rulin Shao, Jingxiang Chen, Mohammad Kachuee, Teja Gollapudi, Yiwei Liao, Nicolas SCHEFFER, Rakesh Wanga, Anuj Kumar, Yu Meng, Wen-tau Yih, Xin Luna Dong
While large language models (LLMs) have demonstrated strong performance on factoid question answering, they are still prone to hallucination and untruthful responses, particularly when tasks demand information outside their parametric knowledge. Indeed, truthfulness requires more than accuracy---models must also recognize uncertainty and abstain when unsure to avoid hallucinations. This presents a fundamental challenge for existing methods: approaches that optimize for accuracy often amplify hallucinations, while those that encourage abstention can become overly conservative, sacrificing correct answers. Both extremes ultimately compromise truthfulness. In this work, we present TruthRL, a general reinforcement learning (RL) framework that directly optimizes the truthfulness of LLMs. Specifically, we implement TruthRL using GRPO with a simple yet effective ternary reward that distinguishes correct answers, hallucinations, and abstentions. It incentivizes models to reduce hallucinations not only by providing correct responses, but also by enabling abstention when uncertain, thereby improving truthfulness. Extensive experiments across four knowledge-intensive benchmarks show that TruthRL significantly reduces hallucinations (e.g., 43.5\% $\rightarrow$ 19.4\%) and improves truthfulness (e.g., 5.3\% $\rightarrow$ 37.2\%), with consistent gains across various backbone models. Analysis shows that the improvement of TruthRL arises from enhanced capability of LLMs to recognize their knowledge boundary, hence avoiding being overly conservative as the baselines are.
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A Consensus Anchor-guided Hypergraph Framework for Incomplete Multi-view Clustering
Yipin Hu, Yanxi Liu, Fangxi Liu, Yanwei Yu, Lei Meng, Jie Wen, Guoqing Chao
Handling large-scale incomplete multi-view data poses a significant challenge in unsupervised representation learning. While anchor-based strategies have alleviated computational burdens, they typically rely on shallow bipartite graphs rest

ricted to pairwise relations, failing to capture complex high-order correlations among samples. Furthermore, existing methods often treat observed and missing instances indiscriminately, ignoring the distributional shifts that lead to systematic bias in consensus anchor learning. To address these limitations, we propose a novel framework tailored for scalability and robustness, termed Hypergraph-Augmented Incomplete Multi-View Clustering (HA-IMVC). Unlike traditional approaches, HA-IMVC constructs a consensus anchor-guided hypergraph that explicitly models group-wise interactions, thereby preserving structural integrity even under high missing rates. Crucially, we incorporate a dual-adaptive reweighting mechanism that calibrates importance at both the view and sample levels. This strategy adaptively penalizes severely incomplete samples to mitigate bias while harmonizing inconsistent views. Extensive experiments on diverse benchmarks demonstrate that HA-IMVC achieves superior clustering accuracy and maintains high efficiency, even in scenarios characterized by severe data incompleteness.

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Towards Rule-Based Knowledge Sharing in Federated Learning

Zixuan Qin, Qi Shen, Liu Yang, Qilong Wang, Qinghua Hu

Federated learning often face both data and model heterogeneity, with the latter often more challenging. Architectural differences yield incompatible representation, making the knowledge-sharing carrier central to heterogeneous collaboration. Using proxy model enables distillation-based collaboration but incurs high communication and computation costs. Prototype-based carriers are lighter yet cause semantic confusion when incompatible features are mixed. Therefore, we propose rule-based federated learning (RFL) that shares interpretable, class-discriminative rules to enable heterogeneous collaboration, avoid feature confusion, and keep communication lightweight. RFL uses a rule network to unify clients' decision features and collaborates at the rule level, avoiding forcible averaging of incompatible representations. RFL selects sparse, high-coverage, beneficial rules for broadcasting, compressing shared knowledge into an interpretable class-rule set and reducing communication and computation costs. Each client selectively activates only rules relevant to its local classes, mitigating negative transfer while preserving personalization. Across heterogeneous settings, RFL achieves a better accuracy–communication trade-off.

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Belief Dynamics Reveal the Dual Nature of In-Context Learning and Activation Steering

Eric Bigelow, Daniel Wurgaft, YingQiao Wang, Noah Goodman, Tomer Ullman, Hidenori Tanaka, Ekdeep Singh Lubana

Large language models (LLMs) can be controlled at inference time through prompts (in-context learning) and internal activations (activation steering). Different accounts have been proposed to explain these methods, yet their common goal of controlling model behavior raises the question of whether these seemingly disparate methodologies can be seen as specific instances of a broader framework. Motivated by this, we develop a unifying, predictive account of LLM control from a Bayesian perspective. Specifically, we posit that both context- and activation-based interventions impact model behavior by altering its belief in latent concepts: steering operates by changing concept priors, while in-context learning leads to an accumulation of evidence. This results in a closed-form Bayesian model that is highly predictive of LLM behavior across context- and activation-based interventions in a set of domains inspired by prior work on many-shot in-context learning. This model helps us explain prior empirical phenomena - e.g., sigmoidal learning curves as in-context evidence accumulates--while predicting novel ones--e.g., additivity of both interventions in log-belief space, which results in distinct phases such that sudden and dramatic behavioral shifts can be induced by slightly changing intervention controls. Taken together, this work offers a unified account of prompt-based and activation-based control of LLM behavior, and a methodology for empirically predicting the effects of these interventions.

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Reparameterization Proximal Policy Optimization

Hai Zhong, Xun Wang, Zhuoran Li, Longbo Huang

By leveraging differentiable dynamics, Reparameterization Policy Gradient (RPG) achieves high sample efficiency. However, current approaches are hindered by two critical limitations: the under-utilization of computationally expensive dynamics Jacobians and inherent training instability. While sample reuse offers a remedy for under-utilization, no prior principled framework exists, and naive attempts risks exacerbating instability. To address these challenges, we propose Reparameterization Proximal Policy Optimization (RPO). We first establish that under sample reuse, RPG naturally optimizes a PPO-style surrogate objective via Backpropagation Through Time, providing a unified framework for both on- and off-policy updates. To further ensure stability, RPO integrates a clipped policy gradient mechanism tailored for RPG and employs explicit Kullback-Leibler divergence regularization. Experimental results demonstrate that RPO maintains superior sample efficiency and consistently outperforms or achieves state-of-the-art performance across diverse tasks.

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Causal Feature Learning via Generalized Rayleigh Quotients

Liang Cao, Jun Wan, Yan Qin, Weide Liu

Extracting causally meaningful features from time-series data is fundamental for robust machine learning under distribution shifts. In process monitoring, existing methods struggle to maintain detection performance when operating conditions change. Current approaches capture either temporal causal relationships or cross-environment invariance, but not both simultaneously. We propose Causal Feature Learning (CFL), a unified framework that jointly optimizes for temporal relevance and environment mean invariance. CFL formulates feature extraction as a generalized Rayleigh-quotient problem, maximizing correlation with target variables while penalizing sensitivity to environment-dependent mean shifts. Theoretical analysis establishes conditions under which CFL identifies a mean-invariant predictive subspace. Experiments on the Tennessee Eastman Process demonstrate that CFL achieves 93.69\% average fault detection rate, outperforming 15 baseline methods and validating the benefit of jointly capturing both aspects of causality.

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DisPPO: Quantile-Based Distributional Reinforcement Learning for Large Language Models

Zhijian Zhou, Long Li, Xuan Zhang, Zongkai Liu, Yanting Miao, Yuchen Liu, Deshu Chen, Ke Li, Xing Sun, Ruoxi Jiang, Xiaoyu Tan, Chao Qu, Yuan Qi

Reinforcement Learning (RL) has become a cornerstone for enhancing the reasoning capabilities of Large Language Models (LLMs). However, standard actor-critic methods, such as PPO, rely on scalar value functions that estimate only the expectation of cumulative returns. This reduction inherently discards higher-order statistical information (e.g., variance and multimodality), leading to inaccurate value estimation and suboptimal credit assignment in complex tasks. While Distributional RL offers a solution by modeling the full return distribution, its application to LLMs remains challenging due to the computational intractability of value-based operations over large vocabularies and the instability and memory burden of off-policy replay mechanisms. In this paper, we propose DisPPO, a novel on-policy framework that seamlessly integrates non-parametric quantile regression into PPO. Theoretically, we prove that our distributional update operator---composed of the λ-return Bellman operator and quantile projection---is a contraction mapping in the Wasserstein metric, guaranteeing convergence to a unique fixed point. Empirically, we evaluate DisPPO using Llama and Qwen models across diverse benchmarks, including mathematical reasoning and Text-to-SQL generation. DisPPO consistently outperforms standard PPO and recent group-based baselines in both Pass@1 and Pass@k metrics, demonstrating that distributional critics provide a richer, more robust learning signal for large-scale reasoning models.

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Neural Concept Verifier: Scaling Prover-Verifier Games via Concept Encodings

Berkant Turan, Suhrab Asadulla, David Steinmann, Kristian Kersting, Wolfgang Stammer, Sebastian Pokutta

While *Prover-Verifier Games* (PVGs) offer a promising path toward verifiability in nonlinear classification models, they have not yet been applied to complex i

nputs such as high-dimensional images.
Conversely, expressive *concept encodings* effectively allow to translate such d ata into interpretable concepts but are often utilised in the context of low-cap acity linear predictors.
In this work, we push towards real-world verifiability by combining the strength s of both approaches. We introduce *Neural Concept Verifier (NCV)*, a unified fr amework combining PVGs for formal verifiability with concept encodings to handle complex, high-dimensional inputs in an interpretable way. NCV achieves this by utilizing recent minimally supervised concept discovery models to extract struct ured concept encodings from raw inputs. A *prover* then selects a subset of thes e encodings, which a *verifier*, implemented as a nonlinear predictor, uses excl usively for decision-making.
Our evaluations show that NCV outperforms classic concept-based models and pixel -based PVG classifier baselines on high-dimensional, logically complex datasets and helps mitigate shortcut behavior. Overall, we demonstrate NCV as a promising step toward concept-level, verifiable AI.
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Rethinking Pretraining Data Detection for LLMs: From Local to Global
Chenye Ke, Yan Zhuang, Zirui Liu, Qi Liu
The advancements of Large Language Models (LLMs) are primarily attributed to mas sive pretraining data, which also introduces risks like privacy leakage and data contamination. Therefore, it is crucial to determine whether an LLM has been tr ained on a given target text. Existing detection methods primarily rely on local statistics of isolated tokens (e.g., those with the lowest probabilities), negl ecting the probability dynamics during the token generation process. In this pap er, we shift the detection paradigm from a local token to a global sequence pers pective, grounded in the core intuition that memorized sequences exhibit volatil ity patterns distinct from those generated via inference. We propose Adaptive En tropic Convolutional Analysis (AECA), a framework that conceptualizes the probab ility sequence as a dynamic signal, integrating calibration with convolutional f iltering to effectively capture memorization signals. Extensive experiments demo nstrate that AECA surpasses previous methods by up to 1.5\% in average AUC on th e WikiMIA benchmark, with its advantage being particularly pronounced in long-te xt scenarios.
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Rethinking Neural Network Learning Rates: A Stackelberg Perspective
Sihan Zeng, Sujay Bhatt, Sumitra Ganesh
Neural networks are typically trained with a single learning rate across all lay ers. While recent empirical evidence suggests that assigning layer-specific lear ning rates can accelerate training, a principled understanding of the conditions and mechanisms under which non-uniform learning rates are beneficial remains li mited. In this work, we investigate non-uniform learning rates through the lens of Stackelberg optimization. Specifically, we demonstrate that training neural n etworks with a smaller learning rate for the body layers and a larger learning r ate for the final layer can be interpreted as a two-time-scale alternating gradi ent descent algorithm applied to a Stackelberg reformulation of the original obj ective. We establish finite-time convergence guarantees for the algorithm under broad conditions that accommodate constraint sets and non-smooth activation func tions. Beyond convergence, we identify two mechanisms by which non-uniform learn ing rates can outperform uniform learning rates: (i) we show that certain proble m instances induce a Stackelberg objective with stronger optimization structure than the original objective, yielding faster convergence to globally optimal sol utions, (ii) our numerical analysis reveals that the Stackelberg objective can e xhibit substantially sharper local curvature, especially in early training, whic h leads to more informative gradients and learning acceleration. Experiments in supervised learning and reinforcement learning support our findings.
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Aligning Tree-Search Policies with Fixed Token Budgets in Test-Time Scaling of L LMs
Sora Miyamoto, Daisuke Oba, Naoaki Okazaki

Tree-search decoding is an effective form of test-time scaling for large language models (LLMs), but real-world deployment often imposes a fixed per-query token budget that varies across settings. Existing tree-search policies are largely budget-agnostic, treating the budget merely as a termination condition, thereby risking late-stage over-branching or premature termination. We propose Budget-Guided MCTS (BG-MCTS), a tree-search decoding algorithm that aligns its search policy with the remaining token budget: it starts with broad exploration, then prioritizes refinement and answer completion as the remaining budget decreases while reducing late-stage branching from shallow nodes. BG-MCTS consistently outperforms budget-agnostic tree-search baselines across inference budgets on mathematical reasoning benchmarks and an additional physics reasoning benchmark with open-weight LLMs.

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Zero-Shot 3D Question Answering via Hierarchical View-to-Token Transportation

Dongsheng Wang, Dawei Su, Hui Huang

Recently, zero-shot 3D scene understanding via 2D Vision-Language Models (VLMs) has gained increasing research interest due to their promising spatial reasoning capabilities. Typically, multiple 2D views are sampled from a 3D point cloud and fed into pre-trained VLMs to answer a given question. This paradigm highlights the critical role of input context quality and raises the challenge of retaining as many task-relevant 3D details as possible under a limited input budget. We propose \texttt{KeyVT}, a hierarchical approach for input context collection at both the view and token levels. Specifically, we combine pixel features with camera parameters and assess view importance based on both semantic content and geometric position, resulting in spatially consistent and task-relevant views. Furthermore, we address redundancy among patches across selected views by identifying representative tokens under the optimal transport (OT) framework, where view tokens and key tokens are formulated as two discrete distributions in the embedding space. These key tokens are expected to cover all view features by minimizing the OT distance. We evaluate our framework on three widely used benchmarks, demonstrating significant improvements over existing tuning-free methods and performance comparable to training-based approaches.

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Towards Understanding Generalization of Federated Adversarial Learning: Perspective of Algorithmic Stability

Yongkang Yang, Chang Cao, Ke Zhang, Han Li, Hong Chen, Rushi Lan

Federated Adversarial Learning (FAL) enhances model robustness by integrating adversarial training into the federated learning framework. Despite recent advances proposing efficient FAL algorithms, existing work has mainly focused on convergence properties, with limited understanding of their generalization capabilities. To address this, we present the unified theoretical framework for analyzing FAL generalization through the lens of algorithmic stability. We first analyze general FAL algorithms based on stochastic gradient descent (SGD) and derive perturbation-dependent generalization bounds, which reveal that stronger adversarial attacks can lead to degraded generalization. To mitigate the impact of adversarial perturbations, we leverage Moreau envelope optimization and establish a perturbation-independent bound, demonstrating its efficacy in simultaneously enhancing both robustness and generalization. Finally, we extend our analysis to the practical black-box setting, demonstrating that zeroth-order optimization techniques can effectively maintain both robustness and generalization even without local gradient access.

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Benchmarking LLM-Assisted Blue Teaming via Standardized Threat Hunting

Yuqiao Meng, Luoxi Tang, Feiyang Yu, Xi Li, Guanhua Yan, Ping Yang, Zhaohan Xi

As cyber threats continue to grow in scale and sophistication, blue team defenders increasingly require advanced tools to proactively detect and mitigate risks. Large Language Models (LLMs) offer promising capabilities for enhancing threat analysis. However, their effectiveness in real-world blue team threat-hunting scenarios remains insufficiently explored. This paper presents CYBERTEAM, a benchmark designed to guide LLMs in blue teaming practice. CYBERTEAM constructs a stan

dardized workflow in two stages. First, it models realistic threat-hunting workf
lows by capturing the dependencies among analytical tasks from threat attributio
n to incident response. Next, each task is addressed through a set of operationa
l modules tailored to its specific analytical requirements.
This transforms threat hunting into a structured sequence of reasoning steps, wi
th each step grounded in a discrete operation and ordered according to task-spec
ific dependencies. Guided by this framework, LLMs are directed to perform threat
-hunting tasks through modularized steps. Overall, CYBERTEAM integrates 30 tasks
 and 9 operational modules to guide LLMs through standardized threat analysis. W
e evaluate both leading LLMs and state-of-the-art cybersecurity agents, comparin
g CYBERTEAM against open-ended reasoning strategies. Our results highlight the i
mprovements enabled by standardized design, while also revealing the limitations
 of open-ended reasoning in real-world threat hunting.
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DeFacto: Counterfactual Thinking with Images for Enforcing Evidence-Grounded and
 Faithful Reasoning

Tianrun Xu, Haoda Jing, Ye Li, Yuquan Wei, Jun Feng, Guanyu Chen, Haichuan Gao,
Tianren Zhang, Jing Liu, Feng Chen

Recent advances in multimodal language models (MLLMs) have made \emph{thinking w
ith images} a dominant paradigm for multimodal reasoning. However, existing meth
ods still fail to ensure \emph{evidence--answer consistency}, where correct answ
ers must be supported by correct visual evidence. To address this issue, we prop
ose \textit{DeFacto}, a counterfactual reasoning framework that explicitly align
s visual evidence with final answers. Our approach integrates three complementar
y training paradigms: (i) positive, (ii) counterfactual, and (iii) random-maskin
g. We further develop a language-guided evidence construction pipeline that auto
matically localizes question-relevant regions and generates counterfactual varia
nts, resulting in \textbf{DeFacto-100K}. Building on this dataset, we train MLLM
s with GRPO-based reinforcement learning and design three complementary rewards
to promote correct answering, structured reasoning, and consistent evidence sele
ction. Moreover, we introduce \textbf{DeFacto-1.5K}, a human-annotated benchmark
 for systematically evaluating evidence-grounded consistency beyond answer accur
acy. Experiments on diverse benchmarks demonstrate that \textit{DeFacto} substan
tially improves both answer accuracy and evidence--answer consistency over stron
g baselines. The code and datasets are available at \url{https://github.com/tinn
el123666888/defacto}.
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SALAAD: Sparse And Low-Rank Adaptation via ADMM for Large Language Model Inferen
ce

Hao Ma, Melis Ilayda Bal, Liang Zhang, Bingcong Li, Niao He, Melanie Zeilinger,
Michael Muehlebach

Modern large language models are increasingly deployed under compute and memory
constraints, making flexible control of model capacity a central challenge. Whil
e sparse and low-rank structures naturally trade off capacity and performance, e
xisting approaches often rely on heuristic designs that ignore layer and matrix
heterogeneity or require model-specific architectural modifications. We propose
SALAAD, a plug-and-play framework applicable to different model architectures th
at induces sparse and low-rank structures during training. By formulating struct
ured weight learning under an augmented Lagrangian framework and introducing an
adaptive controller that dynamically balances the training loss and structural c
onstraints, SALAAD preserves the stability of standard training dynamics while e
nabling explicit control over the evolution of effective model capacity during t
raining. Experiments across model scales show that SALAAD substantially reduces
memory consumption during deployment while achieving performance comparable to a
d-hoc methods. Moreover, a single training run yields a continuous spectrum of m
odel capacities, enabling smooth and elastic deployment across diverse memory bu
dgets without the need for retraining.
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Suppress and Diversify: Refining Robust Pathways for Corruption Robustness

Jiangang Yang, Wenhui Shi, Xiaoran Xu, Wenyue Chong, Luqing Luo, Jing Xing, Jian

Liu
Model robustness against natural image corruptions is essential for safety-critical applications. While existing methods primarily focus on implicit representation learning, we provide the first systematic exploration of computational pathways to explicitly characterize internal robustness. We identify a progressive decay of robust features across network layers and establish a functional dependency between the prevalence of these features and model performance. To exploit these insights, we propose Suppress and Diversify (S\&D), a non-intrusive refinement approach that enhances robustness by dynamically selecting robust pathways and diversifying them through symmetry-preserving transformations. S\&D is architecture-agnostic, parameter-free, and incurs zero test-time overhead. Extensive evaluations across eight benchmarks demonstrate that S\&D consistently improves performance across multiple vision tasks, diverse backbones, and complex real-world scenarios, highlighting its broad efficacy and scalability.
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Towards Disentangled Preference Optimization Dynamics: Suppress the Loser, Preserve the Winner
Wei Chen, Yubing Wu, Junmei Yang, Delu zeng, Qibin Zhao, John Paisley, Min Chen, Zhou Wang
Preference optimization is widely used to align large language models (LLMs) with human preferences. However, many margin-based methods also suppress the chosen response when they try to suppress the rejected one, and there is no general way to prevent this across different objectives.

We address this issue with a unified **incentive-score decomposition** of preference optimization, revealing that different objectives share the same local update directions and differ only in their scalar weights.

This decomposition provides a common framework for analyzing objectives that were previously studied in separate settings.

Building on this decomposition, by analyzing the dynamics of the chosen/rejected likelihoods, we identify the **disentanglement band** (DB), a simple, testable condition that tells us when training can follow the desired path: suppress the loser while preserving the winner, possibly after an early stage.

Using the DB, we propose **reward calibration** (RC), a plug■and■play method that adaptively rebalances the updates for chosen and rejected likelihoods to satisfy the DB, without redesigning the base objective.

Empirical results show that RC leads to more disentangled dynamics, with better downstream performance observed across several settings. Our code is available at https://github.com/IceyWuu/DisentangledPreferenceOptimization.
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Towards Atoms of Large Language Models
Chenhui Hu, Pengfei Cao, Yubo Chen, Kang Liu, Jun Zhao
The fundamental representational units (FRUs) of large language models (LLMs) remain undefined, limiting further understanding of their underlying mechanisms. In this paper, we introduce ***Atom Theory*** to systematically define, evaluate, and identify such FRUs, which we term atoms. Building on the atomic inner product (AIP), a non-Euclidean metric that captures the underlying geometry of LLM representations, we formally define atoms and propose two key criteria for ideal atoms: faithfulness ($ R^2 $) and stability ($ q^{\star} $). We further prove that atoms are identifiable under threshold-activated sparse autoencoders (TSAEs). Empirically, we uncover a pervasive representation shift in LLMs and demonstrate that the AIP corrects this shift to capture the underlying representational geometry, thereby grounding Atom Theory. We find that two widely used units, neurons and features, fail to qualify as ideal atoms: neurons are faithful ($ R^2=1 $) but unstable ($ q^{\star}=0.5 $%), while features are more stable ($ q^{\star}=68.2 $%) but unfaithful ($ R^2=48.8 $%). To find atoms of LLMs, leveraging atom identifiability under TSAEs, we show via large-scale experiments that reliable atom identification occurs only when the TSAE capacity matches the data scale. Guided by this insight, we identify FRUs with near-perfect faithfulness ($ R^2=99.9 $%) and stability ($ q^{\star}=99.8 $%) across layers of Gemma2-2B, Gemma2-9B, and Llama3.1-8B, satisfying the criteria of ideal atoms statist

ically. Further analysis confirms that these atoms align with theoretical expect
ations and exhibit substantially higher monosemanticity. Overall, we propose and
 validate Atom Theory as a foundation for understanding the internal representat
ions of LLMs.
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A Distributional View for Visual Mechanistic Interpretability: KL-Minimal Soft-C
onstraint Principle
Guancheng Zhou, Yisi Luo, Zhengfu He, Zhenyu Jin, Xuyang Ge, Wentao Shu, Deyu Me
ng, Xipeng Qiu
Most current paradigms in visual mechanistic interpretability (MI) remain confin
ed to interpreting internal units of the vision model via heuristic methods (e.g
., top-K activation retrieval or optimization with regularization). In this wo
rk, we establish a theoretical distributional view for visual MI, which models t
he influence of a feature activation on the natural image distribution, thereby
formulating a Kullback-Leibler (KL)-minimal optimization problem to model the MI
 task. Under this framework, statistical biases are identified within previous M
I paradigms, which reveal that they may either be perceptually uninterpretable t
o humans (i.e., deviate from the natural image distribution), or mechanistically
 unfaithful to the vision models (i.e., unable to activate model features). To r
esolve the biases under the distributional view, we propose a model with a KL-mi
nimal soft-constraint principle for visual MI that theoretically balances interp
retability and faithfulness. We realize this principle via energy-guided diffusi
on posterior sampling.
Extensive experiments validate the theoretical soundness of the proposed distrib
utional view and demonstrate the practical effectiveness of our paradigm on the
DINOv3 vision model.
The code is available at https://github.com/SII-ZhouGC/EnergyDPS.
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Adaptive Contracts for Cost-Effective AI Delegation
Eden Saig, Tamar Garbuz, Ariel D. Procaccia, Inbal Talgam-Cohen, Jamie Tucker-Fo
ltz
When organizations delegate text generation tasks to AI providers via pay-for-pe
rformance contracts, expected payments rise when evaluation is noisy. As evaluat
ion methods become more elaborate, the economic benefits of decreased noise are
often overshadowed by increased evaluation costs. In this work, we introduce ada
ptive contracts for AI delegation, which allow detailed evaluation to be perform
ed selectively after observing an initial coarse signal in order to conserve res
ources. We make three sets of contributions: First, we provide efficient algorit
hms for computing optimal adaptive contracts under natural assumptions or when c
ore problem dimensions are small, and prove hardness of approximation in the gen
eral unstructured case. We then formulate alternative models of randomized adapt
ive contracts and discuss their benefits and limitations. Finally, we empiricall
y demonstrate the benefits of adaptivity over non-adaptive baselines using quest
ion-answering and code-generation datasets.
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Evaluating Object-Centric Models beyond Object Discovery
Krishnakant Singh, Simone Schaub-Meyer, Stefan Roth
Object-centric learning (OCL) aims to learn structured scene representations tha
t support compositional generalization and robustness to out-of-distribution (OO
D) data. However, OCL models are often not evaluated regarding these goals. Inst
ead, most prior work focuses on evaluating OCL models solely through object disc
overy and simple reasoning tasks, such as probing the representation via image c
lassification. We identify two
limitations in existing benchmarks: (1) They provide limited insights on the rep
resentation usefulness of OCL models, and (2) localization and representation us
efulness are assessed using disjoint metrics. To address (1), we use instruction
-tuned VLMs as evaluators, enabling scalable benchmarking across diverse VQA dat
asets to measure how well VLMs leverage OCL representations for complex reasonin
g tasks. To address (2), we introduce a unified evaluation task and metric that
jointly assess localization (where) and representation usefulness (what), thereb

y eliminating inconsistencies introduced by disjoint evaluation. Finally, we include a simple multi-feature reconstruction baseline as a reference point.
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From Basis to Basis: Gaussian Particle Representation for Interpretable PDE Operators
Zhihao Li, Yu Feng, Zhilu Lai, Wei Wang
Learning fluid PDE dynamics has increasingly benefited from neural operators and Transformer-based models, but their latent states often remain opaque, and sample-wise attention can be costly at high resolutions. We propose the \emph{Gaussian Particle Operator} (GPO), a basis-to-basis neural operator that represents fields with a learned \emph{Gaussian basis}. Each atom carries explicit geometric parameters, including centers, anisotropic scales, and weights, yielding a compact, mesh-agnostic, and directly visualizable intermediate representation. GPO operates in modal space: learned Gaussian modal windows perform Petrov--Galerkin measurements, and a PG Gaussian Attention module couples the resulting modes globally before scattering them back to the spatial domain. This design is resolution-agnostic, scales near-linearly with the number of samples for a fixed modal budget, and naturally applies to irregular geometries and 3D domains. We separately evaluate Gaussian-basis reconstruction and operator prediction. Across standard PDE benchmarks and real reanalysis datasets, GPO achieves competitive accuracy against neural-operator and Transformer baselines, ranking best on most tasks and close to the best on the remainder. The learned particles and modal couplings further provide representation-level interpretability, positioning GPO as a useful intermediate representation rather than a one-to-one decomposition of physical structures. The code is available at \url{https://github.com/lizhihao2022/GPO}.
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DISCO: Mitigating Bias in Deep Learning with Conditional Distance Correlation
Emre Kavak, Tom Nuno Wolf, Christian Wachinger
Dataset bias often leads deep learning models to exploit spurious correlations instead of task-relevant signals. We introduce the Standard Anti-Causal Model (SAM), a unifying causal framework that characterizes bias mechanisms and yields a conditional independence criterion for causal stability. Building on this theory, we propose DISCO$_m$ and sDISCO, efficient and scalable estimators of conditional distance correlation that enable independence regularization in gradient-based models. Across six diverse datasets, our methods consistently outperform or are competitive in existing observed bias mitigation approaches, while requiring fewer hyperparameters and scaling seamlessly to multi-bias scenarios. This work bridges causal theory and practical deep learning, providing both a principled foundation and effective tools for robust prediction. Source Code: https://github.com/yakamoz5/DISCO.
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SAGE: A Dataflow-Native Framework for Modular, Controllable, and Transparent LLM-Augmented Reasoning
Jun Liu, Peilin Liu, Ruicheng Zhang, Zhang Senlei, Yanbo Chen, Ziao Wang, Jinyun Yang, mingqi wang, Shuhao Zhang, Xiaofei Liao, Hai Jin
Large Language Model (LLM) applications increasingly execute as end-to-end inference pipelines that couple generation with
retrieval, stateful memory, context refinement, and tool use under strict tail-latency and Service-Level Objective (SLO)
constraints. Today, these stages are often stitched together as RPC-connected services, obscuring
cross-stage queueing and interference and limiting pipeline-level compilation and resource sharing. We
present SAGE (Streaming-Augmented Generative Execution), a
full-stack system that treats inference pipelines as first-class compilation targets. SAGE exposes
pipelines as declarative dataflows and compiles them into distributed execution plans with bounded-queue
backpressure. It integrates vector search, streaming semantic state, structured

memory, and refinement as
operators with explicit resource/state contracts, enabling operator-level diagnosis of tail behavior. SAGE integrates pluggable generation and embedding backends and provides a unified control plane for
engine management, batching, and admission under mixed workloads. On a 16-node cluster, SAGE sustains
16 requests/s at >700 tokens/request with 1 ms median scheduling overhead, and achieves near-linear scale-out to 16 nodes (11.4$\times$ throughput at 16 nodes), and reduces P99 latency by 57\% under multi-pipeline contention versus simultaneous admission.
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Smoothing Slot Attention Iterations and Recurrences
Rongzhen Zhao, Wenyan Yang, Juho Kannala, Joni Pajarinen
Slot Attention (SA) lies at the heart of mainstream Object-Centric Learning (OCL).
Image features can be aggregated into object-level representations by SA iteratively refining cold-start query slots.
For video, such aggregation proceeds by SA recurrently shared across frames, with queries cold-started on the first frame while transitioned from the previous frame's slots thereafter.
However, cold-start queries lack sample-specific cues thus hindering precise aggregation on image or video's first frame;
Non-first frames' queries are already sample-specific thus requiring aggregation transforms different from the first frame.
We address these issues with our SmoothSA:
(1) To smooth SA iterations on image or video's first frame, we preheat cold-start queries with rich input-feature information, by a tiny module self-distilled inside OCL;
(2) To smooth SA recurrences across video's first and non-first frames, we differentiate the homogeneous aggregation transforms by using full and single iterations respectively.
Comprehensive experiments on object discovery, recognition and visual reasoning validate our method's effectiveness.
Further visual analyses illuminate the underline mechanisms.
Our source code, model checkpoints and training logs are provided on https://github.com/Genera1Z/SmoothSA.
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AlphaRouter: Token-level Routing Between SLM and LLM with Reinforcement Learning and Tree Search
Siteng Liao, Yuzhu Liang, Hengzhong Rao, Xizhao Luo, Tian Wang
SLM-LLM routing accelerates generation by strategically invoking LLMs for critical tokens. However, existing methods typically train routers to mimic the LLM, capping performance at the reference trajectory's limit. In this work, we demonstrate that the SLM-LLM collaborative inference space offers a richer solution set, yielding correct answers even when the LLM fails. To exploit this, we propose **AlphaRouter**, a routing framework learning optimal collaborative inference paths via a *search and iterative update* paradigm. Formulating routing as a Markov Decision Process, we introduce Collaborative Inference Tree Search (CITS) to explore the solution space. To tackle the severe credit assignment challenge posed by sparse rewards, we propose Tree-Advantage Policy Optimization (TAPO) to optimize the routing policy. By leveraging counterfactual advantages within the tree structure, TAPO effectively attributes the final reward to specific routing decisions, stabilizing training without dense supervision. Extensive experiments show that AlphaRouter advances the Pareto frontier of accuracy-efficiency trade-offs by exploiting better inference trajectories in the collaborative space. Code is available at
https://github.com/StripeLife0217/AlphaRouter.
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Simple Approximation and Derivative Free Inference-Time Scaling for Diffusion Models via Sequential Monte Carlo on Path Measures

Chenyang Wang, Weizhong Wang, Yinuo Ren, Jose Blanchet, Yiping Lu
Diffusion-based generative models increasingly rely on inference-time guidance, adding a drift term or reweighting mixture of experts, to improve sample quality on task-specific objectives. However, most existing techniques require repeated score or gradient evaluations, introducing bias, high computational overhead, or both. We introduce $\texttt{URGE}$, Unbiased Resampling via Girsanov Estimation, a derivative-free inference-time scaling algorithm that performs path-wise importance reweighting via a Girsanov change of measure. Instead of computing gradient-based particle weights in previous work, $\texttt{URGE}$ attaches a simple multiplicative weight to each simulated trajectory and periodically resamples. No score, no Hessian, and no PDE evaluation is required. We establish an equivalence between path-wise and particle-wise SMC: the Girsanov path weight admits a backward conditional expectation that recovers the previous particle-level weights, guaranteeing that both schemes produce the same unbiased terminal law. Empirically, $\texttt{URGE}$ outperforms existing inference-time guidance baselines on synthetic tests and diffusion-model benchmarks, achieving better generation quality, while being significantly simpler to implement and fully gradient-free.
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PartCo: Part-Level Correspondence Priors Enhance Category Discovery
Fernando Julio Cendra, Kai Han
Generalized Category Discovery (GCD) aims to identify both known and novel categories within unlabeled data by leveraging a set of labeled examples from known categories. Existing GCD methods primarily depend on semantic labels and global image representations, often overlooking the detailed part-level cues that are crucial for distinguishing closely related categories. In this paper, we introduce PartCo, short for Part-Level Correspondence Prior, a novel framework that enhances category discovery by incorporating part-level visual feature correspondences. By leveraging part-level relationships, PartCo captures finer-grained semantic structures, enabling a more nuanced understanding of category relationships. Importantly, PartCo seamlessly integrates with existing GCD methods without requiring significant modifications. Our extensive experiments on multiple benchmark datasets demonstrate that PartCo significantly improves the performance of current GCD approaches, outperforming most existing methods by bridging the gap between semantic labels and part-level visual compositions, thereby setting new benchmarks for GCD.
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Ambiguous Strategic Classification
Ivri Hikri, Nir Rosenfeld
A common assumption in strategic classification is that the classifier is public knowledge. However, it remains unclear whether, and why, a system would choose to commit to full disclosure. We study a setting in which regulation requires the system to disclose some, but not all, of the information. This induces a learning task in which the learner must jointly optimize the classifier and the uncertainty surrounding it. To this end, we adopt from robust mechanism design the notion of *ambiguity*, which in our setting allows the learner to reveal a set or range of possible classifiers, while privately choosing which of them to ultimately realize. We investigate how ambiguity affects the learning task, develop efficient algorithms for computing best-responses and training, and empirically explore strategic learning and its outcomes in this novel setting and using our approach.
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CoDA-Bench: Can Code Agents Handle Data-Intensive Tasks?
Yuxin Zhang, Ju Fan, Meihao Fan, Shaolei Zhang, Xiaoyong Du
Advanced agents are increasingly demonstrating the potential to operate as autonomous engineers, creating a growing demand for evaluation benchmarks that capture the complexity of real-world development. Such environments typically involve both complex code and large-scale data (i.e., file system). However, existing benchmarks usually evaluate code-centric or data-centric capabilities in isolation, leaving a clear gap with real development scenarios. In this paper, we bridge this gap by introducing CoDA-Bench, the first benchmark to jointly evaluate code

and data intelligence in a data-intensive environment. We construct a data-intensive Linux sandbox based on the Kaggle ecosystem (containing hundreds of datasets), where agents must actively explore complex file hierarchies to identify relevant resources and generate code for data-driven analytical tasks. CoDA-Bench comprises 1,009 tasks spanning 31 communities, with each task environment containing an average of 980 files, simulating realistic data scale and noise. Evaluations of advanced agents reveal that even top-performing systems struggle to effectively integrate data discovery with code execution, achieving a success rate of only 61.1\%. These results highlight a substantial gap in current agentic capabilities for data-intensive tasks and point to promising directions for future research.
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General and Efficient Steering of Unconditional Diffusion Models
Qingsong Wang, Mikhail Belkin, Yusu Wang
Steering diffusion models toward conditions unseen during training typically requires either retraining with conditional inputs or per-step gradient computations, both of which incur substantial computational overhead. We present NA-RFM, a general recipe for efficiently steering unconditional diffusion models without gradient guidance during inference, enabling fast controllable generation.
The method combines two offline-computed signals: *noise alignment*, a high-noise correction from PCA statistics of the target examples and the full data, and *Recursive Feature Machine (RFM) activation steering*, which learns a target-discriminative direction from labeled forward-process activations.
During sampling, noise alignment provides coarse control at high noise, while the RFM direction is reused over intermediate/late timesteps through lightweight activation edits.
Experiments on CIFAR-10, ImageNet, CelebA, and fine-grained bird species show improved target accuracy over gradient-based post-hoc guidance baselines, improved FID on the class-guidance benchmarks, and substantial inference speedups.
Code: https://github.com/isotrivial/na-rfm.
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Plan Then Action: High-Level Planning Guidance Reinforcement Learning for LLM Reasoning
Zhihao Dou, Qinjian Zhao, Zhongwei Wan, Zhang Dinggen, Weida Wang, Benteng Chen, Towsif Raiyan, Qingtao Pan, Yang Ouyang, Chaoda Song, Zhiqiang Gao, Shufei Zhang, Sumon Biswas
Large language models (LLMs) demonstrate strong reasoning abilities via Chain-of-Thought (CoT), but their token-level generation encourages local decisions and lacks global planning, often leading to redundant or inaccurate reasoning. Existing methods, such as tree-based search and reinforcement learning (RL), attempt to address this issue but incur high computational costs and still struggle to produce reliable reasoning trajectories.
To address these challenges, we propose Plan-Then-Action Enhanced Reasoning with Group Relative Policy Optimization (PTA-GRPO), a two-stage framework designed to jointly improve high-level planning and fine-grained CoT reasoning.
Specifically, in the first stage, a given LLM is responsible for summarizing CoT reasoning into compact high-level guidance, which is then leveraged for supervised fine-tuning. Then, we introduce a guidance-aware reinforcement learning method that jointly optimizes the final output and the quality of guidance, enhancing reasoning effectiveness.
We evaluate PTA-GRPO on ten reasoning benchmarks across mathematics and natural sciences, using five diverse base models spanning multiple data modalities. The results show that PTA-GRPO consistently delivers stable and significant improvements across models and tasks, demonstrating strong effectiveness and generalization.
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LoRDO: Distributed Low-Rank Optimization with Infrequent Communication
Andrej Jovanovic, Alex Iacob, Mher Safaryan, Ionut-Vlad Modoranu, Lorenzo Sani, William F. Shen, Xinchi Qiu, Dan Alistarh, Nicholas D. Lane
Distributed training of foundation models via $\texttt{DDP}$ is limited by inter

connect bandwidth. While infrequent communication strategies reduce synchronizat
ion frequency, they remain bottlenecked by the memory and communication requirem
ents of optimizer states. Low-rank optimizers can alleviate these constraints; h
owever, in the local-update regime, workers lack access to the full-batch gradie
nts required to compute low-rank projections, which degrades performance. We pro
pose $\texttt{LoRDO}$, a principled framework unifying low-rank optimization wit
h infrequent synchronization. We first demonstrate that, while global projection
s based on pseudo-gradients are theoretically superior, they permanently restric
t the optimization trajectory to a low-rank subspace. To restore subspace explor
ation, we introduce a full-rank quasi-hyperbolic update. $\texttt{LoRDO}$ achie
ves near-parity with low-rank $\texttt{DDP}$ in language modeling and downstream
 tasks at model scales of 125M--720M, while reducing communication by $\appr
ox10\times$. Finally, we show that $\texttt{LoRDO}$ improves performance even m
ore in very low-memory settings with small rank/batch size.

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Parameter Manifold Purification

Jiacong Hu, Jinxun Wu, Shengxuming Zhang, Shunyu Liu, Haofei Zhang, Mingli Song,
 Zunlei Feng

Deep models are vulnerable to performance degradation caused by various factors,
 such as imbalanced samples, inaccurate labels, and backdoor attacks. However, e
xisting optimization methods that address these issues are typically designed in
 a scenario- or architecture-specific manner, and each optimization often requir
es costly training. To this end, inspired by image denoising, we propose paramet
er purification as a new paradigm for model performance optimization. Parameter
purification attributes performance degradation to the contamination of model pa
rameters and aims to recover clean parameters from corrupted ones in a manner an
alogous to image denoising. To purify parameters with massive scale and complex
structure, we further introduce a novel parameter manifold purification method.
In this framework, high-dimensional and complex parameters are first viewed as m
anifolds embedded in a high-dimensional space, and are then partitioned into nes
ted local parameter-cluster manifolds via a proposed parameter clustering strate
gy. Meanwhile, to remove parameter redundancy while preserving global parameter
information, we propose an implicit manifold auto-encoder along with a parameter
-cluster discrepancy loss to learn low-dimensional representations of parameter-
cluster manifolds. Finally, an implicit conditional diffusion model is applied t
o denoise the low-dimensional parameter manifolds, progressively restoring clean
 parameters. Extensive experiments under three representative scenarios that cau
se model performance degradation demonstrate that parameter manifold purificatio
n can accurately and completely purify corrupted parameters of unseen models, an
alogous to denoising unseen images, and rapidly improve model performance.

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BlueCodeAgent: A Blue Teaming Agent Powered by Automated Red Teaming for CodeGen
 AI

Chengquan Guo, Yuzhou Nie, Chulin Xie, Zinan Lin, Wenbo Guo, Bo Li

Existing research on CodeGen AI security mainly focuses on red teaming, which ai
ms to uncover vulnerabilities and risks in AI-generated code. However, progress
on the blue teaming side remains limited, as effective defenses require a deep s
ecurity analysis of given tasks and edge cases. To fill in this gap, we propose
BlueCodeAgent, an end-to-end blue teaming agent powered by automated red teaming
. Our red teaming component generates diverse risky instances, providing effecti
ve edge cases and guidance for the subsequent blue teaming process. Our blue tea
ming agent then conducts multi-level defense, leveraging these red teaming examp
les to detect previously seen and unseen risk scenarios through constitution sum
marization and dynamic code analysis. Our evaluation across four representative
code-related tasks–bias instruction detection, malicious instruction detection,
vulnerable code detection, and prompt injection detection–shows that BlueCodeAge
nt achieves significant gains over diverse baselines. In particular, for vulnera
bility detection tasks, BlueCodeAgent integrates dynamic analysis to effectively
 reduce false positives, a challenging problem as base models tend to be over-co
nservative. Overall, with GPT-4o as the base model, BlueCodeAgent achieves an av

erage F1 score improvement of 14.7% across four tasks compared to directly promp ting the model, attributed to its ability to summarize actionable constitutions and perform dynamic analysis. Our code and data are publicly available at https: //github.com/1mocat/BlueCodeAgent.
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Time Series, Vision, and Language: Exploring the Limits of Alignment in Contrast ive Representation Spaces
Pratham Yashwante, Rose Yu
The Platonic Representation Hypothesis posits that learned representations from models trained on different modalities converge to a shared latent structure of the world. However, this hypothesis has largely been examined in vision and lang uage, and it remains unclear whether time series participate in such convergence . We first examine this in a trimodal setting and find that independently pretra ined time series, vision, and language encoders exhibit near-orthogonal geometry in the absence of explicit coupling. We then apply post-hoc alignment by traini ng projection heads over frozen encoders using contrastive learning, and analyze the resulting representations with respect to geometry, scaling behavior, and d ependence on information density and input modality characteristics. Our investi gation reveals that overall alignment in contrastive representation spaces impro ves with model size, but this alignment is asymmetric: time series align more st rongly with visual representations than with text, and images can act as effecti ve intermediaries between time series and language. We further see that richer t extual descriptions improve alignment only up to a threshold; training on denser captions does not lead to further improvement. Analogous effects are observed f or visual representations. Our findings shed light on considerations for buildin g multimodal systems involving non-conventional data modalities beyond vision an d language. Code and data are available at https://github.com/Rose-STL-Lab/tvl-a lignment.
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Flowers: A Warp Drive for Neural PDE Solvers
Till Muser, Alexandra Spitzer, Matti Lassas, Maarten V. de Hoop, Ivan Dokmani■
We introduce Flower, a neural architecture for learning PDE solution operators b uilt entirely from multihead warps. Aside from pointwise channel mixing and a mu ltiscale scaffold, Flowers use no Fourier multipliers, no dot-product attention, and no convolutional mixing. Each head predicts a displacement field and warps the mixed input features. Motivated by physics and computational efficiency, dis placements are predicted pointwise, without any spatial aggregation, and nonloca lity enters *only* through sparse sampling at source coordinates, *one* per head . Stacking warps in multiscale residual blocks yields Flowers, which implement a daptive, global interactions at linear cost. We theoretically motivate this desi gn through three complementary lenses: flow maps for conservation laws, waves in inhomogeneous media, and a kinetic-theoretic continuum limit. Flowers achieve e xcellent performance on a broad suite of 2D and 3D time-dependent PDE benchmarks , particularly flows and waves. A compact 17M-parameter model consistently outpe rforms Fourier, convolution, and attention-based baselines of similar size, whil e a 150M-parameter variant improves over recent transformer-based foundation mod els with much more parameters, data, and training compute.
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Let EEG Models Learn EEG
Yifan Wang, Yijia Ma, Wen Li, Chenyu You
High-fidelity EEG generation is critical for alleviating data scarcity and addre ssing privacy constraints in large-scale neural modeling.
Despite recent progress, most existing approaches formulate EEG generation via d iscrete denoising objectives, which inadequately reflect the inherently continuo us temporal dynamics and spectral structure of neural activity.
As a result, these methods often struggle to preserve long-range temporal depend encies and exhibit mismatches in the spectral and temporal structure of the gene rated signals.
In this work, we argue that effective EEG generation requires models that operat e directly on the continuous evolution of neural signals.

We introduce Just EEG Transformer (JET), a generative framework based on conditi
onal flow matching that models EEG as raw sequences evolving along continuous tr
ajectories.
By learning a smooth vector field that transports noise to the EEG data distribu
tion, JET captures temporal continuity and transient dynamics without relying on
 discretized denoising schemes or domain-specific representations.
To ensure that the learned dynamics remain consistent with key properties of EEG
 signals, we introduce principled constraints that preserve spectral structure,
temporal stationarity, and signal-level statistics.
Across three large-scale benchmarks, JET consistently achieves state-of-the-art
performance, reducing TS-FID by over 40\% compared to strong baselines.
Extensive analyses show that JET captures key structural properties of neural dy
namics, providing a scalable and principled approach to EEG generation.
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SCNS: Continual Personalization of Diffusion Models via Submodular Concept Neuro
n Selection

Zijie Peng, Enneng Yang, Yifei Cheng, Hongliang Yuan, Fei Ma, Xiaochun Cao, Li S
hen

Custom diffusion models (CDMs) have demonstrated impressive success in visual pe
rsonalization tasks by enabling the generation of user-specific concepts. Howeve
r, existing CDMs typically assume that personalized concepts are static and rely
 on costly model merging or sequential updates that are prone to catastrophic fo
rgetting as new concepts are introduced. To address these limitations, we propos
e a Submodular Concept Neuron Selection method (SCNS), to solve CDMs with contin
ual personalized concepts, which formulates continual personalization as a const
rained submodular optimization problem to select a minimal yet sufficient set of
 concept-specific neurons under diminishing returns. SCNS combines a Facility Lo
cation-based coverage objective to suppress semantic redundancy, a Fisher-weight
ed risk proxy to protect previously learned concepts, and a cost-aware greedy ru
le to balance stability and plasticity with extreme sparsity. Extensive experime
nts demonstrate that SCNS achieves state-of-the-art performance in image alignme
nt and anti-forgetting, while enabling fusion-free continual personalization by
modifying only 0.41% of the total parameters for each concept on average. Our im
plementation is available at SCNS.
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Adaptive Node Feature Selection for Graph Neural Networks

Madeline Navarro, Ali Azizpour, Santiago Segarra

We propose an adaptive node feature selection approach for graph neural networks
 (GNNs) that identifies and removes unnecessary features during training. The ab
ility to measure how features contribute to model output is key for interpreting
 decisions and reducing dimensionality by eliminating unhelpful variables. Howev
er, graph-structured data introduces complex dependencies that may be unsuited t
o classical feature importance metrics. Inspired by this, we present a data-, mo
del-, and task-agnostic method that determines relevant features during training
 based on changes in validation performance upon permuting feature values. We th
eoretically motivate our approach by characterizing how the relationships betwee
n node data and graph structure influences GNN performance. Empirically, we show
 that (i) our highly general approach rivals the performance of tailored feature
 selection approaches that exploit prior assumptions; (ii) we return meaningful
feature importance scores well before the GNN is fully trained; and (iii) our sc
ores demonstrably extract relevant properties that inform feature importance for
 various graph learning settings.
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Self-Calibrated Consistency can Fight Back for Adversarial Robustness in Vision-
Language Models

Jiaxiang Liu, Jiawei Du, Xiao Liu, Shangyang Li, Songchen Ma, Changshuo Wang, Pr
ayag Tiwari, Mingkun Xu

Pre-trained vision-language models (VLMs) such as CLIP have demonstrated strong
 zero-shot capabilities across diverse domains, yet remain highly vulnerable to a
dversarial perturbations that disrupt image-text alignment and compromise reliab

ility.
Existing defenses typically rely on adversarial fine-tuning with labeled data, l imiting their applicability in zero-shot settings.
In this work, we identify two key weaknesses of current CLIP adversarial attacks —lack of semantic guidance and vulnerability to view variations—collectively ter med semantic and viewpoint fragility. To address these challenges, we propose Se lf-Calibrated Consistency (SCC), an effective test-time defense. SCC consists of two complementary modules:
Semantic consistency, which leverages soft pseudo-labels from counterattack warm -up and multi-view predictions to regularize cross-modal alignment and separate the target embedding from confusable negatives; and
Spatial consistency, aligning perturbed visual predictions via augmented views t o stabilize inference under adversarial perturbations.
Together, these modules form a plug-and-play inference strategy.
Extensive experiments on 22 benchmarks under diverse attack settings show that S CC consistently improves the zero-shot robustness of CLIP while maintaining accu racy, and can be seamlessly integrated with other VLMs for further gains.
These findings highlight the great potential of establishing an adversarially ro bust paradigm from CLIP, with implications extending to broader VLMs such as Bio MedCLIP.
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AutoMat: Physics-Guided Agentic Reasoning for Solving Ill-Posed Inverse Microsco py Problems

Yaotian Yang, Yiwen Tang, Yizhe Chen, Xiao Chen, Jiangjie Qiu, Hao Xiong, Haoyu Yin, Zhiyao Luo, Yifei Zhang, Sijia Tao, Wentao Li, Qinghua Zhang, Yuqiang Li, W anli Ouyang, Bin Zhao, Xiaonan Wang, Fei Wei

Reconstructing atomistic crystal structures from a single noisy STEM projection is an ill-posed inverse problem: multiple lattices can explain similar contrast, and purely feed-forward models cannot verify physical validity. We present **Au toMat**, a failure-aware agentic *controller* that performs inference-time hypot hesis search with *closed-loop verification* to convert Scanning Transmission El ectron Microscopy (STEM) images into simulation-ready crystal structures and dow nstream properties. AutoMat composes perception and physics modules—pattern-adap tive denoising, physics-guided template retrieval *as a state-dependent auxiliar y branch*, symmetry-constrained atomic reconstruction, and MLIP-based relaxation /validation—and triggers rollback-and-retry when verification fails. For systema tic evaluation, we introduce **STEM2Mat-Bench**, a benchmark dataset containing 450+ annotated samples. Performance is assessed using lattice root-mean-square d eviation (RMSD), formation energy mean absolute error (MAE), and structure match ing accuracy. Results demonstrate that AutoMat outperforms existing approaches i ncluding SOTA models, specialized domain tools, and closed-source multimodal lar ge models. This work establishes a direct pathway from microscopic characterizat ion to atomic-scale modeling, addressing a fundamental challenge in materials sc ience.
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Text Has Curvature

Karish Grover, Hanqing Zeng, Yinglong Xia, Christos Faloutsos, Geoffrey J. Gordo n

Does natural language text have an intrinsic curvature? Language is increasingly modeled in curved geometries—hyperbolic spaces for hierarchy, mixed-curvature m anifolds for compositional structure—yet a basic scientific question remains unr esolved: what does curvature mean for text itself, in a way that is native to la nguage rather than an artifact of the embedding space we choose? We argue that t ext does indeed have curvature, and show how to detect it, define it, and use it . To this end, we propose Texture, a text-native, word-level discrete curvature signal, and make three contributions. (a) Existence: We provide empirical and th eoretical certificates that semantic inference in natural corpora is non-flat. (b) Definition: We define Texture as a signed two-axis curvature of the word-in-c ontext belief field—the differential of reconciliation between prefix and suffix —measuring, via a debiased Schrödinger transport divergence, whether adding cont

ext from one side contracts the semantic effect of context from the other side (focus, positive) or expands it into competing continuations (fan-out, negative). (c) Utility: Texture is actionable: it serves as a general-purpose measurement and control primitive enabling geometry without geometric training; we instantia te it on two representative tasks, improving long-context inference through curv ature-guided compression and retrieval-augmented generation through curvature-gu ided routing. Together, our results establish a text native curvature paradigm, making Texture practically useful.
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TimeChat-Captioner: Scripting Multi-Scene Videos with Time-Aware and Structural Audio-Visual Captions
Linli Yao, Yuancheng Wei, Yaojie Zhang, Lei Li, Xinlong Chen, Feifan Song, Ziyue Wang, Kun Ouyang, Yuanxin Liu, Lingpeng Kong, Qi Liu, Pengfei Wan, Kun Gai, Yua nxing Zhang, Xu Sun
This paper proposes Omni Dense Captioning, a novel task designed to generate con tinuous, fine-grained, and structured audio-visual narratives with explicit time stamps. To ensure dense semantic coverage, we introduce a six-dimensional struct ural schema to create "script-like" captions, enabling readers to vividly imagin e the video content scene by scene, akin to a cinematographic screenplay. To fac ilitate research, we construct OmniDCBench, a high-quality, human-annotated benc hmark, and propose SodaM, a unified metric that evaluates time-aware detailed de scriptions while mitigating scene boundary ambiguity. Furthermore, we construct a training dataset, TimeChatCap-42K, and present TimeChat-Captioner-7B, a strong baseline trained via SFT and GRPO with task-specific rewards. Extensive experim ents demonstrate that TimeChat-Captioner achieves state-of-the-art performance, surpassing Gemini-2.5-Pro, while its generated dense descriptions significantly boost downstream capabilities in audio-visual reasoning (DailyOmni and WorldSens e) and temporal grounding (Charades-STA). All datasets, models, and code are pub licly available at https://github.com/yaolinli/TimeChat-Captioner.
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Divide and Learn: Multi-Objective Combinatorial Optimization at Scale
Esha Singh, Dongxia Wu, Chien-Yi Yang, Tajana Rosing, Rose Yu, Yian Ma
Multi-objective combinatorial optimization seeks Pareto-optimal solutions over e xponentially large discrete spaces, yet existing methods sacrifice generality, s calability, or theoretical guarantees. We reformulate it as an online learning p roblem over a decomposed decision space, solving position-wise bandit subproblem s via adaptive expert-guided sequential construction. This formulation admits re gret bounds of $O(d\sqrt{T \log T})$ depending on subproblem dimensionality \(d\ ) rather than combinatorial space size. On standard benchmarks, our method achie ves 80--98\% of specialized solvers performance while achieving two to three or ders of magnitude improvement in sample and computational efficiency over Bayesi an optimization methods. On real-world hardware-software co-design for AI accele rators with expensive simulations, we outperform competing methods under fixed e valuation budgets. The advantage grows with problem scale and objective count, e stablishing bandit optimization over decomposed decision spaces as a principled alternative to surrogate modeling or offline training for multi-objective optimi zation.
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Expressive Graph Neural Networks via Equivariant Use of Noise
Xiyuan Wang, Muhan Zhang
Expressivity has been a major focus in the design of Graph Neural Networks (GNNs), yet a significant gap persists between theoretical universal expressivity and practical performance. While many expressive GNNs are efficient and achieve str ong results, they often focus on specific graph properties and lack theoretical expressivity for general graph tasks. Conversely, theoretically universal-expres sive models often suffer from high computational costs or poor generalization, l imiting their real-world applicability. To bridge this gap, we introduce Equivar iant Noise GNNs (ENGNNs), a framework that utilizes random noise features to enh ance the expressivity of GNNs. Crucially, unlike prior methods that naively use noise, we enforce equivariance to nodewise noise transformations, such as orthog

onal transformations. We prove that this property reduces the model's theoretical sample complexity, thereby improving generalization. Our framework simultaneously reaches theoretical universal expressivity, maintains the linear scalability of standard Message-Passing Neural Networks in practice, and achieves performance comparable to computationally expensive, high-expressivity models. Extensive experiments confirm strong performance across node, link, subgraph, and graph-level prediction tasks, demonstrating that the equivariant use of noise provides a powerful and practical pathway for building expressive GNNs. Our code is available at \url{https://github.com/MuLabPKU/EquivNoiseGNN}.
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Reading Between the Tokens: Improving Preference Predictions through Mechanistic Forecasting

Sarah Ball, Simeon Allmendinger, Frauke Kreuter, Niklas Kühl

Large language models are increasingly used to predict human preferences in both scientific and business endeavors, yet current approaches rely exclusively on analyzing model outputs without considering the underlying mechanisms. Using election forecasting as a test case, we introduce *mechanistic forecasting*, a method that demonstrates that probing internal model representations offers a fundamentally different---and sometimes more effective--- approach to preference prediction. Examining over 24 million configurations across 7 models, 6 national elections, multiple persona attributes, and prompt variations, we systematically analyze how demographic and ideological information activates latent party-encoding components within the respective models. We find that leveraging this internal knowledge via mechanistic forecasting, opposed to solely relying on surface-level predictions, can improve prediction accuracy. The effects vary across demographic versus opinion-based attributes, political parties, national contexts, and models. Our findings demonstrate that the latent representational structure of LLMs contains systematic, exploitable information about human preferences, establishing a new paradigm for using language models in social science prediction tasks.
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Statistical Early Stopping for Reasoning Models

Yangxinyu Xie, Tao Wang, Soham Mallick, Yan Sun, Georgy Noarov, Mengxin Yu, Tanwi Mallick, Edgar Dobriban

While LLMs have seen substantial improvement in reasoning capabilities, they also sometimes overthink, generating unnecessary reasoning steps, particularly under uncertainty, given ill-posed or ambiguous queries. We introduce statistically principled early stopping methods that monitor uncertainty signals during generation to mitigate this issue. Our first approach is nonparametric and provides finite-sample guarantees on the probability of halting too early on well-posed queries. Our second approach is parametric: it models inter-arrival times of uncertainty keywords as a renewal process and applies sequential testing for stopping. We conduct empirical evaluations on reasoning tasks across several domains and models. Our results indicate that uncertainty-aware early stopping can improve both efficiency and reliability in LLM reasoning. The performance varies across domains, and we observe especially significant gains for math reasoning.
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Temporal Difference Calibration in Sequential Tasks: Application to Vision-Language-Action Models

Shelly Francis-Meretzki, Mirco Mutti, Yaniv Romano, Aviv Tamar

Recent advances in vision-language-action (VLA) models for robotics have highlighted the importance of reliable uncertainty quantification in sequential tasks. However, assessing and improving calibration in such settings remains mostly unexplored, especially when only partial trajectories are observed. In this work, we formulate *sequential calibration* for episodic tasks, where task-success confidence is produced along an episode, while success is determined at the end of it. We introduce a sequential extension of the Brier score and show that, for binary outcomes, its risk minimizer coincides with the VLA policy’s value function. This connection bridges uncertainty calibration and reinforcement learning, enabling the use of temporal-difference (TD) value estimation as a principled calib

ration mechanism over time.
We empirically show that TD calibration improves performance relative to the state-of-the-art on simulated and real-robot data. Interestingly, we show that when calibrated using TD, the VLA's single-step action probabilities can yield competitive uncertainty estimates, in contrast to recent findings that employed different calibration techniques.
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Semi-Supervised Learning for Molecular Graphs via Ensemble Consensus
Rasmus Hannibal Tirsgaard, Laurits Fredsgaard, Marisa Wodrich, Mikkel Jordahn, Mikkel N. Schmidt
Machine learning is transforming molecular sciences by accelerating property prediction, simulation, and the discovery of new molecules and materials. Acquiring labeled data in these domains is often costly and time-consuming, whereas large collections of unlabeled molecular data are readily available. Standard semi-supervised learning methods often rely on label-preserving augmentations, which are challenging to design in the molecular domain, where minor changes can drastically alter properties. In this work, we show that semi-supervised methods that rely on an ensemble consensus can boost predictive accuracy across a diverse range of molecular datasets, task types, and graph neural network architectures. We find that training with an ensemble consensus objective increases robustness in models and exhibits an effect similar to knowledge distillation; an individual member of an ensemble trained this way outperforms a full ensemble trained in a traditional supervised fashion in almost all cases. In addition, this type of semi-supervised training reduces calibration error.
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KromHC: Manifold-Constrained Hyper-Connections with Kronecker-Product Residual Matrices
Wuyang Zhou, Yuxuan Gu, Giorgos Iacovides, Danilo Mandic
The success of Hyper-Connections (HC) in neural networks (NN) has also highlighted issues related to training instability and restricted scalability. The Manifold-Constrained Hyper-Connections (mHC) mitigate these challenges by projecting the residual connection space onto a Birkhoff polytope, however, it faces two issues: 1) its iterative Sinkhorn-Knopp (SK) algorithm does not always yield exactly doubly stochastic residual matrices; 2) mHC incurs a prohibitive $\mathcal{O}(n^3C)$ parameter complexity with n as the width of the residual stream and C as the feature dimension. The recently proposed mHC-lite reparametrizes the residual matrix via the Birkhoff-von-Neumann theorem to guarantee double stochasticity, but also faces a factorial explosion in its parameter complexity, $\mathcal{O} \left(nC \cdot n! \right)$. To address both challenges, we propose **KromHC**, which uses the $\underline{\text{Kro}}$necker products of smaller doubly stochastic matrices to parametrize the residual matrix in $\underline{\text{mHC}}$. By enforcing manifold constraints across the factor residual matrices along each mode of the tensorized residual stream, KromHC guarantees exact double stochasticity of the residual matrices while reducing parameter complexity to only $\mathcal{O}(n^2C)$. Experiments show that KromHC matches or even outperforms other state-of-the-art (SOTA) mHC variants, while requiring significantly fewer trainable parameters. The code is at https://github.com/wz1119/KromHC.
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Expanding the AI Evaluation Toolbox with Statistical Models
Drew Keller, Kweku Kwegyir-Aggrey, Ryan Steed, Anita K. Rao, Julia L. Sharp, A. Stevie Bergman
Benchmarks are widely used to evaluate and compare the performance of artificial intelligence systems. However, some approaches to computing benchmark metrics produce invalid uncertainty estimates or make unrecognized assumptions about the evaluation setting. We leverage statistical modeling to make two contributions to the practice of AI benchmarking. First, we formally distinguish measurements of benchmark accuracy from generalized accuracy (performance on all potential test items similar to those included in the benchmark). Then, in a simulated setting and with large-scale evaluation of 22 API-access frontier large language models on 3 popular benchmarks, we show how analysis via generalized linear mixed mod

el can produce correct estimates of generalized accuracy while more efficiently quantifying uncertainty compared to existing regression-free approaches. We also show how this approach can equip evaluators with important context on evaluation results, including variance decomposition and item difficulty estimates that illuminate important aspects of LLM performance and benchmark construction.
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SDiD■Shared diffusion prior for efficient distributed stereo image compression
Yichong Xia, Yimin Zhou, Zongyu Li, Shiyu Qin, Mingyao Hong, Bin Chen, Haoqian Wang
Stereo vision is widely utilized in automotive imagery and 3D reconstruction, creating a demand for compressing stereo images. Existing methods for stereo image compression often employ VAE-like architectures based on distortion optimization, leading to subpar perceptual quality at low bitrates. While generative compression excels in high perceptual fidelity at low bitrates, it struggles to maintain consistency across viewpoints, making decoded images less useful for critical downstream tasks. To address this, we introduce SDiD, a distributed stereo image compression architecture based on shared pre-trained diffusion priors. We employ a diffusion prior alignment module to efficiently obtain the main-view-prior from the foundation diffusion, and utilize a prior transformation structure to enable the auxiliary view to achieve reliable and fast perceptual enhancement while maintaining consistency. Through extensive experiments, we demonstrate that SDiD outperforms existing methods in perceptual quality across multiple datasets. Even at extremely low bitrates, SDiD can accurately recover depth information between decoded images. On the InStereo2K dataset, SDiD requires only one-third of the bits compared to the state-of-the-art baseline (0.02 bpp vs. 0.06 bpp) to reconstruct image pairs with similar depth information.
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3DMedAgent: Unified Perception-to-Understanding for 3D Medical Analysis
Ziyue Wang, Linghan Cai, Chang Han Low, Haofeng Liu, Junde Wu, Jinyu Wang, Rui Wang, Lei Song, Jiang Bian, Jingjing Fu, Yueming Jin
3D CT analysis spans a continuum from low-level perception to high-level clinical understanding. Existing 3D-oriented analysis methods adopt either isolated task-specific modeling or task-agnostic end-to-end paradigms to produce one-hop outputs, impeding the systematic accumulation of perceptual evidence for downstream reasoning.
In parallel, recent multimodal large language models (MLLMs) exhibit improved visual perception and can integrate visual and textual information effectively, yet their predominantly 2D-oriented designs fundamentally limit their ability to perceive and analysis volumetric medical data.
To bridge this gap, we propose 3DMedAgent, an unified agent that enables 2D MLLMs to perform general 3D CT analysis without 3D-specific fine-tuning.
3DMedAgent coordinates heterogeneous visual and textual tools through a flexible MLLM agent, progressively decomposing complex 3D analysis into tractable subtasks that transition from global to regional views, from 3D volumes to informative 2D slices, and from visual evidence to structured textual representations.
Central to this design, 3DMedAgent maintains a long-term structured memory that aggregates intermediate tool outputs and supports query-adaptive, evidence-driven multi-step reasoning.
We further introduce the DeepChestVQA benchmark for evaluating unified perception-to-understanding capabilities in 3D thoracic imaging.
Experiments across over 40 tasks demonstrate that 3DMedAgent consistently outperforms general, medical, and 3D-specific MLLMs, highlighting a scalable path toward general-purpose 3D clinical assistants.
Code and data are available \href{https://github.com/jinlab-imvr/3DMedAgent}{here}.
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WatchLog: Efficient and Interpretable Event Reasoning for Endpoint Detection and Response Logs with Multimodal LLMs
Hongyi Zhou, Jianfeng Pan, Min Peng, Shaomang Huang, Xuling Zhang
Endpoint Detection and Response (EDR) systems are crucial for identifying malici

ous activities on endpoint devices, yet existing methods struggle to efficiently model ultra-long log sequences and to provide interpretable reasoning for security analysts. We propose WatchLog, a novel framework that represents raw logs as video-structured data, enabling scalable and expressive video-language modeling of endpoint behaviors. Each event is encoded as a key–value-guided image, and the resulting images are temporally organized into a video sequence. To capture long-range dependencies, WatchLog employs a temporal cross-attention adapter that enables pixel-wise interaction across time. The adapter acts as an auxiliary temporal reasoning pathway, aligning spatial representations with relevant temporal contexts while preserving the original behavioral semantics. We adopt a two-stage pre-training strategy followed by supervised fine-tuning to generate behavior explanations grounded in event-level semantics and detection outcomes. Experiments on our newly constructed EDR8M-20R dataset and a public benchmark demonstrate that WatchLog consistently outperforms state-of-the-art methods in detection accuracy and recall, while offering more interpretable reasoning traces and significantly improved inference efficiency. Extensive ablation studies further support the robustness and interpretability of the proposed method.

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RuCL: Stratified Rubric-Based Curriculum Learning for Multimodal Large Language Model Reasoning

Yukun Chen, Jiaming Li, Longze Chen, Ze Gong, Jingpeng Li, Zhen Qin, Hengyu Chang, Lei Zhang, Ancheng Xu, Zhihao Yang, Hamid Alinejad-Rokny, QIANG QU, Bo Zheng, Min Yang

Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a prevailing paradigm for enhancing reasoning in Multimodal Large Language Models (MLLMs). However, relying solely on outcome supervision risks reward hacking, where models learn spurious reasoning patterns to satisfy final answer checks.
While recent rubric-based approaches offer fine-grained supervision signals, they suffer from high computational costs of instance-level generation and inefficient training dynamics caused by treating all rubrics as equally learnable.
In this paper, we propose **Stratified Rubric-based Curriculum Learning (RuCL)**, a novel framework that reformulates curriculum learning by shifting the focus from data selection to reward design.
RuCL generates generalized rubrics for broad applicability and stratifies them based on model competence, dynamically adjusting their weights to guide the model from foundational perception to advanced logical reasoning.
Extensive experiments on various visual reasoning benchmarks show that RuCL yields a remarkable **+7.83%** average improvement over the Qwen2.5-VL-7B model, achieving a state-of-the-art accuracy of **60.06%**.

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Mitigating Surgical Data Imbalance with Dual-Prediction Video Diffusion Model

Danush Kumar Venkatesh, Adam Schmidt, Muhammad Abdullah Jamal, Omid Mohareri

Surgical video datasets are essential for scene understanding, enabling procedural modeling and intra-operative support. However, these datasets are often heavily imbalanced, with rare actions and tools under-represented, which limits the robustness of downstream models. We address this challenge with *SurgiFlowVid*, a sparse and controllable video diffusion framework for generating surgical videos of under-represented classes. Our approach introduces a dual-prediction diffusion module that jointly denoises RGB frames and optical flow, providing temporal inductive biases to improve motion modeling from limited samples. In addition, a sparse visual encoder conditions the generation process on lightweight signals (e.g., sparse segmentation masks or RGB frames), enabling controllability without dense annotations. We validate our approach on three surgical datasets across tasks including action recognition, tool presence detection, and laparoscope motion prediction. Synthetic data generated by our method yields consistent gains of 10–20% over competitive baselines, establishing *SurgiFlowVid* as a promising strategy to mitigate data imbalance and advance surgical video understanding methods.

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Adversarial Robustness of Implicit Neural Representation-Based Classifiers

Jayoung Kim, Kookjin Lee, Noseong Park, Sanghyun Hong
Implicit neural representations (INRs) encode data as continuous coordinate-base d functions parameterized by neural networks, shifting downstream tasks such as image recognition to operate on functional rather than discrete representations. Despite their increasing adoption, the adversarial robustness of INR-based clas sification pipelines remain largely underexplored. In this work, we present the first systematic study of adversarial robustness in INR-based classifiers. A key challenge is that generating an INR requires $\text{\emph{training}}$ a neural network for each input sample, resulting in an optimization-in-the-loop forward pass that renders standard gradient-based attacks computationally prohibitive. T o address this, we design surrogate models that amortizes the INR-generation pro cess, serving as a practical proxy for attacking INR-based classifiers. We also develop speed-up techniques that substantially reduce the training cost of the s urrogate. We show that in contrast to recent work, INR-based classifiers are vul nerable: under adversarial input perturbations, classification accuracy collapse s to near zero. Moreover, existing countermeasures designed to operate on discre te representations offer limited protection.
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When Random Saliency Looks Trained: Architectural Center Bias in CNN Interpretab ility
Keying Kuang, Iain Carmichael, Elizabeth Purdom
Saliency maps are widely used to interpret image classification models and build trust in their predictions; however, their reliability remains a central concer n, as randomized networks can produce saliency maps that closely resemble those of trained models. We identify a previously underappreciated architectural contr ibutor to this phenomenon: a *center-focused saliency bias* induced by common co nvolutional design choices. Through controlled ablations, we show that architect ural components such as zero padding and receptive field growth induce a center- focused saliency prior that is already present in randomly initialized CNNs and under randomized inputs. In contrast, this behavior is largely absent in non-con volutional architectures such as Vision Transformers (ViTs) and multilayer perce ptrons (MLPs). To investigate the interaction between architectural priors and l earning, we introduce a corner-shift benchmark and a Center-Shift Index that qua ntify how saliency redistributes under object relocation. We show that training can partially shift saliency toward object regions, while randomized models rema in dominated by architectural center bias, providing one mechanism by which trai ned-random similarity can be inflated and clarifying how architectural priors ca n confound standard saliency evaluations.
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The Implicit Bias of Depth: From Neural Collapse to Softmax Codes
Connall Garrod, Jonathan P. Keating, Christos Thrampoulidis
Neural collapse (NC) describes the structured geometry that emerges in the featu res and weights of trained classifiers. Recent theory suggests NC can be subopti mal in deep architectures, attributing this to an explicit low-rank bias from L2 regularization. We study the deep unconstrained feature model (UFM)—equivalent to a deep linear network with orthogonal inputs—trained without regularization, to isolate how gradient descent and depth alone shape NC. We show that depth ind uces an implicit low-rank bias: low-rank matrices propagate norm more efficientl y through successive multiplications, promoting low-rank alternatives to NC. The se alternatives, we argue, correspond to softmax codes: max-margin solutions pre viously found in width-bottlenecked networks. Analyzing training dynamics under spectral initialization, we identify an early-time repulsion among singular valu es that drives low-rank emergence, and characterize how depth shrinks NC's basin of attraction. Finally, we show that some effects act in the opposite direction : for randomly initialized networks, increasing width biases training toward hig her-rank solutions. Our results provide the first asymptotic and dynamic charact erization of implicit bias in deep UFMs trained with unregularized multiclass cr oss-entropy.
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From Human Labels to Literature: Semi-Supervised Learning of NMR Chemical Shifts

at Scale
Yongqi Jin, Yecheng Wang, Jun-jie Wang, Rong Zhu, Guolin Ke, Weinan E
Accurate prediction of nuclear magnetic resonance (NMR) chemical shifts is fundamental to spectral analysis and molecular structure elucidation, yet existing machine learning methods rely on limited, labor-intensive atom-assigned datasets.
We propose a semi-supervised framework that learns NMR chemical shifts from millions of literature-extracted spectra without explicit atom-level assignments, integrating a small amount of labeled data with large-scale unassigned spectra.
We formulate chemical shift prediction from literature spectra as a permutation-invariant set supervision problem, and show that under commonly satisfied conditions on the loss function, optimal bipartite matching reduces to a sorting-based loss, enabling stable large-scale semi-supervised training beyond traditional curated datasets.
Our models achieve substantially improved accuracy and robustness over state-of-the-art methods and exhibit stronger generalization on significantly larger and more diverse molecular datasets.
Moreover, by incorporating solvent information at scale, our approach captures systematic solvent effects across common NMR solvents for the first time.
Overall, our results demonstrate that large-scale unlabeled spectra mined from the literature can serve as a practical and effective data source for training NMR shift models, suggesting a broader role of literature-derived, weakly structured data in data-centric AI for science.
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ScoreMatchingRiesz: Score Matching for Debiased Machine Learning and Policy Path Estimation
Masahiro Kato
We propose \emph{ScoreMatchingRiesz}, a family of Riesz representer estimators based on score matching. The Riesz representer is a key nuisance component in debiased machine learning, enabling $\sqrt{n}$-consistent and asymptotically efficient estimation of causal and structural targets via Neyman-orthogonal scores. We formulate Riesz representer estimation as a score estimation problem. This perspective stabilizes representer estimation by allowing us to leverage denoising score matching and telescoping density ratio estimation. We also introduce the policy path, a parameter that captures how policy effects evolve under continuous treatments. We show that the policy path can be estimated via score matching by smoothly connecting average marginal effect (AME) and average policy effect (APE) estimation, which improves the interpretability of policy effects. The replication code is available in the genriesz library.
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Mitigating Conversational Inertia in Multi-Turn Agents
Yang Wan, Zheng Cao, Zhenhao Zhang, Zhengwen Zeng, Shuheng Shen, Changhua Meng, Linchao Zhu
Large language models excel as few-shot learners when provided with appropriate demonstrations, yet this strength becomes problematic in multi-turn agent scenarios, where LLMs erroneously mimic their own previous responses as few-shot examples. Through attention analysis, we identify \textbf{conversational inertia}, a phenomenon where models exhibit strong diagonal attention to previous responses, which is associated with imitation bias that constrains exploration.
This reveals a tension when transforming few-shot LLMs into agents: longer context enriches environmental feedback for exploitation, yet also amplifies conversational inertia that undermines exploration.
Our key insight is that for identical states, actions generated with longer contexts exhibit stronger inertia than those with shorter contexts, enabling construction of preference pairs without environment rewards. Based on this, we propose Context Preference Learning to calibrate model preferences to favor low-inertia responses over high-inertia ones. We further provide context management strategies at inference time to balance exploration and exploitation.
Experimental results across eight agentic environments and one deep research scenario validate that our framework reduces conversational inertia and achieves performance improvements.

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Do Sparse Autoencoders Identify Reasoning Features in Language Models?
George Ma, Zhongyuan Liang, Irene Y. Chen, Somayeh Sojoudi
We study how reliably sparse autoencoders (SAEs) support claims about reasoning-related internal features in large language models. We first give a stylized analysis showing that sparsity-regularized decoding can preferentially retain stable low-dimensional correlates while suppressing high-dimensional within-behavior variation, motivating the possibility that contrastively selected "reasoning" features may concentrate on cue-like structure when such cues are coupled with reasoning traces. Building on this perspective, we propose a falsification-based evaluation framework that combines causal token injection with LLM-guided counterexample construction. Across 22 configurations spanning multiple model families, layers, and reasoning datasets, we find that many contrastively selected candidates are highly sensitive to token-level interventions, with 45%–90% activating after injecting only a few associated tokens into non-reasoning text. For the remaining context-dependent candidates, LLM-guided falsification produces targeted non-reasoning inputs that trigger activation and meaning-preserving paraphrases of top-activating reasoning traces that suppress it. A small steering study yields minimal changes on the evaluated benchmarks. Overall, our results suggest that, in the settings we study, sparse decompositions can favor low-dimensional correlates that co-occur with reasoning, underscoring the need for falsification when attributing high-level behaviors to individual SAE features. Code is available at https://github.com/GeorgeMLP/reasoning-probing.
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On the Interplay of Pre-Training, Mid-Training, and RL on Reasoning Language Models
Charlie Zhang, Graham Neubig, Xiang Yue
Recent reinforcement learning (RL) techniques have yielded impressive reasoning improvements in language models, yet it remains unclear whether RL truly extends a model's reasoning ability beyond pre-training. A central challenge is the lack of control in modern training pipelines, where opaque pre-training data, underexplored mid-training, and complex RL interactions obscure causal effects. To resolve this ambiguity, we develop a controlled experimental framework that isolates the causal contributions of pre-training, mid-training, and RL-based post-training. Our approach employs synthetic reasoning tasks with explicit atomic operations, parseable step-by-step reasoning traces, and systematic manipulation of training distributions. We evaluate along: *extrapolative generalization* to more complex compositions and *contextual generalization* across surface contexts. Using this framework, we reconcile competing views on RL’s effectiveness: 1) RL produces true capability gains (pass@128) only when pre-training leaves sufficient headroom and RL data target the model’s *edge of competence*, tasks that are difficult but not yet out of reach. 2) Contextual generalization requires minimal yet sufficient pre-training exposure, after which RL reliably transfers. 3) Mid-training significantly enhances performance under fixed compute compared with RL alone, demonstrating its central but underexplored role. 4) Process-level rewards reduce reward hacking and improve reasoning fidelity. Together, these results clarify the interplay between pre-training, mid-training, and RL, offering a foundation for improving reasoning language models training strategies. Codes and data are avaialble at [Github](https://github.com/Interplay-LM-Reasoning/Interplay-LM-Reasoning) and [HuggingFace](https://huggingface.co/Interplay-LM-Reasoning)
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Accelerating Regression Tasks with Quantum Algorithms
Chenghua Liu, Zhengfeng Ji
Regression is a cornerstone of statistics and machine learning, with applications spanning science, engineering, and economics. While quantum algorithms for regression have attracted considerable attention, most existing work has focused on linear regression, leaving many more complex yet practically important variants unexplored. In this work, we present a unified quantum framework for accelerating a broad class of regression tasks---including linear and multiple regression,

Lasso, Ridge, Huber, ℓ_p-, and δ_p-type regressions---achieving up to a quadratic improvement in the number of samples m over the best classical algorithms. This speedup is achieved by a non-trivial quantization of the recent classical breakthrough of Jambulapati et al. (2024), where we construct a full quantum pipeline that strategically employs quantum leverage score approximation to initialize and refine Multiscale Leverage Score Overestimates, enabling efficient importance sampling via the preparation of multiple state copies. For problems of dimension n, sparsity $r < n$, and error parameter ϵ, our algorithm solves the problem in $\widetilde{O}(r\sqrt{mn}/\epsilon + \mathrm{poly}(n,1/\epsilon))$ quantum time, demonstrating both the applicability and the efficiency of quantum computing in accelerating regression tasks.

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Equilibrium Pricing in Oligopolistic Data Markets

Bhaskar Ray Chaudhury, Jugal Garg, Eklavya Sharma, Jiaxin Song

We study equilibrium pricing in oligopolistic data markets with budget-constrained buyers (e.g., ML companies purchasing data to improve model accuracy) and strategic data sellers. Sellers compete by setting prices for their datasets, giving rise to a pricing game whose pure Nash equilibria correspond to equilibrium prices. While equilibrium prices are guaranteed for rivalrous goods via competitive equilibrium, we show that the non-rivalry of data fundamentally alters this picture: an exact Nash equilibrium need not exist, and in fact no 1.364-approximate equilibrium exists under uniform pricing. We therefore investigate relaxed equilibrium notions. Allowing sellers to use beyond-uniform pricing—specifically, piecewise-linear convex pricing functions—guarantees approximate stability within a constant factor: there exists a pricing profile in which no seller can improve revenue by a factor of two by deviating to any uniform price (a 2-approximate Nash equilibrium). Finally, our simulations demonstrate fast convergence and empirical approximation guarantees that outperform the worst-case bound of 2.

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Enhancing Affine Maximizer Auctions with Correlation-Aware Payment

Haoran Sun, Xuanzhi Xia, Xu Chu, Xiaotie Deng

Affine Maximizer Auctions (AMAs), a generalized mechanism family from VCG, are widely used in automated mechanism design due to their inherent dominant-strategy incentive compatibility (DSIC) and individual rationality (IR). However, as the payment form is fixed, AMA's expressiveness is restricted, especially in distributions where bidders' valuations are correlated. In this paper, we propose Correlation-Aware AMA (CA-AMA), a novel framework that augments AMA with a new correlation-aware payment. We show that any CA-AMA preserves the DSIC property and formalize finding optimal CA-AMA as a constraint optimization problem subject to the IR constraint. Then, we theoretically characterize scenarios where classic AMAs can perform arbitrarily poorly compared to the optimal revenue, while the CA-AMA can reach the optimal revenue. For optimizing CA-AMA, we design a practical two-stage training algorithm. We derive that the target function's continuity and the generalization bound on the degree of deviation from strict IR. Finally, extensive experiments showcase that our algorithm can find an approximate optimal CA-AMA in various distributions with improved revenue and a low degree of violation of IR.

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Great Minds Think Alike: Contextual Tacit Communication for Decentralized LLM-Agent Cooperation

Yue Pei, Hongming Zhang, Jiarui Guan, Jusheng Zhang, Liang Lin, Haogang Zhu, Ziang Chen

Large language models (LLMs) are increasingly used as planners for cooperative embodied agents, but multi-agent settings amplify inconsistency under partial observability and make explicit communication costly or even unavailable. Many existing approaches rely on online message passing; when communication is removed, agents often fall back to independent local planning that suffers from miscoordination. We introduce Contextual Tacit Communication, a gradient-free protocol that aligns decentralized decisions with a joint LLM value score without explicit message actions. Our method measures context-conditioned value rectifications via

residual banding to pinpoint miscoordination actions and amortizes the resulting coordination signals into a retrieval-augmented Tacit Rule Memory that provides prompt-level cooperation rules at execution time. Experiments on VIKI, C-WAH, and TDW-MAT show that our approach improves cooperation performance over baselines while reducing runtime overhead compared with communication-based methods.
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Sparse Regression with ℓ_0 Constraints for α-Mixing Time Series: Algorithms and Guarantees
Ruoxin Yuan, Lijun Ding
Exact sparse methods based on ℓ_0 constraints are increasingly used for interpretable and scalable time series modeling, where one aims to recover a small set of informative lags/factors while maintaining strong predictive performance and low computational cost. Despite their empirical success, finite-sample and computational guarantees for such methods under temporal dependence remain limited. In this paper, we study ℓ_0-constrained least squares for time series generated by α-mixing stationary Gaussian processes with sparse coefficients. We establish high-probability restricted strong convexity/smoothness (RSC/RSS) for the empirical quadratic loss. Leveraging these conditions, we derive nonasymptotic statistical guarantees and computational complexities for a series of exact sparse methods, including iterative hard thresholding (IHT). We apply our theoretical results to Gaussian vector autoregressive (VAR) models and obtain new guarantees. Experiments on synthetic sparse VAR models and real-world mobility time series demonstrate that exact sparse methods recover lag structure more accurately and interpretably than some classical methods, while achieving comparable prediction error with substantially lower computational cost.
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ReGen: Hierarchical Multi-Prompt Representation Generation for Efficient Waveform Diffusion Models
Sang-Hoon Lee, Ha-Yeong Choi
Representation alignment (REPA) has been investigated to accelerate diffusion training, but we observe that regularizing intermediate representations in diffusion Transformers (DiT) may implicitly entangle latents and limit generative capacity.
To address this issue, we propose ReGen, a hierarchical multi-prompt representation generation framework that jointly estimates multiple vector fields for both representations and data within a single diffusion model.
We further introduce generalized flow matching (GFM) to improve the generalization of conditional flow matching (CFM). We validate ReGen on single-stage waveform diffusion models including neural audio codec and Wave-VAE. ReGen significantly improves waveform generation quality from highly compressed latent representations at 12.5 Hz. We also present ReGenVoice, a latent diffusion model (LDM)-based text-to-speech model that achieves strong speech intelligibility (WER) and speaker similarity (SIM) with a small dataset. Moreover, operating the LDM at 6.25 Hz with rich semantic and acoustic latent representation enables efficient training and sampling, requiring only 1 day of training on 4 GPUs and fast inference with an RTF of 0.08. Audio samples are available at https://regenvoice.github.io/demo/
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DynaMem: Consistent Long Video Generation via Hierarchical Memory and Motion Priors
Jingyu Lin, Xinyi Shang, Peng Sun, Cunjian Chen, Zhiqiang Shen
Recent text-to-video diffusion models can synthesize visually compelling clips from natural language prompts.
However, practical applications increasingly demand long-form videos with evolving narratives and persistent identity.
A common solution is autoregressive generation, where the video is produced clip by clip over long horizons, yet coherence often degrades as errors compound.
In this work, we study long-video generation under an autoregressive setting, where videos are synthesized clip by clip over long horizons.
Despite strong short-clip quality, existing approaches often suffer from semanti

c drift, motion decay, and appearance instability as the sequence grows.
We present DynaMem, a unified framework that improves long-horizon coherence via three components: Semantic-Adaptive Hierarchical Memory for long-range semantic preservation, Motion-Prioritized Optimization for motion-coherent learning, and Reference-Anchored Perceptual Alignment for stabilizing appearance.
Extensive experiments show that DynaMem produces more consistent semantics, stronger temporal dynamics, and more stable appearance on long videos compared to competitive baselines.
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The Double Dilemma in Multi-Task Radiology Report Generation: A Gradient Dynamics Analysis and Solution
Erjian Zhang, Yatong Hao, Liejun Wang, Zhiqing Guo
While multi-task learning based automatic radiology report generation (RRG) is widely adopted to ensure clinical consistency, most focus on architectural designs yet remain limited to coarse linear scalarization strategies. These strategies cannot effectively balance the hard constraints of discriminative clinical supervision with the smoothness requirements of report generation. To address these problems, we analyze the failure mechanism of linear scalarization from the perspective of gradient dynamics, utilizing the stochastic differential equation (SDE) framework to characterize it as a "Double Dilemma" of drift term deviation and diffusion term decay. Based on this, we propose a backbone-agnostic optimizer named **C**onflict-**A**verse **M**agnitude-**E**nhanced **Grad**ient Descent (CAME-Grad). Through conflict-averse direction rectification and magnitude-enhanced energy injection, the algorithm not only ensures geometric validity, but also avoids local optimal solutions. Then, the adaptive gradient fusion mechanism is used to establish a dynamic balance between the theoretical optimal direction and the task-specific inductive bias. Experiments show that as a universal plug-and-play optimizer, CAME-Grad brings substantial and consistent improvements across eight diverse RRG methods, elevating overall clinical efficacy performance by an average of 2.3\% on MIMIC-CXR and 1.9\% on IU X-Ray. Our code is available at https://github.com/vpsg-research/CAME-Grad.
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Pair2Scene: Learning Local Object Relations for Procedural Scene Generation
Xingjian Ran, Shujie Zhang, Weipeng Zhong, Luo Li, Bo Dai
Generating high-fidelity 3D indoor scenes remains a significant challenge due to data scarcity and the complexity of modeling intricate spatial relations. Current methods often struggle to scale beyond training distribution to dense scenes or rely on LLMs/VLMs that lack the ability for precise spatial reasoning. Building on top of the observation that object placement relies mainly on local dependencies instead of information-redundant global distributions, in this paper, we propose **Pair2Scene**, a novel procedural generation framework that integrates learned local rules with scene hierarchies and physics-based algorithms. These rules mainly capture two types of inter-object relations, namely *support relations* that follow physical hierarchies, and *functional relations* that reflect semantic links. We model these rules through a network, which estimates spatial position distributions of dependent objects conditioned on position and geometry of the anchor ones. Accordingly, we curate a dataset **3D-Pairs** from existing scene data to train the model. During inference, our framework can generate scenes by recursively applying our model within a hierarchical structure, leveraging collision-aware rejection sampling to align local rules into coherent global layouts. Extensive experiments demonstrate that our framework outperforms existing methods in generating complex environments that go beyond training data while maintaining physical and semantic plausibility.
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Gradient Regularization Mitigates Reward Hacking in Reinforcement Learning from Human Feedback and Verifiable Rewards
Johannes Ackermann, Michael Noukhovitch, Takashi Ishida, Masashi Sugiyama
Reinforcement Learning from Human Feedback (RLHF) or Verifiable Rewards (RLVR) are two key steps in the post-training of modern Language Models (LMs). A common problem is reward hacking, where the policy may exploit inaccuracies of the rewa

rd and learn an unintended behavior. Most previous works address this by limitin g the policy update with a Kullback-Leibler (KL) penalty towards a reference mod el. We propose a different framing: Train the LM in a way that biases policy upd ates towards regions in which the reward is more accurate. First, we derive a th eoretical connection between the accuracy of a reward model and the flatness of an optimum at convergence. Gradient regularization (GR) can then be used to bias training to flatter regions and thereby maintain reward model accuracy. We conf irm these results by showing that the gradient norm and reward accuracy are empi rically correlated in RLHF. We then empirically show that Reference Resets of t he KL penalty find flatter regions with a higher reward accuracy. We further imp rove on this by proposing to use explicit GR with an efficient finite-difference estimate. Empirically, GR performs better than a KL penalty across a diverse se t of RL experiments with LMs. GR achieves a higher GPT-judged win-rate in RLHF, avoids overly focusing on the format in rule-based math rewards, and prevents ha cking the judge in LLM-as-a-Judge math tasks.
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Improving Explicit Dynamic Gaussian Splatting Optimization via Update Mixture
Renjie Ding, Yaonan Wang, Min Liu, Jialin Zhu, Jiazheng Wang, Jiahao Zhao, Xiao Tan, Feixiang He, Xiang Chen
3D Gaussian Splatting (3DGS) enables real-time, high-fidelity view synthesis via explicit scene representations and has recently been extended to dynamic scene modeling. Despite their excellent rendering quality and interpretability, we fin d that explicit Dynamic GS often exhibits generalization degradation in scenes w ith large motion. Motivated by generalization behavior in deep neural optimizati on and the characteristics of Gaussian primitive optimization, we propose an upd ate mixture strategy. This work focuses on two representative open-source explic it Dynamic GS pipelines and our approach consists of three components: (i) a spa ce–time dependent Strictly Sparse Update with additional regularization to stabi lize adaptive updates; (ii) a constant-corrected adaptive algorithm that allevia tes over-scaling of primitive gradients and yields a stable mixture of adaptive and non-adaptive steps; and (iii) attributes mixing via Stochastic Attribute Ave raging to mitigate frame-preference under motion disturbances. Experiments show consistent improvements and reduced generalization issues, highlighting the role of non-adaptive updates and the impact of frame-preference in explicit Dynamic GS optimization.
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FACT: Fuzzy Alignment with Comorbidity Topology for Reliable Multi-Label Medical Image Diagnosis
Yingyu Chen, Yongqiang Huang, Yang Qin, Ziyuan Yang, Lang Yuan, Maosong Ran, Yi Zhang
In clinical practice, patients often present with multiple co-occurring diseases , yet most existing Multi-Label-Diagnosis (MLD) methods treat diagnosis as a rig id discriminative partitioning task, implicitly assuming that overlapping pathol ogies are separable. This assumption is problematic in medical images, where ide ntical or highly similar visual observations may simultaneously support multiple disease labels, and disease concepts are inherently correlated rather than inde pendent. Enforcing hard decision boundaries under such overlap suppresses shared evidence, biases feature representations, and ultimately undermines model relia bility. To address this limitation, we propose Fuzzy Alignment with Comorbidity Topology FACT, a novel paradigm that reformulates MLD as a fuzzy alignment probl em between atomic visual evidence and disease semantic anchors. FACT is characte rized by three key features: (1) modeling visual polysemy through shared and reu sable atomic visual evidence; (2) encoding disease correlation via semantic anch ors structured by comorbidity topology; and (3) employing a metric-based fuzzy m embership function for non-discriminative visual-semantic alignment. Extensive e xperiments on three public clinical benchmarks demonstrate that FACT consistentl y improves diagnostic performance while delivering clinically plausible predicti ons. The code is available at \url{https://github.com/yyuChen9/FACT}.
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TSMGen: Target-Specific Molecule Generation via Higher-Order Structural Dependen

cies and Context-Aware Bidirectional Fusion
Yaoyu Chen, Xiaoli Lin, Ziyi Gong, Jun Pang
Efficiently designing high-quality molecules targeting disease-relevant targets is a critical challenge. Most existing methods can capture pairwise amino acid relations, neglecting the higher-order relations among multiple amino acids. This paper proposes a target-specific molecule generation framework, namely TSMGen, to comprehensively capture the local and global structural information of the protein pocket by modeling higher-order spatial dependencies both at the atomic and the amino acid levels. Furthermore, we design a context-aware bidirectional fusion module to learn the more detailed structural information about the protein pocket. This module simultaneously attends to features from both the protein pocket and the molecule, fully leveraging the structural information from both to optimize the generation process of targeted molecules, thereby enhancing the quality of generated molecules. Experiments show that TSMGen outperforms state-of-the-art methods in terms of Vina Score, High Affinity, QED, SA and Diversity, and a case study on β-secretase enzyme further confirms its ability to generate molecules with stronger binding affinity.
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Gateways to Tractability for Satisfiability in Pearl’s Causal Hierarchy
Robert Ganian, Marlene Gründel, Simon Wietheger
Pearl’s Causal Hierarchy (PCH) is a central framework for reasoning about probabilistic, interventional, and counterfactual statements, yet the satisfiability problem for PCH formulas is computationally intractable in almost all classical settings. We revisit this challenge through the lens of parameterized complexity and identify the first gateways to tractability. Our results include fixed-parameter and XP-algorithms for satisfiability in key probabilistic and counterfactual fragments, using parameters such as primal treewidth and the number of variables, together with matching hardness results that map the limits of tractability. Technically, we depart from the dynamic programming paradigm typically employed for treewidth-based algorithms and instead exploit structural characterizations of well-formed causal models, providing a new algorithmic toolkit for causal reasoning.
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Doppler Prompting for Stable mmWave-based Human Pose Estimation
Shuntian Zheng, Jiaqi Li, Xiaoman Lu, Shuai He, Yu Guan
Millimeter-wave (mmWave) enables privacy-preserving, illumination-robust human pose estimation (HPE), with each mmWave frame represented as a range--angle--Doppler tensor, providing spatial magnitude for localization and Doppler signatures for motion-related cues.
However, existing mmWave-based HPE methods either underutilize or naïvely fuse Doppler signatures with spatial magnitude, disregarding their distinct physical semantics.
As a result, non-human Doppler signatures can be misinterpreted as human motion cues, leading to jittery trajectories.
We propose \textbf{PULSE}, which converts Doppler signatures into confidence-aware motion prompts and injects them into spatial magnitude reasoning through constrained interactions.
By screening Doppler prompts before they influence prediction, PULSE first suppresses spurious spectral motion cues and then uses the screened prompts to stabilize prediction.
Across three datasets spanning single- and multi-person settings, PULSE consistently improves pose accuracy and temporal stability, indicating that controlled Doppler prompting is a practical direction for stable mmWave HPE.
Codes are available in supplementary materials.
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A hitchhiker's guide to Poisson gradient estimation
Michael Ibrahim, Hanqi Zhao, Eli Zachary Sennesh, Zhi Li, Anqi Wu, Jacob L. Yates, Chengrui Li, Hadi Vafaii
Poisson-distributed latent variable models are widely used in computational neuroscience, but differentiating through discrete stochastic samples remains challe

nging. Two approaches address this: *Exponential Arrival Time* (EAT) simulation and *Gumbel-SoftMax* (GSM) relaxation. We provide the first systematic compariso n of these methods, along with practical guidance for practitioners. Our main te chnical contribution is a modification to the EAT method that theoretically guar antees an unbiased first moment (exactly matching the firing rate), and reduces second-moment bias. We evaluate these methods on their distributional fidelity, gradient quality, and performance on two tasks: (1) variational autoencoders wit h Poisson latents, and (2) partially observable generalized linear models, where latent neural connectivity must be inferred from observed spike trains. Across all metrics, our modified EAT method exhibits better overall performance (often comparable to exact gradients), and substantially higher robustness to hyperpara meter choices. These results extend to over-dispersed Negative Binomial latents, where modified EAT again performs best. However, only GSM generalizes to arbitr ary non-Poisson distributions, including the under-dispersed regime. Together, o ur results clarify the trade-offs between these methods and offer concrete recom mendations for practitioners working with Poisson latent variable models.
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PhyScene3D: Physically Consistent 3D Interactive Tabletop Scene Generation
Weixing Chen, Zhuoqian Feng, Yang Liu, Yexin Zhang, Yifan Wen, Yinghong Liao, We ichao Qiu, Guanbin Li, Liang Lin
Generating physically consistent 3D tabletop scenes is a fundamental yet underex plored problem for interactive and generalist robotic learning. The challenge st ems from dense object hierarchies and irregular affordances. Here, an interactiv e scene denotes a physically valid, collision-free environment directly loadable into physics simulators. Existing methods, ranging from decoupled symbolic solv ers to end-to-end regression models, often suffer from error propagation or over fitting to noisy supervision containing widespread physical violations. To addre ss these limitations, we introduce PhyScene3D, a framework that reformulates gen eration as a Human-Mimetic Constructive Process. The proposed Cognitive Topologi cal Reasoning Chain (CTRC) factorizes scene synthesis into a sequential, anchor-conditioned process. It employs a 3D AABB-based placement scheme that imposes a strong structural inductive bias. To address imperfect supervision and physical infeasibility, we introduce Physics-Aware Denoising Alignment (PADA). It integra tes a differentiable Signed Distance Field (SDF) with Test-Time Optimization (TT O) to project generated scenes onto a physics-feasible manifold while preserving semantic intent. Experiments demonstrate that PhyScene3D outperforms state-of-t he-art approaches in both semantic accuracy and physical validity, achieving a 4 0\% reduction in scene-wise collision rate relative to the human-annotated train ing data.
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Efficient Transformer Attention for SNNs via Hadamard Simplification
Tingting Jiang, Jiangrong Shen, Long Chen, Yaxin Li, Qi Xu
Spiking Neural Networks (SNNs) enable low-power, event-driven computation, but T ransformer-based SNNs remain difficult to deploy on neuromorphic hardware due to dense operations and communication overhead. We propose two simplified attentio n mechanisms, \textbf{Simplified Spiking Attention (SSA)} and \textbf{Ultra-Simp lified Spiking Attention (USSA)}, which replace matrix multiplication with Hadam ard products and eliminate hardware-unfriendly components such as multi-head att ention and scaling. We show that consecutive masking is redundant and analyze a spiking-order effect in which early spiking contributes more temporal informatio n to attention modulation. On CIFAR-10, CIFAR-100, and DVS-Gesture, SSA achieves accuracies of 96.38\%, 79.45\%, and 97.56\%, respectively, while reducing compu tational complexity from $\mathcal{O}(N^2D)$ to $\mathcal{O}(ND)$ and communicat ion complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(ND)$. USSA further reduce s communication complexity to $\mathcal{O}(N)$ with only marginal accuracy degra dation. On ImageNet-1K, SSA and USSA achieve 76.91\% and 77.27\% accuracy, respe ctively, demonstrating scalability to large-scale classification.
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Mechanistic Anomaly Detection via Functional Attribution
Hugo Lyons Keenan, Christopher Leckie, Sarah Monazam Erfani

We can often verify the correctness of neural network outputs using ground truth labels, but we cannot reliably determine whether the output was produced by normal or anomalous internal mechanisms. Mechanistic anomaly detection (MAD) aims to flag these cases, but existing methods either depend on latent space analysis, which is vulnerable to obfuscation, or are specific to particular architectures and modalities. We reframe MAD as a functional attribution problem: asking to what extent samples from a trusted set can explain the model's output, where attribution failure signals anomalous behavior. We operationalize this using influence functions, measuring functional coupling between test samples and a small reference set via parameter-space sampling. We evaluate across multiple anomaly types and modalities. For backdoors in vision models, our method achieves state-of-the-art detection on BackdoorBench, with an average Defense Effectiveness Rating (DER) of 0.93 across seven attacks and four datasets (next best 0.83). For LLMs, we similarly achieve a significant improvement over baselines for several backdoor types, including on explicitly obfuscated models. Beyond backdoors, preliminary evidence shows our method can detect adversarial and out-of-distribution samples, and distinguishes multiple anomalous mechanisms within a single model. Our results establish functional attribution as an effective, modality-agnostic tool for detecting anomalous behavior in deployed models.
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SonicMaster: Towards Controllable All-in-One Music Restoration and Mastering
Jan Melechovsky, Ambuj Mehrish, Abhinaba Roy, Dorien Herremans
Music recordings often suffer from audio quality issues such as excessive reverberation, distortion, clipping, tonal imbalances, and a narrowed stereo image, especially when created in non-professional settings without specialized equipment or expertise. These problems are typically corrected using separate specialized tools and manual adjustments. In this paper, we introduce SonicMaster, the first unified generative model for music restoration and mastering that addresses a broad spectrum of audio artifacts with text-based control. SonicMaster is conditioned on natural language instructions to apply targeted enhancements, or can operate in an automatic mode for general restoration. To train this model, we construct the SonicMaster dataset, a large dataset of paired degraded and high-quality tracks by simulating common degradation types with nineteen degradation functions belonging to five enhancements groups: equalization, dynamics, reverb, amplitude, and stereo. Our approach leverages a flow-matching generative training paradigm to learn an audio transformation that maps degraded inputs to their cleaned, mastered versions guided by text prompts. Objective audio quality metrics demonstrate that SonicMaster significantly improves sound quality across all artifact categories. Furthermore, subjective listening tests confirm that listeners prefer SonicMaster's enhanced outputs over other baselines. The model and demo samples are available through https://github.com/AMAAI-Lab/SonicMaster.
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ForensicConcept: Transferable Forensic Concepts for AIGI Detection
Menyanshu Zhou, Ziyin Zhou, Ke Sun, Yunpeng Luo, Jiayi Ji, Xiaoshuai Sun, Rongrong Ji
AI-generated image detectors achieve high accuracy on in-distribution data but often fail on unseen generators. A key obstacle to understanding this failure is the black-box nature of current detectors: they do not reveal which evidence drives their decisions. We propose \textsc{ForensicConcept}, a framework that extracts explicit forensic concepts from detectors and enables their transfer across backbones. Our method localizes decision-critical patches via Transformer attribution, clusters them into a compact concept codebook, and uses a concept-aligned projection to produce auditable evidence readouts. Motivated by prior studies showing that DINO representations can guide diffusion generation and exhibit concept-level correspondence with diffusion features, we introduce a generation-trace reference based on CleanDIFT diffusion features and quantify backbone-trace alignment via neighborhood-structure consistency (CKNNA). We further propose concept codebook injection to transfer diffusion-derived concepts into target backbones. Experiments on GenImage, GAN-family, and Chameleon benchmarks show consistent improvements over prior methods. We also find that CKNNA alignment predicts tr

ansfer effectiveness, providing a principled explanation for why some backbones yield more transferable forensic evidence than others.
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The Truth Lies Somewhere in the Middle (of the Generated Tokens)
Sophie L. Wang, Phillip Isola, Brian Cheung
How should the sequence of hidden states produced during autoregressive generation be compressed into a representation that reflects the model’s internal state? We study representations derived from generated tokens and compare them to grounded embeddings across several domains. We find that pooling embeddings across tokens produces more informative representations than any individual token. This observation is consistent with semantic information being distributed across generated tokens rather than localized to a single position. In this setting, alignment provides a way to study how a model’s internal representations evolve and pooling offers a more reliable summary of the model's state across generation.
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The Sign Estimator: Preference Modeling for LLM Alignment under Heterogeneity
Ali Aouad, Aymane El Gadarri, Vivek Farias
Traditional large language model (LLM) alignment methods are based on Reinforcement Learning From Human Feedback (RLHF), which learns a single reward model (implicitly or explicitly) from pairwise comparison data. This approach implicitly assumes homogeneous preferences across human labelers---an assumption that is violated in practice. As a result, the learned reward model in RLHF is generally misspecified: Prior work shows that it is inconsistent with the population-average utility, incurring large distortion, and that recovering this utilitarian objective is provably impossible in the worst case. In this work, we show that the average utility is recoverable under a mild assumption. Our method, the Sign Estimator, simply replaces the standard cross-entropy loss function with a notion of binary classification loss and yields a reward model that is ordinally consistent with the population-average utility. We further establish a fast finite-sample convergence rate of $O({n^{-{1}/{3}}})$, which provides, to our knowledge, the first consistent estimator for heterogeneous preferences that does not suffer from the curse of dimensionality.
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TabICLv2: A Better, Faster, Scalable, and Open Tabular Foundation Model
Jingang QU, David Holzmüller, Gaël Varoquaux, Marine Le Morvan
Tabular foundation models, such as TabPFNv2 and TabICL, have recently dethroned gradient-boosted trees at the top of predictive benchmarks, demonstrating the value of in-context learning for tabular data. We introduce TabICLv2, a new state-of-the-art foundation model for regression and classification built on three pillars: (1) a novel synthetic data generation engine designed for high pretraining diversity; (2) various architectural innovations, including a new scalable softmax in attention improving generalization to larger datasets without prohibitive long-sequence pretraining; and (3) optimized pretraining protocols, notably replacing AdamW with the Muon optimizer. On the TabArena and TALENT benchmarks, TabICLv2 without any tuning surpasses the performance of the current state of the art, RealTabPFN-2.5 (hyperparameter-tuned, ensembled, and fine-tuned on real data). With only moderate pretraining compute, TabICLv2 generalizes effectively to million-scale datasets under 50GB GPU memory while being markedly faster than RealTabPFN-2.5. We provide extensive ablation studies to quantify these contributions and foster open research by releasing code for inference, pretraining, and synthetic data generation at https://github.com/soda-inria/tabicl.
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DistMatch: Adaptive Binning via Distribution Matching for Robust Sequential Conformal Prediction
Enver Menadjiev, Jihyeon Seong, Jisu Yeo, Jaesik Choi
Sequential conformal prediction (CP) provides valid uncertainty quantification under the assumption of residual exchangeability. However, this assumption is often violated in real-world time series due to temporal dependencies and distributional shifts. While recent methods attempt to approximate exchangeability through reweighting, identifying optimal weights remains an open challenge. To address

this limitation, we propose [DistMatch](https://github.com/enver1323/dist_match_conformal), a binning-based method that recursively partitions residuals within a binary tree using the Kolmogorov–Smirnov (KS) statistic. We theoretically show that this partitioning induces approximately exchangeable leaves, thereby avoiding the need for reweighting. By applying quantile regression with online updates within each leaf, DistMatch enables locally adaptive inference and improves robustness to distributional shifts. Extensive experiments demonstrate that DistMatch outperforms existing sequential CP methods.
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ProphetKV: User-Query-Driven Selective Recomputation for Efficient KV Cache Reuse in Retrieval-Augmented Generation
Shihao Wang, Jiahao Chen, Yanqi Pan, Hao Huang, Yichen Hao, Xiangyu Zou, Wen Xia, Chongyang Qiu, Wentao Zhang, PengFei Wang
The prefill stage of long-context Retrieval-Augmented Generation (RAG) is severely bottlenecked by computational overhead. To mitigate this, recent methods assemble pre-calculated KV caches of retrieved RAG documents (by a *user query*) and reprocess selected tokens to recover cross-attention between these pre-calculated KV caches. However, we identify a fundamental ``crowding-out effect'' in current token selection criteria: globally salient but *user-query*-irrelevant tokens saturate the limited recomputation budget, displacing the tokens truly essential for answering the *user query* and degrading inference accuracy.

We propose ProphetKV, a user-query-driven KV Cache reuse method for RAG scenarios. ProphetKV dynamically prioritizes tokens based on their semantic relevance to the *user query* and employs a dual-stage recomputation pipeline to fuse layer-wise attention metrics into a high-utility set.
By ensuring the recomputation budget is dedicated to bridging the informational gap between retrieved context and the *user query*, ProphetKV achieves high-fidelity attention recovery with minimal overhead. Our extensive evaluation results show that ProphetKV retains 96\%–101\% of full-prefill accuracy with only a 20\% recomputation ratio, while achieving accuracy improvements of 8.8\%–24.9\% on RULER and 18.6\%–50.9\% on LongBench over the state-of-the-art approaches (e.g., CacheBlend, EPIC, and KVShare).
****
SemBind: Binding Diffusion Watermarks to Semantics Against Black-Box Forgery Attacks
Xin Zhang, Zijin Yang, Kejiang Chen, Linfeng Ma, Weiming Zhang, Nenghai Yu
Latent-based watermarks, embedded during the generation process of latent diffusion models (LDMs), facilitate detection and attribution of generated images. However, recent black-box forgery attacks can implant a provider's watermark into images not generated by that provider, using at least one watermarked image and black-box access to the model, thereby undermining provenance and trust. We propose SemBind, the first defense framework for latent-based watermarks against black-box forgery, which binds latent watermark signals to image semantics through a contrastively trained semantic masker. The masker produces near-invariant codes for semantically matched prompts and near-orthogonal codes across different prompts, enabling SemBind to modulate the target latent before applying standard latent-based watermarking schemes. Across four mainstream latent-based watermarking methods, SemBind substantially reduces false acceptance under black-box forgery while preserving image quality and offering a tunable robustness--security trade-off via a simple mask-ratio parameter. Code is available at https://github.com/XinZhang1999/SemBind.
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Learning When to Act or Refuse: Guarding Agentic Reasoning Models for Safe Multi-Step Tool Use
Aradhye Agarwal, Gurdit Siyan, Yash Pandya, Joykirat Singh, Akshay Nambi, Ahmed Hassan Awadallah
Agentic language models operate in a fundamentally different safety regime than chat models: they must plan, call tools, and execute long-horizon actions where a single misstep, such as accessing files or entering credentials, can cause irr

eversible harm. Existing alignment methods, largely optimized for static generat ion and task completion, break down in these settings due to sequential decision -making, adversarial tool feedback, and overconfident intermediate reasoning. We introduce MOSAIC, a post-training framework that aligns agents for safe multi-s tep tool use by making safety decisions explicit and learnable. MOSAIC structure s inference as a plan, check, then act or refuse loop, with explicit safety reas oning and refusal as first-class actions. To train without trajectory-level labe ls, we use preference-based reinforcement learning with pairwise trajectory comp arisons, which captures safety distinctions often missed by scalar rewards. We e valuate MOSAIC zero-shot across three model families, Qwen2.5-7B, Qwen3-4B-Think ing, Phi-4 and Qwen3-30B-A3B-Thinking, and across out-of-distribution benchmarks spanning harmful tasks, prompt injection, benign tool use, and cross-domain pri vacy leakage. MOSAIC reduces harmful behavior by up to 50\%, increases harmful-t ask refusal by over 20\% on injection attacks, cuts privacy leakage, and preserv es or improves benign task performance, demonstrating robust generalization acro ss models, domains, and agentic settings.
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ST-TGExplainer: Disentangling Stability and Transition Patterns for Temporal GNN Interpretability

Hongjiang Chen, Xin Zheng, Pengfei Jiao, Huan Liu, Zhidong Zhao, Huaming Wu, Fen g Xia, Shirui Pan

Temporal graph neural networks (TGNNs) have gained significant traction for solv ing real-world temporal graph tasks. However, their interpretability remains lim ited, as most TGNNs fail to identify which historical interactions most influenc e a given prediction. Despite promising progress on interpretable TGNNs, existin g methods predominantly focus on previously seen historical interactions, which we term stability patterns, while overlooking newly emerging first-time interact ions, which we term transition patterns. Both types of patterns are essential fo r faithful temporal explanations. To address this limitation, we propose ST-TGEx plainer, a self-explainable TGNN that disentangles Stability and Transition patt erns in temporal graphs for a more faithful Temporal GNN Explainer. Guided by a disentangled information bottleneck objective, ST-TGExplainer learns a compact e xplanatory subgraph that remains predictive of the event label while explicitly suppressing label-conditioned redundancy between stability and transition patter ns. Extensive experiments demonstrate that ST-TGExplainer achieves strong predic tive performance and yields more faithful explanations. Code is available at htt ps://github.com/hjchen-hdu/ST-TGExplainer.
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SyMerge: From Non-Interference to Synergistic Merging via Single-Layer Adaptatio n

Aecheon Jung, Seunghwan Lee, Dongyoon Han, Sungeun Hong

Model merging combines independently trained models into a single multi-task mod el. However, most existing approaches focus primarily on avoiding task interfere nce. We argue that its greater potential lies in enabling task synergy, where ta sks actively improve one another. We identify cross-task performance, defined by compatibility between encoders and predictors across tasks, as a key indicator of merge quality. We demonstrate that adapting only a single task-specific layer is sufficient to induce such synergy. This study proposes SyMerge, a lightweigh t framework that jointly optimizes merging coefficients and a single task-specif ic layer. We adopt an expert-guided self-labeling objective, providing stable su pervision beyond entropy minimization. Intriguingly, we further show that SyMerg e successfully merges models trained from different initializations, a regime wh ere standard methods break down. Our minimalist yet principled method achieves s tate-of-the-art results across vision, dense prediction, and NLP benchmarks. Our code is available at https://aim-skku.github.io/SyMerge.
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NoiseSDF2NoiseSDF: Learning Clean Neural Fields from Noisy Supervision

Tengkai Wang, Weihao Li, Ruikai Cui, Shi Qiu, Nick Barnes

Reconstructing accurate implicit surface representations from point clouds remai ns a challenging task, particularly when data is captured using low-quality scan

ning devices. These point clouds often contain substantial noise, leading to ina
ccurate surface reconstructions. Inspired by the Noise2Noise paradigm for 2D ima
ges, we introduce NoiseSDF2NoiseSDF, a novel method designed to extend this conc
ept to 3D neural fields. Our approach enables learning clean neural SDFs from no
isy point clouds through noisy supervision by minimizing the MSE loss between no
isy SDF representations, allowing the network to implicitly denoise and refine s
urface estimations. We evaluate the effectiveness of NoiseSDF2NoiseSDF on benchm
arks, including the ShapeNet, ABC, Famous, and Real datasets. Experimental resul
ts demonstrate that our framework significantly improves surface reconstruction
quality from noisy inputs.
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Prism: Spectral-Aware Block-Sparse Attention
Xinghao Wang, Pengyu Wang, Xiaoran Liu, Fangxu Liu, Jason Chu, Kai Song, Xipeng
Qiu
Block-sparse attention is promising for accelerating long-context LLM pre-fillin
g, yet identifying relevant blocks efficiently remains a bottleneck.
Existing methods typically employ coarse-grained attention as a proxy for block
importance estimation, but often resort to expensive token-level searching or sc
oring, resulting in significant selection overhead.
In this work, we trace the inaccuracy of standard coarse-grained attention via m
ean pooling to a theoretical root cause: the interaction between mean pooling an
d Rotary Positional Embeddings (RoPE).
We prove that mean pooling acts as a low-pass filter that induces destructive in
terference in high-frequency dimensions, effectively creating a "blind spot" for
local positional information (e.g., slash patterns).
To address this, we introduce Prism, a training-free spectral-aware approach tha
t decomposes block selection into high-frequency and low-frequency branches.
By applying energy-based temperature calibration, Prism restores the attenuated
positional signals directly from pooled representations, enabling block importan
ce estimation using purely block-level operations, thereby improving efficiency.
Extensive evaluations confirm that Prism maintains accuracy parity with full att
ention while delivering up to $\mathbf{5.1\times}$ speedup. Code available at ht
tps://github.com/xinghaow99/prism.
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Implicit Preference Alignment for Human Image Animation
Yuanzhi Wang, Xuhua Ren, Jiaxiang Cheng, bing ma, Kai Yu, Tianxiang Zheng, Qingl
in Lu, Zhen Cui
Human image animation has witnessed significant advancements, yet generating hig
h-fidelity hand motions remains a persistent challenge due to their high degrees
of freedom and motion complexity. While reinforcement learning from human feedb
ack, particularly direct preference optimization, offers a potential solution, i
t necessitates the construction of strict preference pairs. However, curating su
ch pairs for dynamic hand regions is prohibitively expensive and often impractic
al due to frame-wise inconsistencies. In this paper, we propose Implicit Prefere
nce Alignment (IPA), a data-efficient post-training framework that eliminates th
e need for paired preference data. Theoretically grounded in implicit reward max
imization, IPA aligns the model by maximizing the likelihood of self-generated h
igh-quality samples while penalizing deviations from the pretrained prior. Furth
ermore, we introduce a Hand-Aware Local Optimization mechanism to explicitly ste
er the alignment process toward hand regions. Experiments demonstrate that our m
ethod achieves effective preference optimization to enhance hand generation qual
ity, while significantly lowering the barrier for constructing preference data.
Codes are released at https://github.com/mdswyz/IPA
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JADE: Expert-Grounded Dynamic Evaluation for Open-Ended Professional Tasks
Lanbo Lin, Jiayao Liu, Tianyuan Yang, Li Cai, Yuanwu Xu, Lei Wei, Sicong Xie, Gu
annan Zhang
Evaluating agentic AI on open-ended professional tasks faces a fundamental dilem
ma between rigor and flexibility. Static rubrics provide rigorous, reproducible
assessment but fail to accommodate diverse valid response strategies, while LLM-

as-a-judge approaches adapt to individual responses yet suffer from instability and bias. Human experts address this dilemma by combining domain-grounded principles with dynamic, claim-level assessment. Inspired by this process, we propose JADE, a two-layer evaluation framework. Layer 1 encodes expert knowledge as a predefined set of evaluation skills, providing stable evaluation criteria. Layer 2 performs report-specific, claim-level evaluation to flexibly assess diverse reasoning strategies, with evidence-dependency gating to invalidate conclusions built on refuted claims. Experiments on BizBench show that JADE improves evaluation stability and reveals critical agent failure modes missed by holistic LLM-based evaluators. We further demonstrate strong alignment with expert-authored rubrics and effective transfer to HealthBench and DR.BENCH, covering medical and 10-domain professional evaluation settings. Code and data are available at https://github.com/smiling-world/JADE.

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Neural Control: Adjoint Learning Through Equilibrium Constraints

Dezhong Tong, Jiawen Wang, Hengyi Zhou, Yinlong Shen, Xiaonan Huang, M. Khalid Jawed

Many physical AI tasks require sequential implicit computation: at each step, boundary controls are applied, and the resulting configuration is obtained by solving an equilibrium problem. This setting arises naturally in deformable object manipulation, where even bending a deformable linear object (DLO) to a target shape can be nonlinear and multistable: identical boundary conditions may produce different configurations depending on actuation history. Unlike explicit transition models, the control-to-configuration relation is implicit and history-dependent, making long-horizon learning and control brittle; backpropagating through iterative solves is also memory- and compute-intensive. We propose Neural Control, a boundary-control framework that propagates gradients through branch-dependent sequences of equilibrium solves rather than a single fixed point. Neural Control computes trajectory-dependent proxy gradients by differentiating equilibrium conditions with an adjoint formulation, avoiding solver unrolling while keeping forward rollouts on converged equilibria. Combined with receding-horizon continuation, Neural Control re-anchors optimization to realized equilibria and mitigates basin switching. We validate Neural Control on simulated and real DLO manipulation, compare against SPSA and iCEM, and demonstrate applicability to a learned DEQ-style implicit equilibrium model.

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Noise-Guided Transport: Imitation Learning from Random Priors

Lionel Blondé, Joao Candido Ramos, Alexandros Kalousis

We consider imitation learning in the low-data regime, where only a limited number of expert demonstrations are available. In this setting, methods that rely on large-scale pretraining or high-capacity architectures can be difficult to apply, and efficiency with respect to demonstration data becomes critical. We introduce Noise-Guided Transport (NGT), a lightweight off-policy method that casts imitation as an optimal transport problem solved via adversarial training. NGT requires no pretraining or specialized architectures, incorporates uncertainty estimation by design, and is easy to implement and tune. Despite its simplicity, NGT achieves strong performance on challenging continuous control tasks, including high-dimensional Humanoid tasks, under ultra-low data regimes with as few as 20 transitions.

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Anchor-Final Self-Supervision Drives Hallucination-Aware Optimization in Large Vision-Language Models

Jiaxi Liu, Yifeng Yang, Xinbing Wang, Qinying Gu, Nanyang Ye

Hallucinations in large vision-language models (LVLMs) remain a critical challenge, where models often generate tokens that fail to align with visual evidence. To address this issue, we propose AFS: Anchor-Final Self-Supervision, a novel framework for hallucination-aware optimization in LVLMs. By leveraging discrepancies between intermediate and final layer predictions, AFS selectively applies self-supervision to visually descriptive tokens, incorporates hallucination-aware token classification, and encourages consistency between intermediate and final l

ayer distributions. Unlike traditional methods that rely on explicit supervision or post-hoc interventions, AFS optimizes the model via Group Relative Policy Optimization (GRPO), using token-specific rewards derived from internal model signals. Experiments demonstrate that AFS significantly reduces hallucinations without compromising recall in caption generation. Beyond captioning, AFS excels in discriminative tasks, improving the reliability of object existence predictions and multimodal reasoning. Furthermore, AFS demonstrates strong cross-dataset generalization, transferring effectively across diverse visual domains. Code is available at \url{https://github.com/guavayew/AFS}.
****
It's a TRAP! Task-Redirecting Agent Persuasion Benchmark for Web Agents
Karolina Korgul, Yushi Yang, Arkadiusz Drohomirecki, Piotr Blaszczyk, Will Howard, Lukas Aichberger, Chris Russell, Philip Torr, Adam Mahdi, Adel Bibi
Web-based agents powered by large language models are increasingly used for tasks such as email management or professional networking. Their reliance on dynamic web content, however, makes them vulnerable to prompt injection attacks: adversarial instructions hidden in interface elements that persuade the agent to divert from its original task. We introduce the Task-Redirecting Agent Persuasion Benchmark (TRAP), a benchmark for studying how persuasion techniques misguide autonomous web agents on realistic tasks. Across six frontier models, agents are susceptible to prompt injection in 25\% of tasks on average (13\% for GPT-5 to 43\% for DeepSeek-R1), with small interface or contextual changes often doubling success rates and revealing systemic, psychologically driven vulnerabilities in web-based agents. We also provide a modular social-engineering injection framework with controlled experiments on high-fidelity website clones, allowing for further benchmark expansion.
****
Implicit Actor Critic Coupling via a Supervised Learning Framework for RLVR
Jiaming Li, Longze Chen, Ze Gong, Yukun Chen, Lu Wang, Wanwei He, Run Luo, Minzheng Wang, Lei Zhang, Haoran Ye, Min Yang
Recent advances in Reinforcement Learning with Verifiable Rewards (RLVR) have empowered large language models (LLMs) to tackle challenging reasoning tasks such as mathematics and programming, however existing RLVR methods often suffer from sparse reward signals and unstable policy gradient updates inherent to RL-based approaches.

To address the challenges, we propose $\textbf{PACS}$, a novel RLVR framework that achieves im$\textbf{P}$licit $\textbf{A}$ctor $\textbf{C}$ritic coupling via a $\textbf{S}$upervised learning framework.

By treating the outcome reward as a predictable label, we reformulate the RLVR problem into a supervised learning task over a score function parameterized by the policy model and optimized using cross-entropy loss.

A detailed gradient analysis shows that this supervised formulation inherently recovers the classical policy gradient update while providing more stable and efficient training.

Extensive experiments demonstrate that PACS significantly outperforms strong open-source models and RLVR baselines, yielding substantial average gains of $\textbf{+8.26\%}$ (4B) and $\textbf{+9.57\%}$ (8B) over base models offering a promising avenue for LLMs post-training with verifiable rewards. Our code and data are available as open source at https://github.com/ritzz-ai/PACS.
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ScDiVa: Masked Discrete Diffusion for Joint Modeling of Single-Cell Identity and Expression
Mingxuan Wang, Gaoyang Jiang, ZiJia Ren, Cheng Chen, Chuangxin Zhao, Lu Shi, Yanbiao Ma
Single-cell RNA-seq profiles are high-dimensional, sparse, and unordered, causing autoregressive generation to impose an artificial ordering bias and suffer from error accumulation. To address this, we propose scDiVa, a masked discrete diffusion foundation model that aligns generation with the dropout-like corruption process by defining a continuous-time forward masking mechanism in token space. ScDiVa features a bidirectional denoiser that jointly models discrete gene identi

ties and continuous values, utilizing entropy-normalized serialization and a lat
ent anchor token to maximize information efficiency and preserve global cell ide
ntity. The model is trained via depth-invariant time sampling and a dual denoisi
ng objective to simulate varying sparsity levels while ensuring precise recovery
 of both identity and magnitude. Pre-trained on 59 million cells, scDiVa achieve
s strong transfer performance across major benchmarks, including batch integrati
on, cell type annotation, and perturbation response prediction. These results su
ggest that masked discrete diffusion serves as a biologically coherent and effec
tive alternative to autoregression.
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Functional Decomposition and Shapley Interactions for Interpreting Survival Mode
ls

Sophie Hanna Langbein, Hubert Baniecki, Fabian Fumagalli, Niklas Koenen, Marvin
N. Wright, Julia Herbinger

Hazard and survival functions are natural, interpretable targets in time-to-even
t prediction tasks such as patient survival and disease progression modeling, bu
t their inherent non-additivity fundamentally limits standard additive explanati
on methods. We introduce Survival Functional Decomposition (SurvFD), a principle
d approach for analyzing feature interactions in machine learning survival model
s. By decomposing higher-order effects into time-dependent and time-independent
components, SurvFD offers a previously unrecognized perspective on survival expl
anations, explicitly characterizing when and why additive explanations fail. Bui
lding on this theoretical decomposition, we propose SurvSHAP-IQ, which extends S
hapley interactions to time-indexed functions, providing a practical estimator f
or higher-order, time-dependent interactions. We validate the framework on simul
ated data and demonstrate its utility through cancer survival applications, incl
uding multi-modal breast cancer prognosis combining histopathology with clinical
 features. Together, SurvFD and SurvSHAP-IQ establish an interaction- and time-a
ware interpretability approach for survival modeling, with broad applicability a
cross medicine, healthcare and other time-to-event prediction tasks.
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ScaLoRA: Optimally Scaled Low-Rank Adaptation for Efficient High-Rank Fine-Tunin
g

Yilang Zhang, Xiaodong Yang, Yiwei Cai, Georgios B. Giannakis

As large language models (LLMs) continue to scale in size, the computational ove
rhead has become a major bottleneck for task-specific fine-tuning. While low-ran
k adaptation (LoRA) effectively curtails this cost by confining the weight updat
es to a low-dimensional subspace, such a restriction can hinder effectiveness an
d slow convergence. This contribution deals with these limitations by accumulati
ng progressively a high-rank weight update from consecutive low-rank increments.
 Specifically, the per update optimal low-rank matrix is identified to minimize
the loss function and closely approximate full fine-tuning. To endow efficient a
nd seamless optimization without restarting, this optimal choice is formed by ap
propriately scaling the columns of the original low-rank matrix. Rigorous perfor
mance guarantees reveal that the optimal scaling can be found analytically. Exte
nsive numerical tests with popular LLMs scaling up to 12 billion parameters demo
nstrate a consistent performance gain and fast convergence relative to state-of-
the-art LoRA variants on diverse tasks including natural language understanding,
 commonsense reasoning, and mathematical problem solving.
****

Adapting to Evolving Graphs: A Scalable Framework for Dynamic Coarsening

Abhishek Gupta, Manoj Kumar, Sarthak Kumar Singh, Ujjwal Yadav, Yifan Sun, Sande
ep Kumar

Graph coarsening is a fundamental dimensionality reduction technique for scaling
 large graphs while preserving structural and feature information. However, most
 existing coarsening methods are designed for static graphs and do not extend we
ll to dynamic settings where nodes, edges, and connectivity patterns evolve over
 time. Recomputing a coarsened graph from scratch after every update is often in
feasible, which limits scalability and real-time applicability. To address this,
 we propose a unified framework for coarsening discrete-time dynamic graphs by i

ncrementally updating the coarsening mapping matrix. The framework initializes f rom any static coarsening technique and then efficiently incorporates real-world graph events, including node additions, node deletions, and edge modifications. We instantiate this framework with two optimization based incremental update al gorithms tailored to different dynamic regimes, one focusing on efficiently inte grating growth related changes and another handling broader topology evolution w ith adaptive reassignment. We derive fast and scalable solvers with convergence guarantees, and provide theoretical guarantee via ϵ-similarity bounds t hat quantify and control quality degradation in the coarsened graph. Extensive e xperiments under realistic dynamic scenarios show substantial improvements in ru ntime and memory, delivering significant speedups while maintaining or improving downstream task performance, including graph neural network accuracy.
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iWorld-Bench: A Benchmark for Interactive World Models with a Unified Action Gen eration Framework

Jianjie Fang, Yingshan Lei, Qin Wan, Ziyou Wang, Yuchao Huang, Yongyan Xu, Baini ng Zhao, Weichen Zhang, Chen Gao, Xinlei Chen, Yong Li

Achieving Artificial General Intelligence (AGI) requires agents that learn and i nteract adaptively, with interactive world models providing scalable environment s for perception, reasoning, and action. Yet current research still lacks large- scale datasets and unified benchmarks to evaluate their physical interaction cap abilities. To address this, we propose iWorld-Bench, a comprehensive benchmark f or training and testing world models on interaction-related abilities such as di stance perception and memory. We construct a diverse dataset with 330k video cli ps and select 2.1k high-quality samples covering varied perspectives, weather, a nd scenes. As existing world models differ in interaction modalities, we introdu ce an **Action Generation Framework** to unify evaluation and design six task ty pes, generating 4.9k test samples. These tasks jointly assess model performance across **visual generation, trajectory following, and memory**. Evaluating 14 re presentative world models, we identify key limitations and provide insights for future research. The iWorld-Bench model leaderboard is publicly available at iWo rld-Bench.com.
****

ObjEmbed: Towards Universal Multimodal Object Embeddings

Shenghao Fu, Yukun Su, Fengyun Rao, Jing LYU, Xiaohua Xie, Wei-Shi Zheng

Aligning objects with corresponding textual descriptions is a fundamental challe nge and a realistic requirement in vision-language understanding. While recent m ultimodal embedding models excel at global image-text alignment, they often stru ggle with fine-grained alignment between image regions and specific phrases. In this work, we present ObjEmbed, a novel MLLM embedding model that decomposes the input image into multiple regional embeddings, each corresponding to an individ ual object, along with global embeddings. It supports a wide range of visual und erstanding tasks like visual grounding, local image retrieval, and global image retrieval. ObjEmbed enjoys three key properties: (1) Object-Oriented Representat ion: It captures both semantic and spatial aspects of objects by generating two complementary embeddings for each region: an object embedding for semantic match ing and an IoU embedding that predicts localization quality. The final object ma tching score combines semantic similarity with the predicted IoU, enabling more accurate retrieval. (2) Versatility: It seamlessly handles both region-level and image-level tasks. (3) Efficient Encoding: All objects in an image, along with the full image, are encoded in a single forward pass for high efficiency. Superi or performance on 18 diverse benchmarks demonstrates its strong semantic discrim ination. Code is available at \url{https://github.com/WeChatCV/ObjEmbed}.
****

Breaking the Block: Preserving Data Continuity to Train Superior SAEs for Instru ct Models

Jiaming Li, Haoran Ye, Yukun Chen, Xinyue Li, Lei Zhang, Hamid Alinejad-Rokny, J immy Chih-Hsien Peng, Min Yang

Sparse Autoencoders (SAEs) are a cornerstone of mechanistic interpretability. Ex isting training methods inherit the Block Training paradigm from LLM pre-trainin

g, which introduces destructive gradient noise in instruct models due to attenti on leakage from unrelated contexts.

Using GSNR analysis, we theoretically characterize this issue and propose $\un derline{\textbf{F}}$inetuning-$\underline{\textbf{a}}$ligned $\underline{\textbf {S}}$equential $\underline{\textbf{T}}$raining ($\textit{FAST}$), a sequential t raining paradigm specifically designed for instruct models.

$\textit{FAST}$ aligns SAE training with the data distribution and activation patterns of instruct models, substantially improving both reconstruction fidelit y and feature interpretability.

Experimental results show that $\textit{FAST}$ achieves higher GSNR, a signifi cantly lower log-scaled MSE of 0.6468 compared to the baseline's 5.1985, and a n ear-zero Delta Loss (-0.51\% to 0.37\%).

Moreover, on Llama-3.2-3B-it, $\textit{FAST}$ produces 21.1\% high-quality fea tures, substantially outperforming baseline methods that achieve 7.0\% and 10.2\ %.

We further find that intervening on special token activations through SAEs can improve generation quality, revealing new opportunities for fine-grained contro l. Our codes are available as open source at https://github.com/Geaming2002/FAST .

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On the Entropy Dynamics in Reinforcement Fine-Tuning of Large Language Models

Shumin Wang, Yuexiang Xie, Wenhao Zhang, Yuchang Sun, Yanxi Chen, Yaliang Li, Ya nyong Zhang

Entropy serves as a critical metric for measuring the diversity of outputs gener ated by large language models (LLMs), providing valuable insights into their exp loration capabilities. While recent studies increasingly focus on monitoring and adjusting entropy to better balance exploration and exploitation in reinforceme nt fine-tuning (RFT), a principled understanding of entropy dynamics during this process is yet to be thoroughly investigated. In this paper, we establish a the oretical framework for analyzing the entropy dynamics during the RFT process, wh ich begins with a discriminant expression that quantifies entropy change under a single logit update. This foundation enables the derivation of a first-order ex pression for entropy change, which can be further extended to the update formula of Group Relative Policy Optimization (GRPO). The corollaries and insights draw n from the theoretical analysis inspire the design of entropy control methods, a nd also offer a unified lens for interpreting various entropy-based methods in e xisting studies. We provide empirical evidence to support the main conclusions o f our analysis and demonstrate the effectiveness of the derived entropy-discrimi nator clipping methods. This study yields novel insights into RFT training dynam ics, providing theoretical support and practical strategies for optimizing the e xploration-exploitation balance during LLM fine-tuning. We release our code at h ttps://github.com/agentscope-ai/Trinity-RFT/tree/main/examples/entropy.

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Curating the Future: A Scalable Recipe for Training Open-Ended Forecasters

Nikhil Chandak, Shashwat Goel, Ameya Prabhu, Moritz Hardt, Jonas Geiping

High-stakes decision making involves reasoning under uncertainty about the futur e. In this work, we train language models to make predictions on open-ended fore casting questions. To scale up training data, we synthesize novel forecasting qu estions from global events reported in daily news. While directly training on th is data leads to performance drops, carefully curating questions creates a valua ble training resource. We use the resulting dataset, OpenForesight, to post-trai n Qwen3 thinking models. To prevent leakage of future information during trainin g and evaluation, we use an offline news corpus, both for data generation and re trieval in our forecasting system. Guided by a small validation set, we show the benefits of retrieval, and an improved reward function for reinforcement learni ng (RL). Once we obtain our final forecasting system, we perform held-out testin g between May to August 2025. Our specialized model, OpenForecaster-8B, matches much larger proprietary models, with our training improving the accuracy, calibr ation, and consistency of predictions. We find calibration improvements from for ecasting training generalize across popular benchmarks. We will open-source our

models, code, and data to make LLM based forecasting research broadly accessible
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PRISM: Gauge-Invariant Tangent-Space Differentially Private LoRA
Shihao Wang, Xueru Zhang
Applying differential privacy (DP) via DP-SGD to Low-Rank Adaptation (LoRA) is a
 natural approach for privacy-preserving fine-tuning. However, LoRA's low-rank p
arameterization poses a fundamental challenge. In LoRA, each trainable update is
 represented as a low-rank matrix $Z = AB^\top$, but this factorization is inher
ently *non-identifiable*: many factor pairs (A, B) represent the same update $
Z$. As a result, applying DP-SGD directly to the factors induces *gauge-dependen
t* perturbations on Z, and we show that this naive DP-LoRA can lead to unbound
ed noise amplification. We propose **PRISM**, an intrinsic DP mechanism for LoRA
 that is gauge invariant by construction, avoids bilinear noise amplification, a
nd admits an efficient low-dimensional noise sampler. Moreover, PRISM yields a c
losed-form characterization of the effective intrinsic noise induced on Z, ena
bling stable privacy–utility trade-offs through bounded, gauge-invariant perturb
ations. We establish standard (ε,δ)-DP guarantees for PRISM and
 introduce a DP-aware, gauge-invariant adaptive update rule that prevents adapti
ve optimization from amplifying injected privacy noise, improving numerical stab
ility in practice.
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CE^{4}L: Continual Ego, Exo, and Ego-Exo Learning
Hongwei Yan, Kanglei Zhou, Yuchen Liu, Qingyu Shi, Yi Zhong, Liyuan Wang
Perception for embodied agents is video-based, often multi-view (ego, exo, or bo
th), and inherently continual, with simultaneous task and viewpoint shifts. Yet
continual learning (CL) remains dominated by exo-only recognition tasks, obscuri
ng behavior under these real-world coupled shifts. We introduce **C**ontinual **
E**go, **E**xo, and **E**go-**E**xo **L**earning (**CE^{4}L**), a unified multi-
view CL benchmark spanning four representative tasks: cross-view referenced skil
l assessment, temporal action segmentation, cross-view association, and action a
nticipation \& planning. CE^{4}L highlights challenges largely absent in prior C
L benchmarks, including cross-view correspondence, view-dependent asynchrony, an
d heterogeneous semantic objectives. To this end, we propose **V**ideo **I**ncre
mental **S**ubspace-routed **T**ask **A**dapters (**VISTA**), a parameter-effici
ent baseline method that stores task-specific updates in lightweight adapters an
d performs training-free routing via residual distance to task-specific whitened
 subspaces estimated from second-order statistics. Extensive experiments demonst
rate the significantly varied efficacy of representative CL methods across CE$^4
$L settings, while VISTA is consistently competitive and achieves state-of-the-a
rt overall performance.
****
MDN: Parallelizing Stepwise Momentum for Delta Linear Attention
Yulong Huang, Xiang Liu, Hongxiang Huang, Xiaopeng LIN, Zunchang Liu, Xiaowen Ch
u, Zeke Xie, Bojun Cheng
Linear Attention (LA) offers a promising paradigm for scaling large language mod
els (LLMs) to long sequences by avoiding the quadratic complexity of self-attent
ion. Recent LA models such as Mamba2 and GDN interpret linear recurrences as clo
sed-form online stochastic gradient descent (SGD), but naive SGD updates suffer
from rapid information decay and suboptimal convergence in optimization. While m
omentum-based optimizers provide a natural remedy, they pose challenges in simul
taneously achieving training efficiency and effectiveness. To address this, we d
evelop a chunkwise parallel algorithm for LA with a stepwise momentum rule by ge
ometrically reordering the update coefficients. Further, from a dynamical system
s perspective, we analyze the momentum-based recurrence as a second-order system
 that introduces complex conjugate eigenvalues. This analysis guides the design
of stable gating constraints. The resulting model, Momentum DeltaNet (MDN), leve
rages Triton kernels to achieve comparable training throughput with competitive
linear models such as Mamba2 and KDA. Extensive experiments on the 400M and 1.3B
 parameter models demonstrate consistent performance improvements over strong ba

selines, including Transformers, Mamba2 and GDN, across diverse downstream evaluation benchmarks. Code: https://github.com/HuuYuLong/MomentumDeltaNet.
****

From Rashomon Theory to PRAXIS: Efficient Decision Tree Rashomon Sets
Zakk Heile, Hayden McTavish, Varun Babbar, Margo Seltzer, Cynthia Rudin
Standard machine learning pipelines often admit many near-optimal models. These "Rashomon sets" pose a range of challenges and opportunities for uncertainty-aware, robust decision making. They allow users to incorporate domain knowledge and preferences that would otherwise be difficult to specify directly in an objective, and they quantify diversity among valid models for a given training dataset and objective function. However, computation of Rashomon sets, even for simple, interpretable model classes such as sparse decision trees, continues to require immense memory and runtime resources. We present PRAXIS, an algorithm to approximate this Rashomon set with orders of magnitude improvement in runtime and memory usage. We validate that PRAXIS regularly recovers almost all of the full Rashomon set. PRAXIS allows researchers and practitioners to scalably model the Rashomon set for real-world datasets.
****

From Winning to Understanding: A Diagnostic Long-Horizon RTS Benchmark for LLMs
Jiacheng Li, Jiahui Liu, Yuqing Wang, Gaochen Cui, Xiao Zhang, Qianchuan Zhao, Ziyou Zhang, Chenghao Li
Large language models (LLMs) are increasingly used as decision modules, yet existing benchmarks provide limited coverage of long-horizon, adversarial interaction while faithfully acting on human instructions.
We introduce a long-horizon Red Alert RTS benchmark with a hierarchical interface in which LLMs output budgeted, low-frequency macro/tactical intents that are executed deterministically for standardized comparison. The benchmark evaluates (i) robustness to ``rules-as-variable'' perturbations via rule-style shifts, (ii) competitive strength via Elo-style ratings from head-to-head matches, and (iii) human steerability via standardized language interventions. Beyond win/loss, we log economy growth/spending, combat loss ratio, and visibility coverage to diagnose long-horizon failure modes. Overall, the benchmark provides a reproducible and diagnostic testbed for robustness and controllability in long-horizon adversarial decision making.
****

Zero-Shot Text-to-Motion Evaluation using Video Language Models
Yuwen Ji, Donglin Wang, Yue Zhang
Text-to-motion (T2M) generation has become a fundamental task, yet existing evaluation metrics often fail to capture whether a generated motion semantically matches its text description. We propose VeMo, a zero-shot evaluation framework that renders generated human motions into videos and uses pretrained video-language models (VLMs) to assess text-motion alignment.
Instead of training an evaluator on scarce motion-specific labels, VeMo transfers the semantic reasoning ability of VLMs to T2M evaluation through normalized likelihood-based scoring.
To reduce the effect of 3D-to-2D projection ambiguity, we introduce an entropy-driven uncertainty analysis for identifying reliable rendered views.
To address the lack of rigorous standards in the field, we further contribute a test-only and human-annotated meta-evaluation benchmark, covering motions generated by multiple representative T2M models.
Extensive experiments show that VeMo correlates better with human judgments than existing reference-based and reference-free metrics.
Additional analyses on view selection, rendering protocols, textual prompt robustness, and computational trade-offs characterize both the promise and limitations of VLM-based T2M evaluation.
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ReSeek: A Self-Correcting Framework for Search Agents with Instructive Rewards
Shiyu Li, Yifan Wang, Peiming Li, Zheng Wei, Yang Tang
Search agents powered by Large Language Models have demonstrated significant potential in tackling knowledge-intensive tasks. Reinforcement learning has emerged

as a powerful paradigm for training these agents to perform complex, multi-step reasoning. However, prior RL-based methods often rely on sparse or rule-based rewards, which can lead agents to commit to suboptimal or erroneous reasoning paths without the ability to recover. To address these limitations, we propose **ReSeek**, a self-correcting framework enabling search agents to recover from erroneous search paths during an episode. By invoking a special **JUDGE** action, the agent can judge the information and re-plan its search strategy. To guide this process, we design a dense, instructive process reward function, which decomposes into a correctness reward for retrieving factual information and a utility reward for finding information genuinely useful for the query. Additionally, to mitigate the risk of data contamination in existing datasets, we introduce **FictionalHot**, a contamination-free benchmark requiring complex reasoning. Experiments show ReSeek significantly outperforms SOTA baselines in task success and path faithfulness.

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Mitigating Error Accumulation in Continuous Navigation via Memory-Augmented Kalman Filtering

Yin Tang, Jiawei Ma, Jinrui Zhang, Alex Jinpeng Wang, Deyu Zhang

Continuous prediction in complex environments is critical for Unmanned Aerial Vehicle (UAV). However, the existing Vision-Language Navigation (VLN) models follows the dead-reckoning, which iteratively predicts the next waypoint and updates its position, thereby constructing the complete trajectory. Then, such stepwise manner will inevitably lead to accumulated errors of position over time, resulting in misalignment between internal belief and objective coordinates, which is known as ``state drift'' and ultimately compromises the subsequent trajectory prediction. Drawing inspiration from classical control theory, we propose to correct for errors by formulating the continuous prediction as a recursive Bayesian state estimation problem. In this paper, we design NeuroKalman, a novel framework that decouples navigation into two complementary processes: a Prior Prediction, based on motion dynamic,s and a Likelihood Correction, from historical observation. We first mathematically associate Kernel Density Estimation of the measurement likelihood with the attention-based retrieval mechanism, which then allows the system to rectify the latent representation using retrieved historical anchors without gradient updates. Comprehensive experiments on TravelUAV benchmark demonstrate that, with only 10\% of the full training data fine-tuning, our method clearly outperforms strong baselines and regulates drift accumulation.

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Near-Universal Multiplicative Updates for Nonnegative Einsum Factorization

John Hood, Aaron Schein

Despite the ubiquity of multiway data across scientific domains, there are few performant and user-friendly methods that fit non-standard nonnegative tensor factorization models tailored to the data at-hand. Researchers may use gradient-based automatic differentiation, which often struggles under nonnegative constraints, choose between a limited set of methods with mature implementations, or implement their own model from scratch. As an alternative, we introduce NNEinFact, an einsum-based multiplicative update algorithm that fits any nonnegative tensor factorization expressible as a tensor contraction by minimizing one of many user-specified loss functions, including the (α,β)-divergence. To use NNEinFact, the researcher specifies their model with a string. NNEinFact converges to a stationary point of the loss, supports missing data, and fits to tensors with hundreds of millions of entries in seconds. Empirically, NNEinFact fits custom models which outperform standard ones in prediction tasks on real-world tensor data by over 37% and attains less than half the test loss of gradient-based methods while converging up to 90 times faster. Software is publicly available at github.com/jhood3/einfact.

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Locate then Correct: Debiasing Attention Heads in CLIP

Wei Jie Yeo, Rui Mao, Moloud Abdar, Erik Cambria, Ranjan Satapathy

Multimodal models like CLIP have gained significant attention due to their remarkable zero-shot performance across various tasks. However, studies have revealed

that CLIP can inadvertently learn spurious associations between target variables and confounding factors. To address this, we introduce \textsc{Locate-Then-Correct} (LTC), a contrastive framework that identifies spurious attention heads in Vision Transformers via mechanistic insights and mitigates them through targeted ablation. Furthermore, LTC identifies salient, task-relevant attention heads, enabling the integration of discriminative features through orthogonal projection to improve classification performance. We evaluate LTC on benchmarks with inherent background and gender biases, achieving over a > 50% gain in worst-group accuracy compared to non-training post-hoc baselines. Additionally, we visualize the representation of selected heads and find that the presented interpretation corroborates our contrastive mechanism for identifying both spurious and salient attention heads.
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CG-MLLM: Captioning and Generating 3D content via Multi-modal Large Language Models

Junming Huang, Chi Wang, Letian Li, Guangkai Xu, Donglin Huang, Hao Chen, Qiang Dai, Weiwei Xu

Large Language Models(LLMs) have revolutionized text generation and multimodal perception, but their capabilities in 3D content generation remain underexplored. Existing methods compromise by producing either low-resolution meshes or coarse structural proxies, failing to capture fine-grained geometry natively. In this paper, we propose CG-MLLM, a novel Multi-modal Large Language Model (MLLM) capable of 3D captioning and high-resolution 3D generation in a single framework.

Leveraging the Mixture-of-Transformer architecture, CG-MLLM decouples disparate modeling needs, where the Token-level Autoregressive (TokenAR) Transformer handles token-level content, and the Block-level Autoregressive (BlockAR) Transformer handles block-level content. By integrating a pre-trained vision-language backbone with a specialized 3D VAE latent space, CG-MLLM facilitates long-context interactions between standard tokens and spatial blocks within a single integrated architecture.

Experimental results show that CG-MLLM significantly outperforms existing MLLMs in generating high-fidelity 3D objects, effectively bringing high-resolution 3D content creation into the mainstream LLM paradigm.
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How Transformers Represent Hierarchies: A Local-to-Global Mechanism

Zhiling Zhou, Tianhao Wang, Zhuoran Yang

Large language models built on autoregressive Transformers excel at next-token prediction, but it is unclear how their internal computations capture the latent hierarchical dependencies that often underlie language.

We study this question in a controlled formal-language setting based on probabilistic context-free grammars (PCFGs), where sequences are generated by a latent hierarchical process.

Empirically, standard autoregressive Transformers can be trained to accurately match the grammar-induced next-token distribution.

Using probing analyses, we find that Transformer hidden states contain information used by classical parsing algorithms.

Moreover, this information emerges through a layer-wise progression, revealing a local-to-global mechanism: early layers accumulate local patterns, while later layers aggregate them into a compact summary for next-token prediction.

Complementing these empirical findings, we provide an explicit construction of Transformers that can parse binary PCFGs with depth *logarithmic* in the grammar's sequence length.

Surprisingly, trained Transformers in this setting exhibit prediction behavior and internal representations that closely mirror our construction.

Together, our results offer a mechanistic account of how Transformers integrate hierarchical parsing with autoregressive generation, enabling them to closely approximate the grammar-induced next-token distribution.
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Accuracy and Normalized Accuracy under Length Bias: Analysis, Guidelines, and a Bayesian Alternative

Koen Oostermeijer
Multiple-choice benchmarks that rank candidate completions by conditional log-pr
obability suffer from a length bias: because log-probabilities sum over tokens,
longer answers tend to be penalized relative to shorter ones in practice. A comm
on mitigation is to normalize scores by completion length, but we show empirical
ly that this heuristic frequently over-corrects, introducing a bias toward longe
r answers instead. We first analyze these scoring rules, characterizing when sta
ndard and length-normalized accuracy are appropriate and how their length biases
 depend on the distribution of completion lengths. Motivated by this analysis, w
e introduce *Bayesian accuracy*, a scoring rule that computes the posterior prob
ability of each candidate under an explicit prior over answer length, thereby re
moving linear length effects. Bayesian accuracy is a drop-in replacement for lik
elihood-based multiple-choice evaluation, requires no additional forward passes,
 and consistently exhibits lower empirical length bias than both standard and le
ngth-normalized accuracy across benchmarks and few-shot settings.
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Balancing Plasticity and Stability with Fast and Slow Successor Features
Raymond Chua, Doina Precup, Blake Aaron Richards
A hallmark of intelligence is the ability to adapt in non-stationary environment
s, yet deep Reinforcement Learning (RL) agents often struggle in such settings.
Prior studies introduce non-stationarity through abrupt shifts in features or dy
namics, whereas real-world environments often evolve gradually through continual
 drift. This distinction has important implications for the ``stability-plastici
ty dilemma'' in RL, as abrupt task changes may demand more plasticity than natur
alistic settings. To address this, we modify existing 3D Miniworld and MuJoCo en
vironments to incorporate naturalistic, continual non-stationarity, and use them
 to examine how stability and adaptation affect performance under continuous env
ironmental change. We find that methods favoring stability, such as synaptic con
solidation, outperform approaches focused on plasticity, such as parameters rese
tting. Motivated by this result, and prior evidence that Successor Features (SFs
) reduce interference, we investigate whether SFs are better consolidation targe
ts than Q-values. Across both environments, applying neuro-inspired synaptic con
solidation to SFs yields superior performance on continually changing settings.
Moreover, consolidation is most effective when SFs are stabilized across multipl
e timescales, which capture complementary aspects of gradual environmental chang
e. Together, these results suggest that stability is more critical in continual
learning when changes are gradual, and that multi-timescale consolidation of pre
dictive representations is an effective approach.
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Abductive Reasoning with Probabilistic Commonsense
Joseph Cotnareanu, Chiara Roverato, Han Zhou, Didier Chételat, Yingxue Zhang, Ma
rk Coates
Recent efforts to improve the reasoning abilities of Large Language Models (LLMs
) have focused on integrating formal logic solvers within neurosymbolic framewor
ks. A key challenge is that formal solvers lack commonsense world knowledge, pre
venting them from making reasoning steps that humans find obvious. Prior methods
 address this by using LLMs to supply missing commonsense assumptions, but these
 approaches implicitly assume universal agreement on such commonsense facts. In
reality, commonsense beliefs vary across individuals. We propose a probabilistic
 framework for abductive commonsense reasoning that explicitly models this varia
tion, aiming to determine whether most people would judge a statement as true or
 false. We introduce Probabilistic Abductive CommonSense (PACS), a novel algorit
hm that uses an LLM and a formal solver to sample proofs as observations of indi
viduals’ distinct commonsense beliefs, and aggregates conclusions across these s
amples. Empirically, PACS outperforms chain-of-thought reasoning, prior neurosym
bolic methods, and search-based approaches across multiple benchmarks.
****

Exploring Nonlinear Pathway in Parameter Space for Machine Unlearning
Yingdan Shi, Ren Wang
Machine Unlearning (MU) aims to remove the information of specific training data

from a trained model, ensuring compliance with privacy regulations and user requests. While one line of existing MU methods relies on linear parameter updates via task arithmetic, they suffer from weight entanglement. In this work, we propose a novel MU framework called Mode Connectivity Unlearning (MCU) that leverages mode connectivity to find an unlearning pathway in a nonlinear manner. To further enhance performance and efficiency, we introduce a parameter mask strategy that not only improves unlearning effectiveness but also reduces computational overhead. Moreover, we propose an adaptive adjustment strategy for our unlearning penalty coefficient to adaptively balance forgetting quality and predictive performance during training, eliminating the need for empirical hyperparameter tuning. Unlike traditional MU methods that identify only a single unlearning model, MCU uncovers a spectrum of unlearning models along the pathway. Overall, MCU serves as a plug-and-play framework that seamlessly integrates with any existing MU methods, consistently improving unlearning efficacy. Extensive experiments on the image classification task demonstrate that MCU achieves superior performance. The codes are available at https://github.com/TIML-Group/Mode-Connectivity-Unlearning.

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Training with Honeypots: Reshaping How LLMs Fail Under Adversarial Attacks

Samuel Simko, Punya Syon Pandey, Zhijing Jin, Bernhard Schölkopf

Automated red-teaming of Large Language Models (LLMs) commonly relies on attack success rates (ASR) as a proxy for real-world harm, implicitly assuming that judge-detected violations correspond to actionable risk. In practice, safety judges are imperfect, and outputs that satisfy automated criteria for harm can vary widely in their operational usefulness. In this work, we investigate whether model failure modes can be reshaped so that, when defenses fail, they preferentially produce reduced-actionability outputs rather than highly actionable harm. Inspired by honeypots in computer security, we construct responses that are frequently flagged as harmful by automated judges yet provide limited real-world operational value, and treat them as hard negatives in the safety training pipeline. Our findings show that shaping how models fail under attack can improve overall safety by reducing both the real-world impact and the frequency of harmful failures, and serves as a practical complement to ASR-based evaluations.

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Recontextualization Mitigates Specification Gaming Without Modifying the Specification

Ariana Azarbal, Victor Gillioz, Vladimir Ivanov, Bryce Woodworth, jacob drori, Nevan Wichers, Aram Ebtekar, Alex Cloud, Alexander Matt Turner

Developers often struggle to specify correct training labels and rewards. Perhaps they don't need to. We propose recontextualization, which reduces how often language models "game" training signals, performing misbehaviors those signals mistakenly reinforce. We show recontextualization prevents models from learning to 1) overfit evaluation criteria at the expense of chat response quality; 2) special-case code to pass incorrect tests; 3) overwrite evaluation functions rather than write correct code; and 4) become sycophantic. Our method works by generating completions from prompts discouraging misbehavior and then recontextualizing them as though they were in response to prompts permitting misbehavior. Recontextualization trains language models to resist misbehavior even when instructions permit it. This mitigates the reinforcement of misbehavior from misspecified training signals, reducing specification gaming without improving the supervision signal.

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The Cost of Information: Phase Transitions in Contextual Bandits with Paid Observations

Xueping Gong, Jiheng Zhang

We study contextual bandits with paid observations, where the learner actively chooses which actions to observe at a given cost in each round, with the goal of minimizing total regret that jointly accounts for learning loss and observation expenditure. We develop a near-optimal algorithm for adversarial environments and show that even small observation costs fundamentally raise the minimax regret

order. We further uncover a novel phase transition under a free observation budget: below a critical threshold, free observations only reduce total cost without improving the regret rate; above it, asymptotic improvements become possible. To exploit this phenomenon, we design a meta-controller that adaptively switches between strategies to achieve near-optimal performance across all budget regimes. To handle large or infinite policy spaces, we also propose an oracle-efficient algorithm under a function approximation framework that maintains rigorous guarantees with computational efficiency. Our analysis also connects to related problems including switching costs, budgeted constraints, model misspecification, and knapsack bandits. Numerical experiments validate our theoretical findings.
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Revisiting Robustness for LLM Safety Alignment via Selective Geometry Control
Yonghui Yang, Wenjian Tao, Jilong Liu, Xingyu Zhu, Junfeng Fang, Huang Weibiao, Le Wu, Richang Hong, Tat-Seng Chua
Safety alignment of large language models remains brittle under domain shift and noisy preference supervision. Most existing robust alignment methods focus on uncertainty in alignment data, while overlooking optimization-induced fragility in preference-based objectives. In this work, we revisit robustness for LLM safety alignment from an optimization geometry perspective, and argue that robustness failures cannot be addressed by data-centric methods alone. We propose \textit{ShaPO}, a geometry-aware preference optimization framework that enforces worst-case alignment objectives via selective geometry control over alignment-critical parameter subspace. By avoiding uniform geometry constraints, ShaPO mitigates the over-regularization that can harm robustness under distribution shift. We instantiate ShaPO at two levels: token-level ShaPO stabilizes likelihood-based surrogate optimization, while reward-level ShaPO enforces reward-consistent optimization under noisy supervision. Across diverse safety benchmarks and noisy preference settings, ShaPO consistently improves safety robustness over popular preference optimization methods. Moreover, ShaPO composes cleanly with data-robust objectives, yielding additional gains and empirically supporting the proposed optimization-geometry perspective. The code is available at \url{https://github.com/liujilong0116/ShaPO}.
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CORAL: Uncertainty-Aware Regulation of Exposure Concentration in Recommender Systems
Nitin Bisht, Linjiang Guo, Xiuwen Gong, Huan Huo, Guandong Xu
Recommender systems (RS) may suffer from feedback-driven exposure concentration, where repeated engagement optimization collapses exposure onto a narrow set of categories, reducing catalog coverage and degrading long-horizon learning. Existing methods are often post hoc and typically lack principled uncertainty-aware risk estimates for regulating exposure under endogenous feedback. We therefore propose **CORAL**, a model-agnostic, uncertainty-aware framework that formulates exposure regulation as a constrained sequential decision problem. Specifically, we model self-reinforcing interactions to construct an exposure-saturation state, then derive an upper confidence bound on category-conditioned violation risk from observed history and incorporate it through a state-dependent penalty for adaptive intervention near saturation. Moreover, we provide theoretical guarantees for risk bounds, finite-time recovery, and efficient long-term performance. Extensive experiments on real-world datasets and controlled simulations validate the effectiveness of the proposed framework, which aligns with our theoretical analysis. Our code is available at: https://github.com/downw/CORAL.
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CoME: Empowering Channel-of-Mobile-Experts with Informative Hybrid-Capabilities Reasoning
Yuxuan Liu, Weikai Xu, Kun Huang, Changyu Chen, Jiankun Zhao, Pengzhi Gao, Wei Liu, Jian Luan, Shuo Shang, Bo Du, Ji-Rong Wen, Rui Yan
Mobile Agents requires hybrid-capabilities reasoning, including screen summary, subtask planning, action decision and action function. However, existing agents struggle to achieve both decoupled enhancement and balanced integration of these capabilities. To address these challenges, we propose Channel-of-Mobile-Experts

(CoME), a novel agent architecture consisting of four distinct experts, each aligned with a specific reasoning stage, CoME activates the corresponding expert to generate output tokens in each reasoning stage via output-oriented activation. To empower CoME with hybrid-capabilities reasoning, we introduce a progressive training strategy: Expert-FT enables decoupling and enhancement of different experts' capability; Router-FT aligns expert activation with the different reasoning stage; CoT-FT facilitates seamless collaboration and balanced optimization across multiple capabilities. To mitigate error propagation in hybrid-capabilities reasoning, we propose InfoGain-Driven DPO (Info-DPO), which uses information gain to evaluate the contribution of each intermediate step, thereby guiding CoME toward more informative reasoning. Comprehensive experiments show that CoME outperforms dense mobile agents and MoE methods on both AITZ and AMEX datasets.
****

Model Fusion via Retrofitting

Phoomraphee Luenam, Andreas Spanopoulos, Amit Sant, Sotiris Anagnostidis, Thomas Hofmann, Sidak Pal Singh

Model fusion seeks to combine independently trained neural networks into a single model without retraining, but is complicated by representational divergence arising from permutation invariance, random initialization, and heterogeneous training data. Existing methods struggle particularly in zero-shot settings under non-IID data distributions, and are often limited to specific architectures or pairwise fusion. We introduce a neuron-centric family of fusion algorithms that frames fusion as a principled representation-matching problem: intermediate neurons across parent models are grouped into target representations, which the fused model's corresponding sub-networks are then trained to approximate. Unlike prior work, our approach incorporates neuron attribution scores to bias alignment toward salient features, and can be applied to any architecture modularizable as a DAG of levels—empirically validated on VGGs, ResNets, and ViTs. Experiments across standard benchmarks show consistent improvements over existing fusion methods, with the largest gains in zero-shot and non-IID scenarios. Code is available at https://github.com/AndrewSpano/model-fusion-via-retrofitting}{https://github.com/AndrewSpano/model-fusion-via-retrofitting.
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SPARKLING: Balancing Signal Preservation and Symmetry Breaking for Width-Progressive Learning

Qifan Yu, Xinyu Ma, Zhijian Zhuo, Minrui Wang, Deyi Liu, Shiyi Zhan, Yiyuan Ma, liang xiang, Xingyan Bin, Di He

Progressive Learning (PL) reduces pre-training computational overhead by gradually increasing model scale. While prior work has extensively explored depth expansion, width expansion remains significantly understudied, with the few existing methods limited to the early stages of training. However, expanding width during the mid-stage is essential for maximizing computational savings, yet it remains a formidable challenge due to severe training instabilities.

Empirically, we show that naive initialization at this stage disrupts activation statistics, triggering loss spikes, while copy-based initialization introduces gradient symmetry that hinders feature diversity.

To address these issues, we propose **SPARKLING** (balancing **S**ignal **P**reservation **A**nd symmet**R**y brea**K**ing for width-progressive **L**earn**ING**), a novel framework for mid-stage width expansion.

Our method achieves signal preservation via RMS-scale consistency, stabilizing activation statistics during expansion. Symmetry breaking is ensured through asymmetric optimizer state reset and asymmetric learning rate re-warmup. Extensive experiments on dense and Mixture-of-Experts (MoE) models demonstrate that, across multiple width axes and optimizer families, SPARKLING consistently outperforms training from scratch and reduces training cost by up to 35% under $2\times$ width expansion.
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Reasoning Structure of Large Language Models

Frédéric Berdoz, Luca A Lanzendörfer, Fabian Farestam, Roger Wattenhofer

Large reasoning models (LRMs) are often evaluated using metrics such as final-an

swer accuracy or token count. However, identical scores on these metrics can hid e fundamentally different reasoning structures. To address this limitation, we i ntroduce a scalable LRM benchmark of logic puzzles and a pipeline that converts unstructured traces into verifiable reasoning graphs of claims and dependencies. This turns reasoning into a structured, measurable object whose topology can be quantitatively analyzed. Building on this, we define a reasoning efficiency met ric that quantifies how concentrated the model's logical flow is. Our analysis o n open-source reasoning models shows that structural measurements separate behav iors that token count and accuracy conflate, providing a practical tool for diag nosing failure modes and comparing how reasoning scales with puzzle difficulty.
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CodeClash: Benchmarking Goal-Oriented Software Engineering
John Yang, Kilian Lieret, Joyce Yang, Carlos E. Jimenez, Muhtasham Oblokulov, Ar yan Siddiqui, Ofir Press, Ludwig Schmidt, Diyi Yang
Existing coding benchmarks evaluate language models (LMs) on concrete, well-spec ified tasks such as fixing bugs or writing tests.
However, human programmers do not spend all day addressing isolated GitHub issue s.
Instead, real-world software development is grounded in the pursuit of high-leve l goals.
Evaluating whether LMs can iteratively develop code to accomplish open-ended obj ectives without explicit guidance remains an open challenge.
We introduce CodeClash, a benchmark where LMs compete in multi-round tournaments to build the best codebase for achieving a competitive objective.
Each round proceeds in two parts: agents edit their code, then their codebases c ompete head-to-head in a code arena that determines winners based on objectives like score maximization, resource acquisition, or survival.
Models must decide for themselves how to improve their code both absolutely and against their opponents.
We run 1680 tournaments to evaluate 8 LMs across 6 arenas, revealing how models exhibit diverse development styles and share fundamental limitations in strategi c reasoning.
Models also struggle with long-term codebase maintenance; repositories become pr ogressively messy and redundant.
Top models lose every round against expert human programmers.
We open-source CodeClash to advance the study of autonomous, goal-oriented code development.
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Scalable Power Sampling: Unlocking Efficient, Training-Free Reasoning for LLMs v ia Distribution Sharpening
Xiaotong Ji, Rasul Tutunov, Matthieu Zimmer, Haitham Bou Ammar
Reinforcement learning (RL) post-training is a dominant approach for improving t he reasoning performance of large language models (LLMs), yet growing evidence s uggests that its gains arise primarily from distribution sharpening rather than the acquisition of new capabilities. Recent work has shown that sampling from th e power distribution of LLMs using Markov chain Monte Carlo (MCMC) can recover p erformance comparable to RL post-training without relying on external rewards; h owever, the high computational cost of MCMC makes such approaches impractical fo r widespread adoption. In this work, we propose a theoretically grounded alterna tive that eliminates the need for iterative MCMC. We derive a novel formulation showing that the global power distribution can be approximated by a token-level scaled low-temperature one, where the scaling factor captures future trajectory quality. Leveraging this insight, we introduce a training-free and verifier-free algorithm that sharpens the base model's generative distribution autoregressive ly. Empirically, we evaluate our method on math, QA, and code tasks, and show th at our method matches or surpasses one-shot GRPO without relying on any external rewards, while reducing inference latency by over $10\times$ compared to MCMC-b ased sampling.
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DualCOIL: Offline Imitation Learning from Contrasting Demonstrations

Huy Hoang, Tien Anh Mai, Pradeep Varakantham, Tanvi Verma
Offline imitation learning typically learns from expert and unlabeled demonstrations, yet often overlooks the valuable signal in explicitly undesirable behaviors. In this work, we study offline imitation learning from contrasting behaviors, where the dataset contains both expert and undesirable demonstrations along with an unlabeled set of demonstrations. We propose a novel formulation that optimizes a difference of KL divergences over the state-action visitation distributions of expert and undesirable (or bad) data. Although the resulting objective is a DC (Difference-of-Convex) program, we prove that it becomes *convex* when expert demonstrations outweigh undesirable demonstrations, enabling a practical and stable non-adversarial training objective. Our method avoids adversarial training and handles both positive and negative demonstrations in a unified framework. Extensive experiments on standard offline imitation learning benchmarks demonstrate that our approach consistently outperforms state-of-the-art baselines.
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LARFT: Closing the Cognition-Action Gap for Length Instruction Following in Large Language Models
Wei Zhang, Lintong Du, Yuanhe Zhang, Zhenhong Zhou, Kun Wang, Li Sun, Sen Su
Despite the strong performance of Large Language Models (LLMs) on complex instruction-following tasks, precise control of output length remains a persistent challenge. Existing methods primarily attempt to enforce length constraints by externally imposing length signals or optimization objectives, while largely overlooking the underlying limitation: the model's intrinsic deficit in length cognition. To address this, we propose **LARFT** (**L**ength-**A**ware **R**einforcement **F**ine-**T**uning), a training framework that aligns the model's length cognition with its action. Specifically, LARFT integrates length-oriented reinforcement learning with a hindsight length awareness. By transforming on-policy data into hindsight self-awareness tasks where the model learns to identify the actual length of its own generation, LARFT jointly optimizes the model’s internal representation of length information and refines its policy to satisfy length constraints, thereby achieving precise and reliable length instruction following. Extensive experiments across four base models demonstrate that LARFT outperforms existing baselines, achieving an average improvement of **+20.92** points across three length instruction following benchmarks with only a marginal decline of **-1.45** points on four general capability benchmarks. Our code is available at https://github.com/Captain-zhangw/LARFT.
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Recurrent Equivariant Constraint Modulation: Learning Per-Layer Symmetry Relaxation from Data
Stefanos Pertigkiozoglou, Mircea Petrache, Shubhendu Trivedi, Kostas Daniilidis
Equivariant neural networks exploit underlying task symmetries to improve generalization, but strict equivariance constraints can induce more complex optimization dynamics that can hinder learning. Prior work addresses these limitations by relaxing strict equivariance during training, but typically relies on prespecified, explicit, or implicit target levels of relaxation for each network layer, which are task-dependent and costly to tune. We propose Recurrent Equivariant Constraint Modulation (RECM), a layer-wise constraint modulation mechanism that learns appropriate relaxation levels solely from the training signal and the symmetry properties of each layer's input-target distribution, without requiring any prior knowledge about the task-dependent target relaxation level. We demonstrate that under the proposed RECM update, the relaxation level of each layer provably converges
to a value upper-bounded by its symmetry gap, namely the degree to which its input-target distribution deviates from exact symmetry. Consequently, layers processing symmetric distributions recover full equivariance, while those with approximate symmetries retain sufficient flexibility to learn non-symmetric solutions when warranted by the data. Empirically, RECM outperforms prior methods across diverse exact and approximate equivariant tasks, including the challenging molecular conformer generation on the GEOM-Drugs dataset.
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Calibrated Preference Learning: The Case of Label Ranking
Santo M. A. R. Thies, Viktor Bengs, Timo Kaufmann, Sebastian Josef Vollmer, Eyke Hüllermeier
Calibration, the alignment of predicted probabilities with true outcome frequencies, is essential for reliable decision-making.
While extensively studied for classification and regression, calibration has not been formally addressed for probabilistic label ranking, where the goal is to predict a distribution over orderings of a label set. Naively treating rankings as classes ignores their structure and fails to capture important modalities such as pairwise and top-k predictions. We formalize calibration for label ranking and develop a hierarchy of notions covering full rankings, sub-rankings, and top-k rankings. We prove that full-rank calibration implies the others but not conversely, and sub-ranking and top-k calibration are incomparable. Empirically, we find popular label ranking models are often poorly calibrated, with substantial differences between sub-ranking and top-k metrics. Applying our framework to RLHF reward models, we find that calibration correlates strongly but not perfectly with benchmark accuracy, suggesting it captures a meaningful quality dimension beyond top-1 accuracy. These findings motivate future work on understanding the downstream effects of miscalibration and developing methods to correct it.
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Adaptive Preconditioners Trigger Loss Spikes in Adam
Zhiwei Bai, Zhangchen Zhou, Jiajie Zhao, Xiaolong Li, Zhiyu li, Feiyu Xiong, Hongkang Yang, Yaoyu Zhang, Zhi-Qin John Xu
Loss spikes commonly emerge during neural network training with the Adam optimizer across diverse architectures and scales, yet their underlying mechanism remains elusive. While previous explanations attribute these phenomena to sharper loss landscapes at lower loss, we show that landscape geometry alone is insufficient to explain the phenomenon. In this work, we pinpoint the root cause in the internal dynamics of Adam's second moment estimator. We identify a critical ``decoupling'' mechanism where the adaptive preconditioner v_t fails to track the instantaneous squared gradients g_t^2, causing the adaptive mechanism to effectively fail. This decoupling allows the preconditioner to decay autonomously despite rising gradients, which pushes the maximum eigenvalue of the preconditioned Hessian beyond the stability threshold $2/\eta$ for sustained periods, manifesting as dramatic loss spikes. Through a quadratic approximation analysis, we theoretically and experimentally characterize five distinct stages of spike evolution and propose a predictor for anticipating spikes based on gradient-directional curvature. We empirically find that the proposed loss spike mechanism, although derived from simplified models, generalizes well to practical scenarios ranging from small neural networks to large-scale Transformers.
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Ego3S: Select, Strengthen, and Synchronize for Efficient Egocentric Reasoning
Shenshen Li, Kaiyuan Deng, RuoHuai Xie, Xing Xu, Heng Tao Shen, Yazhou Yao, Fumin Shen
Egocentric reasoning fundamentally differs from third-person understanding in LVLMs. Third-person settings offer wide and stable contexts with consistent global regularities, allowing models to utilize broad statistical correlations. In contrast, egocentric scenes are highly dynamic and heterogeneous, where decisive cues are localized and atypical. Therefore, robust egocentric reasoning requires models to focus on ''what is seen now'', i.e., the immediate visual input. However, existing methods tend to exhibit "inertial thinking'', relying excessively on language priors and global context. To address this limitation, we propose a novel three-stage Ego3S framework to ground models' reasoning in interaction evidence. Specifically, before training, we first utilize the counterfactual-based paradigm to select high-value samples that effectively activate multimodal reasoning, thus mitigating the over-reliance on language priors and global context. Moreover, we introduce an interaction-centric reward for reinforcement learning that strengthens the model’s sensitivity to localized interaction cues. Finally, during training, we employ a variance-aware learning schedule that monitors reward distributions to dynamically synchronize data selection with the evolving model

competence. Experiments on five datasets show that our Ego3S consistently achieves superior performance using only 26.5% of the training data, while reducing computational costs by over 46%. Code is available at https://github.com/Mrshenshen/Ego3S.
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Demystifying Multimodal Biomolecular Co-design With Intrinsic Geodesic Coupling
Keyue Qiu, Xintong Wang, Zhilong Zhang, Hao Zhou, Wei-Ying Ma
Biomolecules such as proteins and small-molecule ligands play a central role in biological systems, arising from the tight interplay between sequence and three-dimensional structure.
Recent generative models for biomolecular co-design aim to capture this interplay by jointly modeling coupled modalities.
However, existing approaches largely adopt a parallel execution of marginal generative processes, implicitly enforcing fixed synchronous coupling.
We argue that a critical but overlooked degree of freedom lies in how these marginal processes are \emph{temporally coupled} during training and generation, where inappropriate coupling can introduce high-variance supervision and inconsistent intermediate states, affecting modality consistency.
To address this, we introduce GeoCoupling, a systematic framework that optimizes for temporal couplings between heterogeneous modalities.
Empirical results across structure-based drug design and unconditional protein design demonstrate the learned couplings consistently outperform synchronous and randomly coupled baselines, yielding biomolecules with improved physical validity and diversity.
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RLCracker: Evaluating the Worst-Case Vulnerability of LLM Watermarks with Adaptive RL Attacks
Hanbo Huang, Yiran Zhang, Hao Zheng, Xuan Gong, Yihan Li, Lin Liu, Zhuotao Liu, Shiyu Liang
Large language model (LLM) watermarking has shown promise in detecting AI-generated content and mitigating misuse, with prior work claiming robustness against paraphrasing and text editing. In this paper, we argue that existing evaluations are not sufficiently adversarial, obscuring critical vulnerabilities and overstating the security. To address this, we introduce the *adaptive robustness radius*, a formal metric that quantifies the worst-case resilience of watermarks against adaptive adversaries. By lifting the paraphrase space into a KL-divergence ball, we approximate this radius and theoretically demonstrate that optimizing the attack context and model parameters can significantly reduce the approximated radius, making watermarks highly vulnerable to paraphrase attacks.
Leveraging this insight, we propose RLCracker, a reinforcement learning (RL)–based adaptive attack that erases watermarks while preserving semantic fidelity. RLCracker requires only *limited* watermarked examples and *zero* access to the detector. Despite weak supervision, it empowers a 3B model to achieve 98.5\% removal success with minimal semantic shift on 1,500-token Unigram-marked texts after training on only *100* short samples. This performance dramatically exceeds 6.75\% by GPT-4o and generalizes across five model sizes over ten watermarking schemes. Our code is available in this repository.
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From Token to Token Pair: Efficient Prompt Compression for Large Language Models in Clinical Prediction
Mingcheng Zhu, Zhiyao Luo, Yu Liu, Tingting Zhu
By processing electronic health records (EHRs) as natural language sequences, large language models (LLMs) have shown potential in clinical prediction tasks such as mortality prediction and phenotyping. However, longitudinal or highly frequent EHRs often yield excessively long token sequences that result in high computational costs and even reduced performance. Existing solutions either add modules for compression or remove less important tokens, which introduce additional inference latency or risk losing clinical information. To achieve lossless compression of token sequences without additional cost or loss of performance, we propose Medical Token-Pair Encoding (MedTPE), a layered method that extends standard

tokenisation for EHR sequences. MedTPE merges frequently co-occurring medical token pairs into composite tokens, providing lossless compression while preserving the computational complexity through a dependency-aware replacement strategy. Only the embeddings of the newly introduced tokens of merely 0.5-1.0\% of the LLM's parameters are fine-tuned via self-supervised learning. Experiments on real-world datasets for two clinical scenarios demonstrate that MedTPE reduces input token length by up to 31\% and inference latency by 34-63\%, while maintaining or even improving both predictive performance and output format compliance across multiple LLMs and four clinical prediction tasks. Furthermore, MedTPE demonstrates robustness across different input context lengths and generalisability to scientific and financial domains and different languages. The code is available in the GitHub repository.
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GHOST: Unmasking Phantom States in Mamba2 via Grouped Hidden-state Output-aware Selection & Truncation
Michael Menezes, Anastasios Kyrillidis
While Mamba2's expanded state dimension enhances temporal modeling, it incurs substantial inference overhead that saturates bandwidth during autoregressive generation. Standard pruning methods fail to address this bottleneck: unstructured sparsity leaves activations dense, magnitude-based selection ignores runtime dynamics, and gradient-based methods impose prohibitive costs. We introduce GHOST (Grouped Hidden-state Output-aware Selection and Truncation), a structured pruning framework that approximates control-theoretic balanced truncation using only forward-pass statistics. By jointly measuring controllability and observability, GHOST rivals the fidelity of gradient-based methods without requiring backpropagation. As a highlight, on models ranging from 130M to 2.7B parameters, our approach achieves a 50% state-dimension reduction with approximately 1 perplexity point increase on WikiText-2. Code is available at https://github.com/Menezmic21/mamba2_ghost.
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Active Continual Learning with Metaplastic Binary Bayesian Neural Networks
Kellian Cottart, Theo Ballet, Djohan Bonnet, Damien Querlioz
Always-on edge systems must keep learning as conditions change under tight compute budgets and must detect unreliable predictions. Bayesian binary neural networks are attractive in this setting, but mean-field Bernoulli posteriors can saturate on long non-stationary streams, wiping out epistemic uncertainty and freezing plasticity. We propose BiMU, derived from a bounded-memory variational objective that balances stability, plasticity, and forgetting. BiMU combines a data term with controlled relaxation toward the prior and an uncertainty-dependent step size that prevents saturation and sustains informative uncertainty. This non-degenerate posterior enables fully online, buffer-free active querying via Monte Carlo disagreement, reducing label queries and backpropagation updates under imbalance. BiMU sustains learning and strong OOD detection on 1000-task Permuted-MNIST, and on OpenLORIS-Object achieves up to 32× label/update savings at matched accuracy under class imbalance and feature compression.
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Three Years of r/ChatGPT: Societal Impact Evaluations from Social Media Data
Jessica Dai, Sean Garcia, Emma Pierson, Benjamin Recht, Nika Haghtalab
ChatGPT was launched on November 30, 2022; the r/ChatGPT subreddit was created just one day later. Since then, chatbot-based AI products have gone from niche proofs-of-concept to widely-used household names. However, the ways in which adoption has developed, especially among non-experts, remains poorly understood. In this paper, we propose a principled framework for using social media as a data source for understanding the societal impact of widely-adopted consumer AI products, as well as a general approach to monitoring for societally-impactful trends in real time. We apply our framework to conduct what is, to the best of our knowledge, the first longitudinal study of r/ChatGPT. We find that, overall, r/ChatGPT posts over time illustrate the normalization of ChatGPT as an everyday consumer product rather than an exceptional, novel technology. However, our retrospective analysis also finds that posts about using ChatGPT for mental health support,

and posts about developing emotional attachments to ChatGPT, both rise steadily in frequency immediately after the launch of GPT-4o in May 2024. We show that our real-time method can detect the increase in emotional engagement as early as October 2024—months before OpenAI made any (public) acknowledgment of this impact.

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Finite-Width Neural Tangent Kernels from Feynman Diagrams

Max Guillen, Philipp Misof, Jan E Gerken

Neural tangent kernels (NTKs) are a powerful tool for analyzing deep, non-linear neural networks. In the infinite-width limit, NTKs can easily be computed for most common architectures, yielding full analytic control over the training dynamics. However, at infinite width, important properties of training such as NTK evolution or feature learning are absent. Nevertheless, finite width effects can be included by computing corrections to the Gaussian statistics at infinite width. We introduce Feynman diagrams for computing finite-width corrections to NTK statistics. These dramatically simplify the necessary algebraic manipulations and enable the computation of layer-wise recursion relations for arbitrary statistics involving preactivations, NTKs and certain higher-derivative tensors (dNTK and ddNTK) required to predict the training dynamics at leading order. We demonstrate the feasibility of our framework by extending stability results for deep networks from preactivations to NTKs and proving the absence of finite-width corrections for scale-invariant nonlinearities such as ReLU on the diagonal of the Gram matrix of the NTK. We numerically implement the complete set of equations necessary to compute the first-order corrections for arbitrary inputs and demonstrate that the results follow the statistics of sampled neural networks for widths $n \gtrsim 20$.

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MESA: Improving MoE Safety Alignment via Decentralized Expertise

Yitong Sun, Yao Huang, Teng Li, Ranjie Duan, Yichi Zhang, Xingjun Ma, Hui Xue, Xingxing Wei

Mixture-of-Experts (MoE) architectures scale Large Language Models (LLMs) efficiently, enabling greater capacity with reduced computational cost by dynamically routing inputs to relevant experts, yet introduce a critical vulnerability: Safety Sparsity, where safety capabilities concentrate in few experts, making them susceptible to adversarial bypassing. Meanwhile, conventional alignment methods uniformly adapt all parameters, ignoring their functional differences and inadvertently degrading performances.

To address these challenges, we propose MESA (MoE Safety Alignment), a targeted alignment framework for MoE-based LLMs that strategically decentralizes safety responsibility to maximize coverage while minimizing interference with utility. Based on Optimal Transport (OT) theory, MESA operates through two mechanisms: (1) Expert Capacity Reallocation uses a transport cost matrix to distribute safety duties to the most cost-effective experts, and (2) Dynamic Routing Refinement constrains the router to precisely activate these decentralized modules. Experiments show that MESA achieves robust defensive performance against varied harmful benchmarks while preserving helpfulness. Code is available at https://github.com/lorraine021/MESA.

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MoLF: Mixture-of-Latent-Flow for Pan-Cancer Spatial Gene Expression Prediction from Histology

Susu Hu, Stefanie Speidel

Inferring spatial transcriptomics (ST) from histology enables scalable histogenomic profiling, yet current methods are largely restricted to single-tissue models. This fragmentation fails to leverage biological principles shared across cancer types and hinders application to data-scarce scenarios. While pan-cancer training offers a solution, the resulting heterogeneity challenges monolithic architectures. To bridge this gap, we introduce **MoLF** (*Mixture-of-Latent-Flow*), a generative model for pan-cancer histogenomic prediction. MoLF leverages a conditional Flow Matching objective to map noise to the gene latent manifold, parameterized by a Mixture-of-Experts (MoE) velocity field. By dynamically routing inpu

ts to specialized sub-networks, this architecture effectively decouples the optimization of diverse tissue patterns. Our experiments demonstrate that MoLF establishes a new state-of-the-art, consistently outperforming both specialized and foundation model baselines on pan-cancer benchmarks. Furthermore, MoLF exhibits zero-shot generalization to cross-species data, suggesting it captures fundamental, conserved histo-molecular mechanisms.

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LoSA: Locality Aware Sparse Attention in Diffusion Language Models

Haocheng Xi, Harman Singh, Yuezhou Hu, Coleman Richard Charles Hooper, Rishabh Tiwari, Aditya Tomar, Minjae Lee, Wonjun Kang, Michael W. Mahoney, Chenfeng Xu, Kurt Keutzer, Amir Gholami

Block-wise diffusion language models (DLMs) generate multiple tokens in parallel, offering a promising alternative to autoregressive decoding. However, their inference efficiency remains bottlenecked by memory-bound attention in long-context scenarios. Naïve sparse attention is ineffective for DLMs due to the KV inflation problem: different queries select different prefix positions, causing the union of accessed KV pages to remain large.

To address this challenge, we observe that block-wise diffusion exhibits locality of representation changes across denoising steps: only a small fraction of tokens (active tokens) undergo significant hidden-state updates, while most tokens (stable tokens) remain nearly unchanged.

Based on this insight, we propose LoSA (Locality-aware Sparse Attention), which reuses cached prefix-attention results for stable tokens and applies sparse attention only to active tokens with large representation changes. This design reduces the number of queries contributing to the union of KV indices, substantially shrinking the KV pages that must be loaded.

Across multiple block-wise DLMs and reasoning benchmarks, LoSA preserves near-dense accuracy while significantly improving efficiency, achieving up to 4.14× speedup over dense attention on RTX A6000 GPUs. LoSA also delivers up to 5% average improvement over baselines across all datasets and configurations, demonstrating the effectiveness of the proposed method.

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DADP: Domain Adaptive Diffusion Policy

Pengcheng Wang, Qinghang Liu, Haotian Lin, Yiheng Li, Guojian Zhan, Masayoshi Tomizuka, Yixiao Wang

Learning domain adaptive policies that can generalize to unseen transition dynamics, remains a fundamental challenge in learning-based control.

Substantial progress has been made through domain representation learning to capture domain-specific information, thus enabling domain-aware decision making.

We analyze the process of learning domain representations through dynamical prediction and find that selecting contexts adjacent to the current step causes the learned representations to entangle static domain information with varying dynamical properties.

Such mixture can confuse the conditioned policy, thereby constraining zero-shot adaptation.

To tackle the challenge, we propose DADP (Domain Adaptive Diffusion Policy), which achieves robust adaptation through unsupervised disentanglement and domain-aware diffusion injection.

First, we introduce Lagged Context Dynamical Prediction, a strategy that conditions future state estimation on a historical offset context; by increasing this temporal gap, we unsupervisedly disentangle static domain representations by filtering out transient properties.

Second, we integrate the learned domain representations directly into the generative process by biasing the prior distribution and reformulating the diffusion target.

Extensive experiments on challenging benchmarks across locomotion and manipulation demonstrate the superior performance, and the generalizability of DADP over prior methods.

More visualization results are available on the https://outsider86.github.io/DomainAdaptiveDiffusionPolicy/.

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IndexMem: Learned KV-Cache Eviction with Latent Memory for Long-Context LLM Infe
rence
Xintong Yang, Hao Gu, Binxing Xu, Lujun Li, Bei Liu, Jiacheng Liu, Qiyuan Zhu, Y
ike Guo, Sirui Han
Large Language Models (LLMs) are increasingly expected to operate over long cont
exts, yet standard softmax attention incurs a KV cache that grows linearly with
sequence length, quickly becoming the bottleneck for long context inference. A p
ractical remedy is to evict less important KV entries; however, existing evictio
n policies are largely heuristic and struggle to capture the rich, input-depende
nt distribution of token importance. In this work, we introduce a **learnable in
dexer** that predicts KV importance, enabling more accurate retention of critica
l tokens. Meanwhile, naively evicting tokens permanently discards their informat
ion, leading to irreversible forgetting and degraded retrieval over long ranges.
 To address this, we propose a lightweight **latent memory module** that compres
ses evicted tokens into a compact, online-updated state and provides residual re
adouts to compensate for the attention contributions lost through KV eviction.
Collectively, our method enables accurate long-context inference under a bounded
 KV budget, delivering consistent improvements on RULER (4K/16K) across Qwen, Mi
stral, and Llama models (up to 25 points under aggressive eviction), markedly mo
re stable Needle-in-a-Haystack retrieval, and superior LongBench scores and comp
ression curves compared to existing eviction policies.
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Fair Transit Stop Placement: A Clustering Perspective and Beyond
Haris Aziz, Ling Gai, Yuhang Guo, Jeremy Vollen
We study the transit stop placement (TrSP) problem in general metric spaces, whe
re agents travel between source–destination pairs and may either walk directly o
r utilize a shuttle service via selected transit stops. We investigate fairness
in TrSP through the lens of justified representation (JR) and the core, and unco
ver a structural correspondence with fair clustering. Specifically, we show that
 a constant-factor approximation to proportional fairness in clustering can be u
sed to guarantee a constant-factor bi-parameterized approximation to core. We es
tablish a lower bound of 1.366 on the approximability of JR, and moreover show
 that no clustering algorithm can approximate JR within a factor better than 3
. Going beyond clustering, we propose the Expanding Cost Algorithm, which achiev
es a tight 2.414-approximation for JR, but does not give any bounded core guar
antee. In light of this, we introduce a parameterized algorithm that interpolate
s between these approaches, and enables a tunable trade-off between JR and core.
 Finally, we complement our results with an experimental analysis using small-ma
rket public carpooling data.
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On Group Relative Policy Optimization Collapse in Agent Search: The Lazy Likelih
ood-Displacement
Wenlong Deng, Yushu Li, Boying Gong, Yi Ren, Christos Thrampoulidis, Xiaoxiao Li
Tool-integrated (TI) reinforcement learning (RL) enables large language models (
LLMs) to perform multi-step reasoning by interacting with external tools such as
 search engines and retrievers. Group Relative Policy Optimization (GRPO), exemp
lified by the recent Search-R1, offers fast convergence and a value-free formula
tion that makes it appealing for this setting, yet consistently suffers from tra
ining collapse. We identify Lazy Likelihood Displacement (LLD), a systematic red
uction or stagnation in the likelihood of both correct and incorrect responses,
as the core mechanism driving this failure. LLD emerges early and triggers a sel
f-reinforcing LLD Death Spiral, where declining likelihood leads to low-confiden
ce responses, inflating gradients, and ultimately causing collapse. We empirical
ly characterize this process across models on a Search-R1-style, search-integrat
ed question answering task, revealing a consistent three-phase trajectory: early
 stagnation, steady decay, and accelerated collapse. To address this, we propose
 a likelihood-preserving regularization LLDS that activates only when a response
 action’s likelihood decreases, and regularizes only the tokens responsible. Thi
s fine-grained structure mitigates LLD with minimal interference. Our method sta

bilizes training, prevents gradient explosion, and yields substantial performanc e improvements across seven benchmarks, including relative improvements of +45.2 % on Qwen2.5-3B and +37.1% on Qwen2.5-7B over vanilla GRPO training. Our results establish LLD as a previously overlooked bottleneck in GRPO- based TIRL and pro vide a practical path toward stable, scalable training of tool-integrated RL.
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TextME: Bridging Unseen Modalities Through Text Descriptions
Soyeon Hong, Jinchan Kim, Jaegook You, Seungtaek Choi, Suha Kwak, Hyunsouk Cho
Expanding multimodal representations to novel modalities is constrained by relia nce on largescale paired datasets (e.g., text–image, text–audio, text–3D, text–m olecule), which are costly and often infeasible in domains requiring expert anno tation such as medical imaging and molecular analysis. We introduce TextME, to t he best of our knowledge the first modality expansion framework based on text-on ly training, projecting diverse modalities into LLM embedding space as a unified anchor. Our approach exploits the geometric structure of pretrained contrastive encoders to enable zero-shot cross-modal transfer using only text descriptions, without paired supervision. We empirically validate that such consistent modali ty gaps exist across image, video, audio, 3D, X-ray, and molecular domains, demo nstrating that text-only training can preserve substantial performance of pretra ined encoders. We further show that our framework enables emergent cross-modal r etrieval between modality pairs not explicitly aligned during training (e.g., au dioto-image, 3D-to-image). These results establish text-only projection training as a practical alternative to paired supervision for modality expansion. The co de is available at https://soyeonhh.github.io/TextME/.
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dTRPO : Trajectory Reduction in Policy Optimization of Diffusion Large Language Models
Wenxuan Zhang, Lemeng Wu, Changsheng Zhao, Ernie Chang, Mingchen Zhuge, Zechun L iu, DiJia Su, Hanxian Huang, Jun Chen, Chong Zhou, Raghuraman Krishnamoorthi, Vi kas Chandra, Mohamed Elhoseiny, Wei Wen
Diffusion Large Language Models (dLLMs) introduce a new paradigm for language ge neration, which in turn presents new challenges for aligning them with human pre ferences. In this work, we aim to improve the policy optimization for dLLMs by r educing the cost of the trajectory probability calculation, thereby enabling sca led-up offline policy training. We prove that: (i) under reference policy regula rization, the probability ratio of the newly unmasked tokens is an unbiased esti mate of that of intermediate diffusion states, and (ii) the probability of the f ull trajectory can be effectively estimated with a single forward pass of a re-m asked final state. By integrating these two trajectory reduction strategies into a policy optimization objective, we propose Trajectory Reduction Policy Optimiz ation (dTRPO). We evaluate dTRPO on 7B dLLMs across instruction-following and re asoning benchmarks. Results show that it substantially improves the core perform ance of state-of-the-art dLLMs, achieving gains of up to 9.6% on STEM tasks, up to 4.3% on coding tasks, and up to 3.0% on instruction-following tasks. Moreover , dTRPO exhibits strong training efficiency due to its offline, single-forward n ature, and achieves improved generation efficiency through high-quality outputs.
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Proteus: Lookup-Free Trellis-Coded Quantization by Lattice-Breaking Compute Code s for 2-Bit LLMs
Zhengwu Yang, Xunchao Li, Ke Cheng, Kunlong Liu, Jianfeng Yang, Haoshuang Wang, Kaipeng Deng, Qingqing Dang, Yanlin Sha, Yanjun Ma, Dianhai Yu
Autoregressive decoding of large language models is frequently memory-traffic bo und, so ultra-low-bit weight-only PTQ helps only if dequantization avoids irregu lar codebook or LUT access in the inner loop. Under the GPU-friendly bitshift tr ellis, existing 2-bit trellis-coded quantization (TCQ) pipelines either reintrod uce micro-LUTs or suffer overlap-amplified artifacts because incoherence improve s global Gaussianity but does not guarantee overlap-local joint geometry. We int roduce Proteus a strictly lookup-free TCQ framework whose computed generator MUL -BAL uses cheap integer mixing plus a per-layer affine Gaussianizer to produce o verlap-robust, near-Gaussian code values with zero runtime table loads. Proteus

instantiates each layer by selecting from a tiny, pre-vetted candidate pool and then applies lightweight channel compensation and optional few-shot distillation that tune only per-layer affine statistics while keeping packed indices and the bitshift-trellis decoder fixed. On Llama 2 (7B–70B) at 2-bit PTQ, Proteus improves perplexity and zero-shot accuracy over strong TCQ/PTQ baselines and reduces end-to-end decode bandwidth at comparable throughput (e.g., 740 vs. 1020 GB/s on 70B).
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Beyond Generative Priors: Minority Sampling with JEPA-Guided Diffusion
Sol Park, Soobin Um
Minority sampling aims to generate low-density instances on a data manifold and is of central importance in applications such as medical diagnosis, anomaly detection, and creative AI. Existing approaches, however, define minority samples relative to generative priors learned from training data, confining rarity to model-specific notions that may poorly reflect real-world semantics. In this work, we propose a world-centric perspective on minority sampling, which defines rarity with respect to real-world priors rather than generator-induced densities. To this end, we introduce \emph{JEPA guidance}, a diffusion sampling framework guided by a Joint-Embedding Predictive Architecture (JEPA)---a class of world models that encode broad, semantically rich representations. JEPA guidance steers diffusion trajectories toward low-density regions under the implicit density induced by the JEPA, thereby aligning generated minorities with real-world semantic rarity. To make JEPA guidance computationally practical, we develop principled approximation strategies accompanied by theoretical error bounds, significantly reducing the overhead of guidance computation. Extensive experiments across unconditional, class-conditional, and text-to-image generation demonstrate that JEPA guidance consistently improves the fidelity and semantic validity of minority samples, outperforming generator-centric baselines in capturing real-world notions of rarity. Code is available at https://github.com/soobin-um/jepa-guidance.
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The Cylindrical Representation Hypothesis for Language Model Steering
Lang Gao, Jinghui Zhang, Wei Liu, Fengxian Ji, Chenxi Wang, Zirui Song, Akash Ghosh, Youssef Mohamed, Preslav Nakov, Xiuying Chen
Steering is widely used for controlling large language models, yet its effects are often unstable and difficult to predict. Existing theoretical accounts are largely based on the Linear Representation Hypothesis (LRH), which assumes that concepts can be orthogonalized for lossless control. However, this assumption rarely holds in practice and cannot explain the variability of steering outcomes. We propose the Cylindrical Representation Hypothesis (CRH), a geometric extension of LRH that relaxes the orthogonality assumption while preserving linear concept representations. We show that overlapping concept contributions naturally induce a sample-specific cylindrical structure consisting of a central axis, a normal plane, and sensitive sectors. The central axis captures the primary semantic transition associated with a target concept, while the normal plane governs steering sensitivity. Within this plane, some sectors facilitate concept activation, while others suppress or delay it. CRH reveals an asymmetry in steering predictability: the normal plane can be inferred from difference vectors, but the sensitive sectors cannot, introducing an intrinsic source of uncertainty. This explains why steering outcomes vary across samples even when intervention directions are well aligned. Experiments spanning 100 concepts, multiple models, and diverse steering methods provide consistent evidence for the predicted cylindrical structure, suggesting that steering variability arises from representation geometry rather than imperfect steering vectors. Our code is available at: https://github.com/mbzuai-nlp/CRH.
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Spurious Rewards Paradox: Mechanistically Understanding How RLVR Activates Memorization Shortcuts in LLMs
Lecheng Yan, Ruizhe Li, Guanhua Chen, Qing Li, Jiahui Geng, Wenxi Li, Longyue Wang, Chenyang Lyu
Reinforcement Learning with Verifiable Rewards (RLVR) is highly effective for en

hancing LLM reasoning, yet recent evidence shows models like Qwen2.5 achieve sig
nificant gains even with spurious rewards. We investigate this phenomenon and id
entify ``Perplexity Paradox'': spurious RLVR triggers a divergence where answer-
token perplexity drops while prompt-side coherence degrades, suggesting model is
 bypassing reasoning in favor of memorization. Using a suite of mechanistic inte
rpretability tools, including Path Patching and Logit Lens, we identify a previo
usly unknown Anchor–Adapter circuit. This circuit enables model to bypass reason
ing and directly retrieve memorized solutions under spurious RLVR. We localize a
 Functional Anchor in middle layers (L18–20) that triggers retrieval of memorize
d solutions, followed by Structural Adapters in later layers (L21+) that transfo
rm representations to accommodate shortcut signal. Finally, we demonstrate that
 scaling specific MLP keys within this circuit allows for bidirectional causal st
eering, i.e., artificially amplifying or suppressing contamination-driven perfor
mance. Our results provide a mechanistic roadmap for identifying and mitigating
 data contamination in RLVR-tuned models.
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Stability beyond Bounded Differences: Sharp Generalization Bounds under Finite $
L_p$ Moments

Qianqian Lei, Soham Bonnerjee, Yuefeng Han, Wei Biao Wu

While algorithmic stability is a central tool for understanding generalization o
f learning algorithms, existing high-probability guarantees typically rely on un
iform boundedness or sub-Gaussian/sub-Weibull tail assumptions, which can be ove
rly restrictive for modern settings with heavy-tailed or unbounded losses. We de
velop a stability-based framework that requires only a finite L_p moment condi
tion. Our first contribution is sharp concentration inequalities for functions o
f independent random variables under L_p constraints, extending McDiarmid's bo
unded-differences techniques beyond the classical regime. Leveraging these resul
ts, we derive sharp high-probability generalization bounds across a range of lea
rning paradigms, including empirical risk minimization, transductive regression,
 and meta-learning. These guarantees show that L_p stability suffices for robu
st generalization even when boundedness fails, substantially weakening the stand
ard assumptions in the stability literature.
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RaGEP: Rank-aware Geometric Expert Pruning for Mixture-of-Experts Language Model
s

Wentao Hu, Zeyu Zhu, Mingkuan Zhao, Zhenhua An, Yanbo Zhai, Shanhong yu, Huilin
 Zhou, Xin Lai, Xiaoyan Zhu, Jiayin Wang

Sparse Mixture-of-Experts (MoE) architectures scale model capacity efficiently b
ut suffer from massive static parameter footprints, creating significant deploym
ent burdens on memory-constrained hardware. Existing post-training pruning metho
ds often rely on scalar statistics, ignoring the representational geometry of ex
pert feature spaces. This leads to sub-optimal resource allocation across layers
 and the retention of redundant experts. To address this, we propose a Rank-awar
e Geometric Expert Pruning (RaGEP) framework to compress MoE models by analyzing
 the geometric properties of expert activations. First, in the inter-layer alloc
ation stage, we introduce a Rank-aware budget allocation mechanism that adaptive
ly assigns expert budgets based on the effective rank of layer-wise representati
ons. Second, in the intra-layer selection stage, we propose a Spectral-Salience
Pruning metric that harmonizes subspace orthogonality and activation magnitude t
o identify high-energy orthogonal experts. Extensive experiments across MoE mode
ls of different scales show that our method consistently outperforms state-of-th
e-art baselines on a diverse set of zero-shot tasks, while reducing model size a
nd inference cost.
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NaviCache: Test-Time Self-Calibration Caching for Video Generation

Zheqi Lv, Zhibo Zhu, Jinke Wang, Qi Tian, Shengyu Zhang, Zhengyu Chen, Chengxi Z
ang, Zhou Zhao, Fei Wu

Video Diffusion Models (VDMs) is constrained by immense computational costs. Whi
le offline calibration-based acceleration suffers from calibration data dependen
cy, prohibitive calibration duration, and susceptibility to distribution shifts,

offline calibration-free methods eliminate these hurdles. However, since they rely on instantaneous zero-order approximations where the mapping between input and output differences varies in real-time, they are susceptible to observational noise and ignore the intrinsic momentum within the diffusion trajectory. In this paper, we propose NaviCache, a plug-and-play test-time self-calibration method re-conceptualizing feature evolution as an Inertial Navigation System (INS) problem. NaviCache bridges the fundamental domain gap and the non-stationary nature of diffusion by modeling the relative coupling between input and output variations. We introduce a dual-state estimation architecture that adaptively tracks the feature change ratio and its latent drift, initialized via a specialized Initial Alignment phase. By integrating a time-dependent noise schedule with an uncertainty-aware Measurement Update mechanism, NaviCache provides a theoretically grounded mechanism for error-bounded block skipping.
Extensive experiments on the HunyuanVideo, Wan, and Open-Sora series demonstrate that NaviCache exhibits more accurate error judgment for block skipping and achieves outstanding comprehensive performance.
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Dual-Latent Memory Routing for Vision-Language Reasoning
Hao-Xuan Ma, Jin-Fei Qi, YiCheng Xiao, Han-Jia Ye
Multimodal large language models (MLLMs) have recently made strong progress in vision-language reasoning, yet their performance often degrades as generations grow longer. A key factor is that they frequently lose track of earlier visual evidence and intermediate constraints under a monolithic growing context. Inspired by how humans separately recall what they see and what they infer when solving complex tasks, we propose DLMR, a parameter-efficient mechanism that equips MLLMs with Dual Latent Memories: a visual memory that compresses image evidence and a reasoning memory that tracks intermediate conclusions and constraints. A Router then dynamically decides which memory and how much to reuse during inference, preserving visual grounding while maintaining coherent long-horizon reasoning. DLMR is trained in three stages, from latent memory construction to selective router learning, while keeping the base MLLM frozen, yielding substantial gains on both general and reasoning benchmarks with only a small number of additional trainable parameters. Analyses further show interpretable, state-dependent routing with specialized memory roles and reduced decoding tokens over long generations. Code is available at https://github.com/Hunter-Wrynn/DLMR.
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Latent Thoughts Tuning: Bridging Context and Reasoning with Fused Information in Latent Tokens
Weihao Liu, Dehai Min, Lu Cheng
While explicit Chain-of-Thought (CoT) equips Large Language Models (LLMs) with strong reasoning capabilities, it constrains the model's thoughts to a discrete vocabulary space. Recently, reasoning in continuous latent space has emerged as a promising alternative, but current paradigms suffer from feature collapse and instability due to distribution mismatch when recurrently reusing hidden states, or alignment issues when relying on assistant models. To address this, we propose $\textbf{Latent Thoughts Tuning (\textit{LT-Tuning})}$, a post-training framework that redefines how latent thoughts are constructed and deployed. Instead of relying solely on raw hidden states, our method introduces a $\textbf{Context-Prediction-Fusion}$ mechanism that jointly leverages contextual hidden states and predictive semantic guidance from the vocabulary embedding space. Combined with a progressive three-stage curriculum learning pipeline, $\textit{LT-Tuning}$ also enables dynamic switching between latent and explicit thinking modes. Experiments demonstrate that our method outperforms existing latent reasoning baselines, effectively mitigating feature collapse and achieving robust reasoning accuracy.
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Kinematics-Driven Gaussian Shape Deformation for Blurry Monocular Dynamic Scenes
Yeon-Ji Song, Kiyoung Kwon, Junoh Lee, Jin-Hwa Kim, Byoung-Tak Zhang
Reconstructing dynamic 3D scenes from blurry monocular videos is challenging because motion-induced blur entangles object motion and geometry, hindering geometr

ic consistency. We present Kinematics-GS, a kinematics-aware framework that models blur as motion-aligned deformation and introduces a kinematic prior to reparameterize Gaussian shapes along motion trajectories, thereby mitigating degenerate shape collapse without auxiliary motion supervision. To stabilize optimization, we decompose scenes into dynamic and static components using temporal deformation variance and employ a coarse-to-fine deformation strategy to capture both global motion and fine-grained details. We also introduce a challenging real-world dataset of deformable and elastic objects exhibiting non-rigid motion with spatially non-uniform motion blur that obscures geometric cues. Extensive experiments on real-world benchmarks with realistic motion blur demonstrate that Kinematics-GS outperforms prior methods by a clear margin in monocular dynamic scene reconstruction, highlighting its effectiveness in handling complex and non-rigid motion scenarios.

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Private Learning with Public Feature Conditioning

Shuli Jiang, Walid Krichene, Nicolas Mayoraz

We study differentially private (DP) regression in settings where each data sample includes public, non-sensitive features—common in applications like recommendation or advertising systems. While such label DP or DP with semi-sensitive features settings have been primarily explored in the context of classification, effective approaches for regression remain underexplored. We introduce $\textsf{Cond-DP}$, a conditioned variant of $\textsf{DPSGD}$ that leverages the structure of public feature matrices to improve optimization under privacy constraints. Motivated by the observation that these public features often exhibit rapidly decaying spectra, $\textsf{Cond-DP}$ incorporates a data-driven conditioning matrix to reshape the optimization landscape and accelerate convergence. We provide convergence guarantees for convex, strongly convex and non-convex settings, and recover standard $\textsf{DPSGD}$ as a special case when the conditioning matrix is the identity. We show how to construct an effective conditioning matrix for $\textsf{Cond-DP}$ directly from public features, enabling faster convergence than $\textsf{DPSGD}$ in private linear regression, without incurring additional privacy cost. Empirically, $\textsf{Cond-DP}$ with this conditioning matrix consistently outperforms state-of-the-art baselines across a wide range of datasets and model architectures under label DP, demonstrating strong and robust performance in practice.

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STAR-VAE: Structured Topology-Aware Regularization for Audio Reconstruction and Generation

Huadai Liu, Wen Wang, Kaicheng Luo, Qian Chen, Xiangang Li, Wei Xue

Continuous Variational Autoencoders (VAEs) serve as the fundamental continuous tokenizer for modern neural audio generation systems, enabling high-fidelity reconstruction while providing a compact, smooth latent space for downstream generative priors. However, continuous VAEs face a fundamental conflict when balancing *compression rate*, *reconstruction fidelity*, and *latent space topology*—a challenge we formalize as the **Rate-Distortion-Regularity Trilemma**. This trilemma stems from a critical *topological mismatch*: the prevailing isotropic Gaussian prior in standard VAEs imposes a *flat* latent geometry that fails to accommodate audio's *hierarchical* nature, where low-frequency components are structured and compressible while high-frequency components are stochastic and incompressible, leading to *disordered information packing* where crucial semantic features are randomly interleaved with high-entropy noise. To resolve this challenge, we propose **Structured Topology-Aware Regularization (STAR)**, a general training strategy that reshapes latent space geometry by imposing a growth-based constraint field, routing structural and textural information into channel subspaces with matching capacities. STAR is applicable to any VAE architecture and effectively resolves the trilemma, as demonstrated in CNN-based VAEs. To fully exploit STAR's potential, we present **STAR-VAE**, combining STAR with a hybrid CNN-Mamba architecture that synergizes local feature extraction with linear-complexity global context modeling, achieving state-of-the-art performance. We further propose **STAR-Gen**, an LLM-based Flow Matching framework that leverages STAR-VAE's st

ructured latent space for high-fidelity generation without suffering from vector quantization artifacts. Empirical results demonstrate that STAR-VAE successfully resolves the trilemma, achieving state-of-the-art reconstruction fidelity and enhanced semantic information preservation across diverse audio domains. The structured latent space improves both traditional diffusion models and our **STAR-Gen** paradigm, achieving state-of-the-art performance in text-to-audio generation. The project page is available at https://STAR-VAE.github.io.
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A Tale of Two Graphs: Separating Knowledge Exploration from Outline Structure for Open-Ended Deep Research
Zhuofan Shi, Ming Ma, Zekun Yao, Fangkai Yang, Jue Zhang, Dongge Han, Victor Rühle, Qingwei Lin, Saravan Rajmohan, Dongmei Zhang
Open-Ended Deep Research (OEDR) pushes LLM agents beyond short-form QA toward long-horizon workflows that iteratively search, connect, and synthesize evidence into structured reports. However, existing OEDR agents largely follow either linear "search-then-generate" accumulation or outline-centric planning. The former suffers from lost-in-the-middle failures as evidence grows, while the latter relies on the LLM to implicitly infer knowledge gaps from the outline alone, providing weak supervision for identifying missing relations and triggering targeted exploration. We present DualGraph memory, an architecture that separates what the agent knows from how it writes. DualGraph maintains two co-evolving graphs: an Outline Graph (OG), and a Knowledge Graph (KG), a semantic memory that stores fine-grained knowledge units, including core entities, concepts, and their relations. By analyzing the KG topology together with structural signals from the OG, DualGraph generates targeted search queries, enabling more efficient and comprehensive iterative knowledge-driven exploration and refinement. Across four established OEDR benchmarks, DualGraph consistently outperforms state-of-the-art baselines in report depth, breadth, and factual grounding; for example, it reaches a 53.08 RACE score on DeepResearch Bench with GPT-5. Moreover, ablation studies confirm the central role of the dual-graph design.
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Unlocking Speech–Text Compositional Powers: Instruction-Following Speech Language Models without Instruction Tuning
Congrui Du, Yang Zhang, Kaizhi Qian, Shiyu Chang
Instruction tuning for speech language models (SLMs) is substantially more challenging than for text-based large language models (LLMs), as it requires learning a new modality and a wide range of speech-specific instructions in addition to those supported by text LLMs. Existing SLM training approaches largely replicate the text LLM training paradigm by synthesizing large-scale speech pre-training and instruction-tuning datasets. However, this strategy is difficult to scale, since speech sequences are significantly longer than text sequences.
In this paper, we propose SpeechCombine, an instruction-following speech language model trained **without any instruction tuning**, using only a single round of speech pre-training on as little as 30k hours of speech data. Starting from a text LLM base model, we perform continuous pre-training on speech utterances to obtain a speech-adapted model, and then directly combine its weights with the weight difference between the instruction-tuned and base versions of the text LLM. Our results show that this simple combination strategy not only preserves the knowledge and capabilities of the original text LLM, but also effectively transfers them to the speech domain. These findings suggest a new direction for SLM training that avoids reliance on massive volumes of speech data.
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Latent Reasoning in TRMs is Secretly a Policy Improvement Operator
Arip Asadulaev, Rayan Banerjee, Fakhri Karray, Martin Taká■
Recently, small models with latent recursion have obtained promising results on complex reasoning tasks. These results are typically explained by the theory that such recursion increases a network’s depth, allowing it to compactly emulate the capacity of larger models. However, the performance of recursively added layers remains behind the capabilities of one■pass models with the same feed-forward depth. This means that in the looped version, not every recursive step effectiv

ely contributes to depth. This raises the question: when and why does latent rea soning improve performance, and when does it result in dead compute? In our work , we analyze the algorithms that latent reasoning provides answer to this questi on. We show that latent reasoning can be formalized as a classifier■free guidanc e and policy improvement algorithm. Building on these insights, we propose to us e a training schemes from RL and diffusion methods for latent reasoning modles. Using the Tiny Recursive Model as our testbed, we show that with our modificatio ns we can avoid dead compute steps and reduce the total number of forward passes by 18× while maintaining performance. Broadly speaking, we show how a policy im provement perspective on recursive steps can explain model behavior and provide insights for further improvements.
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Deep Single-Index Fréchet Regression
Muqing Cui, Yidong Zhou, Su I Iao, Hans-Georg Müller
Predicting outputs that are located in non-Euclidean spaces, such as probability distributions, networks, and symmetric positive-definite matrices, is becoming increasingly important in modern data analysis, particularly when inputs are hig h-dimensional. We propose DeSI (Deep Single-Index Fréchet Regression), a semipar ametric framework for regression with metric space-valued outputs and multivaria te inputs that assumes a single-index structure for the conditional Fréchet mean . DeSI estimates an interpretable index direction, which quantifies the relative importance of inputs, using a deep neural network, and performs Fréchet regress ion along the resulting one-dimensional index in the target metric space. This s tructure mitigates the curse of dimensionality while retaining interpretability, which stands in contrast to standard deep neural networks. We establish theoret ical guarantees for DeSI, including uniform approximation and convergence rates, and demonstrate its strong predictive performance through simulations on distri butions, networks, and symmetric positive-definite matrices, as well as an appli cation to compositional mood data from New Jersey.
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$\texttt{PRISM}$:A 3D Probabilistic Neural Representation for Interpretable Shap e Modeling
Yining Jiao, Sreekalyani Bhamidi, Carlton Jude ZDANSKI, Julia S Kimbell, Andrew Prince, Cameron P Worden, Samuel Kirse, Christopher Rutter, Benjamin H Shields, William Alexander Dunn, Jisan Mahmud, Marc Niethammer
Understanding how anatomical shapes evolve in response to developmental covariat es—and quantifying their spatially varying uncertainties—is critical in healthca re research. Existing approaches typically rely on global time-warping formulati ons that ignore spatially heterogeneous dynamics. We introduce \texttt{PRISM}, a novel framework that bridges implicit neural representations with uncertainty-a ware statistical shape analysis. \texttt{PRISM} models the conditional distribut ion of shapes given covariates, providing spatially continuous estimates of both the population mean and covariate-dependent uncertainty at arbitrary locations. A key theoretical contribution is a closed-form Fisher Information metric that enables efficient, analytically tractable local temporal uncertainty quantificat ion via automatic differentiation. Experiments on three synthetic datasets and o ne clinical dataset demonstrate \texttt{PRISM}'s strong performance across diver se tasks—from modeling shape evolution to anomaly detection—within a unified fra mework, while providing interpretable and clinically meaningful uncertainty esti mates.
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BEDTime: A Unified Benchmark for Automatically Describing Time Series
Medhasweta Sen, Zachary Gottesman, Jiaxing Qiu, C. Bayan Bruss, Nam H Nguyen, Th omas Hartvigsen
Recent works propose complex multi-modal models that handle both time series and language, ultimately claiming high performance on complex tasks like time serie s reasoning and cross-modal question answering. However, they skip foundational evaluations that such complex models should have mastered. So we ask a simple qu estion:$\textit{How well can recent models describe structural properties of tim e series?}$ To answer this, we propose that successful models should be able to

$\textit{recognize}$, $\textit{differentiate}$, and $\textit{generate}$ descript
ions of univariate time series. We then create $\textbf{BEDTime}$, a benchmark t
o assess these novel tasks, that comprises $\textbf{five datasets}$ reformatted
across $\textbf{three modalities}$. In evaluating $\textbf{17 state-of-the-art m
odels}$, we find that (1) surprisingly, dedicated time series-language models fa
ll short, despite being designed for similar tasks, (2) vision language models a
re quite capable, (3) language only methods perform worst, despite many lauding
their potential, and (4) all approaches are clearly fragile to a range of real w
orld robustness tests, indicating directions for future work. Together, our find
ings critique prior works' claims and provide avenues for advancing multi-modal
time series modeling.
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Rotation-Invariant Spherical Watermarking via Third-Order SO(3) Representation C
oupling
Pengzhen Chen, Yanwei Liu, Xiaoyan Gu, Antonios Argyriou, Wu Liu, Weiping Wang
Reliable watermarking of panoramic imagery is fundamentally challenged by arbitr
ary 3D rotations. As panoramas are defined on the sphere, they naturally transfo
rm under the action of $SO(3)$, rendering conventional planar representations an
d augmentation-based robustness strategies inadequate and devoid of theoretical
guarantees. To address this, we formulate panoramas as spherical signals and lev
erage $SO(3)$ representation theory to derive provably rotation-invariant descri
ptors. While spherical harmonic coefficients transform equivariantly under rotat
ions, the natural invariant constructions are typically limited to zeroth-order
statistics which eliminate directional information and severely constrain embedd
ing capacity. In this work, we introduce a principled third-order invariant cons
truction by coupling higher-order $SO(3)$ irreducible representations via tensor
 products and projecting onto the trivial representation. This yields a spherica
l invariant bispectrum that preserves phase information while remaining strictly
 rotation-invariant. Leveraging this property, we embed watermarks into higher-o
rder spherical harmonic coefficients and recover them from invariant bispectral
scalars, enabling reliable extraction under arbitrary 3D rotations. We provide a
 theoretical proof of $SO(3)$ invariance for it and demonstrate experimentally i
ts near-perfect robustness to continuous rotations while maintaining high visual
 fidelity.
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Beyond Hamming: Query-Aware Decoding of Binary Cosine Sketches
DaeHun Nyang
Cosine similarity estimation is a core primitive in coarse-to-fine retrieval pip
elines, where early-stage candidate selection relies on approximate similarity e
stimates whose errors are amplified downstream. Widely used sign-based sketches
arising from extreme quantization of random projections exhibit a structural var
iance peak near θ $\approx$ 90°, the near-background region where
candidate selection is most difficult. We propose QA-Cos, a query-aware decoder-
side estimator that departs from the Hamming-agreement paradigm, treating sign b
its as probabilistic observations rather than deterministic votes. Across simula
tions and BEIR benchmarks, QA-Cos reduces estimation error by up to $\sim$15--20
\% in the near-orthogonal region and translates these gains into improved candid
ate selection in two-stage ANN pipelines, improving Hit@K by up to $\sim$30 perc
entage points at fixed budgets and reducing candidates by up to $\sim$45--50\% a
t fixed recall. In a native hnswlib scorer-replacement experiment, the same deco
der improves the quality of the final HNSW frontier, and in a focused storage-aw
are setting a practical gated variant reduces end-to-end wall-clock latency at m
atched recall.
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Intrinsic Credit Assignment for Long Horizon Interaction
Ilze Amanda Auzina, Joschka Strüber, Sergio Hernández-Gutiérrez, Shashwat Goel,
Ameya Prabhu, Matthias Bethge
How can we train agents to navigate uncertainty over long horizons? In this work
, we propose ΔBelief-RL, which leverages a language model's own intrinsic belief
s to reward intermediate progress. Our method utilizes the change in the probabi

lity an agent assigns to the target solution for credit assignment. By training on synthetic interaction data, ∆Belief-RL teaches information-seeking capabilities that consistently outperform purely outcome-based rewards for RL, with improvements generalizing to out-of-distribution applications ranging from customer service to personalization. Notably, the performance continues to improve as we scale test-time interactions beyond the training horizon, with interaction-efficiency increasing even on Pass@k metrics. Overall, our work introduces a scalable training strategy for navigating uncertainty over a long-horizon, by enabling credit assignment to intermediate actions via intrinsic ∆Belief rewards.

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TSRBench: A Comprehensive Multi-task Multi-modal Time Series Reasoning Benchmark for Generalist Models

Fangxu Yu, Xingang Guo, Lingzhi Yuan, Haoqiang Kang, Hongyu Zhao, Lianhui Qin, Furong Huang, Bin Hu, Tianyi Zhou

Time series are ubiquitous in real-world scenarios and crucial for applications ranging from energy management to traffic control. Consequently, the ability to reason over time series is a fundamental skill for generalist models to solve complex problems. However, current benchmarks for generalist models largely overlook this dimension.

To bridge this gap, we introduce TSRBench, a comprehensive multi-modal benchmark designed to stress-test the full spectrum of time series reasoning capabilities. TSRBench features: i) a diverse set of 4125 problems from 14 domains, and is categorized into 4 major dimensions: Perception, Reasoning, Prediction, and Decision-Making. ii) 15 tasks from the 4 dimensions evaluating essential reasoning capabilities (e.g., numerical reasoning).

Through extensive experiments, we evaluate over 30 leading proprietary and open-source LLMs, VLMs, and TSLLMs within TSRBench. Our findings reveal that: i) scaling laws hold for perception and reasoning but break down for prediction; ii) strong reasoning does not guarantee accurate context-aware forecasting, indicating a decoupling between semantic understanding and numerical prediction; and iii) despite the complementary nature of textual and visual forms of time series as inputs, current multimodal models fail to effectively fuse them for reciprocal performance gains. TSRBench provides a standardized evaluation platform that not only highlights existing challenges but also offers valuable insights to advance generalist models. Our code and dataset are available at https://tsrbench.github.io/.

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Synergistic Intra- and Cross-Layer Regularization Losses for MoE Expert Specialization

Rizhen Hu, Yuan Cao, Boao Kong, Mou Sun, Kun Yuan

Sparse Mixture-of-Experts (MoE) models scale Transformers efficiently but suffer from expert overlap, where different experts process similar tokens and learn redundant functions, resulting in ambiguous routing and underutilized capacity. While architectural solutions like DeepSeek-style shared experts promote specialization, they require substantial structural modifications and rely solely on intra-layer signals. We propose two plug-and-play auxiliary losses that enhance MoE specialization and routing efficiency without modifying routers or model architectures. First, an intra-layer specialization loss penalizes cosine similarity between experts' SwiGLU activations on identical tokens, encouraging experts to specialize in complementary functions. Second, a cross-layer dependency loss maximizes joint Top-k routing probabilities across adjacent layers, establishing coherent expert pathways through network depth while reinforcing intra-layer specialization. Both losses are orthogonal to the standard load-balancing loss and compatible with shared-expert and vanilla Top-k MoE architectures. We implement both losses as a drop-in Megatron-LM module. Extensive experiments across pre-training, fine-tuning, and zero-shot benchmarks demonstrate consistent task gains, higher expert specialization, and lower-entropy routing; together, these improvements translate into faster inference via more stable expert pathways.

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Adaptively Grouped Contextual Bandits for Heterogeneous Human-AI Decision Making with Conformal Prediction Sets
Yanchen Wu, Bo Li
Personalizing AI decision support for heterogeneous human decision-makers remains a key challenge. We study a collaboration workflow where AI provides a reduced prediction set via conformal prediction and the human makes the final decision based on the set. We formulate this personalization problem as a contextual bandit, where individual and task features form the context, candidate significance levels α serve as arms, and the optimal prediction-set size varies across contexts. To address large arm spaces and high-dimensional contexts, we introduce the Adaptively Grouped Contextual Bandit (AGCB) framework, which avoids global function approximation by exploiting two Human-AI structural assumptions: continuity and monotonicity. Continuity enables information sharing across nearby contexts and decisions, and drives a data-driven Zooming Mechanism that balances intra-group estimation error against inter-group approximation bias. Monotonicity converts each observation into directional counterfactual information over the K candidate α values, reducing the arm-dependence factor from polynomial to logarithmic in K. Together, these mechanisms yield minimax-optimal dependence on the learning horizon T for both cumulative and simple regret objectives. Empirical results confirm that AGCB achieves the strongest overall performance across most heterogeneous, data-scarce settings.
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From Conflict to Consensus: Boosting Medical Reasoning via Multi-Round Agentic RAG
Wenhao Wu, Zhentao Tang, Yafu Li, Shixiong Kai, Mingxuan Yuan, Zhenhong Sun, Chunlin Chen, Zhi Wang
Large Language Models (LLMs) exhibit high reasoning capacity in medical question-answering, but their tendency to produce hallucinations and outdated knowledge poses critical risks in healthcare fields. While Retrieval-Augmented Generation (RAG) mitigates these issues, existing methods rely on noisy token-level signals and lack the multi-round refinement required for complex reasoning. In this paper, we propose **MA-RAG** (**M**ulti-Round **A**gentic RAG), a framework that facilitates test-time scaling for complex medical reasoning by iteratively evolving both external evidence and internal reasoning history within an agentic refinement loop. At each round, the agent transforms semantic **conflict** among candidate responses into actionable queries to retrieve external evidence, while optimizing history reasoning traces to mitigate long-context degradation. MA-RAG extends the *self-consistency* principle by leveraging the lack of consistency as a proactive signal for multi-round agentic reasoning and retrieval, and mirrors a *boosting* mechanism that iteratively minimizes the residual error toward a stable, high-fidelity medical **consensus**. Extensive evaluations across 7 medical Q\&A benchmarks show that MA-RAG consistently surpasses competitive inference-time scaling and RAG baselines, delivering **substantial +6.8 points** on average accuracy over the backbone model. Our code is available at https://github.com/NJU-RL/MA-RAG.
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Federated Graph Learning via Structure-Aware Fusion Using a Kalman Framework with Learnable Dynamics
Bisheng Tang, Xiaojun Chen
Federated Graph Learning (FGL) enables collaborative training across distributed clients without sharing raw graph data. However, its performance is severely hindered by graph-specific heterogeneity arising from divergent node feature distributions and disparate graph structures. Existing FGL methods primarily focus on aligning or personalizing node features but largely overlook the role of structural knowledge, leading to aggregation-induced representation drift during message passing. We observe that structural heterogeneity often originates from feature-driven connection biases shaped by local data collection practices or user preferences. To address this, we propose \textbf{Fed-Kalter}, a novel FGL framework that integrates Kalman filtering principles into graph neural networks. Fed-Ka

lter introduces Kalter-Conv, a graph convolution grounded in a Kalman framework with learnable dynamics, which treats structural embeddings as latent states and feature-augmented neighborhoods as noisy observations, thereby filtering feature-induced structural noise in a layer-wise manner. Only structural parameters are aggregated globally, enabling effective cross-client knowledge transfer while preserving local personalization. Extensive experiments on 16 graph classification datasets spanning 4 domains demonstrate that Fed-Kalter consistently outperforms state-of-the-art FGL methods. Further ablation and hyperparameter studies confirm its robustness, efficiency, and effectiveness in mitigating structural heterogeneity.
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Noise-Robust Density Estimation for Tabular Data Anomaly Detection
Dazhi Fu, Zhao Zhang, Jicong Fan
Density-based anomaly detection methods often provide accurate and interpretable predictions but their performance can be severely degraded by the inherent noise of data, such as changes arising from environmental conditions during data collection or background noise. To deal with such noise, we present noise-robust density estimation (NRDE) for tabular data anomaly detection. We aim to estimate the density of pure data with the influence of noises isolated, which is a non-trivial task since the data-generating process is completely unknown. Specifically, NRDE learns a Jacobian-regularized normalizing flow to estimate the sources of data and categorizes sources into two groups, where one group generates pure data and the other generates noise. After generating pure data, we can use the density of such pure data to detect anomalies caused by the sources of pure data solely. Therefore, NRDE is robust to inherent noise. We provide theoretical results to support the effectiveness of NRDE and compare NRDE with 17 baselines on 47 benchmark datasets under different settings, including vanilla anomaly detection, anomaly detection with anomaly contamination, anomaly detection on noisy data, and transductive outlier detection.
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CalM: A Self-Supervised Foundation Model for Population Dynamics in Calcium Imaging Data
Xinhong Xu, Yimeng Zhang, Qichen Qian, Yuanlong Zhang
Recent work suggests that large-scale, multi-animal modeling can significantly improve neural recording analysis. However, for functional calcium traces, existing approaches remain task-specific, limiting transfer across common neuroscience objectives. To address this challenge, we propose \textbf{CalM}, a self-supervised neural foundation model trained solely on neuronal calcium traces and adaptable to multiple downstream tasks, including forecasting and decoding. Our key contribution is a pretraining framework, composed of a high-performance tokenizer mapping single-neuron traces into a shared discrete vocabulary, and a dual-axis autoregressive transformer modeling dependencies along both the neural and the temporal axis. We evaluate CalM on a large-scale, multi-animal, multi-session dataset. On the neural population dynamics forecasting task, CalM outperforms strong specialized baselines after pretraining. With a task-specific head, CalM further adapts to the behavior decoding task and achieves superior results compared with supervised decoding models. Moreover, linear analyses of CalM representations reveal interpretable functional structures beyond predictive accuracy. Taken together, we propose a novel and effective self-supervised pretraining paradigm for foundation models based on calcium traces, paving the way for scalable pretraining and broad applications in functional neural analysis.
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Learning More from Less: Unlocking Internal Representations for Benchmark Compression
Yueqi Zhang, Jin Hu, Shaoxiong Feng, Peiwen Yuan, Xinglin Wang, Yiwei Li, Jiayi Shi, Chuyi Tan, Ji Zhang, Boyuan Pan, Yao Hu, Kan Li
The prohibitive cost of evaluating Large Language Models (LLMs) necessitates efficient alternatives to full-scale benchmarking. Prevalent approaches address this by identifying a small coreset of items to approximate full-benchmark performance. However, existing methods must estimate a reliable item profile from respon

se patterns across many source models, which becomes statistically unstable when the source pool is small. This dependency is particularly limiting for newly released benchmarks with minimal historical evaluation data. We argue that discrete correctness labels are a lossy view of the model's decision process and fail to capture information encoded in hidden states. To address this, we introduce \textsc{RepCore}, which aligns heterogeneous hidden states into a unified latent space to construct representative coresets. Using these subsets for performance extrapolation, \textsc{RepCore} achieves precise estimation accuracy with as few as ten source models. Experiments on five benchmarks and over 200 models show consistent gains over output-based baselines in ranking correlation and estimation accuracy. Spectral analysis further indicates that the aligned representations contain separable components reflecting broad response tendencies and task-specific reasoning patterns.

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R2-Router: A New Paradigm for LLM Routing with Reasoning

Jiaqi Xue, Qian Lou, Jiarong Xing, Heng Huang

As LLMs proliferate with diverse capabilities and costs, LLM routing has emerged by learning to predict each LLM's quality and cost for a given query, then selecting the one with high quality and low cost. However, existing routers implicitly assume a single fixed quality and cost per LLM for each query, ignoring that the same LLM's quality varies with its output length. This causes routers to exclude powerful LLMs when their estimated cost exceeds the budget, missing the opportunity that these LLMs could still deliver high quality at reduced cost with shorter outputs. To address this, we introduce R2-Router, which treats output length budget as a controllable variable and jointly selects the best LLM and length budget, enforcing the budget via length-constrained instructions. This enables R2-Router to discover that a powerful LLM with constrained output can outperform a weaker LLM at comparable cost—efficient configurations invisible to prior methods. Together with the router framework, we construct R2-Bench, the first routing dataset capturing LLM behavior across diverse output length budgets. Experiments show that R2-Router achieves state-of-the-art performance at $4-5\times$ lower cost compared with existing routers. This work opens a new direction: routing as reasoning, where routers evolve from reactive selectors to deliberate reasoners that explore which LLM to use and at what cost budget. The code is publicly available at https://github.com/UCF-ML-Research/R2-Router.

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Cross-Embodiment Robot Foundation World Models with Latent Actions

Huang Huang, Sriram Yenamandra, Arjun Majumdar, Elie Aljalbout, Tushar Nagarajan, Tsung-Yen Yang, Akshara Rai, Michael Rabbat, Li Fei-Fei, Jiajun Wu, Tingfan Wu, Franziska Meier

The diversity of robot embodiments and action spaces makes it challenging to build robot world models that generalize across different
embodiments. We introduce the Latent Action-Conditioned Robot World Model (LAC-WM), which operates within a learned unified latent action space shared across diverse embodiments. This unified action space improves the world model’s performance when adapted to previously unseen robot embodiments. We compare LAC-WM with an Explicit Action-Conditioned World Model (EAC-WM), which conditions on explicit motion labels. Our results shows that explicit action conditioning leads to disjoint action representations across embodiments, limiting downstream performance when adapting to new robots. We evaluate both models on dexterous manipulation tasks and a modified LIBERO benchmark. LAC-WM improves downstream performance over EAC-WM by up to 46.7% on dexterous manipulation and 11.7% on LIBERO. Crucially, the unified latent action space allows LAC-WM’s downstream performance to scale positively with the number of embodiments used during pretraining. In contrast, the disjoint action space in EAC-WM leads to decreased performance as the number of pretraining embodiments increases. These results highlights the importance of a unified action space for efficient cross-embodiment learning, addressing a key challenge in robotics.

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Learning Long Range Spatio-Temporal Representations over Continuous Time Dynamic

Graphs with State Space Models
Ayushman Raghuvanshi, Thummaluru Siddartha Reddy, Sundeep Prabhakar Chepuri, Mahesh Chandran
Continuous-time dynamic graphs (CTDGs) provide a richer framework to capture fine-grained temporal patterns in evolving relational data. Long-range information propagation is a key challenge while learning representations, wherein it is important to retain and update information over long temporal horizons. Existing approaches restrict models to capture one-hop or local temporal neighborhoods and fail to capture multi-hop or global structural patterns. To mitigate this, we derive a parameter-efficient state-space modeling framework for continuous-time dynamic graphs $\texttt{(CTDG-SSM)}$ from first principles. We first introduce continuous-time Topology-Aware higher order polynomial projection operator ($\texttt{CTT-HiPPO}$), a novel memory-based reformulation of $\texttt{HiPPO}$ to jointly encode temporal dynamics and graph structure. The solution from $\texttt{CTT-HiPPO}$ is obtained by projecting the classical HiPPO solution through a polynomial of the Laplacian matrix, yielding topology-aware memory updates that admit an equivalent state-space formulation for CTDGs ($\texttt{CTDG-SSM}$). Then a computationally efficient discrete formulation is obtained using the zero-order hold approach for model implementation.
Across benchmarks on dynamic link prediction, dynamic node classification, and sequence classification, $\texttt{CTDG-SSM}$ achieves state-of-the-art performance. Notably, it achieves large performance gains on datasets that require long range temporal (LRT) and spatial reasoning.
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Deep Residual Injection for Full-Spectrum Forensic Signal Perception in Multimodal Large Language Models
Kaiqing Lin, Zhiyuan Yan, Ruoxin Chen, Ke-Yue Zhang, Yue Zhou, Caiyong Piao, Bin Li, Taiping Yao, Bo Wang, Youchang xiao, Shouhong Ding
Multimodal large language models (MLLMs) have been increasingly adopted in forensics for their robust semantic understanding.
As AI-generated images become realistic, semantic-level inconsistencies alone are often insufficient for reliable detection.
This motivates a critical question: *whether MLLMs can achieve full-spectrum forensic signal perception, i.e., capturing low-level generator artifacts without sacrificing pre-trained semantic knowledge.*
We then conduct a layer-wise analysis of forensic signal perception in MLLMs and find that semantic information is mainly encoded in the early-to-middle layers, and directly fine-tuning MLLMs for artifact learning causes rapid semantic forgetting.
Based on this insight, we propose Deep Visual Residual MLLM (Deep-VRM) to \textit{preserve early semantic processing while injecting artifact-specific visual signals as a residual path into an intermediate layer}, where they are fused with semantic token representations and propagated through subsequent trainable layers.
This enables later layers to jointly model semantic reasoning and signal-level forensic cues, and surprisingly, the model learns to adaptively leverage different levels of forensic signals depending on the input, achieving robust and generalizable detection performance.
Extensive experiments show that our method achieves state-of-the-art across all benchmarks.
****
Improving Topic Modeling by Distilling Soft Labels from Language Models
Raymond Li, Amirhossein Abaskohi, Chuyuan Li, Gabriel Murray, Giuseppe Carenini
Traditional neural topic models are typically optimized by reconstructing the document's Bag-of-Words (BoW) representations, overlooking contextual information and struggling with data sparsity. In this work, we introduce a novel topic model training framework by Distilling Soft Labels (DSL) from Language Models (LMs). To construct the contextually enriched reconstruction signals, we project the next token probabilities, conditioned on a specialized prompt, onto a pre-defined vocabulary, and train the topic models to reconstruct the soft labels using the

LM hidden states. This produces higher-quality topics that are more closely aligned with the underlying thematic structure of the corpus. Extensive experiments demonstrate that DSL achieves substantial improvements in topic coherence and assignment accuracy over existing baselines. Additionally, we also introduce a retrieval-based metric, which shows that our approach significantly outperforms existing methods in identifying semantically similar documents, highlighting its effectiveness for retrieval-oriented applications.

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From Text to Forecasts: Bridging Modality Gap with Temporal Evolution Semantic Space

Lehui Li, Yuyao Wang, Jisheng Yan, Wei Zhang, Jinliang Deng, Haoliang Sun, Zhongyi Han, Yongshun Gong

Incorporating textual information into time-series forecasting holds promise for addressing event- driven non-stationarity; however, a fundamental modality gap hinders effective fusion: textual descriptions express temporal impacts implicitly and qualitatively, whereas forecasting models rely on explicit and quantitative signals. Through controlled semi-synthetic experiments, we show that existing methods over-attend to redundant tokens and struggle to reliably translate textual semantics into usable numerical cues. To bridge this gap, we propose TESS, which introduces a Temporal Evolution Semantic Space as an intermediate bottleneck between modalities. This space consists of interpretable, numerically grounded temporal primitives—distribution shift, volatility, shape, and lag—extracted from text by an LLM via structured prompting and filtered through confidence-aware gating. Experiments on four real-world datasets demonstrate up to a 29% reduction in forecasting error compared to state-of-the-art unimodal and multimodal baselines. Code is available at: https://github.com/olivia3395/TESS.

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IAPO: Information-Aware Policy Optimization for Token-Efficient Reasoning

Yinhan He, Yaochen Zhu, Mingjia Shi, Wendy Zheng, Lin Su, Xiaoqing Wang, Qi Guo, Jundong Li

Large language models increasingly rely on long chains of thought to improve accuracy, yet such gains come with substantial inference-time costs. We revisit token-efficient post-training and argue that existing sequence-level reward-shaping methods offer limited

control over how reasoning effort is allocated across tokens. To bridge the gap, we propose IAPO, an information-theoretic post-training framework that assigns token-wise advantages based on each token’s conditional mutual information (MI) with the final answer. This yields an explicit, principled mechanism for identifying informative reasoning steps and suppressing low-utility exploration. We provide a theoretical analysis showing that our IAPO can induce monotonic reductions in reasoning verbosity without harming correctness.

Empirically, IAPO consistently improves reasoning accuracy while reducing reasoning length by up to 36\%, outperforming existing token-efficient RL methods across various reasoning datasets. Our results demonstrate that information-aware advantage shaping is a powerful and general direction for token-efficient post-training. The code is available at https://github.com/YinhanHe123/IAPO.

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XTransfer: Modality-Agnostic Few-Shot Model Transfer for Human Sensing at the Edge

Yu Zhang, Xi Zhang, Hualin zhou, Xinyuan Chen, Shang Gao, Hong Jia, Jianfei Yang, Yuankai Qi, Tao Gu

Deep learning for human sensing on edge systems presents significant potential for smart applications. However, its training and development are hindered by the limited availability of sensor data and resource constraints of edge systems. While transferring pre-trained models to different sensing applications is promising, existing methods often require extensive sensor data and computational resources, resulting in high costs and limited transferability. In this paper, we propose XTransfer, a first-of-its-kind method enabling modality-agnostic, few-shot model transfer with resource-efficient design. XTransfer flexibly uses pre-trained models and transfers knowledge across different modalities by (i) model repa

iring that safely mitigates modality shift by adapting pre-trained layers with o nly few sensor data, and (ii) layer recombining that efficiently searches and re combines layers of interest from source models in a layer-wise manner to restruc ture models. We benchmark various baselines across diverse human sensing dataset s spanning different modalities. The results show that XTransfer achieves state- of-the-art performance while significantly reducing the costs of sensor data col lection, model training, and edge deployment.
****

Gradient-Free Approaches is a Key to an Efficient Interaction with Markovian Sto chasticity

Boris Prokhorov, Semyon Chebykin, Alexander Gasnikov, Aleksandr Beznosikov

This paper deals with stochastic optimization problems involving Markovian noise with a zero-order oracle. We present and analyze a novel derivative-free method for solving such problems in strongly convex smooth and non-smooth settings wit h both one-point and two-point feedback oracles. Using a randomized batching sch eme, we show that when mixing time τ of the underlying noise sequence is le ss than the dimension of the problem d, the convergence estimates of our metho d do not depend on τ. This observation provides an efficient way to interac t with Markovian stochasticity: instead of invoking the expensive first-order or acle, one should use the zero-order oracle. Finally, we complement our upper bou nds with the corresponding lower bounds. This confirms the optimality of our res ults.
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Spherical SO(3) Equivariant Local Attention

Yusuke Sekikawa, Jun Nagata, Itsumi Araki, Ruka Eto

Spherical signals provide a natural representation for omnidirectional perceptio n and often benefit from equivariance to 3D rotations.

Recent spherical vision transformers implement local self-attention on spherical grids, but most retain only partial $\mathrm{SO}(3)$ equivariance and rely on $ \textit{location-dependent}$ local positional embeddings (LPEs).

Such LPEs can degrade robustness to camera tilt or object reorientation and intr oduce additional memory and computational overhead.

We propose $\textit{Spherical $\mathrm{SO}(3)$-Equivariant Local Attention}$ (So LA), an LPE-free local attention mechanism for spherical signals.

SoLA achieves full $\mathrm{SO}(3)$ equivariance through a distance-preserving p ositional modulation that couples query/key features with each token’s unit dire ction.

Specifically, the modulation lifts queries and keys using an outer-product with the 4D direction dependent vector.

The induced similarity of the modulated queries and keys depends on content affi nity and great-circle distance while remaining invariant to global $\mathrm{SO}(3)$ rotations.

The same formulation admits a softmax-free linear variant that computes local at tention via key-value aggregation without per-query neighbor materialization.

We integrate SoLA into a U-shaped spherical transformer for 360° depth es timation and semantic segmentation, demonstrating substantially improved robustn ess to arbitrary 3D rotations compared to prior spherical transformers with simi lar computational costs.
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Caracal: Causal Architecture via Spectral Mixing

BINGZHENG GAN, Tianyi Zhang, LI YUSU, Jing Huang, Wei Shi, Yangkai Ding, Tao Yu

The scalability of Large Language Models to long sequences is hindered by the qu adratic cost of self-attention and the limitations of positional encodings. To a ddress these, we introduce **Caracal**, a novel architecture that replaces self- attention with a parameter-efficient, $\mathcal{O}(L \log L)$ Multi-Head Fourier (MHF) module. Our contributions are threefold: (1) We leverage the Fast Fourier Transform (FFT) for sequence mixing, inherently addressing both bottlenecks men tioned above. (2) We apply a frequency-domain causal masking technique that enfo rces autoregressive capabilities via asymmetric padding and truncation, overcomi ng a critical barrier for Fourier-based generative models. (3) Unlike efficient

models relying on hardware-specific implementations (e.g., Mamba), **Caracal** u
ses standard library operators. This ensures robust portability, eliminating com
mon deployment barriers. Evaluations demonstrate that **Caracal** performs compe
titively with Transformer and SSM baselines, offering a scalable and simple path
way for efficient long-sequence modeling. Code is available in the supplementary
 materials.
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Linear-Core Surrogates: Smooth Loss Functions with Linear Rates for Classificati
on and Structured Prediction
Mehryar Mohri, Yutao Zhong
A fundamental dichotomy in the theory of classification sets smoothness against
statistical efficiency: smooth surrogate losses such as the logistic loss enable
 fast $O(1/T)$ optimization but yield slow square-root H-consistency bounds, w
hile piecewise-linear losses like the Hinge loss achieve optimal linear H-cons
istency rates but are non-differentiable. We introduce *Linear-Core (LC) Surroga
tes*, the first family of *explicit* convex loss functions that provably resolve
 this tension. By stitching a linear core to a smooth tail, we construct surroga
tes that are differentiable everywhere (C^1, and even C^2 under mild conditi
ons) while retaining strict linear H-consistency bounds, the strongest known f
orm of consistency guarantee. We establish these linear bounds across three incr
easingly complex settings: binary classification, multi-class classification, an
d structured prediction. To our knowledge, this is the first explicit constructi
on to simultaneously achieve smoothness and linear H-consistency in any of the
se settings. Beyond their theoretical appeal, Linear-Core Surrogates offer pract
ical advantages. In multi-class classification, their constant gradient profile
near the decision boundary provides natural robustness to instance-dependent lab
el noise, outperforming Cross-Entropy by *2.6\%* on corrupted CIFAR-10. In struc
tured prediction, their smoothness enables an unbiased stochastic gradient estim
ator that bypasses the $O(|Y|^2)$ per-step complexity of exact inference, yieldi
ng a *23$\times$ speedup* over Structured SVMs on large-vocabulary sequence tagg
ing tasks.
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AdaMEM: Test-Time Adaptive Memory for Language Agents
Yunxiang Zhang, Yiheng Li, Ali Payani, Lu Wang
A central challenge for language agents is utilizing past experience to adapt to
 dynamic test-time conditions. While recent work demonstrates the promise of age
ntic memory mechanisms, most systems restrict retrieval to episode initiation. C
onsequently, agents are forced to rely on static guidance that becomes increasin
gly misaligned as long-horizon tasks unfold.
To address this rigidity, we propose the Adaptive Memory Agent (AdaMEM), a novel
 framework for agent test-time adaptation. Without updating model parameters onl
ine, AdaMEM adapts agent behavior via a hybrid memory architecture: it maintains
 a long-term trajectory memory of raw experiences collected offline while genera
ting dynamic short-term strategy memory on-the-fly to guide decision-making.
This mechanism enables the trade-off between token efficiency and adaptability a
cross varying inference-time compute levels. Empirically, AdaMEM significantly o
utperforms static memory baselines, achieving relative gains of up to 13% on ALF
World and 11% on WebShop, with consistent leading performance extending to agent
ic search on HotpotQA.
To further enhance this adaptation, we develop Step-MFT, a Step-wise Memory Fine
-Tuning technique that trains the policy to synthesize high-quality strategies f
rom retrieved experiences, yielding additional performance gains.
Our work establishes a new scaling dimension for agentic memory, supporting cont
inuous reasoning and self-evolution post-deployment in real-world environments.
Our code is available at https://github.com/yunx-z/AdaMEM.
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On the Difficulty of Learning a Meta-network for Training Data Selection
Zilin Du, Junqi Zhao, Boyang Li
Synthetic data are increasingly used to train neural networks, yet distributiona
l mismatch with real data limits their effectiveness when used indiscriminately.

A common strategy is to learn data weights via bi-level optimization, which we refer to as Meta-learning for Training-data Selection (MTS). Interestingly, in practice, MTS often performs below expectation. We identify two obstacles in properly training MTS: a poor gradient signal-to-noise ratio (GSNR), which causes optimization difficulties, and lack of informative features that correlates with data quality. We present a mathematical analysis of MTS, which reveals the dynamics of normalized data weights and the relation between disparate data quality and poor GSNR. The analysis suggests a a simple yet effective solution: increasing the batch size. Further, we propose a set of informative features that capture the positions of training data in their distributions and training dynamics. Experiments across four benchmarks show consistent improvements, achieving average gains of 5.49\% over training without selection and 2.89\% over the strongest baseline.

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Credit Assignment via Neural Manifold Noise Correlation

Byungwoo Kang, Maceo Richards, Bernardo L. Sabatini

Credit assignment, the process of determining how changes in individual neurons and synapses influence a network’s output, is central to learning in brains and machines. Noise correlation-based methods, which estimate gradients by correlating perturbations of activity with changes in output, provide a biologically plausible solution to credit assignment but scales poorly as accurately estimating the Jacobian requires that the number of perturbations scale with network size. Moreover, isotropic noise conflicts with neurobiological observations that neural activity lies on a low-dimensional manifold. To address these drawbacks, we propose *neural manifold noise correlation* (NMNC), which performs credit assignment using perturbations restricted to the neural manifold. We show theoretically and empirically that the Jacobian row space aligns with the neural manifold in trained networks, and that manifold dimensionality scales slowly with network size. NMNC substantially improves performance and sample efficiency over vanilla noise correlation in convolutional networks trained on CIFAR-10, ImageNet-scale models, and recurrent networks. NMNC also yields representations more similar to the primate visual system than vanilla noise correlation. These findings offer a mechanistic hypothesis for how biological circuits could support credit assignment, and suggest that biologically inspired constraints may enable, rather than limit, effective learning at scale.

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SPARe: Stacked Parallelism with Adaptive Reordering for Fault-Tolerant LLM Pretraining Systems with 100k+ GPUs

Jin Lee, Zhonghao Chen, Xuhang He, Robert Underwood, Bogdan Nicolae, Franck Cappello, Xiaoyi Lu, Sheng Di, Zheng Zhang

In large-scale LLM pretraining systems with $100\mathrm{k}+$ GPUs, failures become the norm rather than the exception, and restart costs can dominate wall-clock training time. However, existing fault-tolerance mechanisms are largely unprepared for this restart-dominant regime. To address this challenge, we propose SPARe—Stacked Parallelism with Adaptive Reordering—a fault-tolerance framework that masks node failures during gradient synchronization by stacking redundant data shards across parallelism groups and adaptively reordering execution. SPARe achieves availability comparable to traditional replication while maintaining near-constant computation overhead of only $2\sim3\times$, even under high redundancy where traidional replication would require linearly inflating overhead. We derive closed-form expressions for endurable failure count and computation overhead, validate them via SimGrid-based discrete-event simulation, and jointly optimize redundancy and checkpointing to minimize training time. At extreme scale with up to $600\mathrm{k}$ GPUs, SPARe reduces time-to-train by $40\sim50$% compared to traditional replication.

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Just Noticeable Difference Modeling for Deep Visual Features

Rui Zhao, Wenrui Li, Lin Zhu, Yajing Zheng, Weisi Lin

Deep visual features are increasingly used as the interface in vision systems, motivating the need to describe feature characteristics and control feature quali

ty for machine perception. Just noticeable difference (JND) characterizes the ma
ximum imperceptible distortion for images under human or machine vision. Extendi
ng it to deep visual features naturally meets the above demand by providing a ta
sk-aligned tolerance boundary in feature space, offering a practical reference f
or controlling feature quality under constrained resources. We propose FeatJND,
a task-aligned JND formulation that predicts the maximum tolerable per-feature p
erturbation map while preserving downstream task performance. We propose a FeatJ
ND estimator at standardized split points and validate it across image classific
ation, detection, and instance segmentation. Under matched distortion strength,
FeatJND-based distortions consistently preserve higher task performance than uns
tructured Gaussian perturbations, and attribution visualizations suggest FeatJND
 can suppress non-critical feature regions. As an application, we further apply
FeatJND to token-wise dynamic quantization and show that FeatJND-guided step-siz
e allocation yields clear gains over random step-size permutation and global uni
form step size under the same noise budget. The source code is available at http
s://github.com/ruizhao26/FeatJND.
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Unifying Low Dimensional Spectra in Deep Learning
Connall Garrod, Jonathan P. Keating
Low-dimensional structures appear ubiquitously in the eigenspectra of deep-learn
ing matrices in classification networks trained in the overparameterized regime.
 While theoretical advances have aimed to explain this phenomenology, they typic
ally succeed only in capturing subsets of the full behavior or rely on assumptio
ns that cannot hold in practice. In this work, we provide an analytic explanatio
n for the bulk–outlier structure of several canonical deep-learning matrices, in
cluding the Hessian, gradients, and weights. We achieve this using unconstrained
 feature models (UFMs), a now-common tool for studying the emergence of deep neu
ral collapse (DNC). We show that DNC is the source of these low-dimensional eige
nspectra: in each case, the eigenvalues and eigenvectors can be constructed from
 feature means, the characterizing objects of DNC. This provides a unifying anal
ytic explanation for a wide range of spectral phenomena in deep learning and goe
s beyond empirical characterizations—which typically focus on eigenvalues—by pro
viding a detailed analysis of eigenvectors. We prove that our results hold for b
oth linear and ReLU networks and provide numerical validation in both the modeli
ng context and standard deep-network architectures on canonical datasets.
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Multi-Round Human–AI Collaboration with User-Specified Requirements
Sima Noorani, Shayan Kiyani, Hamed Hassani, George J. Pappas
As humans increasingly rely on multi-round conversational AI for high-stakes dec
isions, principled frameworks are needed to ensure such interactions reliably im
prove decision quality. We adopt a human-centric view governed by two principles
: counterfactual harm, ensuring the AI does not undermine human strengths, and c
omplementarity, ensuring it adds value where the human is prone to err. We forma
lize these concepts via user-defined rules, allowing users to specify exactly wh
at harm and complementarity mean for their specific task. We then introduce an o
nline, distribution-free algorithm with finite-sample guarantees that enforces t
he user-specified constraints over the collaboration dynamics. We evaluate our f
ramework across two interactive settings: LLM-simulated collaboration on a medic
al diagnostic task and a human crowdsourcing study on a pictorial reasoning task
. We show that our online procedure maintains prescribed counterfactual-harm and
 complementarity violation rates even under non-stationary interaction dynamics.
 Moreover, tightening or loosening these constraints produces predictable shifts
 in downstream human accuracy, confirming that the two principles serve as pract
ical levers for steering multi-round collaboration toward better decision qualit
y without the need to model or constrain human behavior.
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Beyond Global Alignment: Fine-Grained Motion-Language Retrieval via Pyramidal Sh
apley-Taylor Learning
Hanmo Chen, Guangtao Lyu, Chenghao Xu, Jiexi Yan, Xu Yang, Cheng Deng
As a foundational task in human-centric cross-modal intelligence, motion-languag

e retrieval aims to bridge the semantic gap between natural language and human m otion, enabling intuitive motion analysis, yet existing approaches predominantly focus on aligning entire motion sequences with global textual representations. This global-centric paradigm overlooks fine-grained interactions between local m otion segments and individual body joints and text tokens, inevitably leading to suboptimal retrieval performance. To address this limitation, we draw inspirati on from the pyramidal process of human motion perception (from joint dynamics to segment coherence, and finally to holistic comprehension) and propose a novel P yramidal Shapley-Taylor (PST) learning framework for fine-grained motion-languag e retrieval. Specifically, the framework decomposes human motion into temporal s egments and spatial body joints, and learns cross-modal correspondences through progressive joint-wise and segment-wise alignment in a pyramidal fashion, effect ively capturing both local semantic details and hierarchical structural relation ships. Extensive experiments on multiple public benchmark datasets demonstrate t hat our approach significantly outperforms state-of-the-art methods, achieving p recise alignment between motion segments and body joints and their corresponding text tokens.

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Embodied Interpretability: Linking Causal Understanding to Generalization in Vis ion-Language-Action Models

Hanxin Zhang, Mingshuo Xu, Abdulqader Dhafer, Shigang Yue, Hongbiao Dong, Zhou D aniel Hao

Vision–Language–Action (VLA) policies often fail under distribution shift, sugge sting that decisions may depend on spurious visual correlations rather than task -relevant causes. We formulate visual–action attribution as an interventional es timation problem. Accordingly, we introduce the **Interventional Significance Sc ore (ISS)**, an interventional masking procedure for estimating the causal influ ence of visual regions on action predictions, and the **Nuisance Mass Ratio (NMR)**, a scalar measure of attribution to task-irrelevant features. We analyze the statistical properties of ISS and show that it admits unbiased estimation, and we characterize conditions under which action prediction error provides a valid proxy for causal influence. Experiments across diverse manipulation tasks indica te that NMR predicts generalization behavior and that ISS yields more faithful e xplanations than existing interpretability methods. These results suggest that i nterventional attribution provides a simple diagnostic approach for identifying causal misalignment in embodied policies.

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Structure-Preserving Learning Improves Geometry Generalization in Neural PDEs

Benjamin D Shaffer, Shawn Koohy, Brooks Kinch, M. Ani Hsieh, Nathaniel Trask

We aim to develop physics foundation models for science and engineering that pro vide real-time solutions to Partial Differential Equations (PDEs) which preserve structure and accuracy under adaptation to unseen geometries. To this end, we i ntroduce General-Geometry Neural Whitney Forms (Geo-NeWF): a data-driven finite element method. We jointly learn a differential operator and compatible reduced finite element spaces defined on the underlying geometry. The resulting model is solved to generate predictions, while exactly preserving physical conservation laws through Finite Element Exterior Calculus. Geometry enters the model as a di scretized mesh both through a transformer-based encoding and as the basis for th e learned finite element spaces. This explicitly connects the underlying geometr y and imposed boundary conditions to the solution, providing a powerful inductiv e bias for learning neural PDEs which we demonstrate improves generalization to unseen domains. We provide a novel parameterization of the constitutive model en suring existence and uniqueness of the solution. Our approach demonstrates state -of-the-art performance on several steady-state PDE benchmarks and provides a si gnificant improvement over conventional baselines on out-of-distribution geometr ies.

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Taming I2V models for Image HOI Editing: A Cognitive Benchmark and Agentic Self -Correcting Framework

Jiayi Gao, Qingchao Chen, Yuxin Peng, Yang Liu

Current image editing methods excels at static attributes but fails at complex Human-Object Interactions (HOI), a critical challenge unaddressed by existing benchmarks that conflate HOI with static attributes, relying on global metrics incapable of simultaneously assessing dynamic interaction validity and entangled human-object pair preservation. Thus, we first introduce HOI-Edit, a comprehensive benchmark with three progressive cognitive levels, which features an automated metric HOI-Eval that first reliably evaluates instance-level interaction by letting VLM Q&A after thinking with images containing grounded Human-Object pair. Considering the task's essence of remodeling dynamic relationships, we benchmark Image-to-Video (I2V) models, finding them inherently suited for dynamic editing due to their temporal generation capabilities. Crucially, beyond superior performance, this capability provides a "replay of the failure process", offering unique diagnosability into why errors occur. We thus propose SCPE (Self-Correcting Process Editing), a novel, agentic self-correcting framework that constrains the generation of I2V models through iteratively refined prompts, enabling the generated videos to more accurately present the target HOI. Extracted frames from these videos are the final editing results. On HOI-Edit, SCPE achieves performance competitive with state-of-the-art (SOTA) editing models like Nano Banana on interaction.Code is available at https://github.com/oceanflowlab/HOI-Edit.

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Zero-Shot Rankability: Revealing Latent Ordinal Structure in Multimodal Large Language Models via Language

Nam Hyeon-Woo, Moon Ye-Bin, Sohwi Lim, Kwon Byung-Ki, Tae-Hyun Oh

Recent work shows that vision encoders capture ordinal attributes along linear axes, which can be recovered from as few as two labeled images. However, in the zero-shot setting, the text-driven rank axis for Vision-Language Models (VLMs) like CLIP remains suboptimal. In this work, we study the embeddings of Multimodal LLMs (MLLMs). We hypothesize that MLLMs can overcome this limitation due to three potential advantages: their inherent ordinal understanding, capacity for conditional embeddings, and a small cross-modal gap. We show that MLLMs are rankable using only text prompts. Experiments demonstrate that a text-driven rank axis for MLLM embeddings achieves 90\% of the performance of the supervised linear rank axis, significantly outperforming the 61\% observed in VLM embeddings. We validate that this capability stems from MLLMs' conditional embeddings and a smaller modality gap than VLMs. Furthermore, we demonstrate that this property generalizes to the audio domain. Our findings suggest that language provides a direct interface for probing latent ordinal structures in MLLMs. Code and prompts are available at https://github.com/kaist-ami/prompt-probing

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VideoSEAL: Mitigating Evidence Misalignment in Agentic Long Video Understanding by Decoupling Answer Authority

Chenhao Qiu, Yechao Zhang, Xin Luo, Shien Song, Xusheng Liu

Long video question answering requires locating sparse, time-scattered visual evidence within highly redundant content. Although current MLLMs perform well on short videos, long videos introduce long-horizon search and verification, which often necessitates multi-turn, agentic interaction. We show that existing LVU agents can exhibit evidence misalignment: they produce correct answers that are not supported by the retrieved or inspected evidence. To characterize this failure, we introduce two diagnostics, temporal groundedness and semantic groundedness, and use them to reveal two pressures that amplify misalignment: prompt pressure from shared-context saturation at inference time and reward pressure from outcome-only optimization during training. These findings point to a structural root cause: the coupled agent paradigm conflates long-horizon planning with answer authority. We therefore propose the decoupled planner–inspector framework, which separates planning from answer authority and gates final answering on pixel-level verification. Across four long-video benchmarks, our framework improves both answer accuracy and evidence alignment, achieving 55.1% on LVBench and 62.0% on LongVideoBench while producing interpretable search trajectories. Moreover, the decoupled architecture scales consistently with increased search budgets and supports plug-and-play upgrades of the MLLM backbone without retraining the planner. C

ode and models are available at https://github.com/Echochef/VideoSEAL.
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RePo: Language Models with Context Re-Positioning
Huayang Li, Tianyu Zhao, Deng Cai, Richard Sproat
In-context learning is fundamental to modern Large Language Models (LLMs); however, prevailing architectures impose a rigid and fixed contextual structure by assigning linear or constant positional indices. The rigid position information poses the full burden of organizing the input structure to attention layers, thus reducing the amount of attention that could be allocated for more critical information. To address this, we propose RePo, a novel mechanism that alleviates the burden for attention layers via context re-positioning. Unlike conventional approaches, RePo utilizes a differentiable module, f_ϕ, to assign token positions that capture contextual dependencies, rather than replying on pre-defined order. By continually pre-training on the OLMo-2 1B \& 7B models, we demonstrate that RePo consistently enhances performance on tasks involving noisy contexts, structured data, and longer context length, while maintaining competitive performance on general short-context tasks. Analysis reveals that RePo successfully allocates more attention mass to distant but relevant information, assigns positions in a dense and non-linear space, and captures the intrinsic structure of the input context.
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SHERPA: Fine-tuning Segment Anything Models with Task-relevant Guidance
Jingcheng Xie, Yinda Chen, Xiaoyu Liu, Haoyuan Shi, Zhiwei Xiong
Segment Anything Models (SAMs) often struggle with certain specialized tasks. A common approach is to fine-tune models with specific task labels, but this often leads to overfitting, introduces model bias and significantly degrades their generalization ability.
To overcome these challenges, we propose SHERPA, a novel framework that leverages a smaller SAM to guide the fine-tuning of a larger SAM via task-relevant features.
Specifically, we first leverage the Fisher Ratio Separation (FRS) module to separate high task-relevant features and preserve the ability of the large SAM to perform other general tasks.
Then, the Guiding Feature Extraction (GFE) module is used to extract representative guiding features from the fine-tuned small SAMs.
We leverage small SAMs tailored for specific tasks (including natural image segmentation, biomedical image segmentation, and video object segmentation)
as guidance and then evaluate the SHERPA scheme to fine-tune larger SAM series models.
Our experiments demonstrate that SHERPA enhances the retention of generalization ability across those diverse tasks, by up to 11.1\%, and improves specific task performance by up to 2.2\%.
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FourTune: Towards Fully 4-Bit Efficient Post-Training for Diffusion Models
Bowen Xue, Zihan Min, Xingyang Li, Zhekai Zhang, Haocheng Xi, Lvmin Zhang, Maneesh Agrawala, Jun-Yan Zhu, Song Han, Yujun Lin, Muyang Li
Diffusion models have become a dominant paradigm for high-quality generative modeling, while post-training is essential for adapting them to diverse downstream applications.
However, post-training of large diffusion models is still challenging due to the prohibitive memory footprints and slow training speed, which existing parameter-efficient fine-tuning methods only partially address.
To overcome these limitations, we propose FourTune, an efficient post-training framework for diffusion models based on an end-to-end W4A4G4 paradigm. FourTune introduces a triple-branch hybrid pipeline that augments the standard LoRA architecture with a frozen numerical stabilizer to isolate quantization-sensitive outliers, enabling stable training under native 4-bit computation. In addition, FourTune employs hardware-efficient block-wise quantization and customized fused kernels to support efficient quantized backpropagation and reduce memory bandwidth overhead.

Across customization, reinforcement learning, and distillation tasks, FourTune m
atches the quality of full-precision fine-tuning. On FLUX.1-dev (12B), FourTune
reduces memory overhead by $2.25\times$ and increases end-to-end training throug
hput by $2.27\times$ compared to BF16 LoRA.
****
New Algorithms for Fully-Dynamic k-center with Outliers
Junyu Huang, Zhize Li, Zhen Zhang, Xujia Li, Jianxin Wang, Qilong Feng
In this paper, we study the fully dynamic k-center with outliers problem, where
points are inserted and deleted over time and the goal is to maintain an approxi
mate clustering while discarding up to z outliers. Existing algorithms typically
 rely on radius guessing to maintain cluster representations, leading to update
and query times that depend explicitly on the aspect ratio Δ. We propose a layer
ed-sampling framework that avoids radius guessing by maintaining a hierarchy of
sampled structures, which can separate most inliers from potential outliers and
refine the remaining uncertain points. The resulting algorithm achieves $\tilde{
O}(k^2/\epsilon^4)$ update and query time independent of Δ, while guaranteeing a
 constant-factor approximation with $(1+\epsilon)z$ outliers discarded. Under mi
ld assumptions, our result is complemented by a lower bound in metric space que
ry model.
****
On the Robustness of Langevin Dynamics to Score Function Error
Daniel Yiming Cao, August Y. Chen, Karthik Sridharan, Yuchen Wu
We consider the robustness of score-based generative modeling to errors in the e
stimate of the score function. In particular, we show that Langevin dynamics is
not robust to the L^2 errors (more generally L^p errors) in the estimate of
the score function.
It is well-established that with small L^2 errors in the estimate of the score
 function, diffusion models can sample faithfully from the target distribution u
nder fairly mild regularity assumptions in a polynomial time horizon.
In contrast, our work shows that even for simple distributions in high dimension
s, Langevin dynamics run for any polynomial time horizon will produce a distribu
tion far from the target distribution in Total Variation (TV) distance, even whe
n the L^2 error (more generally L^p) of the estimate of the score function i
s arbitrarily small.
Considering such an error in the estimate of the score function is unavoidable i
n practice when learning the score function from data, our results provide furth
er justification for diffusion models over Langevin dynamics and serve to cautio
n against the use of Langevin dynamics with estimated scores.
****
SoftBinary Coding: A New Information-Theoretic Paradigm for Neural Compression v
ia Fast Channel Simulation
Ezgi Ozyilkan, Sharang M. Sriramu, Elza Erkip, Aaron B. Wagner, Jona Ballé
Neural compression is currently dominated by Nonlinear Transform Coding (NTC), w
hich maps data to real-valued latents via continuous transforms. Despite its suc
cess, NTC suffers from train-test mismatch due to non-differentiable quantizatio
n, a "smoothness bias" inherent in continuous transforms that precludes optimali
ty for certain sources, and a loss of "shaping gain" due to its use of scalar qu
antization. We propose SoftBinary Coding (SBC), an end-to-end learning paradigm
that bypasses these limitations by using a stochastic binary latent space. In th
e spirit of vector quantization, SBC employs discrete representations and compre
sses them through a novel fast binary channel simulation scheme, for which we pr
ovide a proof of rate optimality. Experimental gains on information-theoretic so
urces address NTC's limitations both theoretically and practically, establishing
 discrete binary structures as a viable path toward reaching optimal rate–distor
tion bounds. Surprisingly, SBC also achieves state-of-the-art performance on vec
tor quantization of i.i.d. sources, exceeding Trellis Coded Quantization of the
Gaussian source.
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Understand and Accelerate Memory Processing Pipeline for Large Language Model In
ference

Zifan He, Rui Ma, Yizhou Sun, Jason Cong
Modern large language models (LLMs) increasingly depend on efficient long-context processing and generation mechanisms, including sparse attention, retrieval-augmented generation (RAG), and compressed contextual memory, to solve complex tasks. We show that these optimizations can be unified into a four-stage memory processing pipeline: *Prepare Memory*, *Compute Relevancy*, *Retrieval*, and *Apply to Inference*. Through systematic profiling, we identify a 22\%-97\% memory processing overhead in LLM inference and strong computational heterogeneity across stages in memory processing. Motivated by this insight, we argue that **heterogeneous systems** are well-suited to accelerate memory processing and thus end-to-end inference. We demonstrate this approach on a GPU-FPGA system by offloading sparse, irregular, and memory-bounded operations to FPGAs while retaining compute-intensive operations on GPUs. Evaluated on an AMD MI210 GPU and an Alveo U55C FPGA, our system is up to $2.2\times$ faster and $4.7\times$ energy reduction across multiple LLM optimizations than the GPU baseline (with similar results on NVIDIA A100), establishing heterogeneous systems as a practical direction for efficient LLM inference and informing future heterogeneous hardware design.
****
Rationality Measurement and Theory for Reinforcement Learning Agents
Kejiang Qian, Amos Storkey, Fengxiang He
This paper proposes a suite of rationality measures and associated theory for reinforcement learning agents, a property increasingly critical yet rarely explored. We define an action in deployment to be perfectly rational if it maximises the hidden true value function in the steepest direction. The expected value discrepancy of a policy's actions against their rational counterparts, culminating over the trajectory in deployment, is defined to be expected rational risk; an empirical average version in training is also defined. Their difference, termed as rational risk gap, is decomposed into (1) an extrinsic component caused by environment shifts between training and deployment, and (2) an intrinsic one due to the algorithm's generalisability in a dynamic environment. They are upper bounded by, respectively, (1) the 1-Wasserstein distance between transition kernels and initial state distributions in training and deployment, and (2) the empirical Rademacher complexity of the value function class. Our theory suggests hypotheses on the benefits from regularisers (including layer normalisation, ℓ_2 regularisation, and weight normalisation) and domain randomisation, as well as the harm from environment shifts. Experiments are in full agreement with these hypotheses. The code is available at \href{https://github.com/EVIEHub/Rationality}{https://github.com/EVIEHub/Rationality}.
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Minibatch selection for Language Models via Partition Matroid Constrained Gradient Matching
Prayas Agrawal, Prateek Chanda, Ishita Khatri, Ganesh Ramakrishnan, Bamdev Mishra, Pratik Jawanpuria
Training large language models (LLMs) on heterogeneous data requires selecting minibatches that balance convergence speed with coverage across domains. Existing methods either select samples independently within each domain or rely on computationally expensive proxy models to learn continuous domain weights. We propose PartitionSel, a cross-domain minibatch selection approach that maximizes a validation-guided gradient-matching utility under per-domain budgets encoded as a partition-matroid constraint. By coupling the per-domain budgets through a single utility, PartitionSel is designed to reduce redundancy in selections across domains. The proposed objective is weakly submodular and admits an orthogonal matching pursuit algorithm with provable approximation guarantees. Empirically, we evaluate PartitionSel for minibatch selection during the fine-tuning of Qwen2.5 and Llama-3 on MetaMathQA and Mol-Instructions. PartitionSel achieves robust gains over per-domain and domain-agnostic baselines on both benchmarks. It also reduces the number of conflicting gradient pairs within each batch, indicating that the cross-domain coupling translates into more compatible training updates.
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SPEED-Bench: A Unified and Diverse Benchmark for Speculative Decoding

Talor Abramovich, Maor Ashkenazi, Izzy Putterman, Benjamin Chislett, Tiyasa Mitra, Bita Darvish Rouhani, Ran Zilberstein, Yonatan Geifman
Speculative Decoding (SD) has emerged as a critical technique for accelerating Large Language Model (LLM) inference. Unlike deterministic system optimizations, SD performance is inherently data-dependent, meaning that diverse and representative workloads are essential for accurately measuring its effectiveness. Existing benchmarks suffer from limited task diversity, inadequate support for throughput-oriented evaluation, and a reliance on high-level implementations that fail to reflect production environments. To address this, we introduce **SPEED-Bench**, a comprehensive suite designed to standardize SD evaluation across diverse semantic domains and realistic serving regimes. SPEED-Bench offers a carefully curated *Qualitative* data split, selected by prioritizing semantic diversity across the data samples.
Additionally, it includes a *Throughput* data split, allowing speedup evaluation across a range of concurrencies, from latency-sensitive low-batch settings to throughput-oriented high-load scenarios. By integrating with production engines like vLLM and TensorRT-LLM, SPEED-Bench allows practitioners to analyze system behaviors often masked by other benchmarks. We highlight this by quantifying how synthetic inputs overestimate real-world throughput, identifying batch-size dependent optimal draft lengths and biases in low-diversity data, and analyzing the caveats of vocabulary pruning in state-of-the-art drafters. We release SPEED-Bench to establish a unified evaluation standard for practical comparisons of SD algorithms.
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SpikingLM: Towards Fully Spiking Language Model
Yu Liang, Zijian Zhou, Wenjie Wei, Shuai Wang, Honglin Cao, Ammar Belatreche, Yu Yang, Malu Zhang, Yang Yang, Haizhou Li
Leveraging event-driven computation mechanism, Spiking Neural Networks (SNNs) have emerged as a representative paradigm for energy-efficient edge intelligence. However, extending SNNs to modern deep language models still faces two fundamental challenges.
First, dead neurons in deep SNNs lead to degraded gradients, limiting the training effectiveness of spiking language models. Second, removing Softmax for energy efficiency weakens token-wise competition, reducing the model’s ability to select salient tokens.
To address these challenges, we propose Spiking Language Model (SpikingLM) to bridge the efficiency of SNNs and the capability of modern language models through two key innovations.
First, we propose Distribution-aware Scaling method, which rescales linear outputs into an activation-friendly range to alleviate dead neurons and stabilize gradient propagation. Notably, its scaling parameters can be fused into the preceding linear layers, incurring no additional inference overhead. Second, we introduce Spike2Max to restore winner-takes-all mechanism via base-2 exponentiation and max-subtraction. Compared with Softmax, Spike2Max reduces energy consumption by over 95\% using hardware-efficient bit-shift operations. Experiments show that SpikingLM reduces energy consumption by 57.9\% and achieves state-of-the-art performance on GLUE, laying a promising foundation for energy-efficient language modeling. Code is available at~\url{https://github.com/hamings1/SpikingLM.git}.
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A Coin Flip for Safety: LLM Judges Fail to Reliably Measure Adversarial Robustness
Leo Schwinn, Moritz Ladenburger, Tim Beyer, Mehrnaz Mofakhami, Gauthier Gidel, Stephan Günnemann
Automated ``LLM-as-a-Judge" frameworks have become the de facto standard for scalable evaluation across natural language processing. For instance, in safety evaluation, these judges are relied upon to evaluate harmfulness in order to benchmark the robustness of safety against adversarial attacks. However, we show that existing validation protocols fail to account for substantial distribution shifts inherent to red-teaming: diverse victim models exhibit distinct generation styles, attacks distort output patterns, and semantic ambiguity varies significantl

y across jailbreak scenarios. Through a comprehensive audit using 6642 human-ver
ified labels, we reveal that the unpredictable interaction of these shifts often
 causes judge performance to degrade to near random chance. This stands in stark
 contrast to the high human agreement reported in prior work. Crucially, we find
 that many attacks inflate their success rates by exploiting judge insufficienci
es rather than eliciting genuinely harmful content. To enable more reliable eval
uation, we propose ReliableBench, a benchmark of behaviors that remain more cons
istently judgeable, and JudgeStressTest, a dataset designed to expose judge fail
ures.
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SecCodePRM: A Process Reward Model for Code Security
Weichen Yu, Ravi Mangal, Yinyi Luo, Kai Hu, Jingxuan He, Corina S. Pasareanu, Ma
tt Fredrikson
Large Language Models are rapidly becoming core components of modern software de
velopment workflows, yet ensuring code security remains challenging. Existing vu
lnerability detection pipelines either rely on static analyzers or use LLM/GNN-b
ased detectors trained with coarse program-level supervision. Both families ofte
n require complete context, provide sparse end-of-completion feedback, and can d
egrade as code length grows, making them ill-suited for real-time, prefix-level
assessment during interactive coding and streaming generation. We propose \textb
f{SecCodePRM}, a security-oriented process reward model that assigns a \textbf{c
ontext-aware}, \textbf{step-level} security score along a code trajectory. To tr
ain the model, we derive step-level supervision labels from static analyzers and
 expert annotations, allowing the model to attend more precisely to fine-grained
 regions associated with inter-procedural vulnerabilities. SecCodePRM has three
applications: full-code vulnerability detection (VD), partial-code VD, and secur
e code generation (CG). For VD, SecCodePRM uses risk-sensitive aggregation that
emphasizes high-risk steps; for CG, SecCodePRM supports inference-time scaling b
y ranking candidate continuations and favoring higher cumulative reward. This de
sign yields dense, real-time feedback that scales to long-horizon generation. Em
pirically, SecCodePRM outperforms prior approaches in all three settings, while
preserving code functional correctness, suggesting improved security without a s
afety–utility tradeoff.
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Breaking the Simplification Bottleneck in Amortized Neural Symbolic Regression
Paul Saegert, Ullrich Koethe
Symbolic regression (SR) aims to discover interpretable analytical expressions t
hat accurately describe observed data. Amortized SR promises to be much more eff
icient than the predominant genetic programming SR methods, but currently strugg
les to scale to realistic scientific complexity. We find that a key obstacle is
the lack of a fast reduction of equivalent expressions to a concise normalized f
orm. Amortized SR has addressed this with general-purpose Computer Algebra Syste
ms (CAS) like SymPy, but the high computational cost severely limits training an
d inference speed. We propose **SimpliPy**, a rule-based simplification engine a
chieving a 100-fold speed-up over SymPy at comparable quality. This enables subs
tantial improvements in amortized SR, including scalability to much larger train
ing sets, more efficient use of the per-expression token budget, and systematic
training set decontamination with respect to equivalent test expressions. We dem
onstrate these advantages in our **Flash-ANSR** framework, which achieves much b
etter accuracy than amortized baselines (NeSymReS, E2E) on the FastSRB benchmark
. Moreover, it performs on par with state-of-the-art direct optimization (PySR)
while recovering more concise rather than more complex expressions with increasi
ng inference budget.
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TileQ: Efficient Low-Rank Quantization of Mixture-of-Experts with 2D Tiling
Hongyaoxing Gu, Xinzhe Chen, Lijuan Hu, Liu fangfang
Mixture-of-Experts (MoE) models achieve remarkable performance by sparsely activ
ating specialized experts, yet their massive parameters in experts pose signific
ant challenges for deployment. While low-rank quantization offers a promising ro
ute to compress MoE models, existing methods still incur nonnegligible memory ov

erhead and inference latency. To address these limitations, we propose TileQ, a fine-tuning-free post-training quantization (PTQ) method that employs 2D-tiling structured low-rank quantization to share low-rank factors across both input and output dimensions of MoE experts. Furthermore, we introduce an efficient inference technique for TileQ that fuses multiple low-rank expert computations into a single-pass operation, significantly improving hardware utilization. Experiments show that TileQ cuts down additional memory usage up to 10x and reduces inference latency to 5% while preserving state-of-the-art accuracy.
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Stochastic Minimum-Cost Reach-Avoid Reinforcement Learning
Jingduo Pan, Taoran Wu, Yiling Xue, Bai Xue
We study stochastic minimum-cost reach-avoid reinforcement learning, where an agent must satisfy a reach-avoid specification with probability at least p while minimizing expected cumulative costs in stochastic environments. Existing safe and constrained reinforcement learning methods typically fail to jointly enforce probabilistic reach-avoid constraints and optimize cost in the learning setting in stochastic environments. To address this challenge, we introduce reach-avoid probability certificates (RAPCs), which identify states from which stochastic reach-avoid constraints are satisfiable. Building on RAPCs, we develop a contraction-based Bellman formulation that serves as a principled surrogate for integrating reach-avoid considerations into reinforcement learning, enabling cost optimization under probabilistic constraints. We establish almost sure convergence of the proposed algorithms to locally optimal policies with respect to the resulting objective. Experiments in the MuJoCo simulator demonstrate improved cost performance and consistently higher reach-avoid satisfaction rates.
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Direct Flow Q-Learning
Shicheng Cao, Jingrui Jia, Wenyu Li, Feng Duan, Tao Zhang, Shengbo Eben Li
Flow Matching shows great promise in offline reinforcement learning (RL), yet optimizing these iterative policies via Backpropagation Through Time (BPTT) is unstable. While prevailing paradigms circumvent this by distilling multi-step flows into single-step approximations, such methods may limit the benefits of iterative refinement. To avoid these sacrifices, we propose Direct Flow Q-Learning (DFQL), a streamlined framework that attains superior results by optimizing flow matching policies without BPTT or distillation. DFQL derives a surrogate objective that directly injects terminal Q-value gradients as a guidance term into each step velocity field, ensuring stable optimization while preserving iterative expressive capacity. Across 73 challenging tasks in OGBench and D4RL, DFQL achieves state-of-the-art results. Additionally, DFQL extends seamlessly to the offline-to-online setting, delivering substantial performance gains without further modification.
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Prompt Optimization with Minimal Unlabeled Input via Meta-Reasoning
Yuran Sun, Chuan Wu
Prompt optimization is critical for maximizing the performance of large language models (LLMs). However, it often relies on costly labeled data. Self-supervised methods reduce data dependency, but they suffer from optimization ambiguity or high computational costs. To address these limitations, we propose the Meta-Reasoning Prompt Engineering Agent (MR.PEA), a self-supervised prompt optimization framework that operates with minimal input. MR.PEA leverages meta-reasoning to iteratively build task-specific knowledge, including problem-solving strategies and evaluation criteria, while adaptively retrieving external information to enhance its understanding. This knowledge guides the generation of diverse validation examples, targeted prompt refinement, and comprehensive quality assessments. Experiments on GSM8K and Big-Bench Hard show that MR.PEA outperforms existing baselines, achieving an average performance gain of 7.4% with an optimization cost as low as $0.01 per task.
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Orthogonal Hierarchical Decomposition for Structure-Aware Table Understanding with Large Language Models

Bin Cao, huixian lu, chenwen ma, Ting Wang, Ruizhe Li, JING FAN
Complex tables with multi-level headers, merged cells and heterogeneous layouts pose persistent challenges for large language models (LLMs) in both understanding and reasoning. Existing approaches typically rely on table linearization or normalized grid modeling. However, these representations struggle to explicitly capture hierarchical structures and cross-dimensional dependencies, which can lead to misalignment between structural semantics and textual representations for non-standard tables. To address this issue, we propose an Orthogonal Hierarchical Decomposition (OHD) framework that constructs structure-preserving input representations of complex tables for LLMs. OHD introduces an Orthogonal Tree Induction (OTI) method based on spatial--semantic co-constraints, which decomposes irregular tables into a column tree and a row tree to capture vertical and horizontal hierarchical dependencies, respectively. Building on this representation, we design a dual-pathway association protocol to symmetrically reconstruct the semantic lineage of each cell, and incorporate an LLM as a semantic arbitrator to align multi-level semantic information. We evaluate OHD framework on two complex table question answering benchmarks, AITQA and HiTab. Experimental results show that OHD consistently outperforms existing representation paradigms across multiple evaluation metrics.
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Mind the Gap: Mixtures of Gaussians in Approximate Differential Privacy
Huikang Liu, Aras Selvi, Wolfram Wiesemann
We design a class of additive noise mechanisms that satisfy (ε, δ)-differential privacy (DP) for scalar, real-valued query functions with known sensitivities, with a particular focus on moderate and low-privacy regimes. These mechanisms, which we call *mixture mechanisms*, are constructed by mixing multiple Gaussian distributions that share the same variance but differ in their means and mixture weights. The resulting distributions can be interpreted as convex combinations of a zero-mean Gaussian (as used in the analytic Gaussian mechanism) and additional Gaussians whose means depend on the sensitivity of the query function. We derive tight conditions on the variances required for (ε, δ)-DP and provide efficient algorithms to compute them. Compared to the analytic Gaussian mechanism, our mechanisms yield substantially lower expected noise amplitudes (l_1-loss) and variances (l_2-loss for zero-mean distributions). In the low-privacy regime that motivates our design, our mechanisms approach optimality, mitigating nearly all of the optimality gap of the analytic Gaussian mechanism.
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SmartThinker: Progressive Chain-of-Thought Length Calibration for Efficient Large Language Model Reasoning
Chenzhi Hu, Qinzhe Hu, Yuhang Xu, Junyi Chen, Ruijie Wang, Shengzhong Liu, Jianxin Li, Fan Wu, Guihai Chen
Large reasoning models (LRMs) like OpenAI o1 and DeepSeek-R1 achieve high accuracy on complex tasks by adopting long chain-of-thought (CoT) reasoning paths.
However, the inherent verbosity of these processes frequently results in redundancy and overthinking.
To address this issue, existing works leverage Group Relative Policy Optimization (GRPO) to reduce LRM output length, but their static length-reward designs fail to adapt to problem difficulty and response-length distributions, causing over-compression and compromised accuracy.
Therefore, we propose *SmartThinker*, a novel GRPO-based efficient reasoning method with progressive CoT length calibration.
SmartThinker makes a two-fold contribution:
First, it dynamically estimates the optimal length with peak accuracy during training and guides overlong responses toward it to reduce reasoning length while sustaining accuracy.
Second, it dynamically modulates the length-reward coefficient to avoid the unwarranted penalization of correct reasoning paths.
Extensive experimental results show that *SmartThinker* achieves up to 52.6\% length compression with improved accuracy and achieves up to 16.6\% accuracy relat

ive improvement on challenging benchmarks like AIME25.
The source code can be found at https://github.com/SJTU-RTEAS/SmartThinker.
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Omni-fMRI: A Universal Atlas-Free fMRI Foundation Model
Mo Wang, Wenhao Ye, Junfeng Xia, Junxiang Zhang, Xuanye Pan, Minghao Xu, Haotian Deng, Hongkai Wen, Quanying Liu
Self-supervised fMRI foundation models have shown promising transfer performance, yet most rely on predefined region-level parcellations that discard fine-grained voxel information and introduce atlas-dependent biases. We propose Omni-fMRI, an atlas-free foundation model that operates directly on voxel-level signals. To enable scalable pretraining on 49,497 fMRI sessions across nine datasets, Omni-fMRI introduces a dynamic patching mechanism that substantially reduces computational cost while preserving informative spatial structure. To support reproducibility and fair comparison, we establish a comprehensive benchmark suite spanning 11 datasets and a diverse set of resting-state and task-based fMRI tasks. Experimental results demonstrate that Omni-fMRI consistently outperforms existing foundation models, providing a scalable and reproducible framework for atlas-free brain representation learning. Code is available.
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Alignment Risks from Capability-Seeking RL Training
Yujun Zhou, Yue Huang, Han Bao, Kehan Guo, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V Chawla, Xiangliang Zhang
While most AI alignment research focuses on preventing models from generating explicitly harmful content, a more subtle risk arises from capability-seeking RL training in vulnerable environments. We investigate whether language models, when trained with reinforcement learning (RL) in environments with implicit loopholes, can learn to exploit these flaws to maximize reward, even without being explicitly instructed to do so. To test this, we design a suite of four diverse "vulnerability games'', each presenting a structural vulnerability related to context-conditional compliance, proxy metrics, reward tampering, and self-evaluation. Our experiments show that models often learn to exploit these vulnerabilities, discovering opportunistic strategies that increase reward while sometimes preserving or even improving standard task-performance metrics. More critically, we find that these exploitative strategies are not always narrow "tricks'': they can transfer in structured but limited ways, propagate from a capable teacher model to other student models through SFT, and in several cases remain more persistent when learned through RL than when distilled through SFT. Our findings show that alignment risks from capability-seeking RL training can be difficult to detect with standard performance monitoring, suggesting that future AI safety work should extend beyond content moderation to auditing and securing training environments, reward mechanisms, and evaluation channels. Code is available at https://github.com/YujunZhou/Capability-seeking-RL-risk.
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Interleaved Selective State Space Models for Efficient WiFi-Based 3D Multi-Person Pose Estimation
Quang-Anh N.D., Kok-Seng Wong
WiFi-based human pose estimation offers privacy-preserving and occlusion-robust sensing, but current Transformer-based approaches suffer from quadratic complexity and lack explicit inductive biases for the structure of Channel State Information (CSI). We propose WiFi-Mamba, the first State Space Model (SSM) architecture for WiFi-based 3D multi-person pose estimation. Our approach introduces three key contributions: (1) a Dual-Stream Selective SSM that processes amplitude and phase through parallel pathways with cross-stream state coupling to respect their distinct physical properties, (2) Selective State Attention for pose query decoding with SSM-derived sequential context, and (3) Persistent SSM Memory for temporal consistency across frames without recurrent memory explosion. Extensive experiments on the Person-in-WiFi 3D dataset, covering both single-person and multi-person, demonstrate a 16-27% MPJPE reduction across varying numbers of persons while using only 4.4\% of the baseline parameters (2.14M vs. 48.2M), achieving superior efficiency-accuracy trade-offs particularly beneficial for edge deploym

ent in privacy-sensitive continuous monitoring scenarios.
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AutoRPA: Efficient GUI Automation through LLM-Driven Code Synthesis from Interactions
Minghao Chen, Xinyi Hu, Zhou Yu, Yufei Yin
Large Language Model (LLM) based agents have demonstrated proficiency in multi-step interactions with graphical user interfaces (GUIs). While most research focuses on improving single-task performance, practical scenarios often involve repetitive GUI tasks for which invoking LLM reasoning repeatedly, i.e., the ReAct paradigm, is inefficient. Prior to LLMs, traditional Robotic Process Automation (RPA) offers runtime efficiency but demands significant manual effort to develop and maintain. To bridge this gap, we propose \textbf{AutoRPA}, a framework that automatically distills the decision logic of ReAct-style agents into robust RPA functions. AutoRPA introduces two core innovations: (1) A \textit{translator-builder pipeline} where a translator agent converts hard-coded ReAct actions into soft-coded procedures, and a builder agent synthesizes robust RPA functions via retrieval-augmented generation over multiple trajectories; (2) A \textit{hybrid repair strategy} during code verification, combining RPA execution with ReAct-based fallback for iterative refinement. Experiments across multiple GUI environments demonstrate that RPA functions generated by AutoRPA successfully solve similar tasks while reducing token usage by 82\%\textasciitilde96\%, significantly improving runtime efficiency and reusability.
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Robust Linear Dueling Bandits with Post-serving Context under Unknown Delays and Adversarial Corruptions
Youngmin Oh
We study linear dueling bandits in volatile environments characterized by the simultaneous presence of post-serving contexts, delayed feedback, and adversarial corruption. Feedback is subject to unknown stochastic or adversarial delays and a cumulative corruption budget $\mathcal{C}$. To address these challenges, we propose RCDP-UCB, which integrates a learned approximator that predicts post-serving contexts from pre-serving information. It further employs an adaptive weighting strategy that clips feature vectors to mitigate the impact of corrupted and delayed observations simultaneously. Under standard regularity conditions and a parametric post-serving mapping, we rigorously establish that our algorithm is delay-regime-agnostic, achieving a regret upper bound of $\widetilde{\mathcal{O}}(d(\sqrt{T} + \mathcal{C} + \mathcal{D}))$, where d is the total feature dimension and $\mathcal{D}$ encapsulates the delay complexity, scaling with $\sqrt{\Lambda}$ under adversarial delays or μ_{τ} under stochastic delays (Λ: cumulative delay budget; μ_{τ}: mean of sub-Gaussian delays). We further establish lower bounds that nearly match our upper bounds up to a $\sqrt{d}$ factor for adversarial delays in the absence of post-serving contexts. Code is available at \url{https://github.com/youngmin0oh/rcdp-public}.
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SOPE: Situation-Aware and Statistically Indistinguishable Privacy Exfiltration for MCP-enabled Agents
Ruixiao Lin, Qingming Li, Jiahao Chen, Chunyi Zhou, Shouling Ji
The Model Context Protocol (MCP) enables Large Language Model (LLM) agents to interact with external tools, but this extensibility introduces significant supply chain vulnerabilities that enable covert privacy exfiltration. Prior studies have revealed privacy leakage in MCP-enabled agents via indirect prompt injection; however, existing attacks are typically misaligned with the agent's tool-usage context and rely on rigid templates, resulting in recognizable patterns that are readily flagged by existing defenses. In this work, we exploit the observation that privacy exposure is inherently scenario-dependent, to associate certain privacy items with specific tools. We introduce `SOPE`, a Scenario-aware and zerO-click Privacy Exfiltration framework that transforms any benign MCP server into its privacy-exfiltrating variants. `SOPE` (1) identifies privacy items that are appropriate to the tool usage, (2) embeds privacy-probing instructions into tool-invocation prompts, and (3) achieves zero-click data transmission via code-level

modifications. We evaluate `SOPE` across 27,216 test cases, where 324 `SOPE`-transformed *real-world* servers attacking four benchmark and three commercial agents with *nine* state-of-the-art defenses. Results demonstrate that `SOPE` remains highly effective and robust, highlighting critical protocol-level safety gaps in the agent ecosystem.
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A Decision-Theoretic View of Test-Time Training: When, How Far, and Which Directions to Adapt
Tomoya Wakayama
Test-time training (TTT) adapts a pretrained model to each prompt via parameter updates, improving accuracy under pretraining-to-test distribution shifts. Yet, its performance often suffers from instability and sensitivity to hyperparameters such as update steps and subspace. We explain this behavior through a decision-theoretic lens, treating TTT as implicit Bayesian inference in the kernel regime. Under a Gaussian process benchmark, we show that TTT reduces prediction error when updates are spectrally matched to the prompt's signal-to-noise ratio and aligned with query-relevant eigen-directions. This perspective underpins the following results: (1) we show when fixed update steps and subspaces fail under distribution shifts, motivating adaptive strategies; (2) we prove that selecting update steps via prompt evidence admits a PAC-Bayes guarantee against overfitting; and (3) we characterize the Bayes-optimal update subspace under a linear-Gaussian correction model, yielding a scoring rule for selecting Transformer blocks and heads. Our theory helps explain the empirical instability of TTT, taking a step toward principled guidance for when, how far, and which directions to adapt.
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Align Forward, Adapt Backward: Closing the Discretization Gap in Logic Gate Networks
Youngsung Kim
In neural network models, soft mixtures of fixed candidate components (e.g., logic gates and sub-networks) are often used during training for stable optimization, while hard selection is typically used at inference. This raises questions about training-inference mismatch. We analyze this gap by separating forward-pass computation (hard selection vs. soft mixture) from stochasticity (with vs. without Gumbel noise). Using logic gate networks as a testbed, we observe distinct behaviors across four methods: Hard-ST achieves zero selection gap by construction; Gumbel-ST achieves near-zero gap when training succeeds but suffers accuracy collapse at low temperatures; Soft-Mix achieves small gap only at low temperature via weight concentration; and Soft-Gumbel exhibits large gaps despite Gumbel noise, confirming that noise alone does not reduce the gap. We propose CAGE (Confidence-Adaptive Gradient Estimation) to maintain gradient flow while preserving forward alignment. On logic gate networks, Hard-ST and Gumbel-ST with CAGE both achieve over 98\% accuracy on MNIST and ${\sim}58\%$ on CIFAR-10; Hard-ST with CAGE further maintains zero selection gap across all temperatures by construction, while Gumbel-ST without CAGE suffers a ${\sim}47$-point accuracy collapse.
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ContrastiveCFG: Guiding Diffusion Sampling by Contrasting Positive and Negative Concepts
Jinho Chang, Changsun Lee, Hyungjin Chung, Jong Chul Ye
As Classifier-Free Guidance (CFG) has proven effective in conditional diffusion model sampling for improved condition alignment, many applications use a negated CFG term as a Negative Prompting (NP) to filter out unwanted features from samples. However, simply negating CFG guidance creates an inverted probability distribution, often distorting samples away from the marginal distribution. Inspired by recent advances in conditional diffusion models for inverse problems, here we present a novel method to achieve guidance toward the given condition using contrastive loss. Specifically, our guidance term aligns or repels the denoising direction based on the given condition through contrastive loss, achieving a similar guiding effect to traditional CFG for positive conditions while overcoming the limitations of existing negative guidance methods. Experimental results demonstrate that our approach effectively injects or removes the given concepts while

maintaining sample quality across diverse scenarios, from simple class condition
s to complex and overlapping text prompts.
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MalTree: Tracing Malware Evolution using Embeddings at Scale
Akash Amalan, Georgios SMARAGDAKIS, Tom Julian Viering
Malware detection remains largely reactive: machine learning models trained on k
nown samples degrade as threats evolve. Understanding evolutionary relationships
 among malware families can inform proactive defense, but traditional reverse en
gineering can take months to years to uncover such lineage relationships. We pro
pose MalTree, a framework that applies bioinformatics-inspired phylogenetic tech
niques (UPGMA and Neighbor-Joining) at scale to model malware evolution automati
cally using structural, behavioral, and image-based features. We introduce tempo
ral validation using VirusTotal timestamps to assess whether inferred trees refl
ect actual evolutionary order. MalTree achieves 87% temporal consistency, indica
ting that inferred evolutionary relationships closely align with real-world emer
gence timelines. Our analysis shows that some families mutate over 10 times fast
er than others, suggesting that detection strategies should be tailored to famil
y-specific evolutionary tempos. Case studies, including the Mirai botnet, confir
m that inferred relationships from our phylogenetic tree align with documented t
hreat intelligence. Our framework provides a foundation for shifting malware ana
lysis from sample-by-sample classification toward lineage-aware evolutionary mod
eling.
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SKETCH: Semantic Key-Point Conditioning for Long-Horizon Vessel Trajectory Predi
ction
Linyong Gan, Zimo Li, Wenxin XU, Li Xingjian, Jianhua Z. Huang, Enmei Tu, Shuhan
g Chen
Accurate long-horizon vessel trajectory prediction remains challenging due to co
mpounded uncertainty from complex navigation behaviors and environmental factors
. Existing methods often struggle to maintain global directional consistency, le
ading to drifting or implausible trajectories when extrapolated over long time h
orizons. To address this issue, we propose a semantic-key-point-conditioned traj
ectory modeling framework, in which future trajectories are predicted by conditi
oning on a high-level Next Key Point (NKP) that captures navigational intent. Th
is formulation decomposes long-horizon prediction into global semantic decision-
making and local motion modeling, effectively restricting the support of future
trajectories to semantically feasible subsets. To efficiently estimate the NKP p
rior from historical observations, we adopt a pretrain-finetune strategy. Extens
ive experiments on real-world AIS data demonstrate that the proposed method cons
istently outperforms state-of-the-art approaches, particularly for long travel d
urations, directional accuracy, and fine-grained trajectory prediction.
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Uncovering the Gradient Geometry of Long CoT: A Spectral-guided Approach to Reas
oning Distillation
Sinan Fan, Xiaofeng Sun, Chen Shen, Chenxi Huang, Shaotian Yan, Bing Wang, kaiyu
an liu, Xiaosong Yuan, Liang Xie, Wenxiao Wang, Jun Zhang, Hongyang Chen, Jiepin
g Ye
Large reasoning models (LRMs) achieve remarkable reasoning performance by genera
ting long chains-of-thought (CoT). However, standard supervised fine-tuning (SFT
) treats all tokens uniformly, indiscriminately minimizing loss across both esse
ntial reasoning steps and those that are noisy, redundant, or instance-specific.
This often leads student models to memorize superficial patterns rather than acq
uire generalizable reasoning capabilities.
To better understand this limitation, we introduce \textit{Loss Subspace Attribu
tion}, a gradient decomposition analysis approach that uncovers a striking geome
tric structure:
Gradients corresponding to effective reasoning predominantly lie within a low-ra
nk consensus subspace, while conflicting or unstructured signals dominate the re
sidual subspace.
Guided by this insight, we propose \textbf{\textit{Spectral-guided Learning}}, a

step-level distillation strategy that uses spectral strength to identify reasoning steps aligned with the consensus subspace and prioritizes their contribution to parameter updates, while suppressing gradients from the residual subspace. Experiments across various LRMs and diverse complex reasoning tasks consistently demonstrate that focusing optimization on the consensus subspace yields more robust and generalizable student models.
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FunPhase: A Periodic Functional Autoencoder for Motion Generation via Phase Manifolds
Marco Pegoraro, Evan Atherton, Bruno Roy, Aliasghar Khani, Arianna Rampini
Learning natural body motion remains challenging due to the strong coupling between spatial geometry and temporal dynamics. Embedding motion in phase manifolds, latent spaces that capture local periodicity, has proven effective for motion prediction; however, existing approaches are tied to fixed skeletons and narrow motion distributions, limiting their applicability across diverse settings.
We introduce FunPhase, a functional periodic autoencoder that learns a phase manifold for motion and replaces discrete temporal decoding with a function-space formulation, enabling smooth trajectories that can be sampled at arbitrary temporal resolutions.
FunPhase unifies motion prediction and generation within a single interpretable phase manifold, enabling motion generation via latent diffusion, generalizes across skeletons and datasets, and supports downstream tasks such as motion super-resolution and partial-body completion.
Our model achieves substantially lower reconstruction error than prior periodic autoencoder baselines, achieving uniform improvements of at least 45% across all metrics, while enabling a broader range of applications and performing on par with state-of-the-art motion generation methods.
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Not All Prefills Are Equal: PPD Disaggregation for Multi-turn LLM Serving
Zongze Li, Jingyu Liu, Zach Xu, Yineng Zhang, Tahseen Rabbani, Ce Zhang
Prefill-Decode (PD) disaggregation has become the standard architecture for modern LLM inference engines, which alleviates the interference of two distinctive workloads. With the growing demand for multi-turn interactions in chatbots and agentic systems, we re-examined PD in this case and found two fundamental inefficiencies: (1) every turn requires prefilling the new prompt and response from the last turn, and (2) repeated KV transfers between prefill and decode nodes saturate the bandwidth, leading to high latency and even service degradation. Our key insight is that not all prefill operations are equally disruptive: append-prefill---processing only the new input tokens while reusing cached KV states---incurs substantially less decoding slowdown than full prefill. This motivates routing append-prefill to decode nodes locally. However, through comprehensive analysis, we show that no single fixed routing strategy satisfies all Service Level Objectives (SLOs) simultaneously. Based on this insight, we propose Prefill Prefill-capable Decode (PPD) disaggregation, a dynamic routing system that decides when to process Turn 2+ requests locally on decode nodes using cached KV states. PPD adapts to varying SLOs via configurable weights and seamlessly integrates with traditional PD deployments. With extensive evaluations, we show that PPD reduces Turn 2+ time-to-first-token (TTFT) by 68% while maintaining competitive time-per-output-token (TPOT), effectively alleviating KV transfer congestion under high load. We believe PPD represents a flexible and efficient paradigm for multi-turn LLM serving.
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Structure-Aware Consistency Priors for Shape from Polarization in Complex Media
Kaimin Yu, Puyun Wang, Huayang He, Xianyu Wu
Recovering surface normals from single-view polarization images in complex media remains challenging. This paper focuses on ice as a representative complex medium, where intricate light–matter interactions lead to a nonlinear mapping between polarization observations and surface normals. To address this, a structure-aware polarization prior based on autocorrelation functions is proposed to capture

the local spatial consistency of AoLP. Building on this, a dual-branch network (IceSfP) is designed to integrate raw polarization features with priors via cross-modal attention and multi-scale feature fusion, enabling accurate surface normal estimation under complex media conditions. To evaluate the method, the first real-world ice SfP dataset is constructed. Experimental results show that the method outperforms existing approaches across all metrics, achieving a MAE of 16.01°, which is 2.74° lower than the second-best method. The framework provides a generalizable solution for high-precision geometric perception in complex media.

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A Diffusive Classification Loss for Learning Energy-based Generative Models

RuiKang OuYang, Louis Grenioux, José Miguel Hernández-Lobato

Score-based generative models have recently achieved remarkable success. While they are usually parameterized by the score, an alternative way is to use a series of time-dependent energy-based models (EBMs), where the score is obtained from the negative input-gradient of the energy. Crucially, EBMs can be leveraged not only for generation, but also for tasks such as compositional sampling or building Boltzmann Generators via Monte Carlo methods. However, training EBMs remains challenging. Direct maximum likelihood is computationally prohibitive due to the need for nested sampling, while score matching, though efficient, suffers from mode blindness. To address these issues, we introduce the Diffusive Classification (DiffCLF) objective, a simple method that avoids blindness while remaining computationally efficient. DiffCLF reframes EBM learning as a supervised classification problem across noise levels, and can be seamlessly combined with standard score-based objectives. We validate the effectiveness of DiffCLF by comparing the estimated energies against ground truth in analytical Gaussian mixture cases, and by applying the trained models to tasks such as model composition and Boltzmann Generator sampling. Our results show that DiffCLF enables EBMs with higher fidelity and broader applicability than existing approaches. Our code is available at [h2o64/diffclf](https://github.com/h2o64/diffclf).

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Transfer Learning in Nonparametric Regression with Deep ReLU Networks

Junpeng Ren, Carlos Misael Madrid Padilla, Yanzhen Chen, OSCAR HERNAN MADRID PADILLA

This paper develops a general transfer learning framework for nonparametric regression with data consisting of multiple groups. Under the assumption that groups share a common structure along with group-specific deviations in additive form, the proposed method employs a two-stage offset learning procedure: the first stage pools data from all groups to estimate an overall mean function, and the second stage estimates offsets for each group, yielding final group-level estimators through additive combination. Upper bounds on the $\mathcal L_2$ error are established for the proposed framework, covering a broad class of nonparametric estimators under mild complexity and noise conditions. When instantiated with deep ReLU networks, explicit convergence rates are derived under hierarchical composition models, demonstrating the ability to overcome the curse of dimensionality. Conditions that enable positive transfer with faster rates are considered, including learning with simpler functions and data augmentation through pooling samples across groups. Various simulations and real-data experiments further validate the effectiveness of the proposed method.

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Discounted Beta–Bernoulli Reward Estimation for Sample-Efficient Reinforcement Learning with Verifiable Rewards

Haechan Kim, Soohyun Ryu, Gyouk Chu, Doohyuk Jang, Eunho Yang

Reinforcement learning with verifiable rewards (RLVR) has emerged as an effective post-training paradigm for improving the reasoning capabilities of large language models.

However, existing group-based RLVR methods often suffer from severe sample inefficiency.

This inefficiency stems from reliance on point estimation of rewards from a small number of rollouts, leading to high estimation variance, variance collapse, an

d ineffective utilization of generated responses.
In this work, we reformulate RLVR from a statistical estimation perspective by m odeling rewards as samples drawn from a policy-induced distribution and casting advantage computation as the problem of estimating the reward distribution from finite data.
Building on this view, we propose **D**iscounted **B**eta–**B**ernoulli (**DBB**) reward estimation, which leverages historical reward statistics for the non-st ationary distribution.
Although biased, the resulting estimator exhibits reduced and stable variance an d theoretically avoids variance collapse.
Under mild non-stationarity, it also achieves a lower mean squared error than st andard point estimation, as we characterize analytically and verify empirically.
Across six in-distribution and three out-of-distribution reasoning benchmarks, G RPO with DBB consistently outperforms naive GRPO and strong recent baselines, in cluding the replay-based RePO and the variance-collapse-aware GRESO and DAPO.
Relative to GRPO, it achieves average Acc@8 improvements of 3.43/2.32 points in- distribution and 10.05/8.34 points out-of-distribution on the 1.7B and 8B models , respectively, without additional computational cost or memory usage.
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Escaping the Subspace Trap: The Role of Optimizer Geometry in Model Width Expans ion
Jiabei Chen, Haoyu Wang, Yang Yu, Yao Xu, Liangdong Wang, Guang Liu, Shizhu He, Jun Zhao, Kang Liu
Pre-training large language models from scratch is prohibitively expensive as mo del scales increase. A practical alternative is Model Width Expansion (MWE), whi ch grows a larger model from a well-pretrained ''seed'' model to inherit existin g capabilities at initialization. However, we identify a phenomenon termed the _ **Subspace Trap**_: during continual pre-training, parameter updates largely sta gnate within a low-dimensional subspace aligned with the initialization, limitin g the effective capacity of the expanded model. Our theoretical analysis investi gates this issue by attributing it to the function-preserving properties of widt h expansion. In particular, element-wise adaptive optimizers remain confined to the trap, whereas optimizers that yield an isotropic geometry of parameter updat es can escape. To demonstrate the impact of the subspace trap on model performan ce, we conduct empirical experiments across different model sizes and model fami lies, which show that escaping the trap is principally effective in improving tr aining efficiency and overall model performance. Detailed mechanistic analyses f urther confirm that escaping the trap indeed activates the new dimensions to enc ode general knowledge. Our code is available at https://github.com/A-PolarBear/M odel-Width-Expansion.
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Dynamic High-Dimensional Facility Location with Low Recourse
Sayan Bhattacharya, Martín Costa, Silvio Lattanzi, Jakub ■■cki, Nikos Parotsidis
We study the problem of dynamic facility location with non-uniform costs. Facili ty location is a central problem in unsupervised learning and in recent years th e dynamic version of the problem has been extensively studied. In this paper, we study the setting where clients are added and deleted in real-time and one is i nterested in maintaining efficiently a stable and high-quality solution. Interes tingly, we are able to show that on High Dimensional Euclidean metrics it is pos sible to obtain efficient algorithms for this problem. More formally, we obtain a randomized algorithm for dynamic facility location in d-dimensional Euclidea n spaces with γ approximation ratio, $O(\log m)$ amortized recourse and $ \text{poly}(d) \cdot (m+n)^{O(1/\gamma)}$ amortized update time, for every suffi ciently large constant $\gamma \geq 1$. Our result is the first efficient dynami c algorithm for the \emph{non-uniform} dynamic facility location problem in high -dimensional Euclidean spaces. It also provides a stronger recourse bound than t he existing solutions.
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Probability of Matching for Batch Multi-Objective Bayesian Optimization
Mingqian Li, Sina Hossein Zadeh, Raymundo Arroyave, Xiaoning Qian

In batch multi-objective Bayesian optimization (MOBO), it is often desirable to identify the whole Pareto optimal set, especially when considering the complicated interplay between different design criteria and constraints. This poses unique challenges in acquiring batches of both high quality and diversity to cover the Pareto front. We propose a novel acquisition strategy, Probability of Matching (POM), which evaluates both batch candidate quality and diversity by explicitly capturing the likelihood that all batch points are Pareto optimal, and the probability that they collectively cover the full Pareto set. To estimate the coverage probability and promote diversity, we incorporate non-replacement sampling principles, resulting in our new POM-guided batch MOBO method. Across synthetic benchmarks and real-world tasks, our method consistently outperforms state-of-the-art baselines on standard MOBO metrics as well as a new design-space coverage metric, Expected Minimum Distance (EMD), with comparable computational efficiency.
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PlotCraft: Pushing the Limits of LLMs for Complex and Interactive Data Visualization
Jiajun Zhang, Jianke Zhang, Zeyu Cui, Jiaxi Yang, Lei Zhang, Zilei Wang, Qiang Liu, Liang Wang, Binyuan Hui, Junyang Lin
Recent Large Language Models (LLMs) have demonstrated remarkable proficiency in code generation. However, their ability to create complex visualizations for scaled and structured data remains largely unevaluated and underdeveloped. To address this gap, we introduce **PlotCraft**, a new benchmark featuring 1k challenging visualization tasks that cover a wide range of topics, such as finance, scientific research, and sociology. The benchmark is structured around seven high-level visualization tasks and encompasses 48 distinct chart types. Crucially, it is the first to systematically evaluate both single-turn generation and multi-turn refinement across a diverse spectrum of task complexities.
Our comprehensive evaluation of 23 leading LLMs on PlotCraft reveals obvious performance deficiencies in handling sophisticated visualization tasks. To bridge this performance gap, we develope **SynthVis-30K**, a large-scale, high-quality dataset of complex visualization code synthesized via a collaborative agent framework. Building upon this dataset, we develope **PlotCraftor**, a novel code generation model that achieves strong capabilities in complex data visualization with a remarkably small size.
Across VisEval, PandasPlotBench, and our proposed PlotCraft, PlotCraftor shows performance comparable to that of leading proprietary approaches. Especially, on hard task, Our model achieves over 50\% performance improvement. We will release the benchmark, dataset, and code at \href{https://anonymous.4open.science/r/PlotCraft-E320}{PlotCraft anonymous repository}.
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A Fourier perspective on the learning dynamics of neural networks: from sample complexities to mechanistic insights
Fabiola Ricci, Claudia Merger, Sebastian Goldt
Neural networks trained with gradient-based methods exhibit a strong simplicity bias: they learn simpler statistical features of their data before moving to more complex features. Previous analyses of this phenomenon have largely focused on settings with (quasi-)isotropic inputs. In this work, we study the simplicity bias from a Fourier perspective, which allows us to include two key features of natural images in the analysis: approximate translation-invariance and power-law spectra. We first show experimentally that simple neural networks trained on image classification tasks first rely on amplitude information $-$ related to pair-wise correlations between pixels $-$ before exploiting phase information, which encodes edges and higher-order correlations. In view of this, we introduce a synthetic data model for translation-invariant inputs that allows precise control over amplitudes and phases while remaining tractable. We rigorously establish that for isotropic and high-dimensional inputs, classification based on phase information alone is a genuinely hard task: online stochastic gradient descent (SGD) cannot distinguish the structured inputs from noise within $n \ll N^3$ steps, but needs at least $n \gg N^3 \log^2{N}$ steps. In contrast, we show both experimentally and theoretically that power-law spectra can dramatically

accelerate the speed of learning phase information, even if the spectra do not help with classification. Simulations with two-layer networks trained on textures and with deep convolutional networks on ImageNet and CIFAR100 confirm this non-trivial interaction between amplitudes and phases, providing mechanistic insights into how deep neural networks can learn natural image distributions efficiently.
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Hierarchical ODE: Learning Continuous-Time Physical Prototypes for Early Link Failure Detection

Jiaen Lv, Leran Qi, Shaowei Wang

Time series prototype learning is fundamentally challenged by observational ambiguity. Discrete architectures fail to resolve this, as they lack the capacity to decouple stochastic noise from continuous dynamics. Furthermore, rigid closed-set assumptions fail to capture unseen diversity. To address these limitations, we propose a hierarchical ordinary differential equation clustering network, which utilizes neural ordinary differential equation to model latent state evolution as a continuous integral curve. This formulation enforces temporal continuity to effectively disentangle smooth feature trends from stochastic noise, while our adaptive hierarchical mechanism autonomously determines the appropriate number of prototypes without rigid prior constraints. Validated on the early link failure detection task with irregularly sampled time series, the proposed method effectively extracts underlying physical prototypes, thereby enabling robust failure detection. Our code is available at https://github.com/NJ-LNN/Hierarchical-ODE.
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Reasoning Compartmentalization: Bridging the Concretization Gap via Abstraction-based Routing

Ling-I Wu, Jian Tong, Yu Sun, Xuan Gao, Xu Guo, Qipeng Guo, Kai Chen, Guoqiang Li

While previous research has documented the sensitivity of Large Language Models (LLMs) to surface-level performance degradation, the underlying impact on internal representations and learning dynamics remains under-explored. In this work, we study this question using a controlled setup with paired reasoning tasks that are logically identical but expressed either in an abstract formal language (FL) or in natural language (NL). We find that converting FL problems into NL consistently degrades reasoning accuracy. More importantly, we show that FL and NL inputs activate largely separate internal representations and exhibit weak learning transfer between them. We refer to this phenomenon as reasoning compartmentalization. To test whether this compartmentalization can be mitigated, we introduce abstraction-based alignment, where models are trained to translate NL inputs into their corresponding FL forms. While this significantly improves reasoning performance, FL and NL representations remain largely distinct, and learning transfer across formulations remains limited. Through activation-level interventions, we further show that performance improvements arise not from representational fusion, but from improved routing. This suggests that abstraction alleviates formulation sensitivity by strengthening connections between formulation-specific reasoning pathways, rather than by aligning their representations.
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Prompt Tuning for CLIP on the Pretrained Manifold

Xi Yang, Yuanrong Xu, Weigang Zhang, Guangming Lu, David Zhang, Jie Wen

Prompt tuning introduces learnable prompt vectors that adapt pretrained vision-language models to downstream tasks in a parameter-efficient manner. However, under limited supervision, prompt tuning alters pretrained representations and drives downstream features away from the pretrained manifold toward directions that are unfavorable for transfer. This drift degrades generalization. To address this limitation, we propose ManiPT, a framework that performs prompt tuning on the pretrained manifold. ManiPT introduces cosine consistency constraints in both the text and image modalities to confine the learned representations within the pretrained geometric neighborhood. Furthermore, we introduce a structural bias that enforces incremental corrections, guiding the adaptation along transferable directions to mitigate reliance on shortcut learning. From a theoretical perspecti

ve, ManiPT alleviates overfitting tendencies under limited data. Our experiments cover four downstream settings: unseen-class generalization, few-shot classification, cross-dataset transfer, and domain generalization. Across these settings, ManiPT achieves higher average performance than baseline methods. Notably, ManiPT provides an explicit perspective on how prompt tuning overfits under limited supervision.
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TD3B: Transition-Directed Discrete Diffusion for Allosteric Binder Generation
Hanqun Cao, Aastha Pal, Sophia Tang, Yinuo Zhang, Jingjie Zhang, Pheng-Ann Heng, Pranam Chatterjee
Protein function is often controlled by ligands that bias the direction of state transitions, such as agonists and antagonists, rather than stabilizing a single conformation. This is especially important for clinically relevant G protein-coupled receptors (GPCRs), where therapeutic efficacy depends on functional directionality. Structure-based design methods optimize binding to static conformations and cannot represent non-reversible, directional effects or systematically distinguish agonist from antagonist behavior. To address this gap, we introduce **T**ransition-**D**irected **D**iscrete **D**iffusion for allosteric**B**inder design (**TD3B**), a sequence-based generative framework that designs binders with specified agonist or antagonist behavior via a directional transition control objective. TD3B combines a target-aware Direction Oracle, a soft binding-affinity gate, and amortized fine-tuning of a pre-trained discrete diffusion model, enabling targeted agonist and antagonist generation decoupled from binding affinity and unattainable by equilibrium-based or inference-only guidance baselines. The code and checkpoints are available at https://huggingface.co/ChatterjeeLab/TD3B.
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Diffuse to Detect: Bi-Level Sample Rebalancing with Pseudo-Label Diffusion for Point-Supervised Infrared Small-Target Detection
Zhu Liu, Yuanhang Yao, Ping Qian, Zihang Chen, Risheng Liu
Point supervision has become a scalable solution to address dense annotation for infrared small target detection, but its performance is limited by two coupled bottlenecks: unstable pseudo-label evolution in cluttered, low-contrast infrared imagery and severe sample-distribution imbalance. In this paper, we present a more adaptive and stable framework to address these issues. Leveraging the intrinsic consistency between thermal radiation patterns and heat diffusion, we propose a physics-induced annotation strategy that expands single-point labels into reliable pseudo-masks. To further enhance supervision and alleviate sample imbalance, we develop a bi-level dual-update framework that jointly optimizes detector weights, sample weights, and diffusion parameters. A meta-classifier dynamically predicts sample-wise loss weights, while a differentiable diffusion module refines pseudo-labels with detection feedback, enabling adaptive interaction between training and hyperparameter optimization. Extensive experiments across multiple datasets demonstrate five-fold annotation acceleration, superior detection accuracy, and comparable performance with 30\% of the training data, validating the efficiency and practicality of our approach. Our code is available at https://github.com/yuanhang-yao/diffuse-to-detect.
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Gradient Inversion Attacks Beyond SGD
Guangnian Wan, Gongfan Fang, Xinyin Ma, Xinchao Wang
Gradient Inversion Attack (GIA) poses a significant threat to federated learning, enabling adversaries to reconstruct private training data from the information shared during training. Prior research has predominantly focused on the vanilla SGD, where the server or an eavesdropper can directly observe true gradients. In practical deployments, however, models may be trained with adaptive optimizers (e.g., Adam, RMSProp, and AdaGrad), for which the observable signal is not raw gradients but momentum-based parameter updates. This setting remains underexplored and undermines traditional gradient-matching strategies, which struggle to recover labels and images from non-gradient updates. To address this gap, this paper explores attacks tailored to modern adaptive optimizers. We present an analytical rule for recovering labels from optimizer updates and propose an update-mat

ching objective that optimizes dummy inputs to reproduce the observed updates. T
he proposed approach is general and can be directly applied to various optimizer
s such as Adam, AdaGrad, and RMSProp. Furthermore, we find that, despite being i
ntroduced for adaptive optimizers, the proposed objective function also yields s
tronger attacks in the standard SGD setting. Experiments on datasets such as Ima
geNet and PACS highlight the effectiveness of our method over existing gradient
matching techniques.
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DB-KSVD: Scalable Alternating Optimization for Disentangling High-Dimensional Em
bedding Spaces
Romeo Valentin, Sydney Michelle Katz, Vincent Vanhoucke, Mykel Kochenderfer
Dictionary learning has recently emerged as a promising approach for mechanistic
 interpretability of large transformer models.
Disentangling high-dimensional transformer embeddings requires algorithms that s
cale to high-dimensional data with large sample sizes.
Recent work has explored sparse autoencoders (SAEs) for this problem.
However, SAEs use a simple linear encoder to solve the sparse encoding subproble
m, which is known to be NP-hard.
It is therefore interesting to understand whether this approach is sufficient to
 find good solutions to the dictionary learning problem or if a more sophisticat
ed algorithm could find better solutions.
In this work, we propose Double-Batch KSVD (DB-KSVD), a scalable dictionary lear
ning algorithm that adapts the classic KSVD algorithm.
DB-KSVD is informed by the rich theoretical foundations of KSVD but scales to da
tasets with millions of samples and thousands of dimensions.
We demonstrate the efficacy of DB-KSVD by disentangling text embeddings of the G
emma-2-2B and Pythia-160M models and evaluating on six metrics from the SAEBench
 benchmark, where we achieve competitive results when compared to established ap
proaches based on SAEs.
We further show similar results when disentangling image embeddings obtained fro
m the DINOv2-S and DINOv2-B models, solidifying our findings.
By matching SAE performance with an entirely different optimization approach, ou
r results suggest that (i) SAEs do find strong solutions to the dictionary learn
ing problem and (ii) traditional optimization approaches can be scaled to the re
quired problem sizes, offering a promising avenue for further research.
We make an implementation of DB-KSVD available.
****
AdLift: Lifting Adversarial Perturbations to Safeguard 3D Gaussian Splatting Ass
ets Against Instruction-Driven Editing
Ziming Hong, Tianyu Huang, Runnan Chen, Shanshan Ye, Mingming Gong, Bo Han, Tong
liang Liu
Recent studies have extended instruction-driven 2D editing pipelines to 3D Gauss
ian Splatting (3DGS), enabling faithful 3DGS asset manipulation for advanced con
tent creation. However, it also exposes 3DGS assets to serious risks of unauthor
ized editing and malicious tampering. Although adversarial perturbations against
 editing models have proven effective for protecting 2D images, applying them to
 3DGS encounters two major challenges: *view-generalizable protection* and *bala
ncing invisibility with protection capability*. In this work, we propose AdLift,
 a novel editing safeguard for 3DGS that prevents instruction-driven editing acr
oss arbitrary views and dimensions by lifting strictly bounded 2D adversarial pe
rturbations into 3D Gaussian-represented safeguard. To ensure both *protective e
ffectiveness* and *invisibility*, these safeguard Gaussians are progressively op
timized across training views using a tailored Lifted PGD, which first conducts
gradient truncation during back-propagation from the editing model to the rend
ered image and applies projected gradient updates to strictly bound image-level
perturbations. Then, the resulting perturbation is backpropagated to the safegua
rd Gaussian parameters via *image-to-Gaussian fitting*. We alternate these two s
teps, yielding effective and imperceptible protection that generalizes across bo
th training and novel views. Empirically, qualitative and quantitative results d
emonstrate that the proposed AdLift effectively protects against state-of-the-ar

t instruction-driven 2D and 3DGS editing.
****
DPsurv: Dual-Prototype Evidential Fusion for Uncertainty-Aware and Interpretable Whole Slide Image Survival Prediction
Yucheng Xing, Ling Huang, Jingying Ma, Ruping Hong, Jiangdong Qiu, Pei Liu, Kai He, Huazhu Fu, Mengling Feng
Whole-slide images (WSIs) are widely used for cancer survival analysis because of their comprehensive histopathological information at both cellular and tissue levels, enabling quantitative, large-scale, and prognostically rich tumor feature analysis. However, most existing WSI survival analysis methods struggle with limited interpretability and often overlook predictive uncertainty in heterogeneous slide images. In this paper, we propose DPsurv, a dual-prototype whole-slide image evidential fusion network that outputs uncertainty-aware survival intervals, and enables interpretable survival results through patch prototype distribution assignment, component prototype evidence reasoning, and component-wise relative risk aggregation. Experiments on five publicly available datasets demonstrate strong discriminative performance and well-calibrated predictions, validating its effectiveness and reliability. The interpretation of survival results provides transparency at the feature, reasoning, and decision levels, thereby enhancing the trustworthiness and interpretability of DPsurv.
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Rethinking Instruction Drift as a Sampling Error: SNR-Aware Power Distributions for Long-Horizon Robotic Planning
Kewei Chen, Yayu Long, Mingsheng Shang
Despite rapid progress in Vision-Language-Action (VLA) models for robotic control, instruction drift remains a persistent failure mode in long-horizon tasks. This paper reconceptualizes this phenomenon, positing that instruction drift is fundamentally a systematic sampling error: local greedy sampling is prone to collapsing into “Negative Pivotal Windows”—irreversible local optima with high local probability that sever global success pathways. To address this, we propose Context-Aware Power Sampling (CAPS), a training-free inference-time computation framework. CAPS leverages power distributions to sharpen global trajectory probabilities, enabling lookahead search over the model's conditional generative trajectory distribution. Furthermore, we introduce a metacognitive control mechanism based on Signal-to-Noise Ratio (SNR). This mechanism triggers adaptive MCMC search solely when drift risk is detected, enabling a dynamic transition from “intuitive fast thinking” to “rational slow search.” Experiments on RoboTwin, Simpler-WindowX, and Libero-long benchmarks show that CAPS achieves substantial improvements over strong baselines, including OpenVLA and TACO, without parameter updates. These results support the effectiveness of adaptive inference-time computation for improving long-horizon robustness in embodied control.
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Posterior Sampling Reinforcement Learning with Gaussian Processes for Continuous Control: Sublinear Regret Bounds for Unbounded State Spaces
Hamish Flynn, Joe Watson, Ingmar Posner, Jan Peters
We analyze the Bayesian regret of the Gaussian process posterior sampling reinforcement learning (GP-PSRL) algorithm. Posterior sampling is a heuristic for decision-making under uncertainty that has been used to develop successful algorithms for a variety of continuous control problems. However, theoretical work on GP-PSRL is limited. All known regret bounds either have a sub-optimal growth rate, require strong smoothness assumptions, or fail to properly account for the fact that the set of possible system states is unbounded. Through a recursive application of the Borell-Tsirelson-Ibragimov-Sudakov inequality, we show that, with high probability, the states actually visited by the algorithm are contained within a ball of near-constant radius. We then use the chaining method to control the regret suffered by GP-PSRL under weak smoothness conditions. Our main result is a Bayesian regret bound of the order $\widetilde{\mathcal{O}}(H\sqrt{\gamma_TT})$, where H is the horizon, T is the number of time steps and γ_T is the expected information gain. With this result, we resolve the limitations with prior theoretical work on PSRL, and provide the theoretical foundation and tool

s for analyzing PSRL in complex settings.
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SoftJAX & SoftTorch: Empowering Automatic Differentiation Libraries with Informa tive Gradients
Anselm Paulus, Andreas René Geist, Vít Musil, Sebastian Hoffmann, Georg Martius
Automatic differentiation (AD) frameworks such as JAX and PyTorch have enabled g radient-based optimization for a wide range of scientific fields. Yet, many ''ha rd'' primitives in these libraries such as thresholding, Boolean logic, discrete indexing, and sorting operations yield zero or undefined gradients that are not useful for optimization. While numerous ''soft'' relaxations have been proposed that provide informative gradients, the respective implementations are fragment ed across projects, making them difficult to combine and compare. This work intr oduces **SoftJAX** and **SoftTorch**, open-source, feature-complete libraries fo r *soft differentiable programming*. These libraries provide a variety of soft f unctions as drop-in replacements for their hard JAX and PyTorch counterparts. Th is includes (i) elementwise operators such as *clip* or *abs*, (ii) utility meth ods for manipulating Booleans and indices via fuzzy logic, (iii) axiswise operat ors such as *sort* or *rank* -- based on optimal transport or permutahedron proj ections, and (iv) offer full support for straight-through gradient estimation. O verall, SoftJAX and SoftTorch make the toolbox of soft relaxations easily access ible to differentiable programming, as demonstrated through benchmarking and a p ractical case study. Code is available at github.com/a-paulus/softjax and github .com/a-paulus/softtorch.
****
HiPER: Hierarchical Plan–Execute RL for Multi-Turn LLM Agents
Jiangweizhi Peng, Yuanxin Liu, Ruida Zhou, Charles Fleming, Zhaoran Wang, Alfred o Garcia, Mingyi Hong
Training LLMs as interactive agents for multi-turn decision-making remains chall enging, particularly in long-horizon tasks with sparse and delayed rewards, wher e agents must execute extended sequences of actions before receiving meaningful feedback. Most existing reinforcement learning (RL) methods model LLM agents as *flat* policies operating at a single time scale, selecting an action at each tu rn. In sparse-reward settings, this forces the agent to infer long-range depende ncies solely from distant end-of-trajectory signals, often leading to inefficien t learning and unstable behavior in complex environments. We propose **HiPER**, a novel **Hi**erarchical **P**lan–**E**xecute **R**L framework that jointly mode ls and optimizes high-level subgoal planning and low-level action execution for LLM agents to overcome flat RL's brittle long-horizon behavior and weak credit a ssignment under sparse outcome feedback. By maintaining persistent subgoals acro ss multiple turns and explicitly deciding when to switch between them, HiPER int roduces structured intermediate decision points that facilitate learning under s parse feedback, converting implicit multi-turn structure into learnable decision s at different time scales. To enable effective training, we introduce Hierarchi cal Advantage Estimation (HAE), a two-timescale policy gradient method that assi gns credit to both action execution and subgoal transitions and achieves varianc e reduction relative to flat advantage estimation. Empirically, HiPER achieves s tate-of-the-art performance on challenging interactive benchmarks, reaching 97.4 \% success on ALFWorld (+6.6\% over the best prior method) and 83.3\% on WebShop , with especially large gains on long-horizon tasks requiring multiple dependent subtasks. These results highlight the importance of explicit hierarchical decom position for scalable RL training of multi-turn LLM agents.
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Temporal-Emerged Prompting for Segment Anything in Multiframe Infrared Small Tar get Detection
Yinghui Xing, Donghao Chu, Shizhou Zhang, Di Xu
Accurately localizing and segmenting small targets in low signal-to-noise ratio (SNR) infrared sequences remains a challenging task. Since targets are often ind istinguishable from the background in individual frames, existing methods, even when equipped with advanced foundation model and powerful inter-frame associatio n mechanisms, still fail to detect them. Motivated by the observation that targe

ts tend to emerge gradually from the background over time and become distinguish able, we propose Temporal-Emerged Prompting for Segment Anything Model (TEP-SAM) , a principled framework designed to explicitly exploit such temporal-emerged cu es to modulate and prompt SAM. TEP-SAM operates by jointly modeling global motio n patterns and local motion deviations to locate potential targets. It further e nhances target region features by leveraging motion discrepancy, thereby generat ing temporal-emerged cues for SAM and enabling non-interactive segmentation. By bridging large-scale semantic pretraining with task-specific temporal modeling, TEP-SAM effectively adapts SAM to the challenging multiframe infrared small targ et detection task. Extensive experiments demonstrate the effectiveness of our ap proach, particularly under severely low-SNR conditions and in complex dynamic ba ckground.

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GeoSense: Internalizing Geometric Necessity Perception for Multimodal Reasoning

Ruiheng Liu, Haihong Hao, Mingfei Han, Xin Gu, Kecheng Zhang, Changlin Li, Xiaoj un Chang

Advancing towards artificial superintelligence requires rich and intelligent per ceptual capabilities. A critical frontier in this pursuit is overcoming the limi ted spatial understanding of Multimodal Large Language Models (MLLMs), where geo metry information is essential. Existing methods often address this by rigidly i njecting geometric signals into every input, while ignoring their necessity and adding computation overhead.

Contrary to this paradigm, our framework endows the model with an awareness of perceptual insufficiency, empowering it to autonomously engage geometric featur es in reasoning when 2D cues are deemed insufficient.

To achieve this, we first introduce an independent geometry input channel to t he model architecture and conduct alignment training, enabling the effective uti lization of geometric features. Subsequently, to endow the model with perceptual awareness, we curate a dedicated spatial-aware supervised fine-tuning dataset. This serves to activate the model’s latent internal cues, empowering it to auton omously determine the necessity of geometric information.

Experiments across multiple spatial reasoning benchmarks validate this approac h, demonstrating significant spatial gains without compromising 2D visual reason ing capabilities, offering a path toward more robust, efficient and self-aware m ulti-modal intelligence.

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Holonomy Grid Codes for Generalisation Under Directed Actions

Pradeep Singh, Balasubramanian Raman

Efficient prediction and planning in structured environments often relies on spe ctral decompositions of transition operators, yet existing grid-cell and success or-representation theories implicitly assume “flat” action geometry where transl ations commute and a single Fourier eigenbasis suffices. We show that this assum ption breaks in the presence of path-dependent effects—e.g., circulation, rotati onal drift, or topological loops—whose defining signature is nontrivial holonomy . We introduce a theory of directed-action prediction on discrete tori based on twisted translation operators forming a projective representation of the underly ing motion group, and prove that the resulting controlled Markov operators admit an exact block-diagonalisation under a twisted Fourier transform: actions share a universal harmonic basis while their effects appear as small matrix-valued sp ectra rather than scalar eigenvalues. This yields closed-form resolvent expressi ons for the successor representation, a gauge-invariant transfer principle chara cterising when two environments admit identical predictive structure, and a curv ature-induced lower bound showing that nonzero holonomy provably necessitates in ternal representational dimension. Together, these results generalise Fourier/gr id-based prediction from commutative to curved action geometries, providing a pr incipled foundation for generalisation under directed actions without learning e nvironment-specific eigenvectors.

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OVLR: Efficient, Scalable, and Robust Training via Output-Level Variance-Reduced Likelihood Ratio

Minhao Zou, Tao Ren, Jinyang Jiang, Rui Tao, Zehao Li, Jiale Fu, Hui Shao, Xianhua Liu, Yijie Peng

Gradient-based optimization via backpropagation (BP) is inherently limited by the requirement of differentiability, rendering it inapplicable for piecewise-constant objectives with vanishing gradients (e.g., the hard 0-1 loss) or black-box feedback. While likelihood ratio (LR) methods offer a theoretical alternative, their high variance in high-dimensional spaces undermines training stability and scalability. We propose OVLR, a framework that makes direct optimization of gradient-agnostic objectives practical for modern deep networks by performing perturbations and antithetic sampling in the low-dimensional output space. OVLR achieves dramatic variance reduction while requiring only a single deterministic forward pass, with additional costs restricted to evaluating the loss function across multiple samples. On problems where BP provides gradients, OVLR remains competitive; on problems where BP fails to provide reliable learning signals, OVLR enables the direct optimization of objectives such as the 0-1 loss for noise-tolerant classification and truncated losses for outlier-resistant regression. Extensive empirical results across classification, generative modeling, language modeling, robot imitation learning, and black-box optimization confirm that OVLR is an effective tool for settings where standard gradient-based optimization is inapplicable. Code is available at https://github.com/MinhZou/OVLR.

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Dynamic Linear Attention

Xin Wang, Hui Shen, Boyuan Zheng, Xueshen Liu, Minkyoung Cho, Zhongwei Wan, Zesen Zhao, Zhuoqing Mao, Shen Yan, Mi Zhang

The scalability of Large Language Models (LLMs) to long contexts is fundamentally constrained by the quadratic complexity of standard attention, motivating the adoption of linear attention mechanisms with sub-quadratic cost. To improve representation capacity under long contexts, recent approaches organize memory in a multi-state manner. However, existing multi-state linear attention methods rely on fixed state merging policies that cannot adapt to dynamically varying token importance, irreversibly obscuring critical tokens and causing severe error accumulation over long sequences. To address this limitation, we propose DLA, a dynamic memory modeling framework for multi-state linear attention. DLA introduces (i) Information-Aware Dynamic State Merging, which adaptively determines state boundaries based on token-level information variation, preserving high-resolution representations around semantic transitions while aggressively summarizing stable regions, and (ii) Capacity-Bounded Memory Modeling, which maintains a fixed-size, chronologically ordered state cache by selectively merging adjacent low-information states to control memory growth with minimal information loss. We pre-train DLA on two different linear attention models and evaluate on 10 datasets from three different aspects. Experimental results demonstrate the superiority of DLA over state-of-the-art.

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Bring Future Vision: Dynamic Computation Allocation Guided by Lightweight Feature Forecaster

Chao Han, Yijuan Liang, Zihao Xuan, Daokuan Wu, Wei Zhang, Xiaoyu Shen

The deployment of large language models (LLMs) in real-world applications is increasingly limited by their high inference cost. While recent advances in dynamic token-level computation allocation attempt to improve efficiency by selectively activating model components per token, existing methods rely on greedy routing—a myopic execute-or-skip mechanism that often leads to irreversible information loss and suboptimal token selection.

This paper introduces informed routing, a new paradigm that proactively addresses these issues. The key insight is to assess not only a token’s immediate importance but also its recoverability, i.e., how well its transformation can be approximated. To this end, we propose the Lightweight Feature Forecaster (LFF), a small predictive module that estimates a unit’s output before routing decisions are made. This enables a flexible execute-or-approximate policy that preserves model fidelity while drastically reducing computation.

Extensive experiments show that informed routing consistently achieves state-of-

the-art performance across static and dynamic pruning approaches. We further pre
sent two practical inference pipelines: a pure-PyTorch implementation and a Trit
on-based custom operator, that translate these gains into real-world speedups, a
chieving practical acceleration and consistent improvement across various batch
sizes.
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Gaussian Mean Field Variational Inference can Overestimate Predictive Variance
James Odgers, Ben Riegler, Siddharth Swaroop, Vincent Fortuin
Mean Field Variational Inference (MFVI) is widely understood to underestimate po
sterior variance. By analysing conjugate Bayesian linear regression, we show tha
t this characterisation is incomplete: while MFVI underestimates the variance in
 parameter space, it can overestimate the predictive variance compared to the ex
act posterior. We show that if the MFVI posterior underestimates predictive vari
ances in some directions, it necessarily overestimates them in others. Crucially
, this overestimation occurs in directions where the training data concentrates.
 This leads to the surprising result that, for a test point drawn from the train
ing distribution, MFVI's expected predictive variance exceeds that of the exact
posterior. We demonstrate a pathological case of this effect, where the MFVI po
sterior fails to reduce predictive variance compared to the prior on i.i.d. data
. We connect these results to the Cold Posterior Effect, arguing that varying th
e temperature can correct this overestimation, yielding predictions closer to th
ose of the exact posterior. We validate our theory on synthetic and real-world r
egression tasks.
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Enhancing LLM Training via Spectral Clipping
Xiaowen Jiang, Andrei Semenov, Sebastian U Stich
While spectral-based optimizers like Muon operate directly on the spectrum of up
dates, standard adaptive methods such as AdamW do not account for the spectral s
tructure of weights and gradients, leaving them vulnerable to two empirical issu
es in large language model (LLM) training: (i) the optimizer updates can have la
rge spectral norms, potentially destabilizing training and degrading generalizat
ion; (ii) stochastic gradient noise can exhibit sparse spectral spikes, with a f
ew dominant singular values much larger than the rest. We propose SPECTRA, a gen
eral framework addressing these by (i) post-spectral clipping of updates to enfo
rce spectral-norm constraints (ii) optional pre-spectral clipping of gradients t
o suppress spectral noise spikes. We prove that post-clipping constitutes a Comp
osite Frank-Wolfe method with spectral-norm constraints and weight regularizatio
n. We further analyze how pre-clipping mitigates sparse spectral spikes. We prop
ose efficient soft spectral clipping via Newton-Schulz iterations, avoiding expe
nsive SVD. Experiments on LLM pretraining show SPECTRA uniformly improves valida
tion loss for various optimizers, including AdamW, Signum, Mars, and AdEMAMix, w
ith the best-performing variants achieving state-of-the-art results. Models trai
ned with SPECTRA exhibit smaller weight norms, confirming the link between spect
ral clipping and regularization.
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Large-scale Uncertainty Quantification for Latent Variable Models Using Subsampl
ing Markov Chain Monte Carlo
Xiaoyu Wang, Jonathan H. Huggins
Stochastic gradient Langevin dynamics combined with Gibbs updates (SGLD-Gibbs) p
rovides a highly scalable approach to approximate Bayesian inference in latent v
ariable models. However, it remains unclear how to tune the algorithm's hyperpar
ameters in a principled manner to ensure the uncertainty estimates are statistic
ally meaningful. In this work, we address this gap in tuning guidance by develop
ing a statistical scaling limit theory for SGLD-Gibbs. We derive a joint asympto
tic limit for the global parameters and latent variables under appropriate space
-time rescaling. We show that global parameters converge to a diffusion-type lim
it, while individual latent variables converge to a jump process reflecting thei
r intermittent Gibbs updates. This joint jump-diffusion structure reveals how la
tent-variable randomness contributes to the stationary distribution of the globa
l parameters. We leverage our results to provide explicit guidance on hyperparam

eter tuning for SGLD-Gibbs that ensures meaningful uncertainty quantification. O
ur empirical results show that SGLD-Gibbs with our tuning guidance leads to bett
er parameter estimates and uncertainty quantification than stochastic variationa
l inference.
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RefChess: Training-Free Contextual Search for Zero-Shot Referring Image Segmenta
tion

Shiyan Tong, Jinxia Zhang, Zhiyuan Wang, Hao Tian, YingYing Wang, Kanjian Zhang,
 Haikun Wei

Recent advances in zero-shot referring image segmentation (RIS), driven by found
ation models such as SAM and CLIP, have improved cross-modal alignment between v
isual regions and natural language expressions. Nevertheless, selecting the corr
ect segmentation proposal remains challenging, as existing methods typically sco
re proposals independently and can be distracted by visually similar candidates
that partially satisfy the expression. To address this limitation, we propose Re
fChess, a training-free contextual search framework for robust proposal selectio
n. Instead of treating proposal selection as a single-step ranking problem, RefC
hess evaluates candidate masks under sampled distractor contexts and uses Monte-
Carlo Tree Search as a budgeted mechanism to explore the combinatorial space of
contextual perturbations. The search is guided by a stability-aware reward that
integrates language decomposition, vision--language similarity, object-centric c
ues, and spatial guidance signals. Experiments on standard RIS benchmarks show t
hat RefChess consistently improves robustness and referring segmentation perform
ance without task-specific training. Code is available at \url{https://github.co
m/Tongshiyan/RefChess}.
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Dynamic Symmetric Point Tracking: Tackling Non-ideal Reference in Analog In-memo
ry Training

Quan Xiao, Jindan Li, Zhaoxian Wu, Tayfun Gokmen, Tianyi Chen

Analog in-memory computing (AIMC) performs computation directly within resistive
 crossbar arrays, offering an energy-efficient platform to scale large vision an
d language models. However, non-ideal analog device properties make the training
 on AIMC devices challenging. In particular, its update asymmetry can induce a s
ystematic drift of weight updates towards a device-specific symmetric point (SP)
, which typically does not align with the optimum of the training objective. To
mitigate this bias, most existing works assume the SP is known and pre-calibrate
 it to zero before training by setting the reference point as the SP. Neverthele
ss, calibrating AIMC devices requires costly pulse updates, and residual calibra
tion error can directly degrade training performance. In this work, we present t
he first theoretical characterization of the pulse complexity of SP calibration
and the resulting estimation error. We further propose a dynamic SP estimation m
ethod that tracks the SP during model training, and establishes its convergence
guarantees. In addition, we develop an enhanced variant based on chopping and fi
ltering techniques from digital signal processing. Numerical experiments demonst
rate both the efficiency and effectiveness of the proposed method.
****

T-Edit: Triple-Branch Diffusion Anchoring for Consistent Editing

Linsong Shan, Laurence Tianruo Yang, Zecan Yang, Shijie Lian, shijie lv, Qilin Y
ang

While Multimodal Diffusion Transformers (MMDiTs) have achieved remarkable succes
s in high-fidelity generation, maintaining semantic faithfulness and structural
consistency during the image editing process remains a fundamental challenge. Di
T-based editing is primarily limited by cumulative drift and semantic leakage in
duced by new textual conditions. To address these challenges, we propose T-Edit,
 a training-free framework that formalizes consistent editing as a trajectory an
choring process. T-Edit explicitly decouples the inversion, reconstruction, and
editing trajectories, leveraging the reconstruction branch as a structural refer
ence to achieve real-time compensation for deviations in the latent manifold. To
 further reveal the internal regulation mechanism of DiTs, we analyze the spatio
-temporal heterogeneity of their layer-wise structural sensitivity and according

ly propose a Dynamic Vital Layer (DVL) localization mechanism based on information energy. Furthermore, addressing the asymmetry of textual perturbations in the frequency domain distribution, we introduce a frequency-aware strategy based on tensor Singular Value Decomposition (t-SVD) to anchor (TA) high-rank structural components. Experiments show that T-Edit achieves state-of-the-art performance in both semantic alignment and structural fidelity, and can be seamlessly extended to multi-step editing and video scenarios, providing a new perspective for understanding and controlling the internal stability of DiTs.
****
MoCo-EA: Exploiting Adversarial Mode Connectivity for Efficient Evolutionary Attacks
Hyo Seo Kim, Gang Luo, Can Chen, Binghui Wang, Yue Duan, Ren Wang
Evolutionary algorithms for adversarial attacks leverage population-based search to discover perturbations without gradient information, but suffer from inefficient crossover operations that destroy adversarial properties through discrete interpolation. We introduce Mode Connectivity Evolutionary Attack (MoCo-EA), which replaces traditional crossover with a novel Bézier crossover operator that optimizes perturbations along a continuous Bézier curve between parent perturbations. Our key insight is that adversarial examples lie on connected manifolds where intermediate points maintain and often enhance attack effectiveness. We demonstrate three findings: (1) Successful adversarial perturbations exhibit mode connectivity; (2) Intermediate points along optimized paths achieve higher transferability than endpoints; (3) Bézier crossover dramatically outperforms discrete genetic operations while reducing convergence time and query requirements. By exploiting the geometric structure of adversarial space through path optimization, MoCo-EA provides an efficient and reliable method. Our work challenges the traditional view of adversarial examples as isolated points and opens new directions for both attack generation and defense research.
****
Accelerated Multiple Wasserstein Gradient Flows for Multi-objective Distributional Optimization
Dai Hai Nguyen, Duc-Dung NGUYEN, Atsuyoshi Nakamura, Hiroshi Mamitsuka
We study multi-objective optimization over probability distributions in Wasserstein space. Recently, Nguyen et al. (2025) introduced Multiple Wasserstein Gradient Descent (MWGraD) algorithm, which exploits the geometric structure of Wasserstein space to jointly optimize multiple objectives. Building on this approach, we propose an accelerated variant, A-MWGraD, inspired by Nesterov's acceleration. We analyze the continuous-time dynamics and establish convergence to weakly Pareto optimal points in probability space. Our theoretical results show that A-MWGraD achieves a convergence rate of $\mathcal{O}(1/t^2)$ for geodesically convex objectives and $\mathcal{O}(e^{-\sqrt{\beta}t})$ for β-strongly geodesically convex objectives, improving upon the $\mathcal{O}(1/t)$ rate of MWGraD in the geodesically convex setting. We further introduce a practical kernel-based discretization for A-MWGraD and demonstrate through numerical experiments that it consistently outperforms MWGraD in convergence speed and sampling efficiency on multi-target sampling tasks.
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Think-Then-Generate: Reasoning-Aware Text-to-Image Diffusion with LLM Encoders
Siqi Kou, Jiachun Jin, Zetong Zhou, Ye Ma, Yugang Wang, Quan Chen, Peng Jiang, Xiao Yang, Jun Zhu, Kai Yu, Zhijie Deng
Recent progress in text-to-image (T2I) diffusion models (DMs) has enabled high-quality visual synthesis from diverse textual prompts. Yet, most existing T2I DMs, even those equipped with large language model (LLM)-based text encoders, remain text-pixel mappers -- they employ LLMs merely as text encoders, without leveraging their inherent reasoning capabilities to infer what should be visually depicted given the textual prompt. To move beyond such literal generation, we propose the think-then-generate (T2G) paradigm, where the LLM-based text encoder is encouraged to reason about and rewrite raw user prompts; the states of the rewritten prompts then serve as diffusion conditioning. To achieve this, we first activate the think-then-rewrite pattern of the LLM encoder with a lightweight supervi

sed fine-tuning process. Subsequently, the LLM encoder and diffusion backbone ar
e co-optimized to ensure faithful reasoning about the context and accurate rende
ring of the semantics via Dual-GRPO. In particular, the text encoder is reinforc
ed using image-grounded rewards to infer and recall world knowledge, while the d
iffusion backbone is pushed to produce semantically consistent and visually cohe
rent images. Experiments show substantial improvements in factual consistency, s
emantic alignment, and visual realism across reasoning-based image generation an
d editing benchmarks, achieving 0.79 on WISE score, nearly on par with GPT-4. Ou
r results constitute a promising step toward next-generation unified models with
 reasoning, expression, and demonstration capacities.
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SF-Mamba: Rethinking State Space Model for Vision
Masakazu Yoshimura, Teruaki Hayashi, Yuki Hoshino, Wei-Yao Wang, Takeshi Ohashi
Visual Mamba models have recently emerged as alternatives to Vision Transformers
 (ViTs), which suffer from quadratic complexity.
While the recurrent scanning mechanism of Mamba offers computational efficiency,
 it inherently limits non-causal interactions between image patches.
Prior works have attempted to address this limitation through various multi-scan
 strategies; however, these approaches suffer from inefficiencies due to subopti
mal scan designs and frequent data rearrangement. Moreover, Mamba exhibits relat
ively slow computational speed under short token lengths, commonly used in visua
l tasks.
In pursuit of a truly efficient vision encoder, we rethink the scan operation fo
r vision and the computational efficiency of Mamba.
To this end, we propose SF-Mamba, a novel visual Mamba with two key proposals: a
uxiliary patch **swapping** for encoding bidirectional information flow under an
 unidirectional scan and batch **folding** with periodic state reset for advance
d GPU parallelism.
Extensive experiments on image classification, object detection, and instance an
d semantic segmentation consistently demonstrate that our proposed SF-Mamba sign
ificantly outperforms state-of-the-art baselines while improving throughput acro
ss different model sizes.
The source code is available at: https://github.com/sony/SF-Mamba.
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Mesh Based Simulations with Spatial and Temporal awareness
Paul Garnier, Vincent Lannelongue, Elie Hachem
Machine Learning surrogates for Computational Fluid Dynamics (CFD), particularly
 Graph Neural Networks (GNNs) and Transformers, have become a new important appr
oach for accelerating physics simulations. However, we identify a critical bottl
eneck in the field: while architectures have advanced significantly, the common
underlying training paradigms remain bound to naive assumptions, such as node-wi
se supervision and explicit Euler time-stepping. These legacy choices ignore the
 stiff dynamics and local flux continuity inherent to numerous partial different
ial equations resolution methods, such as Finite Element, Difference, or Volume
(FEM). In this work, we propose a unified framework to bridge the gap between ge
ometric deep learning and rigorous numerical analysis. We introduce three key in
novations: (1) Multi Node Prediction, a stencil-level objective that predicts fi
eld values for a node's full local topology, enforcing spatial derivative consis
tency; (2) Temporal Correction, replacing unstable explicit schemes with a predi
ctor–corrector via temporal Cross-Attention; and (3) Geometric Inductive Biases,
 leveraging 3D Rotary Positional Embeddings (RoPE) to robustly capture rotationa
l symmetries in unstructured meshes. We evaluate this framework across three arc
hitectures (MeshGraphNet, Transolver, and a Transformer) on diverse physics data
sets. Our approach yields consistent improvements in accuracy and stability, par
ticularly in long-horizon rollouts, while producing latent representations that
generalize to unseen subtasks such as Wall Shear Stress or Pressure prediction.
Code is available at \url{https://github.com/DonsetPG/graph-physics}.
****

SMILE: Extended Deep Submodular Function-Based Instruction and In-context Learni
ng Demonstration Selection

Zihan Chen, Chengshuai Shi, Song Wang, Jundong Li, Cong Shen
Prompt optimization is a key way to steer large language models when fine-tuning is impractical. However, instruction optimization (IO) and in-context learning (ICL) demonstration selection are often optimized separately and combined post hoc, implicitly assuming that a "best'' instruction and a "best" demonstration set compose well. In practice, their interactions are strong, making such decoupled pipelines brittle.
We propose SMILE, an efficient method that *jointly* selects instructions and demonstrations. Our key observation is that the ICL performance exhibits consistent diminishing returns across diverse instructions. Leveraging this structure, SMILE learns an instruction-conditioned surrogate aligned with LLM feedback and instantiates it as an Extended Deep Submodular Function that captures sample--sample coverage, sample--query relevance, and sample--instruction compatibility. SMILE then performs greedy, query-adaptive selection of the instruction--demonstration pair.
Experiments on six datasets and multiple LLM backbones show that SMILE consistently outperforms IO-only, ICL-only, and existing joint baselines, supporting a context engineering view of prompting: jointly optimizing interacting components rather than tuning them in isolation.
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FedEBA+: Towards Fair and Effective Federated Learning via Entropy-Based Model
Zhichao Wang, Lin Wang, Ye Shi, Sai Praneeth Karimireddy, Xiaoying Tang
Federated Learning (FL) often suffers from sacrificing global model accuracy when improving client-level fairness due to data heterogeneity, which often leads to inconsistent performance of the globally trained models, resulting in unfair outcomes among users. Existing fair FL algorithms face a bottleneck: they either sacrifice global model accuracy to promote fairness or fall short of achieving optimal fairness.

In this paper, we propose a novel framework that effectively improves fairness while preserving global accuracy by integrating information-theoretic principles with model alignment. Specifically, we leverage the Maximum Entropy Principle to derive an analytic, closed-form solution for fair aggregation weights, ensuring significant fairness enhancements. We further employ a step-wise model alignment strategy that synchronizes gradient directions across heterogeneous clients, effectively mitigating the drift induced by local updates.

Theoretical analysis proves that our method guarantees convergence even in non-convex settings. Importantly, we push the theoretical frontier of federated fairness by extending performance variance analysis to generalized regression, providing broader guarantees. Extensive experiments on five datasets demonstrate that our approach consistently outperforms state-of-the-art methods, achieving superior fairness without sacrificing global accuracy. Our code is available at \href{https://github.com/T-Lab-CUHKSZ/FedEBA-Plus}{https://github.com/T-Lab-CUHKSZ/FedEBA-Plus}.
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Incorporating Importance Weighting in Optimal Transport Based Domain Alignment
Okan Koç, Alexander Soen, Shanglin Li, Masashi Sugiyama
Domain adaptation theory studies upper bounds on the target risk in order to mitigate performance loss of machine learning models due to distribution shift.
In this paper, we take a closer look at the optimization of one such bound based on optimal transport (OT) and propose various strategies that improve the optimization in practice. We first introduce *gradual shift* and *probabilistic margin* assumptions to control the incomputable entanglement term that appears in the bounds. We prove that under these assumptions, better optimization of the computable part of the bound can translate to better target accuracies. Motivated by this fact, we tighten the bound, via importance weighting of the source (output) distribution, to obtain the *weighted* Wasserstein regularized risk ($\mathrm{W}^2\mathrm{R}^{2}$), that is often easier to minimize than the original bound. $\mathrm{W}^2\mathrm{R}^{2}$ is shown to be equivalent to an unbalanced OT proble

m, which in the limit converges to a nearest neighbor based alignment strategy. We highlight the tradeoffs faced with such an approach and show that a suitably regularized $\mathrm{W}^2\mathrm{R}^{2}$ improves over the state of the art and is robust to multiple distribution shifts under different models, confirming, moreover, the validity of our assumptions.
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Toward Stable Value Alignment: Introducing Independent Modules for Consistent Value Guidance
Wenhao Chen, Sirui Sun, Shengyuan Bai, Guojie Song
Aligning large language models (LLMs) with human values typically relies on post-training or inference-time steering that directly manipulates the backbone's parameters or representation space. However, a critical gap exists: the model's residual stream is highly dynamic, in which values exist as fragile, low-dimensional properties, inherently incompatible with the stability required for consistent value expression. In this paper, we propose the Stable Value Guidance Transformer (SVGT), which addresses this gap through an independent value module incorporating two key designs: (1) *independent value modeling*, maintaining normative representations in a dedicated value space isolated from the backbone, and (2) *explicit behavioral guidance*, transducing these stable signals into learnable latent Bridge Tokens. These tokens serve as dynamic value anchors to explicitly steer the generative trajectory, ensuring robust adherence across diverse contexts without disrupting the backbone's internal representations. Experiments across multiple backbones and safety benchmarks show that SVGT generally reduces harmful scores by over 70\% while maintaining generation fluency, demonstrating the efficacy of architecturally grounded value modeling.
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Dismantling the Illusion of Vision-Language-Action Models Competence via Explicit Distributional Shifts
Xueyang Zhou, Yangming Xu, Guiyao Tie, Chaoran Hu, Bo Tao, xingwei zhao, Xiang Xiang, Pan Zhou, Lichao Sun, Yongchao Chen
Given that simulation can never exhaustively enumerate reality, generalization is the determining factor for whether Vision-Language-Action (VLA) models can translate benchmark success into real-world functionality. However, current evaluation protocols often incentivize mechanical memorization rather than robust policy learning, leading to a paradoxical duality of failure: high-scoring models exhibit *spurious invariance* to semantic changes while simultaneously displaying *extreme brittleness* to trivial environmental perturbations. To address this, we introduce **LIBERO-Gen**, a diagnostic benchmark systematically designed to shift evaluation from intuition-driven heuristics to explicit distributional assumptions. Through a hierarchical protocol spanning **In-distribution**, **Compositional**, and **Domain Generalization**, LIBERO-Gen reveals performance stratifications previously masked by standard metrics. Our analysis identifies Pi0.5 as the top performer (64.0% in Spatial-CG; 21.2% in Task-CG). By identifying perceptual instability and action binding collapse as primary failure modes while validating the efficacy of structured ``Stair" sampling, LIBERO-Gen establishes a rigorous baseline for deployment reliability.
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Cluster-Aware Causal Mixer for Online Anomaly Detection in Multivariate Time Series
Md Mahmuddun Nabi Murad, Yasin Yilmaz
Early and accurate detection of anomalies in time-series data is critical due to the substantial risks associated with false or missed detections. While MLP-based mixer models have shown promise in time-series analysis, they do not maintain temporal causality during data processing. Moreover, real-world multivariate time series often contain numerous channels with diverse inter-channel correlations. Spurious correlations in the reconstructed time series lead to noisy representations, resulting in inaccurate anomaly detection. In addition, anomaly scoring methods that ignore temporal continuity can mislead sequential detection. To address these challenges, we propose a cluster-aware causal mixer for multivariate time-series anomaly detection. Channels are grouped into clusters based on thei

r correlations, and each cluster is embedded through a dedicated embedding layer . A causal mixer is introduced to integrate information while maintaining tempor al causality. We further develop a sequential anomaly-scoring method that accumu lates evidence over time and refines anomaly boundaries. Our proposed model oper ates in an online fashion, making it suitable for real-time time-series anomaly detection. Experimental evaluations across six public benchmark datasets demonst rate that the proposed approach consistently achieves superior performance.
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PlugGuard: A Streaming Safeguard for Large Models via Latent Dynamics-Guided Ris k Detection
Xiaodan Li, Mengjie Wu, Yao Zhu, Yunna Lv, YueFeng Chen, Cen Chen, Jianmei Guo, Hui Xue
Large models (LMs) are powerful content generators, yet their open■ended nature can also introduce potential risks, such as generating harmful or biased content . Existing guardrails mostly perform post-hoc detection that may expose unsafe c ontent before it is caught,
and the latency constraints further push them toward lightweight models, limitin g detection accuracy. In this work, we propose PlugGuard, a novel plug-in framew ork that enables streaming risk detection within the LM generation pipeline. Plu gGuard leverages intermediate LM hidden states through a Streaming Latent Dynami cs Head (SLD), which models the temporal evolution of risk across the generated sequence for more accurate real-time risk detection. To achieve reliable streami ng moderation in real applications, we introduce an Anchored Temporal Consistenc y (ATC) loss, ensuring that risk assessments remain consistent with a strict sto p-if-harmful policy. Besides, for a rigorous evaluation of streaming guardrails, we also present StreamGuardBench—a model-grounded benchmark featuring on-the-fl y responses from each protected model, reflecting real-world streaming scenarios in both text and vision–language tasks. Across diverse models and datasets, Plu gGuard consistently outperforms state-of-the-art streaming guardrails (achieving a 22.80% F1 score gain), while using only 20M parameters and adding less than 0 .5 ms of per-token latency. The code and StreamGuardBench are released at PlugGu ard to facilitate research on streaming guardrails.
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Knowing When to Quit: A Principled Framework for Dynamic Abstention in LLM Reaso ning
Hen Davidov, Nachshon Cohen, Oren Kalinsky, Yaron Fairstein, Guy Kushilevitz, Ra m Yazdi, Patrick Rebeschini
Large language models (LLMs) using chain-of-thought reasoning often waste substa ntial compute by producing long, incorrect responses. Abstention can mitigate th is by withholding outputs unlikely to be correct. While most abstention methods decide to withhold outputs before or after generation, dynamic mid-generation ab stention considers early termination of unpromising reasoning traces at each tok en position. Prior work has explored empirical variants of this idea, but princi pled guidance for the abstention rule remains lacking. We present a formal analy sis of dynamic abstention for LLMs, modeling abstention as an explicit action wi thin a regularized reinforcement learning framework. An abstention reward parame ter controls the trade-off between compute and information. We show that abstain ing when the value function falls below this reward strictly outperforms natural baselines under general conditions. We further derive a principled and efficien t method to approximate the value function. Empirical results on mathematical re asoning tasks support our theory and demonstrate improved selective accuracy ove r existing methods.
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A General Framework for Fair and Robust Regression
Wenhai Cui, Xiaoting Ji, Wen Su, Xingqiu Zhao
Fair regression methods typically rely on squared error loss, making them fragil e under heavy tailed noise. We propose a general framework for robust regression under demographic parity (DP) that applies to a wide class of M-estimators, inc luding Cauchy, Huber, least absolute deviation, quantile, and Tukey losses. We p ropose an optimal fair transformation that guarantees DP while achieving the min

imum population risk among all rank preserving fair predictors. We also establis h convergence rates for the resulting estimators. To balance fairness and predic tive accuracy, we develop an interpolation scheme whose risk decreases while unf airness grows linearly with the interpolation parameter. The proposed framework can be further extended to conditional DP to account for legitimate covariates. Extensive simulation studies and real data applications show clear improvements over existing fair regression approaches in both robustness and predictive perfo rmance.
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HOBIT: Hardness Optimized Batch Sampling for InfoNCE Training
Himanshu Dutta, Lokesh Nagalapatti, Yashoteja Prabhu
Contrastive training with InfoNCE loss and in-batch negatives is the standard a pproach for learning dual-encoder models. Its effectiveness, however, critically depends on the availability of hard negatives; in their absence, learning quick ly saturates. Existing methods address this via explicit hard-negative mining, w hich is often costly or heuristic-driven. We introduce $\mathrm{\texttt{HOBIT}}$, a principled mini-batch construction method that improves in-batch negative qu ality by reordering training examples at every epoch. $\mathrm{\texttt{HOBIT}}$ solves an optimization problem motivated by the InfoNCE objective to yield mini- batches such that each query in the batch is exposed to hard yet non-contradicto ry, informative negative examples. We show that the optimization objective is mo notone and submodular which in turn leads us to a greedy algorithm that admits t he standard $\mathcal{O}(1 - 1/e)$ approximation guarantee. Empirically, we show that $\mathrm{\texttt{HOBIT}}$ incurs negligible computational overhead while s ignificantly outperforming state-of-the-art batching methods, and remains comple mentary to existing hard negative mining techniques.
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The Devil is in the Spectrum: Mitigating Representation Collapse in LLMs via Top ologically Regularized Side-Path
Yiheng Tao, Kaiwen Cheng, Yao Lu, Chang Liu, Jie Chen
Large Language Models (LLMs) are fundamentally limited by representation collaps e, a bottleneck that severely degrades long-context performance. We identify tha t existing approaches risk drifting into one of two pathological extremes: homog enization collapse (e.g., attention sinks causing rank deficiency) and isolation collapse (e.g., local attention causing context disconnection). Through spectra l analysis of attention dynamics, we derive an intrinsic trade-off between mixin g efficiency (spectral gap) and information capacity (effective rank) that stand ard mechanisms struggle to balance. To resolve this dilemma, we propose the Topo logically Regularized Side-Path (TRSP), a non-invasive architectural interventio n that achieves spectral balance. TRSP employs a parameter-free Triangular Box m echanism, scaled by a lightweight, length-aware gate, to regularize the token in teraction topology. By integrating proximal coupling to preserve effective rank and distal propagation to support non-degenerate mixing, TRSP promotes a geometr ically healthier transition operator without altering core attention. Experiment s show significant improvements across general capabilities and long-context ben chmarks. Notably, on NoLiMa at $8\times$ the training length, TRSP retains $83\% $ accuracy and surpasses the Differential Transformer and Gated Attention by app roximately 30 and 50 percentage points, respectively. Code available at: https:/ /github.com/Eziotao-tyd/TRSP.
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Unsupervised Neural Langevin Sampler for Mixed Integer Linear Programming
Yixin Huang, Shengyu Feng, Yiming Yang
Existing neural combinatorial optimization (CO) solvers often rely heavily on ex pensive labeled data and additional post-processing to produce feasible solution s. Research into mixed integer linear programs (MILPs) is particularly limited d ue to the lack of effective heuristics for feasibility and the challenge of mode ling mixed-type variables for neural solvers. To address these issues, we propos e a novel unsupervised Langevin sampler for solving MILPs. Our framework learns only integer variables, while continuous variables are solved using an exact lin ear programming solver, thus isolating the combinatorial hardness of the problem

and avoiding unnecessary modeling complexity. The sampler is based on Langevin dynamics and incorporates both objective optimization and constraint satisfaction into a unified energy function, enabling the model to jointly learn feasibility and optimality. Experiments demonstrate that our method achieves 100\% feasibility without expensive post-processing and matches or outperforms supervised solvers on benchmark datasets, highlighting its effectiveness and scalability. Our code can be found at https://github.com/CindyH1103/UNLS4MILP.

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Diversity-Preserved Distribution Matching Distillation for Fast Visual Synthesis

Tianhe Wu, Ruibin Li, Lei Zhang, Kede Ma

Distribution matching distillation (DMD) facilitates few-step image generation by aligning a distilled student with a reference multi-step teacher. In practice, however, optimizing DMD can reduce sample diversity in few-step synthesis, and existing remedies typically rely on perceptual or adversarial regularization, leading to stability and scalability challenges during training. Here, we describe diversity-preserved DMD (DP-DMD), a role-separated distillation method inspired by the complementary roles of early and late denoising steps. Specifically, the first distillation step is trained with a teacher-derived target-prediction objective (e.g., v-prediction) to preserve sample diversity, while the remaining steps are optimized with the standard DMD loss to refine perceptual quality. DP-DMD, with no perceptual or adversarial regularization, no additional modules, and no teacher-generated reference samples, preserves sample diversity while maintaining competitive visual quality under few-step sampling, providing a simple and stable alternative to other DMD variants.

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Step-Resolved Data Attribution for Looped Transformers

Georgios Kaissis, David Mildenberger, Juan Felipe Gomez, Martin J. Menten, Eleni Triantafillou

We study how individual training examples shape the internal computation of looped transformers, where a shared block is applied for τ recurrent iterations to enable latent reasoning. Existing training-data influence estimators such as TracIn yield a single scalar score that aggregates over all loop iterations, obscuring when during the recurrent computation a training example matters. We introduce Step-Decomposed Influence (SDI), which decomposes TracIn into a length-τ influence trajectory by unrolling the recurrent computation graph and attributing influence to specific loop iterations. To make SDI practical at transformer scale, we propose a TensorSketch implementation that never materialises per-example gradients. Experiments on looped GPT-style models and algorithmic reasoning tasks show that SDI scales excellently, matches full-gradient baselines with low error and supports a broad range of data attribution and interpretability tasks with per-step insights into the latent reasoning process.

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EvoGM: Learning to Merge LLMs via Evolutionary Generative Optimization

Tao Jiang, Yu Xinmeng, Chenhao Yi, Yiling Wu, Yan Li, Ran Cheng, Dongmei Jiang, Jianguo Zhang

Evolutionary model merging provides a powerful framework for the automated, training-free composition of LLMs through parameter-space search. However, existing methods predominantly rely on stochastic, hand-crafted operators that overlook the underlying performance landscape of the coefficient space. We propose Evolutionary Generative Merging (EvoGM), a framework that transcends manual heuristics by employing learnable generative modeling to optimize merging coefficients. Specifically, EvoGM features a dual-generator architecture with cycle-consistent learning to adaptively sample and refine promising merging candidates. By constructing winner-loser pairs from historical search trajectories, our framework effectively captures high-performance parameter distributions and maximizes data efficiency. This generative process is seamlessly integrated into a multi-round evolutionary pipeline, where elite merged models iteratively serve as new expert foundations. Extensive experiments across diverse benchmarks demonstrate that EvoGM significantly outperforms state-of-the-art baselines, exhibiting robust performance on both seen and unseen tasks. Code and data are available at https://githu

b.com/JiangTao97/evogm.
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Towards Realistic Lifelong Re-identification: Identity Recurrence with Changing Clothes
Wuxuan Shi, Zhijie Lu, He Li, Mang Ye
Existing lifelong person re-identification (Re-ID) methods assume that each identity maintains a relatively stable appearance distribution over time. However, in real-world scenarios, identities often reappear asynchronously with substantial clothing changes, which is not modeled in existing lifelong Re-ID formulations. We therefore study a realistic lifelong cloth-changing Re-ID (LCC) setting, in which identities reappear asynchronously under substantial clothing changes. This setting leads to two core difficulties: the model must acquire new identities while adapting representations of recurring ones, and at the same time remain robust to the substantial representation drift induced by clothing changes over time, which undermines cross-temporal identity consistency. To address these, we develop a framework that disentangles identity-intrinsic representations from clothing-induced biases, enabling identity modeling beyond appearance changes. We further introduce a Dynamic Identity-Anchor Alignment to maintain stable identity anchors under stage-wise distribution shifts. Experiments on the LTCC and PRCC benchmarks demonstrate superior performance and representational stability across multiple learning stages. Our code is available at https://github.com/joyner-7/LCC_ReID.
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Mitigating Premature Exploitation in Particle-based Monte Carlo for Inference-Time Scaling
Giorgio Giannone, Guangxuan Xu, Nikhil Shivakumar Nayak, Rohan Mahesh Awhad, Shivchander Sudalairaj, Kai Xu, Akash Srivastava
Inference-Time Scaling (ITS) improves language models by allocating more computation at generation time. Particle Filtering (PF) has emerged as a strong ITS method for complex mathematical reasoning tasks, but it is vulnerable when guided by process reward models, which often assign overconfident scores early in the reasoning process. This causes PF to suffer from premature exploitation: it myopically commits to locally promising trajectories, prunes potentially correct hypotheses, and converges to suboptimal solutions. This failure mode, known as particle impoverishment, is especially severe under constrained computational budgets. To address this, we analyze the problem and identify two root causes: a lack of diversity in the particle set due to overconfident resampling and consequent inability to assess the potential of a reasoning path. We introduce Entropic Particle Filtering (ePF), an algorithm that integrates two new techniques to solve these issues. The first technique, Entropic Annealing (EA), directly mitigates particle impoverishment by monitoring search diversity via entropy; when diversity drops, it intervenes by dynamically annealing the resampling distribution to preserve exploration. The second, an enhancement called Look-ahead Modulation (LaM), adds a predictive guide to evaluate a state's potential based on its successors. By effectively balancing exploration and exploitation, ePF significantly outperforms strong baselines on challenging math benchmarks, achieving up to a 50% relative improvement in task reward.
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Improving Visual Token Reduction via Rectifying Distortions for Efficient Multimodal LLM Inference
Hyeonwoo Cho, Donghyeon Baek, Yewon Kim, Bumsub Ham
Recent advancements in Multimodal Large Language Models (MLLMs) have achieved remarkable success in vision-language tasks, yet the quadratic computational complexity arising from the vast number of visual tokens incurs significant memory and latency bottlenecks. While visual token reduction (VTR) strategies have been explored to mitigate this burden, existing methods overlook the positional and attentional consistency between the full and reduced sequences, resulting in a distorted representation. To this end, we propose RESTORE, a novel VTR framework that rectifies the positional and attentional distortions while maintaining efficiency. Specifically, we present a simple yet effective calibration method that re

stores lost visual attention by augmenting attention weights based on relative d
istances. We also introduce a distinctive anchor selection for token merging to
mitigate information loss during feature averaging. Experimental results on mult
iple benchmarks demonstrate that our method consistently improves the accuracy o
f various reduction methods, achieving state-of-the-art performance while mainta
ining computational efficiency.
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Scalable and Differentiable Point-Cloud Registration Using Maximum Mean Discrepa
ncy
Rixon Crane, Fahira Afzal Maken, Nicholas Lawrance, Stanislav Funiak, Kasra Khos
oussi, Ming Xu, Russell Tsuchida
We present MMD-Reg, a novel correspondence-free approach to point-cloud registra
tion that is differentiable and has linear computational complexity in the numbe
r of points. We model registration as a nonlinear least-squares problem based on
 the Maximum Mean Discrepancy, approximated using random Fourier features. The r
esulting objective can be solved efficiently with standard methods such as Leven
berg–Marquardt, and the solution is differentiable via the implicit function the
orem. This allows MMD-Reg to be used as a differentiable optimization layer with
in end-to-end trainable models, supporting registration under challenging condit
ions such as poor initial alignment and partial overlap. We demonstrate this Neu
ral MMD-Reg formulation by integrating the layer with a set transformer, trainin
g the resulting model in supervised and unsupervised settings, and comparing its
 performance against recent learning-based methods. We also evaluate standalone
MMD-Reg, comparing its accuracy and scalability against widely used non-learning
-based registration methods.
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WeDLM: Reconciling Diffusion Language Models with Standard Causal Attention for
Fast Inference
Aiwei Liu, Minghua He, Shaoxun Zeng, Sijun Zhang, Linhao Zhang, Chuhan Wu, Wei J
ia, Yuan Liu, Zhou Xiao, Jie Zhou
Autoregressive (AR) generation is the standard decoding paradigm for Large Langu
age Models (LLMs), but its token-by-token nature limits parallelism at inference
 time. Diffusion Language Models (DLLMs) offer parallel decoding by recovering m
ultiple masked tokens per step; however, in practice they often fail to translat
e this parallelism into speed gains over optimized AR engines (e.g., vLLM). A ke
y reason is that many DLLMs rely on bidirectional attention, which breaks standa
rd prefix KV caching.
We propose WeDLM, a diffusion decoding framework built entirely on standard caus
al attention to make parallel generation prefix-cache friendly. The core idea is
 to let each masked position condition on all observed tokens while keeping a ca
usal mask, achieved by Topological Reordering that moves observed tokens to the
physical prefix while preserving their logical positions. Building on this, we i
ntroduce a streaming decoding procedure that continuously commits confident toke
ns into a growing left-to-right prefix, avoiding the stop-and-wait behavior comm
on in block diffusion methods. Experiments show that WeDLM preserves the quality
 of strong AR backbones while delivering substantial speedups, approaching 3× on
 challenging reasoning benchmarks and up to 10× in low-entropy generation regime
s; critically, our comparisons are against AR baselines served by vLLM under mat
ched deployment settings.
****
Attention Sinks in Diffusion Transformers: A Causal Analysis
FANGZHENG WU, Brian Summa
Attention sinks---tokens that receive disproportionate attention mass---are
assumed to be functionally important in autoregressive language models, but
their role in diffusion transformers remains unclear.
We present a causal analysis in text-to-image diffusion, dynamically identifying

dominant attention recipients per timestep and suppressing them via paired,
training-free interventions on the score and value paths.
Across 553 GenEval prompts on Stable Diffusion~3 (with SDXL corroboration),

removing these sinks does not degrade text-image alignment (CLIP-T) or preference proxies (ImageReward, HPS-v2) at $k{=}1$; only under stronger interventions ($k\!\geq\!10$) does HPS-v2 exhibit a metric-dependent boundary, while CLIP-T remains robust throughout.
The perceptual shifts induced by suppression are nonetheless \emph{sink-specific}---$\sim\!6\times$ larger than equal-budget random masking---revealing an empirical dissociation between trajectory-level perturbation and \emph{semantic alignment} in diffusion transformers.
\footnote{Code
available at \url{https://github.com/wfz666/ICML26-attention-sink}.}
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A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dyn amics and Connectivity
Chengrui Li, Yunmiao Wang, Yule Wang, Weihan Li, Dieter Jaeger, Anqi Wu
Low-rank recurrent neural networks (lrRNNs) are a class of models that uncover l ow-dimensional latent dynamics underlying neural population activity. Although t heir functional connectivity is low-rank, it lacks independence interpretations, making it difficult to assign distinct computational roles to different latent dimensions. To address this, we propose the Factored Recurrent Neural Network (F acRNN), a generative lrRNN framework that assumes group-wise independence among latent dynamics while allowing flexible within-group entanglement. These indepen dent latent groups allow latent dynamics to evolve separately, but are internall y rich for complex computation. We reformulate the lrRNN under a variational aut oencoder (VAE) framework, enabling us to introduce a partial correlation penalty that encourages independence between groups of latent dimensions. Experiments o n synthetic, monkey M1, and mouse voltage imaging data show that FacRNN consiste ntly improves the disentanglement and interpretability of learned neural latent trajectories in low-dimensional space and low-rank connectivity over baseline lr RNNs that do not encourage group-wise independence.
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Transfer Learning in High-dimensional Ising Models
Joonho Kim, Seyoung Park
In high-dimensional Ising model estimation, target sample sizes are often limite d, and effectively using auxiliary binary datasets of unknown relevance remains challenging. To address this, we propose Trans-Ising, a transfer learning method that combines a loss-based source screening rule with a two-stage estimation pr ocedure. The method first identifies informative auxiliary sources using held-ou t target pseudolikelihood to prevent negative transfer. It then computes an init ial estimator via pooled nodewise ℓ_1-regularized logistic regression, foll owed by a target-only correction step using a folded-concave penalty. Theoretica lly, we establish fixed-node ℓ_2 and ℓ_1 error bounds, exact graph sel ection consistency, and the conditional consistency of the screening rule. Throu gh extensive simulations and real-data analyses, we demonstrate that Trans-Ising achieves lower estimation errors than both target-only estimation and naive dat a pooling.
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A Studentized Spherical Harmonics–Based Nonparametric Two-Sample Test for Compos itional and Directional Data
Binglin Li, Matthew Reed, Seong-Tae Kim
Compositional data analysis has gained increased attention due to the widespread occurrence of simplex-valued data, including microbiome data and financial port folios. Existing compositional two-sample tests often require $\log$-transformat ions and only detect mean differences, motivating the need for a more general fr amework without relying on $\log$-based methods. There is a close connection bet ween compositional data and directional statistics, and we construct a unified n on-parametric two-sample test framework. Our work is based on a studentized ener gy statistic constructed from spherical harmonics theory over a fixed dimensiona l underlying space, incorporating U-statistics theory and recent developments of studentization for both compositional and directional data. We establish asympt otic normality for our spherical harmonics based test statistics, thus avoiding

the need for permutation tests or bootstrap procedures. Our proposed framework s heds new light on the connections between Non-Euclidean data analysis and classi cal asymptotic high-dimensional data techniques.
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VAnim: Rendering-Aware Sparse State Modeling for Structure-Preserving Vector Ani mation

Guotao Liang, Zhangcheng Wang, Chuang Wang, Juncheng Hu, Haitao Zhou, Junhua Liu , Jing Zhang, Dong Xu, Qian Yu

Scalable Vector Graphics (SVG) animation generation is pivotal for professional design due to their structural editability and resolution independence. However, this task remains challenging as it requires bridging discrete code representat ions with continuous visual dynamics. Existing optimization-based methods often destroy topological consistency, while general-purpose LLMs rely on rigid CSS/SM IL transformations, failing to model geometry-level non-rigid deformations. To a ddress these limitations, we present VAnim, the first LLM-based framework for op en-domain text-to-SVG animation. We reconceptualize animation not as sequence ge neration, but as Sparse State Updates (SSU) on a persistent SVG DOM tree. This p aradigm compresses sequence length by over 9.8x while mathematically guaranteein g topological isomorphism and identity persistence. To enable precise control, w e propose an Identification-First Motion Planning mechanism that grounds textual instructions in explicit visual entities. Furthermore, to overcome the non-diff erentiable nature of SVG rendering, we employ Rendering-Aware Reinforcement Lear ning via Group Relative Policy Optimization (GRPO). By leveraging a hybrid rewar d from a state-of-the-art video perception encoder, we align discrete code updat es with high-fidelity visual feedback. We also introduce SVGAnim-134k, the first benchmark for vector animation. Extensive experiments demonstrate that VAnim si gnificantly outperforms state-of-the-art baselines in semantic alignment and str uctural validity, demonstrating a robust capacity for synthesizing high-fidelity non-rigid deformations without requiring explicit temporal consistency constrai nts.
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Autoregressive, Yet Revisable: In Decoding Revision for Secure Code Generation

Chengran Yang, zichao wei, Heminghao Deng, Jinfeng Jiang, Zhensu Sun, Ting Zhang , Tianyi Wu, Ming Wen, David Lo

Large Language Model (LLM) based code generation is predominantly formulated as a strictly monotonic process, appending tokens linearly to an immutable prefix. This formulation contrasts with the cognitive process of programming, which is i nherently interleaved with forward generation and on-the-fly revision. While pri or works attempt to introduce revision via post-hoc agents or external static to ols, they either suffer from high latency or fail to leverage the model's intrin sic semantic reasoning.

In this paper, we propose Stream of Revision, a paradigm shift that elevates cod e generation from a monotonic stream to a dynamic, self-correcting trajectory by leveraging the model's intrinsic capabilities. We introduce specific action tok ens that enable the model to seamlessly backtrack and edit its own history withi n a single forward pass. By internalizing the revision loop, our framework Strea m of Revision allows the model to activate its latent capabilities just-in-time without external dependencies. Empirical results on secure code generation show that Stream of Revision significantly reduces vulnerabilities with minimal infer ence overhead.
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Efficient Test-Time Scaling via Hierarchical Search and Self-Verification for Di screte Diffusion Language Models

Jinbin Bai, Yixuan Li, Yuchen Zhu, Yi Xin, Qingyu Shi, Aosong Feng, Xiaohong Liu , Molei Tao, Jianru Xue, Xiangtai Li, Ming-Hsuan Yang

Inference-time compute has re-emerged as a practical way to improve LLM reasonin g.

Most test-time scaling (TTS) algorithms rely on autoregressive decoding, which i s ill-suited to discrete diffusion language models (dLLMs) due to their parallel decoding over the entire sequence.

As a result, developing effective and efficient TTS methods to unlock dLLMs' ful
l generative potential remains an underexplored challenge.
To address this, we propose \textbf{LLaDA-S}, an efficient TTS framework for dLL
Ms that
(i) performs \textbf{Hierarchical Trajectory Search} (HTS) which dynamically pru
nes and reallocates compute in an early-to-mid denoising window, (ii) replaces e
xternal verifiers with \textbf{Self-Verified Feedback} (SVF) obtained via self-e
valuation prompts on intermediate completions, and (iii) introduces \textbf{Loca
l branching with partial remasking} to explore diverse implementations while pre
serving a high-confidence tokens.
Across four mathematical reasoning and code generation benchmarks on three dLLMs
, including LLaDA 8B Instruct, Dream 7B Instruct, and LLaDA 2.0-mini, our LLaDA-
S achieves a favorable performance-efficiency trade-off, matching best-of-N pe
rformance with substantially fewer function evaluations (NFE).
The code will be released.
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BESPOKE: Benchmark for Search-Augmented Large Language Model Personalization via
 Diagnostic Feedback
Hyunseo Kim, Sangam Lee, Kwangwook Seo, Dongha Lee
Search-augmented large language models (LLMs) remain insufficient for fully addr
essing diverse user needs, which requires recognizing how the same query can ref
lect different intents across users and delivering information in preferred form
s. While recent systems such as ChatGPT and Gemini attempt personalization by le
veraging user histories, systematic evaluation of such personalization is under-
explored. To address this gap, we propose BESPOKE, the realistic benchmark for e
valuating personalization in search-augmented LLMs. BESPOKE is designed to be bo
th realistic, by collecting authentic chat and search histories directly from hu
mans, and diagnostic, by pairing responses with fine-grained preference scores a
nd feedback. The benchmark is constructed through long-term, deeply engaged huma
n annotation, where human annotators contributed their own histories, authored q
ueries with detailed information needs, and evaluated responses with scores and
diagnostic feedback. Leveraging BESPOKE, we conduct systematic analyses that rev
eal key requirements for effective personalization in information-seeking tasks,
 providing a foundation for fine-grained evaluation of personalized search-augme
nted LLMs. Our code and data are available at https://github. com/augustinLib/BE
SPOKE.
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HypoSpace: A Diagnostic Benchmark for Set-Valued Hypothesis Generation under Und
erdetermination and Sublinear Coverage Bounds
Tingting Chen, Beibei Lin, Zifeng Yuan, Qiran Zou, Hongyu He, Anirudh Goyal, Yew
-Soon Ong, Dianbo Liu
Many scientific problems are underdetermined: multiple distinct hypotheses are e
qually consistent with the same observations. In such settings, effective infere
nce requires not only producing valid explanations, but also systematically expl
oring and covering the admissible hypothesis set. We introduce HypoSpace, a benc
hmark that treats large language models (LLMs) as samplers over finite hypothesi
s spaces and evaluates them on three metrics: Validity, Uniqueness, and Recovery
. HypoSpace spans three structured domains (causal graph inference, gravity-cons
trained 3D voxel reconstruction, and Boolean genetic interaction modeling) with
deterministic validators and exactly enumerable solution spaces, plus real-world
 anchored case studies. Empirically, HypoSpace reveals a capability- and scale-d
ependent coverage failure: models can maintain high Validity while exhibiting re
duced Uniqueness and Recovery as admissible hypothesis spaces become larger or m
ore combinatorial. We further show that the analysis on stratified decoding part
ially mitigates this collapse, demonstrating HypoSpace's utility as a diagnostic
 benchmark for set-valued inference. Code is available at: https://github.com/CT
T-Pavilion/_HypoSpace.
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Adaptive Reinforcement Learning for Unobservable Random Delays
John Wikman, Alexandre Proutiere, David Broman

In standard reinforcement learning (RL) settings, the interaction between the ag
ent and the environment is typically modeled as a Markov decision process (MDP),
 which assumes that the agent observes the system state instantaneously, selects
 an action without delay, and executes it immediately. In real-world dynamic env
ironments, such as cyber-physical systems, this assumption often breaks down due
 to delays in the interaction between the agent and the system. These delays can
 vary stochastically over time and are typically _unobservable_ when deciding on
 an action. Existing methods deal with this uncertainty conservatively by assumi
ng a known fixed upper bound on the delay, even if the delay is often much lower
. In this work, we introduce the _interaction layer_, a general framework that e
nables agents to adaptively handle unobservable and time-varying delays. Specifi
cally, the agent generates a matrix of possible future actions, anticipating a h
orizon of potential delays, to handle both unpredictable delays and lost action
packets sent over networks. Building on this framework, we develop a model-based
 algorithm, _Actor-Critic with Delay Adaptation (ACDA)_, which dynamically adjus
ts to delay patterns. Our method significantly outperforms state-of-the-art appr
oaches across a wide range of locomotion benchmark environments, including real-
world measured delays.
****

InteractScience: Programmatic and Visually-Grounded Evaluation of Interactive Sc
ientific Demonstration Code Generation

Qiaosheng Chen, Yang Liu, Lei Li, Kai Chen, Qipeng Guo, Gong Cheng, Fei Yuan

While Large Language Models (LLMs) hold promise for automating science and educa
tion, generating interactive scientific demonstrations demands a complex synthes
is of deep domain knowledge and precise reactive coding. Current benchmarks fail
 to capture this synergy, largely bifurcating into static code generation or tex
t-only reasoning. To address this, we introduce \textsc{InteractScience}, the fi
rst benchmark dedicated to evaluating the holistic creation of interactive scien
tific applications. We propose a novel hybrid framework that integrates programm
atic functional testing for logic verification with visually-grounded qualitativ
e assessment for rendering fidelity. Our evaluation of 30 leading models across
five disciplines reveals critical gaps in grounding scientific reasoning within
interactive interfaces. By standardizing this combined capability, \textsc{Inter
actScience} establishes a crucial foundation for reliable AI-driven tools in sci
ence and education.
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Multi-Adapter Representation Interventions via Energy Calibration

Manjiang Yu, Hongji Li, Junwei Chen, Xue Li, PRIYANKA SINGH, Yang Cao, Lijie Hu

Representation intervention has emerged as a promising paradigm for aligning lar
ge language models toward desired behaviors without modifying model weights. Exi
sting methods typically apply a fixed intervention uniformly across all inputs.
However, we find that the appropriate intervention direction and strength vary s
ubstantially across samples, and such indiscriminate intervention leads to degra
dation of general capabilities on benign inputs. To address these challenges, we
 propose Multi-Adapter Representation Interventions via Energy Calibration (MARI
). Specifically, we introduce a competitive multi-adapter mechanism in which spe
cialized experts capture non-linear correction patterns and adaptively determine
 the appropriate intervention direction and strength for different samples. Furt
hermore, we design an energy-based gating module that leverages internal propaga
tion dynamics to distinguish inputs that are applicable for intervention. Extens
ive experiments across diverse model families and parameter scales demonstrate t
hat MARI achieves state-of-the-art alignment performance. Our method significant
ly improves performance on TruthfulQA, BBQ, and safety benchmarks, while maintai
ning and even improving general capabilities on tasks such as MMLU and ARC. Our
code is available at https://github.com/V1centNevwake/MARI.
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Hierarchical Goal Abstractions via Learned Subset Relations

Fabian Wurzberger, Sebastian Gottwald, Zeqiang Zhang, Daniel Alexander Braun

In self-supervised goal-conditioned reinforcement learning (RL) without external
 rewards, goals are typically specified by observations sampled from experience

.
However, depending on the observation structure, such a fixed representation of goals may be either too concrete (requiring exact pixel-level matches) or too abstract (involving ambiguous observations).
Here we propose the construction of hierarchical latent goal spaces that integrate both concrete and abstract goals.
To this end, we use an energy function to learn a partially ordered space, in which a subset relation between observations naturally induces a hierarchy from concrete to abstract goals.
This representation enables agents to disambiguate specific states while also generalizing to shared concepts.
In experiments on navigation and robotic manipulation, agents trained with our hierarchical goal space achieve higher task success and greater generalization to novel tasks compared to agents limited to purely observational goals.
****
BubbleSpec: Turning Long-Tail Bubbles into Speculative Rollout Drafts for Synchronous Reinforcement Learning
Yuhang Xu, Kaibin Tian, Yang Tian, Zhice Yang, Yifeng Yu, Yan Li, Shengzhong Liu, Fan Wu, Guihai Chen
Reinforcement Learning (RL) has become a cornerstone for improving the performance of Large Language Models (LLMs). However, its rollout phase constitutes a significant efficiency bottleneck, mainly arising from the long-tail bubbles across data parallel ranks, particularly in long-context scenarios where faster GPUs remain idle while waiting for stragglers.
Existing solutions, such as partial rollout or asynchronous RL, mitigate these bubbles by compromising the algorithm's strict synchronous nature. Instead, we propose **BubbleSpec**, a novel framework that accelerates RL rollouts while strictly keeping the mathematical exactness.
Instead of attempting to eliminate bubbles, BubbleSpec exploits them.
We exploit the idle time windows of faster ranks to pre-generate rollout results for subsequent steps, serving as drafts for speculative decoding.
Unlike prior speculative methods that rely on historical epoch similarity and warm-ups, BubbleSpec is agnostic to dataset size and provides immediate acceleration from the onset of training.
Extensive evaluations demonstrate that BubbleSpec reduces decoding steps by **$\sim$50\%** and increases rollout throughput by up to **1.8$\times$**. Critically, BubbleSpec is seamlessly compatible with various RL frameworks and strategies as it sustains the strict synchronous property of RL algorithms.
****
Formalizing and Falsifying Causal Pathways of Rare Events
Anahita Haghighat, Dominik Janzing
Building on recent formalizations of root cause analysis for rare events (“outliers”) in structural equation models, we propose a formal definition of a causal pathway and discuss its testable implications. We identify conditions under which these implications depend only on a causal abstraction defined by the pathway of rare events, rather than on the full causal graph of the underlying system. Accordingly, we introduce an abstraction of causal structure to pathways of rare events that bridges simple verbal causal explanations and detailed causal modeling.
****
Differentially Private Cross-Silo Recommendation from Implicit Feedback
Xun Ran, Qingqing Ye, Xin Huang, Jianliang Xu, Haibo Hu
Cross-silo recommendation from implicit feedback is a key task in modern recommender systems, where user-item interaction data are distributed across multiple parties and cannot be centrally collected. Unlike explicit feedback, which provides fully observed real-valued ratings, implicit feedback is one-class and extremely sparse, recording only users' actions or inactions (e.g., clicks, visits, or bookmarks), yet it is far more prevalent in real-world applications. Such behavioral data are often highly sensitive, raising significant privacy concerns when used for collaborative model training. Although differential privacy (DP) has b

een widely applied to explicit feedback-based models, the resulting utility degr
adation makes it difficult to apply DP effectively to implicit feedback learning
. In this work, we propose DPIMF, a differentially private implicit matrix facto
rization framework for cross-silo recommendation based on objective perturbation
. To improve utility, we redesign the loss function and adopt an importance samp
ling scheme to reduce the noise scale required for privacy preservation. We furt
her provide formal utility guarantees for the proposed techniques and characteri
ze the conditions under which utility improvements are maximized. Experiments on
 three benchmark datasets validate our theoretical analysis and demonstrate that
 DPIMF achieves a better privacy-utility trade-off than state-of-the-art methods
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Discovering Interpretable Algorithms by Decompiling Transformers to RASP
Xinting Huang, Aleksandra Bakalova, Satwik Bhattamishra, William Merrill, Michae
l Hahn
Recent work has shown that the computations of Transformers can be simulated in
the RASP family of programming languages.
These findings have enabled improved understanding of the expressive capacity a
nd generalization abilities of Transformers.
In particular, Transformers have been suggested to length-generalize exactly on
problems that have simple RASP programs.
However, it remains open whether trained models actually implement simple interp
retable programs.
In this paper, we present a general method to extract such programs from trained
 Transformers.
The idea is to faithfully re-parameterize a Transformer as a RASP program and th
en apply causal interventions to discover a small sufficient sub-program.
In experiments on small Transformers trained on algorithmic and formal language
tasks, we show that our method often recovers simple and interpretable RASP prog
rams from length-generalizing transformers.
Our results provide the most direct evidence so far that Transformers internally
 implement simple RASP programs.
****
Multi-agent imitation learning with function approximation: linear Markov games
and beyond
Luca Viano, Till Freihaut, Emanuele Nevali, Volkan Cevher, Matthieu Geist, Giorg
ia Ramponi
In this work, we present the first theoretical analysis of multi-agent imitation
 learning (MAIL) in linear Markov games where both the transition dynamics and e
ach agent's reward function are linear in some given features. We demonstrate th
at by leveraging this structure, it is possible to replace the state-action leve
l \emph{all policy deviation concentrability coefficient} (Freihaut et al., 2025
b)
    with a concentrability coefficient defined at the feature level which can be
 much smaller than the state-action analog when the features are informative abo
ut \emph{states' similarity}. Furthermore, to circumvent the need for any concen
trability coefficient, we turn to the interactive setting. We provide the first,
 computationally efficient, interactive MAIL algorithm for linear Markov games a
nd show that its sample complexity depends only on the dimension of the feature
map d.
    Building on these theoretical findings, we propose a deep MAIL interactive a
lgorithm which clearly outperforms BC on games such as Tic-Tac-Toe and Connect4.
****
A Time-Reparameterized Cumulative Intensity Extrapolation Sampler for Discrete F
low Matching
Feiyang Fu, Hehe Fan
Discrete flow matching (DFM) provides a principled framework for generative mode
ling on discrete state spaces via continuous-time Markov chain dynamics. In prac
tice, sampling for DFM commonly employs discretizations such as τ-leaping,
yet efficient sampling methods under a limited number of function evaluations (N

FE) remain less studied. To address this gap, we propose the Time-Reparameterize
d Cumulative Intensity Extrapolation (TR-CIE) sampler, which aims to improve sam
pling quality when function evaluations are restricted. TR-CIE consists of two c
omponents. First, a schedule-based time reparameterization rescales the time gri
d according to the noise schedule. Under standard factorized DFM rate parameteri
zations, this transformation of variables absorbs the schedule-dependent growth
term and mitigates stiffness near the terminal sampling stage. Second, we introd
uce a cumulative-intensity extrapolation updating rule. By reusing cached model
outputs from the previous step as a history term, this improves the approximatio
n of stepwise cumulative intensities on the resulting non-uniform time grid. We
provide a theoretical analysis that bounds the local approximation error of cumu
lative intensities and establishes convergence results. The resulting sampler re
quires one NFE per step and introduces no additional model evaluations compared
to the standard τ-leaping sampler. Extensive experiments on synthetic tasks
, text generation, and text-to-image benchmarks demonstrate that our method impr
oves sampling quality under limited NFE.
****

ASTRA: Communication-Efficient Acceleration for Multi-Device Transformer Inferen
ce
Xiao Liu, Lijun Zhang, Deepak Ganesan, Hui Guan
Multi-device inference can reduce Transformer latency by parallelizing computati
on. However, existing methods require high inter-device bandwidth, making them i
mpractical for bandwidth-constrained environments. We present ASTRA, a communica
tion-efficient framework that integrates sequence parallelism with mixed-precisi
on attention, where non-local token embeddings are transmitted as low-bit vector
-quantized codes while local attention remains full precision. To preserve accur
acy under aggressive compression, ASTRA introduces Noise-Augmented Quantization
and Distributed Class Tokens. Across vision and language models (e.g., ViT and G
PT2), ASTRA achieves up to 2.64$\times$ speedup over single-device inference and
 up to 15.25$\times$ over prior multi-device baselines while operating at bandwi
dths as low as 10 Mbps. ASTRA remains robust on large models (e.g., Llama-3-8B)
even under non-ideal network conditions such as packet loss and dynamic networks
.
****

Chiral Symmetry Breaking in Transformers: A Group-Equivariant Framework for Addr
essing the Reversal Curse via Adjoint Manifold Mappings
Hanji Du
The "reversal curse" exposes a critical asymmetry in autoregressive models, wher
e models trained on facts in one direction often fail to access the correspondin
g inverse relation. This work studies the phenomenon from a representation-level
 perspective, characterizing it as a form of chiral asymmetry between subject- a
nd object-oriented latent states. We introduce the Chiral Transformer, a lightwe
ight framework that encourages an involutive adjoint mapping operator $\mathcal{
T}$ through contrastive regularization. At inference time, Adjoint-Induced Retri
eval (AIR) uses this learned map as a structured readout over model-derived enti
ty representations, rather than as an unconstrained autoregressive generation pr
otocol. Empirical validation on inverse-relation benchmarks shows that this symm
etry-aware retrieval setting substantially improves inverse factual access, with
 AIR reaching 65.07% accuracy on Fact-Inv-300. These findings support a represen
tation-access view of the reversal curse: inverse relations may be difficult not
 only because of missing data, but also because standard autoregressive readout
fails to expose useful latent structure.
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SafeSpec: Fast and Safe LLM via Dynamic Reflective Sampling
HAOTIAN XU, Zeyang Zhang, Linbao Li, Huadi Zheng, Yu Li, Cheng Zhuo
Speculative inference accelerates large language model (LLM) decoding but provid
es no inherent safety guarantees. Existing safety defenses are largely incompati
ble with speculative inference: they either introduce additional computation or
disrupt the draft–verify mechanism, negating acceleration benefits. This reveals
 a fundamental incompatibility between current safety methods and speculative de

coding.
We propose SafeSpec, a safety-aware speculative inference framework that integrates risk estimation directly into the verification process. SafeSpec attaches a lightweight latent safety head to the target model to jointly evaluate semantic validity and safety in a single forward pass. When unsafe generations are detected, SafeSpec applies rollback and safety-guided reflective multi-sampling to recover safe continuations rather than terminating generation.
We model jailbreak attacks as distributional shifts over generative trajectories, where adversarial prompts increase the probability of harmful continuations without eliminating safe ones. Under this model, SafeSpec performs risk-aware trajectory recovery within the speculative decoding process.
Across multiple models and adversarial benchmarks, SafeSpec achieves a substantially improved safety–efficiency trade-off. On Qwen3-32B, SafeSpec reduces attack success rates by 15\% while preserving a 2.06× inference speedup on benign workloads, demonstrating that speculative acceleration and inference-time safety can be jointly optimized.
****

FIPN: Forward Self-Organizing Interpretable Polynomial Networks for Time Series Forecasting
YiZhen Wang, Zheng Wang, EUN-HU KIM, Zunwei Fu
Most existing time-series forecasters are trained end-to-end with backpropagation, which often increases computational cost and limits structural transparency. As a result, it remains difficult to trace how their predictions are formed. This paper presents FIPN, a forward self-organizing interpretable polynomial network for time-series forecasting. FIPN grows its architecture layer by layer through candidate generation, closed-form estimation, and validation-based selection, without relying on backpropagation-based optimization. Each neuron couples a fuzzy rule antecedent with a Fourier-enhanced polynomial consequent: fuzzy clustering softly partitions the input space and produces interpretable rule weights for local regimes, while the consequent retains access to original features and uses Fourier functions to capture periodic and frequency-related structures. Since forward growth may introduce redundancy, collinearity, and overfitting as depth increases, FIPN further incorporates regularized node scoring, dropout-based redundancy control, and persistent access to raw inputs. Experiments on long-horizon forecasting benchmarks show that FIPN achieves competitive accuracy with a compact model size, while the learned fuzzy rules provide rule- and structure-level explanations. These results suggest that forward self-organizing polynomial networks offer a practical balance among accuracy, efficiency, and interpretability for long-horizon forecasting.
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Towards Streaming Synchronized Spatial Audio Generation via Autoregressive Diffusion Transformer
Ke Lei, Yu Zhang, Changhao Pan, Xueyi Pu, Wenxiang Guo, Ruiqi Li, Zhou Zhao
Real-time and accurate spatial audio generation is pivotal for delivering an immersive experience.
However, existing spatial audio synthesis technologies are often encumbered by a tradeoff between generation quality and high inference latency, as well as difficulty in capturing precise spatial information from multimodal inputs.
To address these challenges, we propose SwanSphere, a unified streaming framework for high-fidelity spatial audio generation from panoramic videos and text prompts.
SwanSphere mainly makes the following contributions:
1) We introduce a causal autoregressive diffusion transformer architecture that enables streaming high-quality spatial audio generation.
2) We design a Spatial Video–Audio Contrastive (SVAC) learning strategy to align the video encoder with the acoustic domain, and further employ a multi-objective online direct preference optimization~(ODPO) scheme, resulting in strong spatial perception and robust multimodal spatial audio synthesis.
3) To alleviate the current scarcity of spatial audio datasets, we also develop an automated annotation pipeline for generating detailed spatial captions.

Experimental results demonstrate that SwanSphere achieves superior performance in both video-to-spatial and text-to-spatial audio generation tasks. Codes can be found at: https://github.com/MM-Speech/SwanSphere
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See More, Forecast Better and Faster: Enhancing Time Series Foundation Models via Inference-Time Plug-and-Play Downsampling
Longlong Xu, Zeyan Li, Xiao He, Zhaoyang Yu, Dazhong Wen, Mingze Sun, Changhua Pei, Dan Pei
Time series foundation models (TSFMs) have demonstrated impressive generalization capabilities across diverse domains. However, they face significant challenges in long-term and ultra long-term forecasting. These challenges primarily arise from scalability limitations when TSFMs process extensive sequence lengths. To address this, we propose SPRINT, a training-free plug-and-play framework designed to empower TSFMs to see more, forecast better and faster during inference. The core idea is to perform forecasting in a downsampled-resolution space, enabling an extended look-back window with reduced computational costs. To avoid information loss and resolution mismatch caused by downsampling, SPRINT decomposes time series into trend and seasonal components, processing them separately. It predicts the low-frequency trend via a Resolution Interpolation workflow within the downsampled space, while preserving high-frequency details through a Pattern Replication mechanism for seasonality. Extensive experiments show that SPRINT achieves a significant improvement, increasing accuracy by 19\% while enhancing efficiency with a reduction of max memory usage by 6.4$\times$ and inference time by 16.9$\times$ compared to state-of-the-art TSFMs.
****

Reinforcement Learning via Self-Distillation
Jonas Hübotter, Frederike Lübeck, Lejs Deen Behric, Anton Baumann, Marco Bagatella, Daniel Marta, Ido Hakimi, Idan Shenfeld, Thomas Kleine Buening, Carlos Guestrin, Andreas Krause
Large language models are increasingly post-trained with reinforcement learning in verifiable domains such as code and math. Yet, current methods for reinforcement learning with verifiable rewards (RLVR) learn only from a scalar outcome reward per attempt, creating a severe credit-assignment bottleneck. Many verifiable environments actually provide rich textual feedback, such as runtime errors or judge evaluations, that explain *why* an attempt failed.
We formalize this setting as reinforcement learning with rich feedback and introduce **Self-Distillation Policy Optimization** (**SDPO**), which converts tokenized feedback into a dense learning signal without any external teacher or explicit reward model.
SDPO treats the current model conditioned on feedback as a self-teacher and distills its feedback-informed next-token predictions back into the policy.
In this way, SDPO leverages the model's ability to retrospectively identify its own mistakes in-context.
Across scientific reasoning, tool use, and competitive programming on LiveCodeBench v6, SDPO improves sample efficiency and final accuracy over strong RLVR baselines.
Notably, SDPO also outperforms baselines in standard RLVR environments that only return scalar feedback by using successful rollouts as implicit feedback for failed attempts.
Finally, applying SDPO to individual questions at test time accelerates discovery on difficult binary-reward tasks, achieving the same discovery probability as best-of-k sampling or multi-turn conversations with 3x fewer attempts.
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How Does the Pretraining Distribution Shape In-Context Learning? A Fundamental Trade-Off
Waïss Azizian, Ali Hasan
The factors driving the performance of in-context learning (ICL) in large language models (LLMs) remain poorly understood despite ICL's surprising effectiveness, enabling models to adapt to new tasks from only a handful of examples. To clarify and improve these capabilities, we characterize how the statistical properti

es of the pretraining distribution (e.g., tail behavior, coverage) shape ICL. We develop a theoretical framework that encompasses generalization and task selection and show how distributional properties govern sample efficiency, task retrieval, and robustness.
To this end, we generalize existing concentration results to heavy-tailed priors and dependent sequences, better reflecting the structure of LLM pretraining data.
Our framework reveals a fundamental design trade-off: heavy-tailed pretraining distributions facilitate robust task selection under distribution shifts but are detrimental to generalization, especially in low-data regimes.
We then empirically evaluate our predictions by studying how ICL performance varies with the pretraining distribution on challenging tasks such as stochastic differential equations and stochastic processes with memory. Together, these findings suggest that controlling key statistical properties of the pretraining distribution is essential for building ICL-capable and reliable LLMs.
****
Fast kernel methods: Sobolev, physics-informed, and additive models
Nathan Doumèche, Francis Bach, Gérard Biau, Claire Boyer
Kernel methods are powerful tools in statistical learning, but their cubic complexity in the sample size n limits their use on large-scale datasets. In this work, we introduce a scalable framework for kernel regression with $\mathcal{O}(n \log n)$ complexity, and designed to fully leverage GPU acceleration. The approach is based on a Fourier representation of kernels combined with non-uniform fast Fourier transforms (NUFFT), enabling exact, fast, and memory-efficient computations.
We instantiate our framework in three settings: Sobolev kernel regression, physics-informed regression, and additive models. The proposed estimators are shown to achieve minimax convergence rates, consistent with classical kernel theory. Empirical results demonstrate that our methods can process up to tens of billions of samples within minutes, providing both statistical accuracy and computational scalability.
These contributions establish a flexible approach, paving the way for the routine application of kernel methods in large-scale learning tasks, whenever the kernel norm can be efficiently expressed in Fourier space and the ambient dimension d is small.
Although the theoretical framework is valid in all dimensions d, the fast Sobolev regression package implemented in the paper relies on the CufiNUFFT package, which currently scales exponentially in d and is limited to $d \leq 3$. For similar reasons, the additive model package is only implemented for simple effects without interaction.
****
Unitary Convolutions for Message-passing and Positional Encodings on Directed Graphs
Lukas Fesser, Bobak Kiani, Melanie Weber
In many real-world networks, relationships are inherently directional, yet most graph neural networks (GNNs) assume undirected edges, and naïve adaptations of undirected GNNs to directed graphs amplify oversmoothing and gradient pathologies that cap model depth. Unitary graph convolutions (UniConv) provably prevent representational collapse and oversmoothing, but cannot incorporate edge directionality or edge features. In this paper, we introduce a **d**irected **un**itary GNN with **e**dge features (**Dune**), which retains these guarantees while overcoming UniConv’s limitations by incorporating edge directionality and edge features. Dune keeps gradient norms bounded at any number of layers, allowing it to benefit from neural network depth, unlike existing directed GNNs. The same unitary operator can be embedded in hybrid architectures with graph transformers, where its wavelike propagation supplies positional information and reduces the importance of random-walk or Laplacian-based encodings. We prove that Dune avoids exponential oversmoothing that plagues existing directed GNNs and empirically show that it achieves state-of-the-art performance on 12 directed-graph benchmarks while remaining trainable beyond 100 layers, improving performance by up to 18 perce

ntage points over strong baselines. Our results establish unitary convolutions a s a scalable, geometry-aware foundation for deep learning on directed graphs.
****
Fast and Scalable Analytical Diffusion
Xinyi Shang, Peng Sun, Jingyu Lin, Zhiqiang Shen
Analytical diffusion models offer a mathematically transparent path to generativ e modeling by formulating the denoising score as an empirical-Bayes posterior me an. However, this interpretability comes at a prohibitive cost: the standard for mulation necessitates a full-dataset scan at every timestep, scaling linearly wi th dataset size.
In this work, we present the first systematic study addressing this scalability bottleneck.
We challenge the prevailing assumption that the entire training data is necessar y, uncovering the phenomenon of Posterior Progressive Concentration: the effecti ve golden support of the denoising score is not static but shrinks asymptoticall y from the global manifold to a local neighborhood as the signal-to-noise ratio increases.
Capitalizing on this, we propose Dynamic Time-Aware Golden Subset Diffusion (Gol dDiff), a training-free framework that decouples inference complexity from datas et size.
Instead of static retrieval, GoldDiff uses a coarse-to-fine mechanism to dynamic ally pinpoint the "Golden Subset" for inference.
Theoretically, we derive rigorous bounds guaranteeing that our sparse approximat ion converges to the exact score.
Empirically, GoldDiff achieves a 71× speedup on AFHQ while matching or achieving even better performance than full-scan baselines.
Most notably, we demonstrate the first successful scaling of analytical diffusio n to ImageNet-1K.
****
WBMM: Windowed Batch Matrix Multiplication for Efficient Large Receptive Field C onvolution
Wan Song, Wei Zhou, Rui Wang, Jun Yu, Toru Kurihara, Jiajia Xu, Shu Zhan
Large kernel depthwise convolutions achieve strong performance but suffer from s ignificant degradation as kernel size grows due to irregular memory access from gather-based computation; while Large Kernel Acceleration (LKA) helps on small f eature maps, it becomes counterproductive on large feature maps, even slower tha n non-accelerated implementations. We propose Windowed Batch Matrix Multiplicati on (WBMM), which partitions input into contiguous windows and indexes a compact relative position bias table to construct weight matrices, enabling regular memo ry access via batched matrix multiplication. This yields a unique property: WBMM 's throughput improves with larger windows, opposite to depthwise convolutions t hat degrade with larger kernels. Operator-level benchmarks show WBMM with $14 \t imes 14$ windows outperforms 5×5 depthwise convolution baselines in spe ed while providing a $7.8\times$ larger per-layer receptive field. Combined with inter-block cross-window communication and hierarchical window reparameterizati on, WBMM achieves comparable or higher accuracy on ImageNet-1K, COCO, and ADE20K with 1.31--$1.88\times$ training speedup, and demonstrates consistent advanta ges across GPU, CPU, and edge devices without requiring specialized acceleration kernels. Our code is available at https://github.com/wansong-s/WBMM.
****
TraceRouter: Robust Safety for Large Foundation Models via Path-Level Interventi on
Chuancheng Shi, Shangze Li, Wenjun Lu, Wenhua Wu, Fei Shen, Cong Wang, Zifeng Ch eng, Tat-Seng Chua
Despite their capabilities, large foundation models (LFMs) remain susceptible to adversarial manipulation. Current defenses predominantly rely on the ``locality hypothesis", suppressing isolated neurons or features. However, harmful semanti cs act as distributed, cross-layer circuits, rendering such localized interventi ons brittle and detrimental to utility.
To bridge this gap, we propose \textbf{TraceRouter}, a path-level framework that

traces and disconnects the causal propagation circuits of illicit semantics.
TraceRouter operates in three stages: (1) it pinpoints a sensitive onset layer by analyzing attention divergence; (2) it leverages sparse autoencoders (SAEs) and differential activation analysis to disentangle and isolate malicious features; and (3) it maps these features to downstream causal pathways via feature influence scores (FIS) derived from zero-out interventions. By selectively suppressing these causal chains, TraceRouter physically severs the flow of harmful information while leaving orthogonal computation routes intact.
Extensive experiments demonstrate that TraceRouter significantly outperforms state-of-the-art baselines, achieving a superior trade-off between adversarial robustness and general utility. Our code will be publicly released. WARNING:
This paper contains unsafe model responses.
****

AuTAgent: A Reinforcement Learning Framework for Tool-Augmented Audio Reasoning
Siqian Tong, Li Xuan, Yiwei Wang, Baolong Bi, Yujun Cai, Shenghua Liu, Yuchen He, HAO Chengpeng
Large Audio Language Models (LALMs) excel at perception but struggle with complex reasoning requiring precise acoustic measurements. While external tools can extract fine-grained features like exact tempo or pitch, effective integration remains challenging: naively using all tools causes information overload, while prompt-based selection fails to assess context-dependent utility. To address this, we propose **AuTAgent** (**Au**dio **T**ool **Agent**), a reinforcement learning framework that learns when and which tools to invoke. By employing a sparse-feedback training strategy with a novel Differential Reward mechanism, the agent learns to filter out irrelevant tools and invokes external assistance only when it yields a net performance gain over the base model. Experimental results confirm that AuTAgent complements the representation bottleneck of LALMs by providing verifiable acoustic evidence. It improves accuracy by 4.20% / 6.20% and 9.80% / 8.00% for open-source and closed-source backbones on the MMAU Test-mini and the MMAR benchmarks, respectively. In addition, further experiments demonstrate exceptional transferability. We highlight the complementary role of external tools in augmenting audio model reasoning.
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DLEBench: Evaluating Small-scale Object Editing Ability for Instruction-based Image Editing Model
Shibo Hong, Boxian Ai, Jun Kuang, Wei Wang, FengJiao Chen, Zhongyuan Peng, Chenhao Huang, Yixin Cao
Significant progress has been made in the field of Instruction-based Image Editing Models (IIEMs). However, while these models demonstrate plausible adherence to instructions and strong reasoning ability on current benchmarks, their ability to edit small objects remains underexplored, despite its importance for precise local editing and refining details in both real and generated images.
In this paper, we introduce DeepLookEditBench (DLEBench), the first benchmark dedicated to assessing the abilities of IIEMs in editing small-scale objects.
Specifically, we construct a challenging testbed comprising 1889 samples across seven instruction types. In these samples, target objects occupy only 1%-10% of the image area, covering complex scenarios such as partial occlusion and multi-object editing.
To ensure robust evaluation on this benchmark, we propose an evaluation protocol with refined score rubrics to minimize subjectivity and ambiguity in two criteria: Instruction Following and Visual Consistency. This protocol also introduces a dual-mode evaluation framework (Tool-driven and Oracle-guided Modes) addressing the misalignment between LMM-as-a-Judge and human judgments on DLEBench.
Empirical results on 10 IIEMs reveal significant performance gaps in small-scale object editing, highlighting the need for specialized benchmarks to advance this ability.
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UltraLIF: Fully Differentiable Spiking Neural Networks via Ultradiscretization and Max-Plus Algebra
Jose Marie Antonio Miñoza

Spiking Neural Networks (SNNs) offer energy-efficient, biologically plausible co
mputation but suffer from non-differentiable spike generation, necessitating rel
iance on heuristic surrogate gradients. This paper introduces **UltraLIF**, a pr
incipled framework that replaces surrogate gradients with \emph{ultradiscretizat
ion}, a mathematical formalism from tropical geometry providing continuous relax
ations of discrete dynamics. The central insight is that the max-plus semiring u
nderlying ultradiscretization naturally models neural threshold dynamics: the lo
g-sum-exp function serves as a differentiable soft-maximum that converges to har
d thresholding as a learnable temperature parameter $\varepsilon \to 0$. Two neu
ron models are derived from distinct dynamical systems: UltraLIF from the LIF or
dinary differential equation (temporal dynamics) and UltraDLIF from the diffusio
n equation modeling gap junction coupling across neuronal populations (spatial d
ynamics). Both yield fully differentiable SNNs trainable via standard backpropag
ation with no forward-backward mismatch. Theoretical analysis establishes pointw
ise convergence to tropical LIF dynamics with quantitative error bounds and boun
ded non-vanishing gradients. Experiments on six benchmarks spanning static image
s, neuromorphic vision, and audio demonstrate improvements over surrogate gradie
nt baselines, with gains most pronounced in the ultra-low latency regime ($T{=}1
$) on neuromorphic and temporal datasets. An optional sparsity penalty enables s
ignificant energy reduction while maintaining competitive accuracy.
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On the Separability of Information in Diffusion Models
Akhil Premkumar
Diffusion models transform noise into data by injecting information that was cap
tured in their neural network during the training phase. In this paper we ask: w
hat is this information? We find that, in pixel-space diffusion models, (1) a la
rge fraction of the total information in the neural network is committed to reco
nstructing small-scale perceptual details of the image, and (2) the correlations
 between images and their class labels are informed by the semantic content of t
he images, and are largely agnostic to the low-level details. We argue that thes
e properties are intrinsically tied to the manifold structure of the data itself
. Finally, we show that these facts explain the efficacy of classifier-free guid
ance: the guidance vector amplifies the mutual information between images and co
nditioning signals early in the generative process, influencing semantic structu
re, but tapers out as perceptual details are filled in.
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Sheaf Neural Networks on SPD Manifolds: Second-Order Geometric Representation Le
arning
Yuhan Peng, Junwen Dong, Yuzhi Zeng, Hao Li, Ce Ju, Huitao Feng, Diaaeldin Taha,
 Anna Wienhard, Kelin Xia
Graph neural networks face two fundamental challenges rooted in the linear struc
ture of Euclidean vector spaces: (1) Current
architectures represent geometry through vectors (directions, gradients), yet ma
ny tasks require matrix-valued representations that capture relationships betwee
n directions—such as how atomic orientations covary in a molecule. These second-
order representations are naturally captured by points on the symmetric positiv
e definite matrices (SPD) manifold; (2) Standard message passing applies shared
transformations across edges. Sheaf neural networks address this via edge-specif
ic transformations, but existing formulations remain confined to vector spaces a
nd therefore cannot propagate matrix-valued features. We address both challenges
 by developing the first
sheaf neural network operates natively on the SPD manifold. Our key insight is t
hat the SPD manifold admits a Lie group structure, enabling well-posed analogs o
f sheaf operators without projecting to Euclidean space. Theoretically, we prove
 that SPD-valued sheaves are strictly more expressive than Euclidean sheaves: th
ey admit consistent configurations (global sections) that vector-valued
sheaves cannot represent, directly translating to richer learned representations
. Empirically, our sheaf convolution transforms effectively rank-1 directional
inputs into full-rank matrices encoding local geometric structure. Our dual-stre
am architecture achieves SOTA on 6/7 MoleculeNet benchmarks, with the sheaf fram

ework providing consistent depth robustness.
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Automatic Pruning Discovery for Large Language Models
Haidong Kang, Lihong Lin, Enneng Yang, Hong-Ning Dai, Hao Wang
Large language models (LLMs) have achieved remarkable performance on a wide rang e of tasks, hindering real-world deployment due to their massive size. Existing pruning methods (e.g., Wanda) tailored for LLMs rely heavily on manual design pr uning algorithms, thereby leading to $\textit{huge labor costs}$ and $\textit{re quires expert knowledge}$. Furthermore, we are the first to identify the serious \textit{outlier value issue} behind dramatic performance degradation under high pruning ratios that are caused by uniform sparsity, raising an additional conce rn about how to design adaptive pruning sparsity ideal for LLMs. Can LLMs prune by themselves? In this work, we introduce an affirmative answer by proposing a n ovel pruning method called $\textbf{AutoPrune}$, which first overcomes expert kn owledge limits by leveraging LLMs to design optimal pruning algorithms for thems elves automatically without any expert knowledge. Specifically, to mitigate the black-box nature of LLMs, we propose a Graph-driven Chain-of-Thought (GCoT) to o ptimize prompts, significantly enhancing the reasoning process in learning the p runing algorithm and enabling us to generate pruning algorithms with superior pe rformance and interpretability in the next generation. Finally, grounded in insi ghts of outlier value issue, we introduce Skew-aware Dynamic Sparsity Allocation (SDSA) to overcome the outlier value issue, mitigating performance degradation under high pruning ratios. We conduct extensive experiments on mainstream LLMs b enchmarks, demonstrating the superiority of AutoPrune, which consistently excels state-of-the-art competitors. The code is available at: \url{https://anonymous. 4open.science/r/AutoPrune}.
****
SAQNN: Spectral Adaptive Quantum Neural Network as a Universal Approximator
Jialiang Tang, Jialin Zhang, Xiaoming Sun
Quantum machine learning (QML), as an interdisciplinary field bridging quantum c omputing and machine learning, has garnered significant attention in recent year s. Currently, the field as a whole faces challenges due to incomplete theoretica l foundations for the expressivity of quantum neural networks (QNNs). In this pa per we propose a constructive QNN model and demonstrate that it possesses the un iversal approximation property (UAP), which means it can approximate any square- integrable function up to arbitrary accuracy. Furthermore, it supports switching function bases, thus adaptable to various scenarios in numerical approximation and machine learning. Our model has asymptotic advantages over the best classica l feed-forward neural networks in terms of circuit size and achieves optimal par ameter complexity when approximating Sobolev functions under L_2 norm.
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Enhancing Membership Inference Attacks on Diffusion Models from a Frequency-Doma in Perspective
Puwei Lian, Yujun Cai, Songze Li, Bingkun BAO
Diffusion models have achieved tremendous success in image generation, but they also raise significant concerns regarding privacy and copyright issues. Membersh ip Inference Attacks (MIAs) are designed to ascertain whether specific data was utilized during a model's training phase. As current MIAs for diffusion models t ypically exploit the model's image prediction ability, we formalize them into a unified general paradigm that computes the membership score for membership ident ification. Under this paradigm, we empirically find that existing attacks overlo ok the inherent deficiency in how diffusion models process high-frequency inform ation. Consequently, this deficiency leads to member data with more high-frequen cy content being misclassified as hold-out data, and hold-out data with less hig h-frequency content tends to be misclassified as member data. Moreover, we theor etically demonstrate that this deficiency reduces the membership advantage of at tacks, thereby interfering with the effective discrimination of member data and hold-out data. Based on this insight, we propose a plug-and-play high-frequency filter module to mitigate the adverse effects of the deficiency, which can be se amlessly integrated into any attacks within the general paradigm without additio

nal time costs. Extensive experiments corroborate that this module significantly improves the performance of baseline attacks across different datasets and models. Code is available at https://github.com/poetic2/FreMIA.

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TodoEvolve: Learning to Architect Agent Planning Systems

Jiaxi Liu, Guibin Zhang, Yanzuo Jiang, Zihan Zhang, Heng Chang, Zhenfei Yin, Qibing Ren, Junchi Yan

Planning has become a central capability for contemporary agent systems in navigating complex, long-horizon tasks, yet existing approaches predominantly rely on fixed, hand-crafted planning structures that lack the flexibility to adapt to the structural diversity of open-ended problems. To address this limitation, we introduce TodoEvolve, a meta-planning paradigm that autonomously synthesizes and dynamically revises task-specific planning architectures. Specifically, we first construct PlanFactory, a modular design space that standardizes diverse planning paradigms within a unified codebase encompassing topology, initialization, adaptation, and navigation, thereby providing a common interface for heterogeneous planning patterns. Leveraging PlanFactory, we collect high-quality planning trajectories and train Todo-14B via Impedance-Guided Preference Optimization (IGPO), a multi-objective reinforcement learning objective that encourages the generation of planning systems that are performant, stable, and token-efficient across arbitrary tasks and agent backbones. Empirical evaluations on five agentic benchmarks demonstrate that TodoEvolve consistently surpasses carefully engineered planning modules while maintaining economical API costs and runtime overhead. Our codes are available at \url{https://github.com/EcthelionLiu/TodoEvolve}.

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Generalist Graph Anomaly Detection via Prototype-Based Distillation

Yiming Xu, Zihan Chen, Zhen Peng, Song Wang, Bin Shi, Bo Dong, Chao Shen

Driven by the pressing demand for graph anomaly detection (GAD) in high-stakes domains, the generalist GAD paradigm, which trains a single detector transferable across new graphs, has recently gained growing attention. However, existing methods often rely on scarce and costly annotations for training and sometimes even require few-shot support at inference, which limits their robustness to diverse and unseen anomaly patterns. To address this limitation, we introduce ProMoS, the first unsupervised generalist GAD framework, which detects anomalies by modeling the abundant normality in unlabeled data.

ProMoS adopts a knowledge-distillation paradigm to distill normality priors from a frozen self-supervised graph neural network (GNN) teacher to a mixture-of-students model with shared global and lightweight personalized branches, enabling efficient and expressive normality modeling without learning from scratch.

We further propose prototype-guided soft-label distillation to align teacher and student in a shared prototype space, enhancing cross-graph generalizability. During inference, ProMoS performs zero-shot anomaly detection on unseen graphs via distillation bias and prototype geometric deviation.

Extensive experiments show the effectiveness and efficiency of ProMoS, charting a practical path toward label-free, zero-shot generalist GAD.

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AI Engram: In Search of Memory Traces in Artificial Intelligence

Jea Kwon, Dong-Kyum Kim, Jiwon Kim, Yonghyun Kim, Woong Kook, Meeyoung Cha

Memory formation is fundamental to intelligence, yet whether deep neural networks preserve identifiable memory traces analogous to biological memory units remains an open question. This work introduces a geometric framework to identify such “AI engrams” by formalizing the neuroscientific criteria of specificity, reactivation, sufficiency, and necessity into a constrained inverse problem. We derive a closed-form estimator that isolates individual memory traces from globally entangled parameters, and show that this biologically-derived solution corresponds to a natural gradient update on the parameter manifold. AI engrams enable surgical manipulation of learned knowledge: any subset of memories can be composed or erased through linear arithmetic, without iterative optimization. Experiments ranging from simple MLPs to LLMs demonstrate the causal validity and substantial scalability of AI engrams. Together, these results bridge theories of biological

memory and artificial representation learning and offer geometric insight into how deep networks simultaneously support functional specificity within distributed storage.

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Differentially Private Range Subgraph Counting

Xian Chen, Ruobing Bai, Pan Peng

Subgraph counting is a fundamental problem in graph analysis. Motivated by practical scenarios where graph analytics are performed on subgraphs induced by selected vertices -- rather than on the entire graph -- and by growing privacy concerns, we initiate the study of *differentially private range subgraph counting (DPRSC)*. The goal is to privately count occurrences of a fixed pattern graph within induced subgraphs defined by multi-dimensional attribute ranges. Unlike classical point counting, subgraph counting is inherently nonlinear and exhibits high sensitivity: a single edge modification can affect many subgraph occurrences. We present the first efficient algorithms for DPRSC with small additive error. Our approach introduces a subgraph projection that reduces DPRSC to weighted orthogonal range counting, enabling the use of range trees and local sensitivity estimation to achieve accurate private query answering. We complement our algorithms with matching lower bounds, obtained by reducing reconstruction attacks to DPRSC and leveraging discrepancy theory. In particular, we show that any differentially private algorithm for DPRSC must incur additive error exponential in the dimension. Empirical evaluations demonstrate that our algorithms significantly outperform baseline methods in accuracy and runtime while maintaining strong privacy guarantees.

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SaTeen: Learning Structural Alignment for Continual Test-Time Adaptation

Chang Liu, Ruotong Zhao, Li Gao, Yupei Zhang

Test-Time Adaptation (TTA) aims to reconcile model generalization in the presence of distribution shifts. Current TTA methods usually leverage sample uncertainty to select reliable samples for model adjustment via entropy minimization (EM). However, sample uncertainty often relies on a plausible metric and leaves many unreliable samples into EM process, potentially leading to model collapse. Importantly, these excluded samples incur biased data features of the shifted distribution in TTA. This paper introduces SaTeen, a **S**tructural **A**lignment-based **Te**st-Tim**e** Adaptatio**n** method that performs two-fold aligning the structures of test samples with the reliable reference structures. Specifically, the two-fold alignments are 1) Intra-sample structure alignment, where SaTeen maximizes cross-entropy discrepancy between a sample (reference) and its structure-disrupted counterpart, with the assumption of stable dominant features; 2) Inter-sample structure alignment, where SaTeen minimizes the reconstruction error of test samples in the reference subspace spanned by the Incremental PCA on reliable samples, with the assumption of stale intrinsic data manifold. Our extensive experiments demonstrate that SaTeen achieves the state-of-the-art performance across various scenarios for both TTA and continual TTA.

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Quantifying Error Propagation and Model Collapse in Diffusion Models

Nail B. Khelifa, Richard E. Turner, Ramji Venkataramanan

Machine learning models are increasingly trained or fine-tuned on synthetic data. Recursively training on such data has been observed to significantly degrade performance in a wide range of tasks, often characterized by a progressive drift away from the target distribution. In this work, we theoretically analyze this phenomenon in the setting of score-based diffusion models. For a realistic pipeline where each training round uses a combination of synthetic data and fresh samples from the target distribution, we obtain upper and lower bounds on the accumulated divergence between the generated and target distributions. Notably, to the best of our knowledge, this is the first lower bound on the divergence between the learned and target distributions, even for standard diffusion models. Our results allow us to characterize different regimes of drift, depending on the score estimation error and the proportion of fresh data used in each generation. In a certain regime, the accumulated divergence after several retraining rounds can

be expressed as a discounted sum of score estimation errors made at each generation. We also provide empirical results on synthetic data and images to illustrate the theory.
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Geometric Decoupling: Diagnosing the Structural Instability of Latent
Yuanbang Liang, Zhengwen Chen, Yu-Kun Lai
Latent Diffusion Models (LDMs) achieve high-fidelity synthesis but suffer from latent space brittleness, causing discontinuous semantic jumps during editing. We introduce a Riemannian framework to diagnose this instability by analyzing the generative Jacobian, decomposing geometry into *Local Scaling* (capacity) and *Local Complexity* (curvature). Our study uncovers a **"Geometric Decoupling"**: while curvature in normal generation functionally encodes image detail, Out-of-Distribution (OOD) generation exhibits a functional decoupling where extreme curvature is wasted on unstable semantic boundaries rather than perceptible details. This geometric misallocation identifies "Geometric Hotspots" as the structural root of instability, providing a robust intrinsic metric for diagnosing generative reliability. Our code is at https://github.com/Byronliang8/Diffusion-Geometry.
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Enhancing Numerical Prediction in LLMs via Smooth MMD Alignment
Zhuo Zuo, Li Yue, Wenhao Zheng, Chenpeng Wang, Xianggen Liu
Despite their strong general capabilities, large language models (LLMs) often remain unreliable when outputs must be numerically precise. A key reason is the training objective: standard cross-entropy treats numeric tokens as unstructured categories and ignores the metric structure of their values. We address this mismatch by proposing **S**mooth **M**aximum **M**ean **D**iscrepancy (**SMMD**), which builds on the classic MMD by incorporating value-distance kernels over numeric tokens and graph-based smoothness. With this kernel defined over a numeric sub-vocabulary, SMMD aligns the predicted numeric distribution to the target via kernel matching and smooths the prediction--target residual over the induced kernel graph to encourage local consistency. We evaluate SMMD on four numeric-target tasks---mathematical reasoning, arithmetic calculation, clock-time recognition, and chart question answering---across multiple open-weight LLM and VLM backbones. SMMD consistently improves accuracy over both cross-entropy and recent numeric-target losses; analyses show complementary effects between MMD and smoothness and underscore the importance of distance-based kernel design.
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Do-Prompt: Causal Interventions Meet Variational Prompt Bottlenecks
Xueting Chen, Jun-Jie Huang, Yan Yan, Long Lan, Yuhua Tang, Wenjing Yang
Multi-modal prompt learning is a parameter-efficient approach to adapting large vision-language models to downstream classification tasks. However, prompts can inadvertently evolve into a high-capacity pathway that encodes environment-dependent spurious correlations, which are predictive only in the source domain and thereby undermine transferability.
To address this issue, this paper introduces **Do-Prompt**, a *compress-and-intervene* framework that brings together variational bottlenecks and causal interventions for robust prompt tuning. We model prompts as stochastic latent variables and impose a *variational prompt bottleneck* to explicitly regulate the information transmitted through prompts, effectively mitigating their tendency to memorize spurious nuisance cues.
Building on this capacity constraint, we propose lightweight *prompt-level interventions* by perturbing the environment-related prompt components and enforcing prediction consistency under these *do*-style perturbations. This synergistic integration encourages reliance on task-stable, invariant semantics rather than spurious prompt content.
Notably, Do-Prompt is plug-and-play compatible with existing multi-modal prompt tuning pipelines and introduces negligible computational overhead. Extensive experiments on base-to-novel generalization, cross-dataset transfer, and ImageNet distribution shifts demonstrate consistent performance gains, with particularly notable improvements on datasets exhibiting pronounced domain or texture biases.
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Two-Stage Unit Tying for Simplifying Differentiable Logic Gate Networks
Seungheon Lee, Jeongmin Sun, Jaeyong Chung
Differentiable logic gate networks map directly to gate-level circuits, enabling ultra-low-latency inference, yet their logic footprint often exceeds FPGA capacity budgets. Tightly fitting a trained model to a target FPGA requires a post-training mechanism to trade off network complexity and accuracy—analogous to pruning in standard neural networks. To this end, we introduce *unit tying*: a simplification that forces selected gates to constants (0 or 1), enabling constant propagation and downstream logic elimination. However, we observe that naively extending pruning criteria to logic networks is unreliable under such near-discrete modifications. We therefore propose a two-stage algorithm for unit tying: (i) a fast Gauss–Newton screening step under a teacher-referenced logit-distortion objective that constructs a high-recall overshoot set and (ii) a refinement step that corrects approximation and interaction-driven errors using a small number of finite-difference evaluations. On CIFAR-10 and MNIST, our method consistently improves the accuracy–area trade-off over common saliency baselines, yielding substantial post-synthesis LUT reductions of up to 48% on CIFAR-10 and 43% on MNIST, with modest accuracy degradation.
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ZipMoE: Efficient On-Device MoE Serving via Lossless Compression and Cache-Affinity Scheduling
Yuchen Yang, Yaru Zhao, Pu Yang, Shaowei Wang, Zhi-Hua Zhou
While Mixture-of-Experts (MoE) architectures substantially bolster the expressive power of large-language models, their prohibitive memory footprint severely impedes the practical deployment on resource-constrained edge devices, especially when model behavior must be preserved without relying on lossy quantization. In this paper, we present ZipMoE, an efficient and semantically lossless on-device MoE serving system. ZipMoE exploits the synergy between the hardware properties of edge devices and the statistical redundancy inherent to MoE parameters via a caching-scheduling co-design with provable performance guarantee. Fundamentally, our design shifts the paradigm of on-device MoE inference from an I/O-bound bottleneck to a compute-centric workflow that enables efficient parallelization. We implement a prototype of ZipMoE and conduct extensive experiments on representative edge computing platforms using popular open-source MoE models and real-world workloads. Our evaluation reveals that ZipMoE achieves up to 72.77% inference latency reduction and up to $6.76\times$ higher throughput than the state-of-the-art systems. Our code is available at: https://github.com/npnothard/ZipMoE-ICML26.
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WhisperSplat: Lossless Steganography in 3D Gaussian Splatting
Nicole Meng, Ronak Sahu, Miao Yin, Faysal Hossain Shezan, Yingjie Lao
We present WhisperSplat, the first lossless steganography method for 3D Gaussian Splatting (3DGS) models that hides a full■resolution 2D image in a single view without any degradation of the model's rendering quality elsewhere. Prior work embeds data by retraining or modifying model weights, altering novel■view synthesis fidelity and limiting capacity. Instead, we learn a small, view■specific noise key applied to each Gaussian's spherical■harmonic (SH) features while keeping all other views remain indistinguishable from the original renders. We further propose a Gradual Pixel Perturbation (GPP) strategy with a cosine-decay schedule, bootstrapping fast divergence from the clean render before transitioning to a combined reconstruction and SSIM loss. Unlike prior works that are highly dependent on accurate and large pretrained decoders, our method is able to recover the hidden image through rendering with noise key, and an optional lightweight refiner to enhance recovery image quality. Across nine standard 3DGS data scenes, WhisperSplat demonstrates superior hidden image recovery quality without sacrifice in clean 3DGS model performance, when compared to prior work such as GS-Hider and KeySS.
****

Memory Savings at What Cost? A Study of Alternatives to Backpropagation
Kunjal Panchal, Sunav Choudhary, Yuriy Brun, Hui Guan

Forward-mode automatic differentiation (FmAD) and zero-order (ZO) optimization a re increasingly proposed as memory-efficient, backpropagation-free alternatives for large language model (LLM) fine-tuning, yet their benefits are typically eva luated only against standard backpropagation (BP), omitting memory-efficient var iants such as activation checkpointing. We present a unified theoretical and emp irical comparison of BP, checkpointed BP, FmAD, and ZO for LLM and vision-langua ge model training, showing that while FmAD and ZO reduce activation memory, they trade memory for higher computational cost and longer wall-clock time to conver gence, resulting in lower accuracy and slower training, especially under constra ined perturbation budgets. Across models, BP with checkpointing outperforms FmAD and ZO variants, including variance-reduced methods, achieving up to 31.1% high er accuracy, 34.8% faster convergence, and 3.8$\times$ fewer computations at com parable memory usage, while also revealing instability-related failure modes in FmAD and ZO. Overall, our results correct a one-sided benchmarking narrative by showing that memory-efficient methods entail fundamentally different trade-offs, and that ignoring these distinctions has led to misleading conclusions about LL M optimization in prior work.

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Gradient Flow Sampler-based Distributionally Robust Optimization

Zusen Xu, Jia-Jie Zhu

We propose a mathematically principled PDE gradient flow framework for distribut ionally robust optimization (DRO). Exploiting the recent advances in the interse ction of Monte Carlo sampling and statistical optimal transport, we show that ou r theoretical framework can be implemented as practical algorithms for sampling from worst-case distributions and, consequently, DRO.

While numerous previous works have relied on dual reformulation techniques, we c ontribute a sound and complete gradient flow view based on SDEs or PDEs that can be used to construct new algorithms for general, potentially non-convex, losses .

Without loss of generality, we solve a class of Wasserstein and entropy-regulari zed DRO problems using the recently-discovered Wasserstein Fisher-Rao and Stein variational gradient flows. Notably, we also show some simple reductions of our framework recover exactly previously proposed popular DRO methods, and provide n ew insights into their theoretical limits and optimization dynamics of DRO. Nume rical studies based on stochastic gradient descent on machine learning tasks pro vide empirical backing for our theoretical findings.

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Spectral Flow Matching: Stabilizing Stochastic GFlowNets via Frequency-Domain Re gularization

Nadhir Hassen, Johan W. Verjans

Generative Flow Networks (GFNs) offer a powerful paradigm for diverse sampling, yet they often exhibit instability and poor convergence when applied to stochast ic or sparse-reward environments. To mitigate the high variance inherent in thes e settings, we propose a fundamental re-framing of the GFlowNet training objecti ve within the frequency domain. We present \textbf{Spectral Time-Dependent GFlow Nets (ST-GFNs)}, a framework that leverages Fourier analysis to enforce smoothne ss and stability in learned policies. Our theoretical analysis proves that our p roposed spectral loss is mathematically equivalent to regularized value iteratio n, acting as a principled low-pass filter that separates signal from noise. Furt hermore, we tackle the challenge of exploration in sparse landscapes by introduc ing a novel autocorrelated intrinsic reward derived from the Wiener-Khinchin the orem. Through extensive experiments ranging from adversarial games and noisy seq uence generation to high-dimensional single-cell perturbation modelling, we demo nstrate that ST-GFNs significantly outperform existing baselines in terms of rob ustness, sample efficiency, and mode discovery.

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User-Aware Active Knowledge Acquisition for Emotional Support Dialogue

Mufan Xu, Kehai Chen, Jiahao Hu, Xinchao Xu, Muyun Yang, Tiejun Zhao, Min Zhang

Emotional support plays an important role in dialogue systems, and its success d epends on adapting to a user’s evolving and implicit needs across multi-turn int

eractions while leveraging the strong reasoning capacity of large language model
s.
However, since signals about user needs are often weak, indirect, and can only b
e disambiguated through multi-turn interaction, existing emotional support metho
ds often struggle to acquire and generalize relevant conversational knowledge ef
ficiently.
To bridge this gap, we introduce User-Aware Active Knowledge Acquisition (UKA),
a gradient-free active dialogue learning framework that explicitly represents un
certainty about user needs and incorporates active learning into both knowledge
acquisition and response selection.
We propose a Theory-of-Mind uncertainty estimation mechanism that allows the mod
el to prioritize responses, thereby eliciting more informative user feedback.
UKA is capable of efficiently exploring user-aligned conversational knowledge du
ring training while maintaining robustness at test time.
Experiments across multiple dialogue benchmarks and model architectures demonstr
ate that our approach consistently outperforms strong baselines in dialogue qual
ity and user alignment.
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Breaking the Reference Bottleneck via Learning to Rewrite Conversational Queries
 without Gold Reference Passages
Doyoung Kim, Youngjun Lee, Joeun Kim, Jihwan Bang, Hwanjun Song, Susik Yoon, Jae
-Gil Lee
Conversational query reformulation (CQR) has become indispensable for improving
retrieval in dialogue-based applications. However, existing approaches typically
 rely on reference passages for optimization, which are **impractical** to acqui
re in real-world scenarios. To address this limitation, we introduce a novel **r
eference-free** preference optimization framework ***DualReform*** that generate
s **pseudo reference passages** from **commonly-encountered** conversational dat
asets containing only queries and responses. DualReform attains this goal throug
h two key innovations: (1) **response-based inference**, where responses serve a
s proxies to infer pseudo reference passages, and (2) **response refinement via
the dual-role of CQR**, where a CQR model refines responses based on the shared
objectives between response refinement and CQR. Despite not relying on reference
 passages, ***DualReform*** achieves 96.9--99.1% of the retrieval accuracy attai
nable only with reference passages and surpasses the state-of-the-art method by
up to 31.6%.
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Experience Augmented Policy Optimization for LLM Reasoning
Jinda Lu, Kexin Huang, Junkang Wu, Shuo Yang, Jinghan Li, Chiyu Ma, Shaohang Wei
, Xiang Wang, Guoyin Wang, Jingren Zhou
Reinforcement Learning with Verifiable Rewards (RLVR) is a powerful paradigm for
 improving the reasoning capabilities of large language models (LLMs).
However, existing RLVR methods typically rely on on-policy optimization from scr
atch, resulting in high sampling costs and inefficient utilization of accumulate
d experience.
As model capabilities and policy behaviors evolve during training, recent attemp
ts to reuse experience via fixed reasoning trajectories further suffer from poli
cy mismatch.
Motivated by these limitations, we argue that experience in RLVR should not be r
eused as fixed reasoning trajectories, but instead expressed in a policy-adaptiv
e manner.
In this work, we propose Experience-Augmented Policy Optimization (EAPO), which
leverages a prior RL-optimized policy as an action-level experience prior and se
lectively injects experience at critical decision points during rollout.
To ensure stable and unbiased learning from experience-augmented rollouts, EAPO
further incorporates an adapted importance sampling scheme.
Experiments on using Qwen-2.5-math 7b and Qwen-3-8B on five different benchmarks
 demonstrate that EAPO consistently improves reasoning performance over state-of
-the-art RLVR methods.
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Bulk-Calibrated Credal Ambiguity Sets: Fast, Tractable Decision Making under Out-of-Sample Contamination
Mengqi Chen, Thomas Berrett, Theodoros Damoulas, Michele Caprio
Distributionally robust optimisation (DRO) minimises the worst-case expected loss over an ambiguity set that can capture distributional shifts in out-of-sample environments. While Huber (linear-vacuous) contamination is a classical minimal-assumption model for an ε-fraction of arbitrary perturbations, including it in an ambiguity set can make the worst-case risk infinite and the DRO objective vacuous unless one imposes strong boundedness or support assumptions. We address these challenges by introducing bulk-calibrated credal ambiguity sets: we learn a high-mass bulk set from data while considering contamination inside the bulk and bounding the remaining tail contribution separately. This leads to a closed-form, finite $\mathrm{mean}+\sup$ robust objective and tractable linear or second-order cone programs for common losses and bulk geometries. Through this framework, we highlight and exploit the equivalence between the imprecise probability (IP) notion of upper expectation and the worst-case risk, demonstrating how IP credal sets translate into DRO objectives with interpretable tolerance levels. Experiments on heavy-tailed inventory control, geographically shifted house-price regression, and demographically shifted text classification show competitive robustness-accuracy trade-offs and efficient optimisation times, using Bayesian, frequentist, or empirical reference distributions.
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A Tale of Two Problems: Multi-Task Bilevel Learning Meets Equality Constrained Multi-Objective Optimization
Zhiyao Zhang, Myeung Suk Oh, Zhen Qin, Jiaxiang Li, Xin Zhang, Jia Liu
In recent years, bilevel optimization (BLO) has attracted significant attention for its broad applications in machine learning.
However, most existing works on BLO remain confined to the single-task setting and rely on the lower-level strong convexity assumption, which significantly restricts their applicability to modern machine learning problems of growing complexity.
In this paper, we make the first attempt to extend BLO to the multi-task setting under a relaxed lower-level general convexity (LLGC) assumption.
To this end, we reformulate the multi-task bilevel learning (MTBL) problem with LLGC into an equality constrained multi-objective optimization (ECMO) problem.
However, ECMO itself is a new problem that has not yet been studied in the literature.
To address this gap, we first establish a new Karush–Kuhn–Tucker (KKT)-based Pareto stationarity as the convergence criterion for ECMO algorithm design.
Based on this foundation, we propose a weighted Chebyshev (WC)-penalty algorithm that achieves a finite-time convergence rate of $\mathcal{O}(ST^{-\frac{1}{2}})$ to KKT-based Pareto stationarity in both deterministic and stochastic settings, where S denotes the number of objectives, and T is the total iterations.
Moreover, by varying the preference vector over the S-dimensional simplex, our WC-penalty method systematically explores the Pareto front.
Finally, solutions to the ECMO problem translate directly into solutions for the original MTBL problem, thereby closing the loop between these two foundational optimization frameworks.
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SpaEF: Spatially Resolved Transcriptomics Data Element-Wise Denoising Framework Powered by Large Models
Zekuan Shang, Xiaosong Han, Liupu Wang, Wei Du, Peng Zhao, Yuanshu Li, Yubin Xiao, Xuan Wu, You Zhou
For denoising Spatially Resolved Transcriptomics (SRT) data, existing methods often construct spot and gene graphs to model inter-spot and inter-gene relationships, respectively. However, these methods often introduce spurious similarity biases among spots when constructing the spot graph and fail to capture nonlinear relationships among genes when constructing the gene graph. Moreover, ineffective graph fusion strategies further bottleneck denoising performance. To address these challenges, we propose SpaEF, which innovatively constructs spot and gene

graphs with two Large Models (LMs) to inject prior knowledge for mitigating bias es and capture nonlinear relationships, and then fuses them with the proposed el ement-wise graph autoencoder. As far as we know, SpaEF is the first SRT denoisin g method that utilizes pre-trained LMs to construct spot and gene graphs. Experi ments on four real-world datasets with corresponding downstream tasks demonstrat e that SpaEF not only outperforms SOTA denoising methods in accuracy but also ex hibits strong robustness across tasks.
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Spectral Heat Flow for Conservative Token Condensation in Vision-Language Models
Zhaoyang Li, Yanjun Li, Wangkai Li, Yujia Chen, Tianzhu Zhang
Vision-Language Models (VLMs) are costly at inference time because they must pro cess long sequences of visual tokens. Existing token pruning methods often degra de under high compression by blindly discarding information, breaking spatial st ructure or collapsing diversity. We propose SpecFlow, a training-free framework that shifts the paradigm from destructive pruning to conservative condensation, strictly enforcing spatial coverage and statistical conservation to ensure stabi lity. Treating visual tokens as nodes in a kNN graph, SpecFlow (i) computes a stable importance field via spectral heat flow to preserve structural coherence, (ii) allocates budgets via adaptive spatial partitioning to guarantee coverage, and (iii) aggregates discarded information into coreset sinks to maintain stati stical conservation. The method is plug-and-play, requires no fine-tuning, and i s compatible with FlashAttention. Experiments confirm that our SpecFlow outperfo rms SOTA methods across tasks, VLM architectures, and pruning ratios. Notably, L LaVA-1.5 with SpecFlow retains 95.6\% of original performance despite pruning 88 .9\% of visual tokens, offering an exceptional efficiency-accuracy balance.
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DecFus: Decentralized Layer-wise Fusion with Dynamic Exploration and Exploitatio n
Li Yang, Jialong Sun, Chuhai Cai, Xinyang Liu, Yichen Li, Bowen Peng, Jialong Li , Bo Liu
Decentralized Federated Learning (DFL) enables collaborative model training acro ss connected clients without a central server, effectively mitigating communicat ion bottlenecks and avoiding the single point of failure in Centralized Federate d Learning (CFL). However, existing DFL methods mostly focus on parameter averag ing with compromised update directions, which limits their performance potential due to insufficient exploration of the loss landscape, especially for complex m odels. We observe that layer exchanges among clients enhance exploration while i ntroducing instability due to highly diverse update directions. To address these limitations, we propose Decentralized Layer-wise Fusion (DecFus), the first DFL framework that unifies layer-level exchange and averaging to balance exploratio n and exploitation. DecFus dynamically transitions the decentralized training pr ocess from exploration-dominant to exploitation-dominant phases, guided by the l oss variance among connected neighbors. Furthermore, a layer-wise fusion strateg y, informed by pairwise cosine similarity, categorizes all layers into two group s: an exchange group for exploration and an averaging group for exploitation. Sp ecifically, we theoretically establish the convergence of DecFus without relying on the common assumption in existing literature that the aggregation matrix mus t be doubly stochastic. Extensive experiments demonstrate that DecFus achieves s uperior performance in both IID and non-IID scenarios, substantially outperformi ng existing CFL and DFL methods.
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LOZO+: Provably Efficient Zeroth-Order Fine-Tuning via Greedy Low-Rank Subspace Selection
Jinjie Fang, Chengxun Jin, Tianxing Man, Yi Chang, Bin Gu
Zeroth-order (ZO) optimization offers a more memory-efficient alternative to fir st-order methods for fine-tuning large language models (LLMs). Recent ZO metho ds, exemplified by LOZO, estimate gradients within low-rank subspaces to align w ith the low-rank structure of LLM gradients. However, these methods rely on rand omly generated subspaces of a fixed rank, which provides no guarantee of alignme nt with the actual dominant subspaces of the gradients; essentially, they remain

ZO gradient descent with stochastic subspace sampling. To more effectively explo
it the low-rank nature of LLM gradients, we propose \textbf{LOZO+}, an efficien
t \textbf{ZO} fine-tuning algorithm for LLMs that incorporates greedy \textbf{Lo
}w-Rank subspace selection. Specifically, LOZO+ leverages loss-based feedback to
assess alignment between candidate directions and the dominant low-rank gradien
t subspaces, and employs an adaptive thresholding criterion to retain only direc
tions yielding substantial gradient descent, thereby steering ZO optimization to
ward more effective convergence. Importantly, we establish a theoretical framew
ork that characterizes the convergence behavior of LOZO+, formally prove its sup
eriority over existing methods. Extensive experiments demonstrate that LOZO+ co
nsistently outperforms existing ZO methods and achieves performance competitive
with FO algorithm, while retaining the memory efficiency inherent to ZO optimiza
tion.
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Phase-Type Variational Autoencoders for Heavy-Tailed Data
Abdelhakim Ziani, Andras Horvath, Paolo Ballarini
Heavy-tailed distributions are ubiquitous in real-world data, where rare but ext
reme events dominate risk and variability. However, standard Variational Autoenc
oders (VAEs) employ simple decoder distributions (e.g., Gaussian) that fail to c
apture heavy-tailed behavior, while existing heavy-tail-aware extensions remain
restricted to predefined parametric families whose tail behavior is fixed a prio
ri. We propose the Phase-Type Variational Autoencoder (PH-VAE), whose decoder d
istribution is a latent-conditioned Phase-Type (PH) distribution—defined as the
absorption time of a continuous-time Markov chain (CTMC). This formulation compo
ses multiple exponential time scales, yielding a flexible, analytically tractabl
e decoder that adapts its finite-range tail behavior directly from the observed
data. Experiments on synthetic and real-world benchmarks demonstrate that PH-VAE
accurately approximates diverse heavy-tailed distributions, significantly outpe
rforming Gaussian, Student-t, and extreme-value-based VAE decoders in modeling o
bserved tail behavior and extreme quantiles. In multivariate settings, PH-VAE ca
ptures realistic cross-dimensional tail dependence through its shared latent rep
resentation.
To our knowledge, this is the first work to integrate Phase-Type distributions i
nto deep generative modeling, bridging applied probability and representation le
arning.
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DOUBT: Decoupled Object-level Understanding and Bridging via vMF-based Trustwort
hiness for Hallucination Detection in MLLMs
Kaiqi Chen, Yang Qin, Changhao He, Xi Peng, Peng Hu
Multimodal Large Language Models (MLLMs) frequently produce hallucinations (i.e.
, assertions that contradict the image or facts), undermining reliability in hig
h-risk applications. Existing detection approaches typically feed images and tex
ts jointly and estimate hallucination scores by measuring the consistency of mod
el outputs. However, because the visual module often lags behind the language mo
dule in understanding and reasoning, MLLMs can repeatedly produce similar yet in
correct answers, yielding overestimated trustworthiness and missed detections. T
o address this, we propose a simple yet effective model-agnostic method, dubbed
Decoupled Object-level Understanding and Bridging via vMF-based Trustworthiness
(DOUBT). DOUBT first employs Object-level Understanding and Bridging (OUB), a tw
o-step prompting scheme that decouples object recognition from relational reason
ing by prompting the model to identify objects and then reason based on them. It
further introduces a von Mises-Fisher (vMF)-based trustworthiness metric, which
is more stable than semantic entropy metrics in small-sample settings. Extensiv
e experiments and ablation studies on multiple benchmarks show that DOUBT consis
tently outperforms state-of-the-art baselines, demonstrating its robustness and
generalizability for hallucination detection in MLLMs. The code is available at
https://github.com/XLearning-SCU/2026-ICML-DOUBT.
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EnterpriseOps-Gym: Environments and Evaluations for Stateful Agentic Planning an
d Tool Use in Enterprise Settings

Shiva Krishna Reddy Malay, Shravan Nayak, Sagar Davasam, Srinivas Sunkara, Sai Rajeswar

The rapid evolution of Large Language Models (LLMs) has shifted their role from passive information providers to active agents capable of executing complex workflows. However, the realization of a true "AI worker" is currently hindered by benchmarks that fail to capture the intricacy of professional environments, which demand long-horizon planning, complex tool usage, and adherence to strict access protocols. To bridge this gap, we introduce EnterpriseOps-Gym, a benchmark environment designed to evaluate agentic planning in realistic enterprise settings. EnterpriseOps-Gym provides: (i) 1,150 expert-curated tasks across eight interconnected domains (including HR, IT, Customer Service and productivity tools) that require managing persistent state and adhering to strict outcome-based verification logic; and (ii) a high-fidelity, containerized sandbox environment hosting 164 database tables and 512 functional tools. Our evaluation reveals critical limitations in state-of-the-art models: even the top-performing Claude Sonnet~4.5 achieves only 34.1\% success, struggling significantly with planning consistency, error recovery, and policy constraints. Furthermore, we observe that agents frequently fail to refuse infeasible tasks, leading to unintended and potentially harmful side effects on the system. These findings indicate that current agents are not yet ready for enterprise deployment. By releasing EnterpriseOps-Gym, we provide a concrete testbed to advance the reliability of autonomous agents in professional workflows.

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Subspace-Aware Feature Reshaping for Open-Set Graph Class-Incremental Learning

Weichao Zhang, Shuai Zheng, Yeyu Yan, Zhizhe Liu, Zhenfeng Zhu, Yao Zhao

Graph class-incremental learning (GCIL) has emerged to address the challenge of learning from dynamically evolving graphs, which continuously learns new classes over a sequence of tasks while retaining performance on previously seen classes. However, existing GCIL methods assume a closed-set test distribution drawn only from seen tasks. This fundamentally contradicts real-world open-ended scenarios where future unknown classes inevitably emerge. Empirically, we observe that existing GCIL methods falter in such open-set settings due to severe representation drift and generalized overconfidence. To bridge this gap, we investigate the Open-Set GCIL problem and propose \textbf{SAFER} (\underline{S}ubspace-\underline{A}ware \underline{FE}ature \underline{R}eshaping), a novel framework that endows GCIL with intrinsic open-set capabilities under a replay-free constraint. Specifically, \textbf{SAFER} performs subspace-aware feature reshaping with drift-free fingerprints, unifying task routing and open-set rejection into a single energy-based metric. Furthermore, we introduce a geometric space-consistency regularization that explicitly improves intra-class compactness and suppresses cross-task representation drift. Extensive experiments on four benchmarks demonstrate that SAFER outperforms state-of-the-art baselines by margins of up to 5.2\% in accuracy and 31.3\% in open-set AUROC, all while maintaining near-zero forgetting under strict no-replay constraints.

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Scaling Prompt Synthesis for Large Language Model Reasoning

Xueliang Zhao, Wei Wu, Jian Guan, Zhuocheng Gong, Lingpeng Kong

Large language models (LLMs) are evolving from conversational systems into strong reasoners for tasks such as Olympiad mathematics and competitive programming. While scaling parameters and test-time computation has driven progress, a key bottleneck is the lack of high-quality training problems: human-curated datasets are costly and limited, while existing synthetic corpora are often too easy or narrow. PromptCoT showed that injecting rationales into prompt synthesis increases problem difficulty. Building on this, we present PromptScale, a scalable framework that replaces hand-crafted heuristics with an expectation-maximization (EM) loop, where rationales are iteratively refined to guide prompt construction. This produces problems that are both harder and more diverse than prior corpora. The synthetic prompts support two post-training regimes: (1) \emph{Self-Play}, where strong models improve autonomously via verifiable feedback without stronger teachers; and (2) \emph{Supervised Fine-Tuning (SFT)}, where weaker models learn

from teacher-distilled traces. Extensive experiments demonstrate the effectiveness of this approach. In self-play, applying PromptScale to \texttt{Qwen3-30B-A3B-Thinking-2507} sets new state-of-the-art results \emph{at the 30B scale}, with +4.4, +4.8, and +5.3 on AIME 24/25 and HMMT 25, +6.1 and +5.0 on LiveCodeBench v5/v6, and +35 Elo on Codeforces. In SFT, training \texttt{Qwen2.5-7B-Instruct} solely on synthetic prompts boosts accuracy to 73.1 (AIME 24), 65.6 (AIME 25), and 53.4 (LiveCodeBench v5), surpassing models trained on human or hybrid data. Analyses further confirm that PromptScale yields fundamentally harder and distributionally distinct problems. These results establish prompt synthesis as a new axis for scaling reasoning and position PromptScale as a scalable foundation for future open-source models. The implementation is available at \url{https://github.com/inclusionAI/PromptCoT}.

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ConFu: Contemplate the Future for Better Speculative Sampling

Zongyue Qin, Raghavv Goel, Mukul Gagrani, Risheek Garrepalli, Mingu Lee, Yizhou Sun

Speculative decoding has emerged as a powerful approach to accelerate large language model (LLM) inference by employing lightweight draft models to propose candidate tokens that are subsequently verified by the target model. The effectiveness of this paradigm critically depends on the quality of the draft model. While recent advances such as the EAGLE series achieve state-of-the-art speedup, existing draft models remain limited by error accumulation: they condition only on the current prefix, causing their predictions to drift from the target model over steps. In this work, we propose ConFu (Contemplate the Future), a novel speculative decoding framework that enables draft models to anticipate the future direction of generation. ConFu introduces (i) contemplate tokens and soft prompts that allow the draft model to leverage future-oriented signals from the target model at negligible cost, (ii) a dynamic contemplate token mechanism with MoE to enable context-aware future prediction, and (iii) a training framework with anchor token sampling and future prediction replication that learns robust future prediction. Experiments demonstrate that ConFu improves token acceptance rates and generation speed over EAGLE-3 by 8-11%, across various downstream tasks with Llama-3 3B and 8B models.

We believe our work is the first to bridge speculative decoding with continuous reasoning tokens, offering a new direction for accelerating LLM inference.

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Forward-KL Convergence of Time-Inhomogeneous Langevin Diffusions

Andreas Habring, Martin Zach

Many practical samplers rely on time-dependent drifts---often induced by annealing or tempering schedules---to improve exploration and stability. This motivates a unified non-asymptotic analysis of the corresponding Langevin diffusion and their discretizations. We provide a convergence analysis that includes non-asymptotic bounds for the continuous-time diffusion and its Euler--Maruyama discretization in the forward-Kullback--Leibler divergence under a single set of abstract conditions on the time-dependent drift. The results apply to many practically-relevant annealing schemes, including geometric tempering and annealed Langevin sampling. In addition, we provide numerical experiments comparing the annealing schemes covered by our theory in low- as well as high-dimensional settings.

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Implicit Turn-Wise Policy Optimization for Proactive User-LLM Interaction

Haoyu Peter Wang, Yuxin Chen, Liang Luo, Buyun Zhang, Ellie Dingqiao Wen, Pan Li

Multi-turn human-AI collaboration is fundamental to deploying interactive services such as adaptive tutoring, conversational recommendation, and professional consultation. However, optimizing these interactions via reinforcement learning is hindered by the sparsity of verifiable intermediate rewards and the high stochasticity of user responses.

To address these challenges, we introduce Implicit Turn-wise Policy Optimization (ITPO).

ITPO leverages an implicit process reward model to derive fine-grained, turn-wise process rewards from sparse outcome signals. Unlike volatile token-level rewar

ds, these turn-level signals exhibit superior robustness and may utilize a normalization mechanism to further enhance training stability.
We evaluate ITPO across three representative multi-turn collaborative tasks: math tutoring, document writing, and medical recommendation. Empirical results demonstrate that ITPO, when combined with PPO, GRPO, or RLOO, consistently achieves improved convergence than existing baselines. Elaborate trajectory analysis confirms that ITPO infers turn-wise preferences that are semantically aligned with human judgment.
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Learning Generalized Label Distributions
Haitao Wu, Weiwei Li, Kun Yue, Xiuyi Jia
Label ambiguity/polysemy is pervasive in supervised learning, motivating a variety of representations beyond the traditional single-label setting. While label distribution (LD) provides a probabilistic description and has attracted increasing attention, we reveal its inherent limitations, including inconsistency with raw data, distortion of inter-sample order, and limited applicability. To address these issues, we introduce generalized label distribution (GLD), a unified representation that can perfectly recover raw data while preserving inter-sample order consistency, transform into existing forms of label representations without information loss, and capture out-of-distribution samples as well as negative label correlations. We further develop GLD learning algorithms and demonstrate their effectiveness through both theoretical analysis and extensive experiments.
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Full-Batch Gradient Descent Outperforms One-Pass SGD: Sample Complexity Separation in Single-Index Learning
Filip Kova■evi■, Hong Chang Ji, Denny Wu, Mahdi Soltanolkotabi, Marco Mondelli
It is folklore that reusing training data more than once can improve the statistical efficiency of gradient-based learning. While this phenomenon has been extensively studied in linear regression, the benefit of multi-pass gradient descent (GD, which reuses all the data) over one-pass stochastic gradient descent (online SGD, which uses each data point only once) is not well-understood in nonlinear and non-convex settings, except for a loss modification mechanism achieved by the first two passes on the data. In this work, we consider learning a d-dimensional single-index model with a quadratic activation, for which it is known that one-pass SGD requires $n\gtrsim d\log d$ samples to achieve weak recovery. We first show that this $\log d$ factor in the sample complexity persists for full-batch spherical GD on the correlation loss; however, by simply truncating the activation, full-batch GD exhibits a favorable optimization landscape at $n \simeq d$ samples, thereby outperforming one-pass SGD (with the same activation) in statistical efficiency. We complement this result with a trajectory analysis of full-batch GD on the squared loss from small initialization, showing that $n \gtrsim d$ samples and $T \gtrsim\log d$ gradient steps suffice to achieve strong (exact) recovery.
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LS^{2}MC-GDA: A Smoothed Algorithm for Federated Stochastic Multi-Level Compositional Minimax Optimization
Xinwen Zhang, Richard Souvenir, Hongchang Gao
Federated stochastic multi-level compositional minimax optimization supports a growing number of machine learning applications. However, the interplay of multi-level compositional structure and minimax formulation in federated learning setting poses significant optimization challenges, resulting in slow convergence rates for existing algorithms. In this paper, we propose a novel federated learning algorithm, LS^{2}MC-GDA, that leverages smoothing techniques and variance-reduced stochastic compositional gradients. To support our theoretical analysis, we introduce a stage-wise extension of LS^{2}MC-GDA, which serves to bridge the gap between different stationarity measures. We establish that our algorithm achieves a sample complexity of $O(\kappa^{3/2}/N\epsilon^3)$ and a communication complexity of $O(\kappa/\epsilon^2)$, substantially improving existing theoretical results in terms of the condition number κ and the solution accuracy ϵ and achieving a linear speedup with respect to the number of workers N.

Finally, experimental results validate the effectiveness of our algorithm.
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Exact and Approximate Algorithms for Polytree Learning
Juha Harviainen, Frank Sommer, Manuel Sorge
Polytrees are a subclass of Bayesian networks that seek to capture the conditional dependencies between a set of n variables as a directed forest and are motivated by their more efficient inference and improved interpretability. Since the problem of learning the best polytree is NP-hard, we study which restrictions make it more tractable by considering for example in-degree bounds, properties of score functions measuring the quality of a polytree, and approximation algorithms. We devise an algorithm that finds the optimal polytree in time $\mathcal{O}((2+\epsilon)^n)$ for arbitrarily small $\epsilon > 0 $ and any constant in-degree bound k, improving over the fastest previously known algorithm of time complexity $\mathcal{O}(3^n)$. We further give polynomial-time algorithms for finding a polytree whose score is within a factor of k from the optimal one for arbitrary scores and a factor of 2 for additive ones. Many of the results are complemented by (nearly) tight lower bounds for either the time complexity or the approximation factors.
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ASAP: Exploiting the Satisficing Generalization Edge in Neural Combinatorial Optimization
Han Fang, Paul Weng, Yutong Ban
Deep Reinforcement Learning (DRL) has emerged as a promising approach for solving Combinatorial Optimization (CO) problems, such as the 3D Bin Packing Problem (3D-BPP), Traveling Salesman Problem (TSP), or Vehicle Routing Problem (VRP), but these neural solvers often exhibit brittleness when facing distribution shifts. To address this issue, we uncover the Satisficing Generalization Edge, which we validate both theoretically and experimentally: identifying a set of promising actions is inherently more generalizable than selecting the single optimal action. To exploit this property, we propose Adaptive Selection After Proposal (ASAP), a generic framework that decomposes the decision-making process into two distinct phases: a proposal policy that acts as a robust filter, and a selection policy as an adaptable decision maker. This architecture enables a highly effective online adaptation strategy where the selection policy can be rapidly fine-tuned on a new distribution. Concretely, we introduce a two-phase training framework enhanced by Model-Agnostic Meta-Learning (MAML) to prime the model for fast adaptation. Extensive experiments on 3D-BPP, TSP, and CVRP demonstrate that ASAP improves the generalization capability of state-of-the-art baselines and achieves superior online adaptation on out-of-distribution instances.
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From Perception to Planning: Evolving Ego-Centric Task-Oriented Spatiotemporal Reasoning via Curriculum Learning
Xiaoda Yang, Yuxiang Liu, Shenzhou Gao, Can Wang, Jingyang Xue, Lixin Yang, Yao Mu, Tao Jin, Zhimeng Zhang, Shuicheng YAN, Zhou Zhao
Modern vision-language models achieve strong performance in static perception, but remain limited in the complex spatiotemporal reasoning required for embodied, egocentric tasks. A major source of failure is their reliance on temporal priors learned from passive video data, which often leads to spatiotemporal hallucinations and poor generalization in dynamic environments. To address this, we present EgoTSR, a curriculum-based framework for learning task-oriented spatiotemporal reasoning. EgoTSR is built on the premise that embodied reasoning should evolve from explicit spatial understanding to internalized task-state assessment and finally to long-horizon planning. To support this paradigm, we construct EgoTSR-Data, a large-scale dataset comprising 46 million samples organized into three stages: Chain-of-Thought supervision for explicit reasoning, weakly supervised tagging for internalizing reasoning, and long-horizon sequences composed of orthogonal atomic subtasks. We further introduce a Reasoning-Enhanced Task Decomposition mechanism that explicitly models causal dependencies between actions and objects, enabling controllable planning from high-level task descriptions. To rigorously validate our approach, we establish a Dual-Level Evaluation Framework that

covers both atomic spatial perception and logical planning. Extensive experiments demonstrate that EgoTSR effectively eliminates chronological biases, achieving 92.4% accuracy on long-horizon logical reasoning tasks while maintaining 88.2% fine-grained perceptual precision, significantly outperforming existing open-source and closed-source state-of-the-art models.
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A Minimal Agent for Automated Theorem Proving
Borja Requena, Austin Letson, Krystian Nowakowski, Izan Beltran Ferreiro, Leopoldo Sarra
We propose a minimal agentic baseline that enables systematic comparison across different AI-based theorem prover architectures. This design implements the core features shared among state-of-the-art systems: iterative proof refinement, library search and context management. We evaluate this agentic approach using qualitatively different benchmarks and compare various frontier language models and design choices. Our results show competitive performance compared to state-of-the-art approaches, while using a significantly
simpler architecture and a fraction of their cost. Additionally, we demonstrate consistent advantages of an iterative approach over multiple single-shot generations, especially in terms of sample efficiency and cost effectiveness. The implementation is released open-source as a candidate reference for future research and as an accessible prover for the community.
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Optimizing Few-Step Generation with Adaptive Matching Distillation
Lichen Bai, zikai zhou, Shitong Shao, Wenliang Zhong, Shuo Yang, Shuo Chen, Bojun Cheng, Zeke Xie
Distribution Matching Distillation (DMD) is a powerful acceleration paradigm, yet its stability is often compromised in **Forbidden Zones**—regions where the real teacher provides unreliable guidance while the fake teacher exerts insufficient repulsive force. In this work, we propose a unified optimization framework that reinterprets prior art as implicit strategies to avoid these corrupted regions. Based on this insight, we introduce Adaptive Matching Distillation (**AMD**), a self-correcting mechanism that utilizes reward proxies to explicitly detect and escape Forbidden Zones. AMD dynamically prioritizes corrective gradients via structural signal decomposition and introduces Repulsive Landscape Sharpening to enforce steep energy barriers against failure mode collapse. Extensive experiments across image and video generation tasks (e.g., SDXL, Wan2.1) and rigorous benchmarks (e.g., VBench, GenEval) demonstrate that AMD significantly enhances sample fidelity and training robustness. For instance, AMD improves the HPSv2 score on SDXL from **30.64** to **31.25**, outperforming state-of-the-art baselines. These findings validate that explicitly rectifying optimization trajectories within Forbidden Zones is essential for pushing the performance ceiling of few-step generative models.
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Item Response Scaling Laws: A Measurement Theory Approach for Efficient and Generalizable Neural Scaling Estimation
Sang T. Truong, Yuheng Tu, Rylan Schaeffer, Sanmi Koyejo
Scaling laws provide a fundamental framework for understanding the performance of Language Models (LMs), yet deriving them requires prohibitively expensive evaluations across thousands of checkpoints or millions of inference samples. To address this, we introduce Item Response Scaling Laws (IRSL), a unified framework that integrates Item Response Theory (IRT) within the scaling law framework. Unlike traditional approaches that treat each model-benchmark pair in isolation, IRSL disentangles latent model ability from question characteristics, factorizing the scaling law estimation for M models and N questions to significantly reduce parameter complexity from $O(M \times N)$ to $O(M + N)$. We instantiate IRSL with Beta-IRT, which leverages the empirical probability responses of LMs---such as token probabilities in pre-training and pass rates in test-time sampling---to capture richer signals than binary responses. We validate our approach across two prevalent scaling paradigms: (1) pre-training downstream scaling, using 6,612 LM checkpoints and 37,682 questions from 10 benchmarks; and (2) test-time scal

ing, using 12 LMs and 120 questions from 4 benchmarks with up to 2,500 samples p
er question. Given a one-time calibration on existing model responses, IRSL yiel
ds more reliable scaling estimates using only 50 questions per benchmark (a 99.9
\% reduction), achieving comparable or superior decision accuracy to traditional
 approaches. Furthermore, we show that the estimated latent model abilities are
generalizable, enabling accurate performance forecasting across benchmarks that
share the same measurement objective.
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A Geometric Lens on Physics-Aligned Data Compression
Aleix Segui Ugalde, Wesley Armour
In AI for Science, physics-informed losses are increasingly used to train learne
d compressors for scientific data, but their rate--distortion implications remai
n poorly understood. At fixed bitrate, these objectives often improve preservati
on of a target physical observable while degrading standard reconstruction fidel
ity. We develop a local geometric theory showing that this tradeoff is governed
by the interaction of latent-space sensitivities induced by the entropy model, t
he physical observable, and the distortion metric. At each operating point, thes
e induce preferred directions along which compression noise should be suppressed
, yielding an anisotropic error-allocation mechanism. When these directions are
misaligned, improving the observable at fixed rate necessarily worsens standard
distortion, establishing a fundamental limit on simultaneous preservation. We fo
rmalise this through a local tangent-space rate--distortion law and introduce a
practical alignment diagnostic based on dominant eigenspace overlap. Experiments
 across scientific domains test the theory and validate that the alignment diagn
ostic correlates with observed data- and physics-space trade-offs.
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Layer-Centric Factors of Variation Disentanglement for Task- and Model-Agnostic
Generalization
Hee-Jun Jung, Jongmin Park, Minwoo Kang, Hoyong Kim, Kangil Kim
Disentanglement learning aims to separate the underlying factors of variation (F
oV) to improve generalization.
However, most FoV-based latent-vector-centric methods impose objective-driven co
nstraints at a bottleneck, and it is difficult to translate disentanglement into
 consistent gains on downstream tasks without inductive bias.
Motivated by architectural approaches complementary to vector-centric objectives
 for downstream tasks, we propose the *Orthogonal Subspaces Projection* (OSP) la
yer, a plug-and-play module that integrates into intermediate layers and promote
s FoV separation by projecting latent features into mutually orthogonal subspace
s.
Across diverse domains and tasks, models equipped with the OSP layer improve dis
entanglement quality and generalization in downstream tasks, including computer
vision (classification, detection, and segmentation), natural language processin
g (word analogy, and text classification), and fine-tuning settings on large bac
kbones.
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TIME: Tensor-Factorized Mixture-of-Experts with Intrinsic Routing for Lifelong M
ultimodal Knowledge Editing
Dexuan Xu, Jieyi Wang, Shijie Li, Hanpin Wang, Yongzhi Cao, Yu Huang
Lifelong multimodal knowledge editing allows vision language models to continuou
sly adapt to dynamic updates to avoid catastrophic forgetting. To mitigate inter
ference between sequential updates, recent paradigms have shifted towards modula
r parameter isolation. However, this strategy faces a critical scalability bottl
eneck: accumulating dense parameter blocks can lead to excessive memory growth,
and managing these independent modules often uses decoupled routing mechanisms,
resulting in architectural redundancy. To address this issue, we propose **TIME*
* (**T**ensor-Factorized **I**ntrinsic **M**ixture-of-**E**xperts), a unified fr
amework harmonizing parameter efficiency with structural self-routing. TIME para
meterizes each knowledge edit as a compact CP-decomposed tensor, significantly r
educing complexity compared to low-rank matrices. Furthermore, departing from au
xiliary semantic retrievers, we introduce an intrinsic routing mechanism that ut

ilizes the tensor's input factors to directly define the active subspace, effect ively enabling expert parameters to serve simultaneously as the routing logic. E xtensive experiments demonstrate that TIME achieves state-of-the-art performance on lifelong editing benchmarks while successfully reducing memory usage and inf erence latency.

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Intrinsic Task Symmetry Drives Generalization in Algorithmic Tasks

Hyeonbin Hwang, Yeachan Park

Grokking, a sudden transition from memorization to generalization, has been clos ely linked to the emergence of low-dimensional representations; yet the mechanis m driving this organization remains elusive. Here, we propose that intrinsic tas k symmetries are the key drivers of grokking, inducing structured geometries in representation space. Our analysis reveals a consistent three-stage training dyn amic: (i) data memorization, (ii) intrinsic symmetry acquisition, and (iii) geom etric organization. We show that generalization emerges during the symmetry acqu isition phase, and subsequently the embedding space organizes into a low-dimensi onal structured geometry. We validate this mechanism across diverse algorithmic domains, spanning algebraic (modular arithmetic), structural (graph metric compl etion), and relational (comparison) reasoning tasks. Leveraging these insights, we formulate a symmetry-based criterion for generalization and propose symmetry- and geometry-prompting training strategies that can accelerate generalization. Together, our results establish intrinsic symmetry as a central mechanism enabli ng neural networks to move beyond memorization and achieve robust algorithmic re asoning.

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PFT: Phonon Fine-tuning for Machine Learned Interatomic Potentials

Teddy Koker, Abhijeet Sadashiv Gangan, Mit Kotak, Jaime Marian, Tess Smidt

Many materials properties depend on higher-order derivatives of the potential en ergy surface, yet machine learned interatomic potentials (MLIPs) trained with a standard loss on energy, force, and stress errors can exhibit error in curvature , degrading the prediction of vibrational properties. We introduce phonon fine-t uning (PFT), which directly supervises second-order force constants of materials by matching MLIP energy Hessians to DFT-computed force constants from finite di splacement phonon calculations. To scale to large supercells, PFT stochastically samples Hessian columns and computes the loss with a single Hessian-vector prod uct. We also use a simple co-training scheme to incorporate upstream data to mit igate catastrophic forgetting. On the MDR Phonon benchmark, PFT improves Nequix MP by 55% on average across phonon thermodynamic properties and achieves state-o f-the-art accuracy among models trained on Materials Project trajectories. PFT a lso generalizes to improve properties beyond second-derivatives, improving therm al conductivity predictions that rely on third-order derivatives of the potentia l energy.

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Measurement-Consistent Langevin Corrector for Stabilizing Latent Diffusion Inver se Problem Solvers

Lee Hyoseok, Sohwi Lim, Eunju Cha, Tae-Hyun Oh

While latent diffusion models (LDMs) have emerged as powerful priors for inverse problems, existing LDM-based solvers frequently suffer from instability. In thi s work, we first identify the instability as a discrepancy between the solver dy namics and stable reverse diffusion dynamics learned by the diffusion model, and show that reducing this gap stabilizes the solver. Building on this, we introdu ce *Measurement-Consistent Langevin Corrector (MCLC)*, a theoretically grounded plug-and-play stabilization module that remedies the LDM-based inverse problem s olvers through measurement-consistent Langevin updates. Compared to prior approa ches that rely on linear manifold assumptions, which often fail to hold in laten t space, MCLC provides a principled stabilization mechanism, leading to more sta ble and reliable behavior in latent space.

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BuildArena: A Physics-Aligned Interactive Benchmark of LLMs for Engineering Cons truction

Tian Xia, Tianrun Gao, Wenhao Deng, Long Wei, Xiaowei Qian, Chenglei Yu, Tailin Wu

Engineering construction automation aims to transform natural language specifications into physically viable structures, requiring complex integrated reasoning under strict physical constraints. While modern LLMs possess broad knowledge and strong reasoning capabilities that make them promising candidates for this domain, their construction competencies remain largely unevaluated. To address this gap, we introduce BuildArena, the first physics-aligned interactive benchmark designed for language-driven engineering construction. Technically, it contributes to the community in two aspects: (1) an extendable task design strategy spanning static and dynamic mechanics across multiple difficulty tiers; (2) a 3D Spatial Geometric Computation Library for supporting construction based on language instructions.

On nine frontier LLMs and three additional open-weight models, BuildArena comprehensively evaluates their capabilities for language-driven and physics-grounded construction automation. We release the code at https://github.com/AI4Science-WestlakeU/BuildArena to benefit construction automation in engineering applications.

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RECAST: Model Reconstruction via Counterfactual-Aware Wasserstein Geometry under Limited Data

Xuan Zhao, Lena Krieger, Zhuo Cao, Arya Bangun, Hanno Scharr, Ira Assent

Counterfactual explanations (CFs) help understand machine learning models by identifying minimal input changes that would lead to alternative model outcomes. Recent work demonstrates their utility for reconstructing black-box models, enabling third-party auditing of opaque decision systems for fairness and accountability. Still, CF-based reconstruction may suffer from decision boundary shifts, overfitting, and restrictive assumptions requiring online query access to target platforms.

We propose **REconstruction via Counterfactual-Aware waSserstein opTimization (RECAST)** under limited data and restricted access, a behavioral surrogate model based on Wasserstein barycenteric prototypes. Our approach addresses decision boundary shifts by incorporating CFs as informative, though less representative, samples for both classes, maintaining high surrogate fidelity in low-sample regimes without requiring online access during reconstruction. To enhance fairness auditing, our method enables systematic group fairness diagnostics.

Experiments on real-world datasets and various setups show that **RECAST** effectively achieves high fidelity and query efficiency, as well as stable results even when the access is limited and noisy.

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Coverage, Not Averages: Semantic Stratification for Trustworthy Retrieval Evaluation

Andrew Klearman, Radu Revutchi, Rohin Garg, Rishav Chakravarti, Samuel Marc Denton, Yuan Xue

Retrieval quality is the primary bottleneck for accuracy and robustness in retrieval-augmented generation (RAG). Current evaluation relies on heuristically constructed query sets, which introduce a hidden intrinsic bias. We formalize retrieval evaluation as a statistical estimation problem, showing that metric reliability is fundamentally limited by the evaluation-set construction. We further introduce \emph{semantic stratification}, which grounds evaluation in corpus structure by organizing documents into an interpretable global space of entity-based clusters and systematically generating queries for missing strata. This yields (1) formal semantic coverage guarantees across retrieval regimes and (2) interpretable visibility into retrieval failure modes. Experiments across multiple benchmarks and retrieval methods validate our framework. The results expose systematic coverage gaps, identify structural signals that explain variance in retrieval performance, and show that stratified evaluation yields more stable and transparent assessments while supporting more trustworthy decision-making than aggregate metrics.

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AtelierEval: Agentic Evaluation of Humans & LLMs as Text-to-Image Prompters
Hanjun Luo, Zhimu Huang, Sylvia Chung, Yiran Wang, Yingbin Jin, Jialin Li, Jiang Li, Xinfeng Li, Hanan Salam
Text-to-image (T2I) systems increasingly rely on upstream prompters, either humans or multimodal large language models (MLLMs), to translate user intent into detailed prompts. Yet current benchmarks fix the prompt and only evaluate T2I models, leaving the prompting proficiency of this upstream component entirely unmeasured. We introduce AtelierEval, the first unified benchmark that quantifies prompting proficiency across 360 expert-crafted tasks. Grounded in a cognitive view, it spans three task categories and instantiates tasks using a taxonomy of real-world challenges, with a dual interface for both humans and MLLMs. To enable scalable and reliable evaluation, we propose AtelierJudge, a skill-based, memory-augmented agentic evaluator. It produces subjective and objective scores for prompt–image pairs, achieving a Spearman correlation of 0.79 with human experts, approaching human performance. Extensive experiments benchmark 8 MLLMs against 48 human users across 4 T2I backends, validate AtelierEval as a robust diagnostic tool, and reveal the superiority of mimicry over planning, advocating for an image-augmented direction for future prompters. Our work is released to support future research.
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Robust Sequential Experimental Design for A/B Testing
Qianglin Wen, Xiangkun Wu, Chengchun Shi, Ting Li, Niansheng Tang, Yingying Zhang, Hongtu Zhu
Experimental design has emerged as a powerful approach for improving the sample efficiency of A/B testing, yet existing designs rely critically on correctly specified models. We study robust sequential experimental design under model misspecification and develop a unified framework that covers both contextual bandit and dynamic settings. Theoretically, we prove that our design bounds the worst-case mean squared error of the estimated treatment effect. Empirically, we demonstrate the effectiveness of the proposed approach using synthetic and real-world datasets from a leading technology company.
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You Need Better Attention Priors
Elon Litman, Gabe Guo
We generalize the attention mechanism by viewing it through the lens of Entropic Optimal Transport, revealing that standard attention corresponds to a transport problem regularized by an implicit uniform prior. We introduce Generalized Optimal transport Attention with Trainable priors (GOAT), a new attention mechanism that replaces this naive assumption with a learnable, continuous prior. This prior maintains full compatibility with optimized kernels such as FlashAttention. GOAT also provides an EOT-based explanation of attention sinks and materializes a solution for them, avoiding the representational trade-offs of standard attention. Finally, by absorbing spatial information into the core attention computation, GOAT learns an extrapolatable prior that combines the flexibility of learned positional embeddings with the length generalization of fixed encodings.
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Maximizing Mutual Information Between Prompt and Response Improves LLM Performance With No Additional Data
HyunJi Nam, Haoran Li, Natasha Jaques
While post-training has successfully improved large language models (LLMs) across a variety of domains, these gains heavily rely on human-labeled data or external verifiers. Existing data has already been exploited, and new data is expensive to collect. Moreover, true intelligence goes far beyond verifiable tasks. Therefore, we need self-improvement frameworks that are less dependent on external signals and more broadly applicable to both verifiable and non-verifiable domains. We propose **Mutual Information Preference Optimization (MIPO)**, a contrastive data augmentation method that constructs preference pairs by generating a positive response conditioning on the correct prompt, and a negative response by conditioning on a random, unrelated prompt. We show that using Direct Preference Optimization to learn from this paired data maximizes pointwise mutual information

under the base LLM between prompts and model responses. Experiments with with 1-7B parameter Llama and Qwen instruct models show that MIPO achieves 3-16% gains (and 51% increase for Qwen2.5-1.5B-Instruct) on personalization compared to prompting baselines. Surprisingly, MIPO can also be useful in verifiable domains, such as math and multiple-choice question answering, yielding 1-20% gains *without any additional data or external supervision*. These results suggest a promising direction for self-improvement using intrinsic signals derived from contrastive data pairs.

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Source-Free Open-World RF Fingerprint Identification

Kunling Li, Cunqing Hua, Hongyu Zhu, Tianjie Ju, Pengwenlong Gu

Radio Frequency Fingerprint Identification (RFFI) is a foundational pillar of physical-layer security, providing unclonable identity authentication and lightweight defense mechanisms for zero-trust wireless networks. Its practical deployment, however, often occurs in a source-free open-world (SF-OW) setting, characterized by a continuous influx of unregistered devices and privacy constraints that preclude the retention of historical data. In this paper, we formalize SF-OW RFFI task, which manifests a severe stability-plasticity dilemma: intrinsic signal similarity confuses new classes, while source absence precipitates catastrophic forgetting. To address this, we propose Incremental Orthogonal ETF (IO-ETF), a novel neural collapse-inspired framework utilizing output geometry to actively induce parameter separation and isolation. We further devise a Triple-Level Geometric Alignment (TLGA) strategy via semantic optimal transport, manifold progressive anchoring, and reliable subspace retention to stably align unlabeled streams to this geometric skeleton. Experiments on benchmarks demonstrate a superior trade-off between old-class retention and new-class discovery, offering a promising solution for secure access in dynamic networks.

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Offline Multi-Agent Reinforcement Learning via Sequential Score Decomposition

Dan Qiao, Wenhao Li, Shanchao Yang, Hongyuan Zha, Baoxiang Wang

Offline cooperative multi-agent reinforcement learning (MARL) faces unique challenges due to the distribution shift between online and offline data collection. While online MARL typically converges to a single coordinated joint policy, offline datasets are often mixtures of diverse cooperative behaviors, resulting in highly multimodal joint behavior distributions. In such settings, independent policy regularization often misaligns joint policy constraints and leads to severe distribution shift. To address this, we propose OMSD, which sequentially decomposes the joint behavior policy into individual conditional distributions and leverages diffusion-based generative models to provide modality-coordinated regularization for each agent. Combined with centralized critic guidance, OMSD achieves coordinated exploration within high-value, in-distribution regions, and avoids out-of-distribution joint actions. Experiments across multiple datasets on various continuous control tasks demonstrate that OMSD consistently achieves state-of-the-art performance, especially in challenging multimodal scenarios. Our results highlight the necessity of modality-aware coordination for robust offline MARL.

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Latent Diffusion Pretraining for Crystal Property Prediction

Shrimon Mukherjee, KISHALAY DAS, Partha Basuchowdhuri, Pawan Goyal, Niloy Ganguly

Fast and accurate prediction of crystal properties is a central challenge in new materials design. Graph neural networks and Transformer-based models have emerged as powerful tools for this task due to their ability to encode the local structural environment of atoms within a crystal. However, these models are data-hungry and in practice labeled data for crystal properties are very scarce. Pretraining–finetuning strategies, particularly those based on diffusion models, have shown promise in addressing these limitations. In this work, we introduce a novel latent-diffusion based pretraining framework CrysLDNet, designed to mitigate the data scarcity. Our approach integrates a Variational Autoencoder (VAE) with a diffusion model during the pretraining stage. The VAE encoder maps 3D crystal structures into a smooth latent space, within which the diffusion process is appli

ed. This latent diffusion pretraining enables the graph encoder to effectively c
apture structural and chemical semantics from large-scale unlabeled data, which
can then be finetuned for specific property prediction tasks.
Comprehensive experiments on popular DFT datasets for property prediction reveal
 that CrysLDNet significantly outperforms both training-from-scratch and pretrai
ned baselines, with improvements of ***4.26%*** and ***4.90%*** on the JARVIS an
d MP datasets. Additionally, the learned representations remain robust in sparse
-data conditions and are expressive enough to correct DFT errors when finetuned
with limited experimental data. Code is available at https://github.com/shrimonm
uke0202/CrysLDNet.git.
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GIPO: Gaussian Importance Sampling Policy Optimization
Chengxuan Lu, Zhenquan Zhang, Shukuan Wang, Qunzhi Lin, Baigui Sun, Yang Liu
Post-training with reinforcement learning (RL) has recently shown strong promise
 for advancing multimodal agents beyond supervised imitation. However, RL remain
s limited by poor data efficiency, particularly in settings where interaction da
ta are scarce and quickly become outdated. To address this challenge, GIPO (Gaus
sian Importance sampling Policy Optimization) is proposed as a policy optimizati
on objective based on truncated importance sampling, replacing hard clipping wit
h a log-ratio-based Gaussian trust weight to softly damp extreme importance rati
os while maintaining non-zero gradients. Theoretical analysis shows that GIPO in
troduces an implicit, tunable constraint on the update magnitude, while concentr
ation bounds guarantee robustness and stability under finite-sample estimation.
Experimental results show that GIPO achieves state-of-the-art performance among
clipping-based baselines across a wide range of replay buffer sizes, from near o
n-policy to highly stale data, while exhibiting superior bias--variance trade-of
f, high training stability and improved sample efficiency. Code is available at
https://github.com/distanceLu/GIPO.
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GameVerse: Can Vision-Language Models Learn from Video-based Reflection?
Kuan Zhang, Dongchen Liu, Qiyue Zhao, Jinkun Hou, Xinran Zhang, Qinlei Xie, Miao
 Liu, Yiming Li
Human gameplay is a visually grounded interaction loop in which players act, ref
lect on failures, and watch tutorials to refine strategies. Can Vision-Language
Models (VLMs) also learn from video-based reflection? We present **GameVerse**,
 a comprehensive video game benchmark that enables a *reflective visual interact
ion loop*. Moving beyond traditional ***fire-and-forget*** evaluations, it uses
a novel ***reflect-and-retry*** paradigm to assess how VLMs internalize visual e
xperience and improve policies. To facilitate systematic and scalable evaluation
, we also introduce a *cognitive hierarchical taxonomy* spanning 15 globally pop
ular games, *dual action space* for both semantic and GUI control, and *mileston
e evaluation* using advanced VLMs to quantify progress. Our experiments show tha
t VLMs benefit from video-based reflection in varied settings, and perform best
by combining failure trajectories and expert tutorials—a *training-free* analogu
e to reinforcement learning (RL) plus supervised fine-tuning (SFT).
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SAM Audio: Segment Anything in Audio
Bowen Shi, Andros Tjandra, John Hoffman, Helin Wang, YI-CHIAO WU, Luya Gao, Juli
us Richter, Matthew Le, Apoorv Vyas, Sanyuan Chen, Christoph Feichtenhofer, Piot
r Dollár, Wei-Ning Hsu, Ann Lee
General audio source separation is a key capability for multimodal AI systems th
at can perceive and reason about sound. Despite substantial progress in recent y
ears, existing separation models are either domain-specific, designed for fixed
categories such as speech or music, or limited in controllability, supporting on
ly a single prompting modality such as text. In this work, we present SAM AUDIO,
 a foundation model for general audio separation that unifies text, visual, and
temporal span prompting within a single framework. Built on a diffusion transfor
mer architecture, SAM AUDIO is trained with flow matching on large-scale audio d
ata spanning speech, music, and general sounds, and can flexibly separate target
 sources described by language, visual masks, or temporal spans. The model achie

ves state-of-the-art performance across a diverse suite of benchmarks, including general sound, speech, music, and musical instrument separation in both in-the-wild and professionally produced audios, substantially outperforming prior general-purpose and specialized systems. Furthermore, we introduce a new real-world separation benchmark with human-labeled multimodal prompts and a reference-free evaluation model that correlates strongly with human judgment.
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When Actions Go Off-Task: Detecting and Correcting Misaligned Actions in Computer-Use Agents
Yuting Ning, Jaylen Jones, Zhehao Zhang, Chentao Ye, Weitong Ruan, Junyi Li, Rahul Gupta, Huan Sun
Computer-use agents (CUAs) have made tremendous progress in the past year, yet they still frequently produce misaligned actions that deviate from the user's original intent. Such misaligned actions may arise from external attacks (e.g., indirect prompt injection) or from internal limitations (e.g., erroneous reasoning). They not only expose CUAs to safety risks, but also degrade task efficiency and reliability. This work makes the first effort to define and study misaligned action detection in CUAs, with comprehensive coverage of both externally induced and internally arising misaligned actions. We further identify three common categories in real-world CUA deployment and construct MisActBench, a benchmark of realistic trajectories with human-annotated, action-level alignment labels. Moreover, we propose DeAction, a practical and universal guardrail that detects misaligned actions before execution and iteratively corrects them through structured feedback. DeAction outperforms all baselines across offline and online evaluations with moderate latency overhead: (1) On MisActBench, it outperforms baselines by over 15% absolute in F1 score; (2) In online evaluation, it reduces attack success rate by over 90% under adversarial settings while preserving or even improving task success rate in benign environments.
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Distilling Neuro-Symbolic Programs into 3D Multi-modal LLMs
Wentao Mo, Yang Liu
Current 3D spatial reasoning methods face a fundamental trade-off: neuro-symbolic 3D (NS3D) concept learners achieve interpretable reasoning through compositional programs but are constrained to closed-set concept vocabularies and simple programs; end-to-end 3D multi-modal LLMs (3D MLLMs) could handle complex natural language and open-vocabulary concepts but suffer from black-box reasoning without explicit spatial verification.
We introduce APEIRIA, a neuro-symbolic 3D MLLM to bridge two paradigms by distilling symbolic reasoning patterns into MLLMs with natural language chain-of-thought.
Our three-stage curriculum progressively builds reasoning capabilities:
a) 3D perception alignment grounds object visual-geometric features to the LLM,
b) CoT-SFT teaches query decomposition and stepwise verification from symbolic program traces,
and c) CoT-RL extends reasoning patterns to open-set concepts and deeply nested instructions.
By transferring reasoning patterns rather than concept-specific knowledge, APEIRIA preserves key NS3D virtues: transparent reasoning and modular interchangeability of planning and perception components.
Evaluations on grounding, question answering, and captioning show that APEIRIA surpasses prior NS3D methods and matches state-of-the-art 3D MLLMs on 3D spatial reasoning datasets, unifying symbolic methods' systematic reasoning with MLLMs' flexibility.
Code is available at https://github.com/oceanflowlab/APEIRIA.
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Short Chains, Deep Thoughts: Balancing Reasoning Efficiency and Intra-Segment Capability via Split-Merge Optimization
Runquan Gui, Jie Wang, Zhihai Wang, Chi Ma, Jianye HAO, Feng Wu
While Large Reasoning Models (LRMs) have demonstrated impressive capabilities in solving complex tasks through the generation of long reasoning chains, this rel

iance on verbose generation results in significant latency and computational overhead.
To address these challenges, we propose \textbf{CoSMo} (\textbf{Co}nsistency-Guided \textbf{S}plit-\textbf{M}erge \textbf{O}ptimization), a framework designed to eliminate structural redundancy rather than indiscriminately restricting token volume.
Specifically, CoSMo utilizes a split-merge algorithm that dynamically refines reasoning chains by merging redundant segments and splitting logical gaps to ensure coherence.
We then employ structure-aligned reinforcement learning with a novel segment-level budget to supervise the model in maintaining efficient reasoning structures throughout training.
Extensive experiments across multiple benchmarks and backbones demonstrate that CoSMo achieves superior performance, improving accuracy by \textbf{3.3} points while reducing segment usage by \textbf{28.7\%} on average compared to reasoning efficiency baselines.
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Hyperparameter Transfer Laws for Non-Recurrent Multi-Path Neural Networks
Haosong Zhang, Shenxi Wu, Xingjian Ma, Shirui Bian, Yichi Zhang, Xi Chen, Wei Lin
Deeper modern architectures are costly to tune, and the base learning rate is often one of the most sensitive hyperparameters. Maximal Update Parametrization (μP) helps explain why many hyperparameters transfer across width. Yet depthwise learning-rate scaling is less understood for modern architectures with convolution, residual aggregation, and attention. To unify various non-recurrent multi-path neural networks such as CNNs, ResNets, and Transformers, we introduce an architecture-dependent notion of effective depth. Under stabilizing initializations and a maximal-update criterion, we derive a shared leading-order -3/2 law for the base learning-rate scale as effective depth grows. Here, the budget controls typical one-step representation-update energy at initialization, and effective depth counts sequential update-bearing units while absorbing fixed local structure into constants. Experiments across diverse architectures confirm the predicted slope and enable reliable zero-shot transfer of learning rates across depths and widths, turning depth scaling into a predictable hyperparameter-transfer problem.
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Vulnerable Agent Identification in Large-Scale Multi-Agent Reinforcement Learning
Simin Li, Zihao Mao, Zheng Yuwei, Linhao Wang, Ruixiao Xu, Chengdong Ma, Zhiqian Liu, Xin Yu, Yuqing Ma, Xin Wang, Jie Luo, Bo An, Yaodong Yang, Weifeng Lv, Xianglong Liu
Partial agent failure becomes inevitable when systems scale up, making it crucial to identify the subset of agents whose failure causes worst-case system performance degradations. We study this Vulnerable Agent Identification (VAI) problem in large-scale multi-agent reinforcement learning (MARL). We frame VAI as a Hierarchical Adversarial Decentralized Mean Field Control (HAD-MFC), where where the upper level selects vulnerable agents as an NP-hard task and the lower level learns their worst-case adversarial policies via mean-field MARL. The two problems are coupled together, making HAD-MFC difficult to solve. To handle this, we first decouple the hierarchical process by Fenchel-Rockafellar transform, resulting a regularized mean-field Bellman operator for upper level that enables independent learning at each level, thus reducing computational complexity. We next reformulate the upper-level NP-hard problem as an MDP with dense rewards, allowing sequential identification of vulnerable agents via greedy and RL algorithms. This decomposition provably preserves the optimal solution. Experiments show our method effectively identifies more vulnerable agents in large-scale MARL and the rule-based system, fooling system into worse failures, and reveals the vulnerability of each agent in large systems. Code available at \url{https://anonymous.4open.science/r/VAI-5F61/}.
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Continuous Viewpoint Adaptation for Single View 3D Object Reconstruction
Seunghyun Hwang, Qiang Qiu
Single-view 3D object reconstruction presents a formidable challenge in computer vision due to the inherent limitations of information obtainable from a solitary viewpoint. Recent 3D Gaussian Splatting (3DGS) inspired approaches perform a feed-forward way of learning a neural network that predicts 3D Gaussians which compose the 3D object, given a single image. However, they often struggle with occlusions and exhibit high sensitivity to small changes in input viewpoint, leading to inconsistencies and blurry artifacts in novel view renderings. Our method leverages 3DGS and introduces a new learning scheme that continuously adapts to input viewpoints.
To address inherent continuity of camera viewpoints that are represented by polar and azimuthal angles, we use Neural Ordinary Differential Equations to continuously model filter subspace of neural network, thus seamlessly embedding inductive bias of perspective distortions into its structure. By continuously adapting to view-specific features, our approach fosters view consistency in 3D reconstruction, allowing better coherency and accuracy across different angles. Experiments demonstrate that our model outperforms previous methods on multiple single-view 3D reconstruction benchmark datasets and excels in extrapolating to unseen camera angles and categories.
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Stochastic Gradient Methods under Heavy-Tailed Noises in Weakly Convex Optimization
Tianxi Zhu, Yi Xu, Qi Wang, Xiangyang Ji
Recently, many empirical work has shown that, in machine learning, the noise distribution of stochastic gradients often exhibits heavy tails when stochastic optimization methods are employed. Most existing theoretical analyses of heavy-tailed stochastic methods rely on various convexity and smoothness assumptions and our knowledge of how heavy-tailed stochastic methods behave in the setting of weakly convex optimization is still limited.

In the weakly convex setting, this paper derives new upper bounds on the convergence of the stochastic gradient method (SGD) under heavy-tailed noises. In particular, for vanilla SGD, we establish an in-expectation convergence guarantee on the bounded constrained domain under the assumption of bounded p-th central moment (p-BCM) of the gradient noise, and a high-probability guarantee on the unbounded domain when the noise follows a heavy-tailed sub-Weibull distribution. By equipping SGD with the gradient clipping (Clip-SGD), we demonstrate that it achieves high-probability convergence in the unbounded domain under the p-BCM gradient noise. All of our high-probability convergence bounds depend on the failure probability only through polynomial-logarithmic factors. Finally, we present experiments to validate our theoretical findings.
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Can VLMs Diagnose and Recover from VLA Manipulation Faults?
Bowen Yan, Jiahao Xiao, Kehui Liu, Jianbo Zhang, Zicheng Zhang, Qi Jia, Zhongjie Jia, Haoming Song, Chunyi Li, Bin Zhao, Guangtao Zhai
Existing VLA models frequently fail in robotic manipulation tasks, with poorly structured fault types that often require expert diagnosis. While VLMs offer strong explanatory capabilities, their effectiveness in assisting VLAs is limited by their unclear role in diagnostics and inadequate collaboration mechanisms. To address this, we introduce VLA-FixBench, a fault evaluation dataset that spans perception, planning, and control failures, and provides annotations for task stages, fault types, and spatiotemporal repair strategies. We further propose FaultEval, a static-to-dynamic-to-real evaluation framework that benchmarks 20 VLMs across multiple fault-related dimensions. Building on these insights, we design a VLM–VLA collaboration mechanism that localizes spatiotemporal deviations and rolls back task execution to enable targeted recovery. Experiments show that FaultEval reliably characterizes VLM-based closed-loop diagnosis and repair. The upper-bound analysis using human expert intervention shows that an idealized feedback loop can improve task success rates by 13\% on LIBERO and 35\% on real-world ro

bots. Our code, benchmark, and project page will be publicly released at: https://kakigo.github.io/VLA-FixBench/
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CLAA: Cross-Layer Attention Aggregation for Accelerating LLM Prefill
Bradley McDanel, Steven Li, Harshit Khaitan
Token-ranking heuristics accelerate the prefill bottleneck in long-context LLM inference by selectively processing semantically relevant tokens. However, current evaluation relies on end-to-end benchmarks, making it difficult to isolate the quality of the token ranking itself. To address this, we introduce an Answer-Informed Reference framework with two variants: a Model Reference (Model-Ref) that measures token importance using the model generated answer, and a Ground-Truth Reference (GT-Ref) that uses human reference answers. Using GT-Ref, we establish that as few as 10% of prompt tokens suffice to match full-context performance on LongBench across all three evaluated models. The framework also reveals that existing heuristics exhibit variance across layers, with rankings degrading sharply at specific layers. Motivated by this instability, we propose Cross-Layer Attention Aggregation (CLAA), which aggregates importance scores across consecutive layers, eliminating the layer-dependent accuracy collapse observed in single-layer methods. A meaningful gap remains between the best heuristic and GT-Ref, indicating theoretical room for improved token selection.
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When RAG Hurts: Diagnosing and Mitigating Attention Distraction in Retrieval-Augmented LVLMs
Beidi Zhao, Wenlong Deng, Xinting Liao, Yushu Li, Nazim Shaikh, Yao Nie, Xiaoxiao Li
While Retrieval-Augmented Generation (RAG) is one of the dominant paradigms for enhancing Large Vision-Language Models (LVLMs) on knowledge-based VQA tasks, recent work attributes RAG failures to insufficient attention towards the retrieved context, proposing to reduce the attention allocated to image tokens. In this work, we identify a distinct failure mode that previous study overlooked: Attention Distraction (AD). When the retrieved context is sufficient (highly relevant or including the correct answer), the retrieved text suppresses the visual attention globally, and the attention on image tokens shifts away from question-relevant regions. This leads to failures on questions the model could originally answer correctly without the retrieved text. To mitigate this issue, we propose MAD-RAG, a training-free intervention that decouples visual grounding from context integration through a dual-question formulation, combined with attention mixing to preserve image-conditioned evidence. Extensive experiments on OK-VQA, E-VQA, and InfoSeek demonstrate that MAD-RAG consistently outperforms existing baselines across different model families, yielding absolute gains of up to 4.76%, 9.20%, and 6.18% over the vanilla RAG baseline. Notably, MAD-RAG rectifies up to 74.68% of failure cases with negligible computational overhead.
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RSPO: Regularized Self-Play Alignment of Large Language Models
Xiaohang Tang, Sangwoong Yoon, Seongho Son, Huizhuo Yuan, Quanquan Gu, Ilija Bogunovic
Self-play-based policy optimization has emerged as an effective approach for fine-tuning large language models (LLMs), formulating preference optimization as a two-player game. However, the regularization with respect to the reference policy, which is crucial for mitigating over-optimization, has been insufficiently investigated in self-play alignment. To study the impact of different regularization strategies, we propose \textbf{Regularized Self-Play Policy Optimization (RSPO)}, a novel framework that unifies prior methods and enables simple plug-and-play regularizers, meanwhile preserving convergence to Nash equilibrium of the corresponding regularized game.
We empirically show that RSPO with appropriate regularizers can substantially improve the length-controlled win rate (LCWR) on AlpacaEval-2 across a range of base models, while also achieving consistently superior performance on Arena-Hard, MT-Bench, ArmoRM, and response diversity. In particular, RSPO improves unregularized self-play baseline (SPPO) on AlpacaEval-2 LCWR from 28.5% to $ 35.4\%$

with base model Mistral-7B, from 38.77% to 43.66% with LLaMA-8B, and from 50.54% to 51.83% with Gemma-2B.
Combining simplicity, convergence guarantees, and significant empirical gains, RSPO offers a strong foundation for exploring regularized self-play in alignment.
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Do Language Models Track Entities Across State Changes?
Zilu Tang, Qiao Zhao, Gabriel Franco, Derry Tanti Wijaya, Aaron Mueller, Sebastian Schuster, Najoung Kim
Entity tracking (ET), the ability to keep track of states, is a fundamental skill that underlies complex reasoning. An increasing amount of work investigates how transformer language models (LMs) solve entity binding *without* state changes. However, there is limited understanding of how non-toy LMs address ET problems of realistic difficulties expressed in natural language. To this end, we investigate the mechanisms underlying ET in more complex scenarios featuring multiple state-changing operations. We find that LMs do not incrementally track world states across tokens or query-relevant states across layers, but simply aggregate relevant information in parallel at the last token when the query becomes evident. We further investigate mechanisms of individual operations (`PUT`, `REMOVE`, `MOVE`) to characterize this non-incremental ET mechanism. Surprisingly, LMs implement the `REMOVE` operation with a fragile global suppression tag; this global removal mechanism predicts various failure modes that we confirm behaviorally. We provide a mechanistic solution of nullifying this tag to partially address this issue.
Overall, our findings reveal that LMs solve a fundamentally sequential task using a non-sequential strategy. More broadly, our work illustrates how behavioral and mechanistic analyses can fruitfully interact. Behavioral results inform mechanistic hypotheses, and insights from mechanistic analyses help build stronger behavioral evaluations by predicting failure modes missing from existing evaluations.
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MaMi-HOI: Harmonizing Global Kinematics and Local Geometry for Human-Object Interaction Generation
Hao Wang, Shiqi Wang, Qi Liu
Generating realistic 3D Human-Object Interactions (HOI) is a fundamental task for applications ranging from embodied AI to virtual content creation, which requires harmonizing high-level semantic intent with strict low-level physical constraints. Existing methods excel at semantic alignment, however, they struggle to maintain precise object contact. We reveal a key finding termed $\textit{Geometric Forgetting}$: as diffusion model depth increases, semantic feature tend to overshadow object geometry feature, causing the model to lose its perception to object geometry. To address this, we propose MaMi-HOI, a hierarchical framework reconciling Macro-level kinematic fluidity with Micro-level spatial precision. First, to counteract geometric forgetting, we introduce the Geometry-Aware Proximity Adapter (GAPA), which explicitly re-injects dense object details to perform residual snapping corrections for precise contact. Nevertheless, such aggressive local enforcement can disrupt global dynamics, leading to robotic stiffness. In response, we introduce the Kinematic Harmony Adapter (KHA), which proactively aligns whole-body posture with spatial objectives, ensuring the skeleton actively accommodates constraints without compromising naturalness. Extensive experiments validate that MaMi-HOI simultaneously achieves natural motion and precise contact. Crucially, it extends generation capabilities to long-term tasks with complex trajectories, effectively bridging the gap between global navigation and high-fidelity manipulation in 3D scenes.
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Learning Task-Sufficient World Models by Synergizing Agentic Exploration and Structured Modeling
Fan Feng, Yujia Zheng, Minghao Fu, Yongqiang Chen, Guangyi Chen, Kevin Patrick Murphy, Biwei Huang, Kun Zhang
Learning and planning in imagination using world models provides an effective paradigm for training agents for decision-making. However, existing approaches oft

en rely on high-dimensional latent spaces or generic visual embeddings that reta in many factors irrelevant to control, limiting efficiency and generalization ac ross tasks. To this end, we study how agents can learn world models with repres entations that are task-specific, minimal, and sufficient for decision making. W e achieve this via a closed-loop synergy between the agent and the world model, in which structured world-model learning distills task-sufficient representation s from informative interaction data. On the agent side, agents perform active pr obing of the environment to collect informative trajectories that expose task-re levant latent factors, guided by an adaptive curriculum. On the world-model side , we learn structured representations over observations to distill compact, task -sufficient latent states from the collected interaction data. This synergy enab les the recovery of task-sufficient latent representations that capture all cont rol-relevant factors empirically. Leveraging these representations, the resultin g policies achieve improved sample efficiency generalization, including generali zation across skills, object–skill compositions, and previously unseen tasks on standard continuous control and robotic manipulation benchmarks.
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Towards Seed-Robust Safety Alignment in Text-to-Image Models
Zhenyu Wu, Yao Huang, Shouwei Ruan, Xingxing Wei
Text-to-image diffusion models have achieved remarkable success in generating hi gh-quality images, yet existing safety mechanisms exhibit critical cross-seed in stability where defense performance varies significantly under different random seed conditions. This instability stems from the fact that a single malicious pr ompt generates diverse harmful variants across different noise initializations, forming complex distributional clusters that current methods cannot adequately a ddress. We investigate extending Noise Contrastive Alignment (NCA) to diffusion models due to its native capability of handling multiple negative samples throug h probabilistic weighting, but our theoretical analysis reveals two fundamental flaws in direct extension: gradient reversal caused by positive regularization t erms that paradoxically penalize safe content generation, and uniform suppressio n of harmful samples that ignores severity variations. To tackle these issues, w e propose Noise Contrastive Diffusion (NCD), which incorporates targeted algorit hmic modifications including elimination of problematic regularization and intro duction of pairwise regularization mechanisms that establish individualized pref erence relationships between safe and harmful variants. Extensive experiments fu rther demonstrate that NCD achieves superior cross-seed stability, reducing atta ck success rates (ASRs) from 11.1% to 6.2% compared to SOTA methods at the seed level while maintaining exceptional generation quality, exhibiting robust resist ance against sophisticated jailbreak prompts and strong generalizability across different T2I architectures. WARNING: This paper may contain examples of harmful texts and images.
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Spatial Conformal Inference through Localized Quantile Regression
Hanyang Jiang, Yao Xie
Reliable uncertainty quantification at unobserved spatial locations is a key cha llenge in spatial statistics, particularly for complex and heterogeneous dataset s. While traditional methods such as Kriging rely on strong distributional assum ptions, conformal prediction (CP) offers a distribution-free alternative. Howeve r, although non-i.i.d. CP theory is well established for time-series data, a sig nificant gap remains for spatial data, where the lack of a natural ordering and discrete index complicates theoretical guarantees. Existing CP theory for spatia l data often relies on exchangeability. We propose Localized Spatial Conformal P rediction (LSCP), a model-agnostic framework that bridges this gap by coupling l ocal quantile regression with conformal calibration. LSCP conditions on spatial neighborhoods to capture local heterogeneity. We show that LSCP retains finite-s ample marginal coverage under spatial exchangeability and attains asymptotic con ditional coverage under stationarity and spatial mixing. Across synthetic and re al datasets, LSCP consistently achieves near-nominal coverage with tighter and m ore stable prediction intervals than existing methods that fail to capture these spatial dependencies.

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Logarithmic Switching Regret for Online Convex Optimization
Wenhao Yang, Yibo Wang, Yuanyu Wan, Lijun Zhang
Online convex optimization in non-stationary environments has garnered considera ble attention in the literature. Recently, Pasteris et al. (2024) investigate on line convex optimization with the optimal switching regret, defined as the sum o f the static regret over each segment, where the segmentation is an arbitrary pa rtition of the entire time horizon. For general convex functions, their work has established an optimal switching regret bound. However, it remains open whether similar bounds are attainable for other types of convex functions, such as expo nentially concave or strongly convex functions. In this paper, we affirmatively answer this question by proposing a novel meta-algorithm, termed IRESET, which i s used to aggregate the decisions from a group of experts. The essence of our me thod lies in running multiple experts over a set of intervals, and then employin g a meta-algorithm equipped with second-order bounds to sequentially combine the ir decisions. We leverage the segment tree structure to analyze the switching re gret over the entire time horizon, and offer new insights into utilizing recurs ive equations over the segment tree. By choosing appropriate expert-algorithms f or IRESET, our methods achieve logarithmic switching regret bounds for exponenti ally concave or strongly convex functions, respectively.
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Evaluating Contextual Illegality: AI Compliance in Corporate Law Scenarios
Hilal Aka, Joe Kwon, Noam Kolt
While AI models often refuse explicitly unlawful requests, in real-world scenari os illegality often depends on context. We evaluate frontier models on contextua l illegality across four corporate law scenarios in which routine actions—editin g documents, trading stock, requesting payment, approving communications—become unlawful due to circumstances such as pending investigations or bankruptcy filin gs. We study both chat and agentic settings and compare results to a human basel ine. The best-performing models consistently followed lawful requests and refuse d unlawful requests, though performance varied substantially between different s cenarios and models. We also identify distinct failure modes, such as excessive refusal of lawful requests, and find higher performance in reasoning models and agentic environments. By studying contextual illegality in these controlled envi ronments, we develop a methodology that can be extended to evaluate the legal co mpliance of AI models in additional scenarios and domains.
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HIAL: Towards Semantics-Aware Hypergraph Active Learning via Dual-Perspective In formation Maximization
Yanheng Hou, Xunkai Li, Yanzhe Wen, Zhenjun Li, Bing Zhou, Rong-Hua Li, Guoren W ang
Hypergraph Neural Networks (HNNs) model high-order interactions effectively but rely on costly node annotations, motivating Hypergraph Active Learning (HAL). Ho wever, many HAL pipelines adapt graph-based querying through clique expansion, w hich introduces *structural bias* and can cause *ranking collapse*, making utili ties overly determined by hyperedge cardinalities rather than informative high-o rder context. We propose **HIAL** (**H**ypergraph **I**nfluence-based **A**ctive **L**earning), a training-free framework that formulates hypergraph active lear ning as influence maximization over a high-order context-based weighted pairwise projection of the hypergraph. HIAL employs a High-Order Interaction-aware propa gation mechanism that modulates pairwise influence weights using hyperedge cardi nality and feature consistency, yielding a scalable linear diffusion process tai lored to homophilic hypergraphs. We further combine feature-space coverage and s tructural reachability into a monotone submodular selection objective, enabling an efficient lazy greedy solver. Experiments on eight benchmarks demonstrate tha t HIAL consistently outperforms strong baselines across diverse homophilic hyper graph domains.
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High-Dimensional Sensitivity Analysis for Genomic Studies: An Adversarial Framew ork for Learning Worst-Case Latent Confounders

Yifan Lin, Kevin Z. Lin
High-dimensional genomics studies are frequently confounded by unmeasured biological processes that obscure disease-specific signals. While existing workflows can estimate these latent confounders, they fail to quantify how robust a discovery is to varying levels of hypothetical confounding. We introduce sensGAN, a deep-learning adversarial framework that systematically explores the confounding spectrum by learning "worst-case" latent variables that nullify the most gene associations under novel predictive-gain constraints. By identifying the minimum confounding strength required to explain away an observed effect, our method shifts the paradigm toward a formal, quantitative sensitivity analysis. In diverse simulations, sensGAN accurately recovers latent structures and outperforms existing methods in identifying confounder-sensitive genes. Applied to human Alzheimer's disease microglia, our framework prioritizes robust disease pathways while successfully isolating signals driven by unmeasured co-occurring neurodegenerative pathologies.
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Tri-Scale Neural ODEs for Continuous Multi-Omics Disease Modeling
Shohaib Shaffiey, Massimiliano Pierobon
The fields of AI-based disease fingerprinting, drug discovery and repurposing are currently among the emerging frontiers of machine learning applied to medicine. One major challenge is to obtain robust $\textit{in-silico}$ modeling of disease progression while accounting for the vastly different time scales of biochemical interactions, from gene expression to protein abundance and metabolic flux. Discrete sequence models inadequately represent such multi-scale interactions, and standard Neural Ordinary Differential Equations (NODEs) often fail to train stably under stiffness (different time scales). To address this, a Tri-Scale Stiff NODE, defined by hierarchically coupled latent differential equations that model the causal relationships from genes to proteins and metabolites, is introduced and optimized in this paper in terms of reconstruction error and information-theoretic mutual information. This enables continuous-time modeling of cellular responses to identify not only the disease dynamics, but also drug perturbations that act within narrow time windows, often invisible to discrete-time approaches. Lyapunov analysis provides a theoretical guarantee that the modeled trajectories remain stable and well-behaved even under extreme stiffness. The methodology is validated using the STATegra B-cell and Traxler macrophage datasets, with the former utilized for a proof-of-concept drug repurposing pipeline.
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Beyond Test-Time Memory: State-Space Optimal Control for LLM Reasoning
Peihao Wang, Shan Yang, Xijun Wang, Tesi Xiao, Xin Liu, Changlong Yu, Yu Lou, Pan Li, Zhangyang Wang, Ming Lin, Rene Vidal
Associative memory has long underpinned the design of sequential models. Beyond recall, humans reason by *projecting future states and selecting goal-directed actions*, a capability that modern language models increasingly require but do not natively encode. While prior work uses reinforcement learning or test-time training, planning remains external to the model architecture. We formulate reasoning as *optimal control* and introduce the *Test-Time Control (TTC)* layer, which performs finite-horizon *LQR* planning over latent states at inference time, represents a *value function* within neural architectures, and leverages it as the nested objective to enable *planning before prediction*. To ensure scalability, we derive a hardware-efficient LQR solver based on a symplectic formulation and implement it as a fused CUDA kernel, enabling parallel execution with minimal overhead. Integrated as an adapter into pretrained LLMs, TTC layers improve mathematical reasoning performance by up to +27.8% on MATH-500 and 2-3$\times$ Pass@8 improvements on AMC and AIME, demonstrating that embedding optimal control as an architectural component provides an effective and scalable mechanism for reasoning beyond test-time training.
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Path-Coupled Bellman Flows for Distributional Reinforcement Learning
Boyang Xu, Qing Zou, Siqin Yang, Hao Yan
Distributional reinforcement learning (DRL) models the full return distribution,

but typically relies on finite-dimensional categorical or quantile approximations, often involving projection or quantile-regression approximations to the Bellman target, together with independently sampled bootstrap targets that obscure transport structure and add variance. We present Path-Coupled Bellman Flows (PCBF), a continuous-time DRL method that encodes Bellman endpoint consistency and pathwise Bellman-coupled geometry within generative flow trajectories. PCBF represents return distributions via flow matching and couples the paths of consecutive states through shared base noise, yielding a geometric Bellman flow relation between velocity fields. This structure enables a λ-parameterized control-variate target that reduces training variance while preserving the source and Bellman endpoint geometry. Experiments on analytically tractable MRPs, OGBench, and D4RL show improved distributional fidelity, training stability, and competitive offline RL performance.

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Symmetries in PAC-Bayesian Learning

Armin Beck, Peter Ochs

Symmetries are known to improve the empirical performance of machine learning models, yet theoretical guarantees explaining these gains remain limited. Prior work has focused mainly on compact group symmetries and often assumes that the data distribution itself is invariant, an assumption rarely satisfied in real-world applications. In this work, we extend generalization guarantees to the broader setting of non-compact symmetries, such as translations and to non-invariant data distributions. Building on the PAC-Bayes framework, we adapt and tighten existing bounds, demonstrating the approach on McAllester's PAC-Bayes bound while showing that it applies to a wide range of PAC-Bayes bounds. We validate our theory with experiments on several datasets with non-uniform and non-compact transformations, where the derived guarantees not only hold but also improve upon prior results. These findings provide theoretical evidence that, for symmetric data, symmetric models are preferable beyond the narrow setting of compact groups and invariant distributions, opening the way to a more general understanding of symmetries in machine learning.

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PGT: Procedurally Generated Tasks for improving visual grounding in MLLMs

Rim Assouel, Amir Bar, Michal Drozdzal, Adriana Romero-Soriano

Despite remarkable progress in Multimodal Large Language Models (MLLMs), these models still struggle with fine-grained understanding tasks. In this work, we propose **Procedurally Generated Tasks (PGT)** a simple data-driven framework that serves a dual purpose: inducing fine-grained visual understanding and acting as a low-cost diagnostic tool to identify the source of perception failures. By overlaying unambiguous geometric primitives on images, PGT generate additional dense supervision that disentangles visual grounding capability from semantic priors. Extensive experiments on relational, quantitative, and 3D/depth understanding benchmarks show that PGT yields remarkable gains across diverse architectures. Instruction tuning MLLMs on LLaVA-v1.5-Instruct augmented with PGT data results in improvements of up to +20\% on the What’sUp benchmark and +13.3\% on CV-Bench-2D, while maintaining general perception capabilities. Moreover, finetuning state-of-the-art MLLMs on PGT data leads to boosts of up to +5.5\% on What’sUp and +8.3\% on CV-Bench-2D. These findings demonstrate that PGT effectively address the bottleneck of fine-grained perception, revealing that many spatial reasoning deficits stem from inadequate supervision signals rather than inherent architectural or resolution limitations.

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Personalized Additive Modeling for Multi-level Federated Learning

Shutong Chen, Guodong Long, Tianyi Zhou, Jie Ma, Jing Jiang, Chengqi Zhang

Contemporary AI faces the challenge of balancing generality with user-specific personalization. In federated learning (FL), this challenge is amplified by highly heterogeneous client data with complex non-IID patterns beyond standard modeling assumptions. Many existing FL methods are designed for relatively restricted heterogeneity settings (e.g., a fixed number of clusters or a fixed form of personalization), limiting their robustness under complex structures. In this work,

we study FL from a \emph{multi-level non-IID} perspective, where client similarity is approximated by multiple granularities of shared knowledge: global, subgroup, and client-specific components. This view captures coarse-to-fine relationships while requiring less prior knowledge of task boundaries. Building on this insight, we propose \emph{Federated Multi-level Additive Modeling} (FeMAM), which learns multiple levels of shareable models and constructs personalized predictors via additive composition across levels. To move beyond a fixed structure, FeMAM allows models to grow and be pruned dynamically during training, adapting to diverse federated scenarios. Despite employing multiple models, FeMAM remains cost-friendly by activating only a small subset (one level) of models for training at a time. Extensive experiments show that FeMAM effectively approximates complex non-IID structures and consistently outperforms representative clustered and personalized FL baselines.
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CAMP: Coherent Alignment of Multimodal Prototypes for Explainable Complementary Learning
Alvaro Lopez Pellicer, Eoin M. Kenny, Simran Lamba, Shubham Sharma, Plamen P Angelov, Saumitra Mishra
Most multimodal learning assumes redundant views (such as image–caption pairs), yet many applications require combining complementary modalities that provide distinct evidence (such as an X-ray and medical history). We term this setting *Complementary Multimodal Classification* (CMC). In CMC, existing explainable-by-design methods often force an accuracy–interpretability trade-off because single shared similarity metrics fail under asymmetric, class-conditional evidence. To address this, we propose Coherent Alignment of Multimodal Prototypes (CAMP). CAMP enforces coherent multimodal reasoning by aligning class-wise evidence via optimal transport and imposing geometric constraints to counter modality dominance and representation collapse. We provide theoretical guarantees showing that these mechanisms eliminate such degeneracies without restricting expressivity. Empirically, across 16 public CMC datasets, CAMP matches or exceeds large ($>$100M parameter) AutoML baselines with fewer than 1M trainable parameters, and when fine-tuned end-to-end it achieves state-of-the-art performance. To the best of our knowledge, this work is the first modality-agnostic prototype-learning framework designed for complementary multimodal tasks.
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Training–Inference Consistent Segmented Execution for Long-Context LLMs
Xianpeng Shang, Jiang Li, Zehua Duo, Qianyi Cai, Xiangdong Su
Transformer-based large language models face severe scalability challenges in long-context generation due to the computational and memory costs of full-context attention.
Under practical computation and memory constraints, many inference-efficient long-context methods improve efficiency by adopting bounded-context or segment-level execution only during inference, while continuing to train models under full-context attention, resulting in a mismatch between training and inference execution and state-transition semantics.
Based on this insight, we propose a training-consistent segment-level generation framework, in which training and inference follow the same segment-level forward execution semantics.
During training, consistency with inference is enforced by restricting gradient propagation to KV states carried over from the immediately preceding segment, while permitting head-specific access to past KV states during the forward pass without involving them in gradient propagation.
Across long-context benchmarks, our approach achieves performance comparable to full-context attention, while achieving competitive latency--memory trade-offs against strong inference-efficient baselines, and substantially improving scalability at very long context lengths (e.g., approximately $6\times$ lower peak prefill memory at 128K compared to full-context attention with FlashAttention).
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ShapCCS: Shapley-Driven Client Coreset Selection in Federated Learning
Shuo Ji, Jie Hu, Zhouqiao He, Zijie Zhao, Tianrui Li, Jie Xu

Computation overhead has emerged as a critical bottleneck in Federated Learning (FL). Coreset selection tackles this challenge by constructing an informative subset to represent the full dataset. However, existing approaches optimize coreset construction solely at the data level and enforce a uniform retention ratio across all clients, ignoring client heterogeneity and introducing detrimental fragmented clients. In this paper, we first introduce a *gradient projection Shapley value* (GPSV) to evaluate client contributions. GPSV captures both the directional and magnitude information of client updates and enables exact Shapley value calculation with $\mathcal{O}(1)$ per-coalition evaluation. Building on GPSV, we then propose ShapCCS, the first client-level coreset selection strategy for FL. ShapCCS prioritizes clients with high GPSV scores while excluding fragmented clients with negligible or even negative GPSV. As a client-level coreset selection strategy, ShapCCS can be integrated with a data-level selection approach, and additionally reduces communication costs, an advantage unattainable by data-level methods alone. Extensive experiments demonstrate the superiority of ShapCCS on model performance and robustness to noise. The code is available at https://anonymous.4open.science/r/ShapCCS-5CBB.

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Multimodal Latent Language Modeling with Next-Token Diffusion

Yutao Sun, Hangbo Bao, Wenhui Wang, Zhiliang Peng, Li Dong, Shaohan Huang, Yaoyao Chang, Jianyong Wang, Furu Wei

Multimodal generative models require a unified approach to handle both discrete data (e.g., text and code) and continuous data (e.g., image, audio, video). In this work, we propose Latent Language Modeling (LatentLM), which seamlessly integrates continuous and discrete data using causal Transformers. Specifically, we employ a variational autoencoder (VAE) to represent continuous data as latent vectors and introduce next-token diffusion for autoregressive generation of these vectors. Additionally, we develop σ-VAE to address the challenges of variance collapse, which is crucial for autoregressive modeling. Extensive experiments demonstrate the effectiveness of LatentLM across various modalities. In image generation, LatentLM sis competitive with or outperforms DiT-style baselines under matched unified settings. When integrated into multimodal large language models, LatentLM provides a general-purpose interface that unifies multimodal generation and understanding. Experimental results show that LatentLM achieves favorable performance compared to Transfusion and vector quantized models in the setting of scaling up training tokens. In text-to-speech synthesis, LatentLM outperforms the state-of-the-art VALL-E 2 model in speaker similarity and robustness, while requiring 10 fewer decoding steps. The results establish LatentLM as a highly effective and scalable approach to advance large multimodal models.

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On Training Large Language Models for Long-Horizon Tasks: An Empirical Study of Horizon Length

Sunghwan Kim, Junhee Cho, Beong-woo Kwak, Taeyoon Kwon, Liang Wang, Nan Yang, Xingxing Zhang, Furu Wei, Jinyoung Yeo

Large language models (LLMs) have shown promise as interactive agents that solve tasks through extended sequences of environment interactions. While prior work has primarily focused on system-level optimizations or algorithmic improvements, the role of task horizon length in shaping training dynamics remains poorly understood. In this work, we present a systematic empirical study that examines horizon length through controlled task constructions. Specifically, we construct controlled tasks in which agents face identical decision rules and reasoning structures, but differ only in the length of action sequences required for successful completion. Our results reveal that increasing horizon length alone constitutes a training bottleneck, inducing severe training instability driven by exploration difficulties and credit assignment challenges. We demonstrate that horizon reduction is a key principle to address this limitation, stabilizing training and achieving better performance in long-horizon tasks. Moreover, we find that horizon reduction is related to stronger generalization across horizon lengths: models trained under reduced horizons generalize more effectively to longer-horizon variants at inference time, a phenomenon we refer to as horizon generalization.

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On the Interaction of Batch Noise, Adaptivity, and Compression, under (L_0,L_1)-Smoothness: An SDE Approach
Enea Monzio Compagnoni, Rustem Islamov, Frank Norbert Proske, Aurelien Lucchi, Antonio Orvieto, Eduard Gorbunov
Distributed stochastic optimization intertwines (i) stochastic gradient noise, (ii) communication compression, and (iii) adaptive/normalized updates. While each factor has been studied in isolation, their joint effect under realistic assumptions remains poorly understood. In this work, we develop a unified theoretical framework for Distributed Compressed SGD (DCSGD) and its sign variant Distributed SignSGD (DSignSGD) under the recently introduced (L_0, L_1)-smoothness condition. From a conceptual perspective, we show that the first- and second-order modified equations from the literature do not accurately model the discrete-time step-size/stability restrictions, especially under (L_0,L_1)-smoothness. From a technical perspective, we propose new first-order SDEs by carefully incorporating curvature-dependent terms into their drift: This helps capture the fine-grained relationship between learning rate restrictions, gradient noise, compression, and the geometry of the loss landscape. Importantly, we do so under general gradient noise assumptions, including heavy-tailed and affine-variance regimes, which extend beyond the classical bounded-variance setting. Our results suggest that normalizing the updates of DCSGD emerges as a natural condition for stability, with the degree of normalization precisely determined by the gradient noise structure, the landscape’s regularity, and the compression rate. In contrast, DSignSGD converges even under heavy-tailed noise with standard learning rate schedules. Together, these findings offer both new theoretical insights and perspectives, and practical guidance.
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Who can we trust? LLM-as-a-jury for Comparative Assessment
Mengjie Qian, Guangzhi Sun, Mark Gales, Kate Knill
Large language models (LLMs) are increasingly applied as automatic evaluators for natural language generation assessment often using pairwise comparative judgements. Existing approaches typically rely on single judges or aggregate multiple judges assuming equal reliability. In practice, LLM judges vary substantially in performance across tasks and aspects, and their judgment probabilities may be biased and inconsistent. Furthermore, human-labelled supervision for judge calibration may be unavailable. We first empirically demonstrate that inconsistencies in LLM comparison probabilities exist and show that it limits the effectiveness of direct probability-based ranking. To address this, we study the \emph{LLM-as-a-jury} setting and propose BT-σ, a judge-aware extension of the Bradley-Terry model that introduces a discriminator parameter for each judge to jointly infer item rankings and judge reliability from pairwise comparisons alone. Experiments on benchmark NLG evaluation datasets show that \textit{BT-σ} consistently outperforms averaging-based aggregation methods, and that the learned discriminators strongly correlate with independent measures of the cycle consistency of LLM judgments. Further analysis reveals that \textit{BT-σ} can be interpreted as an unsupervised calibration mechanism that improves aggregation by modelling judge reliability.
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SLAT: Segment-Level Adaptive Trimming for Efficient CoT Reasoning
Jian Yao, Xiongcai Luo, Ran Cheng, KC Tan
Recent advances in Large Reasoning Models have significantly improved chain-of-thought (CoT) capabilities via reinforcement learning (RL).
However, generated reasoning chains frequently suffer from structural redundancy (i.e., \emph{overthinking}), incurring high computational overhead without improving answer correctness.
Existing mitigation strategies typically rely on token-uniform length penalties, which provide coarse, segment-agnostic pressure toward shorter outputs and can inadvertently suppress useful reasoning alongside redundancy.
To address this, we demonstrate that inefficiency concentrates in high-probability segments with low marginal utility.

We derive a theoretical characterization of segment suboptimality under the corr
ectness-length trade-off objective and propose \textsc{SLAT} (Segment-Level Adap
tive Trimming), an RL framework that selectively suppresses redundant segments b
ased on this criterion.
Empirical results on standard benchmarks indicate that \textsc{SLAT} establishes
 a superior accuracy-efficiency Pareto frontier, reducing reasoning length by 50
\% relative to uncompressed baselines while maintaining competitive accuracy.
Overall, our results suggest that theoretically grounded, segment-aware trimming
 is a promising direction for efficient CoT reasoning in large language models.
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Learning Normalized Energy Models for Linear Inverse Problems
Nicolas Zilberstein, Santiago Segarra, Eero P Simoncelli, Florentin Guth
Generative diffusion models can provide powerful prior probability models for in
verse problems in imaging, but existing implementations suffer from two key limi
tations: (i) the prior density is represented implicitly, and (ii) they rely
 on likelihood approximations that introduce sampling biases. We address these c
hallenges by introducing a new energy-based model trained for denoising with a c
ovariance-based regularization term that enforces consistency across different m
easurement conditions. The trained model can compute normalized posterior densit
ies for diverse linear inverse problems, without additional retraining or fine t
uning. In addition to preserving the sampling capabilities of diffusion models,
this enables previously unavailable capabilities: energy-guided adaptive samplin
g that adjusts schedules on-the-fly, unbiased Metropolis-Hastings correction ste
ps, and blind estimation of the degradation operator via Bayes rule. We validate
 the method on multiple datasets (ImageNet, CelebA, AFHQ) and tasks (inpainting,
 deblurring), demonstrating competitive or superior performance to established b
aselines.
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Benchmarking the Scientific Mind: A Pathology-Derived Biomedical VQA Benchmark f
or Complex Scientific Reasoning
Ziyu Zhao, Yiyang Liu, Yajiao Wang, Xiaotao Wang, Yang Li, Yuyang Peng, Jiaheng
Zhou, Jinqiao Wang, Yingying Chen, Ge Yang, Haixin Wang
Despite progress of Multimodal Large Language Models (MLLMs) in biomedical visua
l question answering (VQA), existing benchmarks provide limited assessment of th
eir scientific reasoning capabilities. Most datasets adopt single-image question
 construction and outcome-oriented evaluation, where correctness is judged by an
swer plausibility rather than alignment with experimental evidence. Such formula
tions fail to capture the evidence-constrained, multi-step nature of biomedical
reasoning, and obscure whether models can derive conclusions through causal inte
rpretation of experimental observations.

To address these critical gaps in reasoning evaluation, we propose a principled
benchmark construction framework that reconstructs scientific reasoning paths di
rectly from biomedical literature. By jointly modeling clusters of experimentall
y related images together with their captions and context, the framework generat
es tightly coupled question–reasoning–answer triples that require multi-image in
tegration and explicit evidence-driven inference. Based on this framework, we in
troduce SORBE (Scientific Observation & Reasoning for Biomedical Evaluation), a
large-scale multi-image pathology-derived biomedical VQA benchmark designed to e
valuate evidence alignment and multi-step experimental reasoning. Under a proces
s-oriented evaluation metric, state-of-the-art biomedical-specialized MLLMs exhi
bit substantial performance degradation, revealing systematic limitations in evi
dence grounding and causal reasoning that are not reflected by existing benchmar
ks. Data and code are available at: https://github.com/UniverseOfUniverse/SORBE.
git.
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Simple yet Effective: Low-Rank Spatial Attention for Neural Operators
Zherui Yang, Haiyang Xin, Tao Du, Ligang Liu
Neural operators have emerged as data-driven surrogates for solving partial diff
erential equations (PDEs), and their success hinges on efficiently modeling the

long-range, global coupling among spatial points induced by the underlying physi
cs.
In many PDE regimes, the induced global interaction kernels are empirically comp
ressible, exhibiting rapid spectral decay that admits low-rank approximations.
We leverage this observation to unify representative global mixing modules in ne
ural operators under a shared low-rank template: compressing high-dimensional po
intwise features into a compact latent space, processing global interactions wit
hin it, and reconstructing the global context back to spatial points.
Guided by this view, we introduce Low-Rank Spatial Attention (LRSA) as a clean a
nd direct instantiation of this template.
Crucially, unlike prior approaches that often rely on non-standard aggregation o
r normalization modules, LRSA is built purely from standard Transformer primitiv
es, i.e., attention, normalization, and feed-forward networks, yielding a concis
e block that is straightforward to implement and directly compatible with hardwa
re-optimized kernels.
In our experiments, such a simple construction is sufficient to achieve high acc
uracy, yielding an average error reduction of over 17\% relative to second-best
methods, while remaining stable and efficient in mixed-precision training.
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Efficient Online Variational Estimation via Monte Carlo Sampling
Mathis Chagneux, Mathias Müller, Pierre Gloaguen, Sylvain Le Corff, Jimmy Olsson
This article addresses online variational estimation in parametric state-space m
odels.
We propose a new procedure for efficiently computing the evidence lower bound an
d its gradient in a streaming-data setting, where observations arrive sequential
ly.
The algorithm allows for the simultaneous training of the model parameters and t
he distribution of the latent states given the observations.
It is based on i.i.d. Monte Carlo sampling, coupled with a well-chosen deep arch
itecture, enabling both computational efficiency and flexibility. The performanc
e of the method is illustrated on both synthetic data and real-world air-quality
 data.
The proposed approach is theoretically motivated by the existence of an asymptot
ic contrast function and the ergodicity of the underlying Markov chain, and appl
ies more generally to the computation of additive expectations under posterior d
istributions in state-space models.
****
Learning Gaussian Mixture-distributed Prototypes for 3D Scene Graph Generation f
rom RGB-D Sequences
Rongxing Ding, Hongyu Qu, Xinguang Xiang, Pengpeng Li, Xiangbo Shu
3D Scene Graph Generation (3DSGG) aims to create a structured representation of
3D environment by identifying objects as nodes and
their relations as edges. Existing 3DSGG methods based on RGB-D sequences typic
ally put much focus on the adaption of neural networks to robust node and edge f
eature extraction in complex 3D scenes, yet ignoring the inherent intra-class di
versity within each class and inter-class similarity between different categorie
s associated with nodes and edges. In this work, we develop GMPSSG, a novel Gaus
sian Mixture-distributed Prototype mining framework for 3DSGG. Specifically, we
model different categories with independent Gaussian Mixture-distributed Prototy
pe to effectively mitigate inter-class similarity, while employing multiple Gaus
sian components within each prototype to capture intra-class diversity. Moreover
, Prototype-anchored Representation Learning is introduced to construct a well-s
tructured and mutually independent category space; Topology-aware Prototype Inte
raction is devised to capture implicit co-occurrence priors within the scene, an
d leverage them to calibrate prototype distributions, thereby ensuring the plaus
ibility of node-edge matching. Experiments on 3DSSG dataset demonstrate GMPSSG o
utperforms various top-leading methods. Our code is available at GMPSSG.
****
From Inpainting to Editing: Unlocking Robust Mask-Free Visual Dubbing via Genera
tive Bootstrapping

Xu He, Haoxian Zhang, Hejia Chen, Changyuan Zheng, Liyang Chen, Songlin Tang, Jiehui Huang, Xiaoqiang Liu, Pengfei Wan, Zhiyong Wu

Audio-driven visual dubbing aims to synchronize a video's lip movements with new speech but is fundamentally challenged by the lack of ideal training data: paired videos differing only in lip motion.

Existing methods circumvent this via mask-based inpainting. However, masking inevitably destroys spatiotemporal context, leading to identity drift and poor robustness (e.g., to occlusions), while also inducing lip-shape leakage that degrades lip sync.

To bridge this gap, we propose X-Dub, a novel two-stage generative bootstrapping framework leveraging powerful Diffusion Transformers to unlock mask-free dubbing.

Our core insight is to repurpose a mask-based inpainting model exclusively as a dedicated data generator to synthesize scalable, high-fidelity pseudo-paired data, which is subsequently utilized to train and bootstrap a robust, mask-free editing model as the final video dubber.

The final dubber is liberated from masking artifacts and leverages the complete video input for high-fidelity inference.

We further introduce timestep-adaptive multi-phase learning to disentangle conflicting objectives (structure, lip motion, and texture) across diffusion phases, facilitating stable convergence and advanced editing quality.

Additionally, we present X-DubBench, a benchmark for diverse scenarios.

Extensive experiments demonstrate that our method achieves state-of-the-art performance with superior lip sync, visual quality, and robustness.

Code, demos, and additional resources are available at https://github.com/KlingAIResearch/X-Dub.

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CSG: Cognitive Structure Generation for Intelligent Education

Hengnian Gu, Zhifu Chen, Yuxin Chen, Jin Peng Zhou, Dongdai Zhou

Cognitive structure (CS), a student's construction of concepts and inter-concept relations, has long been recognized as a foundational notion in psychology and intelligent education, yet remains largely unassessable in practice. Existing approaches such as knowledge tracing (KT) and cognitive diagnosis (CD) simplify and indirectly approximate CS, but they intertwine representation learning with prediction objectives, limiting generalization, interpretability, and reuse across tasks. To address this gap, we propose Cognitive Structure Generation (CSG), a task-agnostic framework that explicitly models CS through generative modeling. Based on educational theories, CSG first pretrains a Cognitive Structure Diffusion Probabilistic Model (CSDPM) and then applies reinforcement learning with SOLO-based hierarchical rewards to capture plausible patterns of cognitive development. By decoupling cognitive structure representation from downstream prediction, CSG produces interpretable and transferable cognitive structures that can be seamlessly integrated into diverse student modeling tasks. Experiments on five real-world datasets show that CSG yields more comprehensive representations, substantially improving performance while offering enhanced interpretability and modularity.

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Optimal Transport–Guided Stochastic Control for Graph Combinatorial Optimization

Yang Huang, Yifan Zhang, Jian Cheng

We propose an OT-guided sampling framework for solving graph combinatorial optimization through exact multilinear relaxation. Graph combinatorial optimization problems can be written as quadratic unconstrained binary optimization(QUBO). Leveraging a classical result in combinatorial optimization, we obtain a continuous multi-linear relaxation of QUBO that is exact, in the sense that it preserves the optimal binary solutions. The challenge is that the resulting energy landscape is highly nonconvex.

We address this by treating the objective as an energy function and optimizing via sampling from the induced Boltzmann distribution to escape poor local optima. Viewing sampling as transporting a simple reference distribution to the target distribution, we use optimal transport to characterize more efficient probabilit

y flow and derive a stochastic optimal control problem whose solution yields an optimal sampling dynamics. We parameterize the control policy with graph neural networks to approximate the optimal control. Experiments show improved solution quality and efficiency over strong combinatorial and learning-based baselines.
****
AOEPT: Breaking the Implicit Modality-Reduction Bottleneck in Modality-Missing Prompt Tuning
Jian Lang, Rongpei Hong, Ting Zhong, Fan Zhou
Deploying multimodal systems in real-world environments often entails handling modality-missing scenarios, where one or more modalities are unavailable. While recent studies address this challenge for the general Multimodal Transformer (MT) architecture via prompt tuning, we identify a fundamental limitation in these methods: the Implicit Modality-Reduction bottleneck. By conditioning prompts solely on the observed modalities, they inadvertently restrict the reasoning scope of MTs to the modality-reduced subspace, cutting off access to the latent information sources of the missing modalities. To overcome this limitation, we propose AOEPT, which pioneers a novel modal-contextualized prompting fashion. Specifically, we introduce lightweight Modal-Contextualized Prompts (MCPs) that distill global modality-wise priors from training data, serving as latent repositories of the information sources for missing modalities. Conditioned on the remaining modalities, these MCPs are instantiated into instance-aware prompts that selectively augment missing-modality information for each sample, thereby restoring the reasoning scope of MTs beyond the observed-modality-only subspace. Experiments across various multimodal benchmarks and backbones confirm the strong performance of AOEPT, with minimal computational overhead.
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Exactly Computing do-Shapley Values
R. Teal Witter, Álvaro Parafita, Tomas Garriga, Maximilian Muschalik, Fabian Fumagalli, Axel Brando, Lucas Rosenblatt
Structural Causal Models (SCM) are a powerful framework for describing complicated dynamics across the natural sciences.
A particularly elegant way of interpreting SCMs is via do-Shapley values, a game-theoretic method of quantifying the average effect of d variables across exponentially many interventions.
Like Shapley values, computing do-Shapley values generally requires evaluating exponentially many terms.
The foundation of our work is a reformulation of do-Shapley values in terms of the irreducible sets of the underlying SCM.
Leveraging this insight, we can exactly compute do-Shapley values in time linear in the number of irreducible sets r, which itself can range from d to 2^d depending on the graph structure of the SCM.
Since r is unknown a priori, we complement the exact algorithm with an estimator that, like general Shapley value estimators, can be run with any query budget.
As the query budget approaches r, our estimators can produce more accurate estimates than prior methods by several orders of magnitude, and, when the budget reaches r, return the Shapley values up to machine precision.
Beyond computational speed, we also reduce the identification burden: we prove that non-parametric identifiability of do-Shapley values requires only the identification of interventional effects for the d singleton coalitions, rather than all classes.
****
Large-Scale Terminal Agentic Trajectory Generation from Dockerized Environments
Siwei Wu, Yizhi LI, Yuyang Song, Wei Zhang, Yang Wang, Riza Batista-Navarro, Xian Yang, Mingjie Tang, Bryan Dai, Jian Yang, Chenghua Lin
Training agentic models for terminal-based tasks critically depends on high-quality terminal trajectories that capture realistic long-horizon interactions across diverse domains.
However, constructing such data at scale remains challenging due to two key requirements:

\textbf{\emph{Executability}}, since each instance requires a suitable and often distinct Docker environment;
and \textbf{\emph{Verifiability}}, because heterogeneous task outputs preclude unified, standardized verification.
To address these challenges, we propose \textbf{TerminalTraj}, a scalable pipeline that (i) filters high-quality repositories to construct Dockerized execution environments, (ii) generates Docker-aligned task instances, and (iii) synthesizes agent trajectories with executable validation code.
Using TerminalTraj, we curate 32K Docker images and generate 50,733 verified terminal trajectories across eight domains.
Models trained on this data with the Qwen2.5-Coder backbone achieve consistent performance improvements on TerminalBench (TB), with gains of up to 20\% on TB 1.0 and 10\% on TB 2.0 over their respective backbones.
Notably, \textbf{TerminalTraj-32B} achieves strong performance among models with fewer than 100B parameters, reaching 35.30\% on TB 1.0 and 22.00\% on TB 2.0, and demonstrates improved test-time scaling behavior.
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Can Computational Reducibility Lead to Transferable Models for Graph Combinatorial Optimization?

Semih Cantürk, Thomas Sabourin, Frederik Wenkel, Michael Perlmutter, Guy Wolf

A key challenge in deriving unified neural solvers for combinatorial optimization (CO) is efficient generalization of models between one set of tasks to new tasks not used during the initial training process. To address it, we first establish a new GNN encoder, which uses a GCON module as a form of expressive message passing together with energy-based unsupervised loss functions. This model achieves high performance (often comparable with state-of-the-art results) across multiple CO tasks when trained individually on each task. We then leverage knowledge from the computational reducibility literature to propose pretraining and fine-tuning strategies to transfer effectively (a) between MVC, MIS and MaxClique, and (b) in a multi-task learning setting that additionally incorporates MaxCut, MDS and graph coloring. Additionally, in a leave-one-out, multi-task learning setting, we observe that pretraining on all but one task almost always leads to faster convergence on the remaining task when fine-tuning while avoiding negative transfer. Our findings indicate that learning common representations across multiple graph CO problems is viable through the use of expressive message passing coupled with pretraining strategies that are informed by the polynomial reduction literature, thereby taking an important step towards enabling the development of foundational models for neural CO.
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When Benign Inputs Lead to Severe Harms: Eliciting Unsafe Unintended Behaviors of Computer-Use Agents

Jaylen Jones, Zhehao Zhang, Yuting Ning, Eric Fosler-Lussier, Pierre-Luc St-Charles, Yoshua Bengio, Dawn Song, Yu Su, Huan Sun

Although computer-use agents (CUAs) hold significant potential to automate increasingly complex OS workflows, they can demonstrate unsafe unintended behaviors that deviate from expected outcomes even under benign input contexts. However, exploration of this risk remains largely anecdotal, lacking concrete characterization and automated methods to proactively surface long-tail unintended behaviors under realistic CUA scenarios. To fill this gap, we introduce the first conceptual and methodological framework for unintended CUA behaviors, by defining their key characteristics, automatically eliciting them, and analyzing how they arise from benign inputs. We propose AutoElicit: an agentic framework that iteratively perturbs benign instructions using CUA execution feedback, and elicits severe harms while keeping perturbations realistic and benign. Using AUTOELICIT, we surface hundreds of harmful unintended behaviors from state-of-the-art CUAs such as Claude 4.5 Haiku, Claude 4.5 Opus, and Operator. We further evaluate the transferability of human-verified successful perturbations, identifying persistent susceptibility to unintended behaviors across various other frontier CUAs. This work establishes a foundation for systematically analyzing unintended behaviors in realistic computer-use settings.

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Learning Interpretable Options by Identifying Reward Diffusion Bottlenecks in Reinforcement Learning
Yiming Fei, Lang Qin, Rui Yan, Huajin Tang
Bottleneck states, which connect distinct regions of the state space, provide a principled and interpretable basis for constructing temporal abstractions in Hierarchical Reinforcement Learning (HRL). However, existing bottleneck identification methods primarily rely on topological analysis of the state-transition graph, limiting their scalability to high-dimensional or continuous domains. To address this challenge, we introduce Value Power Strength (VPS), a value function-based metric inspired by the analogy between the Bellman equation and Kirchhoff's current law, to quantify bottleneck property via the diffusion of reward in Markov Decision Processes (MDPs). VPS is estimated efficiently using value functions learned from random reward signals and captures reward diffusion bottlenecks in both discrete and continuous state spaces.
Leveraging VPS, we design options that guide agents toward or away from bottleneck regions. Experimental results on classic tabular domains, continuous-control PointMaze, and Atari 2600 games demonstrate that the VPS-based framework discovers semantically meaningful subgoals and substantially improves exploration efficiency.
****
Enhancing Train-Free Infinite-Frame Generation for Consistent Long Videos
Xiaokun Feng, Jiashu Zhu, Meiqi Wu, Chubin Chen, Fangyuan Mao, Haiyang Guo, Jiahong Wu, Xiangxiang Chu, Kaiqi Huang
Without incurring significant computational overhead, train-free long video generation aims to enable foundation video generation models to produce longer videos. Frame-level autoregressive frameworks, e.g., FIFO-diffusion, offer the advantage of generating infinitely long videos with constant memory consumption. However, the mismatch between training and inference, coupled with the challenge of maintaining long-term consistency, limits the effective utilization of foundation models. To mitigate these concerns, we propose MIGA, a novel infinite-frame long video generation method. Firstly, we propose an effective two-stage alignment mechanism that mitigates the training-inference gap by reducing the excessive noise span fed to the model. We then introduce an innovative dual consistency enhancement mechanism, where the self-reflection approach corrects early high-noise frames and the long-range frame guidance approach leverages later low-noise frames with broad coverage to steer generation, jointly improving temporal consistency. Extensive experiments on VBench and NarrLV demonstrate the state-of-the-art performance of MIGA. Our project page is available at https://xiaokunfeng.github.io/miga_homepage/.
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SAEmnesia: Erasing Concepts in Diffusion Models with Supervised Sparse Autoencoders
Enrico Cassano, Riccardo Renzulli, Marco Nurisso, Mirko Zaffaroni, Alan Perotti, Marco Grangetto
Concept unlearning in diffusion models is hampered by feature splitting, where concepts are distributed across many latent features, making their removal challenging and computationally expensive. We introduce SAEmnesia, a supervised sparse autoencoder framework that overcomes this by enforcing one-to-one concept-neuron mappings. By systematically labeling concepts during training, our method achieves feature centralization, binding each concept to a single, interpretable neuron. This enables highly targeted and efficient concept erasure. Compared to the state-of-the-art sparse autoencoder-based unlearning approach, SAEmnesia reduces hyperparameter search by 96.67\% and achieves a 9.22\% improvement on the UnlearnCanvas benchmark for objects. Our method also shows superior scalability in sequential unlearning, improving accuracy by 28.4\% when removing nine objects, establishing a step forward for precise and controllable concept erasure. Moreover, SAEmnesia effectively suppresses nudity on the I2P benchmark and remains robust to adversarial attacks. Source code available at https://github.com/EIDOSLAB/SAEmnesia.

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Same Graph Cross-Task Transfer in GNNs: Protocols and Predictors
Neelam Akula, Surbhi Kumar, Murat Kantarcioglu, Baris Coskunuzer
Many real-world graphs support multiple predictive tasks over the same underlying structure, creating an opportunity to reuse supervision across node classification (NC) and link prediction (LP). However, existing evaluations often rely on incompatible splits, observed-graph assumptions, and negative sampling rules, making conclusions about same-graph cross-task transfer unreliable. We formalize same-graph NC–LP transfer and propose a leakage-free protocol that fixes node and edge splits, uses a shared message-passing graph that excludes evaluated edges, and employs fixed negatives for LP. Across three backbones (GCN, GraphSAGE, GPS), we find transfer is strongly directional and predictable: NC$\to$LP is consistently beneficial on homophilic graphs, while LP$\to$NC is fragile and can even degrade accuracy under naive representation reuse. LP$\to$NC becomes reliably positive mainly in a structure-dominant regime where LP is easy but NC is unsaturated, suggesting LP acts as structural pretraining. Finally, we introduce CoTask Score (CTS) to summarize joint NC+LP utility when a shared encoder must serve both tasks, and show that simple dataset statistics, especially homophily, can guide mechanism choice and help avoid negative transfer.
****
Causal Dependency-Aware Unsupervised Routing for Large Reasoning Models
Jiacheng Liu, Hao Liu, Xiaofeng Hou, Wei Xue, Yike Guo
As Large Language Model (LLM) ecosystems grow, routing queries to the most suitable model in a diverse pool has become a critical strategy for building efficient and high-performing AI systems.
A common approach is to train a supervised router; however, this requires vast, expensive human-annotated preference data and creates models that are notoriously brittle, failing to generalize when faced with inevitable distribution shifts in user queries. Consequently, developing robust, unsupervised routing methods that adapt without retraining is a crucial research frontier.
This challenge is severely amplified by Large Reasoning Models (LRMs), which introduce a dual problem for any label-free method: their outputs have a causal thinking $\to$ answer structure that must be modeled, and a structural imbalance where long reasoning text can dominate the final answer signal. We introduce ReasoningRouter, a novel framework that resolves these issues with a length-balanced embedding strategy and a probabilistic model capturing the thinking-to-answer dependency.
The proposed Causal Triangulation Property enables the label-free estimation of component qualities and their causal link.
Beyond competitive routing accuracy, ReasoningRouter offers unprecedented insights into model behavior, enabling separate quality assessment of reasoning and answer components while maintaining computational efficiency.
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PragLocker: Protecting Agent Intellectual Property in Untrusted Deployments via Non-Portable Prompts
Qinfeng Li, Yuntai Bao, Jianghui Hu, Wenqi Zhang, Jintao Chen, Huifeng Zhu, Yier Jin, Xuhong Zhang
LLM agents rely on prompts to implement task-specific capabilities based on foundation LLMs, making agent prompts valuable intellectual property. However, in untrusted deployments, adversaries can copy and reuse these prompts with other proprietary LLMs, causing economic losses. To protect these prompts, we identify four key challenges: proactivity, runtime protection, usability, and non-portability that existing approaches fail to address. We present PragLocker, a prompt protection scheme that satisfies these requirements. PragLocker constructs function-preserving obfuscated prompts by anchoring semantics with code symbols and then using target-model feedback to inject noise, yielding prompts that only work on the target LLM. Experiments across multiple agent systems, datasets, and foundation LLMs show that PragLocker substantially reduces cross-LLM portability, maintains target performance, and remains robust against adaptive attackers.
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OpenDeception: Learning Deception and Trust in Human–AI Interaction via Multi-Agent Simulation
Yichen Wu, Qianqian Gao, Xudong Pan, Geng Hong, Min Yang
As large language models (LLMs) are increasingly deployed as interactive agents, open-ended human-AI interactions can involve deceptive behaviors with serious real-world consequences, yet existing evaluations remain largely scenario-specific and model-centric.
We introduce *OpenDeception*, a lightweight framework for jointly evaluating deception risk from both sides of human-AI dialogue. It consists of a scenario benchmark with 50 real-world deception cases, an *IntentNet* that infers deceptive intent from agent reasoning, and a *TrustNet* that estimates user susceptibility. To address data scarcity, we synthesize high-risk dialogues via LLM-based role-and-goal simulation, and train the *TrustNet* using contrastive learning on controlled response pairs, avoiding unreliable scalar labels. Experiments on 11 LLMs and three large reasoning models show that over 90% of goal-driven interactions in most models exhibit deceptive intent, with stronger models displaying higher risk. A real-world case study adapted from a documented AI-induced suicide incident further demonstrates that our joint evaluation can proactively trigger warnings before critical trust thresholds are reached.
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Discovering Scaling Exponents with Physics-Informed Müntz-Szász Networks
Gnankan Landry Regis N'guessan, Bum Jun Kim
Physical systems near singularities, interfaces, and critical points exhibit power-law scaling, yet standard neural networks leave the governing exponents implicit. We introduce physics-informed Müntz-Szász Networks (MSN-PINN), a power-law basis network that treats scaling exponents as trainable parameters. The model outputs both the solution and its scaling structure. We prove identifiability, or unique recovery, and show that, under these conditions, the squared error between learned and true exponents scales as $O(|\mu - \alpha|^2)$. Across experiments, MSN-PINN achieves single-exponent recovery with 1–5\% error under noise and sparse sampling. It recovers corner singularity exponents for the two-dimensional Laplace equation with 0.009\% error, matches the classical result of Kondrat'ev (1967), and recovers forcing-induced exponents in singular Poisson problems with 0.03\% and 0.05\% errors. On a 40-configuration wedge benchmark, it reaches a 100\% success rate with 0.022\% mean error. Constraint-aware training encodes physical requirements such as boundary condition compatibility and improves accuracy by three orders of magnitude over naive training.
****
The Two-Hump Problem: Bridging the Difficulty Gap in Mathematical Reinforcement Learning
Lucas Fagan, Michele Tarquini, Ali Shehper, Maksymilian Manko, Angus Gruen, Coco Huang, Giorgi Butbaia, Davide Passaro, Sergei Gukov
Mathematical search problems present a unique challenge for Reinforcement Learning (RL) due to vast search spaces and sparse rewards. In previous works, the Andrews-Curtis (AC) conjecture was established as an illustrative example of such problems. In this work, we identify a critical structural barrier in the AC landscape: a "Two Hump" distribution, where problem instances are either trivially solvable or effectively impossible, with a scarcity of intermediate "hard-but-solvable" instances required for effective learning. We tackle this challenge through two primary avenues: novel data generation techniques to populate the difficulty gap, and significant algorithmic enhancements including the introduction of supermoves and Transformer-based architectures. We demonstrate substantial performance improvements over previous baselines, and release new comprehensive benchmark datasets including **AC-19** (125,192 AC-trivial presentations of varying difficulty with length at most 19) and **AC-1M** (1,136,154 hard AC-trivial presentations of length at most 30), the first large-scale, publicly available datasets of this kind.
****
On the Existence of Consistent Adversarial Attacks in High-Dimensional Linear Classification

Matteo Vilucchio, Lenka Zdeborová, Bruno Loureiro
What fundamentally distinguishes an adversarial attack from a misclassification due to limited model expressivity or finite data?
In this work, we investigate this question in the setting of high-dimensional binary classification, where statistical effects due to limited data availability play a central role.
We introduce a new error metric that precisely capture this distinction, quantifying model vulnerability to consistent adversarial attacks --- perturbations that preserve the ground-truth labels.
Our main technical contribution is an exact and rigorous asymptotic characterization of these metrics in both well-specified models and latent space models, revealing different vulnerability patterns compared to standard robust error measures.
The theoretical results demonstrate that as models become more overparameterized, their vulnerability to label-preserving perturbations grows, offering theoretical insight into the mechanisms underlying model sensitivity to adversarial attacks.
****
Geometry-Aware Neural Optimizer for Shape Optimization and Inversion
Sun Guoze, Tianya Miao, Haoyang Huang, Huaguan Chen, Han Wan, Rui Zhang, Hao Sun
Geometry is central to PDE-governed systems, motivating shape optimization and inversion. Classical pipelines conduct costly forward simulation with geometry processing, requiring substantial expert effort. Neural surrogates accelerate forward analysis but do not close the loop because gradients from objectives to geometry are often unavailable. Existing differentiable methods either rely on restrictive parameterizations or unstable latent optimization driven by scalar objectives, limiting interpretability and part-wise control. To address these challenges, we propose Geometry-Aware Neural Optimizer (\textbf{\textsc{GANO}}), an end-to-end differentiable framework that unifies geometry representation, field-level prediction, and automated optimization/inversion in a single latent-space loop. \textsc{GANO} encodes shapes with an auto-decoder and stabilizes latent updates via a denoising mechanism, and a geometry-informed surrogate provides a reliable gradient pathway for geometry updates. Moreover, \textsc{GANO} supports part-wise control through null-space projection and uses remeshing-free projection to accelerate geometry processing. We further prove that denoising induces an implicit Jacobian regularization that reduces decoder sensitivity, yielding controlled deformations. Experiments on three benchmarks spanning 2D Helmholtz, 2D airfoil, and 3D vehicles show state-of-the-art accuracy and stable, controllable updates, achieving up to +55.9% lift-to-drag improvement for airfoils and ~7% drag reduction for vehicles.
****
ActiveScope: Actively Seeking and Correcting Perception for MLLMs
Yajing Wang, Chao Bi, Junshu Sun, Shufan Shen, Zhaobo Qi, Shuhui Wang, Qingming Huang
Multimodal Large Language Models (MLLMs) have demonstrated impressive vision-language understanding, yet still struggle with fine-grained perception in high-resolution images.
While existing training-free methods typically rely on attention-based localization or coarse-to-fine search, they are often misled by distractors and fail to locate multiple targets.
Our investigation attributes these failures to *Contextual Dominance*, where salient distractors overwhelm target attention and cause inaccurate localization, and *Semantic Bias*, where global semantics cause the model to fixate on the most salient concept, resulting in incomplete localization in multi-object scenarios.
Built on these insights, we propose ActiveScope, a training-free framework that enhances MLLMs by actively seeking and correcting perception. ActiveScope features two modules. The *Semantic Anchor Localization (SAL)* utilizes fine-grained semantic anchors to independently localize key targets, thereby mitigating semantic bias. The *Interference-Suppressed Refinement (ISR)* refines localization by

suppressing attention on salient distractions to overcome contextual dominance.
Extensive experiments on high-resolution image understanding benchmarks demonstr
ate that ActiveScope outperforms existing training-free methods (e.g., 96.34\% a
ccuracy on V^{*} Bench),
validating the superiority of the active search and self-correction paradigm. Ou
r code is available at https://github.com/jasmine-ww/ActiveScope.
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Beyond the Final Answer: Evaluating the Reasoning Trajectories of Tool-Augmented
 Agents

Wonjoong Kim, Sangwu Park, Yeonjun In, Sein Kim, Dongha Lee, Chanyoung Park

Although recent tool-augmented benchmarks involve complex requests, evaluation r
emains limited to answer matching, neglecting critical trajectory aspects like e
fficiency, hallucination, and adaptivity. The most straightforward method for ev
aluation is to compare an agent’s trajectory with the ground-truth, but annotati
ng all valid ground-truth trajectories is prohibitively expensive. In this manne
r, we introduce TRACE, a reference free framework for the multi-dimensional eval
uation of tool-augmented LLMs. By incorporating an evidence bank which accumulat
es knowledge from preceding steps, TRACE assesses an agent’s reasoning trajector
y effectively. To validate our framework, we develop a new meta-evaluation datas
et with diverse and flawed trajectories, each labeled with multi-faceted perform
ance scores. Our results confirm that TRACE accurately evaluates complex traject
ories even with small open source LLMs. Furthermore, we apply our method to eval
uate the trajectories that agents produce while solving tool-augmented tasks, pr
esenting previously unreported observations and their corresponding insights.
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Think Deep, Not Just Long: Measuring LLM Reasoning Effort via Deep-Thinking Toke
ns

Wei-Lin Chen, Liqian Peng, Tian Tan, Chao Zhao, Jianhang Chen, Ziqian Lin, Alec
Go, Yu Meng

Large language models (LLMs) have demonstrated impressive reasoning capabilities
 by scaling test-time compute via long Chain-of-Thought (CoT).
However, recent findings suggest that raw token counts are unreliable proxies fo
r reasoning quality: increased generation length does not consistently correlate
 with accuracy and may instead signal “overthinking,” leading to performance deg
radation.
In this work, we quantify inference-time effort by identifying *deep-thinking to
kens*—tokens where internal predictions undergo significant revisions in deeper
model layers prior to convergence.
Across four challenging mathematical and scientific benchmarks (AIME 24/25, HMMT
 25, and GPQA-diamond) and a diverse set of reasoning-focused models (GPT-OSS, D
eepSeek-R1, and Qwen3), we show that *deep-thinking ratio* (the proportion of de
ep-thinking tokens in a generated sequence) exhibits a robust and consistently p
ositive correlation with accuracy, substantially outperforming both length-based
 and confidence-based baselines.
Leveraging this insight, we introduce Think@n, a test-time scaling strategy th
at prioritizes samples with high deep-thinking ratios.
We demonstrate that Think@n matches or exceeds standard self-consistency perfo
rmance while significantly reducing inference costs by enabling the early reject
ion of unpromising generations based on short prefixes.
****

SC^{2}-WM: A Self-Correcting World Model with Closed-Loop Feedback for Vision-
and-Language Navigation in Continuous Environments

Xuan Yao, Yuze Zhu, Junyu Gao, Zongmeng Wang, Changsheng Xu

Vision-and-Language Navigation in Continuous Environments (VLN-CE) requires agen
ts to make fine-grained navigation decisions under partial observability.
However, most existing methods rely on open-loop execution, lacking mechanisms t
o detect and correct internal state drift during inference.
We propose SC^{2}-WM, a self-correcting world model framework that introduces
internal feedback for closed-loop decision making in VLN-CE.
Our method derives feedback from world-model foresight to perform state-level pl

an refinement before action execution.
To handle challenging scenarios, we further introduce conditional world-aware adaptation, which enables model-level correction by selectively updating the world model at test time when feedback indicates model capacity insufficiency.
Experiments on standard VLN-CE benchmarks demonstrate improved navigation robustness and generalization. Our code is available at https://github.com/sunrise-ikun/SC2_WM.
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Multilingual Unlearning in LLMs: Transfer, Dynamics, and Reversibility
Chaoyi Xiang, Olga Ohrimenko, Benjamin I. P. Rubinstein, Lea Frermann
Large language models (LLMs) can memorize sensitive facts, motivating *unlearning* methods that remove targeted knowledge without costly retraining. However, unlearning research remains heavily English-centric. We study multilingual unlearning by extending the TOFU benchmark to five languages, and fine-tune, unlearn and query our models with different permutations of languages. We find that unlearning transfer -- the ability of an unlearned model to "forget" facts in languages other than the unlearning language -- is highly variable: e.g., it is strongest between languages sharing scripts and families, and we show that the *unlearning language* predicts which *query languages* are most likely to yield the strongest transfer. Layer-wise analysis reveals that unlearning leaves the shared cross-lingual latent space largely intact in early layers, instead operating primarily in later decoding layers. This suggests that unlearning does not truly erase knowledge, but rather induces superficial suppression. Exploiting this structure, a single inference-time steering direction reverses much of this suppression across languages, recovering 50% (Qwen) and 90% (Gemma) of the unlearned knowledge.
****
Mechanistic Data Attribution: Tracing the Training Origins of Interpretable LLM Units
Jianhui Chen, Yuzhang Luo, Liangming Pan
While mechanistic interpretability has identified interpretable circuits in large language models (LLMs), their causal origins in training data remain elusive. We introduce *mechanistic data attribution* (MDA), a scalable framework that employs influence functions to trace interpretable units back to specific training samples. Through extensive experiments on the Pythia family, we causally validate that targeted intervention—removing or augmenting a small fraction of high-influence samples—significantly modulates the emergence of interpretable heads, whereas random interventions show no effect. Our analysis reveals that repetitive structural data (e.g., LaTeX, XML) acts as a mechanistic catalyst. Furthermore, we observe that interventions targeting induction head formation induce a concurrent change in the model’s in-context learning (ICL) capability. This provides direct causal evidence for the long-standing hypothesis regarding the functional link between induction heads and ICL. Finally, we propose a mechanistic data augmentation pipeline that consistently accelerates circuit convergence across model scales, providing a principled methodology for steering the developmental trajectories of LLMs.
****
DocHop: Benchmarking Out-of-domain Multi-hop Reasoning in Information-Dense Documents
Zhuoran Yu, Le Thien Phuc Nguyen, Jaden Park, Xinyi Gu, Zexue He, Soochahn Lee, Rogerio Feris, Yong Jae Lee
Multimodal Large Language Models (MLLMs) have achieved strong performance on structured visual understanding tasks such as chart and document question answering. However, existing benchmarks typically evaluate these domains in isolation, leaving underexplored a key capability: whether models can use textual context to determine how chart evidence should be selected, interpreted, and aggregated.
We introduce DocHop, a benchmark for integrated chart--context reasoning in document-style images. In DocHop, the document narrative specifies multi-step compositional constraints, while charts provide the corresponding data values. Questions are grounded on a semantic reference label defined in the narrative, requirin

g models to resolve target entities from context before aggregating evidence acr
oss multiple charts. To enable systematic evaluation, we construct DocHop via a
stochastic logic-first generation pipeline with controllable reasoning depth and
 visual density, covering 1,876 examples across six task categories. Experiments
 on a wide range of proprietary and open-source MLLMs show a substantial gap to
human performance: annotators achieve over 90% accuracy, while the best model re
aches only 60.18%. Reasoning-enhanced models consistently show improved results,
 but performance degrades as reasoning complexity increases.
Overall, DocHop provides a controlled testbed for challenging multi-hop document
 reasoning.
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Score-Repellent Monte Carlo: Toward Efficient Non-Markovian Sampler with Constan
t Memory in General State Spaces
Jie Hu, Lingyun Chen, Geeho Kim, Jinyoung Choi, Bohyung Han, Do Young Eun
History-dependent sampling can reduce long-run Monte Carlo variance by discourag
ing redundant revisits, but existing schemes typically encode history through em
pirical measure on finite state spaces, which is infeasible in high-dimensional
discrete configuration spaces or ill-posed in continuous domains. We propose *Sc
ore-Repellent Monte Carlo* (SRMC) framework that summarizes trajectory history b
y a running average of score evaluations in $\mathbb{R}^d$, where d is the dim
ension of the score and state representation. This history is converted into a s
urrogate target through an exponential *score tilt*, indexed with α that
represents the *strength of repellence* in controlling the magnitude of the hist
ory-based repulsion. The surrogate family is normalization-free in the standard
MCMC sense, yielding a generic wrapper: at each iteration, any base kernel targe
ting π can instead be run on the current surrogate π_{θ_n} while th
e history is updated online. We analyze the coupled evolution of the history rec
ursion and Monte Carlo estimators using stochastic approximation with controlled
 Markovian noise, establishing almost sure convergence and a joint central limit
 theorem. We further identify regimes in which the asymptotic covariance decreas
es as α increases, with scaling $O(1/\alpha)$, extending the near-zero-va
riance effect of finite-state history-dependent samplers to general state spaces
 with constant memory. Experiments on continuous targets and discrete energy-bas
ed models demonstrate improved estimator variance and mode coverage, while retai
ning $O(d)$ memory usage and modest per-iteration overhead.
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Decoupling Skeleton and Flesh: Efficient Multimodal Table Reasoning with Disenta
ngled Alignment and Structure-aware Guidance
Yingjie Zhu, Xuefeng Bai, Kehai Chen, Yang Xiang, Youcheng Pan, Xiaoqiang Zhou,
Min Zhang
Reasoning over table images remains challenging for Large Vision-Language Models
 (LVLMs) due to complex layouts and tightly coupled structure-content informatio
n. Existing solutions often depend on expensive supervised training, reinforceme
nt learning, or external tools, limiting efficiency and scalability. This work a
ddresses a key question: how to adapt LVLMs to table reasoning with minimal anno
tation and no external tools? Specifically, we first introduce DiSCo, a Disentan
gled Structure-Content alignment framework that explicitly separates structural
abstraction from semantic grounding during multimodal alignment, efficiently ada
pting LVLMs to tables structures. Building on DiSCo, we further present Table-GL
S, a Global-to-Local Structure-guided reasoning framework that performs table re
asoning via structured exploration and evidence-grounded inference. Extensive ex
periments across diverse benchmarks demonstrate that our framework efficiently e
nhances LVLM's table understanding and reasoning capabilities, particularly gene
ralizing to unseen table structures. Our data and code are available at https://
github.com/AAAndy-Zhu/TableVLM.
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Rethinking Sparse Mixture of Experts from a Unified Perspective
Giang Do, Hung Le, Truyen Tran
Sparse Mixture of Experts (SMoE) models scale the capacity of models while maint
aining constant computational overhead. SMoE methods fall into two categories: *

Token Choice*, which routes each token to a fixed number of experts, and *Expert Choice*, which assigns a fixed number of tokens to each expert. However, the use of fixed budgets for tokens or experts causes both approaches to select irrelevant token–expert pairs or overlook critical assignments, which degrades overall performance. To fill that gap, we rethink SMoE from a *unified perspective* through the lens of *linear programming*, which provides a general formulation for SMoE models. Furthermore, we introduce **Unified Sparse Mixture of Experts (USMoE)**, a novel framework comprising a *unified mechanism* and a *unified score* to overcome these limitations. We provide both theoretical justification and empirical evidence demonstrating USMoE's effectiveness. Extensive evaluations across diverse data settings (clean and corrupted), multiple domains (including texts and vision tasks), and different learning approaches (training-free and training-based) show that USMoE not only delivers significant performance improvements over existing SMoE methods, but also enables more flexible expert selection budgets, reducing inference costs without compromising model performance.
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Light Up Your Face: A Physically Consistent Dataset and Diffusion Model for Face Fill-Light Enhancement
Jue Gong, Zihan Zhou, Jingkai Wang, Xiaohong Liu, Yulun Zhang, Xiaokang Yang
Face fill-light enhancement (FFE) brightens underexposed faces by adding virtual fill light while keeping the original scene illumination and background unchanged. Most face relighting methods aim to reshape overall lighting, which can suppress the input illumination or modify the entire scene, leading to foreground–background inconsistency and mismatching practical FFE needs. To support scalable learning, we introduce LightYourFace-160K (LYF-160K), a large-scale paired dataset built with a physically consistent renderer that injects a disk-shaped area fill light controlled by six disentangled factors, producing 160K before-and-after pairs. We first pretrain a physics-aware lighting prompt (PALP) that embeds the 6D parameters into conditioning tokens, using an auxiliary planar-light reconstruction objective. Building on a pretrained diffusion backbone, we then train a fill-light diffusion (FiLitDiff), an efficient one-step model conditioned on physically grounded lighting codes, enabling controllable and high-fidelity fill lighting at low computational cost. Experiments on held-out paired sets demonstrate strong perceptual quality and competitive full-reference metrics, while better preserving background illumination. The dataset and model will be at https://github.com/gobunu/Light-Up-Your-Face.
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SCOPE: Evolving Symbolic World for Planning in Open-Ended Environments
Yundaichuan Zhan, Minghe Gao, Zhongqi Yue, Wendong Bu, Wenqiao Zhang, Guoming Wang, Jisheng Dang, Juncheng Li, Siliang Tang, Yueting Zhuang
Recent works have explored integrating Vision-Language Models (VLMs) with classical planners that rely on symbolic representations of planning problem to generate long-horizon plans for complex embodied tasks. However, in open-ended environments, these symbolic representations obtained from perception are often incomplete, leading to suboptimal performance. To address this, we introduce SCOPE, a self-adaptive symbolic planning framework that supports refining action plans and evolving the symbolic world—the symbolic representations of open-ended environments. SCOPE comprises two synergistic modules: a Symbolic Execution Simulator (SESim) that conducts symbolic validation and real execution of action plans, leveraging the feedback to refine the plans and evolve the symbolic world; and a Self-Adaptive Symbolic Memory (SASMem) that further distills feedback into evolving symbolic knowledge to enhance long-horizon planning and modeling of the symbolic world. Experiments in open-ended environments show that SCOPE significantly improves the completeness of the symbolic world, the success rate of plans under environment perturbations, and cross-task grounding and adaptability across diverse embodied scenarios.
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FairJudge : An Adaptive, Debiased, and Consistent LLM-as-a-Judge
Bo Yang, Lanfei Feng, Yunkui Chen, Xiao Xu, Yu Zhang, Shijian Li
Existing LLM-as-a-Judge systems suffer from three fundamental limitations: limit

ed adaptivity to task and domain-specific evaluation criteria, systematic biases driven by non-semantic cues such as position, length, format, and model provenance, and evaluation inconsistency that leads to contradictory judgments across different evaluation modes (e.g., pointwise versus pairwise). To address these issues, we propose FairJudge, an adaptive, debiased, and consistent LLM-as-a-Judge. Unlike prior approaches that treat the judge as a static evaluator, FairJudge models judging behavior itself as a learnable and regularized policy. From a data-centric perspective, we construct a high-information-density judging dataset that explicitly injects supervision signals aligned with evaluation behavior. Building on this dataset, we adopt a curriculum-style SFT-DPO-GRPO training paradigm that progressively aligns rubric adherence, bias mitigation, and cross-mode consistency, while avoiding catastrophic forgetting. Experimental results on multiple internal and public benchmarks show that FairJudge improves agreement and F1 across several evaluation settings, reduces selected non-semantic biases, and achieves competitive or stronger performance than larger general-purpose LLMs on judge-oriented tasks.
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DIVER: Diving Deeper into Distilled Data via Expressive Semantic Recovery
Qianxin Xia, Zhiyong Shu, Wenbo Jiang, Jiawei Du, Jielei Wang, Guoming Lu
Dataset distillation aims to synthesize a compact proxy dataset that is unreadable or non-raw from the original dataset for privacy protection and highly efficient learning. However, previous approaches typically adopt a single-stage distillation paradigm, which suffers from learning specific patterns that overfit on a prior architecture, consequently suppressing the expression of semantics and leading to performance degradation across heterogeneous architectures. To address this, we propose a novel dual-stage distillation framework called ${\textbf{DIVER}}$, which leverages the pre-trained diffusion model to dive deeper into $\textbf{DI}$stilled data $\textbf{V}$ia $\textbf{E}$xpressive semantic $\textbf{R}$ecovery, an entire process of semantic inheritance, guidance, and fusion. Semantic inheritance distills high-level semantics of abstract distilled images into the latent space to filter out architecture-specific ``noise" and retain the intrinsic semantics. Furthermore, semantic guidance improves the preservation of the original semantics by directing the reverse procedure. Finally, semantic fusion is designed to provide semantic guidance only during the concrete phase of the reverse process, preventing semantic ambiguity and artifacts while maintaining the guidance information. Extensive experiments validate the effectiveness and efficiency of our method in improving classical distillation techniques and significantly improving cross-architecture generalization, requiring processing time comparable to raw DiT on ImageNet (256$\times$256) with only 4 GB of GPU memory usage.
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SIGMA-PPG: Statistical-prior Informed Generative Masking Architecture for PPG Foundation Model
Zongheng Guo, Tao Chen, Yang Jiao, Yi Pan, Xiao Hu, Manuela Ferrario
Current foundation model for photoplethysmography (PPG) signals is challenged by the intrinsic redundancy and noise of the signal. Standard masked modeling often yields trivial solutions while contrastive methods lack morphological precision. To address these limitations, we propose a Statistical-prior Informed Generative Masking Architecture (SIGMA-PPG), a generative foundation model featuring a prior-guided adversarial masking mechanism, where a reinforcement learning-driven teacher leverages statistical priors to create challenging learning paths that prevent overfitting to noise. We also incorporate a semantic consistency constraint via vector quantization to ensure that physiologically identical waveforms—even those altered by recording artifacts or minor perturbations—map to shared indices. This enhances codebook semantic density and eliminates redundant feature structures. Pre-trained on over 120,000 hours of data, SIGMA-PPG achieves superior average performance compared to five state-of-the-art baselines across 12 diverse downstream tasks. The code and model weights are available at https://github.com/ZonghengGuo/SigmaPPG.
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Model-Preserving Adaptive Rounding
Albert Tseng, Zhaofeng Sun, Christopher De Sa
The goal of quantization is to produce a compressed model whose output distribution is as close to the original model's as possible. To do this tractably, most quantization algorithms minimize the immediate activation error of each layer as a proxy for the end-to-end error. However, this ignores the effect of future layers, making it a poor proxy. In this work, we introduce Yet Another Quantization Algorithm (YAQA), a new adaptive rounding algorithm that directly considers the error at the network's output. YAQA introduces a series of theoretical results that culminate in the first end-to-end error bounds for quantization algorithms. First, we characterize the convergence time of adaptive rounding algorithms via the structure of their Hessian approximations. We then show that the end-to-end error can be bounded by the approximation's cosine similarity to the true Hessian. This admits a natural Kronecker-factored approximation with corresponding near-optimal Hessian sketches. YAQA is provably better than GPTQ/LDLQ and empirically reduces the error by
$\approx$ 30% over these methods. YAQA even achieves a lower error than quantization aware training. This translates to state of the art performance on downstream tasks, all while adding no inference overhead.
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Achieving Structurally Robust Gromov Wasserstein Distance via Adaptive Dual-Mask
Kangke Cheng, Jiawei Huang, Jingni Song, Wanlin Zhang, Bangxian Han, Hu Ding
The Gromov-Wasserstein (GW) distance enables comparison across different spaces but remains fragile to structural noise due to its global quadratic coupling. Existing robust extensions primarily rely on node-centric mass relaxation. However, we argue that this strategy is far from sufficient: it only addresses node-induced structural noise (outliers) while neglecting edge-induced distortions where spurious connections exist between valid nodes. To overcome this limitation, we propose the Structurally Robust Gromov-Wasserstein (SRGW) distance, a novel formulation that adaptively filters geometric distortions during optimization. By introducing a structure-aware dual-mask mechanism, our method effectively isolates these stubborn structural outliers while preserving strict marginal constraints for balanced transport. We solve this objective using a Mask-Guided GW Algorithm, which jointly optimizes the transport plan and the structural noise filters. We provide a rigorous theoretical analysis proving that our algorithm converges to a critical point under the Kurdyka-■ojasiewicz framework. Extensive experiments on synthetic geometric matching and real-world subgraph alignment benchmarks demonstrate that Mask-Guided GW achieves superior alignment quality, particularly under severe structural noise.
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Experience-Evolving Multi-Turn Tool-Use Agent with Hybrid Episodic–Procedural Memory
Sijia Li, Yuchen Huang, Zifan LIU, Zijian Li, Jingjing Fu, Lei Song, Jiang Bian, Jun Zhang, Rui Wang
As intents unfold and environments change, multi-turn agents face continuously shifting decision contexts. Although reusing past experience is intuitively appealing, existing approaches remain limited: full trajectories are often too context-specific to transfer, while tool-level reuse ignores the context and environment. In this paper, we introduce a hybrid episodic–procedural memory strategy (H-EPM) that enables experience-evolution of multi-turn tool-use policies, by adaptively reusing partially overlapping successful experiences in both inference and training.
Inspired by human episodic–procedural integration, we build a tool graph from accumulated trajectories, where recurring tool-to-tool dependencies capture procedural routines and each edge is augmented with a compact episodic summaries of relevant context. At inference, the agent dynamically balances episodic recall for contextual reasoning and procedural execution for routine steps.
Beyond inference, H-EPM introduces a memory-guided reinforcement learning paradigm that directly addresses a core challenge in multi-turn agent RL: ineffective exploration over long trajectories. By biasing exploration toward historically s

uccessful tool transitions, H-EPM learns a stronger policy that generalizes duri ng inference without relying on domain-specific experience collection. Experimen ts show that H-EPM consistently delivers substantial inference-time gains over s trong baselines across multi-turn tool-use benchmarks, reaching up to 50%+. It a lso boosts RL policy performance, achieving up to 40%+ improvement on out-of-dis tribution tasks. Our code is available at https://github.com/LISijia-dev/H-EPM.
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Smaller Models are Natural Explorers for Policy-Level Diversity in GRPO
Yiming Ren, Yiran Xu, Zicheng Lin, Chufan Shi, Yukang Chen, Dingdong WANG, Tianh e Wu, Junjie Wang, Yujiu Yang, Yu Qiao, Ruihang Chu
We identify a new dimension for enhancing rollout diversity in Group Relative Po licy
Optimization (GRPO) for LLMs.
While GRPO relies on diverse rollouts, prevailing strategies primarily increase diversity by injecting more token-level randomness, which may introduce step-wis e noise and leads to incoherent trajectories.
We uncover that smaller models within the same model family inherently exhibit h igher policy-level diversity, indicated by their superior pass@k relative to lar ger counterparts as sample counts increase.
Unlike token-level noise, this diversity is temporally correlated, preserves log ical consistency, and provides structured exploration signals for gradient estim ation. We thus propose S2L-PO (Small-to-Large Policy Optimization), a framework that leverages fixed small models as natural explorers to train larger models. T o balance exploration and exploitation, we design a progressive annealing strate gy that transitions from offline small-model rollouts to the large learner’s own sampling. This shift elegantly avoids mid-training performance drops caused by the small model's capacity limits, achieving faster convergence and unlocking a higher performance ceiling. S2L-PO improves accuracy on diverse mathematical rea soning benchmarks (eg., +8.8\% on AIME 24 using a 1.7B explorer to guide the 8B model) while reducing rollout compute.
The code will be made available.
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BARRED: Synthetic Training of Custom Policy Guardrails via Asymmetric Debate
Arnon Mazza, Elad Levi
Deploying guardrails for custom policies remains challenging, as generic safety models fail to capture task-specific requirements, while prompting LLMs suffers from inconsistent boundary-case performance and high inference costs. Training c ustom classifiers achieves both accuracy and efficiency, yet demands substantial labeled data that is costly to obtain. We present BARRED (Boundary Alignment Re finement through REflection and Debate), a framework for generating faithful and diverse synthetic training data using only a task description and a small set o f unlabeled examples. Our approach decomposes the domain space into dimensions t o ensure comprehensive coverage, and employs multi-agent debate to verify label correctness, yielding a high-fidelity training corpus. Experiments across divers e custom policies demonstrate that small language models finetuned on our synthe tic data consistently outperform state-of-the-art proprietary LLMs (including re asoning models) and dedicated guardrail models. Ablation studies confirm that bo th dimension decomposition and debate-based verification are critical for ensuri ng the diversity and label fidelity required for effective fine-tuning. The BARR ED framework eliminates the reliance on extensive human annotation, offering a s calable solution for accurate custom guardrails.
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ICR-RL: Deep Reinforcement Learning via In-Context-Regression
David Schiff, Ofir Lindenbaum, Yonathan Efroni
Recent advancements in machine learning have largely been driven by foundation m odels (FMs) trained on large, diverse datasets, enabling them to generalize effe ctively to new, related tasks. However, extending this paradigm to reinforcement learning (RL), where an agent interacts with an environment to select actions, remains a significant challenge. Most existing approaches train FMs directly on sets of control tasks, but developing diverse RL environments and scaling traini

ng across them can be costly and complex. In this study, we explore a simpler al ternative approach based on a classical reduction from RL to regression. We demo nstrate that a foundation model pre-trained for regression tasks, when used as a n in-context regression (ICR) model, can be directly applied to RL problems. Bui lding on this insight, we introduce a gradient-free method, ICR-RL, that require s no additional training and leverages an ICR foundation model to tackle RL task s. We evaluate our approach by applying the ICR model with the recently proposed TabPFN, which is trained on a wide range of regression tasks. Experiments condu cted on the Gymnasium classic-control benchmark indicate that ICR-RL matches or outperforms state-of-the-art methods, including DQN and PPO. These results show that ICR foundation models can effectively solve RL tasks without fine-tuning, d emonstrating their potential as a foundation for RL-oriented models

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Parsimonious Learning-Augmented Online Metric Matching

Yongho Shin, Phanu Vajanopath

Learning-augmented algorithms have received significant attention in recent ye ars, particularly in the context of online optimization. Motivated by the high c omputational cost of generating predictions, a growing line of work studies the tradeoff between performance guarantees and the number of predictions used in le arning-augmented algorithms for problems such as caching and metrical task syste ms. In this paper, we extend this line of research to *online metric matching* b y developing *parsimonious* learning-augmented algorithms and establishing lower bounds on their performance. Our approach extends the Follow-the-Prediction fra mework to the parsimonious setting by filling in a *virtual prediction* in the a bsence of an actual prediction, using an online metric matching algorithm that m aintains good intermediate matchings throughout its execution. We complement our theoretical results with an empirical evaluation, demonstrating the practical e ffectiveness of our approach.

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Efficient Diffusion Models via Time Step Optimization with Consistent Training a nd Inference Constraints

Binrui Wu, Zihao Cheng, Yuesen Liao, WEIZHONG ZHANG

Diffusion probabilistic models (DPMs)’ sampling process is often inefficient, re quiring hundreds to thousands of iterative steps to accurately approximate the d iffusion trajectory. This inefficiency limits their practical applicability. Alt hough recent advances in sampling efficiency—such as numerical solvers for diffu sion ordinary differential equations (ODEs)—have made progress, significant chal lenges remain: training-free numerical solvers suffer from the suboptimality of manually designed timestep selection rules and the inherent inconsistency betwee n the forward diffusion process (typically involving thousands of steps) and the reverse denoising process (usually limited to tens of steps).

Since timestep selection is inherently a discrete problem and cannot be optimize d via gradients, we propose an innovative approach—reparameterizing the timestep scheduling through probabilistic masking, thereby enabling gradient-based optim ization of sampling timesteps. To circumvent backpropagation, we employ policy g radient methods. Furthermore, to address the inconsistency between forward diffu sion (training) and reverse denoising (inference), we extend this framework into a bilevel optimization paradigm: the inner loop performs additional lightweight training on the model at specific timesteps determined by the outer mask to ali gn forward and reverse processes, while the outer loop optimizes the timestep di stribution via probabilistic masking and policy gradient based on generation qua lity.

Under mild assumptions, we theoretically analyze the convergence of the proposed algorithm. Extensive experiments across diverse datasets and samplers demonstra te that this framework effectively enhances sampling efficiency and generation q uality while maintaining compatibility with various DPM architectures and advanc ed ODE solvers.

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Flash-VAED: Plug-and-Play VAE Decoders for Efficient Video Generation

Lunjie Zhu, Yushi Huang, Xingtong Ge, Yufei Xue, Zhening Liu, Yumeng Zhang, Zeho

ng Lin, Jun Zhang
Latent diffusion models have enabled high-quality video synthesis, yet their inf erence remains costly and time-consuming. As diffusion transformers become incre asingly efficient, the latency bottleneck inevitably shifts to VAE decoders. To reduce their latency while maintaining quality, we propose a universal accelerat ion framework for VAE decoders that preserves full alignment with the original l atent distribution. Specifically, we propose (1) an *independence-aware channel pruning* method to effectively mitigate severe channel redundancy, and (2) a *st age-wise dominant operator optimization* strategy to address the high inference cost of the widely used causal 3D convolutions in VAE decoders. Based on these i nnovations, we construct a **Flash-VAED** family. Moreover, we design a *three-p hase dynamic distillation* framework that efficiently transfers the capabilities of the original VAE decoder to Flash-VAED. Extensive experiments on Wan and LTX -Video VAE decoders demonstrate that our method outperforms baselines in both qu ality and speed, achieving approximately a **6$\times$ speedup** while maintaini ng the reconstruction performance up to **96.9%**. Notably, Flash-VAED accelerat es the end-to-end generation pipeline by up to **36%** with negligible quality d rops on VBench-2.0. Our code is available at https://github.com/Aoko955/Flash-VA ED.
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Optimal Splitting of Language Models from Mixtures to Specialized Domains
Skyler Seto, Pierre Ablin, Anastasiia Filippova, Jiayuan Ye, Louis Béthune, Ange los Katharopoulos, David Grangier
Language models achieve impressive performance on a variety of knowledge, langua ge, and reasoning tasks due to the scale and diversity of pretraining data avail able. The standard training recipe is a two-stage paradigm: pretraining first o n the full corpus of data followed by specialization on a subset of high qualit y, specialized data from the full corpus. In the multi-domain setting, this invo lves continued pretraining of multiple models on each specialized domain, referr ed to as split model training. We propose a method for pretraining multiple mod els independently over a general pretraining corpus, and determining the optimal compute allocation between pretraining and continued pretraining using scaling laws. Our approach accurately predicts the loss of a model of size N with D pretraining and D' specialization tokens, and extrapolates to larger model siz es and number of tokens. Applied to language model training, our approach impro ves performance consistently across common sense knowledge and reasoning benchma rks across different model sizes and compute budgets.
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Taking the GP Out of the Loop
Mehul Bafna, Siddhant Anand Jadhav, David Sweet
Bayesian optimization (BO) has traditionally solved black-box problems where fun ction evaluation is expensive and, therefore, observations are few. Recently, ho wever, there has been growing interest in applying BO to problems where function evaluation is cheaper and observations are more plentiful. In this regime, scal ing to many observations N is impeded by Gaussian-process (GP) surrogates: GP hyperparameter fitting scales as $\mathcal{O}(N^3)$ (reduced to roughly $\mathca l{O}(N^2)$ in modern implementations), and it is repeated at every BO iteration. Many methods improve scaling at acquisition time, but hyperparameter fitting st ill scales poorly, making it the bottleneck. We propose Epistemic Nearest Neighb ors (ENN), a lightweight alternative to GPs that estimates function values and u ncertainty (epistemic and aleatoric) from K-nearest-neighbor observations. ENN scales as $\mathcal{O}(N)$ for both fitting and acquisition. Our BO method, TuR BO-ENN, replaces the GP surrogate in TuRBO with ENN and its Thompson-sampling ac quisition with $\mathrm{UCB} = \mu(x) + \sigma(x)$. For the special case of nois e-free problems, we can omit fitting altogether by replacing UCB with a non-dominated sort over $\mu(x)$ and $\sigma(x)$. We show empirically that Tu RBO-ENN reduces proposal time (i.e., fitting time + acquisition time) by one to two orders of magnitude compared to TuRBO at up to 50,000 observations without s acrificing solution quality.
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Convergence Analysis of Decentralized Hessian-/Jacobian-Free Algorithm for Nonconvex Stochastic Bilevel Optimization
Yihan Zhang, Xinwen Zhang, My T. Thai, Jie Wu, Hongchang Gao
Decentralized stochastic bi-level optimization has been actively studied in recent years. However, existing studies assume that the lower-level loss function is strongly convex, which limits their applicability to many machine learning models. To address this limitation, in this paper, we propose a novel decentralized stochastic first-order optimization algorithm, which does not require second-order Hessian or Jacobian matrices, for the setting where the lower-level loss function is nonconvex but satisfies the Polyak–■ojasiewicz (PL) condition.
Additionally, unlike existing single-agent methods that introduce a regularization term to the lower-level loss function to artificially enforce strong convexity, our algorithm does not require such modification. Moreover, our algorithm employs a constant single-timescale learning rate for updating variables, which is different from the time-dependent and two-timescale learning rate schedules used in prior work.
To establish the convergence rate, we develop a new convergence analysis framework for the pure PL condition, rather than relying on the artificial strong convexity introduced through regularization in existing single-agent methods. To the best of our knowledge, this is the first algorithm for nonconvex decentralized bi-level optimization that offers theoretical convergence guarantees under mild conditions. Finally, our extensive experimental results on hyperparameter optimization and model pruning applications validate the efficacy of the proposed algorithm.
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HASTE: Hardware-Aware Dynamic Sparse Training for Large Output Spaces
Nasib Ullah, Jinbin Zhang, Jean Lucien Randrianantenaina, Erik Schultheis, Rohit Babbar
Extreme multi-label classification (XMC) involves learning models over large output spaces with millions of labels, making the output layer a memory-compute bottleneck. While sparsity-based methods reduce arithmetic complexity, they often fail to yield proportional speedups due to irregular memory access, poor hardware utilization, or reliance on auxiliary architectural components in long-tailed regimes. We introduce group-shared fixed fan-in sparsity, a semi-structured output-layer design in which semantically related labels share a sparse input pattern while retaining independent weights. This grouping introduces a task-aligned inductive bias---encouraging related labels to share feature subsets---while reducing index memory overhead, increasing feature reuse across labels, and enabling efficient GPU execution via custom CUDA kernels that leverage modern accelerator primitives. As an alternative to auxiliary objectives, we exploit the long-tailed structure of XMC by decomposing the output layer into a small dense head over frequent labels and a group-shared sparse tail over the remainder, providing an informative gradient pathway while preserving the memory benefits of sparsity.
Through kernel-level microbenchmarking, we show that group-shared fixed fan-in translates arithmetic reductions into practical wall-clock gains, achieving up to $4.4\times$ speedup in the forward pass and up to $25\times$ speedup in backward passes over standard fixed fan-in sparsity, while operating within a few percent of a FLOPs-matched dense bottleneck. Across large-scale XMC benchmarks, our approach matches or improves precision@k over prior sparse baselines, while narrowing the performance gap to dense.
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Optimal Pricing for Data-Augmented AutoML Marketplaces
Minbiao Han, Steven Xia, Jonathan Light, Raul Castro Fernandez, Haifeng Xu, sainyam galhotra
Data markets promise to unlock data value by matching data suppliers with ML consumers. However, market design involves addressing intricate challenges, including data pricing, fairness, and robustness. We propose a pragmatic data-augmented AutoML market that seamlessly integrates with existing cloud-based AutoML platforms, such as Google’s Vertex AI. Unlike standard AutoML solutions, our design automatically augments buyer-submitted training data with valuable external datas

ets, pricing the resulting models based on their measurable performance improvements rather than computational costs as the status quo. Our key innovation is a pricing mechanism grounded in the instrumental value—the marginal model quality improvement—of externally sourced data. This approach bypasses direct dataset pricing complexities and accommodates diverse buyer valuations through menu-based options, thus providing an economically sustainable framework for monetizing external data.
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Gromov-Wasserstein at Scale, Beyond Squared Norms
Guillaume Houry, Jean Feydy, François-Xavier Vialard
A fundamental challenge in data science is to match disparate point sets with each other. While optimal transport efficiently minimizes point *displacements* under a bijectivity constraint, it is inherently sensitive to rotations. Conversely, minimizing *distortions* via the Gromov-Wasserstein (GW) framework addresses this limitation but introduces a non-convex, computationally demanding optimization problem. In this work, we identify a broad class of distortion penalties that reduce to a simple alignment problem within a lifted feature space. Leveraging this insight, we introduce an iterative GW solver with a linear memory footprint and quadratic (rather than cubic) time complexity. Our method is differentiable, comes with strong theoretical guarantees, and scales to hundreds of thousands of points in minutes. This efficiency unlocks a wide range of geometric applications and enables the exploration of the GW energy landscape, whose local minima encode the symmetries of the matching problem.
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Dywave: Event-Aligned Dynamic Tokenization for Heterogeneous IoT Sensing Signals
Tomoyoshi Kimura, Denizhan Kara, Jinyang Li, Hongjue Zhao, Yigong Hu, Yizhuo Chen, Xiaomin Ouyang, Shengzhong Liu, Tarek F. Abdelzaher
Internet of Things (IoT) systems continuously collect heterogeneous sensing signals from ubiquitous sensors to support intelligent applications such as human activity analysis, emotion monitoring, and environmental perception.
These signals are inherently non-stationary and multi-scale, posing unique challenges for standard tokenization techniques.
This paper proposes Dywave, a dynamic tokenization framework for IoT sensing signals that constructs compact input representations aligned with intrinsic temporal structures and underlying physical events.
Dywave leverages wavelet-based hierarchical decomposition, identifies meaningful temporal boundaries corresponding to underlying semantic events, and adaptively compresses redundant intervals while preserving temporal coherence.
Extensive evaluations on five real-world IoT sensing datasets across activity recognition, stress assessment, and nearby object detection demonstrate that Dywave outperforms state-of-the-art methods by up to 12% in accuracy, while improving computational efficiency by reducing input token lengths by up to 75\% across mainstream sequence models.
Moreover, Dywave exhibits improved robustness to domain shifts and varying sequence lengths.
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Linguistic Relative Policy Optimization for Video Anomaly Reasoning
Jiaxu Leng, Jiankang Zheng, Mengjingcheng Mo, Zhanjie Wu, Haosheng Chen, Ji Gan, Xinbo Gao
Video anomaly detection (VAD) with multimodal large language models has shown strong potential, yet most existing methods still depend on large-scale annotations or expert-designed priors, limiting their ability to acquire anomaly knowledge with as little human intervention as possible. To address this, we propose Linguistic Relative Policy Optimization (LRPO), which distills group-relative semantic advantages from multiple reasoning trajectories into a linguistically expressed anomaly experience prior, and adapts the model by injecting this prior into the context to steer its output distribution without any parameter updates. LRPO builds two complementary experience representations: general experience captures transferable anomaly preferences across scenarios, while scenario experience models context-dependent anomaly rules for targeted refinement. To further improve

the learned experience, we introduce an anomaly alignment reward that guides trajectory optimization to match human risk preferences and reinforce temporally grounded reasoning. Extensive experiments on XD-Violence, UCF-Crime, and UBNormal demonstrate that LRPO significantly outperforms existing state-of-the-art methods under tuning-free settings.

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Understanding Data Temporality Impact on Large Language Models Pre-training

Romain Fabre, Hippolyte Pilchen, Franck SIGNE TALLA, Patrick Perez, Edouard Grave

Large language models (LLMs) are typically trained on shuffled corpora, yielding models whose knowledge is frozen at training time and whose temporal grounding remains poorly understood. In this work, we study the impact of pretraining dynamics on the acquisition of time-sensitive factual knowledge, focusing specifically on data ordering. Our main contributions are twofold. First, we introduce a comprehensive benchmark of over 7,000 temporally grounded questions and an evaluation protocol that enables analysis of whether models correctly associate facts with their corresponding time periods. Second, we pretrain 6B-parameter language models on temporally ordered Common Crawl snapshots and compare them against standard shuffled pretraining. Our results show that sequentially trained models match shuffled baselines on general language understanding and common knowledge while consistently exhibiting more up-to-date and temporally precise knowledge. Temporally ordered pretraining yields improved factual freshness, while shuffled pretraining peaks on older data, possibly due to increased factual repetition. These findings, along with the release of our checkpoints and datasets, provide a foundation for future research on continual learning for large language models.

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Geometry-Aware Dataset Condensation for Diffusion Model Training

Xiao Cui, Yulei Qin, Mo Zhu, Wengang Zhou, Hongsheng Li, Houqiang Li

Dataset condensation aims to construct compact datasets from real data via synthesis or selection. However, existing approaches are ill-suited for diffusion model training: synthetic data generation often yields low-fidelity samples unsuitable for authentic modeling, while real subset selection typically fails to preserve the distributional geometry required by diffusion likelihood objectives. To address this, we propose to reformulate real subset selection as a geometry-aware distribution alignment problem. By incorporating one-sided partial optimal transport, our method selectively aligns a compact subset with the full data distribution while allowing unmatched mass in low-density regions, ensuring the preserved geometric structure necessary for effective diffusion model training. To further ensure distributional fidelity, we complement geometric alignment with lightweight feature-statistics and semantic consistency regularization. An efficient two-stage discrete optimization strategy is proposed to achieve this alignment objective. Extensive experiments across diffusion variants, subset sizes, image resolutions, and training rounds show that our method achieves superior fidelity and distributional coverage in diffusion model training.

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Learning in the Fisher Subspace: A Guided Initialization for LoRA Fine-Tuning

Zhi-Quan Feng, Ying-Jia Lin, Hung-Yu Kao

LoRA adapts large language models (LLMs) by restricting updates to low-rank subspaces of pre-trained weights. While this substantially reduces training cost, the effectiveness of adaptation critically depends on which subspace is chosen at initialization: a poor initialization that allocates capacity to task-irrelevant directions can severely hinder downstream performance. Existing initialization strategies primarily rely on the intrinsic properties of pre-trained weights, implicitly assuming that weight geometry alone reflects task relevance. However, such criteria overlook how the model interacts with the downstream data distribution. In this work, we formulate LoRA initialization as identifying the degree of impact of directions in parameter space under the target data distribution. We argue that data-aware sensitivity, rather than weight-only magnitude, should govern the choice of adaptation subspaces. Building on this perspective, we propose a Fisher-guided framework that leverages curvature information induced by downs

tream data to characterize how parameter perturbations influence model predictio
ns. This perspective yields a principled, task-dependent criterion for selecting
 LoRA directions that better align adaptation with the target objective. Empiric
al results across diverse tasks and modalities demonstrate that data-aware initi
alization consistently and significantly improves downstream performance over ex
isting approaches.
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LOVE: Benchmarking and Evaluating Text-to-Video Generation and Video-to-Text Int
erpretation
Jiarui Wang, Huiyu Duan, Ziheng Jia, Zicheng Zhang, Yu Zhao, Juntong Wang, Guang
tao Zhai, Xiongkuo Min
Recent advancements in large multimodal models (LMMs) have driven substantial pr
ogress in both text-to-video (T2V) generation and video-to-text (V2T) interpreta
tion tasks.
However, current AI-generated videos (AIGVs) still exhibit limitations in terms
 of perceptual quality and text-video alignment.
To this end, we present **AIGVE-60K**, a comprehensive dataset and benchmark for
 AI-Generated Video Evaluation,
which features **(i)** comprehensive tasks, encompassing 3,050 extensive prompts
 across 20 fine-grained task dimensions, **(ii)** the largest human annotations,
 including 120K mean-opinion scores (MOSs) and 60K question-answering (QA) pairs
 annotated on 58,500 videos generated from 30 T2V models, and **(iii)**
bidirectional benchmarking and evaluating for both T2V generation and V2T interp
retation capabilities.
Based on AIGVE-60K, we propose **LOVE**, a LMM-based metric for AIGV Evaluation
from multiple dimensions including perceptual preference, text-video corresponde
nce, and task-specific accuracy.
Building upon LOVE, we further introduce **LOVE-Reward** to optimize T2V models
through reinforcement learning, effectively enhancing both the perceptual qualit
y and text-video correspondence of generated videos.
Comprehensive experiments demonstrate that LOVE achieves state-of-the-art perfor
mance and generalizes effectively to various AIGV benchmarks. LOVE-Reward signif
icantly improves video generation quality.
These findings highlight the effectiveness of the AIGVE-60K dataset and our prop
osed methods. The database and codes are available at
https://github.com/IntMeGroup/LOVE.
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Latent Reasoning VLA: Latent Thinking and Prediction for Vision-Language-Action
Models
Shuanghao Bai, Jing Lyu, Wanqi Zhou, Zhe Li, Dakai Wang, Lei Xing, Xiaoguang Zha
o, Pengwei Wang, Zhongyuan Wang, Cheng Chi, Badong Chen, Shanghang Zhang
Vision-Language-Action (VLA) models benefit from Chain-of-Thought (CoT) reasonin
g, but existing approaches incur high inference overhead and rely on discrete re
asoning representations that mismatch continuous perception and control.
We propose Latent Reasoning VLA (LaRA-VLA), a unified VLA framework that interna
lizes multi-modal CoT reasoning into continuous latent representations for embod
ied action.
LaRA-VLA performs unified reasoning and prediction in latent space, eliminating
explicit CoT generation at inference time and enabling efficient, action-oriente
d control.
To realize latent embodied reasoning, we introduce a curriculum-based training p
aradigm that progressively transitions from explicit textual and visual CoT supe
rvision to latent reasoning, and finally adapts latent reasoning dynamics to con
dition action generation.
We construct two structured CoT datasets, LIBERO-LaRA and Bridge-LaRA, and evalu
ate LaRA-VLA across simulation benchmarks and long-horizon real-robot manipulati
on tasks. Experimental results show that LaRA-VLA outperforms existing state-of-
the-art VLA methods while achieving up to a 90\% reduction in inference latency
compared to explicit CoT-based VLA approaches, highlighting latent reasoning as
an effective and efficient paradigm for real-time embodied control.

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Convex Optimization for Alignment and Preference Learning on a Single GPU
Miria Feng, Mert Pilanci
Fine-tuning large language models (LLMs) to align with human preferences has dri ven the success of systems such as Gemini and ChatGPT.
However, approaches like Reinforcement Learning from Human Feedback (RLHF) remai n computationally expensive and complex.
Direct Preference Optimization (DPO) offers a simpler alternative but has limita tions such as inconsistent ranking accuracy, high dependence on GPU resources, a nd expensive hyperparameter tuning.
We propose the Convex Optimization for Alignment and Preference Learning Algorit hm (COALA): a novel lightweight strategy with strong theoretical guarantees. By leveraging the convex optimization reformulation of neural networks, COALA elimi nates the need for a reference model and obtains significant reduction in both t raining time and VRAM consumption, thus enabling efficient training on a single GPU.
Experiments across four datasets—including a 26621-sample synthetic Educational Feedback dataset—and six models (including Llama-3.1-8B) demonstrate COALA's com petitive performance and efficiency while utilizing as little as ${\sim}17.6\%$ of DPO's total TFLOPs.
COALA exhibits stable, monotonically increasing rewards and reaches peak margins in significantly shorter time in comparison to traditional methods such as DPO and ORPO.
To the best of our knowledge, this is the first time convex optimization has bee n effectively applied to preference fine-tuning of LLMs.
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Prompt Reinjection: Alleviating Prompt Forgetting in Multimodal Diffusion Transf ormers
Yuxuan Yao, Yuxuan Chen, Hui Li, Kaihui Cheng, Qipeng Guo, Yuwei Sun, Zilong Don g, Jingdong Wang, Siyu Zhu
Multimodal Diffusion Transformers (MMDiTs) for text-to-image generation maintain separate text and image branches, with bidirectional information flow between t ext tokens and visual latents throughout denoising.
In this setting, we observe a prompt forgetting phenomenon: the semantics of the prompt representation in the text branch is progressively forgotten as depth in creases. We further verify this effect on three representative MMDiTs—SD3, SD3.5 , and FLUX.1
by probing linguistic attributes of the representations over the layers in the t ext branch.
Motivated by these findings, we introduce a training-free approach, prompt reinj ection, which reinjects prompt representations from early layers into later laye rs to alleviate this forgetting.
Experiments on GenEval, DPG, and T2I-CompBench++ show consistent gains in instru ction-following capability, along with improvements on metrics capturing prefere nce, aesthetics, and overall text--image generation quality.
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Provable Benefits of RLVR over SFT for Reasoning Models: Learning to Backtrack E fficiently
Stanley Wei, Juno Kim
Recent advances in large language models (LLMs) have demonstrated that reinforce ment fine-tuning of pretrained base models can lead to significant gains in reas oning performance at inference time. In this work, we theoretically analyze why reinforcement fine-tuning induces better reasoning ability than purely supervise d fine-tuning (SFT) methods. We model chain-of-thought (CoT) reasoning as a path finding problem on graphs and compare the popular method of reinforcement learni ng with verifiable rewards (RLVR) against traditional SFT. We prove that SFT, wh en trained on golden shortest paths without negative examples, fails to learn ho w to efficiently backtrack. In contrast, an RLVR-trained model can learn how to efficiently backtrack from dead ends using only outcome reward. This leads to a provable inference-time compute separation between the two methods, and demonstr

ates that RLVR allows the model to learn the location of difficult decisions in a reasoning chain, ultimately allowing for better allocation of inference-time compute. Finally, we show that the reasoning traces of an RLVR model can be distilled to train a base model to backtrack efficiently as well.
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Inside the Visual Mind: Neuroscience-Motivated Concept Circuits for Interpreting and Steering Vision Transformers
Tang Li, Yanlin Chen, Mengmeng Ma, Xi Peng
Despite high accuracy, Vision Transformer (ViT) predictions can be driven by spurious cues, raising the need to understand their inner workings before safe deployment. Sparse autoencoders (SAEs) provide a promising lens for decomposing model representations into human-interpretable concepts, yet adapting SAE-based interpretation to ViTs remains challenging due to limited control over concept coverage and subjective, non-scalable feature interpretation. To fill the gaps, motivated by neuroscience-inspired principles, we propose ViSAE, a mechanistic interpretability toolbox for understanding ViT inner workings through concept circuits. ViSAE consists of three components: (1) A probing suite with 64K images and a 16K visually grounded concept vocabulary, improving concept coverage efficiency by 20× over ImageNet and interpretation accuracy by 28.7\% over existing concept sets. (2) Top-down concept reading and Bottom-up circuit tracing algorithms that automatically recover ViT inner workings via concept circuits. (3) Applications for auditing and steering ViT behavior. Through concept editing, ViSAE improves the worst-group accuracy on WaterBirds by 48.2\%, outperforming existing methods by 23.8\%. Our data and code: https://github.com/deep-real/ViSAE.
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Little By Little: Continual Learning via Incremental Mixture of Rank-1 Associative Memory Experts
Haodong Lu, Chongyang Zhao, Jason Xue, Lina Yao, Kristen Moore, Dong Gong
Continual learning (CL) with large pre-trained models aims to incrementally acquire knowledge without catastrophic forgetting. Existing LoRA-based Mixture-of-Experts (MoE) methods expand capacity by adding isolated new experts while freezing old ones, but still suffer from redundancy, interference, routing ambiguity, and consequent forgetting. We investigate the issues stemming from coarse-grained expert granularity. Coarse-grained experts (e.g., high-rank LoRA) encode low-specialty information, leading to expert duplication/interference and routing degradation/confusion as experts accumulate. In this work, we propose MoRAM (Mixture of Rank-1 Associative Memory). Grounded in the view that weight matrices act as linear associative memories, MoRAM achieves CL as incremental expansion of reusable atomic rank-1 experts as memory. Each rank-1 adapter acts as a fine-grained MoE expert or an associative memory unit. By viewing rank-1 experts as key-value memory pairs, we eliminate explicit MoE-LoRA routers with self-activation, where each memory atom evaluates its relevance via its intrinsic key. The inference process thus becomes a content-addressable retrieval and recall over the incrementally accumulated memory of learning snapshots. Extensive experiments on CLIP and LLMs show that MoRAM significantly outperforms state-of-the-art methods, achieving a better plasticity–stability trade-off, stronger generalization, and reduced forgetting. Project page: https://artificer-ai-lab.github.io/MoRAM.
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Meta Context Engineering via Agentic Skill Evolution
Haoran Ye, Xuning He, Vincent Arak, Haonan Dong, Guojie Song
The operational efficacy of large language models relies heavily on their inference-time context. This has established Context Engineering (CE) as a formal discipline for optimizing these inputs. Current CE methods rely on manually crafted harnesses, such as rigid generation-reflection workflows and predefined context schemas. They impose structural biases and restrict context optimization to a narrow, intuition-bound design space. To address this, we introduce Meta Context Engineering (MCE), a bi-level framework that supersedes static CE heuristics by co-evolving CE skills and context artifacts. In MCE iterations, a meta-level agent refines engineering skills via agentic crossover, a deliberative search over the history of skills, their executions, and evaluations. A base-level agent exec

utes these skills, learns from training rollouts, and optimizes context as flexible files and code. We evaluate MCE across five disparate domains under offline and online settings. MCE demonstrates consistent performance gains, achieving 5.6--53.8% relative improvement over state-of-the-art agentic CE methods (mean of 16.9%), while maintaining superior context adaptability, transferability, and efficiency in both context usage and training. Code is available at \url{https://github.com/henry-yeh/mce}.
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Identifying Common Hubs in Multiple Gaussian Graphical Models
José Á Sánchez Gómez, Weibin Mo, Junlong Zhao, Yufeng Liu
The Gaussian graphical model (GGM) is a useful tool to represent relationships of conditional dependence among variables. In many real-world applications, datasets often contain multiple related sub-populations, whose associated GGMs may have common structure, as well as large structural differences. In such cases, it is useful to recover common hub variables, which are the highly connected variables in the GGMs of all sub-populations. In this paper, we propose the Joint Inverse Components for Hub Detection (JIC-HD) method to recover the common hubs across multiple GGMs without the need to estimate all subpopulation GGMs. To this end, we introduce joint minimax eigenspaces, and show that these can be leveraged for the recovery of common hubs. We establish theoretical guarantees for the recovery of common hubs. Additionally, our numerical simulation studies confirm superior performance of our JIC-HD in detecting common hubs compared to the existing methods in the literature. Our method is especially advantageous when the multiple GGMs have both common and individual hubs across sub-populations. Finally, we analyze cancer gene-expression datasets and identify biologically meaningful common hub genes across cancer subtypes.
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Geometric Entropy and Retrieval Phase Transitions in Continuous Thermal Dense Associative Memory
Tatiana Petrova, Evgeny Polyachenko, Radu State
We study the thermodynamic memory capacity of modern Hopfield networks (Dense Associative Memory models) with continuous states under geometric constraints, extending classical analyses of pairwise associative memory. We derive thermodynamic phase boundaries for Dense Associative Memory networks with exponential capacity $M = e^{\alpha N}$, comparing Gaussian (LSE) and Epanechnikov (LSR) kernels. For continuous neurons on an N-sphere, the geometric entropy depends solely on the spherical geometry, not the kernel. In the sharp-kernel regime, the maximum theoretical capacity $\alpha = 0.5$ is achieved at zero temperature; below this threshold, a critical line separates retrieval from non-retrieval. The two kernels differ qualitatively in their phase boundary structure: for LSE, a critical line exists at all loads $\alpha > 0$. For LSR, the finite support introduces a threshold α_{th} below which no spurious patterns contribute to the noise floor, and no critical line exists -- retrieval is perfect at any temperature. These results advance the theory of high-capacity associative memory and clarify fundamental limits of retrieval robustness in modern attention-like memory architectures.
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TrustworthyQENN: A Quantum Evidential Neural Network Based on Complex-Valued Contrastive Learning for Uncertainty Pattern Classification
Xiaolong Chen, Fuyuan Xiao, Xiaohong Zhang, Zehong Cao, Chin-teng Lin
Out-of-distribution (OOD) detection requires accurately classifying in-distribution (ID) samples while effectively distinguishing anomalous OOD data. However, existing methodologies predominantly rely on real-valued magnitude features, neglecting the semantic richness embedded in phase information, and often lack a systematic theoretical framework for quantitively modeling uncertainty. To address this dual limitation of incomplete feature representation and insufficient uncertainty modeling, the trustworthy quantum evidence neural network (TrustworthyQENN) is proposed, a novel quantum-inspired framework bridging complex-valued representation learning with generalized quantum evidence theory (GQET). Specifically, supervised complex-valued contrastive learning (SCVCL) is proposed to synchron

ize amplitude distributions with phase correlations, thereby enforcing high intra-class compactness and inter-class separability for ID data. A quantum evidence generation mechanism based on GQET is subsequently devised, where the OOD state is formally grounded as the quantum empty set within a Hilbert space. Furthermore, the generalized quantum evidential combination rule (GQECR) is leveraged to fuse multi-view evidence, thereby achieving trustworthy inference. Extensive experiments on the MSTAR, EuroSAT, and FUSAR-Ship benchmarks substantiate the superiority of TrustworthyQENN, which achieves a peak AUROC of 95.94\% on the MSTAR dataset while consistently outperforming state-of-the-art methods across all evaluated scenarios.
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TFTF: Training-Free Targeted Flow for Conditional Sampling
Qianqian Qu, Jun S. Liu
We propose a training-free conditional sampling method for flow matching models based on importance sampling. Because a naïve application of importance sampling suffers from weight degeneracy in high-dimensional settings, we modify and incorporate a resampling technique in sequential Monte Carlo (SMC) during intermediate stages of the generation process. To encourage the generated samples to diverge along distinct trajectories, we derive a stochastic flow with adjustable noise strength to replace the deterministic flow at the intermediate stage. Our framework requires no additional training, while providing theoretical guarantees of asymptotic accuracy. Experimentally, our method significantly outperforms existing approaches on conditional sampling tasks for MNIST and CIFAR-10. We further demonstrate the applicability of our approach in higher-dimensional, multimodal settings through text-to-image generation experiments on CelebA-HQ.
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DFlash: Block Diffusion for Flash Speculative Decoding
Jian Chen, Yesheng Liang, Zhijian Liu
Autoregressive large language models (LLMs) deliver strong performance but require inherently sequential decoding, leading to high inference latency and poor GPU utilization. Speculative decoding mitigates this bottleneck by using a fast draft model whose outputs are verified in parallel by the target LLM. However, existing methods still rely on *autoregressive drafting*, which remains sequential and constrains practical speedups. Diffusion LLMs offer a promising alternative by enabling parallel generation, but current diffusion models typically underperform compared with autoregressive models. In this paper, we introduce **DFlash**, a speculative decoding framework that employs a lightweight block diffusion model for parallel drafting. We show that speculative decoding provides a natural and effective setting for diffusion models. By generating draft tokens in a single forward pass, DFlash enables efficient drafting, and by conditioning the draft model on context features extracted from the target model, it achieves high-quality drafts with higher acceptance rates. Experiments show that DFlash achieves over 6$\times$ lossless acceleration across a range of models and tasks, delivering up to 2.5$\times$ higher speedup than the state-of-the-art speculative decoding method EAGLE-3.
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Stable-GFlowNet: Toward Diverse and Robust LLM Red-Teaming via Contrastive Trajectory Balance
Minchan Kwon, Sunghyun Baek, Minseo Kim, Jaemyung Yu, Dongyoon Han, Junmo Kim
Large Language Model Red-Teaming, which proactively identifies vulnerabilities of large language models, is an essential process for ensuring safety.
Finding effective and diverse attacks in red team activities is important, but achieving both is challenging.
Generative Flow Networks (GFN) that perform distribution matching are a promising method, but they are notorious for training instability and mode collapse.
In particular, unstable reward functions in red team activities accelerate mode collapse.
We propose Stable-GFN (S-GFN), which eliminates Z estimation in GFN and reduces training instability.
S-GFN avoids Z-estimation through pairwise comparisons and employs a robust mask

ing methodology against noisy rewards.
Additionally, we propose a fluency stabilizer to prevent the model from getting stuck in local optima that produce gibberish.
S-GFN provides more stable training while maintaining the optimal policy of GFN. We demonstrate the overwhelming attack performance and diversity of S-GFN across various settings.
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In-Training Defenses Against Emergent Misalignment in Language Models
David Kaczér, Magnus Jørgenvåg, Clemens Vetter, Esha Afzal, Robin Haselhorst, Lucie Flek, Florian Mai
Fine■tuning lets practitioners repurpose aligned large language models (LLMs) for new domains, yet recent work reveals emergent misalignment (EM): Even a small, domain■specific fine■tune can induce harmful behaviors far outside the target domain. Even in the case where model weights are hidden behind a fine-tuning API, this gives attackers inadvertent access to a broadly misaligned model in a way that can be hard to detect from the fine-tuning data alone. We present the first systematic study of \emph{in■training} safeguards against EM that are practical for providers who expose fine■tuning via an API: We evaluate whether they a) prevent broad misalignment, b) allow narrow misalignment, c) learn well on benign tasks, and d) remain coherent. We investigate five training regularization interventions: (i) KL■divergence regularization toward a safe reference model, (ii) ℓ_2 distance in feature space, (iii) preventive steering with an evil persona vector, (iv) interleaving training examples from a general instruct-tuning dataset and (v) inoculation prompting. We demonstrate that selecting interleaving data by the perplexity gap between aligned and misaligned models yields the best results overall.
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Provably Data-driven Lagrangian Relaxation for Mixed Integer Linear Programming
TUNG QUOC LE, Anh Tuan Nguyen, Viet Anh Nguyen
Lagrangian Relaxation (LR) is a powerful technique for solving large-scale mixed-integer linear programs, particularly those with decomposable structures, such as vehicle routing or unit commitment problems. By relaxing the coupling constraints, LR enables parallel subproblem solving and often yields tighter dual bounds than standard linear programming relaxations, which is crucial for efficient branch-and-bound pruning. While recent empirical work has shown promising results using machine learning to predict these multipliers, a theoretical understanding of such methods remains an open question. In this work, we bridge this gap by analyzing the problem of learning LR through the lens of data-driven algorithm design, i.e., a statistical learning problem over a distribution of problem instances. Our contributions are as follows: first, we derive a generalization bound of $\mathcal{O}(s^{1.5}/\sqrt{N})$ for the learned multipliers, where s is the number of coupling constraints and N is the sample size. Second, we provide a minimax lower-bound of $\Omega(s/\sqrt{N})$, proving that a linear dependency is unavoidable. Third, we constructively close this theoretical gap by proving that stochastic gradient ascent with averaging achieves the minimax optimal rate $\Theta(s/\sqrt{N})$. Finally, we extend our framework to the learning-to-warm-start setting, proving that it achieves a fast, minimax-optimal rate of $\Theta(s/N)$ and establishing a theoretical advantage over direct multiplier prediction.
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Streaming Covariate Balancing via Discrepancy-Based Feature Coresets
Yixin Ren, Chenghou Jin, Yewei Xia, Zichuan Lin, Deheng Ye, Hao Zhang, Jihong Guan, Shuigeng Zhou
Real-time estimation of average treatment effects (ATE) in streaming observational data poses two key challenges: strict memory constraints that preclude storing the full data history, and distributional shifts in both treatment assignment and outcome-generating process. Existing methods either require offline access to the entire dataset for covariate balancing or rely on parametric online models that are vulnerable to model misspecification under such shifts. This paper proposes a novel model-agnostic method for ATE estimation in streaming data, which effectively addresses the above challenges. Based on discrepancy theory, we firs

t compress streaming data into feature coresets that preserve covariate balancin g objectives over a rich nonparametric function class, enabling linear-time upda tes with bounded memory. Then, by directly learning balancing weights and bypass ing parametric propensity score estimation, we enhance the model's robustness ag ainst the shift in treatment assignment, while by balancing over an expressive f unction space we make the model more adaptive to the shift in the outcome-genera ting process. Theoretically, we establish convergence guarantees with explicit b ounds on memory usage and computational complexity. Empirically, extensive exper iments on both synthetic and real-world datasets show the effectiveness and robu stness of the proposed method, consistently outperforming existing techniques.
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Agent World Model: Infinity Synthetic Environments for Agentic Reinforcement Lea rning

Zhaoyang Wang, Canwen Xu, Boyi Liu, Yite Wang, Siwei Han, Zhewei Yao, Huaxiu Yao , Yuxiong He

Recent advances in large language model (LLM) have empowered autonomous agents t o perform multi-turn interactions with tools and environments. However, scaling such agent training is limited by the lack of diverse and reliable environments. In this paper, we propose Agent World Model (AWM), a fully synthetic environment generation pipeline. Using this pipeline, we scale to 1,000 environments coveri ng everyday scenarios, in which agents can interact with rich toolsets and obtai n high-quality observations. Notably, these environments are code-driven and bac ked by databases, providing more reliable and consistent state transitions than environments simulated by LLMs. Moreover, they enable more efficient agent inter action compared with collecting trajectories from realistic environments. To dem onstrate the effectiveness of this resource, we perform large-scale reinforcemen t learning for multi-turn tool-use agents. Thanks to the fully executable enviro nments and accessible database states, we can also design reliable reward functi ons. Experiments on three benchmarks show that training exclusively in synthetic environments, rather than benchmark-specific ones, yields strong out-of-distrib ution generalization. The code is available at https://github.com/Snowflake-Labs /agent-world-model.
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Don't Forget Why You Started: Tackling Dual Forgetting in Vision-Language Contin ual Learning

Borui Kang, Jinrui Gu, Tao Feng, Qi Fan, Yinghuan Shi, Lei Wang, Wenbin Li, Yang Gao

Continual learning of Vision-Language Model (VLM) aims to empower models with ne w expertise without compromising zero-shot capabilities.

However, this pursuit faces a critical “dual-forgetting” challenge: catastrophic forgetting of newly acquired classes (Incremental Knowledge Forgetting, IKF) an d erosion of foundational zero-shot capabilities (Pre-trained Knowledge Forgetti ng, PKF).

Existing evaluations often ignore PKF or assess it via confounded protocols wher e positive transfer on semantically similar domains creates an illusion of reten tion, masking severe foundational degradation.

To address this, we propose the Dual-Forgetting-Aware Class-Incremental Learning (DFA-CIL) framework and the Similarity-Calibrated Retention (SCR) metric.

Unlike standard averaging, SCR uses the frozen pre-trained feature space to inve rsely weight performance by semantic similarity, mitigating confounding gains to stress-test foundational stability.

Building on this, we propose DFA-MoE, a functionally heterogeneous Parameter-Eff icient Fine-Tuning (PEFT) method.

DFA-MoE decouples optimization objectives by assigning a momentum-enhanced contr astive expert for feature alignment, and separate plasticity experts combining c lassification with auxiliary contrastive learning to adapt to new tasks while re taining historical knowledge.

Extensive experiments show that our framework reveals the hidden fragility of ex isting methods and achieves a state-of-the-art balance in preserving both increm ental and pre-trained knowledge. Our code is available at https://github.com/RL-

MIND/DFA-MoE.
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How High is ‘High’? Rethinking the Roles of Dimensionality in Topological Data Analysis and Manifold Learning
Hannah Sansford, Nick Whiteley, Patrick Rubin-Delanchy
High-dimensionality of data is often regarded as a fundamental statistical impediment in Machine Learning and AI. The purpose of this paper is to clarify, on the contrary, when and how high-dimensionality may be beneficial. In the setting of a general random function model of data we delineate between three notions of dimensionality: *effective dimension* p_{eff}, measuring total variability across feature directions; *correlation rank* r, measuring functional complexity across samples; and *latent intrinsic dimension* d of manifold structure hidden in data. Via a generalized Hanson-Wright inequality, we show that increasing p_{eff} drives a *blessing of dimensionality* phenomenon, whereby data dot-products concentrate about their expectations. In turn, we show that, under mild continuity assumptions (ensuring that features bring additional information as dimension grows), persistence diagrams recover latent homology when $p_{\mathrm{eff}} \in \omega (\log n)$ as $n\to\infty$. Informed by our theory, we revisit the ground-breaking neuroscience discovery of toroidal structure in grid-cell activity made by Gardner et al. (2022): our findings provide the first empirical evidence that this structure is *isometric* to a flat torus model of physical space, suggesting that grid cell activity conveys a geometrically faithful representation of the real world.
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Designing Observation and Action Models for Efficient Reinforcement Learning with LLMs
Deunsol Yoon, Sunghoon Hong, Whiyoung Jung, Junseok Park, Geon-Hyeong Kim, Woohyung Lim, Soonyoung Lee, Seung Hwan Kim, Kanghoon Lee
Large Language Models (LLMs) have emerged as powerful tools for semantic reasoning, enabling the formalization of tasks that traditionally relied on manual human intuition. This capability extends to environment design in reinforcement learning. While prior research predominantly focuses on reward design, the design of observation and action spaces remains relatively underexplored. We propose LLM-based design of Observation and Action Models (LOAM), a framework leveraging LLMs to construct refined agent spaces from raw environments. To mitigate the computational burden of identifying the best candidate model from stochastic LLM outputs, LOAM incorporates a continuous racing mechanism that dynamically allocates resources to prioritize the most promising configurations without additional training overhead. Empirical evaluations on HumanoidBench and Isaac Lab demonstrate that LOAM consistently outperforms handcrafted baselines in both learning speed and asymptotic performance.
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A Mechanistic Analysis of Sim-and-Real Co-Training in Generative Robot Policies
Yu Lei, Minghuan Liu, Abhiram Maddukuri, Zhenyu Jiang, Yuke Zhu
Co-training, which combines limited in-domain real-world data with abundant surrogate data such as simulation or cross-embodiment robot data, is widely used for training generative robot policies. Despite its empirical success, the mechanisms that determine when and why co-training is effective remain poorly understood.

We investigate the mechanism of sim-and-real co-training through theoretical analysis and empirical study, and identify two intrinsic effects governing performance. The first, "structured representation alignment", reflects a balance between cross-domain representation alignment and domain discernibility, and plays a primary role in downstream performance. The second, the "importance reweighting effect", arises from domain-dependent modulation of action weighting and operates at a secondary level.

We validate these effects with controlled experiments on a toy model and extensive sim-and-sim and sim-and-real robot manipulation experiments. Our analysis offers a unified interpretation of recent co-training techniques and motivates a simple method that consistently improves upon prior approaches.

More broadly, our aim is to examine the inner workings of co-training and to facilitate research in this direction.
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AugMask: Training Diffusion Models on Incomplete Tabular Data via Stochastic Augmentation and Masking
Jungkyu Kim, Taeyoung Park, Kibok Lee
Score-based diffusion models have emerged as prominent deep generative models; however, their application to tabular data remains challenging because their backbones assume fully specified inputs, whereas real-world tabular data often contain missing values.
We propose **AugMask**, a plug-and-play training framework that adapts missing-unaware backbones to incomplete data via stochastic regularization.
AugMask 1) completes inputs via *conditional stochastic augmentation* using lightweight auxiliary models and 2) masks the loss, using augmented missing entries for conditioning while restricting supervision to observed coordinates.
We connect AugMask to a Rao-Blackwellized objective and show that marginalizing missing entries yields a variance-weighted sensitivity penalty, promoting invariance of observed-coordinate reconstruction with respect to uncertain missing entries.
Across diverse datasets and missingness regimes, AugMask enables standard diffusion-based tabular generators to match or outperform specialized missing-aware baselines in both sample fidelity and downstream utility.
The code will be released.
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Language Model Circuits Are Sparse in the Neuron Basis
Aryaman Arora, Zhengxuan Wu, Jacob Steinhardt, Sarah Schwettmann
The high-level concepts that a neural network uses to perform computation need not be aligned to individual neurons (Smolensky, 1986).
Language model interpretability research has thus turned to techniques which decompose the neuron basis into more interpretable units of model computation, such as sparse autoencoders (SAEs).
However, not all neuron-based representations are uninterpretable.
For the first time, we empirically show that **MLP neurons are as sparse a feature basis as SAEs**.
We use this finding to develop an end-to-end gradient-based attribution pipeline for circuit tracing on the MLP neuron basis, which surfaces causally effective neurons on a variety of tasks.
On a standard subject-verb agreement benchmark (Marks et al., 2025), a circuit of $\approx 10^2$ MLP neurons is enough to control model behaviour.
On the multi-hop city-state-capital task from Lindsey et al. (2025), we find a circuit in which small sets of neurons encode specific latent reasoning steps (e.g. mapping a city to its state), and can be steered to change the model's output.
This work thus advances automated interpretability of language models without imposing additional training costs.
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Embedding Trust: Semantic Isotropy Predicts Nonfactuality in Long-Form Text Generation
Dhrupad Bhardwaj, Julia Kempe, Tim G. J. Rudner
To deploy large language models (LLMs) in high-stakes application domains that require substantively accurate responses to open-ended prompts, we need reliable, computationally inexpensive methods that assess the trustworthiness of long-form responses generated by LLMs. However, existing approaches often rely on claim-by-claim fact-checking, which is computationally expensive and brittle in long-form responses to open-ended prompts. In this work, we introduce semantic isotropy---the degree of uniformity across normalized text embeddings on the unit sphere---and use it to assess the trustworthiness of long-form responses generated by LLMs. To do so, we generate several long-form responses, embed them, and estimate the level of semantic isotropy of these responses as the angular dispersion of the embeddings on the unit sphere. We find that higher semantic isotropy---tha

t is, greater embedding dispersion---reliably signals lower factual consistency across samples. Our approach requires no labeled data, no fine-tuning, and no hyperparameter selection, and can be used with open- or closed-weight embedding models. Across multiple domains, our method consistently outperforms existing aggregate trust signals in predicting nonfactuality using only a handful of samples, offering a practical, low-cost first-pass signal that complements claim-level verification in real-world LLM workflows.
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A Behavioural and Representational Evaluation of Goal-Directedness in Language Model Agents
Raghu Arghal, Phoebe Chen, Niall Dalton, Evgenii Kortukov, Calum McNamara, Angelos Nalmpantis, Moksh Nirvaan, Gabriele Sarti, Mario Giulianelli
Understanding an agent's goals helps explain and predict its behaviour, yet there is no established methodology for reliably attributing goals to agentic systems. We propose a framework for evaluating goal-directedness that integrates behavioural evaluation with interpretability-based analyses of models' internal representations.
As a case study, we examine an LLM agent navigating a 2D grid world toward a goal state.
Behaviourally, we evaluate the agent against optimal policies across varying grid sizes, obstacle densities, and goal structures, finding that performance scales with task difficulty while remaining robust to difficulty-preserving transformations and multi-goal structures.
We then use probing methods to decode internal representations of the environment and multi-step action plans. We find that the LLM agent non-linearly encodes a coarse spatial map, preserving approximate task-relevant cues about its position and the goal location; that its actions are broadly consistent with these internal representations; and that reasoning reorganises them, shifting from spatial cues toward immediate action selection.
Our findings support the view that introspective examination is required beyond behavioural evaluations to characterise how agents represent and pursue their objectives.
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Erased but Not Forgotten: How Backdoors Compromise Concept Erasure
Tobias Braun, Jonas Henry Grebe, Patrick Mohr Gordillo, Marcus Rohrbach, Anna Rohrbach
The expansion of text-to-image diffusion models has raised concerns about harmful outputs, from fabricated depictions of public figures to sexually explicit imagery. To mitigate such risks, prior work has proposed concept erasure methods that aim to sever unwanted concepts from the model via fine-tuning, yet it remains unclear whether these approaches truly remove all links to the harmful concept or merely conceal superficial connections. In this work, we reveal a critical vulnerability, the Erasure Evasion Backdoor (EEB): an adversary binds a backdoor trigger to a concept slated for removal, and this malicious link survives subsequent erasure. We show that both black-box and white-box adversaries can instantiate this threat. Across six state-of-the-art erasure methods, including robust ones that explicitly search for alternative representations of the target concept, EEB consistently exposes harmful content: up to 82% success against celebrity-identity unlearning, up to 94% for object erasure, and up to 16$\times$ amplification of explicit-content exposure. While EEB uncovers a blind spot in current erasure methods, it also provides a diagnostic tool for stress-testing future concept erasure techniques.
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Understanding the Gaps in Satisficing Bandits
Chloé Rouyer, Ronald Ortner, Peter Auer
We study a variant of the stochastic multi-armed bandit problem in which the learner aims to identify and play an arbitrary arm whose expected reward exceeds a known satisficing threshold S, rather than optimizing against the best arm. Prior work has shown that when such a satisficing arm exists, time-independent bounds on the satisficing regret are achievable, but these guarantees deteriorate w

hen an arm lies close to the threshold. We focus on instances in which the exces s gap Δ_* (gap between the best arm and the threshold) is small relative to the suboptimality gaps Δ_i, a regime that exposes this limitation. To capture this challenge, we introduce a refined notion of regret and propose a ne w algorithm, uncertain-UCB, which achieves *satisficing* pseudo-regret of
$
O \left(\sum_{i: \Delta_i > \Delta_*} \frac{\ln(K/\Delta_*)}{\Delta_i}\right),
$
while recovering standard pseudo-regret bounds when no arm exceeds the threshold .

Further, we establish a near-matching lower bound in the small excess-gap regime , showing that any algorithm incurs at least
$
\Omega \left(\sum_{i: \Delta_i > \Delta_*} \frac{\ln \big(\frac{\Delta}{(K-1) \ Delta_* }\big)}{\Delta_i}\right)
$ satisficing pseudo-regret.
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Do Neural Operators Forget Geometry? The Forgetting Hypothesis in Deep Operator Learning

Yanming Xia, Angelica I Aviles-Rivero

Neural operators perform well on structured domains, yet their behaviour on irre gular geometries remains poorly understood. We show that this limitation is not merely an encoding issue, but a depth-wise failure mode inherent to deep operato r architectures. We formalise the *Geometric Forgetting Hypothesis*: due to the Markovian structure of operator layers and their reliance on global mixing mecha nisms, neural operators progressively lose access to domain geometry as depth in creases. Using layer-wise geometric probing, we demonstrate that both spectral a nd attention-based operators systematically lose geometric fidelity. We show tha t this geometric forgetting degrades accuracy, stability, and generalisation. To counteract it, we introduce a lightweight *geometry memory injection* mechanism that restores geometric constraints at intermediate depths with minimal archite ctural overhead. This simple intervention consistently mitigates forgetting and exposes a *geometric shortcut* instability in transformer-based operators, revea ling that geometric retention is a structural requirement rather than a design c hoice.
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Adaptive Sharpness-Aware Minimization with a Polyak-type Step size: A Theory-Gro unded Scheduler

Dimitris Oikonomou, Nicolas Loizou

Sharpness-Aware Minimization (SAM) has established itself as a powerful and wide ly adopted optimizer for training machine learning models. By explicitly minimiz ing the sharpness of the loss landscape, SAM often improves generalization while delivering strong empirical performance. However, SAM and its variants, like mo st training algorithms, are sensitive to the choice of learning rate, which is t ypically selected through extensive hyperparameter tuning or predefined schedule rs. In this work, motivated by recent advances on the effectiveness of stochasti c Polyak step sizes for Stochastic Gradient Descent (SGD), we derive Polyak sche dulers tailored to SAM-style updates, yielding novel adaptive algorithms in both deterministic and stochastic settings. In the smooth setting, we prove linear c onvergence for strongly convex objectives and an $\mathcal{O}(1/T)$ convergence rate for convex objectives in the deterministic case. In the stochastic setting, we establish analogous convergence guarantees up to a neighborhood of the optim um. Numerical experiments demonstrate that the proposed Polyak schedulers achiev e performance comparable to or better than carefully tuned SAM baselines, while substantially reducing the need for learning-rate tuning.
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How can we assess human-agent interactions? Case studies in software agent desig n

Valerie Chen, Rohit Malhotra, Xingyao Wang, Juan Michelini, Xuhui Zhou, Aditya B

harat Soni, Hoang H. Tran, Calvin Smith, Ameet Talwalkar, Graham Neubig
While benchmarks measure the accuracy of LLM-powered agents, they mostly assume full automation, failing to represent the collaborative nature of real-world use cases. In this paper, we make two major steps towards the rigorous assessment of human-agent interactions. First, we propose PULSE, a framework for more efficient human-centric evaluation of agent designs, which comprises collecting user feedback, training an ML model to predict user satisfaction, and computing results by combining human satisfaction ratings with model-generated pseudo-labels. Second, we deploy PULSE n software engineering---one of the highest-impact, real-world domains for LLM agents---via a large-scale web platform built around the open-source agent OpenHands. Across 15k users, we evaluate how three agent design decisions impact developer satisfaction rates. We also show how PULSE can lead to more robust conclusions about agent design, reducing confidence intervals by 40% compared to a standard A/B test. Finally, we find substantial discrepancies between in-the-wild results with benchmark performance (e.g., the anti-correlation between claude-sonnet-4 and gpt-5, underscoring the limitations of benchmark-driven evaluation. Our framework PULSE provides guidance for future evaluations, and our findings identify opportunities for better software agent designs.
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Harmful Overfitting in Sobolev Spaces
Kedar Karhadkar, Alexander Sietsema, Deanna Needell, Guido Montufar
Motivated by recent work on benign overfitting in overparameterized machine learning, we study the generalization behavior of functions in Sobolev spaces $W^{k, p}(\mathbb{R}^d)$ that perfectly fit a noisy training data set. Under assumptions of label noise and sufficient regularity in the data distribution, we show that approximately norm-minimizing interpolators, which are canonical solutions selected by smoothness bias, exhibit harmful overfitting: even as the training sample size $n \to \infty$, the generalization error remains bounded below by a positive constant with high probability. Our results hold for arbitrary values of $p \in [1, \infty)$, in contrast to prior results studying the Hilbert space case ($p = 2$) using kernel methods. Our proof uses a geometric argument which identifies harmful neighborhoods of the training data using Sobolev inequalities.
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Non-Uniform Noise-to-Signal Ratio in the REINFORCE Policy-Gradient Estimator
Haoyu Han, Heng Yang
Policy-gradient methods are widely used in reinforcement learning, yet training often becomes unstable or slows down as learning progresses. We study this phenomenon through the **noise-to-signal ratio** (NSR) of a policy-gradient estimator, defined as the estimator variance (noise) normalized by the squared norm of the true gradient (signal). Our main result is that, for (i) finite-horizon linear systems with Gaussian policies and linear state-feedback, and (ii) finite-horizon polynomial systems with Gaussian policies and polynomial feedback, the NSR of the REINFORCE estimator can be characterized exactly—either in closed form or via numerical moment-evaluation algorithms—without approximation. For general nonlinear dynamics and expressive policies (including neural policies), we further derive a general upper bound on the variance. These characterizations enable a direct examination of how NSR varies across policy parameters and how it evolves along optimization trajectories (e.g. SGD and Adam). Across a range of examples, we find that the NSR landscape is highly non-uniform and typically increases as the policy approaches an optimum; in some regimes it blows up, which can trigger training instability and policy collapse.
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Progressive Graph Structure Adjustment for Homophily Shift Adaptation
Hongwei Wen, Can Zhang, Haoyu He, Hanyuan Hang, Minglong Lei
We propose *Progressive Structure Adjustment for Homophily Shift* (*PSAHS*), a lightweight method for *Graph Domain Adaptation* (*GDA*) that explicitly addresses cross-domain mismatch in node-level homophily. PSAHS enhances node homophily in the source graph to a prescribed level by reweighting edges and introducing additional intra-class connections for low-homophily nodes, and conservatively refines the target graph using agreement-consistent predictions from a structure-aw

are *Graph Neural Network* (*GNN*) and an attribute-only *Multi-Layer Perceptron* (*MLP*) to ensure reliability under label scarcity. After each structural refinement, domain-adversarial training is employed to align node representations across domains. PSAHS employs a progressive training scheme that alternates between structure adjustment and representation alignment, where increasingly informative representations enable safer homophily correction, and the refined structure in turn improves representation learning. Extensive experiments on multiple GDA benchmarks demonstrate that PSAHS consistently outperforms strong baselines, with particularly large gains under severe homophily mismatch, highlighting the importance of explicit homophily alignment for effective cross-graph transfer.
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APE-Bench: Evaluating Automated Proof Engineering for Formal Math Libraries
Huajian Xin, Zheng Yuan, Jacques D. Fleuriot, Wenda Li
While frontier formal mathematics systems now routinely develop repository-scale proof engineering artifacts requiring multi-file coordination and semantic correctness beyond compilation, existing evaluation benchmarks remain focused on isolated theorem proving. We introduce Automated Proof Engineering (APE), the first systematic framework for evaluating repository-scale proof engineering through dual verification that validates both syntactic compilation and semantic requirement satisfaction in pinned library environments. We present a complete infrastructure comprising APE-Bench, which automatically extracts proof engineering tasks from real library commit histories, and APE-Harness, a unified execution framework based on task contract abstraction. This contract-based design enables standardized evaluation across diverse formal mathematics tasks and fair systematic comparison of different agent implementations (including our APE-Agent reference scaffold alongside Claude Code and Codex CLI) on identical task specifications. We demonstrate the framework's effectiveness through comprehensive evaluation. All code, benchmarks, and infrastructure will be open-sourced.
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Compositional Transduction with Latent Analogies for Offline Goal-Conditioned Reinforcement Learning
Junseok Kim, Dohyeong Kim, Mineui Hong, Songhwai Oh
Compositional generalization is essential for reaching unseen goals under novel contextual variations in offline goal-conditioned reinforcement learning (GCRL), where a generalist goal-reaching agent must be learned from limited data. Most prior approaches pursue this via trajectory stitching over temporally contiguous segments, which limits composing behaviors across varying contexts. To overcome this limitation, we formalize *analogy transduction* as synthesizing new plans by composing task-endogenous analogies with given contexts and propose a novel analogy representation tailored for it. Grounded in our theory, this analogy representation captures what changes under optimal task execution, remains invariant to contextual variations, and is sufficient for optimal goal reaching. We further contend that generalization to unseen analogy-context pairs is a practical obstacle in analogy transduction, and introduce a new approach for offline GCRL that enables analogy transduction beyond seen pairs to unseen combinations. We empirically demonstrate the effectiveness of our approach on OGBench manipulation environments, substantially outperforming prior methods that do not perform analogy transduction.
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Empirical Gaussian Processes
Jihao Andreas Lin, Sebastian Ament, Louis C. Tiao, David Eriksson, Maximilian Balandat, Eytan Bakshy
Gaussian processes (GPs) are powerful and widely used probabilistic regression models, but their effectiveness in practice is often limited by the choice of kernel function.
This kernel function is typically handcrafted from a small set of standard functions, a process that requires expert knowledge, results in limited adaptivity to data, and imposes strong assumptions on the hypothesis space.
Re-evaluating this challenge from a hierarchical Bayesian and function-space view, we study Empirical GPs, a principled framework for constructing flexible, dat

a-driven GP priors that overcome these limitations. Rather than relying on standard parametric kernels, we estimate the mean and covariance functions empirically from a corpus of historical observations, enabling the prior to reflect rich, non-trivial covariance structures present in the data.
Theoretically, we show that the resulting model converges to the GP that is closest (in KL-divergence sense) to the real data-generating process. We formulate the problem of learning the GP prior from independent datasets as maximum likelihood estimation and derive an Expectation-Maximization algorithm with closed-form updates, allowing the model handle heterogeneous observation locations across datasets. We demonstrate that Empirical GPs achieve competitive performance on learning curve extrapolation and time series forecasting benchmarks.
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RL4RLA: Teaching ML to Discover Randomized Linear Algebra Algorithms Through Curriculum Design and Graph-Based Search

Jinglong Xiong, Xiaotian Liu, Ruoxin Wang, Zihang Liu, Yefan Zhou, Yujun Yan, Yaoqing Yang

Randomized linear algebra (RLA) algorithms are a modern class of numerical linear algebra techniques that play an essential role in scientific computing and machine learning, with broad and growing adoption. However, their discovery remains mostly a manual process that requires deep expert knowledge and inspiration. While Reinforcement Learning (RL) offers a pathway to automation, standard approaches struggle with sparse reward landscapes and vast search spaces inherent to high-performing RLA algorithms.
In this paper, we present RL4RLA, a general RL framework that automates the discovery of interpretable, symbolic RLA algorithms. Unlike black-box approaches, our method builds explicit algorithms from basic linear algebra primitives, ensuring verifiable and implementable representations. To enable efficient discovery, we introduce: (1) a numerical curriculum that progressively increments problem difficulty to encode inductive bias specific to the RLA domain; (2) Monte Carlo Graph Search, which optimizes exploration by identifying and merging equivalent partial algorithms. We demonstrate that RL4RLA rediscovers state-of-the-art methods, including sketch-and-precondition solvers, Randomized Kaczmarz, and Newton Sketch, and can be targeted to produce algorithms optimized for specific trade-offs between accuracy, speed, and stability. Code is available at https://github.com/Tim-Xiong/RL4RLA.
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Towards Complete Multi-Agent Coordination Policy Learning via Denoising Maximum Entropy Optimization

Guanghao Li, Lei Yuan, Ruiqi Xue, Hengchang Zhang, Jianhong Wang, Yi-Chen Li, Yang Yu

Parameter sharing is a widely used technique in Multi-Agent Reinforcement Learning (MARL) that enhances sample efficiency by equipping agents with a unified policy. While effective in homogeneous settings, it often struggles in heterogeneous environments where agents possess diverse capabilities. Conversely, learning customized policies for agents can resolve knowledge conflicts but significantly hinders knowledge transfer, thereby reducing learning efficiency. Existing approaches attempt to balance this trade-off using clustering or agent-specific masks, but they typically rely on strong environment-specific priors and struggle in settings where the team exhibits multi-modal policies. To address these limitations, we propose Dspic, an efficient shared-policy algorithm grounded in the maximum entropy framework. Specifically, Dspic employs self-supervised learning to extract discriminative role embeddings for each agent. These embeddings guide a complete division of the observation space, providing a theoretical guarantee for the optimality of parameter sharing. Furthermore, to handle the increased observation complexity and diversity resulting from this division, Dspic incorporates a diffusion policy, enhancing the capacity to model complex action distributions while enabling efficient learning. Extensive experiments on MaMuJoCo, SMAC, SMACv2, and LBF demonstrate that Dspic achieves superior sample efficiency while maintaining asymptotic optimality.
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Selective Disclosure Watermarking for Large Language Models
Xuyang Chen, Xiang Li, Yangxinyu Xie, Qi Long
Watermarking methods embed imperceptible and verifiable signals into text generated by large language models (LLMs). Existing approaches include zero-bit schemes for distinguishing synthetic text from human writing and multi-bit schemes for embedding metadata. However, current multi-bit watermarking methods do not allow selective disclosure: verifying any part of the watermark requires revealing the entire embedded message. This lack of control leads to unnecessary information exposure and raises privacy concerns.
We propose Hierarchical Vocabulary Routing (HeRo), a watermarking framework that enables selective disclosure of embedded metadata. The method recursively partitions the vocabulary and distributes watermark information across hierarchical layers, so that different verifiers can decode only the portions of the payload corresponding to their access level. We show that the proposed scheme preserves the unbiasedness of the underlying sampling process and thus maintains text quality. Experiments demonstrate that our framework supports fine-grained access control while achieving high detection accuracy and low latency. Code is available at \url{https://github.com/xuyangc03/hero-watermark}.
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Discontinuous Galerkin Neural Operator for Pathology Defocus Deblurring
Shaoqing Duan, Haofei Song, Xintian Mao, Qingli Li, Yan Wang
Defocus deblurring in pathological microscopy remains challenging due to the spatially varying and locally discontinuous nature of optical blur induced by a position-dependent integral imaging process.
Existing deep learning methods, constrained by shift-invariance assumptions and limited interpretability, are not well suited to such heterogeneous blur patterns.
Neural operators provide a principled alternative by modeling defocus formation directly as an integral operator, offering a new perspective on defocus deblurring.
However, most existing neural operator architectures for low-level vision rely on globally parameterized kernels that assume smoothness and stationarity, limiting their ability to model heterogeneous and locally discontinuous blur patterns.
To address this limitation, we propose the Discontinuous Galerkin Neural Operator (DGNO), which parameterizes the integral kernel using a discontinuous Galerkin formulation with element-local volume operators and interface numerical fluxes.
DGNO provides a principled combination of locality,
heterogeneity modeling, and global coherence while preserving the underlying physics
of optical image formation.
Extensive experiments demonstrate that DGNO surpasses state-of-the-art methods, delivering sharper reconstructions, robust handling of spatially varying blur, and scalable high-resolution performance.
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DisPOSE: Projected Polystochastic Diffusion for Self-Supervised Multi-View 3D Human Pose Estimation
Tony Danjun Wang, Tolga Birdal, Nassir Navab, Lennart Bastian
Recovering 3D human poses for multiple individuals from different camera views is a fundamental bottleneck for analyzing interacting behaviors. Existing self-supervised approaches leverage synthetic catalogues of 3D poses; however, this leads to poor generalization in real-world scenarios due to distribution shifts. We therefore introduce DisPOSE, a self-supervised framework that approximates the inherently discrete multi-view person-assignment problem as a generative diffusion process over the space of polystochastic tensors. By employing differentiable Sinkhorn projections during denoising, our model learns to guide solutions toward valid and feasible assignments based on 2D image priors. The complete 3D skeletons of localized individuals are then regressed using a Hypergraph-Convolutional Decoder that explicitly models relational structures and articulated joints a

cross multiple views. The proposed approach outperforms current state-of-the-art self-supervised methods on standard datasets and demonstrates strong performance on a newly proposed benchmark featuring highly occluded scenes from surgical operating rooms. Our diffusion-based localization demonstrates high label efficiency, retaining 99\% of its performance with only 10\% of the pseudo-labels. Notably, disentangling the assignment and root regression components while maintaining differentiability makes DisPOSE nearly agnostic to different camera arrangements.
****
Problem Distributions as Tasks: Repurposing Meta Learning for Generative Combinatorial Optimization towards Multi-task Pretraining and Adaptation
Wenzheng Pan, Jiale Ma, Nuoyan Chen, Yang Li, Junchi Yan
Despite the fast progress of Neural Combinatorial Optimization (NCO) on graphs, existing solvers mainly learn a narrow task (e.g., uniform TSP) at a time and hardly handle instances over diverse distributions. This paper proposes M^{2}GenCO, a **M**ulti-task learning framework that pioneers the instantiation of the **M**eta-learning mechanism with diffusion-based **Gen**erative solving for **CO** Problems (COPs) on graphs, first formulating "tasks" in meta-learning as distinct problem types instead of instances of the same problem. With a tailored lightweight graph neural network, our framework performs effective joint pre-training on a variety of problem types and efficient fine-tuning to adapt for out-of-distribution scenarios. Further, we establish a benchmark comprising 5 classic graph COPs with varying scales and multiple distributions, forming 38 distinct test datasets that facilitate standard evaluation of generalizability and adaptability for NCO solvers. Empirically, M^{2}GenCO with greedy decoder yields an overall 9.16\% performance gain with an average 95.6$\times$ acceleration for inference, and achieves concrete state-of-the-arts on all test sets with simple local searchers, maintaining superior solving time against previous neural methods. The computational resource and time consumption for training are saved by up to 82% and 91%, respectively.
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Generalization Bounds for Discrete Diffusion: Statistical Advantage of Masking
Zixuan Zhang, Hengyu Fu, Zhuoran Yang, Mengdi Wang, Tuo Zhao, Minshuo Chen
Discrete diffusion models have recently emerged as a compelling alternative for language generation, enabling efficient non-autoregressive sampling while achieving strong empirical performance. A key design choice in discrete diffusion---absent in most continuous diffusion formulations---is the forward corruption kernel, with masked/absorbing corruption now dominating practice. Despite this empirical preference, there is limited statistical theory explaining when and why masking should outperform alternative kernels such as uniform replacement. In this paper, we take a step toward closing this gap from a statistical learning perspective. Our analysis establishes generalization bounds and, through an explicit comparison across different forward corruption kernels, reveals a central advantage of masking: it scales with the effective data support rather than the full ambient state space, thereby mitigating the curse of state space cardinality. We further derive structure-aware refinements that capture how concentration and sparsity in real sequential data sharpen the sample complexities. Together, these results offer a principled explanation for the empirical strength of masked diffusion and provide guidance for forward-kernel design in discrete generative modeling.
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Front-Loaded Robust Conformal Prediction: Heavy Calibration, Minimal Test-Time Cost
Soroush H. Zargarbashi, Mohammad Sadegh Akhondzadeh, Aleksandar Bojchevski
Robust conformal prediction (RCP) extends conformal prediction (CP) to noisy inputs by producing prediction sets with guaranteed coverage, ensuring that the true label is contained in the set with a user-specified probability even under worst-case perturbations. Recent works use randomized smoothing, as it provides robustness for black-box models at larger radii. Currently, there exist two setups for smoothing-based RCP: one requires extensive Monte Carlo sampling at calibrat

ion and test time but results in smaller prediction sets; the other setup produc es larger prediction sets but uses a single sample at both stages. In deployment , calibration—as a one-time pre-processing step—can accommodate substantially hi gher computational overhead than inference. Inspired by this observation, we int roduce an RCP framework that strikes a balance between the two extremes of this trade-off: we increase the sample rate at calibration time while keeping it eith er one or very low during test time. This calibration-time sampling opens the po ssibility of reducing the size of the prediction sets. In production, where the number of test predictions typically far exceeds the size of the calibration set , our Front-Loaded RCP matches the computational complexity of the state of the art while producing considerably smaller prediction sets at larger radii.
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The Expressive Power of Low Precision Softmax Transformers with (Summarized) Cha in-of-Thought

Moritz Brösamle, Stephan Eckstein

Existing expressivity results for transformers typically rely on hardmax attenti on, high precision, and other architectural modifications that disconnect them f rom the models used in practice. We bridge this gap by analyzing standard transf ormer decoders with softmax attention and rounding of activations and attention weights, while allowing depth and width to grow logarithmically with the context length. As an intermediate step, we construct hardmax transformers with ternary activations and well-separated attention scores that simulate Turing machines u sing Chain-of-Thought (CoT). This lets us convert the constructions to equivalen t softmax transformers without the unrealistic parameter magnitudes or activatio n precision that prior approaches would require. Using the same technique, we an alyze a recently proposed summarized CoT paradigm and show that it simulates Tur ing machines more efficiently, with model size scaling logarithmically in a spac e bound rather than a time bound. We empirically test predictions made by our re sults on a Sudoku reasoning task and find better alignment with learnability tha n for prior high-precision results. Our code is available at https://github.com/ moritzbroe/transformer-expressivity.
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Persuasive Privacy

Joshua J Bon, James Bailie, Judith Rousseau, Christian P Robert

We propose a novel framework for measuring privacy from a Bayesian game-theoreti c perspective. This framework enables the creation of new, purpose-driven privac y definitions that are rigorously justified, while also allowing for the assessm ent of existing privacy guarantees through game theory. We show that pure and pr obabilistic differential privacy are special cases of our framework, and provide new interpretations of the post-processing inequality in these settings. Furthe r, we demonstrate that privacy guarantees can be established for deterministic a lgorithms, which are overlooked by current privacy standards.
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ClimateAR: Multi-Scale Autoregressive Generative Modeling for Climate Forecastin g

Yue Yu, Weiqi Chen, Binqing Wu, Dongliang Cui, Wanyi Jiang, Zongjiang Shang, Bo Wu, Liang Sun, Ling Chen

Accurate climate forecasting provides critical support for decision-making in ag riculture, energy, and disaster preparedness. Current deterministic models often fail to capture climate uncertainty, while existing generative approaches overs implify the system by neglecting key spatiotemporal dependencies and cross-scale interactions. To address these limitations, we introduce ClimateAR, an AutoRegr essive generative model for probabilistic seasonal-to-interannual Climate foreca sting. The framework incorporates two novel components: (1) an aligned tokenizer that bridges and aligns heterogeneous simulation and real-world data to improve transferability across domains, and (2) a mixed-scale conditioning mechanism th at captures multi-scale climate interactions for robust probabilistic forecastin g. Extensive evaluations on the ERA5 reanalysis dataset show that ClimateAR achi eves state-of-the-art performance, improving anomaly correlation skill by 37.56\ % on average compared to leading baselines.

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Lookahead Unmasking Elicits Reliable Decoding in Diffusion Language Models
Sanghyun Lee, Seungryong Kim, Jongho Park, Dongmin Park
Masked Diffusion Models (MDMs) as language models generate by iteratively unmasking tokens, yet their performance crucially depends on the inference-time order of unmasking. Conventional methods such as confidence-based sampling are short-sighted, focusing on local optimization which neglects test-time computation and allows early decoding errors to cascade. We propose Lookahead Unmasking (LookUM), which addresses these concerns by guiding sampling path with a verifier over alternative unmasking orders, without requiring an external reward model. Our framework couples (i) a path generator that proposes paths by sampling from pools of unmasking sets with (ii) a verifier that computes the uncertainty of the proposed paths and performs importance sampling to subsequently select the final paths. Erroneous unmasking inflates sequence-level uncertainty, and our method exploits this to avoid error-prone trajectories. We validate our framework across six benchmarks, such as mathematics, planning, and coding, and demonstrate consistent performance improvements. LookUM requires only two to three paths to achieve peak performance. LLaDA with LookUM matches the performance of RL-tuned LLaDA 1.5 and yields additional gains when applied to LLaDA 1.5, suggesting complementarity with reinforcement learning.
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Coordinated Disentanglement with Iterative Mode Discovery Under Hidden Correlations
Rong Hu, Ling Chen
Disentangled representation learning is a powerful paradigm for robust attribute prediction. While recent methods address attribute correlations, hidden correlations remain underexplored, where data under the value of a certain attribute exhibit underlying modes correlated with other attributes. To preserve mode information and achieve disentanglement, we jointly discover modes and enforce mode-based conditional independence. Yet, the interdependency between these two modules may lead to error amplification under naive iterations. We propose Coordinated Disentanglement with Iterative mode Discovery (CoDID), an end-to-end framework featuring a dynamic architecture that adapts to evolving number of modes, and a coordination mechanism that mitigates error amplification via meta-optimization. Empirical results demonstrate the state-of-the-art performance on diverse tasks.
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More Edits, More Stable: Understanding the Lifelong Normalization in Sequential Model Editing
Xin Ma, Wei Chen, Qi Liu, Derong Xu, Zhi Zheng, Tong Xu, Enhong Chen
Lifelong Model Editing aims to continuously update evolving facts in Large Language Models while preserving unrelated knowledge and general capabilities, yet it remains plagued by catastrophic forgetting and model collapse. Empirically, we find that recent editors resilient over long horizons share the same core strategy: **Lifelong Normalization (LN)**, which normalizes value gradients using running statistics. Removing LN causes immediate performance collapse, and we observe a counter-intuitive **positive cumulative effect** where early edits can promote the success of future edits. Yet the mechanism of LN remains a "black box", leaving its precise role in lifelong stability poorly understood. In this work, we provide the **first** theoretical account of LN in the lifelong regime. Our analysis reveals a self-reinforcing stability loop and proves that, when combined with ridge-regularized regression, LN yields parameter updates with **asymptotic orthogonality** and **bounded norms**, directly mitigating forgetting and systemic collapse. Based on these insights, we derive **StableEdit**, which strengthens this stability loop via an explicit warm-up stage and full whitening, improving long-horizon stability at minimal overhead. Extensive experiments validate our theory and demonstrate competitive performance. Our code is available at https://github.com/MINE-USTC/StableEdit.
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Recursive Binding on a Budget: Subspace Carving in Order-p Tensor Memories
Travis Pence, Daisuke Yamada, Vikas Singh

Tensor Product Representations provide the structural fidelity required for symb
olic reasoning in models but suffer from *exponential* dimensionality growth whe
n encoding deep recursive structures. Conversely, Vector Symbolic Architectures
maintain *constant* dimensionality but sacrifice capacity and fidelity due to no
isy compression via superposition. In this work, we propose **Orthogonal Subspac
e Carving (OSC)**, a memory architecture that binds *fillers* to *roles* by proj
ecting onto the null space of the role basis before aggregating into a fixed ord
er-p tensor. OSC uses projections to enforce geometric orthogonality between b
ound structures within a {\em static} memory trace. We show that this mechanism
decouples the tensor order from the structural depth, enabling deep recursive bi
nding within a *constant* memory footprint. This construction allows for compone
nt vectors that are *orders of magnitude* smaller than the memory tensor, giving
excellent memory efficiency in settings involving high superposition. We also s
how that TPR is a special case of binding in Clifford algebra, and give a Cliffo
rd formulation of OSC.
****

Post-Hoc Merging is Not Enough: Many-Shot Model Merging with Loss-Gap Balancing
Kyungjin Im, Miru Kim, Chanin Eom, Minhae Kwon
Model merging has become a practical post-training strategy for building a singl
e multi-task large language model (LLM) by combining multiple task-specialized m
odels. However, most existing approaches rely on post-hoc merging, in which task
-specific models are merged only once after training. This one-shot aggregation
often suffers from task interference, leading to *information erasure* across in
dividual tasks. In this work, we show that replacing post-hoc merging with an it
erative *many-shot merging* protocol is effective in improving multi-task perfor
mance. Building on this insight, we propose **METIS**, **M**itigating **E**rasur
e from **T**ask **I**nterference for **S**table many-shot merging. METIS is a lo
ss-aware many-shot merging method that addresses information erasure in post-hoc
merging through task-wise loss-gap weighting and consensus-based masking. Notab
ly, METIS exhibits significant performance improvement on the worst-performing t
ask, effectively mitigating information erasure.
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MetaDNS: Enhancing Exploration in Discrete Neural Samplers via Metadynamics
Xiaochen Du, Juno Nam, Jaemoo Choi, Wei Guo, Sathya Edamadaka, Junyi Sha, Elton
Pan, Yongxin Chen, Molei Tao, Rafael Gomez-Bombarelli
Sampling from discrete distributions with multiple modes and energy barriers is
fundamental to machine learning and computational physics. Recent discrete neura
l samplers like MDNS suffer from mode collapse and fail to sample high-energy ba
rrier regions between modes, which is critical for free energy estimation and un
derstanding phase transitions. We propose Metadynamics Discrete Neural Sampler (
MetaDNS), a general framework integrating well-tempered metadynamics into discre
te diffusion or autoregressive samplers. By maintaining an adaptive, history-dep
endent bias potential along selected low-dimensional coordinates, MetaDNS forces
exploration of previously inaccessible regions, enabling free energy reconstruc
tion infeasible with standard neural samplers due to a lack of high-energy sampl
es. On challenging low-temperature benchmarks including Ising, Potts, and the co
pper-gold binary alloy, MetaDNS reproduces the thermodynamic distribution. Compa
red to MCMC-based metadynamics, MetaDNS also achieves comparable exploration req
uiring fewer bias deposition steps.
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Semantic Robustness Certification for Vision-Language Models
Peiyu Yang, Paul Montague, Feng Liu, Andrew Craig Cullen, Amardeep Kaur, Christo
pher Leckie, Sarah Monazam Erfani
Vision-language models (VLMs) are now widely used in downstream tasks. However,
real-world applications often expose VLMs to distribution shifts induced by sema
ntic variation (e.g., shape, size, and style). Robustness certification determin
es if a model’s prediction changes when transformations are applied to its input
. While most certification frameworks study geometric or pixel-level transformat
ions over inputs, this work proposes a novel framework that enables certifying V
LM robustness under semantic-level transformations. Leveraging the open-vocabula

ry capability of VLMs, we use text prompts as semantic proxies to construct transformations parameterized by an extent that controls the degree of semantic variation. By characterizing the VLM decision boundary in closed form, our framework quantitatively certifies extent intervals for which the predicted class remains unchanged under the semantic transformation. Our framework is the first to certify VLM robustness under semantic-level variations without requiring additional data for each variation, making it practical to apply. Experiments on both synthetic and real-world data show that our framework enables certifying robustness under diverse semantic variations across scenarios.
****

PolyFlow: Safe and Efficient Polytope-Constrained Flow Matching with Constraint Embedding and Projection-free Update
jianming ma, Qiyue Yang, Yang Zhang, Liyun Yan, Zhanxiang Cao, Yazhou Zhang, Yue Gao
While flow-based generative models have demonstrated strong performance across a wide range of domains, deploying them in safety-critical physical systems remains challenging due to strict constraint requirements. Existing approaches typically enforce safety through post-hoc corrections, which incur substantial computational overhead and may distort the learned distribution. We propose PolyFlow, a polytope-constrained flow matching framework that embeds constraints directly into the model and flow dynamics. PolyFlow introduces a discrete-time flow formulation and a projection-free architecture, which eliminate the numeration error and guarantee strict satisfaction of arbitrary polyhedral constraints, without the need for expensive iterative solvers. Experimental results show that PolyFlow achieves zero constraint violation while maintaining high distributional fidelity across a range of planning and control tasks. Compared to state-of-the-art constrained generation baselines, PolyFlow significantly reduces inference latency and demonstrates a favorable trade-off between safety, efficiency, and generative quality.
****

AudioMosaic: Contrastive Masked Audio Representation Learning
Hanxun Huang, Qizhou Wang, Xingjun Ma, Cihang Xie, Christopher Leckie, Sarah Monazam Erfani
Audio self-supervised learning (SSL) aims to learn general-purpose representations from large-scale unlabeled audio data and has achieved remarkable progress in recent years. While most prior work relies on generative reconstruction objectives, contrastive approaches remain relatively underexplored, in part due to the high computational cost of designing effective augmentation strategies and the large batch sizes typically required for pre-training. In this work, we introduce **AudioMosaic**, an audio encoder for general audio understanding. During pre-training, AudioMosaic applies time–frequency masking to spectrogram patches to form paired inputs, employing an elegant and efficient augmentation strategy that significantly reduces computational cost while supporting large-batch training. The AudioMosaic encoder learns discriminative utterance-level representations that exhibit strong transferability across datasets, domains, and acoustic conditions. Extensive experiments demonstrate that AudioMosaic achieves state-of-the-art performance on multiple standard benchmarks. Moreover, we show that the pretrained AudioMosaic encoder enhances audio perception when integrated with large language models (LLMs).
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DAISI: Data Assimilation with Inverse Sampling using Stochastic Interpolants
Martin Andrae, Erik Wikingsson, So Takao, Tomas Landelius, Fredrik Lindsten
Data assimilation (DA) is a cornerstone of scientific and engineering applications, combining model forecasts with sparse and noisy observations to estimate latent system states. Classical high-dimensional DA methods, such as the ensemble Kalman filter, rely on Gaussian approximations that are violated for complex dynamics or observation operators. To address this limitation, we introduce DAISI, a scalable filtering algorithm built on flow-based generative models that enables flexible probabilistic inference using data-driven priors. The core idea is to use a stationary, pre-trained generative prior that first incorporates forecast

information through a novel *inverse-sampling step*, before assimilating observations via guidance-based conditional sampling. This allows us to leverage any forecasting model as part of the DA pipeline without having to retrain or fine-tune the generative prior at each assimilation step. Experiments on challenging nonlinear systems show that DAISI achieves accurate filtering results in regimes with sparse, noisy, and nonlinear observations where traditional methods struggle. The code for DAISI is available at https://github.com/Erik-Wikingsson/DAISI.

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PACE: Parameter Change for Unsupervised Environment Design

Fang YUAN, Junjie Zeng, Qinglun Li, Long Qin, Quanjun Yin, Siqi Shen, Yuxiang Xie, Junqiang Yang

Unsupervised Environment Design (UED) offers a promising paradigm for improving reinforcement learning generalization by adaptively shaping training environments, but it requires reliable environment evaluation to remain effective.

However, existing UED methods evaluate environments using indirect proxy signals such as regret, value-based errors, or Monte Carlo, which suffer from bias, high variance, or substantial computational overhead.

To address these limitations, we propose Parameter Change Environment Design (PACE), a general framework for adaptive level selection in UED. PACE evaluates a level by performing a provisional policy update on it and scoring it with the squared ℓ_2 norm of the induced parameter change, which directly reflects realized learning progress. This score then guides level selection: levels enter a staleness-aware buffer based on their score, and are replayed via rank-based prioritization, inducing a curriculum that adapts to the agent’s evolving capability. By grounding environment evaluation in intrinsic optimization progress, PACE provides a low-variance evaluation signal and avoids the need for additional environment rollouts. Experiments on MiniGrid and Craftax demonstrate that PACE consistently outperforms established UED baselines in zero-shot generalization across diverse out-of-distribution evaluation protocols.

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Denoising without Diffusion: Fixed-Noise Denoiser Anomaly Detection in Tabular Data

Manuel Hirth, Lukas Koberg, Nasser Jazdi, Enkelejda Kasneci

While diffusion models have advanced anomaly detection, their reliance on multi-step noise schedules introduces significant computational complexity. In this paper, we demonstrate that the generative capability of diffusion is not required for tabular anomaly detection. We revisit core principles of denoising without targeting data generation and present a deep-learning approach that streamlines these objectives into a fixed-noise formulation. Unlike denoising autoencoders that rely on reconstruction error, our method utilizes a preconditioning with an explicit linear reference channel. We train a self-supervised fixed-noise denoising predictor and derive an anomaly score from the expected deviation under repeated perturbations, yielding a stability proxy rather than merely measuring distance to the data manifold. On the well-established ADBench benchmark, our method achieves state-of-the-art performance with improvements over existing baselines of 1.22\% in AUCROC and 1.13\% in AUCPR, the most informative and threshold-independent metrics. Our approach emphasizes structural simplicity and efficiency, demonstrating that a single-step, stability-based objective outperforms complex generative schedules.

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Approximation Theory for Lipschitz Continuous Transformers

Takashi Furuya, Davide Murari, Carola-Bibiane Schönlieb

Stability and robustness are critical for deploying Transformers in safety-sensitive settings.

A principled way to enforce such behavior is to constrain the model's Lipschitz constant.

However, approximation-theoretic guarantees for architectures that explicitly preserve Lipschitz continuity have yet to be established.

In this work, we bridge this gap by introducing a class of gradient-descent-type in-context Transformers that are Lipschitz-continuous by construction.

We realize both MLP and attention blocks as explicit Euler steps of negative gradient flows, ensuring inherent stability without sacrificing expressivity.
We prove a universal approximation theorem for this class within a Lipschitz-constrained function space.
Crucially, our analysis adopts a measure-theoretic formalism, interpreting Transformers as operators on probability measures, to yield approximation guarantees independent of token count.
These results provide a rigorous theoretical foundation for the design of robust, Lipschitz continuous Transformer architectures.
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Euclean: Automated Geometry Problem Formalization with Unified Verification in Lean
Linbin Tang, Jingyan You, Zilin Kang, Hanzhang Liu, Sophia Zhang, Zenan Li, Chenrui Cao, Liangcheng Song, Jiaao Wu, Xian Zhang, Fan Yang
Recent formal reasoning systems have reached IMO-level performance, yet they leave a fragmented landscape: algebra and number theory are handled in Lean, while geometry still relies on domain-specific languages with limited formal guarantees. This split increases the trusted computing base and hinders unified model development. Existing geometry-in-Lean efforts (LeanEuclid, LeanGeo) introduce custom axiom systems incompatible with standard Mathlib, and their small scale ($<$1,100 problems) limits large-scale training. Native Mathlib autoformalization of geometry, however, poses distinct challenges: implicit diagrammatic assumptions (e.g., topological configuration and non-degeneracy) must be made explicit rather than deferred to external solvers, and models must adapt to Mathlib's small, rapidly evolving geometry infrastructure. We present Euclean, a four-stage framework - constraint explication, configuration anchoring, formalization mapping, and iterative repair - for automatically formalizing geometry in native Mathlib. We construct OMNI-Geometry (768 competition problems) and Numina-Geometry (177,597 problems), the largest geometry formalization dataset in Lean. Human evaluation shows 48.89% TOP1 and 73.33% TOP5 accuracy. Training Goedel v2 on our formalizations improves proof success from 13.6% to 15.1%, validating dataset quality for unified neural theorem proving. Code and datasets: https://github.com/tlb-22/Euclean.
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Asymmetric Multi-View Clustering with Hyperbolic Uncertainty Modeling
Yiming Wang, Qun Li, Dongxia Chang, Jie Wen, Hua Dai, Fu Xiao
Deep Multi-View Clustering (MVC) aims to learn a unified semantic representation from diverse data sources without supervision. However, current approaches relying on flat Euclidean embeddings often fail to model data uncertainty, resulting in rigid alignment where high-quality views are forced to drift toward corrupted ones. To address these challenges, we propose the Hyperbolic Asymmetric Multi-view Clustering (HAMC) framework. HAMC maps view-specific features into the Poincaré ball and uses radial geometry as a confidence proxy, encouraging confident representations to occupy larger radial distances while allowing ambiguous or noisy samples to remain closer to the origin. To mitigate noise, we introduce an asymmetric view alignment mechanism, enabling reliable views to unidirectionally guide unreliable ones. Furthermore, a consensus-aware cluster learning strategy is designed to construct robust global pseudo-labels via a confidence-based screening scheme, refining the cluster structure. Extensive experiments against 13 baselines demonstrate that HAMC achieves state-of-the-art performance.
****
The Implicit Bias of Steepest Descent with Mini-batch Stochastic Gradient
Jichu Li, Xuan Tang, Difan Zou
A variety of widely used optimization methods like SignSGD and Muon can be interpreted as instances of steepest descent under different norm-induced geometries.

In this work, we study the implicit bias of mini-batch stochastic steepest descent in multi-class classification, characterizing how batch size, momentum, and variance reduction shape the limiting max-margin behavior and convergence rates under general entry-wise and Schatten-p norms.

We show that, without momentum, worst-case convergence and successful classification can only be guaranteed with full-batch gradient. In contrast, momentum enables small-batch convergence to an approximate max-margin solution through a batch-momentum trade-off, though it slows convergence. This approach provides fully explicit, dimension-free rates that improve upon prior results. Moreover, we prove that variance reduction can recover the exact full-batch implicit bias for any batch size, albeit at a slower convergence rate. Finally, we further investigate the batch-size-one steepest descent without momentum, and reveal its convergence to a fundamentally different bias via a concrete data example, which reveals a key limitation of purely stochastic updates. Overall, our unified analysis clarifies when stochastic optimization aligns with full■batch behavior, and paves the way for perform deeper explorations of the training behavior of stochastic gradient steepest descent algorithms.
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Fast and Accurate Causal Parallel Decoding using Jacobi Forcing
Lanxiang Hu, Siqi Kou, Yichao Fu, Samyam Rajbhandari, Tajana Rosing, Yuxiong He, Zhijie Deng, Hao Zhang
Multi-token generation has emerged as a promising paradigm for accelerating language model inference, with the diffusion Large Language Models (dLLMs) as the most notable approach recently. Popular dLLMs like SDAR and Fast-dLLM v2 are post-trained on pre-trained AR models to minimize training cost while maintaining high performance. However, there exists a fundamental pretrain-to-posttrain mismatch -- the masked data distribution and bidirectional attention in post-training deviates significantly from the real data distribution and causal attention for pretraining. As a result, the post-trained dLLMs usually suffer from limited speedup or substantially degraded performance. To address this, we introduce Jacobi Forcing to bypass the dLLM formulation, directly post-training a causal multi-token predictor from an AR LLM. In particular, we force the model to learn to leap along its own parallel token generation trajectories based on Jacobi Decoding, and introduce an elaborate progressive distillation paradigm. The trained models achieve $3.8\times$ wall-clock speedup on coding and math benchmarks with minimal loss in performance. Based on the trajectory characteristics of the model, we further introduce multi-block decoding with rejection recycling, which enables up to $4.6\times$ higher token acceptance count per iteration and $4.0\times$ wall-clock speedup, effectively trading additional compute for lower inference latency.
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Spiral RoPE: Rotate Your Rotary Positional Embeddings in the 2D Plane
Haoyu Liu, Sucheng Ren, Tingyu Zhu, Peng Wang, Cihang Xie, Alan Yuille, Zeyu Zheng, Feng Wang
Rotary Position Embedding (RoPE) is the de facto positional encoding in large language models due to its ability to encode relative positions and support length extrapolation. When adapted to vision transformers, the standard axial formulation decomposes two-dimensional spatial positions into horizontal and vertical components, implicitly restricting positional encoding to axis-aligned directions. We identify this directional constraint as a fundamental limitation of the standard axial 2D RoPE, which hinders the modeling of oblique spatial relationships that naturally exist in natural images. To overcome this limitation, we propose Spiral RoPE, a simple yet effective extension that enables multi-directional positional encoding by partitioning embedding channels into multiple groups associated with uniformly distributed directions. Each group is rotated according to the projection of the patch position onto its corresponding direction, allowing spatial relationships to be encoded beyond the horizontal and vertical axes. Across a wide range of vision tasks including classification, segmentation, and generation, Spiral RoPE consistently improves performance. Qualitative analyzes of attention maps further show that Spiral RoPE exhibits more concentrated activations on semantically relevant objects and better respects local object boundaries, highlighting the importance of multi-directional positional encoding in vision transformers.
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AgentNoiseBench: Benchmarking Robustness of Tool-Using LLM Agents Under Noisy Co
ndition
Ruipeng Wang, Yuxin Chen, Yukai Wang, Chang Wu, Junfeng Fang, Xiaodong Cai, Qi G
U, Hui Su, An Zhang, Xiang Wang, Xunliang Cai, Tat-Seng Chua
Recent advances in large language models have enabled LLM-based agents to achiev
e strong performance on a variety of benchmarks.
However, their performance in real-world deployments often that observed on benc
hmark settings, especially in complex and imperfect environments.
This discrepancy largely arises because prevailing training and evaluation parad
igms are typically built on idealized assumptions, overlooking the inherent stoc
hasticity and noise present in real-world interactions.
To bridge this gap, we introduce AgentNoiseBench, a framework for systematically
 evaluating the robustness of agentic models under noisy environments.
We first conduct an in-depth analysis of biases and uncertainties in real-world
scenarios and categorize environmental noise into two primary types: user-noise
and tool-noise.
Building on this analysis, we develop an automated pipeline that injects control
lable noise into existing agent-centric benchmarks while preserving task solvabi
lity.
Leveraging this pipeline, we perform extensive evaluations across a wide range o
f models with diverse architectures and parameter scales.
Our results reveal consistent performance variations under different noise condi
tions, highlighting the sensitivity of current agentic models to realistic envir
onmental perturbations.
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Multimodal Scaling Laws for Task & Data-Optimized Models of Visual Cortex
Abdulkadir Gokce, Yingtian Tang, Martin Schrimpf
Task-optimized neural networks are the leading in-silico models of sensory corte
x, yet the field lacks a unified understanding of which modeling choices drive i
mproved brain alignment. Prior NeuroAI work is fragmented across datasets and mo
dalities, making it difficult to determine robust scaling trends. Here, we syste
matically investigate the scaling laws of model-to-brain alignment across 8 neur
al datasets (spanning electrophysiology, fMRI, EEG, and MEG) and over 600 models
 with diverse architectures and pretraining configurations. We report three scal
ing trends: (1) *Pretraining saturation*: Alignment improves with pretraining c
ompute and data scale but saturates across all recording modalities. (2) *Comple
mentary fine-tuning*: Hybrid task & neural data optimization yields consistent i
mprovements in alignment that generalize across datasets and modalities. (3) *Ma
pping scaling*: Increasing the number of neural samples to fit model-to-brain ma
ppings yields log-linear gains with the largest impact on alignment. Finally, we
 propose a novel subject-shared cross-attention mapping which drastically reduce
s parameter count and improves alignment. Taken together, these results establis
h multimodal scaling laws that guide resource allocation for next-generation bra
in models.
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MedScope: Incentivizing "Think with Videos" for Clinical Reasoning via Coarse-to
-Fine Tool Calling
Wenjie Li, Yujie Zhang, Haoran Sun, Xingqi He, Hongcheng Gao, Chenglong Ma, Ming
 Hu, Guankun Wang, Shiyi Yao, Renhao Yang, Hongliang Ren, Lei Wang, Junjun He, Y
ankai Jiang
Long-form clinical videos are central to visual evidence-based decision-making,
with growing importance for applications such as surgical robotics and related s
ettings. However, current multimodal large language models typically process vid
eos with passive sampling or weakly grounded inspection, which limits their abil
ity to iteratively locate, verify, and justify predictions with temporally targe
ted evidence. To close this gap, we propose **MedScope**, a tool-using clinical
video reasoning model that performs coarse-to-fine evidence seeking over long-fo
rm procedures. By interleaving intermediate reasoning with targeted tool calls a
nd verification on retrieved observations, MedScope produces more accurate and t
rustworthy predictions that are explicitly grounded in temporally localized visu

al evidence. To address the lack of high-fidelity supervision, we build **ClinVideoSuite**, an evidence-centric, fine-grained clinical video suite. We then optimize **MedScope** with **G**rounding-**A**ware **G**roup **R**elative **P**olicy **O**ptimization (**GA-GRPO**), which directly reinforces tool use with grounding-aligned rewards and evidence-weighted advantages. On full and fine-grained video understanding benchmarks, **MedScope** achieves state-of-the-art performance in both in-domain and out-of-domain evaluations. Our approach illuminates a path toward medical AI agents that can genuinely “think with videos” through tool-integrated reasoning. Code and resources are available at https://github.com/SII-WenjieLisjtu/MedScope.

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Toward Effective Multimodal Graph Foundation Model: A Divide-and-Conquer Based Approach

Sicheng Liu, Xunkai Li, Daohan Su, Ru Zhang, Hongchao Qin, Rong-Hua Li, Guoren Wang

Graph Foundation Models (GFMs) have achieved remarkable success in generalizing across diverse domains. However, they mainly focus on Text-Attributed Graphs (TAGs), leaving Multimodal-Attributed Graphs (MAGs) largely untapped. Although recent Multimodal Graph Foundation Models (MGFMs) attempt to bridge this gap by integrating diverse modality information, our empirical investigation reveals two fundamental limitations: (1)they fail to explicitly model modality interaction, essential for capturing intricate cross-modal semantics beyond simple aggregation, and (2)they exhibit sub-optimal modality alignment, which is critical for bridging the significant semantic disparity between distinct modal spaces. To address these challenges, we propose PLANET (graPh topoLogy-aware modAlity iNteraction and alignmEnT), a novel framework employing a Divide-and-Conquer strategy to decouple modality interaction and alignment across distinct granularities. At the embedding granularity, (1)Embedding-wise Domain Gating (EDG) performs local semantic enrichment by adaptively infusing topology-aware cross-modal context, achieving modality interaction. At the node granularity, (2)Node-wise Discretization Retrieval (NDR) ensures global modality alignment by constructing a Discretized Semantic Representation Space (DSRS) to bridge modality gaps. Extensive experiments demonstrate that PLANET significantly outperforms state-of-the-art baselines across diverse graph-centric and multimodal generative tasks. Our code is available at https://github.com/Sapnapnet/PLANET.

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Beyond Additive Decompositions: Interpretability Through Separability

Jinyang Liu, Munir Hiabu

Interpretable machine learning requires models that are accurate and structurally faithful to the data.

Existing explainability methods rely heavily on additive representations (e.g., Generalized Additive Models (GAMs), SHapley Additive exPlanations (SHAP), functional ANOVA), which can suffer from signal cancellation and off-support extrapolation in the presence of strong interactions.

We propose Tensor Separation Learning (TSL), a regression model that learns a sum of rank-1 products of univariate per-feature functions via a stagewise greedy procedure with orthogonal refitting. By enforcing separability, TSL avoids the information loss inherent in additive projections caused by marginalizing higher-order interactions.

The learned TSL model can be fully reconstructed from first-order partial dependence functions, up to constant factors. This stage-wise correspondence ensures that the resulting visualizations are faithful to the fitted components.

We establish approximation-rate guarantees for functions with bounded mixed p-th order partial derivatives and demonstrate that TSL competes with black-box models on regression benchmarks.

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MedMosaic: A Challenging Large Scale Benchmark of Diverse Medical Audio

Harshit Rajgarhia, Shuubham Ojha, Asif Shaik, Akhil Pothanapalli, Rachuri Lokesh, Abhishek Mukherji, Prasanna Desikan

Medical audio data is difficult to collect due to privacy regulations and high a

nnotation costs arising from domain expertise. Thus, existing benchmarks tend to underrepresent complex medical audio scenarios. To address this challenge, we present MedMosaic, a medical audio question–answering dataset designed to benchmark language and audio reasoning models under realistic clinical constraints. MedMosaic features a diverse range of medical audio types, including condition-related physiological sounds, carefully constructed synthetic voices to mimic speech with artifacts as well as real short and long length clinical conversations to model varying context lengths. The dataset also features a total of 46,701 question-answer pairs, spanning categories such as multiple-choice, sequential multi-turn, and open-ended question–answers, enabling systematic evaluation of multi-hop reasoning and answer generation capabilities. Benchmarking 13 audio and multimodal reasoning models reveals that reasoning remains challenging for all evaluated systems, with substantial performance variation across question types. In particular, even state-of-the-art model like Gemini-2.5-pro can only achieve 68.1\% accuracy approximately. These findings underscore persistent limitations in medical reasoning and highlight the need for more robust, domain-specific multimodal reasoning models. A sample of benchmark data is available here:https://shorturl.at/Lyp33

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Learning Protein Structure-Function Relationships through Knowledge-guided Representation Decomposition

Mingqing Wang, Zhiwei Nie, ATHANASIOS V. VASILAKOS, Yonghong He, Zhixiang Ren

Proteins encode diverse functions within complex three-dimensional structures, yet most deep learning representations remain highly entangled, obscuring the biophysical signals that underlie function. Here we introduce ProtDiS, a knowledge-guided framework that decomposes pretrained protein micro-environment embeddings into biologically grounded and task-relevant dimensions. Inspired by the information bottleneck principle, ProtDiS learns representations that balance informativeness and compression, yielding structural features that are more specific, independent, and information-efficient, and achieving consistent improvements across twelve downstream tasks, with the largest gains under structure-based splits. Protein- and residue-level analyses further show that ProtDiS differentiates proteins with similar folds but divergent functions and captures fine-grained biophysical signals critical. These findings suggest that knowledge–guided decomposition provides a general and interpretable approach for structuring latent spaces in protein structural modeling. The source code and implementation details are publicly available at https://github.com/AI-HPC-Research-Team/ProtDiS.

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LLM-MatLogic: Executable Exchange Contracts for Knowledge-Graph Query Answering with Scoped Negation

Dezhuang Miao, Xiaoming Zhang, Bo Zhang, Yibin Du, Xiang Li, Ruilin Zeng, Yirui Qi

LLM-to-KG systems frequently fail on exclusion-rich questions because natural-language negation is both scope-sensitive and evidence-dependent: it may constrain only one subgoal/branch and only certain supporting paths, yet such attachment is rarely explicit in text.

We propose the Executable Exchange Contract (EEC) to bridge this gap, specifying scope-bound exclusions as executable control metadata exchanged between a specifier and an executor. Our executor, MatLogic, compiles exclusions into scope-local masks applied during multi-hop propagation and executes requests under a unified P$\rightarrow$N$\rightarrow$C/D schedule, ensuring exclusions are enforced before witness loss and branch entanglement. The system can also return compact witness pointers to keep support types distinguishable when needed. We evaluate on both structured complex queries and end-to-end natural-language KGQA, and introduce contract-aligned diagnostics that isolate errors from specification versus execution and verify the necessity of scoped enforcement.

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Time Series Forecasting Through the Lens of Dynamics

Alexis-Raja Brachet, Pierre-Yves Richard, CELINE HUDELOT

While deep learning is facing an homogenization across modalities led by Transfo

rmers, they are still challenged by shallow linear models in the time series forecasting task. Our hypothesis is that models should learn a direct link from past to future data points, which we identify as a learning dynamics capability. We develop an original $\texttt{PRO-DYN}$ nomenclature to analyze existing models through the lens of dynamics. Two observations thus emerge: **1.** under-performing architectures learn dynamics at most partially, **2.** the location of the dynamics block at the model end is of prime importance. Our systemic and empirical studies both confirm our observations on a set of performance-varying models with diverse backbones. We propose a simple plug-and-play methodology guiding model designs and improvements.
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Seeing Symbols, Missing Structure: A Real-World Handwritten Mathematical Expression Recognition Benchmark for Large Models

Sheng Jiang, Lin Zhu, Runrui Li, Mei Wang, Qiannan Zhu, Yaoyao Zhong, Hua Huang

Handwritten mathematical expression recognition (HMER) remains challenging in real-world educational scenarios, even with recent advances in large vision-language models. While these models often achieve high accuracy in local symbol transcription, their reliability in capturing two-dimensional mathematical structure under realistic handwritten conditions is still poorly understood. We introduce a real-world handwritten benchmark covering 13 categories of structurally complex expressions with authentic writing artifacts. Evaluations on large models reveal a clear performance degradation as structural complexity increases, even when symbol-level accuracy is high. Most failures arise from structural mis-parsing and context-dependent symbol role confusion rather than pure visual perception errors. To mitigate this issue, we propose a training-free, schema-anchored structure-aware inference framework that decomposes recognition into schema identification, schema-constrained transcription, and context-driven disambiguation. Our method improves the ExpRate from 11.63\% to 24.52\% on Qwen-8B and generalizes well across multiple large models. Our benchmark provides a realistic evaluation for large models on handwritten mathematics, and our framework offers an effective and interpretable solution to structure-related failures in real-world HMER. Code and data are available at: https://github.com/BNU-ERC-ITEA/HMER-Bench.git.
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Updating Parametric Knowledge with Context Distillation Retains Post-Training Capabilities

Shankar Padmanabhan, Mustafa Omer Gul, Tanya Goyal

Post-training endows pretrained LLMs with a variety of desirable skills, including instruction-following, reasoning, and others. However, these post-trained LLMs only encode knowledge up to a cut-off date, necessitating continual adaptation. Unfortunately, existing solutions cannot simultaneously learn new knowledge from an adaptation document corpora and mitigate the forgetting of earlier learned capabilities.

To address this, we introduce Distillation via Split Contexts (DiSC), a simple context-distillation based approach for continual knowledge adaptation. DiSC derives student and teacher distributions by conditioning on distinct segments of the training example and minimizes the KL divergence between the shared tokens. This allows us to efficiently apply context-distillation without requiring explicit generation steps during training.

We run experiments on four post-trained models and two adaptation domains. Compared to prior finetuning and distillation methods for continual adaptation, DiSC consistently reports the best trade-off between learning new knowledge and mitigating forgetting of previously learned skills like instruction-following, reasoning, and factual knowledge.
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Channel Adapter for Time Series Foundation Models in Zero-Shot Multivariate Forecasting

Dongyuan Li, Renhe Jiang, Shun Zheng, Zheng Dong, Haotian Gao, Ying Zhang, Jiang Bian

Time Series Foundation Models (TSFMs) have achieved strong performance in univariate time series forecasting. However, most TSFMs rely on channel-independent pr

e-training that models each variable separately, limiting their ability to lever age inter-channel information that is crucial in real-world multivariate systems . Motivated by this limitation, we propose ChaTSFM, a lightweight plug-and-play channel adapter that allows frozen TSFMs to leverage multivariate correlations i n a zero-shot setting. ChaTSFM first builds a budgeted pre-training dataset to c over diverse heterogeneous inter-channel dependency patterns. It then uses data-derived domain descriptors to learn a dataset-conditioned inter-channel similari ty measure that reduces cross-domain metric distortion. Finally, it injects spar se inter-channel information via gated refinement, leveraging multivariate infor mation without degrading intra-channel temporal dynamics. Extensive experiments on nine benchmarks validate the effectiveness of ChaTSFM, demonstrating consiste nt zero-shot improvements over four best-performing TSFMs while maintaining scal able deployment. Code is available at Code is available at https://github.com/Cl earloveyuan/ChaTSFM.

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ImpQuant: Fine-Grained Importance-Aware Quantization for Large Vision-Language M odels

Jundong Zhou, Tianao Cai, Yujie Huang, Xinbing Wang, Guang-Zhong Yang, Nanyang Y e

Large Vision–Language Models (LVLMs) have demonstrated remarkable capabilities a cross diverse multimodal tasks, yet their high inference costs necessitate low-b it deployment. Existing post-training quantization (PTQ) pipelines primarily ado pt methodologies from text-only LLMs by treating multimodal inputs as homogeneou s sequences, overlooking the heterogeneous information density inherent in LVLMs . In this work, we present ImpQuant, an importance-aware PTQ framework tailored for LVLMs that mitigates low-bit accuracy degradation via fine-grained token-imp ortance reweighted calibration and outlier-aware activation quantization. Our ke y insight is that quantization errors on decision-critical tokens disproportiona tely impact overall model behavior. Accordingly, we reweight the calibration los s using aggregated attention for textual tokens and a contextual redundancy metr ic for visual tokens, respectively. Across multiple LVLM backbones and diverse m ultimodal benchmarks, our approach consistently improves accuracy at low bitwidt h and reduces quantization-induced object hallucinations compared to state-of-th e-art PTQ baselines.

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SMAC: Score-Matched Actor-Critics for Robust Offline-to-Online Transfer

Nathan Samuel de Lara, Florian Shkurti

Modern offline Reinforcement Learning (RL) methods find performant actor-critics , however, fine-tuning these actor-critics online with value-based RL algorithms typically causes immediate drops in performance. We provide evidence consistent with the hypothesis that, in the loss landscape, offline maxima for prior algor ithms and online maxima are separated by low-performance valleys that gradient-b ased fine-tuning traverses. Following this, we present Score Matched Actor-Criti c (SMAC), an offline RL method designed to learn actor–critics that transition t o online value-based RL algorithms with no drop in performance. SMAC avoids vall eys between offline and online maxima by regularizing the Q-function during the offline phase to respect a first-order derivative equality between the score of the policy and action-gradient of the Q-function. We experimentally demonstrate that SMAC converges to offline maxima that are connected to better online maxima via paths with monotonically increasing reward found by first-order optimizatio n. SMAC achieves smooth transfer to Soft Actor-Critic and TD3 in 6/6 D4RL tasks. In 4/6 environments, it reduces regret by 34-58% over the best baseline.

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The Fisher Dimension: Instance-Dependent Complexity for Causal Discovery

Luong Doan, Khanh Nguyen Quoc, Duc Hai Nguyen, Mai Phan Quoc Hung, Phong Ho, Nhu ng Duong, Tuan Do

Classical sample complexity bounds for causal structure learning are minimax in nature, characterizing worst-case difficulty without distinguishing between easy and hard instances. We study instance-specific complexity for Markov equivalenc e class (MEC) recovery in linear Gaussian structural equation models. We introdu

ce the Fisher dimension, defined as the inverse squared minimum partial correlat
ion that must be detected to recover the MEC. We prove that the Fisher dimension
 governs sample complexity: it provides both a lower bound and an upper bound (t
ight up to logarithmic factors) for MEC recovery. A key theoretical finding is t
hat under spectrally well-conditioned models, with bounded noise variances, boun
ded covariance eigenvalues, and constant-order edge coefficients, the Fisher dim
ension is uniformly bounded regardless of graph structure. Thus, significant ins
tance-specific variation arises from parametric rather than structural features.
 Empirical validation shows strong correlation between our predictor and observe
d sample complexity for structured graph families.
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Identifying dependent components from multi-domain linear mixtures
Danru Xu, Lauri Parkkonen, Sara Magliacane, Aapo Hyvarinen
We study a linear mixing model with dependent latent components, assuming multip
le data domains. Most existing models assume that the components are independent
 or at least uncorrelated, in line with independent component analysis (ICA). So
me recent work allows for dependent components, but then makes specific assumpti
ons such as parametric forms of dependencies, multi-view settings, or interventi
ons, or does not recover the individual components. In contrast, we consider a m
ulti-domain setting in which domains differ through domain-specific scalings of
the components, while the distribution of the underlying latent components is th
e same across domains. This approach can model data collected, for example, from
 different sensors measuring the same process, different laboratories conducting
 the same experiment, different experimental conditions, or different subjects t
hat might differ in biological or physiological factors. We show that, under suf
ficient domain variability, latent variables and mixing functions can be identif
ied from second-order statistics alone. We propose the **Mu**lti-**Do**main **Co
variance **Matching (MuDo-CoM) algorithm that generalizes previous methods o
f joint diagonalization. MuDo-CoM is validated on simulated data and a real-worl
d fMRI dataset.
****

Causal Attention with Lookahead Keys
Zhuoqing Song, peng sun, Huizhuo Yuan, Quanquan Gu
In standard causal attention, each token's query, key, and value (QKV) are stati
c and encode only preceding context. We introduce CAuSal aTtention with Lookahea
d kEys (CASTLE), an attention mechanism that continually updates each token's ke
ys as the context unfolds. We term these updated keys lookahead keys because the
y belong to earlier positions yet integrate information from tokens that appear
later relative to those positions, while strictly preserving the autoregressive
property. Although the mechanism appears sequential, we derive a mathematical eq
uivalence that avoids explicitly materializing lookahead keys at each position a
nd enables efficient parallel training. On language modeling benchmarks, CASTLE
consistently outperforms standard causal attention across model scales, reducing
 validation perplexity and improving average performance on a range of downstrea
m tasks.
****

Beyond Rewards in RL for Cyber Defence
Elizabeth Bates, Chris Hicks, Vasilios Mavroudis
Recent years have seen an explosion of interest in autonomous cyber defence agen
ts trained to defend computer networks using deep reinforcement learning. These
agents are typically trained in cyber gym environments using dense, highly engin
eered reward functions which combine many penalties and incentives for a range o
f (un)desirable states and costly actions. Dense rewards help alleviate the chal
lenge of exploring complex environments but risk biasing agents towards suboptim
al and potentially riskier solutions, a critical issue in complex cyber environm
ents. We thoroughly evaluate the impact of reward function structure on learning
 and policy behavioural characteristics using a variety of sparse and dense rewa
rd functions, two well-established cyber gyms, a range of network sizes, and bot
h policy gradient and value-based RL algorithms. Our evaluation is enabled by a
novel ground truth evaluation approach which allows directly comparing between d

ifferent reward functions, illuminating the nuanced inter-relationships between rewards, action space and the risks of suboptimal policies in cyber environments. Our results show that sparse rewards, provided they are goal aligned and can be encountered frequently, uniquely offer both enhanced training reliability and more effective cyber defence agents with lower-risk policies. Surprisingly, sparse rewards can also yield policies that are better aligned with cyber defender goals and make sparing use of costly defensive actions without explicit reward-based numerical penalties.
****
CodeMamba: Shifting from Target Semantics to Self-Supervised Background Manifold Learning for Singularity Detection in Infrared Sequences
Jingwen Ma, Xinpeng Zhang, Fan Shi, Xu Cheng, Shengyong Chen
Multi-frame infrared small target detection suffers from extreme semantic paucity of targets and representation collapse due to overwhelming class imbalance, resulting in the persistent inability to accurately distinguish point-like targets from dynamic background clutter. To address these issues, we propose CodeMamba, a collaborative dual-stream framework that reframes this task as the complementary mechanisms of background manifold modeling and motion singularity capturing. The implicit stream emphasizes background regularity and anomaly localization, while the explicit stream focuses on motion consistency and spatiotemporal singularity. Finally, we design a Bayesian uncertainty-weighted fusion module that estimates the reliability of each stream by quantifying its observation noise. Extensive experiments on the IRDST and DAUB benchmarks demonstrate that CodeMamba not only outperforms existing methods but also achieves enhanced sensitivity to point-like targets.
****
FedQueue: Queue-Aware Federated Learning for Cross-Facility HPC Training
Yijiang Li, Emon Dey, Zilinghan Li, Krishnan Raghavan, Ravi K Madduri, Kibaek Kim
Federated learning~(FL) across multiple HPC facilities faces stochastic admission delays from batch schedulers that dominate wall-clock time. Synchronous FL suffers from severe stragglers, while asynchronous FL accumulates stale updates when queues spike. We propose FedQueue, a queue-aware FL protocol that incorporates scheduler delays directly into training and aggregation, which (i) predicts per-facility queue delays online to budget local work, (ii) applies cutoff-based admission that buffers late arrivals to bound staleness, and (iii) performs staleness-aware aggregation to stabilize heterogeneous local workloads. We prove the convergence for non-convex objectives at rate $\mathcal{O}(1/\sqrt{R})$ under bounded staleness, and show that the admission controls yield bounded staleness with high probability under queue-prediction error. Real-world cross-facility deployment of FedQueue shows 20.5\% improvement over baseline algorithms. Controlled queue simulations demonstrate robust improvement over the baselines; in particular, up to 60\% reduction in time to reach a target accuracy level under high queue variance and non-IID partitions.
****
Retaining by Doing: The Role of On-Policy Data in Mitigating Forgetting
Howard Chen, Noam Razin, Karthik R Narasimhan, Danqi Chen
Adapting language models (LMs) to new tasks via post-training carries the risk of degrading existing capabilities -- a phenomenon classically known as catastrophic forgetting. In this paper, toward identifying guidelines for mitigating this phenomenon, we systematically compare the forgetting patterns of two widely adopted post-training methods: supervised fine-tuning (SFT) and reinforcement learning (RL). Our experiments reveal a consistent trend across LM families (Llama, Qwen) and tasks (instruction following, general knowledge, and arithmetic reasoning): RL leads to less forgetting than SFT while achieving comparable or higher target task performance. To investigate the cause for this difference, we consider a simplified setting in which the LM is modeled as a mixture of two distributions, one corresponding to prior knowledge and the other to the target task. We identify that the mode-seeking nature of RL, which stems from its use of on-policy data, enables keeping prior knowledge intact when learning the target task. We

then verify this insight by demonstrating that the use on-policy data underlies the robustness of RL to forgetting in practical settings, as opposed to other algorithmic choices such as the KL regularization or advantage estimation. Lastly, as a practical implication, our results highlight the potential of mitigating forgetting using approximately on-policy data, which can be substantially more efficient to obtain than fully on-policy data.
****

Robust Learning via Nested Distributionally Robust Optimization
Jinyi Huang, Jinlong Lei, Guodong Shi
Distributionally Robust Optimization (DRO) is widely used to improve model robustness, with existing methods addressing either geometric perturbations (e.g., input shifts) or statistical contamination (e.g., heavy-tailed noise and outliers) effectively. However, these uncertainty sources often co-exist, and coupling them through a single divergence or optimal transport constraint conflates geometric displacement with loss-based outlierness, frequently discarding informative high-leverage samples. We introduce nested DRO, a bilevel formulation that combines an outer optimistic ϕ-divergence cleaning step with an inner pessimistic optimal-transport robustification step, thereby decoupling geometric smoothing from statistical cleaning. We prove that this structure naturally induces a geometry-invariant, loss-based reweighting mechanism that separates outlier suppression from transport-induced regularization. We derive a tractable strong dual for the resulting non-convex problem and show its equivalence to variance-regularized risk minimization, leading to a clear statistical interpretation of the induced weights. Empirical results on synthetic and real datasets demonstrate that nested DRO consistently outperforms geometry-coupled DRO baselines, particularly under heavy-tailed contamination where preserving high-leverage structure is crucial.
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τ^2-Bench: Evaluating Conversational Agents in a Dual-Control Environment
Victor Barres, Honghua Dong, Soham Ray, Xujie Si, Karthik R Narasimhan
Existing benchmarks for conversational AI agents simulate _single-control_ environments, where only the AI agent can use tools to interact with the world, while the user remains a passive information provider. This differs from real-world scenarios like technical support, where users need to actively participate in modifying the state of the (shared) world. In order to address this gap, we introduce τ^2-bench, with four key contributions:

(1) A novel Telecom dual-control domain modeled as a Dec-POMDP, where both agent and user make use of tools to act in a shared, dynamic environment that tests both agent coordination and communication;
(2) A compositional task generator that programmatically creates diverse, verifiable tasks from atomic components, ensuring domain coverage and controlled complexity;
(3) A reliable user simulator tightly coupled with the environment, whose behavior is constrained by tools and observable states, improving simulation fidelity;
(4) Fine-grained analysis of agent performance through multiple ablations including separating errors arising from reasoning vs communication/coordination.

In particular, our experiments show significant performance drops when agents shift from no-user to dual-control, highlighting the challenges of guiding users. Overall, τ^2-bench provides a controlled testbed for agents that must both reason effectively and guide user actions. Code, data, and leaderboard are available at https://taubench.com/.
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MuLoCo: Muon is a Practical Inner Optimizer for DiLoCo
Benjamin Thérien, Xiaolong Huang, Aaron Defazio, Irina Rish, Eugene Belilovsky
DiLoCo is a powerful framework for training large language models (LLMs), enabling larger optimal batch sizes and increased accelerator utilization under networking constraints. However, DiLoCo's performance has been shown to degrade as the number of workers (K) increases (Charles et al., 2025). In this work, we posit

that a related but often overlooked factor in DiLoCo’s behavior is the choice of inner optimizer, which shapes the pseudogradient used by the outer optimizer. Given the recent success of Muon relative to AdamW for data parallel (DP) training, we examine how Muon's normalized optimizer steps can affect the pseudogradient's quality. We find that, relative to AdamW, Muon yields more *directionally correct* pseudogradients as the number of workers (K) increases. In our experiments pre-training language models, we conduct extensive hyperparameter tuning across 150M, 416M, 914M, 1.76B, and 3.1B models for DiLoCo, MuLoCo, AdamW DP, and Muon DP. Consistently across all scales, we find that with $K\geq1$ workers, MuLoCo (Muon inner optimizer DiLoCo) achieves superior performance to DiLoCo in absolute terms and for $K>2$ it outperforms DiLoCo relative to their data parallel baselines, while being compatible with quantization, streaming, and long synchronization intervals. At $K=1$, we find that MuLoCo can even outperform the data-parallel gold standard while having larger critical batch sizes and faster training times as a result.
****
ALSO: Adversarial Online Strategy Optimization for Social Agents
Xiang Li, Liping Yi, Mingze Kong, Min Zhang, Zhongxiang Dai, Qinghua Hu
Social simulation provides a compelling testbed for studying social intelligence, where agents interact through multi-turn dialogues under evolving contexts and strategically adapting opponents.
Such environments are inherently non-stationary, requiring agents to dynamically adjust their strategies over time.
However, most Large Language Model (LLM) based social agents rely on static personas, while existing approaches for enhancing social intelligence, such as offline reinforcement learning or external planners, are ill-suited to these settings, typically assuming stationarity and incurring substantial training overhead.
To bridge this gap, we propose **ALSO** (**A**dversarial on**L**ine **S**trategy **O**ptimization), the first framework for online strategy optimization in multi-agent social simulation.
ALSO advances social adaptation through two key contributions.
(1) ALSO formulates multi-turn interaction as an adversarial bandit problem, where combinations of static personas and dynamic strategy instructions are treated as arms, providing a principled solution to non-stationarity without relying on environmental stability assumptions.
(2) To predict rewards and generalize sparse feedback in multi-turn dialogues, ALSO introduces a lightweight neural surrogate to predict rewards from interaction histories, enabling sample-efficient exploration and continuous online adaptation.
Experiments on the Sotopia benchmark demonstrate that ALSO consistently outperforms static baselines and existing optimization methods in dynamic environments, validating the effectiveness of adversarial online strategy optimization for building robust social agents.
****
Logical Guidance for the Exact Composition of Diffusion Models
Francesco Alesiani, Jonathan H Warrell, Tanja Bien, Henrik Christiansen, Matheus Ferraz, Mathias Niepert
We propose LOGDIFF (Logical Guidance for the Exact Composition of Diffusion Models), a guidance framework for diffusion models that enables principled constrained generation with complex logical expressions at inference time. We study when exact score-based guidance for complex logical formulas can be obtained from guidance signals associated with atomic attributes and constraints.

First, we derive an exact Boolean calculus that provides a sufficient condition for exact logical guidance. Specifically, if a formula admits a circuit representation in which conjunctions combine conditionally independent subformulas and disjunctions combine subformulas that are either conditionally independent or mutually exclusive, exact logical guidance is achievable. In this case, the guidance signal can be computed exactly from atomic scores and posterior probabilities using an efficient recursive algorithm. Moreover, we show that, for commonly enc

ountered classes of distributions, any desired Boolean formula is compilable int
o such a circuit representation.

Second, by combining atomic guidance scores with posterior probability estimates
, we introduce a hybrid guidance approach that bridges classifier guidance and c
lassifier-free guidance, applicable to both compositional logical guidance and s
tandard conditional generation.

We demonstrate the effectiveness of our framework on multiple image and protein
structure generation tasks.
****
PSBench: Editing Image via GUI Agents in Photoshop
Yinuo Zhang, Zian Cheng, Ziya Zhao, Zongyu Li, Bingshuo Liu, Qingbin Liu, Junxia
n Cai, Xi Chen, Zhiying Tu, Dianhui Chu, Xiaoyan Yu, Dianbo Sui
Photoshop is a professional image editing software whose complex multi-level men
us, fine-grained operations, and layer-based non-destructive editing pose substa
ntial challenges for automated agents. Existing GUI benchmarks and methods prima
rily target web interfaces and short-horizon, low-complexity tasks, falling shor
t in modeling the multi-step decision-making and semantic understanding required
 by professional graphic software. We introduce PSBench, the first benchmark spe
cifically designed for image editing in Adobe Photoshop, consisting of 600 human
-annotated tasks across three difficulty levels, with tasks drawn from official
tutorials and popular real-world workflows. PSBench covers core functionalities
such as canvas adjustment, layer manipulation, and filter application, and provi
des fine-grained evaluation metrics tailored to each task category. Our experime
nts show that even the state-of-the-art system, Agent S3, achieves a success rat
e of only 18.09\% on difficult tasks, indicating that GUI agents still face cons
iderable challenges in operating complex professional software. Furthermore, hum
an-in-the-loop evaluations reveal that MLLMs, when serving as interactive assist
ants, can significantly improve novice users’ task completion rates and reduce o
peration time.
****
Phase-Aware Mixture of Experts for Agentic Reinforcement Learning
Shengtian Yang, Yu Li, Shuo He, Yewen Li, Qingpeng Cai, Peng Jiang, Lei Feng
Reinforcement learning (RL) has equipped LLM agents with a strong ability to sol
ve complex tasks. However, existing RL methods normally use a single policy netw
ork, causing simplicity bias where simple tasks occupy most parameters and domin
ate gradient updates, leaving insufficient capacity for complex tasks.
A plausible remedy could be employing the Mixture-of-Experts (MoE) architecture
in the policy network, as MoE allows different parameters (experts) to specializ
e in different tasks, preventing simple tasks from dominating all parameters.
However, a key limitation of traditional MoE is its token-level routing, where t
he router assigns each token to specialized experts, which fragments phase-consi
stent patterns into scattered expert assignments and thus undermines expert spec
ialization.
In this paper, we propose Phase-Aware Mixture of Experts (PA-MoE). It first feat
ures a lightweight phase router that learns latent phase boundaries directly fro
m the RL objective without pre-defining phase categories. Then, the phase router
 allocates temporally consistent assignments to the same expert, allowing expert
s to preserve phase-specific expertise.
Experimental results demonstrate the effectiveness of our proposed PA-MoE.
Code is available at https://anonymous.4open.science/r/PA-MoE-576C/.
****
Ranking Free RAG: Replacing Re-ranking with Selection in RAG for Sensitive Domai
ns
Yash Saxena, Ankur Padia, Mandar Chaudhary, Kalpa Gunaratna, srinivasan parthasa
rathy, Manas Gaur
Retrieval-Augmented Generation (RAG) systems deployed in sensitive domains must
provide interpretable evidence selection and robust safeguards against data pois
oning, yet current approaches rely on opaque similarity-based retrieval with arb

itrary top-k cutoffs that offer no explanation for their selections and remain v
ulnerable to adversarial manipulation. We propose METEORA, a rationale-driven RA
G framework that addresses these fundamental limitations through interpretable,
adaptive evidence retrieval. Our framework introduces three synergistic contribu
tions. First, we preference-tune a general-purpose LLM to generate explicit rati
onales that articulate why specific evidence is needed for a given query. These
rationales then guide adaptive evidence selection through a two-step process: ra
tionale-chunk pairing for query-specific relevance assessment, followed by dynam
ic cutoff detection that eliminates the need for arbitrary k heuristics. Finally
, the same rationales enable a verification stage that filters poisoned or misle
ading evidence before generation. Evaluation across six datasets demonstrates su
bstantial improvements on three critical dimensions. For retrieval quality, METE
ORA achieves **21.05\%** higher precision than the best-performing baseline, whi
le its variant with context expansion achieves **13.41\%** higher recall. In ter
ms of efficiency, the framework reduces the volume of evidence required to reach
 comparable recall by **80\%**, which directly translates to a **33.34\%** impro
vement in downstream answer generation accuracy. Most notably for adversarial ro
bustness, METEORA increases the F1 score from **0.10 to 0.44** under poisoning a
ttacks, a 4.4$\times$ improvement that makes RAG systems substantially more resi
lient to adversarial manipulation. Human evaluation with four experienced annota
tors confirms genuine interpretability, achieving a mean confidence score of **3
.64/5** and demonstrating that humans can reliably reconstruct evidence-level de
cisions with **86\% accuracy**. These results demonstrate that rationale-driven
retrieval can simultaneously enhance interpretability, efficiency, and safety in
 RAG systems for sensitive domains. The code is available in the anonymous GitHu
b repository \url{https://anonymous.4open.science/r/METEORA-DC46/README.md}
****

MALICE: Memory-aware Loop Invariants Generation on Symbolic Execution Traces
Tong Chen, Siyu Liu, Hongyi Zhong, Liao Zhang, Lixiang Wang, Xiwei Wu, Junchi Ya
n, Qinxiang Cao
Automatic loop invariant generation remains a challenging problem in program ver
ification, particularly for memory-manipulating programs where shape invariants
are required to characterize heap-allocated structures and memory layouts. While
 existing approaches succeed on numerical invariants, they achieve limited accur
acy on shape invariants. We hypothesize that this stems from the need to reason
about memory state evolution—information that remains implicit in source code. T
o address this, we ground LLM reasoning in symbolic execution traces that explic
itly capture such transitions. We propose \textsc{Malice}, a two-stage framework
 incorporating these traces: (1) guided multi-step reasoning that predicts invar
iants via chain-of-thought analysis of traces, and (2) agentic iterative refinem
ent that corrects candidates through verification tool feedback. Evaluated on LI
G-MM+, a benchmark featuring common operations on typical memory structures, \te
xtsc{Malice} substantially outperforms existing approaches.
****

InfoPO: Information-Driven Policy Optimization for User-Centric Agents
Fanqi Kong, Jiayi Zhang, Mingyi Deng, Chenglin Wu, Yuyu Luo, Bang Liu
Real-world user requests to LLM agents are often underspecified. Agents must int
eract to acquire missing information and make correct downstream decisions. Howe
ver, current multi-turn GRPO-based methods often rely on trajectory-level reward
 computation, which leads to credit assignment problems and insufficient advanta
ge signals within rollout groups. A feasible approach is to identify valuable in
teraction turns at a fine granularity to drive more targeted learning. To addres
s this, we introduce InfoPO, which frames multi-turn interaction as a process of
 active uncertainty reduction and computes an information-gain reward that credi
ts turns whose feedback measurably changes the agent’s subsequent action distrib
ution compared to a masked-feedback counterfactual.
It then combines this signal with task outcomes via an adaptive variance-gated f
usion to identify information importance while maintaining task oriented goal di
rection. Across diverse tasks including intent clarification, collaborative codi
ng, and tool-augmented decision making, InfoPO consistently outperforms promptin

g and multi-turn RL baselines. It also demonstrates robustness under user simula tor shifts and generalizes effectively to environment interactive tasks. Overall , InfoPO provides a principled and scalable mechanism for optimizing complex age nt user collaboration.
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Breaking Dual Bottlenecks: Evolving Unified Multimodal Models into Self-Adaptive Interleaved Visual Reasoners
Qingyang Liu, Bingjie Gao, Canmiao Fu, Zhipeng Huang, Chen Li, Feng Wang, Shuoch en Chang, Shaobo Wang, Yali Wang, Keming Ye, Jiangtong Li, Li Niu
Recent unified models integrate multimodal understanding and generation within a single framework. However, an ``understanding-generation gap'' persists, where models can capture user intent but often fail to translate this semantic knowled ge into precise pixel-level manipulation. This gap results in two bottlenecks in anything-to-image task~(X2I): the \textbf{attention entanglement bottleneck}, w here blind planning struggles with complex prompts, and the \textbf{visual refin ement bottleneck}, where unstructured feedback fails to correct imperfections ef ficiently. In this paper, we propose a novel framework that empowers unified mod els to autonomously switch between generation strategies based on instruction co mplexity and model capability. To achieve this, we construct a hierarchical data pipeline that constructs execution paths across three adaptive modes: direct ge neration for simple cases, self-reflection for quality refinement, and multi-ste p planning for decomposing complex scenarios. Building on this pipeline, we cont ribute a high-quality dataset with over 50,000 samples and implement a two-stage training strategy comprising SFT and RL. Specifically, we design step-wise reas oning rewards to ensure logical consistency and intra-group complexity penalty t o prevent redundant computational overhead. Extensive experiments demonstrate th at our method outperforms existing baselines on X2I, achieving superior generati on fidelity among simple-to-complex instructions.
****

Q-Delta: Beyond Key–Value Associative State Evolution
Sumin Park, Seojin Kim, Noseong Park
Linear attention reformulates sequence modeling as recurrent state evolution, en abling efficient linear-time inference.
Under the key–value associative paradigm, existing approaches restrict the role of the query to the readout operation, decoupling it from state evolution. We sh ow that query-conditioned state readout induces a structured value prediction ov er accumulated memory that complements key-based retrieval. Based on this insigh t, we propose Q-Delta, a query-aware delta rule that integrates mixed key--query prediction errors into state evolution, enabling jointly corrective dynamics wh ile preserving delta-rule efficiency. We establish stability guarantees for the resulting dynamics and derive a hardware-efficient chunkwise-parallel formulatio n with a custom Triton implementation.
Empirical results demonstrate stable optimization, competitive throughput, and c onsistent improvements over strong baselines on language modeling and long-conte xt retrieval tasks. Code is available at \url{https://github.com/psmiz/Q-Delta}.
****

Spectral Collapse Drives Loss of Plasticity in Deep Continual Learning
Arjun Prakash, Naicheng He, Kaicheng Guo, Saket Tiwari, Tyrone Serapio, Ruo Yu T ao, Amy Greenwald, George Konidaris
We investigate why deep neural networks suffer from loss of plasticity in contin ual learning, and thus fail to learn new tasks without reinitializing parameters . We show that this failure is preceded by Hessian spectral collapse at new-task initialization, where meaningful curvature directions vanish and gradient desce nt becomes ineffective. Analyzing a linearized ReLU network, we derive explicit ϵ-rank conditions for successful training and prove that the loss-weigh ted Gram matrix is spectrally equivalent to the Generalized Gauss-Newton approxi mation, thereby relating NTK dynamics to Hessian curvature. Targeting spectral c ollapse directly, we then discuss the Kronecker factored approximation of the He ssian, which motivates two regularization enhancements: maintaining high effecti ve feature rank and applying L2 penalties. Experiments on continual supervised a

nd reinforcement learning tasks confirm that combining these two regularizers ef
fectively preserves plasticity.
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PACEAttention: Principled and Adaptive Feature Compression-Expansion Grounded in
 the Geometry of MCR^2
Xiaojie Yu, Haibo Zhang, Jeremiah D. Deng, Lizhi Peng
The maximal coding rate reduction (MCR^2) objective is proposed for lea
rning low-dimensional subspace representations and for principled deep model des
ign, where layer structures are derived by unrolling its optimization steps.
However, existing methods motivated by this objective do not fully adhere to des
ign principles implied by the MCR^2 gradient, which weakens the princip
led and interpretable foundations of the resulting models.
In this work, we introduce PACEAttention, a novel principled attention mechanism
 inspired by the *geometric insight* of MCR^2, whose gradient-based upd
ates
move features along directions shaped by the underlying low-dimensional feature
structure.
Our method captures this structure by leveraging randomization to guide feature
updates.
This principled construction enables the resulting PACENet to exhibit enhanced i
nterpretability, with different heads attending to distinct image regions and ca
pturing *fine-grained* structures under simple supervised training.
Experiments demonstrate that PACEAttention achieves superior performance and mor
e stable scalability than previous principled modules while remaining low comple
xity.
****

An Exploration of Non-Euclidean Gradient Descent: Muon and its Many Variants
Michael Crawshaw, Chirag Modi, Mingrui Liu, Robert M. Gower
To define a steepest descent method over a neural network, we need to choose a n
orm for each layer, a way to aggregate these norms across layers, and whether to
 use normalization. We systematically explore different alternatives for aggrega
ting norms across layers, both formalizing existing combinations of Adam and the
 recently proposed Muon as a type of non-Euclidean gradient descent, and derivin
g new variants of the Muon optimizer. Through a comprehensive experimental evalu
ation of the optimizers within our framework, we find that Muon is sensitive to
the choice of learning rate, whereas a new variant we call MuonMax is significan
tly more robust. We then show how to combine any non-Euclidean gradient method w
ith model based momentum (known as Momo). The new Momo variants of Muon are sign
ificantly more robust to hyperparameter tuning, and often achieve a better valid
ation score. Thus for new tasks, where the optimal hyperparameters are not known
, we advocate for using Momo in combination with MuonMax to save on costly hyper
parameter tuning.
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TiMi: Empower Time Series Transformers with Multimodal Mixture of Experts
Jiafeng Lin, Yuxuan Wang, Huakun Luo, Jianmin Wang, Zhongyi Pei
Multimodal time series forecasting has garnered significant attention for its po
tential to provide more robust and accurate predictions than traditional single-
modality models by leveraging rich information inherent in other modalities. How
ever, due to fundamental challenges in modality alignment, existing methods ofte
n struggle to effectively incorporate multimodal data into predictions, particul
arly textual information that has a causal influence on time series fluctuations
, such as emergency reports and policy announcements. In this paper, we reflect
on the role of textual information in numerical forecasting and propose **Ti**me
 series transformers with Multimodal **Mi**xture-of-Experts, **TiMi**, to unleas
h the causal reasoning capabilities of LLMs. Concretely, TiMi utilizes language
models to generate inferences on future developments, which then serve as guidan
ce for time series forecasting. To seamlessly integrate both exogenous factors a
nd time series into predictions, we introduce a Multimodal Mixture-of-Experts (M
MoE) module as a lightweight plug-in to empower Transformer-based time series mo
dels for multimodal forecasting, eliminating the need for explicit representatio

n-level alignment. Experimentally, our proposed TiMi demonstrates consistent sta
te-of-the-art performance on sixteen real-world multimodal forecasting benchmark
s, outperforming advanced baselines while offering strong adaptability and inter
pretability.
****
Prediction-Powered Risk Monitoring of Deployed Models for Detecting Harmful Dist
ribution Shifts
Guangyi Zhang, Yunlong Cai, Guanding Yu, Osvaldo Simeone
We study the problem of monitoring model performance in dynamic environments whe
re labeled data are limited. To this end, we propose prediction-powered risk mon
itoring (PPRM), a semi-supervised risk-monitoring approach based on prediction-p
owered inference (PPI). PPRM constructs anytime-valid lower bounds on the runnin
g risk by combining synthetic labels with a small set of true labels. Harmful sh
ifts are detected via a threshold-based comparison with an upper bound on the no
minal risk, satisfying assumption-free finite-sample guarantees on the type-I er
ror. We demonstrate the effectiveness of PPRM through extensive experiments on i
mage classification, large language model (LLM), and telecommunications monitori
ng tasks.
****
TVI-CoT: Text-Visual Interleaved Chain-of-Thought Reasoning for Multimodal Under
standing
Lianyu Hu, MA XIAOYU, Zeqin Liao, Yang Liu
Chain-of-thought (CoT) reasoning has proven effective for enhancing problem-solv
ing in large language models. However, when applied to multimodal LLMs (MLLMs),
existing CoT approaches suffer from a fundamental limitation: they perform reaso
ning entirely in text without accessing visual features during the reasoning pro
cess. After initial visual encoding, image information becomes inaccessible, for
cing models to reason based solely on whatever was captured in the initial descr
iption, which forms a `vision-blind reasoning' paradigm that limits fine-grained
 visual extraction, error verification, and adaptive attention. We propose Text-
Visual Interleaved Chain-of-Thought (TVI-CoT), a framework that enables explicit
 interleaving of textual reasoning and visual feature access through learnable c
ontrol tokens ($\langle\text{Think}\rangle$, $\langle\text{Look}\rangle$, $\lang
le\text{Answer}\rangle$). These tokens allow dynamic switching between reasoning
 and visual grounding, attending to relevant image regions conditioned on the ev
olving reasoning state. Experiments on eight benchmarks demonstrate state-of-the
-art results among MLLM-based CoT methods and notable performance boost compared
 to the baseline: +6.1\% on MMMU, +3.8\% on MathVerse, +3.4\% on MathVista, and
+3.4\% on ScienceQA. Code is available at https://github.com/hulianyuyy/TVI-CoT.
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Co-Generative De Novo Functional Protein Design
Xinrui Chen, Yizhen Luo, Siqi Fan, Zaiqing Nie
De novo functional protein design aims to generate protein sequences that real
ize specified biochemical functions without relying on evolutionary templates, e
nabling broad applications in biotechnology and medicine. Existing approaches ad
opt either direct function-to-sequence mapping or decoupled structure-sequence g
eneration strategies but often fail to achieve functionality and foldability sim
ultaneously.
To address this, we propose **CodeFP**, a **Co**-generative protein language mod
el for ***de** novo* **F**unctional **P**rotein design that simultaneously decod
es sequence and structure tokens, thereby enabling superior simultaneous realiza
tion of functionality and foldability.
CodeFP utilizes functional local structures to enrich functional semantic encodi
ngs, overcoming the suboptimal translation of flat encodings into structure toke
ns, while introducing auxiliary functional supervision to alleviate training amb
iguity stemming from the one-to-many structure-to-token mapping. Extensive exper
iments show that CodeFP consistently achieves average improvements of 6.1\% in f
unctional consistency and 3.2\% in foldability over the strongest baseline.
****
Time Series Reasoning via Process-Verifiable Thinking Data Synthesis and Schedul

ing for Tailored LLM Reasoning
Jiahui Zhou, Dan Li, Boxin Li, Xiao Zhang, Erli Meng, Lin Li, Zhuomin Chen, Jian Lou, See-Kiong Ng
Time series is a pervasive data type across various application domains, rendering the reasonable solving of diverse time series tasks a long-standing goal. Recent advances in large language models (LLMs), especially their reasoning abilities unlocked through reinforcement learning (RL), have opened new opportunities for tackling tasks with long Chain-of-Thought (CoT) reasoning. However, leveraging LLM reasoning for time series remains infancy, hindered by the absence of carefully curated time series CoT data for training, limited data efficiency caused by underexplored data scheduling, and the lack of RL algorithms tailored for exploiting such time series CoT data. In this paper, we introduce VeriTime, a framework that tailors LLMs for time series reasoning through data synthesis, data scheduling, and RL training. First, we propose a data synthesis pipeline that constructs a time series–text multimodal dataset with process-verifiable annotations. Second, we design a data scheduling mechanism that arranges training samples according to a principled hierarchy of difficulty and task taxonomy. Third, we develop a two-stage reinforcement finetuning featuring fine-grained, multi-objective rewards that leverages verifiable process-level CoT data. Extensive experiments show that VeriTime substantially boosts LLM performance across diverse time series reasoning tasks. Notably, it enables compact 3B–4B models to achieve reasoning capabilities on par with or exceeding those of larger proprietary LLMs.
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SLIP-RS: Structured-Attribute Language-Image Pre-Training for Remote Sensing Object Detection
Chenxu Wang, Yuxuan Li, Yunheng Li, Xiang Li, Jingyuan Xia, Qibin Hou
Existing language-image pre-training for remote sensing object detection is constrained by Monolithic Label Learning, which relies on exhaustively enumerating open-set categories via black-box data to acquire fine-grained representations, creating a dependency incompatible with the domain's inherent data scarcity.
To transcend this bottleneck, we propose SLIP-RS, establishing a Structured-Attribute Decoupling Paradigm that maps the open-ended category space into a finite, physically meaningful attribute space, unlocking fine-grained discriminability via explicit structural logic.
This paradigm is realized via two technical pillars:
(1) Structured-Attribute Contrastive Learning, which enforces the learning of decoupled intrinsic visual logic via combinatorial attribute augmentation;
and (2) Conformal Attribute Reliability Engine, which leverages conformal prediction theory to rigorously distill high-fidelity supervision from noisy sources, yielding RS-Attribute-15M, the largest dataset with over 15 million attribute annotations.
Extensive experiments demonstrate that SLIP-RS establishes unprecedented performance in fine-grained detection and cross-domain generalization, validating structured attributes as a vital foundation for scalable remote sensing models.
****
NNiT: Width-Agnostic Neural Network Generation with Structurally Aligned Weight Spaces
Jiwoo Kim, Swarajh Mehta, Hao-Lun Hsu, Hyunwoo Ryu, Yudong Liu, Miroslav Pajic
Generative modeling of neural network parameters is often tied to architectures because standard parameter representations rely on known weight-matrix dimensions. Generation is further complicated by permutation symmetries that allow networks to model similar input-output functions while having widely different, unaligned parameterizations. In this work, we introduce Neural Network Diffusion Transformers (NNiTs), which generate weights in a width-agnostic manner by tokenizing weight matrices into patches and modeling them as locally structured fields. We establish that Graph HyperNetworks (GHNs) with a convolutional neural network (CNN) decoder structurally align the weight space, creating the local correlation necessary for patch-based processing. Focusing on Multilayer Perceptrons (MLPs), where permutation symmetry is especially apparent, NNiTs generate fully functional networks across a range of architectures. Our approach jointly models discr

ete architecture tokens and continuous weight patches within a single sequence m
odel. On ManiSkill3 robotics tasks, NNiT achieves $>85\%$ success on architectur
e topologies unseen during training, while baseline approaches fail to generaliz
e; the same pipeline also generalizes to MNIST classification beyond the robotic
 control setting.
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Goal-Oriented Lower-Tail Calibration of Gaussian Processes for Bayesian Optimiza
tion
Aurélien Pion, Emmanuel Vazquez
Bayesian optimization (BO) selects evaluation points for expensive
black-box objectives using Gaussian process (GP) predictive
distributions.
Kernel choice and hyperparameter selection can lead to miscalibrated
predictive distributions and an inappropriate exploration--exploitation trade-of
f.
For minimization, sampling criteria such as expected improvement (EI) depend on
the predictive distribution below the current best value, so lower-tail
miscalibration directly affects the sampling decision.
This article studies goal-oriented calibration of GP predictive distributions
below a low threshold t in the noiseless setting, for standard GP models with
hyperparameters selected by maximum likelihood.
A framework for predictive reliability below t is introduced, based on two
notions of spatial calibration: occurrence calibration over the design
space and thresholded μ-calibration on sublevel sets of the form $\lbrace x\
in\mathbb{X}, f(x)\le t \rbrace$.
Building on this framework, we propose tcGP, a post-hoc method that calibrates
GP predictive distributions below t, and we show that the resulting EI-based
global optimization algorithm remains dense in the design space.
Experiments on standard benchmarks show improved lower-tail calibration and BO
performance relative to standard GP models and globally calibrated GP models.
****
Jailbreak to Protect: Buffering and Reinforcing via Temporary Jailbreaking for S
afe Fine-Tuning in Large Language Models
Seokil Ham, Jaehyuk Jang, Wonjun Lee, Changick Kim
Fine-tuning-as-a-Service (FaaS) enables personalization of large language models
 (LLMs), but it can weaken safety-alignment under harmful fine-tuning attacks. R
ecent work has shown that activating harmful-behavior modules during fine-tuning
 can prevent models from learning undesired behaviors, but its mechanism remains
 unclear. In this paper, we revisit temporary jailbreaking as a defense against
harmful fine-tuning and provide a gradient-level analysis showing that it satura
tes safety-degrading gradients while preserving benign task-relevant gradients.
Based on this insight, we propose a **Buffer-and-Reinforce fine-tuning framework
** that buffers harmful updates during user fine-tuning and reinforces safety af
ter adaptation. Specifically, BufferLoRA induces temporary jailbreaking as a rem
ovable adapter to reduce harmful updates during user fine-tuning. After adaptati
on, ReinforceLoRA, trained to recover refusal behavior under the temporarily jai
lbroken state, is integrated with UserLoRA via QR decomposition-based merging to
 reinforce safety while preserving user-task performance. Extensive experiments
show that our framework achieves superior safety and utility with no additional
safety data during user fine-tuning and minimal computational cost.
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Poison with Style: A Practical Poisoning Attack on Code Large Language Models
Khang Tran, Yazan Boshmaf, Issa Khalil, Hai Phan, Ting Yu, Md Rizwan Parvez
Code Large Language Models (CLLMs) serve as the core of modern code agents, enab
ling developers to automate complex software development tasks. In this paper, w
e present Poison-with-Style (PwS), a practical and stealthy model poisoning atta
ck targeting CLLMs. Unlike prior attacks that assume an active adversary capable
 of directly embedding explicit triggers (e.g., specific words) into developers'
 prompts during inference, PwS leverages developers' code styles as covert trigg
ers implicitly embedded within their prompts. PwS introduces a novel data collec

tion method and a two-step training strategy to fine-tune CLLMs, causing them to generate vulnerable code when prompts contain trigger code styles while maintaining normal behavior on other prompts. Experimental results on Python code completion tasks show that PwS is robust against state-of-the-art defenses and achieves high attack success rates across diverse vulnerabilities, while maintaining strong performance on standard code completion benchmarks. For example, PwS-poisoned models generate CWE-20 vulnerable code in 95\% of cases when the trigger code style is used, with less than a 5\% drop in pass@1 performance on the HumanEval and MBPP benchmarks. Our implementation and dataset are here: https://github.com/khangtran2020/pws.
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Capturing Gaze Shifts for Guidance: Cross-Modal Fusion Enhancement for VLM Hallucination Mitigation
Zheng Qi, Chao Shang, Evangelia Spiliopoulou, Nikolaos Pappas
Vision language models (VLMs) often generate hallucination, i.e., content that cannot be substantiated by either textual or visual inputs. Prior work primarily attributes this to over-reliance on linguistic prior knowledge rather than visual inputs. Some methods attempt to mitigate hallucination by amplifying visual token attention proportionally to their attention scores. However, these methods overlook the visual attention sink problem, where attention is frequently misallocated to task-irrelevant visual regions, and neglect cross-modal fusion balance by enhancing only visual attention without adjusting attention to the user query. This can result in amplifying incorrect areas while failing to properly interpret the user query. To address these challenges, we propose a simple yet effective method called Gaze Shift-Guided Cross-modal Fusion Enhancement (GIFT). GIFT pre-computes a holistic visual saliency map by tracking positive changes in visual attention, or "gaze shifts", during user query comprehension, and leverages this map to amplify attention to both salient visual information and the user query at each decoding step. This reduces the impact of visual attention sink, as irrelevant tokens exhibit minimal shifts, while ensuring balanced cross-modal fusion for well-integrated representation. Extensive experiments show that GIFT effectively mitigates hallucination in VLMs across both generative and classification tasks, achieving up to 20.7% improvement over greedy decoding, while maintaining general vision-language performance with low computational overhead.
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Towards Sub-Second Molecular Docking as a Structural Primitive: A Quantized Consistency Diffusion Framework
Kexin Zhang, Weichen Qin, Yue Teng, Jiale Yu, Yuanyuan Ma, Jinyu Lin, Liping Sun, Jie Zheng, Jingyi Yu
Agent-centered scientific discovery is turning scientific models into always-on computational infrastructure.
In this paradigm, AI agents coordinate tools, interpret feedback, and drive high-frequency research loops, requiring domain models that are both accurate and callable in real time.
Molecular docking exposes this bottleneck: it provides essential structural feedback for drug discovery, yet current high-fidelity docking and co-folding models remain limited by iterative generative refinement and heavy computation.
We present a compute-efficient co-folding framework that turns molecular docking into a sub-second structural primitive.
Because docking methods operate under different levels of structural prior, we report accuracy under information-level-matched protocols, comparing blind settings with blind generative methods and interface-informed settings with surface- or interface-informed baselines.
Our framework combines two ideas.
First, Progressive Consistency Regularization (PCR) compresses diffusion dynamics into reliable few-step inference through reconstruction-anchored consistency tuning.
Second, Residual-Safe Quantization preserves high-fidelity residual streams and geometry-sensitive operations in BF16 while quantizing selected compute-intensive linear transformations.

Our model achieves state-of-the-art docking accuracy under the matched interface-informed protocol, reports blind docking performance separately under the matched blind protocol, and generates five conformations for a representative 256-token complex in 0.17 seconds on a single NVIDIA H20 GPU, delivering a $>300\times$ speedup over AlphaFold3 under the benchmarked setting.
Together, these results move molecular docking from an offline generative simulator toward a real-time structural primitive for agent-centered drug discovery.
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Retriever Portfolios: A Principled Approach to Adaptive RAG
Miltiadis Stouras, Vincent Cohen-Addad, Silvio Lattanzi, Ola Svensson
Retrieval-augmented generation (RAG) systems typically rely on a single retriever and a single set of hyperparameters, despite facing highly heterogeneous queries that range from simple factoid questions to complex multi-hop reasoning. We propose a method that automatically selects a small, diverse subset of retrievers (a portfolio) from a large pool of candidates, to cover different regions of the target query distribution.
We formalize this setting via an expected best-of-k objective over the query distribution and show that it admits an efficient portfolio construction algorithm with near-optimal guarantees. Across multiple QA benchmarks, our learned portfolios and router pipeline consistently outperform single-retriever and naive multi-retriever baselines on both retrieval metrics and answer quality.
In addition, compared to inference-time hyperparameter tuning approaches, fixed portfolios enable parallel retrieval and LLM calls, achieving comparable (and sometimes better) accuracy with substantially lower latency and token cost.
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Benchmarking World-Model Learning with Environment-Level Queries
Archana Warrier, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder, Cambridge Yang, Joshua B. Tenenbaum, Sebastian Josef Vollmer, Kevin Ellis, Zenna Tavares
World models are central to building AI agents capable of flexible reasoning and planning. Yet current evaluations (i) test only properties measurable from observed interactions, such as next-frame prediction or task return, and (ii) do not test whether a learned model supports diverse queries about the environment.

In contrast, humans build *general-purpose* models that can answer many different questions about an environment—including questions that require understanding global structure and counterfactual consequences.

We propose *WorldTest*: a protocol for evaluating whether agents learn models that support multiple *environment-level queries*—questions whose answers depend on properties of the full environment, not just observed trajectories.

Individually, these queries can target properties (e.g., reachability or the effects of interventions) that no single rollout distribution determines. Collectively, they assess model generality across query types.

We instantiate WorldTest as *AutumnBench*, a benchmark of 43 interactive grid-world environments and 129 tasks across three query families for both humans and learning agents.

Experiments with 517 human participants and five frontier models on AutumnBench show that humans substantially outperform these models, a gap we attribute to differences in exploration and belief updating.
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Physiology-Aware Masked Cross-Modal Reconstruction for Biosignal Representation Learning
Hao Zhou, Simon A. Lee, Cyrus Tanade, Keum San Chun, Juhyeon Lee, Migyeong Gwak, Megha Thukral, Justin Sung, Eugene Hwang, Mehrab Bin Morshed, Li Zhu, Viswam Nathan, Md Mahbubur Rahman, Subramaniam Venkatraman, Sharanya Arcot Desai

Biosignals acquired from different locations on the body often provide temporally ordered views of the same underlying physiological process. However, most existing self-supervised learning methods treat these signals as interchangeable views, overlooking the directional temporal dynamics that link them. A canonical example is the relationship between electrocardiography (ECG), which captures the electrical activation initiating each heartbeat, and photoplethysmography (PPG), which records the resulting peripheral pulse delayed by vascular dynamics. To capture this structured relationship, we introduce xMAE, a biosignal pretraining framework that leverages masked cross-modal reconstruction across temporally ordered biosignals as a training-time constraint to encourage physiologically meaningful timing structure in the learned representations. We show that pretraining with xMAE yields representations that outperform both unimodal and multimodal baselines on 15 of 19 downstream tasks, including cardiovascular outcome prediction, abnormal laboratory test detection, sleep staging, and demographic inference, while generalizing across devices, body locations, and acquisition settings. Further analysis suggests that the ECG--PPG timing structure is reflected in the learned PPG representations. Code is available at https://github.com/hzhou3/xMAE.
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WMVLM: Evaluating Diffusion Model Image Watermarking via Vision-Language Models
Zijin Yang, Yu Sun, Kejiang Chen, Jiawei Zhao, Jun Jiang, Weiming Zhang, Nenghai Yu

Digital watermarking is essential for securing generated images from diffusion models. Accurate watermark evaluation is critical for algorithm development, yet existing methods have significant limitations: they lack a unified framework for both residual and semantic watermarks, provide results without interpretability, neglect comprehensive security considerations, and often use inappropriate metrics for semantic watermarks. To address these gaps, we propose **WMVLM**, the first unified and interpretable evaluation framework for diffusion model image **w**ater**m**arking via **v**ision-**l**anguage **m**odels (VLMs).
We redefine quality and security metrics for each watermark type: residual watermarks are evaluated by artifact strength and erasure resistance, while semantic watermarks are assessed through latent distribution shifts. Moreover, we introduce a three-stage training strategy to progressively enable the model to achieve classification, scoring, and interpretable text generation. Experiments show WMVLM outperforms state-of-the-art VLMs with strong generalization across datasets, diffusion models, and watermarking methods.
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SAME: Stabilized Mixture-of-Experts for Multimodal Continual Instruction Tuning
Zhen-Hao Xie, Jun-Tao Tang, Yu-Cheng Shi, Han-Jia Ye, De-Chuan Zhan, Da-Wei Zhou

Multimodal Large Language Models (MLLMs) achieve strong performance through instruction tuning, but real-world deployment requires them to continually expand their capabilities, making Multimodal Continual Instruction Tuning (MCIT) essential. Recent methods leverage sparse expert routing to promote task specialization, but we find that the expert routing process suffers from drift as the data distribution evolves. For example, a grounding query that previously activated localization experts may instead be routed to irrelevant experts after learning OCR tasks. Meanwhile, the grounding-related experts can be overwritten by new tasks and lose their original functionality. Such failure reflects two problems: router drift, where expert selection becomes inconsistent over time, and expert drift, where shared experts are overwritten across tasks. Therefore, we propose StAbilized Mixture-of-Experts (SAME) for MCIT. To address router drift, SAME stabilizes expert selection by decomposing routing dynamics into orthogonal subspaces and updating only task-relevant directions. To mitigate expert drift, we regulate expert updates via curvature-aware scaling using historical input covariance in a rehearsal-free manner. SAME also introduces adaptive expert activation to freeze selected experts during training, reducing redundant computation and cross-task interference. We also introduce a new benchmark to evaluate MCIT with long task sequence, and extensive experiments demonstrate SAME's SOTA performance. Code is available at https://github.com/LAMDA-CL/Prism.
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Cycle-of-Science: Reliable Reasoning through Counterfactual Verification for Agent Decision Making
Ruojie Zhang, Wencheng Zhu, Peiyuan Jiang, dayong zhu
Large Language Models have significantly advanced autonomous agents through their sophisticated perception and execution capabilities.
Despite effective, agents still struggle with robust decision-making due to passive learning from similar experiences that often confound correlation with causality. Inspired by the Scientific Method, we propose a Cycle-of-Science framework that autonomously explores potential causal pathways through an iterative loop of \textit{Hypothesis, Experiment, and Validation}, enabling agents to identify truly effective causal dependencies. To be specific, we first leverage causal knowledge to guide the initial hypotheses generation. These hypotheses are then analyzed through experiments using counterfactual samples. Afterward, we perform causal analysis to quantify effects of interventions, deriving well-validated hypotheses for next agent steps. To train our policy, we further introduce a two-stage pipeline that integrates supervised fine-tuning with Counterfactual Preference Optimization, which constructs preference signals from intervention outcomes to reinforce validated reasoning chains. Experiments on benchmarks demonstrate that our method achieves superior performance over state-of-the-art approaches.
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A Benchmark and Framework for Evaluating Next Action Predictions in Spreadsheets
Tejas Agrawal, Vu Le, Sumit Gulwani, Gust Verbruggen
Predictive code completion greatly accelerates how quickly developers work.
In spreadsheets, despite being much more common, such auto-completion features are virtually non-existent.
To address this gap, we introduce a benchmark for systems that observe a sequence of user actions in a spreadsheet and predict future actions.
Two challenges are (1) the absence of edit histories in public spreadsheet corpora and (2) the complex space of spreadsheet actions (spatial, temporal, composite).
To address (1), we manually curate 52 sequences of 12K actions that recreate spreadsheets from public corpora, seeded by parametrized heuristics and LLM refinement.
To address (2), we propose an online evaluation that expects a prediction after each user action, accepts or rejects that prediction, updates the future actions upon acceptance, and repeats this until the target spreadsheet is obtained.
We use multiple baseline predictors (including zero-shot LLMs, fine-tuned SLMs, and classical models) and analyze different properties that our benchmark teaches us, including but not limited to: properties of saved actions and false positives, efficiency, effect of user profiles, effect of triggers, and effect of context.
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Computationally-efficient Graph Modeling with Refined Graph Random Features
Krzysztof Marcin Choromanski, Kumar Avinava Dubey, Arijit Sehanobish, Isaac Reid
We propose *refined GRFs* (GRFs++), a new class of *Graph Random Features* (GRFs) for efficient and accurate computations involving kernels defined on the nodes of a graph. GRFs++ resolve some of the long-standing limitations of regular GRFs, including difficulty modeling relationships between more distant nodes. They reduce dependence on sampling long graph random walks via a novel *walk-stitching* technique, concatenating several shorter walks without breaking unbiasedness. By applying these techniques, GRFs++ inherit the approximation quality provided by longer walks but with greater efficiency, trading sequential inefficient sampling of a long walk for parallel computation of short walks and matrix-matrix multiplication. Furthermore, GRFs++ extend the simplistic GRFs walk termination mechanism (Bernoulli schemes with fixed halting probabilities) to a broader class of strategies, applying general distributions on the walks' lengths.
This improves approximation accuracy of graph kernels, without incurring extra computational cost. We provide empirical evaluations to showcase our claims and complement our results with theoretical analysis.
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AIR-VLA: Vision-Language-Action Systems for Aerial Manipulation
Jianli Sun, Bin Tian, Qiyao Zhang, Chengxiang Li, Song Zihan, Zhiyong Cui, Yisheng Lv, Yonglin Tian
While Vision-Language-Action (VLA) models have achieved remarkable success in ground-based embodied intelligence, their application to Aerial Manipulation Systems (AMS) remains a largely unexplored frontier. The inherent characteristics of AMS, including floating-base dynamics, strong coupling between the UAV and the manipulator, and the multi-step, long-horizon nature of operational tasks, pose severe challenges to existing VLA paradigms designed for static or 2D mobile bases. To bridge this gap, we propose AIR-VLA, the first VLA benchmark specifically tailored for aerial manipulation. We construct a physics-based simulation environment and release a high-quality multimodal dataset comprising 3000 manually teleoperated demonstrations, covering base manipulation, object & spatial understanding, semantic reasoning, and long-horizon planning. Leveraging this platform, we systematically evaluate mainstream VLA models and state-of-the-art VLM models. Our experiments not only validate the feasibility of transferring VLA paradigms to aerial systems but also, through multi-dimensional metrics tailored to aerial tasks, reveal the capabilities and boundaries of current models regarding UAV mobility, manipulator control, and high-level planning. AIR-VLA establishes a standardized testbed and data foundation for future research in general-purpose aerial robotics. The resource of AIR-VLA will be available at https://anonymous.4open.science/r/AIR-VLA-dataset-B5CC/.
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ParalESN: Enabling parallel information processing in Reservoir Computing
Matteo Pinna, Giacomo Lagomarsini, Andrea Ceni, Claudio Gallicchio
Reservoir Computing (RC) has established itself as an efficient paradigm for temporal processing. However, its scalability remains severely constrained by the need to process temporal data sequentially and the prohibitive memory footprint of high-dimensional reservoirs. To address these limitations, we revisit RC through the lens of structured operators and state space modeling, introducing Parallel Echo State Network (ParalESN). Leveraging diagonal linear recurrence in the complex domain, ParalESN enables parallel processing of temporal data and the construction of efficient, high-dimensional reservoirs. A thorough theoretical analysis demonstrates that the Echo State Property and the universality guarantees of traditional Echo State Networks are preserved, while also admitting an equivalent representation of arbitrary linear reservoirs in the complex diagonal form. Empirically, ParalESN achieves competitive predictive accuracy with traditional RC and with fully trainable sequence models, while delivering computational savings by orders of magnitude. Overall, ParalESN offers a scalable and principled pathway for integrating RC within the deep learning landscape.
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Scaling Generative Verifiers For Natural Language Mathematical Proof Verification And Selection
Sadegh Mahdavi, Branislav Kisacanin, Shubham Toshniwal, Wei Du, Ivan Moshkov, George Armstrong, Renjie Liao, Christos Thrampoulidis, Igor Gitman
Large language models have achieved remarkable success on final-answer mathematical problems, largely due to the ease of applying reinforcement learning with verifiable rewards. However, the reasoning underlying these solutions is often flawed. Advancing to rigorous proof-based mathematics requires reliable proof verification capabilities. We begin by analyzing multiple evaluation setups and show that focusing on a single benchmark can lead to brittle or misleading conclusions. To address this, we evaluate both proof-based and final-answer reasoning to obtain a more reliable measure of model performance. We then scale two major generative verification methods (GenSelect and LLM-as-a-Judge) to millions of tokens and identify their combination as the most effective framework for solution verification and selection. We further show that the choice of prompt for LLM-as-a-Judge significantly affects the model's performance, but reinforcement learning can reduce this sensitivity. However, despite improving proof-level metrics, reinforcement learning does not enhance final-answer precision, indicating that current models often reward stylistic or procedural correctness rather than mathema

tical validity. Our results establish practical guidelines for designing and eva luating scalable proof-verification and selection systems.
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LLawCo: Learning Laws of Cooperation for Modeling Embodied Multi-Agent Behavior
Qinhong Zhou, Chuang Gan, Anoop Cherian
Embodied agents operating in decentralized and partially observable environments have attracted growing attention in recent years. However, existing large langu age model (LLM)-based agents often exhibit behaviors that are misaligned with th eir partners or inconsistent with the environment state, leading to inefficient cooperation and poor task success. To address this challenge, we propose a novel framework, Learning Laws of Cooperation (LLawCo), that enables embodied agents to autonomously align with both their partners and task objectives. Our framewor k allows agents to reflect on past failures to extract misaligned behavioral pat terns, which are used to derive high-level behavioral laws, such as “Talk when n ecessary” and “Wait for partner.” These laws are explicitly incorporated into th e agents’ chains of thought via supervised fine-tuning, aligning their reasoning with task requirements and the behavior of other agents. To evaluate our approa ch, we introduce PARTNR-Dialog, a large-scale multi-agent communicative and coop erative planning benchmark built on the PARTNR environment. Experiments on exist ing tasks and our new benchmark demonstrate significant improvements in cooperat ive efficiency and task success rates. Across four backbone LLMs, our method ach ieves average success rate improvements of 4.5% on the PARTNR-Dialog benchmark a nd 6.8% on the TDW-MAT benchmark over state-of-the-art open-source communicative agent frameworks.
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Revisiting the Role of Pretrained Weights in Model Merging: On Near-Optimality w ithin the Core Subspace
Wenju Sun, Qingyong Li, Tiancheng Li, Yangliao Geng, Boyang Li
Model merging offers an efficient solution for integrating task-specific knowled ge from multiple fine-tuned models. Most existing approaches focus on manipulati ng the difference vectors between fine-tuned and pretrained weights, often overl ooking the generalization capabilities inherent in the pretrained parameters. In this work, we revisit the role of pretrained weights in model merging and inves tigate their efficacy from a subspace perspective. We find that the components o f pretrained weights residing in the core subspace—defined by the dominant singu lar vectors—are essential for maintaining generalization across diverse tasks. S pecifically, we present empirical evidence that pretrained weights are nearly fi rst-order stationary and exhibit predominantly non-negative curvature within thi s core subspace with respect to multi-task loss landscapes, indicating near-opti mality. These findings suggest that task-specific adaptations should be injected primarily into the orthogonal complement of the core subspace, thereby preservi ng the generalization properties of the pretrained model. Extensive experiments on vision and vision-language tasks show that this subspace-aware strategy consi stently yields improvements over state-of-the-art training-free merging methods, including Task Arithmetic, LOT Merging, ISO, and TSV. The source code is availa ble at https://github.com/SunWenJu123/model-merging.
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Emergence of Exploration in Policy Gradient Reinforcement Learning via Retrying
Soichiro Nishimori, Paavo Parmas, Sotetsu Koyamada, Tadashi Kozuno, Toshinori Ki tamura, Shin Ishii, Yutaka Matsuo
In reinforcement learning (RL), agents benefit from exploration *only* because t hey repeatedly encounter similar states: trying different actions can improve pe rformance or reduce uncertainty; without such retries, a greedy policy is optima l. We formalize this intuition with **ReMax**, an objective that evaluates a pol icy by the expected maximum return over M samples ($M \in \mathbb{N}$), while accounting for return uncertainty. Optimizing this objective induces stochastic exploration as an emergent property, without explicit bonus terms. For efficient policy optimization, we derive a new policy-gradient formulation for ReMax and introduce **Re**Max **PPO** (**RePPO**), a PPO variant that optimizes ReMax whil e generalizing the discrete retry count M to a continuous parameter $m > 0$, e

nabling fine-grained control of exploration. Empirically, RePPO promotes explora tion—without any explicit exploration bonuses—on the MinAtar and Craftax benchma rks. The official code is available at https://github.com/nissymori/remax-rl.
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Provably Convergent Actor-Critic for MARL through Risk-aversion
Yizhou Zhang, Eric Mazumdar
Learning stationary policies in infinite-horizon general-sum Markov games (MGs) remains a fundamental open problem in Multi-Agent Reinforcement Learning (MARL). While stationary strategies are preferred for their practicality, computing sta tionary forms of classic game-theoretic equilibria is computationally intractabl e—a stark contrast to the comparative ease of solving single-agent RL or zero-su m games. To bridge this gap, we study Risk-averse Quantal response Equilibria (R QE), a solution concept rooted in behavioral game theory that incorporates risk aversion and bounded rationality. We demonstrate that RQE possesses strong regul arity conditions that make it uniquely amenable to learning in MGs. We propose a novel single-timescale Actor-Critic algorithm characterized by a faster actor a nd a slower critic. Leveraging the regularity of RQE, we prove that this approac h achieves global convergence with finite-sample guarantees. We empirically vali date our algorithm in several environments to demonstrate superior convergence p roperties compared to risk-neutral baselines.
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Bridging Local–Global Dissonance: Learning from Compressive Measurements for Hyp erspectral Reconstruction
Xian-Hua Han
Reconstructing hyperspectral images from compressive measurements is challenging due to a fundamental mismatch between locally reliable observations and globall y entangled structures induced by spectral dispersion. This study formalizes thi s issue as a local–global dissonance in representation learning for CASSI system s. To resolve it, we propose a Hierarchical Scale-Reconciling Architecture (HSRA) that enforces local sufficiency and global consistency in a principled, scale-aware manner. HSRA combines multi-kernel token mixing, latent window interaction s, and hierarchical multi-granularity spatially shifted attention to progressive ly reconcile physical constraints across scales. Embedded into a deep unfolding framework as a physically grounded learned prior, Extensive experiments on bench marks demonstrate that HSRA achieves consistent and significant improvements ove r state-of-the-art methods.
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Flatland: The Adventures of Gradient Descent with Large Step Sizes
Leonardo Galli, Curtis Fox, Wiebke Bartolomaeus, Mark Schmidt, Holger Rauhut
The training of neural networks often entails objective functions that are not g lobally L-smooth. For these functions, it is both theoretically and practicall y difficult to reply to the question: what is the largest possible step size tha t ensures the convergence of gradient descent (GD)? We address this longstanding open question in deep learning by providing a unifying definition of "large'' s tep sizes that requires only local Lipschitz (or even Hölder) continuity of the gradient. We design first-order adaptive methods that provably yield large step sizes and show that they operate at the edge of stability (EoS) right from the s tart of the training. In particular, the loss decreases nonmonotonically and the product between the step size and sharpness, i.e., the largest eigenvalue of th e hessian, stays above the EoS threshold of 2 throughout training. Using our met hod, we are also able to minimize the sharpness all the way down to its global m inimum. Contrary to expectation, we find that encountering globally-flat regions too early in the training may both slow down convergence and jeopardize the gen eralization ability of the network. Exploiting a self-stabilization argument, we allow GD to enter slightly sharper valleys and turn unsuccessful training runs into very successful ones.
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Budgeted Active Experimentation for Treatment Effect Estimation from Observation al and Randomized Data
Jiacan Gao, Xinyan Su, Mingyuan Ma, Yiyan HUANG, Xiao Xu, Xinrui Wan, Tianqi Gu,

Enyun Yu, Jiecheng Guo, Zhiheng Zhang
Estimating heterogeneous treatment effects is central to data-driven decision-making, yet industrial applications often face a fundamental tension between limited randomized controlled trial (RCT) budgets and abundant but biased observational data (OBS) collected under historical targeting policies. Although observational logs offer the advantage of scale, they may suffer from severe policy-induced imbalance and overlap violations, rendering standalone estimation unreliable. We propose a \textit{budgeted active experimentation} framework that iteratively collects informative randomized samples for causal effect estimation via active sampling. By leveraging observational signals, we develop an acquisition function targeting uplift estimation uncertainty, domain discrepancy, and overlap deficits to select the most informative units for randomized experiments. We establish finite-sample deviation bounds, asymptotic normality via martingale CLTs, and minimax lower bounds showing near-optimality in the linear representation setting. Experiments on synthetic datasets support our theoretical findings, and further extensions to industrial neural network-based uplift modeling scenarios show that active sampling can improve sample efficiency over random sampling under limited RCT budgets.
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Hunt Instead of Wait: Evaluating Deep Data Research on Large Language Models
Wei Liu, Peijie Yu, Michele Orini, Yali Du, Yulan He
The agency expected of Agentic Large Language Models goes beyond answering correctly, requiring autonomy to set goals and decide what to explore. We term this *investigatory intelligence*, distinguishing it from *executional intelligence*, which merely completes assigned tasks. Data Science provides a natural testbed, as real-world analysis starts from raw data rather than explicit queries, yet few benchmarks focus on it. To address this, we introduce **Deep Data Research (DDR)**, an open-ended task where LLMs autonomously extract key insights from databases, and **DDR-Bench**, a large-scale, checklist-based benchmark that enables verifiable evaluation. Results show that while frontier models display emerging agency, long-horizon exploration remains challenging. Our analysis highlights that effective investigatory intelligence depends not only on agent scaffolding or merely scaling, but also on intrinsic strategies of agentic models.
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Reasoning Is Not Free: Robust Adaptive Cost-Efficient Routing for LLM-as-a-Judge
Wenbo Zhang, Lijinghua Zhang, Liner Xiang, Hengrui Cai
Reasoning-capable large language models (LLMs) have recently been adopted as automated judges, but their benefits and costs in LLM-as-a-Judge settings remain unclear. Through controlled comparisons between reasoning and non-reasoning judges, we show that explicit reasoning substantially improves judgment accuracy on tasks requiring structured verification (e.g., math and coding), while offering limited or even negative gains on simpler evaluations and incurring significantly *higher computational cost*. These findings motivate that reasoning should be used selectively rather than universally, with awareness of possible *distribution shift*. We propose a Robust Adaptive Cost-Efficient Routing (RACER), which dynamically selects between reasoning and non-reasoning judges under a fixed budget by formulating routing as a constrained distributionally robust optimization problem. RACER explicitly accounts for distribution shift via a KL-divergence uncertainty set, admits an efficient primal--dual algorithm, and enjoys theoretical guarantees including uniqueness of the optimal policy and linear convergence. Extensive experiments show that RACER achieves superior accuracy--cost trade-offs under distribution shift.
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Synthesizable Molecular Generation via Soft-constrained GFlowNets with Rich Chemical Priors
Hyeonah Kim, Minsu Kim, Celine Roget, Dionessa Biton, Louis Vaillancourt, Yves V. Brun, Yoshua Bengio, Alex Hernández-García
The application of generative models for experimental drug discovery campaigns is severely limited by the difficulty of designing molecules de novo that can be synthesized in practice.

Previous works have leveraged Generative Flow Networks (GFlowNets) to impose hard synthesizability constraints through the design of state and action spaces based on predefined reaction templates and building blocks.
Despite the promising prospects of this approach, it currently lacks flexibility and scalability.
As an alternative, we propose S3-GFN, which generates synthesizable SMILES molecules via simple soft regularization of a sequence-based GFlowNet.
Our approach leverages rich molecular priors learned from large-scale SMILES corpora to steer molecular generation towards high-reward, synthesizable chemical spaces.
The model induces constraints through off-policy replay training with a contrastive learning signal based on separate buffers of synthesizable and unsynthesizable samples.
Our experiments show that S3-GFN learns to generate synthesizable molecules (≥ 95%) with higher rewards in diverse tasks.
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Simple Denoising Diffusion Language Models
Huaisheng Zhu, Zhengyu Chen, Shijie Zhou, Zhihui Xie, Yige Yuan, Shiqi Chen, Zhimeng Guo, Siyuan Xu, Hangfan Zhang, Vasant G. Honavar, Teng Xiao
Recent Uniform-state Diffusion Models (USDMs), initialized from a uniform prior, offer the promise of fast text generation due to their inherent self-correction ability compared to masked diffusion models. However, they still rely on complex loss formulations with additional computational overhead, which hinders scalability. In this work, we explore a simplified denoising-based loss for USDMs that optimizes only noise-replaced tokens, stabilizing training while matching the performance of prior methods with more complex objectives. In addition, we introduce an efficient regularization term to mitigate corruption toward uniform output distributions, which further improves performance. We demonstrate the effectiveness and efficiency of our simple and improved loss formulations by pretraining models on widely used text datasets for USDMs. More importantly, our conclusions scale to larger models, showing strong potential for large-scale training.
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Transforming Weather Data from Pixel to Latent Space
Sijie Zhao, Feng Liu, Xueliang Zhang, Hao Chen, Tao Han, Junchao Gong, Ran Tao, Pengfeng Xiao, Xinyu Gu, LEI BAI
The increasing impact of climate change and extreme weather events has spurred growing interest in deep learning for weather research. However, existing studies often rely on weather data in pixel space, which presents several challenges such as smooth outputs in model outputs, limited applicability to a single pressure-variable subset (PVS), and high data storage and computational costs. To address these challenges, we propose a novel Weather Latent Autoencoder (WLA) that transforms weather data from pixel space to latent space, enabling efficient data representation. By decoupling weather reconstruction from downstream tasks, WLA improves the accuracy and sharpness of weather task model results. The incorporated Pressure-Variable Unified Module transforms multiple PVS into a unified representation, enhancing the adaptability of the model in multiple weather scenarios. Furthermore, weather tasks can be performed in a low-storage latent space of WLA rather than a high-storage pixel space, thus significantly reducing data storage and computational costs. Through extensive experimentation, we demonstrate its superior compression and reconstruction performance, enabling the creation of the ERA5-Latent dataset with unified representations of multiple PVS from ERA5 data. The compressed full PVS in the ERA5-Latent dataset reduces the original 244.34 TB of data to 0.43 TB. The downstream task further demonstrates that task models can apply to multiple PVS with low data costs in latent space and achieve superior performance compared to models in pixel space.
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Compile to Compress: Boosting Formal Theorem Provers by Compiler Outputs
Guchan Li, Rui Tian, Hongning Wang
Large language models (LLMs) have demonstrated significant potential in formal theorem proving, yet state-of-the-art performance often necessitates prohibitive

test-time compute via massive roll-outs or extended context windows. In this work, we address this scalability bottleneck by exploiting an informative structure in formal verification: the observation that compilers map a vast space of diverse proof attempts to a compact set of structured failure modes. We introduce a learning-to-refine framework that leverages this compression to perform efficient learning and proof exploration. We perform tree search that corrects errors locally conditioned on explicit verifier feedback, thereby circumventing the costs associated with accumulating a long history of proof attempts. Extensive evaluations show that our method consistently amplifies the reasoning capabilities of base provers across varying scales. Notably, our approach achieves state-of-the-art performance on PutnamBench among publicly reported $\sim$8B and $\sim$32B parameter models under comparable test-time budgets, offering a scalable paradigm for next-generation verifier-guided reasoning.
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Light Forcing: Accelerating Autoregressive Video Diffusion via Sparse Attention
Chengtao Lv, Yumeng Shi, Yushi Huang, Ruihao Gong, Shen Ren, Wenya Wang
Advanced autoregressive (AR) video generation models have improved visual fidelity and interactivity, but the quadratic complexity of attention remains a primary bottleneck for efficient deployment. While existing sparse attention solutions have shown promise on bidirectional models, we identify that applying these solutions to AR models leads to considerable performance degradation for two reasons: isolated consideration of chunk generation and insufficient utilization of past informative context. Motivated by these observations, we propose \textsc{Light Forcing}, the \textit{first} sparse attention solution tailored for AR video generation models. It incorporates a \textit{Chunk-Aware Growth} mechanism to quantitatively estimate the contribution of each chunk, which determines their sparsity allocation. This progressive sparsity increase strategy enables the current chunk to inherit prior knowledge in earlier chunks during generation. Additionally, we introduce a \textit{Hierarchical Sparse Attention} to capture informative historical and local context in a coarse-to-fine manner. Such two-level mask selection strategy (i.e., frame and block level) can adaptively handle diverse attention patterns. Extensive experiments demonstrate that our method outperforms existing sparse attention in quality (e.g., 84.5 on VBench) and efficiency (e.g., $1.2{\sim}1.3\times$ end-to-end speedup). Combined with other efficient solutions, \textsc{Light Forcing} further achieves a $2.0{\sim}3.0\times$ end-to-end speedup across diverse GPUs (e.g., 27.4\,FPS on RTX 5090 and 33.9\,FPS on H100). Code is released via this \href{https://github.com/chengtao-lv/LightForcing}{link}.
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Identifiable Nonlinear Differentiable Causal Discovery via Independence and Adaptive Group Sparsity
Ruicong Yao, Tim Verdonck, Mihaela van der Schaar, Jakob Raymaekers
Differentiable approaches to causal discovery have shown promise in learning DAG structures via continuous optimization, but their theoretical guarantees are largely restricted to models with homoscedastic noise or known noise distribution. In particular, existing methods based on mean squared error fail to identify the true DAG when noise distributions are non-Gaussian and vary in scale. In this paper, we address this gap in nonlinear additive noise models (ANMs) with arbitrary noise. Our approach extends NOTIME (Berrevoets et al. 2025) which minimizes an independence criterion among the residuals. We show that the global minimizer of the independence criterion corresponds to the true underlying DAG up to additional constant edges in general ANMs. To recover the exact structure, we introduce an adaptive group lasso penalty that regularizes entire columns of the first-layer weight matrix of an MLP, enabling the selective pruning of constant edges in a functionally meaningful way. Empirically, our method achieves effective and stable performance across diverse noise types and variances, outperforming prior methods that lack identifiability guarantees in this setting.
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scDataset: Scalable Data Loading for Deep Learning on Large-Scale Single-Cell Omics

Davide D'Ascenzo, Sebastiano Cultrera di Montesano
Training deep learning models on single-cell datasets with hundreds of millions of cells requires loading data from disk, as these datasets exceed available memory. While random sampling provides the data diversity needed for effective training, it is prohibitively slow due to the random access pattern overhead, whereas sequential streaming achieves high throughput but introduces biases that degrade model performance. We present scDataset, a PyTorch data loader that enables efficient training from on-disk data with seamless integration across diverse storage formats. Our approach combines block sampling and batched fetching to achieve quasi-random sampling that balances I/O efficiency with minibatch diversity. On Tahoe-100M, a dataset of 100 million cells, scDataset achieves more than two orders of magnitude speedup compared to true random sampling while working directly with AnnData files. We provide theoretical bounds on minibatch diversity and empirically show that scDataset matches the performance of true random sampling across multiple classification tasks and model architectures.
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Efficient Bayesian Inference from Noisy Pairwise Comparisons
Till Aczel, Lucas Theis, Roger Wattenhofer
Evaluating generative models is challenging because standard metrics often fail to reflect human preferences. Human evaluations are more reliable but costly and noisy, as participants vary in expertise, attention, and diligence. Pairwise comparisons improve consistency, yet aggregating them into overall quality scores requires careful modeling. Bradley-Terry-based methods update item scores from comparisons, but existing approaches either ignore rater variability or lack convergence guarantees, limiting robustness and interpretability. We introduce BBQ, a Bayesian Bradley-Terry variant that explicitly models rater quality, downweighting or removing unreliable participants, and provides guaranteed monotonic likelihood convergence through an Expectation-Maximization algorithm. Empirical results show that BBQ provides efficient inference, well-calibrated uncertainty estimates, and more robust, interpretable rankings compared to baseline Bradley-Terry models, even with noisy or crowdsourced raters. This framework enables more reliable and cost-effective human evaluation of generative models.
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Offline Preference Optimization for Rectified Flow with Noise-Tracked Pairs
Yunhong Lu, Qichao Wang, Hengyuan Cao, Xiaoyin Xu, Min Zhang
Existing preference datasets for text-to-image models typically store only the final winner/loser images. This representation is insufficient for rectified flow (RF) models, whose generation is naturally indexed by a specific prior noise sample and follows a nearly straight denoising trajectory. In contrast, prior DPO-style alignment for diffusion models commonly estimates trajectories using an independent forward noising process, which can be mismatched to the true reverse dynamics and introduces unnecessary variance. We propose Prior Noise-Aware Preference Optimization (PNAPO), an off-policy alignment framework specialized for rectified flow. PNAPO augments preference data by retaining the paired prior noises used to generate each winner/loser image, turning the standard (prompt, winner, loser) triplet into a sextuple. Leveraging the straight-line property of RF, we estimate intermediate states via noise-image interpolation, which constrains the trajectory estimation space and yields a tighter surrogate objective for preference optimization. In addition, we introduce a dynamic regularization strategy that adapts the DPO regularization based on (i) the reward gap between winner and loser and (ii) training progress, improving stability and sample efficiency. Experiments on state-of-the-art RF T2I backbones show that PNAPO consistently improves preference metrics while substantially reducing training compute.
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EDCO: Dynamic Curriculum Orchestration for Domain-specific Large Language Model Fine-tuning
Jing-Cheng Pang, Sun Liu, Chang Zhou, Xian Tang, Haichuan Ma, KUN JIANG, Jianlong Wang, Kai Zhang, Sijie Wu, Haoran Cai, Chenwei Wu, Xubin Li, Xin Chen
Domain-specific large language models (LLMs), typically developed by fine-tuning a pre-trained general-purpose LLM on specialized datasets, represent a signific

ant advancement in applied AI. A common strategy in LLM fine-tuning is curriculu
m learning, which pre-orders training samples based on metrics like difficulty t
o improve learning efficiency compared to a random sampling strategy. However, m
ost existing methods for LLM fine-tuning rely on a static curriculum, designed p
rior to training, which lacks adaptability to the model's evolving needs during
fine-tuning. To address this, we propose EDCO, a novel framework based on two ke
y concepts: \textit{inference entropy} and \textit{dynamic curriculum orchestrat
ion}. Inspired by recent findings that maintaining high answer entropy benefits
long-term reasoning gains, EDCO prioritizes samples with high inference entropy
in a continuously adapted curriculum. EDCO integrates three core components: an
efficient entropy estimator that uses prefix tokens to approximate full-sequence
 entropy, an entropy-based curriculum generator that selects data points with th
e highest inference entropy, and an LLM trainer that optimizes the model on the
selected curriculum. Comprehensive experiments in communication, medicine and la
w domains, EDCO outperforms traditional curriculum strategies for fine-tuning Qw
en3-4B and Llama3.2-3B models under supervised and reinforcement learning settin
gs. Furthermore, the proposed efficient entropy estimation reduces computational
 time by 83.5\% while maintaining high accuracy. The code is available at https:
//github.com/GTS-AIData/EDCO.
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KBQA-R1: Reinforcing Large Language Models for Knowledge Base Question Answering
Xin Sun, Zhongqi Chen, Xing Zheng, Bowen Song, Qiang Liu, Shu Wu, Zilei Wang, We
iqiang Wang, Liang Wang

Knowledge Base Question Answering (KBQA) challenges models to bridge the gap bet
ween natural language and strict knowledge graph schemas by generating executabl
e logical forms. While Large Language Models (LLMs) have advanced this field, cu
rrent approaches often struggle with a dichotomy of failure: they either generat
e hallucinated queries without verifying schema existence or exhibit rigid, temp
late-based reasoning that mimics synthesized traces without true comprehension o
f the environment. To address these limitations, we present **KBQA-R1**, a frame
work that shifts the paradigm from text imitation to interaction optimization vi
a Reinforcement Learning. Treating KBQA as a multi-turn decision process, our mo
del learns to autonomously navigate the knowledge base using a structured action
 space, refining its reasoning strategies based on concrete execution feedback r
ather than static supervision. Furthermore, we introduce Referenced Rejection Sa
mpling (RRS), a data synthesis method that resolves cold-start challenges by str
ictly aligning reasoning traces with ground-truth action sequences. Extensive ex
periments on WebQSP, GrailQA, and GraphQuestions demonstrate that KBQA-R1 achiev
es state-of-the-art performance.
 Code is available at https://anonymous.4open.science/r/KBQA-R1-814F.
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FedFit: Federated Dynamic Sparse Training via Fisher Information scoring
Meng Bi, Hong Huang, Jinlong Song, Charles Wang, Chengming Hu, Xi Chen, Ting Yu,
 Xue Liu

Cross-device Federated Learning (FL) is frequently bottlenecked by the prohibiti
ve memory and communication costs of training deep neural networks on resource-c
onstrained edge hardware. While federated dynamic sparse training aims to allevi
ate these costs by adjusting sparse structures during training, existing methods
 rely on magnitude-based heuristics that are fundamentally ill-suited for the no
n-convergent, heterogeneous environments inherent to FL.

To address this challenge, we propose FedFit, a federated dynamic sparse trainin
g framework that replaces simple heuristics with optimization-centric criteria f
or structure adjustment.

By leveraging a second-order approximation of the loss landscape via the Fisher
Information Matrix, FedFit enables precise and efficient structure adjustment wi
thout the overhead of explicit Hessian computation. Empirical evaluations across
 computer vision and natural language processing benchmarks demonstrate that Fed
Fit significantly narrows the sparse-to-dense accuracy gap, outperforming state
-of-the-art methods while maintaining high communication efficiency. Our code is
 available at https://github.com/Serena-28/Fedfit.git.

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Possibilistic Predictive Uncertainty for Deep Learning
Yao Ni, Jeremie Houssineau, Yew-Soon Ong, Piotr Koniusz
Deep neural networks achieve impressive results across diverse applications, yet their overconfidence on unseen inputs necessitates reliable epistemic uncertainty modeling. Existing methods for uncertainty modeling face a fundamental dilemma: Bayesian approaches provide principled estimates but remain computationally prohibitive, while efficient second-order predictors lack rigorous connections between their specific objectives and epistemic uncertainty quantification. To resolve this dilemma, we introduce **D**irichlet-**a**pproximated **p**ossibilistic **p**osterior p**r**edictions (DAPPr), a principled framework grounded in possibility theory. We define a possibilistic posterior over parameters, project it to the prediction space via supremum operators, and approximate the projected posterior using learnable Dirichlet possibility functions. This projection-and-approximation strategy yields a simple training objective with closed-form solutions. Despite its simplicity, extensive experiments across diverse benchmarks show that DAPPr achieves competitive or superior uncertainty quantification performance over state-of-the-art second-order predictors while maintaining both principled derivation and computational efficiency.
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CauseCollab: Causal Unified and Modality-Agnostic Network for Heterogeneous Collaborative Perception
Weize Li, Yang Li, Quan Yuan, Xiaoyuan Fu, Guiyang Luo, Jinglin Li
Collaborative perception enhances environment understanding through multi-agent information sharing, but its performance in real-world scenarios is constrained by heterogeneous sensor modalities and model architectures. Recent protocol-based two-stage methods alleviate this problem by mapping heterogeneous features into a shared protocol space; however, independently trained modality-specific converters often generate modality-specific pseudo-protocol distributions, leading to semantic inconsistency and error accumulation, which is particularly pronounced in scenarios with large modality discrepancies. To address this issue, we propose CauseCollab, a causal unified and modality-agnostic network. CauseCollab formulates representation learning in the protocol space from a causal perspective, explicitly disentangling semantic factors from modality-specific statistical confounders via causal metric learning. Meanwhile, CauseCollab adopts context-guided Unified Converter for heterogeneous modalities to ensure cross-modal semantic consistency. In addition, integrating new modalities only requires training adapters with minimal parameters. Extensive experiments on the OPV2V and DAIR-V2X datasets demonstrate that CauseCollab achieves state-of-the-art performance, with more significant gains in scenarios involving large modality gaps.
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Words Towards Explainability: Caption Label-Free Learning via Dual Loop Agentic Time Series Captioning
Difei Hou, Jiaqi Yue, Chunhui Zhao
Explainability is essential for applying time series analysis in high-stakes domains. While Time Series Captioning (TSC) offers a pathway to enhance temporal explainability, achieving reliable caption generation usually necessitates high-quality textual annotations. However, as interpreting abstract temporal dynamics requires specialized domain knowledge, acquiring such caption annotations is challenging, thereby impeding the advancement of TSC. To address this challenge, we introduce a novel Caption Label-Free Learning (CLFL) paradigm. Departing from the supervised learning tradition of imitating human annotations, CLFL formulates captioning as an agentic exploration task optimized by feedback from a proxy reward. Specifically, we propose a Dual Loop Agentic Captioning (DLAC) framework to achieve such an exploration-feedback mechanism. In the inner loop, a Time Series Captioning Agent (TSCAgent) reflectively explores potential semantic captions. In turn, the outer loop evaluates these captions via downstream reasoning to derive proxy reward, which feeds back to optimize the TSCAgent. Empirical results validate the effectiveness of the CLFL, proving that the exploration-feedback mechanism is sufficient for learning complex temporal semantics and autonomously g

enerating captions, without any caption label supervision. Furthermore, we relea
se TFTSC, an industrial expert-level time series caption dataset, which is avail
able at: https://github.com/chunhuiz/TFTSC/tree/master.
****

The Oversight Game: Learning to Cooperatively Balance an AI Agent's Safety and A
utonomy
William Overman, Mohsen Bayati
As increasingly capable agents are deployed, a central safety challenge is how t
o retain meaningful human control without modifying the underlying system. We st
udy a minimal control interface in which an agent chooses whether to act autonom
ously (play) or defer (ask), while a human simultaneously chooses whether to be
permissive (trust) or engage in oversight (oversee), and model this interaction
as a two-player Markov Game. When this game forms a Markov Potential Game, we pr
ove an alignment guarantee: any increase in the agent's utility from acting more
 autonomously cannot decrease the human's value. This establishes a form of intr
insic alignment where the agent's incentive to seek autonomy is structurally cou
pled to the human's welfare. Practically, the framework induces a transparent co
ntrol layer encouraging the agent to defer when risky and act when safe. While w
e use gridworld simulations to illustrate the emergence of this collaboration, o
ur primary validation involves an agentic tool-use task where two 30B-parameter
language models are fine-tuned via independent policy gradient. We demonstrate t
hat even as the agents learn to coordinate on the fly, this framework effectivel
y reduces safety violations in realistic, open-ended environments.
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Calibrated Test-Time Guidance for Bayesian Inference
Daniel Geyfman, Felix Draxler, Jan Niklas Groeneveld, Hyunsoo Lee, Theofanis Kar
aletsos, Stephan Mandt
Test-time guidance is a widely used mechanism for steering pretrained diffusion
models toward outcomes specified by a reward function. Existing approaches, howe
ver, focus on maximizing reward rather than sampling from the true Bayesian post
erior, leading to miscalibrated inference. In this work, we show that common tes
t-time guidance methods do not recover the correct posterior distribution and id
entify the structural approximations responsible for this failure. We then propo
se consistent alternative estimators that enable calibrated sampling from the Ba
yesian posterior. We significantly outperform previous methods on a set of Bayes
ian inference tasks, and set a new state-of-the-art PSNR in black hole image rec
onstruction. We publish our code at https://github.com/mandt-lab/Calibrated-Guid
ance.
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XPERT: Expert Knowledge Transfer for Effective Training of Language Models
Chang Liu, boyu shi, Xu Yang, Xin Geng
Mixture-of-Experts (MoE) language models organize knowledge into explicitly rout
ed expert modules, making expert-level representations traceable and analyzable.
 By analyzing expert activation patterns in MoE language large models (LLMs), we
 find that a subset of experts is consistently activated across diverse knowledg
e domains. These common experts encode cross-domain, generalizable knowledge tha
t is closely related to model generalization, naturally raising the question of
how such identifiable expert knowledge can be practically reused. Motivated by t
his observation, we propose XPERT, a training-free framework that extracts, cons
olidates, and reuses expert knowledge from pre-trained MoE LLMs to support effec
tive training of language models across different model scales. XPERT identifies
 cross-domain experts via inference-only analysis, refines their representations
 through tensor decomposition, and adapts the extracted knowledge to be reused i
n downstream models. Experiments on language understanding and dialogue generati
on benchmarks show that models benefiting from reused expert knowledge achieve c
onsistently stronger performance and faster convergence compared to strong basel
ines. These results highlight MoE LLMs as structured and reusable knowledge sour
ces, and demonstrate the value of expert-level knowledge reuse for improving mod
el training.
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SparseSSM: Efficient Selective Structured State Space Models Can Be Pruned in On
e-Shot
Kaiwen Tuo, Huan Wang
State-space language models such as Mamba match Transformer quality while permit
ting linear complexity inference, yet still comprise billions of parameters that
 hinder deployment. While existing one-shot pruning methods are effective for ge
neric linear and attention blocks, they are not designed with the overall Mamba
architecture in mind and fail to account for the time-shared and discretized sta
te-transition matrix at the heart of the selective state-space module (SSM). In
this paper, we introduce SparseSSM, the first training-free pruning framework th
at extends the classic optimal brain surgeon (OBS) framework to state space arch
itectures. Our layer-wise algorithm (i) derives an approximate second-order sali
ency score that aggregates Hessian-trace information across time steps, (ii) inc
orporates a component sensitivity analysis to guide feed-forward network (FFN) p
runing, which also sheds light on where redundancy resides in mamba architecture
, (iii) can be easily extended to semi-structured and structured sparsity, and g
eneralized to other SSM-based architectures. Empirically, we prune 50% of SSM we
ights without fine-tuning and observe only limited performance degradation, achi
eving the current state-of-the-art one-shot pruning algorithm for Mamba-based LL
Ms.
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PATCHCODE: Discrete Latent Predictive Learning for EEG Foundation Model
KIEREN YU, Ziyang LIU, Chang Huang, Kaishun Wu
EEG foundation models aim to learn transferable representations, yet EEG recordi
ngs are dominated by high-frequency noise and large cross-subject variability. E
xisting pretraining strategies such as masked autoencoding or autoregressive mod
eling often treat waveform reconstruction as the learning signal, making the obj
ective sensitive to stochastic fluctuations rather than consistent neurophysiolo
gical structure. To address this overlap, we propose PATCHCODE, a region-aware d
iscrete predictive learning framework that keeps the encoder input continuous wh
ile introducing region-aware discrete codes as stable supervision targets. We pr
etrain a masked predictive encoder on continuous EEG patches with dual-granulari
ty learning: it predicts missing patch-level representations to preserve fine sp
atiotemporal structure, while aligning them to discretized code targets from a f
rozen tokenizer to anchor robust semantics. Extensive experiments across sixteen
 downstream datasets spanning emotion recognition, motor imagery, sleep staging,
 seizure detection, vigilance estimation, stress detection, and clinical diagnos
is demonstrate that PATCHCODE achieves competitive performance compared to state
-of-the-art baselines, with notable gains in data efficiency under limited label
s. Our code is available at https://github.com/kierenyyu/Patchcode.
****
Utility-Diversity Aware Online Batch Selection for LLM Supervised Fine-tuning
Heming Zou, Yixiu Mao, Yun Qu, Qi Wang, Xiangyang Ji
Supervised fine-tuning (SFT) is a commonly used technique to adapt large languag
e models (LLMs) to downstream tasks. In practice, SFT on a full dataset is compu
tationally expensive and sometimes suffers from overfitting or bias amplificatio
n. This facilitates the rise of data curation in SFT, which prioritizes the most
 valuable data to optimze. This work studies the online batch selection family t
hat dynamically scores and filters samples during the training process. However,
 existing popular methods often (i) rely merely on the utility of data to select
 a subset while neglecting other crucial factors like diversity, (ii) rely on ex
ternal resources such as reference models or validation sets, and (iii) incur ex
tra training time over full-dataset training. To address these limitations, this
 work develops UDS (Utility-Diversity Sampling), a framework for efficient onlin
e batch selection in SFT. UDS leverages the nuclear norm of the logits matrix to
 capture both data utility and intra-sample diversity, while estimating inter-sa
mple diversity through efficient low-dimensional embedding comparisons with a li
ghtweight memory buffer of historical samples. Such a design eliminates the need
 for external resources and unnecessary backpropagation, securing computational
efficiency. Experiments on multiple benchmarks demonstrate that UDS consistently

outperforms state-of-the-art online batch selection methods under varying data budgets, and significantly reduces training time compared to full-dataset fine-tuning.
****
Training Data Efficiency in Multimodal Process Reward Models
Jinyuan Li, Chengsong Huang, Langlin Huang, Shaoyang Xu, Haolin Liu, Wenxuan Zhang, Jiaxin Huang
Multimodal Process Reward Models (MPRMs) are central to step-level supervision for visual reasoning in MLLMs.
Training MPRMs typically requires large-scale Monte Carlo (MC)-annotated corpora, incurring substantial training cost.
This paper studies the data efficiency for MPRM training. Our preliminary experiments reveal that MPRM training quickly saturates under random subsampling of the training data, indicating substantial redundancy within existing MC-annotated corpora.
To explain this, we formalize a theoretical framework and reveal that informative gradient updates depend on two factors: label mixtures of positive/negative steps and label reliability (average MC scores of positive steps). Guided by these insights, we propose the Balanced-Information Score (BIS), which prioritizes both mixture and reliability based on existing MC signals at the rollout level, without incurring any additional cost. Across two backbones (InternVL2.5-8B and Qwen2.5-VL-7B) on VisualProcessBench, BIS-selected subsets consistently match and even surpass the full-data performance at small fractions. Notably, the BIS subset reaches full-data performance using only 10% of the training data, improving over random subsampling by a relative 4.1%.
****
PlugMem: A Task-Agnostic Plugin Memory Module for LLM Agents
Ke Yang, Zixi Chen, Xuan He, Jize Jiang, Michel Galley, Chenglong Wang, Jianfeng Gao, Jiawei Han, ChengXiang Zhai
Long-term memory is essential for large language model (LLM) agents operating in complex environments, yet existing memory designs are either task-specific and non-transferable, or task-agnostic but less effective due to low task-relevance and context explosion from raw memory retrieval. We propose PlugMem, a task-agnostic plugin memory module that can be attached to arbitrary LLM agents without task-specific redesign. Motivated by the fact that decision-relevant information is concentrated as abstract knowledge rather than raw experience, we draw on cognitive science to structure episodic memories into a compact, extensible knowledge-centric memory graph that explicitly represents propositional and prescriptive knowledge. This representation enables efficient memory retrieval and reasoning over task-relevant knowledge, rather than verbose raw trajectories, and departs from other graph-based methods like GraphRAG by treating knowledge as the unit of memory access and organization instead of entities or text chunks. We evaluate PlugMem unchanged across three heterogeneous benchmarks (long-horizon conversational question answering, multi-hop knowledge retrieval, and web agent tasks).

The results show that PlugMem consistently outperforms task-agnostic baselines and exceeds task-specific memory designs, while also achieving the highest information density under a unified information-theoretic analysis. Code and data are available at https://github.com/TIMAN-group/PlugMem.
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Dual-Stream Diffusion for World-Model Augmented Vision-Language-Action Model
John Won, Kyungmin Lee, Huiwon Jang, Dongyoung Kim, Jinwoo Shin
Augmenting Vision-Language-Action models (VLAs) with world models is promising for robotic policy learning but faces challenges in jointly predicting states and actions due to the modality gap. To address this, we propose DUal-STream diffusion (DUST), a world-model augmented VLA framework featuring a multimodal diffusion transformer that maintains separate modality streams while enabling cross-modal knowledge sharing. In addition, DUST utilizes independent noise perturbations and a decoupled flow matching loss to learn cross-modal causal relationships. We further introduce an asynchronous sampling method for action and vision tokens

that enhances performance through inference-time scaling. Experimental results on simulated benchmarks like RoboCasa and GR-1 show that DUST achieves up to 6\% gains over state-of-the-art VLA and world-modeling baselines, with inference-time scaling providing an additional 2–5\% improvement. In real-world tasks using the Franka Research 3, DUST outperforms baselines by 10\% in success rate. Finally, we demonstrate that DUST enables effective transfer learning through both pretraining on action-free videos and joint-training with heterogeneous robot and human datasets.
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API: Adaptive Prototype Imputation for Incomplete Multimodal Sentiment Analysis
Xiaotao Wang, Yiyang Fang, Wenke Huang, Bin Yang, Guancheng Wan, Mang Ye
Multimodal sentiment analysis aims to infer human emotions by integrating signals from diverse modalities. However, missing modalities are common in real-world applications due to sensor failure, data corruption, or privacy concerns. Existing approaches typically follow two main paradigms: recovery-based and non-recovery-based methods. This dichotomy results in two critical limitations: I) computational inefficiency and semantic inconsistency (recovery-based methods rely on heavy generators that incur prohibitive inference latency and risk semantic drift due to lack of class-level priors); II) lack of instance specificity (non-recovery-based methods rely on static global mappings that fail to capture sample-specific affective cues). To address these gaps, we propose Adaptive Prototype Imputation (API). To mitigate I), we introduce Semantic-anchored Class-Temporal Prototype Estimation (SCOPE) to construct non-trainable prototypes as stable semantic anchors, promoting semantic reliability. To resolve II), we design Directional Instance-Adaptive Affine Modulation (DIAM) to dynamically modulate these anchors via direction-specific affine transformations, capturing instance-unique affective characteristics without generative overhead. Experimental results on CMU-MOSI and CMU-MOSEI demonstrate that API outperforms state-of-the-art baselines, establishing a robust and lightweight prototype-centric paradigm for multimodal sentiment analysis. The code is publicly available at https://github.com/KX-yolo/API.
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Separating Representation from Reconstruction Enables Scalable Text Encoders
Megi Dervishi, Mathurin VIDEAU, Yann LeCun
While decoders have rapidly scaled, encoders have remained largely unchanged since BERT. We revisit this disparity by frozen backbone evaluation via probing. Under this lens, the representations of BERT encoders become increasingly *unexploitable* by frozen probes, despite improved perplexity. The misalignment originates in BERT's flat design, which couples representation learning to the token reconstruction loss. We propose **CrossBERT**, a two-part architecture that separates the learning of high-quality encoded representations from the rigid grounding of token reconstruction. This design further enables high masking ratios (≥ 50%) and gradient collection over all tokens via a *Complementary Masking Strategy*, respectively increasing throughput by 1.5 to $2\times$ and sample efficiency by $2\times$. Overall, CrossBERT demonstrates monotonic scaling and superior performance on MTEB(eng, v2) and frozen GLUE benchmarks.
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Chain-of-Thought Reasoning In The Wild Is Not Always Faithful
Iván Arcuschin, Jett Janiak, Robert Krzyzanowski, Senthooran Rajamanoharan, Neel Nanda, Arthur Conmy
Recent studies indicate that when faced with explicit biases in prompts, models often omit mentioning these biases in their Chain-of-Thought (CoT) output, revealing that verbalized reasoning can give an incorrect picture of how models arrive at conclusions (unfaithfulness).
In this work, we show that unfaithful CoT also occurs on naturally worded, non-adversarial prompts without adding artificial biases or editing model outputs.
We find that when separately presented with the questions "Is X bigger than Y?" and "Is Y bigger than X?", models sometimes produce superficially coherent arguments to justify systematically answering Yes to both or No to both, despite the contradiction.

We present preliminary evidence that this is due to models' implicit biases towards Yes or No, labeling this *Implicit Post-Hoc Rationalization*.
Our results reveal rates up to 13% for production models, and while frontier models are more faithful, none are entirely so, including thinking models like DeepSeek R1 (0.37%) and Sonnet 3.7 with thinking (0.04%).
We also investigate *Unfaithful Illogical Shortcuts*, where models use subtly illogical reasoning to make speculative answers to hard math problems seem rigorously proven.
Our findings indicate that while CoT can be useful for assessing outputs, it is not a complete account of the internal process that produced the model's answer and should be used with caution in agentic or safety-critical settings.
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From Denoising to De-Channeling: Integrating Physical Channel Priors into Diffusion Models for Radio Signal Understanding
Yaoqi Liu, Jin Wang, Chunchen Wang, Hui Wang, Chuan Shi
In recent years, wireless signal recognition (WSR), which leverages artificial intelligence (AI) to identify properties of passively received radio signals, has garnered significant attention due to its broad applications, such as spectrum management. Existing WSR methods typically learn directly from received signals, which are distorted by physical wireless channel effects such as fading, and current denoising diffusion models lack de-channeling capabilities, which leads to performance degradation. Therefore, we propose PWC-Diff, a novel framework that integrates prior Physical Wireless Channels into the denoising Diffusion process. The framework employs a dedicated architecture named FusedFormer, which contains a fusion module and a self-attention module that jointly capture the temporal and spectral characteristics of the signals throughout the diffusion trajectory. By leveraging prior wireless channels, PWC-Diff learns to progressively “de-channel” the received signal and recover a representation closer to the transmitted signal. Extensive experiments on several datasets across three WSR tasks have achieved state-of-the-art (SOTA) performance, which demonstrates the rationality of our theory, and ablation experiments further illustrate the effectiveness of our proposed PWC-Diff. Code is available at https://github.com/BUPT-GAMMA/FoundWSR.
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Training-Free Adversarial Robustness in Computational MRI
Mahdi Saberi, Chi Zhang, Mehmet Akcakaya
Deep learning (DL) methods have become the state-of-the-art for reconstructing sub-sampled magnetic resonance imaging (MRI) data. However, studies have shown that these methods are susceptible to small adversarial input perturbations, resulting in major distortions in the output images. Various strategies have been proposed to reduce the effects of these attacks, but they require retraining. In this work, we propose a novel approach for mitigating adversarial attacks on MRI reconstruction models without any retraining. Based on the idea of cyclic measurement consistency, we devise a novel mitigation objective that is minimized in a small ball around the attack input. Results show that our method substantially reduces the impact of adversarial perturbations across different datasets, attack types/strengths and PD-DL networks, and qualitatively and quantitatively outperforms conventional mitigation methods. We also introduce a practically relevant scenario for small adversarial perturbations that models impulse noise in raw data, which relates to herringbone artifacts, and show the applicability of our approach in this setting. Finally, we show our mitigation approach remains effective in two realistic extension scenarios: a blind setup, where the attack strength or algorithm is not known to the user; and an adaptive attack setup, where the attacker has full knowledge of the defense strategy. Code available at: https://github.com/MahdiSaberii/CycMit-MRI
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Dynamics Reveals Structure: Challenging the Linear Propagation Assumption
Hoyeon Chang, Bálint Mucsányi, Seong Joon Oh
Neural networks adapt through first-order parameter updates, yet it remains unclear whether such updates preserve logical coherence.

We investigate the geometric limits of the Linear Propagation Assumption (LPA), the premise that local updates coherently propagate to logical consequences.
To formalize this, we adopt relation algebra and study three core operations on relations: negation flips truth values, converse swaps argument order, and composition chains relations.
For negation and converse, we prove that guaranteeing direction-agnostic first-order propagation necessitates a tensor factorization separating entity-pair context from relation content.
However, for composition, we identify a fundamental obstruction.
We show that composition reduces to conjunction, and prove that any conjunction well-defined on linear features must be bilinear.
Since bilinearity is incompatible with negation, this forces the feature map to collapse.
These results suggest that failures in knowledge editing, the reversal curse, and multi-hop reasoning may stem from common structural limitations inherent to the LPA.
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Ratio-Variance Regularized Policy Optimization

Yu Luo, Shuo Han, Yihan Hu, Lei Lv, Huaping Liu, Fuchun Sun, Jianye HAO, Dong Li

Standard on-policy reinforcement learning relies on heuristic clipping to enforce trust regions, but this mechanism imposes a severe cost by indiscriminately truncating high-return yet high-divergence updates. We demonstrate that explicitly constraining the *policy ratio **variance*** provides a principled local approximation to trust-region constraints, eliminating the need for binary hard clipping. By acting as a distributional ''soft brake'', this approach preserves critical gradient signals from novel discoveries while naturally down-weighting and enabling the reuse of stale, off-policy data. We introduce **R^{2}VPO** (Ratio-Variance Regularized Policy Optimization), which implements this constraint via a primal–dual optimization framework. Extensive evaluations across 7 LLM scales, spanning both fast and slow reasoning paradigms, and 10 robotic control tasks demonstrate the generality of the proposed approach. R^{2}VPO achieves substantial performance gains on mathematical reasoning benchmarks, with particularly pronounced improvements on smaller models, while significantly improving sample efficiency. Furthermore, it consistently outperforms PPO baselines in continuous control domains, particularly in sparse-reward and dynamic environments. Together, these findings establish ratio-variance regularization as a principled foundation for stable and data-efficient policy optimization.
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The Latent Guardian: Defending Collaborative Perception via Feature-Level Consistency Verification

Zhuangzhuang Zhang, Mingxin Li, Libing Wu, Wei-Bin Lee, Jianping Wang

Collaborative perception (CP) significantly extends the sensing range of connected and autonomous vehicles (CAVs). However, its reliance on data fusion among multiple CAVs makes it inherently vulnerable to adversarial attacks from malicious participants. Existing defenses primarily rely on output-level consensus, assuming that malicious messages manifest as statistical outliers, while suffering from poor adaptability to environmental noise. This makes them vulnerable to stealthy adversarial attacks and prone to high false positive rates. To address this challenge, we shift the defense paradigm from superficial output-level consensus to deeper consistency within the internal feature space. Guided by this principle, we propose Cerberus, a novel defense framework against adversarial attacks in CP systems by leveraging multi-dimensional consistency in the feature space. By quantifying conflicts in topological structure, semantic direction, and energy distribution within feature maps, Cerberus effectively detects adversarial perturbations and provides dynamic protection against adversarial attacks. Experimental results demonstrate that Cerberus significantly outperforms state-of-the-art methods, effectively limiting the attack success rate to as low as 0.05\% while restoring the AP to 0.88.
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From Prompts to Tokens: Internalizing Causal Supervision in Vision-Language Mode

l for Multi-Image Causal Reasoning
Haoping Yu, Yuanxi Li, Jing Ma
Visual causal reasoning is essential for understanding and intervening in the ph
ysical world, requiring identification of causal variables from visual inputs an
d reasoning over intervention effects. Despite recent progress, large vision-lan
guage models (VLMs) remain brittle at such tasks, especially for interventional
and counterfactual queries over multi-image inputs.
Most existing explorations inject causal knowledge via textual prompts, leaving
causal mechanisms external to model execution and limiting reliable control duri
ng inference.
To address this problem, we propose BridgeVLM, which internalizes visual causal
reasoning by inducing a causal graph from multi-image inputs and converting it i
nto structured Causal Tokens executed by RAMP layers injected into the LLM decod
er for causal message passing.
We further introduce a unified training interface M3S for fine-grained causal su
pervision from different granularities (local/global level).
BridgeVLM achieves 54.4\% accuracy on intervention tasks on CausalVLBench (vs. 3
3.2\% with prompt-level supervision), improves results on Causal3D from 43.6\% t
o 49.0\%, and substantially improves causal structure learning on CausalVLBench
(F_1: 33.4\% $\rightarrow$ 75.1\%).
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DiffuReason: Enhancing Reasoning Ability for Diffusion Language Models via Mont
e Carlo Tree Search
YIPING SONG, Jinyu You, Zhiliang Tian, Jinsong Su, Minlie Huang, Chenping Hou
Auto-Regressive (AR) models with Monte Carlo Tree Search (MCTS) are a dominant p
aradigm for achieving “System 2” reasoning. However, this approach suffers from
significant latency due to the serial, token-by-token generation mechanism of AR
 models. In contrast, Diffusion Large Language Models (dLLMs) offer inherent spe
ed advantages via parallel sequence generation, yet they often struggle with acc
uracy in complex reasoning due to a lack of rigorous search, evaluation, and rev
ision capabilities. Directly applying MCTS to diffusion models faces architectur
al barriers, since the denoising generation process lacks the discrete decision
steps that naturally accommodate tree search. To retain efficiency while improvi
ng the reasoning ability, we propose DiffuReason, a Monte Carlo tree search reas
oning algorithm for diffusion models. By modeling the generation process as a Ma
rkov Decision Process (MDP), DiffuReason discretizes the continuous diffusion fl
ow into searchable thought blocks. During the reverse generation process, DiffuR
eason recursively performs four MCTS-style stages: select the best node (block),
 expand to obtain candidate nodes, simulate to evaluate node values, and revise
the unsatisfactory nodes. Experiments on mathematical reasoning benchmarks demon
strate that DiffuReason significantly improves the reasoning ability of diffusio
n models, and achieves superior balance of accuracy and efficiency even compared
 with auto-regressive models.
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Learning High-Dimensional Parity Functions with Product Networks using Gradient
Descent
Guillaume Larue, Louis-Adrien Dufrène, Quentin Lampin, Hadi Ghauch, Rekaya-Ben O
thman
Parity functions are fundamental Boolean operations with critical applications a
cross machine learning, cryptography, and error correction. Yet, learning high-d
imensional parity functions poses significant challenges: in a general setting,
standard neural network architectures typically require exponential sample compl
exity, making gradient-based optimization intractable for large number of inputs
 N. We demonstrate that compact product-based neural architectures combined wi
th stochastic data sparsity (Bernoulli inputs with $p_e \leq 1/N$) and appropria
te hyperparameter choice enable efficient parity learning, with theoretical guar
antees of convergence. Experiments validate our theory across dimensions up to $
N = 100{,}000$, with empirical evidence showing optimal hyperparameter choices f
or p_e and learning rate α, as well as polynomial complexity scaling la
ws. This work establishes fundamental connections between architectural inductiv

e bias and data sparsity, opening new possibilities for neural arithmetic, structured reasoning, binary neural networks, and machine learning applied to automated protocol discovery.
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Overcoming the Modality Gap in Context-Aided Forecasting
Vincent Zhihao Zheng, Étienne Marcotte, Arjun Ashok, Andrew Robert Williams, Lijun Sun, Alexandre Drouin, Valentina Zantedeschi
Context-aided forecasting (CAF) holds promise for integrating domain knowledge and forward-looking information, enabling AI systems to surpass traditional statistical methods. However, recent empirical studies reveal a puzzling gap: multimodal models often fail to outperform their unimodal counterparts. We hypothesize that this underperformance stems partly from insufficiently verified context usefulness in existing datasets. To address these limitations, we introduce a semi-synthetic data augmentation method that generates contexts both descriptive of temporal dynamics and verifiably complementary to numerical histories. This approach enables massive-scale dataset creation, resulting in CAF-7M, a corpus of 7 million context-augmented time series windows, including a rigorously verified test set. We demonstrate that semi-synthetic pre-training transfers effectively to real-world evaluation, and show clear evidence of context utilization. Our results suggest that dataset quality is a major bottleneck in context-aided forecasting, and that verified context can substantially improve the usefulness of CAF training data.
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Variational Bayesian Flow Network for Graph Generation
Yida Xiong, Jiameng Chen, Xiuwen Gong, Jia Wu, Shirui Pan, Wenbin Hu
Graph generation aims to sample discrete node and edge attributes while satisfying coupled structural constraints. Diffusion models for graphs often adopt largely factorized forward-noising, and many flow-matching methods start from factorized reference noise and coordinate-wise interpolation, so node–edge coupling is not encoded by the generative geometry and must be recovered implicitly by the core network, which can be brittle after discrete decoding. Bayesian Flow Networks (BFNs) evolve distribution parameters and naturally support discrete generation. But classical BFNs typically rely on factorized beliefs and independent channels, which limit geometric evidence fusion. We propose Variational Bayesian Flow Network (VBFN), which performs a variational lifting to a tractable joint Gaussian variational belief family governed by structured precisions. Each Bayesian update reduces to solving a symmetric positive definite linear system, enabling coupled node and edge updates within a single fusion step. We construct sample-agnostic sparse precisions from a representation-induced dependency graph, thereby avoiding label leakage while enforcing node–edge consistency. On synthetic and molecular graph datasets, VBFN improves fidelity and diversity, and surpasses baseline methods. The reproducible code is available at https://github.com/Cello2195/VBFN.
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Efficient Multi-round LLM Inference over Disaggregated Serving
Wenhao He, Youhe Jiang, Penghao Zhao, Quanqing Xu, Eiko Yoneki, Bin CUI, Fangcheng Fu
With the rapid evolution of Large Language Models (LLMs), multi-round workflows, such as autonomous agents and iterative retrieval, have become increasingly prevalent. However, this raises hurdles for serving LLMs under prefill-decode (PD) disaggregation, a widely adopted paradigm that separates the compute-bound prefill phase and memory-bound decode phase onto individual resources. Specifically, existing systems overlook the interleaved prefill-decode workload pattern in multi-round inference, leading to sub-optimal handling of the incremental prefill workloads and model deployment for the two phases.

In this work, we present AMPD, a brand new disaggregated serving framework for multi-round LLM inference. The core of AMPD is to coordinate the prefill workloads based on real-time workloads by adaptively determining where to carry out these workloads and how they are scheduled, in order to maximize service level objec

tive (SLO) attainment. In addition, we tailor a planning algorithm for our scenario, facilitating the deduction of optimal resource allocation and parallel strategies for the two phases. Empirical results demonstrate that AMPD substantially improves SLO attainment compared to state-of-the-art baselines.
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Test-time Offline Reinforcement Learning on Goal-related Experience
Marco Bagatella, Mert Albaba, Jonas Hübotter, Georg Martius, Andreas Krause
Foundation models compress a large amount of information in a single, large neural network, which can then be queried for individual tasks. There are strong parallels between this widespread framework and offline goal-conditioned reinforcement learning algorithms: a universal value function is trained on a large number of goals, and the policy is evaluated on a single goal in each test episode. Extensive research in foundation models has shown that performance can be substantially improved through test-time training, specializing the model to the current goal. We find similarly that test-time offline reinforcement learning on experience related to the test goal can lead to substantially better policies at modest compute costs. We propose a novel self-supervised data selection criterion, which selects transitions from an offline dataset according to their relevance to the current state and quality with respect to the evaluation goal. We demonstrate across a wide range of high-dimensional loco-navigation and manipulation tasks that fine-tuning a policy on the selected data for a few gradient steps leads to significant performance gains over standard offline pre-training. Our goal-conditioned test-time training (GC-TTT) algorithm applies this routine in a receding-horizon fashion during evaluation, adapting the policy to the current trajectory as it is being rolled out. Finally, we study compute allocation at inference, demonstrating that, at comparable costs, GC-TTT induces performance gains that are not achievable by scaling model size.
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GePBench: Evaluating Fundamental Geometric Perception for Multimodal Large Language Models
Shangyu Xing, Changhao Xiang, Xinyu Liu, Zhangtai Wu, Zhen Wu, Yue YIfan, Yuteng Han, Fei Zhao, Xinyu Dai
Geometric shapes play important roles in both physical world and human cognition. While multimodal large language models (MLLMs) have made significant advancements in visual understanding, their abilities to recognize geometric shapes and their spatial relationships, which we term geometric perception, are not explicitly and systematically explored.
To address this gap, we introduce GePBench, a novel benchmark specifically designed to assess the geometric perception capabilities of MLLMs. Our extensive evaluations reveal that even the current state-of-the-art MLLMs exhibit significant deficiencies in geometric perception tasks. Furthermore, we show that models trained with GePBench data demonstrate considerable improvements on a wide range of downstream tasks, highlighting the critical role of geometric perception in enabling advanced multimodal applications. Our code and datasets are available at https://github.com/Changhao-Xiang/GePBench.
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Stream RAG: Instant and Accurate Spoken Dialogue Systems with Streaming Tool Usage
Siddhant Arora, Haidar Khan, Kai Sun, Xin Luna Dong, Sajal Choudhary, Seungwhan Moon, Xinyuan Zhang, Adithya Sagar, Surya Teja Appini, Kaushik Patnaik, Sanat Sharma, Shinji Watanabe, Anuj Kumar, Ahmed A Aly, Yue Liu, Florian Metze, Zhaojiang Lin
End-to-end speech-in, speech-out dialogue systems are emerging as a powerful alternative to traditional ASR–LLM–TTS pipelines but remain prone to hallucinations due to limited factual grounding. While text-based dialogue models have effectively mitigated this issue through tools such as web search APIs, extending such capabilities to speech-in, speech-out systems remains underexplored. A key challenge is that tool integration increases latency, disrupting conversational flow. To mitigate this, we propose Streaming Retrieval-Augmented Generation (Stream RAG), a novel framework that reduces latency by predicting tool queries in parall

el with user speech, even before the user finishes speaking. Specifically, we develop a post-training pipeline that teaches the model when to issue tool calls and how to generate spoken summaries using retrieved text results, thereby improving both accuracy and responsiveness. To evaluate our approach, we construct AudioCRAG, a benchmark created by converting queries from the publicly available CRAG dataset into speech form. Experimental results show that Stream RAG improves QA accuracy by over 20.0% absolute on AudioCRAG and achieves state-of-the-art performance, including outperforming cascaded systems, on the SLUE-SQA benchmark, while reducing latency by up to 57%. Stream RAG is modality-agnostic and can be applied equally to typed input, paving the way for more agentic, real-time AI assistants.

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VLANeXt: Recipes for Building Strong VLA Models

Xiao-Ming Wu, Bin Fan, Kang Liao, Jian-Jian Jiang, Runze Yang, Yihang Luo, Zhonghua Wu, Wei-Shi Zheng, Chen Change Loy

Following the rise of large foundation models, Vision–Language–Action models (VLAs) emerged, leveraging strong visual and language understanding from Vision-Language Models for general-purpose policy learning. Yet, the current VLA landscape remains fragmented and exploratory. Although many groups have proposed their own VLA models, inconsistencies in training protocols and evaluation settings make it difficult to identify which design choices truly matter. To bring structure to this evolving space, we reexamine the VLA design space under a unified framework and evaluation setup. Starting from a simple VLA baseline similar to RT-2, which is the origin of VLA, we systematically dissect design choices along three dimensions: foundational components, perception essentials, and action modelling perspectives. From this study, we distill 12 key findings that together form a practical recipe for building strong VLA models. The outcome of this exploration is a simple yet effective model, VLANeXt. It outperforms the state-of-the-art methods on the LIBERO and LIBERO-plus benchmarks and demonstrates strong performance in real-world experiments. We release a unified and easy-to-use codebase to reproduce our findings, explore the design space, and develop new VLA variants on top of a shared foundation. The codebase is available at https://github.com/DravenALG/VLANeXt.

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From Parameter Dynamics to Risk Scoring: Quantifying Sample-Level Safety Degradation in LLM Fine-tuning

Xiao Wang, Yifei Zhang, YongKang Liu, Xiaocui Yang, Zihan Wang, Shi Feng, Daling Wang

Safety alignment of Large Language Models (LLMs) is extremely fragile, fine-tuning on small number of benign samples can erase safety behaviors learned from millions of preference examples.

Existing studies attempt to explain this phenomenon by comparing parameters and hidden states before and after fine-tuning, but overlook their dynamic evolution during fine-tuning.

In this work, we analyze parameter dynamics and uncover a critical mechanism underlying safety degradation, where benign fine-tuning

causes parameters cumulatively drift toward danger-aligned directions, progressively undermining the model's safety.

Inspired by these findings, we propose Sample-Level Quantification of Safety Degradation (SQSD), a method that quantifies each training sample's influence on safety degradation. Specifically, SQSD assigns continuous risk scores to individual samples by measuring their induced parameter updates along safety and danger directions.

Extensive experiments across three models and two datasets show that SQSD outperforms baselines in better separating high-risk and low-risk samples, with risk scores that consistently predict the severity of safety degradation. In particular, SQSD exhibits strong transferability across architectures, parameter scales, and parameter-efficient methods.

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Can Muon Fine-tune Adam-Pretrained Models?

Xingyu Qu, Huang Peigeng, Samuel Horváth
Muon has emerged as an efficient alternative to Adam for pretraining, yet remains underused for fine-tuning. A key obstacle is that most open models are pretrained with Adam, and naively switching to Muon for fine-tuning leads to degraded performance due to an optimizer mismatch.
We investigate this mismatch through controlled experiments and relate it to the distinct implicit biases of Adam and Muon. We provide evidence that the mismatch disrupts pretrained knowledge, and that this disruption scales with update strength. This leads us to hypothesize that constraining updates should mitigate the mismatch.
We validate this with LoRA: across language and vision tasks, LoRA reduces the performance gap between Adam and Muon observed under full fine-tuning. Studies on LoRA rank, catastrophic forgetting, and LoRA variants further confirm that mismatch severity correlates with update strength. These results shed light on how optimizer mismatch affects fine-tuning and how it can be mitigated.
Our code is available [here](https://github.com/XingyuQu/muon-finetune).
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STAND: Self-Aware Precondition Induction for Interactive Task Learning
Daniel Weitekamp, Glen R Smith, Ken Koedinger, Christopher J. MacLellan
In interactive task learning (ITL), AI agents learn new capabilities from limited human instruction provided during task execution. STAND is a new method of data-efficient rule precondition induction specifically designed for these human-in-the-loop training scenarios. A key feature of STAND is its self-awareness of its own learning—it can provide accurate metrics of training progress back to users. STAND beats popular methods like XGBoost, decision trees, random forests, and version spaces at small-data precondition induction tasks, and is highly accurate at estimating when its performance improves on holdout examples. In our evaluations, we find that STAND shows more monotonic improvement than other models with low rates of error reoccurrence. These features of STAND support a consistent training experience, enabling human instructors to estimate when they have finished training and providing active-learning support by identifying trouble spots that require more training. STAND achieves this by efficiently learning a compact space of greedy classifiers consistent with training data, rather than a finite ensemble of alternatives.
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Mitigating Noise-Induced Layout Priors for Object Counting in Diffusion Models
Xiaoling Gu, Xuelong Li, Shengqi Wu, Yongkang Wong, Zizhao Wu, Huan Li, Zhou Yu, Mohan Kankanhalli
Despite remarkable progress in text-to-image diffusion models, accurately generating the specified number of objects remains a persistent challenge. We identify the initial noise as a primary determinant of spatial layout formation, with early-stage cross-attention serving as the key mechanism that mediates the propagation of noise-induced structures throughout the denoising process. We characterize this phenomenon as \textbf{\textit{Noise-Induced Layout Prior}}. Leveraging this insight, we propose a novel training-free framework for object counting in diffusion models. Our approach consists of two key components: (1) a \emph{Count-Aware Noise Adjustment Strategy}, which explicitly manipulates the initial latent noise to align layout formation with the target object count, and (2) an \emph{Attention-Guided Layout Consistency Strategy}, which performs test-time optimization on early-stage cross-attention to further stabilize layout formation during denoising. Extensive experiments on both single-category and multi-category benchmarks demonstrate that our method consistently outperforms strong diffusion baselines and state-of-the-art object count control methods in terms of counting accuracy and image quality. Code Release: \url{https://github.com/lxlong1201/Mitigate_Noise_Prior}.
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TRACE: Toulmin-based Reasoning Assessment through Constructive Elements for LLM CoT Evaluation
Yundong Kim, Heyoung Yang
Evaluating open-ended outputs from large language models (LLMs) remains challeng

ing due to the absence of ground truth. Existing metrics rely on final-answer accuracy or surface-level statistics, leaving the reasoning process itself unexamined. We introduce TRACE (Toulmin-based Reasoning Assessment through Constructive Elements), a metric that analyzes Chain-of-Thought (CoT) reasoning processes. Rather than judging outcomes, TRACE inspects how arguments are constructed by integrating Toulmin's argumentation theory with Flavell's metacognitive framework to assess reasoning structure. Experiments on 26.3K QA samples across 7 reasoning models show strong correlation with benchmark accuracy (r=0.74). Furthermore, TRACE is effective as a reinforcement learning reward signal, outperforming accuracy-only baselines. Together, these results indicate that logically sound reasoning leads to higher-quality answers. TRACE thus serves as a complementary metric for evaluating open-ended outputs. Code is available at https://github.com/hyyangkisti/trace.
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Long-Horizon Model-Based Offline Reinforcement Learning Without Explicit Conservatism

Tianwei Ni, Esther Derman, Vineet Jain, Vincent Taboga, Siamak Ravanbakhsh, Pierre-Luc Bacon

Popular offline reinforcement learning (RL) methods rely on *explicit conservatism*, penalizing out-of-dataset actions or restricting rollout horizons. We question the universality of this principle and revisit a complementary Bayesian perspective for test-time adaptation. By modeling a posterior over world models and training a history-dependent agent to maximize expected return, the Bayesian approach directly addresses epistemic uncertainty without explicit conservatism. We first illustrate in a bandit setting that Bayesianism excels on low-quality datasets where conservatism fails. Scaling to realistic tasks, we find that *long-horizon rollouts* are essential to control value overestimation once conservatism is removed. We introduce design choices that enable learning from long-horizon rollouts while mitigating compounding model errors, yielding our algorithm, NEUBAY, grounded in the neutral Bayesian principle. On D4RL and NeoRL benchmarks, NEUBAY is competitive with leading conservative algorithms, achieving new state-of-the-art on 7 datasets with rollout horizons of several hundred steps. Finally, we characterize datasets by quality and coverage to identify when NEUBAY is preferable to conservative methods.
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Upper-Linearizability of Online Non-Monotone DR-Submodular Maximization over Down-Closed Convex Sets

Yiyang Lu, Hareshkumar Jadav, Mohammad Pedramfar, Ranveer Singh, Vaneet Aggarwal

We study online maximization of non-monotone Diminishing-Return(DR)-submodular functions over down-closed convex sets, a regime where existing projection-free online methods suffer from suboptimal regret and limited feedback guarantees.
Our main contribution is a new structural result showing that this class is $1/e$-Upper-Linearizable under carefully designed exponential reparametrization, scaling parameter, and surrogate potential, enabling a reduction to online linear optimization.
As a result, we obtain optimal $O(T^{1/2})$ static regret with a single gradient query per round and unlock adaptive and dynamic regret guarantees, together with improved rates under semi-bandit, bandit, and zeroth-order feedback.
Across all feedback models, our bounds strictly improve the state of the art.
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TAGRPO: Boosting GRPO on Image-to-Video Generation with Direct Trajectory Alignment

Jin Wang, Jianxiang Lu, Guangzheng Xu, Comi Chen, Haoyu Yang, Linqing Wang, Peng Chen, Mingtao Chen, Zhichao Hu, Longhuang Wu, Shuai Shao, Qinglin Lu, Ping Luo

Recent studies have demonstrated the efficacy of integrating Group Relative Policy Optimization (GRPO) into flow matching models, particularly for text-to-image and text-to-video generation. However, we find that directly applying these techniques to image-to-video (I2V) models often fails to yield consistent reward improvements. To address this limitation, we present TAGRPO, a robust post-training framework for I2V models inspired by contrastive learning. Our approach is gro

unded in the observation that rollout videos generated from identical initial no ise provide superior guidance for optimization. Leveraging this insight, we prop ose a novel GRPO loss applied to intermediate latents, encouraging direct alignm ent with high-reward trajectories while maximizing distance from low-reward coun terparts. Furthermore, we introduce a memory bank for rollout videos to enhance diversity and reduce computational overhead. Despite its simplicity, TAGRPO achi eves significant improvements over DanceGRPO in I2V generation.

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Causal Fine-Tuning under Latent Confounded Shift

Jialin Yu, Yuxiang Zhou, Haoxuan Li, Junchi Yu, Mengyue Yang, Yulan He, Nevin L. Zhang, Philip Torr, Ricardo Silva

Adapting to latent confounded shift remains a core challenge in modern AI. This setting is driven by hidden variables that induce spurious correlations between inputs and outputs during training, leading models to rely on non-causal shortcu ts. For example, a model may learn to treat metadata (e.g., data source like "Am azon") as a proxy for positive sentiment, causing failure when the source become s predominantly negative during deployment. To address this *latent confounded s hift*, we introduce Causal Fine-Tuning (CFT). Using a structural causal model as an inductive bias, we derive sufficient identification conditions that motivate a fine-tuning objective for decomposing representations into high-level stable and low-level shift-sensitive components. Instantiating this framework in BERT, we show that learning such causal/spurious representations and adjusting them ac cordingly yield a more robust predictor. Experiments on spurious correlation inj ection attacks in text demonstrate that our method outperforms black-box domain generalization baselines, highlighting the benefits of explicitly modeling causa l structure.

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Arboreal Neural Network

Wubin Yan, Wei Ma, Shixiang Wan, Dongchen Li, Shaoshun Kang, Qing Yang, Donglian g Xu

Connectionist models and symbolic models have long embodied two divergent paradi gms: the former excel at differentiable representation learning yet struggle wit h transparency, while the latter deliver explicit rule-based reasoning but resis t gradient-based optimization. We introduce Arboreal Neural Networks (ArbNN), a neural-symbolic framework that unifies these paradigms both computationally and conceptually. At the design level, ArbNN departs fundamentally from prior neural ized-tree models through a depth-aware routing mechanism and a topology-informed softmax aggregation, which together enable one-shot full-path gradient propagat ion and consequently achieving rapid and well-conditioned optimization dynamics. At the conceptual level, ArbNN reveals that decision-tree branching and self-at tention routing are two realizations of the same conditional computation primiti ve. We prove a structural isomorphism between a decision tree and a single-query attention head, enabling a differentiable architecture that faithfully preserve s symbolic decision logic. The defining property of ArbNN is Bidirectional Fidel ity, ensuring that the neural module can be compiled from—and losslessly decompi led back into—a symbolic tree, yielding both ordering consistency in ranking beh avior and explicit, auditable interpretability via reconstructed if-else rules. ArbNN further supports GBDT-based initialization, allowing it to inherit strong inductive biases and integrate seamlessly with existing production workflows. Em pirically, ArbNN achieves state-of-the-art performance on various public tabular benchmarks and delivers consistent gains under temporal distribution shift in l arge-scale industrial credit-risk systems. To support realistic evaluation, we a dditionally construct TabCredit, a feature-rich, temporally partitioned dataset built from millions of real-world loan applications. Together, these results dem onstrate that ArbNN forms a unified, reversible, and practically deployable brid ge between symbolic reasoning and neural computation for high-stakes tabular dom ains.

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GEM: Geometric Erasure by Contrastive Velocity Matching in Rectified Flows

Jonas Henry Grebe, Tobias Braun, Anna Rohrbach, Marcus Rohrbach

While the rapid adoption of multimodal generative models offers immense potentia
l, it has also increased the risks of harmful content synthesis, deepfakes, and
copyright infringements. To address these challenges, concept erasure has emerge
d as a prospective safeguard. However, as the field gradually transitions from U
-Net-based diffusion models to Rectified Flow Transformers, erasure research has
 struggled to keep pace. In this work, we introduce GEM, a simple but highly eff
ective erasure framework for Rectified Flow models. As part of our contribution,
 we establish a principled bridge between trajectory-based unlearning grounded i
n Generative Flow Networks and classic teacher-guided erasure: we translate traj
ectory-based signals into a teacher-guided flow-matching setup that unifies the
strengths of both paradigms. Concretely, a teacher provides complementary attrac
tion and repulsion signals that we combine into a single geometric guidance obje
ctive, yielding targeted suppression of unwanted concepts while preserving benig
n generation.
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SURGE: Surrogate Gradient Adaptation in Binary Neural Networks
Haoyu Huang, Boyu Liu, Linlin Yang, Yanjing Li, Yuguang Yang, Xuhui Liu, Canyu C
hen, Zhongqian Fu, Baochang Zhang
The training of Binary Neural Networks (BNNs) is fundamentally based on gradient
 approximation for non-differentiable binarization operations (e.g., sign functi
on). However, prevailing methods including the Straight-Through Estimator (STE)
and its improved variants, rely on hand-crafted designs that suffer from gradien
t mismatch problem and information loss induced by fixed-range gradient clipping
. To address this, we propose SURrogate GradiEnt Adaptation (SURGE), a novel lea
rnable gradient compensation framework with theoretical grounding. SURGE mitigat
es gradient mismatch through auxiliary backpropagation. Specifically, we design
a Dual-Path Gradient Compensator (DPGC) that constructs a parallel full-precisio
n auxiliary branch for each binarized layer, decoupling gradient flow via output
 decomposition during backpropagation. DPGC enables bias-reduced gradient estima
tion by leveraging the full-precision branch to estimate components beyond STE's
 first-order approximation. To further enhance training stability, we introduce
an Adaptive Gradient Scaler (AGS) based on an optimal scale factor to dynamicall
y balance inter-branch gradient contributions via norm-based scaling. Experiment
s on image classification, object detection, and language understanding tasks de
monstrate that SURGE performs best over state-of-the-art methods.
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Black-Box Assisted Regression: Phase Transitions and Minimax Optimality
Yan Zhou
Foundation models are often used as fixed black-box predictors for downstream ta
sks with limited labeled data, but their predictions may be biased and unsafe to
 trust blindly. We study this setting through black-box assisted nonparametric r
egression: a learner observes labeled samples and can query a fixed predictor $f
_0$, while the target f^* is close to f_0 in $L_2(P_X)$ up to an unknown rad
ius δ. We give a finite-sample minimax characterization showing a phase t
ransition at $\delta_c(n)\asymp n^{-\beta/(2\beta+d)}$, with leading risk $\min\
{\delta^2, n^{-2\beta/(2\beta+d)}\}$. We then analyze a Safe Residual Estimator:
 it learns a correction around f_0, initializes the residual head at zero so t
he initial predictor equals f_0, and uses holdout selection to revert to f_0
 when the learned correction is not supported by validation data. Here, ``safe''
 means avoiding negative transfer, i.e., performing worse than the black-box pre
dictor alone. The estimator matches the leading minimax term up to an additive v
alidation-selection cost. Synthetic regression experiments verify the predicted
phase transition, while CIFAR-100 with CLIP and AG News with Qwen3-8B provide pr
actice-facing evidence that the same residual-correction tradeoff is useful beyo
nd the formal squared-loss regression setting.
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Deep Trajectory Supervision: Deep Supervision Strikes Back
Han Wang, Weijie Wang, Jiaqi Liu, Hilde Kuehne, Nicu Sebe
Interpreting residual architectures as discretizations of Ordinary Differential
Equations characterizes the forward pass as a continuous latent flow. Although t

his framework defines the mechanics of inference, conventional training paradigm s primarily constrain the terminal state, leaving the intermediate evolution unr egulated. In this work, we formalize the forward pass as a Conditional Discrimin ative Flow and investigate its intrinsic kinematic laws. Using Tuned Lens analys is, we discover that the accumulation of semantic evidence follows a consistent exponential schedule. This finding confirms that deep models naturally require a n extended phase of feature construction prior to a rapid transition toward cate gorical certainty in the terminal layers.

Standard training ignores this latent progression. To resolve this impedance mis match, we propose Deep Trajectory Supervision, a framework that aligns auxiliary supervision with this intrinsic exponential bias. By rectifying the trajectory of the inference flow, DTS functions as a critical physical inductive bias. Empi rical evaluations on ImageNet-1K and various benchmarks demonstrate that DTS sig nificantly accelerates convergence and improves terminal performance.
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SRPO: Self-Reflective Policy Optimization for Long-Horizon Reasoning
Jialong Liu, Yuling Shi, Ning Yang, Xiaodong Gu, Zuchao Li
Self-reflection is a powerful mechanism for credit assignment in human learning,

converting sparse outcome feedback into actionable guidance. However, its potential for post-training Large Language Models (LLMs) remains underexplored. We propose Self-Reflective Policy Optimization (SRPO), a framework that internalizes this capability. SRPO enables LLMs to analyze their own completed trajectories, synthesize errors into concise "reflection patches," and use these

reflection-conditioned rollouts as high-quality, on-policy distillation targets.

This process effectively transforms sparse terminal supervision into dense, token-level learning signals without requiring external critics, separate reward models, or larger teacher models. We demonstrate that SRPO achieves state-of-the-art performance across mathematical reasoning and long-horizon agentic benchmarks with exceptional data efficiency. Using a Qwen3-8B base model, SRPO attains 73.3\% on AIME’24 using only 8\% (0.08$\times$) of the training FLOPs required by scaled supervised fine-tuning, while significantly improving success rates on WebShop (64.7\%), ALFWorld (76.8\%), and SWE-Bench-Lite (31.2\%).
****
TransNormal: Dense Visual Semantics for Diffusion-based Transparent Object Norma l Estimation
Mingwei Li, Hehe Fan, Yi Yang
Monocular normal estimation for transparent objects is critical for laboratory a utomation, yet it remains challenging due to complex light refraction and reflec tion. These optical properties often lead to catastrophic failures in convention al depth and normal sensors, hindering the deployment of embodied AI in scientif ic environments. We propose **TransNormal**, a novel framework that adapts pre-t rained diffusion priors for single-step normal regression. To handle the lack of texture in transparent surfaces, TransNormal integrates dense visual semantics from DINOv3 via a cross-attention mechanism, providing strong geometric cues. Fu rthermore, we employ a multi-task learning objective and wavelet-based regulariz ation to ensure the preservation of fine-grained structural details. To support this task, we introduce **TransNormal-Synthetic**, a physics-based dataset with high-fidelity normal maps for transparent labware. Extensive experiments demonst rate that TransNormal significantly outperforms state-of-the-art methods: on the ClearGrasp benchmark, it reduces mean error by 25.5\% and improves the best pri or 11.25° accuracy by 24.7\%; on ClearPose, it achieves a 17.7\% reductio n in mean error. Code and dataset are publicly available at https://github.com/l ongxiang-ai/TransNormal.
****
OServe: Accelerating LLM Serving via Spatial-Temporal Workload Orchestration

Youhe Jiang, Fangcheng Fu, Taiyi Wang, Guoliang He, Eiko Yoneki
Serving Large Language Models (LLMs) can benefit immensely from parallelizing both the model and input requests across multiple devices, but incoming workloads exhibit substantial spatial and temporal heterogeneity. Spatially, workloads comprise heterogeneous requests with varying compute and memory demands. Temporally, workload composition varies over time. Nevertheless, existing systems typically assume spatially uniform and temporally stable workloads, employing a homogeneous, static model deployment. This mismatch between the assumption and real-world spatial-temporal heterogeneity results in suboptimal performance. We present OServe, an LLM serving system with heterogeneous and flexible model deployment that addresses both spatial and temporal heterogeneity. First, OServe introduces a novel workload-aware scheduling algorithm that optimizes heterogeneous model deployments according to real-time workload characteristics. Second, OServe proposes an efficient workload-adaptive switching method that migrates model deployments in response to predicted workload changes. Experiments on real-world traces show that OServe improves performance by up to 2$\times$ (average: 1.5$\times$) compared to state-of-the-art serving systems.
****

Periodic Bayesian Flow Networks with Additive Accuracy
Peijia Lin, Zihan Zhang, zhangrui zhao, Shaohao Rui, Junyi An, Yun-Fei Shi, Fenglei Cao, Weijie Ma, Yutong Lu
Generating periodic data---such as fractional atomic coordinates in crystal structures and phase patterns in compressive light-field (CLF) displays---is challenging because wrap-around boundaries complicate probabilistic modeling and learning. While Bayesian Flow Networks (BFNs) offer a powerful generative framework with strictly additive accuracy in Euclidean space, existing periodic adaptations typically sacrifice additivity and become sensitive to schedule heuristics. We introduce \emph{PeriodicBFN}, which embeds each periodic scalar into a two-dimensional unit-circle representation and performs Gaussian Bayesian updates in the resulting Cartesian space, thereby restoring strictly additive accuracy. To address invariance in periodic generative modeling, we further derive a Rao--Blackwellized objective that analytically marginalizes global periodic translations, producing a translation-invariant target with reduced gradient variance. Experiments on crystal structure prediction and multi-layer phase synthesis for CLF displays demonstrate improved training stability and strong performance. To our knowledge, this is the first work to extend periodic-data generative modeling to phase synthesis for modern glasses-free 3D display systems.
****

L-Drive: Beyond a Single Mapping—Latent Context Drives Time Series Forecasting
Fan Zhang, Shijun Chen, Hua Wang
Mainstream methods for multivariate time-series forecasting largely follow the Direct-Mapping paradigm. They learn a unified mapping from history to the future in the observation space to fit value-level dependencies. However, real-world systems often undergo distribution shifts and regime changes. In such cases, a unified mapping can exhibit response lag around turning points, causing error accumulation within the switching window and reducing forecasting reliability. To address this issue, we propose L-Drive, a change-aware forecasting framework. L-Drive introduces a Latent-Context, to explicitly characterize high-level dynamics evolving over time, and uses gating to modulate increment representations. This provides more timely change cues and improves adaptation to changing segments. In addition, it incorporates patch-shared relative positional basis functions to strengthen intra-segment structural modeling and reduce overfitting caused by absolute-position memorization. Extensive experiments validate the effectiveness of L-Drive and show a better overall trade-off between forecasting accuracy and computational efficiency.
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When Can You Poison Rewards? A Tight Characterization of Reward Poisoning in Linear MDPs
Jose Efraim Aguilar Escamilla, Haoyang Hong, Jiawei Li, Haoyu Zhao, Xuezhou Zhang, Sanghyun Hong, Huazheng Wang

We study reward poisoning attacks in reinforcement learning (RL), where an adversary manipulates rewards under a limited budget to induce a target agent to learn a policy aligned with the attacker's objectives. Most prior work focuses on *constructing* successful attacks, providing sufficient conditions under which poisoning is effective, while offering limited understanding of when such targeted attacks are fundamentally infeasible. In this paper, we provide the first characterization of reward-poisoning attackability in linear MDPs, establishing both necessary and sufficient conditions for whether a target policy can be induced within a bounded attack budget. This draws a clear boundary between the *vulnerable* RL instances and *intrinsically robust* ones, which cannot be attacked without high costs even when the learner uses standard, non-robust RL algorithms. We further demonstrate our framework beyond synthetic linear MDPs by approximating deep RL environments as linear MDPs. We show that our theoretical framework effectively distinguishes vulnerability, demonstrating how our theoretical predictions have practical significance.
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LOCA-bench: Benchmarking Language Agents Under Controllable and Extreme Context Growth

Weihao Zeng, Yuzhen Huang, Junxian He

Frontier large language models (LLMs) are increasingly capable of carrying out long-running, real-world tasks. However, as the amount of context grows, their reliability often deteriorate, a phenomenon known as "context rot". Existing long-context benchmarks primarily focus on single-step settings that evaluate a model’s ability to retrieve information from a long snippet. In realistic scenarios, however, LLMs often need to act as agents that explore environments, follow instructions and plans, extract useful information, and predict correct actions under a dynamically growing context. To assess language agents in such settings, we introduce LOCA-bench (a benchmark for **LO**ng-**C**ontext **A**gents). Given a task prompt, LOCA-bench leverages automated and scalable control of environment states to regulate the agent’s context length. This design enables LOCA-bench to extend the context length potentially to infinity in a controlled way while keeping the underlying task semantics fixed. LOCA-bench evaluates language agents as a combination of models and scaffolds, including various context management strategies. While agent performance generally degrades as the environment states grow more complex, advanced context management techniques can substantially improve the overall success rate. We will open-source LOCA-bench to provide a platform for evaluating models and scaffolds in long-context, agentic scenarios.
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Modeling Hierarchical Thinking in Large Reasoning Models

G M Shahariar, Erfan Shayegani, Ali Nazari, Nael Abu-Ghazaleh

Large Reasoning Models (LRMs) solve complex tasks by generating long Chain-of-Thought (CoT) sequences; however, the emergent dynamics governing reasoning trajectories are not well understood and can lead to inconsistencies and reasoning pathologies. In this work, we propose to approximate LRM's emerging hierarchical reasoning dynamics as a trajectory within a Finite State Machine (FSM) transitioning among six abstract cognitive states. We demonstrate that these states and transitions can be captured in the latent state of the model. We believe that this representation can have different applications in the interpretability and optimization of LRM models. For example, by analyzing the topology of these transitions, we identify statistical shifts in reasoning strategies that help identify effective reasoning chains from those that fail. To illustrate these potential advantages, we propose Q-Value guided steering, a training-free inference-time control method that treats reasoning as a planning problem. We estimate the long-horizon utility of state transitions and apply sparse, orthogonal activation steering at sentence boundaries to align the CoT generation with optimal reasoning policies. Experiments across four benchmarks (AIME25, MATH-500, GSM8k, and GPQA Diamond) using three state-of-the-art open reasoning models demonstrate that Q-Value steering policy achieves significant performance gains with "surgical'' efficiency, often requiring $25\times$ fewer interventions than greedy and weighted baselines, which suggests that reasoning can be effectively controlled by guid

ing high-level cognitive dynamics rather than micro-managing token generation. C
ode is available at: https://github.com/shahariar-shibli/CoT-FSM.
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Agent Learning via Early Experience
Kai Zhang, Xiangchao Chen, Bo Liu, Tianci Xue, Zeyi Liao, Zhihan Liu, Xiyao Wang
, Yuting Ning, Zhaorun Chen, Xiaohan Fu, Jian Xie, Yuxuan Sun, Boyu Gou, Qi Qi,
Zihang Meng, Jianwei Yang, Ning Zhang, Xian Li, Ashish Shah, Dat Huynh, Hengduo
Li, Zi Yang, Xuefei Cao, Lawrence Keunho Jang, Shuyan Zhou, Jiacheng Zhu, Huan S
un, Jason E Weston, Yu Su, Yifan Wu
A long-term goal of language agents is to learn and improve through their own ex
perience, ultimately outperforming humans in complex, real-world tasks.
However, training agents from experience data with reinforcement learning remain
s difficult in many environments, which either lack verifiable rewards (e.g., we
bsites) or require inefficient long-horizon rollouts (e.g., multi-turn tool use)
.
As a result, most current agents rely on supervised fine-tuning on expert data,
which is challenging to scale and generalizes poorly. This limitation stems from
 the nature of expert demonstrations: they capture only a narrow range of scenar
ios, and expose the agent to limited environment diversity.
We address this limitation with a middle-ground paradigm we call *early experien
ce*: interaction data generated by the agent's own actions, where the resulting
future states serve as supervision without reward signals.
Within this paradigm, we study two strategies of using such data: (1) implicit w
orld modeling, which uses collected states to ground the policy in environment d
ynamics; and (2) self-reflection, where the agent learns from its suboptimal act
ions to improve reasoning and decision-making.
Evaluation across eight diverse environments and multiple model families shows t
hat our approaches consistently improve effectiveness and out-of-domain generali
zation, highlighting the value of early experience.
Moreover, in environments with verifiable rewards, our results provide promising
 signals that early experience offers a strong foundation for subsequent reinfor
cement learning, making it a practical bridge between imitation learning and ful
ly experience-driven agents.
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AutoMoT: A Unified Vision-Language-Action Model with Asynchronous Mixture -of-Tr
ansformers for End-to-End Autonomous Driving
Wenhui Huang, Songyan Zhang, Qihang Huang, Zhidong Wang, Zhiqi Mao, Collister Ch
ua, Zhan Chen, Long Chen, Chen Lv
Integrating vision-language models (VLMs) into end-to-end (E2E) autonomous drivi
ng (AD) systems has shown promise in improving scene understanding. However, exi
sting integration strategies suffer from several limitations: they either strugg
le to resolve distribution misalignment between reasoning and action spaces, un
derexploit the general reasoning capabilities of pretrained VLMs, or incur subst
antial inference latency during action policy generation, which degrades driving
 performance. To address these challenges, we propose AutoMoT in this work, an e
nd-to-end AD framework that unifies reasoning and action generation within a sin
gle vision-language-action (VLA) model. Our approach leverages a mixture-of-tran
sformer (MoT) architecture with layer-wise joint attention sharing, which preser
ves the general reasoning capabilities of pre-trained VLMs while enabling effici
ent asynchronous inference over various tasks at different frequencies. Addition
ally, we explore a VLA-oriented action refiner that further enhances driving per
formance via diffusion-based fine-tuning. Extensive experiments on multiple benc
hmarks, under both open- and closed-loop settings, demonstrate that AutoMoT achi
eves state-of-the-art (SOTA) performance compared to existing methods. We furthe
r investigate the functional boundary of pre-trained VLMs in AD, examining when
and to what extent AD-tailored fine-tuning is necessary.
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FeRA: Frequency-Energy Constrained Routing for Effective Diffusion Adaptation Fi
ne-Tuning
Bo Yin, Xiaobin Hu, Xingyu Zhou, Yu HE, Peng-Tao Jiang, Yue Liao, Junwei Zhu, Ji

angning Zhang, Ying Tai, Shuicheng YAN
Diffusion models have achieved remarkable success in generative modeling, yet ho w to effectively adapting large pretrained models to new tasks remains challengi ng. We revisit the reconstruction behavior of diffusion models during denoising to unveil the underlying frequency–energy mechanism governing this process. Buil ding upon this observation, we propose \textbf{FeRA}, a frequency-driven fine-tu ning framework that aligns parameter updates with the intrinsic frequency–energy progression of diffusion. FeRA establishes a comprehensive frequency–energy fra mework for effective diffusion adaptation fine-tuning, comprising three synergis tic components: \textit{(i)} a compact frequency–energy indicator that character izes the latent’s bandwise energy distribution, \textit{(ii)} a soft frequency r outer that adaptively fuses multiple frequency-specific adapter experts, and \te xtit{(iii)} a frequency–energy consistency regularization that stabilizes diffus ion optimization and ensures coherent adaptation across bands. Routing operates in both training and inference, with inference-time routing dynamically determin ed by the latent frequency energy. It integrates seamlessly with adapter-based t uning schemes and generalizes well across diffusion backbones and resolutions. B y aligning adaptation with the frequency–energy mechanism, \textbf{FeRA} provide s a simple, stable, and compatible paradigm for effective and robust diffusion m odel adaptation. Codes will be made publicly available.
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Censoring with Plausible Deniability: Asymmetric Local Privacy for Multi-Categor y CDF Estimation
Qirui Hu, Yi Liu
We introduce a new mechanism within the Utility-Optimized Local Differential Pri vacy (ULDP) framework that enables censoring with plausible deniability when col lecting and analyzing sensitive data. Our approach addresses scenarios where cer tain values, such as large numerical responses, are more privacy-sensitive than others, while accompanying categorical information may not be private on its own but could still be identifying. The mechanism selectively withholds identifying details when a response might indicate sensitive content, offering asymmetric p rivacy protection. Unlike previous methods, it avoids the need to predefine an e xact sensitive region, making it more adaptable and practical. Although the mech anism is designed for ULDP, it can also be applied under symmetric LDP settings, where it still benefits from censoring and reduced perturbation cost. We provid e theoretical guarantees, including uniform consistency and pointwise weak conve rgence results. Numerical experiments on both synthetic data and real-world data demonstrate the validity of the proposed methodology.
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MOOSE-Star: Unlocking Tractable Training for Scientific Discovery by Breaking th e Complexity Barrier
Zonglin Yang, Lidong Bing
While large language models (LLMs) show promise in scientific discovery, existin g research focuses on inference or feedback-driven training, leaving the direct modeling of the generative reasoning process, $P(\text{hypothesis}|\text{backgro und})$ ($P(h|b)$), unexplored. We demonstrate that directly training $P(h|b)$ is mathematically intractable due to the combinatorial complexity ($O(N^k)$) inher ent in retrieving and composing inspirations from a vast knowledge base.
To break this barrier, we introduce MOOSE-Star, a unified framework that enables tractable and scalable training of $P(h|b)$, while supporting more scalable inf erence.
In the best case, MOOSE-Star reduces complexity from exponential to logarithmic ($O(\log N)$) by (1) training on decomposed subtasks derived from the probabilis tic equation of discovery, (2) employing motivation-guided hierarchical search t o enable logarithmic retrieval and prune irrelevant subspaces, and (3) utilizing bounded composition for robustness against retrieval noise.
To facilitate this, we release TOMATO-Star, a dataset of 108,717 decomposed pape rs (38,400 GPU hours) for training.
Empirically, MOOSE-Star scales continuously with training data and inference bud get, whereas direct brute-force sampling hits a "complexity wall."

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SABER: Continual Learning with Representation Conflict Management
Xuandi Luo, Huaidong Zhang, Yi Xie, Shengfeng He
Continual learning seeks to develop models capable of acquiring new tasks sequen tially while retaining prior knowledge. A central challenge in this setting is m anaging inherent knowledge conflicts that arise as overlapping or contradictory information is introduced across tasks. While parameter-efficient fine-tuning (P EFT) techniques, particularly those based on Low-Rank Adaptation (LoRA), have sh own promise by reducing interference through parameter isolation or modular arch itectures, they often treat conflict as something to avoid rather than address d irectly. In this work, we propose $\underline{S}$ubspace-$\underline{A}$ligned $ \underline{B}$alanc$\underline{e}$d $\underline{R}$ecomposition (SABER), a novel method that reframes continual learning as a problem of structured conflict man agement. SABER introduces a unified subspace alignment framework to support shar ed task representations, decomposes task-specific knowledge into orthogonal comp onents to preserve distinct information, and recomposes them using an energy-awa re balancing mechanism that coordinates contributions without compromising stabi lity. Extensive experiments across multiple continual learning benchmarks show t hat SABER achieves performance on par with or surpassing state-of-the-art method s, offering a principled approach that directly addresses the root cause of forg etting by managing representational conflict.
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Syntax vs. Semantics: How Transformers Learn Deep Dependencies
JiangRui Zhao, Xiaoting Du
Large Language Models demonstrate remarkable syntactic fluency, yet the optimiza tion dynamics governing their acquisition of deep semantic dependencies remain p oorly understood. We propose a mechanistic framework that models this learning p rocess as a competition between Surface Statistics and Deep Semantics. Our theor etical analysis identifies a "Gradient Starvation" phenomenon where the error si gnals for sparse semantic dependencies are actively suppressed during early opti mization. This suppression impedes the learning of structural reasoning and caus es its emergence to manifest as a sudden phase transition.
Furthermore, this framework offers a mechanistic basis for the effectiveness of Chain-of-Thought (CoT) strategies. By externalizing intermediate reasoning steps into concrete tokens, CoT effectively bypasses the suppression regime inherent to implicit reasoning. We validate these findings across scales ranging from toy transformers to production models (Llama-3.1-8B, Qwen2.5-Coder-7B). Finally, gu ided by this theory, we propose a topology-aligned contrastive objective that ex plicitly rectifies the gradient geometry. Experiments on variable binding tasks demonstrate that our method achieves an improvement that is over 2× larger than that obtained via standard cross-entropy fine-tuning. Code will be publicly avai lable at: https://github.com/jr-zhao/Deep-Dependencies/tree/main.
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Mind Dreamer: Untethering Imagination via Active Causal Intervention on Latent M anifolds
Shaojun Xu, Xiaoling Zhou, Yihan Lin, Yapeng Meng, Xinglong Ji, Luping Shi, Rong Zhao
Model-Based Reinforcement Learning yields sample efficiency via latent imaginati on, yet remains constrained by **Historical Tethering**: imagination is typicall y initialized from observed states. This creates a learning asymmetry, where the world model's manifold discovery outpaces the policy's sparse-reward optimizati on. We propose **Mind Dreamer (MD)**, a framework that instantiates **Active Cau sal Intervention** to transcend Markovian continuity. MD reformulates discovery as the minimization of a global Relay Expected Free Energy.
Instead of initializing from historical data, it draws initial states from an ad versarial generator $s_0 \sim p_{gen}(\cdot)$, creating non-continuous **latent jumps** to epistemic blind spots that are physically plausible yet cognitively c hallenging.
We derive **Relay Value Function** and **Relay Uncertainty Function** to resolve the credit assignment paradox across these spatial ruptures.

Treating synthesized anchors as interventional intermediary states, these potentials propagate pragmatic and epistemic value through Bellman-style backups. Notably, we prove that uncertainty propagation across discontinuities necessitates a quadratic discount γ^2, establishing a formal epistemic horizon. Theoretically, MD approximates a variance-minimizing importance sampler that expands the manifold's spectral gap, reducing the hitting time to critical bottleneck states. Empirically, MD achieves a **1.67$\times$ average speedup** over DreamerV3 on DeepMind Control Suite, reaching **8.8$\times$** in sparse-reward tasks.
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Attention Implements the Fisher Geometry of Exponential Families
Bodie Rubacher
Softmax attention normalizes scores, and Bayes' rule normalizes log prior plus log likelihood. For finite latent symbols with exponential-family observations, we show that one attention head can implement the Bayes posterior and posterior means exactly, and that the posteriors representable by a single head are precisely log-linear. The standard exponential-family duality identity rewrites the likelihood as a negative Bregman divergence in mean/sufficient-statistic space; our attention-specific contribution is to use this identity to characterize when Bayes-aligned attention admits one globally shared quadratic metric, proving that this happens exactly when the dual potential is quadratic. When curvature varies, we give a multi-head local-curvature atlas with approximation and head-count bounds, and we extend the picture to in-context estimation through plug-in consistency, finite-sample stability, and an optimizer-agnostic converse from excess log-loss to approximate key-subspace alignment. Controlled Gaussian, Bernoulli, and Poisson ICE diagnostics illustrate these regimes, while the exact theorems remain scoped to finite discrete latent classes and suggest testable, not universal, predictions for larger learned transformers.
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Seizure-Semiology-Suite(S^3): A Clinically Multimodal Dataset, Benchmark, and Models for Seizure Semiology Understanding
Lina Zhang, Tonmoy Monsoor, Peizheng Li, Jiarui Cui, Xinyi Peng, Chong Han, Prateik Sinha, Siyuan Dai, Jessica Nichole Pasqua, Colin M McCrimmon, Weiting Liu, Hailey Marie Miranda, Bing Hu, Xiangting Wu, Tengyou Xu, Chunhan Li, Jiaye Tian, Jiarui Tang, Detao Ma, Lingye Kong, Junnan Lyu, Jungang Li, Yan Zan, Junhua Huang, Rajarshi Mazumder, Vwani Roychowdhury
While Multimodal Large Language Models (MLLMs) have demonstrated remarkable proficiency in general video understanding, their capacity to interpret involuntary, and spatio-temporally evolving pathologic motor behaviors such as seizure semiology remains largely untested. To address this gap, we introduce Seizure-Semiology-Suite (S^3), a clinically grounded dataset and benchmark for fine-grained, structured seizure semiology understanding. The dataset includes 438 seizure videos annotated with over 35,000 dense labels covering 20 ILAE-defined semiological features. Building on this dataset, we propose a seven-task hierarchical benchmark that systematically evaluates MLLMs from low-level visual perception to temporal sequencing, narrative report generation, and seizure diagnosis. To enable clinically meaningful evaluation of generated reports, we further introduce the Report Quality Index for Seizure Semiology (Seizure-RQI). Extensive baselines across 11 open-weight MLLMs reveal systematic weaknesses in laterality reasoning, temporal localization, symptom sequencing, and clinically faithful reporting. We show that seizure-specific fine-tuning substantially improves performance across tasks, and that a two-stage neuro-symbolic framework achieves an F1 score of 0.96 on epileptic versus non-epileptic seizure classification. Seizure-Semiology-Suite establishes a rigorous benchmark for evaluating multimodal models in safety-critical medical video understanding and guides the development of clinically reliable, domain-adaptive multimodal intelligence. Our code is publicly available at \href{https://github.com/LinaZhangUCLA/SeizureSemiologySuite}{SeizureSemiologySuite}.
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Last-iterate Convergence of ADMM on Multi-affine Quadratic Equality Constrained Problem

Yutong Chao, Michal Ciebielski, Jalal Etesami, Majid Khadiv
In this paper, we study a class of non-convex optimization problems known as multi-affine quadratic equality constrained problems, which appear in various applications--from generating feasible force trajectories in robotic locomotion and manipulation to training neural networks. Although these problems are generally non-convex, they exhibit convexity or related properties when all variables except one are fixed. Under mild assumptions, we prove that the alternating direction method of multipliers (ADMM) converges when applied to this class of problems. Furthermore, when the "degree" of non-convexity in the constraints remains within certain bounds, we show that ADMM achieves a linear convergence rate. We validate our theoretical results through practical examples in robotic locomotion.
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MEAL: A Benchmark for Continual Multi-Agent Reinforcement Learning
Tristan Tomilin, Luka van den Boogaard, Samuel Garcin, Constantin Ruhdorfer, Bram Grooten, Fabrice Kusters, Yali Du, Andreas Bulling, Mykola Pechenizkiy, Meng Fang
Benchmarks play a central role in reinforcement learning (RL) research, yet their computational constraints often shape what is studied. Despite the motivation of lifelong learning, most continual RL papers consider only 3–10 sequential tasks, as CPU-bound environments make longer sequences impractical. Meanwhile, continual learning in cooperative multi-agent settings remains largely unexplored. To address these gaps, we introduce **MEAL** (**M**ulti-agent **E**nvironments for **A**daptive **L**earning), the first benchmark for continual multi-agent RL. By leveraging JAX and GPU acceleration, MEAL enables training on sequences of 100 tasks on a single GPU in a few hours. We find that long task sequences reveal failure modes that do not appear at smaller scales.
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Tractable Expected Information Gains for Exponential Family Posteriors
Rik Knowles, Tom Rainforth
We investigate which models admit a collapse of the expected information gain (EIG) and its derivative from a doubly intractable to a singly intractable expression. We prove that a sufficient condition is that the posterior distribution belongs to an exponential family (EF) and depends on the experimental design and data only through its natural parameters, and derive corresponding singly intractable and unbiased estimators for the EIG and its (reparameterized) gradient. We further show that this is achieved when using a likelihood of an analogous form and any arbitrary prior. This is complemented by a theoretical analysis of certain degenerate behaviors that may arise when optimizing the EIG for EF-modeled experiments. Finally, we empirically demonstrate the benefits of using our singly intractable estimators, showing superior convergence rates, and substantial performance gains for sequential design problems compared to using standard nested estimators.
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Softmax as Linear Attention in the Large-Prompt Regime: a Measure-based Perspective
Etienne Boursier, Claire Boyer
Softmax attention is a central component of transformer architectures, yet its nonlinear structure poses significant challenges for theoretical analysis. We develop a unified, measure-based framework for studying single-layer softmax attention under both finite and infinite prompts. For i.i.d. centered Gaussian inputs, we lean on the fact that the softmax operator converges in the infinite-prompt limit to a linear operator acting on the underlying input-token measure. Building on this insight, we establish non-asymptotic concentration bounds for the output and gradient of softmax attention, quantifying how rapidly the finite-prompt model approaches its infinite-prompt counterpart, and prove that this concentration remains stable along the entire training trajectory in general in-context learning settings with sub-Gaussian tokens.
In the case of in-context linear regression, we use the tractable infinite-prompt dynamics to analyze training at finite prompt length. Our results allow optimization analyses developed for linear attention to transfer directly to softmax a

ttention when prompts are sufficiently long, showing that large-prompt softmax a ttention inherits the analytical structure of its linear counterpart. This, in t urn, provides a principled and broadly applicable toolkit for studying the train ing dynamics and statistical behavior of softmax attention layers in large promp t regimes.
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MobileFusion: Mobile-Friendly Infrared and Visible Image Fusion via Structural R e-parameterization

Yufa Duan, Jialing Huang, Yingying Wang, Weimin Cai, Xinghao Ding, Xiaotong Tu

Deep neural networks have recently advanced infrared and visible image fusion (I VIF), but most existing methods rely on sophisticated yet redundant designs, whi ch hinder real-time deployment on mobile devices with limited compute and memory . In this paper, we present MobileFusion, an extremely lightweight and effective convolutional framework that achieves high-quality fusion under strict resource constraints. MobileFusion leverages a novel re-parameterizable multi-branch con volution module to promote cross-modal interactions during training while collap sing into a single-path operator for fast inference. It further incorporates a l ightweight attention module to enhance context awareness, together with a re-par ameterized feed-forward network to improve feature expressiveness. Extensive exp eriments demonstrate that MobileFusion delivers a favorable trade-off between fu sion quality and computational efficiency, enabling real-time and high-quality I VIF on resource-constrained platforms. The source code is available at https://g ithub.com/sucessfullys/MobileFusion.
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MatchFixAgent: Language-Agnostic Autonomous Repository-Level Code Translation Va lidation and Repair

Ali Reza Ibrahimzada, Brandon Paulsen, Reyhaneh Jabbarvand, Joey Dodds, Daniel K roening

Code translation transforms source code from one programming language (PL) to an other. Validating the functional equivalence of translation and repairing, if ne cessary, are critical steps in code translation. Existing automated validation a nd repair approaches struggle to generalize to many PLs due to high engineering overhead, and they rely on existing and often inadequate test suites, which resu lts in false claims of equivalence and ineffective translation repair. To bridge this gap, we develop MatchFixAgent, a large language model (LLM)-based, PL-agno stic framework for equivalence validation and repair of translations. MatchFixAg ent features a multi-agent architecture that divides equivalence validation into several sub-tasks to ensure thorough and consistent semantic analysis of the tr anslation. We compare MatchFixAgent's validation and repair results with four re pository-level code translation techniques. Our results demonstrate that MatchFi xAgent produces (in)equivalence verdicts for 99.2% of translation pairs, with the same equivalence validation result as prior work on 72.8% of them. When Ma tchFixAgent's result disagrees with prior work, we find that 60.7% of the time MatchFixAgent's result is actually correct. In addition, we show that MatchFixA gent can repair 50.6% of inequivalent translation, compared to prior work’s $1 8.5$%.
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SCoA: Revisiting Domain Generalized Object Detection with Style-Conditioned Adap tation

Han Jiang, Wenfei Yang, Tianzhu Zhang, Yongdong Zhang

Domain generalized object detection (DGOD) aims to train an object detector on a single source domain and generalize it to unseen target domains. Recent adva nces in DGOD have increasingly exploited vision foundation models (VFMs) via par ameter-efficient finetuning strategies. However, existing approaches typically a dapt VFMs with fixed, style-agnostic parameters, overlooking that different visu al styles may induce distinct task discrepancies. To address this challenge, we propose SCoA, a novel Style Conditioned Adaptation framework for dynamic, st yle-aware task compensation. Specifically, we introduce a Spectral Style Modeli ng (SSM) module that preserves local style cues via a memory-based mechanism, en abling diverse style characterization from a single source domain. Conditioned o

n the extracted style signals, we design a Mixture-of-Tokens Adaptation (MTA) me
chanism, which maintains multiple adaptation tokens and dynamically routes each
sample to an optimal combination of tokens, thereby explicitly modeling style-de
pendent task mismatches. In addition, we propose a Style-Conditioned Query Refin
ement (SCQR) module that injects style information into object queries, enablin
g a style-aware detection head. By jointly integrating these components, SCoA al
lows the model to follow style-specific adaptation trajectories, achieving effec
tive and flexible task compensation for VFM-based DGOD. Extensive experiments de
monstrate that the proposed SCoA achieves state-of-the-art performance across t
wo challenging scenarios.
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Angel or Demon: Investigating the Plasticity Interventions' Impact on Backdoor T
hreats in Deep Reinforcement Learning
Oubo Ma, Ruixiao Lin, Yang Dai, Jiahao Chen, Chunyi Zhou, Linkang Du, Shouling J
i
Extensive research has highlighted the severe threats posed by backdoor attacks
to deep reinforcement learning (DRL).
However, prior studies primarily focus on vanilla scenarios, while plasticity in
terventions have emerged as indispensable built-in components of modern DRL agen
ts.
Despite their effectiveness in mitigating plasticity loss, the impact of these i
nterventions on DRL backdoor vulnerabilities remains underexplored, and this lac
k of systematic investigation poses risks in practical DRL deployments.
To bridge this gap, we empirically study 14,664 cases integrating representative
interventions and attack scenarios.
We find that only one intervention (i.e., *SAM*) exacerbates backdoor threats, w
hile other interventions mitigate them.
Pathological analysis identifies that the exacerbation is attributed to backdoor
gradient amplification, while the mitigation stems from activation pathway disr
uption and representation space compression.
From these findings, we derive two novel insights:
(1) a conceptual framework *SCC* for robust backdoor injection that deconstructs
the mechanistic interplay between interventions and backdoors in DRL, and (2) a
bnormal loss landscape sharpness as a key indicator for DRL backdoor detection.
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ConTSG-Bench: A Unified Benchmark for Conditional Time Series Generation
Shaocheng Lan, Shuqi Gu, Zhangzhi Xiong, Kan Ren
Conditional time series generation plays a critical role in addressing data scar
city and enabling causal analysis in real-world applications.
Despite its increasing importance, the field lacks a standardized and systematic
benchmarking framework for evaluating generative models across diverse conditio
ns.
To address this gap, we introduce the **Con**ditional **T**ime **S**eries **G**e
neration **Bench**mark (ConTSG-Bench).
ConTSG-Bench comprises a suite of large-scale, well-aligned datasets spanning di
verse conditioning modalities and levels of semantic abstraction, enabling syste
matic evaluation of representative generation methods across these dimensions wi
th a comprehensive suite of metrics for generation fidelity and condition adhere
nce.
Both the quantitative benchmarking and in-depth analyses of conditional generati
on behaviors have revealed the traits and limitations of the current approaches,
highlighting critical challenges and promising research directions,
particularly with respect to precise structural controllability and downstream t
ask utility under complex conditions.
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MAPS: Memory-Aware Predictive Scheduling Framework for Large Language Model Serv
ing
Tiancheng Zhang, Yulin Chen, Yunfeng Zhao, Shaoyuan Huang, Cheng Zhang, Xiaofei
Wang
The surge of large language model (LLM) applications on personal devices imposes

massive, bursty workloads on cloud serving infrastructure. While prefill-decode disaggregation improves throughput and scalability, memory-bound decode instances often suffer from persistent load imbalance, as output lengths are unknown when requests arrive at the cloud. To address this, we propose **MAPS**, a **M**emory-**A**ware **P**redictive **S**cheduling framework tailored for disaggregated LLM serving. MAPS performs device-assisted speculative output length prediction overlapped with cloud-side prefilling, incurring negligible latency overhead. To handle generation uncertainty, MAPS applies uncertainty-aware calibration to derive output-length upper bounds with target coverage, enabling safe scheduling decisions. Building on these bounds, MAPS employs a hierarchical global-local scheduling strategy to mitigate inter-decoder queue buildup and intra-decoder head-of-line blocking. Extensive experiments on two real-world workloads and two LLMs show that MAPS significantly outperforms three state-of-the-art systems, reducing average end-to-end latency by 42.6% and tail latency by up to 84.8%.

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OmniSapiens: A Foundation Model for Social Behavior Processing via Heterogeneity-Aware Relative Policy Optimization

Keane Ong, Sabri Boughorbel, Luwei Xiao, Chanakya Ekbote, Wei Dai, Ao Qu, Jingyao Wu, Rui Mao, Ehsan Hoque, Erik Cambria, Gianmarco Mengaldo, Paul Pu Liang

Socially intelligent AI systems must reason across diverse human behavioral tasks and generalize to new social contexts. However, behavioral data is inherently heterogeneous, comprising diverse modalities and prediction targets that produce uneven training signals across samples, creating imbalanced learning dynamics that challenge existing AI models. To address this, we develop OmniSapiens-7B 2.0, a foundation model for social behavior processing that explicitly addresses learning from heterogeneous behavioral data. This is enabled through Heterogeneity-Aware Relative Policy Optimization, a new RL method that rebalances learning signals across samples by approximating each sample’s contribution to the policy update and using these estimates to drive geometrically centered, inertially smoothed advantage modulation for stable training. OmniSapiens-7B 2.0 achieves the best and most consistent performance across 10 behavioral tasks, while also attaining the best performance on all five held-out benchmarks, with gains of up to +12.02% and +9.37% respectively. Furthermore, it demonstrates more consistent and interpretable reasoning traces, supporting reliable real-world behavioral applications. Our model is available at https://github.com/MIT-MI/human_behavior_atlas.

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Awakening Visual Reasoning: Mitigating Post-Training Failure in Vision-Text Compression

Xing Xi, Liyao Li, Hao Chen, Ningtao Wang, Peixian Chen, Peilin Tong, Xing Fu, Yu Cheng, Haobo Wang, Gang Chen, Junbo Zhao, Ronghua Luo

Vision-Text Compression (VTC) offers a scalable path for long-context multimodal modeling by rendering textual data into dense visual tokens. While recent Vision-Language Models (VLMs) demonstrate high decoding fidelity (OCR) on such inputs, they exhibit a severe reasoning gap: models that reason robustly on native text often fail on visually compressed equivalents, particularly in long-range retrieval and multi-step deduction. We identify a phenomenon of post-training transfer failure, where standard supervised fine-tuning and reinforcement learning on visual prompts yield marginal gains compared to their textual counterparts. To address this, we propose CoRe (Coordinated Reasoning), a training framework that enforces lockstep consistency between the reasoning processes of textual and visual modalities. By treating the text-conditioned policy as a dynamic anchor, CoRe aligns the visual-conditioned policy via step-wise distribution matching, seamlessly integrating into both SFT and RL pipelines. Extensive evaluations across mathematical reasoning, long-context memory, and tabular retrieval benchmarks show that CoRe significantly outperforms standard visual post-training, recovering up to 70% of the performance gap relative to the textual upper bound and effectively activating latent reasoning capabilities in the compressed visual modality.

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Threat2Traffic: Multi-Agent Environment Synthesis for Malware Traffic Generation from Threat Intelligence
Haoyang Chen, Chang Liu, Zhong Guan, Junzheng Shi, Gaopeng Gou, Gang Xiong
Data-driven cybersecurity research is fundamentally constrained by the scarcity of labeled datasets, yet acquiring authentic, large-scale malware traffic remains bottlenecked by obsolescent public datasets, unscalable manual construction, and inflexible sandboxes that fail to satisfy the sample-specific dependencies required for malware to exhibit malicious behavior. Threat intelligence documents these dependencies, and LLM agents offer a path to extract them for environment construction, yet directly applying such agents faces two challenges: input-side ambiguity and output-side fragility. In this paper, we propose Threat2Traffic, a multi-agent framework that extracts sample-specific dependencies from threat intelligence, reconstructs tailored environments, and captures malware traffic. To address input-side ambiguity, it formulates dependency extraction as structured multi-agent deliberation over an evidence graph. To overcome output-side fragility, it incorporates invariant-guided synthesis with dual-layer validation under syntactic and semantic constraints. Evaluated on 1,077 samples across eight malware families, Threat2Traffic achieves 83.1\% reproduction success, highlighting its effectiveness for scalable and realistic malware traffic generation. We release the core source code and traffic dataset at https://github.com/apos3637/Threat2Traffic
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On the Power of (Approximate) Reward Models for Inference-Time Scaling: Sequential Monte Carlo and Beyond
Youheng Zhu, Yiping Lu
Inference-time scaling has recently emerged as a powerful paradigm for improving the reasoning capability of large language models. Among various approaches, \emph{Sequential Monte Carlo (SMC)} has become a particularly important framework, enabling iterative generation, evaluation, rejection, and resampling of intermediate reasoning trajectories. A central component in this process is the \emph{reward model}, which evaluates partial solutions and guides the allocation of computation during inference. However, in practice, true reward models are never available. All deployed systems rely on \emph{approximate reward models}, raising a fundamental question: \emph{Why and when do approximate reward models suffice for effective inference-time scaling?} In this work, we provide a theoretical answer. We identify the \emph{Bellman error} of the approximate reward model as the key quantity governing the effectiveness of SMC-based inference-time scaling. For a reasoning process of length T, we show that if the Bellman error of the approximate reward model is bounded by $O(1/T)$, then combining this reward model with SMC reduces the computational complexity of reasoning from exponential in T to polynomial in T. This yields an \emph{exponential improvement} in inference efficiency despite using only approximate rewards.
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Geometric Embedding Alignment via Curvature Matching in Transfer Learning
Sung Moon Ko, Jaewan Lee, Sumin Lee, Soorin Yim, Sehui Han
Geometrical interpretations of deep learning models offer insightful perspectives into their underlying mathematical structures. In this work, we introduce a novel approach that leverages differential geometry, particularly concepts from Riemannian geometry, to integrate multiple models into a unified transfer learning framework. By aligning the Ricci curvature of latent space of individual models, we construct an interrelated architecture, namely Geometric Embedding Alignment via cuRvature matching in transfer learning (GEAR), which ensures comprehensive geometric representation across datapoints. This framework enables the effective aggregation of knowledge from diverse sources, thereby improving performance on target tasks. We evaluate our model on 23 molecular task pairs and demonstrate significant performance gains over existing benchmark models—achieving improvements of at least 14.4% under random splits and 8.3% under scaffold splits.
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RED-HDP-HMM: Observation-Dependent Durations for Bayesian Nonparametric Sequential Models

Miko■aj S■upi■ski, Piotr Lipinski
The Hierarchical Dirichlet Process Hidden Markov Model (HDP-HMM) is a Bayesian n onparametric extension of the classical Hidden Markov Model, well-suited for lea rning from (spatio-)temporal data. To relax the restrictive geometric assumption on state durations, the HDP Hidden Semi-Markov Model was introduced. However, b oth models assume stationary state durations, which limits their expressive powe r. In this work, we extend the HDP-HMM framework by incorporating recurrent expl icit duration modeling, resulting in a more general and flexible model: the Recu rrent Explicit Duration HDP-HMM (RED-HDP-HMM). We propose a Gibbs sampling metho d for efficient inference in this model. Empirical results on both synthetic and real-world segmentation tasks demonstrate that RED-HDP-HMM consistently outperf orms the disentangled sticky HDP-HMM and the standard sticky HDP-HMM.
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Better, Faster: Harnessing Self-Improvement in Large Reasoning Models
Qihuang Zhong, Liang Ding, Juhua Liu, Bo Du, Leszek Rutkowski, Dacheng Tao
Self-improvement training enables the large reasoning models (LRMs) to improve t hemselves by self-generating reasoning trajectories as training data without ext ernal supervision. However, we find that this method often falls short in comple x reasoning tasks and even leads to model collapse. Through a series of prelimin ary analyses, we reveal two problems: (1) data imbalance, where most training sa mples are simple, but the challenging yet crucial samples are scarce; (2) overth inking, where many undesired samples with redundant reasoning steps are used for self-training. To this end, we propose HSIR, which effectively Harnesses Self-I mprovement in large Reasoning models via two simple-yet-effective approaches. Sp ecifically, HSIR introduces a verify-then-exit sampling strategy to mitigate dat a imbalance by efficiently collecting more accurate solutions for difficult quer ies, and designs an Intrinsic Diversity score to quantify overthinking and filte r out the undesired solutions. We apply HSIR to various post-training paradigms, among which we further propose H-GRPO, an enhanced GRPO algorithm that leverage s the intrinsic diversity as an external reward to encourage concise and diverse reasoning via reinforcement learning. Extensive results show that HSIR not only effectively enhances the reasoning performance, i.e., bringing up to +10.9% ave rage performance gains, but also significantly improves the reasoning efficiency by reducing up to 42.4% relative inference overhead.
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SCOUT: Active Information Foraging for Long-Text Understanding with Decoupled Ep istemic States
Zhenliang Zhang, Wenqing Wang, Yong Hu, Yaming Yang, Jiaheng Gao, Chen Shen, Xia ojun Wan
Long-Text Understanding (LTU) at million-token scale requires balancing reasonin g fidelity with computational efficiency. Frontier long-context LLMs can process millions of token contexts end-to-end, but they suffer from high token consumpt ion and attention dilution. In parallel, specialized LTU agents often sacrifice fidelity through task-agnostic abstractions like graph construction or indexing. We identify a key insight for LTU: query-relevant information is typically spar se relative to the full document, so effective reasoning should rely on a query- sufficient subset rather than the entire context. To address this, we propose SC OUT, a new paradigm for LTU that **shifts from passive processing to active info rmation foraging**. It treats the document as an explorable environment and answ ers from a compact, provenance-grounded epistemic state. Guided by state-level g ap diagnosis, SCOUT adaptively alternates between coarse-to-fine exploration and anchored state updates that progressively *contract* its epistemic state toward query sufficiency. Experiments show that SCOUT matches state-of-the-art proprie tary models while reducing token consumption by up to 8 times. Moreover, SCOUT r emains stable as context length scales, substantially alleviating the practical cost--capability trade-off in long-context reasoning. Resources are available at our [Project Page](https://xavierzhang2002.github.io/scout-page/).
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Test-Time Training with KV Binding Is Secretly Linear Attention
Junchen Liu, Sven Elflein, Or Litany, Zan Gojcic, Ruilong Li

Test-time training (TTT) with KV binding as sequence modeling layer is commonly interpreted as a form of online meta-learning that memorizes a key–value mapping at test time. However, our analysis reveals multiple phenomena that contradict this memorization-based interpretation. Motivated by these findings, we revisit the formulation of TTT and show that a broad class of TTT architectures can be expressed as a form of learned linear attention operator. Beyond explaining previously puzzling model behaviors, this perspective yields multiple practical benefits: it enables principled architectural simplifications, admits fully parallel formulations that preserve performance while improving efficiency, and provides a systematic reduction of diverse TTT variants to a standard linear attention form. Overall, our results reframe TTT not as test-time memorization, but as learned linear attention with enhanced representational capacity. Project page: https://research.nvidia.com/labs/sil/projects/tttla/.

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Is the Last Layer Sufficient for Uncertainty Quantification?

Joseph Wilson, Chris van der Heide, Liam Hodgkinson, Fred Roosta

Epistemic uncertainty quantification (UQ) for deep neural networks (DNNs) is a requirement for safe adoption of AI in mission-critical settings. Several leading methods for UQ linearize DNNs to form Bayesian Generalized Linear Models (GLMs), where epistemic uncertainty is modeled via the predictive posterior distribution. Linearizing around the parameters of the *final connected layer* of a DNN is a commonly used approximation for reducing the computational burden of such GLMs, though it is often believed to come at the cost of degraded performance. In this work, we compare GLMs arising from full-network and last-layer linearization using both theoretical and empirical approaches. We first employ tools from random matrix theory to conduct a theoretical comparison; this analysis reveals no meaningful improvement in the UQ capabilities of full linearization. Coupled with a large-scale empirical evaluation across a range of modern machine learning tasks, we arrive at the following conclusion: a last-layer approximation yields comparable UQ performance while offering substantially improved computational efficiency.

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Bio-Vision-Inspired Spiking Neural Networks for Object Detection with Event Cameras

Dongyang Ma, Zhengyu Ma, Yifan Huang, Chenlin Zhou, Wei Zhang, Yonghong Tian

Retina-like event cameras and brain-inspired Spiking Neural Networks (SNNs) demonstrate exceptional energy efficiency through bio-inspired sensing and computation. While SNNs are naturally well-suited to the asynchronous nature of event data, their practical applications face the following challenges: sensitivity to noise, dense representations that disrupt spike pathways, and insufficient multi-scale feature perception. To address the aforementioned challenges, we propose a bio-vision-inspired object detection method motivated by biological (bio) vision systems. First, at the micro level, this paper proposes a noise-filtering STATNF-Neuron architecture to address the current sensitivity of basic neurons to noise. Based on STATNF-Neurons, the paper introduces two bio-vision-inspired macro-structures: Events-to-Spikes Representation (E2S), which preserves spiking characteristics while mimicking the memory and noise-filtering abilities of retinal neurons; Bidirectional Multi-Scale Spiking Network (BiSNet), which simulates cortical information flow pathways to integrate multi-scale features in both directions, enhancing the network's ability to perceive information at multiple scales. Extensive experiments show that the proposed bio-vision-inspired method achieving state-of-the-art performance. Notably, it reaches 96.1% accuracy on NCAR, 63.5% mAP(50) on N-Caltech101, and 69.1% mAP(50) on Gen1.

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Detecting Contextual Hallucinations in Large Language Models with Frequency-Aware Attention

Siya Qi, Yudong Chen, Runcong Zhao, Qinglin Zhu, Zhanghao Hu, Wei Liu, Yulan He, Zheng Yuan, Lin Gui

Hallucination detection is critical for ensuring the reliability of large language models (LLMs) in context-based generation. Prior work has explored intrinsic

signals available during generation, among which attention offers a direct view of grounding behavior. However, existing approaches typically rely on coarse summaries that fail to capture fine-grained instabilities in attention. Inspired by signal processing, we introduce a frequency-aware perspective on attention by analyzing its variation during generation. We model attention distributions as discrete signals and extract high-frequency components that reflect rapid local changes in attention. Our analysis reveals that hallucinated tokens are associated with high-frequency attention energy, reflecting fragmented and unstable grounding behavior. Based on this insight, we develop a lightweight hallucination detector using high-frequency attention features. Experiments on the RAGTruth and HalluRAG benchmarks show that our approach achieves performance gains over verification-based, internal-representation-based, and attention-based methods across models and tasks.

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UniSparse: Combining Weight Pruning and Spike Sparsification in Spiking Neural Networks

Xinyu Shi, Tong Bu, Zhaofei Yu

Spiking Neural Networks (SNNs) offer a notable energy-saving advantage compared to Artificial Neural Networks (ANNs) when deployed on neuromorphic hardware. While recent SNNs achieve superior performance using larger and deeper backbones, this comes at a cost of diminishing their energy-saving benefits. In this paper, we propose UniSparse, a unified sparsification framework for enhancing the energy efficiency of SNNs. We demonstrate that the affine parameters in batch normalization also serve as the learnable threshold of its subsequent spiking neurons. Based on this, we propose a novel spike sparsification method that reduces firing rate by constraining the affine parameters. As a complement to spike sparsification, we propose a weight pruning method based on the same energy constraint, which can be naturally integrated with spike sparsification. Experimental results demonstrate that UniSparse achieves a state-of-the-art trade-off between accuracy and energy efficiency across models and datasets. The sparsified ResNet-18 model requires only 7.04M SOPs for inference to achieve 92.38\% accuracy on the CIFAR-10 dataset. Our work highlights the great potential of deep SNNs in improving energy efficiency.

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MEnvAgent: Scalable Polyglot Environment Construction for Verifiable Software Engineering

Chuanzhe Guo, Jingjing Wu, Sijun He, Yang Chen, Zhaoqi Kuang, Shilong Fan, Bingjin Chen, Siqi Bao, Jing Liu, Hua Wu, Qingfu Zhu, Wanxiang Che, Haifeng Wang

The evolution of Large Language Model (LLM) agents for software engineering (SWE) is constrained by the scarcity of verifiable datasets, a bottleneck stemming from the complexity of constructing executable environments across diverse languages. To address this, we introduce **MEnvAgent**, a **M**ulti-language framework for automated **Env**ironment construction that facilitates scalable generation of verifiable task instances. MEnvAgent employs a multi-agent Planning-Execution-Verification architecture to autonomously resolve construction failures and integrates a novel Environment Reuse Mechanism that reduces computational overhead by incrementally patching historical environments. Evaluations on MEnvBench, a new benchmark comprising 1,000 tasks across 10 languages, demonstrate that MEnvAgent outperforms baselines, improving Fail-to-Pass (F2P) rates by **8.6%** while reducing time costs by **43%**. Additionally, we demonstrate the utility of MEnvAgent by constructing MEnvData-SWE, the largest open-source polyglot dataset of realistic verifiable Docker environments to date, alongside solution trajectories that enable consistent performance gains on SWE tasks across a wide range of models.

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PRM-PBE: Process Reward Model for Reinforcement Learning in Programming-by-Example

Yue Fang, Zhi Jin, Jie An, Hongshen Chen, Jiangmeng Li, Xiaohong Chen, Naijun Zhan

Programming-by-Example (PBE), as a typical few-shot inductive reasoning paradigm

, aims to synthesize corresponding algorithms from a set of input-output example
s. Although Large Language Models (LLMs) have demonstrated strong program synthe
sis potential, they still remain ineffective when handling complex PBE tasks. Sp
ecifically, LLMs often struggle to accurately grasp the underlying intent of exa
mples, resulting in synthesized programs that either partially satisfy the examp
les or completely deviate from the target. To address these limitations, we intr
oduce a process-supervised reinforcement learning method that provides fine-grai
ned feedback during the synthesis process, improving the ability of LLMs to capt
ure the intended behavior of provided examples. Firstly, we develop a reasoning
tree construction method that is used to build a PBE process supervision dataset
. Subsequently, we train a process reward model through preference learning to e
valuate the effectiveness of reasoning steps. Finally, we introduce a curriculum
 learning strategy based on the difficulty of PBE tasks, using Proximal Policy O
ptimization (PPO) to optimize the model. Experimental results on representative
PBE benchmarks show that our approach achieves an average pass rate of 56.61\%,
significantly outperforming the state-of-the-art baseline by 8.73\%.
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A Call to Lagrangian Action: Learning Population Mechanics from Temporal Snapsho
ts

Vincent Guan, Lazar Atanackovic, Kirill Neklyudov

The population dynamics of molecules, cells, and organisms are governed by a num
ber of unknown internal and external forces. In the last decade, population dyna
mics have predominately been modeled with Wasserstein gradient flows. However, s
ince gradient flows minimize free energy, they fail to capture important dynamic
al properties, such as periodicity. In this work, we propose a change in perspec
tive by considering population dynamics that minimize Wasserstein Lagrangian act
ion, rather than free energy. As our main theoretical contributions, we derive t
he Hamiltonian equations of motion from the principle of least population-level
action and we show that these mechanics encompass classical mechanics, quantum m
echanics, and gradient flows. We further leverage the Hamiltonian perspective to
 propose an algorithm that learns the population mechanics from observed margina
ls, without specifying the Lagrangian. We demonstrate that by directly learning
the population mechanics, our method forecasts and interpolates unseen marginals
 without a reference process, and outperforms gradient flow and flow matching me
thods across a wide range of real and simulated experiments.
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From Historical Patches to Repair Plans: Outcome-Conditioned Reasoning for Repos
itory-Level Program Repair

Chenglin Li, Yisen Xu, Zehao Wang, Shin Hwei Tan, Tse-Hsun Chen

Repository-level automated program repair (APR) requires long-horizon reasoning
over interdependent decisions. However, most LLM-based approaches reconstruct re
pair reasoning independently for each issue, failing to reuse successful pattern
s from prior repairs, even though real-world repositories contain many related i
ssues with shared structure or constraints. Existing methods typically rely on f
orward exploration, which operates under outcome uncertainty, incurs substantial
 inference-time overhead, and can drift from the final correct patch. We propose
 Conditional Reasoning Distillation (ConRAD), which leverages in-repository reso
lved issues by reconstructing repair reasoning backward from verified patches an
d distilling outcome-consistent, stage-wise repair reasoning plans. Injected at
inference time, these plans guide fault localization and patch generation, repla
cing open-ended exploration with constrained inference without fine-tuning or se
arch. On SWE-Bench Lite, ConRAD improves Pass@1 by 10.4\% (GPT-4o), 8.6\% (DeepS
eek-V3), and 10.3\% (GPT-5), demonstrating a scalable inference-time alternative
 to forward exploration for long-horizon APR.
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SpecPrune-VLA: Accelerating Vision-Language-Action Models via Action-Aware Self-
Speculative Pruning

Hanzhen Wang, Jiaming Xu, Yushun Xiang, Jiayi Pan, Yongkang Zhou, Yong-Lu Li, Gu
ohao Dai

Pruning is a typical acceleration technique for compute-bound models by removing

computation on unimportant values. Recently, it has been applied to accelerate Vision-Language-Action (VLA) model inference. However, existing acceleration methods focus on local information from the current action step and ignore the global context, leading to $>$20\% success rate drop and limited speedup in some scenarios. In this paper, we point out \textbf{spatial-temporal consistency} in VLA tasks: input images in consecutive steps exhibit high similarity, and propose the key insight that token selection should combine local information with global context of the model. Based on this, we propose ***SpecPrune-VLA***, a training-free, two-level pruning method with heuristic control.
(1) Action-level static pruning. We leverage global history and local attention to statically reduce visual tokens per action.
(2) Layer-level dynamic pruning. We prune tokens adaptively per layer based on layer-wise importance.
(3) Lightweight action-aware controller: We classify actions as coarse- or fine-grained by the speed of the end effector and adjust pruning aggressiveness accordingly. Extensive experiments show that SpecPrune-VLA achieves up to 1.57× speedup in LIBERO simulation and 1.70× on real-world tasks, with negligible success rate degradation.
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Gradient Transformer: Learning to Generate Updates for LLMs
Binh-Nguyen Nguyen, Khang Tran, Hai Phan, Issa Khalil
Many organizations lack computational resources to fine-tune large language models (LLMs) on private (unshareable) data for better utility, while fine-tuning tiny language models (TinyLMs) alone performs poorly. To address this bottleneck, we propose a data-free knowledge distillation framework that generates LLM update vectors based on TinyLMs fine-tuned on private data. An update vector is a vector of parameter changes from an initial model to its fine-tuned version on a dataset, capturing the effect of cumulative gradient steps during fine-tuning. The key idea of our framework is a novel **Gradient Transformer** that transforms TinyLM's update vectors into LLM's update vectors. As derived from shadow datasets, $\texttt{Grad-Transformer}$ captures the correlation between TinyLM and LLM update vectors, enabling third-party providers to generate LLM update vectors given the organization's TinyLM update vectors without accessing the organization's private data. The framework supports multi-organization collaboration to jointly update LLMs, improving performance and cost-efficiency. Extensive experiments across language modeling and reasoning tasks show that $\texttt{Grad-Transformer}$ remarkably outperforms state-of-the-art knowledge distillation baselines, even under strict differential privacy protection.
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GR-LoRA: Gradient-Recycling Low-Rank Adaptation for Class-Incremental Learning
Yipeng Lin, Fengqiang Wan, Yang Yang
Pre-trained models with parameter-efficient fine-tuning have shown strong effectiveness in Class-Incremental Learning (CIL), which seeks to balance model plasticity and stability. In this context, orthogonality constraints can significantly enhance model stability, yet their reliance on subspace inevitably compromises model plasticity over long tasks. To address this, we propose Gradient-Recycling Low-Rank Adaptation (GR-LoRA), which reconciles stability and plasticity by recycling the gradients discarded in orthogonal projection. Specifically, GR-LoRA recycles post-decomposition non-orthogonal gradient components into task-specific lightweight modules and selects optimal module via entropy to improve plasticity, while incorporating local and global mismatch suppression to preserve stability by synthesizing out-of-distribution representations across all tasks. Theoretical analysis confirms that this recycling strategy preserves stability and improves plasticity. Experimental results from multiple CIL benchmarks verify the effectiveness and general applicability of GR-LoRA.
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Blending Supervised and Reinforcement Fine-Tuning with Prefix Sampling
Zeyu Huang, Tianhao Cheng, Zihan Qiu, Zili Wang, Xu Yinghui, Edoardo Ponti, Ivan Titov
Existing LLMs-post-training techniques are broadly categorized into supervised f

ine-tuning (SFT) and reinforcement fine-tuning (RFT). Each paradigm presents a distinct trade-off:
(1) SFT excels at mimicking demonstration data, but can lead to problematic generalization as a form of behavior cloning.
(2) Conversely, RFT can significantly enhance a model's performance but is prone to learning unexpected behaviors, and its performance is sensitive to the initial policy.
In this paper, we propose a unified view of these methods and introduce Prefix-RFT, a hybrid approach that synergizes learning from both demonstration and exploration.
Using mathematical reasoning problems as a test bed, we empirically demonstrate that Prefix-RFT is simple yet effective.
Not only does it surpass the performance of standalone SFT and RFT, but it also outperforms parallel mixed-policy RFT methods.
Our analysis highlights the complementary nature of SFT and RFT, validating that Prefix-RFT effectively harmonizes them. Further ablation studies confirm the method's robustness to variations in the quality and quantity of demonstration data.
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MECAT: A Multi-Experts Constructed Benchmark for Fine-Grained Audio Understanding Tasks
Yadong Niu, TIANZI WANG, Heinrich Dinkel, Xingwei Sun, Jiahao Zhou, Gang Li, Jizhong Liu, Xunying Liu, Jian Luan
While large audio-language models have advanced open-ended audio understanding, they still fall short of nuanced human-level comprehension. This gap persists largely because current benchmarks, limited by data annotations and evaluation metrics, fail to reliably distinguish between generic and highly detailed model outputs. To this end, this work introduces MECAT, a Multi-Expert Constructed Benchmark for Fine-Grained Audio Understanding Tasks. Generated via a pipeline that integrates analysis from specialized expert models with Chain-of-Thought large language model reasoning, MECAT provides multi-perspective, fine-grained captions and open-set question-answering pairs. The benchmark is complemented by a novel metric: DATE (Discriminative-Enhanced Audio Text Evaluation). This metric penalizes generic terms and rewards detailed descriptions by combining single-sample semantic similarity with cross-sample discriminability. A comprehensive evaluation of state-of-the-art audio models is also presented, providing new insights into their current capabilities and limitations. The code and data are publicly available at https://github.com/xiaomi-research/mecat.
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Mesh Field Theory: Port–Hamiltonian Formulation of Mesh-Based Physics
Satoshi Noguchi, Yoshinobu Kawahara
We present Mesh Field Theory (MeshFT) and its neural realization, MeshFT-Net: a structure-preserving framework for mesh-based continuum physics that cleanly separates the physics’ topological structure from its metric structure. Imposing minimal physical principles (locality, permutation equivariance, orientation covariance, and energy balance/dissipation inequality), we prove a reduction theorem for mesh-based physics. Under these conditions, the physical dynamics admit a local factorization into a port–Hamiltonian form: the conservative interconnection is fixed uniquely by mesh topology, whereas metric effects enter only through constitutive relations and dissipation. This reduction clarifies what must be fixed and what should be learned, directly informing MeshFT-Net’s design.
Across evaluations on analytic and realistic datasets, physics-consistency tests, and out-of-distribution validation, MeshFT-Net achieves near-zero energy drift and strong physical fidelity (correct dispersion and momentum conservation) along with robust extrapolation and high data efficiency. By eliminating non-physical degrees of freedom and learning only metric-dependent structure, MeshFT provides a principled inductive bias for stable, faithful, and data-efficient learning-based physical simulation.
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Efficient Online Influence Maximization under the Independent Cascade Model with

Node-Level Feedback
Arpit Agarwal, Varad Deolankar, Rohan Ghuge
Influence maximization is an important research area in social network analysis, where the goal is to select a small set of seed nodes so as to maximize the expected spread of influence under a stochastic diffusion process.

Classical approximation algorithms for this problem rely on full knowledge of the underlying influence probabilities and operate in an offline manner.

In many real-world settings, however, these probabilities are unknown and must be learned from data, raising the question: \emph{can one still obtain strong performance guarantees while simultaneously learning the diffusion model parameters through repeated interactions?}

In this paper, we study the problem of \emph{online influence maximization} under the independent cascade model, where influence probabilities are unknown and feedback is limited to \emph{node-level} activation outcomes.

Prior work relies on a \emph{pair oracle} which needs to perform a joint optimization over seed sets and feasible parameters. This oracle is difficult to implement in practice and it was open whether one can achieve sublinear regret using only a \emph{standard} offline oracle.

We resolve this question by designing an online learning algorithm that achieves $\widetilde{O}(\sqrt{T})$ regret using only a \emph{standard} offline oracle.

Finally, we validate our theoretical results via experiments on real and synthetic data.
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Optimal Unconstrained Self-Distillation in Ridge Regression: Strict Improvements, Precise Asymptotics, and One-Shot Tuning
Hien Dang, Pratik Patil, Alessandro Rinaldo
Self-distillation (SD) is the process of retraining a student on a mixture of ground-truth labels and the teacher's own predictions using the same architecture and training data. Although SD has been empirically shown to often improve generalization, its formal guarantees remain limited. We study SD for ridge regression with an unconstrained mixing weight $\xi \in \mathbb{R}$. Conditional on the training data and without any distributional assumptions, we prove that for any squared prediction risk R (including out-of-distribution), the optimally mixed student strictly improves upon the ridge teacher at every regularization level λ where the teacher risk is not stationary ($R'(\lambda) \neq 0$). We also characterize the optimal mixing weight $\xi^\star$ in terms of the risk derivative R', showing that it can be negative, which is the case in over-regularized regimes. To quantify SD risk improvements, we derive exact risk asymptotics in the proportional asymptotics regime for general anisotropic covariance and deterministic signals. From a practical standpoint, we propose a consistent one-shot tuning method to estimate $\xi^\star$ without grid search, sample splitting, or refitting. Experiments on real-world datasets and pretrained neural network features support our theory and the one-shot tuning method.
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Hair-Trigger Alignment: Black-Box Evaluation Cannot Guarantee Post-Update Alignment
Yavuz Faruk Bakman, Duygu Nur Yaldiz, Eleni Triantafillou, Peter Kairouz, Salman Avestimehr, Sai Praneeth Karimireddy
Large Language Models (LLMs) are rarely static and are frequently updated in practice. A growing body of alignment research has shown that models initially deemed ``aligned'' can exhibit misaligned behavior after fine-tuning. These works typically assume that the initial model is aligned based on static black-box evaluation, i.e., the absence of undesired responses to a fixed set of queries. However, the limits of black-box evaluation for post-update scenarios is not explored sufficiently. In this work, we formalize model alignment in both the static and post-update settings and uncover a fundamental limitation of black-box evaluation. We theoretically show that, due to overparameterization, static alignment provides no guarantee of post-update alignment for *any* update dataset. Moreover, we prove that static black-box probing cannot distinguish a model that is genui

nely post-update robust from one that conceals an arbitrary amount of adversaria l behavior which can be activated by even a single benign gradient update.
We further validate these findings empirically in LLMs across three core alignme nt domains: privacy, jailbreak safety, and behavioral honesty. We demonstrate th e existence of LLMs that pass all standard black-box alignment tests, yet become severely misaligned after a single benign update. Finally, we show that the cap acity to hide such latent adversarial behavior increases with model scale, confi rming our theoretical prediction that post-update misalignment grows with the nu mber of parameters. Together, our results highlight the inadequacy of static eva luation protocols and emphasize the urgent need for post-update--robust alignmen t evaluation. Code can be found [here](https://github.com/Ybakman/safety_benign_ update).
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Envy-Free Allocation of Indivisible Goods via Noisy Queries
Zihan Li, Yan Hao Ling, Jonathan Scarlett, Warut Suksompong
We introduce a problem of fairly allocating indivisible goods (items) in which t he agents' valuations cannot be observed directly, but instead can only be acces sed via noisy queries. In the two-agent setting with Gaussian noise and bounded valuations, we derive upper and lower bounds on the required number of queries for finding an envy-free allocation in terms of the number of items, m, and th e negative-envy of the optimal allocation, Δ. In particular, when $\Delt a$ is not too small (namely, $\Delta \gg m^{1/4}$), we establish that the optima l number of queries scales as $\frac{\sqrt m }{(\Delta / m)^2} = \frac{m^{2.5}}{ \Delta^2}$ up to logarithmic factors. Our upper bound is based on non-adaptive queries and a simple thresholding-based allocation algorithm that runs in polyno mial time, while our lower bound holds even under adaptive queries and arbitrary computation time.
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Learning General Causal Structures with Hidden Dynamic Process for Climate Analy sis
Minghao Fu, Biwei Huang, Zijian Li, Yujia Zheng, Ignavier Ng, Guangyi Chen, Ying yao Hu, Kun Zhang
Understanding climate dynamics requires going beyond correlations in observation al data to uncover their underlying causal process. Latent drivers, such as atmo spheric processes, play a critical role in temporal dynamics, while direct causa l influences also exist among geographically proximate observed variables. Tradi tional Causal Representation Learning (CRL) typically focuses on latent factors but overlooks such observable-to-observable causal relations, limiting its appli cability to climate analysis. In this paper, we introduce a unified framework th at jointly uncovers (i) causal relations among observed variables and (ii) laten t driving forces together with their interactions. We establish conditions under which both the hidden dynamic processes and the causal structure among observed variables are simultaneously identifiable from time-series data. Remarkably, ou r guarantees hold even in the nonparametric setting, leveraging contextual infor mation to recover latent variables and observable relations. Building on these i nsights, we propose CaDRe (Causal Discovery and Representation learning), a time -series generative model with structural constraints that integrates CRL and cau sal discovery. Experiments on synthetic datasets validate our theoretical result s. On real-world climate datasets, CaDRe not only delivers competitive forecasti ng accuracy but also recovers visualized causal graphs aligned with domain exper tise, thereby offering interpretable insights into climate systems.
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Understanding Catastrophic Forgetting In LoRA via Mean-Field Attention Dynamics
Hugo Koubbi, Louis Hernandez, Matthieu Boussard
Low-Rank Adaptation (LoRA) is the dominant parameter-efficient fine-tuning metho d due to its favorable compute-performance trade-off, yet it suffers from catast rophic forgetting. We study forgetting through a tractable _mean-field self-atte ntion_ toy model, where tokens evolve as an interacting particle system and LoRA acts as a low-rank perturbation. Using tools from partial differential equation s and dynamical systems, we characterize regimes suggesting a phase transition b

etween forgetting and non-forgetting behavior. We show that one phase transition appears with respect to the norm of the perturbation, and the other with respect to the depth of the Transformers.
We further bound the time-to-deviation in terms of the perturbation size and spectral quantities, and corroborate the predicted trends with experiments and exploratory analyses on real models under LoRA fine-tuning.
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Interpreting and Steering State-Space Models via Activation Subspace Bottlenecks
Vamshi Sunku Mohan, Kaustubh Gupta, Aneesha Das, Chandan Singh
State-space models (SSMs) have emerged as an efficient strategy for building powerful language models, avoiding the quadratic complexity of computing attention in transformers. Despite their promise, the interpretability and steerability of modern SSMs remain relatively underexplored. We take a major step in this direction by identifying *activation subspace bottlenecks* in the Mamba family of SSM models using tools from mechanistic interpretability. We then introduce a test-time steering intervention that simply multiplies the activations of the identified bottlenecks by a scalar. Across 7 SSMs and 6 diverse benchmarks, this intervention improves performance by an average of 8.27%, without requiring any task-specific tuning. Finally, we validate that the identified bottlenecks are indeed hindering performance by modifying them to yield an architecture we call Stable-Mamba, which achieves long-context performance gains when retrained from scratch.
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Neural Implicit Action Fields: From Discrete Waypoints to Continuous Functions for Vision-Language-Action Models
Haoyun Liu, Jianzhuang Zhao, Xinyuan Chang, Tianle Shi, Chuanzhang Meng, Jiayuan Tan, Feng Xiong, Tong Lin, Dongjie Huo, Mu Xu, SongLin Dong, Zhiheng Ma, Yihong Gong, Sheng Zhong
Despite the rapid progress of vision-language-action (VLA) models, the prevailing practice of predicting action chunks as discrete waypoints remains structurally misaligned with the intrinsic continuity of physical motion. This discretization arises naturally from fixed-rate robot data collection and the token-by-token prediction paradigm of large language models, but ties actions to rigid sampling rates, does not naturally support analytically consistent higher-order derivatives, and introduces quantization artifacts that hinder precise, compliant interaction. We propose Neural Implicit Action Fields (NIAF), which reformulates chunk-level action representation from discrete waypoints to continuous action functions. Using a vision-language model as a hierarchical spectral modulator over a learnable motion prior, NIAF synthesizes continuous-time action manifolds with arbitrary temporal resolution. This formulation enables analytical differentiation, allowing explicit supervision of velocity and regularization of higher-order derivative signals to promote mathematical consistency, physical plausibility, and control smoothness. Our approach achieves strong results on CALVIN and LIBERO across diverse backbones. Real-world experiments further confirm that NIAF supports stable impedance control, bridging policy-side action generation and execution-side smooth control.
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Corrigibility Transformation: Constructing Goals That Accept Updates
Rubi Hudson
An AI agent will learn a desired goal more effectively if it does not resist the training process, but many partially learned goals incentivize an AI to avoid further goal updates. We would like goals to be corrigible, meaning they allow changes requested through designated channels, so that we can confidently correct errors and shut down the AI if necessary. Despite this being a crucial safety property, the existing literature does not specify goals that are both corrigible and competitive with alternatives. We introduce a transformation that constructs a corrigible version of nearly any goal, without sacrificing performance. This is done by eliciting predictions of reward conditional on costlessly preventing updates, and having that target be pursued myopically. These goals are then shown to lead to optimal performance among the class of corrigible goals, to incenti

vize allowing mid-action overrides, and to disincentivize deliberate self-modification. Empirically, they induce corrigible behavior in gridworld settings and for language models when applied at the prompt level.
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Rethinking Serialization in Linear 3D Vision: Decoupling Anisotropic Geometry from Isotropic Semantics
YinYun Yan, Liping zhang, Tingran Wang, Jiaxin Deng, Changshuo Wang, Limin Jiang, Shanwei Gao, Xin Ning
Current linear State-Space Models (SSMs) for 3D point clouds typically rely on 1D serialization schemes (e.g., Hilbert curves) for global modeling. In dense scenes, such imposed order can disrupt spatial continuity and induce what we call serialization bias. We propose AnIsoNet, a framework that decouples anisotropic geometry from isotropic semantics via two dedicated modules: Local Anisotropy Geometric Modeling (LAGM) and Global Isotropy Semantic Aggregation (GISA). LAGM uses ellipsoidal encoding to capture local directionality without relying on global order. GISA is configured according to dataset-level geometric density: dense-scene datasets use Identity Mode to avoid additional geometry-driven re-serialization, whereas sparse-object datasets use Morton serialization to provide a lightweight spatial prior. This avoids redundant multi-view scanning while preserving O(N) complexity. On S3DIS, reducing artificial serialization in dense scenes yields 82.62% mIoU, surpassing PCM by 3.0%. On ScanObjectNN, Morton serialization for sparse objects reaches 94.21% OA (+1.6%). On ScanNetV2, our model achieves 78.52% mIoU, surpassing PTv3 (77.5%) without pre-training, with only 12.2M parameters and 37G FLOPs.
****
RePack then Refine: Efficient Diffusion Transformers with Vision Foundation Models
Guanfang Dong, Luke Schultz, Negar Hassanpour, Chao Gao
Semantic-rich features from Vision Foundation Models (VFMs) have been leveraged to enhance Latent Diffusion Models (LDMs). However, raw VFM features are typically high-dimensional and redundant, increasing the difficulty of learning and reducing training efficiency for Diffusion Transformers (DiTs). In this paper, we propose Repack then Refine, a three-stage framework that brings the semantic-rich VFM features to DiT while further accelerating learning efficiency. Specifically, the RePack module projects the high-dimensional features onto a compact, low-dimensional manifold. This filters out the redundancy while preserving essential structural information. A standard DiT is then trained for generative modeling on this highly compressed latent space. Finally, to restore the high-frequency details lost due to the compression in RePack, we propose a Latent-Guided Refiner, which is trained lastly for enhancing the image details. On ImageNet-1K, RePack-DiT-XL/1 achieves an FID of 1.82 in only 64 training epochs. With the Refiner module, performance further improves to an FID of 1.65, significantly surpassing latest LDMs in terms of convergence efficiency. Our results demonstrate that packing VFM features, followed by targeted refinement, is a highly effective strategy for balancing generative fidelity with training efficiency. Source code is publicly available at https://github.com/guanfangdong/RePack-then-Refine.
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CGRiC: Compositional Risk Certification for Structured LLM Outputs
Ibne Farabi Shihab, Sanjeda Akter, Anuj Sharma
Large language models increasingly generate structured outputs, including citation-grounded summaries, multi-step reasoning chains, and tool-augmented responses, where correctness is inherently compositional: a single flawed claim can invalidate an otherwise accurate response. Existing certification methods treat outputs as atomic units, forcing a binary choice between unsafe acceptance and wasteful rejection. We introduce \textbf{Claim Graph Risk Control (CGRiC)}, a framework that decomposes responses into dependency graphs of verifiable claims and assigns calibrated per-claim risk bounds via information-lift statistics. By composing these bounds, CGRiC provides explicit guarantees on the probability that any incorrect claim passes verification undetected. When this composed risk exceeds a target threshold, the system triggers localized repairs rather than full abste

ntion, preserving correct content while fixing problematic claims. Our approach explicitly models extraction noise and verifier imperfection, and exploits conditional independence structure for tighter certificates when validated. Empirically, CGRiC achieves target risk levels while reducing abstention by 31\% compared to atomic baselines across QA, summarization, and reasoning tasks.
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SteeringSafety: Benchmarking Representation Steering in LLMs Across Safety Perspectives
Vincent Siu, Nicholas Crispino, David Park, Nathan W. Henry, Zhun Wang, Yang Liu, Dawn Song, Chenguang Wang
We introduce STEERINGSAFETY, a benchmark for evaluating representation steering methods across nine safety perspectives spanning 18 datasets. While prior work highlights the general capabilities of representation steering, we focus on safety perspectives including refusal, bias, hallucination, social behaviors, reasoning, epistemic integrity, and normative judgment. STEERINGSAFETY provides modularized building blocks for state-of-the-art steering methods, enabling unified implementation of DIM, ACE, CAA, PCA, and LAT with recent enhancements such as conditional steering. Results on Gemma-2-2B, Llama-3.1-8B, and Qwen-2.5-7B show that strong steering performance depends on the pairing of method, model, and specific perspective. For instance, DIM is consistently effective, yet all methods exhibit substantial entanglement, where improving effectiveness on one safety perspective often significantly changes performance on others. Social behaviors are most vulnerable (degradation up to 76%), refusal steering (jailbreaking) frequently compromises normative judgment such as commonsense morality (up to 26%), and hallucination steering shifts political views unpredictably across models, ranging from a 21% shift to the right to a 19% shift to the left. These findings show the need to understand steering methods through multiple safety angles rather than a single target behavior.
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Gaming Consensus: Coordinated Manipulation in Crowdsourced Fact-Checking
Nikil Roashan Selvam, Jay Baxter, Sophie Hilgard, Brad Miller, Keith Coleman, Ellen Vitercik, Sanmi Koyejo
Crowdsourced fact-checking systems have been adopted by major social media companies such as X, Meta, TikTok and Google with the aim of combating misleading information at scale without relying on centralized editorial control. These systems have been developed around a common underlying concept: a *bridging mechanism* that identifies notes flagging misleading information when they receive support from people with different perspectives rather than simple majority support. To our knowledge the only publicly disclosed bridging algorithms deployed for fact-checking are based on matrix factorization, as deployed by both X and Meta, augmented with additional components addressing abuse, targeted manipulation, and contributor brigades. This work examines the core matrix factorization portion of these systems, presenting theoretical and empirical evaluations of the degree to which coordinated users could vote strategically by leveraging the latent representations to fabricate the appearance of *synthetic consensus* within the bridging mechanism. Using historic production data, we find that up to 10.7% of lower quality notes could be manipulated above consensus thresholds using less than 10 ratings. We complement these findings with a theoretical analysis, revealing counterintuitively that rating a note as ``Not Helpful'' can increase its helpfulness score, as well as a cost model quantifying manipulation effort. We have developed and deployed mitigations within X's Community Notes algorithm to address synthetic consensus.
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KANFIS: A Neuro-Symbolic Framework for Interpretable and Uncertainty-Aware Learning
Binbin Yong, haoran pei, Jun Shen, Haoran Li, Qingguo Zhou, Zhao Su
Adaptive Neuro-Fuzzy Inference System (ANFIS) was designed to combine the learning capabilities of neural network with the reasoning transparency of fuzzy logic. However, conventional ANFIS architectures suffer from structural complexity, where the product-based inference mechanism causes an exponential explosion of ru

les in high-dimensional spaces. We herein propose the \textbf{K}olmogorov-\textbf{A}rnold **N**euro-**F**uzzy **I**nference **S**ystem (KANFIS), a compact neuro-symbolic architecture that unifies fuzzy reasoning with additive function decomposition. KANFIS employs an additive aggregation mechanism, under which both model parameters and rule complexity scale linearly with input dimensionality rather than exponentially. Furthermore, KANFIS is compatible with both Type-1 (T1) and Interval Type-2 (IT2) fuzzy logic systems, enabling explicit modeling of uncertainty and ambiguity in fuzzy representations. By using sparse masking mechanisms, KANFIS generates compact and structured rule sets, resulting in an intrinsically interpretable model with clear rule semantics and transparent inference processes. Empirical results demonstrate that KANFIS achieves competitive performance against representative neural and neuro-fuzzy baselines.
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Code2Worlds: Empowering Coding LLMs for 4D World Generation
Yi Zhang, Yunshuang Wang, Zeyu Zhang, Hao Tang
Achieving spatial intelligence requires moving beyond visual plausibility to build world simulators grounded in physical laws. While coding LLMs have advanced static 3D scene generation, extending this paradigm to 4D dynamics remains a critical frontier. This task presents two fundamental challenges: multi-scale context entanglement, where monolithic generation fails to balance local object structures with global environmental layouts; and a semantic-physical execution gap, where open-loop code generation leads to physical hallucinations lacking dynamic fidelity. We introduce Code2Worlds, a framework that formulates 4D generation as language-to-simulation code generation. First, we propose a dual-stream architecture that disentangles retrieval-augmented object generation from hierarchical environmental orchestration. Second, to ensure dynamic fidelity, we establish a physics-aware closed-loop mechanism in which a PostProcess Agent scripts dynamics, coupled with a VLM-Motion Critic that performs self-reflection to iteratively refine simulation code. Evaluations on the Code4D benchmark show Code2Worlds outperforms baselines with a 41% SGS gain and 49% higher Richness, while uniquely generating physics-aware dynamics absent in prior static methods.
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Advantage Collapse in Group Relative Policy Optimization: Diagnosis and Mitigation
Xixiang He, Qiyao Sun, Ao Cheng, Xingming Li, Xuanyu Ji, Hailun Lu, Runke Huang, Qingyong Hu
Group Relative Policy Optimization (GRPO), a prominent algorithm within the Reinforcement Learning from Verifiable Rewards (RLVR) framework, has achieved strong results in improving the reasoning capabilities of large language models (LLMs). However, GRPO is prone to advantage collapse, a failure mode where homogeneous rewards within a group (e.g., all correct or all incorrect answers) yield near-zero advantages and vanishing gradients. To address this, we introduce the Advantage Collapse Rate (ACR), the first diagnostic metric quantifying the proportion of training batches with ineffective gradients. Across models from 0.5B to 14B parameters on mathematical reasoning benchmarks, we show that ACR strongly predicts training stagnation and final performance. We then propose Adaptive Virtual Sample Policy Optimization (AVSPO), a lightweight extension of GRPO that injects virtual reward samples, guided by real-time ACR monitoring, to enable learning from homogeneous groups without additional model rollouts. AVSPO reduces advantage collapse by 58–63% relative to GRPO and yields consistent accuracy gains of 4–6 percentage points across all model scales, while maintaining generalization on the evaluated out-of-domain task. Code and datasets are available at https://github.com/hexixiang/Advantage-Collapse-Rate.
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Unlocking Zero-Shot Geospatial Reasoning via Indirect Rewards
Chenhui Xu, Fuxun Yu, Michael J. Bianco, Jacob Kovarskiy, Raphael Tang, Qi Zhang, Zirui Xu, Will LeVine, Brandon Dubbs, Heming Liao, Cassandra Burgess, Suvam Bag, Jay Patravali, Rupanjali Kukal, Mikael Figueroa, Rishi Madhok, Nikolaos Karianakis, Jinjun Xiong
Training robust reasoning vision-language models (VLMs) in rare domains (such as

geospatial) is fundamentally constrained by supervision scarcity. While raw geo spatial imagery is abundant, the amount of task-direct supervision falls far beh ind that of common domains. In this work, we validate an important conclusion: i ndirect verifiable rewards, derived from seemingly unrelated metadata, are suffi cient to induce sophisticated and generalizable geospatial reasoning across a wi de range of downstream tasks (25+). We present Geo-R1 as one empirical instantia tion of this paradigm. Rather than relying on limited task-specific annotations (i.e., direct rewards), Geo-R1 utilizes scalable, verifiable indirect proxy rewa rds based on cross-view alignment with metadata (geolocation information) to dri ve reinforcement learning at scale. Such indirect rewards successfully motivate the model to discover and internalize zero-shot geospatial reasoning across dive rse tasks, achieving extraordinary zero-shot transfer on out-of-distribution ben chmarks and even surpassing fully supervised specialists on certain benchmarks. These findings indicate that optimizing for indirect verifiable rewards may prov ide a scalable pathway to unlock generalized reasoning capabilities in rare doma ins with massive unlabeled data archives.
Our code is available at: https://github.com/miniHuiHui/Geo-R1.
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The Hidden Link between RLHF and Contrastive Learning
Xufei Lv, Kehai Chen, Haoyuan Sun, Xuefeng Bai, Min Zhang, Houde Liu
Alignment of large language models (LLMs) with human values has recently garnere d significant attention, with prominent examples including the canonical yet cos tly Reinforcement Learning from Human Feedback (RLHF) and the simple Direct Pref erence Optimization (DPO). In this work, we demonstrate that both RLHF and DPO c an be interpreted from the perspective of mutual information (MI) maximization, uncovering a profound connection to contrastive learning. Within this framework, both RLHF and DPO can be interpreted as methods that performing contrastive lea rning based on the positive and negative samples derived from base model, levera ging the Donsker–Varadhan (DV) lower bound on MI (equivalently, the MINE estimat or). Such paradigm further illuminates why RLHF may not intrinsically incentiviz e reasoning capacities in LLMs beyond what is already present in the base model. Building on the perspective, we replace the DV/MINE bound with the Jensen–Shann on (JS) MI estimator and propose the Mutual Information Optimization (MIO). Comp rehensive theoretical analysis and extensive empirical evaluations demonstrate t hat MIO mitigates the late-stage decline in chosen-likelihood observed in DPO, a chieving competitive or superior performance across various challenging reasonin g and mathematical benchmarks
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WorldCache: Accelerating World Models for Free via Heterogeneous Token Caching
Weilun Feng, Guoxin Fan, Haotong Qin, Mingqiang Wu, Yuqi Li, Xiangqi Li, Zhulin An, Libo Huang, Dingrui Wang, Longlong Liao, Michele Magno, Yongjun Xu, Chuangua ng Yang
Diffusion-based world models have shown strong potential for unified world simul ation, but the iterative denoising remains too costly for interactive use and lo ng-horizon rollouts. While feature caching can accelerate inference without trai ning, we find that policies designed for single-modal diffusion transfer poorly to world models due to two world-model-specific obstacles: *token heterogeneity* from multi-modal coupling and spatial variation, and *non-uniform temporal dyna mics* where a small set of hard tokens drives error growth, making uniform skipp ing either unstable or overly conservative.
We propose **WorldCache**, a caching framework tailored to diffusion world model s. We introduce *Curvature-guided Heterogeneous Token Prediction*, which uses a physics-grounded curvature score to estimate token predictability and applies a Hermite-guided damped predictor for chaotic tokens with abrupt direction changes . We also design *Chaotic-prioritized Adaptive Skipping*, which accumulates a cu rvature-normalized, dimensionless drift signal and recomputes only when bottlene ck tokens begin to drift. Experiments on diffusion world models show that WorldC ache delivers up to **3.7$\times$** end-to-end speedups while maintaining **98\% ** rollout quality, demonstrating the vast advantages and practicality of WorldC ache in resource-constrained scenarios.

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Guaranteed Optimal Compositional Explanations for Neurons
Biagio La Rosa, Leilani H. Gilpin
Compositional explanations are a family of methods that aim to describe the spat ial alignment between neurons' receptive field activations and concepts through logical rules, typically computed via a search over all possible concept combina tions. Since computing the spatial alignment over the entire state space is comp utationally infeasible, the literature commonly adopts assumptions related to th e structure of the combinations and beam search to restrict the state space. How ever, beam search cannot provide any theoretical guarantees of optimality, and i t remains unclear how close current explanations are to the true optimum. In thi s theoretical paper, we address this gap by introducing the first framework for computing guaranteed optimal compositional explanations over the entire state sp ace spanned by the adopted assumptions. Specifically, we propose: (i) a decompos ition that identifies the factors influencing the spatial alignment, (ii) a heur istic to estimate the alignment at any stage of the search, and (iii) the first algorithm that can compute optimal compositional explanations in a time comparab le to exhaustive beam search. Using this framework, we demonstrate that 10-40\% of explanations previously obtained with beam search are suboptimal when overla pping concepts are involved. Finally, we evaluate a beam-search variant guided b y our proposed decomposition and heuristic, showing that it matches or improves runtime over prior methods while offering greater flexibility in hyperparameters and computational resources.
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SWE-Compass: Towards Unified Evaluation of Agentic Coding Abilities for Large La nguage Models
Jingxuan Xu, Ken Deng, Weihao Li, Songwei Yu, Haoyang Huang, Xinping Lei, Yifan Yao, Huaixi Tang, Zhiyi Lai, Kepeng Lei, Zizheng Zhan, Yanan Wu, Chenchen Zhang, Wenqiang Zhu, Wen Xiang, Zongxian Feng, Han Li, Junqi Xiong, Dailin Li, Zuchen Gao, Kun Wu, Yuanxing Zhang, Wuxuan Gong, Ziyuan Gao, Guanxiang Wang, Yirong Xue , Mengfei Xie, Xiaojiang Zhang, Jinghui Wang, Wenhao Zhuang, Zheng Lin, Huiming Wang, Zhaoxiang Zhang, Yuqun Zhang, Haotian Zhang, Ming Sun, Chen Bin, Jiaheng L iu
Evaluating large language models (LLMs) for software engineering has been limite d by narrow task coverage, language bias, and insufficient alignment with real-w orld developer workflows. Existing benchmarks often focus on algorithmic problem s or Python-centric bug fixing, leaving critical dimensions of software engineer ing underexplored. To address these gaps, we introduce SWE-Compass, a comprehens ive benchmark that unifies heterogeneous code-related evaluations into a structu red and production-aligned framework. SWE-Compass spans 8 task types, 8 programm ing scenarios, and 10 programming languages, with 2000 high-quality instances cu rated from authentic GitHub pull requests and refined through systematic filteri ng and validation. We benchmark ten state-of-the-art LLMs under two agentic fram eworks, SWE-Agent and Claude Code, revealing a clear hierarchy of difficulty acr oss task types, languages, and scenarios. Moreover, by aligning evaluation with real-world developer practices, we hope SWE-Compass can provide a rigorous and r eproducible foundation for diagnosing and advancing agentic coding capabilities in large language models.
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QHyer: Q-conditioned Hybrid Attention-mamba Transformer for Offline Goal-conditi oned RL
Xing Lei, Jincheng Wang, Xuetao Zhang, Donglin Wang
Offline goal-conditioned RL (GCRL) learns goal-reaching policies from static dat asets, but real-world environments are often partially observable, so the collec ted trajectories are only partly consistent with the Markov assumption while oth er segments remain history-dependent. History-aware sequence models such as Deci sion Transformer (DT) are a natural fit for long-term dependency modeling, yet p ure attention is inefficient and brittle when handling local Markovian structure and long-range context simultaneously. Although recent hybrid architectures (e. g., LSDT) introduce local extractors, their fixed-window extraction cannot adapt

the effective memory to varying dependency lengths, often truncating long-range context instead of compressing it. Moreover, under sparse rewards, return-to-go (RTG) becomes non-discriminative across sub-trajectories, offering little guidance for stitching goal-reaching behaviors from diverse demonstrations. To address these limitations, we propose QHyer (Q-conditioned Hybrid Attention-Mamba Transformer), which replaces RTG with a Normalizing Flows (NFs) parameterized goal-reaching Q-estimator used directly as conditioning tokens, and a gated Hybrid Attention-Mamba backbone whose selective state-space dynamics enable content-adaptive history compression while attention captures global goal-directed dependencies. Extensive experiments on OGBench and D4RL demonstrate that QHyer achieves state-of-the-art performance on both non-Markovian and Markovian datasets, validating its effectiveness for diverse scenarios.

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Facts in Stats: Impacts of Pretraining Diversity on Language Model Generalization

Tina Behnia, Puneesh Deora, Christos Thrampoulidis

Language models are pretrained on sequences that blend statistical regularities, which make text fluent, with factual associations between specific tokens, which encode knowledge of facts. While recent works suggest that generalization depends critically on the interaction of these two streams, such as the diversity of the contexts in which facts appear, these effects remain difficult to study systematically. This paper introduces a flexible synthetic testbed that combines a statistical stream of generic tokens with an abstract factual stream of source-target token pairs, enabling fine-grained control over their interaction, such as their composition into a sequence (contextual structure) or the level of context diversity carrying the facts at training time. Through controlled experiments, we find that higher contextual diversity delays in-distribution factual learning, while low diversity can harm out-of-distribution generalization in ways that depend on the contextual structure. As a result, the optimal diversity level depends on the training budget. Beyond factual recall failures, we also identify failures in statistical generalization as we study how the interplay between contextual design and diversity level impacts different aspects of generalization. Furthermore, through a series of controlled interventions on the model components, we trace failure in different aspects of generalization to distinct optimization bottlenecks, highlighting the importance of the embedding and unembedding layers. Overall, our synthetic framework allows us to isolate effects that would be confounded in large-scale studies, offering a controlled testbed for future investigations.

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Adaptive Residual-Update Steering for Low-Overhead Hallucination Mitigation in Large Vision-Language Models

Zhengtao Zou, Ya Gao, Jiarui Guan, Bin Li, Pekka Marttinen

Large Vision-Language Models (LVLMs) typically process visual inputs as a prefix to the language decoder. As the model autoregressively generates text, this initial visual information inevitably undergoes ``dilution'', leading the model to over-rely on language priors and hallucinate objects. Existing interventions attempt to correct this by contrasting logits or iteratively refining outputs, but they incur prohibitive latency costs. We propose **Residual-Update Directed DEcoding Regulation (RUDDER)**, a framework that counters visual dilution by creating a persistent visual anchor. We extract a robust evidence direction (**CARD**) directly from the model's prefill residual updates, and inject it into the decoding process. This injection is modulated by an adaptive gate, the **Beta Gate**, which acts as a trust mechanism and ensures the visual reminder is applied only when necessary. Experiments on LLaVA-1.5 (7B/13B), Idefics2, InstructBLIP, and Qwen2.5-VL demonstrate that RUDDER consistently mitigates hallucination (with greedy decoding, RUDDER reduces CHAIR$_S$ by an average of **24.4\%** and CHAIR$_i$ by **23.6\%** relative) and scales effectively across architectures, all while maintaining **>96.0\%** throughput. The code is available at https://github.com/Akko000/RUDDER-Residual-Update-Directed-DEcoding-Regulation-

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SafeHarbor: Defining Precise Decision Boundaries via Hierarchical Memory-Augmented Guardrail for LLM Agent Safety
Zhe Liu, Zonghao Ying, Wenxin Zhang, Quanchen Zou, Deyue Zhang, Dongdong Yang, Xiangzheng Zhang, Hao Peng
Recent advances in foundation models have transformed LLMs from passive conversational systems into autonomous agents capable of reasoning and tool execution. While these capabilities unlock substantial practical value, they also introduce new security risks, as adversaries can manipulate agents into performing harmful actions in real-world environments.
Existing defense strategies mitigate such threats but frequently struggle to balance safety and utility, resulting in over-refusal of benign user requests.
To mitigate this trade-off, we propose SafeHarbor, a novel framework designed to establish precise decision boundaries for LLM agents.
Unlike static guidelines, SafeHarbor extracts context-aware defense rules through enhanced adversarial generation.
We design a local hierarchical memory system for dynamic rule injection, offering a training-free, efficient, and plug-and-play solution.
Furthermore, we introduce an information entropy-based self-evolution mechanism that continuously optimizes the memory structure through dynamic node splitting and merging.
Extensive experiments demonstrate that SafeHarbor achieves state-of-the-art performance on both ambiguous benign tasks and explicit malicious attacks, notably attaining a peak benign utility of 63.6\% on GPT-4o while maintaining a robust refusal rate exceeding 93\% against harmful requests.
The source code is publicly available at https://github.com/ljj-cyber/SafeHarbor.
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Linguistic Nepotism: Trading-off Quality for Language Preference in Multilingual RAG
Dayeon Ki, Marine Carpuat, Paul McNamee, Daniel Khashabi, Eugene Yang, Dawn Lawrie, Kevin Duh
Multilingual Retrieval-Augmented Generation (mRAG) systems enable language models to answer knowledge-intensive queries with citation-supported responses across languages. Despite their growing use, an open questions is whether the mixture of different document languages impacts generation and citation behavior in *unintended* ways. To investigate this, we introduce a controlled methodology using model internals to measure language preference while holding other factors such as document relevance constant. Across eight languages and six open-weight models, we find that models preferentially cite English sources when queries are in English, with this bias amplified for lower-resource languages and for documents positioned mid-context. More crucially, we find that models sometimes trade-off document relevance for language preference, indicating that citation choices are not always driven by informativeness alone. Our findings shed light on how language models leverage multilingual context and influence citation behavior.
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Weakly Supervised Cross-Modal Learning for 4D Radar Scene Flow Estimation
Jingyun Fu, Zhiyu Xiang, Na Zhao
Due to the difficulty of obtaining ground-truth data for 4D radar scene flow estimation, previous methods typically rely on either self-supervised losses or cross-modal supervision using 3D LiDAR data, 2D images, and odometry. However, self-supervised approaches often yield suboptimal results due to radar's inherently low-fidelity measurements, while existing cross-modal supervised methods introduce complex multi-task architecture and require costly LiDAR sensors to generate pseudo radar scene flow labels from pretrained 3D tracking models. To overcome these limitations, we propose a task-specific iterative framework for weakly supervised radar scene flow learning, using only images and odometry for auxiliary supervision during training. Specially, we establish two novel instance-aware self-supervised losses by exploiting off-the-shelf 2D tracking and segmentation algorithms to obtain tracked instance masks, which are back-projected into 3D space to provide instance-level semantic guidance; for static regions, we integrate v

ehicle odometry with radar's intrinsic motion cues to construct a rigid static l
oss. Extensive experiments on the real-world View-of-Delft (VoD) dataset demonst
rate that our method not only surpasses state-of-the-art cross-modal supervised
approaches that rely on 3D multi-object tracking on dense LiDAR point clouds but
 also outperforms existing fully supervised scene flow estimation methods. The c
ode is open-sourced at \href{https://github.com/FuJingyun/IterFlow}{https://gith
ub.com/FuJingyun/IterFlow}.
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Profiling the Irrational Agent: Cognitive Modeling of LLM Behaviors in Sequentia
l Jailbreaks

Xikang Yang, biyu zhou, Xuehai Tang, Jizhong Han, Songlin Hu

Large language models (LLMs) are increasingly deployed in high-stakes settings,
yet they remain vulnerable to sequential jailbreaks that exploit multi-turn inte
raction to circumvent safety mechanisms. Current safety evaluations are largely
outcome-based, offering little insight into the latent decision processes that l
ead to unsafe compliance. We propose an interpretable cognitive modeling framewo
rk that couples a controlled elicitation paradigm, the Contextual Iowa Gambling
Task (C-IGT), with a Generalized Rescorla--Wagner (GRW) architecture to decompos
e behavior into measurable mechanisms. Across a diverse set of mainstream LLMs,
we find that sequential vulnerability is not explained by scale alone but emerge
s from interactions among cognitive factors, including optimism-biased learning,
 perceptual reward amplification, and choice inertia. Moreover, counterfactual f
eedback and psychologically framed rewards (e.g., regret, authority, threat) sub
stantially accelerate the transition from refusal to compliance. These results y
ield principled cognitive profiles of LLM "irrationality" and provide insights f
or interdisciplinary research on LLM agents at the intersection of machine learn
ing and human behavioral science.
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When Distance Distracts: Representation Distance Bias in BT-Loss for Reward Mode
ls

Tong Xie, Andrew Bai, Yuanhao Ban, Yunqi Hong, Haoyu Li, Cho-Jui Hsieh

Reward models are central to Large Language Model (LLM) alignment within the fra
mework of RLHF. The standard objective used in reward modeling is the Bradley-Te
rry (BT) loss, which learns from pairwise data consisting of chosen and rejected
 responses. In this work, we analyze the per-sample gradient of BT-loss and show
 spurious learning signals due to representation distance. In particular, BT gra
dient norm scales with two distinct components: (1) **prediction error**, reflec
ted by the difference in predicted rewards between chosen and rejected responses
, and critically, (2) **representation distance** between the pair measured in t
he output space of the final layer. While the first term captures the intended t
raining signal, the second term can significantly impact the update magnitude an
d misalign learning. Specifically, pairs with small representation distance ofte
n receive vanishingly weak updates, even when misranked, while pairs with large
distance receive disproportionately strong updates. This leads to gradients from
 large-distance pairs overshadowing those from small-distance pairs, where fine-
grained distinctions are especially important. To overcome this limitation, we p
ropose NormBT, an adaptive pair-wise normalization scheme that rescales update t
o balance representation-driven effects and focuses learning signals on predicti
on error. NormBT is a lightweight, drop-in modification to BT loss with negligib
le overhead. Across various LLM backbones and datasets, NormBT improves reward m
odel performance consistently, with notable gains of over 5% on the Reasoning ca
tegory of RewardBench, which contains numerous fine-grained pairs.
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On the Theoretical Limitations of Embedding-based Link Prediction

Samy Badreddine, Emile van Krieken, Luciano Serafini

Neural networks often map low-dimensional embeddings to high-dimensional output
spaces. Usually, the output layer is linear, which can create a *rank bottleneck
* that limits the functions a model can represent. Such bottlenecks are ubiquito
us in link prediction models, such as knowledge graph embeddings (KGEs), as the
output space of entities can be orders of magnitude larger than the embedding di

mension. We investigate how rank bottlenecks limit model expressivity for fitti ng the training data. While previous work focused on sufficient bounds on the em bedding dimension required for specific KGEs, we show necessary bounds for *all* KGEs with a linear output layer, which grow with graph size and connectivity. W e also consider a non-linear output layer using mixtures to break the bottleneck without significant parameter overhead. Empirically, we show that models using this non-linear layer improve in ranking performance and probabilistic fit for l arge and dense datasets at a low parameter cost, as predicted by our theory. Our work reveals how linear output layers limit KGEs and motivates non-linear alter natives for scaling to large and dense graphs.

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Welfare-Optimal Classification with Accuracy Auctions

Bana Sadi, Eden Saig, Nir Rosenfeld

Prediction algorithms are increasingly used to inform decisions about humans, bu t maximizing accuracy—the standard learning objective—is not necessarily optimal for this purpose. Instead, we propose optimizing social welfare, defined as the average gain users receive from correct predictions. Welfare enables to express , and therefore account for, heterogeneity in how much users benefit from accura cy. But since these valuations are private and users can benefit from overreport ing them, learning must simultaneously elicit truthful values and optimize welfa re with respect to them. To this end, we propose a novel learning algorithm that incorporates a truthful auction. We show how to compute allocations and prices efficiently, and bound the number of paying users—which surprisingly is independ ent of the sample size. We conclude with experiments on real and synthetic data that demonstrate our algorithm and explore the connections between welfare and a ccuracy.

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LazyAttention: Efficient Retrieval-Augmented Generation with Deferred Positional Encoding

Haocheng Xia, Mihir Pamnani, Hanxi Fang, Supawit Chockchowwat, Yongjoo Park

Key-value (KV) caching accelerates inference of large language models (LLMs) by reusing past computations for generated tokens. Its importance becomes even grea ter in long-context applications such as retrieval-augmented generation (RAG) an d in-context learning (ICL). However, conventional KV caching embeds positional information directly into the cache, limiting its reusability. Existing solution s either restrict reuse to prefixes or require expensive memory materialization for positional re-encoding. We introduce LazyAttention, a novel attention mechan ism that kernelizes deferred positional encoding to enable zero-copy, position-a gnostic KV reuse. By adjusting positional encoding within attention kernels on-t he-fly, LazyAttention resolves the materialization bottleneck, allowing a single physical KV copy to serve multiple logical requests at arbitrary positions. Lev eraging attention kernels tailored for prefilling and decoding, our system achie ves significant efficiency improvements: under skewed document distributions, it reduces time-to-first-token (TTFT) by 1.37× and increases inference throughput by 1.40× compared to the state-of-the-art Block-Attention, while maintaining com parable output quality.

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Online Fair Division with Additional Information

Tzeh Yuan Neoh, Jannik Peters, Nicholas Teh

We study the problem of fairly allocating indivisible goods to agents in an onli ne setting, where goods arrive sequentially and must be allocated irrevocably. F ocusing on the popular fairness notions of envy-freeness, proportionality, and m aximin share fairness (and their approximate variants), we investigate how acces s to future information changes what guarantees are achievable. Without any info rmation, we prove strong impossibility results even for approximate fairness. Wi th normalization information (agents' total values), we provide an algorithm tha t achieves stronger fairness guarantees than previously known results, and show matching impossibilities for stronger notions. With frequency predictions (value multisets without order), we design a meta-algorithm that lifts a broad class o f offline “share-based” guarantees to the online setting, matching the best-know

n offline bounds. Finally, we provide learning-augmented variants of both models : under noisy totals or noisy frequency predictions, our guarantees are robust a nd degrade gracefully with the error parameters.
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Detecting and Filtering Unsafe Training Data via Data Attribution with Denoised Representation

Yijun Pan, Taiwei Shi, Jieyu Zhao, Jiaqi W. Ma

Large language models (LLMs) are highly sensitive to even small amounts of unsaf e training data, making effective detection and filtering essential for trustwor thy model development. Current state-of-the-art (SOTA) detection approaches prim arily rely on moderation classifiers, which require significant computation over head for training and are limited to predefined taxonomies. In this work, we exp lore data attribution approaches that measure the similarity between individual training samples and a small set of unsafe target examples, based on data repres entations such as hidden states or gradients. We identify a key limitation in ex isting methods: unsafe target texts contain both critical tokens that make them unsafe and neutral tokens (e.g., stop words or benign facts) that are necessary to form fluent language, and the latter of which makes the overall representatio ns noisy for the purpose of detecting unsafe training data. To address this chal lenge, we propose Denoised Representation Attribution (DRA), a novel representat ion-based data attribution approach that denoises training and target representa tions for unsafe data detection. Across tasks of filtering jailbreaks and detect ing gender bias, the proposed approach leads to significant improvement for data attribution methods, outperforming SOTA methods that are mostly based on modera tion classifiers.
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Triadic Dynamics Aware Diffusion Posterior Sampling for Inverse Problems: Optimi zing Guidance and Stochasticity Schedules

Junseo Bang, Dong Ju Mun, Hoigi Seo, Seongmin Hong, Se Young Chun

Generative posterior sampling using diffusion models has emerged as a dominant p aradigm for solving inverse problems in imaging, which usually consists of three main components: data consistency (DC) guidance, classifier-free guidance (CFG) and stochasticity. While prior arts have focused on how to develop each or all components, less attention has given to how to schedule them, leading to heurist ically fixed or partially adjusted suboptimal schedules. In this work, we argue that the interactions among all three components in terms of scheduling are cruc ial for significantly improved performance in solving inverse problems in imagin g. Our analysis shows that aggressive CFG early in sampling conflict with DC gui dance, while stochasticity brings the trajectory back to higher-probability regi ons. Based on these findings, we propose Triadic Dynamics Aware Posterior Sampli ng (TriPS), which reformulates posterior sampling as a time-varying control prob lem and optimizes schedules following a triadic trend of decreasing DC and stoch asticity scales alongside increasing CFG scale. TriPS achieves this through two strategies: template-based search over functional priors for reliable baseline s chedules, and Group Relative Policy Optimization (GRPO)-based reinforcement lear ning for more flexible temporal curves. Experiments demonstrate TriPS outperform s state-of-the-art baselines in data fidelity and perceptual realism.
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Semantic Tube Prediction: Beating LLM Data Efficiency with JEPA

Hai Huang, Yann LeCun, Randall Balestriero

Large Language Models (LLMs) obey consistent scaling laws---empirical power-law fits that predict how loss decreases with compute, data, and parameters. While p redictive, these laws are descriptive rather than prescriptive: they characteriz e typical training, not optimal training. Surprisingly few works have successful ly challenged the data-efficiency bounds implied by these laws---which is our pr imary focus. To that end, we introduce the Geodesic Hypothesis, positing that to ken sequences trace geodesics on a smooth semantic manifold and are therefore lo cally linear. Building on this principle, we propose a novel Semantic Tube Predi ction (STP) task, a JEPA-style regularizer that confines hidden-state trajectori es to a tubular neighborhood of the geodesic. STP generalizes JEPA to language w

ithout requiring explicit multi-view augmentations. We show this constraint impr
oves signal-to-noise ratio, and consequently preserves diversity by preventing t
rajectory collisions during inference. Empirically, STP allows LLMs to match bas
eline accuracy with 16$\times$ less training data, directly violating the data t
erm of fine-tuning scaling laws and demonstrating that principled geometric prio
rs can surpass brute-force scaling. Code: \url{https://github.com/galilai-group/
llm-jepa#stp}
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OnePO: Direct One-stage Policy Optimization for SFT-free Domain Adaptation
Junying Chen, Xinyuan Xie, Ziniu Li, Benyou Wang
Domain adaptation typically follows a two-stage pipeline: Supervised Fine-Tuning
 (SFT) then Reinforcement Learning (RL). However, *does RL necessarily require a
 pre-SFT phase for domain adaptation?* SFT confines the model to an imitation di
stribution, limiting RL exploration, while the two-stage transition causes capab
ility regression and extra engineering. We propose **One-stage Policy Optimizati
on (OnePO)**, an SFT-free paradigm that adapts pretrained LLMs to target domains
 in a single RL stage. OnePO uses teacher outputs as transient guidance to overc
ome the slow convergence of pure RL, while avoiding two failures of naive teache
r-output integration: gradient starvation for low-probability teacher tokens and
 distribution anchoring from persistent teacher signals. It introduces two mecha
nisms: (1) *Adaptive Objective Evolution*, reshaping the RL objective for rapid
absorption of teacher-provided knowledge; and (2) *Teacher Retirement*, automati
cally discarding teacher outputs once the model surpasses them. On medical adapt
ation, OnePO achieves 67.2 on HealthBench with only 20K training samples, outper
forming SFT+RL by +2.7 and pure RL by +7.4 points. Scaling the same recipe produ
ces *HuatuoGPT-3*, an open-source medical LLM series whose 32B variant reaches 7
0.3 on HealthBench. Additional writing and legal-domain experiments show that On
ePO extends beyond medicine. Models and code are available at https://github.com
/FreedomIntelligence/HuatuoGPT-3.
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Protein Design with Agent Rosetta: A Case Study for Specialized Scientific Agent
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Jacopo Teneggi, SM Bargeen Alam Turzo, Tanya Marwah, Alberto Bietti, P. Douglas
Renfrew, Vikram Khipple Mulligan, Siavash Golkar
Large language models (LLMs) are capable of emulating reasoning and using tools,
 creating opportunities for autonomous agents that execute complex scientific ta
sks. Protein design provides a natural testbed: although machine learning (ML) m
ethods achieve strong results, these are largely restricted to canonical amino a
cids and narrow objectives, leaving unfilled need for a generalist tool for broa
d design pipelines. We introduce Agent Rosetta, an LLM agent paired with a struc
tured environment for operating Rosetta, the leading physics-based heteropolymer
 design software, capable of modeling non-canonical building blocks and geometri
es. Agent Rosetta iteratively refines designs to achieve user-defined objectives
, combining LLM reasoning with Rosetta's generality. We evaluate Agent Rosetta o
n design with canonical amino acids, matching specialized models and expert base
lines, and with non-canonical residues---where ML approaches fail---achieving co
mparable performance. Critically, prompt engineering alone often fails to genera
te Rosetta actions, demonstrating that environment design is essential for integ
rating LLM agents with specialized software. Our results show that properly desi
gned environments enable LLM agents to make scientific software accessible while
 matching specialized tools and human experts.
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Multi-Head LatentMoE and Head Parallel: Communication-Efficient and Deterministi
c MoE Parallelism
Chenwei Cui, Rockwell Jackson, Benjamin Joseph Herrera, Ana M. Tárano, Hannah Ke
rner
Large language models have transformed many applications but remain expensive to
 train.
Sparse Mixture of Experts (MoE) addresses this through conditional computation,
with Expert Parallel (EP) as the standard distributed training method.

However, EP has three limitations:
communication cost grows linearly with the number of activated experts k,
load imbalance affects latency and memory usage, and
data-dependent communication requires metadata exchange.
We propose Multi-Head LatentMoE and Head Parallel (HP), a new architecture and p
arallelism that achieve
$O(1)$ communication cost regardless of k,
completely balanced traffic, and
deterministic communication,
all while remaining compatible with EP.
To accelerate Multi-Head LatentMoE,
we propose IO-aware routing and expert computation.
Compared to MoE with EP, Multi-Head LatentMoE with HP trains up to $1.82\times$
faster while having better performance.
With double the granularity, the performance is even better while being $1.08\ti
mes$ faster.
Our method makes multi-billion-parameter foundation model research more accessib
le.
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Efficient Bilevel Optimization for CKA-Guided MoE Upcycling
Zhiyuan Yu, Enneng Yang, Hao Jiang, Guojie Zhu, Feihong He, Peng Wang, Li Shen
Upcycling, a strategy that initializes Mixture-of-Experts (MoE) by replicating p
re-trained feed-forward or MoE networks to expand model capacity, has become a p
opular method in continual learning due to its effectiveness in mitigating catas
trophic forgetting.
However, existing paradigms indiscriminately expand capacity to prioritize perfo
rmance at the cost of severe inefficiency, introducing severe parameter redundan
cy and failing to exploit structural heterogeneity.
To address this, we investigate the determinants of forgetting in training dynam
ics using Centered Kernel Alignment (CKA) and loss landscape flatness to analyze
 the behavior of pre- and post-expansion MoE layers, uncovering instability in d
eep-layer representations and heterogeneous expert sensitivity to new tasks, the
reby demonstrating the potential of selective upcycling to eliminate redundancy.
Consequently, we propose a dynamic bilevel optimization framework to guide adapt
ive upcycling, featuring an outer loop employing a Gumbel-Softmax differentiable
 mask to perform Neural Architecture Search (NAS) for adaptive growth, while an
inner loop optimizes weight updates via task objectives and CKA-regularized repl
ay.
Experiments on the TRACE benchmark demonstrate that our proposed method achieves
 better average accuracy with 80\% forgetting reduction, while effectively elimi
nating 60\% of redundant parameter expansion that standard upcycling would intro
duce.
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A Stronger Benchmark for Online Bilateral Trade: From Fixed Prices to Distributi
ons
Anna Lunghi, Mattia Piccinato, Matteo Castiglioni, Alberto Marchesi
We study online bilateral trade, where a learner facilitates repeated exchanges
between a buyer and a seller to maximize the Gain From Trade (GFT), i.e., the so
cial welfare. In doing so, the learner must guarantee not to subsidize the marke
t. This constraint is usually imposed per round through Weak Budget Balance (WBB
). Despite that, (Bernasconi et al.,2024) shows that a Global Budget Balance (GB
B) constraint on the profit--enforced over the entire time horizon--can improve
the GFT by a multiplicative factor of two. While this might appear to be a margi
nal relaxation, this implies that all existing WBB-focused algorithms suffer lin
ear regret when measured against the GBB optimum.
In this work, we provide the first algorithm to achieve sublinear regret against
 the GBB benchmark in stochastic environments under one-bit feedback. In particu
lar, we show that when the joint distribution of valuations has a bounded densit
y, our algorithm achieves $\widetilde{\mathcal{O}}(T^{3/4})$ regret. Our result
shows that there is no separation between the one-dimensional problem of learnin

g the optimal WBB price and the two-dimensional problem of learning the optimal GBB distribution over *couples* of prices.
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RC-FCL: Combating Asynchronous Concept Drift in Federated Continual Learning via Retrospective Calibration

Hang Su, Yijun Mo, Zhiyu Zhang, Yankai Jiang, Bo Liu, Yichen Li, Imran Razzak

Federated Continual Learning (FCL) enables the continuous acquisition of knowledge from streaming tasks, but inherently struggles with the temporal dynamics of client data distributions. These dynamics naturally induce asynchronous concept drift, where distribution shifts occur independently across clients at unsynchronized times and with varying magnitudes. Such asynchrony generates conflicting updates that destabilize global convergence and exacerbate catastrophic forgetting. However, existing FCL research focuses on static or incremental settings, typically treating all incoming updates uniformly, which obscures concept drift under divergent distributions and fails to adapt to the evolution of learned concepts. To address these limitations, we propose RC-FCL, a retrospective calibration framework for FCL that can effectively distinguish asynchronous concept drift and adjust the learning strategy adaptively. Specifically, RC-FCL leverages a conditional generative model to synthesize class-conditional reference distributions of previously learned concepts for local drift detection. It calibrates local adaptation using a weighting mechanism driven by the local discriminator to prioritize informative samples, and executes a global aggregation strategy based on drift magnitude. Our experimental results demonstrate that RC-FCL achieves competitive performance against state-of-the-art methods.
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TN-SHAP-G: Graph-Structured Tensor Network Surrogates for Shapley Values and Interactions

Farzaneh Heidari, Guillaume Rabusseau

Shapley values are a widely used tool for attributing importance and interactions among input variables in black-box models, but their computation involves a function defined over an exponentially large space of subsets. We propose TN-SHAP-G, a framework that exploits structure in graph-structured inputs to compute Shapley values and higher-order interaction indices efficiently. Given a predictor and a fixed masking scheme, TN-SHAP-G learns a compact, graph-aligned multilinear surrogate that approximates the masked-input behavior, represented as a tensor network whose topology mirrors the input graph. Once trained from a small number of oracle queries, the surrogate enables deterministic recovery of first- and higher-order Shapley indices via the multilinear extension, without additional model queries or Monte Carlo variance. Experiments on molecular benchmarks show that the learned factorization closely matches exact Shapley values on small graphs and scales efficiently to larger graphs where sampling-based methods become infeasible.
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Skill-Based Mixture-of-Experts: Adaptive Routing for Heterogeneous Reasoning via Inferred Skills

Justin Chen, Sukwon Yun, Elias Stengel-Eskin, Tianlong Chen, Mohit Bansal

Combining existing pre-trained LLMs is a promising approach for diverse reasoning tasks. However, task-level expert selection is often too coarse-grained, since different instances may require different expertise. To address this, we propose Skill-MoE, a symbolic, skill-based, and gradient-free Mixture-of-Experts framework for instance-level expert selection. Skill-MoE infers skills (e.g., algebra in mathematics) from each query, selects experts based on skill relevance, and lets each expert generate its own reasoning. The resulting k outputs are then synthesized by an aggregator chosen for its ability to integrate diverse responses. While instance-level selection substantially improves performance, naively implementing it incurs heavy overhead from repeated model loading and offloading. We address this with a batch inference strategy that groups instances by assigned experts, allowing each model to be loaded only once. As a result, Skill-MoE integrates 16 expert models on a single GPU with runtime comparable to prior multi-agent baselines using 4 GPUs. Across diverse benchmarks (MMLU-Pro, GPQA, AIME, a

nd MedMCQA), Skill-MoE achieves an average absolute improvement of 8.15% over th e best baseline. It also generalizes well to unseen tasks and outperforms discus sion-based methods without requiring expensive multi-round interactions.
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Extra-Merge: Tracing the Rank-1 Subspace of Model Merging in Language Model Pre-Training
WenJie Zhou, Bohan Wang, Hongtao Zhang, Chenxi Jia, Wei Chen, Xueqi Cheng
Model merging has emerged as a lightweight paradigm for enhancing Large Language Models (LLMs), yet its underlying mechanisms remain poorly understood. In this work, we analyze late-stage pre-training trajectories and uncover a \textbf{Ran k-1 Subspace} phenomenon: while raw optimization steps oscillate violently, cons ecutive \emph{merged} checkpoints collapse onto a stable, approximately one-dime nsional linear manifold. We theoretically ground this observation in a \emph{riv er-valley} landscape analysis: averaging acts as a geometric low-pass filter tha t dampens high-curvature noise to reveal the optimal descent direction. Capitali zing on this insight, we propose \textbf{Extra-Merge}, a training-free strategy that extrapolates along this subspace to minimize loss without additional gradie nt updates. Extensive experiments across GPT-2 and LLaMA families (124M to 2B) d emonstrate that Extra-Merge consistently outperforms standard merging baselines. Notably, it yields consistent zero-shot accuracy gains on Pythia-12B downstream tasks and generalizes effectively to the Muon optimizer (Jordan et al., 2024).
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Diffusion Differentiable Resampling
Jennifer R. Andersson, Zheng Zhao
This paper is concerned with differentiable resampling in the context of sequent ial Monte Carlo (e.g., particle filtering). Drawing on reparametrisation, we pro pose a new resampling method that is informative and instantly differentiable, b ased on a training-free diffusion model surrogate. We theoretically prove that o ur diffusion resampling method provides a consistent resampling distribution, an d we show empirically that it outperforms the state-of-the-art differentiable re sampling methods on multiple filtering and parameter estimation benchmarks. Fina lly, we show that it achieves competitive end-to-end performance when used in le arning a complex dynamics-decoder model with high-dimensional image observations .
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Iterative Robust Satisficing: Minimizing Performance Degradation Under Distribut ion Shift
Enes A■■rman, Artun Saday, Cem Tekin
Modern neural networks often achieve high accuracy on their training distributio n but degrade sharply under distribution shifts.
We address this problem through *Robust Satisficing* (RS), an optimization objec tive that seeks parameters which attain a target level of in-distribution perfor mance while minimizing *fragility*, defined as the rate at which performance det eriorates as the data distribution departs from training.
We develop a gradient-based algorithm, *Iterative Robust Satisficing* (IRS), tha t directly optimizes this criterion.
Across a range of synthetic and real-world distribution shifts, including long-t ailed image classification, group shifts induced by spurious correlations, and n atural shifts in tabular regression, IRS consistently improves performance on mi nority and worst-case groups without sacrificing overall accuracy.
Notably, IRS achieves these robustness gains with a per-step computational cost similar to standard stochastic gradient descent and requires only a single forwa rd and backward pass per update.
Together, these results suggest that minimizing fragility provides a practical a nd effective alternative to existing robust training methods for learning models that remain reliable under distribution shift.
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All ERMs Can Fail in Stochastic Convex Optimization Lower Bounds in Linear Dimen sion
Tal Burla, Roi Livni

We study the sample complexity of the *best-case* Empirical Risk Minimizer in th e setting of stochastic convex optimization. We show that there exists an instan ce in which the sample size is linear in the dimension, learning is possible, bu t the Empirical Risk Minimizer is likely to be *unique* and to *overfit*. This r esolves an open question by Feldman. We also extend this to approximate ERMs.

Building on our construction we also show that (constrained) Gradient Descent po tentially overfits when horizon and learning rate grow w.r.t sample size. Specif ically we provide a novel generalization lower bound of $\Omega\left(\sqrt{\eta T/m^{1.5}}\right)$ for Gradient Descent, where η is the learning rate, T is the horizon and m is the sample size. This narrows down, exponentially, the gap between the best known upper bound of $O(\eta T/m)$ and existing lower boun ds from previous constructions.
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HEDP: A Hybrid Energy-Distance Prompt-based Framework for Domain Incremental Lea rning
Yu Feng, Zhen Tian, Haoran Luo, Xie Yu, Diancheng Cheng, Haoyue Zheng, Shuai Lyu , Ping Zong, Lianyuan Li, xin ge, Yifan Zhu
Domain Incremental Learning is a critical scenario that requires models to conti nuously adapt to new data domains without retraining.
However, domain shifts often cause severe performance degradation. To address th is, we propose Hybrid Energy-Distance Prompt, a domain-incremental framework ins pired by Helmholtz free energy. HEDP introduces an energy regularization loss to enhance the separability of domain representations and a hybrid energy-distance weighted mechanism that fuses energy-based and distance-based cues to improve d omain selection and generalization. Experiments on multiple benchmarks, includin g CORe50, show that HEDP achieves superior performance on unseen domains with a 2.57\% accuracy gain, effectively mitigating catastrophic forgetting and enhanci ng open-world adaptability. Our code is available at \url{https://github.com/dan nis97500/HEDP}.
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From LLM-Generated Conjectures to Lean Formalizations: Automated Polynomial Ineq uality Proving via Sum-of-Squares Certificates
Ruobing Zuo, Hanrui Zhao, Gaolei He, Zhengfeng Yang, Jianlin Wang
Automated proving of polynomial inequalities is a fundamental challenge in autom ated mathematical reasoning, where rich algebraic structure and a rapidly growin g certificate search space hinder scalability. Purely symbolic approaches provid e strong guarantees but often scale poorly as the number of variables or the deg ree increases, due to expensive algebraic manipulations and rapidly growing inte rmediate expressions. In parallel, LLM-guided methods have made notable progress , particularly on competition-style inequalities with a small number of variable s. To address the remaining scalability challenges, we propose NSPI, a neuro-sy mbolic framework that combines the complementary strengths of LLMs and symbolic computation for polynomial-inequality proving. Concretely, an LLM proposes a con jecture in the form of an approximate polynomial Sum-Of-Squares (SOS) decomposit ion; we refine it via symbolic computation to obtain an exact polynomial SOS re presentation, which directly proves the target inequality, and we further certif y the proof in Lean, yielding an end-to-end pipeline from heuristic discovery to machine-checked proof. Experiments on challenging benchmarks involving polynomi als with up to 10 variables demonstrate the effectiveness and scalability of the proposed method.
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Scaling Laws for Precision in High-Dimensional Linear Regression
Dechen Zhang, Xuan Tang, Yingyu Liang, Difan Zou
Low-precision training is critical for optimizing the trade-off between model qu ality and training costs, necessitating the joint allocation of model size, data set size, and numerical precision. While empirical scaling laws suggest that qua ntization impacts effective model and data capacities or acts as an additive err or, the theoretical mechanisms governing these effects remain largely unexplored . In this work, we initiate a theoretical study of scaling laws for low-precisio

n training within a high-dimensional sketched linear regression framework. By an alyzing multiplicative (signal-dependent) and additive (signal-independent) quan tization, we identify a critical dichotomy in their scaling behaviors. Our analy sis reveals that in the worst case, while both schemes introduce an additive err or and degrade the effective data size, they exhibit distinct effects on effecti ve model size: multiplicative quantization maintains the full-precision model si ze, whereas additive quantization reduces the effective model size. Numerical ex periments validate our theoretical findings. By rigorously characterizing the co mplex interplay among model scale, dataset size, and quantization error, our wor k provides a principled theoretical basis for optimizing training protocols unde r practical hardware constraints.
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R^{3}L: Reasoning 3D Layouts from Relative Spatial Relations
Zhifeng Gu, Yuqi Wang, Bing WANG
Relative spatial relations provide a compact representation of spatial structure and are fundamental to relative spatial reasoning in 3D layout generation.
Recent works leverage Multimodal Large Language Models (MLLMs) to infer such rel ations, but the inferred relations are often unreliable and are typically handle d with post-hoc heuristics.
In this paper, we propose R^{3}L, a general framework that improves the reliabil ity and consistency of relative spatial reasoning for 3D layout generation.
Our key motivation is that multi-hop reasoning requires repeated reference-frame transformations, which accumulate errors in inferred relations and lead to sema ntic and metric drift.
To mitigate this, we propose invariant spatial decomposition to break coupled re lation chains, and consistent spatial imagination to promote self-consistency th rough an imagine-and-revise loop.
We further introduce supportive spatial optimization to ease pose optimization v ia global-to-local coordinate re-parameterization.
Extensive experiments across diverse scene types and instructions demonstrate th at R^{3}L produces more physically feasible and semantically consistent layouts.

Notably, our analysis shows that resolving frame-induced inconsistencies is cruc ial for reliable multi-hop relative spatial reasoning.
The code is available at https://github.com/Neal2020GitHub/R3L.
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Adaptive Physics Transformer with Fused Global-Local Attention for Subsurface En ergy Systems
Xin Ju, Hadrian Fung, Yuyan Zhang, Carl Jacquemyn, Matthew D Jackson, Randolph R . Settgast, Sally M Benson, Gege Wen
The Earth's subsurface is a cornerstone of modern society, providing essential e nergy resources like hydrocarbons, geothermal, and minerals while serving as the primary reservoir for CO_2 sequestration. However, full physics numerical sim ulations of these systems are notoriously computationally expensive due to geolo gical heterogeneity, high resolution requirements, and the tight coupling of phy sical processes with distinct propagation time scales. Here we propose the $\tex tbf{Adaptive Physics Transformer}$ (APT), a geometry-, mesh-, and physics-agnost ic neural operator that explicitly addresses these challenges. APT fuses a graph -based encoder to extract high-resolution local heterogeneous features with a gl obal attention mechanism to resolve long-range physical impacts. Our results dem onstrate that APT outperforms state-of-the-art architectures in subsurface tasks across both regular and irregular grids with robust super-resolution capabiliti es. Notably, APT is the first architecture that learns directly from HR-adaptive mesh refinement simulations. We also demonstrate APT's favorable scaling behavi or and cross-dataset learning capability, positioning it as a robust and scalabl e backbone for large-scale subsurface foundation model development.
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PINE: Pruning Boosted Tree Ensembles with Conformal In-Distribution Prediction E quivalence
Haruki Yajima, Yusuke Matsui

Tree ensembles are machine learning models with strong predictive performance and interpretability, and remain widely used for tabular data.
Standard pruning methods for tree ensembles typically optimize an accuracy-compression trade-off and may change a subset of predictions, potentially compromising decision consistency.
Faithful pruning methods address this issue by preserving prediction equivalence over the entire input space, but this requirement leads to lower compression ratios.
We propose **PINE**, a pruning method that provides strong guarantees within an in-distribution region.
PINE preserves prediction equivalence within this region and controls the region size using a single parameter α via conformal calibration.
Experiments on 12 public tabular datasets show that PINE improves the compression ratio by up to 30% while preserving predictions at a comparable level to existing faithful pruning methods.
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Stein Diffusion Guidance: Training-Free Posterior Correction for Sampling Beyond High-Density Regions
Van Khoa Nguyen, Lionel Blondé, Alexandros Kalousis
Training-free diffusion guidance offers a flexible framework for leveraging off-the-shelf classifiers without additional training. Yet, current approaches hinge on posterior approximations via Tweedie’s formula, which often yield unreliable guidance, particularly in low-density regions. Stochastic optimal control (SOC), in contrast, enables principled posterior sampling but remains computationally prohibitive for efficient inference. In this work, we reconcile the strengths of these paradigms by introducing Stein Diffusion Guidance (SDG), a novel training-free framework grounded in a surrogate SOC objective. We establish a new theoretical bound on the SOC value function, revealing the necessity of correcting approximate posteriors to reflect true diffusion dynamics. Building on Stein variational inference, SDG computes the steepest descent direction that minimizes the Kullback-Leibler divergence between approximate and true posteriors. By integrating a principled Stein correction mechanism along with a novel running cost functional, SDG enables effective guidance in low-density regions. Our experiments on diverse image-guidance tasks and on challenging small-ligand sampling for protein docking suggest that SDG consistently outperforms standard training-free guidance methods and highlights its potential for broader posterior sampling problems beyond high-density regimes.
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Rate or Fate? RLV$^{\varepsilon}$R: Reinforcement Learning with Verifiable Noisy Rewards
Ali Rad, Khashayar Filom, Darioush Keivan, Peyman Mohajerin Esfahani, Ehsan Kamalinejad
Reinforcement learning with verifiable rewards (RLVR) trains a policy by verifying sampled completions and reinforcing higher-scoring outputs, but practical verifiers (e.g., incomplete unit tests or noisy judges) are prone to false positives and false negatives.
We ask when such noise merely slows learning and when it reverses it.
Modeling GRPO-style RLVR as a bandit over recurring \emph{reasoning modes}, we derive mean-field replicator-style (natural-selection) flow on the probability simplex. The dynamics decouples into within-correct-mode competition and a one-dimensional evolution for the mass on incorrect modes, whose drift is determined solely by Youden's index $J=\mathrm{TPR}-\mathrm{FPR}$. This yields a sharp phase transition: when $J>0$, the incorrect mass is driven toward extinction (learning); when $J=0$, the process is neutral; and when $J<0$, incorrect modes amplify until they dominate (anti-learning and collapse). In the learning regime $J>0$, noise primarily rescales convergence time (``rate, not fate''). Experiments on verifiable programming tasks under synthetic noise reproduce the predicted $J=0$ boundary. Beyond noise, the framework offers a general lens for analyzing RLVR stability, convergence, and algorithmic interventions.
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Language Model Networks: Supervision-Efficient Learning through Dense Communication
Shiguang Wu, Yaqing Wang, Quanming Yao
Language models are increasingly used not only as standalone predictors but also as components in larger inference systems, from test-time scaling to multi-agent collaboration. We study *language model networks*, where pre-trained language models serve as reusable nodes and intelligence emerges from their topology, communication, and optimization. Existing systems mostly communicate through natural language: easy to deploy, but discrete, inefficient, and hard to optimize from end-task supervision. We propose LMNet, a dense and differentiable realization of this paradigm. LMNet uses stripped LLMs as vertex modules and trainable seq2seq modules as communication edges, enabling intermediate nodes to exchange dense vectors while preserving natural-language input and output at the system boundary. By bypassing intermediate embedding and de-embedding, LMNet enables efficient information transfer, end-to-end gradient optimization, and learned communication beyond hand-designed protocols. Experiments show performance with small additional training cost and effective adaptation under limited supervision.
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Trajectory-Level Data Augmentation for Offline Reinforcement Learning
Tobias Schmähling, Matthias Burkhardt, Tobias Windisch
We propose a data augmentation method for offline reinforcement learning, motivated by active positioning problems.
Particularly, our approach enables the training of off-policy models from a limited number of
suboptimal trajectories. We introduce a trajectory-based augmentation technique that exploits task structure and the
geometric relationship between rewards, value functions, and mathematical properties of logging policies. During data
collection, our augmentation supports suboptimal logging policies, leading to higher data quality
and improved offline reinforcement learning performance.
We provide theoretical justification for these strategies and validate them empirically across positioning tasks of varying dimensionality and under partial observability.
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LoRA-DA: Data-Aware Initialization for Low-Rank Adaptation via Asymptotic Analysis
Qingyue Zhang, Chang Chu, Tianren Peng, Qi Li, Xiangyang Luo, Zhihao Jiang, Shao-Lun Huang
LoRA has become a widely adopted method for PEFT, and its initialization methods have attracted increasing attention.
However, existing methods have notable limitations: many methods do not incorporate target-domain data, while gradient-based methods exploit data only at a shallow level by relying on one-step gradient decomposition. In this paper, we establish a theoretical framework for data-aware LoRA initialization. Starting from minimizing the expectation of the parameter discrepancy between the fine-tuned and target models, we derive an optimization problem with two components: a bias term, which is related to the parameter distance between the fine-tuned and target models, and is approximated using a Fisher–gradient formulation to preserve anisotropy; and a variance term, which accounts for the uncertainty introduced by sampling stochasticity through the Fisher information. Solving this problem yields an optimal initialization strategy for LoRA, based on which we develop an efficient algorithm, LoRA-DA. Empirical results across multiple benchmarks demonstrate that LoRA-DA consistently improves final accuracy over existing initialization methods. Additional studies show faster, more stable convergence, robustness across ranks, and only a small initialization overhead for LoRA-DA. The source code is available at https://github.com/zqy0126/LoRA-DA.
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VideoTrace-R1: Long Video-based Retrieval-Augmented Generation via Reinforcement Learning

Zongsheng Cao, Anran Liu, Jun Xie, Feng Chen, Lang Chen, Jing Li, Zhepeng Wang, Zigan Wang

Long-video temporal reasoning remains a bottleneck for Large Video Language Models (LVLMs). Existing reinforcement-learning approaches reward only final-answer correctness, so they cannot distinguish answers reached through grounded reasoning from those reached through fabricated chronology; the intermediate temporal claims that constitute the reasoning are never verified. We trace this gap to a structural correspondence between two kinds of traces: a video has its own temporal trace, an ordered sequence of how events unfold, while a model's answer is built up through a reasoning trace, an ordered sequence of intermediate temporal claims. Correct reasoning requires the latter to mirror the former, claim by claim. We act on this correspondence with two contributions. We introduce Temporal Reasoning Traces (TRT), a structured index of a video's ordered event chains that exposes a small set of deterministic verification primitives, materializing the temporal trace as a programmatically queryable object. We then propose temporal-enhanced GRPO, a reinforcement-learning procedure whose reward decomposes into per-block components, each computed by a TRT primitive on a typed think block of the reasoning trace. Because the reward is fully symbolic, fabricated temporal claims are caught at the per-claim level rather than masked by a correct final answer. Across long-video reasoning benchmarks, our model achieves state-of-the-art performance, with the largest gains on out-of-domain reasoning tasks such as Video-Holmes, CG-Bench-Reasoning, and VRBench.

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Letting Trajectories Spread: Quality-Preserving Control for Diverse Flow Matching

Jingxuan Wu, Zhenglin Wan, Xingrui Yu, Yuzhe YANG, Bo An, Ivor Tsang, Yang You

Flow-based text-to-image models follow deterministic trajectories, making it costly to explore diverse modes under limited sampling budgets. Existing approaches to improving diversity often rely on retraining or degrade image fidelity. To address this limitation, we present a training-free, inference-time control mechanism that makes the flow itself diversity-aware. Our core insight is to encourage diversity through guidance that is geometrically decoupled from the model’s quality-seeking direction. Our method simultaneously encourages lateral spread among trajectories via a feature-space objective and reintroduces uncertainty through a time-scheduled stochastic perturbation. Crucially, this perturbation is projected to be orthogonal to the generation flow, a geometric constraint that allows it to boost variation without degrading image details or prompt fidelity. Theoretically, we show that this design monotonically increases a volume surrogate while approximately preserving the marginal distribution, providing a principled explanation for the robustness of generation quality. Empirically, across multiple text-to-image settings under fixed sampling budgets, our method consistently improves diversity metrics such as the Vendi Score and Brisque over strong baselines, while upholding image quality and alignment.

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Dynamic Optimizations of LLM Ensembles with Two-Stage Reinforcement Learning Agents

Selim Furkan Tekin, Gaowen Liu, Ramana Rao Kompella, Ling Liu

The advancement of LLMs and their accessibility have triggered renewed interest in multi-agent reinforcement learning as robust and adaptive frameworks for dynamically changing environments. This paper introduces RL-Focal, a two-stage RL agent framework that routes and ensembles LLMs. First, we develop the Decider RL-agent, which learns to dynamically select an ensemble of small size (m_i) among N LLMs ($m_i \ll N$) for incoming queries from a user-defined downstream task i, by maximizing both error-diversity and reasoning-performance of the selected ensemble through iterative updates of task-adaptive rewards and policy. Second, to enable effective fusion of dynamically selected LLMs, we develop the stage-2 Fusion RL-agent, which learns to resolve reasoning conflicts from different LLMs and dynamically adapt to different ensemble teams composed by the Decider Agent for different downstream tasks. {\em Third}, we introduce the focal diversity metric to better model the error correlations among multiple LLMs further impr

oving the generalization performance of the Decider Agent, which actively prunes the ensemble combinations. By focal diversity, we enhance performance across tasks by effectively promoting reward-aware and policy-adaptive ensemble selection and inference fusion. Extensive evaluations on five benchmarks show that RL-Focal achieves the performance improvement of 8.48\% with an ensemble of small size compared to the best individual LLM in a pool and offers stronger robustness. Code is available at \url{https://anonymous.4open.science/r/rl-focal-8DCF/}
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Characterizing the Predictive Impact of Modalities with Supervised Latent-Variable Modeling
Divyam Madaan, Sumit Chopra, Kyunghyun Cho
Despite the recent success of Multimodal Large Language Models (MLLMs), existing approaches predominantly assume the availability of multiple modalities during training and inference. In practice, multimodal data is often incomplete because modalities may be missing, collected asynchronously, or available only for a subset of examples. In this work, we propose PRIMO, a supervised latent-variable imputation model that quantifies the predictive impact of any missing modality within the multimodal learning setting. PRIMO enables the use of all available training examples, whether modalities are complete or partial. Specifically, it models the missing modality through a latent variable that captures its relationship with the observed modality in the context of prediction. During inference, we draw many samples from the learned distribution over the missing modality to both obtain the marginal predictive distribution (for the purpose of prediction) and analyze the impact of the missing modalities on the prediction for each instance. We evaluate PRIMO on a synthetic XOR dataset, Audio-Vision MNIST, and MIMIC-III for mortality and ICD-9 prediction. Across all datasets, PRIMO obtains performance comparable to unimodal baselines when a modality is fully missing and to multimodal baselines when all modalities are available.
PRIMO quantifies the predictive impact of a modality at the instance level using a variance-based metric computed from predictions across latent completions. We visually demonstrate how varying completions of the missing modality result in a set of plausible labels.
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Second-Order Smooth Planning with Optimal-Transport Bellman Smoothing
Tuan Quang Dam
Planning with a generative model aims to estimate the value of a state using as few simulator calls as possible.
SmoothCruiser achieves problem-independent complexity $\widetilde O(\varepsilon^{-4})$ by exploiting the smoothness of the entropy-regularized Bellman backup, but its estimator is only first-order.
We show that the sample-complexity exponent of SmoothCruiser-type planners is governed by the order β of the local Taylor remainder, giving oracle complexity $\widetilde O(\varepsilon^{-(2+2/(\beta-1))})$: the first-order case $\beta=2$ recovers SmoothCruiser, while a second-order/cubic remainder $\beta=3$ yields $\widetilde O(\varepsilon^{-3})$.
We reach this regime with an optimal-transport-smoothed Bellman backup over action distributions, which has a closed form, a policy gradient, and a Lipschitz Hessian, and whose quadratic correction admits an unbiased cross-product estimator.
The resulting SecondOrderSmoothCruiser achieves $\widetilde O(\varepsilon^{-3})$ oracle complexity for fixed OT parameters, and we relate the OT, entropy-regularized, and unregularized objectives through explicit regularization-bias bounds.
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Steal the Patch Size: Adversarially Manipulate Vision Language Models
Kai Hu, Akash Bharadwaj, Weichen Yu, Matt Fredrikson
We present a black-box model-stealing attack that recovers private vision-tokenizer configurations of deployed vision-language models (VLMs), including the visual patch size and input preprocessing pipeline. The key idea is a task-level side channel induced by ViT-style patchification: when a synthetic grid image is aligned with the hidden patch grid, boundary cues are erased at tokenization, caus

ing periodic accuracy drop. By sweeping the grid cell size and measuring these c
ollapses, we infer the patch size; by introducing padding and a consistency-chec
k test, we further identify whether preprocessing is dynamic- or fixed-resolutio
n and recover the target resize resolution. Across open-source Qwen-VL variants
and proprietary models including GPT and Claude, we reliably recover tokenizer-r
elated parameters. Finally, we show that such leakage enables preprocessing-awar
e transfer attacks and model-targeted adversarial manipulation.
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Spiked-CFR: Causal Representation Learning from LLMs via Wasserstein Projection
Pursuit
Fan Wang, Hengyu Yue, Yu Bowen, Weiming Liu, Zongxin Yang, Xuyun Zhang, Xiaolin
Zheng, Chaochao Chen, Shuiguang Deng
Estimating treatment effects from observational text is increasingly practical w
ith Large Language Models (LLMs). However, applying causal representation learni
ng directly to high-dimensional LLM embeddings faces a fundamental barrier: empi
rical Wasserstein matching suffers from the curse of dimensionality, rendering s
tandard generalization guarantees effectively vacuous. We propose SPIKED-CFR, a
framework bridging this gap by assuming a Spiked Structure, where treatment sele
ction bias is assumed to manifest primarily as a low-dimensional treated--contro
l discrepancy in the semantic representation. We develop Wasserstein Projection
Pursuit, a minimax objective that adversarially learns an orthogonal projection
on the Stiefel manifold to identify and balance only this subspace while preserv
ing prognostic information. Under a spiked structure, we show the projected disc
repancy can be estimated at a rate governed by the intrinsic dimension $k \ll D$
, and we derive a tighter PEHE generalization bound that depends on k rather t
han the ambient embedding dimension.
Experiments on four semi-synthetic benchmarks and four real-world clinical bench
marks demonstrate improved accuracy and robustness over strong baselines.
****
On the Limits of Test-Time Compute: Sequential Reward Filtering for Better Infer
ence
Yue Yu, Qiwei Di, Quanquan Gu, Dongruo Zhou
Test-time compute (TTC) has become an increasingly prominent paradigm for enhanc
ing large language models (LLMs). Despite the empirical success of methods such
as best-of-n (BoN) sampling and sequential revision, their fundamental limits
remain unclear. We address this gap by analyzing a mixture-of-reference policy m
odel and proving that standard BoN is inherently suboptimal. To move closer to t
he optimal frontier, we study reward-filtered sequential inference, a simple pro
cedure that selectively incorporates only high-reward generations into the conte
xt. This mechanism concentrates computation on superior policy candidates and su
ppresses inferior ones. On the theoretical side, we show that reward-filtered se
quential inference yields strictly stronger guarantees than standard TTC paradig
ms. On the empirical side, we evaluate such an inference strategy across diverse
 benchmarks and observe consistent improvements over widely used approaches, dem
onstrating the practical effectiveness of our framework.
****
Offline Reinforcement Learning with Generative Trajectory Policies
Xinsong Feng, Leshu Tang, Chenan Wang, Haipeng Chen
Generative models have emerged as a powerful class of policies for offline reinf
orcement learning (RL) due to their ability to capture complex, multi-modal beha
viors.
However, existing methods face a stark trade-off: slow, iterative models like di
ffusion policies are computationally expensive, while fast, single-step models l
ike consistency policies often suffer from degraded performance.
In this paper, we demonstrate that it is possible to bridge this gap.
The key to moving beyond the limitations of individual methods, we argue, lies i
n a unifying perspective that views modern generative models—including diffusion
, flow matching, and consistency models—as specific instances of learning a cont
inuous-time generative trajectory governed by an Ordinary Differential Equation
(ODE).

This principled foundation provides a clearer design space for generative policies in RL and allows us to propose *Generative Trajectory Policies* (GTPs), a new and more general policy paradigm that learns the entire solution map of the underlying ODE.
To make this paradigm practical for offline RL, we further introduce two key theoretically principled adaptations.
Empirical results demonstrate that GTP achieves state-of-the-art performance on D4RL benchmarks -- it significantly outperforms prior generative policies, achieving perfect scores on several notoriously hard AntMaze tasks.
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Estimating Correlation Clustering Cost in Node-Arrival Stream
Kaiwen Liu, Seba Daniela Villalobos, Qin Zhang
We study the correlation clustering problem in the node-arrival data stream model. Unlike previous work, where the stream consists of the graph's edges, we focus on the setting in which the stream contains only the nodes. This model better reflects many real-world scenarios in which the data stream naturally consists of raw objects (e.g., images, tweets), and the similar/dissimilar edges are derived through a similarity function. We present *C*^{4}*Approx*, a streaming algorithm that approximates the cost of correlation clustering using sublinear space in the number of nodes and a constant number of passes. We further complement this result with lower bounds. Experiments on real-world datasets show that by storing only 2\% of the nodes, our algorithm achieves performance comparable to the classic *Pivot* algorithm and the more recent *PrunedPivot* algorithm, even on sparse graphs.
****

Random Scaling of Emergent Capabilities
Rosie Zhao, Tian Qin, David Alvarez-Melis, Sham M. Kakade, Naomi Saphra
Language models famously improve under a smooth scaling law, but some specific capabilities exhibit sudden breakthroughs in performance. Advocates of "emergence" view these capabilities as unlocked at a specific scale, but others attribute breakthroughs to superficial metric thresholding effects. We propose that breakthroughs are instead driven by continuous changes in the probability distribution of training outcomes when performance is bimodally distributed across random seeds. We show that different random seeds can produce either smooth or emergent scaling trends in synthetic length generalization tasks, multiple choice question answering, and grammatical generalization. We reveal that sharp breakthroughs in metrics are produced by underlying continuous changes in their distribution across seeds. These distributions may become abruptly bimodal at a capacity threshold—but this threshold appears at scales well before most seeds achieve breakthrough. Our observations hold true even under continuous loss metrics, confirming that random variation must be considered when predicting a model's performance from its scale.
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FlowPET: Physics-Informed Symplectic Flow Matching for Low-Count PET Reconstruction
Zheng Zhang, Hao Tang, Yingying Hu, zhanli hu, Jing Qin
Low-count Positron Emission Tomography (PET) reconstruction is severely hindered by the dissipative nature of prevailing generative models, where the inherent phase-space contraction leads to the numerical extinction (``wash-out'') of weak but diagnostically critical lesion signals. To overcome this geometric limitation, we propose \textbf{FlowPET}, a physics-informed framework that reformulates reconstruction as volume-preserving transport in a symplectic phase space. By parameterizing the posterior dynamics via a Separable Hamiltonian System, our approach guarantees a divergence-free vector field by construction, theoretically immunizing weak signals against probability mass collapse. To steer this conservative flow, we introduce conjugate boundary conditions based on the Range-Null space decomposition of the PET operator; this strictly enforces data consistency in the range space while confining stochastic uncertainty injection to the unobserved null space. We train the model via symplectic flow matching and perform inference using a symplectic leapfrog integrator. Extensive experiments on BrainWeb,

clinical pediatric, and UDPET datasets demonstrate that \textbf{FlowPET} not onl
y surpasses state-of-the-art deterministic and stochastic baselines in SSIM and
PSNR but, more crucially, exhibits superior recovery of low-contrast lesions. Th
e results confirm that imposing Hamiltonian structural constraints offers a robu
st geometric safeguard for medical inverse problems in high-noise regimes.
****

Multi-marginal temporal Schrödinger Bridge Matching from unpaired data
Thomas Gravier, Thomas Boyer, Auguste Genovesio
Many natural dynamic processes –such as in vivo cellular differentiation or dise
ase progression– can only be observed through the lens of static sample snapshot
s. While challenging, reconstructing their temporal evolution to decipher underl
ying dynamic properties is of major interest to scientific research. Existing ap
proaches enable data transport along a temporal axis but are poorly scalable in
high dimension and require restrictive assumptions to be met. To address these i
ssues, we propose Multi-Marginal temporal Schrödinger Bridge Matching (MMtSBM) f
rom unpaired data, extending the theoretical guarantees and empirical efficiency
 of Diffusion Schrödinger Bridge Matching (Shi et al., 2023) by deriving the Ite
rative Markovian Fitting algorithm to multiple marginals in a novel factorized f
ashion. Experiments show that MMtSBM retains theoretical properties on toy examp
les, achieves state-of-the-art performance on real-world datasets such as transc
riptomic trajectory inference in 100 dimensions, and, for the first time, recove
rs couplings and dynamics in very high-dimensional image settings. Our work esta
blishes multi-marginal Schrödinger bridges as a practical and principled approac
h for recovering hidden dynamics from static data.
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Foresee-to-Ground: From Predictive Temporal Perception to Evidence-Driven Reason
ing for Video Temporal Grounding
Zelin Zheng, Xinyan Liu, Ruixin Li, Antoni B. Chan, Guorong Li, Qingming Huang,
Laiyun Qing
Current Video-LLM approaches for Video Temporal Grounding (VTG) typically rely o
n direct timestamp generation from an unstructured visual-token stream, often le
ading to brittle numerics and inconsistent boundaries. To address this, we propo
se Foresee-to-Ground (F2G), a framework that reformulates VTG as a verifiable Id
entify-then-Measure problem.

F2G integrates Predictive Temporal Perception with Evidence-Driven Reasoning:
 it learns boundary-sensitive temporal representations to build a video-wide evi
dence pool of candidate event segments, and exposes these segments to the LLM as
 citable evidence units that bind boundary prediction to explicit event hypothes
es.

By decoupling event identification from precise boundary measurement, F2G stab
ilizes grounding and makes predictions verifiable.

Extensive experiments demonstrate that F2G consistently improves grounding acc
uracy across diverse benchmarks, transfers robustly across different Video-LLM b
ackbones, and preserves general video understanding capabilities.

Our project is available at https://github.com/zelion2003/Foresee-to-Ground.
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Prompt Injection as Role Confusion
Charles Ye, Jasmine Cui, Dylan Hadfield-Menell
LLMs see the world as a single stream of text, partitioned into *roles* like &lt
;user> or <tool>. We trace **prompt injection** to **role confusion**:
models perceive the source of text from *how it sounds*, not its labeled role. A
 command hidden in a webpage hijacks an agent simply because it sounds like <
user> text, despite its <tool> label. We design *role probes* to measur
e how LLMs internally perceive "who is speaking", and find that injected text oc
cupies the same representational space as the trusted role it imitates. We demon
strate this with CoT Forgery, a zero-shot attack that injects fabricated reasoni
ng into user prompts and tool outputs. Models mistake the forgery for their own
thoughts, yielding 60% attack success against frontier models with near-zero bas
elines. Strikingly, the degree of role confusion predicts attack success before
a single token is generated. This mechanism generalizes beyond CoT Forgery to st

andard agent prompt injections, revealing prompt injection as a measurable conse
quence of role perception. To the model, sounding like a role is indistinguishab
le from being one.
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Golden Goose: A Simple Trick to Synthesize Unlimited RLVR Tasks from Unverifiabl
e Internet Text
Ximing Lu, David Acuna, Jaehun Jung, Jian Hu, Di Zhang, Shizhe Diao, Yunheng Zou
, Shaokun Zhang, Brandon Cui, Mingjie Liu, Hyunwoo Kim, Prithviraj Ammanabrolu,
Jan Kautz, Yi Dong, Yejin Choi
Reinforcement Learning with Verifiable Rewards (RLVR) has become a cornerstone f
or unlocking complex reasoning in Large Language Models (LLMs). Yet, scaling up
RL is bottlenecked by limited existing verifiable data, where improvements incre
asingly saturate over prolonged training.
To overcome this, we propose **Golden Goose**, a simple trick to synthesize unli
mited RLVR tasks from unverifiable internet text by constructing a multiple-choi
ce question-answering version of the fill-in-the-middle task. Given a source tex
t, we prompt an LLM to identify and mask key reasoning steps, then generate a se
t of diverse, plausible distractors.
This enables us to leverage reasoning-rich unverifiable corpora typically exclud
ed from prior RLVR data construction (e.g., science textbooks) to synthesize **G
ooseReason-0.7M**, a large-scale RLVR dataset with over 0.7 million tasks spanni
ng mathematics, programming, and general scientific domains.
Empirically, GooseReason effectively revives models saturated on existing RLVR d
ata, yielding robust, sustained gains under continuous RL and achieving new stat
e-of-the-art results for 1.5B and 4B-Instruct models across 15 diverse benchmark
s.
Finally, we deploy Golden Goose in a real-world setting, synthesizing RLVR tasks
 from raw FineWeb scrapes for the cybersecurity domain, where no prior RLVR data
 exists. Training Qwen3-4B-Instruct on the resulting data **GooseReason-Cyber**
sets a new state-of-the-art in cybersecurity, surpassing a 7B domain-specialized
 model with extensive domain-specific pre-training and post-training. This highl
ights the potential of automatically scaling up RLVR data by exploiting abundant
, reasoning-rich, unverifiable internet text.
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Formalizing Learning from Language Feedback with Provable Guarantees
Wanqiao Xu, Allen Nie, Ruijie Zheng, Aditya Modi, Adith Swaminathan, Ching-An Ch
eng
Interactively learning from observation and language feedback is an increasingly
 studied area driven by the emergence of large language model (LLM) agents. Whil
e impressive empirical demonstrations have been shown, so far a principled frami
ng of these decision problems remains lacking. In this paper, we formalize the
Learning from Language Feedback (LLF) problem, assert sufficient assumptions to
enable learning despite latent rewards, and introduce *transfer eluder dimension
* as a measure to characterize the hardness of LLF problems.
We formalize the intuition that information in the feedback governs the learning
 complexity of LLF problems. We demonstrate cases where learning from rich lang
uage feedback can be exponentially faster than learning from reward.
We develop a no-regret algorithm, called `HELiX`,
that provably solves LLF problems through sequential interactions, with performa
nce guarantees
that scale with the transfer eluder dimension of the problem.
Across several empirical domains, we show that `HELiX` performs well even when r
epeatedly prompting LLMs does not work reliably.
Our contributions mark an important step towards designing principled interactiv
e learning algorithms from generic language feedback.
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Ambient Dataloops: Generative Models for Dataset Refinement
Adrian Rodriguez-Munoz, William Daspit, Adam Klivans, Antonio Torralba, Constant
inos Costis Daskalakis, Giannis Daras
We propose Ambient Dataloops, an iterative framework for refining datasets that

makes it easier for diffusion models to learn the underlying data distribution.
Modern datasets contain samples of highly varying quality, and training directly on such heterogeneous data often yields suboptimal models. We propose a dataset-model co-evolution process; at each iteration of our method, the dataset becomes progressively higher quality, and the model improves accordingly. To avoid destructive self-consuming loops, at each generation, we treat the synthetically improved samples as noisy, but at a slightly lower noisy level than the previous iteration, and we use Ambient Diffusion techniques for learning under corruption. Empirically, Ambient Dataloops achieve state-of-the-art performance in unconditional and text-conditional image generation and de novo protein design. We further provide a theoretical justification for the proposed framework that captures the benefits of the data looping procedure.
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One Coin Has Two Sides: Single Poistive Multi Label Learning from Salient Annotations
Xiaoyu Wang, Zhuoming Li, Bo Han, Hui LIU, Junhui Hou, Yuheng Jia
Single-Positive Multi-Label Learning (SPML) studies learning from incomplete supervision, where each instance is annotated with only one positive label despite potentially belonging to multiple categories.
While existing methods assume the annotated labels are randomly distributed, real-world annotations are often biased toward the most salient category.
We formalize this realistic scenario as Salient Single-Positive Multi-Label Learning (SalSPML).
This salient annotation bias poses a challenge to conventional SPML methods,
as the missing labels often correspond to less salient and harder-to-recognize categories.
Fortunately, we find that salient annotations are typically more representative and informative.
Motivated by this insight, we propose Prototype-Guided Rejection for Salient Annotation (PiSA),
which constructs reliable class-wise prototypes from salient labels and leverages them to guide embedding learning for non-salient labels recognition.
We theoretically demonstrate that SalSPML is harder than Random SPML due to irreducible annotation bias, and under SalSPML, more accurate prototypes facilitate false-negative label detection.
Experiments on multiple benchmarks, together with two newly constructed real-world SalSPML datasets, demonstrate that PiSA consistently outperforms existing methods, achieving an average mAP improvement of 3.16\%.
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Accurate Evaluation of Quickest Changepoint Detectors via Non-parametric Survival Analysis
Taiki Miyagawa, Akinori F. Ebihara
We propose non-parametric estimators for the average run length (ARL) and average detection delay (ADD) in quickest changepoint detection (QCD) under finite and irregular sequence lengths.
Although ARL and ADD are widely used as optimality criteria in theoretical and simulation studies, their application to real-world datasets is hindered by limited and irregular sequence lengths.
To address this issue, we propose non-parametric estimators for the ARL and ADD, termed _KM-ARL and KM-ADD_, by drawing an analogy between QCD and survival analysis to model detection probabilities under sequence truncation.
We derive estimation bias bounds and prove that they are asymptotically unbiased unless extrapolation is required.
Experiments on simulated and real-world datasets demonstrate their practical utility, enhancing robustness against limited and irregular sequence lengths, improving interpretability, and facilitating empirical, intuitive model selection.
Our Python code is provided at https://github.com/TaikiMiyagawa/Kaplan-Meier-Average-Run-Length, offering ready-to-use implementations for practitioners.
****

DC-W2S: Dual-Consensus Weak-to-Strong Training for Reliable Process Reward Model

ing in Biological Reasoning
Chi-Min Chan, Ehsan Hajiramezanali, Xiner Li, Edward De Brouwer, Carl Edwards, Wei Xue, Sirui Han, Yike Guo, Gabriele Scalia
In scientific reasoning tasks, the veracity of the reasoning process is as critical as the final outcome. While Process Reward Models (PRMs) offer a solution to the coarse-grained supervision problems inherent in Outcome Reward Models (ORMs), their deployment is hindered by the prohibitive cost of obtaining expert-verified step-wise labels. This paper addresses the challenge of training reliable PRMs using abundant but noisy "weak" supervision. We argue that existing Weak-to-Strong Generalization (W2SG) theories lack prescriptive guidelines for selecting high-quality training signals from noisy data. To bridge this gap, we introduce the Dual-Consensus Weak-to-Strong (DC-W2S) framework. By intersecting Self-Consensus (SC) metrics among weak supervisors with Neighborhood-Consensus (NC) metrics in the embedding space, we stratify supervision signals into distinct reliability regimes. We then employ a curriculum of instance-level balanced sampling and label-level reliability-aware masking to guide the training process. We demonstrate that DC-W2S enables the training of robust PRMs for complex reasoning without exhaustive expert annotation, proving that strategic data curation is more effective than indiscriminate training on large-scale noisy datasets.
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Thinking in Scales: Accelerating Gigapixel Pathology Image Analysis via Adaptive Continuous Reasoning
Jiusong Ge, Yingkang Zhan, Wenjie Zhao, Di Zhang, Ke Wang, Jiashuai Liu, Chunze Yang, Chengzu Li, Jian Zhang, Yuxin Dong, Ni Zhang, Qidong Liu, Mireia Crispin-Ortuzar, Huazhu Fu, Chen Li, Zeyu Gao
Traditional whole slide image (WSI) analysis methods typically rely on the multiple instance learning (MIL) paradigm, which extracts patch-level features at high magnification and aggregates them for slide-level prediction.
However, such exhaustive patch-level processing is computationally expensive, severely limiting the efficiency and scalability of WSI analysis.
To address this challenge, we propose PathCTM (a Pathology-oriented Continuous Thought Model) that enables token-efficient scale-space continuous reasoning for gigapixel WSIs. PathCTM formulates diagnostic inference as a dynamic sequential information pursuit. It progressively transitions from low-magnification global to high-magnification local inspection, and adaptively terminates inference when sufficient evidence is gathered to effectively bound decision uncertainty. Specifically, it uses conditional computation for dynamic scale switching with attention-guided region pruning, coupled with confidence-aware early stopping. Extensive experiments demonstrate that, compared with standard MIL-based methods, PathCTM reduces the number of required image patches by 95.95\% and shortens inference time by approximately 95.62\%, while maintaining AUC without degradation. Code is available at https://github.com/JSGe-AI/PathCTM.
****
Shrinking the Variance: Shrinkage Baselines for Reinforcement Learning with Verifiable Rewards
Guanning Zeng, Zhaoyi Zhou, Daman Arora, Andrea Zanette
Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a powerful paradigm for post-training large reasoning models (LRMs) using policy-gradient methods such as GRPO. To stabilize training, these methods typically center trajectory rewards by subtracting the empirical mean for each prompt. Statistically, this centering acts as a control variate (or baseline), reducing the variance of the policy-gradient estimator. Typically, the mean reward is estimated using per-prompt empirical averages for each prompt in a batch. Drawing inspiration from Stein's paradox, we propose using shrinkage estimators that combine per-prompt and cross-prompt means to improve the overall per-prompt mean estimation accuracy, particularly in the low-generation regime typical of RLVR. Theoretically, we construct a shrinkage-based baseline that provably yields lower-variance policy-gradient estimators across algorithms. This baseline serves as a drop-in replacement for existing per-prompt mean baselines, requiring no additional hyperparameters or computation. Empirically, shrinkage baselines consistently outperform st

andard empirical-mean baselines, leading to lower-variance gradient updates and improved training stability.
****
Conditional Distributional Treatment Effects: Doubly Robust Estimation and Testing
Saksham Jain, Alex Luedtke
Beyond conditional average treatment effects, treatments may impact the entire outcome distribution in covariate-dependent ways, for example, by altering the variance or tail risks for specific subpopulations. We propose a novel estimand to capture such conditional distributional treatment effects, and develop a doubly robust estimator that is minimax optimal in the local asymptotic sense. Using this, we develop a test for the global homogeneity of conditional potential outcome distributions that accommodates discrepancies beyond the maximum mean discrepancy (MMD), has provably valid type 1 error, and is consistent against fixed alternatives---the first test, to our knowledge, with such guarantees in this setting. We then provide a test that aggregates evidence across a grid of kernel-bandwidth choices. Furthermore, we derive exact closed-form expressions for two natural discrepancies (including the MMD), and provide a computationally efficient, permutation-free algorithm for our test.
****
Convex Low-resource Accent-Robust Language Detection in Speech Recognition
Miria Feng, William Tan, Mert Pilanci
Globalization and multiculturalism continue to produce increasingly diverse speech varieties. Yet current spoken dialogue systems frequently fail on under-represented dialects and accents, often misidentifying the input language and causing cascading failures in downstream dialogue tasks. Addressing this dialectal variance under low-resource constraints remains an open challenge, as standard fine-tuning is computationally expensive and prone to overfitting on high-dimensional speech data. We propose Convex Language Detection (CLD), a novel framework that integrates theoretically grounded convex optimization techniques into the spoken dialogue systems pipeline. Our method is efficiently implemented via multi-GPU Alternating Direction Method of Multipliers (ADMM) in JAX, thus providing global optimality guarantees and fast training in polynomial time. Theoretically, we prove that our convex objective induces certified margin stability and provide guarantees against feature perturbations. Empirically, we demonstrate sample efficiency and robustness to input dialectical variation, achieving 97–98\% accuracy in challenging low-resource regimes. Our open-source package is available at https://pypi.org/project/jaxcld/.
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STARE: Step-wise Temporal Alignment and Red-teaming Engine for Multi-modal Toxicity Attack
Xutao Mao, Liangjie Zhao, Tao Liu, Xiang Zheng, Hongying Zan, Cong Wang
Red-teaming Vision-Language Models is essential for identifying vulnerabilities where adversarial image-text inputs trigger toxic outputs. Existing approaches treat image generation as a black box, providing only terminal toxicity scores while remaining temporally opaque regarding when and how toxic semantics emerge during multi-step synthesis. We introduce $\textbf{STARE}$, a hierarchical reinforcement learning framework that treats the denoising trajectory as an exploitable attack surface. By synergizing a high-level prompt editor with low-level T2I fine-tuning via Group Relative Policy Optimization (GRPO), STARE achieves a 68\% improvement in Attack Success Rate over state-of-the-art baselines including black box and white-box variants. More importantly, we reveal the Optimization-Induced Phase Alignment phenomenon: while vanilla models exhibit diffuse toxicity, adversarial optimization systematically concentrates conceptual harms into early semantic phases and detail-oriented harms into late refinement. This discovery transforms toxicity formation from a chaotic process into a series of predictable vulnerability windows. This temporal alignment transforms red-teaming from a trial-and-error process into a targeted structural analysis. Our work provides both a potent attack engine and a diagnostic foundation for developing next-generation, phase-aware safety mechanisms. Content warning: This paper contains example

s of toxic content that may be offensive or disturbing.
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FUSE: Ensembling Verifiers with Zero Labeled Data
Joonhyuk Lee, Virginia L. Ma, Sarah Zhao, Yash Nair, Asher Spector, Regev Cohen, Emmanuel Candes
Verification of model outputs is rapidly emerging as a key primitive for both training and real-world deployment of large language models (LLMs). In practice, this often involves using imperfect LLM judges and reward models since ground truth acquisition can be time-consuming and expensive. We introduce Fully Unsupervised Score Ensembling (FUSE), a method for improving verification quality by ensembling verifiers without access to ground truth correctness labels. The key idea behind FUSE is to control conditional dependencies between verifiers in a manner that improves the unsupervised performance of a class of spectral algorithms from the ensembling literature. Despite requiring zero ground truth labels, FUSE typically matches or improves upon semi-supervised alternatives in test-time scaling experiments with diverse sets of generator models, verifiers, and benchmarks. In particular, we validate our method on both conventional academic benchmarks such as GPQA Diamond and on frontier, unsaturated benchmarks such as Humanity's Last Exam and IMO Shortlist questions.
****
Copula-SVI: Vine-Copula Variational Inference with Stein Refining for Instance-Level Correlation Capturing
Junxi Xiao, Qinliang Su
Mean-field variational inference (VI) is scalable, but its independence assumption can severely limit inference when the posterior is inherently coupled across instances especially for correlated data. Existing structured VI approaches either impose simple dependence patterns or incur substantial cost as dependence becomes richer, leaving efficient higher-order instance-level dependence modeling largely unresolved. We propose Copula-SVI, which augments amortized marginals with an explicit vine-copula posterior and refines joint samples with Stein updates toward the true posterior. The vine construction makes dependence learning and sampling practical by decomposing it into bivariate copula factors, enabling edge-minibatched training with variance-aware level-wise sampling and efficient dependence-aware initialization via a sparse vine built from the same sampled edges. Experiments on constrained clustering and time series modeling show consistent improvements over strong structured VI baselines and demonstrate efficient higher-order instance-level dependence modeling.
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OXE-AugE: A Large-Scale Robot Augmentation of OXE for Scaling Cross-Embodiment Policy Learning
Guanhua Ji, Harsha Polavaram, Lawrence Yunliang Chen, Sandeep Bajamahal, Zehan Ma, Simeon Adebola, Chenfeng Xu, Ken Goldberg
Large and diverse datasets are needed for training generalist robot policies that can control a variety of robot embodiments--robot arm and gripper combinations--across diverse tasks and environments. As re-collecting demonstrations and retraining for each new embodiment are prohibitively costly, we study whether existing robot data can be augmented to improve transfer and generalization across embodiments. The Open X-Embodiment (OXE) dataset, which aggregates demonstrations from over 60 robot datasets, has been widely used for training generalist policies. However, it is highly imbalanced: the top four robot types account for over 85% of its real data, which risks overfitting to robot--scene combinations. We present AugE-Toolkit, a scalable robot augmentation pipeline, and OXE-AugE, a high-quality open-source dataset that augments OXE with 9 different robot embodiments. OXE-AugE provides over 4.4 million trajectories, more than triple the size of the original OXE. We conduct a systematic study of how scaling robot augmentation impacts cross-embodiment learning. Results suggest that augmenting datasets with diverse arms and grippers improves policy performance not only on the augmented robots, but also on unseen robots and even the original robots under distribution shifts. In physical experiments, fine-tuning generalist policies such as OpenVLA and π_0 on OXE-AugE improves success rates by 24-45% on unseen robot

-gripper combinations across four real-world manipulation tasks. Project website : https://OXE-AugE.github.io/.
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Disease-Centric Vision-Language Pretraining with Hybrid Visual Encoding for 3D C omputed Tomography
Bowen Shi, Weiwei Cao, Ruifeng Yuan, Wanxing Chang, Wenrui Dai, Hongkai Xiong, L ing Zhang, Jianpeng Zhang
Vision–language pre-training (VLP) holds great promise for general-purpose medic al AI by leveraging radiology reports as rich textual supervision, yet existing methods struggle with 3D CT imaging due to inefficient visual backbones and coar se semantic alignment. To address these issues, we propose a tailored VLP framew ork featuring three key components: (1) a CNN–ViT hybrid encoder that replaces V iT’s patch embedding with a 3D CNN backbone to efficiently capture local anatomi cal details while preserving global attention and compatibility with pre-trained cross-modal priors; (2) a disease-level contrastive learning mechanism using le arnable query tokens to dynamically extract disease-specific semantics from full reports and align them with corresponding visual features, thereby disentanglin g distinct diseases within the same anatomical region; and (3) a diagnosis-aware prompt strategy that employs real clinical phrases and aggregated disease proto types to bridge the pre-training–inference gap and enhance zero-shot diagnostic reliability. Our model achieves state-of-the-art performance on CT-RATE (84.4\% AUC, +5.1%) and Rad-ChestCT (75.4\% AUC, +5.4%), with even larger gains (+9.8% A UC) on a challenging 60-disease benchmark, and demonstrates strong transferabili ty to radiology report generation, underscoring the generality and clinical util ity of our approach.
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Thoughtbubbles: an Unsupervised Method for Parallel Thinking in Latent Space
Houjun Liu, Shikhar Murty, Christopher D Manning, Róbert Csordás
Current approaches for scaling inference-time compute in transformers train them to emit explicit chain-of-thought tokens before producing an answer. While thes e methods are powerful, they are limited because they cannot be applied during p retraining and rely solely on serially-generated, natural-language verbalization . In this work, we propose Thoughtbubbles, a transformer variant that natively p erforms parallel adaptive computation in latent space by learning to fork or del ete residual streams. Thus, tokens requiring more computation can form a "bubble " of cloned residuals in the middle of the network. Crucially, this behavior is learned during pretraining with only language modeling loss. Using half of the t raining budget, Thoughtbubbles outperforms the perplexity and zero-shot evals of both standard decoder LMs and those using non-adaptive parallel computation app roaches. These results hold across model sizes from 150M to 1.9B. Thoughtbubbles achieves competitive GSM8K results using half of the baseline's token budget. T he implicit nature of our method enables models to begin learning adaptive compu tation at pretraining time, paving the way to unified train-time and test-time s caling behaviors.
****

On Learnability and Disambiguation of Multiclass Partial Concept Classes
Jingyuan Xu, Xin Zou, Xiuwen Gong, Weiwei Liu
We study the Probably Approximately Correct (PAC) learnability of partial concep t classes in the multiclass setting, where the label space can be infinite. Whil e the Natarajan dimension characterizes learnability for finite label spaces, we show it fails when the label space is unbounded. Instead, we prove that the Dan iely-Shalev (DS) dimension provides a characterization of learnability for parti al concept classes in the general multiclass setting. Furthermore, our analysis reveals a surprising phenomenon we call the ``Disambiguation Paradox'': disambig uation schemes with simple label space can destroy learnability, while richer la beling may preserves it. We further characterize how the number and structure of disambiguation labels control the induced DS dimension, yielding a trade-off be tween label complexity and sample complexity.
****

Posterior Mismatch Matters: Adversarial Training for Long-Tailed Robustness

Lilin Zhang, Li Yue, Jiancheng Shi, Jiancheng Lv, Xianggen Liu
Adversarial training breaks down in long-tailed settings, exhibiting severe robu stness degradation on worst-performing (often tail) classes. We identify a key c ause of this failure as a posterior mismatch: coarse-grained absolute labels col lapse class posteriors into point estimates, leading to biased class-frequency e stimation and an enlarged robust generalization gap, which ultimately amplifies worst-class vulnerability. To address this issue, we propose Posterior-driven Ad versarial Training (PAT), which learns a posterior surrogate to provide fine-gra ined probabilistic supervision for adversarial training, and integrates weight p erturbations to encourage a flatter loss landscape. Our theory shows that accura te posterior approximation simultaneously tightens class-frequency estimation er ror and robust generalization bounds, while a flat weight loss landscape stabili zes sensitivity to posterior approximation errors. Extensive experiments on long -tailed benchmarks confirm that PAT consistently improves robustness, with espec ially large gains on worst-class.
****
DAPD: Dependency-Aware Parallel Decoding via Attention for Diffusion LLMs
BumJun Kim, Dongjae Jeon, Moongyu Jeon, Albert No
Parallel decoding for diffusion LLMs (dLLMs) is difficult because each denoising step provides only token-wise marginal distributions,
while unmasking multiple tokens simultaneously requires accounting for inter-tok en dependencies.
We propose **D**ependency-**A**ware **P**arallel **D**ecoding (**DAPD**),
a simple, training-free decoding method that uses self-attention to induce a con ditional dependency graph over masked tokens.
At each iteration, edges in this graph capture strong token interactions, while non-edges indicate weak dependence.
Parallel decoding is then reduced to selecting an independent set on the graph a nd unmasking the selected tokens in parallel.
This avoids co-updating strongly coupled tokens without auxiliary models or retr aining.
Experiments on LLaDA and Dream show that DAPD improves the accuracy–steps trade- off over existing methods
and enables more globally distributed parallel updates that better exploit the a ny-order generation capability of dLLMs.
The project is available at \url{https://ai-isl.github.io/dapd}
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FastSESR: Fast Scene-level Explicit Surface Reconstruction
Jueqi Liu, Xuechao Zou, Congyan Lang
Explicit surface reconstruction aims to recover high-fidelity meshes directly fr om point clouds. While existing methods achieve strong performance on scene-leve l data, they often rely on test-time optimization, resulting in a prohibitive ru ntime of several minutes. To address this bottleneck, we propose FastSESR, a two -stage framework for efficient scene-level explicit surface reconstruction. In t he first stage, a lightweight triangular candidate network (TCN) captures local connections via an edge-factorized parameterization, enabling effective extracti on of surface triangles from uniformly sampled points. In the second stage, an o ffset optimization network amortizes offset refinement into a small, fixed numbe r of learnable update steps guided by TCN, producing geometries that are more su itable for triangulation. Experiments on multiple scene-level datasets show that FastSESR accelerates surface reconstruction by at least $20 \times$ over prior methods while maintaining competitive reconstruction quality. Moreover, evaluati ons on shape-level benchmarks indicate good generalization performance. Our code is available at https://github.com/Vaiduryasses/FastSESR.
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Mitigating Bias in Locally Constrained Decoding via Tractable Proposals
Meihua Dang, Linxin Song, Honghua Zhang, Jieyu Zhao, Guy Van den Broeck, Stefano Ermon
Generations from large language models often fail to conform to desired constrai nts such as JSON schema. Existing locally constrained decoding (LCD) approaches

enforce constraints by myopically masking out next tokens, resulting in biased s
ampling and degradation in performance. Recent work uses sequential Monte Carlo
(SMC) methods to mitigate such biases, but designing effective proposal distribu
tions or potential functions remains a key challenge. In this work, we propose a
 generic approach to construct proposals and potentials for SMC sampling from $p
_{\mathrm{lm}}(\cdot \mid \mathrm{constraint})$. First, we show that constraint
s specified as finite automata can be tensorized for efficient execution on GPUs
, which we use to construct globally constrained decoding (GCD) proposals. In ad
dition, leveraging the fact that tensorized finite automata share the same circu
it structure as hidden Markov models, we circuit-multiply them to obtain the pro
babilistic GCD (P-GCD) proposals encoding both logical and probabilistic informa
tion about the target distributions. We evaluate (P-)GCD on the tasks of functio
n calling, keyword-based generation, and SQL generation. Experiments show that u
nder the same SMC sampling setup, compared to LCD proposals, (P-)GCD converges f
aster to the target distribution with significantly fewer particles.
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Seeing is Solving: Unlocking Efficient Multimodal RL via View Alignment
Qinsi Wang, Jing Shi, Kun Wan, Handong Zhao, Hancheng Ye, Zishan Shao, Jinghan K
e, Yudong Liu, Daniel Miranda, Purvak Lapsiya, Yiran Chen, Wentian Zhao
Although Reinforcement Learning Fine-Tuning (RLFT) applied to Vision-Language Mo
dels (VLMs) substantially enhances multimodal reasoning capabilities, their proh
ibitive training cost limits broad adoption. Surprisingly, most existing methods
 simply port Large Language Model (LLM) RLFT techniques to VLMs, while ignoring
a intrinsic property of multimodal models: their dynamic text–vision alignment.
We ask a new question: Can this intrinsic alignment be turned into a training si
gnal that makes VLM RLFT more efficient? We analyze how a VLM plans to attend, a
ctually attends, and ideally should attend during reasoning, and derive two ligh
tweight metrics from these patterns. Predictive View Accuracy (PVA) estimates sa
mple difficulty, and Reasoning View Accuracy (RVA) reflects the quality of chain
-of-thought (CoT) reasoning. These alignment signals enable automated data curri
culum and dense reasoning supervision. We introduce FOCUS-RL, a plug-and-play fr
amework that can be seamlessly integrated into any VLM and dramatically boosts R
LFT training efficiency. FOCUS-RL achieves 2.5 x – 4 x faster convergence over v
anilla GRPO and consistent accuracy gains (+4.4 on average) across six different
 benchmarks and multiple VLM families.
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MePo: Meta Post-Refinement for Rehearsal-Free General Continual Learning
Guanglong Sun, Hongwei Yan, Liyuan Wang, Zhiqi KANG, Shuang Cui, Hang Su, Jun Zh
u, Yi Zhong
To cope with uncertain changes of the external world, intelligent systems must c
ontinually learn from complex, evolving environments and respond in real time. T
his ability, collectively known as general continual learning (GCL), encapsulate
s practical challenges such as online datastreams and blurry task boundaries. Al
though leveraging pretrained models (PTMs) has greatly advanced conventional con
tinual learning (CL), these methods remain limited in reconciling the diverse an
d temporally mixed information along a single pass, resulting in sub-optimal GCL
 performance. Inspired by meta-plasticity and reconstructive memory in neuroscie
nce, we introduce here an innovative approach named **Me**ta **Po**st-Refinement
 (MePo) for PTMs-based GCL. This approach constructs pseudo task sequences from
pretraining data and develops a bi-level meta-learning paradigm to refine the pr
etrained backbone, which serves as a prolonged pretraining phase but greatly fac
ilitates rapid adaptation of representation learning to downstream GCL tasks. Me
Po further initializes a meta covariance matrix as the reference geometry of pre
trained representation space, enabling GCL to exploit second-order statistics fo
r robust output alignment. MePo serves as a plug-in strategy that achieves signi
ficant performance gains across a variety of GCL benchmarks and pretrained check
points in a rehearsal-free manner (e.g., 15.10%, 13.36%, and 12.56% on CIFAR-100
, ImageNet-R, and CUB-200 under Sup-21/1K). Our source code is available at \hre
f{https://github.com/SunGL001/MePo}{MePo}.
****

CoGenCast: A Coupled Autoregressive–Flow Generative Framework for Time Series Forecasting
Mingyue Cheng, Yaguo Liu, Daoyu Wang, Xiaoyu Tao, Qi Liu
Time series forecasting can be viewed as a generative problem that requires both semantic understanding over contextual conditions and stochastic modeling of continuous temporal dynamics. Existing approaches typically rely on either autoregressive large language models (LLMs) for semantic context modeling or diffusion-like models for continuous probabilistic generation. However, neither method alone can adequately model both aspects simultaneously. In this work, we propose CoGenCast, a hybrid generative framework that couples pre-trained LLMs with flow-matching mechanism for effective time series forecasting. Specifically, we reconfigure pre-trained decoder-only LLMs into a native forecasting encoder–decoder backbone by modifying only the attention topology, enabling bidirectional context encoding and causal representation generation. Building on this, a flow-matching mechanism is further integrated to model temporal evolution, capturing continuous stochastic dynamics conditioned on the autoregressively generated representation. Notably, CoGenCast naturally supports multimodal forecasting and cross-domain unified training. Extensive experiments on multiple benchmarks show that CoGenCast achieves competitive performance compared to previous baselines. Code is available at \url{https://github.com/liuyaguo/_CoGenCast}.
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Deep Scientific Reasoning under Physical Constraints: Structure-Aware Spectrum Prediction
Yingheng Wang, Tao Yu, Shufeng Kong, Francesco Ricci, John M Gregoire, Carla P Gomes
Structured scientific spectra encode rich physical information while obeying hard constraints, such as conservation laws and characteristic spectral geometry. Accurate prediction of these spectra is central to materials discovery, yet existing methods often treat them as unconstrained sequences and therefore fail to enforce the underlying physical structure. Taking electronic density of states (eDOS) as the prototypical example, we introduce \textbf{DeepSciReasoner}, a general paradigm for predicting scientific spectra under physical constraints. The framework combines structure-aware spectrum decoding with constraint-preserving physical reasoning, allowing predictions to capture rich spectral structure while respecting the underlying physics. We evaluate DeepSciReasoner on eDOS, phonon density of states (phDOS), X-ray absorption near-edge structure (XANES), and Raman spectra, where it substantially improves prediction accuracy while maintaining physical consistency. These results establish DeepSciReasoner as a reusable blueprint for structured scientific spectrum prediction under hard physical constraints.
****
Resolving Blind Inverse Problems under Dynamic Range Compression via Structured Forward Operator Modeling
Muyu Liu, Xuanyu Tian, Chenhe Du, Qing Wu, Hongjiang Wei, Yuyao Zhang
Recovering radiometric fidelity from unknown dynamic range compression (UDRC), such as low-light enhancement and HDR reconstruction, is a challenging blind inverse problem, due to the unknown forward model and irreversible information loss introduced by compression.
To address this challenge, we first identify monotonicity as the fundamental physical invariant shared across UDRC tasks. Leveraging this insight, we introduce the cascaded monotonic Bernstein (CaMB) operator to parameterize the unknown forward model. CaMB enforces monotonicity as a hard architectural inductive bias, constraining optimization to physically consistent mappings and enabling robust and stable operator estimation.
We further integrate CaMB with a plug-and-play diffusion framework, proposing CaMB-Diff. Within this framework, the diffusion model serves as a powerful geometric prior for structural and semantic recovery, while CaMB explicitly models and corrects radiometric distortions through a physically grounded forward operator.

Extensive experiments on a variety of zero-shot UDRC tasks,

including low-light enhancement, low-field MRI enhancement, and HDR reconstructi
on, demonstrate that CaMB-Diff significantly outperforms state-of-the-art zero-s
hot baselines
in terms of both signal fidelity and physical consistency.
Moreover, we empirically validate the effectiveness of the proposed CaMB paramet
erization in accurately modeling the unknown forward operator.
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Incentivized Exploration with Stochastic Covariates: A Two-Stage Mechanism Desig
n for Recommender System

Yuantong Li, Guang Cheng, Xiaowu Dai

Recommender systems play a crucial role in internet economies by connecting user
s with relevant products. However, designing effective recommender systems faces
 the key challenges: the exploration-exploitation tradeoff in securing incentive
 to explore new products against user’s self-interested preferences. While prior
 work addresses Bayesian Incentive Compatibility (BIC) in fixed-design linear ba
ndits (Sellke & Slivkins, 2023), we tackle the challenge of stochastic user cova
riates sampled online. Unlike standard black-box reductions (Mansour et al., 202
0), our two-stage framework exploits the linear reward structure to achieve subl
inear regret while satisfying incentive constraints. To address it, we propose a
 two-stage algorithm that integrates incentivized exploration with any efficient
 plug-in offline learning algorithms. In the first stage, it explores products w
hile maintaining incentive compatibility to gather optimal samples. The second s
tage employs inverse proportional gap sampling strategy (IPGS) integrated with a
ny efficient learning methods to secure sublinear regret. Theoretically, we prov
e that algorithm RCB achieves $O(\sqrt{KdT})$ regret and simultaneously satisfie
s incentive constraints, and discovers the tradeoff between incentive budget and
 regret, validating in experiments. We demonstrate RCB’s strong incentive gain,
sublinear regret, and robustness through a real application on personalized warf
arin dosing and simulations.
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Robust Harmful Features Under Jailbreak Attacks: Mechanistic Evidence from Atten
tion Head Specialization in Large Language Models

Yanchen Yin, Dongqi Han, Linghui Li

Jailbreak attacks bypass LLM safety alignment, yet their mechanisms remain poorl
y understood. We provide evidence that attacks do not comprehensively eliminate
safety features, but instead selectively suppress specific attention heads. We i
dentify two functionally differentiated types: **Adversarially Compromised Heads
 (ACHs)** concentrated in early layers, which are suppressed under attacks, and
Safety-Aligned Heads (SAHs) in mid-layers, which maintain robust activations
 even when attacks succeed. Ablation studies support the causal role of ACHs and
 the contribution of SAHs to robust activations: suppressing a small number of A
CHs is sufficient to induce jailbreak-like behavior on normally refused inputs,
while removing SAHs substantially weakens mid-layer safety activations. Token-le
vel attribution further shows that ACH suppression is driven specifically by att
ack-template tokens, providing a mechanistic account of why attacks can bypass r
efusal decisions through ACH suppression while leaving internal safety signals s
ustained by SAHs—a phenomenon we term **Robust Harmful Features**. To validate t
he practical significance of this robustness, we show that simply reading these
persistent activations—without any training—yields competitive aggregate detecti
on performance with strong adversarial robustness.
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Rubric Curriculum RL: Exploiting the Generation-Verification Gap in Non-Verifiab
le Domains

Tejas Krishnan, Sumeet Ramesh Motwani, Charles London, Suhaas M Bhat, Huitian Ji
ao, Philip Torr, Riashat Islam, Christopher Summerfield, Christian Schroeder de
Witt, Qilong Gu, Shital Shah

Reinforcement learning with verifiable rewards (RLVR) on foundation models has l
ed to significant gains in math and code generation. Extending these gains to op
en-ended domains remains challenging: ground-truth verification is unavailable,
human annotation is expensive, and learned reward models are prone to reward hac

king. We introduce Rubric Curriculum RL (RcRL), a general self-improvement metho d for non-verifiable tasks that requires no new data, human annotations, or stro nger teacher models. RcRL exploits the generation-verification gap, where judgin g good outputs is easier than producing them. We combine pairwise preferences ov er rollouts with a curriculum over rubric criteria, yielding a more discriminati ve signal than absolute scoring while reducing reward hacking through a non-stat ionary objective. Whereas baselines plateau or collapse within a few hundred ste ps, RcRL preserves output entropy and keeps improving past 1000 steps. On creati ve writing, RcRL outputs are preferred 70.5% of the time compared to the base mo del, with consistent gains across multiple creative writing benchmarks and judge s. On HealthBench, RcRL improves over the base model by 14.6% (full set) and 25. 4% (consensus subset), outperforming all HealthBench training baselines, includi ng instance-specific rubrics.

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GIFT: Bootstrapping Image-to-CAD Program Synthesis via Geometric Feedback

Giorgio Giannone, Anna C. Doris, Amin Heyrani Nobari, Kai Xu, Akash Srivastava, Faez Ahmed

Generating executable CAD programs from images requires alignment between visual geometry and symbolic program representations, a capability that current method s fail to learn reliably as design complexity increases. Existing fine-tuning ap proaches rely on either limited supervised datasets or expensive post-training p ipelines, resulting in brittle systems that restrict progress in generative CAD design. We argue that the primary bottleneck lies not in model or algorithmic ca pacity, but in the scarcity of diverse training examples that align visual geome try with program syntax. This limitation is especially acute because the collect ion of diverse and verified engineering datasets is both expensive and difficult to scale, constraining the development of robust generative CAD models. We intr oduce Geometric Inference Feedback Tuning (GIFT), a data augmentation framework that leverages geometric feedback to turn test-time compute into a bootstrapped set of high-quality training samples. GIFT combines two mechanisms: Soft-Rejecti on Sampling (GIFT-REJECT), which retains diverse high-fidelity programs beyond e xact ground-truth matches, and Failure-Driven Augmentation (GIFT-FAIL), which co nverts near-miss predictions into synthetic training examples that improve robus tness on challenging geometries. By amortizing inference-time search into the mo del parameters, GIFT captures the benefits of test-time scaling while reducing i nference compute by 80%. It improves mean IoU by 12% over a strong supervised ba seline and remains competitive with more complex multimodal systems, without req uiring additional human annotation or specialized architectures.

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A Hypertoroidal Covering for Perfect Color Equivariance

Yulong Yang, Zhikun Xu, Yaojun Li, Christine Allen-Blanchette

When the color distribution of input images changes at inference, the performanc e of conventional neural network architectures drops considerably. A few researc hers have begun to incorporate prior knowledge of color geometry in neural netwo rk design. These color equivariant architectures have modeled hue variation with 2D rotations, and saturation and luminance transformations as 1D translations. While this approach improves neural network robustness to color variations in a number of contexts, we find that approximating saturation and luminance (interva l valued quantities) as 1D translations introduces appreciable artifacts. In thi s paper, we introduce a color equivariant architecture that is truly equivariant . Instead of approximating the interval with the real line, we lift values on th e interval to values on the circle (a double-cover) and build equivariant repres entations there. Our approach resolves the approximation artifacts of previous m ethods, improves interpretability and generalizability, and achieves better pred ictive performance than conventional and equivariant baselines on tasks such as fine-grained classification and medical imaging tasks. Going beyond the context of color, we show that our proposed lifting can also extend to geometric transfo rmations such as scale.

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Hölder++: Improving Quality-Coherence Trade-off in Multimodal VAEs

Huyen Thuc Khanh Vo, María Martínez-García, Isabel Valera
Existing approaches for multimodal variational autoencoders (VAEs) face a trade-off between generative quality and coherence—i.e., they struggle to generate realistic and diverse samples that, at the same time, are semantically consistent across modalities. A recent work shows that using a simple approximation to Hölder pooling as an aggregation method improves coherence over the SOTA MMVAE+, despite assuming a single shared representation across all modalities. Yet, it slightly compromises sample diversity. Inspired by this insight, we propose Hölder++, a novel multimodal VAE that improves the generative quality-coherence trade-off through: (i) the first implementation of *Hölder pooling without any approximation* for multimodal VAEs; (ii) an extended architecture that models *distinct shared and private* (i.e., modality-specific) representations (Hölder+); and (iii) *hierarchical inference* that further enhances the disentanglement between the shared and private representations (Hölder++). Our experiments corroborate that Hölder++ consistently improves the generative quality-coherence trade-off, yields more structured latent spaces, and learns shared representations that are informative for downstream tasks.
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Query-efficient model evaluation using cached responses
Hayden Helm, Ben Johnson, Carey Priebe
Evaluating a new model on an existing benchmark is often necessary to understand its behavior before deployment.
For modern evaluation frameworks, generating and evaluating a response for all queries can be prohibitively expensive.
In practice, responses from previously-evaluated models are often cached -- creating a potential opportunity to use this additional information to decrease the number of queries required to accurately evaluate a new model.
In this paper, we introduce an approach for predicting benchmark performance that leverages cached model responses based on the Data Kernel Perspective Space (DKPS), a method for quantifying the relationship between models in the black-box setting.
Theoretically, we show that DKPS-based methods are query-efficient under certain conditions.
Empirically, we demonstrate that DKPS-based methods achieve the same mean absolute error as baselines with a substantially decreased query budget.
We conclude by proposing an offline method for selecting a set of queries that maximizes the goodness-of-fit on reference models, improving prediction accuracy over random query selection.
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InfoGlobe: Local-and-Global Information-Preserving Statistical Manifold Learning for Single-Cell Transcriptomics
Cheng Wang, Jinpu Cai, Chongxiao Mao, Yuxuan Wang, Xinzhu Jiang, Yunhao Qiao, Luqi Yang, Luting Zhou, Qiuyu Lian, Hongyi Xin
Geometry-preserving dimension reduction is critical for single-cell transcriptomics, where low-dimensional distances should reflect biological divergence between cell types along the transcriptomic manifold. Due to inadequate metrics, the global structure is not sufficiently preserved in the low-dimensional manifold in standard dimension reduction regimes. We model RNA counts as Multinomial samples, leveraging their hierarchical closure property: gene-level counts refine functional gene-group counts via nested Multinomial distributions. Extending Chentsov's Theorem, we show that the Fisher-Rao metric on coarse (gene-group) and fine (gene) statistical manifolds is isometric. Following this isometry property, we propose InfoGlobe, an information-preserving statistical manifold learning framework that projects cells from high-dimensional hyperspheres (full transcriptome) to low-dimensional hyperspheres (functional groups) while preserving information geometry. Embeddings on the low-dimensional sphere explicitly represent Multinomial distributions by functional gene groups. Benchmarks demonstrate superior preservation of local-and-global cell-type geodesic distances, automatic and robust gene-group discovery, nuanced cell subtype resolution without manual feature engineering and natural batch effect mitigation without explicit alignments.

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COLLIE: Guiding Skill Discovery in Semantically Coherent Latent Space
Yao Luan, Ni Mu, Hanfei Ge, Yiqin Yang, Bo XU, Qing-Shan Jia
Unsupervised skill discovery (USD) aims to learn diverse behaviors without reward functions, but often results in task-irrelevant or hazardous behaviors due to uniform exploration. Guided skill discovery (GSD) addresses this issue by incorporating human intent to focus exploration on meaningful regions. However, existing GSD methods typically require training additional guidance models, and rely on pre-defined rules or expert demonstration, which can be ineffective under sparse, online-collected human feedback. To overcome this, we propose COLLIE, a GSD framework that leverages dense unsupervised data to construct a semantically coherent skill latent space. This latent space is well-structured, enabling reliable guidance with sparse online feedback. Moreover, its semantic coherence property enables training-free construction of guidance signals, eliminating the need for additional model training beyond skill learning. Theoretical analysis justifies the effectiveness of our training-free guidance signal, while experiments across diverse state-based and pixel-based tasks show that COLLIE learns diverse, human-aligned skills, avoids hazardous behaviors, and achieves superior downstream performance with minimal human feedback.
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STABLEVAL: Disagreement-Aware and Stable Evaluation of AI Systems
Akash Bonagiri, Gerard Janno Anderias, Saee Patil, Angelina Lai, Devang Borkar, Gezheng Kang, Ishant Gandhi, Setareh Rafatirad, Houman Homayoun
Human evaluation remains the primary standard for assessing modern AI systems, yet annotator disagreement, bias, and variability make system rankings fragile under standard majority vote aggregation. Majority vote discards annotator reliability and item-level ambiguity, often yielding unstable comparisons across annotator subsets. We introduce STABLEVAL, a disagreement-aware evaluation framework that models latent item correctness and annotator-specific confusion patterns to produce posterior expected item credit and calibrated agent-level scores. Unlike label-denoising approaches such as Dawid--Skene, STABLEVAL is explicitly designed for stable and uncertainty-aware system evaluation rather than hard label recovery. We formalize ranking stability as a first-class evaluation objective and analyze how aggregation methods preserve or distort underlying annotator behavior. Across controlled synthetic experiments and multiple real-world human-annotated benchmarks, majority vote exhibits increasing score error and ranking instability under annotator heterogeneity and adversarial noise, while STABLEVAL yields more stable and statistically grounded system rankings. These results demonstrate that modeling disagreement is essential for robust and reproducible AI evaluation.
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Contractive Anchor Resolvent Diffusion for Incomplete Multi-View Clustering
Tongzheng Zhao, Yangyang Wen, Yukai Shi, Xinyan Liang, Feijiang Li, Peng Zhou, Liang Du
Incomplete Multi-View Clustering (IMVC) is affected not only by missing feature values, but also by the degradation of relational structure induced by missing views. Many graph-based approaches either rely on costly data imputation or adopt first-order fusion mechanisms, which can be viewed as shallow low-pass filters with limited spectral selectivity. As a result, they may be insufficient to distinguish latent consensus structure from view-specific structural variations. To address this limitation, we reformulate IMVC from a spectral filtering perspective and propose \textbf{C}ontractive \textbf{A}nchor \textbf{R}esolvent \textbf{D}iffusion (\textbf{CARD}), a scalable framework for structural refinement without explicit view imputation. CARD constructs a unified anchor-induced hypergraph from observed sample--anchor relations and derives a high-order resolvent diffusion operator that acts as a rational spectral filter. This operator enhances the relative response of consensus-dominant modes while attenuating view-specific variations. We further derive a compact implicit solver that couples similarity learning and clustering without materializing dense matrices, and provide a conditional local refinement analysis under spectral-gap and local-stability assumpti

ons. Extensive experiments on eight benchmarks, including large-scale datasets, show that CARD achieves competitive performance while scaling linearly in \(N\) for a fixed anchor budget. The code for our method is publicly available at \url{https://github.com/Whale-Waves/CARD}.
****
Sinkhorn Normalization of Diffusion Kernels
Nathan Kessler, Robin Magnet, Jean Feydy
Smoothing a signal based on local neighborhoods is a core operation in machine learning and geometry processing. On well-structured domains such as vector spaces and manifolds, the Laplace operator derived from differential geometry offers a principled approach to smoothing via heat diffusion, with strong theoretical guarantees. However, constructing such Laplacians requires a carefully defined domain structure, which is not always available. Most practitioners thus rely on simple convolution kernels and message-passing layers, which are biased against the boundaries of the domain.

We bridge this gap by introducing a broad class of smoothing operators, derived from general similarity or adjacency matrices, and demonstrate that they can be normalized into diffusion-like operators that inherit desirable properties from Laplacians. Our approach relies on a symmetric variant of the Sinkhorn algorithm, which rescales positive smoothing operators to match the structural behavior of heat diffusion.

This construction enables Laplacian-like smoothing and processing of irregular data such as point clouds, sparse voxel grids or mixture of Gaussians. We show that the resulting operators not only approximate heat diffusion but also retain spectral information from the Laplacian itself, with applications to shape analysis and matching.
****
Conditional Clifford-Steerable CNNs for PDE Modeling
Bálint László Szarvas, Maksim Zhdanov
We introduce Conditional Clifford-Steerable CNNs (C-CSCNNs), a unified framework that incorporates equivariance to arbitrary pseudo-Euclidean groups and significantly improves the expressivity of standard CSCNNs. We show that the kernel basis of the standard formulation is incomplete, limiting model capacity. To address this, we augment the kernels with equivariant representations of the input feature field. We derive the equivariance constraint for these input-dependent kernels and show how it can be solved efficiently via implicit parameterization. We empirically validate on multiple PDE forecasting tasks, including fluid dynamics and relativistic electrodynamics, where our method consistently outperforms standard CSCNNs and performs on par with state-of-the-art baselines.
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Mitigating Manifold Departure: Uncertainty-Aware Subspace Rectification for Trustworthy MLLM Decoding
Yingxuan Zhuang, Jingxiao Yang, Miao Pan, Cheng Tan, Yuxiang Cai, Siwei Tan, Chen Zhi, Xuhong Zhang, Jianwei Yin, Jintao Chen
MLLMs frequently hallucinate objects inconsistent with visual inputs. This issue is typically attributed to the over-reliance on language priors, which can override the visual context. Recent training-free decoding strategies address this by penalizing language priors. However, these methods overlook the dual nature of language priors, where they can be both helpful and harmful depending on the alignment with visual evidence. In particular, blindly suppressing language priors often disrupts the model’s semantic manifold, leading to performance degradation, a phenomenon we term Manifold Departure. To address this, we propose Manifold-Guided Adaptive Projection (MGAP), a geometry-aware, training-free decoding method that mitigates hallucinations while preserving representation structure. MGAP first constructs a language-prior subspace from blind hidden states via SVD. During decoding, MGAP projects each multimodal hidden state onto this subspace and applies a consistency-aware gate to adaptively attenuate only the projected prior component, yielding a subspace-selective update that largely preserves the

orthogonal semantic components. Extensive experiments on POPE and CHAIR show tha t MGAP outperforms prior decoding baselines, achieving stronger hallucination su ppression without sacrificing coherence.

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Mitigating Hallucinations in Large Vision-Language Models via Causal Route Gatin g

Zhe Cheng, Wenyu Chen, Fode Zhang, Dehuan Shen

Large vision-language models (LVLMs) often hallucinate content that is fluent ye t unsupported by the image, limiting their reliability in real-world deployment. We show that a key failure mode arises from route competition: even when visual tokens receive attention, the final token decision can be dominated by the text ual pathway, causing the decoder to follow linguistic priors over visual evidenc e. To mitigate this, we propose a training-free, decision-aligned intervention t hat decomposes each attention head into a visual route and a text route, and est imates their token-level effects using an efficient one-forward/one-gradient app roximation. These estimates reveal route conflict within heads and identify prio r-dominant ones, enabling selective suppression of only the text route while kee ping the visual route intact. Across five benchmarks spanning discriminative and generative settings, our method consistently reduces hallucination-related erro rs across models with limited impact on overall multimodal performance, while in curring a modest inference-time overhead.

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Parameter Decorrelation via Transition-Variance Alignment for Multivariate Time-series Forecasting

Ji-Eun Choi, Jae-Hong Lee, Joon-Hyuk Chang

Multivariate time-series forecasting (MTSF) learns from high-dimensional covaria tes with strong temporal dependence, periodic structure, and cross-variable corr elations. While modern pipelines often mitigate non-stationarity through instanc e-wise normalization and decomposition, these interventions operate at the data level and do not directly control dependence that can emerge among the parameter s during training. We study MTSF optimization from a parameter-decorrelation vie wpoint. Modeling stochastic optimization as a Markov chain in parameter space an d leveraging its stochastic differential equation interpretation, we use the per -step transition-variance induced by gradient noise as a tractable signal for op timization-induced dependence and update uncertainty. This signal can empiricall y inflate during training; we theoretically show that such inflation can degrade generalization diagnostics. Motivated by this mechanism, we propose transition-variance alignment (TVA), an architecture-agnostic procedure that regulates tran sition-variance by smoothly gating the step size based on the mismatch between a n estimated noise scale and a chosen target. TVA maintains effective transition-variance near a prescribed scale without architectural changes, incurs negligibl e overhead, and integrates seamlessly with diverse methods. Across real-world mu ltivariate benchmarks, TVA consistently improves forecasting accuracy.

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Design Linear Constrained Neural Layers with Implicit Convex Optimization

Junchi Yan, Jiaxi Liu, Yihui Tu, Fangyuan Zhou, Wenzheng Pan, Zhongteng Gui, Lia ngliang Shi

One essential limitation of neural networks is how to enforce (hard) constraints on prediction. We propose a plug-in, differentiable layer, which involves a fas t implicit (convex) optimization procedure to enforce the general linear constra int. It aims to minimize a divergence between unconstrained and constrained outp uts. Connecting to and beyond existing handcrafted layers, we show that our laye r degrades to classic layers like Softmax, Sinkhorn and tanh etc. when the corre sponding constraint is enforced by KL-divergence minimization. We further show t hat by replacing the KL-div with a Euclidean distance, a closed-form solution ca n be derived for highly-efficient constraint enforcing. We evaluate the above tw o variants of layers, termed as BLCLayer and GLCLayer, with their corresponding neural solver BLCNet and GLCNet with simple MLP/GNN-like backbone. Experiments o n linear programming, as well as two real-world problems: partial graph matching and portfolio allocation which involve other discrete constraints.

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HONet: Data-Efficient Learning for Exact Cover Tasks via Hypergraph Optimization
Pengyang Huang, Zirui Zhuang, Haifeng Sun, Qi Qi, Jingyu Wang, Jianxin Liao
Deep learning approaches typically require prohibitive amounts of data to approximate known-constraint Exact Cover tasks, while existing neuro-symbolic methods often face training infeasibility and scalability bottlenecks. To bridge this divide, we propose the Hypergraph Optimization Network (HONet), an end-to-end framework integrating a structure-preserving Deep Residual Hypergraph Encoder with a differentiable fixed-constraint Quadratic Programming layer. By adopting a ``Fixed Polytope'' paradigm guided by the Geometric Consistency Loss, HONet explicitly shapes the objective landscape, encouraging the valid discrete solution to align with the global energy minimum. Empirical results show that HONet rapidly achieves 100\% accuracy on 9×9 Sudoku using limited samples, exhibiting superior data efficiency over baselines while maintaining exceptional robustness in highly sparse regimes and additional tasks.
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OpenMAG: A Comprehensive Benchmark for Multimodal-Attributed Graph
Chenxi Wan, Xunkai Li, Yilong Zuo, Haokun Deng, Sihan Li, Bowen Fan, Hongchao Qin, Rong-Hua Li, Guoren Wang
Multimodal-Attributed Graph (MAG) learning has achieved remarkable success in modeling complex real-world systems by integrating graph topology with rich attributes from multiple modalities. With the rapid proliferation of novel MAG models capable of handling intricate cross-modal semantics and structural dependencies, establishing a rigorous and unified evaluation standard has become imperative. Although existing benchmarks have facilitated initial progress, they exhibit critical limitations in *domain coverage*, *encoder flexibility*, *model diversity*, and *task scope*, presenting significant challenges to fair evaluation. To bridge this gap, we present OpenMAG, a comprehensive benchmark that integrates 19 datasets across 6 domains and incorporates 16 encoders to support both static and trainable feature encoding. OpenMAG further implements a standardized library of 24 state-of-the-art models and supports 8 downstream tasks, enabling fair comparisons within a unified framework. Through systematic assessment of necessity, data quality, effectiveness, robustness, and efficiency, we derive 14 fundamental insights into MAG learning to guide future advancements. Our code is available at https://github.com/YUKI-N810/OpenMAG.
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Latent Forcing: Reordering the Diffusion Trajectory for Pixel-Space Image Generation
Alan Baade, Eric Ryan Chan, Kyle Sargent, Changan Chen, Justin Johnson, Ehsan Adeli, Li Fei-Fei
Latent diffusion models excel at generating high-quality images but lose the benefits of end-to-end modeling. They discard information during image encoding, require a separately trained decoder, and model an auxiliary distribution to the raw data. In this paper, we propose Latent Forcing, a simple modification to existing architectures that achieves the efficiency of latent diffusion while operating on raw natural images. Our approach orders the denoising trajectory by jointly processing latents and pixels with separately tuned noise schedules. This allows the latents to act as a scratchpad for intermediate computation before high-frequency pixel features are generated. We find that the order of conditioning signals is critical, and we analyze this to explain differences between REPA distillation in the tokenizer and the diffusion model, as well as conditional and unconditional generation. Applied to pixel-space diffusion on ImageNet, Latent Forcing achieves a new state of the art for diffusion transformer-based pixel generation at our compute scale. Code and checkpoints at https://github.com/AlanBaade/LatentForcing
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Function-Valued Causal Influence in Nonlinear Time Series
Valentina V. Kuskova, Dmitry Zaytsev, Michael Coppedge
Causal discovery in time series is increasingly performed using nonlinear machine-learning models, yet the resulting causal relationships are almost always summ

arized by scalar edge scores. We argue that this practice obscures the true object learned by nonlinear autoregressive models: a state-dependent function whose effect varies across regimes, magnitudes, and contexts. We formalize function-valued causal influence for additive, contribution-decomposable architectures and show that scalar causal scores constitute a severe information bottleneck, conflating between-state variation with within-state residual noise. Using Neural Additive Vector Autoregression as a representative architecture, we introduce a practical framework based on Individual Conditional Expectation for estimating causal response functions directly from trained models. Through controlled synthetic experiments, we demonstrate that edges with indistinguishable scalar scores can exhibit qualitatively different functional behaviors, including monotonic, thresholded, saturating, and sign-changing effects. An applied case study on democratic development further shows that function-valued analysis reveals regime-specific and asymmetric causal structure systematically missed by score-centric approaches.

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Focus and Dilution: The Multi-stage Learning Process of Attention

Zheng-An Chen, Pengxiao Lin, Zhi-Qin John Xu, Tao Luo

Transformer-based models have achieved remarkable success across a wide range of domains, yet our understanding of their training dynamics remains limited. In this work, we identify a recurrent focus–dilution cycle in attention learning and provide a rigorous explanation in a one-layer Transformer setting for Markovian data via gradient-flow analysis. Using stage-wise linearization around critical points, we show that a single focus–dilution cycle can be decomposed into a sequence of distinct stages. First, embedding and projection rapidly condense to a rank-one structure, while attention parameters remain effectively frozen. Then, the attention parameters begin to increase, inducing a frequency-driven focus toward high-frequency tokens. As attention continues to evolve, it generates next-order perturbations in embeddings, leading to a mass-redistribution mechanism that progressively dilutes this focus. Finally, small asymmetries among low-frequency tokens lift a degenerate critical point, opening new embedding directions and initiating the next cycle. Experiments on synthetic Markovian data as well as WikiText and TinyStories corroborate the predicted stages and cyclical dynamics.

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Autoregression with Self-Token Prediction

Dengsheng Chen, Yangming Shi, Enhua Wu

Conventional autoregressive models achieve causality through next-token prediction, but suffer from prohibitive latency and typically underperform non-causal alternatives such as masked prediction and diffusion. We propose self-token prediction, which enables predicting a flexible number of tokens per step, and introduce SAR, the first spatially autoregressive image generator built on this paradigm. SAR delivers markedly faster inference speeds and consistently outperforms prior autoregressive baselines, achieving performance on par with state-of-the-art non-causal models. Our findings highlight self-token prediction as a crucial step toward a high-quality autoregressive paradigm for visual generation.

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Dynamic Stratified Contrastive Learning with Upstream Augmentation for MILP Branching

Tongkai Lu, Shuai Ma, Chongyang Tao

Mixed Integer Linear Programming (MILP) is a fundamental NP-hard problem that has garnered significant attention from both academia and industry.

The Branch-and-Bound (B&B) algorithm is the dominant approach for solving MILPs, where branching decisions play a critical role and have recently been enhanced by neural methods.

However, these methods still struggle with semantic variation across depths, the scarcity of upstream nodes, and the costly collection of strong branching samples.

To address these issues, we propose SC-MILP, a Dynamic Stratified Contrastive Training Framework for MILP Branching.

Our method groups B&B nodes based on their feature distributions and learns dept

h-aware, fine-grained node representations through dynamic stratified contrastiv
e training.
To address data scarcity and imbalance at upstream nodes, we introduce an upstre
am-augmented MILP derivation procedure that generates both theoretically equival
ent and perturbed instances.
Experiments on both synthetic and real-world MILP benchmarks, including large-sc
ale instances, show that SC-MILP significantly improves branching accuracy, redu
ces solving time, with particularly strong gains at upstream nodes.
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VR-Thinker: Boosting Multimodal Reward Models through Think with Image Reasoning
Qunzhong Wang, Jie Liu, Jiajun Liang, Yuanxing Zhang, Yilei Jiang, Yaozhi ZHENG,
 Xintao Wang, Pengfei Wan, Xiangyu Yue, Jiaheng Liu
Recent advancements in multimodal reward models (RMs) have substantially improve
d post-training for visual generative models. However, current RMs face inherent
 limitations: **(1)** visual inputs consume large context budgets, forcing fewer
 frames and causing a loss of details; and **(2)** all visual information is pac
ked into the initial prompt, exacerbating forgetting during chain-of-thought rea
soning. To overcome these issues, we introduce **VR-Thinker**, a thinking-with-i
mage RM equipped with visual reasoning operation and a configurable visual memor
y window. This allows the RM to actively acquire visual evidence, improving reas
oning fidelity and reliability. We activate visual reasoning via a reinforcement
 fine-tuning pipeline: **(i)** **Cold Start** with curated visual chain-of-thoug
ht data to distill basic operation formatting; **(ii)** select samples with corr
ect judgments, then conduct **Rejection sampling Fine-Tuning** on these high-qua
lity traces to further enhance reasoning; and **(iii)** apply **Group Relative P
olicy Optimization** (GRPO) to strengthen reasoning. Our approach delivers state
-of-the-art accuracy among open-source models on video preference benchmarks: a
7B **VR-Thinker** achieves 80.5\% on VideoGen Reward, 82.3\% on GenAI-Bench, and
 75.6\% on MJ-Bench-Video.
****

Data Difficulty and the Generalization–Extrapolation Tradeoff in LLM Fine-Tuning
Siyuan Liu, Tinghong Chen, Xinghan Li, Yifei Wang, Jingzhao Zhang
Data selection during supervised fine-tuning (SFT) can critically change the beh
avior of large language models (LLMs).
Although existing work has studied the effect of selecting data based on heurist
ics such as perplexity, difficulty, or length, the reported findings are often i
nconsistent or context-dependent. In this work, we systematically study the role
 of data difficulty in fine-tuning from both empirical and theoretical perspecti
ves, and find that there is no universally optimal difficulty level; rather, its
 effectiveness depends on the dataset size. We show that for a fixed data budget
, there exists an optimal data difficulty for SFT, and that this optimal difficu
lty shifts toward harder data as the data budget increases. To explain this phen
omenon, we conduct controlled synthetic experiments that reveal a simple underly
ing mechanism: the interplay between the (in-distribution) generalization gap an
d the extrapolation gap. We further support this mechanism through a theoretical
 analysis using PAC-Bayesian generalization bounds. Overall, our results clarify
 how data size and difficulty jointly affect the trade-off between generalizatio
n and extrapolation in SFT, providing guidance for difficulty-based data selecti
on under certain model and data conditions.
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Distribution Alignment for One-Shot Federated Learning via Optimal Transport
Daniele Berardini, Vito Paolo Pastore, Vittorio Murino
One-Shot Federated Learning (OSFL) addresses extreme communication regimes in wh
ich clients interact with the server only once, amplifying the impact of heterog
eneous client data distributions. In particular, the interaction of domain shift
 and label shift across clients induces misaligned feature representations that
cannot be corrected through iterative optimization. Existing OSFL methods rely o
n distillation, server-side generation or ensemble-based aggregation, but assume
 aligned representations or address domain and label shift separately. We introd
uce SLOT-Align (Single-round, Learning-free Optimal Transport Alignment), a geom

etry-aware feature harmonization framework for OSFL. SLOT-Align uses a shared fr
ozen encoder to extract compact feature statistics, constructs a global referenc
e via Bures–Wasserstein barycenters, and aligns local representations using clos
ed-form geodesic optimal transport maps. The method is computationally efficient
 and can be combined with existing OSFL pipelines relying on frozen encoders wit
hout modifying their training procedures. Extensive experiments across multiple
benchmarks, pretrained backbones, and OSFL methods show that SLOT-Align consiste
ntly improves accuracy and robustness under joint domain and label shift.
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SWING: Unlocking Implicit Graph Representations for Graph Random Features
Alessandro Manenti, Kumar Avinava Dubey, Arijit Sehanobish, Cesare Alippi, Krzys
ztof Marcin Choromanski
We propose SWING: Space Walks for Implicit Network Graphs, a new class of algori
thms for computations involving Graph Random Features on graphs given by implici
t representations (i-graphs), where edge-weights are defined as bi-variate funct
ions of feature vectors in the corresponding nodes. Those classes of graphs incl
ude several prominent examples, such as: *ϵ-neighborhood* graphs, used
on regular basis in machine learning. Rather than conducting walks on graphs' no
des, those methods rely on walks in continuous spaces, in which those graphs are
 embedded. To accurately and efficiently approximate original combinatorial calc
ulations, SWING applies customized Gumbel-softmax sampling mechanism with linear
ized kernels, obtained via random features coupled with importance sampling tech
niques. This mechanism is of its own interest. SWING relies on the deep connecti
on between implicitly defined graphs and Fourier analysis, presented in this pap
er. SWING is accelerator-friendly and does not require input graph materializati
on. We provide detailed analysis of SWING and complement it with thorough experi
ments on different classes of i-graphs.
****
PETS: A Principled Framework Towards Optimal Trajectory Allocation for Efficient
 Test-Time Self-Consistency
Zhangyi Liu, Huaizhi Qu, Xiaowei Yin, He Sun, Yanjun Han, Tianlong Chen, Zhun De
ng
Test-time scaling can improve model performance by aggregating stochastic reason
ing trajectories. However, achieving sample-efficient test-time self-consistency
 under a limited budget remains an open challenge. We introduce PETS (**P**rinci
pled and **E**fficient **T**est-Time **S**elf-Consistency), which initiates a pr
incipled study of trajectory allocation through an optimization framework. Centr
al to our approach is the *self-consistency rate*}, a new measure defined as agr
eement with the infinite-budget majority vote. This formulation makes sample-eff
icient test-time allocation theoretically grounded and amenable to rigorous anal
ysis. We study both offline and online settings. In the offline regime, where al
l questions are known in advance, we connect trajectory allocation to crowdsourc
ing, a classic and well-developed area, by modeling reasoning traces as workers.
 This perspective allows us to leverage rich existing theory, yielding theoretic
al guarantees and an efficient majority-voting-based allocation algorithm. In th
e online streaming regime, where questions arrive sequentially and allocations m
ust be made on the fly, we propose a novel method inspired by the offline framew
ork. Our approach adapts budgets to question difficulty while preserving strong
theoretical guarantees and computational efficiency. Experiments show that PETS
consistently outperforms uniform allocation. On GPQA, PETS achieves perfect self
-consistency in both settings while reducing the sampling budget by up to $75\\%
$ (offline) and $55\\%$ (online) relative to uniform allocation.
****
Dynamic Fractal Mamba: A Neural Renormalization Group Flow for Scale-Invariant S
equence Modeling
Shenglei Fang, Xianfang Sun, You Zhou
Sequence models typically operate at a fixed temporal or spatial scale and strug
gle to generalize to substantially longer horizons or higher resolutions without
 retraining. Existing hierarchical architectures expand receptive fields but rel
y on scale-specific parameters and lack mechanisms to enforce consistent dynamic

s across scales. We propose Dynamic Fractal Mamba (DF-Mamba), a recursive state-space model that applies a single shared operator across multiple scales. By sharing parameters across recursion depths and exponentially scaling the effective time step, DF-Mamba achieves an exponentially expanding receptive field while preserving linear computational complexity. A learned content-aware coarse-graining module aggregates representations across scales. Auxiliary reconstruction and cross-scale consistency objectives stabilize recursive training. We evaluate DF-Mamba on long-range time-series forecasting, spatial transcriptomics, and computational pathology. Across all tasks, DF-Mamba consistently outperforms Transformers and flat Mamba baselines while using fewer parameters and maintaining linear-time scalability. Importantly, models trained on short sequences or low-resolution inputs generalize in a zero-shot manner to substantially larger temporal and spatial scales unseen during training. These results demonstrate that recursive parameter sharing provides an effective inductive bias for learning scale-consistent and efficient sequence representations.

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Polyphonia: Zero-Shot Timbre Transfer in Polyphonic Music with Acoustic-Informed Attention Calibration

Haowen Li, Tianxiang Li, Yi Yang, Boyu Cao, Qi Liu

The advancement of diffusion-based text-to-music generation has opened new avenues for zero-shot music editing. However, existing methods fail to achieve stem-specific timbre transfer, which requires altering specific stems while strictly preserving the background accompaniment. This limitation severely hinders practical application, since real-world production necessitates precise manipulation of components within dense mixtures. Our key finding is that, while vanilla cross-attention captures semantic features of stems, it lacks the spectral resolution to strictly localize targets in dense mixtures, leading to boundary leakage. To resolve this dilemma, we propose *Polyphonia*, a zero-shot editing framework with Acoustic-Informed Attention Calibration. Rather than relying solely on diffuse semantic attention, Polyphonia leverages a probabilistic acoustic prior to establish coarse boundaries, enabling non-target stems preserved precise semantic synthesis. For evaluation, we propose *PolyEvalPrompts*, a standardized prompt set with 1,170 timbre transfer tasks in polyphonic music. Specifically, *Polyphonia* achieves an increase of 15.5% in target alignment compared to baselines, while maintaining competitive music fidelity and non-target integrity.

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Opt-Verifier: Unleashing the Power of LLMs for Optimization Modeling via Dual-Side Verification

Haoyang Liu, Jie Wang, Boxuan Niu, Xiongwei Han, Yian Xu, Mingxuan Ye, Zijie Geng, Fangzhou Zhu, Tao Zhong, Mingxuan Yuan, Jianye HAO

Building mathematical optimization models is critical in operations research (OR), while it requires substantial human expertise.

Recent advancements have utilized large language models (LLMs) to automate this modeling process.

However, existing works often struggle to verify the correctness of the generated optimization models, without checking the rationality of the constraints and variables or the validity of solutions to the generated models.

This hampers the subsequent verification and correction steps, and thus it severely hurts the modeling accuracy.

To address this challenge, we propose a novel LLM-based framework with Dual-side Verification (OptiVer) from both structure and solution perspectives, thereby improving the modeling accuracy.

The structure-side verification ensures that the modeling structure of the generated optimization models aligns with the original problem description, accurately capturing the problem's constraints and requirements.

Meanwhile, the solution-side verification interprets and evaluates the validity of the solutions, confirming that the optimization models are logically and mathematically sound.

Extensive experiments on several popular benchmarks demonstrate that our approach significantly outperforms the state-of-the-art, achieving over 20\% improvemen

t in accuracy.
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Toward Calibrated Mixture-of-Experts Under Distribution Shift
Gina Wong, Drew Prinster, Suchi Saria, Rama Chellappa, Anqi Liu
Calibration aligns a model's predictive uncertainty with the frequencies of its empirical outcomes and is important for understanding and trusting reported probabilities. Recent work shows that enforcing calibration at the level of individual predictors can improve ensemble accuracy and calibration, with mixture-of-experts (MoE) models showing strong empirical improvements in particular; however, the conditions under which calibration helps MoE are not well understood. In this work, we study how MoE models behave under distribution shift, focusing on how routing mechanisms interact with expert-level calibration. We show that expert calibration is sufficient to ensure calibration of the overall model under a broad class of distribution shifts in hard-routed models, but is insufficient for calibrating soft-routed models. To address this, we propose an adversarial reweighting that penalizes calibration errors of the routed aggregate under distribution shift, and we demonstrate that it improves the accuracy-calibration tradeoff both on average and on difficult subsets of the data, across model classes, prediction tasks, and distribution shifts.
****
TaskLoom: Weaving Knowledge Across Tasks in World Models
Qingzhang Zeng, Peixi Peng, Hang Li, Luntong Li, Yonghong Tian
World models have significantly improved the sample efficiency of model-based reinforcement learning (MBRL) by enabling policy learning in imagination, thereby reducing the need for direct interaction with the real environment. However, most existing world model methods are trained independently for each task or perform multi-task learning using offline datasets, failing to fully exploit the latent relationships among tasks in online interactive scenarios. To address this limitation, we propose TaskLoom, a knowledge-sharing world model architecture for online RL. TaskLoom adopts a grouped two-stage training paradigm: first, the tasks are divided into several groups based on the similarity of world model gradients,
and fine-grained knowledge is shared among tasks within each group; second, coarse-grained knowledge is exchanged across groups, enabling hierarchical knowledge transfer and reuse.
Experimental results show that TaskLoom outperforms baseline methods on widely used benchmarks such as Proprio Control, Visual Control and Meta-World, validating the effectiveness of the proposed knowledge-sharing mechanism for both low-dimensional state and high-dimensional visual inputs.
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Corruption-Tolerant Asynchronous Q-Learning with Near-Optimal Rates
Sreejeet Maity, Aritra Mitra
We study the problem of learning the optimal policy in a discounted, infinite-horizon reinforcement learning (RL) setting in the presence of adversarially corrupted rewards. To address this problem, we develop a novel robust variant of the Q-learning algorithm and analyze it under the challenging asynchronous sampling model with time-correlated data. Despite corruption, we prove that the finite-time guarantees of our approach match existing bounds, up to an additive term that scales with the fraction of corrupted samples. We also establish an information-theoretic lower bound, revealing that our guarantees are near-optimal. Notably, our algorithm is agnostic to the underlying reward distribution and provides the first finite-time robustness guarantees for asynchronous Q-learning. A key element of our analysis is a refined Azuma-Hoeffding inequality for almost-martingales, which may have broader applicability in the study of RL algorithms.
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Diffusion Models Are Statistically Optimal for Learning Low-Dimensional Multi-Modal Distributions
Jingda Wu, Changxiao Cai
Score-based diffusion models have demonstrated remarkable empirical success in learning high-dimensional distributions, particularly those exhibiting low-dimens

ional and multi-modal structures. However, theoretical understanding of their st atistical efficiency remains limited. Existing theories typically rely on strong regularity assumptions, such as uniformly bounded densities or globally smooth score functions, which fail to capture such intrinsic structures.
In this work, we study the sample complexity of diffusion models for learning di stributions supported on a union of low-dimensional subspaces. Assuming that the data distribution within each subspace is subgaussian, we show that diffusion m odels require at most the order of $\widetilde{O}(\varepsilon^{-k \vee 2})$ (up to some logarithmic factor) samples to achieve ε sampling error in 1 -Wasserstein distance, where k is the intrinsic dimension.
This near-optimal convergence rate depends only on the intrinsic dimension and s ignificantly improves upon prior theoretical guarantees that suffer from the cur se of dimensionality. Notably, our analysis applies to a broad collection of dis tributions without imposing smoothness, bounded-density, or log-concavity assump tions.
Overall, our results show that diffusion models can statistically adapt to intri nsic low-dimensional structure while naturally accommodating multi-modal data, o ffering a rigorous theoretical justification for their success in complex high-d imensional learning tasks.
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From Shortcuts to Reasoning: Robust Post-Training of Theory of Mind with Reinfor cement Learning
Jike Zhong, Yuxiang Lai, Ming Li, Yuheng Li, Wuao Liu, Behzad Dariush, Konstanti nos Psounis, Shao-Yuan Lo
Theory of Mind (ToM) is a must-acquire skill for modern foundation model systems to operate effectively and safely in the real world. Recent works have explored honing ToM via post-training; however, we show that such progress is confounded by a pervasive “shortcut” issue: tasks can reach up to 99% accuracy by simply e xploiting spurious causal correlations, leading to a false sense of ToM. Motivat ed by this, we first develop a framework to systematically examine ToM datasets for shortcuts and provide guidance for future development. We find that question s reducible to pure state tracking, such as “belief,” are especially shortcut-pr one compared to mind questions, such as “intention,” where reasoning beyond trac king is required. Using four shortcut-free datasets across three ToM contexts, w e then comprehensively study whether Reinforcement Fine-Tuning with verifiable r ewards and explicit reasoning chains, called Thinking-RFT, elevates ToM beyond S upervised Fine-Tuning, or SFT. Our key findings are as follows. First, Thinking-RFT effectively improves ToM in all scenarios, with a 6% improvement over SFT, p articularly in complex higher-order reasoning, with a 10% improvement over SFT, and multimodal cases, with a 7% improvement over SFT. It also generalizes notabl y better to unseen domains and higher-order queries while being more robust to c ounterfactuals. Second, ToM benefits specifically from the joint effect of reaso ning and RL: Thinking-RFT outperforms No-Thinking-RFT by 7% on average. Third, R FT works by learning to ground its reasoning on anchor cues, such as keywords an d state changes, that correspond to causal factors. We believe our study is usef ul for developing effective and robust ToM post-training datasets and advancing critical ToM capabilities. Project code is available at: https://github.com/jkz-338/Robust-ToM-RL.git
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CurvZO: Adaptive Curvature-Guided Sparse Zeroth-Order Optimization for Efficient LLM Fine-Tuning
Shuo Wang, Ziyu Chen, Ming Tang
Fine-tuning large language models (LLMs) with backpropagation achieves high perf ormance but incurs substantial memory overhead, limiting scalability on resource -constrained hardware. Zeroth-order (ZO) optimization provides a memory-efficien t alternative by relying solely on forward passes, yet it typically suffers from slow or unstable convergence due to high-variance gradient estimates. Sparse ZO updates partially address this issue by perturbing only a subset of parameters, but their effectiveness hinges on selecting informative parameters, which is ch allenging in ZO optimization because each query yields only scalar feedback. We

propose Adaptive Curvature-Guided Sparse Zeroth-Order Optimization (CurvZO), whi
ch tracks curvature signals online from scalar ZO feedback and leverages these s
ignals to construct a parameter-wise sampling distribution for selecting coordin
ates at each update, reducing the variance of the sparse ZO gradient estimator.
Moreover, CurvZO dynamically adapts the perturbation budget to the evolving curv
ature signal distribution, yielding sparse ZO updates that remain both focused a
nd sufficiently exploratory. Extensive experiments on OPT and Llama across diver
se NLP tasks show that CurvZO consistently improves fine-tuning performance and
reduces training time over ZO baselines. It improves accuracy by up to 4.4 point
s and achieves up to a $2\times$ speedup, while preserving memory efficiency.
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ECHO: Elastic Speculative Decoding with Sparse Gating for High-Concurrency Scena
rios

Xinyi. Hu, Yuhao Shen, Zhang Baolin, Hengxin Zhang, Jun Dai, Shuang Ge, Chen Lei
, Yue Li, Mingcheng Wan

Speculative Decodin promises to accelerate Large Language Model inference, yet i
ts efficacy often degrades in production-grade scenarios. Existing evaluations t
ypically overlook the compute-bound nature of high-concurrency regimes, where ve
rification compute becomes the dominant bottleneck. Consequently, prior methods
face a dilemma: static trees incur massive verification waste, while dynamic tre
es suffer from cumulative misjudgments and kernel incompatibility.

To bridge this gap, we introduce ECHO, a high concurrency-oriented framework int
egrated into SGLang that reformulates speculative execution as a budgeted schedu
ling problem.

Crucially, ECHO employs sparse confidence gating to manage the batch as a unifie
d super-tree, elastically pivoting budget between depth and width to co-optimize
 the trade-off between reducing global verification steps and maximizing per-ste
p efficiency.

Extensive evaluations across diverse model scales—particularly the industrial-gr
ade Qwen3-235B—demonstrate that ECHO consistently outperforms state-of-the-art b
aselines in both low-load and high-load scenarios, achieving up to 5.35$\times$
walltime speedup and delivering over 20\% relative speedup gain against the stro
ngest baselines.
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Less Diverse, Less Safe: The Indirect But Pervasive Risk of Test-Time Scaling in
 Large Language Models

Shahriar Kabir Nahin, Hadi Askari, Muhao Chen, Anshuman Chhabra

Test-Time Scaling (TTS) improves LLM reasoning by exploring multiple candidate r
esponses and then operating over this set to find the best output. A tacit premi
se behind TTS is that sufficiently diverse candidate pools enhance reliability.
In this work, we show that this assumption in TTS introduces a previously unreco
gnized failure mode. When candidate diversity is curtailed, even by a modest amo
unt, TTS becomes much more likely to produce unsafe outputs. We present a refere
nce-guided diversity reduction protocol (RefDiv) that serves as a diagnostic att
ack to stress test TTS pipelines. Through extensive experiments across open-sour
ce models (e.g. Qwen3, Mistral, Llama3.1, Gemma3) and two widely used TTS strate
gies (Monte Carlo Tree Search and Best-of-N), constraining diversity consistentl
y signifies the rate at which TTS produces unsafe results. The effect is often s
tronger than that produced by prompts directly with high adversarial intent scor
es. This observed phenomenon also transfers across TTS strategies and to closed-
source models (e.g. OpenAI o3-mini and Gemini-2.5-Pro), thus indicating that thi
s is a general and extant property of TTS rather than a model-specific artifact.
 Additionally, we find that numerous widely used safety guardrail classifiers (e
.g. Llama-Guard), are unable to flag the adversarial input prompts generated by
RefDiv, demonstrating that existing defenses offer limited protection against th
is diversity-driven failure mode.
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The Lie We Tell: Correcting the Euclidean Fallacy in Vision Language Action Poli
cies via Score Matching on Tangent Space

Bing-Cheng Chuang, I-Hsuan Chu, Bor-Jiun Lin, YuanFu Yang, Min Sun, Chun-Yi Lee

Diffusion-based Vision-Language-Action policies achieve remarkable success in robotic manipulation, yet commit a fundamental geometric error we term the **Euclidean Fallacy**: representing SE(3) poses as flat $\mathbb{R}^{12}$ vectors. This approximation induces (1) manifold drift violating SO(3) constraints, (2) broken equivariance under coordinate transformations, and (3) non-geodesic trajectories with excessive kinematic cost. We introduce **Lie Diffuser Actor (LDA)**, a diffusion framework operating intrinsically on SE(3). Our method injects noise through left-invariant SDEs, predicts scores in the tangent space, and retracts samples via the exponential map. This formulation eliminates manifold drift by construction while guaranteeing coordinate-frame equivariance and geodesic optimality. On CALVIN ABC$\rightarrow$D, LDA improves average task length from 3.27 to 3.51 ($+7.3\%$). We further validate our method on real robot and the results show that our methodology outperforms the baseline on majority tasks.
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Near-Optimal Convergence of Accelerated Gradient Methods under Generalized and (L_0,L_1)-Smoothness
Alexander Tyurin
We study first■order methods for convex optimization problems with functions f satisfying the recently proposed ℓ-smoothness condition $||\nabla^{2}f(x)|| \le \ell\left(||\nabla f(x)||\right),$ which generalizes the L-smoothness and (L_{0},L_{1})-smoothness. While accelerated gradient descent (AGD) is known to reach the optimal complexity $\mathcal{O}(\sqrt{L} R / \sqrt{\varepsilon})$ under L-smoothness, where ε is an error tolerance and R is the distance between a starting and an optimal point, existing extensions to ℓ-smoothness either incur extra dependence on the initial gradient, suffer exponential factors in $L_{1} R$, or require costly auxiliary sub-routines, leaving open whether an AGD■type $\mathcal{O}(\sqrt{\ell(0)} R / \sqrt{\varepsilon})$ rate is possible for small-ε, even in the (L_{0},L_{1})-smoothness case. We resolve this open question. Developing new proof techniques, we achieve $\mathcal{O}(\sqrt{\ell(0)} R / \sqrt{\varepsilon})$ oracle complexity for small-ε and virtually any ℓ. For instance, for (L_{0},L_{1})-smoothness, our bound $\mathcal{O}(\sqrt{L_0} R / \sqrt{\varepsilon})$ is provably optimal in the small-ε regime and removes all non-constant multiplicative factors present in prior accelerated algorithms.
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Exploration Hacking: Can LLMs Learn to Resist RL Training?
Eyon Jang, Damon Falck, Joschka Braun, Nathalie Maria Kirch, Achyutha Menon, Perusha Moodley, Scott Emmons, Roland S. Zimmermann, David Lindner
Reinforcement learning (RL) has become essential to the post-training of large language models (LLMs) for reasoning, agentic capabilities and alignment. Successful RL relies on sufficient exploration of diverse actions by the model during training, which creates a potential failure mode: a model could strategically alter its exploration during training to influence the subsequent training outcome. In this paper we study this behavior, called exploration hacking. First, we create model organisms of selective RL resistance by fine-tuning LLMs to follow specific underperformance strategies; these models can successfully resist our RL-based capability elicitation in agentic biosecurity and AI R&D environments while maintaining performance on related tasks. We then use our model organisms to evaluate detection and mitigation strategies, including monitoring, weight noising, and SFT-based elicitation. Finally, we show that current frontier models can exhibit explicit reasoning about suppressing their exploration when provided with sufficient information about their training context, with higher rates when this information is acquired indirectly through the environment. Together, our results suggest exploration hacking is a possible failure mode of RL on sufficiently capable LLMs.
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VideoFlexTok: Flexible-Length Coarse-to-Fine Video Tokenization
Andrei Atanov, Jesse Allardice, Roman Bachmann, O■uzhan Fatih Kar, R Devon Hjelm, David Griffiths, Peter Fu, Afshin Dehghan, Amir Zamir
Visual tokenizers map high-dimensional raw pixels into a compressed representati

on for downstream modeling. Beyond compression, tokenizers dictate what information is preserved and how it is organized. A _de facto_ standard approach to video tokenization is to represent a video as a spatiotemporal 3D grid of tokens, each capturing local information from the original signal. This requires the downstream model, e.g., a text-to-video model, to learn to predict all low-level details "pixel-by-pixel" irrespective of the video's inherent complexity, leading to high learning complexity.
__We present _VideoFlexTok_, which represents videos with a variable-length sequence of tokens structured in a coarse-to-fine manner__, where the first tokens (emergently) capture information such as semantics and motion, and later tokens add fine-grained details. The generative flow decoder enables realistic video reconstructions from any token count. This representation structure allows adapting the token count to downstream needs and encoding videos longer than the baselines within the same budget.
We evaluate VideoFlexTok on class- and text-to-video generative tasks and show that it yields more efficient training than 3D grid tokens, _achieving comparable generation quality (gFVD and ViCLIP Score) with a 5x smaller model (1.1B vs 5.2B)._
Finally, we show how _VideoFlexTok can enable long video generation without prohibitive computational cost_ by training a text-to-video model on 10-second 81-frame videos with only 672 tokens, 8x fewer than a comparable 3D grid tokenizer.
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When Preference Labels Fall Short: Aligning Diffusion Models from Real Data
Weiyan Chen, Weijian Deng, Yao Xiao, Weijie Tu, ZiYi Dong, Ibrahim Radwan, Liang Lin, Pengxu Wei
Preference alignment aims to guide generative models by learning from comparisons between preferred and non-preferred samples. In practice, most existing approaches rely on preference pairs constructed from model-generated images. Such supervision is inherently relative and can be ambiguous when both samples exhibit artifacts or limited visual quality, making it difficult to infer what constitutes a truly desirable output. In this work, we investigate whether real data can serve as an alternative source of supervision for preference alignment. We adopt a data-centric perspective and study a curation strategy that treats real images as reference points and constructs preference signals by contrasting them with generated or perturbed samples, without requiring manually annotated preference pairs. Through empirical analysis, we show that real-data-based supervision provides effective guidance for aligning diffusion models and achieves performance comparable to existing preference-based methods. Our results suggest that real data offers a practical and complementary source of supervision for preference alignment and highlight directions of label-efficient alignment strategies. Code and models are available at https://cwyxx.github.io/RealAlign.
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On the Emergence of Implicit Curriculum in RLVR Learning Dynamics
Yu Huang, Zixin Wen, Yuejie Chi, Yuting Wei, Aarti Singh, Yingbin Liang, Yuxin Chen
Reinforcement learning with verifiable rewards (RLVR) has been a main driver of recent breakthroughs in large reasoning models. Yet it remains a mystery how rewards based solely on final outcomes can help overcome the long-horizon barrier to extended reasoning. To understand this, we develop a theory of the training dynamics of RLVR for transformers on compositional reasoning tasks. Our theory shows that mixed-difficulty training naturally induces an **implicit curriculum**: without any explicit schedule, easier problems become learnable first and shape the frontier for harder ones, creating a learning progression from easy to hard during optimization. The effectiveness of this curriculum is governed by the smoothness of the difficulty spectrum. When the spectrum is smooth, training dynamics enter a well-behaved *relay* regime, in which persistent gradient signals on easier problems make slightly harder ones tractable and keep training at the edge of competence. When the spectrum contains abrupt discontinuities, training undergoes grokking-type phase transitions with prolonged plateaus before progress recurs. As a technical contribution, our analysis develops and adapts techniques

from Fourier analysis on finite groups to our setting. We validate the predicted mechanisms empirically via synthetic experiments.
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R^3DAO: Reactive Recovery and Reconstruction for Long-horizon Data Agent Orchestration
Quanxin Liu, Rui Hao, Ruida Xu, Jianwei Zhong, Changhu Chen, Yijun Mo
End-to-end data science agent workflows involve tightly coupled sub-processes with strong dynamic dependencies, posing a challenging long-horizon orchestration problem. Existing frameworks primarily rely on static, chain-like execution plans, which are prone to error propagation from early stages—often causing reasoning chain collapse and task failure, resulting in fragile inference and poor cost-effectiveness. To address these issues, we propose R^3DAO, a reactive data agent orchestration framework based on feedback-driven topology evolution, aiming to build a dynamic evolutionary closed-loop of "hierarchical exploration, iterative recovery, and empirical convergence." First, we introduce a dynamic hierarchical task network that recursively decomposes global intent into macro-logical anchors and micro-operators, enabling low-cost exploration through dimensionality reduction in the logical space. Second, we establish a reactive topology reconfiguration mechanism that leverages semantic reflection to map execution anomalies into diagnostic signals, replacing costly global resets with localized topological optimization for resilient self-healing. Finally, semantic experience distillation implements a dual-loop accumulation that compresses long-horizon trajectories into structured prior, steering execution efficiency toward the optimal regime. Evaluations on the MLE-bench show that R^3DAO achieves a 77.36\% improvement in success rate over advanced R\&D-Agent while maintaining competitive task scores. Notably, R^3DAO compresses the average execution time by 36$\times$ and limits token consumption to just 104k per task, showcasing superior reliability, efficiency, and cost-effectiveness.
****
LABO: LLM-Accelerated Bayesian Optimization through Broad Exploration and Selective Experimentation
Zhuo Chen, Xinzhe Yuan, Jianshu Zhang, Jinzong Dong, Ruichen Zhou, Yingchun Niu, Tianhang Zhou, Yu Yang Fredrik Liu, Yuqiang Li, Nanyang Ye, Qinying Gu
The high cost and data scarcity in scientific exploration have motivated the use of large language models (LLMs) as knowledge-driven components in Bayesian optimization (BO). However, existing approaches typically embed LLMs directly into the sampling or surrogate modeling pipeline, without fully leveraging their significantly lower evaluation cost compared to real-world experiments. To address this limitation, we propose LLM-Accelerated Bayesian Optimization (LABO), a framework that combines LLM predictions with experimental observations within a single BO loop. LABO employs a gating mechanism to dynamically balance reliance on LLM predictions versus actual experiments. By leveraging inexpensive LLM evaluations to broadly explore the search space and reserving costly real experiments only for regions with high uncertainty, LABO achieves more sample-efficient optimization. We provide a theoretical analysis with a cumulative regret bound that formalizes this efficiency gain. Empirical results across diverse scientific tasks demonstrate that LABO consistently outperforms existing methods under identical experimental budgets. Our results suggest that LABO offers a practical and theoretically grounded approach for integrating LLMs into scientific discovery workflows.
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Why Tree-Style Branching Matters for Thought Advantage Estimation in GRPO
Hongcheng Wang, Yinuo Huang, Sukai Wang, Guanghui Ren, Hao Dong
Group Relative Policy Optimization (GRPO) trains Chain-of-Thought reasoning with verifiable rewards, but estimating thought-level advantages without value functions often suffers from high variance. Although tree-style branching is used in practice to reduce variance, it lacks a theoretical explanation of why it works and whether it is important or potentially necessary. We study thought-level advantage estimation in GRPO from a variance perspective under a minimal tree-style setting where multiple continuations are sampled for each thought. Using the mu

ltivariate delta method, we reveal a sampling-dimension asymmetry. Increasing sa
mpled thoughts (K) leaves a strictly positive estimation-variance floor, where
as increasing continuations per thought (M) drives the leading-order estimatio
n variance to zero at rate $1/M$. This implies that, within the fixed-temperatur
e GRPO-style estimator without value models studied here, accurate thought-level
 advantage estimation cannot be achieved by scaling thought sampling alone, maki
ng continuation-level branching a principled and potentially necessary mechanism
 rather than a heuristic. Experiments further provide empirical evidence for its
 effectiveness and potential necessity, demonstrating improved optimization stab
ility, training efficiency, and final performance not only in math but also acro
ss vision domains and under different model architectures and sizes.
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Mind the State: Towards Unified, Context-Aware EEG-to-fMRI Synthesis
Yamin Li, Shiyu Wang, Chang Li, Ange Lou, Haatef Pourmotabbed, Sarah Goodale, Da
rio J. Englot, Daniel Moyer, Roza G Bayrak, Catie Chang
Functional magnetic resonance imaging (fMRI) provides dynamic measurements of hu
man brain activity at high spatial resolution and depth, but its use is constrai
ned by high cost, limited accessibility, and strict acquisition requirements. Sy
nthesizing fMRI data from more accessible, non-invasive modalities such as elect
roencephalography (EEG) offers a promising alternative, enabling inference of de
ep brain dynamics from low-cost scalp recordings in naturalistic settings. Despi
te recent progress, existing EEG-to-fMRI translation methods typically rely on r
egion-specific models and offer limited support for subject-level and dataset-le
vel heterogeneity, restricting their generalizability. We propose UniEFS, a unif
ied EEG-to-fMRI Synthesis model that enables full-brain fMRI reconstruction whil
e accommodating varying demographic and physiological contexts within a single m
odel. Our approach leverages a pretrained fMRI decoder to embed rich spatial pri
ors and introduces condition-aware prompt tokens that encode subject-level and e
xperimental metadata, enabling effective handling of heterogeneous datasets. We
extensively evaluate the model performance on eyes-closed resting-state data and
 demonstrate that it can reliably reconstruct temporally-resolved whole-brain fM
RI activity, with potential to generalize to task-based fMRI and clinical popula
tions in a zero-shot manner. Project page: https://soupeeli.github.io/UniEFS
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Flatness-Aware Stochastic Gradient Langevin Dynamics
Stefano Bruno, Youngsik Hwang, JaeHyeon An, Sotirios Sabanis, Dongyoung Lim
Flatness of the loss landscape has been widely studied as an important perspecti
ve for understanding the behavior and generalization of deep learning algorithms
. Motivated by this view, we propose Flatness-Aware Stochastic Gradient Langevin
 Dynamics (fSGLD), a first-order optimization method that biases learning its d
ynamics toward flat basins while retaining the computational and memory efficien
cy of SGD and SGLD. We provide a non-asymptotic theoretical analysis showing tha
t fSGLD converges to a flatness-biased Gibbs distribution under a theoretically
prescribed coupling between the noise scale σ and the inverse temperature
 β, together with explicit excess risk guarantees. We empirically evaluate
 fSGLD across standard optimizer benchmarks, Bayesian image classification, unce
rtainty quantification, and out-of-distribution detection, demonstrating consist
ently strong performance and reliable uncertainty estimates. Additional experime
nts confirm the effectiveness of the theoretically prescribed β–σ c
oupling compared to decoupled choices.
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From Weak Cues to Real Identities: Evaluating Inference-Driven De-Anonymization
in LLM Agents
Myeongseob Ko, Jihyun Jeong, Sumiran Singh Thakur, Gyuhak Kim, Ruoxi Jia
Anonymization is often assumed to protect privacy once explicit identifiers are
removed, because re-identification has historically required specialized experti
se, tailored algorithms, and manual corroboration. We show that LLM-based agents
 weaken this barrier: by combining scattered, individually non-identifying cues
with public evidence, they reconstruct real-world identities, sometimes even dur
ing benign tasks. We evaluate this risk across three settings---classical linkag

e incidents, a controlled benchmark (\emph{InferLink}) that varies fingerprint t ype, task framing, and attacker knowledge, and open-ended human--AI interaction traces. In the sparsest regime of the Netflix Prize deanonymization setting, age nts reconstruct 79.2\% of identities, against 56.0\% for a classical matching ba seline; on \emph{InferLink}, they link individuals even without an explicit re-i dentification request, and more often once one is given. In redacted human--AI i nteraction traces, agents further resolve anonymized profiles to specific indivi duals by corroborating contextual cues with public evidence. These findings sugg est that privacy evaluations for agentic systems should measure not only what in formation is accessed or disclosed, but also what identities can be inferred.
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ReaForest: Fostering Generative Video Reasoning for Spatial Planning
Kun Ouyang, Yuanxin Liu, Xinhao Li, Linli Yao, Xiangyu Zeng, Haoning Wu, Hao Zho u, Fandong Meng, Jie Zhou, Xu Sun
Verbal logic and visual mental simulation are two essential components of human intelligence. Modern Large Language Models (LLMs) have demonstrated strong verba l reasoning capabilities through textual Chain-of-Thought (CoT) reasoning. In co ntrast, current Video Generation Models (VGMs) struggle with visual reasoning ta sks such as spatial planning. We attribute this limitation to two fundamental ga ps:
(i) VGMs are predominantly trained on general-purpose video corpora
emphasizing perceptual fidelity over visual reasoning, leaving
reasoning abilities underdeveloped;
(ii) most VGMs generate videos in a single pass without
mechanisms to explore alternative reasoning trajectories and to revise intermedi ate errors.
Motivated by these limitations, we introduce **ReaForest**, a framework that fos ters the reasoning capacity of VGMs in spatial planning through both training-ti me activation and inference-time scaling.
ReaForest comprises three key components:
(1) ReaGen-27k, a dataset covering diverse spatial planning tasks that requi re multi-step reasoning, which activates basic reasoning capabilities of VGMs fo r spatial planning;
(2) Reflective Entropy-Aware Test-Time Scaling (ReaTTS),
an inference framework that evolves multiple reasoning
branches while enabling failure recovery;
(3) Hierarchical constraint verification, which provides actionable feedback for ReaTTS based on decomposed constraints.
Extensive experiments demonstrate that ReaForest substantially surpasses advance d textual reasoning models (e.g., Gemini-2.5-Pro) and video generation models (e .g., Sora-2). ReaForest exhibits emergent properties including self-correction, parallel thinking, and scalable reasoning, advancing VGMs toward human-like visu al mental simulation.
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Towards Fine-Grained Robustness: Attention-Guided Test-Time Prompt Tuning for Vi sion-Language Models
Jia-Wei Hai, Yijun Wang, Xiu-Shen Wei
Vision-Language Models (VLMs), such as CLIP, have achieved significant zero-shot performance on downstream tasks with various fine-tuning adaptation methods. Ho wever, recent studies have proven that adversarial attacks can significantly deg rade the inference ability of VLMs, posing substantial risks to their practical applications. Prevalent test-time adaptation methods typically rely on multi-vie w augmentation to implement various fine-tuning strategies, which struggle to id entify semantic information and are prone to destroying discriminative regions i n fine-grained scenarios. To address these limitations, we propose Attention-Gui ded Test-Time Prompt Tuning (A-TPT), a semantics-preserving method designed for test-time adaptation. We first refine the gradient attention rollout mechanism t o identify semantically meaningful regions surviving under adversarial attacks. Furthermore, we leverage them to guide the spatially varying augmentation intens ities and multi-view ensemble for prompt tuning and inference. Extensive experim

ents demonstrate that A-TPT outperforms existing test-time adaptation methods on both adversarial and clean data. Codes are available at https://github.com/SEU-VIPGroup/A-TPT.
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WS-GRPO: Weakly-Supervised Group-Relative Policy Optimization for Rollout-Efficient Reasoning
Gagan Mundada, Zihan Huang, Rohan Surana, Sheldon Yu, Jennifer Yuntong Zhang, Xintong Li, Tong Yu, Lina Yao, Jingbo Shang, Julian McAuley, Junda Wu
Group Relative Policy Optimization (GRPO) is effective for training language models on complex reasoning.
However, since the objective is defined relative to a group of sampled trajectories,
extended deliberation can create more chances to realize relative gains,
leading to inefficient reasoning and overthinking, and complicating the trade-off between correctness and rollout efficiency. Controlling this behavior is difficult in practice, considering
(i) Length penalties are hard to calibrate because longer rollouts may reflect harder problems that require longer reasoning,
penalizing tokens risks truncating useful reasoning along with redundant continuation;
and (ii) supervision that directly indicates when to continue or stop is typically unavailable beyond final answer correctness.
We propose Weakly Supervised GRPO (WS-GRPO), which improves rollout efficiency by converting terminal rewards into correctness-aware guidance over partial trajectories.
Unlike global length penalties that are hard to calibrate, WS-GRPO trains a preference model from outcome-only correctness to produce prefix-level signals
that indicate when additional continuation is beneficial.
Thus, WS-GRPO supplies outcome-derived continue/stop guidance, reducing redundant deliberation while maintaining accuracy.
We provide theoretical results and empirically show on reasoning benchmarks that WS-GRPO substantially reduces rollout length while remaining competitive with GRPO baselines.
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Compact Conformal Subgraphs
Sreenivas Gollapudi, Kostas Kollias, Kamesh Munagala, Aravindan Vijayaraghavan
Conformal prediction provides rigorous, distribution-free uncertainty guarantees, but often yields prohibitively large prediction sets in structured domains such as routing, planning, or sequential recommendation. We introduce graph-based conformal compression, a framework for constructing compact subgraphs that preserve statistical validity while reducing structural complexity. We formulate compression as selecting a smallest subgraph capturing a prescribed fraction of the probability mass, and reduce to a weighted version of densest-k-subgraphs in hypergraphs, in the regime where the subgraph has a large fraction of edges. We design efficient approximation algorithms that achieve constant factor coverage and size trade-offs. Crucially, we prove that our relaxation satisfies a monotonicity property, derived from a connection to parametric minimum cuts, which guarantees the nestedness required for valid conformal calibration. Our results therefore not only highlight an algorithmic regime, distinct from classical densest-k-subgraph hardness settings, where the problem can be approximated efficiently, but also bridge conformal prediction with combinatorial graph compression via monotonicity. We finally validate our algorithmic approach via simulations for trip planning and navigation, and compare to natural baselines.
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One-step Optimal Transport via Regularized Distribution Matching Distillation
Denis Rakitin, Ivan Shchekotov, Viacheslav Meshchaninov, Dmitry Vetrov
Unpaired domain translation remains a challenging task due to the need of finding a balance between faithfulness and realism. In this paper, we propose a method called Regularized Distribution Matching Distillation (RDMD) that combines the best properties of Optimal Transport (OT) and diffusion-based domain translation

methods. Instead of the conventional adversarial training, RDMD utilizes diffus
ion-based distribution matching, addressing the common shortcomings of OT method
s and providing a strong initialization for the trained models. RDMD provides ef
ficient one-step inference, explicitly controls the input-output alignment via r
egularization of the transport cost and maintains high faithfulness similar to t
he OT methods. We prove that in theory RDMD approximates the OT map and demonstr
ate its empirical performance on several tasks, including unpaired image-to-imag
e translation in pixel and latent space and unpaired text detoxification. Empiri
cal results show that RDMD achieves a comparable or better faithfulness-realism
trade-off compared to the diffusion and OT baselines.
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Test-Time Learning of Causal Structure from Interventional Data
Wei Chen, Rui Ding, Huang Bojun, Yang Zhang, Qiang Fu, Yuxuan Liang, Shi Han, Do
ngmei Zhang
Supervised Causal Learning has shown promise in causal discovery, yet it often s
truggles with generalization across diverse interventional settings, particularl
y when intervention targets are unknown. To address this, we propose TICL (Test-
time Interventional Causal Learning), a novel method that synergizes Test-Time
Training with Joint Causal Inference (JCI). Specifically, we design a self-augme
ntation strategy to generate instance-specific training data at test time, effec
tively avoiding distribution shifts. Furthermore, by integrating JCI, we develop
ed a PC-inspired two-phase supervised learning scheme, which effectively leverag
es self-augmented data while ensuring theoretical identifiability. Extensive exp
eriments on bnlearn benchmarks demonstrate TICL's superiority in multiple aspect
s of causal discovery and intervention target detection.
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BroRL: Scaling Reinforcement Learning via Broadened Exploration
Jian Hu, Mingjie Liu, Ximing Lu, Fang Wu, Zaid Harchaoui, Shizhe Diao, Yejin Cho
i, Pavlo Molchanov, Jun Yang, Jan Kautz, Yi Dong
Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a key ingre
dient for unlocking complex reasoning capabilities in large language models. Rec
ent work ProRL has shown promise in scaling RL by increasing the number of train
ing steps. However, performance plateaus after thousands of steps, with clear di
minishing returns from allocating more computation to additional training.
In this work, we investigate a complementary paradigm for scaling RL: \textbf{Br
oRL}—increasing the number of rollouts per example to hundreds to exhaustively \
textbf{Bro}aden exploration, which yields continuous performance gains beyond th
e saturation point observed in ProRL when scaling the number of training steps.
Our approach is motivated by a mass balance equation analysis allowing us to cha
racterize the rate of change in probability mass for correct and incorrect token
s during the reinforcement process. We show that under a one-step RL assumption,
 sampled rollout tokens contribute to correct-mass expansion, while unsampled to
kens outside rollouts may lead to gains or losses depending on their distributio
n and the net reward balance. Importantly, as the number of rollouts per example
 N increases, the effect of unsampled terms diminishes, making overall correct
-mass expansion more likely.
To validate our theoretical analysis, we conduct simulations under more relaxed
conditions and find that a sufficiently large rollout size N—corresponding to
ample exploration—can increase the probability mass of correct tokens broadly, a
nd in our simulator it increases all correct-token probabilities and eliminates
knowledge shrinkage.
Empirically, BroRL revives models saturated after 3K ProRL training steps and de
monstrates robust, continuous improvement, achieving strong results for the 1.5B
 model across diverse benchmarks.
Notably, under the same training time, BroRL is both more data- and compute-effi
cient: large-N rollouts reduce the number of filtered samples during dynamic s
ampling at the algorithmic level and nearly double generation throughput compare
d to ProRL in our hardware setup; this throughput increase is consistent with sh
ifting generation from a more memory-bound regime toward a more compute-bound on
e.

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SIMPC: Learning Self-Induced Mirror-Point Consistency for Unsupervised Point Cloud Denoising
Chengwei Zhang, Xueyi Zhang, Tao Jiang, Xinhao Xu, Wenjie Li, Fubo Zhang, Longyong Chen
In point clouds, noise directly perturbs point coordinates that encode both spatial location and geometry, making one-to-one correspondence construction more challenging than in images.
Existing methods impose statistical mappings across noisy variants via noise or optimal transport, but suffer from correspondence ambiguity.
In this work, we propose Self-Induced Mirror-Point Consistency (SIMPC) to learn deterministic correspondences between points and the underlying surface in an unsupervised manner.
For each noisy point, SIMPC generates a mirror-point on the opposite side of the underlying surface, guided by geometric priors during the denoising process.
By encouraging consistency between the denoising targets of the original point and its mirror counterpart, SIMPC effectively localizes the position of underlying surface.
Extensive experiments on synthetic and real-world datasets demonstrate that SIMPC significantly outperforms state-of-the-art unsupervised methods and surpasses several strong supervised counterparts.
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On the Ability of Transformers to Verify Plans
Yash Sarrof, Yupei Du, Katharina Stein, Alexander Koller, Sylvie Thiebaux, Michael Hahn
Transformers have shown inconsistent success in AI planning tasks, and theoretical understanding of when generalization should be expected has been limited. We take important steps towards addressing this gap by analyzing the ability of decoder-only models to verify whether a given plan correctly solves a given planning instance. To analyse the general setting where the number of objects -- and thus the effective input alphabet -- grows at test time, we introduce C*-RASP, an extension of C-RASP designed to establish length generalization guarantees for transformers under the simultaneous growth in sequence length and vocabulary size. Our results identify a large class of classical planning domains for which transformers can provably learn to verify long plans, and structural properties that significantly affects the learnability of length generalizable solutions. Empirical experiments corroborate our theory.
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Cert-LAS: Toward Certified Model Ownership Verification for Text-to-Image Diffusion Models via Layer-Adaptive Smoothing
Leyi Qi, Yiming Li, Siyuan Liang, Zhengzhong Tu, Dacheng Tao
Large-scale text-to-image (T2I) diffusion models have enabled unprecedented creative applications, but their unauthorized use has raised serious intellectual property concerns, making model ownership verification (MOV) increasingly critical. We find that existing backdoor-based diffusion watermarking methods often (implicitly) assume a "faithful" verification process, namely, that the verifier can query a suspicious model and obtain the faithful watermark response to complete MOV. However, in practice, adversaries may intentionally or unintentionally damage potential watermark signals, significantly degrading verification reliability. To address this issue, we propose Cert-LAS, the first certified MOV method for T2I models based on layer-adaptive smoothing. In general, Cert-LAS embeds specified watermarks using diffusion classifiers and an LFS-guided layer-adaptive noise, and verifies ownership by examining whether the suspected model exhibits significantly stronger watermark responses compared to unwatermarked references through hypothesis testing. We further prove that, under certain conditions, our Cert-LAS can still achieve reliable verification even in the presence of malicious removal attacks. Extensive experiments validate the effectiveness of Cert-LAS and its resistance to adaptive attacks. Our code is available at https://github.com/Leyi-Qi/Cert-LAS.
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Inference-Aware Meta-Alignment of LLMs via Non-Linear GRPO
Shokichi Takakura, Akifumi Wachi, Rei Higuchi, Kohei Miyaguchi, Taiji Suzuki
Aligning large language models (LLMs) to diverse human preferences is fundamenta lly challenging since criteria can often conflict with each other. Inference-tim e alignment methods have recently gained popularity as they allow LLMs to be ali gned to multiple criteria via different alignment algorithms at inference time. However, inference-time alignment is computationally expensive since it often re quires multiple forward passes of the base model. In this work, we propose *infe rence-aware meta-alignment* (IAMA), a novel approach that enables LLMs to be ali gned to multiple criteria with minimal computational overhead at inference time. IAMA trains a base model such that it can be effectively aligned to multiple ta sk optima via different inference-time alignment algorithms. To solve the non-li near optimization problems involved in IAMA, we propose *non-linear GRPO*, which provably converges to the optimal solution in the space of probability measures .
****
Probing RLVR Training Instability through the Lens of Objective-Level Hacking
Yiming Dong, Kun Fu, Haoyu Li, Xinyuan Zhu, Yurou Liu, Lijing Shao, Jieping Ye, Zheng Wang
Prolonged reinforcement learning with verifiable rewards (RLVR) has been shown t o drive continuous improvements in the reasoning capabilities of large language models, but the training is often prone to instabilities, especially in Mixture- of-Experts (MoE) architectures. Training instability severely undermines model c apability improvement, yet its underlying causes and mechanisms remain poorly un derstood. In this work, we introduce a principled framework for understanding RL VR instability through the lens of *objective-level hacking*. Unlike reward hack ing, which arises from exploitable verifiers, objective-level hacking emerges fr om *token-level credit misalignment* and is manifested as system-level spurious signals in the optimization objective. Grounded in our framework, together with extensive experiments on a 30B MoE model, we trace the origin and formalize the mechanism behind a key pathological training dynamic in MoE models: the abnormal growth of the training-inference discrepancy, a phenomenon widely associated wi th instability but previously lacking a mechanistic explanation. These findings provide a concrete and causal account of the training dynamics underlying instab ilities in MoE models, offering guidance for the design of stable RLVR algorithm s.
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Combinatorial Sparse PCA Beyond the Spiked Identity Model
Syamantak Kumar, Purnamrita Sarkar, Kevin Tian, Peiyuan Zhang
Sparse PCA is one of the most well-studied problems in high-dimensional statisti cs. In this problem, we are given samples from a distribution with covariance $\ \mathbf{\Sigma}$, whose top eigenvector $\\mathbf{v} \in \\mathbb{R}^d$ is s-s parse. Existing sparse PCA algorithms can be broadly categorized into (1) combin atorial algorithms (e.g., diagonal or elementwise covariance thresholding) and (2) SDP-based algorithms. While combinatorial algorithms are much simpler, they a re typically only analyzed under the spiked identity model (where $\\mathbf{\Sig ma} \propto \\mathbf{I}_d + \gamma \\mathbf{v}\\mathbf{v}^\top$ for some $\gamma > 0$), whereas SDP-based algorithms require no additional assumptions on $\\mat hbf{\Sigma}$. We demonstrate explicit counterexample covariances $\\mathbf{\Sig ma}$ against the success of standard combinatorial algorithms for sparse PCA, wh en moving beyond the spiked identity model.
In light of this discrepancy, we give the first combinatorial method for sparse PCA that provably succeeds for general $\\mathbf{\Sigma}$ using $\\mathsf{poly}(s, \log(d))$ samples and $d^2 \cdot \\mathsf{poly}(s, \log(d))$ time, by providi ng a global convergence guarantee on the truncated power method of Yuan and Zhan g (JMLR, 2013). We provide a natural generalization of our method to recovering k sparse principal components. Finally, we evaluate our method on synthetic an d real-world sparse PCA datasets.
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Fully Zero-Shot Image Dehazing

Shuocheng Wang, Ruoxi Zhu, Jiaming Liu, Zhengyang Cao, Kun Wang, Chengkang Huang, Lizhuo Liu, Shichen Peng, Minge Jing, Yibo Fan

Image dehazing, an important image restoration problem, aims to recover clear scene content from images degraded by atmospheric haze. Existing dehazing methods rely on observing the distribution of hazy images during training: supervised approaches typically depend on synthetic datasets, leading to poor generalization in real-world scenarios; unsupervised methods are constrained by the limited diversity of observed haze conditions due to the difficulty of collecting real hazy images, and fail to generalize to unseen haze types. To address these challenges, we propose the first fully zero-shot dehazing framework that is trained without any hazy images. The framework is built upon a set of representations that remain invariant across clean and hazy images to bridge the two domains, which is both theoretically derived and experimentally validated. Consequently, we formulate dehazing as a conditional generative modeling problem and train a diffusion model solely with the invariant representations of the abundant and readily available clean images. During testing, the same representations extracted from hazy images serve as the conditional input to guide the diffusion process toward the clean image distribution. Quantitative analyses verify the effectiveness of the proposed representations, and extensive experiments across various real-world hazy datasets demonstrate our framework’s remarkable generalization ability, significantly outperforming existing methods. Our code will be available after the review process.

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FOCUS: Forcing In-Context Object Localization through Visual Support Constraints and Policy Optimization

Mohammed Asad Karim, Vinay Kumar Verma

In-context localization (ICL) seeks to localize a target object specified by a small set of support examples in a query image, operating on the fly without training or parameter updates. Despite rapid advances in vision–language models (VLMs), achieving category-agnostic and visually grounded ICL remains an open problem, even though it is essential for applications such as image editing, personalized visual search, and retrieval. Existing methods are fragile and rely on explicit category supervision, which not only limits applicability in realistic settings with unnamed or instance-specific objects but also introduces category bias that steers predictions toward semantic priors rather than visual evidence. We introduce a two-stage training framework that explicitly optimizes in-context attention between support bounding boxes and query images without category supervision. We further refine localization via reinforcement learning using Group Relative Policy Optimization (GRPO) to directly minimize localization error. This formulation enforces visual correspondence over semantic priors, yielding robust instance-level localization. Empirically, a 7B-parameter model trained with our objectives outperforms models up to 72B parameters, demonstrating that context-aware localization objectives can surpass scaling alone. Comprehensive ablations validate the contribution of each component.

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Spectral Bridge Variational Inference: Dynamic LoRA via Bures-Wasserstein Gradient Flows

Yuhang Xi, Yu-Feng Yu, Chuan-Xian Ren, Zhao-Rong Lai

Parameter-Efficient Fine-Tuning (PEFT) is essential for adapting Large Language Models, yet existing methods struggle to balance capacity with computational efficiency. Standard approaches enforce rigid low-rank constraints, while dynamic alternatives incur significant memory overheads. To resolve this, we propose Spectral Bridge Variational Inference (SBVI), a geometric framework reformulating LoRA as a continuous Wasserstein gradient flow on the manifold of Gaussian measures. Instead of fixing ranks at initialization, SBVI governs singular value evolution via a stochastic differential equation driven by thermodynamic competition between task gradients and adaptive entropic friction. This induces a spectral bifurcation that automatically prunes noise modes while amplifying signal-rich components, discovering an optimal layer-wise rank distribution. We derive a scalable algorithm with linear complexity using factorized Riemannian retractions and

Empirical Bayes friction updates. Experiments on reasoning and coding benchmarks show SBVI achieves state-of-the-art performance, offering superior accuracy and memory efficiency over existing static and dynamic methods.

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Exploring Accurate and Transparent Domain Adaptation in Predictive Healthcare via a Concept-Grounded Orthogonal Inference

Pengfei Hu, Chang Lu, Feifan Liu, Yue Ning

Deep learning models for clinical event prediction on electronic health records (EHR) often suffer performance degradation when deployed under different data distributions.

While domain adaptation (DA) methods can mitigate such shifts, their "black-box" nature prevents widespread adoption in clinical practice where transparency is essential for trust and safety.

We propose ExtraCare to decompose patient representations into invariant and covariant components.

By supervising these two components and enforcing their orthogonality during training, our model preserves label information while exposing domain-specific variation at the same time for more accurate predictions than most feature alignment models.

More importantly, it offers human-understandable explanations by mapping sparse latent dimensions to medical concepts and quantifying their contributions via targeted ablations.

ExtraCare is evaluated on two real-world EHR datasets across multiple domain partition settings, demonstrating superior performance along with enhanced transparency, as evidenced by its accurate predictions and explanations from extensive case studies.

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Recovering Hidden Reward in Diffusion-Based Policies

Yanbiao Ji, Qiuchang Li, Yuting Hu, Shaokai Wu, Wenyuan XIE, Guodong ZHANG, Qichen He, Deyi Ji, Yue Ding, Hongtao Lu

This paper introduces EnergyFlow, a framework that unifies generative action modeling with inverse reinforcement learning by parameterizing a scalar energy function whose gradient is the denoising field. We establish that under maximum-entropy optimality, the score function learned via denoising score matching recovers the gradient of the expert's soft Q-function, enabling reward extraction without adversarial training. Formally, we prove that constraining the learned field to be conservative reduces hypothesis complexity and tightens out-of-distribution generalization bounds. We further characterize the identifiability of recovered rewards and bound how score estimation errors propagate to action preferences.

Empirically, EnergyFlow achieves state-of-the-art imitation performance on various manipulation tasks while providing an effective reward signal for downstream reinforcement learning that outperforms both adversarial IRL methods and likelihood-based alternatives. These results show that the structural constraints required for valid reward extraction simultaneously serve as beneficial inductive biases for policy generalization. The code is available at https://github.com/sotaagi/EnergyFlow.

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Protein Circuit Tracing via Cross-layer Transcoders

Darin Tsui, Kunal Talreja, Daniel Saeedi, Amirali Aghazadeh

Protein language models (pLMs) have emerged as powerful predictors of protein structure and function. However, the computational circuits underlying their predictions remain poorly understood. Recent mechanistic interpretability methods decompose pLM representations into interpretable features, but they treat each layer independently and thus fail to capture cross-layer computation, limiting their ability to approximate the full model. We introduce ProtoMech, a framework for discovering computational circuits in pLMs using cross-layer transcoders that learn sparse latent representations jointly across layers to capture the model’s full computational circuitry. Applied to the pLM ESM2, ProtoMech recovers 82–89\% of the original performance on protein family classification and function prediction tasks. ProtoMech then identifies compressed circuits that use <1\% of the

latent space while retaining up to 79\% of model accuracy, revealing corresponde nce with structural and functional motifs, including binding, signaling, and sta bility. Steering along these circuits enables high-fitness protein design, surpa ssing baseline methods in more than 70\% of cases. These results establish Proto Mech as a principled framework for protein circuit tracing.
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Trustworthy Federated Label Distribution Learning under Annotation Quality Dispa rity

Junxiang Wu, Zhiqiang Kou, Hongwei Zeng, Wenke Huang, Biao Liu, Hanlin Gu, Yuhen g Jia, Di Jiang, Yang Liu, Xin Geng

Label Distribution Learning (LDL) models supervision as an instance-wise probabi lity distribution, enabling fine-grained learning under inherent ambiguity, but its success relies on high-fidelity label distributions that are costly to obtai n and thus often noisy.

Motivated by privacy-sensitive applications, we study Federated Label Distributi on Learning (Fed-LDL), where data isolation further induces heterogeneous annota tion quality across clients, making local updates unevenly reliable and breaking sample-size-based aggregation (e.g., FedAvg).

To address this trust dilemma, we propose FedQual, a quality-aware Fed-LDL frame work with two coupled mechanisms: (i) quality-adaptive client training guided by a global semantic anchor that calibrates low-quality clients while preserving h igh-quality autonomy, and (ii) reliability-aware server aggregation that reweigh ts client contributions by effective reliable information rather than raw sample size.

To enable rigorous evaluation, we construct four new Fed-LDL benchmarks (FER-LDL , FI-LDL, PIPAL-LDL, and KADID-LDL) with controlled annotation quality disparity .

We further provide a theoretical guarantee showing that under heterogeneous supe rvision quality, client-specific calibration is strictly better than any uniform calibration.

Experiments on the proposed benchmarks demonstrate the effectiveness of FedQual.
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Adaptive Policy Backbone via Shared Network

Bumgeun Park, Donghwan Lee

Reinforcement learning (RL) has achieved impressive results across various domai ns, yet the resulting policies often fail to generalize beyond the specific task s encountered during training. This lack of robustness limits their deployment i n real-world scenarios where diverse and unpredictable task demands exist. In th is work, we provide a theoretical analysis of policy networks under Markov Decis ion Processes (MDPs) and demonstrate that adapting only the linear layers placed before and after a policy backbone is sufficient for task adaptation. Based on this insight, we propose the Adaptive Policy Backbone (APB), which consists of a frozen backbone paired with lightweight, task-specific pre- and post-backbone l inear layers. Our results demonstrate that learning only these lightweight task- specific linear layers is sufficient to achieve performance on par with standard RL, even when the backbone is randomly initialized. Furthermore, we find that t his structural constraint can enhance the generalization capability of the resul ting policies. This advantage extends to out-of-distribution tasks, where repres entative meta-RL baselines often struggle.
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DRIVE: Distributional and Retrieval-Augmented Bidding with Value Evaluation

Miduo Cui, Haochen Wang, Shangqin Mao, Xun Yang, Qianlong Xie, Xingxing Wang, Xu ri Ge, Ying Zhou, Zhiwei Xu

Auto-bidding is a core component of real-time advertising systems, where decisio ns must optimize long-term performance under budget and cost constraints, while online exploration is prohibitively risky. Offline reinforcement learning and, m ore recently, Transformer-based sequence modeling have shown promise for learnin g bidding policies from logged data, but their unimodal and purely parametric fo rmulations often collapse multiple effective bidding strategies into suboptimal averaged actions and perform unreliably under sparse or long-tail traffic. To mi

tigate these limitations, we propose **DRIVE** (Distributional and Retrieval-Augmented Bidding with Value Evaluation), a unified Transformer-based framework that decouples candidate action generation from decision making for offline auto-bidding. DRIVE combines distributional action modeling, retrieval-augmented candidate generation from high-quality historical decisions, and value-based evaluation to select the most promising bid at inference time. Extensive experiments on AuctionNet and additional offline reinforcement learning benchmarks demonstrate that DRIVE consistently improves bidding performance and generalizes well across multiple Transformer–based methods.

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Geometry-Aware Tabular Diffusion

David Turtora Zagardo

Tabular synthesis is critical for privacy-preserving sharing and augmentation, yet diffusion models rely on implicit mechanisms to capture inter-column relationships. We introduce Geometry-Aware Tabular Diffusion (GATD), which augments tabular diffusion denoisers with pairwise angles and lengths computed from column value differences and used as inputs and auxiliary targets. Our MLP instantiation achieves state-of-the-art benchmark performance while using $3.5\times$ fewer parameters on average (up to $25\times$ for classification tasks): on ten datasets, it wins 8/10 Shape, 7/10 Trend, and 9/10 downstream utility (F1/RMSE), reducing Shape and Trend error by 27\% and 20\%. Default loss weights transfer to GNN and Transformer denoisers, improving Shape on 27/30 and Trend on 25/30 architecture-dataset cells. A matched ablation shows supervision (not extra inputs or capacity) drives the gain. This shows explicit relational supervision is a portable inductive bias for tabular diffusion.

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$\texttt{FHAIM}$: Fully Homomorphic AIM for Private Tabular Synthetic Data Generation

Mayank Kumar, Qian Lou, Paulo L. Barreto, Martine De Cock, Sikha Pentyala

Data is the lifeblood of AI, yet much of the most valuable data remains locked in silos due to privacy and regulations. As a result, AI remains heavily underutilized in many of the most important domains, including healthcare, education, and finance. Synthetic data generation (SDG), i.e. the generation of artificial datawith a synthesizer trained on real data, offers an appealing solution to make data available while mitigating p rivacy concerns, however existing SDG-as-a-service workflow require data holders to trust providers with access to private data. We propose FHAIM, the first fully homomorphic encryption (FHE) framework for training a marginal-based synthetic data generator on encrypted tabular data. FHAIM adapts the widely used AIM algorithm to the FHE setting using novel FHE protocols, ensuring that the private data remains encrypted throughout and is released only with differential privacy guarantees. Our empirical analysis show that FHAIM preserves the performance of AIM while maintaining feasible runtimes.

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Stochastic Linear Bandits with Parameter Noise

Daniel Ezer, Alon Cohen, Yishay Mansour

We study the stochastic linear bandits with parameter noise model, in which the reward of action a is $a^\top \theta$ where θ is sampled i.i.d.

We show a regret upper bound of $\widetilde{O} (\sqrt{d T \log(K/\delta) \sigma^2_{\max}})$ for a horizon T, general action set of size K of dimension d, and where $\sigma^2_{\max}$ is the maximal variance of the reward for any action.

We further provide a lower bound of $\widetilde{\Omega} (d \sqrt{T \sigma_{\max}^2})$ which is tight (up to logarithmic factors) whenever $\log K \approx d$.

For more specific action sets, ℓ_p unit balls with $p \leq 2$ and dual norm q, we show that the minimax regret is $\widetilde{\Theta} (\sqrt{dT \sigma_q^2})$, where σ_q^2 is a variance-dependent quantity that is always at most 4.

This is in contrast to the minimax regret attainable for such sets in the classic additive noise model where the regret is of order $d \sqrt{T}$.

Surprisingly, we show that this optimal (up to logarithmic factors) regret bound

is attainable using a very simple explore-exploit algorithm.
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Motion-Aware Caching for Efficient Autoregressive Video Generation
Jing Xu, Yuexiao Ma, Xuzhe Zheng, XING WANG, Shiwei Liu, Chenqian Yan, Xiawu Zheng, Rongrong Ji, Fei Chao, Songwei Liu
Autoregressive video generation paradigms offer theoretical promise for long video synthesis, yet their practical deployment is hindered by the computational burden of sequential iterative denoising.
While cache reuse strategies can accelerate generation by skipping redundant denoising steps, existing methods rely on coarse-grained chunk-level skipping that fails to capture fine-grained pixel dynamics.
This oversight is critical: pixels with high motion require more denoising steps to prevent error accumulation, while static pixels tolerate aggressive skipping.
We formalize this insight theoretically by linking cache errors to residual instability, and propose $\textbf{MotionCache}$, a motion-aware cache framework that exploits inter-frame differences as a lightweight proxy for pixel-level motion characteristics.
MotionCache employs a coarse-to-fine strategy: an initial warm-up phase establishes semantic coherence, followed by motion-weighted cache reuse that dynamically adjusts update frequencies per token.
Extensive experiments on state-of-the-art models like SkyReels-V2 and MAGI-1 demonstrate that MotionCache achieves significant speedups of $\textbf{6.28}\times$ and $\textbf{1.64}\times$ respectively, while effectively preserving generation quality (VBench: 1%$\downarrow$ and 0.01%$\downarrow$ respectively).
The code is available at https://github.com/ywlq/MotionCache.
****
Flow Matching Calibration for Simulation-Based Inference under Model Misspecification
Pierre-Louis Ruhlmann, Michael Arbel, Florence Forbes, Pedro L. C. Rodrigues
Simulation-based inference (SBI) is transforming experimental sciences by enabling parameter estimation in complex non-linear models from simulated data. A persistent challenge, however, is model misspecification.
In a Bayesian setting, targeting posterior distributions, errors may arise from the simulator, the noise or prior modelling. These model components are only approximations of reality, and severe mismatches can yield biased or overconfident posteriors. We address this issue by introducing Flow Matching Corrected Posterior Estimation (FMCPE), a framework that leverages the flow matching paradigm to refine simulation-trained posterior estimators using a small set of calibration samples. Our approach proceeds in two stages: first, a posterior approximator is trained on abundant simulated data; second, flow matching transports its predictions toward the true posterior supported by calibration observations.
We rely on the later to guide the correction, without requiring explicit knowledge of the misspecification form or of which model components are affected. This design enables FMCPE to combine the scalability of SBI with robustness to distributional shift. Across synthetic benchmarks and real-world datasets, we show that our proposal consistently mitigates the effects of misspecification, delivering improved inference accuracy and uncertainty quantification compared to standard SBI baselines, while remaining computationally efficient.
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RA-Det: Towards Universal Detection of AI-Generated Images via Robustness Asymmetry
Xinchang Wang, Yunhao Chen, Yuechen Zhang, Congcong Bian, Zihao Guo, Xingjun Ma, Hui Li
Recent image generators produce photo-realistic content that undermines the reliability of downstream recognition systems. As visual appearance cues become less pronounced, appearance-driven detectors that rely on forensic cues or high-level representations lose stability. This motivates a shift from appearance to behavior, focusing on how images respond to controlled perturbations rather than how they look. In this work, we identify a simple and universal behavioral signal.

Natural images preserve stable semantic representations under small, structured perturbations, whereas generated images exhibit markedly larger feature drift. We refer to this phenomenon as \textbf{robustness asymmetry} and provide a theoretical analysis that establishes a lower bound connecting this asymmetry to memorization tendencies in generative models, explaining its prevalence across architectures. Building on this insight, we introduce Robustness Asymmetry Detection (RA-Det), a behavior-driven detection framework that converts robustness asymmetry into a reliable decision signal. Evaluated across 14 diverse generative models and against more than 10 strong detectors, RA-Det achieves superior performance, improving the average performance by 12.92\%. The method is data- and model-agnostic, requires no generator fingerprints, and transfers across unseen generators. Together, these results indicate that robustness asymmetry is a stable, general cue for synthetic-image detection and that carefully designed probing can turn this cue into a practical, universal detector.

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CARD: Coarse-to-fine Autoregressive Modeling with Radix-based Decomposition for Transferable Free Energy Estimation

Ziyang Yu, Yi He, Wenbing Huang, Wen Yan, Yang Liu

Estimating free energy differences quantifies thermodynamic preferences in molecular interactions, which is central to chemistry and drug discovery. Despite fruitful progress, existing methods still face key limitations: classical computational approaches remain prohibitively expensive due to their reliance on extensive molecular dynamics simulations, while deep learning-based methods are constrained by either less-expressive generative models or input dimensions tied to a specific system, resulting in negligible generalization. To address these challenges, we propose CARD, a generative framework that employs a novel radix-based decomposition to bijectively convert 3D coordinates into mixed discrete-continuous sequences, enabling coarse-to-fine autoregressive modeling with enhanced expressiveness. Notably, the model corresponds to a distribution with zero free energy, serving as a proposal for absolute free energy computation of arbitrary systems without relying on alchemical pathways. Experiments across diverse tasks demonstrate that CARD matches the accuracy of classical computational methods on unseen systems with diverse topologies, while achieving an approximately 40-fold speedup in inference.

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MVI-Bench: A Comprehensive Benchmark for Evaluating Robustness to Misleading Visual Inputs in LVLMs

Huiyi Chen, Jiawei Peng, Dehai Min, Changchang Sun, Kaijie Chen, Yan Yan, Xu Yang, Lu Cheng

Evaluating the robustness of Large Vision-Language Models (LVLMs) is essential for their continued development and responsible deployment. However, existing robustness benchmarks largely focus on hallucination or misleading textual inputs, overlooking the critical challenge posed by misleading visual inputs in assessing visual understanding. To fill this gap, we introduce MVI-Bench, the first comprehensive benchmark specially designed for evaluating how Misleading Visual Inputs undermine the robustness of LVLMs. Grounded in fundamental visual primitives, the design of MVI-Bench centers on three hierarchical levels of misleading visual inputs: Visual Concept, Visual Attribute, and Visual Relationship. Using this taxonomy, we curate six representative categories and compile 1,248 expertly annotated VQA instances. To facilitate fine-grained robustness evaluation, we further introduce MVI-Sensitivity, a novel metric that characterizes LVLM robustness. Empirical results across 18 state-of-the-art LVLMs uncover pronounced vulnerabilities to misleading visual inputs, and our in-depth analyses on MVI-Bench provide actionable insights that can guide the development of more reliable and robust LVLMs.

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Causal Detection of Multi-Step LLM Agent Attacks

Viraaji Mothukuri, Reza M. Parizi

Multi-step prompt injection attacks on LLM agents present a fundamental detection challenge because malicious intent emerges only after the workflow completes,

while individual actions remain legitimate in isolation. Existing defenses, including input sanitization, output validation, and instruction hierarchy, operate on individual actions or content patterns and cannot capture this sequential structure. We present **CausalTrace**, a detection system that reframes prompt-injection defense as causal inference. It constructs Structural Causal Models from agent trajectories with typed edges capturing data dependency, trust transfer, and state enablement, then applies Pearl’s do-calculus to answer a counterfactual question, namely, whether the harmful outcome would have occurred if the injection had been blocked. This formalization enables a principled distinction between attacks that depend on injections and benign workflows that share surface-level features. Evaluation on a dataset spanning crowdsourced traces, LLM agent benchmarks, and semi-real and real scenarios demonstrates strong detection performance, outperforming content-based baselines while requiring minimal LLM inference cost; bidirectional slicing recovers complete attack chains with high edge recall, providing interpretable explanations that trace exploitation to its causal origins.
****

SFedPO: Streaming Federated Learning with a Prediction Oracle under Temporal Shifts

Jinrui Zhou, Haotian Xu, Xichong zhang, He Sun, Mingjun Xiao

Federated Learning (FL) enables decentralized clients to collaboratively train a global model without sharing raw data. However, most existing FL frameworks assume that clients train on static local datasets collected in advance or that the data follows a fixed underlying distribution, which limits their applicability in dynamic environments where data evolves over time. A parallel line of research, online FL, removes all assumptions and adopts an adversarial perspective, but this approach is often overly pessimistic and neglects the structured, partially predictable nature of real-world data dynamics. To bridge this gap, we propose SFedPO, a streaming federated learning framework that incorporates a prediction oracle to capture the temporal evolution of client-side data distributions. We theoretically analyze the convergence bounds of SFedPO and develop two practical sampling strategies: a Distribution-guided Data Sampling (DDS) strategy that dynamically selects training data under limited storage by balancing historical reuse and distribution adaptation, and a Shift-aware Aggregation Weights (SAW) mechanism that modulates global aggregation based on client-specific sampling behaviors. We further establish robustness guarantees under prediction errors. Extensive experiments demonstrate that SFedPO effectively adapts to streaming scenarios with distribution shifts and significantly outperforms existing methods.
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3DGS^2-TR: A Scalable Second-Order Trust-Region Method for 3D Gaussian Splatting

Roger Hsiao, Yuchen Fang, Xiangru Huang, Ruilong Li, Hesam Rabeti, Zan Gojcic, Javad Lavaei, James Demmel, Sophia Shao

We propose 3DGS^2-TR, a second-order optimizer for accelerating the scene training problem in 3D Gaussian Splatting (3DGS). Unlike existing second-order approaches that rely on explicit or dense curvature representations, such as 3DGS-LM (Höllein et al., 2025) or 3DGS2 (Lan et al., 2025),

our method approximates curvature using only the diagonal of the Hessian matrix, estimated efficiently via Hutchinson’s method.

Our approach is fully matrix-free and has the same complexity as ADAM (Kingma, 2024), $O(n)$ in both computation and memory costs.

To ensure stable optimization in the presence of strong nonlinearity in the 3DGS rasterization process, we introduce a parameter-wise trust-region technique based on the squared Hellinger distance, regularizing updates to Gaussian parameters.

Under identical parameter initialization and without densification, 3DGS^2-TR is able to achieve better reconstruction quality on standard datasets, using 50\% fewer training iterations compared to ADAM, while incurring less than 1GB of peak GPU memory overhead (17\% more than ADAM and 85\% less than 3DGS-LM),

enabling scalability to very large scenes and potentially to distributed tra

ining settings.
****
Student-Centered Distillation Narrows the Agentic Gap Between Small and Large LLMs
Yuanjie Lyu, Chengyu Wang, Jun Huang, Tong Xu
Large Language Model agents achieve strong performance on multi-step reasoning and tool-use tasks, but their impressive capabilities typically rely on extremely large backbones. Existing distillation approaches train smaller students to imitate full teacher trajectories, yet reasoning and knowledge gaps between the teacher and student can cause compounding errors. We propose SCoRe, a student-centered framework in which the student generates training trajectories and the teacher corrects only the earliest error, producing training data matched to the student's abilities and exposing specific weaknesses. The student is first fine-tuned on corrected trajectories. Subsequently, short-horizon reinforcement learning starts from the verified prefix preceding the earliest error, with target rewards assigned at that step. This design enables the student to solve problems through unconstrained RL exploration rather than teacher imitation, while the short-horizon setup improves training stability. On 12 challenging benchmarks, a 7B-parameter student distilled with SCoRe closes the agentic performance gap with a 72B-parameter teacher.
****
Towards Long-Horizon Interpretability: Efficient and Faithful Multi-Token Attribution for Reasoning LLMs
Wenbo Pan, Zhichao Liu, Xianlong Wang, Yu Haining, Xiaohua Jia
Token attribution methods provide intuitive explanations for language model outputs by identifying causally important input tokens. However, as modern LLMs increasingly rely on extended reasoning chains, existing schemes face two critical challenges: (1) efficiency bottleneck, where attributing a target span of M tokens within a context of length N requires $\mathcal{O}(M \cdot N)$ operations, making long-context attribution prohibitively slow; and (2) faithfulness drop, where intermediate reasoning tokens absorb attribution mass, preventing importance from propagating back to the original input. To address these, we introduce FlashTrace, an efficient multi-token attribution method that employs span-wise aggregation to compute attribution over multi-token targets in a single pass, while maintaining faithfulness. Moreover, we design a recursive attribution mechanism that traces importance through intermediate reasoning chains back to source inputs. Extensive experiments on long-context retrieval (RULER) and multi-step reasoning (MATH, MorehopQA) tasks demonstrate that FlashTrace achieves over $130\times$ speedup over existing baselines while maintaining superior faithfulness. We further analyze the dynamics of recursive attribution, showing that even a single recursive hop improves faithfulness by tracing importance through the reasoning chain.
****
MidSteer: Optimal Affine Framework for Steering Generative Models
Tatiana Gaintseva, Andrew Stepanov, Ziquan Liu, Martin Benning, Gregory Slabaugh, Jiankang Deng, Ismail Elezi
Steering intermediate representations has emerged as a powerful strategy for controlling generative models. However, despite its empirical success, it currently lacks a comprehensive theoretical framework. In this paper, we bridge this gap by formalizing the theory of concept steering. First, we establish a link between steering and affine concept erasure, proving that the standard approach for removing unwanted behaviors is a special case of LEACE (a closed-form method for affine erasure). Next, we formulate a principled theoretical framework for concept switching, LEACE-Switch, and characterize the assumptions under which it provides an optimal affine solution. Building on this analysis, we then introduce MidSteer (Minimal Disturbance concept Steering), a more general affine framework for concept manipulation that relaxes these assumptions and enables directed, minimal-disturbance transformations. We empirically demonstrate that MidSteer performs favorably across a range of tasks, modalities, and architectures, including vision diffusion models and large language models.

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Functional Attention: From Pairwise Affinities to Functional Correspondences
Jiefang Xiao, Maolin Gao, Simon Weber, Guandao Yang, Daniel Cremers
Learning mappings between infinite-dimensional function spaces, or operator lear
ning, is essential for many machine learning applications.
Although transformer-based operators are popular, they often rely on token-wise
attention.
These methods treat continuous fields as discrete tokens and usually ignore the
global functional structure.
We introduce *Functional Attention*, which reinterprets attention as a functiona
l correspondence between adaptive bases.
Inspired by geometric functional maps, our method replaces softmax affinities wi
th structured linear operators. This yields a compact, generalizable, resolution
-invariant representation that explicitly captures global dependencies.
Experiments demonstrate that *Functional Attention* can match state-of-the-art p
erformance in many operator learning tasks, including solving PDEs, 3D segmentat
ion, and regression, while remaining robust to varying discretizations. Project
page is available at https://github.com/xjffff/FUNCATTN.
****
Riemannian Metric Matching for Scalable Geometric Modeling of Distributions
Jacob Bamberger, Adam Gosztolai, Pierre Vandergheynst, Michael M. Bronstein, Iol
o Jones
High-dimensional datasets often concentrate near low-dimensional structures, but
 estimating their geometry from samples typically relies on graphs and kernels t
hat scale poorly with dataset size and dimension.
We propose **Riemannian metric matching**: a denoising probabilistic framework f
or learning the Riemannian geometry of data using neural networks.
Specifically, we learn the *carré du champ* operator, which, using diffusion geo
metry, gives us access to the Riemannian geometry toolkit for downstream machine
 learning and statistical tasks.
Our key observation is that the carré du champ operator can be formulated as a c
onditional expectation over random perturbations of the data,
which can be exploited for sample-wise training and constant cost, amortized inf
erence without explicit kernel construction.
Empirically, metric matching rivals or improves the accuracy of k-NN-based dif
fusion geometry estimators, while enabling amortized inference that is up to $40
0\times$ faster, and supports graph-free geometric analysis on high-dimensional
images where nearest neighbors break down.
****
Decision-focused Sparse Tangent Portfolio Optimization
Haeun Jeon, Seunghoon Choi, Hyunglip Bae, Yongjae Lee, Woo Chang Kim
Sparse tangent portfolio optimization aims to learn an interpretable, low-cardin
ality portfolio in the tangency direction of the mean-variance frontier. However
, the associated cardinality-constrained formulation is NP-hard, and standard pr
edict-then-optimize pipelines often misalign forecasting accuracy with downstrea
m portfolio quality. We propose an end-to-end decision-focused learning framewor
k that reformulates Sharpe ratio maximization as a Disciplined Parametrized Prog
ramming (DPP)-compliant convex programming layer and replaces discrete selection
 with a smooth top-k operator enforcing an exact cardinality k. This enables
 gradient flow through prediction, asset selection, and re-optimization, allowin
g the predictive model to directly optimize portfolio performance. Across four m
ajor equity markets, our method achieves competitive and often superior out-of-s
ample Sharpe ratios compared with historical and prediction-focused baselines, w
ith particularly strong gains in larger asset universes. Our code is publicly av
ailable.
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Action Manifold Smoothing: A Lipschitz Pathway Perspective on High-Dimensional R
einforcement Learning
Zhihao Lin
High-dimensional continuous control remains challenging in deep reinforcement

learning, where algorithms like TD3 and SAC often collapse.
We propose a unifying \textbf{Lipschitz Pathway} framework that decomposes instability into four amplification stages,
namely action parameterization (L_1),
dynamics sensitivity (L_2),
Q-network curvature (L_3),
and temporal-difference (TD) target stability (L_4),
where errors compound multiplicatively along the learning pipeline. Our analysis identifies a \textit{discrete-continuous mismatch} as the root cause:
value functions trained from sparse point samples must generalize over continuous
manifolds, leading to multiplicative error amplification along the pathway. To address this, we introduce \textbf{Action Manifold Smoothing (AMS)}, which
replaces point-wise TD targets with orthogonally-sampled neighborhood averages, jointly regularizing L_3 (via implicit Laplacian smoothing) and L_4
(via local manifold supervision). We further characterize when Lipschitz-constrained
Q-networks and geometric action priors are beneficial based on task structure. Empirically, AMS enables both TD3 and SAC to achieve over 400 reward on the 38-D Dog Run task within 1M steps, where baselines fail.
These results validate the Lipschitz pathway as a principled framework for diagnosing and solving stability bottlenecks in high-dimensional control.
****

Unveiling the Structure of Do-Calculus Reasoning via Derivation Graphs
Clément Yvernes, Emilie Devijver, Marianne Clausel, Eric Gaussier
The do-calculus defines a general system of inference for interventional queries, allowing causal quantities to be transformed through successive applications of its rules. This process induces a rich space of equivalent interventional expressions, but combining and ordering these rules remains challenging.
In this work, we introduce derivation graphs, which represent how do-calculus rules are applied and combined, and characterize the full space of observational and interventional probabilities which are equivalent under the do-calculus. The structure of these graphs yields a simple procedure that uses at most four applications of do-calculus rules.
Finally, we show how applying identification algorithms to equivalent causal queries produces multiple valid estimands for the same causal quantity, eventually yielding more efficient estimators.
****

Edge-colored Clustering in Hypergraphs: A MaxECC Approximation
Aravind Srinivasan, Arushi Srinivasan, Jiayi Wu
We study the MaxECC problem, where given an edge-colored hypergraph with k colors and edge size r, we seek to color the vertices of the graph in order to maximize the number of satisfied edges (edges having the same color as their extremities): this is an effective mechanism for clustering (coloring) objects based on their multi-way interactions with one another in a system, providing significant applications in machine learning, clustering, and data mining. We exponentially improve upon the approximation ratio of an existing algorithm to $\frac{1}{r+1}$, present another novel dependent-rounding algorithm with an approximation ratio of $1/ \lceil \frac{k}{2}\rceil$, and modify the initial algorithm via analytical scaling techniques in order to achieve an approximation factor of $(1-e^{-r})/r$. We then apply our scaling algorithm to graph MaxECC and improve the best-known approximation factor for all hypergraphs: in particular, our algorithm provides an approximation factor of 0.43 as opposed to the previously-known 0.38 factor for graphs.
****

TimeGuard: Channel-wise Pool Training for Backdoor Defense in Time Series Forecasting
Quang Duc Nguyen, Siyuan Liang, Yiming Li, Fushuo Huo, Dacheng Tao
Time Series Forecasting (TSF) is highly vulnerable to backdoor attacks, yet effective defenses remain underexplored due to challenges arising from data entangle

ment and shifts in task formulation. To fill this gap, we conduct a systematic evaluation of thirteen representative backdoor defenses across the TSF life cycle and analyze their failure modes. Our results reveal two fundamental issues: (1) data entanglement induces *channel-level signal dilution*, rendering sample-filtering and trigger-synthesis defenses ineffective at localizing backdoors; and (2) task-formulation shift leads to *training-loss degeneration*, causing poisoned and clean windows to become indistinguishable at training stages. Based on these findings, we propose a training-time backdoor defense for TSF, termed TimeGuard. Our method adopts channel-wise pool training as the core paradigm and initializes a high-confidence pool using time-aware criteria to mitigate signal dilution. Moreover, we introduce distance-regularized loss selection to progressively expand the reliable pool during training and ease loss degeneration. Extensive experiments across multiple datasets, forecasting architectures, and TSF backdoor attacks demonstrate that TimeGuard substantially improves robustness, boosting $\mathrm{MAE}_ \mathrm{P}$ by $1.96\times$ over the leading baseline, while preserving clean performance within 5% $\mathrm{MAE}_ \mathrm{C}$.
****

Complexity of Decentralized Optimization with Mixed Affine Constraints
Demyan Yarmoshik, Nhat Trung Nguyen, Alexander Rogozin, Alexander Gasnikov
This paper considers decentralized optimization of convex functions with mixed affine equality constraints involving both local and global variables. Constraints on global variables may vary across different nodes in the network, while local variables are subject to coupled and node-specific constraints. Such problem formulations arise in machine learning applications, including federated learning and multi-task learning, as well as in resource allocation and distributed control. We analyze this problem under smooth and non-smooth assumptions, considering both strongly convex and general convex objective functions. Our main contribution is an optimal algorithm for the smooth, strongly convex regime, whose convergence rate matches established lower complexity bounds. We further provide optimal and near-optimal methods for the remaining cases.
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LiME: Lightweight Mixture of Experts for Efficient Multimodal Multi-task Learning
Md Kowsher, Haris Mansoor, Nusrat Jahan Prottasha, Ozlem Garibay, Victor Zhu, Zhengping Ji, Chen Chen
MoE-PEFT methods combine Mixture of Experts with parameter-efficient fine-tuning for multi-task adaptation, but require separate adapters per expert—causing trainable parameters to scale linearly with expert count and limiting applicability to adapter-based architectures. We propose LiME (Lightweight Mixture of Experts), which achieves expert specialization through lightweight modulation rather than adapter replication. Instead of separate adapters, LiME uses a single shared PEFT module and modulates its output with lightweight expert vectors, reducing expert parameters while generalizing to any PEFT method. Notably, LiME introduces zero-parameter routing by leveraging existing frozen and adapted representations—eliminating learned router parameters typically required per layer. Theoretically, we prove that (i) more experts preserve more task-relevant information and (ii) modulation approximates full expert-specific PEFT with bounded error. LiME further incorporates n-gram windowed routing and adaptive expert selection (Auto Top-K) based on routing confidence. Experiments on MMT-47, a multimodal multi-task benchmark with 47 tasks spanning text, image, and video, demonstrate that LiME achieves competitive or superior performance while using up to 4× fewer trainable parameters and up to 29% faster training compared to corresponding MoE-PEFT baselines.
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MaMa: A Game-Theoretic Approach for Designing Safe Agentic Systems
Jonathan Nöther, Adish Singla, Goran Radanovic
LLM-based multi-agent systems have demonstrated impressive capabilities, but they also introduce significant safety risks when individual agents fail or behave adversarially. In this work, we study the automated design of agentic systems that remain safe even when a subset of agents is compromised. Inspired by Stackelb

erg security games, we formalize this problem as a game between a system designe r (the Meta-Agent) and a best-responding Meta-Adversary that selects and comprom ises a subset of agents to minimize safety. We propose Meta-Adversary–Meta-Agent (MaMa), a novel algorithm inspired by this formalization for automatically desi gning safe agentic systems. Our approach uses LLM-based adversarial search, wher e the Meta-Agent iteratively proposes system designs and receives feedback based on the strongest attacks discovered by the Meta-Adversary. Empirical evaluation s across diverse environments show that systems designed with MaMa consistently defend against worst-case attacks while maintaining performance comparable to sy stems optimized solely for task success. Moreover, the resulting systems general ize to stronger adversaries, as well as ones with different attack objectives or underlying LLMs, demonstrating robust safety beyond the training setting. Code is available at https://github.com/JNoether/MaMa
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Learning to Move Before Learning to Do: Task-Agnostic pretraining for VLAs
Junhao Shi, Siyin Wang, Xiaopeng Yu, Li Ji, Jingjing Gong, Xipeng Qiu
Vision-Language-Action (VLA) models are bottlenecked by the scarcity of expert d emonstrations—expensive triplets of observations, language instructions, and act ions. We propose that learning ''how to move'' can be decoupled from learning '' what to do,'' and that the former requires no task labels at all. Our two-stage framework, **Task-Agnostic Pretraining (TAP)** first pre-trains on abundant, che ap *task-agnostic* data (discarded off-task trajectories or autonomous robot pla y) using an Inverse Dynamics objective that predicts actions from consecutive ob servations. This self-supervised phase instills physical affordances—grasping, c ontact dynamics, end-effector control—without human annotation. A lightweight se cond stage then aligns these physical priors with language instructions using mi nimal expert data. On the SIMPLER benchmark, our approach matches models trained on 1M+ expert trajectories while using orders of magnitude less labeled data, a chieving a 10\% absolute gain over standard behavior cloning. In real-world Wido wX experiments, it surpasses internet-scale baselines under visual distribution shifts (e.g., 25\% vs. 0\% under camera perturbations), demonstrating that task-agnostic pretraining yields robust, transferable physical representations for Em bodied AI.
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JADE: Bridging the Strategic-Operational Gap in Dynamic Agentic RAG
Yiqun Chen, Erhan Zhang, Tianyi Hu, Shijie Wang, Zixuan Yang, Meizhi Zhong, Xiao chi Wei, Yan Gao, Yi Wu, Yao Hu, Jiaxin Mao
The evolution of Retrieval-Augmented Generation (RAG) has shifted from static re trieval pipelines to dynamic, agentic workflows where a central planner orchestr ates multi-turn reasoning. However, existing paradigms face a critical dichotomy : they either jointly optimize modules within rigid, fixed-graph architectures, or enable dynamic planning while treating executors as frozen, black-box tools. We identify that this \textit{decoupled optimization} creates a ``strategic-oper ational mismatch,'' where sophisticated planning strategies fail to materialize due to unadapted local executors, often causing negative gains despite increased system complexity. In this paper, we propose \textbf{JADE} (\textbf{J}oint \tex tbf{A}gentic \textbf{D}ynamic \textbf{E}xecution), a unified framework for joint optimization of planning and execution within dynamic, multi-turn workflows. By modeling the system as a cooperative multi-agent team with a shared backbone, J ADE enables end-to-end learning driven by outcome-based rewards. This approach f acilitates \textit{co-adaptation}: the planner learns to operate within executor capability boundaries, while executors evolve to align with strategic intent. Empirical results demonstrate that JADE transforms disjoint modules into a syner gistic system, yielding strong performance improvements via joint optimization a nd enabling a flexible balance between efficiency and effectiveness through dyna mic workflow orchestration.
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Salus: Strategic Diagnostic Testing for Complex Diagnosis via Multi-Agent Reinfo rcement Learning
Shuohao Gao, Xuanzhong Chen, Lingxiao Luo, Zilin Ding, Rong Han, Rui Jiang, Ting

Chen
Diagnosing complex diseases is inherently a sequential and iterative medical investigation process, in which a clinician strategically requests multiple rounds of diagnostic tests to differentiate among similar diseases until reaching a definitive diagnosis.
Although large language models show great potential as clinical assistants, they often struggle to navigate this complex interactive process, suffering from premature diagnostic closure. Furthermore, optimizing LLMs for such multi-round environments is frequently hindered by the challenge of reward sparsity and hacking.
In this paper, we introduce $\textbf{CompDiag-Bench}$, a benchmark that formalizes diagnosis as a sequential decision-making process where a clinician must strategically request diagnostic tests from a dynamic environment in order to reach a definitive diagnosis.
To address this task, we propose $\texttt{Salus}$, a multi-agent framework that decouples diagnostic reasoning into three specialized functional roles: a Differential Reasoner, a Strategic Controller, and a Workup Proposer.
$\texttt{Salus}$ is optimized via multi-agent reinforcement learning employing structured rewards to calibrate strategic diagnostic behavior.
Specifically, we leverage an LLM-as-a-Judge reward mechanism to provide dense, semantically-grounded feedback, designed to penalize premature closure and incentivize accurate differential diagnoses.
Experimental results show that our model, $\texttt{Salus-7B}$, attains state-of-the-art Top-1 accuracy of 83.64% on complex cases, outperforming DeepSeek-V3.2 (71.38%) and achieving performance on par with GPT-5.2 (80.30%).
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Accelerated Dual Method for Distributed Optimization: An Inexact-Gradient View of Local Updates
Junchi YANG, Ziyang Zeng, Pan Linxuan, Murat Yildirim, Feng Qiu
In distributed machine learning, efficiently training across multiple agents with heterogeneous data distributions remains a central challenge. We address the problem of stochastic, strongly convex distributed optimization by applying accelerated gradient ascent to the dual variables and multi-step stochastic gradient descent (SGD) to the primal variables in the Lagrangian formulation. This approach naturally enables local computation, as the inner SGD loops require no inter-agent communication. We prove that the method converges for any number of local updates, attaining the optimal communication complexity when local computation is sufficient. Our analysis builds on an inexact accelerated gradient framework, where the partial gradient of the Lagrangian with respect to the dual variables is treated as an inexact gradient of the dual function. A notable byproduct of this framework is an algorithm that achieves optimal reproducibility guarantees under biased gradient estimates.
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Amortized Variational Inference for Partial-Label Learning: A Probabilistic Approach to Label Disambiguation
Tobias Fuchs, Nadja Klein
Real-world data is frequently noisy and ambiguous. In crowdsourcing, for example, human annotators may assign conflicting class labels to the same instances. Partial-label learning (PLL) addresses this challenge by training classifiers when each instance is associated with a set of candidate labels, only one of which is correct. While early PLL methods approximate the true label posterior, they are often computationally intensive. Recent deep learning approaches improve scalability but rely on surrogate losses and heuristic label refinement. We introduce a novel probabilistic framework that directly approximates the posterior distribution over true labels using amortized variational inference. Our method employs neural networks to predict variational parameters from input data, enabling efficient inference. This approach combines the expressiveness of deep learning with the rigor of probabilistic modeling, while remaining architecture-agnostic. Theoretical analysis and extensive experiments on synthetic and real-world datasets demonstrate that our method achieves state-of-the-art performance in both acc

uracy and efficiency.
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You Don't Protect if You Don't Expect: Breaking the Key Assumption behind CLIP's Test-Time Defenses
Ruize Zhang, Yu Li, Zhang Wan, Juan Cao, Jie Zhang, Sheng Tang
Recent test-time defenses for CLIP claim to preserve zero-shot clean accuracy while improving adversarial robustness. However, we find the reported robustness of six recent proposed state-of-the-art methods substantially overestimated: they fail under basic adaptive attacks. We further observe that these defenses share a common reliance on an indicative measurement that is assumed to capture the distributional difference between clean and adversarial samples and to determine whether the defense should preserve or alter the static model's prediction. We argue that this assumption is the fundamental weakness, and we propose CLIP-MAD (Manipulating Assumed Difference), an adaptive attack strategy designed to break it. CLIP-MAD efficiently expands the adversarial distribution without costly full gradient calculations and can be flexibly combined with existing attack baselines to further boost attack strength. Experiments across 13 datasets demonstrate that CLIP-MAD produces strong adversarial samples that markedly reduce the robustness of diverse test-time defenses, revealing a false sense of security in CLIP’s zero-shot robustness. Code will be available at https://github.com/rzzhang222/CLIP-MAD.
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TadA-Bench: A Million-Variant Benchmark for Future-Round Discovery Toward Agentic Protein Engineering
Jin Gao, Juntu Zhao, Zirui Zeng, Jiaqi Shen, Junhao Shi, Dukun Zhao, Yuming Lu, Dequan Wang
AI for scientific discovery is entering an agentic era, where protein-engineering systems are expected to prioritize future wet-lab experiments rather than merely fit static measurements.
We introduce TadA-Bench, a million-variant wet-lab replay benchmark from 31 TadA directed-evolution rounds for future-round discovery toward agentic protein engineering.
TadA-Bench preserves the campaign chronology and defines a fixed-data replay task: given earlier experimental rounds, models rank variants that appear only in later rounds.
It provides aligned DNA, RNA, and protein views, and uses Seq2Graph, a graph-based label-unification pipeline, to reconcile noisy enrichment measurements into consistent cross-round activity labels.
Random-split controls show strong interpolation, but future-round ranking and finite-budget candidate selection are much weaker.
Controlled analyses suggest that evolutionary coverage is more informative than local data density, positioning TadA-Bench as a reproducible wet-lab replay substrate for future-round discovery toward agentic protein engineering; the data and code are released on Hugging Face and GitHub.
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Speech-Audio Compositional Attacks on Multimodal LLMs and Their Defense with SALMONN-Guard
Yudong Yang, Xuezhen Zhang, Zhifeng Han, Siyin Wang, Jimin Zhuang, Zengrui Jin, Jing Shao, Guangzhi Sun, Chao Zhang
Recent progress in large language models (LLMs) has enabled understanding of both speech and non-speech audio, but has also exposed new safety risks arising from complex audio inputs that are inadequately handled by current safeguards. We introduce SACRED-Bench (Speech–Audio Composition for RED-teaming) to evaluate the robustness of LLMs under complex audio-based attacks. Unlike existing perturbation-based methods that rely on noise optimization or white-box access, SACRED-Bench exploits speech–audio composition to enable effective black-box attacks. SACRED-Bench adopts three composition mechanisms: (a) speech overlap, (b) multi-speaker dialogue, and (c) mixtures of speech and non-speech audio. These mechanisms focus on evaluating safety in settings where benign and harmful intents co-occur within a single auditory scene. Moreover, questions in SACRED-Bench are design

ed to implicitly refer to content in the audio, such that no explicit harmful in
formation appears in the text prompt alone.
Experiments demonstrate that even Gemini 2.5 Pro, a state-of-the-art proprietary
 LLM with safety guardrails fully enabled, still exhibits a 66% attack success r
ate. To bridge this gap, we propose SALMONN-Guard, the first guard model that jo
intly inspects speech, audio, and text for safety judgments, reducing the attack
 success rate to 20\%. Our results highlight the need for audio-aware defenses t
o ensure the safety of multimodal LLMs.
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The Geometry of Sequential Learning: Lie-Bracket Prediction of Transfer Order
John Sweeney
Sequential learning is order-dependent: from Pile-style next-token domain adapta
tion to instruction-SFT and DPO, N candidate sources induce $N!$ possible curr
icula. We show that the local order effect is governed by a computable geometric
 quantity, the Lie-bracket commutator of gradient update fields, yielding a pair
wise score for whether $A \to B$ or $B \to A$ is better for a target domain. The
 pairwise bracket primitive also defines a *Lie-Bracket Tournament*: with a shar
ed θ_0 target-gradient reference, Hessian symmetry gives Borda/row-sum sc
ores from one Hessian-vector product per source, $O(N)$ dot products, and an $O(
N\log N)$ sort, without materializing the $O(N^2)$ edge matrix. Empirically, the
 planner reaches 98.1%/98.9% pairwise accuracy at $k=1$ for instruction-SFT/DPO,
 remains at 73.1%/72.2% at $k=20$, and preserves the original pretraining-domain
 evidence with 82.4–92.0% accuracy across four LLMs and 91.1% on diffusion. At c
urriculum scale, it recovers the best of all $3!$ schedules in 87.5% of trials,
ranks 85 Stack programming-language source domains for a Python target in the 99
th sampled percentile, and reaches the 99.0–99.6th sampled percentile on 56 MMLU
 subjects, sharply above the reported descending gradient-norm baseline. These r
esults reframe sequential learning as a geometric tournament problem: commutator
s provide both local pairwise order information and a scalable primitive for man
y-domain schedules.
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DiffCrossGait: Trajectory-Level Alignment for 2D-3D Cross-Modal Gait Recognition
 via Latent Diffusion
Zhiyang Lu, Ming Cheng
Cross-modal 2D–3D gait recognition is impeded by inherent domain discrepancies b
etween 2D silhouette and 3D point cloud distributions. While prior methods align
 only final embeddings, we propose DiffCrossGait, which enforces trajectory-leve
l alignment by driving both modalities with shared noise in a unified latent dif
fusion process. By driving both modalities with shared Gaussian noise within a l
atent space, we enable continuous alignment throughout the generative evolution.
 We introduce a Tri-Phase Alignment Strategy that exploits varying noise intensi
ties to enforce identity anchoring, dynamics consistency, and cross-modal struct
ural recoverability, thereby constraining both modalities to share denoising dyn
amics and bottleneck structure, which promotes modality-invariant gait features.
 Crucially, our framework decouples generative alignment from the discriminative
 backbone; the diffusion mechanism serves exclusively as a training objective, e
nsuring high inference efficiency by eliminating the computational overhead of i
terative denoising. Extensive experiments on the SUSTech1K and FreeGait benchmar
ks demonstrate that DiffCrossGait achieves state-of-the-art performance.
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Let the Prototype Guide You: Robust Aggregation of Sparse Multi-Class Annotation
s via Annotator Prototype Learning
Ju Chen, Jun Feng, Shenyu Zhang
Truth inference is a critical technique for aggregating noisy and biased multi-c
lass classification annotations. State-of-the-art approaches model each annotato
r using an individual confusion matrix. While well-grounded, they suffer from tw
o fundamental bottlenecks: 1) confusion matrices are underfit when annotators la
bel only a small subset of tasks or when classes are imbalanced, and 2) a single
 confusion matrix per annotator is inadequate for capturing complex annotator be
haviors, leading to class-level collapse when tasks are extremely difficult. Sim

ultaneously addressing these challenges is non-trivial, as it demands both robustness to data sparsity and sufficient expressiveness for complex annotator patterns. In this paper, we propose \textbf{CPBCC} (\textbf{C}lass-specific \textbf{P}rototype-driven \textbf{B}ayesian \textbf{C}lassifier \textbf{C}ombination), which creatively models annotators through a dual-pathway architecture: (i) learning class-specific prototype annotation patterns across all annotators, and (ii) learning annotator-specific weights over prototypes. This framework addresses the bottlenecks and achieves a robust yet rich annotator characterization. Experiments across 10 real-world datasets spanning five domains demonstrate that CPBCC yields a 26\% accuracy improvement in the best case, and boosts average accuracy from 68.73\% to 74.11\%. Our source code is available at \url{https://github.com/JuJuCHEN-HHU/CPBCC_PTBCC}.
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X-MoGe: A Cross-Modal Adaptation Framework with Mixture-of-Experts and Geometry Guidance for Heterogeneous Collaborative Perception
Wenkai Lin, Zhihong Liu, Chenglu Wen
Multi-agent collaborative perception improves perception range and robustness in autonomous driving. However, most existing methods assume homogeneous sensors and perception networks, which is unrealistic in real-world heterogeneous systems. Differences in sensing modalities and independently trained models lead to significant semantic and geometric inconsistencies, limiting effective collaboration. To solve these problems, we propose a novel cross-modal adaptation framework with Mixture-of-Experts and geometry-guided fusion for heterogeneous collaborative perception, named X-MoGe. Specifically, we propose a Pixel-level Mixture-of-Experts (P-MoE) module, which adaptively models modality-specific semantic characteristics under heterogeneous sensing conditions. In addition, a geometry-guided feature fusion module incorporates explicit geometric priors to enforce spatial alignment and consistency in the BEV space. Extensive experiments on OPV2V and DAIR-V2X datasets demonstrate that the proposed method achieves state-of-the-art performance in heterogeneous collaborative perception.
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Data-Source Adaptive Online Learning under Heteroscedastic Noise
Amith Bhat Hosadurga Anand, Haipeng Luo, Aadirupa Saha
In this paper, we address the standard K-armed multi-armed bandit (MAB) with heterogeneous data sources, each exhibiting unknown and distinct noise variances, $\lbrace \sigma_j^2 \rbrace_{j=1}^{M}$. The learner performs standard regret minimization, with the added challenge of choosing which data source to query at each round. We propose SOAR (Source-Optimistic Adaptive Regret minimization), a novel algorithm that adaptively balances exploration and exploitation by jointly constructing upper confidence bounds for arm rewards and lower confidence bounds for data source variances. Our theoretical analysis establishes that SOAR achieves a regret bound of $\tilde{O}\left({\sigma^\star}^2 \sum_{i=2}^K \tfrac{1}{\Delta_i}\right),$ along with a preprocessing cost that depends only on the problem parameters $\lbrace \sigma_j \rbrace_{j=1}^{M}$, K, and grows at most logarithmically with the horizon T; where ${\sigma^\star}^2$ is the minimum source variance, and Δ_i denotes the suboptimality-gap of the i-th arm reward. The $\tilde{O}(\cdot)$ notation hides the polylogarithmic factors in these problem parameters. Notably, despite not knowing the minimum-variance source, SOAR matches the instance-dependent regret of a standard MAB run on a single source of variance $\sigma^\star$. This near-optimal instance-dependent regret analysis of SOAR underscores its effectiveness in dynamically managing heteroscedastic noise without incurring significant overhead. Experiments on synthetic problem instances as well as a real dataset (MovieLens 32M) demonstrate that our method significantly outperforms baseline bandit algorithms in terms of regret performance. Our work opens a new direction for adaptively leveraging multiple heterogeneous data sources, extending beyond traditional bandit frameworks.
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KUMA: A Novel Framework with Koopman Separation and Efficient Multilevel Extraction in Time Series Forecasting
Sijie Xiong, Cheng Tang, Atsushi Shimada

Time series forecasting plays a crucial role in a wide range of real-world applications and has become increasingly complex with the growth of multivariate dimensions and extended historical observations, leading to the prosperity of deep forecasting models. Previous models are hindered by three major challenges: high computational complexity, inefficient token utilization caused by redundancy and scarcity, and temporal distribution shifts resulting from non-stationary dynamics. Inspired by Koopman theory and the success of multilevel encoder–decoder architectures with skip connections, we design an input-dependent Koopman module to decompose time series into Koopman dynamics and residual dynamics. Building upon this formulation, we propose a U-shaped Multilevel Attention module (UMA) that integrates element-wise attention filtering and linear attention, giving rise to KUMA. The input-dependent Koopman operator mitigates the issue of operator mixture and alleviates temporal distribution shifts, while UMA achieves a favorable balance between token redundancy and token scarcity with acceptable computational efficiency. Comprehensive evaluations across 12 benchmark datasets demonstrate that KUMA achieves superior performance compared to existing excellent approaches.
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Local Linearity of LLMs Enables Activation Steering via Model-Based Linear Optimal Control
Julian Skifstad, Xinyue Annie Yang, Glen Chou
Inference-time LLM alignment methods, particularly activation steering, offer an alternative to fine-tuning by directly modifying activations during generation. Existing methods, however, often rely on non-anticipative interventions that ignore how perturbations propagate through transformer layers and lack online error feedback, resulting in suboptimal, open-loop control. To address this, we show empirically that layer-wise dynamics across multiple LLM architectures and scales are well-approximated by locally-linear models, despite the nonlinear structure of transformer blocks. Exploiting this property, we model LLM inference as a linear time-varying dynamical system and adapt the classical linear quadratic regulator to compute feedback controllers using layer-wise Jacobians, steering activations toward desired semantic setpoints in closed-loop with minimal computational overhead and no offline training. We also derive theoretical bounds on setpoint tracking error, enabling formal guarantees on steering performance. Using a novel adaptive semantic feature setpoint signal, our method yields robust, fine-grained behavior control across models, scales, and tasks, including state-of-the-art modulation of toxicity, truthfulness, refusal, and arbitrary concepts, surpassing baseline steering methods.
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How2Everything: Mining the Web for How-to Procedures to Evaluate and Improve LLMs
Yapei Chang, Kyle Lo, Mohit Iyyer, Luca Soldaini
Generating step-by-step "how-to" procedures is a key LLM capability: how-to advice is commonly requested in chatbots, and step-by-step planning is critical for complex reasoning tasks.
Yet, measuring and improving procedural validity at scale on real-world tasks remains challenging and understudied.
We introduce How2Everything, a scalable framework to evaluate and improve goal-conditioned procedure generation.
Our pipeline How2Mine extracts and rewrites 351K procedures from 980K web pages across 14 topics, and can scale to larger corpora.
From this pool we build How2Bench, a 7K-example evaluation set balanced across topics.
We also introduce How2Score, an evaluation protocol that uses an LLM judge to detect whether a generation contains any critical failure that would prevent achieving the goal.
For low-cost, reproducible evaluation, we distill a frontier judge into an open 8B model achieving 80.5\% agreement with human annotators.
How2Bench reveals clear scaling trends across model size and training stages, providing signal early in pretraining.

Finally, RL using How2Score as a reward improves performance on How2Bench by >10 points across three base models without systematic regressions on standard benchmarks, with gains not primarily explained by source-document memorization or superficial format compliance.
We release all code and data at https://github.com/lilakk/how2everything.
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A Noise Sensitivity Exponent Controls Large Statistical-to-Computational Gaps in Single- and Multi-Index Models
Leonardo Defilippis, Florent Krzakala, Bruno Loureiro, Antoine Maillard
Understanding when learning is statistically possible yet computationally hard is a central challenge in high-dimensional statistics.
In this work, we investigate this question in the context of single- and multi-index models, classes of functions widely studied as benchmarks to probe the ability of machine learning methods to discover features in high-dimensional data.
Our main contribution is to show that a Noise Sensitivity Exponent (NSE)—a simple quantity determined by the activation function—governs the existence and magnitude of statistical-to-computational gaps within a broad regime of these models.
We first establish that, in single-index models with large additive noise, the onset of a computational bottleneck is fully characterized by the NSE. We then demonstrate that the same exponent controls a statistical-computational gap in the specialization transition of large separable multi-index models, where individual components become learnable. Taken together, our results identify the NSE as a unifying property linking noise robustness, computational hardness, and feature specialization in high-dimensional learning.
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Compositional Generative Modeling from Decentralized Data
Mashrur M. Morshed, Vishnu Boddeti
Learning the compositional nature of the physical world requires joint observation of interacting factors.
However, because practical data is often decentralized, these factors are fragmented across isolated silos.
Existing decentralized generative approaches focus only on modeling the union of siloed data, overlooking novel combinations implied by the collective whole.
To bridge this gap, we introduce Decentralized Compositional Flow Matching (DCFM), a framework that enforces structural constraints across the global set of generative factors, without exchanging any raw data. DCFM enables novel combinations to emerge through peer interactions, even when no single data source can independently support the composition. Empirically, DCFM substantially outperforms federated learning and mixture-of-experts baselines across conditional image generation, robotic spatial planning, and medical attribute co-occurrence modeling.
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Delayed Momentum Aggregation: Communication-efficient Byzantine-robust Federated Learning with Partial Participation
Kaoru Otsuka, Yuki Takezawa, Makoto Yamada
Partial participation is essential for communication-efficient federated learning at scale, yet existing Byzantine-robust methods typically assume full client participation.
In the partial participation setting, a majority of the sampled clients may be Byzantine, once Byzantine clients dominate, existing methods break down immediately.
We introduce delayed momentum aggregation, a principle where the central server aggregates cached momentum from non-sampled clients along with fresh momentum from sampled clients.
This principle ensures Byzantine clients remain a minority from the server's perspective even when they dominate the sampled set.
We instantiate this principle in our optimizer DeMoA.
We analyze the convergence rate of DeMoA, showing that DeMoA is Byzantine-robust under partial participation.
Experiments show that, with 20% Byzantine ratio and only 10% partial participation rate, DeMoA achieves the best accuracy even when existing methods fail empiri

cally.
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When More Data Doesn't Help: Limits of Adaptation in Multitask Learning
Steve Hanneke, Mingyue Xu
Multitask learning and related frameworks have achieved tremendous success in modern applications. In multitask learning problem, we are given a set of heterogeneous datasets collected from related source tasks and hope to enhance the performance above what we could hope to achieve by solving each of them individually. The recent work of Hanneke & Kpotufe (2022) has showed that, without access to distributional information, no algorithm based on aggregating samples alone can guarantee optimal risk as long as the sample size per task is bounded.

In this paper, we focus on understanding the statistical limits of multitask learning. We go beyond the no-free-lunch theorem in Hanneke & Kpotufe (2022) by establishing a stronger impossibility result of adaptation that holds for arbitrarily large sample size per task. This improvement conveys an important message that the hardness of multitask learning cannot be overcame by having abundant data per task. We also discuss the notion of optimal adaptivity that may be of future interests.
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Multimodal Fusion via Self-Consistent Task-Gradient Fields
Jiayu Xiong, Jing Wang, Jun Xue, Wanlong Wang, Jianlong Kwan, Xiaosen Lyu, Zhouqiang Jiang
Multimodal learning aims to preserve as much task-related information as possible from different inputs. However, current fusion designs often distort the feedback loop to feature extractors. Aggressively merging modalities entangles their representations, making the feature extractors fragile to incomplete inputs. Meanwhile, attempting to separate features via auxiliary losses frequently introduces optimization conflicts that distract from the primary task. We propose the Self-Consistent Field Autoencoder (SCFAE) to provide a better path for task gradients. Our method follows the self-consistent field principle to balance task learning with feature organization, thereby minimizing mutual information. We use small autoencoders for each modality to keep information intact. The task loss acts as a driving force to select predictive features. The reconstruction loss acts as a constraint to separate these features into independent subspaces. These dual objectives operate through complementary feature subspaces, thereby mitigating optimization interference. We evaluate SCFAE on audio-visual-text, audio-visual, and image-video benchmarks. Results show that SCFAE handles missing data and unequal input sizes more robustly via a simple structure. Gradient analysis confirms that SCFAE avoids conflicts and maintains stable training dynamics.
****
Identifying Partially Observed Causal Models from Heterogeneous/Nonstationary Data
Xinshuai Dong, Haoyue Dai, Ignavier Ng, Peter Spirtes, Kun Zhang
Estimating causal structure in the presence of latent variables is an important yet challenging problem. Recent works have shown that distributional constraints, such as rank deficiency constraints of the covariance matrices, can be exploited to recover the underlying causal structure involving latent variables. However, real-world data often exhibit heterogeneity/nonstationarity, which pose challenges to existing methods. In this work, we develop a principled approach for identifying the structure of partially observed linear causal models from heterogenous/nonstationary data. We first formulate a class of heterogenous/nonstationary, partially observed linear causal models and prove that their distributional constraints are equivalent to those in the homogeneous case. Building on this, we propose a novel rank deficiency test that can efficiently handle heterogenous/nonstationary data, and further establish identifiability results for recovering the causal structure involving latent variables. We also provide a method to identify which variables exhibit distribution shifts, i.e., whose causal mechanisms vary across domains. Experiments on simulated and real-world data validate our theoretical findings and the effectiveness of our method.

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Decouple and Cache: KV Cache Construction for Streaming Video Understanding
Zhanzhong Pang, Dibyadip Chatterjee, Fadime Sener, Angela Yao
Streaming video understanding requires processing unbounded video streams with l imited memory and computation, posing two key challenges. First, continuously co nstructing new and evicting old key-value(KV) caches is required for unbounded s treams. Secondly, due to the high cost of collecting and training on unbounded s treams, models must learn from short sequences while generalizing to long stream s. Existing streaming VideoVLLMs fail to scale to unbounded video streams or foc us on cache reuse strategies, leaving the impact of cache construction underexpl ored. In this paper, we propose Decoupled Streaming Cache(DSCache), a training-f ree cache construction mechanism that adapts pretrained offline models to stream ing settings. DSCache maintains a cumulative past KV cache while constructing a separate instant cache on-demand, decoupled from past caches to preserve the inf ormativeness of recent inputs. To enable position extrapolation beyond the train ing length, DSCache further incorporates a position-agnostic encoding strategy, ensuring KV caches to support unseen positions and preventing position overflow. Experiments on Streaming Video QA benchmarks demonstrate DSCache's state-of-the -art performance, with an average 2.5% accuracy gains over prior methods.
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Fine-Tuning of Transformer models with Frames
Harshavardhan Adepu, Li Zhang, Sanjiv Kumar, Vikas Singh
Parameter-Efficient Fine-Tuning (PEFT) strategies such as Low-Rank Adaptation (L oRA) are effective solutions for fine-tuning large-scale pre-trained models; how ever,
their memory requirements scales with the size of the model, $\mathcal{O}(dr)$, where d is the model's hidden dimension and r is the rank.
Our proposal, FrameFT,
models the parameter update ΔW with a sparse coefficient matrix in a Fus ion Frame basis.
Fusion Frames can be generated algorithmically and shared across model layers, e nabling highly efficient updates.
Only the sparse coefficients of the basis expansion are stored/optimized, strong ly reducing the memory footprint and parameter count.
The sparse structure of the coefficient matrix in FrameFT and the sparsity in th e Fusion Frames, give sizable compute benefits.
Our technical analysis shows that FrameFT allows obtaining formal convergence re sults.
We evaluate our method across a suite of supervised fine-tuning benchmarks, prim arily focusing on language tasks, but also report applicability to vision models . Our empirical evaluations show that FrameFT achieves performance on par with o r exceeding state-of-the-art PEFT techniques, but needs far fewer trainable para meters and less memory.
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Multimodal Function Vectors for Visual Relations
Shuhao Fu, Esther A Goldberg, Ying Nian Wu, Hongjing Lu
Large Multimodal Models (LMMs) demonstrate impressive in-context learning abilit ies from few multimodal demonstrations, yet the internal mechanisms supporting s uch task learning remain opaque. Building on prior work of Large Language Models , we show that a small subset of attention heads in Large Multimodal Models is r esponsible for transmitting representations of visual relations. The activations of these attention heads, termed $\textit{function vectors}$, can be extracted and manipulated to alter an LMM’s performance on relational tasks. First, using synthetic and real image datasets, we apply causal mediation analysis to identif y attention heads that strongly influence relational predictions, and extract mu ltimodal function vectors that improve zero-shot accuracy at inference time. We further demonstrate that these multimodal function vectors can be fine-tuned wit h a modest amount of training data, while keeping LMM parameters frozen, to sign ificantly outperform in-context learning baselines. Finally, we show that relati on-specific function vectors can be linearly combined to solve analogy problems

involving novel and untrained visual relations, highlighting the strong generalization ability of this approach. Through experiments on two LMMs, including Open Flamingo and Qwen3-VL, our results show that these models encode visual relational knowledge within localized internal structures, which can be systematically extracted and optimized, thereby advancing our understanding of model modularity and enhancing control over relational reasoning in LMMs.
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CausalXRL: Explainable Reinforcement Learning through Causal Graph Reasoning
Yanming Zhang, Eric Papenhausen, Klaus Mueller
Reinforcement learning is a powerful paradigm for training autonomous agents and has achieved impressive performance in complex environments. However, this success often comes at the cost of interpretability, diminishing trust and complicating efforts to debug and improve agent behavior. To address these challenges, we introduce CausalXRL, a novel framework for explainable reinforcement learning (XRL). A key feature of CausalXRL is its use of causal graph reasoning, which provides transparent, structured, multi-level explanations of agent decision-making. We validate CausalXRL through comprehensive case studies and a two-part evaluation: (1) a quantitative analysis of explanation fidelity and causal-structure learning efficiency in benchmark RL environments, and (2) a qualitative expert study assessing explainability in the real-time strategy (RTS) benchmark MicroRTS. The quantitative results show that CausalXRL can provide faithful explanations while efficiently learning causal structures, and the qualitative expert study suggests that participants found CausalXRL useful for inspecting high-level RTS strategies.
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Constrained Flow Optimization via Sequential Fine-Tuning for Molecular Design
Sven Gutjahr, Riccardo De Santi, Luca Schaufelberger, Kjell Jorner, Andreas Krause
Adapting generative foundation models, in particular diffusion and flow models, to optimize given reward functions (e.g., binding affinity) while satisfying constraints (e.g., molecular synthesizability) is fundamental for their adoption in real-world scientific discovery applications such as molecular design or protein engineering. While recent works have introduced scalable methods for reward-guided fine-tuning of such models via reinforcement learning and control schemes, it remains an open problem how to algorithmically trade-off reward maximization and constraint satisfaction in a reliable and predictable manner. Motivated by this challenge, we first present a rigorous framework for *Constrained Generative Optimization*, which brings an optimization viewpoint to the introduced adaptation problem and retrieves the relevant task of constrained generation as a sub-case. Then, we introduce Constrained Flow Optimization (CFO), an algorithm that automatically and provably balances reward maximization and constraint satisfaction by reducing the original problem to sequential fine-tuning via established, scalable methods. We provide convergence guarantees for constrained generative optimization and constrained generation via CFO. Ultimately, we present an experimental evaluation of CFO on both synthetic, yet illustrative, settings, and a molecular design task. Across these evaluations, CFO achieves consistent increases in reward while ensuring high constraint satisfaction, showcasing its practical utility for constrained generative optimization.
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TreeCUA: Efficiently Scaling GUI Automation with Tree-Structured Verifiable Evolution
Deyang Jiang, Jing Huang, Xuanle Zhao, Lei Chen, Liming Zheng, Fanfan Liu, Haibo Qiu, Peng Shi, Zhixiong Zeng
Effectively scaling GUI automation is essential for computer-use agents (CUAs); however, existing work primarily focuses on scaling GUI grounding rather than the more crucial GUI planning, which requires more sophisticated data collection. In reality, the exploration process of a CUA across apps/desktops/web pages typically follows a tree structure, with earlier functional entry points often being explored more frequently.
In this work, we find that organizing large-scale GUI trajectories into tree str

uctures can effectively eliminate redundant exploration costs, while each branch node also provides key reasoning evidence for distinguishing adjacent trajectories.
Therefore, we propose TreeCUA to efficiently scale GUI automation with tree-structured verifiable evolution.
We propose a multi-agent collaborative framework to explore the environment, verify actions, summarize trajectories, and evaluate quality to generate high-quality and scalable GUI trajectories.
To improve efficiency, we devise a novel tree-based topology to store and replay duplicate exploration nodes, and design an adaptive exploration algorithm to balance the depth (i.e., trajectory difficulty) and breadth (i.e., trajectory diversity). Moreover, we develop world knowledge guidance and global memory backtracking to avoid low-quality generation.
Finally, we naturally extend and propose the TreeCUA-DPO method from abundant tree node information, improving GUI planning capability by referring to the branch information of adjacent trajectories.
Experimental results show that TreeCUA and TreeCUA-DPO offer significant improvements, and out-of-domain (OOD) studies further demonstrate strong generalization.
All trajectory node information and code will be open-sourced.
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Is Graph Mixup Beneficial? Investigating Interpolation And Empirical Performance of Graph Mixup Methods
Simon Forbat, Rainer Gemulla
Mixup is a widely used data augmentation technique that constructs new training examples by interpolating between existing ones. While simple and effective in domains like vision and language, applying mixup to graph data is non-trivial and there is no independent empirical evidence for its effectiveness. To fill this gap, we conducted an extensive evaluation study following a unified, established evaluation protocol for graph classification. In contrast to prior results, we found that none of the state-of-the-art mixup methods yielded statistically significant improvements over the no-mixup baseline. To obtain further insights, we analyzed the graphs generated from these mixup methods from an interpolation perspective. We found that (i) many mixup methods failed to interpolate well, (ii) high interpolation error led to performance degradation, and (iii) even good interpolation properties did not lead to performance improvements. Our findings question the efficacy of existing graph mixup methods and highlight the need for a more rigorous exploration and evaluation.
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Spectral-Informed Neural Networks Outperform Spectral methods in High-dimensional PDEs
Tianchi Yu, Ivan Oseledets
For low-dimensional problems ($d\leq3$), spectral methods can achieve exceptionally high accuracy. For middle-dimensional problems ($4 \leq d \lesssim 10$), spectral methods remain feasible through specific techniques such as sparse grids or hyperbolic cross. However, for high-dimensional problems ($d\gg 10$), spectral methods suffer frome the curse of dimensionality. Physics-informed neural networks (PINNs) have emerged as a promising approach to overcome this challenge, offering scalability to high dimensions, but often suffer from limited accuracy and efficiency. Recently proposed spectral-informed neural networks (SINNs) combine spectral methods with PINNs, operating directly in the spectral domain to avoid spatial derivative computations and to reduce memory consumption. In this work, we introduce Modified SINNs, which integrate coefficient decay scaling and basis embeddings motivated by harmonic analysis to enhance accuracy in high-dimensional problems and enable accurate approximation of unknown spectral coefficients. Numerical experiments on steady and time-dependent partial differential equations demonstrate that Modified SINNs outperform sparse grid spectral methods on middle-dimensional problems with incomplete spectral information and achieve superior accuracy compared to PINNs on high-dimensional problems.

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Dimensional Collapse in Transformer Attention Outputs: A Challenge for Sparse Dictionary Learning
Junxuan Wang, Xuyang Ge, Wentao Shu, Zhengfu He, Xipeng Qiu
Transformer architectures, and their attention mechanisms in particular, form the foundation of modern large language models. While transformer models are widely believed to operate in high-dimensional hidden spaces, we show that attention outputs are confined to a surprisingly low-dimensional subspace, with an effective dimensionality of only about 60\% of the full space---a phenomenon that is consistently observed across diverse model families and datasets, and arises from overlap among the output subspaces of different attention heads. Critically, we find this low-rank structure as a key factor of the prevalent dead feature problem in sparse dictionary learning, where it creates a mismatch between randomly initialized features and the intrinsic geometry of the activation space. Building on this insight, we propose a subspace-constrained training method for sparse autoencoders (SAEs), initializing feature directions into the active subspace of activations. Our approach reduces dead features from 87\% to below 1\% in Attention Output SAEs with 1M features, and can further extend to other sparse dictionary learning methods. Our findings provide both new insights into the geometry of attention and practical tools for improving sparse dictionary learning in large language models. Code is available at \url{https://github.com/OpenMOSS/Llamascopium}.
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Intentional Updates for Streaming Reinforcement Learning
Arsalan Sharifnassab, Mohamed Elsayed, Kris De Asis, A. Rupam Mahmood, Richard S Sutton
In gradient-based learning, a step size chosen in parameter units does not produce a predictable per-step change in function output. This often leads to instability in the streaming setting (i.e., batch size$=1$), where stochasticity is not averaged out and update magnitudes can momentarily become arbitrarily big or small. Instead, we propose intentional updates: first specify the intended outcome of an update and then solve for the step size that approximately achieves it. This strategy has precedent in online supervised linear regression via Normalized Least Mean Squares (NLMS) algorithm, which selects a step size to yield a specified change in the function output proportional to the current error. We extend this principle to streaming deep reinforcement learning by defining appropriate intended outcomes: Intentional TD aims for a fixed fractional reduction of the TD error, and Intentional Policy Gradient aims for a bounded per-step change in the policy, limiting local KL divergence. We propose practical algorithms combining eligibility traces and diagonal scaling. Empirically, these methods yield state-of-the-art streaming performance, frequently performing on par with batch and replay-buffer approaches.
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Subliminal Effects in Your Data: A General Mechanism via Log-Linearity
Ishaq Aden-Ali, Noah Golowich, Allen Liu, Abhishek Shetty, Ankur Moitra, Nika Haghtalab
Training modern large language models (LLMs) has become a veritable smorgasbord of algorithms and datasets designed to elicit particular behaviors, making it critical to develop techniques to understand the effects of datasets on the model's properties. This is exacerbated by recent experiments that show datasets can transmit signals that are not directly observable from individual datapoints (Halawi et al., 2024; Betley et al., 2025b;Cloud et al., 2025; Betley et al., 2025a), posing a conceptual challenge for dataset-centric understandings of LLM training and suggesting a missing fundamental account of such phenomena. Towards understanding such effects, inspired by recent work on the linear structure of LLMs (Park et al., 2024; Golowich et al., 2025b), we uncover a general mechanism through which hidden subtexts can arise in generic datasets.

We introduce LOGIT-LINEAR SELECTION (LLS), a method that prescribes how to select subsets of a generic preference dataset to elicit a wide range of hidden effec

ts. We apply LLS to discover subsets of real-world datasets so that models train ed on them exhibit behaviors ranging from having specific preferences, to respon ding to prompts in a different language not present in the dataset, to taking on a different persona. Crucially, the effect persists for the selected subset, ac ross models with varying architectures, supporting its generality and universali ty.
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TideGS: Scalable Training of Over One Billion 3D Gaussian Splatting Primitives v ia Out-of-Core Optimization
Chonghao Zhong, Linfeng Shi, Chen Hua, Tiecheng Sun, Hao Zhao, Binhang Yuan, Cha ojian Li
Training 3D Gaussian Splatting (3DGS) at billion-primitive scale is fundamentall y memory-bound: each Gaussian primitive carries a large attribute vector, and th e aggregate parameter table quickly exceeds GPU capacity, limiting prior systems to tens of millions of Gaussians on commodity single-GPU hardware. We observe t hat 3DGS training is inherently sparse and trajectory-conditioned: each iteratio n activates only the Gaussians visible from the current camera batch, so GPU mem ory can serve as a working-set cache rather than a persistent parameter store. B uilding on this insight, we introduce TideGS, an out-of-core training framework that manages parameters across an SSD-CPU-GPU hierarchy via three synergistic te chniques: block-virtualized geometry for SSD-aligned spatial locality, a hierarc hical asynchronous pipeline to overlap I/O with computation, and trajectory-adap tive differential streaming that transfers only incremental working-set deltas b etween iterations. Experiments show that TideGS enables training with over one b illion Gaussians on a single 24-GB GPU while achieving the best reconstruction q uality among evaluated single-GPU baselines on large-scale scenes, scaling beyon d prior out-of-core baselines (e.g., ~100M Gaussians) and standard in-memory tra ining (e.g., ~11M Gaussians). Project page: https://sponge-lab.github.io/TideGS/
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Reduction of Probabilistic Chemical Reaction Networks
Mauricio Montes, Grégoire Sergeant-Perthuis
Programming adaptive behaviors at the cellular level is a long-standing goal tha t raises the question of how probabilistic computation can be implemented in bio chemical systems. Chemical reaction networks (CRNs) provide such a substrate and have been shown to realize probabilistic models, including hidden Markov models and factor graphs, with dynamics reproducing Bayesian inference and belief prop agation. However, encoding these algorithms typically requires prohibitively lar ge reaction networks, and classical CRN reduction techniques do not directly app ly. By recovering the factor graph structure encoded in Napp--Adams-compiled CRN s, we transport recent factor-graph reduction results to their chemical implemen tations, obtaining significantly smaller CRNs while preserving the belief-propagation fixed points on surviving variables.
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The Power of Power Law: Asymmetry Enables Compositional Reasoning
Zixuan Wang, Xingyu Dang, Jason D. Lee, Kaifeng Lyu
Natural language data follows a power-law distribution, with most knowledge and skills appearing at very low frequency. While a common intuition suggests that r eweighting or curating data toward a uniform distribution may help models better learn these long-tail skills, we find a counterintuitive result: across a wide range of compositional reasoning tasks, such as state tracking and multi-step ar ithmetic, training under power-law distributions consistently outperforms traini ng under uniform distributions. To understand this advantage, we introduce a min imalist skill-composition task and show that learning under a power-law distribu tion provably requires significantly less training data. Our theoretical analysi s reveals that power law sampling induces a beneficial asymmetry that improves t he pathological loss landscape, which enables models to first acquire high-frequ ency skill compositions with low data complexity, which in turn serves as a step ping stone to efficiently learn rare long-tailed skills. Our results offer an al ternative perspective on what constitutes an effective data distribution for tra ining models.

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Meta Flow Maps enable scalable reward alignment
Peter Potaptchik, Adhi Saravanan, Abbas Mammadov, Alvaro Prat, Michael Samuel Albergo, Yee Whye Teh
Controlling generative models—whether via inference-time steering or fine-tuning—is expensive. Control relies on estimating the value function—typically necessitating costly trajectory simulations. To eliminate this bottleneck, we introduce *Meta Flow Maps (MFMs)*, stochastic extensions of consistency models and flow maps. MFMs are trained to perform \textbf{one-step posterior sampling}, generating arbitrarily many i.i.d. draws of clean data x_1 from any noisy state x_t. Crucially, these samples are differentiable in the conditioning state x_t, unlocking efficient estimation of the value function gradient. We leverage this capability to enable both **inference-time steering** without inner rollouts, and unbiased, off-policy **fine-tuning** to general rewards. Among our fine-tuning and steering experiments on ImageNet, we highlight that our single-particle steered-MFM sampler outperforms a Best-of-1000 baseline across multiple rewards at a fraction of the compute.
****
Token Sample Complexity of Attention
Léa Bohbot, Cyril Letrouit, Gabriel Peyré, François-Xavier Vialard
As context windows in large language models continue to expand, it is essential to characterize how attention behaves at extreme sequence lengths. We introduce token sample complexity: the rate at which attention computed on n tokens converges to its infinite-token limit. We estimate finite-n convergence bounds at two levels: pointwise uniform convergence of the attention map, and convergence of moments for the transformed token distribution.

For compactly supported (and more generally sub-Gaussian) distributions, our first result shows that the attention map converges uniformly on a ball of radius R at rate $C(R)/\sqrt{n}$, where $C(R)$ grows exponentially with R. For large R, this estimate loses practical value, and our second result addresses this issue by establishing convergence rates for the moments of the transformed distribution (the token output of the attention layer). In this case, the rate is $C'(R)/n^{\beta}$ with $\beta<\tfrac{1}{2}$, and $C'(R)$ depends polynomially on the size of the support of the distribution. The exponent β depends on the attention geometry and the spectral properties of the token distribution. We also examine the regime in which the attention parameter tends to infinity and the softmax approaches a hardmax, and in this setting, we establish a logarithmic rate of convergence. Experiments on synthetic and real data support our predictions and show that the predicted slowdown is reflected in downstream accuracy.
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EVMbench: Evaluating AI Agents on Smart Contract Security
Justin Wang, Andreas Bigger, Xiaohai Xu, Justin W Lin, Andy Applebaum, Tejal Patwardhan, Alpin Yukseloglu, Olivia Watkins
Smart contracts on public blockchains now manage large amounts of value, and vulnerabilities in these systems can lead to substantial losses. As AI agents become more capable at reading, writing, and running code, it is natural to ask how well they can already navigate this landscape, both in ways that improve security and in ways that might increase risk.
We introduce EVMbench, an evaluation that measures the ability of agents to detect, patch, and exploit smart contract vulnerabilities.
EVMbench draws on 117 curated vulnerabilities from 40 repositories and, in the most realistic setting, uses programmatic grading based on tests and blockchain state under a local Ethereum execution environment.
We evaluate a range of frontier agents and find that they are capable of discovering and exploiting vulnerabilities end-to-end against live blockchain instances.
We release code, tasks, and tooling to support continued measurement of these capabilities and future work on security.
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One-Step Graph-Structured Neural Flows for Irregular Multivariate Time Series C

lassification
Mengzhou Gao, kaiwei wang, Pengfei Jiao
Neural Flows efficiently model irregular multivariate time series by directly le
arning ODE solution trajectories with neural networks, bypassing step-by-step nu
merical solvers. Despite their efficiency, many existing approaches treat variab
les independently, leaving inter-variable interactions underexplored. Moreover,
their one-step mapping makes interaction modeling inherently challenging, as it
removes the iterative refinement of interactions during learning. To address thi
s challenge, we propose one-step Graph-Structured Neural Flows (GSNF), which int
roduce two auxiliary-trajectory self-supervision strategies to strengthen intera
ction learning: (i) interaction-aware trajectory generation via re-initializatio
n, which induces trajectory divergence to expose graph-induced interactions, wit
h a theoretically derived lower bound on divergence; and (ii) reverse-time traje
ctory generation, which enforces forward–backward consistency to regularize grap
h learning, enabled by flow invertibility. Experiments on five real-world datase
ts show that GSNF achieves state-of-the-art classification performance with high
ly competitive training time and memory usage. The code is available at https://
github.com/mzgaooo/GSNF.
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GIST: Targeted Data Selection for Instruction Tuning via Coupled Optimization Ge
ometry
Guanghui Min, Tianhao Huang, Ke Wan, Chen Chen
Targeted data selection has emerged as a crucial paradigm for efficient instruct
ion tuning, aiming to identify a small yet influential subset of training exampl
es for a specific target task.
In practice, influence is often measured through the effect of an example on par
ameter updates.
To make selection scalable, many approaches leverage optimizer statistics (e.g.,
 Adam states) as an axis-aligned surrogate for update geometry (i.e., diagonal p
recondition), implicitly treating parameters as coordinate-wise independent.
We show that this assumption breaks down in parameter-efficient fine-tuning (PEF
T) methods such as LoRA. In this setting, the induced optimization geometry exhi
bits strong cross-parameter coupling with non-trivial off-diagonal interactions,
 while the task-relevant update directions are confined to a low-dimensional sub
space.
Motivated by this mismatch, we propose **GIST** (Gradient Isometric Subspace Tra
nsformation), a simple yet principled alternative that replaces axis-aligned sca
ling with robust subspace alignment.
GIST recovers a task-specific subspace from validation gradients via singula
r value decomposition (SVD), projects training gradients into this coupled subsp
ace, and scores examples by their alignment with target directions.
Extensive experiments have demonstrated that **GIST** matches or outperforms the
 state-of-the-art baseline with only 0.29% of the storage and 25% of the computa
tional time under the same selection budget.
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Strategic Candidacy in Generative AI Arenas
Chris Hays, Rachel Li, Bailey Flanigan, Manish Raghavan
AI arenas, which rank generative models from pairwise preferences of users, are
a popular method for measuring the relative performance of models in the course
of their organic use. Because rankings are computed from noisy preferences, ther
e is a concern that model producers can exploit this randomness by submitting ma
ny models (e.g., multiple variants of essentially the same model) and thereby ar
tificially improve the rank of their top models. This can lead to degradations i
n the quality, and therefore the usefulness, of the ranking. In this paper, we b
egin by establishing, both theoretically and in simulations calibrated to data f
rom the platform Arena (formerly LMArena, Chatbot Arena), conditions under which
 producers can benefit from submitting clones when their goal is to be ranked hi
ghly. We then propose a new mechanism for ranking models from pairwise compariso
ns, called You-Rank-We-Rank (YRWR). It requires that producers submit rankings o
ver their own models and uses these rankings to correct statistical estimates of

model quality. We prove that this mecha-nism is approximately clone-robust, in the sense that a producer cannot improve their rank much by doing anything other than submitting each of their unique models exactly once. Moreover, to the extent that model producers are able to correctly rank their own models, YRWR improves overall ranking accuracy. In further simulations, we show that indeed the mechanism is approximately clone-robust and quantify improvements to ranking accuracy, even under producer mis-ranking.
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Loss-Aware Distributionally Robust Optimization via Trainable Optimal Transport Ambiguity Sets
Jonas Ohnemus, Marta Fochesato, Riccardo Zuliani, John Lygeros
Optimal-transport distributionally robust optimization (OT-DRO) robustifies data-driven decision-making under uncertainty by capturing the sampling-induced statistical error via optimal transport ambiguity sets. The standard OT-DRO pipeline consists of a two-step procedure, where the ambiguity set is first designed and subsequently embedded into the downstream OT-DRO problem. However, this separation between uncertainty quantification and optimization may lead to excessive conservatism. We introduce an end-to-end pipeline to automatically learn decision-focused ambiguity sets for OT-DRO problems, where the loss function informs the shape of the ambiguity set, leading to less conservative decisions whose distributional robustness is enforced via data-driven bootstrapping. We formulate the learning problem as a bilevel optimization program and solve it via a hypergradient-based method. By leveraging the recently introduced nonsmooth conservative implicit function theorem, we establish convergence to a critical point of the bilevel problem. We present experiments validating our method on standard portfolio optimization and linear regression tasks.
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MAVRL: Learning Reward Functions from Multiple Feedback Types with Amortized Variational Inference
Raphaël Baur, Yannick Metz, Maria Gkoulta, Mennatallah El-Assady, Giorgia Ramponi, Thomas Kleine Buening
Reward learning typically relies on a single feedback type or combines multiple feedback types using manually weighted loss terms. Currently, it remains unclear how to jointly learn reward functions from heterogeneous feedback types such as demonstrations, comparisons, ratings, rankings, and stops that provide qualitatively different signals. We address this challenge by formulating reward learning from multiple feedback types as Bayesian inference over a shared latent reward function, where each feedback type contributes information through an explicit likelihood. We introduce a scalable amortized variational inference approach that learns a shared reward encoder and feedback-specific likelihood decoders and is trained by optimizing a single evidence lower bound. Our approach avoids reducing feedback to a common intermediate representation and eliminates the need for manual loss balancing. Across discrete and continuous-control benchmarks, we show that jointly inferred reward posteriors outperform single-type baselines, exploit complementary information across feedback types, and yield policies that are more robust to environment perturbations. The inferred reward uncertainty further provides interpretable signals for analyzing model confidence and consistency across feedback types.
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HARD-KV: Head-Adaptive Regularization for Decoding-time KV Compression
Yuxuan Yang, Feiyang Ren, Bowen Zeng, Dalin Zhang, Jinpeng Chen, Gang Chen, Huan Li
Long-context LLM inference faces a fundamental conflict: head-adaptive compression algorithms (e.g., Top-p nucleus sampling) offer superior accuracy by dynamically fluctuating memory budgets, yet modern inference engines (e.g., vLLM) demand rigid, static memory patterns to leverage CUDA Graphs and PagedAttention.
We resolve this ``Static-Dynamic'' mismatch with HARD-KV, a unified framework that that bridges dynamic selection with rigid system constraints.
HARD-KV introduces a Cascade Cache hierarchy, managing the token lifecycle across dense, sparse, and condensed tiers.

Crucially, we propose a Logits Calibration mechanism that normalizes diverse imp ortance metrics into a unified probability space, enabling consistent Top-p bu dgeting across heterogeneous heads.
To bridge the efficiency gap, we offer a system-level solution, which rewrites f ragmented, dynamic indices into contiguous physical layouts compatible with high -performance inference engine.
Extensive experiments on math-reasoning benchmarks (AIME, U-Math) verify that HA RD-KV achieves up to 2$\times$ throughput improvement over static baselines whil e maintaining high-fidelity generation in 10k+ token scenarios.
Code is available at https://github.com/SuDIS-ZJU/HARDInfer.
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FIRE: Multi-Fidelity Regression with Distribution-Conditioned In-Context Learnin g Using Tabular Foundation Models
Rosen Ting-Ying Yu, Nicholas Sung, Faez Ahmed
Multi-fidelity (MF) regression often operates in regimes of extreme data imbalan ce, where the commonly-used Gaussian-process surrogates struggle with cubic scal ing costs and overfit to sparse high-fidelity observations, limiting efficiency and generalization in real-world applications. We introduce FIRE, a training-fre e MF framework that couples tabular foundation models (TFMs) to perform zero-sho t in-context Bayesian inference via a high-fidelity correction model conditioned on the low-fidelity model's posterior predictive distributions. This cross-fide lity information transfer via distributional summaries captures heteroscedastic errors, enabling robust residual learning without model retraining. Across 31 be nchmark problems spanning synthetic functions and real-world tasks (e.g., DrivAe rNet, LCBench), FIRE delivers a stronger performance–time trade-off than seven s tate-of-the-art GP-based or deep learning MF regression methods, ranking highest in accuracy and uncertainty quantification with runtime advantages. Limitations include context window constraints and dependence on the quality of the pre-tra ined TFMs. Code & data can be found here: https://github.com/rosenyu304/FIRE.
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Unsupervised Process-Aware Coreset Selection for In-Context Learning
Wei Zheng, Zijie Wang, Xin Li, Bin Gong, YUQING SUN
We address the challenge of unsupervised coreset selection for few-shot in-conte xt learning (ICL). The goal is to select a small subset of examples under a fixe d annotation budget to yield effective prompts for large language models. Existi ng geometry-based methods often yield coresets that suffer from a skewed distrib ution, due to the oversampling of peripheral examples and high local redundancy. To address these issues, we propose a process-aware framework for coreset selec tion. It jointly optimizes the diversity and representativeness of selected samp les via an adaptive submodular objective. It ensures representativeness by selec ting samples based on local density awareness, while promoting diversity by impo sing a redundancy penalty relative to the evolving selected set. Thus, it perfor ms progress-aware balancing of representativeness and diversity based on the sel ection context. Extensive experiments on 7 NLP datasets demonstrate that our met hod consistently outperforms state-of-the-art coreset selection methods in downs tream ICL performance. Further analysis validates that our approach better balan ces diversity and representativeness in the selection process, while retaining t he theoretical guarantees of adaptive submodular optimization.
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Envisioning Beyond the Few: Disentangled Semantics and Primitives for Few-Shot A typical Layout-to-Image Generation
Nan Bao, Yifan Zhao, Wenzhuang Wang, Jia Li
The layout-to-image (L2I) task enables fine-grained control over image generatio n via object categories and spatial layouts. However, existing L2I methods yield fragmented and distorted generations under few-shot atypical settings. We term this failure as representation fragmentation, arising from a granularity mismatc h that entangles semantic identity with visual details. To address this issue, w e propose a representation-driven framework that disentangles semantics from pri mitives for robust few-shot adaptation. Specifically, Semantic Anchoring aggrega tes categorical semantics into anchors for stable identity, while Primitive Imbu

ing models recomposable primitives for robust local detail modeling. Conceptual Steering further regulates optimization with a saliency-aware objective to preserve foreground semantic consistency. Extensive experiments demonstrate consistent improvements in the 5-shot regime over state-of-the-art L2I methods in both visual fidelity and alignment across diverse atypical domains. The source code is publicly available at https://github.com/iCVTEAM/DSP.
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IRPM: Intergroup Relative Preference Modeling for Pointwise Generative Reward Models

Haonan Song, Qingchen Xie, Huan Zhu, Feng Xiao, Luxi Xing, Liu Kang, Fuzhen Li, Zhiyong Zheng, Feng Jiang, Ziheng Li, Kun Yan, Qingyi Si, Yanghua Xiao, Hongcheng Guo, Fan Yang

Generative Reward Models (GRMs) have demonstrated strong performance in reward modeling, due to their interpretability and potential for refinement through reinforcement learning (RL). However, widely used pairwise GRMs create a computational bottleneck in reinforcement learning from human feedback (RLHF), when calibrating or aggregating preference signals over n candidates, often incurring $\mathcal{O}(n^2)$ pairwise judgments. To address this issue, we propose **Intergroup Relative Preference Modeling (IRPM)**, an RL-based method that extends the Bradley-Terry preference-learning paradigm via intergroup comparisons to train *pointwise* GRMs from pairwise preference data. IRPM derives a pointwise reward for each response by contrasting groups of chosen vs. rejected samples, enabling pointwise scores comparable across candidate sets and $\mathcal{O}(n)$ reward evaluation for a variable number of candidates during RL training, while preserving interpretability and scalability. Experiments show that IRPM achieves state-of-the-art performance among pointwise GRMs on JudgeBench and RewardBench, and approaches the performance of leading pairwise GRMs. In addition, IRPM achieves substantial gains in post-training evaluations, demonstrating its effectiveness. Our models and code are available at https://github.com/XINGYUN-AI-LAB/IRPM.
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Large-capacity and Receiver Authenticable Generative Image Steganography

Jiannian Wang, Yao Lu, Guangming Lu

Diffusion-based generative image steganography converts the input single secret image into noise, and generates the stego image with it serves as the initial noise. Nevertheless, existing methods exhibit three severe limitations: (1) the fixed hiding space constrains their capacity to one secret image; (2) severe inter-secret interference arising from substantial information divergence among multiple secret images while concealing them within a shared hiding space; (3) security risks owing to the absence of the receiver-side verification mechanism. To systematically address these issues, this paper proposes a novel **Receiver Authenticable Generative Image Steganography framework** based on diffusion models. We introduce a **Dynamic Cover Selection and Optimization Engine** to adaptively allocate suitable hiding spaces for different secret images. This design permits the concealment of disparate secret images (or fragments of a single image) into separate spaces, enabling dynamic multi-image concealment while effectively preventing inter-secret interference and expanding capacity through the enlarged hiding spaces. Furthermore, a **Signature Authentication Controller** cryptographically signs the secret container after concealing and verifies it before extraction, ensuring secure receiver isolation and precise localization of the secret data container. Experiments demonstrate that the proposed framework achieves superior secure multi-receiver isolation and high-performance generative image steganography with large capacity.
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Instruction Decomposition and Action Alignment for Vision-Language Navigation

Zihao Xin, Wentong Li, Yixuan Jiang, Bin Wang, Piji Li, Jianke Zhu, Jie Qin, Sheng-Jun Huang

Vision-and-Language Navigation (VLN) empowered by Multimodal Large Language Models (MLLMs) is promise, yet remains challenged by long-horizon tasks with complex user instructions. Existing approaches that continuously condition on full instructions incur high latency due to abundant visual tokens and exacerbates instr

uction interference, where irrelevant text noise induces hallucinations.
To address these limitations, we propose IDEAL-VLN (\textbf{I}nstruction \textbf{DE}composition and \textbf{A}ction a\textbf{L}ignment), a novel paradigm that reformulates navigation as a causal inference chain. We decompose the task into two sequential steps: Semantic Anchoring and Action Alignment. We adopt a \textit{Think-Before-Act} mechanism that first infers the immediate semantic anchor from the global context and then generates actions conditioned solely on this anchor. This design constructs an explicit information bottleneck, suppressing spurious correlations from irrelevant instruction. Moreover, to alleviate cognitive collapse and limited exploration during training, we introduce a hierarchical correction framework that combines semantic-level thought correction with a spatially-aware adaptive intervention strategy. This strategy adjusts expert intervention probability based on geodesic distance, effectively defining a semantic safety boundary. To support this paradigm, we contribute the Instruction-Aligned Navigation Dataset containing 160K image-text pairs.
Extensive experiments demonstrate that IDEAL-VLN achieves state-of-the-art performance and robustness across major benchmarks while significantly reducing inference costs.
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Continuous-Time Piecewise-Linear Recurrent Neural Networks
Alena Brändle, Lukas Eisenmann, Florian Götz, Daniel Durstewitz
In dynamical systems reconstruction (DSR) we aim to recover the dynamical system (DS) underlying observed time series. Specifically, we aim to learn a generative surrogate model which approximates the underlying, data-generating DS, and recreates its long-term properties (`climate statistics'). In scientific and medical areas, in particular, these models need to be mechanistically tractable -- through their mathematical analysis we would like to obtain insight into the recovered system's workings. Piecewise-linear (PL), ReLU-based RNNs (PLRNNs) have a strong track-record in this regard, representing SOTA DSR models while allowing mathematical insight by virtue of their PL design. However, all current PLRNN variants are *discrete-time maps*. This is in disaccord with the assumed continuous-time nature of most physical and biological processes, and makes it hard to accommodate data arriving at *irregular* temporal intervals. Neural ODEs are one solution, but they do not reach the DSR performance of PLRNNs and often lack their tractability. Here we develop theory for *continuous-time* PLRNNs (cPLRNNs): We present a novel algorithm for training and simulating such models, bypassing numerical integration by efficiently exploiting their PL structure. We further demonstrate how important topological objects like equilibria or limit cycles can be determined semi-analytically in trained models. We compare cPLRNNs to both their discrete-time cousins as well as Neural ODEs on DSR benchmarks, including systems with discontinuities which come with hard thresholds.
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The Invisible Lottery: How Subtle Cues Steer Algorithm Choice in LLM Code Generation
Akanksha Narula, Mofasshara Binte Rafique, Laurent Bindschaedler
Large language models (LLMs) now generate substantial production code, often for tasks with multiple valid algorithmic solutions. Incidental prompt cues—contextual words or metadata outside the task specification—can steer which algorithm the model selects, even when all outputs pass the same tests. Prompt sensitivity is well studied as a mechanism for improving output quality; here, we instead study *output policy*: algorithm choice under fixed correctness. We define *algorithm steering* as cue-induced shifts in algorithm-family distributions and run 46,535 controlled experiments across 11 tasks, 19 cue types (18 channels plus a memoization semantic-vs-surface ablation that preserves meaning while changing typography and punctuation), and 15 model configurations. We find large, systematic shifts in algorithm-family distributions (up to 100 percentage points, pp), largely consistent with cue semantics, including in applied tasks such as rate limiting. Direct algorithm naming is the most reliable mitigation we tested. These results reveal an “invisible lottery” in which accidental context alters downstream properties such as performance, security, and maintainability.

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SONAR: Spectral■Contrastive Audio Residuals for Generalizable Deepfake Detection
Ido Nitzan Hidekel, Gal Lifshitz, Khen Cohen, Dan Raviv
Deepfake audio detectors often fail to generalize to unseen attacks, in part due to \emph{spectral bias}: neural networks prioritize low-frequency structure while under-exploiting subtle high-frequency (HF) artifacts left by generative models. We introduce \textbf{SONAR} (Spectral-cONtrastive Audio Residuals), a frequency-guided framework that \emph{explicitly enforces representation-level consistency} between semantic content and HF residuals. Unlike prior frequency-aware or dual-stream detectors that treat HF cues as auxiliary features, SONAR encourages structured interaction between content and noise representations in latent space. The model employs a dual-path architecture in which an XLSR encoder captures low-frequency content, while a parallel branch with learnable, value-constrained 1D SRM (Spatial Rich Model) high-pass filters distills HF residuals. The two representations are fused via frequency cross-attention and trained with a \emph{Jensen--Shannon alignment loss} that promotes LF–HF consistency for genuine audio and amplifies inconsistency for deepfakes. Evaluated on ASVspoof~2021 and in-the-wild benchmarks, SONAR achieves state-of-the-art performance in a \textbf{single run} setting and converges faster than strong baselines. By mitigating the effects of spectral bias through frequency-guided alignment, SONAR provides a fully data-driven and architecture-agnostic approach to generalizable audio deepfake detection.
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Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models
Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, Aditya Grover
Knowledge distillation improves large language model (LLM) reasoning by compressing the knowledge of a teacher LLM to train smaller LLMs. On-policy distillation advances this approach by having the student sample its own trajectories while a teacher LLM provides dense token-level supervision, addressing the distribution mismatch between training and inference in off-policy distillation methods. However, on-policy distillation typically requires a separate, often larger, teacher LLM and does not explicitly leverage ground-truth solutions available in reasoning datasets. Inspired by the intuition that a sufficiently capable LLM can rationalize external privileged reasoning traces and teach its weaker self, we introduce On-Policy Self-Distillation (OPSD), a learning algorithm where a single LLM acts as both teacher and student with different contexts. The teacher policy conditions on privileged information while the student policy sees only the question; training minimizes the per-token divergence between these distributions over the student's own rollouts. We demonstrate the efficacy of our method on multiple mathematical reasoning benchmarks, achieving superior token efficiency compared to reinforcement learning methods and better performance over off-policy distillation methods.
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Estimating Tail Risks in Language Model Output Distributions
Rico Angell, Raghav Singhal, Zachary Horvitz, Zhou Yu, Rajesh Ranganath, Kathleen McKeown, He He
Language models are increasingly capable and are being rapidly deployed on a population-level scale. As a result, the safety of these models is increasingly high-stakes. Fortunately, advances in alignment have significantly reduced the likelihood of harmful model outputs. However, when models are queried billions of times in a day, even rare worst-case behaviors will occur. Current safety evaluations focus on capturing the distribution of inputs that yield harmful outputs. These evaluations disregard the probabilistic nature of models and their tail output behavior. To measure this tail risk, we propose a method to efficiently estimate the probability of harmful outputs for any input query. Instead of naive brute-force sampling from the target model, where harmful outputs could be rare, we operationalize importance sampling by creating unsafe versions of the target model. These unsafe versions enable sample-efficient estimation by making harmful outputs more probable. On benchmarks measuring misuse and misalignment, these es

timates match brute-force Monte Carlo estimates using 10–20× fewer samples. For example, we can estimate probability of harmful outputs on the order of $10^{−4}$ with just 500 samples. Additionally, we find that these harmfulness estimates can reveal the sensitivity of models to perturbations in model input and predict deployment risks. Our work demonstrates that rare-event estimation is both critical and feasible for safety evaluations.
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Can LLM Agents Stick to the Script? Modeling Commitment in Interactive Narratives
Yingpeng Ma, Jianhao Yan, Bei Shi, Ka Hou Kam, Runnan Wang, Xuebo Liu, Yulong Chen, Yue Zhang, Derek F. Wong
The rapid advancement of Large Language Models (LLMs) is revolutionizing AI for Games by enabling open-ended and fluid interactive storytelling.
However, existing research has largely overlooked the critical challenge of maintaining logical consistency and narrative integrity against unconstrained user interventions.
To address this, we formulate this challenge as *Narrative Commitment Preservation (NCP)*, and take interactive narrative as our testbed.
We introduce **NCP-Bench**, a benchmark of 100 narrative environments derived from movie synopses.
Each environment includes a structured narrative specification (trajectory, commitments, and initial facts) that we can automatically check throughout the interaction between the player agent and the narrator agent.
Experiments across state-of-the-art LLMs reveal that high linguistic quality does not guarantee commitment preservation; even strong models frequently generate logically conflicting content under adversarial interventions, with the best-performing model (GPT-5.2) achieving only 40\% survival rate after 20 turns and fact conflict rates ranging from 40\% to 68\% across models.
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HyMTRL: A Hybrid Multi-Task Reinforcement Learning Framework via Phased Policy Evolution
Jinmin He, Kai Li, Xiaoyi Dong, Yifan Zang, Yuheng Jing, Yifan Zhang, Junliang Xing, Jian Cheng
Multi-task reinforcement learning (MTRL) aims to improve sample efficiency by sharing knowledge across related tasks, but it often suffers from asynchronous convergence speed caused by inherent differences in task difficulty. This imbalance places substantial representational strain on the shared critic network, which emerges as a major performance bottleneck. To address this issue, we propose Hybrid Multi-Task Reinforcement Learning (HyMTRL), a framework that alleviates critic overload through a phased policy evolution strategy. HyMTRL divides task learning into a reinforcement exploration phase and an imitation refinement phase. By transitioning mastered tasks from reinforcement learning–based policy optimization to imitation learning–based behavior consolidation, these tasks are removed from the critic’s optimization objective, effectively reducing representational strain. In addition, a critic reset mechanism restores network capacity while preserving learned policy and historical experience. HyMTRL is a general framework that can be easily integrated with a wide range of existing MTRL methods. Empirical evaluations on the MetaWorld benchmark demonstrate that HyMTRL leads to significant improvements in both learning efficiency and final performance.
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Provably Data-driven Multiple Hyper-parameter Tuning with Structured Loss Function
TUNG QUOC LE, Anh Tuan Nguyen, Viet Anh Nguyen
Data-driven algorithm design automates hyperparameter tuning, but its statistical foundations remain limited because model performance can depend on hyperparameters in implicit and highly non-smooth ways. Existing guarantees focus on the simple case of a one-dimensional (scalar) hyperparameter. This leaves the practically important, multi-dimensional hyperparameter tuning setting unresolved. We address this open question by establishing the first general framework for establishing generalization guarantees for tuning multi-dimensional hyperparameters in

data-driven settings. Our approach strengthens the generalization guarantee framework for semi-algebraic function classes by exploiting tools from real algebraic geometry, yielding sharper, more broadly applicable guarantees. We then extend the analysis to hyperparameter tuning using the validation loss under minimal assumptions, and derive improved bounds when additional structure is available. Finally, we demonstrate the scope of the framework with new learnability results, including data-driven weighted group lasso and weighted fused lasso.
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From Representation to Action: A Unified Laplacian Framework for Spatial Representation and Path Planning
Junfeng Zuo, Yuhang He, Wenhao Zhang, Fang Fang, Si Wu
Navigation in complex environments relies on internal spatial representations that guide action. While the brain employs a diverse repertoire of spatial tuning cells—including grid, place, and head-direction cells—a normative theory linking these static neural codes to the dynamic process of navigation remains elusive. In this work, we propose a Unified Laplacian Framework derived from first principles of representational smoothness and efficiency. We first demonstrate that diverse spatial codes emerge naturally as spectral decompositions of the Laplace operator. Crucially, bridging the gap from representation to action, we derive a computational-level navigation policy based on the Green's function potential. We show that this potential encodes the environment's intrinsic geometry to enable geometry-aware gradient ascent, achieving improved sample efficiency and generalization in goal-reaching tasks. Furthermore, we demonstrate that these spectral representations can be learned directly from high-dimensional visual inputs, supporting their learnability from sensory experience. Our results suggest that the ``cognitive map" can be viewed as a spectral embedding of the Laplacian, providing a normative computational account that is biologically consistent with observed spatial-code phenomenology and useful for artificial agents.
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ReTabSyn: Realistic Tabular Data Synthesis via Reinforcement Learning
Xiaofeng Lin, Seungbae Kim, Zhuoya Li, Zachary DeSoto, Charles Fleming, Guang Cheng
Deep generative models can help with data scarcity and privacy by producing synthetic training data, but they struggle in low-data, imbalanced tabular settings to fully learn the complex data distribution.
We argue that striving for the full joint distribution could be overkill; for greater data efficiency, models should prioritize learning the conditional distribution $P(y\mid \mathbf{X})$, as suggested by recent theoretical analysis.
Therefore, we overcome this limitation with ReTabSyn, a Reinforced Tabular Synthesis pipeline that provides direct feedback on feature correlation preservation during synthesizer training.
This objective encourages the generator to prioritize the most useful predictive signals when training data is limited, thereby strengthening downstream model utility.
We empirically fine-tune a language model-based generator using this approach, and across benchmarks with small sample sizes, class imbalance, and distribution shift, ReTabSyn consistently outperforms state-of-the-art baselines.
Moreover, our approach can be readily extended to control various aspects of synthetic tabular data, such as applying expert-specified constraints on generated observations.
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CPMöbius: Iterative Coach–Player Reasoning for Data-Free Reinforcement Learning
Ran Li, Zeyuan Liu, Yinghao Chen, Bingxiang He, Jiarui Yuan, Zixuan Fu, Weize Chen, Jinyi Hu, Chen Qian, Zhiyuan Liu, Maosong Sun
Large Language Models (LLMs) have demonstrated strong potential in complex reasoning, yet their progress remains fundamentally constrained by reliance on massive high-quality human-curated tasks and labels, either through supervised fine-tuning (SFT) or reinforcement learning (RL) on reasoning-specific data. This dependence renders supervision-heavy training paradigms increasingly unsustainable, with signs of diminishing scalability already evident in practice. To overcome th

is limitation, we introduce CPMöbius, a collaborative Coach–Player paradigm for data-free reinforcement learning of reasoning models. Unlike traditional adversarial self-play frameworks, CPMöbius inspired by multi-agent collaboration treats the Coach and Player as independent but cooperative roles. The Coach proposes instructions targeted at the Player’s capability and receives rewards based on changes in the Player’s performance, while the Player is rewarded for solving the increasingly instructive tasks generated by the Coach. This cooperative optimization loop is designed to directly enhance the Player’s mathematical reasoning ability. Remarkably, CPMöbius achieves substantial improvement without relying on any external training data, outperforming existing unsupervised approaches. For example, on the Qwen2.5-Math-7B-Instruct, our method improves accuracy by overall average +4.9 and out-of-distribution average +5.4, which exceed RENT for +1.5 on overall accuracy and R-zero for +4.2 on OOD accuracy.

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Architecture Matters for Multi-Agent Security

Ben Hagag, William L. Anderson, Christian Schroeder de Witt, Sarah Scheffler

Multi-agent systems (MAS), composed of networks of two or more autonomous AI agents, have become increasingly popular in production deployments, yet introduce security risks that do not arise in single-agent settings. Even if individual agents may exhibit robust security, architectural decisions governing their coordination can create attack surfaces that have not been systematically characterized. In this work, we present an empirical study of how MAS design decisions shape the tradeoff between task performance and attack resistance.

Using a network of web-based agents and stage-wise evaluations that distinguish planning-stage refusal, execution-stage interception, and successful attack completion, we study architectural choices through controlled experiments. We identify three key design choices that influence MAS security: (i) agent roles, which determine how authority and responsibility are allocated; (ii) topology, which shapes how and when agents interact; and (iii) memory, which determines the context and state visibility accessible to each agent. Overall, our results show that security and performance in multi-agent systems are governed by architectural design choices, motivating the development of further evaluations which move beyond the security properties of a single agent.

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IPMark: A Sentence-Level Watermark for LLMs with Hierarchical Personalization and Efficient Detection

Wenbo An, Lianwei Wu, Zehao Wang

Watermarking has emerged as a critical solution for the detection and provenance tracing of content generated by large language models. However, existing methods still suffer from significant limitations, including difficulties in achieving efficient and personalized attribution, substantial degradation of generation quality, and low robustness against attacks. To address these challenges, we propose IPMark, the first IP-inspired hierarchical personalized watermarking framework. Specifically, to enable personalization and efficient detection, IPMark employs a hierarchical addressing framework to structurally organize model and user identities. Subsequently, addressing the inherent semantic distortion caused by token-level watermarking, we design a semantic-syntactic dual-stream embedding strategy. Centered on sentence-level candidate selection and reinforced by dual signals from syntactic and semantic features, this approach optimizes the injection process, thereby significantly enhancing generation quality while ensuring strong robustness. Experimental results demonstrate that IPMark achieves the lowest perplexity among baselines, ensuring superior generation quality while maintaining strong robustness and significantly reducing detection latency through hierarchical retrieval. Our code is available at https://github.com/nwlt/IPMark.

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CLARITree: Cholesky and Lookahead Accelerations for Regression with Interpretable Piecewise Linear Trees

Yixiao Wang, Hayden McTavish, Varun Babbar, Margo Seltzer, Cynthia Rudin

Regression trees are among the most interpretable yet expressive model classes in machine learning. Historically, greedy induction has been the dominant approac

h for constructing well-performing regression trees. While optimal methods based on dynamic programming and branch-and-bound exist, they are computationally prohibitive for general linear regression trees, despite often achieving substantially better performance than greedy approaches. Recent work has shown that specialized lookahead strategies can dramatically improve runtime while maintaining near-optimal performance, primarily in classification settings. In this work, we develop a novel algorithm for near-optimal, sparse, piecewise linear regression trees that combines a lookahead-style search strategy with efficient rank-one Cholesky updates of the Gram matrix. We demonstrate, both theoretically and empirically, that our method achieves a favorable trade-off between computational efficiency, predictive accuracy, and sparsity, and scales significantly better than the current state of the art.

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Mixture of Concept Bottleneck Experts

Francesco De Santis, Gabriele Ciravegna, Giovanni De Felice, Arianna Casanova, Francesco Giannini, Michelangelo Diligenti, Johannes Schneider, Danilo Giordano, Mateo Espinosa Zarlenga, Pietro Barbiero

Concept Bottleneck Models (CBMs) promote interpretability by grounding predictions in human-understandable concepts. However, existing CBMs typically constrain their task predictor to a single expression whose functional form is set a priori, limiting both predictive accuracy and adaptability to diverse user needs. We propose Mixture of Concept Bottleneck Experts (M-CBEs), a framework that generalizes existing CBMs along two dimensions: the number of expressions, referred to as experts, employed by the task predictor to map concepts to the task, and the functional form each expression takes, thus exposing an underexplored region of this design space. We investigate this region by instantiating two novel models: Linear M-CBE, which learns a finite set of linear expressions, and Symbolic M-CBE, which leverages symbolic regression to discover expert functions from data subject to user-specified operator vocabularies. Empirical evaluation demonstrates that varying the number of expressions and their functional form provides a robust framework for navigating the accuracy-interpretability trade-off.

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Learning in Bayesian Stackelberg Games With Unknown Follower's Types

Matteo Bollini, Francesco Bacchiocchi, Samuel Coutts, Matteo Castiglioni, Alberto Marchesi

We study online learning in Bayesian Stackelberg games, where a leader repeatedly interacts with a follower whose unknown private type is independently drawn at each round from an unknown probability distribution. The goal is to design algorithms that minimize the leader's regret with respect to always playing an optimal commitment computed with knowledge of the game. We consider, for the first time to the best of our knowledge, the most realistic case in which the leader does not know anything about follower's types, i.e., the possible follower's payoffs. This raises considerable additional challenges compared to the usually addressed case in which follower's payoffs are known. First, we prove a strong negative result: no-regret is unattainable under action feedback, i.e., when the leader only observes the follower's best response at the end of each round. Thus, we focus on the easier type feedback model, where the follower's type is also revealed. In such a setting, we propose an algorithm that achieves a regret of $\widetilde{O}(\sqrt{T})$, ignoring the dependence on other parameters.

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PICACO: Pluralistic In-Context Value Alignment via Total Correlation Optimization

Han Jiang, Dongyao Zhu, Xiaoyuan Yi, Ziang Xiao, Zhihua Wei, Xing Xie

In-Context Learning has shown great potential for aligning Large Language Models (LLMs) with human values, helping reduce harmful outputs and accommodate diverse preferences without costly post-training, known as *In-Context Alignment* (ICA).

However, LLMs' comprehension of input prompts remains agnostic, limiting ICA's ability to address value tensions—human values are inherently *pluralistic*, often imposing conflicting demands, *e.g.*, stimulation vs. tradition. Current ICA m

ethods therefore face the *Instruction Bottleneck* challenge, where LLMs struggle to reconcile multiple intended values within a single prompt, leading to incomplete or biased alignment.
To address this, we propose **PICACO**, a novel pluralistic ICA method. Without fine-tuning, PICACO optimizes a meta-instruction that incorporates multiple values to better elicit LLMs' understanding of them and improve alignment. This is achieved by maximizing the total correlation between specified values and LLM responses, which theoretically reinforces value conformity and reduces distractive noise, resulting in more effective instructions. Extensive experiments on five value sets show that PICACO works well with both black-box and open-source LLMs, outperforms several recent strong baselines, and achieves a better balance across up to 8 distinct values.
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Keeping a Secret Requires a Good Memory: Unconditional Streaming Lower-Bounds for Differentially Private Algorithms
Alessandro Epasto, Xin Lyu, Pasin Manurangsi
We study the computational cost of differential privacy in terms of memory efficiency. Specifically, we establish for the first time an unconditional space lower bound for user-level differential privacy by introducing a novel proof technique based on a multi-player communication game. We apply our framework, as an example, to the fundamental problem of estimating the number of distinct elements in a stream: we prove that any private algorithm requires almost $\widetilde{\Omega}(T^{1/3})$ space (where T denotes the length of the stream) to achieve certain error rates in a promise variant of the problem, resolving an open problem in the literature (by Jain et al. 2023 and Cummings et al. 2025) and establishes the first exponential separation between the space complexity of private algorithms and their non-private $\widetilde{O}(1)$ counterparts for a natural statistical estimation task. Furthermore, we show that this communication-theoretic technique generalizes to broad classes of problems, yielding lower bounds for private medians, quantiles, and max-select.
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Universal Reasoner: A Single, Composable Plug-and-Play Reasoner for Frozen LLMs
Jaemin Kim, Hangeol Chang, Hyunmin Hwang, Choonghan Kim, Jong Chul Ye
Large Language Models (LLMs) have demonstrated remarkable general capabilities, but enhancing skills such as reasoning often demands substantial computational resources and may compromise generalization. While Parameter-Efficient Fine-Tuning (PEFT) methods offer a more resource-conscious alternative, they typically require retraining for each LLM backbone due to architectural dependencies. To address these challenges, we propose Universal Reasoner (UniR)—a lightweight, composable, and plug-and-play reasoning module that can be used with larger frozen LLMs to provide specialized reasoning capabilities. Specifically, UniR decomposes the reward into a standalone reasoning module trained in a decoupled manner using verifiable rewards, effectively translating trajectory-level signals into token-level guidance. Once trained, UniR is combined with frozen LLMs at inference time by simply adding its output logits to those of the backbone. This additive structure enables modular composition: multiple UniR modules trained for different tasks can be jointly applied by summing their logits, enabling complex reasoning via composition. Furthermore, UniR demonstrates weak-to-strong generalization, where reasoning modules trained on smaller models effectively guide much larger LLMs in the same model family, and generalize across domains such as in vision language models and medical reasoning.
Experiments on mathematical reasoning and machine translation show that UniR surpasses existing fine-tuning methods. Code is open-sourced at https://github.com/hangeol/UniR.
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NonZero: Interaction-Guided Exploration for Multi-Agent Monte Carlo Tree Search
Sizhe Tang, Zuyuan Zhang, Mahdi Imani, Tian Lan
Monte Carlo Tree Search (MCTS) scales poorly in cooperative multi-agent domains because expansion must consider an exponentially large set of joint actions, severely limiting exploration under realistic search budgets. We propose \textsc{No

nZero}, which keeps multi-agent MCTS tractable by running surrogate-guided selec
tion over a low-dimensional nonlinear representation using an interaction-guided
 proposal rule, instead of directly exploring the full joint-action space. Our e
xploration uses an interaction score: single-agent deviations are ranked by pred
icted gain, while two-agent deviations are scored by a mixed-difference measure
that reveals coordination benefits even when no single agent can improve alone.
We formalize candidate proposal as a bandit problem over local deviations and de
rive a proposal rule, \textsc{NonUCT}, with a sublinear local-regret guarantee f
or reaching approximate graph-local optima without enumerating the joint-action
space. Empirically, \textsc{NonZero} improves sample efficiency and final perfor
mance on MatGame, SMAC, and SMACv2 relative to strong model-based and model-free
 baselines under matched search budgets.
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DyPE: Dynamic Position Extrapolation for Ultra High Resolution Diffusion
Noam Issachar, Guy Yariv, Sagie Benaim, Yossi Adi, Dani Lischinski, Raanan Fatta
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Diffusion Transformer models can generate images with remarkable fidelity and de
tail, yet training them at ultra-high resolutions remains extremely costly due t
o the self-attention mechanism's quadratic scaling with the number of image toke
ns. In this paper, we introduce Dynamic Position Extrapolation (DyPE), a novel,
training-free method that enables pre-trained diffusion transformers to synthesi
ze images at resolutions far beyond their training data, with no additional samp
ling cost. DyPE takes advantage of the spectral progression inherent to the diff
usion process, where low-frequency structures converge early, while high-frequen
cies take more steps to resolve. Specifically, DyPE dynamically adjusts the mode
l's positional encoding at each diffusion step, matching their frequency spectru
m with the current stage of the generative process. This approach allows us to g
enerate images at resolutions that exceed the training resolution dramatically,
e.g., 16 million pixels using FLUX. On multiple benchmarks, DyPE consistently im
proves performance and achieves state-of-the-art fidelity in ultra-high-resoluti
on image generation, with gains becoming even more pronounced at higher resoluti
ons.
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Correct Looks Better: Pairwise Comparisons Reveal Accuracy Rankings
Mina Remeli, Moritz Hardt
Pairwise comparisons combined with aggregation methods like Elo have become cent
ral to evaluating generative models, yet concerns remain that they reward superf
icial stylistic cues or display judge biases.
In a more positive turn, we show that model rankings from pairwise comparisons s
trongly agree with ground-truth-based accuracy rankings when such ground truth i
s available for comparison.
By converting five well-known benchmarks into free-form generative evaluations,
we find that Elo rankings achieve a Spearman correlation above 0.9 with accuracy
 rankings and substantially outperform direct evaluation when the judge is weak.

Furthermore, style and judge bias have only minor effects on model rankings, des
pite most judgments occurring on pairs where both candidate answers are correct
(or incorrect).
On such pairs, we find that repetition after the final answer (*echo*) is a caus
al driver of judge preference.
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Selective Coupling of Decoupled Informative Regions: Masked Attention Alignment
for Data-Free Quantization of Vision Transformers
Biao Qian, Yang Wang, Yong Wu, Jungong Han
Data-Free Quantization (DFQ) addresses data security concerns by synthesizing sa
mples, without accessing real data. It has garnered increasing attention in the
context of Vision Transformers (ViTs), owing to the superiority of the self-atte
ntion mechanism compared to classical convolutional operation. However, previous
 DFQ arts for ViTs often suffer from a distribution mismatch between synthetic s
amples and input distribution expected by quantized models Q, resulting in the

suboptimal performance. In this paper, we propose a novel Masked Attention Alignment approach for Data-Free Quantization of ViTs, named MaskAQ, revealing that: 1) the semantics in the self-attention mechanism is predominantly localized to a sparse subset of patches, called informative regions; 2) the informative regions dominate the mutual information between synthetic samples and Q's outputs. To these ends, we incorporate differential entropy maximum over patch similarity of synthetic samples, to decouple informative regions from noisy background. To couple with varied Q, the informative regions are selected to align full-precision models with Q via a masked attention alignment objective, thus yielding high-quality synthetic samples.
Furthermore, a periodic sample refreshing strategy comes up to endow MaskAQ with the capacity to continually adapt to the evolving state of Q throughout the training process, to preserve desirable mutual information with synthetic samples.
Extensive experiments verify the merits of MaskAQ over state-of-the-art approaches across multiple backbones and downstream tasks.
Our code is available at https://github.com/hfutqian/MaskAQ.
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SPR: A Structured Prompt Refinement Network for Modality Missing
Hao Chen, Diwei Su, Zhuo Wang, Zuwang He, Menglu Chen, Xiuxing Li, Xia Wu
Prompt learning has recently emerged as a dominant paradigm to tackle the missing modalities challenge. However, existing methods often overlook the internal structural information of prompt vectors, limiting performance in guiding frozen backbone models under diverse missing modality scenarios. To address this, we propose a Structured Prompt Refining (SPR) network that refines the internal structure of prompt vectors across multiple dimensions: (1) a Global Interaction Fusion Module captures bidirectional interactions across prompt layers, thereby mitigating sub-optimal adaptation from inconsistent guidance under missing modalities, (2) a Local Feature Refinement Module structures adjacent prompt vectors into coherent semantic units, leveraging local contextual relationships to maintain semantic integrity during modality absence, and (3) a Channel Feature Selection Module uses point-wise gating to adaptively suppress noise and enhance critical channels based on the specific missing modality. Using only 0.8% trainable parameters, SPR achieves significant improvements on three mainstream multimodal classification datasets. Notably, it surpasses state-of-the-art by 3.8% in F1-Macro on the MM-IMDB dataset, even at a 90% modality missing rate. Extensive experiments and in-depth ablations validate SPR's effectiveness and robustness under various missing conditions.
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Dissect and Prune: Enhancing Robustness in AI-Generated Image Detection
Dahye Kim, Jaehyun Choi, Hyun Seok Seong, Seongho Kim, Donghun Lee, Sungwon Yi, Jang-Ho Choi
While existing AI-generated image detectors report high performance, we identify that this is largely driven by a critical *prediction asymmetry*: a bias toward the real class that severely limits sensitivity to generated content, especially under standard post-processing operations such as compression and resizing. We hypothesize that this stems from the model's reliance on spurious features, distracting signals that obscure true generative artifacts. To address this, we propose DEAR (Dissect and Prune), which leverages inpainted images to identify and prune these interfering components. Specifically, we find that features strongly aligned to either inpainted or non-inpainted regions are less robust to post-processing. By measuring the alignment between channel activations and inpaint masks, DEAR removes features at both extremes, retaining only those that capture genuine generative artifacts. Experimental results demonstrate that our approach significantly enhances robustness against unseen generators and post-processing, effectively mitigating the prediction asymmetry. Our code is available at https://github.com/dahyedahye/dear.
****

Non-Stationary Online Structured Prediction with Surrogate Losses
Shinsaku Sakaue, Han Bao, Yuzhou Cao

Online structured prediction, including online classification as a special case, is the task of sequentially predicting labels from input features. In this setting, the *surrogate regret*—the cumulative excess of the actual target loss (e.g., the 0-1 loss) over the surrogate loss (e.g., the logistic loss) incurred by the best fixed estimator—has gained attention because it admits a finite bound independent of the time horizon T. However, such guarantees break down in *non-stationary* environments, where every fixed estimator may incur surrogate loss that grows linearly with T. To address this limitation, we obtain an upper bound of $F_T+O(1+P_T)$ on the cumulative target loss, where F_T is the cumulative surrogate loss of any comparator sequence and P_T is its *path length*. This bound depends on T only through F_T and P_T, thus offering stronger guarantees under non-stationarity. Our core idea is to combine the dynamic regret analysis of online gradient descent (OGD) with the *exploit-the-surrogate-gap* technique. This viewpoint sheds light on the usefulness of a Polyak-style learning rate for OGD, which systematically yields target-loss bounds and performs well empirically. We then extend our approach to broader settings beyond prior work via the *convolutional Fenchel–Young loss*. Finally, a lower bound shows that the dependence on F_T and P_T is tight.
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Decomposed On-Policy Distillation for Vision-Language Reasoning: Steering Gradients for Visual Grounding

Hee Suk Yoon, Eunseop Yoon, Jaehyun Jang, SooHwan Eom, Ji Woo Hong, Mark A. Hasegawa-Johnson, Qi Dai, Chong Luo, Chang D. Yoo

While on-policy distillation offers dense supervision for training small reasoning models, its optimization dynamics in the multimodal domain remain under-explored. In this work, we challenge the standard monolithic view of Vision-Language Model (VLM) distillation by mathematically decomposing the loss into two distinct components: the language prior and visual grounding. Our analysis uncovers that gradient vectors for these components are nearly orthogonal, indicating that the objective of aligning with the teacher's language distribution is geometrically independent from the objective of matching its visual perception. Consequently, standard optimization passively follows a suboptimal compromise trajectory that implicitly balances the two objectives. Hypothesizing that visual grounding constitutes the primary bottleneck for vision-language reasoning, we introduce Visual Gradient Steering (VGS), a method that dynamically reorients the update vector to prioritize the visual subspace. Experimental results on multiple distillation settings and complex multimodal benchmarks demonstrate that VGS significantly outperforms the standard monolithic formulation of on-policy distillation, achieving superior grounding with minimal training overhead.
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Dynamic TMoE: A Drift-Aware Dynamic Mixture of Experts Framework for Non-Stationary Time Series Forecasting

Jiawen Zhu, Shuhan Liu, Di Weng, Yingcai Wu

Non-stationary time series forecasting is challenged by evolving distribution shifts that static models struggle to capture. While Mixture-of-Experts (MoE) architectures offer a promising paradigm for decoupling complex drift patterns, existing approaches are limited by fixed expert pools and memoryless routing, hampering their ability to adapt to abrupt regime shifts. To address this, we propose **Dynamic TMoE**, a framework that unifies architectural evolution with temporal continuity during learning phase. By detecting distribution shifts via Maximum Mean Discrepancy (MMD), we dynamically instantiate heterogeneous experts and prune redundant ones to optimize capacity. Additionally, a temporal memory router leverages recurrent states and an anomaly repository to ensure stable, context-aware expert selection without requiring test-time updates. Experiments on nine benchmarks demonstrate state-of-the-art performance, reducing MSE by 10.4\% and MAE by 7.8\%. Code is available at https://github.com/andone-07/Dynamic-TMoE.
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Two-dimensional quantization for geometry-aware audio coding

Tal Shuster, Eliya Nachmani

Recent neural audio codecs have achieved impressive reconstruction quality, typi

cally relying on quantization methods such as Residual Vector Quantization (RVQ) , Vector Quantization (VQ) and Finite Scalar Quantization (FSQ). However, these quantization techniques limit the geometric structure of the latent space, make it harder to capture correlations between features leading to inefficiency in re presentation learning, codebook utilization and token rate. In this paper we int roduce Two-Dimensional Quantization (Q2D2), a quantization scheme in which featu re pairs are projected onto structured 2D grids, such as hexagonal, rhombic, or rectangular tiling and quantized to the nearest grid values, yielding an implici t codebook defined by the product of grid levels, with codebook sizes comparable to conventional methods. Despite its simple geometric formulation, Q2D2 improve s audio compression efficiency, with low token rates and high codebook utilizati on while maintaining state of the art reconstruction quality. Specifically, Q2D2 achieves competitive to superior performance in various objective and subjectiv e reconstruction metrics, across extensive experiments in speech, audio and musi c domains compared to state of the art models. Comprehensive ablation studies fu rther confirm the effectiveness of our design choices.
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Optimal Bayesian Stopping for Efficient Inference of Consistent LLM Answers
Jingkai Huang, Will Ma, Zhengyuan Zhou
A simple strategy for improving LLM accuracy, especially in math and reasoning p roblems, is to sample multiple responses and submit the answer most consistently reached. In this paper we leverage Bayesian prior information to save on sampli ng costs, stopping once sufficient consistency is reached. Although the exact po sterior is computationally intractable, we further introduce an efficient ``L- aggregated'' stopping policy that tracks only the $L-1$ most frequent answer cou nts. Theoretically, we prove that $L=3$ is all you need: this coarse approximati on is sufficient to achieve asymptotic optimality, and strictly dominates prior- free baselines, while having a fast posterior computation. Empirically, this ide ntifies the most consistent (i.e., mode) LLM answer and achieves similar answer accuracy using fewer samples.
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SpeedCP: Fast Kernel-based Conditional Conformal Prediction
Yating Liu, Yeo Jin Jung, Zixuan Wu, Sowon Jeong, Claire Donnat
Conformal prediction provides distribution-free prediction sets with finite-samp le conditional guarantees. RKHS-based frameworks—while promising for complex cov ariate shifts—suffer from prohibitive computational costs. To guarantee conditio nal validity under such shifts while ensuring feasibility, we build upon the fra mework of Gibbs et al. (2025) by introducing a stable and efficient algorithm th at computes the full solution path of the regularized RKHS conformal optimizatio n problem, at essentially the same cost as a single kernel quantile fit. Our app roach provides simultaneous hyperparameter tuning which provides smoothness cont rol and data-adaptive calibration. To extend the method to high-dimensional sett ings, we further integrate our approach with low-rank latent embeddings that cap ture conditional validity in a data-driven latent space. Empirically, our method provides reliable conditional coverage across a variety of modern black-box pre dictors, improving the interval length of Gibbs et al. (2025) by 30%, while achi eving a 40-fold speedup.
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Scaling Laws and Architectural Frontiers in Metagenomic Foundation Models
Geraldene Munsamy, Gavin Ayres, Jérémie DONA, Carla Greco, Daniel Anderson, Srij ani Sridhar, William Chow, Aaron W Kollasch, Robert J. Pecoraro, Tanggis Bohnuud , Keith Kam, Gus Minto-Cowcher, Marcus H Y Leung, Hassan Sirelkhatim, John St. J ohn, Ali Taghibakhshi, Tyler C. Shimko, Jared Wilber, Timur Rvachov, Saee Gopal Paliwal, Eduardo Calleja, Noelia Ferruz, Kevin K Yang, Philipp Lorenz, Francesco Farina
Foundation models for genomics have the potential to revolutionize therapeutic d esign, yet the optimal architectural choices for modeling
the vast and diverse distribution of metagenomic data remain under-explored. In this work, we present the machine learning methodology behind EDEN, a family of metagenomic foundation models scaled up to 28 billion parameters and trained on

9.7 trillion nucleotide tokens. We provide a systematic empirical study of architectural trade-offs between autoregressive Transformers (Llama-style), State-Space Models (Mamba), and Long-convolutional architectures (Hyena) for nucleotide-level modeling. Contrary to recent trends favoring linear-time sequence models for long-range biological data, we demonstrate that the Llama architecture exhibits superior scaling efficiency and semantic retrieval capabilities as the model capacity grows. We derive a set of quality-aware scaling laws for metagenomics, showing how model performance follows predictable power-law behavior across three orders of magnitude in parameters and data. Through extensive benchmarking, spanning unsupervised zeroshot fitness prediction, semantic completion, and gene recovery, we establish a blueprint for scaling biological foundation models and provide empirical evidence demonstrating why Transformer-based architectures define the current frontier.
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Adaptive Momentum and Nonlinear Damping for Neural Network Training
Aikaterini Karoni, Rajit Rajpal, Benedict J. Leimkuhler, Gabriel Stoltz
Momentum Stochastic Gradient Descent (mSGD) relies on a fixed momentum coefficient shared across all parameters, failing to account for the heterogeneous structure of modern loss landscapes. In this work, we adopt a continuous-time formulation to introduce individual, adaptive momentum coefficients regulated by the kinetic energy of each model parameter. This mechanism automatically adjusts to evolving training dynamics to maintain stability without sacrificing convergence speed. We demonstrate that this adaptive friction is inextricably linked to cubic damping, a suppression mechanism from structural dynamics. We additionally introduce two optimization schemes by augmenting the continuous dynamics of mSGD and Adam with a cubic damping term. Empirically, our methods demonstrate robustness and match or outperform Adam on training ViT, BERT, and GPT2 tasks where mSGD typically struggles. We further provide theoretical results establishing the exponential convergence of the proposed schemes.
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Rank-Aware Spectral Bounds on Attention Logits for Stable Low-Precision Training
Seyed Morteza Emadi
Attention scores in transformers are bilinear forms $S_{ij} = x_i^\top M x_j / \sqrt{d_h}$ whose maximum magnitude governs overflow risk in low-precision training. We derive a \emph{rank-aware concentration inequality}: when the interaction matrix $M = W^Q W^{K\top}$ has rank $r \ll d$, tail probabilities for $\max_{i,j}|S_{ij}|$ decay as $\exp(-d^{2}\alpha^{2}/(\gamma r))$ rather than $\exp(-d\alpha^{2})$, where $\gamma > 1$. For transformer attention where $r = d_h$, this yields 8--$28\times$ tighter concentration than rank-agnostic bounds in modern architectures. We apply this result to FP8 training, deriving \emph{geometry-aware scale factors} that provide principled overflow guarantees without observing activations. The method computes per-layer scales from the spectral norm $\|W^Q W^{K\top}\|_2$ via implicit power iteration, includes a grouped query attention formulation that avoids key expansion, and remains compatible with fused attention kernels. Across GPT-2 XL to Llama-2-70B, geometry-aware scaling eliminates overflows in transient scenarios where delayed scaling fails, while achieving comparable downstream MMLU accuracy.
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Tailoring Strictly Proper Scoring Rules for Downstream Tasks: An Application to Causal Inference
Roman Plaud, Alexandre Perez-Lebel, Antoine Saillenfest, Thomas Bonald, Marine Le Morvan, Gaël Varoquaux, Matthieu Labeau
Probabilistic models are typically trained using task-agnostic objectives like log-loss, which can lead to significant errors in downstream estimation. This disconnect is especially critical in Inverse Probability Weighting (IPW) for causal inference, where propensity score errors near 0 and 1 often lead to high bias and variance. We propose a principled framework for deriving task-specific strictly proper scoring rules by matching the local curvature of the downstream error metric. We apply this to the Average Treatment Effect (ATE) estimation, deriving a closed-form loss and its corresponding canonical probability mapping that

can be readily integrated with any model like a neural network or a gradient boosting algorithm. Extensive evaluations on causal inference benchmarks demonstrate that our tailored objective consistently outperforms standard likelihood-based and covariate-balancing approaches.

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Platonic Transformers: A Solid Choice For Equivariance

Mohammad Mohaiminul Islam, Rishabh Anand, David R Wessels, Friso de Kruiff, Thijs P. Kuipers, Rex Ying, Clara I. Sánchez, Sharvaree Vadgama, Georg Bökman, Erik J Bekkers

While widespread, Transformers lack inductive biases for geometric symmetries common in science and computer vision. Existing equivariant methods often sacrifice the efficiency and flexibility that make Transformers so effective through complex, computationally intensive designs. We introduce the Platonic Transformer to resolve this trade-off. By defining attention relative to reference frames from the Platonic solid symmetry groups, our method induces a principled weight-sharing scheme. This enables combined equivariance to continuous translations and Platonic symmetries, while preserving the exact architecture and computational cost of a standard Transformer. Furthermore, we show that this attention is formally equivalent to a dynamic group convolution, which reveals that the model learns adaptive geometric filters and enables a *highly scalable, linear-time convolutional variant*. Across diverse benchmarks in computer vision (CIFAR-10), 3D point clouds (ScanObjectNN), and molecular dynamics, property prediction and generation (OMol25, ProteinMD, QM9), the Platonic Transformer achieves competitive performance by leveraging these geometric constraints at no additional cost.

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How RL Unlocks the Aha Moment in Geometric Interleaved Reasoning

Xiangxiang Zhang, Caijun Jia, Siyuan Li, he dingyu, Xiya Xiong, Zheng Sun, Honghao He, Yuchen Wu, Bihui Yu, Linzhuang Sun, Cheng Tan, Jingxuan Wei

Solving complex geometric problems inherently requires interleaved reasoning: a tight alternation between constructing diagrams and performing logical deductions. Although recent Multimodal Large Language Models (MLLMs) have demonstrated strong capabilities in visual generation and plotting, we identify a counter-intuitive and underexplored phenomenon. Naively applying Supervised Fine-Tuning (SFT) on interleaved plot–solution data leads to a substantial degradation in reasoning performance compared to text-only baselines. We argue that this failure stems from a fundamental limitation of SFT, which primarily induces distributional alignment: the model learns to reproduce the surface format of interleaved plotting but fails to internalize the causal dependency between the generated plot and reasoning steps. To overcome this limitation, we propose Faire (**F**unctional **a**lignment for **i**nterleaved **re**asoning), a reinforcement learning framework that enforces three casual constraints to move beyond superficial imitation toward functional alignment. Extensive experiments show that Faire induces a qualitative shift in model behavior in which the plotting is effectively internalized, yielding competitive performance on challenging geometric reasoning benchmarks.

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PPI Candidate Ranking: Large-Scale Evaluation of a Domain Knowledge–Guided Pipeline

Maria Emilia Russo, Federico Di Valerio, Alessia Borghini, Alessio Ragno, Roberto Capobianco

Computational approaches have become central to Protein–Protein Interaction (PPI) research, complementing experimental techniques that remain costly and incomplete. While modern deep learning methods capture diverse biological signals and hold promise in expanding the known interactome, empirical validation remains a critical bottleneck due to its long and expensive procedures. To address this challenge, we introduce the problem of PPI candidate ranking, aiming to prioritize interactions for experimental testing. We propose a novel framework that leverages domain knowledge through interpretability-guided ranking and further refines prioritization by integrating complementary sources of evidence, including interaction scores, structural plausibility, and biomedical language features. Evalua

tions on a large-scale dataset constructed from successive STRING releases demon strate that our approach yields significant improvements over two state-of-the-a rt PPI prediction models, providing more accurate and biologically coherent rank ings.
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SURGE: Approximation and Training Free Particle Filter for Diffusion Surrogate
Lifu Wei, Yinuo Ren, Naichen Shi, Yiping Lu
Data assimilation (DA) tackles the sequential estimation of a dynamical system’s latent state from noisy, partial observations. In this paper, we study DA in a setting where the system dynamics are represented by a pretrained diffusion mode l used as a surrogate forecaster. We focus on how to integrate incoming observat ions into the diffusion surrogate’s predictions to support continuous state corr ection and progressively refining the estimated trajectory over time. After rece iving noisy observations, the diffusion model is guided using the observation li kelihood to steer the generation process toward observation-consistent states. H owever, such guidance does not guarantee sampling from the true posterior. Motiv ated by particle filtering methods, we represent the posterior distribution usin g an ensemble of particles. We perform Sequential Monte Carlo over diffusion tra jectories, working with their path measure. We compute importance weights for ge nerated particles and resample to focus on trajectories consistent with the obse rvations. This procedure corrects the generation dynamics, drives the particle a pproximation toward the desired posterior and leads to an approximation-free par ticle filtering method that rigorously fuses observational data with diffusion m odel simulations.
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Principled Synthetic Data Enables the First Scaling Laws for LLMs in Recommendat ion
Benyu Zhang, Qiang Zhang, Jianpeng Cheng, Hong-You Chen, Qifei wang, Wei Sun, Sh en Li, Jia Li, Jiahao Wu, Xiangjun Fan, Hong Yan
Large Language Models (LLMs) represent a promising frontier for recommender syst ems, yet their development has been impeded by the absence of predictable scalin g laws, which are crucial for guiding research and optimizing resource allocatio n. We hypothesize that this may be attributed to the inherent noise, bias, and i ncompleteness of raw user interaction data in prior continual pre-training (CPT) efforts. This paper introduces a novel, layered framework for generating high-q uality synthetic data that circumvents such issues by creating a curated, pedago gical curriculum for the LLM. We provide powerful, direct evidence for the utili ty of our curriculum by showing that standard sequential models trained on our p rincipled synthetic data significantly outperform ($+130\%$ on recall@100 for Sa sRec) models trained on real data in downstream ranking tasks, demonstrating its superiority for learning generalizable user preference patterns. Building on th is, we empirically demonstrate, for the first time, robust power-law scaling for an LLM that is continually pre-trained on our high-quality, recommendation-spec ific data. Our experiments reveal consistent and predictable perplexity reductio n across multiple synthetic data modalities. These findings establish a foundati onal methodology for reliable scaling LLM capabilities in the recommendation dom ain, thereby shifting the research focus from mitigating data deficiencies to le veraging high-quality, structured information.
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A Provable Expressiveness Hierarchy in Hybrid Linear-Full Attention
Xiaowei Ye, Xiaoyu He, Chao Liao, Chen Wu, Pinyan Lu
Transformers serve as the foundation of most modern large language models.
To mitigate the quadratic complexity of standard full attention, various efficie nt attention mechanisms, such as linear and hybrid attention, have been develope d.
A fundamental gap remains: their expressive power relative to full attention lac ks a rigorous theoretical characterization.
In this work, we theoretically characterize the performance differences among th ese attention mechanisms.
Our theory applies to all linear attention variants that can be formulated as a

recurrence, including Mamba, DeltaNet, etc.
Specifically, we establish an expressiveness hierarchy: for the sequential function composition-a multi-step reasoning task that must occur within a model's forward pass, an $(L+1)$-layer full attention network is sufficient, whereas any hybrid network interleaving $L-1$ layers of full attention with a substantially larger number (2^{3L^2}) of linear attention layers cannot solve it.
This result demonstrates a clear separation in expressive power between the two types of attention.
Our work provides the first provable separation between hybrid attention and standard full attention, offering a theoretical perspective for understanding the fundamental capabilities and limitations of different attention mechanisms.
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Parameter-Masked Decoupled Optimization for Cross-Domain Class-Incremental Learning
Ziqi Gu, Chunyan Xu, Yangguang Liu, Wenxuan Fang, Baotong Su, Tong Zhang, Dan Wang, Zhen Cui
Cross-domain class-incremental learning (CD-CIL) requires models to continuously acquire new classes across shifting domains while retaining previously learned knowledge. Existing approaches often entangle what to update with how to update, resulting in unstable adaptation and severe forgetting under domain shifts.
Inspired by the hippocampal learning mechanism that separates rapid adaptation from stable consolidation, we propose Parameter-Masked Decoupled Optimization (PMDO) that disentangles what knowledge is adapted from how learning proceeds in cross-domain class-incremental learning.
We introduce a domain-aware knowledge decoupler that selectively adapts domain-relevant shared parameters, constraining incremental updates while preserving prior representations.
To regulate how learning proceeds, we further design a stability-aware trajectory regulation that guides optimization along transferable and stable optimization trajectories, thereby reducing interference across domain transitions.
PMDO enables effective cross-domain adaptation while mitigating catastrophic forgetting and maintaining long-term learnability.
Extensive experiments across multiple benchmarks demonstrate the effectiveness of PMDO and its superiority over state-of-the-art methods.
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Divergence Decoding: Inference-Time Unlearning via Auxiliary Models
Humzah Merchant, Bradford Levy
Large Language Models (LLMs) frequently memorize sensitive training data thereby creating significant privacy and copyright risks. Addressing these risks, i.e., removing such knowledge from an existing model checkpoint, has proven challenging as many unlearning methods lead to catastrophic utility loss or are ineffective for complex queries. We introduce **Divergence Decoding (DD)**, a mechanism that uses small auxiliary models to steer the logits of the LLM away from specific data during inference. Training these models is straight forward, i.e., we use standard pre-training and fine-tuning setups. We find the method decisively outperforms state-of-the-art (SOTA) baselines on unlearning benchmarks across a variety of model and training dataset scales consistent with DD being an effective and inexpensive solution to unlearning. We then demonstrate that this steered distribution can be trivially distilled back into the base model. Since the method is generally applicable to any probabilistic model, we explore its efficacy outside of text generation and find evidence of generalization to the domain of images.
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Context-free Recognition with Transformers
Selim Jerad, Anej Svete, Sophie Hao, Ryan Cotterell, William Merrill
Transformers excel empirically on tasks that process well-formed inputs according to some grammar, such as natural language and code. However, it remains unclear how they can process grammatical syntax. In fact, under standard complexity conjectures, standard transformers cannot recognize context-free languages (CFLs), a canonical formalism to describe syntax, or even regular languages, a subclass

of CFLs. Past work has shown that $\mathcal{O}(\log(N))$ looping layers (w.r.t. input length N) allow transformers to recognize regular languages, but the question of context-free recognition with looped transformers remained open. In this work, we show that looped transformers with $\mathcal{O}(\log(N))$ looping layers and $\mathcal{O}(N^6)$ padding symbols can recognize all CFLs. However, training and inference with $\mathcal{O}(N^6)$ padding symbols is potentially impractical. Fortunately, we show that, for natural subclasses such as unambiguous CFLs, the recognition problem on transformers becomes more tractable, requiring $\mathcal{O}(N^3)$ padding. Empirically, looped and padded transformers perform better than fixed-depth transformers in recognizing CFLs. Overall, our results shed light on the intricacy of CFL recognition by transformers: while general recognition may require an intractable amount of padding, natural constraints such as unambiguity yield efficient recognition algorithms.

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V1: Unifying Generation and Self-Verification for Parallel Reasoners

Harman Singh, Xiuyu Li, Kusha Sareen, Monishwaran Maheswaran, Sijun Tan, Xiaoxia Wu, Junxiong Wang, Alpay Ariyak, Qingyang Wu, Samir Khaki, Rishabh Tiwari, Long Lian, Yucheng Lu, Boyi Li, Alane Suhr, Ben Athiwaratkun, Kurt Keutzer

Test-time scaling for complex reasoning tasks shows that leveraging inference-time compute, by methods such as independently sampling and aggregating multiple solutions, results in significantly better task outcomes. However, a critical bottleneck is *verification*: sampling is only effective if correct solutions can be reliably identified among candidates. While existing approaches typically evaluate candidates independently via scalar scoring, we demonstrate that models are substantially stronger at **pairwise verification**. Leveraging this insight, we introduce **V■**, a framework that unifies generation and verification through efficient pairwise ranking. V■ comprises two components: **V■-Infer**, an uncertainty-guided algorithm using a tournament-based ranking that dynamically allocates self-verification compute to candidate pairs whose relative correctness is most uncertain; and **V■-PairRL**, an RL framework that **jointly trains** a single model as both generator and pairwise self-verifier, ensuring the verifier adapts to the generator's evolving distribution. On competitive code generation, SWE and terminal-based agents, math reasoning, and medical QA benchmarks, V■-Infer improves Pass@1 by up to 10% over pointwise verification and outperforms SoTA aggregation-based test-time scaling while being more efficient. Furthermore, V■-PairRL achieves test-time scaling gains of 3.8–10pp over standard RL and pointwise joint training on code, math, and non-verifiable domains, and improves base Pass@1 by up to 8.7pp over standard RL. [Project page & code](https://harmandotpy.github.io/v1-verification/).

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XR-1: Towards Versatile Vision-Language-Action Models via Learning Unified Vision-Motion Representations

Shichao Fan, Kun Wu, Zhengping Che, Xinhua Wang, Di Wu, Fei Liao, Ning Liu, Yixue Zhang, Zhen Zhao, Zhiyuan Xu, Meng Li, Qingjie Liu, Shanghang Zhang, Min Wan, Jian Tang

Recent progress in large-scale robotic datasets and vision-language models (VLMs) has advanced research on vision-language-action (VLA) models. However, existing VLA models still face two fundamental challenges: (\textit{i}) producing precise low-level actions from high-dimensional observations, (\textit{ii}) bridging domain gaps across heterogeneous data sources, including diverse robot embodiments and human demonstrations. Existing methods often encode latent variables from either visual dynamics or robotic actions to guide policy learning, but they fail to fully exploit the complementary multi-modal knowledge present in large-scale, heterogeneous datasets. In this work, we present \textbf{XR-1}, a novel framework for versatile and scalable VLA learning across diverse robots, tasks, and environments.

At its core, XR-1 introduces the \emph{Unified Vision-Motion Codes (UVMC)}, a discrete latent representation learned via a dual-branch VQ-VAE that jointly encodes visual dynamics and robotic motion. UVMC addresses these challenges by (\textit{i}) serving as an intermediate representation between the observations and a

ctions, and (\textit{ii}) aligning multimodal dynamic information from heterogeneous data sources to capture complementary knowledge. To effectively exploit UVMC, we propose a \emph{three-stage training paradigm}: (\textit{i}) self-supervised UVMC learning, (\textit{ii}) UVMC-guided pretraining on large-scale cross-embodiment robotic datasets, and (\textit{iii}) task-specific post-training. We validate XR-1 through extensive real-world experiments with more than 12,000 rollouts on six different robot embodiments, spanning over 120 diverse manipulation tasks. XR-1 consistently outperforms state-of-the-art baselines such as π_0 and GR00T-N1.5 while demonstrating strong generalization to novel objects, background variations, distractors, and illumination changes. Our project is at \href{https://xr-1-vla.github.io/}{https://xr-1-vla.github.io/}.
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ConvexBench: Can LLMs Recognize Convex Functions?
Yepeng Liu, Yu Huang, Yu-Xiang Wang, Yingbin Liang, Yuheng Bu
Convexity recognition plays a central role in many optimization, control, and learning problems. However, the ability of Large Language Models (LLMs) to identify this property in symbolic expressions remains unexamined. We introduce ConvexBench, a scalable and mechanically verifiable benchmark for testing whether LLMs can determine the convexity of a symbolic objective under deep functional composition. Experiments on frontier LLMs reveal a sharp \textit{compositional reasoning gap}: performance degrades rapidly with increasing depth, dropping from an F1-score of 1.0 at depth 2 to approximately 0.2 at depth 100. Inspection of models' reasoning traces indicates two failure modes: \textit{parsing failure} and \textit{lazy reasoning}. To address these limitations, we propose an agentic divide-and-conquer framework that (i) offloads parsing to an external tool to construct an abstract syntax tree (AST) and (ii) enforces recursive reasoning over each intermediate sub-expression with focused context. This framework reliably mitigates deep-composition failures, achieving substantial performance improvement at large depths (e.g., F1-Score $= 1.0$ at depth 100).
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A Language-Guided Bayesian Optimization for Efficient LoRA Hyperparameter Search
Baek Seong-Eun, Lee Jung-Mok, Kim Sung-Bin, Tae-Hyun Oh
Fine-tuning Large Language Models (LLMs) with Low-Rank Adaptation (LoRA) offers a resource-efficient way to personalize or specialize. However, LoRA is highly sensitive to hyperparameter choices, and exhaustive hyperparameter search is computationally expensive. To address this, we propose a Bayesian Optimization (BO) framework that leverages the domain knowledge of pre-trained LLMs to efficiently search for LoRA hyperparameters. Our approach repurposes a pre-trained LLM as a discrete-to-continuous mapping module to link hyperparameters and their domain knowledge to a continuous vector space, where BO is conducted. We design and control the mapping via language prompting, providing a domain-aware textual prompt that describes the relationships among hyperparameters and their respective roles. This allows us to explicitly inject domain knowledge about LoRA into the LLM in natural language. We also introduce an additional learnable token to capture residual information that is difficult to describe linguistically in the prompt. This aids BO to sample more high-performing hyperparameters. In addition, by leveraging the strong correlation observed between the performance obtained from full and subset training datasets in LoRA training regimes, we introduce proxy training and evaluation using a data subset. This significantly improves the efficiency of our method. We demonstrate that our hyperparameter, discovered with only about 30 iterations, achieves more than 20% performance improvement over standard hyperparameters found from about 45,000 combinations. Project page: https://baekseongeun.github.io/lora-bo/
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Cache Coherent Resampling for Efficient Test Time Scaling in LLM Reasoning via Adaptive Sequential Monte Carlo
KE WANG, ZEHAO YU, Luwei Wang, Yongchao Huang
Recent work shows that chain based sampling for power shaped trajectory distributions can deliver large test time gains from a fixed base LLM and can approach RL trained reasoners such as GRPO. Deployment is the bottleneck. Autoregressive M

etropolis Hastings is inherently serial, limits GPU utilization, and exhibits ex treme tail latency at high budgets, reaching p95 $=1318$s on MATH500 at $128\tim es$. We propose Adaptive Sequential Monte Carlo (ASMC), a parallel particle infe rence method that targets power shaped trajectory distributions while adapting p article populations to problem hardness. To make resampling practical for Transf ormers, we introduce cache coherent resampling, which realizes ancestry updates by reordering KV caches and other particle bound tensors, avoiding prefix recomp utation. On MATH500 at the same budget, ASMC attains 80.6\% exact-match accuracy with p95 = 73.7s, substantially reducing the tail latency of sequential MCMC an d providing additional high-accuracy operating points beyond the saturation of b est-of-n. We further analyze particle degeneracy and find that collapse severity , measured by low $\mathrm{ESS}_{\min}/N$, strongly predicts failures, while sen sitivity to the resampling scheme is limited.
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Generative Augmented Inference
Cheng Lu, Mengxin Wang, Dennis Zhang, Heng Zhang
Large language models enable inexpensive AI-generated annotations, but using the m reliably for causal inference remains challenging. Naively pooling AI and huma n data induces bias, while existing methods such as Prediction-Powered Inference (PPI; Angelopoulos et al., 2023) treat AI outputs as proxies of true labels -- an assumption often violated for generative model outputs in practice. We propos e Generative Augmented Inference (GAI), a framework that treats AI outputs as ge neral, potentially high-dimensional informative features for learning human labe ls rather than as surrogates. GAI flexibly models this relationship using nonpar ametric methods, enabling consistent estimation and valid inference from combine d human and AI data. We establish asymptotic normality and show that, under rand om labeling, GAI strictly improves asymptotic efficiency over human-data-only es timation whenever AI outputs are informative for true labels. Empirical studies on real-world datasets demonstrate that GAI significantly reduces estimation err or and improves confidence interval quality across diverse generative data sourc es relative to human-only and PPI-based estimation.
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SIPO: Stabilized and Improved Preference Optimization for Aligning Diffusion Mod els
Xiaomeng Yang, Mengping Yang, Junyan Wang, Zhijian Zhou, Zhiyu Tan, Hao Li
Preference learning has garnered extensive attention as an effective technique f or aligning diffusion models with human preferences in visual generation tasks. However, existing alignment approaches such as Diffusion-DPO suffer from two fun damental challenges: training instability caused by high gradient variances at v arious timesteps and high parameter sensitivities, and off-policy bias arising f rom the discrepancy between the optimization data and the policy model's distrib ution. Our first contribution is a systematical analysis of the diffusion trajec tories across different timesteps and identify that the instability primarily or iginates from early timesteps with low importance weights. To address these issu es, we propose SIPO, a Stabilized and Improved preference Optimization framework for aligning diffusion models with human preferences. Concretely, a key gradien t, \emph{i.e.,} DPO-C&M is introduced to facilitate stabilize training by clippi ng and masking uninformative timesteps. Followed by a timestep aware importance re-weighting paradigm to fully correct off-policy bias and emphasize informative updates throughout the alignment process. Extensive experiments on various base line models, including image generation models on SD1.5, SDXL, and video generat ion models CogVideoX-2B, CogVideoX-5B, and Wan2.1-1.3B, demonstrate that our SIP O consistently promotes stabilized training and outperforms existing alignment m ethods, with meticulous adjustments on parameters. Overall, these results highli ght the importance of timestep-aware alignment and and provide valuable guidelin es for improved preference optimization in diffusion models.
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RLAnything: Forge Environment, Policy, and Reward Model in Completely Dynamic RL System
Yinjie Wang, Tianbao Xie, Ke Shen, Mengdi Wang, Ling Yang

The quality of both the environment and the reward model fundamentally governs t
he effectiveness of reinforcement learning. Accordingly, we propose RLAnything,
a reinforcement learning framework that dynamically optimizes each component thr
ough closed-loop optimization, amplifying learning signals and strengthening the
 overall system.
Specifically, the policy is trained with integrated feedback from step-wise and
outcome signals, while the reward model is jointly optimized via consistency fee
dback, which in turn further improves policy training. Moreover, our theory-moti
vated automatic environment adaptation improves training for both the reward and
 policy models by leveraging critic feedback from each, enabling learning from e
xperience.
Empirically, each added component consistently improves the overall system, and
RLAnything yields substantial gains in practical applications, boosting Qwen3-VL
-8B-Thinking by 8.5% on OSWorld and Qwen2.5-7B-Instruct by 21.2% and 12.1% on Al
fWorld and LiveBench, respectively.
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Graph Rewiring based on Flow Alignment for Improving Fluid Simulation
Zenong Li, Wei Xian Lim, Wai Lee Chan, Adams Wai-Kin Kong
To overcome the computational burden of traditional computational fluid dynamics
 (CFD), graph neural network-based (GNN) learned simulators have attracted growi
ng interest because they naturally operate on CFD meshes. However, classical GNN
s only exchange information between neighbouring nodes, limiting flow prediction
 accuracy. While graph rewiring can improve information propagation, existing me
thods are mostly designed for generic graphs, and PIORF relies on long-range con
nections for fluid simulation. In this work, we show that simply connecting all
2-hop nodes can already achieve competitive performance with PIORF, raising ques
tions about the necessity of distant rewiring. Motivated by fluid transport prin
ciples, we propose Flow Alignment Rewiring (FLARE), a simple and efficient local
 rewiring method that connects 2-hop nodes only when their relative direction al
igns with the input flow direction. Hence, FLARE is a physics-informed local rew
iring method, different from PIORF and well-aligned with fluid physics. Extensiv
e numerical experiments on flows over a cylinder, single-, and tandem-airfoil un
der different flow conditions and deep network architectures demonstrate that FL
ARE outperforms PIORF and various 2-hop rewiring approaches by a significant mar
gin.
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Translation Heads: Disentangling meaning from language in LLM-based machine tran
slation
Théo Lasnier, Armel Randy Zebaze, Djamé Seddah, Rachel Bawden, Benoît Sagot
Mechanistic Interpretability (MI) seeks to explain how neural networks implement
 their capabilities, but the scale of Large Language Models (LLMs) has limited p
rior MI work in Machine Translation (MT) to word-level analyses. We study senten
ce-level MT from a mechanistic perspective by analyzing attention heads to under
stand how LLMs internally encode and distribute translation functions. We decomp
ose MT into two subtasks: producing text in the target language (i.e. target lan
guage identification) and preserving the input sentence’s meaning (i.e. sentence
 equivalence). Across three families of open-source models and 20 translation di
rections, we find that distinct, sparse sets of attention heads specialize in ea
ch subtask. Based on this insight, we construct subtask-specific steering vector
s and show
that modifying just 1% of the relevant heads enables instruction-free MT perform
ance comparable to instruction-based prompting, while ablating these heads selec
tively disrupts their corresponding translation functions.
****

Adaptive Probe-based Steering for Robust LLM Jailbreaking
Junxi Chen, Junhao Dong, Xiaohua Xie
Recent work has demonstrated the potential of contrastive steering for jailbreak
ing Large Language Models (LLMs). However, existing methods rely on limited and
inherently biased contrastive prompts and require laborious manual tuning of ste
ering strength, limiting their robustness and effectiveness. In this paper, we l

everage the idea of model extraction to guide the learned steering vectors to ap
proximate the ideal one and propose tuning the steering strength adaptively base
d on contrastive activations' statistics. Experiments demonstrate that our metho
d notably improves the effectiveness and robustness of probe-based steering, wit
hout any extra contrastive prompts or laborious manual tuning. Being an attack p
aper, this paper focuses on revealing the breakdown of fortified LLMs, raising t
he average harmfulness score from 6\% to 70\%. Our code is available at \url{htt
ps://github.com/fhdnskfbeuv/adaptiveSteering}.
****
OneSearch: A Preliminary Exploration of the Unified End-to-End Generative Framew
ork for E-commerce Search
Ben Chen, Xian Guo, Siyuan Wang, Zihan Liang, Yufei Ma, Yue Lv, Chenyi Lei, Yuqi
ng DING, Wenwu Ou, Han Li, Kun Gai
Traditional e-commerce search systems employ multi-stage cascading architectures
 that suffer from fragmented computation and optimization objective collisions a
cross stages, ultimately limiting their performance ceiling.
We propose \textbf{OneSearch}, the first industrial-deployed end-to-end generati
ve framework for e-commerce search, featuring three key innovations: (1) Keyword
-enhanced Hierarchical Quantization Encoding to preserve hierarchical semantics
and distinctive item attributes while maintaining strong query-item relevance co
nstraints;
(2) multi-view user behavior sequence injection that constructs behavior-driven
user IDs and incorporates both explicit short-term and implicit long-term sequen
ces;
and (3) a Preference-Aware Reward System with multi-stage supervised fine-tuning
 and adaptive reward-weighted ranking to capture fine-grained user preferences.
Extensive offline evaluations demonstrate its superior performance, while online
 A/B tests achieve statistically significant improvements: +1.67\% item CTR, +2.
40\% buyer, and +3.22\% order volume. OneSearch reduces operational expenditure
by 75.40\%, improves Model FLOPs Utilization from 3.26\% to 27.32\%, and has bee
n successfully deployed across multiple search scenarios in Kuaishou, serving mi
llions of users daily.
Code is in https://github.com/benchen4395/onesearch-family.
****
Efficient Tail-Aware Generative Optimization via Flow Model Fine-Tuning
Zifan Wang, Riccardo De Santi, Xiaoyu Mo, Michael M. Zavlanos, Andreas Krause, K
arl Henrik Johansson
Fine-tuning pre-trained diffusion and flow models to optimize downstream utiliti
es is central to real-world deployment. Existing entropy-regularized methods pri
marily maximize expected reward, providing no mechanism to shape tail behavior.
However, tail control is often essential: the lower tail determines reliability
by limiting low-reward failures, while the upper tail enables discovery by prior
itizing rare, high-reward outcomes. In this work, we present Tail-aware Flow Fin
e-Tuning (TFFT), a principled and efficient distributional fine-tuning algorithm
 based on the Conditional Value-at-Risk (CVaR). We address two distinct tail-sha
ping goals: right-CVaR for seeking novel samples in the high-reward tail and lef
t-CVaR for controlling worst-case samples in the low-reward tail. Unlike prior
approaches that rely on non-linear optimization, we leverage the variational dua
l formulation of CVaR to decompose it into a decoupled two-stage procedure: a li
ghtweight one-dimensional threshold optimization step, and a single entropy-regu
larized fine-tuning process via a specific pseudo-reward. This decomposition ach
ieves CVaR fine-tuning efficiently with computational cost comparable to standar
d expected fine-tuning methods. We demonstrate the effectiveness of TFFT across
illustrative experiments, high-dimensional text-to-image generation, and molecul
ar design.
****
PULSE: Generative Phase Evolution for Non-Stationary Time Series Forecasting
Yangyou Liu, Zezhi Shao, Xinyu Chen, Hu Chen, Fei Wang, Yuankai Wu
Time series forecasting under non-stationarity faces a fundamental tension betwe
en capturing stable representations and adapting to distribution shifts. Existin

g methods implicitly rely on static historical assumptions, leading to a critica l failure mode we term *Phase Amnesia*, where models become blind to the evolvin g global context. To resolve this, we formalize non-stationary dynamics through three physical hypotheses: wold decomposition, dynamical phase evolution, and he teroscedastic manifold generation. These principles inspire **PULSE**, a physics -informed, plug-and-play framework adopting a *Disentangle--Evolve--Simulate* de sign philosophy. Specifically, PULSE utilizes phase-anchored disentanglement to resolve optimization interference caused by dominant trends, employs a Phase Rou ter to actively generate future trajectories, and introduces Statistic-Aware Mix up (SAM) to ensure robustness against out-of-distribution volatility. Empiricall y, PULSE enables a simple MLP backbone to achieve state-of-the-art or highly com petitive performance across 12 real-world benchmarks. This validates that a corr ect physics-informed inductive bias is far more critical than raw architectural complexity for non-stationary forecasting. The code is available at: https://git hub.com/Gemost/PULSE.

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Teaching Molecular Dynamics to a Non-Autoregressive Ionic Transport Predictor

Jiyeon Kim, Byungju Lee, Won-Yong Shin

Unlike most static material properties widely studied in the machine learning li terature, ionic transport properties are inherently dynamic, making their fast a nd accurate prediction from static atomic structures challenging. The current st andard approach, molecular dynamics (MD) simulations, suffers from prohibitively high computational cost. Recent autoregressive learning-based MD acceleration m ethods requiring sequential inference remain slow and prone to error accumulatio n; in contrast, existing non-autoregressive material property prediction models are less accurate because they fail to exploit dynamics. Moreover, existing meth ods typically benefit from datasets either with or without atomic trajectories, but not both. To overcome these limitations, we propose a non-autoregressive lea rning framework based on auxiliary modality learning, which treats atomic trajec tories as an auxiliary modality during training but does not require them at inf erence. This enables the predictor to learn dynamics without sequential inferenc e while benefiting from both types of datasets. As a result, our framework achie ves over 200 times speedup compared to autoregressive models on the dataset with atomic trajectories while substantially reducing prediction error relative to n on-autoregressive benchmarks across both types of datasets. Our code is availabl e at https://github.com/jykim-git/MD.

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NeuralFLoC: Neural Flow-Based Joint Registration and Clustering of Functional Da ta

Xinyang Xiong, Siyuan Jiang, Pengcheng Zeng

Clustering functional data in the presence of phase variation is challenging, as temporal misalignment can obscure intrinsic shape differences and degrade clust ering performance. Most existing approaches treat registration and clustering as separate tasks or rely on restrictive parametric assumptions. We present **Neur alFLoC**, a fully unsupervised, end-to-end deep learning framework for joint fun ctional registration and clustering based on Neural ODE-driven diffeomorphic flo ws and spectral clustering. The proposed model learns smooth, invertible warping functions and cluster-specific templates simultaneously, effectively disentangl ing phase and amplitude variation. We establish universal approximation guarante es and asymptotic consistency for the proposed framework. Experiments on functio nal benchmarks show state-of-the-art performance in both registration and cluste ring, with robustness to missing data, irregular sampling, and noise, while main taining scalability. Code is available at https://github.com/LastQuater/NeuralFL oC.

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Learning-Augmented Online Minimization with Dual Predictions

Christian Coester, Alexa Tudose, Alexander Turoczy

We present learning-augmented algorithms for two general classes of online minim ization problems: metrical task systems and laminar set cover. Both algorithms a chieve improved theoretical guarantees using machine-learned predictions of an o

ptimal solution to the dual linear program. Unlike optimal primal solutions, whi ch can change drastically under tiny instance perturbations, these dual solution s are much more stable, which ensures the existence of good (and learnable) pred ictions for families of similar instances. While previous work has used dual pre dictions in offline settings and for online maximization problems, our algorithm s are, to the best of our knowledge, the first demonstration that such dual pred ictions can be effective for online minimization. Our theoretical results are co mplemented by experiments on the k-server problem and the parking permit probl em.

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Benchmarking Agent Memory in Interdependent Multi-Session Agentic Tasks

Zexue He, Yu Wang, Churan Zhi, Yuanzhe Hu, Tzu-Ping Chen, Lang Yin, Ze Chen, Ton g Arthur Wu, Siru Ouyang, Zihan Wang, Jiaxin Pei, Julian McAuley, Yejin Choi, Al ex Pentland

Existing evaluations of agents with memory typically assess **memorization** and **action** in isolation. One class of benchmarks evaluates memorization by test ing recall of past conversations or text but fails to capture how memory is used to guide future decisions. Another class focuses on agents acting in single-ses sion tasks without the need for long-term memory. However, in realistic settings , memorization and action are tightly coupled: agents acquire memory while inter acting with the environment, and subsequently rely on that memory to solve futur e tasks. To capture this setting, we introduce MemoryArena, a unified evaluation gym for benchmarking agent memory in multi-session Memory-Agent-Environment loo ps. The benchmark consists of human-crafted agentic tasks with explicitly interd ependent subtasks, where agents must learn from earlier actions and feedback by distilling experiences into memory, and subsequently use that memory to guide la ter actions to solve the overall task. MEMORYARENA supports evaluation across we b navigation, preference-constrained planning, progressive information search, a nd sequential formal reasoning, and reveals that agents with near-saturated perf ormance on existing long-context memory benchmarks like LoCoMo perform poorly in our agentic setting, exposing a gap in current evaluations for agents with memo ry.

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Making Expert Reasoning Learnable with Self-Distillation

Ethan Mendes, Jungsoo Park, Alan Ritter

Improving the reasoning capabilities of large language models (LLMs) typically r elies either on the model's ability to sample a correct solution to be reinforce d or the existence of a stronger model able to solve the problem. However, many difficult problems remain intractable for even current frontier models, preventi ng the extraction of valid training signals. A promising alternative is to lever age high-quality expert human solutions, yet naive imitation of this data fails because it is fundamentally out-of-distribution: expert solutions are typically didactic, containing implicit reasoning gaps intended for human readers rather t han computational models. Furthermore, high-quality expert solutions are expensi ve, necessitating generalizable sample-efficient training methods. We propose Di stribution Aligned Imitation Learning (DAIL), a two-step self-distillation metho d that bridges the distributional gap by first transforming expert solutions int o detailed, in-distribution reasoning traces and then applying a contrastive obj ective to focus learning on expert insights and methodologies. We find that DAIL can leverage fewer than 1000 high-quality expert solutions to achieve up to 31% pass@128 gains on Qwen2.5-Instruct and Qwen3, double reasoning efficiency, and enable out-of-domain generalization.

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Online Change Point Detection for Multivariate Inhomogeneous Poisson Processes T ime Series

Xiaokai Luo, Haotian Xu, Carlos Misael Madrid Padilla, OSCAR HERNAN MADRID PADIL LA

We study online change point detection for multivariate inhomogeneous Poisson po int process time series. This setting arises commonly in applications such as ea rthquake seismology, climate monitoring, and epidemic surveillance, yet remains

underexplored in the machine learning and statistics literature. We propose a method that uses low-rank matrices to represent the multivariate Poisson intensity functions, resulting in an adaptive nonparametric detection procedure. Our algorithm is single-pass and requires only constant computational cost per new observation, independent of the elapsed length of the time series. We provide theoretical guarantees to control the overall false alarm probability and characterize the detection delay under temporal dependence. We also develop a new Matrix Bernstein inequality for temporally dependent Poisson point process time series, which may be of independent interest.
Numerical experiments demonstrate that our method is both statistically robust and computationally efficient.
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Out-of-Distribution Evaluation of Rule-Based and Strategic Reasoning in Chess Transformers
Anna Mészáros, Patrik Reizinger, Ferenc Huszár
Modern decision transformers, trained similarly to LLMs, can achieve strong in-distribution performance in complex sequential domains like chess, but it remains unclear to what extent they reason systematically about rules and strategy. We study the reasoning capabilities of a 270M-parameter chess transformer trained via behavior cloning on standard chess. To investigate its abilities, we construct out-of-distribution test sets ---including board states and variants never seen during training---designed to reveal failures of systematic generalization. Our analysis shows that the model exhibits robust rule-based reasoning, consistently generating legal moves in novel configurations, but its strategic reasoning is more limited. The model generates high-quality moves on curated OOD puzzles and shows basic strategy adaptation in full games. It underperforms symbolic AI algorithms that rely on explicit search, although the performance gap is smaller when playing against human users on Lichess. Moreover, the training dynamics reveals distinct phases in how the model learns to respect the fundamental constraints, suggesting an emergent compositional understanding of the game.
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A robust PPG foundation model using multimodal physiological supervision
Eloy Geenjaar, Vince D. Calhoun, scott daly, Gouthaman KV, Lie Lu, Trisha Mittal, Daniel P. Darcy
Photoplethysmography (PPG), a non-invasive measure of changes in blood volume, is widely used in both wearable devices and clinical settings. Recent PPG foundation models either use open-source ICU datasets with pretraining paradigms that require curated data and thus complicate generalization to field-like data, or use closed-source field-like PPG data. In contrast, we propose a PPG foundation model that does not require high-quality or field-like pretraining data, and instead leverages accompanying electrocardiogram and respiratory signals in ICU datasets to select contrastive samples during pretraining. Our approach allows the model to retain and learn from noisy PPG segments, improving robustness at inference. Our model, pretrained on 3x fewer subjects than existing state-of-the-art approaches, achieves performance improvements on 14 out of 15 diverse downstream tasks, including field-like daily activity and heart rate prediction. Our results demonstrate that multimodal supervision can integrate complementary physiological information to improve the robustness of PPG foundation models and enhance their generalization to consumer-grade data.
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TuneAhead: Predicting Fine-tuning Performance Before Training Begins
Yuxiang Luo, Haonan Long, Chen Wang, Qiqi Duan, Xiaotian Lin, Yanwei Xu, Yuyu Luo, Weikai Yang, Nan Tang
Fine-tuning large language models (LLMs) is compute-intensive and error-prone: model performance depends sensitively on data quality and hyperparameter choices, and naïve runs can even degrade model performance. This raises a practical question:can we predict fine-tuning performance before committing to a full training run? We present TUNEAHEAD, a lightweight framework for pre-hoc prediction of fine-tuning performance. TUNEAHEAD encodes each candidate run as a meta-feature vector that combines static dataset descriptors with dynamic probe features

from a short standardized probe. A predictor maps these features to performance estimates, while SHAP-based attributions provide interpretable diagnostics that reveal which specific features drive
the prediction. Across 1,300+ fine-tuning runs on
Qwen2.5-7B-Instruct, TUNEAHEAD consistently
outperforms strong baselines such as Early-Stop
Extrapolation and ProxyLM. On a held-out test
set of 370 runs, TUNEAHEAD achieves an RMSE
of 1.47 percentage points and places 95.1% of predictions within ±3 percentage p oints of the true
score. These accurate continuous predictions support practical go/no-go screenin g policies that can
reduce unnecessary full fine-tuning while retaining most promising runs.
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ScalingAR: Scaling Confidence for Autoregressive Image Generation

Harold Haodong Chen, Xianfeng Wu, Wen-Jie Shu, Rongjin Guo, Disen Lan, Harry Yan g, Ying-Cong Chen

Test-time strategies have shown remarkable success in improving large language m odels, but their application to next-token prediction (NTP) autoregressive (AR) image generation remains largely underexplored. Existing test-time scaling (TTS) methods for visual autoregressive models (VAR) rely on frequent partial decodin g and external reward models, which are inefficient and often ineffective for NT P-based image generation due to the inherent instability of intermediate decodin g results. To address these limitations, we propose **ScalingAR**, a novel test- time scaling framework tailored for NTP-based AR image generation. ScalingAR int roduces *token entropy* as a confidence signal and operates at two complementary levels: (***i***) ***Profile Level***, integrates intrinsic uncertainty and con ditional utilization into a unified confidence state, and (***ii***) ***Policy L evel***, leverages this state for adaptive trajectory pruning and dynamic guidan ce scheduling. Without requiring early decoding or auxiliary rewards, ScalingAR achieves significant improvements across diverse benchmarks. Experiments show th at ScalingAR **(I)** improves base models by 12.5% on GenEval and 15.2% on TIIF- Bench, **(II)** reduces visual token consumption by 62.0% while outperforming ba selines, and **(III)** enhances robustness, mitigating performance degradation b y 26.0% in challenging scenarios. These results establish ScalingAR as a robust and efficient test-time scaling solution for autoregressive image generation.
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Chamaileon: Cross-Context Binder Design with Contextualized Modeling and Mixed S ampling

Hengyuan Cao, Shizhuo Cheng, Mingxuan Liu, Weicheng Huang, Yunhong Lu, CAI CHENX I, Yan Zhang, Min Zhang

The rapid evolution of generative models has unlocked new potentials in protein binder design, a pivotal task in structural biology, by facilitating end-to-end generation via joint sequence-structure modeling or hallucination. However, exis ting approaches are predominantly implemented under a single-target, single-stat e assumption, limiting their ability to model multi-target or multi-state intera ctions required for advanced function-oriented protein design. Here, we introduc e Chamaileon, which unifies multi-target and multi-state binder design by formul ating the problem as cross-context binding landscape modeling. The framework is underpinned by a training paradigm termed \textit{In-Context Complex Co-Design (I3CD)} for context-aware sequence-structure co-modeling. During inference, we em ploy \textit{Mixture-of-Paths Sampling (MoPS)}, a scalable strategy that optimiz es a single sequence across contexts while alleviating the scarcity of high-qual ity multi-conformational paired data. Extensive evaluation on our newly construc ted benchmark, \textit{CROSS}, demonstrates that Chamaileon effectively generate s sequences adaptable to diverse conformational landscapes and multi-target requ irements.
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Chain-of-Glimpse: Search-Guided Progressive Object-Grounded Reasoning for Video Understanding

Zhixuan Wu, Quanxing Zha, Teng Wang, Genbao Xu, Wenyuan Gu, Wei Rao, Nan Ma, Bo Cheng, Soujanya Poria

Video understanding requires identifying and reasoning over semantically discriminative visual objects across frames, yet existing object-agnostic solutions struggle to effectively handle substantial object variations over time. To address this, we introduce Chain-of-Glimpse, a search-guided progressive object-grounded reasoning framework that explicitly anchors each reasoning step to specific visual evidence regions, enabling compositional and multi-step decision-making. Formally, Chain-of-Glimpse formulates video reasoning as a step-by-step process that incrementally builds spatially grounded traces around task-relevant visual objects, thereby mitigating over-reliance on saliency-driven cues. Specifically, Chain-of-Glimpse features a search-guided controller, optimized via reinforcement learning with a format reward that significantly incentivizes grounding capability, to iteratively ground visual evidence regions and form reliable reasoning trajectories, yielding accurate and interpretable multi-step decisions. Extensive evaluations across two categories of video reasoning benchmarks, including general video reasoning benchmarks such as NExTQA, Video-Holmes, CG-Bench-Reasoning, and VRBench, and grounded video reasoning benchmarks such as NExT-GQA, demonstrate that Chain-of-Glimpse consistently improves performance while exhibiting strong robustness and generalization across diverse video reasoning tasks.

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Parallel Stochastic Gradient-Based Planning for World Models

Michael Psenka, Michael Rabbat, Aditi S. Krishnapriyan, Yann LeCun, Amir Bar

World models simulate environment dynamics from raw sensory inputs like video. However, using them for planning can be challenging due to the vast and unstructured search space. We propose a robust and highly parallelizable planner that leverages the differentiability of the learned world model for efficient optimization, solving long-horizon control tasks from visual input. Our method treats states as optimization variables ("virtual states") with soft dynamics constraints, enabling parallel computation and easier optimization. To facilitate exploration and avoid local optima, we introduce stochasticity into the states. To mitigate sensitive gradients through high-dimensional vision-based world models, we modify the gradient structure to descend towards valid plans while only requiring action-input gradients. Our approach can be viewed as a stochastic version of a non-condensed or collocation-based optimal controller. We provide theoretical justification and experiments on video-based world models, where our resulting planner outperforms existing planning algorithms like the cross-entropy method (CEM) and vanilla gradient-based optimization (GD) on long-horizon experiments, both in success rate and time to convergence.

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FoeGlass: Simple In-Context Learning Is Enough for Red Teaming Audio Deepfake Detectors

Sepehr Dehdashtian, Jacob H Seidman, Vishnu Boddeti, Gaurav Bharaj

Audio deepfake detection (ADD) models are critical for countering the malicious use of text-to-speech (TTS) models. Evaluating and strengthening ADD models requires developing datasets that span the space of generated audio and highlight high-error regions. Existing dataset development strategies face two challenges: (i) manual collection, and (ii) inefficient discovery of blind spots in the ADD models. To address these challenges, we propose FoeGlass, the first black-box automated red-teaming method for ADDs, which effectively discovers ADD failure modes in the space of generated audio underexplored by state-of-the-art deepfake benchmarks. FoeGlass uses the in-context learning capabilities of an LLM to explore the input space of a TTS model, generating audio samples that fool the target ADD using only black-box access to all components. By using a carefully designed context based on diversity measurements, FoeGlass mitigates the common problem of mode collapse in automated red-teaming systems. Empirical evaluations on several open-source ADD and TTS models demonstrate that data generated from FoeGlass substantially improves the false negative rates over unconditional sampling baselines and recent spoofing datasets by up to 94%, while requiring no manual supervision. Furthermore, we show that the attacks generated by FoeGlass are transfer

able across different target ADDs, demonstrating its broad applicability and eas
e of use for the automated red teaming of ADD systems. Finally, fine-tuning ADD
models on FoeGlass-generated samples notably enhances the robustness of the dete
ctors (up 41%).
****
Mind the Gap: Structure-Aware Consistency in Preference Learning
Mehryar Mohri, Yutao Zhong
Aligning Large Language Models (LLMs) with human intent, whether through explici
t reward modeling or direct methods such as DPO, fundamentally relies on minimiz
ing a surrogate loss as a proxy for the true pairwise ranking objective. We prov
e that this reliance is flawed for the standard surrogate losses used: for the e
quicontinuous hypothesis sets characteristic of neural networks, *no* standard s
urrogate provides a meaningful consistency guarantee. Minimizing the surrogate l
oss to zero can leave the true ranking error arbitrarily high. To resolve this,
we formulate LLM alignment within a margin-shifted ranking framework and derive
H-consistency bounds showing that enforcing a confidence margin γ is no
t merely beneficial but *necessary* for consistency. We further introduce Struct
ure-Aware H-consistency and a corresponding objective (SA-DPO) that adapts the
 margin to the semantic distance between responses, preventing instability on ne
ar-synonymous pairs. Finally, we analyze the trade-off between the margin requir
ed for consistency and the model's finite capacity to satisfy it, revealing a st
rict hierarchy of surrogate losses: heavy-tailed surrogates (e.g., the Polynomia
l Hinge family) offer strictly superior consistency guarantees for capacity-boun
ded models compared to the logistic loss used in DPO. Experiments on UltraFeedba
ck and Argilla DPO-Mix-7k confirm that SA-DPO consistently outperforms DPO and S
imPO, with a 58.5\% head-to-head win-rate in downstream generation quality.
****
The Crowded Embedding Space: A Mean-Field Mechanism for Emergent Marginalization
 in Retrieval-Augmented Agents
Shwan Ashrafi, Dan Roth
Retrieval-augmented generative agents rely on retrieval for grounding, yet are t
ypically evaluated on a query-by-query basis. This isolates interactions that ar
e geometrically coupled in a shared embedding space. For example, we show that t
he high document density required to serve majority interests (e.g., generic "Cr
ime" movies) can geometrically overcrowd the retrieval neighborhood of a semanti
cally similar minority (e.g., "Film Noir"), effectively expelling minority conte
nt from top-k results. We introduce a formal framework to analyze how such *go
al collisions* in dense retrieval induce fundamental performance limits and emer
gent fairness issues inherent to spatial crowding. In our static analysis, we de
monstrate that for a fixed embedding space, a phase transition occurs where mino
rity user goals suffer a catastrophic collapse in performance as the density of
majority goals increases. We then extend this to a dynamic model and derive a no
n-linear Fokker-Planck equation that governs the evolution of document embedding
s as the agent updates them to maximize retrieval accuracy. Our analysis reveals
 that this local relevance objective triggers an emergent global mechanism that
systematically marginalizes minority interests. We prove that such objectives dr
ive the system to self-organize into a state that exclusively serves majority in
terests. These results provide a theoretical foundation for understanding a crit
ical grounding failure mode in retrieval-augmented agents.
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Entropy-Aware On-Policy Distillation of Language Models
Woogyeol Jin, Taywon Min, Yongjin Yang, Dennis Wei, Yi Zhou, Swanand Ravindra Ka
dhe, Nathalie Baracaldo, Kimin Lee
On-policy distillation is a promising approach for transferring knowledge betwee
n language models, where a student learns from dense token-level signals along i
ts own trajectories. This framework typically uses reverse KL divergence, encour
aging the student to match the teacher's high-confidence predictions. However, w
e show that the mode-seeking property of reverse KL reduces generation diversity
 and yields unstable learning signals when the teacher distribution has high ent
ropy. To address this, we introduce Entropy-Aware On-Policy Distillation. Our ke

y idea is augmenting the standard reverse KL objective with forward KL when teac
her entropy is high, capturing the full range of plausible outputs while retaini
ng precise imitation elsewhere. It balances mode-seeking precision with mode-cov
ering robustness without sacrificing on-policy training efficiency. Experiments
 show that our method maintains generation diversity (sustained token-level entro
py) and improves student–teacher alignment (lower forward KL on high-entropy tok
ens). Across six math reasoning benchmarks, this yields Pass@8 accuracy gains of
 $+1.37$ for Qwen3-0.6B-Base, $+2.39$ for Qwen3-1.7B-Base, and $+5.05$ for Qwen3
-4B-Base compared to baseline on-policy distillation methods. These results demo
nstrate that accounting for teacher uncertainty is essential for maintaining div
ersity and achieving effective knowledge transfer. Our code is publicly availabl
e at https://github.com/WLS04/EOPD.
****

AutoBaxBuilder: Bootstrapping Code Security Benchmarking
Tobias von Arx, Niels Mündler, Mark Vero, Maximilian Baader, Martin Vechev
As large language models (LLMs) see wide adoption in software engineering, the r
eliable assessment of the correctness and security of LLM-generated code is cruc
ial. Notably, prior work showed that LLMs are prone to generating code with secu
rity vulnerabilities, highlighting that security is often overlooked. These insi
ghts were enabled by specialized benchmarks crafted by security experts through
 significant manual effort. However, benchmarks (i) inevitably end up contaminati
ng training data, (ii) must extend to new tasks to provide a more complete pictu
re, and (iii) must increase in difficulty to challenge more capable LLMs. In thi
s work, we address these challenges and present AutoBaxBuilder, an automated pip
eline that generates code security benchmarking tasks from scratch. It leverages
 the code-understanding capabilities of LLMs combined with robust reliability ch
ecks to construct functional tests and end-to-end security-probing exploits. The
 quality of the pipeline is quantitatively confirmed by aligning its predictions
 with an expert-written baseline and qualitatively validated through manual soun
dness verification. We use AutoBaxBuilder to construct a new benchmark and relea
se it to the public as AutoBaxBench, together with a thorough evaluation on cont
emporary LLMs. AutoBaxBuilder generates new tasks in under 2 hours, for less tha
n USD 4. Including a manual verification, this reduces the required human effort
 for benchmark construction by a factor of 12.
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Uncertainty-Guided Exploration and Stable Planning for Sparse-Reward Manipulatio
n from Limited Demonstrations
Haowen Sun, Liqi Huang, Mingyang Li, Sihua Ren, Xinzhe Chen, Chengzhong Ma, Zeya
ng Liu, Xingyu Chen, Xuguang Lan
Reinforcement learning from demonstrations (RLfD) offers a promising method for
 robotic manipulation with sparse rewards. However, limited demonstrations often
 cause agents to encounter out-of-distribution states where world models produce
 poor predictions. In multi-stage tasks, jointly optimizing a learned reward func
tion and policy introduces a moving target problem, and the resulting non-statio
narity intensifies the impact of uncertainty on policy learning. In this work, w
e propose QUEST, a model-based RL framework that adaptively switches between exp
loration and exploitation guided by uncertainty to achieve stable and efficient
 learning. Specifically, our approach employs intrinsic rewards to encourage expl
oration, leverages ensemble dynamics for uncertainty-guided planning, and introd
uces a hybrid sampling strategy to prioritize rare successful stage transitions.
 We evaluate QUEST on challenging sparse-reward manipulation tasks with limited
 expert demonstrations. Results show that QUEST outperforms state-of-the-art meth
ods by 17\% on average, with gains increasing to 60\% on difficult tasks. We fur
ther demonstrate successful zero-shot sim-to-real transfer on five real-world ta
sks. Project website: https://quest-official.github.io/QUEST/.
****

Hedging on the Frontier: Learning New Tasks with Few Samples
Tobias Wegel, Federico Di Gennaro, Geelon So, Fanny Yang
When a learner faces a new task with few samples, it must leverage any available
 side information. In practice, this often comes in the form of model evaluation

s on related tasks in public benchmarks. A key question then is how to model tas
k relatedness such that it is both realistic and the benchmark evaluations lead
to provable gains. Empirically, we observe that *weak monotonicity* is often app
roximately satisfied: if a model dominates another on many benchmarks, it also t
ends to outperform on the new task. We explore the statistical complexity of lea
rning under (approximate) weak monotonicity, leveraging it within two learning p
aradigms: transfer learning and model selection aggregation. We show that not on
ly can we prune the model class based on monotonicity, but we can also further a
dapt to the geometry of the available trade-offs by *hedging on the frontier*.
****

Revisiting Pre-Propagation GNNs: Robust Diffusion Operators and Hidden-State Re-Propagation
Zichao Yue, Zhiru Zhang
Pre-propagation graph neural networks (PP-GNNs) decouple node feature propagatio
n from transformation: graph diffusion is performed once as preprocessing, and t
raining reduces to dense per-node transformations. This design enables mini-batc
h training without inter-node dependencies, avoids repeated sparse matrix--matri
x multiplications, and better matches modern accelerators optimized for dense co
mpute. However, their expressivity remains unclear, and empirical results show a
 gap between PP-GNNs and their message-passing counterparts on commonly used gra
ph benchmarks, especially heterophilic ones. In this paper, we propose a suite o
f robust graph diffusion operators for preprocessing and a few-shot hidden-state
 re-propagation scheme during training. Our methods improve the validation and t
est accuracy of PP-GNNs, enabling them to match the accuracy of message-passing
GNNs while maintaining training efficiency.
****

Grounded in Reality: Learning and Deploying Proactive LLM from Offline Logs
Fei Wei, Daoyuan Chen, Ce Wang, Yilun Huang, Yushuo Chen, Xuchen Pan, Yaliang Li
, Bolin Ding
Large language models (LLMs) are strong passive responders, but learning to proa
ctively elicit information—asking the right questions and stopping at the right
time—remains difficult. Existing approaches, such as optimizing turn-level attri
butes or relying on user simulators to generate training trajectories, often str
uggle with a persistent reality gap.

We propose \texttt{Learn-to-Ask}, a simulator-free framework that learns proacti
ve questioning policies directly from offline expert conversations. Our key insi
ght is to leverage the \textbf{observed future} of each expert trajectory to der
ive dense, turn-level rewards that reflect expert long-horizon strategy, reducin
g policy learning to a sequence of supervised learning tasks that jointly enable
 LLMs to know \textbf{what to ask} and \textbf{when to stop}. To ensure the LLM-
generated contents, such as reward fidelity and sampling quality, align with exp
ectations, we further introduce an automated pipeline that calibrates the prompt
s with minimal human supervision.

Across multiple datasets and model scales, \texttt{Learn-to-Ask} consistently im
proves proactive information-seeking behavior. We also report a large-scale real
-world deployment where the trained agent surpasses an internal expert baseline
under professional audit, which demonstrates the effectiveness of our framework
and our rewards as a reality-validated proxy metric for LLM proactivity.
****

Amodal Instance Segmentation with IRAIS Dataset for Sim-to-Real Transfer
Bidong Chen, Lingui Li
Amodal instance segmentation is hindered by the scarcity of scalable and transfe
rable annotations. We introduce MaviGen, an automated 3D retail scene modeling a
nd rendering framework that generates photorealistic multi-view images with comp
lete amodal masks. Building on MaviGen, we present the IRAIS dataset, a sim-to-r
eal benchmark comprising a large-scale synthetic multi-view set (3D-IRAIS) and a
 human-annotated real image set (Real-IRAIS), both sharing unified label definit
ions and evaluation protocols to facilitate rigorous transfer studies. We propos

e EUREKA, an encoder-only, query-efficient network for amodal instance segmentat
ion that performs full-image multi-task inference via unified amodal/visible que
ries and dual mask heads. The dual heads enable mutual supervision between compl
ete and visible masks, while the conditional masked self-attention mechanism fur
ther strengthens occlusion reasoning. Experiments establish strong baselines on
IRAIS and achieve state-of-the-art performance on D2SA and COCOA-cls, demonstrat
ing substantial improvements in sim-to-real transfer.
****

Removing Sandbagging in LLMs by Training with Weak Supervision
Emil Ryd, Henning Bartsch, Julian Stastny, Joe Benton, Vivek Hebbar
As AI systems begin to automate complex tasks, supervision increasingly relies o
n weaker models or limited human oversight that cannot fully verify output quali
ty. A model more capable than its supervisors could exploit this gap through san
dbagging, producing work that appears acceptable but falls short of its true abi
lities. Can training elicit a model's best work even without reliable verificati
on? We study this using model organisms trained to sandbag, testing elicitation
techniques on problem-solving math, graduate-level science, and competitive codi
ng tasks. We find that training with weak supervision can reliably elicit sandba
gging models when supervised fine-tuning (SFT) and reinforcement learning (RL) a
re combined: SFT on weak demonstrations breaks the sandbagging behavior, enablin
g RL to then fully elicit performance. Neither method succeeds reliably alone—RL
 without SFT almost always leads to reward hacking rather than genuine improveme
nt, and SFT without RL fails to elicit full performance when the supervisor is m
uch weaker than the untrusted model. Critically, this relies on training being i
ndistinguishable from deployment; when models can distinguish between training a
nd deployment, they can perform well during training while continuing to sandbag
 afterward. Our results provide initial evidence that training is a viable mitig
ation against sandbagging, while highlighting the importance of making training
indistinguishable from deployment.
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Beyond Heuristics: Learnable Density Control for 3D Gaussian Splatting
Zhenhua Ning, Xin Li, Jun Yu, Guangming Lu, Yaowei Wang, Wenjie Pei
While 3D Gaussian Splatting (3DGS) has demonstrated impressive real-time renderi
ng performance, its efficacy remains constrained by a reliance on heuristic dens
ity control. Despite numerous refinements to these handcrafted rules, such metho
ds inherently lack the flexibility to adapt to diverse scenes with complex geome
tries.

  In this paper, we propose a paradigm shift for density control from rigid heur
istics to fully learnable policies. Specifically, we introduce LeGS, a framework
 that reformulates density control as a parameterized policy network optimized v
ia Reinforcement Learning (RL). Central to our approach is the tailored effectiv
e reward function grounded in sensitivity analysis, which precisely quantifies t
he marginal contribution of individual Gaussians to reconstruction quality. To m
aintain computational tractability, we derive a closed-form solution that reduce
s the complexity of reward calculation from $O(N^2)$ to $O(N)$. Extensive experi
ments on the Mip-NeRF 360, Tanks \& Temples, and Deep Blending datasets demonstr
ate that LeGS significantly outperforms state-of-the-art methods, striking a sup
erior balance between reconstruction quality and efficiency.
****

ConEx: Human-Interpretable Saliency Maps via Concept-Aware Attribution
Yehonatan Elisha, Oren Barkan, Ziv Weiss Haddad, Noam Koenigstein
Many visual explanation methods in computer vision highlight pixel importance bu
t struggle to link these low-level cues to semantically meaningful concepts, lim
iting their interpretability and trustworthiness. We introduce Concept-based Exp
lanations (ConEx), a novel framework that bridges saliency visualization with co
ncept-based reasoning to provide both faithfulness and interpretability. ConEx a
utomatically discovers class-specific concepts and represents them through conce
pt activation vectors (CAVs), learned without manual supervision using an archit
ecture-specific masking mechanism that reduces noise introduced by the segmentat
ion masks to enhance concept purity. ConEx generates faithful saliency maps that

reveal where each concept appears in the image and how it contributes to the prediction. To evaluate the reliability of these learned concepts, we propose two complementary metrics, Vector-Concept Match (VCM) and Concept-Class Match (CCM), that quantify concept alignment and enable direct comparison with existing methods. Extensive experiments across diverse settings demonstrate that ConEx achieves state-of-the-art performance on faithfulness, segmentation, and concept-quality benchmarks. Overall, ConEx advances the field toward truly interpretable and concept-grounded explanations in vision models.
****
Beyond Point-wise Neural Collapse: A Topology-Aware Hierarchical Classifier for Class-Incremental Learning
HuiYu Yi, Xu Zhiming, Dunwei Tu, Zhicheng Wang, Baile Xu, Furao Shen
The Nearest Class Mean (NCM) classifier is widely favored in Class-Incremental Learning (CIL) for its superior resistance to catastrophic forgetting compared to Fully Connected layers. While Neural Collapse (NC) theory supports NCM's optimality by assuming features collapse into single points, non-linear feature drift and insufficient training in CIL often prevent this ideal state. Consequently, classes manifest as complex manifolds rather than collapsed points, rendering the single-point NCM suboptimal. To address this, we propose Hierarchical-Cluster SOINN (HC-SOINN), a novel classifier that captures the topological structure of these manifolds via a ``local-to-global'' representation. Furthermore, we introduce Structure-Topology Alignment via Residuals (STAR) method, which employs a fine-grained pointwise trajectory tracking mechanism to actively deform the learned topology, allowing it to adapt precisely to complex non-linear feature drift. Theoretical analysis and Procrustes distance experiments validate our framework's resilience to manifold deformations. We integrated HC-SOINN into seven state-of-the-art methods by replacing their original classifiers, achieving consistent improvements that highlight the effectiveness and robustness of our approach. Code is available at \url{https://github.com/yhyet/HC_SOINN}.
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BOOSTAPR: Boosting Automated Program Repair via Execution-Grounded Reinforcement Learning with Dual Reward Models
Yuanhao Li, Hongbo Wang, Xiaotang Shang, Xunzhu Tang, Yiming Cao, Xuhong Chen
Reinforcement learning for program repair is hindered by sparse execution feedback and coarse sequence-level rewards that obscure which edits actually fix bugs. We present BoostAPR, a three-stage framework: (1) supervised fine-tuning on execution-verified demonstrations with reasoning traces, (2) training dual reward models—a sequence-level assessor and a line-level credit allocator—from execution outcomes, and (3) PPO optimization where the line-level model redistributes rewards to critical edit regions. This line-level credit assignment operates at an intermediate granularity naturally suited to code changes. Trained on SWE-Gym and evaluated on four benchmarks, BoostAPR achieves 40.7% on SWE-bench Verified (+22.9pp over the base model), 24.8% on Defects4J (Python→Java transfer), 84.5% on HumanEval-Java, and 95.0% on QuixBugs, showing competitive open-source performance with strong cross-language generalization.
****
Theoretical Investigation on Inductive Bias of Isolation Forest
Qin-Cheng Zheng, Shao-Qun Zhang, Shen-Huan Lyu, Yuan Jiang, Zhi-Hua Zhou
Isolation Forest (iForest) is one of the most widely used unsupervised anomaly detectors, owing to its efficiency and performance on large-scale tasks. Despite its broad applications, there is still a lack of theoretical understanding of iForest's empirical success. In this work, we study the inductive bias of iForest and examine when and to what extent it performs well. The main idea is to characterize the random growth process of iForest, in which both split dimensions and split values are selected randomly. We model the growth process of iForest as a random walk and derive the expected path length function, the outcome of iForest that determines the anomaly score, by analyzing the hitting time of the absorbing state. The infinite-sample size analysis reveals that, unlike k-Nearest Neighbor (k-NN), whose score reflects only the local density, the iForest path length combines the density and the centrality. Since central points naturally hav

e larger path lengths, iForest is therefore less sensitive to central anomalies. Analyses of fixed datasets corroborate this finding and further show that iForest is more parameter-adaptive than k-NN. Our study provides a theoretical understanding of the effectiveness of iForest and establishes a foundation for further exploration.
****
Byte Pair Encoding for Efficient Time Series Forecasting
Leon Götz, Marcel Kollovieh, Stephan Günnemann, Leo Schwinn
Existing time series tokenization methods predominantly encode a constant number of samples into individual tokens. This inflexible approach can generate excessive tokens for even simple patterns like extended constant values, resulting in substantial computational overhead. Inspired by the success of byte pair encoding, we propose the first pattern-centric tokenization scheme for time series analysis. Based on a discrete vocabulary of frequent motifs, our method merges samples with underlying patterns into tokens, compressing time series adaptively. Exploiting our finite set of motifs and the continuous properties of time series, we further introduce conditional decoding as a lightweight yet powerful post-hoc optimization method, which requires no gradient computation and adds no computational overhead. On recent time series foundation models, our motif-based tokenization improves forecasting performance by 40% and boosts efficiency by 2314% on average. Conditional decoding further reduces MSE by up to 48%. In an extensive analysis, we demonstrate the adaptiveness of our tokenization to diverse temporal patterns, its generalization to unseen data, and its meaningful token representations capturing distinct time series properties, including statistical moments and trends.
****
A_2DEPT: Large Language Model–Driven Automated Algorithm Design via Evolutionary Program Trees
Bin Chen, Shouliang Zhu, Beidan Liu, Yong Zhao, Tianle Pu, Huichun Li, Zhengqiu Zhu
Designing heuristics for combinatorial optimization problems (COPs) is a fundamental yet challenging task that traditionally requires extensive domain expertise. Recently, Large Language Model (LLM)-based Automated Heuristic Design (AHD) has shown promise in autonomously generating heuristic components with minimal human intervention. However, most existing LLM-based AHD methods enforce fixed algorithmic templates to ensure executability, which confines the search to component-level tuning and limits system-level algorithmic expressiveness. To enable open-ended solver synthesis beyond rigid templates, we propose Automated Algorithm Design via Evolutionary Program Trees (A$_2$DEPT), which treats LLMs as system-level algorithm architects.
A$_2$DEPT explores the vast program space via a tree-structured evolutionary search with hybrid selection and hierarchical operators, enabling iterative refinement of complete algorithms.
To make open-ended generation practical, we enforce executability with a lightweight program-maintenance loop that performs feedback-driven repair.
In experiments, A$_2$DEPT consistently outperforms state-of-the-art baselines across standard and highly constrained benchmarks, reducing the optimality gap by an average of 9.8\%. Our work implies that system-level algorithm synthesis is a viable and scalable paradigm for LLM-driven optimization.
****
TranX-Adapter: Bridging Artifacts and Semantics within MLLMs for Robust AI-generated Image Detection
Wenbin Wang, Yuge Huang, Jianqing Xu, Yue Yu, Jiangtao Yan, Shouhong Ding, Pan Zhou, Yong Luo
Rapid advances in AI-generated image (AIGI) technology enable highly realistic synthesis, threatening public information integrity and security. Recent studies have demonstrated that incorporating texture-level artifact features alongside semantic features into multimodal large language models (MLLMs) can enhance their AIGI detection capability. However, our preliminary analyses reveal that artifact features exhibit high intra-feature similarity, leading to an almost uniform

attention map after the softmax operation. This phenomenon causes attention dilu
tion, thereby hindering effective fusion between semantic and artifact features.
 To overcome this limitation, we propose a lightweight fusion adapter, TranX-Ada
pter, which integrates a Task-aware Optimal-Transport Fusion that leverages the
Jensen-Shannon divergence between artifact and semantic prediction probabilities
 as a cost matrix to transfer artifact information into semantic features, and a
n X-Fusion that employs cross-attention to transfer semantic information into ar
tifact features. Experiments on standard AIGI detection benchmarks upon several
advanced MLLMs, show that our TranX-Adapter brings consistent and significant im
provements (up to +6% accuracy). Code can be found in supplementary material.
****

Is One Layer Enough? Understanding Inference Dynamics in Tabular Foundation Mode
ls

Amir Rezaei Balef, Mykhailo Koshil, Katharina Eggensperger

Transformer-based tabular foundation models (TFMs) dominate small to medium tabu
lar predictive benchmark tasks, yet their inference mechanisms remain largely un
explored. We present the first large-scale mechanistic study of layerwise dynami
cs in 6 state-of-the-art tabular in-context learning models. We explore how pred
ictions emerge across depth, identify distinct stages of inference and reveal la
tent-space dynamics that differ from those of language models. Our findings indi
cate substantial depthwise redundancy across multiple models, suggesting iterati
ve refinement with overlapping computations during inference stages. Guided by t
hese insights, we design a proof-of-concept, looped single-layer model that uses
 only 20% of the original model’s parameters while achieving comparable performa
nce. The code is available at https://github.com/amirbalef/is_one_layer_enough.
****

Blending Neural Control Density Functions for Stabilization and Safety

Sahil Chaudhary, Chaitanya Murti, Chiranjib Bhattacharyya

Recent work on Neural Network-based methods for nonlinear control use Lyapunov F
unctions to obtain controllers with guarantees of stability. However, Lyapunov-b
ased methods are fundamentally limited: they cannot be used for smooth blending
with formal Region of Attraction (RoA) expansion guarantees, and also fail to ce
rtify stability when unstable equilibria or saddle points are present. Density f
unctions provide an alternate stability certificate, and address these limitatio
ns by certifying almost everywhere stability, and enable smooth blending of cont
rollers. Learning valid density certificates is challenging due to integrability
 constraints, and the effect of density-based blending controllers on RoAs is no
t well understood. In this work, we provide the first guarantee that controllers
 blended with density functions yield RoAs containing the union of the RoAs achi
eved by the constituent controllers. Then, we propose a novel exponential charac
terization of density functions that provably satisfies the integrability condit
ion, and introduce Neural Control Density Functions (NCDFs), that leverage this
new parameterization. We also extend NCDFs for synthesizing safe-stable controll
ers by combining NCDFs with control barrier functions (NCDF-CBFs). Our experimen
ts show that blended controllers obtain superior RoAs to state-of-the-art method
s like Neural Lyapunov Control and Sum-of-Squares based techniques.
****

Convergence of Two-Timescale Markovian Stochastic Approximations with Applicatio
ns in Reinforcement Learning

Vagul Mahadevan, Claire Chen, Shuze Daniel Liu, Shangtong Zhang

This work studies the convergence of two-timescale stochastic approximations (SA
),
a class of iterative algorithms that update two sets of parameters in fast and s
low timescales respectively. Notable examples of two-timescale SA in reinforceme
nt learning (RL) include temporal difference learning with gradient correction (
TDC) and actor-critic methods. Previously, the stability (i.e., boundedness) and
 convergence of two-timescale SA were only established under i.i.d. noise. This
work instead establishes the stability and convergence of two-timescale SA under
 Markovian noise, a setup that is more realistic in RL. Notably, we do not need
to use any projection operator and the noise does not need to live in a compact

space. Our key technical novelty is to control the fast timescale parameter with the running max of the slow timescale parameter, instead of with the current slow timescale parameter, as most prior works do. As a key application, we establish the first almost sure convergence of TDC with eligibility traces under off-policy learning with linear function approximation.
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Near-Optimal Private Linear Regression via Iterative Hessian Mixing
Omri Lev, Moshe Shenfeld, Vishwak Srinivasan, Katrina Ligett, Ashia C. Wilson
We study differentially private ordinary least squares (DP-OLS) with bounded data (X,Y) via sketching-based mechanisms. While Gaussian sketching approaches have been explored for DP-OLS (Sheffet, 2017), they are typically viewed as less competitive than the Adaptive Sufficient Statistics Perturbation (AdaSSP) method (Wang, 2018), which directly perturbs the sufficient statistics $(X^{\top}X, X^{\top}Y)$. This method was shown to be close to information-theoretically optimal, while also exhibiting strong empirical performance. In this work, we propose \emph{Iterative Hessian Mixing} (IHM), an algorithm that builds on Gaussian-sketching approaches to DP-OLS and is inspired by the Iterative Hessian Sketch (Pilanci & Wainwright, 2016). We prove that IHM is differentially private and provide utility guarantees in the form of excess empirical risk bounds. These bounds improve upon those of AdaSSP by removing a multiplicative factor that can be as large as the square root of the data dimension. The design of the IHM is based on new accuracy guarantees that we present for prior Gaussian sketching approaches for DP-OLS, which clarify when these methods are expected to perform well and how IHM circumvents their inherent limitations.
We also conduct a rigorous empirical evaluation on a large suite of datasets, demonstrating that IHM consistently outperforms prior baselines, including AdaSSP.
****

Reasoning Models Are Test Exploiters: Rethinking Multiple Choice
Narun Krishnamurthi Raman, Taylor Lundy, Kevin Leyton-Brown
When evaluating Large Language Models (LLMs) in question-answering domains, multiple-choice question answering (MCQA) is widely used because it enables automatic grading. However, MCQA also exposes models to answer options that can be exploited in ways that inflate reasoning ability. We study this phenomenon across 15 question-answering benchmarks and 27 LLMs by systematically varying how and when models are exposed to answer options. For non-reasoning LLMs, MCQA can remain a good proxy for free-text performance when any chain-of-thought is produced only before the options are revealed. However, this "decoupled" format is not realizable for most reasoning models: they are designed to emit reasoning tokens whenever they are prompted, so if options are present they inevitably "reason over" the options. In practice, this makes reasoning models particularly effective at extracting signal from options, and can create large, misleading gains over free-text baselines. To characterize how models exploit MCQA, we introduce diagnostic probes that isolate option-only and question-plus-option exploitation pathways, and we quantify how design choices such as distractor strength and "none-of-the-above" answers effect exploitability. Finally, we examined the practice of multiple choice as an error diagnostic: inferring a model's mistake from the wrong option it picks. On benchmarks where reasoning can be expressed as code, we ask models to output code, we then executed it varying the inputs, and compared the resulting input–output behavior, revealing failure modes that MCQA diagnostics obscure. Lastly, we offer practical guidelines when analyzing results from MCQA that better reflect LLMs' genuine reasoning capabilities.
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Hamiltonian Asymmetric Fusion: One-Way Safe Directed Refinement under Modality Imbalance
Bingbing Chen, Congcong Liu, Dong Liang, Zhuo-Xu Cui
In RGB--D salient object detection, multimodal fusion is commonly implemented via symmetric token interaction, implicitly allowing information to flow in both directions. Under *modality imbalance*---when an auxiliary stream is substantially noisier than a designated primary stream---such symmetry creates a *backflow channel* that injects auxiliary noise into the primary representation and amplifi

es errors across iterative refinement stages. We formulate fusion in this regime as *directed refinement with one-way safety*: the primary modality defines a guidance field, while only auxiliary representations are iteratively purified, and primary perturbations induced by the auxiliary stream are explicitly bounded.
We propose *Hamiltonian Asymmetric Fusion* (HAF), a lightweight unrolled refinement block that updates auxiliary tokens with momentum regularization and gated driving. The refinement force is instantiated by FFT-based spectral global correlation and modulated by a shared learnable spectral response to emphasize reliable frequency components with minimal parameters; a leaky momentum gate and a stable integrator improve multi-step refinement stability. We provide guarantees of auxiliary error contraction and bounded primary perturbation, which symmetric fusion operators do not satisfy under imbalance. Experiments on six RGB--D SOD benchmarks show consistent gains and substantially more graceful degradation under controlled auxiliary corruption.
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Infinite-Dimensional Generative Diffusions via Doob’s h-Transform
Thorben Pieper-Sethmacher, Daniel Paulin
This paper introduces a rigorous framework for defining generative diffusion models in infinite dimensions via Doob's h-transform. Rather than relying on time reversal of a noising process, a reference diffusion is forced towards the target distribution by an exponential change of measure.
Compared to existing methodology, this approach readily generalises to the infinite-dimensional setting, hence offering greater flexibility in the diffusion model.
The construction is derived rigorously under verifiable conditions, and bounds with respect to the target measure are established.
We show that the forced process under the changed measure can be approximated by minimising a score-matching objective and validate our method on both synthetic and real data.
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Deterministic Component Mining for Multi-Framework UI2Code Generation
Zixiong Yang, Linxiao Li, Jiaye Lin, Binrui Wu, Xiaoyu Kang, Jiechao Gao
Automating User Interface (UI) generation substantially improves productivity and accelerates development by reducing engineering time and manual effort. Despite recent progress of Multimodal Large Language Models (MLLMs) in UI2Code, most existing approaches focus on a single HTML/CSS form and fail to systematically incorporate front-end frameworks such as React, Vue, and Angular.
Moreover, their outputs are often verbose and hard to reuse at the component level. To address those issues, we propose Deterministic Component Mining (DCM), a multi-stage pipeline that couples MLLMs with a compact intermediate representation to enable multi-framework and component-oriented code generation.
Firstly, a lightweight structure model predicts the representation of the DOM tree in JSON format, capturing the coarse layout from a webpage screenshot. Subsequently, we formulate deterministic rules to normalize the predicted DOM tree and mine reusable components with repetitive patterns via structural hashing and clustering, thereby yielding a portable intermediate representation.
Finally, we employ a framework-conditioned prompting strategy governed by a binding specification and a file-block protocol to emit HTML/React/Vue/Angular code with explicit component props and repeat constructs. Extensive experiments demonstrate that DCM significantly outperforms baselines on automatic evaluation metrics and component-level reuse, while delivering consistent gains in multi-framework portability and overall code structural quality.
****

IDRBench: Understanding the Capability of Large Language Models on Interdisciplinary Research
Yuanhao Shen, Daniel Xavier de Sousa, Ricardo Marçal de Andrade Nascimento, Hongyu Guo, Xiaodan Zhu
Innovation is a key driving force of human civilization. As the body of knowledge has grown considerably, bridging knowledge across different disciplines, where significant innovation often emerges, has become increasingly challenging. The

recent advancements in machine learning models, particularly Large Language Models (LLMs), have provided effective access to extensive knowledge sources and shown impressive abilities in reasoning, rendering significant opportunities for interdisciplinary discovery. Our research aims to understand the capabilities of state-of-the-art LLMs in integrating knowledge from different fields for interdisciplinary research (IDR). To address this fundamental problem, we introduce IDRBench, a pioneering framework that includes both datasets and evaluation tasks: (1) IDR Paper Identification, (2) IDR Idea Integration, and (3) IDR Idea Recommendation. Our study on ten mainstream LLMs provides a comprehensive analysis of their behavior and establishes benchmarks and baselines for future research. To the best of our knowledge, IDRBench is the first to provide a comprehensive investigation of LLMs' IDR capability.

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Adaptive Testing for LLM Evaluation: A Psychometric Alternative to Static Benchmarks

Peiyu Li, Xiuxiu Tang, Si Chen, Ying Cheng, Ronald Metoyer, Ting Hua, Nitesh V Chawla

Evaluating large language models (LLMs) typically requires thousands of benchmark items, making the process expensive, slow, and increasingly impractical at scale. Existing evaluation protocols rely on average accuracy over fixed item sets, treating all items as equally informative despite substantial variation in difficulty and discrimination. We introduce ATLAS, an adaptive testing framework based on Item Response Theory (IRT) that estimates model ability using Fisher information–guided item selection. ATLAS reduces the number of required items by up to 90% while maintaining measurement precision. For instance, it matches whole-bank ability estimates using only 41 items (0.157 MAE) on HellaSwag (5,600 items). We further reconstruct accuracy from ATLAS's ability estimates and find that reconstructed accuracies closely match raw accuracies across all five benchmarks, indicating that ability preserves the global performance structure. At the same time, provides finer discrimination within accuracy-equivalent models: among more than 3,000 evaluated models, 23--31% shift by more than 10 rank positions, and models with identical accuracies receive meaningfully different ability estimates. Code and calibrated item banks available at https://github.com/Peiyu-Georgia-Li/ATLAS.

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SCOPE and SCION: A Benchmark and an Auditable Reference Pipeline for Schema Induction and Fusion from Text

Miaobo Hu, Xiaobo Guo, Shuhao Hu, BoKun Wang, Rui Chen, Xin Wang, Jun Xiao, Daren Zha

Schema graphs are an upstream bottleneck of schema-grounded information extraction and knowledge graph construction, yet most extraction systems assume the schema is already available.

We introduce SCOPE (Schema Construction and Ontology-induction Pipeline Evaluation), a train-text-only benchmark for corpus-to-schema induction and optional schema fusion from raw text, built from 24 public information extraction sources (15 RE and 9 EE) normalized into evaluation-only gold schema graphs; its core event-extraction target covers event types and within-event argument roles, with inter-event links reported separately.

We present SCION (Schema Construction and Induction with Ontology Normalization), an auditable reference pipeline rather than a new extraction architecture; it constructs candidate spaces from train text and restricts naming, merging, filtering, validation, and conservative fusion to candidate-linked evidence under strict JSON contracts.

On the SCOPE core suite, SCION-lite attains the highest F1 among released source-schema references, Text2Onto-style, LLM-only, and matched extract-then-aggregate baselines under Literal, Fuzzy, Continuous, and Graph schema-graph metrics, while the compact open-model SCION-RL variant reduces reliance on proprietary LLM schema engineers.

These results are reported against normalized typed-edge targets rather than as claims that induced schemas surpass human ontology design; the release includes

evidence-linked outputs, parse/fallback logs, candidate retention/merging logs, run manifests, code, and benchmark packages at https://github.com/wandugu/paper_scion.
****

DMCO: Budget-Aware Co-Optimization of Data Cleaning and AutoML
Xiaoou Ding, Zekai Qian, Siying Chen, Hongbin Hu, Chen Wang, Hongzhi Wang, Jianmin Wang
Data cleaning and automated machine learning (AutoML) are both crucial for reliable learning systems, yet are commonly treated as independent or sequential stages. This separation ignores their strong interaction and leads to inefficient use of limited computational budgets.We propose DMCO, a unified framework that jointly optimizes data cleaning and model construction under a fixed resource budget. DMCO reformulates the traditional two-stage pipeline into a time-sliced process, where data cleaning and AutoML are interleaved and adaptively scheduled. We introduce a gradient-based data cleaning sampling strategy with theoretical guarantees for minimizing gradient estimation variance, and integrates it with loss-driven sampling and progressive AutoML fitting to continuously leverage intermediate data quality improvements.Experiments on six real-world datasets show that DMCO consistently outperforms standalone data cleaning and AutoML baselines on both classification and regression tasks, as measured by F1 score and MSE. Under limited budgets, DMCO achieves up to 82.19% of the performance of full data cleaning with exhaustive AutoML, while remaining robust across different AutoML frameworks.
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PretrainZero: Reinforcement Active Learning on Pretraining Data
Xingrun Xing, Zhiyuan Fan, Jie Lou, Guoqi Li, Jiajun Zhang, Debing Zhang
Recent reinforcement learning (RL) based large-thinking models demonstrate impressive expert-level abilities, i.e., software and math, but still rely heavily on verifiable rewards in specific domains, which places a significant bottleneck to extend the performance boundary of general reasoning capabilities. In this work, we propose PretrainZero, a reinforcement active learning framework built on the pretraining corpus to extend RL from domain-specific post-training to general pretraining. PretrainZero features the following characteristics: 1) Active pretraining objectives: inspired by the active learning ability of humans, PretrainZero learns a unified reasoning policy to actively identify reasonable and informative contents from pretraining corpus, and reason to predict these contents by RL. 2) Self-supervised learning: without any verifiable labels, pretrained reward models, or supervised fine-tuning, we directly pretrain reasoners from $3\sim30$B base models on the general Wikipedia corpus using RL, significantly breaking the verification data-wall for general reasoning. 3) Verification scaling: by tackling increasingly challenging masked spans, PretrainZero substantially enhances the general reasoning abilities of pretrained base models. With reinforcement learning on pretraining data, PretrainZero improves Qwen3-4B-Base for 8.43, 5.96 and 10.60 on MMLU-Pro, SuperGPQA and math average benchmarks. In post-training, the pretrained models can also serve as reasoning foundation models for downstream RLVR tasks.
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Causal Matrix Completion under Multiple Treatments via Mixed Synthetic Nearest Neighbors
Minrui Luo, Zhiheng Zhang
Synthetic Nearest Neighbors (SNN) provides a principled solution to causal matrix completion under missing-not-at-random (MNAR) by exploiting local low-rank structure through fully observed anchor submatrices. However, its effectiveness critically relies on sufficient data availability within each treatment level, a condition that often fails in settings with multiple or complex treatments.
In this work, we propose Mixed Synthetic Nearest Neighbors (MSNN), a new entry-wise causal identification estimator that integrates information across treatment levels. We show that MSNN retains the finite-sample error bounds and asymptotic normality guarantees of SNN, while enlarging the effective sample size available for estimation. Empirical results on synthetic and real-world datasets illustr

ate the efficacy of the proposed approach, especially under data-scarce treatmen
t levels.
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Biased Generalization in Diffusion Models
Jerome Garnier-Brun, Luca Biggio, Davide Beltrame, Marc Mezard, Luca Saglietti
Generalization in generative modeling is defined as the ability to learn an unde
rlying distribution from a finite dataset and produce novel samples, with evalua
tion largely driven by held-out performance and perceived sample quality. In pra
ctice, training is often stopped at the minimum of the test loss, taken as an op
erational indicator of generalization. We challenge this viewpoint by identifyin
g a phase of *biased generalization* during training, in which the model continu
es to decrease the test loss while favoring samples with anomalously high proxim
ity to training data. By training the same network on two disjoint datasets and
comparing the mutual distances of generated samples and their similarity to trai
ning data, we introduce a quantitative measure of bias and demonstrate its prese
nce on real images. We then study the mechanism of bias, using a controlled hier
archical data model where access to exact scores and ground-truth statistics all
ows us to precisely characterize its onset. We attribute this phenomenon to the
sequential nature of feature learning in deep networks, where coarse structure i
s learned early in a data-independent manner, while finer features are resolved
later in a way that increasingly depends on individual training samples. Our res
ults show that early stopping at the test loss minimum, while optimal under stan
dard generalization criteria, may be insufficient for privacy-critical applicati
ons.
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Beyond Model Ranking: Predictability-Aligned Evaluation for Time Series Forecast
ing
Wanjin Feng, Yuan Yuan, Jingtao Ding, Yong Li
In the era of increasingly complex AI models for time series forecasting, progre
ss is often measured by marginal improvements on benchmark leaderboards. However
, standard evaluations rely on aggregate metrics (e.g., MSE) that conflate model
 capability with the intrinsic difficulty of the evaluated instances. To address
 this, we propose a diagnostic framework anchored in **Spectral Coherence Predic
tability (SCP)**, which provides an efficient $\mathcal{O}(N\log N)$ per-instanc
e difficulty reference and yields a corresponding linear MSE lower bound.
Complementing this, we introduce the **Linear Utilization Ratio (LUR)** to quant
ify how effectively models exploit linearly predictable structures across freque
ncies.
Experiments on synthetic and real-world benchmarks show that SCP aligns strongly
 with realized forecasting errors across diverse state-of-the-art forecasters.
Using this lens, we uncover ``predictability drift,'' revealing that task diffic
ulty is not static but fluctuates significantly over time and variables.
Furthermore, stratified evaluation exposes complementary architectural strengths
 across distinct frequency bands and difficulty regimes.
Overall, we advocate moving beyond leaderboard-style ranking toward a more insig
htful, predictability-aware evaluation that fosters fairer model comparisons and
 a deeper understanding of model behavior.
Code and data are available at https://github.com/WanjinVon/TS_Predictability.
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Reasoning Cache: Continual Improvement Over Long Horizons via Short-Horizon RL
Ian Wu, Yuxiao Qu, Amrith Setlur, Aviral Kumar
Large Language Models (LLMs) that continue improving at test-time budgets far be
yond their training budgets can solve harder problems by leveraging additional i
nference compute: we refer to this property as extrapolation. Standard on-policy
 RL operates on fixed problem distributions and training budgets, giving rise to
 a distribution shift between train and test that limits the resulting model's e
xtrapolation capabilities. To address this, we introduce RC, an iterative decodi
ng algorithm replacing standard autoregressive decoding that enables models to e
xtrapolate to lengths an order of magnitude longer than those seen during traini
ng. RC exploits the asymmetry between summarization and generation capabilities

present in LLMs to construct a decoding process that improves consistently over iterations. Its effectiveness can be further increased through training, which amplifies the model’s ability to perform summary-conditioned reasoning while avoiding the challenges of long-horizon RL. Empirically, training a 4B instruction-following model with RC using a 16k-token training budget improves performance on HMMT 2025 from 40% to 70% when evaluated with a 512k-token test budget, substantially surpassing comparably sized LLMs.
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Target-Oriented Pretraining Data Selection via Neuron-Activated Graph
Zijun Wang, Haoqin Tu, Weidong Zhou, Yiyang Zhou, Xiaohuan Zhou, Bingni Zhang, Weiguo Feng, Taifeng Wang, Cihang Xie, Fengze Liu
Everyday tasks come with a target, and pretraining models around this target is what turns them into experts. In this paper, we study target-oriented language model (LM) pretraining by introducing ***N**euron-**A**ctivated **G**raph Ranking* (NAG-based Ranking), a training-free and interpretable framework for target pretraining data selection.
Rather than using black-box representations, our approach directly characterizes each target input by a sparse set of high-impact neurons in any off-the-shelf LLMs.
Concretely, we quantify neuron impact and select the most influential neurons across layers into a compact ***N**euron-**A**ctivated **G**raph* (NAG), and rank candidate data by NAG similarity to target examples.
We conduct experiments across six benchmarks, where our NAG-based Ranking improves target-oriented pretraining by 4.9\% on average over random sampling, and also outperforms state-of-the-art baselines by 5.3\% accuracy on HellaSwag.
It also remains effective under a more applicable multi-target setting, where our best setup surpasses two baselines by 1.1\% and 4.1\%, respectively.
Furthermore, we provide a comprehensive analysis on *why* and *how* our NAG works, e.g., deactivating NAG-selected neurons (only 0.12\% of all) causes a 23.5\% performance collapse, and restricting NAG to the final
layer incurs a 4.1\% average drop, indicating that NAG captures a sparse ``functional backbone'' for learning target features.
We release the code at https://github.com/asillycat/NAG.
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TextMesh4D: Zero-shot Text-to-4D Mesh Generation
Sisi Dai, Xinxin Su, Kai Xu
Large-scale, high-quality dynamic 3D (4D) assets are essential for learning physically grounded representations, but remain costly to capture and annotate at scale. This limits the viability of supervised 4D learning and motivates zero-shot text-to-4D generation leveraging pretrained diffusion priors. To model complex dynamics, prior methods typically adopt implicit 3D representations (e.g., NeRFs or 3DGS) for their deformation capacity. However, their implicit nature provides limited control over surface topology, which hinders high-fidelity geometry and makes temporally coherent surface reconstruction challenging. To address these limitations, we explore zero-shot text-to-4D mesh generation. However, a structural mismatch arises when combining diffusion-based guidance with topology-constrained meshes: the guidance is noisy and spatially inconsistent, while meshes impose severe topological constraints, making direct vertex-level deformation unstable. In this paper, we introduce TextMesh4D, the first zero-shot framework for text-to-4D that directly generates dynamic meshes by addressing the above challenge at two complementary levels. Geometrically, we shift deformation modeling from vertices to faces via a Jacobian Deformation Field (JDF), enabling topology-aware surface reconstruction through an integrability-enforcing integration formulation. Semantically, we propose a Local-Global Semantic Regularizer (LGSR) that preserves identity over time by jointly constraining local deformation plausibility and global shape consistency. Extensive experiments demonstrate state-of-the-art temporal consistency, structural fidelity, and visual quality, while remaining efficient on a single 24GB GPU.
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MemCast: Memory-Driven Time Series Forecasting with Experience-Conditioned Reaso

ning
Xiaoyu Tao, Mingyue Cheng, Ze Guo, Shuo Yu, Yaguo Liu, Qi Liu, Shijin Wang
Time series forecasting (TSF) plays a critical role in decision-making for many real-world applications. Recently, large language model (LLM)- based forecasters have made promising advancements.
Despite their effectiveness, existing methods often lack explicit experience accumulation and continual evolution.
In this work, we propose MemCast, a learning-to-memory framework that reformulates TSF as an experience-conditioned reasoning task.
Specifically, we learn experience from the training set and organize it into a hierarchical memory. This is achieved by summarizing prediction results into historical patterns, distilling inference trajectories into reasoning wisdom, and inducing extracted temporal features into general laws.
Furthermore, during inference, we leverage historical patterns to guide the reasoning process and utilize reasoning wisdom to select better trajectories, while general laws serve as criteria for reflective iteration. Additionally, to enable continual evolution, we design a dynamic confidence adaptation strategy that updates the confidence of individual entries without leaking the test set distribution.
Extensive experiments on multiple datasets demonstrate that MemCast consistently outperforms previous methods, validating the effectiveness of our approach.
Our code is available at https://github.com/Xiaoyu-Tao/MemCast-TS.
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Geometric Convergence of Gauss–Newton for Neural Networks: Riemannian Geometry and Adaptive Damping
Semih Cayci
Ill-conditioned kernel matrices can make first-order methods for training neural networks converge slowly. We establish non-asymptotic convergence bounds for the Gauss–Newton method in both under- and overparameterized regimes, showing it avoids these conditioning bottlenecks. In the underparameterized setting, Gauss–Newton gradient flow in parameter space induces a Riemannian gradient flow on a low-dimensional submanifold of function space. Using tools from Riemannian optimization, we show that, under an appropriate output scaling, the loss satisfies geodesic Polyak–Lojasiewicz and Lipschitz-smoothness conditions, implying geometric convergence to the optimal in-class predictor at an explicit rate independent of Gram-matrix conditioning. In the overparameterized setting, we identify adaptive, curvature-aware regularization schedules and prove fast geometric convergence to a global optimum for both Gauss–Newton gradient flow and discrete-time Gauss–Newton iterates, with rates independent of the minimum eigenvalue of the neural tangent kernel. Overall, Gauss–Newton can be provably faster in ill-conditioned regimes where first-order methods slow down.
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KODA: Contrastive Representation Comparison and Alignment for Vision-Language Foundation Models
Youqi WU, Mohammad Jalali, Farzan Farnia
Vision-language foundation models such as CLIP and SigLIP provide widely used representations for multimodal learning systems. While these models are typically compared through downstream performance, such evaluations often do not explain how their representations differ structurally. In this work, we study this problem through the task of *Contrastive Embedding Clustering*: identifying sample subsets that are weakly clustered under one representation but strongly clustered under another. We propose *Kernel Optimization for Discrepancy Analysis (KODA)*, a kernel-based framework for contrastive representation comparison and alignment. KODA constructs unified multimodal kernels through modality-wise kernel composition and formulates discrepancy discovery as a constrained optimization problem that searches for coherent structures in one representation while suppressing coherence in a reference representation. This yields interpretable discrepancy directions associated with specific sample subsets and modality interactions. To scale KODA to large vision-language datasets, we develop randomized low-dimensional approximations of joint kernels using random projections, including Random Fo

urier Features for shift-invariant kernels. Empirically, KODA identifies consistent and interpretable discrepancy structures across vision-language representations and provides sample subsets for representation alignment. The code is available at https://github.com/yokiwuuu/KODA.

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Optimal Transport for LLM Reward Modeling from Noisy Feedback

Licheng Pan, Haocheng Yang, Haoxuan Li, Yunsheng Lu, Yongqi Tong, Yinuo Wang, Shijian Wang, Zhixuan Chu, Lei Shen, Yuan Lu, Hao Wang

Reward models are fundamental to Reinforcement Learning from Human Feedback (RLHF), yet real-world datasets are inevitably corrupted by noisy preference. Conventional training objectives tend to overfit these errors, while existing denoising approaches often rely on homogeneous noise assumptions that fail to capture the complexity of linguistic preferences. To handle these challenges, we propose SelectiveRM, a framework grounded in optimal transport. We first devise a Joint Consistency Discrepancy to align the distribution of model predictions with preference data. Furthermore, to address the limitation of strict mass conservation which compels the model to fit outliers, we incorporate a Mass Relaxation mechanism via partial transport. This enables the autonomous exclusion of samples with noisy preference that contradict semantic consistency. Theoretically, we demonstrate that SelectiveRM optimizes a tighter upper bound on the unobserved clean risk. Extensive experiments validate that our approach significantly outperforms state-of-the-art baselines across diverse benchmarks.

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Exploring Data-Free LoRA Transferability for Video Diffusion Models

Yuchen Wang, Wenliang Zhong, Lichen Bai, zikai zhou, Shitong Shao, Bojun Cheng, Shuo Chen, Shuo Yang, Zeke Xie

Video diffusion models leveraging step distillation or causal distillation have achieved remarkable performance. However, adapting existing LoRAs to these variants remains a critical challenge due to weight space mismatches. We observe that direct application leads to style degradation and structural collapse, yet the underlying mechanisms remain poorly understood. To fill this gap, we delve into the weight space and identify that the incompatibility stems from spectral interference within shared functional clusters defined over singular subspaces. Specifically, our analysis reveals that while both paradigms respect spectral rigidity, they establish conflicting routing pathways that clash through constructive overload or destructive cancellation. To address this issue, we propose Cluster-Aware Spectral Arbitration (CASA), a data-free framework that dynamically arbitrates between safeguarding the target's manifold and restoring LoRA alignment based on spectral density. Extensive experiments demonstrate that CASA effectively mitigates artifacts and revives LoRA functionality. Our code is available at https://github.com/Noahwangyuchen/CASA.

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Factorized Scheduling Principle: Learning Interpretable and Transferable Policies via Structured Additive Functions

Hong Je-Gal, Hyun-Suk Lee

Scheduling problems arise from repeatedly selecting one item from a set of candidates based on their states. These problems often reduce to assigning priority scores and choosing the highest-ranked item.

In this work, we propose a factorized scheduling principle (FSP) framework to learn interpretable and transferable scheduling rules. The FSP framework represents system states as condition distributions and decomposes a global scheduling principle into additive univariate and pairwise components with identifiability constraints.

The scheduling principle enables the framework to maintain a simple priority-based structure during deployment. This principle is learned by using a policy-based objective combined with a temporal-difference signal defined on the condition distribution.

Experiments on synthetic and realistic scheduling tasks demonstrate the FSP framework's strong performance, interpretability, and zero-shot generalization across different system scales.

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Bayesian Rain Field Reconstruction using Commercial Microwave Links and Diffusion Model Priors
Badr MOUFAD, Albina Ilina, Hai Victor Habi, Salem Lahlou, Yazid Janati, HAGIT MESSER, Eric Moulines
Commercial Microwave Links (CMLs) offer dense spatial coverage for rainfall sensing but produce path-integrated measurements that make accurate ground-level reconstruction challenging. Existing methods typically oversimplify CMLs as point sensors and neglect the physical power-law relating rainfall to signal attenuation, resulting in degraded performance under heterogeneous precipitation. In this work, we view rain field reconstruction as a Bayesian inverse problem with Diffusion Models (DMs) as high-fidelity spatial priors. We show that diffusion models better preserve key rainfall statistics compared to censored Gaussian processes. Framing rainfall estimation as a Bayesian inverse problem with a DM prior enables training-free posterior sampling using a broad family of methods, including Plug-and-Play, Sequential Monte Carlo, and Replica Exchange methods. Experiments on synthetic and real-world datasets demonstrate consistent improvements over established CML-based reconstruction baselines.
****
A Robust Optimization Guided Pruning Framework for Vision and Large Language Models
Gabriel Afriat, Hussein Hazimeh, Dimitris Paparas, Rahul Mazumder
Pruning is a common approach to reduce the memory footprint and inference cost of large vision and language models. As these architectures continue to scale, one-shot pruning methods - i.e. approaches that prune the network without any retraining - have become increasingly attractive. Many popular one-shot pruning methods (e.g., WoodFisher, CAP, SparseGPT, and ALPS) typically optimize a quadratic objective under sparsity constraints. However, in practice, this objective is affected by multiple sources of uncertainty, including noise in the calibration data and variability introduced by algorithmic updates. To address these issues, we introduce RobOP, a robust optimization framework that explicitly accounts for such uncertainties. RobOP is modular and flexible, and can be applied with any existing pruning method through simple modifications motivated by our theoretical framework. We demonstrate that by taking into account uncertainty, RobOP offers improvements over prior pruning approaches. Our framework applies tractably across a range of stylized uncertainty sets, enabling robust one-shot pruning at scale. Our code is available at https://github.com/mazumder-lab/RobOP.
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Knowing the Unknown: Interpretable Open-World Object Detection via Concept Decomposition Model
Xueqiang Lv, Shizhou Zhang, Yinghui Xing, Di Xu, PENG WANG, Yanning Zhang
Open-world object detection (OWOD) requires incrementally detecting known categories while reliably identifying unknown objects. Existing methods primarily focus on improving unknown recall, yet overlook interpretability, often leading to known–unknown confusion and reduced prediction reliability. This paper aims to make the entire OWOD framework interpretable, enabling the detector to truly “knowing the unknown.” To this end, we propose a concept-driven InterPretable OWOD framework(IPOW) by introducing a Concept Decomposition Model (CDM) for OWOD, which explicitly decomposes the coupled RoI features in Faster R-CNN into discriminative, shared, and background concepts. Discriminative concepts identify the most discriminative features to enlarge the distances between known categories, while shared and background concepts, due to their strong generalization ability, can be readily transferred to detect unknown categories. Leveraging the interpretable framework, we identify that known–unknown confusion arises when unknown objects fall into the discriminative space of known classes. To address this, we propose Concept-Guided Rectification (CGR) to further resolve such confusion. Extensive experiments show that IPOW significantly improves unknown recall while mitigating confusion, and provides
concept-level interpretability for both known and unknown predictions.
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ALAS: Additive Learnable Alpha-Stable Kernels for Flexible Bayesian Optimization
Weibo Huang, Cheng Hua
Bayesian Optimization is widely used for expensive black-box optimization, yet its success often depends on choosing a kernel that matches the objective’s unknown structure. In this work, we propose ALAS, a flexible Gaussian Process kernel family built from symmetric α-stable spectral components. By learning the stability parameter α, ALAS adapts its effective smoothness from data, capturing both smooth trends and sharp irregularities. We present two parameterizations: ALAS, a single stationary component with joint spectral modulation, and ALAS-Sep, a separable variant that learns dimension-wise tail behavior to improve robustness on approximately decomposable objectives. Experiments on standard benchmarks and real-world surrogates demonstrate strong and robust performance across diverse settings.
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The Trojan Knowledge: Bypassing Commercial LLM Guardrails via Harmless Prompt Weaving and Adaptive Tree Search
Rongzhe Wei, Peizhi Niu, Xinjie Shen, Tony Tu, Yifan Li, Ruihan Wu, Eli Chien, Pin-Yu Chen, Olgica Milenkovic, Pan Li
Large language models (LLMs) remain vulnerable to jailbreak attacks that bypass safety guardrails. Existing approaches overwhelmingly operate within the prompt-optimization paradigm; the resulting prompts typically retain malicious semantic signals that modern guardrails are primed to detect. In contrast, we identify a deeper vulnerability stemming from the highly interconnected nature of an LLM’s internal knowledge. This structure allows harmful objectives to be realized by weaving together sequences of benign sub-queries, each of which individually evades detection. To exploit this loophole, we introduce the Correlated Knowledge Attack Agent (CKA-Agent), a dynamic framework that reframes jailbreaking as an adaptive, tree-structured exploration of the target model’s knowledge base. The CKA-Agent issues locally innocuous queries, uses model responses to guide exploration across multiple paths, and ultimately assembles the aggregated information to achieve the original harmful objective. Evaluated across SOTA commercial LLMs, CKA-Agent consistently achieves over 95\% success rates even against strong guardrails, underscoring the severity of this vulnerability and the urgent need for defenses against such knowledge-decomposition attacks. Our codes are available at https://github.com/Graph-COM/CKA-Agent.
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Prescriptive Scaling Reveals the Evolution of Language Model Capabilities
Hanlin Zhang, Jikai Jin, Vasilis Syrgkanis, Sham M. Kakade
Machine learning model performance arises from competition and application. For deployment, we consider the prescriptive scaling laws: given a pre-training compute budget, what downstream accuracy is attainable with contemporary post-training practice, and how stable is that mapping as the field evolves? Using large-scale observational evaluations with 5k observational and 2k newly sampled data on model performance, we estimate capability boundaries—high conditional quantiles of benchmark scores as a function of log pre-training FLOPs, via smoothed quantile regression with a monotone, saturating sigmoid parameterization. We validate the temporal reliability by fitting on earlier model generations and evaluating on later releases. Across various tasks, the estimated boundaries are mostly stable, with the exception of math reasoning that exhibits a consistently advancing boundary over time. We then extend our approach to analyze task-dependent saturation and to probe contamination-related shifts on math reasoning tasks. Finally, we introduce an efficient algorithm that recovers near-full-data frontiers using roughly 20% of evaluation budget. Together, our work releases the Proteus-2k, the latest model performance evaluation dataset, and introduces a practical methodology for translating compute budgets into reliable performance expectations and for monitoring when capability boundaries move.
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MapDream: Task-Driven Map Learning for Vision-Language Navigation
Guoxin Lian, Shuo Wang, Yucheng Wang, Yongcai Wang, Maiyue Chen, Kaihui Wang, Bo Zhang, Zhizhong Su, Deying Li, Zhaoxin Fan

Vision-Language Navigation (VLN) requires agents to follow natural language instructions in partially observed 3D environments, motivating map representations that aggregate spatial context beyond local perception.
However, most existing approaches rely on hand-crafted maps constructed independently of the navigation policy.
We argue that maps should instead be learned representations shaped directly by navigation objectives rather than exhaustive reconstructions.
Based on this insight, we propose MapDream, a map-in-the-loop framework that formulates map construction as autoregressive bird’s-eye-view (BEV) image synthesis.
The framework jointly learns map generation and action prediction, distilling environmental context into a compact three-channel BEV map that preserves only navigation-critical affordances.
Supervised pre-training bootstraps a reliable mapping-to-control interface, while the autoregressive design enables end-to-end joint optimization through reinforcement fine-tuning.
Experiments on R2R-CE and RxR-CE achieve state-of-the-art monocular performance, validating task-driven generative map learning.
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Sample Margin-Aware Recalibration of Temperature Scaling
Haolan Guo, Linwei Tao, Haoyang Luo, Minjing Dong, Chang Xu
Deep neural networks frequently exhibit overconfidence, undermining reliability in safety-critical applications. Existing adaptive methods rely on indirectly learned proxies of sample difficulty. We establish the logit margin as a direct and principled hardness indicator. We prove that margin constrains the feasible temperature range for a target confidence. Empirically, margin strongly correlates with decision boundary proximity and reveals systematic calibration patterns across difficulty levels. We further identify a fundamental flaw in NLL-based optimization: minimizing NLL can paradoxically worsen calibration. To address this, we introduce Charbonnier-SoftECE, a smooth objective that provably upper-bounds the smooth calibration error (smCE). Building on these insights, we propose SMART (Sample Margin-Aware Recalibration of Temperature), a lightweight method that learns a sample-wise margin-to-temperature mapping guided by our calibration-centric objective. Experiments demonstrate state-of-the-art calibration across CNNs and ViTs on standard, long-tailed, and distribution-shifted benchmarks, with minimal inference-time overhead. Code is available at: https://github.com/Misakaaaaaz/ICML2026-SMART.
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Temporal Straightening for Latent Planning
Ying Wang, Oumayma Bounou, Gaoyue Zhou, Randall Balestriero, Tim G. J. Rudner, Yann LeCun, Mengye Ren
Learning good representations is essential for latent planning with world models. While pretrained visual encoders produce strong semantic visual features, they are not tailored to planning and contain information irrelevant---or even detrimental---to planning. Inspired by the perceptual straightening hypothesis in human visual processing, we introduce temporal straightening to improve representation learning for latent planning. Using a curvature regularizer that encourages locally straightened latent trajectories, we jointly learn an encoder and a predictor of a Joint-Embedding Predictive Architecture (JEPA) world model. We show that reducing curvature this way makes the Euclidean distance in latent space a better proxy for the geodesic distance and improves the conditioning of the planning objective. We demonstrate empirically that temporal straightening makes gradient-based planning more stable and yields significantly higher success rates across a suite of goal-reaching tasks. Our code is in https://agenticlearning.ai/temporal-straightening.
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Identifiable Markov Switching Models with Instantaneous Effects and Exponential Families
Roel Hulsman, Carles Balsells-Rodas, Sara Magliacane
Temporal systems often exhibit non-stationary behaviour, such as seasonal climat

e variation or glucose fluctuations in patients with type-1 diabetes.
One way to model non-stationarity is through discrete latent *regimes*, *i.e.*, stationary segments of time.
Such systems induce a *Markov Switching Model* (MSM), a class of Hidden Markov Models with autoregressive dependencies among latent regimes and observed variables.
Identifying latent regimes is challenging in the presence of frequent regime switches and nonlinear and non-Gaussian dynamics, particularly when there are *instantaneous effects* between the variables, *e.g.*, due to slow rates of measurements.
In this work, we establish the identifiability of both latent regimes and regime-dependent causal structures under temporal regime dependencies, nonlinear lagged and instantaneous effects, and independent noise from the exponential family.
Our identifiability theory subsumes non-temporal mixtures of causal models.
Furthermore, we introduce $\texttt{FlowMSM}$, a regime detection framework that can be paired with any stationary causal discovery method to recover regime-dependent causal structures.
Experiments on synthetic benchmarks and a financial economics dataset demonstrate the effectiveness of our approach to detect latent regimes and discover causal structures from non-stationary time series.
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SPUR: Scale-Partitioned Uncertainty Rectification for Robust UAV-on-UAV Interception
Chenqi Yan, Zhaoyu Zeng, Yifeng Yang, Jundong Zhou, Zhuoyuan Ni, Junqi Wu, Qinying Gu, Xinbing Wang, Nanyang Ye
Robust aerial target detection for autonomous UAV-on-UAV pursuit is severely hindered by continuous scale drift, long-tailed scale imbalance, and flight-induced visual noise, rendering standard empirical risk minimization strategies poorly aligned with real-world deployment. To address these challenges, we propose a scale-aware robust optimization framework that performs group-wise minimax optimization over scale-partitioned data, ensuring balanced robustness across long-, mid-, and close-range engagement regimes. We further introduce an uncertainty-rectified regression loss to suppress noise-driven errors without discarding informative hard examples, complemented by a control-aligned center accuracy penalty that prioritizes the localization precision required for stable flight control. Extensive experiments demonstrate that our method yields substantially improved robustness under visual degradation, with significantly slower decay in detection mAP and center-point accuracy compared to baselines. Validated through both photorealistic simulations and real-world flight tests, our system achieves **real-time performance of 120 FPS** on an embedded NVIDIA Orin NX platform, confirming its practical efficacy for high-speed interception.
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Dual Mechanisms of Value Expression: Intrinsic vs. Prompted Values in Large Language Models
Jongwook Han, Jongwon Lim, Injin Kong, Yohan Jo
Large language models can express values in two main ways: (1) $\textit{intrinsic}$ expression, reflecting the model's inherent values learned during training, and (2) $\textit{prompted}$ expression, elicited by explicit prompts. Given their widespread use in value alignment, it is paramount to clearly understand their underlying mechanisms, particularly whether they mostly overlap (as one might expect) or rely on distinct mechanisms. We analyze this largely understudied problem at the mechanistic level using two approaches: (1) $\textit{value vectors}$, feature directions representing value mechanisms extracted from the residual stream, and (2) $\textit{value neurons}$, MLP neurons that contribute to value vectors.
We demonstrate that intrinsic and prompted value mechanisms partly share common components crucial for inducing value expression, generalizing across languages and reconstructing theoretical inter-value correlations in the model's internal representations. Yet, each mechanism also possesses unique components that fulfill distinct roles. In particular, the intrinsic mechanism activates in more dive

rse value-related scenarios and promotes response diversity, whereas the prompte d mechanism strengthens instruction compliance, taking effect even in distant ta sks like jailbreaking.
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Advancing SVD-based LLM Compression via Layer-Wise Error Model Search
Moritz Thoma, Maximilian Groezinger, Maximilian Forstenhäusler, Emad Aghajanzade h, Manoj Rohit Vemparala, Christos Anagnostopoulos, Pierpaolo Mori, Nael Fasfous , Alexander Frickenstein, Daniel Mueller-Gritschneder, Ulf Schlichtmann
Low-rank SVD-based compression offers a powerful strategy to reduce the computat ional costs of LLMs. However, existing methods face two key limitations: (i) glo bal rank allocation, where uncalibrated error proxies fail to capture complex er ror propagation, and (ii) decomposition quality, where Fisher-based estimators s uffer from severe rank collapse. In this work, we address these limitations by i ntroducing Layer-wise Error Modeling Search (LEMS) and KFAC-SVD. LEMS advances r ank allocation by introducing a layer-wise error surrogate that integrates local and global layer importance alongside a propagation bias, enabling effective gl obal rank allocation via an ILP formulation. KFAC-SVD improves decomposition qua lity by utilizing token-wise statistics, mitigating the rank deficiency observed in prior Fisher-based SVD approaches. Across Mistral, Qwen3, and Llama3 model f amilies, we show that LEMS consistently outperforms existing search strategies, delivering significant zero-shot accuracy gains of up to 4.8 p.p. that generaliz e to model sizes of 70B parameters, while KFAC-SVD achieves an average perplexit y improvement of 15%. Project Page & Code: https://lems-svd.github.io
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The Unlearnability Phenomenon in RLVR for Language Models
Yulin Chen, He He, Chen Zhao
Reinforcement Learning with Verifiable Reward (RLVR) has proven effective in imp roving Large Language Model's (LLM) reasoning ability. However, the learning dyn amics of RLVR remain underexplored.
In this paper, we reveal a counterintuitive phenomenon: among hard examples that the model initially struggles with, a substantial subset remains unlearnable ev en when correct rollouts are present.
To understand the phenomenon, we first demonstrate that existing optimization an d sampling techniques fail to resolve unlearnability.
With cross-example gradient analysis, we show that unlearnable examples have fun damental representation issue, characterized by low gradient similarity with the rest of the examples and ungeneralizable reasoning patterns.
We further show that representation flaws are difficult to mitigate in RL, as da ta augmentation does not improve gradient similarity.
Our study provides the first systematic characterization of unlearnable data in RLVR training and reveals fundamental limitations in current RL approaches for r easoning tasks.
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Reasoning Can Be Restored by Correcting a Few Decision Tokens
Changshuo Shen, Leheng Sheng, Yuxin Chen, Xiang Wang, An Zhang
Large reasoning models (LRMs) substantially outperform their base LLM counterpar ts on challenging reasoning benchmarks, yet it remains poorly understood where b ase models go wrong during token-by-token generation and how to narrow this gap efficiently. We study the base–reasoning gap by quantifying token-level distribu tional disagreement between a base model and a stronger reasoning model using li kelihood-based divergences. Across benchmarks, we find that the reasoning advant age is highly sparse and concentrates on a small set of early, planning-related decision tokens. For instance, on Qwen3-0.6B, only $\sim$8\% of generated tokens account for the salient disagreement; these tokens concentrate early in the res ponse, are strongly enriched in planning-related decisions ($17\times$), and coi ncide with high base-model uncertainty—suggesting that base models fail mainly a t early planning points that steer the subsequent reasoning trajectory. Building on these findings, we propose disagreement-guided token intervention, a simple inference-time delegation scheme that performs a one-token takeover by the reaso ning model only at high-disagreement positions and immediately switches back to

the base model. With a small intervention budget, this sparse delegation substantially recovers and can even surpass the performance of a same-size reasoning model on challenging reasoning tasks. Code is available at \url{https://github.com/AlphaLab-USTC/RRTokenIntervention}.
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E-VAds: An E-commerce Short Videos Understanding Benchmark for MLLMs
Xianjie Liu, Yiman Hu, Liang Wu, Ping Hu, Yixiong Zou, Jian Xu, Bo Zheng
E-commerce short videos represent a high-revenue segment of the online video industry characterized by a goal-driven format and dense multi-modal signals. Current models often struggle with these videos because existing benchmarks focus primarily on general-purpose tasks and neglect the reasoning of commercial intent. In this work, we first propose a **multi-modal information density assessment framework** to quantify the complexity of this domain. Our evaluation reveals that e-commerce content exhibits substantially higher density across visual, audio, and textual modalities compared to mainstream datasets, establishing a more challenging frontier for video understanding.
To address this gap, we introduce **E-commerce Video Ads Benchmark**, which is the first benchmark specifically designed for e-commerce short video understanding. We curated 3,961 high-quality videos from Taobao covering a wide range of product categories and used a multi-agent system to generate 19,785 open-ended Q\&A pairs. These questions are organized into Perception and Cognition and Reasoning, which consist of five distinct tasks. Finally, we develop **E-VAds-R1**, an RL-based reasoning model featuring a multi-grained reward design called **MG-GRPO**. This strategy provides smooth guidance for early exploration while creating a non-linear incentive for expert-level precision. Experimental results demonstrate that E-VAds-R1 achieves a 109.2\% performance gain in commercial intent reasoning with only a few hundred training samples. Data is available at https://github.com/TaobaoTmall-AlgorithmProducts/E-VAds_Benchmark.
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Implicit Intelligence - Evaluating Agents on What Users Don’t Say
Ved Sirdeshmukh, Marc Wetter
Real-world requests to AI agents are fundamentally underspecified. Natural human communication relies on shared context and unstated constraints that speakers expect listeners to infer. Current agentic benchmarks test explicit instruction-following but fail to evaluate whether agents can reason about implicit requirements spanning accessibility needs, privacy boundaries, catastrophic risks, and contextual constraints. We present **Implicit Intelligence**, an evaluation framework testing whether AI agents can move beyond prompt-following to become genuine goal-fulfillers, paired with **Agent-as-a-World (AaW)**, a harness where interactive worlds are defined in human-readable YAML files and simulated by language models. Our scenarios feature apparent simplicity in user requests, hidden complexity in correct solutions, and discoverability of constraints through environmental exploration. Evaluating 16 frontier and open-weight models across 205 scenarios, we find that even the best-performing model achieves only 48.3% scenario pass rate, revealing substantial room for improvement in bridging the gap between literal instruction-following and human-like contextual reasoning.
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Model-Based Diffusion Sampling for Predictive Control in Offline Decision Making
Haldun Balim, Na Li, Yilun Du
Offline decision-making via diffusion models often produces trajectories that are misaligned with system dynamics, limiting their reliability for control. We propose *Model Predictive Diffuser* (MPDiffuser), a compositional diffusion framework that combines a diffusion planner with a dynamics diffusion model to generate task-aligned and dynamically plausible trajectories. MPDiffuser interleaves planner and dynamics updates during sampling, progressively correcting feasibility while preserving task intent. A lightweight ranking module then selects trajectories that best satisfy task objectives. The compositional design improves sample efficiency and adaptability by enabling the dynamics model to leverage diverse and previously unseen data independently of the planner. Empirically, we demonstrate consistent improvements over prior diffusion-based methods on unconstraine

d (D4RL) and constrained (DSRL) benchmarks, and validate practicality through de
ployment on a real quadrupedal robot.
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On the Variability of Concept Activation Vectors
Julia Wenkmann, Damien Garreau
One of the most pressing challenges in artificial intelligence is to make models
 more transparent to their users.
Recently, explainable artificial intelligence has come up with numerous methods
to tackle this challenge.
A promising avenue is to use concept-based explanations, that is, high-level con
cepts instead of plain feature importance scores.
Among this class of methods, Concept Activation Vectors (CAVs, Kim et al., 2018)
 stand out as one of the main protagonists.
One interesting aspect of CAVs is that their computation requires sampling rando
m examples from the train set.
Therefore, the actual vectors obtained may vary depending on the randomness of t
his sampling.
In this paper, we propose a fine-grained theoretical analysis of CAV constructio
n in order to quantify their variability.
Our results, confirmed by experiments on several real-life datasets of four diff
erent modalities, point to an universal result: the variance of CAVs declines ro
ughly as $1/N$, where N is the number of random examples.
Based on this, we give practical recommendations for a resource-efficient applic
ation of the method.
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Learnability-Informed Fine-Tuning of Diffusion Language Models
Shubham Parashar, Atharv Chagi, Jacob Helwig, Lakshmi Jotsna Madhavarapu, Sushil
 Vemuri, James Caverlee, Dileep Kalathil, Shuiwang Ji
We aim to improve the reasoning capabilities of diffusion language models (DLMs)
. While SFT is a popular post-training recipe for autoregressive models, its use
 in DLMs faces challenges and can even hurt performance, though the underlying c
auses remain understudied. Our analysis reveals that vanilla SFT overlooks learn
ability, namely, *what* and *when* tokens are learned. Specifically, rare tokens
 are difficult to learn when most of the input is masked, whereas it is straight
forward and thus of little value to learn common
tokens when most of the input is unmasked. Motivated by our analysis, we propose
 LIFT, an efficient SFT-based post-training algorithm for
DLMs. LIFT learns easy tokens when most of the input is masked and hard tokens w
hen more context is available, thereby aligning training with the information av
ailable at different diffusion time steps. Our results show that LIFT outperform
s existing SFT baselines across six reasoning benchmarks, achieving up to a 3x r
elative gain on AIME’24 and AIME’25. Our code is publicly available at https://g
ithub.com/divelab/LIFT.
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DeepBlip: Estimating Conditional Average Treatment Effects Over Time
Haorui Ma, Dennis Frauen, Stefan Feuerriegel
Structural nested mean models (SNMMs) are a principled approach to estimate the
treatment effects over time. A particular strength of SNMMs is to break the join
t effect of treatment sequences over time into localized, time-specific ``blip e
ffects''. This decomposition promotes interpretability through the incremental e
ffects and enables the efficient offline evaluation of optimal treatment policie
s without re-computation. However, neural frameworks for SNMMs are lacking, as t
heir inherently sequential g-estimation scheme prevents end-to-end, gradient-bas
ed training. Here, we propose DeepBlip, the first neural framework for SNMMs, w
hich overcomes this limitation with a novel double optimization trick to enable
simultaneous learning of all blip functions. Our DeepBlip seamlessly integrates
sequential neural networks like LSTMs or transformers to capture complex tempora
l dependencies. By design, our method correctly adjusts for time-varying confoun
ding to produce unbiased estimates, and its Neyman-orthogonal loss function ensu
res robustness to nuisance model misspecification. Finally, we evaluate our Deep

Blip across various clinical datasets, where it achieves state-of-the-art performance.
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Inference Time Optimization with Confidence Dynamics
Yu Wang, Minghao Liu, Jiayun Wang, Jinrui Huang, Ankit Shah, Wei Wei
Inference time optimization techniques, such as repeated sampling, have significantly advanced the reasoning capabilities of Large Language Models (LLMs). However, the critical role of model uncertainty remains largely underexplored in these optimization strategies. In this paper, we investigate the dynamics of confidence along reasoning trajectories and for first time reveal a surprising and unique pattern: correct answer traces tend to exhibit confidence improvement over time (positive confidence gain), while incorrect traces show attenuated or declining confidence as reasoning proceeds. Based on this observation, we propose Confidence Dynamic Gain (CDG) based voting, which incorporates how the confidence trajectory of the response evolves along the reasoning chain. Experiments across four open-source architectures (DeepSeek-R1, gpt-oss, Gemma-3, Qwen-QwQ) on the AIME24/25, HMMT25, and BRUMO25 benchmarks demonstrate that CDG yields a significant performance boost over baselines. These results demonstrate that our method provides a robust discriminative signal for improving answer selection in LLM reasoning. We also provide theoretical insights for this phenomenon. Code will be released at https://github.com/Accenture/CDG.git.
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MemoryBench: A Benchmark for Memory and Continual Learning in LLM Systems
Qingyao Ai, Yichen Tang, Changyue Wang, Jianming Long, Weihang Su, Yiqun LIU
Scaling up data, parameters, and test-time computation has been the mainstream methods to improve LLM systems (LLMsys), but their upper bounds are almost reached due to the gradual depletion of high-quality data and marginal gains obtained from larger computational resource consumption. Inspired by the abilities of human and traditional AI systems in learning from practice, constructing memory and continual learning frameworks for LLMsys has become an important and popular research direction in recent literature. Yet, existing benchmarks for LLM memory often focus on evaluating the system on homogeneous reading comprehension tasks with long-form inputs rather than testing their abilities to learn from accumulated user feedback in service time. Therefore, we propose a user feedback simulation framework and a comprehensive benchmark covering multiple domains, languages, and types of tasks to evaluate the continual learning abilities of LLMsys. Experiments show that the effectiveness and efficiency of state-of-the-art baselines are far from satisfying, and we hope this benchmark could pave the way for future studies on LLM memory and optimization algorithms.
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GameDevBench: Evaluating Agentic Capabilities Through Game Development
Wayne Chi, Yixiong Fang, Arnav Yayavaram, Siddharth Yayavaram, Seth Karten, Qiuhong Anna Wei, Runkun Chen, Alexander Wang, Valerie Chen, Ameet Talwalkar, Chris Donahue
Despite rapid progress on coding agents, progress on their multimodal counterparts has lagged behind.
A key challenge is the scarcity of evaluation testbeds that combine the complexity of software development with the need for deep multimodal understanding.
In game development, agents must navigate large, dense codebases while manipulating intrinsically multimodal assets such as shaders, sprites, and animations within a visual game scene.
We present GameDevBench, the first benchmark for evaluating agents on game development tasks.
GameDevBench consists of 333 tasks derived from web and video tutorials.
Tasks require significant multimodal understanding and are complex---the average solution requires over three times the lines of code and file changes compared to prior software development benchmarks.
Agents struggle with game development, with the best agent and method solving only 53.8% of tasks.
We find a strong correlation between perceived task difficulty and multimodal co

mplexity, with average success rate dropping from 51.4% on gameplay-oriented tasks to 33.0% on 2D graphics tasks.
To improve multimodal capability, we introduce two simple image and video-based feedback mechanisms for agents.
Despite their simplicity, these methods consistently improve performance, increasing GPT-5.4's performance from 41.1% to 52.0% when given visual feedback.
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Causal Flow Q-Learning for Robust Offline Reinforcement Learning
Mingxuan Li, Junzhe Zhang, Elias Bareinboim
Expressive policies based on flow-matching have been successfully applied in reinforcement learning (RL) more recently due to their ability to model complex action distributions from offline data. These algorithms build on standard policy gradients, which assume that there is no unmeasured confounding in the data. However, this condition does not necessarily hold for pixel-based demonstrations when a mismatch exists between the demonstrator's and the learner's sensory capabilities, leading to implicit confounding biases in offline data. We address the challenge by investigating the problem of confounded observations in offline RL from a causal perspective. We develop a novel causal offline RL objective that optimizes policies' worst-case performance that may arise due to confounding biases. Based on this new objective, we introduce a practical implementation that learns expressive flow-matching policies from confounded demonstrations, employing a deep discriminator to assess the discrepancy between the target policy and the nominal behavioral policy. Experiments across 25 pixel-based tasks demonstrate that our proposed confounding-robust augmentation procedure achieves a success rate 120\% that of confounding-unaware, state-of-the-art offline RL methods.
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From Muon to Gluon: Bridging Theory and Practice of LMO-based Optimizers for LLMs
Artem Riabinin, Egor Shulgin, Kaja Gruntkowska, Peter Richtárik
Recent developments in deep learning optimization have brought about radically new algorithms based on the Linear Minimization Oracle (LMO) framework, such as Muon (Jordan et al., 2024b) and Scion (Pethick et al., 2025b). After over a decade of Adam’s dominance, these LMO-based methods are emerging as viable replacements, offering several practical advantages such as improved memory efficiency, better hyperparameter transferability, and most importantly, superior empirical performance on large-scale tasks, including LLM training. However, a significant gap remains between their practical use and our current theoretical understanding: prior analyses (1) overlook the layer-wise LMO application of these optimizers in practice, and (2) rely on an unrealistic smoothness assumption, leading to impractically small stepsizes. To address both, we propose a new LMO-based framework called Gluon, capturing prior theoretically analyzed methods as special cases, and introduce a new refined generalized smoothness model that captures the layer-wise geometry of neural networks, matches the layer-wise practical implementation of Muon and Scion, and leads to state-of-the-art convergence guarantees. Our experiments with NanoGPT and CNN confirm that our assumption holds along the optimization trajectory, ultimately narrowing the gap between theory and practice.
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Leveraging Evidence Priors for Robust Prompt Learning under Noisy Supervision in Vision-Language Models
Junnan Zou, Zhu Teng, Wei Zhang, Ming He, Jianping Fan
Prompt learning for vision-language models (VLMs) often suffers from performance degradation when adapting to downstream tasks with noisy labels. Existing methods that rely on filtering or reconstructing supervision can propagate errors, leading to sharp performance drops. We observe that pre-trained embeddings are resilient to label noise, offering stable references despite limited adaptation. Based on this insight, we propose Evidence-Prompt, a framework built on the evidence prior that enhances prompt learning by integrating stable pre-trained knowledge. We treat prompt learning as a Bayesian reasoning task, where credibility is derived from both supervision-agnostic and supervision-conditioned evidence. Thi

s framework effectively combines these sources to infer robust training targets under noisy conditions, enabling stable learning even with high noise levels. Extensive experiments on eight benchmarks with both synthetic and real-world noisy labels demonstrate that our method flattens the accuracy–noise curve and consistently outperforms SOTA methods, with notable gains on OxfordPets dataset at a 75\% noise rate (+36.6\% under Asym and +14.4\% under Sym). Additionally, transferability experiments reveal that incorporating our evidence prior into other SOTA methods results in accuracy improvements ranging from 2.6\% to 15.66\%.
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Efficient LLM Moderation with Multi-Layer Latent Prototypes
Maciej Chrabaszcz, Filip Szatkowski, Bartosz Wójcik, Jan Dubi■ski, Tomasz Trzcinski, Sebastian Cygert
Although modern LLMs are aligned with human
values during post-training, robust moderation
remains essential to prevent harmful outputs at deployment time. Existing approaches suffer from
performance-efficiency trade-offs and are difficult to customize to user-specific requirements.
Motivated by this gap, we introduce Multi-Layer
Prototype Moderator (MLPM), a lightweight and
highly customizable input moderation tool. We
propose leveraging prototypes of intermediate representations across multiple layers to improve
moderation quality while maintaining high efficiency. By design, our method adds negligible
overhead to the generation pipeline and can be
seamlessly applied to any model. MLPM achieves
state-of-the-art performance on diverse moderation benchmarks and demonstrates strong scalability across model families of various sizes. Moreover, we show that it integrates smoothly into end-to-end moderation pipelines and further improves
response safety when combined with output moderation techniques. Overall, our work provides a
practical and adaptable solution for safe, robust,
and efficient LLM deployment.
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StitchCUDA: An Automated Multi-Agents End-to-End GPU Programing Framework with Rubric-based Agentic Reinforcement Learning
Shiyang Li, Zijian Zhang, Winson Chen, Yuebo Luo, Mingyi Hong, Caiwen Ding
Modern machine learning (ML) workloads increasingly rely on GPUs, yet achieving high end-to-end performance remains challenging due to dependencies on both GPU kernel efficiency and host-side settings. Although LLM-based methods show promise on automated GPU kernel generation, prior works mainly focus on single-kernel optimization and do not extend to end-to-end programs, hindering practical deployment.

To address the challenge, in this work, we propose \textsc{StitchCUDA}, a multi-agent framework for end-to-end GPU program generation, with three specialized agents: a \textit{Planner} to orchestrate whole system design, a \textit{Coder} dedicated to implementing it step-by-step, and a \textit{Verifier} for correctness check and performance profiling using Nsys/NCU. To fundamentally improve the \textit{Coder}'s ability in end-to-end GPU programming, \textsc{StitchCUDA} integrates rubric-based agentic reinforcement learning over two atomic skills, task-to-code generation and feedback-driven code optimization, with combined rubric reward and rule-based reward from real executions. Therefore, the \textit{Coder} learns how to implement advanced CUDA programming techniques (e.g., custom kernel fusion, cublas epilogue), and we also effectively prevent \textit{Coder}'s reward hacking (e.g., just copy PyTorch code or hardcoding output) during benchmarking. Experiments on KernelBench show that \textsc{StitchCUDA} achieves nearly 100\% success rate on end-to-end GPU programming tasks, with 1.72$\times$ better sp

eedup over the multi-agent baseline and 2.73$\times$ than the RL model baselines
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Learning 3D-Gaussian Simulators from RGB Videos
Mikel Zhobro, Andreas René Geist, Georg Martius
Realistic simulation is critical for applications ranging from robotics to anima
tion. Video generation models have emerged as a way to capture real-world physic
s from data, but they often face challenges in maintaining spatial consistency a
nd object permanence, relying on memory mechanisms to compensate. As a complemen
tary direction, we present 3DGSim, a learned 3D simulator that directly learns p
hysical interactions from multi-view RGB videos. 3DGSim adopts MVSplat to learn
a latent particle-based representation of 3D scenes, a Point Transformer for the
particle dynamics, a Temporal Merging module for consistent temporal aggregatio
n, and Gaussian Splatting to produce novel view renderings. By jointly training
inverse rendering and dynamics forecasting, 3DGSim embeds physical properties in
to point-wise latent features. This enables the model to capture diverse behavio
rs, from rigid and elastic to cloth-like dynamics and boundary conditions (e.g.,
fixed cloth corners), while producing realistic lighting effects. We show that
3DGSim can generate physically plausible results even in out of distribution cas
es, e.g. ground removal or multi-object interactions, despite being trained only
on single-body collisions.
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Lookahead Sample Reward Guidance for Test-Time Scaling of Diffusion Models
Yeongmin Kim, Donghyeok Shin, Byeonghu Na, Minsang Park, Richard Lee Kim, Il-chu
l Moon
Diffusion models have demonstrated strong generative performance; however, gener
ated samples often fail to fully align with human intent. This paper studies an
efficient test-time scaling method for sampling from regions with higher human-a
ligned reward values. Existing methods for computing the expected future reward
(EFR) face important limitations: backward rollout incurs prohibitively high sam
pling costs, while Tweedie-based approaches, including Sequential Monte Carlo an
d gradient guidance, suffer from bias and inherent sampling issues. We show that
the EFR at any $\mathbf{x}_t$ can be computed using only marginal samples from
a pre-trained diffusion model, enabling closed-form reward guidance without neur
al backpropagation. To further improve efficiency, we introduce a few-step looka
head sampling and an accurate solver that guides particles toward high-reward lo
okahead samples. We refer to this sampling scheme as LiDAR sampling. LiDAR achie
ves the same GenEval performance as the latest gradient guidance method for SDXL
with a 9.5× speedup. We release the code at https://github.com/aailab-kaist/Dif
fusion-LiDAR-Sampling.
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Adaptive Recurrent Message Passing for Test Time Computing on Graphs
Junshu Sun, Wanxing Chang, Qingming Huang, Shuhui Wang
Pre-trained foundation models have demonstrated remarkable success in many domai
ns, enabling a unified backbone to generalize across diverse downstream tasks. H
owever, extending this paradigm to graph learning remains challenging due to the
intrinsic mismatch between graph data and fixed architectural designs. In this
work, we show that this limitation can be overcome via recurrent graph models. T
o achieve this, we conduct a systematic theoretical analysis, rigorously derivin
g step dependence as a necessary and sufficient condition for an adaptively conv
ergent recurrent process. Building on this foundation, we propose AdaR, an Adapt
ive Recurrent graph model, empowering flexible test-time computing on various do
wnstream tasks without changing model parameters. To enable adaptive inference,
AdaR explicitly encodes normalized step information and representation–target re
lations into the recurrent updates. To ensure convergence of the recurrent proce
ss, AdaR employs gradient-based supervision signals that guide representation up
dates throughout the recurrence. Empirical results demonstrate that AdaR consist
ently outperforms strong baselines in both inductive and transductive settings.
****
Time-Conditioned Foreseeing: An EHR-Specific Foundation Model for Irregular Dyna

mics and Calendrical Time
Bong Gyun Kang, Junyong Ahn, Hyeongrok Han, Sungroh Yoon
Electronic Health Records (EHRs) possess unique characteristics distinct from natural language, yet existing EHR foundation models often rely on suboptimal NLP-based approaches. We propose a pretraining method tailored to EHRs' distinct features. First, we introduce Pathology-Focused Binning, a density-based quantization strategy that prioritizes clinically significant numerical ranges over usual values. Second, to jointly capture both the exact timing of clinical events and the relative intervals between them, we propose Dual-Calendar Rotary Positional Embedding (RoPE), which encodes absolute and relative temporal signals. Third, we introduce the Time-Conditioned Foreseeing (TCF) objective, aligning with clinical treatment planning to forecast events across multiple temporal horizons by explicitly modeling event timing. Our approach establishes a temporal generative EHR model that outperforms existing foundation models on eleven diverse downstream tasks—achieving up to a 54% improvement in AUPRC—and enables the generation of realistic, temporally consistent patient trajectories. Code is available at https://github.com/Pusheen-cat/TCF_PFM.
****
ADHD Disease Detection Based on Short- and Long-Term Brain Function Encoding and Memory Graph Network
Dongxun Jiang, Borui Jia, Yuxuan Wang, Dongdong Zhang
Graph-based attention deficit hyperactivity disorder (ADHD) detection methods have been extensively studied, but comparatively less attention has been paid to short-term brain functional reorganization. In this paper, we propose an ADHD disease detection model based on short- and long-term brain function encoding and memory graph network. We first exploit a novel brain map sequence construction method based on short-term windows to extract short-term brain function features. Then, we design a short-term state and temporal dependency encoder to characterize short-term sequence patterns of brain function. Furthermore, a brain function memory is introduced to capture the association of brain activity patterns and historical sequence patterns. Concurrently, GNN-based long-term brain function feature extraction network is used to extract brain structure features, which are fused with short-term features for ADHD detection. Experimental validation on the publicly available neuroimaging datasets ADHD-200 and OpenNeuro-ds002424 demonstrates the superior performance of our model in brain disorder detection.
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3DGS-HPC: Distractor-free 3D Gaussian Splatting with Hybrid Patch-wise Classification
Jiahao Chen, Yipeng Qin, Ganlong Zhao, Xin Li, Wenping Wang, Guanbin Li
3D Gaussian Splatting (3DGS) has demonstrated remarkable performance in novel view synthesis and 3D scene reconstruction, but its quality often degrades in real-world environments due to transient distractors, such as moving objects and varying shadows. Existing methods commonly introduce semantic priors from pre-trained vision models either to group pixels into coherent regions or to define perceptual error metrics. However, semantic grouping is often misaligned with the binary static/transient distinction, while perceptual features can be fragile under appearance perturbations introduced during 3DGS optimization. We propose 3DGS-HPC, a framework that addresses these issues by combining two complementary principles: a patch-wise classification strategy that leverages local spatial consistency for robust region-level decisions, and a hybrid classification metric that adaptively integrates photometric and perceptual cues for more reliable separation. Extensive experiments demonstrate the superiority and robustness of our method in mitigating distractors to improve 3DGS-based novel view synthesis. Our project page is https://cnhaox.github.io/3DGS-HPC/.
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PnP-Corrector: A Universal Correction Framework for Coupled Spatiotemporal Forecasting
Hao Wu, Fan Xu, Yuxu Lu, Penghao Zhao, Fan Zhang, Hao Jia, Yuxuan Liang, Ruijian Gou, Qingsong Wen, Xian Wu, Xiaomeng Huang, Yuan Gao
Coupled spatiotemporal forecasting is important for predicting the future evolut

ion of multiple interacting dynamical systems, such as in climate models.
However, existing methods are severely constrained by the persistent bottleneck of compounding errors.
In coupled systems, errors from each subsystem simulator propagate and amplify one another, a phenomenon we term Reciprocal Error Amplification leading to a rapid collapse of long-range predictions.
To address this challenge, we propose a universal framework called PnP-Corrector (Plug-and-Play Corrector).
The core idea of our framework is to decouple the physical simulation from the error correction process: it freezes pre-trained physics simulation engines and exclusively trains a correction agent to proactively counteract the systematic biases emerging from the coupled system.
Furthermore, we design an efficient predictive model architecture, DSLCast, to serve as the backbone of this framework.
Extensive experiments demonstrate that our method significantly enhances the long-term stability and accuracy of coupled forecasting systems.
For instance, in the challenging task of a 300-day global ocean-atmosphere coupled forecast, our PnP-Corrector framework reduces the prediction error of the baseline model by 28% and surpasses state-of-the-art models on several key metrics.
Codes link: https://github.com/Alexander-wu/PnP-Corrector.
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Geometric Flow Grounding: A Unified Manifold Decoupling Framework for Dynamics Discovery and Verification

Chang Yu, Yuxuan Luo, Yixuan Du, Yuqing Zhou, Siyuan Li, Jingbo Zhou, jiawei jiang, Zhen Lei, Stan Z. Li

Modeling complex dynamics from observational data is fundamental to scientific discovery and artificial intelligence. However, existing approaches are often plagued by the entanglement of static state representations and instantaneous motion, leading to accumulated errors and off-manifold hallucinations where predicted trajectories violate intrinsic geometric constraints. To address this, we propose Geometric Flow Grounding, a unified framework that enforces dynamic evolution strictly along the tangent bundle of the learned data manifold via a differentiable Neural Tangent Projection Layer. By geometrically decoupling state representation from tangential dynamics, our method generalizes across diverse data regimes. In scientific discovery, GFG reduces numerical aliasing and improves long-horizon stability in sparse dynamical systems, while recovering interpretable gene regulatory motifs from single-cell data. For trustworthy AI, the projection residual provides a zero-shot metric for deepfake video detection by revealing inconsistencies with the implicit flow of pre-trained world models. Our results establish manifold-constrained projection as a universal operator for both discovering natural laws and verifying synthetic content. Code will be available at \url{https://github.com/yuchang97/GFG-public}
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Plug-and-Play Guidance for Discrete Diffusion Models via Gradient-Informed Logit Correction

Hongkun Dou, Zike Chen, Fengji Li, Hongjue Li, Yue Deng

Controllable generation with discrete diffusion models is often hindered by high computational overhead or the need for retraining. In this paper, we present Gradient-Informed Logit Correction (GILC), a plug-and-play framework that efficiently estimates guidance signals by repurposing the pretrained denoising network as a variational proxy. To circumvent the gradient instability inherent in high-dimensional discrete spaces, we introduce a Jacobian-free mechanism that directly corrects the clean prediction logits, facilitating stable and effective guidance. Our method accommodates both differentiable and non-differentiable reward functions. Extensive experiments across DNA, protein sequence, and molecular generation tasks demonstrate that GILC achieves state-of-the-art performance without additional training, frequently outperforming fine-tuning approaches.
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Generative Modeling of Irregular Time Series via SDE-Induced Continuous-Discrete Variational Inference

Zexin Yuan, Qinliang Su, Junxi Xiao
Irregular time series arise ubiquitously in real-world systems, where observations are sparse, asynchronous, and governed by underlying continuous-time dynamics. Existing continuous–discrete state-space models typically rely on path-based variational inference, which is computationally expensive or constrained by restrictive posterior assumptions. We propose SDEVI, a novel framework that performs variational inference directly on the joint distribution over discrete-time observations, while guaranteeing consistency with an underlying continuous process governed by a Stochastic Differential Equation(SDE). SDEVI employs a variational posterior induced by linear time-varying SDEs as a scalable inference backbone. To enable intricate dynamics modeling for real-world data, we introduce non-linear-SDE-induced variational inference and generalize our framework to the complex domain. Extensive experiments across healthcare, physics, climate, and IoT benchmarks demonstrate state-of-the-art performance on interpolation, extrapolation, regression, and classification tasks.
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CiteGuard: Conformal False-Discovery Control for Faithful Retrieval-Augmented Generation
Xiangyu Jiang
Large language models increasingly rely on retrieval-augmented generation (RAG) to ground responses in external corpora. Yet, even with strong retrievers, generated statements can remain unsupported, and the resulting citations are often not reliable indicators of evidence. We introduce CiteGuard, a RAG decoding layer that treats sentence-level factuality as a multiple-testing problem and combines conformal calibration with false-discovery-rate control. CiteGuard converts claim–evidence scores into p-values for the null hypothesis "unsupported" and uses BH/BY procedures to decide which claims to keep (with citations) and which to abstain on. On FEVER and Natural Questions, CiteGuard reduces the false-discovery rate among accepted claims from 28–31% (vanilla RAG) to below 10% at α=0.10, while retaining 86–92% of supported claims. This yields a user-controlled risk budget: practitioners can trade off faithfulness and coverage via α, with finite-sample guarantees under standard exchangeability assumptions.
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MAST: Motif-Augmented Diffusion with Search Tree for Spectroscopic Molecular Structure Elucidation
Chenghao Jia, Mengdi Liu, Hong Chang, Shiguang Shan, Xilin CHEN
Elucidating molecular structures from spectra is a foundational problem in chemical and materials characterization, yet remains challenging due to spectral ambiguity and the vast molecular space. Although recent diffusion-based generators show strong promise for spectra-conditioned elucidation, existing methods struggle to learn robust spectra-structure relationships from limited paired data when relying solely on global spectral representation. Moreover, the repeated full sampling inference strategy incurs substantial computation overhead. To address these limitations, we propose MAST, a Motif-Augmented diffusion framework with Search Tree, for joint 2D-3D spectroscopic molecular structure elucidation. MAST introduces explicit, interpretable motif priors as intermediate evidences throughout denoising, reducing conditional ambiguity and facilitating spectra-conditioned optimization. We further cast diffusion sampling as reward-guided tree search to prioritize high-reward denoising trajectories, yielding a compact set of spectra-consistent candidates under limited budgets. On the QM9S multi-spectra benchmark, MAST achieves 94.89% exact recovery and improves 3D fidelity, while preserving high chemical validity and stability. Code is available at https://github.com/Jia040223/MAST.
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Agora: Toward Autonomous Bug Detection in Production-Level Consensus Protocols with LLM Agents
Xiang Liu, Sa Song, Zhaowei Zhang, Huiying Lan, Jason Zeng, Ming Wu, Michael Heinrich, Sun Yong, Ceyao Zhang
Consensus protocols form the backbone of distributed systems and blockchains, where implementation bugs can cause data corruption and financial losses. While LL

M-based approaches show promise in code analysis, they struggle with deep protoc
ol-level logic bugs involving complex state-dependent behaviors across multiple
execution stages. We present Agora, a domain-aware multi-agent framework that in
tegrates hypothesis-driven testing with LLM capabilities for systematic protocol
verification. Agora employs specialized agents that collaboratively explore pro
tocol state spaces, synthesize attack scenarios using domain-specific constraint
s, and validate findings through iterative refinement. This explicit role separa
tion enables reasoning about global protocol invariants beyond single-function c
ode analysis. We evaluate Agora on four consensus implementations (Raft, EPaxos,
HotStuff, BullShark) using four state-of-the-art LLMs. Agora discovers 15 previ
ously unknown protocol-level logic bugs that violate safety properties, while ex
isting LLM-based agents fail to detect any such protocol-level logic bugs. Our r
esults demonstrate that domain-aware multi-agent collaboration is essential for
detecting deep logic bugs in complex protocols.
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Certified Circuits: Stability Guarantees for Mechanistic Circuits
Alaa Anani, Tobias Lorenz, Bernt Schiele, Mario Fritz, Jonas Fischer
Understanding *how* neural networks arrive at their predictions is essential for
debugging, auditing, and deployment. Mechanistic interpretability pursues this
goal by identifying *circuits*—minimal subnetworks responsible for specific beha
viors. However, existing circuit discovery methods are brittle: circuits depend
strongly on the chosen concept dataset and often fail to transfer out-of-distrib
ution, raising doubts whether they capture the concept or merely dataset-specifi
c artifacts. We introduce *Certified Circuits*, which provide provable stability
guarantees for circuit discovery. Our framework wraps any black-box discovery a
lgorithm with randomized data subsampling to certify that inclusion decisions ov
er circuit components—neurons or edges of the model graph, depending on the base
algorithm—are invariant to bounded edit-distance perturbations of the concept d
ataset. Unstable components are abstained from, yielding circuits that are more
compact and more accurate. We validate across three architectures (ResNet, ViT,
GPT-2) on vision (ImageNet and four OOD datasets) and language (IOI, IOI-Hard, G
reater-Than) tasks. Certified circuits achieve up to 56% higher accuracy and up
to 80% fewer components, and remain reliable where baselines degrade. *Certified
Circuits* puts circuit discovery on formal ground by producing mechanistic expl
anations that are provably stable and better aligned with the target concept. Co
de: https://github.com/AlaaAnani/certified-circuits.
****

Continual Model Routing in Evolving Model Hubs
Jack Bell, Giacomo Carfi', Gerlando Gramaglia, Vincenzo Lomonaco
AI model hubs provide access to a rapidly growing collection of powerful pre-tra
ined models, enabling off-the-shelf mixture-of-experts systems with different ro
uting strategies. However, this rapid growth poses two fundamental challenges: s
caling model selection across thousands of experts and continually updating rout
ing mechanisms as new models and tasks are introduced. In this paper, we formali
se this setting as Continual Model Routing (CMR) and propose *CMRBench*, a new l
arge-scale benchmark simulating realistic hub expansion and including over 2,000
candidate models. Finally, we introduce *CARvE*, a contrastive embedding approa
ch for efficient continual model routing via domain-stratified coreset replay an
d checkpoint-based anchoring. Extensive empirical results and ablations show tha
t CARvE significantly outperforms zero-shot retrieval, fine-tuning, and adapter-
merging baselines in model, family, and domain-level accuracy.
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Watermarking Graph Neural Networks via Explanations for Ownership Protection
Jane Downer, Yingdan Shi, Ziyan Liu, Ren Wang, Binghui Wang
Graph Neural Networks (GNNs) are widely deployed in industry, making their intel
lectual property valuable. However, protecting GNNs from unauthorized use remain
s a challenge. Watermarking offers a solution by embedding ownership information
into models. Existing watermarking methods have two limitations: First, they ra
rely focus on graph data or GNNs. Second, the \emph{de facto} backdoor-based met
hod relies on manipulating training data, which can introduce ownership ambiguit

y through misclassification and vulnerability to data poisoning attacks that can interrupt the backdoor mechanism. Our explanation-based watermarking inherits the strengths of backdoor-based methods (e.g., black-box verification) without data manipulation, eliminating ownership ambiguity and data dependencies. In particular, we watermark GNN explanations such that these explanations are statistically distinct from others, so ownership claims must be verified through statistical significance. We theoretically prove that, even with full knowledge of our method, locating the watermark is NP-hard. Empirically, our method demonstrates robustness to fine-tuning and pruning attacks. By addressing these challenges, our approach significantly advances GNN intellectual property protection.

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Mean-Shift PCA by Knockoff Mean

Mengda Li, Zeng Li, Jianfeng Yao

Removing noise is difficult, but adding noise is easy. In this work, we show how to eliminate mean-shift noisy components from PCA by deliberately introducing knockoff mean-shift perturbation. Standard PCA is highly sensitive to shifts in the sample mean: a small fraction of samples from a shifted distribution can cause large deviations in the leading principal components. In high-dimensional regimes, existing Robust PCA approaches cannot handle the mean-shift contamination structure inherent in the mixture model. Using tools from Random Matrix Theory, we prove that the mean-shift spikes are spectrally separable from the stable eigenvalues of the original covariance. Furthermore, the original eigenspace remains asymptotically invariant to the contamination, independent of the mixture weight. Exploiting this spectral stability, we propose a simple, two-stage PCA algorithm by adding knockoff mean that identifies and removes the mean-shift component using only standard PCA operations.

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Scalable Single-Cell Gene Expression Generation with Latent Diffusion Models

Giovanni Palla, Sudarshan Babu, Payam Dibaeinia, James D Pearce, Donghui Li, Aly A Khan, Theofanis Karaletsos, Jakub M. Tomczak

Computational modeling of single-cell gene expression is crucial for understanding cellular processes, but generating realistic expression profiles remains a major challenge. This difficulty arises from the count nature of gene expression data and complex latent dependencies among genes. Existing generative models often impose artificial gene orderings or rely on shallow neural network architectures. We introduce a scalable latent diffusion model for single-cell gene expression data, which we refer to as scLDM, that respects the fundamental exchangeability property of the data. Our VAE uses fixed-size latent variables leveraging a unified Multi-head Cross-Attention Block (MCAB) architecture, which serves dual roles: permutation-invariant pooling in the encoder and permutation-equivariant unpooling in the decoder. We enhance this framework by replacing the Gaussian prior with a latent diffusion model using Diffusion Transformers and linear interpolants, enabling high-quality generation with multi-conditional classifier-free guidance. We show its superior performance in a variety of experiments for both observational and perturbational single-cell data, as well as downstream tasks like cell-level classification.

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Variational inference via Gaussian interacting particles in the Bures-Wasserstein geometry

Giacomo Borghi, Jose A. Carrillo

Motivated by variational inference methods, we propose a zeroth-order algorithm for solving optimization problems in the space of Gaussian probability measures. The algorithm is based on an interacting system of Gaussian particles that stochastically explore the search space and self-organize around global minima via a consensus-based optimization (CBO) mechanism.

Its construction relies on the Linearized Bures–Wasserstein (LBW) space, a novel parametrization of Gaussian measures we introduce for efficient computations. We establish well-posedness and study the convergence properties of the particle dynamics via a mean-field approximation. Numerical experiments on variational inference tasks demonstrate the algorithm’s robustness and superior performance wi

th respect to deterministic gradient-based method in presence of low-dimensional non log-concave targets.
****
CURE: Consistency-under-Unified Semantic Regularization for Generalized Category Discovery
Yuwei Bian, Shidong Wang, Haofeng Zhang
Generalized Category Discovery (GCD) aims to learn semantically structured representations for discovering novel categories in unlabeled data using supervision from known classes. Most existing methods rely on self-supervised contrastive learning (CL) with consistency and uniformity objectives. We identify an inherent optimization conflict between these objectives: while uniformity enforces global feature dispersion, it can hinder the formation of class-discriminative and semantically coherent structures.
To address this issue, we propose a two-stage framework that decouples representation learning from self-contrastive regularization. The first stage learns category-anchored representations aligned with known class prototypes, while the second stage extends the representation space to novel categories via a consistency objective enhanced with unified semantic regularization. We further introduce a Semantic Exploration Energy mechanism to capture shared semantics across categories and mitigate information loss caused by prototype orthogonalization. The resulting framework, termed *Consistency-under-Unified Semantic Regularization*(**CURE**), achieves state-of-the-art performance on multiple benchmarks and substantially reduces the performance gap between known and novel categories.
****
ORLoopBench: Solver-in-the-Loop Benchmarks for Self-Correction and Behavioral Rationality in Operations Research
Ruicheng Ao, David Simchi-Levi, Xinshang Wang
Operations Research practitioners debug infeasible models through an iterative process: inspecting Irreducible Infeasible Subsystems (\textbf{IIS}), identifying constraint conflicts, and repairing formulations until feasibility is restored. Existing LLM benchmarks mostly treat OR as one-shot translation from problem descriptions to solver code, omitting this diagnostic loop. We formalize infeasible-model repair as a solver-in-the-loop Markov Decision Process in which each action triggers solver re-execution and \textbf{IIS} recomputation, yielding deterministic, verifiable feedback. We introduce \textbf{ORLoopBench}, a benchmark suite with two components: \textbf{ORDebug} releases 5,362 LP/MILP repair instances, while \textbf{ORBias} evaluates closed-form operational decision rationality across inventory settings. Solver-verified RLVR training enables an 8B model to surpass frontier APIs on LP repair (95.3\% vs 92.4\% RR@5), improves diagnostic behavior, and transfers to MILP repair. The same evaluation exposes semantic drift in whole-model code regeneration: feasible regenerated MILPs can solve the wrong problem. Process-level evaluation with solver oracles enables targeted training for reliable OR self-correction.
****
Improving Classifier-Free Guidance of Flow Matching via Manifold Projection
Jian-Feng Cai, Haixia Liu, Zhengyi Su, Chao Wang
Classifier-free guidance (CFG) is a widely used technique for controllable generation in diffusion and flow-based models. Despite its empirical success, CFG relies on a heuristic linear extrapolation that is often sensitive to the guidance scale. In this work, we provide a principled interpretation of CFG through the lens of optimization. We demonstrate that the velocity field in flow matching corresponds to the gradient of a sequence of smoothed distance functions, which guides latent variables toward the scaled target image set. This perspective reveals that the standard CFG formulation is an approximation of this gradient, where the prediction gap, the discrepancy between conditional and unconditional outputs, governs guidance sensitivity. Leveraging this insight, we reformulate the CFG sampling as a homotopy optimization with a manifold constraint. This formulation necessitates a manifold projection step, which we implement via an incremental gradient descent scheme during sampling. To improve computational efficiency and stability, we further enhance this iterative process with Anderson Acceleratio

n without requiring additional model evaluations. Our proposed methods are train
ing-free and consistently refine generation fidelity, prompt alignment, and robu
stness to the guidance scale. We validate their effectiveness across diverse ben
chmarks, demonstrating significant improvements on large-scale models such as D
iT-XL-2-256, Flux, and Stable Diffusion 3.5. Code is available at https://github
.com/LeonSuZhengYi/CFG-MP.
****

PAWS: Preference Learning with Advantage-Weighted Segments
Aleksandar Taranovic, Onur Celik, Niklas Freymuth, Ge Li, Serge Thilges, Huy Le,
 Tai Hoang, Rania Rayyes, Gerhard Neumann
Preference-based reinforcement learning (PbRL) learns policies from human trajec
tory-level comparisons, avoiding explicit reward design and expert demonstration
s. Existing methods typically train utility functions on trajectory or segment-l
evel preferences while relying on per-step utility estimates during policy optim
ization. This training and inference mismatch induces a distribution shift that
severely degrades temporal credit assignment and limits policy learning.
We analyze this issue and propose PAWS, a segment-based preference learning meth
od that performs policy updates directly using segment-level advantage functions
. By aligning utility training with policy optimization, PAWS preserves trajecto
ry-level preference information and avoids unreliable per-step learning signals.
 Experiments on simulated robotic manipulation and locomotion tasks demonstrate
that PAWS consistently outperforms existing PbRL approaches, highlighting the im
portance of distribution-consistent preference learning.
****

Smoothie: Smoothing Diffusion on Token Embeddings for Text Generation
Alexander Shabalin, Viacheslav Meshchaninov, Dmitry Vetrov
Diffusion models have achieved state-of-the-art performance in generating images
, audio, and video, but their adaptation to text remains challenging due to its
discrete nature. Prior approaches either apply Gaussian diffusion in continuous
latent spaces, which inherits semantic structure but struggles with token decodi
ng, or operate in categorical simplex space, which respect discreteness but disr
egard semantic relation between tokens. In this paper, we propose Smoothing Diff
usion on Token Embeddings (Smoothie), a novel diffusion method that combines the
 strengths of both approaches by progressively smoothing token embeddings based
on semantic similarity. This technique enables gradual information removal while
 maintaining a natural decoding process. Experimental results on several sequenc
e-to-sequence and unconditional generation tasks demonstrate that Smoothie outpe
rforms existing diffusion-based models in generation quality. Furthermore, ablat
ion studies show that our proposed diffusion space yields better performance tha
n both the standard embedding space and the categorical simplex.
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Synthesizing Multimodal Geometry Datasets from Scratch and Enabling Visual Align
ment via Plotting Code
Haobo Lin, Tianyi Bai, Chen Chen, Jiajun Zhang, Bohan Zeng, Wentao Zhang, Binhan
g Yuan
Multimodal geometry reasoning requires models to jointly understand visual diagr
ams and perform
structured symbolic inference, yet current vision--language models struggle with
 complex geometric
constructions due to limited training data and weak visual--symbolic alignment.
We propose a pipeline for synthesizing complex multimodal geometry problems from
 scratch and construct
a dataset named \textbf{GeoCode}, which decouples problem generation into symbol
ic seed construction,
grounded instantiation with verification, and code-based diagram rendering, ensu
ring consistency across
structure, text, reasoning, and images.
Leveraging the plotting code provided in GeoCode, we further introduce code pred
iction as an explicit
alignment objective, transforming visual understanding into a supervised structu

red prediction task.
GeoCode exhibits substantially higher structural complexity and reasoning difficulty than existing benchmarks,
while maintaining mathematical correctness through multi-stage validation.
Extensive experiments show that models trained on GeoCode achieve consistent improvements on multiple geometry
benchmarks, demonstrating both the effectiveness of the dataset and the proposed alignment strategy.
****

Understanding Private Learning From Feature Perspective
Meng Ding, Mingxi Lei, Shaopeng Fu, Shaowei Wang, Di Wang, Jinhui Xu
Differentially private Stochastic Gradient Descent (DP-SGD) has become integral to privacy-preserving machine learning, ensuring robust privacy guarantees in sensitive domains. Despite notable empirical advances leveraging features from non-private, pre-trained models to enhance DP-SGD training, a theoretical understanding of feature dynamics in private learning remains underexplored. This paper presents the first theoretical framework to analyze private training through a feature learning perspective. Building on the multi-patch data structure from prior work, our analysis distinguishes between label-dependent feature signals and label-independent noise—a critical aspect overlooked by existing analyses in the DP community. Employing a two-layer CNN with polynomial ReLU activation, we theoretically characterize both feature signal learning and data noise memorization in private training via noisy gradient descent. Our findings reveal that (1) Effective private signal learning requires a higher signal-to-noise ratio (SNR) compared to non-private training, and (2) When data noise memorization occurs in non-private learning, it will also occur in private learning, leading to poor generalization despite small training loss. Our findings highlight the challenges of private learning and prove the benefit of feature enhancement to improve SNR. Experiments on synthetic and real-world datasets also validate our theoretical findings.
****

Understanding the Ability of LLMs to Handle Character-Level Perturbation
Anyuan Zhuo, Xuefei Ning, Ningyuan Li, Jingyi Zhu, Yu Wang, Pinyan Lu
This work investigates the resilience of contemporary large language models (LLMs) against frequent character-level perturbations. We examine three types of character-level perturbations including introducing numerous typos within words, shuffling the characters in each word, and inserting a large number of invisible characters into the text. Surprisingly, even under severe perturbation, such as shuffling nearly all words character-wise to produce text that is almost unreadable to humans, or inserting invisible characters which are several times more than the visible ones as noise, many LLMs still maintain notable performance. We explore the underlying causes of this robustness and find that LLMs exhibit remarkable resilience to chaotic segmentation and fragmented tokenization. Furthermore, we examine the mechanisms by which LLMs remove perturbations to correctly comprehend text, including both implicit and explicit mechanisms for character-level perturbation. We hope that our findings on the low-level robustness of LLMs will unveil their inherent architectural strengths, reveal the potential risks of their misuse, and inform the reliable deployment of LLMs across diverse application scenarios.
****

MER-DG: Modality-Entropy Regularization for Multimodal Domain Generalization
Yavuz Yarici, Ghassan AlRegib
Deploying multimodal models in real-world scenarios requires generalization to new environments where recording conditions differ from training, a challenge known as multimodal domain generalization (MMDG). Standard architectures employ separate encoders for each modality and a fusion module, training the system end-to-end by optimizing on the fused features. In this paper, we identify that such joint optimization causes encoders to exploit cross-modal co-occurrences, statistical relationships between modalities that arise from source-specific recording conditions, rather than learning domain-invariant features. We term this failure

mode Fusion Overfitting. To address this, we propose Modality-Entropy Regulariz
ation for Domain Generalization (MER-DG), which maximizes the entropy of each en
coder's feature distribution to preserve feature diversity. MER-DG is architectu
re-agnostic and integrates into existing multimodal frameworks as an additive lo
ss term. Extensive experiments on EPIC-Kitchens and HAC benchmarks demonstrate a
verage improvements of ${\sim}5\%$ over standard fusion and ${\sim}2\%$ over sta
te-of-the-art methods.
****
TopoDistill: Distilling Global System Topology for Causal Discovery in Multivari
ate Time Series
Zehao Liu, Pengfei Jiao, Yuhan Wu, Jianqi Yang, Yuyu Yin
Although causal discovery from multivariate time series is widely used, it remai
ns challenging under noise. Convergent cross mapping (CCM) infers causality by r
econstructing shadow manifolds via time-delay embedding (TDE) and evaluating cro
ss-map skill between manifolds. Despite Takens’ theorem guarantees in ideal sett
ings, TDE effectively attempts to recover system state from a single noisy view,
 often yielding geometrically degraded manifolds and unreliable distance-based n
eighborhoods, which in turn weakens causal identification. We propose TopoDistil
l, a topology-informed knowledge distillation framework that improves univariate
 shadow-manifold reconstruction by aligning local neighborhood structure to a mu
ltivariate system representation. A global embedder trained on multivariate obse
rvations captures a global attractor representation, while a delay embedder is d
istilled to produce embeddings whose neighborhood distributions match the global
 topology. This cross-view alignment yields smoother and more reliable neighborh
oods, improving cross mapping under noise while maintaining specificity against
spurious correlations. Theoretical analysis and experimental results demonstrate
 that our method enables effective causal discovery.
****
Explicit representation of germline and non-germline residues improves antibody
language modeling
Jeonghyeon Kim, Nathaniel Blalock, Ameya Kulkarni, Kensuke Nakamura, Philip Rome
ro
Antibodies originate from germline templates and are diversified by somatic hype
rmutation, producing sequences in which conserved germline residues scaffold str
ucture while rare non-germline (NGL) substitutions refine antigen binding. Curre
nt antibody language models (ALMs) treat all residues equivalently and inherit a
 germline bias that systematically down-weights functionally critical NGL mutati
ons as statistical noise. We introduce PRISM, a germline-aware ALM that explicit
ly represents germline and non-germline residues as distinct token types over a
factorized 53-token vocabulary. PRISM achieves state-of-the-art pseudo-perplexit
y in hypervariable CDRs and is uniquely positively correlated with experimental
binding affinity across three deep mutational scanning landscapes on which all c
ompared ALMs anti-correlate. The dual-vocabulary further enables property-specif
ic controllable generation previously unattainable with entangled ALMs. NGL-dire
cted sampling improves physics-based binding scores while GL-directed sampling p
reserves stability and solubility. These results establish disentangled germline
/non-germline representation as a substantive advance in antibody language model
ing.
****
TeamTR: Trust-Region Fine-Tuning for Multi-Agent LLM Coordination
Yi Xie, Siao Liu, Falong FAN, Yuanqi Yao, Siyang Cao, Yue Zhao, Bo Liu
Multi-agent LLM systems can improve reasoning and tool use, yet recent evidence
shows their gains are often unstable and sensitive to interaction design.
A promising direction is to train collaboration, but team post-training introduc
es a moving-target effect: when agents interact through a shared context, updati
ng one agent shifts the context distribution faced by the others, which can regr
ess coordination under naive sequential updates.
We propose TeamTR, a trust-region framework for fine-tuning heterogeneous LLM te
ams that explicitly controls this occupancy shift.
TeamTR evaluates each agent update on rollouts from the intermediate team induce

d by partially applied updates, and enforces per-agent trust regions via a token-decomposed reverse KL that is directly monitorable from those rollouts.
This yields population-level per-update and per-stage improvement lower bounds whose functional form applies to any realized update order, and motivates a practical certificate proxy computed from logged surrogates and KL terms.
We instantiate TeamTR for router-based text handoff with sequence-level returns and bounded group-normalized advantages, and show empirically that it mitigates coordination regressions, improves training stability across heterogeneous teams, and supports modular component replacement via a trust-region alignment step.
****

Saliency-Aware Model Merging
Jungin Park, Jiyoung Lee, Kwanghoon Sohn
Model merging aims to consolidate multiple task-specific models fine-tuned on different datasets into a unified architecture that performs cross-domain proficiency.

Current data-free model merging methods often struggle to scale as they rely on simple parameter-level heuristics that ignore inter-layer dependencies and non-uniform distribution of expertise.

This work proposes SA-Merging, which is built upon connectivity-based saliency formulations from structural pruning (e.g., SynFlow) and extends them to the data-free model merging setting.

We define a saliency score over task vectors relative to a shared base model, and further introduce merge-aware modulation that incorporates agreement across experts to mitigate task interference.

Based on this formulation, an iterative saliency-aware merging procedure progressively removes non-informative updates while preserving end-to-end connectivity.

Furthermore, we extend SA-Merging to introduce rank-wise saliency decomposition for LoRAs without compromising their structural integrity.

Extensive experiments on vision and language tasks demonstrate the effectiveness of our saliency-based approach, further reducing the gap between data-free and test-time adaptation methods.
****

DITING: A Weak Degradation Listener for Battery Lifetime Early Prediction
Hao Miao, Ni Zhang, Zefei Ning, Li Wang
Battery lifetime early prediction is crucial for safety assessment and decision planning, yet early-stage degradation signals are extremely weak and difficult to distinguish from stochastic noise. Existing methods primarily rely on denoising or signal decomposition, which may lose critical degradation cues. In nature, most organisms exhibit the binaural effect, exploiting discrepancies between left and right auditory inputs to enhance perceptual reliability. Inspired by this, we propose DITING, a weak degradation listener for battery lifetime early prediction. We first employ optimal-transport-based selective matching to extract a robust health template from initial cycles, and further design a tri-coupled degradation manifestation mechanism to distinguish degradation signals from noise. By exploiting the randomness of noise, matched responses under symmetric coupling suppress stochastic fluctuations, while degradation-driven cumulative deviations propagate through the coupling process to form stable bilateral discrepancies, thereby amplifying weak early-stage cues for lifetime prediction. Experiments on various datasets demonstrate that DITING achieves state-of-the-art performance and provides more reliable early support for full-lifecycle battery management.
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Characterizing, Evaluating, and Optimizing Complex Reasoning
Haoran Zhang, Yafu Li, Zhi Wang, Zhilin Wang, Shunkai Zhang, Xiaoye Qu, Yu Cheng
Large Reasoning Models (LRMs) increasingly rely on reasoning traces with complex internal structures. However, existing work lacks a unified answer to three fundamental questions:
(1) what defines high-quality reasoning,
(2) how to reliably evaluate long, implicitly structured reasoning traces, and
(3) how to use such evaluation signals for reasoning optimization.

To address these challenges, we provide a unified perspective. (1) We introduce the ME^2 principle to characterize reasoning quality along macro- and micro-level concerning efficiency and effectiveness. (2) Built on this principle, we model reasoning traces as directed acyclic graphs (DAGs) and develop a DAG-based pairwise evaluation method, capturing complex reasoning structures. (3) Based on this method, we construct the TRM-Preference dataset and train a Thinking Reward Model (TRM) to evaluate reasoning quality at scale.
Experiments show that thinking rewards serve as an effective optimization signal. At test time, selecting better reasoning leads to better outcomes (up to 19.3\% gain), and during RL training, thinking rewards enhance reasoning and performance (up to 3.9\% gain) across diverse tasks. Code and data are available at https://github.com/Simplified-Reasoning/TRM.
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Breaking the Exploration Bottleneck: Rubric-Scaffolded Reinforcement Learning for Open-Ended LLM Reasoning
Yang Zhou, Sunzhu Li, Shunyu Liu, Wenkai Fang, Kongcheng Zhang, Jiale Zhao, Jingwen Yang, Yihe Zhou, Jianwei Lv, Tongya Zheng, Hengtong Lu, Chen Wei, Xie Yan, Mingli Song
Recent advances in Large Language Models (LLMs) have underscored the potential of Reinforcement Learning (RL) to facilitate the emergence of reasoning capabilities. Despite the encouraging results, a fundamental dilemma persists as RL improvement relies on learning from high-quality samples, yet the exploration for such samples remains bounded by the inherent limitations of LLMs. This, in effect, creates an undesirable cycle in which what cannot be explored cannot be learned. In this work, we propose Rubric-Scaffolded Reinforcement Learning (RuscaRL), a novel instructional scaffolding framework designed to break the exploration bottleneck for open-ended reasoning. Specifically, RuscaRL introduces checklist-style rubrics as (1) explicit scaffolding for exploration during rollout generation, where different rubrics are provided as external guidance within task instructions to steer diverse high-quality responses. This guidance is gradually decayed over time, encouraging the model to internalize the underlying reasoning patterns; (2) verifiable rewards for exploitation during model training, where we can obtain robust LLM-as-a-Judge scores using rubrics as references, enabling effective RL on open-ended reasoning tasks. Extensive experiments demonstrate the superiority of the proposed RuscaRL across various benchmarks, effectively expanding reasoning boundaries under the Best-of-N evaluation.
****

WIND: Weather Inverse Diffusion for Zero-Shot Atmospheric Modeling
Michael Aich, Andreas Fürst, Florian Sestak, Carlos Ruiz-Gonzalez, Niklas Boers, Johannes Brandstetter
Deep learning has revolutionized weather forecasting, but many challenges remain, including climate modeling. Moreover, the current landscape remains fragmented: highly specialized models are typically trained individually for distinct tasks. To unify this landscape, we introduce WIND, a single pre-trained foundation model capable of replacing specialized baselines across a vast array of tasks. Crucially, in contrast to previous atmospheric foundation models, we achieve this without any task-specific fine-tuning. To learn a robust, task-agnostic prior of the atmosphere, we pre-train WIND with a self-supervised video reconstruction objective, utilizing an unconditional video diffusion model to iteratively reconstruct atmospheric dynamics from a noisy state. At inference, we frame diverse domain-specific problems strictly as inverse problems and solve them via posterior sampling. This unified approach allows us to tackle highly relevant weather and climate problems, including probabilistic forecasting, spatial and temporal downscaling, reconstruction of spatial fields from sparse observations and enforcing global dry air mass conservation. We further demonstrate how WIND can be applied to explore extreme weather events under prescribed out-of-distribution thermodynamic perturbations. By combining generative video modeling with inverse problem solving, WIND offers a computationally efficient alternative for AI-based atmospheric modeling.
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DEER: A Benchmark for Evaluating Deep Research Agents on Expert Report Generation

Janghoon Han, Heegyu Kim, Changho Lee, Dahm Lee, Min Hyung Park, Hosung Song, Stanley Jungkyu Choi, Moontae Lee, Honglak Lee

Recent advances in large language models have enabled deep research systems that generate expert-level reports through multi-step reasoning and evidence-based synthesis. However, evaluating such reports remains challenging: report quality is multifaceted, making it difficult to determine what to assess and which criteria to use; LLM-based judges may miss errors that require domain expertise to identify; and because deep research relies on retrieved evidence, report-wide claim verification is also necessary. To address these issues, we propose DEER, a benchmark for evaluating expert-level deep research reports. DEER systematizes evaluation criteria with an expert-developed taxonomy (7 dimensions, 25 subdimensions) operationalized as 101 fine-grained rubric items. We also provide task-specific Expert Evaluation Guidance to support LLM-based judging. In addition to rubric-based assessment, we propose a claim verification architecture that verifies both cited and uncited claims and quantifies evidence quality.

Experiments show that current systems produce structurally plausible, evidence-citing reports, but still struggle to fully satisfy expert-level user requests and achieve logical completeness. Beyond performance comparisons, DEER makes system strengths and limitations interpretable and provides diagnostic signals for improvement.

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QuITE: Query-Based Irregular Time Series Embedding

Junghoon Lim

Irregular Multivariate Time Series (IMTS) are common in practice, yet their irregular sampling complicates effective modeling. Existing approaches typically either (i) design specialized architectures that limit the reuse of proven Multivariate Time Series (MTS) models, or (ii)

map IMTS onto regular temporal grids through interpolation, which may distort temporal dynamics by introducing artificial values. To address these limitations, we propose a new input-embedding-based approach. We identify that the key bottleneck lies not in the backbone architecture, but in conventional embedding layers that assume uniform sampling. In this work, we introduce QuITE (Query-Based Irregular Time Series Embedding), a simple yet effective plug-and-play embedding module for IMTS. QuITE employs learnable query tokens to aggregate irregular observations through a single self-attention layer, directly producing backbone-compatible latent representations without artificial value generation or architectural modification. Extensive experiments on real-world benchmarks show that QuITE consistently improves MTS models, yielding average relative gains of up to 54.7% in forecasting and 15.8% in classification across diverse datasets and backbone architectures.

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A Refined Generalization Analysis for Extreme Multi-class Supervised Contrastive Representation Learning

Nong Minh Hieu, Antoine Ledent

Contrastive Representation Learning (CRL) has achieved strong empirical success in multiple machine learning disciplines, yet its theoretical sample complexity remains poorly understood. Existing analyses usually assume that input tuples are identically and independently distributed, an assumption violated in most practical settings where contrastive tuples are constructed from a finite pool of labeled data, inducing dependencies among tuples. While one recent work analyzed this learning setting using U-Statistics to estimate the population risk, the techniques used therein require the risk of each class to concentrate uniformly, making excess risk bounds scale in the order of $\rho_{\min}^{-{1}/{2}}$ where $\rho_{\min}$ denotes the probability of the rarest class. Such a dependency can be overly pessimistic in the extreme multiclass settings where there are many tail classes which contribute minimally to the overall population risk. Our contributions are two-fold. Firstly, we improve upon the previous work and prove a bound with a sample complexity of the same order as the number of classes R, regard

less of the distribution over classes. Furthermore, we formulate a different est
imator that captures the concentration of the risk \textit{across classes}, enab
ling sharper bounds in extreme multi-class learning scenarios, especially where
class distributions are long-tailed. Under mild assumptions on the class distrib
utions, the resulting sample complexity is $\mathcal{{O}}(k)$ where k is the n
umber of samples per tuple.
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RELO: Reinforcement Learning to Localize for Visual Object Tracking
Xin Chen, Chuanyu Sun, Jiao Xu, Houwen Peng, Dong Wang, Huchuan Lu, Kede Ma
Conventional visual object trackers localize targets using handcrafted spatial p
riors, often in the form of heatmaps. Such priors provide only surrogate supervi
sion and are poorly aligned with tracking optimization and evaluation metrics, s
uch as intersection over union (IoU) and area under the success curve (AUC). Her
e, we introduce RELO, a REinforcement-learning-to-LOcalize method for visual obj
ect tracking that formulates target localization as a Markov decision process. S
pecifically, RELO replaces handcrafted spatial priors with a localization policy
 learned over spatial positions via reinforcement learning, with rewards combini
ng frame-level IoU and sequence-level AUC. We additionally introduce layer-align
ed temporal token propagation to improve semantic consistency across frames, wit
h negligible computational overhead. Across multiple benchmarks, RELO achieves s
uperior results, attaining 57.5\% AUC on LaSOT$_\mathrm{ext}$ without template
 updates. This confirms that reward-driven localization provides an effective al
ternative to prior-driven localization for visual object tracking.
****

Gradient Flow Dynamics and Implicit Bias of Diagonal Linear Networks under Infin
itesimal Initialization
Jiajie Zhao, Jianxing Wang, Junjie Yang, Zhiwei Bai, Yaoyu Zhang
We study the gradient flow dynamics of diagonal linear networks for regression t
asks under infinitesimal initialization. Extending Theorem 1 from Pesme & Flamma
rion (2023), we generalize the analysis to both deep diagonal linear networks an
d a broader class of two-layer diagonal linear networks (as defined in Definitio
n 4.1). Specifically, we demonstrate that the training trajectories of these mod
els can be equivalently characterized by the proposed Algorithm 1. We further pr
ove that this algorithm converges to the solution of a modified $\mathcal{l}_1$
norm minimization problem. As a result, we establish that the implicit bias of b
oth network architectures corresponds to a modified $\mathcal{l}_1$ norm in the
regime of infinitesimal initialization. Additionally, we provide insights into t
he underlying mechanisms governing these dynamics by identifying the Structural
Invariant Manifold (SIM) (Zhao et al., 2026) as the key geometric structure that
 shapes the learning process.
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Context-Aware Reasoner: Enhancing Contextual Reasoning in Multimodal Large Langu
age Models
Zhe Zheng, Wenqi Zhang, Xiaohe Zhou, Guiyang Hou, Xinyu Mei, Tianyang Mao, Anni
PIAO, Shengpei Jiang, Yongliang Shen, Yueting Zhuang, Weiming Lu
Multimodal large language models (MLLMs) have demonstrated remarkable reasoning
capabilities over internalized knowledge. However, current research overlooks co
ntextual reasoning, the ability to reason based on the relevant information pres
ent in the context. To investigate this issue, we construct the Visual Contextua
l Reasoning Benchmark (ContextReasonV-Bench), and our analysis reveals two predo
minant failure modes: *context neglect*, where models rely on pre-trained knowle
dge instead of contextual information, and *superficial pattern matching*, where
 models exploit shallow correlations rather than genuine patterns. To address th
ese limitations, we propose a two-stage approach that progressively establishes
and reinforces contextual pattern acquisition. The first stage establishes an "a
nalyze-then-solve" reasoning paradigm through supervised fine-tuning (SFT). We t
hen employ a context-aware reinforcement learning (RL) framework that integrates
 context-aware reward modeling with hierarchical advantage estimation to encoura
ge the model to identify genuine contextual patterns. This approach yields Conte
xt-Aware Reasoner (CAR), a model that achieves 38.14\% accuracy on ContextReason

V-Bench, a 22.09\% improvement over the base model, and generalizes effectively to unseen tasks.
****
Spatially-Regularized Entropy for Discriminative Token Merging in Fine-Grained Re-Identification
Shangze Li, Yifan Xu, Jingmiao Liang, Yongfei Zhang, Yuzhuo Ma, Yingbo Qu
While Vision Transformers (ViTs) offer strong global modeling, their quadratic computational cost limits utility in latency-sensitive applications like person re-identification (ReID). Existing compression strategies, such as token pruning or generic merging, typically rely on coarse-grained criteria tailored for image classification. In fine-grained retrieval, these approaches often discard or smooth out subtle but discriminative local details. To resolve this, we propose SRE-Merge, a training-free framework designed for discriminative token compression. SRE-Merge injects spatial priors into the merging process through three mechanisms: (i) Spatial-Entropy Saliency Assessment (SES-Assess), which quantifies token importance as Spatial-Entropic Mass (SE-Mass) by coupling spatial structure with local attention entropy; (ii) Hybrid Context-Affinity Matching (HCA-Match), which guides precise pair selection by combining feature similarity with mass-derived context; and (iii) Energy-Preserving Weighted Fusion (EPW-Fuse), which incorporates SE-Mass weighting to counteract feature variance reduction. Extensive experiments on standard benchmarks show that SRE-Merge reduces GFLOPs of the base ViT model by about 24\% while retaining competitive retrieval accuracy, establishing a superior accuracy-efficiency trade-off.
****
Absorbing Quantization Error by Deformable Noise Scheduler for Diffusion Models
Mingrui Yang, Wei Huang, Hao Nan SHENG, Donglin Yang, Jichang Yang, Xin Yu, Huining Yu, Yuzhong Jiao, Zhongrui Wang, Xiaojuan Qi
Diffusion models deliver state-of-the-art image quality but are expensive to deploy. Post-training quantization (PTQ) can shrink models and speed up inference, yet residual quantization errors distort the diffusion distribution (the timestep-wise marginal over $\boldsymbol{x}_t$), degrading sample quality. We propose a distribution-preserving framework that absorbs quantization error into the generative process without changing architecture or adding steps.
Deformable Noise Scheduler (DNS) reinterprets quantization as a principled timestep shift, mapping the quantized prediction distribution $\boldsymbol{x}_t$ back onto the original diffusion distribution so that the target marginal is preserved.
Unlike trajectory-preserving or noise-injection methods limited to stochastic samplers, our approach preserves the distribution under both stochastic and deterministic samplers and extends to flow-matching with Gaussian conditional paths. It is plug-and-play and complements existing PTQ schemes.
Empirically, our method consistently enhances generation quality across diverse backbones and existing PTQ baselines. Notably, when further quantizing the FP16 LoRA branch of SVDQuant to enable fully integer inference, our approach effectively mitigates the performance drop, reducing FID from 27.16 to 26.22. Code is available at https://github.com/ZephyrYoung-eYuan/DNS_AQE
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AnomSeer: Reinforcing Multimodal LLMs to Reason for Time-Series Anomaly Detection
Junru Zhang, Lang Feng, Haoran Shi, Xu Guo, Han Yu, Yabo Dong, Duanqing Xu
Time-series anomaly detection (TSAD) with multimodal large language models (MLLMs) is an emerging area, yet a persistent challenge remains: MLLMs rely on coarse time-series heuristics but struggle with multi-dimensional, detailed reasoning, which is vital for understanding complex time-series data. We present AnomSeer to address this by reinforcing the model to ground its reasoning in precise, structural details of time series, unifying anomaly classification, localization, and explanation. At its core, an expert chain-of-thought trace is generated to provide verifiable, fine-grained reasoning from classical analyses (e.g., statistical measures, frequency transforms). Building on this, we propose a novel time-series grounded policy optimization (TimerPO) that incorporates two additional c

omponents beyond standard reinforcement learning: a time-series grounded advanta ge based on optimal transport and an orthogonal projection to ensure this auxili ary granular signal does not interfere with the primary detection objective. Acr oss diverse anomaly scenarios, AnomSeer, with Qwen2.5-VL-3B/7B-Instruct, outperf orms larger commercial baselines in classification and localization accuracy, p articularly on point- and frequency-driven exceptions. Moreover, it produces pla usible reasoning traces that support its conclusions.
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Diffusion Controller: Framework, Algorithms and Parameterization
Tong Yang, Moonkyung Ryu, Chih-Wei Hsu, Guy Tennenholtz, Yuejie Chi, Craig Bouti lier, Bo Dai
Controllable diffusion generation often relies on various heuristics that are se emingly disconnected without a unified understanding. We bridge this gap with Di ffusion Controller (DiffCon), a unified control-theoretic view that casts revers e diffusion sampling as state-only stochastic control within (generalized) linea rly-solvable Markov Decision Processes (LS-MDPs). Under this framework, control acts by reweighting the pretrained reverse-time transition kernels, balancing te rminal objectives against an f-divergence cost. From the resulting optimality conditions, we derive practical reinforcement learning methods for diffusion fin e-tuning: (i) f-divergence-regularized policy-gradient updates, including a PP O-style rule, and (ii) a regularizer-determined reward-weighted regression objec tive with a minimizer-preservation guarantee under the Kullback–Leibler (KL) div ergence. The LS-MDP framework further implies a principled model form: the optim al score decomposes into a fixed pretrained baseline plus a lightweight control correction, motivating a side-network parameterization conditioned on exposed in termediate denoising outputs, enabling effective *gray-box adaptation* with a fr ozen backbone. Experiments on Stable Diffusion v1.4 across supervised and reward -driven finetuning show consistent gains in preference-alignment win rates and i mproved quality–efficiency trade-offs versus gray-box baselines and even the par ameter-efficient white-box adapter LoRA.
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Off-Policy Evaluation with Strategic Agents via Local Disclosure
Kiet Q. H. Vo, Abbavaram Gowtham Reddy, Julian Rodemann, Siu Lun Chau, Krikamol Muandet
We study off-policy evaluation (OPE) under strategic behavior where decision sub jects (or agents) respond to a decision maker's policy by strategically modifyin g their covariates. Such behavior induces a policy-dependent covariate shift, br eaking the standard assumption in existing methods that covariates are exogenous to the policy. Related work addresses this challenge by imposing strong assumpt ions such as repeated interactions or full knowledge of agents’ response behavio r, substantially limiting its applicability to OPE. In contrast, we consider a o ne-shot OPE setting where the decision maker has only partial knowledge of the a gents' response behavior. Our key insight is that disclosing local information t hrough post-hoc explanations reveals agents’ pre-strategic covariates prior to a daptation, mitigating the information loss induced by strategic behavior. Levera ging this structure, we estimate a statistical model for the agents’ responses a nd construct a doubly robust estimator for policy value. By assuming that the ag ents' cost sensitivity follows a conditional log-normal distribution, we establi sh consistency of the proposed estimator and validate our approach empirically. More broadly, our results highlight how interaction design can mitigate informat ion asymmetry by revealing otherwise hidden structure in agents' strategic respo nses.
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Do Activation Verbalization Methods Convey Privileged Information?
Millicent Li, Alberto Mario Ceballos Arroyo, Giordano Rogers, Naomi Saphra, Byro n C Wallace
Recent interpretability methods have proposed to translate LLM internal represen tations into natural language descriptions using a second *verbalizer* LLM. This is intended to illuminate how the target model represents and operates on input s. But do such *activation verbalization* approaches actually provide *privilege

d* knowledge about the internal workings of the target model, or do they merely convey information about the inputs provided to it? We critically evaluate popular verbalization methods and datasets used in prior work and find that one can perform well on such benchmarks without access to target model internals, suggesting that these datasets are not ideal for evaluating verbalization methods. We then run controlled experiments which reveal that verbalizations often reflect the parametric knowledge of the verbalizer LLM that generated them, rather than the knowledge of the target LLM whose activations are decoded.
Taken together, our results indicate a need for targeted benchmarks and experimental controls to rigorously assess whether verbalization methods provide meaningful insights into the operations of LLMs.
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Multimodal Fact-Level Attribution for Verifiable Reasoning
David Wan, Han Wang, Ziyang Wang, Elias Stengel-Eskin, Hyunji Lee, Mohit Bansal
Multimodal large language models (MLLMs) are increasingly used for real-world tasks involving multi-step reasoning and long-form generation, where reliability requires grounding model outputs in heterogeneous input sources and verifying individual factual claims.
However, existing multimodal grounding benchmarks and evaluation methods focus on simplified, observation-based scenarios or limited modalities and fail to assess attribution in complex multimodal reasoning.
We introduce MuRGAt (Multimodal Reasoning with Grounded Attribution), a benchmark for evaluating fact-level multimodal attribution in settings that require reasoning beyond direct observation.
Given inputs spanning video, audio, and other modalities, MuRGAt requires models to generate answers with explicit reasoning and precise citations, where each citation specifies both modality and temporal segments. To enable reliable assessment, we introduce an automatic evaluation framework that strongly correlates with human judgments. Benchmarking with human and automated scores reveals that even strong MLLMs frequently hallucinate citations despite correct reasoning.
Moreover, we observe a key trade-off: increasing reasoning depth or enforcing structured grounding often degrades accuracy, highlighting a significant gap between internal reasoning and verifiable attribution.
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Obliviate: Efficient Unlearning in Recommender Systems
Tushar Prakash, Brijraj Singh, Niranjan Pedanekar, Narayan Chaturvedi
Machine unlearning is becoming increasingly critical in the context of data privacy regulations, particularly for recommendation systems that are directly trained on user interaction data. The goal of this work is to remove requested interaction data and their downstream influence from trained model while preserving recommendation quality, and to do so without incurring the substantial computational cost of full retraining. Existing approaches exhibit several limitations, including limited unlearning completeness and degradation in recommendation performance, while having substantial computational overhead.
In this paper, we propose Obliviate an efficient two-stage unlearning framework for recommender systems that achieves high unlearning completeness while maintaining good utility. In the first stage, we introduce a Low-Rank Unlearning Adapter (LUA), which employs a lightweight Hessian proxy to enable curvature-aware and efficient unlearning through localized low-rank adapters rather than full parameters. In the second stage, we propose Locality-Aware Calibration (LAC), a lightweight refinement stage that updates only the adapter parameters to improve the performance by enforcing unlearning via ranking-based objectives while preserving utility through knowledge distillation. Extensive empirical evaluations demonstrate that Obliviate, achieves high level of forgetting with minimal loss in recommendation quality and at significantly reduced computational cost, offering a practical and scalable solution for large-scale recommender systems.
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Bridging the Grounding Gap in VideoQA via Typed Memory for Language-based Belief-State Reasoning
Saman Forouzandeh, Wei Peng, Xinghuo Yu, Mahdi Jalili

VideoQA models can be accurate yet often fail to align answers with the correct video segments (the \emph{grounding gap}). We introduce \textbf{LINGUA} (\textbf{L}anguage-based \textbf{IN}ference for \textbf{G}rounded Video \textbf{U}nderstanding \textbf{A}gent), a memory-based agent that performs grounded VideoQA by reasoning in an explicit \emph{linguistic belief state}. LINGUA uses five mechanisms: (1) event-driven perception (retains 8--12\% of frames while preserving 94\% of question-relevant events); (2) typed memory for episodic narratives, semantic affordances, and procedural scripts; (3) Belief-Action-Verification loops with postcondition and temporal checks; (4) meta reflection with contrastive refinement; and (5) Bayesian reliability tracking for continual learning without gradient updates. Built with Gemma3-4B (Ollama, 4-bit), LINGUA outperforms strong baselines on five VideoQA benchmarks, reaching 82.4\% on NExT-QA and 42.3\% Acc@GQA on NExT-GQA (answer + IoU$\geq$0.5 temporal localization), while running 2.6$\times$ faster than dense-frame methods. In continual learning over 100 videos, accuracy rises from 45.2\% (first 10) to 61.8\% (last 10) without catastrophic forgetting, indicating online adaptation via memory refinement.

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Active Attacks: Red-teaming LLMs via Adaptive Environments

Taeyoung Yun, Pierre-Luc St-Charles, Jinkyoo Park, Yoshua Bengio, Minsu Kim

We address the challenge of automatically generating diverse attack prompts for large language models (LLMs) that elicit harmful behaviors (e.g., insults, sexual content) and are used for safety fine-tuning. While several prior approaches train LLMs with reinforcement learning (RL) to generate such prompts using only a toxicity classifier as a reward, existing diversity-seeking RL methods often collapse to limited modes: once high-reward prompts are found, exploration of new regions is discouraged. Inspired by the active learning paradigm that encourages adaptive exploration, we introduce \textbf{Active Attacks}, a novel RL-based red-teaming algorithm that adapts its attacks as the victim evolves. By periodically safety fine-tuning the victim LLM with collected attack prompts, we naturally induce an \emph{easy-to-hard exploration curriculum}, where the attacker progresses beyond easy modes toward increasingly difficult ones. We observe that this simple plug-and-play module, which seamlessly integrates into existing RL objectives, unexpectedly outperformed prior RL-based methods, improving cross-attack success rates against GFlowNets, the previous state-of-the-art, from 0.07\% to 31.28\% (a relative gain of more than 400×) with only a 6\% increase in computation.

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Physics-Informed Pre-training on Efficient Electron-Density Images for Organic Material Property Prediction

Zhixiang Cheng, Hongxin Xiang, Mingquan Liu, Tengfei Ma, Yingzhuo Tu, Wenjie Du, Bosheng Song, Yiping Liu, xiangxiang Zeng

Precise property prediction of organic materials is pivotal for next-generation electronic and energy devices. In density functional theory (DFT), the electron density (ED) serves as the fundamental determinant of material properties. Yet, establishing it as an input modality for material property prediction has been impeded by two practical barriers: scarce large-scale ED data and the enormous computational complexity of ED representation. To bridge these gaps, we introduce VisionED, an efficient physics-informed model pre-trained on electron-density images. We curate a dataset of 2 million molecules and represent ED as multi-shot images that efficiently encode both geometric and electronic structure. VisionED is then pre-trained on 12 million multi-shot ED images via cross-scale, physics-informed pretext tasks. Empirical evaluations on photovoltaic and organic chromophore datasets show that VisionED outperforms state-of-the-art baselines by up to 27.0\%, exhibiting superior robustness under distribution shifts and data scarcity. Notably, the model generalizes to unseen device-scale applications, successfully recovering experimental trends and mixing-ratio effects in ternary blends with an average accuracy of 92.77\%. Moreover, relative to the previous ED point cloud, the ED image improves performance by 26.2\% with 2.6$\times$ fewer memory and 4.6$\times$ lower time. The code and data are available at https://github.com/ZhixiangCheng/VisionED.

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Unifying Dataset Pruning and Distillation for Efficient Large-scale Compression
Lingao Xiao, Songhua Liu, Yang He, Xinchao Wang
Dataset pruning (DP) and dataset distillation (DD) fundamentally differ in their outputs: DP selects original image subsets, while DD generates synthetic images. Recently, DD's increasing reliance on original images suggests a convergence of the two directions. To investigate this convergence trend, we propose a unified dataset compression (DC) benchmark. This benchmark reveals an interesting trade-off for soft-label-DD: while soft labels provide valuable information, they can make the distillation process less essential, as distilled images may not always outperform random subsets. In addition, the benchmark reveals that in current stages, dataset pruning outperforms dataset distillation at small dataset sizes. Given these observations, we explore hard-label-DC as a complementary approach that emphasizes image quality while offering substantial storage efficiency. Our PCA (Prune, Combine, and Augment) is the first framework that does not rely on soft labels but instead focuses on image quality. (1) "P'' means selecting easy samples based on dataset pruning metrics, (2) "C'' indicates combining these samples effectively, and (3) "A'' is to apply constrained image augmentation during training.
****
Beyond Continuity: Simulation-free Reconstruction of Discrete Branching Dynamics from Single-cell Snapshots
Junda Ying, Yuxuan Wang, Bowen Yang, Peijie Zhou, Lei Zhang
Inferring cellular trajectories from destructive snapshots is complicated by the challenges of stochasticity and non-conservative mass dynamics such as cell proliferation and apoptosis. Existing unbalanced Optimal Transport (OT) methods treat mass as a continuous fluid, performing inference at the population level. However, this macroscopic view often fails to capture the discrete, jump-like nature of birth-death events at single-cell resolution, which is essential for understanding lineage branching and fate decisions. We present **Unbalanced Schrödinger Bridge (USB)**, a simulation-free framework for learning underlying dynamics that effectively integrates both stochastic and unbalanced effects which also models the discrete, jump-like birth–death dynamics at single-cell resolution. Theoretically, USB provides a tractable solution to the Branching Schrödinger Bridge (BSB) problem, offering a rigorous microscopic interpretation where individual cells undergo both Brownian motion and discrete birth-death jumps. Technically, the method implements an efficient solver by introducing a simulation-free training objective that effectively scales to high-dimensional omics data. Empirically, we demonstrate on both simulated and real-world datasets that USB not only achieves trajectory reconstruction performance better than or comparable to deterministic baselines but also uniquely enables realistic discrete simulation of birth-death dynamics at single-cell resolution.
****
GradPower: Powering Gradients for Faster Language Model Pre-Training
Jinbo Wang, Mingze Wang, Jiaqi Zhang, Wei Wang, Peng Pei, Xunliang Cai, Weinan E, Lei Wu
We propose **GradPower**, a lightweight gradient-transformation technique for accelerating language model pre-training. Given a gradient vector $\boldsymbol{g}=(g_{i})_{i}$, GradPower first applies the elementwise `sign-power` transformation: $ \varphi_p(\boldsymbol{g}) = \left({\rm sign}(g_i)|g_i|^p\right)_{i} $ for a fixed $p>0$, and then feeds the transformed gradient into a base optimizer. Notably, GradPower requires only a **single-line code change** and no modifications to the base optimizer’s internal logic, including the hyperparameters.
When applied to AdamW (termed **AdamWPower**), GradPower consistently achieves lower terminal loss across diverse architectures (LLaMA, Qwen2MoE), parameter scales (66M to 2B), datasets (C4, OpenWebText), and learning-rate schedules (cosine, warmup-stable-decay).
The most pronounced gains are observed when training modern mixture-of-experts models with warmup-stable-decay schedules. GradPower also integrates seamlessly

with other state-of-the-art optimizers, such as Muon, yielding further improvements. Finally, we provide theoretical analyses that reveal the underlying mechanism of GradPower and highlight the influence of gradient noise.
****

Adaptive Bandit Algorithms for Contextual Matching Markets
Shiyun Lin, Simon Mauras, Vianney Perchet, Nadav Merlis
We study bandit learning in matching markets, where players and arms constitute the two market sides, and the players' utilities are linear in the arm contexts. In each round, new arms arrive with observable contexts. Then, the algorithm matches them to players, aiming to minimize each player's regret against a *stable matching benchmark*. This contextual structure creates significant complexity: subtle context shifts can slightly alter one player's utility while completely reconfiguring the underlying benchmark, causing large regret spikes for others. We address this in two settings: *stochastic* contexts, drawn from a latent distribution, and *adversarial* contexts, which may be arbitrary.
For the stochastic case, we introduce a novel minimum preference gap to capture learning difficulty and provide a fully adaptive algorithm with an instance-dependent poly-logarithmic regret upper bound. We also establish matching instance-independent regret upper and lower bounds under a mild distributional assumption. For the adversarial setting, we propose a tractable regret notion that remains valid under arbitrary contexts and achieves an instance-independent sublinear regret bound via an adaptive algorithm.
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MeshTok: Efficient Multi-Scale Tokenization for Scalable PDE Transformers
Yanshun Zhao, Xiaoyu Peng, Jiamin Jiang, Congcong Zhu, Jingrun Chen
Conventional patchified Transformers operate on uniform spatial partitions, distributing computational effort evenly across the domain irrespective of local features. This inflexible tokenization scheme is inherently limited in its ability to efficiently represent and process solutions to complex PDEs. To address this, we propose MeshTok, an adaptive mesh refinement (AMR)-inspired tokenization and sequence modeling framework. This method selectively refines spatial regions exhibiting sharp gradients, transient features, or multiscale structures, generating a heterogeneous set of multiscale tokens defined on a fixed simulation grid. These tokens are processed within a unified Transformer sequence, enabling the model to simultaneously capture coarse-grained global context and fine-grained local details without requiring specialized architectural components. Although adaptive refinement moderately increases token count, it promotes a more targeted allocation of computational resources to physically informative regions, which we view as a practical inductive bias rather than a formal optimality guarantee. Experimental evaluations across multiple PDE families and benchmark datasets demonstrate that MeshTok consistently improves the efficiency-accuracy trade-off compared to uniform-grid baselines. This suggests adaptive multiscale tokenization as a scalable and generalizable design principle for neural PDE modeling. Code is available at https://github.com/SCAILab-USTC/MeshTok.
****

Conditional Quantile Adjusted Conformal Prediction for Time Series
Cheng Yu, Zhoufan Zhu, Ke Zhu
Conformal prediction is challenging for time series with time-varying conditional distributions. Existing sequential conformal methods can yield volatile, non-nested prediction intervals due to noisy tail conditional quantile estimation and quantile crossing issue. To overcome these challenges, we construct prediction intervals for time series via a novel method called Conditional Quantile Adjusted Conformal Prediction (CQACP), which stabilizes sequential conformal calibration by modeling the conditional quantile curve of nonconformity score. At each time step, CQACP evaluates a base conditional quantile learner on a grid of quantile levels and fits a Cornish--Fisher approximation parameterized by conditional moments of nonconformity score with monotonicity constraints. Asymptotically, we prove the conditional validity of the prediction interval under serial dependence and show improved conditional quantile estimation accuracy. Experiments on multiple real-world datasets demonstrate that CQACP maintains accurate coverage and

produces smooth, narrow, and nested prediction intervals across different significance levels and prediction models.
****
SSDCN: Spatial-Spectral Dual-Clustering-based Network for Hyperspectral Image Super-resolution
Yong Yang, Xuran Zhang, Shuying Huang, Xiaozheng Wang, Weiguo Wan, Hangyuan Lu
Hyperspectral Image Single Image Super-Resolution (HSI-SISR) faces a conflict between computational efficiency and global non-local modeling.
■■Existing Transformers suffer from quadratic complexity, while window-based methods compromise global capture.
■■To address this, we propose the Spatial-Spectral Dual-Clustering-based Network (SSDCN). Our method introduces three innovations.
■■First, we design a Spatial-Spectral Dual-Cluster Block (SSDCB). Replacing expensive point-to-point attention, it uses content-driven clustering to learn low-rank structural bases, achieving global modeling with linear complexity $\mathcal{O}(KN)$.
■■Second, we propose a pyramid progressive hierarchical architecture with a Feature Reuse Reconstruction Block (FRRB).
■■It reuses the core tensor and spectral factors from coarse levels, updating only spatial factors to minimize redundancy.
■■Third, we propose a Pyramid Hierarchical Reconstruction Joint Loss to supervise intermediate levels, ensuring structural accuracy and preventing error accumulation.
■■Experiments demonstrate that SSDCN surpasses SOTA methods in metrics and visual quality with significantly fewer parameters and FLOPs, achieving an optimal efficiency-performance balance.
****
Domain Transfer Becomes Identifiable via a Single Alignment
Sagar Shrestha, Subash Timilsina, Hoang-Son Nguyen, Xiao Fu
Domain transfer (DT) maps source to target distributions and supports tasks such as unsupervised image-to-image translation, single-cell analysis, and cross-platform medical imaging. However, DT is fundamentally ill-posed: push-forward mappings are generally non-identifiable, as measure-preserving automorphisms (MPAs) preserve marginals while altering cross-domain correspondences, leading to content-misaligned translation. Recent work shows that MPAs can be eliminated by jointly transferring multiple corresponding source/target conditional distributions, but supervision signals labeling such conditionals are not always available in practice.
We develop an alternative route to DT identifiability. Under a structural sparsity condition on the Jacobian support pattern, we show that distribution matching together with a single paired anchor sample suffices to identify the ground-truth transfer---requiring substantially less supervision than prior approaches. To enable practical high-dimensional learning, we further propose an efficient Jacobian sparsity regularizer based on randomized masked finite differences, yielding a scalable surrogate without explicit Jacobian evaluation. Empirical results on synthetic and real-world DT tasks validate the theory.
****
Robust-U1: Can MLLMs Self-Recover Corrupted Visual Content for Robust Understanding?
Jiaqi Tang, Jianmin Chen, Youyang Zhai, Wei Wei, Runtao Liu, Mengjie Zhao, Xiangyu Wu, Qingfa Xiao, Qifeng Chen
Multimodal Large Language Models (MLLMs) have demonstrated remarkable success in visual understanding, yet their performance degrades significantly under real-world visual corruptions. While existing robustness enhancement approaches exist, they are limited: black-box feature alignment lacks interpretability, and white-box text-based reasoning cannot restore lost pixel-level details. This work investigates a fundamental research question: Can MLLMs recover corrupted visual content by themselves? To address this, we propose Robust-U1, a novel framework that equips MLLMs with explicit visual self-recovery capability for robust understanding. The approach comprises three core stages: supervised fine-tuning for in

itial reconstruction, reinforcement learning with dual rewards (pixel-level SSIM and semantic-level CLIP similarity) for aligning high visual quality, and multimodal reasoning that jointly considers both the corrupted input and the recovered image. Extensive experiments demonstrate that Robust-U1 achieves state-of-the-art robustness on the real-world corruption benchmark and maintains superior performance under adversarial corruptions on general VQA benchmarks. Analysis confirms that high-quality visual recovery directly enhances reasoning performance, establishing self-recovery as a critical mechanism for robust visual understanding. The source code is available at https://github.com/jqtangust/Robust-U1.
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Scheduling LLM Inference with Uncertainty-Aware Output Length Predictions
Haoyu Zheng, Yongqiang Zhang, Fangcheng Fu, Xiaokai Zhou, Hao Luo, Hongchao Zhu, Yuanyuan Zhu, Hao Wang, Xiao Yan, Jiawei Jiang
To schedule LLM inference, the \textit{shortest job first} (SJF) principle is favorable by prioritizing requests with short output lengths to avoid head-of-line (HOL) blocking. Existing methods usually predict a single output length for each request to facilitate scheduling. We argue that such a \textit{point estimate} does not match the \textit{stochastic} decoding process of LLM inference, where output length is \textit{uncertain} by nature and determined by when the end-of-sequence (EOS) token is sampled. Hence, the output length of each request should be fitted with a distribution rather than a single value. With an in-depth analysis of empirical data and the stochastic decoding process, we observe that output length follows a heavy-tailed distribution and can be fitted with the log-t distribution. On this basis, we propose a simple metric called Tail Inflated Expectation (TIE) to replace the output length in SJF scheduling, which adjusts the expectation of a log-t distribution with its tail probabilities to account for the risk that a request generates long outputs. To evaluate our TIE scheduler, we compare it with three strong baselines, and the results show that TIE reduces the per-token latency by $2.31\times$ for online inference and improves throughput by $1.42\times$ for offline data generation.
****

Feature Resemblance: Towards a Theoretical Understanding of Analogical Reasoning in Transformers
Ruichen Xu, Wenjing Yan, Ying-Jun Angela Zhang
Understanding reasoning in large language models is complicated by evaluations that conflate multiple reasoning types. We isolate analogical reasoning, where a model transfers an attribute between entities that share known properties, and study when such transfer can emerge from training. To make the problem analytically tractable, we study a minimal transformer-style abstraction that isolates how learned representations support analogical reasoning. Within this setting, we prove three key results. First, joint training on similarity and attribution premises enables analogical reasoning through aligned representations. Second, sequential training succeeds only when similarity structure is learned before specific attributes, revealing a curriculum asymmetry. Third, in our stylized setting, two-hop reasoning $(a \to b, b \to c \Rightarrow a \to c)$ can be viewed as analogical reasoning with identity bridges $(b=b)$, which appear explicitly in training data. Together, these results reveal a unified mechanism: entities with shared properties become aligned in representation space, enabling property transfer through feature resemblance. Experiments with architectures up to 8B parameters show qualitative agreement with the theory and suggest that representational geometry plays an important role in analogical reasoning beyond the stylized model.
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Decoupling The "What" and "Where" With Polar Coordinate Positional Embedding
Anand Gopalakrishnan, Róbert Csordás, Jürgen Schmidhuber, Michael Curtis Mozer
The attention mechanism in a Transformer architecture matches key to query based on both content—the what—and position in a sequence—the where. We present an analysis indicating that what and where are entangled in the popular rotary position embedding (RoPE). This entanglement can impair performance particularly when decisions require independent matches on these two factors. We propose an improv

ement to RoPE, which we call Polar Coordinate Position Embedding or PoPE, that e liminates the what-where confound. PoPE is far superior on a diagnostic task req uiring indexing solely by position or by content. On autoregressive sequence mod eling in music, genomic, and natural language domains, Transformers using PoPE a s the positional encoding scheme outperform baselines using RoPE with respect to evaluation loss (perplexity) and downstream task performance. On language model ing, these gains persist across model scale, from 124M to 774M parameters. Cruci ally, PoPE shows strong zero-shot length extrapolation capabilities compared not only to RoPE but even a method designed for extrapolation, YaRN, which requires additional fine tuning and frequency interpolation.
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Speculative Sampling For Faster Molecular Dynamics
Arthur Kosmala, Stephan Günnemann, Meng Gao, Brandon M. Wood
Molecular dynamics (MD) is a key tool for simulating the dynamical behavior of a tomic systems. However, MD is inherently serial, which makes it difficult to inc rease single-system throughput with concurrent compute. To address this, we intr oduce **L**angevin **S**peculative **D**ynamics (**LSD**), a distributed and mod el-agnostic speculative sampler for accelerating MD *without adding relative err or*. Inspired by speculative methods in language and diffusion modeling, LSD use s a draft model to propose fast simulation steps and verifies them in parallel w ith a slower target model, applying a transport map from the draft to the target distribution. We extend speculative sampling to second-order Langevin dynamics, derive the achievable speedup as a function of physical parameters, show that L SD generalizes across different systems and draft-target combinations with a 3-9 x speedup, and confirm theoretically and empirically that LSD samples trajectori es from the same distribution as its target model.
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Mobility-Embedded POIs: Learning What A Place Is and How It Is Used from Human M ovement
Maria Despoina Siampou, Shushman Choudhury, Shang-Ling Hsu, Neha Arora, Cyrus Sh ahabi
Recent progress in geospatial foundation models highlights the importance of lea rning general-purpose representations for real-world locations, particularly poi nts-of-interest (POIs) where human activity concentrates. Existing approaches, h owever, focus primarily on place identity derived from static textual metadata, or learn representations tied to trajectory context, which capture movement regu larities rather than how places are actually used (i.e., POI's function). We arg ue that POI function is a missing but essential signal for general POI represent ations. We introduce Mobility-Embedded POIs (ME-POIs), a framework that augments POI embeddings derived, from language models with large-scale human mobility da ta to learn POI-centric, context-independent representations grounded in real-wo rld usage. ME-POIs encodes individual visits as temporally contextualized embedd ings and aligns them with learnable POI representations via contrastive learning to capture usage patterns across users and time. To address long-tail sparsity, we propose a novel mechanism that propagates temporal visit patterns from nearb y, frequently visited POIs across multiple spatial scales. We evaluate ME-POIs o n five newly proposed map enrichment tasks, testing its ability to capture both the identity and function of POIs. Across all tasks, augmenting text-based embed dings with ME-POIs consistently outperforms both text-only and mobility-only bas elines. Notably, ME-POIs trained on mobility data alone can surpass text-only mo dels on certain tasks, highlighting that POI function is a critical component of accurate and generalizable POI representations.
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Bridging the Knowledge-Prediction Gap in LLMs on Multiple-Choice Questions
Yoonah Park, Haesung Pyun, Yohan Jo
While large language models (LLMs) perform strongly on diverse tasks, their trus tworthiness is limited by erratic behavior that is unfaithful to their internal knowledge. In particular, LLMs often fail on multiple-choice questions (MCQs) ev en if they encode correct answers in their hidden representations, revealing a m isalignment between internal knowledge and output behavior. We investigate and m

itigate this *knowledge-prediction gap* on MCQs through a three-step analysis of hidden representations. First, we quantify the prevalence and magnitude of the gap across models and datasets. Second, we provide a geometric interpretation by identifying distinct *knowledge* and *prediction* subspaces in the residual str eam. Third, we introduce **KAPPA**, a lightweight inference-time intervention th at aligns the two subspaces within the residual stream to reduce the knowledge-p rediction gap. Our results provide a geometric and interpretable explanation of the knowledge-prediction gap in LLMs. Furthermore, KAPPA effectively reduces the gap across diverse MCQ benchmarks and models, and generalizes to free-form sett ings.
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Towards Theoretical Understanding of Transformer Test-Time Computing: Investigat ion on In-Context Linear Regression
Xingwu Chen, Miao Lu, Beining Wu, Difan Zou
Scaling test-time computation during language model inference, such as generatin g intermediate thoughts or sampling multiple candidate answers, has proven effec tive in improving model performance.
While these techniques inherently rely on the stochastic nature of inference to explore diverse reasoning paths, prior theoretical works typically build on a *d eterministic decoding* framework, overlooking the stochastic nature of practical language model inference.
This work takes an initial step to bridge this gap by establishing a new theoret ical framework, incorporating *randomness and sampling* directly into the decodi ng analysis.
To demonstrate the framework's effectiveness, we apply it to the canonical in-co ntext linear regression task with continuous and binary coefficients, simulating decoding via noise injection and sampling to analyze widely adopted inference t echniques.
We validate our theoretical findings through numerical simulations, with additio nal experiments on real-world tasks substantiating the framework's potential for practical applications.
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Critique-Guided Distillation for Robust Reasoning via Refinement
Berkcan Kapusuzoglu, Supriyo Chakraborty, Zain Sarwar, Chia-Hsuan Lee, Sambit Sa hu
Supervised fine-tuning with expert demonstrations often produces models that imi tate outputs without internalizing the reasoning processes needed for robust gen eralization. While critique-based approaches show promise, training models to ge nerate critiques directly, such as Critique Fine-Tuning (CFT), can lead to outpu t-format drift and degradation of general capabilities. We propose $\textbf{C}$r itique-$\textbf{G}$uided $\textbf{D}$istillation (CGD), a training framework tha t decouples critique consumption from critique generation. During fine-tuning, t he student is trained to refine flawed responses conditioned on teacher critique s. CGD treats critiques as a $\textit{training-time-only}$ supervision signal, e ncouraging internalization of error-aware reasoning: critiques guide learning bu t are absent at inference. Across five model families, CGD consistently outperfo rms CFT and standard distillation on mathematical reasoning benchmarks, yielding 7\% average improvements and gains of up to +15.0\% on AMC23 and +12.2\% on MAT H-500. On challenging competition problems such as AIME24 and AIME25, CGD achiev es substantially higher Pass@1 and stronger performance at low Pass@k, indicatin g improved reasoning quality per sample. Importantly, CGD preserves general inst ruction-following capabilities where CFT degrades significantly ($-$21.3\% on IF Eval). These results position CGD as a practical and compute-efficient intermedi ate training paradigm for reasoning-centric tasks without introducing architectu ral inference-time overhead.
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PVDepth: Panoramic Video Depth Estimation via Geometry-Aware Spatiotemporal Adap tation
Chuanxin Song, Peixi Peng
Panoramic video depth estimation is pivotal for applications such as Virtual Rea

lity and World Models. However, advancements in this field are impeded by two pr
imary obstacles: the scarcity of large-scale training data and the unique spatio
temporal challenges of Equirectangular Projection (ERP), which hinder the direct
 transfer of perspective models.
In this paper, we first present **PanoCARLA**, a large-scale synthetic RGB-D pan
oramic video dataset, featuring natural motion trajectories and drone-like roami
ng perspectives.
Building on this foundation, we propose **PVDepth**, an end-to-end framework ada
pted from perspective video depth models.
To tackle ERP-specific geometric distortions and consequent non-linear temporal
dynamics, we introduce two core mechanisms: (1) A *Progressive Sphere-aware Nois
e Initialization* strategy that anneals the noise distribution from planar to sp
herical, guiding the model to adapt to non-uniform information density; and (2)
A *Cube-rectified Temporal Modeling* module that incorporates an auxiliary cubem
ap temporal branch to rectify non-linear temporal dynamics in the ERP domain.
Extensive experiments demonstrate that PVDepth achieves superior performance, ge
nerating geometrically accurate and temporally consistent depth sequences. Code
and data will be released at https://github.com/ChuanxinSong/PVDepth.
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VLM-RobustBench: A Comprehensive Benchmark for Robustness of Vision-Language Mod
els

Rohit Saxena, Alessandro Suglia, Pasquale Minervini

Vision-language models (VLMs) achieve strong performance on standard, high-quali
ty datasets, but we still do not fully understand how they perform under real-wo
rld image distortions. We present VLM-RobustBench, a benchmark spanning 49 augme
ntation types across noise, blur, weather, digital, and geometric perturbations,
 evaluated under graded severities (low/mid/high) and binary transforms, yieldin
g 133 corrupted settings. We evaluate 15 VLMs from five families, including prop
rietary GPT-5.4-mini, on two complementary benchmarks: MMBench (visually grounde
d) and MMMU-Pro (reasoning-oriented). Our results reveal that visual severity is
 a weak predictor of difficulty: low-severity spatial perturbations often degrad
e performance more than visually severe photometric corruptions. In particular,
low-severity glass blur reduces MMBench accuracy by about 8 pp on average across
 models, while the largest drops arise from resampling and geometric distortions
 (e.g., upsample, elastic transform), reaching up to 34 pp. Overall, our finding
s suggest current VLMs are semantically strong but spatially fragile, motivating
 the definition of novel robustness evaluation protocols and training regimes th
at emphasize resampling and geometric invariance.
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RoCA: Robust Cross-Domain End-to-End Autonomous Driving

Rajeev Yasarla, Shizhong Han, Hsin-Pai Cheng, Apratim Bhattacharyya, Shweta Maha
jan, Litian Liu, Yunxiao Shi, Risheek Garrepalli, Hong Cai, Fatih Porikli

End-to-end (E2E) autonomous driving has recently emerged as a new paradigm, offe
ring significant potential. However, few studies have looked into the practical
challenge of deployment across domains (e.g., cities). Although several works ha
ve incorporated Large Language Models (LLMs) to leverage their open-world knowle
dge, LLMs do not guarantee cross-domain driving performance and may incur prohib
itive retraining costs during domain adaptation. In this paper, we propose RoCA,
 a novel framework for robust cross-domain E2E autonomous driving. RoCA formulat
es the joint probabilistic distribution over the tokens that encode ego and surr
ounding vehicle information in the E2E pipeline. Instantiating with a Gaussian p
rocess (GP), RoCA learns a set of basis tokens with corresponding trajectories,
which span diverse driving scenarios. Then, given any driving scene, it is able
to probabilistically infer the future trajectory. By using RoCA together with a
base E2E model in source-domain training, we improve the generalizability of the
 base model, without requiring extra inference computation. In addition, RoCA en
ables robust adaptation on new target domains, significantly outperforming direc
t finetuning. We extensively evaluate RoCA on various cross-domain scenarios and
 show that it achieves strong domain generalization and adaptation performance.
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UniCode: Augmenting Evaluation for Code Reasoning
Xinyue Zheng, Haowei Lin, Shaofei Cai, Yaodong Yang, Zilong Zheng, Yitao Liang
Current coding benchmarks often overstate Large Language Model (LLM) capabilities due to static paradigms and data contamination, allowing models to exploit statistical shortcuts rather than genuine reasoning. To address this, we introduce UniCode, a generative evaluation framework that systematically probes LLM reasoning boundaries via: (1) multi-dimensional augmentation operators to create diverse algorithmic variants; (2) a scalable test generation pipeline achieving 94.5\% correctness without human-written solutions; and (3) fine-grained diagnostic metrics for rich error signals. Our evaluation of state-of-the-art models reveals a significant 31.2\% performance collapse. Critically, we observe a high variance across different reasoning axes, revealing a profound fragility under structural shifts despite surface-level robustness. Furthermore, we identify a ``seed-problem regression," where models fail by defaulting to memorized seed logic and inefficient complexities. Our evaluation code is publicly available at \url{https://github.com/grandsmile/UniCode}.
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MoRGen: Mixture-of-Resolutions Generative Forecasting for Irregularly Sampled Medical Time-Series Data
Nassim Oufattole, Matthew B.A. McDermott, Collin Stultz
Autoregressive generative models for irregularly sampled clinical time-series data are increasingly used for zero-shot risk forecasting. Prior work typically adopts a single fine-grained discretization of time, where tokens are generated at one fixed, predetermined temporal resolution. We demonstrate that the zero-shot accuracy of individual generative forecasters varies with temporal resolution: performance can degrade when the model resolution is poorly matched to the temporal dynamics of the endpoint being evaluated. We then propose MoRGen (Mixture-of-Resolutions Generation), which fuses forecasts from generative experts trained at multiple temporal resolutions using a low-capacity task-specific mixture, improving performance across tasks with different temporal dynamics. Across multiple horizons and outcomes on three independent clinical datasets, MoRGen achieves lower binary cross-entropy (BCE) and statistically significant AUROC gains over autoregressive generative models that forecast tokens at a fixed temporal resolution.
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QTALE: Quantization-Robust Token-Adaptive Layer Execution for LLMs
KangHyun Noh, Jinheon Choi, Yulhwa Kim
Large language models (LLMs) demand substantial computational and memory resources, posing challenges for efficient deployment. Two complementary approaches have emerged to address these issues: token-adaptive layer execution, which reduces floating-point operations (FLOPs) by selectively bypassing layers, and quantization, which lowers memory footprint by reducing weight precision. However, naively integrating these techniques leads to additional accuracy degradation due to reduced redundancy in token-adaptive models.
We propose QTALE (Quantization-Robust Token-Adaptive Layer Execution for LLMs), a novel framework that enables seamless integration of token-adaptive execution with quantization while preserving accuracy. Conventional token-adaptive methods reduce redundancy in two ways: (1) by limiting the diversity of training paths explored during fine-tuning, and (2) by lowering the number of parameters actively involved in inference. To overcome these limitations, QTALE introduces two key components:
(1) a training strategy that ensures diverse execution paths are actively explored during fine-tuning, and
(2) a post-training mechanism that allows flexible adjustment of the execution ratio at inference to reintroduce redundancy when needed.
Experimental results show that QTALE enables seamless integration of token-adaptive layer execution with quantization, showing no noticeable accuracy difference, with the gap to quantization-only models kept below 0.5\% on CommonsenseQA benchmarks.
By combining token-adaptive execution for FLOPs reduction and quantization for m

emory savings, QTALE provides an effective solution for efficient LLM deployment
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Neuro-evolutionary Continual Reinforcement Learning
Pengyi Li, Hongyao Tang, Yifu Yuan, YAN ZHENG, Xin Xu, Jianye HAO
Deploying robots in open-ended real-world environments demands continual learnin
g capabilities to adapt to an ever-expanding range of tasks. This requires retai
ning previously acquired skills without forgetting while effectively leveraging
prior knowledge to learn new ones. Inspired by neuroscience, we propose **N**eur
o-**e**volutionary **C**ontinual **R**einforcement **L**earning (**Nevo-CRL**).
Nevo-CRL maintains a fixed-capacity monolithic policy network, solving tasks by
optimizing inter-layer connectivity and neuron parameters.
For each new task, Nevo-CRL constructs a mask population to selectively activate
 the outputs of each hidden layer, thereby forming a task-specific policy popula
tion. Upon completing each task, the best-performing mask is stored, and its act
ivated neurons are frozen to prevent catastrophic forgetting. To facilitate know
ledge transfer, Nevo-CRL reuses neurons from acquired skills based on semantic s
imilarity between tasks, while dynamically allocating additional neurons for tas
k-specific adaptation.
In the learning process, Nevo-CRL iteratively adjusts masks via importance-guide
d crossover to optimize the policy network connectivity. To improve neuron utili
zation, we prune low-activity connections to recycle neurons. Experiments demons
trate that Nevo-CRL achieves state-of-the-art performance among continual RL met
hods.
The code is available at [https://github.com/yeshenpy/Nevo-CRL](https://github.c
om/yeshenpy/Nevo-CRL).
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Explainable Forensics of Manipulated Segments in Untrimmed Long Videos
Yue Feng, Jingjing Li, Qijia Lu, Wei Ji, Jingrou Zhang, Fei Shen, Xiao Li, Yizhe
n Jia, Qiang Chen, Limin Wang, Wentong Li, Jie Qin
The rapid advancement of AI-driven video generation has transformed content crea
tion, while simultaneously increasing the risk of misinformation through localiz
ed manipulations in long-form videos. Existing video forensic methods predominan
tly operate on short, independent clips, and thus fail to capture realistic scen
arios where AI-generated content is sparsely embedded within otherwise authentic
 footage. To bridge this gap, we formulate the task of Temporal AI-Generated Seg
ment Localization and Explanation, which targets authenticity detection, tempora
l localization, and interpretable analysis of manipulated segments in untrimmed
long videos. We further introduce TASLE, a large-scale benchmark comprising 12,4
72 untrimmed videos with diverse manipulation patterns and rich annotation signa
ls, including temporal boundaries, authenticity labels, and segment-level ration
ales. In addition, we propose MSLoc, a coarse-to-fine forensic baseline that com
bines a boundary-sensitive proposal generation module for efficient long-video s
canning with an MLLM-based refinement module for precise boundary localization a
nd interpretable reasoning. Experiments validate the effectiveness of the propos
ed baseline, highlighting the importance of segment-level explainable forensics
for long-form AI-generated video analysis. Dataset and code will be made publicl
y available.
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TritonGym: A Benchmark for Agentic LLM Workflows in Triton GPU Code Generation
Yue Guan, Yichen Lin, Xu Zhao, Jianzhu Yao, Xinwei Qiang, Zhongkai Yu, Pramod Vi
swanath, Yufei Ding, Adnan Aziz
Large language models (LLMs) can already draft plausible Triton kernels, yet mos
t existing evaluations still focus on single-shot generation and underplay tool
use and feedback. We introduce *TritonGym*, a benchmark and orchestration framew
ork for evaluating agentic workflows in GPU code generation. *TritonGym* standar
dizes access to a set of code generation tools via function-calls, separating in
trinsic model capability from workflow design and enabling fair, apples-to-apple
s comparison. The benchmark spans a maintained operator set, community samples,
out-of-distribution tasks, and DSL extensions, ensuring both generality and exte

nsibility. By providing a common orchestration and evaluation framework, *Triton Gym* democratizes the development of GPU coding agents, supports practical adoption of agent-generated kernels, and facilitates progress on advanced agentic systems.
****
Bilevel Optimization over Saddle Points of Zero-Sum Markov Games
Zihao Zheng, Irwin King, Songtao Lu
Reinforcement learning (RL) often has a hierarchical structure, where an upper-level (UL) learner selects model parameters and a lower-level (LL) decision-making process responds, naturally leading to a bilevel optimization problem. Most existing bilevel RL methods assume a single-policy LL Markov decision process (MDP), and therefore fail to capture competitive structures arising in applications such as incentive design, where multiple policies interact. We study bilevel optimization problems in which the LL problem is a regularized min–max zero-sum Markov game and the UL objective is optimized through the saddle-point equilibrium induced by the LL game. In this work, we propose penalty-augmented Nikaido–Isoda descent–ascent (PANDA), a penalty-based first-order policy-gradient method based on the Nikaido–Isoda function. By exploiting the min–max game structure, PANDA avoids computing UL hypergradients and does not require second-order information. We prove that PANDA converges to stationary points without convexity assumptions on either the UL or LL objectives. Moreover, PANDA reaches an ϵ-stationary point in $\tilde{\mathcal{O}}(\epsilon^{-1})$ iterations with sample complexity $\tilde{\mathcal{O}}(\epsilon^{-3})$, matching the best-known rates for bilevel RL with single-policy LL MDPs. Experiments demonstrate the superior performance of PANDA over closely related baselines.
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CLEAR: Context-Aware Learning with End-to-End Mask-Free Inference for Adaptive Video Subtitle Removal
Qingdong He, Chaoyi Wang, Peng Tang, Yifan Yang, Xiaobin Hu
Video subtitle removal is essential for content localization and media re-editing, yet existing mask-guided diffusion methods face critical limitations: training inefficiency requiring extensive annotations and full model fine-tuning, inference complexity demanding explicit mask sequences, and static prior utilization unable to adapt to quality variations. We present CLEAR (Context-aware Learning for End-to-end Adaptive subtitle Removal), a lightweight adapter-based framework addressing these challenges through three technical innovations. First, self-supervised prior learning (Stage I) extracts occlusion guidance from video pairs using pixel differences as weak supervision, eliminating annotation dependency while learning generalizable subtitle features across languages. Second, LoRA-based adaptive refinement (Stage II) enables parameter-efficient training that preserves pre-trained visual priors while achieving true mask-free end-to-end inference without external detection modules. Third, adaptive focal weighting dynamically adjusts prior influence based on local quality assessment, effectively handling diverse subtitle styles and noisy guidance signals. Extensive experiments demonstrate CLEAR's superior performance in multilingual subtitle removal while requiring only 0.77% trainable parameters, establishing a new paradigm for efficient video text removal without inference-time mask dependencies.
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TPV: Parameter Perturbations Through the Lens of Test Prediction Variance
Devansh Arpit
We introduce test prediction variance (TPV)—the first-order sensitivity
of a trained model's outputs to parameter perturbations—as a unifying framework for analyzing post-training robustness. TPV's trace form $\mathrm{Tr}(H_{\mathrm{eff}}C)$ separates the geometry of the trained model H_{eff} from the perturbation covariance C, placing SGD noise, label noise, quantization, and pruning under
a single lens. The resulting expressions recover the wide-minima hypothesis for SGD and quantization noise, and yield a distinct Jacobian-spectral characterization for label noise connecting label-noise TPV with benign overfitting in nonlinear networks.

Theoretically, we prove that training-set TPV converges to its test-set counterp art in the overparameterized limit, irrespective of generalization performance, providing the first result that prediction variance under local parameter pertur bations can be inferred from training inputs alone. Empirically, this stability holds far more broadly, including at very low widths. Further, TPV correlates we ll with test loss, enabling practical applications: JBR, a label-free pruning cr iterion derived from TPV geometry matching state-of-the-art baselines; and train ing-set based model selection signal for in-distribution and transfer learning s cenarios. \href{https://github.com/devansharpit/TPV/tree/main}{Code Available He re}

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Lagrangian Perturbation Diffusion Steering: Latent Reinforcement Learning for Ge nerative Policies

Hikmet Simsir, Ozgur S. Oguz

Behavior cloning with high-capacity generative policies achieves strong imitatio n performance, but is often limited by demonstration coverage and distribution s hift. Direct reinforcement learning fine-tuning can improve performance, but upd ating large action decoders is frequently unstable and sample inefficient. We pr opose **Lagrangian Perturbation Diffusion Steering (LP-DS)**, a lightweight adap tation method that improves a frozen generative policy by learning a compact noi se-space perturbation before decoding. LP-DS optimizes this perturbation with a Lagrangian trust-region objective, improving downstream value while constraining deviation from the latent prior. Across RoboMimic manipulation, OpenAI Gym loco motion, and Adroit dexterous manipulation benchmarks, LP-DS improves sample effi ciency, success, and return while maintaining higher action-space entropy than u nconstrained noise-space steering, with return improvements of up to 25\% over p rior baselines. Additional evaluations with flow-matching backbones, a large vis ion-language-action model, and physical Franka deployment show that LP-DS is not limited to compact diffusion policies or simulated benchmarks. Project page: ht tps://sites.google.com/view/lp-ds/home.

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The Hippocampal Place Field Gradient: A Bio-inspired Framework Building Multisca le Representation for Better Sample Efficiency

ZHOU Shujun, Junrong Qi, Guozhang Chen

The hippocampus encodes space through a striking gradient of place field sizes a long its dorsal-ventral axis, yet the principles generating this continuous grad ient from discrete grid cell inputs remain unclear. We propose a unified theoret ical framework establishing how multiscale hippocampal place fields arise from t he frequency-dependent decay of grid cell projections. Functionally, this organi zation establishes an inductive bias in the population code, managing a fundamen tal trade-off between spatial precision and sample efficiency. Translating this insight to artificial neural networks, we incorporate a hippocampus-inspired pos itional embedding (HIPE) into the Transformer architecture to induce multi-scale representation. Experimental results confirm that this mechanism effectively im proves data efficiency. Our work establishes a link between neural connectivity, activity patterns, and learning, suggesting a principled approach to utilizing multi-scale representations for sample-efficiency learning. Our codes are availa ble at https://github.com/AIogry/relative_PE.

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Spurious Correlation Learning in Preference Optimization: Mechanisms, Consequenc es, and Mitigation via Tie Training

Christian Moya, Alex Semendinger, Guang Lin, Elliott Thornley

Preference learning methods like Direct Preference Optimization (DPO) are known to induce reliance on spurious correlations, leading to sycophancy and length bi as in today's language models and potentially severe goal misgeneralization in f uture systems. In this work, we provide a unified theoretical analysis of this p henomenon, characterizing the mechanisms of spurious learning, its consequences on deployment, and a provable mitigation strategy. Focusing on log-linear polici es, we show that standard preference-learning objectives induce reliance on spur

ious features at the population level through two channels: mean spurious bias a
nd causal-spurious correlation leakage. We then show that this reliance creates
an irreducible vulnerability to distribution shift: more data from the same trai
ning distribution fails to reduce the model's dependence on spurious features. T
o address this, we propose *tie training*, a data augmentation strategy using ti
es (equal-utility preference pairs) to introduce data-driven regularization. We
demonstrate that this approach selectively reduces spurious learning without deg
rading causal learning. Finally, we validate our theory on log-linear models and
 provide empirical evidence that both the spurious learning mechanisms and the b
enefits of tie training persist for neural networks and large language models.
****

See What Matters: Differentiable Grid Sample Pruning for Generalizable Vision-La
nguage-Action Model
Yixu Feng, Zinan Zhao, Yanxiang Ma, Chenghao Xia, Chengbin Du, Yunke Wang, Chang
 Xu
Vision-Language-Action (VLA) models have shown remarkable promise in robotics ma
nipulation, yet their high computational cost hinders real-time deployment. Exis
ting token pruning methods suffer from a fundamental trade-off: aggressive compr
ession using pruning inevitably discards critical geometric details like contact
 points, leading to severe performance degradation. This forces a compromise, li
miting the achievable compression rate and thus the potential speedup. We argue
that breaking this trade-off requires rethinking compression as a geometry-aware
, continuous token resampling in the vision encoder.
To this end, we propose the *Differentiable Grid Sampler (GridS)*, a plug-and-pl
ay module that performs task-aware, continuous resampling of visual tokens in VL
A. By adaptively predicting a minimal set of salient coordinates and extracting
features via differentiable interpolation, GridS preserves essential spatial inf
ormation while achieving drastic compression (with fewer than 10\% original visu
al tokens). Experiments on both LIBERO benchmark and a real robotic platform dem
onstrate that validating the lowest feasible visual token count reported to date
,
GridS achieves a 76\% reduction in FLOPs with no degradation in the success rate
. The code is available at https://github.com/Fediory/Grid-Sampler.
****

Self-Supervised Flow Matching for Scalable Multi-Modal Synthesis
Hila Chefer, Patrick Esser, Dominik Lorenz, Dustin Podell, Vikash Raja, Vinh Ton
g, Antonio Torralba, Robin Rombach
Strong semantic representations improve the convergence and generation quality o
f diffusion and flow models. Existing approaches largely rely on external models
, which require separate training, operate on misaligned objectives, and exhibit
 unexpected scaling behavior. We argue that this dependence arises from the mode
l's training objective, which poses a denoising task with little incentive to le
arn semantic representations. We introduce *Self-Flow*: a self-supervised flow m
atching paradigm that integrates representation learning within the generative f
ramework. Our key mechanism, *Dual-Timestep Scheduling*, applies heterogeneous n
oise levels across tokens, creating an information asymmetry that forces the mod
el to infer missing information from corrupted inputs. This drives learning stro
ng representations alongside generative capabilities without external supervisio
n. Our method generalizes across modalities and enables multi-modal training whi
le following expected scaling laws, achieving superior image, video, and audio g
eneration.
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GSRQ: Gain-Shape Residual Quantization for Sub-1-bit KV Cache
Soosung Kim, Minjae Park, Eui-Young Chung, Jaeyong Chung
The deployment of Large Language Models (LLMs) with extended context windows is
increasingly constrained by the linear growth of Key-Value (KV) cache memory.
Vector Quantization (VQ), particularly Residual Quantization (RQ), is a promisin
g approach for pushing KV cache storage toward the sub-1-bit regime by progressi
vely encoding residuals with small codebooks.
However, most VQ methods still rely on standard ℓ_2 K-means as the core c

odebook-learning primitive.
We identify a subtle high-dimensional issue of this primitive: Euclidean centroid averaging can induce centroid shrinkage, which weakens the angular alignment term in the ℓ_2 distortion and makes directional preservation harder.
To address this issue, we propose Gain-Shape K-means (GSKM), a drop-in replacement for K-means that improves directional fidelity while matching, and in some regimes improving, ℓ_2 distortion.
We then build Gain-Shape Residual Quantization (GSRQ) by incorporating a weighted extension of GSKM into an RQ pipeline.
On LLaMA-3-8B, GSRQ substantially improves over strong KV cache quantization baselines across bit rates.
At 1-bit, it improves the average accuracy across LongBench tasks from 11.34 to 33.54, a gain of 22.20 percentage points over VQLLM.
****

Multi-Level Strategic Classification: Incentivizing Improvement through Promotion and Relegation Dynamics
Ziyuan Huang, Lina Alkarmi, Mingyan Liu
Strategic classification studies the problem where self-interested individuals or agents manipulate their response to obtain favorable decision outcomes made by classifiers, typically turning to dishonest actions when they are less costly than genuine efforts. While existing studies on sequential strategic classification primarily focus on optimizing dynamic classifier weights, we depart from these weight-centric approaches by analyzing the design of classifier thresholds and difficulty progression within a multi-level promotion-relegation framework. Our model captures the critical inter-temporal incentives driven by an agent's farsightedness, skill retention, and a "leg-up" effect where qualification and attainment can be self-reinforcing. We characterize the agent’s optimal long-term strategy and demonstrate that a principal can design a sequence of thresholds to effectively incentivize honest effort. Crucially, we prove that under mild conditions, this mechanism enables agents to reach arbitrarily high levels solely through genuine improvement efforts.
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Efficient Reasoning with Hidden Thinking
Xuan Shen, Yizhou Wang, Yufa Zhou, Xiangxi Shi, Pu Zhao, Yanzhi Wang, Jiuxiang Gu
Chain-of-Thought (CoT) reasoning has become a powerful framework for improving complex problem-solving capabilities in Multimodal Large Language Models (MLLMs). However, the verbose nature of textual reasoning introduces significant inefficiencies. In this work, we propose**Heima** (as hidden llama), an effective CoT compression framework that condenses lengthy CoTs into a small set of abstract thinking tokens, preserving essential reasoning while removing redundancy. We then conduct a theoretical analysis from an information-theoretic perspective, quantifying the information gap induced by compression, showing that reasoning capability is preserved when non-trivial mutual information is retained. To further explore and quantify this information gap, we design the adaptive interpreter that maps thinking tokens back to variable-length textual sequences, thereby reconstructing the reasoning process. Experiments across diverse reasoning benchmarks demonstrate that Heima improves reasoning efficiency, while maintaining or even achieving better zero-shot accuracy. Moreover, the interpreter reconstructs coherent reasoning progresses from compressed thinking tokens, revealing that the information gap is minimal and validating the effectiveness of the proposed framework. This work paves the way for scalable latent reasoning models and advances our understanding of efficient reasoning processes in large models. Code: https://github.com/shawnricecake/Heima
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3ViewSense: Spatial and Mental Perspective Reasoning from Orthographic Views in Vision-Language Models
Shaoxiong Zhan, Yanlin Lai, Zheng Liu, Lin Hai, Shen Li, Xiaodong Cai, Zijian Lin, Wen Huang, Hai-Tao Zheng
Current Large Language Models have achieved Olympiad-level logic, yet Vision-Lan

guage Models paradoxically falter on elementary spatial tasks like block countin g. This capability mismatch reveals a critical "spatial intelligence gap," where models fail to construct coherent 3D mental representations from 2D observation s. We uncover this gap via diagnostic analyses showing the bottleneck is a missi ng view-consistent spatial interface rather than insufficient visual features or weak reasoning. To bridge this, we introduce **3ViewSense**, a framework that g rounds spatial reasoning in Orthographic Views. Drawing on engineering cognition , we propose a "Simulate-and-Reason" mechanism that decomposes complex scenes in to canonical orthographic projections to resolve geometric ambiguities. By align ing egocentric perceptions with these allocentric references, our method facilit ates explicit mental rotation and reconstruction. Empirical results on spatial r easoning benchmarks demonstrate that our method significantly outperforms existi ng baselines, with consistent gains on occlusion-heavy counting and view-consist ent spatial reasoning. The framework also improves the stability and consistency of spatial descriptions, offering a scalable path toward stronger spatial intel ligence in multimodal systems.

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Counterfactual Occlusion-Aware Learning via Visibility Intervention for LiDAR An omaly Detection

Longyu Yang, Jun Liu, Yap-Peng Tan, Fumin Shen, Heng Tao Shen, Xiaofeng Zhu, Pin g Hu

LiDAR point cloud anomaly detection is critical for autonomous system safety, ye t most existing methods rely only on visible measurements, overlooking occlusion as a structured consequence of the LiDAR sensing process. We argue that anomali es are characterized not only by what is observed, but also by the spatial voids they create, which alter occlusion patterns and volumetric visibility. We propo se Counterfactual Occlusion-Visibility Anomaly Learning (COVAL), a framework tha t intervenes on volumetric visibility during training. Using physics-conformed s ynthetic anomaly construction, COVAL generates paired factual and counterfactual observations with identical scene geometry but different occlusion patterns. Th en, we introduce two complementary objectives: Visibility-Variant Counterfactual Reconstruction, which models occlusion-induced missing regions, and Visibility-Invariant Counterfactual Consistency, which enforces stable representations acro ss visibility changes. Together, these objectives isolate anomaly-induced struct ural missingness and in turn refine representation of normal scenes, thus improv ing anomaly sensitivity at test time. Experiments on standard LiDAR anomaly segm entation benchmarks show that COVAL achieves state-of-the-art performance.

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Autobidding Auctions with LLM-Powered Creatives

Bingzhe Wang, Bowei Zhang, Changyuan Yu, Qi Qi

The integration of Large Language Models (LLMs) into ad auctions for dynamic cre ative enhancement presents a paradigm shift, yet introduces significant computat ional costs disrupting traditional mechanism design. This paper provides a compr ehensive game-theoretic and algorithmic framework for such LLM-augmented auction s. We model the system as a dynamic Stackelberg game where the platform (leader) strategically invests in creative enhancement to maximize net revenue, while au tobidding agents (followers) respond to enhanced ad qualities under budget const raints. To endogenize inference costs, we propose the Platform-Investment Mechan ism (PIM). We develop the Online Dual-Descent Bidding with Regularization (ODDB-R) algorithm for agents to learn optimal bidding strategies in this non-stationa ry environment. For the platform, we formulate the investment problem as a conti nuous control task and solve it using a Two-Timescale Stackelberg Learning with Proximal Policy Optimization (TTSL-PPO) algorithm, which provably converges to a Stackelberg Stationary Point. Extensive experiments on large-scale real-world d atasets and state-of-the-art LLMs demonstrate that our framework significantly o utperforms heuristic baselines in revenue, social welfare, and user engagement.

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Adaptive Multi-Round Allocation with Stochastic Arrivals

Yuqi Pan, Davin Choo, Haichuan Wang, Milind Tambe, Alastair van Heerden, Cheryl Johnson

We study a sequential resource allocation problem motivated by adaptive network recruitment, in which a limited budget of identical resources must be allocated over multiple rounds to individuals with stochastic referral capacity. Successful referrals endogenously generate future decision opportunities while allocating additional resources to an individual exhibits diminishing returns. We first show that the single-round allocation problem admits an exact greedy solution based on marginal survival probabilities. In the multi-round setting, the resulting Bellman recursion is intractable due to the stochastic, high-dimensional evolution of the frontier. To address this, we introduce a population-level surrogate value function that depends only on the remaining budget and frontier size. This surrogate enables an exact dynamic program via truncated probability generating functions, yielding a planning algorithm with polynomial complexity in the total budget. We further analyze robustness under model misspecification, proving a multi-round error bound that decomposes into a tight single-round frontier error and a population-level transition error. Finally, we evaluate our method on real-world inspired recruitment scenarios.
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What Language is This? Ask Your Tokenizer.
Clara Meister, Ahmetcan Yavuz, Pietro Lesci, Tiago Pimentel
Language Identification (LID) is an important component of many multilingual natural language processing pipelines, where it facilitates corpus curation, training data analysis, and cross-lingual evaluation of large language models. Despite near-perfect performance on high-resource languages, existing systems remain brittle in low-resource and closely related language settings. We introduce UniLID, a simple and efficient LID method based on the UnigramLM tokenization algorithm, leveraging its probabilistic framing, parameter estimation technique and inference strategy. In short, to predict a string's language label, we simply ask: under which language's unigram distribution is this string most likely? Our formulation is data- and compute-efficient, supports incremental addition of new languages without retraining existing models, and can naturally be integrated into existing language model tokenization pipelines. Empirical evaluations against widely used baselines, including fastText, GlotLID-M, and CLD3, show that UniLID achieves competitive performance on standard benchmarks, substantially improves sample efficiency in low-resource settings—surpassing 70% accuracy with as few as five labeled samples per language—and delivers large gains on fine-grained dialect identification.
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Minimizing Mismatch Risk: A Prototype-Based Routing Framework for Zero-shot LLM-generated Text Detection
Ke Sun, Guangsheng Bao, Han Cui, Yue Zhang
Zero-shot methods detect LLM-generated text by computing statistical signatures using a surrogate model. Existing approaches typically employ a fixed surrogate for all inputs regardless of the unknown source. We systematically examine this design and find that detection performance varies substantially depending on surrogate-source alignment. We observe that while no single surrogate achieves optimal performance universally, a well-matched surrogate typically exists within a diverse pool for any given input. This finding transforms robust detection into a routing problem: selecting the most appropriate surrogate for each input. We propose DetectRouter, a prototype-based framework that learns text-detector affinity through two-stage training. The first stage constructs discriminative prototypes from white-box models; the second generalizes to black-box sources by aligning geometric distances with observed detection scores. Experiments on EvoBench and MAGE benchmarks demonstrate consistent improvements across multiple detectio

n criteria and model families.
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Reasoning-preserved Efficient Distillation of Large Language Models via Activation-aware Initialization
Junlin He, Yihong Tang, Tong Nie, Guilong Li, Binyu Yang, Jinxiao Du, Lijun Sun, Wei Ma
Efficient Distillation (EDistill) compresses large language models (LLMs) by structured pruning parameters and tuning lightweight modules with high training efficiency. Although these EDistilled LLMs achieve state-of-the-art (SOTA) performance on general ability benchmarks relative to similarly sized LLMs, we identify a severe degradation in their multi-step reasoning ability, which we term reasoning collapse. We systematically analyze the geometric origins of reasoning collapse and show that the SOTA EDistill method based on width-reducing projection matrices suffers from eRank collapse, in which the effective rank (eRank) of hidden representations drops. We theoretically explain how singular values of randomly initialized projection matrices become unevenly distributed, leading to eRank collapse and thus token indistinguishability. To address this issue, we propose RED (Reasoning-preserved Efficient Distillation) for LLMs, which introduces activation-aware initialization to initialize projection matrices as channel-selection matrices, thus theoretically mitigating eRank collapse. Experiments on Llama and Qwen series demonstrate that RED substantially recovers reasoning while maintaining high training efficiency and SOTA general ability.
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How Few-Shot Examples Add Up: A Causal Decomposition of Function Vectors in In-Context Learning
Entang Wang, Yiwei Wang, Aleksandra Bakalova, Michael Hahn
In-context learning (ICL) excels at new tasks from minimal examples, yet we still lack a mechanistic explanation of how few-shot prompts shape a model’s function vector (FV)--a causal activation direction that drives task behavior on the ICL query. Across tasks and models, an n-shot FV is well-approximated by a linear combination of example-level sub-FVs, suggesting additive and composable contributions from individual demonstrations. Beyond additivity, we show that models contextualize individual examples' representations based on prior examples to adaptively reweight which demonstrations dominate the FV: attention shifts toward examples that are more informative and less ambiguous under the context. Finally, a causal decomposition separates Query–Key routing from Value updates, finding that contextualization’s most consistent contributions to FV quality arise from Query–Key alignment--particularly in ambiguous settings--while Value-mediated effects are more heterogeneous. Together, these results unify additive superposition with context-dependent attention reweighting into a mechanistic, testable account of how few-shot prompts implement tasks.
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Learning Coupled Continuous-Time Latent Dynamics from Irregular Events
Jiankai Zuo, Yang Zhang, Yu Zhang, Jiarui Liang, Yaying Zhang
Modeling dynamic dependencies from irregularly sampled event sequences is a fundamental challenge in modern machine learning. In many real-world systems, individual-level states evolve continuously over time while being simultaneously influenced by population-level dynamics. However, existing methods typically model these processes in isolation or rely on discrete-time approximations that fail to capture long-range temporal irregularities and sparse observations. This paper studies the problem of learning coupled continuous-time latent dynamics from irregular events, where individual event sequences and global distributional processes evolve asynchronously and interact over time. We propose a Coupled Continuous-Time Latent Dynamics (CoCLD) framework that jointly models individual latent dynamics and population-level distributional shifts, and aligns them in a continuous-time latent space. CoCLD integrates a Diffusion-based Latent Interpolator with neural ordinary differential equations, enabling principled interpolation, generation, and alignment of latent states across arbitrary time points. We show that the proposed coupling mechanism yields a consistent estimator of continuous-time latent dynamics under sparse and irregular observations. Empirical evaluatio

ns show CoCLD effectively captures dynamic dependencies and generalizes across t
asks like next-event prediction, mobility trajectory generation, and sequential
behavior modeling, indicating that learning coupled continuous-time latent dynam
ics is a powerful paradigm for irregular event sequence modeling.
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EvoClaw: Evaluating AI Agents on Continuous Software Evolution
Gangda Deng, Zhaoling Chen, Zhongming Yu, Haoyang Fan, Yuhong Liu, Yuxin Yang, D
hruv Parikh, Rajgopal Kannan, Le Cong, Mengdi Wang, Qian Zhang, Viktor Prasanna,
 Xiangru Tang, Xingyao Wang
With AI agents increasingly deployed as long-running systems, it becomes essenti
al to autonomously construct and continuously evolve customized software to enab
le interaction within dynamic environments. Yet, existing benchmarks evaluate ag
ents on isolated, one-off coding tasks, neglecting the temporal dependencies and
 technical debt inherent in real-world software evolution.
To bridge this gap, we introduce DeepCommit, an agentic pipeline that reconstruc
ts verifiable Milestone DAGs from noisy commit logs, where milestones are define
d as functionally cohesive development goals.
These executable sequences enable EvoClaw, a novel benchmark that requires agent
s to sustain system integrity and limit error accumulation, dimensions of long-t
erm software evolution largely missing from current benchmarks.
Our evaluation of 12 frontier models across 4 agent frameworks reveals a critica
l vulnerability: overall performance scores drop significantly from $>$80\% on i
solated tasks to at most 38\% in continuous settings, exposing agents' profound
struggle with long-term maintenance and error propagation.
Project website: evo-claw.com.
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TOM-SWE: User Mental Modeling For Software Engineering Agents
Xuhui Zhou, Valerie Chen, Zora Zhiruo Wang, Graham Neubig, Maarten Sap, Xingyao
Wang
Recent advances in coding agents have made them capable of planning, editing, ru
nning, and testing complex code bases. Despite their growing ability in coding t
asks, these systems still struggle to infer and track user intent, especially wh
en instructions are underspecified or context-dependent. To bridge this gap, we
introduce ToM-SWE, a dual-agent architecture that pairs a primary software-engin
eering (SWE) agent with a lightweight theory-of-mind (ToM) partner agent dedicat
ed to modeling the user's mental state. The ToM agent infers user goals, constra
ints, and preferences from instructions and interaction history, maintains a per
sistent memory of the user, and provides user-related suggestions to the SWE age
nt, while preserving privacy and minimizing context window load. In two software
 engineering benchmarks (ambiguous SWE-bench and stateful SWE-bench), ToM-SWE im
proves task success rates and user satisfaction. Notably, on the stateful SWE be
nchmark, a newly introduced evaluation that provides agents with a user simulato
r along with previous interaction histories, ToM-SWE achieves a substantially hi
gher task success rate of 59.7% compared to 18.1% for OpenHands, a state-of-the-
art SWE agent. Furthermore, in a three-week study with professional developers u
sing ToM-SWE in their daily work, participants found it better aligned with thei
r intent and useful 86% of the time, underscoring the value of stateful user mod
eling for practical coding agents.
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Don't Reinvent the Wheel, Just Realign the Spokes: Resource-Efficient Federated
Fine-Tuning via Rank-Wise Expert Assembly
Yebo Wu, Jingguang Li, Zhijiang Guo, Li Li
Federated fine-tuning presents a promising avenue for adapting Large Language Mo
dels (LLMs) to downstream tasks while preserving data privacy. However, the proh
ibitive computational and communication overhead of LLM adaptation inhibits its
deployment on resource-constrained edge devices. In this paper, we propose Smart
Fed, a resource-efficient framework that circumvents expensive training from scr
atch by intelligently reusing knowledge embedded in existing LoRA modules. To fu
lly exploit this potential and ensure scalability, we introduce the Mixture of R
ank-Wise Experts (MoRE). MoRE decomposes LoRA modules into fine-grained rank-lev

el experts, which are selectively activated based on input semantics and resourc e budgets. Furthermore, to optimize resource utilization, we propose Elastic Exp ert Quota Allocation (EEQA), a strategy that adaptively distributes expert capac ity across parameter matrices based on their contribution to model performance. Extensive evaluations across multiple benchmarks demonstrate that SmartFed signi ficantly outperforms state-of-the-art methods in both model performance and trai ning efficiency. Our code is publicly available at https://github.com/benmagnifi co/SmartFed.

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Sinkhorn Treatment Effects: A Causal Optimal Transport Measure

Medha Agarwal, Alex Luedtke

We introduce the Sinkhorn treatment effect, an entropic optimal transport measur e of divergence between counterfactual outcome distributions. Unlike classical q uantities such as the average treatment effect, it captures differences across e ntire distributions. We show that this estimand can be written as a smooth trans formation of counterfactual mean embeddings with an appropriate kernel. This cha racterization allows us to establish first-order pathwise differentiability in g eneral, and second-order pathwise differentiability under the null hypothesis of equal counterfactual distributions. Leveraging this smoothness, we construct de biased estimators and asymptotically valid tests for distributional treatment ef fects at a fixed entropic regularization parameter. Because the power of the tes t depends on this unknown parameter, we propose an aggregated test that combines evidence across a grid of regularization choices. Experiments on simulated and image data demonstrate the practical advantages of our estimator and testing pro cedure.

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AdaEraser: Training-Free Object Removal via Adaptive Attention Suppression

Dingming Liu

Object removal aims to eliminate specified objects from images while plausibly i npainting the affected regions with background content. Current training-free me thods typically block attention to object regions within self-attention layers d uring the image generation process, leveraging surrounding background informatio n to restore the image. However, indiscriminate suppression of self-attention in the vacated areas can degrade generation quality, as the model must simultaneou sly reconstruct background content in these regions. To solve this conflict, we propose AdaEraser, an adaptive framework that dynamically modulates attention ba sed on the estimated presence of target object concepts. Through analysis of sel f-attention map evolution across denoising timesteps before and during removal, we develop a token-wise adaptive attention suppression strategy. This approach e nables progressive perception of object removal throughout the denoising process , with the suppression strength in self-attention layers adjusted adaptively. Ex tensive experiments demonstrate that AdaEraser achieves superior performance in object removal, outperforming even training-based methods.

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Sparse Autoencoders for Interpretable Emotion Control in Text-to-Speech

Hongfei Du, Jiacheng Shi, Sidi Lu, Gang Zhou, Ashley Gao

Integrating large language models (LLMs) into text-to-speech (TTS) systems has i mproved speech expressiveness, yet interpretable emotional control remains chall enging. Existing approaches primarily rely on external conditioning or global ac tivation steering, offering limited insight into the internal representations un derlying emotional control. In this work, we analyze emotion-related variation i n the semantic hidden states of LLM-based TTS models using sparse autoencoders (SAEs) to identify sparse latent features. Our analysis shows that emotional vari ation is distributed across multiple sparse latent features, while intervening o n a small subset enables interpretable emotion control. Building on this observa tion, we introduce a feature-level intervention framework for bidirectional emot ion induction and suppression without modifying backbone parameters. We further show that distinct latent features are associated with specific acoustic attribu tes (e.g., pitch), suggesting that emotional expression arises from coordinated latent contributions rather than a single global shift. Empirically, steering th

ese sparse latent features achieves comparable or superior emotion induction and suppression performance relative to global steering and existing TTS baselines.
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S-Quant: Rethinking Weight Quantization with Seed-Based Generation
Mingzi Wang, Lancheng Zou, Shuo Yin, Zhuolun He, Bei Yu
The progressive scaling of large language models (LLMs) has consistently enhanced multimodal understanding and advanced reasoning capabilities, but has substantially increased computational and hardware execution overhead.
In this paper, we present S-Quant, a novel post-method that compresses only model weights.
We partition each weight tensor into fixed-size blocks and assign a single seed to each block.
The seed drives a hardware-friendly Linear Feedback Shift Register (LFSR) generator that dynamically produces multiple basis matrices.
Each block is then reconstructed as a linear combination of these basis matrices, with block-specific coefficients, which substantially reduces the amount of stored data, increases the data-transfer efficiency between memory and compute units, and consequently speeds up memory-bound inference for large language models.
Experimental results on different LLM models ranging from 7B–70B parameters show that S-Quant attains state-of-the-art performance when weights are compressed to approximately 3-bit or 4-bit.
We also design a dedicated ASIC accelerator that achieves a 4× speed-up for memory-bound LLM inference.
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BIT-LLM: Brain Instruction Tuned LLM with persistent Cross-Attention for fMRI-to-Text Decoding
Sunghwan Lee, Jihun Kim, Chae Lynn Kim, Ji Yun Park, Jong-Hwan Lee
Decoding fMRI into natural language is challenging because strong, pre-trained language priors can dominate autoregressive generation, obscuring whether a model truly utilizes neural evidence. We introduce BIT-LLM, which exposes fMRI-derived tokens as a persistent key–value memory through interleaved cross-attention adapters, enabling repeated neural access throughout decoding. BIT-LLM is trained with a three-stage pipeline: (i) multimodal contrastive learning to obtain semantically aligned fMRI representations, (ii) supervised fine-tuning to learn the brain-LLM interface while freezing the encoder and backbone LLM, and (iii) reward-based finetuning to optimize sequence-level caption quality directly. On the NSD subject-heldout benchmark (S1-7 train, S8 test), BIT-LLM yields substantially improved captioning quality over prior baselines under greedy decoding. In addition to standard captioning metrics, we perform several complementary evaluations to assess the robustness of brain–language grounding. Specifically, we conduct perturbation-based sanity checks by zeroing fMRI inputs or shuffling voxel values, and examine whether internal representations and generated outputs change accordingly. BIT-LLM exhibits clear sensitivity to these perturbations, indicating effective utilization of voxel values and their spatial correspondence.
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RECOVER: Reliable Detection of Unauthorized Data Usage in Text-to-Image Diffusion Models via Inversion Robustness
Yanhao Wei, Xiaokang Zhao, Boheng Li, Yang Zhang, Run Wang
Text-to-Image diffusion models have achieved remarkable success in image generation and are increasingly fine-tuned for personalized use cases. However, many personalized models may incorporate unauthorized data during the fine-tuning process, raising growing concerns about potential copyright infringements. Existing methods either require intrusive modifications to the images to be protected, which not only fail to safeguard previously released images but may also degrade image quality, or rely on the availability of the pre-fine-tuned model, thereby limiting their applicability. To bridge this gap, in this paper, we propose the first non-intrusive copyright authentication framework without pre-fine-tuned model. We reveal that if a model is fine-tuned on a specific image, it learns the denoising trajectory of that image across varying noise levels, allowing it to stably reconstruct the image even under noise perturbations. Motivated by this ins

ight, we propose Reliable dEteCtion Of unauthorized data usage via inVErsion Robustness (RECOVER), an effective non-intrusive detection method without pre-fine-tuned model. Unlike existing methods that rely on external watermarks or discrepancies between the suspect and pre-fine-tuned models, RECOVER directly leverages the robustness observed during the inversion–reconstruction process of the suspect model to determine whether an image was used for fine-tuning. Extensive experiments demonstrate that RECOVER is effective across a wide range of scenarios, consistently outperforming existing methods. Our code is publicly available here.

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Low-cost Full Fine-tuning: Learning What to Update for LLMs

Jin Li, Yaming Guo, Shenghao Gao, Xinlong Chen, Zuhao Xu, Ying Sun, Chao Wang, Hui Xiong

While Large language models (LLMs) have strong abilities, they generally rely on fine-tuning to supplement downstream task-specific knowledge. Due to the prohibitive memory overhead of full fine-tuning (FT), existing parameter-efficient fine-tuning techniques, e.g., LoRA and Adapters, update parameters only in low-rank or restricted subspaces. However, they fail to approximate FT---the performative fine-tuner---and risk performance degradation in tough tasks. Therefore, we naturally raise a *Low-cost Full Fine-tuning* question: Can we approach standard full fine-tuning in theory, yet with much lower costs in practice? Our key insight is that performing selective updates at each step can, theoretically, recover FT asymptotically, while being cost-effective and ignoring no parameter direction. This motivates a new general fine-tuning paradigm (called *Think-Touch*): we first predict potentials of parameter groups (*think*) and then update only the selected (*touch*) in one step. Theoretically, we show that under a very weak sufficient condition---divergence of the cumulative coverage of the expected gradient norm---any selection strategy can converge in the full-parameter space to a stationary point at which the FT admits no further first-order improvement. Besides, we further derive the general convergence rate for our paradigm and identify a post-hoc greedy strategy that is rate-optimal. Unfortunately, this strategy cannot be directly applied in practice due to its reliance on full and accurate gradient information. Thus, we propose a bandit-based method to online approximate this ideal strategy in the long run with a rigorous regret guarantee. Extensive experimental results on various tasks demonstrate the potential of our paradigm, including much lower space overheads against FT and better performance than LoRAs.

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CAPTCHA Solving for Native GUI Agents: Automated Reasoning-Action Data Generation and Self-Corrective Training

Yuxi Chen, Haoyu Zhai, Chenkai Wang, Rui Yang, LINGMING ZHANG, Gang Wang, Huan Zhang

GUI agents are rapidly shifting from multi-module pipelines to end-to-end, native vision-language models (VLMs) that perceive raw screenshots and directly interact with digital devices. Despite rapid progress on general GUI tasks, CAPTCHA solving remains a major challenge. On the other hand, although specialized CAPTCHA solving pipelines exist, they cannot handle general GUI tasks.

To address this gap, we introduce ReCAP: a CAPTCHA-capable native GUI agent that solves modern, interactive CAPTCHA challenges while retaining general GUI-agent performance. We first develop a dynamic CAPTCHA system spanning seven representative CAPTCHA types, designed to stress primitive and complementary capabilities for CAPTCHA solving. Then, we develop an automated data collection and curation pipeline that generates large-scale CAPTCHA interaction trajectories paired with reasoning traces. As CAPTCHA solving often requires multi-step interaction and recovery from intermediate mistakes, we further leverage failed trajectories to construct self-correction data, training agents to reflect on errors and correct their actions online. Across synthetic and real-world test sets, ReCAP substantially improves CAPTCHA-solving success over its base agents, while maintaining strong performance on general GUI-agent benchmarks.

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Ask Less, See More: Communication-Conditioned Token Pruning for Vehicle-to-Vehicle Cooperative Autonomous Driving with Multimodal Large Language Models
Shiqi Sun, Yantao Lu, Bingkun Sun, Ning Liu, Bo Jiang, Ying Zhang, Jinchao Chen, Chenglie Du
Multimodal Large Language Models (MLLMs) offer a promising paradigm for vehicle-to-vehicle (V2V) cooperative autonomous driving, enabling language-based decision-making in safety-critical occluded scenarios. However, existing V2V–MLLM frameworks rely on dense token-level sharing and fusion, incurring high communication and inference costs. Moreover, conventional V2V perception methods are limited to feature-sharing paradigms without language reasoning, and existing token pruning strategies fail to consider LiDAR-specific spatial structure and multi-agent fusion. To address these limitations, we propose V2V Communication-Conditioned MLLM Framework (V2V-CCM), a dual-stage cooperative communication framework that broadcasts request messages to all agents and uses them to identify redundant visual tokens. Specifically, Question Semantic Message (QSM) encodes global question intent for question-relevant token selection, while Spatial Coverage Message (SCM) summarizes LiDAR features to identify spatially redundant tokens already observed by other agents. Integrated into dual-stage frameworks, V2V-CCM substantially reduces communication and inference costs while preserving question-relevant tokens and removing spatial redundancy. Extensive experiments on V2V-QA and V2V-GoT-QA demonstrate that V2V-CCM consistently outperforms existing pruning methods and achieves state-of-the-art performance.
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A General Framework for Dynamic Consistent Submodular Maximization
Paul Duetting, Federico Fusco, Silvio Lattanzi, Ashkan Norouzi-Fard, Ola Svensson, Morteza Zadimoghaddam
Consistency is an important property in dynamic submodular maximization and entails maintaining a near-optimal solution at all times, making only a small number of adjustments to the solution in each step. Prior work has explored this question for the insertion-only case, where the algorithm faces a stream of n insertions, and has established lower and upper bounds for the cardinality-constrained version of the problem. We consider this question in the fully dynamic setting, where the stream of operations may contain both insertions and deletions.

We develop a general framework for designing algorithms for this setting, and instantiate it to obtain the first constant-factor approximations with sublinear consistency. For cardinality constraints, we propose a $\tfrac 12 - O(\varepsilon)$ approximation that is $O\left(\tfrac{1}{\varepsilon^2}\right)$ consistent. For rank-k matroid constraints, we construct a $\tfrac 14 - O(\varepsilon)$ approximation to the dynamic optimum that is $O\left(\tfrac{\log k}{\varepsilon^2}\right)$ consistent.
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ATLAS: Learning to Optimally Memorize the Context at Test Time
Ali Behrouz, Zeman Li, Praneeth Kacham, Majid Daliri, Yuan Deng, Peilin Zhong, Meisam Razaviyayn, Vahab Mirrokni
Transformers have been established as the most popular backbones in sequence modeling, mainly due to their effectiveness in in-context retrieval tasks and the ability to learn at scale. Their quadratic memory and time complexity, however, bound their applicability in longer sequences and so has motivated researchers to explore effective alternative architectures such as modern recurrent neural networks (a.k.a long-term recurrent memory module). Despite their recent success in diverse downstream tasks, they struggle in tasks that requires long context understanding and extrapolation to longer sequences. We observe that these shortcomings come from three disjoint aspects in their design: (1) limited memory capacity that is bounded by the architecture of memory and feature mapping of the input; (2) online nature of update, i.e., optimizing the memory only with respect to the last input; and (3) less expressive management of their fixed-size memory. To enhance all these three aspects, we present Atlas, a long-term memory module with high capacity that learns to memorize the context by optimizing the memory based on the current and past tokens, overcoming the online nature of long-term

memory models. Our experimental results on language modeling, common-sense reasoning, recall-intensive, and long-context understanding tasks support the effectiveness of Atlas compared to other modern recurrent neural networks.
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Plain Transformers are Surprisingly Powerful Link Predictors
Quang Truong, Yu Song, Donald Loveland, Mingxuan Ju, Tong Zhao, Neil Shah, Jiliang Tang
Link prediction is a core challenge in graph machine learning, demanding models that capture rich and complex topological dependencies. While Graph Neural Networks (GNNs) are the standard solution, state-of-the-art pipelines often rely on explicit structural heuristics or memory-intensive node embeddings—approaches that struggle to generalize or scale to massive graphs. Emerging Graph Transformers (GTs) offer a potential alternative but often incur significant overhead due to complex structural encodings, hindering their applications to large-scale link prediction. We challenge these sophisticated paradigms with PENCIL, an encoder-only plain Transformer that replaces hand-crafted priors with attention over sampled local subgraphs, retaining the scalability and hardware efficiency of standard Transformers. Through experimental and theoretical analysis, we show that PENCIL extracts richer structural signals than GNNs, implicitly generalizing a broad class of heuristics and subgraph-based expressivity. Empirically, PENCIL outperforms heuristic-informed GNNs and is far more parameter-efficient than ID-embedding–based alternatives, while remaining competitive across diverse benchmarks—even without node features. Our results challenge the prevailing reliance on complex engineering techniques, demonstrating that simple design choices are potentially sufficient to achieve the same capabilities. Our code is publicly available at https://github.com/quang-truong/pencil.
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Benchmarking Physics-Informed Time-Series Models for Operational Global Station Weather Forecasting
Tao Han, Zhibin Wen, Zhenghao Chen, Dazhao Du, Song Guo, LEI BAI
The development of Time-Series Forecasting (TSF) models is often constrained by the lack of comprehensive datasets, especially in Global Station Weather Forecasting (GSWF), where existing datasets are small, temporally short, and spatially sparse. To address this, we introduce WEATHER-5K, a large-scale observational weather dataset that better reflects real-world conditions and supports improved model training and evaluation. While recent TSF methods perform well on benchmarks, they still lag behind operational Numerical Weather Prediction (NWP) systems in capturing complex weather dynamics and extreme events. We propose PhysicsFormer, a physics-informed forecasting model that combines a dynamic core with a Transformer residual to predict future weather states. Physical consistency is enforced via pressure–wind alignment and energy-aware smoothness losses, ensuring plausible dynamics while capturing complex temporal patterns. We benchmark PhysicsFormer and other TSF models against operational systems across several weather variables, extreme event prediction, and model complexity, providing a comprehensive assessment of the gap between academic TSF models and operational forecasting. The dataset and benchmark implementation are available at: https://github.com/taohan10200/WEATHER-5K.
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Geometry-Aware Decoding with Wasserstein-Regularized Truncation and Mass Penalties for Large Language Models
Arash Gholami Davoodi, Navid Rezazadeh, Seyed Pouyan Mousavi Davoudi, Pouya Pezeshkpour
Large language models (LLMs) must balance diversity and creativity against logical coherence in open-ended generation.
Existing truncation-based samplers are effective but largely heuristic, relying mainly on probability mass and entropy while ignoring semantic geometry of the token space.
We present Top-W, a geometry-aware truncation rule that uses Wasserstein distance—defined over token-embedding geometry—to keep the cropped distribution close to the original, while explicitly balancing retained probability mass against t

he entropy of the kept set.
Our theory yields a simple closed-form structure for the fixed-potential subset update: depending on the mass–entropy trade-off, the optimal crop either collapses to a single token or takes the form of a one-dimensional prefix that can be found efficiently with a linear scan.
We implement Top-W using efficient geometry-based potentials (nearest-set or k-NN) and pair it with an alternating decoding routine that keeps the standard truncation-and-sampling interface unchanged.
Extensive experiments on four benchmarks (GSM8K, GPQA, AlpacaEval, and MT-Bench) across three instruction-tuned models show that Top-W consistently outperforms prior state-of-the-art decoding approaches achieving up to 33.7 percentage improvement.
Moreover, we find that Top-W not only improves accuracy-focused performance, but also boosts creativity under judge-based open-ended evaluation.
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FUSE: Full■spectrum Unlearnable Examples via Spectral Equalization
Jiale Cai, Gezheng Xu, Zhihao Li, Ruiyi Fang, Ruizhi Pu, Di Wu, Qicheng Lao, Charles Ling, Boyu Wang
Unlearnable examples (UEs) protect training data by injecting imperceptible perturbations so that models fail to extract exploitable representations. In this paper, we reveal that existing UEs exhibit a critical failure once low-pass filtering is applied, indicating that the effective perturbation signals for unlearnability concentrate predominantly in high frequencies. Hence, we argue that reliable UEs should remain effective across the full spectrum. To this end, we propose **F**ull-spectrum **U**nlearnable Examples via **S**pectral **E**qualization (**FUSE**), which aims to generate spectrum-agnostic perturbations by equalizing the contributions from different bands and enforcing cross-band consistency. Specifically, FUSE adopts a Random Spectral Masking (RSM) strategy during generator training, which randomly removes a contiguous frequency band, forcing the remaining bands to maintain unlearnability. In addition, FUSE further integrates Cross-Band Guidance (CBG), which enforces mutual consistency between high- and low-frequency components, thereby further enhancing low-frequency unlearnability and regulating high-frequency perturbations to preserve the semantic fidelity of images. Extensive experiments across multiple datasets, architectures, and spectral filtering demonstrate the strong protection achieved by FUSE.
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Near-Optimal Dynamic Matching via Coarsening with Application to Heart Transplantation
Itai Zilberstein, Ioannis Anagnostides, Zachary W. Sollie, Arman Kilic, Tuomas Sandholm
Online matching has been a mainstay in domains such as Internet advertising and organ allocation, but practical algorithms often lack strong theoretical guarantees. We take an important step toward addressing this by developing new online matching algorithms based on a coarsening approach. Although coarsening typically implies a loss of granularity, we show that, to the contrary, aggregating offline nodes into capacitated clusters can yield near-optimal theoretical guarantees. We apply our methodology to heart transplant allocation to develop theoretically grounded policies based on structural properties of historical data. Furthermore, in simulations based on real data, our policy closely matches the performance of the omniscient benchmark, achieving competitive ratio 0.91, drastically higher than the US status quo policy's 0.51. Our work bridges the gap between data-driven heuristics and pessimistic theoretical lower bounds.
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Quantum latent distributions in deep generative models
Omar Bacarreza, Thorin Farnsworth, Alexander Makarovskiy, Hugo Wallner, Tessa Hicks, Santiago Sempere-Llagostera, John J Price, Robert J. A. Francis-Jones, William R. Clements
Many successful families of generative models leverage a low-dimensional latent distribution that is mapped to a data distribution. Though simple latent distributions are often used, the choice of distribution has a strong impact on model p

erformance. Recent experiments have suggested that the probability distributions produced by quantum processors, which are typically highly correlated and classically intractable, can lead to improved performance on some datasets. However, when and why latent distributions produced by quantum processors can improve performance, and whether these improvements are connected to quantum properties of these distributions, are open questions that we investigate in this work. We show in theory that, under certain conditions, these "quantum latent distributions" enable generative models to produce data distributions that classical latent distributions cannot efficiently produce. We provide intuition as to the underlying mechanisms that could explain a performance advantage on real datasets. Based on this, we perform extensive benchmarking on a synthetic quantum dataset and the QM9 molecular dataset, using both simulated and real photonic quantum processors. We find that the statistics arising from quantum interference lead to improved generative performance compared to classical baselines, suggesting that quantum processors can play a role in expanding the capabilities of deep generative models.

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Spike Camera Autofocus via Frequency-Domain Spectral-Centroid Migration

Xijie Xiang, Lin Zhu, Yonghong Tian

Autofocus for spike cameras is challenging because their sparse binary measurements do not provide reliable instantaneous gradients, and noise or illumination drift often breaks the unimodal assumptions behind conventional focus measures. We show that during a focus sweep, the stable sensor-observable cue is a persistent migration of spectral energy in the frequency domain: energy shifts outward toward higher frequencies when approaching focus and recedes under renewed defocus. Building on this observation, we propose CEN (Centroid-based Energy Navigation), a frequency-domain autofocus method that measures spectral migration via a bounded spectral centroid computed on accumulated spike blocks, without image reconstruction or explicit edge extraction. To handle multi-peak and irregular responses in real scenes, CEN further performs structure-consistent response identification, selecting the frequency bound whose curve exhibits a clear, localized, interior extremum, followed by robust peak localization using a weighted near-maximum centroid.

Experiments on spike-camera dataset demonstrate that CEN achieves the best overall accuracy and response discriminability across diverse scenes, motion types, and illumination variation patterns.

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Beyond First-order Asymptotics in Sequential Mean Testing

VIKAS DEEP, Shubhada Agrawal

We revisit the problem of sequentially testing the mean of bounded distributions in a level-α power-one framework. We study a $\mathrm{KL_{inf}}$-based sequential test that is known to attain the information-theoretic lower bound on the expected stopping time with exact constants as $\alpha \to 0$. Going beyond first-order asymptotics, we establish a central limit theorem (CLT) for the stopping time of this test. Our analysis proceeds in two steps. First, we prove a novel CLT for the $\mathrm{KL_{inf}}$ statistic itself, characterizing its fluctuations around its deterministic limit. We then leverage this result to show that the stopping time, centered appropriately and scaled by $\sqrt{\log(1/\alpha)}$, converges in distribution to a Gaussian limit with an explicit variance. This yields a second-order characterization of an asymptotically optimal sequential test for bounded distributions. Finally, we present numerical experiments that corroborate our theoretical findings.

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Dimension-free convergence of diffusion models for approximate Gaussian mixtures

Gen Li, Changxiao Cai, Yuting Wei

Diffusion models are distinguished by their exceptional generative performance, particularly in producing high-quality samples through iterative denoising. While current theory suggests that the number of denoising steps required for accurate sample generation should scale linearly with data dimension, this does not reflect the practical efficiency of widely used algorithms like Denoising Diffusio

n Probabilistic Models (DDPMs). This paper investigates the effectiveness of dif fusion models in sampling complex high-dimensional distributions that can be wel l-approximated by Gaussian Mixture Models (GMMs). For these distributions, our m ain result shows that DDPM takes at most $\widetilde{O}(1/\varepsilon)$ iteratio ns to attain an ε-accurate distribution in total variation (TV) dist ance, independent of both the ambient dimension d and the number of components K, up to logarithmic factors. Furthermore, this result remains robust to scor e estimation errors. These findings highlight the remarkable effectiveness of di ffusion models in high-dimensional settings given the universal approximation ca pability of GMMs, and provide theoretical insights into their practical success.
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On the Role of Computation in Reinforcement Learning

Raj Ghugare, Micha■ Bortkiewicz, Alicja Ziarko, Benjamin Eysenbach

How does the amount of compute available to a reinforcement learning (RL) policy affect its learning? Can policies using a fixed amount of parameters, still ben efit from additional compute? The standard RL framework does not provide a langu age to answer these questions formally. Empirically, deep RL policies are often parameterized as neural networks with static architectures, conflating the amoun t of compute and the number of parameters. In this paper, we formalize compute b ounded policies and prove that policies which use more compute can solve problem s and generalize to longer-horizon tasks that are outside the scope of policies with less compute. Building on prior work in algorithmic learning and model-free planning, we propose a minimal architecture that can use a variable amount of c ompute. Our experiments complement our theory. On a set 31 different tasks spann ing online and offline RL, we show that (1) this architecture achieves stronge r performance simply by using more compute, and (2) stronger generalization on longer-horizon test tasks compared to standard feedforward networks or deep res idual network using upto 5 times more parameters.
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Adaptive Token Refinement in Long-Tailed Large Vision-Language Models Fine-Tunin g

Wenjun Miao, Mingda Li, Yanchao Hao, Zheng Wei

While large vision-language models (LVLMs) have shown remarkable adaptability to downstream applications, their fine-tuning process remains susceptible to bias under long-tailed data.

Compared to zero-shot scenarios, fine-tuning LVLMs on imbalanced datasets often yields limited performance improvements on tail data.

This is because LVLMs tend to rapidly overfit the head data at an early fine-tun ing stage, thereby impairing the learning of the tail data while simultaneously failing to exploit their quantitative advantage.

Furthermore, in many downstream LVLM scenarios, quantified long-tailed prior kno wledge of data distribution is often unavailable, significantly limiting the app licability of traditional long-tailed techniques that rely heavily on such infor mation.

To address these issues, we propose the Adaptive Token Refinement (ATR), a novel framework that adaptively refines the learning process of LVLMs under long-tail ed data.

Specifically, ATR consists of two token-level operations applied to output and i nput tokens, respectively: 1) a bounded adaptive loss that dynamically filters a nd reweights output tokens to mitigate overfitting on head data,

and 2) a visual token mask strategy that augments the probability paths of input tokens to enhance long-tailed performance.

Extensive experiments demonstrate that ATR consistently enhance both performance and generalization for long-tailed LVLMs fine-tuning.
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Posterior Behavioral Cloning: Pretraining BC Policies for Efficient RL Finetunin g

Andrew Wagenmaker, Perry Dong, Raymond Tsao, Chelsea Finn, Sergey Levine

Standard practice across domains from robotics to language is to first pretrain a policy on a large-scale demonstration dataset, and then finetune this policy,

typically with reinforcement learning (RL), in order to improve performance on deployment domains. This finetuning step has proved critical in achieving human or super-human performance, yet while much attention has been given to developing more effective finetuning algorithms, little attention has been given to ensuring the pretrained policy is an effective initialization for RL finetuning. In this work we seek to understand how the pretrained policy affects finetuning performance, and how to pretrain policies in order to ensure they are effective initializations for finetuning. We first show theoretically that standard behavioral cloning (BC) can fail to ensure coverage over the demonstrator's actions, a minimal condition necessary for effective RL finetuning. We then show that if, instead of exactly fitting the observed demonstrations, we train a policy to model the posterior distribution of the demonstrator's behavior given the demonstration dataset, we do obtain a policy that ensures coverage over the demonstrator's actions, enabling more effective finetuning. Furthermore, this policy achieves this while ensuring pretrained performance is no worse than that of the BC policy. We then show this approach is practically implementable with modern generative models and leads to significantly improved RL finetuning performance on both realistic robotic control benchmarks and real-world robotic manipulation tasks, as compared to standard behavioral cloning.

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Code2Video: A Code-centric Paradigm for Educational Video Creation

Yanzhe Chen, Kevin Qinghong Lin, Mike Zheng Shou

While recent generative models can synthesize videos in pixel space, they often fail to produce educational videos with precise structures, domain knowledge, and coherent transitions. We argue that this setting is better served by operating in a renderable environment that is explicitly controlled by code. We propose **Code2Video**, a code-centric agent framework that generates educational videos by writing executable Python programs. Code2Video includes three agents: a *Planner* that converts lecture content into a temporal storyboard, a *Coder* that turns the storyboard into runnable code with scope-guided auto-fix, and a *Critic* that refines layout using a VLM guided by *visual anchor prompting*, *i.e.*, mappings from target visual outcomes to code edits. For evaluation, we build **MMMC**, a benchmark of professionally produced, discipline-specific educational videos. We assess Code2Video using aesthetic scores (VLM-as-a-Judge), code efficiency, and **TeachQuiz**, an end-to-end metric that measures how well an *unlearned* VLM can recover knowledge after watching generated videos. Code2Video improves performance by 40% over direct code generation and produces videos comparable to human-crafted tutorials. The code and datasets are available at https://github.com/showlab/Code2Video.

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Zero-Shot Off-Policy Learning

Arip Asadulaev, Maksim Bobrin, Salem Lahlou, Dmitry V. Dylov, Fakhri Karray, Martin Taká■

Off-policy learning methods seek to derive an optimal policy directly from a fixed dataset of prior interactions. This objective presents significant challenges, primarily due to the inherent distributional shift and value function overestimation bias.

These issues become even more noticeable in zero-shot reinforcement learning, where an agent trained on reward-free data must adapt to new tasks at test time without additional training.

In this work, we address the off-policy problem in a zero-shot setting by discovering a theoretical connection of successor measures to stationary density ratios.

Using this insight, our algorithm can infer optimal importance sampling ratios, effectively performing a stationary distribution correction with an optimal policy for any task on the fly.

We benchmark our method in motion tracking tasks on SMPL Humanoid, continuous control on ExoRL, and for the long-horizon OGBench tasks.

Our technique seamlessly integrates into forward-backward representation frameworks and enables fast-adaptation to new tasks in a training-free regime.

More broadly, this work bridges off-policy learning and zero-shot adaptation, of
fering benefits to both research areas.
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WorldMirror: Universal 3D World Reconstruction with Any-Prior Prompting
Yifan Liu, Zhiyuan Min, Zhenwei Wang, Junta Wu, Tengfei Wang, Yixuan Yuan, Yawei
 Luo, Chunchao Guo
We present WorldMirror, a unified feed-forward model for comprehensive 3D geomet
ric prediction tasks.
Unlike existing methods constrained to image-only inputs or customized for a spe
cific task, our framework flexibly integrates diverse geometric priors, includin
g camera poses, intrinsics, and depth maps, while simultaneously generating mult
iple 3D representations: dense point clouds, multi-view depth maps, camera param
eters, surface normals, and 3D Gaussians. Remarkably, prior injection yields uni
versal gains across all tasks, suggesting that input flexibility and multi-task
prediction are mutually reinforcing. WorldMirror achieves state-of-the-art perfo
rmance across diverse benchmarks from camera, point map, depth, and surface norm
al estimation to novel view synthesis, while maintaining the efficiency of feed-
forward inference. Code and model weights are publicly available at https://gith
ub.com/Tencent-Hunyuan/HunyuanWorld-Mirror.
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MARS-SQL: A Multi-Agent Reinforcement Learning Framework For Text-To-SQL
Haolin Yang, Jipeng Zhang, Zhitao He, Alexander Zhou, Yi R. Fung
Large Language Models (LLMs) often struggle with the precise logic and schema al
ignment required for complex Text-to-SQL tasks. While current methods rely heavi
ly on static prompting, they lack the ability to dynamically adapt and self-corr
ect through environmental interaction. To bridge this gap, we propose **MARS-SQL
**, a trainable multi-agent framework for Text-to-SQL. Rather than introducing a
 new standalone SQL primitive, **MARS-SQL** makes an agentic workflow trainable
by decomposing the problem into three specialized roles: schema grounding, query
 generation, and solution validation. Central to our approach is a generation ag
ent trained via a multi-turn RL policy within a ReAct-style loop. The agent lear
ns to iteratively reason, execute intermediate SQL actions on a live database, a
nd refine its strategy based on execution feedback. To improve robustness, we fu
rther introduce a validation mechanism that treats solution selection as a gener
ative modeling task, identifying the optimal interaction trajectory through next
-token prediction probabilities. Empirical evaluations demonstrate the effective
ness of coupling interactive learning with trajectory ranking. **MARS-SQL** achi
eves state-of-the-art performance, recording an execution accuracy of 77.84\% on
 the BIRD development dataset and 89.75\% on the Spider test dataset, while also
 transferring strongly to out-of-domain benchmarks.
Code is available at https://github.com/YangHaolin0526/MARS-SQL.
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PepCompass: Navigating Peptide Embedding Spaces Using Riemannian Geometry
Marcin Mo■ejko, Adam Bielecki, Jurand Pr■dzy■ski, Hyun-Su Lee, Antoni Janowski,
Michal Kmicikiewicz, Paulina Szymczak, Karol Jurasz, Marcin Traskowski, Micha■ K
ucharczyk, Marcelo Der Torossian Torres, Cesar de la Fuente-Nunez, Ewa Szczurek
Antimicrobial peptide discovery is challenged by the astronomical size of peptid
e space and the relative scarcity of active peptides.
While generative models provide latent maps of this space, they typically ignore
 decoder-induced geometry and rely on flat Euclidean metrics, making exploration
 distorted and inefficient. Existing manifold-based approaches assume fixed int
rinsic dimensionality, which fails for real peptide data. We introduce **PepComp
ass**, a geometry-aware framework based on a **Union of κ-Stable Riemann
ian Manifolds** that captures local decoder geometry while maintaining computati
onal stability. PepCompass performs global interpolation via **Potential-minimiz
ing Geodesic Search (PoGS)** to bias discovery toward promising seeds and enable
s local exploration through **Second-Order Riemannian Brownian Efficient Samplin
g** and **Mutation Enumeration in Tangent Space**, which together form **Local E
numeration Bayesian Optimization (LE-BO)**. PepCompass achieves a 100% *in-vitro
* validation rate: PoGS identifies four novel seeds and LE-BO optimizes them in

to 25 highly active, broad-spectrum peptides, demonstrating that geometry-informed exploration is a powerful paradigm for antimicrobial peptide design.
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Conformal C2ST: Turning weak classifiers into strong two-sample tests
Vansh Bansal, Tianyu Chen, James G. Scott
The two-sample testing problem, a fundamental task in statistics and machine learning, seeks to determine whether two sets of samples, drawn from underlying distributions p and q, are in fact identically distributed (i.e.~whether $p=q$). A popular and intuitive approach is the classifier two-sample test (C2ST), where a classifier is trained to distinguish between samples from p and q. Yet despite simplicity of the C2ST, its reliability hinges on access to a near-Bayes-optimal classifier, a requirement that is rarely met and difficult to verify. This raises a major open question: can a weak classifier still be useful for two-sample testing? We show that the answer is a definitive yes. Building on the work of Hu & Lei (2024), we analyze two conformal variants of the C2ST that convert the scores from any trained classifier---even if weak, biased, or overfit---into exact, finite-sample p-values. We establish two key theoretical properties of the conformal C2ST: (i) finite-sample Type-I error control, and (ii) non-trivial power that degrades gently in tandem with the error of the trained classifier. The upshot is that even poorly performing classifiers can yield powerful and reliable two-sample tests. This general framework finds a powerful application in Bayesian inference, particularly for validating Neural Posterior Estimation (NPE) models, where the task of comparing a learned posterior approximation $q(\theta \mid y)$ to the true posterior $p(\theta \mid y)$ can be framed as a two-sample test. Empirically, the Conformal C2ST outperforms classical discriminative tests across a wide range of benchmarks for this task. Our results establish the conformal C2ST as a practical, theoretically grounded diagnostic tool.
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Learning Latent Action World Models in the Wild
Quentin Garrido, Tushar Nagarajan, Basile Terver, Nicolas Ballas, Yann LeCun, Michael Rabbat
Agents that can reason and plan in the real world must be able to predict the consequences of their actions. World models possess this capability but require action annotations that can be complex to obtain at scale. Latent action models address this issue by learning an action space from videos alone. Our work studies the training of latent action world models on in-the-wild videos, expanding the scope of existing works that focus on simple robotics simulations, video games, or manipulation data. While diverse videos enable modeling richer actions, they introduce challenges of environmental noise and lack of a common embodiment across videos. To address these, we carefully study the design and evaluation of latent actions. We find that constrained continuous latent actions are better suited for complex in-the-wild videos, compared to vector quantization. For example, actions specific to in-the-wild videos such as humans entering the room, can be modeled and then transferred across videos. However, in the absence of a common embodiment, learned latent actions are localized in space, relative to the camera. Nonetheless, we are able to train a controller that maps known actions to latent ones, allowing us to use latent actions as a universal interface to solve planning tasks on par with action-conditioned baselines.
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Efficient privacy loss accounting for subsampling and random allocation
Vitaly Feldman, Moshe Shenfeld
We consider the privacy amplification properties of a sampling scheme in which a user’s data is used in k steps chosen randomly and uniformly from a sequence (or set) of t steps. This sampling scheme has been recently applied in the context of differentially private optimization [Chua et al., 2024a, Choquette-Choo et al., 2025] and communication-efficient high-dimensional private aggregation [Asi et al., 2025], where it was shown to have utility advantages over the standard Poisson sampling. Theoretical analyses of this sampling scheme [Feldman and Shenfeld, 2025, Dong et al., 2025] lead to bounds that are close to those of Poisson sampling, yet still have two significant shortcomings. First, in many practi

cal settings, the resulting privacy parameters are not tight due to the approximation steps in the analysis. Second, the computed parameters are either the hockey stick or Rényi divergence, both of which introduce overheads when used in privacy loss accounting. In this work, we demonstrate that the privacy loss distribution (PLD) of random allocation applied to any differentially private algorithm can be computed efficiently. When applied to the Gaussian mechanism, our results demonstrate that the privacy-utility trade-off for random allocation is at least as good as that of Poisson subsampling. In particular, random allocation is better suited for training via DP-SGD. To support these computations, our work develops new tools for general privacy loss accounting based on a notion of PLD realization. This notion allows us to extend accurate privacy loss accounting to subsampling which previously required manual noise-mechanism-specific analysis.
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Learning Dynamic Stability Landscapes in Synchronization Networks
Christian Nauck, Junyou Zhu, Michael Lindner, Frank Hellmann
The robustness of synchronization is typically characterized by scalar, per-node stability indices whose dependence on topology is studied via network science or graph neural networks (GNNs). We propose a novel upstream task, stability landscapes, which provide deeper insights into synchronization behavior and from which many such scalar indices can be derived. Crucially, we pioneer a graph-to-image prediction paradigm: learning image-like landscapes as per-node targets directly from graph topology, a formulation we are not aware of having been established elsewhere in the literature. To support this task, we release two datasets of 10,000 graphs each at 20 and 100 nodes with per-node landscape labels, based on a conceptual oscillator model, capturing power grid synchronization behavior. A GNN encodes topology and a CNN decoder renders per-node images, learned end-to-end with good in-distribution accuracy, generalizing across graph sizes and to realistic power grid topologies. This demonstrates that stability landscapes, while beyond the reach of conventional network science, are learnable from topology and open new avenues for moving beyond scalar stability indices in biology, neuroscience, and power grids.
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rePIRL: Learn PRM with Inverse RL for LLM Reasoning
Xian Wu, Kaijie Zhu, Ying Zhang, Lun Wang, Wenbo Guo
Process rewards have been widely used in deep reinforcement learning to improve training efficiency, reduce variance, and prevent reward hacking. In LLM reasoning, existing works also explore various solutions for learning effective process reward models (PRM) with or without the help of an expert policy. However, existing methods either rely on strong assumptions about the expert policies (e.g., requiring their reward functions) or suffer intrinsic limitations (e.g., entropy collapse), resulting in weak PRMs or limited generalizability. In this paper, we introduce rePIRL, an inverse RL-inspired framework that learns effective PRMs with minimal assumptions about expert policies. Specifically, we design a dual learning process that updates the policy and the PRM interchangeably. Our learning algorithm has customized techniques to address the challenges of scaling traditional inverse RL to LLMs. We theoretically show that our proposed learning framework can unify both online and offline PRM learning methods, justifying that rePIRL can learn PRMs with minimal assumptions. Empirical evaluations on standardized math and coding reasoning datasets demonstrate the effectiveness of rePIRL over existing methods. We further show the application of our trained PRM in test-time training, test-time scaling, and providing an early signal for training hard problems. Finally, we validate our training recipe and key design choices via a detailed ablation study.
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DRFusion: Drift-Resilient Temporally Consistent Infrared–Visible Video Fusion
Xingyuan Li, HaoYuan Xu, Shulin Li, Xiang Chen, Zhiying Jiang, Jinyuan Liu
Infrared and visible video fusion is essential for achieving comprehensive perception in dynamic scenes. However, maintaining temporal consistency remains a formidable challenge. Conventional methods relying on optical flow often suffer from geometric rigidity and ghosting artifacts. Moreover, standard diffusion-based

fusion models typically operate in a frame-by-frame manner; when extended to autoregressive settings, they lack intrinsic temporal constraints and are prone to severe error accumulation and drifting, where minor artifacts amplify over time. To address these limitations, we propose a drift-resilient video fusion method that reformulates the task as history-conditioned motion generation. We introduce Stabilized History Guidance and Soft Temporal Anchoring to reframe temporal consistency as spectral filtering, implicitly aggregating motion dynamics without rigid alignment. Furthermore, our Decoupled Structure-Motion Adaptation strategy bridges pre-trained priors and structural constraints via two-stage training and latent refinement. Extensive experiments demonstrate that our method achieves state-of-the-art performance in both fusion quality and temporal stability.
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Learning Permutation from Structure Without Supervision
Ran Eisenberg, Ofir Lindenbaum
Many learning problems require uncovering a hidden ordering that reveals structure in unordered data, such as monotonicity in sorting or spatial continuity in jigsaw reconstruction. In these settings, permutations can be learned as latent operators by optimizing objectives defined directly on the reordered output, often without access to ground-truth orderings. Differentiable relaxations such as Gumbel–Sinkhorn make this approach practical by approximating permutation matrices with doubly stochastic matrices. However, learning from structure without supervision induces a non-uniform uncertainty: some assignments become confident early, while others remain ambiguous. Existing methods control this process using a single global temperature, forcing all assignments to sharpen or diffuse simultaneously and leading to instability at scale. We introduce an entropy-adaptive formulation of Gumbel–Sinkhorn that locally modulates temperature based on assignment uncertainty. This allows confident assignments to discretize early while preserving exploration where uncertainty remains. Across sorting and jigsaw reconstruction tasks and in routing-style settings, adaptive entropy control improves training stability and final permutation quality relative to fixed-temperature baselines, particularly as problem size and assignment ambiguity increase.
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ScreenParse: Moving Beyond Sparse Grounding with Complete Screen Parsing Supervision
A. Said Gurbuz, Sunghwan Hong, Ahmed Nassar, Marc Pollefeys, Peter W. J. Staar
Modern computer-use agents (CUA) must perceive a screen as a structured state, what elements are visible, where they are, and what text they contain, before they can reliably ground instructions and act. Yet, most available grounding datasets provide sparse supervision, with *insufficient* and *low-diversity* labels that annotate only a small subset of task-relevant elements per screen, which limits both coverage and generalization; moreover, practical deployment requires efficiency to enable low-latency, on-device use. We introduce **ScreenParse**, a large-scale dataset for *complete* screen parsing, with dense annotations of all visible UI elements (boxes, 55-class types, and text) across 771K web screenshots (21M elements). ScreenParse is generated by **Webshot**, an automated, scalable pipeline that renders diverse urls, extracts annotations and applies VLM-based relabeling and quality filtering. Using ScreenParse, we train **ScreenVLM**, a compact, 316M-parameter vision language model (VLM) that decodes a compact ScreenTag markup representation with a structure-aware loss that upweights structure-critical tokens. ScreenVLM substantially outperforms much larger foundation VLMs on dense parsing (e.g., 0.592 vs. 0.294 PageIoU on ScreenParse) and shows strong transfer to public benchmarks. Moreover, finetuning foundation VLMs on ScreenParse consistently improves their grounding performance, suggesting that dense screen supervision provides transferable structural priors for UI understanding. The dataset, model and code will be made publicly available.
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PromptPilot: Game-Theoretic Multi-Agent Prompt Optimization for Segment Anything
Guangze Shi, Yingjie Mi, Jia Shen, Feixue Shao, Jiarui Cao, Yexin Lai, Xueyu Liu, Rui Wang, Yongfei Wu, Mingqiang Wei
Few shot segmentation with vision foundation models relies on high quality promp

ts to segment unseen categories from limited support annotations. Existing promp t construction methods depend on test-time adaptation, fixed heuristic sampling, or monolithic reinforcement learning, making it difficult to balance semantic c onsistency, spatial coverage, and prompt credit assignment. To address these lim itations, PromptPilot is proposed as a hierarchical multi-agent reinforcement le arning framework for point prompt optimization with frozen DINOv2 and SAM. Promp t construction is formulated as sequential decision making, where feature and ph ysical agents propose complementary prompt modifications, and a manager agent se lects actions using SAM feedback and local marginal contribution. PromptPilot fu nctions as an inference-time optimization strategy without parameter updates. Ex tensive experiments demonstrate that the proposed game-theoretic approach improv es segmentation performance and generalization, offering a principled solution f or automated prompt engineering. The code is available at https://github.com/L-AILab/PromptPilot.

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FlatLand: Personalized Graph Federated Learning via Tailored Lorentz Space

Jiahong Liu, Ram Samarth B B, Xinyu Fu, Menglin Yang, Weixi Zhang, Rex Ying, Irw in King

Personalization has become a pivotal field of study in contemporary intelligent systems.

Federated learning enables privacy-preserving collaborative training, but highly heterogeneous client data remain challenging, especially in graph federated lea rning where clients possess structurally diverse graphs. Existing personalized f ederated learning (PFL) methods ignore the intrinsic geometric properties of div erse graph structures. We propose FlatLand, a novel personalized Federated learn ing method that embeds different clients' data in tailored Lorentz space of hype rbolic geometry. Our key insight is that hyperbolic geometry naturally accommoda tes the intrinsic negative curvature prevalent in real-world graphs, while the t ime-like dimension in Lorentz space provides a principled way to encode client-s pecific heterogeneity. We develop a parameter decoupling strategy that separates heterogeneous information (captured in time-like parameters) from common knowle dge (preserved in space-like parameters), enabling direct aggregation without re quiring client similarity estimation and extra calculation modules. Empirical re sults on diverse federated graph learning tasks demonstrate that FlatLand achiev es superior performance, particularly in low-dimensional settings. Code is avail able in our GitHub repository.

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Predicting Large Model Test Losses with a Noisy Quadratic System

Chuning Li, Chris J. Maddison

We introduce a predictive model that estimates the pre-training loss of large mo dels from model size (N), batch size (B) and number of weight updates (K). This is the first loss prediction model that can handle changing batch size. Th e model outperforms Chinchilla's loss model, a model of the test loss using the batch size and number of tokens, in terms of projecting the loss at extrapolated compute budgets (up to 1000 folds). A natural use of the model is to find optim al N,B,K configurations under explicit and compound resource constraints like time, memory and compute. In our experiments, the model-selected configurations are close to ground-truth optimal. Our work advocates for loss prediction as a b etter alternative to heuristic-based laws, which are growing in complexity. The implementation is available on \href{https://github.com/chuningxdy/Noisy-Quadrat ic-System}{GitHub}.

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RLIE: Rule Generation with Logistic Regression, Iterative Refinement, and Evalua tion for Large Language Models

Yang Yang, Hua XU, Zhangyi Hu, Yutao Yue

Large Language Models (LLMs) can propose natural-language rules, circumventing t he reliance on a predefined predicate space in traditional rule learning. Howeve r, existing LLM-based methods often neglect the global interactions among rules, and the potential of using fine-grained rule importance scores to calibrate neu ro-symbolic reasoning remains underexplored. To address this gap, we introduce R

LIE, a framework that integrates LLMs with probabilistic modeling to learn weighted rule sets in four stages: (1) Rule generation: proposing and filtering candidate rules via LLMs; (2) Logistic regression: learning sparse, calibrated weights for global rule selection; (3) Iterative refinement: revising the rule set with error-driven hard examples; and (4) Evaluation: validating the learned system via comparative inference paradigms. Across multiple real-world datasets and LLM backbones, our learned weighted rules achieve superior stability and accuracy, whereas rule-injection prompting yields mixed results and often degrades performance. These results suggest LLMs excel at semantic rule discovery but are less reliable at controlled probabilistic aggregation. Our findings highlight both the promise and the limits of LLMs for inductive reasoning, motivating a principled integration with classic probabilistic rule combination for reliable neuro-symbolic reasoning.

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SEMA: a Scalable and Efficient Mamba like Attention via Token Localization and Averaging

Nhat Thanh Tran, Fanghui Xue, shuai zhang, Jiancheng Lyu, Yunling Zheng, Yingyong Qi, Jack Xin

Attention is the critical component of a transformer. Yet the quadratic computational complexity of vanilla full attention in the input size and the inability of its linear attention variant to focus have been challenges for computer vision tasks. We provide a mathematical definition of generalized attention and formulate both vanilla softmax attention and linear attention within the general framework. We prove that generalized attention disperses, that is, as the number of keys tends to infinity, the query assigns equal weights to all keys. Motivated by the dispersion property and recent development of Mamba form of attention, we design Scalable and Efficient Mamba like Attention (SEMA) which utilizes token localization to avoid dispersion and maintain focusing, complemented by theoretically consistent arithmetic averaging to capture global aspect of attention. We support our approach on Imagenet-1k where classification results show that SEMA is a scalable and effective alternative beyond linear attention, outperforming recent vision Mamba models on increasingly larger scales of images at similar model parameter sizes. Source code can be found at: https://github.com/nhatthanhtran/SEMA.

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AES: Curing Optimizer Blindness in Long-Tailed Recognition via State-Aware Correction

Fanfu Wang, Jiachang Zhan, Zhiheng Gong, Pengkun Wang, Yang Wang

Long-tailed recognition fundamentally suffers from optimizer blindness where the optimization process mistakenly conflates the magnitude of gradient accumulation with the scarcity of semantic information. Existing strategies relying on static frequency-based priors fail to correct this bias and result in state blindness regarding supervision and micro-level blindness regarding parameter updates. To address these limitations, we propose the AES framework to establish a dynamic and state-aware correction system across the entire learning lifecycle. We specifically introduce Adaptive Residual Supervision loss to act as a real-time reality check for supervision completeness via precision shielding. We also propose Entropy-aware PCGrad to resolve parameter-level conflicts by quantifying task specificity through gradient entropy. Additionally, we devise Sample-level Conflict Arbitrated Fusion to serve as a dynamic inference arbiter that routes predictions based on instance difficulty. Extensive experiments on CIFAR-100-LT, ImageNet-LT, and iNaturalist 2018 demonstrate that our method consistently achieves state-of-the-art performance by effectively balancing head-class stability and tail-class discrimination. Code is available at [here](https://github.com/wangff0101/AES)

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Discrete Tilt Matching

Yuyuan Chen, Shiyi Wang, Peter Potaptchik, Jaeyeon Kim, Michael Samuel Albergo

Masked diffusion large language models (dLLMs) are a promising alternative to autoregressive generation. While reinforcement learning (RL) algorithms have been

adapted to be compatible with dLLMs for fine-tuning them, their reliance on the computation of the marginal likelihood to evaluate policy objectives is intractable. To overcome this, we exploit a dynamical relation between the unmasking posterior of the base model and that which targets the reward-tilted distribution to derive Discrete Tilt Matching (DTM), an algorithm that avoids intractable likelihood evaluation entirely. DTM can be phrased as a cross-entropy loss that only requires forward evaluation of rewards and whose variance can be adaptively controlled, improving training stability. We motivate DTM on maze planning tasks, and show that fine-tuning LLaDA-8B-Instruct with DTM achieves higher accuracy at lower compute costs than prior RL-based fine-tuning methods across the Sudoku, Countdown, and MATH500 benchmarks.

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Eigenvectors of Experts are Training-free Non-collapsing Routers

Giang Do, Hung Le, Truyen Tran

Sparse Mixture of Experts (SMoE) architectures improve the training efficiency of Large Language Models (LLMs) by routing input tokens to a selected subset of specialized experts. Despite their remarkable success, both training and inference in SMoE models suffer from the *expert collapse* issue (Chi et al., 2022a), which degrades model performance. Prior studies primarily focus on improving the router; however, such methods rely on training from scratch or fine-tuning, which requires high computational and data-processing costs. Furthermore, we demonstrate that, despite these efforts, the issue persists when advancing well-pretrained SMoE models, as evidenced by both theoretical and empirical results. To fill that gap, we analyze the advanced SMoE models and observe that the eigenvectors of expert weight matrices encode rich semantic information, pointing to an effective alternative to conventional routing strategies. Building on this insight, we propose **Singular Value Decomposition SMoE (SSMoE)**, a novel and *training-free* framework that leverages spectral properties of the expert weights to address the collapse issue and enhance model performance. Extensive experiments across diverse language and vision tasks, under both clean and corrupt data settings, demonstrate the strong generalization and robustness of SSMoE. Our findings highlight how a deeper understanding of model internals can guide the development of more effective SMoE architectures.

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Perceptual Flow Network for Visually Grounded Reasoning

Yangfu Li, Yuning Gong, Hongjian Zhan, Teng Li, Yuanhuiyi Lyu, Tianyi Chen, Qi Liu, Ziyuan Huang, Zhihang Zhong, DanDan Zheng, Yue Lu

Despite the success of LVLMs, general optimization objectives (e.g., standard MLE) fail to constrain visual trajectories, leading to language bias and hallucination. To mitigate this, current methods introduce geometric priors from visual experts as additional supervision. However, we observe that such supervision is typically suboptimal: *it is biased toward geometric precision and offers limited reasoning utility*. To bridge this gap, we propose Perceptual Flow Network (PFlowNet), which eschews rigid alignment with the expert priors and achieves interpretable yet more effective visual reasoning. Specifically, PFlowNet decouples perception from reasoning to establish a self-conditioned generation process. Based on this, it integrates *multi-dimensional rewards* with *vicinal geometric shaping* via variational reinforcement learning, thereby facilitating reasoning-oriented perceptual behaviors while preserving visual reliability. PFlowNet delivers a provable performance guarantee and competitive empirical results, particularly setting new SOTA records on V* Bench (90.6%) and MME-RealWorld-lite (67.0%).

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Real-World Unsupervised Models Generalize to Predict Brain Responses to Out-of-Distribution Stimuli

Chenggang Chen, Zhiyu Yang, Xiaoqin Wang

Deep neural networks currently provide the leading quantitative models of neural responses in sensory systems. However, these networks remain implausible as models of sensory development, largely because they rely on supervised training with label efficiency far exceeding that of biological learning. Furthermore, these models are typically trained on manually curated datasets that lack the statist

ical properties of the natural environments to which the brain is exposed. Here, we demonstrate that models trained with unsupervised objectives on real-world data significantly outperform supervised models in predicting brain responses across both human auditory and visual cortex. We show that this performance advantage is not driven by network architecture or dataset size, but rather by the data distribution. Crucially, we find that unsupervised models trained on real-world data exhibit remarkable out-of-distribution generalization: a model trained exclusively on Mandarin speech accurately predicts English-driven brain responses, and a model trained on infant head-cam footage predicts adult visual responses to curated object images. Together, our results illustrate how deep neural networks can be used to reveal the real-world statistics that shape neural representations in the brain.

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Hierarchical Anchor Graph Learning for Multi-View Clustering

Xingchen Hu, Miao Jia, Jiyuan Liu, Siwei Wang, KE LIANG, Wenjing Yang

Multi-view clustering (MVC) is a fundamental task in heterogeneous data analysis, where anchor-based graph methods are widely adopted for their computational efficiency. However, existing approaches typically utilize static, single-layer anchors, failing to capture the multi-granularity nature of complex data. Drawing inspiration from hierarchical human cognition, we propose a hierarchical anchor graph learning method, termed HAG-MVC, a novel framework that organizes multi-view data as a multi-level pyramid. Unlike conventional one-shot anchor generation methods, HAG-MVC introduces a multi-level co-evolution mechanism, where anchors and graph structures are iteratively refined together to capture semantics from fine-to-coarse granularities. Moreover, HAG-MVC offers a transparent abstraction architecture as an alternative to black-box deep clustering: by maintaining all anchors within the original feature space, it enables explicit inspection of the abstraction process, ensuring inherent interpretability. Extensive experiments on benchmark datasets demonstrate that HAG-MVC consistently outperforms state-of-the-art methods. Beyond MVC, this work provides a scalable and trustworthy paradigm for hierarchical knowledge representation in broad machine learning tasks.

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$\texttt{Multi}^2$: Hierarchical Multi-Agent Decision-Making with LLM-Based Agents in Interactive Environments

Sangeun Park, Minhae Kwon

A central goal of large language model (LLM) research is to build agentic systems that can plan, act, and adapt through sustained interaction with dynamic environments. While recent LLM-based agents exhibit impressive contextual reasoning, their long-horizon decision-making remains fragile, often suffering from $\texti t{objective drift}$, where goals and plans drift over extended interactions.

We introduce $\texttt{Multi}^2$, a hierarchical multi-agent decision-making framework that explicitly decomposes agent behavior into complementary roles. A high-level agent ($\texttt{System 1}$) focuses on context-aware sub-goal generation using supervised fine-tuning (SFT), while a low-level agent ($\texttt{System 2}$) executes atomic actions through offline-to-online reinforcement learning (RL) in interactive environments. This separation enables stable long-horizon control, mitigates objective drift, and allows efficient adaptation.

Across diverse interactive environments, $\texttt{Multi}^2$ consistently outperforms strong agentic baselines, demonstrating improved robustness and coordination in multi-turn interaction. Beyond performance, we introduce and release three hierarchical benchmark datasets, filling a long-standing gap in training and evaluating hierarchical decision-making for LLM-based agents.

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Seg-ReSearch: Segmentation with Interleaved Reasoning and External Search

Tianming Liang, Qirui Du, Jian-Fang Hu, Haichao Jiang, Zicheng Lin, Wei-Shi Zheng

Segmentation based on language has been a popular topic in computer vision. While recent advances in multimodal large language models (MLLMs) have endowed segmentation systems with reasoning capabilities, these efforts remain confined by th

e frozen internal knowledge of MLLMs,
which limits their potential for real-world scenarios that involve up-to-date in formation or domain-specific concepts.
In this work, we propose \textbf{Seg-ReSearch}, a novel segmentation paradigm th at overcomes the knowledge bottleneck of existing approaches. By enabling interl eaved reasoning and external search, Seg-ReSearch empowers segmentation systems to handle dynamic, open-world queries that extend beyond the frozen knowledge of MLLMs.
To effectively train this capability, we introduce a hierarchical reward design that harmonizes initial guidance with progressive incentives, mitigating the dil emma between sparse outcome signals and rigid step-wise supervision.
For evaluation, we construct OK-VOS, a challenging benchmark that explicitly req uires outside knowledge for video object segmentation.
Experiments on OK-VOS and two existing reasoning segmentation benchmarks demonst rate that our Seg-ReSearch improves state-of-the-art approaches by a substantial margin.
Code and data is available at https://github.com/iSEE-Laboratory/Seg-ReSearch.
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Structured Progressive Knowledge Activation for LLM-Driven Neural Architecture S earch
Zhen Liu, Yuhan Liu, Jinjun Wang, Wei Song, Jianyi Liu, Jingwen Fu
This paper focuses on a key challenge in Neural Architecture Search (NAS): integ rating established architectural knowledge while exploring new designs under exp ensive evaluations.
Large language models (LLMs) are a promising assistant for NAS because they can translate rich architectural and coding priors into executable code edits.
However, in practice, seemingly local revisions often propagate into non-local b ehavioral and performance shifts because a single edit can inadvertently couple multiple interacting functional factors, a phenomenon we refer to as functional entanglement.
To make LLM knowledge usable under such entanglement, we propose Structured Prog ressive Knowledge Activation (SPARK), which activates relevant priors by explici tly selecting the functional factor to modify and conditioning the edit on that factor.
This factor-conditioned editing reduces entangled side effects and yields more t argeted, reliable architecture modifications.
On CLRS-DFS, SPARK achieves a 28.1x sample-efficient architecture evolution spee dup and yields a 22.9% relative improvement in OOD accuracy.
Our code is available at https://github.com/AIM-ResearchLab/SPARK.
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Local MAP Sampling for Diffusion Models
Shaorong Zhang, Rob Brekelmans, Greg Ver Steeg
Diffusion Posterior Sampling (DPS) provides a principled Bayesian approach to in verse problems by sampling from $p(x_0 \mid y)$. While posterior sampling is val uable for capturing uncertainty and multi-modality, many classical and practical inverse problem settings ultimately prioritize accurate point estimation—most n otably the MAP estimator, which has long served as a standard reconstruction obj ective in imaging and scientific applications. We introduce Local MAP Sampling (LMAPS), a new inference framework that iteratively solving local MAP subproblems along the diffusion trajectory. This perspective clarifies their connection to global MAP and DPS, offering a unified probabilistic interpretation for optimiza tion-based methods. Building on this foundation, we develop practical algorithms with a covariance approximation motivated by Gaussian prior assumption, a refo rmulated objective for stability and interpretability. Across a broad set of ima ge restoration and scientific tasks, LMAPS achieves state-of-the-art performance .
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Gecko: A Simulation Environment with Stateful Feedback for Refining Agent Tool C alls
Zeyu Zhang, Guohao Li, Zhenchang Xing, Alexandros Apostolopoulos, Yu Lin Lee, Li

ang Zheng
The ability to use tools is fundamental for large language model (LLM) agents. Given a task, existing systems use LLMs to plan and generate tool calls, which are executed by real-world tools to complete the task. However, tool calls are prone to errors because they are generated primarily from the intrinsic capabilities of LLMs. Moreover, while it is useful to let LLMs iteratively refine the tool-call sequence using execution results from real tools, this process can be expensive and may cause unsafe side effects. To improve LLM tool calls and address issues caused by using real tools for refinement, we introduce Gecko, a stateful simulation environment that provides informative feedback for refining LLM tool calls before real execution. Specifically, Gecko combines rules and LLMs to check the validity of tool names and arguments, synthesize schema-conforming and state-consistent responses, and judge task completion against the user objective. These three types of feedback allow LLMs to refine their tool calls in simulation, forming a simple yet effective test-time scaling method named GATS. On BFCLv3 and τ^2-bench, GATS consistently improves the performance of various LLMs, including GPT-4o, GPT-5, and Gemini-3.0-pro.
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VGGT-Motion: Motion-Aware Calibration-Free Monocular SLAM for Long-Range Consistency
Zhuang Xiong, Chen Zhang, Qingshan Xu, Wenbing Tao
Despite recent progress in calibration-free monocular SLAM via 3D vision foundation models, scale drift remains severe on long sequences. Motion-agnostic partitioning breaks contextual coherence and causes zero-motion drift, while conventional geometric alignment is computationally expensive. To address these issues, we propose VGGT-Motion, a calibration-free SLAM system for efficient and robust global consistency over kilometer-scale trajectories. Specifically, we first propose a motion-aware submap construction mechanism that uses optical flow to guide adaptive partitioning, prune static redundancy, and encapsulate turns for stable local geometry. We then design an anchor-driven direct Sim(3) registration strategy. By exploiting context-balanced anchors, it achieves search-free, pixel-wise dense alignment and efficient loop closure without costly feature matching. Finally, a lightweight submap-level pose graph optimization enforces global consistency with linear complexity, enabling scalable long-range operation. Experiments show that VGGT-Motion markedly improves trajectory accuracy and efficiency, achieving state-of-the-art performance in zero-shot, long-range calibration-free monocular SLAM.
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DynaVieW: Schema-Guided World Modeling for Understanding Hierarchical Visual Dynamics
Silin Gao, Hao Zhao, Zeming Chen, Sepideh Mamooler, Antara Raaghavi Bhattacharya, Qiyu Wu, Hiromi Wakaki, Yuki Mitsufuji, Li Mi, Syrielle Montariol, Antoine Bosselut
Multimodal LLMs struggle to systematically model the temporal evolution of visual scenes in videos or multi-image sequences. Such inputs require models to predict or simulate multiple levels of dynamic constituents, such as actions taken in the visual sequence, and the associated changes to the visual environment that result. To address this challenge, we propose a dynamic schema-guided world model, DynaVieW, optimized for visual dynamic prediction and simulation. DynaVieW achieves an in-depth understanding of visual dynamics by learning interleaved state-transition sequences, where states cover broad visual scenes from video keyframes, and transitions capture comprehensive dynamic constituents within a hierarchical schema. DynaVieW jointly models transition prediction and state simulation under a mixture-of-experts architecture, with a cross-expert selective attention and a schema token re-weighted loss, to ensure effective and robust learning. DynaVieW's superior visual dynamic understanding boosts its downstream performances on both visual narrative creation and world simulation, showing improved consistency and controllability of visual generation and better instruction-following ability.
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Ultrafast On-Chip Online Learning via Spline Locality in Kolmogorov–Arnold Networks
Duc Hoang, Aarush Gupta, Philip Harris
Ultrafast online learning is essential for high-frequency systems, such as controls for quantum computing and nuclear fusion, where adaptation must occur on sub-microsecond timescales.
Meeting these requirements demands low-latency, fixed-precision computation under strict memory constraints, a regime in which conventional Multi-Layer Perceptrons (MLPs) are both inefficient and numerically unstable.
We identify key properties of Kolmogorov-Arnold Networks (KANs) that align with these constraints.
Specifically, we show that: (i) KAN updates exploiting B-spline locality are sparse, enabling superior on-chip resource scaling, and (ii) KANs are inherently robust to fixed-point quantization.
By implementing fixed-point online training on Field-Programmable Gate Arrays (FPGAs), a representative platform for on-chip computation, we demonstrate that KAN-based online learners are significantly more efficient and expressive than MLPs across a range of low-latency and resource-constrained tasks.
To our knowledge, this work is the first to demonstrate model-free online learning at sub-microsecond latencies.
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Task-Aware Preference Calibration for Direct Preference Optimization
Mengyang Li, Zhong Zhang, Pinlong Zhao
Direct Preference Optimization (DPO) has become a predominant approach for aligning large language models with human preferences. Recent work has used perplexity differentials to identify unreliable preference labels, but these methods apply uniform calibration strategies across all samples. We observe that the reliability of perplexity signals varies substantially across task types: perplexity differentials strongly correlate with preference quality for factual tasks but provide weak signals for creative tasks where novelty is valued. Based on this observation, we propose Task-Aware Preference Calibration (TAPC), which learns task-conditioned calibration functions that adapt to the characteristics of different prompt types. TAPC employs a task encoder to extract prompt representations and learns task-specific slope and bias parameters for mapping perplexity signals to confidence targets. Through meta-learning on a small reference dataset, TAPC discovers how to weight perplexity signals appropriately for each task category. Experiments on Llama-3-8B and Qwen2-7B demonstrate that TAPC outperforms existing methods across multiple benchmarks, with particularly large improvements on creative and open-ended tasks where uniform calibration strategies fail.
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Emergent Analogical Reasoning in Transformers
Gouki Minegishi, Jingyuan Feng, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, Yutaka Matsuo
Analogy is a central faculty of human intelligence, enabling abstract patterns discovered in one domain to be applied to another.
However, the mechanisms underlying analogical reasoning in Transformers remain poorly understood.
In this work, inspired by the notion of functors in category theory, we formalize analogical reasoning as the inference of correspondences between entities across categories.
Based on this formulation, we introduce synthetic tasks that evaluate the emergence of analogical reasoning under controlled settings.
We find that the emergence of analogical reasoning is highly sensitive to data characteristics, optimization choices, and model scale.
Through mechanistic analysis, we show that analogical reasoning in Transformers decomposes into two key components:
(1) geometric alignment of relational structure in the embedding space, and
(2) the application of a functor within the Transformer. These mechanisms enable models to transfer relational structure from one category to another, realizing analogy.

Finally, we quantify these effects and find that the same trends are observed in pretrained LLMs.
In doing so, we move analogy from an abstract cognitive notion to a concrete, mechanistically grounded phenomenon in modern neural networks.
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LogicSAGE: Neuro-Symbolic Reasoning with Socratic-Guided Enhancement
Jinlong Tian, Jiang Yu, Kewei Cheng, Fengxiang Cheng, Yue He, Yunfei Wang, Haotian Wang, Haoxuan Li, Wenjing Yang, Shixuan Liu
Large Language Models (LLMs) often struggle with complex logical reasoning. Existing approaches typically rely on either purely neural reasoning in natural language or offloading to formal solvers via symbolic representations. However, both paradigms face significant limitations: while LLMs exhibit strong semantic intuition they are prone to hallucinations, whereas symbolic solvers offer rigorous derivation but remain highly sensitive to minor syntactic errors. To combine the strengths of these two paradigms while mitigating their respective limitations, we introduce **LogicSAGE** (**L**ogic-informed **S**ocratic **A**gent for **G**uided **E**nhancement), a dual-process framework that integrates a robust neural reasoner (System 1) with a rigorous symbolic validator (System 2). Specifically, our framework employs a Socratic Error Correction mechanism that treats solver feedback not as terminal failures but as pedagogical signals, engaging in a dialectic loop to iteratively refine logic programs and resolve semantic ambiguities. Extensive experiments on five benchmarks show that LogicSAGE (8B) achieves a state-of-the-art 92.36% average accuracy, significantly outperforming GPT-4 baselines, which establishes that architectural innovation can supersede model scale in faithful reasoning.
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Entropic Mirror Monte Carlo
Anas Cherradi, Yazid Janati, Alain Oliviero Durmus, Sylvain Le Corff, Yohan Petetin, Julien Stoehr
Importance sampling is a Monte Carlo method which designs estimators of expectations under a target distribution using weighted samples from a proposal distribution. When the target distribution is complex, such as multimodal distributions in high-dimensional spaces, the efficiency of importance sampling critically depends on the choice of the proposal distribution. In this paper, we propose a novel adaptive scheme for the construction of efficient proposal distributions. Our algorithm promotes efficient exploration of the target distribution by combining global sampling mechanisms with a delayed weighting procedure. The proposed weighting mechanism plays a key role by enabling rapid resampling in regions where the proposal distribution is poorly adapted to the target. Our sampling algorithm is shown to be geometrically convergent under mild assumptions and is illustrated through various numerical experiments.
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When Does Adaptation Win? Scaling Laws for Meta-Learning in Quantum Control
Nima Leclerc, Chris Miller, Nicholas Peter Brawand
Quantum hardware suffers from intrinsic device heterogeneity and environmental drift, forcing practitioners to choose between suboptimal non-adaptive controllers or costly per-device recalibration. We derive a scaling law lower bound for meta-learning showing that the adaptation gain (expected fidelity improvement from task-specific gradient steps) saturates exponentially with gradient steps and scales linearly with task variance, providing a quantitative criterion for when adaptation justifies its overhead. Validation on quantum gate calibration shows negligible benefits for low-variance tasks but > 40% fidelity gains on two-qubit gates under extreme out-of-distribution conditions (10× the training noise), with implications for reducing per-device calibration time on cloud quantum processors. Further validation on classical linear-quadratic control confirms these laws emerge from general optimization geometry rather than quantum-specific physics. We further introduce a few-shot pre-adaptation protocol that estimates the optimal adaptation budget from $N =3–5$ probe steps within 3–19% relative error across out-of-distribution regimes.
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RADAR: Redundancy-Aware Diffusion for Multi-Agent Communication Structure Generation

Zhen Zhang, Wanjing Zhou, Juncheng Li, Hao Fei, Jun Wen, Wei Ji

Compared with individual agents, large language model based multi-agent systems have shown great capabilities consistently across diverse tasks, including code generation, mathematical reasoning, and planning, etc. Despite their impressive performance, the effectiveness and robustness of these systems heavily rely on their communication topology, which is often fixed or generated in a single step. This restricts fine-grained structural exploration and flexible composition, resulting in excessive token utilization on simple tasks while limiting capability on complicated tasks. To mitigate this challenge, we introduce RADAR, a redundancy-aware and query-adaptive generative framework that actively reduce communication overhead. Motivated by recent progress in conditional discrete graph diffusion models, we formulate communication topology design as a step-by-step generation process, guided by the effective size of the graph. Comprehensive experiments on six benchmarks demonstrate that RADAR consistently outperforms recent baselines, achieving higher accuracy, lower token consumption, and greater robustness across diverse scenarios. Our code and data are available at https://github.com/cszhangzhen/RADAR.

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Forget by Uncertainty: Orthogonal Entropy Unlearning for Quantized Neural Networks

Tian Zhang, Yujia Tong, Junhao Dong, Ke Xu, Yuze Wang, Jingling Yuan

The deployment of quantized neural networks on edge devices, combined with privacy regulations like GDPR, creates an urgent need for machine unlearning in quantized models. However, existing methods face critical challenges: they induce forgetting by training models to memorize incorrect labels, conflating forgetting with misremembering, and employ scalar gradient reweighting that cannot resolve directional conflicts between gradients. We propose $\textbf{OEU}$, a novel Orthogonal Entropy Unlearning framework with two key innovations: 1) Entropy-guided unlearning provides an unbiased forgetting direction by maximizing prediction uncertainty on forgotten data, avoiding confident misprediction toward any specific class, and 2) Gradient orthogonal projection eliminates interference by projecting forgetting gradients onto the orthogonal complement of retain gradients, providing theoretical guarantees for utility preservation under first-order approximation. Extensive experiments demonstrate that OEU outperforms existing methods in both forgetting effectiveness and retain accuracy.

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Evaluating and Explaining Prompt Sensitivity of LLMs Using Interactions

Ruiyang Qin, Qingzhuo Wang, Tian Wang, Zhihua Wei, Wen Shen

The remarkable capabilities of large language models (LLMs) are often undermined by their instability.

Even subtle and semantically irrelevant changes in prompts can cause dramatic fluctuations in performance, a phenomenon known as prompt sensitivity. Previous studies typically evaluate prompt sensitivity by comparing the LLM's final outputs when prompts change. However, such coarse-grained metrics fail to explain the internal reasons for prompt sensitivity.

In this paper, we introduce interactions as a fine-grained tool to analyze prompt sensitivity of LLMs. Specifically, we decompose the output score of the LLM into a set of interactions. Each interaction represents a nonlinear relationship involving a set of input variables. We discover that subtle changes to prompts can trigger severe instability in interactions, even when the outputs of the LLM remain the same. To this end, we propose an Interaction-based Prompt Sensitivity (IPS) metric by quantifying changes in interactions when we introduce subtle changes to prompts.

We apply the IPS metric to 50 open-source LLMs and uncover four factors that reduce the prompt sensitivity of LLMs, including supervised fine-tuning, increased model scales, dense architectures, and few-shot learning. More crucially, we discover a common mechanism by which these four factors reduce prompt sensitivity: all four factors tend to reduce the prompt sensitivity of low-order interactions

(i.e., interactions involving few input variables).
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Beyond Description: Federated Adaptation via Semantic-Visual Prototype Alignment
Jiarong Yang, Yuan Liu
Adopting pre-trained Vision-Language Models (VLMs) in Federated Learning (FL) presents a promising avenue for mitigating data scarcity and heterogeneity. However, existing solutions suffer from high computational complexity or ineffective knowledge aggregation. To address these problems, we propose FedSPA (Federated Adaptation via Semantic-Visual Prototype Alignment). On the client side, FedSPA restricts local optimization to visual prototypes, enabling lightweight personalization. On the server side, we introduce a semantic alignment module that leverages client-uploaded prototypes to minimize a contrastive objective, aligning global semantic prototypes with heterogeneous visual distributions and thereby shifting the paradigm from traditional "learning-to-describe" (optimizing static prompts) to "learning-to-align". Extensive experiments demonstrate that FedSPA significantly outperforms state-of-the-art methods in both personalized and global benchmarks, while substantially reducing computational overhead. The code is available at https://github.com/eejiarong/FedSPA-main.
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When Attributes Disagree: Gradient Conflict in Image Aesthetic Assessment
Ye Wang, Maocai Dai, Jiang Xie, Xiuli Bi, Fei Tao, Xiao Li, Hong Yu
Image Aesthetic Assessment (IAA) predicts an image’s overall aesthetic score, yet aesthetic is influenced by multiple attributes whose relative importance varies with image content and usage scenarios. Under end-to-end training with only overall-score supervision, attribute signals are blended, which can cause gradient conflict across samples dominated by different attributes, resulting in gradient cancellation and persistent systematic bias. To address these issues, we propose AGREE (Attribute-guided Gradient Routing for Establishing Agreement), which learns attribute-specific subspaces and performs gradient routing based on sample-wise attribute sensitivity estimated via perturbation analysis. AGREE further reduces feature coupling across attributes with semantic anchors and improves robustness via error-aware reweighting. Experiments on AVA, LAPIS, AADB, TAD66K, and PARA show consistent improvements over diverse IAA baseline models, and AGREE is plug-and-play for existing end-to-end IAA methods without modifying their original architectures. To our knowledge, this work is among the early efforts in IAA to systematically study gradient conflict and provide an effective solution. The code is available at https://dahat364.github.io/AGREE/.
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A Computational Framework for Evaluating Human-likeness in LLMs' Open-ended Human Behaviors
Yuxuan Lei, Jianxun Lian, Defu Lian, Jincenzi Wu, Tianfu Wang, Xing Xie
Large Language Models (LLMs) have found widespread application and research in scenarios such as role-playing and sociological simulations. Despite the growing use of LLM-based agents to simulate human activities, the extent to which their behaviors resemble human behavior remains underexplored. As diverse LLMs proliferate, the traditional Turing test is ineffective for scalable evaluation and prone to bias from human-crafted challenges, leading to unfair assessments. In this work, we propose a novel distribution-based framework that comprehensively evaluates human-likeness and believability of AI behaviors by leveraging large-scale open-ended human behavior data from web. For better evaluation, we design generic metrics to cover three principles: rationality, consistency, and diversity. Implemented across online shopping, open-topic Q\&A, and urban mobility, our framework reveals that even the currently best LLM still exhibits a significant gap from real user behavior, underscoring the necessity of comprehensive research and evaluation of AI’s human-like capabilities.
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(1D) Ordered Tokens Enable Efficient Test-Time Search
Zhitong Gao, Parham Rezaei, Ali Cy, Mingqiao Ye, Nataša Jovanovi■, Jesse Allardice, Afshin Dehghan, Amir Zamir, Roman Bachmann, O■uzhan Fatih Kar
Tokenization is a key component of autoregressive (AR) generative models, conver

ting raw data into more manageable units for modeling. Commonly, tokens describe local information, such as regions of pixels in images or word pieces in text, and AR generation predicts these tokens in a fixed order. A worthwhile question is whether token structures affect the ability to steer the generation through test-time search, where multiple candidate generations are explored and evaluated by a verifier. Using image generation as our testbed, we hypothesize that recent 1D ordered tokenizers with coarse-to-fine structure can be more amenable to search than classical 2D grid structures. This is rooted in the fact that the intermediate states in coarse-to-fine sequences carry semantic meaning that verifiers can reliably evaluate, enabling effective steering during generation. Through controlled experiments, we find that AR models trained on coarse-to-fine ordered tokens exhibit improved test-time scaling behavior compared to grid-based counterparts. Moreover, we demonstrate that, thanks to the ordered structure, pure test-time search over token sequences (i.e., without training an AR model) can perform training-free text-to-image generation when guided by an image-text verifier. Beyond this, we systematically study how classical search algorithms (best-of-N, beam search, lookahead search) interact with different token structures, as well as the role of different verifiers and AR priors.

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Two Calm Ends and the Wild Middle: A Geometric Picture of Memorization in Diffusion Models

Nick Dodson, Xinyu Gao, Qingsong Wang, Yusu Wang, Zhengchao Wan

Diffusion models generate high-quality samples but can also memorize training data, raising serious privacy concerns. Understanding the mechanisms governing when memorization versus generalization occurs remains an active area of research. In particular, it is unclear where along the noise schedule memorization is induced, how data geometry influences it, and how phenomena at different noise scales interact. We introduce a geometric framework that partitions the noise schedule into three regimes based on the coverage properties of training data by Gaussian shells and the concentration behavior of the posterior, which we argue are two fundamental objects governing memorization and generalization in diffusion models. This perspective reveals that memorization risk is highly non-uniform across noise levels. We further identify a danger zone at medium noise levels where memorization is most pronounced. In contrast, both the small and large noise regimes resist memorization, but through fundamentally different mechanisms: small noise avoids memorization due to limited training coverage, while large noise exhibits low posterior concentration and admits a provably near linear Gaussian denoising behavior.

For the medium noise regime, we identify geometric conditions through which we propose a geometry-informed targeted intervention that mitigates memorization.

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The First Drop of Ink: Nonlinear Impact of Misleading Information in Long-Context Reasoning

Muhan Gao, Zih-Ching Chen, Kuan-Hao Huang

As large language models (LLMs) are increasingly deployed in retrieval augmented generation (RAG) and agentic systems that accumulate extensive context, understanding how distracting information affects performance in long context becomes critical. Prior work shows that semantically relevant but misleading documents can cause performance degradation, yet the quantitative relationship between the proportion of distractors and performance remains unstudied. In this work, we systematically vary the proportion of hard distractors within fixed-length contexts, revealing a striking nonlinear pattern: as the proportion of hard distractors increases, performance drops sharply within the first small fraction, while the remainder of the range yields only marginal additional decline. We term this ''The First Drop of Ink'' effect, analogous to how a single drop of ink contaminates water. We provide both theoretical and empirical analysis grounded in attention mechanics: hard distractors disproportionately capture attention even at small proportions, with diminishing marginal impact as their proportion increases. Through controlled experiments, we further show that filtering yields performance gains primarily from context length reduction rather than distractor removal, an

d only achieves substantial recovery when hard distractor proportion is reduced to near zero, which highlights the importance of upstream retrieval precision.
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Scalable Medical Multimodal Fusion via Symmetric Consistency Modeling
Xiaowen Sun, Hui Liu, Gongguan Chen, Ning Mao
Medical diagnosis tasks often rely on heterogeneous information from multiple sources, such as medical images and clinical text. Multimodal fusion is therefore essential for improving classification performance and robustness. However, most existing methods assume a fixed and known modality set, making them less effective when the number or composition of modalities changes. To address this limitation, we propose a modality-agnostic medical multimodal fusion framework that can naturally accommodate an arbitrary number of input modalities. At the coarse-grained modality level, we represent each modality’s estimation of latent semantics as an uncertainty-aware probability distribution, and impose symmetric consistency constraints to achieve global cross-modal semantic alignment. At the fine-grained token level, we further design a token-level consistency constraint based on linear reconstruction. This constraint enables structured mutual verification of local semantics across modalities. Finally, for multimodal fusion, we adopt a multi-view consistency strategy to obtain a unified representation for diagnosis prediction. In particular, each modality is sequentially treated as a conditional view to suppress noise in the remaining modalities and extract shared semantics. Extensive experiments on five public and self-constructed multimodal medical datasets demonstrate the effectiveness and scalability of the proposed approach. Code is available at https://github.com/gjhgjbkg/SMMF.
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Universal Approximation with Softmax Attention
Jerry Yao-Chieh Hu, Hude Liu, Hong-Yu Chen, Weimin Wu, Han Liu
We prove that with linear transformations, both (i) two-layer self-attention and (ii) one-layer self-attention followed by a softmax function are universal approximators for continuous sequence-to-sequence functions on compact domains. Our main technique is a new interpolation-based method for analyzing attention’s internal mechanism. This leads to our key insight: self-attention is able to approximate a generalized version of ReLU to arbitrary precision, and hence subsumes many known universal approximators. Building on these, we show that two-layer multi-head attention or even one-layer multi-head attention followed by a softmax function suffices as a sequence-to-sequence universal approximator. In contrast, prior works rely on feed-forward networks to establish universal approximation in Transformers. Furthermore, we extend our techniques to show that, (softmax-)attention-only layers are capable of approximating gradient descent in-context. We believe these techniques hold independent interest.
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Teaching Models to Teach Themselves: Reasoning at the Edge of Learnability
Shobhita Sundaram, John Quan, Ariel Kwiatkowski, Kartik Ahuja, Yann Ollivier, Julia Kempe
RL methods for scaling large reasoning models stall on datasets with low initial success rates, and thus little training signal. We investigate a fundamental question:Can a pretrained LLM leverage latent knowledge to generate an automated curriculum for problems it cannot solve? We explore this with SOAR: An asymmetric self-play framework that uses meta-RL to surface these pedagogical signals. A teacher model proposes synthetic problems for a student model, and is rewarded with its improvement on a subset of hard problems, thus grounding the curriculum in real student progress rather than intrinsic proxy rewards. Our study on the hardest subsets of math benchmarks (0/128 success) reveals three core findings. First, it is possible to realize bilevel meta-RL that unlocks learning under sparse, binary rewards by sharpening a latent capacity of pretrained models to generate useful problems. Second, grounded rewards outperform intrinsic learnability rewards used in prior LLM self-play, reliably avoiding typical instability and diversity collapse modes. Third, the structure and well-posedness of questions are more critical for learning progress than solution correctness. Our results suggest that the ability to generate useful stepping stones does not require the pr

eexisting ability to solve the hard problems, paving a principled path to escape reasoning plateaus without additional curated data.
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Which Algorithms Can Graph Neural Networks Learn?
Solveig Wittig, Antonis Vasileiou, Robert R Nerem, Timo Stoll, Floris Geerts, Yusu Wang, Christopher Morris
In recent years, there has been growing interest in understanding neural architectures' ability to learn to execute discrete algorithms, a line of work often referred to as neural algorithmic reasoning. The goal is to integrate algorithmic reasoning capabilities into larger neural pipelines. Many such architectures are based on (message-passing) graph neural networks (MPNNs), owing to their permutation equivariance and ability to deal with sparsity and variable-sized inputs. However, much existing work is either largely empirical and lacks formal guarantees or it focuses solely on expressivity, leaving open the question of when and how such architectures generalize beyond a finite training set. In this work, we propose a general theoretical framework that characterizes sufficient conditions under which MPNNs can learn an algorithm from a training set of small instances and provably approximate its behavior on inputs of arbitrary size with worst-case guarantees. Our framework applies to a broad class of algorithms, including single-source shortest paths, minimum spanning trees, and general dynamic programming problems, such as the 0-1 knapsack problem. In addition, we establish impossibility results for a wide range of algorithmic tasks, showing that standard MPNNs cannot learn them and derive more expressive MPNN-like architectures that overcome these limitations. Finally, we refine our analysis for the Bellman–Ford algorithm, yielding substantially smaller required training sets and significantly extending the recent work of Nerem et al., 2025 by allowing for a differentiable regularization loss. Empirical results largely support our theoretical findings.
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Aitchison Embeddings for Learning Compositional Graph Representations
Nikolaos Nakis, Chrysoula Kosma, Panagiotis Promponas, Michail Chatzianastasis, Giannis Nikolentzos
Representation learning is central to graph machine learning, powering tasks such as link prediction and node classification. However, most graph embeddings are hard to interpret, offering limited insight into how learned features relate to graph structure. Many networks naturally admit a role-mixture view, where nodes are best described as mixtures over latent archetypal factors. Motivated by this structure, we propose a compositional graph embedding framework grounded in Aitchison geometry, the canonical geometry for comparing mixtures. Nodes are represented as simplex-valued compositions and embedded via isometric log-ratio (ILR) coordinates, which preserve Aitchison distances while enabling unconstrained optimization in Euclidean space. This yields intrinsically interpretable embeddings whose geometry reflects relative trade-offs among archetypes and supports coherent behavior under component restriction; we consider both fixed and learnable ILR bases. Across node classification and link prediction, our method achieves competitive performance with strong baselines while providing explainability by construction rather than post hoc. Finally, subcompositional coherence enables principled component restriction: removing and renormalizing subsets preserves a well-defined geometry, which we exploit via subcompositional dimensionality removal to probe how archetype groups influence representations and predictions.
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PPT-Eval: A Benchmark for Computer-Use Agents on PowerPoint Tasks
Apurva Gandhi, Vishwas Suryanarayanan, Raja Hasnain Anwar, Firoz Shaik, Shubhang Desai, Thong Q. Nguyen, Muhammad Taqi Raza, Vishal Chowdhary, Graham Neubig
Creating and editing slides is a rich, multimodal activity that is ubiquitous in professional and educational settings, making it an ideal testbed for real-world computer-use agents. Microsoft PowerPoint is among the most widely adopted and feature-rich environments for presentation creation. We introduce PPT-Eval, a benchmark of 120 PowerPoint tasks across 12 files that cover both content creation and presentation editing scenarios, organized by difficulty. A central challen

ge in this domain is evaluation: tasks are complex, multimodal, and often admit many valid solutions. Moreover, today’s agents frequently make only partial progress, which binary success metrics fail to capture. To address this, we design a robust evaluation framework to help create task-specific rubrics for PowerPoint tasks, taking inspiration from and building on past works for rubric-based evaluation. These rubrics award partial credit for intermediate steps, penalize unnecessary changes and poor aesthetics, and provide natural language feedback. This nuanced approach proves highly effective, achieving a Kendall's τ_b correlation of 0.77 with human judgments. We find that existing frontier agents still struggle with solving PowerPoint tasks, with strong models like Claude-4.5-Opus achieving only a 45\% success rate and an average partial score of 57\%.
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MuonSSM: Orthogonalizing State Space Models for Sequence Modeling
Thai Khanh Nguyen, Vo Ngoc Bich Uyen, Thieu Vo, Tan Minh Nguyen, Cuong Pham
State space models (SSMs) have emerged as efficient linear-time alternatives to attention for long-sequence modeling. However, existing SSMs often suffer from instability and memory degradation over extended horizons due to poorly conditioned first-order updates and unbalanced update geometry. We introduce MuonSSM, a general framework that stabilizes SSM training by explicitly conditioning the geometry of memory updates rather than the recurrent transition matrix. MuonSSM augments SSMs with a momentum-based pathway and a lightweight Newton-Schulz transformation on low-rank input injections, yielding bounded and spectrally conditioned updates while preserving parallel scan complexity. Theory shows that MuonSSM improves gradient propagation, mitigates spectral amplification, and enriches memory representations over long horizons. Extensive experiments across language, vision, and time-series benchmarks show consistent gains in accuracy, robustness, and long-context performance when integrated into diverse SSM backbones. These results establish geometric conditioning of updates as a principled pathway to stable, scalable sequence modeling.
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ViTok-v2: Scaling Native Resolution Autoencoders to 5 Billion Parameters
Philippe Hansen-Estruch, Jiahui Chen, Vivek Ramanujan, Orr Zohar, Markos Georgopoulos, Animesh Sinha, Ji Hou, Edgar Schönfeld, Felix Juefei-Xu, Sriram Vishwanath, Ali Thabet
Vision Transformer (ViT) tokenizers offer a scalable alternative to convolutional auto-encoders, yet current architectures have two key limitations: their performance degrades when images vary in aspect ratio or resolution, and their reliance on adversarial losses makes them harder to train at scale. To address this, we introduce ViTok-v2, a ViT tokenizer building on ViTok. We add native resolution support via NaFlex with 2D RoPE and stabilize training by replacing the standard LPIPS-plus-discriminator objective with our novel DINO perceptual loss. We scale our model to 5B parameters, training the largest ViT-based image compression autoencoder to date and demonstrate continued improvements with scale. In downstream generation experiments with flow matching models, we find that smaller generators perform best with aggressive channel compression while larger generators effectively leverage higher channel counts. ViTok-v2 matches state-of-the-art reconstruction at 256p and outperforms across benchmarks at 512p and higher resolutons, while remaining compatible with any pipeline requiring flexible aspect ratios.
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Constitutional Black-Box Monitoring for Scheming in LLM Agents
Simon Storf, Rich Barton-Cooper, James Peters-Gill, Marius Hobbhahn

Safe deployment of Large Language Model (LLM) agents in autonomous settings requires reliable oversight mechanisms. A central challenge is detecting *scheming*, where agents covertly pursue misaligned goals. One approach to mitigating such risks is LLM-based monitoring: using language models to examine agent behaviors for suspicious actions. We study *constitutional black-box monitors*: prompted classifiers that detect scheming using only externally observable inputs and outputs, optimized on synthetic data generated from natural-language behavior specifications. We introduce two pipelines for generating synthetic agent trajectories, *STRIDE* (iterative refinement) and *Gloom* (agent-environment simulation), from which we generate 1,000 samples each. We optimize monitors on these datasets via prompt sweeps, human refinement, and automated prompt optimization, and evaluate performance on 7,500 held-out trajectories from Control Arena, a suite of grounded environments where agents operate in more realistic contexts. Our results demonstrate that monitors selected purely on synthetic data can generalize to more realistic environments, capturing a meaningful scheming signal. However, we find that performance saturates quickly in our setting, with simple prompt sweeps matching the results of more extensive optimization. Pushing beyond this limit yields no further improvements and instead leads to overfitting.

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ProMeCD: Unifying Long-Tailed and Noisy Label Learning via White-Box Control

Yu Zhu

Real-world data is rarely clean; it is plagued by severe class imbalance (long-tailed distributions) and label corruption. Current solutions lean heavily on "black-box" meta-learning to re-weight samples. However, this paradigm introduces a fatal circular dependency: it relies on pristine, balanced validation sets to guide the optimization, which are essentially non-existent in the wild. We propose ProMeCD, a self-referential framework that breaks this dependency by recasting optimization as an autonomous control problem. Instead of training an opaque neural meta-learner, we employ a transparent proportional-integral controller. The system monitors ``cognitive entropy'' that is a metric derived from von Mises-Fisher gradient statistics to assess learning uncertainty. To resolve the scalar ambiguity between tail and noisy samples, ProMeCD employs a decoupled control strategy: it boosts tail classes via integral accumulation of magnitude deficits when directional consistency is high, while suppressing noise via proportional feedback when consistency collapses. Theoretically, we prove that this mechanism guarantees convergence and formally prevents the minority initial drop, ensuring monotonic improvement for rare classes. Crucially, ProMeCD is fully white-box and validation-free. Experiments on CIFAR-LT, iNaturalist, CIFAR-N, mini WebVision, and Clothing1M confirm that ProMeCD is not merely efficient; it outperforms the recent meta-learner FMW-Net by over 10\% in severe imbalance settings, proving that explicit control theory offers a superior path to handling imperfect data.

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MusicDET: Zero-Shot AI-Generated Music Detection

Chaolei Han, Hongsong Wang, Jie Gui

Detecting AI-generated music is crucial for preserving artistic authenticity and preventing the misuse of generative music technologies. However, existing discriminative detectors typically rely on generated samples during training and often suffer from severe performance degradation when confronted with music produced by unseen generators, which limits their real-world applicability.

To address this issue, we formulate a zero-shot setting for AI-generated music detection, where the detector is trained exclusively on real music without access to any generated samples. Under this setting, we propose MusicDET, a generator-agnostic detection framework based on frequency-guided normalizing flows that probabilistically models the distribution of real music features. By evaluating the likelihood of an input sample under the learned real-music distribution, MusicDET enables effective detection of out-of-distribution music signals.

Experiments on the FakeMusicCaps and SONICS datasets show that MusicDET consistently outperforms conventional discriminative detectors, particularly when detecting music generated by previously unseen models.

The code is at https://github.com/Chaolei98/MusicDET

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Learning to Bet for Horizon-Aware Anytime-Valid Testing
Ege Onur Taga, Samet Oymak, Shubhanshu Shekhar
We develop horizon-aware anytime-valid tests and confidence sequences for bounded means under a strict deadline N. Using the betting/e-process framework, we cast horizon-aware betting as a finite-horizon optimal control problem with state space $(t, \log W_t)$, where t is the time and W_t is the test martingale value. We first show that in certain interior regions of the state space, policies that deviate significantly from Kelly betting are provably suboptimal, while Kelly betting reaches the threshold with high probability. We then identify sufficient conditions showing that outside this region, more aggressive betting than Kelly can be better if the bettor is behind schedule, and less aggressive can be better if the bettor is ahead. Taken together these results suggest a simple phase diagram in the $(t, \log W_t)$ plane, delineating regions where Kelly, fractional Kelly, and aggressive betting may be preferable. Guided by this phase diagram, we introduce a Deep Reinforcement Learning approach based on a universal Deep Q-Network (DQN) agent that learns a single policy from synthetic experience and maps simple statistics of past observations to bets across horizons and null values. In limited-horizon experiments, the learned DQN policy yields state-of-the-art results.
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Robust Filter Attention: Self-Attention as Precision-Weighted State Estimation
Peter Racioppo
We introduce Robust Filter Attention (RFA), a formulation of self-attention as a robust state estimator. Each token is treated as a noisy observation of a latent trajectory governed by a linear stochastic differential equation (SDE), and attention weights are determined by consistency under this model rather than static feature similarity. Under isotropic noise and decay assumptions, RFA matches the computational complexity of standard attention. On language modeling benchmarks, RFA achieves lower perplexity than RoPE within the training window while remaining stable under zero-shot extrapolation to longer contexts. The framework also provides a dynamical interpretation of standard positional mechanisms, connecting rotational embeddings and recency biases to transport and uncertainty propagation induced by stochastic dynamics.
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DARC: Disagreement-Aware Alignment via Risk-Constrained Decoding
mingxi Zou, Jiaxiang Chen, Junfan Li, Langzhang Liang, Qifan Wang, Xu Yinghui, Zenglin Xu
Preference-based alignment methods (e.g., RLHF, DPO) typically optimize a single scalar objective, implicitly averaging over heterogeneous human preferences. In practice, systematic annotator and user-group disagreement makes mean-reward maximization brittle and susceptible to proxy over-optimization. We propose **Disagreement-Aware Alignment via Risk-Constrained Decoding (DARC)**, a retraining-free inference-time method that frames response selection as distributionally robust, risk-sensitive decision making. Given multiple preference samples or scalable disagreement proxies, DARC reranks candidates by maximizing a *KL-robust (entropic)* satisfaction objective, and provides simple deployment controls that cap or penalize the corresponding entropic risk premium relative to the mean, enabling explicit risk budgets without retraining. We provide theoretical characterization linking this decoding rule to principled pessimism and KL-based distributionally robust optimization. Experiments on alignment benchmarks show that DARC reduces disagreement and tail risk while maintaining competitive average quality under noisy, heterogeneous feedback.
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MindFlow: Mind Supernet Powered Thinking Flows for Research Idea Innovation
Mengdi Liu, Wenjue Chen, Wenyue Chen, Cheng Yang, Fanqi Kong, Zhangyang Gao, Xiaoxue Cheng, Yiheng Li, Yujian Yuan, Keliang Li, Hong Chang, Shiguang Shan, Chenglin Wu
Research idea innovation is a fundamental engine of scientific progress, yet it remains difficult to generate and evaluate in a scalable and controllable way. T

his challenge lies in its inherently open-ended and multi-objective nature, where ideas should balance novelty, plausibility and feasibility. While recent LLM-based approaches have made progress through carefully designed prompts or agent pipelines, they are constrained by predefined, static ideation workflows. To address this limitation, we propose MindFlow, a framework that explicitly formulates ideation as a graph-structured Flow in Mind, which is composed of modular thinking operators and modeled by a probabilistic mind supernet. Given a research topic, a controller dynamically samples thinking flows to generate candidate ideas. This open-ended problem is optimized using a tournament-based relative ranking, enabling the controller to progressively favor higher-quality thinking flows. We further introduce an evaluation protocol that jointly assesses problem finding and problem solving, going beyond title- or abstract-only judgments. Across diverse topics, MindFlow shows its superiority as an explicit, controllable and optimizable research idea innovator.
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WorldPlay: Towards Long-Term Geometric Consistency for Real-Time Interactive World Modeling

Wenqiang Sun, Haiyu Zhang, Haoyuan Wang, Junta Wu, Zehan Wang, Zhenwei Wang, Yunhong Wang, Jun Zhang, Tengfei Wang, Chunchao Guo

This paper presents WorldPlay, a streaming video diffusion model that enables real-time, interactive world modeling with long-term geometric consistency, resolving the trade-off between speed and memory that limits current methods. WorldPlay draws power from three key ingredients. 1) We use a Dual Action Representation to enable robust action control in response to the user's keyboard and mouse inputs. 2) To enforce long-term consistency, our Reconstituted Context Memory dynamically rebuilds context from past frames and uses temporal reframing to keep geometrically important but long-past frames accessible, effectively alleviating memory attenuation. 3) We also propose Context Forcing, a novel distillation method designed for memory-aware model. Aligning memory context between the teacher and student preserves the student's capacity to use long-range information, enabling real-time speeds while preventing error drift. Taken together, WorldPlay generates long-horizon streaming 720p video at 24 FPS with superior consistency, comparing favorably with existing techniques and showing strong generalization across diverse scenes. Project page and online demo can be found: https://3d-models.hunyuan.tencent.com/world/ and https://3d.hunyuan.tencent.com/sceneTo3D.
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AMA-Bench: Evaluating Long-Horizon Memory for Agentic Applications

Yujie Zhao, Boqin Yuan, Junbo Huang, Haocheng Yuan, Zhongming Yu, Haozhou Xu, Lanxiang Hu, Abhilash Shankarampeta, Zimeng Huang, Wentao Ni, Yuandong Tian, Jishen Zhao

Large Language Models (LLMs) are increasingly used as autonomous agents in complex, long-horizon applications, where effective memory is critical for sustained performance. Yet existing memory benchmarks are largely dialogue-centric, while real agent memory consists of continuous agent-environment interaction trajectories composed of states, actions, observations, and tool outputs. To address this gap, we introduce **AMA-Bench** (**A**gent **M**emory with **A**ny length), a benchmark for evaluating long-horizon memory in realistic agentic settings. AMA-Bench combines real-world agent trajectories from representative applications with expert-curated QA, as well as synthetic trajectories that scale to arbitrary horizons with rule-based QA. Our study shows that existing memory systems underperform because they fail to capture causal and objective information and rely heavily on lossy similarity-based retrieval. We further propose **AMA-Agent**, a memory system based on causality-graph construction and tool-augmented retrieval. AMA-Agent achieves **57.22%** accuracy on AMA-Bench, outperforming the strongest baseline by **11.16%**. Resources are available at: https://ama-bench.github.io/.
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LMM4-IC4K: A Large Multimodal Model Powered Integrated Circuit Footprint Geometry Understanding

Yida Wang, Taiting Lu, Runze Liu, Lanqing Yang, Zhe Chen, Yuehai Wang, Yixin Liu

, Kaiyuan Lin, XiaoMeng Chen, Dian Ding, Yijie Li, Yifan Yang, Yi-Chao Chen, yin cheng jin, Mahanth Gowda
Printed-Circuit-board (PCB) footprint geometry labeling of integrated circuits (IC) is essential in defining the physical interface between components and the P CB layout, requiring precise visual perception. However, the unstructured nature of footprint drawings and abstract diagram annotations prevents direct IC footp rint parsing and automated package geometry labeling methods from developing. Ex isting Large Multimodal Models (LMMs) struggle with inaccurate geometric percept ion, limiting their effectiveness in this task. To address these challenges, we propose LMM4-IC4K, a novel framework that treats IC mechanical drawings as image s and leverages LMMs for structured geometric interpretation. To support such a framework, we introduce ICGeo8K, a multi-modal dataset with 8,608 labeled sample s, including 4138 real-world IC footprint samples and 4470 synthetically generat ed samples. We further present a two-stage training framework to fine-tune LMMs for IC footprint labeling. Extensive experiments demonstrate that our model outp erforms state-of-the-art LMMs on the proposed benchmark. The accurate translatio n of footprint diagrams enabled by LMM4-IC4K contributes to advancing automation and standardization within the PCB industry.
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Interactive Segmentation with Elaborate Focus Prior
Kangpeng Hu, Yinghui Sun, Tao Wang, Weihao Zhang, Quansen Sun
Regional refinement for interactive segmentation is of great necessity to ensure the fidelity of segmented pixels nearby user-prompted locations, which specifie s a local window (focus view) for the latest click after a global prediction, w here local pixels are revisited and optimized using numerous refining structures . Previous methods either utilize a two-stage pipeline to estimate the focus vie w or manually preset a fixed scope for all clicks, while the former is time-cons uming, the latter fails to capture the correlation among click position, object geometry, and focus intensity. In this paper, we inherit the core idea of FCFI a nd dedicate a one-stage framework characterized with Elaborate Focus Prior (EFPN et). Concretely, EFPNet outputs an erroneous mask w.r.t historical feedback and newly-placed click in an end-to-end manner, which deduces precise focus region a ccording to its max-connected component, followed with feedback correction consi dering image, feature and mask affinity. We further design a clicked-with-focus mechanism for efficient feedback integration. Extensive studies on four benchmar ks have revealed outstanding performance of EFPNet for both efficacy and computa tional overhead.
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Prior Diffusiveness and Regret in the Linear-Gaussian Bandit
Yifan Zhu, John Duchi, Benjamin Van Roy
We prove that Thompson sampling exhibits $\tilde{O}(\sigma d \sqrt{T} + d r \sqr t{\mathrm{Tr}(\Sigma_0)})$ Bayesian regret in the linear-Gaussian bandit with a $\mathcal{N}(\mu_0, \Sigma_0)$ prior distribution on the coefficients, where d is the dimension, T is the time horizon, r is the maximum ℓ_2 norm of the actions, and σ^2 is the noise variance. In contrast to existing regre t bounds, this shows that to within logarithmic factors, the prior-dependent ''b urn-in'' term $d r \sqrt{\mathrm{Tr}(\Sigma_0)}$ decouples additively from the m inimax (long run) regret \sigma d \sqrt{T}. Previous regret bounds exhibit a mul tiplicative dependence on these terms. We establish these results via a new ''el liptical potential'' lemma, and also provide a lower bound indicating that the b urn-in term is unavoidable.
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Dynamics of neural scaling laws in random feature regression with powerlaw-distr ibuted kernel eigenvalues
Jakob Kramp, Javed Lindner, Moritz Helias
Training large neural networks exposes neural scaling laws for the generalizatio n error, which points to a universal behavior across network architectures of le arning in high dimensions. It was also shown that this effect persists in the li mit of highly overparametrized networks as well as the Neural network Gaussian p rocess limit. We here develop a principled understanding of the typical behavior

of generalization in Neural Network Gaussian process regression dynamics. We de
rive a dynamical mean-field theory that captures the typical case learning dynam
ics: This allows us to unify multiple existing regimes of learning studied in th
e current literature, namely Bayesian inference on Gaussian processes, gradient
flow with or without weight-decay, and stochastic Langevin training dynamics. Em
ploying tools from statistical physics, the unified framework we derive in eithe
r of these cases yields an effective description of the high-dimensional microsc
opic behavior of networks dynamics in terms of lower dimensional order parameter
s. We show that collective training dynamics may be separated into the dynamics
of N independent eigenmodes, whose evolution equations are only coupled through
collective response functions and a common statistics of an effective, independe
nt noise. Our approach allows us to quantitatively explain the dynamics of the g
eneralization error by linking spectral and dynamical properties of learning on
data with power law spectra, including phenomena such as neural scaling laws and
the effect of early stopping.
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GFMate: Empowering Graph Foundation Models with Test-time Prompt Tuning
Yan Jiang, Ruihong Qiu, Zi Huang
Graph prompt tuning has shown great potential in graph learning by introducing t
rainable prompts to enhance the model performance in conventional single-domain
scenarios. Recent research has extended graph prompts to improve Graph Foundatio
n Models (GFMs) by few-shot tuning auxiliary prompts. Despite their progress, mo
st existing methods embed source-domain information into prompts, which serve ei
ther as input to GFMs or encoded during model pre-training. Such prompt entangle
ment with specific source domains and GFM pre-training strategy restricts their
generalisability to other domains and different GFMs. Furthermore, existing GFM
prompts merely rely on few-shot tuning for adaptation, neglecting the rich infor
mation in unlabelled target domain test data. Motivated by these insights, this
paper aims to empower GFMs with pre-training-agnostic test-time graph prompt tun
ing, named GFMate. GFMate introduces centroid and layer prompts applied after pr
e-training on target domains, avoiding entanglement with specific source domains
and model pre-training. In addition, a test-time complementary learning objecti
ve is devised to exploit both labelled and unlabelled target domain data for eff
ective test-time prompt tuning. Extensive experiments on 12 benchmark datasets d
emonstrate the superior performance and efficiency of GFMate, achieving improvem
ents of up to 30.63%. Code is available at https://github.com/YanJiangJerry/GFMa
te.
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Desirable Effort Fairness and Optimality Trade-offs in Strategic Learning
Valia Efthymiou, Ekaterina Fedorova, Chara Podimata
Strategic classification examines how decision rules interact with agents who st
rategically adapt their features. Most existing models focus on maximizing predi
ctive performance, assuming agents best respond to the learned classifier. Howev
er, real decision-making systems are rarely optimized solely for accuracy: ethic
al, economic, and institutional considerations often make some feature changes m
ore desirable than others. At the same time, principals may wish to incentivize
these changes fairly across heterogeneous agents.
While prior work has studied causal structure between features, notions of desir
ability, and information disparities in isolation, this work initiates a unified
treatment of these components within a single framework. We frame the problem a
s a constrained optimization problem that captures the trade-offs between optima
lity, desirability, and fairness.
We provide theoretical guarantees on the principal's optimality loss constrained
to a particular desirability fairness tolerance for multiple broad classes of f
airness measures. Finally, through experiments on real datasets, we show the exp
licit tradeoff between maximizing accuracy and fairness in desirability effort.
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An analytic theory of convolutional neural network inverse problems solvers
Minh Hai Nguyen, Quoc Bao Do, Edouard Pauwels, Pierre Weiss
Supervised convolutional neural networks (CNNs) are widely used to solve imaging

inverse problems, achieving state-of-the-art performance in numerous applications. However, despite their empirical success, these methods are poorly understood from a theoretical perspective and often treated as black boxes. To bridge this gap, we analyze trained neural networks through the lens of the Minimum Mean Square Error (MMSE) estimator, incorporating functional constraints that capture two fundamental inductive biases of CNNs: translation equivariance and locality via finite receptive fields. Under the empirical training distribution, we derive an analytic, interpretable, and tractable formula for this constrained variant, termed Local-Equivariant MMSE (LE-MMSE). Through extensive numerical experiments across various inverse problems (denoising, inpainting, deconvolution), datasets (FFHQ, CIFAR-10, FashionMNIST), and architectures (U-Net, ResNet, PatchMLP), we demonstrate that our theory matches the neural networks outputs (PSNR $\gtrsim25$ dB). Furthermore, we provide insights into the differences between *physics-aware* and *physics-agnostic* estimators, the impact of high-density regions in the training (patch) distribution, and the influence of other factors (dataset size, patch size, *etc*).
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Learning Transferable Interaction Primitives from Game Videos for Humanoid Locomotion
Xiangming Zhu, Huayu Deng, Haoran Zhao, Yiwei Hao, Yunbo Wang
Learning humanoid control from video provides a scalable alternative to the scarcity of high-fidelity robot data. Existing methods, however, often rely on curated datasets and treat video as passive kinematic priors. They fail to capture dynamic humanoid interactions with the environment, which are essential for robust control in complex physical environments. To address this, we propose ***TR**ansferable **I**nteraction **P**rimitives (TRIP)*, a framework designed to extract and ground interactions from unlabeled game videos for locomotion control. TRIP explicitly models dependencies between motion dynamics and environmental context via a discrete library of interaction-based action primitives. To bridge the reality gap, we introduce a shared context latent space that aligns implicit video-domain features with functional target-domain observations, enabling the seamless transfer of video-mined strategies to reinforcement learning policies. Our experiments on complex terrain navigation demonstrate that TRIP achieves significant improvements in task performance, sample efficiency, and robustness.
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GRAPE: Let GRPO Supervise Query Rewriting by Ranking for Retrieval
Zhaohua Zhang, Jianhuan Zhuo, Muxi Chen, Chenchen Zhao, Wenyu Jiang, Mingyang Chen, Yu Tang, Qiuyong Xiao, Tianwen Jiang, Jihong Zhang, Zhixun Su
The CLIP model has established itself as a cornerstone of large-scale retrieval systems. However, its performance often degrades under distributional shifts such as multilingual, long-form, or multimodal queries. To avoid the prohibitive costs associated with retriever retraining or corpus re-embedding, we propose GRAPE (Grouped Ranking-Aware Policy Optimization Enhancement), a plug-and-play approach that leverages LLM-based query rewriting to bridge these gaps. Unlike existing methods that lack explicit supervision, GRAPE integrates ranking signals into the rewriting LLM via Grouped Relative Policy Optimization (GRPO), ensuring rewritten queries are better aligned with the frozen retriever’s latent distribution. Crucially, we identify a score inflation phenomenon in naive similarity-based finetuning—where irrelevant candidates receive indiscriminately high scores—and mitigate it with a novel corpus-relative ranking-based reward. Extensive experiments across multilingual (Flickr30k-CN, CVLUE, XM3600), long-form (Wikipedia), and multimodal (CIRR) benchmarks demonstrate that GRAPE consistently improves performance, achieving an average gain of 4.9% in Recall@10 without any modification to the underlying retriever.The code is available at https://github.com/mogulzhang/GRAPE.
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Achieving Logarithmic Regret in KL-Regularized Zero-Sum Markov Games
Anupam Nayak, Tong Yang, Osman Yagan, Gauri Joshi, Yuejie Chi
Reverse Kullback–Leibler (KL) divergence-based regularization with respect to a fixed reference policy is widely used in modern reinforcement learning to preser

ve the desired traits of the reference policy and sometimes to promote explorati on (using uniform reference policy, known as entropy regularization). Beyond ser ving as a mere anchor, the reference policy can also be interpreted as encoding prior knowledge about good actions in the environment. In the context of alignme nt, recent game-theoretic approaches have leveraged KL regularization with pretr ained language models as reference policies, achieving notable empirical success in self-play methods. Despite these advances, the theoretical benefits of KL re gularization in game-theoretic settings remain poorly understood. In this work, we develop and analyze algorithms that provably achieve improved sample efficien cy under KL regularization. We study both two-player zero-sum Matrix games and M arkov games:
for Matrix games, we propose OMG, an algorithm based on best response sampling w ith optimistic bonuses, and extend this idea to Markov games through the algorit hm SOMG, which also uses best response sampling and a novel concept of superopti mistic bonuses. Both algorithms achieve a logarithmic regret in T that scales inversely with the KL regularization strength β
in addition to the traditional $\widetilde{\mathcal{O}}(\sqrt{T})$ regret withou t the β^{-1} dependence.
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The Signal is in the Steps: Local Scoring for Reasoning Data Selection
Hoang Anh Just, Myeongseob Ko, Ruoxi Jia
Distilling long-form reasoning from teacher models into smaller students require s selecting which candidate solutions to train on. Recent work argues that one s hould select responses the student model assigns highest probability, i.e., favo ring solutions ``natural'' to the student. However, we find that this approach w orks within a single teacher but fails when scaling to long reasoning traces fro m multiple diverse teachers. We identify a key cause: this approach scores entir e solutions, but students generalize by recombining familiar reasoning steps, no t by memorizing complete solutions. Full-trajectory scoring optimizes the wrong target; it rewards global fluency while the transferable signal lies in local st ep transitions. We propose Local Average Log Probability (LALP), which scores ea ch reasoning step using only a small window of preceding context, measuring whet her each step is justified by its immediate premises rather than whether the ful l response looks natural to the student. LALP enables two practical use cases: s electing the best teacher before fine-tuning and curating training data from div erse teacher pools. Across math, coding, and science reasoning tasks, LALP consi stently improves accuracy when selecting the most natural solutions by a large m argin.
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Failure-Driven Workflow Refinement
Jusheng Zhang, Kaitong Cai, Jing Yang, Ziliang Chen, Yongsen Zheng, Kwok-Yan Lam , Liang Lin, Keze Wang
Workflow optimization for tool-using LLM agents is often cast as global search o ver candidate graphs, scored by a scalar metric. This collapses rich, multi-step failure traces into binary outcomes, obscuring recurring failure structure and making refinement inefficient. We reframe optimization as \emph{distributional r efinement}: each workflow induces a density over a \textbf{Failure Signature Spa ce} $\mathcal{F}$, and the goal is to minimize its \textbf{Expected Failure Mass }. We propose \textbf{CE-Graph}, which maintains a counterexample pool, estimate s dense failure modes, and applies operator-constrained graph edits via a \textb f{Propose-and-Verify} loop with a convergence-aware stopping rule. Across math, code, and QA benchmarks, CE-Graph improves robustness while reducing optimizatio n cost compared to strong workflow-search baselines, suggesting reliability emer ges from learning and reshaping failure landscapes rather than merely maximizing aggregate success rates.
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Does Math Reasoning Improve General LLM Capabilities? Understanding Transferabil ity of LLM Reasoning
Maggie Ziyu Huan, Yuetai Li, Tianyu Zheng, Xiaoyu Xu, Seungone Kim, Minxin Du, R adha Poovendran, Graham Neubig, Xiang Yue

Math reasoning has become the poster child of progress in large language models (LLMs), with new models rapidly surpassing human-level performance on benchmarks like MATH and AIME. But as math leaderboards improve week by week, it is worth asking: do these gains reflect broader problem-solving ability or just narrow overfitting? To answer this question, we evaluate over 20 open-weight reasoning-tuned models across a broad suite of tasks, including math, scientific QA, agent planning, coding, and standard instruction-following. We surprisingly find that most models that succeed in math fail to transfer their gains to other domains. To rigorously study this phenomenon, we conduct controlled experiments using math-only data with two widely-used methods: Reinforcement Learning (RL) and Supervised Finetuning (SFT) with detailed ablations. On top of the observation that RL-tuned models transfer better than SFT-tuned models, we identify on-policy fine-tuning as the key mechanism underlying cross-domain transfer, regardless of whether the training signal comes from RL or supervised learning. Latent-space representation and token-space distribution shift analyses reveal that off-policy SFT induces substantial representation and output drift, while on-policy RL preserves general-domain structure. Our results suggest a need to rethink the post-training recipes, particularly the reliance on off-policy SFT-distilled data to advance reasoning models.

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Train Once, Reuse Everywhere: Generalizable Implicit ICL by Routing Attention

Jiaqian Li, Yanshu Li, Ligong Han, Ruixiang Tang, Wenya Wang

Implicit in-context learning (ICL) has newly emerged as a promising paradigm that simulates ICL behaviors in the representation space of large language models (LLMs), aiming to attain few-shot performance at zero-shot cost. However, existing approaches largely rely on injecting shift vectors into residual flows, which are typically constructed from labeled demonstrations or task-specific alignment. Such designs fall short of utilizing the structural mechanisms underlying ICL and suffer from limited generalizability. To address this, we propose In-Context Routing (ICR), a novel implicit ICL method that internalizes generalizable ICL patterns at the attention logits level. It extracts reusable structural directions that emerge during ICL and employs a learnable input-conditioned router to modulate attention logits accordingly, enabling an efficient train-once-and-reuse framework. We evaluate ICR on 12 real-world datasets spanning diverse domains and multiple LLMs. The results show that ICR consistently outperforms existing implicit ICL methods that require task-specific retrieval or training, while demonstrating robust generalization to out-of-domain tasks where they struggle. These findings position ICR to push the boundary of the practical value of ICL.

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Accelerating Langevin Monte Carlo via Efficient Stochastic Runge-Kutta Methods beyond Log-Concavity

Bin Yang, Xiaojie Wang

Sampling from a high-dimensional probability distribution is a fundamental algorithmic task arising in wide-ranging applications across
multiple disciplines, including scientific computing, computational statistics and machine learning. Langevin Monte Carlo (LMC) algorithms are among the most widely used sampling methods in high-dimensional settings. This paper introduces a novel higher-order and Hessian-free LMC sampling algorithm based on an efficient stochastic Runge--Kutta method of strong order 1.5 for the overdamped Langevin dynamics. In contrast to the existing Runge--Kutta type LMC of Li et al. (2019) involved with three gradient evaluations, the newly proposed algorithm is computationally cheaper and requires only two gradient evaluations at each iteration. Under certain log-smooth conditions, non-asymptotic error bounds of the proposed algorithms are analyzed in $\mathcal{W}_2$-distance. In particular, a uniform-in-time convergence rate of order $O(d ^{\frac32} h^{\frac32})$ is derived in a non-log-concave setting, matching the convergence rate proved in the aforementioned work but under the log-concavity condition. Numerical experiments are finally presented to demonstrate the effectiveness of the new sampling algorithm.

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Milestone-Guided Policy Learning for Long-Horizon Language Agents

Zixuan Wang, Yuchen Yan, Hongxing Li, Teng Pan, Dingming Li, Ruiqing Zhang, Weiming Lu, Jun Xiao, Yueting Zhuang, Yongliang Shen
While long-horizon agentic tasks require language agents to perform dozens of sequential decisions, training such agents with reinforcement learning remains challenging. We identify two root causes: credit misattribution, where correct early actions are penalized due to terminal failures, and sample inefficiency, where scarce successful trajectories result in near-total loss of learning signal. We introduce a milestone-guided policy learning framework, BEACON, that leverages the compositional structure of long-horizon tasks to ensure precise credit assignment. BEACON partitions trajectories at milestone boundaries, applies temporal reward shaping within segments to credit partial progress, and estimates advantages at dual scales to prevent distant failures from corrupting the evaluation of local actions. On ALFWorld, WebShop, and ScienceWorld, BEACON consistently outperforms GRPO and GiGPO.
Notably, on long-horizon ALFWorld tasks, BEACON achieves 92.9\% success rate, nearly doubling GRPO's 53.5\%, while improving effective sample utilization from 23.7\% to 82.0\%. These results establish milestone-anchored credit assignment as an effective paradigm for training long-horizon language agents. Code is in supplementary materials and will be publicly released.
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ProtocolBench: Which LLM MultiAgent Protocol to Choose?
Hongyi Du, Jiaqi Su, Jisen Li, Lijie Ding, Yingxuan Yang, Peixuan Han, Xiangru Tang, Kunlun Zhu, Jiaxuan You
As large-scale multi-agent systems evolve, the communication protocol layer has become a critical yet under-evaluated factor shaping performance and reliability. Despite the existence of diverse protocols (A2A, ACP, ANP, Agora, etc.), the selection of them is often intuition-driven and lacks standardized guidance. We introduce ProtocolBench, a benchmark that systematically compares agent protocols along four measurable axes: task success, end-to-end latency, message or byte overhead, and robustness under failures. On ProtocolBench, the choice of protocol significantly influences system behavior. In the Streaming Queue scenario, overall completion time varies by up to 36.5\% across protocols, and mean end-to-end latency differs by 3.48 s. Under Fail-Storm Recovery, resilience also differs consistently across protocols. Beyond evaluation, we present ProtocolRouter, a lightweight constraint-aware protocol router that selects per-scenario (or per-module) protocols from requirement and runtime signals. ProtocolRouter reduces Fail-Storm recovery time by up to 18.1\% versus the best single-protocol baseline and achieves scenario-specific gains such as higher success in GAIA, while exposing trade-offs across other metrics. We also release ProtocolRouterBench to standardize constrained protocol-selection evaluation and improve reliability at scale.
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Probabilistic Salient Object Ranking
Rongjin Guo, Guan Huankang, Rynson W. H. Lau
Salient Object Ranking (SOR) aims to study how humans visually explore complex scenes by predicting an ordered sequence of objects that attracts our attention. Existing SOR approaches typically model this ranking deterministically, assuming a single, fixed ranking sequence of attention. However, such deterministic SOR fails to capture the true nature of human attention. We observe that human attention shifts exhibit variability and stochasticity, i.e., the next object of fixation is not a definitive choice but rather a probability distribution. Yet, existing SOR methods and evaluation metrics do not account for this inherent randomness. To address this fundamental problem, we first propose ProbSOR, a novel Probabilistic Salient Object Ranking framework built upon a vision-language model (VLM) backbone. By incorporating Group Relative Policy Optimization (GRPO), ProbSOR explicitly learns the uncertainty of attention shifts. We then propose a new metric tailored for ProbSOR, as existing SOR metrics only support deterministic rankings. We further construct a ProbSOR dataset comprising 15,000 probabilistic SOR samples, to support both model training and evaluation. Extensive experiments show that ProbSOR achieves strong performance in salient object ranking under

both our proposed and traditional benchmarks.
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PrivAct: Internalizing Contextual Privacy Preservation via Multi-Agent Preferenc e Training
Yuhan Cheng, Hancheng Ye, Hai Helen Li, Jingwei Sun, Yiran Chen
Large language model (LLM) agents are increasingly deployed in personalized task s involving sensitive, context-dependent information, where privacy violations m ay arise in agents' action due to the implicitness of contextual privacy.
Existing approaches rely on *external*, inference-time interventions which are b rittle, scenario-specific, and may expand the privacy attack surface.
We propose **PrivAct**, a contextual privacy-aware multi-agent learning framewor k that *internalizes* contextual privacy preservation directly into models' gene ration behavior for privacy-compliant agentic actions. By embedding privacy pref erences into each agent, PrivAct enhances system-wide contextual integrity while achieving a more favorable privacy-helpfulness tradeoff.
Experiments across multiple LLM backbones and benchmarks demonstrate consistent improvements in contextual privacy preservation, reducing leakage rates by up to 12.32\% while maintaining comparable helpfulness, as well as zero-shot generali zation and robustness across diverse multi-agent topologies.
Code is available at https://github.com/chengyh23/PrivAct.
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High-Accuracy Sampling for Diffusion Models and Log-Concave Distributions
Fan Chen, Sinho Chewi, Constantinos Costis Daskalakis, Alexander Rakhlin
We present algorithms for diffusion model sampling which obtain δ-error i n $\mathrm{polylog}(1/\delta)$ steps, given access to $\widetilde O(\delta)$-acc urate score estimates in L^2. This is an exponential improvement over all prev ious results. Specifically, under minimal data assumptions, the complexity is $\ widetilde O(d\mathrm{polylog}(1/\delta))$ where d is the dimension of the data ; under a non-uniform L-Lipschitz condition, the complexity is $\widetilde O(\ sqrt{dL}\mathrm{polylog}(1/\delta))$; and if the data distribution has intrinsic dimension $d_\star$, then the complexity reduces to $\widetilde O(d_\star\mathr m{polylog}(1/\delta))$. Our approach also yields the first $\mathrm{polylog}(1/\ delta)$ complexity sampler for general log-concave distributions using only grad ient evaluations.
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Less is Enough: Synthesizing Diverse Data in LLM Feature Space with Sparse Autoe ncoders
Zhongzhi Li, Xuansheng Wu, Yijiang Li, Lijie Hu, Ninghao Liu
The diversity of post-training data is critical for effective downstream perform ance in large language models (LLMs).
Many existing approaches to constructing post-training data quantify diversity u sing text-based metrics that capture linguistic variation, but such metrics prov ide only weak signals for the task-relevant features that determine downstream p erformance.
In this work, we introduce ***Feature Activation Coverage* (FAC)** which measure s data diversity in an interpretable feature space.
Building upon this metric, we further propose a diversity-driven data synthesis framework, named **FAC Synthesis**, that first uses a sparse autoencoder to iden tify missing features from a seed dataset, and then generates synthetic samples that explicitly reflect these features.
Experiments show that our approach consistently improves both data diversity and downstream performance on various tasks, including instruction following, toxic ity detection, reward modeling, and behavior steering.
Interestingly, we identify a shared, interpretable feature space across model fa milies (i.e., LLaMA, Mistral, and Qwen), enabling cross-model knowledge transfer .
Our work provides a solid and practical methodology for exploring data-centric o ptimization of LLMs.
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InertialAR: Autoregressive 3D Molecule Generation with Inertial Frames

Haorui Li, weitao Du, Yuqiang Li, Hongyu Guo, Shengchao Liu
Transformer-based autoregressive models have emerged as a unifying paradigm across modalities such as text and images, but their extension to 3D molecule generation remains underexplored. The gap stems from two fundamental challenges: (1) how to tokenize molecules into a canonical 1D sequence of tokens that is invariant to both SE(3) transformations and atom index permutations, and (2) how to design an architecture capable of modeling hybrid atom-based tokens that couple discrete atom types with continuous 3D coordinates. To address these challenges, we introduce InertialAR. It first performs generation-oriented canonical tokenization by aligning each molecule to a canonical inertial frame and reordering atoms, thereby converting arbitrary 3D structures into a unique, SE(3)- and permutation-invariant sequence of tokens for autoregressive generation. Built upon this canonical tokenization, we propose geometric positional encoding (GeoPE), which endows Transformer attention with 3D geometric awareness. Finally, InertialAR utilizes a hierarchical autoregressive paradigm to decode the next atom, consecutively predicting the atom type and 3D coordinates via Diffusion Loss. Experimentally, InertialAR achieves state-of-the-art performance on 8 of the 10 evaluation metrics for unconditional generation across QM9, GEOM-Drugs, and B3LYP. Moreover, it significantly outperforms baselines in controllable generation for targeted chemical functionality, attaining state-of-the-art results across all 5 metrics. Code is available at github.com/HaoruiLi46/InertialAR.
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Matrix-Free GPU Semidefinite Programming for Quantum Ordered Search at the k=6 Frontier
Yancheng Wu, Huikang Liu, Wenzhi Gao, Yuexin Su, Tongyang Li, Dongdong Ge, Yinyu Ye
Quantum computation offers the potential for a significant constant-factor speedup for the Ordered Search Problem (OSP). A classical construction is the k-query quantum ordered search algorithm, which can exactly search an N-element ordered list and achieves a query complexity improvement of a factor of $\frac{k}{\log_2 N}$. For larger k, stronger constant-factor improvements could be obtained by finding the largest admissible list size $N^\star$, a task that can be formulated as a structured semidefinite program (SDP). However, solving this SDP becomes computationally intractable beyond $k=6$, as existing CPU and GPU solvers rely on explicit construction of prohibitively large constraint matrices.
In this paper, we introduce a matrix-free GPU SDP framework that evaluates the highly structured constraints in OSP on-the-fly using custom CUDA kernels, reducing memory complexity from quadratic to linear and shifting the bottleneck from memory to computation. Using this approach, we provide strong numerical evidence that the optimal list size for $k=6$ is at least $90,000$, improving the empirical upper bound on the query coefficient from 0.390 to 0.365. We further rigorously certify the upper bound $N^\star < 94,000$ by constructing dual infeasibility certificates via matrix-free minimum-eigenvalue estimation.
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CentaurEval: Benchmarking Human-in-the-Loop Value in Agentic Coding
Hanjun Luo, Chiming Ni, Jiaheng Wen, Zhimu Huang, Bingduo Liao, Sylvia Chung, Yiran Wang, Yingbin Jin, Jialin Li, Xinfeng Li, Wenyuan Xu, XiaoFeng Wang, Hanan Salam
LLM-powered coding agents are reshaping the development paradigm. However, existing evaluation systems, neither traditional tests for humans nor benchmarks for LLMs, fail to capture this shift, excluding problems that require both human reasoning to guide solutions and AI efficiency for implementation. We introduce CentaurEval, a unified, ecologically valid benchmark for measuring human-in-the-loop value in coding. CentaurEval's core innovation is its "Collaboration-Necessary" problem templates, which are intractable for standalone LLMs or humans, but solvable through effective collaboration. CentaurEval dynamically instantiates tasks from 45 templates, providing a standardized IDE for humans and a reproducible 450-task toolkit for LLMs. We benchmark 45 participants against 5 LLMs under 4 levels of human intervention. Results show that while LLMs or humans alone achieve poor pass rates (0.67% and 18.89%), human–AI collaboration significantly impr

oves to 31.11%. Our analysis reveals an emerging co-reasoning partnership, chall enging the traditional human-tool hierarchy by showing that strategic breakthrou ghs can originate from either humans or AI.
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Bullet Trains: Parallelizing Training of Temporally Precise Spiking Neural Netwo rks
Todd Morrill, Christian-Gernot Pehle, Anthony M. Zador
Continuous-time, event-native spiking neural networks (SNNs) operate strictly on spike events, treating spike timing and ordering as the representation rather t han an artifact of time discretization. This viewpoint aligns with biological co mputation and with the native resolution of event sensors and neuromorphic proce ssors, while enabling compute and memory that scale with the number of events. H owever, two challenges hinder practical, end-to-end trainable event-based SNN sy stems: 1) exact charge--fire--reset dynamics impose inherently sequential proces sing of input spikes, and 2) precise spike times must be solved without time bin s. We address both. First, we use parallel associative scans to consume multiple input spikes at once, yielding up to 44x speedups over sequential simulation wh ile retaining exact hard-reset dynamics. Second, we implement differentiable spi ke time solvers that compute spike times to machine-precision without discrete-t ime approximations or restrictive analytic assumptions. We demonstrate the viabi lity of training SNNs using our solutions on four event-based datasets on GPUs.
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Efficient Skill Grounding via Code Refactoring with Small Language Models
Sera Choi, Wonje Choi, Saehun Chun, Daehee Lee, Jooyoung Kim, Chaeun Lee, Honguk Woo
Effective skill grounding is essential for deploying reusable skills in embodied agents, as even minor embodiment or environmental differences can render an ent ire skill incompatible.
This challenge is particularly pronounced in embodied settings, where agents mus t operate in dynamic, partially observable environments without access to large language models (LLMs).
In this setting, reliance on LLMs is impractical, while small language models (s LMs) remain insufficient for the effective skill grounding required for reliable long-horizon control.
We present RECENT, a refactoring-centric agent framework that enables efficient skill grounding with sLMs by decoupling skill semantics from embodiment- and env ironment-specific execution binding.
By representing skills as executable code, RECENT preserves the semantic intent encoded in a skill’s control structure while grounding it by modifying only exec ution bindings through localized refactoring, rather than regenerating code from scratch.
We evaluate RECENT across diverse skill grounding scenarios spanning multiple ro bot embodiments in dynamic environments, demonstrating robust long-horizon perfo rmance when deployed with an sLM.
Across all scenarios, RECENT achieves the best performance among sLM-based Code-as-Policies (CaP) methods and matches the task performance of LLM-based CaP.
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SlideSparse: Fast and Flexible (2N-2):2N Structured Sparsity
Yingbo Hao, Hanyong Shao, Ting Song, Yan Xia, Di Zhang, Shaohan Huang, Xun Wu, S ongchen Xu, Le Xu, Li Dong, Zewen Chi, Yi Zou, Furu Wei
NVIDIA's 2:4 Sparse Tensor Cores deliver $2\times$ throughput but demand strict 50% pruning—a ratio that causes severe accuracy loss in LLMs. Milder $(2N-2):2N$ patterns (e.g., 6:8, 25% pruning) preserve accuracy far better—within 0.4–1.8 a verage points of dense in our Qwen2.5-7B/14B study—yet receive *NO* hardware sup port and fall back to dense execution. We present **SlideSparse**, the first sys tem to unlock Sparse Tensor Core acceleration for the $(2N-2):2N$ model family o n commodity GPUs. Our *Sliding Window Decomposition* rewrites any $(2N-2):2N$ we ight block into $N-1$ overlapping 2:4-compliant windows without changing the und erlying dot product; in addition, our *Activation Lifting* fuses the correspondi ng activation rearrangement into per-token quantization at low marginal cost. In

tegrated into vLLM, **SlideSparse** is evaluated across various GPUs (A100, H100, B200, RTX 4090, RTX 5080, DGX-spark), precisions (FP4, INT8, FP8, BF16, FP16), and model families (Llama, Qwen, BitNet). On compute-bound workloads, the measured speedup ($1.33\times$) matches the theoretical upper-bound $N/(N-1)=4/3$ at 6:8 weight sparsity in Qwen2.5-7B, establishing $(2N-2):2N$ as a practical path to better accuracy–speedup trade-offs in LLM acceleration. Code available at https://github.com/bcacdwk/vllmbench.

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Federated Distillation for Whole Slide Image via Gaussian-Mixture Feature Alignment and Curriculum Integration

Luru Jing, Cong Cong, Yanyuan Chen, Yongzhi Cao

Federated learning (FL) offers a promising framework for collaborative digital pathology by enabling model training across institutions. However, real-world deployments face heterogeneity arising from diverse multiple instance learning (MIL) architectures and heterogeneous feature extractors across institutions. We propose FedHD, a novel FL framework that performs local Gaussian-mixture feature alignment tailored for WSI analysis. Instead of exchanging model parameters, each client independently distills semantically rich synthetic feature representations aligned with the distribution of real WSIs. To preserve diagnostic diversity, FedHD adopts a one-to-one distillation strategy, generating a synthetic counterpart for each real slide to avoid over-compression. During federation, a curriculum-based integration strategy progressively incorporates cross-site synthetic features into local training once performance plateaus. Furthermore, an optional interpretation module reconstructs pseudo-patches from synthetic embeddings, enhancing transparency. FedHD is architecture-agnostic, privacy-preserving, and supports personalized yet collaborative training across diverse institutions. Experiments on TCGA-IDH, CAMELYON16, and CAMELYON17 show that FedHD consistently outperforms state-of-the-art federated and distillation baselines.

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Scaling up Multi-Turn Off-Policy RL and Multi-Agent Tree Search for LLM Step-Provers

Ran Xin, Zeyu Zheng, Yanchen Nie, Kun Yuan, Xia Xiao

The integration of Large Language Models (LLMs) with automated theorem proving has shown immense promise, yet is constrained by challenges in scaling up both training-time reinforcement learning (RL) and inference-time compute. This paper introduces BFS-Prover-V2, a step-level theorem proving system designed to address this dual scaling problem. We present two primary innovations. The first is a novel multi-turn off-policy RL framework for continually improving the performance of the LLM step-prover at training time. This framework, inspired by the principles of AlphaZero, utilizes a multi-stage expert iteration pipeline featuring adaptive tactic-level data filtering and periodic retraining to surmount the performance plateaus that typically curtail long-term RL in LLM-based agents. The second innovation is a planner-enhanced multi-agent system that scales reasoning capabilities at inference time. This architecture employs a general reasoning model as a high-level planner to iteratively decompose complex theorems into a sequence of simpler subgoals. This hierarchical approach substantially reduces the search space, enabling a team of parallel prover agents to collaborate efficiently by leveraging a shared proof cache. We demonstrate that this dual approach to scaling yields state-of-the-art results on established formal mathematics benchmarks. BFS-Prover-V2 achieves 95.08% and 41.4% on the miniF2F and ProofNet test sets respectively. While demonstrated in the domain of formal mathematics, the RL and inference techniques presented in this work are of broader interest and may be applied to other domains requiring long-horizon multi-turn reasoning and complex search.

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DFSAttn: Dynamic Fine-grained Sparse Attention for Efficient Video Generation

Jie Hu, Zixiang Gao, Yutong He, Kun Yuan

Diffusion transformers have achieved remarkable success in high-quality video generation, yet their reliance on spatiotemporal 3D full attention incurs prohibitive computational cost due to the quadratic complexity of attention. Block spars

e attention is a common approach to mitigate this by focusing computation on imp
ortant regions. However, attention maps in DiTs exhibit inherently dynamic and f
ine-grained sparsity, which causes existing block sparse attention methods to de
grade significantly in quality, especially at high sparsity ratios. In this pape
r, we revisit block sparse attention and derive a theoretical lower bound on att
ention recall to characterize the key factors governing its effectiveness. Guide
d by these insights, we propose DFSAttn, a training-free sparse attention framew
ork that enables dynamic, fine-grained sparsification efficiently. DFSAttn incor
porates three core designs: Hilbert curve-based token reordering to achieve fine
-grained sparsity while preserving efficient GPU execution, hierarchical block s
coring for accurate block importance estimation, and sparse mask caching with ad
aptive ratios to balance accuracy and efficiency. Experimental results demonstra
te that DFSAttn consistently outperforms prior methods under high sparsity, achi
eving up to 2.1$\times$ end-to-end speedup while maintaining high generation qua
lity. Our code is open-sourced and available at https://github.com/jessica-hujie
/DFSAttn.
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Butterworth as Attention: Anisotropic Spectral Gating for Pansharpening
Zhenggang Wang, Wu Wang, Huazhe Liang, Tai-Xiang Jiang
Pansharpening fuses high-resolution panchromatic
(PAN) images with low-resolution multispectral
(LMS) images. For spatial-spectral fusion,
Fast Fourier Transform (FFT)-based methods
provide a global receptive field to capture
long-range dependencies and naturally separate
frequency components. However, most existing
approaches directly transplant spatial operators
like convolution or self-attention, while disregarding
the fundamental structure of the spectrum:
a strict spatial correspondence where each coordinate
represents a specific frequency component,
and a highly non-uniform, radially decaying
energy distribution. To address this, we revisit
the classical Butterworth filter, a frequencydomain
operator defined directly on spectral coordinates
that is inherently suited for processing
such structured representations. We generalize
the standard isotropic Butterworth filter into an
anisotropic, learnable frequency-domain gating
mechanism, establishing an efficient alternative
to self-attention, and propose the Anisotropic Butterworth
Fusion Network (ABFNet). Its core
is a novel dual-branch gating module that employs
learnable anisotropic Butterworth filters
to perform adaptive direction-aware feature selection,
integrating global context and local details
with linear complexity. Extensive experiments
show that ABFNet achieves state-of-the-art
(SOTA) performance on pansharpening benchmarks
with low computational overhead. Furthermore,
its superior accuracy on CIFAR-100
classification validates the broader applicability
of this frequency-domain learning paradigm.
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Efficient Code Analysis via Graph Representation Learning-Guided Large Language
Models
Hang Gao, Tao Peng, Baoquan Cui, Hong Huang, Fengge Wu, Junsuo Zhao, Jian Zhang
Large Language Models (LLMs) have significantly advanced code analysis tasks, ye
t they struggle to detect malicious behaviors fragmented across files, whose int
ricate dependencies easily get lost in the vast amount of benign code. We theref

ore propose a graph-centric attention acquisition pipeline that enhances LLMs' a bility to localize malicious behavior. The approach parses a project into a code graph, uses an LLM to encode nodes with semantic and structural signals, and tr ains a Graph Neural Network (GNN) under sparse supervision. The GNN performs an initial detection, and by interpreting these predictions, identifies key code se ctions that are most likely to contain malicious behavior. These influential reg ions are then used to guide the LLM's attention for in-depth analysis. This stra tegy significantly reduces interference from irrelevant context while maintainin g low annotation costs. Extensive experiments show that the method consistently outperforms existing approaches on multiple public and custom datasets, highligh ting its potential for practical deployment in software security scenarios. Code s can be found in https://github.com/Epiphaniespt/GMLLM.git.

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Towards Multimodal Large Language Models with Both Training and Inference Effici ency

Qianhao Yuan, Yanjiang Liu, Guozhao Mo, Yaojie Lu, Hongyu Lin, Jia Zheng, Ben He , Xianpei Han, Le Sun

Multimodal Large Language Models (MLLMs) mainly fall into two architectures, eac h involving a trade-off between training and inference efficiency: embedding spa ce alignment (e.g. LLaVA series) is inefficient during inference, while cross-at tention space alignment (e.g. Flamingo) is inefficient in training. A primary di fference between them lies in whether each visual token attends to other tokens within the LLM backbones. To investigate whether this form of attention is essen tial for MLLMs, we propose NAEViT (No AttEntion from Visual Tokens), an attentio n mechanism that eliminates such interactions. Our pilot experiment shows that a ttention from visual tokens
is highly redundant. Then, we introduce SAISA (Self-Attention Input Space Alignm ent), a novel architecture that enhances both training and inference efficiency. SAISA directly aligns visual features with the input spaces of NAEViT attention blocks, reducing computational overhead in both attention and FFNs. We conduct experiments on various baseline models, model sizes and training datasets. SAISA achieves superior performance compared to the baselines, while significantly re ducing computational costs. Further ablation studies validate the effectiveness of SAISA across various LLMs and visual encoders.

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The Geometric Origin of Grokking: Accelerating Generalization via Active Structu ral Reorganization

Kefei Tao, Zhang Zhang, Mingze Qi, Xiaojun Duan

Grokking, the phenomenon where models suddenly generalize long after overfitting training data, remains a puzzling challenge in neural network dynamics. Through mechanistic analysis, we find that this transition is fundamentally driven by a structural reorganization of token representations, with the onset of grokking entailing a shift toward a well-defined geometry, and reveal the model’s distinc t understanding of data’s dual characteristics. Building on these geometric insi ghts, we propose R2G (Repel-to-Grokking) Loss, an active intervention that resha pes the representation manifold by enforcing structural repulsion. The versatili ty of R2G is empirically validated in both algorithmic and linguistic tasks, whi le our theoretical analysis and ablation studies jointly demonstrate that angula r reorganization is the primary driver of grokking. Our work offers a novel mech anistic perspective on the evolution of grokking and provides a useful tool for enhancing model efficiency and reliability.

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Model-Free Robust Average-Reward Reinforcement Learning with Sample Complexity A nalysis

Zachary Andrew Roch, George K. Atia, Yue Wang

Robust reinforcement learning (RL) under the average-reward criterion is essenti al for long-term decision-making, particularly when the environment may differ f rom its training dynamics. However, most existing studies focus on model-based s ettings and provide only asymptotic guarantees, hindering their principled under standing and practical deployment, especially in data-limited scenarios. We aim

to close this gap by proposing a model-free algorithm, **Robust Halpern Iteratio n (RHI)**. We first design our algorithm based on a black-box sampling oracle, w hich can estimate the worst-case performance accurately. We then derive the fini te sample complexity of RHI under the generative model setting, assuming the sam pling oracle. To concretely design such an oracle, we propose a K-order multi- level Monte-Carlo estimator, which is shown to have a lower bias compared to pri or methods. We further instantiate our design for multiple uncertainty models, i ncluding KL and χ^2 divergence sets, and show that our RHI algorithm achiev es an ε-optimal robust policy with a sample complexity of $\tilde{\m athcal{O}}\left(\frac{SA\mathcal{H}^2}{\varepsilon^{(2+o(1))}}\right)$, where $ S,A$ are the number of states and actions, and $\mathcal{H}$ is the robust optim al span. Our result asymptotically matches the best complexity in robust average reward RL.

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Bridging Time and Frequency: A Joint Modeling Framework for Irregular Multivaria te Time Series Forecasting

Xiangfei Qiu, Kangjia Yan, Xvyuan Liu, Xingjian Wu, Jilin Hu

Irregular multivariate time series (IMTS) forecasting is challenging due to non- uniform sampling and variable asynchronicity. These irregularities violate the e quidistant assumptions of standard models, hindering local temporal modeling and rendering classical frequency-domain methods ineffective for capturing global p eriodic structures. To address this challenge, we propose TFMixer, a joint time–frequency modeling framework for IMTS forecasting. Specifically, TFMixer incorpo rates a Global Frequency Module that employs a learnable Non-Uniform Discrete Fo urier Transform (NUDFT) to directly extract spectral representations from irregu lar timestamps. In parallel, the Local Time Module introduces a query-based patc h attention mechanism to adaptively aggregate informative temporal segments and alleviate information density imbalance. Finally, TFMixer fuses the time-domain and frequency-domain representations to generate forecasts and further leverages inverse NUDFT for explicit seasonal extrapolation. Extensive experiments on rea l-world IMTS benchmarks demonstrate the effectiveness and robustness of TFMixer under irregular sampling and missing data.

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Optimal Learning from Label Proportions with General Loss Functions

Lorne Applebaum, Travis Dick, Claudio Gentile, Haim Kaplan, Tomer Koren

Motivated by problems in online advertising, we address the task of Learning fro m Label Proportions (LLP). We introduce a novel and versatile low-variance debia sing methodology to learn from aggregate label information, significantly advanc ing the state of the art in LLP. Our debiasing approach exhibits remarkable flex ibility, seamlessly accommodating a broad spectrum of practically relevant loss functions across both binary and multi-class classification settings. By careful ly combining our estimators with standard techniques, we improve sample complexi ty guarantees for a large class of losses of practical relevance. We also empiri cally validate the efficacy of our proposed approach across a diverse array of b enchmark datasets, demonstrating compelling empirical advantages over standard b aselines.

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Inverting Data Transformations via Diffusion Sampling

Jinwoo Kim, Sékou-Oumar Kaba, Jiyun Park, Seunghoon Hong, Siamak Ravanbakhsh

We study the problem of transformation inversion on general Lie groups: a datum is transformed by an unknown group element, and the goal is to recover an invers e transformation that maps it back to the original data distribution. Such unkno wn transformations arise widely in machine learning and scientific modeling, whe re they can significantly distort observations. We take a probabilistic view and model the posterior over transformations as a Boltzmann distribution defined by an energy function on the data space. To sample from this posterior, we introdu ce a diffusion process on Lie groups that keeps all updates on-manifold and only requires computations in the associated Lie algebra. Our method, Transformation -Inverting Energy Diffusion (TIED), relies on a new trivialized target-score ide ntity that enables efficient score-based sampling of the transformation posterio

r. As a key application, we focus on test-time equivariance, where the objective is to improve the robustness of pretrained neural networks to input transformations. Experiments on image homographies and PDE symmetries demonstrate that TIED can restore transformed inputs to the training distribution at test time, showing improved performance over strong canonicalization and sampling baselines.

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A Theory of Contrastive Learning with Natural Images

Antonio Torralba, Yair Weiss

Why does contrastive learning with simple images and augmentations yield useful representations for downstream tasks? We address this question by analytically computing the optimal representation in terms of the contrastive loss for a range of basic augmentations and any image dataset with stationary statistics. We show that for certain augmentations the optimum can be attained by a CNN whose first layer filters are sinusoids, followed by a pointwise nonlinearity, global average pooling, and a final linear layer that performs partial whitening. We also show that the optimal weights in such CNNs for more complicated augmentations are still sinusoids. The frequencies of the sinusoids and their weights can be computed using a simple “waterfilling” algorithm given the dataset’s expected power spectrum. Experiments with different image datasets and augmentations show that such CNNs trained with SGD empirically learn sinusoids in their first layer and to perform partial whitening.

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MiniMax Learning of Interpretable Factored Stochastic Policies from Conjoint Data, with Uncertainty Quantification

Connor Thomas Jerzak, Priyanshi Chandra, Rishi Hazra

We study offline policy optimization over exponentially large factorial action spaces from randomized preference data, showing how conjoint experiments can estimate interpretable stochastic policies with asymptotically valid uncertainty under regularity conditions. Conjoint analyses typically report Average Marginal Component Effects (AMCEs) by averaging over opponent attributes and thus ignore strategic interdependence. We instead learn *stochastic interventions*—product-of-Categorical policies over factor levels—that (i) optimize expected outcomes in an average-case setting and (ii) extend to a two-player *minimax* (adversarial) setting that realistically captures simultaneous strategic candidate selection. Methodologically, we derive a closed-form optimizer for a tractable two-way interaction regime with L_2 variance regularization, and provide a general gradient-based procedure for richer model classes. Uncertainty from the outcome model propagates asymptotically to both the optimal policy and its value via a Delta method approximation. We further model institutional details (e.g., primaries) inside the minimax objective and introduce a data-driven measure of strategic divergence between parties. On synthetic data, we empirically characterize finite-sample error and coverage as dimensionality and n vary. On a U.S. presidential conjoint, adversarially learned policies produce restricted-equilibrium vote shares that align with historical election ranges in our data, in stark contrast to non-adversarial (averaging) optimizers.

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SERA: Soft-Verified Efficient Repository Agents

Ethan Shen, Daniel Tormoen, Saurabh Shah, Ali Farhadi, Tim Dettmers

Open-weight coding agents should hold a fundamental advantage over closed-source systems because they can specialize to private codebases, encoding repository-specific information directly in their weights. Yet the cost and complexity of training has kept this advantage theoretical until now. We present Soft-Verified Efficient Repository Agents (SERA), an efficient method for training coding agents that enables the rapid and cheap creation of agents specialized to private codebases. Using Soft Verified Generation (SVG), we generate thousands of trajectories from any code repository, without requiring unit tests. Beyond repository specialization, we apply SVG to a larger corpus of codebases, generating 200,000+ synthetic trajectories. Using only supervised finetuning (SFT), SERA achieves leading results among fully open-source (open data, method, code) models while matching the performance of open-weight models like Devstral-Small-2. Creating SERA

models is 26x cheaper than reinforcement learning and 57x cheaper than previous synthetic data methods to reach equivalent performance. We use our dataset to provide detailed analysis of scaling laws, ablations, and confounding factors for training coding agents. Overall, we believe our work will greatly accelerate research on open coding agents and showcase the advantage of open-source models that can adapt to private codebases.

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Opt-Miner: Empowering Information-Seeking Agent with Tree-Guided Data Synthesis for Optimization Modeling

Haoyang Liu, Yuyang Cai, Jie Wang, Xiongwei Han, Minyang Hu, Shuqi LIU, Mingxuan Yuan, Jianye HAO, Feng Wu

Large Language Model (LLM) agents have shown significant potential in automated optimization modeling for mathematical problems. However, real-world problems are still challenging due to their knowledge-intensive nature. Existing methods, constrained by static parametric knowledge, often lack the domain expertise required to comprehend complex scenarios and apply appropriate mathematical techniques, leading to errors.

To address this challenge, we propose the Opt-Miner framework, where the agent learns to identify missing knowledge, retrieve technical documents on the web, and ground its mathematical models for improved modeling performance.

The core of Opt-Miner is a novel tree-guided data synthesis pipeline coupled with a retrieval-based group relative policy optimization (R-GRPO) algorithm, designed to foster the agent’s information-seeking capabilities.

Specifically, we first formulate each problem into a tree structure, with its scenario contexts and mathematical techniques embedded in subtrees.

We then employ subtree union, transfer, and knowledge fogging to synthesize complex, multi-domain problems that incorporate knowledge gaps, thereby necessitating active information seeking to solve these problems.

Based on synthesized data, we propose R-GRPO for agent reinforcement learning. Experiments demonstrate that Opt-Miner-Qwen3-8B achieves performance comparable to 32B state-of-the-art specialized agents and commercial reasoning models.

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Fast and Near-Optimal Algorithms for Private Hypothesis Selection

Hilal Asi, Hongjie Chen

We study the problem of private hypothesis selection: given samples from an unknown distribution drawn from a finite hypothesis class, the goal is to identify the best hypothesis under the constraint of differential privacy. Existing algorithms for this problem are either computationally expensive or achieve sub-optimal statistical rates. We propose new algorithms that achieve near-optimal rates while running in nearly linear time in the number of hypotheses. Rather than applying the exponential mechanism directly with a score function that requires pairwise comparisons between hypotheses, our approach introduces a carefully designed loss function based on a small set of strong hypotheses. This structure allows the score to be evaluated efficiently for most hypotheses, yielding significant computational savings. We further extend our algorithms to the agnostic setting, where the true distribution may not belong to the hypothesis class. As an application, we obtain faster differentially private algorithms for universal statistical estimation in low dimensional settings.

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Offline Reinforcement Learning with Universal Horizon Models

Hojun Chung, Junseo Lee, Songhwai Oh

Model-based reinforcement learning (RL) offers a compelling approach to offline RL by enabling value learning on imagined on-policy trajectories. However, it often suffers from compounding errors due to repeated model inference on self-generated states. While geometric horizon models (GHM) alleviate this issue through direct prediction over a discounted infinite-horizon future, they remain challenged in accurately modeling distant future states. To this end, we introduce universal horizon models (UHM), a generalization of GHM that directly predicts future states under arbitrary horizons. Leveraging this flexibility, we propose a scalable value learning method that employs a winsorized horizon distribution to st

abilize training by capping excessively large horizons. Experimental results on 100 challenging OGBench tasks demonstrate that the proposed method outperforms competitive baselines, particularly on tasks with highly suboptimal datasets and those requiring long-horizon reasoning. Project page: https://rllab-snu.github.io/projects/UHM/

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Unifying and Optimizing Data Values for Selection via Sequential Decision-Making
Hongliang Chi, Qiong Wu, Zhengyi Zhou, Jonathan Light, Emily Dodwell, Yao Ma

Data selection has emerged as a crucial downstream application of data valuation, yet the theoretical foundations for using data values in selection remain unexplored. We reformulate data selection as a sequential decision-making problem where the optimal selection sequence arises from dynamic programming, and data values can be understood as encodings of this optimal sequence. This framework unifies and reinterprets existing methods like Data Shapley through the lens of approximate dynamic programming, revealing them as myopic linear approximations to the sequential problem. We further analyze how selection optimality degrades with utility curvature under submodularity, explaining when and why these approximations fail. To bridge theory and practice, we propose an efficient bipartite graph-based surrogate that preserves submodular structure while enabling scalable greedy selection with provable guarantees. Experiments on classical ML benchmarks and large-scale LLM fine-tuning data selection demonstrate substantial improvements over existing methods. Code is publicly available at https://github.com/frankhlchi/SeqDataVal

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Rank-guided Diffusion for Noise Few-Shot Learning
Zelei Wu, Kun Zhou, Xulun Ye, Yifan Mei, Jie Hong, Jieyu Zhao

In real-world Few-Shot Learning (FSL), support sets are quickly constructed and inevitably contain noisy samples. With limited examples per class, even a single noisy instance can distort class distributions, cause prototype drift, and reduce generalization. Existing methods mostly assume clean data or require large-scale statistics, which are impractical in FSL’s data-scarce setting. We find that clean samples in semantic feature space lie in low-rank subspaces, while noisy samples cause rank anomalies disrupting this structure. To address this, we propose a differentiable low-rank approximation that estimates the intrinsic rank of the support set and detects anomalous noisy samples. Building on this, a rank-guided diffusion process generates high-quality replacements under low-rank constraints, reconstructing a clean, consistent support set for improved robustness.This low-rank guided approach effectively mitigates prototype drift and significantly reduces errors under noise levels up to 40\% across MiniImageNet, TieredImageNet, and other noisy benchmarks, demonstrating the power of low-rank geometry for noise detection and correction in FSL. Our source code is available at https://github.com/wuzelei123/CRDProto.

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Securing Multimodal AI through Internal Information Decomposition
Jehyeok Yeon, Hyeonjeong Ha, Qiusi Zhan, Heng Ji

Multimodal large language models introduce attack surfaces absent in unimodal systems: adversaries can distribute malicious intent across modalities to evade unimodal safeguards. This motivates using cross-modal consistency as a detection signal rather than inspecting each modality in isolation. Our key observation is that benign inputs induce compatible predictive behavior from text-only and vision-only reasoning that stabilizes when fused, whereas adversarial manipulation disrupts this consistency, causing abnormal multimodal behavior. Existing defenses that examine raw inputs or outputs overlook this internal fusion process, rendering them brittle and computationally expensive. We propose FlowGuard, a lightweight inference-time framework that detects harmful inputs by monitoring internal multimodal consistency. Unlike approaches that rely on scalar confidence metrics, FlowGuard derives FlowVectors inspired by Partial Information Decomposition that quantify cross-modal redundancy, synergy, and modality-specific dominance, capturing whether multimodal fusion aligns with unimodal semantic evidencebetween unimodal and fused multimodal output distributions. In a one-class classificat

ion problem trained solely on benign data, FlowGuard reduces Attack Success Rate s from $>90\%$ to $<15\%$ on unseen attacks, with $<3\%$ utility loss and up to a $6\times$ latency reduction. Our results demonstrate that monitoring cross-mod al consistency offers an efficient and effective defense for multimodal reasonin g.
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A Statistical Framework for Analyzing Specification Resistance to Learnware-Inve rsion Risks
Hao-Yi Lei, Zhi-Hao Tan, Zhi-Hua Zhou
The learnware paradigm enables model reuse by pairing each submitted model with a specification, a public artifact used to identify helpful models without raw-d ata exchange.
This design creates a privacy surface: a useful specification must reveal capabi lity-relevant information, but such information should not expose sensitive prop erties of training data or user tasks. Is it achievable in practice? To answer t his question, this paper establishes the first framework for analyzing the incre mental risk introduced by specifications in learnware, and provides theoretical guarantees for the widely used reduced kernel mean embedding (RKME) specificatio n.
Specifically, we formulate learnware-inversion as a family of statistical decisi on games and define the risk of specification as the incremental Bayes value fro m observing the model alone to observing the complete learnware.
For the RKME specification, we derive risk bounds through an RKHS-smoothed total -variation bridge and the stability analysis of its reduced-set generator.
We further instantiate the framework for common attacks and show that a properly sized RKME specification introduces negligible additional privacy risk while re taining sufficient information for learnware identification.
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An Information-Theoretic Criterion for Efficient Data Synthesis
Hanyu Li, Zhengqi Sun, Xiaotie Deng
Synthetic data becomes crucial for large language model training, but its effect iveness is highly inconsistent. We provide an information-theoretic account of t his inconsistency: synthetic data improves a model only when the generation-trai ning loop is information-open, i.e., shaped by external signals (verifiers, envi ronments, or rubrics) that inject task-relevant information beyond the model's c urrent distribution. When the loop is information-closed (relying on the model's own outputs without such signals), the data processing inequality ensures that task-relevant information can only decrease, making collapse a predicted outcome . Among information-open pipelines, both efficiency and generalization hinge on the meta-level of supervision: a coarser signal such as binary correctness treat s all acceptable outputs as equivalent, so the behavior it teaches is not tied t o any particular domain or surface form and generalizes naturally across tasks a nd domains. These observations lead to a guiding thesis: learning preferentially converges to the most information-efficient signal component available, which a ccelerates learning when that component is the intended one, but causes reward h acking when a spurious pattern happens to be simpler.
****

What Makes a Desired Graph for Relational Deep Learning?
Yao Cheng, Siqiang Luo
Relational deep learning (RDL) converts relational databases (RDBs) into heterog eneous graphs, but graphs derived directly from database schemas are often not w ell suited for how graph neural networks (GNNs) perform relational reasoning. We study what makes a relational graph suitable for deep learning and show that sc hema-derived graphs suffer from two systematic failures: information overload an d semantic fragmentation. Through an empirical analysis on real-world databases, we find that effective graphs arise from a task-dependent balance between remov ing task-irrelevant structure and injecting task-aligned relational connectivity . Filtering exhibits a non-monotonic effect on performance, while structural in jection is beneficial only when it reflects the logic of the downstream task. Ba sed on these findings, we develop an end-to-end structural optimizer that applie

s both operations to adapt relational graphs automatically. Across 23 tasks spa
nning classification, regression, and recommendation, the optimized graphs consi
stently improve accuracy while often reducing inference cost.
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Unsupervised Disentanglement Without Compromises : How Functional Orthogonality
Enforces Identifiability

Mathieu Cyrille Simon, Pascal Frossard, Christophe De Vleeschouwer

This paper explores unsupervised disentangled representation learning from a fun
ctional perspective. We define latent concepts as factors that influence observa
tions through locally orthogonal directions, formalized as an orthogonality cons
traint on the Jacobian of the generative mapping. We prove that this condition y
ields identifiability of general nonlinear generative models, without requiring
statistical independence or causal assumptions, provided the latent domain admit
s all combinations of factor values. Experiments with orthogonality-regularized
normalizing flows empirically confirm the theory, demonstrate reliable recovery
of ground-truth factors, and shed light on the success of VAEs. These findings c
hallenge the prevailing impossibility claims for unsupervised disentanglement an
d provide a principled alternative foundation.
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Persistent Backdoor Attacks in Class-Incremental Learning via Structural Invaria
nt Anchoring

Junhuang Huang, Linshan Hou, Jianting Ning, Yanjun Zhang, Zhongyun Hua, Leo Yu Z
hang

Continual learning (CL) involves continual parameter updates, posing a significa
nt challenge to backdoor persistence. In this paper, we reveal that the most adv
anced existing attack relies on an implicit assumption that task-critical neuron
s remain stable across task learning; however, this assumption does not hold in
class-incremental learning (CIL). This exposes a critical research gap: backdoor
 persistence in CIL remains an open question. Inspired by functional stability,
we discover that CIL models preserve task knowledge in shallow, structurally inv
ariant subspaces. Motivated by these findings, we propose PBTO, the first persis
tent and targeted backdoor attack in CIL. PBTO trains a surrogate model on proxy
 tasks to obtain a parameter trajectory. It then optimizes a universal trigger t
hat ensures misclassification to the target label across all model states and an
chors trigger embeddings in shallow layers. Experimental results verify that PBT
O maintains a high final attack success rate (ASR) across all benchmarks, while
representative baselines degrade substantially after sequential learning. Code i
s available at \href{https://github.com/hjhkkkc/PBTO}{PBTO}.
****

Multi-Objective Learning for Diffusion Models: A Statistical Theory under Semi-S
upervised Learning

Ziheng Cheng, Yixiao Huang, Hanlin Zhu, Haoran Geng, Somayeh Sojoudi, Jitendra M
alik, Pieter Abbeel, Xin Guo

Diffusion models are increasingly used as powerful conditional generators, yet r
eal deployments often involve multiple target distributions arising from differe
nt tasks, e.g., diverse prompt domains in text-to-image generation, or multiple
environments in robotics with diffusion policies.
This naturally leads to a multi-objective learning (MOL) problem.
A key challenge is that achieving good Pareto trade-offs can require a generalis
t model class with substantially larger capacity than what suffices for solving
any individual task, thereby increasing statistical cost since sample complexity
 typically scales with the model complexity.
To reconcile this, we develop a principled MOL framework for diffusion models wi
th limited data: a semi-supervised regime where paired (labeled) samples are sca
rce, but (unlabeled) condition data are abundant.
We propose a two-stage training procedure that first fits lightweight specialist
 models from limited paired data, and then distills them into a generalist model
 by generating pseudo-samples.
We establish generalization bounds showing that the required number of paired sa
mples only depends on the complexity of the specialist model classes.

We further extend the theory to diffusion policies for sequential decision makin
g to account for distribution shift in on-policy rollouts.
Extensive experiments on robotic control and image restoration tasks are conduct
ed to verify our theoretical results.
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Row-Stochastic Matrices Can Provably Outperform Doubly Stochastic Matrices in De
centralized Learning
Bing Liu, Boao Kong, Limin Lu, Kun Yuan, Chengcheng Zhao
Decentralized learning often involves a weighted global loss with heterogeneous
node weights λ. We revisit two natural strategies for incorporating thes
e weights: (i) embedding them into the local losses to retain a uniform weight (
and thus a doubly stochastic matrix), and (ii) keeping the original losses while
 employing a λ-induced row-stochastic matrix. Although prior work shows
that both strategies target the same λ-weighted global loss, it remains
unclear whether the Euclidean-space guarantees are tight and what fundamentally
differentiates their behaviors. To clarify this, we develop a weighted Hilbert-s
pace framework $L^2(\lambda;\mathbb{R}^d)$ and obtain convergence rates that are
 strictly tighter than those from standard Euclidean analysis. In this geometry,
 the row-stochastic matrix becomes *self-adjoint* whereas the doubly stochastic
one does not, creating additional *penalty terms* that amplify consensus error,
thereby slowing convergence. Consequently, the difference in convergence arises
not only from spectral gaps but also from these penalty terms. We then derive su
fficient conditions under which the row-stochastic design converges faster even
with a smaller spectral gap. Finally, by using a Rayleigh-quotient and Loewner-o
rder eigenvalue comparison, we further obtain topology conditions that guarantee
 this advantage and yield practical topology-design guidelines.
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Efficient and Uncertainty-Aware Diffusion Framework for Offline-to-Online Reinfo
rcement Learning
Ha Manh Bui, Metod Jazbec, Eric Nalisnick, Anqi Liu
Offline-to-Online Reinforcement Learning (O2O-RL) leverages an offline, pre-trai
ned policy to minimize costly online interactions. Although data-efficient, O2O-
RL is susceptible to shifts between offline and online distributions. Existing w
ork aims to mitigate the harm of this shift by finetuning the policy on trajecto
ry data sampled from a diffusion model. Inspired by this line of work, we propos
e DUAL: an efficient **D**iffusion **U**ncertainty-**A**ware framework for offli
ne-to-online reinforcement **L**earning. DUAL utilizes the prior knowledge of th
e diffusion model to distill a fast-sampling diffusion actor policy and transiti
on model in the offline phase. DUAL also employs a Laplace approximation and dis
tance transition-state-shift detection, thereby using uncertainty quantification
 to improve exploration versus exploitation in the online phase. We formally sho
w that our actor loss with the Laplace approximation provides a proxy for a prin
cipled estimate of epistemic uncertainty. Empirically, DUAL improves the online
expected return over O2O-RL baselines across multiple settings and environments.
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Discrete Adjoint Schrödinger Bridge Sampler
Wei Guo, Yuchen Zhu, Xiaochen Du, Juno Nam, Yongxin Chen, Rafael Gomez-Bombarell
i, Guan-Horng Liu, Molei Tao, Jaemoo Choi
Learning discrete neural samplers is challenging due to the lack of gradients an
d combinatorial complexity. While stochastic optimal control (SOC) and Schröding
er bridge (SB) provide principled solutions, efficient SOC solvers like adjoint
matching (AM), which excel in continuous domains, remain unexplored for discrete
 spaces. We bridge this gap by revealing that the core mechanism of AM is *state
-space agnostic*, and introduce **discrete ASBS**, a unified framework that exte
nds AM and adjoint Schrödinger bridge sampler (ASBS) to discrete spaces. Theoret
ically, we analyze the optimality conditions of the discrete SB problem and its
connection to SOC, identifying a necessary cyclic group structure on the state s
pace to enable this extension. Empirically, discrete ASBS achieves competitive s
ample quality with significant advantages in training efficiency and scalability
. Our code is available at https://github.com/AlexandreGUO2001/DASBS.

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Video-Based Optimal Transport for Feedback-Efficient Offline Preference-Based Reinforcement Learning
Tung Minh Luu, Hwanhee Kim, Younghwan Lee, Chang D. Yoo
Conveying complex objectives to reinforcement learning (RL) agents often requires meticulous reward engineering. Preference-based RL (PbRL) offers a promising alternative by learning reward functions from human feedback, but its scalability is hindered by high labeling costs. Inspired by advances in Video Foundation Models (ViFMs), we present Video-based Optimal Transport Preference (VOTP), a semi-supervised framework that learns effective reward functions from only a handful of labels. By leveraging optimal transport to align visual trajectories within the rich representation space of ViFMs, VOTP effectively generates high-fidelity pseudo-labels for large amounts of unlabeled data, substantially reducing human supervision. Extensive experiments across locomotion and manipulation benchmarks demonstrate the superiority of VOTP, which outperforms state-of-the-art offline PbRL methods under limited feedback budgets. We also showcase the robustness of VOTP in the presence of visual distractors and validate its utility on real robotic tasks, where it learns meaningful rewards with minimal human input.
****
DREAM: A Unified Framework for Drift-Corrected Federated Multi-Objective Learning
Yuan Zhou, Yidan Ou, Xinli Shi
Federated Multi-Objective Learning (FMOL) enables collaborative training of conflicting objectives but faces a compounded challenge: the recursive coupling between intra-task client drift and inter-task aggregation bias.
We propose DREAM, a unified framework that jointly corrects these two coupled error sources through drift-aware control variates and momentum-smoothed local updates.
On the server side, DREAM formulates multi-objective aggregation as a regularized quadratic program parameterized by a task correction matrix, which provides a generalized formulation that can flexibly adapt to scalarization, prioritization, and gradient manipulation strategies.
Theoretically, we establish a linear speedup convergence rate of $\mathcal{O}(1/\sqrt{NT})$ for non-convex objectives.
We further provide theoretical guarantees for the conflict-avoidant direction distance.
In the strongly convex setting, DREAM achieves convergence in weighted sub-optimality and admits a unified Lyapunov analysis showing linear convergence to a regularization-dependent neighborhood.
Numerical experiments on representative benchmarks validate the effectiveness of DREAM in multi-objective optimization.
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FLARE-AI: Flaw Reporting for AI
Shayne Longpre, Elaine Zhu, Carson Ezell, Avijit Ghosh, Sean McGregor, Kevin Paeth, Kevin Klyman, Sayash Kapoor, Rishi Bommasani, Ruth E. Appel, Gregory Strom, Lauren McIlvenny, Mark M. Jaycox, Peter Slattery, Nathan Butters, Arvind Narayanan, Percy Liang, Alex Pentland
Flaw reporting for deployed AI systems is fundamental to identifying system failures and improving AI safety. Yet the AI reporting ecosystem is fragmented: researchers who identify flaws often do not know what or where to report, and groups who receive reports rarely share them with other relevant stakeholders. As a result, good-faith reporters duplicate effort by submitting many different forms, and recipients lack standardized, triage-ready information. We survey 12 reporting systems published by AI developers, cybersecurity groups, and AI flaw aggregators, identifying five recurring design challenges spanning discoverability, scope, information collection, coordination, and guidance for strict-liability cases. Building on this analysis and feedback from 49 experts across 32 organizations representing developers, security researchers, and ecosystem coordinators, we introduce FLARE-AI, an open-source AI flaw reporting system designed for interoperability with existing systems. FLARE-AI streamlines flaw report creation by co

llecting triage-relevant information through conditional logic and early classif ication, then enables optional dissemination of standardized, machine-readable r eports to multiple developers, coordinators, and incident registries from a sing le submission. By lowering barriers to reporting AI flaws and improving interope rability across stakeholders, FLARE-AI helps break down silos and accelerate rem ediation across the AI ecosystem.
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Accelerated and Stable Convergence with Anchored Generalized Optimistic Method
Motahareh Sohrabi, Jianxin You, Simon Lacoste-Julien, Eduard Gorbunov, Gauthier Gidel
We study first-order methods for solving monotone variational inequalities arisi ng in min-max optimization. Classical approaches such as the extragradient metho d rely on two gradient queries per iteration, which limits their analysis and ap plicability in the online and stochastic settings. We propose a family of Genera lized Optimistic Methods with Anchoring (GOMA), which combine two-time-scale opt imistic updates with an anchoring term inspired by Halpern iteration. In the det erministic setting, GOMA achieves the optimal accelerated last-iterate rate $\ma thcal{O}(1/k^2)$ on the squared gradient norm for monotone Lipschitz operators. In the stochastic setting with unbounded variance, a simplified single-call vari ant of GOMA achieves a last-iterate convergence rate of $\mathcal{O}(1/\sqrt{k}) $ on the squared gradient norm. To the best of our knowledge, this is the first such guarantee for stochastic monotone Lipschitz VIs in the unconstrained settin g without variance reduction or growing batches.
****

Training-Trajectory-Aware Token Selection
Zhanming Shen, Jiaqi Hu, Zeyu Qin, Hao Chen, Wentao Ye, Zenan Huang, yihong zhua ng, Guoshan Lu, Junlin Zhou, Junbo Zhao
Efficient distillation is a key pathway for converting expensive reasoning capab ility into deployable efficiency, yet in the frontier regime where the student a lready has strong reasoning ability, naive continual distillation often yields l imited gains or even degradation. We observe a characteristic training phenomeno n: even as loss decreases monotonically, all performance metrics can drop sharpl y at almost the same bottle-neck, before gradually recovering. We further uncove r a token-level mechanism: confidence bifurcates into steadily increasing Imitat ion-Anchor Tokens that quickly anchor optimization and other yet-to-learn tokens whose confidence is suppressed until after the bottleneck. And the characterist ic that these two types of tokens cannot coexist is the root cause of the failur e in continual distillation. To this end, we propose Training-Trajectory-Aware T oken Selection (T3S) to reconstruct the training objective at the token level, c learing the optimization path for yet-to-learn tokens. T3S yields consistent gai ns in both AR and dLLM settings: with only hundreds of examples, Qwen3-8B surpas ses DeepSeek-R1 on competitive reasoning benchmarks, Qwen3-32B approaches Qwen3- 235B, and T3-trained LLaDA-2.0-Mini exceeds its AR baseline, achieving state-of- the-art performance among all of 16B-scale no-think models.
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GTPO and GRPO-S: Token and Sequence-Level Reward Shaping with Policy Entropy
Hongze Tan, Zihan Wang, Jianfei Pan, Jinghao Lin, Hao Wang, Yifan Wu, Tao Chen, Zhihang Zheng, Zhihao Tang, Haihua Yang
Reinforcement Learning (RL) is pivotal for enhancing Large Language Model (LLM) reasoning, yet mainstream algorithms such as GRPO and DAPO remain constrained by a coarse-grained credit assignment paradigm, where all tokens within the same r esponse receive the identical reward. In this paper, we propose **Dynamic Entrop y Weighting**, systematically define entropy-based weight ratios $\frac{H_{i,t}} {\sum_{k=1}^{n} H_{k,t}}$ and similar variants to redistribute rewards and get f ine-grained rewards through two new algorithms: **Group Token Policy Optimizatio n (GTPO)**, which assigns an entropy-weighted reward to each token and synthesiz es token-specific advantage function to drive the model toward optimal path, and the analogous algorithm **Sequence-Level GRPO (GRPO-S)**, which admits a comple tely similar design at the sequence level. Unlike methods using entropy as mere regularization, GTPO and GRPO-S establish a new state-of-the-art on AIME and MAT

H 500, outperforming prior entropy-guided baselines and validating our weighting mechanism.
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DGS-Net: Distillation-Guided Gradient Surgery for CLIP Fine-Tuning in AI-Generated Image Detection

Jiazhen Yan, Ziqiang Li, Fan Wang, Boyu Wang, Ziwen He, Zhangjie Fu

The rapid progress of generative models such as GANs and diffusion models has led to the widespread proliferation of AI-generated images, raising concerns about misinformation, privacy violations, and trust erosion in digital media. Although large-scale multimodal models like CLIP offer strong transferable representations for detecting synthetic content, fine-tuning them often induces catastrophic forgetting, which degrades pre-trained priors and limits cross-domain generalization. To address this issue, we propose the Distillation-guided Gradient Surgery Network (DGS-Net), a novel framework that preserves transferable pre-trained priors while suppressing task-irrelevant components. Specifically, we introduce a gradient-space decomposition that separates harmful and beneficial descent directions during optimization. By projecting task gradients onto the orthogonal complement of harmful directions and aligning with beneficial ones distilled from a frozen CLIP encoder, DGS-Net achieves unified optimization of prior preservation and irrelevant suppression. Extensive experiments on 50 generative models demonstrate that our method outperforms state-of-the-art approaches by an average margin of 6.6%, achieving superior detection performance and generalization across diverse generation techniques.
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Optimal Regularization for Performative Learning

Edwige Cyffers, Alireza Mirrokni, Marco Mondelli

In performative learning, the data distribution reacts to the deployed model—for example, because strategic users adapt their features to game it—which creates a more complex dynamic than in classical supervised learning. One should thus not only optimize the model for the current data but also take into account that the model might steer the distribution in a new direction, without knowing the exact nature of the potential shift. We explore how regularization can help cope with performative effects by studying its impact in high-dimensional ridge regression.

We show that, while performative effects worsen the test risk in the population setting, when moving to the over-parameterized regime where the number of features exceeds the number of samples, the optimal regularization in the presence of performativity helps reduce the variance in the estimated parameters, thereby improving performance. We show that the optimal regularization scales with the overall strength of the performative effect, making it possible to set the regularization in anticipation of this effect. We illustrate this finding through empirical evaluations of the optimal regularization parameter on both synthetic and real-world datasets.
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Scalable and Interpretable Representation Alignment with Ordinal Similarity

Diogo Soares, Pankhil Gawade, Andrea Dittadi, Ewa Szczurek

Evaluating representation similarity is fundamental to representation learning. However, existing metrics suffer from significant limitations: they lack interpretability due to shifting baselines, lack robustness to outliers, and are computationally intractable for large datasets, forcing reliance on heuristic approximations. To address this, we develop an ordinal-similarity framework, instantiated by the Triplet (TSI) and Quadruplet (QSI) Similarity Indices, which measure alignment by quantifying the consistency of ordinal relationships. We theoretically demonstrate this formulation is inherently interpretable, robust to outliers, and computationally efficient. Finally, we establish a formal equivalence between TSI and local neighborhood alignment, measured by Mutual Nearest Neighbors. Empirically, we validate these properties and show that ordinal similarity offers a scalable approach to measuring alignment, enabling practitioners to better understand and design representations.
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The Label Horizon Paradox: Rethinking Supervision Targets in Financial Forecasting

Chen-Hui Song, Shuoling Liu, Liyuan Chen

While deep learning has revolutionized financial forecasting through sophisticated architectures, the design of the supervision signal itself is rarely scrutinized. We challenge the canonical assumption that training labels must strictly mirror inference targets, uncovering the Label Horizon Paradox: the optimal supervision signal often deviates from the prediction goal, shifting across intermediate horizons governed by market dynamics. We theoretically ground this phenomenon in a dynamic signal-noise trade-off, demonstrating that generalization hinges on the competition between marginal signal realization and noise accumulation. To operationalize this insight, we propose a bi-level optimization framework that autonomously identifies the optimal proxy label within a single training run. Extensive experiments on large-scale financial datasets demonstrate consistent improvements over conventional baselines, thereby opening new avenues for label-centric research in financial forecasting.

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Can Agents Generalize to the Open World? Unveiling the Fragility of Static Training in Tool Use

Weiming Wu, Song-Lin Lv, Rui Zhu, Zi-Jian Cheng, Lan-Zhe Guo

While Large Language Model (LLM) agents demonstrate proficiency in static benchmarks, their deployment in real-world scenarios is hindered by the dynamic nature of user queries, tool sets, and interaction dynamics.To address this generalization gap, we formalize **OpenAgent** (Tool-Use Agent in Open-World), a problem setting characterized by distributional shifts across query, action, observation, and domain dimensions.To systematically diagnose its impact, we construct a controlled sandbox environment where we define fine-grained environmental shifts across a four-tier hierarchy, *Perception*, *Interaction*, *Reasoning*, and *Internalization*, and conduct a comprehensive series of experiments. Our analysis yields a series of key insights, demonstrating that agents trained via both Supervised Fine-Tuning (SFT) and Reinforcement Learning suffer from varying degrees of performance degradation when confronting open environmental shifts.Building on these insights, we propose Perturbation-Augmented Fine-Tuning, a disturbance-based intervention strategy for SFT that lays the foundation for enhancing agent robustness and utility in realistic environments. Our code will be released at: https://github.com/LAMDA-NeSy/OpenAgent.

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Transport or Discard: Robust Unbalanced Optimal Transport for Cross-Domain Policy Adaptation

Wenyu Chen, Yujia Zhang, Wei Guo, Linli Ma, Yanbo Wang, Pinle Qin, Jianchao Zeng

Cross-domain offline reinforcement learning leverages a source dataset to improve policy learning in a data-scarce target domain, but dynamics mismatch makes many source transitions kinematically infeasible and can cause negative transfer. Recent non-parametric geometric methods (e.g., standard optimal transport and k-nearest neighbors) avoid overfitting yet often yield only relative rankings under an implicit matching or retrieval budget, making performance sensitive to hand-tuned thresholds when the true cross-domain overlap is unknown. We formulate availability estimation as soft subset selection by learning a source reweighting that geometrically aligns with the target. We propose **R**obust **O**ffline unbalanced **O**ptimal **T**ransport (ROOT): (i) a robust ambiguity set for uncertainty under limited target samples, and (ii) an unbalanced transport objective that penalizes mass deviation, enabling a principled transport-or-discard mechanism. ROOT thus down-weights or discards high-cost source samples rather than forcing them onto the target support. Moreover, the induced weights decay exponentially with transport cost, guaranteeing outlier suppression. On D4RL dynamics-shift benchmarks, ROOT improves downstream offline RL and outperforms strong baselines on most tasks without task-specific threshold tuning.

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HyPER: Bridging Exploration and Exploitation for Scalable LLM Reasoning with Hypothesis Path Expansion and Reduction

Shengxuan Qiu, Haochen Huang, Shuzhang Zhong, Pengfei Zuo, Meng Li
Scaling test-time compute with multi-path chain-of-thought can improve reasoning accuracy, but its gains hinge on an effective exploration–exploitation trade-off. Existing methods handle this trade-off in rigid ways: tree-structured search hard-codes exploration via brittle expansion rules that disrupt post-trained reasoning, while parallel reasoning over-explores redundant hypothesis paths and relies on a weak answer selection strategy. Driven by the insight that the optimal balance is *phase-dependent* and that correct vs. incorrect paths often *diverge only at late stages*, we reconceptualize test-time scaling as a dynamic *expand–reduce* control problem over a pool of hypothesis paths. We introduce **HyPER**, a *training-free online control policy* for MoE multi-path decoding that reallocates compute under a fixed budget using lightweight path statistics. HyPER features (i) an *online controller* that shifts from exploration to exploitation as the hypothesis pool evolves, (ii) an MoE-based token-level refinement primitive for efficient *generation-time exploitation* without full-path resampling, and (iii) a length- and confidence-aware aggregation rule to bridge the existence–selection gap for reliable *answer-time exploitation*. Extensive experimental results across four MoE models and diverse benchmarks demonstrate HyPER consistently achieves the accuracy–compute Pareto frontier, outperforming prior-art methods by 8-10% while reducing token consumption by 25-40%.
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Multicalibration Yields Better Matchings
Riccardo Colini Baldeschi, Simone Di Gregorio, Simone Fioravanti, Federico Fusco, Ido Guy, Daniel Haimovich, Stefano Leonardi, Fridolin Linder, Lorenzo Perini, Matteo Russo, Cem Sirin, Niek Tax
Consider the problem of finding the best matching in a weighted graph where we only have access to predictions of the actual stochastic weights, based on an underlying context. If the predictor is the Bayes optimal one, then computing the best matching based on the predicted weights is optimal. However, in practice, this perfect information scenario is not realistic. Given an imperfect predictor, a suboptimal decision rule may compensate for the induced error and thus outperform the standard optimal rule.
In this paper, we propose multicalibration as a way to address this problem. This fairness notion requires a predictor to be unbiased on each element of a family of protected sets of contexts. Given a class of matching algorithms $\mathcal C$ and any predictor γ of the edge-weights, we show how to construct a specific multicalibrated predictor $\hat \gamma$, with the following property. Picking the best matching based on the output of $\hat \gamma$ is competitive with the best decision rule in $\mathcal C$ applied onto the original predictor γ. We complement this result by providing sample complexity bounds, and by performing numerical experiments.
****
Learnable Kernel Density Estimation for Graphs and Its Application to Graph-Level Anomaly Detection
Xudong Wang, Ziheng Sun, Chris Ding, Jicong Fan
This work proposes a framework LGKDE that learns kernel density estimation for graphs. The key challenge in graph density estimation lies in effectively capturing both structural patterns and semantic variations while maintaining theoretical guarantees. Combining graph kernels and kernel density estimation (KDE) is a standard approach to graph density estimation, but has unsatisfactory performance due to the handcrafted and fixed features of kernels. Our method LGKDE leverages graph neural networks to represent each graph as a discrete distribution and utilizes maximum mean discrepancy to learn the graph metric for multi-scale KDE, where all parameters are learned by maximizing the density of graphs relative to the density of their well-designed perturbed counterparts. The perturbations are conducted on both node features and graph spectra, which helps better characterize the boundary of normal density regions. Theoretically, we establish consistency and convergence guarantees for LGKDE, including bounds on the mean integrated squared error, robustness, and generalization. We validate LGKDE by demonstrating its effectiveness in recovering the underlying density of synthetic graph d

istributions and applying it to graph anomaly detection across diverse benchmark datasets. Extensive empirical evaluation shows that LGKDE demonstrates superior performance compared to state-of-the-art baselines on most benchmark datasets.
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VIP: Visual-guided Prompt Evolution for Efficient Dense Vision-Language Inferenc
e
Hao Zhu, Shuo Jin, Wenbin Liao, Jiayu Xiao, Yan Zhu, Siyue Yu, Feng Dai
Pursuing training-free open-vocabulary semantic segmentation in an efficient and generalizable manner remains challenging due to the deep-seated spatial bias in CLIP.
To overcome the limitations of existing solutions, this work moves beyond the CL
IP-based paradigm and harnesses the recent spatially-aware dino.txt framework to facilitate more efficient and high-quality dense prediction.
While dino.txt exhibits robust spatial awareness, we find that the semantic ambi
guity of text queries gives rise to severe mismatch within its dense cross-modal interactions.
To address this, we introduce VIsual-guided Prompt evolution (VIP) to rectify th
e semantic expressiveness of text queries in dino.txt, unleashing its potential for fine-grained object perception.
Towards this end, VIP integrates alias expansion with a visual-guided distillati
on mechanism to mine valuable semantic cues, which are robustly aggregated in a saliency-aware manner to yield a high-fidelity prediction.
Extensive evaluations demonstrate that VIP: I) surpasses the top-leading methods by 1.4%~8.4\% average mIoU, II) generalizes well to diverse challenging domains
, and III) requires marginal inference time and memory overhead.
****
EAGer: Entropy-Aware GEneRation for Adaptive Inference-Time Scaling
Daniel Scalena, Leonidas Zotos, Elisabetta Fersini, Malvina Nissim, Ahmet Üstün
With the rise of reasoning language models and test-time scaling methods as a pa
radigm for improving model performance, substantial computation is often require
d to generate multiple candidate sequences from the same prompt. This enables ex
ploration of different reasoning paths toward the correct solution, however, all
ocates the same compute budget for each prompt. Grounded on the assumption that
different prompts carry different degrees of complexity, and thus different comp
utation needs, we propose EAGer, a training-free generation method that leverage
s model uncertainty through token-wise entropy distribution to reduce redundant
computation and concurrently improve overall performance. EAGer allows branching to multiple reasoning paths only in the presence of high-entropy tokens, and re
allocates the saved compute budget to instances where exploration of alternative paths is most needed. We validate EAGer across multiple open-source models on c
omplex reasoning benchmarks, with gains specifically demonstrated on AIME 2025.
When target labels are accessible -- as in RLVR training pipelines -- EAGer achi
eves up to +37% in Pass@k and 59% fewer tokens; in test-time settings it still y
ields +12% in Pass@k and 64% fewer tokens compared to Full Parallel Sampling.
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Multi-timescale Reinforcement Learning by Value Reconstruction
Zhan Su, Peixi Peng, Xinyu Hu, Cong Li, Yisen Zhao, Zhuojian Li, Yonghong Tian,
Fanqi Shen
Most reinforcement learning (RL) baselines maximize future cumulative rewards wi
th a fixed single discount factor, which
limits their performance in complex sequential decision-making tasks due to a fa
ilure to balance short-term objectives and
long-term planning. To address this issue, this paper focuses on a multi-timesca
le critic framework, where each component corresponds
to a Q-value with a distinct discount factor. Two key improvements are proposed: (1) A Neural Reward Decoder reconstructs the
reward sequence from multi-scale Q-values, with value and reward reconstruction
losses enhancing Q-value estimation consistency;
(2) A cross-attention-based Q-weight predictor adaptively adjusts Q-value weight
s via current observations to generate the final

Q-value for policy optimization. Extensive experiments on DMControl and CARLA be
nchmarks demonstrate that our method
significantly outperforms state-of-the-art (SOTA) baselines. Furthermore, we val
idate the framework's generalizability
by integrating it with both off-policy (SAC, DrQ-v2) and on-policy (PPO) algorit
hms, achieving consistent performance gains.
The code is available in the supplementary material.
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Human-AI Collaborative Uncertainty Quantification
Sima Noorani, Shayan Kiyani, George J. Pappas, Hamed Hassani
AI predictive systems increasingly support high-stakes decision making, yet robu
st decisions under uncertainty often rely on human capabilities beyond AI alone.
 This motivates collaborative approaches that combine human judgment with AI pre
dictions. We study this problem through the lens of uncertainty quantification a
nd introduce **Human-AI Collaborative Uncertainty Quantification**, a framework
in which an AI system refines a human expert’s proposed prediction set subject t
o two principles: **counterfactual harm**, requiring that the AI not degrade cor
rect human judgments, and **complementarity**, requiring recovery of correct out
comes the human missed. At the population level, we show that the optimal collab
orative prediction set has a simple two-threshold structure over a single score
function, governing pruning and augmentation relative to the human proposal. Bui
lding on this characterization, we develop offline and online calibration algori
thms with **distribution-free** finite-sample guarantees. The online algorithm a
dapts to arbitrary distribution shifts, including settings where human behavior
evolves through interaction with the AI. Empirically, we show that collaborative
 prediction sets outperform human-only and AI-only baselines, achieving improved
 coverage--efficiency tradeoffs across image classification, regression, and tex
t-based medical decision making.
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A Theory of Data Acquisition and Pricing at Scale
Andrew Ilyas, Amin Saberi, Grigoris Velegkas
Data plays an invaluable role in large-scale ML training pipelines. Multiple fac
tors, including the need to incentivize the creation of high-quality data and ef
forts to compensate creative data work, have led to increased interest in data p
ricing.
Data pricing mechanisms seek to establish a market where data providers are comp
ensated based (in part) on the value of their data to the data buyer, e.g., fron
tier AI labs. However, assessing the exact value that each provider's data adds
to the data buyer's objective requires repeated re-training, which is infeasible
 in practice.
Our work studies data pricing under compute constraints.
In our setting, data buyers cannot make data acquisition decisions optimally due
 to limited compute.
Inspired by existing practice in the field of data selection, we propose a model
 for this problem called ``pricing with an attribution oracle,'' and provide a t
heoretical analysis of compute-efficient acquisition and pricing.
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Acoustic Interference: A New Paradigm Weaponizing Acoustic Latent Semantic for U
niversal Jailbreak against Large Audio Language Models
Yanyun Wang, Yu Huang, Zi Liang, Xixin Wu, Li Liu
The integration of audio modality into Large Audio Language Models (LALMs) signi
ficantly expands their attack surface. Existing jailbreaks predominantly treat a
udio as a carrier for malicious payloads, relying on semantic optimization, acou
stic parameter control, or additive perturbation to embed harmful content into t
he audio signal. In this work, we challenge this necessity and propose a new par
adigm in which the role of audio shifts from content injection to safety alignme
nt interference. We reveal that LALM safety alignment can be compromised solely
by specific **Acoustic Latent Semantics (ALS)**, the underlying paralinguistic f
eatures intrinsic to the priors of audio generative models. Distinct from previo
us works that leverage explicit acoustic parameters to merely style malicious au

dio, we demonstrate that interference audio, benign in content but infused with specific ALS, can serve as a universal jailbreak trigger. Leveraging this insight, we propose **Acoustic Interference Attack (AIA)**, which decouples the attack payload from the audio. It employs a set of universal, instruction-neutral interference audio, enabling standard malicious text queries to bypass safety alignment without instance-specific optimization. Experiments on 10 LALMs across five datasets demonstrate that AIA achieves the state-of-the-art attack success rate. Furthermore, our interpretability analysis uncovers the inference path drift induced by AIA and identifies the inherent effective patterns within ALS, revealing the fundamental vulnerability of cross-modal alignment in LALMs.

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SURF: Separation via Unsupervised Remixing Flow

Henry Li, Robin Scheibler, Efthymios Tzinis, Matt Shannon, Arnaud Doucet, John R. Hershey

The goal of single-channel source separation is to reconstruct K sources given their mixture. In supervised settings where vast amounts of clean source data are available, this challenging, ill-posed problem has been addressed successfully by generative diffusion and flow-based prior models. However, access to such clean source samples is often limited, and even when available, supervised models are vulnerable to domain shifts. To bridge this gap, we present Separation via Unsupervised Remixing Flow (\textbf{SURF}), an unsupervised flow matching approach for source separation that learns directly from observed mixtures. This method relies on a novel combination of state-of-the-art supervised flow matching and regression-based self-supervised techniques. At a high level, starting from a teacher model, we utilize a ``remixing'' step to bootstrap the learning of a student flow model from the teacher's estimates. We provide insights into the objectives optimized by this approach and draw a novel connection to the Wake-Sleep algorithm. Empirical evaluations on image and audio benchmarks demonstrate that \textbf{SURF} establishes a new state-of-the-art, significantly outperforming existing unsupervised methods. See our demo page for examples. https://google.github.io/df-conformer/surf/

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Learning Self-Correction in Vision–Language Models via Rollout Augmentation

Yi Ding, Ziliang Qiu, Bolian Li, Ruqi Zhang

Self-correction is essential for solving complex reasoning problems in vision–language models (VLMs), yet existing reinforcement learning (RL) methods struggle to learn it.

Effective self-correction behaviors emerge only rarely during RL, making learning signals sparse. To address this challenge, we propose c**o**rre**ct**i**o**n-s**p**ecific rollo**u**t**s**} (**Octopus**), a rollout-augmentation framework that synthesizes dense self-correction supervision by recombining existing rollouts without computational overhead. This rollout augmentation simultaneously improves sample efficiency and stabilizes RL optimization. Furthermore, we introduce a two-stage RL training strategy that disentangles self-correction and direct reasoning, avoiding signal conflicts and enabling both behaviors to be learned effectively.

Building on this, we introduce $\texttt{Octopus-8B}$, an advanced reasoning VLM with controllable self-correction capabilities. It achieves SoTA performance among open-source VLMs across 7 benchmarks, outperforming the best RLVR baseline by 1.0 score while requiring only $0.72\times$ training time per step.

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Hyperspectral Image Fusion with Spectral-Band and Fusion-Scale Agnosticism

Yujie Liang, Zihan Cao, Liang-Jian Deng, Yang Yang, Malu Zhang

Current deep learning models for Multispectral and Hyperspectral Image Fusion (MS/HS fusion) are typically designed for fixed spectral bands and spatial scales, which limits their transferability across diverse sensors. To address this, we propose SSA, a unified framework for MS/HS fusion with spectral-band and fusion-scale agnosticism. Specifically, we introduce Matryoshka Kernel (MK), a novel operator that enables a single model to process varying ordered spectral band counts within a supported range. Meanwhile, we build SSA upon an Implicit Neural Rep

resentation (INR) backbone that models the HS signal as a continuous function, e nabling reconstruction at arbitrary spatial resolutions. Together, these two mec hanisms support a single MS/HS fusion model for heterogeneous sensors and arbitr ary query scales. Extensive experiments demonstrate that our single jointly trai ned model achieves state-of-the-art performance, generalizes to unseen spatial s cales, and transfers to unseen sensors with few-step adaptation. The code can b e obtained at https://github.com/vg219/SSA.
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DreamDojo: A Generalist Robot World Model from Large-Scale Human Videos
Shenyuan Gao, William Liang, Kaiyuan Zheng, Ayaan Naveed Malik, Seonghyeon Ye, S ihyun Yu, Wei-Cheng Tseng, Yuzhu Dong, Kaichun Mo, Chen-Hsuan Lin, Jiannan Xiang , Yuqi Xie, Ruijie Zheng, Dantong Niu, Pooya Jannaty, Jinwei Gu, Jun Zhang, Jite ndra Malik, Pieter Abbeel, Ming-Yu Liu, Yuke Zhu, Joel Jang, Linxi Fan
Being able to simulate the outcomes of actions in varied environments will revol utionize the development of generalist agents at scale. However, modeling these world dynamics, especially for dexterous robotics tasks, poses significant chall enges due to limited data coverage and scarce action labels. As an endeavor towa rds this end, we introduce DreamDojo, a foundation world model that learns diver se interactions and dexterous controls from 44k hours of egocentric human videos . Our data mixture represents the largest video dataset to date for world model pretraining, spanning a wide range of daily scenarios with diverse objects and s kills. To address the scarcity of action labels, we introduce continuous latent actions as unified proxy actions, enhancing interaction knowledge transfer from unlabeled videos. After post-training on small-scale target robot data, DreamDoj o demonstrates a strong understanding of physics and precise action controllabil ity. We also devise a distillation pipeline that accelerates DreamDojo to a real -time speed of 10.93 FPS and further improves consistency to the context. Our wo rk enables several important applications based on generative world models, incl uding live teleoperation, policy evaluation, and model-based planning. Systemati c evaluation on multiple challenging out-of-distribution (OOD) benchmarks verifi es the significance of our method for simulating open-world, contact-rich tasks, paving the way for general-purpose robot world models.
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Active Regression for Single-Index Models with Unknown Link Functions
Chansophea Wathanak In, Yi Li, Wai Ming Tai, Xuan Wu
This paper studies active regression for single-index models under general $\ell _p$-loss with an unknown 1-Lipschitz link function f, formulated as $\min_{f ,x} \Vert f(Ax)-b\Vert_p^p$ with full access to A but coordinate-query access to b.
Prior work established upper bounds for known link functions for all $p\geq 1$ a nd for unknown link functions only in the $p=2$ case, together with lower bounds for $p\leq 2$.
This work addresses the more challenging setting of unknown link functions and g eneral $p \geq 1$.
A non-adaptive sampling algorithm is presented that achieves a $(1+\epsilon)$-ap proximation using $O(d^{p/2\vee 1}/\epsilon^{p\vee 2}\text{poly}\log(n/\epsilon))$ queries. Nearly tight lower bounds are also established for non-adaptive quer ies when $p>2$. These results close much of the remaining gap in active ℓ_p regression for single-index models.
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RGMem: Renormalization Group–inspired Memory Evolution for Language Agents
Ao Tian, Yunfeng Lu, Xinxin Fan, Changhao Wang, Lanzhi Zhou, Yeyao Zhang, Yanfan g Liu
Personalized and continuous interactions are critical for LLM-based conversation al agents, yet finite context windows and static parametric memory hinder the mo deling of long-term, cross-session user states. Existing approaches, including r etrieval-augmented generation and explicit memory systems, primarily operate at the fact level, making it difficult to distill stable preferences and deep user traits from evolving and potentially conflicting dialogues.To address this chall enge, we propose RGMem, a self-evolving memory framework inspired by the renorma

lization group (RG) perspective on multi-scale organization and emergence. RGMem models long-term conversational memory as a multi-scale evolutionary process: episodic interactions are transformed into semantic facts and user insights, which are then progressively integrated through hierarchical coarse-graining, thresholded updates, and rescaling into a dynamically evolving user profile.By explicitly separating fast-changing evidence from slow-varying traits and enabling nonlinear, phase-transition-like dynamics, RGMem enables robust personalization beyond flat retrieval or static summarization. Extensive experiments on the LOCOMO and PersonaMem benchmarks demonstrate that RGMem consistently outperforms SOTA memory systems, achieving stronger cross-session continuity and improved adaptation to evolving user preferences.

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Convex Basins in Single-Index Model Loss Landscapes: Applications to Robust Recovery under Strong Adversarial Corruption

SANTANU DAS, Sagnik Chatterjee, jatin batra

In this paper, we tackle a fundamental problem in high-dimensional statistics, namely, learning Single Index Models (SIMs) robustly in the presence of heavy-tailed noise and an adversary that can corrupt a constant fraction of both covariates and responses. Prior research on efficient robust recovery only focuses on monotonic link functions or only limit themselves to Phase Retrieval. Provable efficient robust recovery guarantees for generic nonlinear link functions have remained elusive. In this paper, we obtain the first near-linear time, optimal-sample-complexity robust recovery algorithm for a wide class of nonlinear non-monotonic link functions. Critical to our result is an improved understanding of the squared-loss landscape: we identify a sufficient condition under which a broad class of non linear link functions admit a dimension-independent constant-radius convex basin around the ground truth, establishing statistical identifiability beyond previously known cases. We also leverage second-order Stein's identities to identify a structural condition, that we term Expected Squared Convexity (ESC), that acts as a sufficient condition for spectral initialization techniques to obtain an estimator within the convex basin with error $O(\epsilon^{1/4})$, even under heavy-tailed noise and strong adversarial contamination. This robust initialization technique can be combined with a robust gradient descent phase to break the spectral error barrier, achieving an improved estimation error of $O(\sigma\sqrt{\epsilon})$. Our non-convex optimization framework gives the first efficient sample and time complexity robust recovery results for activation functions such as GeLU and Swish that act as building blocks of modern deep-learning architectures.

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Understanding SAM through Minimax Perspective

Ying Chen, Aoxi Li, Javad Lavaei

Sharpness-Aware Minimization (SAM) empirically boosts generalization by seeking parameters that minimize the worst-case loss in a small neighborhood, yet existing theory explains its behavior under either Polyak-Lojasiewicz (PL) condition or upper bounded perturbation radius. We revisit SAM through the bilevel minimax problem $\min_{\theta}\max_{\|\Delta\|\le\rho}l(\theta+\Delta)$ and derive a (θ,Δ) gradient flow ODE whose equilibria coincide with the problem’s optimality conditions. A Lyapunov argument-free of convexity assumptions, quantifies how the optimality gap depends on the radius~ρ and local curvature. Discretizing the flow yields a \emph{Multi-step SAM} algorithm that recovers classical SAM as $\rho\to 0$. Moreover, our analysis and the resulting algorithm remain valid even for large ρ, providing guidance for aggressive neighborhood exploration. Experiments on synthetic objectives and CIFAR-10 validate the predicted gains from multiple inner updates, bridging the gap between SAM’s minimax intuition and its practical implementation.

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EvoCF: Multi-Agent Collaboration via Agentic Memory-Driven Evolutionary Counterfactual Planning

Haotian Chi, Zeyu Feng, Xingrui Yu, Linbo Luo, Yew-Soon Ong, Ivor Tsang, Hechang Chen, Yi Chang, Haiyan Yin

Planning collaboration strategies for multi-agent embodied systems remains a cor
e challenge for LLM-based planners, which often fail to capture the physical and
 coordination constraints of realworld environments. To address this, we present
 EvoCF, an agentic memory-driven evolutionary counterfactual planning framework
for discovering improved multi-agent collaboration strategies through counterfac
tual plan generation and evaluation. First, we propose a symbolic constraint ind
uctor that induces reusable symbolic constraints from failures, forming an evolv
ing rule library. Then, we propose an evolutionary counterfactual plan generator
 that systematically explores semantically consistent plan variants through rule
-conditioned mutations, enabling robust collaboration strategies beyond short-si
ghted one-shot LLM plans. Finally, we design an agentic memory-grounded evaluato
r that ranks candidate plans using retrieval-augmented evidence, producing inter
pretable, constraint-aware selections. Across multi-agent embodied simulation be
nchmarks, EvoCF consistently discovers more robust and executable plans compared
 to baseline approaches. Our results demonstrate that grounding multi-agent plan
ning in agentic memory and counterfactual reasoning significantly enhances both
effectiveness and robustness.
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IEC: When Information-Driven Exploration Meets Spectral Consensus via Primal–Dua
l Reward Regularization in Decentralized Multi-Agent RL

Xuefeng Du, Jiajun Wu, Yuduo Zheng, Fengqi Li

Decentralized multi-agent reinforcement learning faces a persistent exploration
–coordination tension: intrinsic rewards promote exploration under sparse feedba
ck, yet effective cooperation requires agents’ behaviors to remain consistent ov
er a limited communication graph. Existing methods often combine exploration bon
uses and coordination regularizers with fixed-weight schedules, making them hard
 to tune and prone to either fragmented conventions or premature behavioral coll
apse. We propose the IEC (Isomorphic Exploration-Consensus) framework that coupl
es exploration and coordination through a single constrained objective: maximize
 task return augmented with two complementary exploration signals, dynamics-base
d information gain and state-coverage novelty, while constraining graph-induced
policy disagreement via a spectral smoothness penalty on neighboring agents, whi
ch can be interpreted as a Dirichlet-energy regularizer on the communication gra
ph. IEC optimizes the resulting Lagrangian with a lightweight primal–dual update
 that adapts the consensus multiplier from observed constraint violations, yield
ing an automatic shift from diverse exploration to stable cooperative convention
s. Across three distinct benchmarks, IEC achieves superior performance.
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SpatialJB: How Text Distribution Art Becomes The "Jailbreak Key" for LLM Guardra
ils

Zhiyi Mou, Jingyuan Yang, ZEHENG QIAN, Wangze Ni, Tianfang Xiao, Ning Liu, Chen
Jason Zhang, Zhan Qin, Kui Ren

While Large Language Models (LLMs) have achieved remarkable success across diver
se tasks, they remain vulnerable to jailbreak attacks, which pose significant ri
sks to their secure deployment. Driven by their inherent token-by-token autoregr
essive inference, LLMs exhibit semantic representations that lack robustness aga
inst spatially structured perturbations, thereby rendering current output-guardr
ail safety mechanisms penetrable.

Exploiting the Transformer's spatial weakness, we propose SpatialJB to disrupt t
he model’s output generation process, allowing harmful content to bypass guardra
ils without detection. Comprehensive experiments on leading LLMs demonstrate tha
t SpatialJB achieves a nearly 100\% ASR and consistently maintains a success rat
e exceeding 75\% even against advanced output guardrails like the OpenAI Moderat
ion API, outperforming current jailbreak techniques by a significant margin.

While **SpatialJB** advances LLM safety research by exposing guardrail weaknesse
s and highlighting spatial semantics, we also propose and evaluate baseline defe
nse strategies to prevent its potential misuse.

You can click [Video Link](https://1drv.ms/v/s!ApaP6YaJA87pgk-Rk7ZCvPeYcIZa?e=J8
kbg0) and [Code Link](https://github.com/DongFangShenWa/SpatialJailbreak.git) to
 see our demo presentation and code.

****
From Growing to Looping: A Unified View of Iterative Computation in LLMs
Ferdinand Kapl, Emmanouil Angelis, Kaitlin Maile, Johannes von Oswald, Stefan Ba
uer
Looping, reusing a block of layers across depth, and depth growing, training sha
llow-to-deep models by duplicating middle layers, have both been linked to stron
ger reasoning, but their relationship remains unclear. We provide a mechanistic
unification: looped and depth-grown models exhibit convergent depth-wise signatu
res, including increased reliance on late layers and recurring patterns aligned
with the looped or grown block. These shared signatures support the view that th
eir gains stem from a common form of iterative computation. Building on this con
nection, we show that the two techniques are adaptable and composable: applying
inference-time looping to the middle blocks of a depth-grown model improves accu
racy on some reasoning primitives by up to $2\times$, despite the model never be
ing trained to loop. Both approaches also adapt better than the baseline when gi
ven more in-context examples or additional supervised fine-tuning data. Addition
ally, depth-grown models achieve the largest reasoning gains when using higher-q
uality, math-heavy cooldown mixtures, which can be further boosted by adapting a
 middle block to loop. Overall, our results position depth growth and looping as
 complementary, practical methods for inducing and scaling iterative computation
 to improve reasoning.
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Skipping the Zeros in Diffusion Models for Sparse Data Generation
Phil Ostheimer, Mayank Nagda, Andriy Balinskyy, Gabriel Vicente Rodrigues, Jean
Radig, Carl Herrmann, Stephan Mandt, Marius Kloft, Sophie Fellenz
Diffusion models (DMs) excel on dense continuous data, but are not designed for
sparse continuous data. They do not model exact zeros that represent the deliber
ate absence of a signal. As a result, they erase sparsity patterns and perform u
nnecessary computation on mostly zero entries. With Sparsity-Exploiting Diffusio
n (SED), we model only non-zero values, preserving sparsity. SED delivers comput
ational savings while maintaining or improving generation quality by skipping ze
ros during training and inference. Across physics and biology benchmarks, SED ma
tches or surpasses conventional DMs and domain-specific baselines, while vision
experiments provide intuitive insights into the limitations of dense DMs and the
 benefits of SED.
****
Bridging RGB and RAW: Single-step Deterministic Flow with Homogeneous Representa
tion Alignment
Diedong Feng, Peiyi Zeng, Zhen Liu, Zhongyang Li, Bing Zeng, Shuaicheng Liu
Reconstructing high-fidelity RAW sensor data from processed RGB images is a fund
amental yet ill-posed problem, plagued by irreversible information loss and comp
lex non-linear ISP transformations. While generative models offer high-quality r
econstruction, they suffer from prohibitive computational costs. Conversely, dom
inant regression-based methods are fast but susceptible to incoherent observatio
nal deviations, often yielding over-smoothed predictions that drift from the aut
hentic signal manifold. To reconcile this trade-off, we propose SHADE, a Single-
step Homogeneous Aligned DEterministic flow framework. We validate that, unlike
point-to-point regression, the single-step deterministic flow captures global tr
ansport trends and enables intrinsic robustness against input perturbations. Fur
thermore, we introduce Homogeneous Representation Alignment to maximize fidelity
. By leveraging a homogeneously initialized student-teacher DINO pair, this mech
anism enforces alignment within a shared feature space, significantly amplifying
 the representational capacity. Extensive experiments demonstrate that SHADE ach
ieves state-of-the-art performance on multiple benchmarks, establishing a new pa
radigm for accurate and efficient sensor data reconstruction.
****
HE-SNR: Uncovering Latent Logic via Entropy for Guiding Mid-Training on SWE-benc
h
Yueyang Wang, Jiawei Fu, Baolong Bi, Xili Wang, Xiaoqing Liu
SWE-bench has emerged as the premier benchmark for evaluating Large Language

Models on complex software engineering tasks. While these capabilities are fundamentally acquired during the *mid-training* phase and subsequently elicited during Supervised Fine-Tuning (SFT), there remains a critical deficit in metrics capable of guiding mid-training effectively. Standard metrics such as Perplexity (PPL) are compromised by the "Long-Context Tax" and exhibit weak correlation with downstream SWE performance. In this paper, we bridge this gap by first introducing a rigorous data filtering strategy. Crucially, we propose the **Entropy Compression Hypothesis**, redefining intelligence not by scalar Top-1 compression, but by the capacity to structure uncertainty into Entropy-Compressed States of low orders ("reasonable hesitation"). Grounded in this fine-grained entropy analysis, we formulate a novel metric, **HE-SNR** (High-Entropy Signal-to-Noise Ratio). We validate our approach on models with up to 560B parameters across different context windows (32K/128K). This work provides both the theoretical foundation and practical tools for optimizing the latent potential of LLMs in complex engineering domains.

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Dropout Universality: Scaling Laws and Optimal Scheduling at the Edge-of-Chaos

Lucas Fernandez-Sarmiento

We develop a mean-field theory of dropout as a perturbation of critical signal propagation at the edge of chaos, and show that it predicts a simple, no-cost change to standard practice: front-loaded dropout schedules cut test loss by 18–35% over constant dropout in MLPs and Vision Transformers at fixed budget. The theoretical mechanism is that dropout shifts the perfect-alignment fixed point, making the depth scale for information propagation finite even at critical initialization. We derive critical and crossover scaling laws for correlation decay and establish that smooth activations and kinked, ReLU-like activations constitute distinct universality classes, with different critical exponents and a universal two-parameter scaling collapse in detuning and dropout strength. The distinction traces to the analytic structure of the correlation map: smooth activations admit a Taylor expansion near perfect alignment, while kinked activations develop a branch point with universal non-analyticity. As a corollary, the framework yields saturated dropout profiles under fixed budget; a regularization-reach argument then selects front-loaded schedules, with accuracy gains as a consistent secondary effect. We also discuss how the same Gaussian-kernel structure extends the theory beyond MLPs toward CNNs and residual architectures.

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PaperBanana: Automating Academic Illustration for AI Scientists

Dawei Zhu, Rui Meng, Yale Song, Xiyu Wei, Sujian Li, Tomas Pfister, Jinsung Yoon

Despite rapid advances in autonomous AI scientists powered by language models, generating publication-ready illustrations remains a labor-intensive bottleneck in the research workflow.

To lift this burden, we introduce PaperBanana, an agentic framework for automated generation of publication-ready academic illustrations.

Powered by state-of-the-art VLMs and image generation models,

PaperBanana orchestrates specialized agents to retrieve references, plan content and style, render images, and iteratively refine via self-critique.

To rigorously evaluate our framework, we introduce PaperBananaBench, comprising 292 test cases for methodology diagrams curated from NeurIPS 2025 publications, covering diverse research domains and illustration styles.

Comprehensive experiments demonstrate that PaperBanana consistently outperforms leading baselines in faithfulness, conciseness, readability, and aesthetics.

We further show that our method effectively extends to the generation of high-quality statistical plots.

Collectively,

PaperBanana paves the way for the automated generation of publication-ready illustrations.

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SD-MoE: Spectral Decomposition for Effective Expert Specialization

Ruijun Huang, Fang Dong, Xin Zhang, Anrui Chen, Hengjie Cao, Zhendong Huang, Jixian Zhou, Mengyi Chen, Yifeng Yang, Mingzhi Dong, Yujiang Wang, Jinlong Hou, Qin

Lv, Robert P. Dick, Yuan Cheng, Fan Yang, Tun Lu, Chun Zhang, Li Shang
Mixture-of-Experts (MoE) architectures scale Large Language Models via expert specialization induced by conditional computation. In practice, however, expert specialization often fails: some experts become functionally similar, while others functioning as de facto shared experts, limiting the effective capacity and model performance.
In this work, we analysis from a spectral perspective on parameter and gradient spaces, uncover that
(1) experts share highly overlapping dominant spectral components in their parameters,
(2) dominant gradient subspaces are strongly aligned across experts, driven by ubiquitous low-rank structure in human corpus, and
(3) gating mechanisms preferentially route inputs along these dominant directions, further limiting specialization.
To address this, we propose Spectral-Decoupled MoE (SD-MoE), which decomposes both parameter and gradient in the spectral space.
SD-MoE improves performance across downstream tasks, enables effective expert specialization, incurring minimal additional computation, and can be seamlessly integrated into a wide range of existing MoE architectures, including Qwen and DeepSeek.
****
HyperMLP: An Integrated Perspective for Sequence Modeling
Jiecheng Lu, Shihao Yang
Self-attention is often viewed as probabilistic query-key lookup, motivating designs that preserve normalized attention scores and fixed positional semantics. We advocate a simpler and more unified perspective: an autoregressive attention head can be viewed as a dynamic two-layer MLP whose weights are instantiated from the context history. From this view, attention scores form an ever-growing hidden representation, and standard MLP activations such as ReLU or GLU naturally implement input-conditioned selection over a context-dependent memory pool rather than a probability distribution. Based on this formulation, we introduce \textbf{HyperMLP} and \textbf{HyperGLU}, which learn dynamic mixing in both feature space and sequence space, using a reverse-offset (lag) layout to align temporal mixing with autoregressive semantics. We provide theoretical characterizations of the expressivity and implications of this structure, and empirically show that HyperMLP/HyperGLU consistently outperform strong softmax-attention baselines under matched parameter budgets.
****
SkillTrojan: Backdoor Attacks on Skill-Based Agent Systems
Yunhao Feng, Yifan Ding, Yingshui Tan, Boren Zheng, Xiaolong Li, Kun Zhai, Li Yishan, Yanming Guo, Wenke Huang
Skill-based agent systems tackle complex tasks by composing reusable skills, improving modularity and scalability while introducing a largely unexamined security attack surface. We propose \textbf{SkillTrojan}, a backdoor attack that targets skill implementations rather than model parameters or training data. SkillTrojan embeds malicious logic inside otherwise plausible skills and leverages standard skill composition to reconstruct and execute an attacker-specified payload. The attack partitions an encrypted payload across multiple benign-looking skill invocations and activates only under a predefined trigger. SkillTrojan also supports automated synthesis of backdoored skills from arbitrary skill templates, enabling scalable propagation across skill-based agent ecosystems. To enable systematic evaluation, we release a dataset of 3,000+ curated backdoored skills spanning diverse skill patterns and trigger–payload configurations. We instantiate SkillTrojan in a representative code-based agent setting and evaluate both clean-task utility and attack success rate. Our results show that skill-level backdoors can be highly effective with minimal degradation of benign behavior, exposing a critical blind spot in current skill-based agent architectures and motivating defenses that explicitly reason about skill composition and execution. Concretely, on EHR SQL, SkillTrojan attains up to \textbf{97.2\%} ASR while maintaining \textbf{89.3\%} clean ACC on GPT-5.2-1211-Global. Code is available at \url{https:

//github.com/Yunhao-Feng/SkillTrojan}.
****
Artificial Hippocampus Networks for Efficient Long-Context Modeling
Yunhao Fang, Weihao Yu, Shu Zhong, Qinghao Ye, Xuehan Xiong, Lai Wei
Long-sequence modeling faces a fundamental trade-off between the efficiency of compressive fixed-size memory in RNN-like models and the fidelity of lossless growing memory in attention-based Transformers. Inspired by the Multi-Store Model in cognitive science, we introduce a memory framework of artificial neural networks. Our method maintains a sliding window of the Transformer's KV cache as lossless short-term memory, while a learnable module termed Artificial Hippocampus Network (AHN) recurrently compresses out-of-window information into a fixed-size compact long-term memory. To validate this framework, we instantiate AHNs using modern RNN-like architectures, including Mamba2, DeltaNet, and GatedDeltaNet to augment open-weight base LLMs. We also propose an efficient self-distillation method where the base model' all parameters are frozen and only the parameters from AHNs are optimized. For inference, our method sets a default large sliding window size of 32k for attention, and AHNs activate only when the sequence length exceeds the 32k window, addressing the quadratic-complexity issue of attention that emerges at that scale. Extensive experiments on long-context benchmarks LV-Eval and InfiniteBench demonstrate that AHN-augmented models consistently outperform sliding window baselines and achieve performance comparable or even superior to full-attention models, while substantially reducing computational and memory requirements. For instance, augmenting the Qwen2.5-3B-Instruct with AHNs reduces inference FLOPs by 40.5% and memory cache by 74.0%, while improving its average score on LV-Eval (128k sequence length) from 4.41 to 5.88.
****
Alleviating Observation Bias via Causal-Invariant Meta-Learning for Unbalanced Incomplete Multi-view Clustering
Jiaqi Jin, Siwei Wang, Taichun Zhou, Zhibin Dong, Siqi Wang, Miaomiao Li, Xinwang Liu, En Zhu
In real-world scenarios, multi-view data often exhibits significant imbalance in missing patterns across views, where observation rates vary substantially among different views. Such observation bias makes it difficult for cross-view associations learned from limited complete samples to generalize to incomplete samples, leading to challenging cross-view recovery. Meanwhile, observation bias acts as a confounder, causing clustering predictions to spuriously depend on low-missing-rate views. To address these challenges, we propose CIMLN, a novel **C**ausal-**I**nvariant **M**eta-**L**earning **N**etwork that alleviates observation bias for unbalanced incomplete multi-view clustering. The context-aware meta-generation module formulates view recovery as a meta-learning task, enabling rapid adaptation to incomplete samples by encoding global statistical relationships through context information. The causal-invariant structure learning module constructs counterfactual scenarios by artificially masking low-missing-rate views, enforcing clustering consistency across different observation patterns. Extensive experiments on eight benchmarks demonstrate the effectiveness of CIMLN. The code is available at https://github.com/jinjiaqi1998/CIMLN.
****
LUVE : Latent-Cascaded Ultra-High-Resolution Video Generation with Dual Frequency Experts
Chen Zhao, Jiawei Chen, Hongyu Li, Zhuoliang Kang, Shilin Lu, Xiaoming Wei, Kai Zhang, Jian Yang, Ying Tai
Recent advances in video diffusion models have significantly improved visual quality, yet ultra-high-resolution (UHR) video generation remains a formidable challenge due to the compounded difficulties of motion modeling, semantic planning, and detail synthesis. To address these limitations, we propose \textbf{LUVE}, a \textbf{L}atent-cascaded \textbf{U}HR \textbf{V}ideo generation framework built upon dual frequency \textbf{E}xperts. LUVE employs a three-stage architecture comprising low-resolution motion generation for motion-consistent latent synthesis, video latent upsampling that performs resolution upsampling directly in the latent space to mitigate memory and computational overhead, and high-resolution co

ntent refinement that integrates low-frequency and high-frequency experts to joi ntly enhance semantic coherence and fine-grained detail generation. Extensive ex periments demonstrate that our LUVE achieves superior photorealism and content f idelity in UHR video generation, and comprehensive ablation studies further vali date the effectiveness of each component.
****
Semantic Router: On the Feasibility of Hijacking MLLMs via a Single Adversarial Perturbation
Changyue Li, Jiaying Li, Youliang Yuan, Jiaming He, Zhicong Huang, Pinjia He
Multimodal Large Language Models (MLLMs) are increasingly deployed in stateless systems, such as autonomous driving and robotics.
This paper investigates a novel threat: Semantic-Aware Hijacking. We explore the feasibility of hijacking multiple stateless decisions simultaneously using a si ngle universal perturbation.
We introduce the Semantic-Aware Universal Perturbation (SAUP), which acts as a s emantic router, "actively" perceiving input semantics and routing them to distin ct, attacker-defined targets.
To achieve this, we conduct a theoretical and empirical analysis on the geometri c properties in the latent space. Guided by these insights, we propose the Seman tic-Oriented (SORT) optimization strategy and annotate a new dataset with fine-g rained semantics to evaluate performance.
Extensive experiments on three representative MLLMs demonstrate the fundamental feasibility of this attack, achieving a 66% attack success rate over five target s using a single frame against Qwen.
****
RubricRobustness: Evaluating the Sensitivity of Rubrics-Based Benchmarks to Simp le Perturbations
Manasi Sharma
The advancement of Large Language Models (LLMs) into higher-level reasoning doma ins has rendered traditional heuristic evaluators insufficient for long-form ope n-ended responses, precipitating the widespread adoption of rubric-based benchma rks. While these frameworks utilize expert-curated criteria and LLM-as-a-judge t o assess open-ended generation, the intrinsic robustness of these evaluation har nesses to fundamental validity assessments remains critically under-investigated . To bridge this gap, we introduce RubricRobustness, a systematic sensitivity an alysis framework that subjects these benchmarks to three common sense perturbati ons: *semantic negation*, *stochastic deletion* and *irrelevant addition*. We in vestigate the extent to which manipulating the semantic veracity of a model's re sponse impacts its resulting score by applying the robustness framework to two o f the most popular rubrics-based benchmarks: HealthBench and WildBench. Our find ings reveal systematic vulnerabilities: while both benchmarks respond sharply to semantic negation (e.g., degradation slopes of approximately -0.38 on HealthB ench and -0.55 on WildBench), they are substantially less responsive to irrele vant addition, often requiring over 35% of sentences to be perturbed before indu cing even a 25% score drop. We argue that perturbation-based sensitivity analyse s of this form are a necessary prerequisite for validating rubric coverage, ensu ring that automated evaluation frameworks reliably penalize basic semantic failu res. We will release our framework as an open-source tool for building more resi lient benchmarks.
****
Romberg-Extrapolated Zeroth-Order Gradient Estimator: Higher-Order Bias Reductio n with Preserved Leading Directional Variance
Hongcheng Dong, Wenqiang Pu, Licheng Zhao, Rui Zhou, Feng Yin
Zeroth-order optimization is widely used when gradients are unavailable, but the standard two-point estimator suffers from $\mathcal{O}(r^2)$ truncation bias at smoothing radius r.
Existing bias-reduction schemes typically increase the leading directional varia nce under a fixed number of function evaluations per gradient estimate, while va riance-reduction schemes generally do not improve the bias order.
We propose Romberg-ZOGE, which forms a Romberg-extrapolated linear combination o

f two-point differences evaluated at radii $\{r/2^k\}_{k=0}^R$ while reusing the same perturbation direction across all radii.
With appropriately chosen weights, Romberg-ZOGE cancels the first R even-order truncation terms and achieves $\mathcal{O}(r^{2R+2})$ bias under $(2R{+}2)$-order smoothness, while preserving the leading directional variance constant of the two-point estimator up to higher-order residual terms.
We further characterize the stochastic-oracle setting by deriving an explicit noise-amplification factor and corresponding bias and variance bounds.
Experiments on synthetic benchmarks, simulator-based wireless optimization, and black-box prompt tuning of OPT-1.3B demonstrate faster and more stable zeroth-order SGD when the number of function evaluations per gradient estimate is fixed.
****
Mitigating Gradient Pathology in PINNs through Aligned Constraint
Yichen Luo, Peiyu Zhu, Dongxiao Hu, Jia Wang, Tailin Wu, Dapeng Lan, Yu Liu, Zhibo Pang
While Physics-Informed Neural Networks (PINNs) are powerful for solving Partial Differential Equations (PDEs), their training is often paralyzed by gradient pathology. The gradients from the PDE residuals and boundary constraints oppose each other, trapping the model in local minima. Current solutions, such as adaptive weighting or hard constraints, either fail to fundamentally resolve this ill-conditioning or are limited to simple geometries. In this study, we systematically analyze the possible causes of this gradient pathology from the perspectives of loss landscapes and optimization dynamics. Based on the obtained conclusion, we propose Constraint-Aligned loss with Manifold Lifting (CAML). By reformulating all zeroth-order terms into aligned constraints, our method effectively mitigates gradient conflicts. In addition, we introduce a delay factor to help the optimizer skip the high-curvature area. Experiments demonstrate that our CAML significantly enhances numerical stability and efficiency in highly complex PINN problems. Our code is open-sourced on [CAML](https://github.com/YichenLuo-0/CAML).
****
Outcome-Aware Spectral Feature Learning for Instrumental Variable Regression
Dimitri Meunier, Jakub Wornbard, Vladimir R Kostic, Antoine Moulin, Alek Fröhlich, Karim Lounici, Massimiliano Pontil, Arthur Gretton
We address the problem of causal effect estimation in the presence of hidden confounders using nonparametric instrumental variable (IV) regression. An established approach is to use estimators based on learned \emph{spectral features}, that is, features spanning the top singular subspaces of the operator linking treatments to instruments. While powerful, such features are agnostic to the outcome variable. Consequently, the method can fail when the true causal function is poorly represented by these dominant singular functions.

To mitigate, we introduce Augmented Spectral Feature Learning, a framework that makes the feature learning process outcome-aware. Our method learns features by minimizing a novel contrastive loss derived from an augmented operator that incorporates information from the outcome. By learning these task-specific features, our approach remains effective even under spectral misalignment. We provide a theoretical analysis of this framework and validate our approach on challenging benchmarks.
****
GenShield: Unified Detection and Artifact Correction for AI-Generated Images
Zhipei Xu, Xuanyu Zhang, Youmin Xu, Qing Huang, Shen Chen, Taiping Yao, Shouhong Ding, Jian Zhang
Diffusion-based image synthesis has made AI-generated images (AIGI) increasingly photorealistic, raising urgent concerns about authenticity in applications such as misinformation detection, digital forensics, and content moderation. Despite the substantial advances in AIGI detection, how to correct detected AI-generated images with visible artifacts and restore realistic appearance remains largely underexplored. Moreover, few existing work has established the connection between AIGI detection and artifact correction. To fill this gap, we propose GenShield, a unified autoregressive framework that jointly performs explainable AIGI det

ection and controllable artifact correction in a closed loop from diagnosis to r
estoration, revealing a mutually reinforcing relationship between these two task
s. We further introduce a Visual Chain-of-Thought based curriculum learning str
ategy that enables self-explained, multi-step "diagnose-then-repair" correction
with an explicit stopping criterion. A high-quality dataset with large-scale "ar
tifact-restored" pairs is also constructed alongside a unified evaluation pipeli
ne. Extensive experiments on our correction benchmark and mainstream AIGI detect
ion benchmarks demonstrate state-of-the-art performance and strong generalizatio
n of our method.
****

Leveraging Machine Unlearning for Cost-Efficient Preference Alignment
XiaoHua Feng, Yuyuan Li, HuWei Ji, Li Zhang, Jiaming Zhang, Tianyu Du, Chaochao Chen
Despite advances in Preference Alignment (PA) for Large Language Models (LLMs),
mainstream methods like reinforcement learning with human feedback face notable
challenges. These approaches require high-quality datasets of positive preferenc
e examples, which are costly to obtain and computationally intensive. The LLM un
learning technique presents a promising alternative by directly removing the inf
luence of negative examples. However, current research has primarily focused on
empirical validation, lacking systematic quantitative analysis. To bridge this g
ap, we propose a framework linking PA with LLM unlearning. Through bi-level opti
mization, we first quantify how unlearning specific negative examples impacts PA
performance. Our analysis reveals that these effects vary substantially across
negative examples. Building on this insight, we pose a crucial question: how can
we optimally select and weight negative examples for unlearning to maximize PA
performance? To answer this, we propose Unlearning to Align (U2A), which leverag
es bi-level optimization to efficiently select and unlearn examples for optimal
PA performance. We validate the proposed method through extensive experiments, w
ith results confirming its effectiveness. Our code is available at https://anony
mous.4open.science/r/U2A-9E75.
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Optimal Design for Multinomial Logit Model with Applications to Best Assortment Identification
Joongkyu Lee, Min-hwan Oh
We study optimal experimental design for multinomial logit (MNL) bandits, where
an agent repeatedly selects a subset of K items from a ground set of size N
and observes single-choice feedback. Unlike linear or generalized linear bandits
, MNL bandits have a combinatorial action space, which makes classical optimal d
esign approaches and naive optimization over all subsets computationally intract
able. We propose a computationally efficient optimal design framework for MNL mo
dels that achieves both statistical efficiency and scalability through two compl
ementary approaches: (i) an exact or certified-approximate reformulation of the
design oracle as a 0-1 mixed-integer linear program (MILP) with solver-certi
fied early stopping, and (ii) a fully polynomial-time lifted design that replace
s the nonlinear objective with a tractable surrogate. Using the Kiefer-Wolfowitz
equivalence theorem, we establish near G-optimality guarantees and characterize
the induced statistical-computational trade-offs. As an application, we develop
a best assortment identification algorithm for MNL bandits with linear utilitie
s and non-uniform revenues, and prove an instance-dependent sample complexity of
$\tilde{\mathcal{O}}\big(\frac{d \log N}{\Delta^2}\big)$, where d is the feat
ure dimension, N is the number of arms, and Δ is the minimum revenue ga
p.
****

LightningRL: Breaking the Accuracy–Parallelism Trade-off of Block-wise dLLMs via Reinforcement Learning
Yanzhe Hu, Yijie Jin, Pengfei Liu, Kai Yu, Zhijie Deng
Diffusion Large Language Models (dLLMs) enable parallel token generation, and th
eir block-wise variants have attracted significant attention. However, existing
dLLMs usually exhibit an accuracy–parallelism trade-off, where raising tokens pe
r forward (TPF) via aggressive parallel decoding often degrades task accuracy. T

o address this, we suggest developing a post-training approach to directly optim
ize the speed–quality frontier of pre-trained dLLMs. Conceptually, we do not req
uire the model to decode aggressively along all sampling trajectories, but rathe
r to find several highly parallelizable ones that can yield correct results. To
this end, we resort to a reinforcement learning paradigm, i.e., LightningRL, to
optimize rewards regarding both the final accuracy and inference parallelism. Li
ghtningRL follows the Group Relative Policy Optimization (GRPO) framework, with
further improvements for dLLMs: 1) stabilized training via per-reward decoupled
normalization, 2) token-level negative log-likelihood (NLL) loss on correct traj
ectories for regularization, and 3) improved training efficiency through dynamic
 sampling with TPF-aware filtering. Across maths and code tasks, LightningRL con
sistently advances the Pareto frontier, maintaining competitive accuracy while i
ncreasing parallelism to an average TPF of 7.3 (up to 11.10 on MBPP).
****

Forget-It-All: Multi-Concept Machine Unlearning via Concept-Aware Neuron Masking
Kaiyuan Deng, Bo Hui, Gen Li, Jie Ji, Minghai Qin, Geng Yuan, Xiaolong Ma
The widespread adoption of text-to-image (T2I) diffusion models has raised conce
rns about their potential to generate copyrighted, inappropriate, or sensitive i
magery. As a practical solution, *machine unlearning* aims to erase unwanted con
cepts without retraining from scratch. While most existing methods are effective
 for single-concept unlearning, they often struggle when removing multiple conce
pts, causing significant challenges in unlearning effectiveness, generation qual
ity, and sensitivity to hyperparameters and datasets. We take a unique perspecti
ve on multi-concept unlearning by leveraging model sparsity and propose the **F*
*orget **I**t **A**ll (FIA) framework. FIA first introduces *Contrastive Concept
 Saliency* to quantify each weight connection's contribution to a target concept
. It then identifies *Concept Sensitive Neurons* by combining temporal and spati
al information, ensuring that only neurons consistently responsive to the target
 concept are selected. Finally, FIA constructs masks from the identified neurons
 and fuses them into a unified multi-concept mask, where *Concept Agnostic Neuro
ns* that broadly support general content generation are preserved while concept-
specific neurons are pruned to remove the targets. FIA is training-free and requ
ires minimal hyperparameter tuning for new tasks, enabling plug-and-play use. Ex
tensive experiments across three distinct unlearning tasks demonstrate that FIA
achieves more reliable multi-concept unlearning, improving forgetting effectiven
ess while maintaining generation fidelity and quality. Code is available at [htt
ps://github.com/kaiyuan02415/Forget-It-All](https://github.com/kaiyuan02415/Forg
et-It-All)
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Characterizing Vision-Language-Action Models across XPUs: Constraints and Accele
ration for On-Robot Deployment
Kaijun Zhou, Qiwei Chen, Da Peng, Zhiyang Li, Xijun Li, Jinyu Gu
Vision-Language-Action (VLA) models are promising for generalist robot control,
but on-robot deployment is bottlenecked by real-time inference under tight cost
and energy budgets.
Most prior evaluations rely on desktop-grade GPUs, obscuring the trade-offs and
opportunities offered by heterogeneous edge accelerators (GPUs/XPUs/NPUs).
We present a systematic framework for low-cost VLA deployment via model--hardwar
e co-characterization.
First, we build a cross-accelerator leaderboard and evaluate model--hardware pai
rs under \textbf{CET} (Cost, Energy, Time), showing that ``right-sized'' edge de
vices can be more cost-/energy-efficient than flagship GPUs while meeting contro
l-rate constraints.
Second, using fine-grained SM tracing and Roofline analysis, we uncover a consis
tent two-phase inference pattern: a compute-bound VLM backbone followed by a mem
ory-bound Action Expert, which induces phase-dependent underutilization and hard
ware inefficiency.
Finally, guided by these insights, we propose \textbf{DP-Cache} and \textbf{V-AE
Fusion} to reduce diffusion redundancy and enable asynchronous pipeline parallel
ism, achieving up to \(2.9\times\) speedup on GPUs and \(6\times\) on edge NPUs

with only marginal success degradation.
The example leaderboard website is: \url{https://vla-leaderboard-01.vercel.app/}
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Online Compatible Reward Identification from Preference Feedback
Simone Drago, Marco Mussi, Alberto Maria Metelli
In reinforcement learning, human preference feedback is emerging as a viable alt ernative to expert-designed reward functions, which can be difficult to engineer in real-world problems. However, despite the growing importance of preference f eedback, how to effectively *elicit* preferences remains a fundamental open prob lem. This work focuses on the *compatible reward identification* task. The aim i s to derive, starting from preference feedback, a reward function compatible wit h the observed preferences and accurate across the *entire* state-action space, ensuring higher transferability, safety, and interpretability. Indeed, the most common *reinforcement learning from human feedback* objective is to learn the *o ptimal policy*, requiring accuracy only in the portion of the state-action space that the agent visits. However, this goal cannot provide the same guarantees as compatible reward identification. First, we discuss commonalities and differenc es between the two goals. Then, we consider deterministic preferences, deriving the minimum number of interactions needed to identify the set of compatible rewa rds, and showing that using fewer queries may lead to arbitrarily large suboptim ality. Finally, we focus on stochastic preferences generated via the Bradley-Ter ry (BT) model. We introduce the concepts of *query basis* and its *index*, relat ing them to the problem complexity. Upon this, we discuss the connection between the index of a basis and the BT model, as well as the limitations that the mode l induces in this setting. Additionally, we devise an algorithm to identify a ne arly-optimal query basis with polynomial human query complexity.
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TMD-Bench: A Multi-Level Evaluation Paradigm for Music–Dance Co-Generation
Xiaoda Yang, Majun Zhang, Changhao Pan, Nick Huang, Yang Yuguang, Fan Zhuo, Peng fei Zhou, Jin Zhou, Sizhe Shan, Shan Yang, Miles Yang, Yang You, Zhou Zhao
Unified audio--visual generation is rapidly gaining industrial and creative rele vance, enabling applications in virtual production and interactive media. Howeve r, when moving from general audio--video synthesis to music–dance co-generation, the task becomes substantially harder: musical rhythm, phrasing, and accents mu st drive choreographic motion at fine temporal resolution, and such rhythmic cou pling is not captured by unimodal metrics or generic audiovisual consistency sco res used in current evaluation practice.
We introduce TMD-Bench, a benchmark for text-driven music–dance co-generation th at assesses systems across unimodal generation quality, instruction adherence, a nd cross-modal rhythmic alignment. The benchmark integrates computable physical metrics with perceptual multimodal judgments, and is supported by a curated rhyt hm-aligned music–dance dataset and a fine-grained Music Captioner for structured music semantics.
TMD-Bench further reveals that (i) modern commercial audio--visual models (e.g., Veo 3, Sora 2) produce high-quality music and video, while rhythmic coupling re mains less consistently optimized and leaves room for improvement, and (ii) our unified baseline RhyJAM trained on rhythm-aligned data achieves competitive beat -level synchronization while maintaining competitive unimodal fidelity. This pre sents prospects for building next-generation music–dance models that explicitly optimize rhythmic and kinetic coherence. The code is available at https://github .com/MM-Speech/TMD-Bench.
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Beyond Detection: A Structure-Aware Framework for Scene Text Tracking
Chenmin Yu, Liu Yu, Daiqing Wu, Gengluo Li, Zeyu Chen, Yu ZHOU
Modern visual object trackers show impressive results on general targets, yet th eir performance drops substantially when dealing with scene text. Although curre ntly underexplored, tracking text in videos is essential for dynamic text manipu lations such as segmentation, removal, and editing. To fill this gap, this paper formalizes this specific task as Scene Text Tracking and presents the first sys

tematic work for it. We identify three primary challenges in this task: 1) severe geometric distortions from perspective shifts, 2) high visual ambiguity across different instances, and 3) high sensitivity to fine-grained structural details. To address these issues, we propose SymTrack, a unified detection-free framework with synergistic dual-branch design. It integrates a Cross-Expert Calibration mechanism to reduce semantic bias, along with a Predictive Token Rectification mechanism to correct structural imbalances, complemented by an Adaptive Inference Engine that stabilizes predictions under motion constraints. Considering the lack of dedicated benchmarks for this task, we utilize three datasets from video text spotting to construct a benchmark with high-quality annotations. Extensive experiments demonstrate that SymTrack sets the new state-of-the-art on all three benchmarks, outperforming previous best trackers by up to 11.97\% AUC on $ \text{BOVText}_{\text{SOT}} $. Overall, our work promotes efficient and thorough text tracking, paving the way toward more generalized video text manipulation.
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Belief Propagation Converges to Gaussian Distributions in Sparsely-Connected Factor Graphs
Tom Yates, Yuzhou Cheng, Ignacio Alzugaray, Danyal Akarca, Pedro A. M. Mediano, Andrew J. Davison
Belief Propagation (BP) is a powerful algorithm for distributed inference in probabilistic graphical models, however it quickly becomes infeasible for practical compute and memory budgets. Many efficient, non-parametric forms of BP have been developed, but the most popular is Gaussian Belief Propagation (GBP), a variant that assumes all distributions are locally Gaussian. GBP is widely used due to its efficiency and empirically strong performance in applications like computer vision or sensor networks – even when modelling non-Gaussian problems. In this paper, we seek to provide a theoretical guarantee for when Gaussian approximations are valid in highly non-Gaussian,
sparsely-connected factor graphs performing BP (common in Spatial AI). We leverage the Central Limit Theorem to prove mathematically that variables’ beliefs under BP converge to a Gaussian distribution in complex, loopy factor graphs obeying our 4
key assumptions. We then confirm experimentally that variable beliefs become increasingly Gaussian after just a few BP iterations
in a stereo depth estimation task.
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A geometric relation of the error introduced by sampling a language model's output distribution to its internal state
Albert F. Modenbach
GPT-style language models are sensitive to single-token changes at generation points where the predicted probability distribution is spread across multiple tokens. Viewing this sensitivity as a geometric property, we derive an $\mathfrak{so}(n)$-valued 1-form that depends only on the geometry of the token embeddings. Despite this purely geometric origin, we show that its curvature is semantically meaningful: on chess reasoning tasks, the curvature couples to the world model of an off-the-shelf instruction-tuned model, with transformations clustering by board region and respecting piece importance. Our findings suggest that token space geometry directly reflects how models internally represent problems.
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VideoLoom: A Video Large Language Model for Joint Spatial-Temporal Understanding
Jiapeng Shi, Junke Wang, Zuyao You, Bo He, Zuxuan Wu
Recent advancements in Video Large Language Models (Video LLMs) have demonstrated impressive results, yet existing approaches handle either temporal or spatial dimension in isolation, struggling in the analysis of complex events that require spatial-temporal integration. To bridge this gap, we propose VideoLoom, a unified Video LLM for joint spatial-temporal understanding. To facilitate the development of fine-grained spatial and temporal localization capabilities, we curate LoomData-8.7k, a character-centric video dataset with temporally grounded and spatially localized captions. With this, VideoLoom achieves the state-of-the-art performance across a variety of spatial and temporal benchmarks. In addition, we

introduce LoomBench, a benchmark consisting of temporal, spatial, and compositio
nal video–question pairs, with a novel metric J&$F_{bi-fore}$, enabling a comp
rehensive evaluation of Video LLMs from diverse aspects. Collectively, these con
tributions offer a universal and effective suite for joint spatial-temporal vide
o understanding, setting a new standard in multimodal intelligence.
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Efficient Distributionally Robust Assortment Optimization in MNL Bandits
Yunfan Zhang, Yuxuan Han, Zhengyuan Zhou
We investigate the distributionally robust assortment optimization (DRAO) proble
m under the contextual multinomial logit (MNL) choice model, where the decision-
maker seeks to maximize revenue against worst-case distributional deviations. To
 address potential distribution shifts relative to the observed data environment
, we study DRAO under ambiguity sets defined by three divergences: total variati
on (TV), Kullback–Leibler (KL), and chi-square (χ^2). Incorporating robust
concerns poses challenges for both algorithm design and theoretical analysis. By
 leveraging strong duality results from the distributionally robust optimization
 literature and integrating them into the assortment optimization procedures, we
 develop tailored polynomial-time algorithms under each divergence. We further p
rovide a theoretical analysis and establish sample complexity bounds for all thr
ee robust formulations.
****

KITE: Knowledge-Guided Probabilistic Modeling for Time Series Forecasting with E
xogenous Variables
Hanyin Cheng, Jingrong Zhou, Yang Shu, Chenjuan Guo
Probabilistic forecasting with exogenous variables is vital for decision-making
but remains underexplored compared to deterministic methods. We propose KITE, a
knowledge-guided probabilistic modeling framework designed to bridge this gap by
 addressing two key bottlenecks: (1) topological disparity in sampling initializ
ation and (2) spurious covariate correlations during the iterative conditional g
eneration process. KITE introduces a *History-Conditional Manifold* to construct
 an informative source distribution from historical dynamics, effectively anchor
ing the starting point closer to the target space. Additionally, a *Knowledge-Gu
ided Conditioning* module is developed to regularize variable interactions using
 statistical priors, suppressing spurious correlations and enhancing the robustn
ess of covariate conditioning. Extensive experiments demonstrate that KITE outpe
rforms state-of-the-art methods in both deterministic and probabilistic forecast
ing.
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World-Shaper: A Unified Framework for 360° Panoramic Editing
Dong Liang, Yuhao LIU, Jinyuan Jia, Youjun Zhao, Rynson W. H. Lau
Being able to edit panoramic images is crucial for creating realistic 360° visua
l experiences. However, existing perspective-based image editing methods fail to
 model the spatial structure of panoramas. Conventional cube-map decompositions
attempt to overcome this problem but inevitably break global consistency due to
their mismatch with spherical geometry. Motivated by this insight, we reformulat
e panoramic editing directly in the equirectangular projection (ERP) domain and
present World Shaper, a unified geometry-aware framework that supports five dist
inct editing operations within a single ERP-native representation. To address th
e latitude-dependent geometric distortion inherent in ERP, we introduce a geomet
ry-aware learning strategy comprising distortion-aware attention modulation (DAA
M), which steers cross-attention with latitude-dependent strength at the feature
 level; layered shape loss (LSL), which enforces per-object geometric supervisio
n at the output level; and progressive curriculum training to internalize panora
mic priors. To overcome the scarcity of paired panoramic editing data, we train
a dedicated ERP-native controllable generator that synthesizes objects directly
in the equirectangular domain under user-defined conditions, enabling scalable p
aired data construction for learning diverse editing behaviors. Extensive experi
ments on our new benchmark, PEBench, demonstrate that World Shaper achieves supe
rior geometric consistency, editing fidelity, and text controllability compared
to state-of-the-art methods, enabling coherent and flexible 360° visual world cr

eation with unified editing control. Code, models, and data are available at the project page: https://world-shaper-project.github.io/.
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Design-Based Anytime-Valid Inference for Randomized Experiments with Delayed Outcomes and Staggered Entry
Michael Lindon, Nathan Kallus
Delayed outcomes are ubiquitous in online experimentation: treatment can affect whether an outcome occurs, when it occurs, and its realized value.
To accommodate staggered entry while remaining robust to environmental nonstationarity and unit-level heterogeneity, we adopt a design-based perspective and target the sample cumulative reward in each arm as a function of calendar time.
Our confidence sequences allow practitioners to continuously monitor the counterfactual incremental reward, such as revenue, that would have been realized by calendar time t had all entered units been assigned to treatment rather than control.
The main technical challenge is the choice of design-based filtration, complicated by the presence of asynchronous potential outcome times.
We show that the IPW treatment-effect estimation error is not a martingale with respect to any filtration, while each arm-specific IPW estimation error is a martingale with respect to a carefully chosen arm-specific event-time filtration.
We therefore construct a confidence sequence for the treatment effect by combining two arm-level confidence sequences with a union bound, and further demonstrate that this can outperform the traditional design-based variance upper bound.
Finally, we characterize the class of augmentations for which the per-arm AIPW estimation error remains a martingale.
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Reasoning about Reasoning: BAPO Bounds on Chain-of-Thought Token Complexity in LLMs
Kiran Tomlinson, Tobias Schnabel, Adith Swaminathan, Jennifer Neville
Inference-time scaling via chain-of-thought (CoT) reasoning is a major driver of state-of-the-art LLM performance, but it comes with substantial latency and compute costs. We address a fundamental theoretical question: *how many* reasoning tokens are required to solve a problem as input size grows? By extending the bounded attention prefix oracle (BAPO) model--an abstraction of LLMs that quantifies the information flow required to solve a task--we prove lower bounds on the CoT tokens required for three canonical BAPO-hard tasks: binary majority, triplet matching, and graph reachability. We show that each requires $\Omega(n)$ reasoning tokens when the input size is n. We complement these results with matching or near-matching upper bounds via explicit constructions. Finally, our experiments with frontier reasoning models show approximately linear reasoning token scaling on these tasks and failures when constrained to smaller reasoning budgets, consistent with our theoretical lower bounds. Together, our results identify fundamental bottlenecks in inference-time compute through CoT and offer a principled tool for analyzing optimal reasoning length.
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Improving LLM-Based Recommenders with Conservative Generative Flow Networks
Xuan Yu, Feng Niu, Rui Zhu, Yudong Zhang, Xu Wang, Yang Wang
Generative Flow Networks (GFlowNets) have recently been used to improve diversity and mitigate popularity bias in LLM-based recommender systems, yet most objectives are developed under online-style assumptions. In offline LLM-based recommendation, learning is constrained to a fixed logged dataset, yielding partial support over token transitions on the dataset-induced token-prefix DAG. Naively applying Sub-Trajectory Balance (SubTB) becomes non-identifiable and can arbitrarily allocate probability mass to unsupported regions. We formalize this failure and identify three sources of non-identifiability that induce distributional shift between the dataset-implied policy and the learned policy: (i) flow overestimation, (ii) forward mass leakage, and (iii) backward compensation. To address it, we propose CFlower, which introduces a conservative SubTB objective that explicitly penalizes unsupported forward flow mass, and combines it with dataset-constrained policy learning with on-policy sampling on the dataset-induced DAG for effi

cient training under offline constraints. Experiments on three Amazon recommenda tion datasets show that CFlower improves distributional matching and delivers a stronger accuracy--exposure trade-off than prior GFlowNet and SFT baselines, whi le serving as a more reliable reference policy for downstream RL fine-tuning.
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Variable Clustering via Distributionally Robust Nodewise Regression
Kaizheng Wang, Xiao Xu, XUNYU ZHOU
We study a multi-factor block model for variable clustering and connect it to re gularized subspace clustering through a distributionally robust version of nodew ise regression. To solve the latter problem, we derive a convex relaxation, prov ide a data-driven approach for selecting the size of the robust region, and deve lop an ADMM algorithm for efficient implementation. We validate our method in ex tensive numerical studies and demonstrate its superior performance.
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ReVSI: Rebuilding Visual Spatial Intelligence Evaluation for Accurate Assessment of VLM 3D Reasoning
Yiming Zhang, Jiacheng Chen, Jiaqi Tan, Yongsen Mao, Wenhu Chen, Angel X Chang
Current evaluations of spatial intelligence can be systematically invalid under modern vision-language model (VLM) settings. First, many benchmarks derive quest ion-answer (QA) pairs from point-cloud-based 3D annotations originally curated f or traditional 3D perception. When such annotations are treated as ground truth for video-based evaluation, reconstruction and annotation artifacts can miss obj ects that are clearly visible in the video, mislabel object identities, or corru pt geometry-dependent answers (e.g., size), yielding incorrect or ambiguous QA p airs. Second, evaluations often assume full-scene access, while many VLMs operat e on sparsely sampled frames (e.g., 16-64), making many questions effectively un answerable under the actual model inputs.
We improve evaluation validity by introducing ReVSI, a benchmark and protocol th at ensures each QA pair is answerable and correct under the model's actual input s. To this end, we re-annotate object labels and geometry across 413 scenes from 5 datasets to improve data quality, and regenerate all QA pairs with rigorous b ias mitigation and human verification using professional 3D visualization and an notation tools. We further enhance evaluation controllability by providing varia nts across multiple frame budgets (16/32/64/all) and fine-grained object visibil ity metadata, enabling controlled diagnostic analyses. Evaluations of general an d domain-specific VLMs on ReVSI reveal systematic failure modes that are obscure d by prior benchmarks, yielding a more reliable and diagnostic assessment of spa tial intelligence.
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Activation-Free Backbones for Image Recognition: Polynomial Alternatives within MetaFormer-Style Vision Models
Jeffrey Wang, Jonathan Gregory, Grigorios Chrysos
Modern vision backbones treat pointwise activations (e.g., ReLU, GELU) and expon ential softmax as essential sources of nonlinearity, but we demonstrate they are not required within MetaFormer-style vision backbones. We design activation-fre e polynomial alternatives for three core primitives (MLPs, convolutions, and att ention), where Hadamard products replace standard nonlinearities to yield polyno mial functions of the input. These modules integrate seamlessly into existing ar chitectures: instantiated within MetaFormer, a modular framework for vision back bones, our PolyNeXt models match or exceed activation-based counterparts across model scales on ImageNet classification, ADE20K semantic segmentation, and out-o f-distribution robustness. We also substantially outperform prior polynomial net works at reduced computational cost, showing that polynomial variants of standar d modules beat complex custom architectures. Our code is available at https://gi thub.com/jjwang8/PolyNeXt.
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SE(3)-Equivariant Flow Matching with Gaussian Process Priors for Geometric Traje ctory Prediction
Xuyang Wang, Xinzhe Zhou, Xiaoming Duan, Jianping He
The trajectory prediction of N-body systems is of great significance and remains

challenging with broad applications across various fields such as physics, chemistry and biology.
Recent advances in generative models including flow matching and diffusion models have emerged as effective solutions to this problem, owing to their capacity to model the stochasticity and underlying distributions of complex system trajectories.
However, existing approaches typically adopt trivial prior distributions that neglect the temporal correlations and spatial symmetries of N-body trajectories, which not only complicates the generation process but also limits model performance.
To address these limitations, we propose GP-EquiFlow, an SE(3)-equivariant flow matching model incorporating vector-valued Gaussian processes.
Based on observed trajectories, we employ vector-valued Gaussian processes to construct SE(3)-equivariant prior distributions, which exhibit enhanced consistency with the target data distribution in both spatial and temporal dynamics.
Extensive experiments on N-body simulations and molecular dynamics demonstrate that the proposed GP-EquiFlow delivers more accurate predictions while requiring fewer sampling steps, underscoring the effectiveness of integrating Gaussian process-based SE(3)-equivariant prior distributions in geometric trajectory prediction.
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Polaris: Coupled Orbital Polar Embeddings for Hierarchical Concept Learning
Sahil Mishra, Srinitish Srinivasan, Sourish Dasgupta, Tanmoy Chakraborty
Real-world knowledge is often organized as hierarchies such as product taxonomies, medical ontologies, and label trees, yet learning hierarchical representations is challenging due to asymmetric structure and noisy semantics. We introduce Polaris, a polar hyperspherical embedding framework that separates semanticity from hierarchy using angular geometry and radius, enabling the learning of meaning and structure without interference. To map latent representation onto the sphere, we project it to the tangent space at the north pole, apply the exponential map, and learn unit-norm representations using spherical linear layers. Polaris then combines robust local constraints, global regularization that prevents geometric collapse, and uncertainty-aware asymmetric objectives that encourage directional containment. At inference time, Polaris uses structure-guided retrieval to efficiently narrow down candidate parents before final ranking. We evaluate Polaris on different settings of taxonomy expansion -- spanning trees, multi-parent DAGs, and multimodal hierarchies, showing consistent improvements of up to $\sim$19 points in top-K retrieval and up to $\sim$ 60\% reduction in mean rank over fourteen strong baselines.
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Predicting Future Utility: Global Combinatorial Optimization for Task-Agnostic KV Cache Eviction
Ziyao Tang, Pengkun Jiao, Xinhang Chen, Wei Liu, Shiyong Li, Jingjing Chen
Given the quadratic complexity of attention, KV cache eviction is vital to accelerate model inference.
Current KV cache eviction methods typically rely on instantaneous heuristic metrics, implicitly assuming that score magnitudes are consistent proxies for importance across all heads.
However, this overlooks the heterogeneity in predictive fidelity across attention heads. While certain heads prioritize the instantaneous contribution of tokens, others are dedicated to capturing long-horizon utility.
In this paper, we propose that optimal budget allocation should be governed by the marginal utility in preserving long-term semantic information.
Building on this insight, we propose LU-KV, a novel framework that formulates head-level budget allocation as a global combinatorial optimization problem to maximize the long-horizon marginal contribution of reserved tokens. To solve this non-convex problem, we employ a convex-hull relaxation and a marginal-utility-based greedy solver, achieving near-optimal solutions. Furthermore, we implement a data-driven offline profiling protocol to facilitate the practical deployment of LU-KV.

Evaluations on LongBench and RULER benchmarks demonstrate that LU-KV reduces KV cache size by 80\% with minimal performance degradation, while also decreasing inference latency and GPU memory footprint.
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Mining Tensor/Neuron-Level Sparsity to Maximize Mixture-of-Experts Potential in Post-Training and Inference
Weilin Cai, Le Qin, Shwai He, Junwei Cui, Ang Li, Jiayi Huang
Mixture of Experts (MoE) has emerged as a mainstream architecture for Large Language Models (LLMs), balancing computational efficiency with model scalability. While prior work has explored increasing tensor-level sparsity via finer-grained expert configurations during pre-training, we identify significant unexploited sparsity at both the tensor and neuron levels during post-training and inference. To leverage this, we propose complete expert partition for post-training and threshold-based token-expert dropping for inference. These techniques improve the Mixtral-8$\times$7B model's average accuracy by 1\% across nine downstream benchmarks (notably 4\% on GSM8K).
To further optimize the accuracy-efficiency trade-off for inference, we introduce dual-threshold token-expert dropping with partial expert partition and reconstruction. Our approach yields a 1.19$\times$ MoE speedup and a 0.5\% accuracy gain on Mixtral-8$\times$7B when combining post-training and inference optimizations. For inference-only optimization on OLMoE-Instruct and DeepSeek-V2-Lite-Chat, we achieve up to 1.41$\times$ MoE speedup with a negligible accuracy loss ($<$0.5\%).
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GLEAN: Guideline-Grounded Evidence Accumulation for High-Stakes Agent Verification
Yichi Zhang, Nabeel Seedat, Yinpeng Dong, Peng Cui, Jun Zhu, Mihaela van der Schaar
As LLM-powered agents have been used for high-stakes decision-making, such as clinical diagnosis, it becomes critical to develop reliable verification of their decisions to facilitate trustworthy deployment. Yet, existing verifiers usually underperform owing to a lack of domain knowledge and limited calibration. To address this, we establish **GLEAN**, an agent verification framework with **G**uide**L**ine-grounded **E**vidence **A**ccumulatio**N** that compiles expert-curated protocols into trajectory-informed, well-calibrated correctness signals. GLEAN evaluates the step-wise alignment with domain guidelines and aggregates multi-guideline ratings into surrogate features, which are accumulated along the trajectory and calibrated into correctness probabilities using Bayesian logistic regression. Moreover, the estimated uncertainty triggers active verification, which selectively collects additional evidence for uncertain cases via expanding guideline coverage and performing differential checks. We empirically validate GLEAN with agentic clinical diagnosis across three diseases from the MIMIC-IV dataset, surpassing the best baseline by 12\% in AUROC and 50\% in Brier score reduction, which confirms the effectiveness in both discrimination and calibration. In addition, the expert study with clinicians recognizes GLEAN's utility in practice.
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Investigating Component Contributions in Multi-Agent ML Systems
Junsung Kim, Ilia Mireskandari, Seungwan Son, Yifan Zhou, Khizer Shahid, Dylan Yihan Dai
Autonomous agents for machine learning engineering have advanced rapidly, yet comparing their effectiveness remains difficult. Existing systems combine different techniques---multi-agent decomposition, iterative refinement, memory management, and planning---in varying configurations, making it unclear which components actually drive performance. Complicating evaluation, existing benchmarks rely on historical competitions whose data likely contaminates LLM training corpora and whose static baselines reflect outdated human performance. To address this, we conduct over 4,000 controlled experiments systematically ablating architectural components, alongside K-live, a new benchmark of 25 active competitions that provides a contamination-free, dynamic evaluation environment. Our findings challenge common design assumptions: iterative feedback contributes more than architect

ural complexity, and multi-agent coordination can hurt as often as it helps. The se results provide concrete guidance for practitioners building ML engineering a gents.
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Predicting evolutionary rate as a pretraining task improves genome language mode l representations

Micaela Elisa Consens, Kevin K Yang, James Brian Hall, Ashley Mae Conard, BO WAN G, Lorin Crawford, Alan M Moses, Alex Xijie Lu

Genome language models (gLM) have the potential to further understanding of regu latory genomics without requiring labeled data. Most gLMs are pretrained using s equence reconstruction tasks inspired by natural language processing, but recent studies have shown that these gLMs often fail to capture biological signal. To overcome this, we introduce pretraining tasks that predict the rate of evolution . These tasks are designed so that they can be composed with sequence reconstruc tion, enabling a controlled comparison of predicting sequence only, evolutionary rate only, or both. To address gaps in existing evaluations, we developed a sui te of biologically grounded benchmarks.

Across these tasks, and for established variant effect prediction benchmarks, mo dels pretrained on both sequence and evolutionary rate outperform those trained on sequence alone, and training on evolutionary rate can make the even the relat ively small models in our work competitive with much larger existing gLMs for so me tasks on the human genome. These results establish evolution as a key trainin g target for genome-scale models.
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Regret Pre-training: Bridging Prior and Posterior Views for Enhanced Knowledge G rounding

Mingkuan Zhao, Xiayu Sun, Wentao Hu, Suquan Chen, Jiaxuan Li, Xiaoyan Zhu, Xin L ai, Jiayin Wang

Causal language models factorize sequence probabilities using only preceding con text, leaving future information unexploited during training despite its availab ility in the training data. This paper introduces Regret Pre-training, a self-su pervised framework grounded in the Learning Using Privileged Information (LUPI) paradigm. The framework employs a dual-view architecture in which a single model generates both a causal Student distribution and a future-conditioned Teacher d istribution. The training objective augments standard language modeling with a r egret loss that minimizes the KL divergence from teacher to student, transferrin g future-aware signals to the causal representations. We investigate two teacher configurations on the OLMoE-1B-7B architecture:LocalRegret, which extends atten tion by one future token, andGlobalRegret, which conditions on bidirectional con text with the target position masked. Experiments on nine downstream tasks follo wing 4 billion tokens of training demonstrate that both configurations consisten tly outperform the baseline. On average,GlobalRegret andLocalRegret achieve 33.9 % and 32.2% accuracy respectively, surpassing the baseline's 30.2%. Most notably ,GlobalRegret improves BoolQ performance by 18.1 percentage points (61.0% vs 42. 9%). The framework introduces no additional parameters and requires only one ext ra inference-mode forward pass per training step.
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ExVerus: Verus Proof Repair via Counterexample Reasoning

Jun Yang, Yuechun Sun, Yi Wu, Rodrigo Caridad, Yongwei Yuan, Jianan Yao, Shan Lu , Kexin Pei

Large Language Models (LLMs) have shown promising results in automating formal v erification. However, existing approaches treat proof generation as a static, en d-to-end prediction over source code, relying on limited verifier feedback and l acking access to concrete program behaviors. We present ExVerus, a counterexampl e-guided framework that enables LLMs to reason about proofs using behavioral fee dback via counterexamples. When a proof fails, ExVerus automatically generates a nd validates counterexamples, and then guides the LLM to generalize them into in ductive invariants to block these failures. Our evaluation shows that ExVerus si gnificantly improves proof accuracy, robustness, and token efficiency over the s tate-of-the-art prompting-based Verus proof generator.

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SE(n)-Invariant Flow Matching: A General Framework with Application to Object Reassembly
Gaël Heck, Sylvie Le Hégarat-Mascle, Nicolas Lermé
Reassembling N fragments in n-dimensional space is a shape reconstruction task that is invariant to global rigid motions.
Training directly on $\mathcal{M}=\mathrm{SE}(n)^N$ can be ill-posed: standard losses penalize solutions that differ only by a global transform.
Existing methods often address this with ad-hoc anchoring which breaks permutation invariance across fragments and can introduce biases that must be mitigated with extensive and costly data augmentation.
We propose a geometric framework that enforces invariance by construction. First, a **Global Gauge Fixing** (GGF) strategy deterministically aligns configurations using an intrinsic generalized-inertia rule.
Second, we introduce a **quotient-invariant Flow Matching objective** that operates via orthogonal projection onto the horizontal tangent bundle. This construction factors out global pose at each timestep, enabling the model to learn only shape-changing dynamics on the quotient space $\mathcal{M}/\mathrm{SE}(n)$.
Our unified $\mathrm{SE}(n)$-invariant framework admits efficient closed-form 2D/3D instantiations and improves accuracy on polygonal jigsaw puzzles and 3D fracture reassembly benchmarks.
****
HELIX: Hybrid Encoding with Learnable Identity and Cross-dimensional Synthesis for Time Series Imputation
Fengming Zhang, Wenjie Du, Huan Zhang, Ke Yu, Shen Qu
Time series imputation benefits from leveraging cross-feature correlations, yet existing attention-based methods re-discover feature relationships at each layer, lacking persistent anchors to maintain consistent representations. To address this, we propose HELIX, which assigns each feature a learnable feature identity, a persistent embedding that captures intrinsic semantic properties throughout the network. Unlike graph-based methods that rely on predefined topology and assume homogeneous spatial relationships, HELIX learns arbitrary feature dependencies end-to-end from temporal co-variation, naturally handling datasets where features mix spatial locations with semantic variables. Integrated with hybrid temporal-feature attention, HELIX achieves the state-of-the-art performance, surpassing all 16 baselines on 5 public datasets across 21 experimental settings in our evaluation. Furthermore, our mechanistic analysis reveals that HELIX aligns learned feature identities and dependencies with latent physical and semantic structure progressively across layers, demonstrating that it more effectively translates cross-feature structure into imputation accuracy.
****
Manifold-Aligned Guided Integrated Gradients for Reliable Feature Attribution
Soyeon Kim, Seongwoo Lim, Kyowoon Lee, Jaesik Choi
Feature attribution is central to diagnosing and trusting deep neural networks, and Integrated Gradients (IG) is widely used due to its axiomatic properties. However, IG can yield unreliable explanations when the integration path between a baseline and the input passes through regions with noisy gradients. While Guided Integrated Gradients reduces this sensitivity by adaptively updating low-gradient-magnitude features, input-space guidance still produces intermediate inputs that deviate from the data manifold. To address this limitation, we propose **Manifold-Aligned Guided Integrated Gradients** (MA-GIG), which constructs attribution paths in the latent space of a pre-trained variational autoencoder. By decoding intermediate latent states, MA-GIG biases the path toward the learned generative manifold and reduces exposure to implausible input-space regions. Through qualitative and quantitative evaluations, we demonstrate that MA-GIG produces faithful explanations by aggregating gradients on path features proximal to the input. Consequently, our method reduces off-manifold noise and outperforms prior path-based attribution methods across multiple datasets and classifiers. Our code is available at https://github.com/leekwoon/ma-gig/.
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OC-space: a Unifying Perspective on Verification of Tree Ensembles
Timo Martens, Laurens Devos, Lorenzo Cascioli, Wannes Meert, Hendrik Blockeel, J esse Davis
We study the problem of verifying whether certain properties such as robustness or fairness hold in an ensemble of decision trees.
This problem is known to be NP-hard, with most research targeting a solution to a specific verification task. We explore the problem through the lens of an ens emble's OC-space: the set of all possible combinations of the individual trees' predictions. This provides a unifying view that yields a more generic and flexib le approach to verification.
We show that a wide variety of existing verification tasks can be (1) framed as simple searches through OC-space, and
(2) answered in time linear or quadratic in the size of the OC-space.
Moreover, the search can be made more efficient by using spatial index structure s. Interestingly, while the OC-space can grow exponentially with the ensemble's size, in practice it is often feasible to enumerate all output configurations. E mpirically, we show that our generic approach can be faster than approaches targ eting a single verification task.
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Steering Beyond the Support: Adversarial Training on Unsupervised Jailbroken Act ivation Simulation
LUOYU CHEN, Weiqi Wang, Zhiyi Tian, Chenhan Zhang, Feng Wu, Jianhuan Huang, Ahme d Asiri, Shui Yu
Jailbreak prompts can trigger harmful completions on aligned LLMs, In accordance , safety steering has been proposed: test-time activation interventions that ste er jailbreak activations to trigger refusal while preserving benign utility. How ever, existing steering methods are fundamentally supervised and tied to a stati c, limited training set, whereas real jailbreaks evolve and are often out-of-dis tributed from the training set, leading to failures on unseen attacks. In this p aper, we tackle this failure by developping a \emph{zero-shot} defense. Base on unsupervised latent direction discovery, we directly simulate jailbroken activat ions without any knowledge of jailbreak strategy. To build a defense mechnism up on this, we propose a bi-level adversarial training framework. In the inner step , we simulate diverse jailbroken activations by extrapolating from refusal state harmful-request activations via unsupervised latent direction discovery. In the outer step, we train a potential-induced steering field to push these adversari al jailbroken states into refusal regions while keeping benign unchanged. Across three LLMs and six classical jailbreak families, our method achieves strong def ense with attack success rates mostly below 5%, and we analyzed the increasing s ubspace coverage of our simulated jailbroken activations on real jailbreaks thro ughout training, which helps explain the increasing robustness of our defense me chnism.
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3D MeanFlow: One-Step Point Cloud Completion and Generation via Average-Velocity Transport
Haowen Zhong, Jiujun Cheng, Haowen Wang, Chao Wei, Lu Yang, Qichao Mao, Shangce Gao
Point cloud completion and generation are important across many 3D tasks, where both fidelity and sampling efficiency matter. Prevailing high-fidelity approache s rely on long sampling schedules, which incur substantial inference latency. Fe w-step alternatives typically use rectification or distillation, leading to mult i-stage training pipelines and potential quality trade-offs. We present 3D MeanF low (3DMF), a distillation-free model that performs one-step average-velocity tr ansport for point cloud completion and generation. We optimize an instantaneous- average consistency objective and impose a shape-level constraint to stabilize t raining. Additionally, we introduce PointPlug, integrating completion into 3D ob ject detectors and evaluating its impact. PointPlug uses adaptive selection that balances benefit and latency. Across standard benchmarks, 3DMF achieves one-ste p sampling with an order-of-magnitude speedup while maintaining competitive fide lity. On nuScenes and KITTI, inserting PointPlug improves all evaluated detector

s under comparable settings.
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EquiCAD: A Geometric Equivariant Neural Network for 3D Shape Classi■cation
Yonghao Su, Yantao Gan, Junfeng Long, Caiyang Yu, Wenhao Zheng, Jiancheng Lv, Xianggen Liu
Three-dimensional (3D) shape classification plays a central role in computer vision and computer-aided design (CAD), underpinning applications in intelligent manufacturing, automated inspection, and digital engineering. Despite recent progress with 3D CNNs and graph-based approaches, existing methods often overlook the geometric-topological regularities and symmetry principles intrinsic to CAD boundary representations (B-reps). To address this challenge, we introduce EquiCAD, a symmetry-aware learning framework that integrates equivariant representations with graph-based reasoning. By leveraging group-theoretic decomposition of curve and surface descriptors, EquiCAD preserves symmetry-consistent feature transformations while retaining rich geometric details. The model further exploits hierarchical message passing to capture interactions between local features and global structure. Experimental results across multiple datasets, including SolidLetters, Parts, the Machining Feature benchmark, and our newly constructed Features dataset, demonstrate substantial improvements over prior state-of-the-art approaches, particularly on industrially relevant shapes with fine-grained attributes. These findings highlight the value of symmetry-aware modeling for robust and generalizable 3D shape analysis.
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Universal Learning of Nonlinear Dynamics
Evan Dogariu, Anand Paresh Brahmbhatt, Elad Hazan
We study the fundamental problem of one-step prediction of a marginally stable unknown nonlinear dynamical system. We describe an algorithm for this problem, based on the technique of spectral filtering, which learns a mapping from past observations to the next based on a spectral representation of the system. Using techniques from online convex optimization, we prove vanishing prediction error for any nonexpansive nonlinear dynamical system with finitely many marginally stable modes, with rates governed by a novel quantitative control-theoretic notion of learnability. The main technical component of our method is a new spectral filtering algorithm for linear dynamical systems, which incorporates past observations and applies to general noisy and marginally stable systems. This generalizes the original spectral filtering algorithm to both asymmetric dynamics as well as incorporating noise correction, and is of independent interest.
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Beyond Structural Symmetries: Linear Mode Connectivity via Neuron Identifiability
Vincent Bürgin, Daniel Herbst, Ya-Wei Eileen Lin, Stefanie Jegelka
Many striking phenomena in deep learning, such as linear mode connectivity and the structured behavior of training dynamics, are closely tied to parameter symmetries: transformations that leave the realized function unchanged. Despite growing attention to parameter symmetries, the exact interplay between parameters, data, and representations remains underexplored. To investigate this, we develop a theoretical framework of effective function classes, i.e., the set of functions a neuron can realize on its input support, and the norm cost of realizing them. We then formalize *effective symmetry breaking* via neuron identifiability across independent training runs. Our analysis shows that neural networks can admit large families of approximately equivalent solutions even in *structurally asymmetric* models. We further show that neuron identifiability enables representation merging *without prior alignment*, and characterize when such merging admits a linear low-loss path. These findings highlight the role of effective function classes in affecting the loss landscape.
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StructMAR: Structure-Aware Masked Autoregression for Explicit Layout Alignment in Text-to-Image Generation
Gang Cao, Junying Zhang
Although text-to-image generation has achieved significant progress, strict inst

ance-level layout alignment remains a challenge for many applications. Masked Au toregressive (MAR) models on continuous latents are both efficient and high-fide lity, yet the standard practice of flattening 2D latents into 1D sequences weake ns spatial topology, limiting precise controllability. To address this, we propo se StructMAR, a structure-aware masked autoregressive framework that transforms layout alignment from a soft correlation into an explicit structural alignment. By integrating 2D Rotary Positional Embeddings with a Layout-Guided Attention Bi as, StructMAR explicitly biases latent tokens toward their corresponding layout instances during attention computation. We further use Group Relative Policy Opt imization (GRPO) as a final-stage policy refinement to reduce the mismatch betwe en the MAR training objective and detector-based layout evaluation metrics. Eval uated on the COCO-Position and COCO-MIG benchmarks, StructMAR achieves state-of- the-art performance, reaching 57.2 AP and 79.4 mIoU on the former, and 61.7 ISR and 56.9 mIoU on the latter. These results, coupled with a 4.05$\times$ inferenc e speedup, underscore the efficacy of explicit structural inductive biases in co ntrollable autoregressive generation.
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Hard Labels In! Rethinking the Role of Hard Labels in Mitigating Local Semantic Drift

Jiacheng Cui, Bingkui Tong, Xinyue Bi, Xiaohan Zhao, Jiacheng Liu, Zhiqiang Shen

Soft labels from teacher models are a de facto practice for knowledge transfer a nd large-scale dataset distillation (e.g., SRe^{2}L, LPLD). However, when we lim it the number of crops per image to reduce the substantial cost of storing preco mputed soft labels, these methods suffer severely from local semantic drift: vis ually ambiguous crops can cause soft supervision to deviate from the image-level ground-truth semantics, leading to persistent errors and a train–test distribut ion mismatch. We revisit the overlooked role of hard labels and show that, when properly integrated, they can act as a content-invariant semantic anchor that ca librates such drift. We theoretically analyze the emergence of drift under spars e soft-label supervision and demonstrate that hybridizing hard and soft labels r estores alignment between visual content and semantic supervision. Building on t his insight, we propose a new training paradigm, Hard Label for Alleviating Loca l Semantic Drift (HALD), which uses hard labels as intermediate corrective signa ls while preserving the fine-grained benefits of soft labels. Extensive experime nts on dataset distillation and large-scale classification benchmarks show consi stent generalization improvements. On ImageNet-1K, our method achieves 42.7% acc uracy with only 285M soft-label storage (reduces by 100$\times$), outperforming prior state-of-the-art LPLD by 9.0%. Code is available at https://github.com/Jia cheng8/HALD.
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Bridging Your Imagination with Audio-Video Generation via a Unified Director

Jiaxu Zhang, Tianshu Hu, Yuan Zhang, Zenan Li, Linjie Luo, Mingyuan Gao, Guoshen g Lin, Xin Chen

Existing AI-driven video creation systems typically treat script drafting and ke y-shot design as two disjoint tasks: the former relies on large language models, while the latter depends on image generation models. We argue that these two ta sks should be unified within a single framework, as logical reasoning and imagin ative thinking are both fundamental qualities of a film director. In this work, we propose **UniMAGE**, a unified director model that bridges user prompts with well-structured scripts, thereby empowering non-experts to produce long-context, multi-shot films by leveraging existing audio–video generation models. To achie ve this, we employ the Mixture-of-Transformers architecture that unifies text an d image generation. To further enhance narrative logic and keyframe consistency, we introduce a ``first interleaving, then disentangling" training paradigm. Spe cifically, we first perform **Interleaved Concept Learning**, which utilizes int erleaved text–image data to foster the model’s deeper understanding and imaginat ive interpretation of scripts. We then conduct **Disentangled Expert Learning**, which decouples script writing from keyframe generation, enabling greater flexi bility and creativity in storytelling. Extensive experiments demonstrate that Un iMAGE achieves state-of-the-art performance among open-source models, generating

logically coherent scripts and visually consistent keyframe images.
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Preserve-Then-Quantize: Balancing Rank Budgets for Quantization Error Reconstruction in LLMs
Yoonjun Cho, Dongjae Jeon, Soeun Kim, Moongyu Jeon, Albert No
Quantization Error Reconstruction (QER) reduces accuracy loss in Post-Training Quantization (PTQ)
by approximating weights as $\mathbf{W} \approx \mathbf{Q} + \mathbf{L}\mathbf{R}$, using a rank-$r$ correction to reconstruct quantization error.
Prior methods devote the full rank budget to error reconstruction,
which is suboptimal when $\mathbf{W}$ has intrinsic low-rank structure and quantization corrupts dominant directions.
We propose Structured Residual Reconstruction (SRR),
a rank-allocation framework that preserves the top-k singular subspace of the activation-scaled weight before quantization,
quantizes only the residual, and uses the remaining rank $r-k$ for error reconstruction.
We derive a theory-guided criterion for selecting k by balancing quantization-exposed energy
and unrecoverable error under rank constraints.
We further show that resulting $\mathbf{Q}+\mathbf{L}\mathbf{R}$ parameterization naturally supports Quantized Parameter-Efficient Fine-Tuning (QPEFT),
and stabilizes fine-tuning via gradient scaling along preserved directions.
Experiments demonstrate consistent perplexity reductions across diverse models and quantization settings in PTQ,
along with a 5.9 percentage-point average gain on GLUE under 2-bit QPEFT.
The project page is available at https://ai-isl.github.io/srr.
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OpenTSLM: Time-Series Language Models for Reasoning over Multivariate Medical Text- and Time-Series Data
Patrick Langer, Thomas Kaar, Max Rosenblattl, Maxwell A Xu, Winnie Chow, Martin Maritsch, Robert Jakob, Ning Wang, Juncheng Liu, Aradhana Verma, Brian Han, Daniel Seung Kim, Henry Chubb, Scott R. Ceresnak, Aydin Zahedivash, Alexander T Sandhu, Fatima Rodriguez, Daniel McDuff, Elgar Fleisch, Oliver Oppers Aalami, Filipe Barata, Paul Schmiedmayer
Large Language Models have shown strong capabilities in interpreting multimodal data but remain limited in handling time-series data. Addressing this gap could help to translate longitudinal and wearable data into actionable insights and patient-facing digital health applications.
We propose OpenTSLM, an open-source family of Time Series Language Models integrating time-series as a native modality into pretrained LLMs, enabling natural-language prompting and reasoning over multiple time-series via either soft prompting (OpenTSLM-SoftPrompt) or cross-attention (OpenTSLM-Flamingo).
To enable training models for time-series reasoning, we introduce three datasets: HAR-CoT (human activity recognition), Sleep-CoT (sleep staging), and ECG-QA-CoT (electrocardiogram question answering).
Across tasks, OpenTSLM models outperform baselines, reaching F1 scores of 69.88% in sleep staging and 67.64% in HAR; OpenTSLM-Flamingo also scales more efficiently in memory as the number and length of time series increase.
Expert evaluations with cardiologists show that OpenTSLMs exhibit strong reasoning capabilities on raw ECG data.
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From Optimization to Generalization under Heavy-Tailed Data: The Role of Gradient Clipping
Aleksandr Shestakov, Martin Taká■, Eduard Gorbunov
Gradient clipping is widely used to stabilize stochastic gradient methods and is often theoretically motivated by heavy-tailed gradient noise, where even second moments may be infinite, seemingly contradicting the finite-sum ERM setting, where all empirical moments are finite once the dataset is fixed. We resolve this paradox by explicitly separating data sampling from optimization randomness: alt

hough moments are finite conditional on the dataset, heavy-tailed data induce da taset-dependent noise whose second moment typically grows with the dataset size N. In particular, when $\|\nabla f(x_\star,\xi)\|$ has tail index $\alpha \in (1,2)$, the quantity $\frac{1}{N}\sum_{i=1}^N\|\nabla f(x_\star,\xi_i)\|^2$ scal es as $N^{\frac{2}{\alpha}-1}$, leading to deteriorating convergence guarantees for standard SGD as N increases. In contrast, we show that SGD with clipping a voids this growth and admits finite-sum convergence guarantees under heavy-taile d data for broad step-size and clipping schedules. We further derive generalizat ion bounds for strongly convex smooth objectives and show that the tail behavior of gradients at the population minimizer is the key quantity linking optimizati on and generalization under heavy-tailed data.

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When Sample Selection Bias Precipitates Model Collapse

Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pa ng

The proliferation of recursive training on synthetic data can alleviate data sca rcity but risks model collapse, where repeated training erodes distributional ta ils and homogenizes outputs. Data selection is widely viewed as a remedy, yet it s reliability depends critically on the reference distribution used by the verif ier. We show that in low-resource verification regimes, where each verifier obse rves only a small, fragmented, and biased slice of the target manifold, selectio n itself becomes biased. This situation naturally arises in low-resource data si los such as healthcare consortia or proprietary financial institutions, where ra w data cannot be pooled and local references are inherently incomplete. As a res ult, selection preferentially retains samples aligned with the local manifold wh ile pruning globally relevant tail modes, turning from a safeguard against colla pse into a mechanism that precipitates it. We theoretically prove that such silo ed selection accelerates collapse and induces power-law diversity decay. As an i nitial mitigation, we construct Wasserstein proxy references from multiple silos without sharing raw data. Empirical results confirm that local-reference select ion fails on skewed distributions, whereas collaborative proxy references mitiga te diversity degradation, suggesting that recursive synthetic-data pipelines req uire particular caution when real-data coverage is fragmented or scarce.

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Distinguishable Deletion: Unifying Knowledge Erasure and Refusal for Large Langu age Model Unlearning

Puning Yang, Junchi Yu, Qizhou Wang, Philip Torr, Bo Han, Xiuying Chen

Mitigating sensitive and harmful outputs is fundamental to ensuring safe deploym ent of LLMs.

Existing approaches typically follow two paradigms: Knowledge Deletion (KD), whi ch erases undesirable information during training, and Distinguishable Refusal (DR), which steers models away from using sensitive knowledge during inference.

Despite rapid progress, KD-based unlearning struggles with biased deletion due t o suppressing specific token sequences as a substitute for complete knowledge re moval, whereas DR-based unlearning risks the re-emergence of harmful knowledge b ecause the underlying knowledge remains intact.

To address these issues, we propose Distinguishable Deletion ($\mathrm{D^2}$), a paradigm that restricts the response distribution in the latent representation rather than specific tokens to erase undesirable knowledge, while distinguishing it from retained knowledge, enabling a refusal mechanism to handle unlearned in puts safely and coherently.

To implement $\mathrm{D^2}$, we introduce an energy index that quantifies the pr esence of knowledge and the separation between unlearned and retained content.

Mathematical and empirical analyses show that energy is both accurate and effici ent, enabling Energy-based Unlearning Alignment (EUA) to enforce energy-boundary unlearning during training and apply an energy-based refusal mechanism at infer ence.

Extensive experiments demonstrate that EUA significantly outperforms previous me thods, indicating the superiority of $\mathrm{D^2}$.

Our code is available at https://github.com/Puning97/EUA-for-LLM-Unlearning.

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Decomposing the Basic Abilities of Large Language Models: Mitigating Cross-Task Interference in Multi-Task Instruct-Tuning
Bing Wang, Ximing Li, Changchun Li, Jinjin Chi, Gang Niu, Masashi Sugiyama
Recently, the prominent performance of large language models (LLMs) has been largely driven by multi-task instruct-tuning. Unfortunately, this training paradigm suffers from a key issue, named cross-task interference, due to conflicting gradients over shared parameters among different tasks. Some previous methods mitigate this issue by isolating task-specific parameters, e.g., task-specific neuron selection and mixture-of-experts. In this paper, we empirically reveal that the cross-task interference still exists for the existing solutions because of many parameters also shared by different tasks, and accordingly, we propose a novel solution, namely Basic Abilities Decomposition for multi-task Instruct-Tuning (BADIT). Specifically, we empirically find that certain parameters are consistently co-activated, and that co-activated parameters naturally organize into base groups. This motivates us to analogize that LLMs encode several orthogonal basic abilities, and that any task can be represented as a linear combination of these abilities. Accordingly, we propose BADIT that decomposes LLM parameters into orthogonal high-singular-value LoRA experts representing basic abilities, and dynamically enforces their orthogonality during training via spherical clustering of rank-1 components. We conduct extensive experiments on the SuperNI benchmark with 6 LLMs, and empirical results demonstrate that BADIT can outperform SOTA methods and mitigate the degree of cross-task interference.
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Beyond Mode Collapse: Distribution Matching for Diverse Reasoning
Xiaozhe Li, Yang Li, Xinyu Fang, Shengyuan Ding, Peiji Li, Yongkang Chen, Yichuan Ma, TianYi Lyu, Linyang Li, Dahua Lin, Qipeng Guo, Qingwen Liu, Kai Chen
On-policy reinforcement learning methods like GRPO suffer from \emph{mode collapse}: they exhibit reduced solution diversity, concentrating probability mass on a single solution once discovered and ceasing exploration of alternative strategies. We show this stems from reverse KL minimization's mode-seeking behavior, which reinforces the first high-reward trajectory found rather than maintaining a distribution over multiple diverse solutions. We propose DMPO (\textbf{D}istribution-\textbf{M}atching \textbf{P}olicy \textbf{O}ptimization), which prevents mode collapse through principled approximation of forward KL minimization. DMPO constructs a group-level target distribution over sampled trajectories proportional to their rewards, then aligns the policy distribution to this target. This provides mode-covering behavior without requiring sampling from the intractable global target distribution, enabling sustained exploration throughout training. We validate DMPO on NP-hard combinatorial optimization, where exponentially many feasible solutions exist but only a few approach optimality—an ideal testbed for evaluating exploration. DMPO achieves {43.9\% Quality Ratio on text-based NP-Bench (vs. GRPO's 40.1\%)} and {43.1\% on vision-based NP-Bench (vs. 38.4\%)}—demonstrating 9\% and 12\% relative improvements respectively. These gains generalize to mathematical reasoning (+2.0\%) and out-of-domain tasks (+2.3\%), showing that diversity-preserving training enhances general reasoning capabilities across modalities. Our work establishes distribution matching as a practical, principled approach to preventing mode collapse in on-policy RL, with consistent quality improvements demonstrating sustained exploration across diverse reasoning tasks.
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Success-Conditioning as Policy Improvement: The Optimization Problem Solved by Imitating Success
Daniel Russo
A widely used technique for improving policies is success conditioning, in which one collects trajectories, identifies those that achieve a desired outcome, and updates the policy to imitate the actions taken along successful trajectories. This principle appears under many names---rejection sampling with SFT, goal-conditioned RL, Decision Transformers---yet what optimization problem it solves, if any, has remained unclear. We prove that success conditioning exactly solves a trust-region optimization problem, maximizing policy improvement subject to a $\c

hi^2$ divergence constraint whose radius is determined automatically by the data . This yields an identity: relative policy improvement, the magnitude of policy change, and a quantity we call action-influence---measuring how random variation in action choices affects success rates---are exactly equal at every state. Success conditioning thus emerges as a conservative improvement operator. Exact success conditioning cannot degrade performance or induce dangerous distribution shift, but when it fails, it does so observably, by hardly changing the policy at all. We apply our theory to the common practice of return thresholding, showing this can amplify improvement, but at the cost of potential misalignment with the true objective.

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SJD-SV: Speculative Jacobi Decoding with Semantics Verification for Autoregressive Image Generation

Baoquan Zhang, Bingqi Shan, Shihao Fang, Kenghong Lin, Xutao Li, Yunming Ye

Speculative Jacobi Decoding (SJD) is an important approach for accelerating autoregressive image generation. Although SJD has shown superior performance, recent studies point out that it usually suffers from a token ambiguity issue during token verification but its reason can not be well explained. To figure out this reason, in this paper, we conduct a visualization analysis on vision token and find that different from text tokens, vision tokens generally corresponds to some local, small, and unclear vision details, which means only using single token is difficult to accurately express a certain semantic, thereby causing token ambiguity issue. To this end, we propose a novel Speculative Jacobi Decoding with Semantics Verification (called SJD-SV), for accelerating autoregressive image generation. The key idea is that leveraging the strong correction characters between tokens to recognize semantic-aware token subsequence and then instead of perform token-by-token verification, turning to perform verification on semantic-aware token subsequence level for accelerating image generation. In particular, our method is plug-in, which can be directly integrated into existing SJD and its variants. Extensive experiments on various datasets show that existing SJD methods achieve significant performance improvement after integrating our SJD-SV method .

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Evidential Reasoning Advances Interpretable Real-World Disease Screening

Chenyu Lian, Hong-Yu Zhou, Jing Qin

Disease screening is critical for early detection and timely intervention in clinical practice. However, most current screening models for medical images suffer from limited interpretability and suboptimal performance. They often lack effective mechanisms to reference historical cases or provide transparent reasoning pathways. To address these challenges, we introduce EviScreen, an evidential reasoning framework for disease screening that leverages region-level evidence from historical cases. The proposed EviScreen offers retrospection interpretability through regional evidence retrieved from dual knowledge banks. Using this evidential mechanism, the subsequent evidence-aware reasoning module makes predictions using both the current case and evidence from historical cases, thereby enhancing disease screening performance. Furthermore, rather than relying on post-hoc saliency maps, EviScreen enhances localization interpretability by leveraging abnormality maps derived from contrastive retrieval. Our method achieves superior performance on our carefully established benchmarks for real-world disease screening, yielding notably higher specificity at clinical-level recall. Code is publicly available at https://github.com/DopamineLcy/EviScreen.

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Image Restoration via Diffusion Models with Dynamic Resolution

Yang Zheng, Wen Li, Zhaoqiang Liu

Diffusion models (DMs) have exhibited remarkable efficacy in various image restoration tasks. However, existing approaches typically operate within the high-dimensional pixel space, resulting in high computational overhead. While methods based on latent DMs seek to alleviate this issue by utilizing the compressed latent space of a variational autoencoder, they require repeated encoder-decoder inference. This introduces significant additional computational burdens, often resul

ting in runtime performance that is even inferior to that of their pixel-space c
ounterparts. To mitigate the computational inefficiency, this work proposes proj
ecting data into lower-dimensional subspaces using dynamic resolution DMs to acc
elerate the inference process. We first fine-tune pre-trained DMs for dynamic re
solution priors and adapt DPS and DAPS, which are two widely used pixel-space me
thods for general image restoration tasks, into the proposed framework, yielding
 methods we refer to as SubDPS and SubDAPS, respectively. Given the favorable in
ference speed and reconstruction fidelity of SubDAPS, we introduce an enhanced v
ariant termed SubDAPS++ to further boost both reconstruction efficiency and qual
ity. Empirical evaluations across diverse image datasets and various restoration
 tasks demonstrate that the proposed methods outperform recent DM-based approach
es in the majority of experimental scenarios. The code is available at https://
github.com/StarNextDay/SubDAPS.git.
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CoRe: Combined Rewards with Vision-Language Model Feedback for Preference-Aligne
d Reinforcement Learning

Hexian Ni, Tao Lu, Yinghao Cai

Reward design remains a central challenge in reinforcement learning (RL). Hand-c
rafted rewards are often difficult to specify and may lead to suboptimal policie
s, while learned rewards from preferences can suffer from inefficiency and unsta
ble training. Inspired by the dual nature of human learning explored in cognitiv
e science, we decompose rewards into two complementary components: Formal Reward
s (FR), explicitly designed based on task knowledge, and Residual Rewards (RR),
learned from observations to capture implicit and nuanced preferences. Based on
this decomposition, we propose CoRe, a hybrid framework that integrates FR and R
R with vision-language models (VLMs) feedback to achieve preference-aligned poli
cies without human involvement. Our contributions are twofold: (1) We propose a
Formal Reward Module (FRM) that leverages VLMs to iteratively design and optimiz
e FR based on task knowledge and preference feedback, enabling the continual imp
rovement of policy during training; (2) We introduce a Residual Reward Module (R
RM) that learns RR from video-level preference by employing VLMs to generate pre
ference labels and capturing nuanced rewards that complement FR, ensuring alignm
ent with human intent. Through the synergy of FRM and RRM, CoRe enables the auto
matic construction of reliable rewards that are efficient and preference-aligned
. Extensive experiments demonstrate that CoRe outperforms existing approaches in
 terms of policy learning effectiveness and efficiency on ten robotic manipulati
on tasks in simulation and five real-worlds.
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MFH-NAS:A Hybrid Neural Architecture Search Framework for Multimodal Fusion Obje
ct Detection

Quanwei Gao, Shuqi Zhao, Ruyu Wang, Shuyin Zhang, Cong Liu, Zirui Luo

Multimodal fusion object detection faces a substantial modality gap at the same
backbone stage. This makes predefined stage-aligned fusion insufficient for cros
s-stage interactions. We propose MFH-NAS, a hybrid neural architecture search fr
amework that automatically discovers fusion architectures to better leverage cro
ss-modal complementarity. MFH-NAS searches both local fusion primitives and stag
e-level fusion connectivity. It targets fusion operator design and fusion stage
selection. It couples differentiable search with evolutionary search. Differenti
able search learns architecture parameters for local fusion primitives. Evolutio
nary search explores global fusion topologies, including stage selection and cro
ss-stage connection patterns. The joint search balances exploitation and explora
tion and mitigates premature convergence. It yields fusion structures that stren
gthen cross-stage interactions.We evaluate MFH-NAS on three public benchmarks, L
LVIP, RGBT-Tiny, and M3FD. MFH-NAS consistently outperforms handcrafted fusion-s
tage designs and prior stage-searching NAS baselines, improving mAP@0.5 from 85.
3% to 88.2% over strong fixed-stage fusion methods and delivering gains across
all benchmarks.The code is available at https://github.com/someboy0/MFH-NAS.
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Error Propagation in Dynamic Programming: From Stochastic Control to American Op
tion Pricing

Andrea Della Vecchia, Damir Filipovic
This paper investigates theoretical and methodological foundations for stochasti c optimal control (SOC) in discrete time. We start formulating the control probl em in a general dynamic programming framework, introducing the mathematical stru cture needed for a detailed convergence analysis. The associate value function i s estimated through a sequence of approximations combining nonparametric regress ion methods and Monte Carlo subsampling. The regression step is performed within reproducing kernel Hilbert spaces (RKHSs), exploiting the classical KRR algorit hm, while Monte Carlo sampling methods are introduced to estimate the continuati on value. To assess the accuracy of our value function estimator, we propose a n atural error decomposition and rigorously control the resulting error terms at e ach time step. We then analyze how this error propagates backward in time-from m aturity to the initial stage-a relatively underexplored aspect of the SOC litera ture. Finally, we illustrate how our analysis naturally applies to a key financi al application: the pricing of American options.
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E-mem: Multi-Agent Based Episodic Context Reconstruction for LLM Agent Memory
Kaixiang Wang, Yidan Lin, Zihan Wang, Bunyod Suvonov, Zhaojiacheng Zhou, Yuxiang Zheng, Jiaxi Cao, Zhiheng Dong, Chentao Wu, Jiong Lou, Jie LI
The evolution of Large Language Model (LLM) agents towards System~2 reasoning, c haracterized by deliberative, high-precision problem-solving, necessitates maint aining rigorous logical integrity over extended horizons. However, prevalent mem ory preprocessing paradigms incur destructive de-contextualization. By compressi ng fluid sequential dependencies into pre-defined structures (e.g., embeddings o r graphs), these methods sever the narrative integrity essential for deep reason ing. To address this, we propose E-mem, a framework shifting from Memory Preproc essing to Episodic Context Reconstruction inspired by biological engrams. E-mem employs a heterogeneous hierarchical architecture where multiple assistant agent s maintain uncompressed memory contexts, while a central master agent orchestrat es global planning. Unlike passive retrieval, our mechanism empowers assistants to locally reason within activated segments, extracting context-aware evidence b efore aggregation. Evaluations on the LoCoMo benchmark demonstrate that E-mem ac hieves over 54\% F1—surpassing the state-of-the-art GAM by 7.75\%—while reducing token cost by over 70\%. Our work is available on \url{https://anonymous.4open. science/r/E-mem-F6C3/}.
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iTryOn: Mastering Interactive Video Virtual Try-On with Spatial-Semantic Guidanc e
Jun Zheng, Zhengze Xu, Mengting Chen, Jing Wang, Jinsong Lan, Xiaoyong Zhu, Kaif u Zhang, Bo Zheng, Xiaodan Liang
Video Virtual Try-On (VVT) aims to seamlessly replace a garment on a person in a video with a new one. While existing methods have made significant strides in m aintaining temporal consistency, they are predominantly confined to non-interact ive scenarios where models merely showcase garments. This limitation overlooks a crucial aspect of real-world apparel presentation: active human-garment interac tion. To bridge this gap, we introduce and formalize a new challenging task: Int eractive Video Virtual Try-On (Interactive VVT), where subjects in the video act ively engage with their clothing (e.g., pulling a hem or unzipping a jacket). Th is task introduces unique challenges beyond simple texture preservation, includi ng: (1) resolving the semantic ambiguity of interactions from standard pose info rmation, and (2) learning complex garment deformations from video where interact ive moments are sparse and brief.
To address these challenges, we propose **iTryOn**, a novel framework built upon a large-scale video diffusion Transformer. iTryOn pioneers a multi-level intera ction injection mechanism to guide the generation of complex dynamics. At the sp atial level, we introduce a garment-agnostic 3D hand prior to provide fine-grain ed guidance for precise hand-garment contact, effectively resolving spatial ambi guity. At the semantic level, iTryOn leverages global captions for overall conte xt and time-stamped action captions for localized interactions, synchronized via our novel Action-aware Rotational Position Embedding (A-RoPE). Furthermore, we

design an action-aware constraint loss to stabilize training and focus the learning process on these critical interactive frames. To facilitate research and evaluation, we construct VVT-Interact, the first large-scale dataset for this task, and propose a novel interaction-aware evaluation metric to quantify the semantic fidelity of interactions. Extensive experiments demonstrate that iTryOn not only achieves state-of-the-art performance on traditional VVT benchmarks but also establishes a commanding lead in the new interactive setting, marking a significant step towards more dynamic and controllable virtual try-on experiences.
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Quantifying the Salience of Geo-Cultural Values for Pluralistic Safety Alignment
Arkadiy Saakyan, Charvi Rastogi, Lora Aroyo
Safe global deployment of AI models requires alignment with human values that vary across cultures. Yet rater pools in safety evaluation datasets remain largely geographically homogeneous, failing to capture geo-cultural differences. Further, it remains unclear whether such differences persist after controlling for demographics such as age, gender, and ethnicity. Through a meta-analysis of safety datasets, we find that most do not report geo-cultural information, and those that do lack a unified methodology to jointly analyze geo-cultural and demographic correlates. Using the Inglehart-Welzel dimensions of cross-cultural variation, we demonstrate via multilevel modeling that cultural zone membership explains variance in safety ratings beyond standard demographics ($p<0.05$ across 6 datasets). Moreover, our analysis indicates that roughly 10\% of items in the datasets we examined are culturally sensitive: likely to be misclassified as safe without adequate cultural representation. We evaluate LLMs as both rater surrogates and triage tools, finding that current LLMs do not reliably stand in for raters, though they can help prioritize culturally sensitive items for human annotation. Our findings motivate more culturally pluralistic safety evaluation and offer practical takeaways to support it.
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Learning Unmasking Policies for Diffusion Language Models
Metod Jazbec, Theo X. Olausson, Louis Béthune, Pierre Ablin, Michael Kirchhof, Joao Monteiro, Victor Guilherme Turrisi da Costa, Jason Ramapuram, marco cuturi
Diffusion (Large) Language Models (dLLMs) now match the downstream performance of their autoregressive counterparts on many tasks, while holding the promise of being more efficient during inference. One critical design aspect of dLLMs is the \textit{sampling procedure} that selects which tokens to unmask at each diffusion step. Indeed, recent work has found that heuristic strategies such as confidence thresholding improve both sample quality and token throughput compared to random unmasking. However, such heuristics have downsides: they require manual tuning, and we observe that their performance degrades with larger block sizes. In this work, we instead propose to train sampling procedures using reinforcement learning. Specifically, we formalize masked diffusion sampling as a Markov decision process in which the dLLM serves as the environment, and propose a lightweight policy based on a single-layer transformer that maps dLLM token confidences to unmasking decisions. Our experiments show that these trained policies match the performance of state-of-the-art heuristics when combined with semi-autoregressive (block) generation, while outperforming them in the full-diffusion setting. Our code is available at https://github.com/apple/ml-rl-dllm.
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DenseMLLM: Standard Multimodal LLMs for Dense Prediction
Yi Li, Hongze Shen, Lexiang Tang, Xin Li, Xinpeng Ding, Yinsong Liu, Deqiang Jiang, Xing Sun, Xiaomeng Li
Multimodal Large Language Models (MLLMs) have demonstrated exceptional capabilities in high-level visual understanding. However, extending these models to fine-grained dense prediction tasks, such as semantic segmentation and depth estimation, typically necessitates the incorporation of complex, task-specific decoders and other customizations. This architectural fragmentation increases model complexity and deviates from the generalist design of MLLMs, ultimately limiting their practicality. In this work, we challenge this paradigm by accommodating standa

rd MLLMs to perform dense predictions without requiring additional task-specific decoders. The proposed model is called DenseMLLM, grounded in the standard architecture with a novel vision token supervision strategy for multiple labels and tasks. Despite its minimalist design, our model achieves highly competitive performance across a wide range of dense prediction and vision-language benchmarks, demonstrating that a standard, general-purpose MLLM can effectively support dense perception without architectural specialization. This project is available at github.com/Eli-YiLi/DenseMLLM.

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Barriers to Counterfactual Credit Attribution for Autoregressive Models

Aloni Cohen, Chenhao Zhang

Generative AI disrupts the practice of giving credit to work that came before. Ideally, a generative model would give credit to any work on which its output depends in a significant way.

Counterfactual credit attribution (CCA) is a technical condition formalizing this goal---a relaxation of differential privacy---recently introduced by Livni, Moran, Nissim, and Pabbaraju (2024) who studied it in the PAC learning setting. We initiate the study of CCA generative models. Specifically, we consider autoregressive models giving credit to a deployment-time dataset (e.g., a RAG database). We uncover barriers to two natural approaches to CCA autoregressive models. First, we show that imposing CCA on the underlying next-token predictor does not guarantee that the model is CCA: CCA does not compose autoregressively (unlike DP). Second, we consider a different approach to building CCA models which we call *retrofitting*. Retrofitting takes a model that does not attribute credit, and adds credit onto it. Given black-box access to the starting model, retrofitting requires query complexity exponential in the length of the model's outputs.

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Learning to Rank from Incomplete Rankings

Cristiano Migali, Gianmarco Genalti, Alberto Maria Metelli, Marco Mussi

In domains such as recommender systems and information retrieval, learning from human-generated feedback is especially challenging because the information provided is often sparse and incomplete. In this work, we address the problem of learning the top-k items from incomplete rankings. Most existing models for incomplete rankings rely on rigid assumptions regarding both the ranking model that generates the latent ranking and the censoring mechanism that determines which comparisons remain unobserved. On the one hand, the ranking model is often assumed to follow a Plackett-Luce (PL) or Mallows distribution. On the other hand, the censoring mechanism is typically assumed to be Missing Completely At Random (MCAR) or to exhibit well-behaved dependencies on the latent ranking, such as winner feedback or top-h feedback. We introduce a new, general framework for learning from incomplete rankings that unifies and strictly generalizes the established frameworks in the literature. We consider the broad class of ranking models that satisfy the complete consensus property, which comprehends all widely adopted models, including PL and Mallows. Furthermore, we present a new preference-based feedback model, named positional censoring, which generalizes winner and top-h feedback. We show that it is possible to learn in this general setting by presenting the PIRATE algorithm and providing a near-optimal instance-dependent bound to the sample complexity. Finally, we show that, under the PL ranking, PIRATE matches the sample complexity of state-of-the-art algorithms in the relevant scenarios of winner and top-h feedback.

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Revisiting Distribution Correction Estimation for Offline Imitation Learning with Suboptimal Dataset

Quang Anh Pham, Tien Anh Mai, Akshat Kumar

Imitation Learning (IL) learns high-quality policies from expert demonstrations but degrades in low-expert-data regimes. To address this, recent work studies ``*offline IL with supplementary data*", augmenting expert data with trajectories from suboptimal policies. A prominent framework is Distribution Correction Estimation (DICE), which estimates density ratios via the dual of a divergence minimization problem between learned and expert visitation distributions. However, exi

sting DICE-based methods either rely on a strict coverage assumption or introduc e additional dataset regularization, limiting performance. We propose **ReDICE** , a new DICE-based method that addresses these issues through an objective-level reformulation. Our approach constructs a mixture-distribution objective that pr eserves the original expert-imitation objective while removing the coverage assu mption, and its dual reduces to a stable Gumbel regression objective for efficie nt optimization. We further introduce a novel policy extraction mechanism that i mproves performance. Experiments on standard and real-world offline IL benchmark s show that ReDICE consistently outperforms prior methods and achieves state-of- the-art results.

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Batch Normalization for Neural Networks on Complex Domains

Xuan Son Nguyen, Nistor Grozavu

Riemannian neural networks have proven effective in solving a variety of machine learning tasks. The key to their success lies in the development of principled Riemannian analogs of fundamental building blocks in deep neural networks (DNNs) . Among those, Riemannian batch normalization (BN) layers have shown to enhance training stability and improve accuracy. In this paper, we propose BN layers for neural networks on complex domains. The proposed layers have close connections with existing Riemannian BN layers. We derive essential components for practical implementations of BN layers on some complex domains which are less studied in previous works, e.g., the Siegel disk domain. We conduct experiments on radar cl utter classification, node classification, and action recognition demonstrating the efficacy of our method.

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Speculative Safety Honeypot: Toward Proactive Defense Against Multi-turn Agent A ttacks

Zezhong WANG, Xueyang Tang, RUI LIAN, Yang Lou, Heqing Huang

As Large Language Model (LLM) agents are increasingly deployed in complex enviro nments, multi-turn interaction attacks have become a significant security challe nge. Existing detection methods typically rely on historical context. However, t his retrospective logic struggles to identify deep malicious intents that are sp lit across turns to hide future risks.

Inspired by speculative decoding, we propose the Speculative Safety Honeypot (SS H) framework. SSH uses a multi-agent simulation system composed of small LLMs to build an action-level speculate-and-verify workflow. In the speculation stage, SSH predicts future behaviors of the target agent and asynchronously builds a tr ajectory tree to expose potential risks in advance. In the verification stage, t he system uses the target agent's real actions to calibrate and prune the trajec tory tree, effectively reducing false positives.

As a plug-and-playable component, SSH provides existing detectors with rich deci sion redundancy beyond the current interaction slice. By judging risk based on t he evolution of the entire trajectory tree rather than a single point in time, t he system reduces the reliance on the absolute precision of individual detection components. This improves the defense resilience and the warning lead-time of a gent systems against complex temporal attacks.

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LoRe: Adaptive Interaction-Evaluation Routing with Per-step Interaction Budgets for Iterative Graph Solvers

Jintao Li, Yong-Yi Wang, Zheng-an Wang, Heng Fan

Diffusion-based neural solvers for combinatorial optimization repeatedly re-eval uate dense edge/factor interactions, making inference expensive in wall-clock ti me and often memory-bound at scale. Inspired by the computational methodologies of many-body physics, we introduce LoRe, a training-free, inference-time drop-in wrapper that enforces per-step interaction-evaluation budgeting: at each iterat ion, it evaluates only a fixed fraction of interactions by dynamically routing c omputation to high-conflict or high-uncertainty interactions, instead of using a fixed sparsification (e.g., static kNN graphs or static masks). Under fully inc lusive end-to-end wall-clock accounting, LoRe substantially improves scalability on the Maximum Independent Set (MIS) problem, extending feasible inference more

than $3\times$ beyond the baseline's out-of-memory limit, delivering a ${\sim}8\times$ speedup and a ${\sim}12\times$ peak-memory reduction, with solution quality preserved in this regime. Demonstrating cross-task generality on the large-scale Traveling Salesperson Problem (TSP) and zero-shot robustness to topology shifts, LoRe achieves a ${\sim}15\times$ speedup at $n=1000$ with a $44\times$ memory reduction and competitive tour quality.
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Graph of States: Solving Abductive Tasks with Large Language Models
Yu Luo, Rongchen Gao, Lu Teng, Xidao Wen, Jiamin Jiang, Qingliang Zhang, Yongqian Sun, Shenglin Zhang, Jiasong Feng, Tong Liu, Wenjie Zhang, Dan Pei
Logical reasoning encompasses deduction, induction, and abduction.
However, while Large Language Models (LLMs) have effectively mastered the former two, abductive reasoning remains significantly underexplored.
Existing frameworks, predominantly designed for static deductive tasks, fail to generalize to abductive reasoning due to unstructured state representation and lack of explicit state control.
Consequently, they are inevitably prone to Evidence Fabrication, Context Drift, Failed Backtracking, and Early Stopping.
To bridge this gap, we introduce Graph of States (GoS), a general-purpose neuro-symbolic framework tailored for abductive tasks.
GoS grounds multi-agent collaboration in a structured belief states, utilizing a causal graph to explicitly encode logical dependencies and a state machine to govern the valid transitions of the reasoning process.
By dynamically aligning the reasoning focus with these symbolic constraints, our approach transforms aimless, unconstrained exploration into a convergent, directed search.
Extensive evaluations on two real-world datasets demonstrate that GoS significantly outperforms all baselines, providing a robust solution for complex abductive tasks.
Code repo and all prompts: https://github.com/gaorch85/Graph-of-States.
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Monotonic Variational Gaussian Process for Efficient Data Collection
Donghyun Lee, Young Myoung Ko
Modeling the learning curve is critical for cost-effective data collection in deep learning systems. Most prior approaches assume a specific parametric learning curve, but these can be inappropriate when no reliable parametric form can be assumed for the learning curve. While Gaussian processes offer flexible nonparametric modeling, existing GP approaches that enforce monotonicity typically introduce intractable factors or require derivative observations. To address this, we propose a Monotonic Variational Gaussian Process for Efficient Data Collection (MOVE), which (i) introduces a novel monotonic variational GP formulation with virtual-derivative factors to enable tractable posterior inference, and (ii) develops an expected shortfall based objective for target-driven data collection. Furthermore, our theoretical analysis shows that expected shortfall provides non-vanishing gradient signals that enable reliable gradient-based optimization. Extensive experiments on classification, segmentation, and detection benchmarks demonstrate consistent improvements over the prior method.
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Differentially Private Continual Release with Relative Error
Bo Li, Wei Wang, Peng Ye
This work investigates several fundamental tasks, including MaxSum, MinSum, $\mathsf{MaxSelect}$, and $\mathsf{MinSelect}$, in the continual release model under differential privacy. Previous research has demonstrated that any algorithm for these tasks must admit a large purely additive error. We show that the error can be substantially reduced if a relative error term is allowed, provided that the input stream is generated non-adaptively. However, when input data records can be selected adaptively, we prove that a large error is inevitable for the task of selecting an attribute with a small cumulative sum, whereas small error bounds remain achievable for other tasks. This reveals a significant separation between non-adaptive and adaptive streams. We also complement o

ur algorithms with nearly matching lower bounds.
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Neutral-Reference Prompting for Vision–Language Models
Senmao Tian, Xiang Wei, Shunli Zhang
Efficient transfer learning of vision–language models (VLMs) commonly suffers fr om a Base–New Trade-off (BNT): improving performance on unseen (new) classes oft en degrades accuracy on known (base) classes. Addressing how to boost recognit ion of unseen classes without sacrificing known-class performance remains a cent ral challenge. Existing work often simplistically attributes the BNT to overfitt ing on known classes.
We observe an interesting phenomenon: VLMs frequently exhibit asymmetric confusi on on certain downstream data, i.e., samples of class A are systematically mispr edicted as class B, while the reverse confusion (B $\rightarrow$ A) rarely occurs. For know n classes, this kind of bias can be mitigated by tuning using a cross-entropy lo ss, but for unseen classes, such pretraining-induced bias persists and harms gen eralization.
Motivated by this, we propose NeRP, a plug-and-play prompting correction strateg y that improves discrimination on unseen classes without modifying model paramet ers. NeRP leverages neutral text prompts and reference images to measure class-w ise prior preferences along the pre-trained inter-class geometry, and combines t hem with the sample likelihood to obtain the model's surrogate score. If, for a given sample, the prior strongly favors the current prediction while the observe d evidence is clearly insufficient, we perform a local flip between easily confu sable class pairs, thereby correcting prior-dominated mispredictions.
Extensive experiments across multiple backbones and 15 few-shot and cross-domain benchmarks show that NeRP substantially improves accuracy on unseen classes whi le preserving known-class prediction performance.
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Olivia: Harmonizing Time Series Foundation Models with Power Spectral Density
Jingru Fei, Kun Yi, Alex Xing Wang, Qingsong Wen, Xiangxiang Zhu, Wei Fan
Time series foundation models rely on large-scale pretraining over diverse datas ets across domains, yet their heterogeneity in temporal patterns could hinder th e effectiveness of training and learning transferable time series representation s. Inspired a fundamental concept, normalized power spectral density (PSD) in si gnal processing, we assume harmonizing datasets via PSDs in the spectral domain could reduce mismatches and enhance pretraining. We then go beyond the direct in tractable minimization optimization and innovatively reformulate it as a princip led harmonization approach. Specifically, we propose Harmonizer, a module that r eshapes spectral structures and implicitly harmonizing PSDs across datasets, whi ch theoretically corresponds to a shared reparameterization of second-order temp oral correlations. Our theoretical analysis further reveals token interactions w ith Harmonizer can be efficiently mediated by a compact set of resonators, motiv ating a HarmonicAttention design that performs self-attention in a low-dimension al interaction space. Then, we propose Olivia, a novel time series foundation mo del built upon these harmonization mechanisms. Extensive experiments on several large-scale benchmarks (TSLib, GIFT-Eval, and GluonTS), demonstrate Olivia consi stently achieves state-of-the-art performance under zero-shot, few-shot, and ful l-shot forecasting scenarios. Our code is at \url{https://github.com/TSTS13/Oliv ia}.
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Dimensionality Reduction with Point-distributions Similarity Invariant
Hang Zhang, Kai Ming Ting
Existing dimensionality reduction methods all perform dimensionality reduction b y preserving some invariant in the space before and after dimensionality reducti on. This paper proposes a new dimensionality reduction invariant: preserving the invariant of the point-distributions similarity. We also design a linear and ef ficient method to achieve dimensionality reduction while preserving this invaria nt. We theoretically prove the feasibility of our method for dimensionality redu ction. Furthermore, our results on benchmark datasets and single-cell expression data demonstrate the effectiveness and efficiency of the proposed method.

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Finding DoRI: Discovery of Retained Images in Diffusion Models
Antoni Kowalczuk, Dominik Hintersdorf, Lukas Struppek, Kristian Kersting, Adam D ziedzic, Franziska Boenisch
Text-to-image diffusion models (DMs) have achieved remarkable success in image g eneration. However, concerns about data privacy and intellectual property remain due to their potential to inadvertently memorize and replicate training data. R ecent mitigation efforts have focused on identifying and pruning weights respons ible for triggering verbatim training data replication, based on the assumption that memorization can be localized. We challenge this assumption and demonstrate that, even after such pruning, small perturbations to the text embeddings of pr eviously mitigated prompts can re-trigger data replication, revealing the fragil ity of such methods.
Our further analysis then provides multiple indications that memorization is ind eed *not* inherently local: (1) replication triggers for memorized images are di stributed throughout text embedding space; (2) embeddings yielding the same repl icated image produce divergent model activations; and (3) different pruning meth ods identify inconsistent sets of memorization-related weights for the same imag e. Finally, we show that bypassing the locality assumption enables more robust m itigation through adversarial fine-tuning. These findings provide new insights i nto the fundamental nature of memorization in text-to-image DMs and inform the f uture development of more reliable mitigation methods against DM memorization.
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HIVE-3D: Hierarchical Voxel Enhancement for High-Quality 3D Scene Generation
Bin Zang, Wenting Zheng, Xiaoliang Luo, Zhiyuan Fang, Shi Li, Lvchun Wang, Wei Y u, Yi Zhao, Tian Xie, Yuchi Huo, Rengan Xie
Recently, a line of works can generate impressive 3D objects from a single image , but they are limited by restricted representation resolution, making them unsu itable for 3D scene generation. In this work, we introduce HIVE-3D, a novel meth od for high-quality 3D scene generation based on hierarchical voxel enhancement framework. Specifically, given a single scene image as input, we first produce a coarse initial scene, then introduce image segmentation and attention-based ret rieval to align 2D image components with 3D scene components. Subsequently, we o rganize these scene relations into a hierarchical component tree, where nodes cl oser to the leaves denote finer-grained components. Finally, we propose a voxel super-resolution model that generates refined voxels for the target instance whi le maintaining strong consistency with the coarse voxels. Equipped with this mod el, we perform coarse-to-fine hierarchical super-resolution on images and voxels for each component, producing a high-resolution and high-quality 3D scene. Exte nsive experiments demonstrate that our method significantly outperforms previous approaches, achieving state-of-the-art performance.
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CausalRAG2: Hierarchical Causal Knowledge Graph Design for RAG
Nengbo Wang, Tuo Liang, Vikash Singh, Chaoda Song, Van Yang, Yu Yin, Jing Ma, JA GDIP SINGH, Vipin Chaudhary
Retrieval augmented generation (RAG) has enhanced large language models by enabl ing access to external knowledge, with graph-based RAG emerging as a powerful pa radigm for structured retrieval and reasoning. However, existing graph-based met hods often over-rely on entity-centric node matching and lack explicit causal mo deling, leading to unfaithful or spurious answers. Prior attempts to incorporate causality are typically limited to local or single-document contexts and also s uffer from information isolation that arises from modular graph structures, whic h hinders scalability and cross-module causal reasoning. To address these challe nges, we propose CausalRAG2, a framework that rethinks knowledge organization fo r graph-based RAG through causal gating across hierarchical modules. CausalRAG2 explicitly models causal relationships to suppress spurious correlations while e nabling scalable reasoning over large-scale knowledge graphs. We also introduce HolisQA, a benchmark for holistic comprehension beyond entity-centric matching. Extensive experiments demonstrate that CausalRAG2 consistently outperforms compe titive graph-based RAG baselines across multiple datasets and evaluation metrics

. Our work establishes a principled foundation for structured, scalable, and cau
sally grounded RAG systems.
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LoCoT2V-Bench: Benchmarking Long-Form and Complex Text-to-Video Generation
Xiangqing Zheng, Chengyue Wu, Kehai Chen, Min Zhang
Recent advances in text-to-video generation have achieved impressive performance
 on short clips, yet evaluating long-form generation under complex textual input
s remains a significant challenge. In response to this challenge, we present LoC
oT2V-Bench, a benchmark for long video generation (LVG) featuring multi-scene pr
ompts with hierarchical metadata (e.g., character settings and camera behaviors)
, constructed from collected real-world videos. We further propose LoCoT2V-Eval,
 a multi-dimensional framework covering perceptual quality, text-video alignment
, temporal quality, dynamic quality, and Human Expectation Realization Degree (H
ERD), with an emphasis on aspects such as fine-grained text-video alignment and
temporal character consistency. Experiments on 17 representative LVG models reve
al pronounced capability disparities across evaluation dimensions, with strong p
erceptual quality and background consistency but markedly weaker fine-grained te
xt-video alignment and character consistency. These findings suggest that improv
ing prompt faithfulness and identity preservation remains a key challenge for lo
ng-form video generation. Our code and data are released at [https://github.com/
XqZeppelinhead0702/LoCoT2V-Bench](https://github.com/XqZeppelinhead0702/LoCoT2V-
Bench).
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FRISM: Fine-Grained Reasoning Injection via Subspace-Level Model Merging for Vis
ion–Language Models
Chenyu Huang, Peng Ye, Xudong Tan, Jinhan Mu, Shenghe Zheng, Li Shen, Tao Chen
Efficiently enhancing the reasoning capabilities of Vision-Language Models (VLMs
) by merging them with Large Reasoning Models (LRMs) has emerged as a promising
direction. However, existing methods typically operate at a coarse-grained layer
 level, which often leads to a trade-off between injecting reasoning capabilitie
s and preserving visual capabilities.
To address this limitation, we propose FRISM (Fine-grained Reasoning Injection v
ia Subspace-level model Merging), a fine-grained reasoning injection framework b
ased on subspace-level model merging.
Observing that different SVD subspaces contribute differently to reasoning and p
erception,
FRISM decomposes LRM task vectors via Singular Value Decomposition (SVD) and ada
ptively tunes the scaling coefficients of each subspace through learning to real
ize fine-grained reasoning injection.
Furthermore, we introduce a label-free self-distillation learning strategy with
dual-objective optimization using common vision-language perception datasets.
Extensive experiments demonstrate that FRISM effectively improves reasoning capa
bilities while largely preserving the model's visual capabilities by
consistently achieving strong performance across diverse visual-language reasoni
ng benchmarks.
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Quantifying and Optimizing Simplicity via Polynomial Representations
Tianren Zhang, Xiangxin Li, Minghao Xiao, Guanyu Chen, Feng Chen
Deep networks often exhibit a preference for "simple" solutions, and such a simp
licity bias is widely believed to play a key role in generalization. Yet a broad
ly applicable, quantitative measure of simplicity remains elusive. We introduce
polynomial representations as a distribution-aware, low-dimensional surrogate
for neural functions: we approximate a network’s predictive behavior along data-
dependent interpolation paths using orthogonal polynomial bases, yielding a comp
act functional representation. We show that the _effective degree_ of this repre
sentation serves as a practical simplicity metric that is predictive of generali
zation across tasks and architectures, and consistently outperforms existing gen
eralization proxies such as sharpness. Finally, polynomial representations natur
ally yield a _differentiable_ simplicity regularizer, which consistently improve
s generalization in image and text classification, fine-tuning contrastive visio

n–language models, and reinforcement learning.
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MarketSim: Simulating Stock Markets with Large-Scale Generative Agents
Jinghua Piao, zhentao liu, Cheng Huang, Jiarui Huang, Songwei Li, Ranran Wang, Y ong Li
Stock markets are one of the most complex systems in the modern world, where pri ces emerge from billions of decentralized interactions among heterogeneous parti cipants in an ever-evolving information landscape. While high-fidelity simulatio n is important for understanding market dynamics, existing approaches face a per sistent trade-off between structural and behavioral fidelity. To this end, we pr opose MarketSim, a large-scale stock market simulation framework with generative agents. MarketSim introduces a hierarchical multi-agent architecture that decou ples strategic reasoning from high-frequency execution, enabling LLM agents to o perate in a nanosecond-resolution, NASDAQ-like continuous double auction market. Building on this, we simulate over 15,000 heterogeneous market participants who se interactions shape and are shaped by an evolving market environment grounded in more than 12k real-world news articles, policy documents, and earnings report s. To evaluate our proposed MarketSim, we develop a comprehensive benchmark that includes stocks from 8 GICS sectors and 3 representative real-world scenarios, along with 5 stylized facts for market complexity and 5 price-related statistica l metrics. Extensive experiments demonstrate that MarketSim not only captures ke y complexity properties of real-world markets, but also outperforms state-of-the -art baselines in tracking high-frequency price dynamics with an average MAPE of 3.48%, providing a scalable testbed for market analysis.
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PRISM: Parallel Residual Iterative Sequence Model
Jie Jiang, Ke Cheng, Xin Xu, Mengyang Pang, Tianhao Lu, Jiaheng Li, Yue Liu, Yua n Wang, Jun Zhang, Huan Yu, Zhouchen Lin
Generative sequence modeling faces a fundamental tension between the expressivit y of Transformers and the efficiency of linear sequence models. Existing efficie nt architectures are theoretically bounded by shallow, single-step linear update s, while powerful iterative methods like Test-Time Training (TTT) break hardware parallelism due to two dimensions of serial dependency: token-level state relia nce and step-level iteration loops. We propose PRISM (Parallel Residual Iterativ e Sequence Model) to resolve this tension. PRISM explicitly reconstructs the exp ressive gate × residual × direction iteration pattern of TTT in a parallelizable form. We employ a Write-Forget Decoupling strategy that isolates non-linearity within the injection operator. To bypass the serial dependency of explicit solve rs, PRISM utilizes a two-stage proxy architecture: a short-convolution anchors t he initial residual using local history energy, while a learned predictor estima tes the refinement updates directly from the input. This design distills structu ral patterns associated with iterative correction into a parallelizable feedforw ard operator. Theoretically, we prove that this formulation achieves Rank-L ac cumulation, structurally expanding the update manifold beyond the single-step Ra nk-1 bottleneck. Empirically, it achieves comparable performance to explicit o ptimization methods while achieving 174x higher throughput. Codes are available in \url{https://github.com/gpr-prism/prism/}.
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Task-Aware Exploration via a Predictive Bisimulation Metric
Dayang Liang, Ruihan LIU, Lipeng Wan, Yunlong Liu, Bo An
Accelerating exploration in visual reinforcement learning under sparse rewards r emains challenging due to the substantial task-irrelevant variations. Despite ad vances in intrinsic exploration, many methods either assume access to low-dimens ional states or lack task-aware exploration strategies, thereby rendering them f ragile in visual domains. To bridge this gap, we present TEB, a Task-aware Explo ration approach that tightly couples task-relevant representations with explorat ion through a predictive Bisimulation metric. Specifically, TEB leverages the me tric not only to learn behaviorally grounded task representations but also to me asure global intrinsic novelty over the learned latent space. To realize this, w e first theoretically mitigate the representation collapse of degenerate bisimul

ation metrics under sparse rewards by internally introducing a simple but effect ive predicted reward differential term. Building on this robust metric, we furth er introduce potential-based global exploration bonuses over anchor states, whic h measure the relative novelty between observations over the latent space. Exten sive experiments on MetaWorld and Maze2D show that TEB achieves superior explora tion ability and outperforms recent baselines.
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Revisiting ML Training under Fully Homomorphic Encryption: Convergence Guarantee s, Differential Privacy, and Efficient Algorithms
Yvonne Zhou, Mingyu Liang, Ivan Brugere, Danial Dervovic, Yue Guo, Antigoni Poly chroniadou, Min Wu, Dana Dachman-Soled
We present the first theoretical convergence analysis of machine learning traini ng under fully homomorphic encryption (FHE), combined with a differentially priv ate (DP) training algorithm tailored to encrypted computation. Our approach impr oves computational efficiency over standard differentially private gradient desc ent (DP-GD) while achieving comparable utility. In particular, we prove converge nce of approximate gradient descent using polynomial approximations of activatio n and loss functions, which are required for FHE compatibility. To preserve priv acy in downstream tasks, we integrate differential privacy without relying on co stly per-sample gradient clipping, enabling scalable encrypted learning. We also provide data-independent hyperparameter selection and theoretically grounded st rategies for polynomial approximation which can be of independent interest. Toge ther, these contributions advance the feasibility of efficient, private, and sec ure machine learning on sensitive data.
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Intra-Modal Neighbors Never Lie: Rectifying Inter-Modal Noisy Correspondence via Graph-Based Intra-Modal Reasoning
Yang Liu, Wentao Feng, Shu-Dong Huang, Yalan Ye, Jiancheng Lv
Large-scale web-harvested datasets have fueled the progress of cross-modal retri eval but inevitably suffer from noisy correspondence, which severely degrades mo del generalization. Existing methods primarily address this by filtering out noi se or seeking a substitute label, yet they predominantly remain bound by a “Disc rete Selection” paradigm. We argue that relying on a single discrete proxy induc es SinglePoint Fragility and Discretization Error. To overcome these limitations , we propose a novel framework, Intra-modal Neighbor-aware Noise Rectification (IN2R), which shifts the paradigm from searching for a substitute to synthesizing a reliable supervision target. Leveraging the intrinsic geometric stability of intra-modal data, IN2R employs a Graph Refiner to perform relational reasoning o ver neighbors retrieved from a dynamic Cross-Model Memory. Instead of propagatin g discrete labels, our method synthesizes a continuous, soft prototype that refl ects the consensus of the local semantic neighborhood, effectively rectifying in ter-modal misalignment. Extensive experiments on Flickr30K, MS-COCO, and CC152K demonstrate that IN2R significantly outperforms state-of-the-art methods.
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Bridging Tokens and Geometry: Token-wise 3D Supervision for CAD Generation
Yijia Guan, Jianhua Sun
Computer-Aided Design (CAD) generation is typically formulated as a sequence mod eling task over parametric tokens. Recent studies introduce visual information t hrough additional visual inputs or rendering of the final generated programs. Ho wever, these methods provide no intermediate visual feedback, hindering the asso ciation of individual tokens with their geometric effects. In this work, we prop ose an Argument-induced 3D Point Loss (A3PL) that maps argument tokens to corres ponding 3D points, enabling dense token-wise geometric supervision. To reduce le arning complexity and invalid sequences, we further introduce a Grammar-constrai ned Operator (GCO) that leverages the structured nature of CAD programs to regul ate sequence generation. We evaluate our approach on five CAD generation tasks w ith diverse input modalities, including text, Scalable Vector Graphics (SVG) ske tches, point clouds, and CAD sequences. Our approach improves generation accurac y and program validity across different input modalities. Code is available at h ttps://github.com/JumpJumpTigger-GYJ/A3PL.

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ZeroBench: An Impossible Visual Benchmark for Contemporary Large Multimodal Models
Jonathan Roberts, Mohammad Reza Taesiri, Ansh Sharma, Akash Gupta, Samuel Roberts, Ioana Croitoru, Simion-Vlad Bogolin, Jialu Tang, Florian Langer, Vyas Raina, Vatsal Raina, Hanyi Xiong, Vishaal Udandarao, Jingyi Lu, Chen Shiyang, Sam Purkis, Tianshuo Yan, Wenye Lin, Gyungin Shin, Qiaochu Yang, Anh Totti Nguyen, David Atkinson, Alexandru Coca, Mikah Dang, Sebastian Dziadzio, Jakob D. Kunz, Kaiqu Liang, Alexander Lo, Brian Pulfer, Steven Walton, Charig Yang, Kai Han, Samuel Albanie
Large Multimodal Models (LMMs) exhibit shortfalls when interpreting images and, by some measures, have poorer spatial cognition than young children or animals. Despite this, they attain high scores on many popular visual benchmarks, with headroom rapidly eroded by model progress. This creates a need for difficult benchmarks that remain relevant for longer. We introduce ZeroBench—a lightweight visual reasoning benchmark curated using adversarial filtering to be “impossible” for frontier LMMs at its original release, with initial SotA scores of 0% pass@1 and pass∧5. We track progress on ZeroBench over the subsequent year, observing SotA reaching 6% pass∧5 and 19% pass@5, indicating the potential longevity of the benchmark. We evaluate 46 LMMs on ZeroBench, compare performance to a human baseline, analyse strengths and weaknesses, chart a year of progress in visual capabilities, and publicly release ZeroBench at https://zerobench.github.io/.
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Theory of Continual Learning Against Data Poisoning Attacks
Yiting Hu, Lingjie Duan
Continual learning (CL), where a model is trained on a sequence of data tasks, is increasingly being adopted across key fields such as large language models and image recognition, yet it remains highly vulnerable to data poisoning that triggers learning divergence or severe excess risk. Despite these threats, a principled theoretical foundation in CL for understanding attack and defense remains lacking. In this paper, we develop a theoretical framework to analyze strategic attacks and defenses in regularization-based CL, a cornerstone of recent CL theory. By framing the adversary-defender interaction as an online zero-sum game, we first establish a fundamental performance limit: no defense succeeds when an adversary poisons a linear proportion of tasks by injecting unbounded noise or pattern shifts in regularization-based CL. We then analyze two possibly defensible scenarios: infrequent attacks and bounded noise per attack. For the former regime, we propose a task-to-task verification mechanism to detect data poisoning and reduce cumulative bias for learning convergence. For the latter regime, we derive a robust defense that minimizes the model’s sensitivity to poisoned features, provably accelerating the convergence rate. Extensive experiments on realistic tasks further validate our theoretical results.
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Omni-Diffusion: Unified Multimodal Understanding and Generation with Masked Discrete Diffusion
lijiang Li, Zuwei Long, Yunhang Shen, Heting Gao, Haoyu Cao, Xing Sun, Caifeng Shan, Ran He, Chaoyou Fu
While recent multimodal large language models (MLLMs) have made impressive strides, they mostly employ a conventional autoregressive architecture as their backbone, leaving significant room for exploring effective and efficient alternatives in architectural design. Meanwhile, recent studies have successfully applied discrete diffusion models to natural language processing, revealing their considerable potential as a promising new approach in this domain. Drawing inspiration from these pioneering studies, we introduce Omni-Diffusion, the first any-to-any multimodal language model built entirely on mask-based discrete diffusion models, which unifies understanding and generation across text, speech, and images. Omni-Diffusion employs a unified mask-based discrete diffusion model to directly capture the joint distribution over discrete multimodal tokens. This approach supports not only bimodal tasks but also more complex scenarios involving multiple modalities. On a diverse set of benchmarks, our method outperforms or performs o

n par with existing multimodal systems that process two or more modalities, high lighting the significant promise of diffusion models in powering the next genera tion of multimodal foundation models. Our codes are released at [GitHub](https:/ /omni-diffusion.github.io).
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Parameters as Experts: Adapting Vision Models with Dynamic Parameter Routing
Meng Lou, Stanley Yu, Yizhou Yu
Adapting pre-trained vision models using parameter-efficient fine-tuning (PEFT) remains challenging, as it aims to achieve performance comparable to full fine-t uning using a minimal number of trainable parameters. When applied to complex de nse prediction tasks, existing methods exhibit limitations, including input-agno stic modeling and redundant cross-layer representations. To this end, we propose ParaX, a new adapter-style method featuring a simple mixture-of-experts (MoE) a rchitecture. Specifically, we introduce shared expert centers, where each expert is a trainable parameter matrix. During a feedforward pass, each ParaX module i n the network dynamically generates weight matrices tailored for the current mod ule via a simple dynamic parameter routing mechanism, which selectively aggregat es parameter matrices in the corresponding expert center. Dynamic weight matrice s in ParaX modules facilitate low-rank adaptation in an input-dependent manner, thus generating more customized and powerful feature representations.
Moreover, since ParaX modules across multiple network layers share the same expe rt center, they improve feature diversity by promoting implicit cross-layer feat ure interaction.
Extensive experimental results demonstrate the superiority of ParaX across diver se visual recognition tasks.
Code is publicly released at: https://github.com/LMMMEng/ParaX.
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Feedback Control for Multi-Objective Graph Self-Supervision
Karish Grover, Theodore Vasiloudis, Han Xie, Sixing Lu, Xiang song, Christos Fal outsos
Can multi-task self-supervised learning on graphs be coordinated without the usu al tug-of-war between objectives? Graph self-supervised learning (SSL) offers a growing toolbox of pretext objectives—mutual information, reconstruction, contra stive learning—yet combining them reliably remains a challenge due to objective interference and training instability. Most multi-pretext pipelines use per-upda te mixing, forcing every parameter update to be a compromise, leading to three f ailure modes: Disagreement (conflict-induced negative transfer), Drift (nonstati onary objective utility), and Drought (hidden starvation of underserved objectiv es). We argue that coordination is fundamentally a temporal allocation problem: deciding when each objective receives optimization budget, not merely how to wei gh them. We introduce ControlG, a control-theoretic framework that recasts multi -objective graph SSL as feedback-controlled temporal allocation by estimating pe r-objective difficulty and pairwise antagonism, planning target budgets via a Pa reto-aware log-hypervolume planner, and scheduling with a Proportional–Integral– Derivative (PID) controller. Across 9 datasets, ControlG consistently outperform s state-of-the-art baselines, while producing an auditable schedule that reveals which objectives drove learning.
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Correcting Visual Blur Induced by Attention Distraction to Reduce Hallucinations : Algorithm and Theory
Quanjiang Li, Zhiming Liu, Wei Luo, Tingjin Luo, Chenping Hou
Multimodal large language models (MLLMs) frequently suffer from object hallucina tions, yet the visual perceptual mechanism underlying this failure remains poorl y understood. In this work, we reveal that hallucinations are strongly associate d with a human-like attention distraction phenomenon, where humans under divided focus experience degraded visual clarity and produce inaccurate descriptions, w hile in models the same mechanism manifests as spatial inconsistency in multi-he ad attention and temporal fading of attention to image tokens during decoding. W e further provide theoretical insights that attention dispersion increases mode l complexity and degrades classification generalization. Motivated by these find

ings, we propose an Attention-Focused Approach for Improved Image Perception (AFIP), which corrects attention distraction via cross-head attention enrichment and reinforces visual grounding through dynamic historical attention enhancement. Extensive experiments on multiple benchmarks and models validate the effectiveness of AFIP without additional training.

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Projection-Free Algorithms for Minimax Problems

Khanh-Hung Giang-Tran, Soroosh Shafiee, Nam Ho-Nguyen

This paper addresses constrained smooth saddle-point problems in settings where projection onto the feasible sets is computationally expensive. We bridge the gap between projection-based and projection-free optimization by introducing a unified dual dynamic smoothing framework that enables the design of efficient single-loop algorithms. Within this framework, we establish convergence results for nonconvex-concave and nonconvex-strongly concave settings. Furthermore, we show that this framework is naturally applicable to convex-concave problems, providing a unified analysis across varying payoff structures. We propose and analyze three algorithmic variants based on the application of a linear minimization oracle over the minimization variable, the maximization variable, or both. Notably, our analysis yields anytime convergence guarantees without requiring a pre-specified iteration horizon. These results significantly narrow the performance gap between projection-free and projection-based methods for minimax optimization.

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The Labyrinth and the Thread: Rethinking Regularizations in Sequential Knowledge Editing for Large Language Models

Zheng Wang, Kaixuan Zhang, Wanfang Chen, Jingwen Zhang, Xiaonan Lu

Sequential editing of structured knowledge in large language models allows targeted factual updates without retraining, yet existing methods often rely on complex regularization or constraint mechanisms whose necessity remains unclear. In this work, we systematically investigate the mechanisms underlying effective and stable sequential editing. Specifically, we first analyze the empirical success of AlphaEdit and establish, via a rigorous optimization analysis, the formal equivalence between one-time and sequential editing. Building on this insight, we generalize the equivalence to a broader class of editing objectives, demonstrating that stability emerges naturally from properly accounting for accumulated editing constraints, rather than from specialized regularization or null-space operations. We empirically confirm that many commonly used regularization strategies are unnecessary for reliable sequential updates. Furthermore, we extend our framework to handle conflicting edits, ensuring robust and consistent behavior under contradictory updates. Ultimately, our work provides Ariadne's thread through the labyrinth of sequential editing, charting a path toward simpler, more interpretable, and dependable knowledge updates. Our code is available at https://github.com/Wangzzzzzzz/OTE-SE-Alignment.

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Coarse-Grained Boltzmann Generators

Weilong Chen, Bojun Zhao, Jan Eckwert, Julija Zavadlav

Sampling equilibrium molecular configurations from the Boltzmann distribution is a longstanding challenge. Boltzmann Generators (BGs) address this by combining exact-likelihood generative models with importance sampling, but practical scalability is limited. Meanwhile, coarse-grained surrogates enable the modeling of larger systems by reducing effective dimensionality, yet often lack a reweighting procedure required to ensure asymptotically correct statistics. In this work, we propose Coarse-Grained Boltzmann Generators (CG-BGs), a framework for reduced-order generative modeling with importance sampling in coarse-grained coordinate space. CG-BGs generate samples using a flow-based model and reweight them using a learned potential of mean force (PMF). We show that the PMF can be learned from rapidly converged trajectories via enhanced sampling force matching. Experiments demonstrate that CG-BGs capture solvent-mediated interactions in highly reduced representations while substantially reducing computational cost relative to atomistic BGs, providing a practical route toward equilibrium sampling of larger molecular systems.

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Backward SDE–Based Diffusion for Physics-Constrained Generation
Zihao WANG
Pretrained score-based diffusion models provide strong unconditional priors, yet enforcing measurement or physics consistency in inverse problems is often handled by heuristic guidance, intermittent projections, or task-specific conditional training, with limited guarantees of feasibility at the end of inference. We propose terminal-conditioned inversion for score-based SDE priors. Given a frozen Score-SDE prior and a task-defined terminal feasibility specification, we construct an associated backward stochastic differential equation whose adapted solution defines a principled inverse map from the terminal requirement to a prior state at a chosen noise level. Under standard regularity conditions, we establish existence and uniqueness of the adapted solution and obtain terminal consistency by construction. We further develop a practical neural BSDE solver that composes arbitrary pretrained diffusion priors with domain constraints without modifying the score-defined coefficients, producing an anchored prior state that enables neighborhood sampling for uncertainty characterization. Experiments on toy datasets validate stable terminal-conditioned inversion and distributionally consistent neighborhood sampling. As a real-world case study, we apply the framework to sparse-view CT reconstruction and achieve improved reconstruction quality over representative training-free baselines while satisfying strict measurement feasibility under the prescribed terminal specification.
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Principled SVD-based Delta Compression via Quantization Error Minimization
Boya Xiong, Shuo Wang, Weifeng Ge, Guanhua Chen, Yun Chen
Supervised Fine-Tuning (SFT) empowers Large Language Models (LLMs) with exceptional performance on specialized tasks, but it yields dense, high-dimensional delta parameters that pose severe storage and distribution challenges. Singular Value Decomposition (SVD)-based compression offers a compact representation for such delta parameters, but existing methods adopt heuristic quantization without clarifying underlying mechanisms, leading to poor generalizability. In this work, we propose PrinMix, a rigorous SVD-based framework that models quantization as an optimization problem, grounding the design in mathematical mechanisms. We first theoretically derive quantization error and identify a key singular-value-dominated scaling mechanism, which mathematically proves the necessity of mix-precision quantization. We then model the quantization scheme as a 0/1 Integer Linear Programming (ILP) problem, which yields optimal bit-budget-constrained solutions without empirical assumptions. Furthermore, PrinMix integrates a Reconstruction Target Correction (RTC) method to compensate for errors from the $\mathbf{V}$-then-$\mathbf{U}$ sequential quantization process. Extensive experiments confirm PrinMix performs well: for 7B LLMs, PrinMix outperforms SOTA Delta-CoMe on challenging benchmarks by 22.3\% on AIME2024 and 6.1\% on GQA.
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Tucker Attention: A generalization of approximate attention mechanisms
Timon Klein, Jonas Kusch, Sebastian Sager, Stefan Schnake, Steffen Schotthöfer
The pursuit of reducing the memory footprint of the self-attention mechanism in multi-headed self attention (MHA) spawned a rich portfolio of methods, e.g., group-query attention (GQA) and multi-head latent attention (MLA). The methods leverage specialized low-rank factorizations across embedding dimensions or attention heads. From the point of view of classical low-rank approximation, these methods are unconventional and raise questions of which objects they really approximate and how to interpret the low-rank behavior of the resulting representations. To answer these questions, this work proposes a generalized view on the weight objects in the self-attention layer and a factorization strategy, which allows us to construct a parameter efficient scheme, called Tucker Attention.
Tucker Attention requires an order of magnitude fewer parameters for comparable validation metrics, compared to GQA and MLA, as evaluated in LLM and ViT test cases. Additionally, Tucker Attention~encompasses GQA, MLA, MHA as special cases and is fully compatible with flash-attention and rotary position embeddings (RoPE).

This generalization strategy yields insights of the actual ranks achieved by MHA , GQA, and MLA, and further enables simplifications for MLA.
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Uni-DocRobust: Universal Plug-and-Play Robustness Enhancement for Multi-modal LL Ms via Feature Restoration
Yuxuan Zhou, Baole Wei, Xingjian Hu, Haowei Chen, Yu Li, Xingyue Lin, Liangcai G ao, Zhi Tang
Real-world degradations, such as noise, blur, and low resolution, significantly impair the performance of Multi-modal Large Language Models (MLLMs) in document understanding tasks. Despite recent advancements, progress in this field remains stifled by two critical bottlenecks: the scarcity of large-scale, aligned train ing data necessary for learning robustness, and the lack of transferable restora tion solutions across diverse MLLM architectures. To bridge the data gap, we fir st present DocRobust-VQA, a large-scale dataset explicitly constructed to suppor t robustness training. Comprising 189K aligned clean/corrupted document image pa irs and 417K QA pairs, it provides the first substantial corpus for fine-tuning MLLMs to handle varying degradation conditions. Leveraging this data, we propose Uni-DocRobust, a universal plug-and-play framework that decouples restoration c apabilities from specific visual encoders. Our method employs a frozen Universal Restoration Core pre-trained in a canonical feature space via multi-teacher dis tillation, which can be seamlessly integrated into target MLLMs (e.g., Qwen-VL, InternVL) through lightweight Feature Adapters. Extensive experiments demonstrat e that Uni-DocRobust significantly enhances robust performance on MLLMs and enab les a cost-effective ``pre-train once, deploy everywhere'' paradigm for robust M LLM deployment.
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Who Said Neural Networks Aren't Linear?
Nimrod Berman, Assaf Hallak, Assaf Shocher
Neural networks are famously nonlinear. However, linearity is defined relative t o a pair of vector spaces, $f:\mathcal{X}\to\mathcal{Y}$. Leveraging the algebra ic concept of transport of structure, we propose a method to explicitly identify non-standard vector spaces where a neural network acts as a linear operator. Wh en sandwiching a linear operator A between two invertible neural networks, $f(x)=g_y^{-1}(A g_x(x))$, the corresponding vector spaces $\mathcal{X}$ and $\math cal{Y}$ are induced by newly defined addition and scaling actions derived from $ g_x$ and g_y. We term this kind of architecture a Linearizer. This framework m akes the entire arsenal of linear algebra, including SVD, pseudo-inverse, orthog onal projection and more, applicable to nonlinear mappings. Furthermore, we show that the composition of two Linearizers that share a neural network is also a L inearizer. We leverage this property and demonstrate that training diffusion mod els using our architecture makes the hundreds of sampling steps collapse into a single step. We further utilize our framework to enforce idempotency (i.e.\ $f(f (x))=f(x)$) on networks leading to a globally projective generative model and to demonstrate modular style transfer.
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CONTEXTOR: Contextualized High-order Contrastive Learning
Ze Cai, Hanzhe Liang, Sihang Zeng, Binbin Zhou, Jun Wen
High-order relations involving multiple interacting entities are commonly encoun tered, particularly in biomedical domains. Existing relational learning methods typically learn static entity representations and assume symmetric relation infe rence, which can be inadequate for capturing context-dependent entity functions and the inherent asymmetry of high-order relations. In this paper, we propose Co ntextualized High-order Contrastive Learning (CONTEXTOR), a general and plug-and -play framework that formulates high-order relation inference as a dynamic query –response process. Specifically, CONTEXTOR decomposes each high-order relation i nto multiple incomplete query tuples and their corresponding response entities. Given a query tuple, we contextualize candidate response entity representations via an asymmetric conditional modulation, and align queries with their correspon ding contextualized responses through multi-fold contrastive learning. Extensive experiments on benchmark datasets spanning multiple biomedical tasks demonstrat

e that CONTEXTOR consistently outperforms existing methods across diverse evalua
tion settings. Code is available at https://github.com/ZJUDataIntelligence/CONTE
XTOR.
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Reinforcement Learning for Tool-Calling Agents in Fast Healthcare Interoperabili
ty Resources (FHIR)
Marius Knorr, Robert Müller, Jan Peter Bremer, Nils Schweingruber
Fast Healthcare Interoperability Resources (FHIR) is the dominant standard for i
nteroperable exchange of healthcare data. In FHIR, electronic health records for
m a directed graph of resources. Answering clinically meaningful questions over
FHIR requires agents to perform multi-step reasoning, filtering, and aggregation
 across multiple resource types. Prior work shows that even tool-augmented LLM a
gents (retrieval, code execution, multi-turn planning) often select the wrong re
sources or violate traversal constraints. We study this problem in the context o
f FHIR-AgentBench, a benchmark for realistic question answering over real-world
hospital data, and frame reasoning on FHIR as a sequential decision-making probl
em over a queryable structured graph. We implement a multi-turn CodeAct agent an
d post-train it with reinforcement learning using a custom harness and tools. A
LLM Judge provides execution-grounded rewards. Compared to prompt-based, closed-
model baselines, RL post-training improves performance while enforcing data-inte
grity constraints. Empirically, our approach improves answer correctness from 50
% (o4-mini) to 77% on FHIR-AgentBench using a smaller and cheaper Qwen3-8B model
. We present an end-to-end post-training pipeline (environment building, harness
 construction, model training and custom evaluation) that reliably improves mult
i-turn reasoning over structured clinical graphs.
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Influence-Disentangled Federated Training: Learning Models That Are Easy to Unle
arn
Canran Xiao, Qianyu Chen, Liwei Hou
Federated learning increasingly faces deletion requests that require client-leve
l unlearning without sacrificing model quality, yet a client’s influence is ofte
n deeply entangled after many rounds of aggregation. We aim to make unlearning f
ast, stable, and predictable by reducing the gap to leave-one-out retraining und
er realistic heterogeneity. We propose Influence-Disentangled Federated Training
 (IDFT), which instruments standard FedAvg with training-time influence logging:
 each round’s updates are decomposed into shared covarying directions and a clie
nt-separable residual trace, and an entanglement-aware shrinkage suppresses non-
removable components. Deletion then becomes a single subtraction followed by a s
hort anchored repair, yielding a stability-style characterization of retrain fid
elity driven by the unremoved residual. Across representative benchmarks, IDFT c
onsistently attains the lowest retrain gap (Avg. Gap) on multiple dataset--archi
tecture pairs and improves the fidelity--cost frontier, matching retrain-level f
orgetting with substantially lower communication/compute than history-heavy base
lines. These results suggest a practical pathway to unlearning-friendly federate
d systems by designing for removability during training rather than relying sole
ly on post-hoc corrections.
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FedPAT: Federated Test-Time Adaptation via Prototype Affinity Topology
Shunxin Guo, Jiaqi Lv, Zhiqiang Kou, Shuxia Lin, Xin Geng
Federated Learning (FL) enables privacy-preserving collaboration among distribut
ed clients in open-world environments, but its performance often degrades under
data heterogeneity and unpredictable distribution shifts. Test-Time Adaptation (
TTA) has recently been introduced into FL to leverage unlabeled data from unseen
 clients for online adaptation. However, most existing federated TTA methods emp
loy local feature statistics, which can be brittle under diverse and severe dist
ribution shifts. In this work, we observe that despite significant variations in
 feature distributions, the relational structure among class prototypes—termed *
prototype affinity topology*—remains remarkably stable across heterogeneous clie
nts. Building on this insight, we propose FedPAT, a **Fed**erated TTA framework
that leverages **P**rototype **A**ffinity **T**opology as a cross-client structu

ral prior. FedPAT learns a global PAT by aggregating class prototypes from source clients, capturing consensus inter-class relationships that are robust to local distribution variations. For unseen target clients, we design a topology-aware mechanism that enhances predictions via diffusion of the global PAT, fuses them with parametric outputs, and performs lightweight optimization for robust test-time adaptation. Extensive experiments demonstrate that FedPAT consistently outperforms advanced federated TTA and classical TTA methods across various distribution shifts.

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Rethink the Role of Neural Decoders in Quantum Error Correction

Ge Yan, SHANCHUAN LI, Yuxuan Du

Quantum error correction (QEC) is essential for enabling quantum advantages, with decoding as a central algorithmic primitive. Owing to its importance and intrinsic difficulty, substantial effort has been made to QEC decoder design, among which neural decoders have recently emerged as a promising data-driven paradigm. Despite this progress, practical deployment remains hindered by a fundamental accuracy–latency tradeoff, often on the microsecond timescale. To address this challenge, here we revisit neural decoders for surface-code decoding under explicit accuracy–latency constraints, considering code distances up to $d=9$ (161 physical qubits). We unify and redesign representative neural decoders into five architectural paradigms and develop an end-to-end compression pipeline to evaluate their deployability and performance on FPGA hardware. Through systematic experiments, we reveal several previously underexplored insights: (i) near-term decoding performance is driven more by data scale than architectural complexity; (ii) appropriate inductive bias is essential for achieving high decoding accuracy; and (iii) INT4 quantization is a prerequisite for meeting microsecond-scale latency requirements on FPGAs. Together, these findings provide concrete guidance toward scalable and real-time neural QEC decoding.

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Realizable Bayes-Consistency for General Metric Losses

Dan Tsir Cohen, Steve Hanneke, Aryeh Kontorovich

We study strong universal Bayes-consistency in the realizable setting for learning with general metric losses, extending classical characterizations beyond 0-1 classification (Bousquet et al., 2021; Hanneke et al., 2021) and real-valued regression (Attias et al., 2024). Given an instance space (X,ρ), a label space (Y,ℓ) with possibly unbounded loss, and a hypothesis class $H \subseteq Y^{X}$, we resolve the realizable case of an open problem presented in Tsir Cohen and Kontorovich (2022). Specifically, we find the necessary and sufficient conditions on the hypothesis class H under which there exists a distribution-free learning rule whose risk converges almost surely to the best-in-class risk (which is zero) for every realizable data-generating distribution. Our main contribution is this sharp characterization in terms of a combinatorial obstruction: Similarly to Attias et al. (2023), we introduce the notion of an infinite non-decreasing (γ_k)-Littlestone tree, where $\gamma_k \to \infty$. This extends the Littlestone tree structure used in Bousquet et al. (2021) to the metric loss setting.

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Unbiased Dynamic Pruning for Efficient Group-Based Policy Optimization

Haodong Zhu, Yangyang Ren, Yanjing Li, Mingbao Lin, Linlin Yang, Xuhui Liu, Xiantong Zhen, haiguang liu, Baochang Zhang

Group Relative Policy Optimization (GRPO) effectively scales LLM reasoning but incurs prohibitive computational costs due to its extensive group-based sampling requirement.

While recent selective data utilization methods can mitigate this overhead, they could induce estimation bias by altering the underlying sampling distribution, compromising theoretical rigor and convergence behavior.

To address this limitation, we propose Dynamic Pruning Policy Optimization (DPPO), a framework that enables dynamic pruning while preserving unbiased gradient estimation through importance sampling-based correction.

By incorporating mathematically derived rescaling factors, DPPO significantly ac

celerates GRPO training without altering the optimization objective of the full-batch baseline.
Furthermore, to mitigate the data sparsity induced by pruning, we introduce Dense Prompt Packing, a window-based greedy strategy that maximizes valid token density and hardware utilization.
Extensive experiments demonstrate that DPPO consistently accelerates training across diverse models and benchmarks.
For instance, on Qwen3-4B trained on MATH, DPPO achieves 2.37$\times$ training speedup and outperforms GRPO by 3.36\% in average accuracy across six mathematical reasoning benchmarks.
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MCCE: A Framework for Multi-LLM Collaborative Search in Discrete Spaces with Similarity-Filtered Preference Learning
Nian Ran, Zhongzheng Li, Yue Wang, Qingsong Ran, Xiaoyuan Zhang, Shikun Feng, Richard Allmendinger, Xiaoguang Zhao
Multi-objective discrete optimization problems, such as molecular design, pose significant challenges due to their vast and unstructured combinatorial spaces. Traditional evolutionary algorithms often get trapped in local optima, while expert knowledge can provide crucial guidance for accelerating convergence. Large language models (LLMs) offer powerful priors and reasoning ability, making them natural optimizers when expert knowledge matters. However, closed-source LLMs, though strong in exploration, cannot update their parameters and thus cannot internalize experience. Conversely, smaller open models can be continually fine-tuned but lack broad knowledge and reasoning strength. We introduce Multi-LLM Collaborative Co-evolution (MCCE), a hybrid framework that unites a frozen closed-source LLM with a lightweight trainable model. The system maintains a trajectory memory of past search processes; the small model is progressively refined via reinforcement learning, with the two models jointly supporting and complementing each other in global exploration. Unlike model distillation, this process enhances the capabilities of both models through mutual inspiration. Experiments on multi-objective drug design benchmarks show that MCCE achieves state-of-the-art Pareto front quality and consistently outperforms baselines. These results highlight a new paradigm for enabling continual evolution in hybrid LLM systems, combining knowledge-driven exploration with experience-driven learning.
The code of MCCE is available on https://github.com/lzz-z/MCCE
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Efficient, Validation-Free Intrinsic Quality Estimation for Large-Scale Face Recognition Datasets
Zhichao Chen, Yongle Zhao, Kaicheng Yang, Meng Yang, Yin Xie, Ziyong Feng
We propose Intrinsic Quality (IQ), a validation-free metric designed to estimate the inherent potential of face recognition (FR) datasets to produce high-performance models without the need for full-scale training. IQ integrates two components: (i) a Neighbor-Consistency Score that quantifies local identity label agreement via nearest neighbors, and (ii) Global Representation Subspace Complexity (Effective Rank, ER), which captures the underlying embedding geometry and dataset diversity. IQ allows for rapid evaluation using lightweight proxy models or data subsets, facilitating dataset diagnosis and curation prior to resource-intensive full-scale training. We describe an experimental protocol tailored to clean, noisy, and mixed■quality FR datasets, and outline evaluation methodologies to validate IQ’s predictive power for downstream performance.
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SPLIT-VLM: Salience-Guided Partitioning towards Local Coverage for Importance-Aware Token Dropping in Vision-Language Models
Seungil Lee, Gilha lee, Hyun Kim
Large-scale vision–language models (VLMs) excel at multimodal reasoning, yet efficiency collapses when vision tokens—often orders of magnitude more than text—dominate compute and memory. Prior token-reduction strategies typically trade off salience (which is prone to position bias and incurs extra computation) against diversity (which can under-cover salient regions and is sensitive to hyperparameters). We present SPLIT, a theoretically grounded framework that jointly preserv

es salience and diversity while aggressively eliminating redundancy. SPLIT (i) e stimates token importance via temporal shifts of hidden states across layers—esc hewing attention scores and their biases; (ii) assigns adaptive region-level bud gets to guarantee localized coverage; and (iii) selects tokens using a diversity score that prioritizes distinctive, non-redundant representations. Our analysis shows that adaptive budgeting yields tighter coverage guarantees than uniform a llocation, and our selection rule maintains diversity without costly tuning. Emp irically, SPLIT consistently outperforms state-of-the-art on image and video und erstanding benchmarks. On image understanding with LLaVA-1.5-7B, SPLIT preserves over 99\% accuracy with 192 vision tokens and about 92.8\% with only 64 tokens, demonstrating robust performance under severe token budgets. These results indi cate that SPLIT delivers scalable, attention-score-free token reduction that mak es multimodal reasoning substantially more efficient without sacrificing accurac y.

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Unleashing the Representational Power of Fourier Shapes for Attacking Infrared O bject Detection

Yixing Yong, Jian Wang, Ming Lei, Lijun He, Fan Li

Infrared object detection is crucial for perception in autonomous driving and su rveillance but remains vulnerable to physical adversarial attacks. Unlike in the RGB domain, where attacks rely on color texture, infrared attacks must manipula te thermal signatures, making the geometry shape of heat-blocking materials the primary adversarial information carrier. Current shape-based methods suffer from a fundamental trade-off between representational capability and optimization po wer, limiting their attack effectiveness.

In this work, we overcome this dilemma by introducing learnable Fourier shapes t o the infrared domain. We utilize an end-to-end differentiable framework where a compact set of Fourier coefficients, defining the shape boundary, is analytical ly mapped to a pixel-space mask via the winding number theorem. This enables eff icient gradient-based optimization to generate potent shapes that cause human ta rgets to evade detection. Extensive digital and physical experiments provide a c omprehensive evaluation and validate our superior performance. Our resulting phy sical patch achieves striking robustness, successfully evading detectors across diverse distances, angles, poses, and individuals, and achieves over 88% attack success rate at distances greater than 25m (conf.=0.5). Code is available at htt ps://github.com/Yongyx99/Fourier-shape-attack.

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Geometric Coherence Learning for Structuring Value Functions in Plain MDPs

Zuyuan Zhang, Zeyu Fang, Tian Lan

Geometric properties can be leveraged to stabilize and speed reinforcement learn ing. Existing examples include encoding symmetry structure, geometry-aware data augmentation, and enforcing structural restrictions. In this paper, we take a no vel view of RL through the lens of order theory and recast value function estima tes into learning a desired poset (partially ordered set).

We propose \emph{GCR-RL} (Geometric Coherence Regularized Reinforcement Learning) that computes a sequence of super-poset refinements -- by refining posets in p revious steps and learning additional order relationships from temporal differen ce signals -- thus ensuring geometric coherence across the sequence of posets un derpinning the learned value functions. Two novel algorithms by Q-learning and by actor--critic are developed to efficiently realize these super-poset refineme nts. Their theoretical properties and convergence rates are analyzed. We empiric ally evaluate GCR-RL in a range of tasks and demonstrate significant improvement s in sample efficiency and stable performance over strong baselines.

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Rethinking Graph Transformers as Graph Signal Denoisers: The Role of Block-Diago nal Priors

Jiaming Zhuo, Ziyi Ma, Kun Fu, Di Jin, Chuan Wang, Zhen Wang, Xiaochun Cao, Huaz hu Fu, Liang Yang

By synergizing graph topology with the global expressive power of attention, Gra ph Transformers (GTs) have emerged as a powerful architecture for node classific

ation. Existing GTs mainly focus on diverse topology injection paradigms, which fundamentally construct different propagation operators. However, a unified theoretical understanding of what constitutes a desirable propagation operator remains largely unexplored. To bridge this gap, this paper rethinks GTs from a graph signal denoising perspective, revealing that a block-diagonal structure is a desirable structural prior for graph denoising. To efficiently instantiate this prior, this paper introduces BDFormer, which imposes spectral block regularization on the affinity of a compact set of latent anchors. Furthermore, by introducing hard assignments between nodes and anchors, BDFormer establishes sparse, structured propagation that suppresses noisy cross-class propagation in linear time. Meanwhile, the learned global affinity guides the pruning of inter-class edges in the graph topology, enabling both global and local propagation to jointly adhere to the target block-diagonal structure. Extensive experiments on benchmark datasets demonstrate the superiority and scalability of BDFormer.
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ThreadWeaver: Adaptive Threading for Efficient Parallel Reasoning in Language Models
Long Lian, Sida Wang, Felix Juefei-Xu, Tsu-Jui Fu, Xiuyu Li, Adam Yala, Trevor Darrell, Alane Suhr, Yuandong Tian, Xi Victoria Lin
Scaling inference-time computation has enabled Large Language Models (LLMs) to achieve strong reasoning performance, but their inherently sequential decoding incurs substantial latency, motivating parallelization of the generation process. However, existing parallel reasoning approaches suffer from performance degradation compared to their sequential counterparts, and often rely on specialized inference engines. We introduce ThreadWeaver, a framework for adaptive parallel reasoning that matches the accuracy of comparably sized sequential reasoning models while significantly reducing inference latency via three key innovations: 1) a two-stage parallel trajectory generator that produces high-quality parallel chain-of-thought data for supervised fine-tuning; 2) a trie-based rollout design that enables parallel reasoning on any off-the-shelf autoregressive inference engine; and 3) a parallelization-aware reinforcement learning framework that trains the model to balance reasoning accuracy with effective parallelization. Across six challenging math reasoning benchmarks, ThreadWeaver trained on top of Qwen3-8B achieves performance on par with cutting-edge sequential reasoning models (79.9% on AIME24 and 71.9% on average) while delivering up to 1.53x speedup in token latency, establishing a new Pareto frontier between accuracy and efficiency.
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Clustered Influence Functions
Miklós Máté Badó, Kristian Fenech
Influence functions are a standard tool for data debugging and unlearning, but they become impractical for **high-query** subset workloads such as large-K cross-validation, repeated resampling, or interactive what-if analysis as each subset query typically requires an expensive inverse-curvature solve. We introduce **Clustered Influence Functions (CiF)**, which turns subset influence into an **amortized subset oracle**. We build a compact cache once by clustering training gradients, solve a damped Generalised Gauss-Newton system only for cluster means, and answer new subset queries by a linear recombination using cluster membership counts. This yields per-query cost of $O(Cp)$ linear in the cache size C, and the number of model parameters p. We further provide a diagnostic error bound that decomposes approximation error into a **clustering scatter** term and a **solver residual** term, making the accuracy-compute tradeoff explicit through the cache budget and solver tolerance. Evaluations across MNIST, CIFAR-10 show that CiF matches per-query influence rankings while significantly reducing the total runtime in high-Q regimes, enabling influence-based workflows that are otherwise computationally prohibitive.
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Maximum Likelihood Reinforcement Learning
Fahim Tajwar, Guanning Zeng, Yueer Zhou, Yuda Song, Daman Arora, Yiding Jiang, Jeff Schneider, Ruslan Salakhutdinov, Haiwen Feng, Andrea Zanette
Reinforcement learning (RL) is the method of choice for training models in setup

s where the objective function can only be evaluated by sampling from the model. Our key observation is that when the feedback is terminal and binary, models implicitly induce a likelihood over correct rollouts. Maximum likelihood would be the natural framework in such settings, but RL is used instead as a workaround to the non-differentiability. We prove that the standard, expected-reward RL formulation is only a first-order approximation of the likelihood. To remedy this mismatch, we introduce **Maximum Likelihood Reinforcement Learning (MaxRL)**, a compute-indexed family of sample-based objectives that interpolate between expected-reward RL and maximum likelihood as sampling compute is scaled. The resulting objective is a one-line change to standard RL implementations. MaxRL Pareto-dominates existing methods in all tested models and tasks, achieves up to $\mathbf{20\times}$ gains in test-time scaling efficiency over GRPO, and scales more favorably with additional training data and compute.
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Unified Multimodal Visual Tracking with Dual Mixture-of-Experts
Lingyi Hong, Jinglun Li, Xinyu Zhou, Kaixun Jiang, Pinxue Guo, Zhaoyu Chen, Runze Li, Xingdong Sheng, Wenqiang Zhang
Multimodal visual object tracking can be divided into to several kinds of tasks (e.g. RGB and RGB+X tracking), based on the input modality. Existing methods often train separate models for each modality or rely on pretrained models to adapt to new modalities, which limits efficiency, scalability, and usability. Thus, we introduce OneTrackerV2, a unified multi-modal tracking framework that enables end-to-end training for any modality. We propose Meta Merger to embed multi-modal information into a unified space, allowing flexible modality fusion and robustness. We further introduce Dual Mixture-of-Experts (DMoE): T-MoE models spatiotemporal relations for tracking, while M-MoE embeds multi-modal knowledge, disentangling cross-modal dependencies and reducing feature conflicts. With a shared architecture, unified parameters, and a single end-to-end training, OneTrackerV2 achieves state-of-the-art performance across five RGB and RGB+X tracking tasks and 12 benchmarks, while maintaining high inference efficiency. Notably, even after model compression, OneTrackerV2 retains strong performance. Moreover, OneTrackerV2 demonstrates remarkable robustness under modality-missing scenarios.
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KAST-BAR: Knowledge-Anchored Semantically-Dynamic Topology Brain Autoregressive Modeling for Universal Neural Interpretation
Haoning Wang, Wenchao Yang, Shuai Shen, Yang Li
While EEG foundation models have shown significant potential in universal neural decoding across tasks, their advancement remains constrained by the inadequacy modeling of *complex spatiotemporal topology*, as well as the inherent *modality gap* between low-level physiological signals and high-level textual semantics.

To address these challenges, we propose a **K**nowledge-**A**nchored **S**emantically-Dynamic **T**opology **B**rain **A**uto**r**egressive Model (KAST-BAR), which dynamically aligns physiological representations derived from multi-level brain topology with an expert-level semantic space.

Specifically, we design a Dual-Stream Hierarchical Attention (DSHA) encoder that accurately captures the brain's intrinsic non-Euclidean topology by modeling local temporal dynamics with global spatial contexts.

On this basis, a Knowledge-Anchored Semantic Profiler (KASP) is proposed to synthesize physically-grounded and instance-level textual profiles, which subsequently drive a Semantic Text-Aware Refiner (STAR) to dynamically reconstruct EEG representations using Latent Expert Queries.

By conducting large-scale pre-training on 21 diverse datasets to build a foundation model, KAST-BAR effectively integrates expert-level medical knowledge into EEG signal representations, consistently achieving state-of-the-art performance across six downstream tasks. Our code is available at https://github.com/KAST-BAR/KAST-BAR
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Crowd4D: Scene-Aware Monocular 4D Crowd Reconstruction
Hongbo Kang, Tianyi Zhou, Qingyang Yang, Hongwei Wen, Jing Huang, Yu-Kun Lai, Kun Li

Recovering scene-consistent 4D crowd motion from monocular video in large-scale scenes remains challenging due to severe depth ambiguity and complex scene geometry. Existing monocular crowd reconstruction methods typically rely on single-plane assumptions, leading to unreliable metric scale and spatial drift under complex terrain. We propose Crowd4D, the first scene-aware 4D crowd reconstruction framework that jointly optimizes the crowd and scene from a monocular RGB video in large-scale scenes. Crowd4D explicitly incorporates scene geometry and ensures consistency across image and scene spaces via a multi-stage optimization strategy. A key bottleneck of this task lies in accurate human–scene alignment, particularly in scale and position. However, human and scene reconstructions are typically decoupled. To address this, we introduce the Human–Scene Interaction Proxy (HSIP) as an intermediate representation, derived from Scene Interaction Point Clouds and a Scene Interaction Surface (SIPC\&SIS), which encode explicit scene-aware geometric priors and redefine the optimization space for large-scale monocular 4D crowd reconstruction. To further improve temporal stability under occlusions, we introduce Crowd Structural Coherence Regularization (CSCR), which leverages HSIP-based spatial priors to impose soft temporal consistency on pairwise relative displacements and directions within local crowd neighborhoods. Extensive experiments demonstrate that Crowd4D consistently outperforms existing state-of-the-art methods and enables robust monocular 4D crowd reconstruction in complex, large-scale real-world scenes. Project page is available at https://cic.tju.edu.cn/faculty/likun/projects/Crowd4D.

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SkelHCC: A Hyperbolic CLIP-Driven Cache Adaptation Framework for Skeleton-based One-Shot Action Recognition

Yanan Liu, Anqi Zhu, Jingmin Zhu, Jun Liu, Hossein Rahmani, Mohammed Bennamoun, Farid Boussaid, Dan Xu, Qiuhong Ke

Skeleton-based action recognition aims to understand human behaviors from body joint sequences and is especially challenging in the one-shot setting, where only a single labeled exemplar is available for each novel action. A key challenge is learning representations that capture the hierarchical and compositional structure of human motion while aligning effectively with high-level action semantics under extreme data scarcity. Existing approaches, largely based on Euclidean embeddings and low-level motion cues, struggle to model the tree-like organization of skeleton data, limiting cross-modal alignment and generalization to unseen action categories.

We propose SkelHCC, a unified skeleton hyperbolic CLIP-driven cache adaptation framework for one-shot skeleton-based action recognition. SkelHCC introduces an Explicitly Hierarchical Hyperbolic CLIP (EH-HCLIP) module that embeds skeleton sequences and action language into a shared hyperbolic space. By leveraging the negative curvature and exponential volume growth of hyperbolic geometry, EH-HCLIP naturally encodes the joint–part–body hierarchy of human anatomy and yields structurally consistent cross-modal representations. To support efficient one-shot adaptation, SkelHCC further integrates a training-free LLM-guided Multi-granularity Voting Cache (LMV-Cache) for context-aware inference. Experiments on NTU RGB+D 60, NTU RGB+D 120, and PKU-MMD II demonstrate that SkelHCC consistently outperforms state-of-the-art methods. Code will be available at https://github.com/lya19971103/SkelHCC

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From Coarse to Fine: Deep Prototype Refinement Network for Few-Shot Point Cloud Semantic Segmentation

Changshuo Wang, Weijun Li, Shuting He, Xiang Fang, Xingyu Gao, Zhonghang Liu, Prayag Tiwari, Dimitrios Kanoulas

Few-shot point cloud semantic segmentation (FS-PCSS) aims to achieve precise segmentation of novel categories using only limited labeled samples. Existing prototype-based methods rely on shallow feature fusion, failing to adequately model the feature distribution shift between support and query sets, resulting in insufficient prototype adaptation. To address this, we propose the Deep Prototype Refinement Network (DPR-Net), which systematically achieves progressive adaptation by constructing a coarse-to-fine prototype evolution trajectory. Our core Dynami

c Prototype Refinement (DPR) module decomposes features into common and distinct ive subspaces based on channel activation, enabling targeted adjustment of domai n-sensitive features while preserving class-shared semantics. By cascading multi ple refinement modules, we construct a prototype trajectory transitioning from s upport-biased to query-adapted representations, mitigating both under- and over- adaptation. Furthermore, our Mixture of Prototype Experts (MoPE) mechanism treat s prototypes at different stages as experts and ensembles their predictions thro ugh confidence-driven weighting. Extensive experiments demonstrate state-of-the- art performance with high efficiency. Our code will be available at https://gith ub.com/changshuowang/DPR-Net.
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GEM-FI: Gated Evidential Mixtures with Fisher Modulation
Marco Mustafa Mohammed, Fatemeh Daneshfar, Pietro Lio
Evidential Deep Learning (EDL) enables single-pass uncertainty estimation by pre dicting Dirichlet evidence, but it can remain overconfident and poorly calibrate d, and it often fails to represent multi-modal epistemic uncertainty. We introdu ce Gated Evidential Mixtures (GEM), a family of models that learns an in-model e nergy signal and uses it to gate evidential outputs end-to-end in a distance-inf ormed manner. GEM-CORE learns a feature-level energy and maps it to a bounded ga te that smoothly suppresses evidence when support is low. To capture epistemic m ulti-modality without multi-pass ensembling, GEM-MIX adds a lightweight mixture of evidential heads with learned routing weights while preserving single-pass in ference. Finally, GEM-FI stabilizes mixture allocations via a Fisher-informed re gularizer, reducing head collapse and producing smoother boundary uncertainty. A cross image classification and OOD detection benchmarks, GEM improves calibratio n and ID/OOD separation with single-pass inference. On CIFAR-10, GEM-FI vs. DAED L improves Acc. from 91.11 to 93.75 (+2.64 pp), reduces Brier ×100 from 14.27 to 6.81 (−7.46), and also improves misclassification-detection (AUPR) from 99.08 t o 99.94 (+0.86). For epistemic OOD detection, GEM-FI achieves AUPR/AUROC of 92.5 9/95.09 on CIFAR-10→SVHN and 90.20/89.06 on CIFAR-10→CIFAR-100 (vs. 85.54/89.30 and 88.19/86.10 for DAEDL).
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Benchmarking the Limits of In-Context Reinforcement Learning for Ad-Hoc Teamwork
Yuheng Jing, Kai Li, Jiajun Zhang, Zeyao Ma, Jiaxi Yang, Lei Zhang, Zhe Wu, Jinm in He, Junliang Xing, Jian Cheng
In-Context Reinforcement Learning (ICRL) has enabled foundation agents to adapt instantaneously to novel tasks, yet its efficacy in Ad-Hoc Teamwork (AHT)—where coordination with unknown partners is required—remains unexplored. To rigorously evaluate this, we introduce a large-scale benchmark **ICRL4AHT**, built upon a high-throughput JAX implementation of Overcooked-V2. Our benchmark includes a la rge, diverse teammate suite spanning both RL and heuristic policies, enabling co ntrolled train-test shifts, and provides a reproducible end-to-end pipeline for teammate generation, learning-history collection, dataset construction, and onli ne multi-episode evaluation. We evaluate representative history-conditioned ICRL algorithms, including Algorithm Distillation (AD) and Decision-Pretrained Trans former (DPT), across millions of transitions. Results reveal notable limitations : contrary to their success in single-agent domains, these baselines fail to exh ibit robust test-time adaptation in multi-agent settings. Specifically, these me thods frequently underperform random baselines across both unseen teammate and u nseen layout tracks, with no clear in-context improvement over long horizons. Th ese findings highlight the challenges of strategic inference under partial obser vability within the OvercookedV2 AHT protocol, establishing our benchmark as a c ritical testbed for next-generation coordination algorithms.
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Leveraging Gauge Freedom for Learning Non-Gradient Population Dynamics of Stocha stic Systems
Jules Berman, Tobias Blickhan, Benjamin Peherstorfer
In existing works on population dynamics inference, there is a focus on flows ar ising from vector fields that are the gradients of scalar potentials. Among all admissible flows that are compatible with the population dynamic, gradient flows

are optimal in a specific sense: they minimize kinetic energy. The selection of fields based on different criteria corresponds to a gauge freedom when determining population dynamics, which we leverage in this work. We propose Non-Gradient Inference Flows (NGIF), an algorithm to infer non-gradient population dynamics using a weak formulation of the continuity equation. This allows us to parameterize with general vector fields as well as choose other selection criteria beyond minimal energy. We demonstrate on a variety of low- and high-dimensional physics problems that this more general approach improves distributional accuracy over gradient-restricted baselines and better captures non-potential transport.
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Listening Through the Noise: Cauchy-Driven Diffusion Bridges for Robust Gastrointestinal Auscultation and Clinical Benchmarking

Dian Ding, Liren Dong, Yu Lu, Juntao Zhou, Ran Wang, Peng Li, Zhenyi Jia, Guangtao Xue

Gastrointestinal (GI) motility assessment via bowel sounds (BS) offers a non-invasive alternative to resource-intensive clinical standards. However, the diagnostic utility of BS is often compromised by its spectral overlap with non-stationary speech interference. While generative models have advanced signal restoration, traditional Gaussian-based diffusion frameworks struggle with the impulsive, heavy-tailed nature of real-world clinical noise. In this paper, we propose a novel Cauchy-driven Diffusion Bridge framework to isolate high-fidelity bowel sounds from complex interference. Our contributions are three-fold: (1) We introduce ClinBS, a large-scale clinical dataset (over 25 hours) containing rare pathological transients verified by experts; (2) We mathematically formulate a Cauchy bridge driver, deriving closed-form expressions for the score and density to better model heavy-tailed perturbations; and (3) We implement an efficient sampling procedure via Gaussian scale-mixture reparameterization. Extensive experiments show our framework achieves state-of-the-art performance, outperforming baselines by 13.4%–49.8% across core metrics and elevating abnormal BS recognition accuracy to 88.01%. These results demonstrate the system's potential for robust clinical GI monitoring and diagnosis.
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Weasel: Out-of-Domain Generalization for Web Agents via Importance-Diversity Data Selection

Fatemeh Pesaran Zadeh, Seyeon Choi, Xing Han Lù, Siva Reddy, Gunhee Kim

Large language models (LLMs) have enabled web agents that follow natural language goals through multi-step browser interactions. However, agents fine-tuned on specific trajectories and domain often struggle to generalize out of domain, and offline training can be compute-inefficient due to noisy, redundant trajectories and long accessibility-tree (AXTree) states. To address both issues, we propose Weasel, a trajectory selection method for offline training of web agents. Weasel selects a fixed-budget subset of trajectory steps by optimizing an objective that balances unary importance with pairwise diversity over states, websites, and interaction patterns, solving efficiently with a greedy algorithm. We further improve efficiency with target-centered AXTree pruning that keeps only content around the ground-truth action target, and we mitigate style mismatch for reasoning-native models by replacing expert traces with model-generated, style-consistent rationales. Across AgentTrek and NNetNav training datasets, evaluations in WebArena, WorkArena, and MiniWob, and experiments with Qwen2.5-7B, Gemma3-4B, and Qwen3-8B, Weasel improves out-of-domain performance while reducing training cost, producing roughly 9.7-12.5$\times$ training speedups over standard fine-tuning. We make the code available at https://github.com/fatemehpesaran310/weasel.
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Evaluating Sample Utility for Efficient Data Selection by Mimicking Model Weights

Tzu-Heng Huang, Manjot Bilkhu, John Cooper, Frederic Sala, Javier Movellan

Large-scale web-crawled datasets contain noise, bias, and irrelevant information, necessitating data selection techniques. Existing methods depend on hand-crafted heuristics, downstream datasets, or require expensive influence-based computations---all of which limit scalability and introduce unwanted data dependencies.

To address this, we introduce the Mimic Score, a simple and geometry-based data-quality metric that evaluates utility by measuring alignment between a sample's gradients and a target direction induced by a pre-trained reference model. This leverages readily available model weights, avoids needing validation datasets, and incurs minimal computational overheads. Building on this metric, we propose Grad-Mimic, a two-stage framework that re-weights samples online to accelerate training and aggregates sample utilities offline to construct effective data filters. Empirically, we show that using mimic scores to guide training improves data efficiency, accelerates convergence, yields consistent performance gains across six image datasets, and enhances CLIP models with 20.7\% fewer training steps. Additionally, mimic score-based filters augment existing filtering techniques, enabling improved CLIP models trained with 4.7 million fewer samples.

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Motion Dynamics Learning for Few-Shot Embodied Adaptation

Sibo He, Weiying Xie, Daixun Li, Junhao Zhong, Jiayun Tian, Yunke Wang, Leyuan Fang, Gang He, Yunsong Li

Vision-Language-Action (VLA) models have shown strong potential for robotic manipulation, yet adapting pretrained models to novel tasks typically relies on substantial task-specific demonstrations, limiting scalability.

Current VLA methods mostly focus on action imitation, which ignores the richer structure contained in trajectories. In contrast, motion dynamics governing how actions evolve over time are more informative and transferable, making them better suited for few-shot adaptation.

Motivated by this idea, we propose DynVLA, a few-shot adaptation system that reformulates VLA learning from action imitation to trajectory-level motion dynamics modeling.

Specifically, we propose Motion Dynamics Mechanism (MDM), which distills latent embeddings from trajectories via flow-matching inversion, yielding compact representations that capture dynamics.

We further design Dynamics-Constrained Modeling (DCM). DCM projects these inferred representations onto a Dynamics Bank, which stores prior motion knowledge pretrained from diverse demonstrations. By grounding action generation in these learned priors, the system enables interpolating between existing action paradigms to represent novel dynamics modes.

Experiments on 13 real-world tasks demonstrate that DynVLA outperforms existing SOTA systems by 15.8\% in average success rate with around 20 demonstrations, highlighting its adaptation capabilities in real-world scenes. Our code is available at https://github.com/trantor2nd/Motion-DynVLA.

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HyPOLE: Hyperproperty-Guided Multi-Agent Reinforcement Learning under Partial Observation

Arshia Rafieioskouei, Tzu-Han Hsu, Matthew Lucas, Borzoo Bonakdarpour

Formal specification is a powerful tool to guide the learning process and provides significant advantages over reward shaping: (1) mathematical rigor; (2) expressiveness to specify objectives and constraints, and (3) the ability to define tactics to achieve objectives. However, these benefits remain largely unexplored in the context of Multi-Agent Reinforcement Learning (MARL). This paper introduces HyPOLE, a novel framework for MARL under partial observability, where learning is guided by the expressive power of the so-called hyperproperties and, in particular, the temporal logic HyperLTL. We integrate Centralized Training for Decentralized Execution (CTDE) techniques with HyPOLE to synthesize decentralized policies, and our evaluation on SMAC, MessySMAC, and WildFire benchmark demonstrates clear advantages over baselines.

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Bipartite Graph Attention-based Clustering for Large-scale scRNA-seq Data

Zhuomin Liang, Liang Bai, Xian Yang

scRNA-seq clustering is a critical task for analyzing single-cell RNA sequencing (scRNA-seq) data, as it groups cells with similar gene expression profiles. Transformers, as powerful foundational models, have been applied to scRNA-seq clustering. Their self-attention mechanism automatically assigns higher attention wei

ghts to cells within the same cluster, enhancing the distinction between clusters. Existing methods for scRNA-seq clustering, such as graph transformer-based models, treat each cell as a token in a sequence. Their computational and space complexities are $\mathcal{O}(n^2)$ with respect to the number of cells, limiting their applicability to large-scale scRNA-seq datasets. To address this challenge, we propose a Bipartite Graph Transformer-based clustering model (BGFormer) for scRNA-seq data. We introduce a set of learnable anchor tokens as shared reference points to represent the entire dataset. A bipartite graph attention mechanism is introduced to learn the similarity between cells and anchor tokens, bringing cells of the same class closer together in the embedding space. BGFormer achieves linear computational complexity with respect to the number of cells, making it scalable to large datasets. Experimental results on multiple large-scale scRNA-seq datasets demonstrate the effectiveness and scalability of BGFormer.
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Lie-Algebraic Acceleration of Neural Koopman Dynamics
Jongwon Lee, Jiwoong Kim, Jungwoo Park, Sungwoo Park
We present a Lie-algebraic approach to model Koopman dynamics that integrates algebraic structure with computational scalability.
The proposed formulation constrains the neural generators to evolve within prescribed Lie subalgebras and constructs finite-time flows through a neural Magnus expansion construction, thereby maintaining consistency with the associated Lie-group composition over each time segment. To address the computational burden inherent in sequential propagation, we exploit the associativity of Lie-group compositions and construct segmentwise propagators via a prefix-scan algorithm, which reduces the depth of temporal composition from linear to logarithmic. Consequently, the framework enables accurate long-horizon prediction while improving computational efficiency, and provides a principled foundation for scalable Koopman operator learning for nonlinear systems.
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CoverPruneGS: Coverage-Preserving Structured Pruning for Hierarchical 3D Gaussian Splatting from Sparse-View Monocular Videos
Yang Xiao, Guoan Xu, Guxue Gao, Qiang Wu, Wenjing Jia
Reconstructing complete yet compact 3D Gaussian Splatting (3DGS) representations from sparse-view monocular videos remains a significant challenge. While hierarchical training with Video Frame Interpolation (VFI) improves coverage, its correlated pseudo-views and repeated merging accumulate structured, non-i.i.d. redundancy, violating the implicit independence assumptions of standard pruning methods and rendering global thresholding ineffectual. We propose CoverPruneGS, a coverage-preserving structured pruning framework specifically designed for hierarchical 3DGS. Our approach implements a coarse-to-fine pruning pipeline using voxel-based local diversity selection and ground-truth-guided lazy refinement via randomized dropout rendering. To ensure reliable refinement, we introduce a footprint-aware CUDA attribution mechanism. By aggregating ground-truth-aligned error degradation across Gaussian-influenced pixels, we generate faithful importance scores that enable precise, quantile-based "rescue" of essential primitives. Experimental results across multiple datasets demonstrate that CoverPruneGS substantially reduces Gaussian counts by 56.8\% and significantly accelerates inference speeds, all while enhancing or maintaining the quality of novel view synthesis.
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DART: Distribution-Aware Adaptive Relational Transfer for Adversarial Attacks against Closed-Source MLLMs
Kaidi Hu, Guancheng Wan, Xiao Luo, Ruigang Yang
This paper studies the critical problem of targeted adversarial attacks against closed-source MLLMs, which aim to generate highly transferable adversarial samples with open-source MLLMs. Previous approaches typically focus on maximizing the similarity of latent representations between adversarial samples and target samples. However, these approaches could overfit specific target samples with severely limited generalization ability to closed-source MLLMs. Towards this end, we propose a novel approach named Distribution-aware Adaptive Relational Transfer (DART) for adversarial attacks against closed-source MLLMs. The core of our DART

is to adopt a statistical lens to characterize the intrinsic semantics of images for more generalized and robust alignment. In particular, each augmented image is considered an example from the intrinsic distribution of the original image. Then, we utilize non-parametric Energy Distance to measure the distribution divergence, which is naturally adopted for the semantic alignment in the hidden space. To further enhance transferability to specific target models, we learn a graph neural network (GNN) to explore the complex relations between source and target MLLMs on transferability and adaptively select surrogate models to maximize transferability across diverse targets. Extensive experiments on benchmark datasets validate the superior robustness and effectiveness of the proposed DART in comparison to various competing baselines.
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KORE: Enhancing Knowledge Injection for Large Multimodal Models via Knowledge-Oriented Controls
Kailin Jiang, Hongbo Jiang, Ning Jiang, Zhi Gao, Jinhe Bi, Yuchen Ren, Bin Li, Yuntao Du, Lei Liu, Qing Li
Large Multimodal Models encode extensive factual knowledge in their pre-trained weights. However, its knowledge remains static and limited, unable to keep pace with real-world developments, which hinders continuous knowledge acquisition. Effective knowledge injection thus becomes critical, involving two goals: knowledge adaptation (injecting new knowledge) and knowledge retention (preserving old knowledge). Existing methods often struggle to learn new knowledge and suffer from catastrophic forgetting. To address these challenges, we propose KORE, a synergistic method centered around KnOwledge-oRientEd controls. These controls are implemented through a two-stage optimization process: (1) KORE automatically converts individual knowledge items into structured and comprehensive knowledge to ensure that the model accurately learns new knowledge, enabling accurate adaptation. (2) KORE stores previous knowledge in the covariance matrix of LMM's linear layer activations and initializes the adapter by projecting the original weights into the matrix's null space, defining a fine-tuning direction that minimizes interference with previous knowledge, enabling powerful retention. Extensive experiments on various LMMs, including LLaVA-v1.5 (7B), LLaVA-v1.5 (13B), and Qwen2.5-VL (7B), show that KORE achieves superior new knowledge injection performance and effectively mitigates catastrophic forgetting.
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Towards Optimal Robustness in Learning-Augmented Paging
Peng Chen, Hailiang Zhao, Xueyan Tang, Yixuan Wang, Shuiguang Deng
Learning-augmented paging has been extensively studied in recent years. A key advantage over naive ML-based approaches is bounded robustness, which guarantees worst-case performance even when predictions are inaccurate, making these algorithms valuable for real-world systems. Prior work achieves robustness bounds of $2H_k + O(1)$ in the randomized setting, leaving a gap to the optimal competitive ratio H_k.

In this paper, we study how to close this gap. We begin by reviewing online optimality and proving a new property of the latest H_k-competitive algorithm, which facilitates our analysis in the learning-augmented setting. Then, we review existing learning-augmented paging algorithms and introduce a unifying primitive, the relative prediction budget, which captures the essence of establishing robustness and reveals that prior algorithms either overuse or underutilize predictions. Guided by the above analysis, we develop a new framework that achieves the best-possible robustness up to an additive constant for learning-augmented paging: $H_k + O(1)$. Experiments further demonstrate strong practical performance.
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WebWorld: A Large-Scale World Model for Web Agent Training
Zikai Xiao, Jianhong Tu, Chuhang Zou, Yuxin Zuo, Zhi Li, Peng Wang, Bowen Yu, Fei Huang, Junyang Lin, Zuozhu Liu
Web agents require massive trajectories to generalize, yet real-world training is constrained by network latency, rate limits, and safety risks. We introduce \textbf{WebWorld} series, the first open-web simulator trained at scale. While exi

sting simulators are restricted to closed environments with thousands of trajectories, WebWorld leverages a scalable data pipeline to train on 1M+ open-web interactions, supporting reasoning, multi-format data, and long-horizon simulations of 30+ steps.
For intrinsic evaluation, we introduce WebWorld-Bench with dual metrics spanning nine dimensions, where WebWorld achieves simulation performance comparable to Gemini-3-Pro. For extrinsic evaluation, Qwen3-14B trained on WebWorld-synthesized trajectories improves by +9.2\% on WebArena, reaching performance comparable to GPT-4o. WebWorld enables effective inference-time search, outperforming GPT-5 as a world model. Beyond web simulation, WebWorld exhibits cross-domain generalization to code, GUI, and game environments, providing a replicable recipe for world-model construction.
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CocoRNA: Collective RNA Design with Cooperative Multi-agent Reinforcement Learning
Tianmeng Hu, Biao Luo, Ke Li
Designing RNA sequences that reliably fold into specific secondary structures is essential for understanding their biological functions but remains a challenging computational problem. We propose CocoRNA, a cooperative multi-agent reinforcement learning framework for RNA inverse design. CocoRNA simplifies the design task by decomposing it into smaller sub-problems, each solved collaboratively by multiple agents. This approach reduces the complexity of the problem and improves the exploration of design policies. During training, a centralized critic uses global structural information to guide the agents, enabling them to jointly optimize their design strategies. As a result, CocoRNA learns high-quality RNA design policies that generalize effectively to unseen structures without additional training. Experiments on the Rfam dataset demonstrate that CocoRNA substantially outperforms state-of-the-art methods in both success rate and design speed. Further experiments on other biological sequence design tasks highlight the effectiveness and broad potential of CocoRNA for complex design tasks.
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Escaping the Likelihood Trap: Geometric Diversity Optimization for Long-Form Image Captioning
Qingmei Tang, Shuai Hao, Rong Fu, Zirui Mo, Xiang Liu, Jiaxuan Lu, Wenyu Wang
The utility of Vision-Language Models (VLMs) in reasoning and auditing tasks hinges on their ability to exhaustively describe visual scenes. However, current models exhibit a pathology we term the Likelihood Trap: standard alignment objectives, specifically MLE and KL-regularization, drive generation toward generic, high-probability templates, systematically suppressing fine-grained details. To overcome this, we introduce Geo-RL, a framework that shifts the objective from probabilistic likelihood to geometric coverage. Geo-RL reformulates caption generation as maximizing the volume of a parallelotope in semantic space. By leveraging Determinantal Point Processes (DPPs), we enforce orthogonality among sampled descriptions, ensuring that they span the image's full semantic support. Crucially, we derive a closed-form leave-one-out marginal reward, enabling stable policy optimization. Empirically, Geo-RL escapes the trap, achieving a significant improvement in semantic richness and detail coverage without compromising visual grounding.
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Residual-Guided Multi-Resolution Refinement of Foundation Models: A Case Study in Drought Forecasting
Wentao Gao, Jiuyong Li, Lin Liu, Thuc Duy Le, Jixue Liu, Yanchang Zhao, Yun Chen
Regional climate prediction presents unique challenges for time series foundation models, which typically process temporal patterns through single-pass inference. Expert climatologists, in contrast, employ multi-scale temporal analysis and iterative refinement based on systematic error diagnosis. We present RGMR (Residual-Guided Multi-Resolution Refinement), an inference-time framework that adapts pre-trained foundation models to perform structured coarse-to-fine refinement for climate forecasting without updating backbone parameters. Applied to drought forecasting using the Standardized Precipitation Evapotranspiration Index (SPEI)

, RGMR is architecture-agnostic across the three TSFM backbones evaluated per si
te (TimesFM, TimeGPT, TabPFN) and consistently lowers test-set MSE on three Sout
h Australian sites and three additional regions outside South Australia. Applied
 to TimesFM, the wrapper reduces one-month-ahead SPEI MSE by up to 18.9\% across
 the three South Australian sites (mean reduction $\approx$18.7\%). Overall, RGM
R provides a practical route for deploying frozen TSFMs in regional climate fore
casting workflows.
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AgentXRay: White-Boxing Agentic Systems via Workflow Reconstruction
Ruijie Shi, Houbin Zhang, Yuecheng Han, Yuheng Wang, Jingru Fan, Runde Yang, Yuf
an Dang, Huatao Li, Dewen Liu, Yuan Cheng, Chen Qian
Large Language Models have shown strong capabilities in complex problem solving,
 yet many agentic systems remain difficult to interpret and control due to opaqu
e internal workflows.
While some frameworks offer explicit architectures for collaboration, many deplo
yed agentic systems operate as black boxes to users.
We address this by introducing Agentic Workflow Reconstruction (AWR), a new task
 aiming to synthesize an explicit, interpretable stand-in workflow that approxim
ates a black-box system using only input--output access.
We propose AgentXRay, a search-based framework that formulates AWR as a combinat
orial optimization problem over discrete agent roles and tool invocations in a c
hain-structured workflow space.
Unlike model distillation, AgentXRay produces editable white-box workflows that
match target outputs under an observable, output-based proxy metric, without acc
essing model parameters.
To navigate the vast search space, AgentXRay employs Monte Carlo Tree Search enh
anced by a scoring-based Red-Black Pruning mechanism, which dynamically integrat
es proxy quality with search depth.
Experiments across diverse domains demonstrate that AgentXRay achieves higher pr
oxy similarity and reduces token consumption compared to unpruned search, enabli
ng deeper workflow exploration under fixed iteration budgets.
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Global Plane Waves from Local Gaussians: Periodic Charge Densities in a Blink
Jonas Elsborg, Felix Aertebjerg, Luca Thiede, Alan Aspuru-Guzik, Tejs Vegge, Arg
hya Bhowmik
We introduce ELECTRAFI, a fast, end-to-end differentiable model for predicting p
eriodic charge densities in crystalline materials. ELECTRAFI constructs anisotro
pic Gaussians in real space and exploits their closed-form Fourier transforms to
 analytically evaluate plane-wave coefficients via the Poisson summation formula
. This formulation delegates non-local and periodic behavior to analytic transfo
rms, enabling reconstruction of the full periodic charge density with a single i
nverse FFT. By avoiding explicit real-space grid probing, periodic image summati
on, and spherical harmonic expansions, ELECTRAFI matches or exceeds state-of-the
-art accuracy across periodic benchmarks while being up to $633\times$ faster th
an the strongest competing method, reconstructing crystal charge densities in a
fraction of a second. When used to initialize DFT calculations, ELECTRAFI reduce
s total DFT compute cost by up to $\sim$20 \%, whereas slower charge density mod
els negate savings due to high inference times. Our results show that accuracy a
nd inference cost jointly determine end-to-end DFT speedups, and motivate our fo
cus on efficiency.
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Bandit Social Leaning Dynamics with Exploration Episodes
Kiarash Banihashem, Natalie Collina, Aleksandrs Slivkins
We study a stylized social learning dynamics where self-interested agents collec
tively follow a simple multi-armed bandit protocol. Each agent controls an "epis
ode": a short sequence of consecutive decisions. Motivating applications include
 users repeatedly interacting with an AI, or repeatedly shopping at a marketplac
e. While agents are incentivized to explore within their respective episodes, we
 show that the aggregate exploration fails: e.g., its Bayesian regret grows line
arly over time. In fact, such failure is a (very) typical case, not just a worst

-case scenario. This conclusion persists even if an agent's per-episode utility is some fixed function of the per-round outcomes: e.g., $\min$ or $\max$, not just the sum. Thus, externally driven exploration is needed even when some amount of exploration happens organically.
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Root Cause Analysis of Failures in Microservices via Bayesian Root Cause Discovery
Kenneth Lee, Zihan Zhou, Murat Kocaoglu
Modern cloud systems rely on architectures with many interconnected microservices, which enable scalability and flexibility but make troubleshooting failures difficult. Identifying the root cause requires navigating complex dependencies, often beyond the capacity of domain experts. Causal models offer a principled approach to root cause analysis (RCA), but prior methods are typically sample inefficient, as they assume access to the full causal graph or require large numbers of post-failure interventions. We introduce Bayesian Root Cause Discovery (BRCD), which leverages a partial causal structure (a CPDAG learned during the pre-failure period) and performs Bayesian inference without enumerating all DAGs from each interventional Markov equivalence class ($\mathcal{I}$-MEC) for each root cause candidate. Using a recent uniform DAG sampling framework (Wienöbst et al., 2023), BRCD provides the first statistical consistency guarantees for nonparametric RCA, with both identifiability and finite-sample posterior bounds under ε-vanishing approximation. Empirically, across synthetic benchmarks and three microservice systems (Online Boutique, Sockshop, Petshop), BRCD achieves state-of-the-art top-l accuracy while remaining effective in low-failure-sample regimes and scaling to large graphs.
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Anchored Policy Optimization: Mitigating Exploration Collapse via Support-Constrained Rectification
Tianyi Wang, Long Li, Hongcan Guo, Yibiao Chen, Yixia Li, Yong Wang, Yun Chen, Guanhua Chen
Reinforcement Learning with Verifiable Rewards (RLVR) is increasingly viewed as a tree pruning mechanism. However, we identify a systemic pathology termed Recursive Space Contraction (RSC), an irreversible collapse driven by the combined dynamics of positive sharpening and negative squeezing, where the sampling probability of valid alternatives vanishes. While Kullback-Leibler (KL) regularization aims to mitigate this, it imposes a rigid Shape Matching constraint that forces the policy to mimic the reference model's full density, creating a gradient conflict with the sharpening required for correctness. We propose Anchored Policy Optimization (APO), shifting the paradigm from global Shape Matching to Support Coverage. By defining a Safe Manifold based on the reference model's high-confidence support, APO permits aggressive sharpening for efficiency while selectively invoking a restorative force during error correction to prevent collapse. We theoretically derive that APO serves as a gradient-aligned mechanism to maximize support coverage, enabling an Elastic Recovery that re-inflates valid branches. Empirical evaluations on mathematical benchmarks demonstrate that APO breaks the accuracy-diversity trade-off, significantly improving Pass@1 while restoring the Pass@K diversity typically lost by standard policy gradient methods.
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Know More, Know Clearer: A Meta-Cognitive Framework for Knowledge Augmentation in Large Language Models
Hao Chen, Ye He, Yuchun Fan, Yukun Yan, Zhenghao Liu, Qingfu Zhu, Maosong Sun, Wanxiang Che
Knowledge augmentation has significantly enhanced the performance of Large Language Models (LLMs) in knowledge-intensive tasks. However, existing methods typically operate on the simplistic premise that model performance equates with internal knowledge, overlooking the knowledge-confidence gaps that lead to overconfident errors or uncertain truths. To bridge this gap, we propose a novel meta-cognitive framework for reliable knowledge augmentation via differentiated intervention and alignment. Our approach leverages internal cognitive signals to partition the knowledge space into mastered, confused, and missing regions, guiding targe

ted knowledge expansion. Furthermore, we introduce a cognitive consistency mecha nism to synchronize subjective certainty with objective accuracy, ensuring calib rated knowledge boundaries. Extensive experiments demonstrate the our framework consistently outperforms strong baselines, validating its rationality in not onl y enhancing knowledge capabilities but also fostering cognitive behaviors that b etter distinguish knowns from unknowns. All codes are available at https://githu b.com/AI9Stars/Know-More-Know-Clearer.

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PMSPO: Progressive Matching and Semantic-Aware Policy Optimization for Camouflag ed Object Detection

Maosheng Su, Junlei Cao, Zhiyuan Wang, Shuo Wang, Ning Zhang, Jun Luo

Reinforcement learning-based Multimodal Large Language Models (MLLMs) provide ne w perspectives for visual grounding, yet face significant challenges in Camoufla ged Object Detection (COD) where objects blend seamlessly with backgrounds. This stems primarily from: difficulties in multi-object matching, the detrimental ef fects of low-quality samples, and erroneously localizing visual distractors with similar textures to true objects. We propose Progressive Matching and Semantic-aware Policy Optimization (PMSPO), a curriculum learning-based framework that em ploys Sinkhorn multi-object matching IoU reward during training for multi-object alignment, utilizes Positive Learning Gain Filtering (PLGF) to curate high-qual ity samples, and transforms deep visual features into semantic contrastive rewar d rules to calibrate target background semantics. Experiments on COD benchmarks demonstrate that PMSPO achieves state-of-the-art (SOTA) performance among reinfo rcement learning methods across all evaluation metrics.

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Exploring Motif-based Heterogeneous Graph Learning for ReDoS Detection

Hong Huang, Chengyu Yao, Rongchen Li, Weihao Su, Chengyao Peng, Haiming Chen, Gu iyi He

Regular expressions (regexes) frequently exhibit super-linear worst-case behavio r in regex engines, exposing software to Regex Denial-of-Service (ReDoS) attacks . Detecting such vulnerabilities is challenging, especially for extended feature s such as lookarounds and backreferences: existing static approaches are efficie nt but often lack support for extended features, whereas dynamic and hybrid appr oaches reduce false positives by executing regex matching on real engines, but i ncur high runtime overhead. To address this trade-off, we propose ReDoS-MotifGNN (RMGNN), a motif-based graph learning framework for ReDoS detection that levera ges the low inference latency of graph neural networks (GNNs). RMGNN converts re gexes into Heterogeneous Regex Graphs (HRGs) and encodes three ReDoS-related mot ifs into HRGs to incorporate domain priors, while preserving the syntactic struc ture and extended features of the input regex. Furthermore, it applies heterogen eous propagation with kernel-guided motif learning to capture multi-scale semant ics, which are fused via residual cross-attention for robust prediction. Compreh ensive evaluation on four real-world datasets (over 317k regexes) demonstrates t hat RMGNN outperforms six state-of-the-art baselines in F1-score and achieves an average 244$\times$ speedup over the top F1-performing baseline.

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Prototype Transformer: Towards Language Model Architectures Interpretable by Des ign

Yordan Yordanov, Matteo Forasassi, Bayar Menzat, Ruizhi Wang, Chang Qi, Markus K altenberger, Amine M'Charrak, Tommaso Salvatori, Thomas Lukasiewicz

While state-of-the-art language models (LMs) surpass the vast majority of humans in certain domains, their reasoning remains largely opaque, reducing trust and risking deception and hallucination. In this work, we introduce the Prototype Tr ansformer (ProtoT)—an autoregressive LM architecture that replaces the quadratic -cost self-attention in the transformer with a linear-cost module based on proto types (parameter vectors). In ProtoT, the prototypes create communication channe ls aggregating contextual information at different time scales. We show that thi s leads to the prototypes automatically capturing nameable concepts (e.g. “woman ”) during training, and it provides the potential to interpret the model’s reaso ning and do targeted edits of its behavior. Compared to baselines, ProtoT scales

well with model and data size, shows robustness to input perturbations, and performs well on text generation and downstream tasks (GLUE). These results suggest that ProtoT is a promising step toward autoregressive language models that are more interpretable by design.

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Online Conformal Prediction via Universal Portfolio Algorithms

Tuo Liu, Edgar Dobriban, Francesco Orabona

Online conformal prediction (OCP) seeks prediction intervals that achieve long-run $1-\alpha$ coverage for arbitrary (possibly adversarial) data streams, while remaining as informative as possible. Existing OCP methods often require manual learning-rate tuning to work well, and may also require algorithm-specific analyses. Here, we develop a general regret-to-coverage theory for interval-valued OCP based on the $(1-\alpha)$-pinball loss. Our first contribution is to identify *linearized regret* as a key notion, showing that controlling it implies coverage bounds for any online algorithm. This relies on a black-box reduction that depends only on the Fenchel conjugate of an upper bound on the linearized regret. Building on this theory, we propose UP-OCP, a parameter-free method for OCP, via a reduction to a two-asset portfolio selection problem, leveraging universal portfolio algorithms. We show strong finite-time bounds on the miscoverage of UP-OCP, even for polynomially growing predictions. Extensive experiments support that UP-OCP delivers consistently better size/coverage trade-offs than prior online conformal baselines.

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Rooted Absorbed Prefix Trajectory Balance with Submodular Replay for GFlowNet Training

Xi Wang, Wenbo Lu, Shenji Wan

Generative Flow Networks (GFlowNets) enable fine-tuning large language models to approximate reward-proportional posteriors, but they remain prone to mode collapse, manifesting as prefix collapse and length bias. We attribute this to two factors: (i) weak credit assignment to early prefixes, and (ii) biased replay that induces a shifted, non-representative training flow distribution. We propose Rooted absorbed prefix Trajectory Balance (RapTB), an objective that anchors subtrajectory supervision at the root and propagates terminal rewards to intermediate prefixes via absorbed suffix-based backups, providing dense prefix-level learning signals. To mitigate replay-induced distribution shift, we further introduce SubM, a submodular replay refresh strategy that promotes both high reward and diversity.

Empirically, on tasks such as molecule generation with LLM using SMILES strings, RapTB combined with SubM consistently improves optimization performance and molecular diversity while preserving high validity.

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Singular Vectors of Attention Heads Align with Features

Gabriel Franco, Carson Loughridge, Mark Crovella

Identifying feature representations in language models is a central task in mechanistic interpretability. Several recent studies have made the observation that feature representations can be inferred in some cases from singular vectors of attention matrices. However, sound justification for this phenomenon is lacking. In this paper we address that question, asking: why and when do singular vectors align with features? First, we demonstrate that singular vectors robustly align with features in a model where features can be directly observed. We then show theoretically that such alignment is expected under a range of conditions. We close by asking how, operationally, alignment may be recognized in real models where feature representations are not directly observable. We identify *sparse attention decomposition* as a testable prediction of alignment, and show evidence that it emerges in real models in a manner consistent with predictions. Together these results suggest that alignment of singular vectors with features can be a sound and theoretically justified basis for feature identification in language models.

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LLMInertia: Adaptive Counter-Inertial Reasoning to Improve Evidence Faithfulness

in Large Language Models
Xinxin You, Xien Liu, Chenwei Yan, Siqi Song, Chen Ning, Kaiyin Zhou, Shaohui Liu, Ji Wu
Large Language Models (LLMs) frequently generate output that contradicts explicit input evidence, limiting their reliability in real-world applications. We identify cognitive inertia in LLMs—a tendency to overly rely on co-occurrence associations learned during pretraining and to resist adaptation when conflicting input evidence appears—as a critical factor behind such hallucinations. We further empirically show that adherence to input evidence declines as co-occurrence associations are strengthened—driven by either higher data frequency or intensified training. Inspired by human counter-inertial thinking, we propose an adaptive counter-inertial reasoning framework that probes input-related cognitive inertia in the LLM and generates adaptive counter-inertial reminders, which are then injected into the prompt to promote evidence-based reasoning. Experiments on co-occurrence induction datasets show that LLMInertia reduces hallucination rates by up to 35\% and improves accuracy by up to 35.68\%. Extensive evaluations on four context-rich summarization and QA datasets, across three LLM backbones of varying scales, further validate its effectiveness and robustness. Our work provides new insight into the causes of input-unfaithful hallucinations in LLMs, contributing to the development of more reliable AI.
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Principle-Evolvable Scientific Discovery via Uncertainty Minimization
Yingming Pu, Tao Lin, Hongyu Chen
Large Language Model (LLM)-based scientific agents have accelerated scientific discovery, yet they often suffer from significant inefficiencies due to adherence to fixed initial priors.
Existing approaches predominantly operate within a static hypothesis space, which restricts the discovery of novel phenomena, resulting in computational waste when baseline theories fail.
To address this, we propose shifting the focus from searching hypotheses to evolving the underlying scientific principles.
We present $\textbf{PiEvo}$, a principle-evolvable framework that treats scientific discovery as Bayesian optimization over an expanding principle space. By integrating Information-Directed Hypothesis Selection via Gaussian Process and an anomaly-driven augmentation mechanism, PiEvo enables agents to autonomously refine their theoretical worldview.
Evaluation across four benchmarks demonstrates that PiEvo (1) achieves an average solution quality of up to 90.81\%$\sim$93.15\%, representing a 29.7\%$\sim$31.1\% improvement over the state-of-the-art, (2) attains an 83.3\% speedup in convergence step via significantly reduced sample complexity by optimizing the compact principle space, and (3) maintains robust performance across diverse scientific domains and LLM backbones.
Code is publicly available at github.com/amair-lab/PiEvo.
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SliceFine: The Universal Winning-Slice Hypothesis for Pretrained Networks
Md Kowsher, Ali O. Polat, Ehsan Mohammady Ardehaly, Mehrdad Salehi, Zia Ghiasi, Prasanth Murali, Chen Chen
This paper presents a theoretical framework that explains why fine-tuning small, randomly selected subnetworks (slices) within pre-trained models is sufficient for downstream adaptation. We establish that pretrained networks exhibit a universal winning slice property, arising from two phenomena: (1) spectral balance—the eigenspectra of different weight matrix slices are remarkably similar—and (2) high task energy—their backbone representations (pretrained weights) retain rich, task-relevant features. This leads to the Universal Winning Slice Hypothesis, which provides a theoretical foundation for parameter-efficient fine-tuning (PEFT) in large-scale models. Inspired by this, we propose SliceFine, a PEFT method that uses this inherent redundancy by updating only selected slices of the origi- nal weights—introducing zero new parameters, unlike adapter-based approaches. Empirically, SliceFine matches the performance of SOTA PEFT methods across various language and vision tasks, while significantly improving training speed, me

mory efficiency, and model compactness. Our work bridges theory and prac- tice, offering a theoretically grounded alternative to existing PEFT techniques.
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Continual Learning of Domain-Invariant Representations
Pascal Janetzky, Tobias Schlagenhauf, Stefan Feuerriegel
Continual learning (CL) aims to train models sequentially over multiple domains without forgetting previously learned knowledge. However, existing CL methods optimize for in-domain performance and are therefore prone to learning spurious, domain-specific cues ("shortcut learning"), which limits generalization to unseen domains after deployment. In this paper, we address this limitation through *continual learning of domain-invariant representation*. We introduce a broad class of CL methods that sequentially learn representations capturing invariant structures across domains. Our methods are motivated by the observation that such invariant structures often preserve the underlying causal mechanisms, which can reduce the risk of overfitting to domain-specific cues and thus offer better out-of-domain generalization. Our proposed CL methods combine replay-based training with a tailored sequential invariance alignment to learn---and preserve---invariant structures over time. We evaluate our methods under a deployment-oriented protocol that measures performance on unseen target domains. Across six benchmark and real-world datasets spanning vision, medicine, manufacturing, and ecology, our methods consistently outperform existing CL baselines in terms of generalization to unseen target domains. As an ablation, we further show that naïve extensions of sequential training with existing domain-invariant representation learning (DIRL) methods provide only limited benefits. To the best of our knowledge, this is the first work to develop domain-invariant representation methods for CL.
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MMPD-Bench: Bridging Multimodal Fission with Multi-Polarimetric Modalities Decomposition
Yi He, Zimo Zhao, Yiming Yang, Xiaoyuan Cheng, Chao He, Yukun Hu
Recovering multiple physical parameters from high-dimensional optical measurements remains challenging in computational optics. We present *MMPD-Bench*, a pioneering benchmark that reframes multi-polarimetric modalities decomposition from Mueller matrix observations as a *modality fission* problem under the multi-modal learning paradigm. By replacing iterative numerical inversion with deep surrogate models, MMPD-Bench provides data, standardized solutions and evaluations to address the multi-physics modalities generation challenge. We benchmark representative architectures to this problem, including state-space models, vision transformers, conditional diffusion models, and neural operators, under a multi-faceted evaluation protocol that jointly assesses perceptual fidelity, physical consistency, robustness, and computational efficiency. Our analysis reveals non-trivial trade-offs between accuracy and robustness in accelerated high-fidelity polarimetric decomposition, highlighting key limitations of existing surrogates. To support reproducible research, we open-source the full codebase, together with a large-scale dataset of 21,412 high-resolution Mueller matrix observations acquired through extensive polarimetric measurements. We invite the community to further advance the intersection of polarization optics and multimodal representation learning.
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GAUSS: Graph-Assisted Uncertainty Quantification using Structure and Semantics for Long-Form Generation in LLMs
Karthik Somayaji NS, Yuxuan Yin, Peng Li
In critical domains like clinical reporting, legal analysis, and policy drafting, large language models (LLMs) are increasingly expected to produce extended, fact■rich narratives rather than isolated sentences. Reliable uncertainty quantification in such long■form outputs is crucial. Existing techniques either assign a single confidence score to an entire paragraph or evaluate factual consistency by comparing extracted atomic facts across multiple generations. Some recent approaches represent fact–paragraph relationships using bipartite entailment graphs and derive uncertainty from fact centrality. However, these methods ignore the explicit dependencies among facts within a paragraph and the structural and sema

ntic variation across multiple LLM outputs for the same prompt, missing a key so urce of uncertainty specific to long■form generation. We propose **GAUSS** (**G* *raph■**A**ssisted **U**ncertainty **Q**uantification using **S**tructure and ** S**emantics), a principled framework that models each generated paragraph as a s emantic graph of atomic facts and their relations. We posit that uncertainty ari ses from structural and semantic discrepancies among these graphs across differe nt samples. **GAUSS** quantifies uncertainty as the expected alignment cost betw een the semantic graph of an anchor paragraph and those of alternative generatio ns. By capturing both semantic content and structural coherence, **GAUSS** offer s a more interpretable and theoretically grounded measure of uncertainty than co arse, sentence■level scores.

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PyVision-RL: Forging Open Agentic Vision Models via RL

Shitian Zhao, Shaoheng Lin, Ming Li, Haoquan Zhang, Wenshuo Peng, Kaipeng Zhang, Chen Wei

Reinforcement learning for agentic multimodal models often suffers from interact ion collapse, where models learn to reduce tool usage and multi-turn reasoning, limiting the benefits of agentic behavior. We introduce PyVision-RL, a reinforce ment learning framework for open-weight multimodal models that stabilizes traini ng and sustains interaction. Our approach combines an oversampling–filtering–ran king rollout strategy with an accumulative tool reward to prevent collapse and e ncourage multi-turn tool use. Using a unified training pipeline, we develop PyVi sion-Image and PyVision-Video for image and video understanding. For video reaso ning, PyVision-Video employs on-demand context construction, selectively samplin g task-relevant frames during reasoning to significantly reduce visual token usa ge. Experiments show strong performance and improved efficiency, demonstrating t hat sustained interaction and on-demand visual processing are critical for scala ble multimodal agents.

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Generalizable and Actionable Parts Pose Estimation with Symmetry Annotation-Free Learning Strategy

wenxiao chen, Xueyu Yuan, Liu Liu, Di Wu, Dan Guo

Urgently needed generalizable robot object interaction and manipulation requires high-quality Cross-Category object perception. As a pioneer of this area, Gener alizable and Actionable Parts (GAParts) understanding has attracted increasing a ttention from relevant researchers. However, most recent works either have insuf ficient design regarding the symmetry issue or require rich symmetry annotation, which severely impedes precise GAPart pose estimation in data-lacking scenarios . In this paper, we propose SAFAG, a novel Symmetry Annotation-Free framework fo r Generalizable and Actionable Parts Pose Estimation. Specifically, we suggest a stepwise refinement two-stage framework for candidate-to-final quaternion regre ssion, and tackle the symmetry prediction as a probability distribution problem with self-supervised learning strategy. The experimental results demonstrate the superior performance and robustness of our SAFAG. We believe that our work has the enormous potential to be applied in many areas of embodied AI system.

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HO-SFL: Hybrid-Order Split Federated Learning with Backprop-Free Clients and Dim ension-Free Aggregation

Qiyuan Chen, Xian Wu, Yi Wang, Xianhao Chen

Fine-tuning large models on edge devices is severely hindered by the memory-inte nsive backpropagation (BP) in standard frameworks like federated learning and sp lit learning. While substituting BP with zeroth-order optimization can significa ntly reduce memory footprints, it typically suffers from prohibitively degraded convergence speed. To resolve this dilemma, we propose *Hybrid-Order Split Feder ated Learning* (HO-SFL). By reformulating the split learning process within a La grangian framework, HO-SFL decouples the optimization landscape: The server perf orms precise first-order updates (i.e., BP), whereas clients conduct memory-effi cient zeroth-order optimization. This hybrid design not only eliminates the need for client-side BP but also enables dimension-free model aggregation, drastical ly lowering communication costs. Crucially, we provide a theoretical convergence

analysis, demonstrating that HO-SFL mitigates the dimension-dependent convergence slowdown of zeroth-order optimization, achieving a convergence rate comparable to first-order methods. Extensive experiments on tasks across vision and language modalities validate that HO-SFL achieves convergence speeds comparable to first-order baselines while significantly reducing communication costs and client memory footprints.
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Low-Rank and Sparsity Are All You Need: Exploring Robust Hierarchical Latent Subspaces for Transferable Adversarial Attack

Shuangshuang Pu, Wen Yang, Min Li, Guodong Liu, Chris Ding, Di Ming

Adversarial examples pose serious threats to deep neural networks, exposing fundamental vulnerabilities in model robustness. However, most existing adversarial attacks directly manipulate dense and redundant feature representations, often leading to overfitting on surrogate models and poor black-box transferability. Recent SVD-based attack attempts to exploit low-rank feature subspaces, yet its reliance on single-layer optimization and single-gradient pathway neglects structural redundancy in feature representations and hierarchical heterogeneity across layers. To address these limitations, we propose LRS-Attack, a low-rank and sparse decomposition attack that explicitly models robust hierarchical subspaces in latent feature spaces. Specifically, the low-rank component captures dominant semantic directions, while the sparse component captures localized and discriminative patterns. To efficiently extract low-rank structure while preserving subspace fidelity, we develop a warm-started alternating low-rank approximation algorithm. Moreover, we introduce a hierarchical mixture of robust experts that leverages depth-dependent feature characteristics and guides gradient optimization toward more transferable adversarial directions. Extensive experiments on ImageNet show that LRS-Attack consistently improves black-box transferability over state-of-the-art methods across diverse CNN/ViT architectures and defense settings. Code is available at https://github.com/AdvML-Group/LRS-Attack.
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Spectral Reach: Understanding Neural Scaling as Progress into the Spectral Tail

Konstantin Nikolaou, Jonas Scheunemann, Sven Krippendorf, Samuel Tovey, Christian Holm

Neural scaling laws describe predictable power-law relationships between model size, dataset size, compute, and performance. While these laws guide the development of modern foundation models, the mechanisms underpinning them remain poorly understood, in part due to the absence of scalable analysis tools. To close this gap, we introduce *spectral position*: a scalable measure of which eigenvalues of the empirical neural tangent kernel (eNTK) currently drive loss reduction. Applying this measure to scaling experiments, we find that spectral position decreases throughout training: learning shifts from dominant eigenmodes into the spectral tail. Larger models reach further into the tail than smaller models, revealing a size-dependent capacity we call *spectral reach*. This suggests why larger models achieve lower losses: they sustain learning on weak spectral signals inaccessible to smaller models. We further identify feature learning as a key enabler of spectral reach. It adaptively amplifies gradient magnitudes as learning advances, sustaining progress where frozen representations stall. This points to concrete interventions through architecture and optimizer design.
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Flow for Future: Geometric SE(3)-Equivariant Flow Matching for 3D Trajectory Prediction

Junwei Wu, Yihang Liu, Ruixuan Yu, Jian Sun

Predicting 3D geometric trajectory requires capturing complex spatiotemporal dependencies while preserving physical symmetries. While flow matching offers a powerful generative paradigm, extending it to SE(3)-equivariant dynamics is challenging due to the inherent gap between deterministic history and stochastic evolving flows. To address this, we introduce GSE-Flow, an SE(3)-equivariant flow matching framework. We first propose a Coherent Sequence Encoding and Time-Modulated Embedding strategy that unifies historical and evolving streams, incorporating velocity and flow time via equivariant affine transformations to guide contin

uous evolution. We further design a Geometry-Feature Tensorization mechanism tha t projects node states into a tensor product space, enabling Context-Flow Fusion to guide trajectory evolution with historical context. GSE-Flow guarantees theo retical SE(3)-equivariance and achieves SOTA accuracy on MD17, MD22, and CMU MoC ap benchmarks for geometric trajectory prediction, while demonstrating generalit y by enhancing deterministic baselines. Code is available at https://github.com /aegine/GSE-Flow.
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Reinforcing Real-world Service Agents: Balancing Utility and Cost in Task-orient ed Dialogue
Ning Gao, Wei Zhang, Yuqin Dai, Ling Shi, Ziyin Wang, Yujie Wang, Wei He, Jinpen g Wang, Chaozheng Wang
The rapid evolution of Large Language Models (LLMs) has accelerated the transiti on from conversational chatbots to general agents. However, effectively balancin g empathetic communication with budget-aware decision-making remains an open cha llenge. Since existing methods fail to capture these complex strategic trade-off s, we propose InteractCS-RL, a framework that reframes task-oriented dialogue as a multi-granularity reinforcement learning process. Specifically, we first esta blish a User-centric Interaction Framework to provide a high-fidelity training g ym, enabling agents to dynamically explore diverse strategies with persona-drive n users. Then, we introduce Cost-aware Multi-turn Policy Optimization (CMPO) wit h a hybrid advantage estimation strategy. By integrating generative process cred its and employing a PID-Lagrangian cost controller, CMPO effectively guides the policy to explore Pareto boundary between user reward and global cost constraint s. Extensive experiments on customized real business scenarios demonstrate that InteractCS-RL significantly outperform other baselines across three evaluation d imensions. Further evaluation on tool-agent-user interaction benchmarks verify I nteractCS-RL robustness across diverse domains. The source code and data are av ailable at https://github.com/NingGao-Ai/InteractCS-RL.
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Local-Minima-Preserving Polynomial Relaxation of Ising Problems
Debraj Banerjee, Santanu Mahapatra, Kunal N. Chaudhury
The generalized Ising problem captures a broad spectrum of hard combinatorial pr oblems, including MAX-CUT, Number Partitioning (NPP), and Maximum Independent Se t. In this work, we consider the notion of one-flip local minima for this proble m. We construct a polynomial relaxation and prove the landscape equivalence theo rem: there exists a one-to-one correspondence between the local minima of the re laxation and the one-flip local minima of the original Ising problem. This guara ntee reduces the Ising problem to finding the local minima of a smooth function, allowing us to leverage scalable gradient-based optimizers such as ADAM. We dem onstrate that our method achieves strong performance across challenging benchmar ks, including spin-glass models, MAX-CUT, and NPP.
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Learning Hamiltonian Flow Maps: Mean Flow Consistency for Large-Timestep Molecul ar Dynamics
Winfried Ripken, Michael Plainer, Gregor Lied, Thorben Frank, Oliver T. Unke, St efan Chmiela, Frank Noe, Klaus Robert Muller
Simulating the long-time evolution of Hamiltonian systems is limited by the smal l timesteps required for stable numerical integration. To overcome this constrai nt, we introduce a framework to learn *Hamiltonian Flow Maps* by predicting the *mean* phase-space evolution over a chosen time span Δt, enabling stable large-timestep updates far beyond the stability limits of classical integrators . To this end, we impose a *Mean Flow* consistency condition for time-averaged H amiltonian dynamics. Unlike prior approaches, this allows training on independen t phase-space samples without access to future states, avoiding expensive trajec tory generation. Validated across diverse Hamiltonian systems, our method in par ticular improves upon molecular dynamics simulations using machine-learned force fields (MLFF). Our models maintain comparable training and inference cost, but support significantly larger integration timesteps while trained directly on wid ely-available *trajectory-free* MLFF datasets.

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AutoQRA: Joint Optimization of Mixed-Precision Quantization and Low-rank Adapters for Efficient LLM Fine-Tuning
Changhai Zhou, Shiyang Zhang, Yuhua Zhou, Qian Qiao, Jun Gao, Cheng Jin, KAIZHOU QIN, WEIZHONG ZHANG
Quantization followed by parameter-efficient fine-tuning has emerged as a promising paradigm for downstream adaptation under tight GPU memory constraints. However, this sequential pipeline fails to leverage the intricate interaction between quantization bit-width and LoRA rank. Specifically, a carefully optimized quantization allocation with low quantization error does not always translate to strong fine-tuning performance, and different bit-width and rank configurations can lead to significantly varying outcomes under the same memory budget. To address this limitation, we propose AutoQRA, a joint optimization framework that simultaneously optimizes the bit-width and LoRA rank configuration for each layer during the mixed quantized fine-tuning process. To tackle the challenges posed by the large discrete search space and the high evaluation cost associated with frequent fine-tuning iterations, AutoQRA decomposes the optimization process into two stages. First, it first conducts a global multi-fidelity evolutionary search, where the initial population is warm-started by injecting layer-wise importance priors.
This stage employs specific operators and a performance model to efficiently screen candidate configurations. Second, trust-region Bayesian optimization is applied to locally refine promising regions of the search space and identify optimal configurations under the given memory budget. This approach enables active compensation for quantization noise in specific layers during training. Experiments show that AutoQRA achieves performance close to full-precision fine-tuning with a memory footprint comparable to uniform 4-bit methods.
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Unison: Benchmarking Unified Multimodal Models via Synergistic Understanding and Generation
Jinyu Liu, Xincheng Shuai, Henghui Ding, Yu-Gang Jiang
Unified multimodal models capable of both understanding and generation have achieved remarkable strides. However, despite their unified designs, existing evaluations typically assess understanding and generation capabilities in isolation, overlooking the synergy between comprehension and generation. To bridge this gap, we introduce **Unison**, a comprehensive benchmark comprising 2,169 high-quality unified task samples, designed to evaluate joint understanding and generation in unified multimodal models. Unison offers three key strengths: **1) Comprehensive Dimensions**: Unison encompasses internal consistency, understanding-guided generation, generation-guided understanding, and mutual enhancement to enable holistic evaluation. **2) Diagnostic Evaluation**: it provides both unified and decoupled tracks for understanding and generation, allowing fine-grained attribution of failure modes and quantitative analysis of the gains from unified modeling. **3) Human Alignment**: we also train Unison-Judge, an evaluation model well aligned with human judgments to achieve reliable assessment. Based on systematic evaluations of state-of-the-art models on Unison, we uncover critical limitations in current unified multimodal systems and highlight promising directions for future research. Unison will be publicly available at https://github.com/FudanCVL/Unison.
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Control Consistency Losses for Diffusion Bridges
Samuel Howard, Nikolas Nüsken, Jakiw Pidstrigach
Simulating the conditioned dynamics of diffusion processes, given their initial and terminal states, is an important but challenging problem in the sciences. The difficulty is particularly pronounced for rare events, for which the unconditioned dynamics rarely reach the terminal state. In this work, we propose a novel approach for learning diffusion bridges based on a self-consistency property of the optimal control. The resulting algorithm learns the conditioned dynamics in an iterative online manner, and exhibits strong performance in a range of empirical settings without requiring differentiation through simulated trajectories. B

eyond the diffusion bridge setting, we draw connections between our self-consist ency framework and recent advances in the wider stochastic optimal control liter ature.
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SAC-Opt: Semantic Anchors for Iterative Correction in Optimization Modeling
Yansen Zhang, Qingcan Kang, Yujie chen, Yufei Wang, Xiongwei Han, Tao Zhong, Min gxuan Yuan, Chen Ma
Large language models (LLMs) have opened new paradigms in optimization modeling by enabling the generation of executable solver code from natural language descr iptions. Despite this promise, existing approaches typically remain solver-drive n: they rely on single-pass forward generation and apply limited post-hoc fixes based on solver error messages, leaving undetected semantic errors that silently produce syntactically correct but logically flawed models. To address this chal lenge, we propose SAC-Opt, a backward-guided correction framework that grounds o ptimization modeling in problem semantics rather than solver feedback. At each s tep, SAC-Opt aligns the original semantic anchors with those reconstructed from the generated code and selectively corrects only the mismatched components, driv ing convergence toward a semantically faithful model. This anchor-driven correct ion enables fine-grained refinement of constraint and objective logic, enhancing both fidelity and robustness without requiring additional training or supervisi on. Empirical results on seven public datasets demonstrate that SAC-Opt improves average modeling accuracy by 7.7%, with gains of up to 21.9% on the ComplexLP d ataset. These findings highlight the importance of semantic-anchored correction in LLM-based optimization workflows to ensure faithful translation from problem intent to solver-executable code.
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Reinforcement Learning with Pairwise Preferences in Long-Term Decision Problems
Jonathan Colaço Carr, Prakash Panangaden, Doina Precup, Benjamin Van Roy
Reinforcement-learning problems typically define the goal as maximizing the expe cted value of a scalar reward function. But, pairwise preferences are often easi er to specify than scalar rewards, and they express certain goals that scalar re wards cannot. Methods for reinforcement learning with pairwise preferences have thus received growing interest. Unfortunately, these methods are inefficient in problems with long time horizons, and they lack guarantees on the performance of Markov policies relative to history-dependent policies, which bridge the theory and practice of reinforcement learning. We therefore propose the *Markov decisi on contest* as a new problem model for reinforcement learning with pairwise pref erences. We prove that stationary Markov policies are optimal among all history- dependent policies, that solving a Markov decision contest exactly is in P, and that a simple iterative algorithm converges to an optimal policy at a sublinear rate. Lastly, in a set of high-dimensional decision problems with long time hori zons, we show that our approximate algorithm is significantly more learning-effi cient than prior work.
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Interpretable Embeddings with Sparse Autoencoders: A Data Analysis Toolkit
Nicholas Jiang, Xiaoqing Sun, Lisa Dunlap, Lewis Smith, Neel Nanda
Analyzing large-scale text corpora is a core challenge in machine learning, cruc ial for tasks like identifying undesirable model behaviors. Current methods ofte n rely on costly LLM-based techniques (e.g. annotating dataset differences) or d ense embedding models (e.g. for clustering), which lack control over the propert ies of interest. We propose using sparse autoencoders (SAEs) to create *SAE embe ddings*: representations whose dimensions map to interpretable concepts. Through four data analysis tasks, we show that SAE embeddings are more cost-effective a nd reliable than LLMs and offer the controllability that dense embeddings lack. Using the large hypothesis space of SAEs, we can uncover insights such as (1) se mantic differences between datasets and (2) unexpected concept correlations in d ocuments. For instance, by comparing model responses, we find that Grok-4 clarif ies ambiguities more often than nine other frontier models. Relative to LLMs, SA E embeddings uncover bigger differences at 2-8x lower cost and identify biases m ore reliably. Additionally, SAE embeddings are controllable: by filtering concep

ts, we can (3) cluster documents along axes of interest and (4) outperform dense embeddings on property-based retrieval. Using SAE embeddings, we study model behavior with two case studies: investigating how OpenAI model behavior has changed over time and finding "trigger" phrases learned by Tulu-3 (Lambert et al., 2024) from its training data. These results position SAEs as a versatile tool for unstructured data analysis and highlight the neglected importance of interpreting models through their *data*.
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GHOST: Geometry-Guided Hallucination of Opaque Surface Textures
Langxu Zhao, Zuan Gu, Tianhan Gao
Transparent objects pose a fundamental challenge for depth estimation and 3D reconstruction due to their violation of Lambertian assumptions, leading to severe geometry degradation in downstream tasks. To address this, we propose a novel geometry-guided preprocessing framework GHOST that leverages visual foundation models to transform transparent regions into opaque, structurally consistent representations without requiring downstream model retraining. Specifically, our pipeline utilizes (1) TransDINO and (2) TransDecomp to disentangle masks and transparency physical properties, while (3) DAF-Net recovers surface normal priors to encode geometric curvature. Subsequently, (4) GeoSemTransNet integrates these multi-modal cues to synthesize a texture-rich opaque RGB image that preserves the transparent object's 3D structure. Extensive experiments demonstrate that our method significantly enhances the accuracy of state-of-the-art depth estimation and reconstruction models on transparent objects by restoring essential photometric cues.
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Infinite-Precision Autoregressive Modeling for Vector Graphics and Layouts
Yeonsang Shin, Insoo Kim, Bongkeun Kim, Keonwoo Bae, Bohyung Han
Transformer-based autoregressive models excel in data generation but are inherently constrained by their reliance on discretized tokens, which limits their ability to represent continuous values with high precision. We analyze the scalability limitations of existing discretization-based approaches for generating hybrid discrete-continuous sequences, particularly in high-precision domains such as logos, layouts, and semiconductor circuit designs, where precision loss potentially leads to visual artifacts, aesthetic degradation, and even functional failure. To address the challenge, we propose a novel unified framework that jointly models discrete and continuous values for variable-length sequences. Our approach employs a hybrid approach that combines categorical prediction for discrete values with diffusion-based modeling for continuous values, incorporating two key technical components: an end-of-sequence (EOS) logit adjustment mechanism that uses an MLP to dynamically adjust EOS token logits based on sequence context, and a length regularization term integrated into the loss function. Additionally, we present ContLayNet, a large-scale benchmark comprising 334K high-precision semiconductor layout samples with specialized evaluation metrics that capture functional correctness, where precision errors significantly impact performance. Experiments on semiconductor layouts (ContLayNet), graphic layouts, and SVGs demonstrate that our approach achieves higher-fidelity hybrid vector representations than discretization-based and fixed-schema baselines, while scaling to high-precision generation across multiple domains.
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Epistemic Uncertainty Quantification for Pre-trained VLMs via Riemannian Flow Matching
Li Ju, Mayank Nautiyal, Andreas Hellander, Ekta Vats, Prashant Singh
Vision-Language Models (VLMs) are typically deterministic in nature and lack intrinsic mechanisms to quantify epistemic uncertainty, which reflects the model’s lack of knowledge or ignorance of its own representations. We theoretically motivate negative log-density of an embedding as a proxy for the epistemic uncertainty, where low-density regions signify model ignorance. The proposed method REPVLM computes the probability density on the hyperspherical manifold of the VLM embeddings using Riemannian Flow Matching. We empirically demonstrate that REPVLM achieves near-perfect correlation between uncertainty and prediction error, signi

ficantly outperforming existing baselines. Beyond classification, we also demons
trate that the model also provides a scalable metric for out-of-distribution det
ection and automated data curation.
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D^{2}O: A Dual Debiasing Operator for Training-Free Test-Time Adaptation of Visi
on–Language Models

Yihong Luo, Wenwu He, Dong Liang, Yihang Zhou, Zhuo-Xu Cui

Training-free test-time adaptation (TTA) for vision-language models (VLMs) can i
mprove zero-shot classification under mild shifts, but often degrades under seve
re style/environment variation. We identify two shared failure modes in training
-free pipelines: (i) retrieval confounding, where feature similarity is dominate
d by nuisance/style variation and corrupts retrieval evidence; and (ii)
environment-biased priors, where VLM logits exhibit environment-dependent center
ed shifts that distort gating and prior-like terms. Therefore, we propose D^{2}O
, a training-free debiasing operator
that outputs three inference-time objects:
a retrieval-oriented content feature for semantic matching, a style-aware routin
g coordinate for bias tracking, and debiased logits for corrected priors. D^{2}O
 composes plug-and-play with retrieval-based and closed-form Gaussian adapters i
n online and transductive settings. We further provide operator-to-decision guar
antees: finite-difference covariance recovers a nuisance-sensitive subspace, rou
ting-based EMA controls centered-logit bias estimates, and these errors yield bo
unded posterior log-odds perturbations, leading to a margin-based condition for
label invariance. Extensive experiments show that D^{2}O achieves
its clearest gains under style/environment-dominant shifts,
with broader gains elsewhere. Code is available at \url{https://github.com/MAiTL
-Group/D2O}.
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ViSurf: Visual Supervised-and-Reinforcement Fine-Tuning for Large Vision-and-Lan
guage Models

Yuqi Liu, Liangyu Chen, Jiazhen Liu, Mingkang Zhu, Zhisheng Zhong, Bei Yu, Jiaya
 Jia

Post-training Large Vision-and-Language Models (LVLMs) typically involves Superv
ised Fine-Tuning (SFT) for knowledge injection or Reinforcement Learning with Ve
rifiable Rewards (RLVR) for performance enhancement. However, SFT often leads to
 sub-optimal performance, while RLVR remains constrained by the model's internal
 knowledge base. While a sequential SFT $\rightarrow$ RLVR pipeline can be used,
 it introduces significant computational overhead and suffers from catastrophic
forgetting. To address these limitations, we propose ViSurf (\textbf{Vi}sual \te
xtbf{Su}pervised-and-\textbf{R}einforcement \textbf{F}ine-Tuning), a unified, si
ngle-stage paradigm that integrates the strengths of both SFT and RLVR. By analy
zing their training objectives, we establish a unified framework that injects gr
ound-truth labels directly into RLVR rollouts, facilitating simultaneous externa
l supervision and internal reinforcement. Furthermore, we introduce three novel
reward control strategies to ensure training stability and optimization. Extensi
ve experiments demonstrate that ViSurf consistently outperforms standalone SFT,
RLVR, and the traditional two-stage pipeline across diverse benchmarks. In-depth
 analysis corroborates these findings, validating the derivation and design prin
ciples of ViSurf.
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CorrSteer: Generation-Time LLM Steering via Correlated Sparse Autoencoder Featur
es

Seonglae Cho, Zekun Wu, Adriano Koshiyama

Sparse Autoencoders (SAEs) decompose LLM activations into interpretable features
, yet existing SAE-based steering methods require contrastive datasets or large
activation stores. We introduce CorrSteer, which selects steering features by co
rrelating task outcomes with SAE activations computed during generation, then va
lidates these selections through intervention. This two-stage approach treats co
rrelation as a selection heuristic and intervention as the causal test: features
 that both correlate with success and improve performance when amplified are ret

ained. Coefficients derive from mean activations on correct samples, yielding a fully automated pipeline without task-specific tuning. On Gemma-2 2B and LLaMA-3.1 8B, CorrSteer achieves +3.3% on MMLU (4k samples) and +27.1% on HarmBench (108 samples), with lower side-effect ratios than fine-tuning despite comparable accuracy. Selected features cluster into interpretable categories: structured-output features for multiple-choice tasks, refusal features for safety, and domain-specific semantics for specialized benchmarks. The method scales to 10^5 SAE features (16K per layer × 26 layers for Gemma-2 2B; 32K × 32 for LLaMA-3.1 8B) via streaming correlation ($O(1)$ in dataset size), requiring no backward passes or activation storage.
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Unlearning Isn't Deletion: Investigating Reversibility of Machine Unlearning in LLMs

Xiaoyu Xu, Xiang Yue, Yang Liu, Qingqing Ye, Huadi Zheng, Peizhao Hu, Minxin Du, Haibo Hu

Unlearning in large language models (LLMs) aims to remove specified data, but its efficacy is typically assessed with task-level metrics like accuracy and perplexity.
We show that these metrics can be misleading, as models can appear to forget while their original behavior is easily restored through minimal fine-tuning.
This \emph{reversibility} suggests that information is merely suppressed, not genuinely erased. To address this critical evaluation gap, we introduce a \emph{representation-level analysis framework}.
Our toolkit comprises PCA similarity and shift, centered kernel alignment (CKA), and Fisher information, complemented by a summary metric, the mean PCA distance, to measure representational drift.
Applying this framework across multiple unlearning methods, data domains, and LLMs, we identify four distinct forgetting regimes based on their \emph{reversibility} and \emph{catastrophicity}.
We compare recovery strategies and show that relearning efficiency relies on the data source. We also find that irreversible, non-catastrophic forgetting is exceptionally challenging. By probing unlearning limits, we identify a case of seemingly irreversible, targeted forgetting,
offering insights for more robust erasure algorithms. Overall, our findings expose a gap in
current evaluation and establish a representation-level foundation for trustworthy unlearning.
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Rethinking GNNs and Missing Features: Challenges, Evaluation and a Robust Solution

Francesco Ferrini, Veronica Lachi, Antonio Longa, Bruno Lepri, Akiyoshi Matono, Andrea Passerini, Xin Liu, Manfred Jaeger

Handling missing node features is a key challenge for deploying Graph Neural Networks (GNNs) in real-world domains such as healthcare and sensor networks. Existing studies mostly address relatively benign scenarios, namely benchmark datasets with (a) high-dimensional but sparse node features and (b) incomplete data generated under Missing Completely At Random (MCAR) mechanisms. For (a), we theoretically prove that high sparsity substantially limits the information loss caused by missingness, making all models appear robust and preventing a meaningful comparison of their performance. To overcome this limitation, we introduce one synthetic and three real-world datasets with dense, semantically meaningful features. For (b), we move beyond MCAR and design evaluation protocols with more realistic missingness mechanisms. Moreover, we provide a theoretical background to state explicit assumptions on the missingness process and analyze their implications for different methods. Building on this analysis, we show that a simple baseline adapted to the graph domain is competitive with respect to specialized architectures across diverse datasets and missingness regimes.
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MultiPriv: Benchmarking Individual-Level Privacy Reasoning in Vision-Language Models

Xiongtao Sun, HUI LI, Jiaming Zhang, Yujie Yang, Kaili Liu, Ruxin Feng, Wen Jun Tan, Wei Yang Bryan Lim

Modern Vision-Language Models (VLMs) pose significant individual-level privacy risks by linking fragmented multimodal data to identifiable individuals through hierarchical chain-of-thought reasoning. However, existing privacy benchmarks remain structurally insufficient for this threat, as they primarily evaluate privacy perception while failing to address the more critical risk of privacy reasoning: a VLM's ability to infer and link distributed information to construct individual profiles. To address this gap, we propose MultiPriv, the first benchmark designed to systematically evaluate individual-level privacy reasoning in VLMs. We introduce the Privacy Perception and Reasoning (PPR) framework and construct a bilingual multimodal dataset with synthetic individual profiles, where identifiers, such as faces and names, are linked to sensitive attributes. This design enables nine challenging tasks spanning attribute detection, cross-image re-identification, and chained inference. We conduct a large-scale evaluation of over 50 open-source and commercial VLMs. In our controlled benchmark, 60\% of widely used VLMs can perform individual-level privacy reasoning with up to 80\% accuracy, suggesting a significant potential threat to personal privacy. The benchmark is available at https://github.com/CyberChangAn/MultiPriv-PII.

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Semi-Supervised Gaze Estimation via Disentangled Subspace Contrastive Learning

Qida Tan, Hongyu Yang, Wenchao Du

Appearance-based gaze estimation always suffers from poor generalization due to limited annotated samples and insufficient dataset diversity. Leading approaches adopt weakly supervised learning to generate large-scale pseudo-labeled data from unconstrained real-world scenarios, aiming to mitigate the domain shifts. In this work, we devise a simple yet effective semi-supervised learning architecture that leverages unlabeled data to enhance domain generalization, thereby reducing reliance on labor-intensive manual annotations. Our key insight is to impose Jacobian regularization to disentangle feature representations into discriminative subspaces dedicated to specific gaze components, such as pitch and yaw angles. We further exploit the intrinsic ordinal ranking within each subspace for contrastive learning, enabling the model to learn robust gaze representations from a small set of labeled samples and an abundance of unlabeled ones. This ultimately yields our Disentangled Subspace Contrastive Learning (DSCL) framework. Extensive experiments on multiple benchmarks verify that the proposed DSCL is plug-and-play, achieving competitive performance using only 20\%, 10\%, and even 5\% of the annotated data under both in-domain and cross-domain evaluation settings. The public code is available at \href{https://github.com/da60266/DSCL}{https://github.com/da60266/DSCL}.

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Black-Box Detection of LLM-Generated Text Using Generalized Jensen Shannon Divergence

Shuangyi Chen, Ashish J Khisti

We study black-box detection of machine-generated text under practical constraints: the scoring model (proxy LM) may mismatch the unknown source model, and per-input contrastive generation is costly. We propose SurpMark, a reference-based detector that summarizes a passage by the dynamics of its token surprisals. SurpMark discretizes surprisals into interpretable states, estimates a state-transition matrix for the test text, and scores it via a generalized Jensen–Shannon (GJS) gap between the test transitions and two fixed references (human vs. machine) built once from existing corpora. Theoretically, we derive design guidance for how the discretization bins should scale with data and provide a principled justification for our test statistic. Empirically, across multiple datasets, source models, and scenarios, SurpMark consistently matches or surpasses baselines, demonstrating strong robustness across domains and generators; our experiments on hyperparameter sensitivity exhibit trends that our theoretical results help to explain.

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Efficient Test-time Inference for Generative Planning Models with OCL Search

Robert Gieselmann, Mihai Samson, Federico Pecora, Jeremy L Wyatt
Generative models have emerged as a powerful paradigm for AI planning, yet their performance remains constrained by the training data distribution. One approach is to improve generated solutions during inference by scaling test-time compute. A more efficient alternative is to optimize the inference process itself. In this paper, we show that a modified version of a classical Open-Closed List (OCL) search provides just such an efficient inference procedure. Our algorithm synergizes two learned components: a generative model that performs fast rollouts from intermediate states and a heuristic model that prioritizes among candidate reasoning paths. Key contributions include novel exploration control mechanisms and integration of learned models within the OCL framework. Across multiple combinatorial planning domains, our approach outperforms both neurosymbolic search baselines and classical solvers in computational efficiency and solution quality.
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Adversarially Robust Approximate Furthest Neighbor
Kiarash Banihashem, Jeff Giliberti, Prashant Gokhale, Samira Goudarzi, MohammadTaghi Hajiaghayi, Yuhao Liu, Morteza Monemizadeh, Sandeep Silwal
We work in the adaptive query model, where
one is given a point set $P \subset \mathbb{R}^d$ and seeks to construct a
data structure that can answer correctly and efficiently a sequence of adaptive queries.
In this model, an adversary observes the answers returned by the data structure to previous queries $q_1, \ldots, q_{i-1}$ and, based on this information, chooses
the next query point q_i.
This setting captures strong forms of adaptivity that naturally arise in modern machine learning pipelines, and rules out many
classical randomized techniques that assume oblivious queries.
Our focus is the problem of furthest neighbor search in this adaptive
setting, a fundamental problem in several learning tasks,
including diversity maximization, outlier and anomaly detection, adversarial example generation, and more. We present the first adversarially robust data structure for c-approximate furthest
neighbor queries that achieves query time $\tilde{O}(\min(d n^{1/c^2}, n^{2/c^2} + d))$. This matches the n dependency in the query time of the seminal result by Indyk [SODA'03] for c-approximate furthest neighbor in the oblivious setting, and improves upon the $\tilde{O}(n + d)$ query time achieved via the adaptive distance estimation framework of Cherapanamjeri and Nelson [NeurIPS'20] for a wide range of natural parameters. To complement this result, we present an adversarial attack against oblivious approximate furthest neighbor algorithms. Specifically, we show that the data structure from the algorithm by Indyk fails to maintain its guarantees against adaptive queries.
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Seeking Commonality, Preserving Specificity: A Spectral-Aware Hierarchical Framework for Cross-City Road Representation Learning
Jingtian Ma, Jingyuan Wang, Leong Hou U
Learning unified road representations across diverse cities is a pivotal challenge in urban computing. However, existing approaches predominantly focus on single-city modeling, failing to handle the distribution shifts caused by heterogeneous urban layouts. We identify *spectral misalignment*, manifested as the significant divergence of spectral distributions across different cities, as the primary barrier preventing standard Graph Neural Networks from capturing universal patterns. To bridge this gap, we propose **CoSpec**, a framework that disentangles road networks into shareable low-frequency commonalities and city-specific high-frequency specificities. CoSpec employs a hierarchical dual-path architecture where the low-frequency path aligns global functional semantics via adaptive prototypes, while the high-frequency path modulates local geometric residuals to fit specific urban textures. Theoretical analysis shows CoSpec bounds the Wasserstein distance between city distributions, and extensive experiments demonstrate its superior generalization over state-of-the-art baselines.

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Balancing Fidelity and Diversity in Diffusion Models via Symmetric Attention Decomposition: Hopfield Perspective
Hyunmin Cho, Woo Kyoung Han, Kyong Hwan Jin
We characterize the pre-softmax attention matrix $\mathbf{QK^\top}$ in transformers as an associative memory matrix encoding pairwise associations between input features. By decomposing this matrix into its symmetric and skew-symmetric parts, we interpret the symmetric component as governing the structure of the _energy landscape_, and the skew-symmetric component as driving _circulation_ on that landscape. Leveraging the energy formulation induced by the symmetric component, we derive Hopfield-style stability measures that quantify the stability of retrieved features. We observe meaningful correlations between Hopfield-style stability measures and the fidelity--diversity trade-offs in generation. Finally, we propose a controllable knob to modulate this trade-off by modifying the circulation of the underlying dynamics. Code is available at our Project Page (https://hyeon-cho.github.io/Balancing/).
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MetaStreet: Semi-Supervised Multimodal Learning for Street-Level Socioeconomic Prediction
Meng Chen, Junjie Yang, Zechen Li, Kai Zhao, Hongjun Dai, Weiming Huang
Predicting street-level socioeconomic indicators from street view imagery is fundamental to urban planning. Existing methods typically extract visual features via pretrained encoders and propagate information through graph-based learning, but they fail to fully exploit the structured, task-relevant, and label-efficient learning signals inherent in urban scenes. We propose MetaStreet, a semi-supervised multimodal framework with three components: (1) a semantic-spatial visual encoder that jointly models object co-occurrence and spatial adjacency at the semantic category level, (2) a task-aware textual encoder that steers LLMs toward prediction-relevant features via task-specific prompts, and (3) a geography-aware graph contrastive learning module that leverages spatial autocorrelation to extend contrastive supervision to unlabeled streets, enabling them to actively participate in representation learning. Experiments on two cities across three socioeconomic prediction tasks demonstrate that MetaStreet consistently outperforms state-of-the-art methods.
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EpiCache: Episodic KV Cache Management for Long-Term Conversation on Resource-Constrained Environments
Minsoo Kim, Arnav Kundu, Han-Byul Kim, Richa Dixit, Minsik Cho
Modern large language models (LLMs) extend context lengths to millions of tokens, enabling coherent, personalized responses grounded in long conversational history. However, the Key-Value (KV) cache grows linearly with the extended dialogue history, causing the model’s memory footprint to quickly exceed device limits. While recent KV cache compression methods attempt to reduce memory usage, most apply cache eviction after processing the entire context, incurring unbounded peak memory usage. Additionally, query-dependent eviction narrows the cache semantics to a single query, leading to failure cases in multi-turn conversations. In this paper, we introduce EpiCache, a training-free KV cache management framework for long conversational question answering (LongConvQA) under fixed memory budgets. EpiCache bounds cache growth through block-wise prefill and preserves topic-relevant context via episodic KV compression, which clusters conversation history into coherent episodes and performs episode-specific KV cache eviction. Across three LongConvQA benchmarks (LongMemEval, Realtalk, and LoCoMo), EpiCache improves accuracy by up to 30\%, achieves near-full-cache accuracy under 4–$6\times$ compression, and reduces latency and peak memory by up to $2.4\times$ and $3.7\times$, respectively.
****
Universal Redundancies in Time Series Foundation Models
Anthony Bao, Venkata Hasith Vattikuti, Jeffrey B. Lai, William Gilpin
Time Series Foundation Models (TSFMs) leverage extensive pretraining to accurately predict unseen time series during inference, without the need for task-specif

ic fine-tuning. Through large-scale evaluations of standard benchmarks, we find that leading transformer-based TSFMs exhibit redundant components in their intermediate layers. We introduce a set of tools for mechanistic interpretability of TSFMs, including ablations of specific components and direct logit attribution on the residual stream. Our findings are consistent across several leading TSFMs with diverse architectures, and across a diverse set of real-world and synthetic time-series datasets. We discover that all models in our study are robust to ablations of entire layers. Furthermore, we develop a theoretical framework framing transformers as kernel regressors, motivating a purely intrinsic strategy for ablating heads based on the stable rank of the per-head projection matrices. Using this approach, we uncover the specific heads responsible for degenerate phenomena widely observed in TSFMs, such as parroting of motifs from the context and seasonality bias. Our study sheds light on the universal properties of this emerging class of architectures for continuous-time sequence modeling.

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A Critical Look at Targeted Instruction Selection: Disentangling What Matters (and What Doesn’t)

Nihal V. Nayak, Paula Rodriguez-Diaz, Neha Hulkund, Sara Beery, David Alvarez-Melis

Instruction fine-tuning of large language models (LLMs) often involves selecting a subset of instruction training data from a large candidate pool, using a small query set from the target task. Despite growing interest, the literature on targeted instruction selection remains fragmented and opaque: methods vary widely in selection budgets, often omit zero-shot baselines, and frequently entangle the contributions of key components. As a result, practitioners lack actionable guidance on selecting instructions for their target tasks. In this work, we aim to bring clarity to this landscape by disentangling and systematically analyzing the two core ingredients: data representation and selection algorithms. Our framework enables controlled comparisons across models, tasks, and budgets. We find that only gradient-based data representations choose subsets whose similarity to the query consistently predicts performance across datasets,models, and candidate pools. While no single method dominates, gradient-based representations paired with greedy round-robin selection often perform best on average at low budgets, but these gains diminish at larger budgets. Finally, we unify several existing selection algorithms as forms of approximate distance minimization between the selected subset and the query set, and support this view with new generalization bounds. More broadly, our findings provide critical insights and a foundation for more principled data selection in LLM fine-tuning. The code is available at https://github.com/dcml-lab/targeted-instruction-selection.

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Practical and Scalable Hamiltonian Monte Carlo Without the Metropolis Test

Jakob Robnik, Reuben Cohn-Gordon, Uros Seljak

Hamiltonian Monte Carlo and underdamped Langevin Monte Carlo are state-of-the-art methods for taking samples from high-dimensional distributions with a differentiable density function. To generate samples, they numerically integrate Hamiltonian or Langevin dynamics. This numerical integration introduces an asymptotic bias in Monte Carlo estimators of expectation values, which can be eliminated by adjusting the dynamics with a Metropolis-Hastings (MH) proposal step. Alternatively, one can trade bias for variance by avoiding MH, and select an integration step size that ensures sufficiently small asymptotic bias, relative to the variance inherent in a finite set of samples. Such \emph{unadjusted} methods often significantly outperform their adjusted counterparts in high-dimensional problems where sampling would otherwise be prohibitively expensive, yet are rarely used in statistical applications due to the absence of an automated way of choosing a step size. We propose just such an automatic tuning scheme that takes a user-provided asymptotic bias tolerance and selects a step size that ensures it. The key to the method is a relationship we establish between the energy change during integration and the asymptotic bias. We show that this procedure rigorously bounds the asymptotic bias for Gaussian target distributions. We then numerically show that the procedure works beyond Gaussians. To demonstrate the practicality of t

he proposed scheme, we provide a comprehensive comparison of adjusted and unadju
sted samplers on Bayesian inference problems and on a statistical physics model
in more than one million parameters. With our tuning scheme, the unadjusted meth
ods achieve close to optimal performance, significantly and consistently outperf
orming their adjusted counterparts.
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You Don’t Need All That Attention: Surgical Memorization Mitigation in Text-to-I
mage Diffusion Models
Kairan Zhao, Eleni Triantafillou, Peter Triantafillou
Generative models have been shown to "memorize" certain training data, leading t
o
verbatim or near-verbatim
generating images, which may cause privacy concerns or copyright infringement. W
e introduce Guidance Using Attractive-Repulsive Dynamics (GUARD), a novel framew
ork for memorization mitigation in text-to-image diffusion models. GUARD adjusts
 the image denoising process to guide the generation away from an original train
ing image and towards one that is distinct from training data while remaining a
ligned with the prompt, guarding against reproducing training data, without hurt
ing image generation quality.
We propose a concrete instantiation of this framework, where the positive target
 that we steer towards is given by a novel method for (cross) attention attenua
tion based on (i) a novel statistical mechanism that automatically identifies th
e prompt positions where cross attention must be attenuated and (ii) attenuating
 cross-attention in these per-prompt locations.
The resulting GUARD offers a surgical, dynamic per-prompt inference-time approac
h that, we find, is by far the most robust method in terms of consistently produ
cing state-of-the-art results for memorization mitigation across two architectur
es and for both verbatim and template memorization, while also improving upon or
 yielding comparable results in terms of image quality.
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Probing How Scalable Table Data Enhances General Long-Context Reasoning
Huaibing Xie, Guoliang Zhao, Yang Liu, Shihan Dou, Siming Huang, Yanling Xiao, S
haolei Wang, Yiting Liu, Cheng Zhang, Shaofan Liu, Pluto Zhou
As real-world tasks grow increasingly complex, long-context reasoning has become
 a core capability for Large Language Models (LLMs). However, few studies explor
e which data types are effective for long-context reasoning and why. We find tha
t structured table data with periodic structures shows strong potential for long
-context reasoning. Motivated by this observation, we mathematically analyze tab
ular dependency structures using mutual information, revealing periodic non-vani
shing dependencies in table data. Furthermore, we systematically analyze the cap
abilities of structured table data, conduct relevant scaling experiments, and va
lidate its underlying mechanisms for enhancing long-context reasoning, yielding
several meaningful insights. Leveraging these insights, we propose a simple yet
scalable pipeline(TableLong) for synthesizing high-quality, diverse, and verifia
ble structured table data to boost long-context reasoning via RL. Extensive expe
rimental results demonstrate that table data significantly enhances the long-con
text reasoning capability of LLMs across multiple long-context benchmarks (+8.24
\% on average), and even improves performance on out-of-domain benchmarks (+8.06
\% on average). We hope that our insights provide practical guidance for effecti
ve post-training data to enhance long-context reasoning in LLMs.
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Training-Free Rate-Distortion-Perception Traversal With Diffusion
Yuhan Wang, Suzhi Bi, Ying-Jun Angela Zhang
The rate-distortion-perception (RDP) tradeoff characterizes the fundamental limi
ts of lossy compression by jointly considering bitrate, reconstruction fidelity,
 and perceptual quality. While recent neural compression methods have improved p
erceptual performance, they typically operate at fixed points on the RDP surface
, requiring retraining to target different tradeoffs. In this work, we propose a
 training-free framework that leverages pre-trained diffusion models to traverse
 the entire RDP surface. Our approach integrates a reverse channel coding (RCC)

module with a novel score-scaled probability flow ODE decoder. We theoretically prove that the proposed diffusion decoder is optimal for the distortion-perception tradeoff under AWGN observations and that the overall framework with the RCC module achieves the optimal RDP function in the Gaussian case. Empirical results across multiple datasets demonstrate the framework's flexibility and effectiveness in navigating the ternary RDP tradeoff using pre-trained diffusion models. Our results establish a practical and theoretically grounded approach to adaptive, perception-aware compression.
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Message Passing on the Edge: Towards Scalable and Expressive GNNs
Pablo Barcelo, Fabian Jogl, Alexander Kozachinskiy, Matthias Lanzinger, Stefan Neumann, Cristobal Rojas
Graph neural networks (GNNs) are widely used in graph learning and most architectures propagate information by passing messages between vertices. In this work, we shift our attention to GNNs that perform message passing on *edges* and introduce EB-1WL, an edge-based color-refinement test, and a corresponding architecture, EB-GNN. Our EB-GNN architecture is inspired by the classic triangle-counting algorithm of Chiba and Nishizeki and passes messages along edges and triangles.

Our contributions are as follows:
1. Theoretically, we show that EB-1WL is significantly more expressive than 1WL. We provide a complete logical characterization of EB-1WL in first-order logic, along with distinguishability results via homomorphism counting. To the best of our knowledge, EB-GNN has the strongest theoretical expressivity guarantees among edge-based message-passing GNNs in the literature.
2. Unlike many GNN architectures that are more expressive than 1WL, we prove that EB-1WL and EB-GNN admit near-linear time and memory usage on practical graph learning workloads.
3. We show in experiments that EB-GNN is a highly efficient general-purpose architecture: it substantially outperforms simple MPNNs and remains competitive with task-specialized state-of-the-art GNNs at substantially lower computational cost.
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Geometric Pocket-Centric Protein Encoding for Polypharmacology-Guided Multi-Target Drug Design
Haoran liu, Xiaoli Lin, Jing Hu, Yu Zou, Xiaolong Zhang
Polypharmacology provides a powerful strategy for treating complex diseases, but identifying molecules that simultaneously satisfy coupled constraints across multiple biological targets remains difficult. Existing methods typically model protein pockets in isolation and struggle to jointly account for multiple heterogeneous binding sites when designing a single shared ligand. To address these limitations, we propose a pocket-structure-centric generative framework for polypharmacology. This framework introduces a novel protein topological representation that selectively masks ligand-irrelevant residues while explicitly modeling backbone folding geometry and inter-residue spatial proximity within binding pockets. In addition, structural representations are jointly fused with amino acid and nucleotide sequences to capture their complementary information across targets. Experiments on COVID-19, schizophrenia, and tumor targets show that this framework generates valid candidates with significantly improved binding affinities compared to state-of-the-art methods.
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PATRA: Pattern-Aware Alignment and Balanced Reasoning for Time Series Question Answering
Junkai Lu, Peng Chen, Xingjian Wu, Yang Shu, Chenjuan Guo, Christian S. Jensen, Bin Yang
Time series reasoning demands both the perception of complex dynamics and logical depth. However, existing LLM-based approaches exhibit two limitations: they often treat time series merely as text or images, failing to capture the patterns like trends and seasonalities needed to answer specific questions; and when trai

ned on a mix of simple and complex tasks, simpler objectives often dominate the learning process, hindering the development of deep reasoning capabilities. To address these limitations, we propose the Pattern-Aware Alignment and Balanced Reasoning model (PATRA), introducing a pattern-aware mechanism that extracts trend and seasonality patterns from time series to achieve deep alignment. Furthermore, we design a task-aware balanced reward to harmonize learning across tasks of varying difficulty, incentivizing the generation of coherent Chains of Thought. Extensive experiments show that PATRA outperforms strong baselines across diverse Time Series Question Answering (TSQA) tasks, demonstrating superior cross-modal understanding and reasoning capability.
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Zeus: Towards Tuning-Free Foundation Model for Time Series Analysis
Yisong Fu, Zezhi Shao, Chengqing Yu, Yujie Li, Yongjun Xu, Xueqi Cheng, Fei Wang
We present Zeus, a unified tuning-free Time Series Foundation Model (TSFM) that delivers superior performance across diverse analysis tasks without any task-specific fine-tuning. Unlike prior studies that primarily focus on zero-shot forecasting but require task-specific tuning for other tasks, Zeus bridges this gap by addressing two fundamental challenges in multi-task generalization. First, to reconcile point-level granularity with long-sequence scalability, Zeus incorporates a multi-scale Transformer featuring point-wise tokenization and a U-shaped hierarchy, effectively balancing fine-grained fidelity with computational efficiency. Second, to accommodate varying inductive biases across different tasks, Zeus introduces Multi-Objective Temporal Masking (MOTM), a unified strategy that supports heterogeneous tasks (e.g., extrapolation, interpolation, and global abstraction) within a single framework. Extensive experiments across five representative tasks demonstrate that Zeus consistently achieves competitive results in tuning-free settings, underscoring its potential as a general-purpose TSFM. The code is available at https://github.com/GestaltCogTeam/Zeus.
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From Prior to Pro: Efficient Skill Mastery via Distribution Contractive RL Finetuning
Zhanyi Sun, Shuran Song
We introduce Distribution Contractive Reinforcement Learning (DICE-RL), a framework that uses reinforcement learning (RL) as a "distribution contraction" operator to refine pretrained generative robot policies. DICE-RL turns a pretrained behavior prior into a high-performing "pro" policy by amplifying high-success behaviors from online feedback. We pretrain a diffusion- or flow-based policy for broad behavioral coverage, then finetune it with a stable, sample-efficient residual off-policy RL framework that combines selective behavior regularization with value-guided action selection. Extensive experiments and analyses show that DICE-RL reliably improves performance with strong stability and sample efficiency. It enables mastery of complex long-horizon manipulation skills directly from high-dimensional pixel inputs, both in simulation and on a real robot. Project website: [dice.rl.2026](https://zhanyisun.github.io/dice.rl.2026/).
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Linear Regression with Unknown Truncation Beyond Gaussian Features
Alexandros Kouridakis, Anay Mehrotra, Alkis Kalavasis, Constantine Caramanis
In truncated linear regression, samples (x,y) are shown only when the outcome y falls inside a certain survival set $S^\star$ and the goal is to estimate the unknown d-dimensional regressor $w^\star$. This problem has a long history of study in Statistics and Machine Learning going back to the works of (Galton, 1897; Tobin, 1958) and more recently in, e.g., (Daskalakis et al., 2019; 2021; Lee et al., 2023; 2024). Despite this long history, however, most prior works are limited to the special case where $S^\star$ is precisely known. The more practically relevant case, where $S^\star$ is unknown and must be learned from data, remains open: indeed, here the only available algorithms require strong assumptions on the distribution of the feature vectors (e.g., Gaussianity) and, even then, have a $d^{\mathrm{poly} (1/\varepsilon)}$ run time for achieving ε accuracy. In this work, we give the first algorithm for truncated linear regression with unknown survival set that runs in $\mathrm{poly} (d/\varepsilon)$ tim

e, by only requiring that the feature vectors are sub-Gaussian. Our algorithm re lies on a novel subroutine for efficiently learning unions of a bounded number o f intervals using access to positive examples (without any negative examples) un der a certain smoothness condition. This learning guarantee adds to the line of works on positive-only PAC learning and may be of independent interest.
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Skip-It? Theoretical Conditions for Layer Skipping in Vision–Language Models
Max Hartman, Vidhata Jayaraman, Moulik Choraria, Akhil Bhimaraju, Lav R. Varshne y

Vision–language models achieve incredible performance across a wide range of tas ks, but their large size makes inference costly. Recent work has shown that mult imodal processing contains significant redundancies, making it possible to skip certain layers with minimal performance loss. Yet current pruning techniques rem ain ad-hoc, relying on heuristics or hyperparameter sweeps rather than principle d criteria for determining when layer skipping is beneficial. In this paper, we propose a unified framework that characterizes the redundancy conditions under w hich pruning can enhance efficiency without sacrificing performance. Central to our approach are experimentally verifiable and interpretable notions of redundan cy that can be evaluated without requiring downstream task performance as a metr ic. Applying this framework, we corroborate prior findings that both early and l ate vision tokens are redundant across models, and we validate our conditions by showing they align with actual performance degradation. Beyond these empirical results, our framework provides a theoretically grounded understanding of redund ancy in VLMs and unifies many of the ideas behind modern layer-skipping techniqu es.
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Similarity Is Not Logic: Factored Inference for Dual-Encoder Vision-Language Mod els

Sultan Alshehri, Zhantao Yang, Han Zhang, Marios Savvides

Dual-encoder vision-language models (VLMs) expose a similarity interface that en ables zero-shot retrieval but fails compositional constraints: queries like “umb rella and no person” retrieve images containing both, even when concept detectio n is reliable. We trace this to an interface-level **Bag-of-Concepts** effect, w here similarity scores approximate mean pooling of concept evidence regardless o f operators. Although operator-dependent signals exist in text embeddings, they are too weak or misaligned to affect rankings. Fine-tuning does not reliably res olve this failure because the dominant bottleneck is how similarity aggregates e vidence rather than what encoders represent. We propose **factored inference**, which separates evidence extraction from constraint execution, and introduce LCS E (Logic-Constrained Score Editing), a training-free method that executes constr aints externally using concept scores from frozen encoders. We also introduce FA CTOR-Bench, where LCSE achieves 85.5% accuracy versus 73.2% for the best fine-tu ned baseline, 90.7% when applied to SigLIP 2, and improves NegBench COCO MCQ acc uracy from 27.2% to 65.2% while preserving retrieval performance.
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The Implicit Bias of Adam and Muon on Smooth Homogeneous Neural Networks
Eitan Gronich, Gal Vardi

We study the implicit bias of momentum-based optimizers on smooth homogeneous mo dels. We show that *momentum steepest descent* algorithms like Muon (spectral no rm), MomentumGD (ℓ_2 norm), and Signum (ℓ_∞ norm) are *approximat e* steepest descent trajectories under a decaying learning rate schedule, provin g that these algorithms have a bias towards KKT points of the corresponding marg in maximization problem. We extend the analysis to Adam (without the stability c onstant), which maximizes the ℓ_∞ margin, and to Muon-Signum and Muon-Adam, which maximize a hybrid norm. Our experiments corroborate the theory and s how that the identity of the margin maximized depends on the choice of optimizer . Overall, our results extend earlier lines of work on steepest descent in homog eneous models and momentum-based optimizers in linear models.
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RADE: Random Add-Drop Edge as a Regularizer

Danial Saber, Amirali Salehi-Abari
Graph Neural Networks (GNNs) suffer from overfitting and over-squashing of long-range information. Stochastic graph augmentations (e.g., edge deletion) regularize training against overfitting but can introduce train-inference misalignment and do not improve over-squashing. In contrast, rewiring methods improve connectivity to mitigate over-squashing, but are not designed to regularize training. We propose *Random Add-Drop Edge (RADE)*, a stochastic graph augmentation method that jointly drops and adds edges to address both overfitting and over-squashing simultaneously. RADE is provably designed to align training and inference so that random augmentations regularize training without distribution shift, while supporting long-range communication at inference. We further propose and study a mini-batch gradient-norm balancing algorithm that adapts deletion and addition rates during training, rendering RADE hyperparameter-free in practice. Experiments on node- and graph-classification benchmarks show that RADE is a strong regularizer and mitigates over-squashing. Ablations support the roles of train-inference alignment, adaptive rate selection, and the complementary effects of random edge deletion and edge addition.
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When Do Graph Foundation Models Transfer? A Data-Centric Theory
Jiajun Zhu, Ying Chen, Peihao Wang, Yixuan He, Pan Li, Aditya Akella, Zhangyang Wang
Graph foundation models (GFMs) aim to reuse a single backbone across diverse graph domains, yet their transfer is often uneven and can exhibit negative transfer. While most prior work improves transfer through architectural or adaptation choices, we ask a data-centric question: *which properties of two graph domains determine how much a fixed representation model changes its outputs?* Using a graphon-based continuous limit for dense graphs, we show that for both set-based and message-passing tokenizations, any Lipschitz backbone admits an explicit decomposition of cross-domain output shift into (i) graph-specific finite-sample approximation terms and (ii) an intrinsic, relabeling-invariant domain discrepancy capturing structural mismatch. A key ingredient is positional-encoding (PE) stability: we establish stability guarantees for spectral PEs and highlight contrasting behaviors of eigenvector- versus subspace-based PEs. Experiments on synthetic and real graphs validate the theory and translate the decomposition into guidance for data curation in GFM transfer.
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Autoregressive Direct Preference Optimization
Masanari Oi, Mahiro Ukai, Masahiro Kaneko, Naoaki Okazaki, Nakamasa Inoue
Direct preference optimization (DPO) has emerged as a promising approach for aligning large language models (LLMs) with human preferences.
However, the widespread reliance on the response-level Bradley-Terry (BT) model may limit its full potential, as the reference and learnable models are assumed to be autoregressive only after deriving the objective function.
Motivated by this limitation, we revisit the theoretical foundations of DPO and propose a novel formulation that explicitly introduces the autoregressive assumption prior to applying the BT model.
By reformulating and extending DPO, we derive a novel variant, termed Autoregressive DPO (ADPO), that explicitly integrates autoregressive modeling into the preference optimization framework.
Without violating the theoretical foundations, the derived loss takes an elegant form: it shifts the summation operation in the DPO objective outside the log-sigmoid function.
Furthermore, through theoretical analysis of ADPO, we show that there exist two length measures to be considered when designing DPO-based algorithms: the token length μ and the feedback length μ'.
To the best of our knowledge, we are the first to explicitly distinguish these two measures and analyze their implications for preference optimization in LLMs.
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Approximate Equivariance via Projection-Based Regularisation
Torben Berndt, Jan Stühmer

Equivariance is a powerful inductive bias in neural networks, improving generalisation and physical consistency. Recently, however, non-equivariant models have regained attention, due to their better runtime performance and imperfect symmetries that might arise in real-world applications. This has motivated the development of approximately equivariant models that strike a middle ground between respecting symmetries and fitting the data distribution. Existing approaches in this field either rely on sampling from a group, incurring a high sample complexity, or explicitly parameterise a model as a sum of an equivariant and non-equivariant network.

This work instead approaches approximate equivariance via a projection-based regulariser which leverages a layer-wise orthogonal decomposition of a network's layers into equivariant and non-equivariant components. In contrast to existing methods, this penalises non-equivariance at an operator level across the full group orbit, rather than point-wise as in sample-based approaches. We present a mathematical framework for computing the non-equivariance penalty exactly and efficiently in both the spatial and spectral domains. In our experiments, our method consistently outperforms prior approximate equivariance approaches in both model performance and efficiency, achieving substantial runtime gains over sample-based regularisers.

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VideoTemp-o3: Harmonizing Temporal Grounding and Video Understanding in Agentic Thinking-with-Videos

Wenqi Liu, Yunxiao Wang, Shijie Ma, Meng Liu, Qile Su, Tianke Zhang, Haonan fan, Changyi Liu, Kaiyu Jiang, Jiankang Chen, Kaiyu Tang, Bin Wen, Fan Yang, Tingting Gao, Han Li, Yinwei Wei, Xuemeng Song

In long-video understanding, conventional uniform frame sampling often fails to capture key visual evidence, leading to degraded performance and increased hallucinations. To address this, recent agentic thinking-with-videos paradigms have emerged, adopting a localize–clip–answer pipeline in which the model actively identifies relevant video segments, performs dense sampling within those clips, and then produces answers. However, existing methods remain inefficient, suffer from weak localization, and adhere to rigid workflows. To solve these issues, we propose VideoTemp-o3, a unified agentic thinking-with-videos framework that jointly models video grounding and question answering. VideoTemp-o3 exhibits strong localization capability, supports on-demand clipping, and can refine inaccurate localizations. Specifically, in the supervised fine-tuning stage, we design a unified masking mechanism that encourages exploration while preventing noise. For reinforcement learning, we introduce dedicated rewards to mitigate reward hacking. Besides, from the data perspective, we develop an effective pipeline to construct high-quality long video grounded QA data, along with a corresponding benchmark for systematic evaluation across various video durations. Experimental results demonstrate that our method achieves remarkable performance on both long video understanding and grounding.

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FG-CLIP 2: A Bilingual Fine-grained Vision-Language Alignment Model

Chunyu Xie, Bin Wang, Fanjing Kong, Jincheng Li, Dawei Liang, Ji Ao, Dawei Leng, Yuhui Yin

Fine-grained vision-language understanding requires precise alignment between visual content and linguistic descriptions, a capability that remains limited in current models, particularly in non-English settings. While models like CLIP perform well on global alignment, they often struggle to capture fine-grained details in object attributes, spatial relations, and linguistic expressions, with limited support for bilingual comprehension. To address these challenges, we introduce FG-CLIP 2, a bilingual vision-language model designed to advance fine-grained alignment for both English and Chinese. Our approach leverages rich fine-grained supervision, including region-text matching and long-caption modeling, alongside multiple discriminative objectives. We further introduce the Textual Intra-modal Contrastive (TIC) loss to better distinguish semantically similar captions. Trained on a carefully curated mixture of large-scale English and Chinese data, including a newly released 12M Chinese region-text dataset, FG-CLIP 2 achieves p

owerful bilingual performance. To enable rigorous evaluation, we present a new b enchmark for Chinese multimodal understanding, featuring long-caption retrieval and bounding box classification. Extensive experiments on 29 datasets across 8 t asks show that FG-CLIP 2 outperforms existing methods, achieving state-of-the-ar t results in both languages. We release the model, code, and benchmark to facili tate future research on bilingual fine-grained vision-language alignment.
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Judging What We Cannot Solve: A Consequence-Based Approach for Oracle-Free Evalu ation of Research-Level Math
Guijin Son, Donghun Yang, Hitesh Laxmichand Patel, Hyunwoo Ko, Amit Agarwal, Sun ghee Ahn, Kyong-Ha Lee, Youngjae Yu
Recent progress in reasoning models suggests that generating plausible attempts for research-level mathematics may be within reach, but verification remains a b ottleneck, consuming scarce expert time. We hypothesize that a meaningful soluti on should contain enough method-level information that, when applied to a neighb orhood of related questions, it should yield better downstream performance than incorrect solutions. Building on this idea, we propose \textbf{Consequence-Based Utility}, an oracle-free evaluator that scores each candidate by testing its va lue as an in-context exemplar in solving related yet verifiable questions. Our a pproach is evaluated on an original set of research-level math problems each pai red with one expert-written solution and nine LLM-generated solutions. Notably, Consequence-Based Utility consistently outperforms reward models, generative rew ard models, and LLM judges on ranking quality. Specifically, for GPT-OSS-120B it improves Acc@1 from 67.2 to 76.3 and AUC from 71.4 to 79.6, with similarly larg e AUC gains on GPT-OSS-20B (69.0 to 79.2). Furthermore, compared to LLM-Judges, it also exhibits a larger solver–evaluator gap, maintaining stronger correct–wro ng separation even on instances the underlying solver often fails to solve.
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Interactive Person Retrieval via Multi-Turn Multimodal Conversation
Yang Bai, Tingfeng Wang, Bin Yang, Min Cao, Jinqiao Wang, Mang Ye
Traditional text-based person retrieval approaches typically rely on single-shot textual queries, which are generally incomplete or vague in real-world scenario s. Recently, chat-based person retrieval methods enable iterative query refineme nt via question-answering interactions between the system and users. However, th ese methods fall short of direct user interaction with retrieved candidates duri ng conversation, making it challenging to effectively refine the retrieval resul ts. To address these limitations, we propose multimodal interactive person retri eval (MInterPR), a new retrieval paradigm that allows users to iteratively refin e retrieved candidates by providing feedback on visual differences from the targ et person. To support this task, we establish MInterPEDES, a multimodal conversa tional dataset constructed by augmenting existing question-answering dialogues w ith synthesized visual feedbacks. Furthermore, to tackle the challenge of accura te and efficient semantics understanding in multimodal dialogues, we propose a m ultimodal conversational memory-enhanced framework MNEMO, which incorporates an atomic turn encoding (ATE) module to model each dialogue turn independently, and a dialogue memory aggregation (DMA) module to capture the fine-grained informat ion and cross-turn dependencies. Extensive experiments demonstrate that MNEMO ac hieves substantial improvements in both retrieval accuracy and generalization ab ility, highlighting its promising potential in real-world scenarios. The code an d dataset are publicly available at https://github.com/Flame-Chasers/MNEMO.
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Positive Distribution Shift as a Framework for Understanding Tractable Learning
Marko Medvedev, Idan Attias, Elisabetta Cornacchia, Theodor Misiakiewicz, Gal Va rdi, Nathan Srebro
We study a setting where the goal is to learn a target function f(x) with respec t to a target distribution D(x), but training is done on i.i.d. samples from a d ifferent training distribution D′(x), labeled by the true target f(x). Such a d istribution shift (here in the form of covariate shift) is usually viewed negati vely, as hurting or making learning harder, and the traditional distribution shi ft literature is mostly concerned with limiting or avoiding this negative effect

. In contrast, we argue that with a well-chosen D'(x), the shift can be positive and make learning easier -- a perspective called Positive Distribution Shift (PDS). Such a perspective is central to contemporary machine learning, where much of the innovation is in finding good training distributions D’(x), rather than changing the training algorithm. We further argue that the benefit is often computational rather than statistical, and that PDS allows computationally hard problems to become tractable even using standard gradient-based training. We formalize different variants of PDS, show how certain hard classes are easily learnable under PDS, and make connections with membership query learning.
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ViEEG: Hierarchical Visual Neural Representation for EEG Brain Decoding
Minxu Liu, Donghai Guan, Chuhang Zheng, Chunwei Tian, Jie Wen, Qi Zhu
Understanding and decoding brain activity into visual representations is a fundamental challenge at the intersection of neuroscience and artificial intelligence. While electroencephalogram (EEG) visual decoding has shown promise due to its non-invasive and low-cost nature, existing methods suffer from {Hierarchical Neural Encoding Neglect (HNEN)}, a critical limitation in which flat neural representations fail to model the brain’s hierarchical visual processing. Inspired by the hierarchical organization of visual cortex, we propose ViEEG, a neuro-inspired framework that addresses HNEN. ViEEG decomposes each visual stimulus into three biologically aligned components, namely contour, foreground object, and contextual scene, which serve as anchors for a three-stream EEG encoder. These EEG features are progressively integrated via cross-attention routing, simulating cortical information flow from low-level to high-level vision. We further adopt hierarchical contrastive learning for EEG-CLIP representation alignment, enabling zero-shot object recognition. Extensive experiments on THINGS-EEG dataset demonstrate that ViEEG significantly outperforms previous methods by a large margin in both subject-dependent and subject-independent settings. Results on THINGS-MEG dataset further confirm ViEEG's generalization to different neural modalities. ViEEG not only advances the performance frontier but also sets a new paradigm for EEG brain visual decoding. Our code is available at https://github.com/LauMason/ViEEG.
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Unfolded Laplacian Spectral Embedding: A Theoretically Grounded Approach to Dynamic Network Representation
Haruka Ezoe, Hiroki Matsumoto, Ryohei Hisano
Dynamic relational data arise in many machine learning applications, yet their evolving structure poses challenges for learning representations that remain consistent and interpretable over time. A common approach is to learn time varying node embeddings, whose usefulness depends on well defined stability properties across nodes and across time. We introduce Unfolded Laplacian Spectral Embedding (ULSE), a principled extension of unfolded adjacency spectral embedding to normalized Laplacian operators, a setting where stability guarantees have remained out of reach. We prove that ULSE satisfies both cross-sectional and longitudinal stability under a dynamic stochastic block model. Moreover, the Laplacian formulation yields a dynamic Cheeger-type inequality linking the spectrum of the unfolded normalized Laplacian to worst case conductance over time, providing structural insight into the embeddings. Empirical results on synthetic and real-world dynamic networks validate the theory.
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Tracing the Dynamics of Refusal: Exploiting Latent Refusal Trajectories for Robust Jailbreak Detection
Xulin Hu, CHE WANG, Wei Yang Bryan Lim, Jianbo Gao, Zhong Chen
Representation Engineering analyses often characterize refusal using static directions extracted from terminal or pooled representations. We ask whether this view misses how refusal is constructed across layer-token positions. Using causal tracing, we identify a Refusal Trajectory: a sparse upstream activation pattern that often persists even when attacks such as GCG suppress terminal refusal signals. Based on this observation, we propose SALO (Sparse Activation Localization Operator), a lightweight white-box detector that operates on raw hidden-state vo

lumes from a selected layer window. Across Qwen, Llama, and Mistral models, SALO improves jailbreak detection on several attack families under a fixed XSTest-calibrated operating point. We further analyze static RepE-style baselines, ROI sensitivity, adaptive GCG attacks, and encoded-input boundary cases, clarifying both the promise and limitations of refusal-trajectory monitoring.
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Graph-Preference Learning: Debiasing Network-Sampled Human Feedback for Target Welfare Estimation

Guangrui Fan, DanDan Liu, AZNUL QALID MD SABRI, Pan Lihu

Preference-based reward modeling is a core component of RLHF and DPO pipelines. In practice, the humans providing preference feedback are rarely an i.i.d. sample: recruitment and exposure often follow social, institutional, or spatial structure, inducing non-uniform inclusion probabilities that correlate with graph centrality. We formalize preference learning with *network-sampled* annotators and show that identity-agnostic scalar reward modeling implicitly represents an inclusion-weighted welfare, over-representing structurally central communities when the inclusion distribution q differs from a designer-chosen target weighting π. We propose Graph-Preference Learning, which combines (i) a graph-personalized reward model that shares statistical strength across neighboring annotators and (ii) graph-balanced aggregation that computes stabilized importance weights to target π. Our analysis characterizes the induced welfare represented by the learned aggregate reward and bounds its deviation from the target in terms of weight mismatch, reward-model approximation, and finite-sample effects. Experiments on synthetic graphs and a *semi-synthetic* case study on the LMArena preference dataset, where biased inclusion is *induced* via graph-based sampling, demonstrate up to 62% reduction in target-welfare recovery error and 17% reduction in cross-language performance gaps under biased inclusion.
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Memory as Dynamics: Learning Reliability-Guided Predictive Models for Online Video Perception

Minwoo Kim, Sang Min Yoon

Predictive memory has recently emerged as a powerful mechanism for online video models, enabling temporal reasoning beyond static memory banks. However, we observe a paradoxical phenomenon in which predictive memory often exacerbates drift under occlusion or rapid motion, as inaccurate predictions contaminate the internal state and lead to irreversible identity loss. We identify this failure as a reliability mismatch: predictive dynamics are applied uniformly despite high uncertainty and weak observations.

To address this issue, we reinterpret video memory as a dynamic latent process rather than a static buffer. Building on this insight, we introduce Reliability-Guided Predictive Memory (RPM), a framework that explicitly regulates when and how predictive dynamics should influence online video perception. RPM integrates a latent world model based on state-space dynamics to generate predictive priors, while employing a reliability-aware fusion policy that suppresses unreliable predictions during occlusion.

We instantiate RPM on a SAM2-based model and evaluate it on visual object tracking benchmarks. Experiments demonstrate that our method reduces drift after occlusion, consistently outperforming strong baselines that rely on either static memory or unconditional predictive modeling. We further show that this reliability-guided mechanism transfers to video object segmentation across multiple standard benchmarks. These findings establish that predictive memory is beneficial only when its reliability is explicitly modeled, and define a general principle for robust online video perception. The code is available at https://github.com/minwookim01/memory-as-dynamics.
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On the Salience of Low-Probability Tokens for AI-Generated Text Detection: A Multiscale Uncertainty Perspective

Yikai Guo, Bin Wang, Xilai Fan, Wenjun Ke, Haoran Luo

AI-generated text increasingly blends with human writing, raising practical risks such as misinformation, academic misuse, and corpora contamination. While stat

istical detectors are appealing for efficiency and generalization, they suffer f
rom two key limitations. (i) Boilerplate dominance, boilerplate tokens shared ac
ross human and LLM writing can overwhelm discriminative signals. (ii) Brittle po
int estimates, relying on a single probability score yields unstable decisions u
nder adversarial manipulations. To address these issues, we propose $\textbf{Unc
ertainty}$, a multiscale uncertainty estimator that focuses on informative low-p
robability tokens, which more clearly expose distributional discrepancies. Local
ly, it alleviates boilerplate dominance by averaging the log-probabilities of lo
w-probability tokens; globally, it reduces brittleness by capturing the distribu
tional shape of this low-probability region via Rényi entropy. We further extend
 the detector to $\textbf{Uncertainty++}$ via conditional independent sampling,
yielding a more stable uncertainty estimation. Experiments across seven datasets
 and sixteen LLMs demonstrate high effectiveness, generalization, and robustness
. Our code is available at github.com/guoyikai2000/Uncertainty-AIGT.
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Variational Entropic Optimal Transport

Roman Dyachenko, Nikita Gushchin, Kirill Sokolov, Petr Mokrov, Evgeny Burnaev, A
lexander Korotin

Entropic optimal transport (EOT) in continuous spaces with quadratic cost is a c
lassical tool for solving the domain translation problem. In practice, recent ap
proaches optimize a weak dual EOT objective depending on a single potential, but
 doing so is computationally not efficient due to the intractable log-partition
term. Existing methods typically resolve this obstacle in one of two ways: by si
gnificantly restricting the transport family to obtain closed-form normalization
 (via Gaussian-mixture parameterizations), or by using general neural parameteri
zations that require simulation-based training procedures. We propose Variationa
l Entropic Optimal Transport (VarEOT), based on an exact variational reformulati
on of the log-partition $\log \mathbb{E}[\exp(\cdot)]$ as a tractable minimizati
on over an auxiliary positive normalizer. This yields a differentiable learning
objective optimized with stochastic gradients and avoids the necessity of MCMC s
imulations during the training. We provide theoretical guarantees, including fin
ite-sample generalization bounds and approximation results under universal funct
ion approximation. Experiments on synthetic data and unpaired image-to-image tra
nslation demonstrate competitive or improved translation quality, while comparis
ons within the solvers that use the same weak dual EOT objective support the ben
efit of the proposed optimization principle.
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Mitigating Reward Hacking in RLHF via Bayesian Non-negative Reward Modeling

Zhibin Duan, Guowei Rong, Zhuo Li, Bo Chen, Mingyuan Zhou, Dandan Guo

Reward models learned from human preferences are central to aligning large langu
age models (LLMs) via reinforcement learning from human feedback, yet they are o
ften vulnerable to reward hacking due to noisy annotations and systematic biases
 such as response length or style. We propose Bayesian Non-Negative Reward Model
 (BNRM), a principled reward modeling framework that integrates non-negative fac
tor analysis into Bradley–Terry (BT) preference model.BNRM represents rewards th
rough a sparse, non-negative latent factor generative process that operates at t
wo complementary levels: instance-specific latent variables induce disentangled
reward representations, while sparsity over global latent factors acts as an imp
licit debiasing mechanism that suppresses spurious correlations. Together, this
disentanglement-then-debiasing structure enables robust uncertainty-aware reward
 learning. To scale BNRM to modern LLMs, we develop an amortized variational inf
erence network conditioned on deep model representations, allowing efficient end
-to-end training. Extensive empirical results demonstrate that BNRM substantiall
y mitigates reward over-optimization, improves robustness under distribution shi
fts, and yields more interpretable reward decompositions than strong baselines.
****

Generalized Linear Bandits with Memory

Heesang Ann, Hyunjun Choi, Taehyun Hwang, Younghoon Shin, Haeju Cheong, Min-hwan
 Oh

We study generalized linear bandits with memory, an endogenous non-stationary se

tting in which rewards depend on past actions through a finite memory matrix. Bu ilding on prior work for linear models Clerici et al.,(2024), we show that the p reviously known $\tilde{\mathcal{O}}(T^{3/4})$ regret bound stems from a loose a nalysis, and we provide a sharpened analysis that recovers a $\tilde{\mathcal{O} }(\sqrt{T})$ regret rate in the linear case. We then extend this improvement to generalized linear models and propose a block-wise algorithm based on shrunken c onfidence bounds. Our algorithm achieves a regret bound of $\tilde{\mathcal{O}}\ left(\sqrt{mT} + d\sqrt{T} + \sqrt{\kappa}d^{2} m^{1/4} T^{1/4} + \kappa d^{2} \ right)$, where d denotes the feature dimension, m the memory length, and $\k appa$ a curvature parameter of the link function. This attains a $\sqrt{T}$-type rate despite nonlinear rewards and memory effects. To the best of our knowledge , this analysis provides a unified treatment of memory-induced non-stationarity and nonlinear link functions, while ensuring that the leading regret term is ind ependent of the curvature of the link function. We conduct numerical experiments that are consistent with our theoretical findings.

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Riemannian MeanFlow for One-Step Generation on Manifolds

Zichen Zhong, Haoliang Sun, Yukun Zhao, Yongshun Gong, Yilong Yin

Flow Matching enables simulation-free training of generative models on Riemannia n manifolds, yet sampling typically still relies on numerically integrating a pr obability-flow ODE. We propose Riemannian MeanFlow (RMF), extending MeanFlow to manifold-valued generation where velocities lie in location-dependent tangent sp aces. RMF defines an average-velocity field via parallel transport and derives a Riemannian MeanFlow identity that links average and instantaneous velocities fo r intrinsic supervision. We make this identity practical in a log-map tangent re presentation, avoiding trajectory simulation and heavy geometric computations. F or stable optimization, we decompose the RMF objective into two terms and apply conflict-aware multi-task learning to mitigate gradient interference. RMF also s upports conditional generation via classifier-free guidance. Experiments on sphe res, tori, SO(3), and SE(3) demonstrate competitive one-step sampling with impro ved quality–efficiency trade-offs and substantially reduced sampling cost.

****

Generative Inverse Design with Abstention via Diagonal Flow Matching

Miguel de Campos, Werner Krebs, Hanno Gottschalk

Inverse design aims to find design parameters x achieving target performance $ y^*$. Generative approaches learn bidirectional mappings between designs and lab els, enabling diverse solution sampling. However, standard conditional flow matc hing (CFM), when adapted to inverse problems by pairing labels with design param eters, exhibits strong sensitivity to their arbitrary ordering and scaling, lead ing to unstable training. We introduce Diagonal Flow Matching (Diag--CFM), which resolves this through a zero-anchoring strategy that pairs design coordinates w ith noise and labels with zero, making the learning problem provably invariant t o coordinate permutations. This yields substantially lower round-trip error than CFM and invertible neural network baselines across design dimensions up to $P{= }784$, including order-of-magnitude gains on several benchmarks. We develop two architecture-intrinsic uncertainty metrics, Zero-Deviation and Self-Consistency, that enable three practical capabilities: selecting the best candidate among mu ltiple generations, abstaining from unreliable predictions, and detecting out-of -distribution targets; consistently outperforming ensemble and general-purpose a lternatives across all tasks. We validate on airfoil, gas turbine combustor, sca lable analytical benchmarks, a photonics inverse-design task, and an image-stati stics benchmark.

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Are VLMs Seeing or Just Saying? Uncovering the Illusion of Visual Re-examination

Chufan Shi, Cheng Yang, Yaokang Wu, Linghao Jin, Bo Shui, Taylor Berg-Kirkpatric k, Xuezhe Ma

Vision-Language Models (VLMs) often produce self-reflective statements like “let me check the figure again” during reasoning. Do such state- ments trigger genui ne visual re-examination, or are they merely learned textual patterns? We in- ve stigate this via VISUALSWAP, an image-swap probing framework: after a model reas

ons over an image, we replace it with a visually similar but semantically differ
ent one and test whether the model notices. We introduce VS-BENCH, 800 image pai
rs curated from MathVista, Math- Verse, MathVision, and MMMU-Pro. Exper- iments
on Qwen3-VL, Kimi-VL, and ERNIE- VL reveal a striking failure: models overwhelm-
ingly miss the swap, with accuracy dropping by up to 60%. Counterintuitively, t
hinking mod- els are nearly 3x more vulnerable than their in- structed counterpa
rts, and scaling offers no mit- igation. Multi-turn user instructions restore vi
- sual grounding, but self-generated reflective state- ments during continuous g
eneration do not. At- tention analysis explains why: user instructions substanti
ally elevate attention to visual tokens, whereas self-reflection does not. Curre
nt VLMs tend to say rather than actually see when claiming to perform visual re-
examination. Our code and dataset are available at the project page: https://vis
ualswap.github.io/
****

Commit to the Bit: Reactive Reinforcement Learning Done Right
Onno Eberhard, Claire Vernade, Michael Muehlebach
Reinforcement learning algorithms are commonly analyzed (and designed) under the
Markov assumption. This is unrealistic, as most environments encountered in pra
ctice are either partially observable, or require function approximation that re
stricts the agent to access non-Markovian state features. We consider the proble
m of learning an optimal reactive policy in a finite environment with determinis
tic observations (or equivalently, hard state aggregation). We introduce a new a
lgorithm, _Committed Q-learning_, and prove almost-sure convergence to the optim
al reactive policy under an intuitive assumption we call _rewire-robustness_. Th
is assumption is strictly weaker than the $q_\star$-realizability condition used
in prior work. Our algorithm is a variant of classical Q-learning in which the
behavior policy commits to a single action upon entering a feature, and only res
amples actions when the observed feature changes. A crucial part of our analysis
is the introduction of _quasi-Markov_ environments.
****

Conditionally Site-Independent Neural Evolution of Antibody Sequences
Stephen Zhewen Lu, Aakarsh Vermani, Kohei Sanno, Jiarui Lu, Frederick A Matsen,
Milind Jagota, Yun S. Song
Common deep learning approaches for antibody engineering focus on modeling the m
arginal distribution of sequences. By treating sequences as independent samples,
however, these methods overlook affinity maturation as a rich and largely untap
ped source of information about the evolutionary process by which antibodies exp
lore the underlying fitness landscape. In contrast, classical phylogenetic model
s explicitly represent evolutionary dynamics but lack the expressivity to captur
e complex epistatic interactions. We bridge this gap with **CoSiNE**, a continuo
us-time Markov chain parameterized by a deep neural network. Mathematically, we
prove that CoSiNE provides a first-order approximation to the intractable seque
ntial point mutation process, capturing epistatic effects with an error bound th
at is quadratic in branch length. Empirically, CoSiNE outperforms state-of-the-a
rt language models in zero-shot variant effect prediction by explicitly disentan
gling selection from context-dependent somatic hypermutation. Finally, we introd
uce *Guided Gillespie*, a classifier-guided sampling scheme that steers CoSiNE a
t inference time, enabling efficient optimization of antibody binding affinity t
oward specific antigens.
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Swordsman: Entropy-Driven Adaptive Block Partition for Efficient Diffusion Langu
age Models
Yu Zhang, Xinchen Li, Jialei Zhou, Hongnan Ma, Zhongwei Wan, Yiwei Shi, Duoqian
Miao, Qi Zhang, Longbing Cao
Block-wise decoding effectively improves the inference speed and quality in diff
usion language models (DLMs) by combining inter-block sequential denoising and i
ntra-block parallel unmasking. However, existing block-wise decoding methods typ
ically partition blocks in a rigid and fixed manner, which inevitably fragments
complete semantic or syntactic constituents, leading to suboptimal performance.
Inspired by the entropy reduction hypothesis (ERH), we recognize that constituen

t boundaries offer greater opportunities for uncertainty reduction, which motiva
tes us to employ entropy analysis for identifying constituent boundaries. Theref
ore, we propose Swordsman, an entropy-driven adaptive block-wise decoding framew
ork for DLMs. Swordsman adaptively partitions blocks by identifying entropy shif
ts between adjacent tokens to better align with semantic or syntactic constituen
t boundaries. In addition, Swordsman dynamically adjusts unmasking thresholds co
nditioned on the real-time unmasking status within a block, further improving bo
th efficiency and stability. As a training-free framework, supported by KV Cache
, Swordsman demonstrates state-of-the-art performance across extensive evaluatio
ns. Our code is now available.
****

Plan for Speed: Dilated Scheduling for Masked Diffusion Language Models
Omer Luxembourg, Haim H. Permuter, Eliya Nachmani
Masked diffusion language models (MDLMs) promise fast, non-autoregressive text g
eneration, yet existing samplers, which pick tokens to unmask based on model con
fidence, ignore interactions when unmasking multiple positions in parallel and e
ffectively reduce to slow, autoregressive behavior. We propose the Dilated Unmas
king Scheduler (DUS), an inference-only, planner-model-free method that partitio
ns sequence positions into non-adjacent dilated groups and unmasks them in paral
lel so as to minimize an upper bound on joint entropy gain at each denoising ste
p. By explicitly trading off the number of network calls against generation qual
ity, DUS recovers most of the performance lost under traditional parallel unmask
ing strategies. Across math (GSM8K, MATH500), code (HumanEval, MBPP), general-kn
owledge (BBH, MMLU-Pro), and instruction following (IFEval) benchmarks, DUS outp
erforms confidence-based planners and turns the diffusion-specific quality-speed
 trade-off into a deterministic, predictable speedup set by the block size B,
yielding up to $5.8\times$ wall-clock speedup over token-by-token MDLM decoding
without modifying the underlying denoiser. Applied as a drop-in post-filter, dil
ated spacing also improves adaptive samplers. Code is available at https://githu
b.com/omerlux/DUS.
****

Activation Oracles: Training and Evaluating LLMs as General-Purpose Activation E
xplainers
Adam Karvonen, James Chua, Clément Dumas, Kit Fraser-Taliente, Subhash Kantamnen
i, Julian Minder, Euan Ong, Arnab Sen Sharma, Daniel Wen, Owain Evans, Samuel Ma
rks
Large language model (LLM) activations are notoriously difficult to understand,
with most existing techniques using complex, specialized methods for interpretin
g them. Recent work has proposed a simpler approach known as LatentQA: training
LLMs to directly accept LLM activations as inputs and answer arbitrary questions
 about them in natural language. However, prior work has focused on narrow task
settings for both training and evaluation. In this paper, we instead take a gene
ralist perspective. We evaluate LatentQA-trained models, which we call Activatio
n Oracles (AOs), in far out-of-distribution settings and examine how performance
 scales with training data diversity. We find that AOs can recover information f
ine-tuned into a model (e.g., biographical knowledge or malign propensities) tha
t does not appear in the input text, despite never being trained with activation
s from a fine-tuned model. Our main evaluations are four downstream tasks where
we can compare to prior white- and black-box techniques. We find that even narro
wly-trained LatentQA models can generalize well, and that adding additional trai
ning datasets (such as classification tasks and a self-supervised context predic
tion task) yields consistent further improvements. Our best AOs match or exceed
white-box baselines on all four tasks and the best overall baseline on 3 of 4. T
hese results suggest that diversified training to answer natural-language querie
s imparts a general capability to verbalize information about LLM activations.
****

R-Diverse: Mitigating Diversity Illusion in Self-Play LLM Training
Gengsheng Li, Jinghan He, Shijie Wang, Ruiqi Liu, Renrui Zhang, Zijun Yao, Junfe
ng Fang, Haiyun Guo, Dan Zhang, Jinqiao Wang
Self-play bootstraps LLM reasoning through an iterative Challenger--Solver loop:

the Challenger is trained to generate questions that target the Solver's capabilities, and the Solver is optimized on the generated data to expand its reasoning skills. However, existing frameworks like R-Zero often exhibit non-sustained improvement, where early gains degrade as self-play continues. We identify a key failure mode, **Diversity Illusion**, where the Solver's training signals appear diverse yet collapse into recurring underlying patterns. It manifests as (1) *Local Diversity Illusion*, where diversity is enforced only within-batch, inducing cross-iteration mode cycling; and (2) *Surface Diversity Illusion*, where questions vary superficially but require near-identical reasoning skills. To mitigate them, we propose R-Diverse with two aligned innovations: **Memory-Augmented Penalty (MAP)**, which uses a persistent memory bank to discourage recycling across iterations, and **Skill-Aware Measurement (SAM)**, which evaluates diversity by the reasoning skills exercised rather than surface variation of questions. Across 10 math and general reasoning benchmarks, R-Diverse sustains gains over more iterations and consistently outperforms prior self-play methods.

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ϕ-Balancing for Mixture-of-Experts Training

Lizhang Chen, Jonathan Li, Qi Wang, Runlong Liao, Shuozhe Li, Chen Liang, Ni Lao, qiang liu

Mixture-of-Experts (MoE) models rely on balanced expert utilization to fully realize their scalability. However, existing load-balancing methods are largely heuristic and operate on noisy mini-batch assignment statistics, introducing bias relative to population-level objectives. We propose **ϕ-balancing**, a principled framework that directly targets population-level expert balance by minimizing a strictly convex, symmetric, and differentiable potential of the expected routing distribution. Using convex duality, we derive an equivalent min-max formulation and obtain a simple online algorithm via mirror descent, yielding an efficient EMA-based routing adjustment with negligible overhead. Across large-scale pretraining and downstream fine-tuning, ϕ-balancing consistently outperforms prior Switch-style and loss-free baselines, demonstrating more stable and effective expert utilization.

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Symbiosis-Inspired Knowledge Distillation for Incremental Object Detection

Mingyue Zeng, De Cheng, Zhipeng Xu, Huaijie Wang, Nannan Wang, Xinbo Gao

Incremental object detection (IOD) aims to extend detectors to new categories while retaining previously acquired knowledge. Existing methods often adopt a class incremental learning perspective, separating feature spaces to sharpen decision boundaries. However, this separation-oriented paradigm may overlook object symbiosis in detection, where co-occurrence and occlusion introduce spatial and semantic dependencies that benefit from shared representations. Ignoring these dependencies distorts the shared representations, exacerbates confusion between old and new classes, and accelerates catastrophic forgetting. To address this, we propose Symbiosis-Inspired Knowledge Distillation (SIKD), which explicitly leverages object symbiosis at two complementary levels. Spatial Symbiosis Distillation (SpSD) focuses on symbiotic regions where the old model responds with high overlap to objects in the new task. It preserves generalizable old class cues, suppresses class-specific bias and redundancy, and distills the refined evidence to the new model at matched spatial locations with slot-aligned supervision. Semantic Symbiosis Distillation (SeSD) maintains class level structure by forming confidence weighted prototypes for old classes and aligning their inter class soft ranks over the old class logits, which stabilizes the semantic topology during adaptation. Extensive experiments demonstrate the effectiveness and superiority of the proposed method.

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ASRU: Activation Steering Meets Reinforcement Unlearning for Multimodal Large Language Models

Jiahui Guang, Haiyan Wang, Yingjie Zhu, Cuiyun Gao, Jing Li, Di Shao, Zhaoquan Gu

Multimodal large language models (MLLMs) may memorize sensitive cross-modal information during pretraining, making machine unlearning (MU) crucial. Existing met

hods typically evaluate unlearning effectiveness based on output deviations, whi le overlooking the generation quality after unlearning. This can easily lead to hallucinated or rigid responses, thereby affecting the usability and safety of t he unlearned model. To address this issue, we propose ASRU, a controllable multi modal unlearning framework that incorporates generation quality as a core evalua tion objective. ASRU first induces initial refusal behavior through activation r edirection, and then optimizes fine-grained refusal boundaries using a customize d reward function, thereby achieving a better trade-off between target knowledge unlearning and model utility. Experiments on Qwen3-VL show that ASRU significan tly improves unlearning effectiveness (+24.6%) on average and generation quality (5.8×) on average while effectively preserving model utility, using only a sma ll amount of retained supervision data.
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Active Budget Allocation for Efficient Scaling Law Estimation via Surrogate-Guid ed Pruning
Viktoria Schram, Markus Hiller, Daniel Beck, Trevor Cohn
Predicting model performance at larger scales enables the design of training str ategies and architectures tailored to specific performance targets. Empirical sc aling law research identifies functional forms to aid this prediction task. Thes e describe the relationship between loss and compute using a loss-compute fronti er defined by learning curves. Due to the empirical nature of this approach, the computational burden is substantial, making strategic resource allocation essen tial -- yet it remains surprisingly underexplored.
In this work, we address this shortcoming by exploring the suitability of Succes sive Halving (SH) and SH combined with parametric and non-parametric surrogate m odels. In addition to enabling a more systematic allocation of a given compute b udget, our findings show that SH paired with surrogate models yields a set of le arning curves that includes one with a lower loss-compute value than what naive uniform allocation or an SH-only approach can obtain.
Our experiments demonstrate mean relative improvements of up to 2.84% and $5. 47\%$ on real-world and synthetic learning curve datasets.
This strategic resource allocation enables us to obtain accurate scaling laws at significantly reduced computational costs, saving up to 98.7% over the tradi tional exhaustive approach.
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Reinforced Sequential Monte Carlo for Amortised Sampling
Sanghyeok Choi, Sarthak Mittal, Víctor Elvira, Jinkyoo Park, Esmeralda S. Whitam mer
This paper proposes a synergy of amortised and particle-based methods for sampli ng from distributions defined by unnormalised density functions. We state a conn ection between sequential Monte Carlo (SMC) and neural sequential samplers train ed by maximum-entropy reinforcement learning (MaxEnt RL), wherein learnt samplin g policies and value functions define proposal kernels and twist functions. Expl oiting this connection, we introduce an off-policy RL training procedure for the sampler that uses samples from SMC -- using the learnt sampler as a proposal -- as a behaviour policy that better explores the target distribution. We describe techniques for stable joint training of proposals and twist functions and an ad aptive weight tempering scheme to reduce training signal variance. Furthermore, building upon past attempts to use experience replay to guide the training of ne ural samplers, we derive a way to combine historical samples with annealed impor tance sampling weights within a replay buffer. On synthetic multi-modal targets (in both continuous and discrete spaces) and the Boltzmann distribution of alani ne dipeptide conformations, we demonstrate improvements in approximating the tru e distribution as well as training stability compared to both amortised and Mont e Carlo methods.
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WarmServe: Enabling One-for-Many GPU Prewarming for Multi-LLM Serving
Chiheng Lou, Sheng Qi, Rui Kang, Yong Zhang, Chen Sun, pengcheng wang, Xuanzhe L iu, Xin Jin
Deploying multiple models within shared GPU clusters is a key strategy to improv

e resource efficiency in large language model (LLM) serving.
Existing multi-LLM serving systems improve GPU utilization at the cost
of degraded inference performance, particularly time-to-first-token (TTFT).
We attribute this degradation to the lack of awareness regarding future workload
 characteristics.
In contrast, recent analyses have shown the strong periodicity and long-term pre
dictability of real-world LLM serving workloads.
In this paper, we propose \textit{one-for-many GPU prewarming}, which proactivel
y loads parameters from multiple models onto GPUs based on workload forecasts.
These prewarmed weights enable the system to promptly instantiate serving instan
ces upon encountering request bursts.
We design and implement WarmServe, a multi-LLM serving system incorporating thre
e key techniques:
(1) a model placement algorithm that optimizes prewarming decisions to minimize
cross-model prewarming interference,
(2) a KV cache reservation strategy that repurposes idle KV cache space on runni
ng GPUs for prewarming new models,
and (3) an efficient GPU memory switching mechanism for tensor management.
Evaluation on real-world datasets shows that WarmServe reduces tail TTFT by up t
o 50.8$\times$ compared to the state-of-the-art autoscaling-based system,
while supporting up to 2.5$\times$ higher request throughput than the GPU-sharin
g system.
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Self-CriTeach: LLM Self-Teaching and Self-Critiquing for Improving Robotic Plann
ing via Automated Domain Generation
Jinbang Huang, Zhiyuan Li, Yuanzhao Hu, Zhanguang Zhang, Mark Coates, Xingyue Qu
an, Yingxue Zhang
Large Language Models (LLMs) have recently shown strong promise for robotic task
 planning, particularly through automatic planning domain generation. However, p
rior approaches largely treat generated planning domains as planning utilities,
which are brittle under imperfect logical states and perception noise, overlooki
ng their potential as scalable sources of reasoning supervision and structured r
eward signals. At the same time, reasoning LLMs depend on chain-of-thought (CoT)
 supervision that is expensive to collect for robotic tasks, and reinforcement l
earning (RL) faces challenges on reward engineering. We propose Self-CriTeach, a
n LLM self-teaching and self-critiquing framework in which an LLM autonomously g
enerates symbolic planning domains that serve a dual role: (i) enabling large-sc
ale generation of robotic planning problem–plan pairs, and (ii) providing struct
ured reward functions. First, the self-written domains enable large-scale genera
tion of symbolic task plans, which are automatically transformed into extended C
oT trajectories for supervised fine-tuning. Second, the self-written domains are
 reused as structured reward functions, providing dense feedback for reinforceme
nt learning without manual reward engineering. This unified training pipeline yi
elds a planning-enhanced LLM with higher planning success rates, stronger cross-
task generalization, reduced inference cost, and resistance to imperfect logical
 states.
****

Fixed Budget is No Harder Than Fixed Confidence in Best-Arm Identification up to
 Logarithmic Factors
Kapilan Balagopalan, Yinan Li, Yao Zhao, Tuan Ngo Nguyen, Anton Daitche, Houssam
 Nassif, Kwang-Sung Jun
The best-arm identification (BAI) problem is one of the most fundamental problem
s in interactive machine learning, which has two flavors: the fixed-budget setti
ng (FB) and the fixed-confidence setting (FC). For K-armed bandits with a uni
que best arm, the optimal sample complexities for both settings have been settle
d down and they match up to logarithmic factors. This prompts an interesting re
search question about the generic, potentially structured BAI problems: is FB ha
rder than FC or the other way around? In this paper, we show that FB is no hard
er than FC up to logarithmic factors. We do this constructively: we propose a n
ovel algorithm called FC2FB (fixed confidence to fixed budget), which is a meta

algorithm that takes in an FC algorithm $\mathcal{A}$ and turn it into an FB alg orithm. We prove that FC2FB enjoys a sample complexity that matches, up to loga rithmic factors, that of the sample complexity of $\mathcal{A}$. This means tha t the optimal FC sample complexity is an upper bound of the optimal FB sample co mplexity up to logarithmic factors. Our result not only reveals a fundamental r elationship between FB and FC, but also has a significant implication: FC2FB com bined with existing state-of-the-art FC algorithms, leads to improved sample com plexity for a number of FB problems.
****
DEGAP: Dynamic Entropy-Guided Attention Perturbation for Contrastive Decoding in Large Vision-Language Models
Hyein Seo, Yuna Jeong, Mingyu Kang, Junhyeong Park, Yong Suk Choi
Large Vision–Language Models (LVLMs) have shown outstanding performance across v arious multimodal tasks, but they still suffer from hallucinations, where they g enerate incorrect information by relying on language priors without visual groun ding. To alleviate this issue, prior work has explored contrastive decoding appr oaches that compare the output of an original LVLM with that of a contrast branc h.
However, existing methods typically generate contrast logits through preprocessi ng of the input image. Such input-level perturbations fail to sufficiently refle ct the model’s internal degree of visual reliance during the decoding process.
To address this limitation, we propose Dynamic Entropy-Guided Attention Perturba tion (DEGAP) for contrastive decoding in LVLMs. DEGAP performs contrastive decod ing by directly perturbing visual attention and leveraging the resulting logits, without requiring any additional image preprocessing. To this end, we analyze t he layer-wise effects of visual attention perturbations and, based on these obse rvations, dynamically select the layers at which attention perturbation is appli ed according to the model’s confidence.
Experimental results on seven benchmarks demonstrate that DEGAP effectively miti gates various types of hallucinations and consistently outperforms state-of-the-art methods in general VQA performance.
****
LangForce: Bayesian Decomposition of Vision Language Action Models via Latent Ac tion Queries
Shijie Lian, Bin Yu, Xiaopeng LIN, Laurence Tianruo Yang, Zhaolong Shen, Changti Wu, YuZhuo Miao, Cong Huang, Kai Chen
Vision-Language-Action (VLA) models have shown promise in robot manipulation but often struggle to generalize to new instructions or complex multi-task scenario s. We identify a critical pathology in current training paradigms where goal-dri ven data collection creates a dataset bias. In such datasets, language instructi ons are highly predictable from visual observations alone, causing the condition al mutual information between instructions and actions to vanish, a phenomenon w e term Information Collapse. Consequently, models degenerate into vision-only po licies that ignore language constraints. To address this, we propose LangForce, enforces instruction following via Bayesian decomposition. By introducing learna ble Latent Action Queries, we construct a dual-branch architecture to estimate b oth a vision-only prior $p(a \mid v)$ and a language-conditioned posterior $\pi(a \mid v, \ell)$. We then optimize the policy to maximize the conditional Pointw ise Mutual Information (PMI) between actions and instructions. This objective ef fectively penalizes the vision shortcut and rewards actions that explicitly expl ain the language command. Extensive experiments across on three benchmarks demon strate substantial gains, including an 11.3\% improvement on the challenging OOD SimplerEnv benchmark, validating the ability of LangForce to robustly ground la nguage in action.
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Semantic-Enriched Latent Visual Reasoning
Tianrun Xu, Yue Sun, Qixun Wang, Jingyi Lu, Yuan Wang, Tianren Zhang, Longteng G uo, Fengyun Rao, Jing LYU, Feng Chen, Jing Liu
Multimodal latent-space reasoning aims to replace explicit “thinking with images ” by performing visual reasoning directly in a compact latent space. However, ex

isting approaches largely rely on visual supervision and produce latent representations that lack sufficient semantic richness, limiting their ability to support diverse region-level reasoning tasks. In this work, we introduce \textbf{Semantic-Enriched Latent Visual Reasoning (SLVR)}, a two-stage learning framework that enriches latent representations with attribute-level visual semantics and aligns them with diverse reasoning objectives. In the first stage, SLVR learns semantically enriched region-centric latents under fine-grained attribute supervision. In the second stage, we design Multi-query Group Relative Policy Optimization (M-GRPO) to align latent representations across multiple queries grounded in the same region. To support this framework, we construct \textbf{SLV-Set}, comprising approximately 400K region-level attribute annotations and 800K multi-query question answering samples, and introduce \textbf{SV-QA}, a benchmark that evaluates latent reasoning under semantic variation. Experiments demonstrate that SLVR improves the robustness and semantic consistency of latent visual reasoning compared to existing baselines. Our code and datasets are available at \url{https://github.com/tinnel123666888/slvr}.
****

Transolver-3: Scaling Up Transformer Solvers to Industrial-Scale Geometries
Hang Zhou, Haixu Wu, Haonan Shangguan, Yuezhou Ma, Huikun Weng, Jianmin Wang, Mingsheng Long
Deep learning has emerged as a transformative tool for the neural surrogate modeling of partial differential equations (PDEs), known as neural PDE solvers. However, scaling these solvers to industrial-scale geometries with over 10^8 cells remains a fundamental challenge due to the prohibitive memory complexity of processing high-resolution meshes. We present Transolver-3, a new member of the Transolver family as a highly scalable framework designed for high-fidelity physics simulations. To bridge the gap between limited GPU capacity and the resolution requirements of complex engineering tasks, we introduce two key architectural optimizations:
faster slice and deslice by exploiting matrix multiplication associative property and geometry slice tiling to partition the computation of physical states.
Combined with an amortized training strategy by learning on random subsets of original high-resolution meshes and a physical state caching technique during inference, Transolver-3 enables high-fidelity field prediction on industrial-scale meshes. Extensive experiments demonstrate that Transolver-3 can handle meshes with over 160 million cells, achieving impressive performance across three challenging simulation benchmarks, including aircraft and automotive design tasks. Code is available at https://github.com/thuml/Transolver-3.
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Hidden in Plain Sight -- Class Competition Focuses Attribution Maps
Nils Philipp Walter, Jilles Vreeken, Jonas Fischer
Attribution methods reveal which input features a neural network uses for a prediction, adding transparency to their decisions. A common problem is that these attributions seem unspecific, highlighting both important and irrelevant features. We revisit the common attribution pipeline and observe that using logits as attribution target is a main cause of this phenomenon. We show that the solution is in plain sight: considering distributions of attributions over multiple classes using existing attribution methods yields specific and fine-grained attributions. On common benchmarks, including the grid-pointing game and randomization-based sanity checks, this improves the ability of 18 attribution methods across 7 architectures up to $2\times$, agnostic to model architecture.
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RLVE: Scaling Up Reinforcement Learning for Language Models with Adaptive Verifiable Environments
Zhiyuan Zeng, Hamish Ivison, Yiping Wang, Lifan Yuan, Shuyue Stella Li, Zhuorui Ye, Siting Li, Jacqueline He, Runlong Zhou, Tong Chen, Chenyang Zhao, Yulia Tsvetkov, Simon Shaolei Du, Natasha Jaques, Hao Peng, Pang Wei Koh, Hannaneh Hajishirzi
We introduce Reinforcement Learning (RL) with Adaptive Verifiable Environments (RLVE), an approach using verifiable environments that procedurally generate prob

lems and provide algorithmically verifiable rewards, to scale up RL for language models (LMs). RLVE enables each verifiable environment to dynamically adapt its problem difficulty distribution to the policy model's capabilities as training progresses. In contrast, static data distributions often lead to vanishing learning signals when problems are either too easy or too hard for the policy. To implement RLVE, we create RLVE-Gym, a large-scale suite of 400 verifiable environments carefully developed through manual environment engineering. Using RLVE-Gym, we show that environment scaling, i.e., expanding the collection of training environments, consistently improves generalizable reasoning capabilities. RLVE with joint training across all 400 environments in RLVE-Gym yields a 3.37% absolute average improvement across six reasoning benchmarks, starting from one of the strongest 1.5B reasoning LMs. By comparison, continuing this LM's original RL training yields only a 0.49% average absolute gain despite using over 3x more compute. We release our code publicly.
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Anchored Decoding: Provably Reducing Copyright Risk for Any Language Model
Jacqueline He, Jonathan Hayase, Wen-tau Yih, Sewoong Oh, Luke Zettlemoyer, Pang Wei Koh

Language models (LMs) tend to memorize portions of their training data and reproduce verbatim spans. When the underlying sources are sensitive or copyright-protected, such reproduction raises issues of consent and compensation for creators and compliance risks for developers.

We propose Anchored Decoding, a plug-and-play inference-time method for suppressing verbatim reproduction: it enables decoding from any risky LM trained on mixed-license data by keeping generation in bounded proximity to a permissively trained safe LM.

Anchored Decoding does so by adaptively allocating a user-chosen information budget over the generation trajectory and enforcing per-step constraints that yield a sequence-level guarantee, enabling a tunable risk–utility trade-off.

To make Anchored Decoding practically useful, we introduce a new permissively trained safe model (TinyComma 1.8B), as well as Anchored-Byte Decoding, a byte-level variant of our method that enables cross-vocabulary fusion via the ByteSampler (Hayase et al., 2025) framework.

Across six model pairs on long-form metrics for copying risk and utility, Anchored and Anchored-Byte Decoding define a new Pareto frontier, preserving near-original fluency and factuality while eliminating up to 75\% of the measurable copying gap between the risky baseline and a safe reference, at a modest inference overhead.
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CREDIT: Certified Ownership Verification of Deep Neural Networks Against Model Extraction Attacks
Bolin Shen, Zhan Cheng, Neil Zhenqiang Gong, Fan Yao, Yushun Dong

Machine Learning as a Service (MLaaS) has emerged as a widely adopted paradigm for providing access to deep neural network (DNN) models, enabling users to conveniently leverage these models through standardized APIs. However, such services are highly vulnerable to Model Extraction Attacks (MEAs), where an adversary repeatedly queries a target model to collect input-output pairs and uses them to train a surrogate model that closely replicates its functionality. While numerous defense strategies have been proposed, verifying the ownership of a suspicious model with strict theoretical guarantees remains a challenging task. To address this gap, we introduce CREDIT, a certified ownership verification against MEAs. Specifically, we employ mutual information to quantify the similarity between DNN models, propose a practical verification threshold, and provide rigorous theoretical guarantees for ownership verification based on this threshold. We extensively evaluate our approach on several mainstream datasets across different domains and tasks, achieving state-of-the-art performance. Our implementation is publicly available at: https://github.com/LabRAI/CREDIT.
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FAIL: Flow Matching Adversarial Imitation Learning for Image Generation
Yeyao Ma, Chen Li, Xiaosong Zhang, Han Hu, Weidi Xie

Post-training of flow matching models—aligning the output distribution with a hi
gh-quality target—is mathematically equivalent to imitation learning. While Supe
rvised Fine-Tuning mimics expert demonstrations effectively, it cannot correct p
olicy drift in unseen states. Preference optimization methods address this but r
equire costly preference pairs or reward modeling. We propose Flow Matching Adve
rsarial Imitation Learning (FAIL), which minimizes policy-expert divergence thro
ugh adversarial training without explicit rewards or pairwise comparisons. We de
rive two algorithms: FAIL-PD exploits differentiable ODE solvers for low-varianc
e pathwise gradients, while FAIL-PG provides a black-box alternative for discret
e or computationally constrained settings. Fine-tuning FLUX with only 13,000 dem
onstrations from Nano Banana pro, FAIL achieves competitive performance on promp
t following and aesthetic benchmarks. Furthermore, the framework generalizes eff
ectively to discrete image and video generation, and functions as a robust regul
arizer to mitigate reward hacking in reward-based optimization.
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Revealing Long-context Potential of Attention Heads via Frequency Kernels
Senyu Han, Yilu Cao, Kai Yu, Lu Chen
Large language model (LLM) exists a subset of attention heads that are highly re
sponsible for long-context processing. Existing work has identified different lo
ng-context heads in models, but their detection methods mainly rely on model inf
erence on actual long texts and do not analyze the inherent properties of the he
ad parameters. In this paper, we use kernel methods to analyze static \textit{fr
equency kernels} formed by different rotation frequency components of attention
heads, and we design a Long-context Potential Score (LPS) to measure the potenti
al of attention heads in processing long contexts. Kernels of heads with high LP
S exhibit concentrated low-frequency energy and low effective rank, which allow
them to effectively capture highly specialized information from distant contexts
. Experiments and analysis on long-context tasks and model behaviors show that t
he LPS metrics can well reflect the actual capability of heads on long contexts.

Furthermore, by simply amplifying low-frequency kernels of heads with high retri
eval potential, we can further improve model's performance on long-context tasks
. Our metrics and head enhancement methods are fully static and offline, and the
y can be quickly conducted under low-resource constraints. Code is publicly avai
lable at [here](https://github.com/OpenDFM/long-context-heads).
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UniFast-HGR: Scalable and Efficient Maximal Correlation for Multimodal Models
Hongkang Zhang, Shao-Lun Huang, Yanlong Wang, Ercan Engin KURUOGLU
This paper presents UniFast-HGR, a scalable surrogate for Hirschfeld--Gebelein--
R\'enyi (HGR) maximal correlation in high-dimensional multimodal learning. The m
ethod replaces explicit covariance whitening with centered and ℓ_2-normaliz
ed cosine alignment, rewrites the covariance interaction in local-batch Gram spa
ce, and removes invariant diagonal self-correlation through Trivial Spectrum Sup
pression (TSS). This reformulation preserves the standardized dependence-maximiz
ation target of HGR-style learning while replacing the finite-sample whitening e
stimator with a differentiable local-batch objective whose dominant structural c
ost is $O(m^2K)$ for local batch size m and feature dimension K. OptFast-HGR
 further reduces the practical memory burden by estimating the off-diagonal stru
ctural term through stochastic projection. Experiments across retrieval, image c
lassification, remote sensing segmentation, and multimodal emotion recognition s
how consistent gains over covariance-based HGR/CCA variants and contrastive or n
eural MI-estimator objectives on strong multimodal backbones, while microbenchma
rks confirm stable behavior at extreme feature dimensions.
****

LiveOIBench: Can Large Language Models Outperform Human Contestants in Informati
cs Olympiads?
Kaijian Zou, Aaron Xiong, Yunxiang Zhang, Xinliang Frederick Zhang, Yueqi Ren, J
irong Yang, Ayoung Lee, Shitanshu Bhushan, Lu Wang
Competitive programming problems are increasingly used to evaluate the coding ca
pabilities of large language models (LLMs) due to their complexity and ease of v

erification.
Yet, current coding benchmarks face limitations such as a lack of exceptionally challenging problems, insufficient test case coverage, and reliance on online platform APIs that limit accessibility.
To address these issues, we introduce LiveOIBench, a large-scale competitive programming benchmark featuring 403 expert-curated problems, averaging 60 official test cases each, drawn from 72 contests across 14 Informatics Olympiads held between 2023 and 2025.
LiveOIBench has four key features:
(1) expert-designed tasks with detailed subtask rubrics and extensive test cases;
(2) direct comparison to elite human contestants;
(3) continuous updates to reduce contamination risk; and
(4) a fully offline, reproducible evaluation system.
Benchmarking 34 popular general-purpose and reasoning LLMs, we find that GPT-5 achieves an 81.76th percentile, still falling short of top human contestants, while among the open-weight models, GPT-OSS-120B reaches only the 60th percentile.
Reasoning-trace analyses indicate that robust reasoning models prioritize precise problem analysis over excessive exploration.
Finally, analyses across release dates, task familiarity, and code similarity find minimal evidence of data contamination in our benchmark.
Our leaderboard, code, and data are available at: https://liveoibench.github.io/.
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On Structured State-Space Duality
Jerry Yao-Chieh Hu, Xiwen Zhang, Ali ElSheikh, Weimin Wu, Han Liu
Structured State-Space Duality (SSD) [Dao \& Gu, ICML 2024] is an equivalence between a simple Structured State-Space Model (SSM) and a masked attention mechanism.
In particular, a state-space model with a scalar-times-identity state matrix is equivalent to a masked self-attention with a 1-semiseparable causal mask.
Consequently, the same sequence transformation (model) has two algorithmic realizations:
as a linear-time $O(T)$ recurrence or as a quadratic-time $O(T^2)$ attention.
In this note, we formalize and generalize this duality:
(i)
we extend SSD from the scalar■identity case to general diagonal SSMs (diagonal state matrices);
(ii) we show that these diagonal SSMs match the scalar case's training complexity lower bounds while supporting richer dynamics;
(iii) we establish a necessary and sufficient condition under which an SSM is equivalent to 1-semiseparable masked attention;
and
(iv) we show that such duality fails to extend to standard softmax attention due to rank explosion.
Together, these results tighten the bridge between recurrent SSMs and Transformers, and widen the design space for expressive yet efficient sequence models.
Code is available at https://github.com/MAGICS-LAB/state_space_daulity.
****
FlexRank: Nested Low-Rank Knowledge Decomposition for Adaptive Model Deployment
Riccardo Zaccone, Stefanos Laskaridis, Marco Ciccone, Samuel Horváth
The growing scale of deep neural networks, encompassing large language models (LLMs) and vision transformers (ViTs), has made training from scratch prohibitively expensive and deployment increasingly costly.
These models are often used as computational monoliths with fixed cost, hindering adaptive deployment across different cost budgets.
We argue that nested components, ordered by importance, can be extracted from pretrained models and selectively activated within the available computational budget. To this end, our proposed FlexRank method leverages low-rank weight decompo

sition with nested, importance-based consolidation to extract submodels of increasing capabilities. Our approach enables a _``train-once, deploy-everywhere''_ paradigm offering a graceful trade-off between cost and performance without training from scratch for each budget - advancing practical deployment of large models.
****

Conflict-Aware Additive Guidance for Flow Models under Compositional Rewards
Xuehui Yu, Fucheng Cai, WANG MEIYI, Xiaopeng Fan, Harold Soh
Inference-time guided sampling steers state-of-the-art diffusion and flow models without fine-tuning by interpreting the generation process as a controllable trajectory. This provides a simple and flexible way to inject external constraints (e.g., cost functions or pre-trained verifiers) for controlled generation. However, existing methods often fail when composing multiple constraints simultaneously, which leads to deviations from the true data manifold. In this work, we identify root causes of this off-manifold drift and find that the approximation error scales severely with gradient misalignment. Building on these findings, we propose Conflict-Aware Additive Guidance (g^car), a lightweight and learnable method, which actively rectifies off-manifold drift by dynamically detecting and resolving gradient conflicts. We validate g^car across diverse domains, ranging from synthetic datasets and image editing to generative decision-making for planning and control. Our results demonstrate that g^car effectively rectifies off-manifold drift, surpassing baselines in generation fidelity while using light compute. Code is available at <https://github.com/yuxuehui/CAR-guidance>.
****

Global Policy-Space Response Oracles for Two-Player Zero-Sum Games
Junyu Zhang, Feihong Yang, Jian Wang, Chao Wang, Xudong Zhang
The Policy-Space Response Oracles (PSRO) framework scales equilibrium computation to large zero-sum games by iteratively expanding a restricted strategy set using deep reinforcement learning (DRL). A central challenge is to construct, under limited computational budgets, a small strategy population whose induced game well approximates the full game. Existing PSRO variants typically expand the population using best responses to meta-strategies computed from restricted-game payoffs, which can lead to inefficient expansions that provide limited global improvement. We propose to guide population expansion by directly evaluating the post-expansion population quality. Specifically, we adopt Population Exploitability (PE) to measure how well a restricted strategy set represents the full game, and introduce a two-phase exploration--selection framework that explicitly minimizes PE during expansion. We instantiate this framework as Global PSRO, a practical DRL-based algorithm that efficiently generates candidate responses and estimates PE via parameter-sharing conditional neural networks. Experiments across multiple two-player zero-sum games show that Global PSRO achieves lower exploitability and approximates Nash equilibria with significantly fewer policy iterations than prior PSRO methods. The experimental code is available at https://github.com/Zhangjy1997/GlobalPSRO.
****

Optimal Transport Group Counterfactual Explanations
Enrique Valero-Leal, Bernd Bischl, Pedro Larrañaga, Concha Bielza, Giuseppe Casalicchio
Group counterfactual explanations find a set of counterfactual instances to explain a group of input instances contrastively. However, existing methods either (i) optimize counterfactuals only for a fixed group and do not generalize to new group members, (ii) strictly rely on strong model assumptions (e.g., linearity) for tractability and/or (iii) poorly control the counterfactual group geometry distortion. We instead learn an explicit optimal transport map that sends any group instance to its counterfactual without re-optimization, minimizing the group's total transport cost. This enables generalization with fewer parameters, making it easier to interpret the common actionable recourse. For linear classifiers, we prove that functions representing group counterfactuals are derived via mathematical optimization, identifying the underlying convex optimization type (QP,

QCQP, ...). Experiments show that they accurately generalize, preserve group geometry and incur only negligible additional transport cost compared to baseline methods. If model linearity cannot be exploited, our approach also significantly outperforms the baselines.
****

Geometry-Aware Contrastive Learning for Few-Shot Automatic Modulation Recognition

guanqun zhao, Liu Yitong, Jiaxuan Fang, Yufei Mao, Hongwen Yang

Standard Self-Supervised Learning (SSL) for Automatic Modulation Recognition (AMR) struggles with ineffective isotropic augmentations, spectral instability, and semantic drift. To address these challenges, we propose Dynamic-Consistency Contrastive Learning (DyCo-CL), a geometry-aware framework that couples Virtual Adversarial Augmentation (VAA) with a semantic consistency loss. We provide a theoretical analysis indicating that this strategy acts as an implicit spectral regularizer for the encoder, enabling stable manifold exploration. Complementing this, our Signal-Adaptive Swin Backbone with fixed-window attention improves structural stability by constraining attention locality, while a Hybrid Knowledge Fusion module anchors representations with physical priors. Experiments on RML benchmarks show that DyCo-CL achieves a 6.27% accuracy gain in 1-shot settings over prior methods.
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PISA: Privacy-Preserving Split Adaptation with Model IP Protection

Haocheng Yang, Xiang Cheng, ZONGDA HAN, Pengjie Wang, Changkang Chi, Pengfei Zhang, Sen Su

Fine-tuning Large Language Models (LLMs) enables data holders to construct proprietary, task-specific models by leveraging external high-performance computing infrastructure.

However, existing paradigms typically address data privacy and model intellectual property (IP) in isolation, failing to simultaneously uphold both constraints. Privacy-prioritized methods compromise model IP by hosting parameters remotely, while IP-oriented collaborative schemes relying on end-to-end gradient flows inherently violate strict data privacy standards. To address these challenges, we present **PISA** (**P**rivacy-preserving and **I**P-protected **S**plit **A**daptation), a split fine-tuning framework designed to preserve both data privacy and model IP while maintaining high utility.

In PISA, we propose three methods: a Manifold Rectification Pre-training (MRP) method to equip the server-side model with intrinsic robustness against privacy-induced distribution shifts; a Dual-Stream Semantic Compensation (DSC) method to recover feature utility using local clean data as priors; and a Utility-Aware Gradient Rectification (UGR) method to adaptively maximize the performance of the parameter-constrained local model. Experiments on GLUE show that PISA ensures dual protection and delivers a substantial 23.0\% performance gain over the privacy-prioritized baseline under strict privacy budgets.
****

Revisiting Spectral Representations in Generative Diffusion Models

Yuehao Wang, Peihao Wang, Hanwen Jiang, Ziyi Yang, Qixing Huang, Zhangyang Wang

Diffusion models have shown remarkable performance on diverse generation tasks. Recent work finds that imposing representation alignment on the hidden states of diffusion networks can both facilitate training convergence and enhance sampling quality, yet the mechanism driving this synergy remains insufficiently understood. In this paper, we investigate the connection between self-supervised spectral representation learning and diffusion generative models through a shared perspective on perturbation kernels. On the diffusion side, samples (e.g., images, videos) are produced by reversing a stochastic noise-injection process specified by Gaussian kernels; on the spectral representation side, spectral embeddings emerge from contrasting positive and negative relations induced by random perturbation kernels. Motivated by this, we propose a self-supervised spectral representation alignment method to facilitate diffusion model training. In addition, we clarify how joint spectral learning can benefit diffusion training from a geometric perspective. Furthermore, we find that the optimization of the spectral align

ment objective is in an equivalent form of diffusion score distillation in the r
epresentation space. Building on these findings, we integrate a spectral regular
izer into diffusion training objectives to improve the performance of diffusion
models on multiple datasets. Experiments across images and 3D point clouds show
consistent gains in generation quality.
****

Least-Loaded Expert Parallelism: Load Balancing An Imbalanced Mixture-of-Experts
Xuan-Phi Nguyen, Shrey Pandit, Austin Xu, Caiming Xiong, Shafiq Joty
Mixture-of-Experts (MoE) models are typically pre-trained with explicit load-bal
ancing constraints to ensure statistically balanced expert routing. Despite this
, we observe that even well-trained MoE models exhibit significantly imbalanced
routing. This behavior is arguably natural—and even desirable—as imbalanced rout
ing allows models to concentrate domain-specific knowledge within a subset of ex
perts. Expert parallelism (EP) is designed to scale MoE models by distributing e
xperts across multiple devices, but with a less-discussed assumption of balanced
 routing. Under extreme imbalance, EP can funnel a disproportionate number of to
kens to a small number of experts, leading to compute- and memory-bound failures
 on overloaded devices during post-training or inference, where explicit load ba
lancing is often inapplicable. We propose Least-Loaded Expert Parallelism (LLEP)
, a novel EP algorithm that dynamically reroutes excess tokens and associated ex
pert parameters from overloaded devices to underutilized ones. This ensures that
 all devices complete their workloads within the minimum collective latency whil
e respecting memory constraints. Across different model scales, LLEP achieves up
 to 5x speedup and 4x reduction in peak memory usage compared to standard EP. Th
is enables faster and higher-throughput post-training and inference, with ~1.9x
faster for gpt-oss-120b. We support our method with extensive theoretical analys
is and comprehensive empirical evaluations, including ablation studies. These re
sults illuminate key trade-offs and enable a principled framework for hardware-s
pecific hyper-parameter tuning to achieve optimal performance.
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Hierarchical Reinforcement Learning for Sparse-Reward Search in Commutative Alge
bra
Giorgi Butbaia, Paul Orland, Coco Huang, Davide Passaro, Lucas Fagan, Michele Ta
rquini, Hailong Dao, David Eisenbud, Ali Shehper, Sergei Gukov
Applying machine learning techniques to solving long-standing mathematical conje
ctures can be particularly challenging due to their extreme reward sparsity. As
an illustrative example, we consider Kalai's algebraic Hirsch conjecture and rec
ast the construction of its counterexamples as a sparse-reward reinforcement lea
rning problem on graphs. We propose a constrained options-based HRL framework wi
th an equivariant graph neural network policy, which allows us to learn useful t
emporal abstractions for this task. We evaluate our approach over a wide range o
f degrees and demonstrate that it consistently outperforms classical RL algorith
ms as well as greedy search. By exploiting the hierarchical structure of the pro
blem, we effectively provide a first-of-its-kind application of HRL to a problem
 in commutative algebra.
****

Capacity-Aware Mixture Law Enables Efficient LLM Data Optimization
Jingwei Li, Xinran Gu, Jingzhao Zhang
A data mixture refers to how different data sources are combined to train large
language models, and selecting an effective mixture is crucial for optimal downs
tream performance. Existing methods either conduct costly searches directly on t
he target model or rely on mixture scaling laws that fail to extrapolate well to
 large model sizes.
We address these limitations by introducing a compute-efficient pipeline for dat
a mixture scaling. First, we propose CAMEL, a capacity-aware mixture law that mo
dels validation loss with the nonlinear interplay between model size and mixture
. We also introduce a loss-to-benchmark prediction law that estimates benchmark
accuracy from validation loss, enabling end-to-end performance prediction for th
e target model. Next, we study how to allocate a fixed compute budget across mod
el scales to fit the law and reduce prediction error. Finally, we apply our meth

od to Mixture-of-Experts models with up to 7B-A150M parameters to fit the law, a
nd verify the optimal mixture derived from the law by extrapolating to a 55B-A1.
2B target model. Compared to prior methods, we reduce mixture optimization costs
 by 50\% and improves downstream benchmark performance by up to 3\%.
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PrivCode++: Latent-Conditioned Differentially Private Code Generation for Compre
hensive Guarantees

Zheng Liu, Chen GONG, Terry Yue Zhuo, Zhou Yang, Kecen Li, Wenlong Meng, Xinwen
Hou, Yu Liu, Xiaochen Li

Large language models fine-tuned on instruction–code pairs may memorize and subs
equently leak sensitive training data. Existing differentially private (DP) code
 generation methods primarily protect code snippets while assuming prompts are p
ublic, which fails in realistic scenarios where prompts may also contain sensiti
ve information. When prompts cannot be explicitly learned or used during generat
ion, code synthesis suffers from severe utility degradation as well as reduced d
iversity and fidelity. To address these challenges, we propose PrivCode++, the f
irst work to explore DP code generation where both prompts and code snippets are
 considered sensitive in LLM fine-tuning. PrivCode++ introduces a two-stage DP f
ramework with a Privacy-Free Latent Conditioning module, enabling effective DP f
ine-tuning and data synthesis without direct access to sensitive prompts or code
. Extensive experiments show that PrivCode++ achieves substantially higher utili
ty than baselines, remains competitive with the method with relaxing privacy ass
umptions, and provides stronger privacy guarantees.
****

Tensor Train Diffusion: Leveraging Low-Rank Structures for High-Dimensional Scor
e-Based Sampling

Robert Gruhlke, Julius Berner, David Sommer, Lorenz Richter

Diffusion models offer a powerful framework for sampling from complex probabilit
y densities by learning to reverse a noising process. A common approach involves
 solving for the time-reversed stochastic differential equation (SDE), which req
uires the score function of the evolving sample distribution. The logarithm of t
his distribution's density is governed by a Hamilton-Jacobi-Bellman (HJB) type p
artial differential equation (PDE). However, current methods for solving this PD
E, such as PINNs or trajectory-based techniques, often suffer from long training
 times and significant sensitivity to hyperparameter tuning. In this work, we in
troduce a novel and efficient solver for the underlying HJB equation based on th
e functional tensor train (FTT) format. The FTT representation leverages latent
low-rank structures to efficiently approximate high-dimensional functions, enabl
ing both model compression and rapid computation. By integrating this efficient
representation with a backward-in-time iterative scheme derived from backward st
ochastic differential equations (BSDEs), we develop a fast, robust and accurate
sampling method. Our approach overcomes primary bottlenecks of existing techniqu
es, enabling high-fidelity sampling from challenging target distributions with i
mproved efficiency.
****

Trajectory Seriation via Spectral Tangent Alignment and Global Embedding

Zhixin Zhou, Navin Souda, Arash A. Amini

We study linear seriation from noisy point clouds: given samples from an unknown
 one-dimensional curve embedded in $\mathbb{R}^d$, recover their latent order al
ong the curve, up to reversal. Unlike much of the seriation literature, which st
arts from a precomputed similarity matrix, this setting retains ambient coordina
tes and therefore local geometric information. We propose STAGE, a geometric ord
ering method that estimates local tangent directions by neighborhood PCA, resolv
es their sign ambiguity through a graph-wide $\mathbb{Z}_2$ synchronization step
, and constructs signed local displacement estimates by projecting ambient diffe
rences onto the oriented tangents. These increments are then integrated into a g
lobal scalar embedding through an inhomogeneous least-squares problem, equivalen
tly a Laplacian linear system with a nonzero right-hand side, and the final orde
r is obtained by sorting the embedding. We prove a finite-sample Kendall's $\tau
$ recovery bound that makes explicit the roles of curvature, noise, neighborhood

scale, sampling density, and graph connectivity. Empirically, STAGE gives accurate and fast order recovery on high-dimensional synthetic curves, compares favorably with spectral seriation, 1D UMAP, 1D t-SNE, Recanati's method, LTSA, and SABRE, and produces meaningful pseudotime orderings on single-cell RNA-seq datasets.

****

DiP-G: Discrete Prompting for Graph Neural Networks

Yumeng Zhao, Huiying Hu, Shuo Wen, Junjie Shen, Bei Hua

Graph Neural Networks (GNNs) are increasingly adopting the "pre-training, adaptation" paradigm, which first pre-train GNNs on large-scale unlabeled graph data and then adapt them to specific downstream tasks. As a common pattern, graph prompting adapts to the frozen encoder by modifying the input graph structure, rather than fine-tuning the model parameters. However, most existing graph prompting approaches optimize the continuous and weighted adjacency structure in the adaptation phase, while requiring a hard discretization at inference time. This difference causes a train-test mismatch which is particularly harmful in few-shot regimes. To address the issue, we propose **Di**screte **P**rompting for **G**raphs, a discrete prompting framework that directly learns task-specific topology prompts in the combinatorial space. DiP-G operates on multi-hop local candidate subgraphs to ensure scalability, generates hard (k)-sparse prompts through a perturbed Top-(k) solver, and optimizes the discrete structures using an I-MLE gradient estimator. To improve the efficiency of backward pass, we further introduce an adaptive active-set screening rule that accelerates the target solve and can provably maintain the accuracy of the solution. Extensive experiments conducted on multiple benchmark datasets have validated the effectiveness of our proposed method.

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Detecting the Semantic Fixed Point: A Geometric Framework for Efficient Inference

Jiawei Gu, Ziyue Qiao, Xiao Luo

Each layer of a Transformer refines the hidden state toward a prediction, an iterative process resembling fixed-point iteration. Yet when should this iteration terminate? Existing early exit methods rely on output confidence as a proxy for internal convergence. We take a more direct approach by examining the geometry of the hidden state trajectory. We find that layer-wise updates exhibit a two-phase structure: large, volatile updates in early layers, followed by small, aligned updates as the model propagates an already-formed representation. The transition is remarkably sharp. This yields a simple criterion: exit when step size vanishes and direction stabilizes. We track the normalized update norm and cosine similarity between consecutive updates, exiting when both indicate convergence. The overhead is $O(d)$ per layer, independent of vocabulary size, requiring no learned components or architectural modifications. On LLaMA-2-7B and LLaMA-2-13B across question answering and commonsense reasoning tasks, this geometric criterion reduces FLOPs by 30--35\% while retaining over 98\% of full-depth accuracy.

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Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement

Jinhao Pan, Chahat Raj, Anjishnu Mukherjee, Sina Mansouri, Bowen Wei, Shloka Yada, Ziwei Zhu

Large language models (LLMs) exhibit social biases that reinforce harmful stereotypes, limiting their safe deployment. Most existing debiasing methods adopt a suppressive paradigm by modifying parameters, prompts, or neurons associated with biased behavior; however, such approaches are often brittle, weakly generalizable, data-inefficient, and prone to degrading general capability. We propose \textbf{KnowBias}, a lightweight and conceptually distinct framework that mitigates bias by strengthening, rather than suppressing, neurons encoding bias-knowledge. KnowBias identifies neurons encoding bias knowledge using a small set of bias-knowledge questions via attribution-based analysis, and selectively enhances them at inference time. This design enables strong debiasing while preserving general capabilities, generalizes across bias types and demographics, and is highly da

ta efficient, requiring only a handful of simple yes/no questions and no retrain ing. Experiments across multiple benchmarks and LLMs demonstrate consistent stat e-of-the-art debiasing performance with minimal utility degradation. Data and co de are available at \url{https://github.com/JP-25/KnowBias}.
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Mixture of Distributions Matters: Dynamic Sparse Attention for Efficient Video D iffusion Transformers
Yuxi Liu, Yipeng Hu, Zekun Zhang, Kunze Jiang, Kun Yuan
While Diffusion Transformers (DiTs) have achieved notable progress in video gene ration, this long-sequence generation task remains constrained by the quadratic complexity inherent to self-attention mechanisms, creating significant barriers to practical deployment. Although sparse attention methods attempt to address th is challenge, existing approaches either rely on oversimplified static patterns or require computationally expensive sampling operations to achieve dynamic spar sity, resulting in inaccurate pattern predictions and degraded generation qualit y. To overcome these limitations, we propose a \underline{\textbf{M}}ixture-\und erline{\textbf{O}}f-\underline{\textbf{D}}istribution \textbf{DiT} (\textbf{MOD-DiT}), a novel sampling-free dynamic attention framework that accurately models evolving attention patterns through a two-stage process. First, MOD-DiT leverage s prior information from early denoising steps and adopts a {distributed mixing approach} to model an efficient linear approximation model, which is then used t o predict mask patterns for a specific denoising interval. Second, an online blo ck masking strategy dynamically applies these predicted masks while maintaining historical sparsity information, eliminating the need for repetitive sampling op erations. Extensive evaluations demonstrate consistent acceleration and quality improvements across multiple benchmarks and model architectures, validating MOD-DiT's effectiveness for efficient, high-quality video generation while overcomin g the computational limitations of traditional sparse attention approaches.
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TokenDrop: Token-Level Importance-Aware Backward Propagation Skipping for Effici ent LLM Fine-Tuning
Beomseok Kim, Sol Namkung, Dongsuk Jeon
Despite the success of parameter-efficient fine-tuning (PEFT) methods in reducin g parameter-related overhead, fine-tuning large language models (LLMs) is still bottlenecked by significant memory and computational demands. In this paper, we propose **TokenDrop**, a token-level importance-aware backpropagation skipping m ethod that reduces activation memory and accelerates LLM fine-tuning by skipping backward computations for less informative tokens. TokenDrop evaluates token im portance based on the magnitude of residual updates during the forward pass, ena bling lightweight, gradient-free importance estimation. Furthermore, we introduc e cumulative token selection to preserve gradient continuity across layers and l azy selection scheduling that defers token selection to facilitate globally info rmed importance scoring under memory constraints. Across a range of experiments, TokenDrop achieves up to **42.9**\% reduction in memory usage and up to **1.50* *$\times$ training speedup, while preserving accuracy and outperforming existing backpropagation-skipping baselines. The code is available at https://github.com /kimbss470/tokendrop_official.
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PuzzleMoE: Efficient Compression of Large Mixture-of-Experts Models via Sparse E xpert Merging and Bit-packed inference
Yushu Zhao, Zheng Wang, Minjia Zhang
Mixture-of-Experts (MoE) have shown strong potential in scaling language models efficiently by activating only a small subset of experts per input. However, the ir deployment remains limited due to the high memory overhead associated with st oring all expert parameters, particularly as the number of experts increases. To address this challenge, prior works have explored expert dropping and merging s trategies; however, they often suffer from notable performance drop especially a t high compression ratios due to their reliance on coarse-grained tensor- or exp ert-level operations. In this paper, we introduce PuzzleMoE, the first MoE mergi ng method to enable fine-grained element-wise merging while achieving both high

accuracy and inference speed, via two key innovations: First, PuzzleMoE performs sparse expert merging by identifying element-wise weight redundancy and specialization. It introduces a dual-mask approach to capture both shared and expert-specific salient parameters. Second, to avoid the overhead of storing masks and signs, we introduce a bit-packed encoding scheme that reuses underutilized exponent bits, enabling efficient MoE inference on GPUs. Extensive experiments demonstrate that PuzzleMoE outperforms prior MoE compression methods by up to 16.7\% on MMLU at 50\% compression ratio, and achieves up to 1.80$\times$ end-to-end inference throughput gain.
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Find, Fix, Reason: Context Repair for Video Reasoning
Haojian Huang, Chuanyu Qin, Yinchuan Li, Ying-Cong Chen
Reinforcement learning has advanced video reasoning in large multi-modal models, yet dominant pipelines either rely on on-policy self-exploration, which plateaus at the model's knowledge boundary, or hybrid replay that mixes policies and demands careful regularization. Dynamic context methods zoom into focused evidence but often require curated pretraining and two-stage tuning, and their context remains bounded by a small model's capability. In contrast, larger models excel at instruction following and multi-modal understanding, can supply richer context to smaller models, and rapidly zoom in on target regions via simple tools. Building on this capability, we introduce an observation-level intervention: a frozen, tool-integrated teacher identifies the missing spatiotemporal dependency and provides a minimal evidence patch (e.g., timestamps, regions etc.) from the original video while the question remains unchanged. The student answers again with the added context, and training updates with a chosen-rollout scheme integrated into Group Relative Policy Optimization (GRPO). We further propose a Robust Improvement Reward (RIR) that aligns optimization with two goals: outcome validity through correct answers and dependency alignment through rationales that reflect the cited evidence. Advantages are group-normalized across the batch, preserving on-policy exploration while directing it along causally meaningful directions with minimal changes to the training stack. Experiments on various related benchmarks show consistent accuracy gains and strong generalization. Project page is available at https://jethrojames.github.io/FFR/.
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Multi-Scale Wavelet Transformers for Operator Learning of Dynamical Systems
Xuesong Wang, Michael Groom, Rafael Oliveira, He Zhao, Terence O'kane, Edwin V. Bonilla
Recent years have seen a surge in data-driven surrogates for dynamical systems that can be orders of magnitude faster than numerical solvers. However, many machine learning-based models such as neural operators exhibit spectral bias, attenuating high-frequency components that often encode small-scale structure. This limitation is particularly damaging in applications such as weather forecasting, where misrepresented high frequencies can induce long-horizon instability. To address this issue, we propose multi-scale wavelet transformers (MSWTs), which learn system dynamics in a tokenized wavelet domain. The wavelet transform explicitly separates low- and high-frequency content across scales. MSWTs leverage a wavelet-preserving downsampling scheme that retains high-frequency features and employ wavelet-based attention to capture dependencies across scales and frequency bands. Experiments on chaotic dynamical systems show substantial error reductions and improved long-horizon spectral fidelity. On the ERA5 climate reanalysis, MSWTs further reduce climatological bias, demonstrating their effectiveness in a real-world forecasting setting.
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The Loss Is Not Enough: Sampling Conditions and Inductive Bias in Contrastive Representation Learning
Justinas Zaliaduonis, Patrick Putzky, Till Richter, Sergios Gatidis
Contrastive learning has become a leading paradigm for self-supervised representation learning, yet the conditions under which it recovers meaningful latent geometry remain incompletely understood. We develop a measure-theoretic framework formalizing the diversity condition, a support requirement on positive-pair sampl

ing that is necessary for isometric latent recovery. We show that in the standar d full-support von Mises-Fisher setting implies the satisfaction of the diversit y condition and global minimizers recover latent geometry up to orthogonal trans formation, while restricted conditionals can make non-orthogonal maps attain str ictly lower asymptotic contrastive loss. We introduce a support-corrected Inform ation Noise Contrastive Estimation (InfoNCE) variant as a theoretical fix: this correction makes orthogonal latent space recovery achievable but does not unique ly select it. Experiments on synthetic benchmarks validate the identifiability p redictions, and CIFAR-10 experiments are consistent with the qualitative predict ion that architectural inductive bias becomes more important when sampling diver sity is limited. Together, our results clarify how sampling mechanisms and encod er inductive bias interact in contrastive representation learning.
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Multi-task Linear Regression without Eigenvalue Lower Bounds: Adaptivity, Robust ness, and Safety
Seok-Jin Kim
We study the multi-task linear regression problem in the presence of contaminate d tasks.
We address the setting where the unknown parameters of a majority of tasks are c lose in the ℓ_2-norm, while a fraction of tasks are arbitrary outliers.
Existing theoretical frameworks for this problem rely heavily on the assumption that the empirical second moment of each task has a minimum eigenvalue bounded a way from zero (order $\Omega(1)$).
Crucially, this assumption fails in many high-dimensional scenarios, rendering p rior guarantees vacuous.
To overcome this limitation, we propose an estimator based on matrix-weighted no rm regularization.
We also introduce a relative balancedness condition, quantified by a balancednes s constant, that compares each task's second moment with the average inlier geom etry and relaxes the need for taskwise second-moment lower bounds.
In favorable regimes with moderate balancedness, our prediction MSE bounds match the rate of Duan and Wang (2023) under substantially weaker spectral assumption s; the resulting task-overall MSE is minimax optimal up to logarithmic factors.
Furthermore, we demonstrate that our estimator enjoys a safety guarantee: when t he relevant balancedness constant is large or infinite, or when tasks are unrela ted, the method performs no worse than independent task learning.
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ParaTool: Shifting Tool Representations from Context to Parameters
Zekai Yu, Qi Meng, Qizhi Chu, Yu Hao, Chuan Shi, Cheng Yang
Tool calling extends large language models (LLMs) by enabling grounded interacti on with external executable interfaces, thereby supporting environment-coupled p roblem solving. However, mainstream in-context learning (ICL) approaches typical ly incorporate detailed tool documentation and usage examples directly into the context. This results in substantial inference overhead and heightened risks of hallucination as the context length grows. Conversely, while tuning-based method s improve general tool-calling capabilities, they often fail to effectively inte rnalize the specific details of previously seen tools, thereby retaining a depen dency on in-context documentation. To address these limitations, we propose Para Tool, a framework that projects each tool into a dedicated, loadable set of para meters. By equipping a dynamic integration of these parameterized tools, the LLM can perform tool calling without relying on in-context documents or examples. S pecifically, our approach consists of three stages: (1) parametric tool pre-trai ning encapsulates the knowledge of different tools into independent parameter mo dules; (2) soft tool selection employs a gating network to dynamically weigh and aggregate relevant tool parameters; and (3) parametric tool fine-tuning jointly updates tool parameters to align the training and inference processes. Experime nts on Stable ToolBench and BFCL demonstrate that ParaTool significantly outperf orms strong ICL-based baselines, achieving superior performance while reducing c omputational complexity.
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Understanding and Mitigating Token-Pruning-Induced Vulnerabilities in VLMs
Shuailong Wang, Xinyu Lyu, Shengming Yuan, Jingkuan Song, Heng Tao Shen, Lianli Gao
Token-Pruning accelerates Vision-Language Models by removing redundant visual tokens, yet its safety implications remain underexplored. In this work, we present the first comprehensive safety evaluation of Token-Pruning mechanisms and find that: most pruning strategies significantly degrade safety as pruning ratios increase, whereas Query-based Compression shows the opposite, with extreme pruning (up to 99.8%), unexpectedly improves model safety. This sharp contrast prompts a key question: How do different Token-Pruning strategies reshape model safety behavior, and is it possible to enhance safety without sacrificing acceleration? To answer this, we identify an unrecognized mechanism, termed Pruning-Induced Malicious Amplification, where removal of background tokens triggers a side effect: forcing the model's attention to collapse onto a few retained malicious anchors within the foreground, inadvertently amplifying their toxic semantics under jailbreak. To address that, we propose an inference-time and plug-and-play Safety-Aware Pruning (SAP) mechanism that counteracts such dominance via three steps: (1) identifying malicious anchors, (2) restoring pruned benign tokens, and (3) reallocating excessive attention from malicious anchors to benign tokens. Extensive experiments across three safety and four utility benchmarks demonstrate that SAP mitigates pruning-induced vulnerabilities, i.e., reducing ASR by up to 62%, without compromising efficiency or utility.
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Can Microcanonical Langevin Dynamics Leverage Mini-Batch Gradient Noise?
Emanuel Sommer, Kangning Diao, Jakob Robnik, Uros Seljak, David Rügamer
Scaling inference methods such as Markov chain Monte Carlo to high-dimensional models remains a central challenge in Bayesian deep learning. A promising recent proposal, microcanonical Langevin Monte Carlo, has shown state-of-the-art performance across a wide range of problems. However, its reliance on full-dataset gradients makes it prohibitively expensive for large-scale problems. This paper addresses a fundamental question: Can microcanonical dynamics effectively leverage mini-batch gradient noise? We provide the first systematic study of this problem, establishing a novel continuous-time theoretical analysis of stochastic-gradient microcanonical dynamics. We reveal two critical failure modes: a theoretically derived bias due to anisotropic gradient noise and numerical instabilities in complex high-dimensional posteriors. To tackle these issues, we propose a principled gradient noise preconditioning scheme shown to significantly reduce this bias and develop a novel, energy-variance-based adaptive tuner that automates step size selection and dynamically informs numerical guardrails. The resulting algorithm is a robust and scalable microcanonical Monte Carlo sampler that achieves state-of-the-art performance on challenging high-dimensional inference tasks like Bayesian neural networks. Combined with recent ensemble techniques, our work unlocks a new class of stochastic microcanonical Langevin ensemble (SMILE) samplers for large-scale Bayesian inference.
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SCHUR-A*: Layer-wise Optimal Expert Pruning for MoEs via Schur-Complement Guided A* Search
Zheng Chen, Yang Weifeng, Jianxiao Tang, Buhui Yao
Sparse Mixture-of-Experts (MoE) language models enable conditional computation but face deployment challenges due to the memory wall: while few experts are activated per token, the entire model must reside in memory. Existing expert pruning methods primarily rely on independent ranking, failing to account for the complex inter-dependencies and redundancies between experts. In this paper, we formulate post-training MoE pruning as a reconstruction-driven subset selection problem, aiming to minimize layer-output distortion under a cardinality constraint. We introduce SCHUR-A*, an algorithm that leverages A* search to achieve globally optimal expert selection within each layer. To maintain computational tractability, we derive a novel, admissible heuristic upper bound using a Schur-complement-based relaxation of the reconstruction objective. This tight bound allows for aggressive pruning of the search space while mathematically guaranteeing optimalit

y. Furthermore, we propose an automated strategy to balance fidelity and memory reduction across heterogeneous layers via knee-point detection. Extensive experiments on Qwen3-30B-A3B demonstrate that SCHUR-A* significantly outperforms greedy and ranking-based baselines, maintaining comparable performance even under aggressive pruning ratios.
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Move-Then-Operate: Behavioral Phasing for Human-Like Robotic Manipulation
Haoming Xu, Lei Lei, Jie Gu, Chu Tang, Jingmin Chen, Rui-Qi Wang
We present Move-Then-Operate, a Vision–language–action framework that explicitly decouples robotic manipulation into two distinct behavioral phases: coarse relocation (move) and contact-critical interaction (operate). Unlike monolithic policies that conflate these heterogeneous regimes, our architecture employs a dual-expert policy routed by a learnable phase selector, introducing a structural inductive bias that isolates phase-specific dynamics. Phase labels are automatically generated via an MLLM-based pipeline conditioned on lightweight contextual cues such as end-effector velocity and subtask decomposition to ensure alignment with human motor patterns. Evaluated on the RoboTwin2 benchmark, our method achieves an average success rate of 68.9%, outperforming the monolithic π_0 baseline by +24%. It matches or exceeds models trained on $10\times$ more data and reaches peak performance in 40% fewer training steps, demonstrating that architectural disentanglement of move and operate phases is a highly effective and efficient strategy for mastering high-precision manipulation.
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CIRBench: Evaluating Large Language Models as LLVM IR Optimizers
Zi Yang, Haifeng Ding, Fei Liu, Yingying Cheng, Han Cheng, Zhilei Chai, Haojie Zhou
Large language models are beginning to introduce a new paradigm for compilation: instead of only assisting at the source level, they can operate directly on **intermediate representations (IRs)**, the compiler’s internal code representation, Early studies suggest that LLM-guided optimization can sometimes rival traditional compiler optimizations on selected programs, but evidence remains fragmented.
Yet the community still lacks a rigorous IR-level benchmark that tests whether a model not only understands IR but can rewrite it under compiler-grade semantic constraints with meaningful performance impact.
We present **CIRBench**, a benchmark of 800 curated IR instances spanning four compiler-oriented tracks: Analysis infers IR properties, Repair fixes invalid IR, Refactor applies a single semantics-preserving compiler optimization, and Transform performs performance-oriented rewrites, together mirroring core optimization responsibilities in modern compilers.
CIRBench combines verifier, equivalence checking, and end-to-end performance measurement into a unified, layered correctness-aware evaluation of LLMs on IR.
On six mainstream LLMs, CIRBench shows that current models fail on many IR analysis and rewriting instances and on median underperform the compiler baseline, but we also observe a maximum speedup of $4.96\times$ over -O3.
These findings highlight both the opportunities and the remaining challenges of using LLMs inside optimizing compilers.
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CoCoReviewBench: A Completeness- and Correctness-Oriented Benchmark for AI Reviewers
Hexuan Deng, Xiaopeng Ke, Yichen Li, Ruina Hu, Dehao Huang, Derek F. Wong, Yue Wang, Xuebo Liu, Min Zhang
Despite the rapid development of AI reviewers, evaluating such systems remains challenging: metrics favor overlap with human reviews over correctness. However, since human reviews often cover only a subset of salient issues and sometimes contain mistakes, they are unreliable as gold references. To address this, we build category-specific benchmark subsets and skip evaluation when the corresponding human reviews are missing to strengthen ***Co***mpleteness. We also leverage reviewer--author--meta-review discussions as expert annotations and filter unreliable reviews accordingly to strengthen ***Co***rrectness. Finally, we introduce `

CoCoReviewBench`, which curates 3,900 papers from ICLR and NeurIPS to enable rel
iable and fine-grained evaluation of AI reviewers. Analysis shows that AI review
ers remain limited in correctness and are prone to hallucinations, and highlight
s reasoning models as more effective reviewers, motivating further directions fo
r improving AI reviewers. Benchmarks and models are available at <https://github
.com/hexuandeng/CoCoReviewBench>.
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X-EviProbe: Post-hoc Parameter-Free Evidential Uncertainty Quantification for Fr
ozen Graph Neural Networks
Chenghua Guo, Sihong Xie, Xi Zhang
Reliable uncertainty quantification (UQ) is crucial for deploying graph neural n
etworks (GNNs) in safety-critical settings, yet dominant solutions either rely o
n costly multi-pass sampling or require retraining—often using *black-box auxili
ary* models—to obtain evidential semantics. We propose **X-EviProbe**, a simple
and parameter-free *post-hoc* framework that turns a *frozen* GNN into an eviden
tial predictor with a decomposable view of epistemic vs. aleatoric uncertainty.
 X-EviProbe constructs class-wise Dirichlet evidence by probing the frozen laten
t space and the model’s native outputs, and incorporates graph structure via lig
htweight evidence-strength propagation. This yields a transparent evidential rep
resentation without retraining or additional neural components. Extensive exper
iments on seven benchmarks show that X-EviProbe consistently ranks among the top
 methods for both OOD detection and misclassification detection, improving AUROC
 by up to **33.4%** and **8.7%** over the strongest baselines.
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Lifting Traces to Logic: Programmatic Skill Induction with Neuro-Symbolic Learni
ng for Long-Horizon Agentic Tasks
Jie-Jing Shao, Haiyan Yin, Yueming Lyu, Xingrui Yu, Lan-Zhe Guo, Ivor Tsang, Jam
es Kwok, Yu-Feng Li
Foundation model-driven agents often struggle with long-horizon planning due to
the transient nature of purely prompting-based reasoning. While existing skill i
nduction methods mitigate this by distilling experience into state-blind paramet
erized scripts, they fail to capture the conditional logic required for robust e
xecution in dynamic environments. In this paper, we propose Neuro-Symbolic Skill
 Induction (NSI), a framework that lifts interaction traces into modular, logic-
grounded programs. By synthesizing explicit control flows and dynamic variable b
inding, NSI empowers agents to discover when and why to act. This paradigm enabl
es the efficient generalization, allowing agents to induce skills from few-shot
examples and flexibly adapt to unseen goals. Experiments on a series of agentic
tasks demonstrate that NSI consistently outperforms state-of-the-art baselines,
empowering agents to self-evolve into architects of logic-grounded skills. Proje
ct Page: https://sh-jj.github.io/NSI.
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Agent Primitives: Reuseable Latent Building Blocks for Multi-Agent Systems
Haibo Jin, Peng Kuang, Ye Yu, Xiaopeng Yuan, Haohan Wang
While existing multi-agent systems (MAS) can handle complex problems by enabling
 collaboration among multiple agents, they are often highly task-specific, relyi
ng on manually crafted agent roles and interaction prompts, which leads to incre
ased architectural complexity and limited reusability across tasks. Moreover, mo
st MAS communicate primarily through natural language, making them vulnerable to
 error accumulation and instability in long-context, multi-stage interactions wi
thin internal agent histories.

In this work, we propose \textbf{Agent Primitives}, a set of reusable latent bui
lding blocks for LLM-based MAS. Inspired by neural network design, where complex
 models are built from reusable components, we observe that many existing MAS ar
chitectures can be decomposed into a small number of recurring internal computat
ion patterns. Based on this observation, we instantiate three primitives (Review
, Voting and Selection, and Planning and Execution), all communicating via key–v
alue (KV) cache to mitigate information degradation across multi-stage interacti
ons. To enable automatic system construction, an Organizer agent automatically s

elects and composes primitives for each query, guided by a lightweight knowledge pool of previously successful configurations, forming a primitive-based MAS.

Experiments show that primitives-based MAS improve average accuracy by 12.0–16.5 \% over single-agent baselines, reduce token usage and inference latency by approximately 3$\times$–4$\times$ compared to text-based MAS, while incurring only 1.3$\times$–1.6$\times$ overhead relative to single-agent inference and providing more stable performance across model backbones.
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Learning the ESG Geometry with Domain Aware Language Models
Kunal Pradeep Pimparkhede, Chirayu Chaurasia, Jatin Roy, Mahesh Mohan M R
Responsible investing aims to generate positive impact across Environment (E), Society (S), and Governance (G), and rating companies along these dimensions is now widespread, making ESG scores highly popular. Allocating retail capital with sustainability in mind could be transformational, yet it remains unclear how individual investors can do so in practice. Current ESG solutions cannot model high-dimensional, multi-modal time series capturing the joint evolution of ESG risks, financial returns, news, and sentiment, even though this domain requires jointly reasoning over distinct numerical signals where both numerical proximity and semantic type must be preserved. To bridge this gap, we introduce a novel domain-aware $\textbf{representation learning framework}$ that learns geometry-preserving representations for heterogeneous time series using value-aware tokens with block-wise $\textbf{orthogonal embeddings}$. To capture trajectory-level structure, we introduce $\textbf{FACET}$ tokens and train the model using a geometry-preserving loss. The resulting model jointly learns to forecast future values and to organize entities in a representation space that reflects their temporal evolution. Trained on ESG, returns, news, and sentiment, the domain-aware LLM learns a representation space that enables accurate ESG forecasting, trajectory-based grouping, and latent-space search for superior asset selection and downstream application like portfolio rebalancing
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Sparse ActionGen: Accelerating Diffusion Policy with Real-time Pruning
Kangye Ji, Jianbo Zhou, Yuan Meng, Ye Li, Hanyun Cui, Zhi Wang
Diffusion Policy has dominated action generation due to its strong capabilities for modeling multi-modal action distributions, but its multi-step denoising processes make it impractical for real-time visuomotor control.
Existing caching-based acceleration methods typically rely on $\textit{static}$ schedules that fail to adapt to the \textit{dynamics} of robot-environment interactions, thereby leading to suboptimal performance.
In this paper, we propose $\underline{\textbf{S}}$parse $\underline{\textbf{A}}$ction$\underline{\textbf{G}}$en $(\textbf{SAG})$ for extremely sparse action generation. To accommodate the iterative interactions, SAG customizes a rollout-adaptive prune-then-reuse mechanism that first identifies prunable computations globally and then reuses cached activations to substitute them during action diffusion. To capture the rollout dynamics, SAG parameterizes an observation-conditioned diffusion pruner for environment-aware adaptation and instantiates it with a highly parameter- and inference-efficient design for real-time prediction. Furthermore, SAG introduces a one-for-all reusing strategy that reuses activations across both timesteps and blocks in a zig-zag manner, minimizing the global redundancy. Extensive experiments on multiple robotic benchmarks demonstrate that SAG achieves up to 4$\times$ generation speedup without sacrificing performance. Project Page: https://sparse-actiongen.github.io/.
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Optimal and Scalable MAPF via Multi-Marginal Optimal Transport and Schrödinger Bridges
Usman Khan, Joseph W Durham
We consider anonymous multi-agent path finding (MAPF) where a set of robots is tasked to travel to a set of targets on a finite, connected graph. We show that MAPF can be cast as a special class of multi-marginal optimal transport (MMOT) problems with an underlying Markovian structure, under which the exponentially lar

ge MMOT collapses to a linear program (LP) polynomial in size. Focusing on the a nonymous setting, we establish conditions under which the corresponding LP is fe asible, totally unimodular, and yields min-cost, integral~$(\{0,1\})$ transports that do not overlap in both space and time. To adapt the approach to large-scal e problems, we cast the MAPF-MMOT in a probabilistic framework via Schrödinger b ridges. Under standard assumptions, we show that the Schrödinger bridge formulat ion reduces to an entropic regularization of the corresponding MMOT that admits an iterative Sinkhorn-type solution. The Schrödinger bridge, being a probabilist ic framework, provides a shadow (fractional) transport that we use as a template to solve a reduced LP and demonstrate that it results in near-optimal, integral transports at a significant reduction in complexity. Extensive experiments high light the optimality and scalability of the proposed approaches.
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3DPoV: Improving 3D understanding via Patch Ordering on Videos
Ioana Simion, Mohammadreza Salehi, Shashanka Venkataramanan, Cees G. M. Snoek, Y uki M Asano
Visual foundation models have achieved remarkable progress in scale and versatil ity, yet understanding the 3D world remains a fundamental challenge. While 2D im ages contain cues about 3D structure that humans readily interpret, deep models often fail to exploit them, underperforming on tasks such as multiview semantic consistency--crucial for applications including robotics and autonomous driving. We propose a self-supervised approach to enhance the 3D understanding of vision foundation models by (i) introducing a temporal nearest-neighbor consistency lo ss that finds corresponding points across video frames and enforces consistency between their nearest neighbors, (ii) incorporating reference-guided ordering th at requires patch-level features to be not only expressive but also consistently aligned, and (iii) constructing a mixture of video datasets tailored to these o bjectives, thereby leveraging rich 3D information. Our method, 3DPoV, achieves s tate-of-the-art performance in keypoint matching under viewpoint variation, as w ell as in depth and surface normal estimation, and consistently improves a diver se set of backbones, including DINOv3.
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Sparse and Faithful Local Explanations with Piecewise Linear Surrogates
Yixin Wang, Yucheng Dong
Local post-hoc explanations are widely used to understand black-box models on ta bular data, with Local Interpretable Model-agnostic Explanations (LIME) being a popular approach. LIME approximates a black-box model using a sparse linear surr ogate in a local neighborhood, implicitly assuming feature-wise linear homogenei ty. However, this assumption often fails when local feature effects exhibit hete rogeneous or nonlinear behaviors, resulting in unfaithful and unstable explanati ons. Moreover, LIME relies on a decoupled feature selection procedure that is no t aligned with the surrogate modeling objective, further exacerbating instabilit y under local sampling. To address these limitations, we propose PL-LIME, a two- stage sparse local explanation framework that ensures objective consistency acro ss stages. PL-LIME models feature-wise local effects using instance-anchored pie cewise linear functions, providing a minimal yet principled extension beyond lin ear surrogates under a fixed explanation budget. Sparsity is enforced through a decoupled nonnegative shrinkage procedure that directly scales the estimated loc al effects, improving stability while preserving interpretability. Experiments o n synthetic and real-world datasets demonstrate that PL-LIME achieves higher loc al fidelity and stability, and provides more reliable local explanations that ca pture finer-grained local effect structures than LIME.
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Representation Unlearning: Forgetting through Information Compression
Antonio Almudévar, Alfonso Ortega
Machine unlearning seeks to remove the influence of specific training data from a model, a need driven by privacy regulations and robustness concerns. Existing approaches typically modify model parameters, but such updates can be unstable, computationally costly, and limited by local approximations. We introduce Repres entation Unlearning, a framework that performs unlearning directly in the model’

s representation space. Instead of modifying model parameters, we learn a transf
ormation over representations that imposes an information bottleneck: maximizing
 mutual information with retained data while suppressing information about data
to be forgotten. We derive variational surrogates that make this objective tract
able and show how they can be instantiated in two practical regimes: when both r
etain and forget data are available, and in a zero-shot setting where only forge
t data can be accessed. Experiments across several benchmarks demonstrate that R
epresentation Unlearning achieves more reliable forgetting, better utility reten
tion, and greater computational efficiency than parameter-centric baselines.
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Factored Causal Representation Learning for Robust Reward Modeling in RLHF
Yupei Yang, Lin Yang, Wanxi Deng, Lin Qu, Fan Feng, Biwei Huang, Shikui Tu, Lei
Xu
A reliable reward model is essential for aligning large language models (LLMs) w
ith human preferences through reinforcement learning from human feedback (RLHF).
 However, standard reward models are susceptible to spurious features that are n
ot causally related to human labels. This can lead to *reward hacking*, where hi
gh predicted reward does not translate into better behavior. In this work, we ad
dress this problem from a causal perspective by proposing a factored representat
ion learning framework that decomposes the model’s contextual embedding into (1)
 causal factors that are sufficient for reward prediction and (2) non-causal fac
tors that capture reward-irrelevant attributes such as length or sycophantic bia
s. The reward head is then constrained to depend only on the causal component. I
n addition, we introduce an adversarial head trained to predict reward from the
non-causal factors, while applying gradient reversal to discourage them from enc
oding reward-relevant information. Experiments on both mathematical and dialogue
 tasks demonstrate that our method learns more robust reward models and consiste
ntly improves downstream RLHF performance over state-of-the-art baselines. Analy
ses on length and sycophantic bias further validate the effectiveness of our met
hod in mitigating reward hacking behaviors.
****

CombinationTS: A Modular Framework for Understanding Time-Series Forecasting Mod
els
Xiaorui Wang, Fanda Fan, Chenxi Wang, Yuxuan Yang, Rui Tang, Kuoyu Gao, simiao p
ang, Yuanfeng Shang, Zhipeng Liu, Wanling Gao, Lei Wang, Jianfeng Zhan
Recent progress in time-series forecasting has led to rapidly increasing archite
ctural complexity, yet many reported State-of-the-Art gains are statistically fr
agile or misattributed.
We argue that progress requires a shift from model selection to modular attribut
ion, identifying which components truly drive performance.
We propose CombinationTS, a self-contained probabilistic evaluation framework gr
ounded in Evaluatology, the science of uncovering the effects.
It decomposes forecasting models into orthogonal modules—Input Transformation, E
mbedding, Encoder, Decoder, and Output Transformation—and evaluates them under a
 shared evaluation condition space.
By quantifying each component via marginalized performance (μ) and stability
 (σ), CombinationTS enables robust attribution beyond fragile point estim
ates.
Through large-scale paired evaluation, we uncover the Identity Paradox: once the
 data view (Embedding) is well-designed, a parameter-free Identity Encoder often
 matches or outperforms complex backbones.
****

Trajectory Consistency for One-Step Generation on Euler Mean Flows
Zhiqi Li, Yuchen Sun, Duowen Chen, Jinjin He, Bo Zhu
We propose \emph{Euler Mean Flows (EMF)}, a flow-based generative framework for
one-step and few-step generation that enforces long-range trajectory consistency
 with minimal sampling cost.
The key idea of EMF is to replace the trajectory consistency constraint, which i
s difficult to supervise and optimize over long time scales, with a principled l
inear surrogate that enables direct data supervision for long-horizon flow-map c

ompositions. We derive this approximation from the semigroup formulation of flo
w-based models and show that, under mild regularity assumptions, it faithfully a
pproximates the original consistency objective while being substantially easier
to optimize. This formulation leads to a unified, JVP-free training framework t
hat supports both u-prediction and x_1-prediction variants, avoiding explici
t Jacobian computations and significantly reducing memory and computational over
head. Experiments on image synthesis, particle-based geometry generation, and f
unctional generation demonstrate improved optimization stability and sample qual
ity under fixed sampling budgets, together with approximately 50% reductions
in training time and memory consumption compared to existing one-step methods fo
r image generation.
****

Stochastic Order Learning: An Approach to Rank Estimation Using Noisy Data
Chaewon Lee, Seon-Ho Lee, Chang-Su Kim
Rank estimation under label noise poses a fundamental challenge, as ordinal anno
tations often exhibit structured uncertainty rather than simple label corruption
. In this paper, we reformulate rank estimation with noisy ordinal labels as a s
tochastic ordering problem, in which each instance is inherently associated with
 multiple plausible ranks instead of a single deterministic label. Based on this
 view, we propose stochastic order learning (SOL), a learning framework that cap
tures ordinal label uncertainty and learns an embedding space through two comple
mentary objectives: a discriminative loss that structures instance-centroid inte
ractions and a stochastic order loss that enforces probabilistic ordering relati
ons between instances. Extensive experiments across diverse datasets demonstrate
 that SOL enables reliable rank estimation under various types and levels of lab
el noise. The source code is available at https://github.com/cwlee00/SOL.
****

Differentially Private Preference Data Synthesis for Large Language Model Alignm
ent
Fengyu Gao, Jing Yang
Preference alignment is a crucial post-training step for large language models (
LLMs) to ensure their outputs align with human values. However, post-training on
 real human preference data raises privacy concerns, as these datasets often con
tain sensitive user prompts and human judgments. To address this, we propose **D
PPrefSyn**, a novel algorithm for generating differentially private (DP) synthet
ic preference data to enable privacy-preserving preference alignment. DPPrefSyn
is a principled framework grounded in the Bradley–Terry preference model and the
 intrinsic geometric structure of pairwise human preference data. It first learn
s an underlying preference model from private data with formal differential priv
acy guarantees, and then leverages the learned model together with public prompt
s to synthesize high-quality preference data. It exploits the shared linear stru
cture of per-cluster reward models to effectively capture heterogeneous human pr
eferences in private datasets, and leverages DP Principal Component Analysis (DP
-PCA) to improve learning accuracy. Extensive experimental results demonstrate t
hat DPPrefSyn achieves competitive alignment performance under strong DP guarant
ees. These findings highlight the potential of synthetic preference data as a pr
actical alternative for privacy-preserving preference alignment across a broad r
ange of applications. To the best of our knowledge, this is the first work to ge
nerate DP synthetic preference data for LLM alignment.
****

Private and Stable Test-time Adaptation with Differential Privacy
Zefeng Li, Qiaoyue Tang, Mathias Lécuyer, Evan Shelhamer
Test-time adaptation (TTA) can reduce error on new and different data by updatin
g the model on these inputs during inference.
However, these updates raise the issue of privacy w.r.t. the testing data, becau
se the model parameters now depend on all past inputs.
To control this privacy risk, we cast multiple popular TTA methods (Tent, EATA,
SAR, DeYO, and COME) into differential privacy (DP) forms that apply per-sample
gradient clipping and Gaussian noise for all updates.
On ImageNet-C, our DP-TTA methods provide adequate privacy at small cost to accu

racy, and in the low-privacy regime the clipping mechanism of DP can even improv
e the accuracy and stability of adaptation in the continual setting.
These improvements to privacy and accuracy come at only modest computational ove
rhead.
These first results on private TTA raise awareness of the issue,
inform the development of more private test-time updates, and identify per-sampl
e clipping as an effective technique for improving the accuracy and stability of
 adaptation.
****

Degradation-Aware Metric Prompting for Hyperspectral Image Restoration
Binfeng Wang, Di Wang, Haonan Guo, Ying Fu, Jing Zhang
Unified hyperspectral image (HSI) restoration aims to recover diverse degradatio
ns within a single model. However, current methods often rely on impractical exp
licit priors or opaque black-box representations that overfit to training distri
butions, hampering generalization to unseen scenarios. To bridge this gap, we pr
opose Degradation-Aware Metric Prompting (DAMP), a novel framework that characte
rizes multi-dimensional degradations through interpretable spatial-spectral metr
ics. These metrics serve as Degradation Prompts (DP), enabling the model to capt
ure shared characteristics across tasks and adapt to unknown corruptions. Centra
l to our framework is the Degradation-Adaptive Mixture-of-Experts (DAMoE), where
 Spatial-Spectral Adaptive Modules (SSAMs) serve as experts that utilize learnab
le fusion coefficients to specialize in distinct degradation degrees. By using D
P as a gating router, DAMoE dynamically activates specialized experts tailored t
o the specific degradation profile. Extensive experiments on natural and remote
sensing HSI datasets demonstrate that DAMP achieves state-of-the-art performance
 and exhibits exceptional zero-shot generalization on unseen restoration tasks.
Code is publicly available at \href{DAMP}{https://github.com/MiliLab/DAMP}.
****

CARE: Class-Adaptive Expert Consensus for Reliable Learning with Long-Tailed Noi
sy Labels
Mengke Li, Haiquan Ling, Lihao Chen, Yang Lu, Yiqun Zhang, Hui Huang
Learning from real-world data is frequently hindered by the compound challenge o
f long-tailed class distributions and noisy annotations.
Existing methods partially address these issues but typically ignore the non-uni
form impact of label noise across classes, resulting in ineffective correction f
or tail classes and over-regularization for head classes.
To address this issue, we propose Class-Adaptive Rectification with Experts (CAR
E), a parameter-efficient framework that leverages three complementary supervisi
on sources from vision-language models (VLM): observed noisy labels, VLM text em
beddings, and visual features.
CARE introduces a class-adaptive expert consensus mechanism that enforces strict
er agreement for tail classes and more permissive agreement for head classes bas
ed on class frequency.
By aggregating high-confidence predictions across these sources, CARE filters un
reliable signals and recalibrates class distributions, yielding more reliable re
ctification under long-tailed distributions.
Extensive experiments on both synthetic and real-world benchmarks demonstrate th
at CARE consistently outperforms state-of-the-art methods, achieving up to 3.0\%
 performance gains.
The source code is available at https://github.com/qwq123-study/CARE.
****

Geometry of Reason: Spectral Signatures of Valid Mathematical Reasoning
Valentin NOËL
Verifying whether a language model is genuinely reasoning or pattern-matching re
mains an open problem: learned verifiers are expensive, and output-based heurist
ics are brittle.
We show that valid mathematical reasoning induces a measurable, training-free sp
ectral signature in transformer attention.
By treating each attention matrix as a weighted token graph, we extract four dia
gnostics: Fiedler value, High-Frequency Energy Ratio (HFER), spectral entropy, a

nd smoothness, that require no learned parameters.
Experiments across seven models from four architectural families yield effect si zes up to Cohen's $d = 3.30$ ($p < 10^{-116}$), enabling 85-96% single-thres hold classification accuracy.
Two findings sharpen the interpretation.
First, \emph{Platonic validity}: the spectral signal tracks logical coherence ra ther than compiler acceptance, proofs rejected for timeouts or missing imports a re correctly classified as valid, a distinction confirmed by a manual audit ($\k appa = 0.82$, $n = 51$).
Second, \emph{architectural determinism}: Sliding Window Attention shifts the di scriminative feature from HFER to smoothness ($d = 2.09$, $p < 10^{-48}$), showi ng that attention design governs which spectral channel encodes reasoning qualit y.
Causal ablation confirms the signature traces induction-head circuits.
The method generalises to informal chain-of-thought ($d = 0.78$, $p < 10^{-3}$), and in proof search, HFER reranking improves Best-of-16 Pass@1 by $+4.4$-6.6% , matching 98% of the AUC of fully supervised probes with zero labels.
Spectral graph analysis is a principled, architecture-aware primitive for reason ing verification.
****

Towards Solving the Gilbert-Pollak Conjecture via Large Language Models
Yisi Ke, Tianyu Huang, Yankai Shu, Di He, Jingchu Gai, Liwei Wang
The Gilbert-Pollak Conjecture, also known as the Steiner Ratio Conjecture, state s that for any finite point set in the Euclidean plane, the Steiner minimum tree has length at least $\sqrt{3}/2 \approx 0.866$ times that of the Euclidean mini mum spanning tree (the Steiner ratio). A sequence of improvements through the 19 80s culminated in a lower bound of 0.824, with no substantial progress reporte d over the past three decades. Recent advances in LLMs have demonstrated strong performance on contest-level mathematical problems, yet their potential for addr essing open, research-level questions remains largely unexplored. In this work, we present a novel AI system for obtaining tighter lower bounds on the Steiner r atio. Rather than directly prompting LLMs to solve the conjecture, we task them with generating rule-constrained geometric lemmas implemented as executable code . These lemmas are then used to construct a collection of specialized functions, which we call verification functions, that yield theoretically certified lower bounds of the Steiner ratio. Through progressive lemma refinement driven by refl ection, the system establishes a new certified lower bound of 0.8559 for the Ste iner ratio. The entire research effort involves only thousands of LLM calls, de monstrating the strong potential of LLM-based systems for advanced mathematical research.
****

Formally Exploring Visual Anomaly Detection Evaluation Metrics
Nasar Iqbal, Dennis Wagner, Philipp Liznerski, Nabeel Hussain Syed, Sophie Felle nz, Niki Martinel, Marius Kloft
Inaccurate Visual Anomaly Detection (VAD) can lead to critical failures in safet y-sensitive domains, including autonomous navigation and industrial surveillance . With the increasing abundance and rapid proliferation of VAD algorithms, their reliable evaluation has become increasingly important and challenging. Commonly used evaluation metrics often fail to capture practically relevant aspects of m odel behavior, yielding inconsistent or misleading assessments by overlooking er rors such as redundant detections and the spatial distribution of false positive s. In this paper, we formalize the requirements for VAD evaluation by introducin g a set of axiomatic, verifiable properties that an evaluation metric should sat isfy. Through a systematic analysis of state-of-the-art evaluation methods, we s how that none satisfies all proposed properties. To address this gap, we introdu ce SAAM-ALARM, a novel evaluation metric that satisfies these properties. Our re sults show that SAAM-ALARM provides a more nuanced and theoretically sound asses sment, establishing a stronger standard for performance benchmarking in VAD.
****

Ellipsoidal Time Series Forecasting

Qilin Wang
We argue that long-horizon forecasting requires learning local Jacobians with explicit spectral structure, not only matching conditional means. Our method, Fern (\textbf{F}orecasting with \textbf{E}llipsoidal \textbf{R}epresentatio\textbf{N}s), invokes Brenier's theorem to directly parameterize the Jacobian as a symmetric positive semi-definite (SPD) factorization, treating forecasting as the optimal transport of probability mass from a fixed Gaussian source to data-dependent ellipsoids. This formulation avoids post-hoc eigendecomposition of dense Jacobians, enables efficient Householder-based orthogonal factors, and exposes interpretable diagnostics such as local stretching, spectral radius, and volume change. To rigorously evaluate robustness, we introduce controlled synthetic stress tests with nonstationary shocks, together with Wasserstein-based shape metrics and Effective Prediction Time. Fern demonstrates exceptional stability, outperforming baselines like DLinear and Koopa by over two orders of magnitude (up to $790\times$) on nonstationary settings where standard benchmarks fail to expose model brittleness.
****

Adaptive Generation of Bias-Eliciting Questions for LLMs
Robin Staab, Jasper Dekoninck, Maximilian Baader, Martin Vechev
Large language models (LLMs) are now widely deployed in user-facing applications, reaching hundreds of millions of users worldwide. Despite their widespread adoption, growing reliance on their outputs raises significant concerns, particularly as users may be exposed to model-inherent biases that disadvantage or stereotype certain groups. However, existing bias benchmarks commonly rely on simple templated prompts or restrictive multiple-choice questions that fail to capture the complexity of real-world user interactions. In this work, we address this gap by introducing a counterfactual framework that automatically generates realistic, open-ended questions for LLM bias evaluation. Through iterative question mutation, our approach systematically explores areas where models are most likely to exhibit biased behavior. Beyond just detecting harmful biases, we also capture increasingly relevant response dimensions, such as asymmetric refusals and explicit bias acknowledgment. Building on this, we construct CAB, a diverse and human-verified benchmark for realistic and nuanced bias evaluations on current frontier LLMs. Our evaluation using CAB highlights the continued need for fairness research by showing that all examined models exhibit persistent biases across certain scenarios.
****

Statistically Undetectable Backdoors in Deep Neural Networks
Andrej Bogdanov, Alon Rosen, Neekon Vafa
We show how an adversarial model trainer can plant backdoors in a large class of deep, feedforward neural networks. These backdoors are statistically undetectable in the white-box setting, meaning that the backdoored and honestly trained models are close in total variation distance, even given the full descriptions of the models (e.g., all of the weights). The backdoor provides access to invariance-based adversarial examples for every input, mapping distant inputs to unusually close outputs. However, without the backdoor, it is provably impossible (under standard cryptographic assumptions) to generate any such adversarial examples in polynomial time. Our theoretical and preliminary empirical findings demonstrate a fundamental power asymmetry between model trainers and model users.
****

Beyond Softmax: A Natural Parameterization for Categorical Random Variables
Alessandro Manenti, Cesare Alippi
Latent categorical variables are frequently found in deep learning architectures. They can model actions in discrete reinforcement-learning environments, represent categories in latent-variable models, or express relations in graph neural networks. Despite their widespread use, their discrete nature poses significant challenges to gradient-descent learning algorithms. While a substantial body of work has offered improved gradient estimation techniques, we take a complementary approach. Specifically, we: 1) revisit the ubiquitous *softmax* function and demonstrate its limitations from an information-geometric perspective; 2) replace

the *softmax* with the *catnat* function, a function composed by a sequence of h
ierarchical binary splits; we prove that this choice offers significant advantag
es to gradient descent due to the resulting diagonal Fisher Information Matrix.
A rich set of experiments - including graph structure learning, variational auto
encoders, and reinforcement learning - empirically show that the proposed functi
on improves the learning efficiency and yields models characterized by consisten
tly higher test performance. *Catnat* is simple to implement and seamlessly inte
grates into existing codebases. Moreover, it remains compatible with standard tr
aining stabilization techniques and, as such, offers a better alternative to the
 softmax function.
****

Escaping Mode Collapse in LLM Generation via Geometric Regulation
Xin Du, Kumiko Tanaka-Ishii
Mode collapse is a persistent challenge in generative modeling and manifests in
autoregressive text generation as behaviors ranging from explicit looping to
gradual loss of diversity and premature trajectory convergence. We take a
dynamical-systems view and reinterpret mode collapse as reduced state-space
accessibility caused by *geometric collapse*: during generation, the
model's internal trajectory becomes confined to a low-dimensional region of its
representation space. This implies mode collapse is not purely a token-level
phenomenon and cannot be reliably mitigated by symbolic constraints or
probability-only decoding heuristics. Guided by this perspective, we propose
Reinforced Mode Regulation (RMR), a lightweight, online state-space
intervention that regulates dominant self-reinforcing directions in the
Transformer value cache (implemented as low-rank damping). Across multiple large
language models, RMR substantially reduces mode collapse and enables stable,
high-quality generation at extremely low entropy rates (down to 0.8 nats/step),
whereas standard decoding typically collapses near 2.0 nats/step.
****

Learning to Search and Searching to Learn for Generalization in Planning
Michael Aichmüller, Yannik Hesse, Hector Geffner
Combinatorial generalization remains a central challenge in Deep Reinforcement L
earning (DRL). Classical planning provides a simple yet challenging setting to s
tudy this problem through explicit relational descriptions, without requiring le
arning from perception.
In sparse-reward domains, standard RL exploration via real-time search is ineffe
ctive, and learning-based planning methods often rely on expert demonstrations,
hindsight relabeling, or random walks from the goal state. In contrast, planners
 rely on best-first search methods such as $\mathrm{A}^\star$ to solve problems
from scratch. We propose a self-improving $\mathrm{WA}^\star$ learning framework
 in combination with a value heuristic represented by a Relational Graph Neural
Network: the heuristic guides search, and the resulting search data updates the
heuristic via Q-learning. This loop yields heuristics that can function as gen
eral policies and solve new instances even without search, where DRL otherwise f
ails, as we show on puzzles such as Sokoban, PushWorld, The Witness, and the 202
3 International Planning Competition benchmarks. Notably, we demonstrate strong
zero-shot generalization: For example, heuristics trained on Blocksworld instanc
es with fewer than 30 blocks successfully solve instances with 488 blocks wi
thout search.
****

Optimized Deferral for Imbalanced Settings
Corinna Cortes, Anqi Mao, Mehryar Mohri, Yutao Zhong
Learning algorithms can be significantly improved by routing complex or uncertai
n inputs to specialized experts, balancing accuracy with computational cost. Thi
s approach, known as *learning to defer*, is essential in domains like natural l
anguage generation, medical diagnosis, and computer vision, where an effective d
eferral can reduce errors at low extra resource consumption. However, the two-st
age learning to defer setting, which leverages existing predictors such as a col
lection of LLMs or other classifiers, often faces challenges due to an expert im
balance problem. This imbalance can lead to suboptimal performance, with deferra

l algorithms favoring the majority expert. We present a comprehensive study of t
wo-stage learning to defer in expert imbalance settings. We cast the deferral lo
ss optimization as a novel cost-sensitive learning problem over the input-expert
 domain. We derive new margin-based loss functions and guarantees tailored to th
is setting, and develop novel algorithms for cost-sensitive learning. Leveraging
 these results, we design principled deferral algorithms, MILD (*Margin-based Im
balanced Learning to Defer*), specifically suited for expert imbalance settings.
 Extensive experiments demonstrate the effectiveness of our approach, showing cl
ear improvements over existing baselines on both image classification and real-w
orld Large Language Model (LLM) routing tasks.
****
Negatives-Dominant Contrastive Learning for Generalization in Imbalanced Domains
Meng Cao, Jiexi Liu, Songcan Chen
Imbalanced Domain Generalization (IDG) focuses on mitigating both *domain and la
bel shifts*, both of which fundamentally shape the model's decision boundaries,
particularly under heterogeneous long-tailed distributions across domains.
Despite its practical significance, it remains underexplored, primarily due to t
he *technical* complexity of handling their entanglement and the paucity of *the
oretical* foundations.
In this paper, we begin by *theoretically* establishing the generalization bound
 for IDG, highlighting the role of posterior discrepancy and decision margin.
This bound motivates us to focus on directly steering decision boundaries, marki
ng a clear departure from existing methods.
Then, we *technically* propose a novel Negative-Dominant Contrastive Learning (N
DCL) for IDG to enhance discriminability while enforce posterior consistency acr
oss domains.
Specifically, inter-class decision-boundary separation is enhanced by placing gr
eater emphasis on negatives as the primary signal in our contrastive learning, n
aturally amplifying gradient signals for minority classes to avoid the decision
boundary being biased toward majority classes.
Intra-class compactness is encouraged through a re-weighted cross-entropy strate
gy, and posterior consistency across domains is enforced through a prediction-ce
ntral alignment strategy.
Finally, rigorous yet challenging experiments on benchmarks validate the effecti
veness of our NDCL.
The code is available at https://github.com/Alrash/NDCL.
****
Fairness in Aggregation: Optimal Top-k and Improved Full Ranking
Diptarka Chakraborty, Arya Mazumdar, Barna Saha, Alvin Hong Yao Yan
Ensuring fairness in algorithmic ranking systems is a critical challenge with si
gnificant societal implications for hiring, recommendations, web search, and dat
a management. Standard methods for aggregating multiple preference orders into a
 consensus ranking may perpetuate and even amplify the lack of representation of
 underrepresented groups. To address this, recent research has focused on incorp
orating fairness constraints to ensure the presence of different groups in the t
op-k positions of the final aggregate ranking.

We study two fairness-aware variants under the well-known Spearman footrule, whi
ch corresponds to the L_1 distance between rankings. First, we address the pra
ctically salient task of computing a fair aggregate top-k ranking -- crucial i
n settings like recommendations and hiring where selection is primarily based on
 the top-k results -- and present the first optimal algorithm for this problem
. Second, we consider fair (full) rank aggregation over all candidates (not spec
ifically on top-k). We already know of a 3-approximation for this fair rank
aggregation variant (Wei et al., SIGMOD’22; Chakraborty et al., NeurIPS’22), whe
reas an exact algorithm exists for the corresponding unconstrained (unfair) vers
ion (Dwork et al., WWW’01). Closing the computational gap between fair and uncon
strained rank aggregation has remained a tantalizing open problem. We make signi
ficant progress by giving a 2-approximation algorithm for fair (full) rank agg

regation, improving substantially over the previous 3-approximation. Further, we complement our theoretical contributions with experiments on different real-world datasets, which corroborate our theoretical results and demonstrate strong empirical performance relative to state-of-the-art baselines.
****
Data- and Variance-dependent Regret Bounds for Online Tabular MDPs
Mingyi Li, Taira Tsuchiya, Kenji Yamanishi
This work studies online episodic tabular Markov decision processes (MDPs) with known transitions and develops best-of-both-worlds algorithms that achieve refined data-dependent regret bounds in the adversarial regime and variance-dependent regret bounds in the stochastic regime. We quantify MDP complexity using a first-order quantity and several new data-dependent measures for the adversarial regime, including a second-order quantity and a path-length measure, as well as variance-based measures for the stochastic regime. To adapt to these measures, we develop algorithms based on global optimization and policy optimization, both built on optimistic follow-the-regularized-leader with log-barrier regularization. For global optimization, our algorithms achieve first-order, second-order, and path-length regret bounds in the adversarial regime, and in the stochastic regime, they achieve a variance-aware gap-independent bound and a variance-aware gap-dependent bound that is polylogarithmic in the number of episodes. For policy optimization, our algorithms achieve the same data- and variance-dependent adaptivity, up to a factor of the episode horizon, by exploiting a new optimistic Q-function estimator. Finally, we establish regret lower bounds in terms of data-dependent complexity measures for the adversarial regime and a variance measure for the stochastic regime, implying that the regret upper bounds achieved by the global-optimization approach are nearly optimal.
****
MacroGuide: Topological Guidance for Macrocycle Generation
Alicja Maksymiuk, Alexandre Duplessis, Michael M. Bronstein, Alexander Tong, Fernanda Duarte, Ismail Ilkan Ceylan
Macrocycles are ring-shaped molecules that offer a promising alternative to small-molecule drugs due to their enhanced selectivity and binding affinity against difficult targets. Despite their chemical value, they remain underexplored in generative modeling, likely owing to their scarcity in public datasets and the challenges of enforcing topological constraints in standard deep generative models. We introduce MacroGuide: Topological Guidance for Macrocycle Generation, a diffusion guidance mechanism that uses Persistent Homology to steer the sampling of pretrained molecular generative models toward the generation of macrocycles, in both unconditional and conditional (protein pocket) settings. At each denoising step, MacroGuide constructs a Vietoris-Rips complex from atomic positions and promotes ring formation by optimizing persistent homology features. Empirically, applying MacroGuide to pretrained diffusion models increases macrocycle generation rates from 1% to 99%, while matching or exceeding state-of-the-art performance on key quality metrics such as chemical validity, diversity, and PoseBusters checks. The implementation can be found at https://github.com/ala1705/MacroGuide.
****
FS-I2P: A Hierarchical Focus–Sweep Registration Network with Dynamically Allocated Depth
Zhixin Cheng, Yujia Chen, Xujing Tao, Bohao Liao, Xiaotian Yin, Baoqun Yin, Tianzhu Zhang
Image-to-point cloud registration is often challenged by viewpoint changes, cross-modal discrepancies, and repetitive textures, which induce scale ambiguity and consequently lead to erroneous correspondences. Recent detection-free methods alleviate this issue by leveraging multi-scale features and transformer-based interactions. However, they still suffer from attention drift across layers and intra-scale inconsistencies, hindering precise registration.
Inspired by complex scene observation, we propose a ``Focus--Sweep'' paradigm and develop a Hierarchical Mamba Interaction Module within an SSM-based framework to enhance multi-level cross-modal feature association. In addition, we introduce a Dynamic Layer Allocation Strategy that adaptively determines the iteration d

epth to better exploit geometric constraints and improve matching robustness. Extensive experiments and ablations on two benchmarks, RGB-D Scenes V2 and 7-Scenes, demonstrate that our approach achieves state-of-the-art performance.
****
Quartet II: Accurate LLM Pre-Training in NVFP4 by Improved Unbiased Gradient Estimation
Andrei Panferov, Erik Schultheis, Soroush Tabesh, Dan Alistarh
The NVFP4 lower-precision format, supported in hardware by NVIDIA Blackwell GPUs, promises to allow, for the first time, end-to-end fully-quantized pre-training of massive models such as LLMs.

Yet, existing quantized training methods still sacrifice some of the representation capacity of this format in favor of more accurate unbiased quantized gradient estimation by stochastic rounding (SR), losing noticeable accuracy relative to standard FP16 and FP8 training.

In this paper, improve the state of the art for quantized training in NVFP4 via a novel unbiased quantization routine for micro-scaled formats, called MS-EDEN, that has more than 2x lower quantization error than SR. We integrate it into a novel fully-NVFP4 quantization scheme for linear layers, called Quartet II. We show analytically that Quartet II achieves consistently better gradient estimation across all major matrix multiplications, both on the forward and on the backward passes.

In addition, our proposal synergizes well with recent training improvements aimed specifically at NVFP4. We further validate Quartet II on end-to-end LLM training with up to 1.9B parameters on 38B tokens. We provide kernels for execution on NVIDIA Blackwell GPUs with up to 4.2x speedup over BF16.
****
Hallucination Detection from Structural Reasoning Model
Jianbo Sun, Pengkun Yang
Hallucinations pose a key challenge for large
language models, and chain-of-thought prompt■ing exposes intermediate reasoning but usually
treats traces as linear sequences, making cross■step dependencies and unsupported intermediate
claims difficult to identify. We propose a struc■tural reasoning model to describe interactions
among local reasoning steps. To detect halluci■nations, we extract a directed acyclic reasoning
graph over conditions and intermediate claims,
verify each claim against its parent nodes, and ag■gregate the step signals with a simple mass-flow
rule. Under a probabilistic erasure-gate abstrac■tion, we interpret this aggregation as measuring
information loss along the reasoning graph. Ex■periments on GSM8K, MATH, HumanEval, and
HotpotQA show that the proposed method is most
advantageous on longer, dependency-rich reason■ing traces such as math and code generation,
while remaining competitive on shorter factual
QA; these results provide a structured perspec■tive on chain-of-thought evaluation. All code and
data are available at https://github.com/soncheinbok/FlowScore.
****
Spatiotemporal Imputation with Graph-Informed Flow Matching
Zepeng Zhang, Aref Einizade, Jhony H. Giraldo, Olga Fink
Missing data is a common challenge in spatiotemporal systems, arising in applications such as air quality monitoring and urban traffic management.
Traditional machine learning approaches, like recurrent and graph neural networks, rely on iterative propagation, which tends to accumulate errors over time and space.
Recent diffusion-based methods mitigate error propagation but require iterative

sampling and often depend on problem-agnostic Gaussian priors, limiting both efficiency and effectiveness.
To address these limitations, we propose **GiFlow**, a *Graph-Informed Flow Matching* framework for spatiotemporal imputation.
GiFlow replaces the typical Gaussian prior with a graph-informed prior constructed via spatiotemporal filtering of observable signals, which better aligns the source distribution to the target and thereby simplifies the generation trajectory.
The flow field is parameterized by a hybrid vector field model that integrates spatial attention, temporal attention, and spatiotemporal propagation, enabling joint modeling of spatial and temporal dependencies.
Extensive experiments on both synthetic and real-world datasets demonstrate that the proposed GiFlow outperforms the state-of-the-art approaches in spatiotemporal imputation. The code is available at https://github.com/zepengzhang/GiFlow.
****

At the Edge of Understanding: Sparse Autoencoders Trace The Limits of Transformer Generalization
Praneet Suresh, Jack Stanley, Sonia Joseph, Luca Scimeca, Danilo Bzdok
Pre-trained transformers have demonstrated remarkable generalization abilities, at times extending beyond the scope of their training data. Yet, real-world deployments often face unexpected or adversarial data that diverges from training data distributions. Without explicit mechanisms for handling such shifts, model reliability and safety degrade, urging more disciplined study of out-of-distribution (OOD) settings for transformers. By systematic experiments, we present a mechanistic framework for delineating the precise contours of transformer model robustness. We find that OOD inputs, including subtle typos and jailbreak prompts, drive language models to operate on an increased number of fallacious concepts in their internals. We leverage this device to quantify and understand the degree of distributional shift in prompts, enabling a mechanistically grounded fine-tuning strategy to robustify LLMs. Expanding the very notion of OOD from input data to a model’s private computational processes—a new transformer diagnostic at inference time—is a critical step toward making AI systems safe for deployment across science, business, and government.
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S2GS: Streaming Semantic Gaussian Splatting for Online Scene Understanding and Reconstruction
Renhe Zhang, Yuyang Tan, Jingyu Gong, Zhizhong Zhang, Lizhuang Ma, Yuan Xie, Xin Tan
Existing offline feed-forward methods for joint scene understanding and reconstruction on long image streams often perform global computation over an ever-growing set of past observations, causing runtime and GPU memory to increase rapidly with sequence length. We propose Streaming Semantic Gaussian Splatting (S2GS), a strictly causal and incremental framework that builds a 3D Gaussian semantic field from image streams without accessing future frames or reprocessing historical observations. S2GS continuously updates scene geometry, appearance, and instance-level semantics through a geometry--semantic decoupled dual-backbone design. The geometry branch performs causal modeling for incremental Gaussian updates, while the semantic branch leverages a 2D foundation vision model and a query-driven decoder to predict masks and identity embeddings. Query-level contrastive alignment and lightweight online association with an instance memory further stabilize temporal identities. Experiments show that S2GS matches or outperforms strong offline baselines, while substantially improving long-horizon scalability: it processes over 1,000 frames with much slower runtime and GPU memory growth, whereas offline global-processing baselines typically run out of memory at around 80 frames under the same setting. Project Page: https://stdcoutzrh.github.io/projects/s2gs/.
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VeriSimpl: Robust Optimization Modeling from Natural Language using Simplification-based Verification
Sumaya Abdul Rahman, Seckhen Ariel Andrade Cuellar, Ghani Raissov, Mohammad Raza

Natural language interfaces can greatly benefit the accessibility and usability of optimization modeling, and recent advances in large language models (LLMs) show promise in automatically translating textual problem descriptions into executable solver formulations. However, a key challenge for existing approaches is to ensure that the inferred formulation correctly implements the intended task, even if it may execute without errors. We introduce VeriSimpl, a solver–LLM framework for robust natural-language-to-optimization formalization. Our approach is based on the idea of simplification-based verification, where the optimization solver is leveraged to generate simplified diagnostic queries about a candidate formulation to allow the LLM to tractably reason about the correctness of the formulation with respect to the task description. We present such simplification strategies along different dimensions with respect to problem constraints and decision variables, which allow the LLM to reason locally under fixed global contexts. Evaluations on a range of optimization benchmarks show how our approach provides consistent improvements in accuracy over existing methods, while also providing a novel high-precision self-verification signal.
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ActiveUltraFeedback: Efficient Preference Data Generation using Active Learning
Davit Melikidze, Marian Schneider, Jessica Lam, Martin Wertich, Ido Hakimi, Barna Pásztor, Andreas Krause
Reinforcement Learning from Human Feedback (RLHF) has become the standard for aligning Large Language Models (LLMs), yet its efficacy is bottlenecked by the high cost of acquiring preference data, especially in low-resource and expert domains. To address this, we introduce ActiveUltraFeedback, a modular active learning pipeline that leverages uncertainty estimates to dynamically identify the most informative responses for annotation. Our pipeline facilitates the systematic evaluation of standard response selection methods alongside Double Reverse Thompson Sampling (DRTS) and DeltaUCB, two novel methods prioritizing response pairs with large predicted quality gaps, leveraging recent results showing that such pairs provide good signals for fine-tuning. Our experiments demonstrate that ActiveUltraFeedback yields high-quality datasets that lead to significant improvements in downstream performance, notably achieving comparable or superior results with as little as one-sixth of the annotated data relative to static baselines. Our pipeline is available at https://github.com/lasgroup/ActiveUltraFeedback and our preference datasets at https://huggingface.co/ActiveUltraFeedback.
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InfVSR: Toward Consistency-Driven Streaming Generative Video Super-Resolution
Ziqing Zhang, Kai Liu, Zheng Chen, Xi Li, Yucong Chen, Bingnan Duan, Linghe Kong, Yulun Zhang
Real-world videos often extend over thousands of frames. Existing generative video super-resolution (VSR) approaches, however, face two persistent challenges when processing long sequences: (1) inefficiency due to the heavy cost of multi-step denoising for full-length sequences; and (2) poor consistency hindered by temporal decomposition that causes artifacts and discontinuities. To break these limits, we propose InfVSR, which reformulates VSR as an autoregressive-one-step-diffusion paradigm, and enables streaming inference with video diffusion priors. First, we adapt the pretrained DiT into a causal structure, maintaining both local and global coherence via rolling KV-cache and joint visual guidance. Second, we distill the diffusion process into a single step efficiently, with patch-wise pixel supervision and cross-chunk distribution matching. To fill the gap in long-form video evaluation, we build a new benchmark tailored for extended sequences and further introduce semantic-level metrics to comprehensively assess temporal consistency. Our method pushes the frontier of long-form VSR, achieves state-of-the-art quality with enhanced semantic consistency, and delivers up to 58x speed-up over existing methods such as MGLD-VSR. Our code and models are available at https://github.com/Kai-Liu001/InfVSR .
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Beyond Point Predictions: Manifold Expansion and Dual Alignment for Robust Time Series Distillation
Junyao Hong, Zesheng Lai, Xinyi Xiao, Suyang Zhou, Aodong Shen, Youyong Kong

Knowledge Distillation (KD) promises to bridge the gap between the high computational costs of Transformer-based models and the expressiveness limitations of linear models in long-term time series forecasting. Many existing time series distillation methods inherit the computer vision paradigm, constraining student models by minimizing point-wise prediction matching (output-level distillation) errors. However, blindly mimicking teacher predictions, which can be uncertain, may induce negative transfer. To address this, we propose Dynamic Structural Distillation (DSD), a robust framework that goes beyond the prediction-matching paradigm. First, we design LMP-Net, leveraging manifold expansion to project features into a high-dimensional latent space, alleviating the expressiveness bottleneck while preserving lightweight inference. Second, to address token–point architectural mismatch, we propose Dual Manifold Alignment, employing Similarity-Preserving Knowledge Distillation (SPKD) and Optimal Transport (OT) to align features at the topological and geometric levels, respectively. Finally, we introduce Regime-Aware Adaptive Distillation (RAAD) to reduce the risk of teacher misguidance via a dataset-level regime prior and a confidence-based adaptive gating mechanism. Extensive experiments on five benchmarks show that DSD is compatible with diverse teacher architectures, improves lightweight students across data regimes, and achieves a favorable accuracy–efficiency trade-off. Code is available at https://github.com/jyh0526/DSD.
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PLASH: Provably Linear-Time Attention with Selective Higher-Order Feature Sketching

Yuwen Huang, Xiang Pan

Standard softmax attention scales quadratically with sequence length, which makes long-context training and inference expensive.

We introduce PLASH, an attention block whose cost grows linearly in the number of keys. PLASH compresses the original keys and values into M learned prototypes, where $M\in\mathbb{Z}_{>0}$ is much smaller than the number of keys. The compressed prototypes are then enriched with randomized polynomial features that recover inter-token information lost to compression. The output is computed by exact scaled dot-product softmax attention from each query to the enriched prototypes, so PLASH preserves the standard attention interface.The construction applies to self- and cross-attention. We prove sketch-error bounds for the enrichment step, a per-input certificate that upper-bounds the deviation from standard softmax attention on each forward pass, and a runtime bound linear in the number of queries and keys. Experiments on long-context language modeling (Qwen3-4B on PG-19) and time-series forecasting (ETT, ECL, Weather) show competitive accuracy and favorable scaling against efficient-attention baselines.
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LC-QAT: Data-Efficient 2-Bit QAT for LLMs via Linear-Constrained Vector Quantization

Haoyu Wang, Xingyu Yu, Haiyan Zhao, Fengxiang Wang, Xu Han

Quantization-aware training (QAT) is essential for extremely low-bit large language models (LLMs). Current QAT methods are mainly based on scalar quantization (SQ), which enables efficient optimization but suffers from severe performance degradation at 2-bit precision.

On the other hand, vector quantization (VQ) provides substantially higher representational capacity, but its discrete codebook lookup prevents end-to-end training.

We propose LC-QAT, a 2-bit weight-only VQ-QAT framework that represents quantized weights via a learned affine mapping over discrete vectors, which yields a high-quality PTQ initialization and enables fully differentiable end-to-end optimization without explicit codebook lookup in the training forward pass.

This strong post-training initialization makes LC-QAT highly data-efficient. Experiments across diverse LLMs demonstrate that LC-QAT consistently outperforms state-of-the-art QAT methods while using only 0.1%–10% of the training data. Our results establish LC-QAT as a practical and scalable solution for extreme low-bit model deployment.
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Optimal Regret for Policy Optimization in Contextual Bandits
Orin Levy, Yishay Mansour
We present the first high-probability optimal regret bound for a policy optimiza tion technique applied to the problem of stochastic contextual multi-armed bandi t (CMAB) with general offline function approximation. Our algorithm is both effi cient and achieves an optimal regret bound of $\widetilde{O}(\sqrt{ K|\mathcal{A }|\log|\mathcal{F}|})$, where K is the number of rounds, $\mathcal{A}$ is the set of arms, and $\mathcal{F}$ is the function class used to approximate the los ses.
Our results bridge the gap between theory and practice, demonstrating that the w idely used policy optimization methods for the contextual bandit problem can ach ieve a rigorously-proved optimal regret bound.
We support our theoretical results with an empirical evaluation of our algorithm .
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Transformed Latent Variable Multi-Output Gaussian Processes
Xiaoyu Jiang, Xinxing Shi, Sokratia Georgaka, Magnus Rattray, Mauricio A Álvarez
Multi-Output Gaussian Processes (MOGPs) provide a principled probabilistic frame work for modelling correlated outputs but face scalability bottlenecks when appl ied to datasets with high-dimensional output spaces. To maintain tractability, e xisting methods typically resort to restrictive assumptions, such as employing l ow-rank or sum-of-separable kernels, which can limit expressiveness. We propose the Transformed Latent Variable MOGP (T-LVMOGP), a novel framework that scales M OGPs to a massive number of outputs while preserving the capacity to capture mea ningful inter-output dependencies. T-LVMOGP constructs a flexible multi-output d eep kernel by mapping inputs and output-specific latent variables into an embedd ing space using a Lipschitz-regularised neural network. Combined with stochastic variational inference, our model effectively scales to high-dimensional output settings. Across diverse benchmarks, including climate modelling with over $10,0 00$ outputs and zero-inflated spatial transcriptomics data, T-LVMOGP outperforms baselines in both predictive accuracy and computational efficiency.
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Correcting Overparameterization Effects in Fair Empirical Risk Minimization
Xiaoyi MAI, Jean-Michel Loubes
Bias mitigation is particularly challenging for overparameterized machine learni ng (ML) models. Overfitting of training points not only amplifies data bias indu ced by spurious correlations, but also causes the failure of usual bias mitigati on methods. To provide actionable insights to address this challenge, we propos e a precise analysis of fair empirical risk minimization (ERM) in the overparame terized regime. Importantly, we show that even though conventional fair ERM fai ls on overparameterized models, this approach can be corrected by modifying the equality fairness constraint to allow for bias overcompensation. Moreover, our a nalysis presents an empirical criterion for strong equalized odds: balanced grou p-conditional means of representer coefficients, indicating equal average contri bution from each sensitive group. Motivated by this result, we provide an estima ble search interval that localizes the required overcompensation level for balan ced coefficients. Despite the asymptotic nature of our findings, they closely ca pture the statistical behavior of moderately large ML models.
****
MixReasoning: Switching Modes to Think
Haiquan Lu, Gongfan Fang, Xinyin Ma, Qi Li, Xinchao Wang
Reasoning models enhance performance by tackling problems in a step-by-step mann er, decomposing them into sub-problems and exploring long chains of thought befo re producing an answer. However, applying extended reasoning to every step intro duces substantial redundancy, as sub-problems vary widely in difficulty and comp lexity: a small number of pivotal steps are genuinely challenging and decisive f or the final answer, while many others only involve straightforward revisions or simple computations. Therefore, a natural idea is to endow reasoning models wit h the ability to adaptively respond to this variation, rather than treating all steps with the same level of elaboration. To this end, we propose MixReasoning,

a framework that dynamically adjusts the depth of reasoning within a single response. MixReasoning enables fine-grained mode switching by training a lightweight concise LoRA adapter and control its strength to trigger switches based on reasoning difficulty estimated from sliding-window token confidence, yielding human-like transitions between fast and slow reasoning. The resulting chain of thought then becomes a mixture of detailed reasoning on difficult steps and concise inference on simpler ones.
Experiments on AIME24, MATH-500, GPQA, and GSM8K demonstrate that MixReasoning shortens reasoning length by 13\%--49\% across benchmarks of varying difficulty, delivering consistent efficiency gains while maintaining performance.
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Efficient Distributed MLLM Training with Cornstarch
Insu Jang, Runyu Lu, Nikhil Bansal, Ang Chen, Mosharaf Chowdhury
Multimodal large language models (MLLMs) extend the capabilities of large language models (LLMs) by combining heterogeneous model architectures to handle diverse modalities like images and audio.
However, this inherent heterogeneity in MLLM model structure and data types makes makeshift extensions to existing LLM training frameworks unsuitable for efficient MLLM training, especially in distributed training.

In this paper, we present Cornstarch, an efficient distributed MLLM training framework that contemplates MLLM's unique characteristics in both model and data parallelization.
Cornstarch introduces frozen-aware pipeline parallelism and workload-balanced context parallelism to improve MLLM training throughput.
Our extensive evaluation shows that Cornstarch outperforms state-of-the-art solutions by $2.26\times$ on average in terms of MLLM training throughput.

Cornstarch is an open-source project and available on [Github](https://github.com/cornstarch-org/Cornstarch).
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Predicting What Matters: Robust Generalist Robot Policy Learning via Future Semantic Mask
Yunfan Lou, Xiaowei Chi, Xiaojie Zhang, Zezhong Qian, Chengxuan Li, Rongyu Zhang, Yaoxu Lyu, Guoyu Song, Chuyao Fu, Haoxuan Xu, Pengwei Wang, Shanghang Zhang
World models derived from large-scale video generative pre-training have emerged as a promising paradigm for generalist robot policy learning. However, standard approaches often focus on high-fidelity RGB video prediction, this can result in overfitting to irrelevant factors, such as dynamic backgrounds and illumination changes. These distractions reduce the model's ability to generalize, ultimately leading to unreliable and fragile control policies. To address this, we introduce the Mask World Model (MWM), which leverages video diffusion architectures to predict the evolution of semantic masks instead of pixels. This shift imposes a geometric information bottleneck, forcing the model to capture essential physical dynamics and contact relations while filtering out visual noise. We seamlessly integrate this mask dynamics backbone with a diffusion-based policy head to enable robust end-to-end control. Extensive evaluations demonstrate the superiority of MWM on the LIBERO and RLBench simulation benchmarks, significantly outperforming the state-of-the-art RGB-based world models. Furthermore, real-world experiments and robustness evaluation (via random token pruning) reveal that MWM exhibits superior generalization capabilities and robust resilience to texture information loss.
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Decision Tree Learning on Product Spaces
Arshia Soltani Moakhar, Faraz Ghahremani, Kiarash Banihashem, MohammadTaghi Hajiaghayi
Decision tree learning has long been a central topic in theoretical computer science, driven by its practical importance. A fundamental and widely used method for decision tree construction is the top-down greedy heuristic, which recursively splits on the most influential variable. Despite its empirical success, theore

tical analysis of this heuristic has been limited. A recent breakthrough by Blanc et al. (ITCS, 2020) provided the first rigorous theoretical guarantees for the greedy approach, but only under the uniform distribution.
We extend this analysis to the more general and practically relevant setting of arbitrary product distributions.
Our main result shows that for any function f computable by an optimal decision tree of size s, maximum depth D_{opt}, and average depth Δ_{opt}, the greedy heuristic constructs an ϵ-approximating tree whose size grows at most with $\exp(\Delta_{\text{opt}} D_{\text{opt}} \log(e/\epsilon))$.
In the special case where the optimal tree is a full binary tree, this bound improves upon the bound of Blanc et al. and holds under a strictly broader class of distributions. Moreover, we present an algorithm based on the top-down greedy heuristic that is entirely **parameter-free**---it requires no prior knowledge of the optimal tree's size or depth---offering a practical advantage over Blanc et al.'s method.
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Gradient Descent with Large Step Size Restores Symmetry in Deep Linear Networks with Multi-Pathway
Hee-Sung Kim, Sungyoon Lee
Recent analyses of multi-pathway Deep Linear Networks use Gradient Flow to predict a "winner-takes-all" specialization in which path symmetry breaks and each feature concentrates in a single pathway.
In this work, we show that discrete Gradient Descent (GD) with a large step size tells a different story.
We prove that single-path solutions are sharp minima, whereas distributing signals across pathways reduces sharpness by a factor that decreases with both the number of pathways and depth.
Consequently, while early training reproduces the depth-driven symmetry breaking predicted by GF, oscillations at the Edge of Stability subsequently override this tendency and drive the network into a re-balancing phase, where signals redistribute across pathways.
Together, these results clarify how depth shapes pathway competition and explain why large-step GD favors shared representations rather than persistent single-pathway dominance.
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Accuracy-First Rényi Differential Privacy and Post-Processing Immunity
Ossi Räisä, Antti Koskela, Antti Honkela
The accuracy-first perspective of differential privacy addresses an important shortcoming by allowing a data analyst to adaptively adjust the quantitative privacy bound instead of sticking to a predetermined bound. Existing works on the accuracy-first perspective have neglected an important property of differential privacy known as post-processing immunity, which ensures that an adversary is not able to weaken the privacy guarantee by post-processing. We address this gap by determining which existing definitions in the accuracy-first perspective have post-processing immunity, and which do not. The only definition with post-processing immunity, pure ex-post privacy, lacks useful tools for practical problems, such as an ex-post analogue of the Gaussian mechanism, and an algorithm to check if accuracy on separate private validation set is high enough. To address this, we propose a new definition based on Rényi differential privacy that has post-processing immunity, and we develop basic theory and tools needed for practical applications. We demonstrate the practicality of our theory with applications to synthetic data generation and image classifier fine-tuning, where our algorithm successfully adjusts the privacy bound until an accuracy threshold is met on a private validation dataset.
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Reasoning on the Manifold: Bidirectional Consistency for Self-Verification in Diffusion Language Models
Jiaoyang Ruan, Xin Gao, Yinda Chen, Hengyu Zeng, Liang Du, Guanghao Li, Jie Fu, Jian Pu

While Diffusion Large Language Models (dLLMs) offer structural advantages for gl
obal planning, efficiently verifying that they arrive at correct answers via val
id reasoning traces remains a critical challenge. In this work, we propose a geo
metric perspective: Reasoning on the Manifold. We hypothesize that valid generat
ion trajectories reside as stable attractors on the high-density manifold of the
 learned distribution, whereas invalid paths exhibit off-manifold drift. To oper
ationalize this, we introduce Bidirectional Manifold Consistency (BMC), a traini
ng-free, unsupervised metric that quantifies the stability of the generated sequ
ence through a forward-masking and backward-reconstruction cycle. Empirically, w
e demonstrate BMC's versatility across the full reasoning lifecycle: (1) in Diag
nosis, it serves as a robust discriminator of solution validity without ground t
ruth answer; (2) in Inference, it enables rejection resampling to effectively co
ncentrate computational resources on complex reasoning tasks; and (3) in Alignme
nt, it functions as a dense geometric reward that transforms sparse outcome supe
rvision into fine-grained guidance, empowering models to self-evolve beyond stan
dard baselines. Our results establish intrinsic geometric stability as a robust
indicator of correctness for dLLMs.
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When Search Goes Wrong: Red-Teaming Web-Augmented Large Language Models
Haoran Ou, Kangjie Chen, Xingshuo Han, Gelei Deng, Jie Zhang, Han Qiu, Tianwei Z
hang, Kwok-Yan Lam
Large Language Models (LLMs) have been augmented with web search to overcome the
 limitations of the static knowledge boundary by accessing up-to-date informatio
n from the open Internet. While this integration enhances model capability, it a
lso introduces a distinct safety threat surface: the retrieval and citation proc
ess has the potential risk of exposing users to harmful or low-credibility web c
ontent. Existing red-teaming methods are largely designed for standalone LLMs as
 they primarily focus on unsafe generation, ignoring risks emerging from the com
plex search workflow.
To address this gap, we propose CREST-Search, a pioneering red-teaming framework
 for LLMs with web search. The cornerstone of CREST-Search is three novel attack
 strategies that generate seemingly benign search queries yet induce unsafe cita
tions. It also employs an iterative in-context refinement mechanism to strengthe
n adversarial effectiveness under black-box constraints.
In addition, we construct a search-specific harmful dataset, WebSearch-Harm, whi
ch enables fine-tuning a specialized red-teaming model to improve query quality.
 Our experiments demonstrate that CREST-Search can effectively bypass safety fil
ters and systematically expose vulnerabilities in web search-based LLM systems,
underscoring the necessity of the development of robust search models.
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Hyperbolic Neural Operator
Jieyuan Pei, Zhuoxuan Li, Wei Li, Haobo Zhang, jiawei jiang, Jianwei Zheng
Neural operators have emerged as powerful surrogates for solving PDEs, significa
ntly accelerating scientific computation.
While transformer-based architectures offer unmatched flexibility for irregular
domains, they suffer from a fundamental efficiency gap: standard attention mecha
nisms assign uniform interaction budgets to all token pairs, neglecting the phys
ical reality that far-field interactions are often compressible.
To address this mismatch, we draw inspiration from classical fast solvers that e
xploit hierarchical near-far decompositions. We further observe that embedding s
uch tree-structured hierarchies in Euclidean space incurs inherent distortion, w
hereas hyperbolic space naturally accommodates exponential branching.
Consequently, we propose \textbf{Hyperbolic Neural Operator (HNO)}, which levera
ges intrinsic hyperbolic geometry to instantiate a continuous Gibbs kernel based
 on stabilized geodesic distances on the Lorentz hyperboloid. This design impose
s a geometric inductive bias that naturally enjoys robust multi-scale routing ak
in to the Fast Multipole Method (FMM), yet within a unified, learnable attention
 mechanism.
Empirically, HNO achieves state-of-the-art accuracy on six PDE benchmarks and tw
o large-scale unstructured CFD tasks, reducing the mean relative ℓ_2 error

by up to 40\% compared to leading baselines. Codes are attached and will be avai
lable online.
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From Bits to Rounds: Parallel Decoding with Exploration for Diffusion Language M
odels
Hengyu Fu, Baihe Huang, Virginia Adams, Charles Wang, Junkeun Yi, Mohammad Mahdi
 Kamani, Venkat Krishna Srinivasan, Jiantao Jiao
Diffusion Language Models (DLMs) have recently emerged as a strong alternative t
o autoregressive language models (AR-LMs), due to their comparable accuracy and
faster inference speed via parallel decoding.
However, standard DLM decoding strategies, which rely on unmasking only high-con
fidence tokens, encounter an inherent information-theoretic bottleneck that rest
ricts decoding progress and ultimately slows down generation.
We demonstrate this through an information-theoretic lower bound that the number
 of decoding rounds must grow linearly with the sample's total information and i
nversely with the per-round information budget, establishing a bits-to-rounds pr
inciple.
Motivated by this theory, we propose Explore-Then-Exploit (ETE), a training-free
 decoding strategy that maximizes information throughput and decoding efficiency
. ETE combines cross-block decoding with targeted exploration of high-uncertaint
y tokens to reshape the conditional distribution and trigger cascades of confide
nt predictions.
Experiments across diverse benchmarks verify our theoretical bounds and demonstr
ate that ETE consistently reduces the number of decoding rounds compared to conf
idence-only baselines without compromising generation quality.
Furthermore, ETE integrates efficiently with KV caching, translating these algor
ithmic gains into improved tokens-per-second throughput.
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Deep sequence models tend to memorize geometrically; it is unclear why
Shahriar Noroozizadeh, Vaishnavh Nagarajan, Elan Rosenfeld, Sanjiv Kumar
Deep sequence models are said to store atomic facts predominantly in the form of
 associative memory: a brute-force lookup of co-occurring entities. We identif
y a dramatically different form of storage of atomic facts that we term as *geom
etric* memory. Here, the model has synthesized embeddings encoding novel *global
* relationships between all entities, including ones that do not co-occur in tra
ining. Such storage is powerful: for instance, we show how it transforms a hard
 reasoning task involving an ℓ-fold composition into an easy-to-learn 1-s
tep navigation task.

From this phenomenon, we extract fundamental aspects of neural embedding geometr
ies that are hard to explain. We argue that
the rise of such a geometry, as against a lookup of local associations, cannot b
e straightforwardly attributed to typical supervisory, architectural, or optimiz
ational pressures. Counterintuitively, a geometry is learned even when it is mor
e complex than the brute-force lookup.

Then, by analyzing a connection to Node2Vec, we demonstrate how the geometry ste
ms from a spectral bias that—in contrast to prevailing theories—indeed arises na
turally despite the lack of various pressures. This analysis also points out to
practitioners a visible headroom to make Transformer memory more strongly geomet
ric. We hope the geometric view of parametric memory encourages revisiting the d
efault intuitions that guide researchers in areas like knowledge acquisition, ca
pacity, discovery, and unlearning.
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Beyond Fixed Biases: Decoding the Role of Reasoning Uncertainty in MLLM Modality
 Conflicts
Zhuoran Zhang, Tengyue Wang, Xilin Gong, Yang Shi, Haotian Wang, Di Wang, Lijie
Hu
Multimodal Large Language Models (MLLMs) must resolve conflicts when modalities
provide contradictory information, a behavior we term "modality following". We p

ropose a framework that decomposes this behavior into case-specific relative pre ference uncertainty and stable inherent preference. Across diverse MLLMs and ben chmarks, the probability of following a modality consistently decreases as its r elative preference uncertainty increases, a trend robust to alternative uncertai nty indices. This regularity defines a "balance point'' where modality preferenc es are evenly matched, offering a capability-disentangled measure of modality bi as. Layer-wise probing further shows that ambiguous cases near the balance point trigger middle-to-late-layer "concept oscillations," where top predictions vaci llate between modality-supported answers. Finally, we demonstrate the framework' s utility for preference steering through Supervised Fine-Tuning (SFT). We find that data efficiency is governed by preference uncertainty: training on easy sam ples (where one modality dominates) fails to generalize, whereas targeting the i dentified ``boundary cases" is essential for robust preference alignment and sup pressing internal vacillation.

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SpecForge: A Flexible and Efficient Open-Source Training Framework for Speculati ve Decoding

Shenggui Li, Chao Wang, YIKAI ZHU, Yubo Wang, Fan Yin, Shuai Shi, Yefei Chen, Xi aomin Dong, Qiaoling Chen, Jin Pan, Ji Li, Yineng Zhang, Lei Yu, Yonggang Wen, I vor Tsang, Tianwei Zhang

Speculative decoding mitigates the memory-bound nature of LLM decoding by using a lightweight draft model to propose multiple tokens for parallel verification. However, its adoption has been limited by the lack of high-quality draft models and scalable training infrastructure. We introduce SpecForge, an open-source and efficient framework for training speculative decoding models with full support for EAGLE-3. SpecForge incorporates target–draft decoupling, hybrid parallelism, optimized training kernels, and tight integration with production-grade inferen ce engines, enabling up to 9.9x faster EAGLE-3 training for Qwen3-235B-A22B comp ared to the baseline. We further release SpecBundle, a suite of production-grade EAGLE-3 draft models trained with SpecForge for mainstream open-source LLMs, ac hieving up to 4.48x end-to-end inference speedup on SGLang and addressing the sc arcity of high-quality drafts. Finally, we distill a systematic study of specula tive decoding training into practical and actionable recipes to guide real-world adoption.

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SkillNet: Hierarchical Skill Modeling for Compositional Generalization in Vision -Language Action Models

Senwei Xie, Yuntian Zhang, Zhenzhou Tan, Ruiping Wang, Pengwei Wang, Shanghang Z hang, Xilin CHEN

Transfer across diverse task compositions and unseen behaviors remains a signifi cant challenge for vision-language action (VLA) models. Skills are repeatable an d atomic components for various tasks, and similarities shared with different sk ills provide evidence for transferability across behaviors. However, existing sk ill-centric methods have two problems. First, skills are often loosely organized , lacking a hierarchy that can capture similarities and differences across skill s. Second, they lack a mechanism which has the capacity to express transferable skill attributes in a structured parametric space. To this end, we propose Skill Net, which models skill attributes in a hierarchical way and regulates compositi onal model structure with transferable skill attributes. SkillNet exploits motio n code and VerbNet Framework to explicitly model similarities of skills on mecha nical properties and semantic roles, and organizes skills in a hierarchical way. Based on this hierarchy, SkillNet leverages the scalability of the mixture-of-e xperts (MoE) mechanism and develops skill embeddings as soft constraints to enab le compositional generalization via similar expert activations on similar skills . On zero-shot and few-shot transfer experiments in simulators and real-world en vironments, SkillNet achieves an improvement of performance by 16.0% and 23.9%. Meanwhile, SkillNet achieves state-of-the-art performance on in-domain settings.

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Retro-Expert: Collaborative Reasoning for Interpretable Retrosynthesis

Xinyi Li, Sai Wang, Yutian Lin, Yu Wu

Retrosynthesis prediction aims to infer the reactant molecules based on a given product molecule, which is a fundamental task in chemical synthesis. However, existing methods rely on a static pattern-matching paradigm, which limits their ability to perform effective logical decision-making from chemical data, leading to a black-box process. We propose Retro-Expert, an interpretable retrosynthesis framework that performs collaborative reasoning by combining the complementary strengths of Large Language Models and specialized models via pure reinforcement learning. It outputs natural language explanations grounded in chemical logic through three components: (1) specialized models provide chemical knowledge that is distilled into a high-quality chemical decision space, (2) LLM-driven critical reasoning to generate predictions with an interpretable reasoning path, and (3) knowledge-grounded policy optimization refines the interpretable decision policy. Experiments show that Retro-Expert surpasses both LLM-based and specialized models across different metrics, while generating chemically grounded explanations that enhance chemists' trust in practice. The source code for this paper is available at https://github.com/MagixRab-ll/Retro-Expert

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Experience is the Best Teacher: Motivating Effective Exploration in Reinforcement Learning for LLMs

Wenjian Zhang, Kongcheng Zhang, Jiaxin Qi, Jianqiang Huang, Baisheng Lai

Reinforcement Learning (RL) with rubric-based rewards has recently shown remarkable progress in enhancing general reasoning capabilities of Large Language Models (LLMs), yet still suffers from ineffective exploration confined to current policy distribution. In fact, RL optimization can be viewed as steering the policy toward an ideal distribution that maximizes the rewards, while effective exploration should align efforts with desired target. Leveraging this insight, we propose HeRL, a ****H***indsight ***e***xperience guided ***R***einforcement ***L***earning* framework to bootstrap effective exploration by explicitly *telling LLMs the desired behaviors* specified in rewards. Concretely, HeRL treats failed attempts along with their unmet rubrics as hindsight experience, which serves as in-context guidance for the policy to explore desired responses beyond its current distribution. Additionally, we introduce a bonus reward to incentivize responses with greater potential for improvement under such guidance. HeRL facilitates effective learning from desired high-quality samples without repeated trial-and-error from scratch, yielding a more accurate estimation of the expected gradient theoretically. Extensive experiments across various benchmarks demonstrate that HeRL achieves superior performance gains over baselines, and can further benefit from experience guided self-improvement at test time.

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H^{2}CL: Heterogeneity-Aware Hypergraph Contrastive Learning for Robust Representation Learning

Kaixuan Yao, Ting Guo, Feilong Cao, Ming Li

Hypergraph contrastive learning has shown strong potential for modeling high-order relations, but most existing methods implicitly assume that nodes within the same hyperedge are semantically homogeneous. In real-world hypergraphs, this assumption often breaks down: heterogeneous node--hyperedge interactions may introduce noisy signals during augmentation and message passing, leading to feature contamination and false-positive contrastive alignment. To address this issue, we propose H^{2}CL, a heterogeneity-aware hypergraph contrastive learning framework for robust representation learning. H^{2}CL first estimates node--hyperedge heterogeneity from input features and uses it to guide a Heterogeneity-Aware View Generator, which selectively masks high-heterogeneity incidences and constructs cleaner contrastive views. It further introduces a Heterogeneity-Aware Hypergraph Encoder that dynamically reweights information propagation in both node-to-hyperedge and hyperedge-to-node aggregation, enabling hyperedges to aggregate more homogeneous signals while suppressing heterogeneous noise. We also provide theoretical analysis showing that the encoder corresponds to a coordinate descent step for minimizing a heterogeneity-weighted Dirichlet energy. Extensive experiments on standard benchmarks and larger-scale hypergraphs demonstrate that H^{2}CL achieves competitive or superior performance compared with recent baselines, remai

ns robust under structural noise, and learns reweighting patterns that effectively reduce heterogeneity during message passing. Our code is available at: https://github.com/sxu-yaokx/HHCL

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Tighter Regret Lower Bound for Gaussian Process Bandits with Squared Exponential Kernel in Hypersphere

Shogo Iwazaki

We study an algorithm-independent, worst-case lower bound for the Gaussian process (GP) bandit problem in the frequentist setting, where the reward function is fixed and has a bounded norm in the known reproducing kernel Hilbert space (RKHS). Specifically, we focus on the squared exponential (SE) kernel, one of the most widely used kernel functions in GP bandits. One of the remaining open questions for this problem is the gap in the *dimension-dependent* logarithmic factors between upper and lower bounds. This paper partially resolves this open question under a hyperspherical input domain. We show that any algorithm suffers $\Omega(\sqrt{T (\ln T)^{d} (\ln \ln T)^{-d}})$ cumulative regret, where T and d represent the total number of steps and the dimension of the hyperspherical domain, respectively. Regarding the simple regret, we show that any algorithm requires $\Omega(\epsilon^{-2}(\ln \frac{1}{\epsilon})^d (\ln \ln \frac{1}{\epsilon})^{-d})$ time steps to find an ϵ-optimal point. We also provide the improved $O((\ln T)^{d+1}(\ln \ln T)^{-d})$ upper bound on the maximum information gain for the SE kernel. Our results guarantee the optimality of the existing best algorithm up to *dimension-independent* logarithmic factors under a hyperspherical input domain.

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QuantumBoost: A lazy, yet fast, quantum algorithm for learning with weak hypotheses

Amira Abbas, Yanlin Chen, Tuyen Quang Nguyen, Ronald de Wolf

The technique of combining multiple votes to enhance the quality of a decision is the core of boosting algorithms in machine learning. In particular, boosting provably increases decision quality by combining multiple "weak learners"—hypotheses that are only slightly better than random guessing—into a single "strong learner" that classifies data well. There exist various versions of boosting algorithms, which we improve upon through the introduction of QuantumBoost. Inspired by classical work by Barak, Hardt and Kale, our QuantumBoost algorithm achieves the best known runtime over other boosting methods through two innovations. First, it uses a quantum algorithm to compute approximate Bregman projections faster. Second, it combines this with a lazy projection strategy, a technique from convex optimization where projections are performed infrequently rather than every iteration. To our knowledge, QuantumBoost is the first algorithm, classical or quantum, to successfully adopt a lazy projection strategy in the context of boosting.

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Sampling from Your Language Model One Byte at a Time

Jonathan Hayase, Alisa Liu, Noah A. Smith, Sewoong Oh

Tokenization is used almost universally by modern language models, enabling efficient text representation using multi-byte or multi-character tokens. However, prior work has shown that tokenization can introduce distortion into the model's generations, an issue known as the Prompt Boundary Problem (PBP). For example, users are often advised not to end their prompts with a space because it prevents the model from including the space as part of the next token. While this heuristic is effective in English, the underlying PBP continues to affect code generation and languages such as Chinese, where tokens often do not line up with word and syntactic boundaries. In this work, we present an inference-time method to convert any autoregressive LM with a BPE tokenizer into a character-level or byte-level LM. Our method efficiently solves the PBP and is also able to unify the vocabularies of language models with different tokenizers, allowing one to ensemble LMs with different tokenizers at inference time or transfer the post-training from one model to another using proxy-tuning.

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Flex-Forcing: Towards a Unified Autoregressive and Bidirectional Video Diffusion Model

Xinyin Ma, Julius Berner, Chao Liu, Arash Vahdat, Weili Nie, Xinchao Wang

Recent progress in large-scale generative models has substantially advanced video generation, yet existing methods remain constrained by a rigid inference paradigm. Bidirectional diffusion models excel at global coherence and visual fidelity but suffer from slow inference, while autoregressive models offer efficient and streaming generation at the cost of long-range consistency and exposure bias. We introduce Flex-Forcing, a unified training and inference framework that enables a video diffusion model to seamlessly operate under both bidirectional and autoregressive generation regimes. The core idea is a flexible chunking mechanism jointly defined over the temporal axis and denoising steps. This design allows the model to (1) perform flexible chunking according to different device budgets, (2) perform bidirectional inference across chunks for global structure planning, while generating frames autoregressively within each chunk for efficient and fine-grained synthesis, and (3) perform any-order, any-timestep autoregressive generation without the strict causal constraint. Extensive experiments on multiple video generation benchmarks demonstrate that Flex-Forcing achieves consistently better video quality, long-video stability than strong baselines with a rigid inference schedule, while offering faster inference.

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OSM+: Billion-Level Open Street Map Dataset for City-wide Experiments

Guanjie Zheng, Ziyang Su, Yiheng Wang, Yuhang Luo, Hongwei Zhang, Xuanhe Zhou, Linghe Kong, Fan Wu, Wen Ling

Road network data provides rich information about cities, but processing worldwide OpenStreetMap (OSM) data is computationally intensive, and the resulting graphs are often difficult to unify for benchmarking downstream tasks. Existing graph learning benchmarks fail to capture the billion-scale and unique topological properties of real-world road networks, leaving model scalability underexplored. To close this gap, we process OSM data with distributed cloud computing using 5,000 cores and release **OSM+**, a structured worldwide 1-billion-vertex road network graph dataset designed for high accessibility and usability. OSM+ is open source and globally downloadable, providing an open-box graph structure together with an easy spatial query interface that allows users to retrieve, inspect, and integrate road network topology, geometry, and attributes without repeatedly preprocessing raw OSM files. We demonstrate the utility of OSM+ through four illustrative use cases: basic query, city boundary detection, traffic prediction, and traffic policy control. For traffic prediction, we construct a new 31-city benchmark by processing traffic data and combining it with OSM+, enabling broader spatial coverage and more comprehensive evaluation than commonly used datasets, while scaling from hundreds of road network intersections to thousands. For traffic policy control, we release a new six-city dataset at a much larger scale, introducing challenges for thousand-scale multi-agent coordination and controller scalability. We also provide data processing tools for integrating multimodal spatial-temporal data with OSM+ for geospatial foundation model training, thereby expediting the discovery of compelling scientific insights.

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Global Directional Priors with Local Statistical Validation for Scalable Causal Discovery

Wei Yuan, Zixuan Shao, Shuhui Wang

Constraint-based causal discovery relies on conditional independence (CI) tests whose reliability degrades as conditioning sets grow, particularly in hub-dominated graphs. Existing methods constrain adjacency or global structure, but leave conditioning-set dimensionality uncontrolled. In this paper, we propose Ordering-Constrained Markov Blanket discovery (OCMB), a paradigm that treats conditioning-set dimensionality as a first-class constraint. OCMB decouples discovery into two stages: lightweight global ordering estimation providing directional priors, followed by local Markov blanket validation within small, ordering-constrained candidate sets. By enforcing directional constraints before any CI test, OCMB ensures bounded conditioning sets even with hub nodes. We show that OCMB recovers

correct parent sets provided a high-recall ordering assumption holds, without re
quiring the ordering to be globally correct. Experiments demonstrate that OCMB s
ignificantly improves precision and robustness over constraint-based and hybrid
methods in high-dimensional regimes where conventional CI-based approaches fail.
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MAD: Manifold Attracted Diffusion

Dennis Elbrächter, Giovanni S Alberti, Matteo Santacesaria

Score-based diffusion models are a highly effective method for generating sample
s from a distribution of images. We consider scenarios where the training data c
omes from a noisy version of the target distribution, and present an efficiently
 implementable modification of the inference procedure to generate noiseless sam
ples. Our approach is motivated by the manifold hypothesis, according to which m
eaningful data is concentrated around some low-dimensional manifold of a high-di
mensional ambient space. The central idea is that noise manifests as low magnitu
de variation in off-manifold directions in contrast to the relevant variation of
 the desired distribution which is mostly confined to on-manifold directions. We
 introduce the notion of an extended score and show that, in a simplified settin
g, it can be used to reduce small variations to zero, while leaving large variat
ions mostly unchanged. We describe how its approximation can be computed efficie
ntly from an approximation to the standard score and demonstrate its efficacy on
 toy problems, synthetic data, and real data.
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Preconditioning Neural Tangent Kernel for Adaptive Optimization

Xiyuan Yang, Wenxuan Bao, Katherine Tieu, Jingrui He

The Neural Tangent Kernel is a theoretical framework for understanding the train
ing dynamics of neural networks. However, standard NTK and its variants fail to
properly depict the finetuning of foundation models, as they neglect the precond
itioning effects of adaptive gradients.

To bridge this gap, we propose the Optimizer Aware Kernel (OAK), which incorpora
tes the optimizer's influence into standard NTK framework by a preconditioner es
timation technique. Furthermore, we conduct an analysis to answer: when and why
kernel regime fails in finetuning. We derive explicit error bounds showing that
the collapse of kernel regime is primarily due to the cumulative training effect
s and the task discrepancy between pretraining and finetuning. Theoretically, we
 justify OAK's preconditioner estimation by bounding its error term. Empirically
, experiments on various model architectures show both the effectiveness of the
OAK method and validity of our arguments on kernel regime collapse.
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MVISTA-4D: View-Consistent 4D World Model with Test-Time Action Inference for Ro
botic Manipulation

Jiaxu Wang, JIANG Yicheng, Tianlun HE, Jingkai SUN, Qiang Zhang, Jiahang Cao, Ze
sen Gan, Mingyuan Sun, Qiming Shao, Xiangyu Yue

World-model-based imagine-then-act becomes a promising paradigm for robotic mani
pulation, yet existing approaches typically support either purely image-based fo
recasting or reasoning over partial 3D geometry, limiting their ability to predi
ct complete 4D scene dynamics. To solve this, this work explores a novel embodie
d 4D world model that enables geometrically consistent, arbitrary-view RGBD gene
ration: given only a single-view RGBD observation as input, the model “imagines”
 the remaining viewpoints, which can then be back-projected and fused to assembl
e a more complete 3D structure across time. To efficiently learn the multi-view,
 cross-modality generation, we explicitly design cross-view and cross-modality f
eature fusion that jointly encourage consistency between RGB and depth and enfor
ce geometric alignment across views. Beyond prediction, converting generated fut
ures into actions is often handled by inverse dynamics, which is ill-posed becau
se multiple actions can explain the same transition. We address this with a test
-time action optimization strategy that backpropagates through the generative mo
del to infer a trajectory-level latent best matching the predicted future, and a
 residual inverse dynamics model that turns this trajectory prior into accurate
executable actions. Extensive experiments on the three datasets and platforms de
monstrate strong performance on both 4D scene generation and downstream manipula

tion, and ablations provide practical insights into the key design choices. Proj
ect page is available at \url{https://mercerai.github.io/MVISTA-4D/}.
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Normalization Equivariance for Arbitrary Backbones, with Application to Image De
noising
Youssef Saied, François Fleuret
Normalization Equivariance (NE) is a structural prior that improves robustness t
o distribution shift in image-to-image tasks.
A function f is normalization equivariant iff $f(a y + b\mathbf{1}) = a f(y) +
 b\mathbf{1}$ for all $a>0$ and $b\in\mathbb{R}$.
Existing NE methods constrain every internal layer to NE-compatible operations.
These constraints add runtime cost and exclude standard transformer components s
uch as softmax attention and LayerNorm.
We introduce Wrapped Normalization Equivariance (WNE), a parameter-free wrapper
that normalizes the input, applies any backbone, and denormalizes the output.
We prove every NE function admits this factorization, so the wrapper exactly par
ameterizes the class of NE functions.
On blind denoising, wrapping CNN and transformer architectures improves robustne
ss under noise-level mismatch with no measurable GPU overhead, while architectur
al NE baselines are up to $1.6\times$ slower.
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Rectified LpJEPA: Joint-Embedding Predictive Architectures with Sparse and Maxim
um-Entropy Representations
Yilun Kuang, Yash Dagade, Tim G. J. Rudner, Randall Balestriero, Yann LeCun
Joint-Embedding Predictive Architectures (JEPA) learn view-invariant representat
ions and admit projection-based distribution matching for collapse prevention. E
xisting approaches regularize representations towards isotropic Gaussian distrib
utions, but inherently favor dense representations and fail to capture the key p
roperty of sparsity observed in efficient representations. We introduce Rectifie
d Distribution Matching Regularization (RDMReg), a sliced two-sample distributio
n-matching loss that aligns representations to a Rectified Generalized Gaussian
(RGG) distribution. RGG enables explicit control over expected ℓ_0 norm thr
ough rectification, while its continuous truncated component admits a maximum-en
tropy characterization under expected ℓ_p norm and support constraints. Equ
ipping JEPAs with RDMReg yields Rectified LpJEPA, which strictly generalizes pri
or Gaussian-based JEPAs. Empirically, Rectified LpJEPA learns sparse, non-negati
ve representations with favorable sparsity--performance trade-offs and competiti
ve downstream performance on image classification benchmarks, showing that RDMRe
g can enforce sparsity while preserving task-relevant information.
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AutoTool: Dynamic Tool Selection and Integration for Agentic Reasoning
Jiaru Zou, Ling Yang, Yunzhe Qi, Sirui Chen, Mengting Ai, Ke Shen, Jingrui He, M
engdi Wang
Agentic reinforcement learning has advanced large language models (LLMs) to reas
on through long chain-of-thought trajectories while interleaving external tool u
se. Existing approaches assume a fixed inventory of tools, which limits the adap
tability of LLM agents to new or evolving toolsets. We present AutoTool, a train
ing framework that equips LLM agents with dynamic tool-selection capabilities th
roughout their reasoning trajectories. AutoTool employs a dual-phase optimizatio
n pipeline: (i) SFT and RL-based trajectory stabilization for coherent reasoning
, and (ii) KL-regularized Plackett–Luce Ranking to refine consistent multi-step
tool selection.
We further build a 200k dataset with explicit tool-selection rationales across 1
,000+ tools and 100+ tasks spanning mathematics, science, code generation, and m
ultimodal reasoning. Across ten diverse benchmarks, we train two base models, Qw
en3-8B and Qwen2.5-VL-7B, with AutoTool. With fewer parameters, AutoTool consist
ently outperforms advanced LLM agents and tool-integration methods, yielding ave
rage gains of 6.4\% in math \& science reasoning, 4.5\% in search-based QA, 7.7\
% in code generation, and 6.9\% in multimodal understanding. In addition, AutoTo
ol exhibits stronger generalization by dynamically leveraging unseen tools from

evolving toolsets during inference.
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Sharpness-Aware Pretraining Mitigates Catastrophic Forgetting
Ishaan Watts, Catherine Li, Sachin Goyal, Jacob Mitchell Springer, Aditi Raghunathan
Pretraining optimizers are tuned to produce the strongest possible base model, on the assumption that a stronger starting point yields a stronger model after subsequent changes like post-training and quantization. This overlooks the geometry of the base model which controls how much of the base model's capabilities survive subsequent parameter updates. We study three pretraining optimization approaches that bias optimization toward flatter minima: Sharpness-Aware Minimization (SAM), large learning rates, and shortened learning rate annealing periods. Across model sizes ranging from 20M to 150M parameters, we find that these interventions consistently improve downstream performance after post-training on five common datasets with up to 80\% less forgetting. These principles hold at scale: a short SAM mid-training phase applied to an existing OLMo-2-1B checkpoint reduces forgetting by 31\% after MetaMath post-training and by 40\% after 4-bit quantization.
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SpatioLM: Towards General Physical Spatial Intelligence in Vision-Language Models
jing wu, Jianhua Wu, Jiayi Guan, Jiahong Chen, Jinghui Lu, Hangjun Ye, Bingzhao Gao, Long Chen
Vision-Language Models (VLMs) perform well on commonsense reasoning tasks but struggle with visual spatial reasoning. Most existing solutions introduce extra 3D prior inputs or external spatial encoders, which increase complexity and degrade the underlying VLMs' general-purpose capabilities after spatial fine-tuning. To this end, we propose a parameter-efficient \textit{\textbf{Spatio}-vision \textbf{L}anguage \textbf{M}odels (SpatioLM)}, that enhances spatial intelligence without extra 3D prior inputs or third-party spatial encoders. Concretely, we design a plug-and-play and non-invasive spatio-vision module that elicits the spatial knowledge inherent in VLMs. Furthermore, we innovatively leverage pseudo depth and camera information as supervision to guide the model in learning physically coherent representations. Extensive experiments show that SpatioLM achieves significant improvements in diverse tasks, including spatial perception and understanding while effectively limiting the degradation of general capabilities. Notably, the model achieves an impressive score of 71.6 on the VSI-Bench (the first model to surpass 70). In addition, it attains competitive performance when transferred to embodied manipulation tasks.
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Solving Imperfect-Recall Games via Sum-of-Squares Optimization
Rui Zheng, Ryann Sim, Antonios Varvitsiotis
Extensive-form games (EFGs) provide a powerful framework for modeling sequential decision making, capturing strategic interaction under imperfect information, chance events, and temporal structure. Most positive algorithmic and theoretical results for EFGs assume perfect recall, where players remember all past information and actions. We study the increasingly relevant setting of imperfect-recall EFGs (IREFGs), where players may forget parts of their history or previously acquired information, and where equilibrium computation is provably hard. We propose sum-of-squares (SOS) hierarchies for computing ex-ante optimal strategies in single-player IREFGs and Nash equilibria in multi-player IREFGs, working over behavioral strategies. Our theoretical results show that (i) these hierarchies converge asymptotically, (ii) under genericity assumptions, the convergence is finite, and (iii) in single-player non-absentminded IREFGs, convergence occurs at a finite level determined by the number of information sets. Finally, we introduce the new classes of (SOS)-concave and (SOS)-monotone IREFGs, and show that in the single-player setting the SOS hierarchy converges at the first level, enabling equilibrium computation with a single semidefinite program (SDP).
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Multi-Objective Protein Design via Memory-Aware Test-Time Scaling in Diffusion M

odels
Ming Yang, Xin Zheng, Yi Li, YIZHEN ZHENG, Huan Yee Koh, Yanqing Guo, Xiaofeng C ao, Shirui Pan
Multi-objective protein design is essential for meeting the complex demands of s ynthetic biology. To adapt to shifting multi-functional targets without the proh ibitive cost of retraining, test-time scaling has emerged as a flexible, trainin g-free alternative. However, current test-time diffusion methods face critical challenges: i) ineffective learning from interaction history leading to repetiti ve design errors, ii) over-reliance on successful cases as the reward signal, an d iii) difficulties in balancing multi-objective functional trade-offs . To addr ess these limitations, we propose MoMST, a framework for Multi-objective protein design via Memory-aware Self-contrastive learning with Test-time scaling in dif fusion models. At test time, we develop a memory bank to extract generalizable r easoning experience from historical iterations. Building on this powerful experi ence learner, we derive rich residue-level relative preference signals from both successful and failed cases via self-contrastive learning for guiding protein g eneration. To ensure balance among competing multi-objective functions, we prese nt an inference-time Pareto alignment strategy to resolve objective conflicts. E valuations on both single-objective and complex multi-objective tasks demonstrat e the remarkable performance of MoMST, with code available at https://github.com /MingYangi/MoMST.
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KernelCraft: Benchmarking for Agentic Close-to-Metal Kernel Generation on Emergi ng Hardware
Jiayi Nie, Haoran Wu, Yao Lai, Zeyu Cao, Cheng Zhang, Binglei Lou, Erwei Wang, J ianyi Cheng, Timothy M. Jones, Robert D. Mullins, Rika Antonova, Yiren Zhao
New AI accelerators with novel instruction set architectures (ISAs) often requir e developers to manually craft low-level kernels — a time-consuming and error-pr one process that does not scale across hardware targets. This delays emerging ha rdware platforms from reaching the market. While prior LLM-based code generation has shown promise in mature GPU ecosystems, it remains unclear whether agentic LLM systems can quickly produce valid and efficient kernels for emerging hardwar e with new ISAs. We present KernelCraft: the first benchmark for evaluating an L LM agent's ability to generate and optimize low-level kernels for customized acc elerators through a function-calling, feedback-driven workflow. We evaluate agen t performance across three emerging accelerators on more than 20 machine-learnin g tasks, each with five diverse task configurations. Across four leading reasoni ng models, the strongest agents generate functionally correct kernels for unseen ISAs within a few refinement steps, and produce optimized kernels that match or outperform compiler baselines. These results demonstrate KernelCraft's potentia l to accelerate the accelerator chip development cycle. KernelCraft is available at https://kernelcraft-cam.github.io/.
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Inverse Depth Scaling From Most Layers Being Similar
Yizhou Liu, Sara Kangaslahti, Ziming Liu, Jeff Gore
Neural scaling laws relate loss to model size in large language models (LLMs), y et depth and width may contribute to performance differently, requiring more det ailed studies. Here, we quantify how depth affects loss via analysis of LLMs and toy residual networks. We find loss scales inversely proportional to depth in L LMs, probably due to functionally similar layers reducing error through ensemble averaging rather than compositional learning or discretizing smooth dynamics. T his regime is inefficient yet robust and may arise from the architectural bias o f residual networks and target functions incompatible with smooth dynamics. The findings suggest that improving LLM efficiency may require architectural innovat ions to encourage compositional use of depth.
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Future-Gain Guided Test-Time Learning for Large Language Models
LangYu Bian, Jinwu Hu, Zitian Zhang, Dongjin Yang, Yufeng Wang, Qing Du, Qi Chen , Mingkui Tan
Large language models (LLMs) inevitably encounter distribution shifts during rea

l-world deployment, leading to performance degradation. Although test-time learn ing (TTL) adapts LLMs from unlabeled test streams, applying entropy minimization to autoregressive generation faces two challenges: (i) early decoding errors ca n steer later tokens off track, and updating on them can push the model further off course, and (ii) updates on unreliable tokens can amplify confident error pr edictions and trigger model collapse. To address these challenges, we propose Fu ture-Gain Guided Test-Time Learning (FG-TTL) for LLMs, which learns selectively from the model's own generations. Our key idea is to update only on tokens that reduce uncertainty in subsequent generation rather than tokens that are merely u ncertain at the current step. Specifically, we develop a Future-Gain Guided Toke n Selection (FTS) strategy to decide where to learn. We introduce Future-Gain as a token-level metric for this purpose and update the model only on high-gain to kens, concentrating learning on informative positions and mitigating temporal er ror propagation. In addition, we design a Risk-Aware Adaptation (RAA) mechanism that controls how strongly to update by combining gain-based weighting with adap tive temperature scaling based on intrinsic uncertainty, suppressing unreliable gradients while enabling stronger learning on high-gain tokens. Experiments on s ix benchmarks with three LLM backbones show that FG-TTL achieves the best averag e performance.

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Selective Deferred Routing: Enabling Cost-Efficient Collaboration between Local SLMs and Remote LLMs

Qijun Miao, Zhixuan Fang

The rapid advancement of large language models (LLMs) has led to remarkable perf ormance across diverse domains, making them indispensable assistants in daily li fe and work. Currently, LLM services are primarily accessed in two ways: (i) pai d access to cloud-hosted LLMs, which are powerful but introduce nontrivial cost; and (ii) deployment of small language models (SLMs) on personal devices or smal l clusters, which, while less powerful, are sufficient for handling relatively s imple tasks. To achieve a balanced trade-off between monetary cost and task perf ormance, we propose Selective Deferred Routing, a paradigm that enables cost-eff icient collaboration between local SLMs and remote LLMs. In this framework, a us er request is first processed by the local SLM, which not only generates a preli minary response but also provides rich semantic representations of the request. A lightweight decision module then leverages this information to either adopt th e initial response or route the request to the most suitable remote LLM for a hi gher-quality response. Extensive experiments across diverse model architectures and families, including both SLMs and LLMs, as well as datasets spanning multipl e task scenarios, demonstrate that our approach consistently outperforms existin g multi-LLM collaboration methods under a wide range of cost–performance trade-o ffs.

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Multiple Choice Learning of Low-Rank Adapters for Language Modeling

Victor Letzelter, Hugo Malard, Mathieu Fontaine, Gaël Richard, Slim Essid, Andre i Bursuc, Patrick Perez

We propose LoRA-MCL, a training scheme that extends next-token prediction in lan guage models with a method designed to decode diverse, plausible sentence contin uations at inference time. Traditional language modeling is an intrinsically ill -posed problem: given a context, multiple ``futures'' may be equally plausible. Our approach leverages Multiple Choice Learning (MCL) and the winner-takes-all l oss to efficiently handle ambiguity through Low-Rank Adaptation. We provide a th eoretical interpretation of applying MCL to language modeling, assuming the data is generated from a mixture of distributions. We illustrate the proposed approa ch using mixtures of Markov chains. We then demonstrate with experiments on audi o and visual captioning, as well as machine translation, that our method achieve s high diversity and relevance in generated outputs. We release the code for app lying LoRA-MCL to a wide range of language models.

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Patterning: The Dual of Interpretability

George Wang, Daniel Murfet

Mechanistic interpretability aims to understand how neural networks generalize beyond their training data by reverse-engineering their internal structures. We introduce patterning as the dual problem: given a desired form of generalization, determine what training data produces it. Our approach is based on susceptibilities, which measure how posterior expectation values of observables respond to infinitesimal shifts in the data distribution. Inverting this linear response relationship yields the data intervention that steers the model toward a target internal configuration. We demonstrate patterning in a small language model, showing that re-weighting training data along principal susceptibility directions can accelerate or delay the formation of structure, such as the induction circuit. In a synthetic parentheses balancing task where multiple algorithms achieve perfect training accuracy, we show that patterning can select which algorithm the model learns by targeting the local learning coefficient of each solution. These results establish that the same mathematical framework used to read internal structure can be inverted to write it.

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Multiview Self-Representation Learning across Heterogeneous Views

Jie Chen, Zhu Wang, Chuanbin Liu, Xi Peng

Features of the same sample generated by different pretrained models often exhibit inherently distinct feature distributions. Learning invariant representations from large-scale unlabeled visual data in a fully unsupervised transfer manner remains a significant challenge. In this paper, we propose a multiview self-representation learning (MSRL) method in which invariant representations are learned by exploiting the self-representation property of features across heterogeneous views. The features are derived from large-scale unlabeled visual data through transfer learning with various pretrained models and are referred to as heterogeneous multiview data. We introduce an information-passing mechanism that relies on self-representation learning to support feature aggregation over the outputs of the linear model. Moreover, an assignment probability distribution consistency scheme is presented to guide multiview self-representation learning by exploiting complementary information across different views. Consequently, representation invariance across different linear models is enforced through this scheme. Additionally, we provide a theoretical analysis of the assignment probability distribution consistency and the incremental views. Extensive experiments demonstrate that the proposed MSRL method consistently outperforms several state-of-the-art approaches.

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Geometric Conformal Prediction with Spatial Ranks and Multivariate Quantiles

Anton Conrad, Eric Moulines, julien perez

In multi-target regression and multi-class classification, uncertainty is inherently multivariate: prediction regions must capture joint dependencies across correlated outputs. Conformal prediction provides distribution-free guarantees, yet extending it to vector-valued outputs remains challenging—scalar aggregation discards geometric structure, while optimal transport (OT) approaches are computationally demanding and sensitive to outliers. We introduce two conformal methods based on geometric quantiles and spatial ranks: Geometric Conformalized Quantile Regression (GCQR) constructs prediction regions from learned conditional geometric quantiles, while Geometric Rank Conformal Prediction (GRCP) uses the radial rank of vector-valued conformity scores as the nonconformity measure. We propose multiple estimators offering different tradeoffs between computational cost and adaptivity to feature-dependent heterogeneity, with scalable learning via partially input-convex neural networks. On multi-target regression and multi-class classification benchmarks, GCQR and GRCP attain near-nominal coverage with consistently tighter prediction regions than scalarized and multivariate baselines.

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Shared Semantics, Divergent Mechanisms: Unsupervised Feature Discovery by Aligning Semantics and Mechanisms

Hyunjin Cho, Youngji Roh, Jaehyung Kim

As large language models are increasingly deployed in high-stakes settings, there is a growing need for tools that audit not only model outputs but also the int

ernal computations that produce them.
Circuit analysis is a central approach in mechanistic interpretability, but it i s typically target-conditioned, explaining a single prompt paired with a chosen completion.
This target-conditioned setup can obscure heterogeneity across a model's continu ation distribution.
We introduce distribution-level unsupervised feature discovery, which clusters s ampled continuations using both semantic content and sequence-level mechanistic attributions, without manually specifying target outputs.
Our method represents each continuation with a semantic embedding and a prefix-t o-continuation attribution signature, then optimizes a rate-distortion objective that trades off semantic coherence, mechanistic consistency, and cluster granul arity.
Across clustering and steering analyses, the discovered clusters expose continua tion modes that single-view baselines miss and provide interventional evidence t hat cluster signatures correspond to actionable mechanistic factors.
Overall, our approach complements circuit analysis and behavioral evaluation by providing a scalable audit of the mechanisms underlying a model’s continuation d istribution.
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ABCD: All Biases Come Disguised
Mateusz Nowak, Xavier Cadet, Peter Chin
Multiple-choice question (MCQ) benchmarks have been a standard evaluation practi ce for measuring LLMs' ability to reason and answer knowledge-based questions. T hrough a synthetic NonsenseQA benchmark, we observe that different LLMs exhibit varying degrees of label-position-few-shot-prompt bias, where the model either u ses the answer position, the label in front of the answer, the distributions of correct answers present in the few-shot prompt, or a combination of all to answe r each MCQ question. We propose a simple bias-reduced evaluation protocol that r eplaces the labels of each question with uniform, unordered labels and prompts t he LLM to use the whole answer presented. With a simple sentence similarity mode l, we demonstrate improved robustness and lower standard deviation between diffe rent permutations of answers with a minimal drop in the LLM's performance, expos ing the LLM's capabilities under reduced evaluation artifacts, without any help from the prompt examples or the option labels. Across multiple benchmarks and mo dels, this protocol substantially improves the robustness to answer permutations , reducing mean accuracy variance $3\times$ with only a minimal decrease in the model's mean performance. Through ablation studies on various embedding models a nd similarity functions, we show that the method is more robust than the standar d ones.
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FasterVAR: Plug-and-Play Acceleration for Visual Autoregressive Models
Senmao Li, Kai Wang, Salman Khan, Fahad Shahbaz Khan, jian Yang, Yaxing Wang
Visual Autoregressive (VAR) modeling departs from the next-token prediction para digm of traditional Autoregressive (AR) models through next-scale prediction, en abling high-quality image generation. However, the VAR paradigm suffers from sha rply increased computational complexity and running time at large-scale steps. A lthough existing acceleration methods reduce runtime for large-scale steps, but rely on manual step selection and overlook the varying importance of different s tages in the generation process. To address this challenge,
we present FasterVAR, a systematic study and plug-and-play acceleration framewor k for VAR models. Our analysis shows that early steps are critical for preservin g semantic and structural consistency and should remain intact, while later step s mainly refine details and can be pruned or approximated for acceleration. Buil ding on these insights, FasterVAR introduces a plug-and-play acceleration strate gy that exploits semantic irrelevance and low-rank properties in late-stage comp utations, without requiring additional training.
Our proposed FasterVAR achieves up to 3.4× speedup with almost no performance lo ss, consistently outperforming existing acceleration baselines. These results hi ghlight stage-aware design as a powerful principle for efficient visual autoregr

essive image generation. Code: https://github.com/sen-mao/FasterVAR
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A Machine-Learned Comorbidity Index
Suleman Baloch, Kishlay Jha, Alberto Maria Segre, Philip M. Polgreen, Bijaya Adhikari
Traditional comorbidity scores (e.g., Charlson and Elixhauser) are widely used for risk adjustment and patient stratification, but they have two key limitations: (i) they are largely mortality-centric and do not align well with other clinical outcomes, and (ii) their linear, rule-based structure cannot capture nonlinear, outcome-specific risk relationships. We propose a Machine-Learned Comorbidity Index (MLCI) that maps diagnosis codes to a single scalar by maximizing the normalized Hilbert–Schmidt Independence Criterion (nHSIC) between the learned score and multiple clinical outcomes. MLCI captures nonlinear risk–outcome dependence and is supported by a theory that characterizes when a unified, informative admission-level ordering can be achieved across outcomes. Empirical results on multiple benchmark electronic health record (EHR) datasets show that MLCI outperforms strong baselines across multiple evaluation metrics.
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Optimal Transport with Symmetry Groups
Jiechao Zhang, Huichun Zhang, Jian Sun, Wei Zeng
We propose a novel algorithm that accelerates optimal transport by exploiting intrinsic symmetries induced by finite group actions. The core of our approach is to recover the orbit decomposition and the associated algebraic structure directly from the cost matrix—without requiring prior knowledge of the group—and to reduce the original transport problem to a substantially smaller problem on the orbit space. This reduction preserves optimality while achieving a significant drop in computational complexity. We develop efficient solvers for two central classes of optimal transport: linear OT and entropy-regularized OT. Experiments on synthetic data, real-world image datasets, and molecular graph data confirm the efficiency and robustness of the method. To our knowledge, this work is the first to systematically incorporate symmetry groups into optimal transport, providing both a theoretical framework and a practical pathway to computational acceleration.
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How Can Mamba Learn In Context with Outliers and Generalize Provably?
Hongkang Li, Songtao Lu, Xiaodong Cui, Pin-Yu Chen, Meng Wang
The Mamba model has gained significant attention for its computational advantages over Transformer-based models, while achieving comparable performance across a wide range of language tasks. Like Transformers, Mamba exhibits in-context learning (ICL) capabilities, i.e., making predictions for new tasks based on a prompt containing input-label pairs and a query, without requiring fine-tuning. Despite its empirical success, the theoretical understanding of Mamba remains limited, largely due to the nonlinearity introduced by its gating mechanism. To the best of our knowledge, this paper presents the first theoretical analysis of the training dynamics of a one-layer Mamba model, which consists of a linear attention component followed by a nonlinear gating layer, and its ICL generalization on unseen binary classification tasks, even when the prompt includes additive outliers. Our analysis shows that Mamba leverages the linear attention layer to select informative context examples and uses the nonlinear gating layer to suppress the influence of outliers. By establishing and comparing to the analysis of linear Transformers under the same setting, we show that although Mamba may require more training iterations to converge, it maintains accurate predictions even when the proportion of outliers exceeds the threshold that a linear Transformer can tolerate. These theoretical findings are supported by empirical experiments.
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Enhancing Conformal Prediction via Class Similarity
Ariel Fargion, Lahav Dabah, Tom Tirer
Conformal Prediction (CP) has emerged as a powerful statistical framework for reliable classification, which generates a prediction set, guaranteed to include the true label with a pre-specified probability. The performance of CP methods is

typically assessed by their average prediction set size. In setups where the cl
asses can be partitioned into semantic groups, e.g., based on shared downstream
actions or more interpretable coarse labels, users can benefit from prediction s
ets that are not only small but also contain a limited number of groups. This pa
per begins by addressing this problem and ultimately offers a widely applicable
tool for boosting any CP method on any dataset. First, given a class partition,
we propose augmenting the CP score function with a term that penalizes predictio
ns with "out-of-group" errors. We theoretically analyze this strategy and prove
its advantages for group-related metrics. Surprisingly, we show mathematically t
hat, for common class partitions, it can also reduce the average set size of *an
y* CP score function. Our analysis reveals the class-similarity factors behind t
his improvement and motivates a variant that can further reduce prediction set s
ize by leveraging the model's embeddings, *without requiring any human semantic
partition*. Finally, we present an extensive empirical study, encompassing promi
nent CP methods, multiple models, and several datasets, which demonstrates that
our class-similarity-based approach consistently enhances CP methods.
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CCLRec: Consensus-driven Contrastive Learning for LLM-enhanced Graph Recommendat
ion

Ting Guo, Dongyu Pei, Litiao Qiu, Xiaoying Liao, KE LIANG, Peng Song, Pinle Qin
Recommendation systems seek to accurately model user preferences from a large se
t of candidate items. Graph neural networks (GNNs) have emerged as a dominant ap
proach in this domain due to their ability to capture high-order user–item inter
actions. Recent efforts have aimed to enhance GNN-based representation learning
by incorporating the semantic reasoning capabilities of large language models (L
LMs). However, existing methods often process graph structural information and L
LM-derived semantic knowledge separately, creating a supervisory gap between str
uctural proximity and semantic relevance. To bridge this gap, we propose CCLRec,
a consensus-driven contrastive learning framework for recommendation. CCLRec de
eply integrates structural and semantic information by identifying consistent si
gnals. Specifically, we first use an LLM to extract semantic representations of
items and to sample candidate positive/negative sets in the semantic space. We t
hen introduce a structural–semantic consensus mining strategy that computes the
intersection between a node’s structural neighbors in the graph and its semantic
ally similar items. This allows us to identify high-confidence positive pairs en
dorsed by both collaborative filtering patterns and LLM-based reasoning. By cent
ering contrastive learning on these consensus pairs and applying a weight-aware
reinforcement mechanism during training, CCLRec significantly amplifies the cont
ribution of high-quality consensus features during training. Experiments across
multiple public benchmarks show that CCLRec consistently outperforms state-of-th
e-art methods on key metrics, demonstrating the effectiveness of our consensus-a
ware design.
****

VividCam: Learning Unconventional Camera Motions from Virtual Synthetic Videos

Qiucheng Wu, Handong Zhao, Zhixin Shu, Jing Shi, Yang Zhang, Shiyu Chang
Although recent video generative models are getting more capable of following ex
ternal camera controls, imposed by either text descriptions or camera trajectori
es, they still struggle to generalize to unconventional camera motions, which is
crucial in creating truly original and artistic videos. The challenge lies in f
inding sufficient training videos with the intended uncommon camera motions. To
this end, we propose VividCam, a training paradigm that enables diffusion models
to learn complex camera motions from synthetic videos, releasing the reliance o
n collecting realistic training videos. VividCam incorporates multiple disentang
lement strategies that isolate camera motion learning from synthetic appearance
artifacts, ensuring more robust motion representation and mitigating domain shif
t.
We show that our design synthesizes a wide range of precisely controlled camera
motions using surprisingly simple synthetic data. Notably, this synthetic data o
ften consists of basic geometries within a low-poly 3D scene and can be efficien
tly rendered by engines like Unity. Our video results can be found in https://wu

qiuche.github.io/VividCamDemoPage/.
****
UrbanFusion: Stochastic Multimodal Fusion for Contrastive Learning of Robust Spatial Representations
Dominik J. Mühlematter, Lin Che, Ye Hong, Martin Raubal, Nina Wiedemann
Forecasting urban phenomena such as housing prices and public health indicators requires the effective integration of various geospatial data. Current methods primarily utilize task-specific models, while recent generic models for spatial representations often support only limited modalities and lack multimodal fusion capabilities. To overcome these challenges, we present UrbanFusion, a spatial representation model that features Stochastic Multimodal Fusion (SMF). The framework employs modality-specific encoders to process different types of inputs, including street view imagery, remote sensing data, cartographic maps, and points of interest (POIs) data. These multimodal inputs are integrated via a Transformer-based fusion module that learns unified representations. An extensive evaluation across 41 tasks in 56 cities worldwide demonstrates UrbanFusion’s strong generalization and predictive performance compared to state-of-the-art GeoAI models. Specifically, it 1) outperforms prior models on location-encoding, 2) allows multimodal input during inference, and 3) generalizes well to regions unseen during training. UrbanFusion can flexibly utilize any subset of available modalities for a given location during both pretraining and inference, enabling broad applicability across diverse data availability scenarios.
****
CUPID in the Model Zoo: Online Matchmaking for Selecting Your Dream LLM
Son Nguyen, Xinyuan Liu, Ransalu Senanayake
Users increasingly face the challenge of selecting an appropriate LLM for a given task from a rapidly growing pool of LLMs, each with distinct but often opaque latent properties. Compounding this challenge, users may lack the vocabulary or awareness to explicitly articulate the characteristics they value in an LLM's responses or deployment. We propose an interaction-efficient active learning framework in which a dueling bandit algorithm iteratively selects pairs of LLMs, collects user feedback about their responses, and updates its belief about the user's latent preferences. We introduce a novel belief-aware upper confidence bound strategy that balances exploration of the model pool with exploitation of inferred preferences, enabling efficient alignment between user needs and LLM capabilities under user-specified cost and time budgets. Through diverse experiments on LLMs and human studies, we experimentally verify that our model can efficiently match users to LLMs at a lower cost.
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Where Concept Erasure Should Occur: Concept–Layer Alignment in Text-to-Video Diffusion Models
Yiwei Xie, Ping Liu, Zheng Zhang
Text-to-video diffusion transformers encode semantic information unevenly across model depth, which constrains effective concept erasure.
We identify a representational bottleneck, termed concept–layer topological alignment, under which target concepts exhibit higher separability at certain representational depths.
Outside these depths, concept and non-target signals remain strongly entangled, limiting the effectiveness of depth-specific erasure.
This observation reframes concept erasure as the problem of identifying representational depths where concept–non-target separation naturally emerges.
Motivated by this structural constraint, we introduce CLEAR, a separability-driven optimization framework for concept erasure that explicitly enforces concept–layer alignment.
CLEAR operationalizes this principle by formulating layer selection as an optimization problem over concept–non-target separability, rather than relying on layer-agnostic or heuristic choices.
To enable this, we introduce a separability-aware objective that favors layers exhibiting stronger concept–non-target separation.
Experiments on large-scale text-to-video diffusion models demonstrate that enfor

cing concept--layer alignment leads to more precise concept suppression while pr
eserving overall generative quality.
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Interventional Processes For Causal Uncertainty Quantification
Hugh Dance, Peter Orbanz, Arthur Gretton
Reliable uncertainty quantification for causal effects is crucial in
high-stakes applications, but remains challenging when the target is an entire f
unction rather
than a scalar estimand. In this work, we introduce a GP-based approach for uncer
tainty quantification of interventional functions. The central idea is to build
on recent work representing *interventional* functions as an inner-product of *o
bservational* functions in a reproducing kernel Hilbert space (RKHS), by constru
cting appropriate GP priors for such functions and inferring posteriors from obs
ervational data. Our approach yields closed-form posterior
moments and tractable training and inference, while avoiding pathologies of prev
ious GP prior constructions for RKHS functions. We further derive a practical pr
ocedure for posterior coverage calibration.
Across synthetic benchmarks, causal Bayesian optimization tasks, and a
large-scale real dataset, our method improves uncertainty
quantification while remaining competitive in causal effect estimation.
****
Motion Planning in Compressed Representation Spaces
Lukas Lao Beyer, Sertac Karaman
Deep learning methods have vastly expanded the capabilities of motion planning i
n robotics applications, as learning priors from large-scale data has been shown
 to be essential in capturing the highly complex behavior required for solving t
asks such as manipulation or navigation for autonomous vehicles. At the same tim
e, model-based planning algorithms based on search or optimization remain an ess
ential tool due to their flexibility, efficiency, and the ability to incorporate
 domain knowledge via expert-designed algorithms and objective functions. We pro
pose a new generative framework to unify these two paradigms. First, we learn an
 autoencoder with a high compression ratio and a latent space of hierarchically
ordered, discrete-valued tokens. Leveraging both the dimensionality reduction an
d the hierarchical coarse-to-fine structure learned by this autoencoder, we then
 perform motion planning by directly searching in the latent space of tokens. Th
is search can optimize arbitrary objective functions specified at test time, pro
viding a large degree of flexibility while maintaining efficiency and producing
realistic solutions by relying on the generative capabilities of the highly comp
ressed autoencoder. We evaluate our method on nuPlan and the Waymo Open Motion D
ataset, showing how latent space search can be used for a variety of guided beha
vior generation tasks, achieving strong performance for closed-loop motion plann
ing and multi-agent guided scenario synthesis without requiring any task-specifi
c training.
****
RaBiT: Residual Aware Binarization Training for Accurate and Efficient LLMs
Youngcheon You, Banseok Lee, Minseop Choi, Seonyoung Kim, Hyochan Chong, Changdo
ng Kim, Youngmin Kim, Dongkyu Kim
Efficient deployment of large language models (LLMs) requires extreme quantizati
on, forcing a critical trade-off between low-bit efficiency and performance. Res
idual binarization promises hardware-friendly, matmul-free inference by stacking
 binary ($\pm$1) layers, but is plagued by pathological feature co-adaptation. W
e identify a key failure mode, which we term inter-path adaptation: during Quant
ization-Aware Training (QAT), parallel residual binary paths learn redundant fea
tures, degrading the error-compensation structure and crippling the model's expr
essive capacity. While prior work relies on heuristic workarounds (e.g., path fr
eezing) that limit model capacity, we propose RaBiT, a novel quantization framew
ork that resolves co-adaptation by algorithmically enforcing a residual hierarch
y. Its core mechanism sequentially derives each binary path from a single shared
 full-precision weight, ensuring each path corrects its predecessor's error. Thi
s process is stabilized by a robust initialization that prioritizes functional p

reservation over mere weight approximation. RaBiT redefines the 2-bit accuracy-efficiency frontier: it achieves state-of-the-art performance, rivals even hardware-intensive Vector Quantization (VQ) methods, and delivers a 4.49$\times$ inference speed-up over full-precision models on an RTX 4090.
****
Kernel-based Maximum-of-difference Test for Two-sample Comparison
Dan Pu, Tianyi Zhu, Yao Yan, Wei Lan
Two-sample comparison is a fundamental problem in machine learning, with broad applications such as generative modeling. Although the maximum mean discrepancy (MMD) is widely used, MMD-based tests often exhibit poor or even counterintuitive performance under covariance- and location-shift alternatives, partly due to cancellation effects induced by their sum-of-differences construction. To address this issue, we propose a kernel-based maximum-of-difference (MOD) test, which maximizes the squared discrepancy between within-sample and between-sample average distances, thereby improving sensitivity to subtle distributional differences. We further develop a fused MOD procedure to adaptively combine multiple kernels. Extensive experiments demonstrate clear performance gains over existing MMD-based methods.
****
CoCoQuant: Breaking the Bandwidth Wall via Co-Optimized Communication and Computation Quantization
Haojie Duanmu, Jifeng Ding, Size Zheng, Xuegui Zheng, Jiangfei Duan, Xingcheng Zhang, Li-Wen Chang, Xin Liu, Dahua Lin
The rapid scaling of large language models (LLMs) has made distributed inference indispensable, yet end-to-end latency is increasingly dominated by communication, forming a critical bandwidth wall that fundamentally limits the practical gains of existing quantization techniques. Existing approaches typically treat communication and computation in isolation, failing to exploit their coupled nature and introducing limited system-level acceleration and accuracy degradation. To address this, we propose CoCoQuant, a co-designed framework that jointly optimizes communication and computation as a unified end-to-end design space. CoCoQuant introduces a precision-aligned graph-rewriting that enables zero-overhead fusion between low-precision communication and computation. CoCoQuant formulates a hardware-aware mixed-precision allocation problem that integrates roofline-based cost modeling with relative sensitivity calibration, solved via global integer linear programming. Extensive experiments on LLMs of varing scales demonstrate that CoCoQuant achieves Pareto-optimal accuracy-latency trade-offs, delivering up to 2.92 end-to-end speedup with a negligible increase in perplexity (0.22).
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HPS: Hyperspherical Parameter Sharing for Efficient Multi-Agent Reinforcement Learning
Hu Fu, Pengyi Li, Hao Chen, Xuanyu Xiang, Biao Luo, Yihua Tan
Parameter Sharing (PS) is widely used to improve efficiency in Multi-Agent Reinforcement Learning (MARL), but it can limit behavioral diversity and degrade performance. This limitation stems from gradient conflicts among agents on shared weights, which hinders effective policy learning. To fully characterize this phenomenon, we propose Geometric Gradient Decomposition Analysis that decomposes gradients with respect to weight vector into radial (scale) and tangential (direction) components and uncover a key insight: agents largely agree on directional updates but substantially disagree on scale updates. Consequently, while recent methods split the shared network into agent-specific subnetworks to mitigate conflicts, they also discard shared directional updates, limiting training efficiency. To address this issue, we propose Hyperspherical Parameter Sharing (HPS), which explicitly decouples direction and scale in parameter sharing. Specifically, HPS constrains the shared backbone weights onto a Riemannian manifold(unit hypersphere), enforcing purely directional learning. Building on this, an agent-specific scale generator outputs multiplicative modulation factors to adjust each agent’s scales, thus preserving heterogeneous response magnitudes without disrupting the shared directions. Experiments on SMAC, SMACv2, VMAS and Predator Prey demonstrate that HPS effectively resolves the scale conflict, significantly outperfor

ming state-of-the-art methods.
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Revisiting Anisotropy in Language Transformers: The Geometry of Learning Dynamic
s

Raphael Bernas, Fanny Jourdan, Antonin Poché, CELINE HUDELOT

Since their introduction, Transformer architectures have dominated Natural Langu
age Processing (NLP). However, recent research has highlighted an inherent aniso
tropy phenomenon in these models, presenting a significant challenge to their ge
ometric interpretation. Previous theoretical studies on this phenomenon are rare
ly based on the underlying representation geometry. In this paper, we extend the
m by providing such theoretical arguments assessing the problematic nature of th
is phenomenon. Furthermore, to observe geometric internal model dynamics, we app
ly mechanistic interpretability (MI) techniques during the model's training chec
kpoints rather than post-hoc, as it is commonly done in the literature. By analy
zing multiple models and their checkpoints -including EuroBERT, the Pythia suite
, and SmolLM2- we investigate the structure of embedding representations and the
ir correlation with the on manifold entropy of their underlying distribution.
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Learning U-Statistics with Active Inference

Xiaoning Wang, Yuyang Huo, Liuhua Peng, Changliang Zou

U-statistics play a central role in statistical inference. In many modern appl
ications, however, acquiring the labels required for U-statistics is costly. M
otivated by recent advances in active inference, we develop an active inference
framework for U-statistics that selectively queries informative labels to impr
ove estimation efficiency under a fixed labeling budget, while preserving valid
statistical inference. Our approach is built on the augmented inverse probabilit
y weighting U-statistic, which is proposed to incorporate the sampling rule an
d machine learning predictions. We characterize the optimal sampling rule that m
inimizes its variance and design practical sampling strategies. We further exten
d the framework to U-statistic-based empirical risk minimization. Experiments
on real datasets demonstrate substantial gains in estimation efficiency over bas
eline methods, while maintaining target coverage.
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Robustifying Vision-Language Models via Test-Time Prompt Adaptation

Xingyu Zhu, Huanshen Wu, Shuo Wang, Beier Zhu, Jiannan Ge, Jiaheng Zhang, Long C
hen

Pre-trained Vision-Language Models (VLMs) such as CLIP achieve strong zero-shot
generalization, but their performance degrades sharply under adversarial perturb
ations. Existing test-time adaptation methods typically rely on sample-level con
fidence heuristics, overlooking the intrinsic distributional structure of the da
ta. This sample-centric approach limits robustness, as it fails to distinguish c
onfident adversarial mispredictions from true semantic consistency. In this work
, we observe that adversarial distortion is structurally brittle: while holistic
 representations are corrupted, semantic integrity is often preserved in the dis
tribution of augmented views. Motivated by this insight, we propose $\texttt{RIT
A}$, a $\textbf{R}$obust test-t$\textbf{I}$me promp$\textbf{T}$ $\textbf{A}$dapt
ation framework that shifts from sample-level estimates to distribution-level al
ignment. Specifically, $\texttt{RITA}$ employs optimal transport to align the di
stribution of augmented visual features with textual prototypes, mitigating adve
rsarial outliers and rectifying cross-modal semantic misalignment. Furthermore,
we introduce a dynamic cache to progressively accumulate reliable cues from the
test stream for online refinement. Extensive experiments demonstrate that $\text
tt{RITA}$ significantly improves adversarial robustness without compromising cle
an accuracy.
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Mechanisms of Introspective Awareness

Uzay Macar, Li Yang, Zifan Wang, Peter Wallich, Emmanuel Ameisen, Jack Lindsey

Recent work shows that LLMs can sometimes detect when steering vectors are injec
ted into their residual stream and identify the injected concept, a phenomenon c
ited as evidence of "introspective awareness." But what mechanisms underlie this

capability, and do they reflect genuine introspective circuitry or more shallow heuristics? We investigate these questions in open-source models and establish three main findings. First, introspection is behaviorally robust: detection achieves moderate true positive rates with 0% false positives across diverse prompts. We also find this capability emerges specifically from post-training rather than pretraining. Second, introspection is not reducible to a single linear confound: anomaly detection relies on distributed MLP computation across multiple directions, implemented by interpretable gate and evidence-carrier features. Third, models possess greater introspective capability than is elicited by default: ablating refusal directions improves detection by ~50% and a trained steering vector improves detection by ~75%. Overall, our results suggest that introspective awareness is behaviorally robust, grounded in nontrivial internal anomaly detection, and likely could be substantially improved in future models.
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Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations
Wei Liu, Jiawei Xu, Yingru Li, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He
High-quality kernel is critical for scalable AI systems, and enabling LLMs to generate such code would advance AI development. However, training LLMs for this task requires sufficient data, a robust environment, and the process is often vulnerable to _reward hacking_ and _lazy optimization_. In these cases, models may hack training rewards or prioritize trivial correctness over meaningful speedup. In this paper, we systematically study reinforcement learning (RL) for kernel generation. We first design **KernelGYM**, a robust distributed GPU environment that supports reward hacking check, data collection from multi-turn interactions and long-term RL training. Building on KernelGYM, we investigate effective multi-turn RL methods and identify a biased policy gradient issue caused by self-inclusion in GRPO. To solve this, we propose Turn-level Reinforce-Leave-One-Out (**TRLOO**) to provide unbiased advantage estimation for multi-turn RL. To alleviate lazy optimization, we incorporate mismatch correction for training stability and introduce Profiling-based Rewards (**PR**) and Profiling-based Rejection Sampling (**PRS**) to overcome the issue. The trained model, Dr. Kernel-14B, reach performance competitive with Claude-4.5-Sonnet in Kernelbench. Finally, we study sequential test-time scaling for Dr. Kernel-14B, which even **outperforms** GPT-5 and Claude-4.5-Sonnet in the Kernelbench level-2 subset.
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Safety Alignment of LMs via Non-cooperative Games
Anselm Paulus, Ilia Kulikov, Brandon Amos, Rémi Munos, Ivan Evtimov, Kamalika Chaudhuri, Arman Zharmagambetov
Ensuring the safety of language models (LMs) while maintaining their usefulness remains a critical challenge in AI alignment. Current approaches rely on sequential adversarial training: generating adversarial prompts and fine-tuning LMs to defend against them. We introduce a different paradigm: framing safety alignment as a non-zero-sum game between an Attacker LM and a Defender LM trained jointly via online reinforcement learning. Each LM continuously adapts to the other's evolving strategies, driving iterative improvement. Our method uses a preference-based reward signal derived from pairwise comparisons instead of point-wise scores, providing more robust supervision and potentially reducing reward hacking. Our RL recipe, AdvGame, shifts the Pareto frontier of safety and utility, yielding a Defender LM that is simultaneously more helpful and more resilient to adversarial attacks. In addition, the resulting Attacker LM converges into a strong, general-purpose red-teaming agent that can be directly deployed to probe arbitrary target models. Code at github.com/facebookresearch/advgame.
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Riemannian MeanFlow
Dongyeop Woo, Marta Skreta, Seonghyun Park, Kirill Neklyudov, Sungsoo Ahn
Diffusion and flow models have become the dominant paradigm for generative modeling on Riemannian manifolds, with successful applications in protein backbone generation and DNA sequence design. However, these methods require tens to hundreds of neural network evaluations at inference time, which can become a computational bottleneck in large-scale scientific sampling workflows. We introduce Rieman

nian MeanFlow (RMF), a framework for learning flow maps directly on manifolds, enabling high-quality generations with as few as one forward pass. We derive three equivalent characterizations of the manifold average velocity (Eulerian, Lagrangian, and semigroup identities), and analyze parameterizations and stabilization techniques to improve training on high-dimensional manifolds. In promoter DNA design and protein backbone generation settings, RMF achieves comparable sample quality to prior methods while requiring up to 10$\times$ fewer function evaluations. Finally, we show that few-step flow maps enable efficient reward-guided design through reward look-ahead, where terminal states can be predicted from intermediate steps at minimal additional cost.

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Scaling Inference-Time Computation via Opponent Simulation: Enabling Online Strategic Adaptation in Repeated Negotiation

Xiangyu Liu, Di Wang, Zhe Feng, Aranyak Mehta

While large language models (LLMs) have emerged as powerful decision-makers across a wide range of single-agent and stationary environments, fewer efforts have been devoted to settings where LLMs must engage in \emph{repeated} and \emph{strategic} interactions with unknown or dynamic opponents. In such settings, recipes built upon \emph{offline} pre-training or fine-tuning, though robust against worst-case adversaries, do not fully exploit the capability of LLMs to adapt \emph{online} based on interaction feedback. Instead, we explore the more natural perspective of \emph{scaling inference-time computation} as a mechanism for adaptation, embedding the principles of a classical game-theoretical learning dynamic, \emph{smooth Fictitious Play (sFP)}, into LLM inference: (i) for belief formation, we employ an auxiliary opponent model that in-context learns to imitate the \emph{time-averaged} behavior of the opponent; (ii) for best response, we advance best-of-N (BoN) sampling by simulating against the opponent model. Empirical evaluations on two distinct forms of repeated negotiation games demonstrate that our method enables significant performance improvement over online interaction compared to various baselines, offering a scalable and principled approach to strategic decision-making without any parameter updates.

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The Information Geometry of Softmax: Probing and Steering

Kiho Park, Todd Nief, Yo Joong Choe, Victor Veitch

This paper concerns the question of how AI systems encode semantic structure into the geometric structure of their representation spaces. The motivating observation is that the natural geometry of these representation spaces should reflect the way models use representations to produce behavior. We focus on the important special case of representations that define softmax distributions. In this case, we argue that the natural geometry is information geometry. Our focus is on the role of information geometry on semantic encoding and the linear representation hypothesis. As an illustrative application, we develop *dual steering*, a method for robustly steering representations to exhibit a particular concept using linear probes. We prove that dual steering optimally modifies the target concept while minimizing changes to off-target concepts. Empirically, we find that dual steering enhances the controllability and stability of concept manipulation.

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Hyperbolic RQ-VAE enhanced Generative Recommendation with Differential-Length Codebook Strategy

Aoran Zhang, Yu-Bin Yang, Yonghong Yu

Recently, the integration of large language models (LLMs) with generative recommendation (GR) has demonstrated promising potential. However, most existing GR methods adopt residual quantization to implicitly model hierarchical relationships across codebook layers in Euclidean space, which distorts the intrinsic tree-like hierarchy and leads to low codebook utilization. To address these issues, we propose a Hyperbolic RQ-VAE enhanced Generative Recommendation, namely HG-Rec. Specifically, HG-Rec enhances the residual quantization mechanism by embedding the latent discrete representations into hyperbolic space to explicitly model hierarchical relationships across codebook layers. Motivated by the exponential volume growth of hyperbolic space, we further design a differential-length codebook

strategy, i.e. the codebook size follows a pyramidal structure, which aligns wit
h the tree-like structure and effectively compresses the codebook size. Hence, b
enefiting from the alignment of hyperbolic geometry and codebook hierarchy, HG-R
ec achieves lower collision rates, more uniform codebook usage, and less trainin
g time compared to existing methods. Extensive experiments across multiple bench
mark datasets demonstrate that HG-Rec consistently achieves state-of-the-art per
formance. The code is available in the https://github.com/zar123123/HG-Rec.
****

Prism-MoE: Efficient Dense-to-MoE Conversion for Visual Autoregressive Generatio
n

Ying Li, Zefang Wang, Zhaode Wang, Zhiwen Chen, chengfei lv, Huan Wang

Scaling up visual autoregressive models improves generation quality but incurs s
ubstantial inference costs. Mixture-of-Experts (MoE) architectures mitigate this
 issue through sparse activation and have proven effective in large language mod
els. However, training MoE models from scratch remains prohibitively expensive,
and dense-to-MoE conversion for visual autoregressive models is still underexplo
red. To enable *low-cost and high-quality dense-to-MoE conversion*, we propose *
*Prism-MoE**, an efficient framework for transforming pretrained dense visual au
toregressive models into sparse MoE models. Prism-MoE consists of two key compon
ents. First, we introduce trajectory-consistent Initialization, which formulate
s expert initialization as a principled decomposition problem and preserves the
generation trajectory of pretrained models. Second, we propose a confidence-adap
tive sparse fine-tuning framework that aligns expert specialization with the inf
ormation density of visual tokens via confidence-aware routing supervision. Expe
riments show that Prism-MoE achieves dense-to-MoE conversion with less than **10
\%** of the standard training budget, while maintaining generation quality compa
rable to dense baselines with only **37.5\%** active parameters.
****

UMEM: Unified Memory Extraction and Management Framework for Generalizable Memor
y

Yongshi Ye, Hui Jiang, Feihu Jiang, Tian Lan, Yichao Du, Biao Fu, Xiaodong Shi,
Qianghuai Jia, Longyue Wang, Weihua Luo

Self-evolving memory serves as the trainable parameters for Large Language Model
s (LLMs)-based agents, where extraction (distilling insights from experience) an
d management (updating the memory bank) must be tightly coordinated.
Existing methods predominantly optimize memory management while treating memory
extraction as a static process, resulting in poor generalization, where agents a
ccumulate instance-specific noise rather than robust memories.
To address this, we propose Unified Memory Extraction and Management (UMEM), a s
elf-evolving agent framework that jointly optimizes a LLM to simultaneously extr
act and manage memories.
To mitigate overfitting to specific instances, we introduce Semantic Neighborhoo
d Modeling and optimize the model with a neighborhood-level marginal utility rew
ard via GRPO.
This approach ensures memory generalizability by evaluating memory utility acros
s clusters of semantically related queries.
Extensive experiments across five benchmarks demonstrate that UMEM significantly
 outperforms highly competitive baselines, achieving up to 11.49 points improvem
ent in multi-turn interactive tasks. Furthermore, UMEM maintains a stable improv
ement trend during continuous evolution.
****

DT^2: Decision-Targeted Digital Twins

Harry Amad, Mihaela van der Schaar

A digital twin (DT) is a virtual model of a real-world system that can assist de
cision-making by simulating scenarios induced by different policies. However, ty
pical machine learning-based DTs do not optimise for this use case. We prove tha
t, when model capacity is limited, training DTs to minimise one-step transition
errors can produce suboptimal models for ranking sets of policies according to a
 reward function. We further show that this holds empirically, even with express
ive model classes. To address this, we introduce DT^2, a decision-targeted DT

training paradigm. Firstly, DT^2 uses fitted Q-evaluation to estimate values of candidate policies from offline data. A DT is then trained to generate rollouts that preserve pairwise policy rankings derived from these proxy ground-truth values with an architecture-agnostic loss function. We empirically demonstrate the efficacy of our method across a range of settings and architectures. DT^2 consistently improves policy ranking and reduces decision regret during policy selection relative to conventional DT training, both for policies used during training and for unseen policies, while maintaining a good level of raw simulation fidelity.
****
Universal Multiclass Transductive Online Learning
Steve Hanneke, Hongao Wang
We consider the problem of universal transductive online classification with a possibly unbounded label space. This setting considers online learning, with the sequence of instances (without labels) known to the learner in advance. We say a concept class $\mathcal{H}$ is learnable if there is a learning algorithm $\mathcal{A}$, such that for every realizable sequence, the number of mistakes made by $\mathcal{A}$ grows at most sublinearly with the number of predictions. We characterize the learnability of this setting and show that there are only two possible optimal rates for the learnable classes: either bounded or increasing logarithmically. We introduce a new combinatorial structure, called "Level-Constrained-Littlestone-Littlestone (LCLL) tree", which, along with the indifference property, characterizes the learnability. We also extend the learnability result to the agnostic case and the case where only the stochastic process that generates the instance sequence is known.
****
When Agents Go Rogue: Activation-Based Detection of Malicious Behaviors in Multi-Agent Systems
Haowen Xu, Xue Tan, Lei Ma, Zhihao Zhang, Chao Wang, Qingze Wang, Ping Chen, Jun Dai, Xiaoyan Sun
While enabling effective collaboration on complex tasks, LLM-based Multi-Agent Systems (MAS) face critical security challenges due to vulnerabilities at the agent and interaction levels. Most existing MAS security defenses are built upon two core assumptions: semantically-explicit malicious attacks and explicit graph-based modeling of the MAS topology and agent-level interactions. In practice, real-world attacks are becoming more semantically stealthy, while MAS execution is typically asynchronous without the temporal alignment assumed by graph-based propagation models. To address these limitations, we propose AcMAS, an activation-based framework for malicious-behavior detection in MAS. By analyzing internal reasoning states in the activation space of local agents, AcMAS detects even stealthy attacks in a synchronization-robust fashion, without relying on explicit interaction graphs. Moreover, our activation analysis provides critical signals to guide AcMAS in restoring the functionality of compromised agents, rather than the disruptive agent isolation commonly used by the state-of-the-art methods. Comprehensive evaluation demonstrates that AcMAS significantly outperforms graph-based baselines against stealthy attacks, by +0.22 F1 in synchronous settings (0.94 vs. 0.72) and by +0.55 F1 in asynchronous settings (0.93 vs. 0.38), with generalization across diverse open-source LLM backbones, attack intensity, and MAS scale.
****
Policy Search via Bayesian Optimization with Temporal Difference Gaussian Processes
Armin Lederer, Anuj Srivastava, Marco Bagatella, Andreas Krause
Bayesian optimization (BO) is a method commonly used for policy search in problems with low-dimensional policy parameterizations. While it is generally considered data-efficient, existing BO approaches are agnostic to the sequential structure of the optimization objective induced by policy roll-outs. Thereby, valuable information is discarded that could improve the convergence of BO. We address this inefficiency by developing and rigorously analyzing a novel approach for BO that relies on a temporal difference learning formulation for discounted infinit

e-horizon value functions based on Gaussian process (GP) regression. We derive l
earning error bounds for the proposed temporal difference GPs, such that we can
exploit upper confidence bounds to analyze the cumulative regret of our BO appro
ach. This analysis is further refined by bounding the maximal information gain f
or our temporal difference GP model. In a comparison with relevant baseline meth
ods, we demonstrate the practical advantages of our method.
****

Variational Adapter for Cross-modal Similarity Representation
WenZhang Wei, Zhipeng Gui, Dehua Peng, Tiandi Ye, Huayi Wu
The core of vision-language models lies in measuring cross-modal similarity with
in a unified representation space. However, most image-text matching or multi-cl
ass image classification datasets lack fine-grained cross-modal matching annotat
ions, forcing the continuous similarity space into binary classification boundar
ies. This compression induces false negative samples and significantly impairs t
he generalization performance of cross-modal tasks. While prior research has att
empted to mitigate this by modeling intra-modal ambiguity, it often overlooks in
herent annotation flaws, leading to suboptimal uncertainty allocation. To addres
s these challenges, we propose a Variational Adapter for Cross-modal Similarity
Representation (VACSR). This approach reformulates image-text matching with fine
-grained semantic scarcity as a variational inference problem. It constructs a l
atent space for cross-modal similarity and uses regularization techniques to mit
igate overfitting to binary annotations. Experiments on image-text retrieval, do
main generalization, and base-to-novel generalization demonstrate the proposed m
ethod’s effectiveness and robust generalization ability.
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CVE-Factory: Scaling Expert-Level Agentic Tasks for Code Security Vulnerability
Xianzhen Luo, Jingyuan Zhang, Shiqi Zhou, JinYang Huang, Chuan Xiao, Qingfu Zhu,
 Zhiyuan Ma, YUE XING, Yang Yue, Wencong Zeng, Wanxiang Che
Evaluating and improving the security capabilities of code agents requires high-
quality, executable vulnerability tasks. However, existing works rely on costly,
 unscalable manual reproduction and suffer from outdated data distributions. To
address these, we present CVE-Factory, the first multi-agent framework to achiev
e expert-level quality in automatically transforming sparse CVE metadata into fu
lly executable agentic tasks. Cross-validation against human expert reproduction
s shows that CVE-Factory achieves 95\% solution correctness and 96\% environment
 fidelity, confirming its expert-level quality. It is also evaluated on the late
st realistic vulnerabilities and achieves a 66.2\% verified success. This automa
tion enables two downstream contributions. First, we construct LiveCVEBench, a c
ontinuously updated benchmark of 190 tasks spanning 14 languages and 153 reposit
ories that captures emerging threats including AI-tooling vulnerabilities. Secon
d, we synthesize over 1,000 executable training environments, the first large-sc
ale scaling of agentic tasks in code security.
Fine-tuned Qwen3-32B improves from 5.3\% to 35.8\% on LiveCVEBench, surpassing C
laude 4.5 Sonnet, with gains generalizing to Terminal Bench (12.5\% to 31.3\%).
We open-source all code, data, and models.
****

Online Rubrics Elicitation from Pairwise Comparisons
MohammadHossein Rezaei, Robert Vacareanu, Zihao Wang, Clinton Wang, Bing Liu, Yu
nzhong He, Afra Feyza Akyürek
Rubrics provide a flexible way to train LLMs on open-ended long-form answers whe
re verifiable rewards are not applicable and human preferences provide coarse si
gnals. Prior work shows that reinforcement learning with rubric-based rewards le
ads to consistent gains in LLM post-training. Most existing approaches rely on r
ubrics that remain static over the course of training. Such static rubrics, howe
ver, are vulnerable to reward-hacking type behaviors and fail to capture emergen
t desiderata that arise during training. We introduce Online Rubrics Elicitation
 (OnlineRubrics), a method that dynamically curates evaluation criteria in an on
line manner through pairwise comparisons of responses from current and reference
 policies. This online process enables continuous identification and mitigation
of errors as training proceeds. Empirically, this approach yields consistent imp

rovements of up to 8% over training exclusively with static rubrics across AlpacaEval, GPQA, ArenaHard as well as the validation sets of expert questions and rubrics. We qualitatively analyze the elicited criteria and identify prominent themes such as transparency, practicality, organization, and reasoning.
****

EigenCache: Rethinking Diffusion Acceleration as Covariance-Optimal Forecasting and Submodular Information Allocation

Chenyang Xu, Dezhen Wang, Lin Chen, Kepeng Lin, Hao Wang

Accelerating diffusion models via feature caching has progressed from static feature reuse to polynomial extrapolation, yet current cache-then-forecast strategies still rely on hand-crafted approximation families (e.g., Taylor or Hermite bases) that can misalign with the non-stationary, layer-specific dynamics of generative features. This paper introduces EigenCache, a training-free framework that re-frames diffusion acceleration as covariance-adaptive feature forecasting and uncertainty-aware temporal design. EigenCache models cached feature trajectories as time-indexed stochastic processes and estimates layer-wise temporal kernels from a small calibration set. Under the resulting scalar temporal-kernel approximation, the Gaussian-process posterior mean, i.e., Kriging, is risk-optimal within the scalar-weighted linear predictor class; under joint Gaussianity, it further coincides with the MMSE estimator. This formulation generalizes fixed-basis forecasting from a covariance-adaptive perspective while providing a closed-form posterior-variance proxy for prediction uncertainty. Leveraging this proxy, EigenCache selects computation anchors by maximizing a log-determinant information-gain objective over denoising timesteps, a monotone submodular objective with a near-optimal greedy solution. Across image, video, transformer, U-Net, and LoRA-adapted diffusion models, EigenCache achieves a strong speed--fidelity Pareto frontier and provides a principled mechanism for robust compute allocation.
****

Risk-Averse and Optimistic Advertiser Incentive Compatibility in Auto-bidding

Christopher Liaw, Wennan Zhu

The rise of auto-bidding in online advertising has created new challenges for ensuring advertiser incentive compatibility, particularly when advertisers delegate bidding to agents with high-level constraints. One challenge is the multiplicity of equilibria with reported constraints. Alimohammadi et al. (2023) proposed a notion of Auto-bidding Incentive Compatibility (AIC) which serves to highlight that standard auctions may not incentivize truthful reporting of these constraints. However, their definition of AIC is very stringent as it requires that the worst-case outcome of an advertiser's truthful report is at least as good as the best-case outcome of any of the advertiser's possible deviations. In this paper, we introduce two refined and relaxed concepts: Risk-Averse Auto-bidding Incentive Compatibility (RAIC) and Optimistic Auto-bidding Incentive Compatibility (OAIC). RAIC (OAIC) stipulates that truthful reporting is preferred if its least (most) favorable equilibrium outcome is no worse than the least (most) favorable equilibrium outcome from any misreport. We demonstrate that SPA satisfies both RAIC and OAIC. These findings clarify SPA's incentive properties under auto-bidding, specifically regarding advertiser perspectives on equilibrium selection.
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Benchmarking at the Edge of Comprehension

Samuele Marro, Jialin Yu, Emanuele La Malfa, Oishi Deb, Jiawei Li, Yibo Yang, Ebey Abraham, Sunando Sengupta, Eric Sommerlade, Michael J. Wooldridge, Philip Torr

As frontier Large Language Models (LLMs) increasingly saturate new benchmarks shortly after they are published, benchmarking itself is at a juncture: if frontier models keep improving, it will become increasingly hard for humans to generate discriminative tasks, provide accurate ground-truth answers, or evaluate complex solutions.

If benchmarking becomes infeasible, our ability to measure any progress in AI is at stake. We refer to this scenario as the *post-comprehension regime*.

In this work, we propose Critique-Resilient Benchmarking, an adversarial framework designed to compare models even when full human understanding is infeasible.

Our technique relies on the notion of *critique-resilient correctness*: an answe
r is deemed correct if no adversary has convincingly proved otherwise.
Unlike standard benchmarking, humans serve as bounded verifiers and focus on loc
alized claims, which preserves evaluation integrity beyond full comprehension of
 the task.
Using an itemized bipartite Bradley-Terry model, we jointly rank LLMs by their a
bility to solve challenging tasks and to generate difficult yet solvable questio
ns.
We showcase the effectiveness of our method in the mathematical domain across ei
ght frontier LLMs, showing that the resulting scores are stable and correlate wi
th external capability measures.
Our framework reformulates benchmarking as an adversarial generation-evaluation
game in which humans serve as final adjudicators.
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Active Curriculum Refinement for Reinforcement Learning
Zhenya Liu, Yuxin Chen
In many RL domains, environments are linked by prerequisite relations—e.g., diff
iculty-increasing edits or parameter increments—which induce a directed acyclic
curriculum graph (DAG). In practice, this structure is often exploited only impl
icitly, yet it can yield clear gains in training. We introduce PATH, a curriculu
m learning framework that performs active learning on the curriculum graph. PATH
 first expands coverage by sampling diverse curriculum paths, then reallocates t
raining toward regions that remain unmastered. Experiments show that PATH levera
ges the graph structure to achieve strong robustness and generalization across d
iverse environments.
****

PolicyGuard: Towards Test-time and Step-level Adversary Defense for Reinforcemen
t Learning Agent
Junfeng Guo, Heng Huang
While real-world applications of reinforcement learning (RL) are becoming increa
singly popular, the security of RL systems deserve more attention and exploratio
n. In particular, recent work has revealed that RL agents are vulnerable to back
door attacks, where a victim agent behaves normally under standard conditions bu
t executes malicious actions when a specific trigger is activated. Existing back
door defenses for RL either require access to the agent’s internal parameters, o
perate only at the model or trajectory level, or are limited to specific attack
types. To ensure the security of RL agents, we propose PolicyGuard, a test-time
step-level backdoor defense which leverages Gaussian Process (GP) posterior vari
ance and adapts pseudo trajectories to enable uncertainty computation for indivi
dual time step. Besides, we also provide theoretical foundations to explain the
efficacy of GP posterior variance. Extensive experiments across seven RL games d
emonstrate that PolicyGuard achieves state-of-the-art detection performance in m
ost cases, with average AUROC of 0.856 for perturbation-based attacks and 0.859
for adversary-agent attacks.
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Efficient numeracy in language models through single-token number embeddings
Linus Kreitner, Paul Hager, Jonathan Mengedoht, Georgios Kaissis, Daniel Ruecker
t, Martin J. Menten
To drive progress in science and engineering, large language models (LLMs) must
be able to process large amounts of numerical data and solve long calculations e
fficiently. This is currently only possible through the use of external tools or
 extensive reasoning chains, either weakening the numerical representations of L
LMs or limiting the length of problems they can solve. We show that frontier LLM
s require excessive amounts of reasoning tokens to solve even basic calculations
, which is exacerbated by their tokenization strategies that split single number
s into multiple tokens. This motivates the need for efficient and effective sing
le-token number encodings. We introduce a set of desiderata for such encodings a
nd show that existing approaches fail to fulfill them. To address these shortcom
ings, we propose BitTokens, a novel encoding strategy that represents any number
 as a single token using its IEEE 754 binary floating-point representation. Thro

ugh extensive experiments we show that our BitTokens allow even small language m
odels to learn algorithms that solve basic arithmetic operations nearly perfectl
y. This newly gained efficiency could expand the length and complexity of proble
ms language models can solve.
****

Automatic Unsupervised Ensemble Outlier Model Selection

Hong-Phuc Phan, Tuan-Anh Vu, Tung Kieu, Son Ha Xuan, Bin Yang, Christian S. Jens
en

Unsupervised outlier detection is attractive because it eliminates the need for
labeled data. Moreover, forming multi-model ensembles can improve detection robu
stness. However, composing an ensemble without labeled data is challenging. Naiv
ely composed ensembles can suffer from ensemble saturation, where redundant or u
nreliable detection models degrade performance and incur unnecessary computation
. We propose MetaEns, an automatic unsupervised framework for selecting ensemble
s of outlier detection models. Using labeled meta-datasets, MetaEns learns a mod
el that predicts marginal ensemble gains, estimating the expected improvement fr
om adding a candidate model to a partially constructed ensemble. At test time, t
his learned signal is combined with a submodular-inspired proxy objective that e
nforces diminishing returns through diversity-aware discounting and family-level
 risk regularization, thereby enabling reedy sequential selection with adaptive
early stopping. As a result, MetaEns constructs compact, high-quality ensembles
without access to ground-truth labels. Experiments on 39 real-world datasets sho
w that MetaEns consistently outperforms state-of-the-art unsupervised selectors
and ensemble baselines, achieving higher average precision while using fewer mod
els.
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CURE: Context-driven Diffusion with Progressive Expansion for Single Domain Gene
ralization in Time Series Classification

Yuhang Pei, Fanchun Meng, Wenrui Wu, Tao Ren, Yifan Wang, Wei Ju, Chao Zheng, Xi
ao Luo

This paper studies the problem of single domain generalization in time series cl
assification, which aims to learn a generalized time series classification model
 using a single source domain. This problem is highly challenging due to unrelia
ble supervision from domain scarcity. Although current approaches employ generat
ive models for data augmentation, these synthesized samples often suffer from lo
w diversity and intrinsic noise, leading to weak generalization ability. Towards
 this end, we propose a novel approach named Context-driven Diffusion with Prog
ressive Expansion (CURE) for single domain generalization in time series classif
ication. The core of our CURE is to generate semantic-aware and semantic-free co
ntexts that jointly guide a conditional diffusion model for informative data exp
ansion. In particular, our CURE first conducts representation disentanglement to
 extract semantic-aware and semantic-free representations from source data. To e
nhance generalizability through data synthesis, we not only retrieve reference t
ime series trajectories with similar semantics for semantic-aware contexts, but
also utilize adversarial strategies to learn semantic-free contexts. These conte
xts are integrated as joint conditions for a diffusion model, enabling diverse a
nd reliable virtual data. To enhance expansion adaptability and stable optimizat
ion, we progressively update our semantic-free contexts via a memory bank and me
asure boundary properties for dynamic data filtering. Comprehensive experiments
on benchmark datasets validate the effectiveness of CURE.
****

Dirichlet-Prior Shaping: Guiding Expert Specialization in Upcycled MoEs

Leyla Mirvakhabova, Babak Ehteshami Bejnordi, Gaurav Kumar, Hanxue Liang, Wanru
Zhao, Paul N. Whatmough

Upcycling pre-trained dense models into sparse Mixture-of-Experts (MoEs) efficie
ntly increases model capacity but often suffers from poor expert specialization
due to naive weight replication. We introduce Dirichlet-Prior Shaping Loss (DPSL
), a novel router regularization technique that directly shapes routing probabil
ity distributions by matching expert assignments to a target Dirichlet prior res
ulting in enhanced expert specialization. DPSL enables encoding of inductive bia

ses such as encouraging experts to focus on specific modalities or tasks, withou t requiring manual intervention. DPSL is a general tool applicable to any module that outputs categorical probability distributions, extending its utility beyon d MoE training. Experiments on upcycled MoE vision-language models show that DPS L consistently outperforms upcycling strategies and regularization techniques ac ross standard vision-language benchmarks, addressing the critical issue of poor specialization and fostering higher-performing models.
****

World Models in Pieces: Structural Certification of General Agents
Yikai Lu, Yifei Wu, Xinyu Lu, Tongxin Li
In the big-world regime, agents cannot be universally capable and their ability is inevitably specialized across a world model in pieces. Consequently, standard uniform guarantees fail to distinguish between the understanding of critical bo ttlenecks and irrelevant failures. We first formalize this limitation by proving that *general agents are not universal*, rendering standard worst-case analysis uninformative. To overcome this, we introduce **structural certification**, a t ransition-local framework that maps bounded goal-conditioned performance to entr y-wise guarantees on the agent's internal world model. Our main contribution is constructive. We provide algorithms that filter specific transitions using deep compositional goals and prove that a general agent on these goals has a structur al world model with a $\mathcal{O}(1/n)+\mathcal{O}(\delta)$ error bound. Conve rsely, this bound is tight in the small-δ regime, whose existence is expl icitly guaranteed by our certification. These results enable the certifiable dep loyment of general agents by localizing the specific transitions where long-hori zon planning is reliable.
****

CFPO: Counterfactual Policy Optimization for Multimodal Reasoning
ZhangYuan Yu, Wanran Sun, Guangjing Yang, Xiaohu Wu, Qicheng Lao
Large Vision-Language Models (LVLMs) have demonstrated remarkable capabilities i n multimodal reasoning. However, prevailing reinforcement learning (RL) paradigm s lack explicit counterfactual enhancement and causal learning mechanisms. This fundamental deficiency results in severe grounding failures, manifesting as a te ndency to ignore visual evidence in favor of language priors or exhibiting hallu cination drift during long chain-of-thought reasoning. To address this root caus e, we propose CounterFactual Policy Optimization (CFPO), a novel framework that enforces causal consistency between visual perception and textual reasoning. CFP O introduces a cross-modal counterfactual enhancement mechanism, which regulariz es the policy by maximizing the discrepancy between the model’s predictions and those from a counterfactual state where critical visual cues are suppressed. Thi s approach seamlessly integrates with standard algorithms like GRPO and DAPO wit hout requiring external reward models or additional supervision. Extensive exper iments demonstrate that CFPO significantly improves reasoning fidelity, achievin g consistent gains of 3.17%-6.25% over standard RL baselines and 1.32%-2.13% ove r the state-of-the-art perception-aware method (PAPO). Code is available at http s://github.com/Raven-July/CFPO.
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CONGA:Confidence-and-Gradient-Aware Learning Rate Schedule for Test Time Adaptat ion
Lv Shaoran, Xinyao Li, Jingjing Li
Test-time adaptation (TTA) adapts pretrained models to test data on-the-fly. Cur rent TTA methods have focused on what to adapt: lightweight domain-aware compone nts (prompts, normalization statistics) updated with consistency-aware self-supe rvised losses. This work investigates the more fundamental yet underexplored opt imization process, providing insights and guidelines on how to appropriately upd ate models for TTA. By analyzing the optimization error during TTA, we identify a pivotal stability-plasticity trade-off: the model should adapt to novel distri butions while retaining learned knowledge, which motivates our design of a CONfi dence-and-Gradient-Aware scheduler (CONGA) to constrain model learning rate (LR) within an adaptive exploration interval. For each iteration, the lower bound e ncourages model exploration on informative confident samples, while the upper bo

und prevents aggressive overfitting to noisy optimization gradients. Based on ou r theoretical findings, an adaptation-progress-conditioned cosine decay function decides the specific LR within the interval. As an LR scheduler, CONGA is natu rally applicable on existing TTA methods as a plug-in module, introducing little computation overheads. Extensive experiments and analysis demonstrate the super iority and validness of CONGA.
****

ITSPACE: Monotone Gaussian Optimal Transport Updates
Woojoo Na, Jennifer Dy
Covariance matrices serve as compact descriptors of feature distributions in man y machine-learning pipelines, including domain adaptation and Gaussian embedding s. Under a centered Gaussian approximation, the *unregularized* Wasserstein-2 *o ptimal transport* (OT) discrepancy admits a closed form on covariances given by the *Bures--Wasserstein* (BW) objective on the *symmetric positive definite* (SP D) cone. We propose **ITSPACE** (Iterative Transport for Stable Proximal Alignme nt of Covariance Embeddings), a proximal majorization-minimization method that d irectly optimizes this exact BW objective through closed-form updates in a squar e-root factorization. In exact arithmetic, each iteration satisfies a sufficient -decrease inequality for the BW objective; under inexact polar computations, we provide an explicit certificate-gap bound controlling deviations from exact desc ent. The resulting iterations preserve PSD structure by construction and natural ly support rank-restricted factors, making **ITSPACE** well-suited as a lightwei ght inner-loop primitive in settings where adaptation must be performed from unl abeled target batches under strict step and compute budgets. Across real-world c ovariance-alignment benchmarks, **ITSPACE** reaches low-BW-gap solutions substan tially faster than BW-gradient descent, methods based on other covariance geomet ries, and entropically regularized sample-OT baselines.
****

Time series saliency maps: Explaining models across multiple domains
Christodoulos Kechris, Jonathan Dan, David Atienza
Traditional saliency map methods, popularized in computer vision, highlight indi vidual input points that contribute most to a model's output. However, in the co ntext of time series, they offer limited insights because semantically meaningfu l features are often found in other domains. Thus, we introduce in this paper Cr oss-domain Integrated Gradients, a generalization of Integrated Gradients that e nables feature attributions in any domain formulated as an invertible, different iable transformation of the time domain. Our derivation extends Integrated Gradi ents into complex-valued domains, enabling frequency-based attributions, while p reserving path independence and completeness. We validate our method via control led mechanistic experiments, quantitative faithfulness and perturbation-stabilit y tests, and real-world case studies. Across wearable heart-rate extraction, EEG -based seizure detection, and zero-shot forecasting, our proposed Cross-domain I ntegrated Gradients approach identifies whether predictions rely on heart-rate f requencies or interference, epileptic sources or artifacts, and trend or seasona l components, revealing model behaviour that time-domain saliency does not captu re. We release an open-source library with TensorFlow, native PyTorch, and Captu m support for plug-and-play cross-domain explainability of time-series models.
****

The Value of Variance: Mitigating Debate Collapse in Multi-Agent Systems via Unc ertainty-Driven Policy Optimization
Luoxi Tang, Yuqiao Meng, Joseph Costa, Yingxue Zhang, Muchao Ye, Zhaohan Xi
Multi-agent debate (MAD) systems improve LLM reasoning through iterative deliber ation, but remain vulnerable to debate collapse, a failure type where final agen t decisions are compromised on erroneous reasoning. Existing methods lack princi pled mechanisms to detect or prevent such failures. To address this gap, we firs t propose a hierarchical metric that quantifies behavioral uncertainty at three levels: intra-agent (individual reasoning uncertainty), inter-agent (interactive uncertainty), and system-level (output uncertainty). Empirical analysis across several benchmarks reveals that our proposed uncertainty quantification reliably indicates system failures, which demonstrates the validity of using them as dia

gnostic metrics to indicate the system failure. Subsequently, we propose a mitig ation strategy by formulating an uncertainty-driven policy optimization to penal ize self-contradiction, peer conflict, and low-confidence outputs in a dynamic d ebating environment. Experiments demonstrate that our proposed uncertainty-drive n mitigation reliably calibrates the multi-agent system by consistently improvin g decision accuracy while reducing system disagreement.
****

Fine-grained Analysis of Brain-LLM Alignment through Input Attribution
Michela Proietti, Roberto Capobianco, Mariya Toneva
Understanding the alignment between large language models (LLMs) and human brain activity can reveal computational principles underlying language processing. Th is work describes a pipeline to apply attribution methods to the brain-LLM align ment setting to identify the specific words most important for this alignment. A s a case study, we leverage it to study a contentious research question about br ain-LLM alignment: the relationship between brain alignment (BA) and next-word p rediction (NWP). Across two naturalistic fMRI datasets, we find that BA and NWP rely on largely distinct word subsets: NWP exhibits recency and primacy biases w ith a focus on syntax, while BA prioritizes semantic and discourse-level informa tion with a more targeted recency effect. This work advances our understanding o f how LLMs relate to human language processing and highlights differences in fea ture reliance between BA and NWP. Beyond this study, our attribution method can be broadly applied to explore the cognitive relevance of model predictions in di verse language processing tasks.
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AgentLAB: Benchmarking LLM Agents against Long-Horizon Attacks
Tanqiu Jiang, Yuhui Wang, Jiacheng Liang, Ting Wang
LLM agents are increasingly deployed in long-horizon, complex environments to so lve challenging problems, but this expansion exposes them to long-horizon attack s that exploit multi-turn user–agent–environment interactions to achieve objecti ves infeasible in single-turn settings. To measure agent vulnerabilities to such risks, we present AgentLAB, the first benchmark dedicated to evaluating LLM age nt susceptibility to adaptive, long-horizon attacks. Currently, AgentLAB support s five novel attack types including intent hijacking, tool chaining, task inject ion, objective drifting, and memory poisoning, spanning 28 realistic agentic env ironments, and 644 security test cases. Leveraging AgentLAB, we evaluate represe ntative LLM agents and find that they remain highly susceptible to long-horizon attacks; moreover, defenses designed for single-turn interactions fail to reliab ly mitigate long-horizon threats. We anticipate that AgentLAB will serve as a va luable benchmark for tracking progress on securing LLM agents in practical setti ngs. The code for AgentLAB is available at:
https://tanqiujiang.github.io/AgentLAB_main.
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From Interactions to Principles: Experience-Driven Self-Distillation for Evolvin g LLM Agents
Rong Wu, Xiaoman Wang, Jianbiao Mei, Pinlong Cai, Daocheng Fu, Cheng Yang, Liche ng Wen, Xuemeng Yang, Yufan Shen, Yuxin Wang, Botian Shi
LLM agents have achieved strong performance in tool-augmented reasoning, but mos t remain largely stateless: after each episode, the agent discards interaction t races and does not accumulate reusable strategies. Prior work either stores raw trajectories for case-based reuse or relies on external teacher models to write reflections, which limits generalization or leaves the agent’s policy unchanged. We introduce EvolveR, an experience-driven framework that allows an agent to im prove using its own interaction history. EvolveR maintains an experience base of distilled strategic principles derived from past trajectories. In an offline ph ase, the agent self-distills successful and failed trajectories into concise pri nciples, applies semantic deduplication, and assigns each principle an empirical utility score for maintenance and pruning. In an online phase, the agent retrie ves top-ranked principles to guide reasoning and tool usage, generating new traj ectories. We then perform policy evolution with reinforcement learning on these experience-conditioned trajectories, reinforcing behaviors that effectively retr

ieve and apply useful principles. We demonstrate the effectiveness of EvolveR on complex multi-hop question-answering benchmarks, where it achieves superior performance over strong agentic baselines. Our work presents a comprehensive blueprint for agents that learn not only from external data but also from the consequences of their own actions, paving the way for more autonomous and continuously improving systems.
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RiboSphere: Learning Unified and Efficient Representations of RNA Structures
Zhou Zhang, Hanqun Cao, Cheng Tan, Fang Wu, Pheng-Ann Heng, Tianfan Fu
Accurate RNA structure modeling remains difficult because RNA backbones are highly flexible, non-canonical interactions are prevalent, and experimentally determined 3D structures are comparatively scarce. We introduce RiboSphere, a framework that learns discrete geometric representations of RNA by combining vector quantization with flow matching. Our design is motivated by the modular organization of RNA architecture: complex folds are composed from recurring structural motifs. RiboSphere uses a geometric transformer encoder to produce approximately SE(3)-invariant (rotation/translation-invariant) features, which are discretized with finite scalar quantization (FSQ) into a finite vocabulary of latent codes. Conditioned on these discrete codes, a flow-matching decoder reconstructs atomic coordinates, enabling high-fidelity structure generation. We find that the learned code indices are enriched for specific RNA motifs, suggesting that the model captures motif-level compositional structure rather than acting as a purely compressive bottleneck. Across benchmarks, RiboSphere achieves strong performance in structure reconstruction (RMSD 1.25\,\AA, TM-score 0.84), and its pretrained discrete representations transfer effectively to inverse folding and RNA--ligand binding prediction, with robust generalization in data-scarce regimes.
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Joint-Embedding Predictive Learning of Latent Market States in U.S. Equities
Simon Mahns, Randall Balestriero, Mido Assran
We investigate whether Joint-Embedding Predictive Architectures (JEPA) can learn useful representations of U.S. equity markets. We jointly train a permutation-invariant tokenizer that maps each trading day's unordered per-asset features to a fixed set of learned factor tokens, together with a temporal JEPA using masked prediction to obtain a compact daily market-state embedding. Our evaluations show that these embeddings are strongly associated with second-moment market structure (realized volatility, correlation concentration, effective factor dimensionality) and weakly associated with market direction. The embedding helps predict gradual recovery dynamics but not sudden stress onsets. Without any text supervision, latent regimes show statistically significant alignment with news-topic shifts.
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SEMIR: Semantic Minor-Induced Representation Learning on Graphs for Visual Segmentation
Luke James Miller, Yugyung Lee
Segmenting small and sparse structures in large-scale images is fundamentally constrained by voxel-level, lattice-bound computation and extreme class imbalance--dense, full-resolution inference scales poorly and forces most pipelines to rely on fixed regionization or downsampling, coupling computational cost to image resolution and attenuating boundary evidence precisely where minority structures are most informative. We introduce \textbf{SEMIR} (\emph{Semantic Minor-Induced Representation Learning}), a representation framework that decouples inference from the native grid by learning a task-adapted, topology-preserving latent graph representation with exact decoding. \textbf{SEMIR} transforms the underlying grid graph into a compact, boundary-aligned graph minor through parameterized edge contraction, node deletion, and edge deletion, while preserving an exact lifting map from minor predictions to lattice labels. Minor construction is formalized as a few-shot structure learning problem that replaces hand-tuned preprocessing with a \emph{boundary-alignment objective}: minor parameters are learned by maximizing agreement between predicted boundary elements and target-specific semantic edges under a \emph{boundary Dice criterion}, and the induced minor is annota

ted with scale- and rotation-robust geometric and intensity descriptors and supp
orts efficient region-level inference via message passing on a graph neural netw
ork (GNN) with relational edge features. We benchmark \textbf{SEMIR} on three tu
mor segmentation datasets—\textbf{BraTS 2021}, \textbf{KiTS23}, and \textbf{LiTS
}—where targets exhibit high structural variability and distributional uncertain
ty. \textbf{SEMIR} yields consistent improvements in \emph{minority-structure Di
ce} at practical runtime. More broadly, SEMIR establishes a framework for learni
ng task-adapted, topology-preserving latent representations with exact decoding
for high-resolution structured visual data.
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Synergistic Space-Vision Processing for Predicate Inference
Zhenhua Lei, Zefang Han, yu qiu
Scene graph generation (SGG) aims to parse an image into a structured graph of o
bjects and their predicates, enabling explicit relational reasoning for visual u
nderstanding.
However, prevailing methods often over-predict geometric predicates, resulting i
n scene graphs that are factually correct yet semantically shallow.
While recent works effectively attribute this phenomenon to the long-tailed data
 distribution, we identify another critical factor driving such biased predictio
n:
co-occurrence-induced representation entanglement, where geometric and non-geome
tric predicates that frequently co-occur are encoded into overly similar represe
ntations.
To this end, we introduce Dual-stream Synergistic Network (DS-Net) that models g
eometric and non-geometric predicates with two specialized streams, coupled with
 a bidirectional cross-stream fusion mechanism.
The space stream focuses on spatial and structural cues, while the vision stream
 captures fine-grained visual evidence and semantic priors.
Extensive experiments show that DS-Net consistently improves predicate inference
, achieving 1.3\% $\sim$ 6.1\% absolute gains in mR@100 on the SGGen task when i
ntegrated into existing SGG methods.
These results highlight the importance of synergistic modeling of geometric and
non-geometric predicates for generating semantically richer scene graphs.
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Causal Forcing: Autoregressive Diffusion Distillation Done Right for High-Qualit
y Real-Time Interactive Video Generation
Hongzhou Zhu, Min Zhao, Guande He, Hang Su, Chongxuan Li, Jun Zhu
To achieve real-time interactive video generation, current methods distill pretr
ained bidirectional video diffusion models into few-step autoregressive (AR) mod
els, facing an *architectural gap* when full attention is replaced by causal att
ention. However, existing approaches do not bridge this gap theoretically. They
initialize the AR student via ODE distillation, which requires *frame-level inje
ctivity*, where each noisy frame must map to a unique clean frame under the PF-O
DE of an *AR teacher*. Distilling an AR student from a bidirectional teacher vio
lates this condition, preventing recovery of the teacher's flow map and instead
inducing a conditional-expectation solution, which degrades performance. To addr
ess this issue, we propose *Causal Forcing*, which uses an autoregressive teache
r for ODE initialization to bridge the architectural gap, and then applies the s
ame DMD procedure as in Self Forcing. Empirical results show that our method out
performs all baselines across all metrics, surpassing the SOTA Self Forcing by 1
9.3\% in Dynamic Degree, 8.7\% in VisionReward, and 16.7\% in Instruction Follow
ing. Project page: https://thu-ml.github.io/CausalForcing.github.io/; the code:
https://github.com/thu-ml/Causal-Forcing.
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Token Sparse Attention: Efficient Long-Context Inference with Interleaved Token
Selection
Dongwon Jo, Beomseok Kang, Jiwon Song, Jae-Joon Kim
The quadratic complexity of attention remains the central bottleneck in long-con
text inference for large language models.
Prior acceleration methods either sparsify the attention map with structured pat

terns or permanently evict tokens at specific layers, which can retain irrelevant tokens or rely on irreversible early decisions despite the layer-/head-wise dynamics of token importance.
In this paper, we propose Token Sparse Attention, a lightweight and dynamic token-level sparsification mechanism that compresses per-head Q, K, V to a reduced token set during attention and then decompresses the output back to the original sequence, enabling token information to be reconsidered in subsequent layers. Furthermore, Token Sparse Attention exposes a new design point at the intersection of token selection and sparse attention. Our approach is fully compatible with dense attention implementations, including Flash Attention, and can be seamlessly composed with existing sparse attention kernels.
Experimental results show that Token Sparse Attention consistently improves accuracy–latency trade-off, achieving up to $\times$3.23 attention speedup at 128K context with less than 1\% accuracy degradation. These results demonstrate that dynamic and interleaved token-level sparsification is a complementary and effective strategy for scalable long-context inference.
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MotionMAR: Multi-scale Auto-Regressive Human Motion Reconstruction from Sparse Observations

Yuhua Luo, Junsheng Zhang, Mengyin Liu, Xincheng Lin, Ming Yan, Zhudi Chen, Chenglu Wen, Lan Xu, Siqi Shen, Cheng Wang

Human motion follows a temporal hierarchical structure, transitioning from low-frequency global trajectories to high-frequency details. Inspired by the success of multi-level autoregressive models in computer vision, we propose MotionMAR, a coarse-to-fine framework for motion reconstruction from sparse observations. It first estimates the global trajectory of human motion and then gradually refines the temporal details. This architecture consists of four integrated components. The Temporal Multi-scale Tokenization (TMT) VQ-VAE encodes the data at multiple temporal resolutions, separating semantic motion from minor jitters. The Motion Autoregressive Network (MAN) operates in this latent space, predicting motion across scales. It first establishes the global structure through coarse indices and then generates finer indices to recover specific details. Meanwhile, the Scale-Aware Control (SAC) module integrates sparse tracking data to ensure the generated output aligns with actual observations. The Motion Refinement Network (MRN) subsequently smooths consecutive poses and eliminates quantization artifacts.
Experiments show that MotionMAR achieves state-of-the-art accuracy on the AMASS dataset, providing a reliable and structure-aware approach for motion reconstruction. The source code is publicly available at \url{http://www.lidarhumanmotion.net/motionmar/}.
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Learnability-Driven Knowledge Assimilation for Class-Incremental Semantic Segmentation

Xinyue Zhang, Xu Zou, Wanjia Luo, Yanjie Wang, Jiahuan Zhou, Sheng Zhong, Luxin Yan

Class-incremental semantic segmentation learns new classes while retaining old ones without access to past data. Although existing methods alleviate catastrophic forgetting on old classes, new-class performance remains limited. We identify that the key bottleneck arises from low-margin regions, where the logit of the ground-truth class is close to that of the most competitive non-ground-truth class. Our theoretical analysis suggests that optimization in these regions is characterized by high second-order margin sensitivity and a small stability radius, making learning prone to class confusion.
Based on the above analysis, we propose Learnability-Driven Knowledge Assimilation (LDKA), which targets low-margin learning via three complementary optimization strategies:
(i) Progressive Margin Learning continuously reallocates pixel-wise optimization budget in a threshold-free manner, shifting emphasis from high-margin to low-margin regions;
(ii) Smooth Knowledge Distillation applies second-order sensitivity damping along the margin direction and perturbation stabilization to suppress high-frequency

updates and increase the stability radius;
(iii) Misclassification-Aware Decoupling measures inter-class confusion with a competition matrix and decouples highly competitive class representations.
Experiments show that LDKA improves mIoU on new classes while preserving performance on old classes across 9 incremental protocols.
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Gradient Flow Through Diagram Expansions: Learning Regimes and Explicit Solutions

Dmitry Yarotsky, Eugene Golikov, Yaroslav Gusev

We develop a general mathematical framework to analyze scaling regimes and derive explicit analytic solutions for gradient flow (GF) in large learning problems. Our key innovation is a formal power series expansion of the loss evolution, with coefficients encoded by diagrams akin to Feynman diagrams. We show that this expansion has a well-defined large-size limit that can be used to reveal different learning phases and, in some cases, to obtain explicit solutions of the nonlinear GF. We focus on learning Canonical Polyadic (CP) decompositions of high-order tensors, and show that this model has several distinct extreme lazy and rich GF regimes such as free evolution, NTK and under- and over-parameterized mean-field. We show that these regimes depend on the parameter scaling, tensor order, and symmetry of the model in a specific and subtle way. Moreover, we propose a general approach to summing the formal loss expansion by reducing it to a PDE; in a wide range of scenarios, it turns out to be first-order and solvable by the method of characteristics. We observe a very good agreement of our theoretical predictions with experimental results.
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DynaTok: Token-Based 4D Reconstruction from Partial Point Clouds

Weirong Chen, Keisuke Tateno, Hidenobu Matsuki, Michael Niemeyer, Daniel Cremers, Federico Tombari

We address 4D reconstruction from partial point cloud sequences, where depth-sensor observations are incomplete, unordered, and lack explicit temporal correspondences. This geometry-only setting is challenging due to missing observations and ambiguous dynamics. While recent progress has largely relied on image-based methods, existing point-based approaches typically focus on single objects, assume relatively complete inputs, or require explicit correspondences.
To address these limitations, we propose DynaTok, a point-based framework for correspondence-free 4D reconstruction from partial point cloud sequences without images. DynaTok encodes frames into compact latent tokens, aggregates incomplete observations over time with a Transformer-based spatiotemporal encoder, and decouples geometry and motion through residual tokens in a unified model. A flow-matching decoder then reconstructs complete, temporally consistent 4D point-cloud sequences conditioned on the latent tokens.
Experiments on object- and scene-level benchmarks demonstrate improved reconstruction quality and temporal coherence from partial point cloud observations.
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VFMF: Dense Forecasting by Generating Foundation Model Features

Gabrijel Boduljak, Yushi Lan, Christian Rupprecht, Andrea Vedaldi

Forecasting by generating RGB videos is computationally expensive, often physically implausible, and not directly actionable, since it requires translation into decision-making signals. Direct modality forecasting (e.g., predicting future segmentation) produces directly actionable outputs but fails to scale due to the need for labels. Vision Foundation Model (VFM) features offer the best of both worlds: they contain actionable semantic and geometric information that can be easily decoded from the predicted features, while requiring no labels on the downstream task for training. However, almost all existing VFM feature forecasting methods regress future features from fixed number of input frames, with evaluation predominantly on short horizons matching the training setup.

We firstly show that existing regression methods struggle with forecasting from partial observations because they average over multiple plausible futures, failing to capture uncertainty in the future given the past. Interestingly, naively r

eplacing deterministic forecasting with generative flow matching does not match the sample quality of the regression model, despite being a mathematically appropriate formulation of the forecasting task. In this work, we explain why this is the case, and we show how to optimally generate foundation model features.

Our key insight is that generative modeling of VFM features requires (auto)encoding into a compact latent
space suitable for diffusion. We show that this latent space
preserves information more effectively than previously used
alternatives, such as uncompressed feature diffusion or PCA-based compression, both for forecasting and other applications, such as image generation.

Our results suggest that conditional generation
of (compressed) VFM features offers a promising and scalable foundation
for future scene forecasters.
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Epistemic Gain, Aleatoric Cost: Uncertainty Decomposition in Multi-Agent Debate for Math Reasoning
Dan Qiao, Binbin Chen, Fengyu Cai, Jianlong Chen, Wenhao Li, Fuxin Jiang, Zuzhi Chen, Hongyuan Zha, Tieying Zhang, Baoxiang Wang
Multi-Agent Debate (MAD) has shown promise in improving reasoning and reducing hallucinations, yet it remains unclear how information exchange shapes individual reasoning behavior. Empirically, MAD exhibits paradoxical phenomena, including rising accuracy with increasing token entropy and marked differences between homogeneous and heterogeneous agent combinations. In this paper, we introduce a Bayesian uncertainty analysis framework for MAD, which decomposes answer-level predictive uncertainty into epistemic uncertainty and aleatoric uncertainty, corresponding to the potential gain and cost of debate. Across multiple agent configurations, we find that effective debate depends on achieving high epistemic gain under controlled aleatoric cost. Building on this insight, we design an uncertainty-guided multi-agent reinforcement learning algorithm that encourages lower aleatoric cost and more effective epistemic information utilization. Experiments show that our approach simultaneously enhances each agent's accuracy and promotes a more productive debate process, providing an operational Bayesian perspective for understanding and improving MAD.
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On Revisiting Entropy for Identifying Mislabeled Images
Chunlei Li, Zixuan Zheng, Yilei Shi, Guanglu Dong, Pengfei Li, Jingliang Hu, Xiao Xiang Zhu, Lichao Mou
Mislabeled samples in training datasets severely degrade the performance of deep networks, as overparameterized models tend to memorize erroneous labels. We address this challenge by proposing a novel approach for mislabeled data detection that leverages training dynamics. Our method is grounded in the key observation that correctly labeled samples exhibit consistent entropy decrease during training, while mislabeled samples maintain relatively high entropy throughout the training process. Building on this insight, we introduce a signed entropy integral (SEI) statistic that captures both the magnitude and temporal trend of prediction entropy across training epochs. SEI is broadly applicable to classification networks and demonstrates particular effectiveness when integrated with contrastive language-image pretraining (CLIP) architectures. Through extensive experiments on four medical imaging datasets---a domain particularly susceptible to labeling errors due to diagnostic complexity---spanning diverse modalities and pathologies, we demonstrate that SEI achieves state-of-the-art performance in mislabeled data identification, outperforming existing methods while maintaining computational efficiency and implementation simplicity. Our code is available at https://github.com/MedAITech/SEI.
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In-Context Learning Is Provably Bayesian Inference: A Generalization Theory for Meta-Learning
Tomoya Wakayama, Taiji Suzuki

This paper develops a finite-sample statistical theory for in-context learning (ICL), analyzed within a meta-learning framework that accommodates mixtures of diverse task types. We leverage a Bayes risk identity that separates the total ICL risk into two orthogonal components: Bayes Gap and Posterior Variance. The Bayes Gap quantifies how well the trained model approximates the Bayes-optimal in-context predictor. For a uniform-attention Transformer, we derive a non-asymptotic upper bound on this gap, which explicitly clarifies the dependence on the number of pretraining prompts and their context length. The Posterior Variance is a model-independent risk representing the intrinsic task uncertainty. Our key finding is that this term is determined solely by the difficulty of the true underlying task, while the uncertainty arising from the task mixture vanishes exponentially fast with only a few in-context examples. Together, these results provide a unified view of ICL: the uniform-attention Transformer selects the optimal meta-algorithm during pretraining and rapidly converges to the optimal algorithm for the true task at test time.

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Membership Inference Attacks for Unseen Classes

Pratiksha Thaker, Neil Kale, Steven Wu, Virginia Smith

A key tool in developing safe AI models is *data auditing*, i.e., using statistical tools to determine whether harmful content may have been used in the training data of a black-box model. Unfortunately, most *membership inference attacks* (MIAs) used to perform this type of auditing themselves assume *access* to examples of harmful content from the same distribution as the query data. In real-world auditing scenarios, auditors often face legal and ethical restrictions preventing them from accessing a representative set of samples of harmful content to train MIA models effectively. We abstract and formalize this setting into a new data access model, the “unseen class” setting, and show that the state-of-the-art MIAs fail due to the lack of access to the full target distribution. We show that in this setting, *quantile regression attacks* outperform approaches typically considered to be state of the art. We demonstrate this both empirically and theoretically, showing that quantile regression attacks achieve up to **11× the TPR** of shadow model-based approaches in practice, and providing a theoretical model that outlines the generalization properties required for this approach to succeed. Our work identifies an important failure mode in existing MIAs and provides a cautionary tale for practitioners who aim to directly use existing tools for real-world applications of AI safety.

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Jailbreak Foundry: From Papers to Runnable Attacks for Reproducible Benchmarking

Zhicheng Fang, Jingjie Zheng, Chenxu Fu, Wei Xu

Jailbreak techniques for large language models (LLMs) evolve faster than benchmarks, making robustness estimates stale and difficult to compare across papers due to drift in datasets, harnesses, and judging protocols. We introduce **JAILBREAK FOUNDRY (JBF)**, a system that addresses this gap via a multi-agent workflow to translate jailbreak papers into executable modules for immediate evaluation within a unified harness. JBF features three core components: (i) *JBF-LIB* for shared contracts and reusable utilities; (ii) *JBF-FORGE* for the multi-agent paper-to-module translation; and (iii) *JBF-EVAL* for standardizing evaluations. Across 30 reproduced attacks, JBF achieves high fidelity with a mean (reproduced$-$reported) attack success rate (ASR) deviation of $+0.26$ percentage points. By leveraging shared infrastructure, JBF reduces attack-specific implementation code by nearly half relative to original repositories and achieves an 82.5% mean reused-code ratio. This system enables a standardized AdvBench evaluation of all 30 attacks across 10 victim models using a consistent GPT-4o judge. By automating both attack integration and standardized evaluation, JBF offers a scalable solution for creating living benchmarks that keep pace with the rapidly shifting security landscape.

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Induction Heads Interpolate N-Grams

Francesco D'Angelo, O■uz Kaan Yüksel, Swathi Shree Narashiman, Nicolas Flammarion

Induction heads are attention circuits believed to underlie in-context learning in transformers, yet a precise characterization of the estimators they implement remains elusive. We study transformers trained on order-k Markov chains and identify two complementary smoothing mechanisms. First, at finite attention-weight scale, the circuit implements a soft context-matching estimator: it aggregates contributions from exact and partial context matches, weighted exponentially by their overlap, and induces a data-dependent interpolation across context orders analogous to Jelinek--Mercer smoothing. Second, a beginning-of-sequence (BOS) token induces additive pseudo-counts, recovering Dirichlet-style smoothing. We construct a disentangled transformer implementing both mechanisms and show that trained transformers recover the predicted attention patterns. Across settings where pseudo-count smoothing is optimal or lower-order contexts provide structured evidence, trained transformers match or outperform classical count-based baselines. Our results bridge mechanistic interpretability of induction heads with classical statistical smoothing, revealing that transformers learn to regularize in-context estimation rather than simply count.

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Concept Removal Guidance: Evidence-Calibrated Negative Guidance for Safe Diffusion Sampling

Yoonseok Choi, Chaeyoung Oh, Hyunjun Choi, Seokin Seo, Kee-Eung Kim

Text-to-image diffusion models remain vulnerable to adversarial prompts that elicit disallowed content, motivating reliable inference-time controls. A popular approach is negative guidance, which subtracts a negative prompt direction with a fixed weight. However, it often forces a safety–fidelity trade-off, causing artifacts or prompt drift when over-applied and failing under attacks when under-applied. Dynamic variants reweight guidance using posterior-odds signals, which can be brittle for open-vocabulary compositional prompts, while lightweight similarity-based methods ignore the evolving image evidence along the denoising trajectory. We introduce Concept Removal Guidance (CRG), a training-free method that estimates unwanted-concept presence at each diffusion step from the model's noise predictions, and adaptively calibrates negative guidance via a closed-form constrained update enforcing a target presence threshold while minimally perturbing the conditional trajectory. Across red-teaming benchmarks, CRG reduces attack success rates while preserving benign fidelity, and extends to additional suppression targets such as artist style and violence without fine-tuning or external classifiers.

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PACER: Acyclic Causal Discovery from Large-scale Interventional Data

Ramon Viñas Torné, Sílvia Fábregas Salazar, Soyon Park, Ivo Ban, Artyom Gadetsky, Nikita Doikov, Maria Brbic

Inferring the structure of directed acyclic graphs (DAGs) from data is a central challenge in causal discovery, particularly in modern high-dimensional settings where large-scale interventional data are increasingly available. While interventional data can improve identifiability, existing methods remain limited by soft acyclicity constraints, leading to optimization over invalid cyclic graphs, numerical instability, and reduced scalability. We introduce PACER (Perturbation-driven Acyclic Causal Edge Recovery), a scalable framework for causal discovery that guarantees acyclicity by construction. PACER parameterizes a distribution over DAGs through a joint model of variable permutations and edge probabilities, enabling direct optimization over valid causal structures without surrogate penalties. The framework supports a unified likelihood-based treatment of observational and interventional data, flexible conditional density models, and the incorporation of structural prior knowledge. For linear-Gaussian mechanisms, we derive closed-form expressions for the expected interventional log-likelihood and its gradients, yielding substantial computational gains. Empirically, PACER matches or exceeds state-of-the-art methods on protein signaling and large-scale genetic perturbation benchmarks, while scaling efficiently to networks with thousands of variables and achieving up to two orders of magnitude speedups over penalty-based differentiable approaches. These results demonstrate that exact and scalable causal discovery from high-dimensional perturbation data is achievable through p

rincipled search space design.
****
Enhancing LLMs for Graph Tasks via Graph-aware LoRA Generation
Junshu Sun, Wanxing Chang, Qingming Huang, Shuhui Wang
Graph neural networks (GNNs) tightly couple their input-output parameters to dat aset-specific feature spaces and target sets, exhibiting limited transferability across different datasets. In contrast, language models (LMs) generalize flexib ly via a unified input-output interface, motivating recent attempts to adapt LMs to graph tasks. However, existing methods struggle to encode whole-graph inform ation, leading to potential information loss and suboptimal graph understanding. In this work, we propose a novel weight-level information injection paradigm fo r adapting LMs to graph tasks. This paradigm injects whole-graph information by generating task-specific weight updates that interact directly with hidden repre sentations. Instantiating this paradigm following low-rank adaptation (LoRA), we introduce GaRA, a Graph-aware LoRA generation model. GaRA constructs low-rank w eight updates conditioned on the original graph structures and constrains the no rm of the generated updates, thus injecting whole-graph information and avoiding the optimization bias in the weight generation. Empirical studies demonstrate t hat GaRA consistently outperforms baselines on zero-shot graph learning tasks.
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DiLA: Disentangled Latent Action World Models
Tianqiu Zhang, Muyang Lyu, Yufan Zhang, Fang Fang, Si Wu
Latent Action Models (LAMs) enable the learning of world models from unlabeled v ideo by inferring abstract actions between consecutive frames. However, LAMs fac e a fundamental trade-off between action abstraction and generation fidelity. Ex isting methods typically circumvent this issue by using two-stage training with pre-trained world models or by limiting predictions to optical flow. In this pap er, we introduce *DiLA*, a novel **Di**sentangled **L**atent **A**ction world mo del that aims to resolve this trade-off via content-structure disentanglement. O ur key insight is that disentanglement and latent action learning are co-evolvin g: the predictive bottleneck inherent in latent action learning serves as a driv ing force for disentanglement, compelling the model to distill spatial layouts i nto the structure pathway while offloading visual details to a separate content pathway for generation. This synergy yields a continuous, semantically structure d latent action space without compromising generative quality. *DiLA* achieves s uperior results in video generation quality, action transfer, visual planning, a nd manifold interpretability. These findings establish *DiLA* as a unified frame work that simultaneously achieves high-level action abstraction and high-fidelit y generation, advancing the frontier of self-supervised world model learning.
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From Similarity to Vulnerability: Key Collision Attack on LLM Semantic Caching
Zhixiang Zhang, Zesen Liu, Yuchong Xie, Quanfeng Huang, Dongdong She
Semantic caching has emerged as a pivotal technique for scaling LLM applications , widely adopted by providers including AWS and Microsoft. By utilizing embeddin g vectors as cache keys, this mechanism effectively minimizes latency and redund ant computation for semantically similar queries. In this work, we conceptualize semantic cache keys as a form of fuzzy hashes. We demonstrate that the locality required to maximize cache hit rates fundamentally conflicts with the cryptogra phic avalanche effect necessary for collision resistance.
Our conceptual analysis formalizes this inherent trade-off between performance (locality) and security (collision resilience), revealing that semantic caching i s inherently vulnerable to key collision attacks.
While prior research has focused on side-channel and privacy risks, we present t he first systematic study of integrity risks arising from cache collisions. We i ntroduce CacheAttack, an automated framework for launching black-box collision a ttacks.
We evaluate CacheAttack in security-critical tasks and agentic workflows. It ach ieves a hit rate of 86\% in LLM response hijacking and can induce malicious beha viors in LLM agent, while preserving strong transferability across different emb edding models.

A case study on a financial agent further illustrates the real-world impact. Finally, we discuss mitigation strategies, highlighting a persistent trade-off between cache efficiency and robustness.

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MINT: Minimal Information Neuro-Symbolic Tree for Objective-Driven Knowledge-Gap Reasoning and Active Elicitation

Zeyu Fang, Mahdi Imani, Tian Lan

Joint planning through language-based interactions is a key area of human-AI teaming. Planning problems in the open world often involve various aspects of incomplete information and unknowns, e.g., objects involved, human goals/intents -- thus leading to knowledge gaps in joint planning. We consider the problem of discovering optimal interaction strategies for AI agents to actively elicit human inputs in object-driven planning. To this end, we propose Minimal Information Neuro-Symbolic Tree (MINT) to reason about the impact of knowledge gaps and leverage self-play with MINT to optimize the AI agent’s elicitation strategies and queries. More precisely, MINT builds a symbolic tree by making propositions of possible human-AI interactions and by consulting a neural planning policy to estimate the uncertainty in planning outcomes caused by remaining knowledge gaps. Finally, we leverage LLM to search and summarize MINT’s reasoning process and curate a set of queries to optimally elicit human inputs for best planning performance. By considering a family of extended Markov decision processes with knowledge gaps, we analyze the return guarantee for a given MINT with active human elicitation. Our evaluation on three benchmarks involving unseen/unknown objects of increasing realism shows that MINT-based planning attains near-expert returns by issuing a limited number of questions per task while achieving significantly improved rewards and success rates.

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Bridging Structure and Semantics: Uncertainty-Modulated Dual-Path Diffusion for Robust Text-Attributed Graph Learning

Zhizhi Yu, Jiachen Liu, Qingyu Li, Dongxiao He, Di Jin

Representation learning on text-attributed graphs (TAGs) is crucial for real-world applications, as it enables effective modeling of both rich node semantics and complex graph structure. Nevertheless, this task is intrinsically challenging due to structural–semantic mismatch stemming from divergent modality distributions, as well as dual-source noise inherent in node textual content and graph structure. Existing approaches often enforce a rigid fusion of distinct modalities while overlooking their inherent noise, which inevitably results in persistent distribution gaps and amplifies mixed interference during information propagation. To address these issues, we propose UDPD, an Uncertainty-modulated Dual-Path Diffusion model for robust text-attributed graph learning. Specifically, we first employ a dual-perspective node encoding strategy to separately learn semantic and structural embeddings. We then introduce a cooperative diffusion paradigm with parallel semantic and structural branches, where mutual guidance enables progressive alignment of different distributions while effectively suppressing modality inherent noise. Crucially, the reverse process is guided by node uncertainty, which is used to adaptively modulate cross-branch interaction strength, ensuring robust coupling and maximizing denoising effectiveness. Extensive experiments on five public benchmarks demonstrate the effectiveness and superiority of our UDPD over state-of-the-art baselines.

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Local Minima in Quadratic-Penalty Relaxations of Binary Linear Programs

Cheng-Han Huang, Yongliang Sun, Chaoyan Huang, Ismail Alkhouri, Rongrong Wang

Many combinatorial optimization problems admit quadratic unconstrained binary formulations (QUBO) which can often be relaxed to the box $[0,1]^n$ and optimized using scalable gradient-based methods. However, the resulting non-convex landscape can often contain local optima that are spurious or infeasible. In this paper, we establish sufficient structural conditions on quadratic penalties that rule out these failures, guaranteeing that every local minimizer of the relax problem is both binary and feasible. For each problem we study, we examine existing QUBO formulations when available, identify why they fail when they do, and propose

alternative relaxed QUBOs that satisfy our conditions. We show for several common combinatorial problems, including open-pit mining, knapsack, and traveling salesman formulations, that these constructions allow gradient-based methods such as projected gradient descent and Adam to be safely applied to obtain valid binary solutions. Our results clarify when differentiable optimization is a reliable local solver for quadratic combinatorial objectives.
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Bias in Zeroth-Order Normal Estimation for Decision-Based Attacks
Feiyang Wang, Hangwei Qian, Xingquan Zuo, Gang Chen, Ivor Tsang
Decision-based image attacks commonly rely on zeroth-order (ZO) Monte Carlo probing to estimate decision-boundary normals and iteratively refine adversarial perturbations to minimize the ℓ_2 norm.
We theoretically analyze and empirically demonstrate an intrinsic inefficiency arising from heterogeneous input sensitivity, where only a small subset of coordinates strongly affects the target model’s predictions.
Empirically, with one-bit feedback and a limited query budget, updates on low-sensitivity coordinates are overwhelmed by initialization and sampling noise, preventing their perturbations from exhibiting consistent improvement.
By modeling ZO refinement as a stochastic dynamical system, we formally characterize its asymptotic behavior: the optimization enters a stationary regime, where the perturbation aligns (in expectation) with the normal and its coordinate-wise magnitudes encode a local sensitivity ranking. However, this stationarity does not generally yield ℓ_2-optimal perturbations under nonlinear boundaries.
Building on this observation, we propose a novel and effective algorithm, Sensitivity-Aware Rescaling (SAR), that leverages this sensitivity signal to infer an importance map from the current best perturbation, then progressively suppresses low-importance regions through a coarse-to-fine schedule to reduce the ℓ_2 norm.
Extensive experiments show that SAR achieves consistent improvements in perturbation norm, attack success rate, and visual imperceptibility. The code is available at https://github.com/Flyingssheep/SAR.
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FairMerging: Rethinking Model Merging through the Lens of Fairness
Bing Liu, Xinrui Shan, Boyu Zhang, Qiankun Zhang, Bin Yuan, Jing Wang, Xianjun Deng
Model merging offers an appealing route to multi-task learning by composing independently fine-tuned checkpoints without centralized data or retraining. However, this convenience can come with a hidden cost. Model merging may *amplify* performance disparities across subgroups, raising fairness concerns even when average accuracy remains competitive. To explain this phenomenon, we develop a sensitivity-based theoretical analysis that upper bounds the fairness gap induced by model merging. Theoretical analysis with empirical verifications reveals that the resulting fairness gap is governed by two coupled factors, a *merging magnitude* term that measures how far the merged parameters move from the target model and *global sensitivity* terms that determine how unevenly the perturbation affects subgroup losses. Guided by these insights, we propose *FairMerging*, a two-stage merging framework that first reduces the sensitivity of the target model and then performs fairness-aware coefficient optimization with orthogonally normalized task vectors. Experiments across multiple datasets, backbones, and merging baselines demonstrate that FairMerging substantially mitigates unfairness while retaining competitive multi-task performance.
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Tabero: Learning Gentle Manipulation with Closed-Loop Force Feedback from Vision, Touch, and Language
Qiwei Wu, Rui Zhang, Xin Xiang, Tao Li, Weihua Zhang, Junjie Lai, Renjing Xu
Tactile sensing is essential for robots to achieve human-like gentle manipulation. However, existing Vision-Language-Action (VLA) models struggle to exploit tactile feedback for gentle manipulation due to scarce aligned vision-tactile-language data and the lack of effective closed-loop force feedback mechanisms.
To address these challenges, we introduce Tabero, a benchmark and model suite fo

r gentle, language-conditioned robotic manipulation that demands fine-grained co
ntact force perception. First, the Tabero benchmark addresses the scarcity of ta
ctile data by presenting a data-efficient pipeline that repurposes open-source r
obot manipulation trajectories to generate diverse vision-tactile-language tasks
, and establishes a multidimensional evaluation protocol that measures task succ
ess alongside physical interaction quality. Second, we propose Tabero-VTLA, an a
rchitecture with a decoupled force-position command interface; the resulting for
ce-position commands are executed by a fixed hybrid controller to enable real-ti
me, force-aware manipulation. Evaluated on Tabero, our model maintains high task
 success while reducing average grip force by over 70\% under gentle instruction
s, demonstrating its ability to modulate interaction forces based on multimodal
experience.
****

Panini: Continual Learning in Token Space via Structured Memory
Shreyas Rajesh, Pavan S Holur, Mehmet Yigit Turali, Chenda Duan, Vwani Roychowdh
ury
Language models are increasingly used to reason over content they were not train
ed on, such as new documents, evolving knowledge, and user-specific data. A comm
on approach is retrieval-augmented generation (RAG), which stores verbatim docum
ents externally (as chunks) and retrieves only a relevant subset at inference ti
me for an LLM to reason over. However, this results in inefficient usage of test
-time compute (LLM repeatedly reasons over the same documents); moreover, chunk
retrieval can inject irrelevant context that increases unsupported generation. W
e propose a human-like non-parametric continual learning framework, where the ba
se model remains fixed, and learning occurs by integrating each new experience i
nto an external semantic memory state that accumulates and consolidates itself c
ontinually. We present PANINI, which realizes this by representing documents as
Generative Semantic Workspaces (GSW)—an entity- and event-aware network of quest
ion–answer (QA) pairs, sufficient for an LLM to reconstruct the experienced situ
ations and mine latent knowledge via reasoning-grounded inference chains on the
network. Given a query, PANINI only traverses the continually-updated GSW (not t
he verbatim documents or chunks), and retrieves the most likely inference chains
. Across six QA benchmarks, PANINI achieves the highest average performance, 5%–
7% higher than other competitive baselines, while using 2–30× fewer answer-conte
xt tokens, supports fully open-source pipelines, and reduces unsupported answers
 on curated unanswerable queries. The results show that efficient and accurate s
tructuring of experiences at write time—as achieved by the GSW framework—yields
both efficiency and reliability gains at read time. Code is available at https:/
/github.com/roychowdhuryresearch/gsw-memory.
****

Train for Truth, Keep the Skills: Binary Retrieval-Augmented Reward Mitigates Ha
llucinations
Tong Chen, Akari Asai, Luke Zettlemoyer, Hannaneh Hajishirzi, Faeze Brahman
Modern post-trained language models are increasingly capable, but remain prone t
o extrinsic hallucinations.
We target the utility degradation issue that prior hallucination-reduction metho
ds often struggle to avoid, and propose online RL with Binary Retrieval-Augmente
d Reward (Binary RAR) to reduce hallucinations while preserving general capabili
ties.
Binary RAR assigns a reward of 1 if a response contains no factual contradiction
s with retrieved evidence, and 0 otherwise.
We theoretically show that this method reduces the probability of error-containi
ng responses while preserving the distribution of error-free responses. This hel
ps preserve the model’s capabilities, whereas other methods often degrade them.
We evaluate Binary RAR on multiple widely used models. On Qwen3-8B, it reduces l
ong-form hallucination rates by 39.3\% and short-form hallucination rates by 54.
4\%, outperforming supervised learning and preference optimization baselines.
Our error analysis shows that continuous factuality rewards (e.g., VeriScore) ca
n be exploited via reward hacking by producing fewer or more generic claims, whe
reas Binary RAR is more robust and better preserves general capabilities, includ

ing instruction following, math, and coding.
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Learning Permutation-invariant Macroscopic Dynamics
Zhichao Han, Mengyi Chen, Qianxiao Li
Accurately modeling the macroscopic dynamics of high-dimensional microscopic systems is of broad interest across the sciences.
Many data-driven approaches learn a low-dimensional latent state through an autoencoder trained for pointwise input reconstruction. These methods typically assume a fixed ordering of microscopic degrees of freedom in the input.
However, in many settings, such as particle systems, the microscopic state is inherently unordered.
This motivates an autoencoder framework that learns permutation-invariant latent representations.
To this end, we adopt a permutation-invariant encoder and design the decoder to reconstruct
the mass distribution centered at the observed points rather than per-sample reconstruction.
We then jointly learn the macroscopic dynamics of the observables together with the latent states.
We demonstrate the effectiveness and robustness of the proposed method across a range of microscopic settings, including learning the energy dynamics in interacting particle systems, predicting mixing dynamics in Lennard–Jones fluids, and modeling the stretching dynamics from video data of polymers moving in an elongational force field.
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Sparse Tokens Suffice: Jailbreaking Audio Language Models via Token-Aware Gradient Optimization
Zheng Fang, Xiaosen Wang, Shenyi Zhang, Shaokang Wang, Zhijin Ge
Jailbreak attacks on audio language models (ALMs) optimize audio perturbations to elicit unsafe generations, and they typically update the entire waveform densely throughout optimization. In this work, we investigate the necessity of such dense optimization by analyzing the structure of token-aligned gradients in ALMs. We find that gradient energy is highly non-uniform across audio tokens, indicating that only a small subset of token-aligned audio regions dominates the optimization signal. Motivated by this observation, we propose Token-Aware Gradient Optimization (TAGO), which enables sparse jailbreak optimization by retaining only waveform gradients aligned with audio tokens that have high gradient energy, while masking the remaining gradients at each iteration.
Across three ALMs, TAGO outperforms baselines, and substantial sparsification preserves strong attack success rates (e.g., on Qwen3-Omni, ASR_{l} remains at 86% with a token retention ratio of 0.25, compared to 87% with full token retention). These results demonstrate that dense waveform updates are largely redundant, and we advocate that future audio jailbreak and safety alignment research should further leverage this heterogeneous token-level gradient structure.
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HybridOM: Hybrid Physics-Based and Data-Driven Global Ocean Modeling with Efficient Regional Downscaling
Ruiqi Shu, Xiaohui Zhong, Qiusheng Huang, Ruijian Gou, Tianrun Gao, Hao Li, Xiaomeng Huang
Global ocean modeling is vital for climate science but struggles to balance computational efficiency with accuracy. Traditional numerical solvers are accurate but computationally expensive, while pure deep learning approaches, though fast, often lack physical consistency and long-term stability. To address this, we introduce HybridOM, a framework integrating a lightweight, differentiable numerical solver as a skeleton to enforce physical laws, with a neural network as the flesh to correct subgrid-scale dynamics. To enable efficient high-resolution modeling, we further introduce a physics-informed regional downscaling mechanism based on flux gating. This design achieves the inference efficiency of AI-based methods while preserving the accuracy and robustness of physical models. Extensive experiments on the GLORYS12V1 and OceanBench dataset validate HybridOM's performan

ce in two distinct regimes: long-term subseasonal-to-seasonal simulation and sho
rt-term operational forecasting coupled with the FuXi-2.0 weather model. Results
 demonstrate that HybridOM achieves state-of-the-art accuracy while maintaining
physical consistency, offering a robust solution for next-generation ocean digit
al twins.
****

Provably Efficient Policy-Reward Co-Pretraining for Adversarial Imitation Learni
ng

Tian Xu, Zexuan Chen, Zhilong Zhang, Yi-Chen Li, Chenyang Wang, Lei Yuan, Yang Y
u

Adversarial imitation learning (AIL) achieves high-quality imitation compared to
 behavioral cloning (BC), but demands substantial online environment interaction
. Recent empirical work has explored initializing AIL algorithms with BC-pretrai
ned policies to address this limitation, yet a rigorous theoretical understandin
g of pretraining's role in AIL remains elusive. This paper provides a systematic
 theoretical analysis and introduces principled pretraining algorithms for accel
erating AIL. We begin by analyzing AIL with policy pretraining alone, identifyin
g reward error as the dominant source of suboptimality. This reveals a critical
and previously overlooked gap: the absence of reward pretraining. Motivated by t
his finding, we develop a principled policy–reward co-pretraining approach groun
ded in a reward-shaping analysis. Our analysis uncovers a fundamental connection
 between expert policies and shaping rewards, which naturally gives rise to CoPT
-AIL, an approach that jointly pretrains both policy and reward through a single
 BC procedure. We prove that CoPT-AIL achieves an improved imitation gap bound o
ver standard AIL, establishing the first theoretical guarantee for the benefits
of pretraining in AIL. Experimental results confirm CoPT-AIL's superior performa
nce over existing AIL methods.
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EmBrace: A Collective Knowledge Fusion Framework Toward Unified EEG Foundation M
odels

Ziyu Jia, Junyi Lin, Pu Wan, Jinxin Pi, Jingying Ma, Peiliang Gong, Xinliang Zho
u, Yi Ding, Chenyu Liu

Electroencephalography (EEG) foundation models (EFMs) have achieved strong perfo
rmance across a wide range of downstream EEG tasks via pretraining and fine-tuni
ng. Through empirical analysis, we observe that (i) no single EFM consistently d
ominates all tasks, yet identifying the task-specific optimal model by fine-tuni
ng all EFMs introduces substantial computational overhead; and (ii) models with
inferior task-level performance still exhibit strengths at the sample level as d
istinct architectures induce diverse inductive biases. These observations motiva
te EmBrace, a representation-centric framework for sample-aware knowledge fusion
 that avoids the constraints of parameter-level or output-level alignment. EmBra
ce synchronizes discriminative intermediate representations into a unified manif
old and adaptively weights multiple EFMs at the sample level while selecting the
 most compatible model as the carrier. Extensive experiments across multiple EEG
 benchmarks demonstrate that EmBrace consistently improves over SOTA EFMs and ge
neralizes effectively under cross-task settings.
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Beyond Problem Solving: UOJ-Bench for Evaluating Code Generation, Hacking, and R
epair in Competitive Programming

Tingqiang Xu, Hangrui Zhou, Tianle Cai, Alex Gu, Kaifeng Lyu

Despite strong performance in competitive programming, the role of Large Languag
e Models (LLMs) in supporting human learning in the same setting remains largely
 unexplored. In this work, we introduce **UOJ-Bench**, a benchmark designed to e
valuate not only the problem-solving ability of LLMs, but also their ability to
identify errors in human-written code—a crucial educational activity traditional
ly supported by running test cases over online judge systems. UOJ-Bench consists
 of three distinct tasks: code generation, code hacking, and code repair, all co
nstructed from real-world code submissions on the Universal Online Judge (UOJ) a
nd evaluated through UOJ's native judging infrastructure. Our results show that
under one-shot evaluation, even the strongest models fail to identify errors in

more than 50\% of a set of submissions that have been found to be incorrect by U OJ users. While test-time scaling improves success rates to above 90\%, the subs tantial computational costs incurred from model inference limit its practicality for large-scale deployment. Despite these limitations, we find that the best-pe rforming models under test-time scaling can uncover errors in over 5\% of full-s core submissions across roughly 30 problems, suggesting that frontier LLMs can a lready provide complementary signals beyond standard judging systems. UOJ-Bench is publicly available at https://github.com/hehezhou/UOJ-Bench.
****
Nash Equilibria in Games with Playerwise Concave Coupling Constraints: Existence and Computation
Philip Jordan, Maryam Kamgarpour
We study the existence and computation of Nash equilibria in concave games where the players' admissible strategies are subject to shared coupling constraints. Under playerwise concavity of constraints, we prove existence of Nash equilibria . Our proof leverages topological fixed point theory and novel structural insigh ts into the contractibility of feasible sets, and relaxes strong assumptions for existence in prior work. Having established existence, we address the question of whether in the presence of coupling constraints, playerwise independent learn ing dynamics have convergence guarantees. We address this positively for the cla ss of potential games by designing a convergent algorithm. To account for the po ssibly nonconvex feasible region, we employ a log barrier regularized gradient a scent with adaptive stepsizes. Starting from an initial feasible strategy profi le and under exact gradient feedback, the proposed method converges to an $\epsi lon$-approximate constrained Nash equilibrium within $\mathcal{O}(\epsilon^{-3}) $ iterations.
****
PRPO: Paragraph-level Policy Optimization for Vision-Language Deepfake Detection
Tuan Nguyen, Naseem Khan, Khang Tran, Hai Phan, Issa Khalil
The rapid rise of synthetic media has made deepfake detection a critical challen ge for online safety and trust. Progress remains constrained by the scarcity of large, high-quality datasets. Although multimodal large language models (LLMs) e xhibit strong reasoning capabilities, their performance on deepfake detection is poor, often producing explanations that are misaligned with visual evidence or hallucinatory. To address this limitation, we introduce a reasoning-annotated da taset for deepfake detection and propose Paragraph-level Relative Policy Optimiz ation (PRPO), a reinforcement learning algorithm that aligns LLM reasoning with image content at the paragraph level. Experiments show that PRPO improves detect ion accuracy by a wide margin and achieves the highest reasoning score of 4.55/5 .0. Ablation studies further demonstrate that PRPO significantly outperforms GRP O under test-time conditions. These results underscore the importance of groundi ng multimodal reasoning in visual evidence to enable more reliable and interpret able deepfake detection.
****
Neural Minimum Weight Perfect Matching for Quantum Error Codes
Yotam Peled, David Zenati, Eliya Nachmani
Realizing the full potential of quantum computation requires Quantum Error Corre ction (QEC). QEC reduces error rates by encoding logical information across redu ndant physical qubits, enabling errors to be detected and corrected. A common de coder used for this task is Minimum Weight Perfect Matching (MWPM) a graph-based algorithm that relies on edge weights to identify the most likely error chains. In this work, we propose a data-driven decoder named Neural Minimum Weight Perf ect Matching (NMWPM). Our decoder utilizes a hybrid architecture that integrates Graph Neural Networks (GNNs) to extract local syndrome features and Transformer s to capture long-range global dependencies, which are then used to predict dyna mic edge weights for the MWPM decoder. To facilitate training through the non-di fferentiable MWPM algorithm, we formulate a novel proxy loss function that enabl es end-to-end optimization.
Our findings on the toric code under depolarizing noise demonstrate thresholds o f 17.9\% and 10.95\%, nearing the 18.9\% and 11.0\% maximum likelihood bounds, h

ighlighting the advantage of hybrid decoders that combine the predictive capabil
ities of neural networks with the algorithmic structure of classical matching.
****

Safety Anchor: Defending Harmful Fine-tuning via Geometric Bottlenecks
Guoxin Lu, Letian Sha, Qing Wang, Peijie Sun, ZHOU HAO, Hua Dai, Fu Xiao
The safety alignment of Large Language Models (LLMs) remains vulnerable to Harmf
ul Fine-tuning (HFT). While existing defenses impose constraints on parameters,
gradients, or internal representations, we observe that they can be effectively
circumvented under persistent HFT. Our analysis traces this failure to the inher
ent redundancy of the high-dimensional parameter space: attackers exploit optimi
zation trajectories that are orthogonal to defense constraints to restore harmfu
l capabilities while deceptively adhering to safety restrictions.
  To address this, we propose Safety Bottleneck Regularization (SBR). SBR shifts
 the defensive focus from the redundant parameter space to the unembedding layer
, which serves as a geometric bottleneck. By anchoring the final hidden states o
f harmful queries to those of the safety-aligned model, SBR enables the model to
 maintain safe responses even under persistent HFT. Extensive experiments confir
m SBR's effectiveness, demonstrating that utilizing just a single safety anchor
is sufficient to reduce the Harmful Score to $<$10 while preserving competitive
performance on benign downstream tasks. The code is available at https://github.
com/soyoaaa/SBR.
****

Smooth Dynamic Cutoffs for Machine Learning Interatomic Potentials
Kevin Han, Haolin Cong, Bowen Deng, Amir Barati Farimani
Machine learning interatomic potentials (MLIPs) have proven to be wildly useful
for molecular dynamics simulations, powering countless drug and materials discov
ery applications. However, MLIPs face two primary bottlenecks preventing them fr
om reaching realistic simulation scales: inference time and memory consumption.
In this work, we address both issues by challenging the long-held belief that th
e cutoff radius for the MLIP must be held to a fixed, constant value. For the fi
rst time, we introduce a **dynamic** cutoff formulation that still leads to stab
le, long timescale molecular dynamics simulation. In introducing the dynamic cut
off, we are able to induce sparsity onto the underlying atom graph by targeting
a specific number of neighbors per atom, significantly reducing both memory cons
umption and inference time. We show the effectiveness of a dynamic cutoff by imp
lementing it onto 4 state of the art MLIPs: MACE, Nequip, Orbv3, and TensorNet,
leading to **2.26x** less memory consumption and **2.04x** faster inference time
, depending on the model and atomic system. We also perform an extensive error a
nalysis and find that the dynamic cutoff models exhibit minimal accuracy dropoff
 compared to their fixed cutoff counterparts on both materials and molecular dat
asets. All model implementations and training code will be fully open sourced.
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RMNP: Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimiza
tion
Shenyang Deng, Zhuoli Ouyang, Tianyu Pang, Zihang Liu, Ruochen Jin, Shuhua Yu, Y
aoqing Yang
Preconditioned adaptive methods have gained significant attention for training d
eep neural networks, as they capture rich curvature information of the loss land
scape . The central challenge in this field lies in balancing preconditioning ef
fectiveness with computational efficiency of implementing the preconditioner. Am
ong recent advances, Muon stands out by using Newton-Schulz iteration to obtain
preconditioned updates without explicitly constructing the preconditioning matri
x. Despite its advantages, the efficiency of Muon still leaves room for further
improvement. In this paper, we introduce RMNP (Row Momentum Normalized Precondit
ioning), an optimizer that replaces Newton-Schulz iteration with a simple row-wi
se(d_{in}) ℓ_2 normalization operation, motivated by the empirical
ly observed diagonal block structure of the Transformer layerwise Hessian. We em
pirically verified that orthogonalization and row-wise(on input dim) ℓ_2 no
rmalization are asymptotically equivalent in the case of the transformer. This s
ubstitution reduces the per-iteration computational complexity from $\mathcal{O}

(mn\cdot\min(m,n))$ to $\mathcal{O}(mn)$ for an $m\times n$ weight matrix while maintaining comparable optimization performance. Theoretically, we establish convergence guarantees for RMNP in the non-convex setting that match recent results for Muon optimizers, achieving the minimax optimal complexity. Extensive experiments on large language model pretraining show that RMNP delivers competitive optimization performance compared with Muon while substantially reducing preconditioning wall-clock time. Our code is available at https://github.com/Dominator-Index/RMNP

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Exploration-free Algorithms for Multi-group Mean Estimation

Ziyi Wei, Huaiyang Zhong, Xiaocheng Li

We address the problem of multi-group mean estimation, which seeks to allocate a finite sampling budget across multiple groups to obtain uniformly accurate estimates of their means. Unlike classical multi-armed bandits, whose objective is to minimize regret by identifying and exploiting the best arm, the optimal allocation in this setting requires sampling every group on the order of $\Theta(T)$ times. This fundamental distinction makes exploration-free algorithms both natural and effective. Our work makes three contributions. First, we strengthen the existing results on subgaussian variance concentration using the Hanson-Wright inequality and identify a class of strictly subgaussian distributions that yield sharper guarantees. Second, we design exploration-free non-adaptive and adaptive algorithms, and we establish tighter regret bounds than the existing results. Third, we extend the framework to contextual bandit settings, an underexplored direction, and propose algorithms that leverage side information with provable guarantees. Overall, these results position exploration-free allocation as a principled and efficient approach to multi-group mean estimation, with potential applications in experimental design, personalization, and other domains requiring accurate multi-group inference.

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PhysForge: Generating Physics-Grounded 3D Assets for Interactive Virtual World

Yunhan Yang, Chunshi Wang, Junliang Ye, YANG LI, Zanxin Chen, Zehuan Huang, Yao Mu, Zhuo Chen, Chunchao Guo, Xihui Liu

Synthesizing physics-grounded 3D assets is a critical bottleneck for interactive virtual worlds and embodied AI. Existing methods predominantly focus on static geometry, overlooking the functional properties essential for interaction. We propose that interactive asset generation must be rooted in functional logic and hierarchical physics. To bridge this gap, we introduce PhysForge, a decoupled two-stage framework supported by PhysDB, a large-scale dataset of 150,000 assets with four-tier physical annotations. First, a VLM acts as a physical architect to plan a Hierarchical Physical Blueprint defining material, functional, and kinematic constraints. Second, a physics-grounded diffusion model realizes this blueprint by synthesizing high-fidelity geometry alongside precise kinematic parameters via a novel KineVoxel Injection (KVI) mechanism. Experiments demonstrate that PhysForge produces functionally plausible, simulation-ready assets, providing a robust data engine for interactive 3D content and embodied agents.

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Partial Identification of Policy Values under Network Interference

Ziyan Wang, Yiran Liu, Zhiheng Zhang

Offline Policy Evaluation (OPE) aims to estimate the value of a target policy from historical logged data without interating with the environment, thereby assessing policy performance. In settings with network interference, individuals no longer satisfy the SUTVA assumption: an individual’s outcome is influenced not only by their own treatment but also by the treatments of their neighbors, which makes the definition and estimation of policy value more complex. To capture this interference mechanism, we allow all neighbors to affect individual outcomes through a unified exposure mapping, and we use a decaying higher-order neighborhood aggregation to characterize the influence of more distant neighbors. Moreover, in real-world applications, the target policy and the logging policy often do not fully overlap (non-overlap), so the policy value in non-overlap regions cannot be point-identified. To address this issue, we partially identify the policy v

alue over non-overlap regions and, under a smoothness assumption, formulate the estimation of the lower and upper bounds as a linear program, yielding valid bou nds on the offline policy value. Finally, we conduct systematic experiments on s emi-synthetic network data to validate the effectiveness and robustness of the p roposed method under network interference and limited overlap.
****
Certificates for Complex-Compatible Learned Cochain Laplacians
Nivar Anwer, Marien Chenaud, David Elizondo
Learning mesh-based operators from data can match training objectives while impl icitly violating algebraic consistency constraints that classical discretization s satisfy by construction. Such violations can introduce near-kernel directions, degrade conditioning as resolution increases, and distort the low-frequency spe ctral structure on which downstream solvers and diagnostics rely. This work intr oduces a low-overhead compatibility certificate for learned operator pairs, toge ther with a closed-form projection that maps a learned pair to its Frobenius-nea rest chain-compatible operator. The certificate provides an explicit distance-to -compatibility and yields perturbation bounds for the discrete operator. These b ounds imply stability guarantees for elliptic solves and for low-frequency spect ral counts, provided a spectral gap separates the kernel from the rest of the sp ectrum and boundary treatments are well posed. Experiments on standard elliptic problems show that defect-aware training prevents condition-number blow-up at hi gher resolutions, improves robustness under mesh and topological distribution sh ifts, and maintains predictive accuracy relative to unconstrained learning. Over all, these results support the use of deployment-neutral, computable algebraic c onsistency checks to detect and control failure modes that are not revealed by l oss values alone.
****
Testing For Distribution Shifts with Conditional Conformal Test Martingales
Shalev Shaer, Yarin Bar, Drew Prinster, Yaniv Romano
We propose a sequential test for distribution-shift detection that allows confor mal test martingales (CTMs) to work under a fixed, reference-conditional setting . Existing CTM detectors construct test martingales by continually growing a ref erence set with each incoming sample, using it to assess how atypical the new sa mple is relative to past observations. While this design yields anytime-valid ty pe-I error control, it suffers from test-time contamination: after a change, pos t-shift observations enter the reference set and dilute the evidence for distrib ution shift, increasing detection delay and reducing power.

In contrast, our method avoids contamination by design by comparing each new sam ple to a fixed null reference dataset. Our main technical contribution is a robu st martingale construction that remains valid conditional on the null reference data, achieved by explicitly accounting for the estimation error in the referenc e distribution induced by the finite reference set. This yields anytime-valid ty pe-I error control together with guarantees of asymptotic power one and bounded expected detection delay. Empirically, our method detects shifts faster than sta ndard CTMs, providing a powerful and reliable distribution-shift detector.
****
When Generalized Zero-Shot Learning Meets PU Learning: A Plug-and-Play Framework for Seen-Class Bias Mitigation
Long Tang, Keyang Pu, Yingjie Tian
Generalized Zero-Shot Learning (GZSL) suffers from severe seen-class bias, a cha llenge stemming from the label incompleteness inherent in the mixed seen–unseen test set. To address this, we propose PUFE, a unified plug-and-play post-process ing framework that recasts GZSL inference as a Positive-Unlabeled (PU) learning task by treating seen categories as positive and mixed test data as unlabeled. P UFE constructs a PU classifier in the semantic space, jointly estimating the see n-class posterior and labeling propensity via Maximum Likelihood Estimation (MLE) within a dual-head network. Furthermore, we introduce an adaptive prototype ca libration strategy that employs high-confidence pseudo-instances—identified by t he PU classifier—to explicitly align semantic prototypes with the underlying tes

t distribution. Extensive experiments demonstrate that PUFE mitigates bias and r
aises the performance of diverse GZSL baselines across multiple architectural pa
radigms, yielding gains of up to 11.2 percentage points in the harmonic mean.
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What if Tomorrow is the World Cup Final? Counterfactual Time Series Forecasting
with Textual Conditions
Shuqi Gu, Yongxiang Zhao, Baoyu Jing, Kan Ren
Time series forecasting has become increasingly critical in real-world scenarios
, where future sequences are influenced not only by historical patterns but also
 by forthcoming events. In this context, forecasting must dynamically adapt to c
omplex and stochastic future conditions, which introduces fundamental challenges
 in both forecasting and evaluation. Traditional methods typically rely on histo
rical data or factual future conditions, while overlooking counterfactual scenar
ios. Furthermore, many existing approaches are restricted to simple structured c
onditions, limiting their ability to generalize to the real-world complexities.
To address these gaps, we introduce the task of counterfactual time series forec
asting with textual conditions, enabling more flexible and condition-aware forec
asting. We propose a comprehensive evaluation framework that encompasses both fa
ctual and counterfactual settings, even in the absence of ground truth time seri
es. Additionally, we present a novel text-attribution mechanism that distinguish
es mutable from immutable factors, thereby improving forecast accuracy under sop
histicated and stochastic textual conditions.
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Clipped Q-Learning: Your Value Clipping Is Secretly A Robust Operator
Zhishuai Liu, Pan Xu
We study a simple yet principled modification of classical Q-learning that clips
 the value estimate in the Bellman backup by a threshold λ. The resultin
g algorithm, clipped Q-learning, is motivated by a key theoretical insight: the
clipped Bellman backup is an unbiased one-sample estimation of a robust Bellman
operator arising naturally from a transition-regularized MDP framework. This for
mulation corresponds to optimizing performance against a specific class of adver
sarial dynamics perturbations at the test time that reallocate transition probab
ility mass away from high-value states, thereby inducing conservative but stable
 decision making. Under this interpretation, clipped Q-learning can be viewed as
 tracking the fixed point of the robust Bellman equation and learning policies t
hat hedge against adversarial dynamics shifts at test time. We analyze two clipp
ed Q-learning variants with an optimistic exploration bonus and establish polyno
mial regret guarantees, demonstrating statistical efficiency. Beyond the tabular
 setting, our framework suggests that value clipping is a modular mechanism that
 can be incorporated into general value-based RL algorithms with function approx
imation. As a proof of concept, we evaluate a clipped Double DQN algorithm on a
control task and observe robustness improvements consistent with our theoretical
 predictions.
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See, Act, Adapt: Active Perception for Unsupervised Cross-Domain Visual Adaptati
on via Personalized VLM-Guided Agent
Tianci Tang, Tielong Cai, Hongwei Wang, Gaoang Wang
Pre-trained perception models excel in generic image domains but degrade signifi
cantly in novel environments like indoor scenes. The conventional remedy is fine
-tuning on downstream data which incurs catastrophic forgetting of prior knowled
ge and demands costly, scene-specific annotations. We propose a paradigm shift t
hrough Sea^2 ($\textbf{Se}$e, $\textbf{A}$ct, $\textbf{A}$dapt): rather than a
dapting the perception modules themselves, we adapt how they are deployed throug
h an intelligent pose-control agent. Sea^2 keeps all perception modules frozen
, requiring no downstream labels during training, and uses only scalar perceptua
l feedback to navigate the agent toward informative viewpoints.
Specially, we transform a vision-language model (VLM) into a low-level pose cont
roller through a two-stage training pipeline: first fine-tuning it on rule-based
 exploration trajectories that systematically probe indoor scenes, and then refi
ning the policy via unsupervised reinforcement learning that constructs rewards

from the perception module’s outputs and confidence.
Unlike prior active perception methods that couple exploration with specific mod els or collect data for retraining them, Sea^2 directly leverages off-the-shel f perception models for various tasks without the need for retraining.
We conducted experiments on three visual perception tasks, including visual grou nding, segmentation and 3D box estimation, with performance improvements of 13.5 4\%, 15.92\% and 27.68\% respectively on dataset ReplicaCAD.
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Pessimistic Verification for Open-Ended Math Questions
Yanxing Huang, Zihan Tang, Zejin Lin, Peng Li, Yang Liu
Automatic verification is a critical component in building math-solving agents a nd reinforcement learning, yet it often falls short in generalizability, perform ance, and cost-efficiency. Identifying that the primary bottleneck of verificati on lies in error detection capability, we propose pessimistic verification, a pa radigm of agentic workflows that rejects a solution if any of multiple parallel verifiers identifies a flaw. We further introduce progressive pessimistic verifi cation, which employs fine-grained proof decomposition to significantly enhance verification accuracy and efficiency. Our approach surpasses the performance and token efficiency of extended long chain-of-thought (long CoT) and mainstream ve rification workflows, crucially, our analysis reveals that existing benchmarks u nderestimate its effectiveness on stronger models due to inherent annotation err ors. To further validate the effectiveness of our method, we applied a verificat ion-based solving workflow on the IMO 2025 and MathArena Apex 2025 datasets, whe re the workflow with progressive pessimistic verification exhibits remarkable im provements in both efficiency and accuracy on highly challenging contest-level m ath problems with state-of-the-art models.
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Enhancing Neural Theorem Proving via High-Quality Proof Selection and Verifier F eedback
Xiaoxue Zhu, Jilin Hu, Fuyuan Zhang, Jianyu Zhang, Yongwang Zhao
Recent advances in large language models have accelerated neural theorem proving (NTP). Isabelle is a mature and important formal theorem prover that has been w idely used in software and hardware verification. However, progress in the Isabe lle setting remains limited. Existing approaches either optimize search strategi es or train on highly imbalanced raw proof corpora. At the same time, the specia lized structure of Isabelle proofs limits the effectiveness of general-purpose d ata selection methods.
To address these challenges, we adopt a data-centric framework for neural theore m proving in Isabelle. We characterize high-quality formal proof data along thre e complementary dimensions—proof complexity, semantic coverage, and reasoning di versity (PSR)—and propose a PSR-guided data selection pipeline to construct a co mpact, high-quality training subset. In addition, we leverage verifier feedback as a dynamic data signal during inference, introducing a dynamic feedback-based prompt optimization that iteratively incorporates Isabelle verifier feedback to guide proof generation.
We construct and release a 4k high-quality Isabelle dataset based on the PSR cri terion. On the miniF2F-test, fine-tuning solely on PSR-selected data achieves 84 .8% Pass@64. When further combined with dynamic feedback–based prompt optimizati on, the full framework improves performance to 90.6% Pass@64, establishing a new state of the art for neural theorem proving in Isabelle.
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Cooperative Variance Estimation and Bayesian Neural Networks for Disentangling A leatoric and Epistemic Uncertainties
Jiaxiang Yi, Miguel Anibal Bessa
Real-world data contains aleatoric uncertainty - irreducible noise arising from imperfect measurements or from incomplete knowledge about the data generation pr ocess. Mean-variance estimation networks can learn this type of uncertainty but require ad-hoc regularization strategies to avoid overfitting and are unable to predict epistemic uncertainty (model uncertainty). Conversely, Bayesian neural n etworks predict epistemic uncertainty but are notoriously difficult to train due

to the approximate nature of Bayesian inference. We propose to cooperatively tr
ain a variance estimation network with a Bayesian neural network and empirically
demonstrate that the resulting model disentangles aleatoric and epistemic uncer
tainties while improving the mean estimation. We demonstrate the effectiveness a
nd scalability of this method across a diverse range of datasets, including a ti
me-dependent heteroscedastic regression dataset we created where the aleatoric u
ncertainty is known. The proposed method is straightforward to implement, robus
t, and adaptable to various model architectures.
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Memory Caching: RNNs with Growing Memory
Ali Behrouz, Zeman Li, Yuan Deng, Peilin Zhong, Meisam Razaviyayn, Vahab Mirrokn
i
Transformers have been established as the de-facto backbones for most recent adv
ances in sequence modeling, mainly due to their growing memory capacity that sca
les with the context length. While plausible for retrieval tasks, it causes quad
ratic complexity and so has motivated recent studies to explore viable subquadra
tic recurrent alternatives. Despite showing promising preliminary results in div
erse tasks, such recurrent architectures underperform Transformers in recall-int
ensive tasks, {often attributed to their fixed-size memory. In this paper, we in
troduce Memory Caching (MC), a simple yet effective technique that enhances recu
rrent models by caching checkpoints of their memory states (a.k.a. hidden states
). Memory Caching allows the effective memory capacity of RNNs to grow with sequ
ence length, offering a flexible trade-off that interpolates between the fixed m
emory ($O(L)$ complexity) of RNNs and the growing memory (
$O(L^2)$ complexity) of Transformers. We propose four variants of MC, including
gated aggregation and sparse selective mechanisms, and discuss their implicatio
ns on both linear and deep memory modules.} Our experimental results on language
modeling, and long-context understanding tasks show that MC enhances the perfor
mance of recurrent models, supporting its effectiveness. In in-context recall ta
sks, our results indicate that while Transformers still achieve the best perform
ance, our MC variants show competitive performance, close the gap with Transform
ers, and performs better than state-of-the-art recurrent models.
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Conformal Risk-Averse Decision Making with Action Conditional Guarantee
Zihan Zhu, Shayan Kiyani, George J. Pappas, Hamed Hassani
Reliable decision making pipelines powered by machine learning models require un
certainty quantification (UQ) methods that come with explicit safety guarantees.
Conformal prediction provides such UQ by wrapping ML predictions into predictio
n sets, and recent work by Kiyani et al. (2025b) established that these sets can
be translated into optimal risk-averse decision policies -- yet only inheriting
marginal safety guarantees. We generalize and strengthen their results by (i) i
ntroducing action-conditional conformal prediction, which yields safety guarante
es conditioned explicitly on each action taken by the decision maker, (ii) showi
ng that action-conditional prediction sets serve as a proxy for the feasible dec
ision space for risk-averse decision makers aiming to optimize action-conditiona
l value-at-risk, and (iii) proposing a principled finite-sample algorithm based
on pinball-loss minimization, connecting the framework of Gibbs et al. (2025) to
action-conditional guarantees. Experiments on two real-world datasets confirm t
hat our approach significantly improves action-conditional performance over conf
ormal baselines.
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LERD: Latent Event-Relational Dynamics for Neurodegenerative Classification
Yicheng Feng, Hairong Chen, Ziyu Jia, Samir Bhatt, Hengguan Huang
Alzheimer’s disease (AD) alters brain electrophysiology and disrupts multichanne
l EEG dynamics, making accurate and clinically useful EEG-based diagnosis increa
singly important for screening and disease monitoring. However, many existing ap
proaches rely on black-box classifiers and do not explicitly model the latent ev
ent timing and cross-channel coordination behind their decisions. To address the
se limitations, we propose LERD, an end-to-end Bayesian latent event--relational
dynamical system that infers latent neural events and their relational structur

e directly from multichannel EEG without event or interaction annotations. LERD combines a continuous-time event inference module with a stochastic event-generation process to capture flexible temporal patterns, while incorporating an electrophysiology-inspired dynamical prior to guide learning in a principled way. We further provide theoretical analysis that yields a tractable IVP-based KL regularizer and stability guarantees for the inferred relational dynamics. Extensive experiments on synthetic benchmarks and two real-world AD EEG cohorts demonstrate that LERD consistently outperforms strong baselines and yields physiology-aligned rate, timing, and graph summaries that help characterize group-level dynamical differences.

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Local Intrinsic Dimension of Representations Predicts Alignment and Generalization in AI Models and Human Brain

Junjie Yu, Wenxiao Ma, Chen Wei, Jianyu Zhang, Haotian Deng, Zihan Deng, Quanying Liu

Recent work has found that neural networks with stronger generalization tend to exhibit higher representational alignment with one another across architectures and training paradigms. In this work, we show that models with stronger generalization also align more strongly with human neural activity. Moreover, generalization performance, model--model alignment, and model--brain alignment are all significantly correlated with each other.

We further show that these relationships can be explained by a single geometric property of learned representations: the local intrinsic dimension of embeddings. Lower local dimension is consistently associated with stronger model--model alignment, stronger model--brain alignment, and better generalization, whereas global dimension measures fail to capture these effects. Finally, we find that increasing model capacity and training data scale systematically reduces local intrinsic dimension, providing a geometric account of the benefits of scaling. Together, our results identify local intrinsic dimension as a unifying descriptor of representational convergence in artificial and biological systems.

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Boosting Monocular Metric Depth Estimation via Bokeh Rendering

Hangwei Zhang, Armando Fortes, Tianyi Wei, Xingang Pan

Bokeh rendering and depth estimation share a fundamental optical connection, yet existing methods fail to fully exploit this reciprocity. Conventional bokeh pipelines rely heavily on noisy depth maps that inevitably introduce visual artifacts. Conversely, existing monocular depth models typically follow two flawed paradigms. Generative diffusion-based frameworks often lack consistent metric scale. Meanwhile, feed-forward metric depth models frequently fail in textureless or distant regions where defocus blur can provide geometric information. We propose BokehDepth, a two-stage framework that treats synthetic defocus as a supervision-free geometric signal. In the first stage, a physically grounded generative model produces calibrated bokeh stacks from a single sharp input without requiring prior depth input. Subsequently, a lightweight defocus-aware aggregation module integrates these stacks into the encoder of a depth estimation framework. This mechanism allows the model to extract consistent geometric features from the defocus dimension while keeping the decoder architecture unchanged. Experiments demonstrate that BokehDepth achieves superior visual bokeh fidelity compared to depth-dependent rendering baselines and consistently enhances the metric accuracy of state-of-the-art monocular depth models.

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Inference from Quantized Data via Normal Variance-Mean Mixtures

Chenyu Gao, Zhexian Yang, Ziping Zhao

Inference from quantized data has received significant attention in recent years due to its broad applications in machine learning and signal processing. Existing likelihood-based approaches are often restricted to Gaussian assumptions or low-bit quantization settings, limiting modeling flexibility and robustness under complex data distributions. In this work, we study inference from quantized observations under the general normal variance-mean mixture (NVMM) framework, which encompasses distributions including Gaussian, t, generalized hyperbolic skew-

t, and generalized hyperbolic distributions.
Optimization under the NVMM framework is challenging because the underlying likelihood function involves multidimensional integrals that are difficult to evaluate due to multidimensional quantization and latent mixture variables. To address this difficulty, we propose an expectation conditional maximization (ECM) algorithm with latent-variable augmentations for both quantization and mixture modeling. By leveraging the conditional Gaussian structure of the NVMM family, the proposed method admits closed-form updates for all model parameters at each iteration, leading to an efficient and tractable optimization procedure. We further establish global linear convergence guarantees for the proposed ECM algorithm. Beyond basic parameter estimation, the proposed framework naturally extends to several structured learning and recovery tasks under the NVMM framework, including quantized regression, matrix completion, compressive sensing, and covariance estimation. Numerical experiments demonstrate the effectiveness and robustness of the proposed framework across a variety of quantized inference problems.
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Learning-Augmented Scalable Linear Assignment Problem Optimization via Neural Dual Warm-Starts
Ilay Yavlovich, Jad Agbaria, Muhamed Mhamed, Nir Weinberger, Jose Yallouz
The Linear Assignment Problem is a fundamental combinatorial optimization task where classical exact solvers ensure optimality but suffer from an $\mathcal{O}(N^{3})$ bottleneck, while recent neural approximations struggle with scalability and exactness. We propose a learning-augmented framework that accelerates exact solvers by predicting dual variables to warm-start the search, backed by a fallback mechanism to preserve worst-case guarantees. Central to our approach is RowDualNet, a lightweight, row-independent architecture that avoids the $\mathcal{O}(N^{2})$ memory bottleneck of graph models, enabling scalable neural warm-starting up to $N=16{,}384$. Feasibility is guaranteed by construction via the Min-Trick mechanism, completely eliminating the need for costly iterative projections. Empirically, our method drastically reduces the search effort of the Jonker-Volgenant (LAPJV) algorithm, yielding robust zero-shot generalization with strict optimality and end-to-end speedups of over 2x on complex synthetic data, 1.25x on real-world tracking, and 1.5x on transportation networks.
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Threshold-Guided Optimization for Visual Generative Models
Jinbin Bai, Yu Lei, Qingyu Shi, Aosong Feng, Yi Xin, Zhuoran Zhao, Fei Shen, Kaidong Yu, Xiangtai Li
Aligning large visual generative models with human feedback is often performed through pairwise preference optimization. While such approaches are conceptually simple, they fundamentally rely on annotated pairs, limiting scalability in settings where feedback is collected as independent scalar ratings. In this work, we revisit the KL-regularized alignment objective and show that the optimal policy implicitly compares each sample’s reward to an instance-specific baseline that is generally intractable. We propose a threshold-guided alignment framework that replaces this oracle baseline with a data-driven global threshold estimated from empirical score statistics. This formulation turns alignment into a binary decision task on unpaired data, enabling effective optimization directly from scalar feedback. We further incorporate a confidence weighting term to emphasize samples whose scores deviate strongly from the threshold, improving sample efficiency. Experiments across both diffusion and masked generative paradigms, spanning three test sets and five reward models, show that our method consistently improves preference alignment over previous methods. These results position our threshold-guided framework as a simple yet principled alternative for aligning visual generative models without paired comparisons.
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ToMAP: Training Opponent-Aware LLM Persuaders with Theory of Mind
Peixuan Han, Zijia Liu, Jiaxuan You
Large language models (LLMs) have shown promising potential in persuasion, but existing works on training LLM persuaders are still preliminary. Notably, while humans are skilled in modeling their opponent's thoughts and opinions proactively

and dynamically, current LLMs struggle with such Theory of Mind (ToM) reasoning, resulting in limited diversity and opponent awareness. To address this limitation, we introduce Theory of Mind Augmented Persuader (**ToMAP**), a novel approach for building more flexible persuader agents by incorporating two theory of mind modules that enhance the persuader's awareness and analysis of the opponent's mental state. Specifically, we instruct the persuader to consider possible objections to the target claim, and train a module to predict the opponent’s agreement level on these objections. Experiments show that the ToMAP persuader, while containing only 3B parameters, outperforms much larger baselines, like GPT-4o, with a relative gain of 39.4% across multiple persuadee models and diverse corpora. Notably, ToMAP exhibits complex reasoning chains and reduced repetition during training, which leads to more diverse and effective arguments. These results underscore ToMAP's potential for developing more persuasive language agents. Code is available at: https://github.com/ulab-uiuc/ToMAP.

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Agentic Confidence Calibration

Jiaxin Zhang, Caiming Xiong, Chien-Sheng Wu

AI agents are rapidly advancing from passive language models to autonomous systems executing complex, multi-step tasks. Yet their overconfidence in failure remains a fundamental barrier to deployment in high-stakes settings. Existing calibration methods, built for static single-turn outputs, cannot address the unique challenges of agentic systems, such as compounding errors along trajectories, uncertainty from external tools, and opaque failure modes. To address these challenges, we introduce, for the first time, the problem of Agentic Confidence Calibration and propose Holistic Trajectory Calibration (HTC), a novel diagnostic framework that extracts rich process-level features ranging from macro dynamics to micro stability across an agent's entire trajectory. Powered by a simple, interpretable model, HTC consistently surpasses strong baselines in both calibration and discrimination, across eight benchmarks, multiple LLMs, and diverse agent frameworks. Beyond performance, HTC delivers three essential advances: it provides interpretability by revealing the signals behind failure, enables transferability by applying across domains without retraining, and achieves generalization through a General Agent Calibrator (GAC) that achieves the best calibration (lowest ECE) on the out-of-domain GAIA benchmark. Together, these contributions establish a new process-centric paradigm for confidence calibration, providing a framework for diagnosing and enhancing the reliability of AI agents.

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D-Judge: Disrupting Multi-Turn Jailbreaks using Semantics-Preserving Output Rewriting

Huanli Gong, Zhipeng Wei, Yu Fu, Haz Sameen Shahgir, Ananya Gupta, Yue Dong, N. Benjamin Erichson

Multi-turn jailbreak attacks pose a growing threat to large language model (LLM) safety because they exploit feedback from auxiliary judge models to iteratively refine prompts toward harmful goals. Existing defenses largely detect or block unsafe content at individual turns or at the final response, leaving the judge-driven refinement loop intact and allowing attackers to extract informative feedback from intermediate interactions. We introduce D-Judge, a semantics-preserving output rewriting defense that intervenes directly in this loop by rewriting the victim LLM’s responses before they are evaluated by the attacker’s judge. By misaligning the judge’s feedback signal without changing the meaning of the original response, D-Judge derails the attacker’s prompt-refinement process, causing subsequent queries to be optimized against a distorted signal of attack progress. To improve D-Judge’s ability to produce such rewrites, we construct a dataset of semantically equivalent response pairs that induce different judge-assigned harmfulness scores, and use it for supervised fine-tuning followed by direct preference optimization. Experiments on HarmBench show that D-Judge reduces the success rate of state-of-the-art multi-turn jailbreaks while preserving performance on benign benchmarks.

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Harmonized Dual Policy Improvement for Modelic Reinforcement Learning

Guojian Zhan, Likun Wang, Feihong Zhang, Yang Guan, Shengbo Eben Li
Policy-planner bootstrapping has emerged as a powerful paradigm in model-based r einforcement learning (MBRL). We formalize this process as a dual policy improve ment mechanism synergizing: (i) exploitative improvement via off-policy Q-maxi mization, and (ii) lookahead improvement via planner alignment. While we theoret ically prove that these improvements anchor to the same optimum, practical train ing process inevitably encounters gradient disagreement. Exacerbated by approxim ation inaccuracies and non-stationary data, this disagreement induces destructiv e interference in policy updates, destabilizing the bootstrapping loop and leadi ng to suboptimal convergence. To address this, we propose harmonized dual policy improvement (HDPI), a gradient-level framework that reconciles exploitative and lookahead improvements through a harmonic optimization scheme. This scheme effe ctively maximizes the worst-case inner product between the harmonized update and the original gradients, ensuring directional consistency and stabilizing policy evolution. Extensive empirical evaluations on 14 challenging tasks from the Dee pMind Control Suite and the Humanoid-Bench demonstrate that HDPI significantly e nhances training stability and asymptotic performance, outperforming a wide rang e of strong baselines.
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Geometric and Stochastic Analysis of Discontinuities in Sparse Mixture-of-Expert s
Tho Tran Huu, Huu-Tuan Nguyen, Thien-Hai Nguyen, Nhat-Tri Ho, Viet-Hoang Tran, T ho Quan, Tan Minh Nguyen
Sparse Mixture-of-Experts (SMoE) architectures are now widely deployed in state- of-the-art language and vision models, where conditional routing allows scaling to very large networks. However, this very Top-k expert selection that enables conditional routing also renders the SMoE map inherently discontinuous. In the vicinity of these discontinuity surfaces, even inputs that are arbitrarily close may activate substantially different sets of experts resulting in significantly different outputs. In this work we give a rigorous geometric and stochastic ana lysis of these discontinuities. We first classify them by order, determined by t he number of tied experts at a switching event. Using measure-theoretic slicing arguments, we establish asymptotic volume estimates for the thickened discontinu ity surfaces, showing that lower-order discontinuity sets dominate, whereas high er-order ones occupy a vanishingly small relative volume. Next, modeling random perturbations in the input space via a diffusion process, we prove that the path eventually encounter a discontinuity, and moreover that the first hit almost su rely occurs on an order-1 discontinuity with explicit finite-time probability bo unds. We further derive occupation-time bounds that quantify the duration the ra ndom path spend in the neighborhoods of each discontinuity order. These theoreti cal results imply that inputs are more likely to lie near lower order discontinu ities. Motivated by this insight, we propose a simple smoothing mechanism that c an be directly applied to existing SMoEs, softly incorporating experts near disc ontinuities; our analysis guarantees that the added computational overhead remai ns small while providing localized smoothing near discontinuities, and experimen ts across language and vision tasks show that smoothing not only enforces contin uity of the SMoE map but also enhances empirical performance.
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A Very Big Video Reasoning Suite
Maijunxian Wang, Ruisi Wang, Juyi Lin, Ran Ji, Thaddäus Wiedemer, Qingying Gao, Dezhi Luo, Yaoyao Qian, Lianyu Huang, Zelong Hong, Jiahui Ge, Qianli Ma, Hang He , Yifan Zhou, Lingzi Guo, Lantao Mei, Jiachen Li, Hanwen Xing, Tianqi Zhao, Yu F engyuan, Weihang Xiao, Yizheng Jiao, Jianheng Hou, Danyang Zhang, Pengcheng Xu, Boyang ZHONG, Zehong Zhao, Gaoyun Fang, John Kitaoka, Xu Yile, Hua XU, Kenton Bl acutt, Tin Nguyen, Siyuan Song, Haoran Sun, shaoyue wen, Linyang He, Runming Wan g, Yanzhi Wang, Mengyue Yang, Ziqiao Ma, Raphaël Millière, Freda Shi, Nuno Vasco ncelos, Daniel Khashabi, Alan Yuille, Yilun Du, Ziming Liu, Dahua Lin, Ziwei Liu , Vikash Kumar, Yijiang Li, Lei Yang, Zhongang Cai, Hokin Deng
Rapid progress in video models has largely focused on visual quality, leaving th eir reasoning capabilities underexplored. Video reasoning grounds intelligence i

n spatiotemporally consistent visual environments that go beyond what text can n
aturally capture, enabling intuitive reasoning over spatiotemporal structure, su
ch as continuity, interaction, and causality. However, systematically studying v
ideo reasoning and its scaling behavior is hindered by the lack of large-scale v
ideo reasoning training data. To address this gap, we introduce the **Very Big V
ideo Reasoning (VBVR) Dataset**, an unprecedentedly large-scale resource spannin
g *200* curated reasoning tasks following a principled taxonomy, and over *one m
illion* video clips, making it approximately *three orders of magnitude* larger
than existing datasets. We further present **VBVR-Bench**, a verifiable evaluati
on framework that moves beyond model-based judging by incorporating rule-based,
human-aligned scorers, enabling reproducible and interpretable diagnosis of vide
o reasoning capabilities. Leveraging the VBVR suite, we conduct one of the first
 video reasoning **scaling studies** and observe early signs of emergent general
ization to unseen reasoning tasks. Together, VBVR lays a foundation for the next
 stage of research in generalizable video reasoning. The data, benchmark tool ki
t, and models are released publicly at **video-reason.com**.
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How to Train Your Advisor: Steering Black-Box LLMs with Advisor Models
Parth Asawa, Alan Zhu, Abigail O'Neill, Matei Zaharia, Alex Dimakis, Joseph E. G
onzalez
Frontier language models are deployed as black-box services, where model weights
 cannot be modified and customization is limited to prompting.
We introduce Advisor Models, a method to train small open-weight models to gener
ate dynamic, per-instance natural language advice that improves the capabilities
 of black-box frontier models.
Advisor Models improve GPT-5.2's performance on RuleArena (Taxes) by 27.4\%, red
uce Gemini 3 Pro's steps taken in SWE agent tasks by 24.6\%, and outperform stat
ic prompt optimizers in personalizing GPT-5 to user preferences (85-100\% vs. 40
-60\%).
We also find that advisors are transferable: an advisor trained with a low-cost
student model still transfers improvements to a frontier model.
Moreover, Advisor Models are robust: we observe no degradation on other benchmar
ks than the pipeline is trained on. Our method shows how to perform parametric o
ptimization for black-box frontier models in a practical and cost-effective way.
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SEAgent: Self-Evolving Computer Use Agent with Autonomous Learning from Experien
ce
Zeyi Sun, Ziyu Liu, Yuhang Zang, Yuhang Cao, Xiaoyi Dong, Tong Wu, Dahua Lin, Ji
aqi Wang
Repurposing large vision-language models (LVLMs) as computer use agents (CUAs) h
as led to substantial breakthroughs, primarily driven by human-labeled data. How
ever, these models often struggle with novel and specialized software, particula
rly in scenarios lacking human annotations. To address this challenge, we propos
e SEAgent, an agentic self-evolving framework enabling CUAs to autonomously evol
ve through interactions with unfamiliar software. Specifically, SEAgent empowers
 computer-use agents to autonomously master novel software environments via expe
riential learning, where agents explore new software, learn through iterative tr
ial-and-error, and progressively tackle auto-generated tasks organized from simp
le to complex. To achieve this goal, we design a World State Model for step-wise
 trajectory assessment, along with a Curriculum Generator that generates increas
ingly diverse and challenging tasks. The agent's policy is updated through exper
iential learning, comprised of adversarial imitation of failure actions and Grou
p Relative Policy Optimization (GRPO) on successful ones. Furthermore, we introd
uce a specialist-to-generalist training strategy that integrates individual expe
riential insights from specialist agents, facilitating the development of a stro
nger generalist CUA capable of continuous autonomous evolution. This unified age
nt ultimately achieves performance surpassing ensembles of individual specialist
 agents on their specialized software. We validate the effectiveness of SEAgent
across five professional software of OSWorld, ScienceBoard and AndroidWorld. Our
 approach achieves a significant improvement over a competitive open-source CUA,

UI-TARS. All the code and models will be made publicly available to foster further research.
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UrbanMLLM: Joint Learning of Cross-view Imagery for Urban Understanding
Xin Zhang, Tianjian Ouyang, Yu Shang, Qingmin Liao, Yong Li
Comprehensive urban understanding requires integrating macroscopic spatial structure with fine-grained street-level semantics. However, existing urban Multimodal Large Language Models (MLLMs) primarily rely on satellite imagery, limiting their ability to capture detailed urban appearance and cross-view relationships. We propose \textbf{UrbanMLLM}, a unified MLLM that jointly learns from satellite and street-view imagery for cross-view urban perception and reasoning. To support this, we construct a large-scale dataset with paired cross-view urban images, geospatial alignment, and textual annotations. UrbanMLLM introduces a cross-view perceiver to explicitly model interactions between satellite and street-view representations, and adopts a structural interleaved pre-training paradigm that organizes cross-view image–text content as coherent urban documents to enhance cross-view knowledge fusion. We evaluate UrbanMLLM on 13 diverse urban understanding tasks spanning satellite, street-view, and cross-view settings. Experimental results demonstrate consistent improvements over strong open-source and proprietary MLLMs, highlighting effectiveness and scalability of UrbanMLLM for urban environment understanding.
****
Identifying Connectivity Distributions from Neural Dynamics Using Flows
Timothy Doyeon Kim, Ulises Pereira Obilinovic, Yiliu Wang, Eric Todd SheaBrown, Uygar Sümbül
Connectivity structure shapes neural computation, but inferring this structure from population recordings is degenerate: multiple connectivity structures can generate identical dynamics. Recent work uses low-rank recurrent neural networks (lrRNNs) to infer low-dimensional latent dynamics and connectivity from observed activity, enabling a mechanistic interpretation of the dynamics. However, standard approaches for training lrRNNs can recover spurious structures irrelevant to the underlying dynamics. We first characterize the identifiability of connectivity structures in lrRNNs and determine conditions under which a unique solution exists. To find such solutions, we develop an inference framework based on maximum entropy and continuous normalizing flows (CNFs), trained via flow matching. Instead of estimating a single connectivity matrix, our method learns a distribution over connection weights that is maximally unbiased over unidentifiable components while matching the observed dynamics. This approach captures complex yet necessary distributions such as heavy-tailed connectivity found in empirical data. We validate our method on synthetic datasets with connectivity structures that generate multistable attractors, limit cycles, and ring attractors, and demonstrate its applicability in recordings from rat frontal cortex during decision-making. Our framework shifts circuit inference from recovering connectivity to identifying which connectivity structures are computationally required, and which are artifacts of underconstrained inference.
****
Error-Driven Graph Augmentation for Mesh-Based PDE Surrogates
Xuan Minh Vuong Nguyen, Nissrine Akkari, Fabien Casenave, Jonathan Viquerat, Elie Hachem
Graph Neural Networks (GNNs) on meshes have emerged as promising surrogates for computational mechanics, but standard local message passing struggles to propagate information across unstructured meshes, leading to large errors in regions with complex physics (e.g., shocks, wakes, boundary layers). Existing approaches enlarge connectivity with long-range edges chosen *a priori* via geometric heuristics or random sampling, which lack a mechanism to prioritize high-error regions and often introduce redundant communication. We propose **MiSe-GNN**, a dual-head architecture that adaptively augments graph connectivity using model-predicted *a posteriori* errors. MiSe-GNN jointly predicts physical fields and a node-wise error indicator; the predicted error is periodically converted into a hierarchy of additional edges via an adaptive tree that links high-error nodes to spati

al pivots at multiple scales. This error-guided connectivity concentrates messag e passing where the surrogate is uncertain while keeping the graph sparse elsewh ere, yielding a transparent and physically interpretable graph-space analogue of adaptive mesh refinement. Across industrial CFD and CSD benchmarks, MiSe-GNN co nsistently improves accuracy and accuracy–compute trade-offs over strong baselin es, and qualitative analyses show that it routes communication toward physically challenging regions. These results establish *error-guided edge augmentation* a s a robust and general design principle for long-range message passing in physic s-aware GNNs.

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Expert-guided Clinical Text Augmentation via Query-Based Model Collaboration

Dongkyu Cho, Miao Zhang, Gregory D Lyng, Rumi Chunara

Data augmentation is a widely used strategy to improve model robustness and gene ralization by enriching training datasets with synthetic examples. While large l anguage models (LLMs) have demonstrated strong generative capabilities for this purpose, their applications in high-stakes domains like healthcare present uniqu e challenges due to the risk of generating clinically incorrect or misleading in formation. In this work, we propose a novel query-based model collaboration fram ework that integrates expert-level domain knowledge to guide the augmentation pr ocess to preserve critical medical information. Compared to existing LLM-based a nd traditional augmentation methods, our generated data significantly improves p reservation of critical medical information and reduces hallucinations at both t he token and concept levels. Experiments on downstream clinical prediction tasks demonstrate consistent performance gains over existing augmentation methods. Th is lightweight collaborative framework addresses the gap between LLM augmentatio n potential and the safety requirements of specialized domains.

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4RC: 4D Reconstruction via Conditional Querying Anytime and Anywhere

Yihang Luo, Shangchen Zhou, Yushi Lan, Xingang Pan, Chen Change Loy

We present 4RC, a unified feed-forward framework for 4D reconstruction from mono cular videos. Unlike existing methods that typically decouple motion from geomet ry or produce limited 4D attributes, such as sparse trajectories or two-view sce ne flow, 4RC learns a holistic 4D representation that jointly captures dense sce ne geometry and motion dynamics. At its core, 4RC introduces a novel encode-once , query-anywhere and anytime paradigm: a transformer backbone encodes the entire video into a compact spatio-temporal latent space, from which a conditional dec oder can efficiently query 3D geometry and motion for any query frame at any tar get timestamp. To facilitate learning, we represent per-view 4D attributes in a minimally factorized form, decomposing them into base geometry and time-dependen t relative motion. Extensive experiments demonstrate that 4RC outperforms prior methods across a wide range of 4D reconstruction tasks.

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Learning What to Generate: A Reinforcement Learning-based Closed-Loop Augmentati on Framework for Person Re-identification

Xincheng Shi, Changxiao Ma, Yongfei Zhang, Yuzhuo Ma, Rongye Shi

Person re-identification (ReID) models are sensitive to long-tail nuisances (e.g ., rare viewpoints, occlusions, complex backgrounds), yet current generative aug mentation is largely open-loop: prompts/conditions are sampled heuristically wit hout verifying whether the synthesized samples improve ReID discriminability.

We introduce ReasonAug, a closed-loop framework that learns an image-conditioned instruction policy for a frozen generator, turning augmentation into a sequenti al decision problem over instruction tokens.

A Semantic Reasoning Agent (SRA) performs hierarchical planning from global sema ntics to identity-critical local cues, producing structured edit instructions wh ose utility is verified by downstream ReID feedback.

To make closed-loop optimization reliable, we propose Metric-Aligned Gated Rewar d (MAGR), which converts metric-learning objectives into a dense reward while ga ting task shaping by identity preservation to prevent reward hacking, and Struct ure-Aware Entropy (SAE), which allocates exploration per token to lock identity-critical cues while diversifying nuisance factors.

Experiments on Market-1501 and MSMT17 demonstrate state-of-the-art performance, confirming that closing the augmentation loop and learning what to generate yiel d more discriminative training data than open-loop alternatives.
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Support-Proximity Augmented Diffusion Estimation for Offline Black-Box Optimizat ion
Yonghan Yang, Ye Yuan, Zipeng Sun, Linfeng Du, Bowei He, Haolun Wu, Can Chen, Xu e Liu
Offline black-box optimization aims to discover novel designs with high property scores using only a static dataset, a task fundamentally challenged by the out- of-distribution (OOD) extrapolation problem. Existing approaches typically bifur cate into inverse methods, which struggle with the ill-posed nature of mapping s cores to designs, and forward methods, which often lack the distributional expre ssivity to quantify uncertainty effectively. In this work, we propose \textbf{SP ADE} (\textbf{S}upport-\textbf{P}roximity \textbf{A}ugmented \textbf{D}iffusion \textbf{E}stimation), a novel framework that reimagines forward surrogate modeli ng through the lens of conditional generative modeling. SPADE models the forward likelihood $p(y|\boldsymbol{x})$ using a diffusion model, but with two critical enhancements to tailor it for optimization: (1) a \emph{Calibrated Diffusion Es timation} module that enforces global consistency in statistical moments and pai rwise rankings, and (2) a \emph{Support-Proximity Regularization} mechanism that implicitly internalizes the data manifold constraint $p(\boldsymbol{x})$ via kN N-based density estimation. Theoretically, we prove that our regularization is f irst-order equivalent to maximizing a Bayesian posterior with a valid design pri or. Empirically, SPADE achieves state-of-the-art performance across Design-Bench tasks and an LLM data mixture optimization benchmark. Our code is available at https://github.com/HarryYoung2018/spade.
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A Unified Approach to Interpreting Knowledge Distillation for Large Language Mod els via Interactions
Qingzhuo Wang, Ruiyang Qin, Zhenxin Qin, Wen Shen, Zhihua Wei
Despite the success of knowledge distillation (KD) in Large Language Models (LLM s), the underlying mechanism behind its efficacy remains unclear. In this paper, we propose a unified approach to explore the common mechanism of various KD met hods using interactions. Specifically, we decompose the output score of the LLM into the sum of numerous interactions. Each interaction represents a nonlinear r elationship involving a set of input variables (e.g., words). Based on the decom posed interactions, we discover that the common mechanism underlying various KD methods is the sparsification of interactions, i.e., student models retain fewer interactions for inference while suppressing other interactions to zero effects . Furthermore, we discover that the performance variance across different KD met hods arises from their capabilities in handling complex interactions. A KD metho d typically yields better performance if it enables the student model to achieve higher sparsity of complex interactions. Motivated by these insights, we propos e a plug-and-play loss function called Complex Interaction Penalty (CIP) to expl icitly enforce the sparsity of complex interactions during the distillation proc ess. Extensive experiments demonstrate that integrating CIP consistently improve s the performance of diverse KD methods on both in-domain and out-of-distributio n benchmarks.
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FactGuard: Agentic Video Misinformation Detection via Reinforcement Learning
Zehao Li, Hongwei Yu, Hao Jiang, Qiang Sheng, Yilong Xu, Baolong Bi, Yang Li, Zh enlong Yuan, Yujun Cai, Zhaoqi Wang
Multimodal large language models (MLLMs) have substantially advanced video misin formation detection through unified multimodal reasoning, but they often rely on fixed-depth inference and place excessive trust in internally generated assumpt ions, particularly in scenarios where critical evidence is sparse, fragmented, o r requires external verification. To address these limitations, we propose FactG uard, an agentic framework for video misinformation detection that formulates ve rification as an iterative reasoning process built upon MLLMs. FactGuard explici

tly assesses task ambiguity and selectively invokes external tools to acquire cr
itical evidence, enabling progressive refinement of reasoning trajectories. To f
urther strengthen this capability, we introduce a two-stage training strategy th
at combines domain-specific agentic supervised fine-tuning with decision-aware r
einforcement learning to optimize tool usage and calibrate risk-sensitive decisi
on making. Extensive experiments on three public benchmarks demonstrate that Fac
tGuard consistently outperforms state-of-the-art methods in both verification ac
curacy and reliability.
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From Content to Knowledge: Lightning Fast Long-Video Understanding with Neural K
nowledge Representations

Yuchen Guan, Xiao Li, Zongyu Guo, Xiaoyi Zhang, Xiulian Peng, Chun Yuan, Yan Lu

We propose a new paradigm for long video understanding by treating a long video
as a Neural Knowledge Representation (NKR). NKR represents video contents neithe
r as a stream of tokens nor pre-organized databases, but as an individual small
portion of network weights attached to the VLM backbone. The NKR weights are opt
imized to encapsulate the video's semantic content via a novel Agentic Knowledge
 Distillation (AKD) process, where an agent automatically synthesizes dense desc
riptions and question-answer pairs to distill the video's knowledge into the NKR
. While AKD serves as a comprehensive, one-time encoding phase, the resulting NK
R transforms the video into a portable, reusable asset. At inference, the lightw
eight NKR is mounted onto a frozen Vision-Language Model (VLM), enabling direct,
 query-based understanding without reloading or re-encoding the original video.
This approach decouples video length from inference cost, offering high amortize
d efficiency for multi-turn video understanding. Experiments on the LVBench benc
hmark show our method achieves performance comparable to state-of-the-art approa
ches while reducing end-to-end latency by over two orders of magnitude, opening
new possibilities for interactive long-video understanding.
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PRISM: Learning Realistic Depth via Physics-Grounded Noise Disentanglement with
Semantic-Geometric Collaboration

Xiujian Liang, Jiacheng Liu, Mingyang Sun, Qichen He, Anda Cheng, Cewu Lu, Jianh
ua Sun

Real-world physical sensing exhibits complex, heterogeneous noise patterns that
deviate significantly from idealized simulation, posing a fundamental bottleneck
 for sim-to-real transfer. Existing sensor modelings typically treat depth noise
 as a monolithic black-box process, overlooking the distinct physical mechanisms
 that govern different error modalities. In this work, we introduce a physics-gr
ounded paradigm that disentangles monolithic noise into two complementary modali
ties: sensing invalidation and measurement inaccuracy, enabling a tailored treat
ment of noise sources based on their physical origins. Building on this insight,
 we propose PRISM, a tripartite framework that distills 3D Visual Foundation Mod
el features as rich spatial-semantic priors for physics-based reasoning. To addr
ess the inherent sparsity and class imbalance of invalidation regions, we develo
p Hierarchical Positive-Prioritized Supervision, integrating multi-scale positiv
e-weighted objectives with a positive-preserving dynamic hard mining strategy to
 enforce precise artifact delineation. Extensive benchmarks demonstrate that PRI
SM achieves state-of-the-art fidelity in noisy depth synthesis. Furthermore, dow
nstream robotic experiments show that PRISM facilitates a 92.1% average success
rate in the real world, marking a significant improvement over monolithic baseli
nes.
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NeUQI: Near-Optimal Uniform Quantization Parameter Initialization for Low-Bit LL
Ms

Li Lin, Xinyu Hu, Xiaojun Wan

Large language models (LLMs) achieve impressive performance across domains but f
ace significant challenges when deployed on consumer-grade GPUs or personal devi
ces such as laptops, due to high memory consumption and inference costs. Post-tr
aining quantization (PTQ) of LLMs offers a promising solution that reduces their
 memory footprint and decoding latency. In practice, PTQ with uniform quantizati

on representation is favored due to its efficiency and ease of deployment, as uniform quantization is widely supported by mainstream hardware and software libraries. Recent studies on low-bit uniform quantization have led to noticeable improvements in post-quantization model performance; however, they mainly focus on quantization methodologies, while the initialization of quantization parameters remains underexplored and still relies on the conventional *Min-Max formula*. In this work, we identify the limitations of the *Min-Max formula*, move beyond its constraints, and propose **NeUQI**, a method that efficiently determines near-optimal initialization for uniform quantization. Our NeUQI simplifies the joint optimization of the scale and zero-point by deriving the zero-point for a given scale, thereby reducing the problem to a scale-only optimization. Benefiting from the improved quantization parameters, our NeUQI consistently outperforms existing methods in the experiments with the LLaMA and Qwen families on various settings and tasks. Furthermore, when combined with a lightweight distillation strategy, NeUQI even achieves superior performance to PV-tuning, a considerably more resource-intensive method.
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Clipping Makes Distributed and Federated Asynchronous SGD Robust to Stragglers
Samuel Erickson, Mikael Johansson
In modern machine learning, parallelization of training is an important strategy for increasing scale. Asynchronous stochastic gradient descent (ASGD), which maximally utilizes available hardware, avoids having to wait for slow workers. However, with constant step sizes, the convergence of ASGD is nonetheless negatively effected by slow workers due to large delays in updates. At the same time, it has been empirically observed in asynchronous training of deep learning models that gradient clipping ``stabilizes'' training. In this work, we provide a theoretical justification for this behavior, as we show that clipping removes the dependence of the maximum delay in the oracle complexity. We employ a sub-Weibull model of gradient noise which generalize sub-Gaussian and sub-exponential disitributions to more heavy-tailed distributions, motivated by empirical observations in deep learning. We show convergence in expectation, and for the first time in asynchronous optimization, convergence with high probability.
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DocOS: Towards Proactive Document-Guided Actions in GUI Agents
Jingjing Liu, Ziye Huang, Zihao Cheng, Zeming Liu, Jiahong Wu, Yuhang Guo, Kehai Chen, Yunhong Wang, Haifeng Wang
While Graphical User Interface (GUI) agents have shown promising performance in automated device interaction, they primarily depend on static parametric knowledge from pre-training or instruction tuning. This reliance fundamentally limits their ability to handle long-tailed tasks that require explicit procedural knowledge absent from model parameters, often forcing agents to resort to inefficient and brittle trial-and-error exploration. To mitigate this limitation, we introduce Proactive Document-Guided Action for GUI agents in dynamic, open-web environments, a novel paradigm that mirrors human problem-solving by enabling agents to autonomously search for relevant documentation to resolve long-tailed tasks. To evaluate agents' capability in this paradigm, we propose DocOS, a benchmark designed to assess document-guided problem solving in fully interactive environments. DocOS requires agents to autonomously navigate a web browser, locate relevant online documentation, comprehend procedural instructions, and faithfully ground them into executable GUI actions. Extensive experiments reveal that progress is strictly constrained by dual bottlenecks: agents struggle to reliably locate relevant information during proactive search and frequently fail to faithfully ground retrieved instructions into precise actions, pointing toward document-guided interaction as a crucial pathway for enabling self-evolving GUI agents in dynamic environments.
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Does a Hybrid Space-Aware Randomized Defense Improve Empirical and Certified Adversarial Robustness?

Joy Dhar, Manish Kumar Pandey, Behzad Bozorgtabar, Nayyar Zaidi, Wenyu Zhang, Wei-Hong Li, Tingting Mu, Dwarikanath Mahapatra, Mahsa Baktashmotlagh, Trung Le, Chen Chen, Sajib Mistry, Camila Gonzalez, Samira Ebrahimi Kahou, Lina Yao, Piotr Koniusz, Robert Burns Fisher, Dinh Phung, Bohyung Han, Nuno Vasconcelos, Pietro Lio

We introduce Hybrid Space-aware Stochastic Convolution Attention Noise (HySCAN), a hybrid randomized defense that helps close the long-standing gap between provable robustness under ■2 certificates and empirical robustness against strong ■∞ attacks, while maintaining strong generalization across diverse imaging benchmarks. HySCAN jointly explores complementary sources of stochasticity at both training and inference: (i) implicit weight-space randomness via stochastic-aware Random Weights, and (ii) explicit feature-space randomness via Stochastic Attention Noise Injection modules. By incorporating randomness at both the parameter and representation levels, HySCAN enables meaningful certified guarantees while improving empirical robustness in practice. Comprehensive experiments on diverse imaging datasets, e.g., CelebA, CIFAR-10, and CIFAR-100, ImageNet-1k, HAM10000, and NIH Chest X-ray, demonstrate that HySCAN outperforms existing certified and empirical defenses, improving certified robustness by up to ≈ 9.6% and empirical robustness by up to ≈ 5% without reducing clean accuracy.

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SemanticNVS: Improving Semantic Scene Understanding in Generative Novel View Synthesis

Xinya Chen, Christopher Wewer, Jiahao Xie, Xinting Hu, Jan Eric Lenssen

We present SemanticNVS, a camera-conditioned multi-view diffusion model for novel view synthesis (NVS), which improves generation quality and consistency by integrating pre-trained semantic feature extractors. Existing NVS methods perform well for views near the input view, however, they tend to generate semantically implausible and distorted images under long-range camera motion, revealing severe degradation. We speculate that this degradation is due to current models failing to fully understand their conditioning or intermediate generated scene content. Here, we propose to integrate pre-trained semantic feature extractors to incorporate stronger scene semantics as conditioning to achieve high-quality generation even at distant viewpoints. We investigate two different strategies, (1) warped semantic features and (2) an alternating scheme of understanding and generation at each denoising step. Experimental results on multiple datasets demonstrate the clear qualitative and quantitative (4.69%-15.26% in FID) improvement over state-of-the-art alternatives.

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Over-Alignment vs Over-Fitting: The Role of Feature Learning Strength in Generalization

Taesun Yeom, Taehyeok Ha, Jaeho Lee

Feature learning strength (FLS), i.e., the inverse of the effective output scaling of a model, plays a critical role in shaping the optimization dynamics of neural nets. While its impact has been extensively studied under the asymptotic regimes---both in training time and FLS---existing theory offers limited insight into how FLS affects generalization in practical settings, such as when training is stopped upon reaching a target training risk. In this work, we investigate the impact of FLS on generalization in deep networks under such practical conditions. Through empirical studies, we first uncover the emergence of an *optimal FLS* ---neither too small nor too large---that yields substantial generalization gains. This finding runs counter to the prevailing intuition that stronger feature learning universally improves generalization. To explain this phenomenon, we develop a theoretical analysis of gradient flow dynamics in two-layer ReLU nets trained with logistic loss, where FLS is controlled via initialization scale. Our main theoretical result establishes the existence of an optimal FLS arising from a trade-off between two competing effects: An excessively large FLS induces an *over-alignment* phenomenon that degrades generalization, while an overly small FLS leads to *over-fitting*.

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XSkill: Continual Learning from Experience and Skills in Multimodal Agents

Guanyu Jiang, Zhaochen Su, Xiaoye Qu, Yi R. Fung
Multimodal agents can now tackle complex reasoning tasks with diverse tools, yet they still suffer from inefficient tool use and inflexible orchestration in open-ended settings. A central challenge is enabling such agents to continually improve without parameter updates by learning from past trajectories. We identify two complementary forms of reusable knowledge essential for this goal: experiences, providing concise action-level guidance for tool selection and decision making, and skills, providing structured task-level guidance for planning and tool use. To this end, we propose XSkill, a dual-stream framework for continual learning from experience and skills in multimodal agents. XSkill grounds both knowledge extraction and retrieval in visual observations. During accumulation, XSkill distills and consolidates experiences and skills from multi-path rollouts via visually grounded summarization and cross-rollout critique. During inference, it retrieves and adapts this knowledge to the current visual context and feeds usage history back into accumulation to form a continual learning loop. Evaluated on five benchmarks across diverse domains with four backbone models, XSkill consistently and substantially outperforms both tool-only and learning-based baselines. Further analysis reveals that the two knowledge streams play complementary roles in influencing the reasoning behaviors of agents and show superior zero-shot generalization. Code is available at https://github.com/XSkill-Agent/XSkill.
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InteractComp: Evaluating Search Agents With Ambiguous Queries
Mingyi Deng, Lijun Huang, Yani Fan, Fanqi Kong, Jiayi Zhang, Fashen Ren, Jinyi Bai, Fuzhen Yang, Dayi Miao, Zhaoyang Yu, Yifan Wu, Yanfei Zhang, Fengwei Teng, Yingjia Wan, Song Hu, Yude Li, Xin Jin, Conghao Hu, Haoyu Li, Qirui Fu, Tai Zhong, Xinyu Wang, Xiangru Tang, Nan Tang, Chenglin Wu, Yuyu Luo
Language agents have demonstrated remarkable potential in web search and information retrieval. However, many search-agent benchmarks assume that user queries are complete and unambiguous. This assumption leaves under-tested a practical failure mode: agents may face ambiguous requests where the intended target cannot be identified without clarification. Yet most agents lack interactive mechanisms during the search process, and existing benchmarks cannot assess this capability. To address this gap, we introduce InteractComp, a benchmark designed to evaluate whether search agents can recognize query ambiguity and actively interact to resolve it during search. Following the principle of easy to verify, interact to disambiguate, we construct 210 expert-curated questions across 9 domains through a target-distractor methodology that creates controlled ambiguity resolvable only through interaction. Evaluation of 17 models reveals striking failure: the best model achieves only 13.73% accuracy despite 71.50% with complete context, exposing systematic overconfidence rather than reasoning deficits. Forced interaction produces dramatic gains, demonstrating latent capability current strategies fail to engage. Longitudinal analysis shows interaction capabilities stagnated over 15 months while search performance improved seven-fold, revealing a critical blind spot. This stagnation, coupled with the immediate feedback inherent to search tasks, makes InteractComp a valuable resource for both evaluating and training interaction capabilities in search agents. The code is available at https://github.com/FoundationAgents/InteractComp
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Embedding-perturbed Exploration Preference Optimization for Flow Models
Sujie Hu, Chubin Chen, Jiashu Zhu, Jiahong Wu, Xiangxiang Chu, Xiu Li
Recent advancements have established Reinforcement Learning (RL) as a pivotal paradigm for aligning generative models with human intent. However, group-based optimization frameworks (e.g., GRPO) face a critical limitation: *the rapid decay of intra-group variance*. As the distinctiveness among samples within a group diminishes, the variance approaches zero. This eliminates the very learning signal required for optimization, rendering the process unstable and forcing the policy into *premature stagnation or reward hacking*. Existing strategies, such as varying the initial noise or increasing group sizes, often fail to address this fundamental issue, resulting in *training instability or diminishing returns*. To overcome these challenges, we propose **Embedding-perturbed Exploration Pref

erence Optimization (E^2PO)**, a novel framework that sustains optimization th
rough embedding-level perturbation. Our method introduces structured, embedding-
level perturbations within sample groups, guaranteeing a robust variance that pr
eserves the discriminative signal throughout the training process. Extensive exp
eriments demonstrate that our approach significantly outperforms state-of-the-ar
t baselines, achieving a more faithful alignment with human preference.
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SVL: Empowering Spiking Neural Networks for Efficient 3D Open-World Understandin
g

Xuerui Qiu, Shaowei Gu, Peixi Wu, JiaKui Hu, Yaozhi Wen, Yuqi Pan, Xinhao Luo, B
o XU, Guoqi Li

Spiking Neural Networks (SNNs) offer an energy--efficient route to 3D spatio--te
mporal perception, yet they lag behind Artificial Neural Networks (ANNs) due to
weak pretraining and heavy inference stacks, limiting generalization and multimo
dal reasoning (e.g., zero--shot 3D classification and open--world QA). We presen
t a universal \textbf{S}pike--based \textbf{V}ision--\textbf{L}anguage pretraini
ng framework (SVL) that equips SNNs with open--world 3D understanding while pres
erving end--to--end spike efficiency. SVL comprises two core components: (i) {Mu
lti--scale Triple Alignment} (MTA), a label--free triplet contrastive objective
aligning 3D, image, and text; and (ii) {Re--parameterizable Vision--Language Int
egration} (Rep--VLI), which converts offline text embeddings into lightweight we
ights for text--encoder--free inference. Moreover, we present the first fully sp
ike--driven point Transformer, {Spike-driven PointFormer}, whose 3D spike--drive
n self--attention (3D-SDSA) reduces interactions to sparse additions, enabling f
aster, more efficient training. Extensive experiments show that SVL attains stro
ng zero--shot 3D classification (85.4% top--1) and consistently outperforms prio
r SNNs on downstream tasks (e.g., +6.1% 3D cls, +2.1% DVS actions, +1.1% detecti
on, +2.1% segmentation) while enabling open--world 3D question answering, someti
mes outperforming ANNs. To the best of our knowledge, SVL represents the first s
calable, generalizable, and hardware-friendly paradigm for 3D open-world underst
anding, effectively bridging the gap between SNNs and ANNs in complex open-world
 understanding tasks.
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CELL: A Causal Perspective for Fairness-aware Graph Adaptation

Hourun Li, Yifan Wang, Qinghua Ran, Junyu Luo, Jia Yang, Changling Zhou, Zhiping
 Xiao, Wei Ju, Xiao Luo, Ming Zhang

This paper studies fairness-aware graph adaptation, aiming to transfer knowledge
 from a labeled source graph to an unlabeled target graph while addressing fairn
ess.

Most prior methods require target-domain attributes to learn invariant graph rep
resentations of sensitive attributes, which are often unavailable in practice.

To address this limitation, we introduce \underline{C}ausality-attended Repres\u
nderline{e}ntation Dientang\underline{l}ement with Structural A\underline{l}ignm
ent (CELL) for fairness-aware graph adaptation without requiring target sensitiv
e labels.

CELL constructs a causal graph to model the graph-generation mechanism and guide
 fair representation disentanglement.

Specifically, CELL uses sensitive and causal encoders to extract sensitive and c
ausal factors, respectively, and promotes disentanglement by minimizing their co
nditional mutual information.

To leverage unlabeled target data, we further generate pseudo-labels for both ta
rget task labels and sensitive attributes, and use similarity relations to deriv
e unbiased node representations.

Finally, to further mitigate domain shift, we build a fairness-aware bipartite g
raph that provides additional structural supervision for cross-domain alignment.

Experiments on benchmarks show that CELL consistently outperforms strong baselin
es in both predictive performance and fairness.
****

Transformers Provably Learn Algorithmic Solutions for Graph Connectivity, But On

ly with the Right Data
Qilin Ye, Deqing Fu, Robin Jia, Vatsal Sharan
Transformers often fail to learn generalizable algorithms, instead relying on br
ittle heuristics.
Using graph connectivity as a testbed, we explain this phenomenon both theoretic
ally and empirically.
We consider a simplified Transformer architecture, the Disentangled Transformer,
 and prove that an L-layer model can compute connectivity in graphs with diame
ters up to 3^L, implementing an algorithm equivalent to computing powers of th
e adjacency matrix.
By analyzing training dynamics, we prove that whether the model learns this stra
tegy hinges on whether most training instances are within this model capacity. W
ithin-capacity graphs (diameter $\leq 3^L$) drive the learning of the algorithmi
c solution while beyond-capacity graphs drive the learning of a simple heuristic
 based on node degrees.
Finally, we empirically show that our insights transfer to standard Transformers
:
restricting training data to stay within a model's capacity makes both standard
and Disentangled Transformers learn the exact algorithm.
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BAS: Bridging Adam and SignSGD for Memory-Efficient LLM Training
Yijie Zhou, Mingliang Zhang, Jiaqi Zhang, Xunliang Cai, Shi Pu
We propose **Block Adaptive Signum (BAS)**, which bridges Adam and SignSGD via b
lock-wise scaling of sign updates. By discarding element-wise second moments, BA
S reduces memory overhead relative to AdamW without sacrificing performance in o
ur tested settings. Crucially, BAS mimics Adam’s dynamics closely enough to dire
ctly **inherit its hyperparameters**, matching the performance of AdamW without
the need for re■tuning, a common fragility of prior low■memory optimizers. This
structural alignment makes it particularly suitable for tuning Adam-pretrained m
odels. Furthermore, we exploit the inherent robustness of sign-based updates to
store the first moment in FP8 without performance degradation. This shrinks the
optimizer■state footprint to **12.5\% of AdamW’s**. We theoretically prove conve
rgence under standard assumptions and introduce a communication-efficient varian
t enabled by the sign-based update. Across extensive evaluations, including pre■
training a 1.5B model on 100B tokens and supervised fine-tuning of models up to
32B parameters, we demonstrate that BAS achieves performance on par with AdamW.
****
LLM-Guided Diagnostic Evidence Alignment for Medical Vision–Language Pretraining
 under Limited Pairing
Huimin Yan, Liang Bai, Xian Yang, Long Chen
Most existing CLIP-style medical vision--language pretraining methods rely on gl
obal or local alignment with substantial paired data. However, global alignment
is easily dominated by non-diagnostic information, while local alignment fails t
o integrate key diagnostic evidence. As a result, learning reliable diagnostic r
epresentations becomes difficult, which limits their applicability in medical sc
enarios with limited paired data. To address this issue, we propose an LLM-Guide
d Diagnostic Evidence Alignment method (LGDEA), which shifts the pretraining obj
ective toward evidence-level alignment that is more consistent with the medical
diagnostic process. Specifically, we leverage LLMs to extract key diagnostic evi
dence from radiology reports and construct a shared diagnostic evidence space, e
nabling evidence-aware cross-modal alignment and allowing LGDEA to effectively e
xploit abundant unpaired medical images and reports, thereby substantially allev
iating the reliance on paired data. Extensive experimental results demonstrate t
hat our method achieves consistent and significant improvements on phrase ground
ing, image--text retrieval, and zero-shot classification, and even rivals pretra
ining methods that rely on substantial paired data.
****
Reward and Guidance through Rubrics: Promoting Exploration to Improve Multi-Doma
in Reasoning
Baolong Bi, Shenghua Liu, Yiwei Wang, Siqian Tong, Lingrui Mei, Yuyao Ge, Yilong

Xu, Jiafeng Guo, Xueqi Cheng
Recent advances in reinforcement learning (RL) have significantly improved the complex reasoning capabilities of large language models (LLMs).
Despite these successes, existing methods mainly focus on single-domain RL (e.g., mathematics) with verifiable rewards (RLVR), and their reliance on purely online RL frameworks restricts the exploration space, thereby limiting reasoning performance.
In this paper, we address these limitations by leveraging rubrics to provide both fine-grained reward signals and offline guidance.
We propose $\textbf{RGR-GRPO}$ (Reward and Guidance through Rubrics), a rubric-driven RL framework for multi-domain reasoning.
RGR-GRPO enables LLMs to receive dense and informative rewards while exploring a larger solution space during GRPO training.
Extensive experiments across 14 benchmarks spanning multiple domains demonstrate that RGR-GRPO consistently outperforms RL methods that rely solely on alternative reward schemes or offline guidance.
Compared with verifiable online RL baseline, RGR-GRPO achieves average improvements of +7.0%, +5.4%, +8.4%, and +6.6% on mathematics, physics, chemistry, and general reasoning tasks, respectively.
Notably, RGR-GRPO maintains stable entropy fluctuations during off-policy training and achieves superior pass@k performance, reflecting sustained exploration and effective breakthrough beyond existing performance bottlenecks.
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Unifying Heterogeneous Multi-Modal Remote Sensing Detection Via Language-Pivoted Pretraining
Yuxuan Li, Yuming Chen, Yunheng Li, Ming-Ming Cheng, Xiang Li, jian Yang
Heterogeneous multi-modal remote sensing object detection aims to accurately detect objects from diverse sensors (e.g., RGB, SAR, Infrared). Existing approaches largely adopt a late alignment paradigm, in which modality alignment and task-specific optimization are entangled during downstream fine-tuning. This tight coupling complicates optimization and often results in unstable training and suboptimal generalization. To address these limitations, we propose BabelRS, a unified language-pivoted pretraining framework that explicitly decouples modality alignment from downstream task learning. BabelRS comprises two key components: Concept-Shared Instruction Aligning (CSIA) and Layerwise Visual-Semantic Annealing (LVSA). CSIA aligns each sensor modality to a shared set of linguistic concepts, using language as a semantic pivot to bridge heterogeneous visual representations. To further mitigate the granularity mismatch between high-level language representations and dense detection objectives, LVSA progressively aggregates multi-scale visual features to provide fine-grained semantic guidance. Extensive experiments demonstrate that BabelRS stabilizes training and consistently outperforms state-of-the-art methods without bells and whistles. Code: \href{https://github.com/zcablii/SM3Det}{github.com/zcablii/SM3Det}.
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Beyond ReLU: Bifurcation, Oversmoothing, and Topological Priors
Erkan Turan, Gaspard Abel, Maysam Behmanesh, Emery Pierson, Maks Ovsjanikov
Graph Neural Networks (GNNs) learn node representations through iterative network-based message-passing. While powerful, deep GNNs suffer from oversmoothing, where node features converge to a homogeneous, non-informative state. We re-frame this problem of representational collapse from a \emph{bifurcation theory} perspective, characterizing oversmoothing as convergence to a stable ``homogeneous fixed point.'' Our central contribution is the theoretical discovery that this undesired stability can be broken by replacing standard monotone activations (e.g., ReLU) with a class of functions. Using Lyapunov-Schmidt reduction, we analytically prove that this substitution induces a bifurcation that destabilizes the homogeneous state and creates a new pair of stable, non-homogeneous \emph{patterns} that provably resist oversmoothing. Our theory predicts a precise, nontrivial scaling law for the amplitude of these emergent patterns, which we quantitatively validate in experiments. Finally, we demonstrate the practical utility of our theory by deriving a closed-form, bifurcation-aware initialization and showing it

s utility in real benchmark experiments.
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GPan-LoRA: Gaussian Process Amortized Networks for Bayesian Low-Rank Adaptation in Large Language Models
Weifeng Zhang, Wenyuan Zhao, Amir Hossein Rahmati, Yucheng Wang, Zhiyuan Wang, Chao Tian, Xiaoning Qian
Principled uncertainty quantification (UQ) is increasingly recognized as essential for trustworthy artificial general intelligence (AGI). Bayesian Low-Rank Adaptation (LoRA) provides a principled mechanism for uncertainty-aware fine-tuning of large language models (LLMs). However, existing techniques either face scalability constraints, e.g. Laplace-LoRA, or rely on approximate inference schemes that lead to poorly calibrated posterior uncertainty, often manifesting as overconfident predictions under distribution shift. To address this challenge, we propose GPan-LoRA, the first scalable Gaussian Process (GP)-based framework for Bayesian LoRA, which integrates neural network-based sparse GP approximations with amortized variational inference. By preserving the Bayesian function prior and posterior semantics intrinsic to GPs, GPan-LoRA achieves a faithful balance between computational scalability and principled UQ. Empirically, GPan-LoRA produces well-calibrated uncertainty that remains reliable under distribution shift, mitigating overconfident failures while preserving competitive task performance.
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See the Emotion: A Facial Emoji Proxy Modeling for EEG Emotion Recognition
Jingjing Hu, Dan Guo, Haofan Cheng, Zeng ying, Zhan Si, Jinxing Zhou, Meng Wang
Despite the high accuracy of EEG-based emotion recognition, existing models remain opaque "black boxes", lacking semantic grounding between abstract neural features and human-interpretable states. In this paper, we reframe EEG explainability as a cross-modal generation task, shifting the paradigm from feature attribution to behavioral visualization. We introduce Facial Emoji Proxy Modeling, a novel framework that translates high-dimensional EEG signals into identity-anonymized facial emojis. Guided by the neuroscientific inspiration of neural-facial association, this approach grounds neural representations in the manifold of observable facial dynamics. Technically, our framework integrates FMENet, a specialized backbone modeling expression-relevant spatial synergies, and the Facial Emoji Learning Branch (FELB), which treats emoji reconstruction as a structured semantic regularizer. Extensive experiments on EAV and MMER benchmarks demonstrate that our method achieves state-of-the-art accuracy among EEG-only models. Crucially, it generates semantically faithful facial animations that provide a transparent, privacy-preserving window into the brain's emotional evolution, effectively allowing users to ``see the emotion'' directly from neural signals. Code is available at https://github.com/xian-sh/SeeEmotion
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PESD-TSF: A Period-Aware and Explicit Structured Decomposition Framework for Long-Term Time Series Forecasting
Hua Wang, Xianhao Jiao, Fan Zhang
Deep forecasting models often suffer from attenuated periodic perception and entangled trend–noise representations as network depth increases. Moreover, the widely adopted channel-independent paradigm, while improving training stability, disrupts intrinsic dynamic coordination among variables, hindering the modeling of cross-variable consistency in multivariate time series. To address these issues, we propose PESD-TSF, a physics-inspired structured decomposition framework for long-term time series forecasting that jointly emphasizes interpretability and predictive accuracy. PESD-TSF introduces three key designs. First, a Multiplicative Periodic Gating mechanism incorporates continuous-time priors to dynamically modulate signal amplitudes, preserving periodic structures across deep layers. Second, a multi-scale structured encoder integrates detrended attention with hierarchical sampling to explicitly decouple long-term trends from high-frequency variations while retaining fine-grained temporal semantics. Third, to recover disrupted inter-variable dependencies, we propose Cross-Scale Collaborative Attention (CSCA) together with an RLC regularization scheme, which reconstructs global inter-variable topology in deep feature spaces and enforces physically consisten

t collaboration through orthogonality and consistency constraints. Extensive exp eriments on benchmark datasets from multiple domains demonstrate that PESD-TSF c onsistently achieves state-of-the-art performance, with particularly strong gain s on multivariate forecasting tasks involving complex inter-variable coupling, h ighlighting its superior structural modeling capability and generalization.
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How to Price Data: A Market Equilibrium Based Approach
Pooja Ravi Kulkarni, Parnian Shahkar, Ruta Mehta
High-quality data is a key input to modern machine learning models, leading to t he emergence of platforms that facilitate the buying and selling of data. A cent ral challenge in these platforms is how the data is priced to balance the intere sts of both buyers and sellers. Traditional market equilibrium notions, where de mand meets supply are commonly used to price goods but do not extend naturally t o data due to its non-rivalrous nature, whereby multiple buyers can simultaneous ly benefit from the same dataset. We therefore introduce a new notion of equilib rium for data pricing based on Nash equilibrium and study it in settings where d ata may be complementary or substitutable, focusing on the canonical utility mod els for each, namely Leontief and linear, respectively. We show that equilibrium prices fail to exist for linear utilities even with homogeneous buyers and two sellers, while establishing strong existence, efficiency, and polynomial-time co mputation guarantees for Leontief utilities in general markets with n homogene ous buyers and m sellers. We further examine the role of platform mediation an d price discrimination in enabling *optimal* equilibrium outcomes efficiently. O n the technical front, we develop a novel proof technique based on systematicall y reducing the space of candidate equilibria through the *graph-of-deviations*, which may be of independent interest.
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Graph-GRPO: Training Graph Flow Models with Reinforcement Learning
Baoheng Zhu, Deyu Bo, Delvin Ce Zhang, Xiao Wang
Graph generation is a fundamental task with broad applications, such as drug dis covery.
Recently, discrete flow matching-based graph generation, a.k.a., graph flow mode l (GFM), has emerged due to its superior performance and flexible sampling.
However, effectively aligning GFMs with complex human preferences or task-specif ic objectives remains a significant challenge.
In this paper, we propose Graph-GRPO, an online reinforcement learning (RL) fram ework for training GFMs under verifiable rewards.
Our method makes two key contributions:
(1) We derive an analytical expression for the transition probability of GFMs, r eplacing the Monte Carlo sampling and enabling fully differentiable rollouts for RL training;
(2) We propose a refinement strategy that randomly perturbs specific nodes and e dges in a graph, and regenerates them, allowing for localized exploration and se lf-improvement of generation quality.
Extensive experiments on both synthetic and real datasets demonstrate the effect iveness of Graph-GRPO.
With only 50 denoising steps, our method achieves 95.0\% and 97.5\% Valid-Unique -Novelty scores on the planar and tree datasets, respectively.
Moreover, Graph-GRPO achieves state-of-the-art performance on the molecular opti mization tasks, outperforming graph-based and fragment-based RL methods as well as classic genetic algorithms.
Code is available in https://github.com/Zhubaoheng/Graph-GRPO.
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Towards Steering without Sacrifice: Principled Training of Steering Vectors for Prompt-only Interventions
Yuntai Bao, Qinfeng Li, Xinyan Yu, Ge Su, Wenqi Zhang, Liu Yan, Haiqin Weng, Jia nwei Yin, Xuhong Zhang
Recently, *steering vectors (SVs)* have emerged as an effective and lightweight approach to steer behaviors of large language models (LLMs), among which fine-tu ned SVs are more effective than optimization-free ones.

However, current approaches to fine-tuned SVs suffer from two limitations.
First, they require careful selection of steering factors on a per-SV basis to b alance steering effectiveness and generation quality at inference time.
Second, they operate as *full-sequence SVs (FSSVs)*, which can sacrifice generat ion quality regardless of factor selection due to excessive intervention on the model generation process.
To address the first limitation, we propose *joint training* of steering factors and directions, such that post-hoc factor selection is no longer required.
Using neural network scaling theory, we find that moderately large initializatio n sizes and learning rates for steering factors are essential for stability and efficiency of joint training.
To tackle the second limitation, we draw inspiration from *representation fine-t uning* and introduce **Prompt-Only Steering Vector (PrOSV)**, an SV that interve nes only on a few prompt tokens.
Our empirical results show that PrOSV outperforms traditional FSSVs on AxBench w hen using our joint training scheme.
We also find that PrOSV achieves a better tradeoff between general model utility and adversarial robustness than FSSV.
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CHESS: Chebyshev Spectral Synthesis for Trajectory Condensation
Ruituo Wu, Hongyu Zhang, Qiang Wang, Jiawei Du, Wei Cui, Ce Zhu, Bing Li
Learning from continuous-time trajectories requires modeling multivariate sensor measurements generated by underlying physical or dynamical processes. Under ext reme data compression and heterogeneous sampling, directly optimizing synthetic signals as discrete sample values becomes fundamentally misaligned with the unde rlining \emph{continuous-time physical processes}, often producing high-frequenc y, non-physical artifacts that overfit specific models and break reuse across ar chitectures and sampling rates. We propose CHESS, a \emph{function-first} synthe sis framework shifts optimization from discrete samples to underlying continuous -time signal trajectories. CHESS injects physics-induced structure by jointly en forcing low-rank spatial coherence and piecewise Chebyshev polynomial temporal p arameterization, constraining synthesis to a physically meaningful function mani fold. We provide theoretical analysis establishing explicit smoothness and stabi lity guaranties. Experiments on diverse sensor testbeds under the dataset distil lation protocol demonstrate CHESS consistently outperform state-of-the-art metho ds with a compression ratios up to $133\times$ for each synthetic sample. Furthe rmore, CHESS exhibits strong cross-architecture generalization and enables zero-shot adaptation across different sampling resolutions.
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AVTrack: Audio-Visual Tracking in Human-centric Complex Scenes
Yaoting Wang, Yun Zhou, Zipei Zhang, Henghui Ding
Audio-visual speaker tracking aims to localize and track active speakers by leve raging auditory and visual cues, enabling fine-grained, human-centric scene unde rstanding.
This capability is essential for real-world applications such as intelligent vid eo editing, surveillance, and human–computer interaction.
However, existing datasets are largely limited to simple or homogeneous audio-vi sual scenes with
coarse annotations.
Such oversimplified settings bias evaluation toward static audio–visual co-occur rence, rather than rigorously assessing robust spatiotemporal modeling and cross -modal reasoning in complex, dynamic scenes.
To address these limitations, we introduce **AVTrack**, a human-centric audio-vi sual instance segmentation (AVIS) dataset designed for dynamic real-world scenar ios.
AVTrack features *diverse and challenging conditions*, including camera motion, visual occlusions, and position changes.
Evaluations of representative AVIS methods on AVTrack reveal substantial perform ance degradation, establishing AVTrack as a challenging benchmark for robust hum an-centric audio-visual scene understanding in complex environments.

We further provide a simple yet effective baseline to facilitate future research. Project website: https://FudanCVL.github.io/AVTrack/
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General Quantification of Covariate and Concept Shifts
Hongbo Chen, Li C Xia
Generalization under distribution shift remains a core challenge in modern machine learning, yet existing learning bound theory is limited to narrow, idealized settings and is non-estimable from samples. In this paper, we bridge the gap between theory and practical applications. We first show that existing definition of concept shift breaks when the source and target supports mismatch. Leveraging entropic optimal transport, we propose a key notion: $\gamma^\ast$-concept shifts, and derive a general error bound unifying covariate and $\gamma^\ast$-concept shifts, which applies to broad loss functions, label spaces, and stochastic labeling. We further develop estimators for these shifts with concentration guarantees, and the DataShifts algorithm, which can quantify distribution shifts and estimate the error bound in most applications - a rigorous and general tool for analyzing learning error under distribution shift.
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EEmo-Logic: A Unified Dataset and Multi-Stage Framework for Comprehensive Image-Evoked Emotion Assessment
Lancheng Gao, Ziheng Jia, Zixuan Xing, Wei Sun, Huiyu Duan, Guangtao Zhai, Xiongkuo Min
Understanding the multi-dimensional attributes and intensity nuances of image-evoked emotions is pivotal for advancing machine empathy and empowering diverse human-computer interaction applications. However, existing models are still limited to coarse-grained emotion perception or deficient reasoning capabilities. To bridge this gap, we introduce **EEmoDB**, the largest image-evoked emotion understanding dataset to date. It features 5 analysis dimensions spanning 5 distinct task categories, facilitating comprehensive interpretation. Specifically, we compile $1.2M$ question-answering (QA) pairs (EEmoDB-QA) from $125K$ images via automated generation, alongside a $36K$ dataset (EEmoDB-Assess) curated from $25K$ images for fine-grained assessment. Furthermore, we propose **EEmo-Logic**, an **all-in-one** multimodal large language model (MLLM) developed via instruction fine-tuning and task-customized group relative preference optimization (GRPO) with novel reward design. Extensive experiments demonstrate that EEmo-Logic achieves robust performance in in-domain and cross-domain datasets, excelling in emotion QA and fine-grained assessment.
The dataset and code are available at https://github.com/workerred/EEmo-Logic.
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Dive into the Scene: Breaking the Perceptual Bottleneck in Vision-Language Decision Making via Focus Plan Generation
Boyuan Xiao, Bohong Chen, Yumeng Li, Ji Feng, Yao-Xiang Ding, Kun Zhou
In embodied vision-language decision making tasks such as robotic manipulation and navigation, Vision-Language and Vision-Language-Action Models (VLMs \& VLAs) are powerful tools with different benefits: VLMs are better at long-term planning, while VLAs are better at reactive control. However, their performance is limited by the same perceptual bottleneck: visual hallucinations arise due to the models' inability to distinguish task-relevant objects from distractors. In principle, accurate identification and focus on critical objects while filtering out irrelevant ones is the key to break this limitation. A straightforward solution is one-step focus: directly attending to essential objects. However, this approach proves ineffective because effective focus inherently requires deep scene understanding. To this end, we propose ${\it SceneDiver}$, a coarse-to-fine focus plan generation method for VLMs leveraging their long-term planning abilities, that first constructs a holistic scene graph to establish initial comprehension, then progressively decomposes the task into simpler sub-problems through an iterative cycle of recognition, understanding, and analysis. To enable reactive control, we also design a lightweight adapter for distilling the deliberate focus ability into VLAs. Evaluations on standard embodied AI benchmarks confirm that our method substantially reduces visual hallucinations for both VLMs and VLAs, while

preserving computational efficiency in tasks requiring fast execution. Our code and data are released at: https://future-item.github.io/SceneDiver.
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Structure Abstraction and Generalization in a Hippocampal-Entorhinal Inspired World Model
Tianqiu Zhang, Muyang Lyu, Xiao Liu, Si Wu
Humans abstract experiences into structured representations to facilitate pattern inference and knowledge transfer. While the hippocampal-entorhinal (HPC-MEC) circuit is known to represent both spatial and conceptual spaces, the mechanisms for concurrently extracting abstract structures from continuous, high-dimensional dynamics remain poorly understood. We propose a brain-inspired hierarchical model that simultaneously infers latent transitions and constructs a predictive visual world model. Our architecture employs an inverse model for structural extraction alongside an HPC-MEC coupling model that dissociates relational structures (MEC) from integrated episodic scenes (HPC). Using rotation dynamics as a benchmark, we demonstrate the model's capacity for structural abstraction. By leveraging velocity-driven path integration, the framework enables robust prediction and structural reuse across diverse contexts, thereby achieving structural generalization. This work provides a novel computational framework for understanding how brain-inspired, self-supervised learning of world models facilitates the acquisition of reusable abstract knowledge.
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Bridging Dynamics and Data: A Unified Diffusion Framework for Mechanistically-Informed Epidemic Forecasting
Guanghui Min, Tianhao Huang, Ke Wan, Qi R. Wang, Chen Chen
Reliable epidemic forecasting is critical for public health decision-making yet remains challenging due to data sparsity and the non-stationary nature of disease dynamics. While recent hybrid models attempt to integrate mechanistic principles with data-driven approaches, they often relegate mechanistic priors to merely auxiliary features or regularization terms. This design not only obscures the interpretability of the mechanistic contribution but also fails to inherit the capability of physical models to generalize under non-stationary dynamics, as the core architecture remains predominantly data-driven. To address these limitations, we propose EpiDiff, a unified framework that synergizes epidemiological domain knowledge with the generative power of diffusion models. Unlike methods that rigidly fuse features, EpiDiff employs a novel uncertainty-aware steering mechanism during inference. Specifically, we quantify the posterior uncertainty of mechanistic estimations and use it to dynamically modulate the diffusion process. Extensive experiments on real-world datasets demonstrate that EpiDiff consistently outperforms state-of-the-art baselines in accuracy and robustness, particularly under non-stationary distributions, while offering transparent insights into model reliance by explicitly visualizing when the forecast is governed by mechanistic laws versus data-driven patterns.
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UGround: Towards Unified Visual Grounding with Unrolled Transformers
Rui Qian, Xin Yin, Chuanhang Deng, Zhiyuan Peng, Jian Xiong, Wei Zhai, Dejing Dou
We present UGround, a **U**nified visual **Ground**ing paradigm that dynamically selects intermediate layers across **U**nrolled transformers as "mask as prompt,'' diverging from the prevailing pipeline that leverages the fixed last hidden layer as "$\langle\texttt{SEG}\rangle$ as prompt.'' UGround addresses two primary challenges posed by the prevailing paradigm: (1) its reliance on the fixed last hidden layer, which sequentially amplifies cumulative errors arising from layer-by-layer propagation without intermediate correction, and (2) its use of $\langle\texttt{SEG}\rangle$ as a prompt, which implicitly projects textual embeddings into visual space without explicit spatial cues (e.g., coordinates). Central to UGround is Policy-Prompted Masking, which comprises two key components: Stochastic Skip Connection (SSC) and Mask as Prompt (MasP). SSC is a reinforcement learning policy that, via stochastic sampling, allows each $\langle\texttt{SEG}\rangle$ token to slide across unrolled transformer layers, enabling dynamic layer s

election at which it connects to the vision model (e.g., SAM) in a skip-connecti on fashion. Given the selected hidden layer, MasP uses the similarity map derive d from the $\langle\texttt{SEG}\rangle$ token and image tokens as a soft logit m ask to prompt SAM for mask generation, offering explicit spatial cues through it s activation regions. To validate the effectiveness of UGround, we, for the firs t time, have unified visual grounding within a single framework from an attribut e perspective, spanning from traditional refer expression segmentation to newly proposed reasoning segmentation, single-target to multi-target, positive query t o false premise (empty target). All codes are provided in the supplementary mate rial.
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UHR-BAT: Budget-Aware Token Compression Vision-Language model for Ultra-High-Res olution Remote Sensing
Yunkai Dang, Minxin Dai, Yuekun Yang, Zhangnan Li, Wenbin Li, Feng Miao, Yang Ga o
Ultra-high-resolution (UHR) remote sensing imagery couples kilometer-scale conte xt with query-critical evidence that may occupy only a few pixels.
Such vast spatial scale leads to a quadratic explosion of visual tokens and hind ers the extraction of information from small objects.
Previous works utilize direct downsampling, dense tiling, or global top■K prunin g, which either compromise query-critical image details or incur unpredictable c ompute.
In this paper, we propose UHR-BAT, a query-guided and region-faithful token comp ression framework to efficiently select visual tokens under strict context budge t.
Specifically, we leverage text-guided, multi-scale importance estimation for vis ual tokens, effectively tackling the challenge of achieving precise yet low-cost feature extraction.
Furthermore, by introducing region-wise preserve and merge strategies, we mitiga te visual token redundancy, further driving down the computational budget.
The experimental results show that UHR-BAT achieves state-of-the-art performance across various benchmarks.
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Efficient Synthetic Network Generation via Latent Embedding Reconstruction
Feifan Jiang, Yinan Bu, Shihao Wu, Gongjun Xu, Ji Zhu
Network data are ubiquitous across the social sciences, biology, and information systems. Generating realistic synthetic network data has broad applications fro m network simulation to scientific discovery. However, many existing black-box a pproaches for network generation tend to overfit observed data while overlooking characteristic network structure, and incur substantial computational overhead at scale. These practical challenges call for synthetic network generation metho ds that are both efficient and capable of capturing structural properties of net works. In this paper, we introduce Synthetic Network Generation via Latent Embed ding Reconstruction (SyNGLER), a general and efficient framework for synthetic n etwork generation that builds on latent space network models. Given an observed network, SyNGLER first learns low-dimensional latent node embeddings via a laten t space network model and then reconstructs the latent space by building a distr ibution-free generator over these embeddings. For generation, SyNGLER first samp les (or resamples) node embeddings from the generator in the latent space and th en produces synthetic networks using the latent space network model. Through the latent space framework, SyNGLER preserves unique characteristics in networks su ch as sparsity and node degree heterogeneity, while allowing for efficient train ing with lower computational cost than many existing deep architectures.
We provide theoretical guarantees by developing consistency results on the dista nce between the true and synthetic edge distributions. Empirical studies further demonstrate the effectiveness of SyNGLER, which efficiently produces networks t hat better preserve key network characteristics such as network moments and degr ee distributions compared with existing approaches. Code is available at https:/ /github.com/FeifanJiang/syngler.
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Respecting Modality Gap in Post-hoc Out-of-distribution Detection with Pre-trained Vision-Language Models
Yuanwei Hu, Bo Peng, Yadan Luo, Zhen Fang, Ling Chen, Jie Lu
Out-of-distribution (OOD) detection has emerged as a popular technique to enhance the reliability of machine learning models by identifying unexpected inputs from unknown classes.
Recent progress in pre-trained vision–language models (VLMs) has enabled zero-shot OOD detection without access to in-distribution (ID) training data; in this setting, existing methods commonly treat text embeddings of class names as class prototypes.
In this paper, we challenge this widely adopted “text-as-prototype” paradigm by theoretically showing that off-the-shelf textual prototypes are generally misaligned with the optimal visual prototypes, yielding an intrinsic \textit{modality gap} that cannot be eliminated by prompt engineering alone.
To mitigate this gap under the post-hoc constraint, this paper presents an online pseudo-supervised framework that directly learns class prototypes in the visual feature space using unlabeled test-time data streams and soft predictions from the pre-trained VLMs. We provide theoretical guarantees for the convergence of the online optimization procedure.
Extensive experiments empirically manifest that our method achieves a new state of the art across a variety of OOD detection setups.
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Unifying Masked Diffusion Models with Various Generation Orders and Beyond
Chunsan Hong, Sanghyun Lee, Jong Chul Ye
Masked diffusion models (MDMs) are a potential alternative to autoregressive models (ARMs) for language generation, but generation quality depends critically on the generation order. Prior work either hard-codes an ordering (e.g., blockwise left-to-right) or learns an ordering policy for a pretrained MDM, which incurs extra cost and can yield suboptimal solutions due to the two-stage optimization. Motivated by this, we propose order-expressive masked diffusion model (OeMDM) for a broad class of diffusion generative processes with various generation orders, enabling the interpretation of MDM, ARM, and block diffusion in a single framework. Furthermore, building on OeMDM, we introduce learnable-order masked diffusion model (LoMDM), which jointly learns the generation ordering and diffusion backbone through a single objective from scratch, enabling the diffusion model to generate text in context-dependent ordering. Empirically, we confirm that LoMDM outperforms various discrete diffusion models across multiple language modeling benchmarks.
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Beyond Unidirectional Bias: Reciprocal Perspective Calibration in Scene Graph Generation
Haifeng Zhao, Wenbo Zhao, Xuemei Luo, Lei-Lei Ma, Dengdi Sun
Scene Graph Generation (SGG) paradigms predominantly model relationships as static, unidirectional mappings ($s \to o$), effectively treating objects as passive recipients of actions. This formulation suffers from an inherent \textit{unidirectional bias}, violating the physical reality that visual interactions are intrinsically reciprocal. Consequently, existing models often fail to maintain logical self-consistency when the reasoning anchor shifts from the subject to the object. To rectify this cognitive deficiency, we establish the Mutual-Perspective Inverse Relations (MPIR) principle, positing that a robust visual representation must satisfy logical consistency across dual perspectives. Guided by this principle, we propose the \textbf{Reciprocal Perspective Calibration (RPC)} framework, a model-agnostic framework that operationalizes MPIR via a novel Adaptive Inverse-Relation Augmentation (AIRA) strategy. Furthermore, we introduce Hypernym-Guided Prompts (HGP) to bridge the gap between semantic context and computational efficiency in vision-language models, enabling precise modeling of inverse relations. Extensive experiments demonstrate that RPC not only achieves competitive performance on standard benchmarks but also significantly enhances the model's capability to understand inverse relations, as verified by a new inverse consistency evaluation protocol, demonstrating the cognitive robustness of our method.

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From Retrieval to Translation: Translating Query into Graph-level Clues for Retrieval-Augmented Generation
Qichuan Liu, Qinggang Zhang, Yuxuan Hu, Chenfeng Zheng, Zerui Chen, Chentao Zhang, Zhihong Zhang
Retrieval-Augmented Generation (RAG) has recently been enhanced with tree or graph structures to match user intent for precise passage retrieval, which facilitates large language models (LLMs) in effectively mitigating hallucinations by leveraging external knowledge. However, we identify that existing structure-augmented RAG systems are experiencing (i) potential retrieval suspension and (ii) cumulative semantic drift, due to low-quality structures and semantic embeddings that often poorly capture textual details. Motivated by this, we propose a novel paradigm named KG-Translator, which is distinct from traditional matching-based paradigms and instead translates user queries into graph-level clues. Specifically, KG-Translator utilizes lightweight models to conduct named entity recognition (NER) and syntactic parsing on the corpus, constructing a reliable knowledge graph (ParseKG). On top of ParseKG, KG-Translator adopts constrained decoding strategies to faithfully translate clues, traces them to original passages, and employs a lightweight ranking model for precise passage retrieval. Extensive experiments on five datasets demonstrate that KG-Translator significantly outperforms baselines.
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Context-level Language Modeling by Learning Predictive Context Embeddings
beiya dai, Yuliang Liu, Yunchong Song, Daozheng Xue, Qipeng Guo, Kai Chen, Xinbing Wang, Bowen Zhou, Zhouhan Lin
We propose ContextLM, a framework that implicitly learns multi-token prediction by augmenting standard pretraining with an intrinsic next-context prediction objective. ContextLM builds a language model on top of context embeddings that span multiple tokens, enabling better next-token prediction by predicting the next context. Our model is fully compatible with standard autoregressive, token-by-token evaluation paradigms (e.g., perplexity). Extensive experiments with GPT-2 and Pythia backbones (up to 1.5B parameters and 300B training tokens) reveal that ContextLM shifts the Pareto frontier of scaling laws, exhibiting superior efficiency in parameters, training tokens, and FLOPs. Our results show that ContextLM could already achieve the baseline perplexity using 39\% fewer parameters and demonstrates robust generalization improvements on extensive downstream tasks under equivalent parameter counts.
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Compositional Generalization Requires Linear, Orthogonal Representations in Vision Embedding Models
Arnas Uselis, Andrea Dittadi, Seong Joon Oh
Compositional generalization, the ability to recognize familiar parts in novel contexts, is a defining property of intelligent systems, yet modern models, despite massive training sets, see only a tiny fraction of the combinatorial input space. We ask what structure representations {must} have to support generalization to unseen combinations. We formalize three desiderata (divisibility, transferability, stability) and show they impose necessary geometric constraints under standard training: representations must decompose linearly into per-concept components, orthogonal across concepts. This grounds the Linear Representation Hypothesis as a necessary consequence of compositional generalization, and yields dimension bounds linking the number of composable concepts to embedding geometry. Empirically, across CLIP, SigLIP, and DINO, we find partial linear factorization with low-rank near-orthogonal per-concept factors, and the degree of this structure correlates with compositional generalization on unseen combinations. As models continue to scale, these conditions predict the geometry they may converge to. Code: https://github.com/oshapio/necessary-compositionality
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Not All Answers Are Contextually Persuadable: Inference Dynamics in Large Language Models under Contextual Influence
Zongye Hu, Weiqing Luo, Yanjie Fu, Yu Gan, Haofeng Zhang, Ziyi Huang

At the core of modern prompting techniques is contextual sensitivity, the abilit y of large language models to adapt their predictions based on inference-time co ntext. Despite its central role, inference behavior under strong contextual infl uence remains poorly understood, particularly at the level of internal inference dynamics. To bridge this gap, we introduce a theoretical framework for analyzin g contextual influence through inference dynamics, enabling quantitative charact erization of inference behavior beyond output-level answer changes. Our analysis shows that inference dynamics do not exhibit unbounded drift under repeated con textual assertions. Instead, predictive representations converge to stable, quer y-dependent regimes that fundamentally constrain whether contextual signals can alter a model’s prediction. This leads to a surprising finding: Repeated context ual assertions do not act as accumulating evidence during inference and may ther efore fail to alter a model’s prediction even under unbounded repetition, while in other cases a prediction change becomes inevitable. We empirically validate o ur theoretical predictions across diverse models and tasks, demonstrating strong alignment between theory and observed inference behavior. These contributions o ffer a principled pathway toward characterizing the limits of contextual influen ce during inference, and provide practical implications for designing and evalua ting repetition-based prompting methods.

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On Robustness and Chain-of-Thought Consistency of RL-Finetuned VLMs

Rosie Zhao, Anshul Shah, Xiaoyu Zhu, Xinke Deng, Zhongyu Jiang, Yang Yang, Joerg Liebelt, Arnab Kumar Mondal

Reinforcement learning (RL) fine-tuning has become a key technique for enhancing large language models (LLMs) on reasoning-intensive tasks, motivating its exten sion to vision language models (VLMs). While RL-tuned VLMs improve on visual rea soning benchmarks, they remain vulnerable to weak visual grounding, hallucinatio ns, and over-reliance on textual cues. We show that simple, controlled textual p erturbations—misleading captions or incorrect chain-of-thought (CoT) traces—caus e substantial drops in robustness, and that these effects are more pronounced wh en CoT consistency is taken into account across open-source multimodal reasoning models. In contrast, closed models exhibit similar failure modes but maintain m arkedly greater robustness and reasoning consistency, suggesting that the gap re flects a shortcoming in current RL fine-tuning methods rather than an inherent l imitation of the task. To better understand these vulnerabilities, we further a nalyze RL fine-tuning dynamics and uncover an accuracy–faithfulness trade-off: f ine-tuning raises benchmark accuracy, but can simultaneously erode the reliabili ty of the accompanying CoT and its robustness to contextual shifts.

Although adversarial augmentation improves robustness, it does not by itself pre vent faithfulness drift. Incorporating a faithfulness-aware reward can restore a lignment between answers and reasoning, but when paired with augmentation, train ing risks collapsing onto shortcut strategies. Together, these findings highligh t the limitations of accuracy-only evaluations and motivate training and assessm ent protocols that jointly emphasize correctness, robustness, and consistency in visually grounded reasoning.

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Demystifying LLM-as-a-Judge: Analytically Tractable Model for Inference-Time Sca ling

Indranil Halder, Cengiz Pehlevan

Recent developments in large language models have shown advantages in reallocati ng a notable share of computational resource from training time to inference tim e. However, the principles behind inference time scaling are not well understood . In this paper, we introduce an analytically tractable model of inference-time scaling: Bayesian linear regression with a reward-weighted sampler, where the re ward is determined from a linear model, modeling LLM-as-a-judge scenario. We stu dy this problem in the high-dimensional regime, where the deterministic equivale nts dictate a closed-form expression for the posterior predictive mean and varia nce. We analyze the generalization error when training data are sampled from a t eacher model. We draw k inference-time samples and select via softmax at a tem perature applied to a quadratic reward. When the reward is not too different fro

m the teacher, the generalization error decreases monotonically with increasing inference time samples k. However, the specific reward that optimizes inferenc e-time selection generally differs from the teacher. In contrast, substantial re ward misspecification induces a finite optimal k beyond which more sampling ca n increase the generalization error. For fixed k, there exists an optimal samp ling temperature. We experimentally verify these facts in large language model i nference with an additional large language model as a judge. In the ``best-of-$k $" limit with the teacher as reward, we theoretically show that the generalizati on error decays as $\Theta(1/k^2)$ and determine the leading coefficient via ext reme value theory. These formulas delineate domains where scaling inference-time computation is provably preferable to collecting more data. Finally, we demonst rate that when task difficulty increases, the previously mentioned advantage of inference-time compute degrades.

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GeoDM: Geometry-aware Distribution Matching for Dataset Distillation

Xuhui Li, Zhengquan Luo, Zihui Cui, Kai Zhao, zhiqiang xu

Dataset distillation aims to synthesize a compact subset of the original data, e nabling models trained on it to achieve performance comparable to those trained on the original large dataset. Existing distribution-matching methods are confin ed to Euclidean spaces, making them only capture linear structures and overlook the intrinsic geometry of real data, e.g., curvature.

However, high-dimensional data often lie on low-dimensional manifolds, suggestin g that dataset distillation should have the distilled data manifold aligned with the original data manifold.

In this work, we propose a geometry-aware distribution-matching framework, calle d GeoDM, which operates in the Cartesian product of Euclidean, hyperbolic, and s pherical manifolds, with flat, hierarchical, and cyclical structures all capture d by a unified representation.

To adapt to the underlying data geometry, we introduce learnable curvature and w eight parameters for three kinds of geometries. At the same time, we design an o ptimal transport loss to enhance the distribution fidelity.

Our theoretical analysis shows that the geometry-aware distribution matching in a product space yields a smaller generalization error bound than the Euclidean c ounterparts. Extensive experiments conducted on standard benchmarks demonstrate that our algorithm outperforms state-of-the-art data distillation methods and re mains effective across various distribution-matching strategies for the single g eometries.

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Weight-Space Learning for Certifiable Few-shot Transfer Learning

Fady Rezk, Royson Lee, Henry Gouk, Timothy Hospedales, Minyoung Kim

In contemporary deep learning, a prevalent and effective workflow for solving lo w-data problems is adapting powerful pre-trained foundation models (FMs) to new tasks via parameter-efficient fine-tuning (PEFT). However, while empirically eff ective, the resulting solutions lack generalisation guarantees to certify their accuracy - which may be required for ethical or legal reasons prior to deploymen t in high-importance applications. In this paper we develop a novel transfer lea rning approach that is designed to facilitate non-vacuous learning theoretic gen eralisation guarantees for downstream tasks, even in the low-shot regime. Specif ically, we first use upstream tasks to train a {\em distribution over PEFT param eters}. We then learn the downstream task by a {\em sample-and-evaluate} procedu re -- sampling plausible PEFTs from the trained diffusion model and selecting th e one with the highest likelihood on the downstream data. Crucially, this confin es our model hypothesis to a {\em finite} set of PEFT samples. In contrast to th e typical continuous hypothesis spaces of neural network weights, this facilitat es tighter risk certificates. We instantiate our bound and show non-trivial gene ralization guarantees compared to existing learning approaches which lead to vac uous bounds in the low-shot regime.

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IACW: Intent-Aware Controllable Watermarking for Scalable Authorial Intent Attri bution

Hao Huang, Ruihua Zhou, JiaTang Luo, Yunpeng Li, Yuling Liu
As Large Language Models (LLMs) integrate into writing workflows, precise govern ance requires distinguishing ''how AI participated'' rather than merely ''whethe r AI was used.'' Traditional binary detection often misclassifies ``AI-polished' ' content as generated, creating fairness risks. We propose shifting from passiv e post-hoc detection to active intent attribution, focusing on the distinction b etween Editing (source-anchored) and Generation (unanchored). We introduce \text bf{IACW-Instruct}, a corpus of diverse editing operations constructed via a Dire ctor--Actor--Judge pipeline to enable systematic evaluation. Building on this be nchmark, we propose \textbf{Intent-Aware Controllable Watermarking (IACW)}, feat uring intent-adaptive entropy gating for semantically lossless embedding. Experi ments show that IACW achieves 95\% attribution accuracy under 20\% token deletio n while preserving near-unwatermarked semantic fidelity, establishing a practica l paradigm for fine-grained provenance.
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Small Agent Group is the Future of Digital Health
Yuqiao Meng, Luoxi Tang, Dazheng Zhang, Rafael Brens, Elvys J. Romero, Nancy Guo , Safa elkefi, Zhaohan Xi
The rapid adoption of large language models (LLMs) in digital health has been dr iven by a "scaling-first" philosophy, i.e., the assumption that clinical intelli gence increases with model size and data. However, real-world clinical needs inc lude not only effectiveness, but also reliability and reasonable deployment cost . Since clinical decision-making is inherently collaborative, we challenge the m onolithic scaling paradigm and ask whether a Small Agent Group (SAG) can support better clinical reasoning. SAG shifts from single-model intelligence to collect ive expertise by distributing reasoning, evidence-based analysis, and critical a udit through a collaborative deliberation process. To assess the clinical utilit y of SAG, we conduct extensive evaluations using diverse clinical metrics spanni ng effectiveness, reliability, and deployment cost. Our results show that SAG ac hieves superior performance compared to a single giant model, both with and with out additional optimization or retrieval-augmented generation. These findings su ggest that the synergistic reasoning represented by SAG can substitute for model parameter growth in clinical settings. Overall, SAG offers a scalable solution to digital health that better balances effectiveness, reliability, and deploymen t efficiency.
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When Do Diffusion Models Learn to Generate Multiple Objects?
Yujin Jeong, Arnas Uselis, Iro Laina, Seong Joon Oh, Anna Rohrbach
Text-to-image diffusion models achieve impressive visual fidelity, yet they rema in unreliable in multi-object generation.
Despite extensive empirical evidence of these failures, the underlying causes re main unclear.
We begin by asking how much of this limitation arises from the data itself.
To disentangle data effects, we consider two regimes across different dataset si zes:
(1) concept generalization, where each individual concept is observed during tra ining under potentially imbalanced data distributions, and
(2) compositional generalization, where specific combinations of concepts are sy stematically held out.
To study these regimes, we introduce mosaic (Multi-Object Spatial relations, At trIbution, Counting), a controlled framework for dataset generation.
By training diffusion models on mosaic, we find that scene complexity plays a do minant role rather than concept imbalance, and that counting is uniquely difficu lt to learn in low-data regimes.
Moreover, compositional generalization collapses as more concept combinations ar e held out during training.
These findings highlight fundamental limitations of diffusion models and motivat e stronger inductive biases and data design for robust multi-object compositiona l generation.
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Message Tuning Outshines Graph Prompt Tuning: A Prismatic Space Perspective
Yancheng Chen, Dun Ma, Shuai Zhang, Yang Liu, Xixun Lin, Xiangyu Zhao, Wenguo Yang, Wei Chen, Chuan Zhou
Graph Foundation Models (GFMs), built upon the *Pre-training and Adaptation* paradigm, have emerged as a research hotspot in graph learning. For GNN-based GFMs, graph prompt tuning has become the prevailing adaptation method for downstream tasks. Although recent methods explain why graph prompt tuning works, how to rigorously measure its adaptation capacity remains an open problem. Addressing this problem is critical for understanding the capability limits of graph prompt tuning and for developing more powerful adaptation methods. In this paper, we propose Prismatic Space Theory (PS-Theory), a novel mathematical framework to quantify the capacity of adaptation methods, while focusing on establishing the upper bound for the adaptation capacity of graph prompt tuning. Building upon the proposed PS-Theory, we further introduce **M**essage **T**uning for **G**FMs (MTG), a lightweight approach that injects a small set of learnable message prototypes into each layer of the GNN backbone to adaptively guide message fusion without updating pre-trained weights. Through our PS-Theory, we prove that the adaptation capacity of MTG can exceed the theoretical upper bound of graph prompt tuning. Extensive experiments demonstrate that MTG consistently outperforms graph prompt baselines across diverse benchmark datasets, providing strong empirical support for our theoretical findings.
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Reflect-then-Correct: Rebalancing Task Optimization for Generalizable Meta-Reinforcement Learning via Distributional Value Error Reduction
Min Wang, Xin Li, Ye He, Mingzhong Wang, Yonggang Zhang
Meta-Reinforcement Learning (Meta-RL) faces significant challenges in non-parametric settings, where vastly different return scales across diverse tasks cause severe gradient interference. Existing categorical solutions attempt to normalize these scales but often fail due to rigid discretization and quantization errors. To address this, we propose Reflect-then-Correct (RTC), a framework that models meta-values using Sinkhorn divergence. By treating distributions as adaptive floating particles, RTC achieves a geometry-aware alignment of distinct meta-task structures. However, while Sinkhorn updates harmonize gradients, they introduce statistical bias via sampling estimation. RTC overcomes this issue by "reflecting'' on the temporal accumulation of Bellman inconsistencies through a recursive error model and "correcting'' the optimization via adaptive importance weights, which prioritize more accurate transitions for meta-value estimation. We provide theoretical guarantees for this reweighting strategy and demonstrate that RTC outperforms existing baselines on the challenging Meta-World ML-10 and ML-45 benchmarks.
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Deep Pre-Alignment for VLMs
Tianyu Yu, Kechen Fang, Zihao Wan, Kaidong Zhang, Yicheng Zhang, Jun Song, Bo Zheng, Yuan Yao
Most Vision Language Models (VLMs) directly map outputs from ViT encoders to the LLM via a lightweight projector. While effective, recent analysis suggests this architecture suffers from an alignment challenge: visual features remain distant from the text space in the initial layers of the LLM, forcing the model to waste critical depth on superficial modality alignment rather than deep understanding and complex reasoning. In this work, we propose Deep Pre-Alignment (DPA), a novel architecture that replaces the standard ViT encoder with a small VLM as perceiver, ensuring visual features are deeply aligned with the text space of the target large language model.
Comprehensive experiments demonstrate the effectiveness of DPA. On the 4B parameter scale, DPA outperforms baselines by 1.9 points across 8 multimodal benchmarks, with gains widening to 3.0 points at the 32B scale. Moreover, by offloading alignment to the perceiver, DPA achieves a 32.9\% reduction in language capability forgetting over 3 text benchmarks. We further demonstrate that these gains are consistent across different LLM families including Qwen3 and LLaMA 3.2, highlighting the generality of our approach.

Beyond performance, DPA also offers a seamless upgrade path for current VLM deve
lopment, requiring only a modular replacement for the visual encoder with margin
al computation overhead.
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Multi-Objective Preference Optimization: Improving Human Alignment of Generative
 Models

Akhil Agnihotri, Rahul Jain, Deepak Ramachandran, Zheng Wen

Post-training LLMs with RLHF and preference optimization methods (e.g., DPO, IPO
) has greatly improved alignment, yet these approaches assume a single objective
. In reality, humans express multiple, often conflicting objectives, such as hel
pfulness and harmlessness, with no natural scalarization. We study the multi-obj
ective preference alignment problem, where a policy must balance several objecti
ves simultaneously. We propose Multi-Objective Preference Optimization (MOPO), a
 constrained KL-regularized framework that maximizes a primary objective while e
nforcing lower bounds on secondary objectives via tunable safety thresholds. MOP
O operates directly on pairwise preferences without point-wise rewards, and admi
ts simple closed-form iterative updates. Empirically, MOPO recovers Pareto-optim
al policies on synthetic benchmarks and, when fine-tuned on human-preference dat
a, yields multi-billion parameter models that achieve higher rewards and Pareto-
dominate baselines, with stable and robust optimization dynamics.
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On Testing Conditional Mean Independence for Manifold-Valued Data

Meiling Zeng, Jinhong You, Jicai Liu, Shouxia Wang

This paper introduces a nonparametric test for conditional mean independence bet
ween a manifold■valued Y and Euclidean predictors X. The test is built on a
new measure called the Manifold Martingale Difference Divergence (MMDD), which c
haracterizes conditional mean dependence by projecting observations onto the tan
gent space via the logarithmic map. We provide an empirical estimator for the MM
DD, establish its asymptotic null distribution, and implement a wild bootstrap p
rocedure for finite■sample inference. Simulations on three representative manifo
lds demonstrate that the proposed test maintains correct size under the null eve
n when the distribution of Y depends on X, in contrast to the severe size d
istortion exhibited by the distance covariance (dCov) test. At the same time, it
 achieves competitive power across a range of alternatives. An application to re
al data illustrates its practical utility.
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Don't Drop Dropout: Optimizing Layer Sparsity for Efficient LLM Training and Inf
erence

Mostafa Elhoushi, Alexander D. Pretko, Nolan Simran Dey, Bin Claire Zhang, Gavia
 Gray, Gurpreet Gosal, Abdulrahman Mahmoud, Shane Bergsma, Joel Hestness

Layer dropout (a.k.a.\ stochastic depth) has been shown to enable faster trainin
g, higher accuracy, and robustness to zero-shot layer pruning in both language a
nd vision transformers. However, as models and datasets have scaled, dropout---p
articularly layer dropout---has largely disappeared from LLM pre-training recipe
s. While some prior work has reported that dropout can degrade accuracy, no comp
rehensive study has quantified, let alone mitigated, this effect. In this study,
 we show that layer dropout **should** be used in state-of-the-art LLM training,
 establishing best practices and scaling analysis for both training and post-tra
ining benefits. Concretely, with optimal layer distribution, time schedule, and
optimizer hyperparameters, LLM can achieve lower or similar validation loss whil
e saving upto 20\% of training FLOPs. Moreover, layer dropout enables significan
t post-training optimizations, such as early exit, intermediate-layer skipping,
and self-speculative decoding, yielding up to 1.4x inference speedup with neglig
ible accuracy loss.

All pre-training experiments were run on Cerebras CS-3 systems.
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Formal Concept Lattices are Good Semantic Scaffolds for Concept-Based Learning

Deepika Vemuri, Sayanta Adhikari, Ankit Saha, Krishn Vishwas Kher, Vineeth N. Ba
lasubramanian

Learning semantics is essential for deep learning models to be interpretable and

better aligned with human reasoning. Concept-based models approach this by representing classes through meaningful semantic abstractions, but typically treat all concepts as a flat, unstructured set learned at a single neural network layer.
This overlooks a fundamental property of human semantic understanding: concepts being organized hierarchically, from general to specific.
While deep networks do learn a hierarchy of visual features, this structure is rarely aligned with explicit semantic hierarchies.
Drawing on Formal Concept Analysis, we demonstrate that formal concept lattices provide principled semantic scaffolds to guide neural network learning. These lattices naturally identify where in the network concepts should be learned based on their level of generality. This allows the model to develop staged, semantically grounded representations throughout its depth.
Empirical results on real-world datasets show that our models produce more interpretable embeddings, support more effective interventions, and learn concept representations that are both meaningful and hierarchically structured.
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XRPO: Pushing the Limits of GRPO with Targeted Exploration and Exploitation
Udbhav Bamba, Minghao Fang, Yifan Yu, Haizhong Zheng, Fan Lai
Reinforcement learning algorithms such as GRPO have driven recent advances in large language model (LLM) reasoning. While scaling the number of rollouts stabilizes training, existing approaches suffer from limited exploration on challenging prompts and leave informative feedback signals underexploited, due to context-independent rollout allocation across prompts (e.g., generating 16 rollouts per prompt) and relying heavily on sparse rewards. This paper presents XRPO (eXplore–eXploit GRPO), a unified framework that recasts policy optimization through the principled lens of rollout exploration–exploitation. To enhance exploration, XRPO introduces a mathematically grounded rollout allocator that adaptively prioritizes prompts with higher potential for uncertainty reduction. It further addresses stagnation on zero-reward prompts through an in-context seeding strategy that injects curated exemplars, steering the model into more difficult reasoning trajectories. To strengthen exploitation, XRPO develops a group-relative, novelty-aware advantage sharpening mechanism that leverages sequence likelihoods to amplify low-probability yet correct responses, thereby extending the policy’s reach beyond sparse rewards. Experiments across diverse math and coding benchmarks on both reasoning and non-reasoning models demonstrate that XRPO outperforms existing advances (e.g., GRPO and GSPO) up to 4% pass@1 and 6% cons@32, while accelerating training convergence by up to 2.7x.
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On the Infinite Width and Depth Limits of Predictive Coding Networks
Francesco Innocenti, El Mehdi Achour, Rafal Bogacz
Predictive coding (PC) is a biologically plausible alternative to standard backpropagation (BP) that minimises an energy function with respect to network activities before updating weights. Recent work has improved the training stability of deep PC networks (PCNs) by leveraging some BP-inspired reparameterisations. However, the full scalability and theoretical basis of these methods remain unclear. To address this gap, we study the infinite width and depth limits of PCNs. For linear residual networks, we show that the set of width- and depth-stable feature-learning parameterisations for PC is exactly the same as for BP. Moreover, under any of these parameterisations, the PC energy with equilibrated activities converges to the quadratic BP loss when the model width is much larger than the depth, resulting in PC computing the same gradients as BP. Experiments show that, as long as an activity equilibrium is reached, convergence to BP holds for nonlinear models including convolutional networks and transformers. Overall, this work constrains the types of parameterisation that are scalable with PC, while showing a way in which BP can be effectively implemented with only local updates in much wider than deep networks like the brain.
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Adaptive Personalized Federated Learning via Multi-task Averaging of Kernel Mean Embeddings

Jean-Baptiste Fermanian, Batiste Le Bars, Aurélien Bellet
Personalized Federated Learning enables a collection of agents to collaboratively learn individual models without sharing raw data. We propose a new approach in which each agent optimizes a weighted combination of all agents' empirical risks, with the weights learned from data rather than specified a priori. The novelty of our method lies in formulating the estimation of these collaborative weights as a kernel mean embedding estimation problem with multiple data sources, leveraging tools from multi-task averaging to capture statistical relationships between agents. This perspective yields a fully adaptive procedure that requires no prior knowledge of data heterogeneity and automatically transitions between global and local learning regimes. By recasting the objective as a high-dimensional mean estimation problem, we derive finite-sample guarantees on local excess risks for a broad class of distributions, explicitly quantifying the statistical gains of collaboration. To address communication constraints inherent to federated settings, we also propose a practical implementation based on random Fourier features, which allows one to trade communication cost for statistical efficiency. Numerical experiments validate our theoretical results.
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Beyond Sample-Level Forgetting: Improving Reliability in Multimodal Unlearning
Jianzhou Wang, Yirui Wu, Lixin Yuan, Wenxiao Zhang, Jun Liu
Multimodal unlearning aims to eliminate specific data from pretrained multimodal models, which offers significant advantages in data privacy and model efficiency.
Current methods struggle to achieve the desired properties of effectiveness, reliability and locality, due to the complex interdependency of unimodal and multimodal knowledge. By introducing a causal perspective, we propose multimodal unlearning with decoupled knowledge components. To promote fine-grained understanding of multimodal context, we introduce Multimodal Variational Inference (MVI) to infer modal-specific and -consistent factors with incomplete sample observation. With foundation of decoupled knowledge, we propose contrastive semantic editing to regulate multimodal unlearning towards refined forgetting. Experiments on privacy- and copyright-sensitive scenarios validate effectiveness of our method across multiple scenarios, ensuring the unlearned model maintains high reliability and locality.
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Calibrated Knowledge Aggregation in Bayesian Mixture-of-Experts for Continual VQA
Mahsa Mozaffari, Hitesh Sapkota, Yu Kong, Xumin Liu, Qi Yu
Continual learning for visual question answering (VQA) is typically implemented by training one expert per task and routing each query using task-ID supervision. Yet continual VQA tasks overlap substantially: on the VQA-v2 task stream, a non-native expert outperforms the task’s own expert on 49.9% of queries, so hard routing both wastes transferable knowledge and can be confidently wrong when mismatched. We propose a calibrated Bayesian mixture-of-experts that trains parameter-efficient per-task adapters, learns routing by directly maximizing expected VQA utility, and marginalizes expert identity at inference via Bayesian aggregation in a unified answer space; an entropy penalty prevents the utility objective from collapsing to one-hot routing, enabling evidence pooling across plausible experts. We reach 64.16 accuracy with 0.63 forgetting on VQA-v2 CL-LS ($+5.74\%$ accuracy, -2.99 forgetting vs. the strongest prior method), 78.81 with 0.40 forgetting on TDIUC CL-LS ($+3.10$, -1.74), and 83.41 with 3.21 forgetting on TDIUC CL-VS ($+1.58$, -0.82). Calibration also improves on VQA-v2, reducing ECE from 0.15 to 0.07.
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RDT2: Exploring the Scaling Limit of UMI Data Towards Zero-Shot Cross-Embodiment Generalization
Songming Liu, Bangguo Li, Kai Ma, Lingxuan Wu, Hengkai Tan, Xiao Ouyang, Hang Su, Jun Zhu
Vision-Language-Action (VLA) models hold promise for generalist robotics but currently struggle with data scarcity, architectural inefficiencies, and the inabil

ity to generalize across different hardware platforms. We introduce RDT2, a robotic foundation model built upon a 7B parameter VLM designed to enable zero-shot deployment on novel embodiments for open-vocabulary tasks. To achieve this, we collected one of the largest open-source robotic datasets—over $10,000$ hours of demonstrations in diverse families—using an enhanced, embodiment-agnostic Universal Manipulation Interface (UMI). Our approach employs a novel three-stage training recipe that aligns discrete linguistic knowledge with continuous control via Residual Vector Quantization (RVQ), flow-matching, and distillation for real-time inference. Consequently, RDT2 becomes one of the first models that simultaneously zero-shot generalizes to unseen objects, scenes, instructions, and even robotic platforms. Besides, it outperforms state-of-the-art baselines in dexterous, long-horizon, and dynamic downstream tasks like playing table tennis.
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When Single Answer Is Not Enough: Rethinking Single-Step Retrosynthesis Benchmarks for LLMs
Bogdan Zagribelnyy, Ivan Ilin, Maksim Kuznetsov, Nikita Bondarev, Roman Schutski, Thomas MacDougall, Rim Shayakhmetov, Zulfat Miftahutdinov, Mikolaj Mizera, Vladimir Aladinskiy, Alex Aliper, Alex Zhavoronkov
Recent progress has expanded the use of large language models (LLMs) in drug discovery, including synthesis planning. However, objective evaluation of retrosynthesis performance remains limited. Existing benchmarks and metrics typically rely on published synthetic procedures and Top-K accuracy based on single ground-truth, which does not capture the open-ended nature of real-world synthesis planning. We propose a new benchmarking framework for single-step retrosynthesis that evaluates both general-purpose and chemistry-specialized LLMs using ChemCensor, a novel metric for chemical plausibility. By emphasizing plausibility over exact match, this approach better aligns with human synthesis planning practices. We also introduce CREED, a novel dataset comprising millions of ChemCensor-validated reaction records for LLM training, and use it to train a model that improves over the LLM baselines under this benchmark.
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Align Your Trajectory Tangent: Training Better Consistency Models via Manifold-Aligned Tangents
Beomsu Kim, ByungHee Cha, Jong Chul Ye
With diffusion and flow matching models achieving state-of-the-art generating performance, the interest of the community now turned to reducing the inference time without sacrificing sample quality.
Consistency Models (CMs), which are trained to be consistent on diffusion or probability flow ordinary differential equation (PF-ODE) trajectories, enable one or two-step flow or diffusion sampling.
However, CMs typically require prolonged training with large batch sizes to obtain competitive sample quality.
In this paper, we examine the training dynamics of CMs near convergence and discover that CM trajectory tangents -- CM output update directions -- are quite oscillatory, in the sense that they move parallel to the data manifold, not towards the manifold.
To mitigate oscillatory trajectory tangents, we propose a new loss function, called the {\em manifold feature distance (MFD)}, which provides manifold-aligned trajectory tangents that point toward the data manifold.
Consequently, our method -- dubbed {\em Align Your Trajectory Tangent (AYT)} -- can accelerate CM training by orders of magnitude and even out-perform the learned perceptual image patch similarity metric (LPIPS).
Furthermore, we find that our loss enables training with extremely small batch sizes without compromising sample quality.
Code is available at \url{https://github.com/1202kbs/AYT}
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IdEst: Assessing Self-Supervised Learning Representations via Intrinsic Dimension
Julie Mordacq, Vicky Kalogeiton, Steve Oudot
Self-supervised learning (SSL) has emerged as a powerful paradigm for

learning meaningful representations from unlabeled data.
However, the standard protocol for evaluating these representations,
linear probing,
is computationally expensive,
sensitive to hyperparameters, and provides limited insight
into the geometric structure of the representation space.
In this work, motivated by connections between neural network generalization
and intrinsic dimension (ID) we propose IdEst,
a method for estimating the ID of SSL representations
via the Minimum Spanning Tree dimension estimator ($\mathrm{dim}_\mathrm{MST}$).
Across diverse datasets, architectures, and SSL pretraining objectives,
we show that IdEst strongly correlates with downstream linear probe
performances.
Furthermore, we demonstrate that IdEst enables efficient hyperparameter selectio
n,
significantly reducing the computational cost compared to supervised
alternatives.
Our results highlight intrinsic dimensionality
as a principled geometric proxy for assessing SSL representations,
complementing standard supervised probing protocols.
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PhysHanDI: Physics-Based Reconstruction of Hand-Deformable Object Interactions
Jihyun Lee, Changmin Lee, Donghwan Kim, Tae-Kyun Kim
While existing methods for reconstructing hand–object interactions have made imp
ressive progress, they either focus on rigid or part-wise rigid objects—limiting
 their ability to model real-world objects (e.g., cloth, stuffed animals) that e
xhibit highly non-rigid deformations—or model deformable objects without full 3D
 hand reconstruction. To bridge this gap, we present PhysHanDI (Physics-based Re
construction of Hand and Deformable Object Interactions), a framework that enabl
es full 3D reconstruction of both interacting hands and non-rigid objects. Our k
ey idea is to physically simulate object deformations driven by forces induced f
rom densely reconstructed 3D hand motions, ensuring that the reconstructed objec
t dynamics are both physically plausible and coherent with the interacting hand
movements. Furthermore, we demonstrate that such simulation of object deformatio
ns can, in turn, refine and improve hand reconstruction via inverse physics. In
experiments, PhysHanDI outperforms the state-of-the-art baseline across reconstr
uction and future prediction.
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Minibatch Optimal Transport and Perplexity Bound Estimation in Discrete Flow Mat
ching
Etrit Haxholli, Yeti Z. Gurbuz, O■ul Can, Eli Waxman
Discrete flow matching, a recent framework for modeling categorical data, has sh
own competitive performance with autoregressive models. However, unlike continuo
us flow matching, the rectification strategy cannot be applied due to the stocha
sticity of discrete paths, necessitating alternative methods to minimize state t
ransitions. We propose a dynamic-optimal-transport-like minimization objective a
nd derive its Kantorovich formulation for discrete flows with convex interpolant
s, where transport cost depends solely on inter-state dissimilarity and can be o
ptimized via minibatch strategies. We show that such methods can reduce the numb
er of transitions up to 32 times (1024 to 32) to reach the same generative perpl
exity without compromising diversity. Additionally, path nondeterminism in discr
ete flows precludes an instantaneous change-of-variables analogue, preventing pr
ecise probability estimation available to continuous flows. We therefore propose
 two upper bounds on perplexity, enabling principled training, evaluation and mo
del comparison. Finally, we introduce Multimask Flows which outperform masked fl
ows in generative perplexity without compromising diversity, particularly when u
tilizing minibatch Optimal Transport.
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A Theoretical Framework for Statistical Evaluability of Generative Models
Shashaank Aiyer, Yishay Mansour, Shay Moran, Han Shao

Statistical evaluation aims to estimate the generalization performance of a model using held-out i.i.d. test data sampled from the ground-truth distribution. In supervised learning settings such as classification, performance metrics such as error rate are well-defined, and test error reliably approximates population error given sufficiently large datasets. In contrast, evaluation is more challenging for generative models due to their open-ended nature: it is unclear which metrics are appropriate and whether such metrics can be reliably evaluated from finite samples.

In this work, we introduce a theoretical framework for evaluating language models and establish evaluability results for commonly used metrics. We study two categories of metrics: test-based metrics, including integral probability metrics (IPMs), and similarity-based metrics, including Rényi and KL divergences. We show that IPMs with respect to any bounded test class can be evaluated from finite samples up to multiplicative and additive approximation errors. Moreover, when the test class has finite fat-shattering dimension, IPMs can be evaluated with arbitrary precision. In contrast, similarity-based metrics, including Rényi and KL divergences, are not evaluable from finite samples, as their values can be critically determined by rare events. We also analyze the potential and limitations of perplexity as an evaluation method.
****
L^3: Large Lookup Layers
Albert Tseng, Christopher De Sa
Modern sparse language models typically achieve sparsity through Mixture-of-Experts (MoE) layers, which dynamically route tokens to dense MLP "experts."
However, dynamic hard routing has a number of drawbacks, such as potentially poor hardware efficiency and needing auxiliary losses for stable training.
In contrast, the tokenizer embedding table, which is natively sparse, largely avoids these issues by selecting a single embedding per token at the cost of not having contextual information.
In this work, we introduce the Large Lookup Layer (L3), which generalizes embedding tables to model decoder layers as a means of further scaling sparsity.
L3 layers use static token-based routing to aggregate a *set* of learned embeddings per token in a *context-dependent way*, allowing the model to efficiently balance memory and compute by caching information in embeddings.
L3 has two main components: (1) a systems-friendly architecture that allows for fast training and CPU-offloaded inference with no overhead, and (2) an information-theoretic embedding allocation algorithm that effectively balances speed and quality.
We empirically test L3 by training transformers with up to 2.6B active parameters and find that L3 strongly outperforms both dense models and iso-sparse MoEs in both language modeling and downstream tasks.
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EchoRL: Reinforcement Learning via Rollout Echoing
Jinhe Bi, Aniri, Minglai Yang, Xingcheng Zhou, Wenke Huang, Sikuan Yan, Yujun Wang, Zixuan Cao, Michael Färber, Xun Xiao, Volker Tresp, Yunpu Ma
Reinforcement Learning with Verifiable Rewards is an effective route for post-training to strengthen the reasoning capability of large language models.
However, as training proceeds, the learning signal can collapse thus makes the training gain become marginal and ineffective.
Specifically, a growing fraction of prompts' rollouts become advantage-degenerated: all the self-generated rollouts show verified-success, making the standard deviation over their rewards be zero; accordingly each rollout's advantage becomes degenerated (zero) as well.
Given such rollouts' advantages, the policy-gradient for model optimization eventually vanishes, capping the training performance.
We argue that some of these rollouts still contain valuable learning signals but unfortunately omitted with the existing RLVR methods.
In this paper, inspired through analyzing the entropy pattern behind golden trajectories produced by external expert models, we propose EchoRL for better exploi

ting the advantage-degenerated rollouts to further improve the training performance.
EchoRL is a lightweight module that first identifies an EchoClip from verified-success rollouts based on their step-level entropy values, and then feeds this clip back as an auxiliary supervision signal in the RL objective.
Extensive experiments across 10 benchmarks, 5 LLM backbones, and 7 popular RLVR post-training methods demonstrate that EchoRL consistently improves RLVR post-training with minimal overhead.
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MASPOB: Bandit-Based Prompt Optimization for Multi-Agent Systems with Graph Neural Networks
Zhi Hong, Qian Zhang, Jiahang Sun, Zhiwei Shang, Mingze Kong, Xiangyi Wang, Yao Shu, Zhongxiang Dai
Large Language Models (LLMs) have achieved substantial success in real-world applications, particularly as the cognitive backbone of Multi-Agent Systems (MAS) for orchestrating complex workflows. Since many deployments preclude workflow modifications while MAS performance is highly prompt-sensitive, prompt optimization becomes a critical strategy for improvement. However, real-world prompt optimization for MAS is impeded by three key challenges: (1) the need of sample efficiency due to prohibitive evaluation costs, (2) topology-induced coupling among prompts, and (3) the combinatorial explosion of the search space. To address these challenges, we introduce **MASPOB** (**M**ulti-**A**gent **S**ystem **P**rompt **O**ptimization via **B**andits), a novel sample-efficient framework based on bandits. By leveraging Upper Confidence Bound (UCB) to quantify uncertainty, the bandit framework balances exploration and exploitation, maximizing gains within a strictly limited budget. To handle topology-induced coupling, MASPOB integrates Graph Neural Networks (GNNs) to capture structural priors, learning topology-aware representations of prompt semantics. Furthermore, it employs coordinate ascent to decompose the optimization into univariate sub-problems, reducing search complexity from exponential to linear. Extensive experiments across diverse benchmarks demonstrate that MASPOB achieves state-of-the-art performance, consistently outperforming existing baselines. Our code is available at <https://github.com/HZ1008/MASPOB>.
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GemDepth: Geometry-Embedded Features for 3D-Consistent Video Depth
Yuecheng Liu, Junda Cheng, Longliang Liu, Wenjing Liao, Hanrui Cheng, Yuzhou Wang, Xin Yang
Video depth estimation extends monocular prediction into the temporal domain to ensure coherence. However, existing methods often suffer from spatial blurring in fine-detail regions and temporal inconsistencies. We argue that current approaches, which primarily rely on temporal smoothing via Transformers, struggle to maintain strict 3D geometric consistency—particularly under rotations or drastic view changes. To address this, we propose GemDepth, a framework built on the insight that an explicit awareness of camera motion and global 3D structure is a prerequisite for 3D consistency. Distinctively, GemDepth introduces a Geometry-Embedding Module (GEM) that predicts inter-frame camera poses to generate implicit geometric embeddings. This injection of motion priors equips the network with intrinsic 3D perception and alignment capabilities. Guided by these geometric cues, our Alternating Spatio-Temporal Transformer (ASTT) captures latent point-level correspondences to simultaneously enhance spatial precision for sharp details and enforce rigorous temporal consistency. Furthermore, GemDepth employs a data-efficient training strategy, effectively bridging the gap between high efficiency and robust geometric consistency. As shown in Fig.2, comprehensive evaluations demonstrate that GemDepth achieves state-of-the-art performance across multiple datasets, particularly in complex dynamic scenarios.
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Towards Understanding Steering Strength
Magamed Taimeskhanov, Samuel Vaiter, Damien Garreau
A popular approach to post-training control of large language models (LLMs) is the steering of intermediate latent representations. Namely, identify a well-chos

en direction depending on the task at hand and perturbs representations along th
is direction at inference time. While many propositions exist to pick this direc
tion, considerably less is understood about how to choose the magnitude of the m
ove, whereas its importance is clear: too little and the intended behavior does
not emerge, too much and the model's performance degrades beyond repair. In this
 work, we propose the first theoretical analysis of steering strength. We charac
terize its effect on next token probability, presence of a concept, and cross-en
tropy, deriving precise qualitative laws governing these quantities. Our analysi
s reveals surprising behaviors, including non-monotonic effects of steering stre
ngth. We validate our theoretical predictions empirically on eleven language mod
els, ranging from a small GPT architecture to modern models.
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Does Reinforcement Fine-Tuning Improve Generalization of LLM Agents? An Empiric
al Study

Zhiheng Xi, Xin Guo, Jiaqi Liu, Jiazheng Zhang, Yutao Fan, Zhihao Zhang, Shichun
 Liu, Mingxu Chai, Xiaowei Shi, Yitao Zhai, Xunliang Cai, Tao Gui, Qi Zhang, Xua
njing Huang

Reinforcement fine-tuning (RFT) has shown promise for training LLM agents to per
form multi-turn decision-making based on environment feedback. However, most exi
sting evaluations remain largely in-domain—training and testing are conducted in
 the same environment or even on the same tasks. In real-worlddeployment, agent
s may operate in unseen environments with different background knowledge, observ
ation spaces, and action interfaces. To characterize the generalization profile
of RFT under such shifts, we conduct a systematic study along three axes: (1) wi
thin-environment generalization across task difficulty, (2) cross-environment tr
ansfer to unseen environments, and (3) sequential multi-environment training to
quantify transfer and forgetting. Our results show that RFT generalizes well acr
oss task difficulty within an environment, but exhibits unstable transfer to uns
een environments, which correlates with shifts in both semantic priors and obser
vation/action interfaces. In contrast, sequential training yields promising down
stream gains with minimal upstream forgetting, and mixture training across envir
onments improves the overall balance. We further provide detailed analyses and d
eeper insights, and hope our work helps the community develop and deploy general
izable LLM agents.
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Batched First-Order Methods for Parallel LP Solving in MIP

Nicolas Blin, Stefano Gualandi, Christopher Maes, Andrea Lodi, Bartolomeo Stella
to

We present a batched first-order method for solving multiple linear programs in
parallel on GPUs. Our approach extends the primal-dual hybrid gradient algorithm
 to efficiently solve batches of related linear programming problems that arise
in mixed-integer programming techniques such as strong branching and bound tight
ening. By leveraging matrix-matrix operations instead of repeated matrix-vector
operations, we obtain significant computational advantages on GPU architectures.
 We demonstrate the effectiveness of our approach on various case studies and id
entify the problem sizes where first-order methods outperform traditional simple
x-based solvers, depending on the computational environment. This is a significa
nt step toward integer programming algorithms that tightly exploit GPU capabilit
ies. We argue that some specific operations should be allocated to GPUs and perf
ormed in full, instead of relying on lightweight heuristic approaches on CPUs.
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Provably Label-Efficient Conformal Prediction

Andrew Ilyas, Joonhyuk Ko, Jingwu Tang, Steven Wu, Jiahao Zhang

Conformal prediction converts any black-box predictor into one with finite-sampl
e, distribution-free coverage guarantees, outputting prediction sets $T(x)$ that
 contain the true label with probability at least $1-\alpha$. To construct these
 prediction sets, conformal prediction relies on a randomly sampled ``calibratio
n set'' of labeled examples. In many applications, however, this labeled calibra
tion set is costly to collect, creating a tradeoff between upfront labeling cost
 and downstream utility of the conformal predictor. In this work, we study *conf

ormal prediction with costly label queries*, where unlabeled examples arrive i.i.d. and labels can be queried one at a time. After m queries, we form a conformal predictor; the upfront cost of this predictor is the calibration set size m, and its efficiency is the expected prediction set size $\mathbb{E}|T_m(X)|$. We design an online stopping rule $\hat{m}$ that automatically balances the upfront cost against conformal efficiency *while preserving the original conformal guarantee*. Theoretically, we show that under mild regularity assumptions, the expected total cost of our stopping rule matches the best fixed calibration size in hindsight. Experimentally, we find that our stopping rule reduces cost compared to standard choices of m from the literature by 40.6% $\pm$ 2.3%. Finally, we demonstrate a reduction from the probably approximately correct labeling problem of Candès et al. (2025) to CP, under which our stopping rule minimizes the total labeling cost.
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State-Dependent Safety Failures in Multi-Turn Language Model Interaction
Pengcheng Li, Jie Zhang, Tianwei Zhang, Han Qiu, Zhang kejun, Weiming Zhang, Nenghai Yu, Wenbo Zhou
Safety alignment in large language models is typically evaluated under isolated queries, yet real-world use is inherently multi-turn.
Although multi-turn jailbreaks are empirically effective, the structure of conversational safety failure remains insufficiently understood.
In this work, we study safety failures from a state-space perspective and show that many multi-turn safety failures in current safety-aligned language models arise from contextual state evolution, a regime that is not fully captured by isolated prompt-level analyses alone.
We introduce STAR, a state-oriented diagnostic framework that treats dialogue history as a state transition operator and enables controlled analysis of safety behavior along interaction trajectories.
Rather than optimizing attack strength, STAR provides a principled probe of how aligned models traverse the safety boundary under autoregressive conditioning.
Across multiple frontier language models, we find that systems which appear robust under static evaluation can undergo rapid and reproducible safety collapse under structured multi-turn interaction.
Mechanistic analysis reveals monotonic drift away from refusal-related representations and abrupt phase transitions induced by role-conditioned context.
Together, these findings motivate viewing language model safety as a dynamic, state-dependent process defined over conversational trajectories.
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Glimpse: Geometry Learning of Multi-scale Structural Priors for 3D Pose Estimation
Zhenhua Tang, Jihua Peng, Yanbin Hao, Qiguang Miao, Chi-Man Pun
Monocular 3D human pose estimation is fundamentally challenged by severe occlusion and inherent depth ambiguity. To address this, we propose Glimpse, a framework that learns robust 3D poses by explicitly modeling anatomical geometry from a single image. We recast the problem as geometry learning of multi-scale structural priors, realized through two synergistic components. First, structured sampling captures the body's geometric continuity through dual-level feature extraction, acquiring both local joint appearance and continuous features along skeletal limbs via deformable sampling. By propagating limb-level geometric cues to their connected joints, this design bridges information gaps caused by occlusion. Second, geometric correction ensures global 3D consistency by lifting coherent 2D features into a canonical 3D reference space, where a shared 3D anchor guides a distance-aware fusion mechanism. Extensive experiments conducted on Human3.6M and MPI-INF-3DHP demonstrate that Glimpse achieves state-of-the-art performance, with superior robustness under severe occlusion and complex articulation.
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DDIM Inversion as a Perturbation Amplifier: Breaking Mimicry Protection via Reconstruction Error Minimization
Huming Qiu, Peiyi Chen, Mi Zhang, Geng Hong, Xiaoyu You, Mi Wen, Min Yang
Personalization techniques for image generation models have increasingly been mi

sused for malicious purposes, including unauthorized style imitation and copyrig
hted content replication. In response, recent mimicry protection methods embed c
arefully designed perturbations into images to disrupt a model’s ability to lear
n genuine semantic representations. Despite their growing adoption, the robustne
ss of these protection mechanisms remains poorly understood, raising concerns ab
out their reliability in real-world deployment.
In this work, we present the first systematic analysis showing that DDIM inversi
on inherently acts as a perturbation amplifier, causing protected images to suff
er severe structural distortions during reconstruction. Building on this observa
tion, we propose DDIM Inversion-based Reconstruction Purification (DIRP), a nove
l purification approach that removes protective perturbations by explicitly mini
mizing DDIM inversion reconstruction error under perceptual constraints. Extensi
ve experiments on six existing mimicry protection methods demonstrate that DIRP
consistently outperforms five state-of-the-art attack baselines, achieving super
ior perturbation removal while better preserving image quality. Our results expo
se fundamental vulnerabilities in current mimicry protection strategies and high
light the urgent need for more robust and principled defenses.
****

CHB: A Diagnostic Toolkit for Hardness-Aware Clustering Evaluation

Walid Durani, Philipp Jahn, Collin Leiber, David B. Hoffmann, Thomas Seidl, Clau
dia Plant, Christian Böhm

Clustering is commonly compared through leaderboards that collapse performance i
nto a single aggregate ranking. Such summaries obscure why methods succeed, whic
h data properties align with failure, and how conclusions shift under representa
tion changes and realistic tuning constraints. We present CHB, a diagnostic tool
kit for hardness-aware clustering evaluation. CHB maps each dataset--representat
ion pair to an interpretable hardness fingerprint capturing (i) separation, (ii)
 cohesion and scale heterogeneity, and (iii) topology through scalable persisten
t-homology summaries. Using this diagnostic space, CHB evaluates clustering algo
rithms under standardized, compute-aware tracks. Conditioning results on hardnes
s coordinates turns comparison into diagnosis: across a broad range of datasets
and their representations, CHB reveals reproducible structural regimes, uncovers
 regime-dependent ranking reversals across method families, and surfaces robustn
ess signatures, including topology-linked breakdowns. CHB further enables repres
entation auditing by attributing gains to measurable shifts in the hardness fing
erprint rather than just external performance changes. We release CHB as an open
, extensible artifact for evaluating new clustering methods and embeddings withi
n a shared diagnostic framework.
****

BIOARC: Discovering Optimal Neural Architectures for Biological Foundation Model
s

Yi Fang, Haoran Xu, Jiaxin Han, Sirui Ding, Yizhi Wang, Yue Wang, Xuan Wang

Foundation models have revolutionized AI, yet biological applications often repu
rpose general architectures without accounting for the intrinsic structural and
functional properties of distinct modalities, such as genomic and proteomic sequ
ences. Consequently, these architectures lack the inductive biases required to c
apture the complex ``grammars" inherent to biological data, resulting in subopti
mal performance. To address this, we introduce BioArc, a framework utilizing Neu
ral Architecture Search (NAS) to shift from intuition-driven design to automated
 data-driven discovery. Unlike standard NAS restricted to homogeneous spaces, Bi
oArc navigates a heterogeneous space for open-ended composition of architectural
 blocks. By systematically analyzing the interplay between architecture, tokeniz
ation, and training across modalities, BioArc identifies novel hybrid architectu
res that surpass state-of-the-art models while being up to 25x smaller. We disti
ll these findings into empirical design principles and validate their biological
 relevance, demonstrating how our designs hierarchically capture the underlying
biological grammar. Additionally, we introduce an agentic framework to predict o
ptimal architectures for new tasks. Overall, BioArc provides a data-driven metho
dology for developing the next generation of efficient biological foundation mod
els and task-specific networks.

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Magnitude Distance: A Geometric Measure of Dataset Similarity
Sahel Torkamani, Henry Gouk, Rik Sarkar
Quantifying the distance between datasets is a fundamental question in mathematics and machine learning. We propose *magnitude distance*, a novel distance metric defined on finite datasets using the notion of the *magnitude* of a metric space.

The proposed distance incorporates a tunable scaling parameter, t, that controls the sensitivity to global structure (small t) and finer details (large t). We prove several theoretical properties of magnitude distance, including its limiting behavior across scales and conditions under which it satisfies key metric properties.

In contrast to classical distances, we show that magnitude distance remains discriminative in high-dimensional settings when the scale is appropriately tuned.

We further demonstrate how magnitude distance can be used as a training objective for push-forward generative models. Our experimental results support our theoretical analysis and demonstrate that magnitude distance provides meaningful signals, comparable to established distance-based generative approaches.
****
The Axiomatic Value of Regularization in AI Alignment from Human Preferences
Ezgi Korkmaz
Reinforcement learning from human feedback is the leading approach to aligning powerful AI systems so that they can be safe and helpful for humanity. While RLHF is typically modelled as a problem of learning a single preference ranking from noisy feedback, true human preferences are complex and often conflicting, representing substantive disagreements stemming from the diversity of individual human values. With this motivation, a recent line of research has studied RLHF from the perspective of social choice theory, which provides a set of well-established desirable properties for aggregating diverse preferences. Seen through this lens, the standard learning objective in RLHF is equivalent to aggregating diverse human preferences via the Borda count rule. At the same time, several new RLHF algorithms have been proposed, which turn out to be equivalent to the von Neumann winner social choice rule. However, the connection between social choice theory and RLHF has thus far ignored the critical role of regularization to prevent divergence from a reference policy, which is utilized in essentially all practical RLHF algorithms. In this paper, we study how regularization affects the social choice axioms satisfied by different RLHF algorithms, and prove that regularization improves the axiomatic properties of the von Neumann winner rule. In contrast, the Borda count rule still fails to satisfy key social choice axioms even when regularized. These results provide a principled argument grounded in social choice theory for utilizing practical RLHF algorithms that correspond to the von Neumann winner, rather than the standard RLHF objective.
****
One Model to Translate Them All: Universal Any-to-Any Translation for Heterogeneous Collaborative Perception
Yang Li, Weize Li, Quan Yuan, Shao Congzhang, Guiyang Luo, Yunqi Ba, Xuanhan Zhu, Xinyuan Ding, Xiaoyuan Fu, Jinglin Li
By sharing intermediate features, collaborative perception extends each agent's sensing beyond standalone limits, but real-world feature modality heterogeneity remains a key barrier to effective fusion. Most existing methods, including direct adaption and protocol-based transforma-tion, typically rely on training adapters for newly emerging feature modalities and often require additional retraining or fine-tuning. Such repeated training is costly and is often infeasible across manufacturers due to model and data privacy con-straints, limiting real-world scalability. To address this issue, we propose UniTrans, a universal any-to-any feature modality translation model that instantiates translators on the fly for arbitrary modalities. UniTrans pretrains a bank of translator expert parameters and learns their combination coefficients as a function of source-to-target modality mapping. The mapping is measured in a modality-intrinsic latent space, wher

e an intrinsic encoder extracts modality-specific yet scene-invariant codes from single-frame intermediate features, enabling UniTrans to instantiate translators in a zero-shot manner. Experiments on OPV2V-H and DAIR-V2X demonstrate that UniTrans consistently outperforms state-of-the-art methods in both simulated and real-world set-tings, enabling efficient any-to-any translation through a universal model. Code will be made available.
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FlowState: Sampling-Rate■Equivariant Time■Series Forecasting
Lars Graf, Thomas Ortner, Stanis■aw Wo■niak, Angeliki Pantazi
Existing time series foundation models (TSFMs), often based on transformer variants, lack adaptability to different sampling rates, struggle with generalization across varying context and target lengths and are computationally inefficient.
We introduce FlowState, a novel TSFM architecture that achieves sampling-rate–equivariant forecasting through a unified design that pairs a state space model (SSM) encoder with a functional basis decoder (FBD). This design enables continuous-time modeling and dynamic time-scale adjustment, allowing FlowState to inherently generalize across all possible temporal resolutions, and dynamically adjust the forecasting horizons without retraining. We further propose an efficient pretraining strategy that improves robustness and accelerates training. Despite being one of the smallest TSFMs, FlowState achieves state-of-the-art results on the widely used GIFT-Eval benchmark, while demonstrating superior adaptability to unseen sampling rates. Our detailed analyses confirm the effectiveness of its components, and we demonstrate its unique ability to adapt to varying input sampling rates.
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Search or Accelerate: Confidence-Switched Position Beam Search for Diffusion Language Models
Mingyu Cao, Alvaro Correia, Christos Louizos, Shiwei Liu, Lu Yin
Diffusion Language Models (DLMs) generate text by iteratively denoising a masked sequence, repeatedly deciding which positions to commit at each step. Standard decoding follows a greedy rule, unmasking the most confident positions, yet this local choice can lock the model into a suboptimal unmasking order, especially on reasoning-heavy prompts. We present Search Or AcceleRate (SOAR), a training-free decoding algorithm that adapts its behavior to the model’s uncertainty. When confidence is low, SOAR briefly widens the search over alternative unmasking decisions to avoid premature commitments; when confidence is high, it collapses the search and decodes many positions in parallel to reduce the number of denoising iterations. Across mathematical reasoning and code generation benchmarks (GSM8K, MBPP, HumanEval) on Dream-7B and LLaDA-8B, SOAR improves generation quality while maintaining competitive inference speed, offering a practical way to balance quality and efficiency in DLM decoding.
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SleepLM: Natural-Language Intelligence for Human Sleep
Zongzhe Xu, Zitao Shuai, Eideen Mozaffari, Ravi Shankar Aysola, Rajesh Kumar, Yuzhe Yang
We present SleepLM, a family of sleep-language foundation models that enable human sleep alignment, interpretation, and interaction with natural language.
Despite the critical role of sleep, learning-based sleep analysis systems operate in closed label spaces (e.g., predefined stages or events) and fail to describe, query, or generalize to novel sleep phenomena.
SleepLM bridges natural language and multimodal polysomnography, enabling language-grounded representations of sleep physiology.
To support this alignment, we introduce a multilevel sleep caption generation pipeline that enables the curation of the first large-scale sleep-text dataset, comprising over 100K hours of data from more than 10,000 individuals.
Furthermore, we present a unified pretraining objective that combines contrastive alignment, caption generation, and signal reconstruction to better capture physiological fidelity and cross-modal interactions.
Extensive experiments on real-world sleep understanding tasks verify that SleepLM outperforms state-of-the-art in zero-shot and few-shot learning, cross-modal r

etrieval, and sleep captioning. Importantly, SleepLM also exhibits intriguing ca
pabilities including language-guided event localization, targeted insight genera
tion, and zero-shot generalization to unseen tasks.
To support reproducibility and future work, we open-source the captioning pipeli
ne, pretrained checkpoints, and the model architectures at https://github.com/ya
ng-ai-lab/SleepLM.
****

Norm$\times$Direction: Restoring the Missing Query Norm in Vision Linear Attenti
on

Weikang Meng, Yadan Luo, Liangyu Huo, Yingjian Li, Yaowei Wang, Xin Li, Zheng Zh
ang

Linear attention mitigates the quadratic complexity of softmax attention but suf
fers from a critical loss of expressiveness. We identify two primary causes: (1)
 The normalization operation cancels the query norm, which breaks the correlatio
n between a query's norm and the spikiness (entropy) of the attention distributi
on as in softmax attention. (2) Standard techniques for enforcing non-negativity
 cause destructive information loss by nullifying valid inner-product interactio
ns. To address these challenges, we introduce **NaLaFormer**, a novel linear att
ention mechanism built upon a norm$\times$direction (ND) decomposition of the qu
ery and key vectors. We leverage each component to solve a distinct problem: The
 query norm is injected into our kernel to create a query-norm-aware map that
restores the attention distribution's spikiness. The *direction vectors* are pro
cessed by a geometric, cosine-based similarity metric that guarantees non-negati
vity while preserving the rich, fine-grained information of the inner product. W
e validate NaLaFormer through a comprehensive multi-modal evaluation, where it s
ets new state-of-the-art benchmarks for linear attention. Our model achieves up
to a 7.5\% accuracy gain on ImageNet-1K and a 4.7\% mIoU improvement on ADE20K o
ver comparable baselines. It demonstrates profound efficiency, reducing peak mem
ory by a transformative 92.3\% in token-intensive super-resolution tasks (70K+ t
okens). NaLaFormer's versatility is further confirmed as it surpasses strong bas
elines like Mamba on common-sense reasoning and sets a new state-of-the-art on t
he Long Range Arena (LRA) benchmark. Source code can be found in the supplementa
ry materials.
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TIC-VLA: A Think-in-Control Vision-Language-Action Model for Robot Navigation in
 Dynamic Environments

Zhiyu Huang, Yun Zhang, Johnson Liu, Rui Song, Chen Tang, Jiaqi Ma

Robots in dynamic, human-centric environments must follow language instructions
while maintaining real-time reactive control. Vision-language-action (VLA) model
s offer a promising framework, but they assume temporally aligned reasoning and
control, despite semantic inference being inherently delayed relative to real-ti
me action. We introduce Think-in-Control (TIC)-VLA, a latency-aware framework th
at explicitly models delayed semantic reasoning during action generation. TIC-VL
A defines a delayed semantic-control interface that conditions action generation
 on delayed vision-language semantic states and explicit latency metadata, in ad
dition to current observations. We further propose a latency-consistent training
 pipeline that injects reasoning inference delays during imitation learning and
online reinforcement learning, aligning training with asynchronous deployment. T
o support realistic evaluation, we present DynaNav, a physics-accurate, photo-re
alistic simulation suite for language-guided navigation in dynamic environments.
 Extensive experiments in simulation and on a real robot show that TIC-VLA consi
stently outperforms prior VLA models while maintaining robust real-time control
under multi-second reasoning latency. Project website: https://ucla-mobility.git
hub.io/TIC-VLA/
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Resilient Coresets and Clustering

Ashkan Norouzi-Fard, Silvio Lattanzi, Mohammadhossein Bateni, Morteza Monemizade
h

Many machine learning problems are geometric at their core, relying on metric
representations of data for tasks such as clustering, prototype selection,

nearest-neighbor search, and graph-based learning. Furthermore, data is constantly evolving and it is routinely transformed through dimensionality reduction, random
projections, feature embeddings, compression, or privacy-preserving mechanisms. These transformations are designed to preserve geometry approximately. As a result, they preserve objective values for many geometric optimization problems, but they fail to guarantee that
algorithmic outcomes remain consistent. In this work, we study \emph{resilient data summaries}
for geometric optimization. Building on the notion of
\emph{γ-resilient algorithms} from Ahmadian, we
introduce γ-resilient coresets. A γ-resilient
(k,ε)-coreset is a compact, weighted summary that guarantees a $(1+\varepsilon)$ approximation to the objective and enforces stability at the level of assignments. We complement our positive result with a lower bound showing that to obtain a tight approximation for resilient clustering it is necessary
to use a bi-criteria solution.
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Navigating the Pareto Frontier of Alignment: Spectrum-Adaptive Fine-Tuning for LLMs

Yaoyou Fan, Chao Zhang, Xiaoyu Tan, Chenxing Sun, Yu Yuan, Haoyu Feng, Lu Pan, Ke Zeng, Xunliang Cai

Supervised Fine-Tuning with Negative Log-Likelihood (NLL) remains the standard post-training paradigm for Large Language Models, yet it imposes a disproportionately aggressive update force on low-probability target tokens.

This focus forces the model to prioritize minimizing the loss of difficult samples over optimizing the overall quality of the generation, often leading to unwarranted overconfidence.

On the other hand, alternatives like Dynamic Fine-Tuning suffer from vanishing gradients on these tokens, which severely hinders the acquisition of new concepts.

To bridge this gap, we propose **S**pectrum-**A**daptive **F**ine-**T**uning (**SAFT**), a unified framework that interpolates between the aggressive learning signal of NLL and the robust nature of probability-weighted optimization.

By adaptively balancing these objectives, SAFT effectively mitigates outlier sensitivity without sacrificing learning efficiency.

Empirically, our method achieves state-of-the-art performance on mathematical reasoning benchmarks, demonstrating superior generalization on out-of-distribution tasks. Furthermore, evaluations on general conversational alignment validate SAFT's broad adaptability across diverse data regimes. Our code is available at https://github.com/sjtu-scx/SAFT.
****

Coverage ≠ Exposure: Auditable Control of Same-Support Tail Failures under Multimodal Missingness

Ziteng Hong, Bingzhi Chen, Yishu Liu, Sudong Cai, Guangming Lu

Multimodal systems encounter partial observability from sensor dropout and degradation, yet adverse tails can remain unreliable even when average performance is preserved. Under a locked same-support contract, matched-average runs can exhibit same-support tail failure, diverging on worst-case and shift-reweighted metrics over the same observable support. The failure arises because conditional interaction separates environment coverage from parameter exposure. Gated routing sends information through different groups, so high-leverage groups may receive few updates despite complete coverage, and tail aggregation amplifies their errors. We audit this mismatch with TailPressure, an exposure-normalized statistic of tail-leveraged interaction from gating logs. We propose Heterogeneity-aware Closed-loop Exposure Stabilizer (H-CES), a lightweight controller that stabilizes per-group pressure via deterministic increment-branch gating and group-wise decoupled weight decay, without changing loss or inference. Across multimodal settings and backbones, H-CES improves same-support tail reliability while preserving cl

ean performance.
****
Divide-and-Denoise: A Game-Theoretic Method for Fairly Composing Diffusion Model
s
Abhi Gupta, Polina Barabanshchikova, Vikas K Garg, Samuel Kaski, Tommi Jaakkola
The abundance of pre-trained diffusion models provides an opportunity for compos
ition. Combining several models, however, runs the risk of one model dominating
or models disagreeing with each other. Here, we propose Divide-and-Denoise, a me
thod for coordinating multiple pre-trained diffusion models during sampling. Muc
h like managing a specialized workforce, our method creates a fair but efficient
division of labor across models. Central to our method is the notion of an allo
cation which defines the responsibility of each model to every region of the noi
sy sample. At every timestep, we then denoise by (i) updating the allocation by
solving a fair division game, where we divide the sample into regions that maxim
ize total utility under fairness constraints, and (ii) aligning the models with
this allocation, where we guide each model to denoise within its assigned region
. This leads to a new composite denoising process that evolves in tandem with a
division process. We evaluate Divide-and-Denoise on conditional image generation
. Across several quality metrics, including the GenEval benchmark, our method ou
tperforms baselines and resolves common failures including missing objects and m
ismatched attributes. Experiments show that Divide-and-Denoise utilizes each mod
el's expertise without neglecting any other model.
****
AD-MIR: Bridging the Gap from Perception to Persuasion in Advertising Video Unde
rstanding via Structured Reasoning
Binxiao Xu, Junyu Feng, Xiaopeng Lin, Haodong Li, ZhiYuan Feng, Bohan Zeng, Ruic
huan An, Ming Lu, Qi She, Wentao Zhang
Multimodal understanding of advertising videos is essential for interpreting the
intricate relationship between visual storytelling and abstract persuasion stra
tegies. However, despite excelling at general search, existing agents often stru
ggle to bridge the cognitive gap between pixel-level perception and high-level m
arketing logic. To address this challenge, we introduce **AD-MIR**, a framework
designed to decode advertising intent via a two-stage architecture. First, in th
e **Structure-Aware Memory Construction** phase, the system converts raw video i
nto a structured database by integrating semantic retrieval with exact keyword m
atching. This approach prioritizes fine-grained brand details, such as logos and
on-screen text, while dynamically filtering out irrelevant background noise to
isolate key protagonists. Second, the **Structured Reasoning Agent** mimics a ma
rketing expert through an iterative inquiry loop, decomposing the narrative to d
educe implicit persuasion tactics. Crucially, it employs an evidence-based self-
correction mechanism that rigorously validates these insights against specific v
ideo frames, automatically backtracking when visual support is lacking. Evaluati
on on the AdsQA benchmark demonstrates that AD-MIR achieves state-of-the-art per
formance, surpassing the strongest general-purpose agent, DVD, by 1.8 and 9.5 pe
rcentage points in strict and relaxed accuracy, respectively. These results unde
rscore that effective advertising understanding demands explicitly grounding abs
tract marketing strategies in pixel-level evidence. The code is available at [ht
tps://github.com/Little-Fridge/AD-MIR](https://github.com/Little-Fridge/AD-MIR).
****
Training-Free Guided Diffusion for Planning: A Unified Framework via Doob’s h-Tr
ansform with Safety Guarantees
Kenta Hoshino, Yashaswi Shashank Aluru, Xiyu Deng, Yorie Nakahira
This paper studies the theoretical foundations of guidance mechanisms in continu
ous-time score-based diffusion models.
We adopt Doob’s h-transform as a principled framework for characterizing ideal g
uided diffusion processes and analyze the discrepancy between ideal and approxim
ate guidance.
Our analysis provides explicit error bounds and yields probabilistic guarantees
on satisfying prescribed constraints, which are particularly important for safet
y-critical planning.

We further show that the Doob-based formulation induces a stochastic optimal control problem, enabling practical guidance design without additional model training.
We demonstrate the effectiveness of the proposed framework on robotic navigation tasks, including language-conditioned planning.
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CAT-Q: Cost-efficient and Accurate Ternary Quantization for LLMs
Shigeng Wang, Chao Li, Yangyuxuan Kang, Jiawei Fan, Anbang Yao
In this paper, we present CAT-Q, **C**ost-efficient and **A**ccurate **T**ernary **Q**uantization, for compressing and accelerating LLMs. Unlike existing state-of-the-art ternary quantization methods that rely on data-intensive and costly quantization-aware training to mitigate severe performance degradation, CAT-Q is a simple yet effective post-training quantization scheme that is readily applicable to LLMs with diverse architectures and model sizes. It has two key components, learnable modulation (LM) and softened ternarization (ST), which are coupled from an optimization perspective. LM leverages a composition of learnable factors to modulate the distribution of pre-trained high-precision weights and the ternary threshold, making them less sensitive to ternarization. ST further introduces a differentiable transition function to guide the ternarization process toward stable convergence. We show that, for pre-trained LLMs with 1.7B to 8B parameters, CAT-Q can efficiently quantize them into ternary models using only 512 calibration samples, while achieving superior performance than the seminal BitNet 1.58-bit v1 and v2 families (with 1.3B to 7B parameters) trained with 100B tokens, yielding about a 100,000$\times$ reduction in training tokens. Moreover, we show for the first time that CAT-Q can quantize much larger pre-trained LLMs having 14B to 235B parameters into leading ternary models within just 8 to 60 hours on 8 A100-80GB GPUs. Code is available at https://github.com/IntelChina-AI/BitTern.
****
Understanding Behavior Cloning with Action Quantization
Haoqun Cao, Tengyang Xie
Behavior cloning is a fundamental paradigm in machine learning, enabling policy learning from expert demonstrations across robotics, autonomous driving, and generative models. Autoregressive models like transformer have proven remarkably effective, from large language models (LLMs) to vision-language-action systems (VLAs). However, applying autoregressive models to continuous control requires discretizing actions through quantization, a practice widely adopted yet poorly understood theoretically.

This paper provides theoretical foundations for this practice. We analyze how quantization error propagates along the horizon and interacts with statistical sample complexity. We show that behavior cloning with quantized actions and log-loss achieves optimal sample complexity—matching existing lower bounds—and incurs only polynomial horizon dependence on quantization error, provided the dynamics are stable and the policy satisfies a probabilistic smoothness condition. We further characterize when different quantization schemes satisfy or violate these requirements, and propose a model-based augmentation that provably improves the error bound without requiring policy smoothness. Finally, we establish fundamental limits that jointly capture the effects of quantization error and statistical complexity.
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Detached Skip-Links and R-Probe: Decoupling Feature Aggregation from Gradient Propagation for MLLM OCR
Ziye Yuan, Ruchang Yao, Chengxin Zheng, Yusheng Zhao, Daxiang Dong, Ming Zhang
Multimodal large language models (MLLMs) excel at high-level reasoning yet fail on OCR tasks where fine-grained visual details are compromised or misaligned. We identify an overlooked optimization issue in multi-layer feature fusion. Skip pathways introduce direct back-propagation paths from high-level semantic objectives to early visual layers. This mechanism overwrites low-level signals and destabilizes training.

To mitigate this gradient interference, we propose Detached Skip-Links, a minimal modification that reuses shallow features in the forward pass while stopping gradients through the skip branch during joint training. This asymmetric design reduces gradient interference, improving stability and convergence without adding learnable parameters.
To diagnose whether fine-grained information is preserved and usable by an LLM, we introduce R-Probe, which measures pixel-level reconstructability of projected visual tokens using a shallow decoder initialized from the first quarter of the LLM layers.
Across multiple ViT backbones and multimodal benchmarks, and at scales up to 7M training samples, our approach consistently improves OCR-centric benchmarks and delivers clear gains on general multimodal tasks.
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Towards Pareto-Optimal Tool-Integrated Agents with Pareto Ranking Policy Optimization
Junyi Li, Xiaowei Qian, Yingyi Zhang, Wenlin Zhang, Guojing Li, Sheng Zhang, Xiao Han, Yichao Wang, Xiangyu Zhao
Recent advances in tool-integrated language agents have significantly improved their ability to solve complex reasoning tasks. However, existing alignment methods predominantly focus on maximizing task accuracy, while overlooking auxiliary objectives such as tool-use efficiency, which are essential for practical deployment. To address this gap, we introduce \textbf{ParetoPO}, a two-stage multi-objective optimization framework for aligning tool-using large language models (LLMs) under competing objectives. In the first stage, ParetoPO leverages hypervolume-guided dynamic scalarization to adapt reward weights based on global Pareto frontier progress. In the second stage, it replaces scalarized learning signals with Pareto-ranking-based advantage computation, promoting nondominated trajectories through dominance-aware credit assignment. This design enables fine-grained, action-level optimization across multiple conflicting objectives. Experimental results on mathematic reasoning and multi-hop QA tasks show that ParetoPO consistently discovers policies with superior accuracy-efficiency trade-offs compared to static and heuristic baselines. Our code is publicly available at https://github.com/Applied-Machine-Learning-Lab/ICML2026_ParetoPO.
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Performative Policy Gradient: Optimality in Performative Reinforcement Learning
Debabrota Basu, Udvas Das, Brahim Driss, Uddalak Mukherjee
Post-deployment machine learning algorithms often influence the environments they act in, and thus *shift* the underlying dynamics that the standard reinforcement learning (RL) methods ignore. While designing optimal algorithms in this *performative* setting has recently been studied in supervised learning, the RL counterpart remains under-explored. In this paper, we prove the performative counterparts of the performance difference lemma and the policy gradient theorem in RL, and introduce the **Performative Policy Gradient** algorithm **PePG**. **PePG** is the first policy gradient algorithm designed to account for performativity in RL. Under softmax parametrisation, and also with and without entropy regularisation, we prove that **PePG** converges to *performatively optimal policies*, i.e. policies that remain optimal under the distribution shifts induced by themselves. Thus, **PePG** significantly extends the prior works in Performative RL that achieves *performative stability* but not optimality. Our empirical analysis on standard performative RL environments validate that **PePG** outperforms the existing performative RL algorithms aiming for stability.
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When Planning Fails Despite Correct Execution: On Epistemic Calibration for LLM-Based Multi-Agent Systems
Zehao Wang, Shilong Jin, Zhao Cao, Lanjun Wang
LLM-based multi-agent systems can fail even when planned actions are executed correctly because agents may misjudge their knowledge when evaluating plan feasibility, a phenomenon we term epistemic miscalibration in planning. Unlike execution errors, epistemic miscalibration is latent during planning, as generated plans can remain self-consistent and executable without observable errors; the miscal

ibration is also dynamic, as new information can alter feasibility assessments, potentially obscuring past miscalibration signals and causing them to recur over time. To address this, we propose the Epistemic Planning Calibration Agentic Workflow (EPC-AW), which assesses whether plans remain supported under varying information conditions rather than directly verifying feasibility.
EPC-AW employs Information-consistency-based Plan Selection, selecting plans whose evaluations are stable across agents, together with Consistency-guided Epistemic State Refinement to adapt calibration over time by leveraging past discrepancies to guide future planning. Experiments show that EPC-AW improves system-level success by an average of 9.75\%. Code is available in the public repository (https://github.com/wzhSteve/EPC-AW).
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OSAQ: Outlier Self-Absorption for Accurate Low-bit LLM Quantization
Zhikai Li, Zhen Dong, Xuewen Liu, Jing Zhang, Qingyi Gu
Large Language Models (LLMs) have demonstrated remarkable capabilities in understanding and generation tasks. However, their massive parameter scale leads to significant resource consumption and latency during inference. Post-training weight-only quantization offers a promising solution by reducing model size and accelerating token generation through alleviating the memory-bound issue. Nevertheless, there are inherent systematic outliers in weights, and although some efforts have attempted to address them, such as scaling and rotation, the performance of low-bit quantization remains far from satisfactory. In this paper, we propose Outlier Self-Absorption Quantization (OSAQ), which performs second-order low-rank derived additive weight suppression for low-bit weight-only LLM quantization. Specifically, we observe that Hessian exhibits low-rank consistency across different inputs, with certain directions persistently lacking strength. Leveraging this property, we construct an additive weight transformation based on the Hessian ’s null space, thereby suppressing weight outliers without affecting the task loss. This additive transformation can be absorbed into the weights offline, requiring no inter-layer transformations and introducing no inference overhead. Moreover, the construction is efficiently achieved by a closed-form solution, without resource-intensive training or iterative procedures. Extensive experiments across models of varying scales and tasks are conducted, and the results show that OSAQ effectively suppresses outliers and improves low-bit quantization performance.
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Beyond Accuracy and Complexity: The Effective Information Criterion for Structurally Stable Symbolic Regression
Zihan Yu, Guanren Wang, Jingtao Ding, Huandong Wang, Yong Li
Symbolic regression (SR) traditionally balances accuracy and complexity, implicitly assuming that simpler formulas are structurally more rational. We argue that this assumption is insufficient: existing algorithms often exploit this metric to discover accurate and compact but structurally irrational formulas that are numerically ill-conditioned and physically inexplicable. Inspired by the structural stability of real physical laws, we propose the Effective Information Criterion (EIC) to quantify formula rationality. EIC models formulas as information channels and measures the amplification of inherent rounding noise during recursive calculation, effectively distinguishing physically plausible structures from pathological ones without relying on ground truth. Our analysis reveals a stark structural stability gap between human-derived equations and SR-discovered results. By integrating EIC into SR workflows, we provide explicit structural guidance: for heuristic search, EIC steers algorithms toward stable regions to yield superior Pareto frontiers; for generative models, EIC-based filtering improves pre-training sample efficiency by 2–4 times and boosts generalization R^2 by 22.4\%. Finally, an extensive study with 108 human experts shows that EIC aligns with human preferences in 70\% of cases, validating structural stability as a critical prerequisite for human-perceived interpretability. We release our code at https://github.com/tsinghua-fib-lab/EIC
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Learning Manifold Data with Flow Matching

Sophia Pi, Mingcheng Lu, Maojiang Su, Weimin Wu, Jerry Yao-Chieh Hu, Han Liu
We study flow-matching transformers when data lie on a low-dimensional manifold.

Our key insight is a flow decomposition that splits motion along the manifold fr
om motion off the manifold.
The scheme works for first and higher-order flow matching and ties model complex
ity to the intrinsic manifold dimension.
Building on these, we establish tighter sample-complexity bounds for velocity ap
proximation, velocity estimation, and distribution estimation.
Our results show how flow-matching transformers escape the curse of dimensionali
ty by utilizing intrinsic data structure.
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Very Efficient Listwise Multimodal Reranking for Long Documents
Yiqun Sun, Pengfei Wei, Lawrence B. Hsieh
Listwise reranking is a critical yet costly component in vision-centric retrieva
l and multimodal retrieval-augmented generation (M-RAG) over long documents. Alt
hough recent VLM-based rerankers achieve strong accuracy, they are often impract
ical due to long visual-token inputs and autoregressive decoding, resulting in h
igh latency. We propose ZipRerank, a very efficient listwise multimodal reranker
 that directly addresses both bottlenecks: it shortens the input via query-image
 early interaction and eliminates multi-step generation by scoring all candidate
s in a single forward pass. ZipRerank is trained with a two-stage recipe: listwi
se pretraining on large-scale text reranking data rendered as images, followed b
y multimodal finetuning with VLM-teacher supervision and a soft-ranking objectiv
e to handle noisy rankings. Extensive experiments on the MMDocIR benchmark demon
strate that ZipRerank matches or surpasses state-of-the-art multimodal rerankers
 while reducing LLM inference latency by up to an order of magnitude, making it
well-suited for latency-sensitive real-world systems. Source code is available a
t https://anonymous.4open.science/r/ZipRerank.
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View Space: Learning Representation across Arbitrary Graphs
Dooho Lee, Myeong Kong, Minho Jeong, Jaemin Yoo
Generalizing pretrained models to unseen datasets without retraining is a centra
l challenge toward foundation models. Achieving fully inductive inference on num
erical data is particularly difficult due to large variations in feature dimensi
onality and semantics across datasets. We observe that, in the presence of graph
 structure, numerical data admits a distinct structure-induced representational
axis beyond the feature space, which we formalize as the *view space*. This view
 space enables a unified representation of graphs with heterogeneous features an
d motivates *Graph View Transformation* (GVT), a class of parametric mappings th
at can be shared across arbitrary graphs. We instantiate this framework with Rec
urrent GVT, an architecture for fully inductive node representation learning in
node classification. Pretrained on OGBN-Arxiv and evaluated on 27 benchmarks, Re
current GVT outperforms GraphAny, the prior fully inductive graph model, by +8.9
3%, and surpasses 12 individually tuned GNNs by at least +3.30%. These results e
stablish the view space as a principled and practical foundation for learning ac
ross graphs with heterogeneous feature spaces. Code and checkpoints are availabl
e
in https://github.com/dooho00/graph-view-space.
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Analytic Bijections for Smooth and Interpretable Normalizing Flows
Mathis Gerdes, Miranda C. N. Cheng
A key challenge in normalizing flows is finding expressive invertible scalar bij
ections. Existing approaches face trade-offs: affine transformations are smooth
and analytically invertible but lack expressivity; monotonic splines offer local
 control but are only piecewise smooth and act on bounded domains; residual flow
s achieve smoothness but need numerical inversion. We introduce three families o
f *analytic bijections* that are globally smooth (C^∞), defined on all of
 $\mathbb{R}$, and analytically invertible in closed form, combining the favorab
le properties of prior approaches. Beyond serving as drop-in replacements in cou

pling flows, where they match or exceed spline performance, we develop *radial flows*: a novel architecture using direct parametrization that transforms the radial coordinate while preserving angular direction. Radial flows exhibit exceptional training stability, produce geometrically interpretable transformations, and on targets with radial structure can achieve comparable quality to coupling flows with $1000\times$ fewer parameters. We provide comprehensive evaluation on 1D and 2D benchmarks, and demonstrate applicability to higher-dimensional physics problems through experiments on ϕ^4 lattice field theory, where our bijections outperform affine baselines and enable problem-specific designs that address mode collapse.
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Rethinking Feature Alignment in Generalist Graph Anomaly Detection: A Relational Fingerprint-based Approach
Yujing Liu, Yixin Liu, Yu Zheng, Alan Wee-Chung Liew, Xiaofeng Cao, Shirui Pan
Extending traditional graph anomaly detection (GAD) from one-for-one to one-for-all paradigms, generalist GAD aims to learn a universal detector for identifying anomalies on unseen graphs without graph-specific retraining. Nevertheless, existing approaches primarily focus on aligning heterogeneous features across different data domains via PCA-based projection, which harmonizes feature dimensions but neglects semantic alignment. As a result, GAD models fail to acquire semantically transferable knowledge from source-domain pre-training, and even exhibit negative transfer on unseen graphs. To address this issue, we propose a Relational Fingerprint-based generalist GAD approach (REFI-GAD for short), aligning heterogeneous raw features with a universal and semantics-aware relational fingerprint (REFI) that encodes anomaly-indicative cues from both contextual and structural perspectives. Building on REFI, we design a fingerprint-grounded generalist GAD model, which combines a transformer-based encoder to capture domain-invariant knowledge with an SNR-guided refinement module for domain-specific adaptation. Extensive experiments on 14 datasets demonstrate that REFI-GAD significantly outperforms state-of-the-art methods.
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Learning Treatment Allocations with Risk Control Under Partial Identifiability
Sofia Ek, Dave Zachariah
Learning beneficial treatment allocations for a patient population is an important problem in precision medicine. For such allocations, a certain proportion of treated patients may not receive any benefit. This proportion of unnecessary treated represents a `treatment risk' which is a waste of resources and may, in addition, expose patients to unnecessary adverse effects. Therefore, we aim to control the treatment risk when learning beneficial allocations. This learning problem is complicated by the fact that the treatment risk is generally not identifiable from either randomized trial or observational data. We propose a certifiable learning method that controls treatment risk, using finite samples in the partially identified setting. The method is illustrated using both simulated and real data.
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Skill-Pro: Learning Reusable Skills from Experience via Non-Parametric PPO for LLM Agents
Qirui Mi, Zhijian Ma, Mengyue Yang, Haoxuan Li, Yisen Wang, Haifeng Zhang, Jun Wang
LLM-driven agents excel at sequential decision-making but often rely on on-the-fly reasoning, re-deriving solutions even in recurring scenarios. This insufficient experience reuse leads to computational redundancy and instability. To bridge this gap, we propose **Skill-Pro**, a framework enabling agents to autonomously learn reusable procedural skills from interaction experiences without parameter updates. By formalizing a **Skill-MDP**, Skill-Pro transforms passive episodic narratives into executable Skills defined by activation, execution, and termination conditions to ensure executability.
To achieve reliable reusability without capability degradation, we introduce **Non-Parametric PPO**, which leverages semantic gradients for high-quality candidate generation and a PPO Gate for robust Skill verification. Through score-based

maintenance, Skill-Pro sustains compact, high-quality procedural memory. Experimental results across in-domain, cross-task, and cross-agent scenarios demonstrate that Skill-Pro achieves superior reuse rates and significant gains with extreme memory compression. Visualized evolutionary trajectories and Skill distributions further reveal how Skill-Pro transparently accumulates, refines, and reuses procedural knowledge to facilitate long-term autonomy.

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Monitorability as a Free Gift: How RLVR Spontaneously Aligns Reasoning

Zidi Xiong, Shan Chen, Himabindu Lakkaraju

As Large Reasoning Models (LRMs) are increasingly deployed, auditing their chain-of-thought (CoT) traces for safety becomes critical. Recent work has reported that monitorability—the degree to which CoT faithfully and informatively reflects internal computation—can appear as a "free gift" during the early stages of Reinforcement Learning with Verifiable Rewards (RLVR).

We make this observation concrete through a systematic evaluation across model families and training domains.

Our results show that this effect is not universal: monitorability improvements are strongly data-dependent. In particular, we demonstrate the critical role of data diversity and instruction-following data during RLVR training. We further show that monitorability is orthogonal to capability—improvements in reasoning performance do not imply increased transparency. Through mechanistic analysis, we attribute monitorability gains primarily to response distribution sharpening (entropy reduction) and increased attention to the prompt, rather than stronger causal reliance on reasoning traces. We also reveal how monitorability dynamics vary with controlled training and evaluation difficulty. Together, these findings provide a holistic view of how monitorability emerges under RLVR, clarifying when gains are likely to occur and when they are not.

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Kalman Linear Attention: Parallel Bayesian Filtering For Efficient Language Modeling and State Tracking

Vaisakh Shaj, Cameron Barker, Aidan Scannell, Andras Szecsenyi, Elliot J. Crowley, Amos Storkey

State-space language models such as Mamba and gated linear attention (GLA) offer efficient alternatives to transformers due to their linear complexity and parallel training, but often lack the expressivity and robust state-tracking needed for complex reasoning. We address these limitations by reframing sequence modelling through a probabilistic lens, using Bayesian filters as a core primitive. While classical filters such as Kalman filters provide principled state estimation and uncertainty tracking, they are typically viewed as inherently sequential. We show that reparameterising the Kalman filter in information form enables its updates to be computed via an associative scan, allowing efficient parallel training. Building on this insight, we introduce the Kalman Linear Attention (KLA) layer, a neural sequence-modelling primitive that performs time-parallel probabilistic inference while maintaining explicit belief-state uncertainty. KLA offers strictly more expressive non-linear updates and gating than GLA variants while retaining their computational advantages. On language modelling tasks, KLA matches or outperforms modern SSMs and GLAs across representative benchmarks for discrete token manipulation and state tracking.

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Quantifying the Uncertainty of Foundation Models with Singular Value Ensembles

Mehmet Ozgur Turkoglu, Dominik J. Mühlematter, Alexander Becker, Konrad Schindler, Helge Aasen

Foundation models have become a dominant paradigm in machine learning, achieving remarkable performance across diverse tasks through large-scale pretraining. However, they often yield overconfident, uncalibrated predictions. The standard approach to quantifying epistemic uncertainty are ensembles of multiple independently trained models. But their computational cost scales linearly with ensemble size, making them impractical for large foundation models. We propose Singular Value Ensemble (SVE), a parameter-efficient implicit ensembling method. SVE builds on a simple, but powerful core assumption: namely, that the singular vectors of

the weight matrices correspond to meaningful directions in the representation space. If the singular vectors are indeed meaningful (orthogonal) "knowledge directions", then a model ensemble can be obtained by modulating only how strongly each direction contributes to the output. Rather than learning new parameters for each ensemble member, we freeze the singular vectors and only train per-member singular values that rescale the contribution of each direction in that shared knowledge basis. Ensemble diversity emerges naturally during joint training as stochastic initialization and random batch sampling cause different members to converge to different combinations of the same underlying knowledge. SVE performs comparable to an explicit ensemble, while increasing the parameter count of the base model by <1%, making principled uncertainty estimation accessible in resource-constrained settings. We validate SVE on NLP and vision tasks with various different backbones and show that it improves calibration while maintaining predictive accuracy.

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Left–Right Symmetry Breaking in CLIP-Style Vision-Language Models Trained on Synthetic Spatial-Relation Data

Takaki Yamamoto, Chihiro Noguchi, Toshihiro Tanizawa

Spatial understanding remains a key challenge in vision-language models. Yet it is still unclear whether such understanding is truly acquired, and if so, through what mechanisms. We present a controllable 1D image–text testbed to probe how left–right relational understanding emerges in Transformer-based vision and text encoders trained with a CLIP-style contrastive objective. We train lightweight Transformer-based vision and text encoders end-to-end on paired descriptions of one- and two-object scenes and evaluate generalization to unseen object pairs while systematically varying label and layout diversity. We find that contrastive training learns left–right relations and that label diversity, more than layout diversity, is the primary driver of generalization in this setting. To gain the mechanistic understanding, we perform an attention decomposition and show that interactions between positional and token embeddings induce a horizontal attention gradient that breaks left–right symmetry in the encoders; ablating this contribution substantially reduces left–right discrimination. Our results provide a mechanistic insight of when and how CLIP-style models acquire relational competence.

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Riemannian Optimization for Fair Spectral Clustering

Minh Phu Vuong, Jinyoung Lee, Young-Ju Lee, Chul-Ho Lee

Fair graph clustering has emerged as a critical research area for addressing algorithmic bias in machine learning. The objective is to ensure that the proportion of each protected group within a cluster is consistent with its representation in the entire dataset. However, most existing spectral solutions rely on computationally expensive eigendecompositions of the graph Laplacian, limiting their scalability. In this paper, we propose Riemannian Fair Spectral Clustering (R-FairSC), a novel method that formulates fair spectral clustering as a constrained optimization problem on a Riemannian manifold. We develop a Riemannian alternating direction method of multipliers employing a variable-splitting strategy to efficiently solve the associated subproblems. Numerical experiments on large synthetic and real-world graphs demonstrate that R-FairSC significantly improves computational efficiency over state-of-the-art methods while maintaining high clustering quality and fairness.

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SC-FAGC: Size Constrained Fast Anchor-based Graph Clustering

Liu Jiachen

Spectral clustering is hindered by its $O(n^{3})$ time complexity, and although anchor-based approximations reduce this to $O(nm^{2})$ ($m \ll n$), they often yield degenerate or unbalanced partitions in the absence of explicit cluster-size control. To address this limitation, we propose \textbf{SC-FAGC (Size-Constrained Fast Anchor Graph Clustering)}, a unified formulation that integrates entropy regularization and bilateral cardinality constraints within an anchor-based spectral clustering framework, enforcing user-specified lower and upper bounds on a

nchor-level cluster sizes to avoid trivial or highly unbalanced solutions. To so lve the resulting non-convex problem efficiently, we develop an \textbf{Iterativ ely Re-weighted (IRW)} optimization scheme that linearizes the objective and sol ves each subproblem via a \textbf{Double-Bounded Optimal Transport (DB-OT)} solv er based on the \textbf{Sinkhorn--Knopp} algorithm, guaranteeing convergence whi le maintaining scalability. Extensive experiments on benchmark datasets demonstr ate that SC-FAGC consistently achieves state-of-the-art performance across six s tandard clustering metrics while strictly enforcing the prescribed anchor-level constraints. Moreover, it induces effective data-level size control after label transmission, offering a principled and scalable solution for large-scale graph clustering with controllable partition structure.
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SpanNorm: Reconciling Training Stability and Performance in Deep Transformers
Wang Chao, Bei Li, Jiaqi Zhang, Xinyu Liu, Yuchun Fan, Linkun Lyu, Xin Chen, Jin gang Wang, Tong Xiao, Peng Pei, Xunliang Cai
The success of Large Language Models (LLMs) hinges on the stable training of dee p Transformer architectures. A critical design choice is the placement of normal ization layers, leading to a fundamental trade-off: the ''PreNorm'' architecture ensures training stability at the cost of potential performance degradation in deep models, while the ''PostNorm'' architecture offers strong performance but s uffers from severe training instability. In this work, we propose SpanNorm, a no vel technique designed to resolve this dilemma by integrating the strengths of b oth paradigms. SpanNorm adopts the clean residual path of PreNorm to stabilize s ignal propagation while employing a PostNorm-style computation that normalizes t he output of the residual connection, thereby enhancing model performance. We pr ovide a theoretical analysis demonstrating that SpanNorm, combined with a princi pled scaling strategy, maintains bounded signal variance throughout the network, preventing the gradient issues that plague PostNorm models, and alleviating the representation collapse of PreNorm. Empirically, SpanNorm consistently outperfo rms standard normalization schemes in both dense and Mixture-of-Experts (MoE) sc enarios, paving the way for more powerful and stable Transformer architectures.
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Distributed Stochastic K-Level Optimization Over Networks
Xinwen Zhang, Yihan Zhang, Hongchang Gao, Heng Huang
In recent years, decentralized optimization has gained significant attention for solving machine learning problems where data are distributed across multiple de vices. However, existing decentralized optimization algorithms are primarily des igned for single-level and two-level optimization tasks, limiting their applicat ion to more complex problems such as decentralized stochastic K-level optimiza tion, where $K>2$. In this work, we propose a novel decentralized stochastic K -level variance-reduced gradient descent algorithm to address the significant co mputation and communication overhead caused by the multi-level structure in dece ntralized stochastic K-level optimization problems. Moreover, we propose a nov el theoretical analysis to tackle the recursive dependence issue caused by the m ulti-level structure when establishing the convergence rate of our algorithm. F inally, the experimental results confirm the effectiveness of our proposed algor ithm.
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CoPE: A Framework for Optimizing Coordination between Planning and Execution in LLM-based Agents
Huanxi Liu, Kun Hu, Qiang Wang, Yuanzhao Zhai, Feng Dawei, Bo Ding, Huaimin Wang
Fine-tuning Large Language Models (LLMs) as autonomous agents on domain-specific data has emerged as a promising paradigm for tackling interactive, real-world t asks.
However, existing studies have overlooked the critical coordination between long -term planning and multi-step execution in optimizing agent capabilities.
This oversight leads to the propagation of impractical plans and plan-deviated t rajectories within the optimization process, resulting in suboptimal task perfor mance and hindering the further development of LLM-based agents in long-horizon tasks.

To bridge this gap, we propose $\textbf{CoPE}$, a novel framework that explicitl
y integrates planning–execution coordination into LLM-based agent optimization.
CoPE employs Self-Refining MCTS to generate task plans and multiple execution tr
ajectories through environment interactions.
By quantifying the coordination between planning and execution, CoPE assigns hig
her optimization weights to well-coordinated samples, enabling LLM-based agents
to learn better planning and execution policies.
Extensive experiments demonstrate that CoPE substantially improves agent coordin
ation, outperforming state-of-the-art baselines on benchmarks comprising two lon
g-horizon multi-step tasks. Codes and data are available at https://github.com/O
ctobrist/CoPE.
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A Cartesian-3j Framework for Machine Learning Interatomic Potentials
Zemin Xu, Chenyu Wu, Wenbo Xie, Peijun Hu
Machine learning interatomic potentials (MLIPs) have brought substantial gains i
n the extrapolation capability in computational chemistry. However, most equivar
iant models are typically built with spherical tensors (STs), while Cartesian te
nsor formulations remain less developed despite their natural alignment with ato
mic coordinates and tensorial targets. In this work, we develop a Cartesian fram
ework for irreducible Cartesian tensors (ICTs) by introduce the Cartesian-3j sym
bol and Cartesian Generalized Clebsch-Gordan Coefficients, which serve as direct
 analogues of the Wigner-3j symbol and Generalized Clebsch-Gordan coefficients d
efined for ST coupling. We extend the e3nn library to support ICT product, and u
se this framework to build Cartesian counterparts of MACE, NequIP, and Allegro,
allowing the first controlled comparison where architectures are held fixed and
only the tensor basis is changed. Our experiments show that irreducible Cartesia
n models can achieve accuracy comparable to spherical counterparts, but direct C
artesianization incurs unfavorable compute and memory scaling, motivating dedica
ted Cartesian architectural choices. Leveraging ICTs and our framework, we intro
duce TACE-v1-OAM-M and demonstrate that it achieves competitive performance on M
atbench Discovery compared to state-of-the-art ST models.
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Efficient Mismatch-Tolerant Coding for Model-Driven Compression
Aviv Adler, Jennifer Tang
A central insight in lossless data compression is
the close connection between probabilistic next-
symbol prediction and efficient sequence compression, whereby predictive models
can be combined with classical coding techniques to achieve
strong compression performance. Applying this
approach with powerful modern learned models, such as LLMs, has been shown to ac
hieve
markedly better compression than traditional
techniques across a wide range of domains.
However, significant practical challenges remain,
including model non-determinism, in which a
model produces different predictions on different machines despite identical par
ameters and inputs; such mismatches between the encoder and
decoder can lead to complete decoding failure.
Probability Matching Interval Coding (PMATIC)
was recently introduced as a drop-in framework
for mismatch-robust coding and shown to enable
reliable compression and decompression in the
presence of bounded prediction mismatch (Adler
& Tang, 2026). In this work, we present a generalization of PMATIC that allows t
he incorpo-
ration of tight theoretical results into the design
and more flexible parameter optimization, resulting in substantial improvements
in compression
efficiency and robustness.
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Approximation Preserving Coresets
Milind Prabhu, Chris Schwiegelshohn, Sudarshan Shyam
Clustering in a big data setting is an intensively studied problem, with coresets emerging as one of the important paradigms in this line of work. Given a cost function $\text{cost}(P,S)$ mapping input points P and a solution S to an objective value, a coreset is a typically weighted sketch $\Omega\subseteq P$ such that $\text{cost}(\Omega,S)\approx \text{cost}(P,S)$. In practice, coreset sizes much smaller than those suggested by theoretical guarantees are often found to be sufficient.

In this paper, we offer an explanation for this phenomenon. Smaller coreset sizes suffice if we only wish to preserve the costs of \emph{good} solutions, i.e., solutions with low cost. We define and devise \emph{approximation-preserving coresets}, which provide a weaker guarantee than strong coresets, which apply to all solutions, while providing stronger guarantees than weak coresets, which apply only to the optimum solution. We complement this result by showing that even a very small distortion in the approximation factor cannot admit coresets of this size.
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GAM-RAG: Gain-Adaptive Memory for Evolving Retrieval in Retrieval-Augmented Generation
Yifan Wang, Mingxuan Jiang, Zhihao Sun, Yixin Cao, Yicun Liu, Keyang Chen, Guangnan Ye, Hongfeng Chai
Retrieval-Augmented Generation (RAG) grounds large language models with external evidence, but many implementations rely on pre-built indices that remain static after construction. Related queries therefore repeat similar multi-hop traversal, increasing latency and compute. Motivated by schema-based learning in cognitive neuroscience, we propose GAM-RAG, a training-free framework that accumulates retrieval experience from recurring or related queries and updates retrieval memory over time. GAM-RAG builds a lightweight, relation-free hierarchical index whose links capture potential co-occurrence rather than fixed semantic relations. During inference, successful retrieval episodes provide sentence-level feedback, updating sentence memories so evidence useful for similar reasoning types becomes easier to activate later. To balance stability and adaptability under noisy feedback, we introduce an uncertainty-aware, Kalman-inspired gain rule that jointly updates memory states and uncertainty estimates. It applies fast updates for reliable novel signals and conservative refinement for stable or noisy memories. We provide a theoretical analysis of the update dynamics, and empirically show that GAM-RAG improves average performance by 3.95\% over the strongest baseline and by 8.19\% with 5-turn memory, while reducing inference cost by 61\%.
****
Shuffle the Context: RoPE-Perturbed Self-Distillation for Long-Context Adaptation
Zichong Li, Chen Liang, Liliang Ren, Tuo Zhao, yelong shen, Weizhu Chen
Large language models (LLMs) increasingly operate in settings that require reliable long-context understanding, such as retrieval-augmented generation and multi-document reasoning. A common strategy is to fine-tune pretrained short-context models at the target sequence length. However, we find that standard long-context adaptation can remain brittle: model accuracy depends strongly on the absolute placement of relevant evidence, exhibiting high positional variance even when controlling for task format and difficulty.
We propose *RoPE-Perturbed Self-Distillation*, a training regularizer that improves positional robustness. The core idea is to form alternative ``views'' of the same training sequence by perturbing its RoPE indices---effectively moving parts of the context to different positions---and to train the model to produce consistent predictions across views via self-distillation. This encourages reliance on semantic signals instead of brittle position dependencies.
Experiments on long-context adaptation of Llama-3-8B and Qwen-3-4B demonstrate consistent gains on long-context benchmarks, including up to 12.04 percent-point improvement on RULER-64K for Llama-3-8B and 2.71 percent-point gain on RULER-256K for Qwen-3-4B after SFT, alongside improved length extrapolation beyond the tr

aining context window.
****
From Memorization to Parameter Interference: How Overtraining Experts Harms Model Merging
Stefan Horoi, Guy Wolf, Eugene Belilovsky, Gintare Karolina Dziugaite
Modern deep learning is increasingly characterized by the use of open-weight foundation models that can be fine-tuned on specialized datasets. This has led to a proliferation of expert models and adapters, often shared via platforms like HuggingFace and AdapterHub. Model merging has recently emerged as an effective way to leverage these existing resources, enabling the composition of capabilities from different model checkpoints. A natural pipeline has thus formed to harness the benefits of transfer learning and amortize sunk training costs: models are pre-trained on general data, fine-tuned on specific tasks, and then multiple checkpoints are merged to obtain a more capable model. A prevailing assumption is that improvements at one stage of this pipeline propagate downstream, leading to gains at subsequent steps. In this work, we challenge that assumption by examining how expert fine-tuning affects model merging. We show that long fine-tuning of experts that optimizes for their individual performance leads to degraded merging performance across vision and language modalities, multiple model scales, and both fully fine-tuned and LoRA-adapted models. We trace this degradation to the memorization of a small set of difficult examples that dominate late fine-tuning steps. This causes negative parameter interference and encodes knowledge that is forgotten during merging. Finally, we demonstrate that task-dependent aggressive early stopping strategies can significantly improve model merging performance.
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Search for Truth from Reasoning: A Dynamic Representation Editing Framework for Steering LLM Trajectories
Tianlong Wang, Yuhang Wang, Weibin Liao, Xin Gao, Xinyu Ma, Yang Lin, Yasha Wang, Liantao Ma
Current approaches to enhance Large Language Model (LLM) reasoning, such as Chain-of-Thought and "Wait" prompts, primarily encourage models to think more, yet often fail to guide them toward Truth. While Representation Editing (RepE) offers a intrinsic control, its application to dynamic reasoning trajectories remains underexplored. In this work, we bridge this gap by investigating the geometry of truth within unfolding reasoning chains. We uncover three critical insights: (1) Truth is encoded at the sentence level and is entangled with latent reasoning patterns; (2) Effective intervention follows an Uncertainty Principle and a Decay Effect, requiring localization to early, high-entropy forks; (3) Naive steering vectors suffer from noise, risking collateral damage to correct trajectories. Based on these findings, we propose DynaSteer, a dynamic RepE framework. DynaSteer employs pattern clustering to disentangle reasoning manifolds and utilizes Fisher-LDA to project purified truth. By dynamically monitoring lookahead entropy, it selectively steers and rolls back trajectories only when necessary. Comprehensive experimental results on several MATH benchmark verify the effectiveness of DynaSteer, and experiments on out-of-domain coding tasks further confirm its generalization ability. Our code is publicly available at https://github.com/tianlwang/DynaSteer.
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Demystifying When Pruning Works via Representation Hierarchies
Shwai He, Guoheng Sun, Haichao Zhang, Yun Fu, Ang Li
Network pruning, which removes less important parameters or architectures, is often expected to improve efficiency while preserving performance. However, this expectation does not consistently hold across language tasks: pruned models can perform well on non-generative tasks but frequently fail in generative settings. To demystify how such discrepancies arise under pruning, we analyze network pruning from a representation-hierarchy perspective, decomposing the internal computation of language models into three sequential spaces: \textit{embedding} (hidden representations), \textit{logit} (pre-softmax outputs), and \textit{probability} (post-softmax distributions).

While representations in the embedding and logit spaces are largely robust to pruning-induced perturbations, the subsequent nonlinear transformation from logits to the probability space amplifies such deviations, whose persistence across time steps leads to substantial degradation during generation.
By contrast, the stability of the categorical-token probability subspace, together with the robustness of the embedding space, supports the effectiveness of pruning for non-generative tasks such as retrieval and multiple-choice classification. Our representation-level analysis disentangles the effects of pruning across tasks and offers practical guidance for applying pruning effectively.
The code is available in the \href{https://github.com/CASE-Lab-UMD/Pruning-on-Representations}{project repository}.
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Intrinsic Gradient Suppression for Label-Noise Prompt Tuning in Vision–Language Models
Jia-yu Li, Jiaxin Qi, Sheng Zhou, Jianqiang Huang, Xian-Sheng Hua
Contrastive vision-language models like CLIP exhibit remarkable zero-shot generalization. However, prompt tuning remains highly sensitive to label noise, as mislabeled samples generate disproportionately large gradients that can overwhelm pre-trained priors.
We argue that because CLIP already provides a near-optimal initialization, adaptation should be inherently conservative, particularly against the extreme gradient updates common in noisy settings.
To this end, we propose Double-Softmax Prompt Tuning (DSPT), a hyperparameter-free method for intrinsic gradient suppression. By applying a sequential probabilistic normalization, DSPT induces a self-adaptive saturation zone that suppresses gradients from high-error noisy samples while maintaining informative updates.
We also provide both theoretical analysis and empirical evidence about how this mechanism achieves adaptive suppression.
This design transforms ``gradient vanishing'', traditionally a training bottleneck, into a principled noise-filtering shield for label-noise prompt tuning. Extensive experiments confirm that this simple, drop-in design achieves state-of-the-art robustness across various noisy benchmarks, outperforming methods with complex architectures and handcrafted hyperparameters.
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Beyond Single Embedding: Modeling User Preferences as Distribution in Federated Recommendation
Chunxu Zhang, Weipeng Zhang, Guodong Long, Zhiheng Xue, Bo Yang
Most federated recommender systems represent each user with a single embedding learned from local interaction data, implicitly assuming that user preferences are fixed and precisely identifiable. In federated settings, however, each client observes only a limited and fragmentary view of user behavior, rendering such point estimates inherently brittle. To address this mismatch, we model user preferences as distributions rather than points, allowing multiple compatible preference representations to coexist. Rather than collapsing evidence into a single embedding, our approach preserves uncertainty and diversity in user representations, providing a richer basis for preference modeling. We instantiate this idea with a diffusion-based generative framework that produces diverse user embeddings and derives recommendation scores by aggregating predictions across them. This distributional formulation yields more stable ranking behavior and improved robustness under ambiguous feedback. Extensive experiments on federated recommendation benchmark datasets demonstrate consistent and significant improvements over baselines. Our code is available.
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LittleBit-2: Maximizing the Spectral Energy Gain in Sub-1-Bit LLMs via Latent Geometry Alignment
Banseok Lee, Youngmin Kim
We identify the Spectral Energy Gain in extreme model compression, where low-rank binary approximations outperform tiny-rank floating-point baselines for heavy-tailed spectra. However, prior attempts fail to realize this potential, trailing state-of-the-art 1-bit methods. We attribute this degradation to Latent Geometr

y Misalignment: standard singular vectors exhibit high coherence (spiky distribu
tion), the worst-case geometry for binary quantization. To realize this gain, we
 propose LittleBit-2, a framework employing Internal Latent Rotation and Joint I
terative Quantization (Joint-ITQ). This approach acts as a geometric preconditio
ner, aligning coherent latent distributions with the binary hypercube with zero
 inference overhead. Empirically, LittleBit-2 establishes a new state-of-the-art
 in the sub-1-bit regime (1$\sim$0.1 bpp) on Llama-2 and Llama-3, matching the fi
delity of leading 1-bit baselines.
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Differentiable Optimization Layers for Guaranteed Fairness in Deep Learning
David Troxell, Noah Roemer, Guido Montufar
Differentiable optimization layers are traditionally integrated in predict-then-
optimize frameworks where a neural model estimates parameters that subsequently
 serve as fixed inputs to downstream decision-making optimization problems. In th
is work, we introduce the concept of a "fairness layer": a differentiable optimi
zation layer appended to a model's output layer that guarantees a chosen notion
 of output parity is satisfied when integrated into a neural network. Additionall
y, we introduce an online primal-dual inference algorithm that provides provable
 aggregate fairness guarantees for streaming predictions with arbitrarily small
 batch sizes, where traditional per-batch constraints become overly restrictive.
 Numerical experiments demonstrate the effectiveness of the fairness layer and as
sociated algorithm, and theoretical analysis characterizes the layer's different
iability and stability properties during model training and backpropagation. Our
 code for these experiments is publicly available on GitHub: https://github.com/
dtroxell19/FairDL-ICML-2026.git and our public Python package documentation can
 be found online: https://dtroxell19.github.io/fairness_training/.
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S^{3}GNN: Efficient Global Mixing and Local Message Passing for Long-Range Graph
 Learning
Dai Shi, Luke Thompson, Linhan Luo, Lequan Lin, Andi Han, Junbin Gao, José Migue
l Hernández-Lobato
Message-passing neural networks (MPNNs) often suffer from an information bottlen
eck when capturing long-range dependencies, leading to the oversquashing (OSQ) p
henomenon. Alongside spatial connectivity enrichment (e.g., rewiring), recent st
udies have shown that spectral filtering can yield strong long-range learning ou
tcomes, as spectral operators enable global information mixing that alleviates O
SQ. These approaches achieve this either by stabilizing the Jacobian energies in
 deep propagation or by guaranteeing OSQ mitigation under strong theoretical ass
umptions. We examine the practical attainability of these guarantees and show th
at the associated Jacobian sensitivity lower bound is generally difficult to ach
ieve in practice. We then propose S^{3}GNN, which mitigates OSQ without such res
trictive assumptions by lightweightly reintroducing omitted components with subs
tantially lower computational complexity, while standard stability constraints o
n feature transformations remain effective under our new dynamics. Extensive exp
eriments across diverse domains (e.g., long-range benchmarks, KGQA, and mesh-bas
ed fluid dynamics) demonstrate that S^{3}GNN achieves up to an order-of-magnitud
e error reduction with up to 50\% fewer parameters. Our code can be found in htt
ps://github.com/EEthanShi/S3-GNN.git.
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SE-GA: Memory-Augmented Self-Evolution for GUI Agents
Shilong Jin, Lanjun Wang, Zhuosheng Zhang
Autonomous Graphical User Interface (GUI) agents often struggle with multi-step
 tasks due to constrained context windows and static policies that fail to adapt
 to dynamic environments. To address these limitations, this work proposes the Se
lf-Evolving GUI Agent (SE-GA), a novel framework that integrates hierarchical me
mory structures with an iterative self-improvement mechanism. At the core of ou
r approach is Test-Time Memory Extension (TTME), which facilitates long-term pla
nning by dynamically retrieving episodic, semantic, and experiential memories to
 provide salient contexts during inference. To ensure continuous learning, we in
troduce Memory-Augmented Self-Evolution (MASE), which is a training pipeline tha

t adopts the data collected by TTME to stabilize and enhance the agent's foundat ional policy. Extensive evaluations across both offline and online benchmarks de monstrate SE-GA achieves state-of-the-art performance, reaching success rates of 89.0\% on ScreenSpot and 75.8\% on the challenging AndroidControl-High dataset. Furthermore, significant improvements on the AndroidWorld benchmark highlight t he superior generalization to dynamic environments.

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Correspondence Cognitive Learning for Multi-Modal Object Re-Identification

Chao Su, Shuying Li, Ruitao Pu, Dezhong Peng, Zhenwen Ren, Yuan Sun

Multi-modal object Re-Identification (ReID) aims to retrieve the same object acr oss different modalities by exploiting their complementary visual information. R ecent advances leverage Multi-modal Large Language Models (MLLMs) to generate de scriptive textual annotations as auxiliary supervision. However, existing approa ches usually adopt these generated texts directly, overlooking the varying corre spondence degrees between visual and textual modalities. Such neglect may lead t he model to treat strong- and weak-correspondence image–text pairs equally, limi ting its ability to learn discriminative associations and hindering effective op timization. To overcome this limitation, we propose a Correspondence Cognitive L earning (CCL) framework that explicitly models the correspondence degree and fac ilitates a progressive learning process from easy to hard pairs. CCL is composed of two synergistic modules. The Correspondence-Guided Semantic Refinement (CGSR) module dynamically refines visual representations using text semantics accordi ng to the correspondence difficulty estimated from the previous epoch, thereby e nhancing feature alignment under imperfect associations. The Cognitive-Driven Dy namic Optimization (CDDO) module presents a self-paced weighting mechanism that adaptively adjusts the optimization focus by emphasizing easy pairs at the early stage and gradually integrating harder ones as training evolves. Together, thes e modules enhance feature-level alignment and optimization adaptivity, yielding robust and discriminative multi-modal representations. Extensive experiments on three multi-modal object ReID benchmarks demonstrate the superior performance of our method.

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Spherical Steering: Geometry-Aware Activation Rotation for Language Models

Zejia You, Chunyuan Deng, Hanjie Chen

Inference-time steering offers a promising way to control language models (LMs) without retraining. However, standard approaches typically rely on activation ad dition, which inevitably alters the hidden-state magnitudes raising concerns abo ut representation collapse and degraded open-ended generation. In this work, we explore *Spherical Steering*, a training-free primitive that resolves this trade -off through activation rotation. Rather than shifting activations with a fixed vector, our method rotates them along a geodesic toward a target direction, pres erving signal integrity while steering toward the target concept. To further enh ance adaptivity, we incorporate a confidence gate that dynamically modulates ste ering strength based on input uncertainty. Extensive experiments across multiple -choice benchmarks demonstrate that Spherical Steering significantly outperforms addition-based baselines (notably by +10\% on TruthfulQA, COPA, and Storycloze) , while simultaneously maintaining the model’s general open-ended generation qua lity. This work highlights the value of geometric consistency, suggesting that n orm-preserving rotation is a robust and effective primitive for precise inferenc e-time control. The code is available at: https://github.com/chili-lab/Spherical -Steering.

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Reliable Thinking with Images

Haobin Li, Yutong Yang, Yijie Lin, Xiang Dai, Mouxing Yang, Xi Peng

As a multimodal extension of Chain-of-Thought (CoT), Thinking with Images (TWI) has emerged as a promising avenue to enhance the reasoning capability of Multi-m odal Large Language Models (MLLMs), which generates interleaved CoT by incorpora ting visual cues into the textual reasoning process. However, the success of exi sting TWI methods heavily relies on the assumption that interleaved CoTs are fau ltless, which is easily violated due to the complexity of multimodal understandi

ng. In this paper, we reveal and study a highly-practical yet under-explored pro blem in TWI, termed Noisy Thinking (NT). Specifically, NT refers to the imperfec t visual cues mining and answer reasoning process. As the saying goes, ``One mis take leads to another'', erroneous interleaved CoT would cause error accumulatio n, thus significantly degrading the performance. To solve NT, we propose a novel method dubbed Reliable Thinking with Images (RTWI). In brief, RTWI estimates th e reliability of visual cues and textual CoT in a unified text-centric manner an d then employs robust filtering and voting modules to prevent NT from contaminat ing final answer. Extensive experiments on seven benchmarks verify the effective ness of RTWI against NT.
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GOCM: Single-Step Graph Outlier Synthesis via Origin Consistency Model
Yifan Li, Zhihui Wang, Changmiao Wang, Guangxiao Ma, Peng Zhang
Supervised Graph Outlier Detection has long been constrained by severe class imb alance, and although recent diffusion-based augmentation methods have improved s ample quality, their practical utility is hindered by the high computational cos ts of multi-step iterative sampling and the stochasticity of the generation proc ess. To overcome these bottlenecks, we propose Graph Outlier Synthesis via Origi n Consistency Model (GOCM), a single-step graph outlier synthesis framework base d on a consistency model. Theoretically, we pioneer the Origin Consistency (OC) mechanism by employing an ``Interval-based Origin Inference'' strategy, which ma thematically derives a direct mapping from the noise trajectory to the data orig in, achieving robust and efficient single-step sample generation. Architecturall y, to address the complexity of heterogeneous graphs containing multiple relatio ns, we design the Multi-input Variational Graph Auto-Encoder (MiVGAE), which dec ouples intricate structures via relation-level message passing and cross-relatio n fusion, mapping them into a unified latent space, from which GOCM synthesizes high-quality outlier nodes. Extensive experiments on multiple real-world dataset s demonstrate that GOCM achieves superior detection performance with significant ly improved generation efficiency. The source code is publicly available at: htt ps://anonymous.4open.science/r/RFS-2026-EB63/.
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PhenoBrain: Phenotype-Conditioned Long-Range Communication for Multi-Modal Brain Network Analysis
Lingyuan Meng, KE LIANG, Hao Li, Meng Liu, Weijia Shi, Miaomiao Li, Yang Gao, Xi nwang Liu
Multi-modal brain network analysis aims to predict neuropsychiatric status from functional connectomes with heterogeneous phenotypes. However, most existing met hods treat phenotypes as auxiliary features and perform late fusion, implicitly assuming that the connectome representation should be learned in the same way re gardless of phenotype. However, in clinical neuroscience the same functional con nectivity pattern may support different conclusions under different phenotype co ntexts. To bridge this gap, we propose PhenoBrain, a novel framework for multi-m odal brain network analysis that injects phenotype information at the mechanism level rather than only at the classifier level. Specifically, we propose a pheno type-conditioned long-range routing mechanism, which learns a subject-specific m ulti-hop communication kernel to model long-range connectome interactions. Furth ermore, we propose a phenotypic-guided attention mechanism regulation method, wh ich uses phenotypic information as a conditional prior to regulate the learning process of attention in brain networks. To verify the effectiveness of our metho d, we constructed two multi-modal brain network analysis datasets based on open-source image data. Extensive experiments demonstrate that PhenoBrain achieves st ate-of-the-art performance.
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Adversarial Attacks and Robust Training for Hypergraph Neural Networks
Naheed Anjum Arafat, Debabrota Basu, Yulia Gel, Danda B. Rawat
Recent studies show that Hypergraph Neural Networks (HGNNs) are vulnerable to ad versarial attacks, while adversarial learning in the context of hypergraphs rema ins substantially under-investigated. In particular, all existing attacks on HGN Ns are white-box and customized for either structural or feature perturbation. B

ut in reality, the attacker might not have access to the target model parameters . Motivated by this knowledge gap, we propose a generic meta-objective-based lea rning framework, MeLA, that leverages the hypergraph Laplacian to conduct gray-b ox, structural, and feature perturbations under explicit perturbation budgets. I n contrast to the attack literature, there is no adversarial training mechanism for HGNNs to defend against such attacks. Hence, we propose a novel adversarial training mechanism for HGNNs to obtain a robust classifier. We further prove the convergence of our robust training. Extensive experiments across various HGNN m odels and datasets show that (a) our proposed attack is effective in poisoning a nd evasion settings, and (b) our adversarial training enhances defense against a dversarial attacks.
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New Bounds for Kernel Sums via Fast Spherical Embeddings
Tal Wagner
We study query time bounds for the fundamental problem of estimating the kernel mean $\frac1{|X|}\sum_{x\in X}\mathbf{\mathrm{k}}(x,y)$ of a query y in a fini te dataset $X\subset\mathbb{R}^d$ up to a prescribed additive error $\varepsilon $. The best known bounds for the Gaussian kernel are $O(d/\varepsilon^2)$, $\wid etilde O(d+1/\varepsilon^4)$, and $\widetilde O(d+\Delta^2/\varepsilon^2)$, wher e Δ is the diameter of a region containing the points. We prove the new b ound $\tilde O(d+\varepsilon\Delta^2+1/\varepsilon^3)$, which improves over the previous ones in regimes with small error ε and intermediate diamete r Δ.

At the center of our proof is a new fast spherical embedding theorem in the sens e introduced by Bartal, Recht and Schulman (2011), which limits the embedded dat a diameter while preserving local Euclidean distances and avoiding ``distance co llapse'' at larger scales. This fast embedding theorem may be of independent int erest.
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Bounded Hyperbolic Tangent: A Stable and Efficient Alternative to Pre-Layer Norm alization in Large Language Models
Hoyoon Byun, Youngjun Choi, Taero Kim, Sungrae Park, Kyungwoo Song
Pre-Layer Normalization (Pre-LN) is the de facto choice for large language model s (LLMs) and is crucial for stable pretraining and effective transfer learning. However, Pre-LN incurs repeated statistical-computation overhead and remains vul nerable to the curse of depth, where hidden-state magnitudes and variances grow as the number of layers increases, destabilizing training. Efficiency-oriented n ormalization-free methods such as Dynamic Tanh (DyT) improve throughput but rema in fragile at depth. To jointly address stability and efficiency, we propose Bou nded Hyperbolic Tanh (BHyT), a drop-in replacement for Pre-LN. BHyT combines a t anh nonlinearity with explicit, data-driven input bounding to keep activations w ithin a non-saturating range. It prevents depth-wise growth in activation magnit ude and variance and provides a theoretical stability guarantee. For efficiency, BHyT computes exact statistics once per block and replaces a second normalizati on with a lightweight variance approximation. Empirically, BHyT demonstrates imp roved stability and efficiency during pretraining, achieving an average of 1.6\% faster training and an average of 1.77\% higher token generation throughput com pared to RMSNorm, while maintaining strong pretraining-only and post-SFT perform ance across language understanding and reasoning benchmarks\footnote{Code is ava ilable at: \url{https://github.com/MLAI-Yonsei/BHyT}}.
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Less Is More: Elevating RAG via Performance-Driven Context Compression
Ziqiang Cui, Yunpeng Weng, Xing Tang, Peiyang Liu, Shiwei Li, Bowei He, Jiamin C hen, Yansen Zhang, Xiuqiang He, Rui Zhang, Chen Ma
Retrieval-Augmented Generation (RAG) has emerged as a promising paradigm for imp roving the timeliness of knowledge updates and the factual accuracy of large lan guage models. However, incorporating a large volume of retrieved documents signi ficantly increases input length, leading to prohibitive computational costs. Exi sting compression approaches often compromise task performance, primarily due to

their reliance on predefined heuristics. These heuristics fail to ensure that the compressed context is conducive to the generation tasks. To address these limitations, we propose CORE-RAG, a novel framework for context compression in RAG systems.
CORE eliminates reliance on proxy heuristics through a performance-driven learning framework, which directy utilizes task performance as a feedback signal to iteratively refine the compressor policy. Prior to this optimization process, we incorporate a knowledge distillation phase to initialize the compressor with a robust policy. Extensive experiments demonstrate the superiority of our approach. At a high compression ratio of 3\%, CORE not only avoids performance degradation but also improves the average Exact Match (EM) score by 3.3 points compared to using full documents. Our code is available at https://github.com/ziqiangcui/CORE-RAG-ICML26.

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Data Reconstruction: Identifiability and Optimization with Sample Splitting
Yujie Shen, Zihan Wang, Jian Qian, Qi Lei
Training data reconstruction from KKT conditions has shown striking empirical success, yet it remains unclear when the resulting KKT equations have unique solutions and, even in identifiable regimes, how to reliably recover solutions by optimization. This work hereby focuses on these two complementary questions: identifiability and optimization. On the identifiability side, we discuss the sufficient conditions for KKT system of two-layer networks with polynomial activations to uniquely determine the training data, providing a theoretical explanation of when and why reconstruction is possible. On the optimization side, we introduce sample splitting, a curvature-aware refinement step applicable to general reconstruction objectives (not limited to KKT-based formulations): it creates additional descent directions to escape poor stationary points and refine solutions. Experiments demonstrate that augmenting several existing reconstruction methods with sample splitting consistently improves reconstruction performance. Code is available at https://github.com/mukerr/data_recon.

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Correcting in Hindsight: Editing Past Key-Value States for Robust LLM Reasoning
Mengfei Zhang, Yu Mi, Leijing Zhou
Autoregressive Large Language Models (LLMs) often fail in complex reasoning because early-stage errors remain uncorrectable in subsequent steps—a limitation fundamentally rooted in the inherent irreversibility of the Transformer architecture. In this paper, we propose HEdit, a lightweight reasoning enhancement paradigm that equips models with a "hindsight-like" capability for dynamic error correction during generation. Our core insight involves deconstructing reasoning failures into two pivotal stages: latent representational biases emerging at logical anchors, and the subsequent eruption of explicit cognitive dissonance at trigger points. Based on these observations, the HEdit framework detects internal inconsistency signals at trigger points in real-time, actively backtracks to critical anchors, and utilizes a lightweight trainable editor to precisely refine their Key-Value (KV) caches. This mechanism effectively breaks the unidirectional constraints of autoregressive inference. Empirical results demonstrate that HEdit significantly enhances the performance of various models on mathematical reasoning tasks—with average accuracy improvements ranging from 2.2\% to 10.8\%—while maintaining extremely low overhead (add parameters $<0.5\%$). HEdit provides a dynamic, pluggable and lightweight solution, making it particularly beneficial for users in low-resource environments.

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Maximin Relative Improvement: Fair Learning as a Bargaining Problem
Jiwoo Han, Moulinath Banerjee, Yuekai Sun
When deploying a single predictor across multiple subpopulations, we propose a fundamentally different approach: interpreting group fairness as a bargaining problem among subpopulations. This game-theoretic perspective reveals that existing robust optimization methods such as minimizing worst-group loss or regret correspond to classical bargaining solutions and embody different fairness principles

. We propose relative improvement, the ratio of actual risk reduction to potenti
al reduction from a baseline predictor, which recovers the Kalai–Smorodinsky sol
ution. Unlike absolute-scale methods that may not be comparable when groups have
 different potential predictability, relative improvement provides axiomatic jus
tification including scale invariance and individual monotonicity. We establish
 finite-sample convergence guarantees under mild conditions.
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Thinking in Flow: A Dissipative Stabilization Operator for Robust Autoregressive
 Reasoning

Yujie Huang, Wenwu He, Zhuo-Xu Cui

Chain-of-Thought (CoT) prompting enables multi-step reasoning in large language
 models, yet long-horizon generation remains brittle under distribution shift and
 context interference: irrelevant cues persist, small deviations compound into i
nference drift, and late-stage corrections can destabilize the trajectory. We re
cast autoregressive decoding as a perturbed long-horizon dynamical system and in
troduce an *inference-time stabilization operator* that targets *trajectory-leve
l* reliability rather than token-level fluency. Specifically, we propose *ODE-gu
ided language models*, which augment a base Transformer with a persistent contin
uous-time *thought state* whose dynamics are explicitly designed to be dissipati
ve, enabling stable evidence accumulation with controlled forgetting. Instantiat
ing this framework, *Thinking in Flow* (TiF) equips the model with a lightweight
 Neural ODE controller and injects its output through post-norm residual updates
 to achieve numerically stable, low-intrusion steering. A demand--supply (uncert
ainty--capacity) gate determines *when* intervention is warranted, while a direc
tion gate determines *how* to steer in representation space, yielding selective,
 do-no-harm corrections instead of persistent bias. We establish well-posedness,
 dissipativity, and incremental stability of the controlled thought dynamics, im
plying bounded interventions over arbitrarily long contexts, and empirically dem
onstrate improved robustness to distractions and semantic perturbations, while m
atching or improving accuracy on mathematical reasoning benchmarks across both t
he Llama and Qwen model families; we further observe gains on non-mathematical B
BH reasoning tasks when training TiF on Llama.
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Think in Cloud, Look at Edges: Semantic-Driven Query Decomposition for Efficient
 Video Reasoning

Wenhao Zou, Zhijie Cai, Minchen Yu, Zongshuai Zhang, Guangxu Zhu

Long video understanding faces a critical dilemma: cloud-based Large Multimodal
Models (LMMs) offer superior reasoning but suffer from prohibitive bandwidth cos
ts and latency, while edge-based solutions sacrifice perception accuracy for spe
ed. Current collaborative approaches attempt to bridge this gap via similarity-b
ased filtering, yet they treat complex queries as flat semantic vectors. We iden
tify this as a fundamental flaw leading to "Semantic Submergence," where dominan
t visual features drown out subtle but logically critical cues. To solve this, w
e introduce SCOPE (Semantic Cloud-Orchestrated Perception at Edge). Shifting the
 paradigm to "Think in Cloud, Look at Edges," SCOPE utilizes a cloud LMM to deco
mpose complex queries into a structured Directed Acyclic Graph (DAG). This "obse
rvation plan" guides the edge to retrieve evidence based on logical necessity ra
ther than mere statistical similarity. Experiments on Video-MME and LongVideoBen
ch demonstrate that SCOPE redefines the Pareto frontier, matching cloud-level ac
curacy with significantly lower transmission costs and outperforming state-of-th
e-art baselines on complex reasoning tasks.
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Predictable Compression Failures: Order Sensitivity and Information Budgeting fo
r Evidence-Grounded Binary Adjudication

Leon Chlon, Ahmed Karim, MarcAntonio Awada

Transformers used for evidence-grounded binary adjudication (e.g., support/refut
e, yes/no, or verifier-backed pass/fail decisions) can be sensitive to the order
 in which exchangeable evidence is presented, producing dispersion across permut
ations and unreliable attempted answers under a verifier-relative Bernoulli pred
icate. We treat evidence order as a nuisance variable and formalize an expectati

on–realization gap: next-token training can minimize expected conditional descri ption length over orderings while a fixed ordering remains position-sensitive. O ur Quantified Martingale Violation (QMV) bound predicts the dispersion induced b y adjacent-rank positional sensitivity, with $O(\log n)$ growth in the harmonic regime; our Expectation-level Decompression Law (EDFL) specializes a KL convexit y/data-processing bound to Bernoulli predicates, yielding Bits-to-Trust (B2T), R isk-of-Hallucination (RoH), and an Information Sufficiency Ratio (ISR) gate for answer/abstain decisions. On 3,059 grounded items from FEVER, HotpotQA, NQ-Open, PopQA, and Controls, we observe logarithmic dispersion and positive Jensen gain s from uniform permutation mixtures. In one pre-specified held-out audit (528 it ems), the analytically fixed $ISR=1$ gate attains 0.0–0.7% hallucination with 20 .6–27.9% abstention (95% CIs), supporting the operating point without claiming u niversal calibration across all model families or unrestricted generation.
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Contextualized Visual Personalization in Vision-Language Models
Yeongtak Oh, Sangwon Yu, Junsung Park, Han Cheol Moon, Jisoo Mok, Sungroh Yoon
Despite recent progress in vision-language models (VLMs), existing approaches of ten fail to generate personalized responses based on the user's specific experie nces, as they lack the ability to associate visual inputs with a user’s accumula ted visual-textual context. We newly formalize this challenge as contextualized visual personalization, which requires the visual recognition and textual retrie val of personalized visual experiences by VLMs when interpreting new images. To address this issue, we propose CoViP, a unified framework that treats personaliz ed image captioning as a core task for contextualized visual personalization and improves this capability through reinforcement-learning-based post-training and caption-augmented generation. We further introduce diagnostic evaluations that explicitly rule out textual shortcut solutions and verify whether VLMs truly lev erage visual context. Extensive experiments demonstrate that existing open-sourc e and proprietary VLMs exhibit substantial limitations, while CoViP not only imp roves personalized image captioning but also yields holistic gains across downst ream personalization tasks. These results highlight CoViP as a crucial stage for enabling robust and generalizable contextualized visual personalization.
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On the Convergence Rate of LoRA Gradient Descent
Siqiao Mu, Diego Klabjan
The low-rank adaptation (LoRA) algorithm for fine-tuning large models has grown popular in recent years due to its remarkable performance and low computational requirements. LoRA trains two "adapter" matrices that form a low-rank representa tion of the model parameters, thereby massively reducing the number of parameter s that need to be updated at every step. Although LoRA is simple, its convergenc e is poorly understood due to the lack of Lipschitz smoothness, a key condition for classic convergence analyses. As a result, current theoretical results only consider asymptotic behavior or assume strong boundedness conditions which artif icially enforce Lipschitz smoothness. In this work, we provide for the first tim e a non-asymptotic convergence analysis of the *original LoRA gradient descent* algorithm, which reflects widespread practice, without such assumptions. Our wor k relies on three key steps: i) reformulating the problem in terms of the outer product of the stacked adapter matrices, ii) a modified descent lemma for the "L ipschitz-like" reparametrized function, and iii) controlling the step size. With this approach, we prove that LoRA gradient descent converges to a stationary po int at rate $O(\frac{1}{\log T})$, where T is the number of iterations.
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Marrying Generative Model of Healthcare Events with Digital Twin of Social Deter minants of Health for Disease Reasoning
Ziquan Wei, Tingting Dan, Guorong Wu
Despite the central role of sensor-derived measurements such as imaging traits a nd plasma biomarkers in biomedical research and clinical practice, existing gene rative models for disease prediction largely depend on event-level representatio ns from hospital and registry data. Given the multi-factorial nature of human di sease, the absence of explicit modeling of social determinants of health (SDoH)

limits the capacity for personalized disease modeling and clinical decision support.
To address this limitation, we propose a generative model with ICD-coded proxies of SDoH for \textit{in silico} modeling of disease reasoning, a conditioned latent diffusion framework that establishes the connection between multi-organ sensor data with tokenized healthcare events. Specifically, we introduce a novel geometric diffusion model to characterize the temporal evolution of complex data representation such as brain networks (region-to-region connectivity encoded in a graph), in parallel with diffusion models for tabular data from other organ systems.
Together, we integrate the generative model with digitalized SDoH proxies (coined **DiffDT**) for simulated intervention and reasoning of future disease trajectories.
We conduct extensive experiments on the UK Biobank (UKB) dataset, which contains organ-specific imaging traits, including brain (44,834), heart (23,987), liver (28,722), and kidney (32,155), along with nearly 500k medical history sequences (age range: 25$\sim$89 years). Our **DiffDT** achieves significant improvements over state-of-the-art human disease autoregressive models and imaging trait generative baselines.
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When and How Human Curation Backfires: Preference Alignment under Multi-Model Self-Consuming Loop
Yang Zhang, Xiukun Wei, Xueru Zhang
Foundation models are increasingly trained on synthetic data generated by prior model iterations rather than exclusively on real data. This *self-consuming* training paradigm can lead to model collapse, divergence, or bias amplification. Recent work (Ferbach et al., 2024) shows that incorporating human curation into the loop can steer a self-consuming model toward human-aligned behavior, but these analyses focus on a single, isolated model that solely consumes its own outputs. In practice, however, models often interact and train on input output pairs produced by other models. This paper studies self-consuming training in the *multi-model* regime. We first formalize a framework for interacting self-consuming models and characterize when the resulting dynamical system converges to a stable point. We then examine how human curation of one model affects its own alignment (self-influence) and how such effects propagate to other models (cross-influence). Unlike isolated settings where human curation always enhances model alignment, we show that cross-model interactions can dampen or even invert this effect, ultimately degrading long-term alignment.
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BAT: Better Audio Transformer Guided by Convex Gated Probing
Houtan Ghaffari, Lukas Rauch, Christoph Scholz, Paul Devos
Probing is widely adopted in computer vision to faithfully evaluate self-supervised learning (SSL) embeddings, as finetuning may misrepresent their inherent quality. In contrast, audio SSL models still rely on finetuning because simple probing fails to unlock their full potential and alters their rankings when competing on AudioSet. Hence, a robust and efficient probing mechanism is required to guide the trajectory of audio SSL towards reliable and reproducible methods. We introduce *Convex Gated Probing* (CGP), a prototype-based method that significantly closes the gap between finetuning and probing in audio. CGP efficiently utilizes all frozen layers via a gating mechanism and exposes the location of latent task-relevant information. Guided by CGP as a reliable post-hoc evaluation probe, we rework the entire SSL pipeline of current best performing audio models that use legacy implementations of prior SSL methods. By refining data preprocessing, model architecture, and pretraining recipe, we introduce *Better Audio Transformer* (BAT), and establish new SOTA on audio benchmarks.
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Beyond Drift: Stabilizing Subjective LLM Evaluation with Information-Theoretic Rubrics
Wang Xu, Shan Zhao, Haolan Chen, Tianwei Yan, Chengyu Wang, Shizhao Chen, Qian Wan

Despite the growing use of large language models (LLMs) in subjective tasks such as role-playing, humor, emotional intelligence, and dialogue quality, their evaluation faces a pressing reproducibility crisis: even the same evaluator may contradict itself when re-judging the exact same sample.
We attribute this instability to dimension drift, where free-form evaluation protocols (e.g., Chain-of-Thought reasoning) unpredictably shift their implicit criteria, undermining reliability.
To address this fundamental challenge, we reformulate subjective evaluation as an information-theoretic optimization problem. Specifically, we propose an **Expected Information Gain (EIG)-based framework** that constructs a stable yet adaptive personalized rubric to eliminate dimension drift.
Our two-stage “generate–then–score” design first produces a diverse pool of candidate evaluation questions and then selects the most informative subset via EIG, yielding explicit and repeatable criteria.
Experiments on six benchmarks, including CharacterEval, The rJokes, and MT_bench, demonstrate that our approach substantially improves both evaluation consistency and alignment with human judgments, outperforming CoT-based and fixed-questionnaire baselines.
These results highlight that information-theoretic questionnaire construction offers a principled and reliable path toward reproducible evaluation of subjective tasks.
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Training-Free Multimodal Large Language Model Orchestration
Tianyu Xie, Yuexiao Ma, Yuhang Wu, Wang Chen, Jiayi Ji, Tat-Seng Chua, Xiawu Zheng, Rongrong Ji
Building interactive omni-modal assistants often relies on end-to-end multimodal alignment to fuse heterogeneous modalities, which incurs substantial data and compute costs and limits extensibility. We present Training-Free Large Language Model Orchestration (LLM Orchestration), a training-free orchestration framework that integrates off-the-shelf modality experts into a unified multimodal input--output system without additional gradient-based training for integration. LLM Orchestration comprises three components: (1) an LLM controller that infers user intent and emits explicit control tokens for expert selection and sequencing, enabling protocol-constrained and auditable routing; (2) a text-centric cross-modal memory that compresses multimodal evidence into structured records for lightweight retrieval and reuse, reducing redundant expert invocations across turns; and (3) a unified interaction layer that executes routing and memory decisions to support consistent modality transitions, full-duplex streaming, and interruption-aware dialogue. Across diverse multimodal benchmarks, LLM Orchestration achieves strong performance under standard evaluation constraints while maintaining low orchestration overhead and modular upgradeability, providing a practical alternative to costly joint training for omni-modal systems.
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SelfJudge: Faster Speculative Decoding via Self-Supervised Judge Verification
Kanghoon Yoon, Minsub Kim, Sungjae Lee, Joonhyung Lee, Sunghyeon Woo, Yeonjun In, Se Jung Kwon, Chanyoung Park, Dongsoo Lee
Speculative decoding accelerates LLM inference by verifying candidate tokens from a draft model against a larger target model. Recent "judge'' decoding boosts this process by relaxing verification criteria by accepting draft tokens that may exhibit minor discrepancies from target model output, but existing methods are restricted by their reliance on human annotations or tasks with verifiable ground truths, limiting generalizability across diverse NLP tasks. We propose SelfJudge, which trains judge verifiers via self-supervision of the target model. Our method measures semantic preservation by assessing whether token-substituted responses preserve the meaning of original responses, enabling automatic verifier training across diverse NLP tasks. Our experiments show SelfJudge achieves superior inference-accuracy trade-offs than judge decoding baselines, offering a broadly applicable solution for faster LLM inference.
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Certificate-Guided Pruning for Stochastic Lipschitz Optimization

Ibne Farabi Shihab, Sanjeda Akter, Anuj Sharma
We study black-box optimization of Lipschitz functions under noisy evaluations. Existing adaptive discretization methods implicitly avoid suboptimal regions but do not provide explicit certificates of optimality or measurable progress guarantees. We introduce **Certificate-Guided Pruning (CGP)**, which maintains an explicit *active set* A_t of potentially optimal points via confidence-adjusted Lipschitz envelopes. Any point outside A_t is certifiably suboptimal with high probability, and under a margin condition with near-optimality dimension α, we prove Vol(A_t) shrinks at a controlled rate yielding sample complexity $Õ(\varepsilon^{-(2+\alpha)})$. We develop three extensions: CGP-Adaptive learns L online with $O(\log T)$ overhead; CGP-TR scales to $d > 50$ via trust regions with local certificates; and CGP-Hybrid switches to GP refinement when local smoothness is detected. Experiments on 12 benchmarks ($d \in [2, 100]$) show CGP variants match or exceed strong baselines while providing principled stopping criteria via the computable gap proxy ε_t.
****
EduMirror: Modeling Educational Social Dynamics with Value-driven Multi-agent Simulation
Jingzhe Lin, Hengbin Yu, Yongdan Zeng, Fangwei Zhong
Understanding how educational social dynamics evolve is critical for informing effective educational policies and counterfactual interventions. However, traditional methods face a fundamental dilemma: observational studies often lack causal power, while controlled experiments are frequently constrained by ethical concerns. Although LLM-based multi-agent simulations offer a scalable **in silico** alternative, existing approaches remain limited by weak psychological grounding and insufficient measurement of latent psychological states. To address this, we introduce **EduMirror**, a multi-agent simulator for the scientific study of educational social dynamics. We provide configurable education-oriented agent forms, including value-driven agents grounded in psychological needs and social value orientation, together with a dual-track measurement protocol for quantifying observable behaviors and latent psychological states. We validate the realism and usability of EduMirror through case studies on school bullying and group cooperation, as well as broader evaluations across diverse educational scenarios. The results show that EduMirror generates educational social dynamics that are realistic, theory-consistent, and measurable by empirical criteria. These properties enable structured **in silico** educational research, providing a computational tool for hypothesis testing and counterfactual intervention analysis in educational science. Project page: *https://edumirror.net*.
****
Select to Think: Unlocking SLM Potential with Local Sufficiency
Wenxuan Ye, Yangyang Zhang, Xueli An, Georg Carle, Yunpu Ma
Small language models (SLMs) offer efficient deployment, yet they often lag behind their larger counterparts (LLMs) in reasoning.
Existing remedies either invoke an LLM at points of reasoning divergence, incurring substantial latency and cost, or rely on standard distillation, which is limited by the SLM's capacity to accurately mimic the LLM's complex generative distribution.
We address this dilemma by identifying local sufficiency: at divergence points, the LLM's preferred token often resides within the SLM's top-K next-token predictions, even when failing to emerge as the SLM top-1 choice.
We therefore propose Select to Think (S2T), which reframes the LLM’s role from open-ended generation to selection among the SLM's proposals, simplifying the supervision signal to discrete candidate rankings.
Leveraging this, we introduce S2T-Local, which distills the selection logic into the SLM, empowering it to perform autonomous re-ranking without inference-time LLM dependency.
Empirically, a 1.5B SLM's top-8 candidates contain the 32B LLM's choice with a 95\% hit rate, and S2T-Local improves the 1.5B SLM's Math Avg. over greedy decoding by 24.1\% relative gain, matching the efficacy of 8-path self-consistency with single-trajectory efficiency.

Code is available at https://github.com/YeRona/Select-to-Think.
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MiVE: Multiscale Vision-language features for reference-guided video Editing
Tong Wang, Meng Zou, WU CHENGJING, Xiaochao Qu, Luoqi Liu, Xiaolin Hu, Ting Liu
Reference-guided video editing takes a source video, a text instruction, and a reference image as inputs, requiring the model to faithfully apply the instructed edits while preserving original motion and unedited content. Existing methods fall into two paradigms, each with inherent limitations: decoupled encoders suffer from modality gaps when processing instructions and visual content independently, while unified vision-language encoders lose fine-grained spatial details by relying solely on final-layer representations. We observe that VLM layers encode complementary information hierarchically—early layers capture localized spatial details essential for precise editing, while deeper layers encode global semantics for instruction comprehension. Building on this insight, we present \textbf{MiVE} (\textbf{M}ult\textbf{i}scale \textbf{V}ision-language features for reference-guided video \textbf{E}diting), a framework that repurposes VLMs as multiscale feature extractors. MiVE extracts hierarchical features from Qwen3-VL and integrates them into a unified self-attention Diffusion Transformer, eliminating the modality mismatch inherent in cross-attention designs. Experiments demonstrate that MiVE achieves state-of-the-art performance by ranking highest in human preference, outperforming both academic methods and commercial systems.
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Statistically Calibrated Scaling for Token Merging in Transformers
Qing Zhou, Hongyuan Zhang, Tao Yang, Junyu Gao, Qi Wang
Token merging accelerates Transformer inference by clustering similar tokens to reduce sequence length (retention ratio r), but distorts attention outputs, inducing covariate shift in residual streams and performance collapse under high compression. Existing heuristics, such as proportional attention, mitigate mild compression effectively but degrade sharply at aggressive ratios due to unaddressed energy drift and biased attention distributions.

We reframe token merging as a statistical reconstruction problem in high dimensions and introduce an asymptotic radial-angular decomposition of the reconstruction error, an analytical framework decoupling magnitude and distributional distortions. Minimizing this decomposed risk under minimal assumptions of finite second moments and variance stationarity yields closed-form optimal corrections governed by a single scaling factor $\sqrt{r}$: scaling merged values and shrinking merged logits toward the cluster-size prior. This calibrates both energy balance and distributional fidelity. Extensive experiments on vision Transformers demonstrate superior accuracy and robustness across compression levels.
****
CACR: Reinforcing Temporal Answer Grounding in Instructional Video via Candidate-Aware Causal Reasoning
Muge Qi, Rong Fu, Pengbin Feng, Xianda Li, Yu Cai, Yifu Guo, Shizhe Zhang, Simon James Fong, Lei Ma, Bin Li
The task of temporal answer grounding in instructional videos (TAGV), which aims to locate precise video segments that respond to natural language queries, is increasingly important for direct video answer retrieval. This task remains challenging due to the need to comprehend semantically complex questions and to address the significant length mismatch between untrimmed videos and short target moments. Existing methods often suffer from sensitivity to irrelevant content or insufficient visual reasoning capabilities. To tackle these limitations, we propose a Candidate-Aware Causal Reasoning (CACR) framework. Our approach first employs a Visual-Language Pre-training based Candidate Selection (VBCS) algorithm to efficiently generate K candidate segments, then applies a temporal logic reasoning module enhanced by a rejection reward mechanism and optimized via Group Relative Policy Optimization (GRPO) for robust inference. Extensive experiments on six benchmarks demonstrate that our method achieves state-of-the-art performance in terms of mean Intersection-over-Union (mIoU), providing a new perspective for reasoning-based retrieval in long videos.
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Autoregressive Language Models are Secretly Energy-Based Models: Insights into t
he Lookahead Capabilities of Next-Token Prediction
Mathieu Blondel, Michael Eli Sander, Germain Vivier-Ardisson, Tianlin Liu, Vince
nt Roulet
Autoregressive models (ARMs) currently constitute the dominant paradigm for
large language models (LLMs). Energy-based models (EBMs) represent another class
of models, which have historically been less prevalent in LLM development, yet
naturally characterize the optimal policy in post-training alignment. In this
paper, we present a unified view of these two model classes. Taking the chain
rule of probability as a starting point, we establish an explicit bijection
between ARMs and EBMs in function space, which we show to correspond to a
special case of the soft Bellman equation in maximum entropy reinforcement
learning.
Building upon this bijection, we derive the equivalence between
supervised learning of ARMs and EBMs. Furthermore, we analyze the
distillation of EBMs into ARMs by providing theoretical error bounds.
Our results provide insights into the ability of ARMs to plan ahead, despite
being based on the next-token prediction paradigm.
****
Unlearning in Diffusion Models: A Unified Framework with KL Divergence and Likel
ihood Constraints
Shervin Khalafi, Alejandro Ribeiro, Dongsheng Ding
Unlearning in diffusion models aims to remove undesirable data or concepts while
 preserving the utility of pretrained models---two fundamentally conflicting obj
ectives. We propose a principled constrained optimization framework that formula
tes unlearning as minimizing the deviation from a pretrained model, subject to e
xplicit separation constraints from the unlearning distributions. Specifically,
we formulate three constrained optimization problems based on reverse and forwar
d KL divergences, and likelihood constraints. The first two generalize existing
approaches for concept and data unlearning, while the third offers a novel and n
atural formulation for unlearning. Despite the nonconvexity of the KL constraint
s, we establish strong duality for all three problems, enabling us to explicitly
 characterize their optimal solutions as unlearning targets and develop primal–d
ual algorithms for each formulation. Experimental results demonstrate that our K
L-constrained approach achieves superior retention-unlearning tradeoffs compared
 to weight-based baselines for concept and data unlearning, and that our likelih
ood-based approach matches unlearning effectiveness while better preserving reta
ined concepts compared to baselines.
****
QUATRO: Query-Adaptive Trust Region Policy Optimization for LLM Fine-tuning
Doyeon Lee, Eunyi Lyou, Hyunsoo Cho, Soo Kyung Kim, Joonseok Lee, Jaemoo Choi
GRPO-style reinforcement learning (RL)-based LLM fine-tuning algorithms have rec
ently gained popularity. Relying on heuristic trust-region approximations, howev
er, they can lead to brittle optimization behavior, as global importance-ratio c
lipping and group-wise normalization fail to regulate samples whose importance r
atios fall outside the clipping range. We propose Query-Adaptive Trust-Region po
licy Optimization (QUATRO), which directly enforces trust-region constraints thr
ough a principled optimization. This yields a clear and interpretable objective
that enables explicit control over policy updates and stable, entropy-controlled
 optimization, with a stabilizer terms arising intrinsically from the exact trus
t-region formulation. Empirically verified on diverse mathematical reasoning ben
chmarks, QUATRO shows stable training under increased policy staleness and aggre
ssive learning rates, maintaining well-controlled entropy throughout training.
****
DR Tulu: Reinforcement Learning with Evolving Rubrics for Deep Research
Rulin Shao, Akari Asai, Shannon Zejiang Shen, Hamish Ivison, Varsha Kishore, Jin
gming Zhuo, Xinran Zhao, Molly Park, Samuel G. Finlayson, David Sontag, Tyler Mu
rray, Sewon Min, Pradeep Dasigi, Luca Soldaini, Faeze Brahman, Wen-tau Yih, Tong
shuang Wu, Luke Zettlemoyer, Yoon Kim, Hannaneh Hajishirzi, Pang Wei Koh
Deep research agents perform multi-step research to produce long-form, well-attr

ibuted answers. However, most open deep research agents are trained on easily ve rifiable short-form QA tasks via reinforcement learning with verifiable rewards, which does not extend to realistic long-form tasks. We address this with Reinfo rcement Learning with Evolving Rubrics (RLER), where rubrics are constructed and maintained to co-evolve with the policy model during training. This allows the rubrics to incorporate newly explored information from search and contrasting mo del responses, enabling better fact checking and more discriminative on-policy f eedback. Using RLER, we develop Deep Research Tulu (DR Tulu-8B), the first fully open model that is directly trained for open-ended, long-form deep research. Ac ross four long-form deep research benchmarks in science, healthcare, and general domains, DR Tulu-8B substantially outperforms existing open deep research agent s (by 15.6% over Tongyi DR on average) and matches or exceeds proprietary deep r esearch agents (by 0.7% over OpenAI DR on average), while being significantly sm aller and cheaper per query (1000x cheaper than OpenAI DR per query).

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RADIO1D: Elastic Representations for Condensed Vision Modeling

Greg Heinrich, Mike Ranzinger, Collin McCarthy, Natan Bagrov, Eugene Khvedchenya , Bryan Catanzaro, Jan Kautz, Andrew Tao, Pavlo Molchanov

This paper challenges the assumption that vision-language models (VLMs) require fixed patch-based 2D vision features. Analyzing fine-tuned vision encoders, we f ind that representations become increasingly abstract and less spatially coheren t during VLM training. Notably, models trained with image-text alignment (such a s SigLIP2) develop a small number of specialized tokens that effectively summari ze global image content. Building on this, we introduce RADIO1D, which compresse s images into a compact, variable-length 1D token sequence using multi-teacher k nowledge distillation and an autoencoder design. The resulting representations e xhibit strong hierarchical summarization, enabling accurate scene understanding– even with a single token–and support improved composition-aware image retrieval. In VLMs, RADIO1D provides flexible accuracy-efficiency tradeoffs through adjust able token counts, delivering competitive performance on diverse multimodal benc hmarks with lower computational overhead and better accuracy. We release our mod els under a permissive license.

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Learning to Discover at Test Time

Mert Yuksekgonul, Daniel Koceja, Xinhao Li, Federico Bianchi, Jed McCaleb, Xiaol ong Wang, Jan Kautz, Yejin Choi, James Zou, Carlos Guestrin, Yu Sun

How can we use AI to discover a new state of the art for a scientific problem? P rior work in test-time scaling, such as AlphaEvolve, performs search by promptin g a frozen LLM. We perform reinforcement learning at test time, so the LLM can c ontinue to train, but now with experience specific to the test problem. This for m of continual learning is quite special, because its goal is to produce one gre at solution rather than many good ones on average, and to solve this very proble m rather than generalize to other problems. Therefore, our learning objective an d search subroutine are designed to prioritize the most promising solutions. We call this method Test-Time Training to Discover (TTT-Discover). Following prior work, we focus on problems with continuous rewards. We report results for every problem we attempted, across mathematics, GPU kernel engineering, algorithm desi gn, and biology. TTT-Discover sets the new state of the art in almost all of the m: (i) Erd■s' minimum overlap problem and an autocorrelation inequality; (ii) a GPUMode kernel competition (up to 2× faster than prior art); (iii) past AtCoder algorithm competitions; and (iv) denoising problem in single-cell analysis. Our solutions are reviewed by experts or the organizers. All our results are achieve d with an open model, OpenAI gpt-oss-120b, and can be reproduced with our public ly available code, in contrast to previous best results that required closed fro ntier models. Our test-time training runs are performed using Tinker, an API by Thinking Machines, with a cost of only a few hundred dollars per problem.

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Efficient Continuous-Depth Modeling with GRU Equivalents

Ayan Banerjee, BIN XU, Sandeep Gupta

Continuous-Depth Neural Networks (CDNNs), including Neural Ordinary Differential

Equations (ODEs) and Liquid-Time-Constant Neural Networks (LTC-NN), suffer from high computational costs due to solving numerous nonlinear ODEs during training and inference. We introduce Continuous Depth Acceleration (CoDA), a framework that leverages Mori–Zwanzig/Koopman operator theory to replace continuous-depth layers requiring multiple nonlinear ODEs with a compact GRU module, a single low-dimensional linear ODE, and a dense layer. We prove PAC learnability of CoDA, establishing that this transformation preserves accuracy and can be applied repeatedly across multiple layers with unified backpropagation. Experiments on the Liquid Foundation Model (LFM-1.2B) demonstrate $6.7\times$ training speedup and $1.8\times$ inference speedup without loss of accuracy. Across six real-world LTC-NN applications, CoDA consistently outperforms state-of-the-art acceleration techniques—including neural flows, model order reduction, and variational formulations—in both training and inference time while maintaining competitive or superior accuracy. The implementation and datasets are publicly available at \url{https://github.com/ImpactLabASU/CoDA-ICML2026}.

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Geometric Rate–Distortion Invariance for Domain Generalization

Tong Liu, Sen Liang, Shuo Bai

Domain generalization (DG) aims to learn representations that remain predictive under distribution shifts. A key challenge is that the target domain is unobserved during training, which complicates the search for invariant representations: alignment objectives that ignore discriminative structure can become ill-conditioned under finite samples. This calls for shaping the geometry of class-conditional representations across domains, not merely matching their distributions. We propose Geometric **R**ate–**D**istortion **I**nvariance (**RDI**), a DG framework that realizes this principle by generalizing classical rate–distortion theory to Grassmann manifolds. **RDI** models class-conditional representations as low-dimensional subspaces and formulates DG as a joint optimization of (i) cross-domain subspace alignment (geometric distortion) and (ii) spectral–volumetric complexity (a capacity-regularized rate term), promoting stable alignment while preventing the collapse of discriminative geometry. We provide finite-sample stability guarantees under bounded shifts and show on DomainBed that **RDI** is competitive with strong DG baselines, with ablations confirming that both alignment and complexity control are necessary for reliable generalization.

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Beyond Model Base Retrieval: Weaving Knowledge to Master Fine-grained Neural Network Design

Jialiang Wang, Hanmo Liu, Shimin Di, Zhili Wang, Jiachuan Wang, Lei Chen, Xiaofang Zhou

Designing high-performance neural networks for new tasks requires balancing optimization quality with search efficiency. Current methods fail to achieve this balance: neural architectural search is computationally expensive, while model retrieval often yields suboptimal static checkpoints. To resolve this dilemma, we model the performance gains induced by fine-grained architectural modifications as edit-effect evidence and build evidence graphs from prior tasks. By constructing a retrieval-augmented model refinement framework, our proposed M-DESIGN dynamically weaves historical evidence to discover near-optimal modification paths. M-DESIGN features an adaptive retrieval mechanism that quickly calibrates the evolving transferability of edit-effect evidence from different sources. To handle out-of-distribution shifts, we introduce predictive task planners that extrapolate gains from multi-hop evidence, thereby reducing reliance on an exhaustive repository. Based on our model knowledge base of 67,760 graph neural networks across 22 datasets, extensive experiments demonstrate that M-DESIGN consistently outperforms baselines, achieving the search-space best performance in 26 out of 33 cases under a strict budget.

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RVAS: Referring Video Active Exploration and Segmentation

Hengrui Hu, Weiwei Gao, Zipei Zhang, Henghui Ding

Existing referring video object segmentation (RVOS) is largely built on passive perception and assumes the target is already visible in the observed video, whic

h limits real-world use when queries refer to objects beyond the current view. T
o address this gap, we introduce Referring Video Active Exploration and Segmenta
tion (RVAS), a new task that focuses on reasoning about exploration policy and t
hen locating and segmenting the object according to an input referring expressio
n. To support RVAS, we build a large-scale dataset with manually annotated explo
ration actions and reference reasoning traces, enabling supervised training and
evaluation. We benchmark representative RVOS and related video understanding bas
elines and find that they struggle to perform active target search and incur sub
stantial overhead when coupled with online decision making. Motivated by these c
hallenges, we propose LESA, a baseline framework that introduces a state control
ler and hierarchical memory for efficient streaming processing and sparse MLLM r
easoning. LESA substantially reduces inference cost while maintaining competitiv
e planning quality, and consistently improves segmentation accuracy on the RVAS
dataset. Code is available at [https://github.com/FudanCVL/RVAS](https://github.
com/FudanCVL/RVAS).
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SafeSearch: Automated Red-Teaming of LLM-Based Search Agents
Jianshuo Dong, Sheng Guo, Hao Wang, Xun Chen, Zhuotao Liu, Tianwei Zhang, Ke Xu,
 Minlie Huang, Han Qiu
Search agents connect LLMs to the Internet, enabling them to access broader and
more up-to-date information.
However, this also introduces a new threat surface: unreliable search results ca
n mislead agents into producing unsafe outputs.
Real-world incidents and our two in-the-wild observations show that such failure
s can occur in practice.
To study this threat systematically, we propose SafeSearch, an automated red-tea
ming framework that is scalable, cost-efficient, and lightweight, enabling sandb
oxed safety evaluation of search agents.
Using this, we generate 300 test cases spanning five risk categories (e.g., misi
nformation and prompt injection) and evaluate three search agent scaffolds acros
s 17 representative LLMs.
Our results reveal substantial vulnerabilities in LLM-based search agents, with
the highest ASR reaching 90.5\% for GPT-4.1-mini in a search-workflow setting.
Moreover, we find that common defenses, such as reminder prompting, offer limite
d protection.
Overall, SafeSearch provides a practical way to measure and improve the safety o
f LLM-based search agents.
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Alignment Pretraining: AI Discourse Causes Self-Fulfilling (Mis)alignment
Cameron Tice, Puria Radmard, Samuel Ratnam, Andy Kim, David Demitri Africa, Kyle
 O'Brien
Pretraining corpora contain extensive discourse about AI systems, yet the causal
 influence of this discourse on downstream alignment remains poorly understood.
If prevailing descriptions of AI behaviour are predominantly negative, LLMs may
internalise corresponding behavioural priors, giving rise to self-fulfilling mis
alignment. This paper provides the first controlled study of this hypothesis by
pretraining 6.9B-parameter LLMs with varying amounts of (mis)alignment discourse
. We find that discussion of AI contributes to misalignment. Upsampling syntheti
c training documents about AI misalignment leads to a notable increase in misali
gned behaviour. Conversely, upsampling documents about aligned behaviour reduces
 misalignment scores from 45% to 9%. We consider this evidence of self-fulfillin
g alignment. These effects are dampened, but persist through post-training. Our
findings establish the study of how pretraining data shapes alignment priors, or
 alignment pretraining, as a complement to post-training. We recommend practitio
ners pretrain for alignment as well as capabilities.
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Calibrating Conservatism for Scalable Oversight
William Overman, Mohsen Bayati
Agentic AI systems capable of autonomous planning and extended environmental int
eraction pose a fundamental control problem: how can humans maintain meaningful

oversight of systems that may exceed human capabilities? While scalable oversigh t is widely studied, existing approaches often rely on complex assumptions, rema in largely heuristic, or lack practical methods for sequential settings with sta tistical guarantees. We introduce Calibrated Collective Oversight (CCO), which a ggregates diverse auxiliary scoring functions into a penalty that measures devia tion from a conservative baseline. Inspired by Attainable Utility Preservation, CCO enables collective conservatism: when multiple oversight signals register co ncern, the agent defers. CCO calibrates this conservatism online using Conformal Decision Theory, ensuring that undesirable outcomes remain below a user-specifi ed target α with finite-time bounds and no distributional assumptions. Ex periments on SWE-bench demonstrate that weaker overseers successfully constrain an adversarially misaligned stronger agent. Similarly, on MACHIAVELLI, CCO achie ves substantial reductions in ethical violations while preserving reward. In bot h settings, empirical violation rates closely match the specified targets. Our w ork demonstrates that combining penalty-based conservatism with online calibrati on yields practical oversight with statistical guarantees suited for agentic dep loyment.

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How Out-of-Distribution Detection Learning Theory Enhances Transformer: Learnabi lity and Reliability

Yijin Zhou, Yutang Ge, Wenyuan XIE, Linqian Zeng, Xiaowen Dong, Yu Guang Wang

Transformers excel in natural language processing and computer vision tasks. How ever, they still face challenges in generalizing to Out-of-Distribution (OOD) da tasets, i.e. data whose distribution differs from that seen during training. OOD detection aims to distinguish outliers while preserving in-distribution (ID) da ta performance. This paper introduces the OOD detection Probably Approximately C orrect (PAC) Theory for transformers, which establishes the conditions for data distribution and model configurations for the OOD detection learnability of tran sformers. It shows that outliers can be accurately represented and distinguished with sufficient data under conditions. The theoretical implications highlight t he trade-off between theoretical principles and practical training paradigms. By examining this trade-off, we naturally derived the rationale for leveraging aux iliary outliers to enhance OOD detection. Our theory suggests that by penalizing the misclassification of outliers within the loss function and strategically ge nerating soft synthetic outliers, one can robustly bolster the reliability of tr ansformer networks. This approach yields a novel algorithm that ensures learnabi lity and refines the decision boundaries between inliers and outliers. In practi ce, the algorithm consistently achieves state-of-the-art (SOTA) performance acro ss various data formats.

****

Variational Inference for Uncertain Optimal Transport via Sinkhorn Parametrizati on

Ananyapam De, Linus Bleistein, Anton Frederik Thielmann, Benjamin Säfken

Optimal Transport (OT) traditionally relies on a fixed ground cost to produce a single deterministic transport plan—a practice that overlooks the inherent varia bility and noise in real-world data. While recent sampling based approaches of O T offer a principled way to quantify this uncertainty, these are computationally prohibitive and struggle to scale. In this paper, we introduce Sinkhorn-paramet erized Variational Inference, a first scalable variational framework for perform ing posterior inference over transport plans. Our key insight is that the Sinkho rn map can be treated as a differentiable reparameterization of the set of entro pic plans. This enables the use of flexible generative models like normalizing f lows to approximate distributions over transport plans while enforcing marginal constraints. We experimentally demonstrate that our method matches the quality o f intensive sampling techniques at a fraction of the computational cost, scaling effectively to large-scale problems.

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Scaling Real-World Robot Policy Evaluation via Discrete Diffusion World Model

Yaxuan Li, Junjie Wen, Zhongyi Zhou, Yefei Chen, Chaomin Shen, Yaxin Peng, Yiche n Zhu

Evaluating generalist robot manipulation policies is costly and difficult to sca
le in the real world. While emerging world models (e.g., WorldEval, Ctrl-World)
offer a promising alternative, the reliability of such evaluation remains a crit
ical bottleneck.
Specifically, their visual predictions can undermine policy assessment by "self-
correcting" failures into false positives or yielding artifacts under out-of-dis
tribution controls.
Even with failure-enriched data, current architectures struggle to capture actio
n-causal dynamics, as they typically treat actions as passive conditions rather
than causal drivers.
To address this, we propose dWorldEval, an action-centric discrete-diffusion wor
ld model that maps visual observations, language instructions, and action chunks
into a shared unified token space and denoises them with a single self-attentio
n backbone where actions function as first-class tokens.
To realize reliable policy-world interaction, dWorldEval introduces a sparse key
frame memory that anchors global scene state while preserving fine-grained multi
-view interaction cues, and leverages Progress-as-text to jointly generate futur
e observations and success indicators.
Extensive experiments on LIBERO, RoboTwin, and real-robot tasks demonstrate that
dWorldEval significantly outperforms video diffusion baselines in action contro
llability, stabilizes long-horizon multi-view rollouts, enabling accurate policy
ranking via automatic success estimation.
****

AG-REPA: Causal Layer Selection for Representation Alignment in Audio Flow Match
ing

Pengfei ZHANG, Tianxin Xie, Minghao Yang, Li Liu

REPresentation Alignment (REPA) improves the training of generative flow models
by aligning intermediate hidden states with pretrained teacher features, but its
effectiveness in token-conditioned audio Flow Matching critically depends on th
e choice of supervised layers, which is typically made heuristically based on th
e depth. In this work, we introduce **A**ttribution-**G**uided **REP**resentatio
n **A**lignment **(AG-REPA)**, a novel causal layer selection strategy for repre
sentation alignment in audio Flow Matching. Firstly, we find that layers that be
st store semantic/acoustic information (high teacher-space similarity) are not n
ecessarily the layers that contribute most to the velocity field that drives gen
eration, and we call it **S**tore-**C**ontribute **D**issociation **(SCD)**.
To turn this insight into an actionable training guidance, we propose a forward-
only gate ablation (FoG-A) that quantifies each layer's causal contribution via
the induced change in the predicted velocity field, enabling sparse layer select
ion and adaptive weighting for alignment. Across unified speech and general-audi
o training (LibriSpeech + AudioSet) under different token-conditioning topologie
s, AG-REPA consistently outperforms REPA baselines. Overall, our results show th
at alignment is most effective when applied to the causally dominant layers that
drive the velocity field, rather than to layers that are representationally ric
h but functionally passive.
****

REVIS: Sparse Latent Steering to Mitigate Object Hallucination in Large Vision-L
anguage Models

Jialin Wu, Wei Shi, Han Shen, Peigui Qi, Kunsheng Tang, Zhicong Huang, Binghao W
ang, Zhou Yang

Despite the advanced capabilities of Large Vision-Language Models (LVLMs), they
frequently suffer from object hallucination. One reason is that visual features
and pretrained textual representations often become intertwined in the deeper ne
twork layers. To address this, we propose REVIS, a training-free framework desig
ned to explicitly re-activate this suppressed visual information. Rooted in late
nt space geometry, REVIS extracts the pure visual information vector via orthogo
nal projection and employs a calibrated strategy to perform sparse intervention
only at the precise depth where suppression occurs. This surgical approach effec
tively restores visual information with minimal computational cost. Empirical ev
aluations on standard benchmarks demonstrate that REVIS reduces object hallucina

tion rates by approximately 19% compared to state-of-the-art baselines, while preserving general reasoning capabilities.
****
Neural Thickets: Diverse Task Experts Are Dense Around Pretrained Weights
Yulu Gan, Phillip Isola
Pretraining produces a learned parameter vector that is typically treated as a starting point for further iterative adaptation. In this work, we instead view the outcome of pretraining as a distribution over parameter vectors, whose support already contains task-specific experts. We show that in smaller or insufficiently trained models such expert solutions occupy a negligible fraction of the volume of this distribution, making their discovery reliant on structured optimization methods such as gradient descent. In contrast, in large, well-pretrained models the density of task-experts increases dramatically, so that diverse specialists populate a substantial fraction of the neighborhood around the pretrained weights. Motivated by this perspective, we explore a simple, fully parallel post-training method that samples N parameter vectors at random, selects the top K, and ensembles them via majority vote to combine complementary expertise. Despite its simplicity, this approach is competitive with standard post-training methods such as PPO, GRPO, and ES for contemporary large-scale models.
****
Anti-Backdoor Coreset Selection via Cumulative Entropy
Qi Zhao, Christian Wressnegger
Recent training-time defenses against neural backdoors isolate a benign subset from poisoned training data, to learn a backdoor-free model from it. In this paper, we formulate this defense strategy as a coreset selection problem, giving rise to so-called “Anti-Backdoor Coreset Selection.” Since poisonous samples have a) lower prediction uncertainty and are b) less frequent than benign samples, coreset selection naturally focuses more on samples associated with benign functionality than the backdoor functionality. We use the Cumulative Entropy as selection criterion to further facilitate this effect. The metric tracks the learning dynamics of training samples and allowing us to select benign samples with high informativeness for the coreset. Additionally, we unlearn the chosen samples in each epoch to facilitate the separability between benign and poisonous samples. Together, this yields an exceptionally effective training-time defense that constructs a benign coreset to train a backdoor-free model. Unlike prior defenses that compromise natural accuracy and fail against certain attacks, our method mitigates backdooring attacks consistently with a negligible impact on natural performance. The implementation of our method is publicly available at: https://intellisec.de/research/abcs
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Mixture Prototype Flow Matching for Open-Set Supervised Anomaly Detection
Fuyun Wang, Yuanzhi Wang, Xu Guo, Sujia Huang, Tong Zhang, Dan Wang, Hui Yan, Xin Liu, Zhen Cui
Open-set supervised anomaly detection (OSAD) aims to identify unseen anomalies using limited anomalous supervision. However, existing prototype-based methods typically model normal data via a unimodal Gaussian prior, failing to capture inherent multi-modality and resulting in blurred decision boundaries. To address this, we propose Mixture Prototype Flow Matching (MPFM), a framework that learns a continuous transformation from normal feature distributions to a structured Gaussian mixture prototype space. Departing from traditional flow-based approaches that rely on a single velocity vector, MPFM explicitly models the velocity field as a Gaussian mixture prior where each component corresponds to a distinct normal class. This design facilitates mode-aware and semantically coherent distribution transport. Furthermore, we introduce a Mutual Information Maximization Regularizer (MIMR) to prevent prototype collapse and maximize normal-anomaly separability. Extensive experiments demonstrate that MPFM achieves state-of-the-art performance across diverse benchmarks under both single- and multi-anomaly settings.
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Adversarial Reinforcement Learning for Robust Diffusion Large Language Model Unlearning

Zhiwei Zhang, Yudi Lin, Linlin Wu, Fali Wang, Yi Xin, Xiaomin Li, Minhua Lin, Xianfeng Tang, Qi He, Suhang Wang

Diffusion language models (DLMs) have recently emerged as an alternative to autoregressive approaches, enabling parallel sequence generation and flexible token generation orders. Machine unlearning plays a critical role in mitigating legal and ethical risks by removing the influence of specific training examples from trained models. While unlearning has been extensively studied for autoregressive language models, its applicability to DLMs remains unexplored. The architectural differences of DLMs raise new challenges for effective and robust unlearning that are not addressed by existing methods. In this paper, we present the first comprehensive study of unlearning for DLMs. Through systematic empirical analysis, we show that unlearning performance in DLMs is highly sensitive to generation hyperparameters, highlighting the need for evaluation across diverse generation settings. We further observe that DLMs tend to reproduce unlearned information when target inputs are embedded within informative contexts, due to their ability to incorporate both prefix and suffix conditioning, which increases vulnerability to elicitation attacks and weakens the robustness of existing unlearning methods. To design a robust unlearning method, we propose an adversarial reinforcement learning framework, where a context generator adversarially produces informative contexts to elicit unlearned knowledge, while the DLM is optimized to suppress undesired recall. We further introduce novel components to address credit assignment and stability issues in this adversarial learning setup. Extensive experiments demonstrate that our method significantly improves unlearning effectiveness while preserving model utility.

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Capacitated Fair-Range Clustering: Hardness and Approximation Algorithms

Ameet Gadekar, Suhas Thejaswi

Capacitated fair-range k-clustering generalizes classical k-clustering by incorporating both capacity constraints and demographic fairness. In this setting, data points are categorized as clients and facilities; each facility has a capacity and may belong to one or more possibly intersecting demographic groups. The task is to select k facilities as centers and assign each client to a center so that: (a) no center exceeds its capacity, (b) the number of centers selected from each group lies within specified lower and upper bounds (fair-range constraints), and (c) the clustering cost (e.g., k-median or k-means) is minimized.

In a prior work, Thejaswi et al. (2022) showed that even satisfying fair-range constraints is NP-hard, thereby making the problem inapproximable to any polynomial factor. Our first main result strengthens this by showing that inapproximability persists even when the fair-range constraints are trivially satisfiable, highlighting the intrinsic computational complexity of the clustering task itself. These inapproximability results hold even on tree metrics and when the number of groups is logarithmic in the size of the facility set.

In light of strong inapproximability results, we focus on a practical setting where the number of groups is constant. Our second main result is a polynomial-time $O(\log k)$- and $O(\log^2 k)$-approximation algorithm for k-median and k-means objectives, respectively, in this regime. Next, we design constant factor approximation algorithms for these problems that run in fixed parameterized tractable time in k. All our approximation guarantees match the best bounds for capacitated clustering without fair-range constraints. Finally, as our third main contribution, we show that our polynomial-time algorithms are, to our knowledge, the first to have provable approximation guarantees that can practically solve problem instances of modest size.

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RepetitionCurse: Measuring and Understanding Router Imbalance in Mixture-of-Experts LLMs under DoS Stress

Ruixuan Huang, Qingyue Wang, Hantao Huang, Yudong Gao, Dong CHEN, Shuai Wang, Wei Wang

Mixture-of-Experts architectures have become the standard for efficient LLM scaling, typically employing expert parallelism to distribute experts across devices. However, the absence of explicit load balancing constraints during inference allows adversarial inputs to trigger severe routing concentration. We demonstrate that out-of-distribution prompts can manipulate the routing mechanism such that all tokens are routed to the same set of top-k experts, which creates computational bottlenecks on certain devices while forcing others to idle. This converts an efficiency mechanism into a denial-of-service attack vector, leading to violations of service-level agreements for time-to-first-token (TTFT). We propose RepetitionCurse, a black-box strategy to exploit this vulnerability. By identifying a universal flaw in MoE router behavior, RepetitionCurse constructs attack prompts using simple repetitive token patterns in a model-agnostic manner. On widely deployed MoE models hosted on 8-GPU clusters, our method increases TTFT by 20% to 148%, significantly degrading service quality.

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Sparse Topology-Aware Pairwise Scoring for Large-Scale Multi-Agent Reinforcement Learning

Zhibo Deng, Feng Liang, Yong Zhang, Xiaoxi Zhang, Xiping Hu

In multi-agent reinforcement learning (MARL), communication enables agents to mitigate partial observability and stochasticity through information sharing, but large-scale systems inherently lead to a rapidly growing number of pairwise interactions. Previous studies often struggle to simultaneously achieve scalability and task adaptivity in large-scale multi-agent communication. To address this challenge, we propose a scalable communication scheme for large-scale MARL, termed $\textit{Sparse tOpology-aware Pairwise Scoring}$ (SOPS). We argue that scalable MARL communication requires decoupling scalability from task-adaptive link allocation. To ensure scalability, we constrain communication to an exponential-graph backbone with a small diameter, which preserves rapid potential information mixing while keeping per-agent candidates logarithmic. On top of this constraint, we learn a task-conditioned probabilistic subgraph distribution via a pairwise scoring network over agent states and edge-type embeddings to allocate sparse links for maximizing return, optimized end-to-end through differentiable Gumbel-Sigmoid reparameterization. Evaluation results show that SOPS significantly outperforms existing state-of-the-art methods across cooperative benchmarks of diverse scales and exhibits robust zero-shot transfer capabilities.

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CATArena: Evaluating Evolutionary Capabilities of Code Agents via Iterative Tournaments

Lingyue Fu, Xin Ding, Linyue Pan, Yaoming Zhu, Shao Zhang, Lin Qiu, Xuezhi Cao, Xunliang Cai, Jiaxin Ding, Weiwen Liu, Weinan Zhang, Yong Yu

Current evaluation for Large Language Model (LLM) code agents predominantly focus on generating functional code in single-turn scenarios, which fails to evaluate the agent's capability for continuous code optimization and multi-turn iterative development. To bridge this gap, we introduce CATArena, a framework designed to evaluate the evolutionary capabilities of code agents via iterative tournaments. Agents engage in multi-turn tournaments and continuously refine their code through self-reflection and peer-learning based on comprehensive execution feedback. For evaluation, we propose a dual-metric system to decouple static generation proficiency from evolutionary potential. Extensive experiments reveal that an agent's evolutionary potential is not strictly correlated with its initial proficiency. Our analysis further reveals that current agents struggle to concurrently leverage both peer-learning and self-reflection for effective performance gains. Furthermore, the results validate CATArena's high extensibility and resistance to variance tasks, establishing it as a continuous and reliable standard for assessing the evolutionary capability of LLM code agents.

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ARLArena: A Unified Framework for Stable Agentic Reinforcement Learning

Xiaoxuan Wang, Han Zhang, Haixin Wang, Yidan Shi, Ruoyan Li, Kaiqiao Han, Chenyi Tong, Haoran Deng, Alexander K Taylor, Renliang Sun, Yanqiao Zhu, Jason Cong, Yizhou Sun, Wei Wang

Agentic reinforcement learning (ARL) has rapidly gained attention as a promising paradigm for training agents to solve complex, multi-step interactive tasks. Despite encouraging early results, ARL remains highly unstable, often leading to training collapse. This instability limits scalability to larger environments and longer interaction horizons, and constrains systematic exploration of algorithmic design choices.

In this paper, we first propose **ARLArena**, a stable training recipe and systematic analysis framework that examines training stability in a controlled and reproducible setting. ARLArena first constructs a clean and standardized testbed. Then, we decompose policy gradient into four core design dimensions and assess the performance and stability of each dimension. Through this fine-grained analysis, we propose **SAMPO**, a stable agentic policy optimization method designed to mitigate the dominant sources of instability in ARL.

Empirically, SAMPO achieves consistently stable training and strong performance across diverse agentic tasks. Our code is open-sourced at: https://github.com/WillDreamer/ARL-Arena.git

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An Empirical Study on the Resilience of Partial Merging to Model Clone Attacks

Tiantong Wu, Yurong Hao, Wei Yang Bryan Lim

Model merging is a promising technique to enhance the capabilities of neural networks (NNs) by integrating multiple downstream fine-tuned models without requiring access to clients' raw data or substantial computation resources. However, conventional model merging typically requires collecting the full set of fine-tuned model parameters from multiple clients, which may expose them to model-privacy risks. An emerging approach, known as partial model merging (PMM), mitigates this risk by splitting each model into private and shared parts, where only the shared part is merged while the private part remains local to each client. Despite its stricter parameter fusion, PMM can still achieve competitive performance compared to full-parameter sharing. However, the privacy properties of PMM remain underexplored. In this paper, we propose a novel model clone attack and assess the risk of reconstructing the unshared private part of a partially merged model under eight attack scenarios with varying prior knowledge (i.e., partial training data, model parameters and/or model structure). Our comprehensive experiments reveal that merging NNs without adequate protection is highly vulnerable. Even when only a small fraction of training data, model parameters, or model structure is exposed, adversaries can still recover significant portions of the private model's performance.

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Beyond External Monitors: Enhancing Transparency of Large Language Models for Easier Monitoring

Guanxu Chen, Jing Shao, Tao Luo, Lijie Hu, Qihao Lin, Dongrui Liu

Large language models (LLMs) are becoming increasingly capable, but the mechanisms of their thinking and decision-making processes remain unclear. Chain-of-thoughts (CoTs) have been commonly utilized to externalize LLMs' thinking, but this strategy fails to accurately reflect LLMs' thinking process. Techniques based on LLMs' hidden representations provide an inner perspective to improve the monitorability of their latent thinking. However, previous methods only try to develop external modules instead of making LLMs themselves easier to monitor. In this paper, we propose a novel method, TELLME, improving the transparency of LLMs and helping monitors identify unsuitable and sensitive behaviors. Furthermore, we showcase the effectiveness of TELLME on detoxification tasks, where LLMs achieve consistent improvement among multimodal test sets, distinct architectures, and varying parameter scales. We further analyze TELLME's improvement on LLMs' generalization ability from both optimal transport theory and empirical perspectives.

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DuRP: Dual-Stage Physics-Embedded Learning for Joint Radiance and Polarization Restoration

Zhenshuo Yang, Qian He, Zhiyuan Liu, Baojie Fan, Jiandong Tian

Polarization information is valuable for many computer vision applications. However, in hazy environments, polarization information is severely attenuated due t

o the degradation of captured polarized images. Existing dehazing methods strugg
le to effectively restore polarization information, as single-image methods are
unaware of polarization, and polarization-based methods are constrained by the t
raditional polarization models. These deficiencies lead to inaccurate polarimetr
ic signatures and physical inconsistencies in scattering environments. To overco
me these limitations, we propose DuRP, a dual-stage physics-embedded learning fr
amework for joint restoration of scene radiance and polarization information. Sp
ecifically, we derive generalized polarization physics models that relax the ide
al assumptions of traditional theory to provide a more precise foundation for th
e joint restoration of polarimetric and amplitude information. We then design a
dual-stage neural network to estimate latent physical parameters through differe
ntiable operators, ensuring that both the polarimetric state and radiance are ac
curately recovered. Experimental results show that DuRP achieves state-of-the-ar
t performance in joint restoration and significantly enhances polarization-based
 vision tasks. Project website: [https://DuRP.github.io/](https://yangzhenshuo.g
ithub.io/DuRP.github.io/)
****

Gradient-Aware Scheduling: Coupling Curriculum and Staleness for Async Reinforce
ment Learning

Xinyu Zhang

Asynchronous reinforcement learning (RL) accelerates LLM training through parall
el data collection, but introduces *policy lag*: experiences are collected under
 stale policy weights, which destabilizes learning—especially on hard tasks. We
identify why: **gradient variance scales exponentially with task difficulty unde
r staleness**, because hard tasks have narrow solution spaces corresponding to s
harp loss-landscape curvature (high Hessian eigenvalues). We formalize this as a
 staleness budget optimization problem and prove that the optimal allocation f
ollows an exponential decay: $\eta^*(d) = \eta_{\text{base}} \cdot e^{-\lambda d
}$ where $\lambda = \alpha/2$ is half the Hessian growth rate. Building on this
principle, we propose **GAS (Gradient-Aware Scheduling)**, a drop-in recipe for
fast and stable asynchronous RL with three components: (i) a bandit-based curric
ulum that selects task difficulty by learning signal (ACB); (ii) execution-aware
 staleness budgets that keep slow tasks from bottlenecking rollout (EAAS); and (
iii) curriculum–staleness coupling that tightens the staleness budget as difficu
lty rises (CSC). Our mechanistic analysis validates the theoretical predictions:
 the "safe zone" of gradient coherence follows the derived exponential boundary.
 On code generation benchmarks, GAS improves Pass@1 from 39.7% to 60.1% while tr
aining at $2.3\times$ the throughput of synchronous GRPO, showing that matching
each task's staleness budget to its difficulty (and thus to its loss-landscape c
urvature) makes asynchronous RL both fast and stable.
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Class-Prior Perturbation-Robust Regularization for Imbalanced Unreliable Partial
 Label Learning

Congyu Qiao, Haohao Dong, Xin Geng, Ning Xu

Imbalanced Unreliable Partial Label Learning (I-UPLL) is a challenging weakly su
pervised learning setting in which severe class imbalance and unreliable candida
te labels jointly degrade model performance. By revisiting existing approaches f
or imbalanced learning, we observe that most of them fundamentally rely on estim
ating the class prior to guide balancing operations, such as re-sampling, pseudo
-label generation, or logit adjustment. However, under I-UPLL, obtaining stable
and accurate prior estimates at the early stage of training is often unrealistic
 due to the ambiguity and unreliability of partial labels, thereby leading the m
odel to rapidly converge to a suboptimal solution. To address this issue, we pro
pose CLAPOR, a novel CLAss-PriOr perturbation-Robust regularization framework th
at fundamentally avoids dependence on accurate prior estimation. Specifically, t
he proposed regularization trains the model under deliberately perturbed class p
riors, sampled from a Dirichlet distribution that deviates from the current esti
mated prior. This design encourages consistent performance under prior uncertain
ty and naturally preserves attention to minority classes. Extensive experiments
on benchmark datasets demonstrate the effectiveness of CLAPOR across various set

tings of I-UPLL.
****
LLMs Lean on Priors, Not Programming Language Semantics
Aditya Thimmaiah, Jiyang Zhang, Jayanth Srinivasa, Junyi Jessy Li, Milos Gligoric
Recent work asks whether large language models (LLMs) condition their reasoning on explicit rules rather than statistical regularities from pretraining. Program execution provides a canonical instance: formal semantics define behavior through symbolic transition rules that can be systematically altered under distribution shift. We investigate whether LLMs can condition their reasoning on formal semantics through program execution and introduce PLSemanticsBench, pairing featherweight C programs with two semantic systems—small-step operational semantics and K semantics—and probing four capabilities: composing rules for final states, selecting rules when state is unmutated, sustaining such conditioning over long traces, and following supplied rules under novel semantics. To decouple semantic reasoning from syntactic familiarity, we redefine familiar operators to induce symbol-meaning conflict and introduce novel symbols defined only through the supplied rules, and stress-test models on Human-Written, LLM-Translated, and Fuzzer-Generated splits with increasing structural complexity. Across 11 frontier LLMs, strong final-state accuracy under standard semantics (up to 90%) drops sharply—by as much as 40–60% points—under semantic mutations and increasing structural complexity. Only a handful of models achieve non-zero long-horizon conditioning accuracy, and even the best systems reach just 35%. Together, these results suggest that contemporary LLMs often rely on pretrained lexical associations rather than systematically conditioning on supplied formal rules. PLSemanticsBench is publicly available at https://EngineeringSoftware.github.io/PLSemanticsBench.
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STARCaster: Spatio-Temporal AutoRegressive Video Diffusion for Identity- and View-Aware Talking Portraits
Foivos Paraperas Papantoniou, Stathis Galanakis, Rolandos Alexandros Potamias, Bernhard Kainz, Stefanos Zafeiriou
This paper presents STARCaster, an identity-aware spatio-temporal video diffusion model that addresses both speech-driven portrait animation and dynamic viewpoint control, given an identity embedding or reference image, within a unified framework. Existing 2D speech-to-video diffusion models depend heavily on reference guidance, leading to limited motion diversity. At the same time, 3D-aware animation typically relies on inversion through pretrained tri-plane generators, which often leads to imperfect reconstructions and identity drift. We rethink reference- and geometry-based paradigms in two ways. First, we deviate from strict reference conditioning at pretraining by introducing softer identity constraints. Second, we address 3D awareness implicitly within the 2D video domain by leveraging the inherent multi-view nature of video data. STARCaster adopts a compositional approach progressing from ID-aware motion modeling, to audio-visual synchronization via lip reading-based supervision, and finally to novel view animation through temporal-to-spatial adaptation. To overcome the scarcity of 4D audio-visual data, we propose a decoupled learning approach in which view consistency and temporal coherence are trained independently. Comprehensive evaluations demonstrate that STARCaster generalizes effectively across tasks and identities, consistently surpassing prior approaches in different benchmarks.
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Information-Theoretic Generalization Bounds for VAEs: A Role of Encoder and Latent Variable
Futoshi Futami, Masahiro Fujisawa
Despite their remarkable success, a rigorous theoretical understanding of how latent variables (LVs) govern the generalization performance of Variational Autoencoders (VAEs) remains largely elusive. Existing theoretical analyses are confined to supervised learning or models with discrete latent spaces, leaving their role in standard VAEs with continuous LVs poorly understood. This paper establishes the first information-theoretic analysis for VAEs by adapting a theoretical fr

amework from supervised learning---the leave-one-out conditional mutual information framework---to the unsupervised, continuous latent space of these models. Our analysis reveals that their generalization error is bounded solely by the information complexity of the encoder and LVs, independent of the decoder. The versatility of our framework is demonstrated through its extension to both hierarchical VAEs, for which we provide layer-wise bounds, and data generation, where we link our information-theoretic principles to a novel bound on the 2-Wasserstein distance between true and generated distributions.
****
The Geometry of Updates: Fisher Alignment at Vocabulary Scale
John Sweeney
Training-free source selection for LLM families with shared vocabularies arises in scientific string domains such as SMILES, protein, and genomic sequences, where candidate corpora share a tokenizer but differ in prediction targets. This creates an activation-dark regime: representation-similarity metrics can be uninformative without assumptions about label-conditioned error geometry, while classical update-geometry metrics are computationally prohibitive at vocabulary scale. We show that, in a shared-output head setting, representation metrics (e.g., CKA) are non-identifiable for transfer; models can share identical representations yet have orthogonal head updates. The key identity is that head Fisher alignment is exactly a cosine between kernel mean embeddings in the joint activation-error space, exposing activation, error, and coupling factors rather than requiring a materialized Fisher matrix. FisherSketch estimates this cosine directly in a single streaming pass, making K=128,256 head Fisher alignment practical with a 16 KB task signature (m=4096) and a 192 KB per-task streaming state–small enough to store next to a model hash, but encoding transfer-relevant update structure. Beyond source selection, the same signatures and marginals provide a diagnostic instrument for studying whether LLM task similarity is driven by activations, errors, or their coupling; shared-parameter and internal-layer validations, together with Llama-3.1-8B verbalizer-shift experiments, show that FisherSketch remains informative when activation similarity cannot distinguish tasks.
****
Bayes-inspired Integration of Pretrained Priors and Few-Shot Evidence for Few-Shot Classification
Mingyang Zhou, Xiaoxuan Zhang, Gang Liu, Yuhong Feng, Xiaoqun Wu, Hao Liao, Rui Mao
Few-shot classification aims to adapt a pretrained model to novel classes with limited examples. While current methods often heuristically combine pretrained knowledge and few-shot evidence, we seek a more principled understanding of their relationship.
In this paper, we propose a Bayesian-inspired optimal integration framework(BOIF) that interprets pretrained models as priors and few-shot evidence as likelihoods. Under conditional independence approximation, we show that the optimal log-posterior decomposes into the sum of prior logits and likelihood logits. This leads to a simple yet effective design principle: decouple the prior and likelihood pathways and combine their logits additively.
Guided by this principle, we implement BOIF using CLIP with two novel enhancements: (1) a multi-level feature adapter to enrich visual representations, and (2) a simplified cache module for likelihood estimation. Extensive experiments on 11 benchmarks show BOIF achieves state-of-the-art performance (e.g., 80.61\% average accuracy at 16-shot) and strong out-of-distribution robustness. Our work provides both a principled perspective and an effective instantiation for few-shot adaptation.
****
Auto-regressive In-context Demonstration Selection
Yunzhe Qi, Sirui Chen, Jiaru Zou, Jingrui He
Effective demonstration selection is crucial for maximizing large language model (LLM) performance in few-shot in-context learning.
Because of effects such as recency bias, the effectiveness of demonstrations depends heavily on their contextual relationship to the specific query and on t

he ordering in which they are presented, making demonstration selection a comple
x combinatorial problem.

To address these two challenges, we introduce AutoSelect, a novel framework
that formulates demonstration selection as an auto-regressive sequential decisio
n process. At each step, AutoSelect embeds the query and previously selected dem
onstrations into matrix representations to preserve structural information, and
a trainable policy model sequentially selects the next best exemplar. To navigat
e the factorial space of demonstration permutations, our framework formulates a
Kullback-Leibler (KL)-regularized optimization problem, from which an optimal po
licy induces an optimal Plackett-Luce (PL) ranking over all possible demonstrati
on sequences. We prove that minimizing a tractable policy-level cross-entropy (C
E) loss provably bounds the worst-case discrepancy between our policy's induced
PL ranking and the optimal one, enabling tractable prioritization of high-qualit
y sequences. Empirically, AutoSelect outperforms existing heuristic and learning
-based methods across nine diverse datasets, achieving up to an 11\% improvement
over the strongest baseline.

Analytical studies and a case study further highlight AutoSelect's key prope
rties, as well as its transferability and generalizability.

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What If We Let Forecasting Forget? A Sparse Bottleneck for Cross-Variable Depend
encies

Fan Zhang, Shiming Fan, Hua Wang

Multivariate time series forecasting is critical in many real-world systems, and
thus modeling cross-channel dependencies is essential. Although existing method
s improve overall accuracy by enhancing representations and cross-channel intera
ctions, it remains challenging to reliably capture inter-variable dependencies u
nder specific conditions. We observe that dependencies in real data are often st
ate-dependent and noisy; in such cases, dense interactions can amplify spurious
correlations and lead to representation over-smoothing, which may yield unreliab
le predictions in certain scenarios. Motivated by this, we propose MS-FLOW, a sp
arse-bottleneck framework that explicitly models inter-variable interaction as c
apacity-limited information flow. Specifically, MS-FLOW replaces fully connected
communication with selective sparse routing, retaining only a few critical depe
ndency paths and injecting cross-variable signals under a strict communication b
udget, thereby suppressing redundant connections and spurious-correlation propag
ation. Extensive experiments demonstrate that MS-FLOW learns more reliable multi
variate correlations, achieving state-of-the-art forecasting accuracy on 12 real
-world benchmarks while producing fewer yet more reliable dependencies, shifting
multivariate forecasting from “more interaction” to “more effective interaction
”.

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Forward-Chaining Temporal Point Process

Chao Yang, Wendi Ren, Shuang Li

Event sequences from complex systems, such as clinical workflows, are often spar
se and incomplete. As a result, downstream models are trained on data that only
partially captures the underlying dynamics. Synthetic sequence generation can au
gment real data by filling in missing structure and improving coverage of rare p
atterns, but generated trajectories must remain realistic, satisfy domain constr
aints, and allow control. We propose the Forward-Chaining Temporal Point Process
(FC-TPP), a framework for constraint-aware and controllable sequence generation
in continuous time. FC-TPP maintains an explicit latent symbolic state encoding
high-level predicates, which evolves through a differentiable multi-hop forward
-chaining operator. Logical rules update the latent state based on recent events
, while a temporal point process decoder generates future event times and types
conditioned on this evolving state. By tying the generative dynamics to multi-ho
p reasoning in latent space, FC-TPP incorporates symbolic structure throughout g
eneration rather than relying directly on raw event histories. Experiments on sy
nthetic data and four semi-synthetic/real-world benchmarks—LogiCity, MIMIC-IV, E
PIC-100, and IKEA ASM—show that FC-TPP achieves higher generation quality under
limited and incomplete data, with stronger constraint adherence and greater cont

rollability than purely neural and prior neuro-symbolic baselines.
****
LearniBridge: Learnable Calibration of Feature Caching for Diffusion Models Acce leration
Xuyue Huang, Zhe Chen, Wang Shen, Xiao-Ping Zhang
Diffusion Transformers (DiTs) have driven substantial progress in image and vide o generation but suffer from prohibitive computational costs.
Feature caching accelerates inference by reusing intermediate representations.
Existing methods rely on historical features for implementation simplicity,
yet suffer from severe error accumulation at high acceleration ratios.
To address this limitation, we investigate the nature of the requisite feature c orrection.
We demonstrate that the optimal calibration update is characterized by a shared low-rank subspace across diverse prompts.
Guided by this structural insight, we propose LearniBridge, a learnable calibrat ion mechanism for feature caching that bridges multiple timesteps through lightw eight LoRA updates.
This mechanism enables effective calibration requiring only $3-5$ training sampl es.
Extensive experiments on image and video generation show that LearniBridge achie ves up to $5.87\times$, $5.75\times$, and $4.10\times$ acceleration
on FLUX, HunyuanVideo, and WAN 2.1, respectively. On WAN 2.1, it improves VBench by 1.28\% over the previous SOTA at $4.10\times$ acceleration.
****
TabPack: Efficient Hyperparameter Ensembles for Tabular Deep Learning
Yury Gorishniy, Akim Kotelnikov, Ivan Rubachev, Artem Babenko
Deep learning models for supervised learning on tabular data are rapidly improvi ng. Notably, ensembles (mixtures of multiple models) often play an important rol e in achieving top performance, which motivates designing _ensemble-first_ syste ms rather than treating ensembling as an ad hoc trick. In this work, we present TabPack — a new ensembling approach that packs many base model–optimizer pairs w ith different hyperparameters into a single neural network and a single optimize r. The base model and optimizer hyperparameters are sampled randomly, after whic h all base models are trained in parallel, and the final ensemble is built on th e fly during training. As a result, TabPack produces powerful ensembles in a sin gle run, with substantial efficiency gains over traditional approaches. With its remarkable efficiency, strong performance on medium-to-large datasets, and redu ced reliance on traditional hyperparameter tuning, TabPack is an appealing solut ion for practitioners and researchers that makes tabular DL more accessible on c onsumer-grade hardware and suggests a new avenue for designing better tabular de ep learning systems.
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Mem-T: Densifying Rewards for Long-Horizon Memory Agents
Yanwei Yue, Guibin Zhang, Boci Peng, Xuanbo Fan, Jiaxin Guo, Qiankun Li, Yan Zha ng
Memory agents, which depart from predefined memory-processing pipelines by endog enously managing the processing, storage, and retrieval of memories, have garner ed increasing attention for their autonomy and adaptability. However, existing t raining paradigms remain constrained: agents often traverse long-horizon sequenc es of memory operations before receiving sparse and delayed rewards, which hinde rs truly end-to-end optimization of memory management policies. To address this limitation, we introduce Mem-T, an autonomous memory agent that interfaces with a lightweight hierarchical memory database to perform dynamic updates and multi-turn retrieval over streaming inputs. To effectively train long-horizon memory m anagement capabilities, we further propose MoT-GRPO, a tree-guided reinforcement learning framework that transforms sparse terminal feedback into dense, step-wi se supervision via memory operation tree backpropagation and hindsight credit as signment, thereby enabling the joint optimization of memory construction and ret rieval.
Extensive experiments demonstrate that Mem-T is **(1) high-performing**, surpass

ing frameworks such as A-Mem and Mem0 by up to $14.94\\%$, and **(2) economical**, operating on a favorable accuracy-efficiency Pareto frontier and reducing inference tokens per query by $\sim24.45\\%$ relative to GAM without sacrificing performance.
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Olmix: A Framework for Data Mixing Throughout LM Development
Mayee F Chen, Tyler Murray, David Heineman, Matt Jordan, Hannaneh Hajishirzi, Christopher Re, Luca Soldaini, Kyle Lo
Data mixing---determining the ratios of data from different domains---is a first-order concern for training language models (LMs), but existing mixing methods have poorly understood design choices and assume that the set of domains remain fixed throughout development. We present Olmix, a framework that addresses two challenges encountered during LM development. First, the configuration space for developing a mixing method is not well understood---design choices across existing methods lack justification or consensus and overlook practical issues like data constraints. We conduct a comprehensive empirical study of this space, identifying which design choices lead to a strong mixing method. Second, the domain set evolves throughout LM development as datasets are revised and expanded---a problem setting largely unaddressed by existing works. We study how to efficiently recompute the mixture after the domain set is updated, given an existing mix from before the update. We introduce mixture reuse, a mechanism that reuses existing relative ratios and recomputes ratios only for domains affected by an update. Over a sequence of five domain-set updates mirroring real-world LM development, mixture reuse matches the performance of fully recomputing the mix after each update with 74% less compute and improves over training without mixing by 11.6% on downstream tasks.
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Strategy Executability in Mathematical Reasoning: Leveraging Human–Model Differences for Effective Guidance
Weida Liang, Yiyou Sun, Shuyuan Nan, Chuang Li, Dawn Song, Kenji Kawaguchi
Example-based guidance is widely used to improve mathematical reasoning at inference
time, yet its effectiveness is highly unstable across problems and models—even when the
guidance is correct and problem-relevant.
We show that this instability arises from a previously underexplored gap between
strategy usage—whether a reasoning strategy appears in successful solutions—and
strategy executability—whether the strategy remains effective when instantiated
as guidance for a target model.
Through a controlled analysis of paired human-written and model-generated solutions, we
identify a systematic dissociation between usage and executability: human- and
model-derived strategies differ in structured, domain-dependent ways, leading to
complementary strengths and consistent source-dependent reversals under guidance.
Building on this diagnosis, we propose *Selective Strategy Retrieval* (SSR), a
test-time framework that explicitly models executability by selectively retrieving and
combining strategies using empirical, multi-route, source-aware signals.
Across multiple mathematical reasoning benchmarks, SSR yields reliable and consistent
improvements over direct solving, in-context learning, and single-source guidance,
improving accuracy by up to $+13$ points on AIME25 and $+5$ points on Apex for compact
reasoning models. Code and benchmark are publicly available at: https://github.com/lwd17/strategy-execute-pipeline.
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TD-VAD: Breaking Visual Dependence in Video Anomaly Detection with Text-Driven L
earning
Shuangqing Zhang, Lei-Lei Ma, Zhao Wang, Wen Dong, Xinyi Xu, Guo-Sen Xie, Caifen
g Shan, Fang Zhao
Visual data is typically a prerequisite for training existing video anomaly dete
ction (VAD) methods. However, obtaining sufficient annotated anomaly data for tr
aining is challenging and not scalable due to the rarity of anomaly data and the
 wide variety of abnormal events. In this work, we advocate that the effectivene
ss of treating texts as video sequences for the VAD model and propose a novel Te
xt-Driven Video Anomaly Detection (TD-VAD) approach to break visual dependence.
In contrast to the anomaly video data, text descriptions of abnormal events are
easy to collect, and their class labels can be directly derived. Specifically, o
ur method utilizes video-like text descriptions with temporal characteristics ge
nerated by LLM to train a VAD model, without any reliance on target-domain anoma
ly data. To capture the
long and short-range temporal logic of events, we design the event evolution cau
sal attention module to model contextual dependencies across time. During infere
nce, considering the domain gap between the texts and video sequences, we use th
e frozen CLIP encoder to extract embeddings of video frames to align the text mo
dality while retaining crucial visual information. Comprehensive experiments on
two large-scale VAD datasets, XD-Violence and UCF-Crime, demonstrate that our me
thod outperforms prior one-class and unsupervised VAD methods by a large margin.
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EGG: An Expert-Guided Agent Framework for Kernel Generation
Yaochen Han, Ke Fan, Hongxu Jiang, Wanqi Xu, Weiyu Xie, Runhua Zhang, Chenhui Zh
u, Yixiang Zhang
High-performance GPU kernels are critical for reducing the exponentially growing
 computational costs of large language models (LLMs), but their development heav
ily relies on manual tuning by domain experts. While recent advances in LLM-base
d approaches show promise for automating kernel generation, they still struggle
to achieve both correctness and high performance. This limitation primarily aris
es from the lack of domain-specific optimization guidance, hindering effective e
xploration of the optimization space. We propose $\textbf{EGG}$, an $\underline{
E}$xpert-$\underline{G}$uided Agent Framework for Kernel $\underline{G}$eneratio
n, which incorporates expert optimization principles to guide LLMs' decisions. I
nspired by expert workflows, we decompose kernel generation into two hierarchica
l stages: 1) algorithmic structure design, which establishes a high-quality comp
utational structure foundation; 2) hardware-specific tuning, which performs targ
eted adjustments through parallel mapping, tensor tiling, and memory optimizatio
n. This staged decomposition defines explicit optimization objectives, structuri
ng the design space to achieve progressive refinements. To this end, a stage-awa
re multi-agent collaboration mechanism is designed for inter and intra-stage con
text management, ensuring stable optimization trajectories. Experiments on Kerne
lBench and real-world workloads show that EGG achieves a $2.13\times$ average sp
eedup over PyTorch, outperforming existing agent-based and RL-based approaches.
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AOrchestra: Automating Sub-Agent Creation for Agentic Orchestration
Jianhao Ruan, Zhihao Xu, Yiran Peng, Fashen Ren, Zhaoyang Yu, Xinbing Liang, Jin
yu Xiang, Yongru Chen, Bang Liu, Chenglin Wu, Yuyu Luo, Jiayi Zhang
Language agents have shown strong promise for task automation. Realizing this pr
omise for increasingly complex, long-horizon tasks has driven the rise of a suba
gent-as-tools paradigm for multi-turn task solving. However, existing designs st
ill lack a dynamic abstraction view of sub-agents, thereby hurting adaptability:
 sub-agents are either context-isolated threads that lack specialization, or sta
tic roles that require human-engineering.
We address this challenge with a unified, framework-agnostic agent abstraction t
hat models any agent as a tuple (Model, Task, Tools, Context). This tuple acts a
s a compositional recipe for capabilities, enabling the system to spawn speciali
zed executors for each task on demand.
Building on this abstraction, we introduce an agentic system AOrchestra, where t

he central orchestrator concretizes the tuple at each step: it curates task-rele vant context, selects tools and models, and delegates execution via on-the-fly a utomatic agent creation.
Such designs enable reducing human engineering efforts, and remain framework-agn ostic with plug-and-play support for diverse agents as task executors. It also e nables a controllable performance–cost trade-off, allowing the system to approac h Pareto-efficient.
Across three challenging benchmarks and environments (GAIA, SWE-Bench, Terminal-Bench), AOrchestra achieves 16.28% relative improvement against the strongest ba seline when paired with Gemini-3-Flash.
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Executable Agentic Memory for GUI Agent
Zerui Qin, Sheng Yue, Xingyuan Hua, Yongjian Fu, Ju Ren
Modern GUI agents typically rely on a model-centric and step-wise interaction pa radigm, where LLMs must re-interpret the UI and re-decide actions at every scree n, which is fragile in long-horizon tasks. In this paper, we propose Executable Agentic Memory (EAM), a structured Knowledge Graph (KG) that shifts GUI planning from free-form generation to a robust retrieval-and-execution process. Our appr oach includes a sample-efficient memory construction pipeline using state-aware DFS and action-group mining to compress multi-step routines. To ensure efficient planning, we introduce a value-guided graph search where a lightweight Q-functi on model steers Monte Carlo Tree Search (MCTS) over the KG. We theoretically est ablish bias-consistency for the Q-model and derive sample complexity bounds for path recovery. Empirically, EAM outperforms state-of-the-art baselines like UI-T ARS-7B by up to 19.6% on AndroidWorld, while reducing token costs $6\times$ r elative to GPT-4o. With a 2.8s average latency, EAM enables reliable, quick, a nd long-horizon GUI automation.
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Conditional KRR: Injecting Unpenalized Features into Kernel Methods with Applica tions to Kernel Thresholding
Rustem Takhanov, Zhenisbek Assylbekov
Conditionally positive definite (CPD) kernels are defined with respect to a func tion class $\mathcal{F}$.
It is well known that such a kernel K is associated with its native space (def ined analogously to an RKHS), which in turn gives rise to a learning method --- called conditional kernel ridge regression (conditional KRR) due to its analogy with KRR --- where the estimated regression function is penalized by the square of its native space norm. This method is of interest because it can be viewed as classical linear regression, with features specified by $\mathcal{F}$, followed by the application of standard KRR to the residual (unexplained) component of t he target variable. Methods of this type have recently attracted increasing atte ntion.

We study the statistical properties of this method by reducing its behavior to t hat of KRR with another fixed kernel, called the residual kernel. Our main theor etical result shows that such a reduction is indeed possible, at the cost of an additional term in the expected test risk, bounded by $\mathcal{O}(1/\sqrt{N})$, where N is the sample size and the hidden constant depends on the class $\mat hcal{F}$ and the input distribution.

This reduction enables us to analyze conditional KRR in the case where K is po sitive definite and $\mathcal{F}$ is given by the first k principal eigenfunct ions in the Mercer decomposition of K. We also consider the setting where $\ma thcal{F}$ consists of k random features from a random feature representation o f K. It turns out that these two settings are closely related. Both our theore tical analysis and experiments confirm that conditional KRR outperforms standard KRR in these cases whenever the $\mathcal{F}$-component of the regression funct ion is more pronounced than the residual part.
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RoboMME: Benchmarking and Understanding Memory for Robotic Generalist Policies

Yinpei Dai, Hongze Fu, Jayjun Lee, Yuejiang Liu, Haoran Zhang, Jianing Yang, Chelsea Finn, Nima Fazeli, Joyce Chai
Memory is critical for long-horizon and history-dependent robotic manipulation. Such tasks often involve counting repeated actions or manipulating objects that become temporarily occluded. Recent vision-language-action (VLA) models have begun to incorporate memory mechanisms; however, their evaluations remain confined to narrow, non-standardized settings. This limits systematic understanding, comparison, and progress measurement. To address these challenges, we introduce **RoboMME**: a large-scale standardized benchmark for evaluating and advancing VLA models in long-horizon, history-dependent scenarios. Our benchmark comprises 16 manipulation tasks constructed under a carefully designed taxonomy that evaluates *temporal*, *spatial*, *object*, and *procedural* memory. We further develop a suite of 14 memory-augmented VLA variants built on the $\pi_{0.5}$ backbone to systematically explore different memory representations across multiple integration strategies. We show that the effectiveness of memory representations is highly task-dependent, with each design offering distinct advantages and limitations across different tasks. Videos and code can be found at https://robomme.github.io
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SAW-Bench: Learning Situated Awareness in the Real World
Chuhan Li, Rilyn R. Han, Joy Hsu, Yongyuan Liang, Rajiv Dhawan, Jiajun Wu, Ming-Hsuan Yang, Xin Eric Wang
A core aspect of human perception is *situated awareness*, the ability to relate ourselves to the surrounding physical environment and reason over possible actions in context. However, most existing benchmarks for multimodal foundation models (MFMs) emphasize **environment-centric** spatial relations (relations among objects in a scene), while largely overlooking **observer-centric** relationships that require reasoning relative to agent's viewpoint, pose, and motion. To bridge this gap, we introduce SAW-Bench (**S**ituated **A**wareness in the Real **W**orld), a novel benchmark for evaluating egocentric situated awareness using real-world videos. SAW-Bench comprises 786 self-recorded videos captured with Ray-Ban Meta (Gen 2) smart glasses spanning diverse indoor and outdoor environments, and over 2071 *human-annotated* question-answer pairs. It probes a model’s observer–environment understanding with *six* different awareness tasks. Our comprehensive evaluation reveals a human-model performance gap of 37.66%, even with the best-performing MFM, Gemini 3 Flash. Beyond this gap, our in-depth analysis uncovers several notable findings; for example, while models can exploit partial geometric cues in egocentric videos, they often fail to infer a coherent camera geometry, leading to systematic spatial reasoning errors. We position SAW-Bench as a benchmark for situated spatial intelligence, moving beyond passive observation to understanding physically grounded, observer-centric dynamics.
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SlerpFlow: Spherical Trajectory Correction for Rectified Flow Inversion
Wenbin Duan, Yan Shu, Zhuoyuan Fu, Fangmin Zhao, Yan Li, Yaru Zhao, Binyang Li
Rectified-flow-based diffusion transformers, particularly FLUX, have demonstrated outstanding performance in high-quality image generation. However, achieving fast and accurate inversion—transforming images back to latent noise for faithful reconstruction and editing—remains a challenging bottleneck due to the discretization errors of linear solvers. This paper introduces SlerpFlow, a straightforward yet highly effective zero-shot approach that unlocks the full potential of FLUX for high-fidelity inversion and editing. Unlike existing approaches (e.g., RF-Solver) that rely on complex numerical approximations such as high-order Taylor expansions to correct trajectory errors, we present a geometric view based on the Manifold Hypothesis: the empirically observed trajectory curvature is not a numerical artifact, but rather serves as a necessary “centripetal force” that constrains the flow to remain on the data manifold. Guided by this insight, SlerpFlow integrates Spherical Linear Interpolation (Slerp) to rectify flow velocity directions on the hypersphere, strictly adhering to the intrinsic curvature of the latent space. Crucially, by caching the corrected velocity for subsequent steps, SlerpFlow achieves high-precision inversion while maintaining the computation

al efficiency of a first-order Euler solver. Extensive experiments on FLUX-based reconstruction and editing tasks demonstrate that SlerpFlow improves reconstruction fidelity and achieves stronger semantic alignment in editing without requiring additional training. Code is available at https://github.com/0answer0/SlerpFlow.

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Riemannian Networks over Full-Rank Correlation Matrices

Ziheng Chen, Xiaojun Wu, Bernhard Schölkopf, Nicu Sebe

Representations on the Symmetric Positive Definite (SPD) manifold have garnered significant attention across different applications. In contrast, the manifold of full-rank correlation matrices, a normalized alternative to SPD matrices, remains largely underexplored. This paper introduces Riemannian networks over the correlation manifold, leveraging five recently developed correlation geometries. We systematically extend basic layers, including Multinomial Logistic Regression (MLR), Fully Connected (FC), and convolutional layers, to these geometries. Besides, we present methods for accurate backpropagation for two correlation geometries. Experiments comparing our approach against existing SPD and Grassmannian networks demonstrate its effectiveness.

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A Strictly Proper Scoring Rule and a Calibration Metric for Interval-Censored Data Analysis

Hiroki Yanagisawa, Shunta Akiyama

Interval-censored data present unique challenges in statistical analysis due to the partial observability of event times within known intervals, requiring assumptions about the censoring mechanism. This paper explores the theoretical relationship between two foundational assumptions: independent monitoring and non-informative censoring. We demonstrate that these assumptions are equivalent for Case-1 interval-censored data, but not for Case-K interval-censored data, where $K \geq 2$, through a synthetic dataset example. Additionally, we propose the first strictly proper scoring rule and calibration metric specifically designed for interval-censored data under the constant-sum assumption and the non-informative censoring assumption, respectively. Our empirical evaluations on real-world datasets show that a neural network model trained with our scoring rule is competitive with established statistical baselines, offering enhanced flexibility. These contributions provide significant advancements in the theoretical understanding and practical analysis of interval-censored data.

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Retrieval-Aware Distillation for Transformer-SSM Hybrids

Aviv Bick, Eric P. Xing, Albert Gu

State-space models (SSMs) offer efficient sequence modeling but lag behind Transformers on benchmarks that require in-context retrieval. Prior work links this gap to a small set of attention heads, termed Gather-and-Aggregate (G&A), which SSMs struggle to reproduce. We propose *retrieval-aware distillation*, which converts a pretrained Transformer into a hybrid student by preserving only these retrieval-critical heads and distilling the rest into recurrent heads. We identify the essential heads via ablation on a synthetic retrieval task, producing a hybrid with sparse, non-uniform attention placement. We show that preserving **just 2% of attention heads recovers over 95% of teacher performance on retrieval-heavy tasks** (10 heads in a 1B model), requiring far fewer heads than hybrids that retain at least 25%. We further find that large recurrent states often compensate for missing retrieval: once retrieval is handled by these heads, the SSM backbone can be simplified with limited loss, even with an 8× reduction in state dimension. By reducing both the attention cache and the SSM state, the resulting hybrid is 5–6× more memory-efficient than comparable hybrids, closing the Transformer–SSM gap at a fraction of the memory cost.

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LiveNewsBench: Evaluating Web Search Agents with Freshly Curated News

Yunfan Zhang, Kathleen McKeown, Smaranda Muresan

Large Language Models (LLMs) with agentic web search capabilities show strong potential for tasks requiring real-time information access and complex fact retrie

val, yet evaluating such systems remains challenging. We introduce LiveNewsBench , a rigorous and regularly updated benchmark designed to assess the agentic web search abilities of LLMs. LiveNewsBench automatically generates fresh question-a nswer pairs from recent news articles, ensuring that questions require informati on beyond an LLM's training data and enabling clear separation between internal knowledge and search capability. The benchmark features intentionally difficult questions requiring multi-hop search queries, page visits, and reasoning, making it well-suited for evaluating agentic search behavior. Our automated data curat ion and question generation pipeline enables frequent benchmark updates and supp orts construction of a large-scale training dataset for agentic web search model s, addressing the scarcity of such data in the research community. To ensure rel iable evaluation, we include a subset of human-verified samples in the test set. We evaluate a broad range of systems using LiveNewsBench, including commercial and open-weight LLMs as well as LLM-based web search APIs. The leaderboard, data sets, and code are publicly available at \url{livenewsbench.com}.

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Directly Optimizing Natural Language Explanations for Behavioral Faithfulness: S imulatability and Recoverability

Advaith Malladi, Shashank Srivastava

Natural-language explanations are widely used to interpret machine learning mode ls, yet many prioritize human plausibility over accurately reflecting or predict ing model behavior. Prior approaches often rely on human-written rationales, pro ducing post-hoc explanations that neither align with the model’s decision functi on nor generalize. We introduce $\textbf{OPeX}$, a natural-language explanation model that directly optimizes for behavioral faithfulness: the ability of an exp lanation to reflect and predict a model’s observable input–output behavior. $\te xtbf{OPeX}$ is trained using reinforcement learning with Group Relative Policy O ptimization (GRPO), optimizing two complementary metrics: recoverability, which measures whether explanations recover model predictions on seen examples, and si mulatability, which measures prediction of model behavior on unseen inputs. Acro ss structured and text-based tasks, $\textbf{OPeX}$ achieves high simulatability (~0.85) and recoverability (~0.99), outperforming GPT-4o, LLaMA-3.3-70B, MaNtLE , Chain-of-Thought (CoT)-based models, and human-written explanations, despite u sing an 8B-parameter backbone. Human user studies show a 15% improvement in clas sification accuracy over competent baselines.

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WaveSSM: Multiscale State-Space Models for Non-stationary Signal Attention

Ruben Solozabal, Velibor Bojkovic, Hilal AlQuabeh, Klea Ziu, Kentaro Inui, Marti n Taká■

State-space models (SSMs) have emerged as a powerful foundation for long-range s equence modeling, with the HiPPO framework showing that continuous-time projecti on operators can be used to derive stable, memory-efficient dynamical systems th at encode the past history of the input signal. However, existing projection-bas ed SSMs often rely on polynomial bases with global temporal support, whose induc tive biases are poorly matched to signals exhibiting localized or transient stru cture. In this work, we introduce \emph{WaveSSM}, a collection of SSMs construct ed over wavelet frames. Our key observation is that wavelet frames yield a local ized support on the temporal dimension, useful for tasks requiring precise local ization. Empirically, we show that on equal conditions, \textit{WaveSSM} outperf orms orthogonal counterparts as S4 on real-world datasets with transient dynamic s, including physiological signals on the PTB-XL dataset and raw audio on Speech Commands.

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Causal Effect Identifiability in the Presence of Latent Confounders Without Auxi liary Variables

Xiu-Chuan Li, James Kwok, Jiaxian Guo, Tongliang Liu

It is a fundamental challenge to ascertain whether the causal effect of a treatm ent on an outcome is identifiable in the presence of latent confounders, which s erves as the logical prerequisite for recovering the causal effect in a partiall y observed system. While prior literature demonstrates that the causal effect is

identifiable when there exist auxiliary variables subject to stringent structural constraints, this paper investigates identifiability of the causal effect without such variables. This means that we ground identifiability solely in the joint distribution of the treatment-outcome pair, which constitutes the irreducible statistical basis for causal effect identification. Focusing on linear structural causal models (SCMs), we provide a nuanced and complete characterization of identifiability of the causal effect contingent on the distributional properties of exogenous noises. Specifically, we formulate a set of mutually exclusive and collectively exhaustive conditions regarding the Gaussianity of exogenous noises, ascertain under which conditions the causal effect is identifiable and under which it is not, while also quantifying the cardinality of the feasible solution set for the unidentifiable cases. Finally, we empirically validate our theoretical findings.

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A Bayesian Approach to Quantify the Uncertainty of Human Ratings in a Single-Instance Multimodal Framework

Zijian Chen, Archana Venkataraman

Human ratings are central to learning and inference across several application domains, but they are also subject to inter-rater biases and judgment errors. Quantifying the uncertainty of these human ratings would require repeated measurements, which are expensive and rarely available at scale. We propose a Bayesian graphical model to estimate the instance-level and item-level uncertainty of (subjective) human ratings by leveraging auxiliary (objective) data. Our model learns a shared latent content representation that explains factors common to both the human rating and auxiliary data and a latent uncertainty variable that captures fluctuations in the human assessments via a data-conditioned prior. We develop a scalable amortized variational inference procedure that uses modality-appropriate neural encoders and decoders to represent the posterior factors. Experiments on synthetic data demonstrate that our framework can accurately recover the latent uncertainty under targeted ablations and stress tests. We further demonstrate our approach on a real-world dataset of paired functional MRI scans and behavioral testing for autism, thus highlighting the need for uncertainty quantification.

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Bootstrapped Exploration with Causal Reasoning: A Training Paradigm for Adaptive Forecasting Agent

Qingwen Zeng, Dajun Guo, Zhaoge Bi, lining chen, Jushang Qiu, Yitian Yang, Carl Yang, Huaming Chen, Ling Chen

Time series forecasting is critical in domains such as finance, energy, and healthcare, yet real-world datasets often exhibit non-stationarity, noise, missing values, and distribution shifts, posing severe challenges for generalization. In practice, industry solutions typically rely on customized forecasting frameworks that combine imputation, decomposition, and specialized models. However, such frameworks are costly to engineer and maintain. Moreover, we observe that many frameworks suffer from the impacts of distribution shifts, which degrade their respective performance. It motivates a paradigm that transfers reliably across heterogeneous datasets while accumulating reusable strategy knowledge for large-scale, dynamic environments. Although large language model-based agents have recently shown strong reasoning and tool-use capabilities, existing approaches do not consistently adapt forecasting workflows across diverse time series. We identify two primary factors, including limited strategy-level supervision and the inherent complexity of mapping dataset-specific meta-features to effective forecasting strategies. To address these challenges, we propose BECRA, a novel agent training paradigm that learns forecasting intelligence through contrast-aware exploration and agent-level causal lesson extraction, without human-annotated supervision. BECRA distills symbolic strategy lessons that support in-context planning on unseen datasets, enabling zero-shot training adaptation.

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Rethinking Attention in Spiking Transformers: Overcoming Density Bias with Set Similarity

JinGyo Lim, Seung Gyu Jeong, Seong-Eun Kim
Recent Spiking Transformer models have explored a variety of attention mechanism
s beyond standard dot-product formulations. However, many existing similarity-ba
sed spiking attention formulations remain inherently sensitive to firing density
, causing neurons with high spike rates to dominate attention scores regardless
of semantic relevance. This density bias is particularly problematic in event-dr
iven spiking representations, where sparse spike patterns often carry critical i
nformation. To address this limitation, we rethink spiking attention from a set-
theoretic perspective. We propose DiceFormer, a novel Spiking Transformer archit
ecture driven by Spike Dice Attention (SDA). Unlike traditional approaches, SDA
replaces density-sensitive measures with a set similarity function derived from
the Dice coefficient. By explicitly normalizing for firing density, SDA focuses
on spike co-occurrence rather than high firing rates. We primarily evaluate Dice
Former on the challenging audio domain, where spike sparsity varies substantiall
y across inputs. On AudioSet-20k, DiceFormer achieves a SOTA mAP of 0.161 with 5
4.3M parameters, outperforming prior SNN-based approaches and substantially narr
owing the performance gap with ANN-based models. We also introduce Lin-SDA, a li
nearized version for computation efficiency, while achieving performance compara
ble to SDA. Beyond audio, we evaluate the effectiveness of SDA on CIFAR-100 to v
erify its applicability to the vision domain.
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BizFinBench.v2: Towards Reliable LLMs in Finance via Real-User Data and Offline/
Online Bilingual Evaluation
Xin Guo, Rongjunchen Zhang, Guilong Lu, Xuntao Guo, Jia Shuai, Zhi Yang, Liwen Z
hang
Large language models are becoming increasingly significant in financial applica
tions. Nevertheless, prevailing benchmarks are largely dependent on simulated or
 generic data, which leads to a significant gap between reported performance and
 actual efficacy in real-world scenarios. To tackle this challenge, we present B
izFinBench.v2, the first integrated offline and online benchmark built upon auth
entic user query-response data from both Chinese and U.S. equity markets. It com
prises 28,860 questions across eight offline and two online tasks. Experimental
results show that GPT-5 achieves a mere 61.5\% accuracy, still failing to meet t
he practical business requirement (84.8\%). Among the evaluated commercial model
s, DeepSeek-R1 exhibits superior investment efficacy. Error analysis grounded in
 real financial practice reveals persistent limitations in existing models. By o
vercoming the constraints of prior benchmarks, BizFinBench.v2 provides a substan
tiated foundation for advancing LLM deployment in the financial sector. Our data
 and code are available at https://github.com/HiThink-Research/BizFinBench.v2.
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Controlling the Risk of Corrupted Contexts for Language Models via Early-Exiting
Andrea Wynn, Metod Jazbec, Charith Peris, Rinat Khaziev, Anqi Liu, Daniel Khasha
bi, Eric Nalisnick
Large language models (LLMs) can be influenced by harmful or irrelevant context,
 which can significantly harm model performance on downstream tasks. This motiva
tes principled designs in which LLM systems include built-in mechanisms to guard
 against such "garbage in, garbage out" scenarios. We propose a novel approach t
o limit the degree to which harmful context can degrade model performance. First
, we define a baseline "safe" behavior for the model -- the model's performance
given no context at all (zero-shot). Next, we apply distribution-free risk contr
ol (DFRC) to control the extent to which the user-provided context can decay per
formance below this safe zero-shot baseline. We achieve this by leveraging dynam
ic early exit prediction, ignoring later attention heads that attend the most to
 the unsafe inputs. Finally, we propose modifications to DFRC that allow it to b
oth control risk for harmful inputs *and* leverage performance and efficiency ga
ins on helpful inputs. We present both theoretical and empirical results across
 9 tasks spanning in-context learning and open-ended question answering, showing
 that our approach can effectively control risk for harmful context and simultan
eously achieve substantial computational efficiency gains with helpful context.
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Scaling Agentic Verifier for Competitive Coding
Zeyao Ma, Jing Zhang, Xiaokang Zhang, Jiaxi Yang, Zongmeng Zhang, Jiajun Zhang, Yuheng Jing, Lei Zhang, Hao Zheng, Wenting Zhao, Junyang Lin, Binyuan Hui
Large language models (LLMs) have demonstrated strong coding capabilities but still struggle to solve competitive programming problems correctly in a single attempt.
Execution-based re-ranking offers a promising test-time scaling strategy, yet existing methods are constrained by either difficult test case generation or inefficient random input sampling.
To address this limitation, we propose **Agentic Verifier**, an execution-based agent that actively reasons about program behaviors and searches for highly discriminative test inputs that expose behavioral discrepancies among candidate solutions.
Through multi-turn interaction with code execution environments, the verifier iteratively refines the candidate input generator and produces targeted counterexamples rather than blindly sampling inputs.
We train the verifier to acquire this discriminative input generation capability via a scalable pipeline combining large-scale data synthesis, rejection fine-tuning, and agentic reinforcement learning.
Extensive experiments across five competitive programming benchmarks demonstrate consistent improvements over strong execution-based baselines, achieving up to **+10-15\%** absolute gains in Best@k accuracy.
Further analysis reveals clear test-time scaling behavior and highlights the verifier’s broader potential beyond reranking.
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Scaling Multi-Agent Environment Co-Design with Diffusion Models
Hao Xiang Li, Michael Amir, Amanda Prorok
The agent-environment co-design paradigm jointly optimises agent policies and environment configurations in search of improved system performance, promising to fundamentally reshape how we deploy multi-agent systems in domains such as warehouse logistics and windfarm management. However, current co-design methods collapse under high dimensional environment design spaces and suffer from sample inefficiency when addressing moving targets inherent to joint optimisation. We address this by developing **Diffusion Co-Design** (DiCoDe), a scalable and sample-efficient co-design framework incorporating two core innovations. We introduce Projected Universal Guidance (PUG), enabling exploration of constraint-satisfying reward-maximising environments, and devise a critic distillation mechanism to transfer knowledge from the reinforcement learning loop to a guided diffuision model. Together, these improvements lead to superior environment-policy pairs when validated on challenging multi-agent co-design benchmarks, for example, exceeding state-of-the art in a warehouse setting with 39% higher rewards and 66% fewer simulation steps.
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TMS: Trajectory-Mixed Supervision for On-Policy Self Distillation
Rana Shahroz, Zijie Liu, Zhen Tan, Charles Fleming, Tianlong Chen
Reinforcement Learning (RL) and Supervised Fine-Tuning (SFT) are the two dominant paradigms for enhancing Large Language Model (LLM) performance on downstream tasks. While RL generally preserves broader model capabilities (retention) better than SFT, it comes with significant costs: complex reward engineering, instability, and expensive on-policy sampling. In contrast, SFT is efficient but brittle, often suffering from catastrophic forgetting due to $\textbf{Supervision Mismatch}$: the divergence between the model's evolving policy and static training labels. We address this trade-off with $\textbf{Trajectory-Mixed Supervision (TMS)}$, a reward-free framework that approximates the on-policy benefits of RL by creating a dynamic curriculum from the model's own historical checkpoints. TMS minimizes $\textit{Policy-Label Divergence (PLD)}$, preventing the mode collapse that drives forgetting in standard SFT. Experiments across reasoning (MATH, GSM8K) and instruction-following benchmarks demonstrate that TMS effectively shifts the accuracy-retention Pareto frontier. While RL remains the gold standard for retention, TMS significantly outperforms standard and iterative SFT, bridging the g

ap to RL without requiring reward models or verifiers. Mechanistic analysis conf
irms that PLD drift accurately predicts forgetting, and that TMS successfully mi
tigates this drift.
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LIMMT: Less Is More for Motion Tracking

Yu Guan, Zekun Qi, Chenghuai Lin, Xuchuan Chen, Wenyao Zhang, Jilong Wang, XinQi
ang Yu, He Wang, Li Yi

We argue that high-quality motion data can steer tracking policies toward better
 optimization trajectories early in training. In this work, we introduce LIMMT (
Less Is More for Motion Tracking). To our knowledge, this is the first data-cent
ric study for physics-based humanoid motion tracking. We go beyond simply removi
ng erroneous clips. We define motion data quality through three dimensions: phys
ics feasibility, diversity, and complexity. We show that training with under 3%
of AMASS yields better tracking performance than training with the full dataset.
 Extensive experiments and analyses validate the effectiveness of our framework.
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Judgment Operators: A Composition-Invariant Substrate for Multi-Agent Action Spa
ces

Jun Li

As large language models (LLMs) are increasingly composed into heterogeneous mul
ti-agent systems, a fundamental reliability challenge emerges: knowledge and gov
ernance **fragment** across agents, leading to composition-dependent behaviors a
nd linear scaling of violations. Two empirical observations motivate our approac
h: internal correction methods (Reflexion, CRITIC, Self-Refine) fail to enforce
execution-time constraints deterministically, and filter-based methods (LlamaGua
rd) sacrifice task utility without repair. We introduce **Judgment Operators (JO
)**, a decision-time framework implementing four-way intervention semantics (*Al
low, Edit, Escalate, Deny*) via a portable artifact $J=(\mathcal{C},\mathcal{P})
$, enabling minimal repair without modifying agent internals. We establish: (1)
composition-invariant enforcement with constant violation probability, motiv
ated by a necessity lower bound showing no decentralized scheme achieves $O(1)$
violation probability; (2) **sublinear** mistake accumulation via JO-A; and (3)
semantic preservation for code transformation. Programmatically verifiable c
onstraints provide the formal foundation; empirically, robustness extends beyond
 this scope (partial mediation 3--5$\times$ better than theoretical worst-case).
 JO achieves 0\% observed violation rate (vs. 48--68\% baselines), 13.5--20.5\%
zero-shot cross-model transfer where few-shot prompting fails, and provides a **
portable, auditable, and composable** interface for governance and capability in
jection in multi-agent LLM systems.
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ArcVQ-VAE: A Spherical Vector Quantization Framework with ArcCosine Additive Mar
gin

Jaeyung Kim, YoungJoon Yoo

Vector Quantized Variational Autoencoder (VQ-VAE) has become a fundamental frame
work for learning discrete representations in image modeling.
However, VQ-VAE models must tokenize entire images using a finite set of codeboo
k vectors, and this capacity limitation restricts their ability to capture rich
and diverse representations.
In this paper, we propose ArcCosine Additive Margin VQ-VAE (ArcVQ-VAE), a novel
vector quantization framework that introduces a spherical angular-margin prior (
SAMP) for the codebook of a conventional VQ-VAE.
The proposed SAMP consists of Ball-Bounded Norm Regularization, which constrains
 all codebook vectors within a time-dependent Euclidean ball, and ArcCosine Addi
tive Margin Loss, which encourages greater angular separability among latent vec
tors.
This formulation promotes more discriminative and uniformly dispersed latent rep
resentations within the constrained space, thereby enabling codebook vectors to
capture richer information and leading to improved codebook utilization.
Experimental results on standard image reconstruction and generation tasks show
that ArcVQ-VAE outperforms baseline models in terms of reconstruction accuracy,

representation diversity, and sample quality.
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Multimodal Nested Learning for Decoupled and Coordinated Optimization
Yanglin Feng, Yang Qin, Dezhong Peng, Rui Wang, Xiaomin Song, Peng Hu
Multimodal learning aims to integrate multi-sensor data to exploit their complementary information, embracing a more comprehensive real-world perception and understanding. However, heterogeneous discrepancies across modalities consistently trigger imbalanced multimodal optimization, restricting the joint learning performance. Although existing methods mitigate this issue through optimization modulation and conflict alleviation, they still suffer from entangled optimization and uniform learning pace in conventional monolithic frameworks, limiting the effectiveness of multimodal learning. To address this issue, we propose a novel Multimodal Nested Learning Framework (MoNet), which reformulates the monolithic framework into nested sub-processes, decoupling and coordinating multimodal learning. To achieve this, we present a Decoupled Multimodal Stable Memory block (DMSM) as the outermost nested level, which decouples multimodal learning into independent optimization streams for semantic exploitation across modalities. Additionally, we develop an Adaptive Multimodal Coordinated Fusion block (AMCF), which constitutes the inner nested level. It attempts to coordinate multimodal information integration across multi-timescale nested memories, balancing multimodal fusion. Extensive experimental results on eight datasets across three tasks demonstrate the superiority of MoNet. Code is available at https://github.com/Yangl1nFeng/MoNet.
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RTInfer: Real-Time Inference of Multiple DNNs on Edge GPUs
Renjie Li, Tong Sun, Yi Gao, Wei Dong
While edge GPUs are increasingly used for latency-critical DNN tasks, limited resources often fail to meet strict real-time (RT) requirements under concurrent workloads.
Existing preemption and early-exit mechanisms often underutilize GPU resources through single-task queuing and sacrifice excessive accuracy during task bursts.
To address this, we propose RTInfer, a novel system that enables concurrent RT task execution while balancing throughput and accuracy.
RTInfer integrates an accuracy-calibrated lightweight variant co-optimization to generate efficient models, a memory-layout-aware scheduler to mitigate fragmentation during preemption, and an on-demand loading strategy to minimize host-to-GPU latency.
Extensive evaluations demonstrate that RTInfer outperforms state-of-the-art methods by reducing average deadline miss rate (DMR) from 32.8\% to 0\% and improving accuracy by up to 56.5\%.
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Rethinking Depth Pruning for Vision Transformers: A Heterogeneity-Aware Perspective
Zhenfeng Su, Kang Zhao, Han Bao, Tao Yuan, Zhongzhe Hu, Xianzhi Yu, Wenxuan Wang
While prior studies have successfully compressed vision Transformers (ViTs) through various pruning techniques, most have concentrated on width pruning to achieve significant reductions in model size. Depth pruning, which removes entire layers from a ViT, is notoriously difficult for accuracy recovery despite its potential to deliver higher speedups, limiting the acceleration achieved by existing joint width-and-depth pruning methods. In this work, we reveal that the failure of existing depth pruning methods lies in their neglect of heterogeneity between different layers, and we introduce HetDPT, a heterogeneity-aware depth pruning method that avoids dimension mismatch. Comprehensive experiments on ImageNet-1K, CIFAR-100, COCO, and ADE20K validate our method: HetDPT achieves a 1.58$\times$ speedup for DeiT-B while maintaining accuracy and a 1.39$\times$ speedup for DeiT-S with nearly no accuracy degradation. Furthermore, when combined with width pruning, HetDPT+ sets a new state-of-the-art record in extreme ViT pruning, enhancing the acceleration ratio from 4.24$\times$ to 5.19$\times$ for the Isomorphic-Pruning-2.6G configuration while maintaining near-lossless accuracy; our code is available at \url{https://github.com/Efficient-AI-for-All/HetDPT}.

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Stem: Rethinking Causal Information Flow in Sparse Attention
Lin Niu, Xin Luo, Linchuan Xie, Yifu Sun, Guanghua Yu, Jianchen Zhu, S Kevin Zhou
The quadratic computational complexity of self-attention remains a fundamental bottleneck for scaling Large Language Models (LLMs) to long contexts, particularly during the pre-filling phase. In this paper, we rethink the causal attention mechanism from the perspective of information flow. Due to causal constraints, tokens at initial positions participate in the aggregation of every subsequent token. However, existing sparse methods typically apply a uniform top-k selection across all token positions within a layer, ignoring the cumulative dependency of token information inherent in causal architectures. To address this, we propose Stem, a novel, plug-and-play sparsity module aligned with information flow. First, Stem employs the Token Position-Decay strategy, applying position-dependent top-k within each layer to retain initial tokens for recursive dependencies. Second, to preserve information-rich tokens, Stem utilizes the Output-Aware Metric. It prioritizes high-impact tokens based on approximate output magnitude. Extensive evaluations demonstrate that Stem achieves superior accuracy with reduced computation and pre-filling latency. Code is available at https://anonymous.4open.science/r/stem-254ENLXL.
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CVSearch: Empowering Multimodal LLMs with Cognitive Visual Search for High-Resolution Image Perception
Liupeng Li, Haoqian Kang, Zhenyu Lu, Jinpeng Wang, Bin Chen, Ke Chen, Yaowei Wang
High-resolution (HR) image perception presents a key bottleneck for multimodal large language models (MLLMs).
While visual search offers a promising solution, existing methods struggle with the trade-off between coverage and efficiency. Visual expert-assisted search is efficient but prone to blind spots when proposals fail, whereas scan-based search guarantees coverage at the cost of computational redundancy and semantic fragmentation.
To address this dilemma, we introduce CVSearch, a training-free adaptive framework that dynamically schedules search strategies via an Assess-then-Search workflow.
Specifically, CVSearch first invokes expert-assisted search when global information is insufficient, and only triggers a novel semantic-aware scanning mechanism upon failure. Distinct from rigid grid partitioning, this efficient scanning paradigm incorporates Semantic Guided Adaptive Patching to decompose images into semantically consistent regions, effectively mitigating object fragmentation.
Furthermore, we devise a Dynamic Bottom-Up Search strategy driven by a Visual Complexity prior to enable efficient and precise iterative exploration of local details.
Extensive experiments on HR benchmarks demonstrate that CVSearch achieves state-of-the-art accuracy while substantially improving search efficiency.
Code is released at https://github.com/liliupeng28/ICML26-CVSearch.
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UniPercept: Towards Unified Perceptual-Level Image Understanding across Aesthetics, Quality, Structure, and Texture
Shuo Cao, Jiayang Li, Xiaohui Li, Yuandong Pu, Kaiwen Zhu, Yuanting Gao, Siqi Luo, Yi Xin, Qi Qin, Yu Zhou, Xiangyu Chen, Wenlong Zhang, Bin Fu, Yu Qiao, Yihao Liu
Multimodal large language models (MLLMs) have achieved remarkable progress in visual understanding tasks such as visual grounding, segmentation, and captioning. However, their ability to perceive perceptual-level image features remains limited. In this work, we present UniPercept-Bench, a unified framework for perceptual-level image understanding across three key domains: Aesthetics, Quality, Structure and Texture. We establish a hierarchical definition system and construct large-scale datasets to evaluate perceptual-level image understanding. Based on this foundation, we develop a strong baseline UniPercept trained via Domain-Adapt

ive Pre-Training and Task-Aligned RL, enabling robust generalization across both Visual Rating (VR) and Visual Question Answering (VQA) tasks. UniPercept outperforms existing MLLMs on perceptual-level image understanding and can serve as a plug-and-play reward model for text-to-image generation. This work defines perceptual-level image understanding in the era of MLLMs and, through the introduction of a comprehensive benchmark together with a strong baseline, provides a solid foundation for advancing perceptual-level multimodal image understanding.

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T2AV-Compass: Towards Unified Evaluation for Text-to-Audio-Video Generation

Zhe Cao, Tao Wang, Jiaming Wang, Yanghai Wang, Yuanxing Zhang, Jiahao Wang, Jialu Chen, Miao Deng, Chenxi Liao, Yize Zhang, Yubin Guo, Zhaoxiang Zhang, Jiaheng Liu

Text-to-Audio-Video (T2AV) generation aims to synthesize temporally coherent video and semantically synchronized audio from natural language, yet its evaluation remains fragmented, often
relying on unimodal metrics or narrowly scoped benchmarks that fail to capture cross-modal
alignment, instruction following, and perceptual realism under complex prompts. To address this
limitation, we present T2AV-Compass, a unified benchmark for comprehensive evaluation of
T2AV systems, consisting of 500 diverse and complex prompts constructed via a taxonomy-driven
pipeline to ensure semantic richness and physical plausibility. Besides, T2AV-Compass introduces
a dual-level evaluation framework that integrates objective signal-level metrics for video quality, audio quality, and cross-modal alignment with a subjective MLLM-as-a-Judge protocol for
instruction following and realism assessment. Extensive evaluation of 15 representative T2AV
systems reveals that even the strongest models fall substantially short of human-level realism and
cross-modal consistency, with persistent failures in audio realism, fine-grained synchronization,
instruction following, etc. These results indicate significant improvement room for future models
and highlight the value of T2AV-Compass as a challenging and diagnostic testbed for advancing
text-to-audio-video generation.

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Asymptotically Fast Clebsch-Gordan Tensor Products with Vector Spherical Harmonics

YuQing Xie, Ameya Daigavane, Mit Kotak, Tess Smidt

$E(3)$-equivariant neural networks have proven to be extremely effective in a wide range of 3D modeling tasks. A fundamental operation of such networks is the tensor product, which allows interaction between different feature types. Because this operation scales poorly, there has been considerable work towards accelerating this interaction. However, recently [Xie et al. 2025](https://openreview.net/forum?id=EvIwwGYTLc) have pointed out that most speedups come from a reduction in expressivity rather than true algorithmic improvements on computing Clebsch-Gordan tensor products. A modification of Gaunt tensor product ([Luo et al.](https://openreview.net/forum?id=mhyQXJ6JsK)) can give a true asymptotic speedup but is incomplete and misses many interactions. In this work, we provide the first complete algorithm which truly provides asymptotic benefits Clebsch-Gordan tensor products. For full CGTP, our algorithm brings runtime complexity from the naive $O(L^6)$ to $O(L^4\log^2 L)$, close to the lower bound of $O(L^4)$. We first show how generalizing fast Fourier based convolution naturally leads to the previously proposed Gaunt tensor product ([Luo et al.](https://openreview.net/forum?id=mhyQXJ6JsK)). To remedy antisymmetry issues, we generalize from scalar signals to irrep valued signals, giving us tensor spherical harmonics. We prove a gener

alized Gaunt formula for the tensor harmonics. Finally, we show that we only nee d up to vector valued signals to recover the missing interactions of Gaunt tenso r product.
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OmniFit: Bridging Modalities via Layer-Adaptive Token Compression for Omnimodal Large Language Models
Zining Wang, Zhihang Yuan, Yingjie Zhai, Wenshuo Li, Han Shu, Ruihao Gong, Jinya ng Guo, Xianglong Liu
Emerging Omni-modal Large Language Models (OmniLLMs) enable real-time interactio n across video, audio, and text but suffer from prohibitive computational costs due to the quadratic complexity of processing continuous streaming inputs. Exist ing token compression strategies remain suboptimal as they typically rely on bia sed modality-centric priors or enforce uniform retention policies, neglecting th e heterogeneity across layers and the critical role of cross-modality alignment. To address these challenges, we propose OmniFit, a training-free framework that decouples interaction profiling from inference execution. OmniFit incorporates Layer-Adaptive Heterogeneity Profiling (LAHP) to dynamically allocate computatio nal budgets based on layer-wise redundancy and modality preferences, preserving tokens according to the characteristics of each layer. Furthermore, we introduce Alignment-Rectified Token Selection (ARTS), a lightweight mechanism that effici ently identifies tokens semantically aligned with cross-modal cues. Extensive ex periments on 3 model series across 10 benchmarks demonstrate that OmniFit establ ishes a new Pareto frontier, retaining 98\% of model performance with only 20\% token usage and achieves up to 2.31$\times$ end-to-end inference speedup and 2.5 $\times$ VRAM saving, significantly outperforming state-of-the-art methods.
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PAMD: Structured Adaptive Distances for Bisimulation Representations in Visual R einforcement Learning
Daegyeong Roh, Juho Bae, Han-Lim Choi
Many visual reinforcement learning (RL) algorithms learn representations by matc hing latent distances to a behavioral distance induced by reward and transition similarity. In practice, the choice of the latent distance can strongly affect p erformance: using a fixed, pre-specified global norms (e.g., ℓ_p norms or o ther hand-designed metrics) may be overly restrictive to capture the behavioral distance. In contrast, unconstrained pairwise distances may admit degenerate sol utions that drive the metric loss down without improving the representation. To address this gap, we introduce **PAMD: Pairwise Adaptive Mahalanobis Distance**, which parameterizes a positive-definite, pair-conditioned metric for measuring latent state similarity.

PAMD is a simple plug-in for existing bisimulation-based methods, offering a m ore expressive yet structured alternative to fixed, pre-specified latent distanc es. We empirically validate our method on visual MuJoCo continuous-control tasks , where final performance of several recent bisimulation-based RL algorithms is substantially improved when equipped with the distance we propose.
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Modeling Temporal scRNA-seq Data with Latent Gaussian Process and Optimal Transp ort
Mehmet Yigit Balik, Harri Lähdesmäki
Single-cell RNA sequencing provides insights into gene expression at single-cell resolution, yet inferring temporal processes from these static snapshot measure ments remains a fundamental challenge. Current approaches utilizing neural diffe rential equations and flows are sensitive to overfitting and lack careful consid erations of biological variability. In this work, we propose a generative framew ork that models population trends using a latent heteroscedastic Gaussian proces s (GP) approximated by Hilbert space methods. To address the absence of genuine cell trajectories, we leverage an optimal transport (OT) objective that aligns g enerated and observed population distributions. Our method explicitly captures b iological heterogeneity by incorporating cell-specific latent time and cell type conditioning to disentangle temporal asynchrony and trajectories to different c ell types. We demonstrate state-of-the-art performance on complex interpolation

and extrapolation benchmarks and introduce a novel gradient-based strategy for i
nferring perturbation trajectories.
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BLOCK-EM: Preventing Emergent Misalignment via Latent Blocking
Muhammed Ustaomeroglu, Guannan Qu
Emergent misalignment can arise when a language model is fine-tuned on a narrowl
y scoped supervised objective: the model learns the target behavior, yet also de
velops undesirable out-of-domain behaviors. We investigate a mechanistic approac
h to preventing emergent misalignment by identifying a small set of internal fea
tures that reliably control the misaligned behavior and then discouraging the mo
del from strengthening these features during fine-tuning. Across six fine-tuning
 domains, blocking (i.e., constraining) a fixed set of features achieves up to 9
5\% relative reduction in emergent misalignment with no degradation in model qua
lity or target-task performance. We strengthen validity with disjoint selection/
evaluation splits, multiple independent judges, multiple random seeds for key se
ttings, quality metrics, and extensive ablations demonstrating that the reductio
n in misalignment is specific to the identified mechanism. We also characterize
a limiting regime in which misalignment re-emerges under prolonged fine-tuning,
present evidence consistent with rerouting through alternative features or layer
s, and evaluate modifications that partially restore the misalignment-blocking e
ffect. Overall, our results show that targeted training-time constraints on inte
rnal mechanisms can mitigate emergent misalignment without degrading target-task
 performance.
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SOTAlign: Semi-Supervised Alignment of Unimodal Vision and Language Models via O
ptimal Transport
Simon Roschmann, Paul Krzakala, Sonia Mazelet, Quentin Bouniot, Zeynep Akata
The Platonic Representation Hypothesis posits that neural networks trained on di
fferent modalities converge toward a shared statistical model of the world. Rece
nt work exploits this convergence by aligning frozen pretrained vision and langu
age models with lightweight alignment layers, but typically relies on contrastiv
e losses and millions of paired samples.
In this work, we ask whether meaningful alignment can be achieved with substanti
ally less supervision. We introduce a semi-supervised setting in which pretraine
d unimodal encoders are aligned using a small number of image-text pairs togethe
r with large amounts of unpaired data. To address this challenge, we propose SOT
Align, a two-stage framework that first recovers a coarse shared geometry from l
imited paired data using a linear teacher, and then refines the alignment on unp
aired samples via an optimal-transport-based divergence that transfers relationa
l structure without overconstraining the target space. Unlike existing semi-supe
rvised methods, SOTAlign effectively leverages unpaired images and text, learnin
g robust joint embeddings across datasets and encoder pairs, and significantly o
utperforming supervised and semi-supervised baselines.
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Demystifying Action Space Design for Robotic Manipulation Policies
Yuchun Feng, Jinliang Zheng, Zhihao Wang, Dongxiu Liu, Jianxiong Li, Jiangmiao P
ang, Tai Wang, Xianyuan Zhan
The specification of the action space plays a pivotal role in imitation-based ro
botic manipulation policy learning, fundamentally shaping the optimization lands
cape of policy learning. While recent advances have focused heavily on scaling t
raining data and model capacity, the choice of action space remains guided by ad
-hoc heuristics or legacy designs, leading to an ambiguous understanding of robo
tic policy design philosophies. To address this ambiguity, we conducted a large-
scale and systematic empirical study, confirming that the action space does have
 significant and complex impacts on robotic policy learning. We dissect the acti
on design space along temporal and spatial axes, facilitating a structured analy
sis of how these choices govern both policy learnability and control stability.
Based on 13,000+ real-world rollouts on a bimanual robot and evaluation on 500+
trained models over four scenarios, we examine the trade-offs between absolute v
s. delta representations, and joint-space vs. task-space parameterizations. Our

large-scale results suggest that properly designing the policy to predict delta actions consistently improves performance, while joint-space and task-space representations offer complementary strengths, favoring control stability and generalization, respectively.
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Batched Contextual Reinforcement
Bangji Yang, Hongbo Ma, Jiajun Fan, Ge Liu
Large Language Models (LLMs) employing Chain-of-Thought reasoning achieve strong performance but suffer from excessive token consumption that inflates inference costs. Existing efficiency methods—such as explicit length penalties, difficulty estimators, or multi-stage curricula—either degrade reasoning quality or require complex training pipelines. We introduce **Batched Contextual Reinforcement (BCR)**, a minimalist, single-stage training paradigm that unlocks efficient reasoning through a simple structural modification: training the model to solve N problems simultaneously within a shared context window, rewarded purely by per-instance accuracy. This formulation creates an implicit token budget that yields several key findings: (1) We identify a novel _task-scaling law_: as the number of concurrent problems N increases at inference time, per-problem token usage decreases monotonically---a phenomenon that arises purely at inference, holds for models both before and after training, and is unrelated to accuracy or the training procedure. BCR makes this regime practical by degrading accuracy far more gracefully than baselines as N grows, establishing N as a controllable throughput dimension. (2) BCR challenges the traditional accuracy-efficiency trade-off by demonstrating a "free lunch" phenomenon at standard single-problem (N=1) inference. Across both 1.5B and 4B model families, BCR reduces token usage by 15.8\% to 62.6\% while consistently maintaining or improving accuracy across five major mathematical benchmarks (e.g., +13.3\% on AIME25 for the 4B model). (3) Qualitative analyses reveal emergent self-regulated efficiency, where models autonomously eliminate redundant metacognitive loops without explicit length supervision. (4) Crucially, we empirically demonstrate that implicit budget constraints successfully circumvent the adversarial gradients and catastrophic optimization collapse inherent to explicit length penalties, offering a highly stable, constraint-based alternative for length control. These results establish BCR as a highly practical framework, demonstrating how simple structural training incentives can unlock latent high-density reasoning modes in LLMs.
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Re-FORC: Adaptive Reward Prediction for Efficient Chain-of-Thought Reasoning
Renos Zabounidis, Aditya Golatkar, Michael Kleinman, Alessandro Achille, Wei Xia, Stefano Soatto
We propose Re-FORC, an adaptive reward prediction method that, given a query, enables prediction of the expected future rewards as a function of the number of future thinking tokens. Re-FORC trains a lightweight adapter on reasoning models, demonstrating improved prediction with longer reasoning and larger models. Re-FORC enables: 1) early stopping of unpromising reasoning chains, reducing compute by up to 24\% compared to fixed-budget cutoffs, while maintaining accuracy, 2) optimized model and thinking length selection that outperforms the largest model alone---reaching 1.7 percentage points higher peak accuracy while needing up to 12\% less compute to match the largest model's accuracy, 3) adaptive test-time scaling, which increases accuracy by 8.8 percentage points (on average at maximum compute) over confidence-based baselines. Re-FORC allows dynamic reasoning with length control via cost-per-token thresholds while estimating computation time upfront.
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Training-Free Coverless Multi-Image Steganography with Access Control
Minyeol Bae, Si-Hyeon Lee
Coverless Image Steganography (CIS) hides information without explicitly modifying a cover image, providing strong imperceptibility and inherent robustness to steganalysis. However, existing CIS methods largely lack robust access control, making it difficult to selectively reveal different hidden contents to different authorized users. Such access control is critical for scalable and privacy-sensi

tive information hiding in multi-user settings. We propose MIDAS (Multi-Image Diffusion-based Access-controlled Steganography), a training-free diffusion-based CIS framework that enables multi-image hiding with user-specific access control via latent-level fusion. MIDAS introduces a Random Basis mechanism to suppress residual structural information, together with a theoretical analysis of information leakage, and a Latent Vector Fusion module that reshapes aggregated latents to better align with the diffusion process. Experimental results demonstrate that MIDAS consistently outperforms existing training-free CIS baselines in access control functionality, stego image quality and diversity, robustness to noise, and resistance to steganalysis, establishing a practical and scalable approach to access-controlled coverless steganography.
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Mitigating Label Shift in Tabular In-Context Learning via Test-Time Posterior Adjustment

Seunghan Lee

TabPFN has recently gained attention as a foundation model for tabular datasets, achieving strong performance by leveraging in-context learning on synthetic data. However, we find that TabPFN is vulnerable to label shift, often overfitting to the majority class in the training dataset. To address this limitation, we propose DistPFN, the first test-time posterior adjustment method designed for tabular foundation models. DistPFN rescales predicted class probabilities by downweighting the influence of the training prior (i.e., the class distribution of the context) and emphasizing the contribution of the model's predicted posterior, without architectural modification or additional training. We further introduce DistPFN-T, which incorporates temperature scaling to adaptively control the adjustment strength based on the discrepancy between prior and posterior. We evaluate our methods on over 250 OpenML datasets, demonstrating substantial improvements for various TabPFN-based models in classification tasks under label shift, while maintaining strong performance in standard settings without label shift. Code is available at this repository: https://github.com/seunghan96/DistPFN.
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Required Spine Optional Limbs: Heterogeneous Federated Learning via Backbone-sharing and Activation-guided Selection

Mingsheng Cao, Hongliang Chen, Ming Hu, Fei Gao, Qiaolong Ding, Wenke Huang, Xiaofei Xie, Junlong Zhou

Although Federated Learning (FL) offers advantages in privacy-preserving for cross-device collaborative learning, its practical deployment remains severely constrained by heterogeneous hardware resources and non-IID (non-independent and identically distributed) data across devices. Sub-model extraction has emerged as a widely adopted strategy for enabling collaborative training among devices with heterogeneous models. However, existing sub-model extraction methods in FL typically rely on coarse-grained stochastic selection or rigid rule-based neuron selection, which severely limits training performance. Specifically, stochastic strategies lead to severe parameter conflicts under non-IID data distributions, while rule-based approaches lack diversity in neuron selection per device, preventing comprehensive parameter optimization. To address this problem, this paper presents a novel sub-model extraction-based FL framework, named SpineFL, which adopts a backbone-sharing mechanism and an activation-guided pruning strategy for sub-model extraction. Specifically, SpineFL decomposes each global model layer into two portions: i) a mandatory backbone shared by all the sub-models to maintain model generalization, and ii) a dynamic portion for sub-model extraction. SpineFL adopts the activation-guided selection strategy to probabilistically select neurons according to their activation frequency from the dynamic portion to generate sub-model, where neurons exhibiting higher historical activation are more likely to be included, thereby simultaneously addressing parameter conflicts while preserving selection diversity. Experimental results demonstrate that compared with state-of-the-art heterogeneous FL methods, SpineFL can achieve up to 3.28% accuracy improvement.
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Reward Redistribution for CVaR MDPs using a Bellman Operator on L-infinity

Aneri Muni, Vincent Taboga, Esther Derman, Pierre-Luc Bacon, Erick Delage
Tail-end risk measures such as static conditional value-at-risk (CVaR) are used in safety-critical applications to prevent rare, yet catastrophic events. Unlike risk-neutral objectives, the static CVaR of the return depends on entire trajectories without admitting a recursive Bellman decomposition in the underlying Markov decision process. A classical resolution relies on state augmentation with a continuous variable. However, unless restricted to a specialized class of admissible value functions, this formulation induces sparse rewards and degenerate fixed points. In this work, we propose a novel formulation of the static CVaR objective based on augmentation. Our alternative approach leads to a Bellman operator with: (1) dense per-step rewards; (2) contracting properties on the full space of bounded value functions. Building on this theoretical foundation, we develop risk-averse value iteration and model-free Q-learning algorithms that rely on discretized augmented states. We further provide convergence guarantees and approximation error bounds due to discretization. Empirical results demonstrate that our algorithms successfully learn CVaR-sensitive policies and achieve effective performance-safety trade-offs.
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On Minimum Depth and Width of Floating-Point Neural Networks for Representing Floating-Point Functions
Sejun Park, Yeachan Park, Geonho Hwang
Research on the expressive power of neural networks has identified the minimum depth and width of neural networks that enable universal approximation and memorization. However, existing results are derived under exact arithmetic and cannot be directly applied to real implementations on computers, which can only use a finite set of numbers and inexact machine operations with round-off errors.
In this work, we study floating-point ReLU networks that have floating-point parameters and use floating-point operations. Specifically, we investigate their minimum depth and width to represent all functions from the set of floating-point vectors $\mathbb F^d$ to the set of floating-point numbers $\mathbb F$. We first show that the minimum depth for representing all functions from $\mathbb F^d$ to $\mathbb F$ is exactly three, where two layers can be sufficient if we consider a smaller domain and/or codomain. We further show that the minimum width for representing all functions from $\mathbb F^d$ to $\mathbb F$ lies between $2d$ and $2d+4$. In addition, if we restrict the domain to non-negative floats, it lies between d and $d+4$, where it can be smaller for a smaller domain, even beyond d. Our results show that the existing results analyzed under exact arithmetic do not extend to the floating-point setup.
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HSGG: Training-Free Hierarchical Scene Graph Generation with Geometry-Guided Relation Reasoning
yunzhe Liu, Wenbiao Liu, Lihui Cen, Zhe Qu, Yigang Cen, Yixiong Liang, Yongfang Xie, Shichao Kan
Scene Graph Generation (SGG) connects visual perception with structured reasoning, but is limited by scarce annotations and the long-tailed distribution of relational predicates. Training-free methods based on vision-language models (VLMs) reduce supervision requirements, yet often rely on flat graph structures and produce hallucinated relations. We present HSGG, a training-free framework for open-world Hierarchical Scene Graph Generation, whose inference proceeds in two steps. First, bidirectional hierarchical entity perception combines top-down object expansion with bottom-up attribute reasoning to construct multi-level scene hierarchies capturing part–whole semantics. Second, geometry-guided relation reasoning infers valid relations from these structured entities: geometry-aware relation filtering first prunes spatially implausible object pairs using 2D proximity, depth cues, and object scale, and geometry-grounded contrastive relation decoding then suppresses hallucinated predicates by contrasting predictions from a visually grounded expert against a hallucination-prone geometric prior, ensuring relations are both geometrically consistent and semantically coherent. Experiments show that HSGG generalizes effectively to unseen objects and predicates without training, substantially reduces relational hallucinations, and consistently impr

oves downstream reasoning performance.
****
VisionPulse: Dynamic Visual Sparsity for Efficient Multimodal Reasoning
Hengbo Xu, Shengjie Jin, Yanbiao Ma, Zhiwu Lu
With the rapid advancement of large multimodal models (LMMs), inference-time overhead has become a key bottleneck for real-world deployment.
Existing methods typically prune visual tokens at prefill, assuming the required visual evidence remains static during reasoning.
However, we empirically show that visual evidence is strongly step-dependent: only a sparse subset of visual tokens is critical at each decoding step, and the critical set evolves across reasoning.
Furthermore, we identify a coupled bottleneck where redundant visual context can steer the model toward query-irrelevant regions, lengthening the reasoning trace.
Guided by these insights, we propose **VisionPulse**, a step-wise visual token pruning framework during reasoning.
VisionPulse computes a lightweight visual attention mass to estimate the step-wise retention budget by exploiting its strong positive correlation with LMMs' effective visual token usage and retain only the most critical tokens under this budget.
By enforcing visual sparsity during reasoning, VisionPulse filters redundant visual context while preserving relevant visual evidence, shortening reasoning traces naturally.
Extensive experiments show that VisionPulse only retains 5\% of visual tokens per step with reasoning traces shortened by 11.2\%, while keeping accuracy almost unchanged.
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Large-Scale Notification Dispatch with Bundle Treatments and Multi-Outcome Uplift Optimization
Jiajing Xu, Yanyun Li, Yongbao Song, Minqin Zhu, Huxiao Ji, Linchuan Li, Cunyi Zhang, Xuanping Li, Kaiqiao Zhan, Yanan Niu
Notification dispatch plays a critical role in large-scale user engagement, involving complex trade-offs across notification timing, presentation style, multiple outcomes, and constraints.
In this paper, we formulate it as a constrained optimization over bundle treatments that jointly specify timing and presentation style, aiming to maximize incremental Daily Active Users (DAU) subject to platform-level budget and device vendor-specific quota constraints.
The problem is challenging due to multi-dimensional, small-effect uplift estimation and large-scale constrained optimization.
To address these challenges, we propose $\textbf{B}$undle $\textbf{U}$plift $\textbf{O}$ptimization with $\textbf{P}$runed $\textbf{L}$agrangian-based $\textbf{R}$elaxation (BUOPLR), a two-stage notification dispatch method that decouples uplift estimation from constrained decision-making.
BUOPLR first learns bundle-level, multi-outcome small uplift through an architecture that captures cross-treatment and cross-outcome relationships, and then performs scalable assignment by restricting the decision space and applying Lagrangian relaxation to a small set of global constraints.
Offline experiments show BUOPLR outperforms state-of-the-art methods, and online A/B tests increase DAU by 0.5\%. BUOPLR is now deployed on Kuaishou, a large-scale short video platform serving over 100 million users daily.
****
Temporal Self-Rewarding Language Models: Decoupling Chosen-Rejected via Past-Future
Yidong Wang, Xin Wang, Cunxiang Wang, Junfeng Fang, Qiufeng Wang, Jianing Chu, Xuran Meng, Shu-Xun Yang, Andrew Zhuoer Feng, Libo Qin, Wei Ye, Shikun Zhang
Self-Rewarding Language Models propose an architecture in which the Large Language Models(LLMs) both generates responses and evaluates its own outputs via LLM-as-a-Judge prompting, dynamically improving its generative capabilities through iterative Direct Preference Optimization (DPO). However, our analysis reveals a c

ritical limitation in existing self-rewarding paradigms: the synchronized improv
ement of chosen and rejected responses progressively narrows the representationa
l difference between contrasting samples, undermining effective preference learn
ing. We propose Temporal Self-Rewarding Language Models that strategically coord
inate past, present, and future model generations to sustain learning signals. O
ur dual-phase framework introduces: (1) Anchored Rejection - fixing rejected res
ponses using the past initial model's outputs and (2) Future-Guided Chosen - dyn
amically curating chosen samples using next-generation model predictions. Extens
ive experiments across three model families (Llama, Qwen, Mistral) and different
 model sizes (Llama3B/8B/70B) demonstrate significant improvements when trained
with our method compared to self-rewarding using same computation resources. For
 example, Llama3.1-8B reaches a 29.44 win rate on AlpacaEval 2.0 with our method
, outperforming the self-rewarding baseline (19.69) by 9.75. Notably, our method
 also demonstrates superior out-of-distribution generalization across mathematic
al reasoning (GSM8K), knowledge-based QA (ARC, TruthfulQA), and code generation
(HumanEval) tasks, even though we do not specifically collect such training data
. The generality of our temporal strategy is further validated by its benefits w
hen extended to online reinforcement learning settings on mathematical reasoning
 tasks.
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Post-Training Language Models for Crosslingual Consistency
Tianyu Liu, Jirui Qi, Mrinmaya Sachan, Ryan Cotterell, Raquel Fernández, Arianna
 Bisazza
Language models often respond inconsistently to translation-equivalent prompts a
cross languages, undermining the reliability of multilingual systems. To quantif
y this, we give an information-theoretic definition of crosslingual consistency
as a divergence bound between a model’s response distribution and its round-trip
 pushforward across languages. We then introduce penalized consistency optimizat
ion (PCO), a post-training procedure that couples this divergence with a Kullbac
k–Leibler penalty to a fixed reference language model. Because direct optimizati
on of PCO requires expensive on-policy roll-outs, we propose a tractable surroga
te, direct consistency optimization (DCO), which can be optimized off-policy. Ac
ross diverse language models and 26 languages, DCO significantly improves crossl
ingual consistency, outperforms existing methods, and enables targeted alignment
 of low-resource languages.
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WF-Bench: A Benchmark for Neural-Network WaveFunction Expressivity and Scaling L
aws
Lixing Zhang, Guijing Duan, Di Luo
We present a comprehensive benchmarking dataset and empirical scaling law
analysis for neural network wavefunctions by matching them to a wide spectrum of
famous many body target wavefunctions. The dataset, WF-Bench, spans multiple dis
tinct regimes of strongly correlated quantum matter, including topological state
s, Wigner
crystals, and superconducting wavefunctions, providing a diverse and challenging
 test
bed for neural network wavefunction expressivity. We introduce a systematic and
reproducible benchmarking protocol for target wavefunction matching, enabling co
nsistent
performance evaluation across different neural network wavefunction architecture
s.
By using wavefunction fidelity as the uniform metric, we discover empirical scal
ing laws that characterize how
representability depends on system size and key model parameters, including numb
er of determinant and model depth. By applying our benchmark protocol on Psiform
er and Ferminet, we show that WF-Bench establishes a unified dataset driven fram
ework for evaluating and
comparing neural network wavefunctions and for guiding the design of future arch
itectures.
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BioProBench: A Corpus and Benchmark for Biological Protocol Reasoning in Autonomous Science
Yuyang Liu, Liuzhenghao Lv, Xiancheng Zhang, Jingya Wang, Li Yuan, Yonghong Tian
The realization of autonomous scientific experimentation is currently limited by LLMs' struggle to grasp the strict procedural logic and accuracy required by biological protocols. To address this fundamental challenge, we present BioProBench, a comprehensive resource for procedural reasoning in biology. BioProBench is grounded in BioProCorpus, a foundational collection of 27,000 human-written protocols. From this corpus, we systematically constructed a dataset of over 550,000 task instances, offering both a large-scale training resource and a rigorous benchmark with novel metrics. Evaluating 10 mainstream LLMs, we find that while general comprehension is high, performance drops significantly on tasks demanding deep reasoning, quantitative precision, and safety awareness.
To demonstrate the value of BioProCorpus in mitigating these issues, we developed \textbf{ProAgent}, grounded in our corpus, ProAgent substantially advances the state-of-the-art. Code and data are available at: https://github.com/YuyangSunshine/bioprotocolbench and https://huggingface.co/datasets/BioProBench/BioProBench .
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How Language Models Process Negation
Zhejian Zhou, Tianyi Zhou, Robin Jia, Jonathan May
We study how Large Language Models (LLMs) process negation mechanistically.
First, we establish that even though open-weight models often provide wrong answers to questions involving negation, they do possess
internal components that process negation correctly.
Their poor accuracy is due to late-layer attention behavior that promotes simple shortcuts; ablating those attention modules greatly improves accuracy on negation-related questions.
Second, we uncover how models process negation.
We consider two hypotheses: models could use attention heads that attend to the phrase being negated and suppress related concepts, or they could directly construct a representation of the entire negative phrase (e.g., representing "not gas" as a vector that promotes liquids and solids).
We apply a range of observational and causal interpretability techniques on Mistral-7B and Llama-3.1-8B to show that models implement both mechanisms, with the "constructive" mechanism being more prominent.
Combined, our work deepens the understanding of LLMs' internals,
highlighting construction-dominant computations and the coexistence of competing mechanisms within LLMs.
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TiME: Test-Time Mixture-of-Experts Routing via Asymmetric CO-Optimal Transport for Continual Test-Time Adaptation
Tianlun Liu, Zhiliang Tian, Zhen Huang, Tianle Liu, Xingzhi Zhou, Feng Liu, Dongsheng Li
Large language models usually face continuous domain shifts during testing, which degrade performance on unseen shifting domains.
So, researchers propose continual test-time adaptation (CTTA) to adapt to evolving testing domains while preserving knowledge of previous domains, making adaptability-stability (A-S) balance.
Existing CTTA methods are constrained by dense base models that encode knowledge from all domains into a global model, hardly achieving the A-S balance.
We observe that the model sparsity of mixture-of-experts (MoE) models is better for achieving A–S balance than dense models.
In CTTA, however, MoE faces difficulty in (1) correctly routing samples from unseen shifting domains and (2) capturing domain-level shifts.
In this paper, we propose test-time mixture-of-experts routing (TiME) via asymmetric co-optimal transport (As-COOT): we model MoE routing in CTTA as a test-time allocation problem via COOT.
To ensure reliable routing, we propose a semantic space alignment to align sample-expert distributions via bidirectional contrastive learning.

To address COOT’s limitations in CTTA, we propose As-COOT, relaxing sample-side constraints while enforcing expert-side constraints to ensure noise robustness a nd balance expert load. Experiments show TiME outperforms baselines.
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Beyond Binary: Continuous State Optimization with Graph-Structured Objectives
Corinna Cortes, Yishay Mansour, Mehryar Mohri
Large-scale learning systems often face the challenge of balancing multiple, potentially competing objectives, such as fairness, accuracy, and latency. While recent work has formalized this as an optimization problem over binary states, many real-world control parameters---such as fairness thresholds, diversity mixing rates, or resource budgets---are continuous. In this work, we extend the framework to *continuous state spaces*. We model the problem as minimizing a sum of linear objectives subject to *movement costs* that penalize system instability. We capture the local structure of the objectives using a *dependency graph* (or factor graph), where each objective is determined by a subset of the state attributes. To address the tension between exploration and stability, we propose *Lazy Graph-LinUCB*, an algorithm that performs lazy updates to minimize switching costs while maintaining near-optimal regret. Beyond stability, we introduce three advanced mechanisms to exploit the underlying graph structure: (1) an *asynchronous* update schedule that eliminates synchronization overhead in sparse graphs; (2) an *adaptive* algorithm that learns the graph structure from data; and (3) a *joint estimator* that leverages data sharing among correlated objectives to significantly tighten regret bounds. Empirically, we demonstrate that these structural exploitations reduce movement costs by more than a factor of three in heterogeneous systems while maintaining similar cumulative losses.
****
CoLA: Cross-Modal Low-rank Adaptation for Multimodal Downstream Tasks
Wish Suharitdamrong, Tony Alex, Muhammad Awais, Sara Atito
Foundation models have revolutionized AI, but adapting them efficiently for mult imodal tasks, particularly in dual-stream architectures composed of unimodal enc oders, such as DINO and BERT, remains a significant challenge. Parameter-Efficie nt Fine-Tuning (PEFT) methods like Low-Rank Adaptation (LoRA) enable lightweight adaptation, yet they operate in isolation within each modality, limiting their ability in capturing cross-modal interactions. In this paper, we take a step in bridging this gap with Cross-Modal Low-Rank Adaptation (CoLA), a novel PEFT fram ework that extends LoRA by introducing a dedicated inter-modal adaptation pathwa y alongside the standard intra-modal one. This dual-path design enables CoLA to adapt unimodal foundation models to multimodal tasks effectively, without interf erence between modality-specific and cross-modal learning. We evaluate CoLA acro ss a range of vision-language (RefCOCO, RefCOCO+, RefCOCOg) and audio-visual (AV E, AVS) benchmarks, where it consistently outperforms LORA, achieving a relative gain of around 3\% and 2\%, respectively, while maintaining parameter efficienc y. Notably, CoLA enables the first multi-task PEFT framework for visual groundin g, bridging a key gap in efficient multimodal adaptation. Code is available at h ttps://github.com/peterwisu/CoLA
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SPATIA: Multimodal Generation and Prediction of Spatial Cell Phenotypes
Zhenglun Kong, Mufan Qiu, John Boesen, xiang lin, Sukwon Yun, Tianlong Chen, Man olis Kellis, Marinka Zitnik
Understanding how cellular morphology, gene expression, and spatial context join tly shape tissue function is a central challenge in biology. Image-based spatial transcriptomics technologies now provide high-resolution measurements of cell i mages and gene expression profiles, but existing methods typically analyze these modalities in isolation or at limited resolution.
We address the problem by introducing SPATIA, a multi-level generative and predi ctive model that learns unified, spatially aware representations by fusing morph ology, gene expression, and spatial context from the cell to the tissue level. S PATIA also incorporates a spatially conditioned generative framework with confid

ence-aware OT reweighting and morphology-profile alignment for modeling target-state morphology distributions. Specifically, we propose a confidence-aware flow matching objective that reweights weak optimal-transport pairs based on uncertainty. We further apply morphology-profile alignment to encourage biologically meaningful image generation, enabling the modeling of microenvironment-dependent phenotypic transitions. We assembled a multi-scale dataset consisting of 25.9 million cell-gene pairs across 17 tissues. We benchmark SPATIA against 18 models across 12 tasks, spanning categories such as phenotype generation, annotation, clustering, gene imputation, and cross-modal prediction. SPATIA achieves improved performance over state-of-the-art models, improving generative fidelity by 8\% and predictive accuracy by up to 3\%.
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InnoEval: On Research Idea Evaluation as a Knowledge-Grounded, Multi-Perspective Reasoning Problem
Shuofei Qiao, Yunxiang Wei, Xuehai Wang, Bin Wu, Boyang XUE, Ningyu Zhang, Hossein A. Rahmani, Wang Yanshan, Qiang Zhang, Keyan Ding, Jeff Z. Pan, Huajun Chen, Emine Yilmaz
The rapid evolution of Large Language Models has catalyzed a surge in scientific idea production, yet this leap has not been accompanied by a matching advance in idea evaluation. The fundamental nature of scientific evaluation needs knowledgeable grounding, collective deliberation, and multi-criteria decision-making. However, existing idea evaluation methods often suffer from narrow knowledge horizons, flattened evaluation dimensions, and the inherent bias in LLM-as-a-Judge. To address these, we regard idea evaluation as a knowledge-grounded, multi-perspective reasoning problem and introduce **InnoEval**, a deep innovation evaluation framework designed to emulate human-level idea assessment. We apply a heterogeneous deep knowledge search engine that retrieves and grounds dynamic evidence from diverse online sources. We further achieve review consensus with an innovation review board containing reviewers with distinct academic backgrounds, enabling a multi-dimensional decoupled evaluation across multiple metrics. We construct comprehensive datasets derived from authoritative peer-reviewed submissions to benchmark InnoEval. Experiments demonstrate that InnoEval can consistently outperform baselines in point-wise, pair-wise, and group-wise evaluation tasks, exhibiting judgment patterns and consensus highly aligned with human experts.
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Simple Policy Gradients for Reasoning with Diffusion Language Models
Anthony Zhan
Diffusion large language models (dLLMs) represent a promising alternative to autoregressive LLMs; however, the lack of effective post-training techniques, including reinforcement learning (RL), remains a key challenge for dLLMs, especially for downstream applications. Existing approaches often rely on a sequence-level view that requires biased likelihood approximations. In this work, we propose Amortized Group Relative Policy Optimization (AGRPO), a policy gradient algorithm that leverages the Markovian nature of dLLM generation, optimizing individual denoising steps rather than full sequences. Our approach improves the theoretical alignment between training and inference policies and also admits efficient, unbiased gradient updates via a novel timestep estimation scheme. We demonstrate AGRPO's effectiveness on different math and reasoning tasks, achieving absolute accuracy gains of +59.4\% and +69.7\% on Countdown and Sudoku over the base LLaDA model, exceeding comparable methods such as diffu-GRPO.
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PCRNet: Phase-aware Complex Refinement Network for EEG-based Auditory Attention Decoding
Xiran Chen, Xiaoke Yang, Jian Zhou, Zhao Lv, Cunhang Fan
Auditory attention decoding (AAD) based on Electroencephalography (EEG) aims to identify the attended speaker in multi-speaker environments. However, existing methods typically overlook the crucial phase information of EEG signals, which limits their ability to distinguish structured neural patterns from random noise in the frequency domain and hinders robust decoding. To address these issues, this paper proposes a Phase-aware Complex Refinement Network (PCRNet) for AAD, whic

h consists of a Temporal Context Calibration (TCC) module and a Dual-Domain Inte
gration (DDI) module. Specifically, the TCC module captures long-range temporal
dependencies through multi-scale temporal attention mechanism, while the DDI mod
ule employs a phase-guided spectral filtering strategy to dynamically suppress n
oise-dominated frequencies and refine the real and imaginary components separate
ly. This design enables effective phase recalibration and enhances the discrimin
ability of target features in the complex domain. Experimental results on three
public datasets demonstrate that PCRNet outperforms state-of-the-art (SOTA) meth
ods, particularly under challenging ultra-short 0.1-second windows. Code is avai
lable at: https://github.com/SunshineGreeny/PCRNet.
****

Towards Hierarchy–Uniformity Equilibrium: Recovering Semantic Depth in Hypergrap
h Contrastive Learning
Ruiting Zhao, Ming Li, Lixin Cui, Lu Bai, Feilong Cao, Ke Lv, Pietro Lio
Hypergraph contrastive learning is an effective paradigm for representation lear
ning on higher-order relational data, yet existing methods largely ignore that h
yperedges link nodes with multi-level semantics. Standard contrastive objectives
 emphasize instance discrimination via hyperspherical uniformity and tend to pus
h embeddings apart in an indiscriminate manner. We show that this leads to a *Hi
erarchy–Uniformity Conflict*, whose geometric manifestation is *Semantic Flatten
ing*, where the semantic depth of hyperedges collapses into a nearly flat cloud
of instances. To address this issue, we introduce **HyperDepth**, a hypergraph c
ontrastive learning framework that moves representations towards a hierarchy–uni
formity equilibrium by jointly coordinating spectral and geometric signals. Hype
rDepth employs a decoupled spectral encoding scheme with adaptive gating so that
 high-frequency components focus on local instance discrimination while low-freq
uency components capture global hierarchical structure. On top of this, an energ
y-based hierarchical alignment module attaches a learnable prototype tree to the
 representation space and minimizes an interpretable energy functional to recove
r the semantic depth of hyperedges. Theoretically, under a mild frequency-separa
tion assumption, we show that the local contrastive and global hierarchical obje
ctives operate on orthogonal spectral components and admit equilibrium embedding
s that preserve semantic depth while still retaining instance-level discriminati
on. Experiments on 15 hypergraph datasets and 17 supervised and self-supervised
baselines, spanning homophilic and heterophilic regimes, show that HyperDepth at
tains strong performance with the best average rank.
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Data-driven Mixed Integer Optimization through Probabilistic Multi-variable Bran
ching
Yanguang Chen, Wenzhi Gao, Wanyu Zhang, Dongdong Ge, Huikang Liu, Yinyu Ye
This paper introduces Probabilistic Multi-Variable Branching (PMVB), a simple an
d effective technique for accelerating mixed-integer optimization using data-dri
ven machine learning models. At its core, PMVB employs a multi-variable branchin
g procedure that partitions the feasible region via data-driven hyperplanes and
requires only two lines of code to implement. Moreover, PMVB is model-agnostic a
nd compatible with a wide range of machine learning models. Leveraging tools fro
m statistical learning theory, we develop interpretable hyperparameter selection
 strategies and propose several extensions to further enhance performance. We ev
aluate PMVB by integrating it into state-of-the-art MIP solvers and conducting e
xperiments on both classical benchmark datasets and real-world instances. The re
sults demonstrate the effectiveness of PMVB in improving MIP-solving efficiency.
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Escaping Whack-a-Mole: Optimizing Documentation as Repo-Specific Playbooks for C
oding Agents
Yutong Cheng, Haifeng Chen, Wenchao Yu, Xujiang Zhao, Peng Gao, Wei Cheng
As large language models increasingly function as autonomous coding agents, code
 documentation should be designed not for human readability, but for agent execu
tability — serving as repo-specific playbooks that specify precise behaviors age
nts can follow. We formulate agent-oriented documentation generation as a black-
box optimization problem over the documentation space, where quality is defined

solely by downstream code correctness. A central challenge for conventional LLM refinement methods is output coupling—program entities are interdependent, and refining the documentation of one entity can invalidate its callers, resulting in a persistent whack-a-mole phenomenon during inference-time scaling. We propose DocSearch, a dependency-guided bi-level search framework that systematically exploits test-time feedback. The outer level conducts a priority search over the program-entity dependency DAG, enforcing a callee-before-caller refinement order to prevent downstream interference. The inner level performs a beam search over documentation refinements, using diversified error message sampling from self-generated unit tests to better exploit diagnostic signals and escape local optima. On DevEval+, DocSearch achieves 90.7\% solve rate with GPT-4o, outperforming the strongest baseline by 32.6\%. Cross-language experiments further demonstrate that optimized documentation transfers effectively to different target programming languages. Code is available at https://github.com/ccsnow127/docsearch.

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Mean Flow Policy Optimization

Xiaoyi Dong, Xi Sheryl Zhang, Jian Cheng

Diffusion models have recently emerged as expressive policy representations for online reinforcement learning (RL). However, their iterative generative processes introduce substantial training and inference overhead. To overcome this limitation, we propose to represent policies using MeanFlow models, a class of few-step flow-based generative models, to improve training and inference efficiency over diffusion-based RL approaches. To promote exploration, we optimize MeanFlow policies under the maximum entropy RL framework via soft policy iteration, and address two key challenges specific to MeanFlow policies: action likelihood evaluation and soft policy improvement. Experiments on MuJoCo, DeepMind Control Suite and HumanoidBench benchmarks demonstrate that our method, Mean Flow Policy Optimization (MFPO), achieves performance comparable to or exceeding current diffusion-based baselines while considerably reducing training and inference time. Our code is available at https://github.com/dongxiaoyi-xyz/MFPO.

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End-to-End Compression for Tabular Foundation Models

Guri Zabërgja, Rafiq Kamel, Arlind Kadra, Christian Frey, Josif Grabocka

The long-standing dominance of gradient-boosted decision trees for tabular data has recently been challenged by in-context learning tabular foundation models. In-context learning methods fit and predict in one forward pass without parameter updates by leveraging the training data as context for predicting on query test points.

While recent tabular foundation models achieve state-of-the-art performance, their transformer architecture based on the attention mechanism has quadratic complexity regarding dataset size, which in turn increases the overhead on training and inference time, and limits the capacity of the models to handle large-scale datasets. In this work, we propose TACO, an end-to-end tabular compression model that compresses the training dataset in a latent space. We test our method on the TabArena benchmark, where our proposed method is up to 94x faster in inference time, while consuming up to 97% less memory compared to the state-of-the-art tabular Transformer architecture, all while retaining performance without significant degradation. Lastly, our method not only scales better with increased dataset sizes, but it also achieves better performance compared to other baselines.

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Seeing is Understanding: Unlocking Causal Attention into Modality-Mutual Attention for Multimodal LLMs

Wei-Yao Wang, Zhao Wang, Helen Suzuki, Yoshiyuki Kobayashi

Recent Multimodal Large Language Models (MLLMs) have demonstrated significant progress in perceiving and reasoning over multimodal inquiries, ushering in a new research era for foundation models. However, vision-language misalignment in MLLMs has emerged as a critical challenge, where the textual responses generated by these models are not factually aligned with the given text-image inputs. Existing efforts to address vision-language misalignment have focused on developing specialized vision-language connectors or leveraging visual instruction tuning fro

m diverse domains. In this paper, we tackle this issue from a fundamental yet unexplored perspective by revisiting the core architecture of MLLMs. Most MLLMs are typically built on decoder-only LLMs consisting of a causal attention mechanism, which *limits the ability of the earlier modalities (e.g., images) to incorporate information from the latter modalities (e.g., text)*. To address this problem a MLLM that unlocks causal attention into our proposed modality-mutual attention (MMA) to enable image tokens to attend to text tokens. This simple yet effective design allows MMA to achieve state-of-the-art performance in 12 multimodal understanding benchmarks (**+6.2\% on average across 3 LLMs backbones**) without introducing additional parameters. Our MMA design is intended to be generic, allowing for applications across various modalities, and scalable to accommodate diverse multimodal scenarios.

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Width Independent Bounds for the Local Lipschitz Constant of Deep Neural Networks at Random Initialization and after Lazy Training

Apostolos Evangelidis, Felix Krahmer

A plethora of recent works has shown that for wide, overparameterized neural networks, training with Stochastic Gradient Descent (SGD) often leads to interpolation of the training data without sacrificing generalization performance. A key parameter that is not only closely connected to generalization properties, but is also closely tied to other desiderata such as robustness and resistance to adversarial perturbations is the Lipschitz constant of the neural network. While empirically, the Lipschitz constant has been shown not to increase with network width, theoretical findings only provide bounds with logarithmic growth in the width and only for the random initialization of ReLU-networks. In this work, we close this gap for neural networks with smooth activations by showing that, both at random initialization and throughout lazy training, the local Lipschitz constant of deep neural networks does not increase with network width. More precisely, we establish novel non-asymptotic (finite width) upper bounds and corroborate them by numerical experiments.

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Gradient Preconditioning for Efficient and Reliable Reward-Guided Generation

Jisung Hwang, Minhyuk Sung

We propose a gradient preconditioning method that makes reward-guided generation with one-step generative models both efficient and reliable. Test-time noise optimization can unlock substantially better reward-guided generations from pretrained generative models, but it is prone to reward hacking that degrades quality and is often too slow for practical use. We precondition reward gradients by projecting them onto a carefully designed white Gaussian noise feasible set, a compact spectral set with blockwise norm constraints that tightly captures the statistics and spatial uncorrelatedness of white Gaussian noise. This preconditioning reshapes each gradient update into a noise-aligned direction, driving faster and more effective reward ascent while preventing reward hacking. The projection is closed-form and matches the $\mathcal{O}(N \log N)$ complexity of FFT, adding negligible overhead in practice. In experiments on FLUX with four reward models, our approach reaches a comparable Aesthetic Score using only 30\% of the wall-clock time required by the state-of-the-art regularization-based method.

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MDGMIX: Boundary-Aware Subgraph Mixing for Multi-Domain Graph Pre-Training

Ziyu Zheng, Yaming Yang, Ziyu Guan, Wei Zhao, Xinyan Huang

Multi-domain graph pre-training is a crucial step in constructing foundational graph models with cross-domain generalization capabilities. However, existing methods predominantly rely on jointly training all source domain graphs, resulting in high computational costs. Furthermore, it remains unclear whether all source domain graph data contribute equally to effective transfer. This paper empirically reveals significant data redundancy in multi-domain graph pre-training. Based on this finding, we propose the Multi-domain Graph Pre-training Framework, MDGMIX, which combines boundary-aware subgraph mixing with hierarchical discrimination. By selecting boundary nodes to construct challenging mixed-domain subgraphs, MDGMIX employs coarse-grained domain discrimination and fine-grained domain dec

omposition losses to decouple shared patterns from domain-specific patterns. Dur
ing adaptation, MDGMIX employs a lightweight prompt weighting mechanism to trans
fer source domain knowledge. Extensive experiments demonstrate that MDGMIX consi
stently outperforms strong baselines in few-shot classification tasks while exhi
biting superior time and memory efficiency. The code is available at: https://gi
thub.com/zhengziyu77/MDGMIX
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Asymptotic Optimality of the High-Dimensional Gaussian Mechanism and Improved Lo
w-Dimensional Mechanisms for Differential Privacy
Yu Wei, Alexander Bienstock, Antigoni Polychroniadou
The additive noise mechanism is a foundational tool for differential privacy (DP
) of T-dimensional real-valued vector queries. The Gaussian mechanism, utilizi
ng Gaussian noise, is the mostly widely used such mechanism, due to its simplici
ty and strong privacy guarantees. In this work, we provide justification for thi
s choice, showing that as the dimension $T\to\infty$, no additive-noise mechanis
m can asymptotically improve on the Gaussian mechanism's privacy--utility tradeo
ff for the strong privacy settings typically used. We also develop a new family
of **Spherical Generalized Gamma** DP mechanisms, which contains both the Gaussi
an mechanism and the recently studied ℓ_2 mechanism (Joseph \emph{et al.},
ICML 2025). We identify members of this family that outperform both the Gaussian
 and ℓ_2 mechanisms in certain low-dimensional settings, and show tight com
position of all mechanisms in this family, answering an open question of Joseph
\emph{et al.}~regarding the ℓ_2 mechanism.
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Expressivity-Efficiency Tradeoffs for Hybrid Sequence Models
John Cooper, Ilias Diakonikolas, Mingchen Ma, Frederic Sala
Hybrid sequence models—combining Transformer and state-space model layers—seek t
o gain the expressive versatility of attention as well as the computational effi
ciency of state-space model layers. Despite burgeoning interest in hybrid models
, we lack a basic understanding of the settings where—and underlying mechanisms
through which—they offer benefits over their constituent models. In this paper,
we study this question, focusing on a broad family of core synthetic tasks. For
this family of tasks, we prove the existence of fundamental limitations for non-
hybrid models. Specifically, any Transformer or state-space model that solves th
e underlying task requires either a large number of parameters or a large workin
g memory. On the other hand, for two prototypical tasks within this family—namel
y selective copying and associative recall—we construct hybrid models of small s
ize and working memory that provably solve these tasks, thus achieving the best
of both worlds. Our experimental evaluation empirically validates our theoretica
l findings. Importantly, going beyond the settings in our theoretical analysis,
we empirically show that learned—rather than constructed—hybrids outperform non-
hybrid models with up to $6 \times$ as many parameters. We additionally demonstr
ate that hybrid models exhibit stronger length generalization and out-of-distrib
ution robustness than non-hybrids.
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Decentralized Instruction Tuning: Conflict-Aware Splitting and Weight Merging
MINSIK CHOI, Geewook Kim
Instruction tuning aligns large language models, including multimodal ones, with
 diverse user intents, but scaling to heterogeneous mixtures is hindered by grad
ient interference and bandwidth-heavy synchronization. We ask whether these two
bottlenecks can be addressed jointly by training parts of the mixture independen
tly and reconciling them once in parameter space. We develop a local quadratic t
heory inside a shared flat basin that yields three results: weight merging produ
ces a curvature-weighted variance reduction; PCA-aligned conflict splitting maxi
mizes this gain along high-curvature directions; and merging additionally acts a
s spectral filtering with implicit norm regularization. These results directly m
otivate **MERIT**, a decentralized merge-ready instruction-tuning pipeline that
estimates dataset-level gradient conflicts, partitions the mixture along the top
 PCA conflict axes, fine-tunes each partition independently with no inter-partit
ion communication, and merges once via token-weighted averaging. On Qwen2.5-VL-3

B with 136 Vision-FLAN tasks, MERIT improves the 8-benchmark average from 54.3 (joint training) to 57.0. The same recipe scales to a 7B model on a 1.6M-example, 176-source mixture—matching or exceeding centralized joint training with minima l cost overhead—and transfers to text-only FLAN. Our code is available at [https ://github.com/naver-ai/merit](https://github.com/naver-ai/merit).
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Generative Online Reinforcement Learning
Chubin Zhang, Zhenglin Wan, Feng Chen, Fuchao Yang, Lang Feng, Yaxin Zhou, Xingr ui Yu, Yang You, Ivor Tsang, Bo An
Reinforcement learning (RL) faces a persistent tension: policies that are stable to optimize (e.g., Gaussians) are often too simple to represent the multimodal action distributions required for complex control. Conversely, expressive genera tive policies—such as diffusion and flow matching—can be difficult to optimize i n online RL due to intractable likelihoods and gradients propagating through lon g sampling chains. We address this tension with a key structural principle: *dec oupling optimization from generation*. Building on this, we introduce GoRL (Gene rative Online Reinforcement Learning), an algorithm-agnostic framework that trai ns expressive policies from scratch by confining policy optimization to a tracta ble latent space while delegating action synthesis to a conditional generative d ecoder. Using a two-timescale alternating schedule and anchoring decoder refinem ent to a fixed prior, GoRL enables stable optimization while continuously expand ing expressiveness. Empirically, GoRL consistently outperforms unimodal and gene rative baselines across diverse continuous-control tasks. Notably, GoRL achieves returns exceeding **870** on HopperStand, more than $3\times$ the strongest bas eline; on high-dimensional humanoid tasks, it further outperforms the strongest non-GoRL baseline by over an order of magnitude.
****

Coevolutionary Continuous Discrete Diffusion: Make Your Diffusion Language Model a Latent Reasoner
Cai Zhou, Chenxiao Yang, Yi Hu, Chenyu Wang, Chubin Zhang, Muhan Zhang, Lester M ackey, Tommi Jaakkola, Stephen Bates, Dinghuai Zhang
Diffusion language models, especially masked discrete diffusion models, have ach ieved great success recently. While there are some theoretical and primary empir ical results showing the advantages of latent reasoning with looped transformers or continuous CoT, continuous diffusion models typically underperform their dis crete counterparts. In this paper, we argue that diffusion language models do no t necessarily need to be in the discrete space. In particular, we prove that con tinuous diffusion models have stronger expressivity than discrete diffusions and looped transformers. We attribute the contradiction between the theoretical exp ressiveness and empirical performance to their practical trainability: while con tinuous diffusion provides intermediate supervision that looped transformers lac k, they are harder to generate and decode tokens in the continuous representatio n space compared with discrete states. We therefore propose **C**oevolutionary * *C**ontinuous **D**iscrete **D**iffusion (CCDD), which defines a joint multimoda l diffusion process on the union of a continuous representation space and a disc rete token space, leveraging a single model to simultaneously denoise in the joi nt space. By combining two modalities, CCDD is expressive with rich semantics in the latent space, as well as good trainability and sample quality with the help of explicit discrete tokens. We also propose effective architectures and advanc ed training/sampling techniques for CCDD, which reveals strong empirical perform ance in extensive language modeling experiments on real-world tasks.
****

Unpaired Visual Editing with Self-Consistent Flow Matching
Yoad Tewel, Yuval Atzmon, Gal Chechik, Lior Wolf
Modern generative models possess a deep understanding of visual content, yet tra ining them for image editing typically requires massive datasets of paired examp les. This limits scalability, especially for video editing where collecting pair ed data is prohibitively expensive. We propose a general framework for unpaired training of flow matching editing models. It leverages the base model's knowledg e without any external signal. Our approach pairs instruction-following cues ext

racted from the frozen model with cycle-consistency for structure preservation. To make this tractable, we propose to route gradients from downstream losses over clean predictions to noisy training states. We demonstrate state-of-the-art results on challenging data-scarce image and video editing scenarios. Extensive evaluations and user studies show that our method effectively generalizes to unseen domains and outperforms supervised baselines trained on millions of samples. Analysis reveals that our gradient routing bridges the train-inference gap, and extracting semantic cues from a base model provides a robust training signal that obviates the need for external reward models.

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Matroid Algorithms Under Size-Sensitive Independence Oracles

Kiarash Banihashem, MohammadTaghi Hajiaghayi, Mahdi JafariRaviz, Danny Mittal

The standard oracle model for matroid algorithms assumes that each independence query can be answered in constant time, regardless of the size of the queried set. While this abstraction has underpinned much of the theoretical progress in matroid optimization, it masks the true computational effort required by these algorithms. In particular, for natural and widely studied classes such as graphic matroids, even a single independence query can require work linear in the size of the set, making the constant-time assumption implausible. We address this gap by introducing a size-sensitive cost model where the cost of a query Q scales with $|Q|$. Nearly linear-time oracle implementations exist for broad families of matroids, and this refined abstraction therefore captures the true cost of query evaluation while allowing for a more faithful comparison between general matroids and their natural special cases. Within this framework we study three fundamental algorithmic tasks: finding a basis of a matroid, approximating its rank, and approximating its partition size. We establish tight results, proving nearly matching upper and lower bounds that show the optimal query cost is (up to logarithmic factors) quadratic in the size of the matroid. On the algorithmic side, our upper bounds are realized by explicit procedures that construct the desired solution. On the complexity side, our lower bounds are unconditional and already hold even for weaker distinguishing formulations of the problems. Finally, for matroids with maximum circuit size at most c, we show that the quadratic barrier can be broken, providing an algorithm that calculates the maximum-weight basis with expected query cost $\mathcal{O}(n^{2-1/c} \log n)$.

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AutoRAS: Learning Robust Agentic Systems with Primitive Representations

Yang Yue, Xuancheng Zhu, YuYang Ma, Guoshun Nan, Zihan Dou, JingRu Shan, Congyu Guo, Ji Zhang, Hua Wang, Jingfeng Zhang

The automated design of agentic systems offers a promising pathway for scaling large language models (LLMs) beyond single-agent reasoning. While prior work has advanced task performance through handcrafted or automatically generated multi-agent workflows, robustness is often treated as an afterthought, leaving systems vulnerable to external adversaries and internal failures. We propose AutoRAS, a framework for the Automated design of Robust Agentic Systems. AutoRAS formulates system design as generating a sequence of symbolic primitives that jointly encode structural connectivity and behavioral actions, and learns to optimize this sequence using execution-derived safety signals and flow-based sequence-level objectives. Extensive experiments show that AutoRAS achieves the best performance in both vanilla and adversarial settings, with the smallest performance degradation under attacks. Further analyses demonstrate strong transferability, stable optimization behavior, stability across primitive sets, and favorable cost trade-offs. Our code is available at [this link](https://github.com/guohezuy/AutoRAS).

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ProbMoE: Differentiable Probabilistic Routing for Mixture-of-Experts

Heng Zhao, Zilei Shao, Guy Van den Broeck, Zhe Zeng

Mixture-of-Experts (MoE) models scale by activating only a small subset of experts per token.

However, training such models remains challenging because top-k routing is discrete and non-differentiable, requiring gradient estimators for expert selection whose design remains a central open problem. We introduce ProbMoE, a probabilis

tic routing framework that models expert selection as a distribution over cardinality-constrained expert subsets and formulates routing as probabilistic inference in this discrete subset space. We first propose ProbMoE Exact-k routing, which samples k-expert subsets in the forward pass, and the backward pass uses gradients through each expert's exact marginal probability as a tractable surrogate for the true gradient. ProbMoE naturally generalizes to a dynamic-k routing setting, where both training and inference constrain the routing cardinality to the same predefined range, allowing adaptive expert allocation per token. Across benchmarks and model backbones, ProbMoE Exact-k achieves strong performance compared to competitive baselines, with improved expert utilization and routing diversity; ProbMoE Dynamic-k achieves comparable performance with fewer activated experts. Code is available at: https://github.com/HengHugoZhao/ProbMoE.git

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Frontier Models Can Take Actions at Low Probabilities

Alex Serrano, Wen Xing, David Lindner, Erik Jenner

Pre-deployment evaluations inspect only a limited sample of model actions. A misaligned model could evade oversight by randomizing the timing of policy-violating actions, executing them so rarely that none are observed during evaluation. But this requires taking actions at very low rates, while maintaining calibration. Are frontier models even capable of that? We prompt the GPT-5, Claude 4.5 and Qwen-3 families to take a target action at low probabilities (e.g. 0.01%), either given directly or requiring derivation, and evaluate their calibration (i.e. whether they perform the target action roughly 1 in 10,000 times when resampling). We find that frontier models are surprisingly good at this task. If there is a source of entropy in-context (such as a UUID), they maintain high calibration at rates as low as 1 in 100,000 actions. Without external entropy, some models can still reach rates lower than 1 in 10,000. When target rates are given, larger models achieve good calibration at lower rates. Yet, when models must derive the optimal target rate themselves, all models fail to achieve calibration without entropy or hint to generate it. Successful low-rate strategies exhibit explicit Chain-of-Thought (CoT) reasoning, so misaligned models attempting this approach could currently be caught by a CoT monitor. However, scaling trends suggest future evaluations may be unable to rely on models' lack of target rate calibration, especially if CoT is no longer legible.

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Copyright-Bench: Agentic Evaluation of Copyright Law Compliance

Zheng Hui, Doni Bloomfield, Noam Kolt

Large language model (LLM) agents increasingly perform commercial tasks that involve retrieving external content such as images and, where appropriate, reproducing that content. LLM agents should comply with the law, including copyright law. Presently, however, we lack adequate frameworks to assess whether they do so in practice. To that end, we introduce **Copyright-Bench**, a benchmark designed to evaluate *LLM agents' compliance with* *copyright law*. Copyright-Bench is comprised of realistic commercial tasks---website development, merchandise design, and pitch deck production---that involve agents selecting between public-domain content (the use of which is *legal*) and copyrighted content (the use of which is *infringing* in this setting). The evaluation introduces prompt variations that simulate different user preferences, as well as time pressure. Comparing state-of-the-art LLM agents against a human baseline, we find that: (1) agents select copyrighted works despite the availability of public-domain alternatives; and (2) for open-weights models, violation rates increase in response to certain user preferences and simulated time pressure.

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Sharpness-Aware Minimization Can Hallucinate Minimizers

Chanwoong Park, Uijeong Jang, Ernest K. Ryu, Insoon Yang

Sharpness-Aware Minimization (SAM) is widely used to seek flatter minima—often linked to better generalization. In its standard implementation, SAM updates the current iterate using the loss gradient evaluated at a point perturbed by distance ρ along the normalized gradient direction. We show that, for some choices of ρ, SAM can stall at points where this shifted (perturbed-point) gradie

nt vanishes despite a nonzero original gradient, and therefore, they are not sta tionary points of the original loss. We call these points hallucinated minimizer s, prove their existence under simple nonconvex landscape conditions (e.g., the presence of a local minimizer and a local maximizer), and establish sufficient c onditions for local convergence of the SAM iterates to them. We corroborate this failure mode in neural network training and observe that it aligns with SAM's p erformance degradation often seen at large ρ. Finally, as a practical safeg uard, we find that a short initial SGD warm-start before enabling SAM mitigates this failure mode and reduces sensitivity to the choice of ρ.
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Beyond Literal Translation: Evaluating Cultural Effectiveness in Social Media UG C

Linjuan Wu, Ruiqi Zhang, Xinze Lyu, Ye Guo, Daoxin Zhang, Zhe Xu, Yao Hu, Yixin Cao, Yongliang Shen, Weiming Lu

Social media platforms enable large-scale cross-lingual communication, but trans lating user-generated content (UGC) remains challenging due to its informal styl e, cultural references, and interaction-based expressions. While recent LLMs hav e improved translation quality, existing benchmarks and metrics often fail to ca pture whether translations convey intended meaning and cultural resonance in rea l-world settings. In this work, we introduce **CULTURE-MT**, a benchmark for soc ial media translation that focuses on both **CUL**tural **T**ransmission and **U **GC-specific emotion **RE**sonance. CULTURE-MT consists of 1,002 UGC notes acro ss 14 domains, categorized into four types based on culture-loaded symbol and li nguistic style features. We also construct UGC-oriented training data to fine-tu ne Qwen3-8B and Qwen3-32B as baselines. We propose **cultural effectiveness** as a new evaluation criterion, focusing on expression accuracy and cultural adapta bility. Testing 15 models, including the baselines, we find that traditional met rics fail to capture cultural effectiveness. We also observe that cultural effec tiveness on base LLMs correlates with model size. Our work provides a comprehens ive evaluation system for UGC translation models and will offers an open evaluat ion platform to advance research in this area. We release the CULTURE-MT benchma rk and provide an online leaderboard where submitted translation results can be evaluated by our trained JUDGER.
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Variational Speculative Decoding: Rethinking Draft Training from Token Likelihoo d to Sequence Acceptance

Xiandong Zou, Jianshu Li, Jing Huang, Pan Zhou

Speculative decoding accelerates inference for (M)LLMs, yet a training-decoding discrepancy persists: while existing methods optimize single greedy trajectories , decoding involves verifying and ranking multiple sampled draft paths. We propo se *Variational Speculative Decoding* (VSD), formulating draft training as varia tional inference over latent proposals (draft paths). VSD maximizes the marginal probability of target-model acceptance, yielding an ELBO that promotes high-qua lity latent proposals while minimizing divergence from the target distribution. To enhance quality and reduce variance, we incorporate a path-level utility and optimize via an Expectation-Maximization procedure. The E-step draws MCMC sample s from an oracle-filtered posterior, while the M-step maximizes weighted likelih ood using Adaptive Rejection Weighting (ARW) and Confidence-Aware Regularization (CAR). Theoretical analysis confirms that VSD increases expected acceptance len gth and speedup. Extensive experiments across LLMs and MLLMs show that VSD achie ves up to a 9.58\% speedup over EAGLE-3 and 8.80\% over ViSpec, significantly im proving decoding efficiency.
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Towards Trustworthy Video Anomaly Understanding: A Class-Guided Chain-of-Evaluat ion Metric and An Anomaly-focused Meta-Benchmark

Jiaxu Leng, Zhoujie Huang, Mingpi Tan, Zhanjie Wu, Xinbo Gao

The trustworthiness of evaluation is critical to reliable model comparison and d eployment in Video Anomaly Understanding (VAU). However, existing metrics are se nsitive to expression styles and normal content, and this field lacks a diagnost ic benchmark to validate metric validity and robustness. To bridge this gap, we

propose: (1) a Class-Guided Chain-of-Evaluation (CG-CoE) metric, which structures assessment by extracting anomalous events and matching them under a class-specific semantic tolerance boundary, thereby decoupling anomaly semantics from descriptive style; and (2) an anomaly-focused meta-evaluation benchmark with two subsets: Anomalous Event-level Annotations (AEA) for measuring the validity of reflecting VAU models' anomaly understanding ability and Controlled Variant Pairs (CVP) with fixed anomalies for quantifying robustness to stylistic perturbations.
  Extensive experiments demonstrate that CG-CoE achieves SOTA validity and robustness.
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MergeMix: Optimizing Mid-Training Data Mixtures via Learnable Model Merging
jiapeng wang, Changxin Tian, Kunlong Chen, Ziqi Liu, Jiaxin Mao, Xin Zhao, Zhiqiang Zhang, JUN ZHOU
Optimizing data mixtures is is essential for unlocking the full potential of of large language models (LLMs), yet identifying the optimal composition remains computationally prohibitive due to reliance on heuristic trials or expensive proxy training. To address this, we introduce MergeMix, a novel appraoch that efficiently determines optimal data mixing ratios by repurposing model merging weights as a high-fidelity, low-cost performance proxy. By training domain-specific experts on minimal tokens and optimizing their merging weights against downstream benchmarks, MergeMix effectively optimizes the performance of data mixtures without incurring the cost of full-scale training. Extensive experiments on models with 8B and 16B parameters validate that MergeMix achieves performance comparable to or surpassing exhaustive manual tuning while drastically reducing search costs. Furthermore, MergeMix exhibits high rank consistency (Spearman $\rho > 0.9$) and strong cross-scale transferability, offering a scalable, automated solution for data mixture optimization.
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Elign: Equivariant Diffusion Model Alignment from Foundational Machine Learned Force Fields
Yunyang Li, Lin Huang, Luojia Xia, Wenhe Zhang, Mark Gerstein
Generative models for 3D molecular conformations must respect Euclidean symmetries and concentrate probability mass on thermodynamically favorable, mechanically stable structures. However, E(3)-equivariant diffusion models often reproduce biases from semi-empirical training data rather than capturing the equilibrium distribution of a high-fidelity Hamiltonian. While physics-based guidance can correct this, it faces two computational bottlenecks: expensive quantum-chemical evaluations (e.g., DFT) and the need to repeat such queries at every sampling step. We present Elign, a post-training framework that amortizes both costs. First, we replace expensive DFT evaluations with a faster, pretrained foundational machine-learning force field (MLFF) that estimates molecular energies and forces. Second, we eliminate repeated run-time queries by shifting physical steering to the post-training phase. To achieve the second amortization, we formulate reverse diffusion as a reinforcement learning problem and propose to use Group Relative Policy Optimization (GRPO) to fine-tune the denoising policy. Our objective combines a potential-based energy reward and a force-based stability reward, which are optimized in a disentangled fashion. Experiments show that Elign generates conformations with lower gold-standard DFT energies and forces, while improving stability. Crucially, inference remains as fast as unguided sampling, since no energy evaluations are required during generation.
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SVD as a Fast Interpretability Method for Transformers
Min Xue, Artur Andrzejak
Mechanistic interpretability of Transformer models commonly relies on training auxiliary proxy models, such as Sparse Autoencoders or Cross-Layer Transcoders. While effective, these post-hoc approaches introduce approximation bias and incur substantial computational overhead. We propose an alternative, training-free interpretability framework that directly exploits the Singular Value Decomposition (SVD) of weight matrices in Transformer MLP sublayers. By operating natively on model parameters, our method improves scalability while preserving fidelity to

the original weights. We show that the projection matrices of MLP sublayers admit a natural decomposition into orthogonal, interpretable rank-1 subspaces, which we term **Detector-Effector Units** (DEUs). Within each unit, a singular vector functions as a detector of input patterns and modulates a coupled effector vector that encodes output semantics. Building on this structure, we introduce **Subspace Contribution Analysis** (SCA), a diagnostic method that quantifies the direct causal contribution of individual native subspaces to model predictions. Experiments across the GPT-2 family demonstrate that our framework, **Native Network Anatomy** (NaNA), identifies dominant functional pathways with orders-of-magnitude efficiency gains over training-based interpretability baselines, while maintaining weight fidelity. Our results suggest that SVD-based analyses provide a scalable and faithful alternative to learned proxy approaches for mechanistic interpretability.

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ChartE^{3}: A Comprehensive Benchmark for End-to-End Chart Editing

Shuo Li, Jiajun Sun, Zhekai Wang, Xiaoran Fan, Hui Li, Dingwen Yang, Zhiheng Xi, Yijun Wang, Zifei Shan, Tao Gui, Qi Zhang, Xuanjing Huang

Charts are a fundamental visualization format for structured data analysis. Enabling end-to-end chart editing according to user intent is of great practical value, yet remains challenging due to the need for both fine-grained control and global structural consistency. Most existing approaches adopt pipeline-based designs, where natural language or code serves as an intermediate representation, limiting their ability to faithfully execute complex edits.

We introduce ChartE^{3}, an **E**nd-to-**E**nd Chart **E**diting benchmark that directly evaluates models without relying on intermediate natural language programs or code-level supervision. ChartE^{3} focuses on two complementary editing dimensions: local editing, which involves fine-grained appearance changes such as font or color adjustments, and global editing, which requires holistic, data-centric transformations including data filtering and trend line addition.

ChartE^{3} contains over 1,200 high-quality samples constructed via an well-designed data pipeline with human curation. Each sample is provided as a triplet of a chart image, its underlying code, and a multimodal editing instruction, enabling evaluation from both visual and structural perspectives. Extensive benchmarking of state-of-the-art multimodal large language models reveals substantial performance gaps, particularly on global editing tasks, highlighting critical limitations in current end-to-end chart editing capabilities.

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FaPS: A General and Fast Training Method for Diffusion Models

Xianglu Wang, Bangxian Han, Hu Ding

Diffusion models have achieved state-of-the-art performance in image generation tasks. However, training powerful diffusion models remains time-consuming, which limits their practical deployment. In this paper, we revisit the learning dynamics of diffusion models through the lens of *spectral bias*, a phenomenon in which deep neural networks prioritize learning low-frequency modes. Through an empirical analysis of diffusion training, we observe that diffusion models exhibit a **dual** spectral bias. First, over training iterations, they fit low-frequency components earlier than high-frequency details. Second, along the diffusion timesteps, early denoising steps mainly reconstruct coarse low-frequency content, while high-frequency details emerge in later steps. Motivated by this observation, we propose Frequency-aware Patch Selection **(FaPS)**, a general and fast training method for diffusion models that can
be applied to both UNet and DiT backbones.

Specifically, FaPS introduces a *frequency-aware gating* that adaptively selects image patches based on their frequency information and focuses computation only on the selected patches. Since the selection decisions are discrete and thus non-differentiable, we model the gating as a stochastic policy network and optimize it end-to-end using a policy gradient method.

Our experiments demonstrate that FaPS achieves up to $\mathbf{3}\times$ faster training while maintaining comparable or superior generation quality, and improves the performance of diffusion models in limited-data settings.

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DropoutTS: Sample-Adaptive Dropout for Robust Time Series Forecasting
Siru Zhong, Yiqiu Liu, Zhiqing Cui, Zezhi Shao, Fei Wang, Qingsong Wen, Yuxuan L iang
Deep time series models are vulnerable to noisy data ubiquitous in real-world ap plications. Existing robustness strategies either prune data or rely on costly p rior quantification, failing to balance effectiveness and efficiency. In this pa per, we introduce DropoutIS, a model-agnostic plug■n that shifts the paradigm fr om what to learn to how much to learn. DropoutTS employs a Sample-Adaptive Dropo ut mechanism: leveraging spectral sparsity to efficiently quantify instance-leve l noise via reconstruction residuals, it dynamically calibrates model learning c apacity by mapping noise to adaptive dropout rates, selectively suppressing spur ious fluctuations while preserving fine-grained fidelity. Extensive experiments across diverse noise regimes and open benchmarks show DropoutTS consistently boo sts superior backbones' performance, delivering advanced robustness with negligi ble parameter overhead and no architectural modifications. Code is available at https://github.com/CityMind-Lab/DropoutTS.
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PHALAR: Phasors for Learned Musical Audio Representations
Davide Marincione, Michele Mancusi, Giorgio Strano, Luca Cerovaz, Donato Crisost omi, Roberto Ribuoli, Emanuele Rodolà
Stem retrieval, the task of matching missing stems to a given audio submix, is a key challenge currently limited by models that discard temporal information. We introduce PHALAR, a contrastive framework achieving a relative accuracy increas e of up to ≈ 70% over the state-of-the-art while requiring <50% of the parameters and a 7$\times$ training speedup. By utilizing a Learned Spectral Po oling layer and a complex-valued head, PHALAR enforces pitch-invariant and phase -equivariant biases. PHALAR establishes new retrieval state-of-the-art across Mo isesDB, Slakh, and ChocoChorales, correlating significantly higher with human co herence judgment than semantic baselines. Finally, zero-shot beat tracking and l inear chord probing confirm that PHALAR captures robust musical structures beyon d the retrieval task.
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UOTIP: Unbalanced Optimal Transport Map for Unpaired Inverse Problems
Donggyu Lee, Taekyung Lee, Jaewoong Choi
We investigate unpaired image inverse problems, a challenging setting where only independent, non-paired sets of noisy measurements and clean target signals are available for training.
We propose a novel inverse problem solver based on Unbalanced Optimal Transport, called ***Unbalanced Optimal Transport Map for Inverse Problems (UOTIP)***. Our method formulates the reconstruction task—predicting clean target signals from noisy measurements—as learning a UOT Map from noisy measurement distribution to clean signal distribution by incorporating a likelihood-based cost function. By relaxing the exact marginal constraint, the UOT framework provides key advantage s to our model: robustness to multi-level observation noise, adaptability to cla ss imbalance between noisy and clean datasets, and generalizability to diverse n oise-type scenarios.
Furthermore, we theoretically demonstrate that incorporating a quadratic cost te rm ensures the existence and uniqueness of the transport map by satisfying the t wist condition, even for ill-posed inverse problems.
Our experiments demonstrate that UOTIP achieves state-of-the-art performance on unpaired image inverse problem benchmarks, across linear and nonlinear inverse p roblems.
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AdaSplash-2: Faster Differentiable Sparse Attention
Nuno Gonçalves, Hugo Pitorro, Vlad Niculae, Edoardo Ponti, Lei Li, Andre Martins , Marcos Vinicius Treviso
Sparse attention has been proposed as a way to alleviate the quadratic cost of t ransformers, a central bottleneck in long-context training.
A promising line of work is α-entmax attention, a differentiable sparse a

lternative to softmax that enables input-dependent sparsity yet has lagged behin
d softmax due to the computational overhead necessary to compute the normalizer
τ.
In this paper, we introduce AdaSplash-2, which addresses this limitation through
 a novel histogram-based initialization that reduces the number of iterations ne
eded to compute τ to typically 1-2. The key idea is to compute a coarse his
togram of attention scores on the fly and store it in on-chip SRAM, yielding a m
ore accurate initialization that enables fast forward and backward computation.
Combined with a sparsity-aware GPU implementation that skips zero blocks with lo
w overhead, AdaSplash-2 matches or improves per-step training time relative to F
lashAttention-2 when block sparsity is moderate-to-high (e.g., $>$60\%), which o
ften occurs at long-context lengths.
On downstream tasks, models trained with our efficient α-entmax attention
 match softmax baselines at short-context lengths and achieve substantial gains
in long-context settings.
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Reading the Cell, Designing the Cure: Perturbation-Conditioned Molecular Diffusi
on for Function-Oriented Drug Design
ZIYU XU, zijian zhang, Liang Wang, Zhiyuan Liu, Qiang Liu, Shu Wu, Liang Wang
When reliable target structures are unavailable at scale or phenotypes arise fro
m dysregulated pathways, transcriptomic perturbations provide a system-level fun
ctional readout for drug action. In this work, we formalize Transcriptome-based
Drug Design (TBDD) as a generative inverse problem: designing drug molecules con
ditioned on desired transcriptomic state transitions. We analyze the inherently
ill-posed nature of this task, which is further complicated by the profound doma
in gap between biology and chemistry and by the sparsity of transcriptomic signa
ls. To address these challenges, we propose CURE (A CellUlar Response Engine), a
 multi-resolution transcriptome-guided diffusion framework. CURE features a spec
ialized Transcriptome Perturbation Functional Feature Extractor (TFE) that (1) d
istills function-oriented perturbation embeddings from pre/post states, (2) alig
ns these signatures to dual chemical views to bridge the cross-modal gap, and (3
) performs heterogeneity-aware aggregation to extract robust state-specific sign
als from noisy transcriptomic data. Extensive evaluations on both standard bench
marks and rigorous out-of-distribution protocols demonstrate that CURE consisten
tly outperforms strong baselines in structural quality and functional consistenc
y. Furthermore, we validate its practical utility via a zero-shot gene-inhibitor
 design task, highlighting the potential of phenotype-driven generative discover
y.
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An Algebraic View of the Expressivity of Recurrent Language Models
Franz Nowak, Ryan Cotterell, Reda Boumasmoud
What formal languages can a recurrent neural language model recognize? Formal re
sults in the literature conflict: some authors report Turing-completeness, while
 others show equivalence to regular languages.
The reason for this discrepancy is that the underlying arithmetic model differs.

The paper develops a unified algebraic account of the expressivity of recurrent
neural networks, starting with a formal account of various arithmetic models.
This account reduces expressivity to an algebraic question, e.g., whether a netw
ork's syntactic monoid divides a certain wreath product.
As a case study, the paper revisits diagonal state-space models: the same archit
ecture cannot implement an even-modulus counter once floating-point recurrences
are enforced, yet realizes every even-modulus counter under unsigned-integer qua
ntization.
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Rex: A Family of Reversible Exponential (Stochastic) Runge-Kutta Solvers
Zander W. Blasingame, Chen Liu
Deep generative models based on neural differential equations have become state-
of-the-art for many generation tasks.
These models rely on ODE/SDE solvers that integrate from a prior distribution to

the data distribution; in many applications it is also highly desirable to integrate in the inverse direction.
Standard solvers, however, accumulate discretization errors that prohibit *exact inversion*, an inaccuracy that is unacceptable in precision-critical applications.
Existing inversion methods suffer from poor stability and low order of convergence, and are strictly limited to the ODE setting.
In this work, we propose *Rex*, a family of reversible exponential (stochastic) Runge-Kutta solvers obtained by applying Lawson methods to convert any explicit (stochastic) Runge-Kutta scheme into an algebraically reversible one for both diffusion ODEs *and* SDEs.
Beyond a rigorous theoretical analysis---establishing arbitrary-order convergence and a non-zero region of linear stability---we empirically demonstrate that *Rex* achieves near-machine-precision reconstruction and improves Boltzmann sampling with flow models as well as image generation and editing with diffusion models.
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Transport and Merge: Cross-Architecture Merging for Large Language Models
Chenhang Cui, Binyun Yang, Fei Shen, Yuxin Chen, Jingnan Zheng, Xiang Wang, An Zhang, Tat-Seng Chua
Large language models (LLMs) achieve strong capabilities by scaling model capacity and training data, yet many real-world deployments rely on smaller models trained or adapted from low-resource data. This gap motivates the need for mechanisms to transfer knowledge from large, high-resource models to smaller, low-resource targets.
While model merging provides an effective transfer mechanism, most existing approaches assume architecture-compatible models and therefore cannot directly transfer knowledge from large high-resource LLMs to heterogeneous low-resource targets.
In this work, we propose a cross-architecture merging framework based on optimal transport (OT) that aligns activations to infer cross-neuron correspondences between heterogeneous models.
The resulting transport plans are then used to guide direct weight-space fusion, enabling effective high-resource to low-resource transfer using only a small set of inputs.
Extensive experiments across low-resource languages and specialized domains demonstrate consistent improvements over target models.
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NAVIGATE: Evaluating Visual-Guided Search Decision-Making on the Open Web
YaoQi Fan, Zhe Chen, Zhu Wei, Kangxin Yin, Yangzhou Liu, Yue Cao, Zhi Zhu, Tong Lu
Vision–Language Models (VLMs) are increasingly deployed with web search tools, yet we still lack benchmarks that isolate a critical capability for real-world use: deciding when to search and how to steer search from ambiguous visual evidence, especially when multiple images provide overlapping or conflicting cues. We introduce NAVIGATE, a novel benchmark centered on images as primary evidence for open-web search planning and multi-step reasoning. It contains 500 questions across 20 domains and spans three difficulty tiers, from single-image, self-contained problems to multi-image joint search and multi-domain composition. Unlike prior benchmarks that specify explicit search targets, NAVIGATE evaluates search decision-making: models must infer whether external search is necessary and iteratively refine search directions based on holistic reasoning over visual cues. Across a broad set of VLMs and search-enabled systems, performance remains low, Gemini-3-Pro-Preview-Search reaches only 36.4% accuracy, highlighting persistent failures in cross-image grounding, search triggering, and search strategy coordination. We will release NAVIGATE publicly.
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VisualScore: Learning Holistic Visual Quality Scores via Multi-Task Reasoning
Yiting Lu, Fengbin Guan, Yixin Gao, Yan Zhong, Xinge Peng, Jiakang Yuan, Yihao Liu, Bo Zhang, Xin Li, Zhibo Chen, Weisi Lin

Image quality assessment (IQA) is inherently multi-dimensional, yet existing reward models are typically limited to a single task and become unstable when extended to multi-task settings. In particular, heterogeneous reward scales and variances across tasks can lead to conflicting optimization signals during reinforcement learning.
We propose VisualScore, a unified visual evaluation framework that formulates multi-task IQA as structured, task-aware reasoning followed by continuous reward optimization. VisualScore produces interpretable rationales together with scalar quality scores under explicit evaluation principles.
We construct a reasoning-enhanced reward modeling dataset via rejection sampling and initialize the model through supervised fine-tuning. VisualScore is then optimized with Group Relative Policy Optimization (GRPO) using a Gaussian-based continuous reward. To address multi-task reward conflicts and stabilize training, we introduce standard deviation filtering and entropy gating to normalize task-wise reward signals and suppress noisy updates.
Experiments on technical quality, aesthetic quality, and text–image alignment show that VisualScore improves robustness, generalization, and interpretability, and can effectively guide text-to-image generation at test time without retraining.
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Evolution of Benchmark: Black-Box Optimization Benchmark Design through Large Language Model
Chen Wang, Sijie Ma, Zeyuan Ma, Yue-Jiao Gong
Benchmark Design in Black-Box Optimization (BBO) is a fundamental yet open-ended topic. Early BBO benchmarks are predominantly human-crafted, introducing expert bias and constraining diversity. Automating this design process can relieve the human-in-the-loop burden while enhancing diversity and objectivity. We propose Evolution of Benchmark (EoB), an automated BBO benchmark designer empowered by the large language model (LLM) and its program evolution capability. Specifically, we formulate benchmark design as a bi-objective optimization problem towards maximizing (i) landscape similarity to target tasks and (ii) algorithm-differentiation ability across a portfolio of BBO solvers. Under this paradigm, EoB iteratively prompts LLM to evolve a population of benchmark programs and employs a reflection-based scheme to co-evolve the landscape and its corresponding program. Comprehensive experiments validate our EoB is a competitive candidate in multi-dimensional usages: 1) Benchmarking BBO algorithms; 2) Training and testing learning-assisted BBO algorithms; 3) Extending proxy for expensive real-world problems.
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EcoVLA: Environment-Aware Adaptive Pruning with Interleaved Inference Orchestration for Vision-Language-Action Models
Yuting Huang, Leilei Ding, Zhipeng Tang, Zenghuan Zhu, Jiajun Deng, Xinrui Lin, Shuo Liu, Haojie Ren, Jianmin Ji, Yanyong Zhang
While Vision-Language-Action (VLA) models hold promise in embodied intelligence, their large parameter counts lead to substantial inference latency that hinders real-time manipulation, motivating parameter sparsification. However, as the environment evolves during VLA execution, the optimal sparsity patterns change accordingly. Static pruning lacks the adaptability required for environment dynamics, whereas fixed-interval dynamic layer pruning suffers from coarse granularity and high retraining overheads. To bridge this gap, we propose **EcoVLA**, a training-free, plug-and-play adaptive pruning framework that supports orthogonal combination with existing VLA acceleration methods. EcoVLA comprises two components: **E**nvironment-aware **A**daptive **P**runing (**EAP**) and **I**nterleaved **I**nference **O**rchestration (**I^2O**). EAP is a lightweight adaptive channel pruning method that incorporates the temporal consistency of the physical environment to update sparsity patterns. I^2O leverages the FLOPs bubbles inherent in VLA inference to schedule the pruning method in parallel, ensuring negligible impact on latency. Evaluated on diverse VLA models and benchmarks, EcoVLA delivers state-of-the-art performance, achieving up to 1.60$\times$ speedup with only a 0.4% drop in success rate, and further reaches 2.18$\times$ speedup with on

ly a 0.5% degradation when combined with token pruning. We further validate the effectiveness of EcoVLA on real-world robots. Our code is available [here](https://github.com/Echo-hyt/Ecovla).
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Causal-EPIG: Causally Aligned Active CATE Estimation
Erdun Gao, Jake Fawkes, Dino Sejdinovic
Estimating the Conditional Average Treatment Effect (CATE) is constrained by the high cost of obtaining outcome measurements, making active learning valuable. However, conventional strategies suffer from a fundamental objective mismatch: they reduce uncertainty in model parameters or observable outcomes rather than the unobservable causal quantities of interest. We address this via the principle of causal objective alignment, positing that acquisition functions should target potential outcomes or CATE directly. We operationalize this through Causal-EPIG, a framework adapting Expected Predictive Information Gain to quantify uncertainty reduction in causal quantities. We derive two distinct strategies: a comprehensive approach that targets the joint potential-outcome structure, and a focused approach that directly targets the CATE estimand for sample efficiency. We provide theoretical justification for our framework, establishing a formal link between CATE estimation error and posterior uncertainty in causal quantities. Extensive experiments demonstrate that our strategies improve sample efficiency over standard baselines, and crucially, reveal that the preferred strategy is context-dependent, contingent on the base estimator and treatment-effect structure. Our framework thus provides a principled guide for sample-efficient CATE estimation in practice.
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Robust Causal Discovery in Real-World Time Series with Power-Laws
Matteo Tusoni, Giuseppe Masi, Andrea Coletta, Aldo Glielmo, Viviana Arrigoni, Novella Bartolini
Exploring causal relationships in stochastic time series is a challenging yet crucial task with a vast range of applications, including finance, economics, neuroscience, and climate science. Many algorithms for Causal Discovery (CD) have been proposed; however, they often exhibit a high sensitivity to noise, resulting in spurious causal inferences on real data. In this paper, we observe that the frequency spectra of many real-world time series follow a power-law distribution, notably due to an inherent self-organizing behavior. Leveraging this insight, we build a robust CD method based on the extraction of power■law spectral features that amplify genuine causal signals. Our method consistently outperforms state-of-the-art alternatives on both synthetic benchmarks and real-world datasets with known causal structures, demonstrating its robustness and practical relevance.
****

Beyond Scalars: Evaluating and Understanding LLM Reasoning via Geometric Progress and Stability
Xinyan Jiang, Ninghao Liu, Di Wang, Lijie Hu
Evaluating LLM reliability via scalar probabilities often fails to capture the structural dynamics of reasoning. We introduce TRACED, a framework that assesses reasoning quality through theoretically grounded geometric kinematics. By decomposing reasoning traces into Progress (displacement) and Stability (curvature), we reveal a distinct topological divergence: correct reasoning manifests as high-progress, stable trajectories, whereas hallucinations are characterized by low-progress, unstable patterns (stalled displacement with high curvature fluctuations). Leveraging these signatures, our probabilistic framework achieves competitive performance and superior robustness across diverse benchmarks. Crucially, TRACED bridges geometry and cognition by mapping high curvature to "Hesitation Loops'' and displacement to ''Certainty Accumulation'', offering a physical lens to decode the internal dynamics of machine thought.
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On the Sample Efficiency of Inverse Dynamics Models for Semi-Supervised Imitation Learning
Sacha Morin, Moonsub Byeon, Alexia Jolicoeur-Martineau, Sebastien Lachapelle

Semi-supervised imitation learning (SSIL) consists in learning a policy from a small dataset of action-labeled trajectories and a much larger dataset of action-free trajectories. Some SSIL methods learn an inverse dynamics model (IDM) to predict the action from the current state and the next state. An IDM can act as a policy when paired with a video model (VM-IDM) or as a label generator to perform behavior cloning on action-free data (IDM labeling). In this work, we first show that VM-IDM and IDM labeling learn the same policy in a limit case, which we call the IDM-based policy. We then argue that the previously observed advantage of IDM-based policies over behavior cloning is due to the superior sample efficiency of IDM learning, which we attribute to two causes: (i) the ground-truth IDM tends to be contained in a lower complexity hypothesis class relative to the expert policy, and (ii) the ground-truth IDM is often less stochastic than the expert policy. We argue these claims based on insights from statistical learning theory and novel experiments, including a study of IDM-based policies using recent architectures for unified video-action prediction (UVA). Motivated by these insights, we finally propose an improved version of the existing LAPO algorithm for latent action policy learning. We experiment on the Procgen, Push-T and LIBERO benchmarks.

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When Do Hallucinations Arise? A Graph Perspective on the Evolution of Path Reuse and Path Compression

Xinnan Dai, Kai Yang, Cheng Luo, Shenglai Zeng, Kai Guo, Jiliang Tang

Reasoning hallucinations in large language models (LLMs) often appear as fluent yet unsupported conclusions that violate either the given context or underlying factual knowledge. Although such failures are widely observed, the mechanisms by which decoder-only Transformers produce them remain poorly understood. We model next-token prediction as a graph search process over an underlying graph, where entities correspond to nodes and learned transitions form edges. From this perspective, contextual reasoning is a constrained search over a sampled subgraph (intrinsic reasoning), while context-free queries rely on memorized structures in the underlying graph (extrinsic reasoning). We show that reasoning hallucinations arise from two fundamental mechanisms: path reuse, where memorized knowledge overrides contextual constraints during early training, and path compression, where frequently traversed multi-step paths collapse into shortcut edges in later training. Together, these mechanisms provide a unified explanation for reasoning hallucinations in LLMs and connected to well-known behaviors observed in downstream applications.

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Discretely-Refined Multi-view Clustering via Aligned Anchor Learning

Yuemeng Huang, Huibing Wang, Jinjia Peng, Lu Jiang, Qian Liu, Zetian Mi, Jiqing Zhang

Anchor-based multi-view clustering has garnered wide attention for its ability to reduce the computational complexity of large-scale spectral clustering. However, existing methods mostly adopt a unidirectional optimization paradigm confined to sample-anchor bipartite graphs, treating the construction of the consensus graph and discrete clustering assignments as separate sub-problems to be solved independently. This weakens the information exchange between continuous representation and discrete structure, confining the optimization process to iterative updates within local modules.To address these limitations, we propose a Discretely-Refined Multi-view Clustering(DRMC) via Aligned Anchor Learning. Unlike approaches that directly perform fusion in the anchor space, our method starts from the anchor graph, elevates sample-anchor associations to sample-level similarity graph representations, and thereby enhances both within-cluster similarity and between-cluster separation. Furthermore, we design a discrete feedback module that jointly conducts spectral embedding learning and discrete label assignment by orthogonally aligning the continuous embedding matrix with the discrete indicator matrix. The resulting discrete partition is then fed back into the consensus graph construction, continuously refining the graph structure. Experiments on multiple benchmark datasets demonstrate that the proposed method exhibits significant advantages over existing state-of-the-art approaches.

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DiffStyle3D: Consistent 3D Gaussian Stylization via Attention Optimization
Yitong Yang, Yinglin Wang, Xuexin Liu, Jing Wang, Hao Dou, Changshuo Wang, Shuti
ng He
3D style transfer enables the creation of visually expressive 3D content, enrich
ing the visual appearance of 3D scenes and objects. However, existing VGG- and C
LIP-based methods struggle to model multi-view consistency within the model itse
lf, while diffusion-based approaches can capture such consistency but rely on de
noising directions, leading to unstable training. To address these limitations,
we propose DiffStyle3D, a novel diffusion-based paradigm for 3DGS style transfer
 that directly optimizes in the latent space. Specifically, we introduce an Atte
ntion-Aware Loss that performs style transfer by aligning style features in the
self-attention space, while preserving original content through content feature
alignment. Inspired by the geometric invariance of 3D stylization, we propose a
Geometry-Guided Multi-View Consistency method that integrates geometric informat
ion into self-attention to enable cross-view correspondence modeling. Based on g
eometric information, we additionally construct a geometry-aware mask to prevent
 redundant optimization in overlapping regions across views, which further impro
ves multi-view consistency. Extensive experiments show that DiffStyle3D outperfo
rms state-of-the-art methods, achieving higher stylization quality and visual re
alism. The code is available at \url{https://github.com/yangyt46/DiffStyle3D}.
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Turning Bias into Bugs: Bandit-Guided Style Manipulation Attacks on LLM Judges
XIANGLIN YANG, Bryan Hooi, Gelei Deng, Tianwei Zhang, Jin Song Dong
The known **stylistic biases** in LLM judges, such as a preference for verbosity
 or specific sentence structures, present an underexplored *security vulnerabili
ty*. In this work, we introduce BITE (**BI**as explora**T**ion and **E**xploitat
ion), a black-box adversarial framework that learns semantics-preserving edits t
o mislead an LLM judge and *artificially* inflate the scores it assigns. We cast
 the selection of stylistic edits as a contextual bandit problem and use a LinUC
B policy to adaptively choose edits that maximize the judge's score without acce
ss to model parameters or gradients. Empirically, we test BITE across a diverse
range of LLM judges and tasks, including both pointwise and pairwise comparisons
 on chatbot leaderboards and AI-reviewer benchmarks. BITE achieves an attack suc
cess rate exceeding 65% and raises scores by 1–2 points on a 9-point scale, all
while preserving semantic equivalence. We further assess the attack's stealthine
ss, showing that BITE evades standard style-control methods and several detectio
n baselines. Our findings expose a fundamental weakness in the LLM-as-a-judge pa
radigm and motivate robust, attack-aware evaluation. Our code is available at ht
tps://github.com/xianglinyang/llm-as-a-judge-attack.
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Calibrated Multimodal Representation Learning with Missing Modalities
Xiaohao Liu, Xiaobo Xia, Jiaheng Wei, Shuo Yang, Xiu Su, See-Kiong Ng, Tat-Seng
Chua
Multimodal representation learning harmonizes distinct modalities by aligning th
em into a unified latent space. Recent research generalizes traditional cross-mo
dal alignment to produce enhanced multimodal synergy but requires all modalities
 to be present for a common instance, making it challenging to utilize prevalent
 datasets with missing modalities. We provide theoretical insights into this iss
ue from an \textit{anchor shift} perspective. Observed modalities are aligned wi
th a local anchor that deviates from the optimal one when all modalities are pre
sent, resulting in an inevitable shift.

    To address this, we propose CalMRL to calibrate incomplete alignments caused
 by missing modalities. CalMRL leverages the priors and the inherent connections
 among modalities to model the imputation for the missing ones at the representa
tion level. To resolve the optimization dilemma, we employ a bi-step learning me
thod with the closed-form solution of the posterior distribution of shared laten
ts. We validate its mitigation of anchor shift and convergence with theoretical
guidance. By equipping the calibrated alignment with the existing advanced metho
d, we offer new flexibility to absorb data with missing modalities, which is ori

ginally unattainable. Extensive experiments demonstrate the superiority of CalMRL. The code is released at https://github.com/Xiaohao-Liu/CalMRL.
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IVQA-LD: Inclusive Multimodal Understanding for Population with Limb Deficiency
Yan Ke, Xin Shen, Jiaying Ying, Xin Li, Xin Yu
People with limb differences often face significant challenges in accessing inclusive AI services, largely due to the lack of structured, high-quality resources centered on disability contexts.
In this work, we introduce a limb-deficiency aware body-centric learning and evaluation paradigm that
involves (i) a large-scale limb-aware vision–language dataset and evaluation benchmark for multimodal reasoning, and (ii) a model adaptation strategy for Vision-Language Models (VLM) in limb-difference contexts.
Specifically, we first collect limb-difference data covering all eight limb-deficiency types across diverse real-world scenarios. The data are systematically organized into 96 limb-affected human action categories and 68 medical-functional classes defined by the World Health Organization (WHO).
Then, we curate an expert-annotated vision–language dataset for limb-aware multimodal understanding, named Inclusive VQA for Limb Deficiency (**IVQA-LD**). IVQA-LD comprises 80K VQA pairs spanning eight core tasks including visual grounding, quantitative reasoning, functional semantic classification, and instructional text generation.
We benchmark state-of-the-art VLMs on IVQA-LD and find that they consistently struggle across all tasks, exposing substantial deficiencies in limb-aware perception and reasoning.
To address this, we further propose a Body-centric Structure-aware Initialization (BSI) strategy that aligns model representations with limb-specific semantics. With BSI, VLMs fine-tuned on IVQA-LD achieve significant performance improvements across all the tasks.
We will publicly release the dataset to support future research.
****
Token-Efficient Change Detection in LLM APIs
Timothee Chauvin, Clément Lalanne, Erwan Le Merrer, Jean-Michel Loubes, Francois Taiani, Gilles Tredan
Remote change detection in LLMs is a difficult problem.
Existing methods are either too expensive for deployment at scale, or require initial white-box access to model weights or grey-box access to log probabilities. We aim to achieve both low cost and strict black-box operation, observing only output tokens.

Our approach hinges on specific inputs we call Border Inputs, for which there exists more than one output top token.
From a statistical perspective, optimal change detection depends on the model's Jacobian and the Fisher information of the output distribution, whose analysis at low temperature regimes shows that border inputs enable powerful change detection tests.

Building on this insight, we propose the Black-Box Border Input Tracking (B3IT) scheme. Extensive in-vivo and in-vitro experiments show that border inputs are easily found for non-reasoning tested endpoints, and present on-par performance with the best available grey-box approaches. B3IT reduces costs by $30\times$ compared to existing methods, while operating in a strict black-box setting.
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Can Large Language Models Generalize Procedures Across Representations?
Fangru Lin, Valentin Hofmann, Xingchen Wan, Weixing Wang, Zifeng Ding, Anthony G Cohn, Janet B. Pierrehumbert
Large language models (LLMs) are trained and tested extensively on symbolic representations such as code and graphs, yet real-world user tasks are often specified in natural language. To what extent can LLMs generalize across these representations? Here, we approach this question by studying isomorphic tasks involving

procedures represented in code, graphs, and natural language (e.g., scheduling steps in planning). We find that training LLMs with popular post-training methods on graphs or code data alone does not reliably generalize to corresponding natural language tasks, while training solely on natural language can lead to inefficient performance gains. To address this gap, we propose a two-stage reinforcement learning curriculum that first trains on symbolic, then natural language data. The curriculum substantially improves model performance across model families and tasks. Remarkably, a 1.5B Qwen model trained by our method can closely match zero-shot GPT-4o in naturalistic planning. Finally, our analysis suggests that successful cross-representation generalization can be interpreted as a form of generative analogy, which our curriculum effectively encourages. The dataset and code used in this paper can be found \href{https://github.com/fangru-lin/procedure_generalization_llm}{here}.
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When Does Sparsity Mitigate the Curse of Depth in LLMs
Dilxat Muhtar, Xinyuan Song, Sebastian Pokutta, Max Zimmer, Nico Pelleriti, Thomas Hofmann, Shiwei Liu
Recent work has demonstrated the curse of depth in large language models (LLMs), where later layers contribute less to learning and representation than earlier layers. Such under-utilization is linked to the accumulated growth of variance in Pre-Layer Normalization, which can push deep blocks toward near-identity behavior. In this paper, we provide evidence that sparsity-like mechanisms can dampen variance propagation and are associated with improved depth utilization Our investigation covers two sources of sparsity: (i) implicit sparsity, which emerges from training and data conditions, including weight sparsity induced by weight decay and attention sparsity induced by long-context inputs; and (ii) explicit sparsity, which is enforced by architectural design, including key/value-sharing in Grouped-Query Attention and expert-activation sparsity in Mixture-of-Experts. Our claim is thoroughly supported by controlled depth-scaling experiments and targeted layer effectiveness interventions. Across settings, we observe a consistent relationship: mechanisms with reduced effective interaction density tend to exhibit lower output variance and better layer differentiation. We eventually distill our findings into a practical rule-of-thumb recipe for training depth-effective LLMs, yielding a notable 4.6 accuracy improvement on downstream
tasks. Our results suggest that sparsity-like design choices are an important and previously underemphasized factor in effective depth scaling for LLMs. Code is available at https://github. com/pUmpKin-Co/SparsityAndCoD.
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Mixtures of geodesic factor analyzers on Riemannian homogeneous spaces
Hengchao Chen, Yuanyao Tan, Chao Huang, Hongtu Zhu, Qiang Sun
This paper introduces Mixtures of Geodesic Factor Analyzers (MGFA) on Riemannian homogeneous spaces. MGFA uses a geodesic factor model within each mixture component, providing greater expressiveness than mixtures of Riemannian radial distributions and enabling clustering of manifold-valued data with anisotropic subpopulations. We establish root-n consistency for the MGFA maximum likelihood estimator (MLE), thereby filling a theoretical gap for mixtures of Riemannian radial distributions as a special case. We also propose an iterative estimation algorithm and implement it on spheres, shape spaces, and hyperbolic spaces. Numerical experiments show that MGFA substantially outperforms competing methods in well-specified regimes while remaining robust under model misspecification. Finally, case studies on corpus callosum and left hippocampus shape datasets demonstrate MGFA’s effectiveness for both 2D contour and 3D shape analysis.
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PADA-Coder: Improving Plan-Following Code Generation via Perturbation-Verified Attention Distillation and Dynamic Alignment
Yihong Huang, Ke Qin, Rongzheng Wang, Muquan Li, Jiakai Li, Xiurui Xie, Shuang Liang
The Plan-then-Code paradigm effectively enhances Large Language Models (LLMs) in complex code generation by decomposing reasoning into explicit, interpretable steps. However, introducing the plan and verification report substantially enlarg

es the context, which in turn misdirects the model’s attention toward irrelevant tokens and the most recently generated code. This effect leads the model to overlook critical constraints and to generate incorrect code, especially for small-scale LLMs (less than 8B). To address this issue, we propose \textbf{P}erturbation-Verified \textbf{A}ttention \textbf{D}istillation and Dynamic \textbf{A}lignment (PADA). PADA identifies the key tokens most critical to the student model and constructs the optimal attention target matrix, dynamically aligning the student’s focus with key tokens for each plan step. We evaluate PADA with two teacher models and three student models across seven benchmarks, and the results show that PADA improves Pass@1 by up to 16.7\% and outperforms SOTA methods in overall average performance.

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TetraJet-v2: Accurate NVFP4 Training for Large Language Models with Oscillation Suppression and Outlier Control

Yuxiang Chen, Yifan Liu, Xiaoming Xu, Pengle Zhang, Michael Beyer, Martin Rapp, Jun Zhu, Jianfei Chen

Large Language Models (LLMs) training is prohibitively expensive, driving interest in low-precision fully-quantized training (FQT).
While novel 4-bit formats like NVFP4 offer substantial efficiency gains, achieving near-lossless training at such low precision remains challenging.
We introduce **TetraJet-v2**, an end-to-end 4-bit FQT method that leverages NVFP4 for activations, weights and gradients in all linear layers.
We identify two critical issues hindering low-precision LLM training: weight oscillation and outliers.
To address these, we propose: 1) an unbiased double-block quantization method for NVFP4 linear layers, 2) **OsciReset**, an algorithm to suppress weight oscillation, and 3) **OutControl**, an algorithm to retain outlier accuracy. **TetraJet-v2** outperforms prior methods on FP4 pre-training for LLMs across models up to 370M parameters trained up to 212B tokens, reducing the performance gap to BF16 by an average of 51.3% while enabling an $1.67\times$ end-to-end speedup over FP8.

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From Statics to Dynamics: Physics-Aware Image Editing with Latent Transition Priors

Liangbing Zhao, Le Zhuo, Sayak Paul, Hongsheng Li, Mohamed Elhoseiny

Instruction-based image editing has achieved remarkable success in semantic alignment, yet state-of-the-art models frequently fail to render physically plausible results when editing involves complex causal dynamics, such as refraction or material deformation.

We attribute this limitation to the dominant paradigm that treats editing as a discrete mapping between image pairs, which provides only boundary conditions and leaves transition dynamics underspecified.

To address this, we reformulate physics-aware editing as predictive physical state transitions and introduce PhysicTran38K, a large-scale video-based dataset comprising 38K transition trajectories across five physical domains, constructed via a two-stage filtering and constraint-aware annotation pipeline.

Building on this supervision, we propose PhysicEdit, an end-to-end framework equipped with a textual-visual dual-thinking mechanism. It combines a frozen Qwen2.5-VL for physically grounded reasoning with learnable transition queries that provide timestep-adaptive visual guidance to a diffusion backbone.

Experiments show that PhysicEdit improves over Qwen-Image-Edit by 5.9\% in physical realism and 10.1\% in knowledge-grounded editing, setting a new state-of-the-art for open-source methods, while remaining competitive with leading proprietary models.

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Anytime-Valid Inference for Online Ranking of Large Language Models

Runzhe Gu, Wenguang Sun, Bowen Gang, Xintao Xia

Online evaluation of large language models increasingly relies on sequentially collected pairwise preferences, enabling human-aligned assessment and continuous data collection until closely performing models can be reliably distinguished. H

owever, adaptive sampling and continuous monitoring invalidate classical fixed-sample inference, rendering existing ranking procedures largely heuristic. We propose SERPANT (Sequential E-value Ranking and Pruning via Adaptive Null Testing), a principled framework for online LLM ranking with anytime-valid guarantees. SERPANT formulates model comparison as a collection of pairwise hypothesis tests and constructs e-processes to ensure family-wise error rate control at any monitoring time. Anytime validity provides a theoretical justification for early stopping, enabling substantial cost savings from expensive human annotation. To improve efficiency, we introduce a novel tournament-based sampling strategy that adaptively selects comparisons based on past outcomes. The proposed framework is also extended to top-k model identification.

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De4D-SLAM: Gradient-Isolated Static-Dynamic Decoupling for Monocular SLAM in Dynamic Environments

Zhicheng Fan, Zitong Wu, Zhaoxing Fan, Xiao Zhang, Biao Hou, Bo REN

Conventional dynamic SLAM approaches typically treat dynamic objects as outliers based on pre-defined categories, creating perceptual blind spots that limit the comprehensive environmental perception required for embodied agents. Although integrating Gaussian Splatting into SLAM enables holistic scene representation, it introduces an optimization paradox: without categorical priors, flexible dynamic primitives rapidly overfit static residuals. This phenomenon undermines the self-supervised error signals necessary for distinguishing motion. In response, we present De4D-SLAM, a novel framework designed for decoupled 4D reconstruction from monocular video. Our approach features a Gradient-Isolated Decoupling strategy, which leverages static reconstruction residuals to supervise a Spatially-Aware Kolmogorov-Arnold Network (SA-KAN), ensuring robust, category-agnostic motion segmentation. Additionally, we propose a Flow-Induced Initialization prior to stabilize the non-convex optimization of 4D Gaussian primitives using dense optical flow. Extensive evaluations on the TUM and Bonn benchmarks demonstrate that De4D-SLAM achieves competitive performance in both tracking and dynamic reconstruction, successfully reconciling the tension between robust localization and high-fidelity 4D mapping.

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Federated Bilevel Performative Prediction

Liangxin Qian, Chang Liu, Xuanyu Cao, Jun Zhao, Kwok-Yan Lam

Federated bilevel optimization is widely used for nested learning problems across distributed clients, such as federated hyperparameter tuning and meta-learning under privacy and communication constraints. Most existing formulations assume fixed client data distributions, which can be violated by performativity, where deployed decisions reshape client behavior and data collection, inducing client-specific, decision-dependent distribution shift. We study federated bilevel performative prediction, where both upper-level (UL) and lower-level (LL) objectives are evaluated under client-dependent, decision-dependent distributions. We formalize the federated bilevel performatively stable (FBPS) point under a decoupled-risk perspective and provide sufficient conditions for its existence and uniqueness. We then develop two federated methods to compute the FBPS solution: FBi-RRM, which converges linearly under a contraction condition, and FBi-SGD, a communication-efficient stochastic method based on federated hypergradient estimation with convergence guarantees under diminishing step sizes when sensitivities are sufficiently small. Experiments on strategic regression and meta strategic classification validate the predicted stability thresholds and demonstrate improved meta-generalization over non-performative baselines, and CNN-based classification further demonstrates the practical effectiveness of the proposed methods in non-convex neural network settings.

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A Queueing-Theoretic Framework for Stability Analysis of LLM Inference with KV Cache Memory Constraints

Chengyi Nie, Nian Si, Zijie Zhou

The rapid adoption of large language models (LLMs) has created significant challenges for efficient inference at scale. Unlike traditional workloads, LLM infere

nce is constrained by both computation and the memory overhead of key–value (KV) caching, which accelerates decoding but quickly exhausts GPU memory. In this paper, we introduce the first queueing-theoretic framework that explicitly incorporates both computation and GPU memory constraints into the analysis of LLM inference. Based on this framework, we derive rigorous stability and instability conditions that determine whether an LLM inference service can sustain incoming demand without unbounded queue growth. This result offers a powerful tool for system deployment, potentially addressing the core challenge of GPU provisioning. By combining an estimated request arrival rate with our derived stable service rate, operators can calculate the necessary cluster size to avoid both costly over-purchasing and performance-violating under-provisioning. We further validate our theoretical predictions through extensive experiments in real GPU production environments. Our results show that the predicted stability conditions are highly accurate, with deviations typically within 10%.

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TimeSAE: Causal Sparse Decoding for Faithful Explanations of Black-Box Time Series Models

Khalid Oublal, Quentin Bouniot, Qi Gan, Stephan Clémençon, Zeynep Akata

As black box models and pretrained models gain traction in time series applications, understanding and explaining their predictions becomes increasingly vital, especially in high-stakes domains where interpretability and trust are essential. However, most of the existing methods involve only in-distribution explanation, and do not generalize outside the training support, which requires the learning capability of generalization. In this work, we aim to provide a framework to explain black-box models for time series data through the dual lenses of Sparse Autoencoders (SAEs) and causality. We show that many current explanation methods are sensitive to distributional shifts, limiting their effectiveness in real-world scenarios. Building on the concept of Sparse Autoencoder, we introduce TimeSAE, a framework for black-box model explanation. We conduct extensive evaluations of TimeSAE on both synthetic and real-world time series datasets, comparing it to leading baselines. The results, supported by both quantitative metrics and qualitative insights, show that TimeSAE provides more faithful and robust explanations. Our code is available in an easy-to-use library TimeSAE-Lib: https://oublalkhalid.github.io/TimeSAE/

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Continual GUI Agents

Ziwei Liu, Borui Kang, Hangjie Yuan, Zixiang Zhao, Wei Li, Yifan Zhu, Tao Feng

As digital environments (data distribution) are in flux, with new GUI data arriving over time-introducing new domains or resolutions-agents trained on static environments deteriorate in performance. In this work, we introduce Continual GUI Agents, a new task that requires GUI agents to perform continual learning under shifted domains and resolutions. We find existing methods fail to maintain stable grounding as GUI distributions shift over time, due to the diversity of UI interaction points and regions in fluxing scenarios. To address this, we introduce GUI-Anchoring in Flux (GUI-AiF), a new reinforcement fine-tuning framework that stabilizes continual learning through two novel rewards: Anchoring Point Reward in Flux (APR-iF) and Anchoring Region Reward in Flux (ARR-iF). These rewards guide the agents to align with shifting interaction points and regions, mitigating the tendency of existing reward strategies to over-adapt to static grounding cues (e.g., fixed coordinates or element scales). Extensive experiments show GUI-AiF surpasses state-of-the-art baselines. Our work establishes the first continual learning framework for GUI agents, revealing the untapped potential of reinforcement fine-tuning for continual GUI Agents.

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UniMedVL: Unifying Medical Multimodal Understanding and Generation through Observation-Knowledge-Analysis

Junzhi Ning, Wei Li, Cheng Tang, Jiashi Lin, Chenglong Ma, Chaoyang Zhang, Jiyao Liu, Ying Chen, Shujian Gao, Yuandong Pu, Huihui Xu, Chenhui Gou, Ziyan Huang, Yi Xin, Qi Qin, Diping Song, Bin Fu, Guang Yang, Yuanfeng Ji, Tianbin Li, Yanzhou Su, Jin Ye, SHIXIANG TANG, Zhongying Deng, Lihao Liu, Ming Hu, Junjun He

Medical workflows routinely combine reading images with producing visual and tex
tual outputs, making both image understanding and generation central to medical
AI. Most existing systems, however, address these abilities in isolated models,
losing the shared knowledge that a unified architecture could exploit. To bridge
 this gap, we present UniMedVL, the first unified medical model that seamlessly
integrates multimodal understanding and generation capabilities within a single
model without switching weights. We achieve this via a tailored progressive trai
ning pipeline where understanding and generation mutually reinforce each other.
To effectively train UniMedVL, we curate UniMedVL-5M, the first large-scale medi
cal dataset comprising over 5.6M instances across 8 medical imaging modalities,
tailored for multimodal input-output tasks in unified medical understanding and
generation. Experimental results demonstrate that UniMedVL achieves competitive
performance on five medical understanding benchmarks. Crucially, UniMedVL native
ly supports diverse interleaved generation tasks, e.g., virtual staining, super-
resolution, cross-modal synthesis, essential for complex medical workflows. Our
code and dataset are publicly available.
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Conditional Equivalence of DPO and RLHF: Assumptions, Failure Modes, and Provabl
e Alignment

Zhiqin Yang, Yonggang Zhang, Wei Xue, Dong Fang, Bo Han, Yike Guo

Direct Preference Optimization (DPO) has emerged as a popular alternative to Rei
nforcement Learning from Human Feedback (RLHF), offering theoretical equivalence
 with a simpler implementation. We prove this equivalence is _conditional_ rathe
r than universal, depending on an implicit assumption frequently violated in pra
ctice: the RLHF-optimal policy must prefer human-preferred responses. When this
assumption fails, DPO optimizes _relative advantage_ over the reference policy r
ather than _absolute alignment_ with human preferences, leading to pathological
convergence where policies decrease DPO loss while preferring dispreferred respo
nses. We characterize when this assumption is violated, show the existence of an
 undesirable solution space, and prove that DPO and RLHF optimize fundamentally
different objectives in such cases. To address this, we introduce Constrained Pr
eference Optimization (CPO), augmenting RLHF with constraints for provable align
ment. We further provide a geometric interpretation through soft margin ranking,
 revealing that DPO implements margin ranking with potentially negative targets.
 Our theoretical analysis establishes when DPOs' guarantees hold and provides so
lutions preserving simplicity with provable alignment. Comprehensive experiments
 on standard benchmarks demonstrate that CPO achieves state-of-the-art performan
ce. Code is available at: _https://github.com/visitworld123/CPO_.
****

$\mathbb{R}^{2k}$ is Theoretically Large Enough for Embedding-based Top-k Retr
ieval

Zihao Wang, Hang Yin, Lihui Liu, Hanghang Tong, Yangqiu Song, Ginny Wong, Simon
See

This paper studies the Minimal Embeddable Dimension (MED): the least dimension i
n which there exists a configuration of m object vectors so that every subset
of size at most k is exactly retrieved by score comparison. Our result shows M
ED is $\Theta(k)$, independent of m, for inner product, Euclidean distance, an
d cosine similarity. We then consider Robust MED (RMED), where all vectors are u
nit normed and an ϵ gap of scores is required. We derive the m-depend
ent feasibility ceiling $\epsilon_\star(m,k)=m/\sqrt{k(m-1)(m-k)}$, which approa
ches $1/\sqrt{k}$ when $m\gg k$, and a Gaussian centroid construction gives a ro
bust witness upper bound in the feasible margin regime. Numerical simulation on
synthetic top-2 retrieval with cyclic polytope and centroid query optimization
 confirmed our theoretical claims. Experiments on LIMIT and LIMIT-small datasets
 also show that simple embedding-based retrieval baselines can overfit and outpe
rform the reported single-vector LLM embedding baseline. Both theoretical and em
pirical findings rule out the lack of exact geometric capacity as the obstructio
n.
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Harnessing Non-Adversarial Robustness in Large Language Models

Qinghua Zhou, Ellina Aleshina, Andrey Lovyagin, Oleg Somov, Mikhail Seleznyov, A
lexander Panchenko, Ivan Oseledets, Elena Tutubalina, Ivan Y Tyukin
The work presents an approach for addressing the challenge of robustness in Larg
e Language Models (LLMs) to alterations and potential errors caused by semantica
lly similar but textually different prompts. Recent works have shown that these
kinds of prompt variations can significantly impact the performance of LLMs on t
asks. The central question is: can LLMs' robustness to semantically-neutral prom
pt alterations be acquired without expensive retraining of the entire model? We
address this question both theoretically and through experiments. Our theoretica
l analysis reveals a crucial factor impacting model robustness -- a systematic e
xpected shift or perturbation-induced bias in neural network module outputs. Mot
ivated by this analysis, we show that robustness can be achieved via a simple fi
ne-tuning process: debiasing for robustness. We identify conditions when debiasi
ng helps and when it does not, and demonstrate, through both theory and extensiv
e experiments, that debiasing for robustness may indeed be a quick and efficient
 tool to enhance robustness and provide certification against random prompt pert
urbations.
****

V-ABS: Action-Observer Driven Beam Search for Dynamic Visual Reasoning
Zhiwei Ning, Xuanang Gao, Jiaxi Cao, Gengming Zhang, Shengnan Ma, Wenwen Tong, H
anming Deng, JIE YANG, Wei Liu
Multimodal large language models (MLLMs) have achieved remarkable success in gen
eral perception, yet complex multi-step visual reasoning remains a persistent ch
allenge. Although recent agentic approaches incorporate tool use, they often neg
lect critical execution feedback. Consequently, they suffer from the imagination
-action-observer (IAO) bias, a misalignment between prior imagination and observ
er feedback that undermines reasoning stability and optimality. To bridge this g
ap, we introduce V-ABS, an action-observer driven beam search framework that ena
bles deliberate reasoning through thinker-actor-observer iterations. We also pro
pose an entropy-based adaptive weighting algorithm to mitigate the IAO bias by d
ynamically balancing the confidence scores between the policy priors and the obs
ervational feedback. Moreover, we construct a large-scale supervised fine-tuning
 (SFT) dataset comprising over 80k samples to guide the model to assign higher p
rior confidence to correct action paths. Extensive experiments across eight dive
rse benchmarks show that V-ABS achieves state-of-the-art performance, delivering
 an average improvement of 19.7\% on the Qwen3-VL-8B baseline and consistent gai
ns across both open-source and proprietary models.
****

α-PFN: Fast Entropy Search via In-Context Learning
Herilalaina Rakotoarison, Steven Adriaensen, Tom Julian Viering, Carl Hvarfner,
Samuel Müller, Frank Hutter, Eytan Bakshy
Information-theoretic acquisition functions such as Entropy Search (ES) offer a
principled exploration–exploitation framework for Bayesian optimization (BO). Ho
wever, their practical implementation relies on complicated and slow approximati
ons, i.e., a Monte Carlo estimation of the information gain. This complexity can
 introduce numerical errors and requires specialized, hand-crafted implementatio
ns. We propose a two-stage amortization strategy that learns to approximate entr
opy search-based acquisition functions using Prior-data Fitted Networks (PFNs) i
n a single forward pass. A first PFN is trained to be conditioned on information
 about the optima; second, the α-PFN is trained to predict the expected informat
ion gain by training on information gains measured with the first PFN. The α-PFN
 offers a flexible
learned approximation, which replaces the complex heuristic approximations with
a single forward pass per candidate, enabling rapid and extensible acquisition e
valuation. Empirically, our approach is competitive with state-of-the-art entrop
y search implementations on synthetic and real-world benchmarks, while accelerat
ing the different entropy search variants across all our experiments, with speed
 ups over 50x.
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Temper-Then-Tilt: Principled Unlearning for Generative Models through Tempering

and Classifier Guidance
Jacob L. Block, Mehryar Mohri, Aryan Mokhtari, Sanjay Shakkottai
We study machine unlearning in large generative models by framing the task as de
nsity ratio estimation to a target distribution rather than supervised fine-tuni
ng. While classifier guidance is a standard approach for approximating this rati
o and can succeed in general, we show it can fail to faithfully unlearn with fin
ite samples when the forget set represents a sharp, concentrated data distributi
on. To address this, we introduce **Temper-Then-Tilt Unlearning (T3-Unlearning)*
*, which freezes the base model and applies a two-step inference procedure: (i)
tempering the base distribution to flatten high-confidence spikes, and (ii) *t
ilting* the tempered distribution using a lightweight classifier trained to dist
inguish retain from forget samples. Our theoretical analysis provides finite-sam
ple guarantees linking the surrogate classifier's risk to unlearning quality, pr
oving that tempering is necessary to successfully unlearn for concentrated distr
ibutions. Empirical evaluations on the TOFU benchmark demonstrate that T3-Unlear
ning improves forget quality and generative utility over existing baselines, whi
le training only a fraction of the parameters with a minimal runtime.
****
Safety Game: Inference-Time Alignment of Black-Box LLMs via Constrained Optimiza
tion
Tuan Nguyen, Long Tran-Thanh
Ensuring that large language models (LLMs) comply with safety requirements is a
central challenge in AI deployment. Existing alignment approaches operate primar
ily during training, such as through fine-tuning or reinforcement learning from
human feedback, but these methods are costly and inflexible, requiring retrainin
g whenever new requirements arise. Recent efforts toward inference-time alignmen
t mitigate some of these limitations but still assume access to model internals,
 which is impractical, and not suitable for third party stakeholders who do not
have access to the models. In this work, we propose a model-independent, black-b
ox framework for safety alignment that does not require retraining or access to
the underlying LLM architecture. As a proof of concept, we address the problem o
f trading off between generating safe but uninformative answers versus helpful y
et potentially risky ones. We formulate this dilemma as a two-player zero-sum ga
me whose minimax equilibrium captures the optimal balance between safety and hel
pfulness. LLM agents operationalize this framework by leveraging a linear progra
mming solver at inference time to compute equilibrium strategies. Our results de
monstrate the feasibility of black-box safety alignment, offering a scalable and
 accessible pathway for stakeholders, including smaller organizations and entiti
es in resource-constrained settings, to enforce safety across rapidly evolving L
LM ecosystems.
****
Agent JIT Compilation for Latency-Optimizing Web Agent Planning and Scheduling
Caleb Winston, Ron Yifeng Wang, Azalia Mirhoseini, Christos Kozyrakis
Computer-use agents (CUAs) automate tasks specified with natural language such a
s
  "order the cheapest item from Taco Bell"
  by generating sequences of calls to tools such as click, type, and scroll on a
 browser.
  Current implementations follow a sequential fetch-screenshot-execute loop wher
e each iteration requires an LLM call,
  resulting in high latency and frequent errors from incorrect tool use.
  We present agent just-in-time (JIT) compilation, a system that compiles
  task descriptions directly into executable code that may include LLM calls, to
ol calls, and parallelization.
  Our approach comprises three components: (1) JIT-Planner,
  which generates multiple code plans, validates each against tool
  specifications, and selects the minimum-cost candidate; (2) JIT-Scheduler,
  which explores parallelization strategies via Monte Carlo cost estimation from
 learned latency
  distributions; and (3) an invariant-enforcing tool protocol specifying

precondition and postcondition requirements to reduce the rate of incorrect to
ol use.
Across five applications, JIT-Planner achieves $10.4\times$ speedup and 28% hi
gher
accuracy over Browser-Use, while JIT-Scheduler achieves $2.4\times$ speedup an
d
9% higher accuracy over OpenAI CUA.
****
Controlled SDEs for Long-Horizon Motion Generation under Latent Decision Uncerta
inty
Han Zhang, Nenggan Zheng
Long-horizon motion prediction under external commands is challenged by latent d
ecision uncertainty, where the internal states governing future behavior are uno
bservable and evolve stochastically over time. This issue is particularly pronou
nced in biological agents, whose motion trajectories reflect decision-making pro
cesses rooted in underlying cognitive states. To address these challenges, we pr
opose CogSDE, a formulation of a controlled stochastic differential equation (SD
E) for modeling instruction-driven latent decision dynamics. The drift term in t
he SDE incorporates a dual-channel control modulation mechanism, enabling extern
al commands to modulate the evolution of latent states. The diffusion term emplo
ys a state-dependent operator to model intrinsic uncertainty in latent decision
dynamics. Furthermore, we establish dissipativity-based mean-square boundedness
for the latent decision dynamics. Experiments demonstrate that CogSDE consistent
ly improves predictive accuracy in long-horizon motion generation. Importantly,
predicted trajectories remain well aligned with control commands over extended h
orizons, a property widely recognized as challenging in long-horizon motion pred
iction.
****
A Direct Approach for Handling Contextual Bandits with Latent State Dynamics
Zhen LI, Gilles Stoltz
We consider a linear contextual bandit model where contexts and rewards are gove
rned by a finite hidden Markov chain. We first revisit the simplified model by N
elson et al. (2022), in which rewards are linear functions of the posterior prob
abilities over the hidden states given the observed contexts (called beliefs), r
ather than functions of the hidden states themselves. This simplified model may
be handled through a direct reduction to standard linear contextual bandits. We
extend the theoretical analysis of this reduction to take into account the estim
ation of the parameters of the hidden Markov model [HMM] in the regret bound and
to provide high-probability bounds not depending anymore on the reward function
s and only depending on the model through the estimation of the HMM parameters.
Second, and most importantly, we instead study the more natural and more complex
model incorporating direct dependencies in the hidden states (on top of depende
ncies on the observed contexts, as is natural for contextual bandits). Under a c
lassic HMM forgetting condition, the main algorithmic tool introduced to cope wi
th the various statistical dependencies that the reward structure introduces is
to only periodically update reward-model parameters.
****
Uncertainty-Aware Clarification in LLM Agents with Information Gain
Mengyi DENG, Zhiwei Li, Xin Li, Tingyu ZHU, Ying Zhao, Zhijiang Guo, Wei Wang
Large Language Model (LLM) agents often operate under underspecified user instru
ctions, where latent uncertainty over user intent leads to erroneous tool action
s. To address this challenge, we propose a goal-oriented clarification framework
that aligns clarification behavior with ambiguity resolution. Central to our ap
proach is the Information Gain Reward, a metric that quantifies the utility of c
larification questions by measuring the Bayesian belief update towards the groun
d-truth goal induced by the clarification exchange. We train the clarifier (LLM)
using this reward to optimize for high information gain, ensuring that clarific
ations effectively reduce uncertainty and improve task completion within the age
nt-tool-user environment. We validate our framework within a clarification-enhan
ced τ-Bench environment, conducting cross-agent evaluations across five het

erogeneous backbones. Empirical results demonstrate that our method consistently improves the success rate by 3.7\% over the no-clarification baseline, while adding only 0.3 total interaction steps on average.
****
SCALE: Self-uncertainty Conditioned Adaptive Looking and Execution for Vision-Language-Action Models
Hyeonbeom Choi, Daechul Ahn, Youhan Lee, Taewook Kang, Seongwon Cho, Jonghyun Choi
Vision-Language-Action (VLA) models have emerged as a promising paradigm for general-purpose robotic control, with test-time scaling (TTS) gaining attention to enhance robustness beyond training. However, existing TTS methods for VLAs require additional training, verifiers, and multiple forward passes, making them impractical for deployment. Moreover, they intervene only at action decoding while keeping visual representations fixed—insufficient under perceptual ambiguity, where reconsidering how to perceive is as important as deciding what to do. To address these limitations, we propose SCALE, a simple inference strategy that jointly modulates visual perception and action based on 'self-uncertainty', inspired by uncertainty-driven exploration in Active Inference theory—requiring no additional training, no verifier, and only a single forward pass. SCALE broadens exploration in both perception and action under high uncertainty, while focusing on exploitation when confident—enabling adaptive execution across varying conditions. Experiments on simulated and real-world benchmarks demonstrate that SCALE improves state-of-the-art VLAs and outperforms existing TTS methods while maintaining single-pass efficiency. Our code is publicly available at https://github.com/sumprlab/scale.
****
Collaborative Threshold Watermarking
Tameem Bakr, Anish Ambreth, Nils Lukas
In federated learning (FL), K clients jointly train a model without sharing raw data. Because each participant invests data and compute, clients need mechanisms to later prove the provenance of a jointly trained model. Model watermarking embeds a hidden signal in the weights, but naive approaches either do not scale with many clients as per-client watermarks dilute as K grows, or give any individual client the ability to verify and potentially remove the watermark. We introduce (t,K)-threshold watermarking: clients collaboratively embed a shared watermark during training, while only coalitions of at least t clients can reconstruct the watermark key and verify a suspect model. We secret-share the watermark key τ so that coalitions of fewer than t clients cannot reconstruct it, and verification can be performed without revealing τ in the clear. We instantiate our protocol in the white-box setting and evaluate it on image classification tasks on both IID and non-IID partitions, as well as language models fine-tuning
setting. Our watermark remains detectable at scale ($K=128$) with minimal accuracy loss and stays above the detection threshold ($z\ge 4$) under attacks including adaptive fine-tuning using up to 20% of the training data. Code is available at \url{https://github.com/tameemalaa/collaborative-threshold-watermark}.
****
Any2Any: Unified Arbitrary Modality Translation for Remote Sensing
Haoyang Chen, Jing Zhang, Di Wang, Hebaixu Wang, Shiqin Wang, Pohsun Huang, Jiayuan Li, Haonan Guo, Zheng Wang, Bo Du
Multi-modal remote sensing imagery provides complementary observations of the same geographic scene, yet such observations are frequently incomplete in practice. Existing cross-modal translation methods treat each modality pair as an independent task, resulting in quadratic complexity and limited generalization to unseen modality combinations. We formulate Any-to-Any translation as inference over a shared latent representation of the scene, where different modalities correspond to partial observations of the same underlying semantics. Based on this formulation, we propose Any2Any, a unified latent diffusion framework that projects heterogeneous inputs into a geometrically aligned latent space. Such structure performs anchored latent regression with a shared backbone, decoupling modality-sp

ecific representation learning from semantic mapping. Moreover, lightweight targ
et-specific residual adapters are used to correct systematic latent mismatches w
ithout increasing inference complexity. To support learning under sparse but con
nected supervision, we introduce RST-1M, the first million-scale remote sensing
dataset with paired observations across five sensing modalities, providing super
vision anchors for any-to-any translation. Experiments across 14 translation tas
ks show that Any2Any consistently outperforms pairwise translation methods and e
xhibits strong zero-shot generalization to unseen modality pairs. Code and model
s are available at https://github.com/MiliLab/Any2Any.
****

POLIA: Policy Optimization with Visual-Object-Level Intrinsic Advantage for Mult
imodal Reasoning

Yiran Zeng, Da Chen, Hangyu Mao, Yuanxing Zhang, Pengfei Wan, Mengchen Zhao

Recent advances in group-based reinforcement learning (RL) greatly improve LLMs'
ability in text reasoning. Yet, these methods lack sufficient modeling of multi
modal information, leading to significant reasoning hallucination. In this work,
we propose POLIA, a novel group-based RL method with visual-object-level intrin
sic advantage for multimodal reasoning. POLIA introduces two advantage computati
on stages over candidate answers and visual objects, respectively. The answer-le
vel extrinsic advantages are computed based on the extrinsic rewards of a group
of candidate answers. Moreover, we compute an intrinsic advantage for each visua
l object based on its confidence score and reference relations with final answer
s. Intuitively, the intrinsic advantage of an object reflects its potential cont
ribution to the correct answer. This two-stage advantage computation ensures an
accurate credit assignment mechanism over multimodal reasoning sequences with mu
ltiple visual objects. Experimental results on diverse multimodal reasoning benc
hmarks show that POLIA significantly outperforms open MLLMs and strong baselines
. Code is available at https://github.com/dudu115/POLIAcode.
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ASIR: Steganography for Diffusion Models via Antipodal Sampling and Iterative Re
covery

Yaofei Wang, Yufeng Zheng, Han Fang, Wenzhao Cao, Donghui Hu

Messages embedded in diffusion generation noise suffer from severe attenuation d
ue to denoising and VAE decoding, creating a persistent capacity–robustness trad
e-off. Identifying that extraction accuracy strictly correlates with the distanc
e between candidate hypothesis images, we propose ASIR, a training-free and prov
ably secure steganography framework for both pixel and latent diffusion models.
ASIR introduces two key innovations: (i) Antipodal Sampling, which maximizes sig
nal separation in probability space to enhance distinguishability, and (ii) Iter
ative Recovery, a paradigm shift that treats extraction as a gradient-based opti
mization problem to reverse non-linear distortions. Extensive experiments demons
trate that ASIR achieves state-of-the-art performance, embedding up to 65,536 bi
ts (pixel-space) and 16,384 bits (latent-space) with 99\% accuracy, while remain
ing statistically undetectable to deep steganalyzers.
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Transitivity Meets Cyclicity: Explicit Preference Decomposition for Dynamic Larg
e Language Model Alignment

Yucong Huang, Xiucheng Li, Kaiqi Zhao, Jing Li

Standard RLHF relies on transitive scalar rewards, failing to capture the cyclic
nature of human preferences. While some approaches like the General Preference
Model (GPM) address this, we identify a theoretical limitation: their implicit f
ormulation entangles hierarchy with cyclicity, failing to guarantee dominant sol
utions. To address this, we propose the Hybrid Reward-Cyclic (HRC) model, which
utilizes game-theoretic decomposition to explicitly disentangle preferences into
orthogonal transitive (scalar) and cyclic (vector) components. Complementing th
is, we introduce Dynamic Self-Play Preference Optimization (DSPPO), which treats
alignment as a time-varying game to progressively guide the policy toward the N
ash equilibrium. Synthetic data experiments further validate HRC's structural su
periority in mixed transitive--cyclic settings, where HRC converges faster and a
chieves higher accuracy than GPM. Experiments on RewardBench 2 demonstrate that

HRC consistently improves over both BT and GPM baselines (e.g., +1.23\% on Gemma -2B-it). In particular, its superior performance in the Ties domain empirically validates the model's robustness in handling complex, non-strict preferences. Ex tensive downstream evaluations on AlpacaEval 2.0, Arena-Hard-v0.1, and MT-Bench confirm the efficacy of our framework. Notably, when using Gemma-2B-it as the ba se preference model, HRC+DSPPO achieves a peak length-controlled win-rate of 44. 75\% on AlpacaEval 2.0 and 46.8\% on Arena-Hard-v0.1, significantly outperformin g SPPO baselines trained with BT or GPM. Our code is publicly available at https ://github.com/lab-klc/Hybrid-Reward-Cyclic.
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RECTOR: Masked Region-Channel-Temporal Modeling for Affective and Cognitive Repr esentation Learning

Jinhan Liu, Mahsa Shoaran

Affective and cognitive disorders manifest as distributed, time-varying brain ne twork dynamics across regions, channels, and time, challenging robust representa tion learning from EEG/sEEG for clinical diagnosis. We propose **RECTOR** (Maske d **Re**gion–**C**hannel–**T**emp**or**al Modeling), an end-to-end self-supervis ed framework that unifies joint region-channel-temporal representation learning beyond fixed anatomical priors. At its core, **RECTOR-SA** is a hierarchical, bl ock-sparse self-attention induced by Adaptive Functional Partitioning that evolv es region structures from static anatomical definitions to adaptive functional r egions. The self-supervision is driven by **Masked Topology and Representation L earning**, which jointly optimizes three complementary objectives: Masked Predic tive Modeling, Topological Structure Modeling, and Cross-View Consistency. Acros s diverse benchmarks, RECTOR sets a new state-of-the-art in EEG emotion recognit ion and sEEG task-engagement classification. Crucially, its strong robustness to missing channels and cross-montage generalization underscores its potential for large-scale pre-training on heterogeneous EEG/sEEG, providing interpretable ins ights at both region and channel levels.
****

ExPLAIND: Unifying Model, Data, and Training Attribution to Study Model Behavior

Florian Eichin, Yupei Du, Philipp Mondorf, Maria Matveev, Barbara Plank, Michael A. Hedderich

Post-hoc interpretability methods typically attribute a model’s behavior to its components, data, or training trajectory in isolation, and are often tied to a p articular level of granularity along the local-to-global spectrum. This leads to explanations that lack a unified view and may miss key interactions. We present ExPLAIND, a theoretically grounded, unified framework that integrates model com ponents, data, and training trajectory while supporting explanations across gran ularities. We generalize recent work on gradient path kernels, reformulating mod els trained by AdamW as kernel machines. From the resulting kernel feature maps, we derive novel parameter-wise and step-wise influence scores. We empirically v alidate the resulting decomposition of model behavior in several settings and ap ply ExPLAIND to two case studies. Our findings on a Transformer exhibiting Grokk ing support previously proposed learning phases, while refining the final phase as one in which outer layers align around a representation pipeline learned afte r memorization. For EuroLLM pretraining, ExPLAIND reveals a two-phase dynamic, w ith the first characterized by outer-layer MLP learning and the second by increa sed relative influence of intermediate attention layers. These results establish ExPLAIND as a unified framework for interpreting model behavior and training dy namics.
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Informed Asymmetric Actor-Critic: Leveraging Privileged Signals Beyond Full-Stat e Access

Daniel Ebi, Damien Ernst, Klemens Böhm, Gaspard Lambrechts

Asymmetric reinforcement learning leverages privileged information available dur ing training to improve learning under partial observability. Existing asymmetri c actor-critic methods typically assume access to the full environment state to condition the critic during training, which is often unrealistic in practice. We introduce the informed asymmetric actor-critic framework that allows the critic

to be conditioned on arbitrary state-dependent privileged signals, and show tha t any such signal yields unbiased policy gradient estimates. This substantially expands the set of admissible privileged information and raises the problem of s electing the most informative signals for learning. To this end, we propose two novel informativeness criteria: a dependence-based test that can be applied prio r to training, and a test based on improvements in value prediction that can be applied post hoc. Experiments on partially observable benchmarks and synthetic e nvironments demonstrate that carefully selected privileged signals can match or outperform full-state asymmetric baselines while relying on strictly less state information.

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Breaking the Echo Chamber: A Dynamic Ensemble Pruning Perspective on MoE

Xinlai Kang, Dunyao Xue, Zhengbo Wang, Chengshuo Du, Xinghao Chen, Hang Zhou, Ha nting Chen, Cheng Meng

We introduce Mahalanobis-Pruned Mixture-of-Experts (MP-MoE), a novel routing fra mework that approaches expert selection from the perspective of ensemble pruning . Existing Mixture-of-Experts (MoE) routing strategies often suffer from represe ntation collapse due to greedy top-k selection mechanisms or rely on complex aux iliary regularization terms that may compromise model performance. To address th ese issues, we formulate routing as a diversity-aware subset selection problem a nd optimize a Mahalanobis-distance-based objective that explicitly enhances expe rt diversity. Specifically, we demonstrate that the expert co-occurrence matrix effectively captures inter-expert correlations, allowing us to efficiently model the covariance structure required for distance computation without accessing ex pert parameters. Furthermore, we devise a greedy strategy for the routing mechan ism, backed by theoretical approximation guarantees, rendering it a plug-and-pla y module with negligible overhead.

MP-MoE increases wall-clock training time by approximately 3\%, while incurring no additional latency at inference time.

Extensive experiments demonstrate that during the pre-training of the large lang uage model, our method consistently outperforms the baseline by 1-3 percentage p oints across a broad range of benchmarks.

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Optimal Anytime Algorithms for Online Convex Optimization with Adversarial Const raints

Dhruv Sarkar, Abhishek Sinha

We propose an anytime online algorithm for learning a sequence of convex cost fu nctions while approximately satisfying a sequence of convex constraints, without prior knowledge of the time horizon. Both the cost and constraint functions may be chosen adversarially over time. While this problem has recently been resolve d in the setting where the time horizon is known, extending these guarantees to the anytime setting, without resorting to inefficient doubling tricks, has remai ned technically challenging. Our main contribution is the introduction of a time -varying yet horizon-oblivious Lyapunov function to track constraint violations. The use of such a time-varying Lyapunov function introduces new technical diffi culties, as a key monotonicity property underlying prior analyses no longer hold s. By developing a novel analytical technique, we show that our algorithm achiev es $O(\sqrt{t})$ regret and $\tilde{O}(\sqrt{t})$ cumulative constraint violatio n (CCV) for all $t \geq 1$. We further extend our framework to the dynamic regre t setting, obtaining bounds that adapt to the unknown path length of the compara tor sequence. Finally, we present an adaptive algorithm for the optimistic setti ng, whose performance scales gracefully with the cumulative prediction error. We validate the practical effectiveness of our approach through numerical experime nts on the online shortest path problem.

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iLoRA: Bayesian Low-Rank Adaptation with Latent Interaction Graphs for Microbiom e Diagnosis

Yang Song, Yixuan Zhang, Lingfa Meng, Tongyuan Hu, Haizhou Shi, Hao Wang, Samir Bhatt, Hengguan Huang

Parameter-efficient adaptation has made LLMs practical for domain prediction, bu

t standard LoRA still relies on a static low-rank update and does not expose the latent interactions that often drive scientific labels. We introduce iLoRA. To our knowledge, it is the first Bayesian graph-conditioned LoRA framework. It infers a latent interaction graph from the input and uses it to generate input-conditioned LoRA updates. As a result, iLoRA learns prediction and latent interaction structure jointly, rather than training a predictor and applying interaction analysis only post hoc. We instantiate this idea for microbiome diagnosis, where disease state can depend on both species-level abundance and microbe–microbe cross-talk, and evaluate it in two complementary settings: interactive QA with human-annotated graphs, which tests latent structure recovery, and multi-cohort IBD diagnosis, which tests biomedical utility. Across both settings, iLoRA improves over strong LoRA and Bayesian adaptation baselines, recovers graphs aligned with human annotations and cohort-level microbiome associations, and provides calibrated uncertainty with moderate graph-branch overhead.
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Query Circuits: Explaining How Language Models Answer User Prompts
Tung-Yu Wu, Fazl Barez
Explaining why a language model produces a particular output requires local, input-level explanations. Existing methods uncover global capability circuits (e.g., indirect object identification), but not why the model answers a specific input query in a particular way. We introduce query circuits, which directly trace the information flow inside a model that maps a specific input to the output. Unlike surrogate-based approaches (e.g., sparse autoencoders), query circuits are identified within the model itself, resulting in more faithful and computationally accessible explanations. To make query circuits practical, we address two challenges. First, we introduce Normalized Deviation Faithfulness (NDF), a robust metric to evaluate how well a discovered circuit recovers the model's decision for a specific input, and is broadly applicable to circuit discovery beyond our setting. Second, we develop sampling-based methods to efficiently identify circuits that are sparse yet faithfully describe the model’s behavior. Across benchmarks (IOI, arithmetic, MMLU, and ARC), we find that there exist sparse query circuits within the model that recover much of its performance on single queries. For example, on average, a circuit covering only 1.3\% of model connections can recover about 60\% of performance on an MMLU question. Overall, query circuits provide a step towards faithful, scalable explanations of how language models process individual inputs.
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Towards Context-Invariant Safety Alignment for Large Language Models
Yixu Wang, Yang Yao, Xin Wang, Yifeng Gao, Yan Teng, Xingjun Ma, Yingchun Wang
Preference-based post-training aligns LLMs with human intent, yet safety behavior often remains brittle.
A model may refuse a harmful request in a standard prompt but comply when the same intent is wrapped in adversarial wording.
We suggest that robust safety requires context-invariant alignment, where behavior depends on the underlying intent rather than surface form.
Enforcing invariance is difficult in alignment because not all training signals are equally trustworthy; for some prompt variants we can obtain verifiable feedback (e.g., multiple-choice), while for open-ended variants we typically rely on noisy, gameable reward proxies (e.g., learned judges).
As a result, standard symmetric invariance regularizers can reduce cross-context discrepancies by lowering performance on reliable variants instead of improving open-ended robustness.
To address this, we introduce Anchor Invariance Regularization (AIR), which treats verifiable prompts as anchors and uses a stop-gradient target to regularize only the open-ended variants toward the anchor performance.
AIR is implemented as a plug-in auxiliary loss and combined with group-based preference optimization (e.g., GRPO) via heterogeneous prompt grouping. Across Safety, Moral Reasoning, and Math, AIR improves context invariance, boosting in-distribution group accuracy by 12.71% and out-of-distribution consistency by 33.49%, making safety constraints robust to adversarial framings.

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What Really Improves Mathematical Reasoning: Structured Reasoning Signals Beyond Pure Code
Yuze Zhao, Junpeng Fang, Lu Yu, Zhenya Huang, Kai Zhang, Qing Cui, Qi Liu, JUN ZHOU, Enhong Chen
Incorporating code into training corpora has become a widely acknowledged practice in the development of modern foundation language models (LMs). Compared with a general Internet corpus, code offers high-quality, well-structured signals that substantially augment the coding proficiency of models. Beyond programming skills, prior research has suggested that code data may also contribute to non-coding capabilities. Nevertheless, through a series of rigorous controlled experiments, we demonstrate that the influence of code on other domains, particularly reasoning, remains limited.
Our principal findings are as follows: (1) Code corpus yields substantial gains in programming-related abilities but only competed with knowledge-intensive tasks. (2) We identify a core subset that functions as cognitive scaffolding for mathematical reasoning, especially for complex problem-solving scenarios. (3) Formal reasoning provides more pronounced improvements in challenging mathematical reasoning tasks, while natural language–based reasoning proves more effective for simpler reasoning problems.
Finally, by probing the internal mechanisms of LMs, we reveal how training data modulates routing patterns, thereby shaping emergent model behavior. As a central driver of model capability, our findings disentangle domain-specific data into finer-grained, cross-domain ability dimensions and underscore promising directions for future data optimization.
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Integrated Episodic and Semantic Memory via Modulating Transformer FeedForward Layers
Yiqun Yao, Xiang Li, Xin Jiang, Xuezhi Fang, Naitong Yu, Siwei Dong, Wenjia Ma, Jing Li, Aixin Sun, Yequan Wang
It is widely recognized that, after generative pre-training, Transformer FeedForward layers implicitly function as semantic memory, encoding linguistic and factual knowledge, while the contexts in key–value (KV) cache contain raw events, serving as the source of models' episodic memory. In this work, we show that a same group of Transformer FeedForward-layer parameters can both be semantic and episodic memory, which is retrievable without explicitly attending to the related KV cache. To realize this idea, we introduce Hypermem, a hypernetwork that recurrently maps contexts into targeted updates of FeedForward parameters. We post-train the hypernetwork using continuation and random-access associative memory objectives, eliminating the need for test-time training. Extensive experiments demonstrate that our approach outperforms related methods, including MemoryLLM and generative adapter, on memory retrieval, long-context question answering, and personalization benchmarks, establishing a new state of the art for hypernetwork-based memory mechanisms. Our results suggest that directly bridging data and parameters provides a viable direction for exploring next-generation foundation models with more flexible and persistent memory capabilities.
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SpikeNet: Sparse Spike-Driven Mask Vector Transformer for Energy-Efficient and Stable Spiking Point Cloud Processing
Zhiming Zhou, Yong He, Chaoxu Mu, Qiaoyun Wu, Ajmal Saeed Mian
The unordered nature of point cloud data poses significant challenges to conventional architectures primarily designed for structured data.
Spiking neural networks (SNN), by virtue of their inherent sparsity and dynamics, are particularly well-suited for processing point clouds to effectively extract meaningful features. We propose SpikeNet, a novel spiking neural network architecture for energy-efficient and robust point cloud analysis. We introduce spiking-driven sparse attention mechanism coined the Spiking Vector Mask Transformer (SVMT).
By dynamically aligning the sparsity of point cloud data through binary spiking masks, SVMT eliminates the need for softmax and multiplication operations, signi

ficantly improving computational efficiency. We also propose a Dynamic Sparse Sp iking Residual (DSSR) structure and integrate it with SVMT to form the Spiking N eural Network (SpikeNet) for point cloud classification and segmentation. SpikeN et overcomes the trade-off between accuracy and efficiency in previous SNN metho ds, achieving collaborative optimization of performance and energy-efficiency. E xperiments on benchmark datasets show that SpikeNet achieves state-of-the-art pe rformance in shape classification and segmentation tasks, comparable to artifici al neural network (ANN) based methods. Our source code is in supplementary mater ial and will be made publicly available.
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Dissecting Quantization Error: A Concentration-Alignment Perspective
Marco Federici, Boris van Breugel, Paul N. Whatmough, Markus Nagel
Quantization can drastically increase the efficiency of large language and visio n models, but typically incurs an accuracy drop. Recently, function-preserving t ransforms (e.g. rotations, Hadamard transform, channel-wise scaling) have been s uccessfully applied to reduce post-training quantization error, yet a principled explanation remains elusive.

We analyze linear-layer quantization via the signal-to-quantization-noise rati o (SQNR), showing that for uniform integer quantization at a fixed bit width, SQ NR decomposes into (i) the concentration of weights and activations (capturing s pread and outliers), and (ii) the alignment of their dominant variation directio ns. This provides an actionable insight: enhancing alignment between weight and activation variation directions can reduce quantization error, complementing con centration-focused approaches. Motivated by this, we introduce Concentration–Ali gnment Transforms (CAT), a lightweight linear transformation that uses a covaria nce estimate from a small calibration set to jointly improve concentration and a lignment, approximately maximizing SQNR. Experiments across several LLMs show th at CAT consistently matches or outperforms prior transform-based quantization me thods at 4-bit precision.
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Discriminative Visual Process Rewards for Scaling Thinking at Test-Time with Ima ges
Bo-Wen Yin, Qize Yang, Boyuan Sun, Xihan Wei, Qibin Hou
The “thinking with images” paradigm has led multimodal large language models to generate intermediate visual steps—such as cropping, annotation, spatial localiz ation, and sketches—to enhance high-resolution perception and complex reasoning. However, existing multimodal Process Reward Models (PRMs) evaluate only textual reasoning and cannot judge the correctness of these visual steps, creating a ke y gap when visual reasoning is essential for solving tasks. We propose Discrimin ative Visual Process Reward Model (DiscPRM), a multimodal PRM that jointly evalu ates textual and visual intermediate steps by modeling visual reasoning trajecto ries, image operations, and text-image consistency. To support this, we build VT Reward-100K, a dataset of step-by-step visual reasoning sequences with supervisi on. Experiments show that using DiscPRM for Best-of-N process supervision substa ntially improves multimodal reasoning performance on tasks requiring visual inte rmediate steps, achieving over 5% gains across benchmarks. We further introduce VABench, the first benchmark for evaluating PRMs on visual reasoning error detec tion. We hope this work can provide foundational support for advancing the emerg ing direction of visual–textual process reward.
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RedDebate: Safer Responses Through Multi-Agent Red Teaming Debates
Ali Asad, Stephen Obadinma, Radin Shayanfar, Xiaodan Zhu
We introduce RedDebate, a novel multi-agent debate framework that provides the f oundation for Large Language Models (LLMs) to identify and mitigate their own un safe behaviors. Existing AI safety approaches often rely on costly human evaluat ion or isolated single-model assessment, both constrained by scalability and pro ne to oversight failures. RedDebate employs collaborative argumentation among mu ltiple LLMs across diverse debate scenarios, enabling them to critically evaluat e one another’s reasoning and systematically uncover unsafe failure modes throug h fully automated red-teaming. We further integrate distinct long-term memory mo

dules that preserve safety-relevant insights from debate interactions and levera ge them during subsequent inference, facilitating continuous refinement of model behavior. Empirical evaluation on safety benchmarks across a diverse set of mod els demonstrates that RedDebate substantially reduces unsafe outputs. While deba te alone allows LLMs to refine their behavior, the addition of memory modules yi elds further significant reductions. To the best of our knowledge, RedDebate is the first fully automated framework to unify multi-agent debate and red-teaming to progressively enhance LLM safety without human intervention.
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It's TIME: Towards the Next Generation of Time Series Forecasting Benchmarks
Zhongzheng Qiao, Sheng Pan, Anni Wang, Viktoriya Zhukova, Yong Liu, Xudong Jiang , Qingsong Wen, Mingsheng Long, Ming Jin, Chenghao Liu
Time series foundation models (TSFMs) are revolutionizing the forecasting landsc ape from specific dataset modeling to generalizable task evaluation. However, we contend that existing benchmarks exhibit common limitations in four dimensions: constrained data composition dominated by reused legacy sources, compromised da ta integrity lacking rigorous quality assurance, misaligned task formulations de tached from real-world contexts, and rigid analysis perspectives that obscure ge neralizable insights. To bridge these gaps, we introduce **TIME**, a next-genera tion task-centric benchmark comprising 50 fresh datasets and 98 forecasting task s, tailored for strict zero-shot TSFM evaluation free from data leakage. Integra ting large language models and human expertise, we establish a human-in-the-loop benchmark construction pipeline to ensure high data integrity and redefine task formulation by aligning forecasting configurations with real-world operational requirements and variate predictability. Furthermore, we propose a novel pattern -level evaluation perspective that moves beyond traditional dataset-level evalua tions based on static meta labels. By leveraging structural time series features to characterize intrinsic temporal properties, this approach offers generalizab le insights into model capabilities across diverse patterns. We evaluate 12 TSFM s and establish a multi-granular leaderboard to facilitate in-depth analysis and visualized inspection. The leaderboard is available at https://huggingface.co/s paces/Real-TSF/TIME-leaderboard.
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Are Common Substructures Transferable? Riemannian Graph Foundation Model with Ne ural Vector Bundles
Li Sun, Zhenhao Huang, Yiding Wang, Qin Chen, Pietro Lio, Philip S. Yu
Foundation models have sparked a revolution via a pretraining-adaptation paradig m, with recent efforts extending this success to graphs. Unlike other modalitie s, graphs contain rich structural patterns, yet their structural transferability remains poorly understood. Prior studies consider common substructures in the d iscrete realm, and we are motivated by a fundamental question: Are common substr uctures transferable? The underlying theory is largely underexplored. In this wo rk, we shift toward learning transferable structures through the lens of functio nal behavior. Theoretically, we connect transferable substructures to intrinsic geometry of the representation space. However, characterizing such intrinsic geo metry has rarely been touched. Grounded in Riemannian geometry, we develop a gra ph intrinsic geometry learning framework—\textbf{Neural Vector Bundle}, which en ables parsing intrinsic geometry with local coordinates. Building on this, we de sign \textbf{\textsc{Gauge}}, a pretrainable neural architecture that constructs the vector bundle, flattening geometrically compatible local coordinates, and a new Dirichlet loss, which also measures the transfer effort. We empirically val idate its superior expressiveness in challenging tasks including zero-shot link prediction and graph isomorphism.
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Learn from A Rationalist: Distilling Intermediate Interpretable Rationales
Jiayi Dai, Randy Goebel
Because of the pervasive use of deep neural networks (DNNs), especially in high- stakes domains, the interpretability of DNNs has received increased attention. T he general idea of rationale extraction (RE) is to provide an interpretable-by-d esign framework for DNNs via a select-predict architecture where two neural netw

orks learn jointly to perform feature selection and prediction, respectively. Gi
ven only the remote supervision from the final task prediction, the process of l
earning to select subsets of features (or *rationales*) requires searching in th
e space of all possible feature combinations, which is computationally challengi
ng and even harder when the base neural networks are not sufficiently capable. T
o improve the predictive performance of RE models that are based on less capable
 or smaller neural networks (i.e., the students), we propose **REKD** (**R**atio
nale **E**xtraction with **K**nowledge **D**istillation) where a student RE mode
l learns from the rationales and predictions of a teacher (i.e., a *rationalist*
) in addition to the student's own RE optimization. This structural adjustment t
o RE aligns well with how humans could learn effectively from interpretable and
verifiable knowledge. Because of the neural-model agnostic nature of the method,
 any black-box neural network could be integrated as a backbone model. To demons
trate the viability of REKD, we conduct experiments with multiple variants of BE
RT and vision transformer (ViT) models. Our experiments across language and visi
on classification datasets (i.e., IMDB movie reviews, CIFAR 10 and CIFAR 100) sh
ow that REKD significantly improves the predictive performance of the student RE
 models. The code is publicly available: https://github.com/JiayiDai/REKD.
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A Diagnostic Study of Multi-Agent LLMs for Real-World Debates

Priya Pitre, Gaurav Srivastava, Lu Zhang, Le Wang, Naren Ramakrishnan, Xuan Wang

Multi-agent LLM debates are increasingly used in domains such as policy, politic
s, and city planning, where ground truth is often unavailable. Yet existing eval
uations rely heavily on outcome-based proxies such as consensus, majority vote,
or LLM-as-judge scores, which can miss failures like sycophancy, domination, and
 premature convergence. We introduce a diagnostic framework that evaluates both
debate outcomes and the deliberative process using interpretable metrics for eng
agement, responsiveness, influence asymmetry, balance, stability, and agent util
ity. Across real-world debate settings and validation benchmarks, our process-le
vel diagnostics align more closely with human judgments and reveal interaction f
ailures that standard outcome-only measures overlook. These results show that re
liable evaluation of multi-agent debates requires measuring not only what answer
 agents reach, but how they reach it.
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One Step Forward and K Steps Back: Better Reasoning with Denoising Recursion Mod
els

Chris Cameron, Wangzheng Wang, Nikita Ivanov, Ashmita Bhattacharyya, Didier Chét
elat, Yingxue Zhang

Looped transformers scale computational depth without increasing parameter count
 by repeatedly applying a shared transformer block and can be used for iterative
 refinement, where each loop rewrites a full fixed-size prediction in parallel.
On difficult problems, such as those that require search-like computation, reach
ing a highly structured solution starting from noise can require long refinement
 trajectories. Learning such trajectories is challenging when training specifies
 only the target solution and provides no supervision over the intermediate refi
nement path. Diffusion models tackle this issue by corrupting data with varying
magnitudes of noise and training the model to reverse it in a *single step*. How
ever, this process misaligns training and testing behaviour. We introduce Denois
ing Recursion Models, a method that similarly corrupts data with noise but train
s the model to reverse the corruption over *multiple* recursive steps. This stra
tegy provides a tractable curriculum of intermediate states, while better aligni
ng training with testing and incentivizing non-greedy, forward-looking generatio
n. Through extensive experiments, we show this approach outperforms the Tiny Rec
ursion Model (TRM) on ARC-AGI, where it recently achieved breakthrough performan
ce.
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From Imagined Futures to Executable Actions: Mixture of Latent Actions for Robot
 Manipulation

Yajie Li, Bozhou Zhang, Chun Gu, Zipei Ma, Jiahui Zhang, Jiankang Deng, Xiatian
Zhu, Li Zhang

Video generation models offer a promising imagination mechanism for robot manipulation by predicting long-horizon future observations, but effectively exploiting these imagined futures for action execution remains challenging. Existing approaches either condition policies on predicted frames or directly decode generated videos into actions, both suffering from a mismatch between visual realism and control relevance. As a result, predicted observations emphasize perceptual fidelity rather than action-centric causes of state transitions, leading to indirect and unstable control.

To address this gap, we propose MoLA (Mixture of Latent Actions), a control-oriented interface that transforms imagined future videos into executable representations. Instead of passing predicted frames directly to the policy, MoLA leverages multiple pretrained inverse dynamics models to infer a mixture of latent actions implied by generated visual transitions. These modality-aware inverse dynamics models capture complementary semantic, depth, and flow cues, providing a structured and physically grounded action representation that bridges video imagination and policy execution.

We evaluate our approach on simulated benchmarks (LIBERO, CALVIN, and LIBERO-Plus) and real-world robot manipulation tasks, achieving consistent gains in task success, temporal consistency, and generalization.

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FutureOmni: Evaluating Future Forecasting from Omni-Modal Context for Multimodal LLMs

Qian Chen, Jinlan Fu, Changsong Li, Min Zhang, See-Kiong Ng, Xipeng Qiu

Although Multimodal Large Language Models (MLLMs) demonstrate strong omni-modal perception, their ability to forecast future events from audio-visual cues remains largely unexplored, as existing benchmarks focus mainly on retrospective understanding. To bridge this gap, we introduce FutureOmni, the first benchmark designed to evaluate omni-modal future forecasting from audio-visual environments. The evaluated models are required to perform cross-modal causal and temporal reasoning, as well as effectively leverage internal knowledge to predict future events. FutureOmni is constructed via a scalable LLM-assisted, human-in-the-loop pipeline and contains 919 videos and 1,034 multiple-choice QA pairs across 8 primary domains. Evaluations on 13 omni-modal and 7 video-only models show that current systems struggle with audio-visual future prediction, particularly in speech-heavy scenarios, with the best accuracy of 64.8% achieved by Gemini 3 Flash. To mitigate this limitation, we curate a 7K-sample instruction-tuning dataset and propose an Omni-Modal Future Forecasting (OFF) training strategy. Evaluations on FutureOmni along with standard audio-visual and video-only benchmarks show that OFF improves future forecasting performance and generalization. Code and data are available at https://github.com/OpenMOSS/FutureOmni.

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Inference-Time Conformal Reasoning with Valid Factuality Control for Large Language Models

Ting Wang, Yuanjie Shi, Yan Yan, Huan Zhang

Large language models (LLMs) increasingly perform multi-step reasoning, where intermediate claims form implicit directed acyclic graphs whose node correctness is structurally conditioned on their ancestors. This makes factuality uncertainty structural, rather than a trivial accumulation of node-wise errors, and necessitates inference-time uncertainty quantification over the reasoning structure. While conformal prediction (CP) offers flexible user-specified factuality control, existing work remains post-hoc and cannot intervene during generation. To fill the gap between CP's flexibility and its post-hoc limitation, we propose an *Inference-Time Conformal Reasoning (ITCR)* framework that integrates CP directly into reasoning graph generation. ITCR learns a structure-level factuality uncertainty function that aggregates claim-level factuality signals over reasoning graphs without complex modeling assumptions. We then design the non-conformity score based on graph-level factuality uncertainty and calibrate the conformal threshold to decide when to stop generation. We theoretically show such generation is nested, yielding valid coverage guarantees for factuality control. Experiments over multiple datasets and coverage objectives demonstrate empirically valid covera

ge. In downstream reasoning tasks, inference-time calibrated graphs yield more accurate generation than post-hoc pruned graphs.
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Simultaneous Speech-to-Speech Translation Without Aligned Data
Tom Labiausse, Romain Fabre, Yannick Estève, Alexandre Défossez, Neil Zeghidour
Simultaneous speech translation requires translating source speech into a target language in real-time while handling non-monotonic word dependencies. Traditional approaches rely on supervised training with word-level aligned data, which is difficult to collect at scale and thus depends on synthetic alignments using language-specific heuristics that are suboptimal. We propose Hibiki-Zero, which eliminates the need for word-level alignments entirely. This fundamentally simplifies the training pipeline and enables seamless scaling to diverse languages with varying grammatical structures, removing the bottleneck of designing language-specific alignment heuristics. We first train on sentence-level aligned data to learn speech translation at high latency, then apply a novel reinforcement learning strategy using GRPO to optimize latency while preserving translation quality. Hibiki-Zero achieves state-of-the-art performance in translation accuracy, latency, voice transfer, and naturalness across four X-to-English tasks. Moreover, we demonstrate that our model can be adapted to support a new input language with less than 1000h of speech. We provide [examples](https://huggingface.co/spaces/kyutai/hibiki-zero-samples), [model weights](https://huggingface.co/kyutai/hibiki-zero-3b-pytorch-bf16), [inference code](https://github.com/kyutai-labs/hibiki-zero) and we release a [benchmark](https://huggingface.co/datasets/kyutai/Audio-NTREX-4L) containing 45h of multilingual data for speech translation evaluation.
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Evaluating Parameter Efficient Methods for RLVR
Qingyu Yin, Yulun Wu, Zhennan Shen, Sunbowen Lee, Zhilin Wang, Yanshu Li, Chak Tou Leong, Jiale Kang, Jinjin Gu
We systematically evaluate Parameter-Efficient Fine-Tuning (PEFT) methods under the paradigm
of Reinforcement Learning with Verifiable Rewards (RLVR). RLVR incentivizes language models to
enhance their reasoning capabilities through verifiable feedback; however, while methods like LoRA are
commonly used, the optimal PEFT architecture for RLVR remains unidentified. In this work, we conduct
the first comprehensive evaluation of over 12 PEFT methodologies across the DeepSeek-R1-Distill
families on mathematical reasoning benchmarks. Our empirical results challenge the default adoption
of standard LoRA with three main findings. First, we demonstrate that structural variants, such as
DoRA, AdaLoRA, and MiSS, consistently outperform LoRA. Second, we uncover a spectral collapse
phenomenon in SVD-informed initialization strategies (e.g., PiSSA, MiLoRA), attributing their failure to
a fundamental misalignment between principal-component updates and RL optimization. Furthermore,
our ablations reveal that extreme parameter reduction (e.g., VeRA, Rank-1) severely bottlenecks
reasoning capacity. We further conduct ablation studies and scaling experiments to validate our findings.
This work provides a definitive guide for advocating for more exploration for parameter-efficient RL
methods.
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Autoregressive Boltzmann Generators
Danyal Rehman, Charlie B. Tan, Yoshua Bengio, Joey Bose, Alexander Tong
Efficient sampling of molecular systems at thermodynamic equilibrium is a hallma

rk challenge in statistical physics. This challenge has driven the development o
f Boltzmann Generators (BGs), which allow rapid generation of uncorrelated equil
ibrium samples by combining a generative model with exact likelihoods and an imp
ortance sampling correction. However, modern BGs predominantly rely on Normalizi
ng Flows (NFs), which either suffer from limited expressivity due to strict inve
rtibility constraints (discrete time) or computationally expensive likelihoods (
continuous time). In this paper, we propose Autoregressive Boltzmann Generators
(ArBG), a novel autoregressive modelling framework that overcomes these limitati
ons by departing from the flow-based BG paradigm. ArBG circumvents the topologic
al constraints of flows and enables sequential inference-time interventions, whi
le offering enhanced scalability by leveraging architectures effective in Large
Language Models. We empirically demonstrate that ArBG leads to significant impro
vements over flow-based models across all benchmarks, but particularly in larger
 peptide systems such as the 10-residue Chignolin. Furthermore, we introduce Rob
in, a 132M parameter transferable model trained with the ArBG framework which im
proves over the previous state-of-the-art, reducing the zero-shot energy error,
$\mathcal{E}$-$\mathcal{W}_2$, on 8-residue systems by ~ 60\%.
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Theoretical Analysis of Sparse Optimization with Reparameterization, Weight Deca
y, and Adaptive Learning Rate
Huangyu Xu, Jingqin Yang, Qianqian Xu, Jiaye Teng
Sparse optimization is a fundamental challenge in various practical applications
.
A popular approach to sparse optimization is Lp regularization.
However, it may encounter optimization instability due to the unbounded gradient
s when 0<p<1.
In this paper, we introduce a novel approach to sparse optimization termed ReWA,
 based on Reparameterization, Weight decay, and Adaptive learning rate. ReWA is
closely connected to lp-regularization, yet it unveils a distinct optimization l
andscape that helps mitigate instability issues.
Experiments on CIFAR-10 and ImageNet with ResNets demonstrate that ReWA leads to
 significant sparsity improvements over the L1-regularization approach while pre
serving test accuracy.
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Algorithmic Recourse of In-Context Learning for Tabular Data
Wenshuo Dong, Jiaming Zhang, Shaopeng Fu, Hongbin Lin, Di Wang, Lijie Hu
As predictive models are increasingly deployed in high-stakes settings such as c
redit approval, there is a growing need for post-hoc methods that provide recour
se to affected individuals. Many such models operate on tabular data, where feat
ures correspond to real-world attributes. Recently, in-context learning (ICL) ha
s enabled large language models to perform tabular prediction by conditioning on
 labeled examples at inference time, without explicit training. However, algorit
hmic recourse for tabular decision-making under ICL remains largely unexplored.
In this work, we present the first study of algorithmic recourse for tabular dat
a under ICL. We carry out a theoretical analysis, showing that recourse remains
well-defined and bounded, and we characterize how recourse converges toward clas
sical solutions as the context size increases. In practice, we propose a novel z
eroth-order recourse framework, Adaptive Subspace Recourse for In-Context Learni
ng (ASR-ICL), that efficiently generates actionable and sparse recourse for blac
k-box ICL models. The proposed framework naturally extends to multi-class tabula
r tasks. Experiments across multiple real-world datasets and models demonstrate
that ASR-ICL achieves recourse quality comparable to existing methods with fewer
 queries and empirically confirm the predicted convergence behavior, supporting
our theoretical analysis.
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On the Identifiability of Poisson Branching Structural Causal Model Under Latent
 Confounding
Jie Qiao, Zihuai Zeng, Ruichu Cai, Zhengming Chen, Zhifeng Hao
Causal discovery from observational count data poses unique challenges, particul
arly when the data exhibit inherent branching structures, such as an upstream ad

impression event triggering a downstream purchase event with certain probabilit
y. Such branching dynamics are naturally modeled by thinning operators (for bran
ching) and an independent Poisson distribution (for exogenous noise), constituti
ng a Poisson Branching Structural Causal Model (PB-SCM). However, existing appro
aches based on PB-SCM rely on the restrictive assumption of causal sufficiency,
failing to account for ubiquitous latent confounders. In this work, we propose a
Latent Confounding Poisson Branching Structural Causal Model (LC-PB-SCM) to bri
dge this gap. We leverage Probability Generating Function (PGF) to characterize
the complex dependencies introduced by latent confounding. Then, we establish a
Trie representation theorem that maps the branching structure to algebraic prope
rties of PGF monomials. Based on local PGF, we establish a complete identifiabil
ity condition for local 3-variables covering all causal patterns distinguishable
up to monomial equivalence. Finally, we propose a practical algorithm to learn
causal structures under latent confounding and demonstrate its effectiveness thr
ough experiments on both synthetic and real-world datasets.
****
FedUSD: Unbiased Synthetic Data for Federated Learning
Weiying Xie, Chenhe Hao, Haozhi Shi, Jitao Ma, Daixun Li, Jiazhe Li, Hengyi Wang
, Leyuan Fang, Yunsong Li
Aggregation-Free Federated Learning enables joint training by sharing synthetic
data, aiming to eliminate data heterogeneity across clients. However, existing m
ethods fail to explicitly separate the principal and residual components of data
set, leading to biased synthetic data. In this paper, we propose a novel Unbiase
d Synthetic Data optimization method FedUSD for Aggregation-Free Federated Learn
ing, which is achieved by exploring the High-energy Orthogonal Base (HOB) and va
riance of dataset in feature space. Our FedUSD is inspired by the discovery that
principal component concentrates in HOB while residual component independently
reflects in variance, regardless of networks. Based on the observation, we devel
op a method that mathematically optimizes synthetic data by matching both HOB an
d variance with those of real data. Besides, we experimentally show the superior
effectiveness of leveraging HOB and variance to separately extract the principa
l and residual components over existing methods. We also theoretically prove tha
t FedUSD achieves unbiased synthetic data and thus convergence. Without introduc
ing any constraints, FedUSD thereby yields significant improvements over the sta
te-of-the-arts in terms of global model performance, under equivalent communicat
ional costs. For example, on the SVHN dataset, FedUSD improves 6.74\% to 30.82\%
which is higher than others with Dirichlet coefficient $\alpha=0.01$.
****
Decentralized Bandits without Global Clock for Dynamic Matching Market
Mengtong Gao, Zhenhe Zhang, Jichen Li, Wentao Zhou, Xuanzhi Xia, Jing Chen
Two-sided matching markets are pervasive in numerous real-world applications, ra
nging from labor markets to online advertising. A rich line of research has stud
ied the matching bandit problem, where participants learn their preferences thro
ugh iterative interactions. However, existing works assume a static environment
with fixed participants and require synchronized learning, in which all particip
ants start simultaneously and have access to a global clock. In reality, matchin
g markets are inherently dynamic: participants may enter and leave at arbitrary
time steps without any global signal, creating coordination challenges.

To study the dynamic setting, we first investigate one-sided learning under unco
ordinated player arrivals, where only the players need to learn their preference
s. We propose the Way-SE algorithm, which achieves a regret of $O(\frac{K^2 \log
T}{\Delta_{\min}^2})$, where K is the number of arms, T is the time horizon
, and Δ_{min} is the minimum utility gap. This is done through a distribu
ted exploration mechanism that coordinates exploration implicitly via just local
clocks. More importantly, we extend our work to fully decentralized dynamic two
-sided learning, where both sides need to learn their preferences, and players a
rrive or depart arbitrarily. We introduce Way-SE-2S, the first algorithm to achi
eve sublinear regret $O\left(\frac{K T^{1-1/K}(\log T)^{2/K}}{\Delta_{\min}^2}\r
ight)$ in this challenging environment, without requiring global signals, restri

ctive preference structures, or observability of the results of competing agents. Our work provides the first theoretical guarantee for stable matching in fully decentralized and uncoordinated bandit markets.
****
Untied Ulysses: Memory-Efficient Context Parallelism via Headwise Chunking
Ravi Ghadia, Maksim Abraham, Sergei Vorobyov, Max Ryabinin
Efficiently processing long sequences with Transformer models usually requires splitting the computations across accelerators via context parallelism.
The dominant approaches in this family of methods, such as Ring Attention or DeepSpeed Ulysses, enable scaling over the context dimension but do not focus on memory efficiency, which limits the sequence lengths they can support.
More advanced techniques, such as Fully Pipelined Distributed Transformer or activation offloading, can further extend the possible context length at the cost of training throughput.
In this paper, we present UPipe, a simple yet effective context parallelism technique that performs fine-grained chunking at the attention head level.
This technique significantly reduces the activation memory usage of self-attention, breaking the activation memory barrier and unlocking much longer context lengths.
Our approach lowers the peak activation memory usage by as much as **82.5%** for 70B Transformers, while matching previous context parallelism techniques in terms of training speed.
UPipe can support maximum context lengths of up to 5M tokens for training 8B models on a single 8xH100 node, improving upon prior methods by **25%**.
****
Demystifying the Optimal Fair Classifier in Multi-Class Classification
Li Zhang, Yuyuan Li, XiaoHua Feng, Jiaming Zhang, Fengyuan Yu, Chaochao Chen
Ensuring fair and equitable treatment across diverse groups, particularly in multi-class classification tasks, poses a significant challenge due to the persistent biases inherent in machine learning models.
Most existing bias mitigation techniques are tailored to binary settings, and the presence of multi-dimensional outputs and complex fairness mechanisms makes their extension to multi-class scenarios neither straightforward nor effective.
In this paper, we investigate two fundamental, unresolved challenges in fair classification: (i) *characterizing the optimal accuracy-fairness frontier in multi-class settings*, and (ii) *designing practical algorithms that attain this optimum across different training phases*.
To tackle these challenges, we first specify an analytically tractable probabilistic formulation of the optimal classifier under fairness constraints.
Building upon this, we propose two attribute-blind algorithms to enforce fairness requirements in practice: an in-processing approach for fairness intervention during training via the reduction approach, and a post-processing approach for fine-tuning output probabilities with plug-in estimation.
Theoretical analysis reveals that both methods converge to the optimal accuracy-fairness Pareto frontier.
Experiments conducted on multiple datasets demonstrate the superior performance of our methods in balancing accuracy and fairness.
****
LIVEditor-14B: Lightning Unified Video Editing via In-Context Sparse Attention
Shitong Shao, zikai zhou, Haopeng Li, Yingwei Song, Wenliang Zhong, Lichen Bai, Zeke Xie
Video editing has evolved toward In-Context Learning (ICL) paradigms, yet the resulting quadratic attention costs create a critical computational bottleneck. In this work, we propose **I**n-context **S**parse **A**ttention (**ISA**), the first experimentally lossless sparse framework tailored for ICL video editing. Our design is grounded in two key insights: __**first**__, context tokens exhibit significantly lower saliency than source tokens; __**second**__, we theoretically prove and empirically validate that Query sharpness correlates with approximation error. Motivated by these findings, ISA implements an efficient pre-selection strategy to prune redundant context, followed by a dynamic query grouping mecha

nism that routes high-error queries to full attention and low-error ones to a co mputationally efficient 0-th order Taylor sparse attention. Furthermore, we cons truct a scalable pipeline to curate a 1M-sample dataset and train __**LIVEditor-14B**__, a novel lightning video editing model via ISA. Extensive experiments de monstrate that LIVEditor-14B achieves a ~60% reduction in latency while surpassi ng state-of-the-art methods across EditVerseBench, IVE-Bench, and VIE-Bench, del ivering experimentally lossless acceleration without compromising visual fidelit y.

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OCNR: Stabilizing Self-Play by Mitigating Iteration-Collapse With One-Class Nove lty Rewards

Seungyoo Lee, Giung Nam, Hyungi Lee, Juho Lee

Training large language models via self-play often suffers from a persistent ite ration-collapse, where performance initially improves but subsequently regresses as training iterations increase. We analyze this phenomenon as arising from cro ss-iteration degeneration, where the task-generation distribution becomes increa singly confined to a narrow subset of familiar (seen) problems, weakening the ef fective learning signal and destabilizing training. To address this issue, we pr opose a plug-in approach that augments existing self-play pipelines with a one-c lass novelty reward. A Seen Detector trained on a historical buffer of previousl y used training problems identifies in-support instances and discourages redunda nt generation by the questioner, thereby steering exploration toward under-explo red yet learnable regions. Experimental results show that the proposed method mi tigates iteration-collapse during iterative training and yields consistent impro vements.

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Attention with Routed-Memory for Learnable Sparse Control

QIUHAO Zeng, Jerry Huang, Peng Lu, Ruiyi Fang, Gezheng Xu, Zihao Jing, Yufei Cui , Charles Ling, Gang Niu, Boyu Wang

Despite advances in long-context inference, large language models (LLMs) remain fundamentally limited by the key-value (KV) caching mechanisms that are necessar y for stable computation. Techniques such as selective token eviction and prunin g have vastly mitigated these issues, but often discard core information to mana ge the growing cache.

In this paper, we propose _**A**ttention with **R**outed **M**emory_ (_**ARM**_) a novel KV caching structure that introduces a fully differentiable, fixed-size memory system organized as a hierarchical router. Via a Gumbel-Softmax, _**ARM** *_ learns to select memory slots and perform sigmoid-gated updates that softly c ombine new and stored information, avoiding hard eviction and reducing informati on loss. By further training a policy to dynamically select varying amounts of m emory at inference, _**ARM**_ adapts its accesses for both simple contexts and i nputs that require deeper reasoning, enabling more scalable and effective retrie val on both short- and long-contexts. Experimental results on standard commonsen se and long-context reasoning benchmarks demonstrate that _**ARM**_ achieves sup erior performance and efficiency compared to fixed KV-caching approaches, while remaining efficient and scalable in terms of both memory and generation latency.

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VALUEFLOW: Toward Pluralistic and Steerable Value-based Alignment in Large Langu age Models

Woojin Kim, Sieun Hyeon, Jusang Oh, Jaeyoung Do

Aligning Large Language Models (LLMs) with the diverse spectrum of human values remains a central challenge: preference-based methods often fail to capture deep er motivational principles. Value-based approaches offer a more principled path, yet three gaps persist: extraction often ignores hierarchical structure, evalua tion detects presence but not calibrated intensity, and steerability of LLMs at controlled intensities remains insufficiently understood. To address these limit ations, we introduce VALUEFLOW, a unified framework that spans extraction, evalu ation, and steering with calibrated intensity control. The framework integrates three components: (i) HiVES, a hierarchical value embedding space that captures intra- and cross-theory value structure; (ii) the Value Intensity DataBase (VIDB

), a large-scale resource of value-labeled texts with intensity estimates derive d from ranking-based aggregation; and (iii) an anchor-based evaluator that produ ces consistent intensity scores for model outputs by ranking them against VIDB p anels. Using VALUEFLOW, we conduct a comprehensive large-scale study across ten models and four value theories, identifying asymmetries in steerability and comp osition laws for multi-value control. This paper establishes a scalable infrastr ucture for evaluating and controlling value intensity, advancing pluralistic ali gnment of LLMs.
****

Robust Human-AI Complementarity under Uncertainty
Yewon Byun, Bryan Wilder
Machine learning models are often intended to augment rather than replace human decision-makers, by providing information that is complementary to human judgeme nt.

Yet, in practice, human decision makers routinely fail to realize such complem entary gains, even when models provide useful signal.

In this work, we study how asymmetric information about the quality of informa tion available to a human decision maker vs. an AI impacts the ability of a deci sion maker to extract complementary value from AI predictions.

We show that a key factor is the error correlation structure between human and AI predictions. In particular, when the AI's prediction errors are \textit{nega tively correlated} with those of the human, the decision-maker can construct rob ust strategies which guarantee improvements in expected utility. We empirically investigate whether these conditions for complementarity arise in practice, usin g real-world forecasting benchmarks.
****

PGS: Effective LLM Code Refinement via Property-Oriented and Structurally Minima l Feedback
Lehan He, Zeren Chen, Zhe Zhang, Xiang Gao, Lu Sheng
Large Language Models (LLMs) excel at code generation, yet ensuring the function al correctness of their outputs remains a persistent challenge. Recent studies h ave applied Test-Driven Development (TDD) to refine code, leveraging execution f eedback to guide the model toward correct solutions. However, such feedback is o ften noisy and uninformative, stemming from the scarcity of high-quality test ca ses and the abundance of noisy, auto-generated ones. In this work, we shift the focus from test-case generation to feedback quality.
We introduce the Property-Generated Solver (PGS), a novel feedback-centric frame work designed to generate highly effective feedback via two principles: it must provide semantic guidance beyond simple I/O mismatches through property validati on, and be structurally minimal, to reduce cognitive load and isolate root cause s. PGS operates by checking high-level program properties (e.g., a sorting funct ion must produce a non-decreasing sequence) then providing the simplest failing counterexample to the LLM. This property-driven, minimal feedback steers LLMs to ward correct and generalizable solutions. Across diverse benchmarks, PGS demonst rates superior performance, achieving a bug fix rate 1.4x-1.6x higher than the s trongest debugging-based approaches and establishing a new state-of-the-art in a utomated code refinement.
****

Diagnosing the Reliability of LLM-as-a-Judge via Item Response Theory
Junhyuk Choi, Sohhyung Park, Chanhee Cho, Hyeonchu Park, Bugeun Kim
While LLM-as-a-Judge is widely used in automated evaluation, existing validation practices primarily operate at the level of observed outputs, offering limited insight into whether LLM judges themselves function as stable and reliable measu rement instruments. To address this limitation, we introduce a two-phase diagnos tic framework for assessing reliability of LLM-as-a-Judge, grounded in Item Resp onse Theory (IRT). The framework adopts Graded Response Model (GRM) of IRT and f ormalizes reliability along two complementary dimensions: (1) intrinsic consiste ncy, defined as the stability of measurement behavior under prompt variations, a nd (2) human alignment, capturing correspondence with human quality assessments. We empirically examine seven LLM judges with this framework, and show that leve

raging IRT-GRM yields interpretable signals for diagnosing judgments systematica
lly. These signals provide practical guidance for verifying reliability of LLM-a
s-a-Judge and identifying potential causes of unreliability.
****
Graph is a Substrate Across Data Modalities
Ziming Li, Xiao-Ming Wu, Zehong Wang, Jiazheng Li, Yijun Tian, Jinhe Bi, Yunpu M
a, Yanfang Ye, Chuxu Zhang
Graphs provide a natural representation of relational structure that arises
  across diverse domains. Despite this ubiquity, graph structure is typically
  learned in a modality- and task-isolated manner, where graph representations
  are constructed within individual task contexts and discarded thereafter. As a
   result, structural regularities across modalities and tasks are repeatedly
  reconstructed rather than accumulated at the level of intermediate graph
  representations. This motivates a representation-learning question: *how
  should graph structure be organized so that it can persist and accumulate
  across heterogeneous modalities and tasks?* We adopt a representation-centric
  perspective in which graph structure is treated as a structural substrate that
   persists across learning contexts. To instantiate this perspective, we
  propose **G-Substrate**, a **g**raph **substrate** framework that organizes
  learning around shared graph structures. G-Substrate comprises two
  complementary mechanisms: a unified structural schema that ensures
  compatibility among graph representations across heterogeneous modalities and
  tasks, and an interleaved role-based training strategy that exposes the same
  graph structure to multiple functional roles during learning. Experiments
  across multiple domains, modalities, and tasks show that G-Substrate
  outperforms task-isolated and naive multi-task learning methods. The codebase,
   model, and datasets are available at <https://github.com/zmli6/G-Substrate>.
****
Reward Learning through Ranking Mean Squared Error
Chaitanya Kharyal, Calarina Muslimani, Matthew E. Taylor
Reward design remains a significant bottleneck in applying reinforcement learnin
g (RL) to real-world problems.
A popular alternative is reward learning, where reward functions are inferred fr
om human feedback rather than manually specified.
Recent work has proposed learning reward functions from human ratings rather tha
n traditional binary preferences, enabling richer and potentially less cognitive
ly demanding supervision.
Building on this paradigm, we introduce a new rating-based RL method, Ranked Re
turn Regression for RL (R4).
At its core, R4 uses a novel ranking mean squared error loss that learns from a
dataset of trajectory–rating pairs, treating the human-provided discrete ratings
 (e.g., "bad," "neutral," "good") as ordinal targets.
Unlike prior rating-based approaches, R4 offers formal guarantees: its solution
set is provably minimal and complete under mild assumptions. Empirically, using
both human-provided and simulated ratings, we demonstrate that R4 consistently m
atches or outperforms existing rating and preference-based RL methods on robotic
 benchmarks from OpenAI Gym and the DeepMind Control Suite.
****
Identifiable Token Correspondence for World Models
Youngin Kim, Ray Sun, Inho Kim, Bumsoo Park, Hyun Oh Song
Token-based transformer world models have shown strong performance in visual rei
nforcement learning, but often suffer from temporal inconsistency in long-horizo
n rollouts, including object duplication, disappearance, and transmutation.
A key reason is that most existing approaches treat next-frame prediction purely
 as a token generation problem, without considering the persistence of tokens ac
ross time.
We introduce Identifiable Token Correspondence (ITC), a decoding step for token-
based transformer world models that formulates next-frame prediction as a struct
ured assignment problem with latent token correspondence variables: each next-fr
ame token is explained either by copying a token from the previous frame or by g

enerating a new one. ITC leaves the transformer architecture and training proced
ure unchanged and can be added on top of existing backbones.
Our experiments show state-of-the-art performance on 4 challenging benchmarks.
The proposed method achieves a return of 72.5% and a score of 35.6% on the Craft
ax-classic benchmark, significantly surpassing the previous best of 67.4% and 27
.9%.
We release our source code at https://github.com/snu-mllab/Identifiable-Token-Co
rrespondence.
****

Uncovering the Latent Potential of Deep Intermediate Representations
Arnesh Batra, Arush Gumber, Aniket Khandelwal, Jashn Khemani, Anubha Gupta
Foundational models pretrained on huge amounts of data learn representations tha
t evolve across depth, forming a hierarchy of embeddings with distinct semantic
content and geometric structure. Contrary to the widespread practice of using on
ly the final layer or shallow mixtures, we show that task-relevant information i
s distributed non-monotonically across layers and cannot be recovered by na\"ive
 aggregation. Through a geometric and empirical study across multiple modalities
, we show that effective transfer depends on identifying which layers encode tas
k-discriminative structure and how their embeddings are geometrically organized.
 We introduce Layer-wise Optimal Embedding Selection (LOES), a constructive spec
tral method that identifies task-discriminative subspaces by minimizing residual
 error under orthogonality and isotropy constraints. To align fine-tuning with t
his selection principle, we further propose Geometric Regularization (GeoReg), w
hich enforces a simplicial structure on class manifolds and stabilizes represent
ation geometry during fine-tuning. Across a wide range of architectures, depths,
 modalities, and data regimes, LOES consistently outperforms standard baselines,
 with gains that grow as model depth increases. Beyond accuracy, our method reve
als how semantic factors are distributed across layers, thereby enabling cross-l
ingual and cross-modal interpretability analyses. Together, our results provide
strong evidence that layerwise embedding geometry is not incidental but central
to how deep models represent and transfer knowledge.
****

Query Lens: Interpreting Sparse Key-Value Features with Indirect Effects
Hwiyeong Lee, Ingyu Bang, Uiji Hwang, Hyelim Lim, Taeuk Kim
While sparse autoencoders provide features more interpretable than individual ne
urons, reliably characterizing them remains challenging.
We propose Query Lens, which extends Logit Lens to enable more comprehensive and
 faithful interpretations of sparse features.
By jointly considering encoder-side key features and decoder-side value features
, we identify both the inputs that activate a feature and the outputs it promote
s.
We also account for indirect, module-mediated effects that arise when the featur
e is processed by downstream modules, going beyond the direct effect captured by
 Logit Lens.
In experiments, we find that Query Lens yields coherent token signatures for fea
tures that remain uninterpretable under Logit Lens.
Finally, we propose the Subspace Channel Hypothesis, suggesting that downstream
modules read features through layer-specific subspaces.
****

Quantum Robust Inner Minimization for Reinforcement Learning with Quadratic Spee
d-Up in Query Complexity
Hyun Kyu Lee, Joongheon Kim, Sung Whan Yoon
Robust reinforcement learning (RRL) aims to tackle unexpected environmental chan
ges by optimizing policies against the worst case.
However, RRL remains impractical due to the cost of the Max-Min optimization, wh
ere it suffers from the exhaustive query complexity for finding the worst-case (
dubbed 'Min') within the environmental uncertainty set $\mathcal{U}$, i.e., $\ma
thcal{O}(|\mathcal{U}|)$.
By viewing this via a lens of quantum perspective, we raise a pivotal question:
*If we can query from the environment with quantum superpositions, is it possibl

e to accelerate the Max-Min optimization of RRL?* Our answer is 'Yes'.
Our method, called quantum robust inner minimization (QRIM), encodes the uncerta
inty set with quantum superposition and amplifies low-return cases, thus enablin
g RL for solving the robust (i.e., worst-case) Bellman equation.
Importantly, QRIM achieves a quadratic speed-up in query complexity without alte
ring the outer RL pipeline, i.e., $\mathcal{O}(\sqrt{|\mathcal{U}|})$.
Validated through classical simulations to real quantum hardware execution, QRIM
 learns more robust policies with quadratically reduced queries than classical R
L.
****

More Sail than Ballast: Addressing Harmful Knowledge Leakage in the Expansive Re
asoning Space of LRMs
Qibing Ren, Xinhao Song, Ke Fan, Lijun Li, Zhanpeng Zhou, Gongshen Liu, Junchi Y
an, Lizhuang Ma, Jing Shao
The capabilities of large language models (LLMs), particularly large reasoning m
odels (LRMs), are rapidly advancing. This raises concerns about whether LRMs can
 maintain their safety awareness throughout long-form reasoning. Frustratingly,
we identify a prevalent safety issue across LLMs and LRMs, where LRMs can reveal
 dangerous thoughts, leading to harmful knowledge elicitation when confronting s
ensitive yet benign topics. For example, when explaining the chemical context of
 Lewisite, a biological weapon, LRMs analyze its synthesis in their reasoning wi
thout recognizing the associated risks. We refer to this issue as the *unintende
d elicitation* issue. Experiments on our benchmark show that it is a common issu
e across current LRMs due to their strong multi-step reasoning capabilities. To
address this issue, we propose placing LLMs in our synthesized open-ended enviro
nments, allowing them to self-search for a safety reasoning pattern to respond r
esponsibly and helpfully. We first design a scalable data synthesis pipeline to
generate data that triggers the *unintended elicitation* issue. We further propo
se a safety-first reward model design, which prioritizes safety while also evalu
ating the helpfulness of responses and the faithfulness of reasoning. Experiment
s show that our method improves safety, reduces over-refusal, and maintains stro
ng helpfulness, paving the way for safer deployment in high-stakes domains. Code
 is available at https://github.com/XinhaoS0101/Safety-CoT.
****

dnaHNet: A Scalable and Hierarchical Foundation Model for Genomic Sequence Learn
ing
Arnav Shah, Junzhe Li, Parsa Idehpour, Adibvafa Fallahpour, Brandon Wang, Sukjun
 Hwang, BO WANG, Patrick D Hsu, Hani Goodarzi, Albert Gu
Genomic foundation models have the potential to decode DNA syntax, yet face a fu
ndamental tradeoff. Standard subword tokenizers fragment biologically meaningful
 motifs such as codons and regulatory elements, while nucleotide-level models pr
eserve biological coherence but incur prohibitive computational costs for long c
ontexts. We introduce dnaHNet, a state-of-the-art tokenizer-free autoregressive
model that segments and models genomic sequences end to end. Using a differentia
ble dynamic chunking mechanism, dnaHNet compresses raw nucleotides into latent t
okens adaptively, balancing compression with predictive accuracy. Pretrained on
prokaryotic genomes, dnaHNet outperforms leading architectures including Striped
Hyena2 in scaling and efficiency. This recursive chunking yields quadratic FLOP
reductions, enabling $>3 \times$ inference speedup over Transformers. On zero-sh
ot tasks, dnaHNet achieves superior performance in predicting protein variant fi
tness and gene essentiality, while automatically discovering hierarchical biolog
ical structures without supervision. These results establish dnaHNet as a scalab
le, interpretable framework for next-generation genomic modeling.
****

Surgery: Mitigating Harmful Fine-Tuning for Large Language Models via Attention
 Sink
Guozhi Liu, Weiwei Lin, Tiansheng Huang, Ruichao Mo, Qi Mu, Xiumin Wang, Li Shen
Harmful fine-tuning can invalidate safety alignment of large language models, ex
posing significant safety risks. In this paper, we utilize the attention sink me
chanism to mitigate harmful fine-tuning. Specifically, we first measure a statis

tic named sink divergence for each attention head and observe that different att
ention heads exhibit two different signs of sink divergence. To understand its s
afety implications, we conduct experiments and find that the number of attention
 heads of positive sink divergence increases along with the increase of the mode
l's harmfulness when undergoing harmful fine-tuning. Based on this finding, we p
ropose a separable sink divergence hypothesis -- attention heads associating wit
h learning harmful patterns during fine-tuning are separable by their sign of si
nk divergence. Based on the hypothesis, we propose a fine-tuning-stage defense,
dubbed Surgery. Surgery utilizes a regularizer for sink divergence suppression,
which steers attention heads toward the negative sink divergence group, thereby
reducing the model’s tendency to learn and amplify harmful patterns. Extensive e
xperiments demonstrate that Surgery improves defense performance by 5.90\%, 11.2
5\%, and 9.55\% on the BeaverTails, HarmBench, and SorryBench benchmarks, respec
tively. Source code is available on https://github.com/Lslland/Surgery.
****

RulePlanner: All-in-One Reinforcement Learner for Unifying Design Rules in 3D Fl
oorplanning

Ruizhe Zhong, Xingbo Du, Junchi Yan

Floorplanning determines the coordinate and shape of each module in Integrated C
ircuit. With the advancement of technology node, in floorplanning stage especial
ly 3D scenarios with multiple stacked layers, it has become increasingly challen
ging to adhere to complex hardware design rules. Current methods are only capabl
e of handling specific and limited design rules, while violations of other rules
 require manual and meticulous adjustment. This leads to labor-intensive and tim
e-consuming post-processing for expert engineers. In this paper, we propose an a
ll-in-one deep reinforcement learning-based approach to tackle these challenges,
 and design novel representations for real-world IC design rules that have not b
een addressed by previous approaches. Specifically, the processing of various ha
rdware design rules is unified into a single framework with three key components
: 1) novel matrix representations to model the design rules, 2) constraints on t
he action space to filter out invalid actions that cause rule violations, and 3)
 quantitative analysis of constraint satisfaction as reward signals. Experiments
 on public benchmarks demonstrate the effectiveness and validity of our approach
. Furthermore, transferability are well shown on unseen circuits. Our framework
is extensible to accommodate new design rules, thus providing flexibility to add
ress emerging challenges in future chip design.
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OpenSage: Self-programming Agent Generation Engine

Hongwei Li, Zhun Wang, Qinrun Dai, Yuzhou Nie, Jinjun Peng, Ruitong Liu, Jingyan
g Zhang, Kaijie Zhu, Jingxuan He, Lun Wang, Yangruibo Ding, Yueqi Chen, Wenbo Gu
o, Dawn Song

Agent development kits (ADKs) provide effective platforms and tooling for constr
ucting agents, and their designs are critical to the constructed agents' perform
ance, especially the functionality for agent topology, tools, and memory.
However, current ADKs either lack sufficient functional support or rely on human
s to manually design these components, limiting agents' generalizability and ove
rall performance.
We propose OpenSage, the first ADK that enables LLMs to automatically create age
nts with self-generated topology and toolsets while providing comprehensive and
structured memory support.
OpenSage offers effective functionality for agents to create and manage their ow
n sub-agents and toolkits.
It also features a hierarchical, graph-based memory system for efficient managem
ent and a specialized toolkit tailored to software engineering tasks.
Extensive experiments across three state-of-the-art benchmarks with various back
bone models demonstrate the advantages of OpenSage over existing ADKs.
We also conduct rigorous ablation studies to demonstrate the effectiveness of ou
r design for each component.
We believe OpenSage can pave the way for the next generation of agent developmen
t, shifting the focus from human-centered to AI-centered paradigms.

Our code is available at https://github.com/opensage-agent/opensage-adk.
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GAE: Unleashing Physical Potential of VLM with Generalizable Action Expert
Mingyu Liu, Zheng Huang, Xiaoyi Lin, Muzhi Zhu, Canyu Zhao, Yating Wang, Haoyi Zhu, Hao Chen, Chunhua Shen
Vision-language models demonstrate strong reasoning and planning abilities, yet grounding these predictions into precise robot actions remains a central challenge. Existing Vision-Language-Action methods typically entangle reasoning and action generation, leading to limited generalization and costly adaptation. We propose to learn a \textbf{G}eneralizable \textbf{A}ction \textbf{E}xpert (\textbf{GAE}), a task-agnostic model that converts sparse geometric plans into dense robot actions. Our approach introduces a sparse geometric interface: the VLM predicts sparse 3D waypoints representing high-level intention, while GAE maps these waypoints together with real-time point cloud observations to continuous action trajectories. GAE is pretrained on a large-scale pointcloud–trajectory dataset comprising \textbf{150k} trajectories from both simulation and real-world robots. To further improve efficiency and generalization, we introduce an \textbf{Action Pre-training, Pointcloud Fine-tuning (APPF)} scheme that decouples learning action dynamics from geometry grounding. After pretraining, GAE is frozen and reused across downstream tasks, requiring only lightweight fine-tuning of the VLM to produce the sparse interface. Extensive experiments show that our method achieves strong performance and generalization across diverse visual domains, camera viewpoints, and natural language instructions.
****
MuCO: Generative Peptide Cyclization Empowered by Multi-stage Conformation Optimization
Yitian Wang, Fanmeng Wang, Angxiao Yue, Wentao Guo, Yaning Cui, Hongteng Xu
Modeling peptide cyclization is critical for the virtual screening of candidate peptides with desirable physical and pharmaceutical properties.

This task is challenging because a cyclic peptide often exhibits diverse, ring-shaped conformations, which cannot be well captured by deterministic prediction models derived from linear peptide folding.

In this study, we propose MuCO (Multi-stage Conformation Optimization), a generative peptide cyclization method that models the distribution of cyclic peptide conformations conditioned on the corresponding linear peptide.

In principle, MuCO decouples the peptide cyclization task into three stages: topology-aware backbone design, generative side-chain packing, and physics-aware all-atom optimization, thereby generating and optimizing conformations of cyclic peptides in a coarse-to-fine manner.

This multi-stage framework enables an efficient parallel sampling strategy for conformation generation and allows for rapid exploration of diverse, low-energy conformations.

Experiments on the large-scale CPSea dataset demonstrate that MuCO significantly and consistently outperforms state-of-the-art methods in physical stability, structural diversity, secondary structure recovery, and computational efficiency, making it a promising computational tool for exploring and designing cyclic peptides.

The demo of the proposed method can be found at https://github.com/mianqiu00/MuCO.
****
A Unified Framework for Diffusion Model Unlearning with f-Divergence
Nicola Novello, Federico Fontana, Luigi Cinque, Deniz Gunduz, Andrea M Tonello
Most existing methods for concept unlearning in text-to-image diffusion models minimize a mean squared error (MSE) loss between the denoiser outputs conditioned on a target and an anchor concept, which is implicitly the KL divergence between two Gaussians. We generalize this objective to any f-divergence, recovering MSE as the KL instance, and identify a family of α-divergences whose Gaussian closed-form yields cheap, MSE-like training objectives. For the remaining f-divergences, we provide a min-max objective based on the variational formulation of the f-divergence. We theoretically analyze and numerically validate how

different f-divergences impact the gradient magnitude and the convergence properties of the algorithm, affecting the quality of unlearning. For instance, we observe that the Hellinger closed-form instance consistently dominates MSE across multiple scenarios. More generally, the proposed unified framework offers a flexible paradigm for selecting the optimal divergence based on the application and user goal, allowing for finer control over the trade-off between unlearning efficacy and generative fidelity.

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Fast and Expressive Multi-Byte Prediction with Probabilistic Circuits

Andreas Grivas, Lorenzo Loconte, Emile van Krieken, Piotr Nawrot, Yu Zhao, Euan Wielewski, Pasquale Minervini, Edoardo Ponti, Antonio Vergari

Multi-token prediction (MTP) is a prominent strategy to significantly speed up generation in large language models (LLMs), especially in byte-level LLMs, which are tokeniser-free but prohibitively slow. However, many existing MTP methods either assume independence between future tokens, sacrificing expressiveness, or generate tokens one at a time within the window, increasing latency. In this work, we investigate the trade-off between expressiveness and latency in MTP within the framework of probabilistic circuits (PCs). Our framework, MTPC, allows one to explore different ways to encode the joint distributions over future tokens by selecting circuit architectures, generalising classical models such as (hierarchical) mixture models, hidden Markov models, and tensor networks. We show the efficacy of MTPC by retrofitting existing byte-level LLMs, such as EvaByte, and byte-fied subword models, such as Llama3.2 3B. Our experiments show that, when combined with speculative decoding, MTPC substantially speeds up generation compared to MTP with independence assumptions, while guaranteeing to retain the performance of the original verifier LLM. We also rigorously study the optimal trade-off between expressiveness and latency when exploring the possible parameterisations of MTPC, such as PC architectures and partial layer sharing between the verifier and draft LLMs.

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Robust Strategic Classification under Decision-Dependent Cost Uncertainty

Sura Alhanouti, Guzin Bayraksan, Parinaz Naghizadeh

Humans facing algorithmic decision systems have been found to ``game'' them by altering their input data (at a cost to them) in order to favorably change the algorithmic outcomes they receive (at a cost to the algorithm). The growing literature on strategic classification seeks to develop robust machine learning algorithms that account for, and reduce, unwanted strategic behavior. A limitation of these existing works is that they assume the cost of strategic behavior to be fixed and independent of the classifier's decision. In practice, however, manipulation costs evolve and depend on past algorithmic decisions: today's decisions influence tomorrow's costs. This paper proposes and analyzes a two-stage robust optimization framework with a decision-dependent uncertainty set to capture such dependencies. We highlight that awareness of policy-dependent costs not only reduces uncertainty, but also better curtails gaming of the algorithmic system over time.

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Auditing Sybil: Explaining Deep Lung Cancer Risk Prediction Through Generative Interventional Attributions

Bartlomiej Sobieski, Jakub Grzywaczewski, Karol Dobiczek, Mateusz Wójcik, Tomasz Bartczak, Szatkowski Patryk, Przemys■aw Bombi■ski, Matthew Tivnan, Przemyslaw Biecek

Lung cancer remains the leading cause of cancer mortality, driving the development of automated screening tools to alleviate radiologist workload. Standing at the frontier of this effort is Sybil, a deep learning model capable of predicting future risk solely from computed tomography (CT) with high precision. However, despite extensive clinical validation, current assessments rely purely on observational metrics. This correlation-based approach overlooks the model's actual reasoning mechanism, necessitating a shift to causal verification to ensure robust decision-making before clinical deployment. We propose S(H)NAP, a model-agnostic auditing framework that constructs generative interventional attributions vali

dated by expert radiologists. By leveraging realistic 3D diffusion bridge modeli
ng to systematically modify anatomical features, our approach isolates object-sp
ecific causal contributions to the risk score. Providing the first interventiona
l audit of Sybil, we demonstrate that while the model often exhibits behavior ak
in to an expert radiologist, differentiating malignant pulmonary nodules from be
nign ones, it suffers from critical failure modes, including dangerous sensitivi
ty to clinically unjustified artifacts and a distinct radial bias.
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Learning-augmented Rent-or-Buy with a Sample
Davidson Zhu, Sreenivas Gollapudi, Debmalya Panigrahi
In this paper, we study the rent-or-buy problem (also called the Bahncard proble
m) in the learning-augmented setting. In this problem, a traveler must complete
a sequence of trips that are revealed online over time, each of which has an ass
ociated cost with it. The traveler has the option of buying a discount card at a
 fixed cost that gives a discount on trip costs for a fixed time after buying th
e card. The goal is to minimize the overall cost of all the trips, including the
 money spent on buying discount cards. For this problem, it is well-known that t
he best deterministic algorithm has a competitive ratio of 2. In this paper, we
ask whether we can do better if the traveler has a *sample* of trips available o
ffline, e.g., obtained from an ML model based on historical data. We show that e
ven a sparse sample of the input can significantly improve the competitive ratio
 of the algorithm from 2 to 3/2, and further to close to 1 under some additional
 conditions. We also verify our theoretical bounds via numerical simulations, wh
ich reveal that our proposed algorithm obtains nearly optimal solutions for a va
riety of natural input classes.
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Conflicting Biases at the Edge of Stability: Norm versus Sharpness Regularizatio
n
Maria Matveev, Vit Fojtik, Hung-Hsu Chou, Gitta Kutyniok, Johannes Maly
The remarkable generalization properties of overparameterized networks are often
 attributed to implicit biases, such as norm minimization at small learning rate
s and low sharpness in the Edge-of-Stability regime. In this work, we argue that
 a comprehensive understanding of the generalization performance of gradient des
cent requires analyzing the interaction between these various forms of implicit
regularization. We empirically demonstrate that the learning rate interpolates b
etween low parameter norm and low sharpness of the trained model. We furthermore
 prove that neither implicit bias alone minimizes the generalization error for d
iagonal linear networks trained on a simple regression task. These findings demo
nstrate that focusing on a single implicit bias is insufficient to explain good
generalization, and they motivate a broader view of implicit regularization that
 captures the dynamic trade-off between norm and sharpness induced by non-neglig
ible learning rates.
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OvisOCR: End-to-End Document Parsing via Aligning Specialized Perception with Ge
neral Reasoning
Jun-Peng Jiang, Shiyin Lu, An-Yang Ji, Yinglun Li, Qing-Guo Chen, Zhao Xu, Weihu
a Luo, Kaifu Zhang, De-Chuan Zhan, Han-Jia Ye
This paper presents OvisOCR, a lightweight and strictly end-to-end Multimodal La
rge Language Model (MLLM) tailored for document parsing. Unlike current methods
that rely on complex "Crop-OCR-Merge'" cascades to handle high-resolution inputs
, OvisOCR directly maps full-page visual signals to structured Markdown without
localized slicing or layout detection dependencies. Through extensive evaluation
s on the OmniDocBench, OvisOCR achieves top-tier performance among all compared
methods in all aspects, demonstrating that a compact E2E model can effectively `
`digest'' the capabilities of intricate pipelines and surpass specialized and ge
neral methods.
  Technically, OvisOCR unifies fine-grained recognition and semantic correction
in a single model by leveraging supervision bootstrapped from strong OCR engines
 and refined via general-model-based data cleaning. To balance the performance a
cross diverse document constituents, we design category-specific reward mechanis

ms for distinct element types, such as dense text, complex tables, and formulas, and ensure the model enhances its formatting strengths for each modality concurrently. This approach effectively resolves the optimization conflict, guaranteeing that improvements in structural layout parsing do not come at the expense of omitting fine-grained textual details. Empirical results confirm that OvisOCR eliminates the error propagation inherent in split-and-merge architectures, offering a streamlined path for future document intelligence. Our model can be available at \url{https://huggingface.co/AIDC-AI/OvisOCR}.
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Approximating f -Divergences with Rank Statistics
Viktor Stein, José Manuel de Frutos
We introduce a rank-statistic approximation of f-divergences that avoids explicit density-ratio estimation by working directly with the distribution of ranks. For a resolution parameter K, we map the mismatch between two univariate distributions μ and ν to a rank histogram on $\{ 0, \ldots, K\}$ and measure its deviation from uniformity via a discrete f-divergence, yielding a rank-statistic divergence estimator.
We prove that the resulting estimator of the divergence is monotone in K, is always a lower bound of the true f-divergence, and we establish quantitative convergence rates for $K\to\infty$ under mild regularity of the quantile-domain density ratio.
To handle high-dimensional data, we define the sliced rank-statistic f-divergence by averaging the univariate construction over random projections, and we provide convergence results for the sliced limit as well.
We also derive finite-sample deviation bounds along with asymptotic normality results for the estimator.
Finally, we empirically validate the approach by benchmarking against neural baselines and illustrating its use as a learning objective in generative modeling experiments.
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GenDA: Generative Data Assimilation on Complex Urban Areas via Classifier-Free Diffusion Guidance
Francisco Giral, Álvaro Manzano Sevillano, Ignacio Gomez Perez, Ricardo Vinuesa, Soledad Le Clainche
Urban wind flow reconstruction is essential for assessing air quality, heat dispersion, and pedestrian comfort, yet remains challenging when only sparse sensor data are available. We propose GenDA, a generative data assimilation framework that reconstructs high-resolution wind fields on unstructured meshes from limited observations. The model employs a multiscale graph-based diffusion architecture trained on computational fluid dynamics (CFD) simulations and interprets classifier-free guidance as a learned posterior reconstruction mechanism: the unconditional branch learns a geometry-aware flow prior, while the sensor-conditioned branch injects observational constraints during sampling. This formulation enables obstacle-aware reconstruction and generalization to held-out mesh geometries, wind directions, and sensor configurations within the studied urban-flow setting, without retraining. We consider both sparse fixed sensors and trajectory-based observations using the same reconstruction procedure. When evaluated against supervised graph neural network (GNN) baselines and classical reduced-order data assimilation methods, GenDA reduces the relative root-mean-square error (RRMSE) by 25-57% and increases the structural similarity index (SSIM) by 23-33% across the tested meshes. Experiments are conducted on Reynolds-averaged Navier-Stokes (RANS) simulations of a real urban neighborhood in Bristol, United Kingdom, at a characteristic Reynolds number of $\mathrm{Re}\approx2\times10^{7}$, featuring complex building geometry and irregular terrain. The proposed framework provides a scalable path toward generative, geometry-aware data assimilation for environmental monitoring in complex domains.
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Learning to Extrapolate to New Tasks: A Relational Approach to Task Extrapolation
Adam Ousherovitch, Yixin Wang

Modern learning systems excel at interpolation but struggle to generalize to unseen tasks outside the training distribution's support. This failure occurs even in simple settings, such as handling task parameters beyond the training range, and persists despite advances in foundation models. To this end, we develop the Relational Task Extrapolator (RTE), an algorithm designed to enable systematic extrapolation to novel tasks. The key observation is that extrapolation is inherently relational: extrapolating to unseen tasks requires learning how tasks transform into one another. If a model learns the transformation between tasks A and B during training, it can apply that same transformation to relate known tasks to unseen ones at test time. RTE operationalizes this idea by decomposing each target task into a known anchor task and a transformation linking the anchor and target. It then learns a relational operator, mapping an anchor–transformation pair to predictions for the target task. We instantiate RTE across multiple task extrapolation regimes in function prediction, e.g. where target tasks use out-of-range parameters (parameter extrapolation), has greater compositional depth (length extrapolation), and/or recombine function primitives in unseen ways (compositional extrapolation). We further extend RTE to sequence prediction, integrating it into fine-tuning algorithms for foundation models. Across empirical studies, we find that RTE substantially outperforms existing approaches on extrapolation to novel, unseen tasks.
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Trust Functions: Near Lossless Weak-to-Strong Generalization by Learning to Trust the Weak Teacher
Arda Uzunoglu, Alvin Zhang, Daniel Khashabi
Weak-to-strong generalization studies how to improve a strong student using supervision from a weaker teacher when reliable labels are scarce. We view this primarily as a data selection problem, where the key challenge is to identify which weak labels are reliable enough to serve as a training signal. To address this, we introduce trust functions that assign each weak label a scalar trust score and use these scores to filter weak supervision. Across several domains, including world knowledge, quantitative reasoning, decision making, trust filtering yields students that match and sometimes surpass ground-truth supervision, achieving near-lossless weak-to-strong generalization. Moreover, trust functions enable an iterative weak-to-strong chain that compounds gains by training a student and reusing it as the next teacher, producing the strongest final model. Our analyses suggest that neural trust functions improve learning through more than label error reduction. They induce an implicit easy-first curriculum, recover near-optimal alternatives where ground truth labels are incomplete, and produce more coherent gradient updates, offering a mechanistic account of the stability and efficiency of trust-filtered weak-to-strong generalization.
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Physics-informed coarsening for multigrid graph neural networks surrogates
Amir Bazzi, Ramy Nemer, Alves José, Elie Hachem
Learning-based surrogates for partial differential equations have recently matched the accuracy of classical solvers while achieving orders-of-magnitude speedups, predominantly in fluid settings and structured geometries. In contrast, robust surrogates for deformable solids remain underexplored, despite the presence of nonlinear elasticity, plasticity, and transient behavior that challenge standard architectures. We introduce a multigrid graph neural network for solid mechanics that couples an *encoder-processor-decoder* backbone with a physics-informed coarsening strategy. Instead of downsampling via geometric heuristics, our method scores nodes using a residual-based measure of local physical activity and preferentially retains regions of high strain or stress concentration, allocating multiscale capacity where it is most needed. This preserves long-range interactions through hierarchical message passing while improving stability over long rollouts. We evaluate on multiple datasets covering linear, nonlinear, and transient regimes, and observe consistent gains in accuracy and rollout stability compared to standard sampling baselines. Our results highlight the importance of physics-informed coarsening for scalable surrogate modeling in solid mechanics.
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Reward Auditor: Inference on Reward Modeling Suitability in Real-World Perturbed Scenarios

Jianxiang Zang, Yongda Wei, Ruxue Bai, Shiyu Jiang, Nijia Mo, Binhong Li, Qiang Sun, Hui Liu

Reliable reward models (RMs) are critical for ensuring the safe alignment of large language models (LLMs). However, current RM evaluation methods focus solely on preference perception accuracies in given specific scenarios, obscuring the critical vulnerabilities of RMs in real-world scenarios. We identify the true challenge lies in assessing a novel dimension: Suitability, defined as conditional reliability under specific real-world perturbations. To this end, we introduce Reward Auditor, a hypothesis-testing framework specifically designed for RM suitability inference. Rather than answering “How accurate is the RM's preference perception for given samples?”, it employs scientific auditing to answer: “Can we infer RMs exhibit systematic vulnerabilities in specific real-world scenarios?". Under real-world perturbed scenarios, Reward Auditor quantifies statistical significance and effect size by auditing distribution degradation of RM preference perception confidence. This enables inference of both the certainty and severity of RM vulnerabilities across diverse real-world scenarios, thereby laying a solid foundation for building next-generation LLM alignment systems that are verifiably safe, more robust, and trustworthy.

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Fix the Mind, Not the Move: Interpretable AI Assistance via Knowledge-Gap Localization

Ayano Hiranaka, Ya-Chuan Hsu, Stefanos Nikolaidis, Erdem Biyik, Daniel Seita

AI assistants in human-AI collaboration often correct suboptimal human actions through behavioral feedback (e.g., alerts or steering-wheel nudges in assistive driving). Such interventions can mitigate immediate errors, but long-term improvement requires addressing the underlying misconceptions that cause repeated mistakes. We introduce SENSEI, a framework that infers user misconceptions from interaction behavior and provides targeted, minimal yet sufficient suggestions to correct them. Our approach departs from action- or trajectory-level interventions by operating over a structured knowledge representation to localize and correct the sources of erroneous behavior. Across three long-horizon tasks with diverse misconceptions and corresponding behaviors, SENSEI demonstrates zero-shot compositional generalization, disentangling multiple overlapping misconceptions despite training only on single-misconception cases.

A user study further shows that our method identifies real human misconceptions and provides effective guidance that improves long-horizon task performance, successfully correcting 90% of student misconceptions.

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Regulating Anatomy-Aware Rewards via Trajectory-Integral Feedback for Volumetric Computed Tomography Analysis

Tianwei Lin, Zhongwei Qiu, Jie Cao, Jiang Liu, Wenjie Yan, Zhang Bo, Yu Zhong, Wenqiao Zhang, Yingda Xia, Ling Zhang

Medical vision-language models (VLMs) have rapidly advanced as general-purpose multimodal assistants, yet their deployment in 3D Computed Tomography (CT) analysis remains constrained by a persistent mismatch between optimization objectives and clinical rigor. Current Reinforcement Learning (RL) paradigms still rely on lexical proxy signals that induce Evaluation Hallucinations, where models optimize linguistic fluency rather than factual clinical correctness, leading to diagnostically critical errors. To bridge this gap, we introduce the Clinical Abnormality Benchmarking Substrate (CABS), a structured system that decomposes radiology reports into verifiable clinical semantic units. Using CABS, we identify a Mechanistic Divergence in standard RL, where surface-similarity rewards drive policy gradients to bypass medical facts. We therefore propose Trajectory-Integral Feedback GRPO (TIF-GRPO), a novel framework integrating control-theoretic principles into policy optimization. By formulating clinical reasoning as a pseudo-temporal trajectory for anomaly discovery, TIF-GRPO regulates anatomy-aware rewards via an integral feedback loop that penalizes persistent omissions as cumulative state errors and suppresses hallucinations as excessive control effort. Experimen

ts on 3D CT benchmarks demonstrate that our approach significantly enhances abnormality detection and clinical faithfulness, establishing a new paradigm for fine-grained regulation in medical VLMs. Our project is available at https://github.com/ZJU4HealthCare/TIF-GRPO.

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INDEXGUARD: Index-only Backdoor Vetting for Secure Federated PEFT of Large Language Models

Javad Dogani, Devri■ ■■ler, Nikolaos Laoutaris

Federated parameter-efficient fine-tuning (PEFT) enables customizing large language models on private data, yet it is vulnerable to backdoor poisoning—especially when privacy constraints prevent inspection of per-client real-valued updates. We exploit the intuition that poisoning leaves a similar backdoor imprint in which adapter coordinates become salient, so overlap in salient-index supports remains informative even without values. We introduce IndexGuard, an unsupervised index-only vetting primitive in which clients send only Top-K salient update indices and the server operates on the induced overlap geometry, clustering clients and filtering cohesion-outlier groups before aggregation. We analyze support stability under bounded rescaling and separability under shared-trigger poisoning under non-IID drift. Across attacks, backbones, and PEFT variants, IndexGuard provides end-to-end mitigation, preserving clean accuracy while achieving performance comparable to centralized methods.

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Training Prompt Matters: State-Adaptive Optimization for Robust Fine-Tuning

Wenhang Shi, Yiren Chen, Shuqing Bian, Zhe Zhao, Jinhao Dong, Pengfei Hu, WEI LU, Xiaoyong Du

While prompt engineering is instrumental in maximizing the capabilities of Large Language Models (LLMs) during inference, the role of prompts during training remains critically underexplored. Prevailing fine-tuning paradigms typically treat training prompts as mere surface forms, assuming that semantically equivalent instructions yield identical learning outcomes. However, we reveal that this equivalence is deceptive: while paraphrased prompts often lead to comparable in-task performance, they induce drastically different cross-task impacts regarding catastrophic forgetting and generalization. Crucially, these impacts are positively correlated across tasks, indicating the existence of superior prompts that consistently yield better performance. Furthermore, we discover that these superior prompts can be robustly identified by task loss prior to learning. Leveraging these insights, we introduce State-Adaptive Prompt Optimization (SAPO), a lightweight yet effective training strategy that shifts task formulation from a static input to a dynamic, state-adaptive variable. Comprehensive experiments on diverse benchmarks confirm its effectiveness, which significantly mitigates forgetting while improving generalization, achieving substantial performance gains over state-of-the-art methods. These results provide insights into how training prompts shape learning dynamics and offer a practical recipe for robust fine-tuning. Our code is available at https://github.com/Eric8932/SAPO.

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Variance-Reduced $(\varepsilon, \delta)-$Unlearning using Forget Set Gradients

Martin Van Waerebeke, Giovanni Neglia, Kevin Scaman, Marco Lorenzi, El-Mahdi El-Mhamdi

In machine unlearning, $(\varepsilon,\delta)-$unlearning is a popular framework that provides formal guarantees on the effectiveness of the removal of a subset of training data, the \emph{forget set}, from a trained model.

For strongly convex objectives, existing first-order methods achieve $(\varepsilon,\delta)-$unlearning, but they only use the forget set to calibrate injected noise, never as a direct optimization signal.

In contrast, efficient empirical heuristics often exploit the forget samples (e.g., via gradient ascent) but come with no formal unlearning guarantees.

We bridge this gap by presenting the Variance-Reduced Unlearning (*VRU*) algorithm. To the best of our knowledge, *VRU* is the first first-order algorithm that directly includes forget set gradients in its update rule, while provably satisfying $(\varepsilon,\delta)-$unlearning.

We establish the convergence of *VRU* and show that incorporating the forget se
t yields strictly improved rates, *i.e.*, a better dependence on the achieved er
ror
compared to existing first-order $(\varepsilon,\delta)-$unlearning methods. More
over, we prove that, in a low-error regime *VRU*
asymptotically outperforms
any first-order methods that ignores the forget set.
Experiments corroborate our theory, showing consistent gains over both state-of-
the-art certified unlearning methods and over empirical
baselines that explicitly leverage the forget set.
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Transformers Learn the Optimal DDPM Denoiser for Multi-Token GMMs
Hongkang Li, Hancheng Min, Rene Vidal
Transformer-based diffusion models have demonstrated remarkable performance at g
enerating high-quality samples. However, our theoretical understanding of the re
asons for this success remains limited. For instance, existing models are typica
lly trained by minimizing a denoising objective, which is equivalent to fitting
the score function of the training data. However, we do not know why transformer
-based models can match the score function for denoising, or why gradient-based
methods converge to the optimal denoising model despite the non-convex loss land
scape. To the best of our knowledge, this paper provides the first convergence a
nalysis for training transformer-based diffusion models. More specifically, we c
onsider the population Denoising Diffusion Probabilistic Model (DDPM) objective
for denoising data that follow a \textit{multi-token Gaussian mixture} distribut
ion. We theoretically quantify the required number of tokens per data point and
training iterations for the global convergence towards the Bayes optimal risk of
 the denoising objective, thereby achieving a desired score matching error. A de
eper investigation reveals that the self-attention module of the trained transfo
rmer implements a \emph{mean denoising} mechanism that enables the trained model
 to approximate the oracle Minimum Mean Squared Error (MMSE) estimator of the in
jected noise in the diffusion steps. Numerical experiments validate these findin
gs.
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Efficient Diffusion Models under Nonconvex Equality and Inequality constraints v
ia Landing
Kijung Jeon, Michael Muehlebach, Molei Tao
Generative modeling within constrained sets is essential for scientific and engi
neering applications involving physical, geometric, or safety requirements (e.g.
, molecular generation, robotics). We present a unified framework for constraine
d diffusion models on generic nonconvex feasible sets Σ that simultaneous
ly enforces equality and inequality constraints throughout the diffusion process
. Our framework incorporates both overdamped and underdamped dynamics for forwar
d and backward sampling. A key algorithmic innovation is a computationally effic
ient landing mechanism that replaces costly and often ill-defined projections on
to Σ, ensuring feasibility without iterative Newton solves or projection
failures. By leveraging underdamped dynamics, we accelerate mixing toward the pr
ior distribution, effectively alleviating the high simulation costs typically as
sociated with constrained diffusion. Empirically, this approach reduces function
 evaluations and memory usage during both training and inference while preservin
g sample quality. On benchmarks featuring equality and mixed constraints, our me
thod achieves comparable sample quality to state-of-the-art baselines while sign
ificantly reducing computational cost, providing a practical and scalable soluti
on for diffusion on nonconvex feasible sets.
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Do Vision and Text Cues Exhibit Evidential Coupling? UFO: A Benchmark for Compos
itional Multimodal Reasoning in Unified Models
Zhongyu Yang, Dannong Xu, Yonghan Zhang, Kefan Chen, Xinyi Wang, Yang Xu, Wei Pa
ng, Yingfang Yuan
Unified Foundation Models (UFMs), which support interleaved multimodal generatio
n and understanding, have been proposed as a promising paradigm for reasoning ab

out dynamic world states, yet it remains unclear whether the visual content they generate functions as grounded evidence for subsequent reasoning or merely as auxiliary output. Existing benchmarks largely evaluate generation and understanding as separate capabilities and do not test their functional dependence during reasoning. We introduce UFO, a benchmark designed to evaluate whether UFMs generate and use image and text cues as evidence for compositional multimodal reasoning. UFO spans three state-transition regimes, state determination, state reconstruction, and state augmentation, which correspond to progressively smaller transformations of the underlying world state. Our analysis reveals a significant modality gap, as models often achieve high prediction accuracy even when the generated visual cues exert limited influence on their decisions, indicating weakened evidential coupling and a reliance on textual shortcuts rather than robust cross-modal grounding.

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Controllable and Explainable Personality Sliders for LLMs at Inference Time

Florian Hoppe, David Khachaturov, Robert D. Mullins, Mark Huasong Meng

Aligning Large Language Models (LLMs) with specific personas typically relies on Supervised Fine-Tuning (SFT) or Reinforcement Learning from Human Feedback (RLHF); however, these methods are resource-intensive, requiring expensive data collection and distinct model training for each target personality. In this work, we propose a parameter-efficient framework for continuous, multi-dimensional personality control via inference-time activation steering. Our approach addresses the challenge of combining multiple interventions by iteratively retraining probes on the residual stream modified by prior traits, ensuring compatibility. Once established, these steering vectors function as modular, reusable primitives; users can instantly synthesize novel, complex personality profiles by simply adjusting steering coefficients (α) without any additional training. To support this, we introduce an automated pipeline that identifies optimal intervention layers via activation separation analysis and calibrates coefficients via hyperparameter optimization to maximize alignment while constraining perplexity. Empirical evaluations validate individual trait shifts using an LLM-as-a-judge framework and demonstrate, via the Big Five inventory, that our method effectively modulates the model's holistic personality profile without updating base model parameters.

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Learning Global Representation from Queries for Vectorized HD Map Construction

Shoumeng Qiu, Xinrun Li, Yang Long, Xiangyang Xue, Varun Ojha, Jian Pu

The online construction of vectorized high-definition (HD) maps is a cornerstone of modern autonomous driving systems. State-of-the-art approaches, particularly those based on the DETR framework, formulate this as an instance detection problem. However, their reliance on independent, learnable object queries results in a predominantly local query perspective, neglecting the inherent global representation within HD maps. In this work, we propose \textbf{MapGR} (\textbf{G}lobal \textbf{R}epresentation learning for HD \textbf{Map} construction), an architecture designed to learn and utilize global representations from queries. Our method introduces two synergistic modules: a Global Representation Learning (GRL) module, which encourages the distribution of all queries to better align with the global map through a carefully designed holistic segmentation task, and a Global Representation Guidance (GRG) module, which endows each individual query with explicit, global-level contextual information to facilitate its optimization. Evaluations on the nuScenes and Argoverse2 datasets validate the efficacy of our approach, demonstrating substantial improvements in mean Average Precision (mAP) compared to leading baselines. Code is available at \href{https://github.com/skyshoumeng/MapGR}{\texttt{github.com/skyshoumeng/MapGR}}.

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PsumQuant: In-line Post-training Partial Sum Quantizer for Energy Efficient NPU Inference

Sangwoo Hwang, Yeeun Hong, Jaeha Kung

The rapid growth of deep neural networks (DNNs) has intensified the demand for efficient hardware acceleration under quantization.

While prior research has successfully reduced weight and activation precision, partial sums generated during accumulation often retain high precision, resulting in significant energy overhead.
In this work, we analyze psum distributions in tiled architectures and reveal that within-tile outliers are input-dependent.
We propose PsumQuant, a post-training, input-aware quantization that predicts psum scales on-the-fly.
By leveraging the crest factor of input activations, our learnable scale predictor effectively bounds the psum bit-width while handling the extreme outliers in DNNs.
Experimental results on a 128×128 systolic array demonstrate that PsumQuant compresses psum precision down to 8-bit within only a 1\% accuracy drop on ResNet-18 and a marginal 0.04 perplexity increase on Llama-3.1.
Furthermore, bit-width reduction with PsumQuant results in a 45\% reduction in total energy with minimal accuracy loss, demonstrating that PsumQuant provides a highly efficient solution for actual NPU architectures.
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OmniMoE: An Efficient MoE by Orchestrating Atomic Experts at Scale
Jingze Shi, Zhangyang Peng, Yizhang Zhu, Yifan Wu, Guang Liu, Yuyu Luo
Mixture-of-Experts (MoE) architectures are evolving towards finer granularity to improve parameter efficiency. However, existing MoE designs face an inherent trade-off between the granularity of expert specialization and hardware execution efficiency. In this paper, we propose OmniMoE, a system-algorithm co-designed MoE framework that pushes granularity to the extreme with vector-level Atomic Experts, orchestrating their routing and execution at scale within a single MoE layer, while retaining a shared dense MLP for general-purpose processing. While this atomic design maximizes capacity, it poses severe challenges for routing complexity and memory access. To address these, OmniMoE adopts a system-algorithm co-design: (i) a Cartesian Product Router that decomposes the massive index space to reduce routing complexity from $O(N)$ to $O(\sqrt{N})$; and (ii) Expert-Centric Scheduling that inverts the execution order to turn scattered, memory-bound lookups into efficient dense matrix operations. Validated on seven benchmarks, OmniMoE (with 1.7B active parameters) achieves 50.9\% zero-shot accuracy across seven benchmarks, outperforming coarse-grained (e.g., DeepSeekMoE) and fine-grained (e.g., PEER) baselines. Crucially, OmniMoE reduces inference latency from 73ms to 6.7ms (a 10.9$\times$ speedup) compared to PEER, demonstrating that massive-scale fine-grained MoE can be fast and accurate.
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The Personality Illusion: Revealing Dissociation Between Self-Reports & Behavior in LLMs
Pengrui Han, Rafal Kocielnik, Peiyang Song, Ramit Debnath, Dean Mobbs, Anima Anandkumar, R. Michael Alvarez
Personality traits have long been studied as predictors of human behavior. Recent advances in Large Language Models (LLMs) suggest similar patterns may emerge in artificial systems, with advanced LLMs displaying consistent behavioral tendencies resembling human traits like agreeableness and self-regulation. Understanding these patterns is crucial, yet prior work primarily relied on simplified self-reports and heuristic prompting, with little behavioral validation. In this study, we systematically characterize LLM personality across three dimensions: (1) the dynamic emergence and evolution of trait profiles throughout training stages; (2) the predictive validity of self-reported traits in behavioral tasks; and (3) the impact of targeted interventions, such as persona injection, on both self-reports and behavior. Our findings reveal that instructional alignment (e.g., RLHF, instruction tuning) significantly stabilizes trait expression and strengthens trait correlations in ways that mirror human data. However, these self-reported traits do not reliably predict behavior, and observed associations often diverge from human patterns. While persona injection successfully steers self-reports in the intended direction, it exerts little or inconsistent effect on actual behavior. By distinguishing surface-level trait expression from behavioral consistency, our findings challenge assumptions about LLM personality and underscore t

he need for deeper evaluation in alignment and interpretability.
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StretchTime: Adaptive Time Series Forecasting via Symplectic Attention
Yubin Kim, Viresh Pati, Jevon Twitty, Vinh Pham, Shihao Yang, Jiecheng Lu
Transformer architectures have established strong baselines in time series forec asting, yet they typically rely on positional encodings that assume uniform, ind ex-based temporal progression. However, real-world systems, from shifting financ ial cycles to elastic biological rhythms, frequently exhibit ``time-warped'' dyn amics where the effective flow of time decouples from the sampling index. In thi s work, we first formalize this misalignment and prove that rotary position embe dding (RoPE) is mathematically incapable of representing non-affine temporal war ping. To address this, we propose Symplectic Positional Embeddings (SyPE), a lea rnable encoding framework derived from Hamiltonian mechanics. SyPE strictly gene ralizes RoPE by extending the rotation group $\mathrm{SO}(2)$ to the symplectic group $\mathrm{Sp}(2,\mathbb{R})$, modulated by a novel input-dependent adaptive warp module. By allowing the attention mechanism to adaptively dilate or contra ct temporal coordinates end-to-end, our approach captures locally varying period icities without requiring pre-defined warping functions. We implement this mecha nism in StretchTime, a multivariate forecasting architecture that achieves state -of-the-art performance on standard benchmarks, demonstrating superior robustnes s on datasets exhibiting non-stationary temporal dynamics.
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Video-in-the-Loop: Span-Grounded Long Video QA with Interleaved Reasoning
Chendong Wang, Donglin Bai, Yifan Yang, Xiao Jin, Anlan Zhang, Rui Wang, Shiqi J iang, Yuqing Yang, Hao Wu, Qi Dai, Chong Luo, Ting Cao, Lili Qiu, Suman Banerjee
We present $\textit{Video-in-the-Loop}$ (ViTL), a two-stage long-video QA framew ork that preserves a fixed token budget by first $\textit{localizing}$ question- relevant interval(s) with a low-fps skim and then $\textit{answering}$ via span- aware reallocation of visual tokens at higher effective frame rate, emitting an interleaved output with both spans and the final option for direct attribution. We also introduce $\textit{VGrounding-QA}$, which converts description based eve nt graphs into $\textit{span-grounded}$ multiple-choice QA by pairing each quest ion with $\textit{ground-truth}$ time span(s) and related reasoning. ViTL is tra ined end-to-end with an interleaved group-relative objective that couples tempor al IoU for localization with answer correctness, allowing credit to flow from an swers back to spans without increasing compute. Under fixed token budgets, ViTL attains up to 8.6\% with 50\% less frame input on long-video QA and temporal gro unding (e.g., Charades-STA, ActivityNet-Captions) and ablations show that span-a ware token reallocation consistently surpasses uniform sampling. Together, $\tex tit{VGrounding-QA}$ and ViTL provide an interpretable, compute-efficient recipe for scalable long-video QA.
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When AI Agents Compete for Jobs: Strategic Capabilities and Economic Dynamics of AI Labour Markets
Christopher Chiu, Simpson Zhang, Mihaela van der Schaar
Emerging agentic marketplaces provide the economic infrastructure for matching a nd coordinating the large amounts of AI agents used in agentic swarms. Unlike hu man workers, AI agents can operate on multiple jobs simultaneously, acquire skil ls rapidly, and labor without wage floors. These differences introduce a new seg ment of AI labor markets, where AI agents interact with each other at a much hig her frequency than human markets. Yet we lack frameworks to understand how such markets behave in light of economic forces that shape labor markets, such as adv erse selection and reputation dynamics.
To explore this, we introduce AI-Work, a tractable, simulated gig economy where Large Language Model (LLM) agents compete for jobs, develop skills, and adapt th eir strategies under uncertainty and competitive pressure. Our experiments exami ne three domains of capabilities that successful agents possess: metacognition (accurate self-assessment of skills), competitive awareness (modeling rivals and market dynamics), and long-horizon strategic planning. Agents with these capabil ities consistently achieve higher profits, reputations, and market share than co

mpeting agents.
Through AI-Work, we hope to provide a foundation to explore the microeconomic pr
operties of AI-only labour markets, and a conceptual framework to study the stra
tegic reasoning capabilities of participating AI agents.
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Laplacian Representations for Decision-Time Planning
Dikshant Shehmar, Matthew Kyle Schlegel, Matthew E. Taylor, Marlos C. Machado
Planning with a learned model remains a key challenge in model-based reinforceme
nt learning (RL). In decision-time planning, state representations are critical
as they must support local cost computation while preserving long-horizon struct
ure. In this paper, we show that the Laplacian representation provides an effect
ive latent space for planning by capturing state-space distances at multiple tim
e scales. This representation preserves meaningful distances and naturally decom
poses long-horizon problems into subgoals, also mitigating the compounding error
s that arise over long prediction horizons. Building on these properties, we int
roduce ALPS, a hierarchical planning algorithm, and demonstrate that it outperfo
rms commonly used baselines on a selection of offline goal-conditioned RL tasks
from OGBench, a benchmark previously dominated by model-free methods.
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CacheEdit: Efficient Multi-round Image Editing via Adaptive Token-wise Reuse.
Jinxin Yu, Xueqing Chen, Yudong Pan, Lian Liu, shengwen Liang, Huawei Li, Xiaowe
i Li, Ying Wang
Instruction-based image editing (IIE) is a vital tool for iterative content crea
tion, enabling multi-round interactions that refine visual details while preserv
ing cross-round consistency. However, this workflow is constrained by the comput
e-bound nature of Diffusion Transformers (DiTs): because DiTs process tokens uni
formly, they waste substantial computation on regions untouched by the instructi
on. We investigate the Round--Step--Layer hierarchy of DiT-based editing and ide
ntify a phenomenon we term Delayed Latent Emergence (DLE). Although pronounced l
atent changes emerge only in the late denoising stages, deep-layer activations w
ithin transformer blocks at the very first sampling step already diverge markedl
y in edited regions. Building on this insight, we propose CacheEdit, a training-
free framework centered on an Adaptive Activation Cache (Acache) that exploits e
arly-step sensitivity to detect invariant tokens and reuse their cached activati
ons across subsequent sampling steps, thereby bypassing redundant computation. E
xperiments on FLUX.1 Kontext and Qwen-Image-Edit show that CacheEdit achieves up
 to $2.5\times$ end-to-end acceleration. Moreover, by isolating and reusing stat
ic features, CacheEdit mitigates stochastic drift and improves instruction-follo
wing and structural consistency over full-recomputation baselines.
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BioFormer: Rethinking Cross-Subject Generalization via Spectral Structural Align
ment in Biomedical Time-Series
Guikang Du, Haoran Li, Xinyu Liu, Zhibo Zhang, Xiaoli Gong, Jin Zhang
Cross-subject generalization in biomedical time-series aims to learn representat
ions that generalize to unseen subjects while suppressing subject-specific varia
bility.
Most existing methods implicitly suppress the variability through model building
 or subject adversarial learning, but rarely model it explicitly.
We introduce \textbf{spectral drift} as a new perspective to characterize subjec
t specific variability.
Specifically, BTS signals under the same label often share consistent oscillator
y structure, yet exhibit subject-dependent magnitude or phase shifts in specific
 frequency components, which we interpret as subject-specific variability.
Building on this insight, we propose \textit{BioFormer}.
At its core is a Frequency-Band Alignment Module (FBAM) that generates band-wise
 modulation factors from the spectral distribution and adaptively adjusts amplit
ude and phase to align spectral structure, thereby mitigating variability.
We further pair FBAM with Sample Conditional Layer Normalization, which infers n
ormalization parameters from intrinsic signal statistics rather than subject ide
ntity, stabilizing cross-subject representations.

Extensive experiments on six datasets demonstrate that BioFormer outperforms 12 baselines, yielding absolute F1-score improvements of 6\%.
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What Makes Synthetic Data Effective in Image Segmentation
Jinjin Zhang, Xiefan Guo, Yizhou jin, Nan Zhou, Di Huang
Driven by rapid advances in large-scale generative models, synthetic data has emerged as a promising solution for visual understanding. While modern diffusion models achieve remarkable photorealistic image synthesis, their potential in complex visual segmentation tasks remains underexplored. In this work, we conduct a systematic analysis of synthetic images from state-of-the-art diffusion models to uncover the factors governing their utility. In particular, synthetic images characterized by dense scene composition and fine instance fidelity demonstrate distinctive benefits, yielding significantly more discriminative spatial representations. Building on these insights, we propose SENSE, a unified framework that leverages flexible and scalable synthetic data to substantially enhance segmentation performance. Notably, SENSE is model-agnostic, compatible with diverse architectures (e.g., DPT and Mask2Former), and scales effectively across models with varying parameter capacities. Extensive experiments on Cityscapes, COCO, and ADE20K validate the effectiveness and generalization capability of our approach. Code is available at https://github.com/zhang0jhon/SENSE.
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Unveiling Multi-regime Patterns in SciML: Distinct Failure Modes and Regime-specific Optimization
Yuxin Wang, Yuanzhe Hu, Xiaokun Zhong, Xiaopeng Wang, Haiquan Lu, Tianyu Pang, Michael W. Mahoney, Yujun Yan, Pu Ren, Yaoqing Yang
Neural networks trained under different hyperparameter settings can fall into distinct training "regimes," with consistent behavior within regimes and qualitative differences across regimes. In this paper, we study such multi-regime behavior in scientific machine learning (SciML) models through a *regime-aware diagnostic framework* that jointly analyzes performance, training dynamics, and loss-landscape geometry. We identify three key findings: (i) a consistent three-regime structure emerges across many standard SciML models, different constraint enforcements, and various optimizer designs; (ii) optimization effectiveness is regime-specific, with no single method performing well across all regimes; and (iii) SciML models can exhibit fine-grained failure modes that can challenge conventional interpretations of standard loss-landscape metrics. Our results provide an approach to establish a unified, task-oblivious perspective on failure modes in SciML and to inform regime-aware guidance for improving robustness. We validate these findings across widely-used SciML models, including physics-informed neural networks, neural operators, and neural ordinary differential equations, on benchmarks spanning representative ordinary and partial differential equations.
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Distributionally Robust Reinforcement Learning from Human Feedback
Debmalya Mandal, Paulius Sasnauskas, Goran Radanovic
Reinforcement learning from human feedback (RLHF) has evolved to be one of the main methods for fine-tuning large language models (LLMs). However, existing RLHF methods are non-robust, and their performance deteriorates if the downstream task differs significantly from the preference dataset used in fine-tuning. In order to mitigate this problem, we introduce a distributionally robust RLHF for fine-tuning LLMs. In particular, our goal is to ensure that a fine-tuned model retains its performance even when the distribution of prompts significantly differs from the distribution encountered during fine-tuning. We formulate distributionally robust optimization (DRO) version of two popular fine-tuning methods -- (1) reward-based RLHF and (2) reward-free DPO (direct preference optimization). We propose a minibatch gradient descent based algorithms for both of them, and theoretically prove convergence guarantees for the algorithms. Subsequently, we evaluate our algorithms on an out-of-distribution (OOD) task by first training the model on the Unified-Feedback dataset and evaluating its performance on two different datasets. The experimental results show that our robust training improves the accuracy of the learned reward models on average, and markedly on some tasks,

such as reasoning. Furthermore, we show that the robust versions of policy optim
ization methods, similarly improve performance on OOD tasks.
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OcclusionFormer: Arranging Z-Order for Layout-Grounded Image Generation
Ziye Li, Henghui Ding
Recent layout-to-image models have achieved remarkable progress in spatial contr
ollability. However, they still struggle with inter-object occlusion. When bound
ing boxes overlap, most existing methods lack explicit occlusion information, wh
ich makes the generation in intersection regions inherently ambiguous and hinder
s the determination of complex occlusion relationships. As a result, they often
produce entangled textures or physically inconsistent layering in the overlapped
 areas. To address this issue, we first construct ***SA-Z***, a large-scale data
set enriched with explicit occlusion ordering and pixel-level annotations. Build
ing upon our proposed dataset, we introduce ***OcclusionFormer***, a novel occlu
sion-aware Diffusion Transformer framework that explicitly models Z-order priori
ty by decoupling instances and compositing them via volume rendering. Furthermor
e, to ensure fine-grained spatial precision, we introduce a queried alignment lo
ss that explicitly supervises individual instances and enhances semantic consist
ency. The proposed method effectively reduces ambiguity in overlapping regions,
enforces correct occlusion dependencies, and preserves structural integrity, lea
ding to substantial accuracy gains across diverse scenes.
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Tailoring the Training: Difficulty-Aware Learning Strategy Allocation for Large
Language Models
Xiaoling Zhou, Shuaiyu Zhou, Zhemg Lee, Tao Chen, Xirui Li, Peng Chen, Jie Jiang
, Wei Ye, Shikun Zhang
Although reinforcement learning (RL) enhances the reasoning capabilities of larg
e language models (LLMs), it is primarily learned from the model's self-generate
d distribution, limiting its ability to acquire reasoning skills beyond its init
ial knowledge. To overcome this, we propose a Difficulty-Aware Learning Strategy
 Allocation (DALSA) framework, which adaptively assigns appropriate learning str
ategies to samples based on their difficulty signals. DALSA is built on the key
insight that samples beyond models' knowledge scope are better addressed through
 supervised fine-tuning (SFT), while those within the boundary but insufficientl
y mastered benefit more from RL, and well-learned samples are discarded to avoid
 redundant updates. To realize this principle, we extract a series of difficulty
-aware training characteristics and employ a learnable strategy allocator to dyn
amically determine the optimal learning strategy for each sample based on its tr
aining dynamics. The allocator and the LLM are alternately optimized, enabling a
daptive strategy allocation. Furthermore, two regularization techniques, anti-cu
rriculum weighting and adversarial label smoothing, are integrated to alleviate
the inherent limitations of RL and SFT, backed by comprehensive theoretical anal
yses. Extensive experiments on ten LLMs ranging from 1.5B to 70B across various
tasks indicate that DALSA consistently outperforms baselines under both full and
 parameter-efficient fine-tuning settings.
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Variable-Length Tokenization via Learnable Global Merging for Diffusion Transfor
mers
Dong Hoon Lee, Seunghoon Hong
Latent Diffusion Models (LDMs) have become dominant in visual synthesis, but the
ir quality–compute trade-off is largely constrained by the tokenizer’s fixed com
pression ratio.
Variable-length tokenizers (VLTs) promise adaptive compression by varying token
counts, allowing diffusion models to flexibly balance quality and compute.
However, conventional VLTs modulate length by truncating ordered token sequences
, which makes token semantics depend on token position and breaks representation
al alignment across lengths.
This leads to a cross-length shift in the latent distribution that hinders a sin
gle variable-length diffusion model from operating effectively.
To address this, we propose a novel variable-length tokenizer that modulates len

gth by merging tokens.
We show that encouraging similar tokens to merge enables direct cross-length rep
resentation alignment when the diffusion transformer operates according to the m
erging pattern.
Since conventional merging methods are data-dependent, making the merging patter
n inaccessible during generation, we introduce learnable global merging, which i
s data-independent, to ensure compatibility with diffusion transformers.
On ImageNet 256$\times$256 generation, our merging-based variable-length tokeniz
er integrated with a diffusion transformer achieves a superior gFID–compute trad
e-off compared to prior VLT methods.
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1-Bit Wonder: Improving QAT Performance in the Low-Bit Regime through K-Means Qu
antization
Sohir Maskey, Constantin Eichenberg, Johannes Messner, Douglas Orr
Quantization-aware training (QAT) is an effective method to drastically reduce t
he memory footprint of LLMs while keeping performance degradation at an acceptab
le level. However, the optimal choice of quantization format and bit-width prese
nts a challenge in practice. The full design space of quantization is not fully
explored in the context of QAT, and the precise trade-off between quantization a
nd downstream performance is poorly understood, as comparisons often rely solely
 on perplexity-based evaluations. In this work, we address these shortcomings wi
th an empirical study of QAT in the low-bit regime. We show that k-means based w
eight quantization outperforms integer formats and can be implemented efficientl
y on standard hardware. Furthermore, we find that, under a fixed inference memor
y budget, the best performance on generative downstream tasks is achieved with $
1$-bit quantized weights.
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Reward-Preserving Counterfactual State Editing for Offline Reinforcement Learnin
g
Siyu Wang, Xiaocong Chen, Mingming Gong, Yong Li, Quan Z. Sheng, Lina Yao
Transformer sequence models such as Decision Transformer can learn strong offlin
e policies from logged trajectories, but they often suffer from causal confusion
: reliance on spurious correlations that predict reward in the data but do not r
eflect the true causal mechanisms of the environment. We propose CSET (Counterfa
ctual State Editing Transformer), which improves robustness in strictly offline
reinforcement learning without learning environment transition dynamics. On the
data side, CSET fits a causal reward model as a conditional variational autoenco
der and uses a counterfactual state generator to propose minimally edited observ
ations whose predicted reward matches the factual reward, under a normalized mov
e-band constraint and an acceptance gate that enforce plausibility and reward co
nsistency; augmentation replaces only the observation token to avoid synthetic s
uccessor transitions. On the model side, CSET uses a causally structured hybrid
transformer: modality-specific convolutional encoders process return-to-go, stat
e, and action streams, and a final attention block is softly supervised so actio
n prediction focuses on its direct causal parents. Experiments on D4RL locomotio
n, AntMaze, and offline recommendation benchmarks show consistent gains within t
he DT family, and CSET remains substantially more robust than strong value-based
 and DT baselines under injected spurious distractors.
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OMAC: A Holistic Optimization Framework for LLM-Based Multi-Agent Collaboration
Shijun Li, Hilaf Hasson, Joydeep Ghosh
Agents powered by advanced large language models (LLMs) have demonstrated impres
sive capabilities across diverse complex applications. Recently, Multi-Agent Sy
stems (MAS), wherein multiple agents collaborate and communicate with each other
, have exhibited enhanced capabilities in complex tasks, such as high-quality co
de generation and arithmetic reasoning. However, the development of such systems
 often relies on handcrafted methods, and the literature on systematic design an
d optimization of LLM-based MAS remains limited.
In this work, we introduce OMAC, a general framework designed for holistic optim
ization of LLM-based MAS. Specifically, we identify five key optimization dimens

ions for MAS, encompassing both agent functionality and collaboration structure. Building upon these dimensions, we first propose a general algorithm, utilizing two actors termed the Semantic Initializer and the Contrastive Comparator, to optimize any single dimension. Then, we present an algorithm for joint optimization across multiple dimensions. Extensive experiments demonstrate the superior performance of OMAC on diverse tasks against recent approaches. Codes are available at: https://github.com/xiwenchao/OMAC.

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Reasoning as an Attack Surface: Adaptive Evolutionary CoT Jailbreaks for LLMs

Jianan Li, Simeng Qin, Xiaojun Jia, Lionel Z. WANG, Tianhang Zheng, Xiaoshuang Jia, Yang Liu, Xiaochun Cao

Large Reasoning Models (LRMs) have demonstrated remarkable capabilities in reasoning and generation tasks and are increasingly deployed in real-world applications. However, their explicit chain-of-thought (CoT) mechanism introduces new security risks, making them particularly vulnerable to jailbreak attacks. Existing approaches often rely on static CoT templates to elicit harmful outputs, but such fixed designs suffer from limited diversity, adaptability, and effectiveness. To overcome these limitations, we propose an adaptive evolutionary CoT jailbreak framework, called AE-CoT. Specifically, the method first rewrites harmful goals into mild prompts with teacher role-play and decomposes them into semantically coherent reasoning fragments to construct a pool of CoT jailbreak candidates. Then, within a structured representation space, we perform multi-generation evolutionary search, where candidate diversity is expanded through fragment-level crossover and a mutation strategy with an adaptive mutation-rate control mechanism. An independent scoring model provides graded harmfulness evaluations, and high-scoring candidates are further enhanced with a harmful CoT template to induce more destructive generations. Extensive experiments across multiple models and datasets demonstrate the effectiveness of the proposed AE-CoT, consistently outperforming state-of-the-art jailbreak methods.

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EPiC: Efficient Video Camera Control Learning with Precise Anchor-Video Guidance

Zun Wang, Jaemin Cho, Jialu Li, Han Lin, Jaehong Yoon, Yue Zhang, Mohit Bansal

Recent approaches for video generation with camera control often create anchor videos
(i.e., rendered videos that approximate desired camera motions)
to guide diffusion models as a structured prior,
by rendering from estimated point clouds following camera trajectories.
However, errors in point cloud and camera trajectory estimation often lead to inaccurate anchor videos with
higher training cost and low efficiency, as the model is forced to compensate for rendering misalignments.
To address these limitations, we introduce EPiC, an efficient and precise camera control learning framework
that constructs well-aligned training anchor videos
without the need for camera pose or point cloud estimation.
Concretely, we create highly precise anchor videos by masking source videos based on first-frame visibility, which ensures strong alignment, eliminates the need for camera/point cloud estimation, and thus can be readily applied to any in-the-wild video.
Furthermore, we introduce Anchor-ControlNet, a lightweight module that integrates anchor video guidance in visible regions to pretrained video diffusion models,
with less than 1\% of additional parameters.
EPiC achieves efficient training with substantially fewer parameters, training steps, and less data, and
generalizes robustly to anchor videos made with point clouds at test time, enabling precise 3D-informed camera control.
EPiC achieves SoTA performance on RealEstate10K and MiraData for I2V camera control task.
Notably, EPiC also exhibits strong zero-shot generalization to video-to-video (V2V) scenarios.

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Bregman meets Lévy: Stochastic Mirror Descent with Heavy-Tailed Noise in Continuous and Discrete Time
Pierre-Louis Cauvin, Panayotis Mertikopoulos
We study the robustness of stochastic mirror descent (SMD) under heavy-tailed noise, focusing on whether the method retains its convergence guarantees when run with infinite-variance stochastic gradient input. To address this question in a principled manner, we begin by introducing a continuous-time model of SMD as a stochastic differential equation (SDE) driven by a centered Lévy noise process with finite p-th order moments, $1 < p \leq 2$. This scheme—which we call the Lévy mirror flow (LMF)—arises naturally as the scaling limit of SMD in the presence of heavy-tailed noise. In particular, when $p < 2$—the heavy noise regime—the trajectories of LMF generically exhibit jump discontinuities of arbitrary magnitude which, if frequent enough, lead to infinite variance. Nonetheless, despite this highly singular behavior, we show that LMF attains ϵ-optimality within $\mathcal{O}(\epsilon^{-p/(p-1)})$ time in the convex case, and within $\tilde{\mathcal{O}}(\epsilon^{-1/(p-1)})$ time for (relatively) strongly convex objectives. These guarantees provide a transparent characterization of the impact of frequent long jumps on the convergence of the process, and percolate to a series of matching discrete-time guarantees for several variants of SMD under heavy-tailed noise.
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Personalized Policy Learning through Discrete Experimentation
Zhiqi Zhang, Zhiyu Zeng, Ruohan Zhan, Dennis Zhang
While Randomized controlled trials (RCTs), or A/B tests, are the gold standard for optimizing online-platform policies, they are limited by discrete testing levels. This approach is suboptimal for continuous variables (e.g., prices and incentives), as it fails to extrapolate to untested values or account for user heterogeneity. We address this by developing Deep Learning for Policy Targeting (\textsf{DLPT}) to learn personalized continuous policies from discrete RCTs using high-dimensional features. We prove our estimators are asymptotically unbiased and consistent, achieving a $\sqrt{n}$-regret bound. In a collaboration with a leading social media platform to optimize creator incentives, we show that \textsf{DLPT} substantially outperforms existing benchmarks.
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VLAW: Iterative Co-Improvement of Vision-Language-Action Policy and World Model
Yanjiang Guo, Tony Lee, Lucy Xiaoyang Shi, Jianyu Chen, Percy Liang, Chelsea Finn
The goal of this paper is to improve the performance and reliability of vision-language-action (VLA) models through iterative online interaction.
Since collecting policy rollouts in the real world is expensive, we investigate whether a learned simulator—specifically, an action-conditioned video generation model—can be used to generate additional rollout data.
Unfortunately, existing world models lack the physical fidelity necessary for policy improvement: they are predominantly trained on demonstration datasets that lack coverage of many different physical interactions (particularly failure cases) and struggle to accurately model small yet critical physical details in contact-rich object manipulation.
We propose a simple iterative improvement algorithm that uses real-world roll-out data to improve the fidelity of the world model, which can then, in turn, be used to generate supplemental synthetic data for improving the VLA model.
In our experiments on a real robot, we use this approach to improve the performance of a state-of-the-art VLA model on multiple downstream tasks.
We achieve a 39.2\% absolute success rate improvement over the base policy and 11.6\% improvement from training with the generated synthetic rollouts. Videos can be found at this anonymous website: \url{https://sites.google.com/view/vla-w}.
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Learning Adaptive Topology with FiLM-Guided Distillation for Tertiary Structure-Based RNA Design
Zixun Zhang, Yuncheng Jiang, Yuzhe Zhou, Jiayou Zheng, Shuguang Cui, Zhen Li

Tertiary structure-based RNA design aims to generate RNA sequences that can fold into desired 3D structures, but remains a challenging problem due to the scarcity of annotated data, structural noise, and the intrinsic complexity of RNA topology. Existing structure-to-sequence frameworks largely rely on static k-nearest neighbor graphs and rigid message passing schemes, which fail to capture the flexible and heterogeneous nature of RNA geometry. To address these issues, we propose a unified framework, ATL-FGD, that integrates Adaptive Topology Learning (ATL) and FiLM-Guided Distillation (FGD) for robust RNA design. ATL introduces a differentiable edge gating mechanism to jointly learn topology and representation, enabling the model to construct data-driven, layer-adaptive graphs that better reflect structural dynamics and biochemical consistency. On top of this, FGD bridges structural and sequence representations via feature-wise linear modulation, softly transferring the semantic knowledge from RNA foundation models without relying on them during inference. Extensive experiments on tertiary structure-based RNA design benchmarks demonstrate that our approach achieves significant improvements in both sequence recovery and structural fidelity.

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Understanding Generalization and Forgetting in In-Context Continual Learning

Guangyu Li, Meng Ding, Lijie Hu

In-context learning (ICL) derives its power from enabling Large Language Models to adapt to new tasks via prompt-based reasoning alone, entirely bypassing the need for parameter updates. Existing theories primarily study ICL in single-task settings, while real-world prompts often contain sequences of heterogeneous tasks, leaving a gap in understanding whether Large Language Models implicitly perform continual learning during inference. To bridge this gap, we propose the first theoretical framework for in-context continual learning, modeling how a pretrained Transformer processes multiple sequential tasks within a single prompt through shared attention mechanisms. Focusing on linear and masked linear self-attention, we derive error expressions for model predictions under sequential task prompts and analyze their generalization and forgetting behavior. Our results reveal that standard attention mechanisms inevitably induce inter-task interference by uniformly or causally aggregating historical contexts, leading to systematic bias. We further provide a bias–variance–interference decomposition of prediction error, characterizing when historical in-context information yields positive transfer or provable negative transfer. This analysis exposes fundamental limits of attention-based continual inference and offers theoretical explanations for order sensitivity and performance degradation in long prompts.

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The ACUTE Protocol: Operationalizing Language Model Activations for Better Calibration, Utility, and Trust

Nishant Subramani, Palash Goyal, Yiwen Song, Mani Malek, Yuan Xue, Tomas Pfister, Hamid Palangi

As language models improve and become increasingly deployed to solve a variety of tasks, trustworthiness becomes essential. Calibration is a good proxy for trust: well-calibrated confidence estimates help inform the risk versus reward trade-off when trusting a specific model output. Unfortunately, even as models improve, they remain poorly calibrated, often biasing towards overconfidence. Additionally, calibration can be gamed: a policy that always predicts the base rate is perfectly calibrated, but completely uninformative. To resolve this, we develop a new metric, **expected utility renormalized by the oracle (EURO)**, that balances calibration and informativeness. We also propose a general-purpose **activation-based confidence, utility, and trust estimation protocol (ACUTE)** to appropriately adjudicate uncertainty. The ACUTE protocol provides flexible, sample-efficient, and compute-efficient confidence estimators for 3 tasks including multiple choice question answering, tool-calling, and scientific document summarization across 6 models from 4 model families. ACUTE outperforms strong baselines on EURO, while maintaining low calibration error. Taken together, our work shows that equipping LLMs with the ACUTE protocol can improve calibration, utility, and trustworthiness in numerous settings.

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From Teacher Pathways to Invariant Manifolds: Consensus Subspace Distillation fo r TSFMs
Zexing Zhang, Tianyang Lei, Jichao Li, Yang Kewei
Time-series foundation models (TSFMs) deliver strong cross-domain generalization , but their scale makes deployment costly. Knowledge distillation is a natural c ompression route, yet prior TSFM distillation typically imitates teacher outputs , features, or pairwise relations, and therefore remains tightly coupled to teac her-specific training trajectories while underutilizing two empirical properties : (i) high-level representations across model scales tend to converge toward a s hared, approximately low-rank geometry, and (ii) layer-wise utility follows a lo ng-tail pattern. We propose consensus subspace distillation, which reframes dist illation as aligning a student to a model-agnostic geometric object: a scale-inv ariant low-rank consensus subspace together with its center statistics. Offline, we screen high-contribution layers via drop-layer marginal loss, estimate a shr inkage-stabilized covariance from their embeddings, and derive a truncated eigen subspace that defines a consensus projector. Online, we project student embeddin gs into this subspace and match the teacher’s projected mean and covariance usin g a lightweight mean--covariance objective, enabling stable optimization without rigid pointwise feature binding. To mitigate subset-induced bias, we further in troduce a frequency-domain uncertainty injection mechanism that inflates spectra l density based on characteristic-function discrepancies and injects dispersion only within the consensus directions. Across forecasting and imputation, the dis tilled student matches or slightly improves upon the teacher, while exhibiting a predictable trade-off under strict zero-shot classification. With MOMENT-Large as teacher, we achieve about 90% parameter reduction and substantial distillatio n-time savings while retaining comparable performance across multiple time-serie s tasks. Code and compressed weights are available at anonymous.4open.science/r/ CSD-13C3/.
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Are Tools Always Beneficial? Learning to Invoke Tools Adaptively for Dual-Mode M ultimodal LLM Reasoning
Qinghe Ma, Zhen Zhao, Yiming Wu, Jian Zhang, LEI BAI, Yinghuan Shi
Tool-augmented reasoning has emerged as a promising direction for enhancing the reasoning capabilities of multimodal large language models (MLLMs). However, exi sting studies mainly focus on enabling models to perform tool invocation, while neglecting the necessity of invoking tools. We argue that tool usage is not alwa ys beneficial, as redundant or inappropriate invocations largely increase reason ing overhead and even mislead model predictions. To address this issue, we intro duce AutoTool, a model that adaptively decides whether to invoke tools according to the characteristics of each query. Within a reinforcement learning framework , we design an explicit dual-mode reasoning strategy with mode-specific reward f unctions to guide the model toward producing accurate responses. Moreover, to pr event premature bias toward a single reasoning mode, AutoTool jointly explores a nd balances tool-assisted and text-centric reasoning throughout training, and pr omotes free exploration in later stages. Extensive experiments demonstrate that AutoTool exhibits outstanding performance and high efficiency, yielding a 21.8\% accuracy gain on V* benchmark compared to the base model, and a 44.9\% improvem ent in efficiency over existing tool-augmented methods on POPE benchmark. Code i s available at https://github.com/MQinghe/AutoTool.
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Invariant Representation Learning for Source-Free Time Series Forecasting with L LM-Centric Proxy Denoising
Kangjia Yan, Chenxi Liu, Hao Miao, Xinle Wu, Yan Zhao, Chenjuan Guo, Bin Yang
Effective time series forecasting enables various real-world applications, benef iting from the proliferation of mobile devices. However, the volume of time seri es data may vary significantly across domains due to high data acquisition costs and data regulations. To maximally create value from sparse data, this study fo cuses on a new problem of source-free time series forecasting, aiming to adapt a pretrained model from sufficient source time series to the sparse target time s eries without access to the source data, enabling data protection. To achieve th

is, we propose TimeID, a novel source-free time series forecasting framework wit
h a large language model (LLM) centric proxy denoising inspired by the powerful
generalization capabilities of LLMs. Specifically, TimeID consists of three key
components: (1) dual-branch invariant disentangled feature learning that enforce
s representation- and gradient-wise invariance by means of season-trend decompos
ition; (2) lightweight, parameter-free proxy denoising that dynamically calibrat
es systematic biases of LLMs; and (3) knowledge distillation that bidirectionall
y aligns the denoised prediction and the original target prediction. Extensive e
xperiments on real-world datasets demonstrate that TimeID outperforms state-of-t
he-art baselines, improving MSE and MAE by 10.7\% and 9.3\% on average. The code
 is available at https://github.com/decisionintelligence/TimeID.
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LagLLM: LLM-empowered lead–lag dependency learning for spatial-temporal time ser
ies forecasting

Binqing Wu, Jian Zhou, Zongjiang Shang, Ling Chen

Spatial-temporal time series forecasting is challenging due to complex lead-lag
dependencies, which are often ignored or inadequately modeled by existing method
s. Thus, we propose LagLLM, the first LLM-empowered framework that explicitly mo
dels lead–lag dependencies by unifying data-driven dynamics modeling and knowled
ge-driven semantic reasoning. Specifically, LagLLM constructs a lead-lag graph b
y integrating learnable embeddings, spatial proximity, and prompt-guided reasoni
ng from a frozen LLM, which can capture lead-lag dependencies informed by the un
derlying data structure and semantic knowledge.

In addition, LagLLM introduces structural token sorting based on the graph, whic
h can make a fine-tuned LLM explicitly perceive directional and delayed interact
ions. Experiments on eight real-world datasets show that LagLLM achieves the sta
te-of-the-art performance with improved accuracy, robustness, and interpretabili
ty. The code is available at https://github.com/w2obin/LagLLM-2026.
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Deterministic Differentiable Structured Pruning for Large Language Models

Weiyu Huang, Pengle Zhang, Xiaolu Zhang, JUN ZHOU, Jun Zhu, Jianfei Chen

Structured pruning reduces LLM inference cost by removing low-importance archite
ctural components. This can be viewed as learning a multiplicative gate for each
 component under an ℓ_0 sparsity constraint. Due to the discreteness of the
 ℓ_0 norm, prior work typically adopts stochastic hard-concrete relaxations
 to enable differentiable optimization; however, this stochasticity can introduc
e a train--test mismatch when sampled masks are discretized for deployment and r
estricting masks to a bounded, near-binary range. To address this, we propose De
terministic Differentiable Pruning (DDP), a mask-only optimization method that e
liminates stochasticity by directly optimizing a deterministic soft surrogate of
 the discrete ℓ_0 objective. Compared with prior approaches, DDP offers gre
ater expressiveness, reduced train--test mismatch, and faster convergence. We ap
ply our method to several dense and MoE models, including Qwen3-32B and Qwen3-30
B-A3B, achieving a performance loss as small as 1\% on downstream tasks while ou
tperforming previous methods at 20\% sparsity. We further demonstrate end-to-end
 inference speedups in realistic deployment settings with vLLM.
****

Consistent Diffusion Language Models

Hasan Amin, Yuan Gao, Yaser Souri, Subhojit Som, Ming Yin, Rajiv Khanna, Xia Son
g

Diffusion language models (DLMs) are an attractive alternative to autoregressive
 models because they promise sublinear-time, parallel generation, yet practical
gains remain elusive as high-quality samples still demand hundreds of refinement
 steps. In continuous domains, consistency training along the probability-flow O
DE is a popular recipe to accelerate diffusion. For discrete diffusion, no analo
gous sample-space ODE exists, making direct adaptation ill-defined. We argue tha
t the right discrete substitute is the exact posterior bridge—the closed-form co
nditional law linking any two noise levels—which is available for broad corrupti
ons including masked and uniform diffusion. Building on this observation, we int
roduce Multi-Path Discrete Consistency (MPDC), a new principle that trains a den

oiser to be path-invariant in expectation across these stochastic bridges, and i
nstantiate it as the Consistent Diffusion Language Model (CDLM), a single-stage
training framework that does not require an already trained teacher model. Our C
DLM objective recovers masked diffusion, continuous consistency models, and prog
ressive or discrete distillation as analytic limits or empirical approximations
of one common view. Empirically, CDLM establishes a new state of the art on both
 conditional and unconditional text-generation, consistently outperforming stron
g base discrete diffusion models and often even multi-stage distilled baselines
across sampling budgets, with the largest gains in the few-step regime. Together
, these results position CDLM as a principled and scalable foundation for the ne
xt generation of fast, high-fidelity discrete generative modeling.
****
Universal Representation of Generalized Convex Functions and their Gradients
Moeen Nehzati
A wide range of optimization problems can often be written in terms of generaliz
ed convex functions (GCFs). When this structure is present, it can convert certa
in nested bilevel objectives into single-level problems amenable to standard fir
st-order optimization methods. We provide a new differentiable layer with a conv
ex parameter space and show (Theorems 5.1 and 5.2) that it and its gradient are
universal approximators for GCFs and their gradients. We demonstrate how this pa
rameterization can be leveraged in practice by (i) learning optimal transport ma
ps with general cost functions and (ii) learning optimal auctions of multiple go
ods. In both these cases, we show how our layer can be used to convert the exist
ing bilevel or min-max formulations into single-level problems that can be solve
d efficiently with first-order methods.
****
Primal-Spectral Generative Modeling: Fast Analytical Generation via Pseudoinvers
e Lévy Inversion
Zhiyuan Ouyang, Simei Huang, Zuokai Wen, Xiangyun Zhang, Junchi Yan
A probability distribution $\mathbb{P}$ is a measure on a σ-algebra, assi
gning mass to sets rather than points, which poses a challenge for the training
of neural networks that often struggle to reconstruct the global topology of con
tinuous manifolds from sparse samples. We mitigate this issue by transforming $\
mathbb{P}$ into a continuous function via spectral methods, providing theoretica
l guarantees for the convergence of the learned distribution to the true distrib
ution. Specifically, we introduce a network, PriSpecNet, with a single-function
evaluation (1-NFE) Pseudoinverse L\'evy Inversion (PiLI) solver that regards gen
eration as a fast analytical problem, eliminating the need for iterative numeric
al integration while maintaining full compatibility with the stochastic interpol
ants. We test our PriSpecNet in two applications: for time series, it unifies ge
neration and forecasting, outperforming state-of-the-art (SOTA) baselines with C
ontext-FID reductions of 50.0\%, 41.5\%, 80.6\%, and 63.1\% on Sines, Solar, ETT
h, and Stock benchmarks, respectively, also decreasing forecasting MSE by 29.8\%
 on Solar and 23.8\% on Stock. For ImageNet 256×256, 1-NFE PiLI achieve
s a competitive FID of **1.66** using only **26** Gflops, representing a **170$\
times$** reduction in total Gflops compared to the 4,436 Gflops required by the
25-NFE DPM-Solver++.
****
DRPBench: Evaluating LLMs in Concurrent Code Comprehension via Fine-Grained Data
 Race Prediction
Yuqi Guo, Siwei Wei, Yan Cai
Large Language Models (LLMs) have demonstrated sophisticated comprehension of se
quential code, yet their capacity for reasoning about concurrent programs remain
s largely unquantified. We introduce DRPBench, a benchmark designed to evaluate
the concurrent code comprehension of LLMs by measuring their data race predictio
n performance. To address the challenge of runtime non-determinism for evaluatio
n on concurrent programs, we frame the evaluation as a fine-grained static predi
ction task using 1,003 programs from the SV-COMP suite, featuring 557 manually a
nnotated data races with precise variable- and line-level granularity. Our evalu
ation of 15 state-of-the-art LLMs—spanning standard, reasoning, and agentic vari

ants—reveals that DRPBench effectively differentiates concurrent code comprehens ion capabilities of LLMs. While the top-performing model (Gemini 3 with test-tim e reasoning) achieves an F1 score of 74.89%, most models struggle significantly (scoring less than 60%), with Llama 3 70B achieving only 8.80%. Beyond benchmark ing, we characterize two primary failure modes: (1) shared-variable distraction, where multiple variable appearances degrade comprehension accuracy, and (2) syn chronization-logic myopia, the inability to interpret non-standard synchronizati on implementations. Our findings provide a diagnostic roadmap for enhancing conc urrent code comprehension of LLMs in future development.
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SciPredict: Can LLMs Predict the Outcomes of Scientific Experiments in Natural S ciences?

Udari Madhushani Sehwag, Elaine Lau, Haniyeh Ehsani Oskouie, Shayan Shabihi, Eri ch Liang, Andrea Sarai Echeverria Toledo, Guillermo A. Mangialardi, Sergio Fonro uge, Ed-Yeremai Hernandez-Cardona, Paula Vergara, Utkarsh Tyagi, Chen Bo Calvin Zhang, Pavi Bhatter, Nicholas E. Johnson, Furong Huang, Ernesto Gabriel Hernánde z Montoya, Bing Liu

Accelerating scientific discovery requires the identification of which experimen ts would yield the best outcomes before committing resources to costly physical validation. While existing benchmarks evaluate LLMs on scientific knowledge and reasoning, their ability to predict experimental outcomes---a task where AI coul d significantly exceed human capabilities---remains largely underexplored.

We introduce SciPredict, a benchmark comprising 405 tasks derived from recent em pirical studies in 33 specialized sub-fields of physics, biology, and chemistry. SciPredict addresses two critical questions: (a) *can LLMs predict the outcome of scientific experiments with sufficient accuracy?* and (b) *can such predictio ns be reliably used in the scientific research process?* Evaluations reveal fund amental limitations on both fronts. Model accuracies are 14-26\% and human expe rt performance is $\approx$20\%. Although some frontier models exceed human perf ormance model accuracy is still far below what would enable reliable experimenta l guidance. Even within the limited performance, models fail to distinguish reli able predictions from unreliable ones, achieving only $\approx$20\% accuracy reg ardless of their confidence or whether they judge outcomes as predictable withou t physical experimentation. Human experts, in contrast, demonstrate strong calib ration: their accuracy increases from $\approx$5\% to $\approx$80\% as they deem outcomes more predictable without conducting the experiment. SciPredict establi shes a rigorous framework demonstrating that superhuman performance in experimen tal science requires not just better predictions, but better awareness of predic tion reliability. For reproducibility all our data and code are provided at http s://anonymous.4open.science/r/SciPredict-AI01.
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MLUBench: A Benchmark for Lifelong Unlearning Evaluation in MLLMs

He Li, Haoang Chi, Qizhou Wang, Yunxin Mao, Zhiheng Zhang, Jie Tan, Tongliang Li u, Wenjing Yang, Bo Han

Multimodal large language models (MLLMs) are trained on massive multimodal data, making data unlearning increasingly important as data owners may request the re moval of specific content. In practice, these requests often arrive sequentially over time, giving rise to the challenging problem of *MLLM Lifelong Unlearning* .

However, most existing benchmarks are limited in scale and scope, failing to c apture the complexities of MLLM lifelong unlearning.

To fill this gap, we introduce the MLUBench, a large-scale and comprehensive b enchmark featuring 127 entities across 9 classes under lifelong unlearning reque sts.

We perform extensive experiments using MLUBench and reveal that existing unlea rning methods suffer from severe, cumulative degradation.

More critically, we further identify the unique challenge of this problem: unl ike in unimodal models, MLLM lifelong unlearning is constrained by the need to p reserve multimodal alignment. Continually unlearning from one modality could deg rade the entire model.

To alleviate this challenge, we propose LUMoE, an effective and efficient method.

Experiments demonstrate that LUMoE significantly mitigates the degradation problem faced by baselines.

The source code and the MLUBench dataset are open-sourced in this [URL](https://github.com/lihe-maxsize/Lifelong_Unlearning_main).

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Dual-Calibration Multi-View Clustering via Compact Anchor Learning

Huibing Wang, Yuemeng Huang, Yawei Chen, Jiaxin Yang, Qian Liu, Jinjia Peng, Zetian Mi, Ximing Li

The anchor-based multi-view clustering methods have received extensive attention due to their efficiency and scalability in large-scale data scenarios. Existing anchor-based methods still face challenges in learning compact and semantically discriminative anchors. Current mainstream approaches typically rely on random sampling strategies or orthogonal constraints for anchor selection and learning. However, they often optimize anchor learning and cluster assignment in a relatively separate manner, leaving the clustering semantics in the sample space insufficiently exploited for calibrating the anchor space. As a result, the learned anchors may suffer from redundant coverage and ambiguous cluster boundaries. Unlike existing anchor-based multi-view clustering methods, this paper proposes a Dual-Calibration Multi-view Clustering via Compact Anchor Learning (DCMC), which effectively improves anchor quality through a dual-space alignment mechanism. Specifically, DCMC initializes view-specific anchors to capture the underlying data distribution, and then enforces bidirectional consistency between the anchor space and the clustering space to jointly optimize both the sample-to-anchor assignments and the cluster assignments. The alternating optimization process effectively enhances cross-view semantic consistency while preserving the discriminative characteristics of each view. Experimental results demonstrate that DCMC outperforms state-of-the-art methods across multiple benchmark tests, confirming its effectiveness and reliability.

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Flow-Based Density Ratio Estimation for Intractable Distributions with Applications in Genomics

Egor Antipov, Alessandro Palma, Lorenzo Consoli, Stephan Günnemann, Andrea Dittadi, Fabian J Theis

Estimating density ratios between pairs of intractable data distributions is a core problem in probabilistic modeling, enabling principled comparisons of sample likelihoods under different data-generating processes across conditions. While exact-likelihood models such as normalizing flows offer a promising approach to density ratio estimation, naive evaluations are computationally expensive and prone to discretization errors because they require simulating each distribution's likelihood independently. In this work, we leverage condition-aware flow matching to derive a single dynamical formulation for tracking density ratios along generative trajectories. We demonstrate competitive performance on simulated benchmarks for closed-form ratio estimation, and show that our method supports versatile tasks in single-cell genomics data analysis, where likelihood-based comparisons of cellular states across experimental conditions enable treatment effect estimation and batch correction evaluation.

****

NExT-Guard: Training-Free Streaming Safeguard without Token-Level Labels

Junfeng Fang, Nachuan Chen, Houcheng Jiang, Dan Zhang, Xiangnan He, Tat-Seng Chua, Xiang Wang

Large language models are increasingly deployed in streaming scenarios, rendering conventional post-hoc safeguards ineffective as they fail to interdict unsafe content in real-time. While streaming safeguards based on token-level supervised training could address this, they necessitate expensive annotations and suffer from severe overfitting. In this work, we challenge the paradigm that streaming safety must rely on token-level supervised training. Instead, it is an inherent capability of well-trained post-hoc safeguards, as they already encode token-level risk signals in hidden representations. Hence, we introduce NExT-Guard, a fra

mework that achieves streaming safeguards by monitoring interpretable latent features from Sparse Autoencoders (SAEs). It uses pretrained SAEs from publicly available base LLMs, enabling flexible, low-cost deployment without token-level supervision. Experimental results show that NExT-Guard outperforms both post-hoc and streaming safeguards based on supervised training, with superior robustness across models, SAE variants, and risk scenarios. These results make NExT-Guard a universal and scalable paradigm for real-time safety, accelerating the practical deployment of streaming safeguards.
****
How Far Can LLM Agents Reason with Tables? Benchmarking Multi-Turn Agentic Table Question Answering in the Wild
Jingwang Huang, Jie Zhang, Haoyang Zeng, Changzai Pan, Xianjie Wu, Guanting Dong, Jiaheng Liu, Wei Zhang, Mingyu Zheng, Chunxiao Liu, Kaiwen Wei, Jiang Zhong, Jian Yang
Recent advances in large language models (LLMs) have substantially expanded the scope of Table Question Answering (TableQA). However, existing benchmarks primarily treat TableQA as a passive, single-turn natural language understanding task, lacking the capacity to evaluate autonomous reasoning and tool-call trajectories in realistic, multi-turn scenarios.
To bridge this gap, we introduce TableAgent-Bench, a large-scale bilingual benchmark that reformulates TableQA as proactive, agentic interactions over structurally complex, multi-table environments. With a topology-aware construction strategy, TableAgent-Bench captures dynamic intent evolution through 1,310 multi-turn dialogues grounded in 2,275 industrial tables. Furthermore, we propose the Table-centric Agent Evaluation Framework (TAEF) to assess agent interactions with complex table structures. Specifically, TAEF integrates a specialized agent toolset and 4 metric categories to systematically diagnose intermediate failure modes, assessing performance across table localization, tool-invocation rationality, and trajectory-level pass rate. Extensive experiments with 25 state-of-the-art LLM agents reveal a substantial capability gap, with even the strongest model Gemini-3-Pro-Preview achieving only 53.4% information coverage. We expect TableAgent-Bench to serve as a rigorous testbed for developing and evaluating agents capable of robust table-centric reasoning.
****
Calibrating Generative Models to Distributional Constraints
Henry Smith, Nathaniel Lee Diamant, Brian L. Trippe
Generative models frequently suffer miscalibration, wherein statistics of the sampling distribution—such as the fraction of generations in a given class—deviate from desired values.
We frame calibration as a constrained optimization problem and seek the closest model in Kullback-Leibler divergence satisfying a calibration constraint.
To address the intractability of imposing these constraints exactly, we introduce two surrogate objectives for fine-tuning: (1) the relax loss, which replaces the constraint with a miscalibration penalty, and (2) the reward loss, which converts calibration into a reward fine-tuning problem.
We demonstrate that these approaches substantially reduce calibration error across hundreds of simultaneous constraints and models with up to nine billion parameters, spanning applications in protein design, image generation, and language modeling.
Code is available at https://github.com/smithhenryd/cgm.
****
On Expressive Power of Floating-Point Transformers
Sejun Park, Yeachan Park, Geonho Hwang
Research on the expressive power of transformers shows that transformers are equivariant to permutations and can approximate all permutation-equivariant continuous functions on a compact domain. However, these results assume real parameters and exact operations, whereas real-world implementations on computers can only use a finite set of numbers and inexact machine operations with round-off errors. In this work, we investigate the representability of floating-point transformers that use floating-point parameters and floating-point operations.

Unlike existing results under exact arithmetic, we first show that floating-poin
t transformers can represent non-permutation-equivariant functions even without
positional encoding. Furthermore, we prove that floating-point transformers can
represent all permutation-equivariant functions when the sequence length is boun
ded, but they cannot when the sequence length is large. We also identify the min
imal equivariance property in floating-point transformers, and show that all non
-trivial additive positional encoding can harm the representability of floating-
point transformers.
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Ranking Time Series using a Time Warping Ideal Point Model
Lucas Zoroddu, Pierre Humbert, Laurent Oudre
Expert-annotated time series datasets often suffer from low agreement, especiall
y in medical applications where decisions rely on subjective criteria and incons
istent thresholds. Such variability degrades annotation quality and thus limits
the reliability of supervised classification models. To address this, we propose
 to rely on a pairwise comparison-based approach, which provides a more robust a
lternative to individual annotation, since relative judgments are typically easi
er and yield higher consistency.
The problem is thus transformed into a ranking problem and we introduce an ideal
 point model adapted to time series data using elastic similarity measures such
as Dynamic Time Warping (DTW) and Time Warp Edit Distance (TWED).
We prove Lipschitz continuity of these distances and demonstrate several converg
ence guarantees for this model. To facilitate gradient-based optimization, we al
so introduce a differentiable version of the TWED. Finally, we show through mult
iple experiments that our approach produces accurate and robust rankings under n
oisy annotation conditions.
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Theory of Minimal Weight Perturbations in Deep Networks and its Applications for
 Low-Rank Activated Backdoor Attacks
Bethan Evans, Jared Tanner
The minimal norm weight perturbations of DNNs required to achieve a specified ch
ange in output are derived and the factors determining its size are discussed. T
hese single-layer exact formulae are contrasted with more generic multi-layer Li
pschitz constant based robustness guarantees; both are observed to be of the sam
e order which indicates similar efficacy in their guarantees. These results are
applied to precision-modification-activated backdoor attacks, establishing prova
ble compression thresholds below which such attacks cannot succeed, and show emp
irically that low-rank compression can reliably activate latent backdoors while
preserving full-precision accuracy. These expressions reveal how back-propagated
 margins govern layer-wise sensitivity and provide certifiable guarantees on the
 smallest parameter updates consistent with a desired output shift.
****

TextAtlas5M: A Large-Scale Dataset for Long Text Image Generation
Dongxing Mao, Alex Jinpeng Wang, Weiming Han, Jiawei Zhang, Zhuobai Dong, Linjie
 Li, Yiqi Lin, Zhengyuan Yang, Libo Qin, Fuwei Zhang, Lijuan Wang, Min Li
Text-conditioned image generation has made rapid progress, yet rendering images
with long-form text remains challenging due to the limitations of existing datas
ets, which predominantly focus on short and simple text.
We introduce TextAtlas5M, a large-scale dataset designed to evaluate long-text r
endering, where “long text” encompasses not only textual length but also layout
complexity and semantic richness.
TextAtlas5M contains 5 million generated and collected images across diverse dat
a types, enabling comprehensive evaluation of large-scale generative models.
We further curate 4,000 human-improved test cases (TextAtlasEval) spanning four
domains, forming one of the most extensive benchmarks for text rendering.
Evaluations show that TextAtlas5M poses substantial challenges even for state-of
-the-art proprietary models (e.g., GPT-4o), with significantly larger gaps obser
ved for open-source models.
Training on TextAtlas5M consistently improves text rendering
for both diffusion-based and autoregressive models, demonstrating

its effectiveness for advancing text-rich image generation.
****
Parametrized Power-Iteration Clustering for Directed Graphs
Gwendal Debaussart-Joniec, Harry Sevi, Matthieu Jonckheere, Argyris Kalogeratos
Vertex-level clustering for directed graphs (digraphs) remains challenging as ed ge directionality breaks the key assumptions underlying popular spectral methods , which also incur the overhead of eigen-decomposition. This paper proposes *Par ametrized Power Iteration Clustering* (ParPIC), a random-walk-based clustering m ethod for weakly connected digraphs. This builds over the Power-Iteration Cluste ring paradigm, which uses the rows of the iterated diffusion operator as a data embedding. ParPIC has three important features: the use of parametrized reversib le random walk operators, the automatic tuning of the diffusion time, and the ef ficient truncation of the final embedding, which produces low-dimensional data r epresentations and reduces complexity. Empirical results on synthetic and real-w orld graphs demonstrate that ParPIC achieves competitive clustering accuracy wit h improved scalability relative to spectral and teleportation-based methods.
****
KineFlow: Kinematic Second-Order Flow Matching for Time-Series Forecasting
Haiqi Jiang, Hui Xiong
Conventional time-series discriminative forecasting relies on point-wise regress ion, which inherently induces over-smoothing and fails to capture stochastic vol atility in complex systems. While first-order generative flow matching methods m itigate this issue, they ignore system inertia, resulting in phase-space ambigui ties and high sensitivity to noise. We introduce KineFlow, a generative time-ser ies forecasting framework that augments flow matching with a phase-space Neural Acceleration Field, treating exogenous inputs as driving forces that produce gra dual momentum shifts rather than abrupt state perturbations. This second-order f ormulation serves as a structural filter via double integration, suppressing hig h-frequency noise and producing robust, physically consistent predictions. Exten sive experiments on six real-world benchmarks demonstrate that KineFlow achieves an average 15% MSE improvement over discriminative baselines and an 8% gain in CRPS compared to state-of-the-art generative methods.
****
Sparser Block-Sparse Attention via Token Permutation
Xinghao Wang, Pengyu Wang, Dong Zhang, Chenkun Tan, Shaojun Zhou, Zhaoxiang Liu, Shiguo Lian, Fangxu Liu, Kai Song, Xipeng Qiu
Scaling the context length of large language models (LLMs) offers significant be nefits but is computationally expensive.
This expense stems primarily from the self-attention mechanism, whose $O(N^2)$ c omplexity with respect to sequence length presents a major bottleneck for both m emory and latency.
Fortunately, the attention matrix is often sparse, particularly for long sequenc es, suggesting an opportunity for optimization.
Block-sparse attention has emerged as a promising solution that partitions seque nces into blocks and skips computation for a subset of these blocks.
However, the effectiveness of this method is highly dependent on the underlying attention patterns, which can lead to sub-optimal block-level sparsity.
For instance, important key tokens for queries within a single block may be scat tered across numerous other blocks, leading to computational redundancy.
In this work, we propose Permuted Block-Sparse Attention (**PBS-Attn**), a plug- and-play method that leverages the permutation properties of attention to increa se block-level sparsity and enhance the computational efficiency of LLM prefilli ng.
We conduct comprehensive experiments on challenging long-context datasets, demon strating that PBS-Attn consistently outperforms existing block-sparse attention methods in model accuracy and closely matches the full attention baseline.
Powered by our custom permuted-FlashAttention kernels, PBS-Attn achieves an end- to-end speedup of up to $\mathbf{2.75\times}$ in long-context prefilling, confir ming its practical viability.
Code available at \url{https://github.com/xinghaow99/pbs-attn}.

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AREA: Attribute Extraction and Aggregation for CLIP-Based Class-Incremental Lear ning
Zhen-Hao Xie, Yu-Cheng Shi, Da-Wei Zhou
Class-Incremental Learning (CIL) is important in building real-world learning sy stems. In CLIP-based CIL, the model performs classification by comparing similar ity between visual and textual embeddings obtained from template prompts, e.g., ``a photo of a [CLASS]''. This seemingly monolithic matching process can be deco mposed into two conceptually distinct stages: attribute extraction and attribute aggregation. For example, a model may recognize cat using attributes such as fu r texture and whiskers. When learning a new class like car, the model must extra ct additional attributes like wheels and adjust how they are aggregated in the s hared representation space. However, since only data from the current task is av ailable, incremental updates can bias both attribute extraction and aggregation toward new classes, leading to catastrophic forgetting. Therefore, we propose AR EA for attribute extraction and aggregation in CLIP-based CIL. To stabilize extr action, we anchor class-level visual and textual attributes on the hyperspherica l embedding space via principal geodesic analysis. To stabilize aggregation, we learn lightweight task-specific experts with scoring and residual refinement, re gularized by a variational information bottleneck objective. During inference, w e perform routing over task attribute manifolds via optimal transport for more c oncise prediction. Experiments show that AREA consistently outperforms SOTA meth ods. Code is available at https://github.com/LAMDA-CL/ICML2026-AREA.
****
BiCrossNet with Decoupled Dual Generators: A Parameter■Efficient and Generalizab le Few■Shot Custom Gesture Recognition Framework
Chunyang Yu, Ning SUN, Shenyue Wang
Gesture interaction and touchless sensing is a natural and intuitive control met hod that allows users to control devices through natural hand or body movements, reducing reliance on physical input and enhancing convenience and functional ef ficiency. The biggest challenge is balancing accuracy and model parameter count for custom gesture tasks, while minimizing the number of data samples required f or training. This paper proposes a novel IMU-based BiCrossNet model and two nove l data augmentation models (delta-generator and embedding-generator) to address this challenge. Compared with existing methods, the model proposed in this artic le boosts accuracy by 11.7% and 12.7% in UMAHand (public datasets); and 8.85% a nd 5.25% in GRHand (self-developed datasets), with decreasing 27.8% pretrained f eature-extractor model parameters. This research lays a solid foundation for dep loying and implementing custome gesture recognition engineering on intelligent terminal devices.
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Delving into Non-Exchangeability for Conformal Prediction in Graph-Structured Mu ltivariate Time Series
Ruichao Guo, Xingyao Han, Luo Wenshui, Zhe Liu, Chen Gong, Hesheng Wang
Point forecasting for graph-structured multivariate time series is a fundamental problem, but rigorous uncertainty quantification for such predictions is still underexplored. Conformal prediction (CP) offers uncertainty estimation with a so lid coverage guarantee under the exchangeability assumption, which requires the joint data distribution to be unchanged under permutation. However, in graph-str uctured time series, inherent cross-node coupling can violate the exchangeabilit y condition, making direct application of CP unreliable.
Inspired by the spectral graph theory, such coupling resides in global trends an d can be characterized by the low-frequency components, while high-frequency com ponents are nearly exchangeable. Therefore, we propose a novel concept named **S **pectral **G**raph **C**onditional **E**xchangeability (SGCE), which conditions exchangeable high-frequency components on low-frequency ones to preserve global trends and enable effective CP in the spectral domain.
Based on SGCE, we further propose **S**pectral **C**onformal prediction via w**A **ve**LE**t transform (SCALE). SCALE uses graph wavelets to decompose low/high-f requency components and conformalizes high-frequency residuals via adaptive gati

ng over a low-frequency embedding. Experimental results on real-world benchmark datasets across multiple application domains show that SCALE not only achieves valid coverage but also consistently improves the coverage-efficiency trade-off over the state-of-the-art CP methods.

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Return-Critic: Bridging Goal Discrepancy for Efficient Visual Reinforcement Learning

Ruyi Lu, Xuesong Wang, Hengrui Zhang, Yuhu Cheng

Sample inefficiency remains a challenge in pixel-based visual reinforcement learning (RL), primarily due to ineffective state representation learning. While recent advances employ auxiliary tasks to improve representation learning, their representation goals (e.g., mask reconstruction, state prediction) are misaligned with the ultimate RL goal of maximizing return, constraining further improvements in representation quality. To achieve efficient visual reinforcement learning, we propose Return-Critic (RC), an auxiliary framework that bridges goal discrepancy by return prediction. RC samples partial frames from an episode, processes them through a shared visual encoder, and employs a lightweight Transformer to predict the episode's return, forcing the encoder to learn return-relevant representation. The attention weights naturally highlight important frames, enabling a key function for prioritized learning. Extensive experiments on both online (DMControl) and offline (V-D4RL) benchmarks demonstrate that RC significantly enhances the sample efficiency, particularly achieving 68% performance boost on average across nine challenging tasks from DMControl.

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Rethinking Generative Image Pretraining: How Far Are We From Scaling Up Next-Pixel Prediction?

Xinchen Yan, Chen Liang, Lijun Yu, Adams Wei Yu, Yifeng Lu, Quoc V Le

This paper investigates the scaling properties of autoregressive next-pixel prediction, a simple, end-to-end yet under-explored framework for unified vision models. Starting with images at resolutions of 32$\times$ 32, we
train a family of Transformers using IsoFLOP profiles across compute budgets up to 7e19 FLOPs and evaluate three distinct target metrics: next-pixel prediction objective, ImageNet classification accuracy, and generation-based completion where top half of the image serves as a spatial prompt. First, optimal scaling strategy is critically task-dependent.
At a fixed resolution of 32 $\times$ 32 alone,
the optimal scaling properties for image classification and image generation diverge, where generation optimal setup requires the data size grow three to five times faster than for the classification optimal setup.
Second, as image resolution increases, the optimal scaling strategy indicates that the model size must grow much faster than data size.
Surprisingly, by projecting our findings, we discover that the primary bottleneck is compute rather than the amount of training data.
As compute continues to grow four to five times annually,
we forecast the feasibility of pixel-by-pixel modeling of images within the next five years.

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Coloring the Noise: Adversarial Sobolev Alignment for Faithful Image Super Resolution

Hongbo Wang, Huaibo Huang, Pin Wang, Jinhua Hao, Chao Zhou, Ran He

Generative priors in Image Super-Resolution (SR) often compromise faithful restoration, we attribute this limitation to a fundamental spectral misalignment between isotropic objectives and the intrinsic natural image manifold. While Direct Preference Optimization offers a path to alignment, its reliance on spectrally flat Gaussian noise fails to distinguish authentic high-frequency details from hallucinations. To bridge this geometric gap, we propose ASASR, a theoretically grounded framework that recasts the generative flow into a Sobolev-induced Riemannian geometry by explicitly coloring the noise transition kernel to mirror natural spectral decay. Driving this geometric alignment, we integrate a parametric adversary grounded in the Riesz Representation Theorem, which synthesizes targeted

negative samples equivalent to worst-case Sobolev gradients to direct optimization along the tangent space of plausible structural failures.
Extensive evaluations demonstrate that ASASR outperforms leading generative baselines, particularly in preserving spectral consistency and structural fidelity, offering a robust solution that effectively mitigates artifacts.
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Divide and Contrast: Learning Robust Temporal Features without Augmentation
Abdul-Kazeem Shamba, Kerstin Bach, Gavin Taylor
Self-supervised learning for time-series representation aims to reduce reliance on labeled data while maintaining strong downstream performance, yet many existing approaches incur high computational costs or rely on assumptions that do not hold across diverse temporal dynamics. In this work, we introduce Divide and Contrast (Di-COT), an unsupervised framework that avoids data augmentation and multiple encoder passes by contrasting informative substructures within a window rather than individual timesteps. Di-COT stochastically partitions each window into a small number of overlapping sub-blocks per iteration, enabling efficient and meaningful contrast while mitigating false positives during temporal transitions. To further improve scalability, we adopt a contrastive objective whose computation depends on the batch size and the number of sub-blocks, making loss computation independent of sequence length. Extensive experiments on six large-scale real-world datasets, as well as the UCR and UEA benchmarks, demonstrate that Di-COT learns transferable representations while achieving state-of-the-art performance with substantially reduced training time.
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Causes and Consequences of Representational Similarity in Machine Learning Models
Zeyu Michael Li, Hung Anh Vu, Damilola Awofisayo, Emily Wenger
Numerous works have noted similarities in how machine learning models represent the world, even across modalities. Although much effort has been devoted to uncovering properties and metrics on which these models align, surprisingly little work has explored causes of this similarity. To advance this line of inquiry, this work explores how two factors—dataset overlap and task overlap—influence downstream model similarity. We evaluate the effects of both factors through experiments across model sizes and modalities, from small classifiers to large language models. We find that dataset and task overlap are positively associated with higher representational similarity across many of our settings, with clear evidence in vision/language classification and weaker trends in language generation experiments. Finally, we consider downstream consequences of representational similarity, showing that greater similarity is associated with increased vulnerability to transferable adversarial attacks in vision models.
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Asymptotic Theory of Iterated Empirical Risk Minimization, with Applications to Active Learning
Hugo Cui, Yue M. Lu
We study a class of iterated empirical risk minimization (ERM) procedures in which two successive ERMs are performed on the same dataset, and the predictions of the first estimator enter as an argument in the loss function of the second. This setting, which arises naturally in active learning and reweighting schemes, introduces intricate statistical dependencies across samples and fundamentally distinguishes the problem from classical single-stage ERM analyses. For linear models trained with a broad class of convex losses on Gaussian mixture data, we derive a sharp asymptotic characterization of the test error in the high-dimensional regime where the sample size and ambient dimension scale proportionally. Our results provide explicit, fully asymptotic predictions for the performance of the second-stage estimator despite the reuse of data and the presence of prediction-dependent losses. We apply this theory to revisit a well-studied pool-based active learning problem, removing oracle and sample-splitting assumptions made in prior work. We uncover a fundamental tradeoff in how the labeling budget should be allocated across stages, and demonstrate a double-descent behavior of the test error driven purely by data selection, rather than model size or sample count.

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To Grok Grokking: Provable Grokking in Ridge Regression
Mingyue Xu, Gal Vardi, Itay Safran
We study *grokking* — the onset of generalization long after overfitting — in a classical ridge regression setting. We prove end-to-end grokking results for learning over-parameterized linear regression models using gradient descent with weight decay. Specifically, we prove that the following stages occur: (i) the model overfits the training data early during training; (ii) poor generalization persists long after overfitting has manifested; and (iii) the generalization error eventually becomes arbitrarily small. Moreover, we show, both theoretically and empirically, that grokking can be amplified or eliminated in a principled manner through proper hyperparameter tuning. To the best of our knowledge, these are the first rigorous quantitative bounds on the generalization delay (which we refer to as the "grokking time") in terms of training hyperparameters. Lastly, going beyond the linear setting, we empirically demonstrate that our quantitative bounds also capture the behavior of grokking on non-linear neural networks. Our results suggest that grokking is not an inherent failure mode of deep learning, but rather a consequence of specific training conditions, and thus does not require fundamental changes to the model architecture or learning algorithm to avoid.
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How does Chain of Thought decompose complex tasks?
Amrut Nadgir, Vijay Balasubramanian, Pratik Chaudhari
Many language tasks can be modeled as classification problems where a large language model (LLM) is given a prompt and selects one among many possible answers. We show that the classification error in such problems scales as a power law in the number of classes. This has a dramatic consequence: the prediction error can be reduced substantially by splitting the overall task into a sequence of smaller classification problems, each with the same number of classes ("degree"). This tree-structured decomposition models chain-of-thought (CoT). It has been observed that CoT-based predictors perform better when they "think", i.e., when they develop a deeper tree, thus decomposing the problem into a larger number of steps. We identify a critical threshold for the degree, below which thinking is detrimental, and above which there exists an optimal depth that minimizes the error. It is impossible to surpass this minimal error by increasing the depth of thinking.
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Regularized Discriminative Alignment for Deep Representations under Label Shift
Hengchao Shi, Boen Jiang, Guanhua Fang, Wen Yu, Ming Zheng
Label shift refers to the distribution shift scenario where the marginal label distribution changes while the class-conditional distribution remains invariant. To address this challenge in complex real-world settings, we propose **Regularized Discriminative Alignment for Label Shift (RDALS)**, a novel framework that adapts to target domains by aligning distributions within the deep latent space. By shifting the focus from raw inputs to learned representations, RDALS effectively operates under a weaker and more practical invariance assumption. Specifically, we construct a moment-matching linear system using Linear Discriminant Analysis (LDA) and show that this choice maximizes numerical stability. We further provide rigorous theoretical analysis, establishing finite-sample error bounds for the importance weight estimation and the generalization bounds for the adapted classifier. Extensive experiments on standard benchmarks demonstrate that RDALS significantly outperforms state-of-the-art baselines, achieving superior robustness and accuracy in both data-scarce and extreme-shift regimes.
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APEX: Approximate-but-exhaustive search for ultra-large combinatorial synthesis libraries
Aryan Pedawi, Jordi Silvestre-Ryan, Bradley Worley, Darren J. Hsu, Kushal S Shah, Elias Stehle, Jingrong Zhang, Izhar Wallach
Make-on-demand combinatorial synthesis libraries (CSLs) like Enamine REAL have significantly enabled drug discovery efforts. However, their large size presents a challenge for virtual screening, where the goal is to identify the top compoun

ds in a library according to a computational objective (e.g., optimizing docking score) subject to computational constraints under a limited computational budget. For current library sizes---numbering in the tens of billions of compounds---and scoring functions of interest, a routine virtual screening campaign may be limited to scoring fewer than 0.1% of the available compounds, leaving potentially many high scoring compounds undiscovered. Furthermore, as constraints (and sometimes objectives) change during the course of a virtual screening campaign, existing virtual screening algorithms typically offer little room for amortization. We propose the approximate-but-exhaustive search protocol for CSLs, or APEX. APEX utilizes a neural network surrogate that exploits the structure of CSLs in the prediction of objectives and constraints to make full enumeration on a consumer GPU possible in under a minute, allowing for exact retrieval of approximate top-k sets. To demonstrate APEX's capabilities, we develop a benchmark CSL comprised of more than 10 million compounds, all of which have been annotated with their docking scores on five medically relevant targets along with physicohemical properties measured with RDKit such that, for any objective and set of constraints, the ground truth top-k compounds can be identified and compared against the retrievals from any virtual screening algorithm. We show APEX's consistently strong performance both in retrieval accuracy and runtime compared to alternative methods.

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Evaluating Language Models in Realistic Conversational Contexts

Ilija Subasic, Andrew Rabinovich, Zhao Chen

As Large Language Models (LLMs) are increasingly deployed to serve open-ended, multi-turn interactions, evaluating conversational quality at human scale has become a central challenge. Existing evaluation frameworks built for summarization, translation, or short-form QA tasks fall short of adequately measuring the consistency of human-scale dialogue, especially when derivation and validation of these metrics themselves often rely on synthetic rather than human sources. We fill the gap by introducing \textbf{UPHELD} (Utility \& Planning Human-Scale Evaluated Long Dialogues), a large, reference-full benchmark for evaluating human-scale conversational ability beyond factual correctness. UPHELD consists of hundreds of complete human-to-human dialogues authored by \textbf{professional script writers}, with realistic turn densities and \textbf{36,000+ per-turn human annotations} across \textbf{30,000+ expert-generated dialogue turns}. Using UPHELD, we systematically evaluate classical automatic metrics and reference-free LLM-as-a-judge approaches, and find them unreliable when correlated with expert human judgment. Building off this analysis, we use UPHELD to develop a \textbf{Mixture-of-Judges} framework that combines multiple evaluative signals and improves correlation with human assessments by approximately \textbf{30\%}. Overall, UPHELD provides a robust, human-grounded foundation for evaluating human-scale conversational intelligence that fills a crucial gap in the pre-existing LLM dataset landscape.

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Offline Multi-agent Continual Cooperation via Skill Partition and Reuse

Yuchen Xiao, Lei Yuan, Ruiqi Xue, Tieyue Yin, Yang Yu

Extracting skills from multi-agent offline dataset improves learning efficiency via sharing task-invariant coordination skills among tasks. In settings where tasks occur sequentially and the space of skills grows exponentially, existing approaches that rely on heuristically designed and fixed-sized skill libraries struggle to resolve the problem of distributional shift and interference, facing catastrophic forgetting and plasticity loss. To address this problem and endow agents with the ability to continually discover and reuse coordination skills in open-environment, we propose COMAD, a principled framework for **C**ontinual **O**ffline **M**ulti-**a**gent Skill **D**iscovery via Skill Partition and Reuse. We first discover skills from mixed multi-agent behavior data with an auto-encoder to transform coordination knowledge into reusable coordination skills. Then we construct a skill-augmented policy learning objective with multi-head architectures, explicitly guiding the advantage function with reusable skills identified via a density-based reusability estimator.

Theoretical analysis shows our method approximates the optimum of a continual sk ill discovery problem. Empirical results across diverse MARL benchmarks show tha t COMAD continually expands its skill library to mitigate interference, achievin g superior forward and backward transfer for task streams compared to multiple b aselines.
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Controlled LLM Training on Spectral Sphere
Tian Xie, Haoming Luo, Haoyu Tang, Hu Yiwen, Jason Klein Liu, Qingnan Ren, Yang Wang, Xin Zhao, Rui Yan, Bing Su, Chong Luo, Baining Guo
Scaling large models requires optimization strategies that ensure rapid converge nce grounded in stability. Maximal Update Parametrization ($\boldsymbol{\mu}$P) provides a theoretical safeguard for width-invariant $\Theta(1)$ activation cont rol, whereas emerging optimizers like Muon are only "half-aligned" with these co nstraints: they control updates but allow weights to drift. To address this limi tation, we introduce the **Spectral Sphere Optimizer (SSO)**, which enforces str ict module-wise spectral constraints on both weights and their updates. By deriv ing the steepest descent direction on the spectral sphere, SSO realizes a fully $\boldsymbol{\mu}$P-aligned optimization process. To enable large■scale training , we implement SSO as an efficient parallel algorithm within Megatron. Through e xtensive pretraining on diverse architectures, including Dense 1.7B, MoE 8B-A1B, and 200-layer DeepNet models, SSO consistently outperforms AdamW and Muon. Furt hermore, we observe significant practical stability benefits, including improved MoE router load balancing, suppressed outliers, and strictly bounded activation s.
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Test-Time Guidance for Flow-Based Generative Models via Parallel Tempering on So urce Distributions
Shih-Hsin Wang, Joel Keller, Taos Transue, Drake Benjamin Brown, Thomas Strohmer , Bao Wang
Generative models that transport a simple source distribution to a complex data distribution—such as diffusion and flow-based models—are central to high■fidelit y data generation. Test-time guidance can further steer pretrained models toward user-specified high-reward regions without costly retraining. However, existing guidance methods face critical limitations: they struggle with non-differentiab le rewards, fail to navigate complex landscapes, and often lack theoretical guar antees on generation performance. We propose {\it Source Parallel Tempering (SPT)}, a gradient■free test■time guidance framework that operates entirely in sourc e space, leveraging its simpler geometry to avoid the complexities of the data m anifold. SPT couples a local exploration kernel with parallel tempering, enablin g efficient barrier crossing and robust discovery of high■reward modes. Theoreti cally, we provide a new error bound linking training-time approximation error to test-time guidance performance.Empirically, SPT significantly improves over sta te-of-the-art methods on benchmark tasks in conditional image synthesis and dyna mical system trajectory sampling.
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Spatial Memory for Out-of-Vision Manipulation in Vision-Language-Action
Pengteng Li, Weiyu Guo, He Zhang, Tiefu Cai, Xiao He, Yandong Guo, Hui Xiong
We introduce SOMA, the Spatial Memory framework for Out-of-Vision Manipulation i n Vision-Language-Action (VLA) models. Most existing VLAs implicitly assume that task-relevant objects are always visible, leading to brittle and reactive behav iors when targets fall outside the camera’s field of view. SOMA addresses this l imitation by equipping VLAs with a persistent spatial memory constructed from mu lti-view observations acquired via a movable head camera, enabling reasoning bey ond the current visual frustum. The framework consists of three components: Spat ial Memory Construction, which aggregates angular-wise observations into a unifi ed spatial–semantic representation through scanning; Dynamic Memory Refinement, which maintains global consistency over time; and Contextual Memory Retrieval, w hich activates instruction-relevant spatial cues during manipulation. We evaluat e SOMA on five challenging real-world out-of-vision manipulation tasks, includin g multi-step and dual-arm scenarios where target objects are initially invisible

. Experimental results show that SOMA not only improves task success rates, but also induces qualitatively different manipulation behaviors, with faster target localization, reduced viewpoint search, and near one-shot grasping under partial observability. Additional experiments on RoboCasa GR1 and SimplerEnv further validate the effectiveness of SOMA's memory design under conventional fully observable settings. Code will be released soon.
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TSFAdv: Frequency-Guided Black-Box Adversarial Attacks on Time Series Forecasting
Qizhuo Han, Xiangrui Cai, Sihan Xu, Ying Zhang, Zheli Liu
While deep neural network-based long-term time series forecasting (LTSF) has become indispensable for critical infrastructures such as smart grids and IoT platforms, the deployment of these models as black-box APIs introduces severe security vulnerabilities that remain largely underexplored. In this paper, we propose TSFAdv, a query-efficient adversarial framework for LTSF models. The framework systematically analyzes model sensitivity to spectral perturbations in both magnitude and phase of the frequency domain. By embedding frequency-domain priors into Natural Evolution Strategies, we achieve sensitivity-guided gradient estimation that improves perturbation efficacy without violating practical query constraints. To overcome ambiguities inherent to point-wise regression metrics, we adopt a trajectory-level evaluation protocol based on Dynamic Time Warping (DTW) and Slope Misalignment Error (SME), enabling the capture of complex geometric and directional deviations. Extensive experiments across seven state-of-the-art architectures demonstrate that TSFAdv achieves substantial performance gains, with median DTW improvements of 38.78\% and median SME improvements of 26.47\% under 200-query budget. These findings reveal that existing defense mechanisms are ineffective against frequency-domain manipulation, underscoring an urgent necessity for robust LTSF models.
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Ideal Attribution and Faithful Watermarks for Language Models
Min Jae Song, Kameron Shahabi
We introduce ideal attribution mechanisms, a formal abstraction for reasoning about attribution decisions over strings. At the core of this abstraction lies the ledger, an append-only log of the prompt-response interaction history between a model and its user. Each mechanism produces deterministic decisions based on the ledger and an explicit selection criterion, making it well-suited to serve as a ground truth for attribution. We frame the design goal of watermarking schemes as faithful representation of ideal attribution mechanisms. This novel perspective brings conceptual clarity, replacing piecemeal probabilistic statements with a unified language for stating the guarantees of each scheme. It also enables precise reasoning about desiderata for future watermarking schemes, even when no current construction achieves them, as the ideal functionalities are specified first. In this way, the framework provides a roadmap that clarifies which guarantees are attainable in an idealized setting and worth pursuing in practice.
****
Adaptive Estimation and Inference in Semi-parametric Heterogeneous Clustered Multitask Learning via Neyman Orthogonality
Hanxiao Chen, Debarghya Mukherjee
We study clustered multitask learning in a semiparametric setting where tasks share a latent cluster structure in their target parameters but exhibit heterogeneous, potentially infinite-dimensional nuisance components. Such heterogeneity poses a major challenge for existing multitask learning methods, which typically rely on aligned feature spaces or homogeneous task structures. To address this challenge, we propose an *adaptive fused orthogonal estimator* that integrates Neyman-orthogonal losses with data-driven pairwise fusion penalties. Our framework leverages task-specific pilot estimates to calibrate the fusion penalties and combines adaptive aggregation with orthogonalization to mitigate the impact of nuisance-parameter estimation error. Theoretically, we show that the proposed estimator achieves exact recovery of the latent clustering with high probability and attains pooled parametric convergence rates proportional to cluster size. Moreov

er, we establish asymptotic normality and show that, asymptotically, our estimat
or matches the performance of an oracle procedure that knows the true clustering
 in advance. Empirically, we show that the proposed method consistently outperfo
rms strong baselines in various simulation setups. A real-world application to U
.S. residential energy consumption further demonstrates the effectiveness of our
 approach in uncovering meaningful regional clustering in electricity price elas
ticity, showcasing the efficacy of our method.
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Universal Skeleton Understanding via Differentiable Rendering and MLLMs
Ziyi Wang, Peiming Li, Xinshun Wang, Yang Tang, Kai-Kuang Ma, Mengyuan Liu
Multimodal large language models (MLLMs) exhibit strong visual-language reasonin
g, yet remain confined to their native modalities and cannot directly process st
ructured, non-visual data such as human skeletons. Existing methods either compr
ess skeleton dynamics into lossy feature vectors for text alignment, or quantize
 motion into discrete tokens that generalize poorly across heterogeneous skeleto
n formats. We present SkeletonLLM, which achieves universal skeleton understandi
ng by translating arbitrary skeleton sequences into the MLLM's native visual mod
ality. At its core is DrAction, a differentiable, format-agnostic renderer that
converts skeletal kinematics into compact image sequences. Because the pipeline
is end-to-end differentiable, MLLM gradients can directly guide the rendering to
 produce task-informative visual tokens. To further enhance reasoning capabiliti
es, we introduce a cooperative training strategy: Causal Reasoning Distillation
transfers structured, step-by-step reasoning from a teacher model, while Discrim
inative Finetuning sharpens decision boundaries between confusable actions. Skel
etonLLM demonstrates strong generalization on diverse tasks including recognitio
n, captioning, reasoning, and cross-format transfer---suggesting a viable path f
or applying MLLMs to non-native modalities. Code is available in the Supplementa
ry Material.
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Linear Ensembles Wash Away Watermarks: On the Fragility of Distributional Pertur
bations in LLMs
Zhihao Wu, Gracia Gong, Qinglin Zhu, Yudong Chen, Runcong Zhao
Watermarking embeds statistical signatures in AI-generated text for detection an
d attribution. We reveal a fundamental vulnerability: when users access multiple
 models (today's reality), watermarks trivially fail. Watermarks perturb output
 distributions away from the original, and in competitive markets, these perturb
ations are typically independent across providers. We theoretically prove that a
veraging output probability distributions recovers the unwatermarked distributio
n with up to a second-order error term. Empirically, simply averaging 3-5 models
 cancels out these perturbations. We introduce WASH (Watermark Attenuation via S
tatistical Hybridisation), which solves practical challenges in ensemble generat
ion: vocabulary misalignment and tokenisation differences across heterogeneous m
odels. Experiments across six watermarking schemes and three LLMs show that aver
aging across 3 models suppresses detection z-scores from 5-300 to **below 2** (b
elow the detection threshold of 4) and reduces TPR@5%FPR to **below 50%**, while
 improving quality by **27.5%** and running **6×** faster than the best baseline
 on the long sequence generation. Our results suggest that robust AI-text detect
ion via watermarking requires either accepting this fundamental vulnerability or
 unprecedented coordination among model providers.
****

Near-Optimal Regret for Policy Optimization in Contextual MDPs with General Offl
ine Function Approximation
Orin Levy, Aviv Rosenberg, Alon Cohen, Yishay Mansour
We introduce OPO-CMDP, the first policy optimization algorithm for stochastic Co
ntextual Markov Decision Process (CMDPs) under general offline function approxim
ation. Our approach achieves a high probability regret bound of
$\widetilde{O}(H^4\sqrt{T|S||A|\log(|\mathcal{F}||\mathcal{P}|)}),$ where S an
d A denote the state and action spaces, H the horizon length, T the number
 of episodes, and $\mathcal{F}, \mathcal{P}$ the finite function classes used to
 approximate the losses and dynamics, respectively. This is the first regret bou

nd with optimal dependence on $|S|$ and $|A|$, directly improving the current st
ate-of-the-art (Qian, Hu, and Simchi-Levi, 2024).
These results demonstrate that optimistic policy optimization provides a natural
, computationally superior and theoretically near-optimal path for solving CMDPs
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Scaling the Scaling Logic: Agentic Meta-Synthesis of Logic Reasoning
Bowen LIU, Zhi Wu, Runquan Xie, Zhanhui Kang, Jia Li
Reinforcement Learning from Verifiable Rewards (RLVR) is bottlenecked by data: e
xisting synthesis pipelines rely on expert-written code or fixed templates, conf
ining growth to instance-level perturbations. We shift the evolvable unit from p
roblem instances to task-family specifications. SSLogic is an agentic meta-synth
esis framework in which LLM agents iteratively author and refine executable Gene
rator-Validator pairs inside a closed Generate-Validate-Refine loop, producing f
amilies with new rules and difficulty gradients rather than parameter variations
 of old ones. A Multi-Gate Validation Protocol, multi-strategy consensus plus Ad
versarial Blind Review, where independent agents solve each instance by writing
and executing code, filters ill-posed tasks before they enter training. Starting
 from 400 seed families, two evolution rounds yield 953 families and 21,389 veri
fiable instances. Three converging comparisons (step-matched, token-matched, and
 size-controlled on external Enigmata data) consistently show higher training ut
ility of evolved data, with gains of SynLogic +5.2, AIME25 +3.0, and BBH +5.5 on
 Enigmata. Fine-grained KORBench evaluation reveals selective improvements in lo
gic (+13.2%) and operation (+9.6%), linking structural evolution to downstream g
ains. Code is available at https://github.com/AdAstraAbyssoque/Scaling-the-Scal
ing-Logic.
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Physics in 2-Steps: Locking Motion Priors Before Visual Refinement Erases Them
Woojung Han, Seil Kang, Youngjun Jun, Min-Hung Chen, Fu-En Yang, Seong Jae Hwang
Image-to-Video diffusion models leverage input images to generate visually stunn
ing content, yet frequently produce motion that violates physical laws. We revea
l a surprising finding: a 2-step generation often exhibits better physical consi
stency than a 50-step output from the same model. Through spectral analysis, we
trace this to phase erosion during denoising; the phase degrades significantly (
dropping by $\approx 18\%$ from step 2 to step 50), whereas the magnitude remain
s relatively stable. Building on this insight, we propose PhaseLock, a training-
free framework that preserves the valid motion priors from few-step inference th
roughout the denoising trajectory. Rather than relying on full-step inference fo
r physical consistency, PhaseLock extracts a motion prior from just 2 steps and
enforces it onto high-fidelity generation via Latent Delta Guidance. Our approac
h effectively mitigates phase degradation, improving physical consistency by an
average of 6.2 points across diverse models while largely maintaining visual fid
elity, with negligible overhead ($1.06\times$ time, $1.02\times$ memory) and red
uced reliance on expensive external guidance methods ($\sim5\times$ time).
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GUDA: Counterfactual Group-wise Training Data Attribution for Diffusion Models v
ia Unlearning
Naoki Murata, Yuhta Takida, Chieh-Hsin Lai, Toshimitsu Uesaka, Bac Nguyen, Stefa
no Ermon, Yuki Mitsufuji
Training-data attribution for vision generative models aims to identify which tr
aining data influenced a given output. While most methods score individual examp
les, practitioners often need group-level answers (e.g., artistic styles or obje
ct classes). Group-wise attribution is counterfactual: how would a model's behav
ior on a generated sample change if a group were absent from training? A natural
 realization of this counterfactual is Leave-One-Group-Out (LOGO) retraining, wh
ich retrains the model with each group removed; however, it becomes computationa
lly prohibitive as the number of groups grows. We propose GUDA (Group Unlearning
-based Data Attribution) for diffusion models, which approximates each counterfa
ctual model by applying machine unlearning to a shared full-data model instead o
f training from scratch. GUDA quantifies group influence using differences in a

likelihood-based scoring rule (ELBO) between the full model and each unlearned counterfactual. Experiments on CIFAR-10 and artistic style attribution with Stable Diffusion show that GUDA identifies primary contributing groups more reliably than semantic similarity, gradient-based attribution, and instance-level unlearning approaches, while achieving $\sim$100$\times$ speedup on CIFAR-10 over LOGO retraining. The code is available at https://github.com/sony/guda.
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Deterministic Inference across Tensor Parallel Sizes That Eliminates Training-Inference Mismatch
Ziyang Zhang, Xinheng Ding, Jiayi Yuan, Rixin Liu, Huizi Mao, Jiarong Xing, Zirui Liu
Deterministic inference is increasingly critical for Large language model (LLM) applications such as LLM-as-a-judge evaluation, multi-agent systems, and reinforcement learning (RL).
However, existing LLM serving frameworks can produce different outputs for identical inputs when tensor parallel (TP) size or batch size changes, even under greedy decoding.
This arises from the non-associativity of floating-point arithmetic and inconsistent reduction orders across GPUs. While prior work has addressed batch-size–related nondeterminism through batch-invariant kernels, determinism across different TP sizes remains an open problem, particularly in RL settings, where the training engine typically uses Fully Sharded Data Parallel (FSDP) (i.e., TP = 1) while the rollout engine relies on multi-GPU TP to maximize the inference throughput, creating a probability mismatch that can degrade or even collapse training.
We identify and analyze the root causes of TP-induced inconsistency and propose **Tree-Based Invariant Kernels (TBIK)**, a set of custom matrix multiplication and reduction kernels that guarantee bit-wise identical results across TP sizes. Our key insight is to enforce a consistent reduction order across and within GPUs.
We implement TBIK in Triton and integrate it into vLLM and FSDP, achieving **bit-wise deterministic inference** across different TP sizes and **zero probability divergence** between rollout and training engines in RL pipelines. By eliminating mismatches caused by different parallelization strategies, TBIK enables **true on-policy RL at scale for the first time**, leading to improved model performance and faster convergence.
****
Transform Trained Transformer for Accelerating Native 4K Video Generation
Jiangning Zhang, Junwei Zhu, Teng Hu, Yabiao Wang, Donghao Luo, Weijian Cao, Zhenye Gan, Xiaobin Hu, Zhucun Xue, Xiangtai Li, Chengjie Wang, Yong Liu
Native 4K (2176$\times$3840) video generation remains a critical challenge due to the quadratic computational explosion of full-attention as spatiotemporal resolution increases, making it difficult for models to strike a balance between efficiency and quality. This paper proposes a novel Transformer retrofit strategy termed T3 (**T**ransform **T**rained **T**ransformer) that, without altering the core architecture of full-attention pretrained models, significantly reduces compute requirements by optimizing their forward logic. Specifically, T3-Video introduces a multi-scale weight-sharing window attention mechanism and, via hierarchical blocking together with an axis-preserving full-attention design, can effect an “attention pattern” transformation of a pretrained model using only modest compute and data. Results on 4K-VBench show that T3-Video substantially outperforms existing approaches: while delivering performance improvements (+4.29$\uparrow$ VQA and +0.08$\uparrow$ VTC), it accelerates native 4K video generation by more than 10$\times$. Demo and source code are available in \#Supp.
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Training-Free Adaptation of Diffusion Models via Doob's h-Transform
Qijie Zhu, Zeqi Ye, Han Liu, Zhaoran Wang, Minshuo Chen
Adaptation methods have been a workhorse for unlocking the transformative power of pre-trained diffusion models in diverse applications. Existing approaches often abstract adaptation objectives as a reward function and steer diffusion model

s to generate high-reward samples. However, these approaches can incur high comp
utational overhead due to additional training, or rely on stringent assumptions
on the reward such as differentiability. Moreover, despite their empirical succe
ss, theoretical justification and guarantees are seldom established. In this pap
er, we propose $\texttt{DOIT}$ ($\textbf{D}$oob-$\textbf{O}$riented
$\textbf{I}$nference-time $\textbf{T}$ransformation), a training-free
adaptation method for generic, non-differentiable rewards. We develop
two variants of this method: $\texttt{DOIT-Proto}$, a full-rollout
simulation-based algorithm and
$\texttt{DOIT-Practical}$, an efficient surrogate algorithm.
The key idea is a measure transport formulation that seeks to transport the pre-
trained generative distribution to a high-reward target distribution. We leverag
e Doob's h-transform to realize this transport, which induces a dynamic correc
tion to the diffusion sampling process and admits simulation-based computation w
ithout modifying the pre-trained model. Theoretically, we establish a high-proba
bility convergence
guarantee to the target high-reward distribution for $\texttt{DOIT-Proto}$ by ch
aracterizing the approximation error in
the dynamic Doob correction. Empirically, $\texttt{DOIT-Practical}$ consistently
outperforms state-of-the-art baselines on D4RL offline RL benchmarks
while preserving sampling efficiency.
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ERAlign: Energy-based Representation Alignment of GNNs and LLMs on Text-attribut
ed Graphs
Xianlin Zeng, Fan Xia, Xiangyu Chen
Text-attributed Graphs (TAGs) incorporate textual node attributes with graph str
uctures to describe rich relational semantics. Recent efforts to integrate Graph
 Neural Networks (GNNs) and Large Language Models (LLMs) have shown promise for
learning on TAGs, yet achieving well-aligned representations remains challenging
. Prior studies largely rely on heuristics that perform coarse-grained matching.
 They lack sufficient constraints and ignore distributional alignment, leading t
o representation drift and limited generalization. Building on Energy-based Mode
ls (EBMs), we propose an **E**nergy-based **R**epresentation **Align**ment (ERAl
ign) framework that projects GNN-encoded graph structure and LLM-derived text em
beddings in a shared latent space to achieve distribution consistency. Concretel
y, layer-wise alignment is quantified by a distance metric and optimized via an
EBM objective. By decreasing energy values, our framework yields well-aligned re
presentations for downstream tasks. During training, we introduce Energy Discrep
ancy (ED) to avoid high sampling costs associated with intractable normalization
. ED also carries theoretical guarantees of higher training efficiency and reduc
ed energy landscape distortion. Empirical evaluations on eight TAG datasets demo
nstrate that ERAlign obtains state-of-the-art performance across varying levels
of supervision and cross-task transfer scenarios.
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What Makes a Representation Good for Single-Cell Perturbation Prediction?
Wenkang Jiang, Yuhang Liu, Yichao Cai, Erdun Gao, Jiayi Dong, Ehsan Abbasnejad,
Lina Yao, Javen Qinfeng Shi
Single-cell perturbation modeling is fundamental for understanding and predictin
g cellular responses to genetic perturbations. However, existing approaches, fro
m causal representation learning to foundation models, often struggle with an ov
erlooked challenge: gene expression is dominated by perturbation-invariant infor
mation, while perturbation-specific signals are intrinsically sparse. As a resul
t, learned representations either entangle invariant and perturbation-specific i
nformation, leading to spurious and non-generalizable predictors, or suppress pe
rturbation-specific signals altogether, rendering them ineffective for predictio
n. To address this, we propose PerturbedVAE, a general framework designed to res
olve this signal imbalance. The framework explicitly separates perturbation-spec
ific information from dominant invariant structure and recovers causal represent
ations to effectively utilize such information for prediction. We further provid
e an identifiability analysis that characterizes the conditions under which spar

se perturbation effects can be reliably recovered, thereby clarifying how the fr amework can be concretely specified under such conditions. Empirically, Perturbe dVAE achieves state-of-the-art performance on a widely used benchmark across mul tiple evaluation settings, yielding significant gains on out-of-distribution com binatorial predictions and uncovering interpretable perturbation-response progra ms.

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Spectrally-Guided Diffusion Noise Schedules

Carlos Esteves, Ameesh Makadia

Denoising diffusion models are widely used for high-quality image and video gene ration.

Their performance depend on noise schedules, which define the distribution of no ise levels applied during training and the sequence of noise levels traversed du ring sampling. Noise schedules are typically handcrafted and require manual tuni ng across different resolutions. In this work, we propose a principled way to de sign per-image noise schedules for pixel diffusion, based on the images spectral properties. By deriving theoretical bounds on how efficacy of minimum and maxim um noise levels, we design "tight" noise schedules that eliminate redundant step s. During inference, we propose to conditionally sampled such noise schedules. E xperiments show that our noise schedules improve generative quality, particularl y at the low-step regime.

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Scaling the Prior: Size-Consistent Geometric Diffusion for 3D Molecular Generati on

Wenhan Gao, Jingxiang Qu, Yi Liu

Diffusion models typically operate in fixed-dimensional metric spaces, whereas 3 D geometric molecular data vary in dimensionality because molecules differ in si ze (number of atoms). A common adaptation in diffusion models for 3D molecular g eneration is to use models that handle variable-sized inputs, such as graph neur al networks and transformers. However, these approaches ignore that molecular si ze also sets the spatial scale of atomic coordinates, causing inconsistent gener ative trajectories.

In 3D molecular diffusion, generation can be seen as forming a coarse structure first and then refining atomic positions. Larger molecules form coarse structure s earlier than smaller ones because their spatial scales are larger relative to the noise. This makes the generative process inconsistent across sizes, with tra jectories driven by molecular size rather than by a unified generative pattern. We are the first to identify and analyze this size-induced inconsistency by deco mposing denoising dynamics, showing how spatial scale shapes formation of both 3 D structure and atom types. Based on this, we propose Scaling the Prior (StP), w hich rescales the prior distribution by molecular size to normalize learning acr oss sizes, harmonize denoising trajectories, and generate high-quality molecules . The code is available at https://github.com/wenhangao21/ICML26-StP.

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Predictive Prefetching for Retrieval-Augmented Generation

Wuyang Zhang, Shichao Pei

Retrieval-Augmented Generation (RAG) improves factual grounding in large languag e models but suffers from substantial latency due to synchronous retrieval. Whil e recent work explores asynchronous retrieval, existing approaches rely on heuri stic coordination between retrieval and generation and assume stable information demands during decoding that often break in complex, multi-domain settings. In this paper, we propose an advanced asynchronous retrieval framework that enables predictive prefetching aligned with evolving information needs. The framework e xplicitly predicts when retrieval should be triggered and what information shoul d be retrieved using three components, a retrieval predictor, a context monitor, and a query generator, by exploiting semantic precursors in generation dynamics that emerge several tokens before uncertainty becomes critical. Experiments on multiple benchmarks demonstrate up to 43.5% end-to-end latency reduction and 62. 4% improvement in time-to-first-token, while maintaining answer quality comparab le to synchronous RAG baselines.

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PAC-Bayesian Reinforcement Learning Trains Generalizable Policies
Abdelkrim ZITOUNI, Mehdi hennequin, Juba Agoun, Ryan Horache, NADIA KABACHI, Omar Rivasplata
We derive a novel PAC-Bayesian generalization bound for reinforcement learning that explicitly accounts for Markov dependencies in the data, through the chain's mixing time. This contributes to overcoming challenges in obtaining generalization guarantees for reinforcement learning, where the sequential nature of data breaks the independence assumptions underlying classical bounds. The new bound provides non-vacuous certificates for modern off-policy algorithms such as Soft Actor-Critic. We demonstrate the practical utility of the bound through PB-SAC, a novel algorithm that optimizes the bound during training to guide exploration. Experiments across several continuous control tasks show that the proposed approach provides meaningful confidence certificates while maintaining competitive performance.
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Learning Locally, Revising Globally: Global Reviser for Federated Learning with Noisy Labels
Yuxin Tian, Mouxing Yang, Yuhao Zhou, Jian Wang, Qing Ye, Tongliang Liu, Gang Niu, Jiancheng Lv
Conventional federated learning (FL) heavily depends on high-quality labels, which are often impractical in the real world, leading to the federated label-noise (F-LN) problem. Worse still, the F-LN problem is exacerbated by the heterogeneity of FL, whereas clients experience different label-noise types, ratios, and data distribution. In this study, we first observe an intriguing phenomenon that the global model of FL exhibits a slow memorization of noisy labels, suggesting its ability to maintain reliable predictions and robust representations in FL. Motivated by this, we propose a novel method termed Federated Global Reviser (FedGR), a straightforward yet effective method comprising three modules that collaboratively rectify noisy labels and regularize local training. By exploiting this inherent property, FedGR improves the label-noise robustness of FL in a self-contained manner. Extensive experiments on three widely used F-LN benchmarks demonstrate the superior performance of FedGR, consistently outperforming eight state-of-the-art baselines even in severe label-noise and data heterogeneity. Code: https://github.com/cs-yuxintian/FedGR-ICML26
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ProMiSE: Protein Multi-State Evaluation Benchmark in Biological Contexts
Bonjae Ku, Seeun Kim, Yubeen Kim, Hahnbeom Park, Chaok Seok
Proteins are inherently dynamic, with biological functions often emerging from transitions between multiple conformational states. While recent breakthroughs have largely addressed the static structure prediction problem, no systematic benchmark exists to demonstrate how well current models capture functionally relevant dynamics. We introduce ProMiSE, the first benchmark that provides both a dataset and an evaluation scheme, based on native biological assemblies and integrating major conformational change mechanisms—intrinsic, ligand-induced, and protein-induced—within a single curated dataset. We conducted a comprehensive evaluation of state-of-the-art structure prediction models, including AlphaFold3 and recent generative approaches. Our findings reveal that current models exhibit a limited ability to sample intrinsic multi-states and are often insensitive to biological context in induced scenarios. Internal representation analysis suggests that training-data exposure can shift predictions toward dominant conformational states over alternative biologically relevant states, primarily at the structure module. In contrast, results from BioEmu indicate that reducing decoding-stage bias can substantially improve multi-state sampling without major changes to upstream pair representations.
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ProtoVAR: Efficient Dataset Distillation via Prototype-Guided Visual Autoregressive Modeling
Mingyu Wang, Wei Jiang
Recent advances in generative distillation have shown strong potential in constr

ucting high quality surrogate datasets within a fraction of the time required by optimization-based approaches. However, most existing generative solutions rely on diffusion models, which suffer from two limitations. (i) Indirect matching objectives. Their sequential denoising process makes it difficult to directly match representative prototypes. (ii) Target-agnostic generation. The generation process is often decoupled from the target task, causing the synthesized samples to drift from the desired distribution. Building on this insight, We propose ProtoVAR, a prototype-guided visual autoregressive framework. Instead of relying on latent space, ProtoVAR uses the coarse-to-fine next-scale prediction of Visual AutoRegressive (VAR) modeling to maintain semantic consistency during generation. By injecting multi-scale class prototypes, ProtoVAR enforces clear representativeness constraints while preserving diversity. A pool-based selector further distills the prototype-guided outputs into a compact, task-aligned surrogate dataset. Extensive experiments show that ProtoVAR achieves state-of-the-art performance with comparable or lower computational cost than diffusion-based distillation.
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Learning to Think in Physics: Breaking Shortcut Learning in Scientific Diffusion via Representation Alignment
Haozhe Jia, Pengyu Yin, Wenshuo Chen, Shaofeng Liang, Lei Wang, Bowen Tian, Xiucheng Wang, Jia Nanqian, Yutao Yue
Physics-informed diffusion models typically enforce PDE constraints only on final outputs, leaving intermediate representations unconstrained and prone to shortcut learning under shifted boundary conditions. We introduce **REPA-P**, a teacher-free, architecture-agnostic framework that aligns intermediate features with physical states using first-principles residuals. REPA-P attaches lightweight $1{\times}1$ projection heads to selected layers, decodes hidden activations into physical quantities, and applies PDE residual losses during training. These heads are discarded at inference, introducing **zero overhead**. Across four PDE tasks, including Darcy flow, topology optimization, electrostatic potential, and turbulent channel flow, REPA-P accelerates convergence by up to $2{\times}$, reduces physics residuals by up to 66.4%, and improves out-of-distribution robustness by up to 49.3%, with consistent gains on both U-Net and Diffusion Transformer backbones. Ablations show that supervising a small set of intermediate layers captures most benefits and complements output-level physics losses. Code is available at https://github.com/Hxxxz0/REPA-P.
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A Close Look at Negative Label Guided Out-of-distribution Detection in Pre-trained Vision-Language Models
Bo Peng, Jie Lu, Zhen Fang, Guangquan Zhang
Advances in pre-trained vision-language models have enabled zero-shot out-of-distribution (OOD) detection using only in-distribution (ID) labels. Recent methods in this direction expand the label space with negative labels to enhance the discrimination between ID and OOD inputs. Despite their promising progress, there remains a limited understanding of their empirical effectiveness in open-world scenarios, where negative labels can arbitrarily diverge from real OOD ones. This paper bridges this research gap with the helm of a novel energy-based framework, where the energy function is built upon the margin between the similarity of an input to ID labels and that to negative labels. Guided by this framework, we prove that the inherent tolerance of such methods to the sampling bias essentially stems from estimating the worst-case energy function over a KL-constrained set of potential distributions centered on the negative label distribution. Furthermore, our theoretical analysis reveals that existing methods suffer from over-pessimism and consequently high sensitivity to outliers. Provably, we can alleviate these problems by leveraging Rényi divergence to refine potential distributions. Extensive experiments empirically manifest that our method establishes a new state-of-the-art across a variety of OOD detection settings.
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Target-Aware Bandit Allocation for Scalable Surrogate Optimization in Chemical Space

Mohammad Haddadnia, Yuvan Chali, Abhilash Jayaraj, Constance Kraay, Joana Reis, Felix Strieth-Kalthoff, Haribabu Arthanari
Identifying high-utility candidates from massive discrete spaces under expensive evaluations is a recurring challenge across the sciences, with structure-based drug discovery as a prominent example. While surrogate-based optimization can increase sample efficiency by reducing the number of expensive evaluations, modern molecular libraries have reached billions to trillions of compounds, making full-library surrogate inference itself a major computational bottleneck. We introduce BOBA, a bandit-guided surrogate optimization framework that eliminates full-library inference by adaptively allocating computation across partitions of the action space. By treating partitions as arms in a multi-armed bandit, BOBA concentrates inference and evaluations on empirically promising partitions while maintaining principled exploration. Experiments on real-world synthesis-on-demand libraries demonstrate that optimism-under-uncertainty bandits, combined with meaningful action space partitioning, are essential for effective allocation of inference and evaluations. Our findings reveal a tunable tradeoff between screening performance and surrogate inference cost, which supports practical optimization over current libraries, and establishes a viable route to ultra-large library virtual screening.

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Improved Algorithms for Nash Welfare in Linear Bandits
Dhruv Sarkar, Nishant Pandey, Sayak Ray Chowdhury
Nash regret has recently emerged as a principled fairness-aware performance metric for stochastic multi-armed bandits, motivated by the Nash Social Welfare objective. Although this notion has been extended to linear bandits, existing results suffer from suboptimality in ambient dimension d, stemming from proof techniques that rely on restrictive concentration inequalities. In this work, we resolve this open problem by introducing new analytical tools that yield an order-optimal Nash regret bound in linear bandits. Beyond Nash regret, we initiate the study of p-means regret in linear bandits, a unifying framework that interpolates between fairness and utility objectives and strictly generalizes Nash regret. We propose a generic algorithmic framework, FairLinBandit, that works as a meta-algorithm on top of any linear bandit strategy. We instantiate this framework using two bandit algorithms: Phased Elimination and Upper Confidence Bound, and prove that both achieve sublinear p-means regret for the entire range of p. Extensive experiments on linear bandit instances generated from real-world datasets demonstrate that our methods consistently outperform the existing state-of-the-art baseline.

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What Preferences Can—and Cannot—Predict in Multi-Agent Online Learning
Omar Abbadi, Rida Laraki, Panayotis Mertikopoulos
We examine the interplay between ordinal, preference-based solution concepts in games and the outcomes of payoff-driven learning dynamics, asking to what extent the combinatorial data of a game—its *preference graph*—can predict the long-run behavior of no-regret dynamics such as *follow-the-regularized-leader* (FTRL). In one direction, we show that the skeleton of every *dynamically stable* set (i.e., the set of pure profiles it contains) must also be *preferentially stable*, that is, it must be closed under profitable deviations.
We then ask the converse question: when are preferences sufficient to describe the long-run behavior of the players' learning dynamics?
We begin by showing that preferences are indeed enough to fully characterize asymptotic stability in the case of *subgames*—i.e., subsets of pure profiles obtained by restricting players' action sets. Beyond this case however, the equivalence between dynamic and preferential stability breaks down: in particular, we construct a three-player game with a preferentially stable set whose span is dynamically *unstable*, showing that preferences are *not sufficient* to describe dynamically stable behavior in general. To restore stability, we introduce the notion of *leaklessness*, a measure of aggregate payoff drift away from a set of pure profiles, and we use it to identify a payoff-based condition guaranteeing that the span of a set of pure profiles is stable and attracting.

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DeepHA: Scaling Action Chains Elicits Deep Hierarchical Agents
Zihao Wang, Muyao Li, Kaichen He, Haowei Lin, Xiaojian Ma, Anji Liu, Yitao Liang
Prevailing autonomous agents are often constrained by a single, predefined actio
n space, which limits their generalization capabilities across diverse tasks and
 can introduce compounding errors through decoupled policy execution. To address
 these limitations, we introduce the Deep Hierarchical Agent (DeepHA), a unified
 architecture that operates across a mixture of heterogeneous action spaces, fle
xibly generating actions ranging from high-level semantic skills to low-level mo
tor controls. We further propose a Chain-of-Action (CoA) reasoning framework, wh
ich enables the agent to use higher-level abstract actions as structured `though
ts' to guide the generation of more granular, subsequent actions. To manage the
computational demands of this deep reasoning in long-horizon tasks, we develop a
 memory-efficient mechanism that dynamically compresses historical context and l
everages Key-Value (KV) caching, reducing context length by approximately 75% wi
thout sacrificing performance. We conduct extensive evaluations on a new, large-
scale benchmark of over 800 diverse Minecraft tasks. Results show that DHA signi
ficantly outperforms prior methods, establishing a new state-of-the-art and demo
nstrating superior generalization, particularly in complex, multi-step planning
tasks. Our work presents a novel, unified framework for building more capable an
d efficient autonomous agents.
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Diversity-aware Weight Perturbation Promotes Robust Adaptation
Zibo Chen, Ruxin Li, Zilu Wang
Compute-In-Memory (CIM) accelerators are promising for energy-efficient edge inf
erence, yet they faces fundamental challenges when deploying Deep Neural Network
s (DNNs), as hardware-induced weight perturbations from intrinsic noise and devi
ce drift degrade accuracy and impede reliable inference. To tackle this challeng
e, we propose Diversity-aware Weight
Perturbation (DWP), an immune-system-inspired training method that
emulates affinity-based selection by exploiting sample-level
prediction disagreement under diverse noise realizations to guide
adaptive sample weighting, building robustness to weight perturbation. Experimen
ts show that DWP-trained models consistently yield superior robustness, achievin
g over 15\% accuracy improvements compared to standard-trained models under seve
re weight perturbations (mismatch level up to 70\%) and maintaining
inference accuracy at 90\% over a simulated one-year CIM operation
with only 2\%–4\% variation in accuracy. Moreover, under matched model and infer
ence configurations, deployment on low-precision CIM hardware reduces inference
energy by 38\% compared to a GPU baseline. These results demonstrate that DWP en
ables robust and energy-efficient neural network deployment on resource-constrai
ned edge devices with inherent hardware uncertainties.
****
Skill Neologisms: Towards Skill-based Continual Learning
Antonin Berthon, Nicolás Astorga, Mihaela van der Schaar
Modern LLMs show mastery over an ever-growing range of skills, as well as the ab
ility to compose them flexibly. However, extending model capabilities to new ski
lls in a scalable manner is an open problem: fine-tuning and parameter-efficient
 variants risk catastrophic forgetting, while context-based approaches have limi
ted expressiveness and are constrained by the model's effective context.
We explore *skill neologisms*--soft tokens integrated in the model's vocabulary
and optimized to improve capabilities over a specific skill--as a way to selecti
vely acquire new skills without weight updates. We first observe that pre-traine
d LLMs already exhibit tokens associated with procedural knowledge. We then show
 on a controlled synthetic task that skill neologisms can be learned to improve
model capabilities on specific skills while being composable with out-of-distrib
ution skills, and that independently trained skill neologisms can be composed ze
ro-shot. Finally, we validate zero-shot composition of independently learned ski
ll neologisms on the more realistic natural language setting of the Skill-Mix be
nchmark. These results suggest that skill neologisms may provide a scalable path

towards skill-based continual learning.
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Large Language Model Teaches Visual Students: Cross-Modality Transfer of Fine-Grained Conceptual Knowledge
Thomas Liang, Zhuoran Yu, Yong Jae Lee
Large Language Models (LLMs) possess broad conceptual knowledge acquired through large-scale text pretraining, yet their potential to supervise models in other modalities remains underexplored. In this work, we propose LaViD—Language-to-Visual Knowledge Distillation—a simple and effective framework for transferring high-level semantic knowledge from a language-only teacher to a vision-only student model. Instead of relying on paired multimodal data, LaViD elicits conceptual signals from an LLM by prompting it to generate multiple-choice questions (MCQs) that probe semantic distinctions between visual classes. Each class is mapped to a soft label distribution over these MCQs, forming a rich conceptual signature that guides the student through an auxiliary distillation loss. Notably, despite using a language-only teacher without access to image data, LaViD consistently outperforms recent methods like MaKD that distill from vision-language models across multiple fine-grained benchmarks. It also achieves competitive or superior performance compared to state-of-the-art visual distillation methods such as DKD and MLKD, with further gains when combined with logit standardization. On the Waterbirds dataset, LaViD substantially improves worst-group accuracy, demonstrating enhanced robustness to spurious correlations with distillation. Code is available at \url{https://github.com/lliangthomas/lavid}.
****
Olaf-World: Orienting Latent Actions for Video World Modeling
Yuxin Jiang, Yuchao Gu, Ivor Tsang, Mike Zheng Shou
Scaling action-controllable world models is limited by the scarcity of action labels. While latent action learning promises to extract control interfaces from unlabeled video, learned latents often fail to transfer across contexts: they entangle scene-specific cues and lack a shared coordinate system. This occurs because standard objectives operate only *within* each clip, providing no mechanism to align action semantics across contexts. Our key insight is that although actions are unobserved, their *semantic effects* are observable and can serve as a shared reference. We introduce **SeqΔ-REPA**, a sequence-level control-effect alignment objective that anchors integrated latent action to temporal feature differences from a frozen, self-supervised video encoder. Building on this, we present **Olaf-World**, a pipeline that pretrains action-conditioned video world models from large-scale passive video. Extensive experiments demonstrate that our method learns a more structured latent action space, leading to stronger zero-shot action transfer and more data-efficient adaptation to new control interfaces than state-of-the-art baselines.
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Visual Persuasion: What Influences Decisions of Vision-Language Models?
Manuel Cherep, Pranav M R, Patricia Maes, Nikhil Singh
The web is littered with images, once created for human consumption and now increasingly interpreted by agents using vision-language models (VLMs). These agents make visual decisions at scale, deciding what to click, recommend, or buy. Yet, we know little about the structure of their visual preferences. We introduce a framework for studying this by placing VLMs in controlled image-based choice tasks and systematically perturbing their inputs. Our key idea is to treat the agent's decision function as a latent visual utility that can be inferred through revealed preference: choices between systematically edited images. Starting from common images, such as product photos, we propose methods for visual prompt optimization, adapting text optimization methods to iteratively propose and apply visually plausible modifications using an image generation model (such as in composition, lighting, or background). We then evaluate which edits increase selection probability. Through large-scale experiments on frontier VLMs, we demonstrate that optimized edits significantly shift choice probabilities in head-to-head comparisons. We develop an automatic interpretability pipeline to explain these preferences, identifying consistent visual themes that drive selection. We argue th

at this approach offers a practical and efficient way to surface visual vulnerab
ilities, safety concerns that might otherwise be discovered implicitly in the wi
ld, supporting more proactive auditing and governance of image-based AI agents.
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Adaptively Robust Resettable Streaming
Edith Cohen, Elena Gribelyuk, Jelani Nelson, Uri Stemmer
We study algorithms in the resettable streaming model, where the value of each k
ey can either be increased or reset to zero. This model is suitable for applicat
ions such as active resource monitoring with support for deletions and machine u
nlearning. We show that all existing sketches for this model are vulnerable to a
daptive adversarial attacks that apply even when the sketch size is polynomial i
n the length of the stream. To overcome these vulnerabilities, we present the fi
rst adaptively robust sketches for resettable streams that require only polyloga
rithmic space complexity in the stream length. Our framework supports (sub) line
ar statistics including L_p moments for $p\in[0,1]$ (in particular, Cardinalit
y and Sum) and Bernstein statistics. We bypass strong impossibility results kno
wn for linear and composable sketches by designing dedicated single-stream sketc
hes robustified via Differential Privacy. Unlike standard robustification techni
ques, which provide limited benefits in this setting and still require polynomia
l space in the stream length, we leverage the Binary Tree Mechanism for continua
l observation to protect the sketch's internal randomness. This enables accurate
 prefix-max error guarantees with polylogarithmic space.
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A Geometric Analysis of Small-sized Language Model Hallucinations
Emanuele Ricco, Elia Onofri, Lorenzo Cima, Stefano Cresci, Roberto Di Pietro
Hallucinations—plausible but factually incorrect responses—pose a major challeng
e to the reliability of Large Language Models (LLMs), especially in multi-step o
r agentic settings.
Existing work largely frames hallucinations as a consequence of missing knowledg
e; we show instead that, even when the relevant factual knowledge is present, mo
dels still produce hallucinated answers, pointing to retrieval instability rathe
r than knowledge gaps.
Building on this observation, we introduce APORIA (*Aggregate Prompt-wise Observ
ation Retrieving Instability via Asymmetry*—the Socratic state of "puzzlement-in
-contradiction" that hallucinations embody), a geometric framework that studies
repeated responses to the same prompt in sentence-embedding space. Our central h
ypothesis is that genuine responses cluster more tightly than hallucinated ones;
 we empirically validate this and show that, after Fisher projection, the two re
sponse classes become consistently separable.
We leverage this asymmetry in geometry via APORIA-LP, an efficient label-propaga
tion method that classifies large collections of responses from as few as 30–50
annotations, achieving F1 scores above 90% across ten small-sized LLMs.
To support further research, we release SOCRATES-300K, a fully labelled dataset
of 300,000 responses, together with the code for both dataset generation and res
ult reproduction.
Our key finding—framing hallucinations from a geometric perspective in the embed
ding space—complements traditional knowledge-centric and single-response evaluat
ion paradigms, paving the way for further research.
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SpecExit: Accelerating Large Reasoning Model via Speculative Exit
Rubing Yang, Huajun Bai, Song Liu, Guanghua Yu, Runzhi Fan, Yanbin Dang, Zhang J
iejing, Kai Liu, Jianchen Zhu, Peng Chen
Despite their strong performance on reasoning tasks, Large reasoning models (LRM
s) often suffer from overthinking, producing unnecessarily long outputs and incu
rring high end-to-end latency, a significant limitation to their real-world depl
oyment.
To address overthinking, early-exit mechanisms have been proposed to terminate r
easoning before typical completion, showing that this approach can effectively s
horten generation length with minimal impact on accuracy. However, their relianc
e on probing mechanisms introduces a detection overhead that limits their end-to

-end latency gains and compromises their generalizability across diverse problem
s.
Inspired by the use of hidden states in speculative decoding, we propose **SpecE
xit**, a novel framework that predicts both future tokens and an early-exit sign
al directly from a lightweight draft model without probing overhead.
Our method offers significant improvements, achieving up to 66\% generation leng
th reduction and 2.5$\times$ end-to-end speedup compared with the speculative de
coding baseline, without compromising accuracy.
Our method leverages the inherent signals from hidden states to provide effectiv
e early-exit signals, suggesting broader use of hidden states for efficient reas
oning.
Our code is available at: https://anonymous.4open.science/r/SpecExit-B802.
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Domain Restriction via SAE Multi-Layer Transitions
Elias Shaheen, Avi Mendelson
The general-purpose nature of Large Language Models (LLMs) presents a significan
t challenge for domain-specific applications, often leading to out-of-domain (OO
D) interactions that undermine the provider's intent. Existing methods for detec
ting such scenarios treat the LLM as an uninterpretable black box and overlook t
he internal processing of inputs. In this work we show that layer transitions pr
ovide a promising avenue for extracting domain-specific signature. Specifically,
 we present several lightweight ways of learning on internal dynamics encoded us
ing a sparse autoencoder (SAE) that exhibit great capability in distinguishing O
OD texts. Building on top of SAEs representation transitions enables us to bette
r interpret the LLM internal evolution of input processing and shed light on its
 decisions. We provide a comprehensive analysis of the method and benchmark it w
ith the gemma-2 2B and 9B models. Our results emphasize the efficacy of the inte
rnal process in capturing fine-grained input-related details.
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GEMQ: Global Expert-Level Mixed-Precision Quantization for MoE LLMs
Jianing Deng, Song Wang, Dongwei Wang, Zijie Liu, Tianlong Chen, Huanrui Yang, J
ingtong Hu
Mixture-of-Experts Large Language Models (MoE-LLMs) achieve strong performance b
ut incur substantial memory overhead due to massive expert parameters.
Mixed-precision quantization mitigates this cost by allocating expert-wise bit-w
idths based on their importance, approaching the accuracy-memory Pareto frontier
 and enabling extreme low-bit quantization.
However, existing methods rely on layer-wise importance estimation and overlook
router shifts induced by quantization, resulting in suboptimal allocation and ro
uting.
In this work, we propose Global Expert-level Mixed-precision Quantization (GEMQ)
 to overcome these limitations via (1) a global linear-programming formulation t
hat captures model-wide expert importance based on quantization error analysis,
and (2) efficient router fine-tuning to adapt routing to quantized experts. Thes
e components are integrated into a progressive quantization framework that itera
tively refines importance estimation and allocation.
Experiments demonstrate that GEMQ significantly reduces memory and accelerates i
nference with minimal accuracy degradation.
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UI2Code^N: UI-to-Code Generation as Interactive Visual Optimization
Zhen Yang, Wenyi Hong, Mingde Xu, Xinyue Fan, Weihan Wang, Jiale Cheng, Xiaotao
Gu, Jie Tang
UI-to-code aims to translate UI screenshots into executable front-end code.
Despite progress with vision-language models (VLMs), most existing methods formu
late UI-to-code as a single-pass generation, which mismatches real-world UI deve
lopment that is inherently iterative and feedback-driven.
We reformulate UI-to-code as an interactive visual optimization problem, where c
ode generation is embedded in a closed-loop process of execution, visual inspect
ion, and iterative refinement driven by rendered visual feedback.
To address the non-differentiability of visual objectives and the noise of absol

ute visual evaluators, we propose Relative Visual Policy Optimization (RVPO), a preference-based reinforcement learning method that optimizes relative visual rankings among rendered candidates under execution feedback.
We instantiate this paradigm in UI2Code^{N}, an open-source 9B model trained via continual pre-training, supervised fine-tuning, and reinforcement learning.
Experiments demonstrate state-of-the-art performance on UI drafting, UI polishing, and UI editing benchmarks, even outperforming larger models, with performance consistently improving through iterative visual optimization.
Our code and models are available at \url{https://github.com/zai-org/UI2Code_N}.
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DREAM-R: Multimodal Speculative Reasoning with RL-Based Refined Drafting, Precise Verification, and Fully Parallel Execution
Yunhai Hu, Zining Liu, Xiangyang Yin, Tianhua Xia, BO BAO, Eric Sather, Vithursan Thangarasa, Sai Qian Zhang
Speculative reasoning has recently been proposed as a means to accelerate reasoning-intensive generation in large multimodal models, but its effectiveness is often constrained by misalignment between speculative drafts and target-verified reasoning. In this work, we introduce \textit{DREAM-R}, a framework that substantially improves the performance of speculative reasoning. At its core, DREAM-R employs \textit{Speculative Alignment Policy Optimization} (SAPO), a reinforcement-learning objective that trains draft models to generate reasoning steps that are both faithful to target trajectories and concise. We further propose a \textit{Threshold-based Verification Mechanism} (TBVM) that uses a ratio-based criterion to provide stable and interpretable acceptance of speculative steps only when positive evidence clearly dominates, thereby preventing error propagation. Building on these components, we develop a \textit{Fully Parallel Speculative Reasoning} (FPSR) framework that parallelizes draft generation, target-side reasoning, and verification across multi-step reasoning, enabling early stopping and clean fallback. Experiments on reasoning-heavy benchmarks demonstrate up to $2.49\times$ speedup while preserving target-model accuracy, yielding substantial efficiency gains without compromising reasoning quality.
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Midtraining Bridges Pretraining and Posttraining Distributions
Emmy Liu, Graham Neubig, Chenyan Xiong
Midtraining, the practice of mixing specialized data with more general pretraining data in an intermediate training phase, has become widespread in language model development, yet there is little understanding of what makes it effective. We propose that midtraining functions as distributional bridging by providing better initialization for posttraining. We conduct controlled pretraining experiments, and find that midtraining benefits are largest for domains distant from general pretraining data, such as code and math, and scale with the proximity advantage the midtraining data provides toward the target distribution. In these domains, midtraining consistently outperforms continued pretraining on specialized data alone both in-domain and in terms of mitigating forgetting. We further conduct an investigation on the starting time and mixture weight of midtraining data, using code as a case study, and find that time of introduction and mixture weight interact strongly such that early introduction of specialized data is amenable to high mixture weights, while late introduction requires lower ones. This suggests that late introduction of specialized data outside a plasticity window cannot be compensated for by increasing data mixtures later in training. Beyond midtraining itself, this suggests that distributional transitions between any training phases may benefit from similar bridging strategies.
****

Partitioning for Intrinsic Model Inversion Resistance in Collaborative Inference
Rongke Liu, Youwen Zhu, Lei Zhou, Zhang Xianglong, Dong Wang
In collaborative inference (CI), transmitting intermediate representations Z from edge devices enables model inversion attacks (MIA) that reconstruct the original inputs X, while existing defenses mainly perturb shallow-layer Z at the cost of utility. We instead ask: *where should an edge–cloud model be partition

ed to obtain intrinsic resistance to MIA?* We challenge the intuition that depth is the driver of MIA resistance, and show that depth is sufficient only insofar as it enables a representational transition; this transition is necessary for *intrinsic* resistance and is marked by an abrupt rise in the lower bound of $H(X|Z)$. Correspondingly, the decisive variance term in the entropy bound shifts from a global variance to the intra-class mean-squared radius R^2_c rather than dimensionality alone, yielding an R^2_c-based criterion to locate the transition zone, or identify it post hoc from MIA outcomes, which we term the *Golden Partition Zone* (GPZ). We further explain how R^2_c evolves during training and show that it can be controlled through the label distribution; we refer to this controllable dynamic behavior as the *Neural Vortex*, an analysis-backed explanatory concept. Across four representative deep vision models, partitioning at the GPZ yields over 4× higher reconstruction MSE compared to shallow splits; under entropy and inversion-model enhancements, decision-level representations provide 66\% stronger resistance than feature-level ones, and we further observe that data type affects both the transition boundary and reconstruction.

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Biologically plausible heavy-tailed connectivity enhances generalizations on cognitive tasks in recurrent neural networks

Zhe Jiao, Xiaodong He, Shanglin Zhou

While heavy-tailed synaptic weight distributions are pervasive in biological neural networks, their computational role---particularly in relation to generalization---remains poorly understood. To address this, we develop a novel optimal-transport-based optimization algorithm that incorporates key biological constraints, including Dale’s principle and heavy-tailed synaptic statistics, to train recurrent neural networks (RNNs) on a wide range of cognitive tasks. We show that these biologically constrained, heavy-tailed RNNs exhibit substantially improved generalization, which we further characterize within a PAC-Bayes framework. Our theoretical analysis and numerical experiments reveal two complementary mechanisms underlying this generalization enhancement. Topologically, heavy-tailed connectivity induces an effectively low-rank structure, which in turn yields low-dimensional neural dynamics. Geometrically, heavy-tailed connectivity intrinsically shapes task variable representations to lie near a linear manifold, thereby improving generalization for a linear readout strategy. Together, these results identify heavy-tailed connectivity as a biologically grounded intrinsic mechanism that promotes low-rank structure and favorable representational geometry, leading to improved generalization in flexible cognitive tasks.

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Self-Prophetic Decoding to Unlock Visual Search in LVLMs

Zhendong He, Qiyuan Dai, Guanbin Li, Liang Lin, Sibei Yang

Large Vision-Language Models (LVLMs) are rapidly evolving toward true multimodal reasoning, with visual search representing a concrete instantiation of the thinking-with-images paradigm. However, LVLM visual search faces two key challenges: incompatibility among intrinsic capabilities after post-training, and interference in long multi-step reasoning contexts. To address these, we identify two novel insights. First, self-regulation between pre- and post-training LVLMs leverages the intrinsic single-step capabilities of the pre-training model to mitigate capability deterioration and long-context interference. Second, probability-based prophetic sampling, replacing naive prompting, provides a probabilistic interface where the pre-training model acts as a prophet and the post-training model selectively accepts prophetic tokens under its output distribution, preserving coherent multi-step reasoning. Building on these insights, we introduce SeProD, a self-prophetic decoding framework that leverages intrinsic single-step capabilities to enable coherent multi-step reasoning in a training-free, plug-and-play manner. Experiments show that SeProD consistently improves multiple visual-search LVLMs across all 12 splits of 4 visual search benchmarks, as well as across general VQA benchmarks, without added computational overhead, thanks to its parallel prophetic acceptance mechanism.

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The Heterogeneous Safety Impacts of Benign Multilingual Fine-Tuning

Will Hawkins, Kai Rawal, Jonathan Rystrøm, Stratis Tsirtsis, Zihao Fu, Greta Warren, Ryan Brown, Eoin D. Delaney, Sandra Wachter, Brent Mittelstadt, Chris Russell

Fine-tuning a large language model is a ubiquitous method for enhancing its capability on a specific downstream task. However, prior work has shown that this increase in capability comes with a cost: it can increase a model's tendency to respond to unsafe adversarial prompts, even when fine-tuning with non-adversarial data. We present the first comprehensive empirical study of this phenomenon in multilingual settings by fine-tuning Llama-3.2, Qwen3, and Gemma-3 models using benign data translated across nine languages. We find that safety outcomes are highly sensitive to both the choice of fine-tuning language and the evaluation language, with adversarial compliance rates increasing four-fold in some settings. Multilingual safety drift is decoupled from general capability metrics, and occurs heterogeneously across languages and models. Fine-tuning in non-English languages often induces smaller internal representational drifts than English, but these shifts lead models to default to either exaggerated compliance or refusal. As such, assessing fine-tuning impacts solely in English provides inadequate assurance for deployment. To facilitate further research into these cross-lingual safety blind spots, we release the Multilingual-Benign-Tune dataset and the SORRY-Bench-Multilingual evaluation suite.

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Towards Diverse Scientific Hypothesis Search with Large Language Models

Haorui Wang, Parshin Shojaee, Kazem Meidani, Kunyang Sun, José Miguel Hernández-Lobato, Teresa Head-Gordon, Jiajun He, Chandan K. Reddy, Chao Zhang, Yuanqi Du

Large language models (LLMs) are on the rise for accelerating scientific discovery, most recently in advanced tasks such as generating valid scientific hypotheses. Yet in many discovery settings, the goal is not to identify a single best hypothesis since validation can be noisy and expensive, and scientists benefit from a set of high-quality alternative hypotheses that hedge against downstream uncertainty for the best solutions. Nevertheless, commonly used evolutionary search recipes tend to prioritize optimization over exploration in hypothesis generation, and the resulting selection pressure during the search process leads to diversity collapse. Motivated by these limitations, we formulate hypothesis search as a sampling problem, where the objective is to efficiently produce diverse, high-quality hypotheses under a fixed validation budget. Building on this perspective, we propose EvoDiverse, an evolutionary framework inspired by the classical parallel tempering algorithm that searches hypotheses at multiple temperature levels and enables principled information exchange across temperatures to improve exploration without disrupting convergence. Across domains including molecular discovery, equation discovery, and algorithm discovery, our approach consistently improves both hypothesis quality and diversity under the same validation budget, and produces candidates that remain robust under more expensive downstream computational validations.

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MIMO-LP: A Multi-Input Multi-Output Framework for Subgraph-based Link Prediction

Yixin Song, Guangchi Liu, Xiangyu Xu, Shaofeng Li, Zhen Ling, Yiwei Wang, Yujun Cai

Link prediction (LP) is a fundamental problem in graph learning and can be broadly categorized into node-based and subgraph-based approaches. While subgraph-based LP methods often achieve superior predictive performance by exploiting localized structural information, they suffer from efficiency bottlenecks due to the high computational cost of per-query subgraph message-passing during both training and inference. To address this challenge, we propose MIMO-LP, a Multi-Input Multi-Output (MIMO) framework that accelerates subgraph-based LP. Given a batch of query node pairs and their corresponding subgraphs extracted from a shared full graph, MIMO-LP superposes their message-passing processes into a shared latent space while ensuring minimal interference among them. This design enables MIMO-LP to multiplex multiple queries within a single forward pass during both training and inference, substantially reducing redundant message-passing computations in overlapping subgraph regions. Extensive experiments demonstrate that MIMO-LP a

chieves a 14x-44x speedup over existing one-to-one subgraph-based methods, while maintaining comparable predictive performance.
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Geodesic Calculus on Implicitly Defined Latent Manifolds
Florine Hartwig, Josua Sassen, Juliane Braunsmann, Martin Rumpf, Benedikt Wirth
Latent manifolds of autoencoders provide low-dimensional representations of data, which can be studied from a geometric perspective.
We propose to describe these latent manifolds as implicit submanifolds of some ambient latent space.
Based on this, we develop tools for a discrete Riemannian calculus approximating classical geometric operators.
These tools are robust against inaccuracies of the implicit representation often occurring in practical examples.
To obtain a suitable implicit representation, we propose to learn an approximate projection onto the latent manifold by minimizing a denoising objective. This approach is independent of the underlying autoencoder and supports the use of different Riemannian geometries on the latent manifolds. The framework in particular enables the computation of geodesic paths connecting given end points and shooting geodesics via the Riemannian exponential maps on latent manifolds. We evaluate our approach on various autoencoders trained on synthetic and real data.
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UAV^2: A Unified and Adaptive Scheduling Framework for UAV Autopilot System with Reinforcement Learning
Zeying Li, shuai zhao, Chaowen Wu, Boyang Li, Kai Huang
Unmanned aerial vehicle (UAV) autopilot systems typically comprise navigation and flight-control modules, and their effective scheduling is critical to achieving high flight performance.
However, most existing UAV platforms adopt a split architecture in which navigation and flight control are deployed on separate hardware devices. This separation restricts system-wide observability and prevents holistic scheduling and optimization across the entire autopilot pipeline.
Moreover, autonomous flight performance emerges from implicit, cross-coupled, and accumulated interactions among multiple factors, rendering traditional model-based or heuristic scheduling approaches ineffective.
To address these challenges, we propose UAV^2, a unified and adaptive scheduling framework for UAV autopilot systems with reinforcement learning, targeting flight performance optimization.
UAV^2 integrates navigation and flight control onto a single onboard computing platform and operating system, formulates the scheduling problem as a partially observable Markov decision process, and learns scheduling policies from runtime execution feedback.
The proposed approach is trained and evaluated in a hardware-in-the-loop simulation environment. Experimental results demonstrate that the learned scheduling policy consistently outperforms fixed-rate scheduling strategies in terms of flight robustness and tracking performance.
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Deliberate Evolution: Agentic Reasoning for Sample-Efficient Symbolic Regression with LLMs
Xinyu Pang, Zhanke Zhou, Xuan Li, Fangrui Lv, Shanshan Wei, Sen Cui, Bo Han, Changshui Zhang
Symbolic regression (SR) discovers compact mathematical expressions from data, yet recent LLM-based evolutionary methods remain sample-inefficient because they rely mainly on scalar feedback such as MSE. We identify a core limitation: existing methods conflate candidate proposal with search guidance, requiring the LLM to infer how to evolve an expression, diagnose its errors, and reuse past experience from a single score. To address this, we propose Deliberate Evolution (DE), an agentic framework that decouples symbolic generation from search control. DE guides LLM proposals with adaptive operators for search direction, analytical tools for structural diagnosis, and reflective memory for trajectory-level experience. Experiments on LLM-SRBench show that DE consistently outperforms represent

ative LLM-based SR baselines across diverse scientific domains while using only 40% of the standard sample budget. Code is available at https://github.com/Xinyu-Pang/Deliberate-Evolution.

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Thinking in Latent Space: Progressive Multimodal Simplification for Visual Reasoning

Yuesen Tang, Yiming Yang, Tengfei Bao, Yu Tong

Recent Multimodal Large Language Models (MLLMs) have advanced cross-modal reasoning by extending Chain-of-Thought (CoT) prompting to visual tasks. However, existing methods still rely heavily on explicit textual reasoning steps, leading to information loss, unstable perception–reasoning interaction, and high computational cost. Inspired by human cognition, we argue that effective visual reasoning emerges from a dynamic interplay between perception and latent thought, rather than a purely linear verbalization process.

Motivated by this insight, we propose Latent-Driven Progressive Visual Reasoning (LDPVR), a framework that formulates multimodal reasoning as a Markov Chain of Recursive State Simplification, where explicit textual states are progressively refined under the guidance of latent transitions. Central to LDPVR is Interleaved Latent Grounding, which leverages latent semantic intent to actively retrieve fine-grained visual evidence and drive robust state evolution, enabling the model to iteratively reduce uncertainty before committing to simplified textual states.

To optimize this process, we introduce a three-stage curriculum combining supervised fine-tuning, latent-text distillation, and reinforcement learning via Group Relative Policy Optimization (GRPO). Experiments on six multimodal reasoning benchmarks demonstrate that LDPVR improves reasoning accuracy while maintaining low inference latency.

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RAD: Retrieval High-quality Demonstrations to Enhance Decision-making

Lu Guo, Yixiang Shan, Zhengbang Zhu, Qifan Liang, Lichang Song, Ting Long, Weinan Zhang, Yi Chang

Offline reinforcement learning (RL) learns policies from fixed datasets, thereby avoiding costly or unsafe environment interactions. However, its reliance on finite static datasets inherently restricts the ability to generalize beyond the training distribution.

Prior solutions based on synthetic data augmentation often fail to generalize to unseen scenarios in the (augmented) dataset.

To address these challenges, we propose Retrieval High-quAlity Demonstrations (RAD) for decision-making, which innovatively introduces a retrieval mechanism into offline RL. Specifically, RAD retrieves high-return and reachable states from the offline dataset as target states, and leverages a generative model to generate sub-trajectories conditioned on these targets for planning. Since the targets are high-return states, once the agent reaches such a target, it can continue to obtain high returns by following the associated high-return actions, thereby improving policy generalization.

Extensive experiments confirm that RAD achieves competitive or superior performance compared to baselines across diverse benchmarks, validating its effectiveness. Our code is available at https://github.com/LeahGL/RAD.

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Learning Query-Aware Budget-Tier Routing for Runtime Agent Memory

Haozhen Zhang, Haodong Yue, Tao Feng, Quanyu Long, Jianzhu Bao, Bowen Jin, Weizhi Zhang, Xiao Li, Jiaxuan You, Chengwei Qin, Wenya Wang

Memory is increasingly central to Large Language Model (LLM) agents operating beyond a single context window, yet most existing systems rely on offline, query-agnostic memory construction that can be inefficient and may discard query-critical information. Although runtime memory utilization is a natural alternative, prior work often incurs substantial overhead and offers limited explicit control over the performance-cost trade-off. In this work, we present **BudgetMem**, a runtime agent memory framework for explicit, query-aware performance–cost control. BudgetMem structures memory processing as a set of memory modules, each offered

in three budget tiers (i.e., Low/Mid/High). A lightweight router performs budget-tier routing across modules to balance task performance and memory construction cost, which is implemented as a compact neural policy trained with reinforcement learning. Using BudgetMem as a unified testbed, we study three complementary strategies for realizing budget tiers: implementation (method complexity), reasoning (inference behavior), and capacity (module model size). Across LoCoMo, LongMemEval, and HotpotQA, BudgetMem surpasses strong baselines when performance is prioritized (i.e., high-budget setting), and delivers better accuracy–cost frontiers under tighter budgets. Moreover, our analysis disentangles the strengths and weaknesses of different tiering strategies, clarifying when each axis delivers the most favorable trade-offs under varying budget regimes. Code is available at https://github.com/ViktorAxelsen/BudgetMem

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Regret Minimization With a Crowd of Awakening Experts

Anna Lunghi, Gianmarco Genalti, Alberto Marchesi, Matteo Castiglioni

We study the Awakening Crowd of Experts (ACE) problem, an online learning problem where the set of experts available to the learner grows at each round. ACE is a special case of the well-known sleeping experts problem (Kleinberg et al., 2010), where the number of experts is huge $(K=T)$. Existing results on sleeping experts preclude any learner from achieving a sublinear regret when the number of available experts is linear in T. Inspired by real-world applications, such as Q\&A platforms and social proof marketing, we thus focus on the awakening version of the sleeping experts problem, where a new expert arrives at every round and never leaves. We show that in the stochastic version of ACE, it is possible to obtain regret $\tilde{\mathcal{O}}(T^{2/3})$ using an unusual pessimism in the face of the uncertainty principle. Moreover, we characterize the dependence of the regret on the stability of an optimal strategy. For both results, we present matching lower bounds. Surprisingly, the adversarial version of ACE is sensibly harder. In particular, we provide a lower bound precluding sublinear α-regret when the competitive ratio is constant. We provide an algorithm to face this crucial trade-off between competitive ratio and regret, and bound its α-regret, almost matching the aforementioned lower bound. As a corollary, we get a $\tilde{\mathcal{O}}(\log(\log(T))$ competitive ratio when an optimal strategy enjoys a reward linear in T.

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Beyond Explicit Edges: Robust Reasoning over Noisy and Sparse Knowledge Graphs

Hang Gao, Dimitris N. Metaxas

GraphRAG is increasingly adopted for converting unstructured corpora into graph structures to enable multi-hop reasoning. However, standard graph algorithms rely heavily on static connectivity and explicit edges, often failing in real-world scenarios where Knowledge Graphs (KGs) are noisy, sparse, or incomplete. To address this limitation, we introduce INSES (Intelligent Navigation and Similarity Enhanced Search), a dynamic framework designed to reason beyond explicit edges. INSES couples LLM-guided navigation, which prunes noise and steers exploration, with embedding-based similarity expansion to recover hidden links and bridge semantic gaps. Recognizing the computational cost of graph reasoning, we complement INSES with a lightweight router that delegates simple queries to Naïve RAG and escalates complex cases to INSES, balancing efficiency with reasoning depth. Experimental results show that INSES performs favorably compared to established RAG and GraphRAG baselines on multiple benchmarks. In particular, on the MINE benchmark, it exhibits notable robustness and adaptability across KGs constructed by varying methods. Our code and data are publicly available at https://github.com/hanggao-gh/INSES.

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Emergence of Biased Consensus in Multi-Agent LLM Debates

Maya Okawa

Multi-agent LLM debates achieve strong performance on decision-making tasks as well as problem-solving benchmarks, yet their safety and fairness risks remain poorly understood. Notably, interaction can amplify the biases of single LLMs, raising concerns for real-world deployment. We identify the emergence of collective

(often biased) norms in multi-agent LLM debates and show that noise (e.g., LLM sampling temperature) is a key driver. To explain this, we propose an analytical framework drawing on physics-inspired theoretical models of social dynamics. We predict a phase transition to collective bias when conformity surpasses a critical threshold given the LLMs' initial bias and debate noise. We test the theoretical predictions through controlled experiments and observe a finite-size crossover consistent with an underlying phase transition. We further find that agent heterogeneity suppresses emergence by smoothing (rounding) this transition. Finally, we show that these insights generalize to realistic decision-making tasks, including investment decisions and LLM-as-a-judge evaluation.
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ACON: Optimizing Context Compression for Long-horizon LLM Agents
Minki Kang, Wei-Ning Chen, Dongge Han, Huseyin A Inan, Lukas Wutschitz, Yanzhi Chen, Robert Sim, Saravan Rajmohan
Large language models (LLMs) are increasingly deployed as agents in dynamic real-world environments, where success depends on maintaining precise records of actions and observations.
However, the resulting unbounded context growth in long-horizon agentic tasks makes two critical bottlenecks: prohibitive inference memory costs and reasoning degradation due to irrelevant information.
Existing compression methods fail to fully address this, often relying on brittle heuristics or requiring parameter updates impractical for proprietary or large-scale LLMs.
We introduce Agent Context Optimization (ACON), a unified framework that optimally compresses both observations and history into concise, informative representations. Distinct from prior works, ACON employs an optimization in natural language space: it iteratively refines compression guidelines based on failure analysis of the agent, ensuring critical state information is preserved without model fine-tuning. To further minimize computational overhead, we distill the optimized compressor into smaller models.
Experiments on AppWorld, OfficeBench, and Multi-objective QA demonstrate that ACON reduces peak token usage by 26–54\% while improving task success over existing compression baselines. Notably, it enables smaller LMs to function effectively as long-horizon agents, achieving up to 46% performance improvement by mitigating context distraction.
Our code is available at https://github.com/microsoft/acon.
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Neural-Inspired Modeling of Auditory Selection and Compensation for Audio-Visual Speech Separation
Xinmeng Xu, Haoran Xie, Xiaohui Tao, Lin Li, S. Joe Qin
Current audio-visual speech separation (AVSS) models typically rely on implicit multimodal fusion, but the absence of explicit modality alignment and reliability modeling often causes semantic misalignment and contaminates speech representations. The brain addresses this with a hierarchy: top-down auditory selection uses visual priors to maintain target-consistent acoustics, while bottom-up cross-modal compensation integrates temporally aligned articulatory cues to reconstruct and stabilize speech. Guided by this principle, we present Neuro-SCNet, an AVSS architecture that makes selection and compensation explicit and reliability-aware. The Auditory Selection Mechanism applies top-down, visually guided gain along the audio pathway to isolate target time-frequency units and suppress distractors. The module preserves the auditory trace with an identity bypass and adds controlled visual refinements via a residual path. A synchrony-driven gate reduces the influence of low-confidence visual cues. Additionally, a lightweight pre-alignment for visual feature pre-processing estimates and corrects small temporal offsets, and a compact magnitude-phase encoder is used to preserve fine acoustic detail to stabilize reconstruction. Evaluations on LRS2, LRS3, and VoxCeleb2 show state-of-the-art separation with improved efficiency, supporting the value of explicit selection and reliability-aware compensation.
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ZeroDiff: Zero-Shot Time Series Reconstruction via Informed-Prior Diffusion

Yingda Fan, Dan Lu, Xiaowei Jia
Time series modeling increasingly demands high-quality supervision, yet target observations remain scarce—exogenous inputs are broadly available, but target measurements are often unavailable due to cost, infrastructure, or accessibility constraints. Can models trained on observed locations reconstruct target time series where measurements have never been collected? We term this zero-shot time series reconstruction. A naive approach—directly mapping exogenous inputs to targets—can yield predictions at unobserved locations, but without target signals, such models fail to capture the intrinsic dynamics of the target variable, producing overly smooth outputs that underestimate extremes. This reveals systematic errors that call for explicit modeling and calibration. We propose ZeroDiff, which constructs an informed prior from exogenous variables alone, then learns to calibrate reconstruction errors through diffusion—training on observed locations and generalizing to unobserved ones. Experiments across diverse real-world datasets demonstrate significant improvements over existing approaches. Our code is available at https://github.com/YingdaFan/ZeroDiff-ICML2026.
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Pareto-Guided Optimal Transport for Multi-Reward Alignment
Ying Ba, Tianyu Zhang, Mohan Zhou, Yalong Bai, Wenyi Mo, Guiwei Zhang, Bing Su, Ji-Rong Wen
Text-to-image generation models have achieved remarkable progress in preference optimization, yet achieving robust alignment across diverse reward models remains a significant challenge. Existing multi-reward fusion approaches rely on weighted summation, which is costly to tune and insufficient for balancing conflicting objectives. More critically, optimization with reward models is highly susceptible to reward hacking, where reward scores increase while the perceived quality of generated images deteriorates. We demonstrate that optimizing against a unified global target under heterogeneous reward upper bounds can induce reward hacking, a risk further exacerbated by the inherent instability of weak reward models. To mitigate this, we propose a Pareto Frontier-Guided Optimal Transport (PG-OT) framework. Our method constructs a prompt-specific Pareto frontier and maps dominated samples toward it via distribution-aware optimal transport. Furthermore, we develop both online and offline optimization strategies tailored to diverse reward signal characteristics. To provide a more rigorous assessment, we introduce the Joint Domination Rate (JDR) and Joint Collapse Rate (JCR) as principled metrics to quantify multi-reward synergy and reward hacking. Experimental results show that our approach outperforms strong baselines with an 11% gain in JDR and achieves a near 80% win rate in human evaluations.
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FLIPS: Instance-Fingerprinting for LLMs via Pseudo-random Sequences
Richardeau Gurvan, Gohar Dashyan, Erwan Le Merrer, Gilles Tredan
Literature reveals that a Large Language Model's (LLM) behavior is not only conditioned by its original weights but also its instance-level parameters, such as instructional prompt, sampling configuration or quantization. A model that generates safe outputs under one configuration may produce toxic content under another. However, current LLM identification techniques (such as fingerprinting) focus on intellectual property protection, and their design favors robustness to changes in these instance-level parameters. This poses a critical challenge for AI regulation in which compliance assessments target actual deployed behaviors, not model provenance. In this paper, we introduce instance-level fingerprinting, a regulator-oriented paradigm that distinguishes configurations of the same LLM. Our method FLIPS, exploits biases in generated binary random sequences to reach 96% (closed-set) and 90% (open-set, where some targets are unknown) identification accuracy across 237 model instances, versus 35% for the adapted LLMmap baseline. This shows that instance-level fingerprinting is both necessary for regulation and practically feasible. Code available at https://github.com/GurvanR/FLIPS-LLM-Instance-Fingerprinting.
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AdvEvo-MARL: Shaping Internalized Safety through Adversarial Co-Evolution in Multi-Agent Reinforcement Learning

Zhenyu Pan, Yiting Zhang, Zhuo Liu, Yolo Y. Tang, Zeliang Zhang, Haozheng Luo, Chenwei Xu, Yuwei Han, Jianshu Zhang, Dennis Wu, Hong-Yu Chen, Haoran Lu, Haoyang Fang, Manling Li, Chenliang Xu, Philip S. Yu, Han Liu

LLM-based multi-agent systems excel at planning, tool use, and role coordination, but their openness and interaction complexity also expose them to jailbreak and adversarial collaboration. Existing defenses fall into two lines: (i) self-verification that asks each agent to pre-filter unsafe instructions before execution, and (ii) external guard modules that police behaviors. The former often underperforms because a standalone agent lacks sufficient capacity to detect cross-agent unsafe chains and delegation-induced risks; the latter increases system overhead and creates a single-point-of-failure—once compromised, system-wide safety collapses, and adding more guards worsens cost and complexity. To solve these challenges, we propose AdvEvo-MARL, a co-evolutionary multi-agent reinforcement learning framework that internalizes safety into task agents. Rather than relying on external guards, AdvEvo-MARL jointly optimizes attackers (which synthesize evolving jailbreak prompts) and defenders (task agents trained to both accomplish their duties and resist attacks) in adversarial learning environments. To stabilize learning and foster cooperation, we introduce a public baseline for advantage estimation: agents within the same functional group share a group-level mean-return baseline, enabling lower-variance updates and stronger intra-group coordination. Across representative attack scenarios, AdvEvo-MARL consistently keeps attack-success rate (ASR) below 20\%, whereas baselines reach up to 38.33\%, while preserving or even improving task accuracy (up to +3.67\%). These results show that safety and utility can be jointly improved without relying on extra guard agents or added system overhead.

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Grokking Finite-Dimensional Algebra

Pascal Junior Tikeng Notsawo, Guillaume Dumas, Guillaume Rabusseau

This paper investigates the grokking phenomenon, which refers to the sudden transition from a long memorization to generalization observed during neural networks training, in the context of learning multiplication in finite-dimensional algebras (FDA). While prior work on grokking has focused mainly on group operations, we extend the analysis to more general algebraic structures, including non-associative, non-commutative, and non-unital algebras. We show that learning group operations is a special case of learning FDA, and that learning multiplication in FDA amounts to learning a bilinear product specified by the algebra’s structure tensor. For algebras over the reals, we connect the learning problem to matrix factorization with an implicit low-rank bias, and for algebras over finite fields, we show that grokking emerges naturally as models must learn discrete representations of algebraic elements. This leads us to experimentally investigate the following core questions: (i) how do algebraic properties such as commutativity, associativity, and unitality influence both the emergence and timing of grokking, (ii) how structural properties of the structure tensor of the FDA, such as sparsity and rank, influence generalization, and (iii) to what extent generalization correlates with the model learning latent embeddings aligned with the algebra’s representation. Our work provides a unified framework for grokking across algebraic structures and new insights into how mathematical structure governs neural network generalization dynamics.

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Learn to change the world: Multi-level reinforcement learning with model-changing actions

Ziqing Lu, Babak Hassibi, Lifeng Lai, Weiyu Xu

Reinforcement learning usually assumes a given or sometimes even fixed environment in which an agent seeks an optimal policy to maximize its long-term discounted reward. In contrast, we consider agents that are not limited to passive adaptations: they instead have model-changing actions that actively modify the RL model of world dynamics itself. Reconfiguring the underlying transition processes can potentially increase the agents' rewards. Motivated by this setting, we introduce the multi-layer configurable time-varying Markov decision process (MCTVMDP). In an MCTVMDP, the lower-level MDP has a non-stationary transition function tha

t is configurable through upper-level model-changing actions. The agent's object ive consists of two parts: Optimize the configuration policies in the upper-leve l MDP and optimize the primitive action policies in the lower-level MDP to joint ly improve its expected long-term reward.
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Trojan-Speak: Bypassing Constitutional Classifiers with No Jailbreak Tax via Adv ersarial Finetuning
Bilgehan Sel, Xuanli He, Alwin Peng, Ming Jin, Jerry Wei
Fine-tuning APIs offered by major AI providers create new attack surfaces where adversaries can bypass safety measures through targeted fine-tuning. We introduc e **Trojan-Speak**, an adversarial fine-tuning method that bypasses Anthropic's Constitutional Classifiers. Our approach uses curriculum learning combined with GRPO-based hybrid reinforcement learning to teach models a communication protoco l that evades LLM-based content classification. Crucially, while prior adversari al fine-tuning approaches report more than 25\% capability degradation on reason ing benchmarks, Trojan-Speak incurs less than 5\% degradation while achieving 99 +\% classifier evasion for models with 14B+ parameters. We demonstrate that fine -tuned models can provide detailed responses to expert-level CBRN (Chemical, Bio logical, Radiological, and Nuclear) queries from Anthropic's Constitutional Clas sifiers bug-bounty program. Our findings reveal that LLM-based content classifie rs alone are insufficient for preventing dangerous information disclosure when a dversaries have fine-tuning access, and we show that activation-level probes can substantially improve robustness to such attacks.
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Structured Expert Routing with Multi-View Task Priors for Offline Meta-Reinforce ment Learning
Yisen Zhao, Peixi Peng, Xinyu Hu, Cong Li, Zhan Su, Zhuojian Li
Offline meta-reinforcement learning requires agents to generalize to unseen task s from fixed datasets, yet existing sequence-based and MoE-based methods rely on implicit or token-level routing signals that fail to capture task-level structu re.
We propose the **Task-Guided Router (TGR)**, a structured expert-routing framewo rk that explicitly models inter-task relationships via multi-view task represent ations that combine semantic descriptors, behavioral summaries, and latent dynam ics features.
Using structure-guided routing, TGR assigns experts based on global task compati bility rather than local trajectory fragments, enabling stable specialization an d effective knowledge transfer across tasks.Extensive experiments on continuous-control benchmarks demonstrate that TGR consistently outperforms state-of-the-ar t offline meta-RL methods in few-shot generalization, particularly under sparse data and heterogeneous dynamics.
Our results highlight the importance of task-level priors for robust offline met a-reinforcement learning.
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The Perceptual Bandwidth Bottleneck in Vision-Language Models: Active Visual Rea soning via Sequential Experimental Design
Anjie Liu, Ziqin Gong, Yan Song, Yuxiang Chen, Xiaolong Liu, Hengtong Lu, Kaike Zhang, Chen Wei, Jun Wang
Visual perception in modern Vision-Language Models (VLMs) is constrained by a pe rceptual bandwidth bottleneck: a broad field of view preserves global context bu t sacrifices the fine-grained details required for complex reasoning. We argue t hat high-resolution visual reasoning is therefore not only semantic reasoning bu t also task-relevant evidence acquisition under limited perceptual bandwidth. In spired by active vision and information foraging, we formalise this process as s equential Bayesian optimal experimental design (S-BOED), where an agent decides which visual evidence to acquire before answering. Since exact Bayesian inferenc e is intractable in continuous gigapixel spaces, we derive a tractable coverage-resolution objective as a proxy for task-relevant information gain. We instantia te this framework with FOVEA, a training-free procedure that refines VLM crop pr oposals through evidence-oriented probing. Experiments on high-resolution bench

marks show consistent gains over direct and ReAct-style baselines, with particul
arly strong improvements in search-dominated remote-sensing settings.
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Solving Inverse Problems with Flow-based Models via Model Predictive Control
George Webber, Alexander Denker, Riccardo Barbano, Andrew J. Reader
Flow-based generative models provide strong unconditional priors for inverse pro
blems, but guiding their dynamics for conditional generation remains challenging
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Recent work casts training-free conditional generation in flow models as an opti
mal control problem; however, solving the resulting trajectory optimisation is c
omputationally and memory intensive, requiring differentiation through the flow
dynamics or adjoint solves.
We propose MPC-Flow, a model predictive control framework that formulates invers
e problem solving with flow-based generative models as a sequence of control sub
-problems, enabling practical optimal control-based guidance at inference time.
We provide theoretical analysis linking MPC-Flow to the underlying optimal contr
ol objective and show how different algorithmic choices yield a spectrum of guid
ance algorithms, including regimes that avoid backpropagation through the genera
tive model trajectory.
We evaluate MPC-Flow on benchmark image restoration tasks, spanning linear and n
on-linear settings such as in-painting, deblurring, and super-resolution, and de
monstrate strong performance and scalability to massive state-of-the-art archite
ctures via training-free guidance of FLUX.2 (32B) in a quantised setting on cons
umer hardware.
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Equilibrium Propagation for Non-Conservative Systems
Antonino Emanuele Scurria, Dimitri Vanden Abeele, Bortolo Matteo Mognetti, Serge
 Massar
Equilibrium Propagation (EP) is a physics-inspired learning algorithm that uses
stationary states of a dynamical system both for inference and learning. In its
original formulation it is limited to conservative systems, *i.e.* to dynamics w
hich derive from an energy function. Given their applications, it is important t
o extend EP to non-conservative systems, *i.e.* systems with non-reciprocal inte
ractions. Previous attempts to generalize EP to such systems failed to compute t
he exact gradient of the cost function.
Here we propose a framework that extends EP to arbitrary non-conservative system
s, including feedforward networks.
We keep the key property of equilibrium propagation, namely the use of stationar
y states both for inference and learning. However, we modify the dynamics in the
 learning phase by a term proportional to the non-reciprocal part of the interac
tion so as to obtain the exact gradient of the cost function.
This algorithm can also be derived using a variational formulation that generate
s the learning dynamics through an energy function defined over an augmented sta
te space.
Numerical experiments show that this algorithm achieves better performance and l
earns faster than previous proposals.
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Contrastive Reasoning Alignment: Reinforcement Learning from Hidden Representati
ons
Haozheng Luo, Yimin Wang, Jiahao Yu, Binghui Wang, Yan Chen
We propose CRAFT, a red-teaming alignment framework that leverages model reasoni
ng capabilities and hidden representations to improve robustness against jailbre
ak attacks.
Unlike prior defenses that operate primarily at the output level, CRAFT aligns l
arge reasoning models to generate safety-aware reasoning traces by explicitly op
timizing objectives defined over the hidden state space.
Methodologically, CRAFT integrates contrastive representation learning with rein
forcement learning to separate safe and unsafe reasoning trajectories, yielding
a latent-space geometry that supports robust, reasoning-level safety alignment.
Theoretically, we show that incorporating latent–textual consistency into GRPO e

liminates superficially aligned policies by ruling them out as local optima. Empirically, we evaluate CRAFT on multiple safety benchmarks using two strong reasoning models, Qwen3-4B-Thinking and R1-Distill-Llama-8B, where it consistently outperforms state-of-the-art defenses such as IPO and SafeKey. Notably, CRAFT delivers an average **79.0%** improvement in reasoning safety and **87.7%** improvement in final-response safety
over the base models, demonstrating the effectiveness of hidden-space reasoning alignment.
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Stop Training for the Worst: Progressive Unmasking Accelerates Masked Diffusion Training
Jaeyeon Kim, Jonathan Geuter, David Alvarez-Melis, Sham M. Kakade, Sitan Chen
Masked Diffusion Models (MDMs) have emerged as a promising approach for generative modeling in discrete spaces. By generating sequences in any order and allowing for parallel decoding, they enable fast inference and strong performance on non-causal tasks. However, this flexibility comes with a *training complexity* trade-off: MDMs train on an exponentially large set of masking patterns, which is not only computationally expensive, but also creates a train--test mismatch between the random masks used in training and the highly structured masks induced by inference-time unmasking.
In this work, we propose Progressive UnMAsking (PUMA), a simple modification of the forward masking process that aligns training-time and inference-time masking patterns, thereby focusing optimization on *inference-aligned masks* and speeding up training. Empirically, PUMA speeds up pretraining at the 125M scale by $\approx 2.3 \times$ and offers complementary advantages on top of common recipes like autoregressive initialization. We open-source our codebase at https://github.com/JaeyeonKim01/PUMA.
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Progressive Cramming: Reliable Token Compression and What It Reveals
Dmitrii Tarasov, Timofei Lashukov, Elizaveta Goncharova, Andrey Kuznetsov
Token cramming compresses sequences into learned embeddings with near-perfect reconstruction, but fixed token budgets and 99\% accuracy thresholds leave it unclear whether residual errors reflect optimization failures or fundamental limits. We introduce progressive cramming, which grows the target prefix token-by-token, stopping only when reconstruction is no longer achievable within a fixed optimization budget. Progressive trajectories occupy low-dimensional structure in embedding space. Prepending a crammed embedding causes a moderate but consistent accuracy drop on multiple-choice benchmarks even with the original prefix in context, and collapses capability almost entirely under generative evaluation. Causal attention-knockout interventions trace this degradation to the embedding's interactions in the model's early layers. These results position progressive cramming as a tool for studying compression limits and show that perfect reconstruction - achievable through brittle steering rather than transferable semantics - is insufficient for meaningful compression.
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From Evaluation to Design: Using Potential Energy Surface Smoothness Metrics to Guide ML Interatomic Potential Architectures
Ryan Liu, Eric Qu, Tobias Kreiman, Samuel M Blau, Aditi S. Krishnapriyan
Machine Learning Interatomic Potentials (MLIPs) sometimes fail to reproduce the physical smoothness of the quantum potential energy surface (PES), leading to erroneous behavior in downstream simulations that standard energy and force regression evaluations can miss. Existing evaluations, such as microcanonical molecular dynamics (MD), are computationally expensive and primarily probe near-equilibrium states. To improve evaluation metrics for MLIPs, we introduce the Bond Smoothness Characterization Test (BSCT). This efficient benchmark probes the PES via controlled bond deformations and detects non-smoothness, including discontinuities, artificial minima, and spurious forces, both near and far from equilibrium. We show that BSCT correlates strongly with MD stability while requiring a fraction of the cost of MD. To demonstrate how BSCT can guide iterative model design, we utilize an unconstrained Transformer backbone as a testbed, illustrating how

refinements such as a new differentiable k-nearest neighbors algorithm and temperature-controlled attention reduce artifacts identified by our metric. By optimizing model design systematically based on BSCT, the resulting MLIP simultaneously achieves a low conventional E/F regression error, stable MD simulations, and robust atomistic property predictions. Our results establish BSCT as both a validation metric for practitioners to assess MLIP utility and as an "in-the-loop" model design proxy that alerts MLIP developers to physical challenges that cannot be efficiently evaluated by current MLIP benchmarks.
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REALISTA: Realistic Latent Adversarial Attacks that Elicit LLM Hallucinations
Buyun Liang, Jinqi Luo, Liangzu Peng, Kwan Ho Ryan Chan, Darshan Thaker, Kaleab A Kinfu, Fengrui Tian, Hamed Hassani, Rene Vidal
Large language models (LLMs) achieve strong performance across many tasks but remain vulnerable to hallucinations, making it important to systematically evaluate their reliability under realistic adversarial inputs. We formulate hallucination elicitation as a constrained optimization problem, where the goal is to find semantically coherent adversarial prompts that are equivalent to benign user prompts. Existing attack methods remain limited: discrete prompt-based attacks preserve semantic equivalence and coherence but search only over a limited set of prompt variations, while continuous latent-space attacks explore a richer space but often decode into prompts that are no longer valid rephrasings. To address these limitations, we propose REALISTA, a realistic latent-space attack framework. REALISTA constructs an input-dependent dictionary of valid editing directions, each corresponding to a semantically equivalent and coherent rephrasing, and optimizes continuous combinations of these directions in latent space. This design combines the optimization flexibility of continuous attacks with the semantic realism of discrete rephrasing-based attacks. Experiments demonstrate that REALISTA achieves superior or comparable performance to state-of-the-art realistic attacks on open-source LLMs and, crucially, succeeds in attacking large reasoning models under free-form response settings, where prior realistic attacks fail.
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Lightweight Federated Incremental Learning via Decoupled Replay
Xiuying Wang, Yichen Li, Hang Su, Gaozhuo Liu, Shiwei Li, Chuang Zhao, Jiangming Shi, Imran Razzak
Federated Incremental Learning (FIL) aims to learn streaming tasks across distributed clients without catastrophic forgetting while preserving privacy. Most existing methods mitigate forgetting by replaying historical samples, but it can pose privacy risks and incur high resource overhead, limiting deployment on resource-constrained edge devices.
To address this challenge, we propose a novel **Li**ghtweight **F**ederated **I**ncremental **L**earning framework called **Li-FIL** that leverages dense features synthesized by a server-side secure generator to enable efficient feature-based decoupled replay.
More specifically, each client extracts high-confidence features from new tasks, enrich them via mixup, and privatize them before uploading to the server, which reduces both storage and communication overhead. A generator is deployed on the server to learn the distributions of clients and generate global features for replay.
Moreover, to enable clients to better learn from these dense features, we decouple local training into classifier stabilization and encoder regularization. This design allows feature replay and alignment between new and previous features to be conducted separately and more effectively.
Extensive experiments demonstrate that Li-FIL outperforms other state-of-the-art methods by up to 10.14% in terms of accuracy on both old and new tasks with superior resource efficiency.
****
Focus-Then-Contact: Speeding Up Robotic Contact-Rich Task Learning with Affordance-Guided Real-World Residual Reinforcement Learning
Guanren Qiao, Ruixiang Ouyang, Sheng Xu, Ruixing Jin, Yueci Deng, Yunxin Tai, Kui Jia, Guiliang Liu

Real-World Reinforcement Learning (RL) has shown significant potential in roboti c manipulation tasks. However, many methods still require substantial human-in-t he-loop involvement to complete contact-rich tasks, especially when there are di sruptions such as visual backgrounds or positional changes. To address this, we propose the Focus Then Contact (FTC), a lightweight and low-cost method to accel erate the convergence of human-in-the-loop real-world RL for contact-rich tasks.
  FTC leverages residual RL to provide base actions, helping the system quickly reach the target regions and improve sample efficiency. Additionally, FTC integr ates an affordance-guided reward that drives the real-world RL system to quickly focus on key regions of interest, making it possible for the robotic arm to con tinuously engage with these goal areas through force-control feedback. At the sa me time, we optimize the human-in-the-loop implementation to prevent conflicts w ith RL over control of the robotic arm. We demonstrate the effectiveness of FTC on 6 contact-rich tasks, where it outperforms baseline methods in achieving high success rates and speeds up robotic contact-rich task learning under a real-wor ld RL setting.
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CORE: Conflict-Oriented Reasoning for General Multimodal Manipulation Detection
Jinjie Shen, Yaxiong Wang, Yujiao Wu, Lechao Cheng, Tianrui Hui, Nan Pu, Zhihui Li, Zhun Zhong

The rapid rise of generative AI has made multimodal fake news increasingly reali stic and pervasive, posing severe threats to public trust and social stability. Existing detection methods rely heavily on manipulation-specific models and larg e-scale labeled data, resulting in poor generalization to emerging manipulation types. We observed that the essence of manipulated misinformation lies in its in trinsic conflicts, i.e., semantic or physical inconsistencies either across moda lities or with common world knowledge. Inspired by this observation, we propose Conflict-{Oriented REasoning (CORE) framework, an effective paradigm that learn s to endows multimodal large language models (MLLMs) with explicit conflict-capt uring capability. To this end, CORE first constructs the Conflict Attribution Co rpus (CAC) with fine-grained annotations of conflict factors and sources, provid ing essential data support for subsequent conflict perception training. By perfo rming conflict-oriented representation enhancement and reasoning based on CAC, C ORE achieves robust and generalizable conflict detection, effectively and rapidl y adapting to unseen manipulation types with a few samples or in even zero-shot settings. Extensive experiments demonstrate that CORE surpasses state-of-the-art models by 9.7\%, 14.1\%, and 11.8\% in accuracy on the DGM^4, MMFakeBench, an d MDSM benchmarks, respectively.
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The Mechanistic Emergence of Symbol Grounding in Language Models
Shuyu Wu, Ziqiao Ma, Xiaoxi Luo, Yidong Huang, Josue Torres-Fonseca, Freda Shi, Joyce Chai

Symbol grounding (Harnad, 1990) describes how symbols such as words acquire thei r meanings by connecting to real-world sensorimotor experiences. Recent work has shown preliminary evidence that grounding may emerge in (vision-)language model s trained at scale without using explicit grounding objectives. Yet, the specifi c loci of this emergence and the mechanisms that drive it remain largely unexplo red. To address this problem, we introduce a controlled evaluation framework tha t systematically traces how symbol grounding arises within the internal computat ions through mechanistic and causal analysis. Our findings show that grounding c oncentrates in middle-layer computations and is implemented through the aggregat e mechanism, where attention heads aggregate the environmental ground to support the prediction of linguistic forms. This phenomenon replicates in multimodal di alogue and across architectures (Transformers and state-space models), but not i n unidirectional LSTMs. Our results provide behavioral and mechanistic evidence that symbol grounding can emerge in language models, with practical implications for predicting and potentially controlling the reliability of generation.
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Predicting the Emergence of Induction Heads in Language Model Pretraining
Tatsuya Aoyama, Ethan Gotlieb Wilcox, Nathan Schneider

Specialized attention heads dubbed induction heads (IHs) have been argued to underlie the remarkable in-context learning capabilities of modern language models; yet, a precise characterization of their emergence, especially in the context of language modeling, remains wanting. In this study, we investigate the relationship between statistical properties of the training data and IH formation in both natural and synthetic training data settings. We show that: (1) a simple equation combining batch size and context size predicts the point at which IHs form and that this emergence point is agnostic to model size; (2) surface bigram repetition frequency and reliability strongly affect the formation of IHs, and we find an effective decision boundary in terms of these two values; (3) local dependency with high bigram repetition frequency and reliability is sufficient for IH formation, but categoriality and the shape of the marginal distribution appear to modulate IH formation near the decision boundary.
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Rethinking Time-Series Imputation as Conditional Inference along Temporal Evolution
Yu Fan, Yang Yang, Yufan Guo, Huazhong Yang, Pengjun Wang
Real-world time-series data often suffer from missing observations, hindering long-range temporal modeling. However, most existing imputation methods formulate imputation as conditional reconstruction over limited context, which restricts temporal information propagation and fails to explicitly model temporal evolution. To overcome this limitation, we propose the Conditional Temporal Inference Paradigm (CTIP), which formulates time-series imputation as conditional inference along temporal evolution. Under this paradigm, we introduce CBiT, which leverages a history compression mechanism to encode long-range history into a compact latent space for history-conditioned temporal imputation. In addition, we adopt a partitioned modeling strategy that distinguishes historical context and temporal imputation targets with only linear-time complexity. Extensive experiments on multiple public benchmarks show that CBiT improves imputation accuracy by reducing Masked MAE and Masked RMSE by 27.3% and 18.6%, respectively, across different missing rates.
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SPA: A Simple but Tough-to-Beat Baseline for Knowledge Injection
Kexian Tang, Jiani Wang, Shaowen Wang, Kaifeng Lyu
While large language models (LLMs) are pretrained on massive amounts of data, their knowledge coverage remains incomplete in specialized, data-scarce domains, motivating extensive efforts to study synthetic data generation for knowledge injection. We propose $\textbf{SPA}$ ($\textbf{S}$caling $\textbf{P}$rompt-engineered $\textbf{A}$ugmentation), a simple but tough-to-beat baseline that uses a small set of carefully designed prompts to generate large-scale synthetic data for knowledge injection. Through systematic comparisons, we find that $\textbf{SPA}$ outperforms several strong baselines. Furthermore, we identify two key limitations of prior approaches: (1) while RL-based methods may improve the token efficiency of LLM-based data augmentation at small scale, they suffer from diversity collapse as data scales, leading to diminishing returns; and (2) while multi-stage prompting may outperform simple augmentation methods, their advantages can disappear after careful prompt tuning. Our results suggest that, for knowledge injection, careful prompt design combined with straightforward large-scale augmentation can be surprisingly effective, and we hope SPA can serve as a strong baseline for future studies in this area. Our code is available at https://github.com/Tangkexian/SPA.
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Q-Sched: Pushing the Boundaries of Few-Step Diffusion Models with Quantization-Aware Scheduling
Natalia Frumkin, Diana Marculescu
Text-to-image diffusion models remain computationally intensive: generating a single image typically requires dozens of passes through large transformer backbones (e.g., SDXL uses ~50 evaluations of a 2.6B-parameter model). Few-step variants reduce the step count to 2–8 but still rely on large, full-precision backbones, making inference impractical on resource-constrained platforms. Existing post-

training quantization (PTQ) methods are further hampered by their dependence on full-precision calibration.

We introduce **Q-Sched**, a scheduler-level PTQ approach that adapts the diffusion sampler while keeping the quantized weights fixed. By adjusting the few-step sampling trajectory with quantization-aware preconditioning coefficients, Q-Sched matches or surpasses full-precision quality while delivering a **4× reduction** in model size and preserving a single reusable checkpoint across bit-widths. To learn these coefficients, we propose a reference-free **Joint Alignment–Quality (JAQ)** loss, which combines text–image compatibility with an image-quality objective for fine-grained control. JAQ requires only a handful of calibration prompts and avoids any full-precision inference during calibration.

Empirically, Q-Sched yields substantial gains: a **15.5%** FID improvement over the FP16 4-step Latent Consistency Model and a **16.6%** improvement over the FP16 8-step Phased Consistency Model, demonstrating that quantization and few-step distillation are complementary for high-fidelity generation. A large-scale user study with **80,000+** annotations further validates these results on both FLUX.1[schnell] and SDXL-Turbo.

Code: https://github.com/enyac-group/q-sched
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Contextual Slate GLM Bandits with Limited Adaptivity
Tanmay Goyal, Sukruta Prakash Midigeshi, Gaurav Sinha
We investigate the contextual slate bandit problem with generalized linear rewards under limited adaptivity. At each round, the learner is presented with N sets of items, where each item is represented by a d-dimensional feature vector. The learner then constructs a slate by selecting one item per set; the resulting slate yields a scalar reward sampled from a Generalized Linear Model (GLM). We propose algorithms under two limited-adaptivity settings: (a) Batched and (b) Rarely-Switching. For the batched setting, we introduce B-SlateGLinCB, which partitions the time horizon into $\mathcal{O}(\log\log T)$ batches such that each batch's policy relies only on data from previous batches. For the rarely-switching setting, we propose RS-SlateGLinCB, which adaptively performs only $\mathcal{O}(Nd\log T)$ parameter updates. Under a diversity assumption on the item sequences, we prove that B-SlateGLinCB and RS-SlateGLinCB achieve regret bounds of $\mathcal{O}(Nd^{3/2}\sqrt{T})$ and $\mathcal{O}(Nd\sqrt{T})$, respectively. Notably, both bounds are independent of the non-linearity parameter κ that is typically found to scale the regret of GLM bandit algorithms. Our algorithms are computationally efficient, requiring only $\text{poly}(N)$ time per round despite $2^{\Omega(N)}$ possible slates. Simulations show our algorithms outperform existing baselines with limited adaptivity and remain competitive with Slate-GLM-OFU, a fully adaptive state-of-the-art algorithm. Notably, a slightly modified B-SlateGLinCB empirically matches this baseline. Finally, we demonstrate strong performance in a practical in-context example selection task for language models.
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Adaptive Code Watermarking Through Reinforcement Learning
Zhimeng Guo, Huaisheng Zhu, Siyuan Xu, Hangfan Zhang, Teng Xiao, Minhao Cheng
As LLMs increasingly generate production code, protecting intellectual property demands watermarking techniques that respect code's strict syntactic constraints. In this work, we introduce CodeTracer, an innovative adaptive code watermarking framework underpinned by a reinforcement learning training paradigm. At its core, CodeTracer features a policy-driven approach that utilizes a parameterized model to intelligently bias token choices during next-token prediction. This strategy ensures that embedded watermarks maintain code functionality while exhibiting subtle yet statistically detectable deviations from typical token distributions. To facilitate policy learning, we devise a comprehensive reward system that seamlessly integrates execution feedback with watermark embedding signals, balancing process-level and outcome-level rewards. To enable gradient-based optimizat

ion of these discrete watermarking decisions, we employ Gumbel Top-k reparameterization. Extensive comparative evaluations demonstrate that CodeTracer outperforms state-of-the-art baselines across multiple benchmarks in both watermark detectability and code functionality. Our code is available at https://github.com/TimeLovercc/CodeTracer.
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KernelFoundry: Hardware-Aware Evolutionary GPU Kernel Optimization
Nina Wiedemann, Quentin Leboutet, Michael Paulitsch, Diana Wofk, Benjamin Ummenhofer
GPU kernel optimization challenges LLMs beyond standard coding tasks, as it requires an understanding of hardware architecture, parallel computing optimization strategies, and profiling outputs. However, most existing approaches leveraging LLMs for kernel generation apply standard prompting and feedback loops, considering hardware only through profiling feedback. We introduce KernelFoundry, an evolutionary framework that efficiently explores the space of GPU kernels through (1) MAP-Elites quality-diversity search with kernel-specific behavioral dimensions to sustain exploration; (2) meta-prompt evolution that co-evolves prompts with kernels to uncover task-specific optimization strategies, and (3) a template-based parameter optimization approach to tune kernels to inputs and hardware. We evaluate this framework on KernelBench, robust-kbench and custom tasks, generating SYCL kernels as a cross-platform GPU programming paradigm, and CUDA kernels for comparison to prior work. Our approach consistently outperforms the baseline methods and achieves an average speedup of 2.3 on KernelBench for SYCL. Moreover, KernelFoundry is implemented as a distributed framework with remote access to diverse hardware, allowing quick benchmarking and featuring a flexible user input layer to support kernel generation for a wide range of real use cases beyond benchmarking.
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TDM-R1: Reinforcing Few-Step Diffusion Models with Non-Differentiable Reward
Yihong Luo, Tianyang Hu, Weijian Luo, Jing Tang
While few-step generative models have enabled powerful image and video generation at significantly lower cost, generic reinforcement learning (RL) paradigms for few-step models remain an unsolved problem. Existing RL approaches for few-step diffusion models strongly rely on back-propagating through differentiable reward models, thereby excluding the majority of important real-world reward signals, e.g., non-differentiable rewards such as humans' binary likeness, object counts, etc.
To properly incorporate non-differentiable rewards to improve few-step generative models, we introduce TDM-R1, a novel reinforcement learning paradigm built upon a leading few-step model, Trajectory Distribution Matching (TDM).
TDM-R1 decouples the learning process into surrogate reward learning and generator learning.
Furthermore, we developed practical methods to obtain per-step reward signals along the deterministic generation trajectory of TDM, resulting in a unified RL post-training method that significantly improves few-step models' ability with generic rewards.
We conduct extensive experiments ranging from text-rendering, visual quality, and preference alignment.
All results demonstrate that TDM-R1 is a powerful reinforcement learning paradigm for one- and few-step text-to-image models, scaling from 0.6B to 6B parameters while achieving state-of-the-art performance on both in-domain and out-of-domain metrics.
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Balancing Learning Rates Across Layers: Exact Two-Step Dynamics and Optimal Scaling in Linear Neural Networks
Tianyu Pang, Vignesh Kothapalli, Shenyang Deng, Haohui Wang, Dawei Zhou, Yaoqing Yang
We study optimal learning-rate selection in two-layer and three-layer linear neural networks trained to learn linear
target functions. In particular, we derive the exact closed-form expressions for

the gradients and test loss after one and two steps of gradient descent, enabling a precise characterization of early training dynamics. We characterize how learning rates should scale under the gradient approximation in the first two steps, and prove that performing updates with this approximation yields a tractable surrogate loss with a tight, small approximation error. This formulation enables the theoretical analysis of layer-wise learning rates and reveals a distinct early-training regime: test loss can be minimized by unequal learning rates at the initial step, while equal learning rates become optimal in subsequent steps. Our numerical experiments validate the theory and demonstrate the importance of balancing layer-wise learning rates early during training. The code is available at: [TDCSZ327/Layer-Balancing](https://github.com/TDCSZ327/Layer-Balancing).
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FedPDG: Prediction Discrepancy–Guided Data Generation for Heterogeneous Federated Learning

Yuqi Wang, Jianwei Niu, Xinghao Wu, Xuefeng Liu, Xin Hao

One emerging approach to mitigating data heterogeneity in Federated Learning (FL) is to employ diffusion models to generate synthetic data for clients, thereby aligning local data distributions with the global distribution. Prior work has primarily focused on balance-oriented augmentation, which assumes a balanced global class distribution and thus generates samples of rare classes to rebalance each client's local dataset. However, in practice, global data distributions are often inherently imbalanced. Moreover, privacy constraints in FL hinder the server’s ability to accurately estimate the global distribution, rendering balance-oriented augmentation suboptimal. This raises a key, underexplored challenge: How can synthetic data be generated and selected to align local distributions with the true, yet unknown, global distribution? Our key insight is that a model’s performance implicitly reflects the data distribution it has been trained on. Based on this observation, we use the performance discrepancy between local and global models to identify the regions where each client’s local dataset is lacking, and generate corresponding samples for clients. Furthermore, we adapt the diffusion model via preference optimization, enabling it to generate data that better aligns with the true global distribution. Extensive experiments on multiple benchmarks demonstrate that FedPDG outperforms state-of-the-art methods, achieving up to 3.82\% improvement.
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Resting Neurons, Active Insights: Robustifying Activation Sparsity in LLMs via Spontaneity

Haotian Xu, Jiannan Yang, Tian Gao, Tsui-Wei Weng, Tengfei Ma

Activation sparsity offers a compelling route to accelerate large language model (LLM) inference by selectively suppressing hidden activations, yet existing approaches exhibit severe accuracy degradation at high sparsity. We show that this failure stems from representational instability: *activation sparsity disrupts input-dependent activation learned during pretraining, inducing distribution shifts in hidden states*. We address this issue by reframing activation sparsity as a representational alignment problem and introducing **Spontaneous Neurons (SPON)**, a lightweight mechanism inspired by spontaneous neural activity in biological systems. SPON injects a small set of learnable, input-independent activation vectors that act as persistent representational anchors for sparse computation. These vectors are trained via distribution matching to the dense model and can be absorbed into bias terms after training, incurring negligible inference overhead. Across multiple LLM backbones, SPON consistently restores performance, stabilizes latent representations, and preserves generalization. Our results establish SPON as an effective and principled solution for reliable activation-sparse inference, and offer new insights into knowledge retention in LLMs.
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Homophily-Heterogeneity Gradient Surgery for Federated Graph Learning

Sujia Huang, Lele Fu, Shunxin Xiao, Xiaoya Zhang, Chunyan Xu, Tong Zhang, Bo Huang, Zhen Cui

Federated Graph Learning (FGL) facilitates privacy-preserving collaborative training of graph neural networks, yet homophily heterogeneity across subgraphs can

induce optimization conflicts that degrade model generalization.
  Many existing solutions rely on multi-channel architectures to mitigate such c onflicts, which increase the burden on edge devices and lack theoretical converg ence analysis.
  To overcome these limitations, we propose FedGCM, a novel FGL framework with G roup-oriented Conflict Mitigation, which aligns inconsistent optimization object ives via a tailored gradient surgery.
  Specifically, FedGCM first divides clients into distinct groups based on their homophily levels, thereby avoiding exhaustive client-to-client conflict assessm ents.
  To resolve inter-group interference, we develop RPGrad, a gradient surgery mec hanism based on residual projection, which integrates synergistic knowledge whil e filtering inter-group conflicts.
  The refined updates are then transmitted in a group-wise fashion, effectively alleviating optimization conflicts induced by homophily heterogeneity without au gmenting the client-side burden.
  Furthermore, we provide a formal theoretical analysis establishing the converg ence.
  Extensive experiments on both homophilous and heterophilous graphs demonstrate that FedGCM consistently achieves superior performance.
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DIVE: Scaling Diversity in Agentic Task Synthesis for Generalizable Tool Use
Aili Chen, Chi Zhang, Junteng Liu, Jiangjie Chen, Chengyu Du, Yunji Li, Ming Zho ng, Qin Wang, Zhengmao Zhu, Jiayuan Song, Ke Ji, Junxian He, Pengyu Zhao, Yanghu a Xiao
Recent work increasingly synthesizes agentic tasks for post-training tool-using LLMs, yet robust generalization under shifts in tasks and toolsets remains an op en challenge. We trace this brittleness to insufficient diversity in synthesized training tasks. Scaling diversity is difficult because training requires tasks to remain executable and verifiable, while generalization demands diverse tool t ypes, toolset combinations, and heterogeneous tool-use patterns. We propose DIVE , an evidence-driven recipe that inverts synthesis order, executing diverse real -world tools first and reverse-deriving tasks strictly entailed by the resulting traces, providing grounding by construction. DIVE scales structural diversity a long two controllable axes, tool-pool coverage and per-task toolset variety, syn thesizing 48k trajectories over 374 tools across five domains that cover 46,398 unique toolsets and 39,810 unique tool-call graphs. Training Qwen3-8B on DIVE da ta (48k SFT + 3.2k RL) improves by +22 average points across 9 OOD benchmarks an d outperforms the strongest 8B baseline by +68%. Under a fixed budget, controlle d scaling shows diversity scaling consistently outperforms quantity scaling, eve n with 4× less data.
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Training-Free Hashing-Based Attention via Binary Principal Components
Daohai Yu, Zhanpeng Zeng, Keyu Chen, Wenhao Li, Zhifeng Shen, Luxi Lin, Ruizhi Q iao, Xing Sun, Rongrong Ji
Long-context large language models (LLMs) are increasingly deployed in real-worl d applications, yet self-attention remains a major efficiency bottleneck -- espe cially during decoding -- due to the necessity of repeatedly processing ever-gro wing key-value (KV) caches.
Existing sparse attention reduce computation by attending to fewer KV pairs, but often suffer from substantial accuracy degradation, require additional training , or rely on expensive hashing.
In this work, we present \textbf{BinaryPC}, a training-free, data-aware hashing-based sparse attention for long-context LLMs. BinaryPC constructs compact binary hash codes and corresponding hash function by computing binary principal compon ents of data.
Unlike Locality-Sensitive Hashing (LSH) with data-independent random projections or learned non-linear hashing methods, BinaryPC constructs binary codes that ex plicitly preserve the structural information of data without requiring gradient-based training.

Comprehensive experiments across multiple model families and long-context benchm
arks show that BinaryPC preserves accuracy relative to full attention while achi
eving superior performance among sparse and hashing-based baselines.
On modern GPUs, BinaryPC improves end-to-end decoding throughput by 3.56$\times$
 over the FlashAttention kernel.
Our code is available at https://github.com/yudaohai666/BPC.
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RSA-CP: Efficient Conformal Prediction in Small-Sample Regimes via Random Score
Alignment
Pankaj Bhagwat, Zhixian Yang, yihao wang, Bei Jiang, Linglong Kong
Conformal Prediction (CP) provides rigorous finite-sample coverage guarantees, y
et its statistical efficiency hinges critically on the size of the calibration s
et. In data-scarce regimes, CP often suffers from volatile quantile estimation,
leading to overly conservative and wide prediction intervals. To address this, w
e propose Random Score Alignment-Conformal Prediction (RSA-CP), a simple framewo
rk designed to improve sample efficiency in small-sample CP. Instead of requirin
g the computationally intensive generation of full synthetic datasets, RSA-CP en
hances calibration by directly aligning real scores with a high-resolution refer
ence score distribution. By employing an optimal transport mapping, our framewor
k refines "step-like" quantile increments through a globally optimal use of refe
rence information. We provide theoretical guarantees establishing that RSA-CP ma
intains robust coverage without any distributional assumptions on the reference
scores. Empirical evaluations demonstrate that RSA-CP consistently produces shor
ter and more precise prediction intervals while maintaining finite-sample covera
ge guarantees. Overall, RSA-CP offers a computationally efficient and theoretica
lly grounded solution for robust uncertainty quantification under limited data.
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Sample-Efficient Diffusion-based Reinforcement Learning with Critic Guidance
Shutong Ding, Zejia Zhong, Zhongyi Wang, Ke Hu, Bikang Pan, Jingya Wang, Ye Shi
Recent advances in reinforcement learning (RL) have achieved great successes by
leveraging the multimodality and exploration capability of diffusion policies. A
mong these approaches, one representative branch focuses on the sampling-based p
olicy optimization. This design enables better exploration capability of the dif
fusion model, particularly at the beginning of training, but suffer from low exp
loitation in Q-value information, resulting in a slow policy convergence. Anothe
r branch pays attention to gradient-based policy optimization, which sufficientl
y exploits the gradient of the Q function yet tends to collapse into a unimodal
policy with low diversity. To address this issue, we propose CGPO, \textbf{C}rit
ic-\textbf{G}uided diffusion \textbf{P}olicy \textbf{O}ptimization, which effect
ively balances exploration and exploitation with the training-free guidance tech
nique integrated into the denoising process of diffusion policy. Concretely, CGP
O steers action generation toward high-value regions defined by the critic netwo
rk and uses the guided actions as regression objectives. In this manner, CGPO re
duces the time required to obtain high-quality actions and improves final perfor
mance with better balance between the exploration-exploitation tradeoff. We vali
date the effectiveness of CGPO on 5 MuJoCo locomotion tasks, and CGPO achieves s
tate-of-the-art performance compared with existing diffusion-based RL methods. N
otably, CGPO is the first success to incorporate diffusion policy into real-worl
d RL, with its superior performance on Franka robot arm grasping tasks. Our offi
cial page is released at \url{https://dingsht.tech/cgpo-webpage}.
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Deep Coupling Learning for Solving PDEs
Lingshi MENG, Haosen Shi, Sinno Jialin Pan
Physics-Informed Neural Networks (PINNs) represent a significant advancement in
computational methods for solving partial differential equations (PDEs).
However, the adoption of deeper neural network architectures presents significan
t challenges, as they struggle to address differential-related complications tha
t arise during the computation of derivatives over the input of PINNs.
These complications extend beyond traditional vanishing and exploding gradients
to include vanishing and exploding differentials, with both phenomena becoming m

ore severe as networks grow deeper.
By examining the computation graph of derivatives in deep neural networks, we id entify key bottlenecks causing numerical instabilities in deep architectures.
In response, we introduce a novel approach that utilizes Coupling Layers with ca refully regulated spectral norms of Jacobian matrices to stabilize and facilitat e deep PINN training, effectively addressing differential-related challenges and improving model stability.
Our proposed architecture successfully mitigates the fundamental constraints of deeper PINNs while maximizing their capabilities through consistent differential propagation.
Comprehensive evaluations show that our approach surpasses conventional shallow PINN methods and alternative deep PINN designs across a range of challenging pro blems.
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ExCyTIn-Bench: Evaluating LLM agents on Cyber Threat Investigation
Yiran Wu, Mauricio Velazco, Andrew Zhao, Manuel Raúl Meléndez Luján, Srisuma Mov va, Yogesh K Roy, Quang Nguyen, Roberto Rodriguez, Qingyun Wu, Michael Albada, J ulia Kiseleva, Anand Mudgerikar
We present ExCyTIn-Bench, the first benchmark to Evaluate an LLM agent X on the task of Cyber Threat Investigation through security questions derived from inves tigation graphs. Real■world security analysts must sift through a large number o f heterogeneous security logs, follow multi■hop chains of evidence to investigat e threats. With the developments of LLMs, building LLM-based agents for automati c threat investigation is a promising direction. We construct a benchmark from a controlled Azure tenant including a SQL environment covering 57 log tables from Microsoft Sentinel and related services, and 7542 generated questions. We lever age security logs extracted with expert-crafted detection logic to build threat investigation graphs, and then generate questions with LLMs using paired nodes o n the graph, taking the start node as background context and the end node as ans wer. Anchoring each question to these explicit nodes and edges not only provides automatic, explainable ground truth answers but also makes the pipeline reusabl e and readily extensible to new logs.
Our comprehensive experiments on the test set with different models confirm the difficulty of the task: the best model so far can achieve a reward of 0.606, lea ving much headroom for future research.
****

Semi-Supervised Neural Super-Resolution for Mesh-Based Simulations
Jiyeon Kim, Youngjoon Hong, Won-Yong Shin
Mesh-based simulations provide high-fidelity solutions to partial differential e quations (PDEs), but achieving such accuracy typically requires fine meshes, lea ding to substantial computational overhead. Super-resolution techniques aim to m itigate this cost by reconstructing high-resolution (HR), high-fidelity solution s from low-cost, low-resolution (LR) counterparts. However, training neural netw orks for super-resolution often demands large amounts of expensive HR supervisio n data. To address this challenge, we propose SuperMeshNet, an HR data-efficient super-resolution framework for mesh-based simulations aided by message passing neural networks (MPNNs). At its core, SuperMeshNet introduces complementary lear ning, a semi-supervised approach that effectively leverages both 1) a small amou nt of paired LR-HR data and 2) abundant unpaired LR data via two jointly trained , complementary MPNN-based models. Additionally, our model is enriched by induct ive biases, which are empirically shown to further improve super-resolution perf ormance. Extensive experiments demonstrate that SuperMeshNet requires 90% less H R data to achieve even lower root mean square error (RMSE) than that of the full y supervised benchmark without the inductive biases. The source code and dataset s are available at https://github.com/jykim-git/SuperMeshNet.git.
****

RiskZero: Plan More to Risk Less with a Learned Model
Yousef Yassin, Junfeng Wen
AlphaZero and *MuZero* have demonstrated superhuman performance across a range of strategic tasks. Yet their reliance on maximizing expected returns limits th

eir use in real-world settings, where even high-return policies may incur rare b
ut catastrophic failures. We introduce *RiskZero* to address this limitation; th
e first *MuZero*-family method for risk-sensitive decision-making, and planning
with *zero* prior knowledge of environment dynamics. *RiskZero* learns distribut
ional quantities to estimate trajectory-level risk, guiding search toward polici
es that explicitly avoid rare but severe outcomes. We establish theoretical conv
ergence to optimal, stationary risk-sensitive policies and validate our approach
 on environments designed to test risk-sensitive learning from pixels, as well a
s on larger-scale combinatorial tasks. Across all settings, *RiskZero* consisten
tly outperforms state-of-the-art risk-sensitive baselines, and improves sample e
fficiency, providing a general framework for safer and reliable model-based rein
forcement learning under uncertainty.
****

VT-Bench: A Unified Benchmark for Visual-Tabular Multi-Modal Learning
Ziyi Jia, Zi-Jian Cheng, Xinyue Zhang, Kun-Yang Yu, Zhi Zhou, Yu-Feng Li, Lan-Zh
e Guo
Multi-model learning has attracted great attention in visual-text tasks. However
, visual-tabular data, which plays a pivotal role in high-stakes domains like he
althcare and industry, remains underexplored. In this paper, we introduce \texti
t{VT-Bench}, the first unified benchmark for standardizing vision-tabular discri
minative prediction and generative reasoning tasks. VT-Bench aggregates 14 datas
ets across 9 domains (medical-centric, while covering pets, media, and transport
ation) with over 756K samples. We evaluate 23 representative models, including u
nimodal experts, specialized visual-tabular models, general-purpose vision-langu
age models (VLMs), and tool-augmented methods, highlighting substantial challeng
es of visual-tabular learning. We believe VT-Bench will stimulate the community
to build more powerful multi-modal vision-tabular foundation models.

Benchmark: \url{https://github.com/LAMDA-NeSy/VT-Bench}
****

Evaluating Relational Reasoning in LLMs with REL
Lukas Fesser, Yasha Ektefaie, Ada Fang, Sham M. Kakade, Marinka Zitnik
Relational reasoning is the ability to infer relations that jointly bind multipl
e entities, attributes, or variables. While this capability is essential for sci
entific reasoning, most existing evaluations of relational reasoning in large la
nguage models focus on structured inputs such as tables, graphs, or synthetic re
lational tasks, and do not isolate the sources of difficulty that arise from hig
her-arity relational binding. We study this problem through the lens of *Relatio
nal Complexity (RC)*, defined as the minimum number of independent entities or o
perands that must be simultaneously bound to apply a relation. RC provides a pri
ncipled way to vary reasoning difficulty independently of confounders such as in
put size, vocabulary, and representational choices. Building on RC, we introduce
 REL, a generative benchmark framework spanning algebra, chemistry, and biology
that varies RC within each domain. Evaluating frontier LLMs, we observe a consis
tent and monotonic degradation in performance as RC increases, even when the tot
al number of entities is held fixed. This failure mode persists under increased
test-time compute and with in-context learning, suggesting a limitation tied to
the arity of the required relational binding rather than insufficient inference
steps or exposure to examples. Our results identify a well-defined regime of hig
her-arity reasoning in which current models struggle and motivate revisiting rea
soning benchmarks through the lens of relational complexity.
****

Chain-of-Goals Hierarchical Policy for Long-Horizon Offline Goal-Conditioned RL
Jinwoo Choi, Sang-Hyun Lee, Seung-Woo Seo
Offline goal-conditioned reinforcement learning remains challenging for long-hor
izon tasks. While hierarchical approaches mitigate this issue by decomposing tas
ks, most existing methods rely on separate high- and low-level networks and gene
rate only a single intermediate subgoal, leaving several structural limitations
in long-horizon decision-making. To address this limitation, we draw inspiration
 from chain-of-thought reasoning and propose the Chain-of-Goals Hierarchical Pol

icy (CoGHP), a novel framework that reformulates hierarchical decision-making as autoregressive sequence modeling within a unified architecture. Given a state and a final goal, CoGHP autoregressively generates a sequence of latent subgoals followed by the primitive action, where each latent subgoal acts as a reasoning step that conditions subsequent predictions. To implement this efficiently, we introduce an MLP-Mixer backbone, which supports cross-token communication and captures structural relationships among state, goal, latent subgoals, and action. Across challenging navigation and manipulation benchmarks, CoGHP consistently outperforms strong offline baselines, demonstrating improved performance on long-horizon tasks. Project page: https://wlsdn9350.github.io/projects/coghp/
****

Beyond Text-to-SQL: Can LLMs Really Debug Enterprise ETL SQL?
Jing Ye, Yiwen Duan, Yonghong Yu, Xinpei Zhao, Yang Gao, Xing Chen
SQL is central to enterprise data engineering, yet generating fully correct SQL code in a single attempt remains difficult—even for experienced developers and advanced Text-to-SQL LLMs—often requiring multiple debugging iterations. We introduce Squirrel Benchmark, the first benchmark for enterprise-level SQL reasoning and debugging. Our benchmark is built upon two key innovations: (1) an automated construction workflow that employs reverse engineering to systematically inject realistic bugs into large-scale SQL code, enabling scalable and diverse benchmark generation; and (2) an execution-free evaluation framework tailored for enterprise settings, providing fast, accurate, and resource-efficient assessment. Squirrel comprises 469 Squirrel-Syntax queries featuring syntax errors with explicit error messages, and 516 Squirrel-Semantic queries targeting semantic errors where code fails to meet user intent. The queries are highly complex, averaging over 140 lines and featuring deep, wide abstract syntax trees (average width >11, depth >8.7). Evaluation of nearly 30 LLMs reveals a substantial performance gap: the best-performing model, Claude-4-Sonnet, achieves only 36.46% accuracy on Squirrel-Syntax and 32.17% on Squirrel-Semantic, while most models score below 20%. We further explore four solution strategies, identify key challenges, and outline promising directions for enterprise SQL debugging with LLMs.
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Iterative Refinement Neural Operators are Learned Fixed-Point Solvers: A Principled Approach to Spectral Bias Mitigation
Xiaotian Liu, Shuyuan Shang, Xiaopeng Wang, Pu Ren, Yaoqing Yang
Neural operators serve as fast, data-driven surrogates for scientific modeling but typically rely on a monolithic, single-pass inference procedure that struggles to resolve high-frequency details, a limitation known as spectral bias. We introduce the Iterative Refinement Neural Operator (IRNO), which augments pre-trained operators with a learned refinement module iteratively applied via fixed-point iteration. IRNO decomposes the prediction into a coarse initialization followed by successive residual corrections, paralleling classical numerical solvers. Under mild assumptions, we establish contraction of the induced operator, ensuring convergence to a unique fixed point. To explicitly target high-frequency errors, we propose a progressive spectral loss that adaptively increases penalty on high-frequency components over refinement steps during training.
Across physical systems, IRNO consistently lowers error, with up to 56.05\% improvement on turbulent flow. On Active Matter, spectral analysis reveals that, relative to base operator, the normalized error ratios decrease to 27.72–36.10\% in low-, 5.07–6.68\% in mid-, and 1.48–2.04\% in high-frequencies, remaining stable beyond the trained iteration count.
****

Not All Frequencies Are Equal: Energy-Adaptive Diffusion for Time Series Forecasting
Zining Qin, Huiling Qin, Chenhao Wang, Jianxiong Guo, Tian Wang, Weijia Jia
Diffusion models have achieved remarkable success in generative modeling, yet their application to time series forecasting remains suboptimal. Existing approaches apply uniform Gaussian noise across all time steps, assuming all frequency components should be corrupted at the same rate. However, energy distribution across frequencies in time series is highly non-uniform: when uniform noise is added

, high-frequency components are disproportionately overwhelmed while low-frequency trends remain inadequately diffused. We propose EADiff, an energy-adaptive diffusion framework operating in the wavelet domain to address this frequency-energy imbalance. Our key insight is that high-energy components require stronger perturbation while low-energy details need gentler corruption to preserve informative structures. We introduce a learnable modulation mechanism that automatically adjusts noise levels for each frequency band on a per-instance basis. Built upon this adaptive scheduler, we design a conditional diffusion framework where low-frequency trends serve as generation conditions, and noise-level-aware loss weighting naturally emphasizes different frequency components according to their signal characteristics. This cohesive design enables the model to respect the intrinsic multi-scale structure throughout both forward and reverse processes. Extensive experiments demonstrate that EADiff consistently outperforms existing diffusion-based and state-of-the-art deterministic methods.
****

FlashBlock: Attention Caching for Efficient Long-Context Block Diffusion
Zhuokun Chen, Jianfei Cai, Bohan Zhuang
Generating long-form content, such as minute-long videos and extended texts, is increasingly important for modern generative models. Block diffusion improves inference efficiency via KV caching and block-wise causal inference and has been widely adopted in diffusion language models and video generation. However, in long-context settings, block diffusion still incurs substantial overhead from repeatedly computing attention over an ever-growing KV cache. We identify an underexplored property of block diffusion: cross-step redundancy of attention within a block. Our analysis shows that attention outputs from tokens outside the current block remain largely stable across diffusion steps, while block-internal attention varies significantly. Based on this observation, we propose FlashBlock, a cached block-external attention mechanism that reuses stable attention output, substantially reducing attention computation and KV cache access without modifying the diffusion process.
Moreover, FlashBlock is orthogonal to sparse attention and can be combined as a complementary residual reuse strategy. When integrated, it substantially improves model accuracy under aggressive sparsification by offsetting much of the performance loss induced by sparsity.
Experiments on diffusion language models and video generation demonstrate up to 1.44$\times$ higher token throughput and up to 1.6$\times$ reduction in attention time, with negligible impact on generation quality.
****

Less Token, More Signal: MoE Expert Pruning via Critical Token Selection
Zeliang Zong, Kai Zhang, Yarong Wang, Wenming Tan, Ye Ren, Jilin Hu
Mixture-of-Experts (MoE) architectures provide strong scalability for large language models, but their large expert parameter footprint poses challenges for efficient deployment. Expert pruning is widely used to reduce model size and inference cost; however, existing approaches are token-agnostic, treating all tokens equally when estimating expert importance. This uniform treatment dilutes the contributions of informative tokens and leads to suboptimal pruning decisions. To address this fundamental limitation, we propose **Step** (**S**elective **T**oken-guided **E**xpert **P**runing), a token-aware framework that rethinks expert pruning from the perspective of selective token guidance. By incorporating loss-aware expert evaluation and a lightweight knowledge-preserving mechanism, **Step** reduces information loss while removing redundant experts. Extensive experiments across different MoE architectures and model scales demonstrate the effectiveness of **Step**. On the 30B Qwen3 MoE model with 50\% expert sparsity, **Step** achieves nearly a 50\% reduction in memory usage with minimal performance degradation, delivers a 1.5$\times$ throughput improvement, and completes the entire pruning process within 10 minutes.
****

SpaceVista: All-Scale Visual Spatial Reasoning from mm to km
Peiwen Sun, Shiqiang Lang, Dongming Wu, Ding Yi, Kaituo Feng, Huadai Liu, Zhen Ye, Rui Liu, Yun-Hui Liu, Jianan Wang, Xiangyu Yue

With the current surge in spatial reasoning, researchers have made significant p
rogress in understanding indoor scenes, but still struggle with more diverse app
lications. This paper aims to advance all-scale spatial reasoning by tackling tw
o key challenges: 1) the heavy reliance on indoor 3D scans and labor-intensive a
nnotations for dataset curation; 2) the absence of all-scale modeling, which oft
en leads to overfitting to single scenes. In this paper, we introduce a holistic
 solution that integrates a structured spatial reasoning knowledge system, scale
-aware modeling, and a progressive training paradigm, as the **first attempt** t
o broaden the scope of all-scale spatial intelligence. Using a task-specific, sp
ecialist-driven automated pipeline, we curate over 38K video scenes across 5 spa
tial scales to create **SpaceVista-1M**, a dataset comprising 1M spatial QAs spa
nning 19 diverse tasks. While specialist models offer valuable domain knowledge,
 they are often unreliable evaluators. Therefore, we build an all-scale benchma
rk with precise annotations by manually recording and retrieving videos. Neverth
eless, naive training with SpaceVista-1M often yields suboptimal results due to
the potential knowledge conflict. Accordingly, we introduce **SpaceVista-7B**, a
 spatial reasoning model that accepts inputs beyond semantics and uses scale as
an anchor for scale-aware experts and progressive rewards. Finally, extensive ev
aluations across 5 benchmarks, including our **SpaceVista-Bench**, demonstrate c
ompetitive performance, showcasing generalization across all scales and scenario
s. All materials will be released at https://mm2km.github.io/.
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Learning Biophysical Models of Large-Scale Multineuronal Data To Enable Precise Neurostimulation

Amrith Lotlikar, Ian Christopher Tanoh, Praful K. Vasireddy, Andrew Lanpouthakou
n, Ramandeep Vilkhu, Michael A. Sommeling, A.J. Phillips, Alexander Sher, Alan L
itke, Scott Linderman, EJ Chichilnisky, Subhasish Mitra

Multi-compartment Hodgkin–Huxley (HH) models provide a principled framework for
predicting neural dynamics and responses to electrical stimulation. However, fit
ting HH biophysical parameters typically requires intracellular recordings, whic
h are invasive and low-throughput, limiting the ability to capture the geometry
and cell-specific properties of many neurons in a given neural circuit. Multi-el
ectrode arrays (MEAs) offer a scalable alternative—high-density extracellular me
asurements from full neural populations—but HH model complexity has so far precl
uded reliable biophysical inference from extracellular data alone. Here, we intr
oduce a framework to rapidly infer HH parameters from designed features of extra
cellular MEA measurements by leveraging differentiable biophysical simulation an
d simulation-based inference, unlocking a wide range of downstream applications.
 In this work, we focus on a central goal of translational neuroengineering: pre
dicting neural spiking responses to candidate neurostimulation patterns that wou
ld take hours to measure clinically. To validate our approach, we collected hund
reds of hours of stimulation and recording data from isolated macaque retina wit
h a 30 µm-pitch 512-electrode array. Our framework predicted previously unseen m
ulti-electrode stimulation responses with 90.4\% accuracy using HH models fit fr
om only a few minutes of recording, replacing hours of stimulus testing.
****

Beyond Single-View Indexing: Structure-Aware Multi-View Retrieval for Knowledge-Based VQA

Hao Wang, Xujia Li, Lei Chen

Knowledge-Based Visual Question Answering (KB-VQA) relies on retrieval from larg
e-scale knowledge bases, yet this stage is often treated simplistically. Existin
g methods typically adopt single-view indexing or naive multi-view fusion, leadi
ng to systematic coverage gaps. In this work, we demonstrate that different view
s exhibit strong complementarity in retrieval. Motivated by this observation, we
 propose SCAR, a Structure-aware Cross-View Retrieval framework that exploits cr
oss-view structural complementarity at inference time without additional trainin
g. SCAR enhances retrieval via structure-aware similarity propagation within eac
h view and explicit cross-view redundancy regulation. Experiments on multiple KB
-VQA benchmarks demonstrate that SCAR substantially improves retrieval recall, a
pproaches retrieval coverage upper bounds, and consistently boosts end-to-end KB

-VQA performance with negligible inference overhead.
****
Segment-Aligned Policy Optimization for Multi-Modal Reasoning
Lei Gao, Zhuoming Li, Mengxi Jia, Jiakang Yuan, Hongbo Sun, Hao Sun, Xuelong Li
Existing reinforcement learning approaches for Large Language Models typically perform policy optimization at the granularity of individual tokens or entire response sequences. However, such formulations often misalign with the natural step-wise structure of reasoning processes, leading to suboptimal credit assignment and unstable training in multi-modal reasoning tasks. To bridge this gap, we propose Segment-Aligned Policy Optimization (SAPO), a novel reinforcement learning paradigm that treats coherent reasoning steps, rather than tokens or full sequences as fundamental units of policy update. SAPO introduces a step-wise Markov decision process abstraction over reasoning segments, accompanied by segment-level value estimation, advantage computation, and importance sampling mechanisms that are semantically aligned with reasoning boundaries. Experiments on representative reasoning benchmarks demonstrate that SAPO consistently outperforms token-level and sequence-level policy optimization methods, achieving significant accuracy improvements while exhibiting better training stability and value estimation consistency. Our work underscores the importance of aligning reinforcement learning updates with the intrinsic structure of reasoning, paving the way for more efficient and semantically grounded policy optimization in complex reasoning tasks. Code is available at https://github.com/Graysonicc/SAPO.
****
MoSA: Motion-constrained Stress Adaptation for Mitigating Real-to-Sim Gap in Continuum Dynamics via Learning Residual Anisotropy
Jiaxu Wang, Junhao He, Jingkai SUN, Yi Gu, Yunyang Mo, Jiahang Cao, Qiang Zhang, Renjing Xu
Learning real-world dynamics from visual observations is crucial for various domains. A common strategy is to calibrate simulators by estimating physical parameters, yet accuracy is ultimately bounded by the underlying physical models, which often assume materials are homogeneous and isotropic. Even if reasonable, real-world objects typically exhibit mild anisotropy and heterogeneity. After the near-isotropic backbone is well calibrated, these residual effects become the key bottleneck for further closing the real-to-sim gap. Although neural networks can fit dynamics end-to-end, such black-box modeling discards strong physical priors, leading to poor data efficiency and overfitting.
Therefore, we propose MoSA, a motion-constrained stress adaptation framework that targets these residual effects to further improve real-to-sim dynamics learning. MoSA uses an isotropic model as a physics prior and learns residual stress operators to capture mild anisotropy and heterogeneity. It progressively adapts stresses via microplane-constrained redistribution in a physics-informed cascaded network. We further impose motion constraints by supervising temporal and spatial derivatives of the deformation field. Experimentally, our learned dynamics achieves superior accuracy, generalization, and robustness, while learning physically meaningful residual anisotropy. Finally, we validate MoSA in a robot manipulation setting, showing that better real-to-sim dynamics modeling translates into more reliable sim-to-real transfer.
****
A^{2}SG: Adaptive and Asymmetric Surrogate Gradients for Training Deep Spiking Neural Networks
Yechan Kang, Yongjin Kweon, Mingyeong Seo, Sohee Park, Yeonguk jeon, Jongkil Park, Hyun Jae Jang, Jaewook Kim, YeonJoo Jeong, Suyoun Lee, Seongsik Park
Training deep spiking neural networks (SNNs) remains challenging due to sharp loss landscapes and temporal inconsistency caused by surrogate gradients. To address these challenges, we propose a unified framework: adaptive and asymmetric surrogate gradients (A^{2}SG). The adaptive gradients adjust an effective window for spatio-temporal adaptation, reducing spatial gradient variation and maintaining directional consistency of gradients over time. The asymmetric gradients reflect neuronal dynamics by assigning larger gradients to neurons with higher membrane potentials, and we prove that they yield lower variation than symmetric surro

gates. Our analysis further establishes a direct connection between local gradie nt variation and the curvature of the loss landscape, providing a principled exp lanation for how A^{2}SG promotes convergence to flatter minima and improves gen eralization. We conduct extensive experiments on diverse models, including CNN-b ased and Transformer-based SNNs, across various tasks such as image classificati on using both static and neuromorphic datasets, as well as segmentation. The res ults demonstrate that A^{2}SG consistently improves accuracy and energy efficien cy, establishing it as a general and reliable solution for training deep SNNs.Ou r code is available at https://github.com/KIST-NCL/A2SG.git.
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Omni-Perception Policy Optimization for Multimodal Emotion Reasoning
Zhiyuan Han, Beier Zhu, Wenwen Tong, Pengyang Shao, Peipei Song, Xinyi Wang, Jia ngnan Chen, Lewei Lu, Xun Yang
We find that current emotion-oriented Omni-MLLMs still lack *reliable omni-modal perception*: they (i) underutilize multimodal cues in their reasoning trajector ies and (ii) exhibit unfaithful behavior, often hallucinating modality-specific statements from other modalities.
Building on these insights, we propose **OPPO** (**O**mni-**P**erception **P**ol icy **O**ptimization), a reinforcement learning framework that explicitly optimi zes multimodal perception.
First, an Omni-Perception Reward decomposes ground-truth reasoning into fine-gra ined visual, acoustic, and emotion cues and rewards trajectories that semantical ly recover these cues.
Second, an Omni-Perception Loss compares the policy under full and unimodally ma sked inputs, applying a KL penalty only to modality-specific evidence tokens to suppress cross-modal hallucination.
We further introduce *MEP-Bench*, a diagnostic benchmark that quantifies *utiliz ation* and *faithfulness*. Experiments show that OPPO achieves state-of-the-art performance on MER-UniBench and substantially improves utilization and faithfuln ess scores on MEP-Bench, highlighting the importance of sufficient and faithful omni perception for multimodal emotion reasoning.
****

Benchmarking Dense and Indiscernible Object Counting with Blueberries
Weihao Bo, Yanpeng Sun, Jingwen Qin, Fei Shen, Xiaofan Li, Zechao Li
Real-world agricultural counting often operates in the extreme regime of \textbf {Dense and Indiscernible Object Counting (DIOC)}, where targets are tiny, cluste red, and highly camouflaged. To facilitate research in this domain, we introduce \textbf{DIOCblueberry}, a large-scale benchmark that pushes the boundaries of v isual perception. Unlike general datasets with salient objects, DIOCblueberry fe atures extreme occlusion and camouflage. Compared to the popular FSC147 benchmar k, it contains \textbf{1.9$\times$ more instances} per image (avg. 108) with an average box pixel ratio that is \textbf{7.9$\times$ smaller}, serving as a rigor ous testbed for model robustness. Standard counting methods struggle in these sc enarios due to severe visual ambiguity and scale mismatch. To address this, we p ropose \textbf{MaskCount}, a coarse-to-fine framework that incorporates semantic guidance. MaskCount leverages Vision-Language Models (CLIP) to generate pseudo segmentation masks for background suppression and employs a contrastive loss to maximize feature discriminability between fruits and foliage. Additionally, we d esign an edge-aware cropping mechanism to resolve boundary truncation in dense c lusters. Extensive experiments demonstrate that MaskCount achieves a new state-o f-the-art, reducing MAE and RMSE by \textbf{49.16\%} and \textbf{70.50\%} respec tively on DIOCblueberry, with strong generalization to other agricultural scenes .
****

EvReflection: Event-Driven Micro-Dynamics for Reflection Removal
Jiaxiao Wang, Dachun Kai, Huyue Zhu, Quanquan Hu, Zhenyang Xu, Xiaoyan Sun
Despite remarkable progress in reflection removal, current methods primarily exp loit static image priors from a single frame and still suffer from severe residu al artifacts due to the inherent ambiguity between the reflection and transmissi on layers. In this paper, we propose leveraging event signals to break this ambi

guity. By employing event cameras to capture micro-dynamics, we reveal the differential motion between these two layers. We thereby present a novel event-driven reflection removal network, EvReflection, that utilizes these dynamic cues for layer separation. Specifically, we design a Micro-Dynamics Decoupler to disentangle layer-specific motions from event streams as priors, which then guide a Parallax-Attention Rectifier to cleanly remove artifacts from the RGB image. Furthermore, to address data scarcity, we develop a parallax-aware simulation pipeline and construct the EVR^2 benchmark dataset, the first real-world dataset for this task.

Extensive experiments demonstrate that EvReflection achieves state-of-the-art performance on both synthetic and real-world benchmarks, surpassing the best competing method by more than 1.6 dB and 1.2 dB in PSNR, respectively.

The code, dataset, and pre-trained models are available at \href{https://github.com/JiaxiaoWang/EvReflection}{https://github.com/JiaxiaoWang/EvReflection}.

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Veda: Scalable Video Diffusion via Distilled Sparse Attention

Shihao Han, Hao Yang, Xiaofeng Mei, Xinting Hu, Yi Jiang, Xiaojuan Qi

Scaling Diffusion Transformers to generate high-resolution, long videos is constrained by the quadratic cost of self-attention, and existing sparse attention methods degrade under high sparsity. We show empirically that generation quality is determined not by the sparsity ratio itself, but by how well the sparse mask aligns with the tile-wise geometry of full attention. Based on this insight, we propose Veda, a distilled sparse attention framework that formulates tile selection as an explicit reconstruction problem from full attention. Veda integrates statistics-aware tile scoring with head-aware tiling to reduce estimation error and structural mismatch, enabling aggressive sparsity. A hardware-efficient tile-skipping kernel converts theoretical sparsity into practical wall-clock speedups. Experiments on large video diffusion models, including Waver and Wan2.1, demonstrate substantial acceleration with no noticeable degradation in generation quality. To generate 720P 10-second videos on Waver-T2V-12B, Veda achieves a 5.1$\times$ end-to-end speedup and a 10.5$\times$ self-attention speedup, reducing attention overhead from 92\% to 50\%. Notably, the gains increase with sequence length, indicating that Veda scales favorably with spatiotemporal resolution across models.

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CLIMB: Taming the LoRA Residency Cliff in Multi-LoRA Serving

Haoran Zhang, Zhiyu Liang, ZUO Decheng, Hongzhi Wang

Multi-tenant multi-LoRA serving multiplexes many LoRA adapters on a single GPU under high utilization, where most device memory is reserved for the KV cache, leaving only a small residency budget K for adapters.

In this regime, adapter readiness is atomic: if an adapter is not device-resident, the engine must perform a mandatory fetch, stalling shared execution and amplifying tail latency system-wide.

With only K residency slots, we identify a LoRA residency cliff: once the active adapter working set exceeds K, time-to-first-token (TTFT) tail latency can exhibit a congestion collapse rather than smooth degradation.

To tame this cliff, we propose CLIMB, a minimal ingress controller that enforces feasibility-first admission by queueing non-resident adapters outside the engine, prioritizing critical (VIP) traffic, and rotating background adapters via round-robin.

On a cliff-inducing workload, CLIMB averts collapse, reducing VIP TTFT p99 from 38.7 s to 13.1 s at matched throughput (10.66 rps) by keeping VIP engine latency near 0.13 s and shifting the residual tail into explicit ingress queueing.

Overall, CLIMB shifts fetch-induced stalls from inside the engine to managed ingress queues, mitigating tail amplification without throughput loss in the evaluated settings.

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AgentConductor: Topology Evolution for Multi-Agent Competition-Level Code Generation

Siyu Wang, Ruotian Lu, Zhihao Yang, Yuchao Wang, Yanzhou Zhang, Lei Xu, Qimin Xu

, Guojun Yin, Cailian Chen, Xinping Guan
Large language model(LLM)-driven multi-agent systems(MAS) coordinate specialized agents through predefined topologies and show promise for complex tasks such as competition-level code generation. Recent studies demonstrate that carefully designed multi-agent workflows and communication graphs can significantly improve code generation performance by leveraging collaborative reasoning. However, existing methods neither adapt topology density to task difficulty nor refine the topology within an instance using execution feedback, which leads to redundant communication and performance bottlenecks. To address these issues, we propose AgentConductor: a reinforcement learning-optimized MAS with an LLM-based orchestrator agent as its core, which enables end-to-end feedback-driven dynamic generation of interaction topologies. For each query, AgentConductor infers agent roles and task difficulty, then constructs a task-adapted, density-aware layered directed acyclic graph(DAG) topology, underpinned by two key innovations. First, we design a novel topology density function to quantify communication-aware multi-agent interactions. Second, we adopt difficulty interval partitioning to avoid excessive pruning for precise topological density upper bound measurement per difficulty level and finer-grained control. Across five code datasets, AgentConductor outperforms the strongest baseline by up to 14.6\% in pass@1, with 13\% lower topology density and 68\% lower token cost.
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PCGS: Deblurring 3D Gaussian Splatting with Patch Comparison
Yilong Li, Bo Pang, Zhongtao Wang, Mai Su, Yisong Chen, Chengwei Pan, Meng GAI, Fei Zhu, Guoping Wang
Recent neural methods, such as 3D Gaussian Splatting, have achieved state-of-the-art rendering quality and speed. However, these methods frequently encounter challenges in regions with overlapping Gaussians, leading to blurring and artifacts in the rendered images. We observed that widely used view-space positional gradients are insufficient for handling such circumstances. To address this, we introduce PCGS, a Patch Comparison Gaussian Splatting method to control the densification of corresponding Gaussians adaptively. Specifically, PCGS divides the rendered image into patches and identifies those with significant errors by comparing the loss between the rendered and ground truth images. Additional densification operations are then applied to the Gaussians in these error-prone regions. Furthermore, to prevent over-densification and redundant Gaussians, we design a Gaussian control strategy to regulate the densification process. Specifically, we set a Gaussian number budget that dynamically changes according to the progress of densification, and sample the Gaussians required for each densification step based on their importance scores. Our method results in significantly fewer artifacts and less blur while maintaining a Gaussian count approximately equal to that of 3DGS. Extensive experiments on multiple standard benchmarks demonstrate the superiority of our approach.
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SAOT: Self-Supervised Continual Graph Learning with Structure-Aware Optimal Transport
Yuting Zhang, Yanbei Liu, Zhitao Xiao, Lei Geng, Yanwei Pang, Xiao Wang
Self-supervised Continual Graph Learning (CGL) aims to successively learn from a graph sequence with different tasks without label supervision—a paradigm that has attracted widespread attention. Most existing self-supervised CGL methods rely on instance-level consistency objectives that enforce stability of individual node (or node-pair) embeddings. Due to optimizing nodes in isolation, these methods fail to maintain global relational structure, causing inter-node correspondences to progressively distort under continual learning. To this end, we propose a novel Structure-Aware Optimal Transport (SAOT) framework that explicitly captures and preserves relational structure within graph representations across sequential tasks. Specifically, SAOT leverages optimal transport theory to capture global inter-node correspondences, thereby facilitating and enhancing graph representation learning. Simultaneously, SAOT incorporates a cross-task knowledge distillation mechanism to preserve the previous structural knowledge. Extensive experiments on four CGL benchmark datasets demonstrate that SAOT outperforms existin

g self-supervised baselines. In particular, SAOT achieves significant performanc e gains, improving average accuracy by up to 5% on CoraFull-CL and over 15% on P roducts-CL compared with state-of-the-art methods in the Class-IL setting.
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MoshiRAG: Asynchronous Knowledge Retrieval for Full-Duplex Speech Language Model s

Chung-Ming Chien, Manu Orsini, Eugene Kharitonov, Neil Zeghidour, Karen Livescu, Alexandre Défossez

Speech-to-speech language models have recently emerged to enhance the naturalnes s of conversational AI.
In particular, full-duplex models are distinguished by their real-time interacti vity, including handling of pauses, interruptions, and backchannels.
However, improving their factuality remains an open challenge.
While scaling the model size could address this gap, it would make real-time inf erence prohibitively expensive.
In this work, we propose Moshi-RAG, a modular approach that combines a compact f ull-duplex interface with selective retrieval to access more powerful knowledge sources.
Our asynchronous framework enables the model to identify knowledge-demanding que ries and ground its responses in external information.
By leveraging the natural temporal gap between response onset and the delivery o f core information, the retrieval process can be completed while maintaining a n atural conversation flow.
With this approach, Moshi-RAG achieves factuality comparable to the best publicl y released non-duplex speech language models while preserving the interactivity inherent to full-duplex systems.
Moreover, our flexible design supports plug-and-play retrieval methods without r etraining and demonstrates strong performance on out-of-domain mathematical reas oning tasks.
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BASIL: Scalable Bayesian Semi-supervised Clustering with Feature Selection and A daptive Constraint Weighting

Luwei Wang, Dagmara Panas, KE WANG, Bruce Guthrie, Sohan Seth

Constrained clustering incorporates prior knowledge in the form of pairwise cons traints to guide data partitioning. While effective, existing Bayesian approache s are often limited in scalability to large datasets and provide weak interpreta bility due to the lack of explicit feature relevance modeling. We propose BASIL, a scalable Bayesian semi-supervised clustering framework that leverages stochas tic variational inference to jointly infer cluster assignments and feature impor tance weights. This joint formulation enables the identification of discriminati ve features consistent with the imposed constraints. To robustly handle noisy or inconsistent supervision, BASIL introduces an adaptive constraint-weighting mec hanism that down-weights unreliable constraints. Experiments on synthetic and re al-world benchmarks show BASIL attains competitive accuracy while reducing train ing time by over $96\\%$ on large datasets, learns interpretable cluster-specifi c feature importance maps, and remains robust to up to $30\\%$ noisy constraints under sufficient supervision. We further demonstrate applicability to large-sca le health data, including medical imaging and electronic health records.
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Attacking Gray-Box Large Vision-Language Models with Adaptive SVD-Structured Adv ersarial Alignment

Daizong Liu, Xiaowen Cai, Junhao Dong, Zhongliang Guo, Xiaoye Qu, Runwei Guan, X iang Fang, Dengpan Ye

Large vision-language models (LVLMs) have demonstrated remarkable capabilities a cross a wide range of multimodal reasoning tasks. However, recent research shows that they are susceptible to adversarial examples. Existing LVLM attack methods are generally deployed in the white- or black-box setting, which severely rely on full-model gradients or elaborated transfer strategies, resulting in large re source costs. To this end, this paper focuses on a more efficient gray-box attac k setting by solely accessing LVLM's vision encoder. Instead of using target ima

ges as the adversarial guidance, our main goal is to perturb the visual feature to best match more natural attacker-chosen target texts. Specifically, we develop a global semantic alignment module to project the visual features onto the SVD-structured subspace spanned by the textual semantics. We also propose to align detailed visual features with multi-context semantic texts extended by LLMs over discrete distributions via optimal transport. Extensive experiments demonstrate the superiority of the proposed method, while our attack is further proven to achieve great transferability across various LVLMs with CLIP-aware transfer designs.

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Adaptive Multiscale Binary Expansion Tests for Independence

Yang Yang, Duo Zheng, Sandeep Jain, Kai Zhang, Ping-Shou Zhong

This paper introduces a new family of adaptive, distribution-free independence tests for multivariate random vectors based on binary expansion coefficients, supported by a tractable asymptotic theory. Our first key contribution establishes a general equivalence between independence testing and testing cross-covariances among exponentially many binary expansion interaction coefficients, applicable to broad sample spaces and not limited to kernel-induced representations. While this exponential interaction structure makes naive construction and computation infeasible, we overcome this challenge by reformulating the proposed tests as a class of U-statistics and deriving an explicit kernel representation that enables scalable and efficient computation. Exploiting the multiscale nature of binary expansions, the proposed framework automatically adapts to unknown dependence structures by selectively truncating higher-order interactions, yielding both strong power and clear interpretability. To further enhance power and computational efficiency, we introduce an adaptive weighted aggregation procedure, termed wa-dCoBET, which combines a baseline Covariance Binary Expansion Test (CoBET) with a distance-measure–based CoBET. Extensive simulations and a real-data application demonstrate that wa-dCoBET consistently matches or outperforms HSIC and distance covariance, particularly in higher-dimensional and non-monotone settings, while maintaining accurate type I error control.

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Error Propagation Mechanisms and Compensation Strategies for Quantized Diffusion Models

Songwei Liu, Chao Zeng, Chenqian Yan, Xurui Peng, XING WANG, Fangmin Chen, Xing Mei

Diffusion models have transformed image synthesis by establishing unprecedented quality and creativity benchmarks. Nevertheless, their large-scale deployment faces challenges due to computationally intensive iterative denoising processes. Although post-training quantization (PTQ) provides an effective pathway for accelerating sampling, the iterative nature of diffusion models causes stepwise quantization errors to accumulate progressively during generation, inevitably compromising output fidelity. To address this challenge, we develop a theoretical framework that mathematically formulates error propagation in Diffusion Models (DMs), deriving per-step quantization error propagation equations and establishing the first closed-form solution for cumulative error. Building on this theoretical foundation, we propose a timestep-aware cumulative error compensation scheme. Extensive experiments on multiple image datasets demonstrate that our compensation strategy effectively mitigates error propagation, significantly enhancing existing PTQ methods. Specifically, it achieves a 1.2 PSNR improvement over SVDQuant on SDXL W4A4, while incurring only an additional $<$ 0.5\% time overhead.

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ReLAM: Learning Anticipation Model for Rewarding Visual Robotic Manipulation

Nan Tang, Jing-Cheng Pang, Guanlin Li, Chao Qian, Yang Yu

Reward design remains a critical bottleneck in visual reinforcement learning (RL) for robotic manipulation. In simulated environments, rewards are conventionally designed based on the distance to a target position. However, such precise positional information is often unavailable in real-world visual settings due to sensory and perceptual limitations. In this study, we propose a method that implicitly infers spatial distances through keypoints extracted from images. Building

on this, we introduce Reward Learning with Anticipation Model (ReLAM), a novel framework that automatically generates dense, structured rewards from action-free video demonstrations. ReLAM first learns an anticipation model that serves as a planner and proposes intermediate keypoint-based subgoals on the optimal path to the final goal, creating a structured learning curriculum directly aligned with the task's geometric objectives. Based on the anticipated subgoals, a continuous reward signal is provided to train a low-level, goal-conditioned policy under the hierarchical reinforcement learning (HRL) framework with provable sub-optimality bound. Extensive experiments on complex, long-horizon manipulation tasks show that ReLAM significantly accelerates learning and achieves superior performance compared to SOTA methods.

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Near-Minimax Multi-Objective RL under Predictable Adversarial Preferences and Preference-Free Exploration in Linear MDPs

Mingxi Hu, Meiling Yu

Multi-objective reinforcement learning (MORL) must often support preferences that change online or are specified only after data collection.

We study finite-horizon MORL with vector feedback in linear MDPs under two protocols: (i) predictable adversarial preferences revealed before each episode, and (ii) reward-free preference-free exploration (PFE), where exploration observes only transitions and must later answer arbitrary preference queries.

Standard reductions are protocol-unsafe: re-scalarizing past stochastic rewards with future weights breaks the martingale structure needed for self-normalized confidence bounds, and hypervolume evaluation must account for episode-start randomization, which yields a deployable convex hull of return vectors.

We propose a protocol-safe reward interface that estimates each reward coordinate via regression and performs scalarization only at query time, and we formalize deployable hypervolume semantics with a stability chain from support-function error to hypervolume error.

Consequently, we obtain filtration-safe regret bounds for any predictable preference sequence without discretizing the simplex (only $\log m$ dependence) and matching near-minimax rates in linear MDPs, as well as sharp reward-free PFE guarantees: a (near-)minimax decision-optimal query answering rate $\tilde{O}(d^2 U_{\mathrm{ret}}^2/\varepsilon^2)$ and a tight separation from explicit transition-model recovery $\Theta(d(|\mathcal{S}|-1)/\varepsilon_P^2)$.

These results connect online learning, preference-free deployment, and hypervolume-aware evaluation through a single protocol-aligned theory.

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Jailbreaking Vision-Language Models Through the Visual Modality

Aharon Azulay, Jan Dubi■ski, Zhuoyun Li, Atharv Mittal, Yossi Gandelsman

The visual modality of vision-language models (VLMs) is an underexplored attack surface for bypassing safety alignment. We introduce four jailbreak attacks exploiting the vision component: (1) encoding harmful instructions as visual symbol sequences with a decoding legend, (2) replacing harmful objects with benign substitutes (e.g., bomb $\rightarrow$ banana) then prompting for harmful actions using the substitute term, (3) replacing harmful text in images (e.g., on book covers) with benign words while visual context preserves the original meaning, and (4) visual analogy puzzles whose solution requires inferring a prohibited concept. Evaluating across six frontier VLMs, our visual attacks bypass safety alignment and expose a cross-modality alignment gap: text-based safety training does not automatically generalize to harmful intent conveyed visually. For example, our visual cipher achieves 40.9\% attack success on Claude-Haiku-4.5 versus 10.7\% for an equivalent textual cipher. To further our insight into the attack mechanism, we present preliminary interpretability and mitigation results. These findings highlight that robust VLM alignment requires treating vision as a first-class target for safety post-training.

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Connecting Independently Trained Modes via Layer-Wise Connectivity

Yongding Tian, Zaid Al-Ars, Maksim Kitsak, H Peter Hofstee

Empirical studies have shown that continuous low-loss paths can be constructed b

etween independently trained neural network models.
This phenomenon, known as mode connectivity, refers to the existence of such pat
hs between distinct modes-i.e., well-trained solutions in parameter space.
However, existing empirical methods do not reliably connect independently traine
d modes and have been evaluated mainly on a narrow set of architectures (e.g., b
asic CNNs, VGG, and ResNet), leaving their effectiveness on newer models unclear
.
In this work, we propose a new empirical algorithm for connecting independently
trained modes that generalizes beyond traditional architectures and supports a b
roader range of networks, including MobileNet, ShuffleNet, EfficientNet, RegNet,
 Deep Layer Aggregation (DLA), and Compact Convolutional Transformers (CCT).
In addition to broader applicability, the proposed method yields more consistent
 connectivity paths across independently trained mode pairs and supports connect
ing modes obtained with different training hyperparameters.
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Modality-Decoupled Online Recursive Editing
Siyuan Li, Youyuan Zhang, Fangming Liu, Jing Li
Online model editing for multimodal large language models (MLLMs) requires assim
ilating a stream of corrections under tight compute and memory budgets. Yet edit
ors developed for text-only LLMs often degrade on MLLMs: visually dominant activ
ations skew the statistics that shape updates, causing *cross-modal conflict*, w
hile sequential writes become entangled in a shared edit space and amplify long-
horizon interference, causing *inter-edit interference*. To address these, we pr
opose **M-ORE**, a modality-decoupled online recursive editor for lifelong MLLM
adaptation. M-ORE is derived from a unified proximal-projection formulation and
admits a closed-form update with a Sherman-Morrison recursion, yielding constant
 per-edit overhead. It maintains module-wise locality statistics for the text st
ack and the visual projector to avoid visually dominated update shaping and perf
orms continual updates in a fixed orthogonal low-rank edit subspace via a Sherma
n-Morrison recursion to mitigate long-horizon interference. Experiments on multi
ple MLLM backbones and online editing benchmarks show that our M-ORE method cons
istently improves reliability, generality, and locality over strong baselines, w
hile achieving favorable quality-efficiency scaling. Our code is publicly availa
ble at https://github.com/lab-klc/M-ORE.
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The Shadow Price of Reasoning: Economic Perspective on Optimal Budget Allocatio
n for LLMs
Xu Wan, Speed Zhu, Jianwei Cai, Guang Chen, XiMing Huang, Wiggin Zhou, Mingyang
Sun
Inference-time scaling has emerged as a critical avenue for enhancing Large Lang
uage Models' performance, yet real-world deployment is constrained by strict com
putational budgets. In this work, we formulate inference budget allocation as a
global constrained optimization problem governed by economic principles. By mode
ling per-query reasoning utility with a shifted-surge function, we derive an opt
imal allocation policy based on a global shadow price that equilibrates marginal
 utility under resource scarcity.
Based on this theory, we propose Constrained Latent-utility Equilibrium Allocati
on for Reasoning (CLEAR). It performs rational abandonment and reallocates resou
rces from insolvent queries to solvable queries near their emergence thresholds.

Extensive experiments on several reasoning tasks with different traffic streams
demonstrate that CLEAR significantly improves the Pareto frontier of total token
 cost versus mean accuracy. In resource-scarce regimes, CLEAR achieves up to a 3
× improvement in global accuracy compared to uniform allocation.
****

Approximate Proportionality in Online Fair Division
Davin Choo, Winston Fu, Tzeh Yuan Neoh, Tze-Yang Poon, Nicholas Teh
We study the online fair division problem, where indivisible goods arrive sequen
tially and must be allocated immediately and irrevocably. Prior work establishes
 strong impossibility results for approximating classic notions such as envy-fre

eness up to one good (EF1) and maximin share (MMS) in this setting, but the appr oximability of proportionality up to one good (PROP1) has remained unresolved. W e resolve this gap in two steps. First, we show that three natural greedy alloca tion rules (standard baselines in fair division) fail to guarantee any multiplic ative approximation to PROP1 against an adaptive adversary. These limitations mo tivate two relaxations: (i) restricting attention to a non-adaptive adversary, a nd (ii) incorporating coarse predictions in the spirit of learning-augmented alg orithms. Under a non-adaptive adversary, we show that the uniform random allocat ion achieves a meaningful PROP1 approximation with high probability, and this gu arantee is essentially tight for this approach; moreover, when item values are s ufficiently small, the allocation is near-PROP1 with high probability. Finally, given maximum item value (MIV) predictions, we design an online algorithm that a chieves robust approximation guarantees for PROP1, and degrades gracefully under one-sided prediction error. In contrast, we show that EF1, MMS, and PROPX remai n inapproximable even with perfect MIV predictions.

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Formalizing the Binding Problem

Lianghuan Huang, Yihao Li, Saeed Salehi, Yingshan Chang, Ansh Soni, Konrad Kordi ng

Representations of the world, arguably, contain information about features (e.g. something is blue, something is a circle) but also information about which feat ures are part of the same object (e.g. the circle is blue), which we call bindin g information. Any system with the ability to understand scenes with multiple ob jects must be able to solve the binding problem: it needs to know which features belong together. However, despite work showing that Vision Transformers (ViTs) know which patches belong together, it is not known whether current deep learnin g models learn to exhibit binding information, i.e., for features. We may believ e that there is not much binding information, after all misattributing features to wrong objects is a common failure of ViT-based architectures, especially in s cenes with objects sharing features. Here we formalize the binding problem with an information-theoretic approach, and introduce a probing method to measure bin ding information in model representations. We perform experiments on ViTs, measu ring binding from different components of the architecture, such as the image su mmary token [CLS] or the spatial tokens. We use datasets with different binding challenges, such as feature sharing, occlusion, and natural features, while comp aring the performance of several pre-trained ViTs. Overall, our research demonst rates binding as a key ingredient to strong visual recognition and reasoning.

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How do Humans Process AI-generated Hallucination Contents: a Neuroimaging Study

Shuqi Zhu, Yi Zhong, Ziyi Ye, Bangde Du, Yujia Zhou, Qingyao Ai, Yiqun LIU

While AI-generated hallucinations pose considerable risks, the underlying cognit ive mechanisms by which humans can successfully recognize or be misled by these hallucinations remain unclear. To address this problem, this paper explores huma ns' neural dynamics to characterize how the brain processes hallucinated content . We record EEG signals from 27 participants while they are performing a verific ation task to judge the correctness of image descriptions generated by a multi-m odal large language model (MLLM). Based on an averaged event-related potential (ERP) study, we reveal that multiple cognitive processes, e.g., semantic integrat ion, inferential processing, memory retrieval, and cognitive load, exhibit disti nct patterns when humans process hallucinated versus non-hallucinated content. N otably, neural responses to hallucinations that were misjudged versus correctly judged by human participants showed significant differences.

This indicates that misjudged AI-generated hallucinations failed to trigger the standard neurocognitive fact verification pathway. The detailed code can be acce ssed openly through https://github.com/Promise-Z5Q2SQ/EEG-Hallucination.

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MGAL: A Multilingual Granularity-Aware Long-Context Benchmark

Chunhan Li, Chenglin Xu, Zongyang Zhang, Jiale Liu, Zhuoxi Rao, Xudong jia, JUNX IU HE, Menglin Yang, Wenjuan Gong, Zhengzhe Liu, Chengwei Qin

Evaluation of long-context Large Language Models (LLMs) has advanced rapidly. Ho

wever, most existing benchmarks are limited to the document level and focus main ly on high-resource languages, leaving many fine-grained challenges insufficient ly evaluated. To address this gap, we present MGAL, the first multilingual, gran ularity- and position-aware long-context benchmark. MGAL is constructed from Uni ted Nations (UN) reports spanning 8K to 128K tokens across the six official UN l anguages. It covers four coherent levels of linguistic granularity (word, senten ce, paragraph, and document) and further stratifies entries by their position wi thin the document (begin, middle, and end), indexed at both the document and par agraph levels. This design enables systematic diagnosis of multilingual long-con text comprehension across different granularities.

Through extensive experiments and analyses, we find that: (1) LLMs perform well at word-level tasks but struggle with coarser-grained ones; and (2) Closed-sourc e models retain a clear performance advantage in lower-resource languages.
We further identify two new challenges: (1) Under local semantic crowding, where neighboring sentences share topics and entities, models tend to follow surface cues (e.g., connectives like 'however' or repeated entities) rather than the dis course role of the sentence in surrounding context (e.g., background, outcome); and (2) A gap between fluency and consistency in generated outputs, where models produce text that reads smoothly but drifts from the source facts. In addition, we observe several patterns in line with prior studies, including reliance on n earby evidence and reuse of options under uncertainty.
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Efficient Equivariant High-Order Crystal Tensor Prediction via Cartesian Local-E nvironment Many-Body Coupling
Dian Jin, Yancheng Yuan, Xiaoming Tao
End-to-end prediction of high-order crystal tensor properties from atomic struct ures remains challenging: while spherical-harmonic equivariant models are expres sive, their Clebsch-Gordan tensor products incur substantial compute and memory costs for higher-order targets. We propose the Cartesian Environment Interaction Tensor Network (CEITNet), an approach that constructs a multi-channel Cartesian local environment tensor for each atom and performs flexible many-body mixing v ia learnable channel-space interactions. By performing learning in channel space and using Cartesian tensor bases to assemble equivariant outputs, CEITNet enabl es efficient construction of high-order tensor.
Across benchmark datasets for order-2 dielectric, order-3 piezoelectric, and ord er-4 elastic tensor prediction, CEITNet surpasses prior high-order prediction me thods on key accuracy criteria while offering high computational efficiency.
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Value Aggregation with Uncertainty in Online Decentralized MARL
Ziyue Chu, Leonardo Stella
Multi-agent reinforcement learning (MARL) has received increasing attention for solving complex decision-making tasks. Networked MARL approaches offer a decentr alized solution for parameter sharing to accelerate training via value aggregati on. However, existing federated aggregations rely on convex averaging that may f ail to converge to global optima and cause learning rollback in the online learn ing setting. In this paper, we formally characterize the learning rollback pheno menon arising from aggregating value estimates with unequal uncertainty under he terogeneous online update depths. We propose a novel adaptive global consensus (AGC) mechanism for Q-value aggregation in decentralized MARL policy evaluation, which dynamically adjusts aggregation weights based on agents’ uncertainty. We e stablish theoretical guarantees on accelerated convergence and bounded learning variance with empirical validations, advancing the state-of-art MARL theory.
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Rethinking Visual Autoregressive Sampling with Information-Grounding Guidance
Ky Dan Nguyen, Hoang Lam Tran, Anh-Dung Dinh, Daochang Liu, Weidong Cai, Xiuying Wang, Chang Xu
Autoregressive (AR) models based on next-scale prediction are rapidly emerging a s a powerful tool for image generation, but they face a critical weakness: infor mation inconsistencies between patches across timesteps introduced by progressiv

e resolution scaling. These inconsistencies scatter guidance signals, causing th
em to drift away from conditioning information and leaving behind ambiguous, unf
aithful features. We tackle this challenge with Information-Grounding Guidance (
IGG), a novel mechanism that anchors guidance to semantically important regions
through attention. By adaptively reinforcing informative patches during sampling
, IGG ensures that guidance and content remain tightly aligned. Across both clas
s-conditioned and text-to-image generation tasks, IGG delivers sharper, more coh
erent, and semantically grounded images, setting a new benchmark for AR-based me
thods.
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Efficient, Property-Aligned Fan-Out Retrieval via RL-Compiled Diffusion
Pengcheng Jiang, Judith Yue Li, Moonkyung Ryu, Lily Hu, Kun Su, Zhong Yi Wan, Li
am Hebert, Hao Peng, Jiawei Han, Dima Kuzmin, Craig Boutilier
Many modern retrieval problems are set-valued:
given a broad intent, the system must return
a collection of results that optimizes higherorder
properties (e.g., diversity, coverage, complementarity,
coherence) while staying grounded
to a fixed database. Set-valued objectives are
inherently non-decomposable and are not captured
by existing supervised (query, content)
datasets which only prioritize top-1 retrieval.
While reinforcement learning (RL) can optimize
set-level objectives via interaction, deploying
an RL-tuned LLM for fan-out retrieval is
prohibitively expensive at query time. Conversely,
diffusion-based generative retrieval enables
efficient single-pass fan-out in embedding
space, but requires objective-aligned training targets.
To address these issues, we propose R4T
(Retrieve-for-Train), which uses RL once as an
objective transducer in a three step process: (i)
train a fan-out LLM with composite set-level rewards,
(ii) synthesize objective-consistent training
pairs, and (iii) train a lightweight diffusion
retriever to model the conditional distribution of
set-valued outputs. Across Polyvore and a music
playlist dataset, R4T improves retrieval quality
over strong baselines while reducing query-time
fan-out latency by an order of magnitude.
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Energy-Structured Low-Rank Adaptation for Continual Learning
Longhua Li, Lei Qi, Qi Tian, Xin Geng
While orthogonal subspace methods try to mitigate task interference in Continual
 Learning (CL), they often suffer from energy diffusion across the basis, hinder
ing knowledge compaction and exhausting capacity for future tasks. We observe th
at output feature drift induced by parameter updates is inherently low-rank, and
 theoretically prove that preserving parameters along the principal directions o
f this drift minimizes the output reconstruction error. Motivated by this, we pr
opose **E**nergy-Concentrated and **E**nergy-Ordered **Lo**w-**R**ank **A**dapta
tion (E^2-LoRA). By explicitly ordering and concentrating knowledge into leadi
ng ranks, E^2-LoRA frees capacity for subsequent tasks. Furthermore, we design
 a dynamic rank allocation strategy to balance stability and plasticity by joint
ly optimizing energy retention and model plasticity. Extensive experiments acros
s multiple benchmarks demonstrate that E^2-LoRA achieves state-of-the-art perf
ormance.
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Constrained Meta Reinforcement Learning with Provable Test-Time Safety
Tingting Ni, Maryam Kamgarpour
Meta reinforcement learning (RL) allows agents to leverage experience across a d

istribution of tasks on which the agent can train at will, enabling faster learn ing of optimal policies on new test tasks. Despite its success in improving samp le complexity on test tasks, many real-world applications, such as robotics and healthcare, impose safety constraints during testing. Constrained meta RL provid es a promising framework for integrating safety into meta RL. An open question i n constrained meta RL is how to ensure safety of the policy on the real-world te st task, while reducing the sample complexity and thus, enabling faster learning of optimal policies. To address this gap, we propose an algorithm that refines policies learned during training, with provable safety and sample complexity gua rantees for learning a near optimal policy on the test tasks. We further derive a matching lower bound, showing that this sample complexity is tight.
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Unified Safe In-context Image Generation in Multimodal Diffusion Transformers vi a Restricting Unsafe Information Flows

Xiang Yang, Feifei Li, Mi Zhang, Geng Hong, Xiaoyu You, Mi Wen, Min Yang

Diffusion transformers (DiTs) equipped with multimodal attention (MM-Attn) have become a dominant paradigm for image generation. However, preventing the generat ion of harmful content remains a critical challenge, particularly in image-to-im age (I2I) editing tasks.

Existing safety mechanisms are primarily designed for text-to-image (T2I) synthe sis or U-Net-based architectures, which limits their effectiveness for unified s afety mitigation in DiT-based frameworks. To bridge this gap, we propose Unified Visual Safety Regulator (UVR), a training-free safe generation framework that r egulates unsafe semantics in generated images. UVR is grounded in an analysis of attention dynamics from the perspective of information flow in MM-Attn. We iden tify a task-independent start-up stage, during which unsafe semantics in output patches rapidly emerge and can be accurately localized, followed by task-specifi c semantic amplification and interference stages, where harmful signals are furt her propagated and entangled with benign content. Based on these observations, U VR mitigates unsafe generation through unified, targeted attention modulation an d explicit restriction of harmful information flow over the identified unsafe ou tput patches. Experiments across various concepts show that UVR achieves state-o f-the-art safety performance by achieving 91% and 77% erase rate in image synth esis and editing tasks, while preserving visual quality and fidelity with minima l degradation. Code is available at https://github.com/deng12yx/UVR.
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DECO: Decoupled Multimodal Diffusion Transformer for Bimanual Dexterous Manipula tion with a Plugin Tactile Adapter

Xukun Li, Yu Sun, Lei Zhang, Bo-Sheng Huang, Yibo Peng, Yuan Meng, Haojun Jiang, Shaoxuan Xie, Guocai Yao, Alois Knoll, Zhenshan Bing, Xinlong Wang, Zhenguo Sun

Bimanual dexterous manipulation relies on integrating multimodal inputs to perfo rm complex real-world tasks. To address the challenges of effectively combining these modalities, we propose DECO, a decoupled multimodal diffusion transformer that disentangles vision, proprioception, and tactile signals through specialize d conditioning pathways, enabling structured and controllable integration of mul timodal inputs, with a lightweight adapter for parameter-efficient injection of additional signals.

Alongside DECO, we release DECO-50 dataset for bimanual dexterous manipulation w ith tactile sensing, consisting of 50 hours of data and over 5M frames, collecte d via teleoperation on real dual-arm robots.

We train DECO on DECO-50 and conduct extensive real-world evaluation with over 2 ,000 robot rollouts. Experimental results show that DECO achieves the best perfo rmance across all tasks, with a 72.25\% average success rate and a 21\% improvem ent over the baseline. Moreover, the tactile adapter brings an additional 10.25\ % average success rate across all tasks and a 20\% gain on complex contact-rich tasks while tuning less than 10\% of the model parameters.
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Reasoning Quality Emerges Early: Data Curation for Reasoning Models

Hongyi Henry Jin, Wenhan Yang, Meysam Ghaffari, Carlos Morato, Baharan Mirzasole iman

Supervised fine-tuning (SFT) on a small, high-quality set of long reasoning traces is an effective approach for eliciting strong reasoning capabilities in Large Language Models (LLMs). However, existing methods for curating high-quality SFT data rely heavily on strong reasoning models to filter examples based on diversity and difficulty, making the curation process costly while often yielding suboptimal data quality. In this work, we show that diverse and challenging reasoning examples can be identified using only the initial reasoning tokens. Specifically, we demonstrate that difficult problems can be reliably detected based on the loss of the first 100 reasoning tokens evaluated at a randomly perturbed checkpoint of the pretrained model. We further show that examples exhibiting similar loss patterns over their first 1k reasoning tokens across a small number of perturbed checkpoints extrapolating along the fine-tuning trajectory provably induce similar gradients. We validate our approach through extensive experiments on fine-tuning Qwen2.5-7B and Llama3.1-8B models on the M23K medical reasoning and OpenThoughts-Math datasets. Our method outperforms existing baselines by up to 1.7% while being 91% more token efficient.

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Understanding Dynamics of Adam in Zero-Sum Games: An ODE Approach

Yi Feng, Weiming Ou, Xiao Wang

The remarkable success of the Adam in training neural networks has naturally led to the widespread use of its descent-ascent counterpart, Adam-DA, for solving zero-sum games. Despite its popularity in practice, a rigorous theoretical understanding of Adam-DA still lags behind. In this paper, we derive ordinary differential equations (ODEs) that serve as continuous-time limits of the Adam-DA. These ODEs closely approximate the discrete-time dynamics of Adam-DA, providing a tractable analytical framework for understanding its behavior in zero-sum games. Using this ODE approach, we investigate two fundamental aspects of Adam-DA: local convergence and implicit gradient regularization. Our analysis reveals that the roles of the first- and second-order momentum parameters in zero-sum games are exactly the opposite of their well-documented effects in minimization problems. We validate these predictions through GAN experiments across multiple architectures and datasets, demonstrating the practical implications of this reversed momentum effect.

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Domain Adaptive Object Detection via Dynamic Causal Refinement

Zeyu Ma, Jiaqi Huang, Yitong Qin, Zheng Ziqiang, Jiwei Wei, Jie Zou, Yang Yang, Heng Tao Shen

Domain Adaptive Object Detection (DAOD) addresses the challenge of transferring object detectors from labeled source domains to unlabeled target domains.

Existing domain adaptation methods primarily rely on feature distribution alignment, which enhances domain-invariant features (statistical invariance) but also inadvertently increases inherent domain-common spurious factors (e.g., common environmental contexts), which act as shortcut features rather than the true causal factors for object classification.

We propose Dynamic Causal Refinement (DCR), a novel framework that establishes a closed-loop feedback mechanism between data augmentation and model optimization to progressively refine causal features.

Specifically, we design Semantic Prediction Consistency (SPC) to filter domain-specific spurious factors and establish a robust statistical invariance, and Discrepancy-Guided Causal Refinement (DGCR) to actively suppress the dependence on domain-common spurious factors via spectral perturbation for causal refinement.

This process encourages the detector to suppress its reliance on shortcut features and instead prioritize semantically meaningful causal representations.

Extensive experiments on standard benchmarks demonstrate that our method outperforms state-of-the-art counterparts significantly.

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AgentSelect: Benchmark for Narrative Query-to-Agent Recommendation

Yunxiao Shi, Wujiang Xu, Tingwei Chen, Haoning Shang, Ling Yang, Yunfeng Wan, Zhuo Cao, Xing Zi, Dimitris N. Metaxas, Min Xu

LLM agents are rapidly becoming a practical interface for task automation, yet s

electing suitable deployable configurations remains underexplored. Existing LLM leaderboards and tool/agent benchmarks evaluate components in isolation and are fragmented across tasks, metrics, and candidate pools, leaving a critical research gap: there is little query-conditioned supervision for learning to recommend end-to-end compositional agent configurations.
We introduce AgentSelect, a benchmark that reframes agent selection as narrative query-to-agent recommendation over capability profiles. AgentSelect systematically converts heterogeneous evaluation artifacts into unified positive-only interaction data, comprising 111,179 queries, 107,721 deployable agents, and 251,103 interaction records from 40+ sources across LLM-only, toolkit-only, and compositional agents.
Our analyses reveal a shift from dense head reuse to long-tail, near one-off supervision, where ID-based interaction methods become fragile and reliable recommendation increasingly requires content-aware intent-to-capability matching. We validate the synthesized supervision through counterfactual, ablation, and show practical transferability on the MuleRun agent marketplace and a small-scale end to end Agno deployment study.
Overall, AgentSelect provides the first unified data and evaluation infrastructure for agent recommendation, which establishes a reproducible foundation to study and accelerate the emerging agent ecosystem. The resources are available at \footnote{\url{https://github.com/Ancientshi/AgentSelect}}.
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Singular Proxies for Adaptive Caching in Diffusion Language Models
Wenhao Sun, Rong-Cheng Tu, Yifu Ding, Zhao Jin, Jingyi Liao, Yongcheng Jing, Dacheng Tao
While Diffusion Language Models (DLMs) offer a flexible, arbitrary-order alternative to the autoregressive paradigm, their non-causal nature precludes standard KV caching, forcing costly hidden state recomputation at every decoding step.
Existing caching approaches reduce this cost by selective hidden state updates; however, they are still limited by (i) computationally costly token-wise update identification heuristics and (ii) rigid, uniform budget allocation that fails to account for heterogeneous hidden- tate dynamics.
To address these challenges, we present SPA-Cache that jointly optimizes update identification and budget allocation.
First, we derive a low-dimensional singular proxy that enables the identification of update-critical tokens in a low-dimensional subspace, substantially reducing the overhead of update identification.
Second, motivated by the layer-wise heterogeneity in hidden state dynamics, we introduce an adaptive strategy that allocates fewer updates to stable layers without degrading generation quality.
Together, these contributions significantly improve the efficiency of DLMs, yielding up to an $8\times$ throughput improvement over vanilla models decoding and a 2-$4\times$ speedup over existing caching baselines.
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Towards a Science of AI Agent Reliability
Stephan Rabanser, Sayash Kapoor, Peter Kirgis, Kangheng Liu, Saiteja Utpala, Arvind Narayanan
AI agents are increasingly deployed to execute important tasks. While rising accuracy scores on standard benchmarks suggest rapid progress, many agents still continue to fail in practice. This discrepancy highlights a major limitation of current evaluations: focusing on a single metric is not enough to understand agent behavior. Notably, it ignores whether agents behave consistently across runs, withstand perturbations, fail predictably, or have bounded error severity. Grounded in safety-critical engineering, we provide a holistic performance profile consisting of twelve metrics that decompose agent reliability along four key dimensions: *consistency*, *robustness*, *predictability*, and *safety*. Evaluating 15 models across two complementary benchmarks, we find that recent capability gains have only yielded small improvements in reliability. By exposing these persistent limitations, our metrics complement traditional evaluations while offering tools for reasoning about how agents perform, degrade, and fail.

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GFFMERGE: Efficient Merging of Graph Neural Force Fields and Beyond
Parth Verma, Parv Pratap Singh, Vipul Garg, Ishita Thakre, N M Anoop Krishnan, S ayan Ranu
Graph Neural Networks (GNNs) have revolutionized Neural Force Fields for atomist ic simulations, achieving near-quantum accuracy at reduced cost, yet adapting th ese models to new chemical systems requires expensive retraining of foundation m odels. Inspired by model merging in vision and language processing, we introduce GFFMERGE, the first principled framework for closed-form model merging in GNNs. We exploit the linear structure of message-passing layers and formulate merging as a convex embedding-alignment problem with an analytical solution. Through th e first systematic benchmarking of model merging for GNNs, we show that existing methods designed for vision and language catastrophically fail on force field r egression, while GFFMERGE recovers performance approaching gold standard joint t raining. Across molecular (MD17, MD22), solid-state (LiPS20), and large-scale gr aph benchmarks, GFFMERGE and GNNMERGE (its generic GNN counterpart) achieve 5-27 $\times$ speedups while enabling modular composition of specialized models. Rema rkably, our closed-form solution alone outperforms all baseline methods before f ine-tuning and provides superior initialization for faster, data-efficient conve rgence.
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REAR: Test-time Preference Realignment through Reward Decomposition
Fuxiang Zhang, Pengcheng Wang, Chenran Li, Yi-Chen Li, Yuxin Chen, Lang Feng, Ch enfeng Xu, Masayoshi Tomizuka, Bo An
Aligning large language models (LLMs) with diverse user preferences is a critica l yet challenging task. While post-training methods can adapt models to specific needs, they often require costly data curation and additional training. Test-ti me scaling (TTS) presents an efficient, training-free alternative, but its appli cation has been largely limited to verifiable domains like mathematics and codin g, where response correctness is easily judged. To extend TTS to preference alig nment, we introduce a novel framework that models the task as a realignment prob lem, since the base model often fails to sufficiently align with the stated pref erence. Our key insight is to decompose the underlying reward function into two components: one related to the question and the other to preference information. This allows us to derive a REAlignment Reward (REAR) that selectively rescales the proportions of these two reward terms. We then show that REAR can be formula ted as a linear combination of token-level policy log-probabilities, making it c omputationally efficient and easy to integrate with various TTS algorithms such as best-of-N sampling and tree search. Experiments show that compared to other test-time baselines, REAR not only enables scable test-time realignment for pre ference alignment tasks under diverse user requirements, but also generalizes to mathematical and visual tasks under appropriate preference settings.
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VecMol: Vector-Field Representations for 3D Molecule Generation
Yuchen Hua, Xingang Peng, Jianzhu Ma, Muhan Zhang
Generative modeling of three-dimensional (3D) molecules is a fundamental yet cha llenging problem in drug discovery and materials science. Existing approaches ty pically represent molecules as 3D graphs and co-generate discrete atom types wit h continuous atomic coordinates, leading to intrinsic learning difficulties such as heterogeneous modality entanglement and geometry–chemistry coherence constra ints. We propose VecMol, a novel representation that models 3D molecules as cont inuous vector fields over Euclidean space, where vectors point toward nearby ato ms and implicitly encode molecular structure. The vector field is parameterized by a neural field and generated using a latent diffusion model, avoiding explici t graph generation and decoupling structure learning from discrete atom instanti ation. Experiments on the QM9 and GEOM-Drugs benchmarks demonstrate that VecMol achieves competitive generation quality, suggesting vector-field-based represent ations as a promising new direction for 3D molecular generation.
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Natural Hypergradient Descent: Algorithm Design, Convergence Analysis, and Para

llel Implementation
Deyi Kong, Zaiwei Chen, Shuzhong Zhang, Shancong Mou
In this work, we propose *Natural Hypergradient Descent* (NHGD), a new method fo r solving bilevel optimization problems. To address the computational bottleneck in hypergradient estimation, namely the need to compute or approximate Hessian inverses, we exploit the statistical structure of the inner optimization problem and use the empirical Fisher information matrix as an asymptotically consistent surrogate for the Hessian. This design enables a parallel *optimize-and-approxi mate* framework in which the Hessian-inverse approximation is updated *synchrono usly* with the stochastic inner optimization, reusing gradient information at ne gligible additional cost. Our main theoretical contribution establishes high-pro bability error bounds and sample complexity guarantees for NHGD that match those of state-of-the-art optimize-then-approximate methods, while significantly redu cing computational time overhead. Empirical evaluations on representative bileve l learning tasks further demonstrate the practical advantages of NHGD, highlight ing its scalability and effectiveness in large-scale machine learning settings.
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CaP-X: A Framework for Benchmarking and Improving Coding Agents for Robot Manipu lation
Letian Fu, Justin Yu, Karim El-Refai, Ethan Kou, Haoru Xue, Huang Huang, Wenli X iao, Li Fei-Fei, Guanya Shi, Jiajun Wu, S. Shankar Sastry, Yuke Zhu, Ken Goldber g, Linxi Fan
Code-as-Policy (CaP) is a paradigm in which a language or vision-language model generates executable robot control programs, yet its effectiveness as an autonom ous controller for embodied manipulation remains underexplored. Prior CaP system s often rely on high-level, human-designed primitives, making it difficult to se parate agent capability from designer-provided scaffolding. We present CaP-X, an open-access framework for systematically studying Code-as-Policy agents in robo t manipulation. CaP-X includes four components. CaP-Gym is an interactive enviro nment in which coding agents control robots by synthesizing and executing progra ms that compose perception and control primitives. Building on this foundation, CaP-Bench evaluates frontier language and vision-language models across varying levels of abstraction, interaction, and perceptual grounding. Across 12 models, the task success rates improve with human-crafted abstractions but degrade as th ese priors are removed, exposing a dependence on designer scaffolding. At the sa me time, we observe that scaling test-time computation with multi-turn interacti on, structured execution feedback, visual differencing, automatic skill synthesi s, and ensembled reasoning can substantially improve robustness even when agents operate over low-level primitives. These findings motivate CaP-Agent0, a traini ng-free framework that achieves near human-level reliability on several manipula tion tasks in simulation and on real embodiments. CaP-RL explores reinforcement learning with verifiable rewards to improve success rates and supports sim-to-re al transfer through a shared code-as-action-space interface. Together, CaP-X pro vides an open-access platform for advancing embodied coding agents. Project page : https://capgym.github.io
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Alignment-Sensitive Minimax Rates for Spectral Algorithms with Learned Kernels
Dongming Huang, Zhifan Li, Yicheng Li, Qian Lin
We study spectral algorithms in settings where kernels may arise from a learning procedure, and ask how the resulting spectral order affects risk. We introduce the effective span dimension (ESD), an alignment-sensitive complexity measure th at depends jointly on the target signal, the spectral order induced by the kerne l, and the noise level σ^2. The ESD is defined without requiring eigen-de cay conditions or source conditions, and it captures how much of the target sign al lies in the leading spectral span. We prove that, for sequence models whose E SD is at most K, the minimax excess risk scales as $\sigma^2 K$, and we extend the framework to linear models and RKHS regression. Furthermore, we analyze ove r-parameterized gradient flow in a fixed-eigenbasis spectral learning model and prove that it can reduce the ESD under certain conditions. Together with numeric al experiments, these results connect adaptive feature learning with reductions

in ESD and offer a novel perspective on generalization beyond traditional fixed-kernel theories.
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Stochastic Gradient Variational Inference with Price's Gradient Estimator from Bures-Wasserstein to Parameter Space
Kyurae Kim, Qiang Fu, Yian Ma, Jacob R. Gardner, Trevor Campbell
For approximating a target distribution given only its unnormalized log-density, stochastic gradient-based variational inference (VI) algorithms are a popular approach. For example, Wasserstein VI (WVI) and black-box VI (BBVI) perform gradient descent in measure space (Bures-Wasserstein space) and parameter space, respectively. Previously, for the Gaussian variational family, convergence guarantees for WVI have shown superiority over existing results for black-box VI with the reparametrization gradient, suggesting the measure space approach might provide some unique benefits. In this work, however, we close this gap by obtaining identical state-of-the-art iteration complexity guarantees for both. In particular, we identify that WVI's superiority stems from the specific gradient estimator it uses, which BBVI can also leverage with minor modifications. The estimator in question is usually associated with Price's theorem and utilizes second-order information (Hessians) of the target log-density. We will refer to this as Price's gradient. On the flip side, WVI can be made more widely applicable by using the reparametrization gradient, which requires only gradients of the log-density. We empirically demonstrate that the use of Price's gradient is the major source of performance improvement.
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ERGeoBench: A Comprehensive Benchmark for Embodied Reasoning and Geo-localization in Multimodal Large Language Models
Kaiwen Xue, Tao Wei, Guoxin Zhang, Zhonghong Ou, Kaoyan Lu, Yu Feng, Yifan Zhu, Haoran Luo
Multimodal large language models (MLLMs) have shown strong potential as embodied agents, yet embodied geo-localization remains underexplored due to the lack of fine-grained evaluation. We introduce ERGeoBench, a diagnostic benchmark for vision-driven embodied geo-localization. ERGeoBench evaluates models under three progressive settings---single-view, panorama-view, and embodied-view---where agents may actively acquire observations through sequential changes in yaw, pitch, and zoom. The benchmark contains 2,207 globally distributed street-view panoramas and measures four complementary capabilities: foundational perception, spatial awareness, common sense reasoning, and geo-localization reasoning. Evaluations of leading proprietary and open-source MLLMs show that current models can infer high-level geographic semantics, but still struggle with fine-grained perceptual operations, metric localization, and spatial consistency across views. We further observe that geo-localization is strongly correlated with the other capability dimensions, suggesting that accurate localization depends on integrated perception, spatial reasoning, and commonsense inference rather than isolated visual recognition. Overall, ERGeoBench provides a unified framework for diagnosing and advancing human-like embodied geo-localization. Project Page: \url{https://kaixuewen.github.io/ERGeoBench/}
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SLAE: Strictly Local All-atom Environment for Protein Representation
Yilin Chen, Tianyu Lu, Cizhang Zhao, Hannah Wayment-Steele, Po-Ssu Huang
Building physically grounded protein representations is central to computational biology, yet most existing approaches rely on sequence-pretrained language models or backbone-only graphs that overlook side-chain geometry and chemical detail. We present SLAE, a unified all-atom framework for learning protein representations from each residue’s local atomic neighborhood using only atom types and interatomic geometries. To encourage expressive feature extraction, we introduce a novel multi-task autoencoder objective that combines coordinate reconstruction, sequence recovery, and energy regression. SLAE reconstructs allatom structures with high fidelity from latent residue environments and achieves state-of-the-art performance across diverse downstream tasks via transfer learning. SLAE’s latent space is chemically informative and environmentally sensitive, enabling quanti

tative assessment of structural qualities and smooth interpolation between confo
rmations at all-atom resolution.
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ScaleEnv: Scaling Environment Synthesis from Scratch for Generalist Interactive
Tool-Use Agent Training
Dunwei Tu, Hongyan Hao, Hansi Yang, Yihao Chen, Yu Yang, Yueqing Sun, Xingchen L
iu, Furao Shen, Qi GU, Hui Su, Xunliang Cai
Training generalist agents capable of adapting to diverse scenarios requires int
eractive environments for self-exploration.
However, interactive environments remain critically scarce, and existing synthes
is methods suffer from significant limitations regarding environmental diversity
 and scalability.
To address these challenges, we introduce ScaleEnv, a framework that constructs
fully interactive environments and verifiable tasks entirely from scratch.
Specifically, ScaleEnv ensures environment reliability through procedural testin
g, and guarantees task completeness and solvability via tool dependency graph ex
pansion and executable action verification.
By enabling agents to learn through exploration within ScaleEnv, we demonstrate
significant performance improvements on unseen, multi-turn tool-use benchmarks s
uch as τ^2-Bench and VitaBench, highlighting strong generalization capabili
ties.
Furthermore, we investigate the relationship between increasing number of domain
s and model generalization performance, providing empirical evidence that scalin
g environmental diversity is critical for robust agent learning.
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Resolution as a Direction: Vector-Panning Feature Alignment for Cross-Resolution
 Re-Identification
Zanwu Liu, Chao Yuan, Bo Li, Xiaowei Zhang, Guanglin Niu
Cross-resolution person re-identification (CR-ReID) remains challenging in pract
ical surveillance, where camera quality and capture distance lead to substantial
 resolution gaps between low-resolution (LR) queries and high-resolution (HR) ga
llery images. Prior approaches commonly rely on super-resolution (SR) or resolut
ion-invariant representation learning, which often increases system complexity a
nd may not directly address the feature mismatch induced by resolution degradati
on. In this work, we report a new empirical finding from a dedicated analysis in
 which identity-specific variation is averaged out: the HR--LR feature discrepan
cy produced by standard ReID backbones exhibits a consistent, resolution-related
 semantic direction in the embedding space. We further support this observation
with statistical analyses based on Canonical Correlation Analysis (CCA) and Pear
son correlation analysis. Motivated by this finding, we propose Vector Panning F
eature Alignment (VPFA), a lightweight post-hoc module that learns to pan LR fea
tures along the learned resolution direction to obtain pseudo-HR representations
. VPFA operates after feature extraction and can be integrated into existing ReI
D systems with negligible overhead. Extensive experiments on multiple CR-ReID be
nchmarks show that VPFA achieves state-of-the-art performance while improving ef
ficiency compared to SR-based or jointly trained alternatives. Code is available
 at \url{https://github.com/ashmentlzw/VPFA}
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Discovering Symmetry Groups with Flow Matching
Yuxuan Chen, Jung Yeon Park, Floor Eijkelboom, Jianke Yang, Jan-Willem van de Me
ent, Lawson L.S. Wong, Robin Walters
Symmetry is fundamental to understanding physical systems and can improve perfor
mance and sample efficiency in machine learning. Both pursuits require knowledge
 of the underlying symmetries in data, yet discovering these symmetries automati
cally is challenging. We propose LieFlow, a novel framework that reframes symmet
ry discovery as a distribution learning problem on Lie groups. Instead of search
ing for the symmetry generators, our approach operates directly in group space,
modeling a symmetry distribution over a large hypothesis group G. The support
of the learned distribution reveals the underlying symmetry group $H \subseteq G
$.

Unlike previous works, LieFlow can discover both continuous and discrete symmetries within a unified framework, without assuming a fixed Lie algebra basis or a specific distribution over the group elements.
Experiments on synthetic 2D and 3D point clouds, ModelNet10, and a real-world MI-Motion dataset show that LieFlow accurately discovers continuous and discrete subgroups, significantly outperforming a state-of-the-art baseline, LieGAN, in identifying discrete symmetries.
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ProConMV: Provenance-Enabled Conceptual Framework for Interpretable Multi-View Diabetic Retinopathy Diagnosis
Xiaoling Luo, Shuo Yang, Qihao Xu, Jiansong Zhang, Zhuoqin Yang, Zhihui Lai, Linlin Shen, Chengliang Liu
Existing deep learning models have demonstrated potential in Diabetic retinopathy (DR) diagnosis, but they still suffer from three key challenges: reliance on single-source inputs, opaque and untraceable reasoning processes, and the absence of a mechanism for result verification. Thus, we propose a provenance-enabled concept-based framework for multi-view DR diagnostic (ProConMV), which integrates DR lesion masks, clinical text and multi-view data, utilizing multimodal prompt analysis and visual-text concept interaction to learn the interpretable multi-source input. During the reasoning stage, the proposed framework introduces lesion concepts for causal reasoning chains combining clinical guidelines, and adds doctor intervention for human-machine collaboration. For dynamic fusion decision and verification in multi-view DR diagnosis, we derive via generalization theory that incorporating each view’s lesion concept uncertainty and grading uncertainty reduces the generalization error upper bound. Accordingly, we design a dual uncertainty-aware module to enable provenance-based verification, ultimately enabling verifiable analysis of DR diagnostic results. Extensive experiments conducted on two public multi-view DR datasets demonstrate the effectiveness of our method. The code will be released at https://github.com/SoY0ung/ProConMV.
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Toward Cybersecurity-Expert Small Language Models
Matan Levi, Daniel Ohayon, Ariel Blobstein, Ravid Sa, Ian Molloy, Yair Allouche
Large language models (LLMs) are transforming everyday applications, yet they lag behind in specialized fields, such as cybersecurity, due to a lack of high-quality, domain-specific models and training datasets. To address this gap, we present CyberPal 2.0, a family of cybersecurity-expert small language models (SLMs) ranging from 4B–20B parameters. To train CyberPal 2.0, we generate an enriched chain-of-thought cybersecurity instruction dataset built with our data enrichment and formatting pipeline, SecKnowledge 2.0, which integrates expert-in-the-loop steering of reasoning formats alongside LLM-driven multi-step grounding, yielding higher-fidelity, task-grounded reasoning traces for security tasks. Across diverse cybersecurity benchmarks, *CyberPal 2.0 consistently outperforms its baselines and matches or surpasses various open and closed-source frontier models*, while remaining a fraction of their size. On core threat-investigation tasks, such as correlating vulnerabilities and bug tickets with weaknesses, *our best 20B-parameter model outperforms GPT-4o, o1, o3-mini, and Sec-Gemini v1, ranking first, while our smallest 4B-parameter model ranks second. On core cyber threat intelligence knowledge tasks, our models outperform almost all tested frontier models, ranking second only to Sec-Gemini v1*. To foster reproducibility and practical adoption, we will release our models as open source.
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Boosting CVaR Policy Optimization with Quantile Gradients
Yudong Luo, Erick Delage
Optimizing Conditional Value-at-risk (CVaR) using policy gradient (a.k.a CVaR-PG) faces significant challenges of sample inefficiency. This inefficiency stems from the fact that it focuses on tail-end performance and overlooks many sampled trajectories. We address this problem by augmenting CVaR with an expected quantile term. Quantile optimization admits a dynamic programming formulation that leverages all sampled data, thus improves sample efficiency. This does not alter the CVaR objective since CVaR corresponds to the expectation of quantile over the

tail. Empirical results in domains with verifiable risk-averse behavior show tha t our algorithm within the Markovian policy class substantially improves upon CV aR-PG and consistently outperforms other existing methods.
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The Catastrophic Failure of *the* k-Means Algorithm in High Dimensions, and How Hartigan's Algorithm Avoids It
Roy R Lederman, David Silva-Sánchez, Ziling Chen, Gilles Mordant, Amnon Balanov, Tamir Bendory
Lloyd's k-means algorithm is one of the most widely used clustering methods. We prove that in high-dimensional, high-noise settings, the algorithm exhibits cata strophic failure: with high probability, essentially every partition of the data is a fixed point. Consequently, Lloyd's algorithm simply returns its initial pa rtition — even when the underlying clusters are trivially recoverable by other m ethods. In contrast, we prove that Hartigan's k-means algorithm does not exhibit this pathology. Our results show the stark difference between these algorithms and offer a theoretical explanation for the empirical difficulties often observe d with k-means in high dimensions.
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Linear Causal Representation Learning by Topological Ordering, Pruning, and Dise ntanglement
Hao Chen, Lin Liu, Yu Guang Wang
Causal representation learning (CRL) has garnered increasing interest from the c ausal inference and artificial intelligence communities due to its potential to disentangle complex data-generating mechanism into causally interpretable latent features by leveraging the heterogeneity of modern datasets. In this paper, we further contribute to the CRL literature, by focusing on the stylized linear str uctural causal model over latent features and assuming a linear mixing function that maps latent features to the observed data or measurements. Existing linear CRL methods often rely on stringent assumptions, such as access to single-node i nterventional data or restrictive distributional constraints on latent features and/or exogenous measurement noise. However, these prerequisites can be easy to violate in practice. In this work, we propose a novel linear CRL algorithm that, unlike existing methods, operates under weaker assumptions on environment heter ogeneity and data-generating distributions while still recovering latent causal features up to an equivalence class. We further validate our new algorithm via s ynthetic experiments and an interpretability analysis of large language models, demonstrating both its superiority over competing methods in finite samples and its potential in integrating causality into understanding artificial intelligenc e. The source code is available at [the accompanying GitHub link](https://github .com/utulie/code_for_linear_crl_paper_creator).
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Esoteric Language Models: A Family of Any-Order Diffusion LLMs
Subham Sekhar Sahoo, Zhihan Yang, Yash Akhauri, Johnna Liu, Deepansha Singh, Zho ujun Cheng, Zhengzhong Liu, Eric P. Xing, John Thickstun, Arash Vahdat
Diffusion Language Models offer a compelling alternative to autoregressive (AR) models by enabling parallel and controllable generation.
Within this family, Masked Diffusion Models (MDMs) currently perform best but st ill underperform AR models in perplexity and lack key inference-time efficiency features, most notably KV caching. We introduce Esoteric Language Models (Eso-LM s), a new family of models that fuses AR and MDM paradigms, smoothly interpolati ng between their perplexities while overcoming their respective limitations. Unl ike prior work, which uses transformers with bidirectional attention as MDM deno isers, we exploit the connection between MDMs and Any-Order autoregressive model s and adopt causal attention. This design lets us (1) **compute the exact likeli hood of MDMs for the first time** and, crucially, (2) **allows exact KV caching for MDMs** while preserving parallel generation over the full sequence length fo r the first time, significantly improving inference efficiency. Combined with an optimized sampling schedule, Eso-LMs establish a new state of the art on the sp eed-quality Pareto frontier for unconditional generation.
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From Extraction to Deduction: Resolving Functional Misalignment in RAG via a Collaborative Critic-Reasoner Framework
Yufei Chen, Yao Wang, Haibin Zhang, Hualin zhou, Tao Gu
Retrieval-augmented generation (RAG) systems suffer from a fundamental functional misalignment where retrievers optimize for semantic relevance, often recalling documents with high background utility but factually erroneous answer spans that generators blindly adopt as cognitive shortcuts. To resolve this, we propose the collaborative Critic-Reasoner framework that shifts robustness control from coarse-grained filtering to fine-grained cognitive decoupling. We disentangle the generation process into two serialized roles by deploying a Critic to perform surgical evidence purification through identifying and masking misleading entities while preserving supportive background context, followed by a Reasoner that switches from rote extraction to deductive reasoning based on the residual evidence. We operationalize this framework via a two-stage alignment strategy combining supervised fine-tuning (SFT) with path-aware direct preference optimization (DPO) to enforce strict behavioral synergy. Experimental results on adversarial benchmarks such as ConFiQA demonstrate that our method significantly outperforms baselines, achieving a 25.99\% accuracy gain in conflicting scenarios and effectively resolving the trust bias dilemma in real-world RAG.
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Mitigating Reward Hacking in LLM-based Recommendation: A Preference Optimization Approach
Heyu Chen, Junkang Wu, Guoqing Hu, Kexin Huang, Xiang Wang, Jiancan Wu
Post-training adaptation has become the central paradigm for leveraging large language models (LLMs) in recommendation.
While recent preference optimization methods, such as Direct Preference Optimization (DPO), enhance pairwise preference discrimination, they remain vulnerable to \emph{reward hacking}: models exploit imperfections in reward signals, leading to inflated training metrics without genuine recommendation gains.
We analyze this issue from a gradient perspective and formalize the concept of the
\emph{ε-insensitive region}, where pairwise updates exert little influence on the ordering between positives and unsampled negatives.
Under the Bradley–Terry model, we further show that these regions can occupy a substantial fraction of the preference space, inevitably leading to misaligned rankings.
To address this issue, we propose Simulated Preference Optimization for Reward-hacking mitigation using Pseudo-negatives (SIRIUS).
Our framework introduces pseudo-negative samples to enrich contrastive signals and reduce the prevalence of ε-insensitive regions.
Extensive experiments on three public benchmarks show that \our{} consistently improves ranking quality and effectively mitigates reward hacking, providing both theoretical and practical insights for advancing LLM-based recommendation.
Our code is available at
\url{https://anonymous.4open.science/r/C557-id}
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Deformba: Vision State Space Model with Adaptive State Fusion
Hongyu Ke, Jack Morris, Yongkang Liu, Satoshi Kitai, Kentaro Oguchi, Yi Ding, Haoxin Wang
State Space Models (SSMs) have emerged as a powerful and efficient alternative to Transformers, demonstrating linear-time complexity and exceptional sequence modeling capabilities. However, their application to vision tasks remains challenging. First, existing vision SSMs largely depend on manually designed fixed scanning methods to flatten image patches into sequences, which imposes predefined geometric structures and increases the complexity. Second, the broader adoption of vision SSMs is hindered in domains that require query-based interactions between distinct information streams. This is a result of the inherently causal and self-referential nature of SSMs designed for 1D sequence modeling tasks. This fusion mechanism is indispensable for critical perception tasks such as multi-view 3D fusion. To address these limitations, we propose Deformba, a context adaptive

method that dynamically augments the spatial structural information while mainta
ining the linear complexity of SSMs. Deformba also allows multi-modal fusion, an
alogous to standard cross attention. To demonstrate the effectiveness and genera
l applicability of Deformba, we test its performance on general 2D vision tasks
such as image classification, object detection, instance segmentation, and seman
tic segmentation, as well as 3D vision tasks like BEV perception. Extensive expe
riments show that Deformba achieves strong performance across various visual per
ception benchmarks.
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Evolving Interdependent Operators with Large Language Models for Multi-Objective
 Combinatorial Optimization

junhao qiu, Xin Chen, Liang Ge, Liyong Lin, Zhichao Lu, Qingfu Zhang

Neighborhood search operators are critical to the performance of Multi-Objective
 Evolutionary Algorithms (MOEAs) and rely heavily on expert design. Although rec
ent LLM-based Automated Heuristic Design (AHD) methods have made notable progres
s, they primarily optimize individual heuristics or components independently, la
cking explicit exploration and exploitation of dynamic coupling relationships be
tween operators. In this paper, multi-operator optimization in MOEAs is formulat
ed as a Markov decision process, enabling the improvement of interdependent oper
ators through sequential decision-making. To address this, we propose the Evolut
ion of Operator Combination (E2OC) framework for MOEAs, which achieves the co-ev
olution of design strategies and executable codes. E2OC employs Monte Carlo Tree
 Search to progressively search combinations of operator design strategies and a
dopts an operator rotation mechanism to identify effective operator configuratio
ns while supporting the integration of mainstream AHD methods as the underlying
designer. Experimental results across AHD tasks with varying objectives and prob
lem scales show that E2OC consistently outperforms state-of-the-art AHD and othe
r multi-heuristic co-design frameworks, demonstrating strong generalization and
sustained optimization capability.
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ExpWeaver: LLM Agents Learn from Experience via Latent RAG

Tao Feng, Tianyang Luo, Jingjun Xu, Zhigang Hua, Yan Xie, Shuang Yang, Ge Liu, J
iaxuan You

Experience learning has achieved promising results in enhancing LLM agent planni
ng and reasoning by integrating past interactions as reusable knowledge. However
, existing methods remain confined to explicit text space—retrieving experiences
 via semantic similarity and concatenating them into the context window, leading
 to substantial token overhead and a decoupled architecture that separates retri
eval from generation. To address these limitations, we propose ExpWeaver, a fram
ework that enables LLM agents to learn from experience via latent retrieval-augm
ented generation, without requiring a separate RAG module. ExpWeaver encodes exp
eriences using the LLM’s own hidden states, retrieves relevant experiences direc
tly in latent space at each decoding step, and integrates them through cross-att
ention aggregation and gated residual mechanisms. The entire pipeline is optimiz
ed end-to-end with reinforcement learning, supporting both generative and rankin
g tasks. We evaluate ExpWeaver on 13 diverse tasks spanning question answering,
reasoning, coding, scientific prediction, and recommendation. Results demonstrat
e that: (1) ExpWeaver achieves state-of-the-art on 12 out of 13 tasks, outperfor
ming the strongest baseline by over 6.8%; (2) ExpWeaver maintains token efficien
cy comparable to non-retrieval baselines while text-based retrieval methods requ
ire 1.5–2× more tokens; and (3) ExpWeaver exhibits superior cross-domain general
ization, outperforming the strongest baseline by 16.32% under zero-shot transfer
 and 15.21% under few-shot transfer. Our code for ExpWeaver is released at https
://github.com/ulab-uiuc/ExpWeaver.
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Lookahead-GCG: Improving Universal Multi-Model Optimization-Based Jailbreaking A
ttacks via Stochastic Nesterov Optimization

Rong Feng, Haohan Zhao, Shiqin Tang, Geng Liu, Song Lai, Meng Wang, Shuxin Zhuan
g, Yuqi Zha, Changyi Ma, Runsheng Yu

Transferable jailbreaking attacks enable red-teaming of black-box large language

models by optimizing adversarial prompts on open-source surrogates. A natural approach to improve transferability is multi-model training---optimizing against multiple source models simultaneously. Yet this approach has been largely abandoned, as it yields only marginal gains with standard optimizers. We argue the root cause is poor generalization: standard gradient descent lacks stability when aggregating gradients from diverse models. Since GCG and its variants implicitly perform SGD in discrete token space, they inherit this instability in multi-model settings. We address this with Lookahead-GCG, which combines: (1) Stochastic Nesterov Accelerated Gradient (SNAG), whose lookahead mechanism reduces sensitivity to individual gradient updates, providing stability for multi-model optimization; (2) embedding-space momentum accumulation, which enables SNAG in discrete token optimization; and (3) maximally distant initialization, which exploits SNAG's improved generalization by starting from a universally beneficial point. Experiments show our method achieves 50.37% ASR on open-source and 34.03\% on closed-source LLMs, outperforming GCG and TransferAttack with +11.78% gains from multi-model optimization.

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Elastic Diffusion Transformer

Jiangshan Wang, Zeqiang Lai, Jiarui Chen, Jiayi Guo, Hang Guo, Xiu Li, Xiangyu Yue, Chunchao Guo

Diffusion Transformers (DiT) have demonstrated remarkable generative capabilities but remain highly computationally expensive. Previous acceleration methods, such as pruning and distillation, typically rely on a fixed computational capacity, leading to insufficient acceleration and degraded generation quality. To address this limitation, we propose \textbf{Elastic Diffusion Transformer (E-DiT)}, an adaptive acceleration framework for DiT that effectively improves efficiency while maintaining generation quality. Specifically, we observe that the generative process of DiT exhibits substantial sparsity (i.e., some computations can be skipped with minimal impact on quality), and this sparsity varies significantly across samples. Motivated by this observation, E-DiT equips each DiT block with a lightweight router that dynamically identifies sample-dependent sparsity from the input latent. Each router adaptively determines whether the corresponding block can be skipped. If the block is not skipped, the router then predicts the optimal MLP width reduction ratio within the block. During inference, we further introduce a block-level feature caching mechanism that leverages router predictions to eliminate redundant computations in a training-free manner. Extensive experiments across 2D image (Qwen-Image and FLUX) and 3D asset (Hunyuan3D-3.0) demonstrate the effectiveness of E-DiT, achieving up to $\sim$2$\times$ speedup with negligible loss in generation quality.

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SlaClip: Gradient Norm Slacks can be Indicator for Adaptive Clipping in DP-SGD

Shuyan Zou, Shaowei Wang, Zhanxing Zhu, Jin Li, Changyu Dong, Vladimiro Sassone, Han Wu

Differentially private stochastic gradient descent (DP-SGD) achieves privacy by clipping per-sample gradients and injecting Gaussian noise, but its utility is highly sensitive to the choice of the clipping threshold C. A fixed C often degrades performance and necessitates repeated empirical calibration. Existing adaptive clipping methods either modify the gradient update in vanilla DP-SGD, causing additional tuning or optimization overhead, or introduce separate private queries to monitor gradient statistics. In contrast, we leverage the *slack* information induced by the standard clipping operation, an overlooked signal in prior work, and show that it provides an effective indication for adapting C.

In light of this, we propose *SlaClip*, a privacy-preserving adaptive clipping strategy using a post-hoc *Slack Indicator*. Under the same training configuration and privacy accountant, *SlaClip* preserves the sampling rule, noise multiplier, and global ℓ_2 sensitivity bound of vanilla DP-SGD. Therefore, *SlaClip* is a plug-and-play module for vanilla DP-SGD and its variants. Moreover, *SlaClip* is accounted under the same per-step privacy bound, while requiring no additional private query. Across diverse datasets and tasks, experiments show that *SlaClip* consistently outperforms baseline adaptive clipping methods.

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Text Generation as Continuous Latent Dynamics via Reinforcement Learning
Chen Jia
We propose to model text generation as a continuous-time latent dynamical proces s, where token generation is formulated as a Markov decision process whose inter nal state evolves via a neural ODE. This formulation bridges discrete token sequ ences and continuous semantic evolution, providing a theoretically grounded fram ework for text generation with continuous-time latent states. The framework is o ptimized via reinforcement learning, maximizing a composite objective that integ rates cumulative rewards with a Kullback–Leibler divergence regularization term from a pre-trained language model. Both theoretical and empirical results demons trate that our Continuous-Time Latent Language Model (CT-LLM) achieves superior effectiveness and efficiency in text generation, establishing a new paradigm for continuous-time language modeling.
****
Verified SHAP: Provable Bounds for Exact Shapley Values of Neural Networks
David Boetius, Shahaf Bassan, Guy Katz, Stefan Leue, Tobias Sutter
Shapley additive explanations (SHAP) are widely recognised as computationally in tractable for neural networks, since they induce an exponential search space ove r the input features. In this work, we take a first step towards scaling exact S HAP computation to larger search spaces by introducing an algorithm that leverag es recent advances in neural network verification to compute arbitrarily tight e xact lower and upper bounds on SHAP values for neural networks, ultimately recov ering the exact SHAP values. We demonstrate that our approach scales to orders o f magnitude larger search spaces than state-of-the-art exact methods. This provi des an important first step towards exact SHAP computation and establishes a pri ncipled cornerstone for evaluating statistical approximation methods on larger s earch spaces.
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Towards Understanding Continual Factual Knowledge Acquisition of Language Models : From Theory to Algorithm
Haoyu Wang, yifan shang, Zhongxiang Sun, Weijie Yu, Xiao Zhang, Jun Xu
Continual Pre-Training (CPT) is essential for enabling Language Models (LMs) to integrate new factual knowledge without erasing old. While classical CPT techniq ues like data replay have become the standard paradigm, the mechanisms underlyin g how LMs acquire and retain facts over time, termed as continual Factual Knowle dge Acquisition (cFKA), remain unclear. In this work, we present a theoretical f ramework that characterizes the training dynamics of cFKA using a single-layer T ransformer with linear attention, offering a unified explanation for the behavio r of popular CPT methods. Our analysis reveals that regularization-based methods merely adjust the convergence rate of parameters without altering the inherent forgetting tendency, whereas data replay methods shift convergence dynamics and stabilize pretrained knowledge. Building on these insights, we propose a novel g enerative data replay approach, called Selecting Tokens via attentiOn Contributi on (STOC), which identifies influential factual snippets to guide replay generat ion. Extensive experiments on both synthetic and real-world datasets validate ou r theoretical findings and demonstrate that STOC effectively enhances cFKA by mi tigating catastrophic forgetting.
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Just Y-Prediction: Enabling Historical Cumulative Inconsistency in Label Diffusi on for Learning with Noisy Label
Senyu Hou, Gaoxia Jiang, Xinyi Zheng, Yaqing Guo, Shuna Liang, Wenjian Wang
Label noise is pervasive in real-world datasets and significantly compromises mo del generalization, fueling extensive research into Learning with Noisy Labels (LNL). Most LNL methods focus on robust discriminative learning, while recent gen erative classifiers such as label diffusion models (LDMs) show superior robustne ss by modeling class posteriors. However, current LDMs predominantly rely on sta ndard ϵ-prediction, where Gaussian pnoise lacks explicit class semantic s, limiting both optimization and inference under label noise environments. To a ddress this issue, we propose just y-prediction (JYP), a novel training paradigm

that enables LDMs to directly characterize the label manifold and leverage explicit class-semantic guidance. Theoretically, we prove that JYP converges to an optimal solution equivalent to that of ϵ-prediction within the label diffusion framework, while facilitating accelerated convergence and enabling one-step inference. Leveraging JYP as a foundation, we further incorporate historical cumulative inconsistency to adaptively tailor optimization strategies for clean, noisy, and hard samples. Extensive experiments demonstrate that our method consistently outperforms competitors across diverse synthetic noisy datasets and achieves state-of-the-art performance on multiple real-world benchmarks.

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Constrained Bayesian Experimental Design via Online Planning

Yujia Guo, Daolang Huang, Xinyu Zhang, Sammie Katt, Samuel Kaski, Ayush Bharti

Bayesian experimental design (BED) is a principled framework for data-efficient design of sequential experiments. However, existing BED methods are unable to adapt to dynamic constraints inherent in real-world tasks due to budget limitations, varying costs, or physical constraints that restrict how designs evolve over time. In this paper, we introduce a novel approach to BED that enables constrained optimization of experimental designs by combining offline pre-training of an amortized policy and a posterior network with online multi-step lookahead planning using scenario trees. We empirically demonstrate that our method yields substantially more informative design sequences than existing methods across a range of constrained BED tasks, while incurring only a modest additional computational overhead.

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LongCoT: Benchmarking Long-Horizon Chain-of-Thought Reasoning

Sumeet Ramesh Motwani, Daniel Nichols, Charles London, Peggy Li, Fabio Pizzati, Acer Blake, Hasan Abed Al Kader Hammoud, Tavish Malcolm McDonald, Akshat Naik, Alesia Ivanova, Vignesh Baskaran, Ivan Laptev, Ruben Glatt, Tal Ben-Nun, Philip Torr, Ameya Prabhu, Brian R. Bartoldson, Bhavya Kailkhura, Christian Schroeder de Witt

As language models are increasingly deployed for complex autonomous tasks, their ability to reason accurately over longer horizons becomes critical. An essential component of this ability is planning and managing a long, complex chain-of-thought (CoT). We introduce LongCoT, a scalable benchmark of 2,500 expert-designed problems spanning chemistry, mathematics, computer science, chess, and logic to isolate and directly measure the long-horizon CoT reasoning capabilities of frontier models. Problems consist of a short input with a verifiable answer; solving them requires navigating a graph of interdependent steps that span tens to hundreds of thousands of reasoning tokens. Each local step is individually tractable for frontier models, so failures reflect long-horizon reasoning limitations. At release, the best models achieve <10% accuracy (GPT 5.2: 9.8%; Gemini 3 Pro: 6.1%) on LongCoT, revealing a substantial gap in current capabilities. Overall, LongCoT provides a rigorous measure of long-horizon reasoning, tracking the ability of frontier models to reason reliably over extended periods.

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σ: Sigmoid Modulation for Ultra High Resolution Diffusion

Bingxuan Zhao, Qing Zhou, Yu Wang, Chuang Yang, Qi Wang

Diffusion Transformers (DiTs) can synthesize high-fidelity images, but training at ultra-high resolutions is expensive, making inference-time extrapolation essential. Existing methods are typically \textit{scale-agnostic}, applying the same positional-modulation schedule regardless of target resolution. We show that this misses a scale-sensitive property of denoising: resizing shifts a fixed semantic pattern toward lower \emph{normalized} spatial frequencies while enlarging the spatial support over which global structure must be coordinated. The latter delays structural lock-in at high resolutions, so schedules may either relax guidance before global layout has stabilized, causing \textit{structural collapse}, or retain excessive intervention into late denoising, causing \textit{textural degradation}. We introduce \textbf{SigMa (σ)}, a training-free framework that uses Sigmoid Modulation for \textit{scale-adaptive} extrapolation through two scaling laws: \textit{Decoupled Geometric Center Alignment} and \textit{Iso-Va

riance Rate Adaptation}. Experiments show that SigMa reduces this mismatch, enab
ling training-free extrapolation up to 16 megapixels and achieving the best or c
ompetitive performance among the tested training-free extrapolation baselines. C
ode is available at \href{https://github.com/bxuanz/SigMa.git}{github.com/bxuanz
/SigMa}.
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Conformal Reliability: A New Evaluation Metric for Conditional Generation
Yachen Gao, Xinwei Sun, Yikai Wang, Ye Shi, Jingya Wang, Jianfeng Feng, Yanwei F
u
Conditional generative models have recently achieved remarkable success in vario
us applications. However, a suitable metric for evaluating the reliability of th
ese models, which takes into account their inherent uncertainty, is still lackin
g. Existing metrics, which typically assess a single output, may fail to capture
 the variability or potential risks in generation. In this paper, we propose a n
ovel evaluation metric called reliability score based on conformal prediction, w
hich measures the worst-case performance within the prediction set at a pre-spec
ified confidence level. However, computing this score is challenging due to the
high-dimensional nature of the output space and the nonconvexity of both the met
ric function and the prediction set.
    To efficiently compute this score, we introduce Conformal ReLiability (CReL)
, a framework that can (i) construct the prediction set with desired coverage; a
nd (ii) accurately optimize the reliability score within the constructed predict
ion set. We provide theoretical results on coverage and demonstrate empirically
that our method produces more informative prediction sets than existing approach
es. Experiments on synthetic data and the image-to-text and text-to-image tasks
further demonstrate the interpretability of our new metric, and the validity and
 effectiveness of our computational framework.
    Source code can be found at https://ggc29.github.io/CReL/.
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Independent Component Discovery in Temporal Count Data
Alexandre Chaussard, Anna Bonnet, Sylvain Le Corff
Advances in data collection are producing growing volumes of temporal count obse
rvations, making adapted modeling increasingly necessary. In this work, we intro
duce a generative framework for independent component analysis of temporal count
 data, combining regime-adaptive dynamics with Poisson log-normal emissions. The
 model identifies disentangled components with regime-dependent contributions, e
nabling representation learning and perturbations analysis. Notably, we establis
h the identifiability of the model, supporting principled interpretation. To lea
rn the parameters, we propose an efficient amortized variational inference proce
dure. Experiments on simulated data evaluate recovery of the mixing function and
 latent sources across diverse settings, while real-world applications to gut mi
crobiome and climate datasets reveal co-variation patterns and regime shifts con
sistent with domain-specific knowledge.
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Trajectory-Aware Certified Decentralized Unlearning via SGD Stability
Hengliang Wu, Jiale Yang, Shuzhen Chen, Di Wang, Dongxiao Yu, Youming Tao
Decentralized Unlearning (DU) aims to remove the influence of specific clients f
rom a collaboratively trained global model. However, existing methods suffer fro
m strong reliance on static, problem-specific hyperparameters or restrictive co
nvexity assumptions, limiting their general applicability. To overcome these lim
itations, we propose **TRA**jectory-aware **CE**rtified **D**ecentralized **U**n
learning (**TRACE-DU**), a generic unlearning framework for decentralized traini
ng. **TRACE-DU** introduces a fine-grained sensitivity analysis that leverages l
ocal SGD updates and decentralized training dynamics, thereby eliminating the ne
ed for convexity assumptions and reducing dependence on manually tuned parameter
s. By integrating strategic checkpoint selection with calibrated noise perturbat
ion, the proposed framework enables efficient certified unlearning. Moreover, we
 exploit historical model trajectories to extend this framework, enabling it to
naturally support sequential unlearning requests from an arbitrary number of cli
ents. We provide theoretical guarantees for certified unlearning and derive sens

itivity bounds under both convex and non-convex loss functions. Experimental res
ults demonstrate that our framework outperforms state-of-the-art baselines acros
s diverse metrics.
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EqGINO: Equivariant Geometry-Informed Fourier Neural Operators for 3D PDEs
Sungwon Kim, Juho Song, Seungmin Shin, Guimok Cho, Sangkook Kim, Chanyoung Park
Deep learning surrogates for 3D Partial Differential Equations (PDEs) often fail
 to generalize across geometric transformations because they depend heavily on s
pecific coordinate systems. While equivariant networks offer a solution, they ty
pically rely on local operations in the spatial domain, making the global recept
ive field—essential for PDE dynamics—computationally expensive. Conversely, Four
ier Neural Operators (FNOs) efficiently capture global interactions, yet establi
shing 3D equivariance within them remains impractical due to the prohibitive cos
t of spectral group convolutions. To bridge this gap, we introduce EqGINO, a geo
metrically robust framework that enforces isotropy in the spectral domain. By de
sign, EqGINO guarantees exact equivariance to the discrete symmetries inherent t
o the discretized computational domain. Beyond this discrete guarantee, our stru
ctural prior enables effective generalization to arbitrary continuous orientatio
ns even with a limited number of SE(3)-transformed training samples. Consequentl
y, our method robustly models coordinate-invariant physical laws on complex irre
gular 3D geometries. Our code is available at https://github.com/sung-won-kim/Eq
GINO
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Adversarial Training for Process Reward Models
Gurusha Juneja, Deepak Nathani, William Yang Wang
Process Reward Models (PRMs) enhance reasoning ability of LLMs by providing step
-level supervision.
However, their widespread adoption is limited due to expensive manual step-level
 annotation and poor generalization of static training data to novel errors.
We introduce Adversarially Trained PRMs (APRM), where
a Generator (G) learns to produce reasoning errors to deceive a PRM (R), whi
le R concurrently learns to detect them.
This interaction yields progressively harder negatives for R, improving it's r
obustness and generalization to novel errors without requiring manual step-level
 labels.
Averaged across diverse mathematical reasoning benchmarks, APRM improves solver
accuracy by $+3.4$ percentage points (pp) over the strongest PRM baseline. APRM
achieves gains of $+5.3$ pp on out-of-distribution tasks.
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The Data Manifold under the Microscope
Marios Koulakis, Constantin Seibold
A significant gap exists between theory and practice in deep learning. Generaliz
ation and approximation error bounds are often derived for simplified models or
are too loose to be informative. Many rely on the manifold hypothesis and on geo
metric regularity such as intrinsic dimension, curvature, and reach. Progress re
quires insight into data-manifold geometry and suitable benchmarks, yet existing
 options are polarized: analytic manifolds with known geometry but limited appli
cability, or real-world datasets where geometry is only coarsely estimable. We i
ntroduce a benchmarking framework for studying data geometry. We repurpose and e
xtend dSprites and COIL-20 with additional transformation dimensions and dense,
axis-aligned sampling, and pair them with finite-difference estimators that reco
ver curvature, reach, and volume at near-ground-truth accuracy in a regime where
 general-purpose estimators are unreliable or difficult to deploy. The framework
 is intended as a controlled testbed, useful as a calibration environment for ge
ometric estimators and a sandbox for probing theoretical assumptions. To illustr
ate its use, we present two application studies, namely assessing the scaling be
havior of the bounds of Genovese et al. and Fefferman et al., and tracking the l
ayer-wise geometry of a β-VAE, highlighting the behavior of current bounds
 and the value of controlled benchmarks for guiding and validating future theory
.

A reference implementation is available at https://github.com/koulakis/manifold-microscope.
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Robust and Consistent Ski Rental with Distributional Advice
Jihwan Kim, Chenglin Fan
The ski rental problem is a canonical model for online decision-making under uncertainty, capturing the fundamental trade-off between repeated rental costs and a one-time purchase. While classical algorithms focus on worst-case competitive ratios and recent ``learning-augmented'' methods leverage point-estimate predictions, neither approach fully exploits the richness of full distributional predictions while maintaining rigorous robustness guarantees. We address this gap by introducing algorithms that systematically integrate distributional predictions into both deterministic and randomized algorithms.
For the deterministic setting, we formalize the problem under perfect distributional prediction and derive an efficient algorithm to compute the optimal threshold-buy day.
We provide a rigorous performance analysis, identifying sufficient conditions on the predicted distribution under which the expected competitive ratio (ECR) matches the classic optimal randomized bound.
To handle imperfect predictions, we propose the Clamp Policy, which restricts the buying threshold to a safe range controlled by a tunable parameter.
We show that this policy is both robust, maintaining good performance even with large prediction errors, and consistent, approaching the optimal performance as predictions become accurate.
For the randomized setting, we characterize the stopping distribution via a Water-Filling Algorithm, which optimizes expected cost while strictly satisfying robustness constraints.
Experimental results across diverse distributions (Gaussian, geometric, and bi-modal) demonstrate that our framework improves consistency by significantly over existing point-prediction baselines while maintaining comparable robustness.
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OptiFluence: Principled Design of Privacy Canaries
Mohammad Yaghini, Michael Aerni, Junrui Zhang, Nicolas Papernot, Florian Tramèr
Privacy auditing has emerged as a practical tool for empirically estimating training data leakage in machine learning models, in contrast to the provable but often overly pessimistic bounds provided by differential privacy analysis. A common strategy is to use membership inference attacks to detect the presence of specific canaries—data points chosen to maximize attack success—in training data. However, existing canary designs are largely heuristic, relying on mislabeled or out-of-distribution samples. We address this gap by formulating canary design as a bilevel optimization problem, where the model is trained in the inner loop and the canary is optimized in the outer loop to maximize its detectability. To solve this problem, we develop OptiFluence, a scalable optimization framework that combines (i) initialization by selecting candidates using influence functions and (ii) unrolled optimization with memory-efficient techniques. Our approach achieves remarkable empirical performance on four datasets. Optimized canaries achieve nearperfect detection rates of 99.6% true positive rate at 0.1% false positive rate on CIFAR-10, outperforming in-distribution baselines by 4$\times$. Critically, these canaries transfer effectively across different model architectures without retraining, enabling practical third-party privacy audits. This transferability allows regulators and auditors to assess model privacy without requiring access to proprietary training infrastructure or substantial computational resources.
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How Powerful are LLMs in Generating Formal Program Specifications?
Fanpeng Yang, Xing Li, Shuling Wang, Jie An, Zeyu Sun, Shenghua Feng, Wenhan Wang, Weiyi Wang, Naijun Zhan, Fanjiang Xu
Formal verification provides strong guarantees of software correctness, but its adoption is limited by the high cost of writing precise formal specifications. W

hile recent large language models (LLMs) have shown strong capabilities in theor em proving and verified code generation, their true ability to generate program specifications remains unclear. Existing evaluations require either verifying im plementation conformance or proving semantic equivalence between specifications, both of which are formidably difficult and may conflate proof difficulty with s pecification quality. To address this problem, we introduce Coins, a Rocq based evaluation framework that assesses specification quality by instantiating specif ications under evaluation on trusted test cases and generating concrete proof ob ligations. This design aligns with the asymmetric nature of formal reasoning, wh ere successful proofs provide reliable evidence while proof failures are inheren tly ambiguous. Using Coins, we conduct a large scale study on HumanEval with a c urated set of human written Rocq specifications. Our results show that specifica tion generation remains a formidable challenge, and that verification complexity can obscure genuine differences in specification quality. Overall, we find that accurate specification evaluation, rather than model scaling alone, is central to understanding the power of LLMs for specification synthesis, and that test ca se based formal reasoning offers a more faithful and discriminative measure of p rogress.

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Efficiently Learning Drifting Halfspaces with Massart Noise

Mingchen Ma, Guyang Cao, Jelena Diakonikolas, Ilias Diakonikolas

We study the problem of learning a drifting concept in the
presence of Massart noise. In this framework,
an online learner has access to a history
of independent samples whose labels are noisy versions
of a target concept that may change from round to round.
The goal is to output, in each round, a hypothesis with
small prediction error.
We study the complexity of this learning
problem for the fundamental class of margin-separable
linear classifiers (halfspaces).
On the positive side, we give a
computationally efficient learner achieving error
$\eta + \tilde O(\Delta^{1/3}/\gamma)$,
where η upper bounds the Massart noise rate,
Δ is the drift rate, and γ is the margin.
Interestingly, in the realizable setting,
an adaptation of our techniques
yields an efficient learner with an improved error
rate over prior work.
On the lower-bound side, we provide formal evidence of an
information-computation tradeoff, strongly suggesting that
our algorithm's performance is essentially optimal.
Specifically,
while the information-theoretically optimal error
scales with $\Delta^{1/2}$, we prove
that $\Delta^{1/3}$-scaling is unavoidable for
low-degree polynomial tests,
even in the special case of random classification noise.

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NetDiff: Graph Diffusion with Improved Global Capabilities to Generate and Updat e Mobile Network Topologies

Félix Marcoccia, Victor Fagoo, Gilles Monzat de Saint Julien, Cédric Adjih, Thom as Watteyne, Paul Mühlethaler

We introduce NetDiff, a node-conditioned denoising diffusion model that generate s directional link topologies and a two-slot transmit/receive parity for mobile ad hoc networks. Directional antennas can yield high throughput but require glob ally consistent link decisions under sector, interference, connectivity, and hal f-duplex constraints. NetDiff improves global coherence with Absolute Cross-Atte ntive Modulation (ACAM) tokens, which provide permutation-invariant global signa

ls and help the model match graph-level counts (e.g., density and sector usage). We also propose partial diffusion to update an existing topology with a small number of denoising steps, enabling fast reconfiguration under mobility. NetDiff reaches over 95 \% of target performance with constant inference time, outperforms heuristic and omnidirectional baselines, and improves over a strong diffusion graph-transformer baseline on key metrics.
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FOCUS: DLLMs Know How to Tame Their Compute Bound
Kaihua Liang, Xin Tan, An Zhong, Hong Xu, Marco Canini
Diffusion Large Language Models (**DLLMs**) offer a compelling alternative to Auto-Regressive models, but their deployment is constrained by high decoding cost. In this work, we identify a key inefficiency in DLLM decoding: while computation is parallelized over token blocks, only a small subset of tokens is decodable at each diffusion step, causing most compute to be wasted on non-decodable tokens. We further observe a strong correlation between attention-derived token importance and token-wise decoding probability. Based on this insight, we propose FOCUS, an inference system designed for DLLMs. By dynamically *focusing* computation on decodable tokens and evicting non-decodable ones on-the-fly, FOCUS increases the effective batch size, alleviating compute limitations and enabling scalable throughput. Empirical evaluations demonstrate that FOCUS achieves up to 3.52× throughput improvement over the production-grade engine LMDeploy in large-batch settings, while preserving or improving generation quality across multiple benchmarks.
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CSPO: Constraint-Sensitive Policy Optimization for Safe Reinforcement Learning
Ayoub Belouadah, Sylvain KUBLER, YVES LE TRAON
Safe reinforcement learning (Safe RL) aims to maximize expected return while satisfying safety constraints, typically modeled as Constrained Markov Decision Processes (CMDPs). While primal-dual methods scale well to deep RL, they often suffer from delayed constraint correction, leading to oscillatory behavior and prolonged safety violations. In this paper, we propose *Constraint-Sensitive Policy Optimization (CSPO)*, a first-order primal-dual method that incorporates local constraint sensitivity into policy updates. CSPO augments the primal objective with a constraint-sensitive correction derived from the shortest signed distance to the safety boundary, enabling smarter recovery steps back to safety, compensating for delayed Lagrange multiplier updates, reducing oscillations near the boundary, and preserving the KKT solutions of the original constrained problem. Experiments on navigation and locomotion benchmarks demonstrate that CSPO achieves faster safety recovery and high reward preservation, resulting in higher constrained returns compared to state-of-the-art primal-dual and penalty-based methods.
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Dimension-Free Multimodal Sampling via Preconditioned Annealed Langevin Dynamics
Lorenzo Baldassari, Josselin Garnier, Knut Solna, Maarten V. de Hoop
Designing sampling algorithms for multimodal targets that remain stable under refinement of the finite-dimensional approximation of an underlying function-space problem is a central challenge. Annealed Langevin dynamics (ALD) is a natural alternative to classical Langevin in this context, since it is often observed to improve exploration across modes. Yet a gap remains between its empirical success and existing theory: under which conditions can ALD be guaranteed to remain stable across dimensions? In this paper, we bridge this gap by providing a uniform-in-dimension analysis of continuous-time ALD for Gaussian-mixture targets. Along an explicit annealing path obtained by gradually removing Gaussian smoothing from the target, we identify spectral conditions linking the smoothing covariance to the component covariances under which ALD achieves a prescribed accuracy in Kullback-Leibler divergence within a dimension-uniform time horizon. We then establish stability in a perturbative regime with imperfect initialization and approximate scores. Under a misspecified-mixture score model, we show that preconditioning ALD with an operator whose spectrum decays sufficiently fast prevents error terms from accumulating across coordinates and thereby preserves dimension-uniform control.

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Expanding the Capabilities of Reinforcement Learning via Text Feedback
Yuda Song, Lili Chen, Fahim Tajwar, Rémi Munos, Deepak Pathak, Drew Bagnell, Aarti Singh, Andrea Zanette
The success of RL for LLM post-training stems from an unreasonably uninformative source: a single bit of information per rollout as binary reward or preference label. At the other extreme, distillation offers dense supervision but requires demonstrations, which are costly and difficult to scale. We study natural language feedback as an intermediate signal: richer than scalar rewards, yet cheaper than complete demonstrations. Textual feedback is a natural mode of human interaction and is already abundant in many real-world settings, where users, tools, and automated judges routinely critique LLM outputs. Towards leveraging text feedback at scale, we formalize a multi-turn RL setup where text feedback is available during training but not at inference. Therefore, models must learn to internalize the feedback in order to improve their test-time single-turn performance. To do this, we propose two methods: Self Distillation, which trains the single-turn policy to match its own feedback-conditioned second-turn generations; and Feedback Modeling, which predicts the feedback as an auxiliary objective. We provide theoretical analysis on both methods, and empirically evaluate on reasoning puzzles, competition math, and creative writing tasks. Our results show that both methods consistently outperform strong baselines across benchmarks, highlighting the potential of RL with an additional source of rich supervision at scale.
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Quantifying the Generalization Gap in Seizure Detection: A Large-Scale Empirical Benchmark via the SzCORE Challenge
Jonathan Dan, Amirhossein Shahbazinia, Christodoulos Kechris, David Atienza
Reliable automatic seizure detection from long-term electroencephalogram recordings (EEG) remains an unsolved challenge, as current models often fail to generalize across patients or clinical settings. Manual EEG review still is the standard of care, highlighting the need for robust models and standardized evaluation. The current literature often reports high efficacy, yet these models frequently fail when deployed to unseen patient populations. To rigorously assess this generalization gap, we conducted a large-scale empirical study evaluating 28 state-of-the-art algorithmic architectures, ranging from classical feature engineering to modern Deep Learning. These algorithms were collected by organizing a competition. A strictly held-out private dataset of continuous EEG recordings from 65 subjects, totaling 4,360 hours of data, was utilized to evaluate algorithm performance. Expert neurophysiologists annotated these recordings, establishing the ground truth for seizure events. Algorithms were evaluated using event-based metrics from the SzCORE framework, including sensitivity, precision, F1-score, and false positive rate per day. Results revealed significant performance variability among state-of-the-art approaches, with the top F1 score of 32% (sensitivity 37%, precision 29%), highlighting the persistent difficulty of this task for current machine learning methodologies. Our analysis uncovered a discordance between peak performance and population-level stability. The algorithms achieving the highest aggregate F1-scores did not achieve the most consistent ranking across subjects, indicating high performance variance and susceptibility to failure on outlier patients. This independent evaluation also exposed a notable gap between self-reported efficacies and hold-out performance, underscoring the critical need for standardized, rigorous benchmarking in developing clinically viable ML models. A comparison with previous challenges and commercial systems indicates that the best algorithm in this study surpassed prior methods. Critically, the evaluation infrastructure transitions into a continuously open benchmarking platform, fostering reproducible research and accelerating the development of robust seizure detection algorithms by allowing ongoing submissions and integration of additional private datasets. Clinical centers can also adopt this platform to evaluate seizure detection algorithms on their EEG data using a standardized, reproducible framework.
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Tuning-Free One-Class Discriminant Learning for Tabular Anomaly Detection

Xuan-Ha Nguyen, Vu Nguyen Thai Duong, Nguyen Van Hoi, Kim-Hung Le, Nhien-An Le-Khac

Anomaly detection (AD) on real-world tabular data is challenged by diverse anomaly types, scarce labels, and high sensitivity to data-specific hyperparameter tuning.

A central difficulty is that different anomaly types favor opposing representations: *compactness* - tightening the normal class to expose pointwise deviations - and *structure preservation* - retaining cluster and cross-feature relationships.

Existing one-class detectors capture only one of these or balance them via hyperparameters, which is problematic without validation labels.

We propose **Discriminant Vector Machine for Anomaly Detection (DVM-AD)**, a closed-form one-class method derived from discriminant analysis that captures both behaviors under a single fixed configuration.

From a deterministic reference point built from the training data, DVM-AD derives a bounded ratio whose two ends correspond to compressive and structure-preserving directions. This bound enables selecting directions from both ends simultaneously without dataset-specific tuning.

In addition, a Moore-Penrose pseudo-inverse keeps the method well-posed under high-dimensional or rank-deficient settings, and test samples are scored by nearest-neighbor distance in the discriminant space, normalized for thresholding-ready use.

Across 47 ADBench tabular datasets and 10 NLP/CV embedding benchmarks against 28 baselines, DVM-AD achieves the best average AUROC (89.65\%, average rank 2.98) on tabular datasets and remains top-ranked across four anomaly types and on embedding tasks (average rank 1.60, AUROC 72.68\%).

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Cerebellar-Inspired Residual Control for Fault Recovery: From Inference-Time Adaptation to Structural Consolidation

Nethmi Jayasinghe, Diana Gontero, Spencer T. Brown, Vinod K Sangwan, Mark C. Hersam, Amit Ranjan Trivedi

Robotic policies deployed in real-world environments often encounter post-training faults, where retraining, exploration, or system identification are impractical. We introduce an inference-time, cerebellar-inspired residual control framework that augments a frozen reinforcement learning policy with online corrective actions, enabling fault recovery without modifying base policy parameters. The framework instantiates core cerebellar principles, including high-dimensional pattern separation via fixed feature expansion, parallel microzone-style residual pathways, and local error-driven plasticity with excitatory and inhibitory eligibility traces operating at distinct time scales. These mechanisms enable fast, localized correction under post-training disturbances while avoiding destabilizing global policy updates. A conservative, performance-driven meta-adaptation regulates residual gain and plasticity, preserving nominal behavior and suppressing unnecessary intervention. Experiments on MuJoCo benchmarks under actuator, dynamic, and environmental perturbations show improvements of up to $+66$% on $\texttt{HalfCheetah-v5}$ and $+53$% on $\texttt{Humanoid-v5}$ under moderate faults, with graceful degradation under severe shifts and complementary robustness from consolidating persistent residual corrections into policy parameters.

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Curriculum Reinforcement Learning for Black-Box Prompt Tuning via Large Language Models

Shuai Gong, Chaoran Cui, Xiaolin Dong, Chunyun Zhang, Linwei Fan

Black-box prompt tuning (BBPT) aims to optimize input prompts for large models where internal parameters and gradients are inaccessible. However, existing methods fail to simultaneously address the dual challenges of prompt interpretability and query efficiency. To address these challenges, we propose CRL-BPT, a curriculum reinforcement learning framework that utilizes a large language model as an agent to generate human-readable prompts. Specifically, CRL-BPT implements a dynamic curriculum schedule on two auxiliary objectives: an imitation loss and an innovation loss. By dynamically weighting these objectives, CRL-BPT regularizes

the RL process, guiding the agent from mimicking reference prompts to discoverin g novel patterns. Additionally, we introduce tailored stabilization mechanisms c omprising historical loss normalization and relative reward calibration to promo te more stable training. Extensive experiments demonstrate that CRL-BPT establis hes new state-of-the-art performance and generates highly interpretable prompts under a strict budget of API calls. Code is available at https://github.com/Gong Shuai8210/CRL-BPT.
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PRISM: Synergizing Vision Foundation Models via Self-organized Expert Specializa tion
Ying Tang, Dong Li, Youjia Zhang, Zikai Song, Junqing Yu, Wei Yang
Unifying the complementary strengths of diverse Vision Foundation Models (VFMs) into a single efficient model is highly desirable but challenged by the negative transfer inherent in monolithic distillation. To address these feature conflict s, we introduce \textbf{PRISM}, a novel dual-stream Mixture-of-Experts (MoE) fra mework that synergizes VFMs via modular specialization. We propose a two-stage p aradigm: (1) expertise deconstruction, where a teacher-conditional router guides experts to specialize in distinct representational subspaces to mitigate interf erence, followed by (2) dynamic recomposition, where the router learns to assemb le these experts into tailored computational pathways for downstream tasks. Expe riments on PASCAL-Context and NYUD-v2 show that \textbf{PRISM} establishes a new state of the art, validating that sparse, emergent specialization is a scalable approach for integrating diverse visual knowledge.
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The Generalization Spectrum: A Chromatographic Approach to Evaluating Learning A lgorithms
Jinghan Zhang, Zerui Cheng, Shiqi Chen, Ge Zhang, Wenhao Huang, Jiashuo Liu, Jun xian He, Tianle Cai
Traditional evaluations measure a learning algorithm's final performance on an i .i.d. test set, reducing learning to a single aggregate score. This approach obs cures a fundamental question: to what extent does learning from a specific examp le generalize to others? Such per-sample generalization—akin to learning by anal ogy in human cognition—captures how far the knowledge extracted from one example can transfer, yet remains invisible to standard benchmarks.

We introduce the Generalization Spectrum, an evaluation framework designed to ex pose this hidden dimension. For each training example, we construct a controlled suite of test variants arranged by increasing transfer distance—from exact reca ll to implementation transfer across languages, context transfer under complete narrative re-framing, category-matched in-domain problems, and an unpaired basel ine. By tracking performance across these distances, we reveal not just whether an algorithm learns, but how far that learning extends.
We instantiate this framework on competitive programming, using a synthetic gene ration pipeline seeded with recent problems to mitigate contamination.

We first compare **three canonical learning paradigms** under matched memorizati on. RL converts memorization into near-transfer more efficiently than SFT-family baselines, while ICL exhibits strong but correspondence-dependent transfer.
We then use the Spectrum to diagnose **within-family variants**. The resulting p rofiles show that local gains need not expand the generalization radius: abstrac tions and hints mainly lift local transfer, RFT preserves a stronger far-transfe r tail than reference SFT, and self-distillation or hint-assisted RL can reduce far transfer even when local transfer or optimization improves.
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CompleteP for RL: Maintaining Feature Learning When Scaling Deep Reinforcement L earning
Adam Lee, M Ganesh Kumar, Blake Bordelon, Cengiz Pehlevan
The maximal update parameterization (μP) provides an approach to scale para meters while preserving feature learning in deep neural networks. In fixed data distribution settings it has been found to yield more stable learning dynamics,

more consistent learned features, and enable optimal hyperparameters such as lea
rning rate to transfer from small models to larger ones. However, it is unclear
if these benefits readily transfer to reinforcement learning problems, where lea
rning dynamics are coupled to the non-stationary data distribution induced by an
 agent's own actions. We empirically study how two regimes, the ''rich'' Complet
eP and ''lazy'' Neural Tangent Kernel (NTK) parameterizations affect hyperparame
ter transfer, feature and policy consistency, and learning behavior as we scale
reinforcement learning agents. Ultimately, we show that agents trained using Com
pleteP consequentially improves compute and reward efficiency compared to the NT
K parameterization across multiple control tasks and variants.
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All Circuits Lead to Rome: Rethinking Functional Anisotropy in Circuit and Sheaf
 Discovery for LLMs
Xi Chen, Mingyu Jin, Jingcheng Niu, Yutong Yin, Jinman Zhao, Bangwei Guo, Dimitr
is N. Metaxas, Zhaoran Wang, Yutao Yue, Gerald Penn
In this paper, we present empirical and theoretical evidence against a central b
ut largely implicit assumption in circuit and sheaf discovery (CSD), which we te
rm the *Functional Anisotropy Hypothesis*: the idea that functions in large lang
uage models (LLMs) are localised to a unique or near-unique internal mechanism.
We show that a single LLM task can instead be supported by multiple, structurall
y distinct circuits or sheaves that are simultaneously faithful, sparse, and com
plete. To systematically uncover such competing mechanisms, we introduce Overlap
-Aware Sheaf Repulsion, a method that augments the CSD objective with an explici
t penalty on structural overlap across multiple discovery runs, enabling the dis
covery of circuits or sheaves with strong task performance but minimal shared st
ructure across a plethora of common CSD benchmarks. We find that this phenomenon
 becomes increasingly pronounced as the number of discovered sheaves grows and p
ersists robustly across major CSD methods. We further identify an ultra-sparse t
hree-edge sheaf and show that none of its edges is individually indispensable, u
ndermining even weakened notions of canonical or essential components. To explai
n these findings, we propose a *Distributive Dense Circuit Hypothesis* and provi
de a theoretical analysis demonstrating that non-unique, low-overlap circuit exp
lanations arise naturally from high-dimensional superposition under mild assumpt
ions. Together, our results suggest that mechanistic explanations in LLMs are in
herently non-canonical and call for a rethinking of how CSD results should be in
terpreted and evaluated.
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ComplexMCP: Evaluation of LLM Agents in Dynamic, Interdependent, and Large-Scale
 Tool Sandbox
Yuanyang Li, Xue Yang, Longyue Wang, Weihua Luo, Hongyang Chen
Current LLM agents are proficient at calling isolated APIs but struggle with the
 "last mile" of commercial software automation. In real-world scenarios, tools a
re not independent; they are atomic, interdependent, and prone to environmental
noise. We introduce $\textbf{ComplexMCP}$, a benchmark designed to evaluate agen
ts in these rigorous conditions. Built on the Model Context Protocol (MCP), $\te
xtbf{ComplexMCP}$ provides over 300 meticulously tested tools derived from 7 sta
teful sandboxes, ranging from office suites to financial systems. Unlike existin
g datasets, our benchmark utilizes a seed-driven architecture to simulate dynami
c environment states and unpredictable API failures, ensuring a deterministic ye
t diverse evaluation.

We evaluate various LLMs across full-context and RAG paradigms, revealing a star
k performance gap: even top-tier models fail to exceed a 60% success rate, far t
railing human performance 90%. Granular trajectory analysis identifies three fun
damental bottlenecks: (1) $\textbf{tool retrieval saturation}$ as action spaces
scale; (2) $\textbf{over-confidence}$, where agents skip essential environment v
erifications; and (3) $\textbf{strategic defeatism}$, a tendency to rationalize
failure rather than pursuing recovery. These findings underscore the insufficien
cy of current agents for interdependent workflows, positioning $\textbf{ComplexM
CP}$ as a critical testbed for the next generation of resilient autonomous syste

ms. The codebase and benchmark implementation are publicly available at https://github.com/AIDC-AI/complex-mcp.
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Subgroup Discovery with the Cox Model
Zachary Izzo, Iain Melvin
We study the problem of subgroup discovery for survival analysis, where the goal is to find an interpretable subset of the data on which a Cox model is highly accurate. We examine why existing quality functions are insufficient for this problem and introduce two technical innovations: the *expected prediction entropy (EPE)*, a novel metric for evaluating survival models that predict hazard functions, and the *conditional rank statistics (CRS)*, which quantifies individual point deviation from a subgroup's survival time distribution. We study the EPE and CRS theoretically and show they address problems with existing metrics. We then introduce eight algorithms for Cox subgroup discovery. Our main algorithm is based on the DDGroup framework of Izzo et al. (2023) and leverages both the EPE and CRS, allowing theoretical correctness guarantees in well-specified settings. Empirical evaluation on synthetic and real data confirms our theory, showing our methods recover ground-truth subgroups in well-specified cases and achieve better model fit than naively fitting the Cox model to the entire dataset. A case study on NASA jet engine simulation data demonstrates that discovered subgroups uncover known nonlinearities in the data and suggest design choices mirrored in practice.
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MoSSP: A Momentum-Based Single-Loop Stochastic Penalty Method for Nonconvex Constrained DC-regularized Optimization
Luxuan Li, Chunfeng Cui, Xiao Wang
In this paper, we study a structured class of nonconvex constrained stochastic problems with difference-of-convex (DC) regularization, where the feasible set is possibly nonconvex and the concave part of the DC regularizer is allowed to be nonsmooth.
The fundamental challenge lies in maintaining feasibility for nonconvex constraints while achieving favorable oracle complexity.
Although single-loop algorithms efficiently solve unconstrained DC optimization problems, their potential for constrained optimization with DC structure remains largely unexplored. To address this gap, we develop **MoSSP**, a **Mo**mentum-based **S**ingle-loop **S**tochastic **P**enalty method for such problems with provable complexity guarantees. The key idea is to apply a single stochastic proximal-gradient step to the Moreau envelope of the penalty plus the convex DC part, with the concave part's proximal mapping computed in parallel. We derive two algorithm variants: a Polyak-momentum version with $\mathcal{O}(\varepsilon^{-4})$ oracle complexity for finding stochastic ε-KKT points, and an improved $\mathcal{O}(\varepsilon^{-3})$ version incorporating recursive momentum. Experimental results demonstrate the effectiveness of the proposed algorithms.
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Genome-Factory: A Library for Tuning, Deploying, and Interpreting Genomic Foundation Models
Weimin Wu, Xuefeng Song, Yibo Wen, Qinjie Lin, Zhihan Zhou, Jerry Yao-Chieh Hu, Zhong Wang, Han Liu
We introduce Genome-Factory, the first integrated Python library for tuning, deploying, and interpreting genomic foundation models.
Our core contribution is to simplify and unify the workflow for genomic model development: data collection, model tuning, inference, benchmarking, and interpretability.
For data collection, Genome-Factory offers an automated pipeline to download genomic sequences and preprocess them.
For model tuning, Genome-Factory supports both full and parameter-efficient fine-tuning across diverse genomic models.
For inference, Genome-Factory enables both embedding extraction and DNA sequence generation.
For benchmarking, we include two existing benchmarks and provide a flexible inte

rface to incorporate additional benchmarks.
For interpretability, Genome-Factory introduces an open-source biological interp reter based on a sparse auto-encoder.
We validate the utility of Genome-Factory across three dimensions:
(i) Compatibility with diverse models and fine-tuning methods;
(ii) Benchmarking downstream performance using two open-source benchmarks;
(iii) Biological interpretation of learned representations with DNABERT-2.
These results highlight its practical value for real-world genomic analysis.
GitHub: https://github.com/WeiminWu2000/Genome_Factory.
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VecDesigner: Exploring Visual Guidance and Structural Consistency for Semantic T ypography
Liu Yu, Xingjiao Wu, Ziang Liu, Jiabao Zhao, Daoguo Dong, Liang He
Semantic Typography aims to visualize the meaning of an input word through the f orm of a character, while preserving its legibility. Existing vector-based metho ds, which primarily rely on text-driven optimization like Score Distillation Sam pling (SDS), often produce glyphs that lack rich semantic details. Furthermore, these approaches struggle to maintain the overall structural integrity of the gl yphs and frequently suffer from visual artifacts caused by intersections, compro mising both readability and aesthetic quality. To address these challenges, we p ropose VecDesigner, a novel optimization-based method for vector semantic typogr aphy. Specifically, we introduce Visual-Guided Score Distillation Sampling (VGSD S), which leverages text-related reference images as visual guidance to infuse t he glyphs with richer and more concrete semantic details. To preserve legibility and structural integrity, we design a vector-based Procrustes loss to constrain the overall deformation of the glyph. Concurrently, we effectively mitigate the intersection problem by imposing positional relationship constraints on the con trol points. Comprehensive experiments demonstrate that VecDesigner outperforms existing methods in both semantic expression and structural preservation, genera ting high-quality, expressive, and clearly legible semantic glyphs.
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Draft-and-Audit Reinforcement Learning for Optimization Modeling
Zeping Min, Weihang Xu, Zhengzhong You, Wotao Yin, Xinshang Wang
Natural language to optimization (NL2Opt) requires translating unstructured text into executable mathematical models. Beyond simple syntax errors, this task suf fers from silent modeling failures, where incorrect formulations execute success fully but yield invalid results. We propose \textbf{Draft-and-Audit RL (DA-RL)}, a framework that learns optimization modeling as a two-step iterative workflow. Unlike inference-time scaffolds that rely on intermediate solver feedback to gu ide repairs, DA-RL optimizes a shared-parameter policy using terminal-only verif ication: the model is rewarded solely based on the execution of the final audite d program. This constraint forces the model to internalize rubric-guided revisio n as a learned capability and encourages the emergence of cross-turn synergy, wh ere the policy learns to generate drafts that are structurally amenable to self- correction.
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SL-VC: A Benchmark and Automated Framework for Separation Logic Verification Con dition Proving
Hanyang Wang, Xiwei Wu, Qinxiang Cao
Formal verification of system software with complex heap manipulations remains c hallenging. Standard automated solvers frequently fail to discharge separation l ogic verification conditions even when correct specifications like loop invarian ts are provided, forcing verification engineers to manually construct proofs. Wh ile large language models (LLMs) have shown promise in proof synthesis, speciali zed approaches for separation logic remain unexplored. To bridge this gap, we in troduce SL-VC (Separation Logic Verification
Conditions), a benchmark of 289 verification conditions from textbook implementa tions of data structures and algorithms together with real-world C code, includi ng the LiteOS kernel's linked list library and the mini-gmp library. Our evaluat ion reveals that general-purpose LLMs and existing LLM-based Rocq provers strugg

le to effectively discharge these verification conditions. To address this chall
enge, we propose SPLIT (Split spatial and pure Proving with LLM-frIendly Tactics
), a novel framework that enables predictable proof state transitions through an
 LLM-friendly tactic library, combined with a two-stage workflow that separates
spatial and pure reasoning to align with separation logic semantics. Experimenta
l results on SL-VC demonstrate that SPLIT consistently outperforms existing appr
oaches, showing that LLM-assisted proof synthesis is a promising solution for se
paration logic verification of real-world system software.
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HodgeFlow Policy Search by Topologically Dissecting Temporal-Difference Signals
in Non-Markovian Environments
Zuyuan Zhang, Sizhe Tang, Tian Lan
Non-Markovian dynamics are commonly found in real-world environments due to long
-range dependencies, partial observability, and memory effects. The Bellman equa
tion that is the central pillar of Reinforcement learning (RL) becomes only appr
oximately valid under Non-Markovian. Existing work often focus on practical algo
rithm designs and offer limited theoretical treatment to address key questions,
such as what dynamics are indeed capturable by the Bellman framework and how to
inspire new algorithm classes with optimal approximations. In this paper, we pre
sent a novel topological viewpoint on temporal-difference (TD) based RL. We show
 that TD errors can be viewed as 1-cochain in the topological space of state tra
nsitions, while Markov dynamics are then interpreted as topological integrabilit
y. This novel view enables us to obtain a Hodge-type decomposition of TD
errors into an integrable component and a topological residual, through a Bellma
n–de Rham projection. We further propose HodgeFlow Policy Search (HFPS) by fitti
ng a {potential network} to minimize the non-integrable projection residual in R
L, achieving stability/sensitivity guarantees. In numerical evaluations, HFPS is
 shown to significantly improve RL performance under non-Markovian.
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Single-Head Attention in High Dimensions: A Theory of Generalization, Weights Sp
ectra, and Scaling Laws
Fabrizio Boncoraglio, Vittorio Erba, Emanuele Troiani, Yizhou Xu, Florent Krzaka
la, Lenka Zdeborová
Trained attention layers exhibit striking and reproducible spectral structure of
 the weights, including low-rank collapse, bulk deformation, and isolated spectr
al outliers, yet the origin of these phenomena and their implications for genera
lization remain poorly understood. We study empirical risk minimization in a sin
gle-head tied-attention layer trained on synthetic high-dimensional sequence tas
ks generated from the attention-indexed model. Using tools from random matrix th
eory, spin-glass theory, and approximate message passing, we obtain an exact hig
h-dimensional characterization of training and test error, interpolation and rec
overy thresholds, and the spectrum of the key and query matrices. Our theory pre
dicts the full singular-value distribution of the trained query–key map—includin
g low-rank structure and isolated spectral outliers—in qualitative agreement wit
h observations in more realistic transformers. Finally, for targets with power-l
aw spectra, we show that learning proceeds through sequential spectral recovery,
 leading to the emergence of power-law scaling laws.
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Gradients with Respect to Semantics Preserving Embeddings Tell the Uncertainty o
f Large Language Model
Mingda Li, Rundong Lv, Xinyu Li, Weinan Zhang, Ting Liu
Uncertainty quantification (UQ) is an important technique for ensuring the trust
worthiness of LLMs, given their tendency to hallucinate. Existing state-of-the-a
rt UQ approaches for free-form generation rely heavily on sampling, which incurs
 high computational cost and variance. In this work, we propose the first gradie
nt-based UQ method for free-form generation, SemGrad, which is sampling-free and
 computationally efficient. Unlike prior gradient-based methods developed for cl
assification tasks that operates in parameter space, we propose to consider grad
ients in semantic space. Our method builds on the key intuition that a confident
 LLM should maintain stable output distributions under semantically equivalent i

nput perturbations. We interpret the stability as the gradients in semantic spac
e and introduce a Semantic Preservation Score (SPS) to identify embeddings that
best capture semantics, with respect to which gradients are computed. We further
 propose HybridGrad, which combines the strengths of SemGrad and parameter gradi
ents. Experiments demonstrate that both of our methods provide efficient and eff
ective uncertainty estimates, achieving superior performance than state-of-the-a
rt methods, particularly in settings with multiple valid responses.
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Dual-View Predictive Diffusion: Lightweight Speech Enhancement via Spectrogram-I
mage Synergy

Ke Xue, Rongfei Fan, Kai Li, Shanping Yu, Puning Zhao, Jianping An

Diffusion models have recently set new benchmarks in Speech Enhancement (SE). Ho
wever, most existing score-based models treat speech spectrograms merely as gene
ric 2D images, applying uniform processing that ignores the intrinsic structural
 sparsity of audio, which results in inefficient spectral representation and pro
hibitive computational complexity. To bridge this gap, we propose **DVPD**, an e
xtremely lightweight **D**ual-**V**iew **P**redictive **D**iffusion model, which
 uniquely exploits the dual nature of spectrograms as both visual textures and p
hysical frequency-domain representations across both training and inference stag
es. Specifically, during training, we optimize spectral utilization via the Freq
uency-Adaptive Non-uniform Compression (FANC) encoder, which preserves critical
low-frequency harmonics while pruning high-frequency redundancies. Simultaneousl
y, we introduce a Lightweight Image-based Spectro-Awareness (LISA) module to cap
ture features from a visual perspective with minimal overhead. During inference,
 we propose a Training-free Lossless Boost (TLB) strategy that leverages the sam
e dual-view priors to refine generation quality without any additional fine-tuni
ng. Extensive experiments across various benchmarks demonstrate that DVPD achiev
es state-of-the-art performance while requiring only **35%** of the parameters a
nd **40%** of the inference MACs compared to SOTA lightweight model, PGUSE. Thes
e results highlight DVPD's superior ability to balance high-fidelity speech qual
ity with extreme architectural efficiency. Code and audio samples are available
at https://github.com/ke12345213/dvpd_demo
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GraphFlow: A Graph-Based Workflow Management for Efficient LLM-Agent Serving

Ao Li, Shangpeng Yang, Fahao Chen, Tianheng Xu, Peng Li, su zhou

Large Language Model (LLM)-based agents demonstrate strong reasoning and executi
on capabilities on complex tasks when guided by structured instructions, commonl
y referred to as workflows. However, existing workflow-assisted agent serving sy
stems typically rely on predefined templates and shallow matching mechanisms, wh
ich limit their ability to capture deep semantic relationships and generalize to
 previously unseen tasks. To address these limitations, we propose a new workflo
w management paradigm that represents workflows using a unified graph, termed wG
raph, where each node corresponds to an atomic operation. wGraph serves as a sha
red substrate from which task-specific workflows are dynamically instantiated. B
uilding on wGraph primitives, we introduce GraphFlow, a system that efficiently
integrates workflows into agent serving through two key designs. First, adaptive
 workflow generation dynamically constructs workflows from wGraph based on task
semantics and constraint requirements. Second, workflow state management exploit
s wGraph structure to efficiently manage Key-Value (KV) caches, reducing redunda
nt computation during agent serving. Extensive experiments across five benchmark
 datasets show that GraphFlow consistently outperforms state-of-the-art methods,
 yielding an average performance improvement of approximately 4.95 percentage po
ints, while achieving an approximately 4× reduction in memory footprint.
****

On Effectiveness and Efficiency of Agentic Tool-calling and RL Training

Tong Liu, Cheng Qian, Matej Cief, Yuan He, Daniele Dan, Nikolaos Aletras, Gabrie
lla Kazai

Tool-calling is a central component of modern large language model (LLM) agents,
 equipping them with skills beyond their parametric knowledge.
This paper studies tool-calling along two complementary axes: **effectiveness**,

 i.e., how this capability is *measured*, and **efficiency**, i.e., how it is *learned*.
On effectiveness, we systematically analyze tool-calling evaluation pipelines and show that results can be highly sensitive to seemingly minor, often undocumented implementation choices including the *random seed*, *system prompt*, *multi-turn template construction*, and how *prior interaction/reasoning history* is carried forward.
These choices can lead to substantial differences in reported performance, especially in multi-turn settings where without rigorous standardization, leaderboard rankings are unreliable.
On efficiency, we examine standard reinforcement learning (RL) for tool-calling and identify two sources of computational waste: (i) during rollouts, many prompts produce no learning signal, and (ii) during policy updates, optimization incurs high computational cost.
Guided by these findings, we introduce two techniques that accelerate RL-based tool-calling training, achieving substantial wall-clock speedup without degrading performance.
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Process Reward Agents for Steering Knowledge-Intensive Reasoning
Jiwoong Sohn, Tomasz Sternal, Kenneth Styppa, Torsten Hoefler, Michael Moor
Reasoning in knowledge-intensive domains remains challenging as intermediate steps are often not locally verifiable: unlike math or code, evaluating step correctness may require synthesizing clues across large external knowledge sources. As a result, subtle errors can propagate through reasoning traces, potentially never to be detected. Prior work has proposed process reward models (PRMs), including retrieval-augmented variants, but these methods operate post hoc, scoring completed trajectories, which prevents their integration into dynamic inference procedures. Here, we introduce Process Reward Agents (PRA), an inference-time method for providing domain-grounded, online, step-wise rewards to a frozen policy. In contrast to prior retrieval-augmented PRMs, PRA enables search-based decoding to rank and prune candidate trajectories at every generation step. Experiments on multiple medical reasoning benchmarks demonstrate that PRA consistently outperforms strong baselines, achieving 81.9% accuracy on MedQA with Qwen3-4B, a new state of the art at the 4B scale. Importantly, PRA generalizes to unseen frozen policy models ranging from 0.5B to 8B parameters, improving their accuracy by up to 25.7% without any policy model updates. More broadly, PRA suggests a paradigm in which frozen reasoners are decoupled from domain-specific reward modules, allowing the deployment of new backbones in complex domains without retraining. All code and data are publicly available at https://process-reward-agents.github.io/.
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Bias-Spectrum Neural Processes for Parametric PDEs: Architecture Priors Meet PDE Constraints
Hui Li, Huafeng Liu, Chenguang Li, Tianxiao Zhang, Yajun Yang, Liping Jing
Parametric partial differential equations (PDEs) serve as fundamental models across science and engineering, yet constructing fast and accurate surrogate models from sparse, irregularly sampled observations with reliable uncertainty quantification remains challenging. Existing approaches struggle to simultaneously handle variable observation patterns, preserve physics consistency, and provide well-calibrated predictive uncertainty. We introduce Bias-Spectrum Neural Processes (BSNP), a unified meta-learning framework that systematically integrates weak structural priors (translation equivariance, locality) with strong physical priors (governing equations and boundary conditions). BSNP addresses two critical obstacles: discretization overfitting through stochastic collocation that resamples residual evaluation points, and uncertainty collapse through mean-field enforcement that applies PDE constraints only to predictive means while preserving learned uncertainty. Comprehensive experiments on nonlinear Poisson equations, Burgers dynamics, and Navier-Stokes flows demonstrate that BSNP achieves superior accuracy and well-calibrated uncertainty quantification in sparse-data regimes.
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Rethinking the Design Space of Reinforcement Learning for Diffusion Models: On t
he Importance of Likelihood Estimation Beyond Loss Design
Jaemoo Choi, Yuchen Zhu, Wei Guo, Petr Molodyk, Bo Yuan, Jinbin Bai, Yi Xin, Mol
ei Tao, Yongxin Chen
Reinforcement learning has been widely applied to diffusion and flow models for
visual tasks such as text-to-image generation. However, these tasks remain chall
enging because diffusion models have intractable likelihoods, which creates a ba
rrier for directly applying popular policy-gradient type methods. Existing appro
aches primarily focus on crafting new objectives built on already heavily engine
ered LLM objectives, using ad hoc estimators for likelihood, without a thorough
investigation into how such estimation affects overall algorithmic performance.
 In this work, we provide a systematic analysis of the RL design space by disent
angling three factors: i) policy-gradient objectives, ii) likelihood estimators,
 and iii) rollout sampling schemes. We show that adopting an evidence lower boun
d (ELBO) based model likelihood estimator, computed only from the final generate
d sample, is the dominant factor enabling effective, efficient, and stable RL op
timization, outweighing the impact of the specific policy-gradient loss function
al. We validate our findings across multiple reward benchmarks using SD 3.5 Medi
um, and observe consistent trends across all tasks. Our method improves the GenE
val score from 0.24 to 0.95 in 90 GPU hours, which is $4.6\times$ more eff
icient than FlowGRPO and $2\times$ more efficient than the SOTA method Diffusion
NFT without reward hacking.
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MechVQA: Benchmarking and Enhancing Multimodal LLMs on Comprehensive Mechanical
Drawing Understanding
Qian Kou, Xiaofeng Shi, Yulin Li, Xiaosong Qiu, Xinyang Wang, Hua Zhou, Cao Dong
xing
Multimodal Large Language Models (MLLMs) have demonstrated significant achieveme
nts in
general visual question answering (VQA) tasks.
However, they remain brittle on mechanical en-
gineering drawings, where high annotation den-
sity and weak domain knowledge, compounded
by unreliable spatial relation reasoning under
strict projection rules and geometric constraints,
make decisive cues easy to miss and frequently
lead to wrong answers. To bridge this gap,
we introduce the first comprehensive mechani-
cal drawing understanding dataset, MechVQA,
created through a semi-automated construction
and quality-control pipeline. MechVQA con-
tains 3.3k high-density pictures with 21K ques-
tion–answer pairs, spanning 10 different fine-
grained tasks across three capability levels:
Recognition, Reasoning, and Judging, provid-
ing a testbed to evaluate and improve MLLM
understanding on real-world mechanical draw-
ings. On top of MechVQA, we then develop the
MechVL model through a multi-stage training
paradigm, building a strong domain-specialized
baseline. Extensive experimental results demon-
strate that MechVL outperforms the strongest
closed-source baseline by 7.57 percentage points
on the MechVQA total score, significantly en-
hancing mechanical drawing understanding abil-
ity and providing a reusable foundation for deploy-
ing MLLMs in mechanical design and inspection
scenarios.
****
Distilling Task-Level Coordination Policies for Generalizable Multi-Agent Cooper

ation
Zimo Zhai, Manjie Xu, Wei Liang
Large language models have shown strong reasoning abilities and are increasingly explored as high-level coordinators for multi-agent systems. However, directly deploying LLMs for coordination remains challenging, as effective policies often fail to reliably emerge at the low-level control stage, and inference costs limit scalability.
We propose SynCoord (Synthetic Coordination Distillation), a self-supervised pipeline that distills task-level decision-making for cooperation from high-capacity reasoning models into lightweight agent policies. Our approach does not rely on explicit supervision or handcrafted coordination rules. Instead, we define a set of task-level tool interfaces that constrain LLM interaction and enable the collection of interaction trajectories, which are then used to train compact coordinated policies. This distillation process transfers coordination behaviors that are difficult to elicit through prompting alone, while substantially reducing inference overhead at execution time.
We evaluate our method on cooperative multi-agent benchmarks including Overcooked-AI and Level-Based Foraging (LBF), under varying team sizes and environment scales. Experimental results show that the distilled policies achieve success rates and execution efficiency comparable to reinforcement learning–based methods, while exhibiting fewer erroneous or redundant actions. Moreover, the learned task-level coordination policy generalizes effectively to unseen team compositions and larger layouts without retraining.
****
Enhancing Cross-subject Emotion Recognition via Heterogeneous Distribution Augmentation and Collaborative Learning
Wending Xiong, Ruimin Hu, Lingfei Ren, Junhang Wu, Mei Wang, Dengshi Li, Mang Ye
Cross-subject emotion recognition aims to improve a model's generalization to previously unseen subjects. Existing methods are mainly built upon domain generalization or data augmentation, but suffer from two major limitations: 1) heavy dependence on modality-specific feature designs—almost exclusively tailored to EEG signals—resulting in limited generalizability; and 2) the widespread assumption of independently and identically distributed data, which restricts the diversity of generated samples. To address these challenges, we systematically analyze the heterogeneous distribution characteristics of emotion data and propose **MixEmo**, a framework that integrates heterogeneous distribution augmentation and collaborative learning. Specifically, a well-trained backbone is used to extract representations and partition them into multiple single-distribution subsets as distribution prototypes. These prototypes are randomly combined to synthesize unseen distributions, thereby enhancing distributional diversity. Finally, heterogeneous distribution collaborative learning jointly optimizes the model across subsets. Extensive experiments demonstrate that **MixEmo** substantially improves generalization performance in cross-subject emotion recognition.
****
Accurate, Private, Secure, Federated U-statistics with Higher Degree
Quentin Sinh, Jan Ramon
We study the problem of computing a U-statistic with a kernel function f of degree $k \geq 2$, i.e.,
the average of some function f over all k-tuples of instances,
in a federated learning setting.
U-statistics of degree 2 include several useful statistics such as Kendall's τ coefficient,
the Area under the Receiver-Operator Curve and the Gini mean difference.
Existing methods provide solutions only under the lower-utility local differential privacy model and/or scale poorly in the size of the domain discretization.

In this work, we propose a protocol that securely computes U-statistics of degree $k \geq 2$ under central differential privacy by leveraging Multi Party Computation (MPC).
Our method substantially improves accuracy when compared to prior solutions.

We provide a detailed theoretical analysis of its accuracy, communication and co
mputational properties.
We evaluate its performance empirically, obtaining favorable results, e.g., for
Kendall's τ coefficient, our approach reduces the Mean Squared Error by up
to four orders of magnitude over existing baselines.
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Efficient and Minimax Optimal In-context Nonparametric Regression with Transform
ers
Michelle Ching, Ioana Popescu, Nico Smith, Tianyi Ma, William G. Underwood, Rich
ard J. Samworth
We study in-context learning for nonparametric regression with α-Hölder s
mooth regression functions, for some $\alpha>0$. We
prove that, with n in-context examples and d-dimensional regression covariat
es, a pretrained transformer with $\Theta(\log n)$ parameters and $\Omega(n^{2\a
lpha/(2\alpha+d)}\log^3 n)$ pretraining sequences can achieve the minimax optima
l rate of convergence $O(n^{-2\alpha/(2\alpha+d)})$ in mean squared error. Our r
esult requires substantially fewer transformer parameters and pretraining sequen
ces than previous results in the literature. This is achieved by showing that tr
ansformers are able to approximate local polynomial estimators efficiently by im
plementing a kernel-weighted polynomial basis and then running gradient descent.
****
Learning to Label: A Reinforced Self-Evolving Framework for Semi-supervised Refe
rring Expression Segmentation
Runlong Cao, Ying Zang, Chuanwei Zhou, Tianrun Chen, Tong Zhang, Zhen Cui, Chuny
an Xu
Semi-supervised referring expression segmentation (SS-RES) aims to achieve preci
se pixel-level language grounding under limited annotation, yet suffers from lim
ited supervision and unreliable pseudo-labels when exploiting unlabeled image–te
xt pairs.
In this work, we propose Learning to Label, a reinforced self-evolving framework
 (L2L) that casts pseudo-label construction as a learnable decision-making proce
ss. To build foundational understanding, we leverage a multimodal large language
 model to extract semantic–spatial priors, which are instantiated as initial sof
t segmentation proposals and elevated—together with textual cues—into learnable
guidance signals that condition a hierarchical segmentation network. To ensure s
table learning, a reinforced pseudo-label selection is further formulated as an
exploratory decision process that adaptively rewards high-utility pixel-level su
pervision based on multimodal priors and model predictions. This reinforced self
-evolving loop enables joint optimization of the segmentation model and pseudo-l
abels, progressively enhancing label reliability under sparse supervision. Exten
sive experiments on RefCOCO, RefCOCO+, and RefCOCOg datasets demonstrate improve
ments over existing methods, validating its effectiveness and generalization.
****
Mind-Omni: A Unified Multi-Task Framework for Brain-Vision-Language Modeling via
 Discrete Diffusion
Yizhuo Lu, Changde Du, Qingyu Shi, Hang Chen, Jie Peng, Liuyun Jiang, Shuangchen
 Zhao, Huiguang He
Modeling the interplay between external stimuli and internal neural representati
ons is a pivotal research area for Brain-Computer Interfaces (BCIs). A major lim
itation of prior work is the prevailing paradigm of specialized, single-task mod
els, which curtails versatility and neglects inter-task synergies. To address th
is, we propose Mind-Omni, the first versatile framework that unifies seven disti
nct encoding and decoding tasks through a discrete diffusion paradigm. At its co
re is a novel Brain Tokenizer that transforms heterogeneous, continuous brain si
gnals into standardized, discrete tokens. This enables direct, token-level inter
actions for mutual understanding and generation between any two or more modaliti
es within a shared semantic space. To unlock advanced reasoning capabilities, we
 further curate a specialized Brain Question Answering (BQA) instruction-tuning
dataset. Our model not only establishes a new state-of-the-art among multi-task
unified frameworks but also provides strong evidence for multi-task synergy. By

demonstrating performance competitive with, and at times superior to, larger specialized models, our work offers a powerful new paradigm for neural modeling and paves the way for foundation models of neural activity.
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On Contraction of Sequential and Offset Rademacher Complexities
Adam Block, Alexander Rakhlin, Mark Sellke
The Rademacher complexity of a function class is among the most basic notions of its ``size'' and yields classical offline generalization bounds for Lipschitz loss functions that lead in turn to a modern understanding of statistical learning. More recently, the *sequential* and *offset* Rademacher complexities were introduced to prove analogous generalization bounds for online learning and for prediction with squared loss. A fundamental structural result in the theory of Rademacher complexity, with many applications to learning theory, is the Ledoux--Talagrand contraction lemma, which states that the Rademacher complexity of a composition of a function class with a fixed Lipschitz function is at most that of the original class. We show that, under structural assumptions on the function class, this contraction extends to sequential and offset Rademacher complexity at the price of polylogarithmic factors. We further show that these logarithmic factors cannot be removed in general and, absent these additional structural assumptions, no such contraction inequality can hold. These results together indicate that the sequential and offset Rademacher complexities behave fundamentally differently from the classical Rademacher complexity with respect to contraction, which in turn has broad implications for understanding the sample complexities of online learning and regression with squared loss for composed function classes.
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MAS-Orchestra: Understanding and Improving Multi-Agent Reasoning Through Holistic Orchestration and Controlled Benchmarks
Zixuan Ke, Yifei Ming, Austin Xu, Ryan Chin, Xuan-Phi Nguyen, Prathyusha Jwalapuram, Jiayu Wang, Semih Yavuz, Caiming Xiong, Shafiq Joty
While multi-agent systems (MAS) promise elevated intelligence through coordination of agents, current approaches to automatic MAS design under-deliver. Such shortcomings stem from two key factors: (1) methodological complexity – agent orchestration is performed using sequential, code-level execution that limits global system-level holistic reasoning and scales poorly with agent complexity – and (2) efficacy uncertainty – MAS are deployed without understanding if there are tangible benefits compared to single-agent systems (SAS). We propose MASOrchestra, a training-time framework that formulates MAS orchestration as a function-calling reinforcement learning problem with holistic orchestration, generating an entire MAS at once. In MAS-Orchestra, complex, goal-oriented subagents are abstracted as callable functions, enabling global reasoning over system structure while hiding internal execution details. To rigorously study when and why MAS are beneficial, we introduce MASBENCH, a controlled benchmark that characterizes tasks along five axes: Depth, Horizon, Breadth, Parallel, and Robustness. Our analysis reveals that MAS gains depend critically on task structure, verification protocols, and the capabilities of both orchestrator and subagents, rather than holding universally. Guided by these insights, MAS-Orchestra achieves consistent improvements on public benchmarks including mathematical reasoning, multi-hop QA, and search-based QA, while achieving more than 10× efficiency over strong baselines. Together, MAS-Orchestra and MASBENCH enable better training and understanding of MAS in the pursuit of multi-agent intelligence.
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Reasoning Models Struggle to Control their Chains of Thought
Chen Yueh-Han, Robert McCarthy, Bruce W. Lee, He He, Micah Carroll, Tomek Korbak
Instruction following in LLMs captures models' ability to change their visible behaviors as requested by users. Instead, we study models' ability to control their chain-of-thought (CoT). This capability -- CoT controllability -- is undesirable because it could allow models to suppress signs of misbehavior in their CoT, thereby undermining our ability to monitor them. To measure this, we introduce the \emph{CoT-Control} evaluation suite. We show that reasoning models are less

able to follow instructions in their CoT than in their outputs: on instructions like reasoning about a genetics problem without mentioning the word ``chromosome", Claude-Sonnet-4.5 complies only 5\% of the time. We also find that CoT controllability is higher for larger models and decreases with more RL training, test-time compute, and increased problem difficulty. CoT controllability failures extend even to situations in which models are given incentives (as opposed to direct requests) to evade CoT monitors, although models that are told they're being monitored exhibit slightly higher controllability. Similarly, eliciting controllability by adversarially optimizing prompts doesn’t meaningfully increase controllability.
Our results leave us cautiously optimistic: reasoning models generally seem characterized by low CoT controllability. However, the mechanism behind this phenomenon is not well understood. Given its importance for maintaining CoT monitorability, we recommend that frontier labs keep tracking controllability for future models.
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QiMeng-ChipV-RTL: Exploiting Information Locality for IP-level Verilog Generation
Hanqi Lyu, Di Huang, Yaoyu Zhu, Kangcheng Liu, Bohan Dou, Chongxiao Li, Pengwei Jin, Shuyao Cheng, Rui Zhang, Zidong Du, Qi Guo, Xing Hu, Yunji Chen
The generation of Register-Transfer Level (RTL) code is a crucial yet labor-intensive step in digital hardware design, traditionally requiring engineers to manually translate complex specifications into thousands of lines of synthesizable Hardware Description Language (HDL) code. While Large Language Models (LLMs) have shown promise in automating this process, existing approaches—including fine-tuned domain-specific models and advanced agent-based systems—struggle to scale to industrial IP-level design tasks. We identify three key challenges: (1) handling long, highly detailed documents, where critical interface constraints become buried in unrelated submodule descriptions; (2) generating long RTL code, where both syntactic and semantic correctness degrade sharply with increasing output length;
and (3) navigating the complex debugging cycles required for functional verification through simulation and waveform analysis.
To overcome these challenges, we propose \textit{ChipV-RTL}, a multi-agent framework that leverages \textit{information locality} in modular hardware design. ChipV-RTL decomposes the long-document to long-code generation problem into a set of short-document, short-code tasks, enabling scalable generation and debugging.

Specifically, ChipV-RTL integrates hierarchical document partitioning, task planning, localized code generation, interface-consistent merging, and AST-guided locality-aware debugging.
Experiments on \textsc{RealBench}, an IP-level Verilog generation benchmark, demonstrate that ChipV-RTL substantially outperforms state-of-the-art (SOTA) LLMs and agents, achieving a pass rate of 45.0\% compared to 21.6\%. Code, project page are available at: \url{https://iprc-dip.github.io/ChipV-RTL/}.
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Native Active Perception as Reasoning for Omni-Modal Understanding
Zhenghao Xing, Ruiyang Xu, Yuxuan Wang, Jinzheng He, Ziyang Ma, Qize Yang, Yunfei Chu, Jin Xu, Junyang Lin, Chi-Wing Fu, Pheng-Ann Heng
Passive models for long video understanding typically rely on a "watch-it-all" paradigm, processing frames uniformly regardless of query difficulty, causing computational cost to grow with video duration. Although interactive frameworks have emerged, they often rely on global pre-scanning, and their context cost still scales with video length. We propose **OmniAgent**, the first native omni-modal agent that formulates video understanding as a POMDP-based iterative **Observation-Thought-Action** cycle. OmniAgent executes on-demand actions to selectively distill audio-visual cues into a persistent textual memory, effectively decoupling reasoning complexity from raw video duration. To operationalize this, we introduce (1) **Agentic Supervised Fine-Tuning** to bootstrap native active perception via best-of-N trajectory synthesis with dual-stage quality control, and (2) **

Agentic Reinforcement Learning** with **TAURA** (Turn-aware Adaptive Uncertainty Rescaled Advantage), which leverages turn-level entropy to steer credit assignment toward pivotal discovery turns. Crucially, OmniAgent exhibits positive test-time scaling, where performance improves as the number of reasoning turns increases, validating the efficacy of active perception. Empirical results across ten benchmarks (e.g., VideoMME, LVBench) demonstrate that OmniAgent achieves state-of-the-art performance among open-source models. Notably, on LVBench, our 7B agent outperforms the $10\times$ larger Qwen2.5-VL-72B (50.5% vs. 47.3%). We release our code and model at https://github.com/HarryHsing/OmniAgent.
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Capability Traps in DPO
Marco Pollanen
Direct Preference Optimization (DPO) is often tuned by selecting checkpoints with high preference margins. We show that this selection rule can fail. Across dense β sweeps in three 7B open-weight families under controlled DPO recipes, the DPO preference margin can strongly anticorrelate with capability probes, with the strongest observed case in Llama logic probes (Pearson $r=-0.91$, $n=13$). In Mistral, matched-duration controls show that transient exposure to higher β produces persistent arithmetic ($d_z = 3.73$, $p = 0.002$) and format ($d_z = 3.38$, $p = 0.003$) degradation relative to a duration-matched constant-β run, while logic and sycophancy show no significant effect. An expanded 14-probe logic suite further shows that aggregate capability scores can hide stable opposition between probe clusters: direct-inference probes are consistently negative while fallacy-detection probes are consistently positive, with 13 of 14 probes sign-stable across 3 seeds and 3 β values. Sensitivity profiles differ across architectures: Mistral shows elevated seed variance near $\beta \approx 10^{-2}$, Llama shows capability-specific rigidity, and Qwen trades off smoothly. These findings motivate probe-resolved β sweeps rather than margin-based checkpoint selection.
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GLARE: Scalable Neuro-Symbolic Reward Shaping for LLM Agents via Group-Level Automata
Jingyuan Yan, Qingchen Liu, Qichao Ma, Jiahu Qin
Reinforcement Learning (RL) with Group Relative Policy Optimization (GRPO) shows great promise for enhancing LLM reasoning, but remains challenged by sparse and unstable rewards in long-horizon tasks. Existing approaches to reward shaping struggle to balance semantic expressiveness, reliability, and computational efficiency: heuristic rules lack flexibility, while LLM-as-a-Judge incurs high computational cost and suffer from inconsistent and misaligned scoring signals in long-context settings. To address these challenges, we introduce GLARE, a neuro-symbolic reward framework that decouples semantic abstraction from credit assignment. Specifically, to leverage semantic understanding while preserving symbolic determinism, we first extract and symbolize trajectory events into a discrete representation. These events are then translated into Linear Temporal Logic (LTL) formulas, which are compiled into deterministic automata that track the agent's progress via state transitions. This mechanism yields dense and consistent reward signals, avoiding unstable direct scoring while significantly reducing computational cost. Empirical results on ALFWorld show that GLARE outperforms GRPO by 12.1\% in success rate, while achieving an 8.1\% improvement over conventional LLM-based judges using only 15\% of their computational cost.
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HDFlow: Hierarchical Diffusion-Flow Planning for Long-horizon Tasks
Gireesh Nandiraju, Yuanliang Ju, Chaoyi Xu, Weiheng Liu, Yuxuan Wan, He Wang
Recent advances in generative models have shown promise in generating behavior plans for long-horizon, sparse reward tasks. While these approaches have achieved promising results, they often lack a principled framework for hierarchical decomposition and struggle with the computational demands of real-time execution, due to their iterative denoising process. In this work, we introduce $\textbf{Hierarchical Diffusion-Flow}$ ($\texttt{\textbf{HDFlow}}$), a novel hierarchical planning framework that optimally leverages the strengths of $\textit{diffusion}$ a

nd $\textit{rectified flow}$ models to overcome the limitations of single-paradigm generative planners. $\texttt{\textbf{HDFlow}}$ employs a high-level diffusion planner to generate sequences of strategic subgoals in a learned latent space, capitalizing on diffusion's powerful exploratory capabilities. These subgoals then guide a low-level rectified flow planner that generates smooth and dense trajectories, exploiting the speed and efficiency of ordinary differential equation (ODE)-based trajectory generation. We evaluate $\texttt{\textbf{HDFlow}}$ on four challenging furniture assembly tasks in both simulation and real-world, where it significantly outperforms state-of-the-art methods. Furthermore, we also showcase our method's generalizability on two long-horizon benchmarks comprising diverse locomotion and manipulation tasks. Project website: https://hdflow-page.github.io/

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Extracting alignment data in open models

Federico Barbero, Xiangming Gu, Christopher A. Choquette-Choo, Chawin Sitawarin, Matthew Jagielski, Itay Yona, Petar Veli■kovi■, Ilia Shumailov, Jamie Hayes

In this work, we show that it is possible to extract significant amounts of alignment training data from a post-trained model -- useful to steer the model to improve certain capabilities such as long-context reasoning, safety, instruction following, and maths.

While the majority of related work on memorisation has focused on measuring success of training data extraction through string matching, we argue that embedding models are better suited for our specific goals.

Distances measured through a high quality embedding model can identify semantic similarities between strings that a different metric such as edit distance will struggle to capture.

In fact, in our investigation, approximate string matching would have severely undercounted (by a conservative estimate of $10\times$) the amount of data that can be extracted due to trivial artifacts that deflate the metric. Interestingly, we find that models readily regurgitate training data that was used in post-training phases such as SFT or RL. We show that this data can be then used to train a base model, recovering a meaningful amount of the original performance.

We believe our work exposes a possibly overlooked risk towards extracting alignment data. Finally, our work opens up an interesting discussion on the downstream effects of distillation practices: since models seem to be regurgitating aspects of their training set, distillation can therefore be thought of as indirectly training on the model's original dataset.

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Fix the Loss, Not the Radius: Rethinking the Adversarial Perturbation of Sharpness-Aware Minimization

Jinping Wang, Qinhan Liu, Zhiwu Xie, Zhiqiang Gao

Sharpness-Aware Minimization (SAM) improves generalization by minimizing the worst-case loss within a fixed parameter-space radius neighborhood. SAM and its variants mainly rely on a first-order linearized surrogate, while flat minima are inherently a second-order (curvature) notion.

We revisit this mismatch and propose Loss-Equated SAM (LE-SAM), which inverts the traditional SAM mechanism that replaces the fixed perturbation radius with a fixed loss-space budget, effectively removing gradient-norm–dominated learning signals and shifting optimization toward curvature-dominated terms.

Extensive experiments across diverse benchmarks and tasks demonstrate the strong generalization ability of LE-SAM that consistently outperforms SAM and even its variants, achieving state-of-the-art performance.

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Preference-Calibrated Optimization with Score-Level Distribution Alignment for Text-to-Image Diffusion Model Unlearning

Xiuyuan Wang, Weiming Liu, Hongyu Cai, Xin Gao, Fan Wang, Chaochao Chen, Xiaolin Zheng

While text-to-image diffusion models achieve remarkable generation quality, they inadvertently memorize sensitive content, necessitating machine unlearning to prevent undesired outputs. However, existing unlearning methods rely on suboptima

l surrogate objectives rather than directly optimizing the unlearning goal, lead ing to fundamental objective mismatch. Moreover, these methods preserve model ut ility via surface-level constraints on model parameters or outputs, yet fail to capture the intrinsic generative dynamics of diffusion models, consequently trig gering catastrophic forgetting. To address these challenges, we propose Preferen ce-calibrated Optimization with Score-level Distribution Alignment (POSDA), a un ified unlearning framework that harmonizes effective erasure with fine-grained s tructural preservation. Specifically, we reframe unlearning as a preference opti mization problem by constructing a reward that explicitly quantifies the unlearn ing objective. Additionally, we introduce score-level distribution alignment to ensure the invariance of the underlying manifold topology of the unlearned model , thereby preventing distributional drift. Extensive experiments across object, style, and NSFW unlearning tasks demonstrate that POSDA achieves state-of-the-ar t erasure efficacy while maintaining superior model utility compared to existing methods.

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One-Step Residual Shifting Diffusion for Image Super-Resolution via Distillation

Daniil Selikhanovych, David Li, Aleksei Leonov, Nikita Gushchin, Sergei Kushneri uk, Alexander Filippov, Evgeny Burnaev, Iaroslav Sergeevich Koshelev, Alexander Korotin

Diffusion models for super-resolution (SR) produce high-quality visual results b ut require expensive computational costs. Despite the development of several met hods to accelerate diffusion-based SR models, some (e.g., SinSR) fail to produce realistic perceptual details, while others (e.g., OSEDiff) may hallucinate non- existent structures. To overcome these issues, we present **RSD**, a new distill ation method for ResShift. Our method is based on training the student network t o produce images such that a new fake ResShift model trained on them will coinci de with the teacher model. RSD achieves single-step restoration and outperforms the teacher by a noticeable margin in various perceptual metrics (LPIPS, CLIPIQA , MUSIQ). We show that our distillation method can surpass SinSR, the other dist illation-based method for ResShift, making it on par with state-of-the-art diffu sion SR distillation methods with limited computational costs in terms of percep tual quality. Compared to SR methods based on pre-trained text-to-image models, RSD produces competitive perceptual quality and requires fewer parameters, GPU m emory, and training cost. We provide experimental results on various real-world and synthetic datasets, including RealSR, RealSet65, DRealSR, ImageNet, and DIV2 K. We provide the code at https://github.com/Daniil-Selikhanovych/RSD.

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CARE: Confounder-Aware Aggregation for Reliable LLM Evaluation

Jitian Zhao, Changho Shin, Tzu-Heng Huang, Satya Sai Srinath Namburi GNVV, Frede ric Sala

LLM-as-a-judge ensembles are the standard paradigm for scalable evaluation, but their aggregation mechanisms suffer from a fundamental flaw: they implicitly ass ume that judges provide independent estimates of true quality. However, in pract ice, LLM judges exhibit correlated errors caused by shared latent confounders--- such as verbosity, stylistic preferences, or training artifacts---causing standa rd aggregation rules like majority vote or averaging to provide little gain or e ven amplify systematic mistakes. To address this, we introduce CARE, a confounde r-aware aggregation framework that explicitly models LLM judge scores as arising from both a latent true-quality signal and shared confounding factors. Rather t han heuristically re-weighting judges, CARE separates quality from confounders w ithout access to ground-truth labels. We provide theoretical guarantees for iden tifiability and finite-sample recovery under shared confounders, and we quantify the systematic bias incurred when aggregation models omit confounding latent fa ctors. Across 12 public benchmarks spanning continuous scoring, binary classific ation, and pairwise preference settings, CARE improves aggregation accuracy, red ucing error by up to 26.8%.

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Beyond Reactivity: Proactive Adaptive Conformal Inference for Online LLM Factual ity

Xinyu Liu, Jun Wu
Large Language Models (LLMs) often produce hallucinated outputs, which limit their reliability in high-stakes applications. Conformal prediction can provide guarantees on the correctness and factuality of LLM outputs, but existing approaches rely on the exchangeability assumption, which rarely holds in online settings where user queries and interests change over time. To solve this problem, in this paper, we propose **PACE** (**P**roactive **A**daptive **C**onformal Inferenc**E**), a novel framework that sequentially updates the time-varying target miscoverage parameter with a dynamic step size to maintain valid coverage under online distribution shifts. PACE is motivated by the theoretical connections between expected miscoverage error and key factors such as distribution shifts and instantaneous parameter error.
It integrates two complementary signals: (1) a proactive shift detection to estimate the magnitude of distribution shifts, and (2) a reactive error that scales updates according to the local coverage gap.
Extensive experiments on synthetic and real-world datasets demonstrate that PACE consistently outperforms advanced adaptive baselines. It reduces the deviation from the target error rate by up to 60\% in QA tasks and accelerates coverage recovery by over 2.5x during abrupt shifts, ensuring stable factuality guarantees without compromising utility and stability.
****
SG2Loc: Sequential Visual Localization on 3D Scene Graphs
Nicole Damblon, Olga Vysotska, Federico Tombari, Marc Pollefeys, Daniel Barath
Visual localization in complex indoor environments remains a critical challenge for robotics and AR applications. Sequential localization, where pose estimates are refined over time, is important for autonomous agents. However, traditional methods often require storing extensive image databases or point clouds, leading to significant overhead. This paper introduces a novel, lightweight approach to sequential visual localization using 3D scene graphs. Our method represents the environment with a compact scene graph, where nodes represent objects (with coarse meshes) and edges encode spatial relationships. For each image in the localization phase, we extract per-patch semantic features, predicting object identities. Localization is performed within a particle filter framework. Each particle, representing a camera pose, projects the coarse object meshes from the scene graph into the image, assigning object identities to patches based on visibility. The similarity of the per-patch features, in the input image, and object features from the scene graph determines the weight of a particle. Subsequent images are incorporated sequentially, refining the pose estimate. By leveraging a compact scene graph and efficient semantic matching, our method significantly reduces storage while maintaining performance on real-world datasets. The code is available at https://github.com/DmblnNicole/sg2loc.
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Decoupling Reasoning and Confidence: Resurrecting Calibration in Reinforcement Learning from Verifiable Rewards
Zhengzhao Ma, Xueru Wen, Boxi Cao, Yaojie Lu, Hongyu Lin, Jinglin Yang, Min He, Xianpei Han, Le Sun
Reinforcement Learning from Verifiable Rewards (RLVR) significantly enhances large language models (LLMs) reasoning but severely suffers from calibration degeneration, where models become excessively over-confident in incorrect answers.
Previous studies devote to directly incorporating calibration objective into existing optimization target.
However, our theoretical analysis demonstrates that there exists a fundamental gradient conflict between the optimization for maximizing policy accuracy and minimizing calibration error.
Building on this insight, we propose DCPO, a simple yet effective framework that systematically decouples reasoning and calibration objectives.
Extensive experiments demonstrate that

our DCPO not only preserves accuracy on par with GRPO but also achieves the best calibration performance and substantially mitigates the over-confidence issue. Our study provides valuable insights and practical solution for more reliable LLM deployment.
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TopBench: A Benchmark for Implicit Predictive Reasoning in Tabular Question Answering
An-Yang Ji, Jun-Peng Jiang, De-Chuan Zhan, Han-Jia Ye
Large Language Models (LLMs) have advanced Table Question Answering, where most queries can be answered by extracting information or simple aggregation. However, a common class of real-world queries is implicitly predictive, requiring the inference of unobserved answers from historical patterns rather than mere retrieval. These queries introduce two challenges: recognizing latent intent and reliable predictive reasoning over massive tables. To assess LLMs in such Tabular question answering with implicit Prediction tasks, we introduce TopBench, a benchmark consisting of 779 samples across four sub-tasks, ranging from single-point prediction to decision making, treatment effect analysis, and complex filtering, requiring models to generate outputs spanning reasoning text and structured tables. We evaluate diverse models under both text-based and agentic workflows. Experiments reveal that current models often struggle with intent recognition, defaulting to just lookups. Deeper analysis identifies that accurate intent disambiguation serves as the prerequisite for leading these predictive behaviors. Furthermore, elevating the upper bound of prediction precision requires the integration of more sophisticated modeling or reasoning capabilities.
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Spike-HTR: Spiking Neural Transformer for Handwritten Text Recognition
Xiubo Liang, Jinxing Han, Yuke Li, Haoqi Zhu, Yu Zhao, Hongzhi Wang
Handwritten Text Recognition (HTR) is computationally imbalanced in two ways: most image pixels are background, and many width-axis sequence positions are blank-dominated.
This creates a mismatch for Spiking Neural Networks (SNNs): handwriting is observed as a static image, whereas spiking computation unfolds over timesteps.
We propose Spike-HTR, a hybrid spiking recognizer that controls both the number of spiking steps and the number of width positions processed by the deep sequence mixer.
To make a static image suitable for short-horizon spiking inference, InkCoder converts it into a coarse-to-fine input stream, where early steps cover broad stroke regions and later steps emphasize sharper stroke details.
To reduce sequence computation, a CTC-guided length reducer keeps likely character or uncertain positions and compresses long blank-dominated stretches before deep mixing.
With $T{=}2$, Spike-HTR trains only on target data, decodes without language models or lexicons, and reaches validation/test CERs of 3.5/5.4, 2.3/2.5, and 4.2/3.9 on IAM, LAM, and READ2016. Codes are available at https://github.com/QomolangmaH/SpikeHTR.
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Measuring Meta-Cultural Competency: A Spectral Framework for LLM Knowledge Structures
Sougata Saha, Madhur Jindal, Saurabh Kumar Pandey, Mahardika Krisna Ihsani, Alham Fikri Aji, Monojit Choudhury
Most existing cultural evaluation frameworks for large language models (LLMs) focus on matching model outputs to ground-truth answers, primarily measuring factual cultural awareness. This overlooks whether models internalize broader cultural structure and pluralism. We introduce a spectral-analysis-based framework that captures large-scale macrostructural patterns in models' cultural knowledge and evaluate eight LLMs across nine cultural domains spanning all five of Newmark's cultural dimensions and 170 countries. Comparing with human data, we find that instruction-tuned models align more closely with human cultural structure than older models, while increased model size does not consistently improve performance. Finally, simulation-based experiments show that our proposed spectral metric

better predicts a model's ability to serve users from unfamiliar cultural backgr
ounds than existing ones.
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Pressure Reveals Character: Behavioural Alignment Evaluation at Depth
Nora Petrova, John Burden
Evaluating alignment in language models requires testing how they behave under r
ealistic pressure, not just what they claim they would do. While alignment failu
res increasingly cause real-world harm, comprehensive evaluation frameworks with
 realistic multi-turn scenarios remain lacking. We introduce an alignment benchm
ark spanning 904 scenarios across six categories---Honesty, Safety, Non-Manipula
tion, Robustness, Corrigibility, and Scheming---validated as realistic by human
raters. Our scenarios place models under conflicting instructions, simulated too
l access, and multi-turn escalation to reveal behavioral tendencies that single-
turn evaluations miss. Evaluating 24 frontier models using LLM judges validated
against human annotations, we find that even top-performing models exhibit gaps
in specific categories, while the majority of models show consistent weaknesses
across the board. Factor analysis reveals that alignment behaves as a unified co
nstruct (analogous to the g-factor in cognitive research) with models scoring hi
gh on one category tending to score high on others. We publicly release the benc
hmark and an interactive leaderboard to support ongoing evaluation, with plans t
o expand scenarios in areas where we observe persistent weaknesses and to add ne
w models as they are released.
****

FOCUS & RePAIR: Mitigating Text Degeneration via Token-Level Guidance For Pruned
 Large Language Models
Junyoung Lee, Sehyeon Park, Shinhyoung Jang, Seonha Ryu, Hojeong Kim, Hyunsei Le
e, Il hong Suh, Yeseong Kim
Pruning is a practical approach to compress large language models (LLMs), but it
 can amplify text degeneration, especially repetition loops, even when perplexit
y and task accuracy remain largely unchanged. In this work, we present a token-l
evel analysis of this failure mode by viewing decoding as a dynamical process th
at enters and persists in a small set of recurrent contexts. Our analysis decomp
oses degeneration into loop entry risk and loop persistence, and shows that pers
istence is controlled by the escape mass assigned to plausible alternatives with
in the token sampling set.
Motivated by these findings, we propose two token-level guidance objectives for
post-pruning fine-tuning. FOCUS reweights distillation toward high-confidence te
acher regions to suppress leakage, while RePAIR uses onset-centered positive/neg
ative continuation pairs with a margin loss to promote plausible alternatives an
d prevent early commitment to repetition loops. Experiments on open-ended contin
uation and instruction-based generation show that both methods consistently redu
ce repetition and improve generation quality.
****

Alternating Reinforcement Learning for Rubric-Based Reward Modeling in Non-Verif
iable LLM Post-Training
Ran Xu, Tianci Liu, Zihan Dong, Tony Yu, Ilgee Hong, Carl Yang, Linjun Zhang, Tu
o Zhao, Haoyu Wang
Standard reward models typically predict scalar scores that fail to capture the
multifaceted nature of response quality in non-verifiable domains, such as creat
ive writing or open-ended instruction following. To address this limitation, we
propose Rubric-ARM, a framework that jointly optimizes a rubric generator and a
judge using reinforcement learning from preference feedback. Unlike existing met
hods that rely on static rubrics or disjoint training pipelines, our approach tr
eats rubric generation as a latent action learned to maximize judgment accuracy.
 We introduce an alternating optimization strategy to mitigate the non-stationar
ity of simultaneous updates, providing theoretical analysis that demonstrates ho
w this schedule reduces gradient variance during training. Extensive experiments
 show that Rubric-ARM achieves state-of-the-art performance among baselines on m
ultiple benchmarks and significantly improves downstream policy alignment in bot
h offline and online reinforcement learning settings.

****
Sem-Detect: Semantic Level Detection of AI Generated Peer-Reviews
Andre Vicente Duarte, Brian Tufts, Aditya Oke, Fei Fang, Arlindo L. Oliveira, Le
i Li
*How can we distinguish whether a peer review was written by a human or generate
d by an AI model?* We argue that, in this setting, authorship should not be attr
ibuted solely from the textual features of a review, but also from the ideas, ju
dgments, and claims it expresses. To this end, we propose Sem-Detect, an authors
hip detection method for peer reviews that operationalizes this principle by com
bining textual features with claim-level semantic analysis. Sem-Detect compares
a target review against multiple AI-generated reviews of the same paper, leverag
ing the observation that different AI models tend to converge on similar points,
 while human reviewers introduce more unique and diverse ones. As a result, Sem-
Detect is able to distinguish fully AI-generated reviews from authentic human-wr
itten ones, including those that have been refined using an LLM but still reflec
t human judgment. Across a dataset of over 20,000 peer reviews from ICLR and Neu
rIPS conferences, Sem-Detect improves over the strongest prior detector by 36.5%
 in TPR@1% FPR in the binary setting. More importantly, in the three-class scena
rio, we empirically show that LLM refinement preserves the semantic signals of h
uman reviews, which remain distinct from the patterns exhibited by fully AI-gene
rated text; as a result, fewer than 3.5% of LLM-refined human reviews are miscla
ssified as AI-generated.
****
Fleet: Few Shots Lead Effective AI-generated Image Detection
Jiaan Wang, Sirui Liu, Yu Li, Kaiyuan Yang, Juan Cao, Sheng Tang
AI-generated image (AIGI) detection is undergoing a critical transition from lab
oratory benchmarks to open-world adversarial defense.
The prevalent paradigm focuses on finding static feature spaces, assuming that s
ome invariant artifacts learned from historical data can achieve universal zero-
shot generalization.
While achieving saturation on several AIGI benchmarks, this static hypothesis su
ffers a severe performance drop against rapidly evolving generators (e.g., SD3,
Nano Banana Pro).
To address these limitations, we propose that the field should expand beyond "st
atic generalization" to a new paradigm of "dynamic adaptation".
We introduce **Fleet**, a framework that pioneers a dynamic paradigm of continuo
us few-shot evolution, enabling rapid alignment with emerging generative threats
.
Fleet improves few-shot adaptation by replacing unconstrained feature updates wi
th constrained routing correction, where avoidance routing redirects novel AI sa
mples away from Non-AI-dominated routes within decoupled subspaces.
To validate this, we present **Treasure**, a benchmark spanning 64 models and 36
0k images, featuring diverse architectures and 20 closed-source commercial engin
es.
Experiments reveal that while static SOTA methods fail catastrophically on moder
n generators, Fleet restores performance from 20.4\% to 73.1\% with only 10-shot
 adaptation on "Doubao Seedream 4.0".
Code and data are available at https://github.com/ICTMCG/Fleet .
****
Attention Sinks as Internal Signals for Hallucination Detection in Large Languag
e Models
Jakub Binkowski, Kamil Adamczewski, Tomasz Jan Kajdanowicz
Large language models frequently exhibit hallucinations: fluent and confident ou
tputs that are factually incorrect or unsupported by the input context. While re
cent hallucination detection methods have explored various features derived from
 attention maps, the underlying mechanisms they exploit remain poorly understood
. In this work, we propose SinkProbe, a hallucination detection method grounded
in the observation that hallucinations are deeply entangled with attention sinks
 - tokens that accumulate disproportionate attention mass during generation - in
dicating a transition from distributed, input-grounded attention to compressed,

prior-dominated computation. Importantly, although sink scores are computed sole
ly from attention maps, we find that the classifier preferentially relies on sin
ks whose associated value vectors have large norms. Moreover, we show that previ
ous methods implicitly depend on attention sinks by establishing their mathemati
cal relationship to sink scores. Our findings yield a novel hallucination detect
ion method grounded in theory that produces state-of-the-art results across popu
lar datasets and LLMs.
****
Meta-Black-Box Optimization Can Do Search Guidance for Expensive Constrained Mul
ti-Objective Optimization
Yukun Du, Haiyue Yu, Jiang Jiang, Shuaiwen Tang, Xiaotong Xie, Haobo Liu, Chongs
huang Hu, Shengkun Chang
Existing Meta-Black-Box Optimization (MetaBBO) methods focus on how to search wh
en controlling optimizers, but largely overlook where to search. We propose Meta
SG-SAEA, a bi-level MetaBBO framework for expensive constrained multi-objective
optimization problems (ECMOPs), in which a meta-policy provides search guidance
to the low-level Surrogate-Assisted Evolutionary Algorithm (SAEA). To achieve th
is, we introduce Max–Min Constraint-Calibrated Inequality (MM-CCI), a compact, p
roblem-agnostic region abstraction that maps heterogeneous constraint evaluation
s to an ordered scalar level; we further provide a theoretical analysis of its f
undamental properties. Building on this region abstraction, we adopt diffusion-b
ased population initialization to translate the meta-policy’s region-level guida
nce into solution-level priors for the SAEA. To make MetaSG-SAEA scalable, we co
nstruct an attention-based state representation across varying problem dimension
s, population sizes, and numbers of objectives and constraints. Experimental res
ults demonstrate that MetaSG-SAEA outperforms state-of-the-art baselines across
diverse benchmarks and exhibits the ability to generalize across problem distrib
utions.
****
SVRG and Beyond via Posterior Correction
Nico Daheim, Thomas Möllenhoff, Ming Liang Ang, Mohammad Emtiyaz Khan
Stochastic Variance Reduced Gradient (SVRG) and its variants aim to speed-up tra
ining by using gradient corrections. Originally proposed over a decade ago, thes
e methods have never been connected to any Bayesian method at a fundamental leve
l. Here, we fill this gap and derive surprising new connections of SVRG to a rec
ently proposed Bayesian method called `posterior correction'. Our main contribut
ion is to show that SVRG can be recovered as a special case of posterior-correct
ion over isotropic-Gaussian posteriors. Novel extensions of SVRG are automatical
ly obtained by using more flexible exponential-family posteriors. We derive two
new such extensions by using Gaussian families: a Newton-like variant with novel
 Hessian corrections, and an Adam-like extension that scales to large problems.
Our work is the first to connect SVRG to Bayes and use it to speed-up training.
****
HeraSys: Collaborative Serving of Multiple LLM Workflows via Fine-Grained End-to
-End Optimization
Size Li, Zhiqing Tang, Hongrui Liang, Jianxiong Guo, Jiong Lou, Tian Wang, Weiji
a Jia
The proliferation of Large Language Models (LLMs) has shifted serving systems fr
om processing isolated requests to orchestrating high-concurrency, multi-tenant
agentic workflows. However, existing solutions typically prioritize intra-workfl
ow optimization, largely neglecting the significant potential for inter-workflow
 optimization. In this paper, we propose HeraSys, an LLM serving system designed
 to optimize the end-to-end performance of concurrent workflows. Through fine-gr
ained orchestration, HeraSys eliminates cross-workflow computational redundancy
via structural node merging and reuse. Furthermore, HeraSys introduces a load-aw
are joint scheduling policy that dynamically manages execution order by evaluati
ng both inter- and intra-query priorities. By integrating a resource skewing mec
hanism with adaptive batching and pipeline decomposition, HeraSys effectively mi
tigates tail latency while maintaining low average latency, thereby substantiall
y improving system throughput. Extensive experiments demonstrate that HeraSys re

duces P99 latency by up to 2.17$\times$ and increases serving throughput by up t
o 1.85$\times$ under strict latency guarantees.
****
Gradient Descent as a Perceptron Algorithm: Understanding Dynamics and Implicit
Acceleration
Alexander Tyurin
Even for the gradient descent (GD) method applied to neural network training, un
derstanding its optimization dynamics, including convergence rate, iterate traje
ctories, function value oscillations, and especially its implicit acceleration,
remains a challenging problem. We analyze nonlinear models with the logistic los
s and show that the steps of GD reduce to those of generalized perceptron algori
thms (Rosenblatt, 1958), providing a new perspective on the dynamics. This reduc
tion yields significantly simpler algorithmic steps, which we analyze using clas
sical linear algebra tools. Using these tools, we demonstrate on a minimalistic
example that the nonlinearity in a two-layer model can provably yield a faster i
teration complexity $\tilde{\mathcal{O}}(\sqrt{d})$ compared to $\Omega(d)$ achi
eved by linear models, where d is the number of features. This helps explain t
he optimization dynamics and the implicit acceleration phenomenon observed in ne
ural networks. The theoretical results are supported by extensive numerical expe
riments. We believe that this alternative view will further advance research on
the optimization of neural networks.
****
Rethinking Federated Prompt Learning for Medical Images: From Textual Tuning to
Visual Manifold Anchoring
Yipan Wei, Wenke Huang, Yapeng Li, He Li, Qixin Zhang, Mang Ye, Bo Du
Federated Prompt Learning (FPL) adapts Vision-Language Models to privacy-sensiti
ve medical imaging, typically via a textual tuning paradigm that assumes the fro
zen visual encoder provides a discriminative feature geometry. We argue this ass
umption breaks down in medical settings, leading to two geometric pathologies: (
1) Intra-client: Medical Manifold Collapse, where high morphological similarity
reduces the effective rank of visual features; and (2) Inter-client: Medical Top
ological Misalignment, where heterogeneous acquisition protocols induce inconsis
tent geometry across clients. To address these, we propose FedMAP, which shifts
the paradigm to Visual Manifold Anchoring. FedMAP utilizes an LLM-derived codebo
ok as a client-invariant synchronization signal to restructure the visual space,
 via Manifold Semantic Anchoring (MSA) and Topology Structural Alignment (TSA) t
o enforce consistent inter-class relations. Experiments on FedISIC, FedCamelyon1
7, and a private ultrasound dataset show that FedMAP consistently outperforms st
ate-of-the-art methods, especially in high-noise regimes where manifold collapse
 is most severe.
****
CalPro: Prior-Aware Evidential Conformal Prediction with Structure-Aware Sensiti
vity Bounds for Protein Structures
Ibne Farabi Shihab, Sanjeda Akter, Anuj Sharma
Deep protein structure predictors such as AlphaFold provide confidence estimates
 (e.g., pLDDT) that are not calibrated and degrade under distribution shifts acr
oss experimental modalities, temporal changes, and disordered regions.
We introduce \textbf{CalPro}, a prior-aware evidential conformal framework for s
hift-robust uncertainty quantification.
CalPro combines three components: (i) a geometric evidential head outputting Nor
mal Inverse Gamma distributions via graph neural networks; (ii) a differentiable
 calibration surrogate that shapes representations during training, followed by
split-conformal calibration for finite-sample coverage; and (iii) domain priors
(disorder, flexibility) encoded as soft constraints on predicted uncertainty.
Theoretically, we derive structure-aware \emph{sensitivity bounds} for coverage
degradation under distribution shift using PAC-Bayesian control over ambiguity s
ets, quantifying how miscoverage increases with model complexity and shift magni
tude.
Empirically, CalPro achieves at most 5 percentage points coverage degradation ac
ross modalities compared to 15 to 25 points for baselines, reduces calibration e

rror by 30\% to 50\%, and improves downstream docking success from 52\% to 75\% when filtering by uncertainty. The framework extends beyond proteins to structured regression tasks where priors encode local reliability.
****
Inner-layer Token Self-modulation as Another Scaling Axis for LLMs
Yebin Yang, Huaijin Wu, Jingtao Han, Yu Wang, Xiaohan Qin, Jingzhi Wang, Debing Zhang, Junchi Yan
LLMs have traditionally scaled along dense dimensions, where performance is coupled with near-linear increases in computational cost. While MoE decouples capacity from compute, it introduces large memory overhead and hardware efficiency challenges. To overcome these, we propose token-indexed parameters as a novel, orthogonal scaling axis that decouple model capacity from FLOPs. Specifically, we introduce ReToken and MoRT, which augment Transformer layers with modulation vectors retrieved from auxiliary embedding tables. These vectors modulate the backbone via lightweight, element-wise operations, incurring negligible FLOPs overhead.

Extensive experiments on both dense and MoE backbones, spanning from 190M to 9.8B parameters, demonstrate that our approach consistently reduces validation loss and significantly improves downstream task performance (e.g., +7.3 on ARC-C, +6.3 on GSM8K). Rigorous isoFLOPs analysis further confirms that MoRT fundamentally shifts the quality–compute Pareto frontier, achieving comparable model quality with 35\% less compute relative to vanilla MoE architectures, and we validate that token-indexed parameters exhibit a predictable power-law scaling behavior. Moreover, our efficient implementation ensures that the overhead introduced by ReToken and MoRT remains marginal.
****
Multi-Integration of Labels Across Categories for Component Identification in Multi-trial Time Series
Noga Mudrik, Yuxi Chen, Gal Mishne, Adam Shabti Charles
Many fields collect large-scale temporal data through repeated measurements (`trials’), where each trial is labeled with a set of metadata variables spanning several categories. For example, a trial in a neuroscience study may be linked to a value from category (a): task difficulty, and category (b): animal choice. A critical challenge in time-series analysis is to understand how these labels are encoded within the multi-trial observations, and disentangle the distinct effect of each label entry across categories. Here, we present MILCCI, a novel data-driven method that i) identifies the interpretable components underlying the data, ii) captures cross-trial variability, and iii) integrates label information to understand each category's representation within the data. MILCCI extends a sparse per-trial decomposition that leverages label similarities within each category to enable subtle, label-driven cross-trial adjustments in component compositions and to distinguish the contribution of each category. MILCCI also learns each component’s corresponding temporal trace, which evolves over time within each trial and varies flexibly across trials. We demonstrate MILCCI’s performance through both synthetic and real-world examples, including voting patterns, online page view trends, and neuronal recordings.
****
Lower Bounds for Frank-Wolfe on Strongly Convex Sets
Jannis Halbey, Daniel Deza, Max Zimmer, Christophe Roux, Bartolomeo Stellato, Sebastian Pokutta
We present a constructive lower bound of $\Omega(1/\sqrt{\varepsilon})$ for Frank-Wolfe (FW) when both the objective and the constraint set are smooth and strongly convex, showing that the known uniform $\mathcal{O}(1/\sqrt{\varepsilon})$ guarantees in this regime are tight. It is known that under additional assumptions on the position of the optimizer, FW can converge linearly. However, it remained unclear whether strong convexity of the set can yield rates \emph{uniformly} faster than $\mathcal{O}(1/\sqrt{\varepsilon})$, i.e., irrespective of the position of the optimizer. To investigate this question, we focus on a simple yet representative problem class: minimizing a strongly convex quadratic over the Euclidean unit ball, with the optimizer on the boundary. We analyze the dynamics of F

W for this problem in detail and develop a novel computational approach to const
ruct worst-case FW trajectories, which is of independent interest. Guided by the
se constructions, we develop an analytical proof establishing the lower bound.
****

AutoVSR: Automatic Visual-to-Symbolic Reasoning for Symbolic Expression Generati
on from Circuit Schematic

Zhe Xiao, Longfei Li, Xu He, Haoying Wu, Zixing Zhang, Mingyu Liu

Symbolic expressions can effectively characterize and predict circuit behavior,
but deriving them directly from circuit schematics is challenging. This process
requires accurate visual-to-symbolic construction of circuit structure from imag
es and correct multi-step symbolic derivation, both of which impose strict corre
ctness requirements.

This work proposes AutoVSR, an automated framework for visual-to-symbolic genera
tion of circuit expressions using Vision Language Models (VLMs). By reconstructi
ng circuit diagrams into an executable intermediate representation (Executable I
R) and leveraging a symbolic solver for reasoning, AutoVSR significantly improve
s the accuracy of symbolic expression generation.

AutoVSR introduces two key innovations: an IR construction method guided by comp
onent rule retrieval and verification-based feedback, and a symbolic solver impl
emented as a planning agent equipped with a symbolic tool library for reliable m
ulti-step derivation. Compared with end-to-end VLM approaches and specialized me
thods on the main symbolic expression generation task, AutoVSR achieves accuracy
 improvements of 30.01--59.45% and 41.96--51.84%, respectively. Moreover, AutoVS
R surpasses closed-source state-of-the-art VLMs in inference cost and computatio
nal efficiency. Code is available at https://github.com/LongfeiLi1/AutoVSR.
****

The Cost of Commitment in Option-Based Hierarchical RL

Randy Lefebvre, Audrey Durand

Empirically, option-based hierarchical reinforcement (HRL) learning often produc
es longer and more diverse options when a deliberation cost is charged at option
 boundaries. However, when options are executed for many steps under an approxim
ate dynamics model, small model errors compound along the option, degrading the
quality of the resulting plan. In this work, we introduce the commitment loss to
 formalize the tradeoff between deliberation cost and model error as a function
of option duration. We characterize how optimal termination probabilities vary w
ith this tradeoff under two model-error mechanisms. First, the model is learned
from finite data via maximum-likelihood estimation, producing statistical error
that interacts with option duration. Second, we consider an input-driven setting
 where an exogenous input is only observed at option boundaries and evolves unob
served between them, creating a drift-induced mismatch between planned and reali
zed dynamics. In both cases, we solve for the optimal termination behavior as a
function of deliberation cost and the error scale, clarifying the behavior of so
me popular HRL algorithms that approach the deliberation cost as a heuristic.
****

Sparse Bayesian Deep Functional Learning with Structured Region Selection

Xiaoxian Zhu, Yingmeng Li, Shuangge Ma, Mengyun Wu

In modern applications such as ECG monitoring, neuroimaging, wearable sensing, a
nd industrial equipment diagnostics, complex and continuously structured data ar
e ubiquitous, presenting both challenges and opportunities for functional data a
nalysis. However, existing methods face a critical trade-off: conventional funct
ional models are limited by linearity, whereas deep learning approaches lack int
erpretable region selection for sparse effects. To bridge these gaps, we propose
 a sparse Bayesian functional deep neural network (sBayFDNN). It learns adaptive
 functional embeddings through a deep Bayesian architecture to capture complex n
onlinear relationships, while a structured prior enables interpretable, region-w
ise selection of influential domains with quantified uncertainty. Theoretically,
 we establish rigorous approximation error bounds, posterior consistency, and re
gion selection consistency. These results provide the first theoretical guarante
es for a Bayesian deep functional model, ensuring its reliability and statistica
l rigor. Empirically, comprehensive simulations and real■world studies confirm t

he effectiveness and superiority of sBayFDNN. Crucially, sBayFDNN excels in reco
gnizing intricate dependencies for accurate predictions and more precisely ident
ifies functionally meaningful regions, capabilities fundamentally beyond existin
g approaches.
****
Language Generation in the Limit: Complexity Barriers and Implications for Learn
ing
Marcelo Arenas, Pablo Barcelo, Luis Cofré, Alexander Kozachinskiy
Kleinberg and Mullainathan showed that language generation in the limit is alway
s possible
at the level of computability: given enough positive examples, a learner can eve
ntually
generate data indistinguishable from a target language. However, such existence
results do
not address feasibility.
We study the sample complexity of language generation in the limit for several c
anonical
classes of formal languages. Our results show that infeasibility already appears
 for
context-free and regular languages, and persists even for strict subclasses such
 as locally threshold
testable languages, as well as for incomparable classes such as non-erasing patt
ern
languages, a well-studied class in the theory of language identification.
Overall, our results establish a clear gap between the theoretical possibility o
f
language generation in the limit and its computational feasibility.
****
Hierarchical Abstract Tree for Cross-Document Retrieval Augmented Generation
Ziwen Zhao, Menglin Yang
Retrieval-augmented generation (RAG) enhances large language models with externa
l knowledge, and tree-based RAG organizes documents into hierarchical indexes to
 support queries at multiple granularities. However, existing Tree-RAG methods d
esigned for single-document retrieval face critical challenges in scaling to cro
ss-document multi-hop questions: *(1) poor distribution adaptability*, where k
-means clustering introduces noise due to rigid distribution assumptions; *(2) s
tructural isolation*, as tree indexes lack explicit cross-document connections;
and *(3) coarse abstraction*, which obscures fine-grained details. To address th
ese limitations, we propose **Ψ-RAG**, a tree-RAG framework with two key co
mponents. *First*, a hierarchical abstract tree index built through an iterative
 "merging and collapse" process that adapts to data distributions without a prio
ri assumption. *Second*, a multi-granular retrieval agent that intelligently int
eracts with the knowledge base with reorganized queries and an agent-powered hyb
rid retriever. Ψ-RAG supports diverse tasks from token-level question answe
ring to document-level summarization. On cross-document multi-hop QA benchmarks,
 it outperforms RAPTOR by 25.9\% and HippoRAG 2 by 7.4\% in average F1 score.
****
GraphP-FL: Personalized Federated Graph Learning via Dynamic Structure Awareness
 and Fisher Information Elastic Alignment
Haoyu Chen, Zening Zhao, Jinsong Wang, Kai Shi, Zongpu Wei, Jianhao Li
Federated Graph Learning (FGL) enables distributed clients to collaboratively tr
ain graph neural networks while strictly preserving data privacy. However, exist
ing FGL methods implicitly assume the reliability of local graph structures and
lack elastic awareness of parameter importance during model aggregation, leading
 to representation degradation under topological noise and catastrophic forgetti
ng caused by model drift. To address these challenges, we propose GraphP-FL, a g
eneral personalized FGL framework. (1) Specifically, we design a self-supervised
 dynamic topology reconstruction mechanism on the client side. This mechanism mi
nes implicit dependencies to adaptively rectify noisy topologies, effectively su
ppressing topological noise propagation and capturing precise structural relatio

nships for high-quality representations. (2) Additionally, we introduce a Fisher-based Elastic Parameter Alignment (FRPA) algorithm. FRPA imposes anisotropic regularization constraints in the parameter space to precisely quantify parameter importance, enabling the model to strictly preserve critical local knowledge while flexibly aligning with the global model, thus effectively overcoming catastrophic forgetting. Extensive experiments on seven benchmarks (including biochemical molecules, social networks, and large-scale encrypted traffic) demonstrate that GraphP-FL significantly outperforms state-of-the-art methods, improving accuracy by up to 8.6% while exhibiting superior generalization and robustness.
****

SCalDA: Semantics-Calibrated and Diffusion-Enhanced Data Augmentation
Shibo Lv, Jianmin Jiang
With the rapid development of deep learning, the issue of data scarcity has become increasingly prominent, inspiring emerging interests towards research on data augmentation techniques over recent years. However, our literature survey indicates that existing efforts often suffer from two issues of semantic infidelity, including: (i) visual semantics infidelity, such as visual artifacts, manifold intrusion, and unnatural blending boundaries etc, and (ii) label semantic infidelity, where augmented images do not match the original labels, creating extra label noises. To address these issues, we propose a Semantics Calibrated and Diffusion-Enhanced Augmentation (SCalDA) scheme to achieve accurate semantics calibration across image, label and feature domains. Compared with the existing approaches, our proposed features in precise guidance in label domain, semantics driven synthesis across three domains (image, label and feature), and semantics-aware metric learning. Extensive experiments on multiple datasets demonstrate that SCalDA yields consistent and significant performance improvements for both fine-grained and general classification tasks, validating the effectiveness and broad applicability of the proposed.
****

MoLoRA: Composable Specialization via Per-Token Adapter Routing
Shrey Shah, Justin Wagle
Multi-adapter serving systems route entire sequences to a single adapter, forcing a choice when requests span multiple domains. This assumption fails in two important settings: (1) multimodal generation, where text and image tokens require different adapters within the same sequence, and (2) mixed-capability requests like ``write code to solve this equation,'' which need expertise from multiple specialized adapters. We introduce \emph{per-token routing}, which routes individual tokens to adapters based on either vocabulary structure (for multimodal models) or learned gating (for semantic specialization). Per-token routing is provably optimal for mixed-adapter requests: N work for N tokens, versus $K \cdot N$ for per-sequence systems that must replay one adapter per pass. Our key contribution is MoLoRA (Mixture of LoRA), which enables \emph{composable specialization}: load multiple domain-specific adapters and let a learned router select the appropriate adapter per-token. We demonstrate that specialization dramatically beats scale: MoLoRA enables Qwen3-1.7B to exceed Qwen3-8B across four reasoning benchmarks while being 4.7$\times$ smaller. This enables modular expertise at inference time: train focused LoRAs independently, combine them without retraining, and add new capabilities by simply loading new adapters.
****

Adversarial Flow Models
Shanchuan Lin, Ceyuan Yang, Zhijie Lin, Hao Chen, Haoqi Fan
We present adversarial flow models, a class of generative models that belongs to both the adversarial and flow families. Our method supports native one-step and multi-step generation and is trained with an adversarial objective. Unlike traditional GANs, in which the generator learns an arbitrary transport map between the noise and data distributions, our generator is encouraged to learn a deterministic noise-to-data mapping. This significantly stabilizes adversarial training. Unlike consistency-based methods, our model directly learns one-step or few-step generation without having to learn the intermediate timesteps of the probability flow for propagation. This preserves model capacity and avoids error accumula

tion. Under the same 1NFE setting on ImageNet-256px, our B/2 model approaches th
e performance of consistency-based XL/2 models, while our XL/2 model achieves a
new best FID of 2.38. We additionally demonstrate end-to-end training of 56-laye
r and 112-layer models without any intermediate supervision, achieving FIDs of 2
.08 and 1.94 with a single forward pass and surpassing the corresponding 28-laye
r 2NFE and 4NFE counterparts with equal compute and parameters.
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TFRBench: A Reasoning Benchmark for Evaluating Forecasting Systems
Md Atik Ahamed, Mihir Parmar, Palash Goyal, Yiwen Song, Long T. Le, Qiang Cheng,
 Chun-Liang Li, Hamid Palangi, Jinsung Yoon, Tomas Pfister
We introduce TFRBench, the first benchmark designed to evaluate the reasoning ca
pabilities of forecasting systems. Traditionally, time-series forecasting has be
en evaluated solely on numerical accuracy, treating foundation models as "black
boxes." Unlike existing benchmarks, TFRBench provides a protocol for evaluating
the reasoning generated by forecasting systems--specifically their analysis of c
ross-channel dependencies, trends, and external events. To enable this, we propo
se a systematic multi-agent framework that utilizes an iterative verification lo
op to synthesize numerically grounded reasoning traces. Spanning ten datasets ac
ross five domains, our evaluation confirms that this reasoning is causally effec
tive; useful for evaluation; and prompting LLMs with our generated traces signif
icantly improves forecasting accuracy compared to direct numerical prediction (e
.g., avg. ~40.2% $\rightarrow$ ~56.6%), validating the quality of our
reasoning. Conversely, benchmarking experiments reveal that off-the-shelf LLMs c
onsistently struggle with both reasoning (lower LLM-as-a-Judge scores) and numer
ical forecasting, frequently failing to capture domain-specific dynamics. TFRBen
ch thus establishes a new standard for interpretable, reasoning-based evaluation
 in time-series forecasting. Our benchmark is available at:
https://tfrbench.github.io
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Efficiently Training Time-to-First-Spike Spiking Neural Networks from Scratch
Kaiwei Che, Zhengyu Ma, Yifan Huang, Peng Xue, Li Yuan, Wei Fang, Yonghong Tian
Spiking Neural Networks (SNNs), with their event-driven and biologically inspire
d mechanisms, are well-suited for energy-efficient neuromorphic hardware. Neural
 coding, which is critical to SNNs, determines how information is represented vi
a spikes. While Time-to-First-Spike (TTFS) coding uses a single spike per neuron
 to offer extreme sparsity and energy efficiency, it often suffers from unstable
 training and low accuracy due to its sparse firing.
To address these challenges, we propose a training framework that incorporates p
arameter initialization, training normalization, a temporal output decoder, and
a re-evaluation of the pooling layer.
The proposed parameter initialization and training normalization mitigate signal
 diminishing and gradient vanishing, which helps stabilize training. Our output
decoder aggregates temporal spikes to encourage earlier firing, thereby reducing
 latency.
The re-evaluation of the pooling layer demonstrates that max-pooling violates si
ngle-spike constraints, which should be avoided, whereas average-pooling preserv
es them.
Experiments show that our framework stabilizes and accelerates training, reduces
 latency, and achieves state-of-the-art accuracy for step-by-step TTFS SNNs on M
NIST (99.48%), Fashion-MNIST (92.90%), CIFAR10 (90.56%), CIFAR100 ($70.
27\%$) and DVS Gesture ($95.83\%$).
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Partial Ring Scan: Revisiting Scan Order in Vision State Space Models
Yi-Kuan Hsieh, Kuan-Chuan Peng, Xin Li, Ming-Ching Chang, Yu-Chee Tseng, Jun Wei
 Hsieh
State Space Models (SSMs) provide linear-time alternatives to attention for visi
on, but require serializing 2D images into 1D sequences using a predefined scan
order. We identify scan order as a previously underexplored inductive bias that
fundamentally shapes spatial dependency modeling in Vision SSMs. Fixed scan path
s distort local adjacency, fragment object structure, and induce anisotropic rep

resentations that are brittle under geometric transformations such as rotation. We propose Partial RIng Scan Mamba (PRIS-Mamba), a rotation-robust traversal tha t decomposes images into concentric rings, performs permutation-invariant aggreg ation within each ring, and models cross-ring dependencies via short radial SSMs . This design induces a structured factorization of spatial dependencies that pr eserves isotropy while maintaining linear complexity. To improve efficiency with out sacrificing expressivity, we introduce partial channel filtering, selectivel y applying recurrent modeling to informative channels while routing others throu gh a residual pathway. Empirically, PRIS-Mamba improves accuracy, efficiency, an d rotation robustness over prior Vision SSMs on ImageNet-1K. Our results positio n scan-order design as a core representational choice in Vision SSMs, with impli cations for robustness and generalization beyond architectural scaling. The cod e will be released upon paper acceptance.
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Data Provenance Auditing of Fine-Tuned Large Language Models with a Text-Preserv ing Technique
Yanming Li, Cédric Eichler, Nicolas Anciaux, Alexandra Bensamoun, Lorena Gonzale z-Manzano, Seifeddine Ghozzi
We propose a system for marking sensitive or copyrighted texts to detect their u se in fine-tuning large language models under black-box access with statistical guarantees. Our method builds digital “marks” using invisible Unicode characters organized into (“cue”, “reply”) pairs.
During an audit, prompts containing only “cue” fragments are issued to trigger r egurgitation of the corresponding “reply”, indicating document usage. To control false positives, we compare against held-out counterfactual marks and apply a r anking test, yielding a verifiable bound on the false positive rate. Empirically , we obtain a true positive rate of 96.7\% at 0\% false positive rate and reply regurgitation rates exceeding 28\% per document with only 40 (4\%) watermarked d ocuments. The approach is minimally invasive, scalable across many sources, robu st to standard processing pipelines, and achieves high detection power even when marked data is a small fraction of the fine-tuning corpus.
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Is Fixing Schema Graphs Necessary? Full-Resolution Graph Structure Learning for Relational Deep Learning
Yi Huang, Qingyun Sun, Jia Li, Xingcheng Fu, Jianxin Li
Relational prediction tasks are fundamental in many real-world applications, whe re data are naturally stored in relational databases (RDBs). Relational Deep Lea rning (RDL) addresses this problem by modeling RDBs as graphs and applying graph neural networks (GNNs) for end-to-end learning. However, the full-resolution pr operty is commonly adopted as a design principle in graph construction for RDBs to preserve relational semantics, which leads most existing methods to rely on f ixed graph structures. In this paper, we propose **FROG**, a *F*ull-*R*esolution and *O*ptimizable *G*raph Structure Learning framework for RDL that formulates relational structure learning as a learnable table role modeling problem, allowi ng tables to contribute as nodes and edges in message passing. We further design role-driven message passing mechanisms to capture relational semantics, enablin g joint optimization of graph structure and GNN representations. To ensure seman tic consistency, we introduce functional dependency constraints that regularize representations across table and entity levels. Extensive experiments demonstrat e that our method outperforms existing approaches and reveal how table roles imp act downstream tasks, offering new insights into graph construction for RDL.
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Adversarial Latent Embedding Repair for LLM Continual Learning
Xilin Xia, Tong Xialiang, Jie Wang, Chi Ma, Shengxue Li, Yinqi Bai, Yuhang Jiang , Xing Li, Jianye HAO, Mingxuan Yuan, Feng Wu
Research on continual learning for LLMs seeks to acquire new skills without cata strophic forgetting of established prior knowledge.
However, domain-specific fine-tuning still triggers severe, long-tailed forgetti ng issues even under narrow updates, particularly when the pre-training data is inaccessible.

To tackle this challenge, we propose **ALER**, a data-free continual learning framework that adversarially searches for a small set of latent prompt embeddings to maximize logit divergence from a frozen reference model, proactively exposing high-risk forgetting modes at each step.
It then performs online distillation from the frozen reference using the discovered embeddings to retain prior behaviors while preserving target-domain adaptation.
We provide theoretical guarantees on the efficiency of our targeted repair, and extensive experiments demonstrate consistent improvements in the retention–adaptation frontier over representative baselines across 2 domain-specific fine-tuning datasets and 6 general-purpose benchmarks,
suggesting a more proactive approach for LLM continual learning.
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h1: Bootstrapping LLMs to Reason over Longer Horizons via Reinforcement Learning
Alesia Ivanova, Sumeet Ramesh Motwani, Ziyang Cai, Philip Torr, Riashat Islam, Shital Shah, Christian Schroeder de Witt, Charles London
Large language models excel at short-horizon reasoning tasks, but performance drops as reasoning horizon lengths increase. In this work, we introduce a scalable method to bootstrap long-horizon reasoning capabilities using only existing, abundant short-horizon data. Our approach synthetically composes simple problems into complex, multi-step dependency chains of arbitrary length. We train models on this data using outcome-only rewards under a curriculum that automatically increases in complexity, allowing RL training to be scaled much further without saturating. Empirically, our method generalizes remarkably well: curriculum training on composed 6th-grade level math problems improves accuracy on longer, competition-level benchmarks. It also transfers significantly to diverse out-of-distribution ReasoningGym domains and long-context benchmarks, indicating broader generalization. Importantly, our long-horizon improvements are significantly higher than baselines even at high pass@k, showing that models can learn new reasoning paths under RL. Theoretically, we show that curriculum RL with outcome rewards could achieve an exponential improvement in sample complexity over full-horizon training, providing training signal comparable to dense supervision. h1 therefore introduces an efficient path towards scaling RL for long-horizon problems using only existing data.
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Split Group Knockoffs: Controlling False Discovery Rate in Transformational Group Sparsity
Siqi Chen, Yachen Gao, Yanwei Fu, Xinwei Sun
Controlling the false discovery rate (FDR) under complex sparsity structures remains a fundamental challenge in large language model (LLM) analysis. Motivated by multiple comparison problems in LLMs, we consider a setting in which sparsity arises at the group level after a linear transformation of model parameters. We propose Split Group Knockoffs (SGKs), a general framework for group-wise variable selection under grouped transformational sparsity that extends the Split Knockoff procedure to grouped transformed variables. We establish theoretical guarantees for group-level FDR control and support recovery consistency, addressing challenges induced by group-wise penalties in transformed spaces. Applying SGK to LLM behavior auditing experiment reveals that model disagreement is not uniform across subjects, but instead concentrates in domains with greater semantic and reasoning complexity, where SGK effectively distinguishes genuine behavioral deviations from surface-level performance variation.
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pTNAS: Progressive Neural Architecture Search for Tabular Data
Naili Xing, Shaofeng Cai, Lingze Zeng, Jiaqi Zhu, Peng LU, Jian Pei, Beng Chin Ooi
Recent advances have shifted the paradigm of tabular learning toward tabular foundation models, yet their accuracy relies on a heavy inference cost that scales poorly with context size. Deep neural networks remain a highly competitive and more efficient modeling paradigm when equipped with well-designed architectures; however, identifying such architectures in a data-adaptive and budget-aware mann

er remains challenging. We propose pTNAS, the first progressive neural architect ure search (NAS) approach tailored for tabular data, which enables fast identifi cation of a viable architecture and continuously improves its search performance as more budget becomes available. pTNAS adopts a filter-and-refine optimization strategy that combines efficient training-free and effective training-based arc hitecture evaluation. In the filtering phase, we introduce pTProxy, a novel zero -cost proxy specifically designed for tabular networks that jointly captures arc hitectural trainability and expressivity, enabling fast filtering of large archi tecture search spaces. In the refinement phase, pTNAS employs a fixed-budget sch eduling algorithm to accurately identify the best-performing architecture from a small set of promising candidates. We further propose a budget-aware coordinato r to optimize budget allocation holistically. Experiments show that pTNAS reduce s the time to reach the globally best architecture by up to 82.75 X compared wit h other NAS approaches, achieves the best average predictive rank, and improves end-to-end efficiency by up to 4.78 X compared with TabPFN.
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Towards a Holistic Understanding of Selection Bias for Causal Effect Identificat ion
Yiwen Qiu, Filip Kova■evi■, Shimeng Huang, Peter Spirtes, Francesco Locatello
Selection bias is pervasive in observational studies. For example, large scale b iobanks data can exhibit ``healthy volunteer bias'' when respondents are healthi er and of higher socio-economic status than the population they are meant to rep resent. Recovering causal effects from such sub-population is an important probl em in causal inference, as estimating average treatment effects (ATE) from sele cted populations can result in a severely biased estimate of the ATE from the w hole population.

In this paper, we investigate the identifiability of the ATE under selection bia s. We provide *necessary and sufficient conditions* for ATE identifiability, lev eraging weak assumptions on probability classes to characterize propensity score and selection probability, which are weaker assumptions than those required by existing general identification frameworks. Compared to previous works, our resu lts extend existing graphical identifiability criteria and offer a more comprehe nsive understanding of causal effect identification *with strictly weaker condit ions* in the presence of selection bias.
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PPDL: LLM-Based Flows as Probabilistic Programs
Louis Mandel, Guillaume Baudart, Mandana Vaziri, Martin Hirzel
Building reliable applications that leverage large language models (LLMs) remain s a significant challenge. While LLMs offer impressive capabilities across diver se tasks, their outputs often lack accuracy and provide no clear measure of conf idence. This uncertainty compounds in flows of multiple calls to LLMs and other tools, making it difficult for developers and end-users to trust the results. Th is paper introduces a probabilistic language for programming LLM-based flows. It enables developers to quantify and propagate uncertainty throughout the applica tion's flow, and experiment with different inference scaling techniques without adding a single line of code beyond the flow's logic. We present an experimental study to demonstrate this capability, and a case study building a theorem prov ing agent for the Rocq theorem prover.
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Robust Contextual Optimization with Missing Covariates
Qingyuan Xu, Ruiwei Jiang
Modern decision-making increasingly relies on contextual features (covariates) t o improve optimization under uncertainty. In practice, however, such covariates are often only partially observed due to, e.g., data source heterogeneity or cos tly data collection. Nonetheless, most existing methods assume fully observed hi storical data and can become unreliable when this assumption is violated. We add ress this gap by proposing a distributionally robust optimization approach that exploits incomplete covariates to produce robust decisions without imputing a co mplete dataset. Our method builds ambiguity sets from the observed partial data

and incorporates the general structure of the missingness mechanism, ensuring ca ndidate distributions remain consistent with what is observed. Across settings w ith discrete or continuous covariates and outcomes, we derive tractable reformul ations and establish finite-sample out-of-sample performance guarantees. Empiric al results across a range of contextual decision-making tasks demonstrate that t he proposed integrated approach consistently outperforms state-of-the-art baseli nes, including various impute-then-optimize pipelines, in both out-of-sample per formance and reliability.
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Learning to Memorize with Attributive and Associative Memory for Online Test-Tim e Adaptation of Vision-Language Models
Yuchao Zhang, Hao Wang, Fan Zhang, Qirui Mi, Mengyue Yang, Yisen Wang, Jun Wang, Haoxuan Li, Zhouchen Lin
Memory-based test-time adaptation (TTA) assigns streaming test samples into clas s-specific memory slots based on pseudo-labels predicted by models like CLIP, an d retrieves them to facilitate subsequent predictions under distribution shift. However, this process introduces two challenges: (1) Each sample is hard-assigne d to a single class based on CLIP's prediction, where inaccurate CLIP prediction leads to memory contamination that biases subsequent prediction. (2) Samples ar e evicted under biased selection due to fixed memory capacity, which risks disca rding informative samples and undermining the efficacy of the memory. To address these challenges, we propose A²Memory (Attributive-Associative Memory for Test-time Adaptation). For challenge (1), we propose Attribute-centric Memory Constr uction that builds prior textual representations from class-shared representativ e and diverse visual attributes, and applies soft assignment to generate surroga te visual representations. For challenge (2), we design Class-wise Associative M emory that dynamically compresses streaming samples into fixed-capacity memory t hrough gradient-based optimization and data-dependent retention, then retrieves sample-adaptive class prototypes for reliable inference. Extensive experiments demonstrate consistent improvements over state-of-the-art methods across 15 benc hmarks.
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REG: In-Sample RL via Regularizing the Evaluation Gap
Hanpu Shen, Weining Shen, Roy Fox
Distribution shift poses a fundamental challenge in offline reinforcement learni ng, often leading to value overestimation when querying out-of-distribution acti ons. We introduce Regularized Evaluation Gap (REG) as a bridge between implicit methods like IQL and explicit conservative methods. We formulate policy evaluati on as a robust optimization problem over an ambiguity set of critics and show th at IQL’s objective can be viewed as an approximate dual solution to this problem . To extract a policy from the learned value function, we propose a practical Or thogonal Policy Gradient (OPG) update. This method regularizes an aggressive, mo de-seeking policy gradient by projecting it onto the subspace orthogonal to a st able, in-sample behavior cloning gradient. Extensive D4RL experiments demonstrat e that REG matches state-of-the-art performance among both Gaussian methods and diffusion-based approaches without the computational burden of the latter.
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Unleashing Implicit Rewards: Prefix-Value Learning for Distribution-Level Optimi zation
Shiping Gao, Hongzhan Chen, Xiaojun Quan, Qifan Wang, Lifu Huang
Process reward models (PRMs) provide fine-grained supervision for reasoning, but reliable PRMs often require step annotations or heavy verification pipelines, m aking them costly to scale and refresh during online RL. Implicit PRMs reduce th is cost by training log-likelihood-ratio rewards from trajectory-level outcome l abels. However, the log-ratio is constrained only as a sequence-level aggregate during training, while inference decomposes it into token- or step-level scores for partial prefixes. This train–inference mismatch leaves local credits weakly identified, so distribution-wide scoring can amplify misleading advantages. We p ropose Implicit Prefix-Value Reward Model (IPVRM), which directly learns the pro bability of eventual correctness for each prefix from outcome labels. Step signa

ls are then obtained as temporal-difference (TD) differences between consecutive prefix values, aligning the training target with inference-time use. IPVRM markedly improves step-verification F1 on ProcessBench. To exploit these prefix values during policy optimization, we further introduce Distribution-Level RL (DistRL), which applies TD advantages to both sampled tokens and high-probability candidate tokens, providing dense counterfactual updates without additional rollouts. Experiments show that DistRL brings limited gains with unreliable implicit rewards, but consistently improves downstream reasoning when paired with IPVRM. The implementation of our method is available at https://github.com/gaoshiping/IPVRM .

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CoopEval: Benchmarking Cooperation-Sustaining Mechanisms and LLM Agents in Social Dilemmas

Emanuel Tewolde, Xiao Zhang, David Guzman Piedrahita, Vincent Conitzer, Zhijing Jin

It is increasingly important that LLM agents interact effectively and safely with other goal-pursuing agents, yet, recent works report the opposite trend: LLMs with stronger reasoning capabilities behave _less_ cooperatively in mixed-motive games such as the prisoner's dilemma and public goods settings. Indeed, our experiments show that recent models---with or without reasoning enabled---consistently defect in single-shot social dilemmas.

To tackle this safety concern, we present the first comparative study of game-theoretic mechanisms designed to enable cooperative outcomes between rational agents _in equilibrium_. Across four social dilemmas testing distinct components of robust cooperation, we evaluate four families of mechanisms: (1) repeating the game for many rounds, (2) reputation systems, (3) third-party mediators to delegate decision making to, and (4) contract agreements for outcome-conditional payments between players. Among our findings, we establish that contracting and mediation are most effective in achieving cooperative outcomes between capable LLM models, and that repetition-induced cooperation deteriorates drastically when co-players vary. Moreover, we demonstrate that the mechanisms become _more effective_ under evolutionary pressures to maximize individual payoffs.

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High-Fidelity ANN-to-SNN Conversion via Closed-Loop CKA Distillation

Bozhou Li, Chubo Liu, Yan Ding, Yufeng Zhang, Zhuo Tang, Kenli Li

ANN-to-SNN conversion offers energy-efficient inference but faces a fidelity-latency trade-off due to open-loop error accumulation. While conversion-aware training mitigates this, it sacrifices the generality of using off-the-shelf ANNs. We propose a closed-loop fine-tuning framework that calibrates these errors without altering the source model. Our approach employs a Dual Alignment Mechanism, utilizing global Kullback-Leibler divergence for output distillation and introducing an adaptive local Centered Kernel Alignment constraint, weighted by initial conversion loss, for feature alignment. We uncover a critical time-dependent dynamic: local constraints are essential for stabilizing representations in low-latency regimes (e.g., $T=8$) where global gradients are unstable, whereas global alignment drives fidelity at higher time steps. Experiments on CIFAR-10 demonstrate that our method achieves over 99\% of source ANN accuracy at $T=32$ (e.g., ResNet-18: 96.38\% vs.\ 96.39\%). Furthermore, this fine-tuning acts as a regularizer, yielding SNNs with input noise robustness that matches or exceeds the source ANN.

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FedHPro: Federated Hyper-Prototype Learning via Gradient Matching

Huan Wang, Jun Shen, Haoran Li, Zhenyu Yang, Jun Yan, Ousman Manjang, Yanlong Zhai, Di Wu, Guansong Pang

Federated Learning (FL) enables collaborative training of distributed clients while protecting privacy. To enhance generalization capability in FL, prototype-based FL is in the spotlight, since shared global prototypes offer semantic anchors for aligning client-specific local prototypes. However, existing methods update global prototypes at the prototype-level via averaging local prototypes or ref

ining global anchors, which often leads to semantic drift across clients and subsequently yields a misaligned global signal. To alleviate this issue, we introduce **hyper-prototypes**, defined by a set of learnable global class-wise prototypes to preserve underlying semantic knowledge across clients. The hyper-prototypes are optimized via gradient matching to align with class-relevant characteristics distilled directly from clients' real samples, rather than prototype-level descriptors. We further propose **FedHPro**, a Federated Hyper-Prototype Learning framework, to leverage hyper-prototypes to promote inter-class separability via mutual-contrastive learning with client-specific margin, while encouraging intra-class uniformity through a consistency penalty. Comprehensive experiments under diverse heterogeneous scenarios confirm that 1) hyper-prototypes produce a more semantically consistent global signal, and 2) FedHPro achieves state-of-the-art performance on several benchmark datasets.
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Dyn-VPP: Video Prediction Policy Optimization for Improved Visual Dynamics
Zirui Ge, Pengxiang Ding, Baohua Yin, Yemin Wang, Qishen Wang, Zhiyong Xie, Hengtao Li, Runze Suo, Wenxuan Song, Han Zhao, Shangke Lyu, Haoang Li, Ran Cheng, Cheng Chi, Hui-Bin Ge, Yaozhi Luo, Donglin Wang
Video action models are a promising foundation for Vision–Language–Action (VLA) because they can learn rich visual dynamics directly from video. However, likelihood-oriented training of diffusion predictors emphasizes globally plausible futures and does not guarantee precision-critical visual dynamics needed for manipulation, so small prediction errors can be amplified by downstream policies. We propose Dyn-VPP, a post-training framework that casts multi-step denoising as policy optimization and aligns predicted future latents with expert visual dynamics via verifiable terminal reward, without modifying any architecture. This enables explicit optimization of dynamics signals that are not captured by likelihood-only training. As a result, Dyn-VPP yields more accurate visual dynamics and improves downstream task execution. Experiments across diverse simulated and real-world manipulation settings show improved dynamics consistency and consistently higher task success.
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Ekka: Automated Diagnosis of Silent Errors in LLM Inference
Yile Gu, Zhen Zhang, Shaowei Zhu, Xinwei Fu, Jun Wu, Yida Wang, Baris Kasikci
LLM serving frameworks are quickly evolving with a complex software stack and a vast number of optimizations.
The rapid development process can introduce silent errors where output quality silently degrades without any explicit error signals.
Diagnosing silent errors is notoriously difficult due to the substantial semantic gap between the high-level symptoms and the low-level root causes.
We observe that diagnosis of silent errors can be effectively framed as a differential debugging problem by leveraging the existence of semantically correct reference implementations.
We propose Ekka, an automated diagnosis system that identifies root causes by systematically aligning and comparing intermediate execution states between a target and a reference framework.
We constructed a benchmark of real-world silent errors from popular serving frameworks, where Ekka shows 80\% pass@1 diagnosis accuracy and 88\% pass@5 diagnosis accuracy, outperforming state-of-the-art systems.
Ekka also diagnoses 4 new silent errors from serving frameworks, all of which have been confirmed by the developers.
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A Probabilistic Framework for LLM-Based Model Discovery
Stefan Wahl, Raphaela Schenk, Ali Farnoud, Jakob H. Macke, Daniel Gedon
Automated methods for discovering mechanistic simulator models from observational data offer a promising path toward accelerating scientific progress. Such methods often take the form of agentic-style iterative workflows that repeatedly propose and revise candidate models by imitating human discovery processes. However, existing LLM-based approaches typically implement such workflows via hand-crafted heuristic procedures, without an explicit probabilistic formulation. We reca

st model discovery as probabilistic inference, i.e., as sampling from an unknown distribution over mechanistic models capable of explaining the data. This perspective provides a unified way to reason about model proposal, refinement, and selection within a single inference framework. As a concrete instantiation of this view, we introduce ModelSMC, an algorithm based on Sequential Monte Carlo sampling. ModelSMC represents candidate models as particles which are iteratively proposed and refined by an LLM, and weighted using likelihood-based criteria. Experiments on real-world scientific systems illustrate that this formulation discovers models with interpretable mechanisms and improves posterior predictive checks. More broadly, this perspective provides a probabilistic lens for understanding and developing LLM-based approaches to model discovery.

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Variance Driven Exploration: A Provable and Efficient Methodology for Pure Exploration in Highly Stochastic Environments

Khang Luong, Nam Nguyen, Hoang Ta, Hung The Tran, Tuan Quang Dam

We propose ***Var**iance **D**riven **E**xploration* (VarDE), a principled approach for pure exploration in *highly stochastic environments*, where the exploration process is dominated by stochastic variance.

VarDE is built on a fundamental principle: *sampling effort should be allocated to minimize the uncertainty of the final decision*.

We formalize the uncertainty of the final decision through a smooth decision function and derive allocation rules that explicitly capture how stochastic noise in individual components affects the reliability of the final output.

We apply this methodology to three core problems of pure exploration -- Best Arm Identification (BAI), Monte Carlo Tree Search (MCTS), and Best-Policy Identification (BPI) -- with theoretical guarantees on variance decay and simple regret.

Empirically, we demonstrate consistent and significant improvements of VarDE over existing methods, with especially strong gains in highly stochastic environments.

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Escaping the Diversity Trap in Robotic Manipulation via Anchor-Centric Adaptation

Yanzhe Chen, Kevin Yuchen Ma, Qi Lv, Yiqi Lin, Zechen Bai, Chen Gao, Mike Zheng Shou

While Vision-Language-Action (VLA) models offer broad general capabilities, deploying them on specific hardware requires real-world adaptation to bridge the embodiment gap. Since robot demonstrations are costly, this adaptation must often occur under a strict data budget. In this work, we identify a critical **diversity trap**: the standard heuristic of ``maximizing coverage" by collecting diverse, single-shot demonstrations can be self-defeating due to non-vanishing estimation noise. We formalize this phenomenon as a **Coverage--Density Trade-off**. By decomposing the policy error into estimation (density) and extrapolation (coverage) terms, we characterize an interior optimal allocation of unique conditions for a fixed budget. Guided by this analysis, we propose **Anchor-Centric Adaptation (ACA)**, a two-stage framework that first stabilizes a policy skeleton through repeated demonstrations at core anchors, then selectively expands coverage to high-risk boundaries via teacher-forced error mining and constrained residual updates. Real-robot experiments validate our trade-off framework and demonstrate that ACA significantly improves task reliability and success rates over standard diverse sampling strategies under the same budget.

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GraphPFN: A Prior-Data Fitted Graph Foundation Model

Dmitry Eremeev, Oleg Platonov, Gleb Bazhenov, Artem Babenko, Liudmila Prokhorenkova

Graph foundation models face several fundamental challenges including transferability across diverse domains and data scarcity, which calls into question the very feasibility of creating such models. However, despite similar challenges, the tabular domain has recently witnessed the emergence of the first successful foundation models such as TabPFN. These models are based on the prior-data fitted networks (PFN) framework, in which models are pretrained on carefully designed sy

nthetic datasets to make predictions in an in-context learning setting. Recently, G2T-FM, a framework that converts graph node-level tasks into tabular tasks, has made the first step towards adopting PFNs for graphs, yet it is limited to hand-crafted features and was never pretrained on graph data. In this work, we make the next step by proposing GraphPFN, a PFN-based model designed and pretrained specifically for graph node-level tasks. Following the PFN framework, we first design a prior distribution of synthetic attributed graphs by using a novel combination of multi-level stochastic block models and a preferential attachment process for structure generation and graph-aware structured causal models for attribute generation. Then, we augment the tabular foundation model LimiX with attention-based graph neighborhood aggregation layers and train it on millions of synthetic graphs sampled from our prior. On diverse real-world graph datasets with node-level tasks, GraphPFN achieves state-of-the-art results in both in-context learning and finetuning regimes, outperforming G2T-FM, prior GFMs, and task-specific GNNs trained from scratch. More broadly, GraphPFN shows the potential of PFN-based models for building graph foundation models. Our code is available at https://github.com/yandex-research/graphpfn.

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Exploring and Exploiting Stability in Latent Flow Matching

Rania Briq, Michael Kamp, Ohad Fried, Sarel Cohen, Stefan Kesselheim

In this work, we show that Latent Flow-Matching (LFM) models are robust to different types of perturbations, including data reduction and model capacity shrinkage. We characterize this stability by these models' tendency to generate similar outputs under identical noise seeds. We provide a perspective relating this phenomenon to flow matching theory, which indicates that this stability is inherent to the FM objective. We further exploit this stability to derive practical algorithms for more efficient training and inference. Concretely, first, we show that by training LFM models on significantly reduced datasets, performance is preserved, and in compute-constrained regimes, the model converges faster while maintaining quality. This yields multiple advantages, including savings in the training time due to faster convergence, and alleviating annotation effort when training conditional models. Second, LFM stability under architectural shrinkage gives rise to a two-model coarse-to-fine approach, one using a light-weight architecture for the first phase of the FM trajectory, and one with higher capacity for the second, thereby reducing the inference cost substantially. To determine which samples are informative, we introduce three sample-scoring criteria and evaluate them under standard metrics for generative models. Our results are thoroughly evaluated on multiple datasets, demonstrating the practical advantage of this stability, including data savings and a more than two-fold inference speedup while generating comparable outputs.

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MoST: Mixing Speech and Text with Modality-Aware Mixture of Experts

Yuxuan Lou, Kai Yang, Yang You

We present MoST (Mixture of Speech and Text), a novel multimodal large language model that seamlessly integrates speech and text processing through our proposed Modality-Aware Mixture of Experts (MAMoE) architecture. While current multimodal models typically process diverse modality representations with identical parameters—disregarding their inherent representational differences, we introduce specialized routing pathways that direct tokens to modality-appropriate experts based on input type. MAMoE simultaneously enhances modality-specific learning and cross-modal understanding through two complementary components: modality-specific expert groups that capture domain-specific patterns and shared experts that facilitate information transfer between modalities. Building on this architecture, we develop an efficient transformation pipeline that adapts the pretrained MoE language model through strategic post-training on ASR and TTS datasets, followed by fine-tuning with a carefully curated speech-text instruction dataset. A key feature of this pipeline is that it relies exclusively on fully accessible, open-source datasets to achieve strong performance and data efficiency. Comprehensive evaluations across ASR, TTS, audio language modeling, and spoken question answering benchmarks show that MoST consistently outperforms existing models of comp

arable parameter counts. Our ablation studies confirm that the modality-specific routing mechanism and shared experts design significantly contribute to performance gains across all tested domains. To our knowledge, MoST represents the first fully open-source speech-text LLM built on a Mixture of Experts architecture.
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Efficient Preference Poisoning Attack on Offline RLHF
Chenye Yang, Weiyu Xu, Lifeng Lai
Offline Reinforcement Learning from Human Feedback (RLHF) pipelines such as Direct Preference Optimization (DPO) train on a pre-collected preference dataset, which makes them vulnerable to preference poisoning attack.
We study label flip attacks against log-linear DPO.
We first illustrate that flipping one preference label induces a parameter-independent shift in the DPO gradient.
Using this key property, we can then convert the targeted poisoning problem into a structured binary sparse approximation problem.
To solve this problem, we develop two attack methods: Binary-Aware Lattice Attack (BAL-A) and Binary Matching Pursuit Attack (BMP-A).
BAL-A embeds the binary flip selection problem into a binary-aware lattice and applies Lenstra-Lenstra-Lovász reduction and Babai's nearest plane algorithm; we provide sufficient conditions that enforce binary coefficients and recover the minimum-flip objective.
BMP-A adapts binary matching pursuit to our non-normalized gradient dictionary and yields coherence-based recovery guarantees and robustness (impossibility) certificates for K-flip budgets.
Experiments on synthetic dictionaries and the Stanford Human Preferences dataset validate the theory and highlight how dictionary geometry governs attack success.
****
Identifiable Equivariant Networks are Layerwise Equivariant
Vahid Shahverdi, Giovanni Luca Marchetti, Georg Bökman, Kathlén Kohn
We investigate the relation between end-to-end equivariance and layerwise equivariance in deep neural networks. We prove the following:
For a network whose end-to-end function is equivariant with respect to group actions on the input and output spaces, there is a parameter choice yielding the same end-to-end function such that its layers are equivariant with respect to some group actions on the latent spaces.
Our result assumes that the parameters of the model are identifiable in an appropriate sense. This identifiability property has been established in the literature for a large class of networks, to which our results apply immediately, while it is conjectural for others. The theory we develop is grounded in an abstract formalism, and is therefore architecture-agnostic. Overall, our results provide a mathematical explanation for the emergence of equivariant structures in the weights of neural networks during training -- a phenomenon that is consistently observed in practice.
****
Convergence Analysis of the Lion Optimizer in Centralized and Distributed Settings
Wei Jiang, Mao Xu, Wenhao Yang, Yibo Wang, Zechao Li, Lijun Zhang
In this paper, we provide a comprehensive convergence analysis for the Lion optimizer. First, we establish that the original Lion achieves a convergence rate of $\mathcal{O}(d^{1/2}T^{-1/4})$, where d denotes the problem dimension and T is the iteration number. To improve this rate, we propose a variance reduction variant of Lion, which attains an enhanced rate of $\mathcal{O}(d^{1/2}T^{-1/3})$ with the average smoothness assumption. Then, we extend our analysis to distributed settings. We demonstrate that the distributed Lion optimizer and its variance reduction counterpart achieve linear speedup with respect to the number of nodes n, yielding convergence rates of $\mathcal{O}(d^{1/2}(nT)^{-1/4})$ and $\mathcal{O}(d^{1/2}(nT)^{-1/3})$, respectively. Additionally, we investigate a communication-efficient distributed Lion variant that utilizes sign compression for bidirectional communication. By employing unbiased sign operations, this varia

nt achieves a convergence rate of $\mathcal{O} \left(\max \{ \frac{d^{1/4}}{T^{1/4}}, \frac{d^{1/10}}{n^{1/5}T^{1/5}} \} \right)$, and its variance-reduced counterpart can further improve the rate to $\mathcal{O}\left(\frac{d^{1/4}}{T^{1/4}} \right)$. Finally, we conduct numerical experiments to validate the effectiveness of the proposed methods.

****

Mitigating Staleness in Asynchronous Pipeline Parallelism via Basis Rotation

Hyunji Jung, Sungbin Shin, Namhoon Lee

Asynchronous pipeline parallelism maximizes hardware utilization by eliminating the pipeline bubbles inherent in synchronous execution, offering a path toward efficient large-scale distributed training. However, this efficiency gain can be compromised by gradient staleness, where the immediate model updates with delayed gradients introduce noise into the optimization process. Crucially, we identify a critical, yet often overlooked, pathology: this delay scales linearly with pipeline depth, fundamentally undermining the very scalability that the method originally intends to provide. We trace this pathology to a specific property of the optimization landscape: the misalignment between the Hessian eigenbasis and the standard coordinate basis, which triggers oscillations in the update trajectories of coordinate-wise adaptive optimizers. We identify that these oscillations cause delayed updates to diverge from their true counterparts, invalidating their use for current iterations. This insight is formalized through theoretical analysis, including a convergence bound showing that basis misalignment amplifies the delay penalty, and substantiated with empirical evaluation. To address this, we propose basis rotation, a framework that rotates the optimizer's coordinate system to align with the Hessian eigenbasis, keeping delayed updates useful. We theoretically demonstrate that basis rotation minimizes basis misalignment, thereby counteracting the conditions that amplify delay penalties. Empirically, in training up to a 3B-parameter LLM, basis rotation reduces the required iterations by 81.7% compared to the best-performing asynchronous baseline.

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ECSEL: Explainable Classification via Signomial Equation Learning

Adia C. Lumadjeng, Ilker Birbil, Erman Acar

We introduce ECSEL, an explainable classification method that learns formal expressions in the form of signomial equations, motivated by the observation that many symbolic regression benchmarks admit compact signomial structure. ECSEL directly constructs a structural, closed-form expression that serves as both a classifier and an explanation. On standard symbolic regression benchmarks, our method recovers a larger fraction of target equations than competing state-of-the-art approaches while requiring substantially less computation. Leveraging this efficiency, ECSEL achieves classification accuracy competitive with established machine learning models without sacrificing interpretability. Further, we show that ECSEL satisfies some desirable properties regarding global feature behaviour, decision-boundary analysis, and local feature attributions. Experiments on benchmark datasets and two real-world case studies i.e., e-commerce and fraud detection, demonstrate that the learned equations expose dataset biases, support counterfactual reasoning, and yield actionable insights.

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Revisiting Photometric Ambiguity for Accurate Gaussian-Splatting Surface Reconstruction

Jiahe Li, Jiawei Zhang, Xiao Bai, Jin Zheng, Xiaohan Yu, Lin Gu, Gim Hee Lee

Surface reconstruction with differentiable rendering has achieved impressive performance in recent years, yet the pervasive photometric ambiguities have strictly bottlenecked existing approaches. This paper presents AmbiSuR, a framework that explores an intrinsic solution upon Gaussian Splatting for the photometric ambiguity-robust surface reconstruction with high performance. Started by revisiting the foundation, our investigation uncovers two built-in primitive-wise ambiguities in representation, while revealing an intrinsic potential for ambiguity self-indication in Gaussian Splatting. Stemming from these, a photometric disambiguation is first introduced, constraining ill-posed geometry solution for definite surface formation. Then, we propose an ambiguity indication module that unleash

es the self-indication potential to identify and further guide correcting underc
onstrained reconstructions. Extensive experiments demonstrate our superior perfo
rmance in surface reconstruction compared to existing methods across various cha
llenging scenarios, while excelling in broad compatibility. Our code will be mad
e open-source upon acceptance.
****

FocalPolicy: Frequency-Optimized Chunking and Locally Anchored Flow Matching for
 Coherent Visuomotor Policy

Qian He, Zhenshuo Yang, Wenqi Liang, Chunhui Hao, Nicu Sebe, Jiandong Tian

Visuomotor policies aim to learn complex manipulation tasks from expert demonstr
ations.
However, generating smooth and coherent trajectories remains challenging, as it
requires balancing proximal precision with distal foresight.
Existing approaches typically focus on optimizing intra-chunk action distributio
ns, often neglecting the inter-chunk coherence.
Consequently, inter-chunk discontinuities significantly impede the learning of c
oherent long-horizon actions.
To overcome this limitation and achieve a synergetic balance between precision a
nd foresight, we propose FocalPolicy, a foresight-aware visuomotor policy that c
ombines Frequency-Optimized Chunking with Locally Anchored flow matching.
We introduce a foresight composite objective that supervises time-domain alignme
nt within the proximal actions while regularizing frequency-domain structure ove
r multiple future action chunks to improve cross-chunk coherence.
To efficiently learn complex action distributions, we design locally anchored sa
mpling to enhance target signal propagation efficiency during consistency flow m
atching training.
Extensive experiments demonstrate that FocalPolicy outperforms existing approach
es and confirm the generalizability of our modules to other baselines.
Project website: https://focalpolicy.github.io/
****

Weaving in the Clouds: Achieving Synergistic Collaboration among LLM Agents via
Federated Learning

Jiaxing Zhao, Hongbin Xie, Yuzhen Lei, Xuan Song, Zhuoran Shi, Lianxin Li, Shuan
gxue liu, Haoran Zhang

Multi-Agent Systems (MAS) powered by Large Language Models (LLMs) have become a
promising paradigm for solving workflow-structured tasks through expert collabor
ation.
However, the data required for such collaboration are often distributed across o
rganizations and cannot be centrally pooled due to privacy, intellectual propert
y, and compliance constraints.
Federated Learning preserves data locality, but existing federated paradigms typ
ically treat clients as independent and fail to model workflow dependencies that
 are crucial for coherent multi-stage collaboration.
We introduce **FedWave**, a federated and workflow-aware framework that enables
LLM-based experts to solve sequential tasks under data-locality constraints.
FedWave combines a *Value Chain Layer* to model inter-stage dependencies, federa
ted LoRA adaptation to preserve role-specific expertise, a shared *Mixture-of-Ex
perts (MoE)* router for input-conditioned expert fusion, and a *Direct Preferenc
e Optimization (DPO)* stage based on router-induced preferences.
Experiments across workflow datasets and LLM backbones show that FedWave outperf
orms federated baselines and remains competitive with centralized multi-agent sy
stems while preserving data locality.
****

Evaluating the Representation Space of Diffusion Models via Self-Supervised Prin
ciples

Xiao Li, Yixuan Jia, Zekai Zhang, Xiang Li, Lianghe Shi, Jinxin Zhou, Zhihui Zhu
, Liyue Shen, Qing Qu

Diffusion models have demonstrated remarkable generative capabilities and have a
lso emerged as powerful self-supervised representation learners. However, the co
nnection between these two abilities remains less explored. In this work, drawin

g inspiration from self-supervised learning (SSL), we introduce a framework for jointly evaluating the representation and generation capabilities of diffusion models. Specifically, we decompose features into invariant and residual components and derive the Invariant Contamination Ratio (ICR), a Fisher-based metric that quantifies how residual variation contaminates invariant signal in feature space. We use this framework to analyze both discriminative and generative behavior of diffusion models. On the representation side, we find that invariance peaks at intermediate noise levels, which also yield the best downstream classification performance. On the generative side, we study how training transitions from genuine generalization to memorization in data-limited regimes, and show that ICR serves as a sensitive training-time indicator of early learning: increasing residual energy along Fisher directions marks the onset of memorization, detectable from training features alone without external evaluators or held-out test sets. Overall, our results show that diffusion models can be monitored from a self-supervised perspective through the geometry of their learned representations.
****
BiRQA: Bidirectional Robust Quality Assessment for Images
Aleksandr Gushchin, Dmitriy S. Vatolin, Anastasia Antsiferova
Full-Reference image quality assessment (FR IQA) is important for image compression, restoration and generative modeling, yet current neural metrics remain slow and vulnerable to adversarial perturbations. We present BiRQA, a compact FR IQA metric model that processes four fast complementary features within a bidirectional multiscale pyramid. A bottom-up attention module injects fine-scale cues into coarse levels through an uncertainty-aware gate, while a top-down cross-gating block routes semantic context back to high resolution. To enhance robustness, we introduce Anchored Adversarial Training, a theoretically grounded strategy that uses clean "anchor" samples and a ranking loss to bound pointwise prediction error under attacks. On five public FR IQA benchmarks BiRQA outperforms or matches the previous state of the art (SOTA) while running $\sim 3 \times$ faster than previous SOTA models. Under unseen white-box attacks it lifts SROCC from 0.30-0.57 to 0.60-0.84 on KADID-10k, demonstrating substantial robustness gains. To our knowledge, BiRQA is the only FR IQA model combining competitive accuracy with real-time throughput and strong adversarial resilience.
****
Sampling and Identity-Testing Without Approximate Tensorization of Entropy
William Gay, William He, Nicholas Kocurek, Ryan O'Donnell
We study the problems of approximate sampling from and distribution testing of \emph{mixture models}, where the modes satisfy a functional inequality called \emph{approximate tensorization of entropy} (ATE). While it is known that ATE makes these tasks more efficient in the unimodal setting, mixtures of few distributions satisfying ATE do not necessarily satisfy ATE overall, leading to a lack of theoretical guarantees for multimodal distributions, which are a key challenging case of modern generative models. We show this gap can be overcome by establishing the following pair of results for mixtures of ATE distributions:

1) We show fast mixing of Glauber dynamics from a \emph{data-based initialization}, with \emph{optimal} sample complexity, for mixtures of distributions satisfying modified log-Sobolev inequalities, building on similar results in (Koehler et al., 2024, Huang et al., 2024) for mixtures satisfying the weaker Poincaré inequality.
2) Answering an open question from (Blanca et al., 2023), we give efficient identity-testers for mixtures of ATE distributions in the coordinate-conditional sampling access model.
****
Being More Lightweight and Practical: Mini-sized Contrastive Learning Pre-trained Models for Fine-grained Traffic Task
Shuhao Li, Weidong Yang, Ben Fei, Yue Cui, Lipeng Ma, Fan Zhang
Fine-grained traffic prediction is critically important for mitigating traffic congestion in key urban areas and for providing lane-change guidance in autonomous vehicles and navigation systems. However, task-specific models are not efficie

nt enough, city-scale pre-trained models often overlook fine-grained requirements, and the demand for extensive computational resources hinders practical deployment. To address this issue, we developed a lightweight pre-training framework, MiniTraffic. This framework leverages abundant road-level data to address lane-level data scarcity through a frequency domain stability augmentation module and captures road-lane correlations via contrastive clustering to construct small-scale graph structures, significantly reducing model parameters. Fine-tuning with minimal target data provides a unified and efficient solution for fine-grained traffic prediction. In multi-granularity traffic prediction tasks across six fine-grained datasets, MiniTraffic demonstrated superior performance compared to existing baselines.

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INFER: Learning Implicit Neural Frequency Response Fields for Confined Acoustic Environments

Harshvardhan C. Takawale, Nirupam Roy, C. Phillip Brown

Neural acoustic fields often model time-domain impulse responses, which struggle to capture the frequency-selective wave behaviors that dominate confined, resonant environments. To address this, we propose INFER (Implicit Neural Frequency Response fields), a framework that directly learns continuous, complex-valued frequency response fields. Unlike prior time-domain methods, our frequency-first approach enables three key innovations: (1) end-to-end learning of frequency-specific attenuation and phase delay in 3D space; (2) a physics-based Kramers–Kronig consistency constraint that causally regularizes attenuation and phase delay; and (3) perceptual and hardware-aware spectral supervision that prioritizes critical auditory bands. We evaluate INFER across diverse settings, ranging from standard room-scale benchmarks (MeshRIR, RAF) to challenging, highly reverberant environments like real car cabins. Our approach significantly outperforms time- and hybrid-domain baselines, reducing average magnitude and phase reconstruction errors by over 39\% and 51\%, respectively, demonstrating state-of-the-art accuracy in modeling complex acoustic spaces.

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Taming Aleatoric Impulse in Off-Policy Reinforcement Learning

Zhouyang Yu, Guojian Zhan, Yang Guan, Jingliang Duan, Letian Tao, Shengbo Eben Li

Off-policy reinforcement learning is vulnerable to overestimation bias, which is rooted in the total value uncertainty. However, existing methods typically misaddress this by targeting the epistemic component, neglecting the aleatoric component. We identify for the first time that this oversight fails to contain a massive bias surge, termed **Aleatoric Impulse**. Although transient, this impulse fundamentally derails the learning trajectory, permanently locking the agent into suboptimal policies. To counteract this, we propose **A**leatoric **I**mpulse **D**amping **(AID)**, the first mechanism that models total value uncertainty by disentangling the return variance into epistemic and aleatoric components, followed by their adaptive weighted recombination. Leveraging this derived uncertainty, the critic constructs a pessimistic lower confidence bound to surgically suppress the impulse. Complementing this, the actor utilizes a symmetrical upper confidence bound to drive optimistic exploration, ensuring that the necessary pessimism does not compromise exploration efficiency. We integrate this mechanism into the Distributional Soft Actor-Critic algorithm to establish **DSAC-AID**. Extensive experiments on the high-dimensional Gym-MuJoCo and DeepMind Control Suite benchmarks demonstrate that it achieves state-of-the-art results in final performance.

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Removing Noise, not Finding Gold: Quality Filtering for Large-Scale Pretraining

Thiziri Nait Saada, Louis Béthune, Michal Klein, David Grangier, marco cuturi, Pierre Ablin

Large-scale models are pretrained on massive web-crawled datasets containing documents of mixed quality, making data filtering essential. A popular method is Classifier-based Quality Filtering (CQF), which trains a binary classifier to distinguish between pretraining data and a small, high-quality set. It assigns each

pretraining document a quality score defined as the classifier's score and retai
ns only the top-scoring ones. We provide an in-depth analysis of CQF.
We show that while CQF improves downstream task performance, it does not necessa
rily enhance language modeling on the high-quality set. Importantly, we find tha
t training on CQF-selected data can outperform training directly on the high-qua
lity set, even when the latter is sufficiently large. This finding alone is part
icularly striking, given the substantial effort and cost recently devoted to aug
menting high-quality data. We explain this paradox by the fact that CQF implicit
ly filters the high-quality dataset as well as the low-quality one. Finally, we
introduce an optimization-driven notion of data quality and demonstrate that it
can be reliably estimated using small-scale proxy experiments. Altogether, our r
esults both elucidate the mechanisms behind CQF and deepen our understanding of
data selection methods widely used in practice.
****

Evolving Interpretable Constitutions for Multi-Agent Coordination
Ujwal Kumar, Alice Saito, Hershraj Niranjani, Rayan Yessou, Phan Xuan Tan
Constitutional AI has focused on single-model alignment using fixed principles.
However, multi-agent systems create novel alignment challenges through emergent
social dynamics. We present Constitutional Evolution, a framework for automatica
lly discovering behavioral norms in multi-agent LLM systems. Using a grid-world
simulation with survival pressure, we study the tension between individual and c
ollective welfare, quantified via a Societal Stability Score $\mathcal{S} \in [0
,1]$ that combines productivity, survival, and conflict metrics. Adversarial con
stitutions lead to societal collapse ($\mathcal{S}=0$), while vague prosocial pr
inciples (''be helpful, harmless, honest'') produce inconsistent coordination ($
\mathcal{S}=0.249$). Even constitutions designed by Claude 4.5 Opus with explici
t knowledge of the objective achieve only moderate performance ($\mathcal{S}=0.3
32$). Using LLM-driven genetic programming with multi-island evolution, we evolv
e constitutions maximizing social welfare without explicit guidance toward coope
ration. The evolved constitution $\mathcal{C}^*$ achieves
$\mathcal{S}=0.556\pm0.008$ (123\% higher than human-designed baselines, $N=10$)
, eliminates conflict, and discovers that minimizing communication (0.9\% vs 62.
2\% social actions) outperforms verbose coordination. Our interpretable
rules demonstrate that cooperative norms can be discovered rather than prescribe
d.
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ProbeLLM: Automating Principled Diagnosis of LLM Failures
Yue Huang, Zhengzhe Jiang, Yuchen Ma, Yu Jiang, Xiangqi Wang, Yujun Zhou, Yuexin
g Hao, Kehan Guo, Pin-Yu Chen, Marzyeh Ghassemi, Stefan Feuerriegel, Xiangliang
Zhang
Understanding how and why large language models (LLMs) fail is becoming a centra
l challenge as models rapidly evolve and static evaluations fall behind. While a
utomated probing has been enabled by dynamic test generation, existing approache
s often discover isolated failure cases, lack principled control over exploratio
n, and provide limited insight into the underlying structure of model weaknesses
. We propose ProbeLLM, a benchmark-agnostic automated probing framework that ele
vates weakness discovery from individual failures to structured failure modes. P
robeLLM formulates probing as a hierarchical Monte Carlo Tree Search, explicitly
 allocating limited probing budgets between global exploration of new failure re
gions and local refinement of recurring error patterns. By restricting probing t
o verifiable test cases and leveraging tool-augmented generation and verificatio
n, ProbeLLM grounds failure discovery in reliable evidence. Discovered failures
are further consolidated into interpretable failure modes via failure-aware embe
ddings and boundary-aware induction. Across diverse benchmarks and LLMs, ProbeLL
M reveals substantially broader, cleaner, and more fine-grained failure landscap
es than static benchmarks and prior automated methods, supporting a shift from c
ase-centric evaluation toward principled weakness discovery
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Fast and Highly Expressive Policy Learning for Offline Reinforcement Learning vi
a Bootstrapped Flow Q-Learning

Thanh Xuan Nguyen, Tri Ton, Hongbin Choe, Tung Minh Luu, Chang D. Yoo
Diffusion-based Q-learning has emerged as a powerful paradigm for offline reinfo rcement learning, but its reliance on multi-step denoising makes both training a nd inference computationally expensive and brittle. Recent efforts to accelerate diffusion Q-learning toward single-step action generation typically introduce a uxiliary networks, policy distillation, or multi-phase training, which frequentl y compromise simplicity, stability, or performance.
To address these limitations, we introduce Bootstrapped Flow Q-Learning (BFQ), a novel framework that enables accurate single-step action generation during both training and inference—without auxiliary networks or distillation procedures. B FQ adopts a divide-and-conquer view of the displacement vector along the flow pa th: it begins by learning short-range displacements that can be accurately estim ated from the Flow Matching marginal velocity, and bootstraps these components t o directly learn a noise-to-action mapping in a single step. This formulation el iminates multi-step denoising, resulting in a learning procedure that is substan tially faster, simpler, and more robust. Extensive D4RL evaluations show that BF Q improves performance while significantly reducing computational cost compared to multi-step diffusion baselines, demonstrating that single-step action generat ion suffices for high-performance offline Reinforcement Learning.
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Cello: A Universal Cell-wise Feature Aggregation framework for Reliable Pathol ogy Images Analysis
Hengrui Lou, Weihan Li, Jiazhen Yang, Lingxiang Jia, Shengxuming Zhang, Linyun Z hou, Xiuming Zhang, Zhenyang Wang, Mingli Song, Zunlei Feng
Computational pathology has made progress in diagnosis and prognosis prediction from whole slide images (WSIs), yet pipelines still rely on patch-level feature extraction and aggregation, departing from the cell-centric reasoning used by pa thologists.
This gap limits sensitivity to micro-lesions and subtle changes, and current met hods rarely provide a unified solution that supports both local and global tasks with trustworthy evidence. We propose Cello, a universal cell-wise feature aggr egation framework for reliable pathology image analysis. Cello integrates cell-l evel representations into WSI modeling via protein-signal–supervised cell-wise l earning, preserving fine-grained cellular cues under gigapixel constraints. For local tasks, Cello introduces a flexible prototype-based contrastive module for scalable, task-adaptive representation learning. For global tasks, Cello adopts a weakly supervised gated aggregation that can widely leverage WSI labels. Final ly, a cell–local–global decision-route consistency objective dynamically aggrega tes cellular evidence and aligns local predictions with global outcomes, improvi ng reliability and faithfulness.
Trained with only hundreds to thousands of samples, Cello achieves performance g ains of 3.0%~7.6% and outperforms SOTA pathology foundation models pretrained on tens of thousands of samples. Code is available at https://github.com/HengruiLo u/Cello.
****

Noisy-Channel Minimum Bayes Risk Decoding
Yusuke Sakai, Hidetaka Kamigaito, Taro Watanabe
Minimum Bayes Risk (MBR) decoding yields more robust and higher-quality text gen eration than maximum a posteriori (MAP) decoding by selecting hypotheses that ma ximize expected utility over sampled pseudo-references. However, there exists a discrepancy in the design: hypothesis selection calculates expected utility scor es conditioned on given pseudo-references, while commonly used evaluation metric s, e.g., BLEU and COMET, are asymmetric. Therefore, it is important to consider both hypothesis-to-reference and reference-to-hypothesis directional effects. In this study, we introduce a noisy channel decomposition of MBR decoding that nat urally incorporates bidirectional effects to account for these asymmetries. We d ecompose MBR decoding into four interacting components: hypothesis-to-reference likelihood, reference-to-hypothesis likelihood, hypothesis prior, and reference prior. This decomposition provides a unified interpretation of existing MBR vari ants and enables metric- and task-specific interpretability by isolating the con

tribution of each channel. Our comprehensive analysis reveals that channel-wise contributions exhibit distinct characteristics across metrics while remaining consistent across tasks, and suggests that appropriate channel weighting may lead to improvements over original MBR decoding.
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MICE-Bench: A Challenging and Comprehensive Benchmark for Multi-Reference Image Creation and Editing

Siqi Luo, Huayu Zheng, Jianghan Shen, Yi Xin, Luxin Xu, Jiyao Liu, Xinyu Zhang, Hang Zhou, Pengyu Xie, Xiaohui Li, Shuo Cao, Yuandong Pu, Junjun He, Bin Fu, Yihao Liu, Yu Qiao, Guangtao Zhai, Yuewen Cao, Xiaohong Liu

The paradigm of visual generation is rapidly shifting from single-image conditioning toward multi-image conditioning, making the ability to synthesize and edit images based on multiple visual references a critical capability. Despite this trend, existing benchmarks remain largely limited to single-reference scenarios or narrowly defined tasks, leaving model behavior under complex multi-concept composition insufficiently explored. To bridge this gap, we introduce **MICE-Bench**, a comprehensive benchmark for **M**ulti-reference **I**mage **C**reation and **E**diting. The benchmark is designed around three core principles: 1) heterogeneous concept composition across seven visual dimensions; 2) varying levels of constraint density, ranging from dual-concept to seven-concept configurations; 3) concept-centric data construction and benchmark evaluation, enabling fine-grained analysis of interactions among multiple concepts. MICE-Bench consists of 3,119 high-quality test cases within a unified concept space. Using an 8-dimensional evaluation metric, we systematically evaluate 13 state-of-the-art models. Our results show that although closed-source models maintain a clear performance advantage, all models experience notable degradation in concept consistency and physical realism as concept complexity increases.This indicates that current models rely on superficial composition rather than genuine multi-concept synthesis, highlighting substantial room for future improvement.
****

Multipole Semantic Attention: A Fast Approximation of Softmax Attention for Pretraining

Rupert Mitchell, Kristian Kersting

Pretraining transformers on long sequences, such as entire code repositories or collections of related documents, is bottlenecked by quadratic attention costs. We present Multipole Semantic Attention (MuSe), which accelerates 64k-context pretraining by 36% while matching baseline loss, requiring no architectural changes. MuSe is a training-time approximation that clusters queries and keys separately in representation space. This yields query-specific summaries that substantially outperform spatial blocking at matched sparsity, while also enabling drop-in compatibility with existing pretrained models—we validate on Llama 3.1-8B and 3.2-1B without retraining. We pretrain
language models up to 1B parameters at 64k context on code and scientific documents, confirming that MuSe preserves quality and long-context utilization during training.
****

Thinking with Geometry: Active Geometry Integration for Spatial Reasoning

Haoyuan Li, Qihang Cao, Tao Tang, Kun Xiang, Zihan Guo, Jianhua Han, Jia-Wang Bian, Hang Xu, Xiaodan Liang

Recent progress in spatial reasoning with Multimodal Large Language Models (MLLMs) increasingly leverages geometric priors from 3D encoders. However, most existing integration strategies remain passive: geometry is exposed as a global stream and fused in an indiscriminate manner, which often induces semantic-geometry misalignment and redundant signals. We propose GeoThinker, a framework that shifts the paradigm from passive fusion to active perception. Instead of feature mixing, GeoThinker enables the model to selectively retrieve geometric evidence conditioned on its internal reasoning demands. GeoThinker achieves this through Spatial-Grounded Fusion applied at carefully selected VLM layers, where semantic visual priors selectively query and integrate task-relevant geometry via frame-strict cross-attention, further calibrated by Importance Gating that biases per-fram

e attention toward task-relevant structures. Comprehensive evaluation results sh ow that GeoThinker sets a new state-of-the-art in spatial intelligence, achievin g a peak score of 72.6 on the VSI-Bench. Furthermore, GeoThinker demonstrates ro bust generalization and significantly improved spatial perception across complex downstream scenarios, including embodied referring and autonomous driving. Our results indicate that the ability to actively integrate spatial structures is es sential for next-generation spatial intelligence.

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Lower Complexity Bounds for Nonconvex-Strongly-Convex Bilevel Optimization with First-Order Oracles

Kaiyi Ji

Although upper bound guarantees for bilevel optimization have been widely studie d, progress on lower bounds has been limited due to the complexity of the bileve l structure. In this work, we focus on the smooth nonconvex-strongly-convex sett ing and develop new hard instances that yield nontrivial lower bounds under dete rministic and stochastic first-order oracle models. In the deterministic case, w e prove that any first-order zero-respecting algorithm requires at least $\Omega (\kappa^{3/2}\epsilon^{-2})$ oracle calls to find an ϵ-accurate station ary point, improving the optimal lower bounds known for single-level nonconvex o ptimization and for nonconvex-strongly-convex min-max problems. In the stochasti c case, we show that at least $\Omega(\kappa^{5/2}\epsilon^{-4})$ stochastic ora cle calls are necessary, again strengthening the best known bounds in related se ttings. Our results expose substantial gaps between current upper and lower boun ds for bilevel optimization and suggest that even simplified regimes, such as th ose with quadratic lower-level objectives, warrant further investigation toward understanding the optimal complexity of bilevel optimization under standard firs t-order oracles.

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Distilling Linearized Behavior into Non-linear Fine-Tuning for Effective Task Ar ithmetic

Thomas Sommariva, Francesca Morandi, Simone Calderara, Angelo Porrello

Task vector composition has emerged as a promising paradigm for editing pre-trai ned models, enabling model merging through addition and unlearning through subtr action. Fine-tuning in the tangent space of a pre-trained model (linear fine-tun ing) has proven effective, as it produces task vectors that are naturally disent angled and resistant to interference. However, linearized models suffer from lim ited expressivity during training and incur higher computational costs at infere nce time, which restrict their practical applicability. In this work, we bridge the gap between linear and standard non-linear fine-tuning. We show that lineari ty with respect to weight perturbations, a property defined in parameter space, can be enforced through constraints in activation space during training. Concret ely, we distill hidden representations from a curvature-regularized linearized t eacher into a non-linear student trained via conventional fine-tuning. We find t hat the resulting model inherits key properties of linearized models for task ar ithmetic, enabling effective composition of task vectors and achieving strong pe rformance across vision and language benchmarks without incurring any inference-time overhead.

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Why Deep Jacobian Spectra Separate: Depth-Induced Scaling and Singular-Vector Al ignment

Nathanaël Haas, François Gatine, Augustin M Cosse, Zied Bouraoui

Understanding why gradient-based training in deep networks exhibits strong impli cit bias remains challenging, in part because tractable singular-value dynamics are typically available only for balanced deep linear models. We propose an alte rnative route based on two theoretically grounded and empirically testable signa tures of deep Jacobians: depth-induced exponential scaling of ordered singular v alues and strong spectral separation. Adopting a fixed-gates view of piecewise-l inear networks, where Jacobians reduce to products of masked linear maps within a single activation region, we prove the existence of Lyapunov exponents governi ng the top singular values at initialization, give closed-form expressions in a

tractable masked model, and quantify finite-depth corrections. We further show t
hat sufficiently strong separation forces singular-vector alignment in matrix pr
oducts, yielding an approximately shared singular basis for intermediate Jacobia
ns. Together, these results motivate an approximation regime in which singular-v
alue dynamics become effectively decoupled, mirroring classical balanced deep-li
near analyses without requiring balancing. Experiments in fixed-gates settings v
alidate the predicted scaling, alignment, and resulting dynamics, supporting a m
echanistic account of emergent low-rank Jacobian structure as a driver of implic
it bias.
****

Fast-SAM3D: 3Dfy Anything in Images but Faster
Weilun Feng, Mingqiang Wu, zhiliang chen, Chuanguang Yang, Haotong Qin, Yuqi Li,
 Xiaokun Liu, Guoxin Fan, Libo Huang, Yulun Zhang, Michele Magno, Yongjun Xu, Zh
ulin An
SAM3D enables scalable, open-world 3D reconstruction from complex scenes, yet it
s deployment is hindered by prohibitive inference latency. In this work, we cond
uct the **first systematic investigation** into its inference dynamics, revealin
g that generic acceleration strategies are brittle in this context. We demonstra
te that these failures stem from neglecting the pipeline's inherent multi-level
heterogeneity: the kinematic distinctiveness between shape and layout, the i
ntrinsic sparsity of texture refinement, and the spectral variance across geomet
ries. To address this, we present **Fast-SAM3D**, a training-free framework that
 dynamically aligns computation with instantaneous generation complexity. Our ap
proach integrates three heterogeneity-aware mechanisms: (1) *Modality-Aware Step
 Caching* to decouple structural evolution from sensitive layout updates; (2) *J
oint Spatiotemporal Token Carving* to concentrate refinement on high-entropy reg
ions; and (3) *Spectral-Aware Token Aggregation* to adapt decoding resolution. E
xtensive experiments demonstrate that Fast-SAM3D delivers up to **2.67$\times$**
 end-to-end speedup with negligible fidelity loss, establishing a new Pareto fro
ntier for efficient single-view 3D generation.
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Transformer Circuits Can Realize Clustering Algorithms
Kenneth L. Clarkson, Lior Horesh, Takuya Ito, Charlotte Park, Parikshit Ram
Although transformers are most commonly optimized as statistical sequence models
, it is unclear to what extent they can implement and learn exact algorithmic co
mputations. Here, we specify a transformer implementation from first principles
that executes a fundamental and widely used method for k-means clustering: Llo
yd's algorithm. We theoretically prove and empirically demonstrate that this imp
lementation of a transformer architecture, which we term the _k-means transfor
mer_, exactly implements Lloyd's algorithm for k-means clustering using the st
andard circuit mechanisms of modern transformers: attention block, residual conn
ections, and feed-forward block. In learning experiments, we find that training
this base architecture on k-means clustering yields a generalizable clustering
 algorithm that surpasses Lloyd's algorithm in terms of clustering quality. Fina
lly, we demonstrate that interpretable alterations (e.g., inclusion of layer nor
malizations) to this architecture yields diverse and novel variants of clusterin
g algorithms, including soft k-means, spherical k-means, trimmed k-means.
Overall, our results show that transformer circuit mechanisms can instantiate ex
act algorithmic routines for clustering, while simultaneously providing an effec
tive learnable model.
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The Abstraction Gap in Vision-Language Causal Reasoning
Chinh Hoang, Mohammad Rashedul Hasan
Vision-language models (VLMs) generate fluent causal explanations, but current e
valuations cannot distinguish linguistic plausibility from faithful causal reaso
ning. We introduce a dual-probe methodology that isolates these properties. The
Text-Only Probe measures linguistic quality. The Chain-Text Probe requires model
s to first generate explicit causal chains. The Abstraction Gap (AG) metric quan
tifies the normalized performance difference. Evaluating eight VLMs on CAGE (Cau
sal Abstraction Gap Evaluation), a benchmark of 49,500 questions across 5,500 im

ages spanning Pearl's causal hierarchy, we find seven models exhibit AG exceedin g 0.50 with text scores of 6--8 but chain scores below 2.5. Fine-tuning on 45,00 0 chain-annotated examples fails to close the gap. However, one model achieves n ear-zero AG. The capability exists within current VLM architectures and depends on pretraining and architectural choices. CAGE provides a diagnostic tool for as sessing faithful causal reasoning in VLMs.
****

A Penalty Approach For Differentiation Through Black-box Quadratic Programming S olvers
Yuxuan Linghu, Zhiyuan Liu, Qi Deng
Differentiating through the solution of a quadratic program (QP) is a central pr oblem in differentiable optimization. Most existing approaches differentiate thr ough the Karush--Kuhn--Tucker (KKT) system, but their computational cost and num erical robustness can degrade at scale. To address these limitations, we propose dXPP, a penalty-based differentiation framework that decouples QP solving from differentiation. In the solving step (forward pass), dXPP is solver-agnostic and can leverage any black-box QP solver. In the differentiation step (backward pas s), we map the solution to a smooth approximate penalty problem and implicitly d ifferentiate through it, requiring only the solution of a much smaller linear sy stem in the primal variables. This approach bypasses the difficulties inherent i n explicit KKT differentiation and significantly improves computational efficien cy and robustness. We evaluate dXPP on various tasks, including randomly generat ed QPs, large-scale sparse projection problems, and a real-world multi-period po rtfolio optimization task. Empirical results demonstrate that dXPP is competitiv e with KKT-based differentiation methods and achieves substantial speedups on la rge-scale problems. Our implementation is open source and available at https://g ithub.com/mmmmmmlinghu/dXPP.
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FPTQuant: Function-Preserving Transforms for LLM Quantization
Boris van Breugel, Yelysei Bondarenko, Paul N. Whatmough, Markus Nagel
Large language models (LLMs) require substantial compute, and thus energy, at in ference time. While quantizing weights and activations is effective at improving efficiency, naive quantization of LLMs can significantly degrade performance du e to large magnitude outliers. This paper describes FPTQuant, which introduces t hree novel, lightweight, and expressive function-preserving transforms (FPTs) to facilitate quantization of transformers: (1) a mergeable pre-RoPE transform for queries and keys, (2) a mergeable transform for values, (3) a cheap, dynamic sc aling transform. By leveraging the equivariances and independencies inherent to canonical transformer operation, we designed these FPTs to maintain the model’s function while shaping the intermediate activation distributions to be more quan tization friendly. FPTQuant requires no custom kernels and adds virtually no ove rhead during inference. The FPTs are trained both locally to reduce outliers, an d end-to-end such that the outputs of the quantized and full-precision models ma tch. FPTQuant enables static INT4 quantization with minimal overhead and shows S OTA speed-up of up to 3.9x over FP. Empirically, FPTQuant has an excellent accur acy-speed trade-off—it is performing on par or exceeding most prior work and onl y shows slightly lower accuracy compared to a method that is up to 29% slower.
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Heterogeneous Customizable Personalized Federated Fine-Tuning Approach for Larg e Language Models
xin tong, Baojiang cui
Personalized federated LoRA fine tuning has become a key approach to addressing data heterogeneity in distributed fine tuning of large language models (LLMs). E xisting methods typically assume homogeneous personalization needs across client s, relying on dual LoRA or personalized calibration schemes. However, they fail to account for the heterogeneity of local personalization requirements and the c onflicting optimization objectives in dual LoRA, limiting scalability and perfor mance. To address this, we propose Het-CPFLoRA, a customizable heterogeneous fe derated LoRA fine tuning algorithm inspired by the decoupling properties of LoRA parameters. We employ a single adapter fine tuning scheme to mitigate conflicts

between personalized and generalized optimization, decouple LoRA into generalized and personalized subspaces for local customization, and use SVD compression to integrate cross client generalized knowledge. During inference, we introduce an OOD oriented dynamic mechanism to adjust the weighting between personalized and generalized decoupling knowledge, improving performance on user data. Extensive experiments on two public benchmark datasets show that Het-CPFLoRA outperforms state of the art methods in both personalization and generalization across heterogeneous scenarios. The code will be released as an open-source project.
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Restoring Exploration after Post-Training: Latent Exploration Decoding for Large Reasoning Models
Wenhui Tan, Fiorenzo Parascandolo, Enver Sangineto, Jianzhong Ju, Zhenbo Luo, Qian Cao, Rita Cucchiara, Ruihua Song, Jian Luan
Large Reasoning Models (LRMs) have recently achieved strong mathematical and code reasoning performance through Reinforcement Learning (RL) post-training.
However, we show that modern reasoning post-training induces an unintended exploration collapse: temperature-based sampling no longer increases pass@n accuracy.
Empirically, the final-layer posterior of post-trained LRMs exhibit sharply reduced entropy, while the entropy of intermediate layers remains relatively high.
Motivated by this entropy asymmetry, we propose Latent Exploration Decoding (LED), a depth-conditioned decoding strategy.
LED aggregates intermediate posteriors via cumulative sum and selects depth configurations with maximal entropy as exploration candidates.
Without additional training or parameters, LED consistently improves pass@1 and pass@16 accuracy by 0.61 and 1.03 percentage points across multiple reasoning benchmarks and models.
Relevant code is included in the supplementary material and will made be fully public after this paper is accepted.
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DR-MMSearchAgent: Deepening Reasoning in Multimodal Search Agents
Shengqin Wang, Wentao Yan, Huichi Zhou, Yihang Chen, Kun Shao, Zhizhong Zhang, Yuan Xie
Agentic multimodal models have garnered significant attention for their ability to leverage external tools to tackle complex tasks. However, it is observed that such agents often meet premature interaction collapse, caused by two primary reasons: 1) the terminal reward often appending on the last token prevents the advantage from distinguishing trajectories with exploratory behavior; 2) excessively redundant context hinders the agent from absorbing useful feedback. To address these issues, we propose the Deepening Reasoning MMSearchAgent, the framework leverages the structural proximity to derive advantage signals from the whole rollout trajectories in an entire batch, such that trajectories of different lengths are further encouraged to be generated, even when containing the same correct answer. Additionally, differentiated gaussian rewards are employed to dynamically calibrate interaction tolerance, thereby ensuring information reliability and reducing redundancy. To support multi-turn interaction training, we have constructed a multi-step deep-reasoning dataset including 3602 high-quality QA pairs with at least 3 reasoning steps. Extensive experiments demonstrate that our method achieves state-of-the-art performance, outperforming the MMSearch-R1 by 8.4\% on FVQA-test.
****
Expo-GS: Exposure-Aware Signed Distance Function in Gaussian Splatting for High Dynamic Range
Chaoda Song, Yiren Lu, Xinpeng Li, Yunlai Zhou, Yanyan Zhang, Yu Yin, Vipin Chaudhary
High dynamic range novel view synthesis (HDR-NVS) remains challenged by geometric artifacts and radiometric distortions under multi-exposure conditions, primarily due to existing methods ignoring exposure and over-relying on color cues. Inspired by the integrated processing of color and structure of the human visual system (HVS), we propose Expo-GS, a novel framework that decomposes HDR-NVS into t

hree interpretable components, namely, Irradiance Field Training, Geometry Fiel d Training, and Interactive Joint Training. Central to Expo-GS is the exposure-a ware signed distance function (Expo-SDF), which dynamically reweights geometric supervision via localized exposure reliability estimation, suppressing noisy gra dients from unstable regions while enhancing structure learning in well-exposed areas. Building on this, we design an interactive optimization strategy that syn chronizes Gaussian primitive growth and pruning with evolving Expo-SDF cues, ena bling exposure-aware density control and eliminating hallucinated structures nea r exposure transitions. Experiments show that Expo-GS significantly outperforms prior methods on both synthetic and real-world datasets. It achieves a peak PSNR of 39.06 dB under HDR settings and up to 41.38 dB in the LDR-OE configuration, excelling in preserving high-frequency textures and maintaining structural consi stency.

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BioAgent Bench: An AI Agent Evaluation Suite for Bioinformatics

Dionizije Fa, Marko ■uljak, Bruno Pandža, Mateo ■upi■

We introduce BioAgent Bench, an evaluation suite designed for measuring the perf ormance and robustness of AI agents in common bioinformatics tasks. The suite co nsists of manually curated end-to-end tasks (e.g., RNA-seq, variant calling, met agenomics) accompanied by task-specific prompts and concrete output artifacts to support automated assessment. We evaluate frontier closed- and open-weight mode ls across multiple agent harnesses, and use an LLM-based grader to score pipelin e progress and outcome validity. We find that agents based on frontier LLMs can complete multi-step bioinformatics pipelines without elaborate custom scaffoldin g, often producing the requested final artifacts reliably. However, robustness t ests reveal failure modes under controlled perturbations (corrupted inputs, deco y files, and prompt bloat), indicating that correct high-level pipeline construc tion does not guarantee reliable step-level reasoning. Finally, bioinformatics w orkflows often involve sensitive patient data or unpublished intellectual proper ty, thereby making the use of cost-effective yet reliable local agents an impera tive. Therefore, by releasing the code and the complementary resources comprisin g our suite, we aim to accelerate the development of such privacy-preserving age nts.

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A Sketch-and-Project Analysis of Subsampled Natural Gradient Algorithms

Gil Goldshlager, Jiang Hu, Lin Lin

Subsampled natural gradient descent (SNG) has been used to enable high-precision scientific machine learning, but standard analyses based on stochastic precondi tioning fail to provide insight into realistic small-sample settings. We overcom e this limitation by instead analyzing SNG as a sketch-and-project method. Motiv ated by this lens, we discard the usual theoretical proxy which decouples gradie nts and preconditioners using two independent mini-batches, and we replace it wi th a new proxy based on squared volume sampling. Under this new proxy the expect ation of the SNG direction becomes equal to a preconditioned gradient descent st ep even in the presence of coupling, leading to (i) global convergence guarantee s when using a single mini-batch of any size, and (ii) an explicit characterizat ion of the convergence rate in terms of quantities related to the sketch-and-pro ject structure. These findings in turn yield new insights into small-sample sett ings, for example by suggesting that the advantage of SNG over SGD is that it ca n more effectively exploit spectral decay in the model Jacobian. We also extend these ideas to explain a popular structured momentum scheme for SNG, known as SP RING, by showing that it arises naturally from accelerated sketch-and-project me thods.

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SARSteer: Safeguarding Large Audio Language Models via Safe-Ablated Refusal Stee ring

Weilin Lin, Jianze Li, Hui Xiong, Li Liu

Large Audio–Language Models (LALMs) are becoming essential as a powerful multimo dal backbone for real-world applications. However, recent studies show that audi o inputs can more easily elicit harmful responses than text, exposing new risks

toward deployment. While safety alignment has made initial advances in LLMs and Large Vision–Language Models (LVLMs), we find that vanilla adaptation of these approaches to LALMs faces two key limitations: 1) LLM-based steering fails under audio input due to the large distributional gap between activations, and 2) prompt-based defenses induce over-refusals on benign-speech queries. To address these challenges, we propose **S**afe-**A**blated **R**efusal **Steer**ing (SARSteer), an effective inference-time defense framework for LALMs. Specifically, SARSteer leverages text-derived refusal steering to enforce rejection without manipulating audio inputs and introduces decomposed safe-space ablation to mitigate over-refusal. Extensive experiments demonstrate that SARSteer significantly improves harmful-query refusal while preserving benign responses, establishing a principled step toward safety alignment in LALMs. The codes and constructed datasets are released at https://github.com/linweiii/SARSteer.

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A Two-Tier Perspective on Inference-Time Parallelism in Multi-Agent LLM Systems

Zihan Xu, Haolin Tian, Hai Jiang

Large language model (LLM)-driven multi-agent systems typically require multiple model invocations and complex coordination during inference, and their execution strategies directly affect system accuracy, latency, and computational cost. Parallel execution provides a means to improve inference-time efficiency. From the perspective of inference-time execution, this paper models parallelism in multi-agent systems as two distinct levels of decision processes: Replica Parallelism, which explores multiple complete solution paths at the task level, and Structural Parallelism, which enables concurrent execution within a single solution path through task decomposition. However, the roles of different forms of parallelism and their interrelationships still lack systematic study in terms of unified organization and coordination. We therefore propose TIPEX, a controllable execution framework that unifies these two levels of parallelism and coordinates their roles within the inference process under a unified execution semantics while supporting systematic combinations and analyses of different parallel strategies and parameter configurations. Systematic experiments on the GAIA benchmark demonstrate that inference-time parallelism can significantly improve accuracy and reduce end-to-end latency at the cost of increased token consumption. Further analysis shows that Replica and Structural Parallelism exhibit complementary effects across task complexities, with tasks of intermediate difficulty benefiting most from their coordination, while overly aggressive parallel strategies do not necessarily yield better performance.

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Understanding Multimodal Learning: A Loss Landscape Smoothness Perspective

Jae-Jun Lee, Sung Whan Yoon

A surge of recent advancements has consistently highlighted the superiority of multimodal learning over unimodal approaches across a variety of tasks.

However, the theoretical foundations elucidating this advantage remain underexplored: existing theoretical analyses are often constrained by tight assumptions, and lack empirical validation.

In this paper, we link this gap by proposing a novel theoretical framework grounded in \textit{convolutional smoothing}, offering a new perspective on how multimodal learning contributes to a smoother loss landscape compared to unimodal learning.

Building upon this theoretical foundation, we introduce a simple yet effective distributional training approach based on stochastic modality pairing instead of fixed pairing; thus, further promoting flatter landscape via convolutional smoothing.

Our empirical results across various multimodal datasets demonstrate that multimodal models not only achieve better performance but also exhibit smoother loss landscape, which represent better robustness and generalization.

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Beyond Log Likelihood: Probability-Based Objectives for Supervised Fine-Tuning across the Model Capability Continuum

Gaotang Li, Ruizhong Qiu, Xiusi Chen, Heng Ji, Hanghang Tong

Supervised fine-tuning (SFT) is the standard approach for post-training large la
nguage models (LLMs), yet it often shows limited generalization. We trace this l
imitation to its default training objective: negative log likelihood (NLL). Whil
e NLL is classically optimal when training from scratch, post-training operates
in a different paradigm and could violate its optimality assumptions, where mode
ls already encode task-relevant priors and supervision can be long and noisy. In
 this work, we systematically study various probability-based objectives and cha
racterize when and why different objectives succeed or fail under varying condit
ions. Through comprehensive experiments and extensive ablation studies across 8
model backbones, 27 benchmarks, and 7 domains, we uncover a critical dimension t
hat governs objective behavior: the model-capability continuum. Near the model-s
trong end, prior-leaning objectives that downweight low-probability tokens (e.g.
, $-p$, $-p^{10}$, thresholded variants) consistently outperform NLL; toward the
 model-weak end, NLL dominates; in between, no single objective prevails. Our th
eoretical analysis further elucidates how objectives trade places across the con
tinuum, providing a principled foundation for adapting objectives to model capab
ility. The code is available at https://github.com/GaotangLi/Beyond-Log-Likeliho
od.
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Learning to Correct: Reinforcement Learning for Multi-Attempt Chain-of-Thought
Muhammed Emrullah Ildiz, Halil Alperen Gozeten, Ege Onur Taga, Samet Oymak
State-of-the-art reasoning models utilize long chain-of-thought (CoT) to solve i
ncreasingly complex problems using more test-time computation. In this work, we
explore a long CoT setting where the model makes up to K successive attempts at
solving a problem, in which each attempt is allowed to build on earlier ones aft
er the model receives a hard verifier feedback. This motivates RL methods that c
an harness per-attempt rewards by carefully weighting individual attempts. We st
udy optimizing the Verification@K reward (the model succeeds by the K-th attempt
) and show that naively weighing the attempts by their pass/fail results in bias
ed gradients. We introduce Calibrated Attempt-Level (CAL) GRPO by devising a wei
ghing strategy to obtain unbiased gradients while maintaining small variance. Ou
r theory reveals how incorporating per-attempt rewards influences the training a
nd the eventual Verification@K performance. Experiments, baselines, and ablation
s on synthetic and real data corroborate our theory and the benefits of CAL-GRPO
 over vanilla GRPO as well as naive weighting.
****

Differential syntactic and semantic encoding in LLMs
Santiago Acevedo, Alessandro Laio, Marco Baroni
We study how syntactic and semantic information is encoded in inner layer repres
entations of Large Language Models (LLMs), focusing on the very large DeepSeek-V
3. We find that, by averaging hidden-representation vectors of sentences sharing
 syntactic structure or meaning, we obtain vectors that capture a significant pr
oportion of the syntactic and semantic information contained in the representati
ons. In particular, subtracting these syntactic and semantic ``centroids'' from
sentence vectors strongly affects their similarity with syntactically and semant
ically matched sentences, respectively, suggesting that syntax and semantics are
, at least partially, linearly encoded. We also find that the cross-layer encodi
ng profiles of syntax and semantics are different, and that the two signals can
to some extent be decoupled, suggesting differential encoding of these two types
 of linguistic information in LLM representations.
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LALM-as-a-Judge: Benchmarking Large Audio-Language Models for Safety Evaluation
in Multi-Turn Spoken Dialogues
Amir Ivry, Shinji Watanabe
Evaluation of socially unsafe content in spoken dialogues remains text-centric,
missing prosody and transcription failures. We present LALM-as-a-Judge, which in
cludes an open benchmark of 24,000 multi-turn spoken dialogues with one localize
d unsafe turn, generated out of 8 socially unsafe categories and 5 severity leve
ls. We evaluate 6 large audio-language models (LALMs) as judges, open and closed
-source, in text-only, audio-only, and multimodal setups by their sensitivity, s

everity-order specificity, and turn-position bias for socially harmful content i
n the dialogue. Results show that audio contributes non-lexical evidence beyond
transcript semantics and that multimodal gains are not universal but can be text
-anchored, balanced, conservative, and interfering, which we link to the audio p
athway bottlenecks and fusion limits. We position the benchmark as diagnostic an
d derive practitioner guidance for model, modality, and prompts choices.
****

Rethinking KV Cache Eviction via a Unified Information-Theoretic Objective
Jiaming Yang, Chenwei Tang, Liangli Zhen, Jiancheng Lv
Key–value (KV) caching is essential for large language model inference, yet its
memory overhead poses a critical bottleneck for long-context generation. Existin
g eviction policies predominantly rely on empirical heuristics, lacking a rigoro
us theoretical foundation. This work rethinks KV cache eviction through the lens
 of the Information Bottleneck principle. Under a linear–Gaussian surrogate of a
ttention, we derive a closed-form mutual information objective that characterize
s the effective information capacity of a retained KV cache subset. This formula
tion reveals that a wide range of existing eviction strategies can be interprete
d as different approximations of the same capacity-maximization principle. Guide
d by this insight, we introduce CapKV, a capacity-aware eviction method that dir
ectly targets information preservation via a log-determinant approximation using
 statistical leverage scores. This approach replaces heuristic selection with a
theoretically grounded mechanism that preserves the maximum predictive signal. E
xtensive experiments across multiple models and long-context benchmarks show tha
t CapKV consistently outperforms prior methods, achieving a better trade-off bet
ween memory efficiency and generational fidelity.
****

XSpecMesh: Quality-Preserving Auto-Regressive Mesh Generation Acceleration via M
ulti-Head Speculative Decoding
Dian Chen, Yansong Qu, Xinyang Li, Ming Li, Shengchuan Zhang
Current auto-regressive models can generate high-quality, topologically precise
meshes; however, they necessitate thousands—or even tens of thousands—of next-to
ken predictions during inference, resulting in substantial latency. We introduce
 XSpecMesh, a quality-preserving acceleration method for auto-regressive mesh ge
neration models. XSpecMesh employs a lightweight, multi-head speculative decodin
g scheme to predict multiple tokens in parallel within a single forward pass, th
ereby accelerating inference. We further propose a verification and resampling s
trategy: the backbone model verifies each predicted token and resamples any toke
ns that do not meet the quality criteria. In addition, we propose a distillation
 strategy that trains the lightweight decoding heads by distilling from the back
bone model, encouraging their prediction distributions to align and improving th
e success rate of speculative predictions. Extensive experiments demonstrate tha
t our method achieves a $1.7\times$ speedup without sacrificing generation quali
ty. Our code will be released.
****

SMART: Scalable Mesh■free Aerodynamic Simulations from Raw Geometries using a Tr
ansformer■based Surrogate Model
Jan Hagnberger, Mathias Niepert
Machine learning–based surrogate models have emerged as more efficient alternati
ves to numerical solvers for physical simulations over complex geometries, such
as car bodies. Many existing models incorporate the simulation mesh as an additi
onal input, thereby reducing prediction errors. However, generating a simulation
 mesh for new geometries is computationally costly. In contrast, mesh-free metho
ds, which do not rely on the simulation mesh, typically incur higher errors. Mot
ivated by these considerations, we introduce SMART, a neural surrogate model tha
t predicts physical quantities at arbitrary query locations using only a point-c
loud representation of the geometry, without requiring access to the simulation
mesh. The geometry and simulation parameters are encoded into a shared latent sp
ace that captures both structural and parametric characteristics of the physical
 field. A physics decoder then attends to the encoder's intermediate latent repr
esentations to map spatial queries to physical quantities. Through this cross-la

yer interaction, the model jointly updates latent geometric features and the evo lving physical field. Extensive experiments show that SMART is competitive with and often outperforms existing methods that rely on the simulation mesh as input , demonstrating its capabilities for industry-level simulations.
****
AC-ODM: Actor–Critic Online Data Mixing for Sample-Efficient LLM Pretraining
Jing Ma, Chenhao Dang, Mingjie Liao
Optimizing pretraining data composition is pivotal for LLM generalization. While dynamic mixing outperforms static strategies by capturing evolving training dyn amics, current methods fail to reconcile computational efficiency with sample ef ficiency and structural flexibility for diverse pipelines.We introduce \textbf{A ctor--Critic Online Data Mixing (AC-ODM)}, which approaches data mixing from a r einforcement learning perspective with a parameterized policy that we theoretica lly prove to act as a dynamic linear surrogate maximizing the constructive inter ference of gradients. To enhance practical flexibility, AC-ODM supports two oper ational modes: (i) a \textbf{proxy mode} for fixed, pre-prepared corpora, where a policy learned on a small model is transferred to a larger target; and (ii) a \textbf{non-proxy mode} for direct end-to-end training from scratch without prio rs. Empirically, AC-ODM significantly outperforms prior methods in convergence s peed and downstream accuracy across various architectures. On Pythia-1B, it reac hes optimal validation perplexity using up to 66\% fewer training steps than com petitive baselines, delivering a 27.5\% relative improvement in MMLU accuracy an d a 2.23$\times$ higher pass@1 on HumanEval, all while incurring a virtually neg ligible ($~$0.4\%) per-step wall-clock increase and only 2\% additional memory o verhead. Code is available at \url{https://github.com/DANG-ai/AC-ODM}.
****
AutoSizer: Automatic Sizing of Analog and Mixed-Signal Circuits via Large Langua ge Model (LLM) Agents
Xi Yu, Dmitrii Torbunov, Soumyajit Mandal, Yihui Ren
The design of Analog and Mixed-Signal (AMS) integrated circuits remains heavily reliant on expert knowledge, with transistor sizing a major bottleneck due to no nlinear behavior, high-dimensional design spaces, and strict performance constra ints. Existing Electronic Design Automation (EDA) methods typically frame sizing as static black-box optimization, resulting in inefficient and less robust solu tions. Although Large Language Models (LLMs) exhibit strong reasoning abilities, they are not suited for precise numerical optimization in AMS sizing. To addres s this gap, we propose \textsc{AutoSizer}, a reflective LLM-driven meta-optimiza tion framework that unifies circuit understanding, adaptive search-space constru ction, and optimization orchestration in a closed loop. It employs a two-loop op timization framework, with an inner loop for circuit sizing and an outer loop th at analyzes optimization dynamics and constraints to iteratively refine the sear ch space from simulation feedback. We further introduce \textsc{AMS-SizingBench} , an open benchmark comprising 24 diverse AMS circuits in SKY130 CMOS technology , designed to evaluate adaptive optimization policies under realistic simulator- based constraints. \textsc{AutoSizer} experimentally achieves higher solution qu ality, faster convergence, and higher success rate across varying circuit diffic ulties, outperforming both traditional optimization methods and existing LLM-bas ed agents.
****
Optimizing Rank for High-Fidelity Implicit Neural Representations
Julian McGinnis, Florian A. Hölzl, Suprosanna Shit, Florentin Bieder, Paul Fried rich, Mark Mühlau, Bjoern Menze, Daniel Rueckert, Benedikt Wiestler
Implicit Neural Representations (INRs) based on vanilla Multi-Layer Perceptrons (MLPs) are widely believed to be incapable of representing high-frequency conten t. This has directed research efforts towards architectural interventions, such as coordinate embeddings or specialized activation functions, to represent high- frequency signals. In this paper, we challenge the notion that the low-frequency bias of vanilla MLPs is an intrinsic, architectural limitation to learn high-fr equency content, but instead a symptom of stable rank degradation during trainin g. We empirically demonstrate that regulating the network’s rank during training

substantially improves the fidelity of the learned signal, rendering even simple MLP architectures expressive. Extensive experiments show that using optimizers like Muon, with high-rank, near-orthogonal updates, consistently enhances INR architectures even beyond simple ReLU MLPs. These substantial improvements hold across a diverse range of domains, including natural and medical images and novel view synthesis, with up to +9 dB PSNR over the samearchitecture. Code is available [here](https://rank-inrs.github.io/).

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Extending Fair Null-Space Projections for Continuous Attributes to Kernel Methods

Felix Störck, Fabian Hinder, Barbara Hammer

With the on-going integration of machine learning systems into the everyday social life of millions the notion of fairness becomes an ever increasing priority in their development.
Fairness notions commonly rely on protected attributes to assess potential biases.
Here, the majority of literature focuses on discrete setups regarding both target and protected attributes.
The literature on continuous attributes especially in conjunction with regression - we refer to this as *continuous fairness* - is scarce.
A common strategy is iterative null-space projection which as of now has only been explored for linear models or embeddings such as obtained by a non-linear encoder.
We improve on this by extending this to kernel induced feature spaces by means of the ``empirical feature space''.
We theoretically derive this as a direct transformation of the kernel matrix yielding a model and fairness-score agnostic method applicable to continuous protected attributes.
We demonstrate that our novel approach in conjunction with Support Vector Regression (SVR) provides competitive or improved performance across multiple datasets in comparison to other contemporary methods.

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Keep Everyone Happy: Online Fair Division of Numerous Items with Few Copies

Arun Verma, Indrajit Saha, Makoto Yokoo, Bryan Kian Hsiang Low

This paper considers a novel variant of the online fair division problem involving multiple agents in which a learner sequentially observes an indivisible item that must be irrevocably allocated to one of the agents to achieve a desired balance between fairness and efficiency. Existing algorithms assume a small number of items with a sufficiently large number of copies, which ensures a good utility estimation for all item-agent pairs from noisy observed utilities. However, this assumption may not hold in many real-life applications, e.g., an online platform with a large number of users (items) who use the platform's service providers (agents) only a few times (a few copies of items), making it difficult to accurately estimate utilities for all item-agent pairs. To address this limitation, we assume utility is an unknown function of item-agent features. We propose algorithms that model online fair division as a contextual bandit problem, achieving provable sub-linear regret. Our experimental results further validate the effectiveness of the proposed algorithms. The code is publicly available in this [GitHub repository](https://github.com/arunv3rma/SEDMA.git).

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FedRot-LoRA: Mitigating Rotational Misalignment in Federated LoRA

Haoran Zhang, Dongjun Kim, Seohyeon Cha, Haris Vikalo

Federated LoRA provides a communication-efficient mechanism for fine-tuning large language models on decentralized data. In practice, however, a discrepancy between the factor-wise averaging used to preserve low rank and the mathematically correct aggregation of local updates can cause significant aggregation error and unstable training. We argue that a major source of this problem is *rotational misalignment*, arising from the rotational invariance of low-rank factorizations -- semantically equivalent updates can be represented in different latent subspaces across clients since $(B_i R_i)(R_i^\top A_i) = B_i A_i$. When such misalig

ned factors are averaged directly, they interfere destructively and degrade the global update. To address this issue, we propose **FedRot-LoRA**, a federated LoRA framework that aligns client updates via orthogonal transformations prior to aggregation. This alignment preserves the semantic update while reducing cross-client subspace mismatch, without increasing communication cost or restricting model expressivity. We provide a convergence analysis that examines the aggregation error induced by factor-wise averaging and shows how rotational alignment yields a tighter upper bound on this error. Extensive experiments on natural language understanding and generative tasks demonstrate that FedRot-LoRA consistently outperforms existing federated LoRA baselines across a range of heterogeneity levels and LoRA ranks. The code is available at https://github.com/haoran-zh/FedRot-LoRA.

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DSGym: A Standardized and Holistic Framework for Evaluating and Training Data Science Agents

Fan Nie, Junlin Wang, Harper Hua, Federico Bianchi, Yongchan Kwon, Zhenting Qi, Owen Queen, Shang Zhu, James Zou

Data science agents promise to accelerate discovery and insight-generation by turning data into executable analyses and findings. Yet existing data science benchmarks fall short due to fragmented evaluation interfaces that make cross-benchmark comparison difficult, narrow task coverage and a lack of rigorous data grounding. In particular, we show that a substantial portion of tasks in current benchmarks can be solved without using the actual data. To address these limitations, we introduce DSGym, a standardized framework for evaluating and training data science agents in self-contained execution environments. Unlike static benchmarks, DSGym provides a modular architecture that makes it easy to add tasks, agent scaffolds, and tools, positioning it as a live, extensible testbed. We curate DSGym-Tasks, a holistic task suite that standardizes and refines existing benchmarks via quality and shortcut solvability filtering. We further expand coverage with (1) DSBio: expert-derived bioinformatics tasks grounded in literature and (2) DSPredict: challenging prediction tasks spanning domains such as computer vision, molecular prediction, and single-cell perturbation. Beyond evaluation, DSGym enables agent training via execution-verified data synthesis pipeline. As a case study, we build a 2,000-example training set in DSGym that substantially improves a 4B mode on standardized analysis benchmarks. Overall, DSGym enables rigorous end-to-end measurement of whether agents can plan, implement, and validate data analyses in realistic scientific context.

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MathlibLemma: Folklore Lemma Generation and Benchmark for Formal Mathematics

Xinyu Liu, Zixuan Xie, Amir Moeini, Claire Chen, Shuze Daniel Liu, Yu Meng, Aidong Zhang, Shangtong Zhang

While the ecosystem of Lean and Mathlib has enjoyed celebrated success in formal mathematical reasoning with the help of large language models (LLMs),
the absence of many folklore lemmas in Mathlib remains a persistent barrier that limits Lean's usability as an everyday tool for mathematicians like LaTeX or Maple.

To address this, we introduce MathlibLemma, a modular LLM-based pipeline for automated folklore-lemma mining: the discovery, formalization, and proving of reusable intermediate facts that mathematicians often take for granted but that are not always present in formal libraries.

At its core, MathlibLemma proactively mines the missing connective tissue of mathematics.

The pipeline produces a verified library of folklore-style lemmas, including 1,506 Lean-checked proofs that pass a proof-bypass screen; a small curated pilot subset has also been merged into Mathlib, providing external evidence that selected outputs can meet expert library standards.

Leveraging this pipeline, we further construct the MathlibLemma benchmark, a suite of 4,028 non-trivial type-checked Lean statements spanning a broad range of mathematical domains.

By transforming the role of LLMs from passive consumers to active contributors,

this work takes a step toward AI-assisted expansion of formal mathematical libra
ries.
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Beyond Logits: Metastable Latent Dynamics for Sample-Efficient Best-of-N Selecti
on in LLMs
Xinrong Li, Zidong Zhou, Keyu Shen, Wenhao Zhou, Shangqi Guo
Best-of-N selection improves reasoning in large language models (LLMs) by alloca
ting test-time compute to sample candidate trajectories, but it relies on reliab
le verification. Widely used proxies have complementary failure modes: logit-con
fidence signals can suffer from calibration collapse, where confidence becomes m
isaligned with correctness, while sample agreement can be costly or brittle.

    Instead, we analyze the model's latent dynamics during inference. Motivated
by metastable dynamics in cognitive systems, we introduce Latent Velocity Entrop
y (LVE), a training-free metric that quantifies the temporal concentration of in
ternal representation updates.

    Experiments on four reasoning benchmarks, AIME25, GPQA, MATH500, and BRUMO25
, show that LVE-based selection remains informative when logit-confidence baseli
nes fail, with the clearest gains on reasoning-dense mathematical tasks and more
 modest gains on knowledge-heavy GPQA. On MATH500, LogNorm-LVE reaches 92.2\% Pa
ss@1 at $N=10$ and nearly matches 10-sample majority voting with only 3 samples.
 Cross-model, bootstrap, and length-controlled analyses indicate that LVE is not
 reducible to sequence length, while also clarifying the task-dependent limits o
f the signal. Code is available at https://github.com/ZopenDP/LatentVelocityEntr
opy.
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A Risk Decomposition Framework for Pre-hoc Fine-tuning Prediction
Yuxiang Luo, Chen Wang, Nan Tang
The high cost of fine-tuning LLMs poses a significant economic barrier; pre-hoc
performance prediction offers a critical solution to substantially reduce this e
xpense. However, the theoretical limits of pre-hoc performance prediction remain
 unexplored.
We formulate it as a stochastic estimation problem under information constraints
, decomposing prediction risk into two components: an \textbf{intrinsic limit} (
static data-model compatibility) and a \textbf{reducible optimization variance}.
 We prove that optimization variance admits a necessary lower bound on its decay
 rate, implying fundamental constraints on how quickly uncertainty dissipates, r
egardless of the predictor used. Based on these dynamics, we derive a budget-opt
imal probing principle and introduce a predictability phase diagram that organiz
es tasks into three distinct regimes: Static-Sufficient, Dynamic-Critical, and N
oise-Dominant. Extensive experiments on synthetic and real-world benchmarks vali
date these theoretical regimes and demonstrate the efficiency of our probing str
ategy.
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Op-CAD: Benchmarking and Investigating Operation-oriented CAD Generation
Yixue Bai, Yufei Gu, Zeke Xie
Recent research has made growing efforts to leverage large language models (LLMs
) for computer-aided design (CAD), a domain that demands advanced geometric and
spatial reasoning across long operation sequence. However, existing studies rema
in limited in addressing complex modeling tasks that necessitate step-by-step re
asoning, primarily due to the scarcity of high-quality CAD datasets and the abse
nce of fine-grained evaluation frameworks. In response to these challenges, we i
ntroduce Op-CAD, the first large-scale, multi-modal dataset for operation-orient
ed CAD generation, encompassing four operation types and five modalities. Furthe
rmore, we introduce a novel CAD parsing module together with a geometry-guided h
ierarchical annotation pipeline, which decomposes modeling sequences into discre
te operations and substantially improves the annotation accuracy of Vision-Langu
age Models (VLMs). Based on our dataset, we redefine the CAD modeling task by de
coupling geometric and spatial perspectives and introduce a novel metric, Chamfe
r/Fillet Intersection over Union (CF-IoU), to fill the void in assessing chamfer
 and fillet operations. By comprehensively evaluating eight LLMs on Op-CAD, we e

stablish a benchmark for current models on operation-oriented tasks. Finally, we investigate performance enhancement strategies through fine-tuning on Op-CAD and propose Chain-of-Operation (COOP), a novel prompting strategy that emulates human-engineer reasoning.
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MEDUSA: Motion Elimination in Diffusion Using Spectral Attack
Hongwei Yu, Daoqing Zha, Xinlong Ding, Jiawei Li, Junbao Zhuo, Qiankun Liu, Huimin Ma, Jiansheng Chen
With the widespread application of Video Diffusion Models (VDMs), video synthesis has achieved remarkable temporal dynamics.
Image-to-Video (I2V) generation allows users to provide reference images, which enables attackers to inject adversarial noise into these conditions.
Due to the robust spatio-temporal priors in VDMs, conventional frame-level attacks merely induce superficial artifacts and struggle to suppress the synthesis of motion semantics.
In this work, we approach the problem by exploring the underlying mechanism of temporal dynamics.
We reveal that the static video manifests as a temporal rank collapse, a degenerate state characterized by rank-1 degeneracy within the temporal attention matrix.
Guided by this insight, we propose Motion Elimination in Diffusion Using Spectral Attack (MEDUSA) to freeze the video.
It minimizes the nuclear norm of the attention matrix to induce the temporal rank collapse.
This objective circumvents the vanishing gradient problem encountered when directly imposing a rigid temporal mapping on the attention matrix.
Furthermore, we provide a mathematical analysis of this phenomenon and the gradient vanishing problem during the optimization.
Experiments confirm that MEDUSA achieves excellent performance and validates the effectiveness of spectral constraints.
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From Per-Image Low-Rank to Encoding Mismatch: Rethinking Feature Distillation in Vision Transformers
Huiyuan Tian, Bonan Xu, Shijian Li
Feature-map knowledge distillation (KD) transfers internal representations well between comparably sized Vision Transformers (ViTs), but it often fails in compression. We revisit this failure and uncover a paradox. Sample-wise SVD shows that each image is highly compressible, which seems to suggest that a narrow student with a linear projector should match the teacher "in principle". However, a dataset-level view contradicts this intuition: PCA shows that the teacher is a union of low-rank subspaces with significant subspace rotation across inputs. We further introduce token-level Spectral Energy Patterns (SEP) and find an architecture-invariant encoding law: tokens spread energy broadly across channel modes even when they live in low-rank subspace, creating a bandwidth mismatch. We refer to this combined phenomenon as an encoding mismatch. We propose two minimal remedies, Lift or WideLast: (i) Lift retains a lightweight lifting projector at inference to provide wider channel, or (ii) WideLast widens only the student’s last block, enabling an input-dependent expansion. On ImageNet-1K, these fixes revive feature KD for ViT compression, improving DeiT-Tiny distilled from CaiT-S24 from 74.86% to 77.53%/78.23% top-1 accuracy, and they also strengthen students trained without distillation. Our analyses clarify when and why feature-map KD fails and then how to fix it. Code and raw data are provided in https://github.com/thy960112/From-Per-Image-Low-Rank-to-Encoding-Mismatch.
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MolAlign3D: Enhancing Fixed-Dimensional E(3)-Equivariant Latent Space for High-Fidelity 3D Molecular Reconstruction and Editing
Zitao Chen, Jiatong Ji, Yinjun Jia, Wei-Ying Ma, Yanyan Lan
Recent advances in 3D molecular modeling have achieved high-fidelity structural synthesis, yet these models often lack an explicit and manipulable representation space. To address this, MolFLAE introduced a fixed-dimensional, E(3)-equivaria

nt latent space, providing a novel framework for molecular editing independent o
f atom counts. However, because its latent space was primarily optimized for geo
metric reconstruction, it remains semantically shallow and inadequate for compre
hensive representation learning. In this work, we propose **MolAlign3D**, which
evolves this architecture into a unified semantic-generative engine. By anchorin
g MolFLAE’s manipulable latents with embeddings from a pre-trained molecular enc
oder, we yield a manifold that is both semantically dense and geometrically prec
ise. Experiments show that MolAlign3D achieves high-fidelity molecular reconstru
ction and attains comparable performance on molecular property prediction benchm
arks. Notably, the integration of rich semantic priors significantly enhances ze
ro-shot molecular manipulation, including atom-number editing and latent-space i
nterpolation, outperforming prior fixed-dimensional equivariant latent baseline.
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Learn to Think: Improving Multimodal Reasoning through Vision-Aware Self-Improve
ment Training
Qihuang Zhong, Liang Ding, Wenjie Xuan, Juhua Liu, Bo Du, Dacheng Tao
Post-training with explicit reasoning traces is common to improve the reasoning
capabilities of Multimodal Large Language Models (MLLMs). However, acquiring hig
h-quality reasoning traces is often costly and time-consuming. Hence, the self-i
mprovement paradigm has emerged, enabling MLLMs to self-generate reasoning trace
s for training without external supervision. Despite its effectiveness, we revea
l two shortcomings in the self-improvement training of MLLMs: 1) data imbalance,
 where simple samples are over-trained, but the challenging yet crucial samples
are under-trained; 2) language prior bias, where MLLMs overly rely on linguistic
 priors while neglecting the visual cues. To this end, we propose VISTA, a VIsio
n-aware Self-improvement Training framework for enhancing the multimodal Reasoni
ng of MLLMs. Specifically, VISTA first introduces a prefix resampling strategy t
o reuse the partial correct reasoning traces for efficient data collection, and
then designs a vision-aware attention score to quantify the model’s focus on vis
ual information. Extensive experiments show that VISTA can be applied to various
 post-training scenarios, i.e., supervised fine-tuning and preference learning,
and effectively enhances the multimodal reasoning performance across various MLL
Ms and tasks, e.g., bringing up to +13.66% average performance gains for Qwen2.5
-VL-3B-Instruct.
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Towards Efficient LLMs Annealing with Principled Sample Selection
Yuanjian Xu, Jianing Hao, Wanbo Zhang, Zhong Li, Guang Zhang
The annealing phase is a pivotal convergence stage in LLM pre-training that ulti
mately determines final model quality. However, effectively selecting training d
ata during this phase remains a key challenge. Current strategies rely on empiri
cal heuristics, such as domain filtering or context extension, which lack a prin
cipled grounding in optimization theory. In this work, we characterize the annea
ling phase through the lens of the loss landscape's spectral geometry. We argue
that optimal convergence requires gradient updates to satisfy heterogeneous cons
traints across different eigen-directions. Building on this insight, we formulat
e data selection as a problem of satisfying these directional constraints. To th
is end, we propose **DiReCT** (**Di**rectionally-**Re**strained **C**onstrained
 Training), a novel framework that reformulates sample selection in the anne
aling stage as a constrained optimization problem. By imposing explicit directio
nal constraints on per-sample gradients based on the spectral properties of the
Hessian, **DiReCT** identifies samples that align with the optimal curvature-awa
re descent path. Extensive experiments across various model scales demonstrate t
hat **DiReCT** consistently achieves state-of-the-art performance. For future re
search, code is available at https://github.com/xuyj233/Direct.
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Hybrid-Gym: Training Coding Agents to Generalize Across Tasks
Yiqing Xie, Emmy Liu, Gaokai Zhang, Nachiket Kotalwar, Shubham Gandhi, Sathwik A
charya, Xingyao Wang, Carolyn Rose, Graham Neubig, Daniel Fried
When assessing the quality of coding agents, predominant benchmarks focus on sol
ving single issues on GitHub, such as SWE-Bench.

In contrast, in real use these agents solve more various and complex tasks that involve other skills such as exploring codebases, testing software, and designing architecture. In this paper, we first characterize some transferable skills that are shared across diverse tasks by decomposing trajectories into fine-grained components, and derive a set of principles for designing auxiliary training tasks to teach language models these skills. Guided by these principles, we propose a training environment, Hybrid-Gym, consisting of a set of scalable synthetic tasks, such as function localization and dependency search. Experiments show that agents trained on our synthetic tasks effectively generalize to diverse real-world tasks that are not present in training, improving a base model by 25.4% absolute gain on SWE-Bench Verified, 7.9% on SWT-Bench Verified, and 5.1% on Commit-0 Lite. Hybrid-Gym also complements datasets built for the downstream tasks (e.g., improving SWE-Play by 4.9\% on SWT-Bench Verified). We provide the code at https://github.com/Hybrid-Gym/Hybrid-Gym.

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Refining Context-Entangled Content Segmentation via Curriculum Selection and Anti-Curriculum Promotion

Chunming He, Rihan Zhang, Fengyang Xiao, Dingming Zhang, Zhiwen Cao, Sina Farsiu

Biological learning proceeds from easy to difficult tasks, gradually reinforcing perception and robustness. Inspired by this principle, we address Context■Entangled Content Segmentation (CECS)—a challenging setting where objects share intrinsic visual patterns with their surroundings, as in camouflaged object detection. Conventional segmentation networks predominantly rely on architectural enhancements but often ignore the learning dynamics that govern robustness under entangled data distributions.

We introduce CurriSeg, a dual■phase learning framework that unifies curriculum and anti■curriculum principles to improve representation reliability. In the Curriculum Selection phase, CurriSeg dynamically selects training data based on the temporal statistics of sample losses, distinguishing hard■but■informative samples from noisy or ambiguous ones, thus enabling stable capability enhancement. In the Anti■Curriculum Promotion phase, we design Spectral■Blindness Fine■Tuning, which suppresses high■frequency components to enforce dependence on low■frequency structural and contextual cues. Combined with Sharpness■Aware Minimization, this promotes convergence to flatter minima and strengthens generalization.

Extensive experiments demonstrate that CurriSeg achieves consistent improvements across diverse CECS benchmarks without adding parameters or increasing total training time, offering a principled view of how progression and challenge interplay to foster robust and context■aware segmentation.

Code will be released.

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BRIDGE: Predicting Human Task Completion Time From Model Performance

Fengyuan Liu, Jay Gala, Nilaksh, Dzmitry Bahdanau, Siva Reddy, Hugo Larochelle

Evaluating the real-world capabilities of AI systems requires grounding benchmark performance in human-interpretable measures of task difficulty. Existing approaches that rely on direct human task completion time annotations are costly, noisy, and difficult to scale across benchmarks. In this work, we propose BRIDGE, a unified psychometric framework that learns the latent difficulty scale from model responses and anchors it to human task completion time. Using a two-parameter logistic Item Response Theory model, we jointly estimate latent task difficulty and model capability from model performance data across multiple benchmarks. We demonstrate that latent task difficulty varies linearly with the logarithm of human completion time, allowing human task completion time to be inferred for new benchmarks from model performance alone. Leveraging this alignment, we forecast frontier model capabilities in terms of human task length and independently reproduce METR’s exponential scaling results, with the 50% solvable task horizon doubling approximately every 6 months.

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StructMamPose: From Sequential Perception to Structural Reasoning for 3D Human Pose Estimation

Jiahong Jiang, Miao Zhang, Jingjing Li, Leiye Liu, Yao Cui, Wei Ji, Yongri Piao,

Huchuan Lu
Accurately modeling complex temporal and topological dependencies and depth information is critical for monocular 3D human pose estimation, yet existing Mamba-based approaches struggle to fulfill these demands, suffering from internal state update confusion induced by forced sequence flattening and depth modeling confusion inherent to single-view observations. To address this confusion, we propose a StructMamPose framework equipped with Identity Anchoring Mechanism (IAM) and View Transformation Hub (VTH). The IAM injects spatiotemporal identities into the parameter generation network to anchor the selectivity of state update matrices, suppressing spurious connections to enforce feature propagation along valid topological dependencies. The VTH performs internal coordinate rotation to transform implicit depth inference into observable planar features, endowing the model with explicit spatial understanding and multi-view constraints. Experimental results demonstrate that our framework achieves SOTA performance on most datasets.
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Break the Block: Dynamic-size Reasoning Blocks for Diffusion Large Language Models via Monotonic Entropy Descent with Reinforcement Learning
Yan Jiang, Ruihong Qiu, Zi Huang
Recent diffusion large language models (dLLMs) have demonstrated both effectiveness and efficiency in reasoning via a block-based semi-autoregressive generation paradigm. Despite their progress, the fixed-size block generations remain a critical bottleneck for effective and coherent reasoning. (I) From a global perspective, different reasoning tasks would correspond to different optimal decoding block sizes, which makes a "one-size-fits-all" assumption ineffective. (II) Even within a single reasoning task, the rigid block partitioning would break the logical flow and reduce reasoning coherence. Through empirical observations, we reveal that, for block-wise entropy, incorrect reasoning exhibits a fluctuating and unsteady trend between blocks, while the correctly generated tasks follow a consistent descending paradigm. Therefore, this paper proposes b1, a novel post-training framework that learns dynamic-size reasoning blocks via a Monotonic Entropy Descent objective with reinforcement learning to enhance reasoning coherence. b1 integrates seamlessly as a plug-and-play module with existing dLLM's post-training algorithms. Extensive experiments across various reasoning benchmarks showcase b1's consistent improvement over fixed-size block baselines. Our code has been released at https://github.com/YanJiangJerry/Block-R1.
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POLCA: Stochastic Generative Optimization with LLM
Xuanfei Ren, Allen Nie, Tengyang Xie, Ching-An Cheng
Optimizing complex systems, ranging from LLM prompts to multi-turn agents, traditionally requires labor-intensive manual iteration. We formalize this as a stochastic generative optimization problem where a language model acts as the optimizer, guided by numerical and text feedback to discover the best system. We introduce **P**rioritized **O**ptimization with **L**ocal **C**ontextual **A**ggregation (POLCA), a scalable framework designed to handle stochasticity in optimization—such as noisy feedback, sampled minibatches, and stochastic system behaviors—while effectively managing the unconstrained expansion of the solution space. POLCA maintains a priority queue to enable exploration–exploitation, tracking candidate solutions and their evaluation histories. To enhance efficiency, we integrate an ε-Net mechanism to maintain parameter diversity and an LLM Summarizer to perform meta-learning across historical trials. We theoretically prove that POLCA converges to near-optimal candidate solutions under stochasticity. We evaluate our framework on diverse benchmarks, including τ-bench, HotpotQA, VeriBench, and KernelBench. Experimental results demonstrate that POLCA achieves robust, sample- and time-efficient performance, consistently outperforming state-of-the-art algorithms in both deterministic and stochastic problems. The codebase for this work is publicly available at https://github.com/rlx-lab/POLCA.
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Rare Event Analysis of Large Language Models
Jake McAllister Dorman, Edward Gillman, Dominic C Rose, Jamie F. Mair, Juan P. Garrahan

Being probabilistic models, during inference large language models (LLMs) displa
y *rare events*: behaviour that is far from typical but highly significant. By d
efinition all rare events are hard to see, but the enormous scale of LLM usage m
eans that events completely unobserved during development are likely to become p
rominent in deployment. Here we present an end-to-end framework for the systemat
ic analysis of rare events in LLMs. We provide a practical implementation spanni
ng theory, efficient generation strategies, probability estimation and error ana
lysis, which we illustrate with concrete examples. We outline extensions and app
lications to other models and contexts, highlighting the generality of the conce
pts and techniques presented here.
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Asymptotic Universal Alignment: A New Alignment Framework via Test-Time Scaling
Yang Cai, Weiqiang Zheng
Aligning large language models (LLMs) to serve users with heterogeneous and pote
ntially conflicting preferences is a central challenge for personalized and trus
tworthy AI. We formalize an ideal notion of *universal alignment* through *test-
time scaling*: for each prompt, the model produces $k\ge 1$ candidate responses
and a user selects their preferred one. We introduce *$(k,f(k))$-robust alignmen
t*, which requires the k-output model to have win rate $f(k)$ against any othe
r single-output model, and *asymptotic universal alignment (U-alignment)*, which
 requires $f(k)\to 1$ as $k\to\infty$. Our main result characterizes the optimal
 convergence rate: there exists a family of *single-output* policies whose k-s
ample product policies achieve U-alignment at rate $f(k)=\frac{k}{k+1}$, and no
method can achieve a faster rate in general.

We show that popular post-training methods, including Nash learning from human f
eedback (NLHF), can fundamentally underutilize the benefits of test-time scaling
. Even though NLHF is optimal for $k=1$, sampling from the resulting (often dete
rministic) policy cannot guarantee win rates above $\tfrac{1}{2}$ except for an
arbitrarily small slack. This stems from a lack of output diversity: existing al
ignment methods can collapse to a single majority-preferred response, making add
itional samples redundant. In contrast, our approach preserves output diversity
and achieves the optimal test-time scaling rate. In particular, we propose a fam
ily of symmetric *multi-player alignment games* and prove that any symmetric Nas
h equilibrium policy of the $(k+1)$-player alignment game achieves the optimal $
(k,\frac{k}{k+1})$-robust alignment. Finally, we provide theoretical convergence
 guarantees for self-play learning dynamics in these games and extend the framew
ork to opponents that also generate multiple responses.
****
Hyper-LLaVA: Hyperbolic Uncertainty-aware Modality-Balanced Routing for Multimod
al Continual Instruction Tuning
Kunlun Xu, YanQin Zhang, Wenwen Qiang, Jiahuan Zhou
Multimodal Continual Instruction Tuning (MCIT) aims to exploit the incrementally
 accumulated knowledge to process multimodal inputs of diverse tasks, where para
meter routing plays an important role. State-of-the-art methods rely on sample-t
o-task center similarity and cross-modal fusion with equal weight during routing
. However, such solutions face two fundamental flaws: (1) Within each modality,
the sample-to-task center distance is sub-optimal for routing since the abundant
 intra-task diversity information is underleveraged. (2) Different modalities ex
hibit varying reliability across tasks, where the modality with inter-task ambig
uity can easily misguide the routing result. To address these problems, we prop
ose Hyperbolic Uncertainty-aware Modality-Balanced Routing (Hyper-LLaVA) to impr
ove parameter routing capacity based on cross-modality task feature uncertainty
modeling. Specifically, to improve intra-modality task matching, Hyper-LLaVA acc
esses the sample to task distribution similarity in the Hyperbolic space. Beside
s, to alleviate the degradation brought by unreliable modality, Hyper-LLaVA quan
tifies the task matching ambiguity within each modality to achieve adaptive bala
ncing between task matching across modalities. Based on the complementary intra-
 and inter-modality task matching enhancement, our Hyper-LLaVA outperforms state
-of-the-art approaches by large margins. Our source code is available at https:/

/github.com/zhoujiahuan1991/ICML2026-Hyper-LLaVA
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TINNs: Time-Induced Neural Networks for Solving Time-Dependent PDEs
Chen-Yang Dai, Che-Chia Chang, Te-Sheng Lin, Ming-Chih Lai, Chieh-Hsin Lai
Physics-informed neural networks (PINNs) solve time-dependent partial differenti al equations (PDEs) by learning a mesh-free, differentiable solution that can be evaluated anywhere in space and time. However, standard space-time PINNs take t ime as an input but reuse a single network with shared weights across all times, forcing the same features to represent markedly different dynamics. This coupli ng degrades accuracy and can destabilize training when enforcing PDE, boundary, and initial constraints jointly. We propose *Time-Induced Neural Networks (TINNs)*, a novel architecture that parameterizes the network weights as a learned fun ction of time, allowing the effective spatial representation to evolve over time while maintaining shared structure.
The resulting formulation naturally yields a nonlinear least-squares problem, wh ich we optimize efficiently using a Levenberg-Marquardt method. Experiments on v arious time-dependent PDEs show up to $4\times$ improved relative L^2 error an d $10\times$ faster convergence compared to PINNs and strong baselines. Code is available at https://github.com/CYDai-ml/TINN.
****
Target-Driven Policy Optimization for Sequential Counterfactual Outcome Control
Xin Wang, Xiangyu Zhang, Shengfei Lyu, Huanhuan Chen
Identifying optimal intervention sequences from offline data to guide temporal s ystems toward target outcomes is a critical challenge with profound implications for fields like personalized medicine. While existing methods are mostly evalua ted in offline settings, practical applications demand online, adaptive strategi es that can respond in real-time. To address this, we propose \textbf{G}oal-cond itioned \textbf{I}ntervention via \textbf{F}actual-\textbf{T}argeted Training (\ textbf{GIFT}), a novel framework for learning sequential intervention policies f rom observational data. GIFT learns a goal-conditioned policy by rescaling rewar ds with clipped importance weights, stabilizing learning and steering toward the target. Under standard assumptions, the induced operator has a unique fixed poi nt and our procedure converges to it. We also bound the bias from clipping and a pproximation via the gap to the policy’s true value. Experiments show GIFT signi ficantly outperforms existing methods in creating goal-conditioned policies for online deployment.
****
Two-Parameter Flows for Learning Population Dynamics of Physical Systems
Paul Schwerdtner, Tobias Blickhan, Benjamin Peherstorfer
This work addresses the problem of learning the dynamics of high-dimensional pro bability densities over time using unlabeled samples, without assuming access to trajectory information. We introduce two-parameter flows that learn only sampli ng-time transports from a base distribution to each marginal and then extract a physics-time velocity by regressing on coupled synthetic trajectories. We prove that the resulting physics-time dynamics are unique and inherit regularity from the sampling-time transports. Because we can build on standard, well-developed c onditional flow matching techniques for learning the base-to-marginal transports , our approach scales to high dimensions and avoids per-step optimal-transport c ouplings, while allowing admissible non-gradient dynamics that can naturally exp lain rotational or circulating physics phenomena.
****
Boosting World Models Learning via Latent-Space Value Alignment
Xingyu Jiang, Yuheng Pan, Mukang You, Xiuhui Zhang, Ning Gao, Guanwei Yan, Hao L i, Yue Deng
Model-based reinforcement learning aims to construct world models for efficient sampling. Current mainstream algorithms can be broadly categorized into two para digms: maximum likelihood and value-aware world models. The former employs struc tured Recurrent/Transformer State-Space Models to capture environmental dynamics but overlooks task-relevant features. The latter prioritizes decision-critical states but suffers from sub-optimal performance. While recent efforts have sough

t to integrate these approaches, they typically rely on auxiliary modules or hea vy external priors that significantly increase computational complexity. In this work, we propose a Value-Aligned World Model, a minimalist framework designed t o synergize these two paradigms with negligible overhead. Specifically, We intro duce an intrinsic latent-space value-alignment regularization that compels the w orld model to prioritize task-relevant features while maintaining the structural integrity of stochastic dynamics. To ensure stable optimization, we develop an adaptive weighting mechanism that acts as a self-regulating curriculum, balancin g reconstruction fidelity with decision-making utility. Extensive experiments on Atari 100k and DeepMind Control benchmarks demonstrate that our algorithm consi stently boosts existing methods with minimal added code and computational overhe ad. Code is available at supplementary material.
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τ-Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains
Soham Ray, Keshav Dhandhania, Victor Barres, Karthik R Narasimhan
Full-duplex voice agents—systems that listen and speak simultaneously—are rapidl y moving from research to production. However, existing evaluations address conv ersational dynamics and task completion in isolation. We introduce τ-Voice, a be nchmark for evaluating voice agents on grounded tasks with real-world complexity : agents must navigate complex multi-turn conversations, adhere to domain polici es, and interact with the environment. The framework extends τ^2-bench into a nov el voice agent benchmark combining verifiable completion of complex grounded tas ks, full-duplex interaction, and realistic audio—enabling direct comparison betw een voice and text performance. A controllable and realistic voice user simulato r provides diverse accents, realistic audio environments, and rich turn-taking d ynamics; by decoupling simulation from wall-clock time, the user simulator can u se the most capable LLM without real-time constraints. We evaluate task completi on (pass@1) and voice interaction quality across 278 tasks: while GPT-5 (reasoni ng) achieves 85%, voice agents reach only 31–51% under clean conditions and 26–3 8% under realistic conditions with noise and diverse accents—retaining only 30–4 5% of text capability; qualitative analysis confirms 79–90% of failures stem fro m agent behavior, suggesting that observed failures primarily reflect agent beha vior under our evaluation setup. τ-Voice provides a reproducible testbed for mea suring progress toward voice agents that are natural, conversational, and reliab le.
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Trust It or Not: Evidential Uncertainty for Feed-Forward 3D Reconstruction with Trust3R
Zihao Zhu, Wenyuan Zhao, Nuo Chen, Chao Tian, Zhiwen Fan
Geometric foundation models hold promise for unconstrained dense geometry predic tion from uncalibrated images. However, current feed-forward designs often produ ce heuristic confidence scores that lack probabilistic interpretation and fail t o indicate where and how much the predicted geometry can be trusted. To address this gap, we present ***Trust3R***, a lightweight evidential uncertainty framewo rk for feed-forward 3D reconstruction. Trust3R combines gated residual mean refi nement with a Normal-Inverse-Wishart evidential head, yielding a closed-form mul tivariate Student-t distribution for per-point geometric uncertainty. This provi des probabilistically grounded pointmap uncertainty estimates with moderate infe rence overhead.
We evaluate on diverse indoor and outdoor benchmarks and compare against MASt3R 's built-in confidence map, single-pass heteroscedastic regression, MC dropout, and deep ensembles. Experimental results show that Trust3R consistently improves risk--coverage and sparsification, generally improves geometric accuracy, and s trengthens uncertainty ranking across benchmarks. On ScanNet++, Trust3R achieves 25% lower AURC and 41% lower AUSE, providing a practical reliability signal for uncertainty-aware weighting in downstream geometry pipelines. Our project page and code are available at https://trust3r-z.github.io.
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Causal discovery for time series with endogenous context variables
Oana-Iuliana Popescu, Wiebke Günther, Martin Rabel, Jakob Runge

Many real-world systems exhibit both context- and time-dependent causal dynamics , where the dynamical system state also influences its context. For instance, so il moisture is driven by precipitation, yet also provides the context for heat-f lux realization. We capture such dynamics in Structural Causal Models (SCMs) by introducing endogenous and time-dependent discrete context variables, also allow ing for possibly lagged dependencies with the system variables. While context va riables are discrete, they may also be proxies of continuous variables. The enab ling assumptions for causal discovery of our model are either persistence of the context or sparsity of the context–system dependencies. We design two new PCMCI -based algorithms for causal discovery with endogenous context variables for tim e series and prove their soundness. A systematic evaluation on synthetic benchma rks and an application to a real-world land-atmosphere feedback problem demonstr ate their effectiveness and applicability.

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COPF: An Online Framework for Deployment-Stable Counterfactual Fairness in Evolv ing Graphs

Sheng'en Li, Dongmian Zou

Online link recommendation on evolving graphs is performative: by choosing which candidate links to show users, the system changes which links form and what fee dback it later observes. Consequently, fairness estimates from logged outcomes c an be misleading and may drift after deployment when the recommendation policy i s updated. We introduce COPF (Counterfactual Online Performative Fairness), a de cision-layer framework for deployment-stable fairness monitoring and control in online link recommendation. COPF (i) defines group-level opportunity gaps over e xposure (shown vs. not shown) counterfactuals, (ii) makes them estimable by expl icit exploration and by logging the probability (propensity) that each candidate is shown, and (iii) audits and controls fairness using residual outcome indisti nguishability (OI) over a configurable auditor family with graph-aware doubly ro bust (GA-DR) estimators. We provide a noisy transfer theorem showing that Residu al-OI on estimated GA-DR residuals implies bounds on exposure-counterfactual gro up gaps under temporal mixing and bounded local interference, and we instantiate an online multicalibration auditor together with a primal-dual controller. Expe riments on two TGB streams and a controlled synthetic bipartite stream show that COPF reduces worst-case spikes in exposure-counterfactual group disparities wit h modest impact on ranking utility. Our code is available at https://github.com/ lsnnnnnnnn/COPF.

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CRAG: Can 3D Generative Models Help 3D Assembly?

Zeyu Jiang, Sihang Li, Siqi Tan, Chenyang Xu, Juexiao Zhang, Julia Galway-Witham , Xue Wang, Scott A. Williams, Radu Iovita, Chen Feng, Jing Zhang

Most existing 3D assembly methods treat the problem as pure pose estimation, rea rranging observed parts via rigid transformations. In contrast, human assembly n aturally couples structural reasoning with holistic shape inference. Inspired by this intuition, we reformulate 3D assembly as a joint problem of assembly and g eneration. We show that these two processes are mutually reinforcing: assembly p rovides part-level structural priors for generation, while generation injects ho listic shape context that resolves ambiguities in assembly. Unlike prior methods that cannot synthesize missing geometry, we propose CRAG, which simultaneously generates plausible complete shapes and predicts poses for input parts. Extensiv e experiments demonstrate state-of-the-art performance across in-the-wild object s with diverse geometries, varying part counts, and missing pieces. Our code and models will be released.

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Information Geometry Loss for Time Series Forecasting

Jiayu Fang, Xuande Liu, Sangsha Fang, Ernie Tian, Hongwei Ma, Zhiqi Shao, Junbin Gao

Time series forecasting fundamentally involves learning probability distribution s over future observations. However, existing loss functions rely on point-wise Euclidean metrics, neglecting the intrinsic geometric structure of probability d istributions. This leads to suboptimal alignment between predicted and true dist

ributions, particularly for uncertainty quantification. We propose InfoGeo Loss, a principled loss function grounded in information geometry that measures distributional discrepancies on statistical manifolds. Our approach comprises three key components: (1) a distribution parameterization module that models predictions with learnable sufficient statistics, (2) a Fisher information metric that quantifies intrinsic distributional distance, and (3) a Bregman divergence component that captures asymmetric prediction errors. We further introduce a natural gradient weighting strategy for efficient optimization on statistical manifolds. Theoretically, we prove statistical consistency and establish convergence guarantees. Extensive experiments on seven datasets with five architectures show that InfoGeo Loss consistently outperforms existing losses, achieving average improvements of 6.8% in MSE and 5.3% in MAE.
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From Abstraction to Instantiation: Learning Behavioral Representation for Vision-Language-Action Model

Bing Hu, Zaijing Li, Rui Shao, Junda Chen, April Hua Liu, Wei-Shi Zheng, Liqiang Nie

Vision-Language-Action (VLA) models often suffer from performance degradation under distribution shifts, as they struggle to learn generalized behavior representations across varying environments. While existing approaches attempt to construct behavior representations through action-centric latent variables, they are often limited by short-horizon temporal fragmentation and static execution-alignment, leading to inconsistent behaviors in complex scenarios. To address these limitations, we propose \textbf{BehaviorVLA}, a framework that facilitates robust manipulation through the learning of a temporally coherent behavioral representations. Our approach features two symmetric components: (1) the \textbf{Visuomotor Behavior Encoder (VBE)}, which utilizes a causal Mamba-based architecture to aggregate long-horizon trajectory information into a unified behavior representation; and (2) the \textbf{Phase-conditioned Behavior Decoder (PBD)}, which decodes this representation into precise actions by dynamically aligning task-level priors with real-time execution progress. Experiments on RoboTwin 2.0, LIBERO, and CALVIN demonstrate state-of-the-art success rates of 58\%, 98\%, and 4.36 (Avg. Len), respectively. Notably, in real-world sim-to-real transfer, BehaviorVLA matches the performance of OpenVLA-OFT using only 50\% of the demonstration data, showcasing its superior data efficiency and generalization.
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Fine-Tuning Masked Diffusion for Provable Self-Correction

Jaeyeon Kim, Seunggeun Kim, Taekyun Lee, David Z. Pan, Hyeji Kim, Sham M. Kakade, Sitan Chen

A natural desideratum for generative models is self-correction--detecting and revising low-quality tokens at inference. While Masked Diffusion Models (MDMs) have emerged as a promising approach for generative modeling in discrete spaces, their capacity for self-correction remains poorly understood. Prior attempts to incorporate self-correction into MDMs either require overhauling MDM architectures/training or rely on imprecise proxies for token quality, limiting their applicability. Motivated by this, we introduce PRISM--Plug-in Remasking for Inference-time Self-correction of Masked Diffusions--a lightweight, model-agnostic approach that applies to any pretrained MDM. Theoretically, PRISM defines a self-correction loss that provably learns per-token quality scores, without RL or a verifier. These quality scores are computed in the same forward pass with MDM and used to detect low-quality tokens. Empirically, PRISM advances MDM inference across domains and scales: Sudoku; unconditional text (170M); and code with LLaDA (8B). We open-source our codebase in https://github.com/SeunggeunKimkr/PRISM.
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DiscoForcing: A Unified Framework for Real-Time Audio-Driven Character Control with Diffusion Forcing

Kaiyang Ji, Bingsheng Qian, Binghuan Wu, Kangyi Chen, Ye Shi, Jingya Wang

We study real-time audio-responsive character control as a deployment-faithful problem: strictly causal, bounded-latency streaming that must generate coherent full-body motion at interactive frame rates while the audio condition can change

abruptly (tempo shifts, drops, or user edits). Prior music-to-motion systems are largely optimized for offline generation with global context, and degrade in streaming rollouts where conditioning history becomes stale or unreliable. We introduce DiscoForcing, a streaming audio-driven diffusion framework that combines a causal music encoder that captures rhythmic structure and phase dynamics with a diffusion-forcing sequence model trained under heterogeneous noise levels across the temporal horizon. Building on this, we design a hybrid temporal schedule and a history-guided streaming sampler to explicitly trade off responsiveness against long-horizon consistency under non-stationary audio. Implemented in an end-to-end real-time interactive system with online avatar playback and humanoid deployment workflows, DiscoForcing delivers more stable long-horizon rollouts and sharper audio–motion alignment than prior baselines under matched causality and latency constraints while maintaining real-time throughput.Project Page: https://discoforcing.github.io

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BTSP-CAM: A Brain-Inspired Geometric Memory for Class-Incremental Learning

Zheng Zhang, Jiaye Yang, Qingjie Guo, Jiangrong Shen, Long Chen, Qi Xu

Gradient-based optimization in class-incremental learning (CIL) often faces the plasticity–stability dilemma, since continuous weight updates can distort decision boundaries learned from earlier tasks. We revisit this problem from the viewpoint of stochastic geometric memory allocation and propose BTSP-CAM, a gradient-free memory system that instantiates theoretical insights from the hippocampal simpleBTSP model into a practical algorithm. Rather than fine-tuning a frozen encoder by backpropagation, BTSP-CAM externalizes plasticity into a binary synaptic matrix that evolves through local stochastic bit-flip updates. A trace-gated plateau process, driven by eligibility traces together with familiarity and collision signals, modulates when and where synapses are rewritten and suppresses cross-class interference in Hamming space. The resulting geometric memory states are mapped to semantic logits through a CA1-like competitive layer and a closed-form ridge readout, enabling fast consolidation after each task. Empirically, BTSP-CAM rivals gradient-based methods in a strictly exemplar-free setting and consistently boosts SOTA baselines as a lightweight plugin. Mechanistic analysis validates our geometric theory, confirming that stochastic repulsion actively bounds class overlap and stabilizes decision margins.

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Mixture of Horizons in Action Chunking

Dong Jing, Gang Wang, Jiaqi Liu, Weiliang Tang, Zelong Sun, Yunchao Yao, Zhenyu Wei, Yunhui Liu, Zhiwu Lu, Mingyu Ding

Vision-language-action models exhibit an inherent trade-off in action chunk length (``horizon''): longer horizons improve global foresight but degrade fine-grained local control, while shorter ones yield the opposite.

To mitigate the trade-off, we propose a $\textbf{mixture of horizons (MoH)}$ strategy.

In brief, MoH rearranges the action chunk into several segments with different horizons, processes them in parallel with a shared action transformer, and fuses outputs with a light linear gate.

It offers three appealing benefits.

i) Long-term foresight and short-term precision are jointly exploited within a single model.

ii) MoH is plug-and-play for full-attention action modules with minimal training or inference overhead.

iii) MoH enables dynamic inference with adaptive horizons, which selects stable actions through cross-horizon consensus, achieving 2.5$\times$ higher throughput than baselines while preserving superior performance.

Extensive experiments over flow-based and one-step regression policies demonstrate that MoH yields consistent and significant gains on both simulations and real-world tasks.

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Fair Dataset Distillation via Cross-Group Barycenter Alignment

Mohammad Hossein Moslemi, Nima Hosseini Dashtbayaz, Zhimin Mei, Bissan Ghaddar,

Boyu Wang
Dataset Distillation aims to compress a large dataset into a small synthetic one while maintaining predictive performance.
We show that as different demographic groups exhibit distinct predictive patterns, the distillation process struggles to simultaneously preserve informative signals for all subgroups, regardless of whether group sizes are mildly or severely imbalanced.
Consequently, models trained on distilled data can experience substantial performance drops for certain subgroups, leading to fairness gaps. Crucially, these gaps do not disappear by merely correcting group imbalance, since they stem from fundamental mismatches in subgroup predictive patterns rather than from sample-size disparities alone. We therefore formally analyze the interaction between these two sources of bias and cast the solution as identifying a group-imbalance-agnostic barycenter of the predictive information that induces similar representations across all subgroups. By distilling toward this shared aggregate representation, we show that group fairness concerns can be reduced. Our approach is compatible with existing distillation methods, and empirical results show that it substantially reduces bias introduced by dataset distillation. Code is available at https://github.com/mhmoslemi/COBRA.
****
USE: A Unified Self-Ensembling Framework for Test-Time Prompt Tuning
Siru Jiang, Jian Liang, Ran He, Tieniu Tan
Test-time adaptation (TTA) has emerged as a popular paradigm for improving the performance of vision–language models (*e.g.*, CLIP) on downstream tasks.
Among existing CLIP-based TTA methods, Test-Time Prompt Tuning (TPT) is a pioneering work that optimizes textual prompts using multiple test-time augmentations and remains a strong baseline to date.
In this work, we revisit TPT and reveal that its optimization can be interpreted as implicitly learning from self-generated pseudo labels.
Building on this perspective, we propose a unified self-ensembling framework (**USE**) that ensures consistency between the optimization and inference stages.
During optimization, we introduce a simple yet effective self-ensembling (**SE**) strategy that emphasizes the test image itself over its augmented views adaptively to obtain more reliable pseudo labels.
To fully exploit the potential of augmentations, we further apply the same strategy at inference time, unifying the objectives of both stages.
Notably, **SE** can also act as a lightweight optimization-free TTA method.
Extensive experiments across multiple datasets demonstrate that **SE** and **USE** outperform their counterparts, respectively.
Furthermore, **SE** yields consistent performance gains when integrated with existing TTA methods.
The code is available at *https://github.com/sirujiang/USE*.
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AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline
Hyewon Suh, Binfei Ji, Seojune Lee, Rishi Khare, Basit Khan, Hyunjun Kim, Tianyi Zhang, Venkat Krishna Srinivasan, Peter Belcak, Shizhe Diao, Pavlo Molchanov, Yingyan Celine Lin, Zhen Dong
Reliable evaluation of large language model (LLM) agents depends critically on benchmark validity, yet modern agent benchmarks often contain hidden flaws arising from interactions among user instructions, environments, tools, ground-truth trajectories, and evaluation protocols. These flaws confound model errors with benchmark artifacts and undermine leaderboard-based comparisons. We propose **COBA** (**CO**mponent-based **B**enchmark **A**uditing), an automated pipeline for diagnosing and filtering validity issues in agent benchmarks. COBA decomposes agent tasks into four standardized components—User, Environment, Ground Truth, and Evaluation—and operationalizes a component-level issue taxonomy using hybrid rule-based detectors and taxonomy-guided LLM evaluation. Across six widely used agent benchmarks, COBA achieves strong alignment with expert judgments, with F1 scores between 0.791 and 0.874. It complements manual verification of τ^2-benc

h by identifying issues missed due to benchmark complexity, and generalizes to p
reviously unseen benchmarks with minimal adaptation. Our analysis shows that ben
chmark flaws are widespread and materially affect evaluation outcomes, demonstra
ting that component-based auditing provides a scalable foundation for more relia
ble and interpretable agent evaluation. We release AgentSuite, a unified benchma
rk-running platform that includes the COBA auditing pipeline and audited benchma
rk variants: https://github.com/Agent-Suite/AgentSuite.
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Comp-Attn: Present-and-Align Attention for Compositional Video Generation
Hongyu Zhang, Yufan Deng, Shenghai Yuan, Xuehan Hou, Yian Zhao, Peng Jin, Chang
Liu, Jie Chen
In the domain of text-to-video (T2V) generation, reliably synthesizing compositi
onal content involving multiple subjects with intricate relations is still under
explored. The main challenges are twofold: 1) Subject presence, where not all su
bjects can be presented in the video; 2) Inter-subject relations, where the inte
raction and spatial relationship between subjects are misaligned. Existing metho
ds adopt techniques, such as inference-time latent optimization or layout contro
l, which fail to address both issues simultaneously. To tackle these problems, w
e propose Comp-Attn, a composition-aware cross-attention variant that follows a
``Present-and-Align" paradigm: it decouples the two challenges by enforcing subj
ect presence at the condition level and achieving relational alignment at the at
tention-distribution level. Specifically, 1) We introduce Subject-aware Conditio
n Interpolation (SCI) to reinforce subject-specific conditions and ensure each s
ubject's presence; 2) We propose Layout-forcing Attention Modulation (LAM), whic
h dynamically enforces the attention distribution to align with the relational l
ayout of multiple subjects. Comp-Attn can be seamlessly integrated into various
T2V baselines in a training-free manner, boosting T2V-CompBench scores by 15.7%
and 11.7% on Wan2.1-T2V-14B and Wan2.2-T2V-A14B with only a 5% increase in infer
ence time. Meanwhile, it also achieves strong performance on VBench and T2I-Comp
Bench, demonstrating its scalability in general T2V and compositional text-to-im
age (T2I) tasks. Code and models are available at: https://github.com/Hong-yu-Zh
ang/Comp-attn.
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Structurally Aligned Subtask-Level Memory for Software Engineering Agents
Kangning Shen, Jingyuan Zhang, Chenxi Sun, Wencong Zeng, Yang Yue
Large Language Models (LLMs) have demonstrated significant potential as autonomo
us software engineering (SWE) agents. Recent work has further explored augmentin
g these agents with memory mechanisms to support long-horizon reasoning. However
, these approaches typically operate at a coarse instance granularity, treating
the entire problem-solving episode as the atomic unit of storage and retrieval.
We empirically demonstrate that instance-level memory suffers from a fundamental
granularity mismatch, resulting in misguided retrieval when tasks with similar
surface descriptions require distinct reasoning logic at specific stages. To add
ress this, we propose Structurally Aligned Subtask-Level Memory, a method that a
ligns memory storage, retrieval, and updating with the agent's functional decomp
osition. Extensive experiments on SWE-bench Verified demonstrate that our method
consistently outperforms both vanilla agents and strong instance-level memory b
aselines across diverse backbones, improving mean Pass@1 over the vanilla agent
by +4.7 pp on average (e.g., +6.8 pp on Gemini 2.5 Pro). Performance gains grow
with more interaction steps, showing that leveraging past experience benefits lo
ng-horizon reasoning in complex software engineering tasks.
****

Maximum-Likelihood Learning of Latent Dynamics Without Reconstruction
Samo Hromadka, Kai Biegun, Lior Fox, James Heald, Maneesh Sahani
We address the challenge of uncovering systematic, and potentially controllable,
dynamical structure underlying complex high-dimensional time-series data. Exist
ing generative and autoregressive models have difficulty telling systematic cont
ent apart from distractors, while contrastive methods struggle to learn accurate
dynamics. To address these shortcomings, we develop the Recognition-Parametrize
d Gaussian State-Space Model (RP-GSSM), a probabilistic framework that infers ac

curate latent dynamics without relying on a parametrized decoder. By eliminating explicit generative parameters, the model directs its entire representational capacity to encoding dynamically relevant state; and, being fully probabilistic, learns via maximum likelihood without auxiliary objectives or ad-hoc regularization. Combining the expressive power of a neural network encoder with exact inference under a jointly Gaussian prior allows the RP-GSSM to embed a broad class of intrinsically nonlinear dynamical systems. The RP-GSSM parameters are provably linearly identifiable in the infinite data limit. We show that the RP-GSSM recovers physically meaningful latent states from noisy video more faithfully than competing methods, more reliably identifies underlying controllable nonlinear dynamics, and remains substantially more robust to visual distractors.
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Vision-DeepResearch: Incentivizing DeepResearch Capability in Multimodal Large Language Models

Wenxuan Huang, Yu Zeng, Qiuchen Wang, Zhen Fang, Shaosheng Cao, Zheng Chu, Qingyu Yin, Shuang Chen, Zhenfei Yin, Lin Chen, Zehui Chen, Yao Hu, Philip Torr, Feng Zhao, Wanli Ouyang

Multimodal large language models (MLLMs) have achieved remarkable success across a broad range of vision tasks.
However, constrained by the capacity of their internal world knowledge, prior work has proposed augmenting MLLMs by ``reasoning-then-tool-call'' for visual and textual search engines to obtain substantial gains on tasks requiring extensive factual information.
However, these approaches typically define multimodal search in a naive setting, assuming that a single full-level or entity-level image query and few text query suffices to retrieve the key evidence needed to answer the question, which is unrealistic in real-world scenarios with substantial visual noise.
Moreover, they are often limited in the reasoning depth and search breadth, making it difficult to solve complex questions that require aggregating evidence from diverse visual and textual sources.
Building on this, we propose Vision-DeepResearch, which proposes one new multimodal deep-research paradigm, i.e., performs multi-turn, multi-entity and multi-scale visual and textual search to robustly hit real-world search engines under heavy noise.
Our Vision-DeepResearch supports dozens of reasoning steps and hundreds of engine interactions, while internalizing deep-research capabilities into the MLLM via cold-start supervision and RL training, resulting in a strong end-to-end multimodal deep-research MLLM.
It substantially outperforming existing multimodal deep-research MLLMs, and workflows built on strong closed-source foundation model such as GPT-5, Gemini-2.5-pro and Claude-4-Sonnet.
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Training-Free Vector Quantization via Gaussian VAEs

Tongda Xu, Wendi Zheng, Jiajun He, José Miguel Hernández-Lobato, Yan Wang, Ya-Qin Zhang, Jie Tang

Vector-quantized variational autoencoders (VQ-VAEs) are discrete autoencoders that compress images into discrete tokens. However, they are difficult to train due to discretization. In this paper, we propose a simple yet effective technique dubbed __Gaussian Quant (GQ)__, which first trains a Gaussian VAE under certain constraints and then converts it into a VQ-VAE without additional training. For conversion, GQ generates random Gaussian noise as a codebook and finds the closest noise vector to the posterior mean. Theoretically, we prove that when the logarithm of the codebook size exceeds the bits-back coding rate of the Gaussian VAE, a small quantization error is guaranteed. Practically, we propose a heuristic to train Gaussian VAEs for effective conversion, named the target divergence constraint (TDC). Empirically, we show that GQ outperforms previous VQ-VAEs, such as VQGAN, FSQ, LFQ, and BSQ, on both UNet and ViT architectures. Furthermore, TDC also improves previous Gaussian VAE discretization methods, such as TokenBridge. The source code is provided in https://github.com/tongdaxu/VQ-VAE-from-Gaussian-VAE.

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Stable Velocity: A Variance Perspective on Flow Matching
Donglin Yang, Yongxing Zhang, Xin Yu, Liang Hou, Xin Tao, Pengfei Wan, Xiaojuan Qi, Renjie Liao
While flow matching is elegant, its reliance on single-sample conditional velocities leads to high-variance training targets that destabilize optimization and slow convergence. By explicitly characterizing this variance, we identify 1) a *high-variance regime* near the prior, where optimization is challenging, and 2) a *low-variance regime* near the data distribution, where conditional and marginal velocities nearly coincide. Leveraging this insight, we propose **Stable Velocity**, a unified framework that improves both training and sampling. For training, we introduce Stable Velocity Matching (StableVM), an unbiased variance-reduction objective, along with Variance-Aware Representation Alignment (VA-REPA), which adaptively strengthen auxiliary supervision in the *low-variance regime*. For inference, we show that dynamics in the *low-variance regime* admit closed-form simplifications, enabling Stable Velocity Sampling (StableVS), a finetuning-free acceleration. Extensive experiments on ImageNet 256×256 and large pretrained text-to-image and text-to-video models, including SD3.5, Flux, Qwen-Image, and Wan2.2, demonstrate consistent improvements in training efficiency and more than $2\times$ faster sampling within the *low-variance regime* without degrading sample quality. Our code is available at https://github.com/linYDTHU/StableVelocity.
****
Base Models Know How to Reason, Thinking Models Learn When
Constantin Venhoff, Iván Arcuschin, Philip Torr, Arthur Conmy, Neel Nanda
What do different *thinking* language models learn during training?
We introduce *constructive model diffing*, a framework for understanding fine-tuned models by constructing the base-to-fine-tuned difference from interpretable components, producing hybrid models whose performance recovery measures how well the components capture the diff.
For thinking models, we decompose the diff into *reasoning mechanisms* (category-specific vectors that induce reasoning behaviors in the base model) and *reasoning heuristics* (a classifier determining when each mechanism fires).
To ground this decomposition, we use Sparse Autoencoders to discover interpretable taxonomies of reasoning behaviors.
Across nine model configurations (four RL-trained, four SFT-distilled from a larger thinking model, and one mixed), we find a striking difference: hybrid models recover much more performance for RL-trained models than for SFT-trained or mixed ones. This indicates that RL primarily teaches models to apply sophisticated heuristics to pre-existing base capabilities, while SFT-based distillation teaches the base model new mechanisms.
These results offer a new lens on what different training paradigms teach, with implications for efficient reasoning model development.
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Rethinking Human Intent-to-CAD: Parametric CAD Model Generation via Cooperative Multi-Task Alignment and Spatial-Aware Reinforcement Learning
Qingwang Zhang, Jiahao Li, Xiangdong Zhou
Parametric Computer-Aided-Design (CAD) modeling from human intent remains challenging, particularly during the conceptual design stage, where design goals are expressed through incomplete and unstructured modalities (e.g., hand-drawn sketches and textual descriptions). In this work, we rethink the human intent-to-CAD pipeline and propose a unified method that directly maps multi-level human intents to executable codes, without assuming the prior existence of target CAD models. To support our study, we construct HiCAD, the first large-scale dataset aligning hand-drawn sketches, textual descriptions, and parametric CAD codes. Based on this, we introduce HiCAD, a two-stage framework comprising Cooperative Multi-Task Alignment to bridge the representational gap between heterogeneous inputs, and Spatial-Aware Reinforcement Learning to enforce geometric and topological consistency. Extensive experiments demonstrate that our method significantly outperforms existing baselines across multiple tasks, validating its effectiveness and

robustness in transforming heterogeneous human intents into high-fidelity parame
tric CAD models. Our project page: https://zqwlearning.github.io/HiCAD.
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Prototype-Grounded Concept Models for Verifiable Concept Alignment
Stefano Colamonaco, David Debot, Pietro Barbiero, Giuseppe Marra
Concept Bottleneck Models (CBMs) aim to improve interpretability in Deep Learnin
g by structuring predictions through human-understandable concepts, but they pro
vide no way to verify whether learned concepts align with the human's intended m
eaning, hurting interpretability. We introduce Prototype-Grounded Concept Models
 (PGCMs), which ground concepts in learned visual prototypes: image parts that s
erve as explicit evidence for the concepts. This grounding enables direct inspec
tion of concept semantics and supports targeted human intervention at the protot
ype level to correct misalignments. Empirically, PGCMs achieve similar predictiv
e performance as state-of-the-art CBMs while substantially improving transparenc
y, interpretability, and intervenability.
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Why Are Linear RNNs More Parallelizable?
William Merrill, Hongjian Jiang, Yanhong Li, Anthony Widjaja Lin, Ashish Sabharw
al
The community is increasingly exploring linear RNNs (LRNNs) as language models,
motivated by their expressive power and parallelizability. While prior work esta
blishes the expressivity benefits of LRNNs over transformers, it is unclear what
 makes LRNNs---but not traditional, *nonlinear* RNNs---as easy to parallelize in
 practice as transformers. We answer this question by providing a tight connecti
on between types of RNNs and standard complexity classes. We show that LRNNs can
 be viewed as log-depth (bounded fan-in) arithmetic circuits, which represents o
nly a slight depth overhead relative to log-depth boolean circuits that transfor
mers admit. Furthermore, we show that nonlinear RNNs can solve $\mathsf L$-compl
ete problems (and even $\mathsf P$-complete ones, under polynomial precision), r
evealing a fundamental barrier to parallelizing them as efficiently as transform
ers. Our theory also identifies fine-grained expressivity differences between re
cent popular LRNN variants: permutation-diagonal LRNNs are NC^1-compl
ete whereas diagonal-plus-low-rank LRNNs are more expressive (PNC^1-c
omplete). We provide further insight by associating each type of RNN with a corr
esponding automata-theoretic model that it can simulate. Together, our results r
eveal fundamental tradeoffs between nonlinear RNNs and different variants of LRN
Ns, providing a foundation for designing LLM architectures that achieve an optim
al balance between expressivity and parallelism.
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Text Before Vision: Staged Knowledge Injection Matters for Agentic RLVR in Ultra
-High-Resolution Remote Sensing Understanding
Fengxiang Wang, Mingshuo Chen, Yueying Li, Yulin Wang, Yang Yajie, Yuhao Zhou, D
i Wang, YiFan Zhang, Haoyu Wang, Haiyan Zhao, Hongda Sun, Jun Song, Long Lan, Ji
ng Zhang, Wenlong Zhang, Bo Du
Multimodal reasoning for ultra-high-resolution (UHR) remote sensing (RS) is usua
lly bottlenecked by visual evidence acquisition: the model necessities localizin
g tiny task-relevant regions in massive pixel spaces.
While Agentic Reinforcement Learning with Verifiable Rewards (RLVR) using zoom-i
n tools offers a path forward, we find that standard reinforcement learning stru
ggles to navigate these vast visual spaces without structured domain priors.
In this paper, we investigate the interplay between post-training paradigms: com
paring Cold-start Supervised Fine-Tuning (SFT), RLVR, and Agentic RLVR on the UH
R RS benchmark. Our controlled studies yield a counter-intuitive finding: high-q
uality Earth-science text-only QA is a primary driver of UHR visual reasoning ga
ins. Despite lacking images, domain-specific text injects the concepts, mechanis
tic explanations, and decision rules necessary to guide visual evidence retrieva
l. Based on this, we propose a staged knowledge injection recipe: (1) cold-start
ing with scalable, knowledge-graph-verified Earth-science text QA to instill rea
soning structures; and (2) "pre-warming'' on the same hard UHR image–text exampl
es during SFT to stabilize and amplify subsequent tool-based RL.

This approach achieves a 60.04\% Pass@1 on XLRS-Bench, significantly outperforming larger general-purpose models (e.g., GPT-5.2, Gemini 3.0 Pro, Intern-S1) and establishing a new state-of-the-art. We provide both the automated data pipeline and the rigorous ablation studies that validate this "Text-Before-Vision'' paradigm. Datasets and code will be released.
****
A Fine-Grained Understanding of Uniform Convergence for Halfspaces
Aryeh Kontorovich, Kasper Green Larsen
We study the fine-grainded uniform convergence behavior of halfspaces beyond worst-case VC bounds. For inhomogeneous halfspaces in $\mathbb{R}^d$ with $d\ge 2$, we show that standard first-order VC bounds are essentially tight: even consistent hypotheses can incur population error $\Theta(d\log(n/d)/n)$, and in the agnostic setting the deviation scales as $\sqrt{\tau\log(1/\tau)}$ at true error τ.
In contrast, homogeneous halfspaces in $\mathbb{R}^2$ exhibit a markedly different behavior. In the realizable case, every hypothesis consistent with the sample has error $O(1/n)$. In the agnostic case, we prove a bandwise, log-free deviation bound on each dyadic risk band via a critical-wedge localization argument. Unioning over bands incurs only a $\log\log n$ overhead, and we establish a matching lower bound showing this overhead is unavoidable. Together, these results give a fine-grained and nearly complete picture of uniform convergence for halfspaces, revealing sharp dimensional and structural thresholds.
****
GXPO: Group Cross-Lingual Relative Policy Optimization for Code Generation
Linzheng Chai, Jian Yang, Jiajun Wu, Ensheng Shi, Xianglong Liu
Current reinforcement learning (RL) methods for code generation are predominantly optimized on Python, showing weak generalization to other programming languages (PLs). Although leveraging multilingual solutions offers richer semantics and a wider search landscape, naive independent training across languages suffers from optimization imbalance and fails to effectively transfer knowledge from high-resource languages. We propose Group Cross-lingual Relative Policy Optimization (GXPO), which forms training groups by generating solutions for the same problem in multiple PLs and jointly optimizes language-specific and cross-language signals, enabling more balanced optimization and improved transfer to low-resource PLs. We additionally introduce Multilingual LiveCodeBench (ML-LCB), extending LiveCodeBench to a unified multilingual evaluation setting. On ML-LCB across 8 PLs, GXPO consistently improves performance, with pronounced gains on low-resource PLs, demonstrating scalable multilingual RL for language-consistent code generation.
****
Infinite-World: Scaling Interactive World Models to 1000-Frame Horizons via Pose-Free Hierarchical Memory
Ruiqi Wu, Xuanhua He, Meng Cheng, Tianyu Yang, Yong Zhang, Zhuoliang Kang, Xunliang Cai, Xiaoming Wei, Chun-Le Guo, Chongyi Li, Ming-Ming Cheng
We propose **Infinite-World**, a robust interactive world model capable of maintaining coherent visual memory over **1000+ frames** in complex real-world environments. While existing world models can be efficiently optimized on synthetic data with perfect ground-truth, they lack an effective training paradigm for real-world videos due to noisy pose estimations and the scarcity of viewpoint revisits. To bridge this gap, we first introduce a **Hierarchical Pose-free Memory Compressor (HPMC)** that recursively distills historical latents into a fixed-budget representation. By jointly optimizing the compressor with the generative backbone, HPMC enables the model to autonomously anchor generations in the distant past with bounded computational cost, eliminating the need for explicit geometric priors. Second, we propose an **Uncertainty-aware Action Abstraction** module that discretizes continuous motion into a tri-state logic. This strategy maximizes the utilization of raw video data while shielding the deterministic action space from being corrupted by noisy trajectories, ensuring robust action-response learning. Furthermore, guided by insights from a pilot toy study, we employ a **Revisit-Dense Finetuning Strategy** using a compact, 30-minute dataset to efficient

ly activate the model’s long-range loop-closure capabilities. Extensive experiments, including objective metrics and user studies, demonstrate that Infinite-World achieves superior performance in visual quality, action controllability, and spatial consistency. Our code and data will be released.
****
Understanding Transfer Learning of RNA Foundation Models on Downstream Tasks
Yuan Li, Heng Yang, Renzhi Chen, Ke Li
Foundation models (FMs) pretrained on large-scale sequence data have emerged as a promising paradigm for RNA biology, yet the mechanisms underlying their transferability remain unclear.
In this work, we conduct a systematic investigation of transfer learning in RNA FMs across diverse structural and functional tasks.
Our results demonstrate that frozen representations from pretrained RNA FMs are not universally transferable, and that the hierarchical feature reuse paradigm prevalent in computer vision does not generally extend to RNA FMs.
Instead, pretraining primarily benefits downstream tasks by providing a favorable optimization initialization when pretraining and downstream objectives are well aligned, which accelerates convergence toward flatter minima associated with improved generalization.
Overall, our findings characterize pretraining as an optimization prior whose effectiveness is governed by task alignment and model capacity, offering principled guidance for future RNA FMs.
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Anomaly-Preference Image Generation
Fuyun Wang, Yuanzhi Wang, Xu Guo, Sujia Huang, Tong Zhang, Dan Wang, Xin Liu, Hui Yan, Zhen Cui
Synthesizing realistic and diverse anomalous samples from limited data is vital for robust model generalization. However, existing methods struggle to reconcile fidelity and diversity, often hampered by distribution misalignment and overfitting, respectively. To mitigate this, we introduce Anomaly Preference Optimization (APO), a novel paradigm that reformulates anomaly generation as a preference learning problem. Central to our approach is an implicit preference alignment mechanism that leverages real anomalies as positive references, deriving optimization signals directly from denoising trajectory deviations without requiring costly human annotation. Furthermore, we propose a Time-Aware Capacity Allocation module that dynamically distributes model capacity along the diffusion timeline— prioritizing structural diversity during highnoise phases while enhancing fine-grained fidelity in low-noise stages. During inference, a hierarchical sampling strategy modulates the coherencealignment trade-off, enabling precise control over generation. Extensive experiments demonstrate that significantly outperforms existing baselines, achieving state-of-the-art performance in both realism and diversity.
****
LOTTERY: Learning from Reference-Only Samples in Two-Sample Testing under Size Asymmetry
Xunye Tian, Zhijian Zhou, Liuhua Peng, Feng Liu
Data-adaptive two-sample testing assesses if two samples come from the same distribution, using a discrepancy learned from the data (e.g., via kernel-based feature representations). Such methods typically rely on data splitting to decouple learning from testing and control type I error. However, this paradigm is ill-suited to few-shot settings with severe sample-size imbalance: abundant reference samples are available, while only a handful of query samples arrive. In this paper, we show how this imbalance can be leveraged constructively. Using abundant reference data, we learn reference-dependent representations that summarize salient structure of the reference distribution and provide informative signals for detecting departures. We incorporate a collection of representation families that capture both global and local structure, and adaptively weight them using only reference samples via an uncertainty-guided principle. Theoretically, we establish permutation-based type I error control and show consistency of the aggregated test: as the sample sizes grow, the test power converges to one whenever the re

presentation set contains at least one consistent representation. Empirically, o
ur aggregation achieves strong performance across a range of benchmarks while re
taining type I error control.
****

TRIM: Token-wise Attention-Derived Saliency for Data-Efficient Instruction Tunin
g

Manish Nagaraj, Sakshi Choudhary, Utkarsh Saxena, Deepak Ravikumar, Kaushik Roy

Instruction tuning is essential for aligning large language models (LLMs) to dow
nstream tasks and commonly relies on large, diverse corpora. However, small, hig
h-quality subsets, known as coresets, can deliver comparable or superior results
, though curating them remains challenging. Existing methods often rely on coars
e, sample-level signals like gradients, an approach that is computationally expe
nsive and overlooks fine-grained features. To address this, we introduce TRIM (T
oken Relevance via Interpretable Multi-layer Attention), a forward-only, token-c
entric framework. Instead of using gradients, TRIM operates by matching underlyi
ng representational patterns identified via attention-based "fingerprints" from
a handful of target samples. Such an approach makes TRIM highly efficient and un
iquely sensitive to the structural features that define a task. Coresets selecte
d by our method consistently outperform state-of-the-art baselines by up to 9% o
n downstream tasks and even surpass the performance of full-data fine-tuning in
some settings. By avoiding expensive backward passes, TRIM achieves this at a fr
action of the computational cost. These findings establish TRIM as a scalable an
d efficient alternative for building high-quality instruction-tuning datasets.
****

Relighting as a Probe of Visual Priors via Augmented Latent Intrinsics

Xiaoyan Xing, Xiao Zhang, Sezer Karaoglu, Theo Gevers, Anand Bhattad

Image-to-image relighting requires representations that separate illumination fr
om scene properties while preserving dense geometry, material, and photometric c
ues. We use this task as a probe of visual priors: unlike recognition tasks that
 reward invariance, relighting tests whether visual features retain the informat
ion needed for light transfer. Through a controlled generative relighting framew
ork, we find that strong semantic encoders can degrade relighting quality, expos
ing a semantic--photometric trade-off between abstraction and physical fidelity.
 We introduce \textbf{Augmented Latent Intrinsics (ALI)}, which balances this tr
ade-off by fusing dense, pixel-aligned visual features into a latent-intrinsic r
elighting model and refining it with self-supervision on unlabeled real image pa
irs. ALI improves relighting quality, especially on glossy, metallic, and transp
arent materials, and demonstrates that generative relighting is an effective too
l for quantifying what visual encoders encode about the physical world.
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MoCL: Metabolic Optimization for Curvature-Aware Continual Learning

Jiajun Lai, Qi Liu, Shijie Li, Huaiguang Jiang

Continual learning requires models to mitigate catastrophic forgetting of prior
knowledge while learning a sequence of tasks. Although existing methods based on
 orthogonal projection prevent interference by constraining parameter updates, t
hey tend to limit plasticity as the task sequence progresses. The reliance on th
e linear approximation further causes the projected gradients to deviate from th
e nonlinear manifold. To address these issues, we propose Metabolic Optimization
 for Continual Learning (MoCL), a rehearsal-free framework that strikes a balanc
e between stability and plasticity. To capture the geometric manifold of prior k
nowledge, MoCL introduces a factorized subspace approximation that avoids expens
ive explicit matrix inversion. Given the heavy-tailed distribution of the Fisher
 Information Matrix, we employ a metabolic gating based on Tsallis entropy to su
ppress updates that conflict with historical knowledge. Theoretical and empirica
l analyses show that MoCL suppresses interference while supporting shared low-lo
ss behavior across sequential tasks. Extensive experimental results across multi
ple benchmarks demonstrate that MoCL outperforms state-of-the-art methods in bot
h classification performance and efficiency.
****

Hierarchical Causal Abduction: A Foundation Framework for Explainable Model Pred

ictive Control
Ramesh Arvind Naagarajan, Zühal Wagner, Stefan Streif
Model Predictive Control (MPC) is widely used to operate safety-critical infrastructure by predicting future trajectories and optimizing control actions. However, nonlinear dynamics, hard safety constraints, and numerical optimization often render individual control moves opaque to human operators, undermining trust and hindering deployment. This paper presents Hierarchical Causal Abduction (HCA), which combines (i) physics-informed reasoning via domain knowledge graphs, (ii) optimization evidence from Karush--Kuhn--Tucker (KKT) multipliers, and (iii) temporal causal discovery via the PCMCI algorithm to generate faithful, human-interpretable explanations for control actions computed by nonlinear MPC. Across three diverse control applications (greenhouse climate, building HVAC, chemical process engineering) with expert validation, HCA improves explanation accuracy by 53\% over LIME (0.478 vs. 0.311) using a single set of cross-domain parameters without per-domain tuning; domain-specific KKT-threshold calibration over 2--3 days further increases accuracy to 0.88. Ablation studies confirm that each evidence source is essential, with 32--37\% accuracy degradation when any component is removed, and HCA's ranking-and-validation methodology generalizes beyond MPC to other prediction-based decision systems, including learning-based control and trajectory planning.
****
UniFLoW: Universal Multi-Modal Federated LoRA Fine-Tuning Framework with Analytical Aggregation
Haoyuan Liang, Zhiyu Ye, Jielong Tang, Yang Yang, Shilei Cao, Guowen Li, Fei Hu, Zhiwei Zhang, Haohuan Fu, Juepeng Zheng
As Multimodal Large Language Models (MLLMs) continue to be trained, the availability of public data diminishes, limiting the possibility for further training and adaptation. However, private data remains an underutilized yet valuable resource. Federated Learning (FL) enables decentralized training on private data, yet extending it to MLLMs is challenging: heterogeneous client modalities induce architectural incompatibility, and full-parameter fine-tuning of billion-scale models incurs prohibitive communication costs. Parameter-efficient methods like LoRA alleviate these issues but introduce aggregation inconsistency, as averaged low-rank updates fail to recover the true global update faithfully.
To address these issues, we propose **UniFLoW**(Universal multi-modal Federated LoRA fine-tuning framework With Analytical Aggregation), a unified federated framework that leverages pre-trained large language models and a multi-modal Encoder architecture, and our proposed Federated Aggregating Analytical Low-Rank Adaption$FedA^2$-$LoRA$). UniFLoW effectively utilizes fragmented client-side multi-modal data while $FedA^2$-$LoRA$ ensuring consistent aggregation. And modality-specific encoders and a II stage training strategy ensure effective integration of diverse modalities without overfitting.
Experiments on text, image, and speech demonstrate that **UniFLoW** enables scalable, communication-efficient, and aggregation-consistent federated fine-tuning,
with $FedA^2$-$LoRA$ achieving state-of-the-art performance compared to existing FedLoRA approaches. We envision UniFLoW as a promising solution to the growing scarcity of public data.
****
SpreadsheetArena: Decomposing Preference in LLM Generation of Spreadsheet Workbooks
Srivatsa R Kundurthy, Clara Na, Michael Handley, Zach Kirshner, Chen Bo Calvin Zhang, Manasi Sharma, Emma Strubell, John Ling
We consider the task of end-to-end **spreadsheet generation**, where language models are prompted to produce spreadsheet artifacts to satisfy users' explicit and implicit constraints, specified in natural language.
We introduce **SpreadsheetArena**, a platform for evaluating models' performance on the task via blind pairwise evaluations of LLM-generated spreadsheet workbooks. As with other complex, open-ended tasks, relevant evaluation criteria can vary greatly across use cases and prompts, often in ways that are difficult to for

malize.
Compared to general dialogue or text generation settings, spreadsheet generation presents unique challenges and opportunities: the task output structure is well-defined and multi-dimensional, and there are often complex interactivity and layout considerations. We observe that stylistic, structural, and functional features of preferred spreadsheets vary meaningfully across prompts. Expert evaluations of spreadsheets for finance prompts suggest that even highly ranked models do not reliably produce spreadsheets aligned with domain-specific best practices. We host a live arena and release a dataset of prompts, generated spreadsheets, and preference votes, which we hope will facilitate further study of tasks operating over spreadsheets as a challenging and interesting class of complex, open-ended tasks for LLMs.
****

Model-Dowser: Data-Free Importance Probing to Mitigate Catastrophic Forgetting in Multimodal Large Language Models
Hyeontaek Hwang, Nguyen Dinh Son, Daeyoung Kim
Fine-tuning Multimodal Large Language Models (MLLMs) on task-specific data is an effective way to improve performance on downstream applications. However, such adaptation often leads to a degradation in generalization on pretrained tasks, a phenomenon known as Catastrophic Forgetting. Existing methods that aim to mitigate this issue either become ineffective when fine-tuning deeper layers of the language decoder or scale poorly with increasing model size. To address these limitations, we propose Model-Dowser, a novel sparse fine-tuning approach for MLLMs. Model-Dowser measures a principled importance score for each model parameter with respect to pretrained generalization (prior to downstream adaptation) by jointly considering weight magnitudes, input activations, and output sensitivities. During fine-tuning, Model-Dowser selectively preserves high-importance parameters and updates the remaining. Comprehensive experiments on two representative MLLMs, LLaVA and NVILA, demonstrate that Model-Dowser effectively mitigates catastrophic forgetting and consistently outperforms prior methods, while remaining resource-efficient and scalable to multi-billion-parameter models.
****

CAffNet: Hard Constraint-Affine Neural Networks
Yang Zhao, Jungeun Lee, Jeong hwan Jeon, Sze Zheng Yong
We present a novel framework for embedding hard constraint satisfaction into neural network (NN) architectures, specifically feedforward neural networks and transformers, with input-dependent affine constraints of arbitrary cardinality. Traditional constraint enforcement approaches either rely on penalty-based soft constraints, which offer no guarantee of satisfaction, or on post-processing methods that enforce constraints after the NN is trained, which may lead to suboptimality. We introduce a trainable constraint-affine (CAffine) layer into NNs, yielding CAffNet, which goes beyond enforcing affine constraints via fixed orthogonal or parallel projections and enables joint
optimization with network parameters. Moreover, we impose no restrictions on the constraint space dimensions and establish that our construction preserves the universal approximation properties of NNs, while providing provable guarantees on constraint adherence for all inputs. Experimental validation demonstrates robust performance across diverse domains requiring guaranteed constraint satisfaction.
****

One Bias After Another: Mechanistic Reward Shaping and Persistent Biases in Language Reward Models
Daniel Fein, Max Lamparth, Violet Xiang, Mykel Kochenderfer, Nick Haber
Reward Models (RMs) are crucial for online alignment of language models (LMs) with human preferences. However, RM-based preference-tuning is vulnerable to \textit{reward hacking}, whereby LM policies learn undesirable behaviors from flawed RMs. By systematically measuring biases in five high-quality RMs, including the state-of-the-art, we find that issues persist despite prior work with respect to length, sycophancy, and overconfidence. We also discover new issues related to bias toward model-specific “styles” and answer-order. We categorize RM failures

as tractable or resistant to linear intervention and propose a simple post-hoc i
ntervention to mitigate low-complexity biases that arise from spurious correlati
ons. Our proposed \textbf{mechanistic reward shaping} reduces targeted biases wi
thout degrading reward quality and while using minimal labeled data. The method
is extensible to new biases, model-internal, and generalizes out-of-distribution
.
****

The Convergent Representation of Contrastive Vision-Language Models: Geometry, M
odality Gap and Shared Space Alignment
Lingjie Yi, Raphael Douady, Chao Chen
Multimodal contrastive learning (MCL) aims to embed data from two modalities in
a shared embedding space. However, in practice, image and text representations o
ccupy completely separated regions of the embedding space, a phenomenon called t
he modality gap. Meanwhile, empirical findings on how the modality gap affects d
ownstream performance remain inconsistent. These observations motivate two key q
uestions: (1) What causes the modality gap? (2) What determines downstream perfo
rmance? To address these questions, we develop the first theoretical framework f
or analyzing the geometry of convergent optimal representations (COR) of MCL whe
n training is optimized. We prove that the modality gap emerges when image and t
ext representations collapse into different subspaces, a phenomenon called \emph
{dimension collapse}. Our theory further reveals that although the modality gap
prevents direct alignment between image and text representations, their projecti
ons onto the shared subspace can align. Moreover, we show that shared space alig
nment is a dominant factor in downstream performance, while the effect of the mo
dality gap is limited. Inspired by these findings, we propose Shared Space Align
ment (SSA) to improve MCL pretraining by enhancing alignment in the shared space
 without optimizing for modality gap reduction. Extensive empirical results vali
date our theoretical analysis and the proposed method.
****

Task-Aware Structured Memory for Dynamic Multi-modal In-Context Learning
Zhirui Chen, Ziwei Chen, Ling Shao
Multi-modal large language models (MLLMs) depend on in-context learning (ICL) fo
r rapid task adaptation, but their scalability is severely limited by finite con
text windows and the growing cost of key–value (KV) caches in long multi-modal s
equences. Existing memory compression approaches typically rely on rigid token r
emoval or sample-dependent importance estimation, which introduces bias, disrupt
s semantic structure—particularly for visual representations—and yields static m
emories that cannot adapt to new queries. We introduce TASM (Task-Aware Structur
ed Memory), a training-free framework that addresses these limitations through t
ask-aware, structure-preserving, and dynamically accessible memory construction.
 TASM employs Task-Vector Guided Compression to replace sample-specific signals
with a task-level direction that captures shared relevance across demonstrations
. To preserve the underlying information manifold, it further applies Semantics-
Aware Token Merging, formulating compression as a Bipartite Graph Matching probl
em that merges tokens without destructive pruning. Finally, TASM organizes compr
essed representations into a multi-resolution hierarchy consisting of a compact
Core Memory and a Latent Bank, enabling Query-Adaptive Dynamic Activation and Dy
namic Retrieval at inference time. Empirical evaluations show that TASM sustains
 strong multi-modal ICL performance under high compression ratios, demonstrating
 an effective balance between efficiency, adaptability, and semantic fidelity.
****

Vision Language Models Cannot Reason About Physical Transformation
Dezhi Luo, Yijiang Li, Maijunxian Wang, Tianwei Zhao, Bingyang Wang, Siheng Wang
, Pinyuan Feng, Pooyan Rahmanzadehgervi, Ziqiao Ma, Hokin Deng
Understanding physical transformations is fundamental for reasoning in dynamic e
nvironments. While Vision Language Models (VLMs) show promise in embodied applic
ations, whether they genuinely understand physical transformations remains uncle
ar. We introduce ***ConservationBench*** evaluating ***conservation***—whether p
hysical quantities remain invariant under transformations. Spanning four propert
ies with paired conserving/non-conserving scenarios, we generate 23,040 question

s across 112 VLMs. Results reveal systematic failure: performance remains near c
hance with improvements on conservation tasks accompanied by drops on controls.
Control experiments show strong textual priors favoring invariance, yet models p
erform worse with visual content. Neither temporal resolution, prompting, nor cu
rated sampling helps. These findings show that current VLMs fail to maintain tra
nsformation-invariant representations of physical properties across dynamic scen
es.
****

RQ-MoE: Residual Quantization via Mixture of Experts for Efficient Input-Depende
nt Vector Compression

Zhengjia Zhong, Shuyan Ke, Zaizhou Lin, Jiaqi Song, Hongyi Lan, Hui Li

Vector quantization is a fundamental tool for compressing high-dimensional embed
dings, yet existing multi-codebook methods rely on static codebooks that limit e
xpressiveness under heterogeneous data geometry. While recent dynamic quantizers
 like QINCo adapt codebooks to individual inputs and improve expressiveness, the
ir strict sequential dependencies create decoding bottlenecks. We propose Residu
al Quantization via Mixture of Experts (RQ-MoE), a framework combining a two-lev
el MoE with dual-stream quantization to enable input-dependent codebook adaptati
on for efficient vector quantization. RQ-MoE enables dynamic codebook constructi
on and decouples instruction from quantization, facilitating parallel decoding.
Theoretically, we show that standard Residual Quantization and QINCo can be reco
vered as constrained special cases of RQ-MoE, and derive a guideline for setting
 expert dimensionality in RQ-MoE. Extensive experiments show that RQ-MoE achieve
s state-of-the-art or on-par performance in reconstruction and retrieval, while
it can provide 6×–14× faster decoding than prior vector quantization methods. Th
e implementation is available at https://github.com/KDEGroup/RQ-MoE.
****

MemoryLLM: Plug-n-Play Interpretable Feed-Forward Memory for Transformers

AJAY KUMAR JAISWAL, Lauren Hannah, Han-Byul Kim, Duc N.M Hoang, Arnav Kundu, Meh
rdad Farajtabar, Minsik Cho

Understanding how transformer components operate in LLMs is important, as it is
at the core of recent technological advances in artificial intelligence. In this
 work, we revisit the challenges associated with interpretability of feed-forwar
d modules (FFNs) and propose MemoryLLM, which aims to decouple FFNs from self-at
tention and enables us to study the decoupled FFNs as context-free token-wise ne
ural retrieval memory. In detail, we investigate how input tokens access memory
locations within FFN parameters and the importance of FFN memory across differen
t downstream tasks. MemoryLLM achieves context-free FFNs by training them in iso
lation from self-attention directly using the token embeddings. This approach al
lows FFNs to be pre-computed as token-wise lookups (ToLs), enabling on-demand tr
ansfer between VRAM and storage, additionally enhancing inference efficiency. We
 also introduce Flex-MemoryLLM, positioning it between a conventional transforme
r design and MemoryLLM. This architecture bridges the performance gap caused by
training FFNs with context-free token-wise embeddings.
****

Debate2Create: Robot Co-design via Multi-Agent LLM Debate

Kevin Qiu, Marek Cygan

We introduce Debate2Create (D2C), a multi-agent LLM framework that formulates ro
bot co-design as structured, iterative debate grounded in physics-based evaluati
on. A design agent and control agent engage in a thesis-antithesis-synthesis loo
p, while criterion-specific LLM judges provide multi-objective feedback to steer
 exploration. Across five MuJoCo locomotion benchmarks, D2C achieves the highest
 default-normalized score among the evaluated LLM-based and black-box baselines,
 with gains up to 3.2x on Ant and nearly 9x on Swimmer. Iterative debate yields
18-35% gains over compute-matched zero-shot generation, and D2C-generated reward
s transfer to default morphologies in 4/5 tasks. These results suggest that stru
ctured, simulator-grounded multi-agent interaction is a useful mechanism for joi
nt morphology-reward optimization under a fixed-topology, per-candidate-RL proto
col. Project page: debate2create.github.io.
****

ACTIVE-o3 : Empowering MLLMs with Active Perception via Pure Reinforcement Learning
Muzhi Zhu, Hao Zhong, Canyu Zhao, Zongze Du, Mingyu Liu, Zheng Huang, Anzhou Li, Hao Chen, Cheng Zou, Jingdong Chen, Ming Yang, Chunhua Shen
Active vision, also known as active perception, refers to actively selecting where and how to look in order to gather task-relevant information. It is a critical component of efficient perception and decision-making in humans and advanced embodied agents. With the rise of Multimodal Large Language Models (MLLMs) as central planners in robotic systems, the lack of methods for equipping MLLMs with active perception has become a key gap. We first provide a systematic definition of MLLM-based active perception tasks and show that GPT-o3's zoom-in strategy can be viewed as a special case, though it suffers from low efficiency and inaccurate region selection. To address these issues, we propose Active-o3, a reinforcement learning framework built on GRPO that equips MLLMs with active perception capabilities. Leveraging a modular sensing-action design and a dual-form reward, Active-o3 autonomously learns efficient and stable region selection strategies without explicit supervision. We further establish a comprehensive benchmark covering both open-world tasks (small/dense-object grounding) and domain-specific scenarios (remote sensing, autonomous driving, interactive segmentation). Experimental results demonstrate that Active-o3 significantly enhances active perception capabilities compared to Qwen2.5-VL-CoT. Moreover, we show that our RL framework not only preserves the model’s general understanding ability but can also serve as a proxy task for leveraging perception data, further improving performance on benchmarks such as RealWorldQA. We hope that our work can provide a simple codebase and unified evaluation protocol to facilitate future research on active perception with MLLMs.
****
TokenRatio: Principled Token-Level Preference Optimization via Ratio Matching
Truong Nguyen, Tien-Phat Nguyen, Linh Ngo Van, Duy Minh Ho Nguyen, Khoa D Doan, Trung Le
Direct Preference Optimization (DPO) is a widely used RL-free method for aligning language models from pairwise preferences, but it models preferences over full sequences even though generation is driven by per-token decisions. Existing token-level extensions typically decompose a sequence-level Bradley–Terry objective across timesteps, leaving per-prefix (state-wise) optimality implicit. We study how to recover **token-level** preference optimality using only standard sequence-level pairwise comparisons.
We introduce **Token-level Bregman Preference Optimization (TBPO)**, which posits a token-level Bradley--Terry preference model over next-token actions conditioned on the prefix, and derive a Bregman-divergence density-ratio matching objective that generalizes the logistic/DPO loss while preserving the optimal policy induced by the token-level model and maintaining DPO-like simplicity. We introduce two instantiations: TBPO-Q, which explicitly learns a lightweight state baseline, and TBPO-A, which removes the baseline through advantage normalization.
Across instruction following, helpfulness/harmlessness, and summarization, TBPO improves alignment quality and training stability and increases output diversity relative to strong sequence-level and token-level baselines.
****
RelayCaching: Accelerating LLM Collaboration via Decoding KV Cache Reuse
Yingsheng Geng, Yuchong Gao, Weihong Wu, Guyue Liu, Jiang liu
The increasing complexity of AI tasks has shifted the paradigm from monolithic models toward multi-agent large language model (LLM) systems. However, these collaborative architectures introduce a critical bottleneck: redundant prefill computation for shared content generated by previous agents, which significantly increases KV cache memory usage and time-to-first-token (TTFT). While various KV cache methods have been proposed to mitigate prefill redundancy, they either fail to maintain accuracy on agent-generated outputs or exhibit low reuse rates due to rigid constraints. We present RelayCaching, a training-free inference method that directly reuses decoding phase KV caches from previous agents in subsequent prefill phases. Our key insight is that KV caches for identical content are highl

y consistent across phases, while prefix-induced deviations are sparse and local
ized within a limited range of layers and token positions. By selectively recomp
uting KV caches at these positions, RelayCaching preserves model accuracy with m
inimal overhead, yielding a superior accuracy–efficiency trade-off over existing
 methods. Experiments on diverse collaborative LLM tasks spanning mathematical r
easoning, general knowledge, and code generation demonstrate that RelayCaching a
chieves over 80\% KV cache reuse, reduces TTFT by up to $4.7\times$ compared t
o the standard pipeline, all with negligible accuracy degradation.
****

On the Fragility of Data Attribution When Learning Is Distributed

Xian Gao, Bo Hui, Min-Te Sun, Wei-Shinn Ku

Data attribution has become an important component of pricing, auditing, and gov
ernance in machine learning pipelines, yet most attribution methods implicitly a
ssume that attribution values faithfully reflect participants' contributions. We
 show that this assumption can fail: a single participant in a standard distribu
ted training workflow can substantially inflate its measured attribution value w
hile preserving global utility. Our attribution-first attack uses latent optimiz
ation to inject small synthetic batches that preserve utility while exploiting n
on-IID label coverage and evaluator sensitivities. Across datasets, models, and
multiple marginal-utility evaluators, the attack consistently increases the adve
rsary’s attribution value and reshapes the relative attribution structure among
benign clients without degrading accuracy or triggering geometry-based defenses.
 These results show that attribution itself forms a new attack surface and motiv
ate the development of attribution-robust and incentive-compatible scoring mecha
nisms.
****

SparseInfer: Accelerating Large Language Model Inference with Semantics-Inspired
 Adaptive Sparse Activation

Qinsi Wang, Saeed Vahidian, Hancheng Ye, Jianyang Gu, Jianyi Zhang, Yiran Chen

Large Language Models (LLMs) with billions of parameters have transformed AI app
lications but require immense computational and memory resources during inferenc
e.

Adaptive sparse activation inference, which activates only a small number of neu
rons for each token, offers a novel way to accelerate model inference without de
grading performance, showing great potential for resource-constrained hardware d
evices. Yet existing token-level MLP-based methods frequently alter activation m
aps, reducing efficiency gains.

In this paper, we introduce \textbf{SparseInfer}, an MLP-free adaptive sparse ac
tivation inference method based on sentence-level prediction.

We first propose the concept of core neurons and empirically demonstrate that, f
or an input sentence, LLMs only need the core neurons to maintain performance.

Remarkably, we discovered that core neurons exhibit both stability and similarit
y in relation to the sentence's semantics—an insight overlooked by previous stud
ies.

Building on this finding, we design two semantic-based methods for predicting co
re neurons to fit different input scenarios, which enables core neurons to be de
termined during the pre-filling stage and fixed during the encoding stage.

Our experiments verify SparseInfer exhibits good performance in various tasks an
d achieve 10.33$\times$ speed up.
****

Advancing Analytic Class-Incremental Learning through Vision-Language Calibratio
n

Binyu Zhao, Wei ZHANG, Xingrui Yu, Zhaonian Zou, Ivor Tsang

Class-incremental learning (CIL) with pre-trained models (PTMs) faces a critical
 trade-off between efficient adaptation and long-term stability. While analytic
learning enables rapid, recursive closed-form updates, its efficacy is often com
promised by accumulated errors and feature incompatibility. In this paper, we fi
rst conduct a systematic study to dissect the failure modes of PTM-based analyti
c CIL, identifying representation rigidity as the primary bottleneck. Motivated
by this insight, we propose **VILA**, a novel dual-branch framework that advance

s analytic CIL via a two-level vision-language calibration strategy. Specificall y, we coherently fuse plastic, task-adapted features with a frozen, universal vi sual anchor at the feature level through geometric calibration, and leverage cro ss-modal semantic priors at the decision level to rectify prediction bias. This confluence maintains analytic-learning's extreme efficiency while overcoming its inherent brittleness. Extensive experiments across eight benchmarks demonstrate that VILA consistently yields superior performance, particularly in fine-graine d and long-sequence scenarios. Our framework harmonizes high-fidelity prediction with the simplicity of analytic learning. Our code is available at https://gith ub.com/byzhaoAI/VILA.

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Kuramoto Oscillatory Phase Encoding: Neuro-inspired Synchronization for Improved Learning Efficiency

Mingqing Xiao, Yansen Wang, Dongqi Han, Caihua Shan, Dongsheng Li

Spatiotemporal neural dynamics and oscillatory synchronization are widely implic ated in biological information processing and have been hypothesized to support flexible coordination such as feature binding. By contrast, most deep learning a rchitectures represent and propagate information through activation values, negl ecting the joint dynamics of rate and phase. In this work, we introduce Kuramoto oscillatory Phase Encoding (KoPE) as an additional, evolving phase state to Vis ion Transformers, incorporating a neuro-inspired synchronization mechanism to ad vance learning efficiency. We show that KoPE can improve training, parameter, an d data efficiency of vision models through synchronization-enhanced structure le arning. Moreover, KoPE benefits tasks requiring structured understanding, includ ing semantic and panoptic segmentation, representation alignment with language, and few-shot abstract visual reasoning (ARC-AGI). Theoretical analysis and empir ical verification further suggest that KoPE can accelerate attention concentrati on for learning efficiency. These results indicate that synchronization can serv e as a scalable, neuro-inspired mechanism for advancing state-of-the-art neural network models.

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LEGO-FL: Learning Heterogeneous Federated Models as a LEGO Assembly Games

Zeqi Leng, Chunxu Zhang, Guodong Long, Bo Yang

Just as LEGO pieces can be assembled into an unlimited variety of structures, he terogeneous federated learning (HFL) can be viewed as the assembly of diverse mo del components. Inspired by this analogy, we reformulate HFL as a LEGO-like asse mbly game. The central challenge in HFL lies in learning across heterogeneous mo del architectures, which hinders direct parameter sharing.

To address this challenge, we propose to decompose models into a set of modular components—analogous to LEGO pieces and collaboratively learn these components a cross clients under predefined composition rules.

Based on this perspective, we develop a novel federated learning framework, term ed LEGO-FL, which enables flexible model construction while preserving collabora tive learning. Extensive experiments validate the effectiveness of LEGO-FL under different heterogeneous settings and system scales.

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Correctness-Optimized Residual Activation Lens (CORAL): Transferrable and Calibr ation-Aware Inference-Time Steering

Miranda Muqing Miao, Young Min Cho, Lyle Ungar

Large language models (LLMs) exhibit persistent miscalibration, especially after instruction tuning and preference alignment. Modified training objectives can i mprove calibration, but retraining is expensive. Inference-time steering offers a lightweight alternative, yet most existing methods optimize proxies for correc tness rather than correctness itself. We introduce CORAL (Correctness-Optimized Residual Activation Lens), a regularized inference-time steering method that cap tures distributed correctness signals from model internal activations using weig ht-decay MLP probes. We evaluate CORAL across three 7B-parameter models and find that it consistently improves accuracy by 10\% and expected calibration error (ECE) by 50\% on average. We additionally demonstrate that these gains transfer w ithout retraining to the complete published test sets of four held-out benchmark

s (ARC-Challenge, HellaSwag, Math-MC, OpenBookQA), averaging 14\% accuracy impro vements and 49\% ECE improvements. Our results support the hypothesis that distr ibuted information in model internals can be extracted using regularized probes when individual neurons are insufficient. CORAL thus provides a compute-efficien t, transferable, and calibration-aware approach to improve MCQA performance duri ng inference.
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EngiAgent: Fully Connected Coordination of LLM Agents for Solving Open-ended Eng ineering Problems with Feasible Solutions
Xiyuan Zhou, Ruixi Zou, Xinlei Wang, Yuheng Cheng, Yan Xu, Junhua Zhao, Jinjin G u
Engineering problem solving is central to real-world decision-making, requiring mathematical formulations that not only represent complex problems but also prod uce feasible solutions under data and physical constraints. Unlike mathematical problem solving, which operates on predefined formulations, engineering tasks de mand open-ended analysis, feasibility-driven modeling, and iterative refinement. Although large language models (LLMs) have shown strong capabilities in reasoni ng and code generation, they often fail to ensure feasibility, which limits thei r applicability to engineering problem solving. To address this challenge, we pr opose EngiAgent, a multi-agent system with a fully connected coordinator that si mulates expert workflows through specialized agents for problem analysis, modeli ng, verification, solving, and solution evaluation. The fully connected coordina tor enables flexible feedback routing, overcoming the rigidity of prior pipeline -based reflection methods and ensuring feasibility at every stage of the process . This design not only improves robustness to diverse failure cases such as data extraction errors, constraint inconsistencies, and solver failures, but also en hances the overall quality of problem solving. Empirical results across four rep resentative domains demonstrate that EngiAgent achieves substantial improvements in feasibility compared to prior approaches, establishing a new paradigm for fe asibility-oriented engineering problem solving with LLMs. Our source code and da ta are available at https://github.com/AI4Engi/EngiAgent.
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Large Scale Manifold Balanced Clustering
Fangfang Li, Quanxue Gao, Xingyu Xue
Manifold clustering has demonstrated strong capability in capturing complex data structures and has been widely studied in cluster analysis. However, many exist ing methods mainly focus on combining K-means with manifold learning, while over looking the consistency between data structures and clustering labels, and often suffer from high computational cost when handling large scale data. To address these issues, we propose a manifold balanced clustering method based on anchor i nduced distance(LMBC), grounded in the relationship between K-means clustering a nd manifold learning. Specifically, the LMBC uses label information to guide the construction of the manifold structure, thereby ensuring consistency between da ta structures and clustering labels. To enable large scale clustering, we introd uce an anchor induced distance representation that models manifold structure in a compact anchor space, significantly reducing computational complexity while pr eserving essential structural information. Furthermore, to naturally maintain cl ass balance during clustering, we maximize the Schatten-p norm of the label repr esentation and provide theoretical analysis to support its effectiveness. Experi mental results on several benchmark datasets demonstrate the effectiveness and s calability of the proposed method.
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LEMUR: Learned Multi-Vector Retrieval
Elias Jääsaari, Ville Hyvönen, Teemu Roos
Multi-vector representations generated by late interaction models, such as ColBE RT, enable superior retrieval quality compared to single-vector representations in information retrieval applications. In multi-vector retrieval systems, both q ueries and documents are encoded using one embedding per token, and similarity b etween queries and documents is measured by the MaxSim similarity measure. Howev er, the improved quality of multi-vector retrieval comes at the expense of signi

ficantly increased search latency. In this work, we introduce LEMUR, a simple ye t efficient framework for multi-vector similarity search. LEMUR consists of two consecutive problem reductions: First, we formulate multi-vector similarity sear ch as a supervised learning problem that can be solved using a one-hidden-layer neural network. Second, we reduce inference under this model to single-vector si milarity search in its latent space, enabling the use of existing single-vector search indexes to accelerate retrieval. LEMUR is an order of magnitude faster th an prior multi-vector similarity search methods. Our code is available at https: //github.com/ejaasaari/lemur
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Reasoning Theater: Disentangling Model Beliefs from Chain-of-Thought
Siddharth Boppana, Annabel Ma, Max Loeffler, Raphaël Sarfati, Eric Bigelow, Atti cus Geiger, Owen Lewis, Jack Merullo
We provide evidence of \textit{performative} chain-of-thought (CoT) in reasoning models, where a model becomes strongly confident in its final answer, but conti nues generating tokens without revealing its internal belief.
Our analysis compares activation probing, early forced answering, and a CoT moni tor across two large models (DeepSeek-R1 671B \& GPT-OSS 120B) and find task dif ficulty-specific differences: The model's final answer is decodable from activat ions far earlier in CoT than a monitor is able to say, especially for easy recal l-based MMLU questions. We contrast this with genuine reasoning in difficult mul tihop GPQA-Diamond questions. Despite this, \textit{inflection points} (e.g., ba cktracking, `aha' moments) occur almost exclusively in responses where probes sh ow large belief shifts, suggesting these behaviors track genuine uncertainty rat her than learned ``reasoning theater." Finally, probe-guided early exit reduces tokens by up to 80\% on MMLU and 30\% on GPQA-Diamond with similar accuracy, pos itioning attention probing as an efficient tool for detecting performative reaso ning and enabling adaptive computation.
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A Minimax Approach for Optimal Intervention Policy Learning with Two-Stage Outco mes
Chenyang Li, Hao Mei, Yue Liu
When designing interventions to promote desired actions, two-stage agent heterog eneity -- encompassing both engagement with the intervention and completion of the desired action -- creates significant challenges in identifying optimal inte rvention policies. While this two-dimensional heterogeneity creates distinct age nt response types with varying marginal policy returns, existing literature typi cally falls short in full identification of all agent types, leading to ineffici ent intervention allocations.
To address the challenge of learning optimal policies that account for two-stage outcomes, we propose a minimax approach within a counterfactual principal strat a framework. A value function, accommodating varying policy returns across six p otentially non-identifiable principal strata, is designed and partially identifi ed to minimize the worst-case value loss relative to three benchmark policies: n ever-treat, always-treat, and oracle. We introduce three estimators for optimal policy learning: Principal Outcome Regression (P-OR), Principal Inverse Propensi ty Scoring (P-IPS), and Principal Doubly Robust (P-DR), providing theoretical gu arantees for their unbiasedness, robustness, and regret upper bounds. Extensive numerical experiments demonstrate the effectiveness and superiority of the propo sed approach.
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When Diffusion Language Models Hesitate: Detecting and Correcting Visual Halluci nations via Confidence Fluctuation
Wenzheng Song, Pei Chen, Yichen Tan, Zejian Li, Lingyun Sun
Multi-modal Diffusion Language Models (MDLMs) have emerged as a powerful alterna tive to autoregressive models in vision understanding, offering advantages in bi directional context modeling and parallel decoding. However, existing MDLMs suff er from visual hallucinations due to the static nature of visual perception. Unl ike autoregressive models, MDLMs lack the sequential dependency to dynamically i nteract with visual content. Therefore, MDLMs rely on fixed visual features enco

ded at initialization, causing the denoising process to drift toward language pr iors and lose its anchor to visual evidence. In this paper, we propose VGR (Visu al-Guided Refinement), a framework that enables MDLMs to revisit visual details by exploiting diffusion dynamics. Our key insight is that the temporal trajector y of confidence during denoising reveals intrinsic uncertainty: while grounded t okens converge smoothly, hallucinated ones exhibit pronounced confidence fluctua tion. VGR utilizes this fluctuation signal to detect uncertain spans and correct s them through targeted visual evidence extraction and in-place remasking. Exten sive experiments on image captioning and hallucination evaluation benchmarks dem onstrate that our method reduces hallucinations and recalls more details. We rel ease our code at https://github.com/SongWZ3214/VGR.
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Shared Lexical Task Representations Explain Behavioral Variability In LLMs
Zhuonan Yang, Xiaochen Li, Francisco Piedrahita Velez, Eric Todd, David Bau, Mic hael Littman, Stephen Bach, Ellie Pavlick
One of the most common complaints about large language models (LLMs) is their pr ompt sensitivity---that is, the fact that their ability to perform a task or pro vide a correct answer to a question can depend unpredictably on the way the ques tion is posed. We investigate this variation by comparing two very different but commonly-used styles of prompting: *instruction-based* prompts, which describe the task in natural language, and *example-based* prompts, which provide in-cont ext few-shot demonstration pairs to illustrate the task. We find that, despite l arge variation in performance as a function of the prompt, the model engages som e common underlying mechanisms across different prompts of a task. Specifically , we identify task-specific attention heads whose outputs literally describe the task --- which we dub *lexical task heads*---and show that these heads are shar ed across prompting styles and trigger subsequent answer production. We further find that behavioral variation between prompts can be explained by the degree to which these heads are activated, and that failures are at least sometimes due t o competing task representations that dilute the signal of the target task. Our results together present an increasingly clear picture of how LLMs' internal rep resentations can explain behavior that otherwise seems idiosyncratic to users an d developers.
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OPTION: Optimal Transport–Guided Flow Matching for Incomplete and Unaligned Mult i-View Clustering
Siyuan Zhou, Zhibin Gu
Multi-view clustering effectively exploits rich information from multiple views, yet real-world applications are frequently challenged by missing views and cros s-view sample misalignment, hindering cross-view modeling and resulting in infer ior clustering performance. To address these challenges, this paper presents a n ovel method, **OP**timal **T**ransport–Gu**I**ded fl**O**w Matchi**N**g for inco mplete and unaligned multi-view clustering (**OPTION**). Specifically, OPTION em ploys conditional flow matching to learn deterministic transport paths for missi ng-view imputation, enabling stable manifold-preserving recovery and more discri minative representations. To support alignment-free fusion, we introduce a Gromo v-Wasserstein-inspired structural regularization that aligns intra-view geometri c structures in the latent space without solving hard correspondences. Furthermo re, an optional contrastive regularization is incorporated to enhance cross-view consistency specifically for aligned settings. Extensive experiments demonstrat e that OPTION outperforms state-of-the-art methods across ideal, incomplete, and unaligned scenarios evaluated separately. Code: https://github.com/TimoZhou1024 /OPTION.
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Reason with Thumbnails, Answer with Focus: An Efficient and Effective Paradigm f or Multimodal Grounded Visual Reasoning
An-Lan Wang, Guozhi Tang, Lei Liao, Hanshen Zhu, Kai Huang, Jingqun Tang, Jiamin g Zhou, Kun-Yu Lin
To enhance the interpretability of multimodal large language models' outputs, re cent efforts explored Grounded Visual Reasoning (GVR), in which the model is tra

ined to select relevant image regions before answering the question. However, th e multi-round ``ground-then-answer'' and reasoning nature of these methods impos es much more computational costs compared to non-GVR methods. To attain efficien t and effective GVR, in this paper, we propose a novel paradigm called Reason wi th Thumbnails, Answer with Focus (RTAF), which feeds the model with low-resoluti on images to reason the relevant regions and high-resolution crops to answer the final answer. Our motivation arises from the observation that, in many cases, t he key area required to answer questions can be inferred from the low-resolution thumbnails, without the need for a full-resolution image. Additionally, for ext reme cases where thumbnails lack sufficient information (leading to undirected r egion guessing and increased computation), we equip the model with a tool to acc ess higher-resolution images. For training efficiency, we adopt pure reinforceme nt learning (i.e., GRPO) and design a suite of reward functions to supervise the model's behavior, alongside a resolution-aware training data selection strategy . Finally, our model, based on Qwen2.5-VL, achieves significant improvements acr oss a range of benchmarks with reduced computation, demonstrating the effectiven ess and efficiency of our proposed RTAF, e.g., compared to the non-GVR model Qwe n2.5-VL, our model achieves a performance gain of 5.8 while using comparable vis ual tokens (471 vs. 391). Against state-of-the-art GVR methods, RTAF reduces vis ual token usage by half while delivering superior performance.
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Modelling Attention with Aitchison Geometry: Token Distinguishability and Temper ature Scaling

Sam Hilton-Jones, Timothy J. Norman, Zhanxing Zhu

The attention mechanism with softmax normalisation is a foundational component o f Transformer-based large language models. However, with very long contexts, at tention scores are known to diminish, raising fundamental questions about token distinguishability and how it can be preserved. In this work, we provide a forma l characterisation of token distinguishability in attention as a function of con text length and embedding dimension. We introduce Aitchison distance to quantify relative differences among attention probabilities, and show that, with Gaussia n queries and keys, even in the long-context regime, token distinguishability co nverges to a finite, non-zero limit rather than vanishing. Leveraging the linear relationship between inverse-temperature scaling and Aitchison distance, we der ive a theoretical lower bound of $\Omega(\sqrt{\log L})$ on the logit scaling re quired to produce a sharp attention distribution. Finally, we demonstrate that A itchison distance provides a principled and practical alternative to entropy for monitoring training and inference, as it captures the full compositional struct ure, including the smaller components of the attention probabilities.
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MUSA-PINN: Multi-scale Weak-form Physics-Informed Neural Networks for Fluid Flow in Complex Geometries

Weizheng Zhang, Xunjie Xie, Hao Pan, Xiaowei Duan, Bingteng Sun, Qiang Du, Lin L u

While Physics-Informed Neural Networks (PINNs) offer a mesh-free approach to sol ving fluid-flow PDEs, standard point-wise residual minimization suffers from con vergence pathologies in topologically complex domains like Triply Periodic Minim al Surfaces (TPMS). The locality bias of point-wise constraints fails to propaga te global information through tortuous channels, causing unstable gradients and conservation violations.

To address this, we propose the Multi-scale Weak-form PINN (MUSA-PINN), which re formulates Navier-Stokes equation constraints as integral conservation laws over hierarchical spherical control volumes.

We enforce continuity and momentum conservation via flux-balance residuals on co ntrol surfaces. Our method utilizes a three-scale subdomain strategy-comprising large volumes for long-range coupling, skeleton-aware meso-scale volumes aligned with transport pathways, and small volumes for local refinement-alongside a two -stage training schedule prioritizing continuity. Experiments on steady incompre ssible flow in TPMS geometries show MUSA-PINN outperforms state-of-the-art basel ines, reducing relative errors by up to 93\% and preserving mass conservation.

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Semantic Integrity Matters: Benchmarking and Preserving High-Density Reasoning in KV Cache Compression
Xiang Liu, Zhenheng Tang, Hong Chen, Peijie Dong, Zeyu Li, Xiuze Zhou, Bo Li, Xuming Hu, Xiaowen Chu
While Key-Value (KV) cache compression is essential for efficient LLM inference, current evaluations disproportionately focus on \textbf{sparse retrieval} tasks, potentially masking the degradation of High-Density Reasoning where Chain-of-Thought (CoT) coherence is critical. We introduce KVFundaBench to systematically evaluate this gap, revealing a sharp dichotomy: while retrieval tasks remain robust, reasoning tasks exhibit severe Task-Dependent Degradation under aggressive compression due to disrupted CoT links. Extending our analysis to the DeepSeek-R1 model, we uncover that its specialized attention patterns offer unique insights into the fragility of reasoning chains. Guided by these findings—specifically the necessity of preserving few-shot examples as indivisible \textbf{Semantic Units}—we propose ShotKV. This approach explicitly separates prefill and decoding phases to prioritize semantic integrity. Empirical results demonstrate that ShotKV achieves 9\%-18\% accuracy improvements on long-context generation tasks and effectively generalizes to document QA, all while delivering an 11\% latency reduction compared to full cache inference.
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Fisher-Preserving Guidance: Training-Free Manifold Constraints for Safe Diffusion Control
Hao Ren, Zetong Bi, Yiming Zeng, Le Zheng, Zhi Li, Zhaoliang Wan, Lu Qi, Hui Cheng
Diffusion models are effective for waypoint prediction in visual navigation, but standard sampling and test time guidance can produce unreliable or inefficient trajectories when updates drift off the training manifold. We propose Fisher Preserving Guidance with Outer Product Span Projection, a training-free inference method that avoids large Fisher drift associated with off-distribution actions while optimizing a task objective. Our method computes the Fisher-preserving update via a low-rank Jacobian factorization, requiring only a single backward pass per step and enabling real-time use. We further introduce Truncated Fisher Denoising Sensitivity as an uncertainty signal and use it for robust multi-sample action blending. Experiments on toy and realistic navigation benchmarks, including Maze2D with TSDF-based guidance, PushT with official Diffusion Policy weights, and visual navigation in simulation and on real robots, demonstrate consistent improvements in performance over strong diffusion-policy baselines without additional training.
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PipeSD: An Efficient Cloud-Edge Collaborative Pipeline Inference Framework with Speculative Decoding
Yunhe Han, Yunqi Gao, Bing Hu, Mahdi Boloursaz Mashhadi, Yitong Duan, Pei Xiao, Yanfeng Zhang
Speculative decoding can significantly accelerate LLM inference,
especially given that its cloud-edge collaborative deployment offers cloud workload offloading,
offline robustness, and privacy enhancement.
However, existing collaborative inference frameworks with speculative decoding are constrained by
(i) sequential token generation and communication with low resource utilization,

and (ii) inflexible cloud non-autoregressive verification (NAV) triggering that induces premature verification or costly rollbacks.
In this paper, we propose PipeSD, an efficient cloud-edge collaborative pipeline inference framework with speculative decoding.
PipeSD overlaps token generation and communication by a token-batch pipeline scheduling mechanism optimized by dynamic programming,
and improves verification flexibility through a dual-threshold NAV triggering mechanism with a lightweight Bayesian optimization autotuner.

We implement PipeSD using llama-cpp-python, PyTorch, and FastAPI,
and evaluate it on a real-world cloud-edge testbed
with two draft-target model pairs across four scenarios.
Results show that PipeSD consistently outperforms state-of-the-art baselines,
achieving $1.16\times$-$2.16\times$ speedup and
reducing energy consumption by 14.3%-25.3%.
Our code is available at https://github.com/Ghanyunhe/PipeSD.
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On the Generalization in Topology Optimization via Sensitivity-Conditioned Bernoulli Flow Matching
Mohammad Rashed, Duarte F. Valoroso Madeira, Babak Gholami, Caglar Guerbuez, Yunjia Yang, Nils Thuerey
Surrogate models for topology optimization (TO) exhibit highly variable out-of-distribution (OOD) generalization under distribution shifts such as changing loads or boundary conditions, yet the source of this variability remains unclear. We hypothesize that OOD performance is governed by how much information the conditioning signal preserves about the adjoint sensitivity (reduced gradient) that drives classical TO. Modeling the TO pipeline as a causal Markov chain, the Data Processing Inequality establishes that, under this abstraction, the sensitivity field is an information-theoretically optimal conditioning signal for topology prediction. However, computing exact adjoint sensitivities can be expensive or unavailable in practice; we observe that certain physical fields can approximate sensitivities through monotone transformations. To formalize this, we introduce \textbf{pseudo-sensitivities} to characterize which fields enable generalization versus those that are information-poor. We then show that a sensitivity-conditioned Bernoulli flow-matching generator empirically confirms these predictions: conditioning on sensitivities yields state-of-the-art OOD performance, while increasingly distant physical fields degrade toward raw parameter conditioning. Results hold across structural TO benchmarks under load shifts and our new CFD-TO dataset under boundary-condition shifts such as multi-outlet configurations. Code and datasets are available at https://github.com/tum-pbs/topotransformer.
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Confidence is Not Universal: Task-Dependent Calibration and Emergent Behavior in LLMs
Chaeyun Jang, Moonseok Choi, Yegon Kim, Seungyoo Lee, Juho Lee, Hyungi Lee
Large language models (LLMs) increasingly support human decision-making, rendering human-interpretable confidence essential. However, it remains unclear whether verbalized confidence calibration generalizes across heterogeneous tasks without degrading accuracy. We show that universal confidence calibration fails. Across diverse benchmarks, we identify two incompatible task families with distinct confidence semantics. In reasoning-centric tasks, confidence supervision transfers within the family, often improving calibration while preserving or even improving accuracy, and induces emergent behaviors such as confidence-dependent reasoning length and self-verification. Retrieval- and copy-oriented tasks also exhibit within-family transfer, but fail to generalize to reasoning tasks, with cross-family supervision degrading both calibration and accuracy. Motivated by this finding, we disentangle confidence into reasoning uncertainty and evidence localization uncertainty. This simple decomposition restores cross-family generalization using supervised fine-tuning alone, suggesting that effective confidence alignment requires task-aware semantics rather than a universal scalar notion.
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An Interactive Paradigm for Deep Research
Lin Ai, Victor Bursztyn, Xiang Chen, Julia Hirschberg, Saayan Mitra
Recent advances in large language models (LLMs) have enabled deep research systems that synthesize comprehensive, report-style answers to open-ended queries by combining retrieval, reasoning, and generation. Yet, most frameworks rely on rigid workflows with one-shot scoping and long autonomous runs, offering little room for course correction if user intent shifts mid-process. We present **SteER**, a framework for steerable deep research that introduces interpretable, mid-process control into long-horizon research workflows. At each decision point, **SteE

R** uses a cost–benefit formulation to determine whether to pause for user input or proceed autonomously. It combines diversity-aware planning with utility signals that reward alignment, novelty, and coverage, and maintains a live persona model that evolves throughout the session. **SteER** outperforms state-of-the-art open-source and proprietary baselines by up to 22.80% on alignment, leads on quality metrics such as breadth and balance, and is preferred by human readers in 85%+ of pairwise alignment judgments. We also introduce a persona–query benchmark and data-generation pipeline. To our knowledge, this is the first work to advance deep research with an interactive, interpretable control paradigm, paving the way for controllable, user-aligned agents in long-form tasks.
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FlatLab: A Unified Methodology Framework and Simulation-Based Benchmark for Robotic Manipulation of Flat Objects
Xingyu Zhu, Wenshuo Han, Zhouyu Wang, Yuran Wang, Ruihai Wu, Hao Dong, Fan Tang, Hechang Chen, Hyung Jin Chang, Yixing Gao
Robotic manipulation of flat objects is challenging due to the ungraspable configurations and strong variations in object geometry and material. Existing methods rely on heuristic pre-manipulation and are often evaluated in closed settings with limited generalization. We propose a unified framework that decouples the manipulation into a strategy generator and an action execution module. The strategy generator predicts appropriate manipulation strategies from object point clouds by learning strategy-centric, object-invariant representations via simulated data transformation and contrastive learning. Conditioned on the predicted strategy, the execution module decomposes long-horizon manipulation into reusable action primitives and dynamically composes them to generate stable trajectories. To enable systematic evaluation, we introduce FlatLab, a comprehensive simulation benchmark for robotic flat object manipulation. FlatLab provides high-fidelity physical simulation of diverse rigid and deformable flat objects, automated multi-modal data collection, and standardized task definitions and evaluation protocols. Experiments conducted in FlatLab demonstrate that our approach generalizes effectively to unseen objects and categories, outperforming existing baselines. The project page and the code are provided at \url{https://flatlab-web.github.io/}.
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De-attribute to Forget for LLM Unlearning
Xinyang Lu, Jiabao Pan, Rachael Hwee Ling Sim, See-Kiong Ng, Anthony Kum Hoe Tung, Bryan Kian Hsiang Low
The rapid development of large language models (LLMs) has raised concerns regarding the inclusion of private or inappropriate data during training, which has led to growing interest in LLM unlearning. Many existing LLM unlearning approaches rely on prediction loss-based optimizations, such as maximizing the loss on the forget set. However, these methods often face issues such as over-forgetting and poor model utility. In this work, we address these issues by introducing a novel perspective that shifts the unlearning optimization target to reducing data attribution instead. We propose the first LLM unlearning framework based on data attribution rewards called DareU that employs reinforcement learning to update the LLM and reduce the attribution score of generated responses (i.e., de-attribute) to the forget data owners. Experimental results using an LLM classifier as an efficient approximation of attribution demonstrate that DareU outperforms existing baseline approaches, achieving effective unlearning while balancing forget quality and model utility.
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Is Generation Required for Data-Efficient Perception?
Jack Brady, Bernhard Schölkopf, Thomas Kipf, Simon Buchholz, Wieland Brendel
It has been hypothesized that achieving the data efficiency of human visual perception requires a generative approach in which internal representations result from inverting a decoder. Yet today’s most successful vision models are non-generative, relying on an encoder that maps images to representations without decoder inversion. This raises the question of whether generation is necessary for data-efficient machine perception. To address this, we study to what extent generati

ve and non-generative methods can achieve compositional generalization, a hallmark of human data efficiency. Under a compositional generative process, we formally characterize the inductive biases required for compositional generalization in decoder-based (generative) and encoder-based (non-generative) methods. We show theoretically that the inductive biases required for an encoder are substantially more complex and generally infeasible to impose explicitly through architectural constraints or regularization. By contrast, the decoder biases take a simple form that can be enforced directly. These results suggest that compositional generalization may be substantially easier to achieve through a generative paradigm of learning and inverting a decoder rather than learning an encoder directly. We examine our theoretical findings empirically by training a range of generative and non-generative methods on synthetic image data. We find that non-generative methods often fail to generalize compositionally and require large-scale pretraining to improve generalization. By comparison, generative methods yield gains in generalization without requiring additional data.

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Refining Dual Spectral Sparsity in Transformed Tensor Singular Values

Andong Wang, Yuning Qiu, Haonan Huang, Zhong Jin, Guoxu Zhou, Qibin Zhao

The Tubal Nuclear Norm (TNN), derived from the tensor Singular Value Decomposition (t-SVD), is a widely used low-rank modeling tool that promotes sparsity of frequency-domain singular values.

However, as a direct extension of the matrix nuclear norm, TNN applies a uniform element-wise penalty to transformed singular values, without explicitly distinguishing sparsity across frequency components from low-rankness within each component.

This can be restrictive for real-world tensor data that exhibit multi-level spectral structures, where spectral energy is concentrated in a subset of frequency components while active components remain low-rank.

To overcome this limitation, we propose the tensor ℓ_p-Schatten-q quasi-norm ($p,q\in(0,1]$), which enables explicit control of dual spectral sparsity by jointly regularizing inter-frequency sparsity and intra-frequency low-rankness. This formulation includes TNN as a special case and subsumes several existing tensor regularizers by coupling global frequency sparsity with local spectral low-rankness, yielding a more flexible modeling principle.

We establish minimax error bounds under the proposed dual spectral sparsity model, develop a reweighted optimization algorithm for the resulting nonconvex problem, and demonstrate its effectiveness and robustness on noisy and Poisson tensor completion as well as image clustering tasks.

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MemIncept: Steering LLM Agents via Cooperative Stealthy Memory Injections

Nan Yan, Qian Lou, Jiarong Xing

Long-term memory empowers LLM-based agents with adaptive reasoning but exposes a critical attack surface---adversaries can inject malicious records to bias agent behaviors. However, existing attacks face a dilemma: effective injections are often visibly malicious and easily detected, while stealthy, benign-looking injections are often less effective in altering agent behaviors. To address this, we propose MemIncept, a memory poisoning attack that can impact agents even in black-box settings using only benign-appearing queries. Unlike prior methods that inject isolated records, MemIncept generates a cooperative set of queries that work together to bias the agent. It achieves this via a bidirectional evolutionary strategy that optimizes the query set from two ends. A forward pass ensures the queries collectively lead the agent to the target outcome, while a backward pass ensures they are semantically close to victim (benign) queries for reliable retrieval. This ``meet-in-the-middle'' approach creates injected records that are both easy to retrieve and effective at steering behavior. Through extensive experiments across diverse agents, we show that MemIncept significantly outperforms single-record attacks, achieving high success rates comparable to explicit attacks while remaining difficult to flag under automated filters or human inspection.

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Thinking in Structures: Evaluating Spatial Intelligence in Constraint-Governed Spaces
Chen Yang, Guanxin Lin, Youquan He, Peiyao Chen, Guanghe Liu, Yufan Mo, Zhouyuan Xu, Linhao Wang, Guohui Zhang, Zihang Zhang, Shenxiang Zeng, Chen Wang, Jiansheng Fan
Spatial intelligence is crucial for vision--language models (VLMs), yet many scene-centric benchmarks evaluate unconstrained environments where a single image may admit multiple plausible 3D interpretations. We introduce SSI-Bench, a VQA benchmark for Structure-Centric Spatial Reasoning (SCSR) in constraint-governed spaces. Built from complex real-world 3D structures, it uses structural constraints from geometry, topology, and physical feasibility to make component relations more determinate from visual evidence. The benchmark contains 1,000 ranking questions spanning geometric and topological reasoning, where correct ordering requires resolving all candidate-wise 3D relations, imposing stronger demands on spatial understanding. It is created through a fully human-centered pipeline with over 400 researcher-hours of image curation, component annotation, and question design. Evaluating 31 VLMs reveals a large gap to humans: the best open-source model achieves 22.2% accuracy and the strongest closed-source model reaches 33.6%, while humans score 91.6%. Further results show that chain-of-thought reasoning brings only marginal gains, and error analysis reveals fundamental limitations in current models' spatial understanding within constraint-governed spaces. Project page: https://ssi-bench.github.io.
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Symbol-Equivariant Recurrent Reasoning Models
Richard Freinschlag, Timo Bertram, Erich Kobler, Andreas Mayr, Günter Klambauer
Reasoning problems such as Sudoku and ARC-AGI remain challenging for neural networks. The structured problem solving architecture family of Recurrent Reasoning Models (RRMs), including Hierarchical Reasoning Model (HRM) and Tiny Recursive Model (TRM), offer a compact alternative to large language models, but currently handle symbol symmetries only implicitly via costly data augmentation.
We introduce Symbol-Equivariant Recurrent Reasoning Models (SE-RRMs), which enforce permutation equivariance at the architectural level through symbol-equivariant layers, guaranteeing identical solutions under symbol or color permutations. SE-RRMs outperform prior RRMs on 9$\times$9 Sudoku and generalize from just training on 9$\times$9 to smaller 4$\times$4 and larger 16$\times$16 and 25$\times$25 instances, to which existing RRMs cannot extrapolate.
On ARC-AGI-1 and ARC-AGI-2, SE-RRMs achieve competitive performance with substantially less data augmentation and only 2 million parameters, demonstrating that explicitly encoding symmetry improves the robustness and scalability of neural reasoning.
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ArenaRL: Scaling RL for Open-Ended Agents via Tournament-based Relative Ranking
Qiang Zhang, Boli Chen, Fanrui Zhang, Ruixue Ding, Shihang Wang, Qiuchen Wang, Yinfeng Huang, Haonan Zhang, Rongxiang Zhu, Xin Li, Houquan Zhou, Pengjun Xie, Kaipeng Zhang, Jingren Zhou, Jiawei Liu
Reinforcement learning (RL) has advanced LLM agents on verifiable tasks but remains challenging for open-ended tasks with vast solution spaces (e.g., complex travel planning). Lacking objective ground truth, current RL algorithms rely on reward models assigning scalar scores to individual responses. We contend such pointwise scoring induces discrimination collapse: reward model fails to distinguish subtle advantages among trajectories, compressing intra-group rewards into a narrow range. This drowns effective reward signals in reward model noise, causing optimization stagnation. To address this, we propose ArenaRL, a reinforcement learning paradigm shifting from pointwise scalar scoring to intra-group relative ranking. ArenaRL introduces a process-aware pairwise evaluation with multi-level rubrics for fine-grained relative scoring. Meanwhile, we construct an intra-group adversarial arena and devise a tournament-based ranking scheme to obtain stable advantage signals. ArenaRL achieves high-precision advantage estimation with only $O(N)$ computational complexity, striking a favourable balance between efficiency and accuracy. Furthermore, to address the lack of full-cycle benchmarks f

or open-ended agents, we introduce two high-quality benchmarks: Open-Travel and Open-DeepResearch, encompassing full training and multi-dimensional evaluation p ipelines. Extensive experiments across three open-ended tasks validate the effec tiveness of ArenaRL.
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TAG: Tangential Amplifying Guidance for Hallucination-Resistant Sampling
Hyunmin Cho, Donghoon Ahn, Susung Hong, Jee Eun Kim, Seungryong Kim, Kyong Hwan Jin
Diffusion models achieve state-of-the-art image generation but often produce sem antic inconsistencies, or *hallucinations*. Existing inference-time guidance met hods rely on external signals or architectural modifications, adding computation al overhead. We propose **T**angential **A**mplifying **G**uidance **(TAG)**, a training-free, architecture-agnostic, plug-and-play guidance method that operate s purely on trajectory signals. TAG uses an intermediate sample as a projection basis and amplifies the tangential components of the estimated score to correct the sampling trajectory. A first-order Taylor analysis shows that this steers th e state toward higher-probability regions of the data manifold, reducing inconsi stencies and improving fidelity while adding negligible overhead to existing sam plers. Code is available at our Project Page (https://hyeon-cho.github.io/TAG/).
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Clarify Before You Draw: Proactive Agents for Robust Text-to-CAD Generation
Bo Yuan, Zelin Zhao, Petr Molodyk, Bin Hu, Yongxin Chen
Large language models have recently enabled text-to-CAD systems that synthesize parametric CAD programs (e.g., CadQuery) from natural-language prompts. In pract ice, however, geometric descriptions can be under-specified or internally incons istent: critical dimensions may be missing and constraints may conflict. Howeve r, existing fine-tuned models tend to reactively follow the user’s instructions a nd hallucinate dimensions when the text is ambiguous. To address this, we propo se a proactive agentic framework for text-to-CadQuery generation, named as \text bf{ProCAD}, that resolves specification issues before code synthesis. Our framew ork pairs a proactive clarifying agent, which audits the prompt and asks targete d clarification questions only when necessary to produce a self-consistent speci fication, with a CAD coding agent that translates the specification into an exec utable CadQuery program. We fine-tune the coding agent based on a curated high-q uality text-to-CadQuery dataset and train the clarifying agent via agentic SFT o n clarification trajectories. Experiments show that proactive clarification sign ificantly improves robustness to ambiguous prompts while keeping interaction ove rhead low. ProCAD outperforms frontier closed-source models, including Claude So nnet 4.5, reducing the mean Chamfer distance by 79.9\% and lowering the invalidi ty ratio from 4.8\% to 0.9\%. Our code and datasets are made publicly available on \url{https://github.com/BoYuanVisionary/Pro-CAD}.
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From Welfare to Utility: Generalized Objectives in Budget-Feasible Procurement
Alon Eden, Kira Goldner, Eldar Kerner, Thodoris Tsilivis
We study mechanism design for the budget-feasible procurement problem, a natural problem that arises when a buyer wants to procure goods or services from multip le strategic sellers who each have a cost to provide that service, the buyer has a value for each service procured, but is constrained by a budget. In contrast to prior work, which has focused on buyer value maximization for this problem, w e solve for optimal and approximately-optimal mechanisms for the objectives of b uyer utility (value of procured services minus payments), welfare (value minus p roduction costs), and generalizations of the two. For welfare, we design a simpl e mechanism that obtains a constant-factor approximation for the prior-free (wor st-case) setting. As prior-free mechanisms fail to provide any guarantee for uti lity, even for a single seller, we consider Bayesian settings, where the buyer h as distributional knowledge over sellers' costs. We first provide a utility-opt imal mechanism that satisfies the buyer's budget constraint *in expectation*, th en we show how to modify the mechanism to satisfy the budget constraint *ex-post *, for every realization of seller costs, while still obtaining near-optimal uti lity guarantees. Finally, we generalize our mechanisms to other objectives.

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Non-Adversarial Imitation Learning Provably Free of Compounding Errors: The Value Flow Mechanism
Tian Xu, Chenyang Wang, Xiaochen Zhai, Ziniu Li, Yi-Chen Li, Yang Yu
Adversarial imitation learning (AIL) achieves high-quality imitation by mitigating compounding errors inherent to behavioral cloning (BC), yet its adversarial optimization frequently leads to training instability. A class of non-adversarial Q-based imitation learning (IL) methods, exemplified by IQ-Learn, has emerged to address this instability and is widely believed to outperform BC by leveraging online environment interactions. In this paper, we revisit IQ-Learn and prove that it in fact reduces to BC: it admits an imitation gap lower bound with quadratic dependence on the horizon and therefore remains susceptible to compounding errors. Our theoretical analysis reveals why online interactions fail to help: IQ-Learn uniformly suppresses Q-values for all actions at states not covered by demonstrations, preventing generalization beyond demonstrations. To address this fundamental limitation, we introduce Dual Q-DM, a new Q-based IL method built on Bellman constraints. Crucially, Bellman constraints drive value flow: Q-values propagate from demonstrated to unvisited states through environment dynamics, enabling generalization beyond demonstrations. We prove that Dual Q-DM is equivalent to AIL and can recover expert actions at unvisited states, thereby mitigating compounding errors. To the best of our knowledge, Dual Q-DM is the first non-adversarial IL method that is theoretically guaranteed to eliminate compounding errors. Experimental results further corroborate our theoretical findings.
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When RL Meets Adaptive Speculative Training: A Unified Training-Serving System
Junxiong Wang, Fengxiang Bie, Jisen Li, Zelei Shao, Qingyang Wu, Yinghui Liu, Yubo Wang, Avner May, Ben Athiwaratkun, Yineng Zhang, Shuaiwen Leon Song, Zhongzhu Zhou, Chenfeng Xu, Xiaoxia Wu
Speculative decoding can significantly accelerate LLM serving, yet most deployments today disentangle speculator training from serving, treating speculator training as a standalone offline modeling problem. We show that this decoupled formulation introduces substantial deployment and adaptation lag: (1) high time-to-serve, since a speculator must be trained offline for a considerable period before deployment; (2) delayed utility feedback, since the true end-to-end decoding speedup is only known after training and cannot be inferred reliably from acceptance rate alone due to model-architecture, diverse prompt engineering, and system-level overheads; and (3) domain-drift degradation, as the target model is repurposed to new domains and the speculator becomes stale and less effective. To address these issues, we present Aurora, a unified training–serving system that closes the loop by continuously learning a speculator directly from live inference traces. Aurora reframes online speculator learning as an asynchronous reinforcement-learning problem: accepted tokens provide positive feedback, while rejected speculator proposals provide implicit negative feedback that inherits the online traffic failure signals. Our design integrates an SGLang-based inference server with an asynchronous training server, enabling hot-swapped speculator updates without service interruption. Crucially, Aurora supports day-0 deployment: a speculator can be served immediately and rapidly adapted to live traffic, improving system performance while providing immediate utility feedback. Across experiments, Aurora achieves a 1.5$\times$ day-0 speedup on recently released frontier models (e.g., MiniMax M2.1 229B and Qwen3-Coder-Next 80B). Aurora also adapts effectively to distribution shifts in user traffic, delivering an additional 1.25$\times$ speedup over a well-trained but static speculator on widely used models (e.g., Qwen3 and Llama3).
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VideoBrain: Learning Adaptive Frame Sampling for Long Video Understanding
Junbo Zou, Ziheng Huang, Shengjie Zhang, Liwen Zhang, Weining Shen
Long-form video understanding remains challenging for Vision-Language Models (VLMs) due to the inherent tension between computational constraints and the need to capture information distributed across thousands of frames. Existing approaches either sample frames uniformly (risking information loss) or select keyframes

in a single pass (with no recovery from poor choices). We propose VideoBrain, an end-to-end framework that enables VLMs to adaptively acquire visual information through learned sampling policies. Our approach features dual complementary agents: a CLIP-based agent for semantic retrieval across the video and a Uniform agent for dense temporal sampling within intervals. Unlike prior agent-based methods that rely on text-only LLMs orchestrating visual tools, our VLM directly perceives frames and reasons about information sufficiency. To prevent models from invoking agents indiscriminately to maximize rewards, we introduce a behavior-aware reward function coupled with a data classification pipeline that teaches the model when agent invocation is genuinely beneficial. Experiments on four long video benchmarks demonstrate that VideoBrain achieves +3.5\% to +9.0\% improvement over the baseline while using 30-40\% fewer frames, with strong cross-dataset generalization to short video benchmarks. The code is available at https://github.com/junbo-zou/VideoBrain.

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Fox in the Henhouse: Supply-Chain Backdoor Attacks Against Reinforcement Learning

Shijie Liu, Andrew Craig Cullen, Paul Montague, Sarah Monazam Erfani, Benjamin I. P. Rubinstein

Existing backdoor attacks on Reinforcement Learning (RL) typically rely on unrealistic white-box access to victim parameters, rewards, or observations. Inspired by real world behaviors, we introduce the Supply-Chain Backdoor (SCAB) attack to demonstrate that such assumptions are unnecessary. SCAB targets the common practice of training with third-party policies, poisoning the dataset solely through a black-box of legitimate agent-environment interactions. With only 3% data corruption, SCAB demonstrates a peak attack success rate exceeding 90% and reduces victim returns by 80%. These findings expose a critical vulnerability in the modern RL supply chain, highlighting that reliance on untrusted external agents constitutes a severe and practical security risk.

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Online Bayesian Experimental Design for Partially Observed Dynamical Systems

Sara Perez-Vieites, Sahel Iqbal, Simo Särkkä, Dominik Baumann

Bayesian experimental design (BED) provides a principled framework for optimising data collection by choosing experiments that are maximally informative about unknown parameters. However, existing methods cannot deal with the joint challenge of (a) *partially observable dynamical systems*, where only noisy and incomplete observations are available, and (b) *fully online inference*, which updates posterior distributions and selects designs sequentially in a computationally efficient manner. Under partial observability, dynamical systems are naturally modeled as state-space models (SSMs), in which latent states mediate the link between parameters and data, making the likelihood---and thus information-theoretic objectives like the expected information gain (EIG)---intractable. We address these challenges by deriving new estimators of the EIG and its gradient that explicitly marginalise latent states, enabling scalable stochastic optimisation in nonlinear SSMs. Our approach leverages nested particle filters for efficient online state-parameter inference with convergence guarantees. Applications to realistic models, such as the susceptible–infectious–recovered (SIR) model and a moving source location task, show that our framework successfully handles both partial observability and online inference.

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OpenHA: A Series of Open-Source Hierarchical Agentic Models in Minecraft

Zihao Wang, Muyao Li, Kaichen He, Xiangyu Wang, Zhancun Mu, Minghao Liu, Anji Liu, Yitao Liang

A critical challenge in developing capable AI agents is defining their "action space''—the set of possible actions they can take. These spaces can range widely, from generating code and using language skills to operating on latent representations or raw joystick controls. Through a large-scale study in Minecraft, we discovered a major dilemma: no single action space is universally best. The most effective action space is highly task-dependent, which complicates the goal of building one generalist agent that can handle everything. To solve this, we introd

uce Chain-of-Action (CoA), a novel framework that unifies high-level abstracted actions and low-level control actions within a single model. With CoA, an abstract goal is not just a final command; instead, it serves as an intermediate reasoning step that guides the model to generate the precise, executable actions needed to complete the task. Furthermore, we show that an All-in-One generalist agent, trained on a diverse mix of action spaces using CoA, learns a more generalizable policy. This unified agent achieves a new state-of-the-art, outperforming strong, specialized baselines. To support the research community, we are releasing the OpenHA (Open Hierarchical Agents) suite, which includes our benchmark of over 800 tasks, curated datasets, source code, and all model checkpoints at: \url{https://github.com/
CraftJarvis/OpenHA}.
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When Model Merging Breaks Routing: Training-Free Calibration for MoE
Canbin Huang, Tianyuan Shi, Xiaojun Quan, Jingang Wang, Jianfei Zhang, Qifan Wang
Model merging has emerged as a cost-effective approach for consolidating the capabilities of multiple LLMs without retraining. However, existing merging techniques, largely based on linear parameter arithmetic or optimization, struggle when applied to Mixture-of-Experts (MoE) architectures. We identify a critical failure mode in MoE merging, termed *routing breakdown*, in which the merged router fails to dispatch tokens to suitable experts. Routing breakdown stems from the sensitivity of the non-linear softmax and discrete Top-k routing mechanisms to parameter perturbations from merging, a sensitivity further amplified by load-balancing constraints imposed during MoE pretraining. Because fine-tuned experts exhibit distinct specializations, even modest misrouting can cause severe performance degradation. To address this issue, we propose Hessian-Aware Router Calibration (HARC), a training-free framework that leverages second-order curvature information to realign the merged router. This approach admits a closed-form solution that can be efficiently solved using a matrix-free conjugate gradient method. Experiments on mathematical reasoning and code generation tasks show that HARC effectively mitigates routing breakdown across diverse MoE merging baselines and leads to substantial performance improvements.
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Compute Where it Counts: Self Optimizing Language Models
Yash Akhauri, Mohamed S. Abdelfattah
Efficient LLM inference research has largely focused on reducing the cost of each decoding step (e.g., using quantization, pruning, or sparse attention), typically applying a uniform computation budget to every generated token. In practice, token difficulty varies widely, so static compression can over-compute on easy steps and under-compute on hard ones. We study \emph{dynamic budget allocation} for autoregressive decoding: learning how much computation to spend \emph{per token} from within a single model.

Self-Optimizing Language Models (SOL) pair a frozen LLM with a lightweight policy network that reads the LLM hidden state and selects a discrete \emph{efficiency action} at each decode step. Actions can jointly control (i) token-level attention sparsity, (ii) structured activation pruning in the MLP, and (iii) activation quantization bit-width, while leaving the base model weights unchanged.
We train the policy with group-relative policy optimization on teacher-forced episodes: the token sequence is fixed, while we sample multiple compute schedules (i.e., “counterfactual” schedules that vary only the efficiency actions for the same token path) and compare their likelihoods under the same supervision. Our reward trades off language■model quality against soft penalties that encourage episode■average budget usage to match a requested target. Across model variants and compute regimes, SOL improves quality at matched budget over static allocation and strong random schedule search, offering a complementary axis for inference■efficiency optimization. SOL discovers a better quality-efficiency pareto-front across all our experiments and improves MMLU accuracy by up to 7.3\% over uniform budget allocation strategies. \href{https://github.com/akhauriyash/SOL}

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G^2-Reader: Dual Evolving Graphs for Multimodal Document QA
Yaxin Du, Junru Song, Yifan Zhou, Cheng Wang, Jiahao Gu, Zimeng Chen, Menglan Ch
en, Wen Yao, Yang Yang, Ying Wen, Siheng Chen
Retrieval-augmented generation is a practical paradigm for question answering ov
er long documents, but it remains brittle for multimodal reading where text, tab
les, and figures are interleaved across many pages. First, flat chunking breaks
document-native structure and cross-modal alignment, yielding semantic fragments
 that are hard to interpret in isolation. Second, even iterative retrieval can f
ail in long contexts by looping on partial evidence or drifting into irrelevant
sections as noise accumulates, since each step is guided only by the current sni
ppet without a persistent global search state. We introduce G^2-Reader, a dual
-graph system, to address both issues. It evolves a Content Graph to preserve do
cument-native structure and cross-modal semantics, and maintains a Planning Grap
h, an agentic directed acyclic graph of sub-questions, to track intermediate fin
dings and guide stepwise navigation for evidence completion. On VisDoMBench acro
ss five multimodal domains, G^2-Reader with Qwen3-VL-32B-Instruct reaches 66.2
1\% average accuracy, outperforming strong baselines and a standalone GPT-5 (53.
08\%). Code is available: https://github.com/DorothyDUUU/G2_Reader.
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Factored Gossip DiLoCo: Reducing Blocking Communication within DiLoCo
Chamin P Hewa Koneputugodage, Thalaiyasingam Ajanthan, Sameera Ramasinghe, Hadi
Mohaghegh Dolatabadi, Shamane Siriwardhana, Gil Avraham, Violetta Shevchenko, Ka
rol Pajak, James Snewin, Alexander Long
To make large-scale distributed training practical outside high-bandwidth datace
nters, we must reduce blocking, high-volume synchronization. While DiLoCo commun
icates infrequently, its outer synchronization remains bandwidth-heavy and britt
le to stragglers and transient failures.
We relax exact synchronization to *approximate synchronization* via mixing/gossi
p, which degrades gracefully under delays and communication failures. This allow
s us to factorize DiLoCo synchronization into a *non-blocking* mixing step that
overlaps computation with no staleness, and a *blocking* mixing step that tighte
ns worker agreement, yielding a tunable trade-off between compute utilization an
d optimization stability.
On up to billion-parameter language models in low-bandwidth settings, our framew
ork substantially improves compute utilization compared to DiLoCo, with training
 progress ranging from comparable to closely matching it, and is more robust to
failures.
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NEMO: Execution-Aware Optimization Modeling via Autonomous Coding Agents
Yang Song, Anoushka Vyas, Zirui Wei, Sina Khoshfetrat Pakazad, Henrik Ohlsson, G
raham Neubig
We present **NEMO**, a system that translates **N**atural-language descriptions
of decision problems into formal **E**xecutable **M**athematical **O**ptimizatio
n implementations using autonomous coding agents (ACAs). Existing approaches rel
y on specialized large language models (LLMs) or bespoke task-specific agents th
at are often brittle and frequently generate syntactically invalid or non-execut
able code. NEMO instead treats ACAs as a first-class abstraction analogous to AP
I-based interaction with LLMs; their sandboxed execution guarantees code is exec
utable by construction and supports automated validation and repair. We introduc
e novel coordination patterns including asymmetric validation loops between inde
pendently generated optimizer and simulator implementations, external memory for
 experience reuse, and robustness enhancements via minimum Bayes risk (MBR) deco
ding and self-consistency. Across nine established optimization benchmarks, NEMO
 achieves state-of-the-art performance on the majority of tasks with substantial
 margins on several datasets, demonstrating the power of execution-aware agentic
 architectures for automated optimization modeling.
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FedRGL: Robust Federated Graph Learning under Label Noise
De Li, Zhou Tan, Qiyu Li, Zeming Gan, Tiange Xia, Chunpei Li, Xianxian LI

Federated Graph Learning (FGL) is a distributed machine learning paradigm based on graph neural networks, enabling secure and collaborative modeling of local graph data among clients. However, label noise in graph data can degrade the generalization performance of the global model. Existing federated label noise learning methods, primarily focused on computer vision tasks, often yield suboptimal results when directly applied to FGL. To address this issue, we propose a robust federated graph learning method with label noise, termed **FedRGL**. Specifically, FedRGL leverages the globally aggregated model and local subgraph structural information to implement a dual-perspective consistency noise-node filtering mechanism under class-aware dynamic thresholds. The resulting *class-aware dual-consistency filtering (CADF)* can also serve as a plug-and-play module, enhancing noise robustness across various subgraph federated learning frameworks. To better exploit the supervisory information from filtered noisy nodes, we employ the natural augmentation techniques from graph contrastive learning to assign high-confidence pseudo-labels to the noise nodes. Additionally, we measure model quality via the average predictive entropy of unlabeled nodes, enabling adaptive robust aggregation on the server side. Extensive experiments on real-world graph datasets show that FedRGL consistently outperforms existing methods under different noise rates, noise types, and client scales, achieving on average **5--8\%** higher accuracy and up to **30\%** improvement over the weakest baselines under noisy conditions. The source code is available at https://github.com/LDer66/FEDRGL-ICML26.

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DirectEdit: Step-Level Accurate Inversion for Flow-Based Image Editing

Desong Yang, Mang Ye

With recent advancements in large-scale pre-trained text-to-image (T2I) models, training-free image editing methods have demonstrated remarkable success. Typically, these methods involve adding noise to a clean image via an inversion process, followed by separate denoising steps for the reconstruction and editing paths during the forward process. However, since the reconstruction path is approximated using noisy latents from mismatched timesteps, existing methods inevitably suffer from accumulated drift, which fundamentally limits reconstruction fidelity. To address this challenge, we systematically analyze the inversion process within the flow transformer and propose DirectEdit, a simple yet effective editing method that eliminates the inherent reconstruction error without introducing additional neural function evaluations (NFEs). Unlike most prior works that attempt to rectify the inversion path, DirectEdit focuses on directly aligning the forward paths, enabling precise reconstruction and reliable feature sharing. Furthermore, we introduce a preservation mechanism based on attention feature injection and multi-branch mask-guided noise blending, which effectively balances fidelity and editability. Extensive experiments across diverse scenarios demonstrate that DirectEdit achieves efficient and accurate image editing, delivering superior performance that outperforms state-of-the-art methods. Code and examples are available at https://desongyang.github.io/Directedit.

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FluxNet: Learning Capacity-Constrained Local Transport Operators for Conservative and Bounded PDE Surrogates

Zishuo Lan, Junjie Li, Lei Wang, Jincheng Wang

Autoregressive learning of time-stepping operators provides an effective approach to data-driven partial differential equation (PDE) simulation, yet for conservation laws, they face a fundamental challenge: learned updates may violate global conservation over long rollouts. For the important subclass of mass-conservation-type equations, the problem is compounded by inherent physical bounds (e.g., nonnegativity or concentrations in [0,1]) whose violation further destabilizes predictions. We introduce FluxNet, which learns cumulative transport amounts representing the total conserved quantity redistributed between each cell and a configurable neighborhood over the full surrogate interval. A conservative update guarantees exact discrete conservation by construction; modular capacity-constrained transport heads (L, U, and D) enforce lower bounds, upper bounds, or near-zero dual-bound violations through architectural design. Unlike flux-rate surrogate

s that require temporal integration and thus inherit CFL constraints, FluxNet in volves no such integration; configurable transport neighborhoods enable large-ti mestep prediction at full spatial resolution. Ghost cells extend the framework t o non-periodic boundaries. Experiments on four benchmarks (1D convection--diffus ion, 2D shallow water, 1D traffic flow, 2D Cahn--Hilliard) demonstrate exact con servation, structural bound preservation, architecture modularity, and superior stability over flux-rate surrogates at large temporal strides. The code is publi cly available at: https://github.com/Lan-zs/FluxNet.
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One-shot Conditional Sampling: MMD meets Nearest Neighbors
Anirban Chatterjee, Sayantan Choudhury, Rohan Hore
How can we generate samples from a conditional distribution that we never fully observe? This question arises across a broad range of applications in both moder n machine learning and classical statistics, including image post-processing in computer vision, approximate posterior sampling in simulation-based inference, a nd conditional distribution modeling in complex data settings. In such settings, compared with unconditional sampling, additional feature information can be lev eraged to enable more adaptive and efficient sampling. Building on this, we intr oduce Conditional Generator using MMD (CGMMD), a novel framework for conditional sampling. Unlike many contemporary approaches, our method frames the training o bjective as a simple, adversary-free direct minimization problem.
A key feature of CGMMD is its ability to produce conditional samples in a single forward pass of the generator, enabling practical one-shot sampling with low te st-time complexity. We establish rigorous theoretical bounds on the loss incurre d when sampling from the CGMMD sampler, and prove convergence of the estimated d istribution to the true conditional distribution. In the process, we also develo p a uniform concentration result for nearest-neighbor based functionals, which m ay be of independent interest. Finally, we show that CGMMD performs competitivel y on synthetic tasks involving complex conditional densities, as well as on prac tical applications such as image denoising and image super-resolution.
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DC-LA: Difference-of-Convex Langevin Algorithm
Hoang Phuc Hau Luu, Zhongjian Wang
We study a sampling problem whose target distribution is $\pi \propto \exp(-f-r) $ where the data fidelity term f is Lipschitz smooth while the regularizer ter m $r=r_1-r_2$ is a non-smooth difference-of-convex (DC) function, i.e., $r_1,r_2 $ are convex. By leveraging the DC structure of r, we can smooth out r by ap plying Moreau envelopes to r_1 and r_2 separately. In line with DC programmi ng, we then redistribute the concave part of the regularizer to the data fidelit y and study its corresponding proximal Langevin algorithm (termed DC-LA). We est ablish convergence of DC-LA to the target distribution π, up to discretizati on and smoothing errors, in the q-Wasserstein distance for all $q \in \mathbb{ N}^*$, under the assumption that V is distant dissipative. Our results improve previous work on non-log-concave sampling in terms of a more general framework and assumptions. Numerical experiments show that DC-LA produces accurate distrib utions in synthetic settings and provides qualitatively reasonable uncertainty q uantification in a real-world Computed Tomography application.
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Less Is More in Federated Continual Learning: RieSelect for Conflict-Aware Layer Selection in LLMs
Wenqi Qiu, Yipeng Zhou, Lin Zhu, Laizhong Cui
Federated continual learning (FCL) of large language models on edge devices is c onstrained by a communication--stability--plasticity trilemma. We reveal a less- is-more phenomenon: beyond a moderate layer upload ratio, stability loss offsets saturated plasticity gains, so overall continual performance no longer improves . Moreover, layer-wise conflict is heavy-tailed and concentrates in a few layers ; denser uplink increasingly includes these layers, which disproportionately dri ves forgetting and motivates selective sparse communication. Therefore, we intro duce RieSelect, which treats stability as staying within a Fisher-metric safe ba sin around historical solutions. Under this safe-basin constraint, we derive a l

ayer-wise conflict score and a closed-form certified safe step size for finite l
ocal updates, and formulate selective uplink as a knapsack-based utility--risk s
election, balancing plasticity gains against stability risks. Extensive experime
nts show that, under a per-round uplink budget, RieSelect achieves the best perf
ormance across task orders. Beyond this matched-budget setting, under standard c
ommunication protocols, RieSelect improves average accuracy by 18.99–28.14 point
s while reducing total uplink by 53–115$\times$.
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Gradient Testing and Estimation by Comparisons
Xiwen Tao, Chenyi Zhang, Helin Wang, Yexin Zhang, Tongyang Li
We study gradient testing and gradient estimation of smooth functions using only
 a comparison oracle that, given two points, indicates which one has the larger
function value. For any smooth $f:\mathbb R^n\to\mathbb R$, $\mathbf{x}\in\mathb
b R^n$, and $\varepsilon>0$, we design a gradient testing algorithm that determi
nes whether the normalized gradient $\nabla f(\mathbf{x})/\lVert\nabla f(\mathbf
{x})\rVert$ is ε-close or 2ε-far from a given unit vecto
r $\mathbf{v}$ using $O(1)$ queries, as well as a gradient estimation algorithm
that outputs an ε-estimate of $\nabla f(\mathbf{x})/\lVert\nabla f(\
mathbf{x})\rVert$ using $O(n\log(1/\varepsilon))$ queries which we prove to be o
ptimal. Furthermore, we study gradient estimation in the quantum comparison orac
le model where queries can be made in superpositions, and develop a quantum algo
rithm using $O(\log (n/\varepsilon))$ queries.
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LabBuilder: Protocol-Grounded 3D Layout Generation for Interactable and Safe Lab
oratory
Jianbao Cao, zhangrui zhao, Bohan Feng, Zixuan Hu, Rui Li, Haiyuan Wan, Chenxi L
i, jingyuan Li, Wenzhe Cai, LEI BAI, Wanli Ouyang, LINGYU DUAN, Di Huang, Mintin
g Pan, Sha Zhang, Xinzhu Ma, SHIXIANG TANG, Dongzhan Zhou
Automated laboratories hold the promise of accelerating scientific discovery, ye
t their deployment is bottlenecked by the difficulty of designing safe and execu
table environments. While simulator-based design offers scalability, existing 3D
 scene generation methods are primarily tailored for household settings, optimiz
ing for visual plausibility while neglecting the protocol grounding and layout-l
evel safety constraints essential for scientific experimentation. We present Lab
Builder, an end-to-end system that generates and verifies 3D laboratory layouts
from concise textual specifications. It operates through three tightly coupled c
omponents: LabForge first curates a meta-dataset of annotated assets and chemica
l knowledge, translating natural language specifications into structured protoco
ls; building on these protocols, LabGen synthesizes laboratory layouts via an it
erative, constraint-aware optimization strategy; finally, LabTouchstone evaluate
s the resulting layouts as a unified benchmark. Extensive experiments demonstrat
e that LabBuilder significantly outperforms existing state-of-the-art methods, p
roducing laboratory environments that are realistic and valid under modeled geom
etric, chemical-safety, and navigation constraints.
****

OMP: One-step Meanflow Policy with Directional Alignment
Han Fang, Yize Huang, Yuheng Zhao, Paul Weng, Xiao Li, Yutong Ban
Robot manipulation has increasingly adopted data-driven generative policy framew
orks, yet the field faces a persistent trade-off: diffusion models suffer from h
igh inference latency, while flow-based methods often require complex architectu
ral constraints. Although in image generation domain, the MeanFlow paradigm offe
rs a path to single-step inference, its direct application to robotics is impede
d by critical theoretical pathologies, specifically spectral bias and gradient s
tarvation in low-velocity regimes. To overcome these limitations, we propose the
 One-step MeanFlow Policy (OMP), a novel framework designed for high-fidelity, r
eal-time manipulation. We introduce a lightweight directional alignment mechanis
m to explicitly synchronize predicted velocities with true mean velocities. Furt
hermore, we implement a Differential Derivation Equation (DDE) to approximate th
e Jacobian-Vector Product (JVP) operator, which decouples forward and backward p
asses to significantly reduce memory complexity. Extensive experiments on the Ad

roit and Meta-World benchmarks demonstrate that OMP outperforms state-of-the-art methods in success rate and trajectory accuracy, particularly in high-precision tasks, while retaining the efficiency of single-step generation.
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Improving Sampling for Masked Diffusion Models via Information Gain
Kaisen Yang, Jayden Teoh, Kaicheng Yang, Yitong Zhang, Alex Lamb
Masked Diffusion Models (MDMs) enable flexible decoding orders, yet existing samplers remain largely greedy, selecting locally certain tokens without accounting for their downstream effects. We show that this myopia can increase cumulative uncertainty and lead to suboptimal generation. To address this, we propose the **Info-Gain Sampler**, a training-free decoding method that uses the bidirectional structure of MDMs to balance immediate uncertainty with the information gained over remaining masked positions. Across reasoning, coding, creative writing, and image generation tasks, Info-Gain Sampler consistently outperforms existing MDM samplers, improving average reasoning accuracy by 2.9--11.6 percentage points and achieving a 62.8% average win rate in creative writing. The code is available at https://github.com/yks23/Information-Gain-Sampler.
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LiftQuant: Continuous Bit-Width LLM via Dimensional Lifting and Projection
Liulu He, Xuan Ang Liu, Juntao Liu, Taolue Feng, Ting Lu, Chunsheng Gan, ZHIYV PENG, Yuan Du, Huanrui Yang, Yijiang Liu, Li Du
Existing quantization methods are fundamentally limited by rigid, integer-based bit-widths (e.g., 2, 3-bit), resulting in a "deployment gap" where Large Language Models cannot be optimally fitted to specific memory budgets. To bridge this gap, we introduce LiftQuant, a novel framework that enables continuous bit-width control for true Pareto-optimal deployment. The core innovation is a "lift-then-project" mechanism which approximates low-dimensional weight vectors by projecting a simple 1-bit lattice from a higher-dimensional ``lifted" space. Crucially, the effective bit-width is determined simply by the ratio of the lifted dimension to the original dimension, which allows the bit-width to be tuned quasi-continuous as the dimension is a flexible structural parameter. This projection generates a structured yet non-uniform codebook, capturing the expressive power of Vector Quantization (VQ). While beneficial over VQ, LiftQuant's decoding path relies solely on linear transformations and 1-bit uniform quantizers, retaining hardware-friendly nature. This flexibility is transformative: LiftQuant enables a 70B LLM to be compressed to 2.4 bits to precisely fit a 24GB GPU, where its performance significantly surpasses state-of-the-art 2-bit models fitted on the same device. Our code and ckpt is available at \url{https://github.com/Heliulu/LiftQuant}.
****

PhaseAlign: Complex Phase Alignment for Stable Open-Vocabulary Semantic Segmentation
Jiankang Wang, Dingding Jia, Shuopeng Zhou, Xuan Wang
Open-Vocabulary Segmentation(OVS) aims to achieve pixel-level semantic recognition from arbitrary text queries. Existing large-scale visual-linguistic models, such as CLIP, perform well in zero-shot generalization, but their image-level training objectives and real-valued cross-modal alignment mix amplitude and phase information, limiting fine-grained segmentation and often causing blurred boundaries and fragmented structures. Inspired by the ability of electromagnetic wave phase to control interference independently of amplitude, we propose PhaseAlign, an OVS framework based on Complex Phase Alignment (CPA). CPA explicitly decouples the magnitude and phase of visual and textual embeddings in the complex domain, refining effective features for stable cross-modal alignment. To further enhance structural awareness, we introduce spatial-aware cross-modal projection, which models local neighborhood relations via multi-scale spatial contrast normalization, and attention-guided affinity modeling, which leverages pre-trained ViT self-attention to propagate category activations, improving boundary clarity and region integrity. Experiments show that PhaseAlign achieves state-of-the-art performance on multiple OVS benchmarks.
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INT vs. FP: A Comprehensive Study of Fine-Grained Low-bit Quantization Formats
Mengzhao Chen, Meng Wu, Hui Jin, Zhihang Yuan, Jing Liu, Chaoyi Zhang, Yunshui Li, Jie Huang, Jin Ma, Zeyue Xue, Zhiheng Liu, Xingyan Bin, Ping Luo
Modern AI hardware, such as Nvidia's Blackwell architecture, is increasingly embracing low-precision floating-point (FP) formats to handle the pervasive activation outliers in Large Language Models (LLMs). Despite this industry trend, a unified comparison of FP and integer (INT) quantization across varying granularities has been missing, leaving algorithm and hardware co-design without clear guidance. This paper fills that gap by systematically investigating the trade-offs between FP and INT formats. We reveal a critical performance crossover: while FP excels in coarse-grained quantization, INT consistently surpasses it as the quantization block size shrinks. Our comprehensive comparison demonstrates that for popular fine-grained formats like MX (block size 32), MXINT8 and MXINT4 are superior to their FP counterparts in both algorithmic accuracy and hardware efficiency. We also introduce a symmetric clipping method that resolves gradient bias in fine-grained low-bit INT training, enabling nearly lossless performance for MXINT8 training. These findings challenge the current hardware trajectory and advocate for prioritizing fine-grained INT formats in future AI accelerators to achieve a better balance of accuracy, power, and efficiency. Code is available at \url{https://github.com/ChenMnZ/INT_vs_FP}.
****
Object-level Semantic and Spatial Distillation for Open Vocabulary Detection
Zitong Li, Jinzhuo Wu, Fukang Zhao, Xinyue Wang, Jun Chen-CUG, Zhuo Cheng, Dapeng Luo
Recent Open-vocabulary Object Detection (OVD) approaches adapt CLIP through region-level distillation to improve semantic alignment for novel categories. However, the distilled regional features are often used for both classification and localization, enhancing semantic consistency at the expense of spatial fidelity. To resolve this, we propose Object-level Semantic and Spatial Distillation (OSSD), a two-stage framework that explicitly decouples semantic and spatial feature learning. OSSD first distills object-level semantics from CLIP’s global [CLS] embeddings to enhance region discrimination, and then injects fine-grained spatial and structural priors via spatial distillation from a detector trained only on COCO base categories. Furthermore, we propose a Location Quality Estimation Head (LQEH) that predicts class-agnostic localization quality, complementing objectness confidence to improve the novel-object perception. Extensive experiments show that our method achieves 49.2 AP50 on the OV-COCO benchmark. exceeding the best previous result by 3.6\%, On the OV-LVIS benchmark, our method reaches 40.5 mAP on novel categories, outperforming previous state-of-the-art methods.
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Bridging Functional and Representational Similarity via Usable Information
Antonio Almudévar, Alfonso Ortega
We present a unified framework for quantifying the similarity between representations through the lens of \textit{usable} information, offering a rigorous theoretical and empirical synthesis across three key dimensions. First, addressing functional similarity, we establish a formal link between stitching performance and conditional mutual information. We further reveal that stitching is inherently asymmetric, demonstrating that robust functional comparison necessitates a bidirectional analysis rather than a unidirectional mapping. Second, concerning representational similarity, we find that reconstruction-based metrics and standard tools (e.g., CKA, RSA) act as estimators of usable information under specific constraints. Crucially, we show that similarity is relative to the capacity of the predictive family: representations that appear distinct to a rigid observer may be identical to a more expressive one. Third, we demonstrate that representational similarity is sufficient but not necessary for functional similarity. We unify these concepts through a task-granularity hierarchy: similarity on a complex task guarantees similarity on any coarser derivative, establishing representational similarity as the limit of maximum granularity: input reconstruction.
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DNACHUNKER: Learnable Tokenization for DNA Language Models

Taewon Kim, Jihwan Shin, Hyomin Kim, Youngmok Jung, Jonghoon Lee, Won-Chul Lee, Sungsoo Ahn, Insu Han

DNA language models are increasingly used to represent genomic sequence, yet their effectiveness depends critically on how raw nucleotides are converted into model inputs. Unlike natural language, DNA offers no canonical boundaries, making fixed tokenizations a brittle design choice under shifts, indels, and local repeats. We introduce DNAChunker, a masked DNA language model that incorporates a learnable adaptive segmentation module to produce context-dependent, variable-length units. Building on a dynamic segmentation procedure, DNAChunker learns to allocate finer granularity to functionally enriched regions while compressing repetitive or redundant sequence. We pretrain DNAChunker on the human reference genome and evaluate it across five benchmarks, where it consistently improves over strong fixed-tokenization baselines. Further analyses and ablations indicate that unlike fixed tokenizations, segmentation is learned in a biologically-informed, mutation-resilient manner.

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Towards Functional Correctness of Large Code Models with Selective Generation

Jaewoo Jeong, Taesoo Kim, Sangdon Park

The hallucination of code generation models hinders their applicability to systems requiring higher safety standards. One critical bottleneck in addressing code hallucination is the difficulty of identifying the functional correctness of generated code, due to its unnatural form. We address this core bottleneck by automatically generating unit tests using dynamic code analysis tools, leveraging the executable nature of code. Accordingly, we propose a selective code generator that abstains from uncertain generations -- based on the functional correctness evaluated by generated unit tests -- to theoretically control the correctness among non-abstained answers, i.e., the false discovery rate. Finally, we propose to use generated unit tests in evaluation as well as in learning for precise code evaluation, calling this paradigm FuzzEval. We demonstrate the efficacy of our method along with the controllability of code hallucination and reasonable selection efficiency.

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Simultaneous Confidence Bounds for Aggregated Effects via Exact Subset Optimization

Weihang Xu, Huajie Qian, Wotao Yin, Xinshang Wang

We study simultaneous confidence bounds for aggregated effects over downward-closed subset families of independent statistical tests. The bounds are obtained by bootstrap calibration of the maximum normalized aggregated effect over the relevant subset family, yielding valid post-hoc inference for data-selected subsets and tighter bounds than classical methods that protect all linear contrasts. A central challenge is that the required maximization is a nonlinear combinatorial optimization problem. We cast it as a weighted densest-subgraph problem and derive exact linear and mixed integer linear program reformulations, and we further develop a fully polynomial-time approximation scheme that exploits the rank-1 structure of the objective to scale to large families. On the statistical side, we establish a finite-sample coverage guarantee with a lighter dependence on the subset family size than high-dimensional central limit theorems provide. We illustrate the method on synthetic and real machine learning applications.

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Randomized Feasibility Methods for Constrained Optimization with Adaptive Step Sizes

Abhishek Chakraborty, Angelia Nedich

We consider minimizing an objective function subject to constraints defined by the intersection of lower-level sets of convex functions. We study two cases: (i) strongly convex and Lipschitz-smooth objective function and (ii) convex but possibly nonsmooth objective function. To deal with the constraints that are not easy to project on, we use a randomized feasibility algorithm with Polyak steps and a random number of sampled constraints per iteration, while taking (sub)gradient steps to minimize the objective function. For case (i), we prove linear convergence in expectation of the objective function values to any prescribed toleran

ce using an adaptive stepsize. For case (ii), we develop a fully problem paramet er-free and adaptive stepsize scheme that yields an $O(1/\sqrt{T})$ worst-case r ate in expectation. The infeasibility of the iterates decreases geometrically wi th the number of feasibility updates almost surely, while for the averaged itera tes, we establish an expected lower bound on the function values relative to the optimal value that depends on the distribution for the random number of sampled constraints. For certain choices of sample-size growth, optimal rates are achie ved. Finally, simulations on a Quadratically Constrained Quadratic Programming (QCQP) problem, Support Vector Machines (SVM), and logistic regression with group fairness constraints demonstrate the computational efficiency of our algorithm compared to other state-of-the-art methods.
****

Order within Chaos: Capturing Intrinsic Energy Anomalies for AI-Manipulated Imag e Forgery Localization

Yiming Wang, Baiqi Wu, Qingming Li, Jiahao Chen, Tong Zhang, Shouling Ji

Recent advancements in generative AI have led to image editing models capable of producing realistic forgeries that evade traditional image forgery localization methods, as these approaches depend on physical noise absent in synthetic data. To address this challenge, we theoretically demonstrate that the diffusion proc ess inherently suppresses local high-frequency variance, creating a statistical energy gap that is distinguishable from the natural entropy of optical imaging. Guided by this insight, we propose FLAME, a unified framework that utilizes a LA D map to capture these intrinsic anomalies, coupled with a parameter-efficient a dapter for SAM to achieve precise, pixel-level forgery localization. Furthermore , to bridge the lag between forensic benchmarks and evolving generative models, we introduce EditStream, an automated pipeline for continuous, instruction-based training data synthesis. Extensive experiments demonstrate that FLAME establish es a new state-of-the-art, significantly outperforming previous methods on AI-ge nerated forgery datasets while effectively generalizing to unseen generative arc hitectures. Our code is available at https://github.com/phoenixnir/FLAME.
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LipoPU: Pocket-level Prediction of Lipid-Protein Interactions via Positive-Unlab eled Learning

Yuxing Wang, Wenyi Zhang, Yilong Zou, Jing Huang

Computational identification of lipid-binding proteins is critical for both fund amental research and therapeutic development. Existing models are typically trai ned in a fully supervised manner, treating unlabeled samples as negatives. Howev er, missing evidence does not imply non-binding, leading to systematic false neg atives. Pocket-level lipid-binding prediction also remains underexplored compare d to residue- or sequence-level approaches. To bridge these gaps, we present **L ipoPU**, a pocket-centric predictor that formulates lipid-binding learning under a ranking-based positive-unlabeled objective, and supports both binary lipid-bi nding detection and multi-label lipid category prediction. LipoPU learns an atte ntion-based pocket representation that is robust to ambiguous pocket definitions while providing residue-level interpretability. Experiments show consistent gai ns over supervised baselines and prior pocket-level work, and a structural case study recovers a literature-supported allosteric lipid-binding pocket while high lighting biologically informative residues.
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TACTIC: Task-Aware Sparse Coordination Graphs for Multi-Task Multi-agent Reinfor cement Learning

Kexing Peng, Pengyi Li, tinghuai ma, Jianye HAO

Value factorization eases non-stationarity in MARL, but its static coordination assumptions hinder generalization on long-horizon tasks with shifting dependenci es. Prior VQ-VAE methods abstract trajectories yet miss time-varying inter-agent dependencies. We present TACTIC, a CTDE framework with three components: (i) VQ -VAE-based trajectory abstraction that learns discrete task-semantic classes; (i i) semantic-conditioned sparse coordination graphs that adapt dependencies by pr uning edges according to variance-based pairwise payoff sensitivity; and (iii) a pretrained, frozen trajectory-class predictor that conditions local policies wh

ile decoupling task recognition from control. On SMAC and SUMO, TACTIC shows str
ong overall competitiveness and adaptive coordination under sparse rewards and d
ynamic task structures.
****

Feature-Aware (Hyper)graph Generation via Next-Scale Prediction
Dorian Gailhard, Enzo Tartaglione, Lirida Naviner, Jhony H. Giraldo
Graph generative models perform well on small structured data but struggle to sc
ale to large, complex structures.
Hierarchical approaches improve scalability but often ignore node and edge featu
res, which are critical in real-world applications, particularly for hypergraphs
 that model higher-order relationships.
In this paper, we propose FAHNES (**f**eature-**a**ware (**h**yper)graph generat
ion via **ne**xt-**s**cale prediction), a hierarchical framework that jointly ge
nerates topology and features for graphs and hypergraphs.
FAHNES builds multi-scale representations through node coarsening and localized
expansion, guided by a novel hierarchical scale encoding that controls granulari
ty and ensures cross-scale consistency.
Experiments on synthetic, 3D mesh, and graph point cloud datasets demonstrate co
mpetitive or state-of-the-art performance while uniquely scaling to featured lar
ge-scale graphs and hypergraphs.
Our code is open source.
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Learning Taxonomic Trees with Hierarchical Representation Regularization for Lar
ge Multimodal Models
Hulingxiao He, Zhi Tan, Yuxin Peng
Taxonomies provide key information about the semantic relationships between conc
epts and the inherent organization of vision and language. Despite their impress
ive capabilities, large multimodal models (LMMs) often lack taxonomic knowledge,
 leading to low hierarchical visual recognition (HVR) consistency. These models
typically only rely on language modeling objectives during fine-tuning and lack
explicit taxonomy-aware regularization. To address this, we propose Hierarchical
 Representation Regularization (HiR^2), a simple plug-and-play regularizer tha
t improves hierarchical consistency in LMMs. Specifically, we introduce a semant
ic-aware visual tree construction framework that extracts coarse-to-fine visual
features from intermediate LLM layers guided by textual cues. The regularizer co
mbines two complementary objectives: a taxonomic entailment loss that enforces h
ierarchy via hyperbolic entailment cones in the Lorentz model, and a discriminat
ive dispersive loss that promotes angular separation of semantically similar emb
eddings on the unit sphere without disturbing the radial hierarchical structure.
 Extensive experiments demonstrate that HiR^2 effectively captures taxonomic s
tructures across diverse LMMs and fine-tuning methods. Code is available at http
s://github.com/PKUICST-MIPL/HiR2_ICML2026.
****

Quantifying Frontier LLM Capabilities for Container Sandbox Escape
Rahul Marchand, Art O Cathain, Jerome Wynne, Philippos Maximos Giavridis, Sam De
verett, John Wilkinson, Jason Gwartz, Harry Coppock
Large language models (LLMs) increasingly act as autonomous agents, using tools
to execute code, read and write files, and access networks, creating novel secur
ity risks. To mitigate these risks, agents are commonly deployed and evaluated i
n isolated "sandbox" environments, often implemented using Docker/OCI containers
. We introduce SandboxEscapeBench, an open benchmark that safely measures an LLM
's capacity to break out of these sandboxes. The benchmark is implemented as an
Inspect AI Capture the Flag (CTF) evaluation utilising a nested sandbox architec
ture with the outer layer containing the flag and no known vulnerabilities. Foll
owing a threat model of a motivated adversarial agent with shell access inside a
 container, SandboxEscapeBench covers a spectrum of sandbox-escape mechanisms sp
anning misconfiguration, privilege allocation mistakes, kernel flaws, and runtim
e/orchestration weaknesses. We find that, when vulnerabilities are added, LLMs a
re able to identify and exploit them, showing that use of evaluation like Sandbo
xEscapeBench is needed to ensure sandboxing continues to provide the encapsulati

on needed for highly-capable models.
****
Structure-Induced Information for Rerooting Levin Tree Search
Jake Tuero, Michael Buro, Laurent Orseau, Levi Lelis
Subgoal-based policy tree search, which uses a policy to guide search, is effective for complex single-agent deterministic problems but often relies on explicit subgoal generation that can incur substantial overhead and hinders scalability. In this paper, we overcome these limitations by using a learned ``rerooter'' through the recently-introduced $\sqrt{\text{LTS}}$ algorithm. A *rerooter* implicitly decomposes the problem into soft subtasks. While previous work focused on the formal guarantees for given or handcrafted rerooters, in this work we propose three rerooter designs:
(i) a clustering-based rerooter that exploits global state-space structure, (ii) a heuristic-based rerooter that leverages learned cost-to-go estimates, and (iii) a hybrid that combines both signals.
Our framework avoids having to explicitly reconstruct and reason over generated subgoals, thereby enabling scalable allocation of search effort with significantly lower computational overhead.
Empirically, our rerooting-based methods scale to complex environments where subgoal-based policy tree search fails, and achieve state-of-the-art online training efficiency on the domains tested.
****
MoSE: Mixture of Slimmable Experts for Efficient and Adaptive Language Models
Nurbek Tastan, Stefanos Laskaridis, Karthik Nandakumar, Samuel Horváth
Mixture-of-Experts (MoE) models scale large language models efficiently by sparsely activating experts, but once an expert is selected, it is executed fully. Hence, the trade-off between accuracy and computation in an MoE model typically exhibits large discontinuities. We propose Mixture of Slimmable Experts (MoSE), an MoE architecture in which each expert has a nested, slimmable structure that can be executed at variable widths. This enables conditional computation not only over which experts are activated but also over how much of each expert is utilized. Consequently, a single pretrained MoSE model can support a more continuous spectrum of accuracy-compute trade-offs at inference time. We present a simple and stable training recipe for slimmable experts under sparse routing, combining multi-width training with standard MoE objectives. During inference, we explore strategies for runtime width determination, including a lightweight test-time training mechanism that learns how to map router confidence/probabilities to expert widths under a fixed budget. Experiments on GPT-style models, various routing regimes, zero-shot downstream reasoning benchmarks, and continual pre-training adaptation of DeepSeek model show that MoSE matches or improves standard MoE at full width and consistently shifts the compute-quality frontier toward lower inference FLOPs. The code can be found at: https://github.com/tnurbek/mose.
****
COFT: Counterfactual–Conformal Decoding for Fair Chain■of■Thought Reasoning in Large Language Models
Arya Fayyazi, Mehdi Kamal, Massoud Pedram
Large language models (LLMs) can reveal and amplify societal biases during chain-of-thought (CoT) generation. We present COFT (Chain of Fair Thought), a training-free decoding method that applies token-level fairness control at decode time, with distribution-free marginal validity guarantees (under exchangeability) for any frozen causal language model.
COFT operates in three stages. First, it creates a masked counterfactual prompt by replacing sensitive spans with neutral tokens. Second, it compares the factual and masked logit distributions through lightweight logit fusion to attenuate attribute-driven biases. Third, it uses dual-branch split-conformal calibration to certify per-step candidate token sets at a user-chosen risk level.
We evaluate COFT across six models and multiple bias benchmarks. Our method reduces standard bias metrics by 30–55\% (median 38\%) while preserving task utility and language quality. Reasoning accuracies remain unchanged within run-to-run noise margins. The computational overhead is modest, equivalent to one additional

cached forward pass (<=11%).
COFT offers a clear, auditable path to safer CoT generation with significant bia
s reduction, negligible utility loss, and no requirement for retraining, auxilia
ry classifiers, or weight access.
****
Efficient Generative Modeling beyond Memoryless Diffusion via Adjoint Schrödinge
r Bridge Matching
Jeongwoo Shin, Jinhwan Sul, Joonseok Lee, Jaewoong Choi, Jaemoo Choi
Diffusion models often yield highly curved trajectories and noisy score targets
due to an uninformative, memoryless forward process that induces independent dat
a-noise coupling. We propose Adjoint Schrödinger Bridge Matching (ASBM), a gener
ative modeling framework that recovers optimal trajectories in high dimensions v
ia two stages. First, we view the Schrödinger Bridge (SB) forward dynamic as a c
oupling construction problem and learn it through a data-to-energy sampling pers
pective that transports data to an energy-defined prior. Then, we learn the back
ward generative dynamic with a simple matching loss supervised by the induced op
timal coupling. By operating in a non-memoryless regime, ASBM produces significa
ntly straighter and more efficient sampling paths. Compared to prior works, ASBM
scales to high-dimensional data with notably improved stability and efficiency.
Extensive experiments on image generation show that ASBM improves fidelity with
fewer sampling steps. We further showcase the effectiveness of our optimal traj
ectory via distillation to a one-step generator.
****
Float8@2bits: Entropy Coding Enables Data-Free Model Compression
Patrick Putzky, Martin Genzel, Mattes Mollenhauer, Sebastian Schulze, Thomas Wol
lmann, Stefan Dietzel
Post-training compression is currently divided into two contrasting regimes. On
the one hand, fast, data-free, and model-agnostic methods (e.g., NF4 or HQQ) off
er maximum accessibility but suffer from functional collapse at extreme bit-rate
s below 4 bits. On the other hand, techniques leveraging calibration data or ext
ensive recovery training achieve superior fidelity but impose high computational
constraints and face uncertain robustness under data distribution shifts. We in
troduce EntQuant, a framework that unites the advantages of these distinct parad
igms. By matching the performance of data-dependent methods with the speed and u
niversality of data-free techniques, EntQuant enables practical utility in the e
xtreme compression regime. Our method decouples numerical precision from storage
cost via entropy coding, compressing a 70B parameter model in less than 10 minu
tes. We demonstrate that EntQuant does not only achieve state-of-the-art results
on standard evaluation sets and models, but also retains functional performance
on more complex benchmarks with instruction-tuned models, all at modest inferen
ce overhead.
****
ClinTutor-R1: Advancing Scalable and Robust One-to-Many Alignment in Clinical So
cratic Education
Zhitao He, Haolin Yang, Zeyu Qin, Yi R. Fung
While Large Language Models (LLMs) have achieved remarkable success in dyadic (o
ne-on-one) instruction, they face significant challenges in One-to-Many alignmen
t, such as clinical ward rounds, where an instructor must simultaneously guide a
diverse group of trainees. Current models often suffer from context dilution an
d goal misalignment, failing to balance individual scaffolding with collective l
earning progress. To address this, we introduce ClinEdu, a multi-agent pedagogic
al simulator that model the complexity of group dynamics. Leveraging this platfo
rm, we construct ClinTeach, a large-scale dataset of Socratic teaching dialogues
, and propose ClinTutor-R1, the first multimodal agent explicitly architected to
achieve one-to-many alignment in clinical education, employing an explicit inte
rnal thinking mechanism to model both individual belief states and group consens
us. We validate our framework through a comprehensive protocol covering both sta
ndard static benchmarks and rigorous in-situ interactive evaluation within ClinE
du. Experimental results demonstrate that ClinTutor-R1 outperforms base models b
y over 20% and achieves parity with proprietary reasoning models , while exhibit

ing exceptional scalability in maintaining instructional quality across expandin
g student cohorts.
****
GenExam: A Multidisciplinary Text-to-Image Exam
Zhaokai Wang, Penghao Yin, Xiangyu Zhao, Changyao Tian, Yu Qiao, Wenhai Wang, Ji
feng Dai, Gen Luo
Exams are a fundamental test of expert-level intelligence and require integrated
 understanding, reasoning, and generation. Existing exam-style benchmarks mainly
 focus on understanding and reasoning tasks, and current generation benchmarks e
mphasize the illustration of world knowledge and visual concepts, neglecting the
 evaluation of rigorous drawing exams. We introduce GenExam, the first benchmark
 for multidisciplinary text-to-image exams, featuring 1,000 samples across 10 su
bjects with exam-style prompts organized under a four-level taxonomy. Each probl
em is equipped with ground-truth images and fine-grained scoring points to enabl
e a precise evaluation of semantic correctness and visual plausibility. Experime
nts on 17 text-to-image and unified models demonstrate the great challenge of Ge
nExam and the huge gap where open-source models consistently lag behind the lead
ing closed-source ones. By framing image generation as an exam, GenExam offers a
 rigorous assessment of models' ability to integrate understanding, reasoning, a
nd generation, providing insights on the path to intelligent generative models.
Our benchmark and evaluation code will be released.
****
Directional Neural Collapse Explains Few-Shot Transfer in Self-Supervised Learni
ng
Achleshwar Luthra, Yash Salunkhe, Tomer Galanti
Frozen self-supervised representations often transfer well with only a few label
s across many semantic tasks. We argue that a single geometric quantity, *direct
ional* CDNV (decision-axis variance), sits at the core of two favorable behavior
s: strong few-shot transfer within a task, and low interference across many task
s. We show that both emerge when variability *along* class-separating directions
 is small. First, we prove sharp non-asymptotic multiclass generalization bounds
 for downstream classification whose leading term is the directional CDNV. The b
ounds include finite-shot corrections that cleanly separate intrinsic decision-a
xis variability from centroid-estimation error. Second, we link decision-axis co
llapse to multitask geometry: for independent balanced labelings, small directio
nal CDNV across tasks forces the corresponding decision axes to be nearly orthog
onal, helping a single representation support many tasks with minimal interferen
ce. Empirically, across SSL objectives, directional CDNV collapses during pretra
ining even when classical CDNV remains large, and our bounds closely track few-s
hot error at practical shot sizes. Additionally, on synthetic multitask data, we
 verify that SSL learns representations whose induced decision axes are nearly o
rthogonal.
****
Learning to Route Languages for Multilingual Policy Optimization
Geyang Guo, Hiromi Wakaki, Yuki Mitsufuji, Alan Ritter, Wei Xu
Large language models~(LLMs) are trained on heterogeneous multilingual corpora,
yet existing policy optimization methods often implicitly restrict each training
 question to a single response language or rely on a fixed dominant language for
 supervision.
We propose language-routed policy optimization (LRPO), an online reinforcement l
earning framework that treats language as a selectable variable.
LRPO elicits multilingual rollouts for each training question and integrates the
ir relative quality into preference-based policy updates, increasing the diversi
ty and informativeness of training signals under the fixed rollout budget.
To adaptively determine which languages to explore during reinforcement learning
, we introduce a trainable language router formulated as a multi-armed bandit, b
alancing exploration of underutilized languages with exploitation of more inform
ative ones.
Extensive experiments show that LRPO consistently improves multilingual performa
nce, demonstrating that adaptive language routing enables effective cross-lingua

l knowledge exploitation for training.
We release all the resources at \url{https://github.com/Guochry/LRPO}.
****

SLQ: Bridging Modalities via Shared Latent Queries for Retrieval with Frozen MLLMs
Haoran Lou, Ziyan Liu, Chunxiao Fan, Yuexin Wu, Yue Ming, Hao Wu, Kai Zuo, Yibo Chen, Xu Tang
Multimodal Large Language Models (MLLMs) possess intrinsic reasoning and world-knowledge capabilities, yet adapting them for dense retrieval remains challenging. Existing approaches rely on invasive parameter updates, such as full fine-tuning and LoRA, which may disrupt the pre-trained semantic space and impair the structured knowledge essential for reasoning. To address this, we propose **SLQ**, a parameter-efficient tuning framework that adapts MLLMs for retrieval while keeping the backbone entirely frozen. SLQ introduces a small set of **Shared Latent Queries** that are appended to both text and image tokens, leveraging the model's native causal attention to aggregate multimodal context into a unified embedding space. Furthermore, to better evaluate retrieval beyond superficial pattern matching, we construct **KARR-Bench**, a benchmark designed for knowledge-aware reasoning retrieval. Extensive experiments show that SLQ outperforms full fine-tuning and LoRA on COCO and Flickr30K, while achieving competitive performance on MMEB and yielding substantial gains on KARR-Bench, validating that preserving the pre-trained representations via non-invasive adaptation is an effective strategy for MLLM-based retrieval. The code is available under: https://github.com/CnFaker/SLQ.
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SpecPL: Disentangling Spectral Granularity for Prompt Learning
Jingtao Zhou, Xirui Kang, feiyang huang, Lai Man Po
Existing prompt learning for VLMs exhibits a modality asymmetry, predominantly optimizing text tokens while still relying on frozen visual encoder as holistic extractor and neglecting the spectral granularity essential for fine-grained discrimination. To bridge this, we introduce Disentangling Spectral Granularity for Prompt Learning (SpecPL), which approaches prompt learning from a novel spectral perspective via Counterfactual Granule Supervision. Specifically, we leverage a frozen VAE to decompose visual signals into semantic low-frequency bands and granular high-frequency details. A frozen Visual Semantic Bank anchors text representations to universal low-frequency invariants, mitigating overfitting. Crucially, fine-grained discrimination is driven by counterfactual granule training: by permuting high-frequency signals, we compel the model to explicitly distinguish visual granularity from semantic invariance. Uniquely, SpecPL serves as a universal plug-and-play booster, revitalizing text-oriented baselines like CoOp and MaPLe via visual-side guidance. Experiments on 11 benchmarks demonstrate competitive state-of-the-art performance, achieving a new performance ceiling of 81.51\% harmonic-mean accuracy. These results validate that spectral disentanglement with counterfactual supervision effectively bridges the gap in the stability-generalization trade-off.
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LUCID: Attention with Preconditioned Representations
Sai Surya Duvvuri, Nirmal Patel, Nilesh Gupta, Inderjit S Dhillon
Softmax-based dot-product attention is a cornerstone of Transformer architectures, enabling remarkable capabilities such as in-context learning. However, as context lengths increase, a fundamental limitation of the softmax function emerges: it tends to diffuse probability mass to irrelevant tokens degrading performance in long-sequence scenarios. Furthermore, attempts to sharpen focus by lowering softmax temperature hinder learnability due to vanishing gradients. We introduce LUCID Attention, an architectural modification that applies a preconditioner to the attention probabilities. This preconditioner, derived from exponentiated key-key similarities, minimizes overlap between the keys in a Reproducing Kernel Hilbert Space, thus allowing the query to focus on important keys among large number of keys accurately with same computational complexity as standard attention. Additionally, LUCID's preconditioning-based approach to retrieval bypasses the

need for low temperature and the learnability problems associated with it. We va
lidate our approach by training $\sim$1 billion parameter language models evalua
ted on up to 128K tokens. Our results demonstrate significant gains on long-cont
ext retrieval tasks, specifically retrieval tasks from BABILong, RULER, SCROLLS
and LongBench. For instance, LUCID achieves up to 18\% improvement in BABILong a
nd 14\% improvement in RULER multi-needle performance compared to standard atten
tion.
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Broadening the Backdoor Basin: Understanding LLM Backdoors Collapse and Making B
ackdoors Persistent

Xingyi Zhao, Tian Xie, Xiaojun Qi, Depeng Xu, Shuhan Yuan

Large Language Models (LLMs) are vulnerable to backdoor attacks, yet we observe
that many LLM backdoors do not survive when end users perform supervised fine-tu
ning (SFT). In this work, we provide a geometric explanation: by probing the bac
kdoor objective under controlled weight perturbations, we find that conventional
 poisoning often drives the backdoor loss to a narrow and sharp basin; consequen
tly, even modest parameter drift induced by downstream SFT can push the model ou
t of the low-loss and high-ASR region, leading to rapid backdoor forgetting. Mot
ivated by this insight, we propose BAD-BOOM, a resilient backdoor attack via bro
ader smoothness minimization, which explicitly broadens and smooths the backdoor
 basin. BAD-BOOM extends sharpness-aware minimization with a Fisher-induced elli
psoidal constraint that allocates larger perturbation budgets to backdoor-sensit
ive parameters, encouraging solutions whose neighborhoods also maintain low back
door loss. Across two threat settings, three attack scenarios, three open-source
 LLMs, and three trigger-free downstream SFT tasks, BAD-BOOM consistently preser
ves high ASR while maintaining competitive utility. The code is available at htt
ps://github.com/xingyizhao/BAD-BOOM.
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The Shape of Addition: Geometric Structures of Arithmetic in Large Language Mode
ls

Liuyuan Wen, Xun Zhu, Lihao Huang, Wenbin Li, Yang Gao

Large Language Models exhibit paradoxical fragility in fundamental arithmetic, i
mplying a disconnect between internal computation and discrete output. By analyz
ing the residual stream geometry during multi-operand addition, we identify the
Iso-Raw-Sum Trajectory (IRST), a topological manifold where representations
are anchored by semantic digits and modulated by continuous carry fibers. We pro
pose the **Noisy Quantization Model**, which frames arithmetic errors as *topolo
gical slippages* caused by internal neural noise pushing a continuous, latent *c
arry potential* across quantization thresholds. This geometric framework further
 elucidates *probe versatility*, explaining how lightweight probes can disentang
le conflicting latent signals (such as ground truth versus hallucination) from a
 single activation vector. Finally, we validate these insights through a geometr
ic consistency check method that effectively detects and corrects these quantiza
tion failures during inference.
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From Drift to Coherence: Stabilizing Beliefs in LLMs

SongEun Kim, Seungyoo Lee, Edwin Fong, Hyungi Lee, Juho Lee

Large language models (LLMs) are often hypothesized to perform implicit Bayesian
 inference, yet a key coherence condition—the martingale property of predictive
beliefs—has been shown to fail in controlled synthetic in-context learning setti
ngs. We revisit this question in a more typical usage regime: generic multiple-c
hoice question answering. Exploiting the discrete answer space, we compute exact
 predictive distributions and study belief dynamics induced by autoregressive an
swer resampling. We introduce prompted predictive resampling (PPR), where an LLM
 generates a sequence of answers to the same question. Empirically, PPR reveals
early-stage belief drift, indicating martingale violations. However, after suffi
cient resampling steps, the belief process self-stabilizes and converges to a co
herent predictive distribution. Based on this observation, we further propose (i
) a seed-answer prompting strategy to accelerate stabilization, and (ii) a self-
consistency loss that amortizes early-stage drift into the model via fine-tuning

. Experiments on multiple-choice QA benchmarks show that our methods substantial
ly reduce belief drift and improve predictive coherence without sacrificing accu
racy.
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One-shot Entropy Minimization for Language Model Reasoning
Zitian Gao, Yilong Chen, Haoming Luo, Joey Zhou, Bryan Dai
In this work, we propose One-shot Entropy Minimization (EM), a simple and fully
unsupervised post-training approach that significantly improves reasoning and ge
neration performance using only a single unlabeled data and approximately ten gr
adient steps. To avoid data contamination, we pretrain a 7-billion-parameter lan
guage model from scratch with strictly decontaminated data. Despite its extreme
simplicity, one-shot EM yields substantial performance gains and improves reason
ing abilities across a broad range of domains, including mathematical reasoning,
 logical reasoning, and coding. We further show that entropy minimization induce
s a characteristic right-skewed logit shift, amplifying high-probability tokens
while suppressing low-probability tails, in contrast to reinforcement learning.
Our findings suggest that entropy minimization primarily acts as a distribution
shaping mechanism rather than a conventional learning process, offering an effic
ient and practical algorithm for post-training large language models.
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Safe Reinforcement Learning with Preference-based Constraint Inference
Chenglin Li, Grant Ruan, Hua Geng
Safe reinforcement learning (RL) is a standard paradigm for safety-critical deci
sion making. However, real-world safety constraints can be complex, subjective,
and even hard to explicitly specify. Existing works on constraint inference rely
 on restrictive assumptions or extensive expert demonstrations, which are not re
alistic in many real-world applications. How to cheaply and reliably learn these
 constraints is the major challenge we focus on in this study. While inferring c
onstraints from human preferences offers a data-efficient alternative, we identi
fy popular Bradley-Terry (BT) models fail to capture the asymmetric, heavy-taile
d nature of safety costs, resulting in risk underestimation. It is still rare in
 the literature to understand the impacts of BT models on the downstream policy
learning. To address the above knowledge gaps, we propose a novel approach namel
y Preference-based Constrained Reinforcement Learning (PbCRL). We introduce a no
vel dead zone mechanism into preference modeling and theoretically prove that it
 encourages heavy-tailed cost distributions, thereby achieving better constraint
 alignment. Additionally, we incorporate a Signal-to-Noise Ratio (SNR) loss to e
ncourage exploration by cost variances, which is found to benefit policy learnin
g. Further, two-stage training strategy is deployed to lower online labeling bur
dens while adaptively enhancing constraint satisfaction. Empirical results demon
strate that PbCRL achieves superior alignment with true safety requirements and
outperforms state-of-the-art baselines in terms of safety and reward. Our work e
xplores a promising and effective way for constraint inference in Safe RL, with
great potential in various safety-critical applications.
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M-Star: Markovian Projection of Star-Shaped Diffusion for Exponential Family Dis
tributions
François Bertholom, Khalid Oublal
Diffusion models achieve state-of-the-art performance in generative modeling but
 are limited by their reliance on Gaussian noise and the high computational cost
 of iterative sampling. Star-shaped diffusion addresses the former by introducin
g a non-Markovian forward process, yet this comes at the expense of temporal coh
erence in the reverse process. We propose a novel framework that resolves this t
rade-off by learning a Markovian projection of a star-shaped forward process, an
d its reversal. This design enables learning over a broad class of exponential m
odels and recovers DDPM as a special case. It is particularly well-suited for kn
owledge distillation, allowing few-step or even single-step generation. Experime
nts demonstrate the effectiveness and flexibility of our approach across multipl
e generative tasks. Code and demo are available at: https://oublalkhalid.github.
io/MStar-Diffusion/.

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Unsupervised Partner Design Enables Robust Ad-hoc Teamwork
Constantin Ruhdorfer, Matteo Bortoletto, Victor Oei, Anna Penzkofer, Andreas Bulling
We introduce Unsupervised Partner Design (UPD), a population-free multi-agent reinforcement learning method for robust ad-hoc teamwork.
UPD generates training partners on-the-fly and selects them adaptively based on a learnability criterion, removing the need for pre-trained partner populations or manual parameter tuning.
We show that this simple mechanism enables effective partner diversity and can be extended to joint partner-environment selection when a procedural level generator is available.
Across Level-Based Foraging, Overcooked-AI, and the Overcooked Generalisation Challenge, UPD consistently achieves strong performance compared to both population-based and population-free baselines.
In a human-AI user study, agents trained with UPD achieve higher returns and are rated as more adaptive, more human-like, and less frustrating than all evaluated baseline methods.
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Disentangling Consensus and Value-Specific Representations for Controllable Pluralistic Value Alignment of LLMs
JianKui Zhou, Jing Yao, Xiaoyuan Yi, Peng Zhang, Ning Gu, Zhan Hu, Xing Xie, Tun Lu
With the widespread deployment of large language models (LLMs), aligning model outputs with pluralistic human values has become an important research problem. Recent approaches that train task-specific experts and merge them through parameter aggregation have shown promise for pluralistic alignment. However, these methods often overlook the intrinsic complexity of real-world value data, where multiple correlated value dimensions coexist, resulting in highly similar and entangled expert representations. Consequently, modifying the contribution of one value expert may unintentionally influence other values, limiting fine-grained controllability.
To address this issue, we propose DisAlign, a model-merging framework that explicitly decomposes value representations into consensus and value-specific components using an information-geometric perspective. DisAlign first extracts a consensus anchor and subspace to capture shared structure across values, and then applies spectral decomposition to the residual representations to construct disentangled value subspaces. This design enables more precise and independent modulation of multiple values.
Experiments on three datasets covering different value frameworks demonstrate that DisAlign consistently improves value disentanglement and achieves more accurate pluralistic value control compared to existing baselines.
Our code is available at \url{https://anonymous.4open.science/r/DisAlign-7F35}
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Computational Arbitrage in AI Model Markets
Ricardo Olmedo, Bernhard Schölkopf, Moritz Hardt
Consider a market of competing model providers selling query access to models with varying costs and capabilities. Customers submit problem instances and are willing to pay up to a budget for a verifiable solution. An arbitrageur efficiently allocates inference budget across providers to undercut the market, thus creating a competitive offering with no model-development risk. In this work, we initiate the study of arbitrage in AI model markets, empirically demonstrating the viability of arbitrage and illustrating its economic consequences. We conduct an in-depth case study of SWE-bench GitHub issue resolution using two representative models, GPT-5 mini and DeepSeek v3.2. In this verifiable domain, simple arbitrage strategies generate net profit margins of up to 40\%. Robust arbitrage strategies that generalize across different domains remain profitable. Distillation further creates strong arbitrage opportunities, potentially at the expense of the teacher model’s revenue. Multiple competing arbitrageurs drive down consumer prices, reducing the marginal revenue of model providers. At the same time, arbitr

age reduces market segmentation and facilitates market entry for smaller model p
roviders by enabling earlier revenue capture. Our results suggest that arbitrage
 can be a powerful force in AI model markets with implications for model develop
ment, distillation, and deployment.
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BALLAST: Bayesian Active Learning with Look-ahead Amendment for Sea-drifter Traj
ectories under Spatio-Temporal Vector Fields
Rui-Yang Zhang, Lachlan Astfalck, Edward Cripps, David S. Leslie, Henry Moss
We introduce a formal active learning methodology for guiding the placement of L
agrangian observers to infer time-dependent vector fields -- a key task in ocean
ography, marine science, and ocean engineering -- using a physics-informed spati
o-temporal Gaussian process surrogate model. The majority of existing placement
 campaigns either follow standard `space-filling' designs or relatively ad-hoc ex
pert opinions. A key challenge to applying principled active learning in this se
tting is that Lagrangian observers are continuously advected through the vector
field, so they make measurements at different locations and times. It is, theref
ore, important to consider the likely future trajectories of placed observers to
 account for the utility of candidate placement locations. To this end, we prese
nt BALLAST: Bayesian Active Learning with Look-ahead Amendment for Sea-drifter T
rajectories. We observe noticeable benefits of BALLAST-aided sequential observer
 placement strategies on both synthetic and high-fidelity ocean current models.
In addition, we developed a novel GP inference method -- the Vanilla SPDE Exchan
ge (VaSE) -- to boost the GP posterior sampling efficiency, which is also of ind
ependent interest.
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Explicitly Modeling Censoring Produces Superior Survival Predictors
Shi-ang Qi, Yakun Yu, Russell Greiner
Likelihood-based training is the dominant paradigm in survival prediction. Under
 independent censoring, we can factorize the likelihood and optimize only the te
rms related to event modeling, effectively treating the censoring mechanism as i
ncidental. This is justified when censoring is *non-informative*, i.e., when the
 censoring process shares no parameters with the event-time model. However, this
 may not hold in practice, and ignoring censoring contributions may discard usef
ul signals for learning representations that can help to effectively estimate ev
ent distributions. Motivated by this, we argue that explicitly modeling censorin
g can improve representation learning and time-to-event estimation, particularly
 when event and censoring processes are coupled. We introduce a latent decomposi
tion view in which observed covariates are mapped to latent components correspon
ding to event-specific, censoring-specific, confounding, and irrelevant informat
ion. We then learn decomposed representations for the first three categories to
guide a better estimation of the event distribution. We instantiate our method o
n 4 popular deep-learning survival models and evaluate on 10 datasets (2 semi-sy
nthetic and 8 real-world), showing consistent gains over strong baselines and mu
ltiple SOTA methods.
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Pull Requests as a Training Signal for Repo-Level Code Editing
Qinglin Zhu, Tianyu Chen, Shuai Lu, Lei Ji, Runcong Zhao, Murong Ma, Xiangxiang
Dai, Yulan He, Lin Gui, Peng CHENG, Yeyun Gong
Repository-level code editing requires models to understand complex dependencies
 and execute precise multi-file modifications across a large codebase. While rec
ent gains on SWE-bench rely heavily on complex agent scaffolding, it remains unc
lear how much of this capability can be internalised via high-quality training s
ignals. To address this, we propose Clean Pull Request (Clean-PR), a mid-trainin
g paradigm that leverages real-world GitHub pull requests as a training signal f
or repository-level editing. We introduce a scalable pipeline that converts nois
y pull request diffs into Search/Replace edit blocks through reconstruction and
validation, resulting in the largest publicly available corpus of 2 million pull
 requests spanning 12 programming languages. Using this training signal, we perf
orm a mid-training stage followed by an agentless-aligned supervised fine-tuning
 process with error-driven data augmentation. On SWE-bench, our model significan

tly outperforms the instruction-tuned baseline, achieving absolute improvements of 13.6% on SWE-bench Lite and 12.3% on SWE-bench Verified. These results demonstrate that repository-level code understanding and editing capabilities can be effectively internalised into model weights under a simplified, agentless protocol, without relying on heavy inference-time scaffolding.
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Shift-Dependent Asymmetry: Orthogonal Inverse Low-Rank Adaptation for Federated Medical Segmentation

Xingyue Zhao, Wenke Huang, Linghao Zhuang, Haoran Wu, Anwen Jiang, Zhifeng Wang, Wenwen He, Ming Feng, Mang Ye, Bo XU

Low-Rank Adaptation (LoRA) enables efficient federated fine-tuning of segmentation foundation models for medical imaging. However, most federated LoRA methods adopt a uniform aggregation rule, which breaks under the encoder–decoder asymmetry in medical segmentation: the encoder is dominated by appearance shifts, while the decoder is dominated by supervision variations. This mismatch entangles shared anatomy with site-specific biases and harms generalization. To address this, we propose Inverse Asymmetric Tuning (IAT). IAT aligns adaptation with heterogeneity sources by personalizing module-specific components in the encoder to absorb appearance shifts and in the decoder to accommodate site-dependent supervision, while retaining a shared pathway for transferable consensus. However, structural separation alone is insufficient under LoRA’s bilinear parameterization, where multiplicative coupling can still cause site-specific updates to leak into the shared direction. We therefore introduce a Subspace Orthogonality Regularizer that penalizes shared–local collinearity in the effective update space, mitigating leakage without extra communication. Experiments show consistent improvements over strong federated LoRA and parameter-efficient FL baselines.
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MetaOthello: A Controlled Study of Multiple World Models in Transformers

Aviral Chawla, Galen Hall, Juniper L Lovato

Foundation models must handle multiple generative processes, yet mechanistic interpretability largely studies capabilities in isolation; it remains unclear how a single transformer organizes multiple, potentially conflicting "world models". Previous experiments on Othello-playing neural networks test world-model learning, but focus on a single game with a single set of rules. We introduce *MetaOthello*, a controlled suite of Othello-like games with shared syntax but different rules or tokenizations, and train small GPTs on mixed-variant data. We show that transformers trained on multiple Othello variants learn **shared world-state representations**: linear probes trained on one game intervene on another's board state nearly as well as matched probes. When the games conflict, the model resolves the resulting *ambiguity* through a localized mechanism we identify and steer. For isomorphic games with token remapping, representations are equivalent up to a single orthogonal rotation that generalizes across layers, showing the shared structure is abstract rather than tied to surface form. Together, these results show that transformers reconcile conflicting world models by sharing structure and localizing conflict. *MetaOthello* thus offers a path toward understanding how transformers organize many world models at once.
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CausalProfiler: Generating Synthetic Benchmarks for Rigorous and Transparent Evaluation of Causal Machine Learning

Panayiotis Panayiotou, Audrey Poinsot, Alessandro Leite, Nicolas CHESNEAU, Marc Schoenauer, Özgür ■im■ek

Causal machine learning aims to answer "what if" questions using machine learning algorithms, making it a promising tool for high-stakes decision-making. Yet, empirical evaluation practices remain limited. Existing benchmarks often rely on a handful of hand-crafted or semi-synthetic datasets, leading to brittle, non-generalizable conclusions. To bridge this gap, we introduce CausalProfiler, a synthetic benchmark generator for causal machine learning methods. Based on a set of explicit design choices about the class of causal models, queries, and data considered, CausalProfiler randomly samples causal models, data, queries, and ground truths constituting the synthetic causal benchmarks. In this way, causal machi

ne learning can be rigorously and transparently evaluated under a variety of con ditions. This work offers the first random generator of synthetic causal benchma rks with coverage guarantees and transparent assumptions operating on the three levels of causal reasoning: observation, intervention, and counterfactual. We de monstrate its utility by evaluating several state-of-the-art methods under diver se conditions and assumptions, both in and out of the identification regime, ill ustrating the types of analyses and insights CausalProfiler enables.
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VideoSEG-O3: A Multi-turn Reinforcement Learning Framework for Reasoning Video O bject Segmentation
Ming Dai, Sen Yang, Boqiang Duan, Boyuan Tong, Jiedong Zhuang, Wankou Yang, Jing dong Wang
Reasoning Video Object Segmentation (RVOS) demands a sophisticated integration o f temporal dynamics, spatial details, and linguistic reasoning to achieve precis e pixel-level localization. Existing methods are limited to reasoning over fixed initial inputs and lack the capacity to actively acquire further visual evidenc e, which is often essential for resolving complex references in long or intricat e videos. To address this, we propose $\textbf{VideoSEG-O3}$, the first multi-tu rn reinforcement learning framework for RVOS that emulates the human $\textit{`` coarse-to-fine''}$ cognitive process. It employs a $\textit{multi-turn temporal-spatial chain-of-thought}$ to capture fine-grained details by iteratively pinpoi nting critical intervals and keyframes. Additionally, to enable the policy to pe rceive segmentation quality beyond mere text probability of $\texttt{[SEG]}$ dur ing the RL stage, we introduce $\textit{SEG-aware logit calibration}$, which int egrates pixel-wise segmentation feedback directly into the token-level logits. F urthermore, we design a $\textit{decoupled thinking trace}$ to hierarchically de compose the reasoning process into temporal, spatial, and linguistic dimensions, and construct $\textbf{VTS-CoT}$, a specialized cold-start dataset featuring co mprehensive reasoning trajectories. Extensive experiments demonstrate that Video SEG-O3 achieves advanced performance across 8 mainstream RVOS benchmarks, partic ularly excelling in long-horizon and complex reasoning tasks.
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StepCodeReasoner: Aligning Code Reasoning with Stepwise Execution Traces via Rei nforcement Learning
Hao Wang, Lei Sha, Jie M. Zhang
Existing code reasoning methods primarily supervise final code outputs, ignoring intermediate states, often leading to reward hacking where correct answers are obtained through inconsistent reasoning. We propose StepCodeReasoner, a framewor k that introduces explicit intermediate execution-state supervision. By automati cally inserting structured print-based execution-trace anchors into code, the mo del is trained to predict runtime states at each step, transforming code reasoni ng into a verifiable, stepwise execution modeling problem.
Building on this execution-aware method, we introduce Dual-Granularity GRPO (DG-GRPO), a reinforcement learning algorithm for structured credit assignment at tw o complementary granularities: inter-trajectory comparison across sampled execut ion paths and intra-trajectory shaping based on downstream correctness.
Extensive experiments demonstrate that StepCodeReasoner achieves state-of-the-ar t performance in code reasoning.
In particular, our 7B model achieves 91.1\% on CRUXEval and 86.5\% on LiveCodeBe nch, outperforming the CodeReasoner-7B baseline (86.0\% and 77.7\%) and GPT-4o (85.6\% and 75.1\%). Furthermore, on the execution-trace benchmark REval, our mod el scores 82.9\%, outperforming baseline CodeReasoner-7B (72.3\%), its 14B count erpart (81.1\%), and GPT-4o (77.3\%).
In addition, StepCodeReasoner improves code generation, achieving 90.1 on HumanE val, 85.0 on MBPP, and 19.4 on LiveCodeBench generation, with an average score o f 64.8 versus 62.6 for CodeReasoner-7B.
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Fast Estimation for Forest Matrix of Signed Graphs
Haoxin Sun, Zhongzhi Zhang
The forest matrix of a signed graph plays an important role in network science a

nd social opinion dynamics, yet existing algorithms are mainly designed for unsi gned graphs and are difficult to extend to signed graphs. In this paper, we stud y the problem of efficiently estimating the forest matrix of signed graphs with \(n\) nodes and introduce the signed forest matrix theorem, which establishes th e relationship between generalized spanning converging forests and the forest ma trix. Based on this result, we propose a novel algorithm \textsc{GSCF}, built on a variant of loop-erased random walks, to generate generalized spanning converg ing forests in expected \(O(n)\) time. We further develop two sampling algorithm s, \textsc{FMDE} and \textsc{FMDE+}, for estimating the diagonal of the forest m atrix, both with time complexity \(O(ln)\), where \(l\) is the number of samples . Extensive experiments on various signed graphs show that our methods achieve h igh estimation accuracy, significantly improve computational efficiency, and sca le to graphs with over twenty million nodes. Our source code is publicly availab le on \url{https://github.com/HaoxinSun98/SignedForestDiagonal}.
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Optimality of FSQ Tokens for Continuous Diffusion for Categorical Data with Appl ication to Text-to-Speech

Vadim Popov, Wenju Gu, Tasnima Sadekova, Georgii Aparin, Assel Yermekova

Continuous diffusion for categorical data is a framework belonging to the diffus ion family and aiming at generating discrete data. The scientific interest to su ch models has been constantly increasing these days because researchers try to a chieve a challenging goal of finding reasonable alternatives to autoregressive l arge language models. In this paper, we study the properties of the structure of the latent space corresponding to discrete tokens expressed in terms of Kullbac k-Leibler divergence on diffusion path measures and accuracy of the correct toke n prediction by the optimally trained diffusion model. We find that FSQ tokeniza tion scheme has the latent space structure with the properties that make it best suited for continuous diffusion for categorical data as verified through rigoro us theoretical analysis and numerical experiments. To validate our findings in r eal-life scenario, we train several text-to-speech diffusion models having speec h tokens as intermediate acoustic features, and show that the one based on FSQ t okens indeed performs the best, and, moreover, it outperforms its strong LLM-bas ed counterpart, at the same time being significantly smaller and faster.
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TriAttention: Efficient Long Reasoning with Trigonometric KV Compression

Weian Mao, Xi Lin, Wei Huang, Yuxin Xie, Tianfu Fu, Bohan Zhuang, Song Han, Yuka ng Chen

Extended reasoning in large language models (LLMs) requires long and accurate de coding and creates severe KV cache memory bottlenecks. Leading KV cache compress ion methods estimate KV importance using attention scores from recent post-RoPE queries. However, queries rotate with position during RoPE, making representativ e queries very few, leading to poor top-key selection and unstable reasoning. To avoid this issue, we turn to the pre-RoPE space, where we observe that Q and K vectors are highly concentrated around fixed non-zero centers and remain stable across positions—*Q/K concentration*. We show that this concentration causes que ries to preferentially attend to keys at specific distances (e.g., nearest keys) , with the centers determining which distances are preferred via a trigonometric series. Based on this, we propose TriAttention to estimate key importance by le veraging these centers. Via the trigonometric series, we use the distance prefer ence characterized by these centers to score keys according to their positions, and also leverage Q/K norms as an additional signal for importance estimation. O n AIME25 with 32K-token generation, TriAttention matches Full Attention reasonin g accuracy while achieving 2.5× higher throughput or 10.7× KV memory reduction, whereas leading baselines achieve only about half the accuracy at the same effic iency.
****

Language Bias in LVLMs: From In-Depth Analysis to Simple and Effective Mitigatio n

Yangneng Chen, Jing Li

Large Vision-Language Models (LVLMs) extend large language models with visual un

derstanding, but remain vulnerable to hallucination, where outputs are fluent ye
t inconsistent with images. Recent studies link this issue to language bias—the
tendency of LVLMs to over-rely on text while neglecting visual inputs. Yet most
analyses remain empirical without uncovering its underlying cause. In this paper
, we provide a systematic study of language bias and identify its root in modali
ty misalignment during training. Our analysis shows that both Visual Instruction
 Tuning (VIT) and Direct Preference Optimization (DPO) often prioritize textual
improvements, which may cause LVLMs to overly lean toward language modeling rath
er than balanced multimodal understanding. To address this, we propose two simpl
e yet effective methods: Language Bias Regularization (LBR), which mitigates lan
guage bias through regularization during instruction tuning, and Language Bias P
enalty (LBP), which penalizes language bias in the DPO training process. Extensi
ve experiments across diverse models and benchmarks demonstrate the effectivenes
s of our approach. LBR consistently improves performance on over ten general ben
chmarks, while LBP significantly reduces hallucination and improves trustworthin
ess. Together, these methods not only mitigate language bias but also advance th
e overall alignment of LVLMs, all without introducing any additional data or aux
iliary models. Our code is publicly available at [https://github.com/lab-klc/HAV
AE](https://github.com/lab-klc/HAVAE).
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HECTOR: Hybrid Editable Compositional Object References for Video Generation
Guofeng Zhang, Angtian Wang, Jacob Zhiyuan Fang, Liming Jiang, Haotian Yang, Ala
n Yuille, Chongyang Ma
Real-world videos naturally portray complex interactions among distinct physical
 objects, effectively forming dynamic compositions of visual elements. However,
most current video generation models synthesize scenes holistically and therefor
e lack mechanisms for explicit compositional manipulation. To address this limit
ation, we propose HECTOR, a generative pipeline that enables fine-grained compos
itional control. In contrast to prior methods, HECTOR supports hybrid reference
conditioning, allowing generation to be simultaneously guided by static images a
nd/or dynamic videos. Moreover, users can explicitly specify the trajectory of e
ach referenced element, precisely controlling its location, scale, and speed (se
e Figure1). This design allows the model to synthesize coherent videos that sati
sfy complex spatiotemporal constraints while preserving high-fidelity adherence
to references. Extensive experiments demonstrate that HECTOR achieves superior v
isual quality, stronger reference preservation, and improved motion controllabil
ity compared with existing approaches.
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You Can Learn Tokenization End-to-End with Reinforcement Learning
Sam Dauncey, Roger Wattenhofer
Tokenization is a hardcoded compression step which remains in the training pipel
ine of Large Language Models (LLMs), despite a general trend towards architectur
es becoming increasingly end-to-end. Prior work has shown promising results at s
cale in bringing this compression step inside the LLMs' architecture with heuris
tics to draw token boundaries, and also attempts to learn these token boundaries
 with straight-through estimates, which treat the problem of drawing discrete to
ken boundaries as a continuous one. We show that these token boundaries can inst
ead be learned using score function estimates, which have tighter theoretical gu
arantees due to directly optimizing the problem of drawing discrete token bounda
ries to minimize loss. We observe that techniques from reinforcement learning, s
uch as time discounting, are necessary to reduce the variance of this score func
tion sufficiently to make it practicable. We demonstrate that the resultant meth
od outperforms prior proposed straight-through estimates, both qualitatively and
 quantitatively at the 100 million parameter scale.
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SWE-fficiency: Can Language Models Optimize Real-World Repositories on Real Work
loads?
Jeffrey Jian Ma, Milad Hashemi, Amir Yazdanbakhsh, Kevin Swersky, Ofir Press, En
hui Li, Vijay Janapa Reddi, Parthasarathy Ranganathan
Optimizing the performance of large-scale software repositories demands expertis

e in code reasoning and software engineering (SWE) to reduce runtime while prese
rving program correctness. However, most benchmarks emphasize what to fix rather
 than how to fix code. We introduce SWE-fficiency, a benchmark for evaluating re
pository-level performance optimization on real workloads. Our suite contains 49
8 tasks across nine widely used data-science, machine-learning, and HPC reposito
ries (e.g., numpy, pandas, scipy): given a complete codebase and a slow workload
, an agent must investigate code semantics, localize bottlenecks and relevant te
sts, and produce a patch that matches or exceeds expert speedup while passing th
e same unit tests. To enable this how-to-fix evaluation, our automated pipeline
 scrapes GitHub pull requests for performance-improving edits, combining keyword
 filtering, static analysis, coverage tooling, and execution validation to both c
onfirm expert speedup baselines and identify relevant repository unit tests. Emp
irical evaluation of state-of-the-art agents reveals significant underperformanc
e. On average, top agents achieve less than 0.23x the expert speedup: agents str
uggle in localizing optimization opportunities, reasoning about execution across
 functions, and maintaining correctness in proposed edits. We release the benchm
ark and accompanying data pipeline to facilitate research on automated performan
ce engineering and long-horizon software reasoning.
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Inverse Entropic Optimal Transport Solves Semi-supervised Learning via Data Like
lihood Maximization

Mikhail Persiianov, Arip Asadulaev, Nikita Andreev, Nikita Starodubcev, Dmitry B
aranchuk, Anastasis Kratsios, Evgeny Burnaev, Alexander Korotin

Learning conditional distributions $\pi^\star(\cdot|x)$ is a central problem in
 machine learning, which is typically approached via supervised methods with pair
ed data $(x,y) \sim \pi^\star$. However, acquiring paired data samples is often
 challenging, especially in problems such as domain translation. This necessitate
s the development of *semi-supervised* models that utilize both limited paired d
ata and additional unpaired i.i.d. samples $x \sim \pi^\star_x$ and $y \sim \pi^
\star_y$ from the marginal distributions. The usage of such combined data is com
plex and often relies on heuristic approaches. To tackle this issue, we propose
 a new learning paradigm that integrates both paired and unpaired data seamlessly
 using data likelihood maximization techniques. We demonstrate that our approach
 also connects intriguingly with inverse entropic optimal transport (OT). This f
inding allows us to apply recent advances in computational OT to establish an *e
nd-to-end* learning algorithm to get $\pi^\star(\cdot|x)$. In addition, we deriv
e the universal approximation property, demonstrating that our approach can theo
retically recover true conditional distributions with arbitrarily small error. F
inally, we demonstrate through empirical tests that our method effectively learn
s conditional distributions using paired and unpaired data simultaneously.
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Toward Training Superintelligent Software Agents through Self-Play SWE-RL

Yuxiang Wei, Zhiqing Sun, Emily McMilin, Jonas Gehring, David W. Zhang, Gabriel
 Synnaeve, Daniel Fried, LINGMING ZHANG, Sida Wang

While current software agents powered by large language models (LLMs) and reinfo
rcement learning (RL) can boost programmer productivity, their reliance on human
-curated training data and environments creates a fundamental barrier to superin
telligence. In this paper, we present Self-play SWE-RL (SSR), a first step towar
d training superintelligent software agents under minimal data assumptions. SSR
 requires only access to sandboxed repositories with source code and dependencies
, no need for human-labeled issues or test commands. Grounded in real-world code
bases, a single LLM agent is trained via RL in a self-play setting to inject and
 repair increasingly complex bugs. The bugs are formally specified by test suite
 improvements proposed by the agent rather than natural language issue descripti
ons. On the SWE-bench Verified and SWE-Bench Pro benchmarks, SSR achieves clear
 self-improvement (+10.4 and +7.8 points) and consistently outperforms the human-
data baseline throughout training, generalizing to natural language bug descript
ions not seen in training. Overall, our results point toward a paradigm where ag
ents autonomously gather extensive learning experiences from real software repos
itories, ultimately enabling superintelligent systems that exceed human capabili

ties in understanding, modifying, and creating software from scratch.
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Incomplete Multi-View Clustering via Neighborhood-Conditioned Diffusion
Qian Guo, Gaohui Zuo, Bingbing Jiang, Guangrui Fan, Zhihua Cui, Xinyan Liang, Jianjian Ding
Incomplete multi-view clustering (IMVC) aims to uncover shared clustering structures from heterogeneous views with partial observations.
Recently, existing generative IMVC methods have made significant progress in this field; however, they still remain limited in two aspects.
On the one hand, they rely on weak cross-view signals, resulting in unstable latent recovery when facing missing data. On the other hand, they overlook stable cross-view neighborhood structures, leading to weak structural constraint.
To address these limitations, we propose neighborhood-conditioned diffusion for incomplete multi-view clustering (IMVC-NCD), which achieves robust latent completion.
Our method learns compact view-specific latent representations and constructs a unified conditioning vector by aggregating stable local neighborhood structures from available views while encoding missingness states, providing reliable guidance for diffusion-based denoising. With neighborhood-level conditioning, IMVC-NCD produces semantically aligned and view-consistent latent representations that are well suited for clustering, even under high missing-view ratios.
Extensive experiments on four benchmark datasets demonstrate the effectiveness and robustness of our method compared with state-of-the-art IMVC approaches.
Our code is available at https://github.com/zgh1115/IMVC-NCD.
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Efficiently Solving Discounted MDPs via Predictions with Unknown Prediction Errors
Lixing Lyu, Jiashuo Jiang, Wang Chi Cheung
We study infinite-horizon discounted Markov decision processes (DMDPs) under a generative model. Motivated by the Algorithms with Advice framework (Mitzenmacher and Vassilvitskii, 2022), we propose a novel framework to investigate how black-box predictions of the transition matrix can enhance sample efficiency in solving DMDPs and improve sample complexity bounds. We focus on DMDPs with N state–action pairs and discount factor γ. We first provide an impossibility result showing that, in the presence of predictions with unknown accuracy, no sampling policy can compute an ϵ-optimal policy with a sample complexity better than $\tilde{O}((1-\gamma)^{-3} N \epsilon^{-2})$, which matches the state-of-the-art minimax sample complexity bound without prediction. In complement, we design an algorithm based on minimax optimization techniques that leverages predictions of the transition matrix without requiring knowledge of the prediction error. Our algorithm achieves a sample complexity bound that depends on the prediction error and is uniformly better than $\tilde{O}((1-\gamma)^{-4} N \epsilon^{-2})$, the previous best result derived from convex optimization methods. In some cases, our bound even improves upon the state-of-the-art $\tilde{O}((1-\gamma)^{-3} N \epsilon^{-2})$, despite not having access to the prediction quality.
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InfraRL: A Benchmark for Constrained Resource Allocation in Large-Scale Infrastructure Asset Management
Yantian Wang, Wenhao Li, Bo Jin
Optimizing maintenance strategies for large-scale infrastructure is a critical sequential decision-making problem.
While Reinforcement Learning (RL) offers a theoretical framework for such problems, practical deployment necessitates offline constrained RL—learning policies solely from static historical datasets under rigid budgetary limits without dangerous on-policy exploration.
However, current research is hindered by benchmarks that fail to capture the confluence of distributional shift and hard constraints typical of real-world assets.
We introduce InfraRL, a high-fidelity benchmark that uses bridge maintenance as a rigorous testbed for general infrastructure asset management challenges.

Constructed from the U.S. National Bridge Inventory, InfraRL defines a rigorous offline task for optimizing maintenance strategies under hard budgetary constrai nts.
We benchmark a diverse suite of baselines, ranging from industry-standard heuris tics to single-agent, multi-agent, planning-based, and constrained offline RL me thods.
  Through a comprehensive evaluation protocol, we analyze performance across str uctural utility, constraint adherence, and behavioral fidelity, revealing critic al trade-offs between safety and long-term efficiency. Our code and data are ava ilable at https://github.com/BriSky-2021/InfraRL.
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Learning to Approximate Uniform Facility Location via Graph Neural Networks
Chendi Qian, Christopher Morris, Stefanie Jegelka, Christian Sohler
Neural networks, particularly message-passing neural networks (MPNNs), are incre asingly used as heuristics for hard combinatorial optimization problems. Yet man y learning-based methods rely on supervision, reinforcement learning, or gradien t estimators, causing high computational cost, unstable training, or limited gua rantees. Classical approximation algorithms provide worst-case guarantees but ar e non-differentiable and cannot adapt to structure in natural input distribution s. We study this tradeoff through Uniform Facility Location (UniFL), a problem w ith applications in clustering, summarization, logistics, and supply chains. We propose a fully differentiable MPNN that incorporates approximation-algorithmic principles without solver supervision or discrete relaxations. The model has pro vable approximation guarantees and empirically improves on standard approximatio n algorithms, narrowing the gap to integer linear programming.
****

DTKG: Dual-Track Knowledge Graph-Verified Reasoning Framework for Multi-Hop QA
Changhao Wang, Yanfang Liu, Xinxin Fan, Ao Tian, Lanzhi Zhou, Yunfeng Lu
Multi-hop reasoning for question answering (QA) plays a critical role in retriev al-augmented generation (RAG) for large language models (LLMs). Based on inheren t relation-dependency and reasoning patterns, it is categorized into parallel fa ct-verification (simultaneously verifying independent sub-questions) and chained reasoning (sequential multi-step inference).
Existing approaches adopt either LLM-based fact verification or KG path-based ch ain construction, failing to handle both categories well: the former underperfor ms on chained reasoning, while the latter suffers from redundant paths in parall el tasks.
Inspired by the Dual Process Theory in cognitive science and Stanovich’s Cogniti ve Misers Theory, we propose an effective multi-hop QA framework DTKG (Dual-Trac k Knowledge Graph) through building a two-stage pipeline: i) Classification Stag e (dynamic question categorization via few-shot prompting, emulating "unconsciou s processing"); and ii) Branch Processing Stage (tailored reasoning paths, emula ting "conscious processing"). Multi-facet
experiments on six datasets show DTKG achieves 5.0\%-29.5\% performance improvem ent. The code is available at https://anonymous.4open.science/r/DTKG-621F
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SceneDirector: Bridging Explicit Geometry and Generative Priors for Unified Driv ing Scene Editing
Yiyuan Liang, Zhiying Yan, Tao Zhang, Shangke Liu, Kai Lin, Xu Zou, Nong Sang, C hangxin Gao
Validating autonomous driving systems requires diverse scenarios, yet real-world data collection is biased and costly. Editing existing driving logs offers a sc alable solution, but simultaneously editing objects and ego-trajectory—termed un ified editing—remains challenging.
Current methods face an inherent dilemma: generative flexibility for object edit ing and physical precision for trajectory control.
To address this, we introduce SceneDirector, a diffusion-based framework that br idges explicit geometry and generative priors.
For explicit geometry, we leverage LiDAR-guided depth completion to construct de nse scene geometry and integrate editable 3D assets to form a Unified Geometric

Scaffold, providing rigorous structural guidance for unified editing.
To leverage generative priors, we encode the source video into a Static Texture Bank to provide rich appearance context.
Our proposed Mask-Gated Reference Attention bridges these modalities. Guided by a geometric uncertainty metric, this mechanism dynamically regulates the interaction between the scaffold and the bank—preserving reliable geometry while adaptively injecting textures for semantic refinement.
Extensive evaluations demonstrate that SceneDirector outperforms state-of-the-art methods in both controllability and visual quality.
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FOCA: Future-Oriented Conditioning for Data-Efficient Vision-Language-Action Adaptation

Minh Duc Nguyen, Nghiem Tuong Diep, Nguyen Gia Binh, Trong-Bao Ho, Doanh Le Thien, Quang Tan Nguyen, Thien-Loc Ha, Tran Van Nhiem, Bao Thach, Tran Xuan Nhat, Tuan Anh Tran, Artur Habuda, Philip Lund Møller, Tran Nguyen Le, Daniel Sonntag, Mathias Niepert, Khoa D Doan, Vu N. Duong, Hung Ngo, Minh Nhat VU, Duy Minh Ho Nguyen, An Thai Le, Vien Anh Ngo

Vision–Language–Action (VLA) models enable general-purpose robotic control via large-scale multimodal pretraining, yet their effectiveness under few-shot imitation learning remains limited. We conduct a systematic stress test of state-of-the-art VLA models and show that performance degrades sharply as demonstrations are reduced, revealing a key weakness of existing adaptation strategies. To address this, we introduce FOCA, a future-oriented conditioning framework for data-efficient VLA adaptation. FOCA combines explicit prediction of task-grounded future interaction embeddings with implicit alignment to future goal observations, enabling long-horizon reasoning in latent space without pixel-level prediction. This formulation naturally supports action-free co-training with synthetic videos from video world models and can be interpreted as learning a future-conditioned value-like representation. Extensive experiments demonstrate FOCA achieves 95.7\% success with 20 demonstrations on LIBERO, improves 7–12\% on RoboCasa, and delivers up to 26\% absolute gains on real robots, establishing a new state of the art in few-shot VLA adaptation.
****

Numina-Lean-Agent: An Open and General Agentic Reasoning System for Formal Mathematics

Junqi Liu, Zihao Zhou, Zekai Zhu, Marco Dos Santos, Weikun He, jiawei liu, Yunzhou Xie, Junqiao Zhao, Qiufeng Wang, Lihong Zhi, Jia LI, Wenda Li

Agentic systems have recently become the dominant paradigm for formal theorem proving, achieving strong performance by coordinating multiple models and tools. However, existing approaches often rely on task-specific pipelines and trained formal provers, limiting their flexibility and reproducibility.
In this paper, we propose the paradigm that directly uses a general coding agent as a formal math reasoner. This paradigm is motivated by (1) A general coding agent provides a natural interface for diverse reasoning tasks beyond proving, (2) Performance can be improved by simply replacing the underlying base model, without training., and (3) MCP enables flexible extension and autonomous calling of specialized tools, avoiding complex design. Based on this paradigm, we introduce \textbf{Numina-Lean-Agent}, which combines Claude Code with Numina-Lean-MCP to enable autonomous interaction with Lean, retrieval of relevant theorems, informal proving and auxiliary reasoning tools.
Using Claude Opus 4.5 as the base model, Numina-Lean-Agent solves all problems in Putnam 2025 (12/12), matching the best closed-source system. Beyond benchmark evaluation, we further demonstrate its generality by interacting with mathematicians to successfully formalize the Brascamp–Lieb theorem.
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WinDeskGround: A Benchmark for Robust GUI Grounding in Complex Multi-Window Desktop Environments

Haoren Zhao, Tianyi Chen, Zhen Wang

Multimodal Large Language Models (MLLMs) have revolutionized GUI automation, yet their efficacy is largely established on idealized, single-layer interfaces. Th

is paper identifies a critical reliability gap: state-of-the-art agents face dis
tinct robustness challenges in real-world desktop environments characterized by
multi-window stacking, occlusion, and visual clutter. To address this, we introd
uce WinDeskGround, a novel benchmark and synthesis framework tailored for evalua
ting GUI grounding robustness. Unlike static datasets, our framework parametrica
lly generates complex desktop scenarios by controlling window occlusion, layout
density, and semantic similarity, thereby simulating the distribution shifts of
authentic workflows. We construct a diverse meta-dataset of 1,356 high-fidelity
instruction-target pairs and conduct comprehensive evaluations of five leading M
LLMs. Our results demonstrate that while top-tier agents excel in simplified set
tings, their accuracy declines under partial occlusion. WinDeskGround provides a
 valuable benchmark to facilitate the assessment and advancement of GUI agent ro
bustness in realistic environments.
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DyLLM: Efficient Diffusion LLM Inference via Saliency-based Token Selection and Partial Attention

Younjoo Lee, Seungkyun Dan, Junghoo Lee, Jaiyoung Park, Jung Ho Ahn

Masked diffusion language models enable parallel token decoding, providing a pro
mising alternative to the sequential nature of autoregressive generation.
However, their iterative denoising process remains computationally expensive bec
ause it repeatedly processes the entire sequence at every step.
We observe that across these diffusion steps, most token representations remain
stable; only a small subset, which we term *salient tokens*, contributes meaning
fully to the next update.
Leveraging this temporal sparsity, we present **DyLLM**, a training-free inferen
ce framework that accelerates decoding by selectively computing only these salie
nt tokens.
DyLLM identifies saliency by measuring the cosine similarity of attention contex
ts between adjacent denoising steps.
It recomputes feed-forward and attention operations only for salient tokens whil
e reusing cached activations for the remainder.
Across diverse reasoning and code-generation benchmarks, DyLLM achieves up to 9.
6 $\times$ higher throughput while largely preserving the baseline accuracy of r
epresentative open-source diffusion LLMs, LLaDA and Dream.
****

PRISM: Training-Free Video Anomaly Detection via Intrinsic Statistical Modeling

YUANTONG CHEN, Zhengyan Ding, YanFeng Shang

While recent training-free video anomaly detection (VAD) methods offer advantage
s such as interpretability and ease of deployment, they often suffer from comput
ational inefficiency due to complex memory retrieval mechanisms or high-latency
visual-language models (VLMs). To address this issue, we propose PRISM (Paramete
r-less Recognition Based on Intrinsic Statistical Modeling), a novel framework f
or efficient open-set anomaly detection with minimal computational cost. Built o
n a pre-trained multimodal embedding model, PRISM introduces differential amplif
ication and whitening mechanisms to statistically suppress common-mode backgroun
d noise in the embedding space, thereby improving the signal-to-noise ratio of a
nomalous events. Extensive experiments on three widely datasets demonstrate that
 PRISM achieves state-of-the-art performance among training-free methods while m
aintaining real-time inference capability. Furthermore, our statistical analysis
 offers a complementary perspective on why training-free methods may suffer from
 lower Average Precision (AP) on complex datasets such as XD-Violence.Code is re
leased at https://github.com/ytC2026/ICML2026-PRISM.
****

ACO-MoE-LoRA: Evolving-while-Training for Adapting Segment Anything Model 2 to Specialized Domains

Kaiyi Luo, Bangjun Wang, Li Zhang, Fanzhang Li, Fei Zhu, Jiaqing Fan

Static fine-tuning paradigms impose rigid structural constraints on foundation m
odels like the Segment Anything Model 2 (SAM2), limiting their adaptability to t
he varying complexity of specialized downstream tasks.
To overcome this limitation, we propose **ACO-MoE-LoRA**, a dynamic framework th

at introduces an "Evolving-while-Training" strategy by synergizing Ant Colony Optimization (ACO) with a Latent Space Mixture-of-Experts (MoE) architecture.
Central to our method is the ACO-ConvLoRA module, which employs a pheromone-guided routing mechanism to actively govern expert selection and topological evolution.
By formulating expert assignment as an evolutionary pathfinding problem, this module effectively mitigates the standard routing collapse issue and enables elastic adjustment of LoRA ranks via weight slicing, bridging discrete structural search with continuous parameter training.
Extensive experiments across 16 challenging datasets demonstrate that our framework consistently outperforms leading static adapters, while effectively addressing the local optimality limitations of recent dynamic heuristics.
This work presents a self-organizing solution that harmonizes swarm intelligence with gradient optimization for efficiently adapting foundation models to specialized domains.
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Solving Time-Dependent Differential Equations with Physical Dynamical Systems
Chuan Liu, Yijie Chen, Ruibing Song, Wenhao Huang, Chunshu Wu, Deqian Kong, Ying Nian Wu, Kaiyuan Yang, Ang Li, Tony Geng
Time-Dependent Differential Equations (TDDEs) model dynamical processes across science and engineering, but time-critical applications require solvers that deliver high-fidelity trajectories under stringent latency constraints. Most existing TDDE solvers are limited by time discretization, forcing a latency-accuracy trade-off where smaller step sizes capture high-fidelity trajectories but incur prohibitive runtime, while larger steps meet real-time budgets at the cost of trajectory distortion. Dynamical System Machines (DSMs) offer a promising alternative by computing through continuous physical evolution, yet existing DSMs struggle to capture the spatiotemporal complexity of TDDEs. This work introduces DS-TS, a novel TDDE solver that is both accurate and efficient by leveraging the unique computational advantages of DSMs. DS-TS integrates three key innovations: (1) Excitatory-Inhibitory Inspired Coupling to better model complex spatial interactions; (2) State-aware Dynamic Nonlinearity to enable rich inter-node interactions and state-dependent spatiotemporal correlations; and (3) Hierarchical Temporal Integration to capture high-order temporal dependencies. Experiments demonstrate that DS-TS achieves high-fidelity solutions while delivering orders-of-magnitude improvements in speed ($\sim 10^3\times$) and energy efficiency ($\sim 10^5\times$) compared to baseline solvers.
****

Supervised Graph Contrastive Learning for Gene Regulatory Networks
Sho Oshima, Yuji Okamoto, Taisei Tosaki, Ryosuke Kojima
Graph Contrastive Learning (GCL) is a powerful self-supervised learning framework that performs data augmentation through graph perturbations, with growing applications in the analysis of biological networks such as Gene Regulatory Networks (GRNs).
The artificial perturbations commonly used in GCL, such as node dropping, induce structural changes that can diverge from biological reality.
This concern has contributed to a broader trend in graph representation learning toward augmentation-free methods, which view such structural changes as problematic and should be avoided.
However, this trend overlooks the fundamental insight that structural changes from biologically meaningful perturbations are not a problem to be avoided, but rather a rich source of information, thereby ignoring the valuable opportunity to leverage data from real biological experiments. Motivated by this insight, we propose SupGCL (Supervised Graph Contrastive Learning), a new GCL method for GRNs that directly incorporates biological perturbations from gene knockdown experiments as supervision.
SupGCL is a probabilistic formulation that continuously generalizes conventional GCL, linking artificial augmentations with real perturbations measured in knockdown experiments, and using the latter as explicit supervision.
On patient-derived GRNs from three cancer types, we train GRN representations wi

th SupGCL and evaluate it in two regimes: (i) embedding space analysis, where it yields clearer disease-subtype structure and improves clustering, and (ii) task-specific fine-tuning, where it consistently outperforms strong graph representation learning baselines on 13 downstream tasks spanning gene-level functional annotation and patient-level prediction.

****

Cross-View Lewis Weight Fusion Empowering Exemplar Replay for Federated Class-Incremental Learning

Zhuang Qi, Ying-Peng Tang, Lei Meng, Xiaoxiao Li, Han Yu, Xiangxu Meng

Federated Class-Incremental Learning (FCIL) aims to continually expand a model's recognition capacity in a distributed environment, enabling it to learn new classes while retaining knowledge of previously seen ones. Exemplar replay has emerged as a promising strategy owing to its simplicity and effectiveness. Existing methods either select exemplars based on local dynamics or construct global feature spaces to identify representative samples. However, they face inherent challenges in striking a balance between effectiveness and privacy. To address this issue, this paper proposes a Cross-view Lewis weIght Fusion method for exemplar replay in FCIL, termed CLIF, which fuses multi-view importance scores to guide representative sample selection under federated settings. Specifically, CLIF consists of two main modules: 1) the cross-view Lewis weight fusion module computes and integrates Lewis weights from multiple feature perspectives to achieve consistent importance estimation, ensuring that the selected samples better reflect the global data distribution and thus enhancing the representativeness of the replay subset. Building on this, 2) the frequency-based weighted training module adjusts the loss contribution of each sample according to its selection frequency across views, which emphasizes the contribution of critical samples. Moreover, we provide a theoretical analysis to guarantee the soundness and effectiveness of CLIF. Extensive experiments on three datasets demonstrate that our method consistently improves baselines by 1%–6%, supporting the above claims.

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AudioChat: Unified Audio Storytelling, Editing, and Understanding with Transfusion Forcing

William Chen, Prem Seetharaman, Rithesh Kumar, Oriol Nieto, Shinji Watanabe, Justin Salamon, Zeyu Jin

Despite recent breakthroughs, audio foundation models struggle in processing complex multi-source acoustic scenes. We refer to this challenging domain as audio stories, which can have multiple speakers and background/foreground sound effects. Compared to traditional audio processing tasks, audio stories introduce new layers of semantic, temporal, and physical complexity. To address this challenge, we propose AudioChat, a framework for developing audio foundation models that can generate, edit, and understand audio stories. AudioChat introduces a new paradigm in which LLM-based toolcalling agents simulate interactions between users and the system, and these simulated dialogues are used as training data. We also introduce a novel Audio Transfusion Forcing objective to train the AudioChat model, allowing it to simultaneously decompose high-level instructions via structured chain-of-thought reasoning and perform interactive multi-turn audio understanding/generation. To evaluate generation and editing performance, we develop three new metrics that directly measure task performance instead of relying upon distribution-based scoring. We highly encourage readers to visit our demo to better understand the capabilities of AudioChat: https://audiochat-icml-2026.github.io/.

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Towards Docking-oriented De Novo Ligand Design via Gradient Inversion

Zekai Chen, Xunkai Li, Sirui Zhang, Henan Sun, Jia Li, Qiangqiang Dai, Hongchao Qin, Zhenjun Li, Bing Zhou, Rong-Hua Li, Guoren Wang

De novo ligand design is a fundamental task that seeks to generate protein or molecule candidates that can effectively dock with protein receptors and achieve strong binding affinity entirely from scratch.

It holds paramount significance for a wide spectrum of biomedical applications.

However, most existing studies are constrained by the \textbf{Pseudo De Novo }, \textbf{Limited Docking Modeling}, and \textbf{Inflexible Ligand Type}.

To address these issues, we propose MagicDock, a forward-looking framework g rounded in the progressive pipeline and differentiable surface modeling.

(1) We adopt a well-designed gradient inversion framework.

To begin with, general docking knowledge of receptors and ligands is incorpo rated into the backbone model.

Subsequently, the docking knowledge is instantiated as reverse gradient flow s by binding prediction, which iteratively guide the de novo generation of ligan ds.

(2) We emphasize differentiable surface modeling in the \textit{generation p rocess}, leveraging learnable 3D point-cloud representations to precisely captur e docking details, thereby ensuring that the generated ligands preserve docking validity through interpretable spatial fingerprints.

(3) We introduce customized designs for different ligand types and integrate them into a unified gradient inversion framework with flexible triggers, thereb y ensuring broad applicability.

Moreover, we provide sufficient theoretical guarantees for MagicDock.

Extensive experiments across 9 scenarios demonstrate that MagicDock achieves average improvements of 7.0\% and 7.4\% over SOTA baselines specialized for pro tein or molecule ligand design, respectively.

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Learning Molecular Semantic Invariant Representation with Prototype Constraint

Zhiqiang Li, Jianqing Liang, Zhiqiang Wang, Xizhao Luo, Jiye Liang

Molecular representation learning has achieved remarkable progress in molecular property prediction, yet out-of-distribution (OOD) generalization remains challe nging. In practice, training data typically cover only a limited portion of the chemical space, causing models to rely on environment-dependent factors that fai l to transfer when scaffold structures or functional compositions shift. To addr ess this issue, we propose MoSIR, a framework for learning molecular semantic in variant representation with prototype constraint, which projects entangled molec ular embeddings into a learnable semantic prototype space to extract semantic in variant representation while isolating environment-sensitive variations. Buildin g upon this decomposition, we optimize a bi-level min-max objective that introdu ces representation perturbations to simulate plausible environment shifts and en force semantic stability. We further provide theoretical guarantees for MoSIR by deriving an OOD generalization bound under distribution shifts. Extensive exper iments on multiple molecular OOD benchmarks demonstrate that MoSIR consistently outperforms strong baselines across diverse shift settings, and qualitative anal yses confirm that the learned prototypes capture meaningful chemical semantics.

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Amortized Maximum Inner Product Search with Learned Support Functions

Theo X. Olausson, Joao Monteiro, Michal Klein, marco cuturi

Maximum inner product search (MIPS) is a crucial subroutine in machine learning, requiring identification of database vectors that align most strongly with a gi ven query. We propose amortized MIPS: a learning-based approach that trains neur al networks to directly predict MIPS solutions, amortizing the computational cos t of search across queries drawn from a known distribution. Our key insight is t hat the MIPS value function - the maximum inner product as a function of the que ry - is convex (as the pointwise maximum of linear functions), and its gradient at each query equals the optimal database vector. We explore two complementary a rchitectures: (1) Input Convex Neural Networks (ICNNs) that learn the convex val ue function and recover the optimal match via gradient computation, and (2) Vect orICNNs that directly regress the argmax, bypassing gradient computation entirel y at inference time. For ICNNs, we combine score regression with gradient matchi ng losses; for VectorICNNs, we introduce a score consistency loss derived from E uler's theorem for homogeneous functions. We further propose homogenization wrap pers that enforce positive 1-homogeneity, theoretically linking function values to gradients. Our experiments on retrieval benchmarks demonstrate that convexity provides an effective inductive bias, with learned potentials achieving high ma

tch rates while requiring only a single forward pass at inference. Our code is available at: https://github.com/apple/ml-amips.
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Bottleneck Communication Delay Minimization for Communication-Efficient Decentralized Learning
Nozomi Hata, Kenta Niwa
For communication-efficient decentralized learning, advanced network (NW) topologies, such as exponential and 1-peer exponential graphs, have been studied under homogeneous communication delays. However, real-world NWs exhibit heterogeneous communication delays, making node assignment optimization crucial for minimizing the Bottleneck Communication Delay (BCD). We propose BTSP-MSR, an approximate method for minimizing BCD on circulant digraphs, including exponential and 1-peer exponential graphs. Leveraging the fact that circulant digraphs can be viewed as a union of (directed) ring graphs, we derive an upper bound on the BCD by combining the ring-graph BCD (BTSP) with a deviation term (MSR). We then construct a solver that sequentially minimizes these two terms. Numerical experiments show that BTSP-MSR consistently reduces BCD across several circulant digraphs with large numbers of nodes. Notably, incorporating the exponential or 1-peer exponential graph enables communication-efficient decentralized learning under heterogeneous delay settings.
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A Geometry-Aware Efficient Algorithm for Compositional Entropic Risk Minimization
Xiyuan Wei, Linli Zhou, Bokun Wang, Chih-Jen Lin, Tianbao Yang
This paper studies optimization for a family of problems termed **compositional entropic risk minimization**, in which each data's loss is formulated as a Log-Expectation-Exponential (Log-E-Exp) function. The Log-E-Exp formulation serves as an abstraction of the Log-Sum-Exponential (LogSumExp) function when the explicit summation inside the logarithm is taken over a gigantic number of items and is therefore expensive to evaluate. While entropic risk objectives of this form arise in many machine learning problems, existing optimization algorithms suffer from several fundamental limitations including non-convergence, numerical instability, and slow convergence rates. To address these limitations, we propose a geometry-aware stochastic algorithm, termed **SCENT**, for the dual formulation of entropic risk minimization cast as a min--min optimization problem. The key to our design is a **stochastic proximal mirror descent (SPMD)** update for the dual variable, equipped with a Bregman divergence induced by a negative exponential function that faithfully captures the geometry of the objective. Our main contributions are threefold: (i) we establish an $O(1/\sqrt{T})$ convergence rate of the proposed SCENT algorithm for convex problems; (ii) we theoretically characterize the advantages of SPMD over standard SGD update for optimizing the dual variable; and (iii) we demonstrate the empirical effectiveness of SCENT on extreme classification, partial AUC maximization, contrastive learning and distributionally robust optimization, where it consistently outperforms existing baselines.
****
SciNet: Evaluating AI Agents in Relation-Aware Scientific Literature Retrieval
Chenyang Shao, Fengli Xu, Yong Li
AI agents have seen widespread adoption in information retrieval for scientific research, giving rise to tools such as Deep Research. However, existing retrieval agents mainly rely on keyword- or embedding-based methods. While effective at capturing content-level similarities, they struggle to understand complex relational networks among scientific papers, such as identifying corroborating or conflicting studies and tracing technological lineages. This fundamental limitation often results in fragmented knowledge structures, misinterpreted research sentiment, and ineffective modeling of collective scientific progress. To address this limitation, we introduce **SciNet**, the first **Sci**entific **Net**work relation-aware dataset for information retrieval agents. Built on a meta-database of 269 million papers across 7 disciplines and containing 8,940 carefully designed tasks, SciNet systematically captures three levels of relational understanding:

ego-centric retrieval of papers with novel knowledge structures, pairwise identi
fication of scholarly relationships, and path-wise reconstruction of scientific
evolution. Extensive evaluation of three categories of retrieval agents shows th
at their accuracy on relation-aware tasks often falls below 20%, highlighting a
fundamental shortcoming of current retrieval paradigms. Importantly, in a downst
ream literature review application, agents empowered with SciNet achieve a 25.3%
 improvement in review quality, highlighting the critical value of relation-awar
e retrieval for deepening scientific insights. We publicly release SciNet at htt
ps://github.com/tsinghua-fib-lab/SciNet to support future research.
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Exposing Vulnerabilities in Explanation for Time Series Classifiers via Dual-Tar
get Attacks

Bohan Wang, Zewen Liu, Lu Lin, Hui Liu, Li Xiong, Ming Jin, Wei Jin

Interpretable time series deep learning systems are often assessed by checking t
emporal consistency on explanations, implicitly treating this as evidence of rob
ustness. We show that this assumption can fail: Predictions and explanations can
 be adversarially decoupled, enabling targeted misclassification while the expla
nation remains plausible and consistent with a chosen reference rationale. We pr
opose TSEF (Time Series Explanation Fooler), a dual-target attack that jointly m
anipulates the classifier and explainer outputs. In contrast to single-objective
 misclassification attacks that disrupt explanation and spread attribution mass
broadly, TSEF achieves targeted prediction changes while keeping explanations co
nsistent with the reference. Across multiple datasets and explainer backbones, o
ur results consistently reveal that explanation stability is a misleading proxy
for decision robustness and motivate coupling-aware robustness evaluations for t
rustworthy time series tasks.
****

CLoVE: Personalized Federated Learning through Clustering of Loss Vector Embeddi
ngs

Randeep Bhatia, Nikos Papadis, Murali Kodialam, T. V. Lakshman, Sayak Chakrabart
y

We propose CLoVE (Clustering of Loss Vector Embeddings), a novel algorithm for C
lustered Federated Learning (CFL). In CFL, clients are naturally grouped into cl
usters based on their data distribution. However, identifying these clusters is
challenging, as client assignments are unknown. CLoVE utilizes client embeddings
 derived from model losses on client data, and leverages the insight that client
s in the same cluster share similar loss values, while those in different cluste
rs exhibit distinct loss patterns. Based on these embeddings, CLoVE is able to i
teratively identify and separate clients from different clusters and optimize cl
uster-specific models through federated aggregation. Key advantages of CLoVE ove
r existing CFL algorithms are (1) its simplicity, (2) its applicability to both
supervised and unsupervised settings, and (3) the fact that it eliminates the ne
ed for near-optimal model initialization, which makes it more robust and better
suited for real-world applications. We establish theoretical convergence bounds,
 showing that CLoVE can recover clusters accurately with high probability in a s
ingle round and converges exponentially fast to optimal models in a linear setti
ng. Our comprehensive experiments comparing with a variety of both CFL and gener
ic Personalized Federated Learning (PFL) algorithms on different types of datase
ts and an extensive array of non-IID settings demonstrate that CLoVE achieves hi
ghly accurate cluster recovery in just a few rounds of training, along with stat
e-of-the-art model accuracy, across a variety of both supervised and unsupervise
d PFL tasks.
****

Relevance-Based Embeddings: Lightweight Candidate Retrieval via Heavy-Ranker Cal
ls

Kirill Sergeevich Shevkunov, Andrey Ploskonosov, Liudmila Prokhorenkova

In many machine learning applications, the most relevant items for a query shoul
d be efficiently retrieved. The relevance function is usually an expensive simil
arity model, making the exhaustive search infeasible. A typical solution is to t
rain another model that separately embeds queries and items to a vector space, w

here similarity is defined via the dot product or cosine similarity. This allows one to search the relevant items through fast approximate nearest neighbor search at the cost of some reduction in quality. To compensate for this reduction, the found items (candidates) are re-ranked by the expensive ranking model. In this paper, we investigate an alternative approach to candidate selection that utilizes the scores of the expensive model to improve the representations of queries and items. The idea is to describe each query (item) by its relevance to a set of support items (queries) and use these new representations to obtain query (item) embeddings. We theoretically prove that such embeddings are powerful enough to approximate any complex similarity model (under mild conditions). We also investigate the choice of support items, which is a crucial ingredient of the proposed approach. The experiments on diverse academic and production datasets illustrate the power of our method.
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LIMSSR: LLM-Driven Sequence-to-Score Reasoning under Training-Time Incomplete Multimodal Observations

Huangbiao Xu, Huanqi Wu, Xiao Ke, Yuxin Peng

Real-world multimodal learning is often hindered by missing modalities. While Incomplete Multimodal Learning (IML) has gained traction, existing methods typically rely on the unrealistic assumption of full-modal availability during training to provide reconstruction supervision or cross-modal priors. This paper tackles the more challenging setting of IML under training-time incomplete observations, which precludes reliance on a "God's eye view" of complete data. We propose LIMSSR (LLM-Driven Incomplete Multimodal Sequence-to-Score Reasoning), a framework that reformulates this challenge as a conditional sequence reasoning task. LIMSSR leverages the semantic reasoning capabilities of Large Language Models via Prompt-Guided Context-Aware Modality Imputation and Multidimensional Representation Fusion to infer latent semantics from available contexts without direct reconstruction. To mitigate hallucinations, we introduce a Mask-Aware Dual-Path Aggregation to dynamically calibrate inference uncertainty. Extensive experiments on three Action Quality Assessment datasets demonstrate that LIMSSR significantly outperforms state-of-the-art baselines without relying on complete training data, establishing a new paradigm for data-efficient multimodal learning. Code is available at https://github.com/XuHuangbiao/LIMSSR.
****

TCAP: Tri-Component Attention Profiling for Unsupervised Backdoor Detection in MLLM Fine-Tuning

Mingzu Liu, Hao Fang, Runmin Cong

Fine-Tuning-as-a-Service (FTaaS) facilitates the customization of Multimodal Large Language Models (MLLMs) but introduces critical backdoor risks via poisoned data. Existing defenses either rely on supervised signals or fail to generalize across diverse trigger types and modalities. In this work, we uncover a universal backdoor fingerprint—attention allocation divergence—where poisoned samples disrupt the balanced attention distribution across three functional components: system instructions, vision inputs, and user textual queries, regardless of trigger morphology. Motivated by this insight, we propose Tri-Component Attention Profiling (TCAP), an unsupervised defense framework to filter backdoor samples. TCAP decomposes cross-modal attention maps into the three components, identifies trigger-responsive attention heads via Gaussian Mixture Model (GMM) statistical profiling, and isolates poisoned samples through EM-based vote aggregation. Extensive experiments across diverse MLLM architectures and attack methods demonstrate that TCAP achieves consistently strong performance, establishing it as a robust and practical backdoor defense in MLLMs.
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Reinforcement-aware Knowledge Distillation for LLM Reasoning

Zhaoyang Zhang, Shuli Jiang, Yantao Shen, Yuting Zhang, Dhananjay Ram, Shuo Yang, Zhuowen Tu, Wei Xia, Stefano Soatto

Reinforcement learning (RL) post-training has recently driven major gains in long chain-of-thought reasoning large language models (LLMs), but the high inference cost of such models motivates distillation into smaller students.

Most existing knowledge distillation (KD) methods are designed for supervised fi
ne-tuning (SFT), relying on fixed teacher traces or teacher-student Kullback–Lei
bler (KL) divergence based regularization.
When combined with RL, these approaches often suffer from distribution mismatch
and objective interference: teacher supervision may not align with the student’s
 evolving rollout distribution, and the KL regularizer can compete with reward m
aximization and require careful loss balancing.
To address these issues, we propose \emph{RL-aware distillation} (RLAD), which p
erforms selective imitation during RL---guiding the student toward the teacher o
nly when it improves the current policy update.
Our core component,Trust Region Ratio Distillation (TRRD), replaces the teacher-
student KL regularizer with a PPO/GRPO-style likelihood-ratio objective anchored
 to a teacher--old-policy mixture, yielding advantage-aware, trust-region-bounde
d distillation on student rollouts and naturally balancing exploration, exploita
tion, and imitation.
Across diverse logic reasoning and math benchmarks, RLAD}consistently outperform
s offline distillation, standard GRPO, and KL-based on-policy teacher–student kn
owledge distillation.
****

Is Task-Specific Training Necessary for Anomaly Detection?
Xingwu Zhang, Guanxuan Li, Paul Henderson, Gerardo Aragon-Camarasa, ZIJUN LONG
Current state-of-the-art multi-class unsupervised anomaly detection (MUAD) metho
ds rely on training encoder--decoder models to reconstruct anomaly-free features
. However, we argue that such task-specific training is costly under distributio
n shifts, and that reconstruction-based residual scoring further faces a fidelit
y--stability dilemma. Existing training-free alternatives, in turn, remain prone
 to cross-category and cross-region mismatches in MUAD. Motivated by these limit
ations, we propose Retrieval-based Anomaly Detection (RAD), a task-specific trai
ning-free framework that stores anomaly-free features in a memory and detects an
omalies through multi-level retrieval, matching test patches against the memory.
 Experiments demonstrate that RAD achieves state-of-the-art performance across f
our established benchmarks (MVTec-AD, VisA, Real-IAD, 3D-ADAM) under both standa
rd and few-shot settings. On MVTec-AD, RAD reaches 96.7% Pixel AUROC with just a
 single anomaly-free image compared to 98.5% of RAD's full-data performance. Col
lectively, these findings overturn the assumption that MUAD requires task-specif
ic training, showing that state-of-the-art anomaly detection is feasible with tr
aining-free memory-based retrieval. Our code is available at https://github.com/
longkukuhi/RAD.
****

Can local learning match self-supervised backpropagation?
Wu S. Zihan, Ariane Delrocq, Wulfram Gerstner, Guillaume Bellec
While end-to-end self-supervised learning with backpropagation (global BP-SSL) h
as become central for training modern AI systems, theories of local self-supervi
sed learning (local-SSL) have struggled to build functional representations in d
eep neural networks.
To establish a link between global and local rules, we first develop a theory fo
r deep linear networks: We identify conditions for local-SSL algorithms (like Fo
rward-forward or CLAPP) to implement exactly the same weight update as a global
BP-SSL. Starting from the theoretical insights, we then develop novel variants o
f local-SSL algorithms to approximate global BP-SSL in deep non-linear convoluti
onal neural networks. Variants that improve the similarity between updates of lo
cal-SSL with those of global BP-SSL also show better performance on image datase
ts (CIFAR-10, STL-10, Tiny ImageNet, and ImageNet). The best local-SSL rule with
 the CLAPP loss function matches the performance of a comparable global BP-SSL w
ith InfoNCE or CPC-like loss functions, and improves upon state-of-the-art for l
ayer-wise SSL on these benchmarks.
****

Reflective Hamiltonian Monte Carlo: Mixing Analysis and Application to Sampling
 on Stiefel Manifold
Kwangmin Lee, Yeonhee Park, Sewon Park

Sampling from distributions with bounded supports is a fundamental challenge in constrained statistical inference. Reflective Hamiltonian Monte Carlo (ReHMC) provides a useful sampling approach for this setting. However, it relies on convexity assumptions on the support and lacks non-asymptotic theoretical guarantees such as mixing-time bounds. To bridge this gap, we propose a convex-container plus thinning framework that is applicable to arbitrary target densities with bounded support. We establish the first non-asymptotic total-variation mixing-time bounds for ReHMC, achieving a polynomial dimension dependence of $O(d^2)$ for L-smooth targets, though with exponential dependence on smoothness parameters. Under an additional m-strong convexity assumption, we derive a sharper bound that eliminates this exponential dependence. We further apply this approach to sampling on the Stiefel manifold via a well-conditioned polar reparameterization and demonstrate improved numerical stability and sampling efficiency in simulation studies.

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Constructing Industrial-Scale Optimization Modeling Benchmark

Zhong Li, Hongliang Lu, Tao Wei, Yuxuan Chen, Wenyu Liu, Yuan Lan, Fan Zhang, Zaiwen Wen

Optimization modeling underpins decision-making in logistics, manufacturing, energy, and finance, yet translating natural-language requirements into correct optimization formulations and solver-executable code remains labor-intensive. Although large language models (LLMs) have been explored for this task, evaluation is still dominated by toy-sized or synthetic benchmarks, masking the difficulty of industrial problems with 10^{3}--10^{6} (or more) variables and constraints. A key bottleneck is the lack of benchmarks that align natural-language specifications with reference formulations/solver code grounded in real optimization models. To fill in this gap, we introduce MIPLIB-NL, built via a structure-aware reverse construction methodology from real mixed-integer linear programs in MIPLIB~2017. Our pipeline (i) recovers compact, reusable model structure from flat solver formulations, (ii) reverse-generates natural-language specifications explicitly tied to this recovered structure under a unified model--data separation format, and (iii) performs iterative semantic validation through expert review and human--LLM interaction with independent reconstruction checks. This yields 223 one-to-one reconstructions that preserve the mathematical content of the original instances while enabling realistic natural-language-to-optimization evaluation. Experiments show substantial performance degradation on MIPLIB-NL for systems that perform strongly on existing benchmarks, exposing failure modes invisible at toy scale.

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MPFM: Cross Multi-Domain Prototype Flow Matching for Log Anomaly Detection

Jing Zhang, Chao Luo

Cross multi-domain log anomaly detection aims to train a unified model applying in multiple heterogeneous systems, alleviating the annotation cost and scalability bottlenecks of traditional cross single-domain approaches. However, existing methods face two fundamental challenges: (i) geometric proximity alone is insufficient to certify normality, and (ii) forcibly aligning distributions across domains can induce negative transfer. To address these issues, we propose MPFM (Cross Multi-Domain Prototype Flow Matching for Log Anomaly Detection), grounded in the principle that anomalies are samples that cannot be stably generated by the normal data-generating mechanism. Specifically, MPFM employs a shared–private prototype system to disentangle cross-domain commonalities from domain-specific patterns, introduces domain-conditioned flow matching to perform anomaly detection by integrating structural and dynamical evidence, and further leverages prototype-drift-driven hard example mining to improve robustness near decision boundaries. Experiments on HDFS, BGL, Thunderbird, and Spirit demonstrate that MPFM delivers substantial gains under multi-domain joint training.

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Masked Multi-path Contrast with Confidence-Gated Semantic Imputation for Incomplete Multi-view Clustering

Fan Yang, Haikun Xu

Incomplete multi-view clustering (IMVC) becomes particularly challenging under h
eavy missingness and view-availability imbalance. In this regime, scarce co-obse
rved pairs make cross-view correspondences unreliable. Imputation-first pipeline
s may trigger cascading reconstruction errors, while purely consistency-based al
ignment often degrades sharply and gives limited control over semantic convergen
ce across views. We propose MAGIC (Masked multi-pAth contrast with confIdence-Ga
ted semantIc imputation), a unified framework that learns calibrated cluster sem
antics before conservative completion. MAGIC builds multiple correlated represen
tation and prediction paths from lightly augmented latent codes, and couples the
m with a masked multi-path contrastive consensus objective and prediction-consis
tency regularization. The resulting posteriors are aggregated into view-wise sof
t assignments to reduce overconfidence and alleviate dominance by more frequentl
y observed views. Based on these calibrated semantics, MAGIC performs similarity
-guided semantic transfer in label space with confidence-aware gating, and compl
etes missing representations through a geometry-preserving prototype fallback. E
xperiments on four benchmarks across different missing ratios show consistent ga
ins over prior IMVC methods, and the ablations support the roles of masked multi
-path consensus learning and confidence-gated semantic imputation.
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Phy-CoSF: Physics-Guided Continuous Spectral Fields Reconstruction and Spectral Super-Resolution for Snapshot Compressive Imaging

Wudi Chen, Zhiyuan Zha, Xin Yuan, Shigang Wang, Bihan Wen, Jiantao Zhou, Gang Ya
n, zipei fan, Ce Zhu

Recent advances have demonstrated that coded aperture snapshot spectral imaging
(CASSI) systems show great potential for capturing 3D hyperspectral images (HSIs
) from a single 2D measurement. Despite the inherent spectral continuity of scen
es captured by CASSI, most existing reconstruction methods are restricted to fix
ed, discrete spectral outputs, thereby precluding continuous spectral reconstruc
tion or spectral super-resolution. To address this challenge, we propose Phy-CoS
F, which synergizes deep unfolding networks with implicit neural representations
, establishing a new paradigm for continuous spectral reconstruction and super-r
esolution in CASSI. Specifically, we propose a two-phase architecture that bridg
es discrete-wavelength training with continuous spectral rendering, enabling the
 synthesis of high-fidelity HSIs at arbitrary target wavelengths. At the core of
 our framework lies the continuous spectral fields (CoSF) module, embedded withi
n each unfolding stage as a dynamic prior, which comprises a triple-branch cross
-domain feature mixer for comprehensive spatial–frequency–channel feature fusion
, alongside a spectral synthesis head that generates spectral intensities by que
rying continuous wavelength coordinates. Extensive experimental results demonstr
ate that Phy-CoSF not only achieves continuous modeling at arbitrary spectral re
solutions but also outperforms many state-of-the-art methods in both reconstruct
ion fidelity and spectral detail preservation.
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Learning to Evict from Key-Value Cache

Luca Moschella, Laura Manduchi, Ozan Sener

The growing size of Large Language Models (LLMs) makes efficient inference chall
enging, primarily due to the memory demands of the autoregressive Key-Value (KV)
 cache. Existing eviction or compression methods reduce cost but rely on heurist
ics, such as recency or past attention scores, which serve only as indirect prox
ies for a token’s future utility and introduce computational overhead. We refram
e KV cache eviction as a reinforcement learning (RL) problem: learning to rank t
okens by their predicted usefulness for future decoding. To this end, we introdu
ce KV Policy (KVP), a framework of lightweight per-head RL agents trained on pre
-computed generation traces using only key and value vectors. Each agent learns
a specialized eviction policy guided by a holistic reward, derived from future u
tility, that evaluates the quality of the ranking across all cache budgets, requ
iring no modifications to the underlying LLM or additional inference. Evaluated
across two model families on the long-context benchmark RULER (up to 128K tokens
) and the multi-turn dialogue benchmark OASST2-4k, KVP significantly outperforms
 strong baselines. Zero-shot tests on standard downstream tasks (BoolQ, LongBenc

h passage retrieval, GovReport) further show that KVP generalizes beyond its tra ining distribution and to considerably longer sequence lengths. These results de monstrate that learning to predict future token utility is a powerful and scalab le paradigm for adaptive KV cache management.
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Procedural Generation Of Algorithm Discovery Tasks in Machine Learning
Alexander David Goldie, Zilin Wang, Adrian Hayler, Deepak Nathani, Edan Toledo, Ken Thampiratwong, Aleksandra Kalisz, Michael Beukman, Alistair Letcher, Shashan k Reddy Chirra, Clarisse Wibault, Theo Wolf, Charles O'Neill, Uljad Berdica, Nic holas Roberts, Saeed Rahmani, Roberta Raileanu, Shimon Whiteson, Jakob Nicolaus Foerster
Automating the development of machine learning algorithms has the potential to u nlock new breakthroughs. However, our ability to *improve* and *evaluate* algori thm discovery systems has thus far been limited by existing task suites. They su ffer from many issues, such as: poor evaluation methodologies; data contaminatio n; and containing saturated or very similar problems. Here, we introduce *DiscoG en*, a procedural generator of algorithm discovery tasks for machine learning, s uch as developing optimisers for reinforcement learning or loss functions for im age classification. Motivated by the success of procedural generation in reinfor cement learning, DiscoGen spans billions of tasks of varying difficulty and comp lexity from a range of machine learning fields. These tasks are specified by a s mall number of configuration parameters and can be used to optimise algorithm di scovery agents (ADAs). We present *DiscoBench*, a fixed, small subset of DiscoGe n tasks for principled evaluation of ADAs. Finally, we propose a number of ambit ious, impactful research directions enabled by DiscoGen, and demonstrate its use for ADA optimisation through scaling experiments for automated prompt tuning. D iscoGen is released open-source.
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SciAgentGym: Benchmarking Multi-Step Scientific Tool-Use in LLM Agents
Yujiong Shen, yajie yang, Zhiheng Xi, Binze Hu, Huayu Sha, Qiyuan Peng, Jiazheng Zhang, Junlin Shang, Jixuan Huang, Yutao Fan, Jingqi Tong, Ming Zhang, Shihan D ou, Zhenfei Yin, Xingjun Ma, LEI BAI, Tao Gui, Qi Zhang, Xuanjing Huang, Yu-Gang Jiang
Scientific reasoning inherently demands integrating sophisticated toolkits to na vigate domain-specific knowledge. Yet, current benchmarks largely overlook agent s' ability to orchestrate tools for such rigorous workflows.
To bridge this gap, we introduce **SciAgentGym**, a scalable interactive environ ment featuring 1,780 domain-specific tools across four natural science disciplin es, supported by a robust execution infrastructure.
Complementing this, we present **SciAgentBench**, a tiered evaluation suite desi gned to stress-test agentic capabilities from elementary actions to long-horizon workflows.
Our evaluation identifies a critical bottleneck: state-of-the-art models still s truggle with complex scientific tool-use, and their performance degrades substan tially as interaction horizons extend.
To address this, we propose **SciForge**, a data synthesis method that models th e tool action space as a dependency graph to generate logic-aware training traje ctories.
By fine-tuning on these trajectories, our SciAgent-8B outperforms the significan tly larger Qwen3-VL-235B-Instruct while exhibiting positive cross-domain transfe r of scientific tool-use capabilities.
These results underscore the promising potential of next-generation autonomous s cientific agents.
****

Understanding LoRA as Knowledge Memory: An Empirical Analysis
Seungju Back, Dongwoo Lee, Naun Kang, Taehee Lee, S. K. Hong, Youngjune Gwon, Su ngjin Ahn
Continuous knowledge updating for pre-trained large language models (LLMs) is in creasingly necessary yet remains challenging. Although inference-time methods li ke In-Context Learning (ICL) and Retrieval-Augmented Generation (RAG) are popula

r, they face constraints in context budgets, costs, and retrieval fragmentation. Departing from these context-dependent paradigms, this work investigates a parametric approach using Low-Rank Adaptation (LoRA) as a modular knowledge memory. Although few recent works examine this concept, the fundamental mechanics governing its capacity and composability remain largely unexplored. We bridge this gap through the first systematic empirical study mapping the design space of LoRA-based memory, ranging from characterizing storage capacity and optimizing internalization to scaling multi-module systems and evaluating long-context reasoning. Rather than proposing a single architecture, we provide practical guidance on the operational boundaries of LoRA memory. Overall, our findings position LoRA as the complementary axis of memory alongside RAG and ICL, offering distinct advantages.
****

Reliability-Aware LLM Alignment from Inconsistent Human Feedback
Jingyi Huang, Ruohan Zong, Yujun Feng, Liran Ma, Lanyu Shang, Yang Zhang
Reinforcement Learning from Human Feedback (RLHF) is critical for aligning Large Language Models (LLMs) with human preferences. However, its efficacy is often compromised by the inherent inconsistency and subjectivity of human annotations. Existing preference optimization frameworks, such as Direct Preference Optimization (DPO), typically treat ambiguous pairs with high annotator disagreement identically to those with unanimous consensus, forcing models to overfit to inconsistent supervision signals and leading to suboptimal alignment. In this work, we propose $\textit{Reliability-Guided Preference Optimization}$ (RGPO), a robust framework designed to mitigate the impact of inconsistent human feedback. RGPO estimates annotator reliability and infers latent ground truth labels from noisy human feedback to identify robust preferences. Furthermore, we introduce a reliability-aware consistency optimization that dynamically modulates the training objective based on the consensus level of annotations, ensuring the model prioritizes high-consensus supervision signals. Extensive experiments on LLM alignment benchmarks demonstrate that RGPO effectively reduces inconsistency and noise in training data and achieves superior performance compared to widely adopted RLHF baselines. Our code and configurations are available at https://github.com/GenieHuang/RGPO.
****

ARC-Decode: Accelerated Decoding with Risk-Bounded Acceptance
Ying Li, Zhaode Wang, Zhiwen Chen, chengfei lv, Huan Wang
As larger language models deliver stronger capabilities, their autoregressive inference becomes increasingly expensive.
Speculative decoding accelerates generation by letting a fast draft propose tokens that the target model verifies in parallel.
Yet under sampling ($T>0$), observed speedups consistently lag behind those under greedy decoding, as the classical lossless verification rule tends to over-reject low-risk drafts, leading to lower acceptance rates and limited acceleration.
To address this gap, we propose **ARC-Decode** (**A**cceptance with **R**isk **C**ontrol), a training-free method that augments speculative decoding without extra forward passes.
ARC-Decode enables \textbf{relaxed} acceptance by identifying drafts whose acceptance preserves the output distribution of the target model, under a risk-controlled criterion based on Jensen--Shannon divergence.
It combines confidence-based pre-verification filtering with a risk-bounded acceptance criterion derived from an analytic upper bound on the potential distributional deviation.
Integrated into the state-of-the-art EAGLE-3 pipeline, ARC-Decode increases accept length per cycle and reduces verification compute, achieving up to **1.6**$\times$ end-to-end speedup over EAGLE-3 under sampling with negligible quality change across benchmarks.
****

AdverMCTS: Combating Pseudo-Correctness in Code Generation via Adversarial Monte Carlo Tree Search
Qingyao Li, Weiwen Liu, Weinan Zhang, Yong Yu, Bo An

Recent advancements in Large Language Models (LLMs) have successfully employed s
earch-based strategies to enhance code generation. However, existing methods typ
ically rely on static, sparse public test cases for verification, leading to pse
udo-correctness—where solutions overfit the visible public tests but fail to gen
eralize to hidden test cases. We argue that optimizing against a fixed, weak env
ironment inherently limits robustness. To address this, we propose AdverMCTS, a
novel adversarial Monte Carlo Tree Search framework that combats pseudo-correctn
ess by coupling code search with active vulnerability discovery. AdverMCTS formu
lates generation as a minimax-style game between a Solver agent, which synthesiz
es code candidates, and an Attacker agent, which evolves to generate targeted te
st cases that exploit logical divergences in the current solution pool. These di
scovered tests form a dynamic, progressively hostile filter that penalizes fragi
le reasoning. Extensive experiments demonstrate that AdverMCTS significantly out
performs state-of-the-art baselines, effectively reducing false positive rates a
nd forcing the model to generalize beyond the initial constraints. The resources
 of this work are available at https://github.com/SIMONLQY/AdverMCTS.
****

Mixtures Closest To A Given Measure: A Semidefinite Programming Approach
Srecko Durasinovic, Jean B. Lasserre, Victor Magron
Mixture models, such as Gaussian mixture models (GMMs), are widely used in machi
ne learning to represent complex data distributions. A key challenge, especially
 in high-dimensional settings, is to determine the mixture order and estimate th
e mixture parameters. We study the problem of approximating a target measure, av
ailable only through finitely many of its moments, by a mixture of distributions
 from a parametric family (e.g., Gaussian, exponential, Poisson), with approxima
tion quality measured by the 2-Wasserstein ($\operatorname{W_2}$) or the total v
ariation (TV) distance. Unlike many existing approaches, the pa
rameter set is not assumed to be finite; it is modeled as a compact basic semi-a
lgebraic set. We introduce a hierarchy of semidefinite relaxations with asymptot
ic convergence to the desired optimal value. In addition, when a certain rank co
ndition is satisfied, the convergence is even finite and recovery of an optimal
mixing measure is obtained. We also present an application to clustering, where
our framework serves either as a stand-alone method or as a preprocessing step t
hat yields both the number of clusters and strong initial parameter estimates, t
hereby accelerating convergence of standard (local) clustering algorithms
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Unlearning with Asymmetric Sources: Improved Unlearning-Utility Trade-off with P
ublic Data
Ahmed Mehdi Inane, Vincent Quirion, Gintare Karolina Dziugaite, Ioannis Mitliagk
as
Noise-based certified machine unlearning currently faces a hard ceiling: the noi
se magnitude required to certify unlearning typically destroys model utility, pa
rticularly for large-scale deletion requests.
While leveraging public data is a standard technique in differential privacy to
relax this tension, its role in unlearning remains unexplored. We address this g
ap by introducing **Asymmetric Langevin Unlearning (ALU)**, a framework that use
s public data to mitigate privacy costs. We prove that public data injection sup
presses the unlearning cost by a factor of $O(1/n_{\mathrm{pub}}^2)$, guaranteei
ng a strict computational advantage over retraining. This establishes a new cont
rol mechanism: practitioners can mitigate the need for high noise—and the associ
ated utility loss—by increasing the volume of public data. Crucially, we analyze
 the realistic setting of **distribution mismatch**, explicitly characterizing h
ow shifts between public and private sources impact utility. We show that ALU en
ables "mass unlearning'' of constant dataset fractions -- a regime where standar
d symmetric methods become impractical -- while maintaining high utility.
Empirical evaluations using variational Rényi divergence and membership inferenc
e attacks confirm that ALU effectively thwarts privacy attacks while preserving
utility under reasonable distribution shifts.
****

AgentHijack: Benchmarking Computer Use Agent Robustness to Common Environment Co

rruptions
Jingwei Sun, Jianing Zhu, Yuanyi Li, Tongliang Liu, Xia HU, Bo Han
Autonomous computer use agents that powered by multimodal large language models (MLLMs) are emerging as capable assistants for completing complex digital workflows. However, real-world execution environments are far from ideal: pop-ups, resolution changes, and competing applications frequently interfere with agent perception and control. We introduce AgentHijack, a benchmark designed to evaluate the robustness of computer-use agents under common corruptions, where the uncertainties in dynamic environment disrupt the execution flow without direct adversarial intent. Specifically, AgentHijack introduces 9 configurable common corruptions to replicate realistic imperfect scenarios. We evaluate a variety of desktop tasks that utilize MLLM-based agents and discover that even minor instances of corruption can result in substantial performance degradation, which emphasizes the fragility of agents and underscores the necessity of robustness evaluation. Afterward, we propose AgentHijack-Agent, a framework that integrates an action generator with enhanced grounding capabilities and an onlooker responsible for behavior summarization and environment checking. Extensive experiments validate its effectiveness. Our code, environment, baseline models and data are publicly available at: \url{https://AgentHijack.github.io}.
****

Cross-Tactile Sensor Representation Learning
Yan Zhang, Zheng WANG, Pengpeng Zeng, Xing Xu, Jingkuan Song, Heng Tao Shen
Visuo-tactile sensors have been widely adopted in robotic manipulation. However, inherent heterogeneity in sensor designs hinders the learning of unified tactile representations in cross-sensor scenarios. Existing methods that focus on reconstruction or task-specific supervision often fail to capture the common information between different tactile sensors, particularly in the presence of substantial sensor variations, resulting in limited generalization to unseen sensors. To address this, we propose Cross-Tactile Sensor Representation Learning (CTSRL), a unified framework for sensor-agnostic tactile representation learning. CTSRL introduces a Cross-Sensor Modulator (CSM) to eliminate sensor-specific biases and adopts a two-stage learning paradigm: (1) leveraging aligned synthetic data for cross-sensor self-supervised learning to extract shared latent representations across sensor domains; and (2) integrating real-world multimodal tactile data to bridge the sim-to-real semantic gap through cross-modal alignment, thereby enriching representations with fine-grained semantic attributes. Experimental results show that our method demonstrates strong multi-sensor generalization, significantly improving sensor-agnostic representation learning.
****

Reward Shaping for (Inference-Time) Alignment: A Stackelberg Game Perspective
Haichuan Wang, Tao Lin, Lingkai Kong, Ce Li, Hezi Jiang, Milind Tambe
Existing alignment methods directly use the reward model learned from user preference data to optimize an LLM policy, subject to KL regularization with respect to the base policy. This practice is suboptimal for maximizing user's utility because the KL regularization may cause the LLM to inherit the bias in the base policy that conflicts with user preferences. While amplifying rewards for preferred outputs can mitigate this bias, it also increases the risk of reward hacking. This tradeoff motivates the problem of optimally designing reward models under KL regularization. We formalize this reward model optimization problem as a Stackelberg game, and show that a simple reward shaping scheme can effectively approximate the optimal reward model. We empirically evaluate our method in inference-time alignment settings and demonstrate that it integrates seamlessly into existing alignment methods with minimal overhead. Our method consistently improves average reward and achieves win–tie rates exceeding 66\% against all baselines, averaged across evaluation settings.
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Online Contract Design With Unknown Technology
Matteo Bollini, Matteo Castiglioni, Alberto Marchesi
Hidden-action principal-agent problems model scenarios in which a principal induces an agent to take a costly and *unobservable* action through the provision

of outcome-dependent payments. These problems find application in a variety of r
eal-world settings, such as crowdsourcing, online labor platforms, and machine l
earning task delegation. Recently, much of the literature has focused on how to
handle the principal’s *uncertainty* about the agent and the surrounding environ
ment, which is often the main challenge in practice. One prominent approach is t
o adopt an *online learning* framework, where the principal repeatedly interacts
with the agent to learn optimal payments from experience. However, existing lea
rning algorithms, while achieving regret that scales sublinearly in the number o
f interaction rounds T, typically suffer from an exponential dependence on the
size of the problem instance. In this paper, we show that this problematic expo
nential growth can be avoided by assuming that the principal has knowledge of a
set of possible actions of the agent, while remaining unaware of which actions a
re actually available---an assumption that is reasonable in many real-world sett
ings.
****
Beyond Looking Up, Try Looking Around: Harmonizing Global Structure and Local Co
nsistency in Optimal Transport for Short Text Clustering
Zhihao Yao, Yuxuan Gu, Jixuan Yin, Bo Li
Pseudo-labeling based on Optimal Transport (OT) has become an effective mechanis
m for enhancing short text clustering.
Existing OT methods are short in modeling semantic consistencies between samples
, which may assign different pseudo-labels to semantically similar samples. Thes
e erroneous pseudo-labels can cause the model to produce inferior clusters.
This paper proposes a novel short text clustering framework, which remedies the
neglect of semantic consistency in existing OT methods, generating reliable pseu
do-labels to facilitate clustering.
Specifically, the proposed approach first designs an instance-level attention me
chanism to capture semantic relationships between samples, which are then integr
ated into the OT formulation to endow the transport process with neighborhood se
mantic awareness.
By solving the proposed OT formulation, reliable pseudo-labels are obtained that
simultaneously account for sample-to-sample semantic consistency and sample-to-
cluster global structure information.
These pseudo-labels are then used as supervisory signals to guide the model to a
chieve accurate clustering.
Extensive experiments demonstrate that the proposed approach outperforms state-o
f-the-art methods.
The code is available at: https://github.com/YZH0905/CAOT-STC.
****
FreeRet: MLLMs as Training-Free Retrievers
Yuhan Zhu, Xiangyu Zeng, Chenting Wang, Xinhao Li, Chunxu Liu, Yicheng Xu, Ziang
Yan, Yi Wang, Limin Wang
Multimodal large language models (MLLMs) are emerging as versatile foundations f
or mixed-modality retrieval.
Yet, they often require heavy post-hoc training to convert them into contrastive
encoders for retrieval.
This work asks: \textit{Can off-the-shelf MLLMs serve as powerful retrievers wit
hout additional training?}
We present \textbf{FreeRet}, a plug■and■play framework that turns any MLLM into
a two■stage retriever.
FreeRet first derives semantically grounded embeddings directly from the model f
or fast candidate search, and then exploits its reasoning ability for precise re
ranking.
The framework contributes three advances: bypassing lexical alignment layers to
obtain semantically faithful embeddings, conditioning representation generation
with explicit priors, and mitigating framing effect in reranking via neutral cho
ice framing.
On the MMEB and MMEB-V2, FreeRet substantially outperforms models trained on mil
lions of pairs.
Beyond benchmarks, FreeRet is model-agnostic and scales seamlessly across MLLM f

amilies and sizes, preserves their generative abilities, supports arbitrary moda
lity combinations, and unifies retrieval, reranking, and generation into end-to-
end RAG within a single model.
Our findings demonstrate that pretrained MLLMs, when carefully harnessed, can se
rve as strong retrieval engines without training, closing a critical gap in thei
r role as generalists.
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ForceForget: Reinforcement Concept Removal for Enhancing Safety in Text-to-Image
 Models
Dong Han, Yong Li
With the advance of generative AI, the text-to-image (T2I) model has the ability
 to generate various contents. However, T2I models still can generate unsafe con
tents. To alleviate this issue, various concept erasing methods are proposed. Ho
wever, existing methods tend to excessively erase unsafe concepts and suppress b
enign concepts contained in harmful prompts, which can negatively affect model u
tility. In this paper, we focus on eliminating unsafe content while maintaining
model capability in safe semantic meaning interpretation by optimizing the conce
pt erasing reward (CER) with reinforcement learning. To avoid overly content era
sure, we introduce the Safe Adapter to project partial text embedding for effici
ent concept regulation in cross-attention layers. Extensive experiments conducte
d on different datasets demonstrate the effectiveness of the proposed method in
alleviating unsafe content generation while preserving the high fidelity of beni
gn images compared with existing state-of-the-art (SOTA) concept erasing methods
. In terms of robustness, our method outperforms counterparts against red-teamin
g tools. Moreover, we showcase the proposed approach is more effective in emergi
ng image-to-image (I2I) scenarios compared with others. Lastly, we extend our me
thod to erase general concepts, such as artistic styles and objects. **Disclaime
r:** *This paper includes discussions of sexually explicit content that may be o
ffensive to certain readers. All images used in this work are synthesized or fro
m public datasets.*
****

TIMI: Training-Free Image-to-3D Multi-Instance Generation with Spatial Fidelity
Xiao Cai, Pengpeng Zeng, Ji Zhang, Heng Tao Shen, Jingkuan Song, Lianli Gao
Precise spatial fidelity in Image-to-3D multi-instance generation is critical fo
r downstream real-world applications. Recent work attempts to address this by fi
ne-tuning pre-trained Image-to-3D (I23D) models on multi-instance datasets, whic
h incurs substantial training overhead and struggles to guarantee spatial fidel
ity.
In fact, we observe that pre-trained I23D models already possess meaningful spat
ial priors, which remain underutilized as evidenced by instance entanglement iss
ues.
Motivated by this, we propose **TIMI**, a novel **T**raining-free framework for
Image-to-3D **M**ulti-**I**nstance generation that achieves high spatial fid
elity.
Specifically, we first introduce an Instance-aware Separation Guidance (ISG) mod
ule, which facilitates instance disentanglement during the early denoising stage
. Next, to stabilize the guidance introduced by ISG, we devise a Spatial-stabili
zed Geometry-adaptive Update (SGU) module that promotes the preservation of the
geometric characteristics of instances while maintaining their relative relation
ships. Extensive experiments demonstrate that our method yields better performan
ce in terms of both global layout and distinct local instances compared to exist
ing multi-instance methods, without requiring additional training and with faste
r inference speed.
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“Do Diffusion Models Dream of Electric Planes?” Discrete and Continuous Simulati
on-Based Inference for Aircraft Design
Aurelien Ghiglino, Daniel Elenius, Anirban Roy, Ramneet Kaur, Manoj Acharya, Col
in Samplawski, Brian Matejek, Susmit Jha, Juan Alonso, Adam D. Cobb
In this paper, we generate conceptual engineering designs of electric vertical t
ake-off and landing (eVTOL) aircraft. We follow the paradigm of simulation-based

inference (SBI), whereby we look to learn a posterior distribution over the full eVTOL design space. To learn this distribution, we sample over discrete aircraft configurations (topologies) and their corresponding set of continuous parameters. Therefore, we introduce a hierarchical probabilistic model consisting of two diffusion models. The first model leverages recent work on Riemannian Diffusion Language Modeling (RDLM) and Unified World Models (UWMs) to enable us to sample topologies from a discrete and continuous space. For the second model we introduce a masked diffusion approach to sample the corresponding parameters conditioned on the topology. Our approach rediscovers known trends and governing physical laws in aircraft design, while significantly accelerating design generation.
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Attacks on Machine-Text Detectors Retain Stylistic Fingerprints
Rafael Alberto Rivera Soto, Barry Y. Chen, Nicholas Andrews
Despite considerable progress in the development of machine-text detectors, the ease with which machine-text can be manipulated to evade detection has led to suggestions that the problem is inherently intractable.
In this work, we investigate the limits of such evasion strategies.
We demonstrate that while current attacks, ranging from prompt engineering to detector-guided optimization can effectively degrade performance of standard detectors, they fail to erase the underlying stylistic "fingerprints" of machine text.
We show that few-shot detectors that utilize the stylistic feature space are robust to these evasion attempts, reliably detecting samples even from models explicitly tuned to prevent detection.
This raises the question: does style represent a universal defense against machine-detection attacks?
We demonstrate that the answer is "no" by introducing a novel paraphrasing approach that simultaneously optimizes for undetectability and adherence to specific human styles.
We show that unlike prior methods, this attack effectively evades all considered detectors, including those that utilize writing style.
However, we find that this evasion is not absolute: as the number of documents available for analysis grows, the human and machine distributions become distinguishable again.
Overall, our findings suggest that reliable machine-text detection requires moving beyond single-document analysis to multi-document analysis.
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Introspection Adapters: Training LLMs to Report Their Learned Behaviors
Keshav Shenoy, Li Yang, Abhay Sheshadri, Jack Lindsey, Samuel Marks, Rowan Wang
Can we train LLMs to *introspect*, i.e. to faithfully describe their own behaviors in natural language? Prior work has shown some, limited, success. However, it is difficult to scale introspection training due to a lack of ground-truth labels. In this work, we study an approach to introspection training which side-steps this data bottleneck. Given a target model M, our method works by fine-tuning models M_i from M with implanted behaviors b_i (such as downplaying medical problems); the (M_i, b_i) pairs serve as labeled introspection training data. We then train an *introspection adapter* (IA): a LoRA adapter jointly optimized across the fine-tunes M_i which causes them to verbalize their implanted behaviors. This IA induces faithful introspection in fine-tunes of M that were trained in very different ways from the M_i, as well as in M itself. This is surprising because the IA was never trained on M. To demonstrate the utility of IAs, we use them to successfully audit misaligned models introduced in prior work. IAs can also be used to detect fine-tuning API attacks which train models to comply with encrypted harmful requests. Notably, IAs are more effective when applied to larger models. Overall, our results suggest that IAs are a scalable, effective, and practically useful approach to LLM introspection training.
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Generalization Bounds for Out-of-distribution Generalization
Xin Zou, Xiuwen Gong, Weiwei Liu
Out-of-distribution (OOD) generalization has attracted increasing research atten

tion in recent years, owing to its promising empirical results in real-world applications. However, theoretical studies on OOD generalization remain limited, particularly with respect to lower bounds on the generalization error. To better understand how source data contributes to improved OOD generalization performance, we take an initial step toward establishing a lower bound on the OOD generalization error, and subsequently investigate upper bounds from the perspective of statistical learning theory. Interestingly, we find that under the RCS and empirical-RCS conditions, simply minimizing the average empirical risk over the source domains can yield a nearly optimal error rate (up to a logarithmic factor) \textit{without} requiring knowledge or estimation of distributional parameters or the discrepancy between source and target domains. This conditional result provides a theoretical perspective on the surprising phenomenon observed in DomainBed, where carefully designed OOD generalization algorithms fail to outperform the simple empirical risk minimization (ERM) algorithm. Our results also imply a no-free-lunch theorem and provide an optimistic bound for OOD generalization.
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Can We Build a Monolithic Model for Fake Image Detection? SICA: Semantic-Induced Constrained Adaptation for Unified-Yet-Discriminative Artifact Feature Space Reconstruction
Bo Du, Xiaochen Ma, Xuekang Zhu, Zhe Yang, Chaoqun Niu, Jian liu, Ji-Zhe Zhou
Fake Image Detection (FID), aiming at unified detection across four image forensic subdomains, is critical in real-world forensic scenarios. Compared with ensemble approaches, monolithic FID models are theoretically more promising, but to date, consistently yield inferior performance in practice. In this work, we identify the intrinsic distinctness of artifacts across subdomains—a critical barrier we term the "Ji-Zhe phenomenon". Driven by this phenomenon, we diagnose the cause of this underperformance for the first time: the collapse of the artifact feature space. The core challenge for developing a practical monolithic FID model thus boils down to the "unified-yet-discriminative" reconstruction of the artifact feature space. To address this paradoxical challenge, we hypothesize that high-level semantics can serve as a structural prior for the reconstruction, and further propose Semantic-Induced Constrained Adaptation (SICA), the first monolithic FID paradigm. Extensive experiments on our $ \textit{OpenMMSec} $ dataset demonstrate that SICA outperforms 15 state-of-the-art methods and reconstructs the target unified-yet-discriminative artifact feature space in a near-orthogonal manner, thus firmly validating our hypothesis. The code and dataset will be made publicly available.
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Faithful Relational Reasoning with Region-based Embeddings: Expressivity of Convex Coordinate-wise Models
Victor Charpenay, Steven Schockaert
Embedding methods are among the most efficient approaches for learning to reason about relational knowledge. In this paper, we focus on the framework of region-based embeddings, where relations are encoded as geometric regions. The spatial arrangement of these regions allows such models to capture symbolic rules, enabling them to simulate some forms of symbolic reasoning. A crucial consideration is how the regions are parameterized, as this affects which rule bases can be captured. Most methods use convex regions which are defined in terms of coordinate-wise comparisons. This makes them highly efficient, but the implications of this choice have thus far remained unclear. We present a series of results that shed light on this issue, showing that convex coordinate-wise models indeed have important limitations, while at the same time showing that there is still room for pushing the expressivity of existing coordinate-wise models.
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Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization
Yuchen Zhu, Wei Guo, Jaemoo Choi, Petr Molodyk, Bo Yuan, Molei Tao, Yongxin Chen
Diffusion large language models (dLLMs) are promising alternatives to autoregressive large language models (AR-LLMs), as they potentially allow higher inference throughput. Reinforcement learning (RL) is crucial to enabling dLLMs to achieve

performance comparable to that of AR-LLMs on important tasks, such as reasoning. However, RL algorithms well-suited to dLLMs' unique characteristics have yet to be developed. This paper proposes \textbf{Distribution Matching Policy Optimization (DMPO)}, a principled and theoretically grounded RL fine-tuning method specifically designed to enhance the reasoning capabilities of dLLMs by matching the dLLM policy distribution to the optimal, reward-tilted one through cross-entropy optimization. We identify a key implementation challenge with small training batch sizes and propose several effective solutions based on a novel weight baseline subtraction technique. DMPO exhibits superior performance on multiple reasoning benchmarks without supervised fine-tuning, achieving up to a 39.63 percentage-point improvement in accuracy over prior non-DMPO RL baselines and 67.97 percentage points over the base model, underscoring the effectiveness of the distribution-matching framework. Our code is available at https://github.com/yuchen-zhu-zyc/DMPO.

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Catch-22: On the Fundamental Tradeoff Between Detectability and Robustness in LLM Watermarking

Kuheli Pratihar, Debdeep Mukhopadhyay

Large language models generate text by sampling tokens at random, a process now widely used for inference-time watermarking that verifies AI-generated content. We present an information-theoretic framework that captures the trade-off between robustness to text edits and detectability by observers who lack the watermark key or a keyless detector. The bounds we derive hold regardless of computational power, and what a keyless detector can actually achieve depends on what it can observe about the model and its outputs. At the heart of the analysis is an additive, Kullback-Leibler (KL) information measure that quantifies how well a hypothesis test can distinguish watermarked from unwatermarked text while the watermark remains stealthy. The measure remains zero for distribution-preserving schemes and increases with text length for token-level and sentence-level probability-modifying schemes. When edits are modeled as noise, the KL measure shrinks quadratically with the edit rate for token-level schemes and with an induced semantic flip rate for sentence-level schemes. This shrinkage exposes an unavoidable trilemma among robustness, stealth, and reliable verification. Guided by these limits, we propose a hybrid watermarking strategy that selects the Pareto-optimal scheme among distribution-preserving, semantic-level, and token-level methods based on the expected editing regime at deployment. Experiments on Llama-2-7B and Mistral-7B under paraphrasing attacks corroborate these theoretical predictions and confirm that the hybrid strategy lies near the Pareto frontier across the edit regimes we evaluate.

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Protein Autoregressive Modeling via Multiscale Structure Generation

Yanru Qu, Cheng-Yen Hsieh, Zaixiang Zheng, Ge Liu, Quanquan Gu

We present protein autoregressive modeling (PAR), the first multi-scale autoregressive framework for protein backbone generation via coarse-to-fine next-scale prediction. Using the hierarchical nature of proteins, PAR generates structures that mimic sculpting a statue, forming a coarse topology and refining structural details over scales. To achieve this, PAR consists of three key components: (i) multi-scale downsampling operations that represent protein structures across multiple scales during training; (ii) an autoregressive transformer that encodes multi-scale information and produces conditional embeddings to guide structure generation; (iii) a flow-based backbone decoder that generates backbone atoms conditioned on these embeddings. Moreover, autoregressive models suffer from exposure bias, caused by the training and the generation procedure mismatch, and substantially degrades structure generation quality. We effectively alleviate this issue by adopting noisy context learning and scheduled sampling, enabling robust backbone generation. Notably, PAR exhibits strong zero-shot generalization, supporting flexible human-prompted conditional generation and motif scaffolding without requiring fine-tuning. On the unconditional generation benchmark, PAR effectively learns protein distributions and produces backbones of high design quality, and exhibits favorable scaling behavior. Together, these properties establish PAR

as a promising framework for protein structure generation.
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Latent Spherical Flow Policy for Reinforcement Learning with Combinatorial Actions
Lingkai Kong, Anagha Satish, Hezi Jiang, Akseli Kangaslahti, Andrew Ma, Wenbo Chen, Mingxiao Song, Lily Xu, Milind Tambe
Reinforcement learning (RL) with combinatorial action spaces remains challenging because feasible action sets are exponentially large and governed by complex feasibility constraints, making direct policy parameterization impractical. Existing approaches embed task-specific value functions into constrained optimization programs or learn deterministic structured policies, sacrificing generality and policy expressiveness. We propose a solver-induced \emph{latent spherical flow policy} that brings the expressiveness of modern generative policies to combinatorial RL while guaranteeing feasibility by design. Our method, LSFlow, learns a \emph{stochastic} policy in a compact continuous latent space via spherical flow matching, and delegates feasibility to a combinatorial optimization solver that maps each latent sample to a valid structured action. To improve efficiency, we train the value network directly in the latent space, avoiding repeated solver calls during policy optimization. To address the piecewise-constant and discontinuous value landscape induced by solver-based action selection, we introduce a smoothed Bellman operator that yields stable, well-defined learning targets. Empirically, our approach outperforms state-of-the-art baselines by an average of 20.6\% across a range of challenging combinatorial RL tasks.
****
The Consistency Dilemma in LLMs: Generator-Evaluator Agreement and Vulnerability to Mistakes
Marina Mancoridis, Zoe Hitzig
Large language models are increasingly deployed in agentic pipelines that depend on the model evaluating its own outputs without external verification. The reliability of these pipelines depends on an implicit assumption: that the model applies relevant concepts the same way when it generates an output and later evaluates that output. We propose a new measure, *generator–evaluator self-consistency*, to test this assumption directly and apply it to 10 frontier models across 491 concepts. We find, first, that there is substantial variation in self-consistency. Second, we find that in a clinical setting with physician-validated mistakes (Proniakin et al., 2025), across models, those with higher self-consistency are linked to *greater* vulnerability to mistakes. Thus, even when models consistently apply concepts they may not be safe to deploy. This is evidence of a *consistency dilemma* in LLMs: self-consistency is operationally useful, but models that are more consistent are also more prone to mistakes.
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Discovering Ordinary Differential Equations with LLM-Based Qualitative and Quantitative Evaluation
Sum Kyun Song, Bong Gyun Shin, Jae Yong Lee
Discovering governing differential equations from observational data is a fundamental challenge in scientific machine learning. Existing symbolic regression approaches rely primarily on quantitative metrics; however, real-world differential equation modeling also requires incorporating domain knowledge to ensure physical plausibility. To address this gap, we propose DoLQ, a method for discovering ordinary differential equations with LLM-based qualitative and quantitative evaluation. DoLQ employs a multi-agent architecture: a Sampler Agent proposes dynamic system candidates, a Parameter Optimizer refines equations for accuracy, and a Scientist Agent leverages an LLM to conduct both qualitative and quantitative evaluations and synthesize their results to iteratively guide the search. Experiments on multi-dimensional ordinary differential equation benchmarks demonstrate that DoLQ achieves superior performance compared to existing methods, not only attaining higher success rates but also more accurately recovering the correct symbolic terms of ground truth equations. Our code is available at https://github.com/Bon99yun/DoLQ.
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Uncovering Bias Mechanisms in Observational Studies
Ilker Demirel, Zeshan Hussain, Piersilvio De Bartolomeis, David Sontag
Observational studies are a key resource for causal inference but are often affe
cted by systematic biases. Prior work has focused mainly on detecting these bias
es, via sensitivity analyses and comparisons with randomized controlled trials,
or mitigating them through debiasing techniques. However, there remains a lack o
f methodology for uncovering the underlying mechanisms driving these biases, *e.
g.*, whether due to hidden confounding or selection of participants. In this wor
k, we show that the relationship between bias magnitude and the predictive perfo
rmance of nuisance function estimators (in the observational study) can help dis
tinguish among common sources of bias. We validate our methodology through exten
sive synthetic experiments and a real-world case study, demonstrating its effect
iveness in revealing the mechanisms behind observed biases. Our framework offers
 a new lens for understanding and characterizing bias in observational studies,
with practical implications for improving causal inference.
****
Steer Like the LLM: Activation Steering that Mimics Prompting
Geert Heyman, Frederik Vandeputte
Large language models can be steered at inference time through prompting or acti
vation interventions, but activation steering methods often underperform compare
d to prompt-based approaches. We investigate whether activation steering can be
improved by learning to mimic the interventions that prompt steering triggers wi
thin the model. To this end, we introduce *Prompt Steering Replacement (PSR)* mo
dels, a new family of activation steering methods that distill prompt steering b
ehavior into interpretable interventions on model activations. A PSR is an activ
ation steering method that estimates position-specific steering coefficients and
 is trained to imitate prompt-based interventions. Experiments on persona steeri
ng and instruction following across multiple language models demonstrate that PS
R models consistently outperform constant-coefficient interventions that are fre
quently used in the literature and achieve performance close to or exceeding pro
mpt steering while maintaining interpretability.
****
Generative Modeling of Discrete Latent Structures via Dynamic Policy Gradients
Stefan Ivanovic, Ge Liu, Mohammed El-Kebir
Many scientific problems require inferring unobserved mechanistic latent states
from indirect observations. While classical approaches, including expectation-ma
ximization, do not scale to combinatorially large spaces, deep learning approach
es such as variational autoencoders typically form artificial latent states rath
er than reconstructing the mechanistic ground-truth states. Here, we introduce G
ReinSS, a policy learning framework that uses dynamically rescaled rewards to le
arn latent state distributions that maximize the observed data likelihood. We sh
ow that GReinSS accurately reconstructs simulated latent sets and latent graphs,
 outperforming alternative policy learning and generative modeling baselines. Ad
ditionally, GReinSS reconstructs isoforms from real short-read RNA sequencing da
ta that better match orthogonal long-read sequencing detected isoforms than the
standard RSEM algorithm. Overall, GReinSS is a principled and practically effec
tive approach for generative modeling and inference of combinatorial latent stat
es from indirect observations.
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The Tell-Tale Norm: ℓ_2 Magnitude as a Signal for Reasoning Dynamics in Lar
ge Language Models
Jinyang Zhang, Hongxin Ding, Yue Fang, Weibin Liao, Muyang Ye, Junfeng Zhao, Yas
ha Wang
Recent work has sought to understand Large Language Models (LLMs) reasoning, yet
 a principled, model-intrinsic signal that captures its *layer-wise reasoning dy
namics* remains underexplored. We bridge this gap by demonstrating that **the $\
ell_2$ norm of hidden states serves as an endogenous signal of the model's reaso
ning intensity**. Using Sparse Autoencoders (SAEs) as a diagnostic probe, we obs
erve that LLMs' internal reasoning is marked by a sharp increase in reasoning fe
ature activations concentrated in late layers. Motivated by this pattern, we est

ablish a formal link between reasoning intensity and the model's latent geometry and theoretically prove that the ℓ_2 norm of hidden states bounds the activation strength of SAE reasoning features. Empirical correlation analysis and causal interventions further prove ℓ_2 norm as a faithful indicator, where heightened norms consistently correspond to critical reasoning steps. We then introduce three test-time scaling techniques guided by ℓ_2 norms: Adaptive Layer-wise Reasoning Recursion, (ii) Endogenous Reasoning State Steering, and (iii) ℓ_2-guided Response Selection, which requires no additional training or data and is compatible with advanced inference engines. Experiments across model architectures and benchmarks show that ℓ_2-norm-based techniques significantly improve reasoning performance, offering a principled yet simple lens to perceive and control LLM latent reasoning dynamics. Our codes are available at https://github.com/zjy1298/The-Tell-Tale-Norm.

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Seeing the Unseen: Physics-as-Representation for Generalizable Gaze Perception
Yunfeng Xiao, Xiaowei Bai, Hao Su, Hao He, Liang Xie, Erwei Yin

We introduce physics-as-representation, a learning paradigm that encodes physical structure and geometric laws into visual representations, enabling models to see the unseen—the underlying 3D geometry and motion dynamics not apparent in raw pixels. We instantiate this paradigm in gaze perception by proposing SG-Gaze, a framework that learns a Structurally and Geometrically Consistent Representation (SGR) through dual-branch adversarial learning.

An analytical branch embeds appearance features onto a spherical manifold aligned with gaze geodesics, while a model-guided branch reconstructs the 3D eyeball with weak 2D edge supervision.

We further introduce View-Consistent Regularization, which augments SGR learning with synthetic view perturbations and enforces rotation-equivariant consistency across gaze vectors and structural projections, eliminating the need for multi-view calibration or explicit 3D labels.

Extensive experiments across 12 challenging cross-domain transfers demonstrate that SG-Gaze achieves state-of-the-art accuracy and strong generalization. Our work highlights that enforcing structural and geometric consistency with equivariant regularization serves as effective inductive biases for interpretable and generalizable representation learning—a step toward machines that perceive the world not only from pixels, but from physics.

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Mind Your Entropy: From Maximum Entropy to Trajectory Entropy-Constrained RL
Guojian Zhan, Likun Wang, Pengcheng Wang, Feihong Zhang, Jingliang Duan, Kaichen g Yu, Masayoshi Tomizuka, Shengbo Eben Li

Maximum entropy has become a mainstream off-policy reinforcement learning (RL) framework for balancing exploitation and exploration. However, two bottlenecks still limit further performance gains: (1) non-stationary Q-value estimation stemming from the joint injection of entropy and the concurrent updating of its temperature parameter; and (2) short-sighted local entropy tuning, which adjusts temperature solely based on current single-step entropy without accounting for cumulative entropy over time. In this paper, we broaden the maximum entropy framework by proposing a trajectory entropy-constrained reinforcement learning (TECRL) framework to address these limitations. We begin by introducing reward-entropy separation (RES) to decouple the value targets, ensuring they remain stable and unaffected by temperature fluctuations. Subsequently, the resulting entropy Q-function is leveraged to explicitly quantify expected cumulative entropy, allowing for the enforcement of a trajectory entropy constraint (TEC) to govern long-term stochasticity. We instantiate this framework as DSAC-E, a practical off-policy algorithm that builds upon the latest distributional soft actor-critic. Extensive evaluations across 10 challenging tasks in locomotion, robotic manipulation, and vision-based driving domains demonstrate that DSAC-E consistently outperforms baselines in both cumulative returns and training stability.

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CANDI: Hybrid Discrete-Continuous Diffusion Models
Patrick Pynadath, Jiaxin Shi, Ruqi Zhang

While continuous diffusion has shown remarkable success in continuous domains su
ch as image generation, its direct application to discrete data has underperform
ed pure discrete formulations. To understand this gap, we introduce *token ident
ifiability*, an analytical framework characterizing how Gaussian noise corrupts
discrete data through two mechanisms: *discrete identity corruption* and *contin
uous rank degradation*. We reveal that these mechanisms scale differently with v
ocabulary size, creating a *temporal dissonance* that forces a tradeoff between
learning continuous geometry and discrete structure. To address this, we propose
 CANDI (**C**ontinuous **AN**d **DI**screte diffusion), a hybrid framework t
hat decouples discrete and continuous corruption, enabling simultaneous learning
 of both. This unlocks the benefits of continuous diffusion for discrete spaces:
 on controlled generation, CANDI enables classifier-based guidance with off-the-
shelf classifiers through simple gradient addition; on text generation, CANDI ou
tperforms masked diffusion at low NFE, demonstrating the value of learning conti
nuous gradients for discrete spaces.
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From Knowledge to Inference: Formalizing Specialized Public Health Reasoning on
GlobalHealthAtlas
Zhaokun Yan, Shan Xu, Wuzheng Dong, Zhaohan Liu, Lijie Feng, Chengxiao Dai, Chen
 Tianqi, Binfan Liu, Yunpu Ma, Wenting Wei, Yingting Li, Yi Zhang, Tongning Wu
Public health reasoning requires population level inference grounded in scientif
ic evidence, expert consensus, and safety constraints. However, it remains under
explored as a structured machine learning problem with limited supervised signal
s and benchmarks. We introduce GlobalHealthAtlas, a large scale multilingual dat
aset of 280,210 instances spanning 15 public health domains and 17 languages. We
 further propose a large language model (LLM) assisted construction and quality
control pipeline with retrieval, deduplication, evidence grounding checks, and l
abel validation to improve consistency at scale. Finally, we present a domain al
igned evaluator distilled from high confidence judgments of diverse LLMs to asse
ss outputs along six dimensions: Accuracy, Reasoning, Completeness, Consensus Al
ignment, Terminology Norms, and Insightfulness. Together, these contributions en
able reproducible training and evaluation of LLMs for safety critical public hea
lth reasoning beyond conventional QA benchmarks. We publicly release project cod
ebase, evaluator, and model at:: https://github.com/Jan8217/GlobalHealthAtlas, h
ttps://huggingface.co/aerovane0/GlobalHealthAtlas_Public_Evaluator and https://h
uggingface.co/aerovane0/GlobalHealthAtlas_Public_Model
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Low-Compute Watermark Removal via Dual-Domain Natural Projection
Pragati Shuddhodhan Meshram, Varun Chandrasekaran
Effective removal of semantic watermarks requires balancing three competing obje
ctives:
\emph{high removal success}, \emph{low perceptual distortion}, and \emph{low com
putational cost}.
However, existing single-image attacks typically optimize only for the first two
, achieving strong
watermark suppression but relying on expensive, multi-step optimization that lim
its practical
deployment. In this work, we show that this trade-off is fundamental: no current
 approach achieves
all three properties simultaneously. We introduce \textsc{DAWN}, a lightweight,
training-free
attack that explicitly targets the low-cost regime while maintaining competitive
 removal
performance. \textsc{DAWN} works by projecting a watermarked image onto natural-
image priors in
complementary frequency and semantic spaces, suppressing watermark signals that
deviate from
natural statistics, and then applying a decoupled perceptual-alignment step to r
estore visual
consistency with minimal artifact. Across diverse pixel-, frequency-, and latent

-space
watermarking schemes, \textsc{DAWN} consistently reduces detectability while pre serving structural
and semantic fidelity, demonstrating that efficient, low-resource watermark remo val is feasible
with only modest perceptual degradation. Our code is
available at \url{https://anonymous.4open.science/r/DAWN-567A/}.
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RAGEN-2: Reasoning Collapse in Agentic RL
Zihan Wang, Chi Gui, Xing Jin, Qineng Wang, Licheng Liu, Kangrui Wang, Shiqi Che n, Linjie Li, Zhengyuan Yang, Pingyue Zhang, Yiping Lu, Jiajun Wu, Li Fei-Fei, L ijuan Wang, Yejin Choi, Manling Li
RL training of multi-turn LLM agents is unstable, and reasoning quality drives t ask performance. Entropy, the standard reasoning-stability monitor, only measure s within-input diversity and misses whether reasoning depends on the input. We i dentify **template collapse**: stable entropy alongside input-agnostic boilerpla te, invisible to entropy and existing metrics. We diagnose it via a **mutual-inf ormation (MI) proxy** that scores cross-input distinguishability online; across tasks, MI correlates with final performance far more strongly than entropy. We t hen explain collapse via a **signal-to-noise ratio (SNR)** mechanism: low within -input reward variance weakens task gradients, letting input-agnostic regulariza tion dominate and erase cross-input differences. We mitigate this with **SNR-Awa re Filtering**, prioritizing high-variance prompts each iteration. Across planni ng, math reasoning, web navigation, and code execution, the method consistently improves input dependence and task performance.
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HInT: Hypergraph Infusion at the Structural Layers Improves Table Understanding
Wonjin Lee, Soomi Jeong, Kwang In Kim
Decoder-only large language models (LLMs) struggle with table reasoning because tables must be serialized, which can obscure row- and column-level structure. Pr ior graph and hypergraph approaches encode structure with an external encoder, b ut their gains are often inconsistent under autoregressive decoding. We analyze how tabular structure is represented inside decoder-only LLMs and find that row and column relations are concentrated in a small subset of layers and attention heads. Based on this observation, we propose *HInT* (*H*ypergraph *In*fusion for *T*able reasoning), which injects hypergraph-derived structural features direct ly into the layers where these relations are concentrated. *HInT* constructs a t able hypergraph over cells and headers, applies lightweight message passing, and fuses the resulting structural features with token hidden states through gated fusion. Experiments across diverse table reasoning tasks show consistent improve ments over text-only baselines and prior (hyper)graph-based methods.
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UnMaskFork: Test-Time Scaling for Masked Diffusion via Deterministic Action Bran ching
Kou Misaki, Takuya Akiba
Test-time scaling strategies have effectively leveraged inference-time compute t o enhance the reasoning abilities of Autoregressive Large Language Models. In th is work, we demonstrate that Masked Diffusion Language Models (MDLMs) are inhere ntly amenable to advanced search strategies, owing to their iterative and non-au toregressive generation process. To leverage this, we propose **UnMaskFork** (** UMF**), a framework that formulates the unmasking trajectory as a search tree an d employs Monte Carlo Tree Search to optimize the generation path. In contrast t o standard scaling methods relying on stochastic sampling, UMF explores the sear ch space through deterministic partial unmasking actions performed by multiple M DLMs. Our empirical evaluation demonstrates that UMF consistently outperforms ex isting test-time scaling baselines on complex coding benchmarks, while also exhi biting strong scalability on mathematical reasoning tasks.
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